SKETCHES FOR EXTENSIONS OF THE YM3 CONSTRUCTION

Abstract. The companion manuscript constructs centred continuum correlations for all field-strength bilinears $\mathrm{tr}[F_{jk}F_{\ell m}]$ in three-dimensional SU(2) Yang–Mills theory. These correlations satisfy the Osterwalder–Schrader axioms at weak coupling on Schwartz tests flat at coincidences. We give detailed sketches of further results: real-time tensor fields, short-distance laws, compact gauge groups, Wilson loops, local nets, dynamics and a confinement criterion. Each sketch describes the construction and the estimates and limit passages needed to establish its conclusion.

Contents

1. Introduction 1
2. Tensor correlations on a fixed torus 2
3. Real-time tensor fields 57
4. Point splitting at two scales 79
5. Plane fields and anisotropic variations 155
6. The two-term short-distance law 254
7. The lattice two-loop coefficient 308
8. Natural scaling and failure of dilation covariance 422
9. Compact gauge groups 456
10. Renormalized Wilson loops 546
11. Centre flux 593
12. Vanishing latent heat 640
13. Infrared bounds 675
14. The Haag net of the action density 718
15. A continuum Langevin process 787
16. Positive-time gradient flow 824
17. A finite-size criterion for confinement 865
18. Further propositions 952
Appendix A. Index of notation and conventions 1324
References 1325

1. Introduction

The curvature-bilinear manuscript [8] constructs the Euclidean correlations of three-dimensional SU(2) Yang–Mills theory and proves their Osterwalder–Schrader axioms on Schwartz tests flat at coincidences. We develop further consequences and extensions of that construction. The first concerns local fields in real time.

Theorem 0 (Real-time tensor fields, sketched). Every clustering component of the curvature-bilinear correlations admits a reconstruction by local tempered tensor fields in $1 + 2$ dimensions. Their energy-momentum spectrum lies in the closed forward cone, and each component has a cyclic vacuum unique among translation-invariant vectors. ■□

The precise setting and reconstruction are given in §3. The Euclidean construction, stress conservation and rotation argument are used from [8]; their proofs are not repeated here.

The remaining sections each develop one principal result. Point splitting relates a close pair of scalar insertions to a curvature tensor. Weighted backgrounds describe anisotropic variations. The two-term short-distance law and its lattice coefficient determine a correction to natural scaling. Other sketches concern compact gauge groups, Wilson loops, centre flux, latent heat, infrared bounds, a Haag net, and stochastic and deterministic dynamics. The finite-size criterion for confinement is followed by further sketches in §18. These include a two-loop pressure expansion, short-distance scalar laws at every order, white-noise fluctuations of the centered action density, adjoint sectors and monopoles, and thermal-slab limits.

The arguments are drawn from [7]. We explain their reductions and estimates, and retain the technical reasoning for the additional results. Each statement headed *Sketch* gives the intended conclusion and the route to it. The general-group construction and continuum gradient flow require the uniform estimates and limiting identifications specified in their sketches; these steps are not proved here. The confinement criterion assumes its stated box inequality along the continuum trajectory.

We use the scalar construction of [1] and the curvature-bilinear theory of [8]. Lattice spacing is $a_K = L^{-K}$, with odd $L \geq 13$, and the physical inverse-coupling label is $b = a_K \beta_K$. Thus a fixed-label sequence has $\beta_K = L^K b$. The reference coupling that controls the renormalization steps is distinct from $b^{-1/2}$. We recall the necessary conventions where each sketch uses them. Full convergence on a fixed torus, subsequential infinite-volume convergence, and selection of a clustering component have different meanings; their order is specified in each argument.

2. Tensor correlations on a fixed torus

The curvature main manuscript constructs tensor correlations by taking subsequences and then constructs their Osterwalder–Schrader spaces. Here the volume is fixed. Comparing two consecutive lattice cutoffs gives a summable error, and hence a limit along the full sequence. The same comparison identifies several lattice representatives of a tensor insertion and controls changes in the physical coupling label. The detailed comparisons below retain the supplied proof sketches; the following statement collects the conclusions they are intended to establish.

Write $T_\ell = (\mathbb{R}/\ell\mathbb{Z})^3$, $a_K = L^{-K}$, and $\Lambda_{K,\ell} = a_K \mathbb{Z}^3/\ell\mathbb{Z}^3$, where the blocking factor L is odd. The physical label is $\mathsf{b} = a_K \beta$; thus a fixed label means Wilson measures at $\beta = L^K \mathsf{b}$. Initially use the reference regime $0 < g \leq \gamma_W^{\mathrm{ref}}$ and

$$\mathsf{b} \in I(g) = (g^{-2} + \bar{B}, X_g - \bar{B}], \qquad \ell \in \mathfrak{L}(g) = \{2L^m : m \geq m_{\mathfrak{L}}(g)\}.$$

Here $B_\sharp$ is the fixed error bound in the main manuscript's first-passage construction, $\bar{B} = B_\sharp L/(L-1)$, $X_g = L(g^{-2} + B_\sharp)$, and $m_{\mathfrak{L}}(g)$ is its admitted-side threshold. In this regime $(K, L^K \mathsf{b})$ is a first-passage datum for every $K \geq 1$. An exact dilation, described below, removes this initial restriction on the positive label and on the side $2L^m$.

Let $V = \mathrm{Sym}_3(\mathbb{R})$ and let $O_K^T(H)$ be the main manuscript's lattice representative of $\frac{1}{2}\int \mathrm{tr}(HT)\,\mathrm{d}x$. Its diagonal terms are the three plaquette-plane forms, and its off-diagonal terms are the three bond forms. For a matrix weight set

$$|H| = \sup_x \max_{i,j} |H_{ij}(x)|, \qquad \|H\|_{\mathrm{Lip}} = |H| + \mathrm{Lip}_\infty H,$$

$$\mathcal{S}_{n;K}(H_1, \dots, H_n) = \langle O_K^T(H_1); \dots; O_K^T(H_n) \rangle_{K,\ell,L^K \mathsf{b}}.$$

The Lipschitz seminorm uses the sup distance on the torus, entry by entry. Support separations in this overview use that distance. For bounded Borel weights, $\bar{O}_K^T(H)$ uses the cell means

$$H^{(K)}(x) = a_K^{-3} \int_{x + a_K[-1/2,1/2)^3} H(y)\,\mathrm{d}y$$

at the least corner x of the plaquette or bond. This specifies the sampling convention even when point values have no meaning.

We also need to compare insertions built from different plaquette corners. For an ordered pair of signed directions $l = (\alpha, \alpha')$, $\alpha' \neq \pm\alpha$, let $W_{x,l}$ be its plaquette loop, $\lambda(l)$ its normal direction, and $s(l) = \operatorname{sgn}(\alpha)\operatorname{sgn}(\alpha')\varepsilon_{\lambda(l)|\alpha||\alpha'|}$ its orientation. Put

$$w(U) = \frac{U - U^{-1}}{2i}, \qquad Q_x(l, l') = \frac{1}{2}\operatorname{tr}[w(W_{x,l})w(W_{x,l'})],$$
$$\mathcal{Q}_{l,l';\xi}(\phi) = a_K^{-1}\sum_x \phi(x + a_K\xi)Q_x(l, l'), \qquad |\xi|_\infty \leq \frac{1}{2}.$$

A class insertion $\mathcal{E}$ is a finite sum of these corner forms and of plane forms $\mathcal{O}_\lambda(\phi) = O_K^T(\phi E_{\lambda\lambda})$, with real Lipschitz weights. Here E_{ij} are the matrix units. Its quadratic symbol $\operatorname{Sym}\mathcal{E}$ is the sum of

$$\phi E_{\lambda\lambda}, \qquad \phi s(l)s(l')(E_{\lambda(l)\lambda(l')} + E_{\lambda(l')\lambda(l)}),$$

respectively; its norm $\|\mathcal{E}\|$ is the sum of $\sup|\phi| + \operatorname{Lip}_\infty\phi$ over its terms. In particular $\operatorname{Sym} O_K^T(H) = H$.

Sketch A (Full fixed-torus limit). Fix $n \geq 2$ and $d > 0$. Suppose the weights H_a vanish off closed sets $\Sigma_a \subset T_\ell$ with pairwise sup distances at least d. For Lipschitz weights, there are C, p, K_1 such that

$$|\mathcal{S}_{n;K+1}(H_1, \dots, H_n) - \mathcal{S}_{n;K}(H_1, \dots, H_n)| \leq C\ell^{3(n+1)}K^pL^{-\theta_n K}\prod_a \|H_a\|_{\mathrm{Lip}},$$
$$\theta_n = \frac{1}{1260n(8n+3)}, \qquad K \geq K_1.$$

Consequently $\mathcal{S}_n^{\ell,\mathsf{b}} = \lim_K \mathcal{S}_{n;K}$ exists along the full sequence, with the tail bounded by the sum of the displayed right-hand side over $K' \geq K$. For $\bar{O}_K^T$ the same assertions hold for bounded Borel weights, with $\prod_a |H_a|$ and a possibly different constant. The two limits agree for Lipschitz weights.

Class insertions $\mathcal{E}_a, \mathcal{E}'_a$ over these sets with equal symbols have cumulants differing by at most

$$C^{\mathrm{u}}K^pL^{-\theta_n K}\prod_a(\|\mathcal{E}_a\| + \|\mathcal{E}'_a\|).$$

Their common limit is $\mathcal{S}_n^{\ell,\mathsf{b}}(\operatorname{Sym}\mathcal{E}_1, \dots, \operatorname{Sym}\mathcal{E}_n)$. The Lipschitz seminorms in this comparison can be omitted when the representatives carry the same weights at the same sites.

The cutoff estimates are uniform for b in any compact interval $J \subset I(g)$. The limits are continuous in the label; if b is an interior point of $I(g)$ and $a_K\beta_K \to \mathsf{b}$, the full cutoff sequence at β_K has the same limit. Exact dilation extends these conclusions to every $\mathsf{b} > 0$ and every side $2L^m$, $m \geq 0$. ■□

2.1. **Comparing two cutoffs through one block field.** The new estimate concerns the relation between the measures at K and $K + 1$. Bounds on each measure separately would only provide compactness. The comparison instead evaluates their insertions over one block field at a common depth. This introduces three errors: replacing each insertion by its value at a background, comparing the two background values, and comparing the two laws of the block field. Prop. 2.1 gives this argument for plane and bond forms; Lem. 2.11 includes corner forms. Expanding a tensor insertion into its six entries then gives 6^n terms.

The two depths have different purposes. Set

$$\sigma = \left\lfloor \frac{K}{1260n} \right\rfloor, \qquad s = \left\lfloor \frac{\sigma}{8n+3} \right\rfloor.$$

At depth s the supports are far enough apart for the separated-source calculus. Each insertion has size at most a constant times ℓ^3L^{4s} times its weight norm. Replacing it by the first source

member evaluated at the background of the depth-σ block field leaves a large-field error, a loss-of-memory error decaying like a polynomial times $L^{-(\sigma-s)/2}$, and a connecting-polymer error decaying exponentially in $L^s d$. The inequality $\sigma - s \geq (8n+2)s$ pays for the growth L^{4ns} of a product of n insertions. After that cost, the slowest decay is L^{-s}.

The background comparison is the substantive local step. In a suitable gauge the fine plaquette variables are curls of a small Lie-algebra-valued field to second order; the coarse variables are their block averages. The variance about an average is controlled by the defect in Jensen's inequality. Summing that defect relates it to the action lost in averaging. The comparison of background actions controls this loss, and the minimality of the coarse background controls the difference between that background and the averaged fine background. This gives, for one insertion, a defect bounded by a constant times

$$\ell^3 L^{4\sigma} K^{1/2} L^{-(K-\sigma)/2} (\sup|\phi| + L^{-\sigma} \operatorname{Lip}_\infty \phi).$$

Finally, the two block-field laws differ by at most a constant times $\ell^3 K^{\mathsf{p}_0} L^{-K/10}$ on bounded gauge-invariant functions. Here p_0 denotes the fixed polynomial exponent of that comparison. The small choice of σ pays for the $L^{4n\sigma}$ growth in both of these errors: their remaining decays are at least $L^{-K/3}$ and $L^{-K/11}$, respectively. Thus $s \geq \theta_n K - 2$ gives the stated summable rate.

2.2. **Sampling, symbols, and changes of label.** Cell means remove the regularity cost at its source. The cell of a coarse site y is the union of the cells of its L^3 fine sites $B(y)$, so

$$\phi^{(K)}(y) = L^{-3} \sum_{x \in B(y)} \phi^{(K+1)}(x).$$

The weights therefore match exactly across one averaging step. Point sampling introduces a mismatch of order $a_K \operatorname{Lip}_\infty \phi$ instead. The uniform sup-norm bound of Lem. 2.10(b) makes this mismatch vanish in the limit. Uniform convergence on sup-norm balls also transfers bounded pointwise continuity from the finite-cutoff cell means to the limit. In each argument separately, the resulting functional is a countably additive measure, vanishes on Lebesgue null sets, and has an L^1 density. This is the precise measure-theoretic conclusion for separated weights.

Equality of quadratic symbols is stronger than agreement of a formal leading expression. To prove it, the comparison replaces each loop by a canonical loop in the same plane, with its orientation sign. Moving its anchor costs $O(a_K \operatorname{Lip}_\infty \phi)$. The plane insertion is reduced to corner products by the exact identity

$$a_p = \frac{1}{2} Q_x(l,l) + \frac{1}{2} a_p^2.$$

The last term is fourth order at the backgrounds and lies within the comparison error. The resulting canonical form depends only on the symbol. Thus bond averages, clovers, and individual corners produce the same separated continuum field, with a quantified error before taking the limit.

Uniformity in the label requires tracking where it enters the comparison. Let $\Theta_\star$ be the fixed threshold for comparing the two backgrounds. On $J \subset I(g)$ the block-law constants and their threshold are common; the background-radius bounds are common; and $L^\sigma \mathsf{b} \geq \Theta_\star$ holds throughout J once it holds at its least label. At fixed K the compact-group integral is continuous in $\beta = L^K \mathsf{b}$. Uniform cutoff convergence therefore gives a continuous limit in b. For $\mathsf{b}_K = a_K \beta_K \to \mathsf{b}$, split the error into the uniform cutoff tail at b_K and continuity of the limiting function at b.

For an initially inadmissible side or label, dilation replaces (ℓ, b, K) by $(L^z \ell, L^z \mathsf{b}, K - z)$, keeps the underlying graph and bare coupling unchanged, and multiplies an n-fold cumulant by L^{zn}. For sufficiently large z, the reference coupling $g' = (L^z \mathsf{b}/2)^{-1/2}$ admits this new side and places the new label in the interior of $I(g')$. This is the exact transfer used in Thm. A.

The same theorem retains translations, signed-permutation covariance, coordinate-plane reflection positivity, and the separated fixed-torus stress identity. Derivatives in that identity act on periodic continuum test functions. Its two-point refinements are stated with their own scale restrictions. Extensions through coinciding supports, their contact ambiguity, and stronger rates under weaker

separation are separate tasks below; they are not consequences of the separated-support convergence estimate.

Sources. Thm. YM3_I:D follows an insertion through the levels as a source under a majorant. A *majorant* on $\Lambda_{K,\ell}$ is a pair (s, S) of a number $\mathsf{s} \geq 0$, its *size*, and a set S of plaquettes; its weight is the plaquette source $\mathsf{s}\mathbf{1}_S$, and $\mathsf{m}_j = 12M^3\alpha_0^2 L^{K-j}\mathsf{s}$, $j \geq 0$, are the numbers that the calculus of Thm. YM3_I:D attaches to that weight at the level j, $\alpha_0 \in (0,1]$ a basic constant of the calculus. A *source under* (s, S) is a triple of: a *first member* F, a continuous function of the configurations of $\Lambda_{K,\ell}$, invariant under the gauge transformations, with $|F(U)| \leq a_K^{-1}\mathsf{s}\sum_{p\in S} a_p(U)$; a local family at the level 0 of norm at most $\mathsf{c}^\flat\mathsf{m}_0$, $\mathsf{c}^\flat$ a basic number of the calculus; and local families at the levels $j \geq 1$ of norms at most $\frac{1}{2}\mathsf{m}_j$; each family vanishes at the regions of its level that hold no plaquette of S, and each sums at its level to F, up to the value at the trivial field.

A *source system under majorants* is a tuple of sources under majorants (s_a, S_a); all the numbers of the calculus are then formed at the weights $\mathsf{s}_a\mathbf{1}_{S_a}$. A source under (s, S) is a source under every (s', S') with $\mathsf{s}' \geq \mathsf{s}$ and $S' \supset S$, the majorant entering only the right members of three inequalities and the two clauses of vanishing; the sum of two sources, member by member, is a source under the sum of the sizes and the union of the sets, and c times a source is one under $|c|$ times the size. For the value at short distance the calculus reads besides, of a real source, the kernel $\mathsf{t}(p,p') = \mathsf{t}(p',p)$ of its first member to the second order at a small field, vanishing unless $p, p' \in S$, with $2\sum_{p'} |\mathsf{t}(p,p')| \leq \mathsf{s}$.

[7, Lem. C.4] says which insertions are sources.

Proposition 2.1 (Two consecutive cutoffs on one torus for the plaquette forms of one plane and the bond forms)**.** Let $0 < g \leq \gamma_W^{\mathrm{ref}}$, $b \in I(g)$, $\ell \in \mathfrak{L}(g)$, $n \geq 2$ and $\mathsf{d} > 0$; by Cor. YM3_I:118.4 the pairs $(K, L^K b)$ are first-passage data at g with label b. For a bounded Borel function ψ on T_ℓ and $x \in \Lambda_{K,\ell}$ put $\psi^{(K)}(x) = a_K^{-3}\int_{x+a_K[-\frac{1}{2},\frac{1}{2})^3}\psi$. A *cell insertion* at the cutoff K is one of the following: the plaquette form $\mathcal{O}(w)$ of one plane $\{\kappa,\kappa'\}$, with $w(p) = \psi^{(K)}(x)$ at the plaquette p spanned at x by $a_K e_\kappa$, $a_K e_{\kappa'}$ and $w = 0$ at the plaquettes of the two other planes; the sum of the three, the action density with the weight $\psi^{(K)}$; and the bond form $Y[\mathsf{w}]$ of one direction μ, with $\mathsf{w}(\langle x, x + a_K e_\mu\rangle) = \psi^{(K)}(x)$ and $\mathsf{w} = 0$ at the bonds of the two other directions. Let $\mathcal{E}_1, \dots, \mathcal{E}_n$ be cell insertions, of any kinds, whose functions ψ_a are real and vanish off closed sets at pairwise sup distances $\geq \mathsf{d}$, and $\kappa_K = \langle \mathcal{E}_1; \dots; \mathcal{E}_n\rangle_{K,\ell,L^K b}$.

(a) There are numbers C, p, formed from the constants, n and b, and a cutoff K_1, formed besides from g, ℓ and d, such that for $K \geq K_1$

$$|\kappa_{K+1} - \kappa_K| \leq C\,\ell^{3(n+1)}K^p L^{-\theta_n K}\prod_a \sup|\psi_a|, \qquad \theta_n = \frac{1}{1260\,n(8n+3)}.$$

So κ_K converges along the full sequence $K \to \infty$, and the limit is n-linear and symmetric. The same holds, with $\sup|\psi_a| + \mathrm{Lip}_\infty\psi_a$ in the place of $\sup|\psi_a|$, for the forms of the three kinds whose weights are real functions ψ_a with $\mathrm{Lip}_\infty\psi_a < \infty$ read at the barycentres of the plaquettes and at the midpoints of the bonds, $w(p) = \psi_a(x(p))$, $\mathsf{w}(\mathfrak{b}) = \psi_a(x_\mathfrak{b})$, under the same separation of the supports; and for such weights the two limits agree.

(b) In this clause the separation of the head is replaced by a weaker one. Let $\psi_1, \dots, \psi_n$ be real Lipschitz functions on T_ℓ, none of them 0, let $\mathcal{E}_1 = Y[\mathsf{w}]$ be the bond form of one direction μ with $\mathsf{w}(\mathfrak{b}) = \psi_1(x_\mathfrak{b})$ at the bonds $\mathfrak{b}$ of the direction μ and $\mathsf{w} = 0$ at the others, let $\mathcal{E}_a = \mathcal{O}(w_a)$, $w_a(p) = \psi_a(x(p))$, $2 \leq a \leq n$, be action densities, and let κ_K be the joint cumulant of these n forms. If the n supports have no common point and their spread is at least $\mathsf{d}_0 > 0$ (the definition preceding Lem. YM3_I:114.6), then from a cutoff on $|\kappa_{K+1} - \kappa_K| \leq C\,K^p L^{-K/120}\prod_a \|\psi_a\|_\sharp$, $\|\psi\|_\sharp = \max\{\sup|\psi|, 40M\mathrm{Lip}\,\psi\}$, with C, p and the cutoff formed from the constants, n, ℓ, b and d_0.

(c) If all the $\mathcal{E}_a$ are action densities with real Lipschitz weights read at the barycentres, $\mathcal{E}_a = \mathcal{O}(w_a)$, $w_a(p) = \psi_a(x(p))$, and κ_K is their joint cumulant, the convergence is given by

Thm. YM3_I:AR, equation YM3_I:(2203), Cor. YM3_I:117.1 and Cor. YM3_I:118.4: from a cutoff on, $|\kappa_{K+1}-\kappa_K| \le C\,\ell^{3(n+1)}K^p L^{-K/10}\prod_a \|\psi_a\|_\sharp$.

Clauses (a) and (b) carry the convergence of Thm. YM3_I:AR over to the plaquette form of one plane, to the bond form and to weights that are means over the lattice cubes; clause (c) is that theorem for the action density. ■

Proof sketch. (a) At the two cutoffs K and $K+1$, at the one label b on the one torus, put

$$\sigma = \lfloor K/(1260\,n)\rfloor, \qquad s = \lfloor \sigma/(8n+3)\rfloor;$$

then $\sigma \le \lceil K/1260\rceil$, $\sigma - s \ge (8n+2)s$, $s \ge \theta_n K - 2$ and $4n\sigma \le K/315$. Every insertion is the first member of a source under a majorant at both cutoffs ([7, Lem. 2.11], which holds on every side with (Sd.1), so on every $\ell \in \mathfrak{L}(g)$ by Prop. YM3_I:74.1): under (s_a, S_a) with $\mathsf{s}_a = \sup|\psi_a|$ for a plaquette form or an action density and $\mathsf{s}_a = 59\sup|\psi_a|$ for a bond form, S_a the plaquettes whose corners lie within $\frac{3}{2}a_K$ of the closed set off which ψ_a vanishes; a source under a majorant is one under every larger majorant, so that these majorants serve, and corners of plaquettes of different S_a are at sup distance $\ge \mathsf{d} - 3a_K \ge \mathsf{d}/2$ once $6a_K \le \mathsf{d}$. The two cumulants are compared through three errors, each bounded by a number formed from the constants and n times the factors written.

(i) *One cutoff.* By Thm. YM3_I:D, on the small fields, the chain with the sources is run up to the level $K-s$, at which the n insertions have sizes at most $4\ell^3 L^{4s}\mathsf{s}_a$; each insertion is replaced by its *read-out*, its first member read at the background of the block field of the depth σ, a function of that block field alone. The cumulant of the n insertions and the cumulant of their n read-outs differ by at most

$$\Big[2L^{4n}V_\ell\,L^{-(4n+2)s} + n\,\epsilon(\sigma-s) + 2e^{3-L^s\mathsf{d}/(2M)}\Big]\prod_a 4\ell^3L^{4s}\mathsf{s}_a.$$

The first term is the large-field term (Prop. YM3_I:115.2), $V_\ell = 4M^{-3}\ell^3 e^{\mathsf{B}(X_g)\ell^3}$ times a number formed from the constants and n. The second is the loss of memory of the progression with sources between the depths s and σ: the sources switched on at the level $K-\sigma$ in the place of the level 0 change the effective terms of the level $K-s$ by at most $\epsilon(\sigma-s)$ times their sizes, $\epsilon(\nu) = C_{\mathrm{m}}(\nu+1)^{2\mathfrak{p}_0+2}L^{-\nu/2}$ with a number C_{m} of the constants, because the linearised steps contract on the small fields, as the flows of Props. YM3_I:114.2, YM3_I:114.3 forget their start. The third bounds the polymers that join two supports, whose linear size is at least $L^s\mathsf{d}/(2M)-2$ in the units of the level $K-s$.

(ii) *The tie at the depth σ.* The read-outs of the two cutoffs are functions of one block field V of the depth σ, vanishing where V has large fields, of sizes at most $\frac{3}{2}B_3^2\ell^3L^{4\sigma}\mathsf{s}_a$; over one such V they differ by at most

$$\tau_a = C_{\mathrm{t}}\,\ell^3L^{4\sigma}K^{1/2}L^{-(K-\sigma)/2}\sup|\psi_a|$$

for weights that are means over the lattice cubes, and by the same with $\sup|\psi_a| + L^{-\sigma}\mathrm{Lip}_\infty\psi_a$ in the place of $\sup|\psi_a|$ for a weight read at the points of the two lattices, C_{t} a number of the constants; so, under one law of V, the cumulants of the read-outs of the two cutoffs differ by at most $\sum_a \tau_a\prod_{b\ne a}\frac{3}{2}B_3^2\ell^3L^{4\sigma}\mathsf{s}_b$. The tie is a statement of its own, the counterpart of equation YM3_I:(2156) for the products of two plaquette loops at a site. Let $(t,\varepsilon_{\mathrm{r}},V)$ be data of the comparison (the setting preceding Lem. YM3_I:113.8), $t = K-\sigma$, U^{A}, U^{B} the backgrounds over V of the bare lattices $\Lambda_{K,\ell}$, $\Lambda_{K+1,\ell}$, and $N_t = \ell^3L^{3\sigma}$ the number of the sites of the level t; for two loops l, l' at a site x let $Q_x(l,l') = \langle w(W_{x,l}), w(W_{x,l'})\rangle$, $\langle X,Y\rangle = \frac{1}{2}\operatorname{tr}XY$, $W_{x,l}$ the plaquette loop of l at x. Then for $\xi \in [-1,1]^3$ and a complex function ϕ on T_ℓ with $\mathrm{Lip}_\infty\phi < \infty$,

$$\Big|L^{t+1}\sum_{x\in\Lambda_{K+1,\ell}}\phi(x+a_{K+1}\xi)\,Q_x(l,l')(U^{\mathrm{B}}) - L^t\sum_{y\in\Lambda_{K,\ell}}\phi(y+a_K\xi)\,Q_y(l,l')(U^{\mathrm{A}})\Big| \le C_{\mathrm{t}}\,\|\phi\|_{(t)}\,\varepsilon_{\mathrm{r}}^2N_t\,t^{1/2}L^{-t/2},$$

$\|\phi\|_{(t)}$ the norm in equation YM3_I:(2154); and for a weight v_{f} on the sites of $\Lambda_{K+1,\ell}$ and its block mean $\mathsf{v}_{\mathrm{c}}(y) = L^{-3}\sum_{x\in B(y)}\mathsf{v}_{\mathrm{f}}(x)$ on $\Lambda_{K,\ell}$, $B(y)$ the L^3 sites of the block of y, the same holds with v_{f},

v_c in the place of the two readings of ϕ and $\sup|\mathsf{v}_\mathrm{f}|$ in the place of $\|\phi\|_{(t)}$. The proof follows that of Prop. YM3_I:113.9: in a gauge on a box about a site of the coarse lattice the background is e^{ia} with a small 1-form a with values in the Lie algebra, the plaquette variables $w(W_{x,l})$ of the fine lattice are the curls of a up to the second order, and those of the coarse lattice their averages (one averaging step with values in the Lie algebra); the variance of the curls about their averages, plaquette by plaquette, is bounded by the defect of Jensen's inequality, whose sum over the plaquettes is $2L^{-2}$ times the difference $L\sum_p a_p - \sum_{p'} a_{p'}$ of the actions of the fine background and of its block average, a difference bounded through the comparison of the two background actions, equation YM3_I:(2155), and the minimality of the background of the coarse lattice; and the background of the coarse lattice is compared with the block average of the fine background, a nearly minimal competitor, through its excess of action.

The bond density is a sum of four such products, $q_\mathfrak{b} = \frac{1}{4}\sum_{\varepsilon,\varepsilon'=\pm}\varepsilon\varepsilon' Q_x((\mu,\varepsilon\lambda),(\mu,\varepsilon'\lambda'))$ at $\xi = \frac{1}{2}e_\mu$, and the plaquette density of one plane is one such product up to a_p^2, by $a_p = \frac{1}{2}Q_x(l,l)+\frac{1}{2}a_p^2$ at the loop l of the plane, a_p^2 being of the fourth order in the field at the backgrounds; the action density is equation YM3_I:(2156) itself. With $a_K^{-1} = L^\sigma L^t$ and $\varepsilon_\mathrm{r} \le 1$, $t \le K$, this gives τ_a; for means over the lattice cubes the weights of the two cutoffs are tied by the block mean, $\psi^{(K)}(y) = L^{-3}\sum_{x\in B(y)}\psi^{(K+1)}(x)$, since the cell of y is the union of the cells of the $x \in B(y)$, and the tie asks no regularity.

(iii) *The laws.* The laws of the block field of the depth $\sigma \le \lceil K/1260\rceil$ at the two cutoffs differ, on the bounded gauge-invariant functions, by at most $\mathsf{C}_\mathrm{bf}\ell^3 K^{\mathsf{p}_0}L^{-K/10}$ times their sup norm for $K \ge K_\mathrm{bf}$ (Cor. 2.8), with the numbers C_bf, K_bf of Def. 2.7 (5) at the label b and $\mathsf{p}_0 = 2\mathfrak{p}_0 + \frac{1}{2}$; for the cumulant of the read-outs this costs $\mathsf{C}_\mathrm{bf}\ell^3 K^{\mathsf{p}_0}L^{-K/10}\prod_a \frac{3}{2}B_3^2\ell^3 L^{4\sigma}\mathsf{s}_a$.

The balance. In (i), $\sigma - s \ge (8n+2)s$ gives $L^{4ns}\epsilon(\sigma-s) \le C_\mathrm{m}(\sigma+1)^{2\mathfrak{p}_0+2}L^{-s}$; and from a cutoff on $L^{4n}V_\ell L^{-s} \le 1$ and $2e^{3-L^s\mathsf{d}/(2M)}L^{(4n+1)s} \le 1$; so (i) is at most $(4\ell^3)^n[3+nC_\mathrm{m}(\sigma+1)^{2\mathfrak{p}_0+2}]L^{-s}\prod_a \mathsf{s}_a$. In (ii) and (iii), $4n\sigma \le K/315$ gives $L^{4n\sigma}L^{-(K-\sigma)/2} \le L^{-K/3}$ and $L^{4n\sigma}L^{-K/10} \le L^{-K/11}$. As $L^{-s} \le L^2 L^{-\theta_n K}$, $\theta_n < \frac{1}{11}$, $\sigma+1 \le K$ and $\mathsf{s}_a \le 59\sup|\psi_a|$, the three errors give (a) with $p = 2\mathfrak{p}_0+2$; the factor $\ell^{3(n+1)}$ comes from the n sizes and the law, and C is formed from the constants, n and C_bf, so from the constants, n and b.

The cutoff K_1 is the least from which $K \ge K_\mathrm{bf}$, $s \ge 1$, $L^s\mathsf{d} \ge 6M$, $6a_K \le \mathsf{d}$, $L^\sigma b \ge \Theta_\star$ (the number of equation YM3_I:(2148)), the two inequalities above and the thresholds of the chain of n sources at the depth s hold; through V_ℓ, the distance and K_bf it reads besides g, ℓ and d, while C and p read neither. The Lipschitz numbers enter at one place, the tie, with the factor $L^{-\sigma}$. The two readings of a Lipschitz weight, at the points and through the means over the lattice cubes, differ by at most $a_K\mathrm{Lip}_\infty\psi_a$, the size of the weight of a source; the bound of the moments of first members of sources at one fixed depth then makes the two cumulants at the cutoff K differ by at most a number formed from the constants, n, g, ℓ and d times $a_K\sum_a \mathrm{Lip}_\infty\psi_a\prod_{b\ne a}\sup|\psi_b|$, which tends to 0; so the two limits agree.

Since $\sum_K K^p L^{-\theta_n K} < \infty$, κ_K converges along the full sequence; multilinearity and symmetry pass to the limit.

(b) Only the bond form needs a tie: the cumulant is the mixed derivative at 0, in the parameters of the action densities, of the mean of the bond form under the measure with these densities as sources; along the chain the derivative of the bond form obeys a linear recursion, whose sizes are those of a member of the level j and of what a step creates, so that no disc in a parameter of the bond form is needed; the polymers that meet all the n supports, which make the contact terms, have at a depth u a linear size of at least $\mathsf{d}_0 L^u/M - 2$ in the units of their level, as in the polymer estimate in Thm. YM3_I:AR, so that the spread, and no pairwise distance, enters; and the comparison of the bond forms of the two backgrounds over one block field, the tie of (ii), gives the rate $L^{-K/120}$. (c) is Thm. YM3_I:AR, equation YM3_I:(2203) with Cor. YM3_I:117.1 and Cor. YM3_I:118.4. □

2.3. **The pointwise source identity.** We use the actual sourced history of [8, Thm. D, Prop. 5.59, §7.2], with its admitted torus, coupling range, source ball and fixed analytic radii. The coefficient sequence is $x_j = L^{-1}x_{j-1} - e_j$; the counterterms e_j are fixed independently of the sources. For two plaquette weights put $w_\lambda = \lambda_1 w_1 + \lambda_2 w_2$, and let $E_\lambda^{(j)}$ be the effective action at height j for these same source parameters. The new point here is an identity at every admitted real chart point, with the whole source parameter left in its exponential, rather than only an almost-everywhere identity for the pushed-forward measure.

We recall the size conventions used in that identity. Set $\omega_n = (n+1)L^{-(n-1)}$. At height j the source-size recursion and the remaining analytic room are

$$s_j = (1+\beta_{\mathrm{s}})\delta(x_j)/\alpha_0, \qquad \theta_j = K_{\mathbf{M}} s_j, \qquad N_{\star,j} = w_\star/(4K_\Lambda s_j),$$
$$m_{j,a} = 12M^3\alpha_0^2 L^{K-j}|w_a|_\infty, \qquad c^\flat = 81 C_P^2 C_{\mathcal{A}}^2/K_\Lambda,$$
$$d_{1,a} = c^\flat\theta_0 m_{0,a}, \qquad n_{j,a} = m_{j,a} + \sum_{i=1}^{j}\omega_{j+1-i}d_{i,a}, \qquad d_{j+1,a} = \theta_j n_{j,a},$$
$$N_j^{\mathrm{rm}} = N_{\star,j} - \left(\sum_{n=1}^{j}\omega_n\right)(E_0 + \tfrac{3}{2}\alpha_0^2 B_\sharp).$$

All constants and radii in these formulas are those of the fixed weighted source construction just cited. The admitted parameter domains are the successive intersections

$$\begin{aligned}\Lambda_0^{\mathrm{w}} &= \{\lambda : c^\flat \sum_a |\lambda_a| m_{0,a} < N_{\star,0}\},\\ \Lambda_j^{\mathrm{w}} &= \{\lambda \in \Lambda_{j-1}^{\mathrm{w}} : \sum_a |\lambda_a| n_{j,a} < N_j^{\mathrm{rm}}\}.\end{aligned} \tag{2.i}$$

We write $\Lambda^{\mathrm{w}} = \Lambda_{K-1}^{\mathrm{w}}$ for the common admitted domain through the required heights. The norm $N_i(g)$ below is the bare-array norm at height i; the map $\mathbb{V}'[g]^{\mathrm{tot}}(P;B)$ is the sum of its local evaluations in the fourth source slot of the actual step, over its finite domain set $\mathcal{D}_i$.

For clarity, write $w_i(P,B)$ for the fluctuation density of this step, including the interaction and the chart Jacobian, on its real cutoff domain Rg_{x_i}. Its normalizer and normalized measure are

$$\Xi(x_i;P) = \int_{\mathsf{Rg}_{x_i}} w_i(P,B)\,\mathrm{d}B, \qquad \nu_i[P](\mathrm{d}B) = \Xi(x_i;P)^{-1} w_i(P,B)\,\mathrm{d}B.$$

For complex P this measure need not be positive. The normalization and insertion identities below first control its total variation; positivity is then used only on the specified real chart class. The regular backgrounds and enlarged cubes are those of [8, §7.2, eq. 7.xxvi], including the stated range $d_* \geq 3$ of the background estimate.

Evaluation radii. Write $\ell = 2L^m$ and $\eta_T = L^{-T}$. Here K is the bare cutoff, T is a regularity level of a real bare configuration U, and $1 \leq j \leq T$ is the level of the pointed local family being evaluated. Regularity at radius ε means $|\partial U - 1| < \varepsilon\eta_T^2$ and $|J^{(T)}(U)| < \varepsilon$, with the current of the main construction. Choose $c_A > 0$ no larger than the regular-family constant of [8, §7.2] and the constants used in the evaluations below; the notation c_{A} in those evaluations denotes this same constant. Throughout we use the repaired radius

$$d_*(j;T,\varepsilon) = \min\left\{\frac{c_A\alpha_0 L^{2(T-j)}}{M\varepsilon}, \frac{L^{m+K-j}-5}{2M}\right\} - 2. \tag{2.ii}$$

The pointwise regular-family estimate is valid also when this radius is negative, with every domain included in its exceptional term. An invocation of the estimate summed over the lower levels retains the additional hypothesis $D = d_*(T;T,\varepsilon) \geq 3$.

Put $\varepsilon(x) = A_0 x^{-1/2} (\log x)^{\mathfrak{p}_0}$ and fix

$$X_{\mathrm{ev}} = \max\left\{\gamma^{-2}, e^{2\mathfrak{p}_0+1}, \left(\frac{10MB_3A_0(4\mathfrak{p}_0)^{\mathfrak{p}_0}}{c_A\alpha_0}\right)^4\right\}, \qquad \gamma_{\mathrm{ev}} = X_{\mathrm{ev}}^{-1/2},$$
$$\gamma_W^{\mathrm{sd}} := \min\{\gamma_{W,0}^{\mathrm{sd}}, \gamma_{\mathrm{ev}}\}. \tag{2.iii}$$

Here $\gamma_{W,0}^{\mathrm{sd}}$ is the originally cited ceiling of Section YM3_I:126, equation YM3_I:(2327). The later ceiling γ_{C} is decreased below γ_{ev} as well. These are restrictions of the reference coupling; they do not change the retained construction constants. The inequality $\log u \le u$ at $u = x^{1/(4\mathfrak{p}_0)}$ gives $\varepsilon(x) \le A_0(4\mathfrak{p}_0)^{\mathfrak{p}_0} x^{-1/4}$, and hence

$$\varepsilon(x) \le \frac{c_A\alpha_0}{10MB_3} \qquad (x \ge \gamma_{\mathrm{ev}}^{-2}).$$

The least-side definition gives $L^m \ge L^2 M$ on $\mathfrak{L}(g)$, because its cube-size function takes values at least L. Thus $L^{m+K-T} \ge L^2 M \ge 10M + 5$ for $T \le K$. At the background radius $2B_3\varepsilon_T$, both entries before the subtraction in eq. 2.ii are at least 5 whenever $g \le \gamma_{\mathrm{ev}}$. Consequently $D \ge 3$ at every working or upper level. The chain-class evaluations below use the pointwise estimate and their displayed tail sums directly, so do not require an additional positivity assumption on their individual radii. Constants in those tail sums are chosen with the repaired radius.

At the canonical source choice we retain the following numerical part of the choice explicitly. Put

$$N_1 = \frac{L}{L-1}(E_0 + \tfrac{3}{2}\alpha_0^2 B_\sharp), \qquad N_\star \ge \frac{2L-1}{L-1}N_1, \qquad K_\Lambda^\omega = (1+\beta_{\mathrm{s}})^4 K_\Lambda,$$
$$K_{\mathbf{M}}(K_\Lambda^\omega)\left(\delta(\gamma^{-2}) + s_\gamma N_1 + e^{-\gamma_2 \mathfrak{b}(\gamma^{-2})^2/2}\right) \le E_0/4, \tag{2.iv}$$
$$s_\gamma \le \frac{w_\star}{4K_\Lambda^\omega N_\star}, \qquad 8K_P s_\gamma \le w_\star.$$

The allowance $N_\star$ is chosen after $E_0, B_\sharp$ and before γ. The last three requirements are upper bounds tending to zero with γ, with all other quantities already fixed. They can therefore be retained in the ordered common choice of Prop. YM3_I:69.1. They are the source-room conditions also imposed in §2.3.1 and Lem. 14.14; a half-size bound alone is not used in the source induction.

Lemma 2.2 (Normalized step integrals)**.** At an actual step of height i in [7, Def. B.16], let $P \in \mathcal{U}_{i+1}^\circ(T_\ell; \alpha_0, \alpha_1)$. The integral defining $\Xi(x_i; P)$ is absolutely convergent and nonzero. Its normalized measure $\nu_i[P]$ has finite total variation and mass one. For every bare array g through level i, the function $B \mapsto \mathbb{V}'[g]^{\mathrm{tot}}(P; B)$ is bounded and continuous on Rg_{x_i}, with

$$|\mathbb{V}'[g]^{\mathrm{tot}}(P; B)| \le \#\mathcal{D}_i \, K_\Lambda s_i N_i(g). \tag{2.v}$$

The first and mixed insertion identities are equation YM3_I:(2298). For a real chart point of norm less than r_α, the normalized measure is a probability measure by Prop. YM3_I:124.5. ■

Proof. The rule sending an array to the fourth potential bounds each of its local members by $K_\Lambda s_i N_i(g)$ on the step domain. Summing over the finitely many domains gives eq. 2.v. Each member is holomorphic in the fluctuation field there, hence continuous on the real box. To evaluate the normalization, insert μg in the source slot and restrict μ so that its array norm is smaller than the remaining room. The whole-torus identity of Thm. YM3_I:V, equation YM3_I:(902), expresses the integral as the exponential of the sum of the localized output, so it is nonzero and absolutely convergent. Here $\tau = 1$ removes the additional quadratic potential. At $\mu = 0$, division by this nonzero integral gives mass one and finite total variation. The bound eq. 2.v justifies differentiation of the normalized integral; the first derivative of its logarithm is the mean of the inserted total, and the mixed derivative is its covariance. This is precisely the argument for equation YM3_I:(2298). Positivity is asserted only at the real chart points covered by Prop. YM3_I:124.5. □

For the tower identity we use the real two-source frame of Lem. 4.5(a), retaining its actual canonical places, flat-top hypotheses and weighted progression. The chart, lowering map and chain classes are those of Section YM3_I:124. In particular, a member of $\mathsf{Cc}_n(r)$ is a real bare potential A satisfying

$$\|v_A^{[n]}\| \le r, \qquad v_A^{[n]} = (A, J^{(n)}(e^{i\eta_n A})), \qquad e^{i\eta_n A} = U^u, \quad U \in \mathcal{C}(\mathbf{B}_n, W), \quad \varpi(W) < \varepsilon_{\mathcal{W}}\alpha_0. \tag{2.vi}$$

Thus the norm condition is accompanied by an actual background condition. The two radii in equation YM3_I:(2285) are denoted here by ϱ_{ch} and ϱ_{fl}.

Lemma 2.3 (The domains of a chain background)**.** In this frame, let $1 \le n \le K$, $r \ge 0$ and $A \in \mathsf{Cc}_n(r)$. The array actions through level n are defined at $\widetilde{U} = e^{i\eta_n A}$. If, in addition, $r < r' \le \varrho_{\mathrm{ch}}$, then $\widetilde{U}$ is regular of level n at radius $3r'$, and

$$P[v_A^{[n]}] \in \mathcal{U}_n^\circ(T_\ell; \alpha_0, \alpha_1), \qquad 0 \le a_p(\widetilde{U}) < \tfrac{9}{2}(r')^2\eta_n^4.$$

If $r = 0$, then $A = 0$ and $\widetilde{U} = \mathbf{1}$. ∎

Proof. Under the additional radius condition the chart norm is at most $r < r'$. Its image lies in the simple domain with radii $(3r', r', r')$, and in the core because $r' \le r_\alpha$. This is the usable-chart construction in equation YM3_I:(933) and the domain verification preceding equation YM3_I:(2287). Hence

$$|\partial\widetilde{U} - \mathbf{1}| < 3r'\eta_n^2, \qquad |J^{(n)}(\widetilde{U})| < r',$$

which gives regularity. To check the earlier array actions, use the background U and block field W in eq. 2.vi. The small-plaquette clause of Prop. YM3_I:52.3 puts W in the actual step domain. At height $n-1$ the canonical sum rule has its array actions defined at every representative of this orbit. This gives all levels through $n-1$; at level n the plain-core clause in the proof of the same proposition supplies the remaining pairs. At $n = 1$ there are no earlier levels to check. The trivial pairs lie in their cores as well. These are exactly the domain conditions for the absolute array actions. Their domains are gauge invariant, so they hold also at $U^u = \widetilde{U}$.

For an $SU(2)$ plaquette, $a_p(\widetilde{U}) = \frac{1}{2}|\partial\widetilde{U}(p) - \mathbf{1}|^2$; the regularity bound gives the asserted estimate. Finally the chart norm dominates $|A|$, so $r = 0$ forces $A = 0$. □

Write $\mathsf{Sum}[f](U)$ for the finite sum over the domains and base points of the family f, evaluated at the pairs of U. For $w_\lambda = \lambda_1 w_1 + \lambda_2 w_2$ and $\lambda \in \Lambda^{\mathrm{w}} = \Lambda^{\mathrm{w}}_{K-1}$, set

$$D_i^\lambda(U) = \mathcal{O}(w_\lambda)(U) + \sum_{j=1}^{i} \mathsf{Sum}[E_\lambda^{(j)} - E^{(j)}](U), \qquad \mathcal{O}(w)(U) = L^K \sum_p w(p) a_p(U),$$

$$\Phi_{a,i} = \partial_a D_i^\lambda|_0, \qquad \mathsf{F}_i = \partial_1\partial_2 D_i^\lambda|_0, \tag{2.vii}$$

$$\Phi_{a,i}^{\mathrm{ir}}(U) = \sum_{j=1}^{i} (\mathsf{Sum}[\dot{E}_a^{(j)}](U) - \mathsf{Sum}[\dot{E}_a^{(j)}](\mathbf{1})).$$

These functions are defined when the array actions through level i are defined at U. Then the relative exponent is holomorphic in λ and $D_i^0 = 0$, by [7, Prop. B.8].

Proposition 2.4 (The two-source tower)**.** In the frame just specified, let $1 \le i < K$ and $r \ge 0$ satisfy

$$r < \varrho_{\mathrm{ch}}, \qquad \delta_i < \varrho_{\mathrm{fl}}, \qquad 6L^{-2}r + 8C_{\mathrm{db}}\delta_i < \varepsilon_{\mathcal{W}}\alpha_0. \tag{2.viii}$$

For $A \in \mathsf{Cc}_{i+1}(r)$ put $\widetilde{U} = e^{i\eta_{i+1}A}$, $v = v_A^{[i+1]}$, $P = P[v]$ and $\nu = \nu_i[P]$. For $B \in \mathsf{Rg}_{x_i}$ put

$$\widetilde{U}^{\mathrm{lw}}(B) = e^{i\eta_i A^{\mathrm{lw}}(v, g_i B)}.$$

Then ν is a probability measure. The relative exponents and their derivatives in eq. 2.vii are defined at $\widetilde{U}$ through level $i+1$ and at $\widetilde{U}^{\mathrm{lw}}(B)$ through level i. Their latter values, as well as $\mathcal{O}(w_a)(\widetilde{U}^{\mathrm{lw}}(B))$, are bounded continuous functions of B. For every $\lambda \in \Lambda^{\mathrm{w}}$,

$$e^{D^\lambda_{i+1}(\widetilde{U})} = \int e^{D^\lambda_i(\widetilde{U}^{\mathrm{lw}}(B))}\,\mathrm{d}\nu(B). \tag{2.ix}$$

Consequently,

$$\begin{aligned} \Phi_{a,i+1}(\widetilde{U}) &= \int \Phi_{a,i}(\widetilde{U}^{\mathrm{lw}})\,\mathrm{d}\nu, \\ \mathsf{F}_{i+1}(\widetilde{U}) &= \int \mathsf{F}_i(\widetilde{U}^{\mathrm{lw}})\,\mathrm{d}\nu + \big\langle \Phi_{1,i}(\widetilde{U}^{\mathrm{lw}}); \Phi_{2,i}(\widetilde{U}^{\mathrm{lw}})\big\rangle_\nu. \end{aligned} \tag{2.x}$$

■

Proof. Prop. YM3_I:124.2 applies under eq. 2.viii. It gives

$$A^{\mathrm{lw}}(v, g_i B) \in \mathsf{Cc}_i(L^{-1}r + \tfrac{11}{10}c_{\mathfrak{h}}\delta_i),$$

and Lem. 2.3 puts P in the level-$(i+1)$ core and verifies the array actions at both backgrounds. Restriction to earlier levels preserves this definedness: it retains a subset of the required core conditions. The chart point is real and has norm less than r_α, so Prop. YM3_I:124.5 gives the asserted positivity and normalization of ν.

The source enters only through the bare array. At height i, the difference $g_\lambda = f_{\lambda,i} - f_{0,i}$ has zero counterterms and equals

$$g_\lambda = ((E^{(j)}_\lambda - E^{(j)}, 0)_{j\le i}, \emptyset) \sqcup \mathsf{W}_{L^{K-i}w_\lambda}.$$

Indeed $E^{(j)}_0 = E^{(j)}$ and the counterterms were kept fixed. The action of a zero-counterterm array is the sum of its pointed totals minus their values at $\mathbf{1}$. Its weight member adds $\eta_i^{-1}\sum_p L^{K-i}w_\lambda(p)a_p(U) = \mathcal{O}(w_\lambda)(U)$. Since the plaquette action vanishes at $\mathbf{1}$, this gives

$$S[g_\lambda](U) = D^\lambda_i(U) - D^\lambda_i(\mathbf{1}).$$

Apply the array sum rule, equation YM3_I:(2289), to g_λ. The constant at $\mathbf{1}$ cancels, yielding

$$\mathbb{V}'[g_\lambda]^{\mathrm{tot}}(P; B) = D^\lambda_i(\widetilde{U}^{\mathrm{lw}}(B)) - D^\lambda_i(\widetilde{U}). \tag{2.xi}$$

The same calculation for the first and mixed derivatives of g_λ, and for its weight member alone, gives the analogous identities for $\Phi_{a,i}$, F_i and $\mathcal{O}(w_a)$. Lem. 2.2 makes these potential totals bounded and continuous on the fluctuation box; hence so are the listed functions. Finally $\Phi^{\mathrm{ir}}_{a,i} = \Phi_{a,i} - \mathcal{O}(w_a) - \sum_{j\le i}\mathsf{Sum}[\dot{E}^{(j)}_a](\mathbf{1})$, which gives the remaining continuity assertion.

Let $\Xi_\lambda(P)$ be the step integral with array $f_{\lambda,i}$ and with all other data unchanged. The finite-source estimate puts this array in the admitted norm ball. The actual one-step theorem therefore gives an absolutely convergent integral and the identity

$$\Xi_\lambda(P) = \exp \mathsf{Sum}[\mathsf{St}_i(f_{\lambda,i})](P) \neq 0. \tag{2.xii}$$

Here the calibration equals the running coefficient, so $\tau = 1$ and the additional quadratic potential, bounded by a constant times $|1-\tau|$, is zero. The whole torus is an admitted localization domain, and the output radii are (α_0, α_1). These are the hypotheses for the whole-torus identity of Thm. YM3_I:V; the same specialization is used in [7, Prop. B.8].

At zero source this integral is precisely the normalization of $\nu_i[P]$, equation YM3_I:(2293). The provision is linear in its bare array, so division of the two integrals and eq. 2.xi give

$$\frac{\Xi_\lambda(P)}{\Xi_0(P)} = e^{-D^\lambda_i(\widetilde{U})}\int e^{D^\lambda_i(\widetilde{U}^{\mathrm{lw}}(B))}\,\mathrm{d}\nu(B).$$

On the other hand the recursion [7, eq. 200] and eq. 2.xii give

$$\frac{\Xi_\lambda(P)}{\Xi_0(P)} = \exp \mathsf{Sum}[E_\lambda^{(i+1)} - E^{(i+1)}](P) = \exp(D_{i+1}^\lambda(\widetilde{U}) - D_i^\lambda(\widetilde{U})).$$

The sum at P equals the sum at its pair, which is the pair of $\widetilde{U}$. Comparing the two quotients proves eq. 2.ix.

To differentiate this identity, the pointwise holomorphy in the sources needs a bound uniform in the fluctuation field. Define the finite numbers

$$K_{i,a} = 2L^K \sum_p |w_a(p)| + \sum_{j\le i} \sum_{X\in\mathcal{D}_j} \#(X \cap \mathbb{T}^{(j)}) d_{j,a},$$

where $d_{j,a}$ are the Lipschitz bounds in [7, eq. 196]. All pairs entering the lowered array actions lie in their cores. The family norm bounds each of their values, and $0 \le a_p(U) \le 2$ on real configurations. Consequently

$$|D_i^\lambda(\widetilde{U}^{\mathrm{lw}}(B))| \le \sum_{a=1}^{2} |\lambda_a| K_{i,a} \quad (\lambda \in \Lambda^{\mathrm{w}},\ B \in \mathsf{Rg}_{x_i}).$$

Thus the exponential is uniformly bounded on each compact source set, independently of B. Cauchy's formula and integration against the probability measure justify differentiation. Since $D_i^0 = 0$, the first derivatives give the first identity in eq. 2.x. The mixed derivative gives the integral of $\mathsf{F}_i + \Phi_{1,i}\Phi_{2,i}$. Subtracting the product of the two first-derivative identities leaves the covariance in its second identity. □

2.3.1. *The calibrated cutoff comparison.* We record the choices used in the comparison of two successive cutoffs. They must be made before fixing the torus or its physical coupling. The setting has three parts, denoted by (Th1)–(Th3).

In (Th1), choose the real construction constants and canonical auxiliary data as in Lem. YM3_I:118.1. Impose the global background, response, localization and actual-domain hypotheses of Section YM3_I:113, and the source and recursion hypotheses of Section YM3_I:114. In particular the fixed pair radii are (α_0, α_1), $\alpha_0 \le a_{\mathrm{H}}$, the lag weights are $L^{-(j-i)}$, and

$$E_0^{\mathrm{s}} = E_0, \qquad N_\star^{\mathrm{s}} = \frac{L}{L-1}(E_0 + \tfrac{3}{2}\alpha_0^2 B_\sharp).$$

The constants $K_{\mathrm{Q}}^\vee, K_{\mathrm{L}}^\vee, \mu_\star, \sigma_{\mathrm{src}}$ and the three smallness inequalities are those in equations YM3_I:(2170) and the surrounding source construction. Take

$$\varrho = \min\left\{\frac{20\mu_\star}{\sqrt{3}K_{\mathrm{L}}^\vee\gamma}, \frac{\sigma_{\mathrm{src}}}{2}(\gamma^{-2} - \bar{B})\right\} > 0, \qquad \mathsf{Src} = \mathsf{Src}_{\mathsf{Bc}}(2\varrho, 2\varrho).$$

Thus the coupling sequence, Gaussian structures, auxiliary data and pair cores are independent of the source.

In (Th2), fix $0 < \eth < 1$ before decreasing γ, and use the tail calibration of Section YM3_I:115:

$$c' = (1-\eth)c_\star, \qquad C_{\mathrm{maj}}^\Sigma = C_{\mathrm{maj}} + (6L^3+1)\log 2, \qquad C_{\mathrm{low}}(x) = c_{\mathrm{low}} + \tfrac{9}{2}\log x.$$

The ceiling γ is chosen so that $\log\gamma^{-2}$ exceeds both the case-II history threshold and the summability threshold specified there. These thresholds, the source smallness conditions, and the background and response conditions are part of the calibration. The proof of Lem. YM3_I:118.1 verifies that they can be imposed simultaneously. None depends on the finite lattice.

For later use put $c_{\log} = \log L/\log\gamma^{-2}$ and

$$c_\rho = \max\left\{2B_3, \frac{C_H A_1}{(1-L^{-3/2})A_0}\right\}.$$

We write $\mathsf{t}_\natural$ for the fixed calibration denoted by $\tau_\natural$ in equation YM3_I:(2134). It is the least $\tau \ge \gamma^{-2}$ at which

$$2c_\rho\varepsilon(\tau) \le \min\{\varepsilon_{\mathcal{W}}\alpha_0, c_2^{\mathrm{SO}}, \varepsilon_{\mathrm{v}}/4\},$$
$$\delta(\tau) \le \varepsilon_{\mathcal{W}}, \qquad 4C_{\mathrm{db}}\delta(\tau) \le \min\{1, r_{\mathrm{B}}, r_{\mathrm{B}}^{\mathrm{s}}/2\}.$$

The thresholds are continuous, decreasing and tend to zero; their positive fixed ceilings therefore make this minimum finite. All three inequalities persist at larger τ.

In (Th3), fix

$$K \ge 2, \quad \mathsf{b}' \ge \gamma^{-2}, \quad m \in \mathbb{N}, \quad MR(\mathsf{b}') \le L^m, \quad \ell = 2L^m, \quad x_j = L^{K-j}\mathsf{b}' \quad (0 \le j \le K).$$

Choose an integer top level $1 \le t_* \le K-1$ such that $x_{t_*} \ge \mathsf{t}_\natural$. These are precisely the finite-lattice data in Section YM3_I:115. The setting (E) retains (Th1), (Th2), the label and the side, allowing K to vary. A compact label interval $J = [b_-, b_+]$ is admissible on this side when $b_- \ge \gamma^{-2}$ and $MR(b_+) \le L^m$, as in equation YM3_I:(2201).

For a comparison involving n insertions, the depths are $u = \lceil K/[180(3n+4)]\rceil$ and $s = \lceil K/9\rceil$. The errors and smallness conditions are those in Section YM3_I:116; the additional support condition is $d_\infty(\underline{f})L^u \ge 6M$ when $n \ge 2$. For bounded functions of block fields one takes $n = 1$ and omits this support condition. §2.3.2 displays all the resulting errors and conditions (k0)–(k4), (k6).

Lemma 2.5. Let $0 < g \le \gamma$ and $\ell = 2L^m \in \mathfrak{L}(g)$. Then

$$J_g = [g^{-2} + B_\sharp - L^{-1}\bar{B},\ X_g + \bar{B}]$$

is an admissible label interval on T_ℓ, and it contains $I(g)$ and every first-passage label at g. The larger interval $\mathsf{J}_g = [\gamma^{-2}, L(X_g + \bar{B})]$ is also admissible on T_ℓ. ■

Proof. The lower endpoint is at least γ^{-2}, since $B_\sharp - L^{-1}\bar{B} = B_\sharp(L-2)/(L-1) \ge 0$. The side floor in equation YM3_I:(1528) gives $L^m \ge LMR(X_g + \bar{B})$, and hence the required room $MR(\max J_g) \le L^m$. The containment of $I(g)$ follows from its endpoints. The first-passage label bound is equation YM3_I:(1531). For the larger interval, the radius scaling inequality in the proof of Prop. YM3_I:74.1 gives $R(LX) \le LR(X)$ for $X \ge \gamma^{-2}$. Apply it at $X = X_g + \bar{B}$ and use the same side floor to obtain $MR(\max \mathsf{J}_g) \le LMR(X_g + \bar{B}) \le L^m$. □

2.3.2. *The laws of the block fields.* The cutoff comparison also controls bounded functions of a fixed block field. We prove the resulting total-variation convergence here. We use the calibrated small-field setting of Section YM3_I:115: all constants, auxiliary data, source radii and tail thresholds are fixed before the cutoff, with $E_0^{\mathrm{s}} = E_0$, $N_\star^{\mathrm{s}} = L(E_0 + 3\alpha_0^2 B_\sharp/2)/(L-1)$, $\mathsf{b} \ge \gamma^{-2}$, $MR(\mathsf{b}) \le L^m$ and $\ell = 2L^m$. This is the setting denoted by (E). The precise calibration, including the source and tail smallness conditions, is that of Prop. YM3_I:115.2, not merely the displayed lower bound on b.

At cutoffs K and $K+1$ use the same physical depth, and put

$$u = \lceil K/1260\rceil, \qquad s = \lceil K/9\rceil, \qquad t_1 = K - u, \qquad t_0 = K - s.$$
$$x_{[d]} = L^d\mathsf{b}, \qquad N_{[d]} = \ell^3 L^{3d}, \qquad y_{[d]} = (\log x_{[d]})^{2\mathfrak{p}_0}.$$

The errors in Prop. YM3_I:116.1 and the discarded mass in Prop. YM3_I:115.2 are

$$\epsilon_{\mathrm{fg}} = C_{\mathrm{old}}C_{\mathrm{fg}}\frac{L}{L-1}E_0 N_{[u]}L^{-(s-u)},$$
$$\epsilon_{\mathrm{cl}} = 4\mathsf{K}_{\mathrm{w}}c_\rho^2 A_0^2(1 + 2\varrho/\mathsf{b})y_{[K]}N_{[s]}\sqrt{K}L^{-(K-s)/2},$$
$$\epsilon_{\mathrm{sh}} = 3(s-u)N_{[s]}\exp[C_{\mathrm{sh}} - \kappa_{\mathrm{b}}A_1^2 y_{[u]}/4],$$
$$\mathsf{R}_{t_1} = 2M^{-3}N_{[u]}\exp[-c'A_0^2 y_{[u]} + (\mathsf{C}_{\mathrm{maj}}^{\Sigma} + 2 + C_{\mathrm{low}}(\mathsf{b}))\ell^3].$$

Here $c' = (1-\eth)c_\star > 0$, with the fixed $0 < \eth < 1$ of that calibration. We retain the following names for the comparison conditions:

$$\text{(k0)}\quad 1 \le u < s < K, \qquad \text{(k1)}\quad L^u \mathsf{b} \ge \Theta_\star,$$
$$\text{(k2)}\quad \epsilon_{\mathrm{fg}} \le 1/8, \qquad \text{(k3)}\quad \epsilon_{\mathrm{cl}}, \epsilon_{\mathrm{sh}} \le 1/20,$$
$$\text{(k4)}\quad 256 \mathsf{K}_u c_\rho^2 A_0^2 N_{[s]} L^{-(K-s)} \le \gamma_{\mathrm{r}} A_1^2, \qquad \text{(k6)}\quad \mathsf{R}_{t_1} \le 1/2.$$

There is no support-separation condition in this measure statement. The constants $\Theta_\star$, K_u, K_{w} and the other comparison constants are those in Sections YM3_I:116 and YM3_I:115, with identical level units. In particular the block lattice at depth d is $\Lambda_{d,\ell}$ on both cutoffs.

For uniformity in b, let $\mathsf{J} = [\mathsf{b}_-, \mathsf{b}_+]$ satisfy $\mathsf{b}_- \ge \gamma^{-2}$ and $MR(\mathsf{b}_+) \le L^m$. Define $\bar{\epsilon}_{\mathrm{sh}}$ by replacing b by b_- in $y_{[u]}$, and define $\bar{\mathsf{R}}$ by that same replacement in its negative exponent and by replacing $C_{\mathrm{low}}(\mathsf{b})$ with $C_{\mathrm{low}}(\mathsf{b}_+)$. They dominate their unbarred counterparts. The uniform threshold K_0 in Thm. YM3_I:AR, taken at $n = 1$, makes (k0)–(k4) and (k6) valid for all $K \ge K_0$. Its proof shows that $K^a e^{a'K}(\bar{\mathsf{R}} + \bar{\epsilon}_{\mathrm{sh}}) \to 0$ for every fixed real a, a'. The reason is $u \ge K/1260$ and $2\mathfrak{p}_0 \ge 2$ in the negative exponents.

We first record the measure-theoretic passage to the limit.

Lemma 2.6 (A summable sequence of probability measures)**.** Let $(\Omega, \mathfrak{B})$ be a measurable space, let $K_* \in \mathbb{N}$, and let μ^K, $K \ge K_*$, be probability measures on $\mathfrak{B}$. Let $q_K \ge 0$ be numbers with $\sum_{K \ge K_*} q_K < \infty$ and

$$\sup_{E \in \mathfrak{B}} \bigl|\mu^{K+1}(E) - \mu^K(E)\bigr| \le q_K \qquad (K \ge K_*).$$

Then $\mu(E) := \lim_{K\to\infty} \mu^K(E)$ exists for every $E \in \mathfrak{B}$,

$$\sup_{E \in \mathfrak{B}} \bigl|\mu^K(E) - \mu(E)\bigr| \le \sum_{k \ge K} q_k \qquad (K \ge K_*),$$

and μ is a probability measure on $\mathfrak{B}$. ∎

Proof. Step 1: the limit and the bound. For $K' > K \ge K_*$ and $E \in \mathfrak{B}$,

$$|\mu^{K'}(E) - \mu^K(E)| \le \sum_{k=K}^{K'-1} q_k \le \sum_{k \ge K} q_k,$$

and the right-hand side tends to 0 as $K \to \infty$. Hence $(\mu^K(E))_K$ is a Cauchy sequence, and its limit $\mu(E)$ exists. Letting $K' \to \infty$ in the display gives the asserted bound, uniformly in E.

Step 2: elementary properties. As limits of the corresponding properties of the μ^K, we have $\mu(E) \ge 0$ for all $E \in \mathfrak{B}$, $\mu(\emptyset) = 0$, $\mu(\Omega) = 1$, and μ is finitely additive.

Step 3: σ-additivity. Let $E_1, E_2, \dots \in \mathfrak{B}$ be pairwise disjoint and put $R_N := \bigcup_{i>N} E_i$. By finite additivity,

$$\mu\Bigl(\bigcup_i E_i\Bigr) = \sum_{i \le N} \mu(E_i) + \mu(R_N).$$

Let $\eta > 0$, and choose $K \ge K_*$ with $\sum_{k \ge K} q_k \le \eta$. By Step 1, $\mu(R_N) \le \mu^K(R_N) + \eta$ for every N. Since μ^K is a measure and R_N decreases to $\emptyset$, we have $\mu^K(R_N) \to 0$ as $N \to \infty$; hence $\limsup_N \mu(R_N) \le \eta$. As $\eta > 0$ is arbitrary and $\mu(R_N) \ge 0$, $\mu(R_N) \to 0$, and $\mu(\bigcup_i E_i) = \sum_i \mu(E_i)$. □

Definition 2.7 (The laws of the block fields)**.** Assume (E), with label b and side $\ell = 2L^m$.

(1) At cutoff K set $\beta = L^K \mathsf{b}$ and

$$\rho_0(U) = e^{-A^{q_K}(\mathsf{b}, U)}, \qquad Z_K = \int \rho_0 \, \mathrm{d}U > 0, \qquad \langle H \rangle_K = Z_K^{-1} \int H \rho_0 \, \mathrm{d}U.$$

Here dU is product Haar probability measure and $A^{a_K}(\mathsf{b},U)=\beta\sum_p a_p(U)$ is the Wilson action.

(2) For $K\ge\max\{d,1\}$ the law at depth d is

$$\mu_d^K(E)=\langle \mathbf{1}_E(\mathcal{M}^{K-d}U)\rangle_K \qquad (E\subset\mathcal{V}(\Lambda_{d,\ell}) \text{ Borel}). \tag{2.xiii}$$

The averaging map is Borel and gauge covariant. Its map from $\Lambda_{d+1,\ell}$ to $\Lambda_{d,\ell}$ is independent of the bare cutoff: its definition uses precisely these two lattices. These properties are proved in Prop. YM3_I:4.5. Thus μ_d^K is a Borel probability measure.

(3) Write $\mathfrak{B}_d$ for the sigma-algebra of invariant Borel sets in $\mathcal{V}(\Lambda_{d,\ell})$. Gauge transformations are bijections, so complements and countable unions of invariant sets remain invariant. The map $\mathcal{M}$ in the consistency identity below is the fixed map from depth $d+1$ to depth d.

(4) For $K\ge 2$, use the errors above and set

$$\mathfrak{o}_K=8\mathsf{R}_{t_1}+12\epsilon_{\mathrm{fg}}+8\epsilon_{\mathrm{cl}}+12\epsilon_{\mathrm{sh}}. \tag{2.xiv}$$

(5) For an interval $\mathsf{J}=[\mathsf{b}_-,\mathsf{b}_+]$ as above, put

$$\mathsf{C}_{\mathrm{bf}}:=2\Big[1+12\,C_{\mathrm{old}}C_{\mathrm{fg}}\tfrac{L}{L-1}E_0\,L^4+32\,\mathsf{K}_{\mathrm{w}}c_\rho^2A_0^2(1+\tfrac{2\varrho}{\mathsf{b}_-})\big(\log(L\mathsf{b}_+)\big)^{2\mathfrak{p}_0}L^{7/2}\Big] \tag{2.xv}$$

and

$$K_{\mathrm{bf}}:=\min\Big\{k\in\mathbb{Z}:\ k\ge K_0,\ \text{and}\ \ 8\,\bar{\mathsf{R}}+12\,\bar{\epsilon}_{\mathrm{sh}}\le\tfrac12\,\mathsf{C}_{\mathrm{bf}}\,\ell^3K^{\mathsf{p}_0}L^{-K/10}\ \ \text{at all integers } K\ge k\Big\},$$

where $\mathsf{p}_0=2\mathfrak{p}_0+1/2$ and K_0 is the uniform threshold specified above. This minimum exists: the bracket defining C_{bf} is at least 1, and the barred errors decay faster than every exponential in K. Its dependencies are only ℓ, the endpoints of J and the fixed constants.

■

The comparison of the two lattices made above also controls the laws of Def. 2.7, and Lem. 2.6 turns this control into their convergence as the cutoff tends to infinity.

Corollary 2.8 (Convergence of the laws of the block fields)**.** Assume the setting (E), and take u,s,t_1,t_0 and the comparison errors defined above.

(a) Let $K\ge 2$ be an integer at which the conditions (k0), (k1), (k2), (k3), (k4) and (k6) hold. Then, for every integer $0\le d\le u$ and every bounded Borel function $F:\mathcal{V}(\Lambda_{d,\ell})\to\mathbb{C}$ invariant under the gauge transformations of $\Lambda_{d,\ell}$,

$$\Big|\big\langle F(\mathcal{M}^{K+1-d}U)\big\rangle_{K+1}-\big\langle F(\mathcal{M}^{K-d}U)\big\rangle_K\Big|\ \le\ \mathfrak{o}_K\ \sup|F|. \tag{2.xvi}$$

If moreover J is an interval of labels under (Tor) on the side ℓ with $\mathsf{b}\in\mathsf{J}$, then at every integer $K\ge K_{\mathrm{bf}}$ the conditions (k0)–(k4) and (k6) hold, and

$$\mathfrak{o}_K\ \le\ \mathsf{C}_{\mathrm{bf}}\,\ell^3\,K^{\mathsf{p}_0}\,L^{-K/10}; \tag{2.xvii}$$

here C_{bf} and K_{bf} are the numbers of Def. 2.7(5), which depend on the label b only through J.

(b) Let K_{bf} be taken at $\mathsf{J}=\{\mathsf{b}\}$. For every integer $d\ge 0$ and every $E\in\mathfrak{B}_d$ the limit $\mu_d(E):=\lim_{K\to\infty}\mu_d^K(E)$ exists, and

$$\sup_{E\in\mathfrak{B}_d}\big|\mu_d^K(E)-\mu_d(E)\big|\ \le\ \sum_{K'\ge K}\mathfrak{o}_{K'}<\infty \qquad \text{for every integer } K\ge K_{\mathrm{bf}} \text{ with } \lceil K/1260\rceil\ge d. \tag{2.xviii}$$

The set function μ_d is a probability measure on $\mathfrak{B}_d$. The family $(\mu_d)_{d\ge0}$ is consistent: if $E\in\mathfrak{B}_d$, then $\mathcal{M}^{-1}(E)\in\mathfrak{B}_{d+1}$ and

$$\mu_{d+1}(\mathcal{M}^{-1}E)=\mu_d(E), \tag{2.xix}$$

where $\mathcal{M}:\mathcal{V}(\Lambda_{d+1,\ell})\to\mathcal{V}(\Lambda_{d,\ell})$ is the averaging map of Def. 2.7(3).

(c) For every integer $d \geq 0$, every integer $K \geq \max\{d, 1\}$ and every Borel set $E \subset \mathcal{V}(\Lambda_{d,\ell})$,

$$\mu_d^K(E) = \int \bar{\mathbf{1}}_E \, \mathrm{d}\mu_d^K, \qquad \bar{\mathbf{1}}_E(V) := \int \mathbf{1}_E(V^v) \, \mathrm{d}v, \tag{2.xx}$$

where dv is the product of one Haar probability measure of G for each site of $\Lambda_{d,\ell}$, a measure on the group of the gauge transformations v of $\Lambda_{d,\ell}$. The function $\bar{\mathbf{1}}_E$ is Borel, takes its values in $[0, 1]$ and is gauge invariant. Consequently the limit in (b), the bound eq. 2.xviii and the identity eq. 2.xix hold for all Borel sets: $\mu_d^K \to \mu_d$ in total variation, that is, uniformly on the Borel subsets of $\mathcal{V}(\Lambda_{d,\ell})$; μ_d is a Borel probability measure on $\mathcal{V}(\Lambda_{d,\ell})$ that is gauge invariant; and the image of μ_{d+1} under $\mathcal{M}$ is μ_d.

■□

Proof. We first compare the normalized means at depth u. On either lattice $\bullet \in \{\mathrm{A}, \mathrm{B}\}$, let $\Lambda_0^\bullet$ be the full unnormalized law at that depth, $\Lambda_1^\bullet$ its small-field chain restricted to $\mathsf{In}_{[u]}$, and $\Lambda_2^\bullet$ the chain restarted from the classical action at depth s, again restricted to $\mathsf{In}_{[u]}$. We regard them as functionals on bounded gauge-invariant Borel functions F of the depth-u field. These are precisely the chain and classical-start measures used in Props. YM3_I:115.2 and YM3_I:116.1, with source zero. All three functionals are positive and have strictly positive mass. For the full law this is immediate. The chain is the image of the nonnegative small-field weight times the bare density, by Prop. YM3_I:115.2. For the classical start, the initial density is positive, the chart kernels preserve positivity, and the final exponential density does not vanish on the open set $\mathsf{In}_{[u]}$ containing the trivial configuration; these are the positivity statements in the proof of Prop. YM3_I:116.1.

We use the following elementary normalization bound. If positive functionals P, Q satisfy $P(1) > 0$ and

$$|Q(F) - P(F)| \leq \epsilon \|F\|_\infty P(1), \qquad 0 \leq \epsilon \leq 1/2,$$

then $Q(1) \geq (1 - \epsilon)P(1) > 0$, and subtraction of the two normalized fractions gives

$$\left| \frac{Q(F)}{Q(1)} - \frac{P(F)}{P(1)} \right| \leq \frac{2\epsilon}{1 - \epsilon} \|F\|_\infty \leq 4\epsilon \|F\|_\infty.$$

The discarded-mass estimate in Prop. YM3_I:115.2 gives $0 \leq \Lambda_0^\bullet - \Lambda_1^\bullet$ on this algebra and

$$(\Lambda_0^\bullet - \Lambda_1^\bullet)(1) \leq \mathsf{R}_{t_1} \Lambda_0^\bullet(1).$$

Consequently their normalized means differ by at most $4\mathsf{R}_{t_1} \|F\|_\infty$.

Forgetting the early array changes the final exponent by at most ϵ_{fg}, by equation YM3_I:2180 and the definition of ϵ_{fg} above. Write the two final densities as $p_1 = e^{c^\bullet + \Phi^\bullet}$ and $p_2 = e^{\hat{c}^\bullet + \hat{\Phi}^\bullet}$ on $\mathsf{In}_{[u]}$. Positivity gives $p_1 = |p_1|$ almost everywhere and $p_2 = |p_2| > 0$ there. Thus, with $\varkappa = |e^{c^\bullet - \hat{c}^\bullet}|$,

$$\frac{p_1}{p_2} \in \varkappa[e^{-\epsilon_{\mathrm{fg}}}, e^{\epsilon_{\mathrm{fg}}}] \quad \text{almost everywhere.}$$

Since $\epsilon_{\mathrm{fg}} \leq 1/8$, $e^{\epsilon_{\mathrm{fg}}} - 1 \leq (5/4)\epsilon_{\mathrm{fg}} \leq 1/2$. Apply the normalization bound to $P = \varkappa\Lambda_2^\bullet$, $Q = \Lambda_1^\bullet$. The factor $\varkappa$ cancels, and these two normalized means differ by at most $5\epsilon_{\mathrm{fg}} \|F\|_\infty$.

The two classical starts can be written on the common depth-s configuration space as $k^\bullet \hat{F}_0^\bullet \, \mathrm{d}W$, followed by the same block averaging down to depth u. The final total-variation estimate in the proof of Prop. YM3_I:116.1 is

$$\int |k^{\mathrm{A}} \hat{F}_0^{\mathrm{A}} - k^{\mathrm{B}} \hat{F}_0^{\mathrm{B}}| \, \mathrm{d}W \leq (2\epsilon_{\mathrm{cl}} + 3\epsilon_{\mathrm{sh}}) \Lambda_2^{\mathrm{A}}(1).$$

We will also use the relative version of the mismatch bound. For every bounded nonnegative gauge-invariant function H of the final block field, the proof of Prop. YM3_I:116.1 gives

$$\int H(\mathcal{M}^{s-u} W) \, k^\bullet \mathbf{1}_{\{k^\circ = 0\}} \hat{F}_0^\bullet \, \mathrm{d}W \leq \epsilon_{\mathrm{sh}} \Lambda_2^\bullet(H).$$

Indeed, at a mismatching test some fluctuation of lattice $\bullet$ exceeds half its threshold. Prop. YM3_I: 114.4 bounds the marked next measure by

$$\exp(C_{\mathrm{sh}} - \kappa_{\mathrm{b}}\mathfrak{b}(x_{(k)})^2/4)$$

times the ordinary next measure. Each later positive cut and image operation preserves this order. Sum over at most $s-u$ steps and $3N_{[s]}$ bonds at each step, and use $\mathfrak{b}(x_{(k)})^2 \geq A_1^2 y_{[u]}$ to obtain the displayed bound. This proves an inequality of measures before testing against H.

The total-variation proof bounds separately the mass where a fluctuation test differs and the difference of the classical action on the common cut. Inserting F in this inequality costs only $\|F\|_\infty$. By (k3), $2\epsilon_{\mathrm{cl}} + 3\epsilon_{\mathrm{sh}} \leq 1/4$, so normalization gives an error $(8\epsilon_{\mathrm{cl}} + 12\epsilon_{\mathrm{sh}})\|F\|_\infty$. Adding the five comparisons

$$\Lambda_0^{\mathrm{A}},\ \Lambda_1^{\mathrm{A}},\ \Lambda_2^{\mathrm{A}},\ \Lambda_2^{\mathrm{B}},\ \Lambda_1^{\mathrm{B}},\ \Lambda_0^{\mathrm{B}}$$

bounds the difference of the normalized full laws by

$$[8\mathsf{R}_{t_1} + 10\epsilon_{\mathrm{fg}} + 8\epsilon_{\mathrm{cl}} + 12\epsilon_{\mathrm{sh}}]\|F\|_\infty \leq \mathfrak{o}_K\|F\|_\infty.$$

For $d \leq u$, apply this to $F \circ \mathcal{M}^{u-d}$. Block averaging is Borel and gauge covariant, and its maps at equal physical depth coincide. Hence this composition remains bounded and gauge invariant, with no increase of its supremum norm. This proves eq. 2.xvi.

For the rate, the ceiling inequalities in Thm. YM3_I:AR give $4u - s \leq -K/10 + 4$ and $(7s-K)/2 \leq -K/9 + 7/2$. Therefore

$$12\epsilon_{\mathrm{fg}} \leq 12C_{\mathrm{old}}C_{\mathrm{fg}}\frac{L}{L-1}E_0L^4\ell^3L^{-K/10},$$

and, since $\log(L^K\mathsf{b}_+) \leq K\log(L\mathsf{b}_+)$,

$$8\epsilon_{\mathrm{cl}} \leq 32\mathsf{K}_{\mathrm{w}}c_\rho^2A_0^2(1+2\varrho/\mathsf{b}_-)(\log(L\mathsf{b}_+))^{2\mathfrak{p}_0}L^{7/2}\ell^3K^{2\mathfrak{p}_0+1/2}L^{-K/9}.$$

Their sum is at most half the right side of eq. 2.xvii; the other half bounds $8\mathsf{R}_{t_1} + 12\epsilon_{\mathrm{sh}}$ by the choice of K_{bf}. This proves the uniform rate.

For (b), fix d and let K_* be the least $K \geq K_{\mathrm{bf}}$ with $\lceil K/1260\rceil \geq d$. For every invariant Borel set E, apply (a) to $F = \mathbf{1}_E$. The resulting differences of the probability measures are bounded by the summable sequence $\mathfrak{o}_K$, since a fixed power of K times $L^{-K/10}$ is summable. Lem. 2.6 gives the limit, the rate and countable additivity. Gauge covariance makes $\mathcal{M}^{-1}E$ invariant, and the exact finite-cutoff identity

$$\mu_{d+1}^K(\mathcal{M}^{-1}E) = \mu_d^K(E)$$

passes to the limit. This proves consistency.

For (c), invariance of the Wilson density and Haar measure makes the bare law gauge invariant. Gauge covariance of block averaging and extension of a coarse site gauge to the fine sites give the same invariance for μ_d^K. The action $(V, v) \mapsto V^v$ is continuous, so Fubini's theorem shows that $\bar{\mathbf{1}}_E$ is Borel. Right invariance of Haar measure gives

$$\bar{\mathbf{1}}_E(V^{v'}) = \int \mathbf{1}_E(V^{vv'})\,\mathrm{d}v = \bar{\mathbf{1}}_E(V).$$

Integrating the invariance of μ_d^K over v proves eq. 2.xx. Now apply (a) to this function, whose values lie in $[0,1]$. The same summable bound holds for every Borel set E, so Lem. 2.6 applies on the full Borel sigma-algebra. The finite-cutoff consistency and gauge invariance pass to these limits. For invariant E, $\bar{\mathbf{1}}_E = \mathbf{1}_E$, so this extension agrees with the measure in (b). □

2.3.3. *Notation and hypotheses. Notation.* All constants depend only on the basic constants (a common real choice as in Prop. YM3_I:69.1, among them $\mathfrak{p}_0$, A_0, $M = L^{m'}$, C_{lat} and the last, γ) and on the arguments shown; a constant is called basic if it depends on the basic constants alone. The blocking factor L is odd; $a_K = L^{-K}$, $\mathbb{L}_K = a_K\mathbb{Z}^3$; T_ℓ is the torus of side $\ell = 2L^m$ and $\Lambda_{K,\ell}$ its lattice of spacing a_K; $\langle\cdot\rangle_{K,\ell,\beta}$ is Wilson's measure on $\Lambda_{K,\ell}$, $\langle\mathcal{A}_1;\dots;\mathcal{A}_n\rangle$ a cumulant, $\mathcal{G}_{K,\beta}$ the set of the periodic Gibbs states on $\mathbb{L}_K$ at β, that is, of the Gibbs states of Wilson's weight $e^{-\beta\sum_p a_p}$ that are invariant under the translations of a subgroup of finite index of $a_K\mathbb{Z}^3$ ([7, §K.1]); by Prop. YM3_I:103.4 every $\nu \in \mathcal{G}_{K,\beta}$ is invariant under $a_K\mathbb{Z}^3 \rtimes W_3$ and reflection positive at the lattice mirrors $\{n\cdot x = c\}$, $n \in \{e_\mu, e_\mu \pm e_\lambda\}$, of the nine families.

On $\mathbb{R}^3$ and on T_ℓ, $|x|$ is the Euclidean norm and $|x|_\infty = \max_\mu |x_\mu|$ the sup norm; the balls $B(x,r)$ (open) and $\bar{B}(x,r)$ (closed), the distances $|x-y|$ of points and the distance dist of sets are Euclidean. Where sets or supports are said to be pairwise $\geq \delta$ apart, to have a diameter $\leq \varrho$, or where a weight vanishes within δ of a set, the sup distance dist_∞ is meant, unless the Euclidean distance is named, as in Thm. B(a), (c). In Lems. [7, Lem. C.15] and [7, Lem. C.16], in [7, Thm. BV](e) and in Thm. B(c), (d) the separations of sets are read with the Euclidean distance $\mathrm{dist}(A,B) = \inf_{x\in A, y\in B}|x-y|$, which these statements name; the factor $1/\sqrt{3}$ between it and the sup distance is paid inside their proofs.

The depth at the level k of the renormalisation is $s = K-k$ ([7, result G.2]); the depth of a distance d is the least $s = s(d) \geq 0$ with $L^s d \geq C_W(s)$, where $C_W(s) = c_W(\log \bar{x}_s)^{2\mathfrak{p}_0}$, $\bar{x}_s = L^{s+1}(g^{-2}+\bar{B})$ and c_W is basic; it depends on g, it does not decrease as d falls or as g^{-2} grows, and $s(d) \to \infty$ as $d \to 0$; if $s(d) \geq 1$, then $C_W(s) \leq L^s d < LC_W(s)$ at $s = s(d)$, so that d is about $C_W(s)$ units of the level of that depth, and, for $0 < d < 1$, $Y_{s(d)} := \log \bar{x}_{s(d)} \leq 2\log\frac{1}{d} + C(g)$. The reference coupling $g > 0$ is that of YM3_I:74, $\mathfrak{d}(g)$ is the set of its first-passage data (K,β) (YM3_I:74, YM3_I:74), g_t is the running coupling at the level t, and $\mathfrak{b} = a_K\beta$ is the label ([7, result G.2]). $B_\sharp$ bounds $|L^{-1}x_{j-1} - x_j|$ as in [7, §G.2], $\bar{B} = B_\sharp L/(L-1)$, $X_g = L(g^{-2}+B_\sharp)$ as in YM3_I:(1526), $I(g) = (g^{-2}+\bar{B}, X_g - \bar{B}]$ as in YM3_I:(1527), and $\mathfrak{L}(g) = \{2L^m : m \geq m_{\mathfrak{L}}(g)\}$ (YM3_I:(1528), YM3_I:(1528)) is the set of the sides at g, ℓ_g its least member.

Every $(K,\beta) \in \mathfrak{d}(g)$ has $g^{-2} < a_K\beta \leq L(g^{-2}+\bar{B})$, and $(K, L^K\mathfrak{b}) \in \mathfrak{d}(g)$ for every integer $K \geq 1$ if $\mathfrak{b} \in I(g)$. (H) is the condition $0 < g \leq \gamma_W^{\mathrm{ref}}$, where $\gamma_W^{\mathrm{sd}} \leq \gamma_W^{\mathrm{ref}} \leq \gamma$ are the thresholds of the choices in equations YM3_I:(2326)–(2327); (T) is (H) together with $\ell \in \mathfrak{L}(g)$ and $\mathfrak{b} \in I(g)$, and then $\langle\cdot\rangle_K := \langle\cdot\rangle_{K,\ell,L^K\mathfrak{b}}$.

The field. $V = \mathrm{Sym}_3(\mathbb{R}) = V_A \oplus V_E \oplus V_T$, the multiples of $\mathbf{1}$, the traceless diagonal matrices and the matrices with zero diagonal, is the splitting under the group W_3 of the 48 signed permutation matrices; $V_{\mathbb{C}} = \mathrm{Sym}_3(\mathbb{C})$; $E_{\lambda\lambda'}$ are the matrix units and $e_{ij} = E_{ij} + E_{ji}$. As $F_{\mu\nu} = \varepsilon_{\mu\nu\lambda}B_\lambda$ in three dimensions, the six fields are the entries of $T_{\lambda\lambda'} = \langle B_\lambda, B_{\lambda'}\rangle$. A weight is a bounded Borel function H on $\mathbb{R}^3$ (or on T_ℓ) with values in V or $V_{\mathbb{C}}$ and compact support, $|H| = \sup_x \max_{\lambda\lambda'}|H_{\lambda\lambda'}(x)|$; an isometry $rx = Rx + v$ acts by $(rH)(x) = RH(r^{-1}x)R^t$, so that a reflection acts on points and on indices.

At a site x the 24 plaquette loops are $W_{x,l}$, $l = (\alpha,\alpha')$ an ordered pair of signed directions with $\alpha' \neq \pm\alpha$; $\lambda(l)$ is the normal of the plane of l and $s(l) = \mathrm{sgn}\,\alpha\,\mathrm{sgn}\,\alpha'\,\varepsilon_{\lambda(l)|\alpha||\alpha'|}$ its orientation. With $w(U) = \frac{U-U^{-1}}{2i}$, $Q_x(l,l') = \frac{1}{2}\mathrm{tr}\,w(W_{x,l})w(W_{x,l'})$, $a_p = 1 - \frac{1}{2}\mathrm{tr}\,U(\partial p)$, and, for a bond $b = \langle x, x+a_Ke_\mu\rangle$ with $\lambda_b < \lambda_b'$ the two other directions, $q_b = \frac{1}{4}\sum_{\epsilon,\epsilon'=\pm}\epsilon\epsilon' Q_x((\mu,\epsilon\lambda_b),(\mu,\epsilon'\lambda_b'))$, the lattice form of $\frac{1}{2}\int\mathrm{tr}(HT)$ is [7, eq. 1] ($x(p)$, x_b the midpoints, $\lambda(p)$ the normal of p). $O_K^T(f\mathbf{1}) = \mathcal{O}(f)$ is the action density, and $\mathcal{O}_\lambda(\phi) = O_K^T(\phi E_{\lambda\lambda}) = a_K^{-1}\sum_{\lambda(p)=\lambda}\phi(x(p))a_p$ are the three plane fields. $\bar{O}_K^T(H)$ is [7, eq. 1] with $H(x(p))$, $H(x_b)$ replaced by the mean of H over the cell $x + a_K[-\frac{1}{2},\frac{1}{2})^3$, x the least corner of p, resp. of b. A corner field is $\mathcal{Q}_{l,l';\xi}(\phi) = a_K^{-1}\sum_x \phi(x+a_K\xi)Q_x(l,l')$, $|\xi|_\infty \leq \frac{1}{2}$; a site form is $a_K^{-1}\sum_x\sum_{l,l'}v(x;l,l')Q_x(l,l')$, of norm $\|v\| = \sup_x\sum_{l,l'}|v(x;l,l')|$; a mixed source is a pair $\mathfrak{s} = (w,\mathsf{w})$ of real weights of the plaquettes and of the bonds, with the lattice form

$\mathcal{O}^T(\mathfrak{s}) = a_K^{-1}[\sum_p w(p)a_p - \sum_b \mathsf{w}(b)q_b]$ and $|\mathfrak{s}| = \sup|w| + 8\sup|\mathsf{w}|$. $\mathfrak{S}(O)$ is the set of the tuples $X = (H_1, \dots, H_n)$, $n \geq 0$, of continuous weights with pairwise disjoint supports in an open set O, and $\mathfrak{F}(O)$ its complex span; rX, θF, $\tau_v F$ and $\bar{F}$ act entry by entry, $F' \cdot F$ concatenates, and $A_K^{\nu'}(X) = \prod_a [O_K^T(H_a) - \nu'(O_K^T(H_a))]$, $A^\nu[\mathcal{A}] = \mathcal{A} - \nu(\mathcal{A})$. A pair at scale d is a pair of real weights vanishing off closed balls $\bar{B}(x_0, d_-)$, $\bar{B}(y_0, d_-)$, $0 < d_- < d$.

Free values and the stress tensor. Let

$$\mathcal{W}(H_1, H_2) = \tfrac{3}{2} \iint \mathrm{tr}[H_1(x)K_3(x-y)H_2(y)K_3(x-y)]\,\mathrm{d}x\,\mathrm{d}y, \qquad K_3(u) = \frac{3\hat{u}\hat{u}^t - \mathbf{1}}{4\pi|u|^3}. \tag{2.xxi}$$

The free values are the separated connected Wick contractions, with no contraction within one insertion, as in [7, Thm. V](f)(i). Use three independent colors with off-diagonal covariance $\langle B_\lambda^c(x) B_\mu^d(y)\rangle = \delta^{cd}\mathsf{b}^{-1}K_3(x-y)_{\lambda\mu}$ for $x \neq y$. The cumulants of the smeared bilinears formally denoted by $\frac{1}{2}\int \mathrm{tr}(H_a T^{\mathrm{free}})$, $T^{\mathrm{free}} = \sum_c B^c(B^c)^t$, at separated supports are: $\mathsf{b}^{-2}\mathcal{W}(H_1, H_2)$ at two points; the kernel $3\mathsf{b}^{-3}k_3$, $k_3(x,y,z) = \mathrm{tr}[K_3(x-y)K_3(y-z)K_3(z-x)]$, for three action densities and $3\mathsf{b}^{-4}k_4$, k_4 the sum over the three rings $(x_{\varsigma(1)}, \dots, x_{\varsigma(4)})$ of $\mathrm{tr}\prod_l K_3(x_{\varsigma(l)} - x_{\varsigma(l+1)})$, for four; and $3\mathsf{b}^{-n}K_{\underline{\lambda}}^{(n)}(\phi)$ for n plane fields $\mathcal{O}_{\lambda_a}(\phi_a)$.

On the lattice, $3\mathsf{b}^{-n}K_K^{(n)}(w_1, \dots, w_n)$ is the cumulant of the plaquette forms $a_K^{-1}\sum_p w_a(p)a_p$ for three free lattice photon fields, that is with a_p replaced by $\frac{1}{2}\sum_c (F_p^c)^2$, the F^c independent Gaussian plaquette variables of covariance $\beta^{-1}C^{\mathrm{free}}$, C^{free} the free lattice covariance; likewise, for mixed sources the free lattice value of the cumulant of $\mathcal{O}^T(\mathfrak{s}_1)$, $\mathcal{O}^T(\mathfrak{s}_2)$ is the cumulant obtained when every a_p and every $Q_x(l,l')$ is replaced by its quadratic part in three such independent fields; it is $\frac{3}{2}\mathsf{b}^{-2}\mathrm{Tr}(\mathsf{W}_1 C^{\mathrm{free}}\mathsf{W}_2 C^{\mathrm{free}})$, W_a the symmetric matrix with $\mathcal{O}^T(\mathfrak{s}_a) = (2a_K)^{-1}\sum_c F^{c\,t}\mathsf{W}_a F^c$ after this replacement. $\mathsf{w}_0 = \frac{1}{12}388^{-6}$.

For $\eta \in C_c^2(\mathbb{R}^3; \mathbb{R}^3)$ put $H^{\mathrm{cons}}[\eta] = (\nabla\cdot\eta)\mathbf{1} - \nabla\eta - (\nabla\eta)^t$, $(\nabla\eta)_{ij} = \partial_i\eta_j$, and $\|\eta\|_{C^2} = \sup|\eta| + \sup|\nabla\eta| + \sup|\nabla^2\eta|$; then $\frac{1}{2}\mathrm{tr}(H^{\mathrm{cons}}[\eta]T) = \sum_{ij}\partial_i\eta_j\Theta_{ij}$ with the stress tensor $\Theta_{ij} = \frac{1}{2}\delta_{ij}\mathrm{tr}\,T - T_{ij}$: insertions of $H^{\mathrm{cons}}[\eta]$ express the conservation of Θ, and $\frac{1}{2}\mathrm{tr}(s_{ij}T) = \Theta_{ij}$ for $s_{ij} = \delta_{ij}\mathbf{1} - e_{ij}$. For a unit vector n, $\Theta_n = \mathbf{1} - 2nn^t$.

Chessboards and limits. A board of side $\mathsf{l} = L^{-i}$, $i \in \mathbb{Z}$, divides space into cubes, its cells, by bond mirrors, its walls; for $\ell' \in \mathfrak{L}(g)$, $\ell' \geq 2\mathsf{l}$, and D a function of the bonds of one cell, $\|D\|_{\mathrm{ch},\ell'}$ is the N-th root of $|\langle\prod_\kappa D_\kappa\rangle_{K,\ell',\beta}|$, the D_κ being the $N = (\ell'/\mathsf{l})^3$ mirror images of D in the cells of $T_{\ell'}$, conjugated after an odd number of reflections as in the main chessboard convention. A Gibbs sequence at g is (K_j, β_j, ν_j) with $(K_j, \beta_j) \in \mathfrak{d}(g)$, $K_j \to \infty$, $\nu_j \in \mathcal{G}_{K_j,\beta_j}$; a limit at g is $\sigma(X) = \lim_j \nu_j(A_{K_j}^{\nu_j}(X))$, $X \in \mathfrak{S}(\mathbb{R}^3)$, along a Gibbs sequence with $a_{K_j}\beta_j \to \mathsf{b}$, its label.

Its trace part $\sigma_0(f_1, \dots, f_n) = \sigma(f_1\mathbf{1}, \dots, f_n\mathbf{1})$ is a scalar limit, to which Thm. YM3_I:AP (the scalar action density on $\mathbb{R}^3$), with Thm. YM3_I:AL applies, with a canonical decomposition $(\mathsf{X}_0, \lambda_0, (\sigma_{0,\zeta})_{\zeta\in\mathsf{X}_0})$. With θ the reflection in $\{x_3 = 0\}$, $\mathcal{H}(\sigma)$ is the Hilbert space of the form $\langle[F'],[F]\rangle = \sigma(\theta\bar{F}'\cdot F)$ on $\mathfrak{F}(\{x_3 > 0\})$, and $\Omega = [\emptyset]$. More generally, let σ have the properties of [7, Thm. BV](c), as every limit at g has, and let u be a unit vector at which σ is reflection positive: $\sigma(\theta_u\bar{F}\cdot F) \geq 0$ for $F \in \mathfrak{F}(\{u\cdot x > 0\})$, θ_u the reflection in $\{u\cdot x = 0\}$, acting on points and on indices by $E \mapsto \Theta_u E \Theta_u$.

The *picture of* u is formed in the same way: the Hilbert space $\mathcal{H}_u$ of the form $\langle[F'],[F]\rangle_u = \sigma(\theta_u\bar{F}'\cdot F)$ on $\mathfrak{F}(\{u\cdot x > 0\})$, $\Omega_u = [\emptyset]_u$, the self-adjoint $H_u \geq 0$ with $e^{-tH_u}[F]_u = [\tau_{tu}F]_u$, $t \geq 0$, the unitary group $\mathcal{U}^u(v)[F]_u = [\tau_v F]_u$, $v \in u^\perp$, which commutes with it, and their joint spectral measure $\mathcal{P}_u$ on $[0,\infty)\times u^\perp$, with $e^{-tH_u}\mathcal{U}^u(v) = \int e^{-tE+iv\cdot q}\,\mathrm{d}\mathcal{P}_u(E,q)$; the picture of e_3 is $(\mathcal{H}(\sigma), H, \mathcal{U}, \mathcal{P})$ of [7, Thm. BW](a). On a torus, $\mathcal{S}_{n;K}(X) = \langle O_K^T(H_1); \dots; O_K^T(H_n)\rangle_K$.

Theorem A (The tori)**.** Assume (T), $n \geq 2$, $d > 0$, $\theta_n = 1/(1260n(8n+3))$, and let the supports of $X = (H_1, \dots, H_n)$ be pairwise $\geq d$ apart.

(a) There are an exponent p depending only on the basic constants (one may take $\mathsf{p} = 2\mathfrak{p}_0 + 2$), a number C^{od} depending only on n and on the constant C_{bf} of the comparison of two cutoffs at the label b, and a threshold K_1 depending besides on g, b, ℓ, d, on the threshold K_{bf} of that comparison and on the number $\Theta_\star$ (C_{bf}, K_{bf} of Def. 2.7(5), formed at a compact interval $J \subset I(g)$ that holds b, so that they read the label only through J (Cor. 2.8(a)), $\Theta_\star$ of Section YM3_I:113), such that for $K \geq K_1$ and Lipschitz weights

$$|\mathcal{S}_{n;K+1}(X) - \mathcal{S}_{n;K}(X)| \leq 6^n C^{\mathrm{od}} \ell^{3(n+1)} K^{\mathsf{p}} L^{-\theta_n K} \prod_a (|H_a| + \operatorname{Lip} H_a). \tag{2.xxii}$$

So $\mathcal{S}_n = \mathcal{S}_n^{\ell,\mathsf{b}} = \lim_K \mathcal{S}_{n;K}$ exists along the full sequence of cutoffs, with $|\mathcal{S}_{n;K}(X) - \mathcal{S}_n(X)| \leq 6^n C^{\mathrm{od}} \ell^{3(n+1)} \sum_{K' \geq K} K'^{\mathsf{p}} L^{-\theta_n K'} \prod_a (|H_a| + \operatorname{Lip} H_a)$; it is n-linear and symmetric, $|\mathcal{S}_n(X)| \leq 60^n K_n^{\mathrm{ub}}(g, \ell, \frac{d}{2}) \prod_a |H_a|$ with the number of Lem. 2.10(b), and the centred moments converge to $\sigma^{\ell,\mathsf{b}}(X) = \sum_\pi \prod_{B \in \pi} \mathcal{S}_{|B|}(X_B)$, the sum over the partitions π of the entries without singletons; $\sigma^{\ell,\mathsf{b}}(\emptyset) = 1$, $\sigma^{\ell,\mathsf{b}}((I)) = 0$. With $\bar{O}_K^T$, eq. 2.xxii holds for bounded Borel weights vanishing off closed sets Σ_a pairwise $\geq d$ apart, with $\prod_a |H_a|$ on the right and a number C^{av} of the same kind as C^{od}; the limit extends $\mathcal{S}_n$, and in each argument $A \mapsto \mathcal{S}_n(\dots, 1_A E, \dots)$, $E \in V$, is a countably additive measure on Σ_a vanishing on Lebesgue null sets, so it is given by a density in $L^1(\Sigma_a; V)$.

(b) $\mathcal{S}_n(rX) = \mathcal{S}_n(X)$ for $rx = Rx + v$, $R \in W_3$, $v \in \mathbb{R}^3$; $\sigma^{\ell,\mathsf{b}}(\theta \bar{F} \cdot F) \geq 0$ for the reflection θ in a plane $\{x_\mu = c\}$, acting on points and indices, and $F \in \mathfrak{F}(\{c < x_\mu < c + \frac{\ell}{2}\})$; at $H_a = f_a \mathbf{1}$, $\mathcal{S}_n$ is the limit of the truncated functions of the action density.

(c) A class insertion over a closed set Σ is a finite sum $\mathcal{E}$ of terms $\mathcal{O}(\phi 1_\lambda) = a_K^{-1} \sum_{\lambda(p)=\lambda} \phi(x(p)) a_p$ and $\mathcal{Q}_{l,l';\xi}(\phi)$ with real Lipschitz ϕ supported in Σ; its symbol $\operatorname{Sym} \mathcal{E}$ is the sum of the weights $\phi E_{\lambda\lambda}$, resp. $\phi\, s(l) s(l') (E_{\lambda(l)\lambda(l')} + E_{\lambda(l')\lambda(l)})$, of its terms, and $\|\mathcal{E}\|$ is the sum of their numbers $\sup|\phi| + \operatorname{Lip}\phi$. There is C^{u} depending only on n, g, ℓ, d such that for $K \geq K_1 + 1$ and class insertions $\mathcal{E}_a$, $\mathcal{E}'_a$ over closed sets pairwise $\geq d$ apart with $\operatorname{Sym} \mathcal{E}_a = \operatorname{Sym} \mathcal{E}'_a$,

$$|\langle \mathcal{E}_1; \dots; \mathcal{E}_n \rangle_K - \langle \mathcal{E}'_1; \dots; \mathcal{E}'_n \rangle_K| \leq C^{\mathrm{u}} K^{\mathsf{p}} L^{-\theta_n K} \prod_a (\|\mathcal{E}_a\| + \|\mathcal{E}'_a\|),$$

and the Lipschitz numbers drop out of the bound if $\mathcal{E}_a$, $\mathcal{E}'_a$ carry the same weights at the same sites. So $\lim_K \langle \mathcal{E}_1; \dots; \mathcal{E}_n \rangle_K = \mathcal{S}_n(\operatorname{Sym} \mathcal{E}_1, \dots, \operatorname{Sym} \mathcal{E}_n)$: the bond form, the clover form and the single corners give one field, and $\operatorname{Sym} O_K^T(H) = H$.

(d) Let (H_1, H_2) be a pair at scale $d < 1$ with $3d \leq |x_0 - y_0| \leq \frac{\ell}{4}$. There are a threshold $s_1(g, \ell)$ on the depth and a number $\mathsf{e}_{g,\ell}^T(d)$, tending to 0 like $(\log \frac{1}{d})^{-2\mathfrak{p}_0}$, such that if $s(d) \geq s_1(g, \ell)$, then for Lipschitz pairs, and for bounded ones with $\bar{O}_K^T$,

$$|\mathcal{S}_2(H_1, H_2) - \mathsf{b}^{-2} \mathcal{W}(H_1, H_2)| \leq \mathsf{e}_{g,\ell}^T(d)\, \mathsf{b}^{-2} |H_1||H_2|, \tag{2.xxiii}$$

and the same holds for class insertions over the two balls, with $\operatorname{Sym} \mathcal{E}_a$ for H_a. At finite cutoff, for $(K, \beta) \in \mathfrak{d}(g)$ with $K \geq s(d) + 2$ and $4a_K \leq d - d_-$, and for continuous pairs, eq. 2.xxiii holds for $\langle O_K^T(H_1); O_K^T(H_2) \rangle_{K,\ell,\beta}$ with $a_K \beta$ for b and the further error $\mathsf{b}^{-2}[c_K |H_1||H_2| + \frac{9}{8} \mathrm{rm}_K(H_1, H_2)]$, where $c_K = 28000\, C_{\mathrm{lat}}(1 + C_{\mathsf{k}})(a_K/d)^2$ with C_{k} basic, and $\mathrm{rm}_K \geq 0$ depends only on K, d, the two weights and the choice of O_K^T or $\bar{O}_K^T$, and tends to 0 as $K \to \infty$. One may take $s_1(g, \ell) \leq \hat{s}(g, \ell) := \max\{s_\flat(g), \lceil 2\Lambda_\ell / \log L \rceil\}$, with a depth $s_\flat(g)$ depending only on g and $\Lambda_\ell = (12 \log \ell / (c' A_0^2))^{1/(2\mathfrak{p}_0)}$, c' basic; from this depth on, the error of the finite-cutoff law against the lattice Wick value of [7, Lem. C.10] is independent of ℓ. Further, under the same conditions on d: (i) $|\mathcal{S}_2(rH_1, rH_2) - \mathcal{S}_2(H_1, H_2)| \leq 10\, \mathsf{e}_{g,\ell}^T(d) \mathsf{b}^{-2} |H_1||H_2|$ for $rx = x_0 + R(x - x_0)$, $R \in O(3)$; (ii) for the reflection θ in a plane $\{x_\mu = c\}$, a point z at distance $\frac{3}{2}d$ from it, a Lipschitz function ϕ with $1_{B(z,d/128)} \leq \phi \leq 1_{B(z,d/64)}$ and $E \in V$: $\mathcal{S}_2(\theta(\phi E), \phi E) \geq (\mathsf{w}_0 - \mathsf{e}_{g,\ell}^T(d)) \mathsf{b}^{-2} \mathrm{tr}\, E^2$; so no non-zero real combination of the

six fields vanishes. And at every d: (iii) for $0 < d_- < d < 1$, $\frac{3}{2}d \le c - x_{0,\mu} \le \frac{\ell}{2} - \frac{3}{2}d$ and real Lipschitz H vanishing off $\bar B(x_0, d_-)$, $0 \le \mathcal{S}_2(\theta H, H) \le 3600\, C_3^*(g,\ell)\mathsf{b}^{-2}|H|^2$, with $C_3^*(g,\ell) \le c_3(g)(1+\log\ell)^8 e^{16\Lambda_\ell}$ independent of d.

(e) For $q \ge 2$, $\eta \in C^2(T_\ell;\mathbb{R}^3)$ and Lipschitz $H_2,\dots,H_q$, the q supports pairwise $\ge d$ apart: $\mathcal{S}_q(H^{\mathrm{cons}}[\eta], H_2,\dots,H_q) = 0$, and $\sigma^{\ell,\mathsf{b}}((H^{\mathrm{cons}}[\eta])\cdot X') = 0$ for $X' \in \mathfrak{S}(T_\ell)$ off $\operatorname{supp}\eta$. There are C, p, K_0 depending only on g,ℓ,q,d such that for all $(K,\beta) \in \mathfrak{d}(g)$ with $K \ge K_0$ and all bounded H_a,

$$|\langle O_K^T(H^{\mathrm{cons}}[\eta]); O_K^T(H_2); \dots; O_K^T(H_q)\rangle_{K,\ell,\beta}| \le C K^{\mathsf{p}} L^{-K/56} \|\eta\|_{C^2} \prod_{a\ge 2} |H_a|,$$

also with $\bar O_K^T$; the bound is uniform in the label, and it stays true, with another C, if $O_K^T(H^{\mathrm{cons}}[\eta])$ is changed by a site form plus a plaquette form, carried within $2a_K$ of $\operatorname{supp}\eta$, whose weights are at most $36 a_K \|\eta\|_{C^2}$. At $H_a = h_a \mathbf{1}$ this is the conservation of the stress tensor among action densities.

(f) The numbers p, C^{od}, C^{av} and K_1 of (a) may be taken the same at all the labels of a compact interval $J \subset I(g)$ (Lem. 2.11(b)), and $\mathsf{b} \mapsto \mathcal{S}_n^{\ell,\mathsf{b}}(X)$ is continuous on $I(g)$. For Lipschitz weights, and for bounded ones with $\bar O_K^T$: if b is an inner point of $I(g)$ and $a_K\beta_K \to \mathsf{b}$, then $\langle O_K^T(H_1);\dots;O_K^T(H_n)\rangle_{K,\ell,\beta_K} \to \mathcal{S}_n^{\ell,\mathsf{b}}(X)$ along the full sequence of cutoffs.

By [7, Lem. C.8], (a)–(f) hold on every torus of side $2L^m$, $m \ge 0$, and at every label $\mathsf{b} > 0$, with the constants of a triple $(g', L^z\mathsf{b}, L^z\ell)$ with (T), taken at the distance $L^z d$, and powers of L^z; for large z one may take $g' = (L^z\mathsf{b}/2)^{-1/2}$, and then $L^z\mathsf{b}$ is an inner point of $I(g')$. ■□

The remaining estimate for overlapping sources. In the setting (T), consider the source derivatives of the step maps on their admitted analytic domains, with the weighted local-family norms and source sizes of the construction. For two bond forms Y_a, Y_b whose weights overlap, one still needs to show that the derivative of order two created at depth u decomposes into a local part $\mathfrak{l}$ of norm at most cL^u and a remainder of norm at most $cK^{\mathsf{p}}L^{u/2}$, in the units of the two source sizes. The local part vanishes at the trivial background. At the real background $U_{u'}$ of a small block field of a coarser level, of depth $u' \le u$, one also needs the evaluated sum estimate

$$\left|\sum_{X,z} \mathfrak{l}(X, U_{u'}, z)\right| \le K^{\mathsf{p}}\mathsf{b}^{-2} N L^{-2(u-u')},$$

where the sum runs over every polymer and base point and N is the number of sites of that coarser level. The family-norm bounds on the decomposition hold on the admitted analytic domains; the last estimate is required at these real backgrounds. The constants may depend on the number of derivatives, the fixed torus and a compact label interval, but not on the cutoff or the depths. The sources in the overlapping cases are the off-diagonal bond forms in clause (b) below.

The cancellation behind the evaluated sum estimate concerns the linear parts of the two insertions in the fluctuation field. If these parts have coefficient vectors r_a, r_b, their pairing through the fluctuation covariance C_u satisfies

$$|\langle r_a, C_u r_b\rangle| \le \|C_u\|_{2\to 2}\|r_a\|_2\|r_b\|_2.$$

Thus a uniform covariance bound and the estimates

$$\|r_a\|_2^2 \le C K^{\mathsf{p}}\mathsf{b}^{-2} N L^{-2(u-u')}\|\phi_a\|_\sharp^2$$

give the required estimate, after absorbing fixed constants and the two source sizes. The proposed gain comes from writing the linear curvature response as a discrete curl and summing by parts: a difference then falls on the product of the weight and the background curvature. Over one lattice spacing at depth u, the local gauge bounds at depth u' give the factor $L^{-(u-u')}$ in the L^2 norm, whose square gives the displayed gain. To complete this argument for the actual off-diagonal step

insertions, one still needs the quantitative decomposition above and the identification and bounds of its linear coefficient vectors, including the remaining nonlinear terms. These are the additional input here; the local gauge estimate and the step-map Cauchy estimate alone do not supply that identification.

Theorem B (Coinciding points, overlapping insertions, more points)**.**

(a) Assume (H), $\ell \in \mathfrak{L}(g)$, $(K,\beta) \in \mathfrak{d}(g)$, $0 < d < 1$ and $s_{\mathrm{th}}(n,g) \le s(d) \le K$, s_{th} a threshold depth depending on n and g. For complex weights whose supports lie within the Euclidean distance $0.87d$ of points x_a and are at pairwise Euclidean distances $\ge 1.26d$, with no bound on their mutual distances,

$$|\langle O_K^T(H_1);\dots;O_K^T(H_n)\rangle_{K,\ell,\beta}| \le 60^n C_n(g,\ell)\big[2\log\tfrac{1}{d}+C(g)\big]^{8n\mathfrak{p}_0}\mathsf{b}^{-n}\prod_a |H_a| :$$

the power counting d^{-3} of an entry, the dimension of $\operatorname{tr} FF$, up to logarithms. On a side that is small over the canonical sequence of (K,β) (equation YM3_I:(2312)), for $g \le \gamma_W^{\mathrm{sd}}$ the least side ℓ_g (Lem. YM3_I:126.3), the number $C_n(g,\ell)$ is free of ℓ; on every other $\ell \in \mathfrak{L}(g)$ it carries the factor $\max\{1,\mathsf{c}(\ell)\}^n$, $\mathsf{c}(\ell) = 2\ell^3 e^{\mathsf{B}(\bar{x}_0)\ell^3}/(c'A_0^2)$, with the function B and the number c' of equation YM3_I:(2311). Read at the reference coupling γ_W^{sd} after a dilation, the bound takes a scale-covariant form: let $\delta_\star = [L((\gamma_W^{\mathrm{sd}})^{-2}+\bar{B})]^{-1}$ and $\beta > X_{\gamma_W^{\mathrm{sd}}}$, let $z \in \mathbb{Z}$ be fixed by the first-passage window of β at γ_W^{sd}, so that $\delta_\star \mathsf{b} \le L^{-z} < (\gamma_W^{\mathrm{sd}})^2\mathsf{b}$, and let $L^z\ell$ be the least side at γ_W^{sd}; then for every $d < \delta_\star\mathsf{b}$ with $s_{\mathrm{th}}(n,\gamma_W^{\mathrm{sd}}) \le s(L^z d) \le K - z$, the depth and the threshold being formed at γ_W^{sd}, the bound holds with $[2\log\frac{\mathsf{b}}{d}+C]^{8n\mathfrak{p}_0}$ in the place of $[2\log\frac{1}{d}+C(g)]^{8n\mathfrak{p}_0}$ and numbers depending only on n and the basic constants, and this statement is invariant under $(K,\ell,\mathsf{b},d) \mapsto (K-1,L\ell,L\mathsf{b},Ld)$.

For supports only at pairwise Euclidean distances $\ge d$ the bound holds at $d'' = \min\{\frac{3}{4}d, L^{-s_{\mathrm{th}}(n,g)}\}$, for $K \ge s(d'')$, with $\sum_Q \sup_Q |H_a|$ in the place of $|H_a|$ and b^{-n} kept, Q running through the boxes of a partition of T_ℓ into at most $(2\ell/d'')^3$ boxes of sides $\le \frac{99}{100}d''$: each box lies within the Euclidean distance $0.87d''$ of its centre and pieces of two weights are at Euclidean distances $\ge d \ge 1.26d''$, so every product of the pieces $H_a\mathbf{1}_Q$ has the geometry of the first display at d''. In the amalgam norm $d''^3\sum_Q \sup_Q |H_a|$ this is the bound with $\mathsf{b}d''^3$ in the place of b: the power d^{-3} of an entry.

At $n=2$ and for $g \le \gamma_W^{\mathrm{sd}}$ the off-diagonal components need no logarithm and no number of the side: for every $\ell \in \mathfrak{L}(g)$, $(K,\beta) \in \mathfrak{d}(g)$, $h > 0$ with $1.31h < 1$ and $s(1.31h) \le K$, and complex weights $\mathsf{w}_1,\mathsf{w}_2$ of the bonds with x_b in closed cubes Q_1,Q_2 of side $\frac{3}{2}h$ at a sup distance $\ge \frac{5}{2}h$ wherever $\mathsf{w}_i(b) \ne 0$, no bound being asked on the distance of the cubes, the bond forms $Y_i = a_K^{-1}\sum_b \mathsf{w}_i(b)q_b$ have $|\langle Y_1;Y_2\rangle_{K,\ell,\beta}| \le 36\, C_\infty(g)\,\mathsf{b}^{-2}\sup|\mathsf{w}_1|\sup|\mathsf{w}_2|$, with the number $C_\infty(g)$ of [7, Lem. C.7], which depends on g alone; so the off-diagonal part of $\mathcal{S}_2^{\ell,\mathsf{b}}$ has the power counting of [7, Prop. C.22] with $q=0$ and a number A free of ℓ.

The bound passes to $\mathcal{S}_n^{\ell,\mathsf{b}}$ for weights vanishing off closed sets in balls of radius $0.86d$ at pairwise Euclidean distances $\ge 1.27d$, with a number depending on ℓ. So $\mathcal{S}_n^{\ell,\mathsf{b}}$ has the power counting of [7, Prop. C.22] with $q = 8n\mathfrak{p}_0$, a number $A = A_n(g,\ell)\mathsf{b}^{-n}$ and $h_0 \in \ell 2^{-\mathbb{N}}$ with $h_0 \le \min\{\frac{\ell}{64n}, \frac{1}{2}L^{-s_{\mathrm{th}}(n,g)}\}$: it extends over all the diagonals to a distribution $\tilde{\mathcal{S}}_n$ of order $\le 3n-2$ with the same power counting and one more logarithm. The lattice functionals $\tilde{u}_K$ formed from the cumulants at the cutoff K with the subtractions of [7, Prop. C.22], the sum running over the products of Whitney cubes of the sides h with $s(1.52h) \le K$ alone, obey $|\tilde{u}_K(F)| \le C_E A\,\mathfrak{l}(\lambda)^{8n\mathfrak{p}_0+1}\|F\|_{(\lambda),3n-2}$, one number for all K, for $8h_K \le \lambda \le h_0$, $h_K \le 2C_W(K)a_K$ the least side admitted at the cutoff K, and every F supported in a product of closed cubes of side 2λ, wherever their centres lie; and $\tilde{u}_K(F) \to \tilde{\mathcal{S}}_n(F)$ for every smooth F.

Within this power counting the extension is unique exactly up to the sums $\sum_{\pi,\alpha} \partial^\alpha \delta_\pi \otimes T_{\pi,\alpha}$ over the partitions π into at most $n-1$ blocks and $|\alpha| \le 3|\pi|$. At two points the real, exchange-symmetric, translation-invariant contact terms in $\mathfrak{E}_2(0)$ form a twelve-dimensional space under W_3 on $\mathbb{R}^3$ or T_ℓ, and a six-dimensional space under $O(3)$ on $\mathbb{R}^3$, by Lem. 3.6(f). $\mathcal{S}_2$ is no measure across the diagonal.

(b) The convergence rates in this clause remain targets until the actual step insertions satisfy the overlapping-source estimates described in §2.3.3. Under (T) let $r \ge 2$ tensor entries $O_K^T(\phi_a E_a)$, $E_a \in V$, and n action densities $\mathcal{O}(f_i)$ be given, all with Lipschitz weights, and let $d > 0$. Their joint cumulants κ_K converge as $K \to \infty$ at a rate: $|\kappa_{K+1} - \kappa_K| \le CK^{\mathsf{p}} L^{-cK} \prod_a \|\phi_a\|_\sharp |E_a| \prod_i \|f_i\|_\sharp$ in the norms $\|\phi\|_\sharp = \max\{\sup|\phi|, 40M \operatorname{Lip}\phi\}$, $|E|$ the largest modulus of the entries of E, for K beyond a threshold, with C, p depending on r, n, ℓ and a compact interval of labels about b and the threshold depending besides on the geometry of the supports (on d in the cases where d is named), in the following cases:
  (i) $c = \frac{1}{120r}$ if the tensor supports are pairwise $\ge d$ apart, nothing being asked of the supports of the f_i, which may meet one another and the tensor supports
  (ii) $c = \frac{1}{120r}$ also if $\phi_a \phi_b = 0$ for $a \ne b$, the tensor weights not overlapping (their supports may touch), provided that of any $r+n$ points, one from each support, the supports of the f_i included, two are $\ge d$ apart, so that no point is common to all the supports (at $n = 0$ the numbers C, p depend on d as well)
  (iii) $c = \frac{1}{360r}$ for off-diagonal E_a, $r \ge 3$, if no three tensor supports share a point, nothing being asked of the f_i
  (iv) $c = \frac{1}{864}$ for $r = 2$, off-diagonal E_a and $n \ge 1$, the two tensor supports overlapping at will, if of any $n+2$ points, one from each support, two are $\ge d$ apart.

(c) The arbitrary-order interacting conclusion remains a target. It requires the remainder estimate described in §5.1.2, with its real plaquette weights, first-creation bound, numerical ranges and error e_n. Keep $36d < \ell$. The cyclic formula is exact for three free lattice photon fields; the corresponding interacting conclusion is the following. For the plane fields, with real Lipschitz ϕ_a in the open balls $B(x_a, d)$, these in a ball of radius $\le \frac{\ell}{4}$, $\operatorname{Lip}\phi_a \le |\phi_a|/(\vartheta_0 d)$ and $|x' - y'| \ge d$ for $x' \in \operatorname{supp}\phi_a$, $y' \in \operatorname{supp}\phi_b$, $a \ne b$ (Euclidean distances, as for the w_a): $|\langle \mathcal{O}_{\lambda_1}(\phi_1); \dots; \mathcal{O}_{\lambda_n}(\phi_n)\rangle_{K,\ell,\beta} - 3\mathsf{b}^{-n} K_{\underline{\lambda}}^{(n)}(\phi)| \le [\mathsf{e}_n + C_n(\frac{a_K}{\vartheta_0 d} + \frac{d^3}{\ell^3})]\mathsf{b}^{-n} \prod_a |\phi_a|$, $C_n = \frac{3}{2} n! [6^n C_{\mathrm{lat}}(1 + 3C_{\mathrm{lat}})^{n-1} + 43 \cdot 11^n]$; the same with $3^n C_n$ for real combinations $\sum_\lambda k_{a,\lambda} \mathcal{O}_\lambda(\phi_a)$, $k_a \in [-1,1]^3$, among them the diagonal entries $\mathcal{O}(\phi) - 2\mathcal{O}_m(\phi) = O_K^T(\phi s_{mm})$ of the stress tensor.

For $2 \le n \le 7$ and $g \le \min\{\gamma_n, \gamma_W^{\mathrm{sd}}\}$, with d below a threshold depending on n and g and the balls $B(x_a, d)$ in one ball of radius $9d$, the lattice inequality in §5.1.2, with $\mathcal{K}_K^{(n)}$ formed on T_{ℓ_g} and with $\varepsilon_n^{\mathrm{rt}}(g,d)$ of [7, Lem. C.16](a) added to e_n, holds also in every periodic Gibbs state and on every torus of $\mathfrak{L}(g)$; $\varepsilon_n^{\mathrm{rt}}(g,d)$ tends to 0 with d like a positive power of d times a power of $\log \frac{1}{d}$.

(d) Let g be below a basic threshold. On the least torus, for $n = 2, 3, 4$ and a label $\mathsf{b} \in I(g)$, the limit $\mathcal{S}_n^{\ell_g, \mathsf{b}}$ of Thm. A(a) has on each set of points named below a Borel kernel: $\mathcal{S}_n^{\ell_g,\mathsf{b}}(f_1\mathbf{1}, \dots, f_n\mathbf{1}) = \int s_n \prod_a f_a$ for continuous f_a whose supports form a product inside that set, s_n the kernel of the truncated n-point function of the action density, and $\mathcal{S}_2^{\ell_g,\mathsf{b}}(f_1 E_{\lambda\lambda}, f_2 E_{\mu\mu}) = \iint \tau_{\lambda\mu} f_1 f_2$ at $0 < |x - y| < d_\sigma(g)$. For a limit σ at g the kernel of three action densities is $s_3(\cdot; \mathbf{1}, \mathbf{1}, \mathbf{1})$ of [7, Thm. BW](d), the third centred moment being the third cumulant, and $\tau_{\lambda\mu} = s_2(\cdot; E_{\lambda\lambda}, E_{\mu\mu})$; for a component σ_ξ they are the kernels of its re-centred σ_ξ^c. On T_{ℓ_g} almost everywhere, and for every limit at g and every re-centred

component at every such triple of points, the kernel s_3 of three action densities obeys

$$|s_3 - 3\mathsf{b}^{-3}k_3| \le \varepsilon_A(g,d)\,\mathsf{b}^{-3}r^{-3}d^{-6} \quad \text{for } 0<|x-y|=r\le d=\operatorname{dist}(z,\{x,y\})\le d_A(g),$$
$$\varepsilon_A \le C(g)\Big[\big(\log\tfrac{1}{d}\big)^{39\mathfrak{p}_0+2}d^{1/2}+\Big(\frac{d}{\ell_g}\Big)^3\Big],$$

uniformly in $\frac{r}{d}\in(0,1]$, $d_A(g)>0$ depending on g. At a cutoff the same law holds for narrow weights, in every periodic Gibbs state and on T_{ℓ_g}: for real plaquette weights w_1,w_2,w_3 carried by sets $S_a\subset B(x_a,\rho)$ with $\operatorname{dist}(S_1,S_2)\ge r$, $|x_1-x_2|\le 8r$, $\operatorname{dist}(S_a,S_3)\ge d$ and $|x_a-x_3|\le 18d$ for $a=1,2$, where $0<\rho\le r\le d$, d below a threshold depending on g, and $s(\rho)\le K$ (the geometry of [7, Lem. C.15](c)), the third cumulant of the forms $a_K^{-1}\sum_p w_a(p)a_p$ is $3\mathsf{b}^{-3}K_K^{(3)}(w_1,w_2,w_3)$ up to $[C(g)Y^{39\mathfrak{p}_0+2}d^{1/2}+C(g)d^{3/2}]\rho^9r^{-3}d^{-6}\mathsf{b}^{-3}\prod_a|w_a|$, $Y=\log\bar{x}_{s(d)}$; for Lipschitz bumps $w_a=f_a\circ x$ with $\operatorname{Lip}f_a\le|f_a|/(\vartheta\rho)$, $0<\vartheta\le 1$, the same holds against $3\mathsf{b}^{-3}\int k_3\prod_a f_a$ with $C(g)(\frac{a_K}{\vartheta\rho}+(\frac{d}{\ell_g})^3)$ more in the bracket. For $d=|c-z|\le\frac12 d_A(g)$ and $0<r\le\frac{d}{2}$, almost everywhere on the torus and at every (c,z) where the kernels are continuous, in particular for limits and re-centred components,

$$\Big|\tfrac{2\pi}{3}\mathsf{b}r^3 s_3\big(c+\tfrac{r}{2}e_\lambda,c-\tfrac{r}{2}e_\lambda,z\big)-t_\lambda(c,z)\Big|\le\Big[\varepsilon'(g,d)+2.7\frac{r^2}{d^2}\Big]\mathsf{b}^{-2}d^{-6},\qquad \varepsilon'(g,d)\le C(g)\big(\log\tfrac{1}{d}\big)^{39\mathfrak{p}_0+2}d^{1/2},$$

$t_\lambda=\sum_\mu(\tau_{\lambda\mu}-\frac13\sum_{\lambda'}\tau_{\lambda'\mu})$, the kernel of the traceless diagonal component $E_{\lambda\lambda}-\frac13\mathbf{1}$ against the action density, formed with the kernels $\tau_{\lambda\mu}$ of the same torus, limit or component: the tensor is recovered, to leading order, from two action densities that approach one another along e_λ, the right member being small against the natural size $\mathsf{b}^{-2}d^{-6}$ as $d\to0$ and $r/d\to0$. Four action densities at two scales are free in the same sense, on the torus and at a cutoff: on T_{ℓ_g}, for $r\le|x_1-x_2|,|x_3-x_4|\le 2r$ and $r\le d=\operatorname{dist}(\{x_1,x_2\},\{x_3,x_4\})\le d'_A(g)$, the kernel s_4 of the truncated four-point function of the action density has $|s_4-3\mathsf{b}^{-4}k_4|\le\varepsilon_4(g,d)\mathsf{b}^{-4}r^{-6}d^{-6}$ almost everywhere, $\varepsilon_4\le C(g)[(\log\frac1d)^{47\mathfrak{p}_0+2}d^{1/2}+(\frac{d}{\ell_g})^3]$, uniformly in $\frac{r}{d}\in(0,1]$; and at a cutoff, in every periodic Gibbs state and on T_{ℓ_g}, for real plaquette weights $w_1,\dots,w_4$ in two such narrow pairs $\{1,2\}$, $\{3,4\}$, with $\operatorname{dist}(S_3,S_4)\ge r$, $|x_3-x_4|\le 8r$, $\operatorname{dist}(S_a,S_b)\ge d$ and $|x_a-x_b|\le 18d$ for $a\le2<b$, d below a threshold depending on g, the fourth cumulant of the plaquette forms is $3\mathsf{b}^{-4}K_K^{(4)}(w_1,\dots,w_4)$ up to $[C(g)Y^{47\mathfrak{p}_0+2}d^{1/2}+C(g)d^{3/2}]\rho^{12}r^{-6}d^{-6}\mathsf{b}^{-4}\prod_a|w_a|$.

■□

Corollary 2.9 (A tensor against the action density)**.** Let σ be a limit at g under (H), or a re-centred component σ^c_ξ. There are real-analytic functions A,B on $(0,\infty)$ with $s_2(x,y;e,\mathbf{1})=\operatorname{tr}\big(e\,[A(r)\mathbf{1}+B(r)\hat{u}\hat{u}^t]\big)$, $u=x-y$, $r=|u|$, for all $e\in V$; $F=3A+B$ is the kernel of two action densities, and

$$B'(r)+\frac{3}{r}B(r)=\frac14F'(r),\qquad\text{so}\qquad B(r)=r^{-3}\Big(c+\frac14\int_1^r t^3F'(t)\,\mathrm{d}t\Big)$$

for one real number c. ■□

Lemma 2.10 (Exact symmetries; separated insertions)**.** (a) Let $rx=Rx+v$, with $R\in W_3$, $Re_\mu=s_\mu e_{\pi(\mu)}$, and $v\in a_K\mathbb{Z}^3$, act on configurations by moving the bonds. Then $rO_K^T(H)=O_K^T(rH)$ for every complex weight, on $\Lambda_{K,\ell}$ and on $\mathbb{L}_K$; in components $ra_p=a_{rp}$ and $rq_b=s_{\lambda_b}s_{\lambda'_b}q_{rb}$. Under a law ν invariant under these maps, as the torus measures and the periodic Gibbs states are (Prop. YM3_I: 103.4), $\nu(q_b)=0$ and $\nu(a_p)=e_\nu$ does not depend on p, so that $\nu(O_K^T(H))=e_\nu a_K^{-1}\sum_p H_{\lambda(p)\lambda(p)}(x(p))$: the off-diagonal components need no subtraction; moments and cumulants are invariant. $O_K^T(H)$ depends only on the bonds within a_K of a point where $H\neq0$.

(b) Assume (H), $\ell \in \mathfrak{L}(g)$, $q \geq 2$, $\delta > 0$. In the calculus of sources of Thm. YM3_I:D let $C_{(m)}$, $m \geq 1$, be the numbers, depending only on m, with which the mixed derivative families of order m of a source system under majorants are bounded at the level t by $C_{(m)}L^{4m(T-t)}$ times the product of the sizes of the sources, T the working level; put $K_r = \sum_\pi \prod_{B\in\pi} C_{(|B|)}$, the sum over the partitions of $\{1,\dots,r\}$; and let $h_q(s)$ be the largest left member, at the depth s, of the smallness conditions that this calculus asks of the families of the orders $2,\dots,q$ for majorants at least $3M$ units of the working level apart, a function of the depth depending only on q and g and tending to 0 as $s \to \infty$.

Let $s_q(\delta)$ be the least $s \geq 0$ with $L^s\delta \geq 3M$ and $h_q(s) \leq 1$; at $s = s_q(\delta)$ put $M_r = K_r[1 + L^{4r}V_\ell(4q)L^{-(4q+1)s}]$, where $V_\ell(b) = 4M^{-3}\ell^3 e^{\mathsf{B}(X_g)\ell^3}\mathsf{C}_{\rm ag}(b+1,\eth)$, with the function B and the number $\mathsf{C}_{\rm ag}$ of equation eq. 4.xxxii at a fixed $\eth \in (0,1)$, so that $K_rL^{4r}V_\ell(b)L^{-(b+1)s}$, $b \geq 4r$, bounds, in units of the sizes, the part of a centred mean of r first members that is carried by the histories in which a large field is created below the working level (Prop. 4.54); and

$$K_q^{\rm ub}(g,\ell,\delta) = (4\ell^3L^{4s})^q \sum_\pi (|\pi|-1)! \prod_{B\in\pi} M_{|B|},$$

the sum over all partitions of $\{1,\dots,q\}$. Let $(K,\beta) \in \mathfrak{d}(g)$ with $K \geq s_q(\delta)+2$ and $3a_K \leq \delta$, and let each $\mathcal{F}_a$ be one of: $O_K^T(H_a)$ or $\bar O_K^T(H_a)$ with H_a vanishing off a closed set Σ_a (size $\mathsf{s}_a = 60|H_a|$); a site form with weights at the sites within $\frac12 a_K$ of Σ_a (size $23\|v_a\|$); $\mathcal{O}(w_a)$ with $w_a(p) = 0$ unless $x(p)$ is within a_K of Σ_a (size $|w_a|$); or a sum of two of these (the sum of the sizes); the Σ_a being pairwise $\geq 2\delta$ apart. Then

$$|\langle \mathcal{F}_1;\dots;\mathcal{F}_q\rangle_{K,\ell,\beta}| \leq K_q^{\rm ub}(g,\ell,\delta)\prod_a \mathsf{s}_a.$$

■

Proof sketch. (a) $\operatorname{tr}U^{-1} = \operatorname{tr}U$ on $SU(2)$ gives $ra_p = a_{rp}$, and $w(U^{-1}) = -w(U)$ gives the signs in rq_b; inserted into [7, eq. 1] they give $rO_K^T(H) = O_K^T(rH)$, as $(rH)_{\pi\lambda,\pi\lambda'}(rx) = s_\lambda s_{\lambda'}H_{\lambda\lambda'}(x)$. The reflection through b normal to λ_b lies in the group, fixes b and reverses q_b; the group is transitive on the plaquettes. (b) By [7, Lem. C.4] each $\mathcal{F}_a$ is the first member of a source, in the calculus of Thm. YM3_I:D, under the majorant made of its size and of the plaquettes of its form (for $O_K^T(H)$ the size is $1+59$, for its plaquette and its bond part), and so, the notion being monotone in the majorant, under the majorant made of its size and of the plaquettes whose corners lie within $\frac32 a_K$ of Σ_a; as $3a_K \leq \delta$, corners of plaquettes of different majorants are $\geq \delta$ apart. At the depth $s = s_q(\delta)$, $L^s\delta$ exceeds the range of the block terms, and the mean of a centred product of r first members is at most $M_r\prod_a 4\ell^3L^{4s}\mathsf{s}_a$, up to constants common to all the subproducts, which cancel in cumulants; the Möbius formula for cumulants gives $K_q^{\rm ub}$; the number grows with ℓ and as $\delta \to 0$, since only the sizes of the sources enter. □

Lemma 2.11 (Two successive cutoffs)**.** Assume (T). An entry is one of $\mathcal{O}(w)$, $\mathcal{O}(w1_\lambda)$,

$$a_K^{-1}\sum_{b\|\mu} w(x_b)q_b$$

and $\mathcal{Q}_{l,l';\xi}(w)$, with a real function w read at every cutoff at the barycentres of the plaquettes, at the midpoints of the bonds, resp. at the shifted sites. (a) eq. 2.xxii holds for n entries with supports pairwise $\geq d$ apart, without the factor 6^n and with $|w_a|$ for $|H_a|$; and for entries whose weights are the cell means of bounded Borel functions, with $\prod_a |w_a|$ on the right. (b) Every compact interval $J \subset I(g)$ is an interval of labels under (Tor) on every side $\ell \in \mathfrak{L}(g)$ (§2.3.1): its labels exceed $g^{-2} \geq \gamma^{-2}$, and $MR(x) \leq L^{m_{\mathfrak{L}}(g)-1}$ for $x \leq L(g^{-2}+\bar B)$ by YM3_I:(1528). With $C_{\rm bf}$ and $K_{\rm bf}$ formed at J (Def. 2.7(5)), which serve the comparison of Cor. 2.8(a) at every label of J, the numbers p, $C^{\rm od}$ and $C^{\rm av}$ of Thm. A(a) serve all labels of J, and so does K_1, formed with $K_{\rm bf}$ of J and at the least label of J. (c) Let (t,ε_r,V) be data of the comparison of the cutoffs K and $K+1$

(the setting preceding Lem. YM3_I:113.8): a level $1 \le t \le K$ of the cutoff K with $L^{K-t}\mathsf{b} \ge \Theta_\star$, the radius

$$\varepsilon_r = 2c_\rho \varepsilon(L^{K-t}\mathsf{b})$$

of that Definition, and a block field V of the depth $K-t$ with $\varpi(V) < \varepsilon_r$; let U^A and U^B be its backgrounds at the level t of the cutoff K and at the level $t+1$ of the cutoff $K+1$, and $\eta_t = L^{-t}$, $N_t = \ell^3 L^{3(K-t)}$. The *corner form* of a weight v on the sites and of loops l, l' at a configuration U of a lattice is $\eta^{-1}\sum_x v(x) Q_x(l,l')(U)$, with $\eta = \eta_{t+1}$ on $\Lambda_{K+1,\ell}$ at U^B and $\eta = \eta_t$ on $\Lambda_{K,\ell}$ at $\mathcal{M}U^B$ and at U^A; as $a_K^{-1} = L^{K-t}\eta_t^{-1}$ and

$$a_{K+1}^{-1} = L^{K-t}\eta_{t+1}^{-1},$$

an entry is L^{K-t} times a corner form, and $L^{K-t}N_t = \ell^3 L^{4(K-t)}$ gives the factor $L^{4\sigma}$ below, $\sigma = K-t$. B_3 is the fixed background regularity constant of Thm. YM3_I:E; its bound at these data is $B_3\varepsilon_r$, as used in the comparison preceding Lem. YM3_I:113.8. For loops l, l' and l_1, l_1' in the same planes and complex v on the sites,

$$\Big|\eta_{t+1}^{-1}\sum_x v(x)\big[s(l)s(l')Q_x(l,l') - s(l_1)s(l_1')Q_x(l_1,l_1')\big](U^B)\Big| \le 2K_0^{\mathrm{cf}} \sup|v|\, \varepsilon_r^2 N_t L^{-t/2},$$

K_0^{cf} basic: at a background the corner forms depend, in the mean, on their planes and signs alone. (d) Let U^A be the background of the coarser lattice and $\mathcal{M}U^B$ the block average of U^B. For weights v_f, v_c on the sites of the fine and of the coarse lattice, with $v_{\max} = \max\{\sup|v_f|, \sup|v_c|\}$ and the mismatch $m = \max_y |L^{-3}\sum_{x\in B(y)} v_f(x) - v_c(y)|$, $B(y)$ the block of y:

$$\Big|\eta_{t+1}^{-1}\sum_x v_f(x) Q_x(l,l')(U^B) - \eta_t^{-1}\sum_y v_c(y) Q_y(l,l')(\mathcal{M}U^B)\Big| \le [4B_3^2 m + K_0^{\mathrm{cf}} v_{\max} L^{-t/2}]\varepsilon_r^2 N_t;$$

and $\mathcal{M}U^B$, a nearly minimal competitor at the level t, has $|\eta_t^{-1}\sum_x v(x)[Q_x(l,l')(\mathcal{M}U^B) - Q_x(l,l')(U^A)]| \le 12\sup|v|\,(C\,t\,A\,\Delta A)^{1/2}$, A its action and ΔA its excess. So an entry with a Lipschitz weight, read at the two cutoffs, changes over one block field of depth σ by at most $C\ell^3 L^{4\sigma} K^{1/2} L^{-(K-\sigma)/2}(\sup|w| + L^{-\sigma}\mathrm{Lip}\, w)$, and an entry whose weights at the two cutoffs are tied by the block mean by the same with $\sup|w|$ alone. ∎

Proof sketch. (a) The entries are first members of sources under majorants ([7, Lem. C.4]), of the sizes $\sup|w|$, $59\sup|w|$, resp. $23\sup|w|$, and the corners of the plaquettes of two majorants are $\ge d - 3a_K \ge d' = \frac{d}{2}$ apart. *First link, at one cutoff.* By Thm. YM3_I:D a source system of n members is carried by its first members read at the backgrounds of the block field of a depth σ up to the error

$$2\mathfrak{K}_n'[2L^{4n}V_\ell(b)L^{-(b+1)s} + n\,\epsilon_{\mathrm{fg}}(\sigma - s) + 2e^{3-L^s d'/M}](4\ell^3 L^{4s})^n \prod_a \mathsf{s}_a$$

at a second depth $s < \sigma$, for every real $b \ge 4n$, with V_ℓ and M_r of Lem. 2.10(b) formed at the depth s and $\mathfrak{K}_n' = \sum_\pi |\pi|! \prod_{B\in\pi} M_{|B|}$: the first term is carried by the histories in which a large field is created below the working level of the depth s (Prop. 4.54); the second is the forgetting of what the members of the sources make below the level $K-\sigma$, $\epsilon_{\mathrm{fg}}(\nu) = C_{\mathrm{fg}}(\nu+1)^{2\mathfrak{p}_0+2}L^{-\nu/2}$ with C_{fg} basic; the third is the decay over the distance d' in the units of the working level; no number of it reads K or β. *Second link, between the cutoffs.* The laws of the block field of the depth σ at the cutoffs K and $K+1$ differ in total variation by at most $C_{\mathrm{bf}}\ell^3 K^{\mathfrak{p}_0} L^{-K/10}$, $\mathfrak{p}_0 = 2\mathfrak{p}_0 + \frac{1}{2}$ the exponent of Def. 2.7(5) (Cor. 2.8(a)); over one block field an entry read at the two backgrounds changes by the defect of (d); and a first member read at a background is at most $\frac{3}{2}B_3^2\ell^3 L^{4\sigma}$ times its size.

With $\sigma = \lfloor K/(1260n)\rfloor$, $s = \lfloor \sigma/(8n+3)\rfloor$ and $b = 4n+1$ the three errors are at most $L^2 L^{-\theta_n K}$, $L^{-K/3}$ and $L^{-K/11}$ times the sizes, up to powers of K and of ℓ: the first as $\sigma - s \ge (8n+2)s$ makes each term of the bracket at most a power of K times $L^{-(4n+1)s}$, and $L^{-s} \le L^2 L^{-\theta_n K}$; the second as $L^{4n\sigma}L^{-(K-\sigma)/2} \le L^{-K/3}$; the third as $L^{4n\sigma}L^{-K/10} \le L^{-K/11}$: this is θ_n. A Lipschitz weight read at

the points of two lattices has the mismatch $m \le \frac{3}{2} a_K \operatorname{Lip} w$, whence the term $L^{-\sigma} \operatorname{Lip} w$; cell means have $m = 0$, as the cell of a coarse site is the union of the cells of the sites of its block, whence sup norms. (b) The first link reads the label only through the data of the depth s of $(K, L^K \mathsf{b})$, whose numbers are uniform on the compact J, the numbers of its error being free of the label.

Otherwise the label enters the proof of (a) only through C_{bf} and K_{bf}, which depend on it only through J (Cor. 2.8(a)), through the condition $L^\sigma \mathsf{b} \ge \Theta_\star$ of the data of the comparison at the depth σ, which holds at every label of J once it holds at the least one, and through the radius ε_r of the backgrounds, which the calibration conditions in equation YM3_I:(2148) bound at the levels where $L^\sigma \mathsf{b} \ge \Theta_\star$, and of which the defects of (c), (d) read only $\varepsilon_r \le 1$. (c) Over a block, in a suitable gauge, each loop at the background is $\pm$ one plaquette variable of the block, depending only on the plane of the loop, up to $O(\varpi^2)$, ϖ the size of the background; the variance of the plaquette variables of a block is bounded, plaquette by plaquette, by the defect of Jensen's inequality, whose sum is the action that one averaging step loses, and this is controlled by the weighted actions of the backgrounds (Prop. YM3_I:113.9). (d) The first display is (c) carried across one averaging step, the block mean tying the weights; for the second, $\mathcal{M} U^B$ is nearly minimal at the level t (Props. YM3_I:113.5, YM3_I:113.7), and a nearly minimal competitor has nearly the corner forms of the background, by its energy; this gives the factor $t^{1/2}$ and, with $t \le K$, the defect. □

Step 1: the tori, (a)–(c) *and* (f). By [7, eq. 1], $O_K^T(H) = \sum_\lambda \mathcal{O}(H_{\lambda\lambda} 1_\lambda) + \sum_{\lambda<\lambda'} O^{\mathrm{od}}_{\lambda\lambda'}(H_{\lambda\lambda'})$, $O^{\mathrm{od}}_{\lambda\lambda'}(\phi) = -a_K^{-1} \sum_{b \| \mu} \phi(x_b) q_b$ for μ the third direction, with weights supported in $\operatorname{supp} H$, of sup norm $\le |H|$ and Lipschitz number $\le \operatorname{Lip} H$; so a cumulant of the $O_K^T(H_a)$ is a sum of 6^n cumulants of entries. Lem. 2.11(a) gives eq. 2.xxii, and summing over $K' \ge K$ the limit with its rate; the bound of $\mathcal{S}_n$ is Lem. 2.10(b) at $\delta = \frac{d}{2}$; the formula for the centred moments holds at every K.

With $\bar{O}_K^T$ the cumulant at fixed K is a polynomial in finitely many cell means of the H_a, each continuous under bounded pointwise convergence, and the rate is uniform on balls of the sup norm: so the limit is continuous under bounded pointwise convergence, which makes it countably additive in each argument, and it vanishes when all the cell means vanish, in particular on null sets; the theorem of Radon–Nikodym gives the density. For (b), Lem. 2.10(a) gives the invariance under $a_K \mathbb{Z}^3 \rtimes W_3$ at every cutoff; a real and a lattice translation of a weight differ by a weight of sup norm $\le a_K \operatorname{Lip} H$, and the bounds are in sup norms. As L is odd, $\{x_\mu = c\}$ with $c \in a_{K_0}(\mathbb{Z} + \frac{1}{2})$ is a bond mirror of all the lattices with $K \ge K_0$, at which Wilson's measure is reflection positive (Cor. YM3_I:85.3, Prop. YM3_I:91.4) and $\theta O_K^T(H) = O_K^T(\theta H)$; such c are dense, and the reflection at another c is conjugate to one of these by a translation.

At $H_a = f_a \mathbf{1}$, $O_K^T(f_a \mathbf{1}) = \mathcal{O}(f_a)$. For (c): every class insertion is, at the rate, its standard form, built from the corner fields with the loops $l_\lambda = (\kappa, \kappa')$, $\kappa < \kappa'$ the two directions other than λ, at the sites. The shift ξ costs a site form of norm $\le \frac{1}{2} a_K \operatorname{Lip} \phi$, bounded by Lem. 2.10(b); loops in one plane are exchanged, with the sign $s(l) s(l')$, inside the rate of Lem. 2.11(a), the difference being tied at the depth σ with the defect of Lem. 2.11(c); and $a_p = \frac{1}{2} Q_x(l_\lambda, l_\lambda) + \frac{1}{2} a_p^2$, whose second term is of fourth order at the backgrounds. The standard form depends on the symbol alone; $O_K^T(\operatorname{Sym} \mathcal{E})$ is a class insertion with the same symbol, and (a) gives the limit; with the same weights at the same sites no shift and no Lipschitz number occur.

For (f), take the numbers C_{bf}, K_{bf} of Def. 2.7(5) at a compact interval $J \subset I(g)$ that holds b in its interior: by Cor. 2.8(a) they read the label only through J, and the comparison of the block fields of two successive cutoffs holds with them at every label of J. The rate of (a) reads the label at these places alone: C_{bf} and K_{bf}; the condition $L^\sigma \mathsf{b} \ge \Theta_\star$, which is monotone in the label; the radius of the backgrounds, which the calibration conditions in equation YM3_I:(2148) bound at the levels where $L^\sigma \mathsf{b} \ge \Theta_\star$; and the data of the depth s of $(K, L^K \mathsf{b})$, whose numbers are uniform on the compact J.

So, by Lem. 2.11(b), one exponent p, one C^{od}, one C^{av} and one threshold K_1 serve all labels of J, for the entries and for the cell means. At a fixed K, $\mathcal{S}_{n;K}$ is continuous in the label, an integral

over a compact group against a density continuous in $\beta = L^K \mathsf{b}$; so $\mathcal{S}_n^{\ell,\cdot}$ is a uniform limit on J of functions continuous in the label, hence continuous on J; with $\mathsf{b}_K = a_K \beta_K \in J$ for large K,

$$|\mathcal{S}_{n;K}(\mathsf{b}_K) - \mathcal{S}_n(\mathsf{b})| \le |\mathcal{S}_{n;K}(\mathsf{b}_K) - \mathcal{S}_n(\mathsf{b}_K)| + |\mathcal{S}_n(\mathsf{b}_K) - \mathcal{S}_n(\mathsf{b})| \to 0,$$

as in Cor. YM3_I:117.1; with $\bar{O}_K^T$ the same.

Step 2: every label and side; all of (H). [7, Lem. C.8](a) gives

$$\langle O_K^T(H_1); \dots; O_K^T(H_n)\rangle_{K,\ell_*,L^K\mathsf{b}_*} = L^{zn}\langle O_{K-z}^T(H_1^{(z)}); \dots; O_{K-z}^T(H_n^{(z)})\rangle_{K-z,L^z\ell_*,L^{K-z}(L^z\mathsf{b}_*)},$$

which carries Thm. A to every label and side, with $L^z d$ for d; with $g' = (L^z \mathsf{b}/2)^{-1/2}$ the label $L^z\mathsf{b}$ is an inner point of $I(g')$, which carries (f). Lems. [7, Lem. C.12] and [7, Lem. C.13], [7, Prop. C.21] for its limits and its growth, and Steps 6, 7 and 10–13 of the proof of [7, Thms. BU–BX] are first carried out at couplings $g \le \gamma_W^{\mathrm{sd}}$; [7, Lem. C.8](b) carries the data, the states, the limits, the Hilbert spaces, the semigroups, [7, eq. 263] and the decomposition to all of (H), and $\sigma = L^{nz}\sigma_z(X^{(z)})$ carries the bounds of [7, Prop. C.21](b), (c) and its growth, the powers of L^z entering C_G and C; the count of [7, Prop. C.21](a) at a cutoff holds on all of (H) with the numbers set in its statement. In particular one may take $d_0(g) = L^{-z}\min_i d_0(g_i')$, with the error $\max_i \mathsf{e}_{g_i'}^{\infty}(L^z d)$ at d, for z large enough at g.

Step 3: two points on one torus. By [7, Lem. C.10](a) and $|K_3(u)_{\lambda\lambda'}| \le (2\pi|u|^3)^{-1}$, $|u| \ge d$ on the supports, the Wick value is a Riemann sum of $\mathcal{W}$: with $a_K^3 \sum_{p,p'} |\mathfrak{t}_H(p,p')| \le 45 d^3 |H|$, by the row sums $2\sum_{p'} |\mathfrak{t}_H(p,p')| \le 3|H|$ of the kernel of $O_K^T(H)$ (Step 14(a)) and the counts of the plaquettes and the bonds in a ball,

$$\Big|6\sum \mathfrak{t}_1\mathfrak{t}_2 CC - \mathcal{W}(H_1,H_2)\Big| \le 12150\, E_1(\tfrac{1}{\pi} + E_1)|H_1||H_2| + \mathrm{rm}_K(H_1,H_2),$$

where rm_K is the difference between a Riemann sum over the anchors, that is the barycentres, the midpoints or the sites of the cells, and the integral $\mathcal{W}$; it depends only on K, d and the weights, and it tends to 0, by the uniform continuity of K_3 on the two balls, for continuous weights and for cell means. With $|\mathcal{W}| \le 54|H_1||H_2|$ ([7, Lem. C.9](a)) and the bounds on v^2, eq. 2.xxiii follows at finite cutoff with

$$\mathsf{e}_{g,\ell}^T(d) = \tfrac{9}{8}\big[3600\,\varepsilon_{g,\ell}^e(s) + 12150\, E^{\infty}(\tfrac{1}{\pi} + E^{\infty})\big] + 162\bar{B}g^2L^{2-s}, \qquad E^{\infty} = E_1 - C_{\mathrm{lat}}(a_K/d)^2,$$

and $\frac{9}{8}\cdot 12150\,[E_1(\frac{1}{\pi}+E_1) - E^{\infty}(\frac{1}{\pi}+E^{\infty})] \le c_K$; the term $162\bar{B}g^2L^{2-s}$ is 54 times $3\bar{B}g^2L^{2-s}$. As $K \to \infty$, (a) gives eq. 2.xxiii; $\mathsf{e}^T \to 0$ like $(\log\frac{1}{d})^{-2\mathfrak{p}_0}$ because $Y_s^{2\mathfrak{p}_0}\varepsilon_{g,\ell}^e(s)$ is bounded and $t = L^s d \ge C_W(s) \to \infty$. The class: (c) at $n = 2$. The bound on $s_1(g,\ell)$ and the independence of ℓ: [7, Lem. C.10](b). (i): $\mathcal{W}$ is invariant ([7, Lem. C.9](a)) and $|rH| \le 3|H|$, so the law at (H_1,H_2) and at (rH_1,rH_2) gives $(1+9)\mathsf{e}^T$. (ii): [7, Lem. C.9](b) and the law at the pair $(\theta(\phi E),\phi E)$, whose centres are $3d$ apart, with $|\phi E|^2 \le \mathrm{tr}\, E^2$. (iii): [7, Lem. C.10](c) at $c \in a_{K_0}(\mathbb{Z} + \frac{1}{2})$ and the limit; a general c by (b).

Step 4: conservation on the torus. Cut η by a partition of unity into pieces of diameter < 1; by multilinearity it suffices to treat one piece. In its slot, by [7, eq. 264], $\mathsf{b}O_K^T(H^{\mathrm{cons}}[\eta]) = \mathrm{St}_K[\eta] - \Delta_K[\eta]$, and $\mathrm{St}_K[\eta]$ may be replaced by $-\mathrm{Rm}_K[\eta]$, the null terms $\mathcal{N}_\eta$, $\mathcal{N}_\eta'$ vanishing against the far entries. This leaves the remainders of [7, Prop. C.17](a): the cubic one $\mathcal{S}[\mathsf{b}\eta]$, the five local forms of J_η, the form $\frac{3}{2}a_K\mathcal{O}(w_\eta)$ and the forms of $\Delta_K[\eta]$, each with a conditional mean, given the block field of depth ϕK, that is constant on T_ℓ up to $K^{\mathfrak{p}}L^{-(\frac{1}{28}-3\phi)K}\|\eta\|_{C^2}$: by [7, Prop. C.17](c) for the cubic one; by power counting at the backgrounds of the level $K - \phi K$, as in the proof of [7, Prop. C.17](b), for the local forms, those of the order 2 (the three of J_η, with weights at most $(3+5\mathsf{b})\|\eta\|_{C^2}$, the form $\frac{3}{2}a_K\mathcal{O}(w_\eta)$ and the two forms of $\Delta_K[\eta]$ with weights at most $60\mathsf{b}a_K \sup|\nabla^2\eta|$) having oscillations at most $CK^{2\mathfrak{p}_0}L^{4\phi K - K}\|\eta\|_{C^2}$, and the two of the orders 3 and 4 of J_η, with weights at most $L^K\mathsf{b}\sup|\nabla\eta|$, oscillations at most $CK^{3\mathfrak{p}_0}L^{4\phi K - \frac{K}{2}}\|\eta\|_{C^2}$; and by the cut at the level $K - \lceil K/14\rceil$ for the form of vanishing symbol of $\Delta_K[\eta]$ alone.

A joint cumulant of entries far apart is at most the product of the oscillations of their conditional means at that level, a spectator costing $L^{4\phi K}$; at $\phi = 1/(56(4q-1))$ the exponent is $-\frac{1}{28}+(4q-1)\phi = -\frac{1}{56}$. (a) gives the limit, and the moments vanish, being sums of products of cumulants of which one holds the slot of $H^{\mathrm{cons}}[\eta]$. The estimate of [7, Prop. C.17](c) is uniform on the interval of labels, which gives the uniformity in the label; $\bar{O}_K^T$ as in Step 1; a change by forms with weights $\le 36 a_K \|\eta\|_{C^2}$ costs, by Lem. 2.10(b), a_K times a number independent of K. At $H_a = h_a \mathbf{1}$ the spectators are action densities.

The rate $L^{-K/36}$ also holds by the argument of [7, proof of Thms. BU–BX, Step 8] on T_ℓ.

Step 14: coinciding points, overlaps and more points. (a): $O_K^T(H)$ is the first member of a source of Thm. YM3_I:D under a majorant of size $60|H|$, carried by the plaquettes p with $H_{\lambda(p)\lambda(p)}(x(p)) \neq 0$ and by the plaquettes through the bonds b with $H_{\lambda_b\lambda'_b}(x_b) \neq 0$ (Lem. 2.10(b)): its plaquette part has the size $|H|$, and its bond part $-a_K^{-1}\sum_b H_{\lambda_b\lambda'_b}(x_b)q_b$ the size $59|H|$. Indeed q_b is a bilinear form in the vector parts $w(W_{x,(\mu_b,\alpha)})$, $\alpha \in \{\pm\lambda_b, \pm\lambda'_b\}$, of the four plaquette loops at the bond $b = \langle x, x + a_K e_{\mu_b}\rangle$; for a form $-a_K^{-1}\sum_b \mathsf{w}(b)q_b$ the members at the height 0 and at every level are carried by the at most six cubes with a common corner that hold the four plaquettes through b, and they are bounded there as those of a plaquette source of the weight $59\sup|\mathsf{w}|$ on these plaquettes; to the second order at a small field the form is a_K^{-1} times the quadratic form $-\sum_b \mathsf{w}(b)\langle \mathsf{u}_{b,\lambda_b}, \mathsf{u}_{b,\lambda'_b}\rangle$ in the plaquette variables, $\mathsf{u}_{b,\lambda}$ linear forms with the coefficients $0, \pm\frac{1}{2}$, whose symmetric kernel has the row sums $2\sum_{p'}|\mathfrak{t}(p,p')| \le 2\sup|\mathsf{w}|$, so that the kernel $\mathfrak{t}_H$ of $O_K^T(H)$ has $2\sum_{p'}|\mathfrak{t}_H(p,p')| \le 3|H|$.

So at the depth $s(d)$ an insertion of size d is a source of size $v|H|$ in the units of the working level, and Thm. YM3_I:D bounds the cumulant of n such sources, wherever they lie, by $60^n k'_n(g,s)\mathsf{b}^{-n}\prod_a |H_a|$ on a side that is small over the canonical sequence of (K,β) (equation YM3_I:(2312)), for $g \le \gamma_W^{\mathrm{sd}}$ the least side ℓ_g (Lem. YM3_I:126.3), the bound reading only the sizes of the sources; $k'_n(g,s) \le C_n(g) Y_s^{8n\mathfrak{p}_0}$, of the $8\mathfrak{p}_0$ of one entry $2\mathfrak{p}_0$ coming from the supremum of the plaquette action, in units of the working level, on the small fields, and $6\mathfrak{p}_0$ from the number $(LC_W(s))^3$ of the unit cubes of the working level in a ball of radius d; and $Y_{s(d)} \le 2\log\frac{1}{d} + C(g)$.

On every other side $\ell \in \mathfrak{L}(g)$ the histories with a large field on T_ℓ, summed against their weights (Prop. 4.54), add the factor $\max\{1, \mathsf{c}(\ell)\}^n$, $\mathsf{c}(\ell) = 2\ell^3 e^{\mathsf{B}(\bar{x}_0)\ell^3}/(c' A_0^2)$, with the function B and the number c' of equation YM3_I:(2311), as V_ℓ does in Lem. 2.10(b): this is the ℓ of $C_n(g,\ell)$. The scale-covariant form is the same bound read at the coupling γ_W^{sd} after the dilation of [7, Lem. C.8] by L^z: $(K-z,\beta)$ is a first-passage datum at γ_W^{sd} with the label $L^z\mathsf{b} \in ((\gamma_W^{\mathrm{sd}})^{-2}, L((\gamma_W^{\mathrm{sd}})^{-2} + \bar{B})]$, which gives $\delta_\star \mathsf{b} \le L^{-z} < (\gamma_W^{\mathrm{sd}})^2 \mathsf{b}$, so that $\log\frac{1}{L^z d}$ differs from $\log\frac{\mathsf{b}}{d}$ by at most a basic number, and the factor L^{zn} of the dilation turns $(L^z\mathsf{b})^{-n}$ into b^{-n}; the depth and the threshold are those of the dilated distance at γ_W^{sd}, and the shift $(K,\ell,\mathsf{b},d) \mapsto (K-1, L\ell, L\mathsf{b}, Ld)$ is one more step of the dilation.

Supports only at pairwise Euclidean distances $\ge d$ are cut into the boxes Q of sides $\le \frac{99}{100}d''$ of a partition of T_ℓ into at most $(2\ell/d'')^3$ boxes, $d'' = \min\{\frac{3}{4}d, L^{-s_{\mathrm{th}}(n,g)}\}$: a piece $H_a 1_Q$ lies within the Euclidean distance $\frac{\sqrt{3}}{2}\cdot\frac{99}{100}d'' < 0.8574\, d'' \le 0.87\, d''$ of the centre of Q, and pieces of different entries are at Euclidean distances $\ge d \ge \frac{4}{3}d'' \ge 1.26\, d''$; the bound at the distance d'' for each product of pieces, summed by the n-linearity, the sum of the products being the product of the sums, gives $\prod_a\big(\sum_Q \sup_Q |H_a|\big)$ in the place of $\prod_a |H_a|$, at the same b^{-n}: that is $(\mathsf{b}d''^3)^{-n}$ times the product of the norms $d''^3\sum_Q \sup_Q |H_a|$, a factor d''^{-3} for each entry.

At $n = 2$, for $g \le \gamma_W^{\mathrm{sd}}$, every $\ell \in \mathfrak{L}(g)$, $1.31h < 1$ and $s(1.31h) \le K$, two off-diagonal bond forms $Y[\mathsf{w}_\iota] = -a_K^{-1}\sum_b \mathsf{w}_\iota(b)q_b$ whose bonds have their midpoints in cubes of side $\frac{3}{2}h$ at the sup distance $\ge \frac{5}{2}h$ have $|\langle Y[\mathsf{w}_1]; Y[\mathsf{w}_2]\rangle_{K,\ell,\beta}| \le 36\, C_\infty(g)\mathsf{b}^{-2}\sup|\mathsf{w}_1|\sup|\mathsf{w}_2|$, with the number $C_\infty(g)$ of [7, Lem. C.7], of g alone. This rests on [7, Lem. C.7], applied at $d = 1.31h$: at every $0 < d < 1$ with $s(d) \le K$, two real bond forms of one direction each, with $|x_b - x_\iota| \le d - 2a_K$ on $\operatorname{supp}\mathsf{w}_\iota$

and $|x_1 - x_2|_\infty \ge 3d + a_K$, with no upper bound on that distance, have $|\langle Y[\mathsf{w}_1]; Y[\mathsf{w}_2]\rangle_{K,\ell,\beta}| \le C_\infty(g)\mathsf{b}^{-2} \sup|\mathsf{w}_1| \sup|\mathsf{w}_2|$.

The means vanish, by Lem. 2.10(a); a bond mirror parts the two forms at the distance $\ge \frac{3}{2}d$ from both points, and reflection positivity there (Cor. YM3_I:85.3, Prop. YM3_I:91.4), that is Schwarz's inequality for the positive form, bounds the cumulant by the square roots of the two mirror pairs $\langle \theta Y[\mathsf{w}_\iota]; Y[\mathsf{w}_\iota]\rangle$. By the transfer operator of the normal direction, positive by reflection positivity, a mirror pair is, as a function of the distance t of the form from the mirror, a sum of exponentials with positive coefficients, symmetric under $t \mapsto \frac{\ell}{2} - t$, as the reflection fixes a second plane of the torus at the distance $\frac{\ell}{2}$; so on $[\frac{3}{2}d, \frac{\ell}{2} - \frac{3}{2}d]$ it is largest at $t = \frac{3}{2}d$, where the form and its image are at the distance $3d$.

The proof of [7, Lem. C.7] bounds such a mirror pair on every torus of $\mathfrak{L}(g)$ and at every $K \ge s(d)$ by a number of g alone times b^{-2} and the sizes, the cutoffs $K = s(d)$ and $K = s(d) + 1$ among them; for $d < d_0(g)$ and $K \ge s(d) + 2$ this is also the range of [7, Lem. C.12]: on every torus of $\mathfrak{L}(g)$ the pair has its Wick value of [7, Lem. C.10](a) on T_{ℓ_g} up to a number of g alone times v^2 and the sizes, and the Wick value of two entries of size d at the distance $3d$ is at most a basic multiple of $v^2 \le \frac{9}{8}\mathsf{b}^{-2}$ times the sizes, by $|C(p,q)| \le Ca_K^3 d^{-3}$ and $a_K^3 \sum_{p,p'} |\mathfrak{t}_H(p,p')| \le 45d^3|H|$, as in [7, Lem. C.10](c); for $d \ge d_0(g)$ the forms are cut into pieces of a size below $d_0(g)$, whose pairs are bounded so, in a number that depends on g alone; and where K lies below the depth that these pieces ask, the crude bound $c\, a_K^{-4}$ of a form serves, as in [7, Lem. C.10](c).

At $d = 1.31h$, x_ι the centres of the cubes: $|x_b - x_\iota| \le 1.2991h \le d - 2a_K$ and $|x_1 - x_2|_\infty \ge 4h \ge 3d + a_K$; and a complex weight of the bonds is the sum of six weights, each i^ϵ times a real weight carried by the bonds of one direction, of sup norm at most that of the weight: thirty-six pairs. The bound passes to $\mathcal{S}_n$ by Thm. A(a) for $\bar{O}_K^T$ and continuity, which gives the power counting of [7, Prop. C.22] with $q = 8n\mathfrak{p}_0$ and $h_0 \in \ell 2^{-\mathbb{N}}$ with $h_0 \le \min\{\frac{\ell}{64n}, \frac{1}{2}L^{-s_{\rm th}(n,g)}\}$; [7, Prop. C.22] gives the extension and the contact terms; the same subtractions made at each cutoff, with the products of cubes restricted to those of a side h with $s(1.52h) \le K$, give lattice functionals that are bounded uniformly in K and converge to the extension.

The counts at two points are those of Lem. 3.6(f): take real, exchange-symmetric, translation-invariant contact terms in $\mathfrak{E}_2(0)$. They are $\sum_{|\alpha|\le 3} \partial^\alpha \delta(x-y)\, c_\alpha(e,e')$ with bilinear c_α, symmetric under the exchange of the two entries; under $O(3)$ on $\mathbb{R}^3$, which contains $-\mathbf{1}$, the odd orders drop, and there remain $\mathrm{tr}\, e\, \mathrm{tr}\, e'$ and $\mathrm{tr}(ee')$ at the order 0 and the contractions of $\partial_i\partial_j$ with $\delta_{ij}\mathrm{tr}\, e\, \mathrm{tr}\, e'$, $\delta_{ij}\mathrm{tr}(ee')$, $e_{ij}\mathrm{tr}\, e' + e'_{ij}\mathrm{tr}\, e$ and $(ee' + e'e)_{ij}$ at the order 2: six; under W_3, on $\mathbb{R}^3$ or T_ℓ, there are twelve. $\mathcal{S}_2$ is no measure across the diagonal: at the scale d a unit cube holds $\gtrsim d^{-3}$ disjoint mirror pairs of balls, each of mass $\ge (\mathsf{w}_0 - \mathsf{e}^T)\mathsf{b}^{-2}\mathrm{tr}\, E^2$ by Thm. A(d)(ii). (b): the action densities stay in the exponent: their weights form the source $w_z = \sum_i z_i f_i$ of the sourced bare density $\rho_0^{w_z} = \exp[-\beta A_K + \sum_i z_i \mathcal{O}(f_i)]$ of equation YM3_I:(10), which Thm. YM3_I:D carries through all the levels as a source at the plaquettes; the tensor entries, in the third and fourth cases the bond forms $Y_a = -a_K^{-1} \sum_{b\|\mu_a} \phi_a(x_b) q_b$ of the off-diagonal labels E_a, enter through their sources of Lem. 2.10(b) and are derivatives at the strength 0: $\kappa_K = \partial_{z_1} \cdots \partial_{z_n} \partial_{\tau_1} \cdots \partial_{\tau_r} \log \int \exp[\sum_a \tau_a Y_a]\, \rho_0^{w_z}\, \mathrm{d}U$ at $z = 0$, $\tau = 0$; and likewise, in the first two cases, where the tensor weights do not overlap, for diagonal labels with the plaquette forms $Y_a = a_K^{-1} \sum_p \phi_a(x(p))(E_a)_{\lambda(p)\lambda(p)} a_p$, and for every $E_a \in V$ with $Y_a = O_K^T(\phi_a E_a)$, by [7, eq. 1] the plaquette form of the diagonal part of E_a plus the bond forms of its off-diagonal part.

Differentiated at the strength 0 along the tensor entries, the step maps of Thm. YM3_I:D give a recursion for the members of each order, each derivative of a step being bounded by Cauchy's estimate in the strengths. The input of §2.3.3 is used at order two: for two bond forms Y_a, Y_b whose weights overlap, one renormalisation step at the depth u creates a part $\mathfrak{l}$ bounded by cL^u and a remainder bounded by $cK^{\mathfrak{p}}L^{u/2}$ in the units of the sizes, and $\mathfrak{l}$, read at a background of a lower depth u', is at most $K^{\mathfrak{p}}\mathsf{b}^{-2}NL^{-2(u-u')}$, N the number of the sites of the level of that depth:

the arrays of order two born above a depth are forgotten below it, and this carries the case of two overlapping entries.

The remaining step is to establish this second-derivative estimate, including its stated forgetting bound, for these bond forms; and it is here that the third and fourth cases use that the labels E_a are off-diagonal. Where the tensor supports are $\geq d$ apart, a member of order ≥ 2 vanishes unless its polymer reaches across the distance d; where only $\phi_a\phi_b = 0$, the arrays of order $m \geq 2$ are at most $K^{\mathsf{p}}L^{-mu/2}$ in the units of the sizes, the weights not overlapping, and with the condition on the $r+n$ supports this gives the same rate. For $r \geq 3$ off-diagonal entries of which any two may overlap and no three supports share a point, each overlapping pair is carried by its arrays of order two, which are forgotten below their depth as for two entries; an array of order $m \geq 3$ joins m entries whose supports have no common point, so, the supports being compact, its polymer must cross a positive distance fixed by their geometry, and it is bounded as where the supports are apart; the depths proportional to K then give $c = \frac{1}{360r}$.

Two cutoffs are then compared as in Lem. 2.11, at depths proportional to K, which gives the rates. (c), with Euclidean distances between sets, here and in (d): at the depth $s(d)$ the n-th cumulant of the quadratic insertions in the Gaussian part is the sum over the cyclic orders, $3\mathsf{b}^{-n}K_K^{(n)}$, the value for three free lattice photon fields, for which the left member vanishes identically; the uniform interacting remainder is the estimate still to be established in §5.1.2, with its explicit first-creation term bounded by the weights ϖ_t; for the plane fields the free lattice cumulants differ from $3\mathsf{b}^{-n}K_{\underline{\lambda}}^{(n)}$ by at most $\frac{1}{3}C_n(\frac{a_K}{\vartheta_0 d} + \frac{d^3}{\ell^3})\prod_a |\phi_a|$, by the comparison of the lattice kernel with the free kernel (equations YM3_I:(2245)–(2246)).

Every periodic Gibbs state and every torus of $\mathfrak{L}(g)$, for $2 \leq n \leq 7$: establishing the remainder estimate on T_{ℓ_g} and [7, Lem. C.16](a), the weights being bounded functions of the plaquettes in balls of radius d within one ball of radius $9d$, give the lattice inequality in §5.1.2 with $\varepsilon_n^{\mathrm{rt}}(g,d)$ added to e_n, and $\varepsilon_n^{\mathrm{rt}}(g,d) \to 0$ with d since some $\sigma \in (0,1)$ has $4 - \frac{n}{2} - 3\sigma > 0$ exactly when $n \leq 7$. (d): [7, Lem. C.15](c) gives the laws at a cutoff on T_{ℓ_g}, the large fields being bounded as in [7, Lem. C.15](a), and the comparison of the free lattice cycles with k_3, k_4, by equations YM3_I:(2245)–(2246), gives the terms $(d/\ell_g)^3$; [7, Lem. C.16](c) carries the laws to every periodic Gibbs state, at three points with the further $C(g)d^{3/2}\rho^9 r^{-3}d^{-6}\prod_a |w_a|$ and at four with $C(g)d^{3/2}\rho^{12}r^{-6}d^{-6}\prod_a |w_a|$, $d^{3/2}$ times the natural sizes; first passing to a continuum subsequential limit at fixed positive widths, then letting the widths tend to 0 and using Lebesgue's differentiation theorem gives the kernels, with estimates uniform in the states, those of the truncated functions; the law of four points is that on the torus and at a cutoff; for limits and re-centred components σ_ξ^c, [7, Lem. C.14] and [7, Thm. BW](d), (f), the kernels being continuous, and for a limit the third centred moment being the third cumulant; the point-split law follows from the three-point law, [7, Lem. C.9](d) and the kernels of [7, Thm. BV](e), the constant 2.7 covering 2.66 and the passage from $\mathrm{dist}(z,\{x,y\})$ to $|c-z|$.

Proof of Cor. 2.9. By [7, Thm. BW](c), (d) the kernel $s_2(x,y;e,\mathbf{1})$ is real-analytic, a function of $u = x - y$, and linear in e, so $s_2(x,y;e,\mathbf{1}) = \mathrm{tr}(e\,\tau(u))$ with $\tau(u) \in V$; invariance under $O(3)$ gives $\tau(Ru) = R\tau(u)R^t$, so $\tau(u)$ commutes with the stabiliser of $\hat{u}$ and is $A(r)\mathbf{1} + B(r)\hat{u}\hat{u}^t$, and $\mathrm{tr}\,\tau = 3A + B = F$. Conservation against one action density, $\sigma((H^{\mathrm{cons}}[\eta]),(f\mathbf{1})) = 0$ for $\mathrm{supp}\,\eta$ off $\mathrm{supp}\,f$, reads $\iint[(\nabla\cdot\eta)(x)F(x-y) - 2\sum_{ij}\partial_i\eta_j(x)\tau_{ij}(x-y)]f(y)\,\mathrm{d}x\,\mathrm{d}y = 0$; integration by parts in x, for all such η and f, gives $2\sum_i \partial_i\tau_{ij} = \partial_j F$ off $u = 0$, that is $2(A' + B' + \frac{2}{r}B) = F'$, as $\sum_i \partial_i(B\hat{u}_i\hat{u}_j) = (B' + \frac{2}{r}B)\hat{u}_j$. With $A = \frac{1}{3}(F - B)$ this is $B' + \frac{3}{r}B = \frac{1}{4}F'$, and $(r^3B)' = \frac{1}{4}r^3F'$. □

Remark 2.12. The limits on $\mathbb{R}^3$ are taken along subsequences of Gibbs sequences, and each theorem speaks of the limit of the sequence chosen; the label, the bounds and the laws at short distance are the same for all of them. The uniqueness of the vacuum in [7, Thm. BW](f) is that of each component σ_ξ, and the vacuum of $\mathcal{H}(\sigma)$ is unique exactly when the index space X is a point. [7, Thm. BW] concerns limits through periodic Gibbs states; of the limits along growing tori [7, Thm. BV](c) is

asserted, with positivity at the coordinate mirrors alone. The laws at short distance are laws to leading order, with errors that tend to 0 with the distance, and Thm. A and Thm. B(a), (b) concern one torus at a time, with constants that depend on its side.

2.3.4. *Notation and hypotheses. Data.* The radius, the constants and the data are those of the main theorems (Prop. YM3_I:69.1 and Section YM3_I:82; the constants chosen in the order of that proof; the uniform free-response geometry, radius and decay conditions in Section YM3_I:52.6, retained in Section YM3_I:120.3; $c_\star$ by equation YM3_I:(1429)): L odd, M a power of L, A_0, A_1, $P = 2\mathfrak{p}_0 \geq 18$, $\eth \in (0,1)$, $c' = (1-\eth)c_\star$, the last constant γ, and the ceilings $\gamma_1 = \gamma_W^{\mathrm{sd}} \leq \gamma_2 = \gamma_W^{\mathrm{ref}} \leq \gamma$ of the choices in equations YM3_I:(2326)–(2327), with γ_W^{sd} further restricted by eq. 2.iii. $B_\sharp \geq 0$ is the flow bound: $|L^{-1}x_{j-1} - x_j| \leq B_\sharp$ at every step ([7, §G.2]), and $\bar{B} = B_\sharp L/(L-1)$ (Section YM3_I:74); both are numbers of the constants alone. T_ℓ, $\ell = 2L^m$, is the torus of side ℓ, $\Lambda_{K,\ell}$ its lattice of spacing $a_K = L^{-K}$, $\mathbb{L}_K = a_K\mathbb{Z}^3$; $|\cdot|$ and $|\cdot|_\infty$ are the Euclidean and the maximum norm, on a torus minimised over lifts; $\hat{u} = u/|u|$.

A first-passage datum (K,β) at the reference coupling g (YM3_I:74.2, YM3_I:74.1) has the running couplings $x_0 = \beta, \dots, x_K$, with $|L^{-1}x_{t-1} - x_t| \leq B_\sharp$ for $1 \leq t \leq K$ (Prop. YM3_I:74.1), so that, the differences telescoping, $|L^{-z}x_{K-z} - x_K| \leq B_\sharp \sum_{0\leq i<z} L^{-i} \leq \bar{B}$ for $0 \leq z \leq K$; $y_t = \log x_t$, $g_t = x_t^{-1/2}$, the large-field threshold $p_0(g_t)^2 = A_0^2 y_t^P$; the first passage $x_K \leq X_g < x_{K-1}$ YM3_I:74 at $X_g = L(g^{-2} + B_\sharp)$, so that $X_g + \bar{B} = L(g^{-2} + \bar{B})$ YM3_I:(1526); and the label $b = a_K\beta \in I^{\mathrm{lab}}(g) := [g^{-2}, \bar{x}_0]$, $\bar{x}_0 = L(g^{-2} + \bar{B}) = X_g + \bar{B}$; for b in the interval $I(g) = (g^{-2} + \bar{B}, X_g - \bar{B}] \subset I^{\mathrm{lab}}(g)$ of Section YM3_I:74, $X_g - \bar{B} = L(g^{-2} + \bar{B}) - 2\bar{B}$, $(K, L^K\mathsf{b})$ is a first-passage datum for every integer $K \geq 1$. $\mathfrak{L}(g) = \{2L^m : m \geq m_{\mathfrak{L}}(g)\}$ is the family of the sides at g (equation YM3_I:(1528)), $\ell' = 2\mathsf{l}$, $\mathsf{l} = L^{m_{\mathfrak{L}}(g)}$, its least member, $\mathsf{l} \geq LM$.

A side $\ell = 2L^m$ is *admissible* if $MR(x_K) \leq L^m$, $R(x)$ the least power of L that is at least $(\log x)^r$, r a constant; it is *small* if it is small over the running couplings in the sense of equation YM3_I:(2312): admissible, $LM \leq L^m$, $\ell \geq N_{\mathsf{H}}$, the lower bound of condition (H), and $\mathsf{B}(x_K)\ell^3 \leq \frac{1}{2}c' p_0(g_K)^2$, B the function equation YM3_I:(2311) of the volume factor $e^{\mathsf{B}\ell^3}$ (equation eq. 4.xxxii); for $g \leq \gamma_1$ the side ℓ' is small for every datum (Lem. YM3_I:126.3). As R does not decrease, $x_K \leq X_g < \bar{x}_0$ and $L^{m_{\mathfrak{L}}(g)} \geq LMR(\bar{x}_0)$ YM3_I:(1528), every side $2L^m \in \mathfrak{L}(g)$ has $LMR(x_K) \leq L^m$ for every datum at g: the sides of $\mathfrak{L}(g)$ are admissible.

Depth. $Y_s = (s+1)\log L + \log(g^{-2} + \bar{B}) = \log \bar{x}_s$, $\bar{x}_s = L^{s+1}(g^{-2} + \bar{B})$ and $C_W(s) = C_{W,0}A_0^2 Y_s^P$ as in equation YM3_I:(2321), with the room constant $C_{W,0} = \max\{1, 4L^2M/A_0^2\}$ (equation YM3_I:(2320)), so that $C_W(s) \geq 4L^2 M Y_s^P$ and $L^{-2}C_W(s) \geq 4MY_s^P \geq M$; $\mathsf{Lm}_s = L^2 C_W(s) Y_s^{3\mathfrak{p}_0}$, $x_s^- = 4L^{s-2}g^{-2}/(5\mathsf{Lm}_s)$. For $0 < d < 1$, $s(d)$ is the least s with $L^s d \geq C_W(s)$; so $Y_{s(d)} \leq 2\log\frac{1}{d} + C(g)$. At the working level $\mathsf{k} = \max\{K - s(d), 2\}$: $y = y_{\mathsf{k}}$, $\hat{d} = d/a_{K-\mathsf{k}}$, $\upsilon^{-1} = a_{K-\mathsf{k}}x_{\mathsf{k}}$, and $\upsilon^2 \leq \frac{9}{8}b^{-2}$; if $s(d) \leq K$, then $\hat{d} \geq L^{s(d)-2}d \geq L^{-2}C_W(s(d)) \geq M$ and $d \geq C_W(s(d))a_{s(d)} \geq 4L^2 M a_K \geq 676 a_K$.

The free template of Section YM3_I:120 has, for $0 \leq n \leq K$, the orthogonal projection $\Pi_n = \Pi_n^{\mathrm{tp}}$ onto the background space of the level n of the exact curvatures, with its kernel $\Pi_n(p,q)$ on the plaquettes (Section YM3_I:120); the free kernel is Π_0, the background kernel of the depth s' is $C_{K,s'}^{\mathrm{bg}} = \Pi_{K-s'}$, $0 \leq s' \leq K$, and the slice of the step j is $\mathcal{C}_{(j)} = \Pi_{j-1} - \Pi_j$, $1 \leq j \leq K$ YM3_I:(2227). So $\Pi_0 = \Pi_n + \sum_{1\leq j\leq n}\mathcal{C}_{(j)}$ for every n, and $\Pi_k - \Pi_T = \sum_{k<j\leq T}\mathcal{C}_{(j)}$ for $k \leq T$ YM3_I:(2227); in particular the free kernel is the sum of the background kernel Π_K of the top level and of all the slices.

With $\eta_n = L^{-n}$ and $\mathsf{r} = |x(p) - x(q)|_\infty$, $|\Pi_n(p,q)| \leq C_{\mathrm{tp}}^{\mathrm{bg}}\eta_n^3$ and $|\mathcal{C}_{(j)}(p,q)| \leq C_{\mathrm{tp}}\eta_{j-1}^3 e^{-\delta_{\mathrm{tp}}\mathsf{r}/a_{K-j+1}}$ (Props. YM3_I:120.5 and YM3_I:120.6). The background kernel of the working level is $C^{\mathrm{bg}} = C_{K,K-\mathsf{k}}^{\mathrm{bg}} = \Pi_{\mathsf{k}}$, and the free kernel is C^{bg} plus the slices $\mathcal{C}_{(j)}$, $j \leq \mathsf{k}$, the slice $\mathcal{C}_{(j)}$ decaying exponentially beyond the scale a_{K-j+1}. $K_3(u) = (3\hat{u}\hat{u}^{\mathrm{T}} - \mathbf{1})/(4\pi|u|^3)$.

Large fields; the chain. The chain has the levels $0 \le j \le K$, $\eta_j = L^{-j}$, local families with the sizes of YM3_I:(759), the geometry of Prop. YM3_I:45.1 and [7, Lem. O.1], and the contraction numbers $\theta^{[j]}$. A history is a sequence of large-field regions up to a step; $Z_t(h)$ is its region at the level t, $\tilde\nu_K(h) = Z^{-1}\int \mathfrak{w}_h \rho_0 \, \mathrm{d}U$ the law of the histories; w_t is the probability that the first large field is created at the step t, so that $\sum_{t'\le t} \mathsf{w}_{t'} = \tilde\nu_K(\{Z_t \ne \emptyset\})$; $\mathsf{W}_T = \sum_{t\le T} \mathsf{w}_t$. $\Phi^{(t)}[X]$ is the conditional mean of X given the block field of the level t on the history without large fields. For sources with majorants (Lem. 2.16) D_j is the least sup distance, in units of a_{K-j}, of corners of plaquettes of different sets S_a. For two sources at the working level, $\mathsf{k} < j_1 \le T$ are the middle and the upper level that go with $(\mathsf{k}, \hat d, y)$ (equation YM3_I:(2316)), $n_T = T - \mathsf{k}$, $L^{n_T} \ge \hat d y^{3\mathfrak{p}_0}$; for $\mathsf{L} \ge 0$, $c > 0$,

$$\mathrm{sm}_T^{\mathsf{L},c} = \max\Big\{x_T^{-1/2},\ x_{\mathsf{k}} e^{-d_*(T)},\ x_{\mathsf{k}} e^{-(\mathsf{D}_{\mathsf{k}}/M-2)},\ \max_{\mathsf{k}\le i<T}(\delta_i + \varepsilon_i^{\mathrm{co}}),\ x_{\mathsf{k}}^2 \mathsf{L} L^{3(K-\mathsf{k})} e^{-cp_0(g_T)^2}\Big\},$$

and sm_T is its value at $c = \frac12 c'$ and $\mathsf{L} = \mathsf{L}_\ell := 2M^{-3}\ell^3$, the number of the weights of a small side: by the rarity of large fields on one torus (Prop. 4.54), $\mathsf{w}_t \le \tilde\nu_K(\{Z_t \ne \emptyset\}) \le 2M^{-3}\ell^3 L^{3(K-t)} e^{-c' p_0(g_t)^2} e^{\mathsf{B}(x_K)\ell^3}$ for $1 \le t \le K$, for $g \le \gamma_2$, $K \ge 3$ and every admissible side, and on a small side $\mathsf{B}(x_K)\ell^3 \le \frac12 c' p_0(g_K)^2 \le \frac12 c' p_0(g_t)^2$, as $x_t \ge x_K$; so $\mathsf{w}_t \le \mathsf{L}_\ell L^{3(K-t)} e^{-\frac12 c' p_0(g_t)^2}$ and $\mathsf{W}_T \le \mathsf{L}_\ell L^{3(K-T)} e^{-\frac12 c' p_0(g_T)^2}$ there. The symbols $\alpha_0, \beta_s, \kappa, K_M, K_\Lambda, C_{\mathcal A}$ retain their meanings in the common choice of Prop. YM3_I:69.1 and Section YM3_I:82 and C_P is the number of Def. YM3_I:51.2; $\delta(t) = A_1 t^{-1/2}(\log t)^{\mathfrak{p}_0}$ and $\delta_i = \delta(x_i)$; $\varepsilon_i^{\mathrm{co}}$ is the error e_i^{co} defined before equation YM3_I:(2302), $d_*(T) = d_*(T; T, 2B_3\varepsilon_T)$ with the radius of eq. 2.ii, M_{pl} that of equation YM3_I:(2285); c_X, c_I, C_{M}, C_i, c_{sb} and c_{tl} are fixed in eq. 2.xxiv–eq. 2.xxv and the preceding paragraph; $\mathfrak{p}_0$ is the exponent of Cor. 2.8; C_{tp}, $C_{\mathrm{tp}}^{\mathrm{bg}}$, δ_{tp} are the numbers of Props. YM3_I:120.5 and YM3_I:120.6, C_{lat} that of equation YM3_I:(2245), and $C_\star^{\mathrm{fl}}$, $C_\flat^{\mathrm{tm}}$ those of equation eq. 6.xxi; c_θ and S_3 are numbers of the constants alone.

The short-distance estimates use the constants of Lems. YM3_I:125.2 and YM3_I:125.5. We write these constants with the font used here: $\mathsf{C}_{\mathrm{sz}} = C_{\mathrm{sz}}$, $\mathsf{C}_{\mathsf{M}} = C_{\mathsf{M}}$, $\mathsf{c}_{\mathrm{up}} = c_{\mathrm{up}}$, $\mathsf{c}_I = c_I$ and $\mathsf{C}_i = C_i$, with the same convention for C_3', C_3''. In particular $\mathsf{c}_X = 405 B_3^2 A_0^2/2$. Fix these choices before the source, lattice and levels, and put

$$\mathsf{c}_{\mathfrak{p}} = \sup_{a\in\mathbb{Z}_{\ge0}} (a+2)^{4\mathfrak{p}_0} L^{-a}, \qquad \mathsf{c}_{\mathrm{sb}} = \mathsf{C}_{\mathrm{sz}} + \mathsf{c}_{\mathrm{up}}, \qquad \mathsf{c}_{\mathrm{tl}} = 2L^8 \mathsf{c}_{\mathfrak{p}} + 1. \tag{2.xxiv}$$

The supremum is finite because exponential decay dominates the fixed power of a. For $\vartheta \in \{1,2\}$ define

$$\begin{aligned} \mathsf{C}_B^{(\vartheta)} &= \mathsf{C}_{\mathsf{M}} + 4\mathsf{c}_X\mathsf{c}_I + 6\mathsf{c}_I^2 \\ &\quad + \mathsf{c}_{\mathrm{tl}}\big[\mathsf{C}_{\mathsf{M}} + (2 + \vartheta^2 + \vartheta^2\mathsf{c}_{\mathrm{tl}})\mathsf{c}_{\mathrm{sb}}^2 + 2\vartheta\mathsf{c}_{\mathrm{sb}}(\mathsf{c}_X + \mathsf{c}_I)\big], \\ \mathsf{C}_{\mathrm{B}} &= \mathsf{C}_7 + 2(1+A_1)^4\mathsf{C}_5 + \mathsf{C}_3' + \mathsf{C}_3'' + \mathsf{C}_6 + \mathsf{C}_B^{(1)}, \\ \mathsf{C}_B' &= \mathsf{C}_B^{(2)}, \qquad \mathsf{C}_B^e = \mathsf{C}_7 + 2(1+A_1)^4\mathsf{C}_5 + \mathsf{C}_3' + \mathsf{C}_3'' + \mathsf{C}_6 + \mathsf{C}_B'. \end{aligned} \tag{2.xxv}$$

These are fixed numbers, and $\mathsf{C}_{\mathrm{B}} \le \mathsf{C}_B^e$.

Lemma 2.13 (The coefficient of the collected error)**.** Under the hypotheses of Cor. YM3_I:125.6, the coefficient C_B in its estimate may be chosen to be C_{B} in eq. 2.xxv. ∎

Proof. Use the notation of that corollary: $p = \mathfrak{p}_0$, $q = \widehat{d}_{\mathsf{k}}$, $y = y_{\mathsf{k}}$, $n = T - \mathsf{k} \ge 1$, and $s_T \le 1$. Divide the errors by $\widehat m_1 \widehat m_2$, and write $\mathfrak{m} = n^4 q^6 y^{15p}$. The first four terms in equation YM3_I:(2319) are the direct terms of the seven-error estimate. Its remaining terms are bounded by

$$[\mathsf{C}_7 + 2(1+A_1)^4\mathsf{C}_5 + \mathsf{C}_3' + \mathsf{C}_3'' + \mathsf{C}_6]\mathfrak{m} s_T.$$

Indeed $x_T^{-1} \le s_T^2 \le s_T$, $W_T \le t_T \le s_T$, and $1 + A_1 y^p \le (1+A_1) y^p$.

In the remainder of equation YM3_I:(2315), use $i_T \le \mathsf{c}_I q^3 y^{3p} s_T$ and $z_T \le \mathsf{c}_X y^{-7p} + i_T$. The separated-support term costs $\mathsf{C}_{\mathsf{M}} \mathfrak{m} s_T$. The cross terms cost $(4\mathsf{c}_X \mathsf{c}_I + 2\mathsf{c}_I^2)\mathfrak{m} s_T$, and the top term is at most

$$2z_T^2 \le 4\mathsf{c}_X^2 y^{-14p} + 4\mathsf{c}_I^2 \mathfrak{m} s_T.$$

For its last line, the size and tail bounds give

$$2L^8 \mathsf{c}_{\mathsf{p}} + n \le \mathsf{c}_{\mathrm{tl}} n, \qquad b_0 \le \mathsf{c}_{\mathrm{sb}} n q^3 x_{\mathsf{k}} y^{2p}, \qquad x_{\mathsf{k}}^2 t_T \le s_T.$$

Consequently the bracket in that line is at most

$$[\mathsf{C}_{\mathsf{M}} + (3 + \mathsf{c}_{\mathrm{tl}})\mathsf{c}_{\mathrm{sb}}^2 + 2\mathsf{c}_{\mathrm{sb}}(\mathsf{c}_X + \mathsf{c}_I)] n^3 q^6 y^{5p} x_{\mathsf{k}}^2.$$

Multiplication by its remaining factor, at most $\mathsf{c}_{\mathrm{tl}} n t_T$, gives the last part of $\mathsf{C}_B^{(1)} \mathfrak{m} s_T$. Adding these coefficients proves the assertion. □

We use this admissible value of the collected coefficient in the depth error of equation YM3_I:(2330). We use the following names for its auxiliary functions:

$$\mathsf{Lm}_s = \mathsf{L}_s = L^2 C_W(s) Y_s^{3\mathfrak{p}_0}, \qquad \mathsf{sm}_g(s) = \sigma_g(s). \tag{2.xxvi}$$

Here L_s is defined in equation YM3_I:(2322), and σ_g, ϵ_g, Y_s^-, x_s^- and $s_1(g)$ are the functions and threshold of equations YM3_I:(2328)–(2332). The proof there gives $s^a \mathsf{sm}_g(s) \to 0$ for every $a \ge 0$ and boundedness of $Y_s^{2\mathfrak{p}_0} \epsilon_g(s)$. Changing the fixed collected coefficient to the admissible value above preserves that proof: its only additional occurrence is a fixed multiple of a polynomial times $\mathsf{sm}_g(s)$.

Laws. $\langle\cdot\rangle_\ell = \langle\cdot\rangle_{K,\ell,\beta}$ is the Wilson measure on T_ℓ, $\tilde{\mu}_\ell$ its periodic extension to $\mathbb{L}_K$, $\mathcal{G}_{K,\beta}^{\mathrm{per}}$ the set of the periodic Gibbs states on $\mathbb{L}_K$; a *law* at (K,β) is a member of $\mathcal{G}_{K,\beta}^{\mathrm{per}}$ or a $\tilde{\mu}_\ell$, $\ell \in \mathfrak{L}(g)$, and for the latter distances are those of T_ℓ. $\langle X_1; \dots; X_n\rangle_\nu$ is the joint cumulant (complex multilinear), $\nu(X;Y) = \langle X; Y\rangle_\nu$, $A[F] = A^\nu[F] = F - \nu(F)$; for a tuple $X = (H_1, \dots, H_n)$, $A^\nu(X) = \prod_a A^\nu[O_K^T(H_a)]$, extended linearly to combinations of tuples; along a sequence of laws ν_j, $A_j = A^{\nu_j}$, and $A(X) = A^\rho(X)$ for the law ρ at hand. $p_\ell = \ell^{-3} \log Z_{K,\ell,\beta}$ is the pressure, and $G_g = C_S + C_g + 2C_{\mathrm{tt}} C'_{\mathrm{tt}} E_0$ is formed with the constants of [7, §N], with which the canonical source energies form a system of localised source energies ([7, Lem. N.1]).

A *mirror* θ is the reflection in a coordinate plane through midpoints of bonds; it acts on points, on functions of the field by $(\theta F)(U) = F(\theta U)$, and on tensor labels. The *nine families* are the planes whose normals are e_μ or $(e_\mu \pm e_\lambda)/\sqrt{2}$, $\mu \ne \lambda$ (eighteen unit normals); every periodic Gibbs state is invariant under the lattice isometries and reflection positive, in the sense of the reflection form below, at the lattice planes of the nine families (Prop. YM3_I:103.4). For a mirror θ_c in $\{x_{\mu_0} = c\}$ and a law ν, $\|F\|_{\nu,c} = \nu(\overline{\theta_c F} F)^{1/2}$ on the functions of the field on $\{x_{\mu_0} > c\}$.

Boards. A board of $\mathbb{L}_K$ has a side $\mathsf{l}_0 = a_i$, $i \le K-1$ an integer (negative i admitted, so that $\mathsf{l} = L^{m_{\mathfrak{L}}(g)}$ is a side of a board), and an offset $o \in a_K \mathbb{Z}^3$; it is the partition of $\mathbb{R}^3$ into the cubes $o + \mathsf{l}_0 \mathbf{n} + [-\frac{1}{2}\mathsf{l}_0, \frac{1}{2}\mathsf{l}_0)^3$, $\mathbf{n} \in \mathbb{Z}^3$, its *cells*, so that the offset is the centre of a cell, and its *walls* are the planes $\{x_\mu = o_\mu + \mathsf{l}_0(t + \frac{1}{2})\}$, $t \in \mathbb{Z}$, $\mu = 1,2,3$. As $\mathsf{l}_0/a_K = L^{K-i}$ is odd, the walls lie in $a_K(\mathbb{Z} + \frac{1}{2})$ and pass through midpoints of bonds: they are planes of mirrors. A *function of a cell* is a function of the variables of the bonds with both ends in the open cell; its images under successive reflections in the walls are its *copies*, with complex conjugation after an odd number of reflections. For a side $\ell_0 \in 2\mathsf{l}_0\mathbb{N}$ the chessboard seminorm $\|\Phi\|_{\mathrm{ch},\ell_0} \ge 0$ of a bounded function Φ of a cell has as its N_0-th power, $N_0 = (\ell_0/\mathsf{l}_0)^3$, the $\tilde{\mu}_{\ell_0}$-mean of the product of the N_0 copies of Φ in a period box.

Field and label conventions. The field is O_K^T of [7, eq. 1], with the same $w(G)$, a_p, q_b and midpoint convention. We use $\langle X, Y\rangle = \frac{1}{2}\operatorname{tr}(XY)$ and the sup norm $|H|_\infty$. In this section $E_{\lambda\lambda'}$ denotes the symmetric label with entries 1 at (λ,λ') and (λ',λ), rather than a matrix unit. A loop is a pair $l = (\alpha, \alpha')$, $\alpha, \alpha' \in \{\pm1, \pm2, \pm3\}$, $|\alpha| \ne |\alpha'|$; $W_x(l)$ is the holonomy from x along e_α, then $e_{\alpha'}$, around the plaquette $p(x,l)$, and $\sigma(l) = \operatorname{sgn}\alpha \operatorname{sgn}\alpha' \operatorname{sgn}(|\alpha'| - |\alpha|)$. We write $S_n^T[\nu](H_1, \dots, H_n)$ for

the joint cumulant of the $O_K^T(H_a)$ under ν. The free covariance is $\mathcal{W}$ in eq. 2.xxi. The test norm used here is

$$N(H) = \max_{\lambda\lambda'} (|H_{\lambda\lambda'}|_\infty + d\ \mathrm{Lip}\, H_{\lambda\lambda'}).$$

Forms. A *cell* is a plaquette, a positively oriented bond with the four loops that start along it, or a site with all its 24 loops. A *form* is $F = a_K^{-1} \sum_p h(p) a_p$ or $F = a_K^{-1} \sum_{\mathsf{c}} \sum_{l,l'} v(\mathsf{c}, l, l') \langle w(W_l), w(W_{l'}) \rangle$, the cells c being bonds, resp. sites, with real weights of finite support (complex weights where this is said); its *size* $|F|_\star$ is $\sup |h|$, $59\|v\|$, resp. $23\|v\|$, $\|v\| = \sup_{\mathsf{c}} \sum_{l,l'} |v(\mathsf{c}, l, l')|$; its *carrier* is the set of the centres of the cells of its support, $S[F]$ the set of the plaquettes of these cells; its *kernel* is $\mathsf{t}_F(p, p') = \frac{1}{2} h(p) \delta_{pp'}$, resp. $\mathsf{t}_F(p', p'') = \frac{1}{2} \sum v(\mathsf{c}, l, l') \sigma(l) \sigma(l') [\delta_{p'p(l)} \delta_{p''p(l')} + \delta_{p'p(l')} \delta_{p''p(l)}]$. For a weight w of the bonds, $Y[\mathsf{w}] = -a_K^{-1} \sum \mathsf{w}(\mathsf{e}) q_{\mathsf{e}}$ is a bond form of size $59 \sup |\mathsf{w}|$; $O_K^T(H)$ is a sum of six forms, each on the plaquettes of one normal or on the bonds of one direction, of total size at most $180|H|_\infty$. The *Gaussian value* of two forms is

$$G(F_1, F_2) = 6v^2 \sum \mathsf{t}_{F_1}(p, p')\, \mathsf{t}_{F_2}(q, q')\, C^{\mathrm{bg}}(p, q)\, C^{\mathrm{bg}}(p', q'),$$

Wick's value for three free photon fields, $2(N^2 - 1) = 6$ at $N = 2$.

Limits. S denotes a volume-first limit, or a continuum limit along growing tori, of the action density, with its defining sequence ([7, §K.1]). A tuple is *separated* if its supports are pairwise disjoint. Along a sequence of laws at which all $S_n^T[\nu_j](X)$ converge, $\sigma^T(X) = \sum_\pi \prod_{I \in \pi} S_{\#I}^T(X_I)$, the sum over the partitions into blocks of at least two, is the limit of the means of the products of the insertions centred in the states, and $\sigma^T(\emptyset) = 1$. A *limit through periodic Gibbs states* is a limit along $\nu_j \in \mathcal{G}^{\mathrm{per}}_{K_j, \beta_j}$, $K_j \to \infty$, that converges at all separated tuples in $C_c^1(\mathbb{R}^3; V)$; H, P_1, P_2, Ω are the Hamiltonian, the momenta and the vacuum reconstructed from σ^T at $\{x_3 = 0\}$ (Prop. YM3_I:94.1; classes $[F]$ of combinations F of tuples in $\{x_3 > 0\}$, $\langle [F'], [F] \rangle = \sigma^T(\theta \bar{F}' \cdot F))$, with $e^{-tH}[F] = [\tau_{te_3} F]$, the unitary group $\mathcal{U}(v)[F] = [\tau_{(v,0)} F]$, $v \in \mathbb{R}^2$, of the horizontal translations, and the joint spectral measure $\mathcal{P}$ of (H, P_1, P_2); and alike at a plane of each of the nine families: the *nine pictures*. τ_v is the translation by v.

Numbers. Each symbol $C.$, $c.$, $k.$, $s.(g)$, and each error function $\varepsilon., \omega., o., e.$, Df. of s, is a function of the constants, of g and of the arguments shown alone; it does not depend on K, β, ℓ, the law, a weight or a position unless these are shown, and error functions tend to 0 as $s \to \infty$. $s_1(g)$ is the first depth of the envelopes of equations YM3_I:(2328)–(2336), with the notation in eq. 2.xxvi, $s_p(g)$ (p even) that of Lem. 2.22.

Remark 2.14 (Comparison of the creation tests). To obtain the stronger rarity exponent, we must prove the following comparison. At every admitted datum and level, form the gauge-averaged local first-creation test functions $0 \le F_Q \le 1$ described in Step 14 below, including each plaquette, witness and, at the first step, density test. We must construct a second admissible choice with the same A_0, $A_1'' = \frac{93}{100} A_1$, and the other parameters entering $c_\star$ unchanged, and show that, for every finite family of these test functions with disjoint witness supports,

$$\left\langle \prod_Q F_Q \right\rangle_{K,\ell,\beta} \le \tilde{\nu}_K''(\{\text{every witness centre belongs to } Z_t''\}).$$

Here the left member is the mean after the gauge averages are undone in the original history sum, and the right member is the history law of the second choice at the same bare datum and level. We must prove the comparison uniformly over the levels, test types, witness families and admitted data used in Step 14. It is the step needed for the strong pair $(\frac{3}{4} c_\star, C_\flat)$ and the consequent pair $(\frac{1}{2} c', C_\flat)$; the pair $(c_2^\natural, C_\natural)$ follows from the separate witness construction below.

Theorem C. Let (K, β) be a first-passage datum at g and $b = a_K \beta$.

(A) Let $0 < g \le \gamma_2$ and let ℓ be small. Then the following hold.

(A1) For $n \ge 2$, $0 < d < 1$, $K \ge K_0'(n,d)$ and forms $F_1, \dots, F_n$ on $\Lambda_{K,\ell}$, with complex weights allowed at bonds and sites, whose carriers are pairwise at distance $\ge d$,

$$|\langle F_1; \dots; F_n\rangle_\ell| \le k^\circ(n,d)\, b^{-n} \prod_a |F_a|_\star,$$

where $d'' = \min\{\frac{3}{4}d, L^{-s_{\rm th}(n)}\}$, $K_0'(n,d) = s(d'')$, $k^\circ(n,d) = (2\ell_{\max}/d'')^{3n} k_n'(g, s(d''))$, $\ell_{\max}^3 = \frac{1}{2}c' A_0^2 Y^P_{s(d'')}$, with $s_{\rm th}$, k_n' of Lem. 2.18: the power d^{-3} for each entry, times a power of $\log \frac{1}{d}$. No upper bound on the distances is asked.

(A2) Let $s_1(g) \le s(d) \le K$, $3d \le |x_1 - x_2| \le 4d$, and let the carrier of F_i lie within $d - 2a_K$ of x_i. Then $|\langle F_1; F_2\rangle_\ell - G(F_1, F_2)| \le \varepsilon_g^\natural(s(d))\, v^2 |F_1|_\star |F_2|_\star$, $\varepsilon_g^\natural$ of Lem. 2.20. For complex tensor weights H_i supported within $d - 2a_K$ of x_i the same holds for $O_K^T(H_1)$, $O_K^T(H_2)$, the Gaussian value being formed with their kernels (Lem. 2.16) and the error being $4 \cdot 60^2 \varepsilon_g^\natural(s(d)) v^2 |H_1|_\infty |H_2|_\infty$. If $H_i = \varphi_i E_{\lambda_i \lambda_i'}$, φ_i real and Lipschitz with support in $B(x_i, d - 2a_K)$, and $\lambda_1 \ne \lambda_1'$, then

$$|\langle O_K^T(H_1); O_K^T(H_2)\rangle_\ell - b^{-2} \mathcal{W}(H_1, H_2)| \le e^{\rm od}\, b^{-2} N(H_1) N(H_2),$$

with the number $e^{\rm od}$ of Step 2, at most $\bar e_g^{\rm od}(s(d))$, $\bar e_g^{\rm od}$ an error function.

(A3) Let $s_1(g) \le s(d) \le K$, and let the carrier of $Y = Y[\mathsf{w}]$, w real on the bonds of one direction, lie within $d - 2a_K$ of a point at a distance between $\frac{3}{2}d$ and $2d$ from the plane of a mirror θ. Then

$$0 \le \langle \theta Y; Y\rangle_\ell \le C^\circ_{\rm od}(g)\, b^{-2} \sup|\mathsf{w}|^2, \qquad C^\circ_{\rm od}(g) = \tfrac{9}{8}[600 C_{\mathsf{k}}^2 + 59^2 \bar\varepsilon_g^\natural], \quad \bar\varepsilon_g^\natural = \sup_{s \ge s_1(g)} \varepsilon_g^\natural(s),$$

C_{k} of Lem. 2.17. For a real form F at bonds, resp. sites, with the weight v, placed in the same way, $0 \le \langle \theta F; F\rangle_\ell \le \frac{9}{8}[6N^2 C_{\mathsf{k}}^2 + c^2 \bar\varepsilon_g^\natural]\, b^{-2} \|v\|^2$, $(N, c) = (30, 59)$, resp. $(10, 23)$. For real tensor weights H_i as in (A2), $|\langle O_K^T(H_1); O_K^T(H_2)\rangle_\ell| \le \frac{9}{8}[12150 C_{\mathsf{k}}^2 + 60^2 \bar\varepsilon_g^\natural]\, b^{-2} |H_1|_\infty |H_2|_\infty$.

(B) Let $g \le \gamma_1$, $s_8(g) \le s(d) \le K$, let $\theta_1', \theta_2', \theta_3'$ be mirrors of $T_{\ell'}$, θ_μ' normal to e_μ, and let Y be as in (A3), carried within $d - 2a_K$ of a point at a distance between $\frac{3}{2}d$ and $2d$ from a plane of θ'_{μ_0} and, up to $a_K/2$, midway between the planes of the two other mirrors. With $\theta'^\eta = \theta_1'^{\eta_1} \theta_2'^{\eta_2} \theta_3'^{\eta_3}$,

$$0 \le M_8 = \Big\langle \prod_{\eta \in \{0,1\}^3} \theta'^\eta Y \Big\rangle_{\ell'} \le C_8\, b^{-8} \sup|\mathsf{w}|^8, \qquad 0 \le M_8 - \langle \theta'_{\mu_0} Y; Y\rangle_{\ell'}^4 \le (\tfrac{9}{8})^4 59^8 \omega_{g,8}^\sharp(s(d))\, b^{-8} \sup|\mathsf{w}|^8,$$

$C_8 = (\frac{9}{8})^4[(106 C_{\mathsf{k}}'^2)^4 + 59^8 \bar\omega_{g,8}]$, $\bar\omega_{g,8} = \sup_{s \ge s_8(g)} \omega_{g,8}(s)$, with C_{k}' of Lem. 2.17 and $\omega_{g,8}$, $\omega_{g,8}^\sharp$ of Lem. 2.22.

(C) Let $g \le \gamma_1$ and let ν, ν' be laws at (K, β).

(C1) For $n \ge 2$, $d, R > 0$, every K, and forms $F_1, \dots, F_n$ on $\mathbb{L}_K$ whose carriers lie in balls of radius R and are pairwise at distance $\ge d$: $|\langle F_1; \dots; F_n\rangle_\nu| \le C_n^T(d; R)\, b^{-n} \prod_a |F_a|_\star$, with C_n^T of Step 4; complex weights cost a factor 2^n. So

$$|S_n^T[\nu](H_1, \dots, H_n)| \le 60^n C_n^T(d; R)\, b^{-n} \prod_a |H_a|_\infty$$

for supports in such balls at mutual distance $\ge d$, and $|\nu(F) - \nu'(F)| \le 2 C_1^T(R)\, b^{-1} |F|_\star$.

(C2) If $s(d) \le K$, $\operatorname{supp} H_i \subset B(x_i, d - 3a_K)$ and $|x_1 - x_2|_\infty \ge 3d + a_K$ (on a torus for all lifts):

$$|S_2^T[\nu](H_1, H_2)| \le 180^2 C_\infty^\diamond(g)\, b^{-2} |H_1|_\infty |H_2|_\infty,$$

$C_\infty^\diamond$ of Step 5. For $Y[\mathsf{w}_i]$, w_i real on the bonds of one direction, carried within $d - 2a_K$ of x_i:

$$|\nu(Y[\mathsf{w}_1]; Y[\mathsf{w}_2])| \le C_\infty^{\rm per}(g)\, b^{-2} \sup|\mathsf{w}_1| \sup|\mathsf{w}_2|,$$

with a number $C_\infty^{\rm per} \ge e^{2G_g|^3} C_8^{1/4}$.

(C3) Let $s_{16}(g) \le s = s(d) \le K$, $3d \le |x_1 - x_2| \le 4d$, let the carriers of real forms F_i lie within $d - 2a_K$ of x_i, $\hat m = \upsilon^2 |F_1|_\star |F_2|_\star$, and let F_i' be the forms on $T_{\ell'}$ with the weights of F_i on the board of side l whose cell holds the pair in its middle. Then

$$|\nu(F_i) - \tilde\mu_{\ell'}(F_i)| \le e^{G_g \mathsf{l}^3} o_{g,16}(s)^{1/8} \upsilon |F_i|_\star, \quad |\nu(F_1;F_2) - \tilde\mu_{\ell'}(F_1;F_2)| \le \mathrm{Df}_g(s)\, \hat m,$$
$$|\nu(F_1;F_2) - G(F_1', F_2')| \le [\omega'_{g,16} + \mathrm{Df}_g](s)\, \hat m,$$

the Gaussian value being formed on $T_{\ell'}$ and Df_g given in Step 5; the error function and the first depth are the same for all the laws. The same holds for $F_i = O_K^T(H_i)$, H_i real tensor weights supported within $d - 2a_K$ of x_i, with $60|H_i|_\infty$ in the place of $|F_i|_\star$.

(D) Let $g \le \gamma_1$, let (K_j, β_j) be first-passage data at g and ν_j laws at (K_j, β_j): all periodic Gibbs states, or all $\tilde\mu_{\ell_j}$ with $\ell_j \to \infty$. Tuples $X = (H_1, \dots, H_n)$ are in $C_c(\mathbb{R}^3; V)$ with pairwise disjoint supports.

(D1) Along a subsequence ($K_j \to \infty$ is not asked) $a_{K_j}\beta_j \to \mathsf{b} \in I^{\mathrm{lab}}(g)$ and $S_n^T[\nu_j](X) \to S_n^T(X)$ for all $n \ge 2$ and all X; S_n^T is symmetric, multilinear, bounded as in (C1) with b for b, and in each entry a V^*-valued Radon measure off the other supports. For laws ν_j' at the same data, $\lim(\nu_j - \nu_j')(O_{K_j}^T(H))$ exists along a further subsequence, of modulus $\le 120 C_1^T(R)\mathsf{b}^{-1}|H|_\infty$ if $\operatorname{supp} H$ lies in a ball of radius R. If each ν_j is a weak limit of measures $\tilde\mu_\ell$ at (K_j, β_j), then for any numbers Λ_j there are sides $\ell_j \ge \max\{\Lambda_j, j\}$ in $\mathfrak{L}(g)$ with $S_n^T[\tilde\mu_{\ell_j}](X) \to S_n^T(X)$ along the full sequence: a limit through limit states of the tori is a limit along growing tori.

(D2) Over every volume-first limit S and every continuum limit along growing tori there is such an S^T, with $S_n^T(f_1\mathbf{1}, \dots, f_n\mathbf{1}) = S_n(f_1, \dots, f_n)$, $f_i \in C_c^1$.

(D3) If $K_j \to \infty$: $S_n^T(rX) = S_n^T(X)$ for $rx = Ax + y$, $A \in W_3$; and for every coordinate plane $\{x_\mu = c\}$, with its reflection θ acting on points and on indices, all finitely many tuples X_i (the empty one included) with supports in $\{x_\mu > c\}$ and all complex c_i: $\sum_{i,k} \bar c_i c_k \sigma^T(\theta X_i \cdot X_k) \ge 0$. For limits through periodic Gibbs states the same holds at the planes with the normals $e_\mu \pm e_\lambda$, the Hilbert space is separable, and $H \ge |P_1|$, $H \ge |P_2|$. The numbers $\mathfrak{c}_j = a_{K_j}^{-4}(\nu_j(a_p) - \tilde\mu_{\ell'}(a_p))$, the same at all plaquettes, obey $|\mathfrak{c}_j| \le C(g)\mathsf{b}^{-1}$ for all large j; along a further subsequence $\mathfrak{c}_j \to \mathfrak{c}$, and along it the moments centred at the means of $\tilde\mu_{\ell'}$ in the place of those of the ν_j converge too, to $\sum_B \prod_{i\in B}(\mathfrak{c} \int \operatorname{tr} H_i)\, \sigma^T(X_{B^c})$, the sum over the subsets B of the entries, a functional with the same properties whose one-point function is $\mathfrak{c} \int \operatorname{tr} H$.

(D4) For $\mathsf{b} \in I(g)$, $\ell \in \mathfrak{L}(g)$ and continuous $H_a : T_\ell \to V$ with disjoint supports, $S_n^{T,(\ell,\mathsf{b})}(H) = \lim_K \langle O_K^T(H_1); \dots; O_K^T(H_n)\rangle_{K,\ell,L^K\mathsf{b}}$ exists along the full sequence of cutoffs (for $g \le \gamma_2$) and obeys the bound of (D1). Along a subsequence of every $\ell_i \to \infty$ these limits converge, and the limit is also the limit of $S_n^T[\tilde\mu_{\ell_i}]$ at suitable cutoffs $K_i \to \infty$: a limit along growing tori, with the properties of (D3) for these limits.

(D5) Let $K_j \to \infty$, $s(d) \ge s_{16}(g)$, $3d \le |x_1 - x_2| \le 4d$, and H_i Lipschitz with compact support in the open ball $B(x_i, d)$. Then

$$|S_2^T(H_1, H_2) - \mathsf{b}^{-2}\mathcal{W}(H_1, H_2)| \le 36\, \bar e_g(s(d))\, \mathsf{b}^{-2} N(H_1) N(H_2),$$

the function $\bar e_g$ being the same for all sequences; the same holds for the limits of (D4), on every ℓ. At $H_i = f_i\mathbf{1}$ the leading term is $\frac{9}{16\pi^2}\mathsf{b}^{-2} \iint f_1(x) f_2(y) |x - y|^{-6}\, \mathrm{d}x\, \mathrm{d}y$, the scalar leading coefficient in Thm. YM3_I:AW.

(E) Let $g \le \gamma_2$, $K \ge 3$, and let $\ell = 2L^m$ be admissible.

(E1) With $\tilde\nu_K$ the law of the histories and Z_t the large-field region of level t, the proved local rarity estimate gives, for $1 \le t \le K$,

$$0 \le \mathsf{w}_t \le \tilde\nu_K(\{Z_t \ne \emptyset\}) \le C_\natural M^{-3} \ell^3 L^{3(K-t)} \exp[-c_2^\natural p_0(g_t)^2],$$

where $c_2^\natural$ is the constant of Def. 4.47 and Prop. 4.52, and $C_\natural$ is a function of the constants. The stronger conclusion is the same bound with $(\frac{3}{4}c_\star, C_\flat)$ for $(c_2^\natural, C_\natural)$. Here $C_\flat$ depends on the two admissible choices alone, not on g, K, β, ℓ, t; $\frac{3}{4}c_\star \geq \frac{1}{2}c'$, and $\frac{3}{4}c_\star \geq c'$ if $\eth \geq \frac{1}{4}$.

(E2) Let (c, C) be a pair for the rarity bound in (E1); $\sigma_c = \frac{2}{\log L}(\frac{6}{cA_0^2})^{1/P}$, $\hat{s}(g,\ell) = \max\{s_\flat(g), \lceil \sigma_c (\log \ell)^{1/P} \rceil\}$, the numbers below being formed at (c, C) and $s_\flat(g)$ being the first depth of the envelopes at (c, C), chosen so that also $2c\mathfrak{p}_0 A_0^2 y^{P-1} \geq 13$ at the working level. Let $\hat{s} \leq s(d) \leq K$, $|x_1 - x_2| \geq 3d$ (no upper bound), and let F_i be sums of forms within $d - 2a_K$ of x_i, of size the sum of the sizes. Then (i) $|\langle F_1; F_2\rangle_\ell - G(F_1, F_2)| \leq \varepsilon_g^\circ(s(d)) v^2 |F_1|_\star |F_2|_\star$, with $Y_s^P \varepsilon_g^\circ(s)$ bounded; (ii) $|\langle F_1; F_2\rangle_\ell| \leq C_\star^\circ(g) b^{-2} |F_1|_\star |F_2|_\star$; (iii) if x_1 is at a distance $\geq \frac{3}{2}d$ from both planes of θ, $0 \leq \langle \theta F_1; F_1\rangle_\ell \leq C_\star^\circ(g) b^{-2} |F_1|_\star^2$; (iv) for all d with $s(d) \leq K$ and F placed as in (iii), $0 \leq \langle \theta F; F\rangle_\ell \leq C_3^\circ(g,\ell) b^{-2} (|F|_\star/59)^2$, which is $C_3^\circ b^{-2} \sup |\mathsf{w}|^2$ at $F = Y[\mathsf{w}]$, w real on the bonds of one direction, and $C_3^\circ(g,\ell) \leq c_3(g)(1 + \log \ell)^8 \exp[16(\frac{6 \log \ell}{cA_0^2})^{1/P}]$, so that $\ell^{-\epsilon} C_3^\circ(g,\ell)$ is bounded for every $\epsilon > 0$; (v) if all points of the carriers of F_1 and F_2 are $\geq r$ apart and $K \geq s(\min\{\frac{r}{8}, L^{-\hat{s}}\})$: $|\langle F_1; F_2\rangle_\ell| \leq (\ell \max\{\frac{8}{r}, L^{\hat{s}}\} + 1)^6 C_\star^\circ(g) b^{-2} |F_1|_\star |F_2|_\star$. On the sides with $\log \ell \leq (s_\flat(g)/\sigma_c)^P$ the first depth is $\hat{s} = s_\flat(g)$, free of ℓ.

(E3) Let $\mathsf{L} < \infty$ and $c'' \geq \frac{1}{2}c'$ be numbers with $\mathsf{w}_t \leq \mathsf{L} L^{3(K-t)} e^{-c'' p_0(g_t)^2}$ for $1 \leq t \leq K$. Then: (i) Lem. 2.18, for sources under majorants, holds on T_ℓ with $\max\{1, M^3\mathsf{L}/(c'A_0^2)\}^n k_n'$ for k_n', under its hypotheses $n \geq 2$ and $s_{\mathrm{th}}(n) \leq s(d) \leq K$ and in its geometry: $|\langle F_1; \dots; F_n\rangle_\ell| \leq \max\{1, M^3\mathsf{L}/(c'A_0^2)\}^n k_n'(g, s(d))\, b^{-n} \prod_a \mathsf{s}_a$, with no factor of a covering; (ii) from (i), by the covering of Step 1, (A1) holds on T_ℓ with 2ℓ for $2\ell_{\max}$ and $\max\{1, M^3\mathsf{L}/(c'A_0^2)\}^n k_n'$ for k_n'. One such pair is available on every admissible side: $\mathsf{L} = 2M^{-3}\ell^3 e^{\mathsf{B}(\bar{x}_0)\ell^3}$, $c'' = c'$, by Prop. 4.54, since B increases and $x_K \leq X_g < \bar{x}_0$. Thus (i) and (ii) hold there with $\max\{1, c(\ell)\}^n k_n'$ for k_n', (ii) with 2ℓ for $2\ell_{\max}$, where $c(\ell) = 2\ell^3 e^{\mathsf{B}(\bar{x}_0)\ell^3}/(c'A_0^2)$. The strong pair of (E1) also gives $\mathsf{L} = C_\flat M^{-3}\ell^3$, $c'' = \frac{3}{4}c_\star \geq \frac{1}{2}c'$, and $c(\ell)$ may then be replaced by $C_\flat \ell^3/(c'A_0^2)$, a power of ℓ. No such replacement follows from $c_2^\natural > 0$ alone: the premise $c'' \geq \frac{1}{2}c'$ must still be checked.

■□

Theorem D (Contact extensions and mirror bounds)**.** Let $g \leq \gamma_1$ unless another range is given, and let $\sigma = \sigma^T$, with cumulants S_n^T, be a limit through periodic Gibbs states.

(F8) (*Coinciding points.*) Let $g \leq \gamma_2$, $\mathfrak{b} \in I(g)$, $\ell \in \mathfrak{L}(g)$, $n \geq 2$. $S_n^{T,(\ell,\mathfrak{b})}$ extends to a symmetric, translation invariant distribution $\tilde{S}_n$ on T_ℓ^n of order $\leq 3n-2$ with

$$|\tilde{S}_n(F)| \leq C\mathfrak{b}^{-n}(1 + \log \tfrac{h_0}{\lambda})^{8n\mathfrak{p}_0+1} \max_{|\gamma| \leq 3n-2} \lambda^{|\gamma|} \sup |\partial^\gamma F|$$

for F supported in a product of closed sup-balls of radius $\lambda \leq h_0$ about any points, coinciding or not (C, h_0 of g, ℓ, n); the lattice cumulants with their Taylor parts at the diagonals subtracted (Lem. 2.26) obey this bound for all K and all $8h_K \leq \lambda \leq h_0$, $h_K \leq 2C_W(K)a_K$ the least scale at the cutoff K, with one number for all cutoffs, and they converge to $\tilde{S}_n$ at every smooth F; two extensions with this bound differ by a sum $\sum_{|\pi| \leq n-1} \sum_{|\alpha| \leq 3|\pi|} \partial_\xi^\alpha \delta_\pi \otimes T_{\pi,\alpha}$ of contact terms, and every such sum may be added; at $n = 2$ the translation invariant, real, symmetric contact terms covariant under the cubic group form a space of dimension twelve, under $O(3)$ on $\mathbb{R}^3$ one of dimension six.

The reconstruction (the Hilbert space, the semigroup, the momenta and the kernels of [7, Lem. D.13]) uses the values of σ at separated tuples alone, so no choice of the extension and no contact term enters it; beyond the separated tuples it rests on the insertion bound of [7, Thm. BZ(F2)(vi)]. For $g \leq \gamma_1$ every limit point of $\langle Y[\varphi_1]; Y[\varphi_2]\rangle_{K,\ell,\beta}$ as $K, \ell \to \infty$, $\ell \in \mathfrak{L}(g)$, in any order, is bounded at separated cubes by $36 C_\infty^{\mathrm{od}}(g)(\inf b)^{-2} \sup|\varphi_1| \sup|\varphi_2|$,

with no power of the distance and no logarithm, and so extends to $\mathbb{R}^6$ as a distribution of order ≤ 4.

(F9) (*Mirror bound in every state; the power* 6.) Let $s(d) \le K$. If $X = A[O_K^T(H)]$ is carried in $B(x_0, d - 2a_K)$ and x_0 is at a distance $\ge \frac{3}{2}d$ from the plane of θ, then $0 \le \nu(\theta X \cdot X) \le C_\infty(g)\, b^{-2} |H|_\infty^2$ in every periodic Gibbs state ν, $C_\infty(g)$ defined in eq. 2.xxvii; for a bond form $Y = Y[\mathsf{w}]$ with a complex weight carried there, $\nu(Y) = 0$ and $0 \le \nu(\overline{\theta Y} Y) \le 36 C_\infty^{\mathrm{od}}(g) b^{-2} |\mathsf{w}|_\infty^2$, with $C_\infty^{\mathrm{od}}(g)$ alone for real weights of one direction. For $H_1, H_2 \in C_c(\mathbb{R}^3; V)$ with disjoint supports and off-diagonal values,

$$|\sigma(H_1, H_2)| \le A_6 C_\infty^{\mathrm{od}}(g)\, \mathsf{b}^{-2} \iint \|H_1(x)\|_1 \|H_2(y)\|_1 \min\{|x-y|_\infty, 3\}^{-6}\, \mathrm{d}x\, \mathrm{d}y, \qquad \|E\|_1 = \sum_{\lambda < \lambda'} |E_{\lambda\lambda'}|,$$

$A_6 = 64 \cdot 36 \cdot ((8/\pi^3) 32^3)^2$; for the whole tensor the same holds with a pure number A in the place of A_6, with $C_\infty(g)$ in the place of $C_\infty^{\mathrm{od}}(g)$ and with $\|E\|_1 = \sum_{\lambda \le \lambda'} |E_{\lambda\lambda'}|$; by (D5) the power 6 is the true one.

■□

Lemma 2.15 (Matrices and corner forms)**.** (a) For complex 2×2 matrices, with the operator norm: $|\langle X, Y\rangle| \le |X||Y|$; if $\det V = \det V' = 1$ and T is invertible, then $w(TVT^{-1}) = Tw(V)T^{-1}$, $w(V^{-1}) = -w(V)$, $|V^{-1} - \mathbf{1}| = |V - \mathbf{1}|$ and $|V^{-1} - V'^{-1}| = |V - V'|$; so $|w(V)| \le |V - \mathbf{1}|$ and $|w(V) - w(V')| \le |V - V'|$.

(b) For loops l, l' of one cell the corner form $Q = \langle w(W_l), w(W_{l'})\rangle$ is a polynomial in the bond variables, 0 at the trivial configuration, gauge invariant, and real at SU(2) configurations, where $|Q| \le a_{p(l)} + a_{p(l')}$. Let $\eta, b_0, b_1 > 0$ be arbitrary. At complex configurations with $|U(\mathsf{e})| \le e^{\eta b_1}$ on the bonds of the cell and $|\partial U(p) - \mathbf{1}| < b_0 \eta^2$ at its two plaquettes, $|Q| \le 4 e^{4t\eta b_1} b_0^2 \eta^4$, $t = 1$ at bonds and $t = 2$ at sites. If $|U(\mathsf{e})| \le e^{1/13}$ at the bonds of $p = p(l)$, $p' = p(l')$ and at their reverses, $|\partial U(p) - \mathbf{1}| \le \zeta_0$ and $|\partial U(p') - \mathbf{1}| \le \zeta_0$ with $e^{4/13}\zeta_0 \le 1$, and B' is a 1-form with $\rho = \max |B'| \le \frac{1}{128}$, then $|Q(e^{iB'}U) - Q(U)| \le 784\rho^2 + 112 e^{2t/13} \rho \zeta_0$. At the small points of a chart of height i, $|Q - \sigma(l)\sigma(l')\langle \mathrm{ang}_{p(l)}, \mathrm{ang}_{p(l')}\rangle| \le 12 M_{\mathrm{pl}}^2 \eta_i^4 \mathsf{rt}^{-3}$, where ang_p is the angle of p and $\mathsf{rt} \ge 2$ is the quotient of the radii of the chart over the point (Lem. YM3_I:124.4). ■

Proof sketch. (a) The trace of a 2×2 matrix is the sum of two entries, each at most the norm. For $\det V = 1$, $V^{-1} = (\mathrm{tr}\, V)\mathbf{1} - V$; so $V^{-1} - \mathbf{1} = (\mathrm{tr}\, E)\mathbf{1} - E$ with $E = V - \mathbf{1}$, and $V^{-1} - V'^{-1} = (\mathrm{tr}\, E)\mathbf{1} - E$ with $E = V - V'$; for 2×2 matrices $(\mathrm{tr}\, E)\mathbf{1} - E = JE^{\mathrm{T}}J^{-1}$ with the unitary $J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$, and $|E^{\mathrm{T}}| = |E|$; finally $w(V) = [(V - \mathbf{1}) - (V^{-1} - \mathbf{1})]/2i$. (b) For $V = \exp(i\phi\, n \cdot \sigma) \in \mathrm{SU}(2)$, $w(V) = \sin\phi\; n \cdot \sigma$ and $\langle w(V), w(V)\rangle = \sin^2\phi \le 2(1 - \cos\phi)$, which with the Schwarz inequality for $\langle \cdot, \cdot \rangle$ gives $|Q| \le a_{p(l)} + a_{p(l')}$. The complex bounds follow from (a) applied to the holonomies of the two plaquettes; the bound on a shift and the second order follow from the expansion of the holonomy in the 1-form and in the chart, $w(W_l) = \sigma(l)\, \mathrm{ang}_{p(l)}$ to the first order; every bound of a vector part at a complex configuration goes through (a). □

Lemma 2.16 (Forms are sources)**.** (a) Let $\bar{k} \ge 1$ be an integer, the *top index*. Call F a *source of top index* $\bar{k}$ *under the majorant* (s, S), $\mathsf{s} \ge 0$ a number and S a set of plaquettes, if F is continuous, gauge invariant, $|F| \le L^K \mathsf{s} \sum_{p \in S} a_p$, and F is the sum, at height 0, of a family in the fourth place over a point (§15.3.1) of size $\le c^\flat \mathsf{m}_0$, $c^\flat = 81 C_P^2 C_{\mathcal{A}}^2 / K_\Lambda$, and, at each height $1 \le j \le \bar{k}$, of a family without base point (Def. YM3_I:49.5) of size $\le \frac{1}{2}\mathsf{m}_j$, $\mathsf{m}_j = 12 M^3 \alpha_0^2 L^{K-j} \mathsf{s}$, both vanishing on the regions that miss S. The top index is that of the data of the chain at which the source is used: the working level k for n sources (Lems. 2.18 and 2.19, where the chain runs up to the level $\mathsf{k} + 1$), and $T - 1$, T the upper level, for two sources and for far pairs (Lems. 2.20 and 2.22), where the one-step

covariance of the level j, $\mathsf{k} < j \le T$, is formed with the members of the height $j-1$; since $n_T \ge 2$, there $T - 1 > \mathsf{k}$.

A source is a source of the top index of the data at hand. Call it *quadratic*, with the kernel t, if besides t is a real symmetric function on $S \times S$ with $2\sum_{p'} |\mathsf{t}(p,p')| \le \mathsf{s}$ such that $|a_K F - \sum \mathsf{t}(p,p')\langle \mathrm{ang}_p, \mathrm{ang}_{p'}\rangle| \le M_{\mathrm{pl}}^2 \eta_i^4 \mathsf{rt}^{-3}(\#S)\mathsf{s}$ at the small points of the chart of height i. Then every form F is a source under $(|F|_\star, S[F])$ of every top index, since it has its member at every height $j \ge 1$ of the chain, linearly in its weights, and it is quadratic with the kernel t_F if it is real; a plaquette form with the weight h is one under $(\sup|h|, \operatorname{supp} h)$, with $\mathsf{t} = \frac{1}{2}h\delta$; for real weights at bonds or sites $2\sum_{p'} |\mathsf{t}_F(p,p')| \le 8\|v\|$; $O_K^T(H)$ is one under $(60|H|_\infty, S_H)$, S_H the plaquettes of its six forms, with $2\sum_{p'} |\mathsf{t}_H(p,p')| \le 3|H|_\infty$.

Under sums the sizes add and the sets unite, and a source under (s, S) is one under (s', S) for every $\mathsf{s}' \ge \mathsf{s}$.

(b) The proof of the two-point law of the action density on a torus of small side (Thm. YM3_I:AU), and the proofs of the statements on which it rests (Thm. YM3_I:AT, Prop. YM3_I:125.1, Lems. YM3_I:125.2, YM3_I:125.4 and YM3_I:125.5; the weighted progression of Prop. YM3_I:122.1; the first step and the term of the history without large-field regions, the first-step part of [7, Lem. B.19] and Prop. YM3_I:123.1; the quotients of the sources and of a step, their tower and the reduction of the covariance to one-step covariances, Prop. YM3_I:123.1 and equation YM3_I:(2268), the quotient definitions in [7, §B.9], the quotient identities preceding equation YM3_I:(2273), equation YM3_I:(2273), [7, Lem. B.20] and Prop. YM3_I:123.2; the carried family and the flat top, Prop. YM3_I:123.3, equation YM3_I:(2283); and the sum over the level of the first creation, equation eq. 4.xxxi), use the source only through the properties of (a) and its boundedness; they therefore apply to every source in the sense of (a) whose top index is that of the data of the statement applied, and to every quadratic source where the kernel is used.

(c) The step map and its Lipschitz bound hold for forms: for weights v, v' at the cells the outputs of one step differ by at most $\theta^{[j]} 8 n_{\mathfrak{c}} e^{4t_{\mathfrak{c}}/13} M^3 \alpha_0^2 \|v - v'\|$ in the norms of the families of Def. YM3_I:49.5, a pure number times the number for plaquette weights ($n_{\mathfrak{c}} = 18$, $t_{\mathfrak{c}} = 1$ at bonds; $n_{\mathfrak{c}} = 7$, $t_{\mathfrak{c}} = 2$ at sites); the step map depends only on its input array. ■

Proof sketch. (a) The cubes of the partition of height k that hold a corner of a plaquette of a cell form a connected region of at most 6 (bond) or 7 (site) cubes with a common corner, and a region of height k serves at most $n_{\mathfrak{c}} M^3 \eta_k^{-3}$ cells. At height 0, with $\mathbf{U}$ the configuration of the bond variables and B' the 1-form in the fourth place, of which the family is a function, the family $\sum_{X_0(\mathsf{c})=Y}[f_{\mathsf{c}}(e^{iB'}\mathbf{U}) - f_{\mathsf{c}}(\mathbf{U})]$, $f_{\mathsf{c}} = \sum_{l,l'} v(\mathsf{c},l,l')\langle w(W_l), w(W_{l'})\rangle$, $X_0(\mathsf{c})$ the region of the cell, is holomorphic and gauge covariant, and Lem. 2.15(b) bounds it at complex configurations; with the count, $n_{\mathfrak{c}} \cdot 3138/(81 \cdot 12) \le 59$, resp. 23, it gives the size $c^\flat \mathsf{m}_0$ with the factor 59, resp. 23, of $|F|_\star$.

At height $j \ge 1$ the member over a region X is $\eta_j^{-1} \sum_{X_j(\mathsf{c})=X} f_{\mathsf{c}}$, of size $\le \frac{1}{2}\mathsf{m}_j$ by the same count. To second order $w(W_l) = \sigma(l)\,\mathrm{ang}_{p(l)}$, whence $\sum \mathsf{t}_F(p,p')\langle \Phi_p, \Phi_{p'}\rangle = \sum v(\mathsf{c},l,l')\sigma(l)\sigma(l')\langle \Phi_{p(l)}, \Phi_{p(l')}\rangle$, with the error of Lem. 2.15(b). Finally $O_K^T(H) = a_K^{-1}\sum_p w_H(p) a_p + Y[\mathsf{w}_H]$ with the plaquette weight $w_H(p) = H_{\lambda(p)\lambda(p)}(x(p))$ and the bond weight $\mathsf{w}_H(\mathsf{e}) = H_{\lambda\lambda'}(x_{\mathsf{e}})$: a plaquette form of size $\le |H|_\infty$ and a bond form of size $\le 59|H|_\infty$, whose sizes add to $60|H|_\infty$. (b) The source enters these proofs at four places.

First, through the clause on F itself: its continuity and its gauge invariance in the first step, where the Wilson density is multiplied by e^{F_λ}, $F_\lambda = \sum_a \lambda_a F_a$ (the first-step part of [7, Lem. B.19]), and in the sum over the level of the first creation (equation eq. 4.xxxi); its pointwise majorant, which bounds F at the backgrounds of the small fields of the level t by $L^K \mathsf{s} \sum_{p\in S} \sup a_p$, the supremum over these backgrounds, that is by the size of the source at the level t, and by $2L^K \mathsf{s} \# S$ at the height 0, in the quotients, their tower, the reduction and the identity of the quotients (Prop. YM3_I:123.1 and

equation YM3_I:(2268), the quotient identities preceding equation YM3_I:(2273), equation YM3_I:(2273) and Prop. YM3_I:123.2); it is the source of the power $2\mathfrak{p}_0$ of Lem. 2.18.

Second, at the height 0, through its family in the fourth place over a point and the size of that family (the height-zero part of [7, Prop. B.8]; here the clause of the height 0 of (a)). Third, at the heights $1 \le j \le \bar{k}$, through its families without base point, their sizes and the action formula ([7, Lem. P.6]), the one-step covariance of the level $j+1$ being formed with the member of the height j ([7, Lem. B.20]); here the clause of the heights of (a), up to the top index. Fourth, at the second order at the small points of a chart, through its kernel (Lem. YM3_I:124.4), in the carried family and the flat top (Props. YM3_I:125.1 and YM3_I:123.3, equation YM3_I:(2283)); here the kernel t of (a).

Every other step of these proofs is stated there for arbitrary bare arrays and local families, for bounded Borel functions of the field, or for numbers. (c) The step map is formed from its input array alone; its Lipschitz constant for the arrays of plaquette weights, with the count of (a) and Lem. 2.15(b), gives the factor $8n_{\mathfrak{c}}e^{4t_{\mathfrak{c}}/13}$. □

Lemma 2.17 (The kernel)**.** Let $s' \le K$, $d \ge a_{s'}$, $\hat{d} = d/a_{s'}$, and $|u| = |x(p) - x(q)| \ge d$. Then

$$|C^{\mathrm{bg}}_{K,s'}(p,q) - \epsilon_p\epsilon_q a_K^3 K_3(u)_{\lambda(p)\lambda(q)}| \le (a_K/d)^3 E_1, \qquad E_1 = C_{\mathrm{lat}}[(a_K/d)^2 + (d/\ell)^3] + C^{\mathrm{fl}}_\star \hat{d}^3 e^{-\delta_{\mathrm{tp}} L\hat{d}/\sqrt{3}},$$

$\epsilon_p = \pm 1$ the orientation of p; so $|C^{\mathrm{bg}}(p,q)| \le C_{\mathsf{k}}(a_K/d)^3$, $C_{\mathsf{k}} = 1 + 3C_{\mathrm{lat}} + C^{\mathrm{fl}}_\star C^{\mathrm{tm}}_\flat$, and in the form in which far pairs use it the same holds with $C'_{\mathsf{k}} = 3^{3/2}C_{\mathrm{tp}}\mathsf{S}_3 + C^{\mathrm{bg}}_{\mathrm{tp}}$. If $8a_K \le d$ and $4d < \ell$, a ball of radius d holds at most $10(d/a_K)^3$ sites, at most as many centres of bonds of one direction, and as many centres of plaquettes of one normal. ■

Proof sketch. The free kernel is the background kernel plus the fluctuation kernel $C^{\mathrm{fl}}_{K,s'} = \Pi_0 - \Pi_{K-s'}$, the sum of the slices of the scales below $a_{s'}$ YM3_I:(2227). The comparison of the lattice free kernel with $a_K^3 K_3$, with the number C_{lat} of equation YM3_I:(2245), gives the first term of E_1, and the exponential decay of the fluctuation kernel beyond the scale $a_{s'}$, $|C^{\mathrm{fl}}_{K,s'}(p,q)| \le C^{\mathrm{fl}}_\star (a_K/a_{s'})^3 e^{-\delta_{\mathrm{tp}} L\mathsf{r}/a_{s'}}$ for $\mathsf{r} = |x(p) - x(q)|_\infty \ge a_{s'}/\sqrt{3}$ (Prop. YM3_I:120.6 at $t_0 = 3^{-1/2}$), with $\mathsf{r} \ge d/\sqrt{3}$, gives the second; condition (H) fixes the constants, and $|K_3(u)_{\lambda\lambda'}| \le (2\pi)^{-1}|u|^{-3}$. The count is a count of lattice points in a ball. □

Lemma 2.18 (n sources on a small torus)**.** Let $0 < g \le \gamma_2$, $n \ge 2$, let ℓ be small, and let $s_{\mathrm{th}}(n) \le s(d) \le K$, where $s_{\mathrm{th}}(n)$ is the least integer $s \ge 3$ with $2T_n(3)\theta(L^{s-2}g^{-2}) \le 1$, $\theta(t) = K_M(1+\beta_s)\delta(t)/\alpha_0$, which does not increase on $[e^P, \infty)$, $\delta(t) = A_1 t^{-1/2}(\log t)^{\mathfrak{p}_0}$, and T_n is the series [7, eq. 3]. Let (s_a, S_a) be majorants with points x_a, $|x(p) - x_a| \le \frac{7}{8}d$ on S_a, $|x(p) - x(q)| \ge \frac{5}{4}d$ for $p \in S_a$, $q \in S_{a'}$, $a \ne a'$, and let F_a be (complex) sources of top index k, the working level, under them. Then

$$|\langle F_1; \dots; F_n\rangle_\ell| \le k'_n(g, s(d))\, b^{-n} \prod_a \mathsf{s}_a, \qquad k'_n(g,s) \le C_n(g) Y_s^{8n\mathfrak{p}_0},$$

so $k'_n(g, s(d)) \le C_n(g)[2\log\frac{1}{d} + C(g)]^{8n\mathfrak{p}_0}$. No upper bound on the distances is asked. ■

Proof sketch. Insert $F_\lambda = \sum_a \lambda_a F_a$ as the source into the weighted progression of [7, eq. 200] (Prop. YM3_I:122.1). A chain rule for the step map in any number of parameters (Cauchy estimates on polydiscs) and locality give, for every set β of $r = \#\beta \ge 2$ sources: the mixed derivative $E^{(j)}_\beta = \partial_\beta E^{(j)}_\lambda|_0$ of the local family of level j vanishes on the polymers that miss some S_a, $a \in \beta$, so that its norm is at most $e^{-4\beta_s\kappa[\mathsf{D}_j/M - 2]_+}$ times its size $\mathsf{d}_{j,\beta}$; splitting into the top block and the rest, by induction on r and on j, $\mathsf{d}_{j+1,\beta} \le c_\theta \hat{G}_r \prod_{a\in\beta} \hat{g}_{j,a}$, $\hat{G}_r$ given by a recursion in r and $\hat{g}_{j,a}$ the sizes of the sources at the level j, as long as $\theta^{[j]}T_n(\mathsf{D}_j/M) \le \frac{1}{2}$; the threshold $s_{\mathrm{th}}(n)$ gives this for $1 \le j \le \mathsf{k}$, since $\mathsf{D}_j/M \ge 3$ and $x_j > L^{s-2}g^{-2}$ there.

The chain runs up to the level $\mathsf{k}+1$, and its top index is the working level k, so that the sources need their families without base point at the heights $1 \le j \le \mathsf{k}$ alone. On the history without

large fields, with e^Ψ the density of the level $t \le \mathsf{k}$, Ψ formed with the source F_λ, the identity $\partial_B e^\Psi = e^\Psi \sum_\pi \prod_{\beta\in\pi} \partial_\beta \Psi$ gives, almost everywhere and for any complex constants c_a,

$$\Phi^{(t)}\Big[\prod_{a\in B}(F_a - c_a)\Big] = \sum_{\pi\in \mathrm{Part}(B)} \prod_{\beta\in\pi} \varkappa_\beta^{(t)}, \qquad \varkappa_{\{a\}}^{(t)} = \partial_a \Psi|_0 - c_a, \qquad \varkappa_\beta^{(t)} = \partial_\beta \Psi|_0 \quad (\#\beta \ge 2),$$

since the constants multiply the density by $e^{-\sum_a \lambda_a c_a}$; the blocks of two or more elements do not depend on the c_a, and all the blocks are bounded by the sizes YM3_I:(759). The sum over the level t of the first creation (equation eq. 4.xxxi) adds the weights $\mathsf{w}_t \le 2M^{-3}\ell^3 L^{3(K-t)} e^{-\frac12 c' p_0(g_t)^2}$ (Lem. YM3_I:125.3), and on a small side $2M^{-3}\ell^3 \le M^{-3} c' A_0^2 Y_s^P$; the Gaussian decay in $K-t$ makes the sum converge. Möbius inversion gives the cumulant from the moments of the $F_a - c_a$. These steps alone give a power of $\frac{1}{d}$: the sizes of the blocks hold the factor $x_{\mathsf{k}}^{r/2} \le \bar{x}_s^{r/2}$, and the sum over t the factor L^{3s}.

Three estimates remove both factors. (i) $\hat{d} \ge 4MY_s^P$ (Lem. YM3_I:126.1, equation YM3_I:(2323)), and corners of plaquettes of different sets S_a are at the sup distance $\ge \frac{5}{4\sqrt{3}} d - a_K$; so $\mathsf{D}_\mathsf{k}/M \ge 2Y_s^P$, as in Lem. YM3_I:126.1, and a block of an order $r \ge 2$ at the working level is at most a number times $\hat{d}^3 (x_{\mathsf{k}}^{1/2} y^{\mathfrak{p}_0})^r e^{-(\mathsf{D}_\mathsf{k}/M - 2)} \prod_{a\in\beta} \upsilon \mathsf{s}_a$, in which $e^{2-2Y_s^P} \bar{x}_s^{r/2} = e^{2-2Y_s^P + rY_s/2}$ is bounded for $r \le n$ (at the levels $t < \mathsf{k}$ the blocks are larger by $L^{(3+r/2)(\mathsf{k}-t)}(\mathsf{k}-t+2)^{r\mathfrak{p}_0}$, and they enter the sum over t alone). (ii) In the sum over t, $y_{\mathsf{k}-m} \ge y + m \log\frac{3L}{4}$ for $m \ge 0$, $(y+u)^P \ge y^P + u^P$, and $u^P \ge u^2$ at $u = m\log\frac{3L}{4}$ give $e^{-\frac12 c' p_0(g_{\mathsf{k}-m})^2} \le e^{-\frac12 c' A_0^2 y^P} e^{-\frac12 c' A_0^2 (m\log\frac{3L}{4})^2}$; the first factor is kept, and as $y > Y_s^- = Y_s - \log(2L^3)$ (Lem. YM3_I:126.1, equation YM3_I:(2324), equations YM3_I:(2328)–(2336), with the notation in eq. 2.xxvi) it is at most $e^{-\frac12 c' A_0^2 (Y_s^-)^P}$, against which $L^{3s}\bar{x}_s^r$ is bounded. (iii) Take for c_a the constant part of the first order of F_a at the working level, $c_a = \sum_{i\le\mathsf{k}} \sum_{(X,z)} \dot{E}_a^{(i)}(X, \mathbf{1}, z)$, $\dot{E}_a^{(i)}$ the derivative in λ_a at 0 of the local family of the level i.

Then the singleton $\varkappa_{\{a\}}^{(\mathsf{k})}$ is the first-order part of F_a at the block field of that level, at most its supremum over the small fields, the size $\mathsf{X}_{\mathsf{k},a} \le \mathsf{c}_X y^P \hat{d}^3 \upsilon \mathsf{s}_a$ of the size definitions preceding and in Prop. YM3_I:123.2 (Lem. YM3_I:125.2, equation YM3_I:(2310), at each pair of the majorants), plus an irrelevant part, smaller by a factor that tends to 0 as $s \to \infty$; neither holds a factor x_{k}. So the leading term of a moment is the product of the singletons, at most $\prod_a C\, Y_s^P (LC_W(s))^3 \upsilon \mathsf{s}_a$; the partitions with a block of an order ≥ 2 and the sum over t add at most a number of n and g times this, and with $\upsilon^2 \le \frac98 b^{-2}$ Möbius inversion gives $k'_n(g,s) \le C_n(g) Y_s^{8n\mathfrak{p}_0}$.

Of the power $8\mathfrak{p}_0$ for each entry, $2\mathfrak{p}_0$ come from the supremum of $x_{\mathsf{k}} a_p$ over the small fields, of order $A_0^2 y^{2\mathfrak{p}_0}$, and $6\mathfrak{p}_0$ from the number $\hat{d}^3 < (LC_W(s))^3$ of the unit cubes of the working level in a ball of radius d. Of the side the proof reads the settings of the chain at the n majorants, the conditions of the sum over the level of the first creation and the bound on the weights w_t, which serves only to bound that sum. The weights enter only the tail term of each block of the Möbius inversion, the sum over the level of the first creation in the moment of that block: linearly in their prefactor, and through the Gaussian sum of the estimate (ii) above, in which $e^{-c'' p_0(g_t)^2}$ with $c'' \ge \frac12 c'$ may stand for $e^{-\frac12 c' p_0(g_t)^2}$.

As the settings and the conditions hold on every admissible side for $K \ge 3$ (Step 14), the lemma holds on every admissible side, for $K \ge 3$ and weights $\mathsf{w}_t \le \mathsf{L} L^{3(K-t)} e^{-c'' p_0(g_t)^2}$, $1 \le t \le K$, $c'' \ge \frac12 c'$, with the factor $\max\{1, M^3\mathsf{L}/(c' A_0^2)\}^n$ on k'_n: as $Y_s \ge 1$, each block gains at most the factor $\max\{1, M^3\mathsf{L}/(c'A_0^2)\}$ over the tail of a small side, and there are at most n blocks (Thm. C(E3)). □

Lemma 2.19 (As many sources as unit cubes)**.** Let $0 < g' \le \gamma_2$, (K', β) a first-passage datum at g', ℓ a side small over its running couplings, $0 < d < 1$ with $s = s(d) \le K'$, the depth, the working level and υ being formed at g', and let n be an integer with $2 \le n \le \ell^3$. Let (s_a, S_a), $1 \le a \le n$, be majorants placed as in Lem. 2.18 and F_a sources of top index k, the working level, under them, and

suppose

$$2Y_s^P \ge u_*(n), \qquad y_{K'} \ge 2\log L + \mathfrak{p}_0 \log 3, \qquad L^{3s} \le x_{K'}^{20},$$

where $u_*(n) = \max\{4, (2/A_\kappa)(n[c_1 + \log(c^* n)] + \log 3)\} \le Cn\log(en)$, $A_\kappa = 4\beta_s\kappa$, $c^* = \max\{4, \frac{1}{2}c^\flat\}$, c_1 a number of the constants. Then there are complex numbers c_a such that for every nonempty $B \subset \{1, \dots, n\}$

$$\Big|\Big\langle \prod_{a\in B}(F_a - c_a)\Big\rangle_{K',\ell,\beta}\Big| \le \prod_{a\in B} \mathsf{A}_n(s)\upsilon\mathsf{s}_a, \qquad \mathsf{A}_n(s) \le Cn^{4+\mathfrak{p}_0}(LC_W(s))^3 \bar{x}_s Y_s^P.$$

■

Proof sketch. It is the proof of Lem. 2.18 with two changes. First, the mixed derivative families of every order $r \le n$ are bounded with the numbers $(c^* r)^r$ in the place of $\hat{G}_r$, by Cauchy estimates on polydiscs of radii proportional to $1/r$; for this it suffices that $T_m(u)(c^* m)^m \le 1$ for $2 \le m \le n$ at $u = \mathsf{D}_j/M$, which holds as soon as $u \ge 4$ and $\frac{1}{2}A_\kappa u \ge m[c_1 + \log(c^* m)] + \log 3$, so for $u \ge u_*(n)$, and the first hypothesis gives $\mathsf{D}_j/M \ge u_*(n)$. Second, the moments are bounded directly, from the conditional cumulants at the upper level and the sum over the partitions, and not through the cumulants of the sources; the sum over the level of the first creation (equation eq. 4.xxxi) is bounded by the count of the cubes of a small side, through the last two hypotheses. Hence a bound in product form, with no factorial in n beyond the power in $\mathsf{A}_n(s)$. □

Lemma 2.20 (Two sources at one scale)**.** (a) Let $g \le \gamma_2$, ℓ small, $s_1(g) \le s(d) \le K$, $3d \le |x_1 - x_2| \le 4d$, and let (F_a, t_a) be real quadratic sources of top index $T-1$ under (s_a, S_a) in the sense of Lem. 2.16(a), with their families without base point at the heights $1 \le j \le T-1$, T the upper level (forms have a member at every height $j \ge 1$), S_a within $d - a_K$ of x_a. Then

$$\Big|\langle F_1; F_2\rangle_\ell - 6\upsilon^2 \sum \mathsf{t}_1(p,p')\mathsf{t}_2(q,q')C^{\mathrm{bg}}(p,q)C^{\mathrm{bg}}(p',q')\Big| \le \varepsilon_g^\natural(s(d))\upsilon^2\mathsf{s}_1\mathsf{s}_2, \qquad |\langle F_1; F_2\rangle_\ell| \le C_g^\sharp(s(d))\upsilon^2\mathsf{s}_1\mathsf{s}_2,$$

where $\varepsilon_g^\natural(s) = \varepsilon_g^\flat(s) + 2(1+A_1)^4\mathsf{C}_5 s^2 (LC_W(s))^6 Y_s^{2P}\mathrm{sm}_g(s)$,

$$\begin{aligned}\varepsilon_g^\flat(s) = {} & (\mathsf{C}_2 + \mathsf{C}_3)(Y_s^-)^{-2\mathfrak{p}_0} + \mathsf{C}_3\, s\,(Y_s^-)^{-5\mathfrak{p}_0} \\ & + \mathsf{C}_6(Y_s^-)^{-6\mathfrak{p}_0} + \mathsf{C}_8\big((LC_W(s))^3 + 1\big)(Y_s^-)^{-9\mathfrak{p}_0} \\ & + 4\mathsf{c}_X^2 (Y_s^-)^{-14\mathfrak{p}_0} \\ & + (\mathsf{C}'_B + \mathsf{C}_7 + \mathsf{C}'_3 + \mathsf{C}''_3 + \mathsf{C}_6)\, s^4 (LC_W(s))^6 Y_s^{11\mathfrak{p}_0}\mathrm{sm}_g(s),\end{aligned}$$

$Y_s^- = Y_s - \log(2L^3)$ and sm_g the functions of equations YM3_I:(2328)–(2336), with the notation in eq. 2.xxvi, C_i the constants of Lem. YM3_I:125.5, with the convention in eq. 2.xxiv–eq. 2.xxv: $\varepsilon_g^\flat$ bounds all the errors of the proof but those of the Gaussian values at the flat top, which the second summand bounds, and both tend to 0 as $s \to \infty$ (the decay argument preceding equation YM3_I:(2332)); and

$$C_g^\sharp(s) = \mathsf{C}_2 (LC_W(s))^6 Y_s^{2P} + 4\mathsf{c}_X^2 + \mathsf{C}'_B s^4 (LC_W(s))^6 Y_s^{6\mathfrak{p}_0}\mathrm{sm}_g(s) \le C(g) Y_s^{16\mathfrak{p}_0}.$$

(b) Let ℓ be admissible, $K \ge 3$, $s = s(d) \le K$, $\mathsf{Lm}_s \le L^{s-3}$, $x_s^- \ge e^P$; then the settings of the chain hold at the levels $\mathsf{k} < j_1 \le T \le K-2$, with $n_T \le s$, $676 a_K \le d$, $20d < \ell$, $M \le \hat{d} < LC_W(s)$. Let $|x_1 - x_2| \ge 3d$, $|L^{-1}x_{t-1} - x_t| \le B_\sharp$ for all t, $B_\sharp$ the flow bound of the running couplings and $\bar{B} = LB_\sharp/(L-1)$ (Section YM3_I:74), $y_K \ge 2\log(1+\bar{B})$; let $\mathsf{L} \ge 0$, $c > 0$ with $\mathsf{w}_t \le \mathsf{L}L^{3(K-t)}e^{-cp_0(g_t)^2}$ $(t \le T)$, $\mathsf{W}_T \le \mathsf{L}L^{3(K-T)}e^{-cp_0(g_T)^2}$, $2c\mathfrak{p}_0 A_0^2 y^{P-1} \ge 13$, and $\mathrm{sm}_T^{\mathsf{L},c} \le 1$. Then on T_ℓ the error in (a) is at most

$$\{(\mathsf{C}_2 + \mathsf{C}_3)y^{-2\mathfrak{p}_0} + \mathsf{C}_3 n_T y^{-5\mathfrak{p}_0} + \mathsf{C}_6 y^{-6\mathfrak{p}_0} + \mathsf{C}_8(\hat{d}^3 + 1)y^{-9\mathfrak{p}_0} + 4\mathsf{c}_X^2 y^{-14\mathfrak{p}_0} + \mathsf{C}_B^e n_T^4 \hat{d}^6 y^{15\mathfrak{p}_0}\mathrm{sm}_T^{\mathsf{L},c}\}\upsilon^2\mathsf{s}_1\mathsf{s}_2,$$

and the remainder Rest'_T of the proof of (a) obeys $|\mathrm{Rest}'_T| \le [4\mathsf{c}_X^2 y^{-14\mathfrak{p}_0} + \mathsf{C}'_B n_T^4 \hat{d}^6 y^{6\mathfrak{p}_0} \mathrm{sm}_T^{\mathsf{L},c}] \upsilon^2 \mathsf{s}_1\mathsf{s}_2$; for two plaquette weights $\mathsf{C}_B^{(1)}$ may replace C'_B.

(c) In the setting of (a), almost everywhere at the upper level T and for all the quotients,

$$\big|\Phi^{(T)}[F_1F_2] - \Phi^{(T)}[F_1]\,\Phi^{(T)}[F_2] - G\big| \le \varepsilon_g(s(d))\,\upsilon^2 \mathsf{s}_1\mathsf{s}_2,$$

$G = 6\upsilon^2 \sum \mathsf{t}_1(p,p')\mathsf{t}_2(q,q') C^{\mathrm{bg}}(p,q) C^{\mathrm{bg}}(p',q')$ the Gaussian value of (a) and $\varepsilon_g = \epsilon_g$ the error function in equation YM3_I:(2330), with the coefficient fixed in Lem. 2.13; and almost everywhere at every level $j \le T$ the conditional covariance $\Phi^{(j)}[F_1F_2] - \Phi^{(j)}[F_1]\Phi^{(j)}[F_2]$ is at most the crude size M'_j of the size definitions preceding and in Prop. YM3_I:123.2, formed at the majorants, which Lem. YM3_I:125.2 bounds by $[\mathsf{m}_\star L^{4n}(n+1)^{2\mathfrak{p}_0} + 2n_T \mathsf{sb}_0^2]\upsilon^2\mathsf{s}_1\mathsf{s}_2$, $n = [\mathsf{k}-j]_+$. ■

Proof sketch. (a) The covariance is a sum of one-step covariances plus a rest, $\langle F_1; F_2\rangle = \sum_{\mathsf{k}<j\le T} \mathrm{qm}_j + \mathrm{Rest}'_T$, $\mathrm{qm}_j = \int \mathrm{Qt}_j[X_{j-1,1}; X_{j-1,2}]\,\mathrm{d}\mathsf{m}_{j|T}$, with the step quotients Qt_j of the quotient definitions in [7, §B.9], their measures $\mathsf{m}_{j|T}$ and the first-order parts $X_{j-1,a}$ of the sources (Prop. YM3_I:123.2, which holds for these sources by Lem. 2.16(b)). The steps $j_1 < j \le T$ are bounded crudely, by $2\mathsf{X}_{j-1,1}\mathsf{X}_{j-1,2}$ each: the error Er_2 of equation YM3_I:(2307). For $\mathsf{k} < j \le j_1$ the one-step covariance is almost everywhere the total of the mixed derivative family $\mathsf{K}^{(j)}$ of the step j ([7, Lem. B.20]); carried to the upper level T (Prop. YM3_I:123.3) and evaluated at the trivial background it is the number $\mathsf{sum}_T^{(j)}$, up to the error $\mathsf{Ft}_T^{(j)}$ of the flat top, which holds at every point of the small-field set of the level T (Prop. YM3_I:123.3, equation YM3_I:(2283), in its sup-norm form); the mass $1 - \mathsf{W}_T$ of $\mathsf{m}_{T|T}$ costs $\mathsf{W}_T(\bar{\mathsf{n}}_j^{\mathrm{G}} + \mathsf{e}_{j,T})$; and $\mathsf{sum}_T^{(j)}$ is the Gaussian number $\mathsf{n}_{j,T}^{\mathrm{G}}$ up to the error $\mathsf{e}_{j,T}$ of equations YM3_I:(2303)–(2305) (Prop. YM3_I:125.1).

With kernels t_a in the place of $\frac{1}{2}w_a\delta$ this proof reads the source at three places alone: the covariance of one step at a triple is the covariance of the first-order variations of the two sources under the fluctuation measure of that step; the second order of a source at a small point of the chart is the quadratic form of its kernel; and the forms of the slices are built with the kernels. So it gives the Gaussian numbers $2(N^2-1)\upsilon_j^2 \sum \mathsf{t}_1(p,p')\mathsf{t}_2(q,q')\mathcal{C}_{(j)}(p,q)\mathcal{C}_{(j)}(p',q') + 4(N^2-1)\sum_{j<i'\le T} \upsilon_j\upsilon_{i'} \sum \mathsf{t}_1(p,p')\mathsf{t}_2(q,q')\mathcal{C}_{(j)}(p',q')\mathcal{C}_{(i')}(p,q)$, with the same errors $\mathsf{e}_{j,T}$ and the same constants, for every real quadratic source, the bond forms under $(59\sup|\mathsf{w}|, S[\mathsf{w}])$ among them (Lem. 2.16(a)).

The Gaussian numbers sum to $6\upsilon^2 \sum \mathsf{t}_1\mathsf{t}_2 DD$, $D = \sum_{\mathsf{k}<j\le T}\mathcal{C}_{(j)} = \Pi_{\mathsf{k}} - \Pi_T$, since the slices telescope and the terms (j,i'), (i',j) agree under the symmetry of the kernels, and $C^{\mathrm{bg}} = \Pi_{\mathsf{k}}$; the Gaussian numbers above j_1, the replacing of the couplings υ_j of the steps by υ and the dropping of Π_T, with $|\Pi_T(p,q)| \le C_{\mathrm{tp}}^{\mathrm{bg}} \eta_T^3$ (Prop. YM3_I:120.5), cost the errors Er_6, Er_7, Er_8 of equation YM3_I:(2307). With the deposit inequality $L^{3a}\mathsf{e}_{\mathsf{k}-a} \le L^{-10a}\mathsf{e}_{\mathsf{k}}$, $\mathsf{e}_k = e^{-\frac{1}{2}c' p_0(g_k)^2}$ (the proof of Lem. YM3_I:125.4; its hypotheses hold by equation YM3_I:(2326) and the logarithmic floors immediately following that equation),

$$|\mathrm{Rest}'_T| \le [4\mathsf{c}_X^2 y^{-14\mathfrak{p}_0} + \mathsf{C}'_B n_T^4 \hat{d}^6 y^{6\mathfrak{p}_0} \mathrm{sm}_T]\upsilon^2\mathsf{s}_1\mathsf{s}_2.$$

Collected, the left member of (a) is at most $|\mathrm{Rest}'_T| + \mathsf{Er}_2 + \sum_{\mathsf{k}<j\le j_1} \mathsf{Ft}_T^{(j)} + \mathsf{W}_T \sum_{\mathsf{k}<j\le j_1} \bar{\mathsf{n}}_j^{\mathrm{G}} + 2\sum_{\mathsf{k}<j\le j_1} \mathsf{e}_{j,T} + \mathsf{Er}_6 + \mathsf{Er}_7 + \mathsf{Er}_8$. By the bounds of the errors at the attached levels (Lem. YM3_I:125.5), with $Y_s^- < y \le Y_s$ (Lem. YM3_I:126.1, equation YM3_I:(2324)), $1 \le n_T \le s$, $\hat{d} < LC_W(s)$ and $\mathrm{sm}_T \le \mathrm{sm}_g(s)$, all these terms but the sum of the $\mathsf{e}_{j,T}$ are together at most $\varepsilon_g^\flat(s)\upsilon^2\mathsf{s}_1\mathsf{s}_2$, and $2\sum_{\mathsf{k}<j\le j_1}\mathsf{e}_{j,T} \le 2(1+A_1)^4\mathsf{C}_5 s^2 (LC_W(s))^6 Y_s^{2P} \mathrm{sm}_g(s)\upsilon^2\mathsf{s}_1\mathsf{s}_2$.

The crude bound is the same sum with the Gaussian numbers bounded by their sizes. (b) In the proof of (a) the side enters only through the weights w_t and the sum over the level of the first creation (equation eq. 4.xxxi), and the distance only through a lower bound; the factor $e^{-cp_0(g_t)^2}$ is used at three places alone: in the deposit inequality at c, in its monotonicity in t, and in the bound

on w_t. For the first, the proof of Lem. YM3_I:125.4 applies with $c'/2$ replaced by c. To make this substitution explicit, put $P = 2\mathfrak{p}_0$, $y = y_{\mathsf{k}}$ and $\Delta = y_{\mathsf{k}-a} - y$ for $1 \le a \le \mathsf{k}$. The flow bound and $y_K \ge 2\log(1+\bar{B})$ give

$$\Delta \ge a\log L + \log(1-\bar{B}/x_{\mathsf{k}}) > 2a, \qquad y(a\log L - \Delta) \le \frac{8}{3e}\frac{\bar{B}}{1+\bar{B}} < 1.$$

Keeping the linear and quadratic terms of the power gives

$$(y+\Delta)^P - y^P \ge Py^{P-1}a\log L + Py^{P-2}\left[\frac{P-1}{2}\Delta^2 - y(a\log L - \Delta)\right] \ge Py^{P-1}a\log L.$$

Thus, for $\mathsf{e}_j = e^{-c\mathfrak{p}_0(g_j)^2}$,

$$L^{3a}\mathsf{e}_{\mathsf{k}-a} \le L^{-(2c\mathfrak{p}_0A_0^2y^{P-1}-3)a}\mathsf{e}_{\mathsf{k}} \le L^{-10a}\mathsf{e}_{\mathsf{k}}.$$

The case $a = 0$ is immediate. This uses exactly the rate condition $2c\mathfrak{p}_0A_0^2y^{P-1} \ge 13$ in (b), at the working level k; no stronger rate condition at K is needed. The settings follow from the arithmetic of the working level under the two depth conditions, the flow bound and the lower bound on y_K. (c) A block of two is the conditional covariance $\Phi_{12}^{(n)} = \Phi^{(n)}[F_1F_2] - \Phi^{(n)}[F_1]\Phi^{(n)}[F_2]$.

Almost everywhere, by the tower of the quotients (equation YM3_I:(2273)), $\Phi_{12}^{(n)} = \mathrm{Qt}_n[\Phi_{12}^{(n-1)}] + \mathrm{Qt}_n[\Phi^{(n-1)}[F_1];\Phi^{(n-1)}[F_2]]$, and $\Phi^{(n-1)}[F_a]$ is a constant plus the first-order part $X_{n-1,a}$ plus an irrelevant part (equation YM3_I:(2267) and Prop. YM3_I:123.1); this gives the crude bound at every level (the proof of Prop. YM3_I:123.2, through Lem. 2.16(b)). For $\mathsf{k} < j \le j_1$ the one-step covariance $\mathrm{Qt}_j[X_{j-1,1};X_{j-1,2}]$ is almost everywhere the total $\mathsf{F}_j^{(j)}$ of the family $\mathsf{K}^{(j)}$ ([7, Lem. B.20]), and $\mathrm{Qt}_{i+1}[\mathsf{F}_i^{(j)}] = \mathsf{F}_{i+1}^{(j)}$ almost everywhere (the proof of Prop. YM3_I:123.3).

So, by induction on n from k to T, the remainder $R_n = \Phi_{12}^{(n)} - \sum_{\mathsf{k}<j\le\min\{n,j_1\}}\mathsf{F}_n^{(j)}$ is almost everywhere at most M_{k}, plus twice the products of the sizes of the first-order and of the irrelevant parts at the levels $\mathsf{k} \le j < n$, plus twice the products of the sizes of the first-order parts at the steps above j_1. At $n = T$ the measure m_T is carried by the small-field set of the level T, at every point of which $\mathsf{F}_T^{(j)}$ is within $\mathsf{Ft}_T^{(j)}$ of the number $\mathsf{sum}_T^{(j)}$ (Prop. YM3_I:123.3, equation YM3_I:(2283), before any integration); $\mathsf{sum}_T^{(j)}$ is $\mathsf{n}_{j,T}^{\mathrm{G}}$ up to $\mathsf{e}_{j,T}$, and the Gaussian numbers give G up to $\mathsf{Er}_6 + \mathsf{Er}_7 + \mathsf{Er}_8$, as in (a).

No sum over the levels of the first creation enters, so that there are no tails and no weight W_T: the error is at most the bracket of Cor. YM3_I:125.6 times $\upsilon^2\mathsf{s}_1\mathsf{s}_2$, and with $Y_s^- < y \le Y_s$, $n_T \le s$, $\hat{d} < LC_W(s)$ and $\mathrm{sm}_T \le \mathrm{sm}_g(s)$ this bracket is at most $\varepsilon_g(s)$, term by term. □

Lemma 2.21 (Symmetries and copies)**.** (a) For a lattice isometry r with $re_\mu = s_\mu e_{\pi(\mu)}$: $q_{(r\mathsf{e})^+}(rU) = s_\lambda s_{\lambda'} q_{\mathsf{e}}(U)$, $\lambda < \lambda'$ the directions other than that of e; so $rO_K^T(H) = O_K^T(rH)$, and there is a linear bijection $r^\sharp$ of the weights, with $\|r^\sharp v\| = \|v\|$, real weights kept real and carriers and sets $S[\cdot]$ carried by r, such that the image of a form is the form of the same kind with the weight $r^\sharp v$; in particular $Y[r^\sharp\mathsf{w}](rU) = Y[\mathsf{w}](U)$ and $\theta Y[\mathsf{w}] = Y[\theta^\sharp\mathsf{w}]$.

(b) A law that is invariant under the reflections in the site planes has $\nu(q_{\mathsf{e}}) = 0$; all laws are such, so $\nu(Y[\mathsf{w}]) = 0$.

(c) For a board of side l_0 and $\ell_0 = 2\mathsf{l}_0$, a form whose carrier lies in a cell at a distance $\ge \rho > a_K$ from the walls is a function of the cell; its copies on T_{ℓ_0} are forms of the same kind, size and mean, with the mirror images as carriers; copies in different cells of a period box are $\ge 2\rho$ apart, two in one cell at the original distance.

(d) If $3d \le |x_0 - y_0| \le 4d$, the midpoint is within $\frac{a_K}{2}$ of the offset of the board of side l in each coordinate, and the weights are carried within $d - 2a_K$ of x_0, y_0, then the corners of their plaquettes lie within the sup distance $3d$ of the centre of the cell, and the sixteen copies on $T_{\ell'}$ form eight pairs,

partners $3d$ to $4d$ apart, different pairs at sup distance $\ge \mathsf{l} - 6d \ge 11M$, the suprema of the weights kept; the same holds for tensor weights and arbitrary majorants.

(e) For majorants with plaquettes within $d - a_K$ of x_0, y_0, $3d \le |x_0 - y_0| \le 4d$, the tents $\tilde f_a = \mathsf{s}_a \sum_{p\in S_a} \max\{0, 1 - 2a_K^{-1}|u - x(p)|_\infty\}$ are Lipschitz, equal $\mathsf{s}_a \mathbf{1}_{S_a}$ at the barycentres, have their supports in $B(x_a, d)$, and $d \le \operatorname{dist}(\operatorname{supp}\tilde f_1, \operatorname{supp}\tilde f_2) \le 6d$. Let besides $0 < g \le \gamma_2$, (K, β) a first-passage datum at g, ℓ small, $0 < d < 1$ and $s_1(g) \le s = s(d) \le K$, so that $\mathsf{Lm}_s \le L^{s-3}$ and $x_s^- \ge e^P$ (equation YM3_I:(2331)). Then $s \ge 3$, $2 \le \mathsf{k} \le K - 1$, $M \le L^{-2} C_W(s) \le \hat d < L C_W(s)$, $\mathsf{D}_\mathsf{k} \ge 3M$ for the pair, $20d \le \ell$, and the middle and the upper level have $\mathsf{k} < j_1 \le T \le K - 2$, $1 \le n_T \le s$, $L^{n_T} < \mathsf{Lm}_s$, $x_T > x_s^-$.

Let $n \ge 1$ pairs of such majorants be given, $p = 2n$, the corners of plaquettes of different pairs at the sup distance $\ge 11M$. Then D_k, the least over all $a \ne a'$, is $\ge 3M$: partners give $3M$, members of different pairs $11ML^{K-\mathsf{k}}$. The settings of the chain for the p sources hold with the top index $T - 1$, with the smallness conditions at the heights $0, \dots, T - 1$ and $d_*(t) \ge 3$ for $1 \le t \le T$; the levels, the lattices, the histories, the measures and the quotients of the chain are the same for all pairs, and for the chain of each pair alone, and do not depend on the sources; and $\theta^{[j]} T_p(\mathsf{D}_j/M) \le \frac12$ for $1 \le j < T$ as soon as

$$\mathrm{th}_{g,p}(s) = 2s^{2+p\mathfrak{p}_0} \mathsf{Lm}_s^{p/2-1} T_p^+ \, \theta(x_s^-) \le 1, \qquad T_p^+ = T_p(3) + (\tfrac43)^{p/2} L,$$

θ the function of Lem. 2.18; $\mathrm{th}_{g,p}(s)$ tends to 0 as $s \to \infty$, being a power of s times $L^{-s/2}$. ■

Proof sketch. (a) r carries the loops at e to those at $r\mathsf{e}$; reversal sends $w(V)$ to $w(V^{-1}) = -w(V)$, conjugation keeps $\langle \cdot, \cdot \rangle$, and the sign is the product of the signs of the two vectors u_λ, $u_{\lambda'}$. (b) The reflection in the site plane through x_e normal to λ flips s_λ alone, so that it changes the sign of q_e; torus measures are invariant under it, periodic Gibbs states by Prop. YM3_I:103.4. (c) Each coordinate of a copy lies between two consecutive walls, at a distance $\ge \rho$ from both, and the mean is kept by the invariance of $\tilde\mu_{\ell_0}$ under the lattice isometries. (d) The corners lie within the sup distance $3d$ of the centre of the cell, and $\mathsf{l} \ge LM$. (e) A tent is carried within $\frac{\sqrt3}{2} a_K$ of its barycentre and vanishes at the other barycentres; the supports need not come near the centres, which is why only $d \le \mathrm{dist} \le 6d$ holds, and nothing uses an upper bound on the distance.

The settings and the numbers come from the arithmetic of the working level (Lem. YM3_I:126.1) and from Lem. YM3_I:126.2, by which the objects of the construction satisfy the settings of the chain; neither involves a source, and of the majorants the settings for the p sources read the weights $\mathsf{s}_a \mathbf{1}_{S_a}$ and the bound $\mathsf{D}_\mathsf{k} \ge 3M$ alone. *The threshold.* $T_p(u) = \sum_{n' \ge 1} (\frac43)^{p/2} (n' + 1)^{1+p\mathfrak{p}_0} L^{1+(p/2-1)n'} e^{-4\beta_s \kappa [L^{n'-1} u - 2]_+}$ does not increase. If $L^q u \ge 3$, $q \ge 1$, the q terms with $n' \le q$ are each at most $(\frac43)^{p/2} (q+1)^{1+p\mathfrak{p}_0} L^{1+(p/2-1)q}$, and the term of the index $n' = q + n''$ is at most $(q+1)^{1+p\mathfrak{p}_0} L^{(p/2-1)q}$ times the term of the index n'' of $T_p(3)$, as $n' + 1 \le (q+1)(n''+1)$ and $L^{n'-1} u \ge 3L^{n''-1}$; so $T_p(u) \le (q+1)^{2+p\mathfrak{p}_0} L^{(p/2-1)q} T_p^+$.

For $1 \le j \le \mathsf{k}$, $\mathsf{D}_j/M \ge 3$ and $T_p(\mathsf{D}_j/M) \le T_p(3) \le T_p^+$; for $\mathsf{k} < j < T$, $q = j - \mathsf{k} \le n_T - 1$ has $L^q \mathsf{D}_j/M = \mathsf{D}_\mathsf{k}/M \ge 3$, $q + 1 \le n_T \le s$ and $L^{(p/2-1)q} < \mathsf{Lm}_s^{p/2-1}$. And $\theta^{[j]} = \theta(x_j) \le \theta(x_s^-)$, since $x_j \ge x_T > x_s^- \ge e^P$ and θ does not increase on $[e^P, \infty)$. So $\theta^{[j]} T_p(\mathsf{D}_j/M) \le \frac12 \mathrm{th}_{g,p}(s) \le \frac12$ for $1 \le j < T$. □

Lemma 2.22 (Far pairs; centred sources)**.** Let $0 < g \le \gamma_2$, ℓ small, $n \ge 1$, $p = 2n$, $s = s(d) \le K$. For pairs $\tau \le n$ let $3d \le |z_{(\tau,1)} - z_{(\tau,2)}| \le 4d$, let the majorants have their plaquettes within $d - a_K$ of z_a, and let corners of plaquettes of different pairs be at sup distance $\ge 11M$; let (F_a, t_a) be real quadratic sources of top index $T - 1$ under the majorants, with their families without base point at the heights $1 \le j \le T - 1$ (as forms have them), $\tilde F_a = F_a - \langle F_a \rangle_\ell$, $\tilde P_J = \prod_{\tau\in J} \tilde F_{(\tau,1)} \tilde F_{(\tau,2)}$, G_τ the Gaussian value of the pair, $\hat m_a = \upsilon \mathsf{s}_a$, $\hat m_\tau = \hat m_{(\tau,1)} \hat m_{(\tau,2)}$, $\hat m_J = \prod_{\tau\in J} \hat m_\tau$. For $s \ge s_p(g)$, a first

depth $\ge s_1(g)$, and all nonempty J and B:

$$|\langle\tilde P_J\rangle_\ell - \textstyle\prod_J G_\tau| \le \omega'_{g,p}(s)\hat m_J, \qquad |\langle\tilde P_J\rangle_\ell - \prod_J\langle\tilde P_\tau\rangle_\ell| \le \omega'^\sharp_{g,p}(s)\hat m_J, \qquad |G_\tau| \le g_0\hat m_\tau;$$

if N_2 pairs lie in B and N_1 meet it in one element,

$$\Big|\Big\langle\prod_B \tilde F_a\Big\rangle_\ell\Big| \le [e_g(s)^{N_1}(g_0+\delta_g(s))^{N_2} + \mathrm{Bl}(p)\mathrm{far}_{g,p}(s)(1+g_0+\varepsilon_g(s))^n + \mathrm{tl}_{g,p}(s)]\prod_B \hat m_a,$$

so that this mean is $\le o_{g,p}(s)\prod_B\hat m_a$, $o_{g,p} = e_g + \mathrm{Bl}(p)\mathrm{far}_{g,p}(1+g_0+\varepsilon_g)^n + \mathrm{tl}_{g,p}$, if B holds no pair, and $\le \mathrm{Mo}_{g,p}(s)\prod_B\hat m_a$ in every case, $\sup_{s\ge s_p(g)}\mathrm{Mo}_{g,p}(s) < \infty$; $Y_s^P\omega'^\sharp_{g,p}(s)$ is bounded. Here

$$\omega'_{g,p} = n\delta_g(1+g_0+\delta_g)^{n-1} + \mathrm{Bl}(p)\mathrm{far}_{g,p}(1+g_0+\varepsilon_g)^n + \mathrm{sm}_g(1+g_0)^n + \mathrm{tl}_{g,p}, \qquad \omega'^\sharp_{g,p} = \omega'_{g,p}[1+n(1+g_0+\omega'_{g,p})^{n-1}],$$

with $g_0 = \frac{3}{2}81^2C_{\mathsf k}'^2$ for arbitrary majorants and $g_0 = 422C_{\mathsf k}'^2$ for majorants of bond weights of one direction. For bond forms $Y[\mathsf w_a]$ with real weights on the bonds of one direction, whose means vanish, $|G_\tau| \le 106C_{\mathsf k}'^2\upsilon^2\sup|\mathsf w_{(\tau,1)}|\sup|\mathsf w_{(\tau,2)}|$, and the same functions at $g_0 = 106C_{\mathsf k}'^2/59^2$, written $\omega_{g,p}$, $\omega^\sharp_{g,p}$, serve in units of $\hat m_a = 59\upsilon\sup|\mathsf w_a|$. Here $\mathrm{far}_{g,p}(s) = C_p^{\mathrm{far}}(LC_W(s))^3\bar x_s^{p/2}Y_s^{p\mathfrak p_0}e^{-11L^{s-2}/\mathsf{Lm}_s}$, ε_g is the error function of equations YM3_I:(2328)–(2336), with the notation in eq. 2.xxvi for a close pair, $\delta_g = \varepsilon_g + e_g^2$, e_g and $\mathrm{tl}_{g,p}$ are the error functions of the singletons and of the tails,

$$e_g(s) = 2\mathsf c_X(Y_s^-)^{-7\mathfrak p_0} + 2(\mathsf c_I + \mathsf c_{\mathrm{tl}}\mathsf c_{\mathrm{sb}})\,s^2(LC_W(s))^3Y_s^{3\mathfrak p_0}\mathrm{sm}_g(s),$$

$$\mathrm{tl}_{g,p}(s) = \mathrm{Bl}(p)\,\mathsf c_\sigma^p\mathsf c_{\mathrm{sb}}^p(1+S_p^{\mathrm{tl}})\,s^{1+3p/2}(LC_W(s))^{3p}Y_s^{2p\mathfrak p_0}\,M^{-3}c'A_0^2Y_s^PL^{3s}\bar x_s^p\,e^{-\frac12c'A_0^2(\log x_s^-)^P},$$

$\mathsf c_\sigma = 4 + \mathsf C_{\mathsf M}^{1/2}/\mathsf C_{\mathrm{sz}} + \sqrt2$, $\mathsf C_{\mathrm{sz}}$ of Lem. YM3_I:125.2 and the constant convention preceding eq. 2.xxiv, $S_p^{\mathrm{tl}} = \sum_{a\ge0}L^{(3+4p)a+4p}(a+2)^{2p\mathfrak p_0}e^{-\frac12c'A_0^2(a\log\frac{3L}{4})^2}$, $\mathrm{Mo}_{g,p} = (1+g_0+\delta_g)^n + \mathrm{Bl}(p)\mathrm{far}_{g,p}(1+g_0+\varepsilon_g)^n + \mathrm{tl}_{g,p}$, $\mathrm{Bl}(p)$ is the number of partitions of p elements, and $s_p(g)$ is the least integer $\ge s_1(g)$ from which $\mathrm{th}_{g,p}$ of Lem. 2.21(e), $\mathrm{far}_{g,p}$ and e_g are at most 1 and $2\mathsf c_{\mathrm{tl}}s\,\mathrm{sm}_g(s) \le 1$; it exists, since these four functions tend to 0. ■

Proof sketch. By Lem. 2.21(e) all pairs share one working level, one upper level T and one tower of quotients, and, as $\mathrm{th}_{g,p}(s) \le 1$ for $s \ge s_p(g)$, $\theta^{[j]}T_p(\mathsf D_j/M) \le \frac12$ for $1 \le j < T$, so that the mixed derivative families of every order $\le p$ are bounded up to the level T (the bound of the proof of Lem. 2.18). At T, almost everywhere, $\Phi^{(T)}[\prod_B\tilde F_a] = \sum_\pi\prod_{\beta\in\pi}\varkappa_\beta^{(T)}$ (the identity of the proof of Lem. 2.18, which holds at every level of the chain in use, here up to T, at $c_a = \langle F_a\rangle_\ell$); the conditional cumulants of order ≥ 2 do not depend on the centring constants.

A block that links two pairs is at most $\mathrm{far}_{g,p}(s)\prod\hat m_a$: by the vanishing of mixed derivatives on polymers that miss a support it is exponentially small in the distance $11M$ measured in units of the level T, that is in $L^{K-T} > L^{s-2}/\mathsf{Lm}_s$. A block of one pair is the conditional covariance $\Phi^{(T)}[F_{(\tau,1)}F_{(\tau,2)}] - \Phi^{(T)}[F_{(\tau,1)}]\Phi^{(T)}[F_{(\tau,2)}]$, the quotients being those of the chain of that pair alone (Lem. 2.21(e)); by Lem. 2.20(c) it is almost everywhere within $\varepsilon_g(s)\hat m_\tau$ of G_τ. A singleton obeys, by the sum over the level of the first creation (equation eq. 4.xxxi) applied to F_a minus its flat-top constant,

$$|\Phi^{(T)}[F_a] - \langle F_a\rangle_\ell| \le 2\mathsf Y_{T,a} + \rho'_a \le e_g(s)\hat m_a,$$

$\mathsf Y_{T,a}$ the size of F_a at the level T and ρ'_a its first-creation sum; for bond forms the mean is 0, and the bound is used only in its last line, so that centring replaces the vanishing of the mean. So the partitions of B without a linking block add up to the product, over the pairs that meet B, of $\varkappa^{(T)}_{\{(\tau,1),(\tau,2)\}} + \varkappa^{(T)}_{\{(\tau,1)\}}\varkappa^{(T)}_{\{(\tau,2)\}} = G_\tau + O(\delta_g(s))\hat m_\tau$ if both members lie in B, and of a singleton if one does. Crude bounds of the conditional moments at all the levels $t \le T$ (for a block of one pair, the crude clause of Lem. 2.20(c)), multiplied by $\mathsf w_t$ and summed, give the tails $\mathrm{tl}_{g,p}$.

Summing over the at most $\mathrm{Bl}(p)$ partitions gives the bounds of the lemma, and $\langle\tilde P_\tau\rangle = G_\tau + O(\delta_g)\hat m_\tau$ gives the second from the first. *The size of G_τ.* For plaquettes $p \in S_{(\tau,1)}$, $q \in S_{(\tau,2)}$, $|x(p)-x(q)| \ge 3d - 2(d-a_K) > d$; so $|C^{\mathrm{bg}}(p,q)| \le C'_{\mathsf k}(a_K/d)^3$ (Lem. 2.17), and $|G_\tau| \le 6C'^2_{\mathsf k}(a_K/d)^6(\sum|\mathsf t_{(\tau,1)}|)(\sum|\mathsf t_{(\tau,2)}|)v^2$. The two sums are counted by volume. The midpoints of the bonds of one direction form a translate of $a_K\mathbb Z^3$. Let $\mathsf w$ be a weight on the bonds of one direction with $S[\mathsf w]$ within $d-a_K$ of z_a.

A bond of its support has its four plaquettes in $S[\mathsf w]$, and its midpoint is the midpoint of the centres of two of them, opposite in one plane, so that it lies within $d-a_K$ of z_a; the open cubes of side a_K about these midpoints are disjoint and lie within $d - a_K + \frac{\sqrt3}{2}a_K < d$ of z_a, in $B(z_a,d)$, a Euclidean ball as $20d \le \ell$; so the support of $\mathsf w$ has at most $\frac{4\pi}{3}(d/a_K)^3$ bonds. For a bond form $Y[\mathsf w]$ on the bonds of one direction, q_b being a quarter of a signed sum of four corner forms, $\sum_{p,p'}|\mathsf t(p,p')| \le \sum_b|\mathsf w(b)| \le \frac{4\pi}{3}(d/a_K)^3\sup|\mathsf w|$, and $6(\frac{4\pi}{3})^2 < 106$. For a majorant of such a form, $\#S \le 4\cdot\frac{4\pi}{3}(d/a_K)^3$, each bond lying in four plaquettes, and $\sum|\mathsf t| \le \frac12(\#S)\mathsf s$, as $2\sum_{p'}|\mathsf t(p,p')| \le \mathsf s$; so $6(\frac{8\pi}{3})^2 < 422$.

For an arbitrary majorant, $\#S \le 81(d/a_K)^3$ (the plaquette count following equation YM3_I:(2300)) and $\sum|\mathsf t| \le \frac12(\#S)\mathsf s$ give $6(\frac{81}{2})^2 = \frac32 81^2$. □

Proposition 2.23 (Pressure, domination, boards, extraction)**.** (a) For $0 < g \le \gamma$, $\ell_1 \le \ell_2$ in $\mathfrak L(g)$ and every datum, $0 \le p_{\ell_1} - p_{\ell_2} \le G_g$; from Thm. YM3_I:D alone, $0 \le p_{\ell_1} - p_{\ell_2} \le G^F_g = C_g + C_{\mathrm{maj}} + 2 + 2C_{\mathrm{tt}}C'_{\mathrm{tt}}E_0 \le 2C_g + 2C_{\mathrm{tt}}C'_{\mathrm{tt}}E_0$.

(b) Let K, $\beta > 0$, a board of side $\mathsf l_0$, a side $\ell_0 \ge 2\mathsf l_0$ and G with $p_{\ell_0} - p_\ell \le G$ on an unbounded set of sides be given. For bounded functions $\Phi_{\mathbf k}$ of distinct cells and every $\nu \in \mathcal G^{\mathrm{per}}_{K,\beta}$ (and for $\tilde\mu_\ell$ if the cells are distinct modulo the period): $|\nu(\prod\Phi_{\mathbf k})| \le \prod e^{G\mathsf l_0^3}\|\Phi_{\mathbf k}\|_{\mathrm{ch},\ell_0}$. No reflection invariance of ν is used.

(c) Let $h > 0$ with $\mathsf l_0 \in h\mathbb N$, as $h = a_K$, or $h = a_{K-t}$ for a level t. If $n(r+2\rho+h) < \mathsf l_0$, any n cubes of side r admit an offset in $h\mathbb Z^3$ with all their points at a distance $\ge\rho$ from the walls, also after translations by $\mathsf l_0\mathbb Z^3$; at $h = a_{K-t}$ the walls lie along faces of the blocks of the level t.

(d) Let functionals $\Lambda^{(j)}_n$ on tuples in $C_c(\mathbb R^3;V)$ with disjoint supports be linear in each entry on the tuples in $B(0,R')$ from an index $j_0(n,d,R,R')$ on, and bounded there by $c_n(d;R)\prod|H_a|_\infty$ at the tuples of type $(d;R)$ (supports in balls of radius R, at mutual distance $\ge d$). Then every subsequence has a subsequence that converges at all tuples, to multilinear limits with the same bounds. ■

Proof sketch. (a) At the source 0 the localised source energies of [7, Lem. N.1] give $E/\ell^3 - C_g \le p_\ell \le E/\ell^3 + C_S$ and $|E_\ell/\ell^3 - E_{\ell'}/\ell'^3| \le 2C_{\mathrm{tt}}C'_{\mathrm{tt}}E_0$, and p_ℓ does not increase with ℓ by the chessboard estimate (Cor. YM3_I:86.6). Without Thms. YM3_I:AJ and YM3_I:AD, $\tilde Z(0) = e^{-E}Z$ with $e^{-C_g\ell^3} \le \tilde Z(0) \le e^{(C_{\mathrm{maj}}+2)\ell^3}$ gives G^F_g. (b) The chessboard estimate on T_ℓ (Prop. YM3_I:86.3) gives $|\langle\prod\Phi_{\mathbf k}\rangle_\ell| \le \prod\|\Phi_{\mathbf k}\|_{\mathrm{ch},\ell}$, and comparing Z_ℓ with Z_{ℓ_0} cell by cell gives $\|\Phi\|_{\mathrm{ch},\ell} \le e^{\mathsf l_0^3(p_{\ell_0}-p_\ell)}\|\Phi\|_{\mathrm{ch},\ell_0}$ (Cor. YM3_I:86.6).

For every periodic Gibbs state ν and every unbounded set of sides, the domination theorem (Prop. YM3_I:91.9) gives $|\nu(\prod\Phi_{\mathbf k})| \le \liminf_\ell\prod\|\Phi_{\mathbf k}\|_{\mathrm{ch},\ell}$, the lower limit taken along the sides of the set; along the sides with $p_{\ell_0} - p_\ell \le G$ the comparison bounds the right member by $\prod e^{G\mathsf l_0^3}\|\Phi_{\mathbf k}\|_{\mathrm{ch},\ell_0}$. (c) In each direction there are $\mathsf l_0/h$ offsets modulo $\mathsf l_0$, and fewer than $n((r+2\rho)/h+1) < \mathsf l_0/h$ of them are bad. (d) The bound gives the equicontinuity $|\Lambda(X) - \Lambda(X')| \le c_n\sum_a|H_a - H'_a|_\infty\prod_{a'\ne a}\max\{|H_{a'}|_\infty, |H'_{a'}|_\infty\}$; a countable set of Lipschitz functions is dense with control of the supports; a diagonal sequence converges on it, hence everywhere. □

Lemma 2.24 (The rate on a fixed torus)**.** Let $g \le \gamma_2$, $\mathsf b \in I(g)$, $\ell \in \mathfrak L(g)$, $q \ge 2$, $d > 0$, $\theta_q = 1/(1260q(8q+3))$, $\mathsf p = \max\{2\mathsf p_0+2, \mathsf p_0, 1\}$, where $\mathsf C_{\mathrm{bf}}\ell^3K^{\mathsf p_0}L^{-K/10}$, $K \ge K_{\mathrm{bf}}$, is the comparison of the laws of the block fields at two cutoffs of Cor. 2.8. There are C, of the constants, q and

C_{bf}, and K_1, besides of g, b, ℓ, d, such that for $K \geq K_1$ and $G_K = \langle \mathcal{E}_1; \dots; \mathcal{E}_q \rangle_{K,\ell,L^K\mathsf{b}}$, the $\mathcal{E}_a$ any of $\mathcal{O}_K(\varphi) = O_K^T(\varphi \mathbf{1})$, $O_K^T(\varphi E_{\lambda\lambda})$ (the plaquettes of the plane normal to λ alone) and $O_K^T(\varphi E_{\lambda\lambda'})$, $\lambda \neq \lambda'$, read at the centres of the plaquettes and the bonds, with real φ_a, Lipschitz in $|\cdot|_\infty$, whose supports are pairwise at sup distance $\geq d$:

$$|G_{K+1} - G_K| \leq C\ell^{3(q+1)} K^{\mathsf{p}} L^{-\theta_q K} \prod_a (\sup|\varphi_a| + \operatorname{Lip}\varphi_a).$$

So $\lim_K G_K$ exists, with the tail of the series as the error; any number of the entries may be off-diagonal, and one pair (C, K_1) serves a compact interval of labels. The same holds with $\prod_a \sup|\psi_a|$ for bounded Borel functions ψ_a that vanish off closed sets at pairwise sup distances $\geq d$, entered through their cell means: at a site x of $\Lambda_{K,\ell}$ the mean $\psi_a^{(K)}(x) = a_K^{-3} \int_{x + a_K[-\frac{1}{2},\frac{1}{2})^3} \psi_a$ is the weight of the plaquette with the corner x and the edges $a_K e_\kappa$, $a_K e_{\kappa'}$, for the component $\lambda\lambda$, $\{\kappa, \kappa', \lambda\} = \{1, 2, 3\}$, and of the bond $\langle x, x + a_K e_\mu \rangle$, for the component $\lambda\lambda'$, $\lambda \neq \lambda'$, μ the third direction, for all six components.

As L is odd, the cells of $\Lambda_{K+1,\ell}$ at the sites of the block $B(y)$ of a site y of $\Lambda_{K,\ell}$ make up the cell of y, so that $\psi^{(K)}(y) = L^{-3} \sum_{x \in B(y)} \psi^{(K+1)}(x)$: the weights of two cutoffs are tied exactly. So the limit exists for test functions of no regularity and agrees with that for Lipschitz functions. For one function evaluated at the points of the lattices of two cutoffs no bound of this kind by the sup norms holds. For one off-diagonal entry among action densities the rate is free of their number: if $n \geq 1$, $\lambda \neq \lambda'$, and $\varphi, f_1, \dots, f_n$ are real Lipschitz functions on T_ℓ, none of them 0, whose $n+1$ supports have no common point, then $\kappa_K = \langle O_K^T(\varphi E_{\lambda\lambda'}); \mathcal{O}_K(f_1); \dots; \mathcal{O}_K(f_n) \rangle_{K,\ell,L^K\mathsf{b}}$ obeys

$$|\kappa_{K+1} - \kappa_K| \leq C K^p L^{-K/120} (\sup|\varphi| + \operatorname{Lip}\varphi) \prod_{i=1}^n (\sup|f_i| + \operatorname{Lip} f_i) \qquad (K \geq K_2),$$

with C, p, K_2 of the constants, n, ℓ, the end points of a compact interval of labels that holds b and the spread d_∞ of the $n+1$ supports (the definition preceding Lem. YM3_I:114.6). ∎

Proof sketch. Let $s < \sigma' \leq \lceil K/1260 \rceil$ be depths, the second at most the depth up to which Cor. 2.8 compares the laws of the block fields, and $\sigma' \geq (8q+3)s$. At each of the cutoffs K, $K+1$ the cumulant is, up to three errors, the cumulant of the first members of the sources, that is of the functions $\mathcal{E}_a$ themselves and not of their families at the heights, evaluated at the background of the block field of the depth σ': the progression with the sources switched on at the level $K - \sigma'$ forgets what lies below, with the error $\epsilon(\sigma' - s)$, $\epsilon(\nu) = C(\nu+1)^{2\mathfrak{p}_0+2} L^{-\nu/2}$; the large fields cost $L^{-(b'+1)s}$, for any $b' \geq 4q$; and the distance of the supports costs $e^{3 - L^s d/M}$.

Each of the three multiplies $(4\ell^3 L^{4s})^q \prod_a (\sup|\varphi_a| + \operatorname{Lip}\varphi_a)$, the bound of the entries at the depth s; as $L^{4qs} L^{-(\sigma'-s)/2} \leq L^{-s}$ for $\sigma' - s \geq (8q+2)s$ and $L^{4qs} L^{-(b'+1)s} \leq L^{-s}$ for $b' \geq 4q$, together they are of the order $K^{2\mathfrak{p}_0+2} L^{-s}$. Over one block field V of the depth σ', with $t = K - \sigma'$, the background U^{A} on $\Lambda_{K,\ell}$ and the background U^{B} on $\Lambda_{K+1,\ell}$ are compared in three links. (1) $U' = \mathcal{M}(U^{\mathrm{B}})$, U^{B} averaged once, is a configuration of the coarse lattice over the same block field, a competitor whose excess energy over U^{A} is at most $C\varepsilon_{\mathrm{r}}^2 \eta_t N_t$, ε_{r} the radius of the small fields of the depth σ' and N_t the number of the cubes of the level t: its energy is at least that of U^{A}, by the minimality of U^{A}, and at most that of U^{B} up to a number of the same order, by the inequality of Jensen of (2); and the energy of U^{B} exceeds that of U^{A} by at most a number of the same order, since the prolongation of U^{A} to the fine lattice is a competitor of U^{B} whose energy exceeds that of U^{A} by at most such a number. (2) The fine background against its average: for a bilinear form of the loops at one site or at one bond, the defect of Jensen's inequality $\sum_p \mathsf{W}(p', p)|g_p|^2 \geq L^{-2}|G'_{p'}|^2$, with the averaging kernel W, g_p the fine and $G'_{p'}$ the averaged curvature, is the variance of the fine curvature on the block of p', and its sum over the blocks is the difference of the two energies; so the curvature is nearly constant in mean square under the shifts of a plaquette in its own plane within a block, with the error $L^{-t/2}$, and the weights enter only through the mismatch of their block means, which is 0

for cell means and at most $\frac{3}{2}a_K \mathrm{Lip}\,\varphi$ for one function read at the centres on the two lattices. (3) The competitor against U^{A}: along the t consecutive levels each pair of backgrounds differs by one gauge transformation and one small 1-form A_k, with $\sum_k L^{t-k}\|A_k\|^2$ at most a number times the excess, which costs $(t\eta_t)^{1/2}$.

So the corner fields $a_K^{-1}\sum_x \varphi(x+a_K\xi)Q_x(l,l')$, $Q_x(l,l')$ the corner forms of Lem. 2.15(b), of which $O_K^T(\varphi E_{\lambda\lambda'})$ is a sum of four with the coefficients $\pm\frac{1}{4}$, agree at the two backgrounds up to $C\ell^3 L^{4\sigma'}K^{1/2}L^{-(K-\sigma')/2}(\sup|\varphi|+L^{-\sigma'}\mathrm{Lip}\,\varphi)$ for Lipschitz weights, and up to the same number with $\sup|\psi|$ for weights tied by the cell mean; for one function evaluated at the points of two lattices the mismatch of the block means is not small in the sup norm, and there the Lipschitz number pays. The block-field laws at K and $K+1$ differ by $\mathsf{C}_{\mathrm{bf}}\ell^3 K^{\mathsf{p}_0}L^{-K/10}$ (Cor. 2.8(a), whose numbers read the label only through an interval that holds it).

With $\sigma'=\lfloor K/(1260q)\rfloor$ and $s=\lfloor\sigma'/(8q+3)\rfloor$ the four errors give the rate θ_q. For one off-diagonal entry among action densities whose supports have no common point another road keeps the action densities as sources in the exponent of the progression, as in the comparison of two cutoffs with sources (Thm. YM3_I:AR), and passes the one off-diagonal entry alone to the block field as a bounded function: it is put into the chain as a member whose tower is split at the depth $\lceil K/36\rceil$; what it creates below the level $K-\lceil K/36\rceil$ is born at a small coupling, and above it the member is carried along the run and forgotten.

No factor $L^{4q\sigma'}$ enters, the rate is free of n, and it is $L^{-K/120}$. □

Proposition 2.25 (What the properties of the limits do not decide)**.** Let σ be a limit through periodic Gibbs states, $g\le\gamma_1$, and for real ζ let Λ_ζ be the linear map of V that is 1 on the diagonal and ζ on the off-diagonal matrices, $\sigma^\zeta(X)=\sigma(\Lambda_\zeta X)$.

(a) σ^ζ is a real symmetric functional, bounded as in (D1) with the factor $\max\{1,|\zeta|\}^n$, invariant under translations and W_3, reflection positive at the nine families; $[F]\mapsto[\Lambda_\zeta F]$ is an isometry of its Hilbert space into that of σ that intertwines the semigroups, the translations and the vacua; so σ^ζ has the cone of [7, Thm. BZ(F4)], the insertion bound of [7, Thm. BZ(F2)] with the factor $\max\{1,|\zeta|\}$ for each entry, and the plateau of [7, Thm. BZ(F6)].

(b) For $\zeta\neq 1$, σ^ζ is not invariant under the rotations about e_3; it is reflection positive at no plane outside the nine families; and it does not obey the free law of [7, Thm. BZ(F1)] at any label. ■

Proof sketch. (a) Λ_ζ commutes with $\rho(w)$ for $w\in W_3$, since a signed permutation carries diagonal matrices to diagonal and off-diagonal to off-diagonal ones, and the mirrors of the nine families act on the labels by elements of W_3; the rest follows from the isometry. (b) Let $D=E_{11}-E_{22}$ and $O=E_{12}$, the symmetric matrix with the entries 1 at $(1,2)$ and $(2,1)$, so that $\rho(R_t)D=\cos 2t\,D+\sin 2t\,O$, $\rho(R_t)O=-\sin 2t\,D+\cos 2t\,O$ for the rotations R_t about e_3, and let $C=E_{33}$. If σ and σ^ζ were both invariant under the R_t, the mixed kernel would obey $(\zeta-1)\sin 2t\;\mathcal{G}(R_tu)(O,C)=0$, so $\mathcal{G}(\cdot)(O,C)\equiv 0$; but by [7, Thm. BZ(F1)] at short distance $\mathcal{G}(u)(O,C)$ is close to b^{-2} times the mixed free kernel $3K_3(u)_{13}K_3(u)_{23}$, which does not vanish identically.

By [7, Prop. D.16](b) σ^ζ is then positive at no further plane. If σ^ζ obeyed the free law at a label b', the law at two diagonal entries, where $\sigma^\zeta=\sigma$, would give $\mathsf{b}'=\mathsf{b}$; and the mixed law depends on ζ with its sign, $\sigma^\zeta(H_1,H_2)=\zeta\sigma(H_1,H_2)$ for H_1 off-diagonal and H_2 diagonal, while the mixed free covariance does not vanish identically; so $\zeta=1$. In the proof of [7, Prop. D.15] every input but conservation holds for σ^ζ: it is conservation that excludes it at all distances, and the mixed free law at short distance. □

Lemma 2.26 (Extension over coinciding points)**.** Let $\mathsf{M}=\mathbb{R}^3$ or T_ℓ, $n\ge 2$, $A,h_0>0$, $q\ge 0$, and let u be a functional of n entries, n-linear on $\prod_a C_c^\infty(O_a;V_{\mathbb{C}})$ for open O_a with pairwise disjoint closures, with the bound

$$|u(f_1,\dots,f_n)|\le A\,(1+\log\tfrac{h_0}{h})^q\prod_a\sup|f_a|$$

for $0 < h \le h_0$ and all f_a supported in closed cubes of side $\frac{3}{2}h$, wherever they lie, at pairwise sup distances $\ge \frac{5}{2}h$; on T_ℓ the number h_0 lies in $\ell 2^{-\mathbb{N}}$ and $h_0 \le \ell/(64n)$.

(a) There is a distribution $\tilde u$ on M^n of order $\le 3n-2$ that agrees with u on separated tensor products and has

$$|\tilde u(F)| \le C_{\mathrm{ex}} A (1+\log \tfrac{h_0}{\lambda})^{q+1} \max_{|\gamma|\le 3n-2} \lambda^{|\gamma|} \sup |\partial^\gamma F|$$

for every F supported in a product of closed sup-balls of radius $\lambda \le h_0$ about any points, C_{ex} depending only on n and q, with the label norm and auxiliary smooth cutoffs fixed; if u is invariant under permutations of the entries, translations and a closed subgroup of $O(3)$ that maps M to itself, $\tilde u$ can be chosen invariant.

(b) Two extensions with this bound differ by $\sum_{|\pi|\le n-1} \sum_{|\alpha|\le 3|\pi|} \partial_\xi^\alpha \delta_\pi \otimes T_{\pi,\alpha}$, π the partitions of the entries, δ_π the diagonal of π, ξ the differences of the points in the blocks, and $T_{\pi,\alpha}$ distributions on $\mathsf{M}^{|\pi|}$ with the bound of (a) at the power $\lambda^{|\alpha|}$; and every such sum can be added. A distribution with the bound of (a) at a power $\lambda^{\kappa'}$, $\kappa' > 3k$, on M^k is 0.

(c) At $n=2$ the translation invariant terms of (b) are $\sum_{|\alpha|\le 3} \int \mathsf{c}_\alpha(\partial^\alpha f_1(x), f_2(x))\,\mathrm{d}x$ with bilinear forms c_α; the real ones that are symmetric in the two entries and covariant under $O(3)$ on $\mathbb{R}^3$ form a space of dimension six, spanned in Fourier space by $\operatorname{tr} H \operatorname{tr} H'$, $\operatorname{tr} HH'$, $p^2 \operatorname{tr} H \operatorname{tr} H'$, $p^2 \operatorname{tr} HH'$, $(p\cdot Hp)\operatorname{tr} H' + (p\cdot H'p)\operatorname{tr} H$, $(Hp)\cdot(H'p)$; covariant under the cubic group on T_ℓ, they form a space of dimension twelve. ■

Proof sketch. (a) Apply Lem. 3.6(a)–(d) at $\kappa = r = 0$, with the stated torus convention. Its construction reads as follows. M^n minus the big diagonal is the union of the admissible products of dyadic cubes of sides $h_j = 2^{-j}h_0$, pairwise at sup distances $\ge 3h_j$, that lie in no larger admissible product (on T_ℓ the dyadic cubes tile the torus, as $h_0 \in \ell 2^{-\mathbb{N}}$, and each cluster lies in a sup ball of radius $13nh_0 < \frac{\ell}{4}$, which lifts to $\mathbb{R}^3$ and defines the differences within that cluster); there is a partition of unity $\varphi_{\mathbf{Q}}$ with $|\partial^\gamma \varphi_{\mathbf{Q}}| \le c_\gamma h(\mathbf{Q})^{-|\gamma|}$, and on each product u extends to the smooth functions with the bound, by an absolutely convergent expansion in products.

The cubes of a product at mutual distances $\le 12h$ form its clusters, a partition $\pi(\mathbf{Q})$ with k blocks, where $k \le n-1$ below the coarsest side h_0; put $\tilde u(F) = \sum_{\mathbf{Q}} u_{\mathbf{Q}}(\varphi_{\mathbf{Q}}(F - P_{\mathbf{Q}}F))$, $P_{\mathbf{Q}}F$ the Taylor polynomial of order $3k$ averaged over the piece of size h of the stratum $\Delta_{\pi(\mathbf{Q})}$ next to $\mathbf{Q}$ when $k \le n-1$, and $P_{\mathbf{Q}} = 0$ otherwise. On the products with $k \le n-1$ the remainder is of order $(h/\lambda)^{3k+1}$, the number of products of side h near a given point is of order $(\lambda/h)^{3k}$, and the sum over the scales converges with one more logarithm; the order of the extension is $3(n-1)+1 = 3n-2$. Choose the grid partition and the Taylor averages covariantly under $h_0\mathbb{Z}^3$. Average the resulting extension over the parameters $t \in [0,h_0)^3$ with normalized Lebesgue measure; periodicity in t makes this mean invariant under every translation. Then average over permutations and the compact orthogonal subgroup, as in Lem. 3.6(d). Its support-radius comparison and the fixed-scale localization in part (c) preserve the stated bound, with a constant depending on n, q. (b) A distribution supported on the big diagonal with the bound of (a) is, stratum by stratum, a sum of transversal derivatives of order $\le 3|\pi|$ of δ_π; the vanishing for $\kappa' > 3k$ is the scaling of the bound. (c) A translation invariant contact term at $n=2$ is a polynomial of order ≤ 3 in the relative momentum p; $-\mathbf{1} \in O(3)$ acts trivially on V and changes the sign of p, so only the orders 0 and 2 remain, and the invariant theory of $O(3)$ on symmetric matrices gives two forms at the order 0 and four at the order 2; under the cubic group $3+9$. □

Proof of Thms. C and D. Step 1: (A1). Cut T_ℓ into at most $(2\ell/d'')^3$ boxes of radius $\le 0.86\,d''$, and each form into its pieces on these boxes. A choice of nonzero pieces, one of each form, has the geometry of Lem. 2.18 at d'', with the majorants of Lem. 2.16: the pieces lie within $\frac{7}{8}d''$ of points, and the plaquettes of pieces of different forms are at least $d - 2a_K \ge \frac{5}{4}d''$ apart, as $d \ge \frac{4}{3}d''$;

$s(d'') \geq s_{\mathrm{th}}(n)$ since $L^{s(d'')} \geq 1/d''$, and $\ell \leq \ell_{\max}$ on a small side. Multilinearity gives

$$|\langle F_1; \dots; F_n\rangle_\ell| \leq (2\ell_{\max}/d'')^{3n} k'_n(g, s(d''))\, b^{-n} \prod_a |F_a|_\star .$$

Step 2: (A2), (A3). (A2) is Lem. 2.20(a) with Lem. 2.16(a): a real form is a real quadratic source under $(|F|_\star, S[F])$ with the kernel t_F, and its plaquettes lie within $d - a_K$ of x_i when its cells lie within $d - 2a_K$; for complex tensor weights the real and imaginary parts give four real pairs, and $O_K^T(H)$ is a source under $(60|H|_\infty, S_H)$. For the continuum form, Lem. 2.17 gives $C^{\mathrm{bg}}(p,q) = \epsilon_p\epsilon_q a_K^3 K_3(x(p) - x(q))_{\lambda(p)\lambda(q)} + O((a_K/d)^3 E_1)$; the sums over plaquettes and bonds are Riemann sums of the integral, with an error $(6/\pi)(a_K/d)$ for each factor at Lipschitz weights; υ^2 against b^{-2} costs $8\bar{B}g^2L^{2-s}$; and

$$\tfrac{3}{2}\mathrm{tr}[E_{\lambda_1\lambda_1'} K E_{\lambda_2\lambda_2'} K] = 3[K_{\lambda_1\lambda_2}K_{\lambda_1'\lambda_2'} + K_{\lambda_1\lambda_2'}K_{\lambda_1'\lambda_2}], \quad (\lambda_1 \neq \lambda_1',\ \lambda_2 \neq \lambda_2'),$$
$$\tfrac{3}{2}\mathrm{tr}[E_{\lambda\lambda'} K E_{\gamma\gamma} K] = 3K_{\lambda\gamma}K_{\lambda'\gamma} \quad (\lambda \neq \lambda')$$

identify the Gaussian value with $b^{-2}\mathcal{W}$. So, with $s = s(d)$,

$$e^{\mathrm{od}} = \tfrac{9}{8}[59^2\varepsilon_g^\natural(s) + 600E_2(\tfrac{1}{\pi} + E_2) + 636/C_W(s)] + 8\bar{B}g^2L^{2-s}, \qquad E_2 = E_1 + \tfrac{6}{\pi}\tfrac{a_K}{d},$$

which is at most an error function of s, since $a_K/d \leq C_W(s)^{-1}$, $d < LC_W(s)L^{-s}$ and $M \leq L^{-2}C_W(s) \leq \hat{d} < LC_W(s)$ (Lem. 2.21(e)), so that also the term $C_\star^{\mathrm{fl}}\hat{d}^3 e^{-\delta_{\mathrm{tp}}L\hat{d}/\sqrt{3}}$ of E_1 is at most an error function of s. For (A3) apply (A2) to $(\theta Y, Y)$, whose points are $3d$ to $4d$ apart. By Lem. 2.17 and its count, at most $10(d/a_K)^3$ centres of bonds of one direction in a ball of radius d,

$$|G(\theta Y, Y)| \leq 6C_{\mathsf{k}}^2(a_K/d)^6\Big(\sum|\mathsf{t}|\Big)^2\upsilon^2 \leq 600C_{\mathsf{k}}^2\upsilon^2 \sup|\mathsf{w}|^2,$$

and with $\upsilon^2 \leq \frac{9}{8}b^{-2}$, $|Y|_\star = 59\sup|\mathsf{w}|$ the bound follows. Since $\langle Y\rangle_\ell = 0$ (Lem. 2.21(b)), $\langle \theta Y; Y\rangle_\ell = \langle \theta Y \cdot Y\rangle_\ell \geq 0$ by reflection positivity at a plane through midpoints of bonds (Cor. YM3_I:85.3); for other forms the centred form $F - \langle F\rangle_\ell$ gives the sign. For forms at bonds and sites the count gives $\sum|\mathsf{t}| \leq N(d/a_K)^3\|v\|$, $N = 30$, resp. 10, and for $O_K^T(H)$ the same count gives the number $12150 = 6\cdot 45^2$, with $60^2\bar{\varepsilon}_g^\natural$ from the size $60|H|_\infty$.

Step 3: (B). By Lem. 2.21(a) the copies $\theta'^\eta Y$ are bond forms of the same size; by Lem. 2.21(d) partners across the plane of θ'_{μ_0} are $3d$ to $4d$ apart and other pairs $\geq 11M$; their means vanish by Lem. 2.21(b). Lem. 2.22 with $n = 4$, for bond forms, gives $M_8 \leq (g_0^4 + \bar{\omega}_{g,8})\hat{m}_J$ with $\hat{m}_J = 59^8\upsilon^8\sup|\mathsf{w}|^8$ and $59^2 g_0 = 106C_{\mathsf{k}}'^2$, so $M_8 \leq C_8 b^{-8}\sup|\mathsf{w}|^8$; and $|M_8 - \prod_\tau\langle P_\tau\rangle| \leq \omega_{g,8}^\sharp\hat{m}_J$ with $\langle P_\tau\rangle = \langle \theta'_{\mu_0}Y; Y\rangle_{\ell'}$ for all four pairs. The signs come from reflection positivity: M_8 is a mean of mirror copies, and the chessboard estimate gives $\langle \theta'_{\mu_0}Y \cdot Y\rangle_{\ell'} \leq M_8^{1/8}M_8^{1/8}$ times the seminorms 1 of the six other cells.

Step 4: (C1). With $\delta = \min\{d, \frac{1}{2}, \mathsf{l}/(4n)\}$ cut each form into its pieces on the cubes of side δ (at most $(2R/\delta + 2)^{3n}$ tuples of pieces); only the boundedness of the weights is used. For each tuple of pieces Prop. 2.23(c) gives a board of side l with all pieces at distance $\frac{\delta}{2}$ from the walls; on a torus, pieces are first moved by periods into one period box. Centre by $c_a = \tilde{\mu}_{\ell'}(F_a)$, write the cumulant through moments, and dominate each moment by Prop. 2.23(b). The eighth power of the seminorm of a product of centred pieces in one cell is a centred moment of at most $8m$ forms on $T_{\ell'}$ at mutual distance $\geq \delta$ (Lem. 2.21(c)), bounded by (A1):

$$\Big\|\prod_a(F_a - c_a)\Big\|_{\mathrm{ch},\ell'} \leq \mathrm{Bl}_{8m}^{1/8}\prod_a b^{-1}|F_a|_\star, \qquad \mathrm{Bl}_N = \sum_{\pi\in\mathrm{Part}_2(N)}\prod_{I\in\pi}k^\circ(\#I, \delta),$$

$\mathrm{Part}_2(N)$ the partitions into blocks of at least two. So

$$C_n^T(d; R) = \max\{(2R/\delta + 2)^{3n}\mathsf{f}_n e^{nG_g\mathsf{l}^3}\mathsf{Bm}_n^n,\ C_n^{\mathrm{cr}}(d; R)\},$$

$\mathsf{f}_n = \sum_\pi (\#\pi - 1)!$, $\mathsf{Bm}_n = \max_{m \le n} \max\{1, \mathrm{Bl}_{8m}^{1/8}\}$; below the cutoff $K_n = \max_{r \le 8n} K_0'(r, \delta)$ bounds by suprema give $C_n^{\mathrm{cr}}(d; R) = \mathsf{f}_n[6\bar{x}_0 L^{K_n}(2RL^{K_n} + 1)^3]^n$. The one-point number $C_1^T(R)$ is the case $n = 1$. Complex weights: real and imaginary parts.

Step 5: (C2), (C3). For (C2) let $D = |x_1 - x_2|_\infty$, on a torus minimised over the lifts, be attained in the coordinate μ_0. If $D > 3$, the supports lie in balls of radius 1 more than 1 apart, and (C1) at $(\frac{1}{2}; 1)$ bounds the covariance. Let $D \le 3$. One board of side l serves both insertions: it has a wall normal to e_{μ_0} at the midpoint of x_{1,μ_0} and x_{2,μ_0}, up to $\frac{a_K}{2}$, and its offset o at $\frac{1}{2}(x_{1,\nu} + x_{2,\nu})$, up to $\frac{a_K}{2}$, in the two coordinates $\nu \ne \mu_0$. The distance t_i of x_i to the wall lies in $[\frac{D}{2} - \frac{a_K}{2}, \frac{D}{2} + \frac{a_K}{2}] \subset [\frac{3}{2}d, 2)$, and $|x_{i,\nu} - o_\nu| < 2$, so that the corners of the plaquettes of either insertion lie within 3 of the middle of its cell in the coordinates $\nu \ne \mu_0$.

Of the eight copies of one insertion on $T_{\ell'}$, the partners across the wall are $2t_i \in [3d', 4d']$ apart, $d' = \max\{d, \frac{t_i}{2}\}$, and copies that differ in a reflection normal to an e_ν, $\nu \ne \mu_0$, are at least $\mathsf{l} - 6 \ge 11M$ apart ($\mathsf{l} \ge LM$, $M \ge 3$); the carriers lie within $d' - 2a_K$ of their points, and $s(d') \le s(d) \le K$. So Lem. 2.22 with $n = 4$ at the distance d', for bond forms, whose means vanish, bounds the eighth power of the seminorm of a bond insertion by $(g_0^4 + \bar{\omega}_{g,8})\hat{m}_J = C_8 b^{-8} \sup |\mathsf{w}|^8$, as in Step 3, on this board, which the two insertions fix together, and its centred one-direction case bounds the seminorm of each of the forms of a tensor insertion; domination bounds the covariance by the product of the two seminorms; if $s(d') < s_8(g)$, (A1) is used at the distance $d_\flat = L^{-s_8(g)}$.

This gives

$$C_\infty^\diamond = \max\{2(c_\infty^\diamond)^2, C_2^T(\tfrac{1}{2}; 1)\}, \qquad c_\infty^\diamond = e^{G_g \mathsf{l}^3} \max\{(\tfrac{9}{8})^{1/2}((422 C_{\mathsf{k}}'^2)^4 + \bar{\omega}_{g,8}^{\mathrm{o}})^{1/8}, \ \mathrm{Bl}_8(d_\flat)^{1/8}, \ 12\bar{x}_0 L^{K_\flat}(2L^{K_\flat} + 1)^3\},$$

$\bar{\omega}_{g,8}^{\mathrm{o}}$ the supremum of the error of Lem. 2.22 for majorants of one direction, $\mathrm{Bl}_8(d_\flat)$ the number of Step 4 at the distance $d_\flat$ and $K_\flat$ its cutoff. For bond forms the mean vanishes in every law, and C_∞^{per} is formed in the same way with $C_8^{1/4}$; complex weights of all three directions cost the factor 36. For (C3) place the pair in the middle of a cell (Lem. 2.21(d)) and use Lem. 2.22 with $n = 8$ for the sixteen copies (in its proof the blocks of one pair rest on the clause of Lem. 2.20(c), almost everywhere at the upper level): for $X_i = F_i - c_i$, $c_i = \tilde{\mu}_{\ell'}(F_i)$, and $m = \tilde{\mu}_{\ell'}(X_1 X_2) = \tilde{\mu}_{\ell'}(F_1; F_2)$,

$$\|X_1 X_2 - m\|_{\mathrm{ch},\ell'}^8 = \sum_J (-m)^{8 - \#J} \langle \tilde{P}_J \rangle_{\ell'} \le \omega_{g,16}'^\sharp(s)(|m| + \hat{m})^8,$$

since $\langle \tilde{P}_J \rangle$ replaced by $m^{\#J}$ gives $(m - m)^8 = 0$ and $|\langle \tilde{P}_J \rangle - m^{\#J}| \le \omega_{g,16}'^\sharp \hat{m}^{\#J}$. For one point, $\|X_i\|_{\mathrm{ch},\ell'}^8$ is a centred moment of eight of the sixteen forms, one from each pair, a set that holds no pair, so it is at most $o_{g,16}(s)(v|F_i|_\star)^8$. Domination now gives the three bounds of (C3), with

$$\mathrm{Df}_g(s) = e^{G_g \mathsf{l}^3}[\omega_{g,16}'^\sharp(s)]^{1/8}(1 + g_0 + \omega_{g,16}'(s)) + e^{2G_g \mathsf{l}^3} o_{g,16}(s)^{1/4},$$

since $|m| \le (g_0 + \omega_{g,16}')\hat{m}$, and $|m - G(F_1', F_2')| \le \omega_{g,16}'\hat{m}$ by Lem. 2.22 at one pair. For $O_K^T(H_i)$ the same, with the majorants $(60|H_i|_\infty, S_{H_i})$.

Step 6: (D1)–(D3). (C1), Prop. 2.23(d), the compactness of $I^{\mathrm{lab}}(g)$ and the theorem of F. Riesz give (D1); the second clause is (C1) at $n = 1$; for the third, the local functions $O_K^T(H)$ and the weak convergence give some $\ell_j \ge \max\{\Lambda_j, j\}$ in $\mathfrak{L}(g)$ at which the first j tuples of a dense countable family are matched within $\frac{1}{j}$. $O_K^T(f\mathbf{1}) = \mathcal{O}_K(f)$ gives (D2), along a subsequence of the defining sequence of S. Lem. 2.21(a) gives $rO_K^T(H) = O_K^T(rH)$; the invariance and the reflection positivity of the laws at the lattice mirrors pass to the limits, since lattice isometries approximate the real ones and the bounds are equicontinuous; periodic Gibbs states are positive at the nine families.

The semigroup of symmetric contractions $[F] \mapsto [\tau_{te_3} F]$ is e^{-tH}, Stone's theorem gives the momenta and the joint spectral measure, and the lattice energy–momentum inequality (Prop. YM3_I: 99.3) gives $H \ge |P_k|$. Centred at the means of $\tilde{\mu}_{\ell'}$, the moments in ν_j are $\sum_B \prod_{i \in B}(\nu_j -$

$\tilde\mu_{\ell'})(O^T_{K_j}(H_i))\,\nu_j(A_j(X_{B^c}))$. The bond parts have mean 0 in both laws, and all plaquettes have one mean in each law (Prop. YM3_I:103.4), so that $(\nu_j-\tilde\mu_{\ell'})(O^T_{K_j}(H)) = \mathfrak{c}_j\, a^3_{K_j}\sum_p H_{\lambda(p)\lambda(p)}(x(p))$, a Riemann sum of $\mathfrak{c}_j\int \mathrm{tr}\, H$, with $\mathfrak{c}_j = a^{-4}_{K_j}(\nu_j(a_p)-\tilde\mu_{\ell'}(a_p))$; (C1) at $n=1$, applied to a plaquette form, bounds $|\mathfrak{c}_j|$ by $C(g)\mathsf{b}^{-1}$ for large j.

So the $\mathfrak{c}_j$ converge along a further subsequence, to $\mathfrak{c}$, and along it the recentred moments converge to the functional of (D3), whose one-point function is $\mathfrak{c}\int \mathrm{tr}\, H$; it is reflection positive, since the means of $\tilde\mu_{\ell'}$ are invariant under the lattice mirrors.

Step 7: (D4). For Lipschitz entries Lem. 2.24 makes the cumulants Cauchy in K, and its form for cell means gives the same for bounded Borel weights; continuous entries are approximated by Lipschitz ones uniformly in K, by the torus form of (C1); Prop. 2.23(d) applies in ℓ to periodised tuples, which are defined and linear once ℓ is large against the supports; a diagonal choice of the cutoffs gives the limits along growing tori.

Step 8: (D5) and [7, Thm. BZ(F1)]. On $T_{\ell'}$, (C3) and the continuum form of Step 2, with $\frac{3}{2}\mathrm{tr}[E_{\lambda\lambda}KE_{\gamma\gamma}K] = \frac{3}{2}K^2_{\lambda\gamma}$, give the law for all 36 pairs of components; then $j\to\infty$. For two off-diagonal components the error is $\bar e_g^{\mathrm{od}} + \frac{9}{8}59^2\mathrm{Df}_g$, for an off-diagonal and a diagonal one $\bar e_g^{\mathrm{od}} + \frac{9}{8}59\,\mathrm{Df}_g$, for two diagonal ones $\frac{9}{8}[\omega'_{g,16} + \mathrm{Df}_g + 150\bar E_1(\frac{1}{\pi}+\bar E_1) + 159/C_W(s)] + 2\bar B g^2 L^{2-s}$, $\bar E_1$ the bound of E_1 by a function of s ((C3) being used with the majorants of the respective kinds); $\bar e_g$ is their maximum. At $H_i = f_i\mathbf{1}$, $\frac{3}{2}\mathrm{tr}\, K_3(u)^2 = \frac{9}{16\pi^2}|u|^{-6}$; the leading term is covariant under $O(3)$. For (F1), Lem. 2.22 with arbitrary majorants, centred, at $n=8$ bounds the means, $|\nu(O^T_K(H)) - \tilde\mu_{\ell'}(O^T_K(H))| \le 60e^{G_g\mathsf{l}^3}\mathrm{mn}_g(s)\upsilon|H|_\infty$, where $\mathrm{mn}_g = [e_g^8 + \mathrm{Bl}(16)\mathrm{far}_{g,16}(1+g_0+\varepsilon_g)^8 + \mathrm{tl}_{g,16}]^{1/8} \le o^{1/8}_{g,16}$ comes from its moment bound at $N_1=8$, so sup norms suffice, and the law holds at a finite cutoff for continuous weights, the Riemann sums of continuous weights giving $r_K\to 0$ and the lattice term of E_1 the term $C(g)(a_K/d)^2$; letting $j\to\infty$,

$$e_g^\infty(d) = \tfrac{9}{8}\Big\{3600[\omega'_{g,16} + e^{G_g\mathsf{l}^3}(\omega'^\sharp_{g,16})^{1/8}(1+g_0+\omega'_{g,16}) + e^{2G_g\mathsf{l}^3}\mathrm{mn}_g^2](s) + 12150E_1^\infty(\tfrac{1}{\pi} + E_1^\infty)\Big\} + 162\bar B g^2 L^{2-s}, \qquad s = s(d),$$

$E_1^\infty = E_1 - C_{\mathrm{lat}}(a_K/d)^2$. The errors are powers of $Y^{-1}_{s(d)}$, the worst the root $[\omega'^\sharp_{g,16}]^{1/8}$, and $Y_{s(d)}\sim\log\frac{1}{d}$. This root can be avoided: in Step 5 the eighth power of the seminorm of X_1X_2-m is an alternating sum of means of products of the sixteen copies, and at the upper level the parts of these means without a block that links two pairs recombine to $\prod_\tau(\Psi_\tau - m)$, Ψ_τ the conditional means of the pairs, with $|\Psi_\tau - m| \le (\delta_g + \omega'_{g,16})\hat m$ almost everywhere at the upper level (the blocks of one pair by Lem. 2.20(c), the singletons as in the proof of Lem. 2.22, and $|G(F'_1,F'_2) - m| \le \omega'_{g,16}\hat m$ as in Step 5); so the seminorm itself is at most a multiple of $(\delta_g+\omega'_{g,16})\hat m$ plus the eighth roots of the terms far and tl, which decay faster than every power of Y_s^{-1}, and the means are bounded with mn_g, so that the second term of Df_g may be taken $e^{2G_g\mathsf{l}^3}\mathrm{mn}_g^2$; as ε_g, $\omega'_{g,16}$ and $1/C_W(s)$ are of order $Y_s^{-2\mathfrak{p}_0}$ and mn_g^2 is of smaller order, the error is at most $C(g)(\log\frac{1}{d})^{-2\mathfrak{p}_0}$; by [7, Lem. D.19] the same holds for $g\le\gamma_2$, as $(\log\frac{1}{L^z d})^{-2\mathfrak{p}_0} \le 2^{2\mathfrak{p}_0}(\log\frac{1}{d})^{-2\mathfrak{p}_0}$ for $d\le L^{-2z}$.

On the tori of $\mathfrak{L}(g)$ the torus limits of (D4) are limits along cutoffs at the labels $L^K\mathsf{b}$, and the same bound passes to them. The law fixes b, since $\mathcal{W}\neq 0$. Since $\mathcal{W}$ is invariant under $O(3)$ and $|rH|_\infty \le 3|H|_\infty$, the law gives $|\sigma(rH_1,rH_2) - \sigma(H_1,H_2)| \le (9+1)e_g^\infty\mathsf{b}^{-2}|H_1|_\infty|H_2|_\infty$. Non-degeneracy: $\mathcal{W}(\theta(\varphi E),\varphi E) \ge w_0\,\mathrm{tr}\,E^2$ for this bump. □

The mirror bound and the power-six kernel estimate. (F9). In a periodic Gibbs state ν, $\langle F', F\rangle_{\nu,c} = \nu(\overline{\theta_c F'}F)$ is a positive semi-definite Hermitian form on the local functions on $\{x_{\mu_0} > c\}$, the translation by $a_K e_{\mu_0}$ is a contraction for it, and so the mirror may be moved away: $\|F\|_{\nu,c} \le \|F\|_{\nu,c'}$ for $c\le c'$, $c'-c\in a_K\mathbb{Z}$. It therefore suffices that x_0 lies at a distance between $\frac{3}{2}d$ and $2d$ from the plane. There the plane is a wall of a board of side l, θX is the copy of X in the neighbouring

cell, and, as $\nu(\theta X\cdot X)=\nu(\theta(F-c)\cdot(F-c))-(\nu(F)-c)^2$ for $F=O_K^T(H)$ and c its mean on $T_{\ell'}$, domination by $T_{\ell'}$ gives $\nu(\theta X\cdot X)\le e^{2G_g\mathsf{l}^3}\|F-c\|^2_{\mathrm{ch},\ell'}$, the eighth power of the seminorm being the mean of eight copies. For bond forms (B) bounds it, and for the whole tensor Lem. 2.22 with arbitrary majorants, centred, at $n=4$; if $s(d)<s_8(g)$, (C1) at the distance $L^{-s_8(g)}$ bounds the covariance directly. So

$$C^{\mathrm{od}}_\infty(g)=\max\{e^{2G_g\mathsf{l}^3}C_8^{1/4},\,59^2C_2^T(L^{-s_8(g)};1)\},\qquad C_\infty(g)=\max\{e^{2G_g\mathsf{l}^3}(C'_8)^{1/4},\,60^2C_2^T(L^{-s_8(g)};1)\},\tag{2.xxvii}$$

$C'_8=(\frac98)^4 60^8[g_0^4+\bar\omega'_{g,8}]$, $g_0=\frac32 81^2C'^2_{\mathsf{k}}$; complex weights of all directions cost 36. For the kernel bound at off-diagonal entries, [7, Prop. D.15](a) bounds the two-point kernel at u by $\Gamma(|u|/2\sqrt3)$, and the mirror bound for bond forms applied to bumps at the height t bounds $\Gamma(t)$ by a multiple of $C^{\mathrm{od}}_\infty(g)\mathsf{b}^{-2}\max\{1,t^{-1}\}^6$; the holomorphy of the two-point function in the relative position and the mean value inequality on a polydisc of radius proportional to $|x-y|$ turn this bound for bumps at the scale $|x-y|$ into the pointwise bound $A_6C^{\mathrm{od}}_\infty(g)\mathsf{b}^{-2}\min\{|x-y|_\infty,3\}^{-6}$ of the kernel. For the whole tensor the same steps, with the mirror bound for $X=A[O_K^T(H)]$ and its number $C_\infty(g)$, give the bound with a pure number A in the place of A_6. By (D5) the kernel is of order $|x-y|^{-6}$ at short distance, so the power is the true one. □

Contact extensions on tori and the mirror bound. (F8). Every side $\ell=2L^m\in\mathfrak{L}(g)$ is admissible over the run of every datum at g: $LM\,R(x_K)\le L^m$ (equation YM3_I:(1533), as in the paragraph *Data* of the Setting).

On T_ℓ the clause (i) of (E3), on Lem. 2.18 (the bound of that Lemma on an admissible side with $\max\{1,c(\ell)\}^n k'_n$ for k'_n, without the factor of the covering), at the geometry of Lem. 2.18 at $d=1.52h$ (a cube of side $\frac32 h$ lies within $1.3h\le\frac78 d$ of its centre, and cubes at sup distance $\ge\frac52 h$ are more than $\frac54 d$ apart), bounds the cumulants of n entries at the cubes of side $\frac32 h$ at sup distances $\ge\frac52 h$, for $0<h\le h_0$, where $h_0\in\ell 2^{-\mathbb{N}}$ and $h_0\le\min\{\ell/(64n),\frac12 L^{-s_{\mathrm{th}}(n)}\}$, so that $1.52h\le 1.52h_0<L^{-s_{\mathrm{th}}(n)}$, by $60^n\max\{1,c(\ell)\}^nC_n(g)[2\log\frac{1}{1.52h}+C(g)]^{8n\mathfrak{p}_0}b^{-n}\prod|H_a|_\infty$ uniformly in the cutoff, which is at most $C(1+\log\frac{h_0}{h})^{8n\mathfrak{p}_0}b^{-n}\prod|H_a|_\infty$; through the cell means of Lem. 2.24 the bound passes to the limits of (D4), and Lem. 2.26 gives the extension, its bound, the contact terms and the counts.

At a cutoff the same construction, with the products of the cubes restricted to the scales h with $s(1.52h)\le K$, gives subtracted lattice cumulants with the bound for all $\lambda\ge 8h_K$, $h_K\le 2C_W(K)a_K$ the least scale, uniformly in K; by Lem. 2.24 they converge to $\tilde S_n$; their difference to the lattice cumulants depends on F only through the derivatives of the orders $\le 3k$ on the strata of k blocks and the values at distances $<6h_K$. On $\mathbb{R}^3$: let two cubes of side $\frac32 h$ lie at sup distance $\ge\frac52 h$, $d=1.31h$, $s(d)\le K$. Their centres are at sup distance $\ge 4h\ge 3d+a_K$, the cubes lie within $d-2a_K$ of their centres, and a mirror plane $\{x_{\mu_0}=c\}$ between them, normal to a coordinate in which the centres are $\ge 4h$ apart, has each centre at a distance $\ge 2h-\frac{a_K}{2}\ge\frac32 d$.

On a torus T_ℓ, $\ell\in\mathfrak{L}(g)$, the mirror θ has the second plane $\{x_{\mu_0}=c+\frac{\ell}{2}\}$, at a distance $\ge\frac{\ell}{4}-\frac{a_K}{2}\ge 2h-\frac{a_K}{2}\ge\frac32 d$ from each centre, as the sup distance $\ge 4h$ of the centres on T_ℓ is at most $\frac{\ell}{2}$. For a bond form Y with a complex weight of all directions, carried within $d-2a_K$ of a point at a distance $\ge\frac32 d$ from both planes, and τ the translation by $a_Ke_{\mu_0}$, the invariance of the torus measure gives $\psi(k):=\langle\overline{\theta\tau^kY}\,\tau^kY\rangle_\ell=\langle\theta\bar Y\,\tau^{2k}Y\rangle_\ell$, and the Schwarz inequality for the positive form of θ on the half torus between the two planes (Cor. YM3_I:85.3) gives $\psi(k)^2\le\psi(k-1)\psi(k+1)$ as long as the translates keep the distance $\ge\frac32 d$ from both planes.

A non-negative sequence with this property takes its largest value at an end of its range, and there the point lies at a distance between $\frac32 d$ and $2d$ from one of the two planes. That plane is a wall of a board of side l, $\frac{\ell}{2}$ being a multiple of l, the cells are distinct modulo the period, and domination by

$T_{\ell'}$ with (B), or (C1) at the distance $L^{-s_8(g)}$ if $s(d) < s_8(g)$, bounds ψ there as in (F9); so the mirror bound of (F9) for bond forms with complex weights of all directions, $0 \le \langle \overline{\theta Y} Y\rangle_\ell \le 36 C^{\mathrm{od}}_\infty(g) b^{-2} |\mathsf{w}|^2_\infty$, holds on the tori of $\mathfrak{L}(g)$. With the Schwarz inequality for the positive form of that mirror, the means of bond forms being 0, this mirror bound gives $|\langle Y[\mathsf{w}_1]; Y[\mathsf{w}_2]\rangle_{K,\ell,\beta}| \le 36 C^{\mathrm{od}}_\infty(g) b^{-2} |\mathsf{w}_1|_\infty |\mathsf{w}_2|_\infty$ on every $\ell \in \mathfrak{L}(g)$: the bound of Lem. 2.26 with $q = 0$ and a number free of the side, which every limit point inherits. □

Sketch of the first-creation and tail refinements. Step 14: (E1)–(E3). (E1): $\tilde\nu_K(\{Z_t \ne \emptyset\}) = \sum_{t' \le t} \mathsf{w}_{t'}$, the weights of the first creations, and the run may be stopped at the level t without changing this number. A first creation forces the failure of a local test in some test cube of the step; averaging the test over the gauge variables of its cube with the density of the gauge fixing of the step gives Borel functions $0 \le F \le 1$ that are local, gauge invariant and covariant under the lattice isometries, and the average is undone exactly inside the sum over the histories of any admissible set of values with the same A_0; so w_t is at most a sum of means of such functions, two for each test cube, its plaquette test and its averaged witness test, and at $t = 1$ one for each density cube too: at most $2N_\tau + N_\delta$ for each cell of the level t, N_τ the number of the test cubes and N_δ that of the density cubes of a cell (N_δ entering at $t = 1$ alone), and at most $(2N_\tau + N_\delta) M^{-3} \ell^3 L^{3(K-t)}$ in all.

For the strong branch we must prove the comparison in remark 2.14, including the local-background, gauge-factor and compatibility-test estimates for these actual test functions. Once this comparison is established, the deposit constant of the second admissible choice satisfies $(\frac{93}{100})^2 c_\star \le c''_\star \le c_\star$. Let s_t be the side of a cell of the level t. If the torus has at least $2L^2$ cells of the level t along a side, take the board of side $L^2 s_t$ whose offset is the centre of the test function F, its *witness*; the chessboard estimate with the mirrors in the walls of this board bounds $\langle F\rangle$ by the N-th root of the mean of the product of the copies of F in its N cells on the torus, and this product asks a large field in every cell of the board.

A creation reaches less than $49 s_t < L^2 s_t$ and so serves one cell alone: such a history has at least N creations. Thm. YM3_I:D, which bounds the density of these histories from above by $\exp[-\frac{9}{10} c''_\star p_0(g_t)^2 N]$, at the fraction $\frac{9}{10}$ of the deposit, and that of the empty history from below, with the volume term $C y_t (L^2 M R(x_t))^3 N$, which $\frac{1}{40} c_\star p_0(g_t)^2 N$ pays, gives $\langle F\rangle \le \exp[-\frac{3}{4} c_\star p_0(g_t)^2]$, as $\frac{9}{10}(\frac{93}{100})^2 - \frac{1}{40} \ge \frac{3}{4}$; with fewer cells of the level t Thm. YM3_I:D is used without mirrors, at the first set of values; a threshold on y_K is removed by stopping the couplings at the last level above it.

The comparison and the union bound over the cells then give $C_\flat$ for the strong branch. The pair $(c^\natural_2, C_\natural)$: if $Z_t \ne \emptyset$, some cell of the level t has its centre in Z_t, and the rarity at a point of Prop. 4.52 with the union bound over the at most $M^{-3}\ell^3 L^{3(K-t)}$ cells gives the bound. (E2): Lem. 2.20(b) with $\mathsf{L} = C M^{-3} \ell^3$; half of the exponent $c p_0(g_t)^2$ pays ℓ^3 once $\frac{1}{2} c A_0^2 (\frac{1}{2} s \log L)^P \ge 3 \log \ell$, which is $s \ge \sigma_c (\log \ell)^{1/P}$; the other half pays $\log C + 3s \log L + 2Y_s$ and the envelopes from $s_\flat(g)$ on; the depth condition $2c\mathfrak{p}_0 A_0^2 y^{P-1} \ge 13$ is the rate condition of the deposit lemma at c. For (iv) there are three cases.

If $s(d) \ge \hat s$, (iv) is (iii). If $s(d) < \hat s$, then $d \ge L^{-s(d)} \ge L^2 d'$, $d' = L^{-\hat s - 1}$; let $s' = s(d') \ge \hat s$. If besides $K \ge s'$, cover by balls of radius d', at which (ii) and (iii) hold; this costs the factor $(4/d' + 2)^6 \le 5^6 L^{6\hat s + 6}$. If $K < s'$, the suprema of the forms give the number $100 L^{8s'} \bar x_0^2$. So C°_3 is the larger of $(4/d' + 2)^6$ times the number of (ii) and (iii), in the units of (iv), and of $100 L^{8s'} \bar x_0^2$; as $s' \le \hat s + 1 + \lceil \log_L C_W (2\hat s + 2)\rceil$, $L^{8s'} \le L^{8\hat s + 16} C_W (2\hat s + 2)^8$, and with $L^{\hat s} \le L^{s_\flat(g)+1} \exp[2(\frac{6 \log \ell}{c A_0^2})^{1/P}]$ this gives the bound of (iv), whose factor $(1 + \log \ell)^8 \exp[16(\frac{6 \log \ell}{c A_0^2})^{1/P}]$ comes from the third case, the covering costing less; for (v) cover by balls of radius $\min\{\frac{r}{8}, L^{-\hat s}\}$. (E3): let $\ell = 2L^m$ be admissible, $K \ge 3$, and $s_{\mathrm{th}}(n) \le s(d) \le K$.

The proof of Lem. 2.18 reads the side at the following places, and each of them holds on an admissible side. The arithmetic of the working level (Lem. YM3_I:126.1) asks nothing of

the side. Lem. YM3_I:126.2 asks (Sd.1) alone, $MR(x_K) \le L^m$, and gives the frame, the order $x_0 > x_1 > \cdots > x_K$ of the run (so $y_K \le y$), (Sd.2) and (Sd.3) ($LM \le L^m$, $\ell \ge N_{\mathsf{H}}$), the data of the construction at every level $\le K$, and the settings of the chain at the levels $\mathsf{k} < j_1 = T = \mathsf{k}+1$ at every pair of the majorants; the setting of the chain for n sources ([7, Prop. B.21] at $p = n$) asks these data and the frame alone, and the threshold $s_{\mathrm{th}}(n)$ reads no side.

The sum over the level of the first creation (equation eq. 4.xxxi) holds at all the levels $\le K$ (Thm. YM3_I:D and Cor. YM3_I:69.4, as in the proof of Lem. YM3_I:126.2), and the rarity on one torus, Prop. 4.54 at the level K, whose hypotheses hold by that canonical construction, bounds the weights before any smallness of the side is used:

$$\mathsf{w}_t \le \tilde\nu_K(\{Z_t \ne \emptyset\}) \le 2M^{-3}\ell^3 e^{\mathsf{B}(x_K)\ell^3} L^{3(K-t)} e^{-c' p_0(g_t)^2}, \qquad \mathsf{B}(x_K) \le \mathsf{B}(\bar x_0).$$

From there on the side enters the proof only through this number $2M^{-3}\ell^3 e^{\mathsf{B}(\bar x_0)\ell^3}$, in the tail of the sum over the level of the first creation, where a small side has $2M^{-3}\ell^3 \le M^{-3}c' A_0^2 Y_s^P$, and through the number of the boxes of the covering. As $Y_s \ge 1$ the quotient of the two numbers is at most $c(\ell)$, each block of a partition takes at most the factor $\max\{1, c(\ell)\}$, and there are at most n blocks: the bound of Lem. 2.18 holds on every admissible side with $\max\{1, c(\ell)\}^n k'_n(g, s(d))$ for $k'_n(g, s(d))$, which is (E3)(i) at this pair; the covering of Step 1, with 2ℓ for $2\ell_{\max}$, gives (E3)(ii). The same argument serves every pair (L, c'') of (E3): the weights enter only the tail term of each block, linearly in L and through a Gaussian sum in which $c'' \ge \frac12 c'$ may stand for $\frac12 c'$, and no smallness of w_t is used; so, as $Y_s \ge 1$, each block takes at most the factor $\max\{1, M^3\mathsf{L}/(c'A_0^2)\}$, which is $\max\{1, c(\ell)\}$ at the pair above, and (E3)(i) and (ii) hold with it. Once the strong bound in (E1) is established, its weights, $\mathsf{w}_t \le C_\flat M^{-3}\ell^3 L^{3(K-t)} e^{-\frac12 c' p_0(g_t)^2}$, give $C_\flat \ell^3/(c'A_0^2)$ in place of $c(\ell)$. The preceding proof with the fixed-torus pair is independent of this additional comparison. □

3. Real-time tensor fields

Let $V = \mathrm{Sym}_3(\mathbb{R})$. The gauge-invariant curvature tensor $T = (T_{ij})$ is normalized by

$$\mathcal{T}(H) = \frac12 \int \mathrm{tr}(H(x)T(x))\,\mathrm{d}x, \qquad H \in C_c^\infty(\mathbb{R}^3; V_{\mathbb{C}}).$$

This notation denotes a smeared bilinear, rather than a pointwise product of separately constructed gauge-dependent curvatures. Its trace gives the action density $\mathcal{O}(f) = \mathcal{T}(f\mathbf{1})$. An orthogonal matrix R acts on tensor labels by $\rho(R)E = RER^{\mathsf{T}}$. Reflections act on these labels as well as on the points.

Fix a clustering component σ of the Euclidean correlations. Reflection in $x_3 = 0$ gives a Hilbert space $\mathcal{H}$, the vacuum Ω, a positive generator H of Euclidean time translations, and spatial momenta P_1, P_2. These objects and the analytic insertion bounds are supplied by [8, §§6,8]. The purpose here is to obtain operator-valued tempered distributions on Minkowski space, including their tensor covariance and locality. The Euclidean theorem on flat tests alone is not being quoted as a theorem about distributions on all ordinary Euclidean tests.

Sketch B (Reconstruction of the tensor fields). The component σ reconstructs a covariant local tensor field on $\mathbb{R}^{1+2}$, defined on a common invariant domain and tempered in its real-time test functions. Its translation spectrum lies in the closed forward cone. The vacuum is cyclic and unique among translation-invariant vectors. Multiplying a component with r Euclidean-time indices by i^r gives its Hermitian real-time component. Their separated Euclidean correlations agree with σ. ■□

3.1. **What the continuation must accomplish.** The input consists of separated Euclidean insertions. Their analytic continuation must do more than assign numbers to imaginary times: it must produce distributions with a common domain, identify adjoints in a real tensor frame, and make products commute at spacelike separation. The bounded insertion operators give a route to these tasks. They keep positive Euclidean time gaps while the spatial and temporal arguments are continued, and their power bounds control removal of those gaps.

The main manuscript has already constructed bounded slab insertions and analytic point kernels. We use those objects without repeating their lattice estimates or the proof of rotation covariance. A slab insertion between two positive time gaps is a bounded operator. Concentrating its test near a point gives the regularized point operators; the uniform L^1 insertion bound controls their dependence on the regularization. Their products, with positive gaps left between consecutive factors, give holomorphic vector-valued kernels.

The spectrum controls the permitted complex translations. For $z \in \mathbb{C}^3$, the tube

$$\operatorname{Re} z_3 > |(\operatorname{Im} z_1, \operatorname{Im} z_2)|$$

keeps the translation operator bounded. Consecutive differences of insertion points must lie in this tube; the absolute positions alone do not specify a product's analytic domain. Rotation covariance identifies the continuations obtained from different choices of the Euclidean time direction. Reflection supplies the adjoint relation. The tensor action in that relation is essential for selecting the Hermitian real-time components.

At the boundary of the tube, the point kernels need not have pointwise limits. Their polynomial bounds give distributional boundary values. Smearing those boundary values defines the fields and their products on the vacuum domain. The same bounds give temperateness and the energy-domain extensions in the detailed argument. Locality follows by comparing the continuations in the two orders on a region of spacelike separation and taking their boundary values there. The conclusion concerns the common invariant domain; passing to a net of bounded local operators is the separate task in §14.

There is also a Euclidean extension issue. The theory on tests flat at coincidences does not select contact terms supported on the diagonals. The additional extension discussion below describes the freedom left by the separated correlations. It does not assert that an arbitrary choice of such terms inherits reflection positivity on ordinary Euclidean tests.

For an open set O, let $\mathfrak{S}_V(O)$ denote the tuples of $C_c^1(O;V)$ entries with pairwise disjoint supports, including the empty tuple, and let $\mathbb{C}\mathfrak{S}_V(O)$ be their complex linear span. We also write $\mathfrak{S}_V^1(O)$ for this space of tuples when distinguishing it from tuples of continuous entries. Put $\mathbb{E}_+ = \{x_3 > 0\}$ and $V_0 = \mathbb{R}\mathbf{1}$. The stress tensor is $\Theta_{ij} = \frac{1}{2}\delta_{ij} \operatorname{tr} T - T_{ij}$; its labels are $\mathsf{m}_{ij} = \delta_{ij}\mathbf{1} - e_i e_j^{\mathsf{T}} - e_j e_i^{\mathsf{T}}$, so $\frac{1}{2}\operatorname{tr}(\mathsf{m}_{ij}T) = \Theta_{ij}$. The notation $\mathcal{T}_K(H)$ below is another name for $O_K^T(H)$. Isometries and linear maps of V act entry by entry. Put $\mathcal{D}_{n,c} = \mathbb{C}\mathfrak{S}_V(\{n \cdot x > c\})$, $\Theta_n = 1 - 2nn^{\mathsf{T}}$, and $(\theta_{n,c}H)(x) = \Theta_n H(\theta_{n,c}x)\Theta_n$, where $\theta_{n,c}x = x - 2(n \cdot x - c)n$. Positivity at (n,c) means $\sigma(\theta_{n,c}\bar{F} \cdot F) \geq 0$ for $F \in \mathcal{D}_{n,c}$.

Definition 3.1 (Data with labels; real time; the frame)**.**

(1) A *symmetric functional on labelled tuples* is a map σ on $\mathfrak{S}_V^1(\mathbb{R}^3)$ with $\sigma(\varnothing) = 1$, invariant under the permutations of the entries and n-linear on $\prod_a C_c^1(O_a;V)$ for open sets O_a with pairwise disjoint closures; it is continued linearly to $\mathbb{C}\mathfrak{S}_V^1$. It is *reconstruction data with labels* with the numbers $C_V, p_V \geq 0$, $c_0 \in (0, \frac{1}{2}]$, if: $(\mathrm{Rec1})_V$ $\sigma(\mathsf{r}X) = \sigma(X)$ for every isometry r; $(\mathrm{Rec2})_V$ $\sigma(\theta\bar{F} \cdot F) \geq 0$ on $\mathbb{C}\mathfrak{S}_V^1(\mathbb{E}_+)$, the space $\mathcal{H}(\sigma)$ is separable, and H, $\mathcal{U}$, $\mathcal{P}$ are formed as in [7, Def. 2.2] (3); $(\mathrm{Rec3})_V$ $\mathcal{P}(\{E < |q|\}) = 0$; $(\mathrm{Rec4})_V$ for $\eta \in (0,1]$, $F, F' \in \mathbb{C}\mathfrak{S}_V^1(\{x_3 > \eta\})$ and $h \in C_c^1(\{|x_3| < c_0\eta\};V)$,

$$|\sigma(\theta\bar{F} \cdot (h) \cdot F')| \leq C_V \eta^{-p_V} \|h\|_{L^1} \, \|[\tau_{-\eta e_3}F]\| \, \|[\tau_{-\eta e_3}F']\|.$$

(Rec5) is the condition $\ker H = \mathbb{C}\Omega$; it is not part of the data.

(2) A point of $\mathbb{R}^{1+2}$ is $x = (t, \vec{x})$; $U(x) = e^{itH}\mathcal{U}(\vec{x})$; $\mathcal{D}_\infty = \bigcap_j \operatorname{dom} H^j$; $\mathcal{S}_V = \mathcal{S}(\mathbb{R}^{1+2}; V_{\mathbb{C}})$ and $f^* = \rho(\theta)\bar{f}$. $\iota(t, \vec{x}) = (\vec{x}, -it)$; $\mathcal{L}$ is the restricted Lorentz group of $\mathbb{R}^{1+2}$, and $\hat{\Lambda} = \iota\Lambda\iota^{-1} \in \mathrm{SO}(3, \mathbb{C})$. For $\mathfrak{z} \in \mathbb{C}^3$, $d_\square(\mathfrak{z}) = \operatorname{Re}\mathfrak{z}_3 - |\operatorname{Im}\mathfrak{z}_1| - |\operatorname{Im}\mathfrak{z}_2|$ and $d_\circ(\mathfrak{z}) = \operatorname{Re}\mathfrak{z}_3 - |(\operatorname{Im}\mathfrak{z}_1, \operatorname{Im}\mathfrak{z}_2)|$.

(3) The *frame* (M1)–(M5) concerns a tensor carrying one label on a space with a scalar field $\phi(f)$, $f \in \mathcal{S}(\mathbb{R}^{1+2})$, a unique vacuum and a representation U of $\mathbb{R}^{1+2} \rtimes \mathcal{L}$; with

$\vartheta_1 = \vartheta_2 = 1$, $\vartheta_3 = -1$: (M1) sesquilinear forms $\mathfrak{q}_{ij} = \mathfrak{q}_{ji}$ on $\mathcal{D}_\infty \times \mathcal{D}_\infty$ with $|\mathfrak{q}_{ij}(\psi', \psi)| \leq c_T \|(1+H)^l \psi'\| \, \|(1+H)^l \psi\|$, $\overline{\mathfrak{q}_{ij}(\psi', \psi)} = \vartheta_i \vartheta_j \mathfrak{q}_{ij}(\psi, \psi')$, $\mathfrak{q}_{ij}(\Omega, \Omega) = 0$, and the operators $T^E_{ij}(F)$ they define, with the real-time components $T_{00} = -T^E_{33}$, $T_{0j} = -iT^E_{3j}$, $T_{jk} = T^E_{jk}$; (M2) continuous kernels $\mathsf{s}_{ij}(x; y_1, \dots, y_m)$ of one tensor entry among m scalar entries, covariant under O(3) and the translations, $\mathsf{s}_{ij}(\mathsf{R}x + b; \mathsf{R}y + b) = \mathsf{R}_{ii'} \mathsf{R}_{jj'} \mathsf{s}_{i'j'}(x; y)$, related to the forms by the product formula of two point operators at different heights, $\mathsf{s}_{ij}^{F',G}(x, y) = \mathfrak{q}_{ij}(\psi'_x, \psi_x)$ as written out in the proof of Lem. 3.5 (c); (M3) $T^E_{ij}(F_1)\phi(f)\psi = \phi(f) T^E_{ij}(F_1)\psi$ on the span $\mathcal{D}_0$ of Ω and the vectors $\phi(f_1) \cdots \phi(f_n)\Omega$, for spacelike separated supports; (M4) (a) $T_{0\nu}(\partial_0 F)\psi = \sum_{k=1,2} T_{k\nu}(\partial_k F)\psi$ on $\mathcal{D}_0$; (M5) $U(\Lambda) T[a, b](F)\Omega = T[\Lambda a, \Lambda b](F \circ \Lambda^{-1})\Omega$, $T[a, b] = a^\mu b^\nu T_{\mu\nu}$.

■

Theorem E (Covariance, reconstruction, locality)**.** Let $0 < g \leq \gamma_W^{\mathrm{sd}}$ and let σ be a tensor limit at g with label b_*.

(a) $\sigma(\mathsf{R}X) = \sigma(X)$ for every $\mathsf{R} \in \mathrm{O}(3)$ and every $X \in \mathfrak{S}_V(\mathbb{R}^3)$.

(b) $\sigma(\theta_{n,c} \bar{F} \cdot F) \geq 0$ for every unit vector n, every $c \in \mathbb{R}$ and every $F \in \mathbb{C}\mathfrak{S}_V(\{n \cdot x > c\})$, where $(\theta_{n,c} H)(x) = \Theta_n H(x - 2(n \cdot x - c)n)\Theta_n$ and $\Theta_n = 1 - 2nn^{\mathrm{T}}$.

(c) σ^1 is reconstruction data with labels at $V = \mathrm{Sym}_3(\mathbb{R})$ with the invariant norm, with $c_0 = \frac{1}{2}$, every $p_V > \frac{17}{2}$ and $C_V = 2^{p_V} C_{p_V} b_*^{-1} \leq 2^{p_V} C_{p_V} g^2$; its invariance holds at all tuples of continuous entries. The map $[F]_{\sigma^1} \mapsto [F]_\sigma$ extends to a unitary map of $\mathcal{H}(\sigma^1)$ onto $\mathcal{H}(\sigma)$ that carries Ω, H, $\mathcal{U}$, $\mathcal{P}$ to Ω, H, $\mathcal{U}$, $\mathcal{P}$.

(d) (i) For every n there is exactly one continuous function $\mathsf{s}_n(x; e)$ on the non-coinciding configurations, n-linear in $e \in V^n$, with $\sigma(H_1, \dots, H_n) = \int \mathsf{s}_n(x; H_1(x_1), \dots, H_n(x_n)) \, \mathrm{d}x$ for all $X \in \mathfrak{S}_V(\mathbb{R}^3)$; it is real-analytic, symmetric when points and labels are permuted together, $\mathsf{s}_n(\mathsf{r}x_1, \dots, \mathsf{r}x_n; \rho(\mathsf{R})e_1, \dots, \rho(\mathsf{R})e_n) = \mathsf{s}_n(x; e)$ for every isometry $\mathsf{r}x = \mathsf{R}x + v$, and $|\mathsf{s}_n(x; e)| \leq \big(C_V (2n^2 / \min\{1, \mathsf{d}(x)\})^{p_V}\big)^n \prod_a |e_a|$ at every point, $\mathsf{d}(x) = \min_{a \neq b} |x_a - x_b|$; so $|\sigma(X)| \leq (C_V (2n^2/d)^{p_V})^n \prod_a \|H_a\|_{L^1}$ at pairwise distances $\geq d$, $0 < d \leq 1$, and $|\mathsf{S}_n(\varphi)| \leq (8\pi^2 3^{\lceil p_V \rceil + 2} (2e^2)^{p_V} C_V)^n (n!)^{2p_V} |\varphi|_{(\lceil p_V \rceil + 4)n}$. (ii) There is a strongly continuous unitary representation U of $\mathbb{R}^{1+2} \rtimes \mathcal{L}$ on $\mathcal{H}(\sigma)$ with $U(x, 1) = e^{itH} \mathcal{U}(\vec{x})$, $U(a, \Lambda)\Omega = \Omega$, $U(a, \Lambda)\mathcal{D}_0 = \mathcal{D}_0$, the joint spectrum in the closed forward cone, and

$$U(a, \Lambda) T(f) U(a, \Lambda)^{-1} = T\big(\hat{\Lambda} f(\Lambda^{-1}(\cdot - a))\hat{\Lambda}^{\mathrm{T}}\big) \quad \text{on } \mathcal{D}_0,$$

with the transpose, not the adjoint. (iii) $T(f)T(f')\psi = T(f')T(f)\psi$ for $\psi \in \mathcal{D}_0$ whenever $f, f' \in \mathcal{S}_V$ have spacelike separated supports: all six components are local with respect to all six.

(e) Let $0 < g \leq \gamma$. Every sequence of first-passage data at g with $K_j \to \infty$ and periodic Gibbs states has a subsequence along which the tensor limit exists. Every such limit σ satisfies (a), (b) and the conclusion of [7, Thm. E] (c), and σ^1 is reconstruction data with labels with $c_0 = \frac{1}{2}$, every $p_V > \frac{17}{2}$ and

$$C_V = 3^{p_V} L^{4z} \max_{k=1,2} 2^{p_V} C_{p_V}[g'_k] \, g'^2_k,$$

where z is the least integer ≥ 1 with $L^z g^{-2} \geq 2(\gamma_W^{\mathrm{sd}})^{-2}$, $g'^{-2}_1 = L^z g^{-2}/2$, $g'^{-2}_2 = L^{z+1} g^{-2}/4$, and $C_p[g']$ is the number C_p of [7, Thm. C] (c) at the coupling g': a number formed from the constants, g and p_V, one triple for all the tensor limits at g; so (c) and (d) hold for it.

■□

Theorem F (Components and Wightman tensor fields)**.** Let σ be the tensor limit of a sequence (K_j, β_j, ν_j) of first-passage data and periodic Gibbs states at g, and $A^\nu_K(X) = \prod_a [\mathcal{T}_K(H_a) - \nu(\mathcal{T}_K(H_a))]$.

(a) *(Decomposition; $0 < g \le \gamma_W^{\mathrm{ref}}$.)* There are a compact metrisable space X, a Borel probability measure λ on X with support X, and real symmetric functionals σ_ξ, $\xi \in \mathsf{X}$, on labelled tuples, with $\xi \mapsto \sigma_\xi(X)$ continuous and $\xi \mapsto \sigma_\xi$ injective, such that σ has iterated plateaux in the sense of [7, Def. 2.2] (5) at every $(X^1, \dots, X^q)$, $q \ge 1$, $X^i \in \mathfrak{S}_V(\mathbb{R}^3)$, and its iterated plateau there is $\int \prod_i \sigma_\xi(X^i)\, \mathrm{d}\lambda(\xi)$.

(b) *(Each component; $0 < g \le \gamma_W^{\mathrm{ref}}$.)* For every $\xi \in \mathsf{X}$: (i) there are translation-invariant events E_j with $\nu_j(E_j) > 0$ such that $\rho_j = \nu_j(\cdot \,|\, E_j) \in \mathcal{G}^{\mathrm{per}}_{K_j,\beta_j}$ and $\sigma_\xi(X) = \lim_j \rho_j(A^{\nu_j}_{K_j}(X))$ along the full sequence, indeed $\|E_{\rho_j}[A^{\nu_j}_{K_j}(X) \,|\, \mathcal{I}] - \sigma_\xi(X)\|_{L^\infty(\rho_j)} \to 0$, $\mathcal{I}$ the invariant σ-field; (ii) $\sigma_\xi((H)) = \frac{c_{\pi(\xi)}}{3} \int \operatorname{tr} H$ with a real number $c_{\pi(\xi)}$, and the recentred functional $\sigma^c_\xi(X) = \sum_{J \subset \{1..n\}} \prod_{a \notin J} (-\sigma_\xi((H_a)))\, \sigma_\xi(X_J)$ is the tensor limit of (K_j, β_j, ρ_j); so it has all the properties of a tensor limit, and $[F] \mapsto [\mathsf{S}_\xi F]$, $\mathsf{S}_\xi X = \sum_J \prod_{a \notin J} \sigma_\xi((H_a)) X_J$, is a unitary map of $\mathcal{H}(\sigma_\xi)$ onto $\mathcal{H}(\sigma^c_\xi)$ that carries Ω_ξ to Ω and intertwines H and $\mathcal{P}$; (iii) $\ker H_\xi = \mathbb{C}\Omega_\xi$; (iv) there is exactly one map π of X onto the space X_0 of the canonical decomposition $(\mathsf{X}_0, \lambda_0, (\sigma_{0,\zeta}))$ of the scalar part $\sigma|_{V_0}$ with $\sigma_\xi(f_1\mathbf{1}, \dots, f_n\mathbf{1}) = \sigma_{0,\pi(\xi)}(f_1, \dots, f_n)$; it is continuous and onto, and $\pi_*\lambda = \lambda_0$.

(c) *(Wightman fields; $0 < g \le \gamma_W^{\mathrm{sd}}$.)* For every ξ, the restriction of σ^c_ξ to C^1_c entries is reconstruction data with labels with $c_0 = \frac{1}{2}$, every $p_V > \frac{17}{2}$ and the numbers of Thm. E (c), and it has (Rec5). Its reconstruction is a Wightman tensor field of six components in $1+2$ dimensions: the fields $T(f)$, $f \in \mathcal{S}_V$, on $\mathcal{D}_0$ are tempered; $\langle \psi, T(f)\psi' \rangle = \langle T(f^*)\psi, \psi' \rangle$, so that T_{33} and T_{jk}, $j,k \le 2$, are Hermitian and T_{3j} is Hermitian after the factor i; $U(a,\Lambda)T(f)U(a,\Lambda)^{-1} = T(\hat\Lambda f(\Lambda^{-1}(\cdot - a))\hat\Lambda^{\mathrm{T}})$; the joint spectrum lies in the closed forward cone; Ω_ξ is the only invariant vector and is cyclic; all components are local with respect to all. The growth condition of [7, Thm. C] (h) holds with the same numbers for all components. The given limit σ is the integral over λ of its components in the sense of (a).

(d) *(The frame over a component; $0 < g \le \gamma_W^{\mathrm{sd}}$.)* On the scalar subspace of $\mathcal{H}(\sigma^c_\xi)$ the frame (M1)–(M5) holds for the tensor formed at the label matrices m_{ij} of the stress tensor, with the forms $q_0(\cdot,\cdot;\mathsf{m}_{ij})$, the kernels $\mathsf{s}_{ij}(x;y) = \mathsf{s}_{1+m}(x,y;\mathsf{m}_{ij},\mathbf{1},\dots,\mathbf{1})$, the scalar field $\phi(f) = T(f\mathbf{1})$ and the restriction to the scalar subspace of the representation U of Thm. E (d) (ii) for σ^c_ξ. The scalar subspace reduces $U(a,\Lambda)$: $\phi(f)$ keeps it; $U(a,\Lambda)\phi(f)U(a,\Lambda)^{-1} = \phi(f(\Lambda^{-1}(\cdot - a)))$ on $\mathcal{D}_0$, since $\hat\Lambda\mathbf{1}\hat\Lambda^{\mathrm{T}} = \mathbf{1}$; $U(a,\Lambda)\Omega = \Omega$; and Ω with the vectors $\phi(f_1)\cdots\phi(f_n)\Omega$ spans a dense subspace of it, by Prop. 3.3 (a) (v) for the restriction of σ^c_ξ to the tuples with entries in V_0, whose space is the scalar subspace and whose fields are the restrictions of the $T(f\mathbf{1})$. In particular the real-time stress tensor is conserved, (M4) (a).

(e) *(The frame on $\mathcal{H}_0$; $0 < g \le \gamma_W^{\mathrm{sd}}$.)* For σ itself, (M1) holds, the kernels $\mathsf{s}_{ij}(x;y)$ of one tensor entry among scalar entries are covariant as in (M2) and continuous at all non-coinciding points, $[T^E_{ij}(F), \phi(f)] = 0$ between scalar vectors at spacelike separation as in (M3), and $\langle \Omega, \phi(f_1)\cdots\phi(f_n)T[t](F)\Omega \rangle$ is invariant under $(f,t,F) \mapsto (f \circ \Lambda^{-1}, \Lambda t, F \circ \Lambda^{-1})$ as in (M5). ■□

Proposition 3.2 (Covariance at one label suffices)**.** Let $0 < g \le \gamma_W^{\mathrm{sd}}$ and let σ be a tensor limit at g. If $\sigma(\mathsf{R}X) = \sigma(X)$ for all $\mathsf{R} \in \mathrm{O}(3)$ at the tuples with exactly one entry outside V_0, then the kernels of (M2) are covariant at all non-coinciding points and (M3), (M5) hold on $\mathcal{H}_0$. ■

By Thm. E (a) every tensor limit at g satisfies the condition of Prop. 3.2, and its conclusion is contained in Thm. F (e). Its proof uses rotational covariance only for tuples with exactly one entry outside V_0. It follows the proofs of Props. YM3_I:108.3, YM3_I:110.2 and YM3_I:110.1, with one labelled insertion.

Proposition 3.3 (Reconstruction for data with labels)**.**

(a) Let σ be reconstruction data with labels with the numbers C_V, $p = p_V$, c_0; (Rec5) is not supposed. Then $\mathcal{H}(\sigma)$ is separable, and there are a strongly continuous unitary representation U of $\mathbb{R}^{1+2} \rtimes \mathcal{L}$ with $U(x,1) = e^{itH}\mathcal{U}(\vec{x})$, operators $\phi(f)$, $f \in \mathcal{S}_V$, on the span $\mathcal{D}_0$ of Ω and of the vectors $\phi(f_1)\cdots\phi(f_n)\Omega$, functions $\mathbf{W}_n(\mathfrak{z}; e)$ holomorphic on $\{d_\circ > 0\}^{n-1}$ and n-linear in e, and vectors $\Psi_n(z; e)$ with $[X] = \int \Psi_n(x; h_1(x_1), \dots, h_n(x_n))\,\mathrm{d}x$ for the tuples in $\mathbb{E}_+$ ordered in the third coordinate, such that: (i) the spectral measure of $U(\cdot, 1)$ is carried by the closed forward cone and $U(a,\Lambda)\Omega = \Omega$; with (Rec5), the invariant vectors are the multiples of Ω; (ii) $\phi(f)\mathcal{D}_0 \subset \mathcal{D}_0$, $U(a,\Lambda)\mathcal{D}_0 = \mathcal{D}_0$, $f \mapsto \langle\psi, \phi(f)\psi'\rangle$ is tempered, and $\langle\psi, \phi(f)\psi'\rangle = \langle\phi(f^*)\psi, \psi'\rangle$; (iii) $U(a,\Lambda)\phi(f)U(a,\Lambda)^{-1} = \phi(D(\Lambda)f(\Lambda^{-1}(\cdot - a)))$ on $\mathcal{D}_0$, where $D(\Lambda) = \rho_{\mathbb{C}}(\hat{\Lambda})$ and $\rho_{\mathbb{C}}$ is the holomorphic continuation of $\rho|_{\mathrm{SO}(3)}$ to $\mathrm{SO}(3,\mathbb{C})$; at $V = \mathrm{Sym}_3(\mathbb{R})$, $D(\Lambda)A = \hat{\Lambda}A\hat{\Lambda}^{\mathrm{T}}$; (iv) $\phi(f)\phi(f')\psi = \phi(f')\phi(f)\psi$ on $\mathcal{D}_0$ at spacelike separated supports, for all components; (v) $\mathcal{D}_0$ is dense; (vi) $|\mathbf{W}_n(\mathfrak{z}; e)| \le (C_V\eta^{-p})^n \prod_k |e_k|$ for $e \in V^n$, $d_\circ(\mathfrak{z}_k) \ge 2\eta$, $0 < \eta \le 1$; $\langle\Omega, \phi(f_1)\cdots\phi(f_n)\Omega\rangle = \lim_{\epsilon\downarrow 0}\int \mathbf{W}_n(\iota(x_2 - x_1 + i\epsilon\alpha_1), \dots; f_1(x_1), \dots, f_n(x_n))\,\mathrm{d}x$ for all α_k in the open forward cone; $\sigma(X) = \int \mathbf{W}_n(x_2 - x_1, \dots; h_1(x_1), \dots)\,\mathrm{d}x$ at the tuples ordered in the third coordinate; (vii) there is exactly one function $\mathbf{S}_n$, holomorphic on the union of the tubes $\{d_\circ(\mathsf{R}^{-1}(z_{\varsigma(k+1)} - z_{\varsigma(k)})) > 0,\ 1 \le k < n\}$ over $\mathsf{R} \in \mathrm{SO}(3)$ and the permutations ς, which on each of them is $\mathbf{W}_n$ written in the frame R at the labels $\rho(\mathsf{R})^{-1}e_{\varsigma(k)}$; its restriction s_n to the real non-coinciding configurations is real-analytic, is the only continuous kernel of σ, is symmetric in points and labels together, and is invariant under every isometry; (viii) Ω and the vectors $\Psi_n(x; e)$ at real x with $0 < x_{13} < \cdots < x_{n3}$ are total, and so are Ω and the classes of the tuples whose entries lie in disjoint slabs.

Clauses (i)–(vi) hold as well for complex symmetric functionals invariant under the orientation-preserving isometries, with $\overline{\sigma(W)} = \sigma(\theta\bar{W})$ and with $(\mathrm{Rec2})_V$–$(\mathrm{Rec4})_V$.

(b) For data as in (a) the two-fold plateau $m_2(X,Y) = \lim_{|v|\to\infty}\sigma(X \cdot \tau_v Y)$ exists, and σ has a Hermitian canonical decomposition: a compact metrisable X, a Borel probability measure λ with support X, and complex symmetric functionals σ_ξ with $\overline{\sigma_\xi(X)} = \sigma_\xi(\theta X)$, continuous and injective in ξ, such that the iterated plateaux of σ ([7, Def. 2.2] (5)) are $\int \prod_i \sigma_\xi(X^i)\,\mathrm{d}\lambda$; it is unique up to exactly one homeomorphism, and its components are real, so that it is a canonical decomposition, if and only if $m_2(F,F) \ge 0$ for the real combinations F of tuples.

Every component has $(\mathrm{Rec2})_V$–$(\mathrm{Rec4})_V$ with the numbers of σ and (Rec5), is reflection positive at every mirror, and $|\sigma_\xi(X \cdot \tau_v Y) - \sigma_\xi(X)\sigma_\xi(Y)| \le K(X,Y)|v|^{-1}$ for $|v| \ge r(X,Y)$, with K, r formed from C_V, p, c_0 and the two tuples, the same for all ξ. Let V_0 be a subrepresentation such that the classes of the tuples with entries in V_0 are total, and let $(\mathsf{X}_0, \lambda_0, (\sigma_{0,\zeta}))$ be the Hermitian canonical decomposition of $\sigma|_{V_0}$, which is reconstruction data with labels in V_0 with the numbers of σ. There is exactly one map $\pi : \mathsf{X} \to \mathsf{X}_0$ with $\sigma_\xi|_{V_0} = \sigma_{0,\pi(\xi)}$ for every ξ; it is continuous and onto, $\pi_*\lambda = \lambda_0$, and $g \mapsto g \circ \pi$ is a unitary map of $L^2(\mathsf{X}_0, \lambda_0)$ onto $L^2(\mathsf{X}, \lambda)$.

So for every tuple X there is a bounded Borel function g_X on X_0 with $\sigma_\xi(X) = g_X(\pi(\xi))$ for λ-almost every ξ, and the iterated plateaux of σ are $\int \prod_i g_{X^i}\,\mathrm{d}\lambda_0$: the decomposition of $\sigma|_{V_0}$ bears that of σ as measure spaces. The map π is a homeomorphism if and only if every g_X agrees λ_0-almost everywhere with a continuous function, in particular if $\dim\ker H < \infty$. ■

Proof sketch. (a) On the circular cone $|e^{-\mathfrak{z}_3 E + i(\mathfrak{z}_1 q_1 + \mathfrak{z}_2 q_2)}| \le e^{-d_\circ(\mathfrak{z})E}$; this gives the contractions $\mathcal{T}(\mathfrak{z})$, holomorphic inside the tube. By $(\mathrm{Rec4})_V$ the forms $(F, F') \mapsto \sigma(\theta\bar{F} \cdot (h) \cdot F')$ are bounded, whence insertion operators $B_\eta(h)$ with $\|B_\eta(h)\| \le C_V\eta^{-p}\|h\|_{L^1}$, covariant under the translations, $B_\eta(\tau_y h') = \mathcal{T}(y + (\eta - \eta')e_3)B_{\eta'}(h')\mathcal{T}(-y + (\eta - \eta')e_3)$. Since translations act on them Lipschitz-continuously (on the cone each derivative of $e^{-(\eta/2 + y_3)E + i\vec{y}\cdot q}$ is at most $4/(e\eta)$), the limits $\mathsf{B}(\eta, \eta; e) =$

$\lim_{\epsilon\downarrow 0} B_\eta(\delta_\epsilon e)$, $\delta_\epsilon = \epsilon^{-3}\chi(\cdot/\epsilon)$ with $\chi \geq 0$ radial of integral 1, exist in norm, and $\mathsf{b}(w|e|w') = \mathcal{T}(w - se_3)\mathsf{B}(s,s;e)\mathcal{T}(w' - se_3)$ are holomorphic point operators with $B_\eta(h) = \int \mathsf{b}(y+\eta e_3|h(y)|-y+\eta e_3)\,\mathrm{d}y$.

Ordered tuples are products of insertion operators and semigroup factors; this gives $\mathbf{W}_n$, Ψ_n and the integral formulas. Any n distinct points are ordered in some frame $\mathsf{R} \in \mathrm{SO}(3)$ whose cap $\{\mathsf{R}e_3 \cdot \hat{v} > \frac{1}{2}\}$ contains a given direction (the zones $\{|u \cdot (x_a - x_b)| < \gamma_N |x_a - x_b|\}$, $\gamma_N = 1/(2N^2)$ with $N \geq n$, cover less than a quarter of the sphere); the pieces in two frames agree at the real points of the intersection of two convex tubes, hence on it, and they form one function $\mathbf{S}_n$. Totality: the map $y \mapsto \Phi(y)$, the limit of the classes of tuples of bumps at y, has the real-analytic Gram kernel $\mathsf{s}_{2n}(\theta y, y')$, so its matrix elements are real-analytic; for a vector orthogonal to the Ψ_n they vanish on the ordered configurations, hence on the connected set of the non-coinciding ones.

The fields are defined by the forms q_0 with $\mathsf{A}_l(e) = \Gamma(l)^{-2}\iint (ss')^{l-1}e^{-s-s'}\mathsf{B}(s,s';e)\,\mathrm{d}s\,\mathrm{d}s'$, $\mathsf{B}(s,s';e) = \mathsf{b}(se_3|e|s'e_3)$, convergent for $l > p$; then $\mathcal{T}(w)\phi(f)\mathcal{T}(w') = \int \mathsf{b}(w + \iota x|f(x)|w' - \iota x)\,\mathrm{d}x$ for w, w' in the open tube, and $\|(1+H)^j\phi(f)\psi\| \leq c_j(f)\|(1+H)^{2l+j}\psi\|$. *Boundary values* (vi): with $c_k = \iota(\frac{i}{2}\epsilon\alpha_k)$, α_k in the open forward cone, the contractions $\mathcal{T}(2c_k)$ commute with H and tend strongly to 1 as $\epsilon \downarrow 0$; inserted between the fields, they turn $\langle\Omega, \phi(f_1)\cdots\phi(f_n)\Omega\rangle$ into $\int \mathbf{W}_n(\iota(x_2 - x_1 + i\epsilon\alpha_1), \ldots; f_1(x_1), \ldots, f_n(x_n))\,\mathrm{d}x$, up to an error that the energy bounds, applied from the right to the left, send to 0.

The density of $\mathcal{D}_0$ (v): let $\psi \perp \mathcal{D}_0$. For a word w in the operators $\phi(f)$ and $\mathcal{T}(w)$, $\psi \perp \mathsf{w}\Omega$, by induction on the number of the letters $\mathcal{T}$: the matrix element is continuous on the closed tube and holomorphic inside in the argument w of the leftmost such letter; it vanishes at $w = \iota a$, $a \in \mathbb{R}^{1+2}$, because $\mathcal{T}(\iota a) = U(a)$ moves the rest of the word into a word with translated test functions and one such letter fewer; so it vanishes on the half planes in time by the reflection principle, and on the tube by the identity theorem in the space variables. As $\mathcal{T}(w)\phi(g_k e)\mathcal{T}(w') \to \mathsf{b}(w|e|w')$ in norm for $g_k \geq 0$ of integral 1 whose supports shrink to a point, $\psi \perp \Psi_n(z;e)$, and $\psi = 0$ by (viii).

The representation D: with the covering $\mathsf{p} : \mathrm{SL}(2,\mathbb{C}) \to \mathrm{SO}(3,\mathbb{C})$, the differential of $\rho \circ \mathsf{p}$ on $\mathrm{SU}(2)$ continues complex-linearly to $\mathfrak{sl}(2,\mathbb{C})$ and integrates to a holomorphic representation of the simply connected $\mathrm{SL}(2,\mathbb{C})$, trivial on ± 1, hence to a holomorphic representation $\rho_{\mathbb{C}}$ of $\mathrm{SO}(3,\mathbb{C})$ that continues ρ; $\Lambda \mapsto \hat{\Lambda}$ is a homomorphism of $\mathcal{L}$ into $\mathrm{SO}(3,\mathbb{C})$, so $D(\Lambda) = \rho_{\mathbb{C}}(\hat{\Lambda})$ is a representation, and $D(\Lambda)e^* = (D(\Lambda)e)^*$. Every $\Lambda \in \mathcal{L}$ is $(1 \oplus \mathsf{R}_1)B(s)(1 \oplus \mathsf{R}_2)$ with rotations $\mathsf{R}_1, \mathsf{R}_2$ of the space plane and the boost $B(s)$: a spatial rotation and a boost send the future unit vector e_0 to Λe_0, and their inverse followed by Λ fixes e_0 and acts as a rotation of its orthogonal spatial plane; $\widehat{1 \oplus \mathsf{R}} = \mathsf{R} \oplus 1$, and $\widehat{B(s)}$ is the rotation in the plane (e_1, e_3) by the imaginary angle is.

Lorentz covariance: $\mathbf{W}_n$ is invariant under the real rotations about e_3 and under those in the plane (e_1, e_3), and this last invariance continues holomorphically in the angle to imaginary angles, the boosts, the labels being moved by $e^{is\,\mathrm{d}\rho(\omega_1)}$, $\omega_1 = E_{13} - E_{31}$, $d\rho$ the differential of ρ; no bound on this factor is needed. So $\mathbf{W}_n(\hat{\Lambda}\mathfrak{z}; D(\Lambda)e) = \mathbf{W}_n(\mathfrak{z}; e)$, and the boundary values at the cone vectors $\Lambda^{-1}\alpha_k$ give $\langle\Omega, \phi(\Lambda f_1)\cdots\phi(\Lambda f_n)\Omega\rangle = \langle\Omega, \phi(f_1)\cdots\phi(f_n)\Omega\rangle$ with $(\Lambda f)(x) = D(\Lambda)f(\Lambda^{-1}x)$.

As $(\Lambda f)^* = \Lambda(f^*)$, the map $\phi(f_1)\cdots\phi(f_n)\Omega \mapsto \phi(\Lambda f_1)\cdots\phi(\Lambda f_n)\Omega$ is isometric on the dense $\mathcal{D}_0$ and extends to a unitary operator $U(\Lambda)$; these form a representation with $U(\Lambda)U(a)U(\Lambda)^{-1} = U(\Lambda a)$, whose strong continuity reads of D only that it is locally bounded. With the spectrum of $U(\cdot, 1)$, the image of $\mathcal{P}$, carried by the closed forward cone by $(\mathrm{Rec3})_V$, this gives (i)–(iii). *Locality*: the componentwise argument is that of Prop. YM3_I:110.2. Fix the labels and compactly supported scalar tests with two adjacent supports spacelike separated. The height deformation used there stays in tilted tubes with every consecutive margin at least 3η. The Cauchy–Riemann equations and integration by parts move derivatives onto the tests, and Lem. YM3_I:109.4 gives a finite smeared bound $Q_e(g)$ uniform along the deformation. For g the product test, the integrals for the two adjacent orders differ by at most $2\eta Q_e(D'g)$, where $D' = \frac{1}{2}(\partial_{t_{k+1}} - \partial_{t_k})$ and $k, k+1$ are the exchanged entries. Their endpoint boundary values therefore agree as $\eta \to 0$. Graph-norm continuity extends the equality to Schwartz tests, as in the cited proof.

These are the proofs for scalar data of Props. YM3_I:108.3, YM3_I:109.1, YM3_I:110.1, YM3_I:110.2 and Thm. YM3_I:AO among the construction results, with a label at each entry and with D in the place of the trivial representation; besides them they read the spectral theorem, the identity theorem, the reflection principle in one variable and the covering of $\mathrm{SO}(3,\mathbb{C})$ by $\mathrm{SL}(2,\mathbb{C})$. They read the reality of σ and its invariance under an orientation-reversing isometry at one place only, the adjoint $B_\eta(h)^* = B_\eta(\theta\bar h)$, where $\overline{\sigma(W)} = \sigma(\theta\bar W)$ is all that is used; this gives (i)–(vi) for the complex functionals of the last sentence of (a). (b) For a combination Y of tuples put $\mathsf{e}(Y) = P_0[\tau_{se_3}Y]$, P_0 the projection onto $\ker H$ and s so large that $\tau_{se_3}Y$ lies over $\mathbb{E}_+$: it does not depend on s, it is invariant under the translations, and these vectors are dense in $\ker H$.

In a frame R, by the cone, $\sigma(A\cdot\tau_w Y) \to \langle \mathsf{e}(\theta\bar A'), \mathsf{e}(Y')\rangle$, $A' = \mathsf{R}^{-1}A$, $Y' = \mathsf{R}^{-1}Y$, uniformly as $|w|\to\infty$ with $\mathsf{R}e_3\cdot w \geq \frac12|w|$; two frames whose caps meet give the same limit, and the caps cover the sphere; so the two-fold plateau exists, $m_2(A,Y) = \langle \mathsf{e}(\theta\bar A), \mathsf{e}(Y)\rangle$. *The rotations are trivial at infinity*: for a unit vector u, $\beta_u(F,G) = m_2(\theta_u\bar F, G)$ is a positive Hermitian form; for $u'\perp u$ the involution $\theta_u\theta_{u'}$ acts on its space as a self-adjoint unitary operator ≥ 0, which is 1; hence $m_2(\theta_u A, Y)$ does not depend on u, $m_2(\mathsf{r}A,Y) = m_2(A,Y)$ and $\mathsf{e}(\mathsf{r}Y) = \mathsf{e}(Y)$ for the orientation-preserving isometries r, and $J\mathsf{e}(Y) = \mathsf{e}(\theta Y)$ defines a unitary involution J of $\ker H$ with $m_2(\bar A, Y) = \langle J\mathsf{e}(A), \mathsf{e}(Y)\rangle$.

The operators at infinity: $M(X)$ is the bounded operator of $\ker H$ with $\langle \mathsf{e}(Z), M(X)\mathsf{e}(Y)\rangle = \lim_{|v|\to\infty} m_2(\theta\bar Z, X\cdot\tau_v Y)$, the limit of $\sigma(\tau_w\theta\bar Z\cdot X\cdot\tau_v Y)$ as w and v go to infinity on opposite sides of X; for X ordered in a frame, $M(X) = P_0\mathsf{b}_X|_{\ker H}$, b_X the product of the insertion operators and semigroup factors of the ordered tuple, and a partition of unity reduces every tuple to such ones. Then $M(F)^* = M(\theta\bar F)$, $M(\mathsf{r}X) = M(X)$ for the orientation-preserving and $M(\mathsf{r}X) = JM(X)J$ for the other isometries, $M(X\cdot\tau_v Y)\to M(X)M(Y)$ strongly as $|v|\to\infty$, and, σ being symmetric, the $M(X)$ commute.

They generate a separable commutative C^*-algebra; its space of characters is X, the vacuum gives λ, whose support is X because Ω is cyclic, hence separating, for the algebra; $\ker H$ is carried unitarily onto $L^2(\mathsf{X},\lambda)$, $\sigma_\xi(X)$ is the Gelfand transform of $M(X)$, so that $\overline{\sigma_\xi(X)} = \sigma_\xi(\theta X)$, and the iterated plateaux of σ are $\langle\Omega, \prod_i M(X^i)\Omega\rangle$. Uniqueness follows from the density of the polynomials in the $\sigma_\xi(X)$ and their conjugates. *Reality*: $m_2(\bar F, F) = \langle J\mathsf{e}(F), \mathsf{e}(F)\rangle$, so $m_2(F,F)\geq 0$ for the real combinations F if and only if $J\geq 0$, that is $J=1$, that is $M(X)^* = M(X)$ for every tuple, that is every σ_ξ is real.

The components: an inequality of σ at the tuples $F\cdot\tau_{tu}Y$, the spectator Y sent to infinity along u, becomes an inequality between elements of the algebra on a dense set of $\ker H$, and, the Gelfand transforms being continuous and $\operatorname{supp}\lambda = \mathsf{X}$, an inequality at every ξ (test it with $\lambda(O)^{-1/2}\mathbf{1}_O$ for small neighbourhoods O of ξ); so every σ_ξ is reflection positive at every mirror, has $(\mathrm{Rec4})_V$ with the numbers of σ, and the translations act on its space by contractions and unitary operators, which gives $(\mathrm{Rec2})_V$, one countable family of tuples making all these spaces separable.

The spectral measures disintegrate, $\mu_F = \int \mu_F^\xi \,\mathrm{d}\lambda(\xi)$ for F over $\mathbb{E}_+$; so $\mu_F^\xi(\{E<|q|\}) = 0$ for λ-almost every ξ, and, this number being lower semicontinuous in ξ, for every ξ: $(\mathrm{Rec3})_V$. At $\{(0,0)\}$ the disintegration gives $\int \|P_0^\xi[F]_\xi\|^2\,\mathrm{d}\lambda = \|P_0[F]\|^2 = \int|\sigma_\xi(F)|^2\,\mathrm{d}\lambda$, while $\|P_0^\xi[F]_\xi\|\geq|\sigma_\xi(F)|$, P_0^ξ the projection onto $\ker H_\xi$: so $\ker H_\xi = \mathbb{C}\Omega_\xi$ for almost every ξ. *The rate*: for data, and for Hermitian data, $|\sigma(X\cdot\tau_v Y) - m_2(X,Y)|\leq K(X,Y)|v|^{-1}$ for $|v|\geq r(X,Y)$, with K, r formed from C_V, p, c_0 and the two tuples alone.

In each frame of a finite family whose caps cover the sphere and in which X and Y are ordered, the difference $f(v)$ is the integral of $e^{-(v_3'-c)E+i\vec v'\cdot q}$, $v' = \mathsf{R}^{-1}v$, against a complex measure off $(0,0)$ whose variation is at most the product of the numbers $C_V\eta^{-p}\|h\|_{L^1}$ of the entries of X and Y. The image κ_v of this measure, with the exponential, under $(E,q)\mapsto s = E^2-|q|^2$ has the moments $\Delta^k f(v)$, which are independent of the frame. Its damping gives $\int e^{a\sqrt s}\,d|\kappa_v|(s)<\infty$ for every $a>0$ below the positive time margin, since $0\leq\sqrt s\leq E$. This square-root exponential condition determines complex measures within this class by their moments. Indeed, the real and imaginary

parts of the difference of two such measures have zero moments, including order zero. Their positive and negative parts have equal masses and moments and inherit the same integrability condition; after normalization when the common mass is nonzero, Hardy's criterion applies to them [2, Thm. 1]. Thus κ_v is independent of the frame, and its masses $f_{\mathsf B}(v) = \kappa_v(\mathsf B)$ on windows $\mathsf B$ are functions outside a ball $\{|v| < r_2\}$, with $\Delta f_{\mathsf B}(v) = \int_{\mathsf B} s\,\kappa_v(ds)$.

For $\mathsf B = [m_1^2, m_2^2)$, $|(\Delta - m_1^2)\operatorname{Re}(e^{i\varphi} f_{\mathsf B})| \le (m_2^2 - m_1^2)\mathsf V_{\mathsf B}$, $\mathsf V_{\mathsf B}$ the variation of the part, and the comparison principle for $\Delta - m_1^2$ outside the ball, with the radial solution $\frac{r_2}{r} e^{-m_1(r-r_2)}$ of the dimension three, gives $|f_{\mathsf B}(v)| \le \mathsf V_{\mathsf B}(\frac{r_2}{|v|} + \frac{m_2^2 - m_1^2}{m_1^2})$; the harmonic part at the mass 0, which tends to 0 at infinity, is at most $\mathsf V_{\{0\}} \frac{r_2}{|v|}$; the sum over the windows $[(1+\epsilon)^j, (1+\epsilon)^{j+1})$, $j \in \mathbb Z$, with $\epsilon \downarrow 0$, gives the rate. One rate serves all the components, which are Hermitian data with the numbers of σ; with the continuity in ξ and $\operatorname{supp}\lambda = \mathsf X$ the bound by $\sigma_\xi(X)\sigma_\xi(Y)$ holds at every ξ, so that $\langle [F]_\xi, e^{-tH_\xi}[F]_\xi\rangle \to |\sigma_\xi(F)|^2$ and $\ker H_\xi = \mathbb C\Omega_\xi$ for every ξ.

The subrepresentation: the defining limits of the operators at infinity of $\sigma|_{V_0}$ and of σ agree at the tuples with entries in V_0; when their classes are total, $\sigma|_{V_0}$ and σ have one space and one $\ker H$, the operators of these tuples generate a C^*-subalgebra with the same unit, for which Ω is cyclic too, π is the restriction of the characters, the only map with $\sigma_\xi|_{V_0} = \sigma_{0,\pi(\xi)}$ because $\zeta \mapsto \sigma_{0,\zeta}$ is injective, $\pi_*\lambda = \lambda_0$ by the vacuum state, π is onto because its image is compact of full measure, and $g \mapsto g \circ \pi$ is unitary because both decompositions carry the one space $\ker H$ unitarily onto their L^2; π is injective if and only if the subalgebra is the whole algebra, by the theorem of Stone and Weierstrass, that is if and only if every g_X agrees almost everywhere with a continuous function; and if $\dim\ker H = N < \infty$, then $\mathsf X$ and $\mathsf X_0$ have N points each. □

Lemma 3.4 (The circular cone from five mirrors; for data)**.** Let σ be a real symmetric multilinear functional on labelled tuples, with $\sigma(\varnothing) = 1$ and bounds independent of the position, invariant under the translations. For a unit vector u at whose mirror $\{u \cdot x = 0\}$ σ is reflection positive, let $\mathcal H_u$, H_u, $\mathcal P_u$ be formed as in [7, Def. 2.2] (3) with u in the place of e_3.

(a) Let u' be a unit vector orthogonal to u. If σ is reflection positive at the mirrors with the normals u and $(u+u')/\sqrt2$, then $H_u \ge |u' \cdot P_u|$.

(b) Let (u, u', u'') be an orthonormal basis and $n_\beta = \cos\beta\, u + \sin\beta\, u'$. If σ is reflection positive at the four planes through the axis $\mathbb R u''$ with the normals $n_0, n_{\pi/4}, n_{\pi/2}, n_{3\pi/4}$ and at one plane with the normal $(n_{\beta_1} + u'')/\sqrt2$, $\beta_1 \in \frac\pi4\mathbb Z$, then $\mathcal P_{n_\beta}(\{E < |q|\}) = 0$ for every $\beta \in \frac\pi4\mathbb Z$.

(c) *(Eight measures in one plane; analysis alone.)* Let $R > 0$, $\kappa > 0$, $C < \infty$, $\mathsf V = \{(E,q) \in \mathbb R^2 : E \ge |q|\}$, and for each of the eight angles $\beta \in \frac\pi4\{0, 1, \dots, 7\}$ let m_β be a complex Borel measure on $\mathsf V$ with $|m_\beta|(\mathsf V) \le C$; for $\rho > 0$ and complex c put

$$f_\beta(c;\rho) = \int_{\mathsf V} \exp\bigl(2RE - \rho[E\cos(c-\beta) - iq\sin(c-\beta)]\bigr)\,\mathrm dm_\beta(E,q)$$

where the integrand is bounded. Suppose that $f_\beta(t;\rho) = f_{\beta'}(t;\rho)$ for any two of the eight angles β, β', all $\rho > 0$ and all real t with $\rho\cos(t-\beta) > 2R$ and $\rho\cos(t-\beta') > 2R$; that $\mu = e^{2RE} m_0$ is a finite positive measure; and that for one of the eight angles, β_1, the measure m_{β_1} is carried by $\{E \ge \kappa\}$. Then $\mu(\{(E,q) \in \mathsf V : E^2 - q^2 < \kappa^2\}) = 0$.

No invariance under W_3, under a reflection or under a rotation is supposed. ■

Proof sketch. (a) is the argument of Prop. YM3_I:94.4, applied at the mirrors with the normals u and $(u+u')/\sqrt2$; the labels ride along, the one number $\sigma(\theta_u \bar F \cdot \tau_w F)$ being read in the two pictures. (b) By the symmetry of σ (the entries commute) one and the same number $\sigma(A \cdot \tau_x B)$ is a matrix element in every picture whose mirror separates A from $\tau_x B$, also when x has gone around A. After a cut $|q \cdot u''| \ge \kappa$ in the momentum normal to the plane of the eight normals n_β, which is common to the eight pictures, this gives eight complex measures m_β, $\beta \in \frac\pi4\mathbb Z$, carried by $\mathsf V$ by (a) in each picture, whose Laplace–Fourier transforms $\int \exp(2RE - \rho[E\cos(c-\beta) - iq\sin(c-\beta)])\,\mathrm dm_\beta$

agree wherever two of them are defined, by the uniqueness of the Fourier transform in the common momentum; the one of the picture u is positive after the weight e^{2RE}, and by (a) at the fifth mirror the one of the picture n_{β_1} is carried by $\{E \geq |q \cdot u''|\} \subset \{E \geq \kappa\}$.

Clause (c) gives $\mu(\{E^2 - q^2 < \kappa^2\}) = 0$ for every $\kappa > 0$, and so the circular cone in each picture. (c) Write $c = t + i\gamma$. On V, $E\cosh\gamma + q\sinh\gamma \geq Ee^{-|\gamma|}$; so on $S_\beta(\rho) = \{\rho\cos(t-\beta)e^{-|\gamma|} > 2R\}$ the integrand of f_β is bounded by 1, and $f_\beta(\cdot;\rho)$ is holomorphic there (Morera), bounded by C and of the period 2π. The sets $S_\beta(\rho)$ cover the band $\{|\gamma| < \Gamma(\rho)\}$, $\Gamma(\rho) = \log(\rho\cos\frac{\pi}{8}/2R)$, and two of the functions agree on real intervals of the overlaps, hence on the overlaps (identity theorem): they glue to one holomorphic function $f(\cdot;\rho)$ on the band, $|f| \leq C$.

Near β_1, $|f(c;\rho)| \leq Ce^{-\kappa\rho_\delta}$ for $|c - \beta_1| \leq \delta$, $\rho_\delta = \rho e^{-\delta}\cos\delta - 2R$, by the carrier of m_{β_1}. With $N(c) = \log|2\sin((c-\beta_1)/2)|$, harmonic off $\beta_1 + 2\pi\mathbb{Z}$, the function $\log|f| - \log C + \kappa\rho_\delta h$, $h = (\mathsf{M} - N)/(\mathsf{M} + \log(2/\delta))$, $\mathsf{M} = \log(2\sinh(\Gamma'/2))$, $\Gamma' = \Gamma(\rho) - 1$, is subharmonic on the band $\{|\gamma| < \Gamma'\}$ without the discs $|c - \beta_1 - 2\pi k| \leq \delta$ and ≤ 0 on its boundary; by the maximum principle $|f| \leq Ce^{-\kappa\rho_\delta h}$ there. At the points $i\gamma_0$, $f(i\gamma_0;\rho) = \int \exp(-\rho(E\cosh\gamma_0 + q\sinh\gamma_0))\,\mathrm{d}\mu$ is the Laplace transform of the positive measure μ, and $h(i\gamma_0) \to 1$ as $\rho \to \infty$. For fixed $\gamma_0 \neq 0$, choose $\delta > 0$ small enough that $i\gamma_0$ lies outside the excluded discs. First letting $\rho \to \infty$ and then $\delta \downarrow 0$ gives $E\cosh\gamma_0 + q\sinh\gamma_0 \geq \kappa$ almost everywhere. Apply this first on a countable dense subset of $\mathbb{R} \setminus \{0\}$ and use continuity in γ_0; the inequality then holds for every real γ_0 on one set of full measure, and the infimum over γ_0 of the left member is 0 if $E = |q|$ and $(E^2 - q^2)^{1/2}$ if $E > |q|$. □

Lemma 3.5 (Conservation of kernels and of operators; for data)**.** Let σ' be scalar reconstruction data with a unique vacuum on $\mathcal{H}$, with a tensor of one label satisfying (M1), (M2), forms $\mathfrak{q}_{ij}$ and kernels s_{ij}.

(a) A subspace $\mathcal{V} \subset \mathcal{D}_\infty$, dense in $\mathcal{H}$ and invariant under $e^{-\tau H}$, $\tau > 0$, is a core of $(1+H)^N$ for every N.

(b) $\mathcal{V}_1$, the span of the $e^{-\tau H}[F']$, and, if $\mathcal{H} \neq \mathbb{C}\Omega$, $\mathcal{V}_2$, the span of the $\mathcal{U}(v)\mathsf{B}_0(s,s')[F]$, B_0 the scalar point operators, are such subspaces.

(c) If $\sum_i \int \partial_i\zeta(x)\,\mathsf{s}_{ij}(x;y)\,\mathrm{d}x = 0$ for all $m \geq 1$, $y \in (\mathbb{R}^3)^m_{\neq}$ and $\zeta \in C^\infty_c$ off the y_k, then the form $CL_j(\psi',\psi) = \mathfrak{q}_{3j}(H\psi',\psi) - \mathfrak{q}_{3j}(\psi',H\psi) - i\sum_k[\mathfrak{q}_{kj}(P_k\psi',\psi) - \mathfrak{q}_{kj}(\psi',P_k\psi)]$, $P_k = \int q_k\,\mathrm{d}\mathcal{P}$, vanishes on $\mathcal{D}_\infty \times \mathcal{D}_\infty$, and $T_{0\nu}(\partial_0 F)\psi = \sum_{k=1,2} T_{k\nu}(\partial_k F)\psi$ on $\mathcal{D}_\infty$. Conversely (M4) (a) gives the identity of the kernels.

■

Proof sketch. (a) A vector orthogonal to $\mathcal{V}$ in the graph norm of $(1+H)^N$ gives a finite measure $(1+E)^N\mu$ that annihilates all the functions $e^{-\tau E}$, hence vanishes by the Stone–Weierstrass theorem. (b) The closure of $\mathcal{V}_2$ reduces H and $\mathcal{U}$ and contains $\phi(f)\mathcal{D}_\infty$, hence all the $\phi(f_1)\cdots\phi(f_n)\Omega$; if it were proper, its orthogonal complement would hold an invariant vector, hence Ω by the unique vacuum, and $\phi(f)\Omega = 0$ would follow. (c) Let $a > 0$, let F', G be scalar tuples over $\mathbb{E}_+$ with smooth entries, $F'(z')$, $G(z)$ the products of their entries at the points z', z, and $P_k = \int q_k\,\mathrm{d}\mathcal{P}$, so that $\mathcal{U}(v) = e^{iv\cdot P}$. For $x \neq y$ in $\{|x_3| < a\}$ with $x_3 < y_3$ put

$$\mathsf{s}_{ij}^{F',G}(x,y) = \int \overline{F'(z')}\,G(z)\,\mathsf{s}_{ij}\big(x;y,\theta(z'+ae_3),z+ae_3\big)\,\mathrm{d}z'\,\mathrm{d}z;$$

the operator formula of (M2), the product of the point operator of $\mathfrak{q}_{ij}$ at x and of the scalar point operator at y between $[F']$ and $[G]$, reads $\mathsf{s}_{ij}^{F',G}(x,y) = \mathfrak{q}_{ij}(\psi'_x,\psi_x)$ with $\psi'_x = \mathcal{U}(-\vec{x})e^{-(a+x_3)H}[F']$ and $\psi_x = \mathcal{U}(-\vec{x})e^{-(h_*-x_3)H}\mathcal{U}(\vec{y})\mathsf{B}_0(y_3-h_*,a-y_3)\mathcal{U}(-\vec{y})[G]$, for any $h_* \in (x_3,y_3)$. As $\partial_{x_3}\psi'_x = -H\psi'_x$, $\partial_{x_3}\psi_x = H\psi_x$, $\partial_{x_k}\psi'_x = -iP_k\psi'_x$ and $\partial_{x_k}\psi_x = -iP_k\psi_x$ for $k = 1,2$, one has $\sum_i \partial_{x_i}\mathsf{s}_{ij}^{F',G}(x,y) = -CL_j(\psi'_x,\psi_x)$, and $|CL_j(\psi',\psi)| \leq 6c_T\|(1+H)^{l+1}\psi'\|\,\|(1+H)^{l+1}\psi\|$.

Given $\tau, s, s' > 0$, $v \in \mathbb{R}^2$ and F, the choice $a = \frac{1}{2}(\tau + s + s')$, $x = (0, \tau - a)$, $y = (v, a - s')$, $G = \tau_{(v,0)}F$ gives $\psi'_x = e^{-\tau H}[F']$ and $\psi_x = \mathcal{U}(v)\mathsf{B}_0(s,s')[F]$. The identity of the kernels supposed in

(c), integrated against $\overline{F'}G$, kills the divergence near x; so $CL_j = 0$ on $\mathcal{V}_1 \times \mathcal{V}_2$, and on $\mathcal{D}_\infty \times \mathcal{D}_\infty$ by (a), (b) and the bound. Read at $U(b)^{-1}\psi'$, $U(b)^{-1}\psi$, $b \in \mathbb{R}^{1+2}$, multiplied by the test function of the operator identity and integrated by parts in b, this is that identity. Conversely, $\mathcal{D}_0$ is invariant under the operators $U(b)$, so it is a core of every $(1+H)^N$, as in (a) with the group in the place of the semigroup, the finite measure $(1+E)^N\mu$ having a vanishing Fourier transform; the bounds $\|T_{\mu\nu}(F_1)\psi\| \le C\|(1+H)^{2l}\psi\|$ of the operators of (M1) carry (M4) (a) from $\mathcal{D}_0$ to $\mathcal{D}_\infty$, and, the test function being arbitrary, $CL_j = 0$ on $\mathcal{D}_\infty \times \mathcal{D}_\infty$.

At a configuration $(x, y_1, \dots, y_m)$ with pairwise different heights take a slab that holds x and the point next to it in height and no other point; the other points are smeared by bumps into G above and, through their mirror images, into F' below; the divergence of $\mathsf{s}_{ij}^{F',G}$ is $-CL_j(\psi'_x, \psi_x) = 0$, and approximate units for the bumps give $\sum_i \int \partial_i\zeta(x)\,\mathsf{s}_{ij}(x;y)\,\mathrm{d}x = 0$ for ζ carried near x. A rotation that makes the heights pairwise different reaches every configuration, by the covariance of the kernels in (M2), and a partition of unity every ζ. □

Lemma 3.6 (Extension over the diagonals; contact terms)**.** Let $\mathbb{M}$ be $\mathbb{R}^3$ or a torus T_ℓ, $n \ge 2$, and V a real vector space of finite dimension with a norm. A *functional of n entries* is a map u on the tuples $(f_1, \dots, f_n)$ of functions $f_a \in C_c^\infty(\mathbb{M}; V_\mathbb{C})$ with pairwise disjoint supports that is n-linear on $\prod_a C_c^\infty(O_a; V_\mathbb{C})$ for all open sets O_a with pairwise disjoint closures; the entries may as well take their values in n different spaces $V_1, \dots, V_n$, with $\bigotimes_a V_{a,\mathbb{C}}$ in the place of $V_\mathbb{C}^{\otimes n}$ below, and nothing else changes (this is used in (e)). We write $\|G\|_{(h),r} = \max_{|\gamma|\le r} h^{|\gamma|} \sup|\partial^\gamma G|$, on $\mathbb{M}$ and on $\mathbb{M}^n$; $\mathsf{d}_\infty(x) = \min_{a\ne b}|x_a - x_b|_\infty$ for $x \in \mathbb{M}^n$; and $\Delta = \{\mathsf{d}_\infty = 0\}$, the union of the diagonals.

(1) *Strata.* For a partition π of $\{1, \dots, n\}$ with $k = |\pi|$ blocks, $B(a)$ the block of a and $a_B = \min B$: $\Delta_\pi = \{x : x_a = x_b \text{ whenever } B(a) = B(b)\}$, and $\Sigma_k = \bigcup_{|\pi|\le k} \Delta_\pi$, so that $\Sigma_{n-1} = \Delta$. The coordinates at π are $y_B = x_{a_B} \in \mathbb{M}$ and $\xi_a = x_a - x_{a_{B(a)}}$ for $a \ne a_{B(a)}$; $\iota_\pi(y)_a = y_{B(a)}$, so that $\Delta_\pi = \{\xi = 0\} = \iota_\pi(\mathbb{M}^k)$. For a distribution T' on $\mathbb{M}^k$ and a multi-index α of ξ, $(\partial_\xi^\alpha \delta_\pi \otimes T')(F) = (-1)^{|\alpha|}\, T'((\partial_\xi^\alpha F) \circ \iota_\pi)$.

(2) *The bound* $(\mathrm{W}; \kappa, q, r)$, $\kappa \in \mathbb{R}$, $q \ge 0$, $r \in \mathbb{N}_0$, with numbers $A, h_0 > 0$ (on T_ℓ: $h_0 \in \ell 2^{-\mathbb{N}}$, $h_0 \le \ell/(64n)$): for every $0 < h \le h_0$, all closed cubes $Q_1^*, \dots, Q_n^* \subset \mathbb{M}$ of the side $\frac{3}{2}h$, wherever they lie, at pairwise sup distances $\ge \frac{5}{2}h$, this lower bound being the only condition on their distances, and all $f_a \in C_c^\infty(\operatorname{int} Q_a^*; V_\mathbb{C})$,

$$|u(f_1, \dots, f_n)| \le A\, h^\kappa\, l(h)^q \prod_a \|f_a\|_{(h),r}, \qquad l(h) = 1 + \log(h_0/h).$$

(3) *The class.* A function F on $\mathbb{M}^n$ is *λ-supported* if $\operatorname{supp} F \subset \prod_a \{|x_a - z_a|_\infty \le \lambda\}$ for some $z \in \mathbb{M}^n$, the z_a coinciding or not. $\mathfrak{E}_n(\kappa)$ is the set of the $(V_\mathbb{C}^{\otimes n})^*$-valued distributions T on $\mathbb{M}^n$ for which there are C, q, m, h_1 with $|T(F)| \le C\lambda^\kappa(1 + \log\frac{h_1}{\lambda})^q \|F\|_{(\lambda),m}$ for all $0 < \lambda \le h_1$ and all λ-supported F: the *power counting* κ, up to logarithms, also where the points meet. On $\mathbb{M} = T_\ell$ the norms $\|\cdot\|_{(\lambda),m}$, the λ-supported functions, the classes $\mathfrak{E}_n(\kappa)$, the coordinates at π and the distributions $\partial_\xi^\alpha \delta_\pi \otimes T'$ of (1) are those of [7, Def. 2.4]. We put $\omega_k = \lfloor 3k - \kappa \rfloor$.

(4) *Whitney products.* A product $Q = \prod_a Q_a$ of half-open cubes $Q_a \subset \mathbb{M}$ of one side $h(Q) \le h_0$ is *admissible* if $\operatorname{dist}_\infty(Q_a, Q_b) \ge 3h(Q)$ for $a \ne b$; $c(Q)$ is its centre, and Q^+, Q^* are the closed concentric products of the sides $\frac{5}{4}h(Q)$ and $\frac{3}{2}h(Q)$, so that the cubes Q_a^* are at pairwise sup distances $\ge \frac{5}{2}h(Q)$. Let $h_j = 2^{-j}h_0$, $j \ge 0$, and let $\mathcal{Q}$ be the set of the admissible products of dyadic cubes of the sides h_j that lie in no larger admissible product of dyadic cubes. For $Q \in \mathcal{Q}$, $\pi(Q)$ is the partition of $\{1, \dots, n\}$ into the connected components of the graph $a \sim b \iff |c(Q)_a - c(Q)_b|_\infty \le 12h(Q)$, and $k(Q) = |\pi(Q)|$.

Then $\mathcal{Q}$ is a partition of $\mathbb{M}^n \setminus \Delta$; on Q, $\mathsf{d}_\infty \ge 3h(Q)$, and if $h(Q) < h_0$, $\mathsf{d}_\infty < 10h(Q)$ and $k(Q) \le n - 1$; there are $\varphi_Q \in C_c^\infty(\operatorname{int} Q^+)$, $\varphi_Q \ge 0$, with $\sum_{Q\in\mathcal{Q}} \varphi_Q = 1$ on $\mathbb{M}^n \setminus \Delta$

and $|\partial^\gamma \varphi_Q| \le c_\gamma h(Q)^{-|\gamma|}$. Fix a smooth μ on $\mathbb{R}^3$ carried by $\{|y|_\infty < 1\}$ with $\int \mu = 1$. For $Q \in \mathcal{Q}$ with $h = h(Q)$, $\pi = \pi(Q)$, $k = k(Q) \le n-1$ and $\omega_k \ge 0$ put $\mu_Q(y') = \prod_{B\in\pi} h^{-3}\mu((y'_B - c(Q)_{a_B})/h)$ for $y' \in \mathbb{M}^k$ and

$$P_Q F(x) = \int \mu_Q(y') \sum_{|\gamma|\le\omega_k} \frac{(x - \iota_\pi(y'))^\gamma}{\gamma!}\, \partial^\gamma F(\iota_\pi(y'))\,\mathrm{d}y',$$

the Taylor polynomial of F of total order ω_k averaged over the piece of the size h of the stratum Δ_π next to Q; $P_Q = 0$ for the other Q.

Let u be a functional of n entries with $(\mathrm{W};\kappa,q,r)$, and put $N_0 = \lfloor \frac{3n}{2} \rfloor + 1 + nr$ and $m = \max\{\omega_{n-1}+1, N_0\}$.

(a) For every admissible product Q, dyadic or not, in particular for every $Q \in \mathcal{Q}$, there is exactly one linear functional u_Q on $C_c^\infty(\operatorname{int} Q^+; V_{\mathbb{C}}^{\otimes n})$ with $u_Q(g_1\otimes\cdots\otimes g_n) = u(g_1,\dots,g_n)$ and

$$|u_Q(G)| \le C_{\mathrm{F}}\, A\, h(Q)^\kappa\, l(h(Q))^q\, \|G\|_{(h(Q)),N_0},$$

C_{F} formed from n, r and the norm of V; it reads (W) at the side $h(Q)$ alone, and two of these functionals agree where both are defined. There is exactly one distribution u° on $\mathbb{M}^n \setminus \Delta$ with $u^\circ(f_1\otimes\cdots\otimes f_n) = u(f_1,\dots,f_n)$ for the tuples of the definition, and $u^\circ = u_Q$ on $C_c^\infty(\operatorname{int} Q^+)$: off the diagonals u is given by exactly one distribution.

(b) The sum

$$\tilde u(F) = \sum_{Q\in\mathcal{Q}} u_Q(\varphi_Q(F - P_Q F))$$

converges absolutely for $F \in C_c^m(\mathbb{M}^n; V_{\mathbb{C}}^{\otimes n})$, and $\tilde u$ is a distribution of order at most m that extends u°. At $\kappa = 0$, $r = 0$: $\omega_k = 3k$ and $m = 3n-2$.

(c) For $0 < \lambda \le h_0$ and every λ-supported F,

$$|\tilde u(F)| \le C_{\mathrm{ext}}\, A\, \lambda^\kappa\, l(\lambda)^{q+\epsilon}\, \|F\|_{(\lambda),m},$$

where $\epsilon = 1$ if $3k-\kappa$ is an integer ≥ 0 for some $k \in \{1,\dots,n-1\}$, $\epsilon = 0$ otherwise, and C_{ext} is formed from n,κ,q,r, the norm of V and μ, and does not depend on u; so $\tilde u \in \mathfrak{E}_n(\kappa)$. *(A range of sides.)* If u has the inequality of $(\mathrm{W};\kappa,q,r)$ at the sides $h_\flat \le h \le h_0$ alone, for some $h_\flat \in (0,h_0]$, then the sum $\tilde u^\flat(F)$ over the $Q \in \mathcal{Q}$ with $h(Q) \ge h_\flat$ has the same bound, with the same C_{ext}, for $8h_\flat \le \lambda \le h_0$ and every λ-supported F.

(d) If u is invariant under a group generated by permutations of the entries, translations of $\mathbb{M}$ acting on all the entries at once, and a closed subgroup of $\mathrm{O}(3)$ that maps $\mathbb{M}$ onto itself, acting by $f \mapsto \rho(\mathsf{R})f(\mathsf{R}^{-1}\cdot)$, then u° has an extension in $\mathfrak{E}_n(\kappa)$ of order at most m that is invariant under this group.

(e) *(Contact terms.)* A distribution T on $\mathbb{M}^n$ with support in Δ lies in $\mathfrak{E}_n(\kappa)$ if and only if

$$T = \sum_{|\pi|\le n-1} \sum_{|\alpha|\le\omega_{|\pi|}} \partial_\xi^\alpha \delta_\pi \otimes T_{\pi,\alpha} \qquad \text{with} \quad T_{\pi,\alpha} \in \mathfrak{E}_{|\pi|}(\kappa + |\alpha|),$$

the sum over the partitions π with at most $n-1$ blocks and the multi-indices α of ξ at π. So two extensions of u° in $\mathfrak{E}_n(\kappa)$ differ by such a sum, and every such sum may be added to an extension. A term with $|\alpha| > \omega_{|\pi|}$ and a coefficient in $\mathfrak{E}_{|\pi|}(\kappa+|\alpha|)$ vanishes, and for every π and every α with $|\alpha| \le \omega_{|\pi|}$ there are nonzero examples when $V \ne \{0\}$. The coefficients in this representation need not be unique. If $\kappa > 3(n-1)$, the extension in $\mathfrak{E}_n(\kappa)$ is unique.

(f) *(Two points.)* Let $n = 2$, $\kappa = 0$, $V = \mathrm{Sym}_3(\mathbb{R})$ with $\rho(\mathsf{R})H = \mathsf{R}H\mathsf{R}^{\mathrm{T}}$, and let $T \in \mathfrak{E}_2(0)$ have its support in $\{x_1 = x_2\}$, be invariant under the translations, real, and symmetric under the exchange of the two entries. Then $T(f_1\otimes f_2) = \sum_{|\alpha|\le 3} \int \mathsf{c}_\alpha(\partial^\alpha f_1(x), f_2(x))\,\mathrm{d}x$

with bilinear forms c_α on $V_{\mathbb{C}}$. If $\mathbb{M} = \mathbb{R}^3$ and T is covariant under $\mathrm{O}(3)$, the symbol $\hat T(p)(H,H') = \sum_\alpha \mathsf{c}_\alpha(H,H')(ip)^\alpha$ is a real combination of the six forms

$$\operatorname{tr} H \operatorname{tr} H', \quad \operatorname{tr} HH', \quad p^2 \operatorname{tr} H \operatorname{tr} H', \quad p^2 \operatorname{tr} HH', \quad (p\cdot Hp)\operatorname{tr} H' + (p\cdot H'p)\operatorname{tr} H, \quad (Hp)\cdot(H'p);$$

if T is covariant under the cubic group W_3, on $\mathbb{R}^3$ or on T_ℓ, then $\hat T$ is a real combination of twelve forms.

■

Proof sketch. (4) The facts on $\mathcal{Q}$: if Q is admissible, so is the product of the next generation that holds a given point; $|x_a - x_b|_\infty \ge \mathsf{d}_\infty(x) - 2h_j \ge 3h_j$ once $5h_j \le \mathsf{d}_\infty(x)$, so every $x \notin \Delta$ lies in exactly one member of $\mathcal{Q}$; if $h(Q) < h_0$ the parent of Q is not admissible, whence $\mathsf{d}_\infty < 10h(Q)$ on Q and two points in one block of $\pi(Q)$ are linked by a chain of centres at most $12h(Q)$ apart. Members of $\mathcal{Q}$ whose sets Q^+ meet have sides in the ratio $\frac12$, 1 or 2, and a point lies in at most $3\cdot 2^{3n}$ sets Q^+; φ_Q is a normalised product of cutoffs of the scale $h(Q)$. A count used below: at most $\mathrm{Bl}(n)(24n)^{3n}(3R/h_j)^{3k}$ members of $\mathcal{Q}$ have the side h_j, k clusters and all the coordinates of their centres within $R \ge h_j$ of those of a given point of $\mathbb{M}^n$, a partition, one cube for each block and the cubes of the other indices within $12(n-1)h_j$ of it being chosen.

(a) Let $h = h(Q)$, $\mathsf{s} = \frac32 h$, $c = c(Q)$. A $G \in C^\infty_c(\operatorname{int} Q^+; V_{\mathbb{C}}^{\otimes n})$, continued periodically with the cell Q^*, is the sum of its Fourier series in the modes $\mathsf{E}_{\mathbf{k}}(x) = \prod_a e^{2\pi i \mathbf{k}_a\cdot(x_a - c_a)/\mathsf{s}}$, $\mathbf{k} \in \mathbb{Z}^{3n}$, with all derivatives, uniformly; with cutoffs $\chi_a \in C^\infty_c(\operatorname{int} Q^*_a)$, equal to 1 on Q^+_a, fixed once at the scale h, $\|\chi_a e^{2\pi i \mathbf{k}_a\cdot(\cdot - c_a)/\mathsf{s}}\|_{(h),r} \le C_r(1+|\mathbf{k}_a|)^r$, and every mode times $\prod_a \chi_a$ is a product of entries carried by the cubes Q^*_a, of the side $\frac32 h$ and at pairwise sup distances $\ge \frac52 h$.

Put $u_Q(G) = \sum_{\mathbf{k}} \hat G_{\mathbf{k}}\, u(\chi_1 e^{2\pi i \mathbf{k}_1\cdot(\cdot - c_1)/\mathsf{s}}, \dots, \chi_n e^{2\pi i \mathbf{k}_n\cdot(\cdot - c_n)/\mathsf{s}})$, componentwise in a basis of V. By (W) the terms are at most $Ah^\kappa l(h)^q C_r^n (1+|\mathbf{k}|)^{nr}|\hat G_{\mathbf{k}}|$; by the Cauchy–Schwarz inequality, with $\sum_{\mathbf{k}}(1+|\mathbf{k}|)^{-2s} < \infty$ for $2s > 3n$, and by Parseval, $\sum_{\mathbf{k}}(1+|\mathbf{k}|)^{nr}|\hat G_{\mathbf{k}}| \le C\|G\|_{(h),N_0}$: this is the origin of $N_0 = \lfloor \frac{3n}{2} \rfloor + 1 + nr$. At a product $G = g_1 \otimes \cdots \otimes g_n$ the coefficients are products, the series of the entries converge in $\|\cdot\|_{(h),r}$, and $u_Q(G) = u(g_1, \dots, g_n)$.

A function carried by a product of open sets O_a is a limit in every C^N, within one compact subset of $\prod_a O_a$, of finite sums of products of functions carried by the O_a (truncated Fourier series times cutoffs); so two functionals that are bounded by a C^N norm and equal to u on products agree, which gives the uniqueness of u_Q and the agreement of u_Q, $u_{Q'}$ on $\operatorname{int} Q^+ \cap \operatorname{int} Q'^+$. Then $u^\circ(G) = \sum_Q u_Q(\varphi_Q G)$ is a finite sum on every compact subset of $\mathbb{M}^n \setminus \Delta$, a distribution; on a tuple with disjoint supports, a partition of unity of $\mathbb{M}$ by dyadic cubes of a side h_j with $6h_j$ below the least distance of the supports and the multilinearity of u give $u^\circ = u$, the products of the cubes that meet the supports being admissible; and u° is unique by the same density.

(b) Four facts on P_Q, at $Q \in \mathcal{Q}$ with $h = h(Q)$, $\pi = \pi(Q)$, $k = k(Q) \le n-1$, $\omega = \omega_k \ge 0$. (P1) For $x \in Q^+$ and y' in the support of μ_Q, $|x_a - y'_{B(a)}|_\infty \le \frac58 h + 12(n-1)h + h \le 12nh$. (P2) $P_Q F = 0$ if F vanishes near Δ_π: it reads the derivatives of F on Δ_π alone. (P3) For $h \le \lambda$ and $|\gamma| \le N_0$, on Q^+, $h^{|\gamma|}|\partial^\gamma(F - P_Q F)| \le C(h/\lambda)^{\omega+1}\|F\|_{(\lambda),m}$: since $\int \mu_Q = 1$ and ∂^γ of the Taylor polynomial of order ω is the Taylor polynomial of order $\omega - |\gamma|$ of $\partial^\gamma F$, Taylor's formula with (P1) gives $(36n^2h)^{\omega+1-|\gamma|}\lambda^{-\omega-1}\|F\|_{(\lambda),\omega+1}$ for $|\gamma| \le \omega$, and for $|\gamma| > \omega$, $\partial^\gamma P_Q F = 0$. (P4) If F is λ-supported at z and $\lambda \le h$, then, in the coordinates at π and after an integration by parts in y', $h^{|\gamma|}|\partial^\gamma P_Q F| \le C\sum_{j\le\omega}(h/\lambda)^{j-3k}\|F\|_{(\lambda),\omega}$ on Q^+, because μ_Q and its derivatives are of the size h^{-3k} and $(\partial^\alpha_\xi F)\circ\iota_\pi$, of the size $\lambda^{-|\alpha|}$, is carried by a set of volume $\le (2\lambda)^{3k}$; and $P_Q F = 0$ unless $|c(Q)_a - z_a|_\infty \le 12nh$ for all a.

Now let $F \in C^m_c$ be carried by a ball of radius $R_F \ge h_0$. By (P3) at $\lambda = h_0$, (a), the count of (4) and $\|\varphi_Q G\|_{(h),N_0} \le C\|G\|_{(h),N_0}$ on Q^+, the terms of the members of the side h_j with k clusters add up to at most $C\,A\,\|F\|_{(h_0),m}((R_F + 12nh_0)/h_j)^{3k} h_j^\kappa l(h_j)^q (h_j/h_0)^{(\omega_k+1)_+}$; the exponent of h_j is

$\varepsilon_k = \kappa - 3k + (\omega_k + 1)_+ > 0$, equal to $\lfloor 3k - \kappa \rfloor + 1 - (3k - \kappa) \in (0, 1]$ if $\omega_k \geq 0$ and to $\kappa - 3k$ if $\omega_k < 0$; for $k = n$ only $h_j = h_0$ occurs. So the series converges absolutely, $\tilde{u}$ is bounded by a C^m norm on every compact set, and by (P2) $\tilde{u} = u^\circ$ off Δ. At $\kappa = 0$, $r = 0$: $\omega_{n-1} + 1 = 3n - 2 \geq \lfloor \frac{3n}{2} \rfloor + 1$.

(c) Let F be λ-supported at z. Only finitely many members with $h(Q) > \lambda$ give a term, and

$$\tilde{u}(F) = \underbrace{\sum_{h(Q) \leq \lambda} u_Q(\varphi_Q(F - P_Q F))}_{\text{I}} + \underbrace{u^\circ\Big(\sum_{h(Q) > \lambda} \varphi_Q F\Big)}_{\text{II}} - \underbrace{\sum_{h(Q) > \lambda} u_Q(\varphi_Q P_Q F)}_{\text{III}}.$$

In I the members that give a term have their centres within $13n\lambda$ of z, and (P3) at the scale λ with the count gives $C\,A\,\lambda^\kappa \|F\|_{(\lambda),m} \sum_k \sum_{h_j \leq \lambda} (h_j/\lambda)^{\varepsilon_k} l(h_j)^q$, which is at most $C\,A\,\lambda^\kappa l(\lambda)^q \|F\|_{(\lambda),m}$, as $l(2^{-i}\lambda) \leq 2(1+i) l(\lambda)$. In II the function is smooth, carried where $\mathsf{d}_\infty \geq \frac{11}{4}\lambda$, with $\|\cdot\|_{(\lambda),N_0} \leq C\|F\|_{(\lambda),N_0}$; a partition of unity by all the products of dyadic cubes of one side $h' \in (\frac{\lambda}{8}, \frac{\lambda}{4}]$ meets its support only in admissible products, at most 19^{3n} of them, and (a) bounds each term by $C\,A\,\lambda^\kappa l(\lambda)^q \|F\|_{(\lambda),N_0}$.

In III, by (P4), at most $(24n+1)^{3n}$ members of each side $h_j > \lambda$ give a term, each at most $C\,A\,\lambda^\kappa l(\lambda)^q (h_j/\lambda)^{\kappa + \omega_k - 3k} \|F\|_{(\lambda),\omega_k}$; the exponent $\kappa + \omega_k - 3k$ is ≤ 0, and it is 0 exactly when $3k - \kappa$ is an integer; so the sum over $\lambda < h_j \leq h_0$ is a convergent geometric series or has at most $1 + \log_2(h_0/\lambda) \leq 2\,l(\lambda)$ equal terms: this is the one logarithm more. *A range of sides:* if (W) holds only at the sides $\geq h_\flat$ and $8h_\flat \leq \lambda$, I loses its members below $h_\flat$ and was bounded term by term; III is unchanged; and II, now $\sum_{h(Q)>\lambda} u_Q(\varphi_Q F)$, is the sum over the products of the side $h' > \frac{\lambda}{8} \geq h_\flat$ of the partition of II, by the agreement in (a) at sides $\geq h_\flat$, and is bounded in the same way.

(d) The grids, so $\mathcal{Q}$ and $\pi(Q)$, and with covariant choices φ_Q, μ_Q, are invariant under the translations by $h_0\mathbb{Z}^3$, so $\tilde{u}$ is; the mean of $\tilde{u}(\tau_t\,\cdot)$ over $t \in [0, h_0)^3$ is invariant under all the translations, and the mean over the permutations and over the Haar measure of the closed subgroup of O(3) keeps this invariance. Each of these maps carries λ-supported functions to $\sqrt{3}\lambda$-supported ones and changes $\|\cdot\|_{(\lambda),m}$ by a factor formed from n and m; so it maps $\mathfrak{E}_n(\kappa)$ to itself with uniform numbers and extensions of the invariant u° to extensions, and means of extensions with one bound are such extensions.

(e) *If:* for λ-supported F the function $g = (\partial_\xi^\alpha F) \circ \iota_\pi$ on $\mathbb{M}^k$ is λ-supported, and $\|g\|_{(\lambda),m'} \leq n^{m'} \lambda^{-|\alpha|} \|F\|_{(\lambda),m'+|\alpha|}$, as $\partial_{y_B} = \sum_{a \in B} \partial_{x_a}$; so $|T_{\pi,\alpha}(g)| \leq C\lambda^{\kappa+|\alpha|}(1 + \log\frac{h_1}{\lambda})^q\, n^{m'} \lambda^{-|\alpha|} \|F\|_{(\lambda),m'+|\alpha|}$, the power counting κ. A distribution $T' \in \mathfrak{E}_k(\kappa')$ with $\kappa' > 3k$ is 0: a test function on $\mathbb{M}^k$ is the sum of at most $C\lambda^{-3k}$ pieces carried by balls of radius λ, each of value $\leq C\lambda^{\kappa'}$ up to logarithms, and $\lambda \to 0$; this is the case $|\alpha| > \omega_k$, where $\kappa + |\alpha| > 3k$.

The Lebesgue measure of $\mathbb{M}^k$ times a fixed nonzero element of the dual lies in $\mathfrak{E}_k(\kappa')$ exactly when $\kappa' \leq 3k$, that is, at $\kappa' = \kappa + |\alpha|$, exactly when $|\alpha| \leq \omega_k$; this supplies a nonzero example for every allowed pair (π, α) when $V \neq \{0\}$. *Only if:* a descent over $k = n-1, \dots, 1$ for $T \in \mathfrak{E}_n(\kappa)$, with its numbers C, q, m, h_1, carried by Σ_k. Off Σ_{k-1} the strata Δ_π with $|\pi| = k$ are disjoint; in the tube $U_\pi = \{|\xi|_\infty < \frac{1}{4}\min\{\mathsf{d}_\infty(y), h_1\}\}$ about the open stratum, which meets no other of them, and with a cutoff χ carried by $\{|\xi|_\infty < 1\}$ and equal to 1 on $\{|\xi|_\infty \leq \frac{1}{2}\}$, the coefficients

$$t_{\pi,\alpha}(g) = (-1)^{|\alpha|}\, T\Big(\chi(\xi/\varepsilon)\,\frac{\xi^\alpha}{\alpha!}\,g(y)\Big), \qquad |\alpha| \leq \omega_k, \quad 0 < \varepsilon \leq \tfrac{1}{4}\min\Big\{\min_{\operatorname{supp} g} \mathsf{d}_\infty, h_1\Big\},$$

do not depend on ε and are distributions on $\mathbb{M}^k \setminus \Delta^{(k)}$ alone, $\Delta^{(k)}$ the diagonals of $\mathbb{M}^k$. *Claim A:* if G is carried by U_π and $\partial_\xi^\alpha G(y, 0) = 0$ for $|\alpha| \leq \omega_k$, then $T(G) = 0$; for $(1 - \chi(\xi/\varepsilon))G$ vanishes near Δ_π and is carried by U_π, which meets the support of T in the open stratum alone, so that $T(G) = T(\chi(\xi/\varepsilon)G)$ for small ε, and a partition of unity at the scale ε in y cuts $\chi(\xi/\varepsilon)G$ into at most $C\varepsilon^{-3k}$ pieces, each 2ε-supported with $\|\cdot\|_{(2\varepsilon),m} \leq C\varepsilon^{(\omega_k+1)_+}$, so that $|T(G)| \leq C\varepsilon^{\kappa - 3k + (\omega_k+1)_+}$ times logarithms, which tends to 0.

Claim B: on $C_c^\infty(\mathbb{M}^n\setminus\Sigma_{k-1})$, by Taylor's formula in ξ and Claim A, $T=\sum_{|\pi|=k}\sum_{|\alpha|\le\omega_k}\partial_\xi^\alpha\delta_\pi\otimes t_{\pi,\alpha}$. The coefficients have a power counting of their own: for cubes R_B^* of the side $\frac{3}{2}h$ at pairwise sup distances $\ge\frac{5}{2}h$, $h\le\frac{1}{4}h_1$, and g carried by $\prod_B R_B^*$, the function $\chi(2\xi/h)\frac{\xi^\alpha}{\alpha!}g(y)$ is $2h$-supported with $\|\cdot\|_{(2h),m}\le Ch^{|\alpha|}\|g\|_{(h),m}$; so $t_{\pi,\alpha}$ is a functional of k entries with $(\mathrm{W};\kappa+|\alpha|,q,m)$, with derivatives of the order m in the norms, which is why (a)–(c) are stated for every r.

For $k\ge2$, (a)–(c) at $r=m$ extend $t_{\pi,\alpha}$ to $T_{\pi,\alpha}\in\mathfrak{E}_k(\kappa+|\alpha|)$; for $k=1$ there are no diagonals in $\mathbb{M}^1$ and $T_{\pi,\alpha}=t_{\pi,\alpha}$ lies in $\mathfrak{E}_1(\kappa+|\alpha|)$ by the same inequality. By Claim B, $T-\sum_{|\pi|=k}\sum_{|\alpha|\le\omega_k}\partial_\xi^\alpha\delta_\pi\otimes T_{\pi,\alpha}$ vanishes off Σ_{k-1}, and it lies in $\mathfrak{E}_n(\kappa)$ by the first part; the descent ends at $\Sigma_0=\varnothing$. If $\kappa>3(n-1)$, all ω_k with $k\le n-1$ are negative and the sum is empty.

(f) At $n=2$ the one stratum has $k=1$, $y=x_1$, $\xi=x_2-x_1$ and $\omega_1=3$; the coefficients T_α are translation-invariant distributions on $\mathbb{M}$, hence constants $\mathsf{c}_\alpha\,\mathrm{d}y$, and an integration by parts gives the form; conversely a constant lies in $\mathfrak{E}_1(\kappa')$ for $\kappa'\le3$. Covariance reads $\hat T(\mathsf{R}p)(\rho(\mathsf{R})H,\rho(\mathsf{R})H')=\hat T(p)(H,H')$, and the exchange $\hat T(-p)(H',H)=\hat T(p)(H,H')$. Since -1 lies in both groups and $\rho(-1)=1$, $\hat T$ is even in p; so only the terms of the orders 0 and 2 in p occur, with real coefficients, $\hat T(p)$ is symmetric in (H,H'), and the number of the independent terms is that of the invariants of $\mathrm{Sym}^2V^*\otimes\mathrm{Sym}^j(\mathbb{R}^3)^*$, $j=0,2$. Under $\mathrm{SO}(3)$, $V=[0]\oplus[2]$, $\mathrm{Sym}^2V=2[0]\oplus2[2]\oplus[4]$ and $\mathrm{Sym}^2\mathbb{R}^3=[0]\oplus[2]$: $2+4=6$; the six forms are invariant. Separating the homogeneous terms of degrees 0 and 2, their independence follows by evaluating at $H=H'\in\{\mathrm{diag}(1,-1,0),\mathrm{diag}(1,0,0)\}$ and $p\in\{e_1,e_2,e_3\}$ within each degree. Under the cubic group $V=A_1\oplus E\oplus T_2$, $\mathrm{Sym}^2V=3A_1\oplus3E\oplus3T_2\oplus T_1$ and $\mathrm{Sym}^2\mathbb{R}^3\cong V$: $3+9=12$; the characters give the same numbers, and 0 for the odd orders. □

Remark 3.7 (The point bound and the bounds of kernels, for data). For reconstruction data with labels the point bound gives pointwise bounds of the kernels by powers of the least distance of the points (Prop. 3.3 and Thm. E (d) (i)). The converse fails: at $V=\mathbb{R}$, $\sigma(f_1,\dots,f_n)=n!\prod_k\int f_k$ is symmetric and has constant kernels, hence every bound of the kind of (E1), but no bound of one insertion between two vectors. The reconstruction uses nothing at coinciding points.

Detailed sketch of tensor reconstruction and locality. Step 3: data (c). $(\mathrm{Rec1})_V$ is the composition of the translations ([7, Thm. C] (b)) with (a); $(\mathrm{Rec2})_V$ is [7, Thm. C] (b), the space being separable by the bounds and a countable dense family of tuples; $(\mathrm{Rec3})_V$ is [7, Thm. C] (e), or follows from (a) and $H\ge|P_1|$ by the unitary rotations about e_3, which turn $\mathcal{P}$; $(\mathrm{Rec4})_V$ is [7, Thm. C] (c). The classes of the C_c^1 tuples are dense, because a continuous entry is the uniform limit of its mollifications and σ is continuous under such limits; so the map of the two spaces is unitary. Since $g^{-2}\le b_*$, the numbers do not depend on σ:

$$C_V=2^{p_V}C_{p_V}b_*^{-1}\le2^{p_V}C_{p_V}g^2.$$

Step 4: kernels, representation, locality (d). Prop. 3.3 (a) at $V=\mathrm{Sym}_3(\mathbb{R})$ applies to σ^1. Its clause (vii) gives the kernels, extended to continuous entries by the continuity of Step 3; for the bound at every point, a direction u with $|u\cdot(x_a-x_b)|\ge|x_a-x_b|/n^2$ for all pairs exists, and the bound of $\mathbf{S}_n$ in the tube of the frame of u at $\eta=\frac{1}{2}\min\{1,\mathsf{d}(x)\}/n^2$ gives

$$|\mathsf{s}_n(x;e)|\le\big(C_V(2n^2/\min\{1,\mathsf{d}(x)\})^{p_V}\big)^n\prod_a|e_a|;$$

integrating gives the bound of σ, and $\mathsf{S}_n(\varphi)=\int\mathsf{s}_n\varphi$ with the Taylor bound of φ near the diagonals gives the growth with the norm $|\cdot|_{(\lceil p_V\rceil+4)n}$. Its clauses (i)–(iii) and (v) give the representation, with $D(\Lambda)A=\hat\Lambda A\hat\Lambda^{\mathrm{T}}$, and (iv) the locality. The fields of Prop. 3.3 are the $T(f)$ of [7, Thm. C] (d): both are determined on $\mathcal{D}_\infty$ by $\langle\psi,T(f)\psi'\rangle=\int\lim_{s\downarrow0}\langle U(x)^{-1}\psi,\mathsf{B}(s,s;f(x))U(x)^{-1}\psi'\rangle\,\mathrm{d}x$, which involves σ alone, and the totality used in Prop. 3.3 (a) (viii) is $\mathcal{H}^{\mathrm{ord}}=\mathcal{H}(\sigma)$, [7, Thm. C] (f).

Step 5: all couplings (e). Let $0<g\le\gamma$. Extract a subsequence with converging labels, $b_j\to b_*$, and let z, g_1', g_2' be as in (e); $g_2'<g_1'\le\gamma_W^{\mathrm{sd}}$, and by [7, Lem. 2.14] (b) the label L^zb_* lies in the

interior of $I(g')$ for one $g' \in \{g'_1, g'_2\}$. By the same Lemma the dilated data are, from an index on, first-passage data at g', [7, Thm. C] (b) at g' gives a further subsequence with a tensor limit σ', and

$$\sigma(X) = L^{zn}\sigma'(X^{(z)})$$

exists. The dilation commutes with O(3), carries mirrors to mirrors and $H^{\mathrm{cons}}[\eta]$ to $H^{\mathrm{cons}}[\eta^{[z]}]$; so (a), (b) and the conservation pass from σ' to σ, and by [7, Lem. 2.14] (b) $(\mathrm{Rec4})_V$ holds with $(3^{p_V}L^{4z}C_V[\sigma'], p_V, \frac{1}{2})$, where $C_V[\sigma'] \le 2^{p_V}C_{p_V}[g']\,g'^2$ by [7, Thm. C] (c) at g'; this is at most the number C_V of (e), which does not depend on the sequence. Then (c), (d) follow as in Steps 3, 4. □

Real-time conservation. The real-time conservation conclusion of [7, Thm. E(d)] follows from Lem. 3.5: the identity of the kernels gives, and is given by,

$$T_{0\nu}(\partial_0 F)\psi = \sum_{k=1,2} T_{k\nu}(\partial_k F)\psi \qquad (\psi \in \mathcal{D}_\infty).$$

□

Fields over a clustering component and their frame. Step 6: Wightman fields (c). By (b) (ii), σ^c_ξ is a tensor limit at g, so Thm. E (c), (d) apply to it with the same numbers; it has (Rec5) by (b) (iii) and the unitary map of (b) (ii). Prop. 3.3 (a) (i) with (Rec5) gives the uniqueness of the invariant vector; the Hermiticity is

$$\langle \psi, T(f)\psi'\rangle = \langle T(f^*)\psi, \psi'\rangle, \qquad f^* = \rho(\theta)\bar{f},$$

under which a component with r indices equal to 3 is Hermitian after the factor i^r; the representation, the spectrum, the locality and the cyclicity are Thm. E (d) and [7, Thm. C] (f) for σ^c_ξ. The growth is [7, Thm. C] (h), whose numbers do not depend on the limit.

Step 7: the frame over a component (d). The frame is formed at the label matrices m_{ij} of the stress tensor, the labels of $\frac{1}{2}\delta_{ij}\operatorname{tr} T - T_{ij}$; (M1), (M2), (M3), (M5) are linear in the label. (M1) and the operator formula of (M2) are [7, Thm. C] (d) (vi) for σ^c_ξ, and the covariance of the kernels is Thm. E (d) (i) with $\rho(\mathsf{R})\mathsf{m}_{ij} = \sum_{i'j'} \mathsf{R}_{i'i}\mathsf{R}_{j'j}\mathsf{m}_{i'j'}$, at $\mathsf{s}_{ij}(x;y) = \mathsf{s}_{1+m}(x,y;\mathsf{m}_{ij},\mathbf{1},\dots,\mathbf{1})$. The scalar subspace reduces H, $\mathcal{U}$, $\mathcal{P}$, and it is invariant under the families $\mathcal{T}(\mathfrak{z})$ and $\mathsf{b}(w|\mathbf{1}|w')$, which are closed under the adjoint; so its projection Π_0 commutes with $\phi(f) = T(f\mathbf{1})$ on $\mathcal{D}_\infty$, and with $T^E_{ij}(F_1) = \Pi_0 T(F_1\mathsf{m}_{ij})\Pi_0$,

$$T^E_{ij}(F_1)\phi(f)\psi = \Pi_0 T(F_1\mathsf{m}_{ij})\phi(f)\psi = \Pi_0\phi(f)T(F_1\mathsf{m}_{ij})\psi = \phi(f)T^E_{ij}(F_1)\psi$$

at spacelike separation by the locality: (M3). The scalar subspace reduces $U(a,\Lambda)$ as well: the restriction of $(\sigma^c_\xi)^1$ to the tuples with entries in V_0 is reconstruction data with the label space V_0, whose space is the scalar subspace, and by Prop. 3.3 (a) (v) the span of Ω and of the vectors $\phi(f_1)\cdots\phi(f_n)\Omega$ is dense in it; as $\hat{\Lambda}\mathbf{1}\hat{\Lambda}^{\mathrm{T}} = \hat{\Lambda}\hat{\Lambda}^{\mathrm{T}} = \mathbf{1}$, Thm. E (d) (ii) gives $U(a,\Lambda)\phi(f)U(a,\Lambda)^{-1} = \phi(f(\Lambda^{-1}(\cdot - a)))$ on $\mathcal{D}_0$, and with $U(a,\Lambda)\Omega = \Omega$ every $U(a,\Lambda)$ maps this span onto itself; U being a unitary representation, the scalar subspace and its orthogonal complement are invariant, and Π_0 commutes with every $U(a,\Lambda)$.

So U restricts to a representation on the scalar subspace, and (M5) is Π_0 applied to the Lorentz covariance on Ω of Thm. E (d) (ii), with the passage from the Euclidean labels to $T_{\mu\nu}$ of Def. 3.1 (3). (M4) (a): conservation passes to σ^c_ξ by [7, Lem. 2.24] (b), so [7, Thm. E] (c), in the second form of [7, eq. 5] with ζ vanishing near the scalar points, gives $\sum_i \int \partial_i\zeta(x)\,\mathsf{s}_{ij}(x;y)\,\mathrm{d}x = \sigma^c_\xi(H^{\mathrm{cons}}[\zeta e_j], \text{the scalar entries}) = 0$; the scalar subspace has a unique vacuum, by (b) (iii); and Lem. 3.5 gives (M4) (a). At the labels $e_ie_j^{\mathrm{T}} + e_je_i^{\mathrm{T}} = \delta_{ij}\mathbf{1} - \mathsf{m}_{ij}$ the identity of the kernels fails: among one scalar entry their divergence is $\partial_{x_j}\mathsf{s}_2(x,y;\mathbf{1},\mathbf{1})$, which does not vanish identically by the free law (E6) of [7, Thm. G]; the components T_{ij} themselves are not conserved.

Step 8: the frame on $\mathcal{H}_0$ (e). For σ itself the same compression applies without the unique vacuum: (M1) is [7, Thm. C] (d) (vi), the covariant kernels come from Thm. E (d) (i), and (M3), (M5) are the compressions of Thm. E (d) (iii), (ii), as identities between scalar vectors, for instance

$$\langle\Omega, \phi(f_1)\cdots\phi(f_n)T[\Lambda t](F\circ\Lambda^{-1})\Omega\rangle = \langle\Omega, \phi(f_1\circ\Lambda)\cdots\phi(f_n\circ\Lambda)T[t](F)\Omega\rangle.$$

□

Proof of Prop. 3.2. The argument follows the proofs of Props. YM3_I:108.3, YM3_I:110.2 and YM3_I:110.1 for scalar data, with one entry carrying a label; its steps are these.

Step 1: the scalar part and one label. Let $e' \in V \setminus V_0$. For a tuple whose entries all lie in V_0, $f_1\mathbf{1} = f_1(\mathbf{1}-e') + f_1e'$ writes it as the sum of two tuples with exactly one entry outside V_0, at which $\sigma(\mathsf{R}X) = \sigma(X)$ is supposed; so the scalar part of σ is invariant under O(3), and with the translations it is invariant under all the isometries. With [7, Thm. C] (b), (c), (e), which give the positivity, the point bound and the circular cone, the space being separable by the bounds and a countable dense family of tuples, its restriction to C^1_c entries is scalar reconstruction data (Rec); Thm. YM3_I:AP gives this as well, as in Step 6 of the proof of [7, Thm. C]. The insertion and point operators with one tensor label and their energy bounds are those of [7, Thm. C] (d), and [7, Prop. 2.18] (c) gives the kernel $\mathsf{s}_{1+m}(x,y;e,\mathbf{1},\dots,\mathbf{1})$ of one tensor entry among m scalar entries, continuous and real-analytic at all non-coinciding points; by continuity, the supposed covariance at one label is $\mathsf{s}_{1+m}(\mathsf{R}x+v,\mathsf{R}y+v;\rho(\mathsf{R})e,\mathbf{1},\dots,\mathbf{1}) = \mathsf{s}_{1+m}(x,y;e,\mathbf{1},\dots,\mathbf{1})$ for $\mathsf{R}\in\mathrm{O}(3)$, $v\in\mathbb{R}^3$.

Step 2: one holomorphic function at one label; (M2). For a frame $\mathsf{R}_u \in \mathrm{SO}(3)$ with $\mathsf{R}_u e_3 = u$ and an order ς of the $1+m$ points along u, the piece on the tube of the configurations ordered along u in the order ς is the function $\mathbf{W}_{1+m}$ of [7, Thm. C] (d) (iii) written in the frame R_u at the label $\rho(\mathsf{R}_u)^{-1}e$; in the components of the label it is

$$\sum_{i',j'} (\mathsf{R}_u)_{ii'}(\mathsf{R}_u)_{jj'}\,\mathbf{W}_{1+m}(\mathsf{R}_u^{-1}\cdot; E_{i'j'},\mathbf{1},\dots,\mathbf{1}),$$

$E_{i'j'}$ the symmetrised matrix units, that is $\rho(\mathsf{R}_u)\mathbf{W}_{1+m}(\mathsf{R}_u^{-1}\cdot)$ at the one label. At the real points of the intersection of two tubes, the configurations ordered in both frames, both pieces are equal to the kernel of Step 1, by the covariance at one label; the intersection is convex and these real points form an open subset of its real section, so the two pieces agree on it, and they form one holomorphic function on the union of the tubes. As $|(\mathsf{R}_u)_{ii'}| \le 1$, its bound is that of $\mathbf{W}_{1+m}$ times at most 9, the number of the terms (i',j'). Its restriction to the real non-coinciding configurations is the kernel; so the kernels $\mathsf{s}_{ij}(x;y) = \mathsf{s}_{1+m}(x,y;\mathsf{m}_{ij},\mathbf{1},\dots,\mathbf{1})$ are covariant at all non-coinciding points, and with the product formula of [7, Thm. C] (d) (vi) this is (M2).

Step 3: locality; (M3). The matrix elements of products of the fields between vectors of $\mathcal{D}_\infty$ are the boundary values of the one function of Step 2 from the tubes of the orders of the real times. For a pair of spacelike separated points, one of them carrying the label, the tubes of the two orders are joined by the tubes of the frames tilted by an angle of the order η towards the direction of the separation, on which the function is one and the same; apply the uniform smeared-bound argument in the proof of Prop. 3.3 (a) to the holomorphic kernel and polynomial bound of Step 2, with its single tensor label fixed and the remaining labels scalar. The Cauchy–Riemann identities and integration by parts give the same bound $2\eta Q_e(D'g)$ for the difference of the two orders, with $Q_e(D'g)$ independent of η; hence the boundary values agree as $\eta\to 0$. So $\langle\psi', [T(F_1\mathsf{m}_{ij}),\phi(f)]\psi\rangle = 0$ at spacelike separation for ψ',ψ in the span of the vectors $e^{-\tau H}[F]$, $\tau>0$, F scalar, which lie in $\mathcal{D}_\infty$, are dense in $\mathcal{H}_0$ and form a core of every power of $1+H$ there (Lem. 3.5 (a), (b), at the span $\mathcal{V}_1$); the bound of (M1) and the commutation of Π_0 with $\phi(f)$ on $\mathcal{D}_\infty$, as in Step 7 of the proof of Thm. F, carry this to $T^E_{ij}(F_1)\phi(f)\psi = \phi(f)T^E_{ij}(F_1)\psi$ on $\mathcal{D}_0$: (M3).

Step 4: Lorentz covariance; (M5). The invariance of $\mathbf{W}_{1+m}$ at one label under the real rotations R_ζ in the plane of e_1, e_3, the label being turned by $\rho(\mathsf{R}_\zeta)$, continues holomorphically in the angle ζ to the imaginary angles $\zeta = is$, which act on the arguments, through ι, as the boosts of rapidity

s in the plane of (t, x_1); the label is then moved by $\rho_{\mathbb{C}}(\mathsf{R}_{is})$, $\rho_{\mathbb{C}}$ the holomorphic continuation of $\rho|_{\mathrm{SO}(3)}$, that is by $e \mapsto \hat\Lambda e \hat\Lambda^{\mathrm{T}}$, and no bound on this factor is needed, the identity being one of holomorphic functions at fixed arguments. With the real rotations in the plane of e_1, e_2, which fix e_3, this gives the invariance of $\langle \Omega, \phi(f_1)\cdots\phi(f_m)T(F)\Omega\rangle$, the tensor entry at the last place, under $(f_k, F) \mapsto (f_k \circ \Lambda^{-1}, \hat\Lambda F(\Lambda^{-1}\cdot)\hat\Lambda^{\mathrm{T}})$ for all $\Lambda \in \mathcal{L}$.

The scalar data of Step 1 carry the representation U of $\mathbb{R}^{1+2} \rtimes \mathcal{L}$ on $\mathcal{H}_0$ with

$$U(\Lambda)\phi(f_1)\cdots\phi(f_m)\Omega = \phi(f_1 \circ \Lambda^{-1})\cdots\phi(f_m \circ \Lambda^{-1})\Omega$$

(Prop. 3.3 (a) at V_0); these vectors are dense in $\mathcal{H}_0$, and the invariance reads $U(\Lambda)\Pi_0 T(F)\Omega = \Pi_0 T(\hat\Lambda F(\Lambda^{-1}\cdot)\hat\Lambda^{\mathrm{T}})\Omega$. At the label matrices m_{ij}, with the passage from the Euclidean labels to $T_{\mu\nu}$ of Def. 3.1 (3), this is (M5). □

The next two propositions use the generic conditions (D0)–(D4) of [7, Def. 6.1, p. 185], distinct from the clauses (D1)–(D5) of theorem C. Condition (D0) is $\sigma(\emptyset) = 1$, symmetry and n-linearity on $\prod_a C^1_c(O_a; V)$ when the open sets O_a have pairwise disjoint closures, with linear extension to combinations. Condition (D1) is the bound

$$|\sigma(H_1, \dots, H_n)| \le C_n(\delta, \varrho_0) \prod_a |H_a|_\infty$$

for $n \ge 1$, supports of sup diameter at most ϱ_0 and pairwise sup distance at least 2δ, uniformly in their positions; complex entries cost a factor 2^n. Conditions (D2) and (D3) are invariance under all translations and under W_3, respectively. Condition (D4) is positivity at (n, c) for every $c \in \mathbb{R}$ and every

$$n \in \mathfrak{N}_{18} = \{\pm e_\mu : 1 \le \mu \le 3\} \cup \{(\pm e_\mu \pm e_\lambda)/\sqrt{2} : \mu < \lambda\}.$$

Here o_{12} has entries 1 at $(1,2)$ and $(2,1)$ and zero elsewhere. If P_A, P_E, P_T are the projections onto scalar, traceless diagonal and off-diagonal matrices, then $Z_z = P_A + P_E + zP_T$; it multiplies just the off-diagonal labels by z.

Proposition 3.8 (Wightman fields from labelled functionals)**.** Let σ be a real functional on $\mathfrak{S}_V(\mathbb{R}^3)$ with (D0), (D1) and $\sigma((H)) = 0$, and suppose:

- (R1) $\sigma \circ \iota = \sigma$ for every isometry ι of $\mathbb{R}^3$;
- (R2) σ is positive at $(e_3, 0)$, and $\mathcal{H}(\sigma)$ is separable;
- (R3) $\mathcal{P}(\{E < |q|\}) = 0$;
- (R4) there are C_V and p with $|\sigma(\theta\bar F \cdot (h) \cdot F')| \le C_V \eta^{-p} \|h\|_{L^1} \|[\tau_{-\eta e_3} F]\| \|[\tau_{-\eta e_3} F']\|$ for $\eta \in (0, 1]$, $F, F' \in \mathcal{D}_{e_3,\eta}$ and $h \in C^1_c(\{|x_3| < \eta/2\}; V)$.

Then:

(i) for $n \ge 2$ there is exactly one continuous function s_n on $\{x \in (\mathbb{R}^3)^n : x_i \ne x_j \ (i \ne j)\} \times V^n$, n-linear in e, with $\sigma(H_1, \dots, H_n) = \int s_n(x; H_1(x_1), \dots, H_n(x_n))\,\mathrm{d}x$ for all tuples, and for the tuples of continuous weights if σ is given there; it is real analytic, symmetric, invariant under the isometries acting on points and labels, and

$$|s_n(x; e)| \le (cC_V)^n \max\{1, 2n^2/\delta(x)\}^{pn} \prod_k |e_k|_\infty, \qquad \delta(x) = \min_{i \ne j} |x_i - x_j|,$$

c a number depending on (V, ρ) and its norm; so $|\sigma(X)| \le (cC_V)^n (2n^2/d)^{pn} \prod_a \|H_a\|_{L^1}$ if the supports are at mutual distances $\ge d$, $0 < d \le 1$. For $\varphi : \mathbb{R}^{3n} \to V_{\mathbb{C}}^{\otimes n}$ whose 6^n components are Schwartz functions vanishing with all their derivatives where two of the n points coincide, $S_n(\varphi) = \int s_n \varphi$ converges absolutely and

$$|S_n(\varphi)| \le (c' C_V)^n (n!)^{2p} |\varphi|_{k_E n}, \qquad k_E = \max\{\lceil p \rceil, 4\}, \quad c' = 6\pi^2 2^p e^{2p} 3^{k_E} c,$$

$|\varphi|_k$ the greatest of the norms $\sup(1 + |x|^2)^{k/2} |\partial^\alpha \varphi_\iota|$, $|\alpha| \le k$, over the components; $S_n(H_1 \otimes \cdots \otimes H_n) = \sigma(H_1, \dots, H_n)$ at smooth entries with compact disjoint supports, and S_n is

the only linear functional bounded for one of the norms with these values. The truncated functions have the kernels that the formula of the cumulants writes in the s_m, $m \le n$.

(ii) Ω and the classes of the tuples whose entries lie in pairwise disjoint slabs $\{t < x_3 < t'\}$ are total in $\mathcal{H}(\sigma)$.

(iii) On $\mathcal{D}_\infty = \bigcap_k \operatorname{dom} H^k$ there are operators $T(f)$, $f \in \mathcal{S}(\mathbb{R}^{1+2}; V_{\mathbb{C}})$, linear in f, with $\langle \psi, T(f)\psi' \rangle = \int \lim_{s\downarrow 0} \langle U(x)^{-1}\psi, B(s,s; f(x)) U(x)^{-1}\psi' \rangle \, \mathrm{d}x$, B the point operators that (R4) provides, and $\langle \psi, T(f)\psi' \rangle = \langle T(\rho(\Theta_{e_3})\bar f)\psi, \psi' \rangle$, so that a component with r indices equal to 3 is Hermitian after the factor i^r; and there is a strongly continuous unitary representation U of the Poincaré group of $\mathbb{R}^{1+2}$ extending $x \mapsto U(x)$, with $U\Omega = \Omega$, the spectrum of the translations in the closed forward cone, and $U(a,\Lambda) T(f) U(a,\Lambda)^{-1} = T(\rho_{\mathbb{C}}(\hat\Lambda) f(\Lambda^{-1}(\cdot - a)))$ on the span $\mathcal{D}_0$ of Ω and of the vectors $T(f_1)\cdots T(f_n)\Omega$.

Here $\hat\Lambda = \iota \Lambda \iota^{-1} \in \mathrm{SO}(3,\mathbb{C})$ for $\Lambda \in \mathrm{SO}_0(1,2)$, with $\iota(t,\vec x) = (\vec x, -it)$: a time t sits at the Euclidean height $x_3 = -it$, as $U(t,\vec x) = \mathsf{E}((\vec x, -it))$; the boost of (t, x_1) by s goes to R_{is}, and a rotation $1 \oplus R$, $R \in \mathrm{SO}(2)$ acting on $\vec x$, to $R \oplus 1$; and $\rho_{\mathbb{C}}(\hat\Lambda) E = \hat\Lambda E \hat\Lambda^{\mathrm{t}}$, so that $\rho_{\mathbb{C}}(\mathsf{R}_{is}) = \exp(is\, \mathrm{d}\rho(E_{13} - E_{31}))$, $\mathrm{d}\rho(\omega) E = \omega E + E\omega^{\mathrm{t}}$.

(iv) $T(f)T(f') = T(f')T(f)$ on $\mathcal{D}_0$ when the supports of f and f' are spacelike separated, all six components with all six; $\mathcal{D}_0$ is dense; the Wightman functions $\langle \Omega, T(f_1)\cdots T(f_n)\Omega \rangle$ are tempered and equal $\lim_{\epsilon\downarrow 0} \int \mathbf{W}_n(\iota(x_2 - x_1 + i\epsilon\alpha_1), \dots; f_1(x_1), \dots) \, \mathrm{d}x$ for all α_k in the forward cone, with the ι of (iii).

(v) The kernels $\mathbf{W}_n$ of the product formula are holomorphic on Tb^{n-1}, with $|\mathbf{W}_n(\zeta; e)| \le (C_V \eta^{-p})^n \prod_k |e_k|_\infty$ for $e \in V^n$ where $\operatorname{Re}\zeta_3^k - |(\operatorname{Im}\zeta_1^k, \operatorname{Im}\zeta_2^k)| \ge 2\eta$ for all k, $0 < \eta \le 1$; $\hat\Lambda$ of (iii) maps $\mathsf{Tb} = \iota(\mathbb{R}^{1+2} + iV_+)$, $V_+ = \{\alpha : \alpha^0 > |\vec\alpha|\}$, onto itself, and $\mathbf{W}_n(\hat\Lambda\zeta^1, \dots, \hat\Lambda\zeta^{n-1}; \rho_{\mathbb{C}}(\hat\Lambda)e_1, \dots, \rho_{\mathbb{C}}(\hat\Lambda)e_n) = \mathbf{W}_n(\zeta; e)$ for every $\Lambda \in \mathrm{SO}_0(1,2)$. The s_n continue to one holomorphic function $\mathbf{S}_n$ on the union of the tubes $\{z \in \mathbb{C}^{3n} : R^{-1}(z_{\varsigma(k+1)} - z_{\varsigma(k)}) \in \mathsf{Tb},\ 1 \le k < n\}$ of the real frames $R \in \mathrm{SO}(3)$ and the permutations ς, on the tube of (R,ς) equal to $\mathbf{W}_n(R^{-1}(z_{\varsigma(2)} - z_{\varsigma(1)}), \dots, R^{-1}(z_{\varsigma(n)} - z_{\varsigma(n-1)}); \rho(R)^{-1}e_{\varsigma(1)}, \dots, \rho(R)^{-1}e_{\varsigma(n)})$; $\mathbf{S}_n$ is symmetric when points and labels are permuted together, $\mathbf{S}_n(Rz_1 + a, \dots, Rz_n + a; \rho(R)e) = \mathbf{S}_n(z; e)$ for the real isometries $x \mapsto Rx + a$, $R \in \mathrm{O}(3)$, and $|\mathbf{S}_n(z;e)| \le (cC_V\eta^{-p})^n \prod_k |e_k|_\infty$ for $e \in V^n$, c the number of (i), where for some (R,ς) the differences $R^{-1}(z_{\varsigma(k+1)} - z_{\varsigma(k)})$ have the margin 2η of the first bound.

If besides $\ker H = \mathbb{C}\Omega$, every vector invariant under the translations is a multiple of Ω, and all the Wightman axioms hold. ∎

Proof. Point operators and kernels. By (R4) the form $h \mapsto \sigma(\theta\bar F \cdot (h) \cdot F')$ is bounded in $\|h\|_{L^1}$; with the covariance under translations this gives point operators with $\|B(s,s;e)\| \le C_V s^{-p} |e|_\infty$ and the product formula of [7, Prop. 6.14](f), whose kernels are holomorphic on Tb^{n-1} by (R3), $\mathsf{E}(\zeta)$ being a contraction on $\overline{\mathsf{Tb}}$; the first bound of (v) is $\prod \|B(\eta,\eta;e_k)\|$. By (R1) the same holds in every frame $R \in \mathrm{SO}(3)$: there the heights are taken along Re_3, the kernel is $\mathbf{W}_n(R^{-1}\cdot; \rho(R)^{-1}e)$, the labels being turned back by $\rho(R)^{-1}$, and the bound holds with $\prod_k |\rho(R)^{-1}e_k|_\infty \le c^n \prod_k |e_k|_\infty$; $c = 3$ serves, as each row of R has $\sum_l |R_{kl}| \le \sqrt{3}$. The kernels of two frames, and of two orders of the points, agree at the common real points of their tubes, as the proof of (i) shows; the intersection of two tubes is convex and holds the real parts of its points, so they agree on it by the identity theorem, and they form the one function $\mathbf{S}_n$ of (v), symmetric as σ is. Its covariance under the real isometries holds at the real points by (i), and on each tube, which an isometry carries onto a tube, by the identity theorem. (i) For a unit vector v the map $u \mapsto u \cdot v$ carries the surface measure of S^2 to the uniform law on $[-1,1]$, so the band $\{|u \cdot v| < n^{-2}\}$ has the fraction n^{-2}; for pairwise different $x_1, \dots, x_n$ the u with $|u \cdot (x_i - x_j)| \ge n^{-2}|x_i - x_j|$ for all $i \ne j$ fill at least the fraction $1 - (n-1)/(2n) > \frac{1}{2}$ of the sphere. In a frame whose third axis is such a u the points are ordered in height with gaps at least $n^{-2}\delta(x)$, and the kernel of that frame is real analytic near x

with the bound, at $\eta = \min\{1, \delta(x)/(2n^2)\}$. For two frames that both order x, both kernels are continuous on a product of small balls about the x_i and have the same integrals against all products $f_1(x_1)\cdots f_n(x_n)$; so they are equal.

The same reason gives uniqueness, symmetry and covariance. For a general tuple, cover the product of the supports by finitely many such products of balls, cut each entry by a partition of unity, and sum by the n-linearity on neighbourhoods with disjoint closures. For S_n: at a point x with $\delta(x) < 1$, φ is flat at the point where the two nearest points coincide, and Taylor's formula of order $m = \lceil pn \rceil$ along the segment gives $\max\{1, 2n^2/\delta(x)\}^{pn}|\varphi(x)| \le (2n^2)^{pn} 3^k |\varphi|_k (1+|x|^2)^{-k/2}$, $k = k_E n \ge 3n+1$; $\int (1+|x|^2)^{-k/2}\,\mathrm{d}x \le \pi^{2n}$ and $n^n \le e^n n!$. (ii) The vectors $\Phi(y)$ of n points of the half space have the Gram kernel s_{2n} at $(\theta y, y')$, real analytic by (i); so for ψ orthogonal to the classes of (ii), $y \mapsto \langle \psi, \Phi(y)\rangle$ is real analytic on the connected set of the non-coinciding n-tuples of the half space, and vanishes where the heights are ordered. (iii), (iv), (v) The fields are built from the point operators and $U(x)$, and the Wightman functions are the boundary values of the $\mathbf{W}_n$ from every direction α of the open cone V_+, tempered by the power p; this follows the reconstruction of Thm. YM3_I:AO, with labels; (ii) gives cyclicity, and $B(s,s';e)^* = B(s',s;\rho(\Theta_{e_3})e)$ gives the adjoints. *The invariance of the* $\mathbf{W}_n$ *in (v).* $\hat\Lambda\iota(x+i\alpha) = \iota(\Lambda x + i\Lambda\alpha)$, and $\Lambda V_+ = V_+$: so $\hat\Lambda \mathsf{Tb} = \mathsf{Tb}$. For a rotation $\Lambda = 1 \oplus R$ of $\vec{x}$, $\hat\Lambda = R \oplus 1$ is a real rotation that keeps the heights; at real $\zeta \in \mathsf{Tb}^{n-1}$, that is at configurations ordered in height, $\mathbf{W}_n = s_n$, and the covariance of s_n gives the identity, which the identity theorem carries to Tb^{n-1}.

For the boost $B(s)$ of (t, x_1), $\hat\Lambda = \mathsf{R}_{is}$: fix a real $\zeta \in \mathsf{Tb}^{n-1}$ and e; there is $\delta > 0$ such that $\mathsf{R}_\tau \zeta^k \in \mathsf{Tb}$ for all k and all complex τ at a distance $< \delta$ from the segment $\{is' : |s'| \le |s|\}$, a convex open set on which $\gamma(\tau) = \mathbf{W}_n(\mathsf{R}_\tau\zeta^1, \dots, \mathsf{R}_\tau\zeta^{n-1}; \rho_{\mathbb{C}}(\mathsf{R}_\tau)e_1, \dots, \rho_{\mathbb{C}}(\mathsf{R}_\tau)e_n)$ is holomorphic; at real τ, $|\tau| < \delta$, the turned points are still ordered in height, and the covariance of s_n under the real rotation R_τ gives $\gamma(\tau) = \gamma(0)$; so γ is constant, and at $\tau = is$ this is the identity at real ζ, which the identity theorem carries to Tb^{n-1}.

The boosts $B(s)$ and the rotations $1 \oplus R$ generate $\mathrm{SO}_0(1,2)$, and $\Lambda \mapsto \hat\Lambda$ is a homomorphism: the identity holds for every Λ. *(iii).* By the substitution $x_k = \Lambda y_k$ and $\iota(\Lambda y + i\epsilon\alpha) = \hat\Lambda\iota(y + i\epsilon\Lambda^{-1}\alpha)$, with $\Lambda^{-1}\alpha \in V_+$, the invariance of the $\mathbf{W}_n$ makes the Wightman functions invariant under $f_k \mapsto \Lambda_a f_k = \rho_{\mathbb{C}}(\hat\Lambda) f_k(\Lambda^{-1}(\cdot - a))$, the translations being had already; as $\bar\iota = \Theta_{e_3}\iota$ and Λ is real, $\rho_{\mathbb{C}}(\hat\Lambda)$ commutes with $e \mapsto \rho(\Theta_{e_3})\bar e$, so the adjoints of (iii) go over too, and $U(a,\Lambda)T(f_1)\cdots T(f_n)\Omega := T(\Lambda_a f_1)\cdots T(\Lambda_a f_n)\Omega$ is well defined and isometric on $\mathcal{D}_0$, which is dense; it extends to a unitary representation, strongly continuous as the fields are continuous in f, which fixes Ω and has the law of (iii) on $\mathcal{D}_0$.

(iv). As $\mathcal{D}_0$ is dense it suffices that a Wightman function does not change when two neighbours f_k, f_{k+1} with spacelike separated supports are exchanged, first for f_j of compact support, to which cut-offs that do not enlarge the supports reduce the general case. Let x run through the product of the supports. At each x let $\hat\xi$ be the unit vector of $\vec{x}_{k+1} - \vec{x}_k$, and tilt e_3 towards it: $u_\vartheta = \cos\vartheta\, e_3 + \sin\vartheta(\hat\xi, 0)$, $\sin\vartheta = A\eta$, a real unit vector. The points $z_j = \iota(x_j) + \eta h_j^\lambda e_3$, $|\lambda| \le 1$, $0 < \eta \le \eta_0$, with heights h_j^λ formed from the supports alone, increasing with j at $\lambda = 1$ and in the order with $k, k+1$ exchanged at $\lambda = -1$, lie in the tube of the real frame with third axis u_ϑ, its differences having the margin 3η: across the pair $k, k+1$ the spacelike separation, $\hat\xi\cdot(\vec{x}_{k+1} - \vec{x}_k) > |t_{k+1} - t_k|$, pays for the exchanged heights, and $|\operatorname{Im}\mathfrak{z} - (u_\vartheta\cdot\operatorname{Im}\mathfrak{z})u_\vartheta| = |t_{j+1} - t_j|\sin\vartheta$ for $\mathfrak{z} = z_{j+1} - z_j$.

There $\mathbf{S}_n$, symmetric in the points with their labels, is bounded by $(cC_V\eta^{-p})^n \prod_k |e_k|_\infty$ by (v). The integral $I(\lambda,\eta)$ of $\mathbf{S}_n(z; f_1(x_1), \dots, f_n(x_n))$ over x has $\partial_\eta I$ and $\partial_\lambda I/\eta$ equal to integrals of the same kind with derivatives of the f_j in the times, after an integration by parts. The one-label smoothing and integration-by-parts argument in the proof of Prop. 3.3, applied to each tensor component, gives Taylor's formula with the integral remainder and bounds all of them uniformly in $\eta \in (0, \eta_0]$. So $|I(1,\eta) - I(-1,\eta)| \le C\eta \to 0$; and $I(1,\eta)$, $I(-1,\eta)$ tend to the Wightman functions

of the two orders, the labels being permuted with the points. *The last sentence.* A vector ψ invariant under the translations has $e^{itH}\psi = U(t,\vec{0})\psi = \psi$ for every real t, by (iii), so $H\psi = 0$ by Stone's theorem; here $\ker H = \mathbb{C}\Omega$ enters and gives $\psi \in \mathbb{C}\Omega$.

With this uniqueness of the vacuum, (iii), (iv) and the separability of $\mathcal{H}(\sigma)$ in (R2) give all the Wightman axioms. □

Proposition 3.9 (Rescaling the off-diagonal labels)**.** Let σ satisfy (D0)–(D4), $z \neq 0$ real, and $\sigma^z = \sigma \circ Z_z$. Then σ^z satisfies (D0)–(D4), with $\max\{1,|z|\}^n C_n(\delta,\varrho_0)$ in (D1), and has the trace part of σ; $[F]_{\sigma^z} \mapsto [Z_z F]_\sigma$ is a unitary map of $\mathcal{H}(\sigma^z)$ onto $\mathcal{H}(\sigma)$ that commutes with the translations and the semigroup, under which the slab and point operators and the kernels of σ^z are those of σ with Z_z applied to the labels. If $\sigma(\varphi_1 E, \varphi_2 E) \neq 0$ for $E = o_{12}$ and some φ_1, φ_2 with disjoint supports, and $z^2 \neq 1$, then σ and σ^z are not both covariant at order 2. For a tensor limit at $g \leq \gamma_W^{\mathrm{sd}}$ the two-point law of [7, Prop. 6.15], at all pairs of weights, holds for σ^z with some label $b' > 0$ in the place of b if and only if $z = 1$; at the pairs of off-diagonal weights alone it holds with the label b of σ if and only if $z^2 = 1$. ■

Proof. Z_z is the identity on V_A and commutes with $\rho(W_3)$, in particular with $\rho(\Theta_n)$ for $n \in \mathfrak{N}_{18}$; this gives the first part. Let R be the rotation by $\pi/4$ about e_3, so that $\rho(R)E = E_{22} - E_{11} \in V_E$. If both were covariant at order 2, then

$$z^2\sigma(\varphi_1 E, \varphi_2 E) = \sigma^z(\varphi_1 E, \varphi_2 E) = \sigma^z(R(\varphi_1 E), R(\varphi_2 E)) = \sigma(R(\varphi_1 E), R(\varphi_2 E)) = \sigma(\varphi_1 E, \varphi_2 E).$$

For the law: at $z = 1$, $\sigma^z = \sigma$. Let σ^z obey the law with an error $\epsilon'(d) \to 0$, at all pairs with a label b', or at the pairs of off-diagonal weights alone with the label b. In the first case, at the pairs of diagonal weights, which Z_z leaves alone, its label is that of σ: there $\sigma^z = \sigma$, and at $(\varphi_1^d\mathbf{1}, \varphi_2^d\mathbf{1})$, with the bumps of [7, Prop. 6.15](a), the kernel of $\mathcal{W}$ is $\frac{3}{2}\operatorname{tr} K_3(u)^2 = 9(4\pi|u|^3)^{-2} > 0$, $u = x - y$, so that $\mathcal{W}(\varphi_1^d\mathbf{1}, \varphi_2^d\mathbf{1})$ is a positive number independent of d, and b'^{-2} and b^{-2} are both the limit of $\sigma(\varphi_1^d\mathbf{1}, \varphi_2^d\mathbf{1})/\mathcal{W}(\varphi_1^d\mathbf{1}, \varphi_2^d\mathbf{1})$ as $d \downarrow 0$: $b' = b$.

With the label b, in both cases, at off-diagonal entries $\sigma^z(H_1, H_2) = z^2\sigma(H_1, H_2)$, and $|z^2 - 1|\, b^{-2}|\mathcal{W}(H_1, H_2)| \leq (z^2\epsilon_g(d) + \epsilon'(d))b^{-2}|H_1|_\infty|H_2|_\infty$ for all small d forces $z^2 = 1$ by the lower bound of $\mathcal{W}$ ([7, Prop. 6.15](a)); conversely at $z = \pm 1$ these pairs are unchanged. At the mixed pair $H_1 = \varphi_1^d\mathbf{1}$, $H_2 = \varphi_2^d o_{12}$, with the bumps of [7, Prop. 6.15](a) placed with $x_2 - x_1 = \frac{7}{2}d(e_1 + e_2)/\sqrt{2}$, one has $\sigma^z(H_1, H_2) = z\,\sigma(H_1, H_2)$, and the kernel of $\mathcal{W}$ is $\frac{3}{2}\operatorname{tr}[K_3(u)o_{12}K_3(u)] = 3(K_3(u)^2)_{12} = 9\hat{u}_1\hat{u}_2(4\pi|u|^3)^{-2}$, with $K_3(u) = (3\hat{u}\hat{u}^{\mathrm{t}} - \mathbf{1})(4\pi|u|^3)^{-1}$, $\hat{u} = u/|u|$, u the difference of the two points; on the supports $\hat{u}$ lies within the angle $\arcsin\frac{2}{7} < \frac{\pi}{4}$ of $\pm(e_1 + e_2)/\sqrt{2}$, so $\hat{u}_1\hat{u}_2 > 0$, and $\mathcal{W}(H_1, H_2) = w_\times > 0$ does not depend on d, $\mathcal{W}$ being invariant under dilations.

So $|z - 1|\, b^{-2}w_\times \leq (|z|\epsilon_g(d) + \epsilon'(d))b^{-2}|\varphi|_\infty^2$ for all small d, and $z = 1$. This is [7, Prop. 6.15](d) at $a = e = 1$. □

Proposition 3.10 (Reconstruction with values in V)**.** Let σ be a real symmetric functional on $\mathfrak{S}(\mathbb{R}^3)$, multilinear at disjoint supports, with $\sigma(\emptyset) = 1$, invariant under all isometries acting on points and indices, positive at $\{x_3 = 0\}$, with separable $\mathcal{H}(\sigma)$, with H, $\mathcal{U}$, $\mathcal{P}$ as in [7, Thm. BW](a), the circular cone $\mathcal{P}(\{E < |q|\}) = 0$, and [7, eq. 263] for some C_V and p_V. Then: (i) there is exactly one continuous s_n at non-coinciding points, n-linear in the labels, with $\sigma(X) = \int s_n(x; H_1(x_1), \dots, H_n(x_n))\,\mathrm{d}x$ for all X; it is real-analytic, symmetric and invariant, and $|s_n(x; e)| \leq (cC_V)^n \max\{1, 2n^2/\delta(x)\}^{p_V n}\prod_k |e_k|$; (ii) the fields $T(f)$ of [7, Thm. BW](e) are defined on $\mathcal{D}_\infty$, with the bound and the adjoint stated there, covariant under the translations, and their vacuum expectations are tempered, the boundary values of the kernels continued to the forward tube; (iii) Ω and the classes of the tuples in disjoint slabs are total in $\mathcal{H}(\sigma)$, and $\mathcal{D}_0$ is dense; (iv) there is a strongly continuous unitary representation of $\mathbb{R}^{1+2} \rtimes \mathcal{L}$ on $\mathcal{H}(\sigma)$ as in [7, Thm. BW](e), acting on the labels by $D(\Lambda)e = \hat{\Lambda}e\hat{\Lambda}^t$, with spectrum in the closed forward cone; (v) the fields commute at spacelike separation.

Nothing is supposed or said of $\ker H$; if $\ker H = \mathbb{C}\Omega$, all Wightman axioms hold. ■

Proof sketch. It is the proof of Thm. YM3_I:AO with values in V. The ordered kernels $\mathbf{W}_n$ of [7, Prop. C.18](c) continue to the circular tube, and [7, eq. 263] gives $|\mathbf{W}_n| \leq (C_V \eta^{-p_V})^n \prod |e_k|$ at distance 2η from its boundary, with no factorial. For n distinct points more than half of the directions u have $|u \cdot (x_a - x_b)| \geq n^{-2}|x_a - x_b|$ for all $a \neq b$ (zones of Archimedes); rotating such a u to e_3, the frames in $SO(3)$ glue the ordered kernels into one continuous, real-analytic function s_n, and the bound at $\eta = \min\{1, \delta(x)/(2n^2)\}$ is (i). The limits $\lim_{s\downarrow 0}\langle\psi, \mathsf{B}(s, s; e)\psi'\rangle$ exist for $\psi, \psi' \in \operatorname{dom} H^l$, $l > p_V$, the power being paid by $(1 + H)^l$; this defines $T(f)$, tempered, with $\langle\Omega, T(f_1)\cdots T(f_n)\Omega\rangle$ the boundary values of the $\mathbf{W}_n$ at ι of the real points, from the directions of the forward cone.

The totality (iii) follows from (i): for $X \in \mathfrak{S}(\{x_3 > 0\})$ cut every entry by a smooth partition of unity in x_3 of mesh ϵ; the terms in which two entries lie in one slab or in neighbouring slabs form a remainder R_ϵ, and $\|[R_\epsilon]\|^2 = \sigma(\theta\bar{R}_\epsilon \cdot R_\epsilon)$ is, by (i), an integral of s_{2n} against the weights of $\theta\bar{R}_\epsilon$ and R_ϵ, where s_{2n} is continuous, so bounded, on the compact product of the supports of the entries of θX and of X, which are pairwise disjoint, while the weights of R_ϵ are carried by a set of measure $O(\epsilon)$; so $[R_\epsilon] \to 0$, and $[X]$ is a limit of sums of classes of tuples in disjoint slabs.

A vector orthogonal to $\mathcal{D}_0$ gives functions holomorphic in the forward tube whose boundary values vanish; their values at imaginary times are its scalar products with the classes of the tuples in disjoint slabs, so by the totality it is 0, and $\mathcal{D}_0$ is dense. The complex rotations R_{is} of the plane (x_3, x_1) are the boosts: invariance gives exact norms along them, so they are unitary on the closed span of the point vectors, which is $\mathcal{H}(\sigma)$ by (iii), and with the translations and the rotations about e_3 they generate $U(a, \Lambda)$ on $\mathcal{H}(\sigma)$, with $U(a, \Lambda)\mathcal{D}_0 = \mathcal{D}_0$; D is a continuous homomorphism of $\mathcal{L}$ into $GL(V_\mathbb{C})$, $\hat{\Lambda}^t\hat{\Lambda} = \mathbf{1}$, with $D(\Lambda)e^* = (D(\Lambda)e)^*$ for $e^* = \Theta_{e_3}\bar{e}\Theta_{e_3}$, as $\overline{\hat{\Lambda}} = \Theta_{e_3}\hat{\Lambda}\Theta_{e_3}$.

Locality: the kernels continue to tubes tilted by an angle of the order of η, and the symmetry of s_n under permutations of points and labels gives the commutation at spacelike separation, all components with all. □

Proposition 3.11 (Fields and kernels from the square cone)**.** Let $0 < g \leq \gamma_W^{\mathrm{sd}}$ and let σ be a limit at g. [7, Thm. BV](a)–(c) and [7, eq. 263], with the square cone $\mathcal{P}(\{E < \max_k |q_k|\}) = 0$ in the place of the circular one, give the following, independently of the invariance under $O(3)$: the part of [7, Thm. BW](d), (e) that rests on the square cone, the one step at which the rotations enter Lorentz covariance, and bounded slab operators for entries side by side. (a) There are bounded operators $\mathsf{b}(w|e|w')$, for w, w' in the open square tube and $e \in V_\mathbb{C}$, linear in e and holomorphic in norm, with $\|\mathsf{b}(w|e|w')\| \leq \sqrt{2}C_V s^{-p_V}|e|$ for $0 < s \leq \min\{1, d_\square(w), d_\square(w')\}$, $d_\square(z) = \operatorname{Re} z_3 - |\operatorname{Im} z_1| - |\operatorname{Im} z_2|$, and $\mathsf{B}_{2\eta}((\tau_{\eta e_3}h)) = \int \mathsf{b}(y + \eta e_3|h(y)| - y + \eta e_3)\,\mathrm{d}y$; and $|\mathbf{W}_n(z; e)| \leq (C_V\eta^{-p_V})^n \prod_k |e_k|$ where $d_\square(z^k) \geq 2\eta$, $0 < \eta \leq 1$. (b) The fields $T(f)$ of [7, Thm. BW](e) are defined on $\mathcal{D}_\infty$, with the same bound and adjoint, covariant under the translations and the dihedral group of order eight that fixes e_3, with joint spectrum in $\{E \geq \max_k |q_k|\}$ and Ω cyclic in $\mathcal{H}(\sigma)$; their vacuum expectations are tempered and are the boundary values of the $\mathbf{W}_n$ from every direction of $\{\alpha^0 > |\alpha^1| + |\alpha^2|\}$; and they commute in the square form: $T(f)T(f') = T(f')T(f)$ on $\mathcal{D}_0$ whenever $\max_k |x^k - x'^k| > |t - t'|$ on the supports. Lorentz covariance needs the rotations at one step only: for one one-parameter group r_t of rotations, the identity $\frac{d}{dt}\mathbf{W}_n(r_t\xi; r_t e_1 r_t^t, \ldots, r_t e_n r_t^t)\big|_{t=0} = 0$ at all points ξ and labels e is equivalent to invariance under $O(3)$. (c) For $m \geq 0$ the kernel $s(x; y_1, \ldots, y_m; e) = \lim_{\epsilon\downarrow 0} \sigma(\delta_\epsilon^x e, \delta_\epsilon^{y_1}\mathbf{1}, \ldots, \delta_\epsilon^{y_m}\mathbf{1})$ of one tensor entry among m action densities, δ_ϵ^z a bump of width ϵ at z, exists at all non-coinciding points, is real-analytic there and represents σ at such tuples; where for one direction μ the coordinate x_μ differs from all $y_{k,\mu}$, it is the matrix element of a point operator between two point vectors of the scalar theory, the points below and above x in that direction. The trace part is data of Thm. YM3_I:AO, and its point vectors exist and are norm continuous at all non-coinciding points of the half space. (d) For $l > 0$ and a tuple $Y = (h_1, \ldots, h_n) \in \mathfrak{S}(\{0 < x_3 < l\})$, its entries side by side or not, let $d(Y) = \min\{1, \operatorname{dist}(\operatorname{supp} Y, \{x_3 = 0\} \cup \{x_3 = l\}), \min_{a\neq b} \operatorname{dist}(\operatorname{supp} h_a, \operatorname{supp} h_b)\}$.

The slab operator $\mathsf{B}_l(Y)[F] = [Y \cdot \tau_{le_3} F]$ is bounded,

$$\|\mathsf{B}_l(Y)\| \le \mathsf{K}_n \big(\sqrt{2} C_V (3/d(Y))^{p_V}\big)^n \prod_a \|h_a\|_{L^1}, \qquad \mathsf{K}_n = (10n)^{\lceil p_V n \rceil n(n-1)/2};$$

and $\sigma(X) = \int s_n(x; h_1(x_1), \dots, h_n(x_n))\, \mathrm{d}x$ for every $X = (h_1, \dots, h_n) \in \mathfrak{S}(\mathbb{R}^3)$, with one function s_n continuous at all non-coinciding points, given at points of pairwise different heights by $\mathbf{W}_n$ at the points ordered in x_3, and $|s_n(x; e)| \le \mathsf{K}_n (\sqrt{2} C_V (3/d_\infty(x))^{p_V})^n \prod_a |e_a|$, $d_\infty(x) = \min\{1, \min_{a \ne b} |x_a - x_b|_\infty\}$. ■

Proof sketch. (a) As in [7, Prop. C.18](c), with [7, eq. 263] in the place of the finite function C_{pt}: covariance moves the point of the plate into the exponentials of the spectral measure, which are contractions on the square tube by the square cone, and the bound is [7, eq. 263] at the distance $d_\square$. (b) As in the proof of Prop. 3.10, with the square tube for the circular one: boundary values as tempered distributions from the directions of the square cone, the translations and the dihedral group of W_3 fixing e_3 acting, and [7, Prop. C.18](b) for the cyclicity. The commutation in the square form: by a partition of unity of f and f' into compact pieces it suffices to treat two pieces, and for them to fix a horizontal direction k with $|x^k - x'^k| - |t - t'| \ge c > 0$ on their supports.

First, at a configuration of n points ordered in x^k with $|\Delta x^k| > |\Delta t|$ for every pair, $\iota(t, \vec{x}) = (\vec{x}, -it)$, the points $\iota(x) + \eta h e_3$, $h = (h_1, \dots, h_n)$ increasing along an order of the points, lie in the e_3-tube of that order for $\eta > 0$, for either order of the two entries that are exchanged, and inside the e_k-tube with a margin free of η, as $\mathrm{Re}\, \Delta z_k = \Delta x^k > |\Delta t| = |\mathrm{Im}\, \Delta z_3|$; the tubes are convex and have a common real open set, on which σ has one kernel, so the two boundary values as $\eta \downarrow 0$ are the one function of the picture e_k at $\iota(x)$. Second, $F(a, b) = \langle T(g_1(\cdot - b_1)) \cdots T(g_m(\cdot - b_m))\Omega, [T(f), T(f')] T(f_1(\cdot - a_1)) \cdots T(f_{m'}(\cdot - a_{m'}))\Omega \rangle$ is continuous where $\mathrm{Im}\, a_1$, the $\mathrm{Im}(a_{i+1} - a_i)$, $\mathrm{Im}\, b_1$ and the $\mathrm{Im}(b_{i+1} - b_i)$ lie in the closed cone $\{\alpha^0 \ge |\alpha^1| + |\alpha^2|\}$, dual to the square cone, and holomorphic in $(a, \bar{b})$ inside, by the spectrum and the bounds of T on $\mathcal{D}_\infty$; it vanishes on the open set of the real shifts far along e_k, by the first step; the edge-of-the-wedge theorem against 0 gives $F = 0$, and Schwartz f, f' follow by density.

The boosts are the complex rotations, and their covariance is the analytic continuation of the rotation covariance from one infinitesimal generator. (c) The trace part has the bounds, the invariance and the positivity required by Thm. YM3_I:AO, whose reconstruction gives the point vectors $\Phi_n(y)$ of the scalar theory. For (x, y) with $x_\mu \ne y_{k,\mu}$ for all k cut by two planes $x_\mu = c_\pm$ next to x_μ, and write the kernel as $\langle \Phi(\text{reflected points below}), \mathcal{U}\, \mathsf{B}_e(x_\mu - c_-, c_+ - x_\mu)\, \mathcal{U}^{-1} \Phi(\text{points above}) \rangle$ in the picture e_μ; the three pictures agree where they overlap, by the uniqueness of continuous kernels.

The remaining points form a null set of codimension three; there holomorphy in each coordinate separately with one bound gives joint holomorphy, by Bernstein's estimate of the Chebyshev coefficients, so the kernel is real-analytic at all non-coinciding points. (d) First the continuation and the bound, then the integral. By the symmetry of σ the class $[Y \cdot \tau_{le_3} F]$ does not depend on the order of the entries. At points x of pairwise different heights let $\mathsf{G}_l(x; e)$ be the product of the operators $\mathsf{b}(w|e|w')$ of (a), taken in the order of the heights, whose arguments are the half gaps of the heights and the distances to the faces of the slab; the spectators enter as two vectors $\mathcal{U}(v)[F']$, $\mathcal{U}(w)[F]$, and $f(x; v, w) = \langle \mathcal{U}(v)[F'], \mathsf{G}_l(x; e) \mathcal{U}(w)[F] \rangle$ is analytic in (v, w).

Take a pair a, b of entries with $|x_a^k - x_b^k| \ge d = d(Y)$ in a horizontal direction k, and let ζ be their relative height, the mean height and the other points being fixed: the two orders of the pair give functions $f_\pm(\zeta; v, w)$, holomorphic in $\pm \mathrm{Re}\, \zeta > 0$, with $|f_\pm| \le \mathsf{M}(|\mathrm{Re}\, \zeta|/2)^{-\mathsf{P}}$ by (a), P the power of the bound of (a) in the half gap of the pair ($\mathsf{P} = 2p_V$ at $n = 2$). For spectators shifted far along e_k, planes $x_k = \text{const}$ part them from the entries; in the picture e_k all is then ordered in x_k, ζ is a translation along the mirror, and the kernel is holomorphic in $|\mathrm{Im}\, \zeta| < |x_a^k - x_b^k|$, equal to f_+ and to f_- at the real ζ of either sign, by the symmetry of σ and the uniqueness of continuous kernels.

For such spectators the Cauchy integral over $|\zeta| = r < d$ of $(\zeta^2 + r^2)^{\mathsf{N}} f(\zeta; v, w)$, $\mathsf{N} = \lceil \mathsf{P} \rceil$, reproduces $(\zeta_0^2 + r^2)^{\mathsf{N}} f(\zeta_0; v, w)$ inside the circle; it is analytic in (v, w), the factor $(\zeta^2 + r^2)^{\mathsf{N}}$ taming f on the circle, and the identity theorem in (v, w) carries the continuation across $\operatorname{Re}\zeta = 0$ to all spectators. On the circle $|\zeta^2 + r^2| = 2r|\operatorname{Re}\zeta|$, and at real ζ_0 inside $|\zeta_0^2 + r^2| \geq r^2$, so the maximum principle gives $|f| \leq 2^{\mathsf{N}}(2/r)^{\mathsf{P}}\mathsf{M}$ there, while where $|\operatorname{Re}\zeta| \geq r$ the bound (a) serves directly; at $n = 2$, $r = \frac{2}{3}d$ gives the factor $2^{\mathsf{N}}(3/d)^{2p_V}\mathsf{M}$.

Made once for each of the $\frac{1}{2}n(n-1)$ pairs, with the total power at most $p_V n$ and radii of the order $d(Y)/n$, this gives $\|\mathsf{G}_l(x; e)\| \leq \mathsf{K}_n(\sqrt{2}C_V(3/d(Y))^{p_V})^n \prod_a |e_a|$ at all points of pairwise different heights, and the continuation of G_l with this bound across the sets of equal heights, where the entries are apart in a horizontal direction. Then the integrand is bounded, and σ charges no set of points with two equal heights: by [7, Prop. C.18](b) the unordered remainder of a cut in x_3 tends to 0, and at ordered tuples the class is the product of the operators of (a) applied to $[F]$, by an induction on the number of the entries; so $[Y \cdot \tau_{le_3} F]$ is the integral, over the points of pairwise different heights, a set of full measure, of $\mathsf{G}_l(x; h_1(x_1), \dots, h_n(x_n))[F]$, which gives the L^1 bound.

So $s_n(x; e) = \langle \Omega, \mathsf{G}_l(x; e)\Omega \rangle$ is continuous at all non-coinciding points, and $\sigma(X) = \langle \Omega, [X] \rangle = \int s_n$ after a translation of X into a slab. □

Step 12: the field; real time. By [7, proof of Thms. BU–BX, Steps 10–11] and [7, Prop. C.18](b), the restriction of σ to C^1 entries has the hypotheses of Prop. 3.10, with p_V and C_V, and the Hilbert spaces of σ and of this restriction are one, the classes of C^1 tuples being dense. Prop. 3.10 gives [7, Thm. BW](e) and the kernels of (d), which represent σ at all continuous entries by the convergence on $\mathfrak{S}(\mathbb{R}^3)$; at supports ordered in x_3 the kernel is the boundary value of $\mathbf{W}_n$ ([7, Prop. C.18](c)). The growth: for φ vanishing with all its derivatives at coinciding points, Taylor's formula at the nearest point with two equal entries, to the order $\lceil p_V n \rceil \leq 9n$, absorbs $\delta(x)^{-p_V n}$; then $(1+|x|^2)^{-k/2}$, $k = 9n > 3n$ the index of the norm below, is integrable, there are 6^n components, and $(2n^2)^{p_V n} \leq (2e^2)^{p_V n}(n!)^{2p_V}$, which at $p_V = 9$ gives $(n!)^{18}|\varphi|_{9n}$.

For [7, Thm. BW(g)], let $\eta \in C_c^2(\mathbb{R}^3; \mathbb{R}^3)$ be carried by the slab $\{|y_3| < a\}$, $a > 0$, and let the spectators be $\tau_{ae_3} F$ and $\theta\tau_{ae_3} F'$, $F, F' \in \mathfrak{F}(\{x_3 > 0\})$, on either side of it. As $H^{\mathrm{cons}}[\eta] = \sum_{ij} \partial_i \eta_j \, s_{ij}$, [7, Thm. BV](f), with the point operators of [7, Thm. BW](b) and an integration by parts in y, gives $\sum_i \partial_{y_i} \langle [F'], \mathcal{U}(\vec{y})\mathsf{B}(a + y_3, a - y_3; s_{ij})\mathcal{U}(-\vec{y})[F] \rangle = 0$ for $|y_3| < a$ and every j; both members are holomorphic in y, and at $y = \iota(x)$, where $y_3 = -it$ and $\partial_{y_3} = i\partial_t$, this is, smeared with f, $-i\,\Theta_{3j}^E(\partial_0 f) = \sum_{k \leq 2} \Theta_{kj}^E(\partial_k f)$ between the vectors $e^{-aH}[F']$ and $e^{-aH}[F]$: the identity of [7, Thm. BW(g)] at $\nu = j \leq 2$, and, multiplied by $-i$, at $\nu = 0$, $j = 3$.

These vectors span a core of every power of $1 + H$, as $(1 + H)^m e^{-aH}$ is bounded for every m, the $[F]$ are dense and the spectral truncations of a vector lie in the range of e^{-aH}; read on the core of $(1 + H)^{2l}$, $l > p_V$, the bound $\|(1 + H)^j T(f)\psi\| \leq c_j(f)\|(1 + H)^{2l+j}\psi\|$ of [7, Thm. BW(e)] at $j = 0$ extends the identity to $\mathcal{D}_0$.

4. Point splitting at two scales

Two nearby scalar insertions retain directional information. To read it, keep the pair a distance r apart and place a spectator at a larger distance d. The limit $r \to 0$ is then taken while d stays positive. There is also a third scale: the support radius of each bump used to approximate a point.

For the coordinate direction λ, write

$$\mathcal{O}_\lambda(\varphi) = a_K^{-1} \sum_{\lambda(p)=\lambda} \varphi(x(p)) a_p, \qquad \mathcal{O} = \sum_\lambda \mathcal{O}_\lambda, \qquad T_\lambda = \mathcal{O}_\lambda - \tfrac{1}{3}\mathcal{O}.$$

Thus T_λ is a diagonal traceless tensor component, not the stress tensor. Let χ_c^ρ be the radial tent of mass one, radius ρ and centre c, and put $f_\pm^c = \chi_{c \pm r e_\lambda/2}^{r/8}$. Denote joint cumulants by $S_{n;K}$. At label

b on the least admissible torus, define

$$A_{r,K} = \frac{2\pi}{3} b r^3 \int \varphi_1(c) S_{3;K}(f_+^c, f_-^c, \varphi_2)\, \mathrm{d}c.$$

Sketch C (The close pair and the tensor insertion). Fix $0 < \beta_H < 1$, a sufficiently small reference coupling, and a label $b \in I(g)$. Take $6d \le \min\{1, b\}$. Suppose that the Lipschitz weights φ_1, φ_2 lie in balls of radius d, with supports at distance at least d and centres at distance at most $3d$. For sufficiently small r, with $128 a_K \le r$ and the point-splitting working level below the cutoff,

$$\left|A_{r,K} - \langle T_\lambda(\varphi_1); \mathcal{O}(\varphi_2)\rangle_K\right| \le C(d) b^{-2} \left[r^{\beta_H/72} + (a_K/r)^{1/18}\right] |\varphi_1|_\infty |\varphi_2|_\infty.$$

Here the constant and the small-radius threshold also depend on the construction, β_H, g, b, and on $d\,\mathrm{Lip}(\varphi_1)/|\varphi_1|_\infty$ when $\varphi_1 \neq 0$; a zero weight gives the identity trivially. At fixed r both expressions have limits as $K \to \infty$. Their difference then tends to zero as $r \to 0$ with d fixed. The construction also gives directional tensor comparisons and three- and four-point laws with their separate scale restrictions. ■□

Sketch of the argument. The Gaussian calculation explains the normalization. The three-point function is a cyclic contraction of three free curvature kernels. When two points approach one another, its singular factor has order r^{-3}. Multiplication by $\frac{2\pi}{3} b r^3$ leaves precisely the free harmonic of the traceless tensor seen by the distant insertion. Plane fields give the diagonal components; the bond bilinears supply the off-diagonal components. Their joint comparison identifies the tensor in every direction.

Agreement of free contractions is insufficient for the limiting identity. Place the close pair and the tensor insertion in one renormalization calculation with sources. Their normalized difference is carried through the same backgrounds and fluctuation laws. Above the scale where the pair is no longer resolved, that difference is nearly constant. Centring removes the constant, while the remaining response and mixed-source errors give the positive power of r in the display. The histories with a first large-field creation are summed with their probability weights. This controls the part of the correlation not accounted for by the explicit Gaussian contractions.

The lattice error has a different role. At finite cutoff the two bumps must remain much farther apart than one mesh spacing. The term $(a_K/r)^{1/18}$ records this restriction. Taking $r \to 0$ first at fixed K would lose the estimate. Removing the cutoff first, or retaining the stated relation between the two scales, makes both errors small.

One must also distinguish kernels from their averaged values. The fixed-torus construction initially gives Borel kernels defined almost everywhere. Narrow-bump comparisons give bounds on their means and hence at their Lebesgue points. They do not select values on an exceptional set. For periodic-Gibbs limits and their components, the independently established continuous kernels turn the corresponding bounds into pointwise assertions.

This point-splitting result identifies a particular tensor insertion beside separated spectators. It does not define every coincident product or supply an unrestricted operator-product expansion. Such a conclusion would require control of additional products, their subtractions and their contact terms.

4.1. **Two short scales and point splitting.** A pair of nearby action-density insertions has a tensor-valued leading term. We compare the close pair with this tensor at a second, larger distance, retaining the dependence on both scales. The three-point comparison, its extension to plane fields, the fixed-distance point-splitting limit and the four-point law have different hypotheses and are stated separately.

The proof begins with the local kernels and their Gaussian contractions, then controls their remainders through the renormalisation steps. The treatment of narrow test functions is needed to distinguish continuous continuum kernels from the almost-everywhere classes obtained on a fixed torus.

4.1.1. *Notation and hypotheses. Constants and conventions.* Fix one common choice of the real and sourced construction constants, as in Prop. YM3_I:69.1 and Section YM3_I:82, at a fixed source radius $2\varrho > 0$. The real construction uses the restriction of this choice, with no change in the value of any retained constant. In particular, the hypotheses of Thms. YM3_I:AJ and YM3_I:AD are imposed at this same choice: the blocking factor $L \geq 13$ is odd, the cube size is $M = L^{m'}$, $a_j = \eta_j = L^{-j}$, and $A_0, A_1, \mathfrak{p}_0 \geq 1$, $\bar{B}$, $B_\sharp$ (with $\bar{B} = B_\sharp L/(L-1)$), α_0, β_s, κ, together with the remaining constants of the real and sourced steps, have these fixed values. By C, c, c', with or without an index, we denote positive numbers made from these constants only, and by $C(g)$, $C(g, \mathsf{b})$, $C(g, \ell)$ numbers that may also depend on the displayed arguments; such a number depends on nothing that is not displayed, in particular not on K, β, ℓ, a level, a length, a position, a direction, a sign or a weight. We put $\delta(t) = A_1(\log t)^{\mathfrak{p}_0} t^{-1/2}$, $\mathfrak{b}(t) = A_1(\log t)^{\mathfrak{p}_0}$, $\varepsilon(t) = A_0(\log t)^{\mathfrak{p}_0} t^{-1/2}$; δ and ε decrease on $[e^{2\mathfrak{p}_0}, \infty)$.

Couplings, data, sides. The reference coupling satisfies $0 < g \leq \gamma$, with the ceiling fixed by Prop. YM3_I:82.2. A first-passage datum (K, β) at g consists of a bare inverse coupling β and the number K of renormalisation steps after which, for the first time, the running inverse coupling that starts at β is at most $L(g^{-2} + B_\sharp)$, as in Prop. YM3_I:74.1. Its run is $x_0 = \beta > x_1 > \cdots > x_K > g^{-2}$; we write $y_j = \log x_j$, $g_j = x_j^{-1/2}$, $\delta_j = \delta(x_j)$, $\mathfrak{b}_j = \mathfrak{b}(x_j)$, $\varepsilon_j = \varepsilon(x_j)$. The label $\mathsf{b} = a_K \beta$ is a length, $g^{-2} < \mathsf{b} \leq \bar{\mathsf{b}} := L(g^{-2} + \bar{B})$. We put $I(g) = (g^{-2} + \bar{B},\, L(g^{-2} + B_\sharp) - \bar{B}]$; for $\mathsf{b} \in I(g)$ the pair $(K, L^K \mathsf{b})$ is a first-passage datum at g with the label b for every $K \geq 1$. $\mathfrak{L}(g)$ is the set of the sides at g, each of the form $\ell = 2L^m$, and $\ell_g = \min \mathfrak{L}(g) \geq 26$ is the least side; T_ℓ is the torus of side ℓ with the lattice $a_K \mathbb{Z}^3$, and the lattice (ℓ, K) is this torus with the coarser lattices of its levels.

States and limits. $\mu_{K,\ell,\beta}$ is the measure of the torus T_ℓ at the cutoff K and the bare coupling β, $\langle \cdot \rangle_{K,\ell,\beta}$ its expectation, $\tilde{\mu}_{K,\ell,\beta}$ its periodic extension to $\mathbb{L}_K = a_K \mathbb{Z}^3$, and $\mathcal{G}^{\mathrm{per}}_{K,\beta}$ is the set of periodic Gibbs states (equation YM3_I:(1819) and the definition of periodic states following it). By a state at (K, β) we mean either a periodic extension $\tilde{\mu}_{K,\ell,\beta}$ with $\ell \in \mathfrak{L}(g)$ or a periodic Gibbs state in $\mathcal{G}^{\mathrm{per}}_{K,\beta}$. The test pairs $\mathfrak{T}_{T_\ell}(r; \vartheta)$ of scale r and shape ϑ are those of [7, Lem. I.1].

Along first-passage data (K_j, β_j) at g with $K_j \to \infty$ and $a_{K_j} \beta_j \to \mathsf{b} > 0$, the torus connected functions converge to $S_n^{(\ell, \mathsf{b})}$ on real Lipschitz tuples with pairwise disjoint supports, independently of the defining sequence, by Thm. YM3_I:A. The first-passage labels lie in $(g^{-2}, \bar{\mathsf{b}}]$ by Prop. YM3_I:74.1; in particular their limiting labels are positive. A law on $\mathbb{R}^3$ is a functional on separated tuples of real C^1_c functions with the properties of the growing-torus or volume-first limits in Props. YM3_I:92.1, YM3_I:92.3 and YM3_I:92.4. Limits of periodic-Gibbs sequences and the components of their canonical decompositions are examples, by Thms. YM3_I:AP and YM3_I:AL.

For limits of periodic-Gibbs sequences and their canonical components, Thm. YM3_I:AP and Prop. YM3_I:108.3 give unique continuous moment kernels, real analytic and invariant under Euclidean isometries. The connected kernels have the same properties by the finite moment–cumulant formula. For an arbitrary growing-torus law, the inherited bounds initially give distributions on separated configurations. Full rotation invariance has the additional Ward criterion of Thm. YM3_I:AQ; the analytic-kernel conclusion is available when the reconstruction hypotheses (Rec) of Prop. YM3_I:108.3 hold. The pointwise assertions below for such a law explicitly assume continuity of the kernels they evaluate. The finite-cutoff, integrated and almost-everywhere assertions retain their stated hypotheses.

Fields. For a plaquette p, $x(p)$ is its barycentre, $\lambda(p) \in \{1, 2, 3\}$ its dual direction (the direction orthogonal to its plane), and $a_p = 1 - \frac{1}{2} \operatorname{tr} \partial U(p)$. For a real weight w on the plaquettes with finite support, $\mathcal{O}(w) = a_K^{-1} \sum_p w(p) a_p$, $|w|_\infty = \max_p |w(p)|$ and $\Sigma(w) = \{x(p) : w(p) \neq 0\}$; for a bounded function f, $\mathcal{O}(f) = \mathcal{O}(f \circ x)$. The plane fields are $\mathcal{O}_\lambda(\varphi) = \mathcal{O}(\mathbf{1}_{\{\lambda(p) = \lambda\}}\, \varphi \circ x)$, with plane weights, so that $\mathcal{O} = \sum_\lambda \mathcal{O}_\lambda$; $T_\lambda = \mathcal{O}_\lambda - \frac{1}{3}\mathcal{O}$ are the diagonal elements of the traceless tensor.

Off the diagonal stand the bond forms. For $V \in SU(2)$ put $\mathrm{w}(V) = (V - V^{-1})/2i$, and $\langle X, Y\rangle = \frac12 \operatorname{tr} XY$. A bond $b = \langle x, x + a_K e_\mu\rangle$ has the base point x, the direction $\mu_b = \mu$ and the midpoint $x_b = x + \frac12 a_K e_\mu$. For signed directions α, α' with $\alpha \neq \pm\alpha'$ let $W_x(\alpha, \alpha')$ be the product of U along the closed path $x \to x + a_K e_\alpha \to x + a_K e_\alpha + a_K e_{\alpha'} \to x + a_K e_{\alpha'} \to x$, where $e_{-\alpha} = -e_\alpha$. For a direction $\alpha \neq \mu_b$ the loops $W_x(\mu_b, \alpha)$ and $W_x(\mu_b, -\alpha)$ run around the two plaquettes of the plane spanned by μ_b and α that contain b, both from the base point x along b first, and we put $u_{b,\alpha} = -\frac12[\mathrm{w}(W_x(\mu_b, \alpha)) - \mathrm{w}(W_x(\mu_b, -\alpha))]$; with $\lambda_b < \lambda_b'$ the two directions other than μ_b, $q_b = \langle u_{b,\lambda_b}, u_{b,\lambda_b'}\rangle$.

For a real weight w on the bonds, $Y[\mathsf{w}] = -a_K^{-1} \sum_b \mathsf{w}(b) q_b$, and $\mathcal{O}^{\mathrm{od}}_{\lambda\lambda'}(\varphi) = Y[\mathsf{w}]$ at $\mathsf{w}(b) = \varphi(x_b)\mathbf{1}_{\{\mu_b = \mu\}}$, where $\{\lambda, \lambda', \mu\} = \{1, 2, 3\}$: the bond weight is taken at the midpoint. Written in the angles F_p of the plaquette variables, the linear part of $u_{b,\lambda}$ is $\sum_p u_{b,\lambda}(p) F_p$, with $u_{b,\lambda}(p) = \frac12 \epsilon_{\lambda\mu_b\lambda'}\epsilon_p$ at the two plaquettes with the barycentres $x_b \pm \frac12 a_K e_\lambda$ and $u_{b,\lambda}(p) = 0$ at all other plaquettes; here $\{\lambda, \mu_b, \lambda'\} = \{1, 2, 3\}$, $\epsilon_{\lambda\mu_b\lambda'}$ is the sign of this permutation and $\epsilon_p = \pm 1$ is the orientation sign of p.

For a unit vector n,

$$T_n(\varphi) = \sum_\lambda n_\lambda^2 \, T_\lambda(\varphi) + \sum_{\lambda < \lambda'} n_\lambda n_{\lambda'} \, \mathcal{O}^{\mathrm{od}}_{\lambda\lambda'}(\varphi).$$

A mixed source is a pair $\mathfrak{s} = (w, \mathsf{w})$ of real weights on the plaquettes and on the bonds, $\mathcal{O}^T(\mathfrak{s}) = \mathcal{O}(w) + Y[\mathsf{w}]$, $|\mathfrak{s}|_\infty = |w|_\infty + 8|\mathsf{w}|_\infty$.

Connected functions and natural sizes. For weights $w_1, \dots, w_n$ and $w_\lambda = \sum_a \lambda_a w_a$, $S_n^T = S^T_{n;K,\ell,\beta}(w_1, \dots, w_n) = \partial_{\lambda_1} \cdots \partial_{\lambda_n} \log \langle e^{\mathcal{O}(w_\lambda)}\rangle_{K,\ell,\beta}|_{\lambda=0}$; for functions f_a it is taken at $w_a = f_a \circ x$; $\langle X; Y\rangle$ denotes S_2^T, and $S_n^T[\mathfrak{s}]$ the same in a state $\mathfrak{s}$. For a function f on $\mathbb{R}^3$, $f^{\langle\ell\rangle}$ is its periodisation. We write $\mathsf{u}_a = \mathsf{b}^{-1}|w_a|_\infty$. Three bumps of width ρ, two of them at the near distance r and the third at the far distance d, are measured against $\mathrm{nat}_3 = \rho^9 r^{-3} \mathsf{d}^{-6} \mathsf{b}^{-3} \prod_{a \le 3} |w_a|_\infty$, and two such pairs against $\mathrm{nat}_4 = \rho^{12} r^{-6} \mathsf{d}^{-6} \mathsf{b}^{-4} \prod_{a \le 4} |w_a|_\infty$ (at functions $\|f_a\|_\infty$ replaces $|w_a|_\infty$); a kernel of three points is measured against $\mathsf{b}^{-3} r_*^{-3} \mathsf{d}_*^{-6}$ and a kernel of two points against $\mathsf{b}^{-2}|x - y|^{-6}$. These are the natural sizes.

Placed weights. Real weights w_1, w_2 are placed at (ρ, r) if there are centres x_a with $\Sigma(w_a) \subset \mathbb{B}(x_a, \rho)$ (closed balls), $\operatorname{dist}(\Sigma(w_1), \Sigma(w_2)) \ge r$ and $|x_1 - x_2| \le 8r$. Weights w_1, w_2, w_3 are placed at (ρ, r, d) if besides $\Sigma(w_3) \subset \mathbb{B}(x_3, \rho)$, $\operatorname{dist}(\Sigma(w_a), \Sigma(w_3)) \ge \mathsf{d}$ and $|x_a - x_3| \le 18\mathsf{d}$ for $a = 1, 2$. Weights $w_1, \dots, w_4$ form two pairs at (ρ, r, d) if (w_1, w_2) and (w_3, w_4) are placed at (ρ, r) and $\operatorname{dist}(\Sigma(w_a), \Sigma(w_b)) \ge \mathsf{d}$, $|x_a - x_b| \le 18\mathsf{d}$ for $a \le 2 < b$. A set β of sources crosses if it meets both clusters, $\{1, 2\}$ and $\{3\}$ (or $\{3, 4\}$), and $c(\beta)$ is the number of whole pairs in β.

Free objects. $\mathbb{K}_3(u) = (4\pi|u|^3)^{-1}(3\hat u\hat u^{\mathsf T} - 1) = (4\pi)^{-1}\nabla\nabla|u|^{-1}$, $\hat u = u/|u|$; $k_3(x, y, z) = \operatorname{tr} \mathbb{K}_3(x - y)\mathbb{K}_3(y - z)\mathbb{K}_3(z - x)$; $k_4(x_1, \dots, x_4) = \frac12 \sum_\varsigma \operatorname{tr} \prod_l \mathbb{K}_3(x_{\varsigma(l)} - x_{\varsigma(l+1)})$, the sum over the six cyclic orders of $\{1, 2, 3, 4\}$ (three rings, each once); $\mathcal{K}_{\lambda\mu}(\varphi_1, \varphi_2) = \iint \varphi_1(x)\varphi_2(y)[\mathbb{K}_3(x - y)_{\lambda\mu}]^2 \, \mathrm{d}x \, \mathrm{d}y$; and for a unit vector n and $X \neq 0$

$$\mathsf{t}^\circ(n, X) = \tfrac32 \mathsf{b}^{-2}[\langle n, \mathbb{K}_3(X)^2 n\rangle - \tfrac13 \operatorname{tr} \mathbb{K}_3(X)^2] = \frac{3}{32\pi^2} \, \mathsf{b}^{-2}|X|^{-6}(3(n \cdot \hat X)^2 - 1).$$

C_K^{free} is the free curvature kernel on the bare plaquettes of the lattice (ℓ, K), $\mathcal{C}_{(j)}$, $1 \le j \le K$, are its slices by level, and $\upsilon_j = (a_{K-j+1} x_{j-1})^{-1}$. Over a base level $k_\flat$ we put $\mathcal{S}_i = \sum_{k_\flat < j \le i} \upsilon_j \mathcal{C}_{(j)}$; for a nonempty set β of sources with $m = |\beta|$, $R_{i;\beta} = \sum w_{a_1} \mathcal{S}_i w_{a_2} \cdots \mathcal{S}_i w_{a_m}$, the sum over the orderings $(a_1, \dots, a_m)$ of β, the weights acting as diagonal matrices; $q_{i;\beta}(F) = \frac{1}{2a_K} \sum_{p,q} R_{i;\beta}(p, q)\langle F_p, F_q\rangle$ for $\mathfrak{su}(2)$-valued functions F of the plaquettes; and $g_{i;\beta} = \frac32 \sum_\varsigma \operatorname{Tr} \prod_l (w_{\varsigma(l)} \mathcal{S}_i)$, the sum over the cyclic orders ς of β. At $\beta = \{1, 2\}$ we write $q_n = q_{n;12}$, so $q_n(F) = a_K^{-1} \sum_{p,q} w_1(p) \mathcal{S}_n(p, q) w_2(q)\langle F_p, F_q\rangle$, and $\mathcal{Q}_i = q_{i+1} - q_i$. The lattice ring of three weights is $3 \operatorname{Tr} \prod_a (w_a \mathsf{b}^{-1} C_K^{\mathrm{free}})$. Finally $\chi_x^\rho = \frac{3}{\pi\rho^3}(1 - |\cdot - x|/\rho)_+$ is the radial tent of radius ρ and mass 1.

The renormalisation group. We use the following signs. The level j is the lattice of spacing a_{K-j}, level 0 being the bare lattice; for a length ϱ, $\hat\varrho_j = \varrho/a_{K-j}$. The chain classes $\mathsf{Cc}_n(r_n)$ have the radii $r_K = 0$, $r_i = L^{-1}r_{i+1} + \frac{11}{10}\mathsf{c}_\mathfrak{h}\delta_i$ ($\mathsf{c}_\mathfrak{h} \ge 1$); for $A \in \mathsf{Cc}_n(r_n)$ we put $\tilde U = e^{i\eta_n A}$ and $\mathsf{ang}_p^{[n]}(A) = \eta_n(\mathrm{d}A)(p)$, and $\mathbf{1}$ is the trivial configuration. The witness depth is $s(\varrho) = \min\{s \ge 0 : L^s\varrho \ge C_W(s)\}$, where $C_W(s) = C_{W,0}A_0^2(\log[L^{s+1}(g^{-2}+\bar B)])^{2\mathfrak{p}_0}$, with the room constant $C_{W,0} = \max\{1, 4L^2M/A_0^2\}$, so that $C_{W,0}A_0^2 \ge 4L^2M$ (equations YM3_I:(2320) and YM3_I:(2322)); the working level is $\mathsf{k}(\varrho) = \max\{K - s(\varrho), 2\}$; $Y_s = (s+1)\log L + \log(g^{-2}+\bar B)$, so that $C_W(s) = C_{W,0}A_0^2Y_s^{2\mathfrak{p}_0}$, $Y_\varrho = Y_{s(\varrho)}$, and Lm_s is a constant times $Y_s^{5\mathfrak{p}_0}$.

For sources $w_1, \dots, w_p$ the weighted progression $E_\lambda^{(j)}$, $1 \le j \le K$, is the sequence of families of local terms of level j that the renormalisation steps make with the source w_λ, for λ in a neighbourhood of 0; $E^{(j)} = E_0^{(j)}$ and $E_\beta^{(j)} = \partial_\beta E_\lambda^{(j)}|_0$. The relative exponent is

$$D_i^\lambda(U) = \mathcal{O}(w_\lambda)(U) + \sum_{i' \le i} \mathsf{Sum}[E_\lambda^{(i')} - E^{(i')}](U), \qquad \mathsf{Sum}[\mathsf{f}](U) = \sum_{(X,z)} \mathsf{f}(X,U,z),$$

the sum over the domains X of the level and their base points z; $G_{i;\beta}(A) = \partial_\beta D_i^\lambda(e^{i\eta_i A})|_0$, the pair functional on configurations is defined by $\mathsf{F}_n(e^{i\eta_n A}) = G_{n;12}(A)$, the jets are $\mathsf{J}_n = G_{K;[n]}(0)$, and $\Delta[\mathsf{f}] = \mathsf{Sum}[\mathsf{f}](\tilde U) - \mathsf{Sum}[\mathsf{f}](\mathbf{1})$. A family f of level i has the norms $\|\mathsf{f}\|_{i,\kappa'} = \sup|\mathsf{f}(X,P,z)|e^{\kappa' d_i(X)}$, d_i the linear size of a domain, and $\lambda_{\mathrm{out}} = (1+4\beta_{\mathrm{s}})\kappa$; π_i is the partition into cubes of side Ma_{K-i}; for a weight w_a, Cr_a is the set of the corners of the plaquettes of $\operatorname{supp} w_a$, $\mathsf{N}_i(w_a)$ the number of the cubes of π_i that hold a site of Cr_a, and D_i^{ab} the least sup distance of Cr_a from Cr_b in level-i units.

The step map $\mathsf{St}^{[j]}$ is the renormalisation step of the height j, viewed as a function from bare arrays $\mathfrak{f} = ((G^{(i)}, e_i)_{i\le j}, (H))$ of level j to families of level $j+1$; $N(\mathfrak{f})$ is the size of an array and $\mathfrak{n}[\mathfrak{f}] \le N(\mathfrak{f})$ the size of the family that it puts in the fourth place of the one-step theorem; $N_\star^{[j]}$ and $\theta^{[j]} = K_{\mathbf{M}}(1+\beta_{\mathrm{s}})\delta_j/\alpha_0$ are numbers of the step j, and W_w is the weight member of a weight w. Further construction objects are defined where they enter.

Definition 4.1 (Ceilings, thresholds, numbers)**.**

(1) The ceiling (Γ1): $0 < g \le \gamma_{\mathrm{jt}}$; $g^{-2} \ge \max\{x_{\mathrm{J}}, e^{2\mathfrak{p}_0}, e^{y_\mathfrak{p}}\}$; $\delta(g^{-2}) \le \min\{1/2c_{\mathrm{rt}}, 1/c_\epsilon, 1/2C_{\mathrm{cu}}\}$; $\theta(g^{-2}) := K_{\mathbf{M}}(1+\beta_{\mathrm{s}})\delta(g^{-2})/\alpha_0 \le \tau^\natural$. The ceiling (Γ2): $0 < g \le \gamma_W^{\mathrm{sd}}$, $g^{-2} \ge x_{\mathrm{J}}$, $\log g^{-2} \ge \mathsf{Y}_{\mathrm{lt}}(\frac12)$; data under (Γ2) have $K \ge 2$. (Γ) is (Γ1) together with (Γ2). Here γ_{jt} is defined in eq. 4.iv, γ_W^{sd} in equation YM3_I:(2327), and x_{J} in [7, eq. 228]; [7, Prop. B.21] verifies the required actual-step hypotheses for every finite number of sources. The least-torus threshold Y_{lt} is specified below; $y_\mathfrak{p}, c_{\mathrm{rt}}, c_\epsilon, C_{\mathrm{cu}}, \tau^\natural$ are numbers formed from the constants in Lems. 4.3, 4.6 and 4.14.

For $0 < \eth < 1$, put $c' = (1-\eth)c_\star$ and $C_{\mathrm{sd}} = 8L^6(\mathsf{C}_{\mathrm{maj}}^\Sigma + 2 + c_{\mathrm{low}} + 9/2)$. Choose $\mathsf{Y}_{\mathrm{lt}}(\eth) \ge 1$ so that all the history thresholds in Prop. 4.54 hold above it and

$$2C_{\mathrm{sd}}y[MR(e^y)]^3 \le \tfrac12 c'A_0^2y^{2\mathfrak{p}_0} \qquad (y \ge \mathsf{Y}_{\mathrm{lt}}(\eth)). \tag{4.i}$$

Such a finite threshold exists: $R(e^y) < Ly^r$ and $2\mathfrak{p}_0 > 3r+2$, as in the proof of Lem. YM3_I:126.3. On the least admissible side that proof gives

$$\mathsf{B}(x_K)\ell_g^3 \le C_{\mathrm{sd}}y_K[MR(e^{y_K})]^3.$$

Thus $\log g^{-2} \ge \mathsf{Y}_{\mathrm{lt}}(\eth)$ implies $2\mathsf{B}(x_K)\ell_g^3 \le c'p_K^2/2$, the stronger small-side condition in Prop. 4.54.

Since δ decreases on $[e^{2\mathfrak{p}_0}, \infty)$, each member is a condition "$g \le$ a number formed from the constants". ($\Gamma^{(4)}$) is (Γ) with $c_\epsilon^{(4)}$ of Lem. 4.14 in the place of c_ϵ. As $c_\epsilon \le c_\epsilon^{(4)}$ (Lem. 4.14(a), where C_{cu}, c_{cu} and the numbers $c_\epsilon^{(n)}$ are formed), (Γ1) with $c_\epsilon^{(4)}$ in the place of c_ϵ contains (Γ1), and ($\Gamma^{(4)}$) contains (Γ): every g under ($\Gamma^{(4)}$) satisfies (Γ). For the bond forms we use

the ceiling $\gamma_{\rm N} = x_{\rm N}^{-1/2}$, where $x_{\rm N}$ is the least real number $\geq x_{\rm jt} = \gamma_{\rm jt}^{-2}$ such that every $t \geq x_{\rm N}$ satisfies $t \geq \max\{x_{\rm J}, e^{2\mathfrak{p}_0}, e^{y_{\mathfrak{p}}}\}$, $\delta(t) \leq \min\{1/2c_{\rm rt}, 1/c_\epsilon\}$, $\theta(t) = K_{\mathbf{M}}(1+\beta_{\rm s})\delta(t)/\alpha_0 \leq \tau^\natural$, $\mathfrak{b}(t) \geq 1$ and $c_{\rm A}\alpha_0 \geq 5M\mathsf{c}_\mathfrak{h}\delta(t)(1+\log t)$, with $c_{\rm A}$ of Lem. 4.8.

So $\gamma_{\rm N} \leq \gamma_{\rm jt}$, and every $0 < g \leq \gamma_{\rm N}$ satisfies (Γ1) but for its member $\delta(g^{-2}) \leq 1/2C_{\rm cu}$.

(2) $r_1(\mathsf{b}) > 0$ is the number of Lem. 4.6, formed from the constants and b; it does not increase in b. $\mathsf{d}_4(g) > 0$ is the number fixed in Prop. 4.19, formed from the constants and g, with $\mathsf{d}_4(g) \leq \min\{e^{-2(15\mathfrak{p}_0+2)}, r_1(\bar{\mathsf{b}})\}$; $\mathsf{d}_5(g) = \min\{2\mathsf{d}_4(g), \min\{1, 16r_1(\bar{\mathsf{b}})\}, \frac{1}{32}\ell_g\}$. For two pairs, $\mathsf{d}_4^{(4)}(g)$ is fixed in the same Proposition, and $\mathsf{d}_5^{(4)}(g) = \min\{2\mathsf{d}_4^{(4)}(g), \frac{1}{48}\ell_g\}$.

(3) $\bar{\mathsf{M}}^{(24)} = 1 + 2L^{24}\mathsf{C}_{\rm ag}(25, \frac{1}{2})$, where $\mathsf{C}_{\rm ag}$ is defined by eq. 4.xxxii; $\Lambda_3 \geq 1$ is the number of Lem. 4.21; $C_{\rm lat}$ is the absolute constant in equations YM3_I:(2245)–(2246), comparing $C_K^{\rm free}$ with $\mathbb{K}_3$ on every side; $C^{(3)} = 3 \cdot 81^3(1+3C_{\rm lat})^2C_{\rm lat} + 2.1 \cdot 10^7$, $C^{(4)} = 3[4 \cdot 81^4C_{\rm lat}(1+3C_{\rm lat})^3 + 1.49 \cdot 10^8]$, $C_{\rm Rs} = 7400 + 729C_{\rm lat}(2+3C_{\rm lat})$, the constants of the lattice sums of Lem. 4.22; $C_{\rm PS} = 62 + 1089C_{\rm lat}(2+3C_{\rm lat})$ and $C_{\rm PS}^{\rm od} = 62 + 729(2+5C_{\rm lat})(1+C_{\rm lat})$, the constants of the lattice sums at two balls on every torus (Lem. 4.33); $C_3^{\rm tr3}(g)$, $C_4^{\rm tr3}(g)$, $C_2^{\rm tr}(g)$ are the numbers of Prop. 4.28, formed from the constants and g.

(4) With the number $C(g)$ of Prop. 4.19: $\epsilon^{\rm nar}(g,t) = C(g)(\log\frac{1}{t})^{39\mathfrak{p}_0+2}t^{1/2}$, $\epsilon_2^{\rm nar}(g,t) = C(g)(\log\frac{1}{t})^{19\mathfrak{p}_0}t^{1/2}$, $\epsilon_4^{\rm nar}(g,t) = C(g)(\log\frac{1}{t})^{47\mathfrak{p}_0+2}t^{1/2}$; and

$$\epsilon_{\rm A}(g,t) = 512\big[\epsilon^{\rm nar}(g,\tfrac{t}{2}) + 3C^{(3)}(5t/\ell_g)^3 + 13\Lambda_3^3(\bar{\mathsf{M}}^{(24)})^3\bar{\mathsf{b}}^3t^{3/2}\big],$$

$$\mathsf{E}_3(g,t) = 512\big[\epsilon^{\rm nar}(g,\tfrac{t}{2}) + 3C^{(3)}(5t/\ell_g)^3 + (13\Lambda_3^3(\bar{\mathsf{M}}^{(24)})^3 + C_3^{\rm tr3}(g))\bar{\mathsf{b}}^3t^{3/2}\big],$$

$$\epsilon_{\rm P}(g,t) = 64\big[\epsilon_2^{\rm nar}(g,\tfrac{t}{2}) + 3\Lambda_3^2(\bar{\mathsf{M}}^{(24)})^2\bar{\mathsf{b}}^2(\tfrac{t}{2})^{3/2} + \tfrac{3}{2}C_{\rm Rs}(t/2\ell_g)^3\big],$$

$$\mathsf{E}_2(g,t) = \epsilon_2^{\rm nar}(g,t) + \big[3\Lambda_3^2(\bar{\mathsf{M}}^{(24)})^2 + C_2^{\rm tr}(g)\big]\bar{\mathsf{b}}^2t^{3/2} + \tfrac{3}{2}C_{\rm Rs}(t/\ell_g)^3, \qquad \epsilon_\sigma(g,t) = 64\sup_{0<t'\leq t}\mathsf{E}_2(g,\tfrac{t'}{2}).$$

So $\epsilon_{\rm A}, \mathsf{E}_3 \leq C(g)[(\log\frac{1}{t})^{39\mathfrak{p}_0+2}t^{1/2} + (t/\ell_g)^3]$ and $\epsilon_{\rm P}, \epsilon_\sigma \leq C(g)[(\log\frac{1}{t})^{19\mathfrak{p}_0}t^{1/2} + (t/\ell_g)^3]$ for small t; all of them are functions of (g,t) alone.

(5) *The threshold $r_\alpha(g,\mathsf{b})$ of Thm. K.* For a label b and $0 < r \leq L^{-4}$ consider every first-passage datum (K,β) at g with $a_K\beta = \mathsf{b}$, $128a_K \leq r$ and $s(\frac{r}{2}) \leq K$, and the near pair $w_{1,2} = \chi^{r/8}_{c\pm\frac{r}{2}e_\lambda} \circ x$ of that Theorem, a test pair of the scale $\frac{r}{2}$; n_1 and n_0 are the least levels with $a_{K-n_1} \geq (\frac{r}{2})^{3/4}$ and $a_{K-n_0} \geq r^{1/2}$. The conditions are:

(α1) $\mathsf{k}(\frac{r}{2}) < n_1 < n_0 < K$, $2(r+a_K) \leq a_{K-n_0}$ and $\frac{27}{8M^3}(r/2a_{K-n_1})^3 \leq 1$;

(α2) for $\mathsf{k}(\frac{r}{2}) \leq i < n_1$, $n_0 \leq n < K$ and $m = n-i-1$: $(4\mathsf{h}_i+4)a_{K-i-1} \leq \ell_g$ and $7\mathsf{h}_iL^{-2m}r_n \leq \delta_i/2\Lambda_i$, where h_i and Λ_i are the numbers of Prop. 4.13 for the near pair at (i,n), formed with $R_w = \frac{5}{4}\hat{r}_{i+1} + 1$ and $\bar{\mathsf{e}} = 2\bar{\mathsf{w}}_1\bar{\mathsf{w}}_2\mathsf{E}_i$, $\bar{\mathsf{w}}_a = 81\hat{r}_i^3|w_a|_\infty/a_{K-i}$;

(α3) $\frac{r}{2} \leq \min\{r_0(\mathsf{b}), r_1(\mathsf{b})\}$;

(α4) $s(\frac{r}{2}) \geq 3$ and $\mathsf{Lm}_{s(r/2)} \leq L^{s(r/2)-3}$;

(α5) $2\sqrt{3}\,\delta_{\rm tp}L(L-1)M(\log\frac{2}{r})^{2\mathfrak{p}_0} \geq 3\log L + \log 2$, so that Lem. 4.17(b) applies at the level $\mathsf{k}(\frac{r}{2})$ to two plaquettes of the two supports of the near pair.

$r_\alpha(g,\mathsf{b})$ is the largest number $r_\star \in (0, L^{-4}]$ such that (α1)–(α5) hold at every $0 < r \leq r_\star$ and every such datum. It is positive. (α1) holds for $r \leq L^{-4}$ at every datum. (α2), (α4) and (α5) hold for r below a number formed from the constants and g: by the working level (Lem. 4.3(c)) and step 6 of the proof of Prop. 4.13, $\mathsf{h}_i \leq C(1+Y)^{2\mathfrak{p}_0}$ and $\Lambda_i \leq C(1+Y)$ with $Y = Y_{r/2} \leq 2\log\frac{2}{r} + C(g)$; $r_n \leq \frac{8}{5}\mathsf{c}_\mathfrak{h}(\frac{5}{4})^{1/2}L^{(n-i)/2}\delta_i$; and $L^{-(n-i)} = a_{K-i}/a_{K-n} < (\frac{r}{2})^{3/4}r^{-1/2} \leq r^{1/4}$; none of these bounds depends on b or on K. (α3) holds for $\frac{r}{2} \leq \min\{\inf_{\mathsf{b}'} r_0(\mathsf{b}'), r_1(\bar{\mathsf{b}})\}$, the infimum over $g^{-2} < \mathsf{b}' \leq \bar{\mathsf{b}}$ being positive by (B.1) and r_1 not increasing. So $r_\alpha(g,\mathsf{b}) \geq r_\alpha^-(g)$ for all $\mathsf{b} \in (g^{-2}, \bar{\mathsf{b}}]$, with a positive number $r_\alpha^-(g)$ formed from the constants and g.

(6) *The ceiling* (Γ3) *and the threshold* $r_\alpha^{\mathcal{W}}(g,\mathsf{b})$. (Γ3) asks: $c_{\mathrm{A}}\alpha_0 \geq 8MB_3\varepsilon(t)(1+\log t)$ for all $t \geq g^{-2}$; $g \leq 2C_{\mathrm{A}}C_{\mathrm{E},3}M^2B_3^2A_0^2/C_{\mathrm{E},1}\alpha_0^2$; and $7488B_3\varepsilon(g^{-2}) \leq \epsilon_{\mathrm{G}}$, $C_{\mathcal{W}}\varepsilon(g^{-2}) \leq \frac{1}{120}$ with $C_{\mathcal{W}} = \max\{720C_{\mathrm{G}}^\sharp, 2\}B_3$. Here c_{A}, C_{A} and $C_{\mathrm{E},1}$ are the numbers of Lem. 4.8 and of its proof, $C_{\mathrm{E},3}$ is the number in [7, Lem. B.10] at $q=2$, $2B_3\varepsilon_n$ is the radius at which the small-field backgrounds are regular (Lem. 4.20(d)), and ϵ_{G}, $C_{\mathrm{G}}^\sharp$ are the numbers of the lemma on a regular gauge on a cube, which precedes Lem. 4.36.

As $\varepsilon(t)(1+\log t) \to 0$ and ε decreases on $[e^{2\mathfrak{p}_0}, \infty)$, each member is a condition "$g \leq$ a number formed from the constants and β_{H}". $r_\alpha^{\mathcal{W}}(g,\mathsf{b})$ is the largest number $r_\star \in (0, r_\alpha(g,\mathsf{b})]$ such that, at every $0 < r \leq r_\star$ and every datum as in (5), the numbers h_i and Λ_i of (α2), formed with $C_{\mathcal{W}}\varepsilon_n$ in the place of r_n, satisfy for $\mathsf{k}(\frac{r}{2}) \leq i < n_1$, $n_0 \leq n \leq K$ and $m = n-i-1$: $(4\mathsf{h}_i+4)a_{K-i-1} \leq \ell_g$, $7\mathsf{h}_iL^{-2m}C_{\mathcal{W}}\varepsilon_n \leq \delta_i/2\Lambda_i$ and $(\mathsf{h}_i+1)L^{-m} \leq 3$, and besides $Lr^{1/2} \leq (2\bar{\mathsf{M}}^{(24)})^{-1/24}$.

The bounds of (5), with $\varepsilon_n = (A_0/A_1)\delta_n$, give $r_\alpha^{\mathcal{W}}(g,\mathsf{b}) \geq r_\alpha^{\mathcal{W}-}(g)$, a positive number formed from the constants, β_{H} and g.

■

4.1.2. *The three lengths and the comparison.* Point splitting asks whether two nearby scalar insertions determine a tensor insertion when tested from a positive distance away. The pair separation is r, the distance to the remaining insertion is d, and the radius of each approximating bump is ρ. These lengths perform different jobs. The free curvature kernel is singular at distance r; the distant insertion sees its direction through a tensor coefficient; and ρ measures how accurately a bump samples a point. The estimates must keep all three lengths visible. An estimate with ρ fixed proportional to d would lose the regime $r/d \to 0$ that defines the point-splitting limit.

For three weights in the geometry of theorem G, the normalization is

$$\mathrm{nat}_3 = \rho^9 r^{-3} d^{-6} b^{-3} \prod_{a=1}^{3} |w_a|_\infty, \qquad \mathrm{nat}_4 = \rho^{12} r^{-6} d^{-6} b^{-4} \prod_{a=1}^{4} |w_a|_\infty \tag{4.ii}$$

for a pair with one spectator and for two separated pairs, respectively. Each bump contributes its volume. Each close pair contributes one curvature line of degree -3, and passage between the two clusters contributes two such lines. These powers explain which errors have to be uniform as ρ/r and r/d decrease. The three-point and four-point conclusions retain different coupling thresholds and different numbers of source derivatives.

The free calculation identifies the tensor and fixes its normalization. Put

$$\mathbb{K}_3(u) = \frac{3\hat{u}\hat{u}^{\mathsf{T}} - 1}{4\pi|u|^3}, \qquad D(n) = 3n \otimes n - 1.$$

The connected three-point kernel is a cyclic product of three $\mathbb{K}_3$ kernels. At the close pair its singular factor is $(4\pi r^3)^{-1}D(n)$. The normalized pair therefore carries the same quadratic form as the traceless tensor in direction n. For $n = e_\lambda$, the tensor weight is $\delta_{\lambda(p)\lambda} - 1/3 = D(e_\lambda)_{\lambda(p)\lambda(p)}/3$. The factor $2\pi br^3/3$ in theorem K makes these two quadratic forms agree. This equality concerns the explicit quadratic terms; the rest of the argument controls their difference in the interacting measure.

4.1.3. *The source derivatives through the levels.* Introduce the close-pair sources and the tensor source in one weighted progression. Let $G_{n;\beta}$ denote its derivative at level n in the source set β. Using one progression is essential: the difference is then formed on the same configurations and with the same conditional measures. For the second source derivative, the chain rule gives the decomposition of lemma 4.7,

$$E_{12}^{(j+1)} = \mathsf{K}^{(j+1)} + \mathsf{R}^{(j+1)}.$$

Here K is obtained by differentiating the step twice in the two original weight members. It supplies the free quadratic kernel. The remaining terms contain an earlier source derivative or the carried mixed derivative. Their norms gain the small factors in lemma 4.9. Both families vanish on a localization domain unless that domain meets both source supports. At levels finer than their separation, this forces a large domain and gives exponential decay.

Above that separation scale, locality alone no longer makes the derivative small. The induction instead compares it with an explicit quadratic form assembled from the curvature slices. In the notation of lemma 4.17, the slice sum obeys

$$\left|\sum_{j\in\mathcal{J}} \upsilon_j \mathcal{C}_{(j)}(p,q)\right| \le Cb^{-1}a_K^3 |x(p)-x(q)|_\infty^{-3}.$$

The flow estimate compares υ_j with b^{-1}, while the exponential decay of each slice makes the lacunary sum finite. The error in replacing the retained slices by the full free kernel has three parts: the change of the running coupling, the coarsest background covariance, and the omitted fine slices. Their separate bounds are retained in lemma 4.17.

For several sources, differentiating the exact conditional integral produces the partitions of the source set. The induction weights in lemma 4.18 are chosen so that a product over such a partition is bounded by the weight of the whole set. A whole close pair needs the factor $q^6u^3\max\{1,u^3\}$, where $q=\rho/r$ and u is r in the working-level units. Cutting that pair into two singletons costs $\hat\rho^6 = q^6u^6$. This explains the extra maximum: u may grow with a power of $\log(1/d)$, so treating it as bounded would invalidate the partition estimate. The resulting jet estimates are uniform in the two ratios in eq. 4.ii.

The jets evaluated at the identity are still not the full connected functions. Lemma 4.20 makes this distinction explicit. Stop each history either at its first large-field creation or at the terminal level if no such creation occurs. The resulting probability measure $\mathbb{P}$ is independent of the sources. If Y_β is the corresponding remainder derivative, then

$$S_n^T - \mathsf{J}_n = \sum_{\pi\in\mathrm{Part}([n])} \kappa_{\mathbb{P}}(\mathsf{Y}_\beta : \beta\in\pi). \tag{4.iii}$$

The bounds on moments of the stopping scale control every term of this finite sum. On the least torus, the first-creation estimates bound these moments uniformly in the cutoff. The dependence on the side is retained when this argument is used on a larger torus.

4.1.4. *Cancellation of the pair against the tensor.* The decisive estimate concerns

$$\Delta_n = (2\pi br^3/3)G_{n;12} - G_{n;T}$$

. The supplied proof first treats chain classes, then actual small-field backgrounds. On a regular cube, a real gauge makes the potential, its first lattice differences and its current small. The lattice Hodge identity applied to a cutoff of the curvature then gives its C^{β_H} estimate. These are estimates for lattice forms at their stated spacing. Their role is to replace the curvature at the two nearby supports by its value at their common centre, with an error tending to zero as r shrinks.

Separate the explicit kernel families created before the pair has been averaged out from all remaining families. The first part is the frozen part in lemma 4.36; its leading quadratic form cancels the tensor weight by the identity involving $D(e_\lambda)$. The remainder has small norm. The exact one-step averaging identity does not increase oscillation modulo constants, so this cancellation can be carried to coarser levels. The source proves first an error of size $r^{\beta_H/36} + (a_K/r)^{1/9}$ for this oscillation.

To attach the distant source, the tower identity of lemma 4.40 produces a covariance of Δ_n with the spectator and terms crossing the two clusters. Bound the covariance in two ways. Its oscillation estimate contains the small pair-minus-tensor error. Its family norm contains the exponential cost

of reaching the distant support. Taking the smaller estimate, and then its geometric-mean bound, retains both advantages. Summing over levels now gives the square root of the former error. This is the source of $\beta_H/72$ and $1/18$ in theorem K; those exponents do not follow from the free contraction alone.

Finally integrate over the pair centre and sum the spectator pieces. Replacing the smeared tensor weight by φ_1 uses its Lipschitz continuity. Consequently the constant depends on $d\,\mathrm{Lip}(\varphi_1)/|\varphi_1|_\infty$ for a nonzero weight. At fixed r, the scalar three-point function has its cutoff limit; the comparison then determines the tensor limit. Only after that limit is taken does $r \to 0$ at fixed d give the point-splitting identity. The lattice error records why the reverse order is unavailable. The kernel statements likewise retain their order: bounds on arbitrarily narrow bumps give a Borel kernel and its almost-everywhere bounds; continuity supplied by reconstruction is the additional fact that permits evaluation at every separated configuration.

4.1.5. *Principal results.*

Theorem G (Three points at two scales)**.** Let g satisfy (Γ).

(A.1) *At a cutoff, on the least torus.* Let (K,β) be a first-passage datum at g, $\mathsf{b} = a_K\beta$, $\ell = \ell_g$; let $0 < \rho \le r \le \mathsf{d} \le \mathsf{d}_4(g)$ and $s(\rho) \le K$, and let w_1, w_2, w_3 be real weights placed at (ρ, r, d), for instance $w_a = f_a \circ x$ with bounded real Borel functions f_a. Then

$$\Big|S_3^T(w_1,w_2,w_3) - 3\,\mathrm{Tr}\prod_a (w_a \mathsf{b}^{-1} C_K^{\mathrm{free}})\Big| \le [\epsilon^{\mathrm{nar}}(g,\mathsf{d}) + 13\Lambda_3^3 \mathsf{M}'^3 \mathsf{b}^3 \mathsf{d}^{3/2}]\mathrm{nat}_3,$$

where $\mathsf{M}' = \int \alpha^{-24} d\mathbb{P} \le \bar{\mathsf{M}}^{(24)}$ (Lem. 4.20). The bound holds at every ratio $r/\mathsf{d} \in (0,1]$.

(A.2) *The kernel on the least torus.* Let b be a limit of labels of first-passage data at g (every $\mathsf{b} \in I(g)$ is one), and $\mathcal{D} = \{(x,y,z) \in T^3_{\ell_g} : 0 < |x-y| = r_* \le \mathsf{d}_* = \mathrm{dist}(z,\{x,y\}) \le \mathsf{d}_5(g)\}$. At every triple of $\mathcal{D}$, for every $0 < \rho < \min\{\frac14 r_*, r_1(\bar{\mathsf{b}})\}$ and all real Lipschitz f_1, f_2, f_3 with supports in balls of radius ρ inside the open balls of radius $\frac14 r_*$ about x, y, z,

$$\Big|S_3^{(\ell_g,\mathsf{b})}(f_1,f_2,f_3) - 3\mathsf{b}^{-3}\int k_3 \prod_a f_a\Big| \le \epsilon_{\mathrm{A}}(g,\mathsf{d}_*)\,\mathsf{b}^{-3} r_*^{-3}\mathsf{d}_*^{-6}\,\rho^9 \prod_a \|f_a\|_\infty.$$

Consequently $S_3^{(\ell_g,\mathsf{b})}$ has on $\mathcal{D}$ a Borel kernel s_3, and at every Lebesgue point of s_3 in $\mathcal{D}$, so almost everywhere on $\mathcal{D}$,

$$|\mathsf{s}_3 - 3\mathsf{b}^{-3}k_3| \le \epsilon_{\mathrm{A}}(g,\mathsf{d}_*)\,\mathsf{b}^{-3} r_*^{-3}\mathsf{d}_*^{-6},$$

uniformly in $r_*/\mathsf{d}_* \in (0,1]$. Choose the closest pair first and call it (x,y). The triple lies in $\mathcal{D}$ whenever the resulting far distance $\mathsf{d}_* = \mathrm{dist}(z,\{x,y\})$ is at most $\mathsf{d}_5(g)$.

(A.3) *Means at every triple.* At every triple of $\mathcal{D}$ and for every $0 < \rho < \min\{\frac14 r_*, r_1(\bar{\mathsf{b}})\}$ the mean of $\mathsf{s}_3 - 3\mathsf{b}^{-3}k_3$ over the product of the three balls of radius ρ about its points is at most $(\frac{3}{4\pi})^3 \epsilon_{\mathrm{A}}(g,\mathsf{d}_*)\mathsf{b}^{-3} r_*^{-3}\mathsf{d}_*^{-6}$ in modulus; with the radial tents χ^ρ about the three points in place of the normalised indicators of the balls the same holds with $(\frac{3}{\pi})^3$ in place of $(\frac{3}{4\pi})^3$.

(A.4) *Every state; limits and components.* (i) Let (K,β) be a first-passage datum at g, $\mathfrak{s}$ a state at (K,β), $0 < \rho \le r \le \mathsf{d} \le 1$, $\rho \le \frac12$, $s(\rho) \le K$, $\rho \le r_1(\mathsf{b})$, and let f_1, f_2, f_3 be bounded real Borel functions on $\mathbb{R}^3$ with $\mathrm{supp}\, f_a \subset \mathbb{B}(x_a,\rho)$, $\mathrm{dist}(\mathrm{supp}\, f_1, \mathrm{supp}\, f_2) \ge r$, $\mathrm{dist}(\mathrm{supp}\, f_a, \mathrm{supp}\, f_3) \ge \mathsf{d}$ for $a = 1,2$, all supports lying in one ball of radius at most $\frac18 \ell_g - 1$. Then

$$|S_3^T[\mathfrak{s}](f_1,f_2,f_3) - S^T_{3;K,\ell_g,\beta}(f_1^{\langle \ell_g\rangle}, f_2^{\langle \ell_g\rangle}, f_3^{\langle \ell_g\rangle})| \le C_3^{\mathrm{tr3}}(g)\,\mathsf{b}^3\mathsf{d}^{3/2}\,\mathrm{nat}_3;$$

so (A.1) holds for $S_3^T[\mathfrak{s}]$ with $C_3^{\mathrm{tr3}}(g)\mathsf{b}^3\mathsf{d}^{3/2}$ more in the bracket. (ii) Let (K_j,β_j) be first-passage data at g with $K_j \to \infty$ and $a_{K_j}\beta_j \to \mathsf{b}$, ν_j a state at (K_j,β_j), and σ' a law on $\mathbb{R}^3$ with $S_3^T[\nu_j] \to S_3^{\sigma'}$ on separated triples. Assume that $S_3^{\sigma'}$ has a continuous kernel on distinct triples,

as it does for a periodic-Gibbs limit or one of its canonical components. Then that kernel s_3 satisfies, at every $(x,y,z)\in(\mathbb{R}^3)^3$ with $0<|x-y|=r_*\le \mathsf{d}_*=\operatorname{dist}(z,\{x,y\})\le \mathsf{d}_5(g)$,

$$|\mathsf{s}_3(x,y,z)-3\mathsf{b}^{-3}k_3(x,y,z)|\le \mathsf{E}_3(g,\mathsf{d}_*)\,\mathsf{b}^{-3}r_*^{-3}\mathsf{d}_*^{-6},$$

uniformly in r_*/d_*. (iii) Every Gibbs sequence at g that converges, with its limit, and the approximants of every component of a canonical decomposition of that limit, with the component, are such data along every subsequence on which the labels converge.

(A.5) *Gibbs sequences at couplings above the ceilings.* Let $0<g_0\le\gamma$ be a reference coupling with $g\le g_0$. There is an integer $z_+\ge 0$ formed from the constants, g_0 and g such that for every Gibbs sequence at g_0 that converges, and every subsequence along which its labels converge, to b_0 say, the continuous kernel s_3 of the limit, and that of every component of a canonical decomposition of it, satisfy at every triple with $0<r_*\le\mathsf{d}_*\le L^{-z_+}\mathsf{d}_5(g)$

$$|\mathsf{s}_3-3\mathsf{b}_0^{-3}k_3|\le \max_{0\le z'\le z_+}\mathsf{E}_3(g,L^{z'}\mathsf{d}_*)\;\mathsf{b}_0^{-3}r_*^{-3}\mathsf{d}_*^{-6}.$$

■□

Theorem H (A close pair seen from the coarser levels)**.** Let $0<g\le\gamma$ be a reference coupling, (K,β) a first-passage datum at g with its run, $\mathsf{b}=a_K\beta$, and $\ell\in\mathfrak{L}(g)$. For a level n and $A\in\mathsf{Cc}_n(r_n)$ put $\tilde U=e^{i\eta_n A}$.

(B.1) *A pair as wide as its distance.* There are numbers $\gamma_{\mathrm O},C,c,c'>0$ formed from the constants, and a number $r_0(\mathsf{b})>0$ formed from these and b, with $\inf\{r_0(\mathsf{b}'):g^{-2}<\mathsf{b}'\le\bar{\mathsf{b}}\}>0$, such that the following holds. Let $g\le\min\{\gamma_{\mathrm{jt}},\gamma_{\mathrm O}\}$; $0<r\le r_0(\mathsf{b})$; $s=s(r)$ with $3\le s\le K$ and $\mathsf{Lm}_s\le L^{s-3}$; $\vartheta\in(0,1)$; $(f_1,f_2)\in\mathfrak{T}_{T_\ell}(r;\vartheta)$ with $w_a=f_a\circ x\not\equiv 0$; $k_0=\max\{K-s,2\}$; and let q_n be formed over the base level k_0. Then for every $k_0<n\le K$ and every $A\in\mathsf{Cc}_n(r_n)$

$$|\mathsf{F}_n(\tilde U)-\mathsf{F}_n(\mathbf{1})-q_n(\mathsf{ang}^{[n]}(A))|\le\Big\{C\mathfrak{b}_n^2\delta_n+C(\log x_{k_0})^c\big[(r/\mathsf{b})^{1/2}+e^{-c'Y_s^{2\mathfrak{p}_0}}\big]\Big\}\,\hat r_n^3\,\mathsf{u}_1\mathsf{u}_2.$$

None of $\gamma_{\mathrm O},C,c,c'$ depends on ϑ, K, ℓ or the level.

(B.2) *A narrow pair.* Let g satisfy (Γ1); let $0<\rho\le r\le e^{-2\sqrt3}$, $s(\rho)\le K$, $\rho\le r_1(\mathsf{b})$; let $w_1,w_2\not\equiv 0$ be real weights placed at (ρ,r), and let q_n be formed over the base level $\mathsf{k}(r)$. Then for every $\mathsf{k}(r)\le n\le K$ and every $A\in\mathsf{Cc}_n(r_n)$

$$|\mathsf{F}_n(\tilde U)-\mathsf{F}_n(\mathbf{1})-q_n(\mathsf{ang}^{[n]}(A))|\le\big[C_{\mathrm N}\mathfrak{b}_n^2\delta_n+\epsilon'_{\mathrm N}(r)\big]\Big(\frac{\rho}{r}\Big)^6\hat r_n^3\,\mathsf{u}_1\mathsf{u}_2,\qquad \epsilon'_{\mathrm N}(r)=C'_{\mathrm N}Y_r^{15\mathfrak{p}_0+2}\big[g\,r^{1/2}+e^{-Y_r^{2\mathfrak{p}_0}/\sqrt3}\big].$$

$C_{\mathrm N},C'_{\mathrm N}$ are formed from the constants, $\epsilon'_{\mathrm N}$ depends on g and r alone; nothing in the bracket depends on ρ, K, ℓ, the position, the direction, the sign or the shape of the weights, and $\epsilon'_{\mathrm N}(r)(\log\frac1r)^N\to0$ as $r\to0$ for every N.

(B.3) *The explicit form.* In both cases $\sup_{A\in\mathsf{Cc}_n(r_n)}|q_n(\mathsf{ang}^{[n]}(A))|\le C\,\mathfrak{b}_n^2(\rho/r)^6\hat r_n^3\mathsf{u}_1\mathsf{u}_2$, with $\rho=r$ in (B.1).

■□

Theorem I (The pair against the plane fields)**.** Let g satisfy (Γ), let b be as in (A.2), and let s_3 be the kernel of (A.2). For points c,z put $\mathsf{d}=|c-z|$, let $0<r\le\frac12\mathsf{d}$, $\mathsf{d}\le\frac{16}{17}\mathsf{d}_5(g)$, $\bar\epsilon_{\mathrm A}=\sup\{\epsilon_{\mathrm A}(g,t):\frac34\mathsf{d}\le t\le\frac{17}{16}\mathsf{d}\}$, and write $(c\pm\frac r2n,z)$ for the triple $(c+\frac r2n,c-\frac r2n,z)$; e_λ is the unit vector of the direction λ, and M_ρ denotes the mean over the balls of radius ρ about the arguments.

(C.1) *Against the free harmonic.* (i) For almost every $(c,h,z)\in(T_{\ell_g})^3$, with $h=rn$, $r=|h|$ in the range above,

$$\big|\tfrac{2\pi}{3}\mathsf{b}r^3\mathsf{s}_3(c\pm\tfrac r2n,z)-\mathsf{t}^\circ(n,c-z)\big|\le\big[12\bar\epsilon_{\mathrm A}+2.7(r/\mathsf{d})^2\big]\mathsf{b}^{-2}\mathsf{d}^{-6}.$$

(ii) For every (c,n,r,z) in the range above the same holds with $\mathsf{s}_3(c\pm\frac{r}{2}n,z)$ replaced by $\limsup_{\rho\to0}$ of its mean $\mathsf{M}_\rho\mathsf{s}_3(c\pm\frac{r}{2}n,z)$; and, for every $0<\rho<\min\{\frac14 r, r_1(\bar{\mathsf{b}})\}$, with $\mathsf{s}_3(c\pm\frac{r}{2}n,z)$ replaced by $\int \mathsf{s}_3\,\chi^\rho_{c+\frac{r}{2}n}\otimes\chi^\rho_{c-\frac{r}{2}n}\otimes\chi^\rho_z$ and $7900\rho/r$ added in the bracket. (iii) For a law σ' as in (A.4)(ii), in particular for every periodic-Gibbs limit at g and every component of its canonical decomposition, the bound of (i) holds for its continuous kernel at every (c,r,z) and every unit vector n, with $12\sup\{\mathsf{E}_3(g,t):\frac34\mathsf{d}\le t\le\frac{17}{16}\mathsf{d}\}$ in place of $12\bar\epsilon_{\mathrm{A}}$.

(C.2) *The plane laws.* (i) At a cutoff: let (K,β) be a first-passage datum at g, $\mathsf{b}=a_K\beta$, $0<\rho\le\mathsf{d}\le\mathsf{d}_4(g)$, $s(\rho)\le K$, $\vartheta\in(0,1]$, and let φ_1,φ_2 be real Lipschitz functions on T_{ℓ_g} with $\operatorname{supp}\varphi_a\subset\mathbb{B}(x_a,\rho)$, $\operatorname{Lip}\varphi_a\le\|\varphi_a\|_\infty/\vartheta\rho$, the balls $\mathbb{B}(x_a,\rho)$ at a distance $\ge\mathsf{d}$ and $|x_1-x_2|\le18\mathsf{d}$; no sign condition. Then for all λ,μ

$$|\langle\mathcal{O}_\lambda(\varphi_1);\mathcal{O}_\mu(\varphi_2)\rangle_{K,\ell_g,\beta}-\tfrac32\mathsf{b}^{-2}\mathcal{K}_{\lambda\mu}(\varphi_1,\varphi_2)|\le\Big[\epsilon_2^{\mathrm{nar}}(g,\mathsf{d})+3\Lambda_3^2\mathsf{M}'^2\mathsf{b}^2\mathsf{d}^{3/2}+\tfrac32 C_{\mathrm{Rs}}\Big(\frac{a_K}{\vartheta\rho}+\Big(\frac{\mathsf{d}}{\ell_g}\Big)^3\Big)\Big]\Big(\frac{\rho}{\mathsf{d}}\Big)^6\mathsf{b}^{-2}\|\varphi_1\|_\infty\|\varphi_2\|_\infty.$$

(ii) Let (K_j,β_j) be first-passage data at g with $K_j\to\infty$, $a_{K_j}\beta_j\to\mathsf{b}$, and $\Omega_{\mathrm{P}}=\{(x,y)\in T_{\ell_g}^2:0<|x-y|<2\mathsf{d}_4(g)\}$. There is a subsequence along which the nine forms $\langle\mathcal{O}_\lambda(\varphi_1);\mathcal{O}_\mu(\varphi_2)\rangle_{K_j,\ell_g,\beta_j}$ converge for all real Lipschitz φ_1,φ_2 with $\operatorname{supp}\varphi_1\times\operatorname{supp}\varphi_2\subset\Omega_{\mathrm{P}}$; the nine limits $T_{\lambda\mu}$ along one such subsequence are a limit point of the plane laws. Every limit point has Borel kernels $\tau_{\lambda\mu}$ on Ω_{P} with

$$|\tau_{\lambda\mu}(x,y)-\tfrac32\mathsf{b}^{-2}[\mathbb{K}_3(x-y)_{\lambda\mu}]^2|\le\epsilon_{\mathrm{P}}(g,|x-y|)\,\mathsf{b}^{-2}|x-y|^{-6}\quad\text{almost everywhere on }\Omega_{\mathrm{P}};$$

before a kernel is named, at every $(x_0,y_0)\in\Omega_{\mathrm{P}}$ with $\mathsf{d}_0=|x_0-y_0|$ and for Lipschitz φ_1,φ_2 supported in balls of radius ρ inside the balls of radius $\frac18\mathsf{d}_0$ about x_0,y_0: $|T_{\lambda\mu}(\varphi_1,\varphi_2)-\frac32\mathsf{b}^{-2}\mathcal{K}_{\lambda\mu}(\varphi_1,\varphi_2)|\le\epsilon_{\mathrm{P}}(g,\mathsf{d}_0)\mathsf{b}^{-2}\mathsf{d}_0^{-6}\rho^6\|\varphi_1\|_\infty\|\varphi_2\|_\infty$. The kernel $t_\lambda=\sum_\mu(\tau_{\lambda\mu}-\frac13\sum_{\lambda'}\tau_{\lambda'\mu})$ of the limit of $\langle T_\lambda;\mathcal{O}\rangle$ satisfies $|t_\lambda(x,y)-\mathsf{t}^\circ(e_\lambda,x-y)|\le6\epsilon_{\mathrm{P}}(g,|x-y|)\mathsf{b}^{-2}|x-y|^{-6}$ almost everywhere.

(C.3) *The pair against the kernel of $\langle T_\lambda;\mathcal{O}\rangle$.* Take one limit point as in (C.2)(ii), along a subsequence of (K_j,β_j); along it $S^T_{3;K_j,\ell_g,\beta_j}\to S_3^{(\ell_g,\mathsf{b})}$. For every c,z,r,λ in the range above

$$\limsup_{\rho\to0}\big|\tfrac{2\pi}{3}\mathsf{b}r^3\mathsf{M}_\rho\mathsf{s}_3(c\pm\tfrac{r}{2}e_\lambda,z)-\mathsf{M}_\rho t_\lambda(c,z)\big|\le\big[12\bar\epsilon_{\mathrm{A}}+6\epsilon_{\mathrm{P}}(g,\mathsf{d})+2.7(r/\mathsf{d})^2\big]\mathsf{b}^{-2}\mathsf{d}^{-6},$$

and the same with the radial tents of mass 1 in place of the means. For almost every (c,h,z), $h=rn$, in the range above,

$$\big|\tfrac{2\pi}{3}\mathsf{b}r^3\mathsf{s}_3(c\pm\tfrac{r}{2}n,z)-t_\lambda(c,z)\big|\le\big[12\bar\epsilon_{\mathrm{A}}+6\epsilon_{\mathrm{P}}(g,\mathsf{d})+2.7(r/\mathsf{d})^2+0.057|n-e_\lambda|\big]\mathsf{b}^{-2}\mathsf{d}^{-6}.$$

(C.4) *Limits and components on $\mathbb{R}^3$.* Let (K_j,β_j), ν_j, σ' be as in (A.4)(ii), now with $S_n^T[\nu_j]\to S_n^{\sigma'}$ for $n=2,3$, assume that $S_3^{\sigma'}$ and $S_2^{\sigma'}$ have continuous kernels s_3 and s_2^T on distinct configurations, and $\Omega_\sigma=\{(x,y)\in(\mathbb{R}^3)^2:0<|x-y|<\mathsf{d}_5(g)\}$. (a) There is a subsequence along which $\langle\mathcal{O}_\lambda(\varphi_1);\mathcal{O}_\mu(\varphi_2)\rangle_{\nu_j}$ converges for all λ,μ and all real Lipschitz φ_1,φ_2 with compact supports, $\operatorname{supp}\varphi_1\times\operatorname{supp}\varphi_2\subset\Omega_\sigma$. (b) For every such subsequence the limits have Borel kernels $\tau^\sigma_{\lambda\mu}$ on Ω_σ with $|\tau^\sigma_{\lambda\mu}(x,y)-\frac32\mathsf{b}^{-2}[\mathbb{K}_3(x-y)_{\lambda\mu}]^2|\le(\frac{3}{\pi})^2\epsilon_\sigma(g,|x-y|)\mathsf{b}^{-2}|x-y|^{-6}$ almost everywhere; $\tau^\sigma_{\lambda\mu}(x,y)$ depends on $x-y$ alone (almost everywhere); and $\sum_{\lambda,\mu}\tau^\sigma_{\lambda\mu}=\mathsf{s}_2^T$ almost everywhere, so that $|\mathsf{s}_2^T(x,y)-\frac{9}{16\pi^2}\mathsf{b}^{-2}|x-y|^{-6}|\le9\epsilon_\sigma(g,|x-y|)\mathsf{b}^{-2}|x-y|^{-6}$ at every $(x,y)\in\Omega_\sigma$. (c) With $t^\sigma_\lambda=\sum_\mu(\tau^\sigma_{\lambda\mu}-\frac13\sum_{\lambda'}\tau^\sigma_{\lambda'\mu})$ there is one null set $\mathcal{N}\subset\Omega_\sigma$, fixed before r is named, such that for every $(c,z)\notin\mathcal{N}$ with $0<\mathsf{d}=|c-z|\le\frac12\mathsf{d}_5(g)$, every

$0<r\le\frac12\mathsf{d}$ and every λ

$$\big|\tfrac{2\pi}{3}\mathsf{b}r^3\mathsf{s}_3(c\pm\tfrac{r}{2}e_\lambda,z)-t^\sigma_\lambda(c,z)\big|\le\big[\epsilon'_\sigma(g,\mathsf{d})+2.7(r/\mathsf{d})^2\big]\mathsf{b}^{-2}\mathsf{d}^{-6},\qquad \epsilon'_\sigma(g,\mathsf{d})=12\sup_{\frac34\mathsf{d}\le t\le\frac{17}{16}\mathsf{d}}\mathsf{E}_3(g,t)+6\epsilon_\sigma(g,\mathsf{d}),$$

and $\epsilon'_\sigma(g,\mathsf{d})\le C(g)(\log\frac{1}{\mathsf{d}})^{39\mathfrak{p}_0+2}\mathsf{d}^{1/2}$; at every (c,z) in this range the same holds with $\sum_\mu(\mathsf{t}_{\lambda\mu}-\frac13\sum_{\lambda'}\mathsf{t}_{\lambda'\mu})(\chi^\rho_c,\chi^\rho_z)$ in place of $t^\sigma_\lambda(c,z)$, $\mathsf{t}_{\lambda\mu}$ the limit forms, for every $0<\rho<\frac18\mathsf{d}$, with $21\rho/\mathsf{d}$ added in the bracket. (d) At a cutoff, for every first-passage datum (K,β) at g, every state $\mathfrak{s}$ at (K,β), points x_0,y_0 with $0<\mathsf{d}_0=|x_0-y_0|\le2\mathsf{d}_4(g)$, $\mathsf{d}=\frac12\mathsf{d}_0$, $s(\rho)\le K$, and bounded real Borel f_1,f_2 on $\mathbb{R}^3$ supported in balls of radius ρ inside the closed balls of radius $\frac18\mathsf{d}_0$ about x_0,y_0:

$$\Big|\langle\mathcal{O}_\lambda(f_1);\mathcal{O}_\mu(f_2)\rangle_{\mathfrak{s}}-\tfrac32\mathsf{b}^{-2}\sum_{\lambda(p)=\lambda,\,\lambda(q)=\mu}f_1(x(p))f_2(x(q))\,C_K^{\mathrm{free}}(\mathrm{pr}\,p,\mathrm{pr}\,q)^2\Big|\le\Big[\mathsf{E}_2(g,\mathsf{d})-\tfrac32C_{\mathrm{Rs}}\Big(\frac{\mathsf{d}}{\ell_g}\Big)^3\Big]\Big(\frac{\rho}{\mathsf{d}}\Big)^6\mathsf{b}^{-2}\|f_1\|_\infty\|f_2\|_\infty,$$

with C_K^{free} of the lattice (ℓ_g,K) and pr the projection onto T_{ℓ_g}.

■□

Theorem J (The traceless tensor in every direction)**.** Let $0<g\le\gamma$ be a reference coupling and $\ell\in\mathfrak{L}(g)$. There are an integer $K_\circ(g)$ and a number $\mathsf{d}_{\mathrm{cl}}(g)>0$ formed from the constants and g, and a finite function $\epsilon^{\mathrm{cl}}(g,\ell,\cdot)$ on $(0,\frac18\ell]$ formed from the constants, g and ℓ, with $\epsilon^{\mathrm{cl}}(g,\ell,\mathsf{d}')\le C(g,\ell)(\log\frac{1}{\mathsf{d}'})^{19\mathfrak{p}_0}\mathsf{d}'^{1/2}$ for $\mathsf{d}'\le\mathsf{d}_{\mathrm{cl}}(g)$, such that the following hold.

(D.1) *The bond forms against the plane fields, on every torus.* Let (K,β) be a first-passage datum at g with $K\ge K_\circ(g)$, $\mathsf{b}=a_K\beta$; $0<16r\le\mathsf{d}'\le\frac18\ell$ with $s(r)\le K$; $x_1,x_2\in T_\ell$ with $|x_1-x_2|=\mathsf{d}'$; $\vartheta\in(0,1]$; φ_1,φ_2 real Lipschitz functions on T_ℓ with $\mathrm{supp}\,\varphi_a\subset\mathbb{B}(x_a,r)$ and $\mathrm{Lip}\,\varphi_a\le\|\varphi_a\|_\infty/\vartheta r$. Then for all $\lambda\ne\lambda'$ and all γ,μ

$$\begin{aligned}&\Big|\langle\mathcal{O}^{\mathrm{od}}_{\lambda\lambda'}(\varphi_1);\mathcal{O}_\gamma(\varphi_2)\rangle_{K,\ell,\beta}\\&\qquad-3\mathsf{b}^{-2}\iint\varphi_1(x)\varphi_2(y)\mathbb{K}_3(x-y)_{\lambda\gamma}\mathbb{K}_3(x-y)_{\lambda'\gamma}\,\mathrm{d}x\,\mathrm{d}y\Big|\\&\le\Big[8\epsilon^{\mathrm{cl}}(g,\ell,\mathsf{d}')+3C^{\mathrm{od}}_{\mathrm{PS}}\frac{a_K}{\vartheta r}+3C^{\mathrm{od}}_{\mathrm{PS}}\Big(\frac{\mathsf{d}'}{\ell}\Big)^3\Big]\Big(\frac{r}{\mathsf{d}'}\Big)^6\mathsf{b}^{-2}\|\varphi_1\|_\infty\|\varphi_2\|_\infty,\\&\big|\langle\mathcal{O}_\lambda(\varphi_1);\mathcal{O}_\mu(\varphi_2)\rangle_{K,\ell,\beta}-\tfrac32\mathsf{b}^{-2}\mathcal{K}_{\lambda\mu}(\varphi_1,\varphi_2)\big|\\&\le\Big[\epsilon^{\mathrm{cl}}(g,\ell,\mathsf{d}')+\tfrac32C_{\mathrm{PS}}\frac{a_K}{\vartheta r}+\tfrac32C_{\mathrm{PS}}\Big(\frac{\mathsf{d}'}{\ell}\Big)^3\Big]\Big(\frac{r}{\mathsf{d}'}\Big)^6\mathsf{b}^{-2}\|\varphi_1\|_\infty\|\varphi_2\|_\infty.\end{aligned}$$

(D.2) *The kernels of $\langle T_n;\mathcal{O}\rangle$ for all directions at once.* Let (K_j,β_j) be first-passage data at g with $K_j\to\infty$ and $a_{K_j}\beta_j\to\mathsf{b}$; $\Omega^\circ=\{(x,y)\in T_\ell^2:0<|x-y|<\frac79\mathsf{d}_{\mathrm{cl}}(g)\}$;

$$\bar\epsilon^{\mathrm{cl}}(t)=\sup\{8\epsilon^{\mathrm{cl}}(g,\ell,\mathsf{d}'')+3\max\{C_{\mathrm{PS}},C^{\mathrm{od}}_{\mathrm{PS}}\}(\mathsf{d}''/\ell)^3:\frac79t\le\mathsf{d}''\le\frac97t\},$$

so that $\bar\epsilon^{\mathrm{cl}}(t)\le C(g,\ell)[(\log\frac1t)^{19\mathfrak{p}_0}t^{1/2}+(t/\ell)^3]$. Along a subsequence all the forms $\langle\mathcal{O}_\lambda(f);\mathcal{O}_\mu(h)\rangle_{K_j,\ell,\beta_j}$ and $\langle\mathcal{O}^{\mathrm{od}}_{\lambda\lambda'}(f);\mathcal{O}_\gamma(h)\rangle_{K_j,\ell,\beta_j}$ converge, at all real Lipschitz f,h with $\mathrm{supp}\,f\times\mathrm{supp}\,h\subset\Omega^\circ$.

For every such subsequence the limits have Borel kernels on Ω° that lie, almost everywhere, within $4.2\,\bar\epsilon^{\rm cl}(|x-y|)\mathsf{b}^{-2}|x-y|^{-6}$ of the free kernels $\frac{3}{2}\mathsf{b}^{-2}[\mathbb{K}_3(x-y)_{\lambda\mu}]^2$ and $3\mathsf{b}^{-2}\mathbb{K}_3(x-y)_{\lambda\gamma}\mathbb{K}_3(x-y)_{\lambda'\gamma}$; and outside one null set of Ω°, for all unit vectors n at once, the kernel t_n of the limit of $\langle T_n(f);\mathcal{O}(h)\rangle$ satisfies

$$|t_n(x,y)-\mathsf{t}^\circ(n,x-y)|\le 38\,\bar\epsilon^{\rm cl}(|x-y|)\,\mathsf{b}^{-2}|x-y|^{-6}.$$

(D.3) *The pair against t_n.* Let moreover g satisfy (Γ) and $\ell=\ell_g$, and take s_3 of (A.2) and a limit point of (D.2). For every c,z,r,n with $0<r\le\frac12\mathsf{d}$, $\mathsf{d}=|c-z|\le\frac{16}{17}\mathsf{d}_5(g)$ and $\mathsf{d}<\frac79\mathsf{d}_{\rm cl}(g)$,

$$\limsup_{\rho\to0}\big|\tfrac{2\pi}{3}\mathsf{b}r^3\mathsf{M}_\rho\mathsf{s}_3(c\pm\tfrac r2 n,z)-\mathsf{M}_\rho t_n(c,z)\big|\le\big[12\bar\epsilon_{\rm A}+38\bar\epsilon^{\rm cl}(\mathsf{d})+2.7(r/\mathsf{d})^2\big]\mathsf{b}^{-2}\mathsf{d}^{-6},$$

the same with the radial tents of mass 1 in place of the means, and the same bound holds for $|\frac{2\pi}{3}\mathsf{b}r^3\mathsf{s}_3(c\pm\frac r2 n,z)-t_n(c,z)|$ at almost every (c,rn,z) in this range.

■□

Theorem K (Pair minus tensor field; the limit $r\to0$)**.** Fix $\beta_{\rm H}\in(0,1)$. There is a number $\gamma_{\rm R}\in(0,\gamma_{\rm jt}]$ formed from the constants and $\beta_{\rm H}$ such that every $0<g\le\gamma_{\rm R}$ satisfies (Γ) and $g\le\gamma_{\rm O}$ of (B.1), and besides the two conditions $\frac32 C_\natural c_\gamma g\le\log2$, where $C_\natural=(L^8+8c_3)L^4$ with c_3 of Lem. 4.6(c) and c_γ of Lem. 4.6(f), and $c_{\rm A}\alpha_0\ge32M\mathsf{c}_\natural\delta(t)(1+\log t)$ for all $t\ge g^{-2}$, which contains the condition $c_{\rm A}\alpha_0\ge5M\mathsf{c}_\natural\delta(t)(1+\log t)$ for all $t\ge g^{-2}$, the last member of the list of $\gamma_{\rm N}$ in Def. 4.1(1), together with $g\le19C_{\rm A}C_{\rm E,3}M^2\mathsf{c}_\natural^2A_1^2/2C_{\rm E,1}\alpha_0^2$, with the numbers of Def. 4.1(6); each of them is a condition "$g\le$ a number formed from the constants", and together they contain the ceilings of all the lemmas used in the proof.

The following holds for $0<g\le\gamma_{\rm R}$ on the least torus $\ell=\ell_g$. Let (K,β) be a first-passage datum at g, $\mathsf{b}=a_K\beta$, λ a direction, $\mathsf{d}>0$ with $6\mathsf{d}\le\min\{1,\mathsf{b}\}$, and φ_1,φ_2 bounded real functions on T_{ℓ_g} with $\operatorname{supp}\varphi_a\subset\mathbb{B}(y_a,\mathsf{d})$ and $\operatorname{dist}(\operatorname{supp}\varphi_1,\operatorname{supp}\varphi_2)\ge\mathsf{d}$. Let $r_\alpha(g,\mathsf{b})>0$ be the number of Def. 4.1(5), formed from the constants, g and b. For every r with

$$128a_K\le r\le\min\{r_\alpha(g,\mathsf{b}),(\mathsf{d}/5LM)^2\},\qquad 3\le s(\tfrac r2)\le K,\qquad \tfrac r2\le\min\{r_0(\mathsf{b}),r_1(\mathsf{b})\},\quad \mathsf{Lm}_{s(r/2)}\le L^{s(r/2)-3},$$

the following holds. For $c\in\operatorname{supp}\varphi_1$ let $w_{1,2}=\chi_{c\pm\frac r2 e_\lambda}^{r/8}\circ x$ and $w_T(p)=(\delta_{\lambda(p)\lambda}-\frac13)\chi_c^{r/8}(x(p))$, so that $\mathcal{O}(w_T)=T_\lambda(\chi_c^{r/8})$; let $\mathsf{c}=\frac{2\pi}{3}\mathsf{b}r^3$; let w_3 be $\varphi_2\circ x$ restricted to the plaquettes with barycentres in a cube of side $\frac r4$ whose centre z has $|c-z|\le6\mathsf{d}$, and $\mu_3=r^3\|\varphi_2\|_\infty$. Let Y_β be the remainders of Lem. 4.20 for the lists (w_1,w_2,w_3) and (w_T,w_3), $\mathsf{Y}_\Delta=\mathsf{c}\mathsf{Y}_{12}-\mathsf{Y}_T$ and $\mathsf{Y}_{\Delta3}=\mathsf{c}\mathsf{Y}_{123}-\mathsf{Y}_{T3}$.

(E.1) *The part that is not explicit.* There is a finite function $\mathsf{C}_{\rm R}(g,\mathsf{b};\cdot)$ on $(0,1]$, formed from the constants, $\beta_{\rm H}$, g and b, such that

$$\Big|\int\mathsf{Y}_{\Delta3}\,\mathrm{d}\mathbb{P}+\kappa_{\mathbb{P}}(\mathsf{Y}_\Delta,\mathsf{Y}_3)\Big|\le\mathsf{C}_{\rm R}(g,\mathsf{b};\mathsf{d})\big[r^{\beta_{\rm H}/72}+(a_K/r)^{1/18}\big]\mathsf{b}^{-2}\mathsf{d}^{-6}\mu_3.$$

If g satisfies besides ($\Gamma3$) of Def. 4.1(6), one may take $\mathsf{C}_{\rm R}(g,\mathsf{b};\mathsf{d})=C(g,\mathsf{b})\,\mathsf{d}^{-5/2}$, and the pair minus the tensor field is nearly constant up to the last level: with $\Delta_n=\mathsf{c}G_{n;12}-G_{n;T}$ and $\Gamma_n=\mathsf{c}G_{n;123}-G_{n;T3}$, formed with one progression of the four sources, with $\varepsilon^{\mathcal{W}}=C(g,\mathsf{b})[r^{\beta_{\rm H}/72}+(a_K/r)^{1/18}]$, and for $r\le r_\alpha^{\mathcal{W}}(g,\mathsf{b})$, there are numbers c_n such that at every level $n\le K$ with $a_{K-n}\ge r^{1/2}$, for m_n-almost every small-field background W of the level n (Lem. 4.20),

$$|\Delta_n(U_n(W))-\mathsf{c}_n|\le\varepsilon^{\mathcal{W}}\varepsilon_n^2a_{K-n}^{-4},\qquad|\mathsf{c}_n-\mathsf{c}_K|\le\varepsilon^{\mathcal{W}}\varepsilon_n^2a_{K-n}^{-4},\qquad|\Gamma_n(U_n(W))-\Gamma_K(\mathbf{1})|\le C(\mathsf{b})\,\mathsf{d}^{-5/2}\varepsilon^{\mathcal{W}}\mathsf{b}^{-2}\mathsf{d}^{-6}\mu_3.$$

(E.2) *The true functions.* There are numbers $C_{\mathrm{A}}(g,\mathsf{b})$, formed from the constants, β_{H}, g and b, and $r_{\mathcal{W}}(g,\mathsf{b},\mathsf{d})>0$, formed besides with d, such that for $r\le r_{\mathcal{W}}(g,\mathsf{b},\mathsf{d})$

$$\big|\mathsf{c}\,S_3^T(w_1,w_2,w_3)-S_2^T(w_T,w_3)\big|\le C_{\mathrm{A}}(g,\mathsf{b})\,\mathsf{d}^{-18}\big[r^{\beta_{\mathrm{H}}/72}+(a_K/r)^{1/18}\big]\mu_3.$$

(E.3) *At a fixed label: the limit in the cutoff, and $r\to 0$.* Let φ_1,φ_2 be as above, Lipschitz, with $|y_1-y_2|\le 3\mathsf{d}$, let $f_\pm^c=\chi^{r/8}_{c\pm\frac{r}{2}e_\lambda}$, and for a first-passage datum (K,β) at g with the label $\mathsf{b}'=a_K\beta$ put

$$\mathsf{A}_{r,K}=\tfrac{2\pi}{3}\mathsf{b}'r^3\int \mathrm{d}c\,\varphi_1(c)\,S^T_{3;K,\ell_g,\beta}(f_+^c,f_-^c,\varphi_2).$$

If either φ_1 or φ_2 is zero, the assertions are immediate by multilinearity; assume both are nonzero in the normalized estimates that follow. At every label $\mathsf{b}'\in(g^{-2},\bar{\mathsf{b}}]$ there are numbers $r_\sharp>0$ and $\mathsf{C}(\mathsf{d})$, formed from the constants, β_{H}, g, b', d and $\mathsf{d}\,\mathrm{Lip}\,\varphi_1/\|\varphi_1\|_\infty$, such that: (i) for $0<r\le r_\sharp$ and every first-passage datum (K,β) at g with the label b', $128a_K\le r$ and $s(\frac{r}{2})\le K$, $|\mathsf{A}_{r,K}-\langle T_\lambda(\varphi_1);\mathcal{O}(\varphi_2)\rangle_{K,\ell_g,\beta}|\le\mathsf{C}(\mathsf{d})[r^{\beta_{\mathrm{H}}/72}+(a_K/r)^{1/18}]\mathsf{b}'^{-2}\|\varphi_1\|_\infty\|\varphi_2\|_\infty$. Let now $\mathsf{b}\in I(g)$ and $\beta_K=L^K\mathsf{b}$, so that (K,β_K) is a first-passage datum at g with the label b for every $K\ge1$; let $\mathsf{A}_{r,K}$ be formed at (K,β_K), and $r_\sharp$, $\mathsf{C}(\mathsf{d})$ at the label b.

Then (ii) the limit $t_\lambda(\varphi_1,\varphi_2)=\lim_{K\to\infty}\langle T_\lambda(\varphi_1);\mathcal{O}(\varphi_2)\rangle_{K,\ell_g,\beta_K}$ exists; (iii)

$$\mathsf{A}_r=\lim_K\mathsf{A}_{r,K}=\frac{2\pi}{3}\mathsf{b}r^3\int\mathrm{d}c\,\varphi_1(c)S_3^{(\ell_g,\mathsf{b})}(f_+^c,f_-^c,\varphi_2)$$

exists and

$$|\mathsf{A}_r-t_\lambda(\varphi_1,\varphi_2)|\le\mathsf{C}(\mathsf{d})r^{\beta_{\mathrm{H}}/72}\mathsf{b}^{-2}\|\varphi_1\|_\infty\|\varphi_2\|_\infty$$

for $0<r\le r_\sharp$; so $\mathsf{A}_r\to t_\lambda(\varphi_1,\varphi_2)$ as $r\to0$ at a fixed d.

(E.4) *Labels that tend to* b. For $\mathsf{b}\in I(g)$, the φ_a of (E.3) and every sequence of first-passage data (K_j,β_j) at g with $K_j\to\infty$ and $a_{K_j}\beta_j\to\mathsf{b}$: $\langle T_\lambda(\varphi_1);\mathcal{O}(\varphi_2)\rangle_{K_j,\ell_g,\beta_j}\to t_\lambda(\varphi_1,\varphi_2)$.

■□

Theorem L (Four points at two scales)**.** Let g satisfy $(\Gamma^{(4)})$, let (K,β) be a first-passage datum at g, $\mathsf{b}=a_K\beta$, $0<\rho\le r\le\mathsf{d}\le\mathsf{d}_4^{(4)}(g)$, $s(\rho)\le K$, and let $w_1,\dots,w_4$ form two pairs at (ρ,r,d).

(a) On every side $\ell\in\mathfrak{L}(g)$ with $64\mathsf{d}\le\ell$,

$$\Big|\mathsf{J}_4-\tfrac{3}{2}\sum_{\varsigma}\mathrm{Tr}\prod_l(w_{\varsigma(l)}\mathsf{b}^{-1}C_K^{\mathrm{free}})\Big|\le\epsilon_4^{\mathrm{nar}}(g,\mathsf{d})\,\mathrm{nat}_4,$$

the sum over the six cyclic orders of $\{1,2,3,4\}$; on the least torus

$$|S_4^T-\mathsf{J}_4|\le77\Lambda_3^4\mathsf{M}'^4\mathsf{b}^4\mathsf{d}^{3/2}\mathrm{nat}_4;$$

and in every state at (K,β), for bounded real Borel functions on $\mathbb{R}^3$ in the geometry of (A.4)(i) for two pairs, the connected function differs from that of the least torus at the periodised functions by at most

$$C_4^{\mathrm{tr3}}(g)\mathsf{b}^4\mathsf{d}^{3/2}\mathrm{nat}_4.$$

(b) If $w_a=f_a\circ x$ with real Lipschitz f_a on T_{ℓ_g} such that $\mathrm{supp}\,f_a\subset\mathbb{B}(x_a,\rho)$, x_a the centres with which the w_a form the two pairs (so that $|x_1-x_2|,|x_3-x_4|\le8r$ and $|x_a-x_b|\le18\mathsf{d}$ for $a\le2<b$), the balls $\mathbb{B}(x_a,\rho)$ are at least r apart in $\{1,2\}$ and in $\{3,4\}$ and at least d apart across, $\mathrm{Lip}\,f_a\le\|f_a\|_\infty/\vartheta\rho$, $\vartheta\in(0,1]$, and all supports lie in one ball of radius $\frac12\mathsf{d}^+$ with $4\mathsf{d}^+<\ell_g$, then

$$\Big|S_4^T(f)-3\mathsf{b}^{-4}\int k_4\prod_a f_a\Big|\le\Big[\epsilon_4^{\mathrm{nar}}(g,\mathsf{d})+77\Lambda_3^4\mathsf{M}'^4\mathsf{b}^4\mathsf{d}^{3/2}+3C^{(4)}\Big(\frac{a_K}{\vartheta\rho}+\Big(\frac{\mathsf{d}^+}{\ell_g}\Big)^3\Big)\Big]\mathrm{nat}_4.$$

Here $8a_K\le\rho$, by the working level at $\varrho=\rho$, and $\mathsf{d}\le\mathsf{d}^+$, the supports of f_1 and f_3 being at least d apart in a ball of diameter d^+; if an f_a vanishes, both members are 0.

(c) Let b be a limit of labels of first-passage data at g and $\mathcal{D}_4 = \{(x_1,\dots,x_4) : r_* \le |x_1 - x_2|, |x_3 - x_4| \le 2r_*,\ r_* \le \mathsf{d}_* = \operatorname{dist}(\{x_1,x_2\},\{x_3,x_4\}) \le \mathsf{d}_5^{(4)}(g)\}$. Then $S_4^{(\ell_g,\mathsf{b})}$ has on $\mathcal{D}_4$ a Borel kernel s_4 with, almost everywhere on $\mathcal{D}_4$ and uniformly in $r_*/\mathsf{d}_* \in (0,1]$,

$$|\mathsf{s}_4 - 3\mathsf{b}^{-4}k_4| \le \epsilon_4(g,\mathsf{d}_*)\,\mathsf{b}^{-4}r_*^{-6}\mathsf{d}_*^{-6}, \qquad \epsilon_4(g,t) = 4096\big[\epsilon_4^{\mathrm{nar}}(g,\tfrac{t}{2}) + 77\Lambda_3^4(\bar{\mathsf{M}}^{(24)})^4\bar{\mathsf{b}}^4 t^{3/2} + 3C^{(4)}(12t/\ell_g)^3\big].$$

■□

Remark 4.2 (The ceilings). Let x_W be the threshold of Section YM3_I:126, equation YM3_I:(2326), and put

$$x_{\mathrm{jt}} = \max\left\{x_W, X_{\mathrm{ev}}, \left(\frac{20M\mathsf{c}_{\mathfrak{h}}A_1(4\mathfrak{p}_0)^{\mathfrak{p}_0}}{c_A\alpha_0}\right)^4\right\}, \qquad \gamma_{\mathrm{jt}} = x_{\mathrm{jt}}^{-1/2}. \tag{4.iv}$$

The notation x_{jt}, γ_{jt} denotes these same numbers. The defining inequalities for x_W include those for x_{J}, so $x_{\mathrm{jt}} \ge x_W \ge \max\{\gamma^{-2}, x_{\mathrm{J}}\}$. For $t > 1$, the inequality $\log u \le u$, applied to $u = t^{1/(4\mathfrak{p}_0)}$, gives

$$\delta(t) = A_1 t^{-1/2}(\log t)^{\mathfrak{p}_0} \le A_1(4\mathfrak{p}_0)^{\mathfrak{p}_0}t^{-1/4}.$$

Consequently $t \ge x_{\mathrm{jt}}$ implies $5\mathsf{c}_{\mathfrak{h}}\delta(t) \le c_A\alpha_0/(4M)$. Also $\gamma_W^{\mathrm{ref}} = x_W^{-1/2}$ and $\gamma_W^{\mathrm{sd}} \le \gamma_W^{\mathrm{ref}}$ by Section YM3_I:126, equation YM3_I:(2327). Hence $\gamma_{\mathrm{jt}} \le \gamma_W^{\mathrm{ref}} \le \gamma$ and $\gamma_W^{\mathrm{sd}} \le \gamma$. Every coupling under (Γ1), (Γ2), (Γ) or ($\Gamma^{(4)}$), or below γ_{N} or γ_{R}, therefore satisfies $0 < g \le \gamma$. For $g \le \gamma_{\mathrm{jt}}$ one also has $g^{-2} \ge e^{2\mathfrak{p}_0}$. All these ceilings are positive numbers depending only on the fixed constants.

4.1.6. *Construction estimates.* We use the construction results at their stated hypotheses, including Thms. YM3_I:D, YM3_I:AJ and YM3_I:AD. The finite construction of Thm. YM3_I:D and its sourced form, Thm. YM3_I:AJ, enters wherever the objects of the theory (the run of a first-passage datum, the effective actions of all levels with their bounds, and the weighted progression with any number of sources) satisfy the actual-step hypotheses of [7, Prop. B.21]; this holds for $0 < g \le \gamma$ on every side of $\mathfrak{L}(g)$, with $g^{-2} \ge x_{\mathrm{J}}$ where sources are present. [7, Prop. B.8] constructs the weighted progression for sources on the bare plaquettes; mixed sources enter through clauses (5) and (6) of the lemma on bond sources, which follows Lem. 4.31: at $0 < g \le \gamma_{\mathrm{N}}$, on every side of $\mathfrak{L}(g)$, the progression and the representation of the torus expectations are formed by the same construction, with the weight member of the mixed source in the source slot of the bare arrays; the frame of the pair functional is read with the hulls of the sources in the place of their supports; and the tower identity holds for a pair of mixed sources in that frame.

Thms. YM3_I:AJ and YM3_I:AD enter through the two-sided stability bound of the torus measures, which together with their reflection positivity gives the chessboard bound of Lem. 4.25. Through this bound they are used in (A.4), (A.5), (C.1)(iii), (C.4) and for the states in Thm. L(a); they are not used in Thm. H, in (A.1)–(A.3), (C.1)(i), (ii), (C.2), (C.3), in Thm. J, or in Thm. L on the tori. Thm. K, like every statement of this section, assumes the construction inputs just listed.

We use the fixed construction setting just specified, the torus limits of Thm. YM3_I:A, and the periodic-Gibbs limits and decompositions of Thms. YM3_I:AP and YM3_I:AL. On one renormalisation step (whose provision is that of Section YM3_I:52.4) and on the progression: Prop. 6.32, with the fourth-family norm eq. 6.xvii; the bracket localisation of the families with its count; the locality and linear-weight dependence of [7, Prop. B.8]; the chart and normalized real measures of Lems. [7, Lem. B.3] and 2.2, with the cumulant bounds of Lem. 4.14; the expansion $|a_p(e^{i\eta_i A}) - \frac{1}{2}|\mathsf{ang}_p^{[i]}(A)|^2| \le \mathsf{M}_{\mathrm{pl}}^2\eta_{i-1}^4(r_i/\varrho_{\mathrm{ch}})^3$ on the chain classes, and $|\mathsf{ang}_p^{[i]}(A)| \le 2\eta_i^2 r_i$; the regular-field evaluation and domain sums of Lems. [7, Lem. B.22] and [7, Lem. B.10]; the lowering maps of the chain classes, by which $A' \in \mathsf{Cc}_{i+1}(r_{i+1})$ and a fluctuation field B give $A^{\mathrm{lw}}(B) \in \mathsf{Cc}_i(r_i)$, and the pair tower identity of Prop. 2.4, extended to several sources in the lemma preceding Prop. 4.16; the history decomposition eq. 4.xxxi, its sourced quotients in [7, Prop. B.15], and the tail bounds of Prop. 4.54.

On the free kernel, Prop. YM3_I:120.6 and equations YM3_I:(2245)–(2246) give the decay of the slices $|\mathcal{C}_{(j)}(p,q)| \le C_{\mathrm{tp}} a_K^3 \mathsf{r}^{-3} t_j^3 e^{-\delta_{\mathrm{tp}} t_j}$, $\mathsf{r} = |x(p)-x(q)|_\infty$, $t_j = \mathsf{r}/a_{K-j+1}$; and the comparison $|\epsilon_p \epsilon_q C_K^{\mathrm{free}}(p,q) - a_K^3 \mathbb{K}_3(u)_{\lambda(p)\lambda(q)}| \le C_{\mathrm{lat}} a_K^3 [a_K^2 |u|^{-5} + \ell^{-3}]$, $|C_K^{\mathrm{free}}(p,q)| \le (1+3C_{\mathrm{lat}}) a_K^3 |u|^{-3}$, for $p \neq q$, u the least lift of $x(p)-x(q)$, $|u| \ge 2^{-1/2} a_K$, $\epsilon_p = \pm 1$ the orientation signs. On the working level (Lem. YM3_I:126.1): for $0 < \varrho < 1$ with $s = s(\varrho) \le K$ and $k = \mathsf{k}(\varrho)$, $Y = Y_s$: $s \ge 3$, $2 \le k \le K-1$, $4MY^{2\mathfrak{p}_0} \le \hat{\varrho}_k < LC_{W,0} A_0^2 Y^{2\mathfrak{p}_0}$, $Y \ge \log \frac{1}{\varrho}$, $y_k \le Y$, $K-k \le s \le Y/\log L$; in particular $a_K \le a_{K-k} = \varrho/\hat{\varrho}_k \le \varrho/4M$, and $Y \ge 2\sqrt{3}$ when $\varrho \le e^{-2\sqrt{3}}$.

The lower bounds come from the definition of $s(\varrho)$ and from $C_{W,0} A_0^2 \ge 4L^2 M$: $L^s \varrho \ge C_W(s) \ge 4L^2 M Y^{2\mathfrak{p}_0} > L^2$ and $\varrho < 1$ give $s \ge 3$; $K-k = \min\{s, K-2\} \ge s-2$ gives $\hat{\varrho}_k = L^{K-k}\varrho \ge L^{-2} C_W(s) \ge 4MY^{2\mathfrak{p}_0}$; and $L^s \varrho \ge C_W(s) \ge 1$ gives $Y \ge s \log L \ge \log \frac{1}{\varrho}$. Besides, $L^m \ge LMy_K$ for the sides $\ell = 2L^m \in \mathfrak{L}(g)$. Finally the dilation of lattices: $x \mapsto L^z x$ carries the lattice (ℓ, K) to the lattice $(L^z \ell, K-z)$ and $\mu_{K,\ell,\beta}$ to $\mu_{K-z, L^z\ell, \beta}$.

4.1.7. *Estimates and proofs.* The argument has eight parts: the calculus of the derivative families (Lems. 4.3–4.10); the kernel families (Lem. 4.11 to Prop. 4.13); the induction over the levels and the jets (Lem. 4.14 to Prop. 4.19); the connected functions on the least torus (Lems. 4.20–4.24); every state (Lem. 4.25 to Lem. 4.30); the bond forms (Lem. 4.31 to Prop. 4.34); pair minus tensor field (the lemma on a regular gauge on a cube, and Lems. 4.36 to 4.41); and the proofs of the theorems. The calculus of the derivative families.

Lemma 4.3 (The run)**.** Let $0 < g \le \gamma$ with $g^{-2} \ge x_{\mathrm{J}}$, which holds for every $0 < g \le \gamma_{\mathrm{jt}}$ (Remark 4.2), and (K, β) a first-passage datum at g with its run and label b.

(a) $x_K > g^{-2} \ge x_{\mathrm{J}} \ge \max\{e^{2\mathfrak{p}_0}, eL, 4\bar{B}\}$, so $y_K \ge \max\{2\mathfrak{p}_0, 1+\log L\}$. For $0 \le i \le j \le K$: $\frac{4}{5}\mathsf{b} \le a_{K-j} x_j \le \frac{4}{3}\mathsf{b}$; $\frac{3}{4} L^{j-i} x_j \le x_i \le \frac{5}{4} L^{j-i} x_j$, in particular $x_j \ge \frac{4}{5} L^{-(j-i)} x_i$; $y_j \le y_i$; $\delta_j \le (\frac{5}{4})^{1/2} L^{(j-i)/2} \delta_i$ and $\delta_i \le \delta_j$; $g^{-2} < \mathsf{b} \le \bar{\mathsf{b}}$; and $\frac{11}{10} \mathsf{c}_{\mathfrak{h}} \delta_i \le r_i \le \frac{8}{5} \mathsf{c}_{\mathfrak{h}} \delta_i$ for $1 \le i < K$, so $x_n r_n^2 \le \frac{64}{25} \mathsf{c}_{\mathfrak{h}}^2 \mathfrak{b}_n^2$. Of these, $g^{-2} < \mathsf{b} \le \bar{\mathsf{b}}$ holds, together with the flow bound $|x_j - L^{-(j-i)} x_i| \le \bar{B}$ $(0 \le i \le j \le K)$, for the label and the run of every first-passage datum at every $0 < g \le \gamma$.

(b) Let $g^{-2} \ge \max\{e^{2\mathfrak{p}_0}, e^{y_{\mathfrak{p}}}\}$ and $\delta(g^{-2}) \le \min\{1/2c_{\mathrm{rt}}, 1/c_\epsilon\}$, where $c_\delta = (\frac{5}{4}L)^{1/2}$, $c_{\mathrm{rt}} = \max\{8\mathsf{c}_{\mathfrak{h}} c_\delta/5\varrho_{\mathrm{ch}}, 1/\varrho_{\mathfrak{fl}}\}$, $c_\epsilon = c_{\mathrm{cu}} + 16 \mathsf{c}_{\mathbf{C}} \mathsf{c}_{\mathfrak{h}} c_\delta / 5 r_\alpha$, $y_{\mathfrak{p}} = \max\{1, (LM)^3 A_1^{-2}(1 + 2\log^+(c_{\mathfrak{p}}/A_1))\}$. Here ϱ_{ch} and $\varrho_{\mathfrak{fl}}$ are the radii denoted by ρ_{ch} and $\rho_{\mathfrak{fl}}$ in equation YM3_I:(2285), and $r_\alpha = a^{\mathrm{us}}(\alpha_0, \alpha_1)$ is the usable chart radius of equation YM3_I:(933). The covariance and fluctuation-tail constants are

$$\mathsf{c}_{\mathbf{C}} = 2/\gamma_r, \qquad c_{\mathfrak{p}} = C_{\mathrm{ex}}((L+2)M)^3 e^{4\lambda_1},$$

as in equations YM3_I:(2294) and YM3_I:(2296). For this regular one-step expansion, $\gamma_2 = 2(LM)^{-3}$ by equation YM3_I:(889); hence its tail factor $e^{-\gamma_2 \mathfrak{b}_i^2/2}$ is $e^{-(LM)^{-3}\mathfrak{b}_i^2}$. Finally, c_{cu} is the constant of Lem. 4.14.

Then at all heights $1 \le i < K$: $2c_{\mathrm{rt}}\delta_i \le 1$, $c_\epsilon \delta_i \le 1$, $y_i \ge y_{\mathfrak{p}}$; hence $2r_i \le \varrho_{\mathrm{ch}}$, $r_{i+1} \le \frac{8}{5} \mathsf{c}_{\mathfrak{h}} c_\delta \delta_i$, $\mathfrak{p}_{\star,i} \le \delta_i$ and $\epsilon_i' \le c_\epsilon \delta_i \le 1$, where $\mathfrak{p}_{\star,i} = c_{\mathfrak{p}} \exp[-(LM)^{-3}\mathfrak{b}_i^2]$ and ϵ_i' is the number of Lem. 4.14; if besides $\delta(g^{-2}) \le 1/2C_{\mathrm{cu}}$, then $\epsilon_i^{\mathrm{cu}} \le 2C_{\mathrm{cu}}\delta_i \le 1$. By the conditions on the constants $\mathfrak{b}_j \ge (LM)^{3/2} \ge 1$, so $g_j \le \delta_j$ and $x_j^{-1} \le \delta_j^2 \le r_j^2$ $(j < K)$; and $\theta^{[j]} \le \theta^{[K-1]} \le \theta(g^{-2})$.

(c) For $0 < \varrho < 1$ with $s(\varrho) \le K$: $\log \frac{1}{\varrho} \le Y_\varrho \le 2\log \frac{1}{\varrho} + C(g)$, $a_{K-\mathsf{k}(\varrho)} \le \varrho/4M$, and $\delta_{\mathsf{k}(\varrho)} \le A_1 Y_\varrho^{\mathfrak{p}_0} (\varrho/\mathsf{b})^{1/2}$.

■

Proof. (a) At every $0 < g \le \gamma$ the run of a first-passage datum has $x_0 > x_1 > \cdots > x_K > g^{-2}$ and $|x_j - L^{-(j-i)} x_i| \le \bar{B}$ for $i \le j$, and its label has $g^{-2} < \mathsf{b} \le \bar{\mathsf{b}}$ (Prop. YM3_I:74.1). The threshold x_{J} is at least $\max\{e^{2\mathfrak{p}_0}, eL, 4\bar{B}\}$ by its definition (the definitions preceding [7, eq. 228]);

so $g^{-2} \ge x_{\mathrm{J}}$ gives $x_j \ge x_K > g^{-2} \ge 4\bar{B}$, hence $\bar{B} \le \frac{1}{4}x_j$ and $\frac{3}{4}x_j \le L^{-(j-i)}x_i \le \frac{5}{4}x_j$, which at $i=0$ gives the bounds of $a_{K-j}x_j$ against $\mathsf{b} = a_K x_0$. As y decreases along the run and $x_i \le \frac{5}{4}L^{j-i}x_j$, $\delta_j/\delta_i = (y_j/y_i)^{\mathfrak{p}_0}(x_i/x_j)^{1/2} \le (\frac{5}{4})^{1/2}L^{(j-i)/2}$.

As $x_i \ge x_j > g^{-2} \ge e^{2\mathfrak{p}_0}$ lie in $[e^{2\mathfrak{p}_0}, \infty)$, where δ decreases, $\delta_i \le \delta_j$. The radii follow from their recursion and these bounds. (b) $x_i > g^{-2} \ge e^{2\mathfrak{p}_0}$ and δ decreases there, so every condition stated at g^{-2} holds at every height; $2r_i \le \varrho_{\mathrm{ch}}$ and $r_{i+1} \le \frac{8}{5}\mathsf{c}_{\mathfrak{h}}c_\delta\delta_i$ by (a). For $y \ge y_{\mathfrak{p}}$: $(LM)^{-3}A_1^2 y^{2\mathfrak{p}_0} \ge \frac{y}{2} + \log^+(c_{\mathfrak{p}}/A_1)$, so $\mathsf{p}_{\star,i} \le A_1 e^{-y_i/2} \le A_1 y_i^{\mathfrak{p}_0} x_i^{-1/2} = \delta_i$; then $\epsilon_i' \le c_{\mathrm{cu}}\delta_i + 2\mathsf{c}_{\mathbf{C}} r_{i+1}/r_\alpha \le c_\epsilon\delta_i$ and $\epsilon_i^{\mathrm{cu}} \le C_{\mathrm{cu}}(\delta_i + \mathsf{p}_{\star,i}) \le 2C_{\mathrm{cu}}\delta_i$. (c) As $K - s \le \mathsf{k}(\varrho) \le K - s + 2$, the working level gives $\varrho L^s \le L^2\hat{\varrho}_k < L^3 C_{W,0}A_0^2 Y^{2\mathfrak{p}_0}$, whence

$$Y = s\log L + \log L(g^{-2} + \bar{B}) < \log\tfrac{1}{\varrho} + 2\mathfrak{p}_0 \log Y + \log[L^4 C_{W,0}A_0^2(g^{-2} + \bar{B})],$$

and $2\mathfrak{p}_0 \log Y \le \frac{1}{2}Y + 2\mathfrak{p}_0 \log 4\mathfrak{p}_0$; further $a_{K-k} = \varrho/\hat{\varrho}_k \le \varrho/4M$ and $\delta_k = A_1 y_k^{\mathfrak{p}_0} x_k^{-1/2} \le A_1 Y^{\mathfrak{p}_0}(5a_{K-k}/4\mathsf{b})^{1/2}$. □

The frame used below includes the identity of the step data with the canonical construction. It also records the numerical room needed to lower a field through the chain. Keeping both requirements explicit allows us to apply the estimates at an arbitrary lower level.

Definition 4.4. Let $\mathfrak{w} = (\ell, K)$, $\ell = 2L^m$, and let $x_0, \dots, x_K$ be a finite running sequence. The pair setting consists of the following conditions.

(Jt.1) Fix $2 \le k_0 < K$, $d \ge a_K$ and real Lipschitz functions f_1, f_2, each supported in a closed ball of radius d. Put $w_a(p) = f_a(x(p))$ and $s_0 = K - k_0$.

(Jt.2) The setting (J) of [7, Def. B.16] holds through the upper level K, with $p = 2$ and the calibration equal to this running sequence. Thus its index k is $K-1$, and its actual step data, including (Pv), are required. Both sampled sources are nonzero, and

$$\widehat{d}_{k_0} := d/a_{K-k_0} \ge M, \qquad D_{k_0} \ge 3M.$$

The running and weighted bounds are

$$\begin{aligned} &|x_j - L^{-(j-i)}x_i| \le \bar{B} \quad (0 \le i \le j \le K), \qquad x_K \ge 4\bar{B}, \qquad y_K \ge \max\{2\mathfrak{p}_0, 1 + \log L\}, \\ &C_0^\flat \mathsf{S}_A \theta^{[K-1]} \le 1, \qquad \varepsilon_j \le c_A\alpha_0/(10MB_3) \quad (1 \le j \le K), \qquad \varepsilon_j \le \varepsilon_\flat \quad (0 \le j < K). \end{aligned} \tag{4.v}$$

The weighted data restrict to every earlier height. At heights $1 \le j < k_0$ they satisfy

$$s^{[j]} \le s^\star/2, \qquad \tfrac{4}{3}C_1^\flat L\theta^{[j]}\mathsf{T}(D_j/M) \le 1.$$

These are the hypotheses for the mixed derivative bound at the lower cut, in addition to the running bounds just stated.

The flat-top radii are $r_K = 0$ and $r_i = L^{-1}r_{i+1} + \frac{11}{10}\mathsf{c}_{\mathfrak{h}}\delta_i$. Set

$$c_R = \max\left\{\tfrac{8}{5}\sqrt{5L/4}\,\mathsf{c}_{\mathfrak{h}}/\varrho_{\mathrm{ch}}, \varrho_{\mathrm{fl}}^{-1}\right\}, \qquad e_i^{\mathrm{KF}} = \epsilon_i^{\mathrm{cu}} + \tfrac{16}{5}\sqrt{5L/4}\,\mathsf{c}_{\mathbf{C}}\mathsf{c}_{\mathfrak{h}}\delta_i/r_\alpha.$$

Require $\ell \ge N_{\mathsf{H}}$ and, for $k_0 \le i < K$,

$$\begin{aligned} c_R\delta_i \le \tfrac{1}{2}, \qquad e_i^{\mathrm{KF}} \le 1, \\ [\tfrac{48}{5}\sqrt{5/4}L^{-3/2}\mathsf{c}_{\mathfrak{h}} + 8C_{\mathrm{db}}]\delta_i < \varepsilon_{\mathcal{W}}\alpha_0. \end{aligned} \tag{4.vi}$$

(Jt.3) The two-source weighted progression is the one of [7, Def. B.7], with $k = K-1$ and $t_j = x_j$. It uses the same canonical data as (Jt.2), when that condition is imposed. Its source is $w_\lambda = \lambda_1 w_1 + \lambda_2 w_2$, $w_\lambda^{[j]} = L^{K-j}w_\lambda$, and $C_0^\flat \mathsf{S}_A \theta^{[K-1]} \le 1$.

(Jt.4) Both sources are nonzero, $s^{[j]} \le s^\star/2$ for $0 \le j < k_0$, and

$$\tfrac{4}{3}C_1^\flat L\theta^{[j]}\mathsf{T}(D_j/M) \le 1 \quad (1 \le j < k_0).$$

(Jt.5) An intermediate integer T is fixed with $k_0 < T \le K$.
(Jt.6) For $k_0 \le i < K$,

$$5\mathsf{c}_\mathfrak{h}\delta_i \le c_A\alpha_0/(4M). \tag{4.vii}$$

The *frame at* k_0 means (Jt.1)–(Jt.3) and (Jt.6). The full pair setting includes (Jt.4) and (Jt.5) as well. ■

Lemma 4.5 (The frame of the pair functional; tents)**.** Let $0 < g \le \gamma_{\mathrm{jt}}$, $g^{-2} \ge x_{\mathrm{J}}$, (K,β) a first-passage datum at g and $\ell \in \mathfrak{L}(g)$.

(a) Let $2 \le k < K$ and let f, h be real Lipschitz functions on T_ℓ, neither $f\circ x$ nor $h\circ x$ identically 0, each with its support in a closed ball of a radius $\bar{\mathsf{d}} \ge a_K$ with $\bar{\mathsf{d}}/a_{K-k} \ge M$, such that the corners of the plaquettes of the two supports are at least $3M$ level-k spacings apart in the sup distance. Then $(f\circ x, h\circ x)$ satisfies, with the lower level k, the four conditions (Jt.1)–(Jt.3) and (Jt.6) of Def. 4.4: the data, the flat top at K, the weighted progression of the two sources, and the smallness of δ_i at the heights $k \le i < K$; we call these four conditions the frame at k.

The two further conditions of that setting, (Jt.4) at the levels below the lower level and (Jt.5) on an intermediate level, hold, together with the other four, for a test pair $(f_1, f_2) \in \mathfrak{T}_{T_\ell}(r;\vartheta)$, $0 < r < 1$, $\vartheta \in (0,1)$, with $f_a \circ x \not\equiv 0$, at the lower level $\max\{K-s, 2\}$ when $3 \le s = s(r) \le K$ and $\mathsf{Lm}_s \le L^{s-3}$ with the intermediate level specified in the proof. The frame at k holds in the same way for the two sampled linear combinations of n functions with pairwise disjoint supports falling into two clusters, provided both combinations are nonzero, the two unions of supports lie in such balls, and the corners of the two clusters are $3M$ level-k spacings apart.

(b) Let real weights w_1, w_2 be placed at (ρ, r), $0 < \rho \le r \le e^{-2\sqrt{3}}$, $s(\rho) \le K$. Then $a_K \le \rho/4M$; with the tent τ of height 1 and radius $\frac{1}{3}a_K$, the functions $f_a = \sum_p w_a(p)\tau(\cdot - x(p))$ are Lipschitz, $f_a \circ x = w_a$, $\operatorname{supp} f_a \subset \mathbb{B}(x_a, 2r)$, and (a) applies at $k = \mathsf{k}(r)$ with $\bar{\mathsf{d}} = 2r$. For $0 \le i \le K$: (P1) $\sum_p |w_a(p)| \le 81(\rho/a_K)^3 |w_a|_\infty$; (P2) $\mathsf{N}_i(w_a) \le (2\hat{\rho}_i/M + 3)^3 \le 108(1+t_i)$, $t_i = \frac{27}{8M^3}\hat{\rho}_i^3$; (P3) $\mathsf{D}_i^{12} \ge \hat{r}_i/\sqrt{3} - 1$; and every corner of a plaquette of $\operatorname{supp} w_1 \cup \operatorname{supp} w_2$ lies within the sup distance $R_w \le 11\hat{r}_{i+1}$, in level-$(i+1)$ units, of a bare site next to x_1. For weights placed at (ρ, r, d), or forming two pairs at (ρ, r, d), with $\mathsf{d} \le e^{-2\sqrt{3}}$, the same holds, (P3) holds across the clusters with $\hat{\mathsf{d}}_i$ for $\hat{r}_i$, and (a) applies to the two clusters at $k = \mathsf{k}(\mathsf{d})$, each lying in a ball of radius $10\mathsf{d}$; for two weights with $\Sigma(w_a) \subset \mathbb{B}(x_a, \rho)$ at a distance $\ge \mathsf{d} \ge \rho$, (a) applies at $k = \mathsf{k}(\mathsf{d})$.

■

Proof. (a) First consider a pair satisfying the four geometric hypotheses in the statement. The first-passage construction gives $x_0 > \cdots > x_K > g^{-2}$ and the running comparison in eq. 4.v; see Prop. YM3_I:74.1. Since $\ell \in \mathfrak{L}(g)$, it also gives $MR(x_K) \le L^m$. The lattice is therefore admissible over every initial segment of the run. In particular $L^m \ge LM$ and $\ell \ge N_{\mathsf{H}}$, by the proof of Lem. YM3_I:126.2.

[7, Prop. B.21], applied with $k = K-1$, supplies the actual setting (J), its weighted progression and the identification (Pv). This is the same canonical datum at every height, with the counterterms of this run. The inequalities defining x_{J} apply at every x_j: they give the half-ceiling and $C_0^\flat \mathsf{S}_A \theta^{[K-1]} \le 1$. For the strengthened regular-background radius, use instead $x_j \ge x_{\mathrm{jt}} \ge X_{\mathrm{ev}}$ and eq. 2.iii, which give $\varepsilon(x_j) \le c_A\alpha_0/(10MB_3)$. The cutoff bound $\varepsilon_j \le \varepsilon_\flat$ is part of the actual history verification in that proposition.

The supports satisfy $\widehat{d}_k \ge M$ and $D_k \ge 3M$ by hypothesis. Their separation at earlier levels is $D_j = L^{k-j}D_k$, so the decreasing function T obeys $\mathsf{T}(D_j/M) \le \mathsf{T}(3)$ for $j < k$. Equation [7, eq. 228] therefore gives all the mixed derivative hypotheses at the lower cut. The threshold inequalities in equation YM3_I:(2326) give eq. 4.vi. The factor 2 in the corresponding lowering-domain inequality

makes its conclusion strict. They also give $y_K \geq \max\{2\mathfrak{p}_0, 1+\log L\}$ and $x_K \geq 4\bar{B}$. Thus (Jt.1)–(Jt.3) hold. Finally $x_i > g^{-2} \geq x_{\mathrm{jt}}$ and the threshold computation in Remark 4.2 give (Jt.6).

This verification uses only the four geometric properties in the statement. It uses neither the filling nor the sign condition of a test pair, nor the formula for the working level. For n functions in two clusters, apply it to the two sums over the clusters, with real coefficients for which both sampled sums are nonzero. Shrinking a support preserves the radius and separation conditions.

For a test pair at $k_0 = \max\{K-s, 2\}$, Lem. YM3_I:126.1 gives $d \geq a_K$, $\widehat{d}_{k_0} \geq M$ and $D_{k_0} \geq 3M$. The preceding argument gives its frame. At all heights below k_0, [7, eq. 228] gives $s^{[j]} \leq s^\star/2$, including $j=0$, and the same separation argument proves (Jt.4). Choose

$$n_T = \left\lceil \log_L(\widehat{d}_{k_0} y_{k_0}^{3\mathfrak{p}_0}) \right\rceil, \qquad T = k_0 + n_T.$$

The room bound $\mathsf{Lm}_s \leq L^{s-3}$ and the final assertion of Lem. YM3_I:126.1 give $k_0 < T \leq K-2$. This proves (Jt.5) and the claimed full pair setting.

(b) The working level at $\varrho = \rho$ gives $a_K \leq a_{K-\mathsf{k}(\rho)} \leq \rho/4M$. Two distinct barycentres are at least $2^{-1/2}a_K$ apart, so the tents are disjoint and $f_a(x(p)) = w_a(p)$.

A corner is at the distance $2^{-1/2}a_K$ from its barycentre: corners of the two supports are $\geq r - \sqrt{2}a_K$ apart, $\geq (r-\sqrt{2}a_K)/\sqrt{3}$ in the sup distance, that is $\geq \hat{r}_i/\sqrt{3} - 1$ level-i units. At $\varrho = r$ the working level gives $2 \leq \mathsf{k}(r) \leq K-1$, $2\hat{r}_{\mathsf{k}(r)} \geq M$ and $\mathsf{D}^{12}_{\mathsf{k}(r)} \geq \frac{4}{\sqrt{3}} M Y_r^{2\mathfrak{p}_0} - 1 \geq 3M$, as $Y_r \geq 2\sqrt{3}$. (P1): the barycentres of one orientation form a translate of $a_K\mathbb{Z}^3$, with at most $(2\rho/a_K+1)^3 \leq 27(\rho/a_K)^3$ points in a ball of radius $\rho \geq a_K$, and there are three orientations. (P2): an interval of length $2\hat{\rho}_i + \eta_i$ meets at most $2\hat{\rho}_i/M + 3$ intervals of length M, and $(2y+3)^3 \leq 4(8y^3+27)$. R_w: all points of $\Sigma(w_1) \cup \Sigma(w_2)$ are within $9r$ of x_1.

The three-point and four-point cases are the same with d for r across the clusters, and $9r + a_K \leq 10\mathsf{d}$. □

Lemma 4.6 (The calculus of the derivative families)**.** Let $0 < g \leq \gamma$ with $g^{-2} \geq x_{\mathrm{J}}$, (K,β) a first-passage datum at g with $K \geq 2$, $\ell \in \mathfrak{L}(g)$, and $w_1, \dots, w_p$ real weights, none identically 0, $p \geq 1$ arbitrary. For $\emptyset \neq \beta \subset [p]$, $m = |\beta|$, $0 \leq j \leq K-1$, put

$$\mathsf{m}_{j,a} = 12M^3\alpha_0^2 L^{K-j}|w_a|_\infty, \quad \varkappa_j = \frac{M^{3/2}}{2\mathsf{C}_{\mathcal{A}}\mathfrak{b}_j}, \quad \gamma_j = \theta^{[j]}\varkappa_j, \quad \mathsf{U}_{j;\beta}$$
$$= \varkappa_j^{m-1}(N_\star^{[j]})^{1-m}\prod_{a\in\beta} 2\mathsf{m}_{j,a}, \quad \bar{\nu}_{j;\beta} = \gamma_j \mathsf{U}_{j;\beta},$$

with the constant $\mathsf{C}_{\mathcal{A}}$ of the one-step theorem; the bare arrays of the weighted progression, $\mathfrak{f}_{\lambda,j} = ((E_\lambda^{(i)}, e_i)_{i\leq j}, (\mathsf{W}_{L^{K-j}w_\lambda}))$ for λ near 0, where the counterterms e_i of the regular progression of the run are real numbers that depend neither on λ nor on the sources, and the weight member W_w is linear in w, so that $E_\lambda^{(j+1)} - E^{(j+1)} = \mathsf{St}^{[j]}(\mathfrak{f}_{\lambda,j}) - \mathsf{St}^{[j]}(\mathfrak{f}_{0,j})$ (equation [7, eq. 200]); their value at $\lambda = 0$, $\mathfrak{f}_{0,j} = ((E^{(i)}, e_i)_{i\leq j}, (0))$; and their derivatives $\mathfrak{f}_{\beta,j} = \partial_\beta \mathfrak{f}_{\lambda,j}|_{\lambda=0} = ((E_\beta^{(i)}, 0)_{i\leq j}, (H_\beta))$ with $H_{\{a\}} = \mathsf{W}_{L^{K-j}w_a}$ and $H_\beta = 0$ for $m \geq 2$; and $\bar{\mathsf{h}}_{j;\beta} = \mathfrak{n}[\mathfrak{f}_{\beta,j}]/\mathsf{U}_{j;\beta}$, $\bar{\Psi}_{j;\beta} = \|E_\beta^{(j+1)}\|_{j+1,\lambda_{\mathrm{out}}}/\bar{\nu}_{j;\beta}$. Then for $1 \leq j \leq K-1$:

(a) *Chain rule.* $E_\beta^{(j+1)} = \sum_{\pi\in\mathrm{Part}(\beta)}(\prod_{\beta'\in\pi}\partial_{s_{\beta'}})\mathsf{St}^{[j]}(\mathfrak{f}_{0,j} + \sum_{\beta'\in\pi} s_{\beta'}\mathfrak{f}_{\beta',j})\big|_{s=0}$; $E_\beta^{(j)}$ is m-linear in $(w_a)_{a\in\beta}$, depends on no other source, and is the same in every progression that holds the sources of β.

(b) *Derivatives of the step.* $N(\mathfrak{f}_{0,j}) \leq \frac{1}{2}N_\star^{[j]}$, and for bare arrays $\mathfrak{f}, \mathfrak{g}_1, \dots, \mathfrak{g}_q$ with $N(\mathfrak{f}) \leq \frac{1}{2}N_\star^{[j]}$

$$\big\|\partial_{s_1}\cdots\partial_{s_q}\mathsf{St}^{[j]}(\mathfrak{f} + \textstyle\sum_l s_l\mathfrak{g}_l)\big|_0\big\|_{j+1,\lambda_{\mathrm{out}}} \leq (2(q-1))^{q-1}\,\theta^{[j]}\varkappa_j^q\,(N_\star^{[j]})^{1-q}\prod_l \mathfrak{n}[\mathfrak{g}_l].$$

(c) *Lags.* If $\mathfrak{g} = ((G^{(i)}, 0)_{i\leq j}, (H))$ and each $G^{(i)}(X,\cdot,z)$ vanishes unless X meets a set Cr of bare sites, then $\mathfrak{n}[\mathfrak{g}] \leq \|H\|_{j,\kappa} + \sum_{i\leq j}\omega_{j-i}^{\mathrm{sp}}(i)\|G^{(i)}\|_{i,\kappa}$, where $\omega_0^{\mathrm{sp}} = 1$, $\omega_l^{\mathrm{sp}}(i) = \varpi_l +$

$L^{-4l}\min\{L^{3l}, c_3\mathsf{n}_i^{\mathrm{cb}}(\mathsf{Cr})\}$, $0 \le \varpi_l \le L^{8-5l}$, $\mathsf{n}_i^{\mathrm{cb}}(\mathsf{Cr})$ is the number of the cubes of π_i that hold a site of Cr, and c_3 is a constant.

(d) *Support.* $E_\beta^{(i)}(X,\cdot,z) = 0$ unless every $\operatorname{supp} w_a$, $a \in \beta$, has a plaquette with its corners in X; and then $d_i(X) \ge [\max_{a,b\in\beta} \mathsf{D}_i^{ab}/M - 2]_+$.

(e) *Start.* $\bar\Psi_{0;\beta} \le m^m(\mathsf{c}^\flat/2)^m$ with a constant $\mathsf{c}^\flat$; and if $\sum_p |w_a(p)| \le 81(\rho/a_K)^3|w_a|_\infty$ for a width $\rho \ge a_K$, then $\|\mathsf{W}_{L^{K-j}w_a}\|_{j,\kappa} \le \varsigma_j \mathsf{m}_{j,a}$, $\varsigma_j = \min\{1, t_j\}$.

(f) *Units.* $\gamma_j = c_\gamma x_j^{-1/2}$, $\bar\nu_{j;\beta} = \bar c_m x_j^{-m/2}\prod_{a\in\beta}(|w_a|_\infty/a_{K-j})$; for $i \le j$: $(\frac45)^{m/2}L^{(j-i)m/2}\bar\nu_{j;\beta} \le \bar\nu_{i;\beta} \le (\frac43)^{m/2}L^{(j-i)m/2}\bar\nu_{j;\beta}$, $\gamma_i \le (\frac43)^{1/2}L^{-(j-i)/2}\gamma_j$, $\sum_{i<j}\gamma_i \le 0.45\gamma_j$; and $\varkappa_j \le \frac16 L^{-3/2}$.

(g) *Recursions.* $\bar\Psi_{j;\beta} \le \sum_{\pi\in\mathrm{Part}(\beta)}(2(|\pi|-1))^{|\pi|-1}\prod_{\beta'\in\pi}\bar{\mathsf{h}}_{j;\beta'}$; and for $m \ge 2$ and every $a_* \in \beta$, with ω^{sp} formed with $\mathsf{n}_i^{\mathrm{cb}} = \mathsf{N}_i(w_{a_*})$,

$$\bar{\mathsf{h}}_{j;\beta} \le \gamma_j(\tfrac43)^{m/2}\sum_{i\le j}\omega_{j-i}^{\mathrm{sp}}(i)\,\sigma_{i;\beta}\,L^{(j-i+1)m/2}\,\bar\Psi_{i-1;\beta}, \qquad \sigma_{i;\beta} = e^{-c_\sigma[\max_{a,b\in\beta}\mathsf{D}_i^{ab}/M-2]_+},$$

$$c_\sigma = \min\{\tfrac12, 4\beta_{\mathrm{s}}\kappa\}.$$

No assumption of (a)–(g) depends on p; the order m enters through the displayed numbers alone. There are numbers $\tau^\natural, \gamma^\natural > 0$ formed from the constants such that $\theta^{[K-1]} \le \tau^\natural$ gives $\gamma_j \le \gamma^\natural$ for all $j < K$ and $2C_\omega\mathsf{c}^\flat\theta^{[0]} \le 1$, C_ω the constant of Lem. 4.9; and a number $r_1(\mathsf{b}) > 0$, formed from the constants and b and not increasing in b, such that every width $0 < \rho \le r_1(\mathsf{b})$ with $s(\rho) \le K$ has $\rho \ge 4Ma_K$ and $E_\sigma(\hat\rho_1/2) \le \varkappa_0^4$, where $E_\sigma(u) = \exp(-c_\sigma[u/\sqrt{3}M - 3]_+)$. ■

Proof. (a) The progression is $E_\lambda^{(j+1)} - E^{(j+1)} = \mathsf{St}^{[j]}(\mathfrak{f}_{\lambda,j}) - \mathsf{St}^{[j]}(\mathfrak{f}_{0,j})$ (equation [7, eq. 200]), and the multilinear derivatives at $\lambda = 0$ of a composition are sums over partitions; the dependence through w_λ alone gives the multilinearity. (b) Within the holomorphic, gauge-invariant input class of Prop. 6.32, the fourth family must vanish at the block field 0 and have finite norm eq. 6.xvii on the polydisc $|B(b)| < \mathsf{C}_{\mathcal{A}}\mathfrak{b}_j$. A family that vanishes at $B = 0$ and is bounded on that polydisc is at most its supremum times $|B|/\mathsf{C}_{\mathcal{A}}\mathfrak{b}_j$ (Schwarz's lemma); so $\mathsf{St}^{[j]}(\mathfrak{f})$ is formed for $\varkappa_j\mathfrak{n}[\mathfrak{f}] \le N_\star^{[j]}$, with

$$|\mathsf{St}^{[j]}(\mathfrak{f}) - \mathsf{St}^{[j]}(\mathfrak{f}')|(X,P,z) \le \theta^{[j]}\varkappa_j\,\mathfrak{n}[\mathfrak{f}-\mathfrak{f}']\,e^{-\lambda_{\mathrm{out}}d_{j+1}(X)},$$

and it is holomorphic in the parameters of $\mathfrak{f} + \sum s_l\mathfrak{g}_l$ on $\{\varkappa_j\mathfrak{n}[\cdot] < N_\star^{[j]}\}$. Dividing the Lipschitz bound by $|s|$ and letting $s \to 0$ gives $\|\mathsf{DSt}^{[j]}(\mathfrak{h})[\mathfrak{g}_1]\|_{j+1,\lambda_{\mathrm{out}}} \le \theta^{[j]}\varkappa_j\mathfrak{n}[\mathfrak{g}_1]$ wherever $\varkappa_j\mathfrak{n}[\mathfrak{h}] < N_\star^{[j]}$; this is (b) at $q = 1$. For $q \ge 2$, as $\varkappa_j\mathfrak{n}[\mathfrak{f}] \le N(\mathfrak{f}) \le \frac12 N_\star^{[j]}$, the point $\mathfrak{f} + \sum_{l\ge2} s_l\mathfrak{g}_l$ stays in that set on the polydisc of the radii $N_\star^{[j]}/2(q-1)\varkappa_j\mathfrak{n}[\mathfrak{g}_l]$ in $s_2,\dots,s_q$; Cauchy's inequality in these $q-1$ variables, applied to the first derivative in the direction $\mathfrak{g}_1$, gives the factor $\prod_{l\ge2}2(q-1)\varkappa_j\mathfrak{n}[\mathfrak{g}_l]/N_\star^{[j]}$, that is (b), with $(2(q-1))^{q-1} = 1, 2, 16, 216$ for $q = 1,\dots,4$.

This factor $1/\mathfrak{b}_j$ for every order in the sources is the whole of the loss $\mathfrak{b}_j^2$ that a bound through the size N would cost. $N(\mathfrak{f}_{0,j}) \le \frac12 N_\star^{[j]}$ is the half-ball estimate in the proof of Prop. YM3_I:122.2: the admitted bound $s_j = \theta^{[j]}/K_{\mathbf{M}} \le s_\omega^\star/2$ gives $N_j(f_{0,j}) \le N_\star \le w_\star/(4K_\Lambda s_\omega^\star) \le w_\star/(8K_\Lambda s_j) = N_{\star,j}/2$. [7, Prop. B.21] verifies this ceiling at the actual run. (c) Use the wide-domain and octet decomposition in the proof of Prop. YM3_I:51.16, with zero counterterms and all relative localization radii equal to 1. Put $l = j - i \ge 1$ and $D_l = \max\{2L^{l-1} - 2, 0\}$. The wide domains have size at least D_l. Their rooted sum gives the factor $\varpi_l = L^{3l}e^{-D_l}$ of equation YM3_I:(884). For $l = 1$ this is L^3; for $l \ge 2$,

$$D_l \ge 2(l-1)(L-1) \ge 8(l-1)\log L, \qquad \varpi_l \le L^{8-5l},$$

where the second inequality uses $L \ge 13$. For a narrow domain, equation YM3_I:(864) bounds the octet term minus its constant by

$$C_{\mathrm{oc}}\|G^{(i)}\|_{i,\kappa}(d_i(X)+2)^2L^{4-4l}e^{-\kappa d_i(X)}.$$

Sum first over domains meeting Cr and then over their base points. The rooted count in [7, Lem. B.10] applies with the marked sites of Cr in place of plaquette corners: each relevant domain contains one of the $\mathsf{n}_i^{\mathrm{cb}}(\mathsf{Cr})$ marked cubes, and $\#(X\cap\mathbb{T}^{(i)})\le 8M^3(d_i(X)+1)$. The remaining sum is bounded by the same convergent series of a polynomial in $d_i(X)$ times its exponential decay. This replaces the L^{3l} possible positions in the unrestricted count by $c_3\mathsf{n}_i^{\mathrm{cb}}(\mathsf{Cr})$. Taking the better of the two counts, with the normalization of equation YM3_I:(885), gives the narrow contribution $L^{-4l}\min\{L^{3l},c_3\mathsf{n}_i^{\mathrm{cb}}(\mathsf{Cr})\}$. The fixed constants, including L^4, are those in the bracket normalization and in c_3. There is no growing-radius factor here: the relative radii are 1. The level-j family and H have no lag and retain their ordinary norm bounds. This proves (c). (d) is the locality of the progression with the linear size of a domain. (e) The one-step theorem applied at the height 0, where the source stands in the fourth place itself, with Schwarz's lemma as in (b); the weight member: $|a_p|<\frac12\alpha_0^2\eta_j^4$ on the cores, and the count of plaquettes. (f) is arithmetic along the run (Lem. 4.3). (g) follows from (a), (b), (e) in the units, and from (c) with $\mathsf{Cr}=\mathsf{Cr}_{a_*}$, (d) and (f).

The number $\gamma^\natural$, formed from the constants, is the smallness that Lem. 4.9 reads: $\gamma^\natural\le(3.6\,C_\omega)^{-1}$, so that $\epsilon_j^{(1)}=C_\omega(0.9\gamma_j+\mathsf{c}^\flat\theta^{[0]})\le\frac34$ in Lem. 4.9(a) once $2C_\omega\mathsf{c}^\flat\theta^{[0]}\le1$; $\gamma^\natural$ is at most the number under which the feedback $\sum_{i'<j}\gamma_{i'}\le0.45\gamma_j$ is absorbed in the recursion of the proof of Lem. 4.9(b); and $\gamma^\natural\le(5C_1L^{7/2}\hat C)^{-1}$, which the proof of Lem. 4.9(e) reads, where $\hat C=\sup_{v>0}(1+v^3)\max\{1,v^{12}\}E_\sigma(v/2)$ is finite, E_σ decaying exponentially, and C_1 is the constant of the bound $\omega_l^{\mathrm{sp}}(i)\le C_1L^{-4l}(1+t_i)$ in that proof.

Then $\tau^\natural=\min\{6L^{3/2}\gamma^\natural,(2C_\omega\mathsf{c}^\flat)^{-1}\}$: for $j<K$, $\delta_j\le\delta_{K-1}$ (Lem. 4.3(a)) gives $\theta^{[j]}\le\theta^{[K-1]}$, in particular $\theta^{[0]}\le\theta^{[K-1]}$, and with $\varkappa_j\le\frac16L^{-3/2}$ of (f), $\theta^{[K-1]}\le\tau^\natural$ gives $\gamma_j=\theta^{[j]}\varkappa_j\le\gamma^\natural$ and $2C_\omega\mathsf{c}^\flat\theta^{[0]}\le1$. For r_1, let $C_7=\sup_{u\ge1}[4\mathfrak{p}_0\log(1+\log u)-c_\sigma u/4\sqrt{3}LM]$, finite, let $\Lambda_1(\mathsf{b})\ge1$ be the least number such that for all $\Lambda\ge\Lambda_1(\mathsf{b})$

$$\frac{c_\sigma}{\sqrt{3}L}\Lambda^{2\mathfrak{p}_0}\ge3c_\sigma+C_7+4\log^+(2\mathsf{C}_{\mathcal{A}}A_1M^{-3/2})+4\mathfrak{p}_0\big[\log(1+\log^+\mathsf{b})+\log(1+\Lambda)\big],$$

and $r_1(\mathsf{b})=e^{-\Lambda_1(\mathsf{b})}$. The left member grows like $\Lambda^{2\mathfrak{p}_0}$, $\mathfrak{p}_0\ge1$, the right one like $\log\Lambda$, so $\Lambda_1(\mathsf{b})$ is finite; it is formed from the constants and b, it holds no K, and since the right member increases with b, r_1 does not increase in b. Let $0<\rho\le r_1(\mathsf{b})$ with $s(\rho)\le K$, $\Lambda=\log\frac1\rho\ge\Lambda_1(\mathsf{b})\ge1$ and $u=\rho/a_K=\hat\rho_0$. The working level at $\varrho=\rho<1$ gives $u\ge\hat\rho_{\mathsf{k}(\rho)}\ge4MY_\rho^{2\mathfrak{p}_0}\ge4M\Lambda^{2\mathfrak{p}_0}\ge4M$, so $\rho\ge4Ma_K$. Next, $1/\varkappa_0=2\mathsf{C}_{\mathcal{A}}A_1M^{-3/2}y_0^{\mathfrak{p}_0}$, and $x_0=\beta=\mathsf{b}/a_K=\mathsf{b}u/\rho$ gives $y_0=\log\mathsf{b}+\log u+\Lambda$, whence $\log y_0\le\log(1+\log^+\mathsf{b})+\log(1+\log u)+\log(1+\Lambda)$, as $x+y+z\le(1+x)(1+y)(1+z)$ for $x,y,z\ge0$. As $\hat\rho_1=u/L$, the bound $E_\sigma(\hat\rho_1/2)\le\varkappa_0^4$ follows from

$$c_\sigma\Big(\frac{u}{2\sqrt{3}LM}-3\Big)\ge4\log^+(2\mathsf{C}_{\mathcal{A}}A_1M^{-3/2})+4\mathfrak{p}_0\log y_0.$$

Halve the first term: by $u\ge4M\Lambda^{2\mathfrak{p}_0}$ one half is at least $\frac{c_\sigma}{\sqrt{3}L}\Lambda^{2\mathfrak{p}_0}$, and the other half is at least $4\mathfrak{p}_0\log(1+\log u)-C_7$; with the bound of $\log y_0$, what is left is the display that defines $\Lambda_1(\mathsf{b})$, at $\Lambda=\log\frac1\rho$. The term $\log(1+\log u)$, through which y_0 grows with K, is beaten by the half of the exponent, so no number in the argument depends on K. □

Lemma 4.7 (The kernel family and the rest family)**.** In the data of Lem. 4.6 with $p\ge2$, let $\dot{\mathfrak{e}}_{a,j}=((E_a^{(i)},0)_{i\le j},(0))$ and $\mathfrak{u}_{a,j}=((0,0)_{i\le j},(\mathsf{W}_{L^{K-j}w_a}))$, so that $\mathfrak{f}_{\{a\},j}=\dot{\mathfrak{e}}_{a,j}+\mathfrak{u}_{a,j}$, and $\mathsf{B}_j(\mathfrak{g}_1,\mathfrak{g}_2)=\partial_{s_1}\partial_{s_2}\mathsf{St}^{[j]}(\mathfrak{f}_{0,j}+s_1\mathfrak{g}_1+s_2\mathfrak{g}_2)|_0$. With the kernel family $\mathsf{K}^{(j+1)}=\mathsf{B}_j(\mathfrak{u}_{1,j},\mathfrak{u}_{2,j})$, for $1\le j\le K-1$

$$E_{12}^{(j+1)}=\mathsf{K}^{(j+1)}+\mathsf{R}^{(j+1)},\qquad \mathsf{R}^{(j+1)}=\mathsf{B}_j(\dot{\mathfrak{e}}_{1,j},\mathfrak{u}_{2,j})+\mathsf{B}_j(\mathfrak{u}_{1,j},\dot{\mathfrak{e}}_{2,j})+\mathsf{B}_j(\dot{\mathfrak{e}}_{1,j},\dot{\mathfrak{e}}_{2,j})+\mathsf{DSt}^{[j]}(\mathfrak{f}_{0,j})[\mathfrak{f}_{12,j}],$$

and $\mathsf{R}^{(j+1)}$ is a pointed family of level $j+1$ that vanishes off Cr_1 and off Cr_2; $\mathsf{K}^{(j+1)}(X,\cdot,z)=0$ unless a plaquette of $\operatorname{supp}w_1$ and one of $\operatorname{supp}w_2$ have their corners in X. ■

Proof. Lem. 4.6(a) at $\beta = \{1,2\}$ has two partitions. B_j is additive in each entry: for bare arrays $\mathfrak{a}, \mathfrak{b}, \mathfrak{g}$ the value $h(t) = \mathsf{St}^{[j]}(\mathfrak{f}_{0,j} + t_1\mathfrak{a} + t_2\mathfrak{b} + t_3\mathfrak{g})(X, P, z)$ is holomorphic on a polydisc about 0, the size being subadditive, and $\partial_{s_1}\partial_{s_2}h(s_1, s_1, s_2)|_0 = (\partial_{t_1} + \partial_{t_2})\partial_{t_3}h(0)$. The first and second derivatives of the step are pointed families with the locality of the step, and Lem. 4.6(d) gives where they vanish. □

Lemma 4.8 (One level at a regular configuration)**.** Let g satisfy (Γ1), (K, β) be a first-passage datum at g and $\ell \in \mathfrak{L}(g)$.

(a) Let the frame at a level k_0 hold for a pair of real Lipschitz functions (Lem. 4.5(a)), and let $k_0 \le n < K$, $1 \le i' \le n$, $A \in \mathsf{Cc}_n(r_n)$, Cr a set of bare sites, $D' \ge 0$, and f a pointed family of level i' that vanishes off Cr and on the domains X with $d_{i'}(X) < D'$. Then

$$|\Delta[\mathsf{f}]| \le C_\Delta\, \mathsf{n}^{\mathrm{cb}}_{i'}(\mathsf{Cr})\, \|\mathsf{f}\|_{i',\lambda_{\mathrm{out}}}\, r_n^2\, L^{-4(n-i')} e^{-D'/2}.$$

(b) Let G be a family of local terms of level i that vanishes unless $\mathcal{P}(X)$, the set of the plaquettes with their corners in X, meets a set $\mathcal{S}$ of plaquettes and $d_i(X) \ge D_i$, and let N_i be the number of the cubes of π_i that hold a corner of a plaquette of $\mathcal{S}$. At every configuration U whose pairs of level i lie in the cores, $\sum_{(X,z)} |G(X,U,z)| \le 8M^3 C_{\mathrm{E},1} \mathsf{N}_i e^{-D_i} \|G\|_{i,\lambda_{\mathrm{out}}}$; if U is besides regular of a level $t' \ge i$ at a radius ε with $4M\varepsilon \le c_{\mathrm{A}}\alpha_0$, then $\sum_{(X,z)} |G(X,U,z) - G(X,\mathbf{1},z)| \le 2C_{\mathrm{f}} L^{-4(t'-i)} \mathsf{N}_i e^{-D_i/2} \|G\|_{i,\lambda_{\mathrm{out}}}$.

The numbers $C_\Delta, C_{\mathrm{f}}, C_{\mathrm{E},1}, c_{\mathrm{A}}$ are formed from the constants. Of (Γ1) the proof of (a) uses only, besides the frame at k_0, the two bounds $2r_n \le \varrho_{\mathrm{ch}}$ and $x_n^{-1} \le r_n^2$ at the heights concerned, and the proof of (b) only $g \le \gamma$ and $g^{-2} \ge x_{\mathrm{J}}$; both use besides the bound $L^m \ge LMy_K$ of the sides $\ell = 2L^m \in \mathfrak{L}(g)$. So (a) holds whenever the frame at k_0 and these two bounds hold, and (b) holds whenever $g \le \gamma$ and $g^{-2} \ge x_{\mathrm{J}}$, with no frame. ■

Proof. (a) Let $k_0 \le n < K$. Then $r_n > 0$ and $2r_n \le \varrho_{\mathrm{ch}}$ (Lem. 4.3(a), (b)); in the frame at k_0, Lem. 2.3, at the radii $r_n < r' = 2r_n \le \varrho_{\mathrm{ch}}$, gives that $\tilde{U}$ is regular of level n at the radius $\varepsilon := 6r_n$ on T_ℓ and that the actions of the bare arrays over the absolute system of height n are formed at it; this is where the frame is read. For one domain X of size d and $\mathsf{E} = \mathsf{f}(X, \cdot, z)$, the evaluation of a local term at a regular configuration gives either $|\mathsf{E}(\tilde{U}) - \mathsf{E}(\mathbf{1})| \le C_{\mathrm{A}} \|\mathsf{f}\| e^{-\lambda_{\mathrm{out}} d} (\varepsilon/\alpha_0)^2 L^{-4(n-i')} M^2 (1+d)^2$, or $d \ge [d_*]_+$ and $|\mathsf{E}(\tilde{U}) - \mathsf{E}(\mathbf{1})| \le 2\|\mathsf{f}\| e^{-\lambda_{\mathrm{out}} d}$, with the depth $d_* = \min\{c_{\mathrm{A}}\alpha_0\tau/6M, (L^{m+K-i'} - 5)/2M\} - 2$, $\tau = L^{2(n-i')}/r_n$.

A contributing X holds one of the $\mathsf{n}^{\mathrm{cb}}_{i'}$ cubes, has $d \ge D'$ and at most $8M^3(d+1)$ base points; the sum over domains through a cube is bounded by [7, eq. 320]. In the tail, $e^{-[d_*]_+/2} \le e[e^{-c_{\mathrm{A}}\alpha_0\tau/12M} + e^{-(L^{m+K-i'}-5)/4M}]$; the first part is $\le C\tau^{-2}$; for the second, $L^m \ge LMy_K$ on the sides of $\mathfrak{L}(g)$, $x_n \le \frac{5}{3} L^{K-n} x_K$ and $x_n^{-1} \le r_n^2$ give $e^{-(L^{m+K-i'}-5)/4M} \le \frac{5}{3} e^{5/(4M)} r_n^2 L^{-4(n-i')}$. So the bound holds on every side $\ell \in \mathfrak{L}(g)$. (b) is the same count at a configuration whose pairs lie in the cores, and, for the difference, at a configuration regular of level t', with $\vartheta = L^{-(t'-i)}$. □

Lemma 4.9 (Sizes at three lengths)**.** Let g satisfy (Γ1), (K, β) be a first-passage datum at g and $\ell \in \mathfrak{L}(g)$, and let real weights be placed at (ρ, r), or at (ρ, r, d), or form two pairs at (ρ, r, d), with $0 < \rho \le r \le \mathsf{d} \le e^{-2\sqrt{3}}$, $s(\rho) \le K$, $\rho \le r_1(\mathsf{b})$. Put $q = \rho/r$, $u_i = \hat{r}_i$, $v_i = \hat{\mathsf{d}}_i$, $\Xi^r_j = q^6 \min\{1, u_j\}^3$, $\mathsf{z}(\beta) = q^{3c(\beta)}(\rho/\mathsf{d})^6$ for crossing β, $\mathsf{D}(u) = [u/\sqrt{3}M - 3]_+$, $C_\sigma = \sup_u \max\{1, u^6\} E_\sigma(u)$ and $\bar{\nu}_j = \bar{\nu}_{j;12}$. There are numbers $C_\omega, \mathsf{A}_{\mathrm{N}}, C_{\mathrm{B}}, \mathsf{A}^\times_m, C^\Psi_m, C_\times$ formed from the constants such that for $1 \le j \le K-1$:

(a) $\mathfrak{n}[\mathfrak{u}_{a,j}] \le \varsigma_j \mathsf{m}_{j,a}$, $\mathfrak{n}[\dot{\mathfrak{e}}_{a,j}] \le \epsilon^{(1)}_j \varsigma_j \mathsf{m}_{j,a}$ with $\epsilon^{(1)}_j = C_\omega(0.9\gamma_j + \mathsf{c}^\flat\theta^{[0]}) \le \frac{3}{4}$, and $\bar{\mathsf{h}}_{j;a}, \bar{\Psi}_{j;a} \le \varsigma_j$;

(b) for a pair, $\bar{\mathsf{h}}_{j;12} \le \mathsf{A}_{\mathrm{N}} \gamma_j \Xi^r_j$ and $\bar{\Psi}_{j;12} \le 2\varsigma_j^2 + \mathsf{A}_{\mathrm{N}} \gamma_j \Xi^r_j$;

(c) $\|\mathsf{K}^{(j+1)}\|_{j+1,\lambda_{\mathrm{out}}} \le \frac{1}{2}\varsigma_j^2 \bar{\nu}_j$ and $\|\mathsf{R}^{(j+1)}\|_{j+1,\lambda_{\mathrm{out}}} \le [\frac{3}{2}\epsilon^{(1)}_j \varsigma_j^2 + \mathsf{A}_{\mathrm{N}} \gamma_j \Xi^r_j] \bar{\nu}_j$;

(d) below the level of the distance: if $1 \le n \le K-1$, $u_n \ge \sqrt{3}M$, and the pairs of all levels $\le n$ of U lie in the cores, then $|\sum_{i \le n} \mathsf{Sum}[E^{(i)}_{12}](U)| \le C_{\mathrm{B}} q^6 e^{-u_n/2\sqrt{3}M} \bar{\nu}_n$;

(e) for crossing β, $m = |\beta| \le 4$, with $\Xi^3_j(\beta) = \mathsf{z}(\beta)\min\{1, v_j\}^{3/2}$: $\bar{\mathsf{h}}_{j;\beta} \le \mathsf{A}^{\times}_m \gamma_j \Xi^3_j(\beta)$, and $\mathsf{N}_i E_\sigma(v_i)\bar{\Psi}_{i-1;\beta} \le 108\, C^{\Psi}_m \mathsf{z}(\beta)\min\{1, v_i\}^{3/2} E_\sigma(v_i/6)$ for $1 \le i \le K$; and at $n = \mathsf{k}(\mathsf{d})$, with $P = v_n$ and $A \in \mathsf{Cc}_n(r_n)$, $|G_{n;\beta}(A)| \le C_\times e^{-P/2\sqrt{3}M}\mathsf{z}(\beta)\bar{\nu}_{n;\beta}$;

(f) *without the placing:* the bounds (a)–(d), and those of (e) but the last one, at the level $\mathsf{k}(\mathsf{d})$, hold with the same numbers for real weights, none identically 0, that form one pair, or a pair and a third source, or two pairs, with numbers $0 < \rho \le r \le \mathsf{d} \le 1$ such that for $0 \le i \le K$: $a_K \le \rho$; (P1) and (P2); the corners of the plaquettes of the supports are at least $u_i/\sqrt{3} - 1$ level-i spacings apart in a cluster and at least $v_i/\sqrt{3} - 1$ across; $\rho \ge \frac{2}{3} M a_K$; $E_\sigma(\hat{\rho}_1/2) \le \varkappa_0^4$; and $\gamma_{K-1} \le \gamma^\natural$. No bound on the distances of the centres and no bound $\mathsf{d} \le e^{-2\sqrt{3}}$ is asked in (f); these serve only the frame, the working level, the number R_w and the rings.

■

Proof. All of it is Lem. 4.6 with (P1)–(P3) of Lem. 4.5, $\rho \ge \frac{2}{3} M a_K$ (so $\varsigma_0 = 1$), $E_\sigma(\hat{\rho}_1/2) \le \varkappa_0^4$, $\gamma_j \le \gamma^\natural$, and three inequalities of arithmetic, $\omega^{\mathrm{sp}}_l(i) L^{l+1}\varsigma_{i-1} \le C_\omega \varsigma_{i+l}$, $\omega^{\mathrm{sp}}_l(i) \le C_1 L^{-4l}(1 + t_i)$ and $(1 + t_i)\varsigma^2_{i-1} \le 2L^6 t_i^2$, which use only $a_K \le \rho$ and (P2). (a) By induction on j, with (c) of Lem. 4.6 at $\mathsf{Cr} = \mathsf{Cr}_a$, (e) and the first of these inequalities; then (b) of Lem. 4.6 at $q = 1$. (b) In the recursion (g) of Lem. 4.6 the support rule and (P3) give $\sigma_{i;12} \le E_\sigma(u_i)$; the kernel parts are $(1 + t_i)2\varsigma^2_{i-1} \le 4L^6 q^6 u_i^6$, and

$$\sum_{i \le j} L^{-3(j-i)} \min\{u_i^6, C_\sigma\} \le 2.002\, C_\sigma \min\{1, u_j\}^3$$

($u_i = L^{j-i}u_j$: two geometric sums); the term $i = 1$ uses $E_\sigma(u_1) \le E_\sigma(u_1/2)E_\sigma(\hat{\rho}_1/2) \le E_\sigma(u_1/2)\varkappa_0^4$; the feedback $\sum_{i'<j} \gamma_{i'} \le 0.45\gamma_j$ is absorbed under $\gamma_j \le \gamma^\natural$. (c) $\mathsf{K}^{(j+1)} = \mathsf{B}_j(\mathfrak{u}_1, \mathfrak{u}_2)$ and (b) of Lem. 4.6 at $q = 2$; for $\mathsf{R}^{(j+1)}$, Lem. 4.7 with (a), (b) and $\frac{1}{2}(2 + \epsilon)\epsilon \le \frac{3}{2}\epsilon$. (d) By the support rule $E^{(i)}_{12}$ vanishes on the domains with $d_i(X) < \mathsf{D}(u_i)$; Lem. 4.8(b) with (b), $\bar{\nu}_{i-1} \le \frac{4}{3} L^{n-i+1}\bar{\nu}_n$, and $\sum_l L^{l+1} e^{-\frac{5}{8}\mathsf{D}(L^l u_n)} \le C e^{-u_n/2\sqrt{3}M}$. (e) Induction on m and, for one β, on the level, through one inequality on the natural sizes $\mu_i(\beta') = \hat{\rho}_i^{3|\beta'|} u_i^{-3c(\beta')} v_i^{-6[\beta' \text{ crosses}]}$, which is $\hat{\rho}_i^3$ at a point, $q^6 u_i^3$ at a whole pair and $\mathsf{z}(\beta')$ at a crossing set, at every level (there $3|\beta'| = 3c(\beta') + 6$): for crossing β and a partition π of it with $|\pi| \ge 2$,

$$\prod_{\beta' \in \pi} \mu_i(\beta') \le \max\{1, v_i^{12}\}\, \mathsf{z}(\beta);$$

the quotient is $u_i^{3s} v_i^{6-6k_\times}$, s the number of whole pairs of β that π cuts and $k_\times$ the number of its crossing blocks (with k blocks that do not cross, $3k + 6k_\times = 6 + 3s$), and $u_i \le v_i$, $k_\times \le 2$, and $k_\times = 2$ forces $s = 2$: at $k_\times = 0$ the quotient is $\le v_i^{3s+6}$, at $k_\times = 1$ (then $s = k \ge 1$) it is $u_i^{3s} \le \max\{1, v_i^6\}$, at $k_\times = 2$ it is $(u_i/v_i)^6 \le 1$. At $v_i < 1$ the display gives the bound $\mathsf{z}(\beta)$ alone (at $k_\times = 2$ the quotient is 1 when $r = \mathsf{d}$); the factor $\min\{1, v_i\}^{3/2}$ comes, at $k_\times \ge 1$, from the crossing blocks, whose factor of the induction is kept, and at $k_\times = 0$ from the quotient.

The blocks. At a level $i - 1 \ge 1$: a point by (a), $\bar{\mathsf{h}}_{i-1;a} \le \varsigma_{i-1} \le t_{i-1} = L^3 t_i \le L^3 \mu_i(\{a\})$; a whole pair c by (b), $\bar{\mathsf{h}}_{i-1;c} \le \mathsf{A}_{\mathrm{N}}\gamma^\natural q^6 \min\{1, Lu_i\}^3 \le \mathsf{A}_{\mathrm{N}}\gamma^\natural L^3 \mu_i(c)$; a crossing proper subset β' by the first bound of (e) at $|\beta'| < m$, $\bar{\mathsf{h}}_{i-1;\beta'} \le \mathsf{A}^{\times}_{|\beta'|}\gamma^\natural \mathsf{z}(\beta')\min\{1, Lv_i\}^{3/2} \le \mathsf{A}^{\times}_{|\beta'|}\gamma^\natural L^{3/2}\mu_i(\beta')\min\{1, v_i\}^{3/2}$. With $\mathsf{a}_{(2)} = \max\{1, \mathsf{A}_{\mathrm{N}}\gamma^\natural\}$ and $\mathsf{a}_{(m+1)} = \max\{\mathsf{a}_{(m)}, \mathsf{A}^{\times}_m\gamma^\natural\}$, every partition π of β with $|\pi| \ge 2$ has

$$\prod_{\beta' \in \pi} \bar{\mathsf{h}}_{i-1;\beta'} \le (L^3 \mathsf{a}_{(m)})^m \min\{1, v_i\}^{3/2} \max\{1, v_i^{12}\}\, \mathsf{z}(\beta) :$$

if π has a crossing block, the factor $\min\{1, v_i\}^{3/2}$ of one of them is kept and the display bounds the rest; if it has none, $k_\times = 0$ and the quotient is $\le v_i^{3s+6} \le \min\{1, v_i\}^{3/2}\max\{1, v_i^{12}\}$, as $s \le 2$. The block $\pi = \{\beta\}$ is the first bound of (e) for β itself at the level $i - 1$: $\bar{\mathsf{h}}_{i-1;\beta} \le \mathsf{A}^{\times}_m\gamma^\natural L^{3/2}\mathsf{z}(\beta)\min\{1, v_i\}^{3/2}$.

The second bound. Put $T_i = (1+t_i)E_\sigma(v_i)\bar\Psi_{i-1;\beta}$ and $\hat C = \sup_v(1+v^3)\max\{1,v^{12}\}E_\sigma(v/2)$. As $t_i \le \hat\rho_i^3 \le v_i^3$, $E_\sigma(v)\le E_\sigma(v/2)^2$ and $E_\sigma(v/2)\le E_\sigma(v/6)^3\le 1$, we have $(1+t_i)\max\{1,v_i^{12}\}E_\sigma(v_i)\le \hat C\,E_\sigma(v_i/6)$; so the first recursion of (g) of Lem. 4.6 gives for $2\le i\le K$

$$T_i \le \hat C\big[L^{3/2}\mathsf{A}_m^\times\gamma^\natural + \mathsf{b}_m(L^3\mathsf{a}_{(m)})^m\big]\mathsf{z}(\beta)\min\{1,v_i\}^{3/2}E_\sigma(v_i/6),\qquad \mathsf{b}_m = \sum_{\pi\in\mathrm{Part}(\beta),\,|\pi|\ge 2}(2(|\pi|-1))^{|\pi|-1} = 2,22,326.$$

At $i=1$, (e) of Lem. 4.6 gives $\bar\Psi_{0;\beta}\le \mathsf{a}_m^\flat := m^m(\mathsf{c}^\flat/2)^m$; as $\rho\ge\frac23 Ma_K$ and $M\ge L$, $v_1\ge u_1\ge\hat\rho_1\ge\frac23$, so that $\mathsf{z}(\beta)=\hat\rho_1^{3m}u_1^{-3c(\beta)}v_1^{-6}\ge(\hat\rho_1/v_1)^{3m}\ge(\frac23)^{12}\max\{1,v_1^{12}\}^{-1}$ and $\min\{1,v_1\}^{3/2}\ge(\frac23)^{3/2}$; hence $T_1\le \mathsf{a}_m^\flat(\frac32)^{27/2}\hat C\,\mathsf{z}(\beta)\min\{1,v_1\}^{3/2}E_\sigma(v_1/6)$. With $C_m^\Psi = 2\hat C[\mathsf{b}_m(L^3\mathsf{a}_{(m)})^m + \mathsf{a}_m^\flat(\frac32)^{27/2}]$, $\mathsf{A}_m^\times = 2.5\,C_1L^2C_m^\Psi$ and $\gamma^\natural\le(5C_1L^{7/2}\hat C)^{-1}$, a smallness of the same kind as the one that absorbs the feedback in (b) and part of the choice of $\gamma^\natural$, the feedback $\hat C L^{3/2}\mathsf{A}_m^\times\gamma^\natural$ is $\le\frac12 C_m^\Psi$, and $T_i\le C_m^\Psi\mathsf{z}(\beta)\min\{1,v_i\}^{3/2}E_\sigma(v_i/6)$ for $1\le i\le K$; with $\mathsf{N}_i\le 108(1+t_i)$ (P2) this is the second bound of (e). *The first bound.* By the second recursion of (g), with $\sigma_{i;\beta}\le E_\sigma(v_i)$ (the corners across are $v_i/\sqrt3-1$ apart, and $[(v_i/\sqrt3-1)/M-2]_+\ge\mathsf{D}(v_i)$), $\omega^{\mathrm{sp}}_{j-i}(i)\le C_1L^{-4(j-i)}(1+t_i)$, $v_{j-l}=L^lv_j$ and $\min\{1,L^lv_j\}^{3/2}\le L^{3l/2}\min\{1,v_j\}^{3/2}$:

$$\bar{\mathsf{h}}_{j;\beta}\le(\tfrac43)^{\frac m2}C_1L^{\frac m2}\gamma_j\sum_{l=0}^{j-1}L^{(-4+\frac m2)l}T_{j-l}\le(\tfrac43)^{\frac m2}C_1L^{\frac m2}C_m^\Psi\,\gamma_j\,\mathsf{z}(\beta)\min\{1,v_j\}^{3/2}\sum_{l\ge0}L^{(-4+\frac m2+\frac32)l};$$

the exponent $-4+\frac m2+\frac32\le-\frac12$ ($m\le4$) gives $\sum_l\le1.39$, and $\bar{\mathsf{h}}_{j;\beta}\le\mathsf{A}_m^\times\gamma_j\Xi_j^3(\beta)$. The induction runs over $m=2,3,4$ and, for one β, over the level: T_i reads the first bounds at the levels $\le i-1$ only. The bound of $G_{n;\beta}$ at the level of d is Lem. 4.8(b) with the factors $e^{-\frac12\mathsf{D}(v_i)}$, summed over $i\le n$. (f): the proofs of (a)–(d) and of the first two bounds of (e) use of the weights only the properties listed in (f), the separation of the corners standing for (P3); the last bound of (e) uses besides the working level $\mathsf{k}(\mathsf{d})$, which asks $\mathsf{d}\le e^{-2\sqrt3}$. □

Lemma 4.10 (One source of any width)**.** Let g satisfy (Γ1), let the frame at k_0 hold for a pair that holds the source w_a, and let w_a satisfy (P1), (P2) at a width ρ with $\frac23 Ma_K\le\rho$. Then for $k_0\le n\le K$ and $A\in\mathsf{Cc}_n(r_n)$, with $q_{n;a}(F)=\frac{1}{2a_K}\sum_p w_a(p)|F_p|^2$,

$$|G_{n;a}(A)-G_{n;a}(0)-q_{n;a}(\mathsf{ang}^{[n]}(A))|\le C_{\mathrm S}\,\mathfrak{b}_n^2\delta_n\,\hat\rho_n^3\,\mathsf{u}_a.$$

■

Proof. At $n=K$, $A=0$ and both members vanish; let $k_0\le n<K$. $G_{n;a}(A)=\mathcal{O}(w_a)(\tilde U)+\sum_{i\le n}\mathsf{Sum}[E_a^{(i)}](\tilde U)$ and $a_p(\mathbf{1})=0$. The weight member: the expansion of a_p on the chain class and (P1) give $|\mathcal{O}(w_a)(\tilde U)-q_{n;a}(\mathsf{ang}^{[n]}(A))|\le Cr_n^3\hat\rho_n^3x_n\mathsf{u}_a\le C\mathfrak{b}_n^2\delta_n\hat\rho_n^3\mathsf{u}_a$ (Lem. 4.3). The families: Lem. 4.8(a), in the frame at k_0, at Cr_a, $D'=0$, with $\|E_a^{(i)}\|\le\varsigma_{i-1}\mathsf{m}_{i-1,a}\mathsf{g}_i$, $\mathsf{g}_i=2\gamma_{i-1}$ ($i\ge2$), $\mathsf{g}_1=\mathsf{c}^\flat\theta^{[0]}$ (Lem. 4.9(a)), and $\mathsf{N}_i\varsigma_{i-1}\le216L^3t_i$; then every level gives the same natural size

$$t_i\,\mathsf{m}_{i-1,a}L^{-4(n-i)}=\tfrac{81}{2}\alpha_0^2L\,\hat\rho_n^3|w_a|_\infty/a_{K-n}$$

times g_i, and $\sum_{i\le n}\mathsf{g}_i\le C\delta_n$, $|w_a|_\infty r_n^2/a_{K-n}\le C\mathfrak{b}_n^2\mathsf{u}_a$. □

The kernel families.

Lemma 4.11 (From a real segment to the second order)**.** Let $R>0$, $0<\mathsf{e}\le\mathsf{M}$, and let ψ be holomorphic on $\{|\zeta|<R\}$ with $|\psi|\le\mathsf{M}$ there, $|\psi(t)|\le\mathsf{e}$ for real $t\in(-R,R)$, and $\psi(0)=\psi'(0)=0$. With $\Lambda=\max\{1,\log(\mathsf{M}/\mathsf{e})\}$, $|\psi(\zeta)|\le4e\,\mathsf{e}\,\Lambda^2|\zeta|^2/R^2$ for $|\zeta|\le R/2\Lambda$. ■

Proof. On the upper half disc $h(\zeta) = \frac{2}{\pi} \arg \frac{R+\zeta}{R-\zeta}$ is harmonic, 0 on the diameter and 1 on the arc; $\log|\psi| - \log \mathsf{e} - h \log(\mathsf{M}/\mathsf{e})$ is subharmonic and bounded above, with nonpositive boundary values but at two points, so it is ≤ 0 (the two-constants theorem); the same on the lower half disc. As $h(\zeta) \le 2|\zeta|/R$, $|\psi| \le e\,\mathsf{e}$ on $|\zeta| \le R/2\Lambda$; $\psi(\zeta)/\zeta^2$ is holomorphic, and the maximum principle on that disc gives the claim. □

Lemma 4.12 (A local gauge)**.** In units of a level, let the bare lattice have the spacing $\xi \le \frac{1}{13}$, let x_0 be a bare site, $N_s = \{x : |x - x_0|_\infty \le s\}$, $\rho \ge 2$ with $4\rho + 4$ at most the side of the torus, and let $v = (A, \mathbf{J})$ be a real pair of $\mathfrak{su}(2)$-valued 1-forms on the bare bonds with $|A| \le \alpha \le \frac{1}{60}$, $|\nabla^\xi A| \le \gamma_\circ$, $|\mathbf{J}| \le \iota$. With $c_\mu = A_\mu(x_0)$, $X(x) = \sum_\mu (x - x_0)_\mu c_\mu$ and $u = e^{iX}$ on $N_{2\rho+1}$, there is a real pair $v' = (A', \mathbf{J}')$ which at the bonds with base point in N_ρ is the gauge transform of v under u, with $\|v'\| \le \max\{\rho(6\gamma_\circ + 19\alpha^2), \iota\}$ in the norm of the chart, $|A'(b)| \le 3s\gamma_\circ + (9s + 8\xi)\alpha^2$ at the bonds with base point in N_s, $s \le \rho$, and $|\nabla^\xi A'| \le \frac{100}{99}(\gamma_\circ + 3\alpha^2)$ on N_ρ. In the sharp first-difference bound, the whole difference stencil must lie in N_ρ; the chart-norm bound is global.

The same holds in a local form, with data on $N_{\rho+1}$ alone: let U be a configuration, $\mathsf{g}\colon N_{\rho+1} \to SU(2)$ and A a real $\mathfrak{su}(2)$-valued 1-form on the bonds $b = \langle b_-, b_+\rangle$ with both end points in $N_{\rho+1}$ such that $\mathsf{g}(b_-)U(b)\mathsf{g}(b_+)^{-1} = e^{i\xi A(b)}$, $|A| \le \alpha \le \frac{1}{60}$ and $|\nabla^\xi A| \le \gamma_\circ$ there, the current $\mathbf{J}(U)$ of U being at most ι there; with c_μ and X as above and $h = e^{iX}\mathsf{g}$ on $N_{\rho+1}$, there is a real pair v' on all the bare bonds which, at the bonds with base point in N_ρ, is the image of U and of $\mathbf{J}(U)$ under the gauge transformation h, and which obeys the three bounds above; and if $\|v'\|$ is less than the radius r_α of the ball of the chart, every local term of the level whose domain lies in $N_{\rho-1}$, formed at U, has the same value at $P[v']$ restricted to that domain. ■

Proof. For hermitian Y, a, c with $|a| + |c| \le \frac{1}{4}$ let $\Psi(Y, a, c) = -i \log(e^{iY} e^{ia} e^{-i(Y+c)})$; matrix calculus gives $|\Psi| \le |a - c| + 3|Y||c| + 2(|a| + |c|)^2$ and a Lipschitz bound in (Y, a). Put $A^u_\mu(x) = \xi^{-1}\Psi(X(x), \xi A_\mu(x), \xi c_\mu)$, so that $e^{i\xi A^u_\mu(x)} = u(x) e^{i\xi A_\mu(x)} u(x + \xi e_\mu)^{-1}$, a real form; $|X| \le 3s\alpha$ and $|A_\mu - c_\mu| \le 3s\gamma_\circ$ on N_s along three segments. Cut off with $\chi(x) = \min\{1, \max\{0, 2 - |x - x_0|_\infty/\rho\}\}$: $A' = \chi A^u$, $\mathbf{J}' = \chi u \mathbf{J} u^{-1}$. The point is that all first difference quotients of A are bounded; lowered by m levels, $|A|$ falls like L^{-m} and $|\nabla A|$ like L^{-2m}, and only the constant part is too large, which a gauge linear in x removes at the price $|c|^2$.

In the local form A^u is formed by the same formula from the potential A of the transform of U under g, so that $e^{i\xi A^u(b)} = h(b_-)U(b)h(b_+)^{-1}$ at the bonds with both end points in $N_{\rho+1}$, and the cut-off $\chi(x) = \min\{1, \max\{0, (\rho + 1 - 2\xi - |x - x_0|_\infty)/(1 - 2\xi)\}\}$, of the width $1 - 2\xi$ and equal to 1 on N_ρ, takes the place of the cut-off of the width ρ: $A' = \chi A^u$, $\mathbf{J}' = \chi h \mathbf{J}(U) h^{-1}$. Both vanish at the bonds whose base point lies at a distance $\ge \rho + 1 - 2\xi$ from x_0, and the other bonds have both end points in $N_{\rho+1}$, where A, h and the bound ι are given. As $\nabla^\xi_\nu(\chi A^u)(x) = \chi(x + \xi e_\nu)\nabla^\xi_\nu A^u(x) + A^u(x)(\chi(x + \xi e_\nu) - \chi(x))/\xi$, $|\chi(x + \xi e_\nu) - \chi(x)| \le \frac{13}{11}\xi$, and, on $N_{\rho+1}$, $|\nabla^\xi A^u| \le \frac{100}{99}(\gamma_\circ + 3\alpha^2)$ and $|A^u| \le 3(\rho + 1)\gamma_\circ + (9\rho + 9.62)\alpha^2$ (the third and the second bound, at $s = \rho + 1$, by the Lipschitz bound of Ψ and the three segments in $N_{\rho+1}$, with $8\xi \le 0.62$), for $\rho \ge 2$

$$|\nabla^\xi A'| \le \tfrac{100}{99}(\gamma_\circ + 3\alpha^2) + \tfrac{13}{11}\left[3(\rho+1)\gamma_\circ + (9\rho + 9.62)\alpha^2\right] \le \rho(6\gamma_\circ + 19\alpha^2), \qquad |A'| \le \rho(6\gamma_\circ + 19\alpha^2),$$

which with $|\mathbf{J}'| \le \iota$ gives the global chart bound; on stencils contained in N_ρ, the cutoff equals 1 and the sharper local bounds are unchanged. The bonds that a local term with its domain in $N_{\rho-1}$ reads have their end points in N_ρ, where $P[v']$ is the image of U under h, and a local term is a gauge invariant function of the pair on its domain. □

Proposition 4.13 (The kernel families)**.** Let g satisfy (Γ1), let the frame at k_0 hold for (w_1, w_2), let the forms q_n be formed over a base level $k_\flat \le k_0$, and let $k_0 \le i < n \le K$, $m = n - i - 1$, $A \in \mathsf{Cc}_n(r_n)$. As $k_\flat \le i$, $\mathcal{S}_{i+1} - \mathcal{S}_i = \upsilon_{i+1}\mathcal{C}_{(i+1)}$, so that $\mathcal{Q}_i = q_{i+1} - q_i$ is the slice $\mathcal{Q}_i(F) = a_K^{-1}\upsilon_{i+1}\sum_{p,q} w_1(p)\mathcal{C}_{(i+1)}(p, q)w_2(q)\langle F_p, F_q\rangle$. At $n = K$, $A = 0$ and the member LM below is 0; the

numbers below are formed for $n < K$. Put $\mathsf{w}_a = a_K^{-1}\eta_i^4 \sum_p |w_a(p)|$; let R_w be such that every corner of a plaquette of $\operatorname{supp} w_1 \cup \operatorname{supp} w_2$ lies within the sup distance R_w, in level-$(i+1)$ units, of one bare site; $\alpha = L^{-m}r_n$, $\gamma_\circ = L^{-2m}r_n$, $\mathsf{c}_i = C_{\mathrm{tp}}e^{-\delta_{\mathrm{tp}}[\mathsf{D}_i^{12}-1]_+}$, $\mathsf{M}_i^{\mathsf{K}} = 8M^3 C_{\mathrm{E},1}\mathsf{N}_{i+1}(w_1)\|\mathsf{K}^{(i+1)}\|_{i+1,\lambda_{\mathrm{out}}}$, $\mathsf{M}_i = 2\mathsf{M}_i^{\mathsf{K}} + 4L^{-4}\mathsf{c}_i g_i^2\delta_i^2\mathsf{w}_1\mathsf{w}_2$, and, with $\varrho_\wedge = \min\{\varrho_{\mathrm{ch}}, \varrho_{\mathrm{fl}}\}$, $\epsilon = \epsilon_i^{\mathrm{cu}}$ and $\mathsf{c}_{\mathbf{C}}$ of Lem. 4.14(a), $\epsilon' = \epsilon + 2\mathsf{c}_{\mathbf{C}}\delta_i/r_\alpha$, the number $C_{\mathrm{tp}}^{\mathrm{row}}$ of Prop. YM3_I:120.6 that bounds the row sums $\sum_b |\mathsf{T}_i^{\mathrm{cv}}(p,b)| \le C_{\mathrm{tp}}^{\mathrm{row}}\eta_i^2$ of the matrix $\mathsf{T}_i^{\mathrm{cv}}$ of step 3 (Prop. YM3_I:120.6, equation YM3_I:(2239)), $\bar X = (4L^{-2}\mathsf{c}_{\mathfrak{h}} + 2\mathsf{c}_{\mathfrak{h}}^2)\delta_i^2$ and $\bar\rho = \mathsf{M}_{\mathrm{pl}}^2\delta_i^3/\varrho_\wedge^3$,

$$\mathsf{E}_i = 12L^{-4}(C_{\mathrm{tp}}^{\mathrm{row}})^2 g_i^2\delta_i^2\epsilon' + 6\sqrt{3}L^{-2}(C_{\mathrm{tp}}^{\mathrm{row}})^3 g_i^3\delta_i\epsilon(1 + 2\mathsf{c}_{\mathbf{C}} + 2\epsilon) + \tfrac{9}{4}(C_{\mathrm{tp}}^{\mathrm{row}})^4 g_i^4\big[\epsilon + 4\epsilon^2 + 4(\mathsf{c}_{\mathbf{C}} + \epsilon)\epsilon^2 + 2\epsilon'(2\mathsf{c}_{\mathbf{C}} + \epsilon)\big] + 4\bar X\bar\rho + 2\bar\rho^2,$$

and for $\delta_i \le \mathsf{r} \le \frac12\varrho_{\mathrm{ch}}$ let $\mathsf{E}_i(\mathsf{r})$ be the same number with $g_i^2\mathsf{r}^2$, $g_i^3\mathsf{r}$ in the places of $g_i^2\delta_i^2$, $g_i^3\delta_i$, with $\epsilon + 2\mathsf{c}_{\mathbf{C}}\mathsf{r}/r_\alpha$ in the place of ϵ', $4L^{-2}\mathsf{c}_{\mathfrak{h}}\mathsf{r}\delta_i + 2\mathsf{c}_{\mathfrak{h}}^2\delta_i^2$ in that of $\bar X$ and $\mathsf{M}_{\mathrm{pl}}^2\max\{\mathsf{r}/\varrho_{\mathrm{ch}}, \delta_i/\varrho_{\mathrm{fl}}\}^3$ in that of $\bar\rho$. All summands are ≥ 0 and the last one is $2\bar\rho^2$, so $2\mathsf{M}_{\mathrm{pl}}^4\varrho_\wedge^{-6}\delta_i^6 \le \mathsf{E}_i$. As $g_i \le \delta_i$ (Lem. 4.3(b)) and $\epsilon_i^{\mathrm{cu}} \le 1$, E_i and, for $\mathsf{r} \le \delta_i + r_{i+1}$, $\mathsf{E}_i(\mathsf{r})$ are at most $C_{\mathsf{E}}\delta_i^4(\delta_i + \epsilon_i^{\mathrm{cu}})$, with a number C_{E} formed from the constants; and as $\epsilon_i^{\mathrm{cu}} \le C_{\mathrm{cu}}(\delta_i + \mathsf{p}_{\star,i})$ (Lem. 4.14(a)) and $\mathsf{p}_{\star,i} \le \max\{1, (c_{\mathsf{p}}/A_1)e^{y_{\mathsf{p}}/2}\}\delta_i$ at every height (the quotient $\mathsf{p}_{\star,i}/\delta_i = (c_{\mathsf{p}}/A_1)y_i^{-\mathfrak{p}_0}\exp[\frac{y_i}{2} - (LM)^{-3}A_1^2 y_i^{2\mathfrak{p}_0}]$ is ≤ 1 for $y_i \ge y_{\mathsf{p}}$ by the proof of Lem. 4.3(b), and $\le (c_{\mathsf{p}}/A_1)e^{y_{\mathsf{p}}/2}$ for $1 \le y_i < y_{\mathsf{p}}$; $y_i > \log g^{-2} \ge \log\gamma^{-2} > 1$ since $\gamma \le \frac12$), they are at most $C\delta_i^5$.

Let $\bar{\mathsf{e}}$ be any number $\ge 2\mathsf{w}_1\mathsf{w}_2\mathsf{E}_i$; $\Lambda_i = \max\{1, \log(\mathsf{M}_i/\bar{\mathsf{e}})\}$, $\mathsf{h}_i = \max\{2, R_w + 1 + M(2 + \Lambda_i + 2\log^+(\delta_i/\gamma_\circ))\}$, $\mathsf{s} = \mathsf{h}_i\gamma_\circ(6 + 19r_n)$. Then $\mathsf{LM} = |\Delta[\mathsf{K}^{(i+1)}] - \mathcal{Q}_i(\mathsf{ang}^{[n]}(A))|$ satisfies:

(i) always $\mathsf{LM} \le 2\mathsf{w}_1\mathsf{w}_2\mathsf{E}_i(\max\{\delta_i, \alpha\})$, where $\max\{\delta_i,\alpha\} \le \min\{\frac12\varrho_{\mathrm{ch}}, \delta_i + r_{i+1}\}$ ($\alpha \le r_{i+1}$, as $r_j \ge L^{-1}r_{j+1}$; $2\delta_i \le \varrho_\wedge$ and $2r_{i+1} \le \varrho_{\mathrm{ch}}$ in the frame, and under (Γ1) by Lem. 4.3(b)), so that $\mathsf{E}_i(\max\{\delta_i,\alpha\}) \le C\delta_i^5$;

(ii) if $\mathsf{s} \le \delta_i/2\Lambda_i$: $\mathsf{LM} \le 8e\,\bar{\mathsf{e}}\Lambda_i^2\mathsf{s}^2/\delta_i^2 + L^{-4}\mathsf{c}_i g_i^2\mathsf{w}_1\mathsf{w}_2[18(\alpha^2 + \alpha'^2)(\gamma_\circ + \gamma') + 48R_w\alpha\gamma_\circ^2]$, with $\alpha' = (R_w + 1)(3\gamma_\circ + 10\alpha^2)$, $\gamma' = \frac{100}{99}(\gamma_\circ + 3\alpha^2)$.

The weights enter only through w_a, R_w, D_i^{12}, $\mathsf{N}_{i+1}(w_1)$ and $\|\mathsf{K}^{(i+1)}\|$; no test pair, filling or sign is asked. For weights placed at (ρ, r) as in (B.2), with $k_0 = \mathsf{k}(r)$, this gives for $\mathsf{k}(r) \le i < n \le K$

$$\mathsf{LM} \le \{C_{\mathrm{K}}\mathfrak{b}_i^4\delta_i\Lambda_\sharp^2\mathsf{h}_\sharp^2 u_i^3 + C'_{\mathrm{K}}\mathsf{c}_i(1+u_i)^2u_i^3\,\mathfrak{b}_n^2 r_n\}(\rho/r)^6\hat r_n^3\mathsf{u}_1\mathsf{u}_2, \qquad \Lambda_\sharp \le CY_r, \quad \mathsf{h}_\sharp \le CY_r^{2\mathfrak{p}_0},$$

and neither $\Lambda_\sharp$ nor $\mathsf{h}_\sharp$ depends on ρ. Of (Γ1) the proofs of (i) and (ii) use only the frame at k_0 and, at the heights concerned, $2r_n \le \varrho_{\mathrm{ch}}$ (so $r_n \le \frac{1}{120}$, as $\varrho_{\mathrm{ch}} \le \frac{1}{60}$ by equation YM3_I:(2285)), $\epsilon_i^{\mathrm{cu}} \le 1$ and $x_n^{-1} \le r_n^2$, besides the bound $\ell \ge 2LMy_K$ of the sides $\ell \in \mathfrak{L}(g)$; (i) and (ii), with the bounds of E_i and $\mathsf{E}_i(\cdot)$ stated above, hold whenever these hold. The narrow display holds under the data of (B.2), where step 6 reads besides Lem. 4.9(c) and $\delta_i \le \delta_n$. ∎

Proof. 1. *Lowering.* $L^{-m}A$, viewed at the level $i+1$, is a real point v of the chart with $|A_m| \le \alpha$, $|\nabla A_m| \le \gamma_\circ$, current $\le \gamma_\circ$, and $\mathsf{ang}^{[i+1]}(A_m) = \mathsf{ang}^{[n]}(A)$; $\mathsf{Sum}[\mathsf{K}^{(i+1)}](\tilde U) = \mathcal{K}_i(v) := \mathsf{Sum}[\mathsf{K}^{(i+1)}](P[v])$, and $\mathcal{Q}_i(v)$ is the third term.

2. *Holomorphy and no first order.* $\mathcal{K}_i$ is holomorphic on the ball of radius r_α of the chart, $|\mathcal{K}_i| \le \mathsf{M}_i^{\mathsf{K}}$, and it is invariant under constant gauge rotations; as $\sum_{k=1}^3 \mathrm{Ad}(i\sigma_k) = -\mathrm{id}$, its first derivative at 0 vanishes.

3. *The real ball.* At the real points $\|v\| \le \delta_i$, $\mathcal{K}_i(v)$ is a covariance under the probability measure $\nu_{i,v}$ of one step of the plaquette terms at the lowered configuration; this rests on a sum rule at every point of the chart, which (a) and (b) give.

(a) *The sum rule at every point.* Prop. YM3_I:51.15 proves the sum rule of one local term at the two admitted background points, in equation YM3_I:(872), and from it the sum rule of the totals at the chart points v_A of the chain classes (Prop. YM3_I:124.2, equation YM3_I:(2289)). The finite-vertex argument in that proof uses only membership in the localization domain $\mathfrak{D}(T_\ell)$ until it

identifies the two endpoint values with background values; so for a term f on a domain X whose radius exceeds 1, as do those of the localisation of bare arrays, and every point $\mathfrak{u} = (P, \mathcal{A}) \in \mathfrak{D}(T_\ell)$, critical or not,

$$\sum_{Y' \in \mathcal{D}_i \colon Y' \supset X^\uparrow} \mathbb{V}[f](Y'; \mathfrak{u}|_{Y'}) = f(\rho_X P \dotplus \mathfrak{h}_{T_\ell}(\mathbf{1}; \mathfrak{u})) - f(\rho_X P).$$

Indeed, with the interpolating function $F(s) = \Phi_{T_\ell}[f](s, 1; \mathfrak{u})$ of equation YM3_I:(869), including its distinct rescaling factors for the potential and the current, the resummation of brackets (eq. 5.xxvi) writes $F(\mathbf{1}) - F(0)$ as the sum of the brackets $F^*[\mathcal{K}|\mathsf{N}_0]$ over the sets $\mathcal{K}$ of fine cubes off the cubes N_0 of $X^\uparrow$; a bracket for which $\mathcal{K} \cup \mathsf{N}_0$ has more than one wall-component vanishes, F being local for the roots N_0 (Prop. YM3_I:51.15 and Lem. 5.25); if it has one, its union is a region $Y' \supset X^\uparrow$ and the bracket is $\mathbb{V}[f](Y'; \mathfrak{u}|_{Y'})$ by the zero-extension consistency in Prop. YM3_I:51.15, and $\mathcal{K} \mapsto Y'$ is a bijection onto these regions; finally $F(\mathbf{1}) = f(\rho_X P \dotplus \mathfrak{h}_{T_\ell}(\mathbf{1}; \mathfrak{u}))$ by the definition of F, and $F(0) = f(\rho_X P)$ by the zero-value statement in Prop. YM3_I:51.15. Identifying the endpoint values with the two background values gives equation YM3_I:(872); the finite-bracket identity itself requires only the stated domain membership.

(b) *The totals at every point of the chart.* The weight member of a weight w is, on a domain X of level i, $\eta_i^{-1} \sum_{p : X(p) = X} w(p) a_p$, every bare plaquette p having one domain $X(p)$; let $w = w_a^{[i]} = L^{K-i} w_a$, so that $\eta_i^{-1} L^{K-i} = L^K$, and let $\mathfrak{f}_i[\mathsf{W}_{w_a^{[i]}}]$ be the bare array whose only member is this weight member. Applied to its members at the point $\mathfrak{u} = (P[v], g_i B)$, with the pair identity in equation YM3_I:(2286) for the configuration of the substituted pair, (a) gives at every point $v = (A, \mathbf{J})$ of the chart of level $i+1$ with $\|v\| < \varrho_{\mathrm{ch}}$, real or not, in a chain class or not, and every $B \in \mathsf{Rg}_{x_i}$ (as $\delta_i < \varrho_{\mathfrak{fl}}$)

$$\mathbb{V}'\big[\mathfrak{f}_i[\mathsf{W}_{w_a^{[i]}}]\big]^{\mathrm{tot}}(P[v]; B) = L^K \sum_p w_a(p) \Big[a_p(e^{i\eta_i A^{\mathrm{lw}}(v, g_i B)}) - a_p(e^{i\eta_{i+1} A}) \Big],$$

This proves at every such chart point the algebraic identity whose real-background form is equation YM3_I:(2290).

(c) *The covariance.* By Lem. 2.2 and equation YM3_I:(2298), $\mathsf{Sum}[\mathsf{K}^{(i+1)}](P)$ is, at every triple P of the core, the covariance under $\nu_i[P]$ of the two totals of (b); the terms $a_p(e^{i\eta_{i+1} A})$ are constants. So at every real v with $\|v\| \le \delta_i$

$$\mathcal{K}_i(v) = L^{2K} \sum_{p,q} w_1(p) w_2(q) \big\langle a_p(\tilde{U}^{\mathrm{lw}}); a_q(\tilde{U}^{\mathrm{lw}}) \big\rangle_{\nu_{i,v}}, \qquad \tilde{U}^{\mathrm{lw}}(B) = e^{i\eta_i A^{\mathrm{lw}}(v, g_i B)},$$

where $\nu_{i,v}$ is a probability measure on Rg_{x_i} and $|\langle F; F' \rangle_{\nu_{i,v}}| \le 2 \sup|F| \sup|F'|$ (Prop. YM3_I:124.5).

(d) *The plaquette terms.* $(v, g_i B)$ is a real point with $\|v\| \le \delta_i$ and $\sup|g_i B| < g_i \mathfrak{b}_i = \delta_i$. The linear part of the lowered angle in the fluctuation is $\mathsf{sg}_i \sum_b \mathsf{T}_i^{\mathrm{cv}}(p, b) \mathcal{A}(b)$, where $\mathsf{T}_i^{\mathrm{cv}} = (\mathsf{T}_i^{\mathrm{cv}}(p, b))$ is the real matrix, indexed by the bare plaquettes and the positively oriented fluctuation bonds, of the curvature response of one step composed with its elimination map (the definitions preceding Lem. YM3_I:124.3), $\mathsf{sg}_i \in \{1, -1\}$ is the sign of Lem. YM3_I:124.3, and $\sum_b |\mathsf{T}_i^{\mathrm{cv}}(p, b)| \le C_{\mathrm{tp}}^{\mathrm{row}} \eta_i^2$ (Prop. YM3_I:120.6, equation YM3_I:(2239)). Lem. YM3_I:124.4 at $r = a = \delta_i$, $\mathsf{rt} = \varrho_\wedge / \delta_i \ge 2$, gives

$$a_p(\tilde{U}^{\mathrm{lw}}(B)) = \tfrac{1}{2} |F_p|^2 + \langle F_p, \xi_p \rangle + \tfrac{1}{2} \langle \xi_p, \xi_p \rangle + \mathsf{o}_p(B), \qquad \xi_p = \mathsf{sg}_i g_i \sum_b \mathsf{T}_i^{\mathrm{cv}}(p, b) B(b),$$

with $F_p = \mathsf{ang}_p^{[i+1]}(A)$, $|\mathsf{o}_p| \le \mathsf{M}_{\mathrm{pl}}^2 \eta_i^4 \mathsf{rt}^{-3} = \eta_i^4 \bar{\rho}$, $|F_p| \le 2 L^{-2} \eta_i^2 \delta_i$ and $|\xi_p| \le 2 \mathsf{c}_{\mathfrak{h}} \eta_i^2 \delta_i$. So $\mathsf{X}_p = \langle F_p, \xi_p \rangle + \frac{1}{2} \langle \xi_p, \xi_p \rangle$ has $\sup|\mathsf{X}_p| \le \eta_i^4 \bar{X}$, and $\langle a_p; a_q \rangle_{\nu_{i,v}}$ differs from $\langle \mathsf{X}_p; \mathsf{X}_q \rangle_{\nu_{i,v}}$ by at most $\eta_i^8 (4 \bar{X} \bar{\rho} + 2 \bar{\rho}^2)$.

(e) *The Gaussian part.* Under $\nu_{i,v}$ the joint cumulants of the orders 1, 3, 4 of the components of B are at most $\epsilon = \epsilon_i^{\mathrm{cu}}$, and those of the order 2 lie within $\epsilon' = \epsilon + 2\mathsf{c}_{\mathbf{C}}\delta_i/r_\alpha$ of $\mathbf{C}_0$, the covariance of the step at $v = 0$ (Prop. YM3_I:124.5, equations YM3_I:(2296)–YM3_I:(2297), at $\|v\| \le \delta_i$); and $\mathbf{C}_0 = (\mathsf{A}_i^{\mathrm{cv}})^{-1} \otimes 1$ with $\mathsf{T}_i^{\mathrm{cv}}(\mathsf{A}_i^{\mathrm{cv}})^{-1}(\mathsf{T}_i^{\mathrm{cv}})^{\mathsf{T}} = \eta_i \mathcal{C}_{(i+1)}$, $\mathsf{A}_i^{\mathrm{cv}}$ the real symmetric matrix of one step at the trivial configuration on the fluctuation bonds (equation YM3_I:(2297)).

A covariance of products is the sum, over the partitions that are no refinements of the two factors, of the products of the cumulants of the blocks. With $\mathbf{C}_0$ in the place of the second cumulants, the part of $\langle\langle F_p, \xi_p\rangle; \langle F_q, \xi_q\rangle\rangle$ is $g_i^2\eta_i\mathcal{C}_{(i+1)}(p,q)\langle F_p, F_q\rangle$, that of $\frac{1}{4}\langle\langle \xi_p, \xi_p\rangle; \langle \xi_q, \xi_q\rangle\rangle$ is $\frac{3}{2}g_i^4\eta_i^2\mathcal{C}_{(i+1)}(p,q)^2$ (three colours, two pairings), the mixed covariances $\langle\langle F_p, \xi_p\rangle; \frac{1}{2}\langle \xi_q, \xi_q\rangle\rangle$ have none, and the rest is at most η_i^8 times the first three summands of E_i, by the row sums.

Multiplied by $L^{2K}w_1(p)w_2(q)$ and summed ($L^{2K}\eta_i^8 \sum_p |w_1(p)| \sum_q |w_2(q)| = \mathsf{w}_1\mathsf{w}_2$, $L^{2K}g_i^2\eta_i = a_K^{-1}\upsilon_{i+1}$), this gives, with $\mathsf{k}_i = \frac{3}{2}g_i^4\eta_i^2 L^{2K}\sum_{p,q} w_1(p)w_2(q)\mathcal{C}_{(i+1)}(p,q)^2$ and $\mathcal{Q}_i(v) = \mathcal{Q}_i(\mathsf{ang}^{[i+1]}(A))$,

$$|\mathcal{K}_i(v) - \mathsf{k}_i - \mathcal{Q}_i(v)| \le \mathsf{w}_1\mathsf{w}_2\mathsf{E}_i.$$

(f) *Why a ball.* A ball of the chart is closed under $v \mapsto \zeta v$, under cut-offs and under small gauge transformations, which steps 4 and 5 use; a chain class is closed under none of the three, and this is why the totals are taken at every point of the chart. The radius δ_i enters only the expansion, the bound of $|F_p|$ and the ratio $\|v\|/r_\alpha$; read at a radius $\delta_i \le \mathsf{r} \le \frac{1}{2}\varrho_{\mathrm{ch}}$ ($a = \delta_i$, $\mathsf{rt} = \min\{\varrho_{\mathrm{ch}}/\mathsf{r}, \varrho_{\mathrm{fl}}/\delta_i\} \ge 2$), the same argument gives $|\mathcal{K}_i(v) - \mathsf{k}_i - \mathcal{Q}_i(v)| \le \mathsf{w}_1\mathsf{w}_2\mathsf{E}_i(\mathsf{r})$ at the real points with $\|v\| \le \mathsf{r}$; at $\mathsf{r} = \max\{\delta_i, \alpha\}$, at the point v of step 1 and at 0, this is (i).

4. *Two constants.* For real v' with $\|v'\| \le \delta_i/2\Lambda_i$ put $\psi(\zeta) = \mathcal{K}_i(\zeta v') - \mathcal{K}_i(0) - \zeta^2\mathcal{Q}_i(v')$: holomorphic for $|\zeta| < \delta_i/\|v'\|$, $\psi(0) = \psi'(0) = 0$, $|\psi| \le \mathsf{M}_i$, $|\psi| \le \bar{\mathsf{e}}$ on the real segment (step 3 at tv' and at 0); Lem. 4.11 gives $|\psi(1)| \le 4e\bar{\mathsf{e}}\Lambda_i^2\|v'\|^2/\delta_i^2$ (if $\bar{\mathsf{e}} > \mathsf{M}_i$, then $|\psi| \le \mathsf{M}_i < \bar{\mathsf{e}}$ on the disc and Schwarz's lemma of the order two gives more). Of the complex ball only the bound M_i is used, through $\log \mathsf{M}_i$; the smallness comes from the real ball, where the step measure is a probability measure, and any $\bar{\mathsf{e}} \ge 2\mathsf{w}_1\mathsf{w}_2\mathsf{E}_i$ serves.

5. *The gauge.* Let first $4\mathsf{h}_i + 4$ be at most the side ℓL^{K-i-1} of the torus in level-$(i+1)$ units. Lem. 4.12 on the box N_{h_i} about the pair gives v' with $\|v'\| \le \mathsf{s}$; the domains in the box take the same values at v and at v', those outside have $d_{i+1}(X) \ge (\mathsf{h}_i - 1 - R_w)/M - 2$ and cost $2\mathsf{M}_i^{\mathsf{K}}e^{-[\cdot]_+} \le \bar{\mathsf{e}}\gamma_\circ^2/\delta_i^2$ by the choice of h_i; replacing $\mathcal{Q}_i(v')$ by $\mathcal{Q}_i(v)$ costs the last term of (ii), since $|\mathcal{C}_{(i+1)}| \le \mathsf{c}_i\eta_i^3$ on the supports. If $4\mathsf{h}_i + 4 > \ell L^{K-i-1}$, then $L^m \le L^{K-i-1} < (4\mathsf{h}_i + 4)/\ell \le \mathsf{h}_i$, as $\ell \ge 8$ and $\mathsf{h}_i \ge 2$; so $\|v\| \le \max\{\alpha, \gamma_\circ\} = \alpha = L^m\gamma_\circ \le \mathsf{h}_i\gamma_\circ \le \mathsf{s}$, and step 4 applies at $v' = v$ itself, with no gauge and no domains outside a box: (ii) holds without its last term.

6. *The narrow pair.* Take $\bar{\mathsf{e}} = 2\bar{\mathsf{w}}_1\bar{\mathsf{w}}_2\mathsf{E}_i$ with $\bar{\mathsf{w}}_a = 81\hat{\rho}_i^3|w_a|_\infty/a_{K-i} \ge \mathsf{w}_a$. By Lem. 4.9(c), $\|\mathsf{K}^{(i+1)}\| \le \frac{1}{2}t_i^2\bar{\nu}_i = Cx_i^{-1}\bar{\mathsf{w}}_1\bar{\mathsf{w}}_2$: the factor $\hat{\rho}_i^6$ cancels in $\mathsf{M}_i/\bar{\mathsf{e}} \le C(1 + \mathsf{N}_{i+1})x_i^{-1}\delta_i^{-6} \le C(1 + u_{\mathsf{k}(r)}^3)x_{\mathsf{k}(r)}^2$ (by the lower bound $2\mathsf{M}_{\mathrm{pl}}^4\varrho_\wedge^{-6}\delta_i^6 \le \mathsf{E}_i$), so $\Lambda_i \le CY_r$ by the working level; $R_w \le 11u_{i+1}$ and $\log^+(\delta_i/\gamma_\circ) \le 2m\log L \le 2Y_r$ give $\mathsf{h}_i \le CY_r^{2\mathfrak{p}_0}$. If $\mathsf{s} \le \delta_i/2\Lambda_i$, (ii) with $\bar{\mathsf{w}}_1\bar{\mathsf{w}}_2x_i^{-1}L^{-4m}r_n^2 \le C\mathfrak{b}_n^2u_i^3(\rho/r)^6\hat{r}_n^3\mathsf{u}_1\mathsf{u}_2$; else $L^{3(n-i)} \le C\mathsf{h}_i^2\Lambda_i^2$, and (i) gives the first term. □

The induction over the levels; the jets.

Lemma 4.14 (One step: cumulants, quadratic polynomials, the Gaussian step)**.**

(a) In the data of one renormalisation step at a height i, for a real point v of the chart of the level $i+1$ with $\|v\| < r_\alpha$, every $n \ge 1$ and components $B^{\alpha_1}(b_1), \dots, B^{\alpha_n}(b_n)$ of the fluctuation field, repetitions allowed, the joint cumulants under the measure $\nu_{i,v}$ of the step are formed and satisfy $|\kappa(B^{\alpha_1}(b_1), \dots)| \le \epsilon_i^{\mathrm{cu},n}$ for $n \ne 2$ and $|\kappa(B^{\alpha_1}(b_1), B^{\alpha_2}(b_2)) - \mathbf{C}_v((b_1,\alpha_1),(b_2,\alpha_2))| \le \epsilon_i^{\mathrm{cu},2}$, where

$$\epsilon_i^{\mathrm{cu},n} = C_{\mathrm{E},0}(K_\Pi\mathsf{n}_i^{\mathbb{V}} + K_P^\Pi\mathsf{p}_{\star,i})(n\mathsf{n}_{\mathrm{f}}/w_\star)^n$$

with construction constants, $\mathsf{n}_i^{\mathbb{V}} \le c_{\mathbb{V}}\delta_i$; $\mathbf{C}_v$ is the covariance of the step, $|\mathbf{C}_v| \le \mathsf{c}_{\mathbf{C}}$ and $|\mathbf{C}_v - \mathbf{C}_0| \le 2\mathsf{c}_{\mathbf{C}}\|v\|/r_\alpha$ entry by entry, $\mathbf{C}_0 = (\mathsf{A}_i^{\mathrm{cv}})^{-1} \otimes 1$, and $\mathsf{T}_i^{\mathrm{cv}}(\mathsf{A}_i^{\mathrm{cv}})^{-1}(\mathsf{T}_i^{\mathrm{cv}})^{\mathsf{T}} = \eta_i \mathcal{C}_{(i+1)}$ (at $n \le 4$ the bounds on the cumulants are those of the proof of Prop. YM3_I:124.5, and the statements on $\mathbf{C}_v$ and $\mathbf{C}_0$ are in Prop. YM3_I:124.5).

Here $\mathsf{A}_i^{\mathrm{cv}}$ is the real symmetric matrix of one step at the trivial configuration on the positively oriented fluctuation bonds $b \in \mathfrak{X}^+$, positive definite as $\mathbf{C}_0$ is (Prop. YM3_I:124.5 at $v = 0$); $\mathsf{T}_i^{\mathrm{cv}} = (\mathsf{T}_i^{\mathrm{cv}}(p,b))$ is the real matrix, indexed by the bare plaquettes and the bonds of $\mathfrak{X}^+$, of the curvature response of one step composed with its elimination map (the definitions preceding Lem. YM3_I:124.3); and there are a sign $\mathsf{sg}_i \in \{1,-1\}$ and the number $C_{\mathrm{tp}}^{\mathrm{row}}$ of Prop. YM3_I:120.6 such that the linear part of the lowered angle in a fluctuation $\mathcal{A}$ is $\mathsf{sg}_i \sum_b \mathsf{T}_i^{\mathrm{cv}}(p,b)\mathcal{A}(b)$ and $\sum_b |\mathsf{T}_i^{\mathrm{cv}}(p,b)| \le C_{\mathrm{tp}}^{\mathrm{row}}\eta_i^2$ (Lem. YM3_I:124.3 and Prop. YM3_I:120.6, equation YM3_I:(2239), on a side ℓ at least the constant N_{H} fixed in Section YM3_I:120.3, as in the frame).

At a real point $(v, g_i B)$, $v = (A, \mathbf{J})$, $B \in \mathsf{Rg}_{x_i}$, with $r \ge \|v\|$ and $\mathsf{rt} = \min\{\varrho_{\mathrm{ch}}/r, \varrho_{\mathrm{fl}}/\delta_i\} \ge 2$, Lem. YM3_I:124.4 (at $a = \delta_i$) splits the angle of the lowered configuration:

$$\mathsf{ang}_p^{[i]}(A^{\mathrm{lw}}(v, g_i B)) = \mathsf{ang}_p^{[i+1]}(A) + \psi_p(B) + \zeta_p(B), \qquad \psi_p(B) = \mathsf{sg}_i g_i \sum_b \mathsf{T}_i^{\mathrm{cv}}(p,b) B(b),$$

$|\psi_p| \le 2\mathsf{c}_{\mathfrak{h}}\eta_i^2\delta_i$, $|\zeta_p| \le 4\mathsf{M}_{\mathrm{gr}}\eta_i^2 \mathsf{rt}^{-2}$, with the number M_{gr} of equation YM3_I:(2285); under the centred Gaussian with covariance $\mathbf{C}_0$, ψ has the covariance $g_i^2\eta_i\mathcal{C}_{(i+1)} = a_K \upsilon_{i+1}\mathcal{C}_{(i+1)}$, as $\mathsf{sg}_i^2 = 1$ and $g_i^2\eta_i = a_K\upsilon_{i+1}$. At $v = v_{A'}$, $A' \in \mathsf{Cc}_{i+1}(r_{i+1})$, one takes $r = r_{i+1}$; under (Γ1) Lem. 4.3(b) gives $\mathsf{rt} \ge 2$ and $|\zeta_p| \le C\eta_i^2\delta_i^2$. We put $\epsilon_i^{\mathrm{cu}} = \epsilon_i^{\mathrm{cu},4}$; with $C_\star = C_{\mathrm{E},0}\max\{K_\Pi c_{\mathbb{V}}, K_P^\Pi\}$, so that $\epsilon_i^{\mathrm{cu},n} \le C_\star(\delta_i + \mathsf{p}_{\star,i})(n\mathsf{n}_{\mathrm{f}}/w_\star)^n$, we take $C_{\mathrm{cu}} = C_\star(4\mathsf{n}_{\mathrm{f}}/w_\star)^4$, $c_{\mathrm{cu}}^{(n)} = 2C_\star(2n\mathsf{n}_{\mathrm{f}}/w_\star)^{2n}$ and $c_{\mathrm{cu}} = c_{\mathrm{cu}}^{(3)}$, so that $\epsilon_i^{\mathrm{cu}} \le C_{\mathrm{cu}}(\delta_i + \mathsf{p}_{\star,i})$ and $\epsilon_i^{\mathrm{cu},2n} \le c_{\mathrm{cu}}^{(n)}(\delta_i + \mathsf{p}_{\star,i})/2$; we put $\epsilon_i'^{(n)} = \epsilon_i^{\mathrm{cu},2n} + 2\mathsf{c}_{\mathbf{C}} r_{i+1}/r_\alpha$ and $\epsilon_i' = \epsilon_i'^{(3)}$, and $c_\epsilon^{(n)}$ is the number c_ϵ of Lem. 4.3(b) with $c_{\mathrm{cu}}^{(n)}$ in place of c_{cu}.

We have $4\mathsf{n}_{\mathrm{f}}/w_\star > 1$ ($\mathsf{n}_{\mathrm{f}} = \frac{1}{2}M^{3/2}$ and $w_\star < 1$: equation YM3_I:(2294) and the proof of Prop. YM3_I:124.5); so with $x = \mathsf{n}_{\mathrm{f}}/w_\star$, $(nx)^n \le (n'x)^n \le (n'x)^{n'}$ for $n \le n'$, $n' \ge 4$, and $\epsilon_i^{\mathrm{cu},n} \le \epsilon_i^{\mathrm{cu},2m}$ for $n \le 2m$, $m = 2,3,4$: at $\|v\| \le r_{i+1}$ the joint cumulants of all the orders $\le 2m$ lie within $\epsilon_i'^{(m)}$ of those of the centred Gaussian with covariance $\mathbf{C}_0$ (at the order two through $|\mathbf{C}_v - \mathbf{C}_0| \le 2\mathsf{c}_{\mathbf{C}} r_{i+1}/r_\alpha$). Besides $\epsilon_i'^{(2)} \le \epsilon_i' \le \epsilon_i'^{(4)}$ and $c_\epsilon^{(2)} \le c_\epsilon \le c_\epsilon^{(4)}$; so ($\Gamma$1) with $c_\epsilon^{(4)}$ in place of c_ϵ contains (Γ1), and ($\Gamma^{(4)}$) contains (Γ).

(b) Let I be finite, ν a probability measure with bounded support on $\mathbb{R}^I$, $\mathbf{C}$ a real symmetric positive semidefinite matrix on I with $|\mathbf{C}(\mathsf{i},\mathsf{j})| \le \mathsf{c}$, $\mathsf{c} \ge 1$, $0 \le \epsilon \le 1$, and let the cumulants of ν of orders $n \le 6$ be within ϵ of those of the centred Gaussian G with covariance $\mathbf{C}$. For quadratic polynomials P_1, P_2, P_3 in the coordinates, with $\|P\|_{\mathrm{pol}}$ the sum of the moduli of their coefficients of orders one and two, and $1 \le m \le 3$: $|\kappa_\nu(P_1,\dots,P_m) - \kappa_{\mathsf{G}}(P_1,\dots,P_m)| \le c_6\epsilon\prod_l \|P_l\|_{\mathrm{pol}}$, $c_6 = 1218(\mathsf{c}+1)^5$. If only the cumulants of the orders $n \le 4$ are within ϵ of those of G, the same holds for $m \le 2$ with $c_4 = 60(\mathsf{c}+1)^3$; if those of the orders $n \le 8$ are, it holds for $m \le 4$ (four sources) with $c_8 = 33120(\mathsf{c}+1)^7$; and $c_4 \le c_6 \le c_8$.

(c) Let $k_{\mathfrak{b}} \le i < K$, $\mathsf{s} = \upsilon_{i+1}\mathcal{C}_{(i+1)}$, and let ψ be a centred Gaussian vector of $\mathfrak{su}(2)$-valued functions of the plaquettes with $\mathbb{E}\psi_p^\alpha\psi_q^{\alpha'} = a_K\delta_{\alpha\alpha'}\mathsf{s}(p,q)$. For every F' and every nonempty set β of sources,

$$\sum_{\pi \in \mathrm{Part}(\beta)} \kappa\big(q_{i;\beta'}(F' + \psi) : \beta' \in \pi\big) = (g_{i+1;\beta} - g_{i;\beta}) + q_{i+1;\beta}(F').$$

■

Proof. (a) The bound on the cumulants of one step (the proof of Prop. YM3_I:124.5) is proved with n insertions of one sort, and depends on the number n only through the factor $(n\mathsf{n}_{\mathrm{f}}/w_\star)^n$; so it holds at every order. The remaining statements in (a) follow from the results cited there, and the comparison of the orders is the arithmetic stated in (a). (b) The cumulant of products $X_{S_1},\dots,X_{S_m}$ is the sum, over the partitions σ of $S=\bigsqcup S_l$ with $\sigma\vee\{S_1,..,S_m\}=\{S\}$, of the products of the cumulants of the blocks (from the expansions of exp and log in $\mathbb{C}[z]/(z_l^2)$). Replacing the factors one by one costs $|\sigma|\epsilon(\mathsf{c}+1)^{|\sigma|-1}\le 2m\epsilon(\mathsf{c}+1)^{2m-1}$, as $|S|\le 2m$, and a set of $2m$ elements has B_{2m} partitions, $B_4=15$, $B_6=203$, $B_8=4140$: so $c_{2m}=2m\,B_{2m}(\mathsf{c}+1)^{2m-1}$ serves, which is $c_6=6\cdot 203(\mathsf{c}+1)^5$ for $m\le 3$, c_4 for $m\le 2$ and c_8 for $m\le 4$. (c) Put $w_\lambda=\sum_a\lambda_a w_a$, $R_i^\lambda=w_\lambda(1-\mathcal{S}_i w_\lambda)^{-1}$, $g_i^\lambda=-\frac{3}{2}\operatorname{Tr}\log(1-\mathcal{S}_i w_\lambda)$ and $q_i^\lambda(F)=\frac{1}{2a_K}\langle F,R_i^\lambda F\rangle$; their coefficients of $\lambda^\beta=\prod_{a\in\beta}\lambda_a$ are $R_{i;\beta}$, $g_{i;\beta}$ and $q_{i;\beta}$.

The Gaussian integral, with the factor 3 from the dimension of $\mathfrak{su}(2)$, is

$$\log\mathbb{E}\,e^{q_i^\lambda(F'+\psi)}=-\tfrac{3}{2}\operatorname{Tr}\log(1-\mathsf{s}R_i^\lambda)+\tfrac{1}{2a_K}\langle F',R_i^\lambda(1-\mathsf{s}R_i^\lambda)^{-1}F'\rangle,$$

and $1-\mathsf{s}R_i^\lambda=(1-\mathcal{S}_{i+1}w_\lambda)(1-\mathcal{S}_i w_\lambda)^{-1}$, $R_i^\lambda(1-\mathsf{s}R_i^\lambda)^{-1}=R_{i+1}^\lambda$, as $\mathcal{S}_{i+1}=\mathcal{S}_i+\mathsf{s}$. The coefficient of λ^β of the left member is the sum over the partitions of the joint cumulants; that of the right member is $(g_{i+1;\beta}-g_{i;\beta})+q_{i+1;\beta}(F')$. This is Wick's rule for the photon fields of one step. □

Lemma 4.15 (The tower identity at several sources)**.** Let $0<g\le\gamma_{\mathrm{jt}}$, $g^{-2}\ge x_{\mathrm{J}}$, (K,β) a first-passage datum at g, $\ell\in\mathfrak{L}(g)$, $n\ge 2$, and let $w_a=f_a\circ x\not\equiv 0$, $a\in[n]$, with real Lipschitz functions f_a on T_ℓ with pairwise disjoint supports (for instance the tents of Lem. 4.5(b)) that fall into two nonempty clusters $[n]=I_1\sqcup I_2$ lying in the frame at a level k. Let $k\le i<K$, $A'\in\mathsf{Cc}_{i+1}(r_{i+1})$, $\tilde U=e^{i\eta_{i+1}A'}$, $v_{A'}=(A',J^{(i+1)}(\tilde U))$ the real point of the chart of level $i+1$ that A' gives, with $\|v_{A'}\|\le r_{i+1}$, $\nu=\nu_{i,A'}:=\nu_{i,v_{A'}}$ the measure of the step, and $B\mapsto A^{\mathrm{lw}}(B)=A^{\mathrm{lw}}(v_{A'},g_iB)$ the lowering map (Prop. YM3_I:124.2). Then for every nonempty $\beta\subset[n]$

$$G_{i+1;\beta}(A')=\sum_{\pi\in\mathrm{Part}(\beta)}\kappa_\nu\big(G_{i;\beta'}\circ A^{\mathrm{lw}}:\beta'\in\pi\big).$$

The same holds for real mixed sources $\mathfrak{s}_a\ne 0$, $a\in[n]$, at $0<g\le\gamma_{\mathrm{N}}$ with $K\ge 2$, the progression being formed with their weight members in the source slot (the lemma on bond sources, which follows Lem. 4.31), when the two unions of the hulls of the clusters I_1, I_2 satisfy the conditions of the frame at k of clause (6) of that lemma. For mixed sources the frame is that of clause (6), read with the hulls in the place of the supports, and the tower identity for a pair is the one stated in that clause. ■

Proof. For λ near 0 in $\mathbb{C}^n$ put $\Phi(\lambda)=\exp D_{i+1}^\lambda(\tilde U)-\int\exp D_i^\lambda(e^{i\eta_i A^{\mathrm{lw}}(B)})\,\mathrm{d}\nu(B)$, formed with the progression of $(w_1,\dots,w_n)$. Each relative exponent is a finite sum of functions that are holomorphic in λ, and the integrand is bounded uniformly in B for λ near 0, as the progression with any number of sources lies in the setting of one step (Thm. YM3_I:D); so Φ is holomorphic near 0. Let $t\in\mathbb{R}^n$ be nonzero on I_1 and on I_2, and put $f^t=\sum_{a\in I_1}t_a f_a$, $h^t=\sum_{a\in I_2}t_a f_a$. These are real Lipschitz functions; $f^t\circ x$ and $h^t\circ x$ do not vanish identically, the supports of the w_a being disjoint; and their supports lie in the two unions of supports.

So (f^t,h^t) lies in the frame at k, and the tower identity among the construction results for a pair (equation eq. 2.ix) gives $\exp D_{i+1}^\mu[f^t,h^t](\tilde U)=\int\exp D_i^\mu[f^t,h^t](e^{i\eta_i A^{\mathrm{lw}}})\,\mathrm{d}\nu$ for μ near 0 in $\mathbb{C}^2$, with the same measure ν and the same lowering map, neither of which depends on the sources. The progression depends on the parameters and the sources only through w_λ; at $\lambda(\mu)=(\mu_1 t|_{I_1},\mu_2 t|_{I_2})$ one has $w_{\lambda(\mu)}=\mu_1 f^t\circ x+\mu_2 h^t\circ x$, so $D^{\lambda(\mu)}=D^\mu[f^t,h^t]$, and at $\mu=(s,s)$: $\Phi(st)=0$ for all small real s. With the expansion $\Phi=\sum_N\Phi_N$ in homogeneous polynomials, $\sum_N s^N\Phi_N(t)=0$ for small s; so $\Phi_N(t)=0$ for every N and every t of a dense subset of $\mathbb{R}^n$, hence $\Phi_N=0$ and $\Phi=0$ near 0.

As $D^0_{i+1}=0$ and $D^0_i=0$, the multilinear jets of the two members of $\exp D^\lambda_{i+1}(\tilde U)=\int \exp D^\lambda_i\,\mathrm{d}\nu$ agree in $\mathbb{C}[\lambda_1,\dots,\lambda_n]/(\lambda_a^2)$, and the logarithm of a moment series is the cumulant series; differentiation under the integral is allowed by Cauchy's formula in λ. The coefficient of λ^β is the claim. At $n=2$ the lemma is the identity among the construction results. For mixed sources the argument is the same, with the weight member of $\mathfrak{s}_\lambda=\sum_a\lambda_a\mathfrak{s}_a$, which is linear in $\mathfrak{s}_\lambda$ (clause (1) of the lemma on bond sources), in the source slot. The $t\in\mathbb{R}^n$ at which $\sum_{a\in I_1}t_a\mathfrak{s}_a$ or $\sum_{a\in I_2}t_a\mathfrak{s}_a$ is 0 form two proper linear subspaces, every $\mathfrak{s}_a$ being nonzero; off them, a dense set, the two sums are nonzero mixed sources whose hulls lie in the two unions of the hulls of the clusters, so they are in the frame of clause (6), and the identity for a pair stated there, with the same measure ν and the same lowering map, stands in the place of equation eq. 2.ix. □

Proposition 4.16 (The induction over the levels)**.** Let $n\in\{2,3,4\}$ and let g satisfy (Γ1), with $c_\epsilon^{(4)}$ in place of c_ϵ if $n=4$; let (K,β) be a first-passage datum at g, $\ell\in\mathfrak{L}(g)$, and let real weights $w_1,\dots,w_n$ with the two clusters $\{1\},\{2\}$, or $\{1,2\},\{3\}$, or $\{1,2\},\{3,4\}$ lie in the frame at a level k; let $k\le k_1<K$, $0\le k_\flat\le k_1$, and form the free objects over the base level $k_\flat$. Let numbers $\mathsf{v}_\beta>0$, c°_β and $\epsilon_\circ\ge0$, $\emptyset\ne\beta\subset[n]$, satisfy

(I1) $|G_{k_1;\beta}(A)-c^\circ_\beta-q_{k_1;\beta}(\mathsf{ang}^{[k_1]}(A))|\le\epsilon_\circ\mathsf{v}_\beta$ for $A\in\mathsf{Cc}_{k_1}(r_{k_1})$;
(I2) $\varrho_{i;\beta}:=\eta_i^4\sum_{p,q}|R_{i;\beta}(p,q)|/(a_Kx_i)\le L^{-3(i-k_1)}\mathsf{v}_\beta$ for $k_1\le i<K$;
(I3) $\prod_{\beta'\in\pi}\mathsf{v}_{\beta'}\le\mathsf{v}_\beta$ for every partition π of β.

Put $Z=Z_0(1+\mathfrak{b}_{k_1})^2$ and $\varpi=\epsilon_\circ+\sum_{i=k_1}^{K-1}L^{-3(i-k_1)}(\delta_i+c_{2n}\epsilon_i'^{(n)})$, with a number Z_0 formed from the constants and the numbers c_{2n} of Lem. 4.14(b), and let $(K-k_1)\varpi\le1$. Then $\varpi\le\epsilon_\circ+\frac65(1+c_{2n}c_\epsilon^{(n)})\delta_{k_1}$, and for $k_1\le i\le K$, $A\in\mathsf{Cc}_i(r_i)$ and every β, $m=|\beta|$,

$$|G_{i;\beta}(A)-c^\circ_\beta-(g_{i;\beta}-g_{k_1;\beta})-q_{i;\beta}(\mathsf{ang}^{[i]}(A))|\le\mathsf{a}_mZ^m\varpi\,\mathsf{v}_\beta,\qquad(\mathsf{a}_1,\mathsf{a}_2,\mathsf{a}_3)=(1,14,285),$$

a_4 a number; in particular $|\mathsf{J}_n-c^\circ_{[n]}-g_{K;[n]}+g_{k_1;[n]}|\le\mathsf{a}_nZ^n\varpi\mathsf{v}_{[n]}$. Of the base level only the relation $\mathcal{S}_{i+1}-\mathcal{S}_i=\upsilon_{i+1}\mathcal{C}_{(i+1)}$, $i\ge k_1$, is used. ■

Proof. Let $\mathsf{e}_{i;\beta}$ be the supremum over $\mathsf{Cc}_i(r_i)$ of the left member. 1. *The tower.* For $A'\in\mathsf{Cc}_{i+1}(r_{i+1})$ and the measure $\nu=\nu_{i,A'}$ of the step, $G_{i+1;\beta}(A')=\sum_\pi\kappa_\nu(G_{i;\beta'}\circ A^{\mathrm{lw}}:\beta'\in\pi)$ (the lemma on the tower identity at several sources, which precedes this proposition, for the two clusters of the frame). 2. *The lowered angle.* The point $v_{A'}$ of the chart is real, with $\|v_{A'}\|\le r_{i+1}<\varrho_{\mathrm{ch}}\le r_\alpha$ by the definition of the chain classes (equation eq. 2.vi); $\nu=\nu_{i,v_{A'}}$ is a probability measure carried by the real B with $|B(b)|<\mathfrak{b}_i$ (Prop. YM3_I:124.5), and $A^{\mathrm{lw}}(B)\in\mathsf{Cc}_i(r_i)$ (Prop. YM3_I:124.2 with the radius recurrence YM3_I:(2291)).

By the splitting of Lem. 4.14(a) at $r=r_{i+1}$, $\mathsf{ang}_p^{[i]}(A^{\mathrm{lw}}(B))=F'_p+\psi_p(B)+\zeta_p(B)$, with $F'=\mathsf{ang}^{[i+1]}(A')$, $\psi_p(B)=\mathsf{sg}_ig_i\sum_b\mathsf{T}_i^{\mathrm{cv}}(p,b)B(b)$ linear in the fluctuation field B, $|\psi_p|\le2\mathsf{c}_\mathfrak{h}\eta_i^2\delta_i$ and $|\zeta_p|\le C\eta_i^2\delta_i^2$ by the smallness of Lem. 4.3(b); besides $|F'_p|\le2\eta_{i+1}^2r_{i+1}\le\mathsf{c}_\mathfrak{h}\eta_i^2\delta_i$. So $Y_\beta=G_{i;\beta}\circ A^{\mathrm{lw}}$ is within $\mathsf{e}_{i;\beta}+Z\delta_iL^{-3(i-k_1)}\mathsf{v}_\beta$ of the quadratic polynomial $P_\beta=c^\circ_\beta+g_{i;\beta}-g_{k_1;\beta}+q_{i;\beta}(F'+\psi)$ in the coordinates $B^\alpha(b)$: the term with ζ is at most $Ca_K^{-1}\eta_i^4\delta_i^3\sum_{p,q}|R_{i;\beta}(p,q)|=C\mathfrak{b}_i^2\delta_i\varrho_{i;\beta}$.

The linear part of P_β has the coefficients $\mathsf{sg}_ig_ia_K^{-1}\sum_{p,q}R_{i;\beta}(p,q)\mathsf{T}_i^{\mathrm{cv}}(q,b)F'^\alpha_p$, its quadratic part the coefficients $\frac12g_i^2a_K^{-1}\sum_{p,q}R_{i;\beta}(p,q)\mathsf{T}_i^{\mathrm{cv}}(p,b)\mathsf{T}_i^{\mathrm{cv}}(q,b')$ of $B^\alpha(b)B^\alpha(b')$; with the row sums $\sum_b|\mathsf{T}_i^{\mathrm{cv}}(p,b)|\le C_{\mathrm{tp}}^{\mathrm{row}}\eta_i^2$, $x_ig_i\delta_i=\mathfrak{b}_i$ and $x_ig_i^2=1$,

$$\|P_\beta\|_{\mathrm{pol}}\le\big[2\mathsf{c}_\mathfrak{h}C_{\mathrm{tp}}^{\mathrm{row}}\mathfrak{b}_i+\tfrac32(C_{\mathrm{tp}}^{\mathrm{row}})^2\big]\varrho_{i;\beta};$$

so, by (I2), $\mathfrak{b}_i\le\mathfrak{b}_{k_1}$ and $|B(b)|<\mathfrak{b}_i$, $\|P_\beta\|_{\mathrm{pol}}$ and $\sup|P_\beta-\mathrm{const}|$ are $\le ZL^{-3(i-k_1)}\mathsf{v}_\beta$, Z_0 being at least $2\mathsf{c}_\mathfrak{h}C_{\mathrm{tp}}^{\mathrm{row}}+\frac32(C_{\mathrm{tp}}^{\mathrm{row}})^2$ and the constant of the term with ζ. 3. *The Gaussian value.* Under the Gaussian with covariance $\mathbf{C}_0$, $\psi=\mathsf{sg}_ig_i\mathsf{T}_i^{\mathrm{cv}}B$ has the covariance $g_i^2\eta_i\mathcal{C}_{(i+1)}=a_K\upsilon_{i+1}\mathcal{C}_{(i+1)}$

(Lem. 4.14(a): $\mathsf{sg}_i^2 = 1$, $\mathsf{T}_i^{\mathrm{cv}}(\mathsf{A}_i^{\mathrm{cv}})^{-1}(\mathsf{T}_i^{\mathrm{cv}})^{\mathsf{T}} = \eta_i \mathcal{C}_{(i+1)}$, $g_i^2\eta_i = a_K \upsilon_{i+1}$), and Lem. 4.14(c) gives $\sum_\pi \kappa_{\mathsf{G}}(P_{\beta'}) = c_\beta^\circ + g_{i+1;\beta} - g_{k_1;\beta} + q_{i+1;\beta}(F')$. 4.

From G *to* ν by Lem. 4.14(a), (b), with $\mathsf{c} = \max\{1, \mathsf{c}_{\mathbf{C}}\}$: as $\|\upsilon_{A'}\| \le r_{i+1}$, all the cumulants of the orders $\le 2n$ lie within $\epsilon = \epsilon_i'^{(n)}$ of those of G, and $\epsilon_i'^{(n)} \le c_\epsilon^{(n)}\delta_i \le 1$ (under (Γ1), as $c_\epsilon^{(2)} \le c_\epsilon$, and with $c_\epsilon^{(4)}$ at $n = 4$); the number is c_{2n}; *from* P *to* Y by $|\kappa(X,Y)| \le \|X\|^\sim\|Y\|^\sim$, $|\kappa(X,Y,W)| \le 2\|X\|^\sim\|Y\|^\sim\|W\|^\sim$, $\|X\|^\sim = \inf_c \sup|X - c|$. 5. *The recursion.* With (I3) every product of sizes over a partition is at most $Z^m \mathsf{v}_\beta$ times pure numbers, and $\varepsilon_{i;\beta} = \mathsf{e}_{i;\beta}/Z^m\mathsf{v}_\beta$ obeys $\varepsilon_{i+1;a} \le \varepsilon_{i;a} + \tau_i$, $\varepsilon_{i+1;ab} \le \varepsilon_{i;ab} + 2\tau_i + 4\varpi^2 + 4\theta_i\varpi$ and the like at three sources, $\tau_i = L^{-3(i-k_1)}(\delta_i + c_{2n}\epsilon_i'^{(n)})$, $\theta_i = L^{-3(i-k_1)}$; summed over at most $K - k_1$ heights with $(K-k_1)\varpi \le 1$ this gives 1, 14 and 285.

At $i = K$, $\mathsf{Cc}_K = \{0\}$ and $q_{K;\beta}(0) = 0$. The bound of ϖ: $\epsilon_i'^{(n)} \le c_\epsilon^{(n)}\delta_i$ (as $\epsilon_i' \le c_\epsilon\delta_i$ in Lem. 4.3(b)), $\delta_i \le (\frac54)^{1/2}L^{(i-k_1)/2}\delta_{k_1}$, $(\frac54)^{1/2}\sum_k L^{-5k/2} \le \frac65$. For two sources Steps 1–5 use only eq. 2.x and the cumulants up to order four. □

Lemma 4.17 (The slices)**.** Let (K,β) be a first-passage datum at $g \le \gamma$, $p \ne q$ plaquettes, $\mathsf{r} = |x(p) - x(q)|_\infty$, $c_L^{(N)} = \sup_\tau \sum_{j\in\mathbb{Z}}(L^j\tau)^N e^{-\delta_{\mathrm{tp}}L^j\tau}$ and $\mathsf{m}_K = \frac54 C_{\mathrm{tp}}c_L^{(3)}$. (a) For every set $\mathcal{J} \subset \{1,\dots,K\}$: $|\sum_{j\in\mathcal{J}} \upsilon_j \mathcal{C}_{(j)}(p,q)| \le \mathsf{m}_K \mathsf{b}^{-1} a_K^3 \mathsf{r}^{-3}$. (b) If $t = \mathsf{r}/a_{K-k}$ satisfies $\delta_{\mathrm{tp}}L(L-1)t \ge 3\log L + \log 2$, then

$$\Big|\sum_{k<j\le K} \upsilon_j \mathcal{C}_{(j)}(p,q) - \mathsf{b}^{-1}C_K^{\mathrm{free}}(p,q)\Big| \le \mathsf{b}^{-1}a_K^3\big[\tfrac54 C_{\mathrm{tp}}c_L^{(2)}\bar{B}\,\mathsf{r}^{-2}\mathsf{b}^{-1} + C_{\mathrm{tp}}^{\mathrm{bg}} + 2L^3 C_{\mathrm{tp}}\mathsf{r}^{-3}t^3 e^{-\delta_{\mathrm{tp}}Lt}\big],$$

with $C_{\mathrm{tp}}^{\mathrm{bg}}$ from Prop. YM3_I:120.5, evaluated at $s = 0$ in equation YM3_I:(2236). ■

Proof. (a) $\upsilon_j = (a_{K-j+1}x_{j-1})^{-1} \le \frac{24}{23}\mathsf{b}^{-1} \le \frac54\mathsf{b}^{-1}$ at every $0 < g \le \gamma$: the flow bound and $g^{-2} < \mathsf{b}$, which Lem. 4.3(a) states for every first-passage datum at every $0 < g \le \gamma$, give $|a_{K-j+1}x_{j-1} - \mathsf{b}| \le a_{K-j+1}\bar{B} \le \bar{B}/L$, and $\bar{B} \le \frac{13}{24}g^{-2} < \frac{13}{24}\mathsf{b}$ at $g \le \gamma$ ([7, Lem. G.3] gives this bound), so $a_{K-j+1}x_{j-1} \ge \frac{23}{24}\mathsf{b}$ as $L \ge 13$; and the decay of the slices: $\sum_j(\mathsf{r}/a_{K-j+1})^3 e^{-\delta_{\mathrm{tp}}\mathsf{r}/a_{K-j+1}} \le c_L^{(3)}$, a lacunary sum. (b) Equation YM3_I:(2227) gives $\sum_{k<j\le K}\mathcal{C}_{(j)} = C_K^{\mathrm{free}} - C_{K,0}^{\mathrm{bg}} - \sum_{j\le k}\mathcal{C}_{(j)}$, and Prop. YM3_I:120.5 bounds $|C_{K,0}^{\mathrm{bg}}(p,q)|$ by $C_{\mathrm{tp}}^{\mathrm{bg}}a_K^3$; $|\upsilon_j - \mathsf{b}^{-1}| = |\mathsf{b} - a_{K-j+1}x_{j-1}|\upsilon_j\mathsf{b}^{-1} \le \frac54\bar{B}a_{K-j+1}\mathsf{b}^{-2}$ by the same bounds of the run, which with the decay gives the first term; the slices of the levels $j \le k$ are at $t_j \ge Lt$ and give the last term. □

Lemma 4.18 (The explicit sizes; the weights of the induction)**.** Let weights be placed at (ρ, r, d), or form two pairs at (ρ, r, d), with $\mathsf{d} \le e^{-2\sqrt{3}}$, $s(\rho) \le K$; let $k_\flat = \mathsf{k}(r)$, $k_1 = \mathsf{k}(\mathsf{d})$, $u = \hat{r}_{k_1}$, $\hat{\rho} = \hat{\rho}_{k_1} = qu$, $P = \hat{\mathsf{d}}_{k_1}$, and I_1, I_2 the clusters. Put $\mathsf{w}(\emptyset) = 1$, $\mathsf{w}(\{a\}) = \hat{\rho}^3$, w(a whole pair) $= q^6u^3\max\{1,u^3\}$, $c_* = \max\{102, 4\cdot 81\cdot 3^{3/2}(1+\mathsf{m}_K)\}$, and

$$\mathsf{v}_\beta = c_*^{|\beta|}\,\mathsf{w}(\beta\cap I_1)\,\mathsf{w}(\beta\cap I_2)\,\mathsf{u}_\beta, \qquad \mathsf{u}_\beta = \prod_{a\in\beta}\mathsf{u}_a.$$

Then (I2) and (I3) of Prop. 4.16 hold; $\mathsf{v}_{[3]} \le c_*^3 P^9 \mathrm{nat}_3$ and $\mathsf{v}_{[4]} \le c_*^4 P^{12}\mathrm{nat}_4$; for crossing β, $\mathsf{z}(\beta)\mathsf{u}_\beta \le P^{-6}\mathsf{v}_\beta$, and if $\mathsf{e}_{\mathrm{far}} = 8e^{-\delta_{\mathrm{tp}}LP/2\sqrt{3}} \le 1$, then $\varrho_{k_1;\beta} \le \mathsf{e}_{\mathrm{far}}\mathsf{v}_\beta$ and $|g_{k_1;[n]}| \le \frac32(n-1)!(81\cdot 3^{3/2}\mathsf{m}_K)^n\mathsf{e}_{\mathrm{far}}^2\mathrm{nat}_n$. The factor $\max\{1,u^3\}$ cannot be dropped: u may be a power of $\log\frac{1}{\mathsf{d}}$. ■

Proof. By Lem. 4.17(a), $|\mathcal{S}_i(p,q)| \le \mathsf{m}_K\mathsf{b}^{-1}a_K^3\mathsf{r}^{-3}$ with $\mathsf{r} \ge r/\sqrt{3}$ in a cluster and $\ge \mathsf{d}/\sqrt{3}$ across. An ordering of β is a path with $m-1$ edges, at most $c(\beta)$ of them inside a whole pair; with (P1),

$$\varrho_{i;\beta} \le \tfrac54 m!\,81^m(3^{3/2}\mathsf{m}_K)^{m-1}\rho^{3m}r^{-3c}\mathsf{d}^{-3(m-1-c)}a_{K-i}^{-3}\mathsf{u}_\beta,$$

and with $a = a_{K-k_1}$ the powers of a cancel: $\rho^{3m}r^{-3c}\mathsf{d}^{-3(m-1-c)}a^{-3} = \hat{\rho}^{3m}u^{-3c}P^{-3(m-1-c)} \le \mathsf{w}(\beta\cap I_1)\mathsf{w}(\beta\cap I_2)$, $a_{K-i}^{-3} = L^{-3(i-k_1)}a^{-3}$: (I2). (I3): either $\beta\cap I_\iota$ lies in one block, or it is a whole

pair cut into two points, and $\hat\rho^6 = q^6u^6 \le q^6u^3\max\{1,u^3\}$. The top sets: $q^6u^3\hat\rho^3 = P^6\rho^9r^{-3}\mathsf{d}^{-6}$. At the level k_1 the slices of the levels $\le k_1$ at distances $\ge \mathsf{d}/\sqrt{3}$ are exponentially small, $|\mathcal{S}_{k_1}(p,q)| \le \mathsf{e}_{\rm far}\mathsf{m}_K\mathsf{b}^{-1}a_K^3\mathsf{r}^{-3}$; a path through a crossing set has a far edge, a ring through all sources two. □

Proposition 4.19 (The jets of narrow bumps)**.** There are numbers $C(g)$ and $\mathsf{d}_4(g), \mathsf{d}_4^{(4)}(g) \in (0, \min\{e^{-2(15\mathfrak{p}_0+2)}, r_1(\bar{\mathsf{b}})\}]$, formed from the constants and g, such that the following holds for g under (Γ1), every first-passage datum (K,β) at g, every side $\ell \in \mathfrak{L}(g)$ with $64\mathsf{d} \le \ell$, and $s(\rho) \le K$.

(a) For real weights placed at (ρ, r, d) with $0 < \rho \le r \le \mathsf{d} \le \mathsf{d}_4(g)$: $|\mathsf{J}_3 - 3\operatorname{Tr}\prod_a(w_a\mathsf{b}^{-1}C_K^{\rm free})| \le \epsilon^{\rm nar}(g,\mathsf{d})\,\mathrm{nat}_3$.

(b) For real weights w_1, w_2 with $\Sigma(w_a) \subset \mathbb{B}(x_a,\rho)$, $\operatorname{dist}(\Sigma(w_1),\Sigma(w_2)) \ge \mathsf{d}$, $|x_1 - x_2| \le 18\mathsf{d}$, $0 < \rho \le \mathsf{d} \le \mathsf{d}_4(g)$: $|\mathsf{J}_2 - \frac{3}{2}\mathsf{b}^{-2}\sum_{p,q} w_1(p)w_2(q)C_K^{\rm free}(p,q)^2| \le \epsilon_2^{\rm nar}(g,\mathsf{d})(\rho/\mathsf{d})^6\mathsf{u}_1\mathsf{u}_2$.

(c) Under (Γ1) with $c_\epsilon^{(4)}$ in place of c_ϵ, for two pairs at (ρ, r, d) with $\mathsf{d} \le \mathsf{d}_4^{(4)}(g)$: $|\mathsf{J}_4 - \frac{3}{2}\sum_\varsigma \operatorname{Tr}\prod_l(w_{\varsigma(l)}\mathsf{b}^{-1}C_K^{\rm free})| \le \epsilon_4^{\rm nar}(g,\mathsf{d})\,\mathrm{nat}_4$.

The normalized error factors and constants are uniform in ρ/r, r/d, K, ℓ, position and direction; in particular they hold for $w_a = f_a \circ x$ with bounded Borel f_a, and for plane weights. ■

Proof. 1. *Levels and data.* $k_1 = \mathsf{k}(\mathsf{d})$, $k_\flat = \mathsf{k}(r) \le k_1$, $P = \hat{\mathsf{d}}_{k_1}$; by the working level $4MY_{\mathsf{d}}^{2\mathfrak{p}_0} \le P < CY_{\mathsf{d}}^{2\mathfrak{p}_0}$, $K - k_1 \le Y_{\mathsf{d}}/\log L$, $\delta_{k_1} \le A_1Y_{\mathsf{d}}^{\mathfrak{p}_0}(\mathsf{d}/\mathsf{b})^{1/2}$. Lem. 4.5(b) puts the two clusters in the frame at k_1; the smallness holds at all heights (Lem. 4.3); (I2), (I3) hold with the weights of Lem. 4.18.

2. *(I1).* At a single source, $c_a^\circ = G_{k_1;a}(0)$ and Lem. 4.10 give the tolerance $C\mathfrak{b}_{k_1}^2\delta_{k_1}\mathsf{v}_a$. At the whole pair, $c^\circ = G_{k_1;12}(0)$: by the multilinearity $G_{k_1;12}$ is the pair functional of the pair alone, and (B.2) at $n = k_1 \ge \mathsf{k}(r)$ (its proof, given below, rests on Lems. 4.3–4.13 alone) gives $[C_{\rm N}\mathfrak{b}_{k_1}^2\delta_{k_1} + \epsilon_{\rm N}'(r)]q^6u^3\mathsf{u}_{12} \le [\dots]\mathsf{v}_{12}$. At a crossing set, $c_\beta^\circ = 0$: Lem. 4.9(e) and Lem. 4.18 give $|G_{k_1;\beta}| \le Ce^{4Y_{\mathsf{d}} - \frac{2}{\sqrt{3}}Y_{\mathsf{d}}^{2\mathfrak{p}_0}}\mathsf{v}_\beta$, and $|q_{k_1;\beta}| \le C\mathfrak{b}_{k_1}^2\mathsf{e}_{\rm far}\mathsf{v}_\beta$. So

$$\epsilon_\circ = C_\circ\{\mathfrak{b}_{k_1}^2\delta_{k_1} + \epsilon_{\rm N}'(r) + e^{4Y_{\mathsf{d}} - \frac{2}{\sqrt{3}}Y_{\mathsf{d}}^{2\mathfrak{p}_0}} + \mathfrak{b}_{k_1}^2\mathsf{e}_{\rm far}\}.$$

Here $C_\circ$ is a number formed from the constants that dominates the constants of these four bounds (of Lem. 4.10, of (B.2), of Lem. 4.9(e) with Lem. 4.18, and of the bound of $q_{k_1;\beta}$), and those of the two bounds of (I1) in step 6. The tolerance of (B.2) falls together with its yardstick; for this reason no relation between r and d is needed, and the bounds hold at every ratio $r/\mathsf{d} \in (0,1]$.

3. *The induction* (Prop. 4.16) at $i = K$: $|\mathsf{J}_n - g_{K;[n]} + g_{k_1;[n]}| \le \mathsf{a}_nZ^n\varpi c_*^nP^{3n}\mathrm{nat}_n$.

4. *The rings.* Lem. 4.17(b) at $k = \mathsf{k}(r)$ for p, q in different supports ($t \ge \hat r_{\mathsf{k}(r)}/\sqrt{3}$) gives $|\mathcal{S}_K - \mathsf{b}^{-1}C_K^{\rm free}| \le \mathrm{br}\,\mathsf{b}^{-1}a_K^3\mathsf{r}^{-3}$ with

$$\mathrm{br} = 25C_{\rm tp}c_L^{(2)}\bar B\,\mathsf{d}/\mathsf{b} + 8000C_{\rm tp}^{\rm bg}\mathsf{d}^3 + 2L^3C_{\rm tp}\sup_{t \ge \frac{4}{\sqrt{3}}MY_r^{2\mathfrak{p}_0}} t^3e^{-\delta_{\rm tp}Lt};$$

telescoping a product of n kernels, $|g_{K;[n]} - \frac{3}{2}\sum_\varsigma \operatorname{Tr}\prod_l(w_{\varsigma(l)}\mathsf{b}^{-1}C_K^{\rm free})| \le C\,\mathrm{br}\,\mathrm{nat}_n$; at $n = 3$ the subtrahend is $3\operatorname{Tr}\prod_a(w_a\mathsf{b}^{-1}C_K^{\rm free})$, two cyclic orders with one trace. With Lem. 4.18 for $g_{k_1;[n]}$:

$$\epsilon_{(n)}^{\rm nar} = \mathsf{a}_nZ^nc_*^nP^{3n}\varpi + C\,[\mathsf{e}_{\rm far}^2 + \mathrm{br}].$$

5. *The threshold* $\mathsf{d}_4(g)$. It is fixed by six conditions: (i) $\mathsf{d} \le e^{-2(15\mathfrak{p}_0+2)}$, so that by Lem. 4.3(c) $\epsilon_{\rm N}'(r) \le C(g)(\log\frac{1}{\mathsf{d}})^{15\mathfrak{p}_0+2}\mathsf{d}^{1/2}$ for all $r \le \mathsf{d}$; (ii) $\frac{4}{\sqrt{3}}M\delta_{\rm tp}L(L-1)(\log\frac{1}{\mathsf{d}})^2 \ge 3\log L + \log 2$; (iii) $\log\frac{1}{\mathsf{d}} \ge \max\{5, \sqrt{3}/\delta_{\rm tp}LM\}$, so that $e^{4Y_{\mathsf{d}} - \frac{2}{\sqrt{3}}Y_{\mathsf{d}}^2} \le \mathsf{d}$, $\mathsf{e}_{\rm far} \le 8\mathsf{d}^2$ and the last member of br is $\le C\mathsf{d}$; (iv) $8\mathsf{d}^2 \le 1$, $\overline{\mathrm{br}} \le 1$; (v) $(Y_{\mathsf{d}}/\log L)[\bar\epsilon_\circ + \frac{6}{5}(1 + c_6c_\epsilon)A_1Y_{\mathsf{d}}^{\mathfrak{p}_0}g\mathsf{d}^{1/2}] \le 1$; (vi) $\mathsf{d} \le r_1(\bar{\mathsf{b}})$, so that $\rho \le r_1(\mathsf{b})$ holds of itself. In (iv) and (v) the numbers br and $\epsilon_\circ$, which read r and the datum, stand replaced

by majorants that read g and d alone. As $r \le \mathsf{d}$, $s(r) \ge s(\mathsf{d})$ and $Y_r \ge Y_{\mathsf{d}}$; with $\mathsf{d}/\mathsf{b} < g^2\mathsf{d}$,

$$\mathrm{br} \le \overline{\mathrm{br}} = 25C_{\mathrm{tp}}c_L^{(2)}\bar{B}\,g^2\mathsf{d} + 8000C_{\mathrm{tp}}^{\mathrm{bg}}\mathsf{d}^3 + 2L^3C_{\mathrm{tp}} \sup_{t \ge \frac{4}{\sqrt{3}}MY_{\mathsf{d}}^{2\mathfrak{p}_0}} t^3 e^{-\delta_{\mathrm{tp}}Lt}.$$

In $\epsilon_\circ$, $\epsilon'_{\mathrm{N}}(r)$ is at most the bound of (i); $\mathfrak{b}_{k_1} \le A_1 Y_{\mathsf{d}}^{\mathfrak{p}_0}$ and $\delta_{k_1} \le A_1 Y_{\mathsf{d}}^{\mathfrak{p}_0} g\mathsf{d}^{1/2}$ (step 1, $\mathsf{b} > g^{-2}$); and $\mathsf{e}_{\mathrm{far}} \le 8\mathsf{d}^2$ by (iii), as $P \ge 4MY_{\mathsf{d}}^{2\mathfrak{p}_0}$. So

$$\epsilon_\circ \le \bar{\epsilon}_\circ = C_\circ\{A_1^3 Y_{\mathsf{d}}^{3\mathfrak{p}_0} g\mathsf{d}^{1/2} + C(g)(\log\tfrac{1}{\mathsf{d}})^{15\mathfrak{p}_0+2}\mathsf{d}^{1/2} + e^{4Y_{\mathsf{d}} - \frac{2}{\sqrt{3}}Y_{\mathsf{d}}^{2\mathfrak{p}_0}} + 8A_1^2 Y_{\mathsf{d}}^{2\mathfrak{p}_0}\mathsf{d}^2\}.$$

Y_{d} is a function of g and d, with $\log\frac{1}{\mathsf{d}} \le Y_{\mathsf{d}} \le 2\log\frac{1}{\mathsf{d}} + C(g)$ (Lem. 4.3(c)). So (i)–(vi) are conditions on (g, d) alone, and they hold for all small d, every majorised member tending to 0 with d. $\mathsf{d}_4(g)$ is half the supremum of the numbers d_* such that (i)–(vi) hold at every $0 < \mathsf{d} \le \mathsf{d}_*$: a positive number formed from the constants and g, which reads no K, β, ℓ, ρ, r; at every $\mathsf{d} \le \mathsf{d}_4(g)$, every $r \le \mathsf{d}$, every datum at g and every label $\mathsf{b} \in (g^{-2}, \bar{\mathsf{b}}]$ the conditions hold with br and $\epsilon_\circ$ themselves.

As $\varpi \le \epsilon_\circ + \frac{6}{5}(1+c_6c_\epsilon)\delta_{k_1}$ at $n=3$ (Prop. 4.16) and $K - k_1 \le Y_{\mathsf{d}}/\log L$, (v) secures $(K-k_1)\varpi \le 1$; at $n=2$ (step 6) the same (v) secures it, as $c_4c_\epsilon^{(2)} \le c_6c_\epsilon$ (Lem. 4.14) and $C_\circ$ dominates the constants of step 6. $\mathsf{d}_4^{(4)}(g)$ is formed in the same way with $n=4$, that is with $c_8c_\epsilon^{(4)}$ in (v) and the tolerance $\epsilon_\circ$ of two pairs, under (Γ1) with $c_\epsilon^{(4)}$. Then $Z^n \le CY_{\mathsf{d}}^{2n\mathfrak{p}_0}$, $P^{3n} \le CY_{\mathsf{d}}^{6n\mathfrak{p}_0}$, $\mathfrak{b}_{k_1}^2\delta_{k_1} \le CY_{\mathsf{d}}^{3\mathfrak{p}_0} g\mathsf{d}^{1/2}$, and the exponents are $8n\mathfrak{p}_0 + 15\mathfrak{p}_0 + 2$: $39\mathfrak{p}_0 + 2$ at $n=3$, $47\mathfrak{p}_0 + 2$ at $n=4$.

6. *Two sources (b).* The same with the base level equal to the start level $k_1 = \mathsf{k}(\mathsf{d})$, $\mathsf{v}_a = c_*\hat{\rho}^3\mathsf{u}_a$, $\mathsf{v}_{12} = c_*^2\hat{\rho}^6\mathsf{u}_{12}$; at the pair $\mathcal{S}_{k_1} = 0$, so $q_{k_1;12} = g_{k_1;12} = 0$, and Lem. 4.9(d) at $n = k_1$ gives $|G_{k_1;12}| \le Ce^{2Y_{\mathsf{d}} - \frac{2}{\sqrt{3}}Y_{\mathsf{d}}^{2\mathfrak{p}_0}}\mathsf{v}_{12}$; then $g_{K;12} = \frac{3}{2}\sum w_1(p)w_2(q)\mathcal{S}_K(p,q)^2$ and Lem. 4.17(b) at $k = k_1$; the logarithms count $4\mathfrak{p}_0 + 12\mathfrak{p}_0 + 3\mathfrak{p}_0 = 19\mathfrak{p}_0$. □

The connected functions on the least torus.

Lemma 4.20 (The mixture; the conditioning stopped at any level)**.** Let $0 < g \le \gamma$ with $g^{-2} \ge x_{\mathrm{J}}$, (K, β) a first-passage datum at g with $K \ge 2$, $\ell \in \mathfrak{L}(g)$, and $w_1, \dots, w_p$ real weights, $p \ge 1$ arbitrary. For $0 \le k \le K$ let $\Omega_k = \mathcal{V}_k \sqcup \bigsqcup_{t=1}^k \mathcal{V}_{t-1}$ and $\mathbb{P}_k = Z_k^{-1}[\mathsf{m}_k \oplus \bigoplus_{t \le k} \mathsf{g}_t^{\mathrm{w}}\mathsf{m}_{t-1}]$, where $\mathcal{V}_t$ is the space of the block fields of the level t, m_t the measure of the never-large history up to the level t, and $\mathsf{g}_t^{\mathrm{w}} \ge 0$ the weight of a first large field at the step t in eq. 4.xxxi; $t' = k$ on the first part and $t' = t-1$ on the part t; $\alpha = a_{K-t'}$; $U_0(W) = W$, and for $t' \ge 1$ let $U_{t'}(W)$ be the background configuration of W in the small-field region $\mathcal{V}_{t'}^{\chi}$. Write $\Omega = \Omega_K$, $\mathbb{P} = \mathbb{P}_K$, and $\mathsf{Y}_\beta(\omega) = \partial_\beta[D_{t'}^\lambda(U_{t'}(W)) - D_K^\lambda(\mathbf{1})]|_{\lambda=0}$ on Ω. Then:

(a) $\mathbb{P}_k$ is a Borel probability measure formed with no source; its parts with $t \le k$ are parts of $\mathbb{P}$; the part t of $\mathbb{P}$ has the mass w_t, and $\int \alpha^{-b}d\mathbb{P} = \mathbb{P}(\mathcal{V}_K) + \sum_t L^{b(K-t+1)}\mathsf{w}_t$ for every real b.

(b) $\langle e^{\mathcal{O}(w_\lambda)}\rangle_{K,\ell,\beta} = \int \exp[D_{t'}^\lambda(U_{t'}(W))]\,\mathrm{d}\mathbb{P}_k$ for λ near 0; coefficientwise, $\langle\prod_{\iota\in\gamma}\mathcal{O}(w_\iota)\rangle_{K,\ell,\beta} = \int[\exp\mathbb{W}]_\gamma\,\mathrm{d}\mathbb{P}$ in $\mathbb{C}[\lambda_\iota]/(\lambda_\iota^2)$, with $\mathbb{W}$ the multilinear jet of $\lambda \mapsto D_{t'}^\lambda(U_{t'}(W))$; each $[\mathbb{W}]_\gamma$ is a bounded Borel function that depends on the sources of γ alone.

(c) $S_n^T - \mathsf{J}_n = \sum_{\pi\in\mathrm{Part}([n])}\kappa_{\mathbb{P}}(\mathsf{Y}_{\beta'} : \beta' \in \pi)$; and with $G_\beta(\omega) = G_{t';\beta}(U_{t'}(W))$ on Ω_k, $S_n^T = \sum_\pi \kappa_{\mathbb{P}_k}(G_{\beta'} : \beta' \in \pi)$.

(d) Where $t' \ge 1$, $U_{t'}(W)$ has its pairs of all levels $\le t'$ in the cores and is regular of the level t' at the radius $2B_3\varepsilon_{t'}$, with $10MB_3\varepsilon_{t'} \le c_{\mathrm{A}}\alpha_0$.

(e) On the least torus, under (Γ2): $\int \alpha^{-b}d\mathbb{P} \le 1 + 2L^b\mathsf{C}_{\mathrm{ag}}(b+1, \frac{1}{2})$ for every $b \ge 0$, so $\mathsf{M}' = \int \alpha^{-24}d\mathbb{P} \le \bar{\mathsf{M}}^{(24)}$; and $\int \alpha^{-b}d\mathbb{P}_k \le a_{K-k}^{-b} + \mathsf{M}'$, $\int_{\{\alpha<\varrho\}} \alpha^{-b}d\mathbb{P}_k \le \varrho^{24-b}\mathsf{M}'$ for $0 \le b \le 24$, $\varrho \le a_{K-k}$. On a side $\ell \in \mathfrak{L}(g)$: $\int \alpha^{-b}d\mathbb{P} \le 1 + 4L^bM^{-3}\ell^3 e^{\mathsf{B}(X_g)\ell^3}\mathsf{C}_{\mathrm{ag}}(b+1, \eth)$, finite and growing with ℓ, with the function B and the letter $\eth$ and $X_g = L(g^{-2} + B_\sharp)$.

■

Proof. Equation eq. 4.xxxi writes the torus expectation of a bounded gauge invariant function F as Z^{-1} times the integral against m_k of its quotient at the level k plus the integrals against $\mathsf{g}_t^{\mathrm{w}}\mathsf{m}_{t-1}$ of its quotients at the levels $t-1$, $t \le k$, at every level $k \le K$. At $F=1$ this makes $\mathbb{P}_k$ a probability measure; at $F = e^{\mathcal{O}(w_\lambda)}$ the quotient at a level t is, on $\mathcal{V}_t^\chi$, $\exp D_t^\lambda(U_t(W))$, by [7, Prop. B.15], applied to the finite progression of [7, Prop. B.8]. Coefficients of the multilinear jets give (b); the logarithm of the moment series is the cumulant series, and constants drop out of the cumulants of orders ≥ 2, which gives (c), at every level k. (d) is [7, Lem. B.17], in the setting verified by [7, Prop. B.21] at $x_K \ge x_{\mathrm{J}}$. (e) The mass w_t is at most the probability that the first large field arises at the step t; on the least torus Prop. 4.54, under $\log g^{-2} \ge \mathsf{Y}_{\mathrm{lt}}(\frac12)$, gives $\sum_t L^{b(K-t+1)}\mathsf{w}_t \le 2L^b\mathsf{C}_{\mathrm{ag}}(b+1,\frac12)$ for every real $b \ge 0$; on a larger torus the number of places costs $\ell^3 e^{\mathsf{B}(X_g)\ell^3}$. The parts $t \le k$ of $\mathbb{P}_k$ are those of $\mathbb{P}$, and on the first part $\alpha = a_{K-k}$. □

Lemma 4.21 (The remainder)**.** Let g satisfy (Γ), (K,β) be a first-passage datum at g, $\ell = \ell_g$, $n \in \{3,4\}$, and let real weights satisfy the conditions of weights placed at (ρ, r, d) (if $n=3$), or of two pairs at (ρ, r, d) (if $n=4$), but for the upper bounds on the distances of the centres, with $0 < \rho \le r \le \mathsf{d} \le 1$, $\rho \le \frac12$, $s(\rho) \le K$, $\rho \le r_1(\mathsf{b})$, all supports in one ball of radius $\le \frac18\ell_g - 1$. There are numbers $\Lambda_3 \ge 1$ and C_{J} formed from the constants such that at $\mathbb{P}$-almost every ω, for every nonempty β,

$$|\mathsf{Y}_\beta(\omega)| \le \Lambda_3\,\alpha^{-6}\,\mathsf{s}(\beta)\prod_{a\in\beta}|w_a|_\infty, \qquad \mathsf{s}(\beta) = \rho^3 \text{ (a point)}, \quad \rho^6 r^{-3} \text{ (a whole pair)},$$
$$(\rho/r)^{3c(\beta)}\rho^6\mathsf{d}^{-9/2} \ (\beta \text{ crossing});$$

hence $|S_n^T - \mathsf{J}_n| \le \mathsf{k}_n\Lambda_3^n\mathsf{M}'^n\mathsf{b}^n\mathsf{d}^{3/2}\mathrm{nat}_n$ with $\mathsf{k}_3 = 13$, $\mathsf{k}_4 = 77$, and $|\mathsf{J}_n| \le C_{\mathrm{J}}(\mathsf{b}/\mathsf{d})^{n/2}\mathrm{nat}_n$, uniformly in $\rho \le r \le \mathsf{d}$ and in K. For two weights w_1, w_2 with $\Sigma(w_a) \subset \mathbb{B}(x_a,\rho)$ whose supports are at a distance $\ge \mathsf{d}$, with $0 < \rho \le \mathsf{d} \le 1$, $\rho \le \frac12$, $s(\rho) \le K$, $\rho \le r_1(\mathsf{b})$ and both supports in one ball of radius $\le \frac18\ell_g - 1$, at $\mathbb{P}$-almost every ω

$$|\mathsf{Y}_a(\omega)| \le \Lambda_3\,\alpha^{-6}\rho^3|w_a|_\infty, \qquad |\mathsf{Y}_{12}(\omega)| \le \Lambda_3\,\alpha^{-6}\rho^6\mathsf{d}^{-3}|w_1|_\infty|w_2|_\infty$$
$$\le \Lambda_3\,\alpha^{-6}\rho^6\mathsf{d}^{-9/2}|w_1|_\infty|w_2|_\infty,$$

and

$$|S_2^T - \mathsf{J}_2| \le 3\Lambda_3^2\mathsf{M}'^2\rho^6\mathsf{d}^{-9/2}|w_1|_\infty|w_2|_\infty;$$

if besides the balls are $\ge \mathsf{d}$ apart,

$$|\mathsf{J}_2| \le C_{\mathrm{J}}(\mathsf{b}/\mathsf{d})\rho^6\mathsf{d}^{-6}\mathsf{b}^{-2}|w_1|_\infty|w_2|_\infty.$$

■

Proof. For $|\beta| \ge 2$, $\mathsf{Y}_\beta = \sum_{i\le t'}(\mathsf{Sum}[E_\beta^{(i)}](U) - \mathsf{Sum}[E_\beta^{(i)}](\mathbf{1})) - \sum_{t'<i\le K}\mathsf{Sum}[E_\beta^{(i)}](\mathbf{1})$, $U = U_{t'}(W)$; at a point $\mathcal{O}(w_a)(U) - \mathcal{O}(w_a)(\mathbf{1})$ comes besides, and for $t' \ge 1$ it is $\le 162B_3^2\varepsilon_{t'}^2\rho^3\alpha^{-4}|w_a|_\infty$ on the regular configuration of Lem. 4.20(d). The first sum is bounded by Lem. 4.8(b) at the regular background of Lem. 4.20(d), the second at $\mathbf{1}$, both with the sizes of Lem. 4.9(f) at the width ρ, whose conditions hold: (P1), (P2) and the separation of the corners follow, as in the proof of Lem. 4.5(b), from the balls that hold the supports and from the distances of the supports alone, and $\rho \ge \frac23 Ma_K$, $E_\sigma(\hat\rho_1/2) \le \varkappa_0^4$ from $\rho \le r_1(\mathsf{b})$, $s(\rho) \le K$ (Lem. 4.6); the sums over the levels on both sides of t' are geometric.

For $t' \ge 1$, with $j = t'-1$, $a_{K-j} = \alpha/L$ and $x_j > 1$: $\bar\nu_{j;\beta} \le \bar c_m(L/\alpha)^m\prod|w_a|_\infty$, $\Xi_j^r \le \rho^6 r^{-3}L^3\alpha^{-3}$, and $\mathsf{z}(\beta)\min\{1,v_j\}^{3/2} \le \mathsf{s}(\beta)L^{3/2}\alpha^{-3/2}$; so $|\mathsf{Y}_\beta| \le C\alpha^{-6}\mathsf{s}(\beta)\prod|w_a|_\infty$ ($\alpha \le 1$, $m \le 4$). The part $t=1$ of Ω has $t'=0$, $\alpha = a_K$, and there $W \in \mathcal{V}_0$ is any configuration and $U_0(W) = W$: $D_0^\lambda = \mathcal{O}(w_\lambda)$, so the first sum is empty; at a point $0 \le a_p \le 2$ and (P1) give $|\mathcal{O}(w_a)(W)| \le 2a_K^{-1}\sum_p|w_a(p)| \le 162\rho^3 a_K^{-4}|w_a|_\infty = 162\rho^3\alpha^{-4}|w_a|_\infty$, and $\mathcal{O}(w_a)(\mathbf{1}) = 0$; the second sum runs over $1 \le i \le K$ at $\mathbf{1}$

and is led by $i=1$, with the sizes above at $j=0$: $\bar\nu_{0;\beta}\le \bar c_m\alpha^{-m}\prod|w_a|_\infty$ $(x_0>1)$, $\Xi_0^r\le\rho^6 r^{-3}\alpha^{-3}$, $\mathsf{z}(\beta)\min\{1,v_0\}^{3/2}\le\mathsf{s}(\beta)\alpha^{-3/2}$.

So the bound holds at every ω of this part, with $\Lambda_3\ge 162$. The weights $\mathsf{v}_1=\mathsf{v}_2=\rho^3r^{-3/2}$, $\mathsf{v}_3=\rho^3\mathsf{d}^{-9/2}$ (and $\mathsf{v}_a=\rho^3r^{-3/2}\mathsf{d}^{-9/8}$ at $n=4$) have $\mathsf{s}(\beta)\le\mathsf{v}^\beta$ and $\mathsf{v}^{[n]}=\mathsf{s}([n])$, so every product over a partition is $\le\mathsf{s}([n])$; Lem. 4.20(c), the bounds $|\kappa_2|\le 2$, $|\kappa_3|\le 6$ times the mean of the product of the bounds, and the moments of α^{-1} give $13=1+3\cdot 2+6$ at $n=3$; finally $\mathsf{s}([3])\prod|w_a|_\infty=\mathsf{b}^3\mathsf{d}^{3/2}\mathrm{nat}_3$. The bound of J_n is Lem. 4.8(b) at $\mathbf{1}$ with the bounds of Lem. 4.9(e) at all the levels, in the form (f).

At two sources whose supports are at a distance $\ge\mathsf{d}$ the two weights form one pair at the widths (ρ,d) in the sense of Lem. 4.9(f): (P1), (P2) and the separation of the corners follow from the balls $\mathbb{B}(x_a,\rho)$ and from the distance $\ge\mathsf{d}$ of the supports, and $\rho\ge\frac23 Ma_K$, $E_\sigma(\hat\rho_1/2)\le\varkappa_0^4$ from $\rho\le r_1(\mathsf{b})$, $s(\rho)\le K$, as above. The proof of the bound above, run for the list of the two weights, gives at a point and at the whole pair, read at $r=\mathsf{d}$, the display at two sources, as $\rho^6\mathsf{d}^{-3}\le\rho^6\mathsf{d}^{-9/2}$ for $\mathsf{d}\le 1$; with $S_2^T-\mathsf{J}_2=\int\mathsf{Y}_{12}d\mathbb{P}+\kappa_{\mathbb{P}}(\mathsf{Y}_1,\mathsf{Y}_2)$ (Lem. 4.20(c)), $|\kappa_2|\le 2$ and the moments of α^{-1} it gives $3=1+2$.

If besides the balls are $\ge\mathsf{d}$ apart, put a dummy bump w_2', the indicator of the plaquettes with barycentres in $\mathbb{B}(x_1+3\rho e,\rho)$, e the unit vector from x_2 to x_1; the three are at the lengths (ρ,ρ,d), and by multilinearity the objects of the pair are those of the list with the dummy: $S_2^T-\mathsf{J}_2=\int\mathsf{Y}_{12}d\mathbb{P}+\kappa_{\mathbb{P}}(\mathsf{Y}_1,\mathsf{Y}_2)$, with $|\mathsf{Y}_a|\le\Lambda_3\alpha^{-6}\rho^3|w_a|_\infty$ and $|\mathsf{Y}_{12}|\le\Lambda_3\alpha^{-6}\rho^6\mathsf{d}^{-9/2}|w_1|_\infty|w_2|_\infty$; and the bound of J_n above, at the crossing set that the two weights form in this list, is the bound of J_2. □

Lemma 4.22 (Lattice sums and continuum integrals)**.** For every side ℓ and every cutoff K, that is on the torus T_ℓ with the lattice $a_K\mathbb{Z}^3$, and with the least lifts of the differences, the following hold.

(a) For $4a_K\le\rho\le r\le\mathsf{d}\le\mathsf{d}^+$, $4\mathsf{d}^+<\ell$, $\vartheta\in(0,1]$ and Lipschitz f_1,f_2,f_3 with $\operatorname{supp}f_a\subset\mathbb{B}(x_a,\rho)$, $\operatorname{Lip}f_a\le\|f_a\|_\infty/\vartheta\rho$, $\operatorname{dist}(\operatorname{supp}f_1,\operatorname{supp}f_2)\ge r$, $\operatorname{dist}(\operatorname{supp}f_a,\operatorname{supp}f_3)\ge\mathsf{d}$, all supports in one ball of radius $\frac12\mathsf{d}^+$:

$$\Big|\operatorname{Tr}\prod_a((f_a\circ x)C_K^{\mathrm{free}})-\int k_3\prod_a f_a\Big|\le C^{(3)}\Big[\frac{a_K}{\vartheta\rho}+\Big(\frac{\mathsf{d}^+}{\ell}\Big)^3\Big]\rho^9r^{-3}\mathsf{d}^{-6}\prod_a\|f_a\|_\infty.$$

(b) For two pairs with the same data ($8a_K\le\rho$, the balls $\ge r$ apart in a pair and $\ge\mathsf{d}$ apart across): $|\frac12\sum_\varsigma\operatorname{Tr}\prod_l((f_{\varsigma(l)}\circ x)C_K^{\mathrm{free}})-\int k_4\prod f_a|\le C^{(4)}[\frac{a_K}{\vartheta\rho}+(\frac{\mathsf{d}^+}{\ell})^3]\rho^{12}r^{-6}\mathsf{d}^{-6}\prod\|f_a\|_\infty$.

(c) For $8a_K\le\rho\le\mathsf{d}$, $64\mathsf{d}\le\ell$, Lipschitz φ_a in $\mathbb{B}(x_a,\rho)$ with $\operatorname{Lip}\varphi_a\le\|\varphi_a\|_\infty/\vartheta\rho$, the balls $\ge\mathsf{d}$ apart and $|x_1-x_2|\le 18\mathsf{d}$:

$$\Big|\sum_{\lambda(p)=\lambda,\,\lambda(q)=\mu}\varphi_1(x(p))\varphi_2(x(q))C_K^{\mathrm{free}}(p,q)^2-\mathcal{K}_{\lambda\mu}(\varphi_1,\varphi_2)\Big|\le C_{\mathrm{Rs}}\Big[\frac{a_K}{\vartheta\rho}+\Big(\frac{\mathsf{d}}{\ell}\Big)^3\Big]\Big(\frac{\rho}{\mathsf{d}}\Big)^6\|\varphi_1\|_\infty\|\varphi_2\|_\infty.$$

■

Proof. Step A. In a ring, and in a square, every sign ϵ_p stands twice, so $C_K^{\mathrm{free}}(p_l,p_{l+1})$ may be replaced by $\epsilon\epsilon C_K^{\mathrm{free}}$ and then, by equations YM3_I:(2245)–(2246), by $a_K^3\mathbb{K}_3(u_l)_{\lambda(p_l)\lambda(p_{l+1})}$; both are $\le(1+3C_{\mathrm{lat}})a_K^3|u|^{-3}$, and telescoping costs $C_{\mathrm{lat}}a_K^3[a_K^2|u_l|^{-5}+\ell^{-3}]$ times the other factors, with at most $81(\rho/a_K)^3$ plaquettes for each source. *Step B.* For each choice of the directions the remaining sum is a Riemann sum over the cubes of side a_K about the barycentres, which for one orientation form a translate of $a_K\mathbb{Z}^3$; summed over the directions, $\sum\prod_l\mathbb{K}_3(\cdot)_{\lambda_l\lambda_{l+1}}=\operatorname{tr}\prod_l\mathbb{K}_3$. On the cubes that count the differences are at least 0.56 times their least values, $\|\mathbb{K}_3(u)\|\le\frac{1}{2\pi}|u|^{-3}$, $\|\partial\mathbb{K}_3(u)\|\le\frac{6}{4\pi}|u|^{-4}$ (Lem. 4.24(b)), and the Lipschitz bounds of the f_a give the error $a_K/\vartheta\rho$ times the natural size; the numbers $C^{(3)},C^{(4)},C_{\mathrm{Rs}}$ are these counts. □

Lemma 4.23 (Kernels of forms bounded on small bumps)**.** Let $U_1,\dots,U_p$ be open subsets of $\mathbb{R}^3$ or of a torus, $\rho_0>0$, $\mathsf{A}\ge 0$, and T a p-linear form on $\prod_a \mathrm{Lip}_c(U_a)$ with $|T(f_1,\dots,f_p)|\le \mathsf{A}\rho^{3p}\prod_a\|f_a\|_\infty$ whenever $\rho\le\rho_0$ and each f_a has its support in a ball of radius ρ in U_a. Then: (a) $T(f)=\int k\prod_a f_a$ with a Borel function k, $|k|\le(\frac{3}{\pi})^p\mathsf{A}$ almost everywhere, and at every point where T has a continuous kernel; (b) T extends to bounded Borel functions with compact supports in the U_a, and for balls $\mathbb{B}(z_a,\rho)\subset U_a$, $\rho\le\rho_0$, the mean of k over $\prod_a\mathbb{B}(z_a,\rho)$ is at most $(\frac{3}{4\pi})^p\mathsf{A}$ in modulus, at every choice of the centres; with the radial tents, $|\int k\prod_a\chi^\rho_{z_a}|\le(\frac{3}{\pi})^p\mathsf{A}$. ■

Proof. Let $k_\rho(x_1,\dots,x_p)=T(\chi^\rho_{x_1},\dots,\chi^\rho_{x_p})$; $\|\chi^\rho\|_\infty=3/\pi\rho^3$ gives $|k_\rho|\le(\frac{3}{\pi})^p\mathsf{A}$. On functions with supports in fixed compact sets T is bounded in the sup norms (a Lipschitz partition of unity of width ρ_0), and the Riemann sums of $f_a*\chi^\rho$ converge uniformly, so $\int k_\rho\prod f_a=T(f_1*\chi^\rho,\dots)\to T(f)$; a weak-$*$ limit point of k_ρ in L^∞ is a kernel. (b) By dominated convergence T extends with the same bound; at $f_a=\mathbf{1}_{\mathbb{B}(z_a,\rho)}$, $|\int_{\prod\mathbb{B}(z_a,\rho)}k|\le\mathsf{A}\rho^{3p}$, which is the bound of the mean. □

Lemma 4.24 (The free template)**.** Let $X\ne0$, $\mathsf{d}=|X|$, $\mathsf{b}>0$, n,n' unit vectors.

(a) $\mathbb{K}_3(X)^2=(16\pi^2\mathsf{d}^6)^{-1}(3\hat X\hat X^{\mathsf T}+1)$; $\frac{3}{2}\mathsf{b}^{-2}\sum_\mu\{[\mathbb{K}_3(X)_{\lambda\mu}]^2-\frac13\sum_{\lambda'}[\mathbb{K}_3(X)_{\lambda'\mu}]^2\}=\mathsf{t}^\circ(e_\lambda,X)$; $\frac{3}{2}\mathsf{b}^{-2}\sum_{\lambda,\mu}[\mathbb{K}_3(X)_{\lambda\mu}]^2=\frac{9}{16\pi^2}\mathsf{b}^{-2}\mathsf{d}^{-6}$; $\sum_\lambda n_\lambda^2\mathsf{t}^\circ(e_\lambda,X)+\sum_{\lambda<\lambda'}n_\lambda n_{\lambda'}3\mathsf{b}^{-2}[\mathbb{K}_3(X)^2]_{\lambda\lambda'}=\mathsf{t}^\circ(n,X)$; and $|\mathsf{t}^\circ(n,X)-\mathsf{t}^\circ(n',X)|\le0.057|n-n'|\mathsf{b}^{-2}\mathsf{d}^{-6}$.

(b) In the operator norm, $\|\mathbb{K}_3(y)\|=\frac{2}{4\pi}|y|^{-3}$, $\|(h\cdot\nabla)\mathbb{K}_3(y)\|\le\frac{6}{4\pi}|h||y|^{-4}$, $\|(h\cdot\nabla)^2\mathbb{K}_3(y)\|\le\frac{24}{4\pi}|h|^2|y|^{-5}$.

(c) For $0<r\le\frac12\mathsf{d}$ and $c-z=X$: $|\frac{2\pi}{3}\mathsf{b}r^3\cdot3\mathsf{b}^{-3}k_3(c+\frac{r}{2}n,c-\frac{r}{2}n,z)-\mathsf{t}^\circ(n,X)|\le2.66\,(r/\mathsf{d})^2\mathsf{b}^{-2}\mathsf{d}^{-6}$; and for $0<\rho<\frac14 r$ the same function averaged against the radial tents χ^ρ about the three points differs from its value by at most $7900\,(\rho/r)\mathsf{b}^{-2}\mathsf{d}^{-6}$.

(d) For $0<\rho\le\frac18\mathsf{d}$: $|\mathcal{K}_{\lambda\mu}(\chi^\rho_c,\chi^\rho_z)-[\mathbb{K}_3(X)_{\lambda\mu}]^2|\le2.28\,(\rho/\mathsf{d})\mathsf{d}^{-6}$.

■

Proof. (a) With $E=\hat X\hat X^{\mathsf T}$, $E^2=E$: $(3E-1)^2=3E+1$; $\sum_\mu[\mathbb{K}_3(X)_{\lambda\mu}]^2=[\mathbb{K}_3(X)^2]_{\lambda\lambda}$ and $\operatorname{tr}\mathbb{K}_3(X)^2=6(16\pi^2\mathsf{d}^6)^{-1}$; the identity in n is the expansion of $\langle n,\mathbb{K}_3(X)^2n\rangle$; the last bound is $\frac{18}{32\pi^2}\le0.057$ times $|n-n'|$. (b) $\mathbb{K}_3=(4\pi)^{-1}\nabla\nabla|y|^{-1}$; the third and fourth derivatives of $|y|^{-1}$ contracted with h, in a frame with $\hat y=e_1$, have eigenvalues bounded by 6 and 24 times $|h|^k|y|^{-3-k}$. (c) With $D(n)=3nn^{\mathsf T}-1$, $h=\frac{r}{2}n$ and $\phi(t)=\operatorname{tr}[D(n)\mathbb{K}_3(X-th)\mathbb{K}_3(X+th)]$: $\mathbb{K}_3(rn)=(4\pi r^3)^{-1}D(n)$, so the first member is $\frac12\mathsf{b}^{-2}\phi(1)$, while $\frac12\mathsf{b}^{-2}\phi(0)=\mathsf{t}^\circ(n,X)$ by (a). ϕ is even, since $\operatorname{tr}[DAB]=\operatorname{tr}[DBA]$ for symmetric matrices, so $|\phi(1)-\phi(0)|\le\frac12\sup|\phi''|$; with $|X\pm th|\ge\frac34\mathsf{d}$, the trace norm 4 of $D(n)$ and (b),

$$|\phi''|\le\frac{4}{(4\pi)^2}(24\cdot2+2\cdot6\cdot6+2\cdot24)\frac{r^2}{4}\Big(\frac43\Big)^8\mathsf{d}^{-8}=10.63\,(r/\mathsf{d})^2\mathsf{d}^{-6},$$

and $\frac14\cdot10.63\le2.66$. The tents: the mean value theorem with (b) for the three factors of k_3. (d) $|\nabla(\mathbb{K}^2_{3,\lambda\mu})|\le\frac{3}{2\pi^2}|u|^{-7}$ on a segment in $\{|u|\ge\frac34\mathsf{d}\}$, and $\frac{3}{2\pi^2}(\frac43)^7\cdot2\le2.28$. □

Every state.

Lemma 4.25 (The chessboard bound)**.** Let $0<g\le\gamma$, (K,β) a first-passage datum at g, $\ell'\in\mathfrak{L}(g)$, and consider the board of $\mathbb{L}_K$ into cubes of side $\frac12\ell'$, the cells, with any offset in $a_K\mathbb{Z}^3$; let $\mathsf{Q}_0=\exp[\frac18\ell'^3(2C_{\mathrm S}+C_g)]$ with the constants $C_{\mathrm S},C_g$ of the two-sided stability bound. For every $\ell\in\mathfrak{L}(g)$ with $\ell\ge\ell'$, every $\nu\in\mathcal{G}^{\mathrm{per}}_{K,\beta}$, every cell and every bounded complex Borel function F of the variables of that cell,

$$|\tilde\mu_{K,\ell,\beta}(F)|,\ |\nu(F)|\le\mathsf{Q}_0\|F\|_{\mathrm{ch}},\qquad \|F\|^8_{\mathrm{ch}}=\tilde\mu_{K,\ell',\beta}\Big(\prod_{\text{the eight cells of }T_{\ell'}}F^{[\cdot]}\Big)\ge0,$$

$F^{[\cdot]}$ are the reflected copies of F, conjugated when an odd number of coordinate reflections is used. ■

Proof. The bound is a chain of four links. The cells are cubes of side $\mathsf{l} = \frac{1}{2}\ell'$; the torus T_ℓ holds $\mathsf{N}_\ell = (2\ell/\ell')^3$ of them and $T_{\ell'}$ eight. For F as in the statement let F_ℓ^{all} be the product of the reflected copies of F, conjugated with the parity convention in the statement, over all the cells of T_ℓ, and $Z_{K,\ell,\beta}[F_\ell^{\mathrm{all}}]$ the integral of the density of T_ℓ with this insertion. (1) *Reflection positivity on T_ℓ.* The torus measures are reflection positive in the planes of the board, and the chessboard estimate ([7, Cor. B.1](b), (c)) gives $|\tilde\mu_{K,\ell,\beta}(F)| \le \|F\|_{\mathrm{ch},\ell}$, where $\|F\|_{\mathrm{ch},\ell}^{\mathsf{N}_\ell} = Z_{K,\ell,\beta}[F_\ell^{\mathrm{all}}]/Z_{K,\ell,\beta} \ge 0$; on $T_{\ell'}$ this is the number of the statement, $\|F\|_{\mathrm{ch},\ell'} = \|F\|_{\mathrm{ch}}$. (2) *Two sides, with the insertions.* As ℓ/ℓ' is a power of L, the subadditivity in the volume of the integrals with the reflected copies inserted (Prop. YM3_I:86.5, Cor. YM3_I:86.6) gives $Z_{K,\ell,\beta}[F_\ell^{\mathrm{all}}]^{1/\mathsf{N}_\ell} \le Z_{K,\ell',\beta}[F_{\ell'}^{\mathrm{all}}]^{1/8}$; with $p_\ell = \ell^{-3}\log Z_{K,\ell,\beta}$, $Z_{K,\ell,\beta}^{1/\mathsf{N}_\ell} = e^{\mathsf{l}^3 p_\ell}$ and $Z_{K,\ell',\beta}^{1/8} = e^{\mathsf{l}^3 p_{\ell'}}$ this is clause (e) of the same corollary:

$$\|F\|_{\mathrm{ch},\ell} \le \|F\|_{\mathrm{ch},\ell'}\, e^{\mathsf{l}^3(p_{\ell'} - p_\ell)}.$$

(3) *Stability, with no insertion.* The two-sided stability bound among the construction results, which rests on Thms. YM3_I:AJ and YM3_I:AD ([7, Lem. N.1]), concerns the partition functions without insertions: $Z_{K,\ell,\beta} = e^{E_{K,\ell,\beta}}\widetilde{Z}_\ell$ with the vacuum constant $E_{K,\ell,\beta}$ and a number $e^{-C_g\ell^3} \le \widetilde{Z}_\ell \le e^{C_{\mathrm{S}}\ell^3}$, and $|E_{K,\ell,\beta}\ell^{-3} - E_{K,\ell',\beta}\ell'^{-3}| \le C_{\mathrm{S}}$ for two sides $\ell, \ell' \in \mathfrak{L}(g)$. So

$$p_{\ell'} - p_\ell \le (E_{K,\ell',\beta}\ell'^{-3} + C_{\mathrm{S}}) - (E_{K,\ell,\beta}\ell^{-3} - C_g) \le 2C_{\mathrm{S}} + C_g,$$

and (1)–(3), with $\mathsf{l}^3 = \frac{1}{8}\ell'^3$, give $|\tilde\mu_{K,\ell,\beta}(F)| \le \mathsf{Q}_0\|F\|_{\mathrm{ch}}$. (4) *Periodic Gibbs states.* For $\nu \in \mathcal{G}_{K,\beta}^{\mathrm{per}}$ the domination theorem (Prop. YM3_I:91.9, for every board of $\mathbb{L}_K$ and every offset) gives $|\nu(F)| \le \liminf_k \|F\|_{\mathrm{ch},\ell_k}$ along every sequence $\ell_k \to \infty$ in $\mathfrak{L}(g)$, and (2), (3) at $\ell = \ell_k$ bound the right member by $\mathsf{Q}_0\|F\|_{\mathrm{ch}}$. No link reads what F is made of, and Q_0 depends neither on $\ell \ge \ell'$ nor on K. □

Lemma 4.26 (The fold)**.** Let $\mathfrak{N}_I = \mathbb{C}[\lambda_\iota : \iota \in I]/(\lambda_\iota^2)$, $[x]_\gamma$ the coefficient of λ^γ, $\mathfrak{m} = \{[x]_\emptyset = 0\}$, and $\|x\|_u = \sum_\gamma |[x]_\gamma|/u^\gamma$ for numbers $u_\iota > 0$.

(a) $\|\cdot\|_u$ is submultiplicative; for $y \in \mathfrak{m}$, $\|e^y - 1\|_u \le (1 + \|y\|_u)^{|I|} - 1$ and $\|\log(1+y)\|_u \le \|y\|_u(1+\|y\|_u)^{|I|-1}$; the logarithm of the moment series is the cumulant series.

(b) In the data of Lem. 4.25 let $f_1, \dots, f_n$ be bounded real functions that vanish off the interior of one cell, w_ι, $\iota \in I = [n] \times \{\text{the eight cells}\}$, the eight mirror images of the $f_a \circ x$ on $T_{\ell'}$, and $(\Omega, \mathbb{P})$ a probability space with a measurable $\mathbb{W}\colon \Omega \to \mathfrak{m}$ such that $\langle \prod_{\iota\in\gamma} \mathcal{O}(w_\iota)\rangle_{K,\ell',\beta} = \int [\exp \mathbb{W}]_\gamma d\mathbb{P}$ for all $\gamma \subset I$. Let $u_{(a,\cdot)} = u_a > 0$, $\mathsf{c} = \sum \mathsf{c}^{(\cdot)}$ a sum of constants of $\mathfrak{m}$ each in the variables of one cell, of any size, and $\mathsf{Z} \ge 0$ measurable with $\|\mathbb{W} - \mathsf{c}\|_u \le \mathsf{Z}$ almost everywhere. Put $\mathsf{y} = \int[(1+\mathsf{Z})^n - 1]d\mathbb{P}$, $\mathsf{h} = \mathsf{y}(1+\mathsf{y})^{n-1}$, $\eta_0 = [\int(1 + \mathsf{Z} + 8\mathsf{h})^{8n} d\mathbb{P}]^{1/8}$, $\eta = (2^n - 1)\mathsf{Q}_0\eta_0$. Then for every nonempty $\beta \subset [n]$, every $\ell \in \mathfrak{L}(g)$ with $\ell \ge \ell'$, and $\mathfrak{s} = \tilde\mu_{K,\ell,\beta}$ or $\mathfrak{s} \in \mathcal{G}_{K,\beta}^{\mathrm{per}}$:

$$|S_{|\beta|}^T[\mathfrak{s}]((f_a)_{a\in\beta}) - S_{|\beta|}^T[\tilde\mu_{K,\ell',\beta}]((f_a)_{a\in\beta})| \le \eta(1+\eta)^{n-1}u^\beta.$$

The bound holds at every size of Z, of the constants c and of η. ■

Proof. (a) is the algebra of exp and log in an algebra with $|I|$ nilpotent generators. (b) Let $\mathsf{k} = \log\sum_\beta \tilde\mu_{K,\ell',\beta}(\prod_{a\in\beta}\mathcal{O}(f_a))\lambda^\beta$ and $\Xi_\beta = [\exp(\sum_a \lambda_a \mathcal{O}(f_a) - \mathsf{k})]_\beta$, a function of the variables of the cell. The mirror images of Ξ_β are the same expressions in the mirrored sources, so by the representation $\|\Xi_\beta\|_{\mathrm{ch}}^8 = \int[\exp(\mathbb{W} - \mathsf{k}^{\mathrm{all}})]_{\beta\times\{\mathrm{cells}\}} d\mathbb{P}$, $\mathsf{k}^{\mathrm{all}}$ the sum of the eight copies of k. Each copy has $e^{\mathsf{k}^{(\cdot)}} = \int \exp(\pi_{\mathrm{cell}}\mathbb{W})d\mathbb{P}$, so $\mathsf{k}^{(\cdot)} - \mathsf{c}^{(\cdot)} = \log(1 + \mathsf{y}^{(\cdot)})$ with $\|\mathsf{y}^{(\cdot)}\|_u \le \mathsf{y}$ and $\|\mathsf{k}^{(\cdot)} - \mathsf{c}^{(\cdot)}\|_u \le \mathsf{h}$; the constants of any size cancel, $\|\mathbb{W} - \mathsf{k}^{\mathrm{all}}\|_u \le \mathsf{Z} + 8\mathsf{h}$, and $|[\exp(\cdot)]_{\beta\times\{\mathrm{cells}\}}| \le (1 + \mathsf{Z} + 8\mathsf{h})^{8n}(u^\beta)^8$.

So $\|\Xi_\beta\|_{\mathrm{ch}} \le \eta_0 u^\beta$, and Lem. 4.25 gives $|\mathfrak{s}(\Xi_\beta)| \le \mathsf{Q}_0\eta_0 u^\beta$. With $\mathsf{b}' = \sum_\beta \mathfrak{s}(\Xi_\beta)\lambda^\beta = \mathsf{M}_\mathfrak{s} e^{-\mathsf{k}}$, $\mathsf{M}_\mathfrak{s}$ the moment series of $\mathfrak{s}$: $\|\mathsf{b}'-1\|_u \le \eta$ and $\|\log \mathsf{M}_\mathfrak{s} - \mathsf{k}\|_u = \|\log \mathsf{b}'\|_u \le \eta(1+\eta)^{n-1}$. □

Lemma 4.27 (Mirror copies)**.** Zero sources are omitted before forming the weighted jet norm; coefficients involving their directions vanish by multilinearity. Assume the remaining source weights are nonzero. Let g satisfy (Γ), $\ell' = \ell_g$, and let $f_1, \dots, f_n$, $n \in \{2,3,4\}$, be as in (A.4)(i) (for $n=4$, two pairs; for $n=2$, two bounded real Borel functions with $\operatorname{supp} f_a \subset \mathbb{B}(x_a,\rho)$, the two supports at a distance $\ge \mathsf{d}$, $0 < \rho \le \mathsf{d} \le 1$, $\rho \le \frac12$, $s(\rho) \le K$, $\rho \le r_1(\mathsf{b})$, both supports in one ball of radius $\le \frac18\ell_g - 1$), placed by the offset in the interior of one cell, so that two mirror copies are at least $\frac14\ell_g \ge 6.5$ apart. Let $\mathbb{W}$ be the multilinear jet of $\lambda \mapsto D^\lambda_{t'}(U_{t'}(W))$ for the $8n$ sources w_ι on the space of Lem. 4.20, $[\mathsf{c}]_\gamma = \partial^\gamma D^\lambda_K(\mathbf{1})|_0$ for γ in one cell and $[\mathsf{c}]_\gamma = 0$ for γ that link two cells, and $u_a = \mathsf{v}_a|w_a|_\infty$ with $\mathsf{v}_1 = \mathsf{v}_2 = \rho^3 r^{-3/2}$, $\mathsf{v}_3 = \rho^3\mathsf{d}^{-9/2}$ $(n=3)$, $\mathsf{v}_a = \rho^3 r^{-3/2}\mathsf{d}^{-9/8}$ $(n=4)$, $\mathsf{v}_a = \rho^3\mathsf{d}^{-9/4}$ $(n=2)$.

Then $\|\mathbb{W}(\omega) - \mathsf{c}\|_u \le \mathsf{Z}_n\alpha(\omega)^{-6}$ almost everywhere, $\mathsf{Z}_n = 8(2^n-1)\Lambda_3 + 2^{8n}\max_m \Lambda^{\mathrm{L}}_m$, where the numbers Λ^{L}_m, formed from the constants, bound the linking jets: for γ linking two copies, $|\sum_{i\le t}\mathsf{Sum}[E^{(i)}_\gamma](U)| \le \Lambda^{\mathrm{L}}_m \mathsf{b}^{-m/2}\prod_{\text{copies}}\mathsf{s}(\gamma\cap\text{copy})\prod_{\iota\in\gamma}|w_\iota|_\infty$ at every U whose pairs of the levels $\le t$ lie in the cores. ■

Proof. Within one cell the coefficients of $\mathbb{W}-\mathsf{c}$ are the Y_β of the copy (by multilinearity the mixture names no source), bounded by Lem. 4.21 and $\mathsf{s}(\beta) \le \mathsf{v}^\beta$; there are $8(2^n-1)$ such sets. At $n=2$ the two bumps are taken as one pair at the widths (ρ,d) in the sense of Lem. 4.9(f), since (P1), (P2) and the separation of the corners follow from the balls $\mathbb{B}(x_a,\rho)$ and from the distance $\ge \mathsf{d}$ of the supports, as in the proof of Lem. 4.21; that proof, with the size $\rho^6\mathsf{d}^{-3}$ of a whole pair at the distance d in the place of $\rho^6 r^{-3}$, gives at $\mathbb{P}$-almost every ω

$$|\mathsf{Y}_a| \le \Lambda_3\alpha^{-6}\rho^3|w_a|_\infty, \qquad |\mathsf{Y}_{12}| \le \Lambda_3\alpha^{-6}\rho^6\mathsf{d}^{-3}|w_1|_\infty|w_2|_\infty \le \Lambda_3\alpha^{-6}\rho^6\mathsf{d}^{-9/2}|w_1|_\infty|w_2|_\infty \quad (\mathsf{d}\le 1),$$

and $\rho^3 \le \mathsf{v}_a$, $\rho^6\mathsf{d}^{-9/2} = \mathsf{v}_1\mathsf{v}_2$. For a set γ that links two copies, the calculus of Lem. 4.6 at $8n$ sources, with the copies as cells and $\mathsf{D}^{\iota\iota'}_i \ge \hat{\mathsf{D}}^\times_i/\sqrt{3} - 1$ across, where $\mathsf{D}^\times \ge \frac14\ell_g$ is the least distance on T_{ℓ_g} of the supports of two different mirror copies and $\hat{\mathsf{D}}^\times_i = \mathsf{D}^\times/a_{K-i}$, gives by the recursion (g), with $\sigma_{i;\gamma} \le E_\sigma(\hat{\mathsf{D}}^\times_i)$, bounds on $\bar{\mathsf{h}}_{j;\gamma}$ and $\bar\Psi_{j;\gamma}$ by products of the one-cell sizes; with Lem. 4.8(b) and the geometric sums this is the bound of the linking jets. The products of natural sizes over a partition do not exceed that of the whole set. The numbers Λ^{L}_m are finite and formed from the constants, two copies being at least $\frac14\ell_g \ge 6.5$ apart; no condition on g beyond (Γ) is needed. □

Proposition 4.28 (Transfer to every state)**.** Let g satisfy (Γ) and (K,β) be a first-passage datum at g.

(a) For $n \in \{3,4\}$ and the data of (A.4)(i) (for $n=4$ two pairs, all supports in one ball of radius $\le \frac18\ell_g - 1$), every $\ell \in \mathfrak{L}(g)$ and every state $\mathfrak{s}$ at (K,β): $|S^T_n[\mathfrak{s}](f) - S^T_{n;K,\ell_g,\beta}(f^{\langle\ell_g\rangle})| \le C^{\mathrm{tr}3}_n(g)\mathsf{b}^n\mathsf{d}^{3/2}\mathrm{nat}_n$, where $C^{\mathrm{tr}3}_n(g) = \eta(1+\eta)^{n-1}$, $\eta = (2^n-1)\mathsf{Q}_0(g)\eta_0$, $\eta_0 = (1+\mathsf{Z}_n + 8\mathsf{h})^n(\bar{\mathsf{M}}^{(48n)})^{1/8}$, $\mathsf{h} = \mathsf{y}(1+\mathsf{y})^{n-1}$, $\mathsf{y} = (1+\mathsf{Z}_n)^n\bar{\mathsf{M}}^{(6n)}$, $\bar{\mathsf{M}}^{(b)} = 1 + 2L^b\mathsf{C}_{\mathrm{ag}}(b+1,\frac12)$ and $\mathsf{Q}_0(g) = \exp[\frac18\ell_g^3(2C_{\mathrm{S}} + C_g)]$.

(b) For two bumps: let $0 < \rho \le \mathsf{d} \le 1$, $\rho \le \frac12$, $s(\rho) \le K$, $\rho \le r_1(\mathsf{b})$, f_1, f_2 bounded real Borel functions on $\mathbb{R}^3$ with $\operatorname{supp} f_a \subset \mathbb{B}(x_a,\rho)$, the two supports at a distance $\ge \mathsf{d}$, both in one ball of radius $\le \frac18\ell_g - 1$, $t_a\colon \{1,2,3\} \to [-1,1]$, $w_a(p) = t_a(\lambda(p))f_a(x(p))$, and $\bar w_a$ the same with $f_a^{\langle\ell_g\rangle}$ on T_{ℓ_g}. Then for every state $\mathfrak{s}$ at (K,β), $|\langle\mathcal{O}(w_1);\mathcal{O}(w_2)\rangle_\mathfrak{s} - \langle\mathcal{O}(\bar w_1);\mathcal{O}(\bar w_2)\rangle_{K,\ell_g,\beta}| \le C^{\mathrm{tr}}_2(g)(\rho/\mathsf{d})^6\mathsf{d}^{3/2}\|f_1\|_\infty\|f_2\|_\infty$, where $C^{\mathrm{tr}}_2(g) = \eta(1+\eta)$ is the number of (a) formed at $n=2$, with Z_2 of Lem. 4.27.

■

Proof. Every periodic Gibbs state is invariant under lattice translations by Prop. YM3_I:103.4; choose the offset of the board so that the tuple lies in the interior of one cell. Lem. 4.26 with $\ell' = \ell_g$, the representation of Lem. 4.20(b) at the $8n$ mirrored sources, $\mathsf{Z} = \mathsf{Z}_n\alpha^{-6}$ by Lem. 4.27, and the moments of Lem. 4.20(e): $\mathsf{y} \le (1+\mathsf{Z}_n)^n \bar{\mathsf{M}}^{(6n)}$ and $\eta_0 \le (1+\mathsf{Z}_n+8\mathsf{h})^n (\bar{\mathsf{M}}^{(48n)})^{1/8}$, so η is formed from the constants and g. With the weights of Lem. 4.27, $u^{[n]} = \mathsf{s}([n]) \prod |w_a|_\infty = \mathsf{b}^n \mathsf{d}^{3/2} \mathrm{nat}_n$. For (b) the same argument runs at $n = 2$, with Lem. 4.27 at $n = 2$, whose bounds read only the balls of radius ρ and the distance $\ge \mathsf{d}$ of the supports; a mirror map carries a_p to a_{rp} with $x(rp) = \mathsf{r}x(p)$ and $\lambda(rp) = \lambda(p)$, so plane weights go to plane weights, and $u_1 u_2 = (\rho/\mathsf{d})^6 \mathsf{d}^{3/2} |w_1|_\infty |w_2|_\infty$. □

Lemma 4.29 (Limits and components)**.**

(a) Every Gibbs sequence at g that converges has subsequences on which the labels converge (they lie in $(g^{-2}, \bar{\mathsf{b}}]$); along every such subsequence the members are states at their data, the limit σ is a law on $\mathbb{R}^3$ with $S_n^T[\nu_j] \to S_n^\sigma$ on separated tuples, and for every component σ_ξ of a canonical decomposition of σ its approximants ρ_j are periodic Gibbs states at the same data with $S_n^T[\rho_j] \to S_n^{\sigma_\xi}$.

(b) For a law with continuous moment kernels through order three, the connected kernel is $\mathsf{s}_3 = \mathsf{s}_3^{\mathrm{m}} - \mathsf{s}_2 \otimes \mathsf{s}_1 - \cdots + 2\mathsf{s}_1 \otimes \mathsf{s}_1 \otimes \mathsf{s}_1$, and $\mathsf{s}_2^T = \mathsf{s}_2 - \mathsf{s}_1 \otimes \mathsf{s}_1$. These kernels are continuous. For the periodic-Gibbs limits and components in (a), they are also real analytic and Euclidean invariant, by Prop. YM3_I:108.3.

(c) Let $\eta(t) = 1$ $(t \le 0)$, $1 - 3t^2 + 2t^3$ $(0 \le t \le 1)$, 0 $(t \ge 1)$, $\psi_\rho(v) = \eta(5|v|/\rho - 4)$ and $f_a^\rho = \psi_\rho(\cdot - z_a)/\int \psi_\rho$. These are C^1 probability densities supported in $\mathbb{B}(z_a, \rho)$ with $\rho^9 \prod_{a \le 3} \|f_a^\rho\|_\infty \le 0.102$; if s_3 is continuous at (z_1, z_2, z_3) and $|\int (\mathsf{s}_3 - 3\mathsf{b}^{-3} k_3) \prod f_a^\rho| \le \mathsf{A}$ for all small ρ, then $|\mathsf{s}_3 - 3\mathsf{b}^{-3} k_3|(z_1, z_2, z_3) \le \mathsf{A}$.

■

Proof. (a) follows from the component approximation in Thm. YM3_I:AP and the canonical decomposition of Thm. YM3_I:AL; a joint cumulant of order ≥ 2 does not change if constants are added to its entries, so the centring of the approximants does not enter. (b) Apply the finite moment–cumulant formula to the assumed continuous moment kernels. For the laws in (a), their analytic and invariance properties follow from Thm. YM3_I:AP and Prop. YM3_I:108.3. For a self-centred limit, $S_1^\sigma = 0$. (c) ψ_ρ is constant on $|v| \le \frac{4}{5}\rho$, and $\mathsf{s}_3 - 3\mathsf{b}^{-3} k_3$ is continuous at the point. □

Lemma 4.30 (One subsequence for all pairs of bumps)**.** Let Ω be an open subset of $\{(x, y) : 0 < |x - y| < \mathsf{d}_\Omega\}$ in T_ℓ^2 or in $(\mathbb{R}^3)^2$, and let bilinear forms $\mathsf{t}_{\lambda\mu}^{(j)}$, $\lambda, \mu \in \{1, 2, 3\}$, $j \in \mathbb{N}$, on bounded real Borel functions be such that for every $(x_0, y_0) \in \Omega$, with closed balls $\bar{U}, \bar{V}$ about x_0, y_0 of a radius $\varrho(x_0, y_0) > 0$ for which $\bar{U} \times \bar{V} \subset \Omega$, there is $\mathsf{B} < \infty$ with $|\mathsf{t}_{\lambda\mu}^{(j)}(f, h)| \le \mathsf{B} \|f\|_\infty \|h\|_\infty$ for all large j and all f, h supported in $\bar{U}, \bar{V}$. Then there is a subsequence along which $\mathsf{t}_{\lambda\mu}^{(j)}(\varphi_1, \varphi_2)$ converges for all λ, μ and all real Lipschitz φ_1, φ_2 with compact supports, $\operatorname{supp}\varphi_1 \times \operatorname{supp}\varphi_2 \subset \Omega$; the limits are bilinear. If on every such product of balls the limit forms satisfy the assumption of Lem. 4.23, they have Borel kernels on Ω, unique almost everywhere. ■

Proof. $\mathrm{Lip}_c(U)$ is a subset of the separable space $C_0(U)$, so it has a countable dense subset in the sup norm. Take a countable dense set of centres in Ω and a diagonal subsequence along which the countably many bounded sequences $\mathsf{t}_{\lambda\mu}^{(j)}(f, h)$, f, h in these dense sets, converge; by the uniform bound and bilinearity the sequences converge at all Lipschitz f, h in the balls. A pair φ_1, φ_2 as in the claim is decomposed by a Lipschitz partition of unity into cubes of a side small against the distances of $\operatorname{supp}\varphi_1 \times \operatorname{supp}\varphi_2$ to the diagonal and to the boundary of Ω: finitely many pairs, each in one product of balls. Two kernels that represent one form on an open product agree almost everywhere (mollification), and a numbering of the countable cover glues the local kernels into one Borel function. □

The bond forms.

Lemma 4.31 (Dilation to two couplings below the ceiling)**.** Let $0 < g \le \gamma$, let $z = z(g) \ge 1$ be the least integer with $L^{z-1}(g^{-2}+\bar B) - \bar B \ge x_{\mathrm N} = \gamma_{\mathrm N}^{-2}$ (Def. 4.1(1)), and define g_1', g_2' by $g_1'^{-2} = L^z(g^{-2}+\bar B) - \bar B$, $g_2'^{-2} = L^{-1}(g_1'^{-2} + 2B_\sharp) - B_\sharp$.

(a) $0 < g_1' < g_2' \le \gamma_{\mathrm N}$, and both depend on g and the constants alone.

(b) For a first-passage datum (K,β) at g with $K \ge z+1$ there is $\iota \in \{1,2\}$ such that $(K-z,\beta)$ is a first-passage datum at g_ι'; its run is $(x_0,\dots,x_{K-z})$, and its label is $L^z\mathsf b$.

(c) $L^z\ell \in \mathfrak L(g_\iota')$ for $\ell \in \mathfrak L(g)$; if $s(t) \ge z$, then $s_\iota'(L^z t) \le s(t) - z$ for the witness depth s_ι' at g_ι', and if $s(t) < z$, then $L^z t > 1$.

(d) $U \mapsto U^{(z)}$, $U^{(z)}(L^z b) = U(b)$, carries $\mu_{K,\ell,\beta}$ to $\mu_{K-z,L^z\ell,\beta}$, with $a_{L^z p}(U^{(z)}) = a_p(U)$ and $q_{L^z b}(U^{(z)}) = q_b(U)$. So $\mathcal O^T(\mathfrak s)(U) = L^z\mathcal O'^T(\mathfrak s^{(z)})(U^{(z)})$ and $\langle \mathcal O^T(\mathfrak s_1); \mathcal O^T(\mathfrak s_2)\rangle_{K,\ell,\beta} = L^{2z}\langle \mathcal O'^T(\mathfrak s_1^{(z)}); \mathcal O'^T(\mathfrak s_2^{(z)})\rangle_{K-z,L^z\ell,\beta}$, primes marking the objects of the torus $T_{L^z\ell}$ at the cut-off $K-z$ and $\mathfrak s^{(z)}(L^z\cdot) = \mathfrak s(\cdot)$; and the numbers $\|\varphi\|_\infty$, $r\,\mathrm{Lip}\,\varphi$, a_K/r, $\mathsf d'/\ell$, $r/\mathsf d'$ and the integrals $\iint \varphi_1\varphi_2 \mathbb K_3(x-y)_{\lambda\gamma}\mathbb K_3(x-y)_{\lambda'\gamma'}$ are the same for $(\varphi_a; r, \mathsf d', \ell, K)$ and for $(\varphi_a^{(z)}; L^z r, L^z\mathsf d', L^z\ell, K-z)$.

Hence each display of (D.1) at (K,ℓ,β) is the same display at $(K-z, L^z\ell, \beta)$, with the same tolerance, both members being multiplied by L^{-2z}. The identity $\mathcal O^T(\mathfrak s)(U) = L^z\mathcal O'^T(\mathfrak s^{(z)})(U^{(z)})$ holds at every configuration U of $\mathbb L_K$ as well, the primes now marking the fields of $\mathbb L_{K-z}$; in particular $\mathcal O(f)(U) = L^z\mathcal O'(f^{(z)})(U^{(z)})$ with $f^{(z)} = f(L^{-z}\cdot)$. So for every law ν on the configurations of $\mathbb L_K$ and its image $\nu^{(z)}$ under the dilation, $S_n^T[\nu](f_1,\dots,f_n) = L^{nz}S_n^T[\nu^{(z)}](f_1^{(z)},\dots,f_n^{(z)})$ for bounded real functions f_a with compact supports, and the joint cumulants of the fields $\mathcal O^T(\mathfrak s_a)$ of mixed sources scale in the same way; and $\nu \mapsto \nu^{(z)}$ is a bijection of $\mathcal G^{\mathrm{per}}_{K,\beta}$ onto $\mathcal G^{\mathrm{per}}_{K-z,\beta}$.

These statements of (d) hold with every integer $0 \le z' < K$ in the place of z.

■

Proof. (a), (b) are arithmetic of the inverse couplings. Put $G = g^{-2}+\bar B$ and $X_{g'} = L(g'^{-2}+B_\sharp)$ for a coupling g'. One step of the run is $x_{j-1} = L(x_j + b_j)$ with $|b_j| \le B_\sharp$ (the flow bound in [7, §G.2]); as $\bar B = B_\sharp L/(L-1)$ sums the geometric series, z steps give $L^z(x_K - \bar B) + \bar B \le x_{K-z} \le L^z(x_K + \bar B) - \bar B$, and with $g^{-2} < x_K \le X_g$ and $X_g + \bar B = LG$:

$$L^z(g^{-2} - \bar B) + \bar B < x_{K-z} \le L^{z+1}G - \bar B.$$

(a) $g_2'^{-2} = L^{z-1}G - \bar B + 2B_\sharp/L \ge L^{z-1}G - \bar B \ge x_{\mathrm N}$ by the choice of z, and $g_1'^{-2} - g_2'^{-2} = (L-1)L^{z-1}G - 2B_\sharp/L > 0$; both read g and the constants alone. (b) $X_{g_1'} = L^{z+1}G - \bar B$ is the upper end above, and $X_{g_2'} = g_1'^{-2} + 2B_\sharp$; so the half-open intervals $I_\iota = (g_\iota'^{-2} + 2B_\sharp, X_{g_\iota'}]$, $\iota = 1,2$, abut, are disjoint, and $I_2 \cup I_1 = (g_2'^{-2} + 2B_\sharp, X_{g_1'}]$. For the lower end, $g_2'^{-2} + 2B_\sharp \le L^{z-1}G + 2\bar B$ and

$$[L^z(g^{-2}-\bar B) + \bar B] - [L^{z-1}G + 2\bar B] = L^{z-1}[(L-1)g^{-2} - (L+1)\bar B] - \bar B \ge (L^{z-1}-1)\bar B \ge 0,$$

since $\bar B \le \frac{13}{24}g^{-2}$ at $g \le \gamma$ ([7, Lem. G.3] gives this bound) gives $(L-1)g^{-2} \ge \frac{24}{13}(L-1)\bar B \ge (L+2)\bar B$, as $11L \ge 50$. So x_{K-z} lies in exactly one I_ι. There $x_{K-z} \le X_{g_\iota'}$ and $x_{K-z-1} = L(x_{K-z} + b_{K-z}) \ge L(x_{K-z} - B_\sharp) > L(g_\iota'^{-2} + B_\sharp) = X_{g_\iota'}$, and the run decreases: $K-z$ is the first index at which the run that starts at β is at most $X_{g_\iota'}$, which is the first passage at the coupling $g_\iota' \le \gamma_{\mathrm N} \le \gamma$. The run of $(K-z,\beta)$ is the same sequence cut at $K-z$, and its label is $a_{K-z}\beta = L^z\mathsf b$.

The margin $2B_\sharp$ in the lower ends of the I_ι is what gives $x_{K-z-1} > X_{g_\iota'}$, the second half of the first passage. (c) Put $G = g^{-2} + \bar B$. Then $g_1'^{-2} + \bar B = L^z G$ and $g_2'^{-2} + \bar B \le L^z G$. The radius-scaling inequality YM3_I:(1534), applied to $X = LG$ and $X' = L(g_\iota'^{-2} + \bar B)$, gives $R(X') \le L^z R(X)$. The admissibility formula YM3_I:(1528) therefore holds with $m + z$ at g_ι' whenever it holds with m

at g; its fixed C_{s} threshold is also preserved. Moreover $C_W^{g'}(s-z) \le C_W^g(s)$ for $s \ge z$, directly from the displayed definition of C_W. Thus a witness at depth $s(t)$ transfers to depth $s(t)-z$. If $s(t) < z$, then $L^{s(t)}t \ge C_W^g(s(t)) \ge 1$, whence $L^z t > 1$. (d) is the dilation; $\mathsf{b}' = L^z\mathsf{b}$, so $\mathsf{b}^{-2} = L^{2z}\mathsf{b}'^{-2}$, and $\mathbb{K}_3(L^z u) = L^{-3z}\mathbb{K}_3(u)$ makes the integrals invariant. The identity of the fields reads of the configuration only $a_{L^z p}(U^{(z)}) = a_p(U)$ and $q_{L^z b}(U^{(z)}) = q_b(U)$, and of the lattice only $x(L^z p) = L^z x(p)$, $x_{L^z b} = L^z x_b$ and $a_{K-z} = L^z a_K$; so it holds at every configuration of $\mathbb{L}_K$, and the joint cumulants under a law are multilinear in the fields.

The dilation maps the bonds, the plaquettes and the finite sets of bonds of $\mathbb{L}_K$ onto those of $\mathbb{L}_{K-z}$. The Gibbs kernel of a finite set of bonds has the exponent $-\beta\sum_p a_p$ over the plaquettes that contain a bond of the set, which reads β and the incidences of the bonds and the plaquettes alone; so the image of a Gibbs state at (a_K, β) satisfies the equations of the kernels at (a_{K-z}, β), and invariance under a subgroup of finite index of $a_K\mathbb{Z}^3$ becomes invariance under its image under $x \mapsto L^z x$, a subgroup of finite index of $a_{K-z}\mathbb{Z}^3$. The inverse dilation carries the periodic Gibbs states at $(K-z, \beta)$ back in the same way, so $\nu \mapsto \nu^{(z)}$ is a bijection of $\mathcal{G}^{\mathrm{per}}_{K,\beta}$ onto $\mathcal{G}^{\mathrm{per}}_{K-z,\beta}$ (equation YM3_I: (1819) and the definition of periodic states following it).

The same arguments hold at every integer $0 \le z' < K$. □

Lemma 4.32 (Bond sources)**.** Let $0 < g \le \gamma_{\mathrm{N}}$, (K, β) a first-passage datum at g with $K \ge 2$, $\ell \in \mathfrak{L}(g)$, and let $\mathfrak{s} = (w, \mathsf{w})$ be a real mixed source whose barycentres $x(p)$, $w(p) \ne 0$, and midpoints x_b, $\mathsf{w}(b) \ne 0$, lie in a ball of a radius $\rho \ge a_K$. Put $\mathsf{m}_j = 12M^3\alpha_0^2 L^{K-j}|\mathfrak{s}|_\infty$ and $\mathsf{u} = \mathsf{b}^{-1}|\mathfrak{s}|_\infty$, let ς_j be as in Lem. 4.6(e) at the width ρ, and let

$$\mathsf{W}_{\mathfrak{s}}(p,q) = w(p)\delta_{pq} - \sum_b \mathsf{w}(b)\big[u_{b,\lambda_b}(p)u_{b,\lambda'_b}(q) + u_{b,\lambda'_b}(p)u_{b,\lambda_b}(q)\big],$$

a symmetric matrix that joins two plaquettes only through a bond of the support, with the coefficients $u_{b,\lambda}(p)$ of the paragraph on the fields. There are numbers C formed from the constants such that:

(1) the weight member of the source $L^{K-j}\mathfrak{s}$ at the level j is linear in $\mathfrak{s}$, vanishes on the domains that hold no corner of a plaquette p with $w(p) \ne 0$ and no site of a bond b with $\mathsf{w}(b) \ne 0$, and its norm in $\|\cdot\|_{j,\kappa}$ is at most $C\varsigma_j\mathsf{m}_j$;

(2) at the first level, $\|E^{(1)}_{\mathfrak{s}}\|_{1,\lambda_{\mathrm{out}}} \le C\theta^{[0]}\mathsf{m}_0$;

(3) for $1 \le n \le K$, $A \in \mathsf{Cc}_n(r_n)$, $\tilde{U} = e^{i\eta_n A}$ and $F = \mathsf{ang}^{[n]}(A)$:

$$\big|\mathcal{O}^T(\mathfrak{s})(\tilde{U}) - \frac{1}{2a_K}\sum_{p,q}\mathsf{W}_{\mathfrak{s}}(p,q)\langle F_p, F_q\rangle\big| \le C\mathfrak{b}_n^2\delta_n\hat{\rho}_n^3\mathsf{u},$$

and $\eta_n^4\sum_{p,q}|\mathsf{W}_{\mathfrak{s}}(p,q)|/(a_K x_n) \le C\hat{\rho}_n^3\mathsf{u}$;

(4) Lem. 4.14(c) and Prop. 4.16 hold with the matrices $\mathsf{W}_{\mathfrak{s}_a}$ in the place of the diagonal matrices w_a in $R_{i;\beta}$, $q_{i;\beta}$ and $g_{i;\beta}$, for two mixed sources in the frame of (6), the tower identity of (6) standing in Step 1 of the proof of Prop. 4.16;

(5) the weighted progression and the representation of the torus expectations through the never-large histories, which name no source, are formed for mixed sources at $g \le \gamma_{\mathrm{N}}$ on every side $\ell \in \mathfrak{L}(g)$, with the weight member of (1) in the source slot of the bare arrays; the lemma on the tower identity at several sources (it precedes Prop. 4.16), for clusters in the frame of (6), the representation of Lem. 4.20 and the almost everywhere bound of Lem. 4.21 on the remainders hold for mixed sources, the moments of α^{-1} being those of Lem. 4.20(e) on that side;

(6) *the frame and the tower identity.* The hull of a mixed source $\mathfrak{s} = (w, \mathsf{w})$ is the set of the plaquettes p with $w(p) \ne 0$ and of the four plaquettes through each bond b with $\mathsf{w}(b) \ne 0$. For $2 \le k < K$, two mixed sources, neither 0, are in the frame at k, that is they satisfy the four conditions (Jt.1)–(Jt.3) and (Jt.6) of the frame at k of Lem. 4.5(a), read with the hulls in the place of the supports and with the members of (1) in the slot, if each hull has its

barycentres in a closed ball of a radius $\bar{\mathsf{d}} \geq a_K$ with $\bar{\mathsf{d}}/a_{K-k} \geq M$ and the corners of the plaquettes of the two hulls are at least $3M$ level-k spacings apart in the sup distance; this holds at $k = \mathsf{k}(\mathsf{d})$ for hulls in balls of radius $\leq \mathsf{d}$ at a distance $\geq \mathsf{d}$, $0 < \mathsf{d} \leq e^{-2\sqrt{3}}$, $s(\mathsf{d}) \leq K$, as in Lem. 4.5(b). In the frame, for $k \leq i < K$, $A' \in \mathsf{Cc}_{i+1}(r_{i+1})$, $\tilde{U} = e^{i\eta_{i+1}A'}$ and λ near 0,

$$\exp D^\lambda_{i+1}(\tilde{U}) = \int \exp D^\lambda_i(e^{i\eta_i A^{\mathrm{lw}}(B)})\, \mathrm{d}\nu_{i,A'}(B),$$

with the measure $\nu_{i,A'}$ of the step and the lowering map; and the one-source Lem. 4.10 holds for each source $\mathfrak{s}$ of such a pair, with $\frac{1}{2a_K}\sum_{p,q} \mathsf{W}_\mathfrak{s}(p,q)\langle F_p, F_q\rangle$ in the place of $q_{n;a}(F)$, its frame read with the hulls and its count (P1) replaced by the second bound of (3).

■

Proof. (1) $\mathcal{O}^T(\mathfrak{s}) = a_K^{-1}\sum_p w(p)a_p - a_K^{-1}\sum_b \mathsf{w}(b)q_b$, and q_b is a gauge invariant function of the variables of the four plaquettes through b; so the source is a local term of the bare level, linear in $\mathfrak{s}$, which lives at the sites of its support. On the cores the plaquette variables are within $C\alpha_0\eta_j^2$ of $\mathbf{1}$, so that $|q_b| \leq |u_{b,\lambda_b}||u_{b,\lambda'_b}| \leq C\alpha_0^2\eta_j^4$ there, as $|a_p| < \frac{1}{2}\alpha_0^2\eta_j^4$; at most $81(\rho/a_K)^3$ bonds have their midpoints in a ball of the radius $\rho \geq a_K$; with $|\mathfrak{s}|_\infty = |w|_\infty + 8|\mathsf{w}|_\infty$ the count of Lem. 4.6(e) gives the bound. (2) The one-step theorem at the height 0, with this member in the fourth place, and Schwarz's lemma, as in Lem. 4.6(e). (3) On the chain class the angle of a plaquette variable of $\tilde{U}$ is $\pm F_p$ up to $C\eta_n^2 r_n^2$, and $\mathrm{w}(e^{i\theta}) = \theta + O(|\theta|^3)$; so $u_{b,\lambda} = \sum_p u_{b,\lambda}(p)F_p$ up to $C\eta_n^2 r_n^2$, and, as $|F_p| \leq 2\eta_n^2 r_n$, $|q_b - \langle\sum_p u_{b,\lambda_b}(p)F_p, \sum_q u_{b,\lambda'_b}(q)F_q\rangle| \leq C\eta_n^4 r_n^3$, the bound that the expansion of a_p on the chain class has.

Summed against w, this gives the bond part of $\frac{1}{2a_K}\langle F, \mathsf{W}_\mathfrak{s}F\rangle$ with the error $Ca_K^{-1}(\rho/a_K)^3|\mathsf{w}|_\infty\eta_n^4 r_n^3$; with the plaquette part as in the proof of Lem. 4.10, $a_{K-n}x_n \geq \frac{4}{5}\mathfrak{b}$ and $x_n r_n^2 \leq C\mathfrak{b}_n^2$, this is the first bound. Each $u_{b,\lambda}$ has two coefficients of modulus $\frac{1}{2}$, so $\sum_{p,q}|\mathsf{W}_\mathfrak{s}(p,q)| \leq \sum_p |w(p)| + 2\sum_b |\mathsf{w}(b)| \leq 81(\rho/a_K)^3|\mathfrak{s}|_\infty$, and $\eta_n^4 a_K^{-4} = a_{K-n}^{-4}$ gives the second. (4) In the proof of Lem. 4.14(c) the sources enter only through $R_i^\lambda = \mathsf{W}_\lambda(1 - \mathcal{S}_i\mathsf{W}_\lambda)^{-1}$, $\mathsf{W}_\lambda = \sum_a \lambda_a \mathsf{W}_{\mathfrak{s}_a}$, which is symmetric as W_λ and $\mathcal{S}_i$ are, and through the Gaussian integral of a quadratic form with a symmetric matrix; the identity $1 - \mathsf{s}R_i^\lambda = (1 - \mathcal{S}_{i+1}\mathsf{W}_\lambda)(1 - \mathcal{S}_i\mathsf{W}_\lambda)^{-1}$ does not use that W_λ is diagonal.

The proof of Prop. 4.16 uses the sources only through the tower identity of its Step 1, which (6) gives for two mixed sources in the frame, through (I1)–(I3), through the sums $\sum_{p,q}|R_{i;\beta}(p,q)|$, which the second bound of (3) controls as (P1) does for plaquette weights, and through the lowered angles at the plaquettes of the hulls. (5) Props. [7, Prop. B.8] and [7, Prop. B.15], together with eq. 4.xxxi, give the weighted progression and the history representation for sources on the bare plaquettes, on every side of $\mathfrak{L}(g)$ with $g^{-2} \geq x_\mathrm{J}$ (Thm. YM3_I:D). For mixed sources the same construction is made with the weight member of (1) in the source slot of the bare arrays; the other data of the progression name no source.

The proofs among the construction results read a source only through that member, and of the member they read these properties: the entry of the objects into the settings of one step, for any number of sources, $K \geq 2$ and every side, reads that it is a local term of the bare level with the bound of (1); the never-large histories and the measures of the mixture are formed with no source; the representation reads that $\mathcal{O}^T(\mathfrak{s})$ is a bounded gauge invariant function, linear in $\mathfrak{s}$ and 0 at $\mathbf{1}$, whose member is that of (1), so that the quotient at a level t of $e^{\mathcal{O}^T(\mathfrak{s}_\lambda)}$ is $\exp D_t^\lambda$; and the tower identity reads the member in the slot of the step map, which takes any member there with the bound of (1).

By (1) a mixed source has these properties. The proofs of the lemma on the tower identity at several sources, which precedes Prop. 4.16, and of Lem. 4.20 use the sources only through the progression and the tower identity for a pair, which (6) gives for mixed sources; the almost everywhere bound of Lem. 4.21 uses them only through (1), (2), the support rule of Lem. 4.6(d)

and the sizes of Lem. 4.9(f), which hold for mixed sources by (1)–(3) with the hulls in the place of the supports; on a side ℓ the moments of α^{-1} are bounded by Lem. 4.20(e). (6) The frame at k is formed by the four conditions (Jt.1)–(Jt.3) and (Jt.6) of the setting of the pair functional among the construction results, read with the hulls in the place of the supports: the proof of Lem. 4.5(a), using Def. 4.4, reads of the two sources only the four geometric properties listed there, now of the hulls, which hold all the plaquettes that the members of the sources read, and the members in the slot.

The tower identity for a pair is that among the construction results, read with the member of (1) in the slot: its proof forms the measure $\nu_{i,A'}$ and the lowering map with no source and reads the sources only through the relative exponents. Hulls at a distance $\geq \mathsf{d}$ have corners at least $(\mathsf{d}-\sqrt{2}a_K)/\sqrt{3}$ apart in the sup distance, which is at least $3M$ level-$\mathsf{k}(\mathsf{d})$ spacings by the working level, as in the proof of Lem. 4.5(b); a ball of radius d has $\mathsf{d}/a_{K-\mathsf{k}(\mathsf{d})} \geq 4M$. In Lem. 4.10 the member of $\mathfrak{s}$ is expanded on the chain class by the first bound of (3), and the families by the count of its proof with the second bound of (3). □

Lemma 4.33 (Lattice sums at two balls on every torus)**.** On every torus T_ℓ with the lattice $a_K\mathbb{Z}^3$, with the least lifts of the differences, let $4a_K \leq r$, $16r \leq \mathsf{d}'$, $8\mathsf{d}' \leq \ell$, $x_1, x_2 \in T_\ell$ with $|x_1 - x_2| = \mathsf{d}'$, $\vartheta \in (0,1]$, and let φ_1, φ_2 be real Lipschitz functions with $\operatorname{supp}\varphi_a \subset \mathbb{B}(x_a, r)$ and $\operatorname{Lip}\varphi_a \leq \|\varphi_a\|_\infty/\vartheta r$. Put $\mathsf{e}_{\mathrm{PS}} = [a_K/\vartheta r + (\mathsf{d}'/\ell)^3](r/\mathsf{d}')^6\|\varphi_1\|_\infty\|\varphi_2\|_\infty$.

(a) For all λ, μ: $\big|\sum_{\lambda(p)=\lambda,\,\lambda(q)=\mu} \varphi_1(x(p))\varphi_2(x(q))\, C_K^{\mathrm{free}}(p,q)^2 - \mathcal{K}_{\lambda\mu}(\varphi_1,\varphi_2)\big| \leq C_{\mathrm{PS}}\,\mathsf{e}_{\mathrm{PS}}$.

(b) Let $\{\lambda, \lambda', \mu\} = \{1,2,3\}$, γ a direction, $\mathsf{w}(b) = \varphi_1(x_b)\mathbf{1}_{\{\mu_b=\mu\}}$, $w_2(q) = \varphi_2(x(q))\mathbf{1}_{\{\lambda(q)=\gamma\}}$, and $C_\alpha(b,q) = \sum_p u_{b,\alpha}(p) C_K^{\mathrm{free}}(p,q)$ for a direction $\alpha \neq \mu$. Then

$$\Big|-\sum_{b,q} \mathsf{w}(b) w_2(q)\, C_\lambda(b,q) C_{\lambda'}(b,q) - \iint \varphi_1(x)\varphi_2(y)\, \mathbb{K}_3(x-y)_{\lambda\gamma}\mathbb{K}_3(x-y)_{\lambda'\gamma}\,\mathrm{d}x\,\mathrm{d}y\Big| \leq C_{\mathrm{PS}}^{\mathrm{od}}\,\mathsf{e}_{\mathrm{PS}}.$$

■

Proof. Points of the two balls are at least $\mathsf{d}' - 2r \geq \frac{7}{8}\mathsf{d}'$ and at most $\mathsf{d}' + 2r \leq \frac{9}{8}\mathsf{d}' < \frac{1}{4}\ell$ apart, so all the least lifts are taken in one ball of radius $\frac{1}{4}\ell$. (a) Steps A and B of the proof of Lem. 4.22 at these distances. In a square every sign ϵ_p stands twice, so equations YM3_I:(2245)–(2246) replace $C_K^{\mathrm{free}}(p,q)^2$ by $a_K^6[\mathbb{K}_3(u)_{\lambda\mu}]^2$ at the cost $C_{\mathrm{lat}}(2+3C_{\mathrm{lat}})a_K^6|u|^{-3}[a_K^2|u|^{-5} + \ell^{-3}]$ for each pair, and a ball of radius $r \geq a_K$ holds at most $27(r/a_K)^3$ barycentres of one orientation; with $|u| \geq \frac{7}{8}\mathsf{d}'$ and $a_K r \leq \frac{1}{4}r^2 \leq \frac{1}{1024}\mathsf{d}'^2$ this costs at most $729(\frac{8}{7})^3 C_{\mathrm{lat}}(2+3C_{\mathrm{lat}})\,\mathsf{e}_{\mathrm{PS}} \leq 1089\, C_{\mathrm{lat}}(2+3C_{\mathrm{lat}})\,\mathsf{e}_{\mathrm{PS}}$.

The Riemann sums over the cubes of side a_K about the barycentres, which for one orientation form a translate of $a_K\mathbb{Z}^3$, with $\|\mathbb{K}_3(u)\| \leq \frac{1}{2\pi}|u|^{-3}$, $\|\partial\mathbb{K}_3(u)\| \leq \frac{6}{4\pi}|u|^{-4}$ (Lem. 4.24(b)) and the Lipschitz bounds of the φ_a, give the rest of $C_{\mathrm{PS}}\mathsf{e}_{\mathrm{PS}}$. (b) $u_{b,\alpha}(p) \neq 0$ only at the two plaquettes $p_\pm$ with the barycentres $x_b \pm \frac{1}{2}a_K e_\alpha$, which lie in the plane of μ and α, and there $u_{b,\alpha}(p_\pm) = \frac{1}{2}\epsilon_{\alpha\mu\alpha'}\epsilon_{p_\pm}$, $\{\alpha,\mu,\alpha'\} = \{1,2,3\}$. By equation YM3_I:(2245), $C_\lambda(b,q)$ is $\epsilon_{\lambda\mu\lambda'}\epsilon_q a_K^3$ times the mean of $\mathbb{K}_3(\cdot)_{\lambda'\gamma}$ at the two differences $x(p_\pm) - x(q)$, and $C_{\lambda'}(b,q)$ is $\epsilon_{\lambda'\mu\lambda}\epsilon_q a_K^3$ times the like mean of $\mathbb{K}_3(\cdot)_{\lambda\gamma}$, up to the terms of the comparison.

As $\epsilon_{\lambda\mu\lambda'}\epsilon_{\lambda'\mu\lambda} = -1$ and $\epsilon_q^2 = 1$, $-C_\lambda(b,q)C_{\lambda'}(b,q)$ is $a_K^6\,\mathbb{K}_3(x_b - x(q))_{\lambda\gamma}\mathbb{K}_3(x_b - x(q))_{\lambda'\gamma}$ up to these terms and to the shifts by $\pm\frac{1}{2}a_K e_\lambda$, $\pm\frac{1}{2}a_K e_{\lambda'}$, which cost $a_K\|\partial\mathbb{K}_3\|$. The midpoints of the bonds of one direction form a translate of $a_K\mathbb{Z}^3$, as the barycentres of the plaquettes of one orientation do, and steps A and B as in (a) give $C_{\mathrm{PS}}^{\mathrm{od}}\mathsf{e}_{\mathrm{PS}}$. The numbers C_{PS} and $C_{\mathrm{PS}}^{\mathrm{od}}$ of Def. 4.1(3) are these counts. □

Proposition 4.34 (Two mixed sources on every side)**.** For $0 < g \leq \gamma_{\mathrm{N}}$ let $t_{\mathrm{N}}(g) \in (0, \frac{1}{2}e^{-2\sqrt{3}}]$ be the threshold, formed from the constants and g, below which the conditions of the induction at two sources with the base level equal to the start level hold at g and at every label in $(g^{-2}, \bar{\mathsf{b}}]$ (the conditions of step 5 of the proof of Prop. 4.19 that its step 6 reads, with the members that $g \leq \gamma_{\mathrm{N}}$ gives), and put $r_g = \frac{1}{2}\min\{r_1(\bar{\mathsf{b}}), t_{\mathrm{N}}(g)\}$. For $\ell \in \mathfrak{L}(g)$ there is a finite function $\epsilon_\sharp^{\mathrm{ev}}(g,\ell,\cdot)$ on $(0, \frac{1}{8}\ell]$, formed from the

constants, g and ℓ (it is written out at the end of the proof), with $\epsilon_\sharp^{\mathrm{ev}}(g,\ell,\mathsf{d}') \le C(g,\ell)(\log\frac{1}{\mathsf{d}'})^{19\mathfrak{p}_0}\mathsf{d}'^{1/2}$ for $\mathsf{d}' \le \frac{7}{6}t_{\mathrm{N}}(g)$, such that the following holds. Let (K,β) be a first-passage datum at g, $\mathsf{b} = a_K\beta$, $x_1, x_2 \in T_\ell$ with $|x_1 - x_2| = \mathsf{d}'$, $0 < 16r \le \mathsf{d}' \le \frac{1}{8}\ell$, $s(\min\{r, r_g\}) \le K$. For real mixed sources $\mathfrak{s}_a = (w_a, \mathsf{w}_a)$ with $\{x(p) : w_a(p) \neq 0\} \cup \{x_b : \mathsf{w}_a(b) \neq 0\} \subset \mathbb{B}(x_a, r)$,

$$\Big|\langle \mathcal{O}^T(\mathfrak{s}_1); \mathcal{O}^T(\mathfrak{s}_2)\rangle_{K,\ell,\beta} - \tfrac{3}{2}\mathsf{b}^{-2}\,\mathrm{Tr}(\mathsf{W}_{\mathfrak{s}_1} C_K^{\mathrm{free}} \mathsf{W}_{\mathfrak{s}_2} C_K^{\mathrm{free}})\Big| \le \epsilon_\sharp^{\mathrm{ev}}(g,\ell,\mathsf{d}')\Big(\frac{r}{\mathsf{d}'}\Big)^6 \mathsf{b}^{-2}|\mathfrak{s}_1|_\infty|\mathfrak{s}_2|_\infty,$$

where $\mathsf{W}_{\mathfrak{s}}(p,q) = w(p)\delta_{pq} - \sum_b \mathsf{w}(b)[u_{b,\lambda_b}(p)u_{b,\lambda_b'}(q) + u_{b,\lambda_b'}(p)u_{b,\lambda_b}(q)]$ and $u_{b,\alpha}(p)$ are the coefficients of the plaquette angles in the linear part of $u_{b,\alpha}$, given in the paragraph on the fields (it is the matrix of the lemma on bond sources, which follows Lem. 4.31). Consequently both displays of (D.1) hold at $g \le \gamma_{\mathrm{N}}$, with $\epsilon_\sharp^{\mathrm{ev}}$ in place of ϵ^{cl}, for every K with $s(\min\{r, r_g\}) \le K$. ∎

Proof. A bond weight enters the progression through its weight member, a local term on the plaquettes of the two planes that contain the bond. Put $\rho = \min\{r, r_g\}$; the working level at ρ ($\rho \le \frac{1}{4}e^{-2\sqrt{3}}$, $s(\rho) \le K$) gives $a_K \le \rho/4M \le \frac{1}{52}\rho$. The hull of $\mathfrak{s}_a$ (clause (6) of the lemma on bond sources, which follows Lem. 4.31) has its barycentres in $\mathbb{B}(x_a, r + \frac{1}{2}a_K)$, so that for p, q in the two hulls

$$\tfrac{6}{7}\mathsf{d}' \le |x(p) - x(q)| \le \tfrac{8}{7}\mathsf{d}' =: \mathsf{d}^+,$$

all these points lying in a ball of radius $< \frac{1}{4}\ell$, where the least lifts are taken. Put $\mathsf{d} = \min\{\frac{6}{7}\mathsf{d}', t_{\mathrm{N}}(g)\}$, the floor of the distances that is read.

The case $r < r_g$. The width $r_T = r + \frac{1}{2}a_K \le \frac{105}{104}r$ has $r_T \le \mathsf{d}$, $s(r_T) \le K$, and $r_T \le \frac{1}{2}\cdot 1.01\, r_1(\bar{\mathsf{b}}) \le r_1(\mathsf{b})$, as r_1 does not increase and $\mathsf{b} \le \bar{\mathsf{b}}$; so the two sources are in the frame at $\mathsf{k}(\mathsf{d})$ (clause (6) of the lemma on bond sources). By the lemma on bond sources the calculus of Lem. 4.6, the sizes of Lem. 4.9(f), the one-source Lem. 4.10 and the induction at two sources (Prop. 4.16, base level equal to start level $\mathsf{k}(\mathsf{d})$, as in step 6 of the proof of Prop. 4.19) apply to the mixed sources, with the explicit form $\frac{1}{2a_K}\langle F, \mathsf{W}_{\mathfrak{s}} F\rangle$ of a source, the orientation signs entering through $u_{b,\lambda}(p)$. Of (Γ1) these use only the members that $g \le \gamma_{\mathrm{N}}$ gives: the member $\delta(g^{-2}) \le 1/2C_{\mathrm{cu}}$ is used only for the kernel families (Prop. 4.13), which do not occur at two sources whose distance fixes the base level. The difference of the two members of the display is at most the sum of three members.

(i) *The remainder.* $S_2^T - \mathsf{J}_2 = \int \mathsf{Y}_{12}\,\mathrm{d}\mathbb{P} + \kappa_{\mathbb{P}}(\mathsf{Y}_1, \mathsf{Y}_2)$ (Lem. 4.20(c) for mixed sources, by clause (5) of the lemma on bond sources). The two sources form one pair at the widths (r_T, d) in the sense of Lem. 4.9(f), so the proof of Lem. 4.21, with the sizes of clauses (1)–(3) of the lemma on bond sources, gives at $\mathbb{P}$-almost every ω the bounds $|\mathsf{Y}_a| \le \Lambda_T\alpha^{-6}r_T^3|\mathfrak{s}_a|_\infty$ and $|\mathsf{Y}_{12}| \le \Lambda_T\alpha^{-6}r_T^6\mathsf{d}^{-9/2}|\mathfrak{s}_1|_\infty|\mathfrak{s}_2|_\infty$, with a number $\Lambda_T \ge 1$ formed from the constants. With the moments of α^{-1} on the side ℓ, $\int \alpha^{-24}d\mathbb{P} \le \bar{\mathsf{M}}_\ell := 1 + 4L^{24}M^{-3}\ell^3 e^{\mathsf{B}(X_g)\ell^3}\mathsf{C}_{\mathrm{ag}}(25, \eth)$ (Lem. 4.20(e)), which grows with ℓ, this member is at most $3\Lambda_T^2\bar{\mathsf{M}}_\ell^2\bar{\mathsf{b}}^2\mathsf{d}^{3/2}\,(r_T/\mathsf{d})^6\mathsf{b}^{-2}|\mathfrak{s}_1|_\infty|\mathfrak{s}_2|_\infty$.

(ii) *The induction.* The Gaussian step of Lem. 4.14(c) with $\mathsf{W}_{\mathfrak{s}}$ in place of w gives at the top the value $g_{K;12} = \frac{3}{2}\,\mathrm{Tr}(\mathsf{W}_{\mathfrak{s}_1}\mathcal{S}_K\mathsf{W}_{\mathfrak{s}_2}\mathcal{S}_K)$, $\mathcal{S}_K = \sum_{\mathsf{k}(\mathsf{d})<j\le K} \upsilon_j\mathcal{C}_{(j)}$, Wick's rule for two quadratic forms of the photon field. Prop. 4.16 and Lem. 4.9(d), as in step 6 of the proof of Prop. 4.19, bound $|\mathsf{J}_2 - g_{K;12}|$ by $\mathsf{E}_{\mathrm{J}}(g,\mathsf{d})(r_T/\mathsf{d})^6\mathsf{b}^{-2}|\mathfrak{s}_1|_\infty|\mathfrak{s}_2|_\infty$, with a function E_{J} of (g,d) alone that is at most $C(g)(\log\frac{1}{\mathsf{d}})^{19\mathfrak{p}_0}\mathsf{d}^{1/2}$ for $\mathsf{d} \le t_{\mathrm{N}}(g)$.

(iii) *The free kernel.* For p, q in the two hulls, $\mathsf{r} = |x(p) - x(q)|_\infty \in [\mathsf{d}/\sqrt{3}, \mathsf{d}^+]$ and $t = \mathsf{r}/a_{K-\mathsf{k}(\mathsf{d})} \ge \hat{\mathsf{d}}_{\mathsf{k}(\mathsf{d})}/\sqrt{3}$, the floor of the working level, at which the condition of Lem. 4.17(b) holds by the conditions that fix $t_{\mathrm{N}}(g)$ (as condition (ii) of step 5 of the proof of Prop. 4.19). So Lem. 4.17(b) at $k = \mathsf{k}(\mathsf{d})$, multiplied by r^3, gives $|\mathcal{S}_K - \mathsf{b}^{-1}C_K^{\mathrm{free}}|(p,q) \le \mathsf{b}^{-1}a_K^3\mathsf{r}^{-3}\,\mathsf{e}_{\mathrm{fr}}(g;\mathsf{d},\mathsf{d}^+)$, with $\mathsf{b}^{-1} < g^2$ and

$$\mathsf{e}_{\mathrm{fr}}(g;\mathsf{d},\mathsf{d}^+) = \tfrac{5}{4}C_{\mathrm{tp}}c_L^{(2)}\bar{B}g^2\,\mathsf{d}^+ + C_{\mathrm{tp}}^{\mathrm{bg}}(\mathsf{d}^+)^3 + 2L^3C_{\mathrm{tp}} \sup_{t \ge \hat{\mathsf{d}}_{\mathsf{k}(\mathsf{d})}/\sqrt{3}} t^3 e^{-\delta_{\mathrm{tp}}Lt}.$$

Telescoping the two kernels in $\mathrm{Tr}(\mathsf{W}_{\mathfrak{s}_1}\cdot\mathsf{W}_{\mathfrak{s}_2}\cdot)$, with Lem. 4.17(a), $|C_K^{\mathrm{free}}(p,q)|\le(1+3C_{\mathrm{lat}})a_K^3|u|^{-3}$, $\mathsf{r}^{-3}\le 3^{3/2}\mathsf{d}^{-3}$ and the second bound of clause (3) of the lemma on bond sources for $\sum_{p,q}|\mathsf{W}_{\mathfrak{s}_a}(p,q)|$, this member is at most $C\,\mathsf{e}_{\mathrm{fr}}(g;\mathsf{d},\mathsf{d}^+)(r_T/\mathsf{d})^6\mathsf{b}^{-2}|\mathfrak{s}_1|_\infty|\mathfrak{s}_2|_\infty$: a member polynomial in d^+, besides the far term at the floor.

The members (i) and (ii), which carry the renormalisation group, read of the distances the floor d alone; the largest distance d^+ enters the one explicit member (iii), the comparison of the sum of the slices with the free kernel, through a bound polynomial in d^+.

The case $r\ge r_g$. Then $\rho=r_g$ and $s(r_g)\le K$. Tile the lift by the half-open cubes Q of side r_g and cut $\mathfrak{s}_a=\sum_Q\mathfrak{s}_a^Q$, $\mathfrak{s}_a^Q$ the restriction of $\mathfrak{s}_a$ to the plaquettes and the bonds whose barycentres or midpoints lie in Q; $|\mathfrak{s}_a^Q|_\infty\le|\mathfrak{s}_a|_\infty$, and at most $(2r/r_g+2)^3\le 64(r/r_g)^3$ pieces of each source are not 0. The hull of a piece is a part of the hull of $\mathfrak{s}_a$ and has its barycentres within $\frac{\sqrt3}{2}r_g+\frac12 a_K\le r_g$ of the centre of its cube; so every pair of pieces, neither 0, is in the data of the first case at the widths (r_g,d), with $r_g\le\mathsf{d}$, $s(r_g)\le K$, $r_g\le r_1(\mathsf{b})$, the centres of the cubes in the place of x_1,x_2, and the same bounds $\frac67\mathsf{d}'\le|x(p)-x(q)|\le\mathsf{d}^+$ on the hulls. Both members of the display are bilinear in $(\mathfrak{s}_1,\mathfrak{s}_2)$, as $\mathcal{O}^T$ and $\mathsf{W}_{\mathfrak{s}}$ are linear in $\mathfrak{s}$, and the sum over the pairs of pieces costs $64^2(r/r_g)^6(r_g/\mathsf{d})^6=4096(r/\mathsf{d})^6$.

Together. $(\mathsf{d}'/\mathsf{d})^6=(\frac76)^6\max\{1,6\mathsf{d}'/7t_{\mathrm{N}}(g)\}^6$ and $4096(\frac76)^6(\frac{105}{104})^6\le 11060$ give the display with

$$\epsilon_\sharp^{\mathrm{ev}}(g,\ell,\mathsf{d}')=11060\max\Big\{1,\frac{6\mathsf{d}'}{7t_{\mathrm{N}}(g)}\Big\}^6\Big[\mathsf{E}_{\mathrm{J}}(g,\mathsf{d})+C\,\mathsf{e}_{\mathrm{fr}}(g;\mathsf{d},\tfrac87\mathsf{d}')+3\Lambda_T^2\bar{\mathsf{M}}_\ell^2\bar{\mathsf{b}}^2\mathsf{d}^{3/2}\Big],\qquad \mathsf{d}=\min\{\tfrac67\mathsf{d}',t_{\mathrm{N}}(g)\},$$

a finite function on $(0,\frac18\ell]$ formed from the constants, g and ℓ. For $\mathsf{d}'\le\frac76 t_{\mathrm{N}}(g)$ one has $\mathsf{d}=\frac67\mathsf{d}'$ and $\mathsf{d}^+=\frac43\mathsf{d}$, and each of the three members is at most $C(g,\ell)(\log\frac{1}{\mathsf{d}'})^{19\mathfrak{p}_0}\mathsf{d}'^{1/2}$: the first by step 6 of the proof of Prop. 4.19, the second as the member br of its step 4 under the conditions of its step 5, the third as $\mathsf{d}^{3/2}\le\mathsf{d}^{1/2}$.

The plane and the bond forms. For Lipschitz bumps the lattice form is turned into the integrals of (D.1) by Lem. 4.33, whose data hold: $4a_K\le r$, $16r\le\mathsf{d}'$, $8\mathsf{d}'\le\ell$. For the plane fields take $\mathfrak{s}_a=(\mathbf{1}_{\{\lambda(\cdot)=\lambda_a\}}\varphi_a\circ x,0)$, $\lambda_1=\lambda$, $\lambda_2=\mu$, so that $|\mathfrak{s}_a|_\infty\le\|\varphi_a\|_\infty$ and the main term is $\frac32\mathsf{b}^{-2}\sum_{p,q}w_1(p)w_2(q)C_K^{\mathrm{free}}(p,q)^2$; Lem. 4.33(a) times $\frac32\mathsf{b}^{-2}$ gives the second display of (D.1). For the bond form take $\mathfrak{s}_1=(0,\mathsf{w})$, $\mathsf{w}(b)=\varphi_1(x_b)\mathbf{1}_{\{\mu_b=\mu\}}$, so that $Y[\mathsf{w}]=\mathcal{O}^{\mathrm{od}}_{\lambda\lambda'}(\varphi_1)$ and $|\mathfrak{s}_1|_\infty\le 8\|\varphi_1\|_\infty$, and $\mathfrak{s}_2=(\mathbf{1}_{\{\lambda(\cdot)=\gamma\}}\varphi_2\circ x,0)$.

For the symmetric matrix $C=C_K^{\mathrm{free}}$, $\mathrm{Tr}(\mathsf{W}_{\mathfrak{s}_1}C\mathsf{W}_{\mathfrak{s}_2}C)=-2\sum_{b,q}\mathsf{w}(b)w_2(q)C_\lambda(b,q)C_{\lambda'}(b,q)$ with $C_\alpha(b,q)=\sum_p u_{b,\alpha}(p)C(p,q)$, so the main term is $-3\mathsf{b}^{-2}\sum_{b,q}\mathsf{w}(b)w_2(q)C_\lambda(b,q)C_{\lambda'}(b,q)$, and Lem. 4.33(b) times $3\mathsf{b}^{-2}$ gives the first display, with $8\epsilon_\sharp^{\mathrm{ev}}$ in the place of $8\epsilon^{\mathrm{cl}}$. □

Pair minus tensor field.

Lemma 4.35 (A regular gauge on a cube; the curvature is Hölder continuous)**.** Fix $\beta_{\mathrm{H}}\in(0,1)$. There are numbers $\epsilon_{\mathrm{G}}\in(0,1]$ and $C_{\mathrm{G}}^\sharp,C_{\mathrm{H}}\ge1$, formed from the constants and β_{H}, such that the following holds for $1\le n\le K$; lengths are in level-n units, $\xi=\eta_n$, and $Q_s(x_0)$ is the set of the bare sites at the sup distance at most s from a bare site x_0.

(a) Let $12a_{K-n}\le\ell$ and let A be a real $\mathfrak{su}(2)$-valued 1-form on the bonds of $Q_2(x_0)$ such that $|A|$, $|\nabla^\xi A|$ and the current of $e^{i\xi A}$ are at most $\mathsf{r}_\circ\le\frac{1}{120}$ there. Then the linear angles $F_p=\xi(\mathrm{d}A)(p)$ satisfy $|F_p|\le 2\xi^2\mathsf{r}_\circ$ and $|F_{p_{\mu\nu}(x)}-F_{p_{\mu\nu}(x')}|\le C_{\mathrm{H}}\,\xi^2\mathsf{r}_\circ|x-x'|^{\beta_{\mathrm{H}}}$ for the plaquettes $p_{\mu\nu}(x)$, $p_{\mu\nu}(x')$ of one orientation with $x,x'\in Q_{1/2}(x_0)$.

(b) Let $W\in\mathcal{V}_n^\chi$, $U=U_n(W)$, and let $\varrho\ge1$ with $(8\varrho+2)a_{K-n}<\ell$ and $(4\varrho)^2(1+4\varrho)\,4B_3\varepsilon_n\le\epsilon_{\mathrm{G}}$. Then there are $\mathsf{g}\colon Q_{4\varrho}(x_0)\to SU(2)$ and a real 1-form A on the bonds of $Q_{2\varrho}(x_0)$ with $\mathsf{g}(b_-)U(b)\mathsf{g}(b_+)^{-1}=e^{i\xi A(b)}$ there and, on $Q_\varrho(x_0)$, $|A|\le 80C_{\mathrm{G}}^\sharp B_3\varrho^2\varepsilon_n$ and $|\nabla^\xi A|\le$

$80C_{\mathrm{G}}^{\sharp}B_3\varrho\,\varepsilon_n$; the plaquette variables and the current of $e^{i\xi A}$ are those of U conjugated by g, and the current has a modulus $< 2B_3\varepsilon_n$.

None of these numbers depends on n, K, ℓ, the spacing, x_0 or the configuration. ■

Proof. (a) The 2-form $f_{\mu\nu} = \xi^{-2}F_{p_{\mu\nu}} = \nabla^{\xi}_{\mu}A_{\nu} - \nabla^{\xi}_{\nu}A_{\mu}$ is antisymmetric and closed, $\mathrm{d}f = 0$, and its lattice codifferential differs from the current of $e^{i\xi A}$ by at most $40|A||\nabla^{\xi}A| + 32|A|^3$ in the sup bounds, by the expansion of the lattice current; so $|f| \le 2\mathsf{r}_{\circ}$ and $|\mathrm{d}^*f| \le 2\mathsf{r}_{\circ}$. For an antisymmetric lattice 2-form f on a cube $Q_R(x_0)$, $R \ge 8\xi$, with $|f| \le m_0$ and $|\mathrm{d}f|, |\mathrm{d}^*f| \le m_1$, the lattice Hodge identity $\Delta(\chi f) = \mathrm{d}^*\mathrm{d}(\chi f) + \mathrm{d}\mathrm{d}^*(\chi f)$, for a cut-off χ equal to 1 on $Q_{R/2}(x_0)$ and written as $\chi f = G_\xi * \Delta(\chi f)$ with the lattice Green function G_ξ of $\xi\mathbb{Z}^3$, and the bounds $|\nabla G_\xi(z)| \le C(|z|+\xi)^{-2}$, $|\nabla^2 G_\xi(z)| \le C(|z|+\xi)^{-3}$, which are uniform in ξ, give $|f(x) - f(x')| \le C(\beta_{\mathrm{H}})(m_0 + Rm_1)(|x-x'|/R)^{\beta_{\mathrm{H}}}$ for $x, x' \in Q_{R/2}(x_0)$. At $R = 1$, $m_0 = m_1 = 2\mathsf{r}_{\circ}$ this is (a). (b) Put $\alpha = 4B_3\varepsilon_n$ and lift the sites of $Q_{4\varrho}(x_0)$ to the box $\Omega = x_0 + \xi\{-M, \dots, M\}^3$, $M = \lfloor 4\varrho/\xi \rfloor$, of side $s = (2M+1)\xi \le 9\varrho$. The room hypothesis makes the lift injective. By Lem. 4.20(d), U is regular of level n at radius $\alpha_0 = 2B_3\varepsilon_n$. The current-to-matrix-divergence calculation in Lem. YM3_I:9.4 gives $|D_U^{\mathsf{T}}\partial U| \le \xi^{-1}(\alpha_0\xi^2)^2 + \xi^2\alpha_0 < \alpha\xi^2$ at every full box bond; its plaquette bound is also $< \alpha\xi^2$. Choose $\epsilon_{\mathrm{G}} \le \min\{1, c_{\square}\}$ and $C_{\mathrm{G}}^{\sharp} \ge \max\{1, B_{\square}\}$, where $c_{\square}, B_{\square}$ are the box-gauge constants in the proof of Thm. YM3_I:F. The stated smallness implies $s\alpha \le c_{\square}$.

That box estimate gives a real gauge and potential on Ω with $|A| < B_{\square}s\alpha \le 36B_{\square}B_3\varrho\varepsilon_n$, and the same bound for first differences at base sites of depth at least $3/2$. Every positive bond with an endpoint in Q_ϱ has base depth at least $3\varrho - 2\xi > 3/2$. Restricting the potential to $Q_{2\varrho}$ gives both displayed bounds, since $\varrho \ge 1$. Plaquette and current moduli are preserved by the real gauge. □

In Lems. 4.36–4.41 the data are those of Thm. K; one progression holds the sources w_1, w_2, w_T, w_3, so that by multilinearity the objects of the two lists (w_1, w_2, w_3) and (w_T, w_3) are obtained from it. We write $\Delta_n = \mathsf{c}G_{n;12} - G_{n;T}$ and $\Gamma_n = \mathsf{c}G_{n;123} - G_{n;T3}$, functions of the configurations of the level n whose pairs lie in the cores, and $\mathrm{nat}_n^{\mathcal{W}} = \varepsilon_n^2 a_{K-n}^{-4}$.

Lemma 4.36 (Pair minus tensor field on all small-field backgrounds)**.** Let $r \le r_\alpha(g, \mathsf{b})$; let n_1 and n_0 be the least levels with $a_{K-n_1} \ge (\frac{r}{2})^{3/4}$ and $a_{K-n_0} \ge r^{1/2}$, $N_0 = n_0 + m_1$, where $m_1 \ge 1$ is the least integer with $L^{m_1/2} \ge C_7 A_0/(0.85\,\mathsf{c}_{\mathfrak{h}}A_1)$, $C_7 = 2B_3 + 69\cdot 1280\,C_{\mathrm{G}}^{\sharp}B_3$, an integer formed from the constants and β_{H} (through $C_{\mathrm{G}}^{\sharp}$ of the lemma on a regular gauge on a cube, which precedes this lemma), $\epsilon = r^{\beta_{\mathrm{H}}/36} + (a_K/r)^{1/9}$, and let $x_\sharp \ge e^{2\mathfrak{p}_0}$ be the least number such that every $x \ge x_\sharp$ has $17408B_3\varepsilon(x) \le \epsilon_{\mathrm{G}}$ and $1280C_{\mathrm{G}}^{\sharp}B_3\varepsilon(x) \le \frac{1}{100}$, with the numbers ϵ_{G}, $C_{\mathrm{G}}^{\sharp}$ of the lemma on a regular gauge on a cube.

Let $L^{N_0-n_1} \ge 2M$, and let $\epsilon_i^{\mathrm{e}} = C_\omega[1.45\gamma_{i-1} + \frac{1}{2}\mathsf{c}^{\flat}\theta^{[0]}] \le 1$ for $\mathsf{k}(\frac{r}{2}) \le i < n_1$; this number bounds the single-source arrays of the near pair at the level i, $\mathfrak{n}[\dot{\mathfrak{e}}_{a,i}] \le 2\epsilon_i^{\mathrm{e}}\varsigma_i\mathsf{m}_{i,a}$ ($a = 1, 2$, ς_i formed at the width $\frac{r}{8}$), as in Lem. 4.9(a). For $N_0 \le n^\sharp < K$ with $x_{n^\sharp} \ge x_\sharp$ there are real numbers c_n, $N_0 \le n \le n^\sharp$, such that for m_n-almost every $W \in \mathcal{V}_n^{\chi}$

$$|\Delta_n(U_n(W)) - \mathsf{c}_n| \le \varepsilon^{\mathcal{W}}\,\mathrm{nat}_n^{\mathcal{W}}, \qquad |\mathsf{c}_n - \mathsf{c}_{n'}| \le 2.01\,\tilde{\varepsilon}\,\mathrm{nat}_n^{\mathcal{W}} \quad (n \le n' \le n^\sharp),$$

with $\varepsilon^{\mathcal{W}} \le \tilde{\varepsilon} \le C(g, \mathsf{b})\,\epsilon$. ■

Proof. (i) We use the regular-field estimate of [7, Lem. B.22] and the insertion identity of Lem. 2.2 on the small-field backgrounds. The first is the evaluation of a local term at a configuration that is regular of a level, with the factor ε_n^2: for $1 \le i \le n \le K$, $W \in \mathcal{V}_n^{\chi}$ and a pointed family f of level i that vanishes unless one of N given cubes of π_i is a cube of the domain, and on the domains with $d_i(X) < D$,

$$\sum_{(X,z)} |\mathsf{f}(X, U_n(W), z) - \mathsf{f}(X, \mathbf{1}, z)| \le C\,\mathsf{N}\,\|\mathsf{f}\|_{i,\lambda_{\mathrm{out}}}\,\varepsilon_n^2\,L^{-4(n-i)}e^{-D/2}.$$

It is [7, eq. 230] at the radius $\varepsilon = 2B_3\varepsilon_n$ at which $U_n(W)$ is regular of the level n (Lem. 4.20(d), $4M\varepsilon \le c_{\mathrm{A}}\alpha_0$), which gives for one domain of size d the factor $(\varepsilon/\alpha_0)^2 L^{-4(n-i)} M^2(1+d)^2$, together with the tails of the proof of Lem. 4.8(a): the domains beyond the depth $d_* = \min\{c_{\mathrm{A}}\alpha_0 L^{2(n-i)}/M\varepsilon, (L^{m+K-i}-5)/2M\} - 2$ have $e^{-[d_*]_+/2} \le C\varepsilon_n^2 L^{-4(n-i)}$, by $e^{-s/2} \le 8s^{-2}$, by $L^{m+K-n} \ge 13My_n$ on the sides $\ell = 2L^m$ of $\mathfrak{L}(g)$ (from $L^m \ge LMy_K$ and Lem. 4.3(a)) and by $x_n^{-1} \le \varepsilon_n^2/A_0^2$; no ceiling beyond those of the data is read. (Lem. 4.8(b) is the same count without the factor ε_n^2.) The second is the averaging identity of one step: the quotient Qt_{j+1} of one step, a positive normalised mean that takes bounded Borel functions of the block fields of the level j to functions of those of the level $j+1$, takes the action $S[\mathfrak{g}](U_j(\cdot))$ of a bare array $\mathfrak{g}$ of the level j to $(S[\mathfrak{g}] + \mathsf{Sum}[\mathsf{DSt}^{[j]}(\mathfrak{f}_{0,j})[\mathfrak{g}]])(U_{j+1}(\cdot))$, m_{j+1}-almost everywhere; so the seminorm $\inf_c \operatorname{ess\,sup}_{\mathsf{m}_n} |\Phi(U_n(W)) - c|$ of such actions does not grow from one level to the next. (ii) *The split.* For $n_0 \le j < K$ let $\mathfrak{x}_j$ be the bare array whose families are c times the kernel families $\mathsf{K}^{(i)}$ of the near pair at the levels $\mathsf{k}(\frac{r}{2}) < i \le n_1$ and whose slot holds minus the weight member of w_T; its action $X^{\mathrm{fz}} = S[\mathfrak{x}_j]$, the frozen part, does not depend on j.

Let $\mathfrak{r}_j$ hold all the other families of Δ of the levels $\le n_0$. Then $\Delta_n = \Delta_n(\mathbf{1}) + X_n + S_n$, where X_n adds to X^{fz} the families that the first derivatives of the steps above n_0 make from $\mathfrak{x}$ alone, and S_n is the action of $\mathfrak{r}_n$ together with the families $\mathsf{s}^{(i)}$, $n_0 < i \le n$, that the first derivatives of the steps make from $\mathfrak{r}$ and from the $\mathsf{s}^{(i')}$ of the lower levels, and with the second derivatives $\mathsf{c}\,\mathsf{B}_{i-1}(\mathfrak{f}_{\{1\},i-1}, \mathfrak{f}_{\{2\},i-1})$ of the steps (Lem. 4.7). The evaluation of (i), the lags of Lem. 4.6(c) and the sizes of Lem. 4.9 give $|S_n| \le C\varepsilon_S \operatorname{nat}_n^{\mathcal{W}}$ at every small-field background of the level n, with $\varepsilon_S = 2\varepsilon_{\mathfrak{r}} + C(r/a_{K-n_0})^3$ and the number $\varepsilon_{\mathfrak{r}}$ of step (i) of the proof of the lemma on the chain classes, which follows this lemma; that step uses the same split on the chain classes. (iii) The explicit parts cancel.

With $D(n) = 3n \otimes n - 1$, the pair, multiplied by c, carries the explicit form with the kernel $\mathsf{b}^{-1}C_K^{\mathrm{free}} \approx \mathsf{b}^{-1}a_K^3\mathbb{K}_3(re_\lambda) = \mathsf{b}^{-1}a_K^3(4\pi r^3)^{-1}D(e_\lambda)$, while the tensor source carries $\frac{1}{2a_K}\sum_p w_T(p)|F_p|^2$ with $\delta_{\lambda(p)\lambda} - \frac{1}{3} = \frac{1}{3}D(e_\lambda)_{\lambda(p)\lambda(p)}$; the moment condition $\sum_k \mathsf{a}_k D(n_k) = \mathsf{a}_T D(e_\lambda)$, here with one pair of direction e_λ and $\mathsf{a}_1 = \mathsf{a}_T = 1$, makes them equal. (iv) *The frozen part on the backgrounds.* Let $N_0 \le n \le n^\sharp$, $W \in \mathcal{V}_n^\chi$, $U = U_n(W)$, x_0 a bare site with $|x_0 - c|_\infty \le a_K$, and let lengths be in level-n units, $\xi = \eta_n$.

The gauge. As $x_n \ge x_\sharp$ and $34a_{K-n} < \ell$, clause (b) of the lemma on a regular gauge on a cube, which precedes this lemma, applies at $\varrho = 4$ and gives g on $Q_{16}(x_0)$ and a potential A on the bonds of $Q_8(x_0)$ with $\mathsf{g}(b_-)U(b)\mathsf{g}(b_+)^{-1} = e^{i\xi A(b)}$, and on $Q_4(x_0)$ the bounds $|A|, |\nabla^\xi A| \le C_6\varepsilon_n \le \frac{1}{100}$, $C_6 = 1280C_{\mathrm{G}}^\sharp B_3$, and a current of modulus $< 2B_3\varepsilon_n$. *The cut-off.* Let $h = \prod_\mu \chi(|x_\mu - x_{0,\mu}|)$ with $\chi = 1$ on $[0,3]$, $\chi = 0$ on $[\frac{15}{4}, \infty)$, $|\chi'| \le 2$ and $\operatorname{Lip}\chi' \le 11$; put $\hat{A} = hA$ on the bonds of $Q_4(x_0)$, $\hat{A} = 0$ on all other bonds of T_ℓ, and $\hat{U} = e^{i\xi\hat{A}}$.

Then $\hat{U}$ is the transform of U by g on every bond with an end in $Q_3(x_0)$, and the product rules for hA with the expansion of the lattice current give

$$\max\{\sup|\hat{A}|,\ \sup|\nabla^\xi\hat{A}|,\ \sup|J(\hat{U})|\} \le (2B_3 + 69\,C_6)\,\varepsilon_n = C_7\varepsilon_n \qquad \text{on all of } T_\ell.$$

The shift of the level. Seen from the level $n^\flat = n - m_1 \ge n_0$, the same configuration has the potential $\hat{A}^\flat = L^{-m_1}\hat{A}$, with $|\hat{A}^\flat| \le L^{-m_1}C_7\varepsilon_n$, differences $\le L^{-2m_1}C_7\varepsilon_n$ and current $\le L^{-3m_1}C_7\varepsilon_n$ in level-$n^\flat$ units; with $\varepsilon_n = (A_0/A_1)\delta_n$, $\delta_n \le (\frac{5}{4})^{1/2}L^{m_1/2}\delta_{n^\flat}$ (Lem. 4.3(a)) and the choice of m_1,

$$L^{-m_1}C_7\varepsilon_n \le (\tfrac{5}{4})^{1/2}\cdot 0.85\,\mathsf{c}_{\mathfrak{h}}\delta_{n^\flat} < \tfrac{11}{10}\mathsf{c}_{\mathfrak{h}}\delta_{n^\flat} \le r_{n^\flat}.$$

This is the norm clause of the chain class of radius $r_{n^\flat}$, and it is all that the steps below read of the field. *The pair.* Step (ii) of the proof of the lemma on the chain classes is read at the level $n^\flat$ with U in the place of $\tilde{U}$, with two changes. (1) The gauge $u = e^{iX}$ of Lem. 4.12 at the pair is formed for $\hat{A}^\flat$ on the bonds within two level-$n^\flat$ units of x_0, where $\hat{U}$ is the transform of U; it gives the angles F', Hölder continuous as in that step, through the estimate of the proof of clause (a) of the lemma

on a regular gauge on a cube on $Q_1(x_0)$ at $m_0 = 2.08\, r_{n^\flat}$ and $m_1 = 3.7\, r_{n^\flat}$, and the plaquette term of the weight member of w_T in this gauge, as $a_p(U) = a_p(\hat U)$ at the plaquettes of $\operatorname{supp} w_T$, which lie in $Q_{1/2}(x_0)$. (2) For the kernel families of the levels $\mathsf{k}(\frac r2) \le i < n_1$, $m = n^\flat - i - 1$: Lem. 4.12 for the real pair of the level $i+1$ that $\hat A^\flat$ and its current give on all of T_ℓ (the potential multiplied by L^{-m}, the current by L^{-3m}), on the box $N_{\rho'}$ about x_0, $\rho' = \min\{\mathsf{h}_i, 3L^{n-i-1} - 1\}$, with the same u, gives a real point v' of the chart with $\|v'\| \le \mathsf{h}_i L^{-2m} r_{n^\flat}(6 + 19 r_{n^\flat}) \le \delta_i/2\Lambda_i$, by ($\alpha$2) at $n^\flat \ge n_0$, the pair having $|L^{-m}\hat A^\flat| \le L^{-m} r_{n^\flat}$ and differences and current $\le L^{-2m} r_{n^\flat}$.

A domain X of the level $i+1$ with all its sites within $\rho' - 1$ of x_0 lies in $Q_3(x_0)$, where $\hat U$ is the transform of U, so that $\mathsf{K}^{(i+1)}(X, U, z) = \mathsf{K}^{(i+1)}(X, P[v']|_X, z)$ by the gauge invariance and the locality of a local term; the other domains have $d_{i+1}(X) \ge 2L^{n-i-1}/M - 2$, or the depth of step 5 of the proof of Prop. 4.13 if $\rho' = \mathsf{h}_i$, and are paid by the norm (Lem. 4.8(b) at U and at $P[v']$). Steps 2–4 of the proof of Prop. 4.13 (holomorphy, the real ball, two constants), which read of the field only the real point v' and its norm, give at v' the kernel step with the angles F'. As $L^{n-i-1} \ge L^{N_0 - n_1} \ge 2M$ and $\epsilon_i^{\mathrm e} \le 1$, the domains that leave the box cost at most $C\mathsf{b}\, r\, \mathrm{nat}_n^{\mathcal W}$ together.

With $r^2_{n^\flat} a^{-4}_{K-n^\flat} \le C\,\mathrm{nat}_n^{\mathcal W}$ (Lem. 4.3(a), $a_{K-n^\flat} = L^{-m_1} a_{K-n}$) this bounds the frozen part: $|X^{\mathrm{fz}}(U)| \le \varepsilon_X^{\mathcal W} \mathrm{nat}_n^{\mathcal W}$, with $\varepsilon_X^{\mathcal W} \le C[\varepsilon_X + \mathsf{b}\, r]$ and the number ε_X of step (ii) of the proof of the lemma on the chain classes. (v) Carried upward with (i) and the lags of Lem. 4.6(c), $\|X_n\| \le [\varepsilon_X^{\mathcal W} + C\gamma_{N_0-1} + C\gamma_n^{8/9}(\varepsilon_X^{\mathcal W})^{1/9}]\mathrm{nat}_n^{\mathcal W}$; the exponents $\frac89$, $\frac19$, and $\frac12$ below, come from interpolations between the oscillation bound and the bound of the sizes, and the choice of the levels, $\mathsf{k}(\frac r2) < n_1 < n_0$ with $a_{K-n_1} \ge (\frac r2)^{3/4}$, costs the exponent $\frac14$.

Together $\varepsilon^{\mathcal W} \le \tilde\varepsilon \le C(g, \mathsf{b})[r^{\beta_{\mathrm H}/36} + (a_K/r)^{1/9}]$ for $r \le r_\alpha(g, \mathsf{b})$, the levels being ordered by (α1)–(α5) of Def. 4.1(5); the constants c_n of two levels differ by at most these oscillations plus the part of the steps between that the sizes of the sources at those levels bound, which is of the same order. □

Lemma 4.37 (Pair minus tensor field on the chain classes)**.** Let $r \le r_\alpha(g, \mathsf{b})$, let n_1 and n_0 be the levels of Lem. 4.36, and $\epsilon = r^{\beta_{\mathrm H}/36} + (a_K/r)^{1/9}$. There are numbers c'_n, $n_0 \le n \le K$, and a number $\varepsilon_\alpha \le C(g, \mathsf{b})\,\epsilon$ such that

$$\sup_{A \in \mathsf{Cc}_n(r_n)} \big|\Delta_n(e^{i\eta_n A}) - \mathsf{c}'_n\big| \le \varepsilon_\alpha\, r_n^2 a_{K-n}^{-4} \qquad (n_0 \le n \le K).$$

■

Proof. (i) *The split* is that of step (ii) of the proof of Lem. 4.36. For $n_0 \le j < K$, $\mathfrak{x}_j$ is the array whose families are c times the kernel families $\mathsf{K}^{(i)}$ of the near pair at the levels $\mathsf{k}(\frac r2) < i \le n_1$ and whose slot holds minus the weight member of w_T; its action X^{fz}, the frozen part, does not depend on j. $\mathfrak{r}_j$ holds all the other families of Δ of the levels $\le n_0$. Lem. 4.6 and the sizes of Lem. 4.9 give $\mathfrak{n}[\mathfrak{x}_j] \le C a_{K-j}^{-4}$ and $\mathfrak{n}[\mathfrak{r}_j] \le \varepsilon_{\mathfrak r} a_{K-j}^{-4}$, where $\varepsilon_{\mathfrak r}$ is a constant times $\gamma_{n_0} + s(\frac r2)\theta^{[0]} + (r/a_{K-n_1})^3$ and terms exponentially small in $\hat r_{\mathsf{k}(r/2)}$ and in r/a_{K-1}, so that $\varepsilon_{\mathfrak r} \le C(g, \mathsf{b})[r^{1/4} + (a_K/r)^{1/4}]$.

On the chain classes $\Delta_n = \Delta_n(\mathbf 1) + X_n + S_n$, where X_n adds to X^{fz} the families that the first derivatives of the steps above n_0 make from $\mathfrak x$ alone, and S_n collects the rest: the action of $\mathfrak r_n$, the families that the first derivatives make from $\mathfrak r$ and from them, and the second derivatives of the steps; Lem. 4.8(a), in the frame at $\mathsf{k}(\frac r2)$ in which Lem. 4.5(a) puts the near pair, a test pair of the scale $\frac r2$, under the conditions $3 \le s(\frac r2) \le K$ and $\mathsf{Lm}_{s(r/2)} \le L^{s(r/2)-3}$ of the data (and $\mathsf{k}(\frac r2) < n_1 < n_0 \le n < K$ by (α1)), and the lags of Lem. 4.6(c) give $|S_n| \le C\varepsilon_S r_n^2 a_{K-n}^{-4}$ with $\varepsilon_S = 2\varepsilon_{\mathfrak r} + C(r/a_{K-n_0})^3$. (ii) *The frozen part.* Let $n_0 \le n \le K$, $A \in \mathsf{Cc}_n(r_n)$, x_0 a bare site with $|x_0 - c|_\infty \le a_K$, let $u = e^{iX}$, X linear, be the gauge of Lem. 4.12 at the pair, about x_0, A' the potential of $\tilde U = e^{i\eta_n A}$ in this gauge and $F' = \eta_n \mathrm d A'$ its angles. Near the pair $|A'|$, $|\nabla A'|$ and

the current are at most a constant times r_n (Lem. 4.12, at $\alpha = \gamma_\circ = r_n$, in the units of the level n): on N_2, $|A'| \le 6r_n + 18.7r_n^2 \le 6.2\, r_n$ and $|\nabla^{\eta_n} A'| \le \frac{100}{99}(r_n + 3r_n^2) \le 1.04\, r_n$, and the current, conjugated by the gauge, is at most r_n. So the 2-form $f = \eta_n^{-2} F'$ is closed, $|f| \le 2.08\, r_n$, and, as its codifferential differs from the current by at most $40|A'||\nabla^{\eta_n} A'| + 32|A'|^3$, $|\mathrm{d}^* f| \le 3.7\, r_n$ on $Q_1(x_0)$, since $2r_n \le \varrho_{\mathrm{ch}}$ (Lem. 4.3(b)) and $\varrho_{\mathrm{ch}} \le \frac{1}{60}$ by equation YM3_I:(2285).

The lemma on a regular gauge on a cube, which precedes Lem. 4.36, is read through the estimate of the proof of its clause (a) on $Q_1(x_0)$, at $m_0 = 2.08\, r_n$ and $m_1 = 3.7\, r_n$: it makes F' Hölder continuous with the exponent β_{H} on $Q_{1/2}(x_0)$, on the scale of the pair. Prop. 4.13, in the frame of step (i), at the levels $\mathsf{k}(\frac{r}{2}) \le i < n_1$, where ($\alpha$2) of Def. 4.1(5) gives its condition in (ii), taken before the last replacement of step 5 of its proof, so that $\mathcal{Q}_i$ stands at F' and the last term of (ii) does not occur, replaces $\mathsf{c} \sum_i (\mathsf{Sum}[\mathsf{K}^{(i+1)}](\tilde U) - \mathsf{Sum}[\mathsf{K}^{(i+1)}](\mathbf{1}))$ by $\mathsf{c}\, q_{n_1}(F')$ with an error at most $C \sum_i \hat r_i^3 \mathsf{b}_i^2 (\delta_i + \epsilon_i^{\mathrm{cu}}) \Lambda_i^2 \mathsf{h}_i^2\, r_n^2 a_{K-n}^{-4} \le C(g, \mathsf{b})\, r^{1/4} r_n^2 a_{K-n}^{-4}$; the weight member of w_T is expanded in the gauge u at the pair.

With $\xi = \eta_n$, the plaquette angles of $\tilde U$ in this gauge are $F'_p + \Psi_p$ with the commutator part $|\Psi_p| \le 24\xi^2 |A'|^2$; Lem. 4.12 at $\alpha = \gamma_\circ = r_n$ gives $|A'(b)| \le 3sr_n + (9s + 8\xi) r_n^2 \le 4sr_n + 8\xi r_n^2$ at the bonds with base point within s of x_0, and $|F'_p| \le 2.08\, \xi^2 r_n$; as $a_p = \frac12 |\theta|^2 + O(|\theta|^4)$ in the plaquette angle θ,

$$\big| a_p(\tilde U) - \tfrac12 |F'_p|^2 \big| \le |F'_p||\Psi_p| + \tfrac12 |\Psi_p|^2 + C|F'_p + \Psi_p|^4 \le C\xi^4 r_n^3 (s + \xi)^2$$

at the plaquettes within $s \le \frac12$ of x_0. The plaquettes of $\operatorname{supp} w_T$ lie within $s \le r/a_{K-n}$ of x_0, so $s + \xi \le 2r/a_{K-n_0}$; summed against w_T, with $\sum_p |w_T(p)| \le C a_K^{-3}$, this replaces the tensor source by $\frac{1}{2a_K} \sum_p w_T(p) |F'_p|^2$ up to $C(r/a_{K-n_0})^2 r_n^2 a_{K-n}^{-4} \le C\, r\, r_n^2 a_{K-n}^{-4}$, as $a_{K-n_0} \ge r^{1/2}$: in the gauge at the pair the angles are small at the scale $s + \xi$ of the tensor source. By the Hölder continuity and the comparison of $\mathcal{S}_{n_1}$ with $\mathsf{b}^{-1} C_K^{\mathrm{free}}$ and of C_K^{free} with $a_K^3 \mathbb{K}_3$ (Lem. 4.17(b), with (α5)), both main terms equal $\frac{1}{6a_K^4} \sum_{\mu,\nu} D(e_\lambda)_{\mu\nu} \langle F^\circ_\mu, F^\circ_\nu \rangle$, F°_μ the angle of the orientation μ at the centre, up to $C[(r/a_{K-n_1})^{\beta_{\mathrm{H}}} + a_K/r]\, r_n^2 a_{K-n}^{-4}$ and terms of the same order; the moment condition of step (iii) of the proof of Lem. 4.36 cancels the main terms. So $\sup_{\mathsf{Cc}_n(r_n)} |X^{\mathrm{fz}}| \le \varepsilon_X r_n^2 a_{K-n}^{-4}$ with $\varepsilon_X \le C(g, \mathsf{b})[r^{\beta_{\mathrm{H}}/4} + a_K/r]$. (iii) *Carrying upward.* On the chain classes the averaging identity of one step holds at every point: for $n_0 \le m < K$, $A' \in \mathsf{Cc}_{m+1}(r_{m+1})$, the measure $\nu = \nu_{m,A'}$ of the step and every bare array $\mathfrak{g}$ of the level m of size below $N_\star^{[m]}$,

$$\int S[\mathfrak{g}](e^{i\eta_m A^{\mathrm{lw}}(B)})\, \mathrm{d}\nu(B) = \big(S[\mathfrak{g}] + \mathsf{Sum}[\mathsf{DSt}^{[m]}(\mathfrak{f}_{0,m})[\mathfrak{g}]]\big)(e^{i\eta_{m+1} A'}) + c_{\mathfrak{g}},$$

with a number $c_{\mathfrak{g}}$ that does not depend on A'. It is the first derivative, in the direction of $\mathfrak{g}$, of the exact step integral in Lem. 2.2 at the real chart point A', which writes the action of the level $m+1$ through the mean under $\nu_{m,A'}$ of the exponential of the action of the level m at the lowered configurations; the measure $\nu_{m,A'}$ and the lowering map name no array, and the step takes any bare array of that size in its fourth place (Lem. 4.6(b)). As $A^{\mathrm{lw}}(B) \in \mathsf{Cc}_m(r_m)$ and ν is a probability measure, the oscillation of the right member over $\mathsf{Cc}_{m+1}(r_{m+1})$ is at most the oscillation of $S[\mathfrak{g}]$ over $\mathsf{Cc}_m(r_m)$.

A family that the steps above n_0 make from $\mathfrak{x}$, first created at the level $m+1$, is bounded in two ways: by the norm of the calculus, with the factor γ_m (Lem. 4.6(b), (c)), and, through this identity at $\mathfrak{g} = \mathfrak{x}_m$, by the oscillation of X^{fz} at the levels m and $m+1$, at most $2\varepsilon_X L^{4(n-m)} r_n^2 a_{K-n}^{-4}$. The smaller of the two, summed over m with $\sum_{l \ge 0} \min\{aL^{-l/2}, bL^{4l}\} \le 2.4 a^{8/9} b^{1/9}$ ($L \ge 13$), bounds the oscillation of X_n by $[\varepsilon_X + C\gamma_{K-1}^{8/9} \varepsilon_X^{1/9}] r_n^2 a_{K-n}^{-4}$. Together the oscillation of Δ_n on $\mathsf{Cc}_n(r_n)$ is at most $\varepsilon_\alpha r_n^2 a_{K-n}^{-4}$ with $\varepsilon_\alpha = \varepsilon_X + C\gamma_{K-1}^{8/9} \varepsilon_X^{1/9} + C\varepsilon_S \le C(g, \mathsf{b})\, \epsilon$, as $\gamma_{K-1} \le 1$; c'_n is the centre of this oscillation.

At $n = K$ the chain class is $\{0\}$ and $r_K = 0$. □

Lemma 4.38 (Pair minus tensor field up to the last level)**.** Let g satisfy besides (Γ3) of Def. 4.1(6), let $r \le r_\alpha^{\mathcal{W}}(g,\mathsf{b})$, let n_0 be the level of Lem. 4.36, and $\varepsilon^{\mathcal{W}} = C(g,\mathsf{b})[r^{\beta_{\rm H}/72} + (a_K/r)^{1/18}]$ with a suitable number $C(g,\mathsf{b}) \ge 1$. There are numbers c_n, $n_0 \le n \le K$, such that for m_n-almost every $W \in \mathcal{V}_n^\chi$

$$|\Delta_n(U_n(W)) - \mathsf{c}_n| \le \varepsilon^{\mathcal{W}}\,\mathrm{nat}_n^{\mathcal{W}}, \qquad |\mathsf{c}_n - \mathsf{c}_K| \le \varepsilon^{\mathcal{W}}\,\mathrm{nat}_n^{\mathcal{W}},$$

and, for $w_3 \not\equiv 0$ and $\mathsf{d}' = |c - z|$, $|\Gamma_n(U_n(W))| + |\Gamma_K(\mathbf{1})| \le C(\mathsf{b})\,\mathsf{d}'^{-5/2}\varepsilon^{\mathcal{W}}\mathsf{b}^{-2}\mathsf{d}'^{-6}\mu_3$. ■

Proof. The proof of Lem. 4.36 is made at every level $n_0 \le n \le K$, with the radius $C_{\mathcal{W}}\varepsilon_n$ of the backgrounds in the place of the radius of a chain class and without the shift of the level. (i) Use the repaired radius $d_*(i;n,2B_3\varepsilon_n)$ of eq. 2.ii. The first two members of (Γ3) and the geometric tail estimate in Lem. 4.8(b), with 5 in place of 4 and the final subtraction 2 in place of 1, bound the domains that leave the cores by $C\varepsilon_n^2 L^{-4(n-i)}$; the changes multiply the former tail constant by at most $e^{1/2+1/(4M)}$; so a pointed family of level i that vanishes off a set of sites changes on the backgrounds of the level n by at most a constant times its norm, the number of the cubes that hold a site of the set, $L^{-4(n-i)}$ and $\mathrm{nat}_n^{\mathcal{W}}$. (ii) Lem. 4.35(b) at $\varrho = 3$, which applies as $7488B_3\varepsilon_n \le \epsilon_{\rm G}$ by (Γ3) and $26a_{K-n} < \ell_g$ (as $\ell_g \ge 2LMy_K \ge 338$), gives about a bare site x_0 next to c a gauge g on $Q_{12}(x_0)$ and a potential A on the bonds of $Q_6(x_0)$ with $\mathsf{g}(b_-)U(b)\mathsf{g}(b_+)^{-1} = e^{i\eta_n A(b)}$ there, $U = U_n(W)$; on $Q_3(x_0)$ the potential, its differences and the current are at most $\mathsf{r}_\circ := C_{\mathcal{W}}\varepsilon_n \le \frac{1}{120}$.

Lem. 4.12 is used in a local form, with data on $N_{\rho+1}$: *in the units of a level, with the bare spacing $\xi \le \frac{1}{13}$, let $\rho \ge 2$ with $4\rho+4$ at most the side of the torus, and let $\mathsf{g}\colon N_{\rho+1} \to SU(2)$ and a real 1-form A on the bonds with both end points in $N_{\rho+1}$ satisfy $\mathsf{g}(b_-)U(b)\mathsf{g}(b_+)^{-1} = e^{i\xi A(b)}$, $|A| \le \alpha \le \frac{1}{60}$ and $|\nabla^\xi A| \le \gamma_\circ$ there, the current of U being at most ι there; with $c_\mu = A_\mu(x_0)$, $X(x) = \sum_\mu (x-x_0)_\mu c_\mu$ and $h = e^{iX}\mathsf{g}$ on $N_{\rho+1}$ there is a real pair v' on all the bare bonds which, at the bonds with base point in N_ρ, is the image of U and of its current under the gauge transformation h, and which obeys the three bounds of Lem. 4.12; if $\|v'\|$ is less than the radius of the ball of the chart, every local term of the level on a domain $X \subset N_{\rho-1}$ that is formed at U has the same value at $P[v']|_X$.* Its proof is that of Lem. 4.12, with the transformed potential formed on $N_{\rho+1}$ and the cut-off $\chi(x) = \min\{1, \max\{0, (\rho+1-2\xi-|x-x_0|_\infty)/(1-2\xi)\}\}$, of the width $1-2\xi$ and equal to 1 on N_ρ, in the place of the cut-off of the width ρ: as $|\chi(x+\xi e_\nu) - \chi(x)| \le \frac{13}{11}\xi$, $|\nabla^\xi A'| \le \frac{100}{99}(\gamma_\circ + 3\alpha^2) + \frac{13}{11}[3(\rho+1)\gamma_\circ + (9\rho+9.62)\alpha^2] \le \rho(6\gamma_\circ + 19\alpha^2)$ for $\rho \ge 2$; and the bonds that a local term on $X \subset N_{\rho-1}$ reads have their end points in N_ρ, where $P[v']$ is the image of U under h, while a local term is a gauge invariant function of the pair on its domain.

Viewed from the level $i+1$, $\mathsf{k}(\frac{r}{2}) \le i < n_1$, $m = n-i-1$, the data on $Q_3(x_0)$ have $|A| \le \alpha = L^{-m}\mathsf{r}_\circ$, $|\nabla A| \le \gamma_\circ = L^{-2m}\mathsf{r}_\circ$ and a current at most $L^{-3m}\mathsf{r}_\circ \le \gamma_\circ$, and the cube N_{h_i+1} of the level $i+1$ lies in $Q_3(x_0)$, as $(\mathsf{h}_i+1)L^{-m} \le 3$ (Def. 4.1(6)). The local form at $\rho = \mathsf{h}_i$, where $(4\mathsf{h}_i+4)a_{K-i-1} \le \ell_g$, gives a real point v' of the chart with $\|v'\| \le \mathsf{h}_i\gamma_\circ(6+19\mathsf{r}_\circ) \le 7\mathsf{h}_i L^{-2m}C_{\mathcal{W}}\varepsilon_n \le \delta_i/2\Lambda_i$ (Def. 4.1(6)), inside the ball of the chart; the linear gauge X is the same function at all the levels i, so that one h serves them all.

Steps 2–4 of the proof of Prop. 4.13 at the real point v', the domains in N_{h_i-1} taking the same values at U and at $P[v']$, and the others, of a linear size at least $D_* = (\mathsf{h}_i - 1 - R_w)/M - 2$, costing $2\mathsf{M}_i^{\mathsf{K}} e^{-[D_*]_+} \le \bar{\mathsf{e}}\gamma_\circ^2/\delta_i^2$ by the choice of h_i, replace $\mathsf{c}\sum_i(\mathsf{Sum}[\mathsf{K}^{(i+1)}](U) - \mathsf{Sum}[\mathsf{K}^{(i+1)}](\mathbf{1}))$ by $\mathsf{c}\,q_{n_1}(F')$, F' the linear angles in the gauge h, up to $C\sum_i \hat{r}_i^3\mathfrak{b}_i^2(\delta_i + \epsilon_i^{\rm cu})\Lambda_i^2\mathsf{h}_i^2\,C_{\mathcal{W}}^2\mathrm{nat}_n^{\mathcal{W}}$. In the gauge h, in the units of the level n, $|A'| \le 4s\mathsf{r}_\circ + 8\eta_n\mathsf{r}_\circ^2$ at the bonds with base point within $s \le 2$ of x_0, $|\nabla A'| \le 1.04\,\mathsf{r}_\circ$ and $|F_p'| \le 2.08\,\eta_n^2\mathsf{r}_\circ$ there, and $|a_p(U) - \frac{1}{2}|F_p'|^2| \le C\eta_n^4\mathsf{r}_\circ^3(s+\eta_n)^2$ at the plaquettes within $s \le \frac{1}{2}$ of x_0; this replaces the tensor source by $\frac{1}{2a_K}\sum_p w_T(p)|F_p'|^2$ up to $C(r/a_{K-n_0})^2 C_{\mathcal{W}}^2\mathrm{nat}_n^{\mathcal{W}}$.

Lem. 4.35(a) is read through the estimate of its proof on $Q_1(x_0)$, at $m_0 = 2.08\,\mathsf{r}_\circ$ and $m_1 = 3.7\,\mathsf{r}_\circ$ (on N_2, $|A'| \le 6.2\,\mathsf{r}_\circ$, and the codifferential of $\eta_n^{-2}F'$ differs from the current by at most $40|A'||\nabla A'| + 32|A'|^3$): it makes F' Hölder continuous with the exponent $\beta_{\rm H}$ on the scale of

the pair. The comparison of $\mathcal{S}_{n_1}$ with $\mathsf{b}^{-1}C_K^{\mathrm{free}}$ and of C_K^{free} with $a_K^3\mathbb{K}_3$ (Lem. 4.17(b), with (α5)) brings both main terms to $\frac{1}{6a_K^4}\sum_{\mu\nu}D(e_\lambda)_{\mu\nu}\langle F_\mu^\circ,F_\nu^\circ\rangle$, F° the angles at the centre, up to $C[(r/a_{K-n_1})^{\beta_{\mathrm{H}}}+a_K/r]C_{\mathcal{W}}^2\mathrm{nat}_n^{\mathcal{W}}$; the moment condition cancels them, and the frozen part on the backgrounds is at most $\varepsilon_X^{\mathcal{W}}\mathrm{nat}_n^{\mathcal{W}}$. (iii) From one level to the next the averaging identity of one step, written for arbitrary bare arrays, gives m_{n+1}-almost everywhere $\Delta_{n+1}=\mathsf{Qt}_{n+1}[\Delta_n]+\mathsf{b}_n$, where Qt_{n+1} is the quotient of the step, a positive normalised mean, and b_n is c times the covariance, under the step, of the quotients of the two single sources of the near pair.

The one-source bound $|G_{n;a}-G_{n;a}(\mathbf{1})|\le Cr^3|w_a|_\infty\mathrm{nat}_n^{\mathcal{W}}$ on the backgrounds gives $|\mathsf{b}_n|\le CA_0^2y_{n_0}^{2\mathfrak{p}_0}(r/a_{K-n_0})^3\mathrm{nat}_n^{\mathcal{W}}\le C(\mathsf{b})\,r\,\mathrm{nat}_n^{\mathcal{W}}$. A quotient does not increase the oscillation, so the centres c_n of the oscillations of two consecutive levels differ by at most twice the oscillation bound plus $C(\mathsf{b})\,r\,\mathrm{nat}_n^{\mathcal{W}}$; as $\mathrm{nat}^{\mathcal{W}}$ falls by the factor $\frac{5}{4}L^{-3}$ from one level to the next, the sum over the levels gives the second bound. (iv) For Γ_n the averaging identity of one step, in its form for two and for three arrays, gives m_{i+1}-almost everywhere, for $n_0\le i<K$, $\Gamma_{i+1}=\mathsf{Qt}_{i+1}[\Gamma_i]+\mathsf{Qt}_{i+1}[\Delta_i;G_{i;3}]+\mathsf{c}\,\mathsf{Sum}[\mathsf{C}^{(i+1)}]$; here $\mathsf{Qt}_{i+1}[\,\cdot\,;\,\cdot\,]$ is the covariance of the quotients of the step, and the family $\mathsf{C}^{(i+1)}$ is made of the mixed second derivatives of the step in the arrays of the sets $\{1,3\}$ and $\{2\}$ and in those of $\{2,3\}$ and $\{1\}$, and of its third derivative in the arrays of the three single sources.

A quotient does not raise the essential supremum, so the bound of Γ_n is that of Γ_{n_0} plus the sum over $n_0\le i<n$ of the bounds of the two other terms. The covariance is at most the smaller of the bound by the oscillations of Δ_i and of the far source on the backgrounds and the bound by the norms, $\mathsf{C}^{(i+1)}$ is bounded by the norms, and Γ_{n_0} by the first display of Lem. 4.39 at $U_{n_0}(W)$, whose pairs of all levels $\le n_0$ lie in the cores (Lem. 4.20(d)); the sum over the levels is that of the proof of Lem. 4.40, with the bound of the oscillation of Δ on the backgrounds in the place of ε_α.

All the numbers so formed are at most $C(g,\mathsf{b})[r^{\beta_{\mathrm{H}}/36}+(a_K/r)^{1/9}]$ for $r\le r_\alpha^{\mathcal{W}}$; with $(\frac{\mathsf{b}}{\mathsf{d}'})^{3/2}(\frac{r}{\mathsf{d}'})^3\le\mathsf{b}^{3/2}\mathsf{d}'^{-2}r^{1/2}$ and $(\frac{\mathsf{b}}{\rho_0})^{3/2}e^{-\mathsf{d}'/8M\rho_0}\le C\mathsf{b}^{3/2}\mathsf{d}'^{-5/2}r^{1/2}$, $\rho_0=a_{K-n_0}$, the claims follow. □

Lemma 4.39 (Crossing functionals at levels below the distance)**.** For $\beta\in\{13,23,123\}$ of the list (w_1,w_2,w_3) and $\beta=T3$ of the list (w_T,w_3), $m=|\beta|$, $1\le i\le K$, and every configuration U whose pairs of the level i lie in the cores,

$$\sum_{(X,z)}|E_\beta^{(i)}(X,U,z)|\le C_\times(\tfrac{8r}{3\mathsf{d}})^6\bar\nu_{i-1;\beta}\,e^{-D_i^\sharp/2},\qquad D_i^\sharp=[\mathsf{d}/2Ma_{K-i}-2]_+.$$

So if $8Ma_{K-n}\le\mathsf{d}$, then for every such U of the level n:

$$|G_{n;\beta}(U)|\le C'_\times\mathsf{b}^{-m/2}a_{K-n}^{-m/2}e^{-\mathsf{d}/4Ma_{K-n}}(\frac{8r}{3\mathsf{d}})^6\prod_{a\in\beta}|w_a|_\infty.$$

■

Proof. By the support rule (Lem. 4.6(d)) a crossing family lives on domains that reach from the pair to the piece, of linear size at least $D_i^\sharp$; the near supports lie in $\mathbb{B}(c,\frac{5}{8}r)$, the piece in $\mathbb{B}(z,\frac{1}{4}r)$, so the corners of the two kinds are at least $\frac{1}{2}\mathsf{d}$ apart in the sup distance; its norm is bounded by the sizes of Lem. 4.9(e), in the form (f), at the width r and a crossing distance $\mathsf{d}_\times\ge\frac{3}{8}\mathsf{d}$, with $(r/\mathsf{d}_\times)^6\le(\frac{8r}{3\mathsf{d}})^6$; Lem. 4.8(b) at U gives the sum with the factor $e^{-D_i^\sharp/2}$, and $\bar\nu_{i-1;\beta}\le\bar c_m(5L/4\mathsf{b})^{m/2}a_{K-i}^{-m/2}\prod|w_a|_\infty$; the sum over $i\le n$ is led by $i=n$. □

Lemma 4.40 (The difference of the jets)**.** Let $w_3\not\equiv0$. Then the centre z of its cube has $(1-\frac{1}{64})\mathsf{d}\le|c-z|\le6\mathsf{d}$, and

$$|\mathsf{c}\,\mathsf{J}_3(w_1,w_2,w_3)-\mathsf{J}_2(w_T,w_3)|\le C(g,\mathsf{b})\,\mathsf{d}^{-12}[r^{\beta_{\mathrm{H}}/72}+(a_K/r)^{1/18}]\mathsf{b}^{-2}\mathsf{d}^{-6}\mu_3.$$

■

Proof. A barycentre in the cube lies in $\operatorname{supp}\varphi_2$, at a distance $\ge \mathsf{d}$ from c and $\le \frac{\sqrt3}{8}r \le \frac{1}{64}\mathsf{d}$ from z, as $r \le (\mathsf{d}/5LM)^2$. Put $\mathsf{d}' = |c-z|$ and $\rho_0 = a_{K-n_0}$; the data give $16r \le \mathsf{d}' \le \min\{1, \mathsf{b}\}$, $4M\rho_0 \le \mathsf{d}'$ and $64\mathsf{d}' \le \ell_g$ ($\mathsf{d}' \le 6\mathsf{d}$, $r \le (\mathsf{d}/5LM)^2$, $\rho_0 < Lr^{1/2}$, $\ell_g \ge 2LMy_K \ge 338$). The corners of the plaquettes of the near supports and of the piece are at least $\frac12\mathsf{d}'$ apart in the sup distance, so Lem. 4.39 holds with d' in the place of d. Lem. 4.9 is used in its form (f), which bounds the crossing distance by 1 in the place of the bound $e^{-2\sqrt3}$ of the placing: at the width r and the crossing distance $\mathsf{d}_\times = \frac38\mathsf{d}' \in [6r, 1]$, with the corners of the two kinds at least $\frac12\mathsf{d}' \ge \mathsf{d}_\times/\sqrt3$ apart, its conditions hold as in the proof of Lem. 4.39. By the support rule (Lem. 4.6(d)) a domain of the level $i+1$ that holds corners of both kinds has a linear size at least $D_{i+1} = [\mathsf{d}'/2Ma_{K-i-1} - 2]_+$.

The tower. Let $n_0 \le i < K$, $A' \in \mathsf{Cc}_{i+1}(r_{i+1})$ and $\nu = \nu_{i,A'}$. The tower identity at several sources (step 1 of the proof of Prop. 4.16; the two clusters, the near sources and the piece, lie in the frame at the level $n_0 - 1$, as $4M\rho_0 \le \mathsf{d}'$), at the five partitions of $\{1,2,3\}$ and the two of $\{T,3\}$, with every $G_{i;\beta}$ taken at $A^{\mathrm{lw}}(B)$, gives

$$\Gamma_{i+1}(A') = \int \Gamma_i \,\mathrm{d}\nu + \kappa_\nu(\Delta_i, G_{i;3}) + \mathsf{c}\big[\kappa_\nu(G_{i;13}, G_{i;2}) + \kappa_\nu(G_{i;23}, G_{i;1}) + \kappa_\nu(G_{i;1}, G_{i;2}, G_{i;3})\big].$$

As $A^{\mathrm{lw}}(B) \in \mathsf{Cc}_i(r_i)$ and ν is a probability measure, the mean is at most $\sup_{\mathsf{Cc}_i(r_i)}|\Gamma_i|$; as $\mathsf{Cc}_K(r_K) = \{0\}$, $|\Gamma_K(\mathbf{1})| = |\mathsf{c}\mathsf{J}_3 - \mathsf{J}_2|$ is at most $\sup_{\mathsf{Cc}_{n_0}(r_{n_0})}|\Gamma_{n_0}|$ plus the sum over $n_0 \le i < K$ of the suprema of the four cumulants.

The covariance of Δ and of the far source. By the oscillations: the explicit parts of Δ_i cancel by the moment condition of step (iii) of the proof of Lem. 4.36, and Lem. 4.37 bounds the oscillation of Δ_i on $\mathsf{Cc}_i(r_i)$ by $2\varepsilon_\alpha r_i^2 a_{K-i}^{-4}$; Lem. 4.10 and $|\mathsf{ang}_p^{[i]}| \le 2\eta_i^2 r_i$ bound that of $G_{i;3}$ by $C\mu_3 r_i^2 a_{K-i}^{-4}$; with $x_i r_i^2 \le C\mathsf{b}_i^2$ and $a_{K-i}x_i \ge \frac45\mathsf{b}$, $|\kappa_\nu(\Delta_i, G_{i;3})| \le C\varepsilon_\alpha \mathsf{b}_i^4 \mathsf{b}^{-2} a_{K-i}^{-6}\mu_3$. By the norms: $\kappa_\nu(\Delta_i, G_{i;3})$ is the value of Sum of the mixed second derivative of the step in the array of Δ_i, of the size at most Ca_{K-i}^{-4} (Lem. 4.6, and Lem. 4.9 in the form (f)), and in the array of the far source; this family vanishes on the domains with $d_{i+1}(X) < D_{i+1}$, and Lem. 4.6(b), the sizes of Lem. 4.9(a), in the form (f), and Lem. 4.8(b) give $|\kappa_\nu(\Delta_i, G_{i;3})| \le C\mathsf{b}^{-1}a_{K-i}^{-7}e^{-D_{i+1}}\mu_3$.

The smaller of the two bounds is at most their geometric mean, $C\varepsilon_\alpha^{1/2}\mathsf{b}_i^2\mathsf{b}^{-3/2}a_{K-i}^{-13/2}e^{-D_{i+1}/2}\mu_3$; with $\mathsf{b}_i = A_1 y_i^{\mathfrak{p}_0}$ and $y_i \le \log(4\mathsf{b}/3a_{K-i})$ the sum over i is led by the levels with a_{K-i} about d' and gives the first term below.

The crossing terms. $\kappa_\nu(G_{i;1}, G_{i;2}, G_{i;3})$ is the value of Sum of the third derivative of the step in the arrays of the three single sources, and $\kappa_\nu(G_{i;13}, G_{i;2})$, $\kappa_\nu(G_{i;23}, G_{i;1})$ are the values of Sum of its mixed second derivatives in the arrays of a crossing set and of a single source; these families vanish on the domains with $d_{i+1}(X) < D_{i+1}$. Lem. 4.6(b) at $q = 3$, the sizes of Lem. 4.9(a), in the form (f), and Lem. 4.8(b) give $\mathsf{c}|\kappa_\nu(G_{i;1}, G_{i;2}, G_{i;3})| \le C\mathsf{c}\,\bar\nu_{i;123}\varsigma_i^3 e^{-D_{i+1}} \le C\mathsf{b}^{-1/2}r^3 a_{K-i}^{-21/2}e^{-D_{i+1}}\mu_3$, the widths of the three sources being of the order of r; the two others are smaller, by a factor γ_i (Lem. 4.9(e), in the form (f)). The sum over i, led by the levels with a_{K-i} about d', gives the second term below.

The start. At the configurations of $\mathsf{Cc}_{n_0}(r_{n_0})$, whose pairs lie in the cores,

$$\Gamma_{n_0} = \sum_{i \le n_0} \mathsf{Sum}[\mathsf{c}E_{123}^{(i)} - E_{T3}^{(i)}];$$

the first display of Lem. 4.39, with

$$\bar\nu_{i-1;\beta} \le \bar c_m (5L/4\mathsf{b})^{m/2} a_{K-i}^{-m/2} \prod_{a\in\beta} |w_a|_\infty$$

and

$$e^{-D_i^\sharp/2} \le e \cdot e^{-\mathsf{d}'/4Ma_{K-i}},$$

summed over $i \le n_0$, where the quotient of two consecutive terms is at most $L^{3/2}e^{-(L-1)} < 1$ as $4M\rho_0 \le \mathsf{d}'$, gives

$$|\Gamma_{n_0}| \le C(\mathsf{b}/\rho_0)^{3/2} e^{-\mathsf{d}'/4M\rho_0} \mathsf{b}^{-2}\mathsf{d}'^{-6}\mu_3,$$

at most the third term below.

The result is

$$|\Gamma_K(\mathbf{1})| \le C\Big[\Big(\frac{\mathsf{b}}{\mathsf{d}'}\Big)^{1/2}\mathsf{L}_{\mathsf{d}'}\varepsilon_\alpha^{1/2} + \Big(\frac{\mathsf{b}}{\mathsf{d}'}\Big)^{3/2}\Big(\frac{r}{\mathsf{d}'}\Big)^3 + \Big(\frac{\mathsf{b}}{\rho_0}\Big)^{3/2} e^{-\mathsf{d}'/8M\rho_0}\Big]\mathsf{b}^{-2}\mathsf{d}'^{-6}\mu_3, \qquad \mathsf{L}_{\mathsf{d}'} = \Big(1 + \log^+\frac{\mathsf{b}}{\mathsf{d}'}\Big)^{2\mathfrak{p}_0},$$

where $\varepsilon_\alpha \le C(g,\mathsf{b})\epsilon$ is the number of Lem. 4.37. Then $\varepsilon_\alpha^{1/2} \le C(g,\mathsf{b})[r^{\beta_{\mathrm{H}}/72} + (a_K/r)^{1/18}]$, $(r/\mathsf{d}')^3 \le \mathsf{d}'^{-3} r^{\beta_{\mathrm{H}}/72}$, $e^{-s} \le 2s^{-2}$ at $s = \mathsf{d}'/8M\rho_0$ with $\rho_0 < Lr^{1/2}$, and $(1-\frac{1}{64})\mathsf{d} \le \mathsf{d}' \le 6\mathsf{d} \le 1$ give the claim. □

Lemma 4.41 (Uniformity in the label)**.** Let $0 < g \le \gamma_{\mathrm{R}}$ and let $\mathcal{K} \subset (g^{-2}, \bar{\mathsf{b}}]$ be compact. Then $C_{\mathrm{A}}(g,\mathsf{b})$, $\mathsf{C}_{\mathrm{R}}(g,\mathsf{b};\mathsf{d})$ and $\mathsf{C}(\mathsf{d})$ of Thm. K are bounded, and $r_\alpha(g,\mathsf{b})$, $r_{\mathcal{W}}(g,\mathsf{b},\mathsf{d})$, $r_\sharp$, $r_0(\mathsf{b})$, $r_1(\mathsf{b})$ are bounded away from 0, uniformly in $\mathsf{b} \in \mathcal{K}$; here $r_\sharp$ and $\mathsf{C}(\mathsf{d})$ are the numbers that steps 2 and 3 of the proof of (E.3) form at a first-passage datum at g with the label b, for every $\mathsf{b} \in (g^{-2}, \bar{\mathsf{b}}]$. ■

Proof. The thresholds. r_1 does not increase in b, so $r_1(\mathsf{b}) \ge r_1(\bar{\mathsf{b}}) > 0$, and the infimum of r_0 over the labels at g is positive by (B.1). The threshold $r_\alpha(g,\mathsf{b})$ is the largest number below which the inequalities (α1)–(α5) of Def. 4.1(5) hold, and the bound $r \le r_\alpha(g,\mathsf{b})$ enters Lem. 4.36 and the lemma on the chain classes through these inequalities; by the bounds given there each of them holds as soon as r is below a number formed from the constants and g, or, for (α3), below $2\min\{\inf_{\mathsf{b}'} r_0(\mathsf{b}'), r_1(\bar{\mathsf{b}})\}$; so $r_\alpha(g,\mathsf{b}) \ge r_\alpha^-(g) > 0$ for every label at g. $r_{\mathcal{W}}(g,\mathsf{b},\mathsf{d})$ adds to r_α the inequalities of step 0 of the proof of (E.2), which depend on b only through $a_\sharp = 4\mathsf{b}/5x_\sharp \ge 4g^{-2}/5x_\sharp$; and $r_\sharp$ is the least of these thresholds and of those of the data. So all the thresholds are bounded below by positive numbers formed from the constants, β_{H}, g and d, for all labels in $(g^{-2}, \bar{\mathsf{b}}]$.

The constants. The label enters the proofs of Lems. 4.36–4.40 and of Thm. K through five kinds of quantities. (1) The run: by Lem. 4.3(a), $\frac{4}{5}\mathsf{b} \le a_{K-j}x_j \le \frac{4}{3}\mathsf{b}$, so that x_j, y_j, δ_j, ε_j, $\mathfrak{b}_j$, $\gamma_j = c_\gamma x_j^{-1/2}$, $\theta^{[j]}$ and r_j lie between two functions of b/a_{K-j}, each continuous in b, and $\frac{3}{4\mathsf{b}} \le \upsilon_j \le \frac{5}{4\mathsf{b}}$. (2) The prefactor $\mathsf{c} = \frac{2\pi}{3}\mathsf{b}r^3$ and the units $\mathsf{u}_a = \mathsf{b}^{-1}|w_a|_\infty$. (3) The slices, through b^{-1} and $\bar{B}/\mathsf{b}$ (Lem. 4.17). (4) The number $a_\sharp = 4\mathsf{b}/5x_\sharp$, through $a_*^{-1} \le L\max\{8M, 5x_\sharp/4\mathsf{b}\}\ell_\epsilon/\mathsf{d}$. (5) The thresholds, bounded below as above.

Member by member: $\gamma_{n_0} \le c_\gamma(5Lr^{1/2}/4\mathsf{b})^{1/2}$; $s(\frac{r}{2})\theta^{[0]}$, with $x_0 = \mathsf{b}/a_K$; $\sum_i \hat{r}_i^3 \mathfrak{b}_i^2(\delta_i + \epsilon_i^{\mathrm{cu}})\Lambda_i^2\mathsf{h}_i^2$, with $x_i \ge \frac{4}{5}\mathsf{b}/a_{K-i}$, while $\Lambda_i \le C(1+Y_{r/2})$ and $\mathsf{h}_i \le C(1+Y_{r/2})^{2\mathfrak{p}_0}$ are formed with g and r; the cost, at most $C\mathsf{b}\,r\,\mathrm{nat}_n^{\mathcal{W}}$, of the domains that leave the cube of the gauge, and the part of the steps between two levels, of the same order (proof of Lem. 4.36); $y_{n_0} \le \log(4\mathsf{b}/3\rho_0)$; the bracket of the proof of Lem. 4.40, through $(\mathsf{b}/\mathsf{d}')^{1/2}$, $(\mathsf{b}/\mathsf{d}')^{3/2}$, $(\mathsf{b}/\rho_0)^{3/2}$ and $\mathsf{L}_{\mathsf{d}'}$; the size $b_\Delta = (C\mathsf{b}+C)\Lambda_3$ of Y_Δ, the terms T_1–T_4 and the moments M_b of the proof of (E.2); the number $\epsilon_0(g,\mathsf{b})$, the reciprocal of the constant in $\sum_i \epsilon_i^{\mathrm{e}} \le C(g,\mathsf{b})\,\epsilon$; the quotients that form C_{R}; and the constant $C(\mathsf{b},\mathsf{d})$ of step 3 of the proof of (E.3).

Each majorant so formed is a sum of products of a factor that does not contain b and of a function of b made by finitely many products, powers, logarithms and maxima from b, b^{-1}, the constants, β_{H}, g, d and the lower bounds of the thresholds; such a function is continuous on the compact interval $[g^{-2}, \bar{\mathsf{b}}]$, where $g^{-2} \ge e^{2\mathfrak{p}_0} > 1$, and so bounded there. Hence $C_{\mathrm{A}}(g,\mathsf{b})$, $\mathsf{C}_{\mathrm{R}}(g,\mathsf{b};\mathsf{d})$ and $\mathsf{C}(\mathsf{d})$ are bounded on all of $(g^{-2}, \bar{\mathsf{b}}]$ by numbers formed from the constants, β_{H}, g, d and $\mathsf{d}\,\mathrm{Lip}\,\varphi_1/\|\varphi_1\|_\infty$, in particular on $\mathcal{K}$. □

Proofs of the theorems.

Proof of Thm. H. (B.2). 1. *The data.* By Lem. 4.5(b) the tents put the pair in the frame at $k_0 = \mathsf{k}(r)$; (Γ1) gives $g \le \gamma_{\mathrm{jt}}$ and $g^{-2} \ge x_{\mathrm{J}}$, so Lem. 4.6 applies, with (P1)–(P3), and $\rho \le r_1(\mathsf{b})$ gives $\rho \ge 4Ma_K$ and $E_\sigma(\hat\rho_1/2) \le \varkappa_0^4$; the smallness holds at all heights (Lem. 4.3(b)). At $n = K$, $A = 0$ and both members vanish; let $\mathsf{k}(r) \le n < K$.

2. *The decomposition.* Differentiate the defining formula for D_n^λ twice: the source term is linear in λ, so $\mathsf{F}_n = \sum_{i\le n} \mathsf{Sum}[E_{12}^{(i)}]$; with Lem. 4.7 and $q_n = \sum_{i=\mathsf{k}(r)}^{n-1} \mathcal{Q}_i$, $q_{\mathsf{k}(r)} = 0$,

$$\mathsf{F}_n(\tilde U) - \mathsf{F}_n(\mathbf{1}) - q_n(\mathsf{ang}^{[n]}(A)) = \underbrace{\sum_{i=1}^{\mathsf{k}(r)} \Delta[E_{12}^{(i)}]}_{\mathrm{I}} + \underbrace{\sum_{j=\mathsf{k}(r)}^{n-1} \Delta[\mathsf{R}^{(j+1)}]}_{\mathrm{II}} + \underbrace{\sum_{i=\mathsf{k}(r)}^{n-1} \left(\Delta[\mathsf{K}^{(i+1)}] - \mathcal{Q}_i(\mathsf{ang}^{[n]}(A))\right)}_{\mathrm{III}}.$$

3. *The passage to the yardstick* $\mathsf{Y}_n = (\rho/r)^6 \hat r_n^3 \mathsf{u}_1\mathsf{u}_2$. For a pointed family f of level $j+1$ that vanishes off Cr_1 and on the domains with $d_{j+1}(X) < D'$, with $\|\mathsf{f}\| \le \Phi\bar\nu_j$, Lem. 4.8(a) in the frame at $k_0 = \mathsf{k}(r)$ of step 1, $\bar\nu_j \le \frac{25}{16}\bar c_2 x_j \mathsf{u}_1\mathsf{u}_2$, $x_j \le \frac53 L^{n-j} x_n$ and $x_n r_n^2 \le C\mathfrak{b}_n^2$ give

$$|\Delta[\mathsf{f}]| \le C_{\mathrm{U}}\, \mathfrak{b}_n^2\, \mathsf{N}_{j+1}(w_1)\, \Phi\, u_j^{-3} e^{-D'/2}\, u_n^3 \mathsf{u}_1\mathsf{u}_2.$$

4. *I, the levels at and below the working level, are cut by the distance.* By the support rule $D' = \mathsf{D}(u_i)$; Lem. 4.9(b) and (P2) give $\mathsf{N}_i\bar\Psi_{i-1;12} \le Cq^6(1+u_i^6)$; with $u_i = L^l u_{\mathsf{k}(r)}$ and $u_{\mathsf{k}(r)}/\sqrt{3}M \ge \frac{4}{\sqrt3} Y_r^{2\mathfrak{p}_0}$ the sum is led by $i = \mathsf{k}(r)$: $|\mathrm{I}| \le C\mathfrak{b}_n^2 e^{-Y_r^{2\mathfrak{p}_0}/\sqrt3} \mathsf{Y}_n$.

5. *II, the rest families.* Step 3 with $D' = 0$ and Lem. 4.9(c); $\mathsf{N}_{j+1}\varsigma_j^2 u_j^{-3} \le 216L^3q^6u_{j+1}^3$, $\mathsf{N}_{j+1}\Xi_j^r u_j^{-3} \le 216q^6$, $\sum_{j<n}\gamma_j \le 0.45\gamma_n \le C\delta_n/\mathfrak{b}_n$, and $u_{\mathsf{k}(r)}^3[\gamma_{\mathsf{k}(r)} + \theta^{[0]}] \le CY_r^{6\mathfrak{p}_0}\delta_{\mathsf{k}(r)}$, $\delta_{\mathsf{k}(r)} \le A_1 Y_r^{\mathfrak{p}_0}(r/\mathsf{b})^{1/2}$ (Lem. 4.3(c)):

$$|\mathrm{II}| \le \{C\mathfrak{b}_n\delta_n + C\mathfrak{b}_n^2 Y_r^{7\mathfrak{p}_0}(r/\mathsf{b})^{1/2}\}\mathsf{Y}_n.$$

6. *III, the kernel families.* Prop. 4.13, narrow form; $\mathsf{c}_i \le Ce^{-\delta_{\mathrm{tp}}u_i/\sqrt3}$, so $\sum_i \mathsf{c}_i(1+u_i)^2u_i^3 \le C$, and $\sum_{i\ge\mathsf{k}(r)} \mathfrak{b}_i^4\delta_i u_i^3 \le 1.12\,\mathfrak{b}_{\mathsf{k}(r)}^4 u_{\mathsf{k}(r)}^3\delta_{\mathsf{k}(r)}$:

$$|\mathrm{III}| \le \{CY_r^{15\mathfrak{p}_0+2}(r/\mathsf{b})^{1/2} + C\mathfrak{b}_n^2 r_n\}\mathsf{Y}_n.$$

7. *The sum.* $\mathfrak{b}_n \le \mathfrak{b}_{\mathsf{k}(r)} \le A_1 Y_r^{\mathfrak{p}_0}$, $r_n \le \frac85 \mathsf{c}_{\mathfrak{b}}\delta_n$ and $(r/\mathsf{b})^{1/2} \le g\, r^{1/2}$ give the bound. The factor $(\rho/r)^6$ costs nothing: F_n is bilinear in (w_1, w_2), every estimate depends on the weights only through $\sum_p |w_a(p)|$ or through a norm that (P1) bounds, and the one place where a sup norm could enter, Λ_i, sees a quotient (Prop. 4.13, step 6). By Lem. 4.3(c), $\log\frac1r \le Y_r \le 2\log\frac1r + C(g)$, whence $\epsilon'_{\mathrm{N}}(r)(\log\frac1r)^N \to 0$.

(B.1). 1. Lem. 4.5(a) and its proof put the test pair in all six conditions of Def. 4.4 at $k_0 = \max\{K-s, 2\}$, under $3 \le s \le K$ and $\mathsf{Lm}_s \le L^{s-3}$, as required in (B.1); there the number $\theta^{[j]}/K_{\mathbf{M}}$ of every step is at most half of the largest value that the one-step theorem admits, $\hat r_{k_0} \ge M$, so $\varsigma_i = 1$ for $i \le k_0$, and $\mathsf{D}_i^{12} \ge \hat r_i/\sqrt3 - 1$. 2. Below k_0, Prop. YM3_I:122.2 gives $\|E_{12}^{(j)}\| \le C\nu_{j-1}$, with ν_j defined in [7, eq. 199]; its running, half-ceiling and separation hypotheses are those verified in the pair frame of Lem. 4.5(a); from k_0 on, Lem. 4.6(b), (c) and the start at k_0 give the rest families with $\|\mathsf{R}^{(j+1)}\|_{j+1,\lambda_{\mathrm{out}}} \le \epsilon_j^{(2)}\varsigma_j\bar\nu_j$,

$$\epsilon_j^{(2)} \le C[x_j^{-1/2}(1+\mathsf{e}_\downarrow) + \theta^{[k_0]}], \qquad \mathsf{e}_\downarrow = \sum_{i\le k_0} e^{-4\beta_{\mathrm{s}}\kappa[\mathsf{D}_i^{12}/M-2]_+}\varkappa_{i-1}^{-2}:$$

no number in it grows with $\log\frac1r$ or with $n-k_0$, as the derivatives of the step gain the factor $1/\mathfrak{b}_j$ per order in the sources (Lem. 4.6(b)) and the lags of Lem. 4.6(c) fall like L^{-4l} and sum geometrically over the levels. 3. The decomposition of step 2 above, over the base level k_0, with Lem. 4.8(a),

in the frame at k_0, for I and II and Prop. 4.13 at the test pair for III, both in the form stated after them, whose conditions are the frame, the smallness at the heights concerned ($2r_n \le \varrho_{\rm ch}$, so that $r_n \le \frac{1}{120}$ as $\varrho_{\rm ch} \le \frac{1}{60}$; $\epsilon_i^{\rm cu} \le 1$; $x_n^{-1} \le r_n^2$) and the side bound $\ell \ge 2LMy_K$: for a test pair at $k_0 = \max\{K-s, 2\}$ these hold for $g \le \min\{\gamma_{\rm jt}, \gamma_{\rm O}\}$ and $r \le r_0(\mathsf{b})$, the frame by Lem. 4.5(a), the side bound as $\ell \in \mathfrak{L}(g)$, and the smallness at the heights $\ge k_0$ being among the conditions that fix $\gamma_{\rm O}$ and $r_0(\mathsf{b})$.

The shape ϑ does not enter, since any $\bar{\mathsf{e}} \ge 2\mathsf{w}_1\mathsf{w}_2\mathsf{E}_i$ is admissible. 4. In the yardstick $\hat r_n^3\mathsf{u}_1\mathsf{u}_2$ the main part of II is $\le C\mathfrak{b}_n^2[\gamma_n(1+\mathsf{e}_\downarrow) + (n-k_0)\theta^{[k_0]}]$, III is $\le C\hat r_{k_0}^3\mathfrak{b}_{k_0}^4(\delta_{k_0}+\epsilon_{k_0}^{\rm cu})\Lambda^2\mathsf{h}^2 + C\mathfrak{b}_n^2 r_n$, by the bound $\mathsf{E}_i \le C_{\mathsf{E}}\delta_i^4(\delta_i+\epsilon_i^{\rm cu})$ of Prop. 4.13, and I is $\le C\mathfrak{b}_n^2 e^{-c'Y_s^{2\mathfrak{p}_0}}$. With $\theta^{[k_0]} \le C(r/\mathsf{b})^{1/2}y_{k_0}^{\mathfrak{p}_0}$, $K-k_0 \le s \le Cy_{k_0}$ and $\mathsf{e}_\downarrow \le Cy_{k_0}^{2\mathfrak{p}_0}e^{-c'Y_s^{2\mathfrak{p}_0}}$, and the smallness that the recursions need holding for $g \le \gamma_{\rm O}$ and $r \le r_0(\mathsf{b})$, this is the bound of (B.1); its last term $C\mathfrak{b}_n^2 r_n \le C\mathfrak{b}_n^2\delta_n$ is the price of writing the form with the linear angles of one chart. 5.

The conditions that fix $r_0(\mathsf{b})$ depend on b through $(r/\mathsf{b})^{1/2}$, through $y_{k_0} \le \log\frac{\mathsf{b}}{r} + C\log\log\frac{\mathsf{b}}{r}$ and through $x_{k_0} \le \frac{24}{23}\mathsf{b}\hat r_{k_0}/r$; on $(g^{-2}, \bar{\mathsf{b}}]$ these are bounded by their values at the two ends, so r_0 has a positive infimum over the labels at g.

(B.3). $|\mathsf{ang}_p^{[n]}(A)| \le 2\eta_n^2 r_n$, Lem. 4.17(a) with $\mathsf{r} \ge r/\sqrt{3}$, and (P1):

$$|q_n(\mathsf{ang}^{[n]}(A))| \le C\,r_n^2\,\mathsf{b}\,a_{K-n}^{-4}\rho^6 r^{-3}\mathsf{u}_1\mathsf{u}_2 = C\,r_n^2\,\mathsf{b}\,a_{K-n}^{-1}(\rho/r)^6\hat r_n^3\mathsf{u}_1\mathsf{u}_2 \le C\mathfrak{b}_n^2(\rho/r)^6\hat r_n^3\mathsf{u}_1\mathsf{u}_2,$$

as $\mathsf{b}\,a_{K-n}^{-1} \le \frac{5}{4}x_n$ and $x_n r_n^2 \le C\mathfrak{b}_n^2$. □

Proof of Thm. G. 1. *(A.1).* Prop. 4.19(a) bounds J_3 against the lattice ring ($64\mathsf{d} \le \ell_g$ as $\ell_g \ge 26$ and $\mathsf{d} \le e^{-34}$; $\rho \le r_1(\mathsf{b})$ by $\mathsf{d} \le \mathsf{d}_4(g) \le r_1(\bar{\mathsf{b}}) \le r_1(\mathsf{b})$), and Lem. 4.21 bounds $S_3^T - \mathsf{J}_3$ by $13\Lambda_3^3\mathsf{M}'^3\mathsf{b}^3\mathsf{d}^{3/2}\mathrm{nat}_3$ (all supports lie within $20\mathsf{d} \le \frac{1}{8}\ell_g - 1$ of x_1).

2. *The geometry of (A.2).* For a triple $(z_1,z_2,z_3) = (x,y,z)$ of $\mathcal{D}$ put $r = \frac{1}{2}r_*$, $\mathsf{d} = \frac{1}{2}\mathsf{d}_*$; bumps f_a with supports in balls of radius $\rho < \min\{\frac{1}{4}r_*, r_1(\bar{\mathsf{b}})\}$ inside the open balls of radius $\frac{1}{4}r_*$ about the z_a are placed at (ρ, r, d): their supports are $\ge r_* - \frac{1}{2}r_* = r$ apart in the pair and $\ge \mathsf{d}_* - \frac{1}{2}r_* \ge \mathsf{d}$ across, with $|x_1 - x_2| \le 3r$, $|x_a - x_3| \le 5\mathsf{d}$, all in $\mathbb{B}(z_1, \frac{9}{4}\mathsf{d}_*)$; this holds for every ratio $r_*/\mathsf{d}_* \in (0,1]$, and $\mathsf{d}_* \le \mathsf{d}_5(g)$ gives $\mathsf{d} \le \mathsf{d}_4(g)$.

3. *Lattice to continuum.* Lem. 4.22(a) with $\mathsf{d}^+ = 5\mathsf{d}_*$ and a Lipschitz number ϑ of the f_a:

$$|3\,\mathrm{Tr}\prod_a((f_a\circ x)\mathsf{b}^{-1}C_K^{\rm free}) - 3\mathsf{b}^{-3}\int k_3\prod f_a| \le 3C^{(3)}[a_K/\vartheta\rho + (5\mathsf{d}_*/\ell_g)^3]\mathrm{nat}_3.$$

4. *The limit.* Along first-passage data with $K_j \to \infty$ and labels $\to \mathsf{b}$, $S_3^T(f) \to S_3^{(\ell_g,\mathsf{b})}(f)$ (Thm. YM3_I:A), $a_{K_j}/\vartheta\rho \to 0$, and the brackets are bounded with $\bar{\mathsf{b}}$ and $\bar{\mathsf{M}}^{(24)}$ (Lem. 4.20(e)); with $r^{-3}\mathsf{d}^{-6} = 512\,r_*^{-3}\mathsf{d}_*^{-6}$ this is the bound of the form in (A.2).

5. *Kernels.* Lem. 4.23(a) on $\prod_a U_a$, with $(\frac{3}{\pi})^3 \le 1$: a Borel kernel, bounded at its Lebesgue points by the bound of the form at the centre; the cover of $\mathcal{D}$ by such products and the glueing of Lem. 4.30 give one kernel on $\mathcal{D}$; the closest pair of a triple is (x,y) after a relabelling. *(A.3)* is Lem. 4.23(b).

6. *(A.4).* (i) is Prop. 4.28(a) at $n=3$. (ii) For a triple with $\mathsf{d}_* \le \mathsf{d}_5(g)$, the bumps of Lem. 4.29(c) about it are in the data of (i) at every large j (the cube of side $\frac{9}{2}\mathsf{d}_* \le \frac{1}{2}\ell_g$ about x holds them, so the projection keeps all distances and the integrals of k_3); (i), step 4 at the label b, and $S_3^T[\nu_j] \to S_3^{\sigma'}$ give $|\int(\mathsf{s}_3 - 3\mathsf{b}^{-3}k_3)\prod f_a^\rho| \le [\epsilon_{\rm A} + 512C_3^{\rm tr3}\mathsf{b}^3\mathsf{d}_*^{3/2}]\mathsf{b}^{-3}r_*^{-3}\mathsf{d}_*^{-6}\cdot 0.102 \le \mathsf{E}_3(g,\mathsf{d}_*)\mathsf{b}^{-3}r_*^{-3}\mathsf{d}_*^{-6}$, and Lem. 4.29(c) gives the bound at the point. (iii) is Lem. 4.29(a).

7. *(A.5).* Let (K_j,β_j,ν_j) be a Gibbs sequence at g_0; its members are periodic Gibbs states, $\nu_j \in \mathcal{G}^{\rm per}_{K_j,\beta_j}$. For large j, β_j exceeds $L(g^{-2}+B_\sharp)$; let K_j' be the first-passage index of β_j at g and $z_j = K_j - K_j'$; as $g \le g_0$, $0 \le z_j \le z_+$ with z_+ formed from g_0, g and the constants. (1) *The states.* For $0 \le z < K$ the dilation $U \mapsto U^{(z)}$, $U^{(z)}(L^z b) = U(b)$, maps the bonds, the plaquettes and the

finite sets of bonds of $\mathbb{L}_K$ onto those of $\mathbb{L}_{K-z}$, with $a_{L^z p}(U^{(z)}) = a_p(U)$. The Gibbs kernel of a finite set of bonds has the exponent $-\beta\sum_p a_p$ over the plaquettes that contain a bond of the set, which reads β and the lattice and not its spacing; so the image $\nu^{(z)}$ of a Gibbs state at (a_K, β) satisfies the equations of the kernels at (a_{K-z}, β), and invariance under a subgroup Γ' of finite index of $a_K\mathbb{Z}^3$ becomes invariance under $L^z\Gamma'$, of finite index in $a_{K-z}\mathbb{Z}^3$.

So the dilation carries $\mathcal{G}^{\mathrm{per}}_{K,\beta}$ onto $\mathcal{G}^{\mathrm{per}}_{K-z,\beta}$ (equation YM3_I:(1819) and the definition of periodic states following it), and $\nu_j^{(z_j)}$ is a periodic Gibbs state at the first-passage datum (K'_j, β_j) at g, with the label $L^{z_j}a_{K_j}\beta_j$. (2) *The fields.* At every configuration U of $\mathbb{L}_K$, $x(L^z p) = L^z x(p)$ and $a_{K-z} = L^z a_K$ give $\mathcal{O}(f)(U) = L^z\mathcal{O}'(f^{(z)})(U^{(z)})$, $f^{(z)} = f(L^{-z}\cdot)$, the prime marking the field at the cutoff $K - z$, as in Lem. 4.31(d), which is the dilation; so in every law ν on $\mathbb{L}_K$, $S_3^T[\nu](f) = L^{3z}S_3^T[\nu^{(z)}](f^{(z)})$. (3) *The approximants* of a component of a canonical decomposition of the limit are periodic Gibbs states at the same data, by the definition of the components and their approximants in Lem. 4.29(a); (1) and (2) apply to them. (4) *The limit.* On a subsequence $z_j = z$ is constant and the labels converge, to b_0 say; along it $S_3^T[\nu_j^{(z)}](f) = L^{-3z}S_3^T[\nu_j](f(L^z\cdot)) \to L^{-3z}S_3^\sigma(f(L^z\cdot))$ on separated triples, σ the limit, and this functional has the continuous kernel $\mathsf{s}_3^{(z)} = L^{-12z}\mathsf{s}_3(L^{-z}\cdot)$; likewise for a component and its approximants.

The proof of (A.4)(ii) uses of the law σ' the convergence $S_3^T[\nu_j] \to S_3^{\sigma'}$ on separated triples and the continuity of the kernel of $S_3^{\sigma'}$ at the triple (Lem. 4.29(c)); so (A.4)(ii) at g applies to the dilated sequence, whose labels tend to $L^z\mathsf{b}_0$, at the triples with $L^z\mathsf{d}_* \le \mathsf{d}_5(g)$. In kernels $\mathsf{s}_3(x) = L^{12z}\mathsf{s}_3^{(z)}(L^z x)$, and the natural size is invariant: $L^{12z}(L^z\mathsf{b}_0)^{-3}(L^z r_*)^{-3}(L^z\mathsf{d}_*)^{-6} = \mathsf{b}_0^{-3}r_*^{-3}\mathsf{d}_*^{-6}$, $k_3(L^z\cdot) = L^{-9z}k_3$. The approximants of a component are periodic Gibbs states at the same couplings and are dilated alike. □

Proof of Thm. I. 1. *(C.1).* At $x, y = c \pm \frac{r}{2}n$: $r_* = r$; the two distances $|c - z \pm \frac{r}{2}n|$ lie in $[\mathsf{d} - \frac{r}{2}, \mathsf{d} + \frac{r}{2}]$ and the sum of their squares is $2\mathsf{d}^2 + \frac{1}{2}r^2$, so $\mathsf{d}_* \in [\frac{3}{4}\mathsf{d}, \frac{17}{16}\mathsf{d}]$, $r \le \mathsf{d}_* \le \mathsf{d}_5(g)$. (A.2) multiplied by $\frac{2\pi}{3}\mathsf{b}r^3$ gives $\frac{2\pi}{3}(\frac{4}{3})^6 \le 12$ times $\bar\epsilon_{\mathrm{A}}\mathsf{b}^{-2}\mathsf{d}^{-6}$, and Lem. 4.24(c) gives $2.66 \le 2.7$; "almost everywhere" is in (c, h, z), a linear change of the variables (x, y, z). (ii) (A.3) at every triple; k_3 is continuous off the diagonals, so its means tend to its values; with tents, (A.3) in the tent form and the second part of Lem. 4.24(c). (iii) (A.4)(ii) at every triple and Lem. 4.24(c).

2. *(C.2)(i).* Prop. 4.19(b) bounds J_2; Lem. 4.21 at two sources gives

$$|S_2^T - \mathsf{J}_2| \le 3\Lambda_3^2\mathsf{M}'^2\mathsf{b}^2\mathsf{d}^{3/2}(\rho/\mathsf{d})^6\mathsf{b}^{-2}|w_1|_\infty|w_2|_\infty$$

; at the plane weights $w_1 = \mathbf{1}_{\{\lambda(p)=\lambda\}}\varphi_1 \circ x$, $w_2 = \mathbf{1}_{\{\lambda(p)=\mu\}}\varphi_2 \circ x$, Lem. 4.22(c) turns the lattice square into $\mathcal{K}_{\lambda\mu}$ ($8a_K \le \rho$ by $s(\rho) \le K$; the assumption $|x_1 - x_2| \le 18\mathsf{d}$ is the one that this step and the rings need).

3. *(C.2)(ii).* For $(x_0, y_0) \in \Omega_{\mathrm{P}}$ and bounded Borel f, h supported in the closed balls of radius $\frac{1}{8}\mathsf{d}_0$ about x_0, y_0, the jets, the remainder, $|C_K^{\mathrm{free}}(p, q)| \le (1 + 3C_{\mathrm{lat}})a_K^3|u|^{-3}$ and the count of plaquettes give

$$|\langle\mathcal{O}_\lambda(f); \mathcal{O}_\mu(h)\rangle_{K_j,\ell_g,\beta_j}| \le \mathsf{B}(g, \mathsf{d}_0)\|f\|_\infty\|h\|_\infty, \qquad \mathsf{B} = 4^{-6}g^4[\tfrac{3}{2}\cdot 729(1 + 3C_{\mathrm{lat}})^2 + \mathsf{E}_2(g, \tfrac{1}{2}\mathsf{d}_0)],$$

with no Lipschitz number (the Riemann sums of step 2 involve $a_K/\vartheta\rho$ and give no bound in the sup norms). Lem. 4.30 gives one subsequence for all pairs of balls; along it step 2, with $a_{K_j}/\vartheta\rho \to 0$, gives the bump form at every centre: the balls of radius $\frac{1}{8}\mathsf{d}_0$ are $\ge \frac{1}{2}\mathsf{d}_0 =: \mathsf{d}$ apart, $|x_1 - x_2| \le \frac{5}{2}\mathsf{d}$, and $(\rho/\mathsf{d})^6 = 64(\rho/\mathsf{d}_0)^6$. Lem. 4.23 at $p = 2$ gives the kernels and the almost everywhere bound, and Lem. 4.24(a) gives $t_\lambda - \mathsf{t}^\circ(e_\lambda, \cdot) = \sum_\mu(\delta_{\lambda\mu} - \frac{1}{3}\sum_{\lambda'}\delta_{\lambda'\mu})$ with $\delta_{\lambda\mu} = \tau_{\lambda\mu} - \frac{3}{2}\mathsf{b}^{-2}[\mathbb{K}_{3,\lambda\mu}]^2$: three terms and a third of nine, so the factor 6.

4. *(C.3).* Along the subsequence of the limit point $S_3^T \to S_3^{(\ell_g,\mathsf{b})}$, since that limit is taken along all data with labels $\to \mathsf{b}$. By (A.3) and by Lem. 4.23(b) applied to the bump form of (C.2)(ii)

at the centre (c,z), the means over balls of radius ρ of $\mathsf{s}_3 - 3\mathsf{b}^{-3}k_3$ and of $\tau_{\lambda\mu} - \frac{3}{2}\mathsf{b}^{-2}[\mathbb{K}_{3,\lambda\mu}]^2$ are bounded at every centre, with the factors $(\frac{3}{4\pi})^3, (\frac{3}{4\pi})^2 \le 1$; k_3 and $[\mathbb{K}_3]^2$ are continuous off the diagonals; adding with Lem. 4.24(a), (c) gives the display, and the tents likewise. The cone form: (C.1)(i) at n, (C.2)(ii) at (c,z) almost everywhere (a null set of (c,z) is a null set of (c,h,z)), and $|\mathsf{t}^\circ(n,X) - \mathsf{t}^\circ(e_\lambda,X)| \le 0.057|n-e_\lambda|\mathsf{b}^{-2}|X|^{-6}$.

5. *(C.4).* (d) is Prop. 4.28(b) with step 2 on the least torus (the projection is injective and keeps distances on $\mathbb{B}(x_0,\frac{3}{2}\mathsf{d}_0)$, so the integrals and the norms are those of the periodised functions); it gives the bound $\mathsf{B}(g,\mathsf{d}_0)$ of step 3 along ν_j for j large. (a) Lem. 4.30 on Ω_σ. (b) The bump form at every centre of Ω_σ with $64\mathsf{E}_2(g,\frac{1}{2}\mathsf{d}_0)\mathsf{b}^{-2}\mathsf{d}_0^{-6}\rho^6$; Lem. 4.23; the bound at a point by centres $\zeta_m \to (x,y)$ of distances $\mathsf{d}_{\zeta_m} < |x-y|$, whence the supremum in ϵ_σ. Every ν_j is invariant under the translations of $\mathbb{L}_{K_j}$ by Prop. YM3_I:103.4, so the limit forms are invariant under all translations, and by Fubini's theorem $\tau^\sigma_{\lambda\mu}(x,y)$ is a function of $x-y$ almost everywhere. $\sum_{\lambda,\mu}\langle\mathcal{O}_\lambda;\mathcal{O}_\mu\rangle = S_2^T$, which converges along the whole sequence to $S_2^{\sigma'}$ with the continuous kernel s_2^T; with Lem. 4.24(a) this gives the bound of s_2^T almost everywhere, and at every point by continuity, $\epsilon_\sigma(g,\cdot)$ being nondecreasing. (c) Let $\mathcal{N}$ be the union of the null sets of (b); it is fixed before r is named.

Off $\mathcal{N}$, (b) and Lem. 4.24(a) give $|t^\sigma_\lambda - \mathsf{t}^\circ(e_\lambda,\cdot)| \le 6(\frac{3}{\pi})^2\epsilon_\sigma\mathsf{b}^{-2}\mathsf{d}^{-6} \le 6\epsilon_\sigma\mathsf{b}^{-2}\mathsf{d}^{-6}$, and (C.1)(iii) at $n=e_\lambda$ holds at every point, since s_3 is continuous; $12\sup\mathsf{E}_3 + 6\epsilon_\sigma = \epsilon'_\sigma$. With tents of radius $\rho < \frac{1}{8}\mathsf{d}$ at (c,z): the bump form of (b) at the centre (c,z), and Lem. 4.24(d) times $\frac{3}{2}$, six times, give $21\rho/\mathsf{d}$. □

Proof of Thm. J. 1. *(D.1) below* γ_N is Prop. 4.34. 2. *(D.1) at every* $g \le \gamma$. Let z, g'_1, g'_2 be as in Lem. 4.31 and $K_\circ(g) = z + \max_\iota s'_\iota(r_{g'_\iota})$. For $K \ge K_\circ(g)$, $(K-z,\beta)$ is a first-passage datum at some $g'_\iota \le \gamma_\mathrm{N}$, on the side $L^z\ell \in \mathfrak{L}(g'_\iota)$, and the conditions of Prop. 4.34 hold at $(L^z r, L^z\mathsf{d}')$; by Lem. 4.31(d) each display at (K,ℓ,β) is that display at $(K-z,L^z\ell,\beta)$. So (D.1) holds with

$$\epsilon^{\mathrm{cl}}(g,\ell,\mathsf{d}') = \max_{\iota=1,2}\epsilon^{\mathrm{ev}}_\sharp(g'_\iota,L^z\ell,L^z\mathsf{d}'), \qquad \mathsf{d}_{\mathrm{cl}}(g) = \tfrac{7}{6}L^{-z}\min_\iota t_\mathrm{N}(g'_\iota),$$

a finite function of (g,ℓ,d'), with the rate for $\mathsf{d}' \le \mathsf{d}_{\mathrm{cl}}(g)$. On the dilated torus, at the distance $L^z\mathsf{d}'$, the members of $\epsilon^{\mathrm{ev}}_\sharp$ that carry the renormalisation group, the remainder and the induction, read the distances through the floor $\min\{\frac{6}{7}L^z\mathsf{d}', t_\mathrm{N}(g'_\iota)\}$ alone; the largest distance $\frac{8}{7}L^z\mathsf{d}'$ enters the one explicit member, the comparison of the sum of the slices with the free kernel (Lem. 4.17(b)), which is polynomial in that distance; and the passage from the floor to $L^z\mathsf{d}'$ costs the factor $\max\{1, 6L^z\mathsf{d}'/7t_\mathrm{N}(g'_\iota)\}^6$ of $\epsilon^{\mathrm{ev}}_\sharp$ (proof of Prop. 4.34). 3.

(D.2). The forms are bounded in the sup norms at every pair of small balls in Ω°, as in step 3 of the proof of Thm. I; Lem. 4.30 gives one subsequence for the plane and the bond forms; (D.1) with $a_{K_j}/\vartheta r \to 0$ gives the bump forms at every centre; Lem. 4.23, with centres in the window $[\frac{7}{9}t, \frac{9}{7}t]$ and $(\frac{3}{\pi})^2(\frac{9}{7})^6 \le 4.2$, gives the kernels. For T_n: the kernel of $\langle T_\lambda;\mathcal{O}\rangle$ is within $3(4.2+4.2)\bar\epsilon^{\mathrm{cl}} = 25.2\,\bar\epsilon^{\mathrm{cl}}$ of $\mathsf{t}^\circ(e_\lambda,\cdot)$, that of $\langle\mathcal{O}^{\mathrm{od}}_{\lambda\lambda'};\mathcal{O}\rangle = \sum_\gamma\langle\mathcal{O}^{\mathrm{od}}_{\lambda\lambda'};\mathcal{O}_\gamma\rangle$ within $12.6\,\bar\epsilon^{\mathrm{cl}}$ of $3\mathsf{b}^{-2}[\mathbb{K}^2_3]_{\lambda\lambda'}$; with $\sum_\lambda n_\lambda^2 = 1$, $\sum_{\lambda<\lambda'}|n_\lambda n_{\lambda'}| \le 1$ and Lem. 4.24(a), $25.2 + 12.6 \le 38$.

The kernel t_n is a quadratic polynomial in n whose coefficients are the nine kernels, so one null set serves all n. 4. *(D.3)* is (C.1) and (D.2) by the triangle inequality: almost everywhere in (c,h,z) by Fubini's theorem; for the means and the tents, the bump forms of (D.1) at the centre (c,z) itself, extended to indicators of balls by Lem. 4.23(b), bound the means of the kernels with factors below those of the almost everywhere bound. □

Proof of Thm. K. (E.2). 0. *The stopping level.* Let $\epsilon = r^{\beta_\mathrm{H}/36} + (a_K/r)^{1/9}$, $\ell_\epsilon = \max\{1, \log\frac{1}{\epsilon}\}$, $a^\flat = \min\{\mathsf{d}/8M\ell_\epsilon, a_\sharp\}$ with $a_\sharp = 4\mathsf{b}/5x_\sharp$, so that $a_{K-n} \le a_\sharp$ gives $x_n \ge x_\sharp$ (Lem. 4.3(a)), n_* the largest level with $a_{K-n_*} \le a^\flat$, and $a_* = a_{K-n_*} \in (a^\flat/L, a^\flat]$. The conditions that define $r_\mathcal{W}(g,\mathsf{b},\mathsf{d})$, namely $r \le \min\{r_\alpha, (\mathsf{d}/5LM)^2, e^{-1}\}$, $8ML^{m_1+2}r^{1/2}(1+\log\frac{1}{r}) \le \mathsf{d}$, $L^{m_1+2}r^{1/2} \le a_\sharp$ and $r^{-1/4} \ge 2M$,

secure $N_0 < n_*$, $x_{n_*} \geq x_\sharp$ and $L^{N_0-n_1} \geq 2M$, as $L^{N_0-n_1} = L^{m_1} a_{K-n_0}/a_{K-n_1} \geq r^{-1/4}$ (so Lem. 4.36 holds with $n^\sharp = n_*$ when $\epsilon_i^{\mathrm{e}} \leq 1$), $8Ma_* \leq \mathsf{d}$, $e^{-\mathsf{d}/4Ma_*} \leq \epsilon^2$ and $a_*^{-1} \leq C\ell_\epsilon/\mathsf{d}$.

1. *The identity.* Lem. 4.20(c) at $k = n_*$, $\mathbb{P}_* = \mathbb{P}_{n_*}$, for the two lists, G_3 being the same function in both:

$$\mathsf{c}S_3^T(w_1,w_2,w_3) - S_2^T(w_T,w_3) = \int \Gamma \,\mathrm{d}\mathbb{P}_* + \kappa_{\mathbb{P}_*}(\Delta, G_3) + \mathsf{c}[\kappa_{\mathbb{P}_*}(G_{13},G_2) + \kappa_{\mathbb{P}_*}(G_{23},G_1)] + \mathsf{c}\,\kappa_{\mathbb{P}_*}(G_1,G_2,G_3) =: T_1+T_2+T_3+T_4,$$

with Δ, Γ, G_β taken at $U_{t'}(W)$.

2. *Sizes.* $r^3|w_{1,2}|_\infty \leq 489$, $|w_T|_\infty \leq \frac{2}{3}|w_1|_\infty$, $\mathsf{c}|w_1|_\infty|w_2|_\infty r^3 \leq C\mathsf{b}$, $|w_3|_\infty \leq \mu_3 r^{-3}$; the bounds of Lem. 4.21 hold at every point of the small-field region, as their proof uses of W only that the background is regular.

3. *T_1, T_3: the crossings.* At $t' \leq n_*$, $8Ma_{K-t'} \leq \mathsf{d}$, and Lem. 4.39 with $e^{-\mathsf{d}/4Ma_*} \leq \epsilon^2$ gives $|T_1| \leq C(\mathsf{b})a_*^{-3/2}\mathsf{d}^{-6}\epsilon^2\mu_3$ and $|T_3| \leq Ca_*^{-1}r^3\mathsf{d}^{-6}\epsilon^2\mu_3\mathsf{M}_6$, $\mathsf{M}_b = \int \alpha^{-b} d\mathbb{P}_*$: no negative power of r is left.

4. T_4. $|T_4| \leq C\mathsf{b}r^3\Lambda_3^3\mathsf{M}_{18}\mu_3$.

5. *T_2: pair minus tensor field.* $T_2 = \int(\Delta - \mathsf{c}_{n_*})(G_3 - \int G_3)d\mathbb{P}_*$, and $|G_3 - \int G_3| \leq \Lambda_3\mu_3(\alpha^{-6} + \mathsf{M}_6)$ by step 2. On the parts with $t' \geq N_0$ each measure is absolutely continuous with respect to $\mathsf{m}_{t'}$, and Lem. 4.36 gives $|\Delta - \mathsf{c}_{n_*}| \leq 3.01\,\tilde{\varepsilon}\,\alpha^{-4}$ almost everywhere. The number c_{n_*}: the first part of $\mathbb{P}_*$, on which $t' = n_*$ and $\alpha = a_*$, has positive mass, as the density of m_{n_*} on $\mathcal{V}_{n_*}^\chi$ is a positive exponential by [7, eq. 224] and [7, eq. 225]. This region is open, its defining tests are strict, and they hold at the trivial field, where all curvature and fluctuation defects vanish. It therefore has positive Haar measure; at almost every point of it $|\Delta - \mathsf{c}_{n_*}| \leq \varepsilon^{\mathcal{W}} a_*^{-4}$ (Lem. 4.36) and $|\mathsf{Y}_\Delta| = |\Delta - \Delta_K(\mathbf{1})| \leq b_\Delta a_*^{-6}$, $b_\Delta = (C\mathsf{b} + C)\Lambda_3$ (Lem. 4.21 at every point, step 2), so that $|\mathsf{c}_{n_*} - \Delta_K(\mathbf{1})| \leq (\varepsilon^{\mathcal{W}} + b_\Delta)a_*^{-6}$.

On the parts with $t' < N_0$, $\alpha < a_{K-N_0} = L^{m_1}a_{K-n_0} < L^{m_1+1}r^{1/2} < a_*$, so that there $|\Delta - \mathsf{c}_{n_*}| \leq |\mathsf{Y}_\Delta| + |\Delta_K(\mathbf{1}) - \mathsf{c}_{n_*}| \leq (2b_\Delta + \varepsilon^{\mathcal{W}})\alpha^{-6}$ and $\mathsf{M}_6 \leq (1 + \mathsf{M}')\alpha^{-6}$; and $\int_{\{\alpha<\varrho\}} \alpha^{-12} d\mathbb{P}_* \leq \varrho^{12}\mathsf{M}'$ (Lem. 4.20(e)) at $\varrho = a_{K-N_0}$ gives a factor Cr^6.

6. *Together.* $\mathsf{M}_b \leq a_*^{-b} + \mathsf{M}' \leq C(g,\mathsf{b})\ell_\epsilon^b\mathsf{d}^{-b}$, $\epsilon^{1/2}\ell_\epsilon^{10} \leq C$ and $r^{3-\beta_{\mathrm{H}}/72}(1 + \log\frac{1}{r})^{18} \leq C$ give $|\mathsf{c}S_3^T - S_2^T| \leq C_{\mathrm{A}}(g,\mathsf{b})\mathsf{d}^{-18}[r^{\beta_{\mathrm{H}}/72} + (a_K/r)^{1/18}]\mu_3$. If the smallness $\epsilon_i^{\mathrm{e}} \leq 1$ of the single-source arrays of the near pair, which Lem. 4.36 asks at the levels $\mathsf{k}(\frac{r}{2}) \leq i < n_1$, fails, then, as $\sum_{\mathsf{k}(r/2)\leq i<n_0}\epsilon_i^{\mathrm{e}} \leq C_\omega[2.2\gamma_{n_0} + \frac{1}{2}s(\frac{r}{2})\mathsf{c}^\flat\theta^{[0]}] \leq C(g,\mathsf{b})\,\epsilon$ (with $\gamma_{n_0} \leq C(\mathsf{b})r^{1/4}$ and $s(\frac{r}{2})\theta^{[0]} \leq C(\mathsf{b})(a_K/r)^{1/4}$), the number ϵ exceeds a number $\epsilon_0(g,\mathsf{b}) > 0$, the bracket is $\geq (\frac{1}{2}\epsilon_0)^{1/2}$, and the bounds of Lem. 4.21 give the claim with a larger C_{A}.

(E.1). By Lem. 4.20(c) for the two lists,

$$[\mathsf{c}S_3^T - S_2^T] - [\mathsf{c}\mathsf{J}_3 - \mathsf{J}_2] = \int \mathsf{Y}_{\Delta 3} d\mathbb{P} + \kappa_{\mathbb{P}}(\mathsf{Y}_\Delta, \mathsf{Y}_3) + \mathsf{E}, \qquad \mathsf{E} = \mathsf{c}[\kappa_{\mathbb{P}}(\mathsf{Y}_{13},\mathsf{Y}_2) + \kappa_{\mathbb{P}}(\mathsf{Y}_{23},\mathsf{Y}_1) + \kappa_{\mathbb{P}}(\mathsf{Y}_1,\mathsf{Y}_2,\mathsf{Y}_3)],$$

and Lem. 4.21 gives $|\mathsf{E}| \leq C\Lambda_3^3\mathsf{M}'\mathsf{b}^3\mathsf{d}^{3/2}r^3 \cdot \mathsf{b}^{-2}\mathsf{d}^{-6}\mu_3$. For $r \leq r_{\mathcal{W}}$ the first bracket is bounded by (E.2) and the second by Lem. 4.40 (if $w_3 \equiv 0$ all members vanish): together $C(g,\mathsf{b})\mathsf{d}^{-12}[r^{\beta_{\mathrm{H}}/72} + (a_K/r)^{1/18}]\mathsf{b}^{-2}\mathsf{d}^{-6}\mu_3$, and $|\mathsf{E}|$ is of this form as $r^3 \leq r^{\beta_{\mathrm{H}}/72}$. For $r > r_{\mathcal{W}}$, Lem. 4.21 bounds the left member of (E.1) by $C(\mathsf{b})\bar{\mathsf{M}}^{(24)}[\mathsf{d}^{-9/2} + 1]\mu_3$, while the bracket is $\geq r_{\mathcal{W}}^{\beta_{\mathrm{H}}/72}$; the larger of the two quotients so formed is $\mathsf{C}_{\mathrm{R}}(g,\mathsf{b};\mathsf{d})$.

(E.1) under (Γ3). The stronger bounds stated in (E.1) are Lem. 4.38, at the levels $n \geq n_0$, that is at the levels with $a_{K-n} \geq r^{1/2}$, with $\mathsf{d}' = |c - z| \in [(1 - \frac{1}{64})\mathsf{d}, 6\mathsf{d}]$. Let $r \leq r_\alpha^{\mathcal{W}}(g,\mathsf{b})$. On the part of Ω with $t' = n \geq n_0$ the measure $\mathbb{P}$ is absolutely continuous with respect to m_n, $\mathsf{Y}_{\Delta 3} = \Gamma_n(U_n(W)) - \Gamma_K(\mathbf{1})$ and, with $\mathsf{c}_* = \mathsf{c}_K - \Delta_K(\mathbf{1})$, $\mathsf{Y}_\Delta - \mathsf{c}_* = \Delta_n(U_n(W)) - \mathsf{c}_K$, so that Lem. 4.38 bounds both by $\varepsilon^{\mathcal{W}}$ times their natural sizes, $\mathrm{nat}_n^{\mathcal{W}} \leq \alpha^{-4}$; $\kappa_{\mathbb{P}}(\mathsf{Y}_\Delta, \mathsf{Y}_3) = \int(\mathsf{Y}_\Delta - \mathsf{c}_*)(\mathsf{Y}_3 - \int \mathsf{Y}_3 d\mathbb{P})\,\mathrm{d}\mathbb{P}$, and $|\mathsf{c}_*| \leq C(g,\mathsf{b})$, since $\mathbb{P}(\alpha < \alpha_1) \leq \frac{1}{2}$ for $\alpha_1 = (2\bar{\mathsf{M}}^{(24)})^{-1/24}$ (Lem. 4.20(e)), while $a_{K-n_0} < Lr^{1/2} \leq \alpha_1$, so that on

the set $\{\alpha \ge \alpha_1\}$, of positive measure, both $|\mathsf{Y}_\Delta - \mathsf{c}_*| \le 2\varepsilon^{\mathcal{W}}\alpha_1^{-4}$ and the bound of Y_Δ by Lem. 4.21 hold.

The parts with $t' < n_0$ have $\alpha < a_{K-n_0} < Lr^{1/2}$, where nothing cancels and the moments of Lem. 4.20(e) give the factors r^6 and r^9; for $r > r_\alpha^{\mathcal{W}}$ Lem. 4.21 bounds the left member while the bracket is $\ge (r_\alpha^{\mathcal{W}})^{\beta_{\mathrm{H}}/72}$. Together the left member of (E.1) is at most $C(g,\mathsf{b})\mathsf{d}^{-5/2}[r^{\beta_{\mathrm{H}}/72} + (a_K/r)^{1/18}]\mathsf{b}^{-2}\mathsf{d}^{-6}\mu_3$.

(E.3). 1. For $\mathsf{b} \in I(g)$ every $(K, L^K\mathsf{b})$ is a first-passage datum at g with the label b. 2. *The pieces.* Cut φ_2 by the indicators of the cubes of side $\frac{r}{4}$; at most $9^3(\mathsf{d}/r)^3$ of them meet its support, and as $|y_1 - y_2| \le 3\mathsf{d}$ the centre z of each has $|c - z| \le 5\mathsf{d} + \frac{\sqrt{3}}{8}r \le 6\mathsf{d}$ for $c \in \operatorname{supp}\varphi_1$, so every piece is a w_3 of the data. By (E.2), summed over the pieces and integrated against $\varphi_1(c)\,\mathrm{d}c$, $\int|\varphi_1| \le \frac{4\pi}{3}\mathsf{d}^3\|\varphi_1\|_\infty$:

$$\Big|\mathsf{A}_{r,K} - \int \mathrm{d}c\,\varphi_1(c)\,S_2^T(w_T^c,\varphi_2)\Big| \le C(g,\mathsf{b})\mathsf{d}^{-12}\big[r^{\beta_{\mathrm{H}}/72} + (a_K/r)^{1/18}\big]\|\varphi_1\|_\infty\|\varphi_2\|_\infty.$$

3. *The smeared tensor.* $\int \mathrm{d}c\,\varphi_1(c)\mathcal{O}(w_T^c) = T_\lambda(\varphi_1 * \chi^{r/8})$ exactly, $\|\varphi_1 * \chi^{r/8} - \varphi_1\|_\infty \le \frac{r}{8}\operatorname{Lip}\varphi_1$, and the two-point bound of Lem. 4.21 in the sup norms (supports at a distance $\ge \frac{1}{2}\mathsf{d}$, after cutting $\varphi_1 * \chi^{r/8} - \varphi_1$ into pieces) gives $|\langle T_\lambda(\varphi_1 * \chi^{r/8} - \varphi_1); \mathcal{O}(\varphi_2)\rangle| \le C(\mathsf{b},\mathsf{d})\,r\operatorname{Lip}\varphi_1\|\varphi_2\|_\infty$; this is (i), with $r_\sharp$ the least of the thresholds of (E.2) and of the data. Steps 2 and 3 read the datum through (E.2) and Lem. 4.21 at it; so (i) holds, with the same proof, at every first-passage datum (K,β) at g whose label $\mathsf{b}' = a_K\beta$ lies in $(g^{-2},\bar{\mathsf{b}}]$, with $\mathsf{A}_{r,K}$ formed at (K,β) with the prefactor $\frac{2\pi}{3}\mathsf{b}'r^3$ and with the numbers $r_\sharp$ and $\mathsf{C}(\mathsf{d})$ formed at b'. 4.

(ii). At a fixed r, $\mathsf{A}_{r,K} \to \mathsf{A}_r$ as $K \to \infty$: for each c, $S^T_{3;K,\ell_g,\beta_K}(f_+^c, f_-^c, \varphi_2) \to S_3^{(\ell_g,\mathsf{b})}(f_+^c, f_-^c, \varphi_2)$ by Thm. YM3_I:A (Lipschitz functions with disjoint supports), with a bound uniform in c and K (Lem. 4.21 and Prop. 4.19), and dominated convergence in c. Hence, by (i),

$$\limsup_{K,K'\to\infty}\big|\langle T_\lambda(\varphi_1);\mathcal{O}(\varphi_2)\rangle_{K,\ell_g,\beta_K} - \langle T_\lambda(\varphi_1);\mathcal{O}(\varphi_2)\rangle_{K',\ell_g,\beta_{K'}}\big| \le 2\mathsf{C}(\mathsf{d})r^{\beta_{\mathrm{H}}/72}\mathsf{b}^{-2}\|\varphi_1\|_\infty\|\varphi_2\|_\infty,$$

and $r \to 0$ shows that the sequence is a Cauchy sequence; $s(\frac{r}{2}) \le K$ and $128a_K \le r$ hold for all large K, so the conditions of (i) on the cutoff hold along the sequence. 5. *(iii)*: $K \to \infty$ in (i). As g satisfies (Γ), the limit points of (C.2) along data with $\beta_j = L^{K_j}\mathsf{b}$ all give the form t_λ on these pairs.

(E.4). Let $\mathsf{b}_j = a_{K_j}\beta_j \to \mathsf{b} \in I(g)$, and let $\mathsf{A}_r^{(j)}$ be $\mathsf{A}_{r,K}$ formed at (K_j,β_j) with $\frac{2\pi}{3}\mathsf{b}_j r^3$. As $g^{-2} < \mathsf{b} < \bar{\mathsf{b}}$, for j large the labels lie in a compact subset of $(g^{-2},\bar{\mathsf{b}}]$. The bound (i), in its form at every label of $(g^{-2},\bar{\mathsf{b}}]$ (step 3 above; a label b_j may lie outside $I(g)$), holds at (K_j,β_j) with the prefactor $\frac{2\pi}{3}\mathsf{b}_j r^3$, and by Lem. 4.41 its constants and thresholds are uniform in j. At a fixed r, $\mathsf{A}_r^{(j)} \to \mathsf{A}_r$: the prefactor $\frac{2\pi}{3}\mathsf{b}_j r^3$ tends to $\frac{2\pi}{3}\mathsf{b}r^3$; for each c, $S^T_{3;K_j,\ell_g,\beta_j}(f_+^c, f_-^c, \varphi_2) \to S_3^{(\ell_g,\mathsf{b})}(f_+^c, f_-^c, \varphi_2)$ by Thm. YM3_I:A, one limit along all data whose labels tend to b; and the bounds of Lem. 4.21 on $|S_3^T - \mathsf{J}_3|$ and on $|\mathsf{J}_3|$, summed over the pieces of φ_2, are uniform in c and in j, as $\mathsf{b}_j \le \bar{\mathsf{b}}$, so that dominated convergence applies in c. So

$$\limsup_j\big|\langle T_\lambda(\varphi_1);\mathcal{O}(\varphi_2)\rangle_{K_j,\ell_g,\beta_j} - t_\lambda(\varphi_1,\varphi_2)\big| \le \limsup_j\big(|\langle T_\lambda;\mathcal{O}\rangle_j - \mathsf{A}_r^{(j)}| + |\mathsf{A}_r^{(j)} - \mathsf{A}_r|\big) + |\mathsf{A}_r - t_\lambda| \le C\,r^{\beta_{\mathrm{H}}/72},$$

and $r \to 0$. □

Proof of Thm. L. (a) Prop. 4.19(c), Lem. 4.21 at $n = 4$, and Prop. 4.28(a) at $n = 4$. (b) The assumptions of (b) give $\operatorname{supp} f_a \subset \mathbb{B}(x_a,\rho)$, with the balls $\mathbb{B}(x_a,\rho)$ at least r apart in each pair and at least d apart across, as Lem. 4.22(b) asks; the $w_a = f_a \circ x$ form two pairs at (ρ, r, d) with the centres x_a, with the bounds $|x_1 - x_2|, |x_3 - x_4| \le 8r$ and $|x_a - x_b| \le 18\mathsf{d}$, $a \le 2 < b$, of the data; and $8a_K \le \rho$ by $s(\rho) \le K$ (Lem. 4.5(b)) and $\mathsf{d} \le \mathsf{d}^+$, as (b) notes. The bounds of (a) on $|S_4^T - \mathsf{J}_4|$ and on $|\mathsf{J}_4 - \frac{3}{2}\sum_\varsigma \operatorname{Tr}\prod_l(w_{\varsigma(l)}\mathsf{b}^{-1}C_K^{\mathrm{free}})|$, the identity $\frac{3}{2}\sum_\varsigma \operatorname{Tr}\prod_l(w_{\varsigma(l)}\mathsf{b}^{-1}C_K^{\mathrm{free}}) =$

$3\mathsf{b}^{-4}\cdot\frac{1}{2}\sum_{\varsigma}\operatorname{Tr}\prod_{l}((f_{\varsigma(l)}\circ x)C_K^{\mathrm{free}})$ and Lem. 4.22(b) give (b). (c) For a point of $\mathcal{D}_4$ take the open balls of radius $\frac{1}{4}r_*$ about the x_a, $r=\frac{1}{2}r_*$, $\mathsf{d}=\frac{1}{2}\mathsf{d}_*$, and bumps f_a supported in balls $\mathbb{B}(x'_a,\rho)$ of radius ρ that lie inside them: these balls are $\geq r$ apart in a pair and, since $r_*\leq\mathsf{d}_*$, $\geq\mathsf{d}$ apart across, as (b) asks, and the $f_a\circ x$ form two pairs at (ρ,r,d) with the centres x'_a (in a pair the centres are $\leq\frac{5}{2}r_*=5r$ apart, across $\leq 11\mathsf{d}$), all inside a ball of radius $6\mathsf{d}_*=\frac{1}{2}\mathsf{d}^+$; then (b), the limit $K\to\infty$ at the label b (Thm. YM3_I:A), Lem. 4.23 at $p=4$ with $(\frac{3}{\pi})^4<1$, and $r^{-6}\mathsf{d}^{-6}=4096\,r_*^{-6}\mathsf{d}_*^{-6}$.

The rings (1234), (1243) of k_4 have the size $r_*^{-6}\mathsf{d}_*^{-6}$, the ring (1324) the size d_*^{-12}. □

4.1.8. *Scope.* Every statement of this section is made at short distances, on one torus or in states at a cutoff and in their limits, and the constants that depend on the side grow with it. The side of the torus enters through conditions that ask it to be large against the lengths in play, and through constants that grow with it, among them: the conditions that the supports lie in a ball small against it; the conditions that the boxes and the cubes of the local gauges do not wrap around the torus, namely $4\rho+4$ at most the side in Lem. 4.12, in the units of the level, the case distinction of step 5 of the proof of Prop. 4.13 on whether its box fits in the torus, which reads $\ell\geq 8$, $(4\mathsf{h}_i+4)a_{K-i-1}\leq\ell_g$ in (α2) and in Def. 4.1(6), and $12a_{K-n}\leq\ell$, $(8\varrho+2)a_{K-n}<\ell$ for the regular gauge on a cube, read at $\varrho=4$ in Lem. 4.36 and at $\varrho=3$ up to $n=K$ in the proof of the bounds of (E.1) under (Γ3), where $26<\ell_g$; the bound $L^m\geq LMy_K$ of the sides of $\mathfrak{L}(g)$, which gives $\ell_g\geq 338$, in the tails of Lem. 4.8, in Prop. 4.13, in Lem. 4.40 and for $26<\ell_g$; the distance $\frac{1}{4}\ell_g\geq 6.5$ of two mirror copies in Lem. 4.27; the membership $\ell\in\mathfrak{L}(g)$ wherever the objects of the theory are entered, as in the frame of Lem. 4.5; the terms in d/ℓ of the comparison of C_K^{free} with $\mathbb{K}_3$; the moments of Lem. 4.20(e); and the constant Q_0 of Lem. 4.25.

In the bond forms the members that carry the renormalisation group, the remainder and the induction of the proof of Prop. 4.34, read the distances through the floor $\min\{\frac{6}{7}\mathsf{d}',t_{\mathrm{N}}(g)\}$; the largest distance $\frac{8}{7}\mathsf{d}'$ enters the one explicit member, the comparison of the sum of the slices with C_K^{free} (Lem. 4.17(b)), which is polynomial in that distance and a part of $\epsilon_{\sharp}^{\mathrm{ev}}$. Thms. YM3_I:AJ and YM3_I:AD enter through the two-sided stability bound of the torus measures and the chessboard bound of Lem. 4.25, in (A.4), (A.5), (C.1)(iii), (C.4) and for the states of Thm. L(a); Thm. K, like every statement of this section, uses Thms. YM3_I:D, YM3_I:AJ and YM3_I:AD.

The crossing functionals of Lem. 4.39 are joint cumulants of the steps given the block fields of a scale below d. The comparisons with t° and with the kernels t_λ and t_n are additive, in the unit $\mathsf{b}^{-2}\mathsf{d}^{-6}$; $\mathsf{t}^\circ(n,X)$ vanishes on the cone $3(n\cdot\hat{X})^2=1$. On a torus a kernel is a class modulo null sets, while at fixed (c,z) the triples $(c\pm\frac{r}{2}n,z)$ form a null set; there (C.1)–(C.3) and (D.3) are statements almost everywhere, or through means, tents and cones of directions. For the laws on $\mathbb{R}^3$ satisfying the continuity hypotheses in (A.4)(ii) and (C.4), including the periodic-Gibbs limits and their canonical components, s_3 and s_2^T are continuous, and the kernels $\tau_{\lambda\mu}^\sigma$ and t_λ^σ are classes modulo null sets: (C.1)(iii) holds at every point and for every unit vector n; (C.4)(c) holds for s_3 itself, without means, for every $0<r\leq\frac{1}{2}\mathsf{d}$ and every λ, at every (c,z) off one null set fixed before r is named, and at every (c,z) with the limit forms at the tents χ_c^ρ,χ_z^ρ, $0<\rho<\frac{1}{8}\mathsf{d}$, in the place of $t_\lambda^\sigma(c,z)$ and $21\rho/\mathsf{d}$ added in the bracket.

The kernels $\tau_{\lambda\mu}$, t_λ, t_n of (C.2)–(C.4) and (D.2) belong to limit points, each chosen once for all pairs of bumps; at a label of $I(g)$, for $g\leq\gamma_{\mathrm{R}}$ and on the least torus, the forms $\langle T_\lambda(\varphi_1);\mathcal{O}(\varphi_2)\rangle$ of (E.3) have one limit along all cutoffs.

Step 9: narrow bumps; the kernels of the plane fields. In this step, as in Lems. [7, Lem. C.15] and [7, Lem. C.16], in [7, Thm. BV](e), in Thm. B(c), (d) and in Step 14(c), (d), the distance of two sets is the Euclidean one, $\operatorname{dist}(A,B)=\inf_{x\in A,y\in B}|x-y|$, and the separations of sets are read with it; the factor $1/\sqrt{3}$ between it and the sup distance is paid inside the proofs of these lemmas, and the numbers $\frac{7}{8}d'$, $\frac{6}{7}d'$, $\frac{3}{2}t$, $\frac{5}{2}t$, $3t$, $(2r/d')^6$ and $E_2(g,\frac{1}{2}d')$ below are those of this reading. Let

first g be below the threshold of [7, Lem. C.15], $\ell \in \mathfrak{L}(g)$, and let the two bumps be read as mixed sources carried by $B(x_a, r)$, $|x_1 - x_2| = d'$, $16r \le d' \le \frac{\ell}{8}$; their balls are $\ge \frac{7}{8}d'$ apart.

For d' below a threshold depending on g, [7, Lem. C.15](a), (b) at the separation $d = \frac{6}{7}d'$, the distance $d' = \frac{7}{6}d$ of the centres lying between d and $18d$, as (a) asks, and below $8d$, as (b) asks: the cumulant of the two bumps is its free lattice value up to $\varepsilon_2(g,d)(r/d)^6 \mathsf{b}^{-2}|\mathfrak{s}_1||\mathfrak{s}_2|$ plus the part of the large fields, which on T_ℓ is at most $C(g,\ell)(r/d)^6 d^{3/2}\mathsf{b}^{-2}|\mathfrak{s}_1||\mathfrak{s}_2|$; and $(r/d)^6 = (\frac{7}{6})^6(r/d')^6$. Distances d' above that threshold, up to $\frac{\ell}{8}$: the argument of [7, Lem. C.15](a), (b) at a fixed small separation $d = t(g)$, a threshold depending on g, asks no ceiling on the actual distance of the two balls but that they lie in a ball of radius $< \frac{1}{4}\ell$, and gives a tolerance that is a finite function of that distance; where the bumps are wider than the widths that the law admits at the separation d, both members being bilinear, the sources are cut into pieces carried by the cubes of a grid of a mesh depending on g, the number of pairs of pieces being at most a basic multiple of the sixth power of the width over the mesh, and the sum of the laws of the pairs of pieces costs, against $(r/d')^6$, the finite factor $(d'/d)^6$ times a basic number.

So $\varepsilon^{\mathrm{cl}}(g,\ell,\cdot)$ is a finite function on $(0, \frac{\ell}{8}]$, and $\varepsilon^{\mathrm{cl}}(g,\ell,d') \le C(g,\ell)(\log\frac{1}{d'})^{19\mathfrak{p}_0} d'^{1/2}$ for small d', the large fields adding $C(g,\ell)d'^{3/2}$. For Lipschitz bumps the lattice sums against the free kernel, $\sum_{\lambda(p)=\lambda,\lambda(q)=\mu}\phi_1(x(p))\phi_2(x(q))C^{\mathrm{free}}(p,q)^2$, differ from $\iint \phi_1\phi_2[K_3]^2_{\lambda\mu}$ by at most $C_\flat(\frac{a_K}{\vartheta r} + \frac{d'^3}{\ell^3})(\frac{r}{d'})^6|\phi_1||\phi_2|$, by the comparison of the lattice kernel with the free kernel (equations YM3_I:(2245)–(2246)), the term $a_K/(\vartheta r)$ being the Lipschitz defect of a Riemann sum of a bump of width r and $(d'/\ell)^3$ the images on the torus; the off-diagonal bond forms likewise with $C'_\flat$, the factor 8 coming from the four corner forms of a bond and the two directions.

Couplings $0 < g \le \gamma$ above the threshold of [7, Lem. C.15] are reached by the dilation of Step 8, whose arithmetic reads $g^{-2} > \bar{B}$ alone and whose two couplings lie below any basic threshold once $L^z g^{-2}$ is large, at $L^z d'$ and $L^z\ell$, both members of the law being multiplied by L^{-2z}; this defines $\varepsilon^{\mathrm{cl}}$ and $K_\circ(g)$. The tolerance depends on d' alone: there is no gain as $r \to 0$ at fixed d'. In every periodic Gibbs state, for g below a basic threshold: put $t = \frac{1}{2}d'$; bumps of radius r in the balls of radius $\frac{1}{8}d'$ about two points at the distance d' have supports $\ge \frac{3}{2}t$ apart, at centres $\le \frac{5}{2}t$ apart, in one ball of radius $3t \le \frac{1}{8}\ell_g - 1$ when d' is below a threshold depending on g.

On T_{ℓ_g}, [7, Lem. C.15](a) at $(\rho, d') = (r, t)$ gives the law of the plaquette forms with $\varepsilon_2(g,t) + C(g)t^{3/2}$ in units of $(r/t)^6\mathsf{b}^{-2}|\phi_1||\phi_2|$; [7, Lem. C.16](b) at $n = 2$, $d_\times = t$, carries it to every periodic Gibbs state with the further $C_2^{\mathrm{tr}}(g)(r/t)^6 t^{3/2}|\phi_1||\phi_2|$, at most $C(g)t^{3/2}$ in these units, as $\mathsf{b} \le L(g^{-2} + \bar{B})$; and the Riemann sums on T_{ℓ_g} add $\frac{3}{2}C_\flat(\frac{a_K}{\vartheta r} + (\frac{t}{\ell_g})^3)$ in the same units. This is the law in every state with $E_2(g,t) = \varepsilon_2(g,t) + C(g)t^{3/2} + \frac{3}{2}C_\flat(t/\ell_g)^3$ at $t = \frac{1}{2}d'$ for $\varepsilon^{\mathrm{cl}}$, and $(r/t)^6 = (2r/d')^6$; $E_2(g,t) \le C(g)(\log\frac{1}{t})^{19\mathfrak{p}_0}t^{1/2}$.

First take the chosen continuum subsequential limit at each fixed positive width; then let the widths tend to 0. The estimates are uniform in the underlying states. The limits have bounded Borel kernels within $(\frac{3}{\pi})^2\varepsilon_\sigma(g,|x-y|)\mathsf{b}^{-2}|x-y|^{-6}$ of the free ones almost everywhere, by Lebesgue's differentiation theorem, functions of $x - y$ alone by the translations; for a limit at g the kernels are continuous ([7, Thm. BW](d)), so the bound holds at every pair; the sum over λ, μ is the scalar kernel, the trace part being the scalar limit.

4.2. **Sources and the first large-field region.** The comparison in Prop. 6.7 needs a source domain that does not shrink when an early large-field region is created. We construct the source dependence with the real Gaussian structure at every step. Its total constant term agrees near zero with that of the weighted progression; the individual localized families need not agree.

Fix the real canonical run $x_0, \dots, x_K$, its counterterms $e_i = L^{-1}x_{i-1} - x_i$, and its all-small-field histories. We retain the global core, actual-sum-rule, source-radius and normalizer hypotheses of Lem. YM3_I:114.1 and Prop. YM3_I:114.2, together with the canonical history tests of

Prop. YM3_I:123.1. In particular $|e_i| \le B_\sharp$, the source potential $\mathbb{W}^{\mathrm{src}}[v]$ has its actual, rather than merely formal, sum rule, and all steps use their prescribed gauge factors and density tests. Write $E_{\mathrm{s}} = E_0^{\mathrm{s}}$ and $N_{\mathrm{s}} = N_\star^{\mathrm{s}}$. The numerical conditions used below are

$$E_{\mathrm{s}} \ge 2E_{\mathsf{N}}, \qquad N_{\mathrm{s}} \ge \frac{L}{L-1}(E_{\mathrm{s}} + \tfrac{3}{2}\alpha_0^2 B_\sharp),$$

$$K_{\mathbf{M}}\big(\delta(\gamma^{-2}) + \mathsf{s}_\gamma N_{\mathrm{s}} + e^{-\gamma_2 \mathfrak{b}(\gamma^{-2})^2/2}\big) \le E_{\mathrm{s}}/4, \qquad 8K_\Lambda \mathsf{s}_\gamma N_{\mathrm{s}} \le w_\star, \qquad 8K_P \mathsf{s}_\gamma \le w_\star,$$

where $\mathsf{s}_\gamma = (1+\beta_{\mathrm{s}})\delta(\gamma^{-2})/\alpha_0$. We also retain $4\beta_{\mathrm{s}}\kappa \ge 1$ and the normalizer decay bound $\delta_{\mathsf{N}} M \ge \kappa+1$. Put

$$K^{\mathrm{d}} = \tfrac{351}{2} L^3 K_\Pi \le K_{\mathbf{M}}^{\mathrm{f}}, \qquad \mu_\star = \min\{w_\star/32, E_{\mathrm{s}}/(16K_{\mathbf{M}}^{\mathrm{f}})\}, \qquad \sigma_{\mathrm{src}} = \mu_\star / K_{\mathrm{Q}}^\vee.$$

Here $K_{\mathbf{M}}^{\mathrm{f}}$ and $K_{\mathrm{Q}}^\vee, K_{\mathrm{L}}^\vee$ are the constants in that source construction. The level-wise source domain is the open unit ball of

$$\|w\|_{\mathrm{lw}} = \max_{0\le j<K} \max\left\{ \frac{|v_j^w|_\infty}{\sigma_{\mathrm{src}}}, \frac{\mathrm{sw}_j(v_j^w)}{x_j^{-1/2}\mu_\star/K_{\mathrm{L}}^\vee} \right\}, \qquad v_j^w = -L^{K-j} w/x_j. \tag{4.viii}$$

The swing compares bare plaquettes whose barycentres have sup distance at most M in level-j units. Since the amplitude is included, this is a norm on the finite-dimensional complex space of bare plaquette weights.

Proposition 4.42 (The sourced history density)**.** On $\|w\|_{\mathrm{lw}} < 1$ there are local pointed families $F_j[w]$, $1 \le j \le K$, jointly holomorphic in the source and in every open core slice, with

$$\|F_j[w]\|_{j,\kappa+1} \le E_{\mathrm{s}}, \qquad \|\Delta F_j[w]\|_{j,\kappa+1} \le 2E_{\mathrm{s}}\|w\|_{\mathrm{lw}}, \qquad \Delta F_j[w] = F_j[w] - F_j[0]. \tag{4.ix}$$

A member indexed by X depends only on the source plaquettes in X. At zero source these are the ordinary progression's families. Define the finite sums

$$\mathcal{E}_t(w) = \sum_{j\le t} \sum_{X,z} \Delta F_j[w](X, \mathbf{1}, z), \qquad \mathcal{E} = \mathcal{E}_K, \tag{4.x}$$

$$\mathsf{R}_t(w, W) = \mathcal{O}(w)(U_t(W)) + \sum_{j\le t} \sum_{X,z} [\Delta F_j[w](X, U_t(W), z) - \Delta F_j[w](X, \mathbf{1}, z)]. \tag{4.xi}$$

At $t = 0$ set $\mathcal{E}_0 = 0$ and $\mathsf{R}_0(w, U) = \mathcal{O}(w)(U)$. If m_t^w is the all-small-field history measure with bare input $e^{\mathcal{O}(w)}\rho_0\,\mathrm{d}U$, and $m_t = m_t^0$, then

$$m_t^w = e^{\mathcal{E}_t(w) + \mathsf{R}_t(w,\cdot)} m_t. \tag{4.xii}$$

For $t \ge 1$ the right side is evaluated on the passing density set and is zero elsewhere. Moreover, on the common neighborhood of zero in any finite set of source parameters, $\mathcal{E}_t$ equals the total source energy of the weighted progression of Prop. YM3_I:122.1. In particular its mixed second derivative at $t = K$ is J_2. ∎

Proof. Starting with the empty array, use the actual old-potential rule and the canonical localized step at x_j, with $\tau = 1$:

$$\mathfrak{f}_j[w] = ((F_i[w], e_i)_{1\le i\le j}, \emptyset), \qquad F_{j+1}[w] = \mathbf{N}^{(j+1)} + \mathbf{F}[\mathbb{V}'[\mathfrak{f}_j[w]] - \mathbb{W}^{\mathrm{src}}[v_j^w]].$$

This is the real-Gaussian source recursion of Prop. YM3_I:114.2, with the counterterms of the running coupling retained. Their contribution to the array size is explicit:

$$N(\mathfrak{f}_j[w]) = \sum_{i=1}^{j} L^{-(j-i)}(\|F_i[w]\|_{i,\kappa} + \tfrac{3}{2}\alpha_0^2|e_i|) \le \frac{L}{L-1}(E_{\mathrm{s}} + \tfrac{3}{2}\alpha_0^2 B_\sharp) \le N_{\mathrm{s}}.$$

The induction therefore has the same ceilings as that proposition. The old potential has norm at most $w_\star/8$. Lem. YM3_I:114.1 and eq. 4.viii give

$$\mathsf{n}^{\mathrm{f}}(\mathbb{W}^{\mathrm{src}}[v_j^w]) \le 2K_{\mathrm{Q}}^\vee |v_j^w|_\infty + K_{\mathrm{L}}^\vee x_j^{1/2}\,\mathrm{sw}_j(v_j^w) \le 3\mu_\star.$$

Thus the fourth potential has norm at most $7w_\star/32$; the three internal potentials together have norm at most $3w_\star/8$. Lem. YM3_I:77.1 applies. The unperturbed output is bounded by $E_{\mathrm{s}}/4$, the normalizer by $E_{\mathrm{s}}/2$, and the perturbation estimate (1601), p. 631, bounds the source contribution by $3K^{\mathrm{d}}\mu_\star \le 3E_{\mathrm{s}}/16$. The new norm is at most $15E_{\mathrm{s}}/16 < E_{\mathrm{s}}$. The inequalities on the decay rates give the norm with rate $\kappa+1$ used here.

The same step is jointly holomorphic in its potentials and open core variables. Since v_j^w is linear in w, induction proves the asserted holomorphy. The local rule for a member on X uses only earlier members and source plaquettes inside X, proving locality. At $w=0$ the source potential vanishes, so the recursion is exactly the ordinary one. Finally, on the complex line through w the holomorphic difference $F_j[\zeta w/\|w\|_{\mathrm{lw}}] - F_j[0]$ vanishes at zero and has norm at most $2E_{\mathrm{s}}$ for $|\zeta|<1$. Schwarz's lemma proves the second bound in eq. 4.ix; the assertion at $w=0$ is immediate.

We next verify that this recursion gives the same history operator used for the weighted progression. With $j=t-1$, its all-small-field step factor is

$$g_t = (\chi_t \circ \mathcal{M})\,\mathsf{q}_j^0 \mathsf{q}_t \mathfrak{G}_j[x_j]\chi_j^{\mathrm{sf}}, \qquad m_t^w = \mathcal{M}_*(g_t m_j^w).$$

The factors are bounded, nonnegative and Borel, and independent of w. Here $\mathsf{q}_j^0, \mathsf{q}_t$ are the compatibility tests of this history, and χ_t is its full density test. For a passing output W, every point $V_W(B)$ of the actual small-field chart satisfies

$$\mathsf{q}_j^0(V_W(B)) = \mathsf{q}_t(V_W(B)) = \chi_j(V_W(B)) = 1.$$

These are the chart and test properties used in the proof of Prop. YM3_I:123.1. Thus, if $F' = e^c F$ on the passing input set, the image of $g_t F'\,\mathrm{d}V$ has the pointwise chart density $\chi_t(W)e^c \mathsf{I}_j[F](W)$. Indeed, write the image integral against a bounded Borel test function, insert $\chi_t(\mathcal{M}V)$, and apply the physical chart formula to $\mathsf{q}_j^0\mathsf{q}_t F'$. On the chart the displayed identities replace this integrand by $e^c F$. This proves the claim for every passing W.

The source-potential sum rule in Lem. YM3_I:114.1 identifies the sum of the fourth potential with the change along that fibre of

$$S[\mathfrak{f}_j[w]](U) - x_j A^{\eta_j}(v_j^w, U).$$

Its subtracted linear terms sum to zero by criticality. Thm. YM3_I:W supplies the other three sum rules, the same Gaussian covariance and the same positive flat normalizing factor $\exp(\mathsf{cs}_j)$, independently of w. The exact cluster identity then gives the next chart integral, exactly as in Prop. YM3_I:114.2. Keeping the counterterms changes only the following action identity:

$$L^t x_t + \sum_{i=1}^{t} L^i e_i = x_0.$$

Consequently, after t steps the pointwise exponent is

$$A_t^w(U) = -x_0 A^{\eta_0}(U) + \mathcal{O}(w)(U) + \sum_{i\le t}\sum_{X,z} F_i[w](X,U,z) + \sum_{j<t} \mathsf{cs}_j.$$

To see this directly, the counterterms combine with $-x_t A^{\eta_t}$ by the displayed identity, and the flat values subtracted in the bare-array action cancel the flat values in the vacuum factor. Induction using the preceding chart identity gives density $\exp A_t^w(U_t(W))$ on the passing set. The backgrounds are in all required cores, and the local functions are invariant on their critical gauge orbits. The chart formula proves Borel measurability; the finite number of bounded core terms proves boundedness. At $w=0$ this density is strictly positive. Dividing it out and adding and subtracting the flat values proves eq. 4.xii.

For completeness, the comparison of source energies uses this pointwise chart identity, rather than a choice of a version of a measure density. Both constructions start with the identical bare density $e^{\mathcal{O}(w)}\rho_0$ and use the identical factors g_t just displayed. The exact weighted sum rule, on its admitted parameter neighborhood, therefore evaluates the same successive chart integrals. At $W = \mathbf{1}$ all density and compatibility tests pass: their tested curvatures and their deviations from the trivial datum are zero. The canonical background is $U_t(\mathbf{1}) = \mathbf{1}$; this also follows from the trivial-triple assertion of Prop. YM3_I:51.9 and the uniqueness of the canonical orbit. Hence $\mathcal{O}(w)(\mathbf{1}) = 0$ and all centered terms vanish. The common, source-independent flat factors cancel, giving

$$e^{\mathcal{E}_t(w)} = e^{\mathcal{E}_t^{\mathrm{weighted}}(w)}.$$

On a connected common neighborhood of zero their difference is holomorphic, takes values in $2\pi i\mathbb{Z}$, and is zero at zero. It is therefore zero throughout that neighborhood. This proves the claimed equality and the identification of its mixed derivative. □

Proposition 4.43 (Envelopes and first creation)**.** Retain the preceding hypotheses and the calibrated real history bounds of Thm. YM3_I:D. Suppose that the source is supported in one physical ball of radius 1. For $1 \le t \le K$ put

$$d_*(t) = \min\left\{\frac{c_A\alpha_0}{2MB_3\varepsilon_t}, \frac{L^{m+K-t}-5}{2M}\right\} - 2. \tag{4.xiii}$$

Assume $M\alpha_0 \le \alpha_1$, $\alpha_0 + \alpha_1 \le a^\sharp$, and $d_*(t) \ge 3$. There are numbers $\mathsf{G}_t, \mathsf{H}_t$ such that

$$|\mathsf{R}_t(w,\cdot)| \le \|w\|_{\mathrm{lw}}\mathsf{G}_t, \qquad |\mathcal{E}(w) - \mathcal{E}_t(w)| \le \|w\|_{\mathrm{lw}}\mathsf{H}_t, \tag{4.xiv}$$

on the passing set, with

$$\begin{aligned}
&\mathsf{G}_0 \le Cb_+L^{4K},\\
&\mathsf{G}_t \le C\left[b_+\varepsilon_t^2L^{4(K-t)} + (\varepsilon_t^2/\alpha_0^2 + e^{-d_*(t)})L^{3(K-t)}\right] && (1 \le t \le K),\\
&\mathsf{H}_t \le CL^{3(K-t)} && (0 \le t \le K).
\end{aligned}$$

Here C depends only on the fixed construction constants, and b_+ is the upper coupling bound used in Prop. 6.7. On the least torus and for fixed g these imply

$$\mathsf{G}_K \le C(g), \qquad \mathsf{G}_{t-1} + \mathsf{H}_{t-1} \le C_{\mathrm{cnt}}(g)L^{4(K-t+1)}, \qquad C_{\mathrm{cnt}}(g) \ge 1. \tag{4.xv}$$

Let $\mathsf{Cr}_t(w)$ be the contribution whose first large-field region is created at step t, and set $\mathsf{w}_t = \mathsf{Cr}_t(0)/Z[0]$. Then

$$Z[w] = m_K^w(\mathcal{V}_K) + \sum_{t=1}^{K}\mathsf{Cr}_t(w), \tag{4.xvi}$$

$$\left|e^{-\mathcal{E}(w)}m_K^w(\mathcal{V}_K)/Z[0]\right| \le e^{\|w\|_{\mathrm{lw}}\mathsf{G}_K},$$

$$\left|e^{-\mathcal{E}(w)}\mathsf{Cr}_t(w)/Z[0]\right| \le e^{\|w\|_{\mathrm{lw}}(\mathsf{G}_{t-1}+\mathsf{H}_{t-1})}\mathsf{w}_t, \tag{4.xvii}$$

$$\sum_{t=1}^{K}L^{8(K-t)}\mathsf{w}_t \le C_8(g) < \infty, \tag{4.xviii}$$

uniformly in the datum at fixed g. All the sourced quantities in these formulas are holomorphic on the level-wise unit ball. ■

Proof. Put $r = \|w\|_{\mathrm{lw}}$. The amplitude part of the norm and the coupling bounds give $|w|_\infty \le Cb_+r$. There are at most CL^{3K} bare plaquettes in the support ball. At a passing output of level t, the background is regular of that level at radius $2B_3\varepsilon_t$. Consequently

$$0 \le a_p(U_t) \le 2B_3^2\varepsilon_t^2L^{-4t}, \qquad |\mathcal{O}(w)(U_t)| \le Cb_+r\varepsilon_t^2L^{4(K-t)}.$$

At $t = 0$ use $a_p \le 2$, giving $Cb_+ rL^{4K}$.

We spell out the sum of centered local terms. The pointwise regular-family estimate of [8, §7.2], with eq. 2.ii, applies to any local pointed member E at level $j \le t$:

$$|E(U_t) - E(\mathbf{1})| \le \|E\|_X \left[C(\varepsilon_t/\alpha_0)^2 L^{-4(t-j)} M^2 (1 + d_j(X))^2 + 2\mathbf{1}_{\{d_j(X) \ge d_*(j)\}}\right],$$

where

$$d_*(j) = \min\left\{\frac{c_A \alpha_0 L^{2(t-j)}}{2MB_3\varepsilon_t}, \frac{L^{m+K-j} - 5}{2M}\right\} - 2.$$

Apply it to $E = \Delta F_j[w]$ and use eq. 4.ix. Locality makes this member zero unless X meets a source plaquette. Anchor such a domain at a level-j cube meeting the support ball; there are at most $CL^{3(K-j)}$ choices. The rooted-domain sum with rate $\kappa + 1$ absorbs the base-point count and the polynomial $(1 + d)^2$, and its tail beyond $d_*(j)$ is at most $Ce^{-d_*(j)}$. These are the same rooted sums used in (2262), p. 827. Since

$$d_*(j) \ge L^{t-j}(d_*(t) + 2) - 2, \qquad L^{3(t-j)} e^{-d_*(j)} \le L^{-(t-j)} e^{-d_*(t)}$$

when $d_*(t) \ge 3$: writing $h = t - j$, the tail gains $e^{-(d_*(t)+2)(L^h-1)} \le L^{-4h}$, since $5(L - 1) \ge 4 \log L$. The two level sums are therefore bounded by

$$CrL^{3(K-t)}(\varepsilon_t^2/\alpha_0^2 + e^{-d_*(t)}) \sum_{j \le t} L^{-(t-j)}.$$

This geometric sum is bounded independently of t and K, proving the stated G_t bound. For the energy tail, the same anchoring and the uncentered family bound give

$$|\mathcal{E} - \mathcal{E}_t| \le Cr \sum_{j > t} L^{3(K-j)} \le CrL^{3(K-t)}.$$

This includes $t = 0$; at $t = K$ the tail is zero.

The radius condition is available at every datum used in Prop. 6.7. Decrease $\gamma_{\mathrm{C}} \le \gamma_{\mathrm{ev}}$, with the explicit positive γ_{ev} of eq. 2.iii. Then

$$\varepsilon(x) \le c_A \alpha_0/(10MB_3) \quad (x \ge \gamma_{\mathrm{C}}^{-2}).$$

The least-side definition gives $L^m \ge L^2 M$, hence $L^{m+K-t} \ge L^2 M \ge 10M + 5$ for $1 \le t \le K$. Both entries of the minimum defining $d_*(t) + 2$ in eq. 4.xiii are therefore at least 5, so $d_*(t) \ge 3$. The first-passage window bounds x_K and b_+ in terms of g, and the regularity radii are bounded by fixed constants throughout the run. This proves eq. 4.xv. The dependence of C_{cnt} on g is allowed here; in particular it absorbs the factor b_+ in G_0.

Let g_t^{w} be the sum of the nonnegative step factors for the other outcomes, after the all-small-field prefix of length $t - 1$. These factors too are source independent. Partition the histories according to their first such outcome. Summing all later extensions leaves the prefix mass, by the normalized test partition. This gives eq. 4.xvi and the explicit formula

$$\mathsf{Cr}_t(w) = \int g_t^{\mathrm{w}} \, \mathrm{d}m_{t-1}^{w}, \qquad \mathsf{w}_t = Z[0]^{-1} \int g_t^{\mathrm{w}} \, \mathrm{d}m_{t-1} \ge 0.$$

This is the decomposition (2268), p. 829. Substitution of eq. 4.xii, followed by eq. 4.xiv, proves eq. 4.xvii. The all-small-field zero-source mass is at most $Z[0]$, since it is one nonnegative summand of the partition. On each finite lattice the factors are bounded and the input source exponential is entire; integration and the preceding finite holomorphic sums prove the holomorphy assertions.

Finally the event of first creation at t is contained in $\{Z_t \ne \emptyset\}$. Lem. YM3_I:115.1 and the partition-function lower bound used immediately after it give, exactly as in (2185), p. 805,

$$\mathsf{w}_t \le 2M^{-3} N_t \exp\{-c' A_0^2 (\log x_t)^{2\mathfrak{p}_0} + (C_{\mathrm{maj}}^{\Sigma} + 2 + C_{\mathrm{low}}(b_+))\ell_g^3\}.$$

The canonical least torus satisfies the side assumptions of that history estimate. Here $N_t = \ell_g^3 L^{3(K-t)}$, $x_t \geq L^{K-t} b^\vee$, and $b^\vee = g^{-2} - \bar{B} > 1$ after decreasing γ_{C}. With $n = K - t$, the sum in eq. 4.xviii is consequently bounded by a fixed multiple, depending on g, of

$$\sum_{n \geq 0} \exp\{11n \log L - c' A_0^2 (\log b^\vee + n \log L)^{2\mathfrak{p}_0}\}.$$

It converges because $2\mathfrak{p}_0 > 1$: from some index depending only on g and the fixed constants the negative term exceeds $12n \log L$, and the remaining tail is bounded by $\sum_n L^{-n}$. This gives $C_8(g)$ uniformly in K. No uniform bound as g tends to zero is required. □

Lemma 4.44 (Packed source amplitudes)**.** Let u_ι be real Lipschitz functions with disjoint nonzero sets, $\|u_\iota\|_\infty \leq A_\iota$, and $\operatorname{Lip} u_\iota \leq A_\iota/(\vartheta d)$. For $w = (\sum_\iota \zeta_\iota u_\iota) \circ x$ and $B = \max_\iota |\zeta_\iota| A_\iota$, one has

$$\|w\|_{\mathrm{lw}} \leq B/\mathsf{A}_\star(d), \qquad \mathsf{A}_\star(d) = \min\left\{\sigma_{\mathrm{src}} b^\vee, \frac{\mu_\star}{K_{\mathrm{L}}^\vee} \frac{b^\vee}{(b^\wedge)^{1/2}} \left(\frac{\vartheta d}{4\sqrt{3}M}\right)^{1/2}\right\},$$

where $b^\vee = g^{-2} - \bar{B}$ and $b^\wedge = b_+$. ■

Proof. Disjointness gives $|w|_\infty \leq B$. The sum of the weighted functions has Lipschitz constant at most $2B/(\vartheta d)$: for points in different nonzero sets, bound each value by its Lipschitz constant times the distance to the complement of that set. Thus, with $n = K - j$,

$$\mathrm{sw}_j(w) \leq 2B \min\{1, \sqrt{3} M L^{-n}/(\vartheta d)\}.$$

The running-coupling bounds $L^n b^\vee \leq x_j \leq L^n b^\wedge$ now imply

$$|v_j^w|_\infty \leq B/b^\vee,$$

$$\begin{aligned} x_j^{1/2} \mathrm{sw}_j(v_j^w) &\leq \frac{2B(b^\wedge)^{1/2}}{b^\vee} \min\{L^{n/2}, \sqrt{3} M L^{-n/2}/(\vartheta d)\} \\ &\leq \frac{2B(b^\wedge)^{1/2}}{b^\vee} \left(\frac{\sqrt{3}M}{\vartheta d}\right)^{1/2}. \end{aligned}$$

The last inequality uses $\min(a, b) \leq \sqrt{ab}$. Insertion in eq. 4.viii proves the claim. □

Corollary 4.45 (Packed sources in finitely many balls)**.** Retain the numerical, canonical-run and actual-history hypotheses of Props. 4.42 and 4.43, and the calibrated rarity hypotheses of Prop. 4.54. Work on the least torus at a first-passage datum at fixed g. Let $d > 0$, $0 < \vartheta \leq 1$ and $\vartheta d \leq 1$. Let I have $q \geq 1$ elements. For $i \in I$, let $F_i = \sum_\iota h_{i,\iota}$ be a finite sum of real Lipschitz functions whose nonzero sets, for all pairs (i, ι), are pairwise disjoint. Suppose

$$\|h_{i,\iota}\|_\infty \leq A_i, \qquad \operatorname{Lip} h_{i,\iota} \leq A_i/(\vartheta d), \qquad A_i > 0,$$

and that every bump vanishes off the union of $N \geq 1$ physical balls of radius 1. Use the amplitude $\mathsf{A}_\star(d)$ of Lem. 4.44, and put

$$u_i = A_i/\mathsf{A}_\star(d), \qquad w_z = \Big(\sum_{i \in I} z_i F_i\Big) \circ x, \qquad \mathbb{P} = \{z : |z_i| u_i < 1 \text{ for every } i\}.$$

Then

$$\begin{aligned} &\Omega(z) = Z[w_z] e^{-\mathcal{E}(w_z)}/Z[0] \quad \text{is holomorphic on } \mathbb{P}, \\ &\Omega(0) = 1, \qquad |[z^B]\Omega| \leq Q_{N,q}(g) \prod_{i \in B} u_i \quad (\emptyset \neq B \subset I). \end{aligned} \tag{4.xix}$$

Here $[z^B]\Omega = (\prod_{i \in B} \partial_{z_i})\Omega(0)$. The number $Q_{N,q}(g) \geq 1$ depends only on N, q, g and the fixed construction data, not on the cutoff, label, balls, packing, d, ϑ or functions. ■□

Proof. Lem. 4.44 gives $\|w_z\|_{\mathrm{lw}} \le \max_i |z_i| u_i$. The bump index may include several functions with the same parameter z_i; the same maximum bounds each of them. Thus w_z is in the level-wise ball on $\mathbb{P}$. The finite lattice partition function is entire in its source, the source energy is holomorphic on this ball, and $Z[0] > 0$. This proves holomorphy and $\Omega(0) = 1$.

In the proof of Prop. 4.43, the support enters only through the number of bare plaquettes and the number of level-j cubes meeting it. For a union of N balls each count is at most N times its one-ball bound. The local-family bound is applied to the whole source; no decomposition of that source in the level-wise norm is needed. Consequently the proof gives envelopes

$$\mathsf{G}_t^{[N]} = N\mathsf{G}_t, \qquad \mathsf{H}_t^{[N]} = N\mathsf{H}_t, \qquad \mathsf{G}_K^{[N]} \le NC(g), \qquad \mathsf{G}_{t-1}^{[N]} + \mathsf{H}_{t-1}^{[N]} \le NC_{\mathrm{cnt}}(g) L^{4(K-t+1)}.$$

Split $\Omega = \Omega_\emptyset + \sum_{t=1}^K \Omega_t$ by eq. 4.xvi. The bounds eq. 4.xvii, with these envelopes, give

$$|\Omega_\emptyset| \le e^{NC(g)} \quad \text{on } \mathbb{P}, \qquad |\Omega_t| \le e\mathsf{w}_t \quad \text{on } \{|z_i| u_i \mathsf{Cnt}_t < 1\}, \qquad \mathsf{Cnt}_t = NC_{\mathrm{cnt}}(g) L^{4(K-t+1)} \ge 1.$$

Cauchy's formula, separately in the variables of B, yields

$$|[z^B]\Omega_\emptyset| \le e^{NC(g)} u^B, \qquad |[z^B]\Omega_t| \le e\mathsf{w}_t \mathsf{Cnt}_t^{|B|} u^B \le e\mathsf{w}_t [NC_{\mathrm{cnt}}(g) L^4]^q L^{4q(K-t)} u^B, \qquad u^B = \prod_{i \in B} u_i.$$

Prop. 4.54, with exponent $4q$ and $t = K$, bounds the sum of the last weights by

$$\sum_{t=1}^K L^{4q(K-t)} \mathsf{w}_t \le C_{4q}(g), \qquad C_{4q}(g) = 4M^{-3} \ell_g^3 e^{\mathsf{B}(X_g) \ell_g^3} \mathsf{C}_{\mathrm{ag}}(4q+1).$$

Indeed $x_K \le X_g$, the function B in eq. 4.xxxii increases, and the remaining factor $e^{-c' p_K^2/2}$ is at most 1. The displayed bound is finite at fixed g. Therefore one may take

$$Q_{N,q}(g) = e^{NC(g)} + e[NC_{\mathrm{cnt}}(g) L^4]^q C_{4q}(g).$$

All factors have the asserted dependence. □

4.2.1. *The lattice spectral inequality at several steps.* The spectral restriction must be used at a fixed physical displacement while the lattice spacing tends to zero. Its one-step form therefore needs the following elementary extension.

Lemma 4.46. Let T be a positive self-adjoint contraction and V a commuting unitary operator. If

$$(1-T)^2 \ge T(2 - V - V^*),$$

then, for every integer $k \ge 1$,

$$(1-T^k)^2 \ge T^k(2 - V^k - V^{*k}).$$

For the lattice transfer operators of Prop. YM3_I:99.3, this implies the five-point inequality

$$u(0) - 4u(te_\mu) + u(2te_\mu) + u(te_\mu + te_\lambda) + u(te_\mu - te_\lambda) \ge 0, \qquad t = ka,$$

where $u(w) = \nu(\overline{\theta F}\, \tau_w F)$ in the corresponding reflection Hilbert space.

Conversely, suppose strongly commuting self-adjoint operators $H \ge 0$ and P have joint spectral measure $\mathcal{P}$. If, on a dense set of vectors, the bounded quadratic form

$$1 - 4e^{-tH} + e^{-2tH} + e^{-tH}(e^{itP} + e^{-itP})$$

is nonnegative for every t in a subset of $(0, \infty)$ with zero as a limit point, then $\mathcal{P}\{(E, q) : E < |q|\} = 0$. ■

Proof. At a point of the joint spectrum of T, V, write $t_0 = e^{-x} > 0$ and $v = e^{iy}$, where $x \ge 0$. The assumed inequality is $\sinh(x/2) \ge |\sin(y/2)|$. The addition formulas, by induction, give

$$\sinh(kx/2) \ge k \sinh(x/2) \ge k|\sin(y/2)| \ge |\sin(ky/2)|.$$

This is the desired inequality at t_0^k, v^k; at $t_0 = 0$ it is immediate. Joint spectral calculus proves the operator inequality. Taking its matrix element at $[F]$ and using the translation identity in Lem. YM3_I:99.1 gives the five displayed correlations.

For the continuum assertion put

$$f_t(E,q) = 4e^{-tE}\big(\sinh^2(tE/2) - \sin^2(tq/2)\big).$$

This is the real bounded function $1 - 4e^{-tE} + e^{-2tE} + 2e^{-tE}\cos(tq)$. Density extends the assumed positivity to the whole Hilbert space, so $\mathcal{P}\{f_t < 0\} = 0$. Choose a sequence $t_j \downarrow 0$ from the indicated set. Outside the union of these countably many null sets,

$$\frac{2}{t_j}\sinh(t_j E/2) \geq \frac{2}{t_j}|\sin(t_j q/2)|$$

for every j. Taking the ordinary numerical limit gives $E \geq |q|$. The spectral projection of the complementary set is therefore zero. □

4.3. **Local large-field probabilities.** The history bounds of Section YM3_I:70 include an exponential of the volume. To bound the probability that a given point belongs to a large-field region, we first attach a gauge invariant test to each possible creation. Reflecting that test across a board forces many creations. The root in the chessboard estimate then removes the volume factor.

We use the real construction of Thm. YM3_I:D, with the constants constructed in Section YM3_I:69 and the final source ceilings of Section YM3_I:82. The couplings obey $x_{j-1} = L(x_j + b_j)$, $|b_j| \leq B_\sharp$, and the lattice is admissible through level K. Write $y_j = \log x_j$, $g_j = x_j^{-1/2}$ and

$$p_j = p_0(g_j) = A_0 y_j^{\mathfrak{p}_0}, \quad \varepsilon_j = g_j p_j, \quad \delta_j = (A_1/A_0)\varepsilon_j, \quad s_j = M\check{R}_j a_{K-j}, \quad w_j = M_2\check{R}_j a_{K-j}.$$

Here $\check{R}_j$ is the least nonnegative integral power of L at least y_j^r, $M_2 = L^2 M_1$, $M \geq L^2 M_2$, $L \geq 13$, and $2\mathfrak{p}_0 > 3r + 2$. A cell of level j has side s_j; a test cube has side Lw_j, and its enlargement by a layers has side $(L + 2a)w_j$. Distances in this section are periodic sup distances. The probability of a history is $\widetilde{\nu}_k(h) = \mathfrak{t}(h)/Z$, where $\mathfrak{t}(h) = \int \mathfrak{w}_h \rho_0 \, \mathrm{d}U$, $\mathfrak{w}_h = \mathbf{1}_h \mathfrak{G}_h$, and ρ_0 is the Wilson density. The exact history sum gives $\sum_h \mathfrak{t}(h) = Z$.

Definition 4.47 (The two choices and their threshold)**.** Put

$$\kappa_{\mathrm{w}} = \min\{(4\Theta_1)^{-1}, \theta_c\}, \qquad \lambda_\natural = \frac{\kappa_{\mathrm{w}}}{2\pi}, \qquad \lambda'_\natural = \frac{\lambda_\natural}{2 + 30L}.$$

A lowered choice keeps the constants before A_1 and the value of A_0, replaces A_1 by $A'_1 = \lambda'_\natural A_1$, and completes the construction with $B'_\sharp \geq B_\sharp$. Primes denote the constants of this second choice. Set

$$\begin{aligned}
c_1^\natural &= \frac{c'_\star}{4 \cdot 49^3}, \qquad C_1^\natural = \mathsf{C}'_{\mathrm{maj}} + c'_{\mathrm{low}} + \frac{13}{2}, \\
c'_{\mathrm{low}} &= C'_5 + 3\log 7 + 24(B'_3)^2, \\
c_2^\natural &= \frac{1}{2}\min\left\{\frac{c_1^\natural}{8}, \frac{\lambda_\natural^2 A_1^2}{2A_0^2}\right\}, \qquad (4.\mathrm{xx}) \\
N_\tau &= \left(\frac{M}{LM_2}\right)^3, \qquad N_\delta = \left(\frac{M}{M_2}\right)^3, \\
C_3^\natural &= 27(N_\tau + N_\delta) + 540 N_\tau \big((L+6)L^2 M_2\big)^3 (2L)^{3/2}.
\end{aligned}$$

The positive constants $c'_\star, \mathsf{C}'_{\mathrm{maj}}, C'_5$ are those of the deposit and history bounds in Section YM3_I:70 at the second choice. Let $\bar{B} = LB_\sharp/(L-1)$, $Y_0 = \max\{\log\max(e, 2B_\sharp), 2\log(2L)/(L^{1/r} - 1)\}$,

and let Y'_{mark} be the threshold below equation YM3_I:(1455) with $w = \mathfrak{d} = 1/2$, $c = c'_\star$, $c_{\mathrm{Dep}} = c'_\star$ and $c_a = 0$, at the second choice. Explicitly it is

$$Y'_{\mathrm{mark}} = \max\left\{(Y^{\mathrm{II}})'\left(c'_\star/4, 48^3(D'_0 + \lambda'_L)\right), (Y^{\mathrm{S}})'(c'_\star/4)\right\}, \qquad D'_0 = \mathsf{C}'_{\mathrm{maj}} + 27(2+\log(2L))C'_\flat.$$

Here $C'_\flat$ is the vacuum surface constant used in equation YM3_I:(1487); Y^{II} and Y^{S} are the explicit functions in Section YM3_I:67. Finally put

$$Y_{\mathrm{pay}}(c, C) = \max\left\{1, \log(1+\bar{B}), \left(\frac{4}{cA_0}\right)^{1/\mathfrak{p}_0}, \left(\frac{4CM^3L^3}{cA_0^2}\right)^{1/(2\mathfrak{p}_0 - 3r - 2)}\right\}.$$

The threshold $\mathsf{Y}^\natural$ is the least number at least 1 such that every $y \geq \mathsf{Y}^\natural$ satisfies

$$\begin{aligned}
&\text{(Y.1)}\quad y \geq \log(\gamma'^{-2}), && \text{(Y.2)}\quad y \geq Y_0,\\
&\text{(Y.3)}\quad y \geq Y'_{\mathrm{mark}}, && \text{(Y.4)}\quad y \geq Y_{\mathrm{pay}}(c_1^\natural, C_1^\natural),\\
&\text{(Y.5)}\quad C_{\mathrm{cu}}\sqrt{2L}(A_0/A_1)e^{-\delta_{\mathrm{lc}}M_2 y^r/4} \leq \lambda_\natural,\\
&\text{(Y.6)}\quad y \geq \frac{\log C_3^\natural + 3r + 3/2}{c_2^\natural A_0^2},\\
&\text{(Y.7)}\quad y \geq 2\log(1+\bar{B}), \qquad 2c_2^\natural \mathfrak{p}_0 A_0^2 y^{2\mathfrak{p}_0 - 1} \geq 4.
\end{aligned} \tag{4.xxi}$$

The constants C_{cu} and δ_{lc} are those of the local background comparison YM3_I:(1012) at the original choice. All these numbers are independent of the lattice and the coupling sequence. We also write $\mathsf{Y}_i^\natural$ for the lower bound on y in (Y.i), taking the maximum with 1 in (Y.6). Thus $\mathsf{Y}^\natural = \max_i \mathsf{Y}_i^\natural$; in particular the four thresholds used for the witness estimate are exactly those in (Y.1)–(Y.4). ■

Lemma 4.48. For a choice made by the ordered construction of Section YM3_I:69, a lowered choice exists by resuming that construction after A_0. If a lattice is admissible at the original choice and $x_K \geq \gamma'^{-2}$, it is admissible at the lowered choice. The lattices, block averages, local background maps and their stable domains, plaquette tests, gauge densities and Wilson measure are the same for both choices. The fluctuation thresholds satisfy $\delta'_j = \lambda'^{\,}_\natural \delta_j$. ■

Proof. In the ordered parameter choice, A_1 is free in $(0, \infty)$ and has no attached condition. The next constant is A_0. Its eight conditions are

$$A_0 \geq c_{\mathrm{im}}A_1, \quad A_0 > c_{\mathrm{ct}}A_1, \quad A_0 \geq C_{\mathrm{sf}}L^3A_1, \quad A_1/A_0 \leq c_{\mathrm{Q}}, \quad A_1/A_0 \leq c_{\mathrm{no}},$$
$$A_0 \geq 4B_3C_HA_1, \quad A_0/A_1 \geq 64L^3, \quad A_0/A_1 \geq 384\pi L^3 B_5 \mathsf{S}(\delta_{0,\mathrm{tw}}).$$

Every other constant in these conditions precedes A_1. Decreasing A_1 while keeping A_0 preserves all eight inequalities. The admissible partial choice can be completed by the following continuation of Section YM3_I:69. Rechoose C_0, C_1, then the radii $\alpha_1, \alpha_0, \alpha_2, \alpha_3, \alpha_4, a_\flat$ in the stated order, retaining the strict reserves for every difference of radii. The lower bounds on θ_c involve only the earlier fixed Θ_1, M and chart constants, so its original value may be retained. Evaluate the normalizer constants, increase E_0 as required, and choose $B'_\sharp$ above both its new lower bound and $B_\sharp$. All these are finite choices before the coupling ceiling. Finally reduce γ' so that the real numerical inequalities hold on $x \geq \gamma'^{-2}$. The last part of Section YM3_I:69 expresses them as finitely many quantities tending to zero, or finite lower thresholds, on that interval. The source conditions of Section YM3_I:82 impose only further such restrictions on this final ceiling. This constructs the two choices actually used here; no extension assertion for arbitrary partially specified lists is required.

The listed common objects depend only on the constants before A_1, on A_0, and on $(K, m, x_\bullet)$. In particular their definitions do not use the later fluctuation thresholds. The formula for δ_j gives the asserted scaling. The data conditions $\varepsilon_j \leq 1$ and $M\check{R}_j \leq L^{m+K-j}$ are unchanged, $\delta'_j \leq \delta_j$, and the same increments satisfy $|b_j| \leq B'_\sharp$. The remaining coupling condition is precisely $x_K \geq \gamma'^{-2}$. □

4.3.1. *Witnesses for a creation.* Write $V_j = \mathcal{M}^j U$. For a test cube Q of step t, set $Y_Q = (Q^{\sim 3})^{(t)}$ and dress a bond of level $t-1$ by

$$V^{\mathrm{dr}}_{t-1}(b) = V_{t-1}([b_-]_t, b_-) V_{t-1}(b) V_{t-1}([b_+]_t, b_+)^{-1}.$$

Define $F_{t,Q}(U) = 0$ exactly when all the following conditions hold: the plaquette test of Q is one; the local background $V^{\mathrm{bg}}_{t-1,Q}$ is stably defined; all contour averages $V_{t-1}(y, x)$, $y \in Y_Q$, $x \in B(y)$, are defined; and

$$|V^{\mathrm{dr}}_{t-1}(b)(V^{\mathrm{bg}}_{t-1,Q}(b))^{-1} - 1| < \lambda_\natural \delta_{t-1} \tag{4.xxii}$$

for every non-pivot bond incident to $Q^{\sim 2}$. Otherwise set $F_{t,Q} = 1$. At step one include the witnesses $F_{0,Q} = 1 - \chi_0(Q)$ for the bare density cubes. Their support cubes are respectively $Q^{\sim 4}$ and $Q^{\sim 1}$.

Lemma 4.49. Under (Y.2), these witnesses are Borel, gauge invariant, and covariant under the lattice isometries that preserve the partitions. Their support cubes have diameter at most $(L+8)w_t \le (L+8)L^{-2}s_k$ when $t \le k$. If also (Y.5) holds, a cell $\mathfrak{c}$ is created at step t and $\mathfrak{w}_h(U) > 0$, then either a witness whose cube meets $\mathfrak{c}$ equals one, or there is a test cube Q meeting $\mathfrak{c}$ and a non-coarse site $x \in (Q^{\sim 3})^{(t-1)}$ such that $[x]_t \in P''_t(h)$ and the defined contour variable satisfies

$$|V_{t-1}([x]_t, x) - 1| > \lambda_\natural \delta_{t-1}. \tag{4.xxiii}$$

Under (Y.1), on every history at the lowered choice with $\mathfrak{w}'_h(U) > 0$, a witness with value one has its centre in Z'_t. ∎

Proof. The stable domains of the local background maps are open, and the maps are continuous there. Contour averages and iterated block averages are Borel on their domains. Thus all four tests defining F are Borel. The local background reads only the inner bonds of $Q^{\sim 4}$. The blocks needed for every contour or block average are contained in that cube, since all its faces are block faces. This proves the stated support. The bare density witness reads only the plaquettes of $Q^{\sim 1}$. The side bounds follow from $M \ge L^2 M_2$ and the monotonicity of s_t in Lem. YM3_I:67.1.

Under a gauge transformation u, the dressed bond transforms with the values $u([b_-]_t), u([b_+]_t)$ at the coarse endpoints. The local background has exactly the same transformation law, so their quotient is conjugated. The plaquette test and the domains of the contour and background maps are gauge invariant. This proves gauge invariance of F. An isometry preserving the block partitions carries each contour, pivot bond, local minimizing orbit and axial representative to the corresponding object of the image cube. It also carries the stable domain onto the image stable domain, because it is a homeomorphism intertwining the two partial maps. Thus it preserves all four tests. In particular these conclusions apply to the isometries with translations in the lattice of spacing s_k/L: every partition used above has side at most $Lw_t \le s_k/L$, an integral power of L.

For the creation alternative, first observe that, with $w_\pm = V_{t-1}([b_\pm]_t, b_\pm)$,

$$|V_{t-1}(b) - V^{\mathrm{bg}}_{t-1,Q}(b)| \le |w_- - 1| + |w_+ - 1| + |V^{\mathrm{dr}}_{t-1}(b) - V^{\mathrm{bg}}_{t-1,Q}(b)|.$$

Consequently a bond discrepancy at least $3\lambda_\natural \delta_{t-1}$ either makes $F = 1$ or forces one endpoint contour to satisfy eq. 4.xxiii. If $Q^{\sim 3} \subset B(P''_t)$, that endpoint lies in the asserted gauge-fixing region. It cannot be a coarse site, where the contour variable is one.

The created cells are the hulls of the failed plaquette cubes, failed Q-test cubes, failed fluctuation cubes, selected intermediate bonds, and failed low-field cells; at step one there are also failed bare density cubes. A failed plaquette or bare density test directly gives a witness. A failed Q-test gives either an undefined local background or a non-pivot discrepancy at least $2\delta_{t-1}$. Its cube satisfies $Q^{\sim 3} \subset B(P''_t)$, so the preceding observation applies.

In each of the remaining three cases a non-pivot bond incident to a test cube $Q \subset \Omega_t^{\sim -1}$ meeting $\mathfrak{c}$ satisfies

$$|\mathcal{A}_{t-1}(b)| \ge \kappa_{\mathrm{w}} \delta_{t-1}.$$

Indeed the respective failed tests have thresholds δ_{t-1}, $\delta_{t-1}/(4\Theta_1)$ and $\theta_c\delta_{t-1}$. The geometry gives $Q^{\sim 4} \subset \Omega_t \subset B(P_t'')$. The regularity and P, Q tests on the support of the history meet the hypotheses of Prop. YM3_I:53.12. Its principal-log estimate gives

$$|V_{t-1}(b) - V_{t-1}^{\mathrm{bg,step}}(b)| \geq (2/\pi)|\mathcal{A}_{t-1}(b)| \geq 4\lambda_\natural\delta_{t-1}.$$

The local comparison YM3_I:(1012) is applicable on this cube: its four-layer enlargement is contained in Ω_t and both endpoints of b belong to $Q^{\sim 3}$. It gives a difference of backgrounds at most $C_{\mathrm{cu}}\varepsilon_t e^{-\delta_{\mathrm{lc}}M_2\check{R}_t/4}$. Since $\varepsilon_t \leq \sqrt{2L}(A_0/A_1)\delta_{t-1}$ and $\check{R}_t \geq y_K^r$, condition (Y.5) bounds this by $\lambda_\natural\delta_{t-1}$. The discrepancy from the cube background is therefore at least $3\lambda_\natural\delta_{t-1}$, as required.

Finally suppose that a witness centre belongs to Ω_t'. Its whole test cube lies there. The lowered plaquette and Q tests are one, the local background is stably defined, and its bond discrepancy is less than $2\delta_{t-1}'$. Prop. YM3_I:53.12, now at the lowered choice, gives all contour averages and bounds them by $15L\delta_{t-1}'$. Hence the dressed discrepancy is less than $(2+30L)\delta_{t-1}' = \lambda_\natural\delta_{t-1}$. Thus $F = 0$. The bare density test is also one on Ω_1'. A witness with value one consequently has its centre outside Ω_t', hence in $Z_t' = (\Lambda_t')^c$. □

4.3.2. *Gauge integration and the chessboard estimate.* For a class function $0 \leq \phi \leq 1$, let $\mathsf{gv}_{t,x}$ equal $\phi(V_{t-1}([x]_t, x))$ when the contour is defined and $[x]_t \in P_t''(h)$, and zero otherwise. Only non-coarse sites are used. The density of one gauge variable is

$$f_j^{\mathrm{gv}}(u) = z_j^{-1}\mathbf{1}_{\{|u-1|<\varepsilon_j\}}e^{-x_j(1-\frac{1}{2}\operatorname{tr} u)}. \tag{4.xxiv}$$

Lemma 4.50. For a finite set J of distinct pairs (t,x), class functions $0 \leq \phi_{t,x} \leq 1$, and a bounded nonnegative gauge invariant Borel function u of the bare field,

$$\sum_h \int \mathfrak{w}_h u \prod_{(t,x)\in J} \mathsf{gv}_{t,x}\,\mathrm{d}U \leq \left(\prod_{(t,x)\in J}\int_G \phi_{t,x} f_{t-1}^{\mathrm{gv}}\,\mathrm{d}v\right)\int u\,\mathrm{d}U. \tag{4.xxv}$$

For the indicator of $|v-1| > \lambda_\natural\delta_j$, its integral q_j satisfies

$$q_j \leq 20\exp\left[\tfrac{3}{2}y_j - \tfrac{1}{2}\lambda_\natural^2(A_1/A_0)^2p_j^2\right]. \tag{4.xxvi}$$

■

Proof. Induct on the last step. Fix a prefix history and the plaquette outcome of its final step. Summing the remaining outcome indicators gives at most one. The remaining last-step gauge factor is a product of the densities eq. 4.xxiv over $B(P_t'') \setminus P_t''$. Average over independent gauge transformations at these sites, extended constantly over each finer block and equal to one at the coarse centres. This preserves the prefix, the plaquette outcome, Haar measure and u. Earlier contour factors are only conjugated, and their class functions are unchanged. A last-step contour becomes $V_{t-1}([x]_t, x)v(x)^{-1}$. If the necessary contour domains fail, its contribution is zero; otherwise Haar translation turns each site factor into $\int \phi_{t,x}f_{t-1}^{\mathrm{gv}}$, or into one at a site not in J. The plaquette indicators sum to the prefix indicator. Applying the induction hypothesis proves the displayed inequality, including $J = \emptyset$.

On SU(2), $1 - \frac{1}{2}\operatorname{tr} v = |v-1|^2/2$. The Haar radial density for $r = |v-1|$ is $(2/\pi)r^2\sqrt{1-r^2/4}\,\mathrm{d}r$. Since $x_j^{-1/2} \leq \varepsilon_j \leq 1$,

$$z_j \geq \frac{\sqrt{3}}{3\pi\sqrt{e}}x_j^{-3/2} > \frac{1}{20}x_j^{-3/2}.$$

On the tail set the exponential is at most $e^{-x_j\lambda_\natural^2\delta_j^2/2}$. Now $x_j\delta_j^2 = (A_1/A_0)^2p_j^2$, which proves the tail bound. □

Lemma 4.51. Under (Y.1)–(Y.4), every witness F of step t satisfies

$$\langle F\rangle \le e^{-c_1^\natural p_t^2}. \tag{4.xxvii}$$

More generally, at a level $k \ge t$, the same bound holds for its chessboard seminorm on any of the eight boards of side s_k with offset $(L-1)s_k\sigma/(2L)$, $\sigma \in \{0,1\}^3$, whose cell contains its support cube. These boards preserve all construction partitions. For a family of witnesses with support cubes at pairwise distance greater than s_k, their joint mean is at most $\prod_F e^{-c_1^\natural p_{t(F)}^2/8}$. ∎

Proof. Use the eight boards of cell side s_k whose offsets are $(L-1)s_k\sigma/(2L)$, $\sigma \in \{0,1\}^3$. In one coordinate the walls of the two families have minimum separation $(L-1)s_k/(2L)$. An interval shorter than this separation lies between two consecutive walls of at least one family. The support diameter $(L+8)s_k/L^2$ is shorter, since $L \ge 13$. Choosing the family in each coordinate puts every witness in a cell of one of these boards. All board reflections and translations preserve its construction, by Lem. 4.49.

Let $N = (\ell/s_k)^3$ be the number of cells and let Π be the product of all reflected copies of F. Each copy is a witness of the same step t. On a history at the lowered choice with $\mathfrak{w}'_h(U) > 0$, $\Pi = 1$ forces all N witness centres into Z'_t. By Prop. YM3_I:67.2, such a centre is within $\rho_{q,t} = 7s_q + 11\sum_{j=q+1}^{t} s_j$ of a creation at some $q \le t$. The centre of its board cell is at most $s_k/2 + \rho_{q,t} + s_q/2$ from the centre of that creation. Since $2\rho_{q,t} + s_q < 48 s_t \le 48 s_k$, a creation can be assigned at most 49^3 board centres. There are therefore at least $N/49^3$ creations.

For the mark $\Theta = (c'_\star/2)\sum_q p_q^2 n_q^{\mathrm{cr}}$, this gives $\Theta \ge 2c_1^\natural p_t^2 N$. Equation YM3_I:(1487), at $w = \eth = 1/2$ and $c_a = 0$, and Prop. YM3_I:70.2 yield

$$\langle \Pi\rangle \le \exp\left[-2c_1^\natural p_t^2 N + C_1^\natural y_k (M\check{R}_k)^3 N\right].$$

The normalization is the same Wilson partition function for the two choices. Condition (Y.3) is exactly the marked-mass hypothesis. Condition (Y.4) and $\check{R}_k < L y_k^r$, $y_k \ge 1$, give $C_1^\natural y_k (M\check{R}_k)^3 \le c_1^\natural p_k^2 \le c_1^\natural p_t^2$. Thus $\langle\Pi\rangle^{1/N} \le e^{-c_1^\natural p_t^2}$. The chessboard estimate of Prop. YM3_I:86.3 gives eq. 4.xxvii and the seminorm assertion.

For several witnesses assign each to one of the eight boards and let Φ_σ be the product in that class. Distinct witnesses in a class occupy distinct cells, by the separation assumption. Since $\Phi_\sigma^8 = \Phi_\sigma$, Hölder's inequality and then the chessboard estimate give $\langle\prod \Phi_\sigma\rangle \le \prod\langle\Phi_\sigma\rangle^{1/8} \le \prod_F e^{-c_1^\natural p_{t(F)}^2/8}$. □

4.3.3. *From witnesses to points.*

Proposition 4.52. Suppose $y_K \ge \mathsf{Y}^\natural$. For any prescribed cell $\mathfrak{c}$ of level $t \le k$, the probability that it is created at step t is at most $e^{-c_2^\natural p_t^2}$. For every $x \in T_\ell$ and $1 \le n \le k \le K$,

$$\widetilde{\nu}_k\{x \in Z_n\} \le 2\cdot 24^3 e^{-c_2^\natural p_n^2}. \tag{4.xxviii}$$

The same bound holds for a prescribed cell contained in Z_n. For $0 \le a < k$ the right side for $n = k - a$ is at most

$$2\cdot 24^3 e^{-c_2^\natural p_k^2} L^{-2c_2^\natural \mathfrak{p}_0 A_0^2 y_k^{2\mathfrak{p}_0 - 1} a}. \tag{4.xxix}$$

∎

Proof. At most $27(N_\tau + N_\delta)$ witness cubes meet $\mathfrak{c}$. Indeed a partition of side $a \le s_t$ has at most $(s_t/a + 2)^3 \le 27(s_t/a)^3$ cubes meeting it. Apply this to the test cubes, and, at $t = 1$, to the bare density cubes; their quotients are bounded respectively by $M/(LM_2)$ and M/M_2. Each test cube has at most $((L+6)LM_2\check{R}_t)^3$ sites in its three-layer enlargement. There are therefore at most $27N_\tau$ times this many possible gauge-tail sites.

Multiply the alternative of Lem. 4.49 by $\mathfrak{w}_h\rho_0$ and sum and integrate over histories. Lem. 4.50, first with $u = F\rho_0$ and then with $u = \rho_0$, and Lem. 4.51 bound the creation probability by

$$e_t = 27(N_\tau + N_\delta)e^{-c_1^\natural p_t^2/8} + 27N_\tau((L+6)LM_2\check{R}_t)^3 q_{t-1}.$$

Here weakening the first exponent to $c_1^\natural/8$ keeps the same constant as the bound for separated creations. The inequalities $p_{t-1} \geq p_t$, $y_{t-1} \leq y_t + \log(2L)$ and $\check{R}_t < Ly_t^r$ imply

$$e_t \leq e^{-2c_2^\natural p_t^2} C_3^\natural e^{(3r+3/2)y_t} \leq e^{-c_2^\natural p_t^2}.$$

The last inequality follows from (Y.6), $y_t \geq y_K \geq 1$ and $2\mathfrak{p}_0 \geq 2$.

If $x \in Z_n$, Prop. YM3_I:67.2 supplies a creation of level $t \leq n$ within distance $\rho_{t,n}$. There are at most $24^3 L^{3(n-t)}$ possible cells: their count is at most $(2\rho_{t,n}/s_t + 2)^3$, and

$$2\rho_{t,n}/s_t + 2 \leq 16 + 22\sum_{j=1}^{n-t} L^j \leq 24L^{n-t}.$$

We record the coupling estimate needed to sum these bounds, including the small defect from an exact geometric progression. Put $P = 2\mathfrak{p}_0$, $\Delta = y_{n-a} - y_n$, and $v = \bar{B}/x_n$. Condition (Y.7) gives $x_n \geq (1+\bar{B})^2 \geq 4\bar{B}$. The affine flow bound gives, for $a \geq 1$,

$$\Delta \geq a\log L + \log(1-v) > 2a, \qquad a\log L - \Delta \leq \tfrac{4}{3}\bar{B}e^{-y_n}, \qquad y_n(a\log L - \Delta) \leq \frac{8}{3e}\frac{\bar{B}}{1+\bar{B}} < 1.$$

The last inequality uses $ze^{-z/2} \leq 2/e$. The first two terms of the power expansion therefore give

$$y_{n-a}^P - y_n^P \geq Py_n^{P-1}a\log L + Py_n^{P-2}\left[\frac{P-1}{2}\Delta^2 - y_n(a\log L - \Delta)\right] \geq Py_n^{P-1}a\log L.$$

By (Y.7), the rate $c_2^\natural PA_0^2 y_n^{P-1}$ is at least four. Thus the union bound over cells and levels is at most

$$24^3\sum_{a=0}^{n-1} L^{3a}e^{-c_2^\natural p_{n-a}^2} \leq 24^3 e^{-c_2^\natural p_n^2}\sum_{a\geq 0} L^{-a} \leq 2\cdot 24^3 e^{-c_2^\natural p_n^2}.$$

A cell event implies the point event at its centre. The same power estimate with $n = k$ proves eq. 4.xxix. □

4.4. **Gaussian restrictions and history tails.**

4.4.1. *Removing the fluctuation boxes.*

Lemma 4.53. Consider a complete Gaussian structure on a torus satisfying Thm. YM3_I:R, with at most three admitted homogeneous quadratic origins of sort Σ_{q} and no origin of sort Σ_{v}. Let ν^{r} be its normalized measure restricted to the fluctuation box of radius $\mathfrak{b}$, and let ν° be the normalized unrestricted measure with the same quadratic origins. For n insertions W_i of sort Σ_{q}, with nonzero norms $a_i = \|W_i\|_{\star,\lambda_V}$, set

$$r_i = \frac{w_\star}{na_i}, \qquad \Pi_{\mathrm{ins}} = \prod_i r_i^{-1}, \qquad N_{\mathrm{ins}} = \sum_{X:\, X \text{ meets every insertion}} e^{-\lambda_{\mathrm{out}}d(X)}.$$

Here "meets" means that X contains the enlarged localization domain of at least one nonzero member of that insertion. Then

$$\left|\langle W_1^{\mathrm{tot}};\dots;W_n^{\mathrm{tot}}\rangle_{\nu^{\mathrm{r}}} - \langle W_1^{\mathrm{tot}};\dots;W_n^{\mathrm{tot}}\rangle_{\nu^\circ}\right| \leq K_P^\Pi \mathsf{p}_\star \Pi_{\mathrm{ins}} N_{\mathrm{ins}}, \tag{4.xxx}$$

where, in the notation of equation YM3_I:(705),

$$\mathsf{p}_\star = \mathsf{C}_{\mathrm{ex}}((L+2)M)^3 e^{4\lambda_1} e^{-\gamma_2\mathfrak{b}^2/2}.$$

If $\lambda_{\mathrm{out}} \geq c_{\mathrm{E}}$, then for $n = 2$ or 4 and single field components at arbitrary bonds, the right side is at most $C_{\mathrm{bx}} e^{-\gamma_2 \mathfrak{b}^2/2}$, where C_{bx} is independent of the side, cutoff and bonds. Both measures are even, so odd joint cumulants of field components vanish. In the unrestricted measure the second cumulant is its covariance and the fourth is zero. ■

Proof. Add the single origin $\sum_i z_i W_i$ on the polydisc $|z_i| < r_i$. Its norm is at most $\sum_i |z_i| a_i < w_\star$, so the system has at most four admitted origins. Prop. YM3_I:47.4 makes its cluster sums holomorphic. The cluster integral identity of Thm. YM3_I:R gives the generating function as

$$\int e^{\sum_i z_i W_i^{\mathrm{tot}}} \, \mathrm{d}\nu^{\mathrm{r}} = \exp\left[\sum_X (E_z(X) - E_0(X))\right].$$

The same identity holds at threshold infinity. All these integrals are absolutely convergent. To justify differentiation, replace the quadratic origins by their real parts and the inserted origin by $\sum_i r_i |W_i|$. Monotonicity of the potential norm leaves it admitted. Its absolutely integrable density majorizes the integrands on the closed source polydisc; Cauchy's formula on a smaller polydisc and Fubini's theorem justify every derivative.

The mixed derivative of $E_z(X)$ vanishes unless X meets every insertion: otherwise one source coordinate changes no potential in X, and the locality clause of Prop. YM3_I:47.4 makes $E_z(X)$ independent of that coordinate. Thus cumulants are the sums of the mixed cluster derivatives on the set in N_{ins}. Equation YM3_I:(724), applied at every z, bounds $|E_z(X) - E_z^\circ(X)|$ by $K_P^\Pi \mathfrak{p}_\star e^{-\lambda_{\mathrm{out}} d(X)}$; there are no Σ_{v} origins, and the inserted origin remains of sort Σ_{q}. At infinite threshold the same structure and norms apply, while $\mathfrak{p}_\star$ becomes zero. Cauchy's estimate with radii increasing to r_i gives the factor Π_{ins}. Summing proves eq. 4.xxx.

A component at one bond may be placed on a fixed elementary localization domain Y_i. Its norm is bounded by $\frac{1}{2} M^{3/2} e^{\lambda_V d(Y_i)}$, since $s \leq \frac{1}{2} M^{3/2}(1 + M^{-3} s^2)$ for $s \geq 0$. The elementary domains have uniformly bounded size. The polymer sum anchored at one such domain is bounded independently of the torus, by the exponential domain-counting estimate YM3_I:(684) of Prop. YM3_I:45.1. Thus $\Pi_{\mathrm{ins}} N_{\mathrm{ins}}$ is uniformly bounded for $n = 2, 4$. The symmetry $B \mapsto -B$ proves evenness. Completing the Gaussian square gives logarithmic generating function $\frac{1}{2} \sum_{i,j} z_i z_j (\varphi_i, C\varphi_j)$ in the unrestricted measure, proving its stated cumulants. □

In the applications to the quadratic systems of Prop. 6.5, the Gaussian structure and the quadratic origin are precisely the admitted one-step system at field scale zero in Prop. 6.27. The complete structure is that of Prop. YM3_I:120.3; its covariance after the quadratic origin is included is M_i^{-1}. Taking $\mathfrak{b} = \mathfrak{b}(x_i) = A_1 (\log x_i)^{\mathfrak{p}_0}$ gives both exponential error bounds used in the field-scale expansions. Here $\lambda_{\mathrm{out}} = (1 + 4\beta_{\mathrm{s}})\kappa > c_{\mathrm{E}}$: the entropy floor in the choice of κ in Section YM3_I:69 already implies $\kappa > c_{\mathrm{E}}$.

4.4.2. *First creation on a fixed torus.* We use the all-small-field history measures m_t, step factors g_t, and conditional quotients Φ_t of Section YM3_I:123. The history law is $\widetilde{\nu}_k(h) = Z^{-1} \int \mathfrak{w}_h \rho_0 \, \mathrm{d}U$; the measure on the all-small-field history is $m_t[F] = (\mathcal{M}^t)_*(F \mathfrak{w}_{h_t} \rho_0 \, \mathrm{d}U)$, and $m_t = m_t[1]$. Its quotient is the bounded density $\Phi_t[F] = dm_t[F]/dm_t$, defined up to an m_t-null set. The exact identities YM3_I:(2269)–(2271) hold for every bounded gauge invariant insertion F, provided the prescribed tests and the ceilings $\varepsilon_j \leq \varepsilon_\flat$ hold through the levels used. In particular

$$\langle F \rangle = Z^{-1} \int \Phi_T[F] \, \mathrm{d}m_T + Z^{-1} \sum_{t=1}^{T} \int \Phi_{t-1}[F] g_t^{\mathrm{w}} \, \mathrm{d}m_{t-1}, \qquad \sum_{t \leq T} w_t = \widetilde{\nu}_K\{Z_T \neq \emptyset\}. \tag{4.xxxi}$$

These formulas are the partition of histories according to their first non-small-field outcome, followed by the exact sum over all extensions of each prefix. They are valid independently of the special source-dependent representation of Φ_t.

Proposition 4.54. Use the actual real history masses of Thm. YM3_I:D on an admissible lattice. Fix $0 < \eth < 1$, put $c' = (1-\eth)c_\star$, and assume the case-II absorption and summability thresholds of Prop. YM3_I:67.6 at $(c_\star, \eth)$, as well as $y_K \geq \max\{Y_1, 2\log(1+\bar{B})\}$, with Y_1 from Lem. YM3_I:67.1. These are the hypotheses used for the calibrated history tail in Lem. YM3_I:115.1; the couplings here may be any admissible running sequence. Define

$$\begin{aligned} \mathsf{B}(x) &= \mathsf{C}^{\Sigma}_{\mathrm{maj}} + 2 + c_{\mathrm{low}} + \tfrac{9}{2}\log x, \qquad \mathsf{C}^{\Sigma}_{\mathrm{maj}} = \mathsf{C}_{\mathrm{maj}} + (6L^3+1)\log 2, \\ \mathsf{C}_{\mathrm{ag}}(q) &= \exp\left[\frac{(3+q)^2(\log L)^2}{2c' A_0^2 (\log(L/2))^2}\right] \qquad (q \geq 0). \end{aligned} \tag{4.xxxii}$$

We also write $\mathsf{C}_{\mathrm{ag}}(q, \eth)$ when the dependence on $\eth$ needs to be displayed. For $1 \leq t \leq k \leq K$, with $N_t = \ell^3 L^{3(K-t)}$,

$$0 \leq w_t \leq \widetilde{\nu}_k\{Z_t \neq \emptyset\} \leq 2M^{-3} N_t e^{-c' p_t^2} e^{\mathsf{B}(x_K)\ell^3}. \tag{4.xxxiii}$$

If $\mathsf{B}(x_K)\ell^3 \leq c' p_K^2/2$, one may replace its last two exponentials by $e^{-c' p_t^2/2}$. Write $P_t = \widetilde{\nu}_K\{Z_t \neq \emptyset\}$. Then $m_t(\mathcal{V}_t)/Z = 1 - P_t$ and, for every $q \geq 0$,

$$\sum_{t'=1}^{t} L^{q(K-t')} P_{t'} \leq 4M^{-3}\ell^3 e^{\mathsf{B}(x_K)\ell^3} \mathsf{C}_{\mathrm{ag}}(q+1) L^{-(K-t)} e^{-c' p_K^2/2}. \tag{4.xxxiv}$$

If the stronger condition $2\mathsf{B}(x_K)\ell^3 \leq c' p_K^2/2$ holds, then the probability in eq. 4.xxxiii is also at most $L^{3(K-t)} e^{-c' p_t^2/2}$; its weighted sum is at most $2\mathsf{C}_{\mathrm{ag}}(q+1) L^{-(K-t)} e^{-c' p_K^2/2}$. Both weighted-sum bounds also hold with $w_{t'}$ in place of $P_{t'}$. All history probabilities of an event determined through level t are unchanged when the terminal level $k \geq t$ is changed. ■

Proof. We give the dependence on the sequence explicitly. Prop. YM3_I:70.1 and equation YM3_I:(1485) bound the real masses by the summation input of Prop. YM3_I:67.6. Its proof and the proof of Prop. YM3_I:67.5 bound the normalized weight for an exact set W_j of creations by

$$e^{\mathsf{C}_{\mathrm{maj}} N_K} \prod_j (2^{-n_j^{\mathrm{bit}}} \zeta_j)^{\#W_j}, \qquad \zeta_j = e^{-c' p_j^2 - 2p_j}.$$

The multiplicity estimate YM3_I:(1447) cancels the binary factors even if a creation cell is prescribed. Since Z_t is nonempty exactly when a creation occurs at some $j \leq t$, the union bound in this finite subset product gives

$$\sum_{h: Z_t \neq \emptyset} e^{-E_{(K)}} \mathfrak{t}(h) \leq \Pi_K(\mathsf{C}_{\mathrm{maj}}, c') \sum_{j \leq t} \#\mathcal{P}_j e^{-c' p_j^2} \leq 2M^{-3} N_t e^{-c' p_t^2} \Pi_K(\mathsf{C}^{\Sigma}_{\mathrm{maj}}, c').$$

The last sum is the stopped summability estimate YM3_I:(1451), under the threshold assumed in the statement. Dividing by the partition lower bound in Prop. YM3_I:70.2, and using $\Pi_K(\mathsf{C}^{\Sigma}_{\mathrm{maj}}, c') \leq e^{(\mathsf{C}^{\Sigma}_{\mathrm{maj}} + 2)N_K}$, proves eq. 4.xxxiii. This proof uses the admissible flow bound, not an exact geometric line. Consistency follows by summing the extensions of a prefix; the small-field ceilings are part of the construction hypotheses. Equation eq. 4.xxxi gives the assertion for w_t.

For $a = K - t$, the inequality $y_t - y_K \geq a\log(L/2)$ and the first three terms of the power expansion, with $y_K \geq 1$ and $P = 2\mathfrak{p}_0 \geq 2$, imply $p_t^2 \geq p_K^2 + A_0^2 a^2 (\log(L/2))^2$. Completing the square gives

$$L^{(3+q)a} e^{-c' p_t^2/2} \leq \mathsf{C}_{\mathrm{ag}}(q) e^{-c' p_K^2/2}. \tag{4.xxxv}$$

Use $q+1$ and sum $\sum_{a \geq K-t} L^{-a} \leq 2L^{-(K-t)}$ to obtain eq. 4.xxxiv. Monotonicity $p_t \geq p_K$ absorbs the volume exponential under the weaker small-side condition. Under the stronger condition use also $2M^{-3} \leq 1$ and $\ell^3 \leq e^{\mathsf{B}(x_K)\ell^3}$, since $\mathsf{B}(x_K) \geq 1$. This gives the final assertions. □

For a first-passage datum at g, $x_K \leq X_g$, so the preceding bounds remain valid with $\mathsf{B}(X_g)$ in place of $\mathsf{B}(x_K)$. The canonical choice takes its coupling ceiling below the fixed thresholds in this proposition. Its admissibility on all levels is proved in Thm. YM3_I:D, and the first-passage and

stopped-data conditions in Prop. YM3_I:74.1 and [7, Lem. G.3]. On a small side the specialization is Lem. YM3_I:125.3; the least admissible side is small below the ceiling of Lem. YM3_I:126.3.

5. Plane fields and anisotropic variations

Plane-dependent changes of the Wilson action probe the diagonal curvature bilinears. Mixed corner terms probe its off-diagonal components. Their derivatives also produce stress insertions. This connects the tensor correlations with anisotropic variations of the model, but the connection must retain the boundary terms of a localized variation.

For the background estimate, fix the axial trees used for lattice gauge fixing. Let Y_0 be the space of block-data variations that vanish on those trees and have zero next linearized block average.

Sketch D (Plane insertions and anisotropic response). At the admitted coupling and finite-geometry scales, separated plane and stress insertions have bounds uniform in the cutoff. The lattice Ward identity passes to the continuum with its stress flux through the cutoff shell. Near a real regular reference background, sufficiently small plane and symmetric corner weights give a unique constrained critical background, holomorphic in the weights and block datum, with localized response. The Hessian in the block datum has a uniform positive lower bound for its real part on $Y_0 + iY_0$. These conclusions retain the regular-reference, region-separation and norm conditions of the construction. ■□

Sketch of the argument. To see the boundary term, let ω be antisymmetric and choose a smooth cutoff χ equal to one near the scalar spectator supports. The displacement is $\chi(x)\omega x$. Write $\mathfrak{B}_{\omega,\chi}$ for the full stress-flux insertion carried by the shell where χ varies, including the lattice correction to its principal term. On a fixed torus, after the cutoff is removed, the identity has the form

$$\sum_i S_n(f_1, \dots, (\omega x)\cdot \nabla f_i, \dots, f_n) = \lim_{K\to\infty} \langle \mathfrak{B}_{\omega,\chi}; \mathcal{O}_K(f_1); \dots; \mathcal{O}_K(f_n)\rangle.$$

The derivatives on the left are continuum derivatives; the identity from which it comes uses lattice differences. Bounds on the shell insertion allow the limit, but do not make the shell flux vanish. In a rotation-invariant volume-first limit it does vanish. For a general limit along growing tori, vanishing of this flux is the remaining condition in this Ward formulation of rotation invariance.

The background calculation takes place on a finite lattice. A real reference background is regular when its local coordinates, constrained inverse and small-field bounds lie within the construction domains. Write a nearby field in exponential coordinates over this reference, fix the lattice gauge slice and prescribe its block average. The variables are finite-lattice bond fields and row data; the continuum derivatives in the Ward identity do not act on these variables.

The critical equation and the constraint are solved together with their multiplier. Their linear part is inverted using the regular-reference propagators. The remaining constraint and action terms have small derivatives in scaled local norms. One multiplier equation has a bounded linear term without a small factor; weighting the multiplier component in the pair norm makes the complete map a contraction. This is why the proof uses a coupled system rather than a scalar implicit equation. Differentiating its fixed point gives the local response bounds. For a perturbation of the block datum at c_0, for example, the response of A decays exponentially with the block distance from c_0.

The norms matter. The background bound contains a scaled first-gradient Hölder seminorm. Uniform control of that response requires the corresponding regularity of the weights; a supremum bound alone cannot replace it. The positive Hessian estimate also retains its real-reference and longer-sequence conditions. Moving between gauges at complex scalar data uses the separate scalar transfer argument.

For the arbitrary-order free short-distance law, the additional task is a uniform interacting n-source estimate at every order. The separated bounds, the specified low-order laws and the

regular-reference background construction do not use this extension. The argument below specifies the required estimate but does not prove it at arbitrary order.

5.1. **Plane fields, stress and anisotropic variations.** We compare the curvature bilinears with plane insertions and the stress tensor obtained from anisotropic changes of the lattice action. Uniform estimates permit the removal of the cutoff; the collapse and remainder estimates identify the conserved stress insertion.

The Ward identities retain their boundary terms and quantitative hypotheses. The two-scale comparisons use the results of §4.1; they do not identify arbitrary coincident-point products without a specified extension.

5.1.1. *Notation and hypotheses. Model, constants and data.* We work in the common real and sourced construction setting of Prop. YM3_I:69.1 and Section YM3_I:82: the Wilson action of SU(2) on $T_\ell = \mathbb{R}^3/\ell\mathbb{Z}^3$, $\ell = 2L^m$, at the spacing $a = a_K = L^{-K}$, $L \geq 13$ odd; $a_p = 1 - \frac{1}{2}\mathrm{tr}\,\partial U(p)$ and $A = \sum_p a_p$; $\langle\cdot\rangle_{K,\ell,\beta}$ and $\langle\cdot\,;\ldots;\cdot\rangle_{K,\ell,\beta}$ are the means and the joint cumulants of $Z^{-1}_{K,\ell,\beta} e^{-\beta A} dU$, and $S^T_{n;K,\ell,\beta}$ are the connected functions of $\mathcal{O}$. For a plaquette p, $x(p)$ is its barycentre, $\lambda(p)$ its normal direction and $P(p)$ the set of its two directions; for a bond b, x_b is its midpoint and μ_b its direction.

The constants are those of the list of constants and conditions among the construction results, fixed in its order, γ last. For a reference coupling $0 < g \leq \gamma$, $\mathfrak{L}(g)$ is the set of the sides at g (equation YM3_I:(1528)), ℓ_g the least of them, and $\mathfrak{D}(g)$ the set of the first-passage data (K,β) at g (Cor. YM3_I:74.2), with the label $\mathsf{b} = a_K\beta$; $X_g = L(g^{-2} + B_\sharp)$, $\bar{B} = B_\sharp L/(L-1)$, $I(g) = (g^{-2} + \bar{B}, X_g - \bar{B}]$. Every label of $\mathfrak{D}(g)$ lies in $[\gamma^{-2}, X_g + \bar{B}]$; $MR(X_g + \bar{B}) \leq L^m$ on every side $2L^m$ in $\mathfrak{L}(g)$, $R(x)$ being the least power of L above $(\log x)^r$ with the exponent r among the construction results; and for $\mathsf{b} \in I(g)$ every pair $(K, L^K\mathsf{b})$, $K \in \mathbb{N}$, is a first-passage datum at g.

The bounds $\gamma^{\mathrm{sd}}_W \leq \gamma^{\mathrm{ref}}_W = x_W^{-1/2} \leq \gamma$ on the reference coupling (the choices in equations YM3_I:(2326)–(2327)), the threshold x_{J}, the radii α_0, α_1 of the cores, the numbers $\mathfrak{p}_0$, A_0, A_1, c', κ, M, C_{lat} and the function $B(\cdot)$ are those among the construction results, and so is the depth function: $s(\rho)$ is the least integer $s \geq 0$ with $L^s\rho \geq C_W(s)$, where $C_W(s) = C_{W,0}A_0^2 Y_s^{2\mathfrak{p}_0}$ and $Y_s = (s+1)\log L + \log(g^{-2} + \bar{B})$. The choice preceding equation YM3_I:(2326) satisfies every inequality defining x_{J} in [7, eq. 228]. Since x_{J} is the least such threshold, $x_W \geq \max\{\gamma^{-2}, x_{\mathrm{J}}\}$, and hence $g^{-2} \geq x_{\mathrm{J}}$ for $0 < g \leq \gamma^{\mathrm{ref}}_W$. $\mathcal{G}^{\mathrm{per}}_{K,\beta}$ is the set of the periodic Gibbs states at (a_K, β); a *state* at (K,β) is one of them or the measure of a torus with a side in $\mathfrak{L}(g)$; for a state ν we write $A^\nu[F] = F - \nu(F)$.

We write $p_\ell = \ell^{-3}\log Z_{K,\ell,\beta}$ for the pressure of T_ℓ per unit of physical volume, formed with the plain partition function $Z_{K,\ell,\beta}$ of the measure above (Section YM3_I:86); at zero source Thms. YM3_I:AJ and YM3_I:AD write $Z_{K,\ell,\beta} = e^{E_{(K)}[\ell]}\widetilde{Z}_\ell(0)$, with the vacuum constant $E_{(K)}[\ell]$ of the side ℓ and the normalised partition function $\widetilde{Z}_\ell(0)$ among the construction results. For $(K,\beta) \in \mathfrak{D}(g)$ and $\ell \geq \ell'$ in $\mathfrak{L}(g)$, $0 \leq p_{\ell'} - p_\ell \leq G_g$, with the number $G_g = C_{\mathrm{S}} + C_g + C_{\mathrm{gap}}$ among the construction results.

The lower bound is reflection positivity: $Z_{K,\ell,\beta} \leq Z^{(\ell/\ell')^3}_{K,\ell',\beta}$, [7, Cor. B.1](e). The upper bound follows from $|E_{(K)}[\ell]\ell^{-3} - E_{(K)}[\ell']\ell'^{-3}| \leq C_{\mathrm{gap}}$, $\widetilde{Z}_{\ell'}(0) \leq e^{C_{\mathrm{S}}\ell'^3}$ and $\widetilde{Z}_\ell(0) \geq e^{-C_g\ell^3}$ (Props. YM3_I:74.4 and YM3_I:74.5, through [7, Cor. B.1], Prop. YM3_I:89.2 and [7, Lem. N.1], which compare two sides; C_{S} and $C_{\mathrm{gap}} = 2C_{\mathrm{tt}}C'_{\mathrm{tt}}E_0$ are the numbers of [7, §N], and C_g that of [7, eq. 288]). Distances are Euclidean unless they are called sup distances; Lip_∞ is the Lipschitz constant for the sup distance, $\|f\|_\sharp = \max\{\sup|f|, 40M\,\mathrm{Lip} f\}$, $\|\eta\|_{C^2}$ is the sum of the suprema of $|\eta|$, $|\nabla\eta|$, $|\nabla^2\eta|$, and $\|\varphi\|_{C^{1,1}} = \max\{\sup|\varphi|, \sup|\nabla\varphi|, \mathrm{Lip}\,\nabla\varphi\}$.

Constants in the statements depend only on the constants among the construction results and on the arguments shown, never on K, β, the state or the test functions.

Lemma 5.1 (A bound on separated torus cumulants)**.** Fix the common construction choice and $0 < g \le \gamma$. Put $\delta_e = \kappa - \log 9216 > 0$ and

$$S(\kappa) = \frac{2^{12}}{1 - e^{-\delta_e}}, \qquad C'_\Sigma = \frac{27}{26}\,5^3 + \left(\frac{6}{e\delta_e}\right)^3 \frac{e^{2\delta_e}}{1 - e^{-\delta_e}}, \qquad c_V = C_{\mathrm{S}}(2 + \log g^{-2}) + 1.$$

For $n \ge 2$ and $d > 0$, define

$$C_n(d) = \left(\frac{n}{2\varrho}\right)^n \left[4^n \log 2 + 81\sqrt{3}\, C_{\mathrm{S}} S(\kappa) C'_\Sigma d^{-3}\right].$$

At every first-passage datum (K, β) at g, on every $\ell \in \mathfrak{L}(g)$, real Lipschitz functions $f_1, \dots, f_n$ with two supports at Euclidean distance at least d satisfy

$$|S^T_{n;K,\ell,\beta}(f_1, \dots, f_n)| \le C_n(d) e^{n c_V \ell^3} \prod_i \|f_i\|_\sharp. \tag{5.i}$$

The constants are independent of K, β, ℓ and the functions; $C_n(d)$ depends only on n, d, ϱ and the fixed construction constants, while c_V may also depend on g. If each support is in a sup-ball of radius $R > 0$ and $\ell \ge 16n(R+1)$, Prop. YM3_I:89.4 instead gives a bound independent of ℓ. ■

Proof. Prop. YM3_I:88.3, with $Q_\ell = \exp[C_{\mathrm{S}}(2 + \log g^{-2})\ell^3]$, gives

$$|S^T_n(f)| \le \left(\frac{n}{2\varrho}\right)^n \left[\log 2\,(2Q_\ell + 2)^n + 81\sqrt{3} C_{\mathrm{S}} S(\kappa) C'_\Sigma \ell^3 d^{-3}\right] \prod_i \|f_i\|_\sharp.$$

Here C'_Σ is the scale-sum constant in equation YM3_I:(1776). Since $Q_\ell \ge 1$, $(2Q_\ell + 2)^n \le 4^n Q_\ell^n$; and $\ell^3 \le e^{n\ell^3}$. Absorbing these two factors proves eq. 5.i. Symmetry of a cumulant allows the separated pair to occupy any two positions. The per-ball assertion is exactly Prop. YM3_I:89.4, under the additional support and side assumptions stated above. □

In this section c_V denotes the displayed enlarged constant. Every use below is through eq. 5.i; no identity with a pressure or a vacuum-energy coefficient is used.

Plane fields and rings. A *weight* is a real or complex function w of the plaquettes; $\mathcal{O}(w) = a^{-1}\sum_p w(p) a_p$, $\mathcal{O}(f) = \mathcal{O}(f \circ x)$, $\mathcal{O}_\lambda(f) = a^{-1}\sum_{\lambda(p)=\lambda} f(x(p)) a_p$, so that $\mathcal{O} = \mathcal{O}_1 + \mathcal{O}_2 + \mathcal{O}_3$, and $\check{D}_m = \mathcal{O} - 2\mathcal{O}_m$. $C^{\mathrm{free}}_K(p, q)$ is the matrix of the orthogonal projection onto the exact lattice curvatures, and

$$\mathcal{K}^{(n)}_K(w_1, \dots, w_n) = \tfrac{1}{2} \sum_\varsigma \sum_{p_1, \dots, p_n} \prod_a w_a(p_a) \prod_{i \in \mathbb{Z}/n} C^{\mathrm{free}}_K(p_{\varsigma(i)}, p_{\varsigma(i+1)}),$$

the sum over the $(n-1)!$ cyclic orders ς; $\mathcal{K}^{(n)}_{K,\ell'}$ is the same sum on $T_{\ell'}$. With $\mathbb{K}_3(u) = (3\hat{u}\hat{u}^{\mathrm{T}} - \mathbf{1})/(4\pi|u|^3)$, the continuum ring $\mathcal{K}^{(n)}_\lambda(\phi)$ is the same expression with $\int \prod_a \phi_a(y_a)\,\mathrm{d}y_a$ for the sum over the plaquettes and $\mathbb{K}_3(y_{\varsigma(i)} - y_{\varsigma(i+1)})_{\lambda_{\varsigma(i)}\lambda_{\varsigma(i+1)}}$ for C^{free}_K. We put $\mathcal{K}_{\lambda\mu}(\phi_1, \phi_2) = \iint \phi_1(x)\phi_2(y)[\mathbb{K}_3(x - y)_{\lambda\mu}]^2\,\mathrm{d}x\,\mathrm{d}y$, $k_3(x, y, z) = \mathrm{tr}[\mathbb{K}_3(x - y)\mathbb{K}_3(y - z)\mathbb{K}_3(z - x)]$ and, for a unit vector n, $\mathsf{t}^\circ(n, X) = \frac{3}{32\pi^2}\mathsf{b}^{-2}|X|^{-6}(3(n \cdot \hat{X})^2 - 1)$.

Since $\mathbb{K}_3(X)^2 = (16\pi^2|X|^6)^{-1}(3\hat{X}\hat{X}^{\mathrm{T}} + \mathbf{1})$, we have $\frac{3}{2}\mathsf{b}^{-2}\sum_{\lambda,\mu}[\mathbb{K}_3(X)_{\lambda\mu}]^2 = \frac{9}{16\pi^2}\mathsf{b}^{-2}|X|^{-6}$, the free two-point kernel of $\mathcal{O}$, and $\frac{3}{2}\mathsf{b}^{-2}\sum_\mu\{[\mathbb{K}_3(X)_{\lambda\mu}]^2 - \frac{1}{3}\sum_{\lambda'}[\mathbb{K}_3(X)_{\lambda'\mu}]^2\} = \mathsf{t}^\circ(e_\lambda, X)$. For plaquettes $p \ne q$ and u a least lift of $x(p) - x(q)$, equations YM3_I:(2245)–(2246) bound the lattice kernel by $|C^{\mathrm{free}}_K(p, q)| \le (1 + 3C_{\mathrm{lat}})a^3|u|^{-3}$ and compare it with $\mathbb{K}_3$, $|\epsilon_p\epsilon_q C^{\mathrm{free}}_K(p, q) - a^3\mathbb{K}_3(u)_{\lambda(p)\lambda(q)}| \le C_{\mathrm{lat}} a^3(a^2|u|^{-5} + \ell^{-3})$, $\epsilon_p = \pm 1$ the orientations (YM3_I:(2246) and YM3_I:(2245) of equations YM3_I:(2245)–(2246)).

Its differences along one plane obey

$$|C^{\mathrm{free}}_K(p, q) - C^{\mathrm{free}}_K(p', q)| \le C_\nabla\, a^3 |x(p) - x(p')|\,|u|^{-4} \qquad \text{for parallel, equally oriented } p,$$
$$p' \text{ with } |x(p) - x(p')| \le \tfrac{1}{4}|u| \text{ and } |u| \ge 8a,$$

$C_\nabla = 256\sqrt{3}\,C_{\rm h}(1+3C_{\rm lat})$, $C_{\rm h}$ the constant of the interior gradient estimate for discrete harmonic functions on $\mathbb{Z}^3$. Indeed the Laplacian dd^*+d^*d of the cubic lattice acts on 2-forms plane by plane as the scalar lattice Laplacian, and it commutes with the projection; so it carries the column $C_K^{\rm free}(\cdot,q)$, the projection of the unit form δ_q, to $dd^*\delta_q$, which vanishes at every plaquette that shares no bond with q, in particular at every plaquette whose barycentre is more than a away from $x(q)$. On each plane the column is therefore discrete harmonic at the distance $\geq 2a$ from $x(q)$. A lattice path from $x(p)$ to $x(p')$ of length $\leq\sqrt{3}|x(p)-x(p')|$ keeps the distance $\frac12|u|$ from $x(q)$; on the balls of radius $\frac14|u|$ about its points the first bound bounds the column by $4^3(1+3C_{\rm lat})a^3|u|^{-3}$, and the gradient estimate on these balls bounds each step by $4^4C_{\rm h}(1+3C_{\rm lat})a^4|u|^{-4}$.

Lattice calculus. $\mathbf{w}(V)=(V-V^{-1})/2i$ and $\langle X,Y\rangle=\frac12{\rm tr}XY$. For a positively oriented bond b and a plaquette $p\ni b$, $W_{b,p}$ is the holonomy of p from the initial point of b, along b; $p(b,\pm\rho)$, $\rho\neq\mu_b$, are the plaquettes through b in the directions $\pm\rho$; $\mathbf{u}_{b,\rho}=-\frac12[\mathbf{w}(W_{b,p(b,\rho)})-\mathbf{w}(W_{b,p(b,-\rho)})]$, $\mathbf{u}_{b,\mu_b}=0$, the displacement of b, and $\mathfrak{d}_{b,p,\rho}=\langle\mathbf{u}_{b,\rho},\mathbf{w}(W_{b,p})\rangle$, the elementary densities; $\mathbf{u}_{b,\rho}$ is the half sum of the field strengths of the two plaquettes at b on the sides $\pm e_\rho$, and $G_\rho(b)=\mathbf{w}(W_{b,p(b,\rho)})+\mathbf{w}(W_{b,p(b,-\rho)})$ is the transported difference of these field strengths. With $p_\pm=p\pm ae_\rho$,

$$\mathfrak{r}_\rho(p)=\sum_{b\subset\partial p}\mathfrak{d}_{b,p,\rho}-\tfrac12(a_{p_+}-a_{p_-}),\qquad \mathfrak{s}_{l\rho}(p)=\sum_{b\subset\partial p}\frac{(x_b-x(p))_l}{a}\,\mathfrak{d}_{b,p,\rho},\qquad \mathfrak{r}^\sharp_\rho(p)=\mathfrak{r}_\rho(p)-\Psi_\rho(p)+\Psi_\rho(p_-),$$

the orbital remainder, the spin densities and the cubic remainder, where the orbital potential is $\Psi_\rho(p)=\frac12 a_{p\cup p_+}-a_p-a_{p_+}+\frac12 a_pa_{p_+}$ for $\rho\in P(p)$, $a_{p\cup p_+}$ being the plaquette term of the 1×2 rectangle, and $\Psi_\rho(p)=-\frac{1}{16}\sum_y\langle Y_y,Y_y\rangle$, $Y_y=\mathbf{w}(W_p^y)-{\rm Ad}(U(y,y+ae_\rho))\mathbf{w}(W_{p_+}^{y+ae_\rho})$, for $\rho=\lambda(p)$, the sum running over the corners y of p and W_p^y being the holonomy of p from y. For a weight v and a vector field η the stress insertion is

$$\Theta_v[\eta]=\sum_p v(p)\,a^{-1}\sum_{b\subset\partial p}\sum_\rho(\eta^\rho(x_b)-\eta^\rho(x(p)))\mathfrak{d}_{b,p,\rho},$$

and $\Theta_\beta[\eta]$ is its value at $v\equiv\beta$. A *field* is a triple η of real functions on the barycentres; $\mathsf{H}(\eta)=\max_\rho\sup|\eta^\rho|+\max_\rho{\rm Lip}_\infty\eta^\rho$, and

$$\mathfrak{K}^{\rm v}[\eta]=a^{-1}\sum_p\sum_\rho\eta^\rho(x(p))\,\mathfrak{r}_\rho(p),\qquad \mathfrak{K}^\sharp[\eta]=a^{-1}\sum_p\eta^{\lambda(p)}(x(p))\,\mathfrak{r}^\sharp_{\lambda(p)}(p)$$

are the orbital insertion and the cubic insertion. We write ${\rm div}^s\eta(y)=\sum_\rho(2a)^{-1}[\eta^\rho(y+ae_\rho)-\eta^\rho(y-ae_\rho)]$, $\nabla^+_\rho\varphi(y)=a^{-1}[\varphi(y+ae_\rho)-\varphi(y)]$, $\nabla^c_\rho\varphi(y)=a^{-1}[\varphi(y+\frac a2e_\rho)-\varphi(y-\frac a2e_\rho)]$, $e^\eta(p)=[\nabla^c_\mu\eta^\mu+\nabla^c_\nu\eta^\nu](x(p))$ with $\{\mu,\nu\}=P(p)$, and ${\rm St}_K[\eta]=\Theta_\beta[\eta]-\mathfrak{b}\,\mathcal{O}({\rm div}^s\eta)$. With v_η the derivation that displaces U along η and ${\rm div}\,v_\eta$ its divergence, the identities ([7, Lem. J.1], with the integration by parts [7, Lem. J.1]) are $v_\eta A(\beta,\cdot)=\mathfrak{K}^{\rm v}[\beta\eta]-\mathfrak{b}\,\mathcal{O}({\rm div}^s\eta)+\Theta_\beta[\eta]$ and ${\rm div}\,v_\eta=-\frac32 a\,\mathcal{O}(e^\eta)$. So the null term $N_\beta[\eta]=v_\eta A(\beta,\cdot)-{\rm div}\,v_\eta$ of Lem. 5.9(a) is ${\rm St}_K[\eta]+\mathfrak{K}^{\rm v}[\beta\eta]+\frac32 a\,\mathcal{O}(e^\eta)$; and since, identically in U, $\mathfrak{K}^{\rm v}[\beta\eta]=-\beta\sum_{p,\rho}(\nabla^+_\rho\eta^\rho)(x(p))\Psi_\rho(p)+\beta\mathfrak{K}^\sharp[\eta]$ (Lem. 5.9(b)), $N_\beta[\eta]=\mathbb{T}_\beta[\eta]+\beta\mathfrak{K}^\sharp[\eta]$ with the *stress functional*

$$\mathbb{T}_\beta[\eta]={\rm St}_K[\eta]-\beta\sum_{p,\rho}(\nabla^+_\rho\eta^\rho)(x(p))\,\Psi_\rho(p)+\tfrac32 a\,\mathcal{O}(e^\eta),$$

which reads η through its differences alone and vanishes at a constant field; the cubic insertion $\beta\mathfrak{K}^\sharp[\eta]$, which completes it to the null term, is its *cubic part*.

Tensor entries. $V={\rm Sym}_3(\mathbb{R})$ and $|H|_\infty=\sup_x\max_{l,r}|H_{lr}(x)|$. For $H\in C_c(\mathbb{R}^3;V)$ we put $\mathcal{T}_K(H)=\mathcal{O}(w_H)-a^{-1}\sum_b H_{\lambda_b\lambda_b'}(x_b)\langle\mathbf{u}_{b,\lambda_b},\mathbf{u}_{b,\lambda_b'}\rangle$, $w_H(p)=H_{\lambda(p)\lambda(p)}(x(p))$, $\lambda_b<\lambda_b'$ being the directions normal to b; formally $\mathcal{T}_K(H)=\frac12\int{\rm Tr}(HT)$ with $T_{lr}={\rm tr}B_lB_r$, and $\mathcal{T}_K(f\mathbf{1})=\mathcal{O}(f)$,

$\mathcal{T}_K(f(\mathbf{1}-2E_{mm}))=\check{D}_m(f)$. We put $H^{\mathrm{cons}}[\eta]=(\operatorname{div}\eta)\mathbf{1}-\nabla\eta-(\nabla\eta)^{\mathrm{T}}$ and $\Delta_K[\eta]=\mathrm{St}_K[\eta]-\mathsf{b}\,\mathcal{T}_K(H^{\mathrm{cons}}[\eta])$.

At smooth fields the density of $\mathrm{St}_K[\eta]$ against $\partial_l\eta^r$ is $\mathsf{b}[\frac12\delta_{lr}|B|^2-\langle B_l,B_r\rangle]$, so that of $\mathsf{b}\,\mathcal{T}_K(H^{\mathrm{cons}}[\eta])$ is the same and the classical density of $\Delta_K[\eta]$ vanishes. A *tensor limit* at g along (K_j,β_j,ρ_j), with $(K_j,\beta_j)\in\mathfrak{D}(g)$, $K_j\to\infty$ and $\rho_j\in\mathcal{G}^{\mathrm{per}}_{K_j,\beta_j}$, is a functional with $\sigma(H_1,\dots,H_n)=\lim_j\rho_j(\prod_a A^{\rho_j}[\mathcal{T}_{K_j}(H_a)])$ for all $H_a\in C_c(\mathbb{R}^3;V)$ with pairwise disjoint supports; the labels need not converge. The stress entries are $T_{lr}(\varphi)=\beta\sum_p\varphi(x(p))[\mathfrak{s}_{lr}(p)-\delta_{lr}a_p]$.

Rotations. For $\omega\in\mathfrak{so}(3)$, $|\omega|=\max|\omega_{\rho\lambda}|$, and ωy is taken in the chart $(-\frac{\ell}{2},\frac{\ell}{2})^3$; $\nabla^s_\omega\varphi(y)=\sum_\rho(\omega y)_\rho(2a)^{-1}[\varphi(y+ae_\rho)-\varphi(y-ae_\rho)]$, and $\mathfrak{S}_\omega[v]=\sum_p v(p)\sum_{\rho,\lambda}\omega_{\rho\lambda}\mathfrak{s}_{\lambda\rho}(p)$ is the spin insertion. A cut-off function at K is a real function χ on $\mathbb{R}^3$ that vanishes outside $(-\frac{\ell}{2}+2a,\frac{\ell}{2}-2a)^3$, regarded as a function on T_ℓ (Section YM3_I:112); it has the margin $\delta_0\ge 2a$ if it vanishes outside $(-\frac{\ell}{2}+\delta_0,\frac{\ell}{2}-\delta_0)^3$; it covers $f_1,\dots,f_n$ if $\chi=1$ at the barycentre and at the bond midpoints of every plaquette at which some $f_i\neq0$; $\mathrm{Var}(\chi)$ is the closed set of the points near which χ is not constant. The cut-off variation density and the flux insertion of Section YM3_I:112 are

$$h_{\omega,\chi}(p)=a^{-1}\sum_{b\subset\partial p}\sum_\rho[\chi(x_b)-\chi(x(p))](\omega x_b)_\rho\,\mathfrak{d}_{b,p,\rho},\qquad \mathfrak{B}_{\omega,\chi}=\mathsf{b}\,\mathcal{O}(\nabla^s_\omega\chi)-\beta\sum_p h_{\omega,\chi}(p)-\tfrac32 a\,\mathcal{O}(v_\chi),$$

with $v_\chi(p)=\sum_{\rho\in P(p)}(\omega x(p))_\rho\nabla^c_\rho\chi(x(p))$; $\mathfrak{B}_{\omega,\chi}$ depends on U near $\mathrm{Var}(\chi)$ only, and $\beta\sum_p h_{\omega,\chi}(p)=\Theta_\beta[\chi\,\omega y]-\beta\,\mathfrak{S}_\omega[\chi]$. With $\mathcal{O}(f.)=\mathcal{O}(f_1);\dots;\mathcal{O}(f_n)$, the bulk term and the cubic remainder are

$$\mathfrak{U}_{\omega,\chi}(f)=-\beta\langle\mathfrak{S}_\omega[\chi]+\mathfrak{K}^{\mathrm{v}}[\chi\,\omega y];\mathcal{O}(f.)\rangle+a^{-1}\sum_i\langle\mathfrak{S}_\omega[f_i]+\mathfrak{K}^{\mathrm{v}}[f_i\,\omega y];(\mathcal{O}(f_l))_{l\neq i}\rangle,$$

$$\mathfrak{U}^\sharp[\eta](f)=-\beta\langle\mathfrak{K}^\sharp[\eta];\mathcal{O}(f.)\rangle+a^{-1}\sum_i\langle\mathfrak{K}^\sharp[f_i\eta];(\mathcal{O}(f_l))_{l\neq i}\rangle,$$

and, for χ covering the f_i, the rotational Ward identity at finite cutoff is

$$\sum_i\langle\mathcal{O}(f_1);\dots;\mathcal{O}(\nabla^s_\omega f_i);\dots;\mathcal{O}(f_n)\rangle=\mathfrak{U}_{\omega,\chi}(f)+\langle\mathfrak{B}_{\omega,\chi};\mathcal{O}(f.)\rangle.$$

The spread of sets $S_0,\dots,S_n$ is $\inf\max_{i<i'}|y_i-y_{i'}|_\infty$ over $y_i\in S_i$; it is at least the sup distance of any two of them.

5.1.2. *Principal results.*

The arbitrary-order interacting comparison. The cyclic Gaussian calculation is explicit. Extending it to the interacting Wilson measure at arbitrary order still requires uniform estimates for the small-field remainder and for histories with a first large-field creation. For a fixed integer $n\ge2$, the quantitative estimate sought would provide a coupling threshold γ_n, depth thresholds $s_n(g)$ and $\mathsf{d}_n(g,\ell)$, and errors $\varepsilon_n(g,s)$ and $\omega_n(g,s;c)$, uniform in the cutoff, label, position and weights, with the properties below. $\varepsilon_n(g,s)\to0$ as $s\to\infty$ and use the explicit large-field majorant

$$\omega_n(g,s;c)=L^{3s}(\mathsf{Z}^\vee_s)^n\big[L^{4n}c_\Sigma+s\big]\exp\big[-cA_0^2(Y^*_s)^{2\mathfrak{p}_0}\big],\qquad c_\Sigma=\sum_{l\ge0}L^{-2l}(l+2)^{2n\mathfrak{p}_0},$$

$Y^*_s=\log(\frac45L^{-t_1(s)}x^-_s)$, $x^-_s=\frac45L^{s-2}g^{-2}(L^2C_W(s)Y_s^{3\mathfrak{p}_0})^{-1}$, $Y_s=\log\bar{x}_s$, and $\mathsf{Z}^\vee_s\ge1$, $t_1(s)\ge1$ are numbers formed with the basic constants, n, g and s: $\mathsf{Z}^\vee_s$ bounds, in the units of the levels above the working level of depth s, the sizes of the n sources carried there by Thm. YM3_I:D, and is at most a power of $\bar{x}_s$ times a polynomial in Y_s, and $L^{t_1(s)}$ is at most a polynomial in Y_s; so $\mathsf{L}_w\omega_n(g,s;c)$ bounds the large-field histories summed against their weights w_t, each with the n-th power of the sizes of the sources at its level.

Let $n\ge2$, $0<g\le\gamma_n$, $(K,\beta)\in\mathfrak{D}(g)$, $\ell\in\mathfrak{L}(g)$, $0<\mathsf{d}<1$, $s_n(g)\le s=s(\mathsf{d})\le K$, and let $c,\mathsf{L}_w>0$ satisfy $\mathsf{w}_t\le\mathsf{L}_wL^{3(K-t)}e^{-cp_0(g_t)^2}$ $(1\le t\le K)$, $\varsigma_K(c)\ge4n+5$ and $\mathsf{L}_w\omega_n(g,s;c)\le\frac18$;

here w_t is the probability of a first creation of a large field at level t, $p_0(g_t) = A_0(\log g_t^{-2})^{\mathfrak{p}_0}$, and $\varsigma_K(c) = 2c\mathfrak{p}_0 A_0^2(\log g_K^{-2})^{2\mathfrak{p}_0-1}$. For real weights w_a and points x_a with $|x(p) - x_a| < \mathsf{d}$ where $w_a(p) \neq 0$ and $|x(p) - x(q)| \geq \mathsf{d}$ where $w_a(p) \neq 0 \neq w_b(q)$, $a \neq b$,

$$\big|\langle \mathcal{O}(w_1); \dots; \mathcal{O}(w_n)\rangle_{K,\ell,\beta} - 3\mathsf{b}^{-n}\mathcal{K}_K^{(n)}(w)\big| \leq \mathsf{e}_n\, \mathsf{b}^{-n} \prod_a \sup|w_a|, \qquad \mathsf{e}_n = \varepsilon_n(g,s) + 2(1 + \bar{B}g^2 L^{2-s})^n \mathsf{L}_w \omega_n(g,s;c).$$

These target estimates are needed with $c = c'$ and $\mathsf{L}_w = 2M^{-3}\ell^3 e^{B(X_g)\ell^3}$ for $\mathsf{d} \leq \mathsf{d}_n(g,\ell)$ and every $K \geq s(\mathsf{d})$, and then $\mathsf{e}_n \to 0$ as $\mathsf{d} \to 0$ (for $n \geq 6$ provided $\log g^{-2} \geq (\frac{4n+5}{2c'\mathfrak{p}_0 A_0^2})^{1/(2\mathfrak{p}_0-1)}$).

The uniform interacting remainder has not been established here. The cyclic Gaussian calculation and the displayed large-field majorant do not prove it. The free-law conclusions described below remain aims of the sketch at the orders for which this comparison is required.

5.1.3. *What a variation measures.* A plane weight changes the coefficient of the plaquette action in one coordinate plane. Differentiating in that weight inserts the corresponding diagonal curvature bilinear. Symmetric products of two corner curvatures provide the mixed components. This interpretation organizes the section: first bound separated insertions, then take their cutoff limits, and finally identify the response of the constrained backgrounds used by the construction. The bounds on arbitrary separated cumulants, the specified low-order short-distance laws and the weighted-background construction are different assertions and use different estimates.

For the separated cumulants, localize each insertion into cells of a reflection board and centres it at its mean on a reference torus. Products of mirror copies are expanded by the moment–cumulant formula. The estimates on one torus then give uniform bounds in other periodic states through the chessboard inequality and domination by conditional torus laws. A diagonal subsequence gives the plane-field limits. Full-sequence convergence on a fixed torus requires the separate consecutive-cutoff estimate in theorem M. That estimate contains the Lipschitz norms of the test functions because a change of mesh samples them at different barycentres. The fixed-cutoff bound itself uses only their suprema.

The same separation matters in the Ward identity. A displacement $\chi(x)\omega x$, with ω antisymmetric, agrees with a rotation near the spectator supports but varies in the shell where χ changes. Haar invariance and integration by parts in the bond variables give the exact finite-lattice identity. Its divergence terms are null inside the relevant expectation; its remaining principal term is the stress insertion, with the cubic and lattice remainders displayed in lemma 5.9. The insertion estimates control those remainders and pass the identity to the continuum. The shell stress flux remains in that identity. Its vanishing requires the stated limiting argument; boundedness of the shell insertion alone does not give vanishing. The stress-conservation and rotation sections explain the subsequent use of this identity.

There is a further proposed short-distance comparison at arbitrary order. For free photons its cyclic Gaussian expression is exact. The large-field estimates also provide the displayed bounds for histories that leave the small-field region. The interacting comparison also requires a uniform interacting small-field remainder at every source order, with the n-dependent constants specified below. The free cyclic calculation and large-field bound give the other parts of the comparison; they do not supply this remainder. The separated cumulant bounds, the stated low-order laws and the regular-reference response just described have their own supplied arguments and do not require that extension.

Theorem M (Plane fields)**.** Let $0 < g \leq \gamma_W^{\mathrm{ref}}$ and $\ell' \in \mathfrak{L}(g)$.

(a) *One torus.* For $r \ge 2$ and $0 < \mathsf{d} < 1$ there are $K_0(r,\mathsf{d})$, depending on g, r, d, and $k(r,\mathsf{d})$, depending besides on ℓ', such that

$$|\langle \mathcal{O}(w_1); \dots; \mathcal{O}(w_r)\rangle_{K,\ell',\beta}| \le k(r,\mathsf{d}) \prod_a \sup|w_a|$$

for all $(K,\beta) \in \mathfrak{D}(g)$ with $K \ge K_0(r,\mathsf{d})$ and all real weights whose sets of barycentres are mutually at distance $\ge \mathsf{d}$; the weights need no regularity and no sign, and they may depend on the plane; with $2^r k(r,\mathsf{d})$ the same holds for complex weights. For every $p > 7$, $k(r,\mathsf{d}) \le C_{r,p}\mathsf{d}^{-pr}$. Besides, the growth in the order is controlled: for every $0 < g \le \gamma$ with $g^{-2} \ge x_{\mathrm{J}}$, every side $\ell \in \mathfrak{L}(g)$, every $q \ge 2$, every $\mathsf{d} \ge a_K$, all planes and all bounded complex functions $u_1, \dots, u_q$ on T_ℓ whose supports are at pairwise sup distances $\ge 2\mathsf{d}$, and for K above a threshold depending on g, q, d,

$$|\langle \mathcal{O}_{\lambda_1}(u_1); \dots; \mathcal{O}_{\lambda_q}(u_q)\rangle_{K,\ell,\beta}| \le A_\ell^q\, (q!)^{\frac{21}{2}+\mathfrak{p}_0}\, e^{36 c_L q^2} \max\{1, \mathsf{d}^{-4q}\} \prod_a \sup|u_a|, \qquad c_L = \frac{(\log L)^2}{2c' A_0^2 (\log \frac{L}{2})^2},$$

with A_ℓ depending on g and ℓ.

(b) *Every state.* For $n \ge 2$ and $\mathsf{d}_0, R > 0$ there is $C_n(\mathsf{d}_0; R)$, depending on $g, \ell', n, \mathsf{d}_0, R$, such that for every $(K,\beta) \in \mathfrak{D}(g)$, all planes and all bounded real f_i supported in closed balls of radius R at mutual distances $\ge \mathsf{d}_0$,

$$|\langle \mathcal{O}_{\lambda_1}(f_1); \dots; \mathcal{O}_{\lambda_n}(f_n)\rangle_s| \le C_n(\mathsf{d}_0; R) \prod_i \sup|f_i|,$$

where s is any member of $\mathcal{G}^{\mathrm{per}}_{K,\beta}$, or the measure of T_ℓ with $\ell = 2L^m \ge L\ell'$ in $\mathfrak{L}(g)$ and the supports in $(-\frac{\ell}{4}, \frac{\ell}{4})^3$. The same holds for bounded weights, and $|\nu'(\mathcal{O}_\lambda(f)) - \nu(\mathcal{O}_\lambda(f))| \le C_1(R)\sup|f|$ for any two such states ν, ν'.

(c) *Limit points.* Let $(K_j, \beta_j) \in \mathfrak{D}(g)$, $K_j \to \infty$, and let s_j be either any periodic Gibbs state at (K_j, β_j), or the measure of T_{ℓ_j} with $\ell_j \in \mathfrak{L}(g)$, $\ell_j \ge L\ell'$, $\ell_j \to \infty$, the cumulants being formed once the supports lie in $(-\frac{\ell_j}{4}, \frac{\ell_j}{4})^3$. Every subsequence has a further subsequence along which the cumulants of (b) converge for all $n \ge 2$, all planes and all real $f_i \in C^1_c(\mathbb{R}^3)$ with disjoint supports. The limits S_n are real, multilinear and symmetric; they obey the bound of (b) and $|S_n| \le B^n (n!)^{4+p} \prod_i \sup|f_i|$ for every $p > 7$, with B depending on $g, \ell', \mathsf{d}_0, R, p$ and not on n; they are invariant under $x \mapsto rx + v$, $r \in W_3$, $v \in \mathbb{R}^3$, the planes being permuted as the axes, and their sum over the planes is the limit of S_n^T.

With $\Pi(J) = \sum_\pi \prod_{I \in \pi} S_{\#I}(I)$, π running over the partitions of a finite family J of pairs (plane, function) into blocks of at least two members, $\sum_{\alpha,\alpha'} \bar z_\alpha z_{\alpha'} \Pi(\theta J_\alpha \cup J_{\alpha'}) \ge 0$ for families J_α on one side of a mirror θ: at every coordinate mirror, the planes being fixed, in both cases; and, for limits of periodic Gibbs states, also at the diagonal mirrors with normal $e_\mu \pm e_\nu$, which exchange the planes μ, ν in the reflected entries.

(d) *The full sequence on a torus.* Let $0 < g \le \gamma$ with $g^{-2} \ge x_{\mathrm{J}}$, $\mathsf{b} \in I(g)$, $\ell \in \mathfrak{L}(g)$, $n \ge 2$, $\mathsf{d} > 0$ and $\theta_n = 1/(1260n(8n+3))$. There are C, depending on n and b, and K_1, depending on $g, \ell, \mathsf{d}, \mathsf{b}, n$, such that for $K \ge K_1$, all planes and all real ϕ_a with $\mathrm{Lip}_\infty \phi_a < \infty$ and supports at pairwise sup distances $\ge \mathsf{d}$, $G_K = \langle \mathcal{O}_{\lambda_1}(\phi_1); \dots; \mathcal{O}_{\lambda_n}(\phi_n)\rangle_{K,\ell,L^K\mathsf{b}}$ obeys

$$|G_{K+1} - G_K| \le C\, \ell^{3(n+1)} K^{2\mathfrak{p}_0+2} L^{-\theta_n K} \prod_a (\sup|\phi_a| + \mathrm{Lip}_\infty \phi_a).$$

So $\lim_K G_K$ exists with the summed rate; for continuous ϕ_a with supports at pairwise sup distances $\ge 2\mathsf{d}$ it exists without a rate; and the same holds with $\mathsf{b}\check D_m$ for an entry. With

one plane field among action densities the rate is better: for real Lipschitz $k, f_1, \dots, f_n$ with supports pairwise $\ge \mathsf{d}$ apart, $K \ge K_1$ and a C depending on $g, \ell, n, \mathsf{d}, \mathsf{b}$,

$$\big|\langle \mathcal{O}_\lambda(k); \mathcal{O}(f_\cdot)\rangle_{K+1,\ell,L^{K+1}\mathsf{b}} - \langle \mathcal{O}_\lambda(k); \mathcal{O}(f_\cdot)\rangle_{K,\ell,L^K\mathsf{b}}\big| \le C K^{2\mathfrak{p}_0(n+2)+2} L^{-K/80} \|k\|_\sharp \prod_i \|f_i\|_\sharp .$$

The Lipschitz constants cannot be dropped from the bound on $|G_{K+1} - G_K|$: it fails with $\prod_a \sup|\phi_a|$ alone for bounded functions sampled at the barycentres, for instance for the indicator of the points of a ball that are barycentres at an even cutoff and at no smaller cutoff, whereas the bound on $|G_K|$ itself needs sup norms only, by (a). One pair (C, K_1) serves a compact interval of labels in $I(g)$; every $\mathsf{b} \ge 2\max\{\gamma^{-2}, x_{\mathrm{J}}\}$ is admitted at $g = (\mathsf{b}/2)^{-1/2}$, and by the dilation $G_K(\phi; \mathsf{b}, \ell) = L^{zn} G_{K-z}(\phi(L^{-z}\cdot); L^z\mathsf{b}, L^z\ell)$ the same holds at every label $\mathsf{b} > 0$ on every side $2L^m$.

(e) *Two points at one scale.* Let $0 < g \le \gamma_W^{\mathrm{sd}}$, $\ell = \ell_g$, $\vartheta \in (0,1)$, $(K,\beta) \in \mathfrak{D}(g)$, $0 < \mathsf{d} \le \mathsf{d}_0^{\mathrm{ref}}(g,\vartheta)$ with $s(\mathsf{d}) \le K$, $\mathsf{d}_0^{\mathrm{ref}}(g,\vartheta)$ being the threshold in equation YM3_I:(2335), and let (f,h) be a test pair of $\mathfrak{T}_{T_\ell}(\mathsf{d};\vartheta)$ ([7, Lem. I.1]). Then, uniformly in K,

$$\Big|\langle \mathcal{O}_\lambda(f); \mathcal{O}_{\lambda'}(h)\rangle_{K,\ell,\beta} - \tfrac{3}{2}\mathsf{b}^{-2}\mathcal{K}_{\lambda\lambda'}(f,h)\Big| \le C(g,\vartheta)\big(\log\tfrac{1}{\mathsf{d}}\big)^{-2\mathfrak{p}_0}\mathsf{b}^{-2}\|f\|_\infty\|h\|_\infty .$$

Since $\mathcal{K}_{\lambda\lambda'}(f,h) \ge \frac{6}{175}(\frac{\vartheta}{20})^{10}\|f\|_\infty\|h\|_\infty$, the nine two-point functions are positive for small d. So is the two-point function of $\mathcal{O}_1 - \mathcal{O}_2$ at the test pairs with $\hat u_1^2 + \hat u_2^2 \le \frac14$ for all $u \in \operatorname{supp} f - \operatorname{supp} h$, for instance at those carried by balls of radius d whose centres are $4\mathsf{d}e_3$ apart: there

$$\begin{aligned}&[\mathbb{K}_3(u)]_{11}^2 - 2[\mathbb{K}_3(u)]_{12}^2 + [\mathbb{K}_3(u)]_{22}^2\\ &\quad = \frac{9(\hat u_1^2 - \hat u_2^2)^2 - 6(\hat u_1^2 + \hat u_2^2) + 2}{16\pi^2|u|^6} \ge \frac{1}{32\pi^2|u|^6},\end{aligned}$$

and, since $|u| \le 10\mathsf{d}$ on $\operatorname{supp} f - \operatorname{supp} h$ ([7, Lem. I.1](a)) and the masses of f, h are at least $\vartheta\mathsf{d}^3\|f\|_\infty$, $\vartheta\mathsf{d}^3\|h\|_\infty$ ([7, Lem. I.1]), its main term is at least

$$\frac{3}{2}\mathsf{b}^{-2}(32\pi^2)^{-1}10^{-6}\vartheta^2\|f\|_\infty\|h\|_\infty,$$

against 4 times the error of the display; so no plane field is a multiple of $\mathcal{O}$. The law holds with 9 times the error for $\mathcal{O} - 2\mathcal{O}_m$. For two entries $\mathsf{b}[\mathcal{O} - 2\mathcal{O}_m]$, both the main term and the error are multiplied by b^2. The two-point law on a torus of small side, Thm. YM3_I:AU in the form of Prop. YM3_I:126.4 against the lattice kernel, holds for all complex weights w_1, w_2 carried by balls of radius $\mathsf{d} - a_K$ about points x_0, y_0 with $3\mathsf{d} \le |x_0 - y_0| \le 4\mathsf{d}$, with 16 times its tolerance (4 times for real weights), whatever the dependence of the weights on the plane.

(f) *Free photons and the conditional continuum comparison.* For three free lattice photon fields the cyclic expression $3\mathsf{b}^{-n}\mathcal{K}_K^{(n)}(w)$ is the exact cumulant. For the interacting plane fields, the aim is to obtain the corresponding comparison from the estimate to be established in §5.1.2, with its stated data and error e_n. For real Lipschitz ϕ_a with $\operatorname{supp}\phi_a \subset B(x_a,\mathsf{d}) \subset B(c,\frac{\ell}{4})$, $\operatorname{Lip}\phi_a \le \|\phi_a\|_\infty/(\vartheta_0\mathsf{d})$, $0 < \vartheta_0 \le 1$, and supports $\ge \mathsf{d}$ apart, the plane fields $\mathcal{O}_{\lambda_a}(\phi_a)$ obey the law with $\mathcal{K}_\lambda^{(n)}(\phi)$ for $\mathcal{K}_K^{(n)}(w)$ and with the bracket $\mathsf{e}_n + C_n'(\frac{a_K}{\vartheta_0\mathsf{d}} + \frac{\mathsf{d}^3}{\ell^3})$, where $C_n' = \frac32 n![6^n C_{\mathrm{lat}}(1 + 3C_{\mathrm{lat}})^{n-1} + 43\cdot 11^n]$; with $3^nC_n'$ the same holds for the combinations $\sum_\lambda k_{a\lambda}\mathcal{O}_\lambda$, $k_a \in [-1,1]^3$, such as $\mathcal{O} - 2\mathcal{O}_m$. Multiplying any of these entries by b multiplies both the main term and its error by b for that entry.

(g) *The law in every state.* (i) Let $0 < g \le \gamma_W^{\mathrm{sd}}$, $n \ge 1$ and $\sigma \in (0,1)$. There are $\mathsf{d}_2 > 0$ and C, c, depending on g, n, σ, such that for $(K,\beta) \in \mathfrak{D}(g)$, $0 < \mathsf{d} \le \mathsf{d}_2$ with $s(\mathsf{d}) \le K$, all bounded real f_a vanishing outside closed balls of radius d inside one ball of radius $9\mathsf{d}$, at mutual

distances $\geq \mathsf{d}$, and every state s at (K,β),

$$|S_n^T[s](f) - S^T_{n;K,\ell_g,\beta}(f)| \leq \mathsf{e}'_n(g,\mathsf{d})\,\mathsf{b}^{-n}\prod_a \|f_a\|_\infty, \qquad \mathsf{e}'_n(g,\mathsf{d}) \leq C(\log\tfrac{1}{\mathsf{d}})^{cn}\mathsf{d}^{\,3(1-\sigma)-[\frac{n}{2}-1]_+};$$

The free law in every state remains a target until the interacting estimate in §5.1.2 is established on T_{ℓ_g}. Once it is established, and $g \leq \min\{\gamma_n, \gamma_W^{\mathrm{sd}}\}$, the preceding comparison and (f) give the free law in every state for $2 \leq n \leq 7$, under the depth hypotheses of $(\mathrm{Q_n})$. (ii) For the arbitrary-order extension, one must establish that interacting estimate on $T_{\ell'}$ at every order $2 \leq m \leq 8n$. Let $n \geq 2$, $0 < g \leq \min_{2\leq m\leq 8n}\gamma_m$ with $\log g^{-2} \geq (\frac{32n+5}{2c'\mathfrak{p}_0 A_0^2})^{1/(2\mathfrak{p}_0-1)}$, and $\ell' \in \mathfrak{L}(g)$. For $0 < \mathsf{d} \leq \min_{2\leq m\leq 8n}\mathsf{d}_m(g,\ell')$, every $(K,\beta) \in \mathfrak{D}(g)$ with $K \geq s(\mathsf{d})$, every $s \in \mathcal{G}^{\mathrm{per}}_{K,\beta}$ and the measure s of every torus T_ℓ, $\ell \geq \ell'$ in $\mathfrak{L}(g)$, and real plaquette weights with the support and separation conditions of §5.1.2, inside a cube of side $\frac{1}{4}\ell'$,

$$\big|\kappa_s(\mathcal{O}(w_1),\dots,\mathcal{O}(w_n)) - 3\mathsf{b}^{-n}\mathcal{K}^{(n)}_{K,\ell'}(w)\big| \leq \mathsf{e}^\sharp_n(g,\ell',\mathsf{d})\,\mathsf{b}^{-n}\prod_a \sup|w_a|,$$

where $\mathsf{e}^\sharp_n$ depends on no state, side or cutoff and tends to 0 with d. The same argument is intended to give the following comparison for every non-empty $B \subset \{1,\dots,n\}$. For $n = 1$ this target requires the corresponding interacting estimates for $2 \leq m \leq 8$, in their coupling and depth ranges:

$$\big|\kappa_s((\mathcal{O}(w_a))_{a\in B}) - \kappa_{T_{\ell'}}((\mathcal{O}(w_a))_{a\in B})\big| \leq \eta^\sharp(1+\eta^\sharp)^{n-1}\mathsf{b}^{-\#B}\prod_{a\in B}\sup|w_a|,$$

$\kappa_{T_{\ell'}}$ being the joint cumulant under the measure of $T_{\ell'}$, with $\eta^\sharp \to 0$ as $\mathsf{d} \to 0$. The same holds for the plane fields and their combinations $\mathcal{O} - 2\mathcal{O}_m$, with the corresponding three-plane coefficient factors. For $\mathsf{b}[\mathcal{O} - 2\mathcal{O}_m]$, multiply the comparison bound by b for every such entry.

(h) *Short distance in every state and in every limit.* Let $0 < g \leq \gamma_\sigma$, where $\gamma_\sigma \leq \gamma_W^{\mathrm{sd}}$ is a constant formed with the constants, $\mathsf{d}_\sigma(g) > 0$ a threshold and $\Omega_\sigma = \{(x,y) : 0 < |x-y| < \mathsf{d}_\sigma(g)\}$; put $E_2(g,t)$, $E_3(g,t)$ for the tolerances at the distance t of the laws of two and of three points at narrow bumps, which Step 4 proves on the least torus and (i) and (iii) below state, functions of g and t formed with the constants, $E_2(g,t) \leq C(g)(\log\frac{1}{t})^{19\mathfrak{p}_0}t^{1/2}$, $E_3(g,t) \leq C(g)[(\log\frac{1}{t})^{39\mathfrak{p}_0+2}t^{1/2} + (t/\ell_g)^3]$.

(i) At a cutoff, in every state: for $(K,\beta) \in \mathfrak{D}(g)$, every state s at (K,β), points x_0, y_0 with $\mathsf{d}_0 = |x_0 - y_0| \leq \mathsf{d}_\sigma(g)$, $\rho > 0$ with $s(\rho) \leq K$, and bounded real f_1, f_2 supported in balls of radius ρ inside $B(x_0, \frac{1}{8}\mathsf{d}_0)$, $B(y_0, \frac{1}{8}\mathsf{d}_0)$,

$$\Big|\langle\mathcal{O}_\lambda(f_1);\mathcal{O}_\mu(f_2)\rangle_s - \tfrac{3}{2}\mathsf{b}^{-2}\sum_{\lambda(p)=\lambda}\sum_{\lambda(q)=\mu} f_1(x(p))f_2(x(q))C_K^{\mathrm{free}}(p,q)^2\Big| \leq E_2(g, \tfrac{1}{2}\mathsf{d}_0)\Big(\frac{2\rho}{\mathsf{d}_0}\Big)^6\mathsf{b}^{-2}\|f_1\|_\infty\|f_2\|_\infty,$$

C_K^{free} being that of T_{ℓ_g}; for real Lipschitz ϕ_a with $\mathrm{Lip}\,\phi_a \leq \|\phi_a\|_\infty/(\vartheta\rho)$ the same holds with $\frac{3}{2}\mathsf{b}^{-2}\mathcal{K}_{\lambda\mu}(\phi_1,\phi_2)$ and with the bracket $E_2 + C\frac{a_K}{\vartheta\rho}$. By the change of units $x \mapsto L^z x$ this narrow form holds at every $0 < g \leq \gamma$, on every side of $\mathfrak{L}(g)$, for distances d_0 below a threshold depending on g.

(ii) Let $(K_j,\beta_j) \in \mathfrak{D}(g)$, $K_j \to \infty$, with labels $\mathsf{b}_j \to \mathsf{b}$, and let s_j be states at (K_j,β_j). Along a subsequence $\langle\mathcal{O}_\lambda(\phi_1);\mathcal{O}_\mu(\phi_2)\rangle_{s_j}$ converges for all real Lipschitz ϕ_a with $\mathrm{supp}\,\phi_1 \times$

$\operatorname{supp}\phi_2 \subset \Omega_\sigma$, and every such limit has Borel kernels $\tau_{\lambda\mu}(x,y) = \hat\tau_{\lambda\mu}(x-y)$ with

$$\begin{aligned}&|\tau_{\lambda\mu}(x,y) - \tfrac{3}{2}\mathsf{b}^{-2}[\mathbb{K}_3(x-y)_{\lambda\mu}]^2| \\ &\qquad\le (\tfrac{3}{\pi})^2\epsilon_\sigma(g,|x-y|)\,\mathsf{b}^{-2}|x-y|^{-6} \\ &\qquad\qquad \text{almost everywhere,}\end{aligned}$$

$$\epsilon_\sigma(g,t) = 64\sup_{0<t'\le t} E_2(g,\tfrac{1}{2}t') \le C(g)(\log\tfrac{1}{t})^{19\mathfrak{p}_0}t^{1/2};$$

before a kernel is named, at every centre and for bumps of radius ρ in $B(x_0,\frac{1}{8}\mathsf{d}_0)$, $B(y_0,\frac{1}{8}\mathsf{d}_0)$, the limit differs from $\frac{3}{2}\mathsf{b}^{-2}\mathcal{K}_{\lambda\mu}(\phi_1,\phi_2)$ by at most

$$64E_2(g,\frac{1}{2}\mathsf{d}_0)\mathsf{b}^{-2}\mathsf{d}_0^{-6}\rho^6\|\phi_1\|_\infty\|\phi_2\|_\infty.$$

If besides, for all $n\ge 2$, the $S_n^T[s_j]$ converge on separated tuples to the connected functions $S_n^{\sigma'}$ of reconstruction data σ' on $\mathbb{R}^3$, let s_n^T be the continuous kernel of $S_n^{\sigma'}$ ($n=2,3$): with the continuous kernels $\mathsf{s}_1,\mathsf{s}_2,\mathsf{s}_3^{\mathrm{m}}$ of the moments of σ' (Prop. YM3_I: 108.3), $\mathsf{s}_2^T = \mathsf{s}_2 - \mathsf{s}_1\otimes\mathsf{s}_1$ and

$$\begin{aligned}\mathsf{s}_3^T(x,y,z) = \mathsf{s}_3^{\mathrm{m}}(x,y,z) - \mathsf{s}_2(x,y)\mathsf{s}_1(z) - \mathsf{s}_2(x,z)\mathsf{s}_1(y) - \mathsf{s}_2(y,z)\mathsf{s}_1(x) \\ + 2\mathsf{s}_1(x)\mathsf{s}_1(y)\mathsf{s}_1(z).\end{aligned}$$

Then $\sum_{\lambda,\mu}\tau_{\lambda\mu} = \mathsf{s}_2^T$ almost everywhere on Ω_σ, so that $|\mathsf{s}_2^T(x,y) - \frac{9}{16\pi^2}\mathsf{b}^{-2}|x-y|^{-6}| \le 9\epsilon_\sigma(g,|x-y|)\mathsf{b}^{-2}|x-y|^{-6}$ on Ω_σ.

(iii) Three points at two scales: in the latter case, at every (x,y,z) with $0<|x-y|=r_*\le \mathsf{d}_* = \operatorname{dist}(z,\{x,y\}) \le \mathsf{d}_\sigma(g)$,

$$|\mathsf{s}_3^T(x,y,z) - 3\mathsf{b}^{-3}k_3(x,y,z)| \le E_3(g,\mathsf{d}_*)\,\mathsf{b}^{-3}r_*^{-3}\mathsf{d}_*^{-6},$$

uniformly in $r_*/\mathsf{d}_* \in (0,1]$.

(iv) Point splitting: off one null set of pairs (c,z) with $0<\mathsf{d}=|c-z|\le\frac{1}{2}\mathsf{d}_\sigma(g)$, for all $0<r\le\frac{1}{2}\mathsf{d}$ and all λ,

$$\begin{aligned}|\tfrac{2\pi}{3}\mathsf{b}r^3\mathsf{s}_3^T(c+\tfrac{r}{2}e_\lambda, c-\tfrac{r}{2}e_\lambda, z) - t_\lambda(c,z)| \le [\epsilon'_\sigma(g,\mathsf{d}) + 2.7(r/\mathsf{d})^2]\mathsf{b}^{-2}\mathsf{d}^{-6}, \qquad \epsilon'_\sigma(g,\\ \mathsf{d}) \le C(g)(\log\tfrac{1}{\mathsf{d}})^{39\mathfrak{p}_0+2}\mathsf{d}^{1/2},\end{aligned}$$

$t_\lambda = \sum_\mu(\tau_{\lambda\mu} - \frac{1}{3}\sum_{\lambda'}\tau_{\lambda'\mu})$ being the kernel of the limit of $\langle\mathcal{O}_\lambda - \frac{1}{3}\mathcal{O};\mathcal{O}\rangle$; and at every such (c,z), for every unit vector n,

$$\begin{aligned}|\frac{2\pi}{3}\mathsf{b}r^3\mathsf{s}_3^T(c+\frac{r}{2}n, c-\frac{r}{2}n, z) - \mathsf{t}^\circ(n, c-z)| \le [12\sup_{\frac{3}{4}\mathsf{d}\le t\le\frac{17}{16}\mathsf{d}} E_3(g,t) \\ + 2.7(r/\mathsf{d})^2]\mathsf{b}^{-2}\mathsf{d}^{-6}.\end{aligned}$$

(v) On the least torus at a fixed label $\mathsf{b}\in I(g)$, for g small enough in terms of the constants. For real Lipschitz ϕ_1,ϕ_2 with disjoint supports, $\lim_K\langle\mathcal{O}_\lambda(\phi_1) - \frac{1}{3}\mathcal{O}(\phi_1);\mathcal{O}(\phi_2)\rangle_{K,\ell_g,L^K\mathsf{b}}$ exists. Let moreover $\mathsf{d}>0$ with $6\mathsf{d}\le\min\{1,\mathsf{b}\}$ and $0<\vartheta_0\le 1$, let ϕ_1,ϕ_2 be real Lipschitz functions each carried by a ball of radius d, with supports $\ge\mathsf{d}$ apart, $|c-z|\le 6\mathsf{d}$ for $c\in\operatorname{supp}\phi_1$ and $z\in\operatorname{supp}\phi_2$, and $\operatorname{Lip}\phi_1 \le \|\phi_1\|_\infty/(\vartheta_0\mathsf{d})$; for $r>0$ put $\rho=\frac{r}{8}$ and let $f_\pm^c(y) = \frac{3}{\pi\rho^3}(1-|y-c\mp\frac{r}{2}e_\lambda|/\rho)_+$ be the radial tents of radius ρ and mass 1 about $c\pm\frac{r}{2}e_\lambda$, of sup norm $\frac{3}{\pi\rho^3} = \frac{1536}{\pi r^3}$. There are $r_*>0$, depending on g,b,d, and $C(g,\mathsf{b},\mathsf{d},\vartheta_0)$ such that for $0<r\le r_*$ and every K with $128a_K\le r$, that is

$16a_K \le \rho$, and $s(\rho) \le K$

$$\Big|\tfrac{2\pi}{3}\mathsf{b}r^3\int \phi_1(c)\, S^T_{3;K,\ell_g,L^K\mathsf{b}}(f^c_+, f^c_-, \phi_2)\,\mathrm{d}c - \big\langle \mathcal{O}_\lambda(\phi_1) - \tfrac{1}{3}\mathcal{O}(\phi_1); \mathcal{O}(\phi_2)\big\rangle_{K,\ell_g,L^K\mathsf{b}}\Big| \le C(g,\mathsf{b},\mathsf{d},\vartheta_0)\big[r^{\beta_H/72} + (a_K/r)^{1/18}\big]\mathsf{b}^{-2}\|\phi_1\|_\infty\|\phi_2\|_\infty,$$

β_H being the Hölder exponent. As the depth function is non-increasing, $s(\rho) \le K$ gives $s(\frac{r}{2}) \le K$; and r_* is taken so that for $0 < r \le r_*$ the depth $s = s(\frac{r}{2})$ obeys $s \ge 3$ and $L^2 C_W(s) Y_s^{3\mathfrak{p}_0} \le L^{s-3}$ (both hold for small r, since $s(\frac{r}{2}) \to \infty$ as $r \to 0$ and the second holds at all large s) and $\frac{r}{2} \le \min\{r_0(\mathsf{b}), r_1(\mathsf{b})\}$, the radii, depending on g and b, up to which Step 4 forms the induction over the levels and the frame of the pair. Here $C(g,\mathsf{b},\mathsf{d},\vartheta_0) = C(g,\mathsf{b},\vartheta_0)\mathsf{d}^{-9/2}$, of which the part of the three-point function beyond its explicit Gaussian part takes $C(g,\mathsf{b})\mathsf{d}^{-5/2}$.

The explicit Gaussian parts of the two members, which take the rest, are compared on the lattice of T_{ℓ_g}: the near leg, between the two tents, against $a_K^3\mathbb{K}_3(re_\lambda)$, by the comparison of the lattice kernel with $\mathbb{K}_3$ in the setting, at the cost $(a_K/r)^2 + (r/\ell_g)^3$, the mean of $\mathbb{K}_3$, whose entries are harmonic off 0, over the two radial tents being its value at re_λ, and $\frac{2\pi}{3}r^3 \cdot 3\mathbb{K}_3(re_\lambda) = \frac{3}{2}(E_{\lambda\lambda} - \frac{1}{3}\mathbf{1})$; the far legs by the bound on the differences of C_K^{free} along one plane of the setting; and the shift of the argument of ϕ_1 by at most r costs $r\,\mathrm{Lip}\,\phi_1 \le r\|\phi_1\|_\infty/(\vartheta_0\mathsf{d})$.

Hence, at this fixed d, the limit in K is the limit as $r \to 0$ of

$$\frac{2\pi}{3}\mathsf{b}r^3\int \phi_1(c)\, S_3^{(\ell_g,\mathsf{b})}(f^c_+, f^c_-, \phi_2)\,\mathrm{d}c.$$

The limit in K exists for all real Lipschitz ϕ_1, ϕ_2 with disjoint supports, and it is identified in this way for all ϕ_1, ϕ_2 as above.

■□

Theorem N (One stress insertion; stress entries; conservation)**.**

(a) *Bounds in every geometry.* Let $0 < g \le \gamma_W^{\mathrm{ref}}$, $\ell' \in \mathfrak{L}(g)$, $n \ge 1$, $0 < \rho < 1$, $\omega \in \mathfrak{so}(3)$, and let χ be a cut-off function at K, Lipschitz on $\mathbb{R}^3$. For every $(K,\beta) \in \mathfrak{D}(g)$,

$$\Big|\Big\langle \beta\sum_p h_{\omega,\chi}(p); \mathcal{O}(f_\cdot)\Big\rangle_{K,\ell',\beta}\Big| \le \mathsf{b}\, C_n^\flat(\rho)\,|\omega|\,\mathrm{Lip}_\infty(\chi)\prod_i \|f_i\|,$$

(i) with $\|f_i\| = \sup|f_i|$ for bounded real f_i (or real weights), $\mathrm{Var}(\chi)$ and the $\mathrm{supp}\, f_i$ being pairwise $\ge \rho$ apart; (ii) with $\|f_i\| = \|f_i\|_\sharp$ for real Lipschitz f_i in any mutual position, $\mathrm{Var}(\chi)$ being at distance $\ge \rho$ from each $\mathrm{supp}\, f_i$. $C_n^\flat(\rho) = 468\ell' k_n^\flat(\rho)$, where $k_n^\flat(\rho)$ depends on g, ℓ', n, ρ, not on χ, on the f_i or on the place of the shell. In (ii) the number may be taken free of the side and of the state: on every torus T_ℓ, $\ell \ge \ell'$ in $\mathfrak{L}(g)$, and in every $\nu \in \mathcal{G}^{\mathrm{per}}_{K,\beta}$, for K above a threshold depending on g, ℓ', n, ρ, the left member is at most $156\,\mathsf{b}\,C_n^e(\rho)|\omega|\,\mathrm{Lip}(\chi)R_1(\chi)N_\chi\prod_i N_i\|f_i\|_\sharp$, where $R_1(\chi) \le \frac{1}{2} + \frac{3}{2}\ell$ is the largest $|x_b|_1$ at the bonds where χ varies, N_χ and N_i are the numbers of the cubes of a mesh $h(n,\rho)$ that meet the shell and the supports, and $C_n^e(\rho)$ depends on g, ℓ', n, ρ; at every label $\mathsf{b} > 0$ on every side $2L^m$ the same holds for all K with a number depending on $\mathsf{b}, \ell, n, \rho$.

Both bounds are cases of one rule for a rough entry F of Lem. 5.7 whose carrier lies, with respect to the f_i, as $\mathrm{Var}(\chi)$ in (i), resp. (ii): the left member with F in the place of $\beta\sum_p h_{\omega,\chi}(p)$ is at most $k_n^\flat(\rho)\|F\|\prod_i\|f_i\|$ on $T_{\ell'}$, and, in (ii), at most $C_n^e(\rho)N_F\|F\|\prod_i N_i\|f_i\|_\sharp$ free of the side and of the state, N_F being the number of the cubes of the mesh that meet the carrier of F; here $-\beta\sum_p h_{\omega,\chi}(p) = \mathsf{b}\,\mathrm{Pf}(k^h)$ with

$$\|\mathrm{Pf}(k^h)\| \le 156|\omega|\,\mathrm{Lip}(\chi)R_1(\chi) \le 468\ell'|\omega|\,\mathrm{Lip}_\infty(\chi)$$

. The same rule gives the bounds for the whole flux insertion $\mathfrak{B}_{\omega,\chi} = \mathsf{b}\,\mathrm{Pf}(k^h) + \mathcal{O}(v^{\mathfrak{B}})$, of norm $\le (158\mathsf{b}+3a)|\omega|\,\mathrm{Lip}(\chi)R'(\chi)$, $R'(\chi) = \max\{R_1(\chi), R'_1(\chi)\}$, where $R'_1(\chi) \le \frac{3}{2}\ell$ is the number of Lem. 5.7, the largest $|x(p)|_1$ at the plaquettes at which χ is not constant on the points $x(p) \pm ae_\mu$, $x(p) \pm \frac{a}{2}e_\mu$, $\mu = 1,2,3$, among which are those where $\nabla^s_\omega\chi$ or v_χ does not vanish, which hold every plaquette at which $v^{\mathfrak{B}} \neq 0$, so that the largest $|x(p)|_1$ at these may stand for $R'_1(\chi)$ throughout: on $T_{\ell'}$ with $\frac{158\mathsf{b}+3a}{156}C^\flat_n(\rho)$ in the place of $\mathsf{b}\,C^\flat_n(\rho)$, and free of the side and of the state with $(158\mathsf{b}+3a)C^e_n(\rho)|\omega|\,\mathrm{Lip}(\chi)R'(\chi)N_\chi$ in the place of $156\,\mathsf{b}\,C^e_n(\rho)|\omega|\,\mathrm{Lip}(\chi)R_1(\chi)N_\chi$.

For $\mathrm{supp}\,\eta$ at distance $\ge\rho$ from each $\mathrm{supp}\,f_i$ (and, in (i), the $\mathrm{supp}\,f_i$ pairwise $\ge\rho$ apart) it gives them for $\Theta_\beta[\eta] = \mathsf{b}\,\mathrm{Pf}(k^\eta)$, of norm $\le 156\,\mathsf{b}\max_\rho \mathrm{Lip}\,\eta^\rho$, and for $\mathfrak{K}^{\mathrm{v}}[\beta\eta]$, which beside fields that do not read the bonds within $2a$ of $\mathrm{supp}\,\eta$ is the rough entry $-\Theta_\beta[\eta] + \mathsf{b}\,\mathcal{O}(\mathrm{div}^s\eta) - \frac{3}{2}a\,\mathcal{O}(e^\eta)$ of norm $\le (159\mathsf{b}+3a)\max_\rho \mathrm{Lip}\,\eta^\rho$ (the null term of Lem. 5.9(a) drops): on $T_{\ell'}$ with $156\,\mathsf{b}\,k^\flat_n(\rho)\max_\rho \mathrm{Lip}_\infty\eta^\rho$, resp. $(159\mathsf{b}+3a)k^\flat_n(\rho)\max_\rho \mathrm{Lip}_\infty\eta^\rho$, in the place of $\mathsf{b}\,C^\flat_n(\rho)|\omega|\,\mathrm{Lip}_\infty(\chi)$, and free of the side and of the state with $156\,\mathsf{b}\,C^e_n(\rho)\max_\rho \mathrm{Lip}\,\eta^\rho N_\eta$, resp. $(159\mathsf{b}+3a)C^e_n(\rho)\max_\rho \mathrm{Lip}\,\eta^\rho N_\eta$, in the place of $156\,\mathsf{b}\,C^e_n(\rho)|\omega|\,\mathrm{Lip}(\chi)R_1(\chi)N_\chi$, N_η being the number of the cubes of the mesh that meet the $\frac{\rho}{2}$-neighbourhood of $\mathrm{supp}\,\eta$.

Bounded fields, or real weights that depend on the plane, may be added in sup norms if they are $\ge\rho$ apart from the shell, from one another and from all the other fields.

(b) *Stress entries.* $T_{lr}(\varphi)$ is b times a rough entry (Lem. 5.7) of norm at most an absolute constant times $\sup|\varphi|$. So for $0 < g \le \gamma_W^{\mathrm{ref}}$ the joint cumulants on one torus of r entries $T_{l_a r_a}(\varphi_a)$ and $\mathcal{O}(w_a)$ with supports pairwise $\ge \mathsf{d}$ apart are bounded uniformly in $K \ge K'_0(r,\mathsf{d})$, for merely bounded φ_a, w_a; for two entries this holds at every $0 < g \le \gamma$: $|\langle T_{lr}(\varphi); \mathcal{O}(f)\rangle| \le C\mathsf{b}\sup|\varphi|\sup|f|$ and $|\langle T_{lr}(\varphi); T_{l'r'}(\varphi')\rangle| \le C\mathsf{b}^2\sup|\varphi|\sup|\varphi'|$, C depending on g, the side and the distance. Along subsequences $\langle T;\mathcal{O}\rangle$ and $\langle T;T\rangle$ converge for continuous entries with disjoint supports, and the limits are symmetric in (l,r), translation invariant, covariant under W_3 with the signs of a tensor, and reflection positive.

(c) *The value, Lipschitz norms.* Let $\mathsf{b} \ge 2x_W$, $g = (\mathsf{b}/2)^{-1/2}$, $\ell' \in \mathfrak{L}(g)$, $\beta = L^K\mathsf{b}$, $n \ge 1$, $0 < \mathsf{d} < 1$ and $c < \frac{1}{14}$. There are C, p, K_*, depending on $\mathsf{b}, \ell', n, \mathsf{d}, c$, such that for $K \ge K_*$, every real C^2 vector field η and all real Lipschitz f_i, the $n+1$ supports being pairwise at sup distance $\ge \mathsf{d}$,

$$|\langle \mathfrak{K}^{\mathrm{v}}[\beta\eta]; \mathcal{O}(f_\cdot)\rangle| + |\langle \Theta_\beta[\eta] - \mathsf{b}\,\mathcal{O}(\mathrm{div}^s\eta); \mathcal{O}(f_\cdot)\rangle| \le CK^pL^{-cK}\|\eta\|_{C^2}\prod_i \|f_i\|_\sharp.$$

By $x \mapsto L^z x$ the same holds at every label $\mathsf{b} > 0$ on every side $2L^m$.

(d) *The value, sup norms.* In the setting of (c): for every field η and all real weights w_a, the $n+1$ carriers being pairwise at sup distance $\ge \mathsf{d}$, $|\beta\langle \mathfrak{K}^\sharp[\eta]; \mathcal{O}(w_1); \dots; \mathcal{O}(w_n)\rangle| \le CK^pL^{-cK}\mathsf{H}(\eta)\prod_a \sup|w_a|$, and the same with rough entries or block forms (Lems. 5.7 and 5.9) in the places of the $\mathcal{O}(w_a)$; for C^2 fields η vanishing within sup distance d of the carriers, the two members of (c) with $\mathcal{O}(w_a)$ for $\mathcal{O}(f_a)$ are $\le CK^pL^{-cK}\|\eta\|_{C^2}\prod_a \sup|w_a|$. At $n = 1$, for every real vector field η, Lipschitz for the sup distance, and every bounded real h (a function or a weight) with $\eta = 0$ within sup distance d of $\{x(p) : h(p) \neq 0\}$,

$$|\langle \Theta_\beta[\eta]; \mathcal{O}(h)\rangle - \mathsf{b}\langle \mathcal{O}(\mathrm{div}^s\eta); \mathcal{O}(h)\rangle| + |\langle \mathfrak{K}^{\mathrm{v}}[\beta\eta]; \mathcal{O}(h)\rangle| \le CK^pL^{-K/200}(\mathsf{H}(\eta) + \max_\rho \mathrm{Lip}_\infty\eta^\rho)\sup|h|.$$

For two fields with carriers $\ge \mathsf{d}$ apart, $|\beta^2\langle \mathfrak{K}^\sharp[\eta]; \mathfrak{K}^\sharp[\eta']\rangle| \le CK^{2p}L^{-2cK}\mathsf{H}(\eta)\mathsf{H}(\eta')$, and, with $\mathsf{H}^+ = \mathsf{H} + \max_\rho \mathrm{Lip}_\infty$,

$$|\langle \mathfrak{K}^{\mathrm{v}}[\beta\eta]; \mathfrak{K}^{\mathrm{v}}[\beta\eta']\rangle| \le CK^pL^{-K/200}\mathsf{H}^+(\eta)\mathsf{H}^+(\eta')$$

and

$$|\langle \mathrm{St}_K[\eta];\mathrm{St}_K[\eta']\rangle| \le CK^pL^{-K/200}\mathsf{b}\,\mathsf{H}^+(\eta)\max_\rho \mathrm{Lip}_\infty\eta'^\rho.$$

All this holds at every fixed label $\mathsf{b}>0$ on every side $2L^m$. At every first-passage datum $(K,\beta)\in\mathfrak{D}(g)$ on every side of $\mathfrak{L}(g)$, with numbers that depend on g and on no label, the clauses on the cubic insertion alone (one $\beta\mathfrak{K}^\sharp[\eta]$ beside n weights, rough entries or block forms, and two of them) hold for $0<g\le\gamma$ with $g^{-2}\ge x_{\mathrm{J}}$, and the other clauses, which also read the bilinear half of the orbital insertion and the term $a\,\mathcal{O}(e^\eta)$, for $0<g\le\gamma_W^{\mathrm{ref}}$, where $g^{-2}\ge x_{\mathrm{J}}$ holds of itself.

(e) *Conservation and collapse in every periodic Gibbs state.* Let $0<g\le\gamma$ and $R,\mathsf{d}>0$. There are C,p,K_0, depending on g,d,R, such that for all $(K,\beta)\in\mathfrak{D}(g)$ with $K\ge K_0$, all $\nu\in\mathcal{G}^{\mathrm{per}}_{K,\beta}$, all $\eta,\eta'\in C^2_c(\mathbb{R}^3;\mathbb{R}^3)$ and $H'\in C_c(\mathbb{R}^3;V)$ supported in a ball of radius R, with $\operatorname{supp}\eta$ at sup distance $\ge\mathsf{d}$ from $\operatorname{supp}\eta'$ and from $\operatorname{supp}H'$, and with $A=A^\nu$:

$$|\nu(A[\mathrm{St}_K[\eta']]A[\mathrm{St}_K[\eta]])|\le CK^pL^{-K/14}\|\eta'\|_{C^2}\|\eta\|_{C^2},\qquad |\nu(\mathfrak{K}^{\mathrm{v}}[\beta\eta])|\le CK^pL^{-K/28}\|\eta\|_{C^2},$$

$$|\nu(A[\mathcal{T}_K(H^{\mathrm{cons}}[\eta'])]A[\mathcal{T}_K(H^{\mathrm{cons}}[\eta])])|\le CK^pL^{-K/20}\|\eta'\|_{C^2}\|\eta\|_{C^2},$$

$$|\nu(A[\mathcal{T}_K(H^{\mathrm{cons}}[\eta])]A[\mathcal{T}_K(H')])|\le CK^pL^{-K/60}\|\eta\|_{C^2}|H'|_\infty,$$

and $|\nu(A[\Delta_K[\eta]]A[Z])|\le CK^pL^{-K/60}\|\eta\|_{C^2}\mathsf{z}$ for $Z=\mathcal{T}_K(H')$, $\mathsf{z}=|H'|_\infty$, and for $Z=\mathrm{St}_K[\eta']$ or $\Delta_K[\eta']$, $\mathsf{z}=\|\eta'\|_{C^2}$. The last bound says that the stress insertion collapses onto $\mathsf{b}\,\mathcal{T}_K(H^{\mathrm{cons}}[\eta])$ in every periodic Gibbs state.

(f) *Any number of entries.* Let $0<g\le\gamma$, $n\ge2$, $1\le n_1\le n$, $R>0$, $0<c<\frac{1}{28}$ and $u=\lceil\frac{1/28-c}{4n-1}K\rceil$. There are C,p, depending on g,n,R,c, and K_0, depending on g,n,c, such that for all $(K,\beta)\in\mathfrak{D}(g)$ with $K\ge K_0$, all $\nu\in\mathcal{G}^{\mathrm{per}}_{K,\beta}$, every d with $\mathsf{d}L^u\ge K^2$, all $\eta_a\in C^2_c(\mathbb{R}^3;\mathbb{R}^3)$ $(a\le n_1)$ and $H_a\in C_c(\mathbb{R}^3;V)$ $(a>n_1)$ with supports in a ball of radius R at pairwise sup distances $\ge\mathsf{d}$, and for X_a any of $\mathcal{T}_K(H^{\mathrm{cons}}[\eta_a])$, $\mathrm{St}_K[\eta_a]$, $\Delta_K[\eta_a]$, $\mathrm{Rm}_K[\eta_a]$ (Lem. 5.9(c)), $\mathfrak{K}^{\mathrm{v}}[\beta\eta_a]$ $(a\le n_1)$ and $X_a=\mathcal{T}_K(H_a)$ $(a>n_1)$,

$$\Big|\nu\Big(\prod_a A^\nu[X_a]\Big)\Big|\le CK^pL^{-cn_1K}\prod_{a\le n_1}\|\eta_a\|_{C^2}\prod_{a>n_1}|H_a|_\infty,$$

and the same holds for the joint cumulant under ν; with $H_a=h_a\mathbf{1}$ the spectators are action densities. At $n_1=1$, $c=\frac{1}{56}$ and a fixed $\mathsf{d}>0$ this holds for $K\ge K_0(\mathsf{d})$.

(g) *Every tensor limit is conserved.* Let $0<g\le\gamma$ and let σ be a tensor limit at g. Then $\sigma(H^{\mathrm{cons}}[\eta],H_2,\dots,H_n)=0$ for every $n\ge2$, $\eta\in C^2_c(\mathbb{R}^3;\mathbb{R}^3)$ and $H_a\in C_c(\mathbb{R}^3;V)$ whose supports are pairwise disjoint and disjoint from $\operatorname{supp}\eta$. In particular $\sigma(H^{\mathrm{cons}}[\theta\eta],H^{\mathrm{cons}}[\eta])=0$ for $\operatorname{supp}\eta\subset\{x_3>0\}$, $\theta\eta=\theta\circ\eta\circ\theta$, θ the mirror $x_3\mapsto-x_3$, so that in the reconstruction of σ the vector of $\mathcal{T}(H^{\mathrm{cons}}[\eta])$ vanishes; and along the sequence $\mathsf{b}_j^2\rho_j(A[\mathcal{T}(H^{\mathrm{cons}}[\theta\eta])]A[\mathcal{T}(H^{\mathrm{cons}}[\eta])])-\rho_j(A[\mathfrak{K}^{\mathrm{v}}[\beta_j\theta\eta]]A[\mathfrak{K}^{\mathrm{v}}[\beta_j\eta]])\to0$.

For $0<g\le\gamma_W^{\mathrm{sd}}$ tensor limits exist: by [7, Thm. C](a) of [7, §2], the centred products of any number of tensor entries at separated supports are bounded in every periodic Gibbs state by numbers that depend on no cutoff, side, state or position, so every sequence of first-passage data at g with $K_j\to\infty$ and of periodic Gibbs states has a subsequence along which all of them converge. Tensor limits are covariant as tensors under the translations and W_3, reflection positive at the nine families of mirrors, and their trace part is the scalar limit, $\sigma(f_1\mathbf{1},\dots,f_n\mathbf{1})$ being the limit of the centred products of the $\mathcal{O}(f_a)$.

■□

Remark 5.2 (Normalisation; the role of the collapse). Conservation at disjoint supports does not fix the normalisation of a tensor: $X_{lr}=|x|^{-2}[F_ln_r+F_rn_l-\delta_{lr}F\cdot n+n_ln_rF\cdot n]$, $n=x/|x|$, is

symmetric, traceless and conserved off 0, with the flux $\frac{16\pi}{3}F$. And Thm. N(g) rests on the collapse in (e): the decorrelation of the orbital insertion alone (the second bound of (e)) does not make the six tensor fields conserved; it is the identification of the stress insertion with $\mathsf{b}\,\mathcal{T}_K(H^{\mathrm{cons}}[\eta])$, up to $\Delta_K[\eta]$, that does.

Theorem O (Ward identities on tori)**.** Let $0 < g \le \gamma$, $\ell \in \mathfrak{L}(g)$, $n \ge 1$, $\mathsf{d} > 0$, $0 < \delta_0 < \frac{\ell}{2}$ and $c < \frac{1}{14}$.

(a) *Rotations on a fixed torus.* There are C, p, depending on $g, \ell, n, \mathsf{d}, \delta_0, c$, such that for all K, $\mathsf{b}' \in I(g)$, $\beta = L^K\mathsf{b}'$, $\omega \in \mathfrak{so}(3)$, every cut-off function χ at K of class $C^{1,1}$ with the margin δ_0, and all real f_i vanishing outside $(-\frac{\ell}{2}+\delta_0, \frac{\ell}{2}-\delta_0)^3$ and covered by χ:

(i) if $n \ge 2$, $f_i \in C^{1,1}$ and the spread of the supports is $\ge \mathsf{d}$ (the supports may touch if no point is common to all of them), $|\mathfrak{U}_{\omega,\chi}(f)| \le C|\omega|K^pL^{-cK}\|\chi\|_{C^{1,1}}\prod_i \|f_i\|_{C^{1,1}}$;

(ii) if $n = 1$, f_1 is Lipschitz and $\chi = 1$ on the sup d-neighbourhood of $\operatorname{supp} f_1$, $|\mathfrak{U}_{\omega,\chi}(f_1)| \le C|\omega|K^pL^{-cK}\|\chi\|_{C^{1,1}}\|f_1\|_\sharp$;

(iii) whatever the supports, if χ and χ' both equal 1 on the d-neighbourhood of $\bigcup_i \operatorname{supp} f_i$, $|\langle\mathfrak{B}_{\omega,\chi} - \mathfrak{B}_{\omega,\chi'}; \mathcal{O}(f_\cdot)\rangle| \le C|\omega|K^pL^{-cK}(\|\chi\|_{C^{1,1}} + \|\chi'\|_{C^{1,1}})\prod_i \|f_i\|_\sharp$.

If $n \ge 2$, the f_i are of class C^3, $\operatorname{dist}_\infty(\operatorname{supp} f_1, \operatorname{supp} f_2) \ge \mathsf{d}$ and a fixed χ covers the f_i at all large K, then $\mathfrak{U}_{\omega,\chi}(f) + \langle\mathfrak{B}_{\omega,\chi}; \mathcal{O}(f_\cdot)\rangle \to \sum_i S_n^{(\ell,\mathsf{b}')}(f_1, \dots, (\omega y)\cdot\nabla f_i, \dots, f_n)$ at the rate $K^pL^{-K/10} + L^{-K}$, and the flux alone converges to the same limit at the rate K^pL^{-cK}; the limit does not depend on χ. With nothing asked of the bulk term, for $\omega = \omega_{13}$, the generator of the turns in the plane of e_1, e_3, $f^t = f \circ e^{-t\omega}$, $f_i \in C^3$ supported in a ball inside the margin with $\operatorname{dist}(\operatorname{supp} f_1, \operatorname{supp} f_2) \ge \sqrt{3}\mathsf{d}$, and χ covering all the f^t, $0 \le t \le \frac{\pi}{2}$:

$$\Big|\int_0^{\pi/2} [\mathfrak{U}_{\omega,\chi}(f^t) + \langle\mathfrak{B}_{\omega,\chi}; \mathcal{O}(f^t_\cdot)\rangle]\,dt\Big| \le Ca_K \prod_i \|f_i\|_{C^3}, \qquad \int_0^{\pi/2} \sum_i S_n^{(\ell,\mathsf{b}')}(f_1^t, \dots, (\omega y)\cdot\nabla f_i^t, \dots, f_n^t)\,\mathrm{d}t = 0.$$

(b) *Rotations along growing tori.* Let $n \ge 2$, $R, \mathsf{d}_0 > 0$, $\chi_R(y) = \chi_1(|y|/R)$ with $\chi_1 \in C^\infty$, $\chi_1 = 1$ on $[0, \frac{4}{3}]$ and $\chi_1 = 0$ on $[\frac{5}{3}, \infty)$. There are C, p, K_0, depending on g, n, R, d_0, c and on no side, such that

$$|\mathfrak{U}_{\omega,\chi_R}(f)_{K,\ell,\beta}| \le C|\omega|K^pL^{-cK}\prod_i \|f_i\|_{C^{1,1}}$$

for all $(K, \beta) \in \mathfrak{D}(g)$ with $K \ge K_0$, every $\ell \in \mathfrak{L}(g)$ with $2R + 2a < \frac{\ell}{2}$ and all smooth f_i supported in $\{|y| < \frac{R}{2}\}$ at mutual distances $\ge \mathsf{d}_0$; without its cubic member $\mathfrak{U}^\sharp[\chi_R\,\omega y](f)$ the bulk term is $\le C|\omega|K^{1+\mathfrak{p}_0}L^{-c_*K}\prod_i \|f_i\|_{C^{1,1}}$ for every $c_* < \frac{1}{2}$. The same bounds hold under the weak limit points of the torus laws. Along every sequence (K_j, ℓ_j, β_j) with $K_j, \ell_j \to \infty$ along which the connected functions converge to S_n, $\langle\mathfrak{B}_{\omega,\chi_R}; \mathcal{O}(f_\cdot)\rangle \to -\frac{d}{dt}S_n(f(e^{-t\omega}\cdot))|_{t=0}$ for every R; so the limit is invariant under rotations if and only if this flux tends to 0.

For the volume-first scalar limits, which are invariant under O(3), the flux through every shell tends to 0. With the limits taken one after the other: at a label $\mathsf{b}' \in I(g)$, on each torus T_ℓ, $\ell \in \mathfrak{L}(g)$, the limit in K of the flux is $\sum_i S_n^{(\ell,\mathsf{b}')}(\dots, (\omega y)\cdot\nabla f_i, \dots)$ through every enclosing shell, by (a). If along sides $\ell_k \in \mathfrak{L}(g)$, $\ell_k \to \infty$, the limits $S_n^\circ(X) = \lim_k S_n^{(\ell_k,\mathsf{b}')}(X)$ exist for all $n \ge 2$ and all tuples X of real Lipschitz functions with compact supports at positive distances, then for smooth f_i with compact supports at positive mutual distances and all cut-off functions χ_k on T_{ℓ_k} of class $C^{1,1}$ that vanish near the seam and cover the f_i (for instance χ_R at a fixed R, or χ_{R_k} with $R_k \to \infty$ and $2R_k < \frac{1}{2}\ell_k - 1$, a far shell)

$$\lim_k \lim_K \langle\mathfrak{B}_{\omega,\chi_k}; \mathcal{O}(f_\cdot)\rangle_{K,\ell_k,L^K\mathsf{b}'} = \sum_i S_n^\circ(f_1, \dots, (\omega y)\cdot\nabla f_i, \dots, f_n) = -\tfrac{d}{dt}S_n^\circ(f(e^{-t\omega}\cdot))\big|_{t=0},$$

through any shell; and there are integers K_k^* such that S_n° is the limit of the connected functions along $(K_k, \ell_k, L^{K_k}\mathsf{b}')$ for every sequence $K_k \geq K_k^*$. For each fixed order $n \geq 2$, there is $c_\ell(n,g) > 0$ such that the same limit is obtained by taking the spacing limit first on each torus whenever $\ell_j \to \infty$ and $\ell_j \leq c_\ell(n,g) K_j^{2\mathfrak{p}_0/3}$. Thus $S_n = \lim_j S_n^{(\ell_j, \mathsf{b}_j)}$ at any first-passage labels, and raising the cutoffs at fixed sides and labels does not change this limit. The constant is specified in the proof. For every $K \geq 1$, the labels of the first-passage data satisfy

$$I(g) \subset \{L^{-K}\beta : (K,\beta) \in \mathfrak{D}(g)\} \subset \big(g^{-2} + B_\sharp - L^{-1}\bar{B},\, L(g^{-2} + \bar{B})\big].$$

(c) *One stress entry: trace and diagonal.* Let $0 < \mathsf{b}_- \leq \mathsf{b} \leq \mathsf{b}_+$, $\beta = L^K \mathsf{b}$, $\ell = 2L^m$ any side, and let two of the $n+1$ supports be $\geq \mathsf{d}$ apart, the others may touch; $\lambda_L = \log(L/2)/\log L$ and $c_n = \lambda_L/(4(6+3n))$. For γ small enough in terms of the other constants, and with C, p depending on $\ell, \mathsf{b}_\pm, n, \mathsf{d}$,

$$|\langle T_{mm}(\varphi) - \mathsf{b}\check{D}_m(\varphi); \mathcal{O}(f_\cdot)\rangle| + |\langle (T_{11}+T_{22}+T_{33})(\varphi) - \mathsf{b}\,\mathcal{O}(\varphi); \mathcal{O}(f_\cdot)\rangle| \leq C K^p L^{-c_n K} \|\varphi\|_\sharp \prod_i \|f_i\|_\sharp,$$

so that $-T_{33} = \mathsf{b}[\mathcal{O}_3 - \mathcal{O}_1 - \mathcal{O}_2]$ inside these functions. For every $\epsilon > 0$, $\langle (T_{lr} - T_{rl})(\varphi); \mathcal{O}(f_\cdot)\rangle = O(a^{1-\epsilon})\mathrm{Lip}\,\varphi$ and $|\langle T_{lr}(\varphi); \mathcal{O}(f_\cdot)\rangle| \leq C_\epsilon a^{-\epsilon} \sup|\varphi| \prod_i \|f_i\|_\sharp$; and $\langle \mathfrak{K}^{\mathrm{v}}[\beta\eta] - \beta\mathfrak{K}^\sharp[\eta]; \mathcal{O}(f_\cdot)\rangle = O(K^p L^{-c_n K})\mathrm{Lip}\,\eta$ for Lipschitz vector fields η, in contact with the f_i or not. At pairwise separated supports and a fixed label $\lim_K \langle T_{mm}(\varphi); \mathcal{O}(f_\cdot)\rangle$ exists.

(d) *All Ward identities with one stress entry.* Let γ be as in (c), $\gamma^{-2} \leq \mathsf{b}_- \leq \mathsf{b} \leq \mathsf{b}_+$, $MR(\mathsf{b}_+) \leq L^m$, $\ell = 2L^m$; let $\xi = e_r$, ωy or y, where for $\xi = \omega y$ and $\xi = y$, taken in $(-\frac{\ell}{2}, \frac{\ell}{2})^3$, the functions χ and f_i vanish within a fixed distance of $\partial(-\frac{\ell}{2}, \frac{\ell}{2})^3$; and either $n \geq 2$, $f_i \in C^2$ with pairwise disjoint supports, $I \subset \{1,\dots,n\}$ and $\chi \in C^2$ equal to 1 near $\mathrm{supp}\, f_i$ for $i \in I$ and to 0 near the others, or $n = 1$, $I = \emptyset$ and $\chi = 0$ near $\mathrm{supp}\, f_1$. Then

$$\lim_K \Big\langle \sum_{l,r} T_{lr}(\xi^r \partial_l \chi); \mathcal{O}(f_\cdot)\Big\rangle_{K,\ell,L^K\mathsf{b}} = -\sum_{i\in I} S_n^{(\ell,\mathsf{b})}(f_1,\dots,\xi\cdot\nabla f_i,\dots,f_n) + \delta_{\xi,y}(|I| S_n^{(\ell,\mathsf{b})}(f) - \mathsf{b}\, S_{n+1}^{(\ell,\mathsf{b})}(\chi, f)),$$

and at the cutoff the two members differ by $\mathfrak{U}^\sharp[\chi\xi](f)$ up to $O(K^p L^{-c_n K})$ and up to the convergence of the right member. Since $|\mathfrak{U}^\sharp[\chi\xi](f)| \leq C K^p L^{-cK}$ for every $c < \frac{1}{14}$ (Lem. 5.11, the spread of $\{\chi\xi \neq 0\}$ and of the supports being positive), the right member converges at the rate $K^p L^{-K/10}$ and $c_n < \frac{1}{36}$, the left member converges at the rate $K^p L^{-c_n K}$, the slowest of the three. For $I = \emptyset$ this is the conservation of the stress tensor beside action densities; for $\xi = e_r$ and χ a periodic function of y_3 alone, $\langle T_{3r}(\chi'); \mathcal{O}(f_\cdot)\rangle \to -\sum_{i\in I} S_n^{(\ell,\mathsf{b})}(\dots,\partial_r f_i,\dots)$. Thus, inside these cumulants on one torus, T_{3r} integrated over the planes $\{y_3 = \mathrm{const}\}$ acts on the enclosed entries as $-\partial_r$, so that $-T_{33}$, the Euclidean energy density, acts as ∂_3.

(e) *Translations.* Let $(K,\beta) \in \mathfrak{D}(g)$, $n \geq 2$, ρ a direction and $\mathsf{d}' > 0$. For real $f_i \in C^2$ with Lipschitz $\partial_\rho^2 f_i$ and $\mathrm{dist}(\mathrm{supp}\, f_1, \mathrm{supp}\, f_2) \geq \mathsf{d}' + 2a$,

$$\Big|\beta\langle \mathfrak{K}^\sharp[e_\rho]; \mathcal{O}(f_\cdot)\rangle - a^{-1}\sum_i \langle \mathfrak{K}^{\mathrm{v}}[f_i e_\rho]; (\mathcal{O}(f_l))_{l\neq i}\rangle\Big| = \Big|\sum_i S^T_{n;K,\ell,\beta}(f_1,\dots,\nabla^s_\rho f_i,\dots,f_n)\Big| \leq (n+1)2^n C_n(\mathsf{d}') e^{n c_V \ell^3} a_K^2 \prod_i N_i,$$

$N_i = \max\{\|f_i\|_\sharp, \|\partial_\rho f_i\|_\sharp, \frac{1}{2}\|\partial_\rho^2 f_i\|_\sharp\}$, with $C_n(\mathsf{d}')$ and c_V as in Lem. 5.1; at $n = 2$ the left member vanishes identically. For $I \subset \{1,\dots,n\}$, the f_i with $i \in I$ of class C^3 with Lipschitz $\partial_\rho^3 f_i$, the others Lipschitz, $\mathrm{dist}(\mathrm{supp}\, f_1, \mathrm{supp}\, f_2) \geq \mathsf{d}' + a$, and χ a real function on T_ℓ with $\chi(x(p)) = \chi(x_b) = 1$ at every plaquette p at which some f_i, $i \in I$, does not vanish and at its

four bonds b, and $\chi(x(p)) = \chi(x_b) = 0$ at every plaquette p at which some f_i, $i \notin I$, does not vanish and at its four bonds b,

$$\langle N_\beta[\chi e_\rho]; \mathcal{O}(f_\cdot)\rangle - a^{-1}\sum_{i\in I}\langle \mathfrak{K}^{\mathrm{v}}[f_i e_\rho]; (\mathcal{O}(f_l))_{l\neq i}\rangle = -\sum_{i\in I} S^T_{n;K,\ell,\beta}(f_1,\dots,\partial_\rho f_i,\dots,f_n) + \mathsf{r},$$

$|\mathsf{r}| \le \frac{a_K^2}{6} C_n(\mathsf{d}') e^{nc_V\ell^3} \sum_{i\in I} \|\partial_\rho^3 f_i\|_\sharp \prod_{l\neq i} \|f_l\|_\sharp$, $N_\beta[\chi e_\rho] = \mathbb{T}_\beta[\chi e_\rho] + \beta\mathfrak{K}^\sharp[\chi e_\rho]$ the null term of Lem. 5.9(a). Thus the stress functional $\mathbb{T}_\beta[\chi e_\rho]$ of a cut-off translation, taken together with its cubic part $\beta\mathfrak{K}^\sharp[\chi e_\rho]$ and with the contact terms $-a^{-1}\sum_{i\in I}\langle \mathfrak{K}^{\mathrm{v}}[f_i e_\rho]; (\mathcal{O}(f_l))_{l\neq i}\rangle$, generates the translation of the enclosed entries up to $O(a^2)$, with connected functions bounded uniformly in K. If the closed a-neighbourhoods of the supports lie in balls of radius R and $\ell \ge 16n(R+1)$, the numbers may be taken free of ℓ.

■□

Remark 5.3 (The spread is needed). For $n \ge 3$ the bound (a)(i) fails at supports with a common point. The symmetric difference is no derivation: by the invariance of the measure under the lattice translations, $\sum_i S^T_n(\dots,\nabla^s_\rho f_i,\dots)$ is a^2 times connected functions of difference quotients of the f_i, at $n = 3$ exactly

$$\sum_{i=1}^3 S^T_3(\dots,\nabla^s_\rho f_i,\dots) = \tfrac{a^2}{2}\, S^T_3(\nabla^+_\rho f_1, \nabla^+_\rho f_2, \nabla^+_\rho f_3),$$

and connected functions of action densities at coinciding points are of the order a^{-3} (Prop. 5.18(c)). For free photons, let the real f_i have compact supports inside the margin and bounded derivatives through order five, and let $\chi = 1$ on a neighbourhood of their supports. Then $a\,\mathfrak{U}_{\omega,\chi}(f) \to 3\mathsf{b}^{-3}\int \sum_\rho (\omega y)_\rho\, \partial_\rho f_1\, \partial_\rho f_2\, \partial_\rho f_3$ at $n = 3$, while the identity with the true generator survives at supports with a common point: $\sum_i S^T_n(\dots,(\omega y)\cdot\nabla f_i,\dots) - \langle \mathfrak{B}_{\omega,\chi}; \mathcal{O}(f_\cdot)\rangle$ is $O(a)$ at $n = 2$ and $O(a\log\frac{1}{a})$ at $n = 3$. For SU(2), at every $g \le \gamma$, on every side $\ell \in \mathfrak{L}(g)$ and for every $n \ge 3$ there are smooth $f_1,\dots,f_n$ with a common point, a rotation ω and a cut-off function χ such that

$$450\,(n-1)!\,\mathsf{b}'^{-n} \le a_K |\mathfrak{U}_{\omega,\chi}(f)| \le 2.3\cdot 10^5\,(n-1)!\,\mathsf{b}'^{-n}$$

for all large K, uniformly in the labels $\mathsf{b}' \in I(g)$. At $n = 1$ the neighbourhood in (a)(ii) is needed: there are a real Lipschitz f_1, a rotation ω and a cutoff $\chi \in C^{1,1}$ covering f_1, equal to one on its support but on no neighbourhood of it, for which the free-photon model has $\liminf_K \mathfrak{U}_{\omega,\chi}(f_1) > 0$.

Theorem P (One plane field between two times)**.** Let $0 < g \le \gamma^{\mathrm{ref}}_W$, $\ell' \in \mathfrak{L}(g)$, $(K_j,\beta_j) \in \mathfrak{D}(g)$ with $K_j \to \infty$, $\nu_j \in \mathcal{G}^{\mathrm{per}}_{K_j,\beta_j}$, and $(X)_j = \prod_i[\mathcal{O}_{K_j}(f_i) - \nu_j(\mathcal{O}_{K_j}(f_i))]$. Every subsequence has a further subsequence along which the following hold.

(i) $\sigma(X) = \lim_j \nu_j((X)_j)$ and $\sigma^{(m)}(u;X) = \lim_j \nu_j((X)_j[\check D_{m,j}(u) - \nu_j(\check D_{m,j}(u))])$ exist for all real u, $f_i \in C^1_c(\mathbb{R}^3)$ with disjoint supports; they are real and multilinear, $\sum_m \sigma^{(m)}(u;X) = \sigma(u\cdot X)$, the product with one more scalar factor, they are translation invariant and covariant under W_3: a quarter turn that carries e_3 to e_i carries $\sigma^{(3)}$ to $\sigma^{(i)}$.

(ii) On the Hilbert space $\mathcal{H}(\sigma)$ of the reconstruction data of the limit, with the Hamiltonian $H \ge 0$, the reflection θ, the translations $U(\vec x)$, whose generators P_1, P_2 commute with H, and the classes $[G]$, the joint spectrum lies in the cones $H \ge |P_k|$, $k = 1,2$: every periodic Gibbs state obeys the five-point inequality ([7, eq. 241]) of Lem. 4.46 at the mirror normal to e_3 and the direction e_k, which passes to the limit and puts the spectral measure of every class on $\{\cosh(tE) + \cos(tq_k) \ge 2\}$ for a dense set of $t > 0$, hence on $\{E \ge |q_k|\}$. So $H \ge 0$ and $H \ge |P_k|$ are the hypotheses of Lem. 5.13 in the direction x_3 and, in its last sentence, in the directions x_1, x_2; and there is a unique family of bounded operators $B^{(m)}(s,s')$, $s,s' > 0$, with $B^{(m)}(s,s')^* = B^{(m)}(s',s)$, $e^{-tH}B^{(m)}(s,s')e^{-t'H} = B^{(m)}(s+t,s'+t')$ and

$$\sigma^{(m)}(u;\theta\overline{\tau_{se_3}G'}\cdot\tau_{s'e_3}G) = \int u(x)\,\langle [G'], U(\vec x) B^{(m)}(s+x_3, s'-x_3) U(-\vec x)[G]\rangle\,\mathrm{d}x$$

for u supported in $\{-s < x_3 < s'\}$ and G, G' supported in $\{x_3 > 0\}$.

(iii) $\|B^{(m)}(s,s')\|$ is bounded by a finite function of $\min\{s,s',1\}$, g and ℓ', and for u supported in $\{\gamma' - s \le x_3 \le s' - \gamma'\}$ the left member of (ii) is at most $c(\gamma')\|u\|_{L^1}\|[G']\|\,\|[G]\|$. For $0 < g \le \gamma_W^{\mathrm{sd}}$ every such limit with converging labels arises, along a further subsequence, from a tensor limit σ_T (Thm. N(g)): $\sigma(X) = \sigma_T(f_1\mathbf{1}, \dots, f_n\mathbf{1})$ and $\sigma^{(m)}(u;X) = \sigma_T(u(\mathbf{1} - 2E_{mm}), f_1\mathbf{1}, \dots, f_n\mathbf{1})$. Let $\mathcal{H}(\sigma_T)$ be the reconstruction of σ_T at θ, spanned by the classes $[F]$ of the tuples F of entries in $C_c(\{x_3 > 0\}; V)$ with $\langle [F'], [F]\rangle = \sigma_T(\theta \bar{F}' \cdot F)$, with its Hamiltonian and its translations, whose joint spectrum lies in the same cones by the same five-point inequality; its *scalar subspace* $\mathcal{H}_0$, the closed span of the classes of the tuples of scalar entries $f_a\mathbf{1}$, is $\mathcal{H}(\sigma)$ and reduces the Hamiltonian and the translations, and Π_0 is the projection onto it.

For $e \in V$ the *point operator* of the tensor entry at e is the family $B^e(s,s')$ of bounded operators on $\mathcal{H}(\sigma_T)$ that Lem. 5.13 gives for $u \mapsto \sigma_T(ue, \cdot)$, with $\sigma_T(ue, \theta\overline{\tau_{se_3}F'} \cdot \tau_{s'e_3}F) = \int u(x)\langle [F'], U(\vec{x})B^e(s + x_3, s' - x_3)U(-\vec{x})[F]\rangle \, \mathrm{d}x$ as in (ii); it is linear in e, and $\|B^e(s,s')\| \le C|e| \min\{s,s',1\}^{-p}$ for every $p > \frac{17}{2}$, $|e|$ the Euclidean norm of e and C depending on g, p and the limit label and not on the side, by the bound on the chessboard seminorm of a centred tensor entry. Then $B^{(m)}(s,s') = \Pi_0 B^{\mathbf{1}-2E_{mm}}(s,s')\Pi_0$ on $\mathcal{H}_0$, the compression to the scalar subspace of the point operator at $\mathbf{1} - 2E_{mm}$, so that $\|B^{(m)}(s,s')\| \le C \min\{s,s',1\}^{-p}$; and for $Z = (u;X)$ and R_α the turn by the angle α in the plane of e_i, e_3,

$$\sigma^{(3)}(Z \circ R_\alpha) + \sigma^{(3)}(Z \circ R_{-\alpha}) = 2\cos^2\alpha\, \sigma^{(3)}(Z) + 2\sin^2\alpha\, \sigma^{(i)}(Z).$$

Thus the $\check{D}_m$ behave under turns as the diagonal of a symmetric tensor; moreover $\alpha \mapsto \sigma^{(3)}(Z \circ R_\alpha)$ is of the form $c_0 + c_1\cos 2\alpha + c_2 \sin 2\alpha$ with numbers c_0, c_1, c_2, and its second derivative at $\alpha = 0$ is $2[\sigma^{(i)}(Z) - \sigma^{(3)}(Z)]$.

■□

One step of the renormalisation group, at every level and with its never-large term, runs with three complex plane couplings, about its prescribed background with free centres (Thm. Q); on the labels of histories with large-field regions the moved backgrounds, the centres of the steps and the counterterms are those of Thm. R. For a triple $P = (U, A', J)$ at the level n, $\mathbf{U}_P = e^{i\eta_n A'}U$, $\eta_n = L^{-n}$, is its complex configuration (Def. YM3_I:49.1); on a region X let $\mathrm{hol}_n(P;X)$ be the Hölder-$\frac{1}{2}$ number, transports included, of $L^{2n}(\partial\mathbf{U}_P - 1)$ at distances ≤ 1 in level-n units; the *core with the third radius* $\mathcal{U}_n(X)$ is the set of triples in the hereditary core of Def. YM3_I:49.1 at the radii (α_0, α_1), with all the tower conditions retained as in §5.2, with $\mathrm{hol}_n(P;X) < \alpha_2$; $\rho_Z^{k\to j}(U, A', J) = (U, \lambda A', \lambda^3 J)|_Z$, $\lambda = L^{-(k-j)}$; $\mathcal{D}_n$ and d_n are as in Section YM3_I:45.

A region $Z \in \mathcal{D}_j$ is *short* at k if $d_j(Z) + 4 \le L^{k-j}$. The *box* Z^{box} of a short Z is the least box of level-j cubes that holds a lift of Z (a wall-connected union of cubes of the covering that the covering map carries one to one, cube by cube, onto Z; such unions exist, Z being wall-connected, and any two of them differ by a period), enlarged by one level-j cube on each of its six sides and carried back by the covering map: it does not depend on the lift, its sides are at most $d_j(Z) + 4 \le L^{k-j}$ level-j cubes, it lies in $\mathcal{D}_j$, holds Z, and holds both end points of every bond with an end point in Z. Then X_Z is the union of the level-k cubes that hold a level-j cube of Z^{box}: one or two along each axis, so at most eight, with $X_Z \in \mathcal{D}_k$ and $d_k(X_Z) = 0$.

For every $Z \in \mathcal{D}_j$, Z^{up} is the union of the level-k cubes that hold a level-j cube of Z. A *term* on $\mathcal{U}_j(Z)$ is a bounded function on it that is holomorphic on its slices, a function of the pair $(\mathbf{U}_P, J)$, and gauge invariant.

The extended list. Thms. Q and R are stated at the choices of the following extension of the list $\mathcal{T}^{\mathrm{S}}[2\varrho]$ of Thms. YM3_I:AJ and YM3_I:AD (the common real and sourced choice of Section YM3_I:82, including eq. 2.iv). The base choice also retains [7, eq. 194] and the radius, decay and block-number conditions of Lem. 5.34. These are imposed when choosing the response parameters in the

proof of Prop. YM3_I:69.1, before the polymer-window size M: the finitely many local constants here depend only on L, so the response block number $\mathfrak{M}$ can be raised and its radii and decay rate reduced at that stage. For Lem. 5.19 take this same response block number at least M_{ps} and M_H, the floors in Thm. YM3_I:I and Lem. 5.34. Enlarge w_{lin} to at least $\max\{g_{\mathrm{p}}, w_S\}$, where w_S is from Prop. YM3_I:29.1, and reduce α_{lin} so that $2\alpha_{\mathrm{lin}} \le a_Q$ and the four local-system radii are at most a_H. These choices depend only on L. The separated setting has $R_1 \ge L$ and terminal torus side at least $6R_1\mathfrak{M}$, hence at least $6LM_{\mathrm{ps}}$. On a real regular label, $F'(0) = J(U)$ at full bonds; the scalar part of $D_U^{\mathsf{T}}\partial U$ is bounded by $\xi^{-1}\max_{p\ni b}|\partial U(p) - I|^2$. Thus equation YM3_I:(579) gives the full matrix-divergence bound at twice the regularity radius. The ceiling on $C_m\varepsilon$ in part (c) supplies this radius and the four local-system radii, while its width ceiling supplies the enlarged w_{lin}. These are conditions on the common response choice; they are not new analytic assumptions. It adds 108 entries, in this order: four lower bounds on chosen constants among the construction results, each by a function of earlier constants; 85 further determined constants, each a positive explicit formula in earlier constants, placed before γ (four abbreviations are used among the formulas); and 15 further upper bounds on γ. No condition is laid on L, on E_0 or on a window constant.

The following constants retain their meanings in the real construction and its sourced extension (the proof of Prop. YM3_I:69.1 and Section YM3_I:82):

$$\mathsf{C}_{\mathrm{ex}}, \mathsf{C}_{\mathrm{cs}}, \mathsf{C}_{\mathrm{mn}}, \lambda_{\mathrm{s}}, \gamma_{\mathrm{r}}, c_{\mathrm{r}}, \mathsf{u}_{\mathrm{G}}, \lambda_{\mathrm{r,gs}}, \bar{c}_{\mathrm{P}}, \beta_{\mathrm{s}}, \kappa, \kappa'', \bar{r}_H, \bar{\kappa}, \lambda_V, \vartheta_*, \mathsf{c}_{\mathrm{os}}, \mathsf{c}_{\mathfrak{h}}, \varepsilon_{\mathfrak{h}}, r_P^{\circ}, \mathsf{C}_{\mathcal{A}}, \mathsf{s}^{\star}, K_\Lambda, N_\star, K_P,$$
$$w_\star, \varepsilon_{\mathcal{W}}, \mathsf{o}, B^{(\mathrm{cg})}, A_0, A_1, \mathfrak{p}_0, \alpha_0, \alpha_1, B_3, E_0, M, L'$$

the plaquette constant $\mathsf{c}_{\mathrm{pl}} = 600$ and the number C_P of equation YM3_I:(799), $K_{\mathrm{c}} = 96(1+2\mathsf{c}_{\mathrm{pl}})^2 C_{\mathrm{E}}$, $C_{\mathrm{E}} = 4096$, the numbers b_{D}, $\mathsf{B}_\sharp(0)$, $\mathsf{B}_\sharp(1/2)$ and $\mathsf{B}_\sharp = \mathsf{B}_\sharp(0)$ of Prop. 5.35, and $a^\sharp$, $K^\sharp$ of the tower hypotheses of §5.2.

Here $\lambda_{\mathrm{r,gs}}$ denotes the canonical history Gaussian gap, written λ_{r} in equation YM3_I:(1137) with the specialization of equation YM3_I:(1161):

$$\lambda_{\mathrm{r,gs}} = \min\left\{\frac{\gamma_0}{2}, \left(1 + 2(B_{\mathrm{nm}}^2/\gamma_0)^2\right)^{-1}\right\}.$$

The symbols B_{nm} and γ_0 have the values in that specialization. This history gap is distinct in notation from the plain-step gap $\lambda_{\mathrm{r}}^{\mathrm{pv}}$ defined below.

The entries are:

1. the lower bound $\mathsf{C}_{\mathrm{ex}} \ge \max\{2\mathsf{C}_{\mathrm{cs}}, 8\}$;
2. directly behind $\bar{c}_{\mathrm{P}}$: $\lambda_{\mathrm{r}}^{\mathrm{pv}} := \min\{\frac{1}{2}\gamma_{\mathrm{r}}, (1+2c_{\mathrm{r}}^2)^{-1}\}$ and $c_q := \max\{\bar{c}_{\mathrm{P}} + \mathsf{C}_{\mathrm{mn}}^3/\lambda_{\mathrm{r,gs}} + 2,\ \mathsf{C}_{\mathrm{mn}}^3(\frac{1}{2}\mathsf{u}_{\mathrm{G}} + c_{\mathrm{r}} + 3)/\lambda_{\mathrm{r}}^{\mathrm{pv}} + 2\}$;
3. with the abbreviation $E_q := 1 + 2\lambda_{\mathrm{s}} + \mathsf{C}_{\mathrm{cs}}(c_q + 2)$, the lower bounds $(1 + 4\beta_{\mathrm{s}})\kappa \ge E_q$ and $\frac{1}{2}\kappa \ge 2\lambda_{\mathrm{s}} + 8\log 2 + \frac{\mathsf{C}_{\mathrm{cs}}}{L}E_q$;
4. directly behind $\bar{r}_H$: $r_H^q := \kappa'' + \lambda_{\mathrm{s}} + \mathsf{C}_{\mathrm{cs}}(c_q + 2)$, $\kappa' := \kappa'' + 1$, $r_{\mathrm{w}}^q := \max\{\frac{\mathsf{C}_{\mathrm{cs}}}{L}r_H^q + \lambda_{\mathrm{s}}, \frac{8\kappa'}{L}\}$, $r_t := \lambda_{\mathrm{s}}$, $r^q := r_{\mathrm{w}}^q + \lambda_{\mathrm{s}}$, $r_1 := \kappa'' - 8\log 2$, $r_1' := \kappa'' - 16\log 2$;
5. the lower bounds $\bar{\kappa} \ge 1 + \mathsf{C}_{\mathrm{cs}} r_H^q + \log \mathsf{C}_{\mathrm{cs}}$ and $\bar{\kappa} \ge 3 + 16\kappa' + 2\lambda_{\mathrm{s}} + 2\lambda_V$;
6. the lower bound $\mathsf{C}_{\mathrm{mn}}(LM)^3 \ge c_q$;
7. directly before γ, in this order: $c_K := 2/\gamma_{\mathrm{r}} + 4(c_{\mathrm{r}} + \frac{1}{2})^2/\gamma_{\mathrm{r}}^2$; $B' := 6 \cdot 10^5 c_K (LM)^3 e^{16\kappa'}$; $S' := 4096 e^{8\kappa'}$; $b_A := 6.1201^2 M^3$; $K_T^{\mathrm{pl}} := \max\{12L^4M^3K_{\mathrm{c}}, 2b_A, 112b_A + \frac{1}{2}M^3\}$; $a_Z := \mathsf{c}_{\mathrm{os}}\alpha_0$; $r^{\mathrm{pl}} := \min\{r_P^{\circ}, \varepsilon_{\mathfrak{h}}/\sqrt{2}, 1/(3\mathsf{c}_{\mathfrak{h}}), a_Z/(2\mathsf{c}_{\mathfrak{h}})\}$; $\delta' := \min\{1, \varepsilon_{\mathfrak{h}}/\mathsf{C}_{\mathcal{A}}, \mathsf{s}^{\star}\alpha_0/(1+\beta_{\mathrm{s}}), A_1(2\mathfrak{p}_0/e)^{\mathfrak{p}_0}\}$; $r_2^{\mathrm{pl}} := \mathsf{C}_{\mathcal{A}}\delta'$; $r_3^{\mathrm{pl}} := \min\{r^{\mathrm{pl}}, r_2^{\mathrm{pl}}\}$; $\bar{u}_f := K_\Lambda \mathsf{s}^{\star} N_\star$; $\varrho^{\mathrm{pl}} := \min\{\frac{1}{4}r^{\mathrm{pl}}, (r^{\mathrm{pl}})^3/(256B'K_T^{\mathrm{pl}}S')\}$; $\sigma_3 := \min\{(r^{\mathrm{pl}})^2/(64B'K_T^{\mathrm{pl}}S'),\ r^{\mathrm{pl}}\varrho^{\mathrm{pl}}/(4B'K_T^{\mathrm{pl}}S')\}$; $\varepsilon_1 := [8\mathsf{C}_{\mathrm{cs}}^2 L^3 e^{\mathsf{C}_{\mathrm{cs}}(\kappa''+c_q+r_H^q)}]^{-1}$; $v_0 := (2^{17}L^3)^{-1}$; $v_d := (2^{17}L^3)^{-1}$; $\hat{\rho} := \varepsilon_1/(8e)$; $\eta := 1$; $\gamma_2^q := (LM)^{-3}$; $\theta_m := \frac{1}{2}\vartheta_*$; $\varepsilon_m^{\mathrm{pl}} := \min\{\frac{1}{2}\vartheta_*, (LM)^{-3/2}\}$; $\theta_1' := \frac{1}{2}\theta_m$; $\theta_1'' := \frac{1}{4}(1/\varepsilon_m^{\mathrm{pl}} + 4\varepsilon_m^{\mathrm{pl}}/\theta_m^2)^{-1}$; $v_{\mathrm{ce}} := \mathsf{C}_{\mathcal{A}}$; $v^+ := \sqrt{2}\,\mathsf{C}_{\mathcal{A}} + 1$; $a_{\mathrm{cap}} := \varepsilon_1 v_0/(96e)$; $c_1^+ := \min\{\frac{1}{4}r_3^{\mathrm{pl}},\ a_{\mathrm{cap}}(r^{\mathrm{pl}})^3/(128K_T^{\mathrm{pl}}M^3)\}$; $K_Z := 4B'K_T^{\mathrm{pl}}S'/r^{\mathrm{pl}}$; $B_0 :=$

$6{\cdot}10^5 c_K(LM)^3$; $K_6 := 1296M^3\max\{1,2C_P\}B_3L^{-2}$; with the abbreviation $\theta_* := \min\{\theta_1', \theta_1''\}$,

$$\sigma_e := \min\{\sigma_3,\ a_{\rm cap}(r^{\rm pl})^2/(16K_T^{\rm pl}M^3),\ c_1^+/(2K_Z),\ a_Z\theta_*/(4\sqrt{2}K_Z\mathsf{c}_\hbar v^+L),\ \theta_*A_1/(4B_0K_6(2L)^{1/2}A_0)\};$$

$\bar K_{\rm or} := \mathsf{C}_{\rm cs}(LM)^3 e^{\mathsf{C}_{\rm cs}(c_q+r_H^q+\kappa'')}$; $Q^{\rm pl} := e\bar K_{\rm or}/v_d$; $Q' := e^{\kappa-r^q}$; $\bar K_{\rm w} := \frac12(6(LM)^3)^{1/2}$; $c^{\rm pl} := \mathsf{C}_{\rm mn}^{-1}\gamma_2^q(1-\theta_m)^2A_1^2/4$;

8. then, in this order: $\theta_{\rm w} := \min\{1, E_0e^{-(\kappa''-\kappa)}\}$; $c_\hbar^H := \max\{54C_P\mathsf{b}_{\rm D}e^{8\bar\kappa}\mathsf{B}_\sharp(\frac12)/\mathsf{B}_\sharp,\ 9\mathsf{c}_\hbar\}$; $c_0'' := 5200(c_\hbar^H+\mathsf{c}_\hbar)$; $\alpha_2^{\rm pl} := 2c_0''\varepsilon_\hbar$, the third radius α_2 of the cores above; $c_\hbar'' := c_0'' + 102\,\mathsf{c}_\hbar L^{-5/2}\alpha_2^{\rm pl}$; $K_H^0 := 1.02\cdot 746$; $K_H := \max\{2K_H^0B_3, 10\mathsf{c}_\hbar M^3\}$; $C_{\rm gs} := \max\{\alpha_0/\varepsilon_\hbar, 2\mathsf{c}_\hbar/\mathsf{c}_{\rm os}, 1200\mathsf{c}_\hbar\}$; with the abbreviations $m_0 := \min\{\alpha_2^{\rm pl},\alpha_0\}$ and $m_1 := \max\{\alpha_2^{\rm pl},\alpha_0\}$: $c'' := \max\{C_{\rm gs}, c_\hbar''\}$; $c_3^a := L^{-5/2} + c''r^{\rm pl}/m_0$; $c_4^a := \max\{c''/m_0, 1/r^{\rm pl}\}$; $C_V^{\rm pl} := 2^{10}LC_{\rm E}(1-1/L)^{-1}$;

$$K_J^0 := \max\{1,\ 10\mathsf{c}_\hbar M^3L^{-5/2}m_1,\ 8(1+84C_P(M/L)^3)L^{-5/2}m_1\};$$

$K_J := 2K_J^0$; $K_{\rm c}^a := 2B'S'K_J$;

$$K_{\rm ta} := 2^8\cdot 24(LM)^3[K_JK_{\rm c}^a + 2K_T^{\rm pl}(K_{\rm c}^a)^2/(r^{\rm pl})^2];$$

$a_Z' := \min\{a_Z, \mathsf{c}_\hbar\varepsilon_\hbar\}$; $\sigma_5' := a_Z'\theta_*/(4\sqrt{2}K_Z\mathsf{c}_\hbar v^+L)$; $\sigma_*^a := \min\{\sigma_3,\sigma_e,\sigma_5',1\}$; $c_g^{\rm pl} := 2\max\{1/\varepsilon_{\mathcal W}, K_H\}$;

$$w_5 := \min\{K_J^0/(1.01\,C_V^{\rm pl}c_4^aK_{\rm ta}),\ \min\{\varrho^{\rm pl}, \tfrac18 r^{\rm pl}, K_Z\sigma_*^a\}/K_{\rm c}^a,\ \sigma_*^aK_T^{\rm pl}/(6144M^3\alpha_0^2),\ (\sigma_*^aK_T^{\rm pl}/(2C_V^{\rm pl}(c_3^a)^2K_{\rm ta}))^{1/2},\ \sigma_*^a,\ m_0\theta_*A_1/(2K_{\rm c}^ac_g^{\rm pl}(2L)^{1/2}A_0)\};$$

$\sigma_5 := \frac12 w_5$; $\tau_5 := \frac14 w_5$; $b_r := \mathsf{o}LMB^{(\rm cg)}\alpha_0$; $a_3 := 8.4(\alpha_1(1+2b_r)+b_r)$; $C_2^Q := 29.4a_3^2 + 14.7a_3$; $\bar C_2 := \max\{C_2^Q, 1\}$; $a_4 := 7.4a_3 + C_2^Q$; $C_H^Q := 18$; $c_s^Q := \min\{1, \mathsf{o}L, 1/(12Ma_3)\}$, the constant c_s of Thm. Q(a); $c_3^Q := \min\{\frac14 a^\sharp, \alpha_0/(4K^\sharp), \alpha_1, \frac14, \alpha_2^{\rm pl}/18\}$;

$$\lambda_Q := \min\{1/(4a_3),\ a_3/C_2^Q,\ \alpha_1/a_4,\ a^\sharp/(2\alpha_0+a_4),\ [\alpha_0/(K^\sharp(2\alpha_0+a_4^2))]^{1/2}\};$$

$$C_Q := \max\{2\lambda_Q^{-4},\ 32(c_s^Q)^{-4},\ 72M^2\bar C_2^2(c_3^Q)^{-2}\};$$

$K_{\rm Lf} := 2C_{\rm E}(140C_Q+1)$; $u^c := 2^9\cdot 129\,K_{\rm Lf}L/(L-1)$; $\bar u_f^c := \bar u_f + u^c$; $c_J^c := 2^{21}L^3K_{\rm c}^a(7M^3/r_P^\circ + \bar u_f^c/r_3^{\rm pl})$; $w_6 := \min\{w_5, 1/c_J^c\}$, the constant $c_{\rm w}$ of Thm. Q(c); $\sigma_6 := \frac12 w_6$; $\tau_6 := \frac14 w_6$;

9. fifteen upper bounds on γ, each of the form $\sup_{x\ge\gamma^{-2}} F(x) \le \Phi$ with $F(x)\to 0$ as $x\to\infty$ and $\Phi > 0$ formed with earlier constants; with the thresholds $\delta(x) = A_1x^{-1/2}(\log x)^{\mathfrak p_0}$, $\varepsilon(x) = A_0x^{-1/2}(\log x)^{\mathfrak p_0}$ of YM3_I:(634) and

$$\xi(x) := \exp(-\mathsf{C}_{\rm mn}^{-1}\gamma_2^q(1-\theta_m)^2A_1^2(\log x)^{2\mathfrak p_0})$$

they are: $3K_P\delta(x) \le a_{\rm cap}$; $\bar u_f^c\mathsf{C}_{\mathcal A}\delta(x)/r_3^{\rm pl} \le a_{\rm cap}$; $\frac{64}{3}(7M^3+\bar u_f^c)c_1^+\mathsf{C}_{\mathcal A}\delta(x)/(r_3^{\rm pl})^2 \le \frac13 a_{\rm cap}$; $\mathsf{C}_{\mathcal A}\delta(x) \le \frac12 r_3^{\rm pl}$; $(\theta_1'+\theta_1'')\delta(x) \le \frac12 c_1^+$; $6(LM)^3\xi(x) \le 1$; $2(LM)^3\xi(x)^{1/2} \le \varepsilon_1L^3$; $[(1-\theta_m)A_1(\log x)^{\mathfrak p_0}]^{-1} \le 1$; $2Q^{\rm pl}Q'L^{-2c^{\rm pl}\log(x/2)} \le 1$; $Q'\bar K_{\rm w}e^{-c^{\rm pl}(\log(x/2))^2} \le \hat\rho v_d$; $\mathsf{c}_\hbar v^+\delta(x) \le \frac12 a_Z'$; $2B_3\varepsilon(x) \le 1$; $2\varepsilon(x) \le \varepsilon_{\mathcal W}\alpha_0$; $K_H\varepsilon(x) \le \frac12\alpha_2^{\rm pl}$; $8K_\Lambda N_\star(1+\beta_{\rm s})\delta(x)/\alpha_0 \le w_\star$.

Each formula names only constants that stand before it, and none reads K, m, a level, a sequence of couplings, a weight or a field. The list so extended is an order table of admissible shape; choices of it exist; the restriction of a choice to the constants of $\mathcal T^{\rm S}[2\varrho]$ is a choice of that list, and its restriction to the constants of the list $\mathcal T^{\rm F+}$ of Thm. YM3_I:D (the real choices in Prop. YM3_I:69.1) is a choice of that list. Thm. R lowers γ further where it says so.

Theorem Q (Three plane couplings)**.** Let the constants be a choice of the extended list stated before this theorem, which adds to the list of Thms. YM3_I:AJ and YM3_I:AD four lower bounds on chosen constants, each by a function of earlier constants, further determined constants placed before γ and further upper bounds on γ, and no new condition on L itself; its choices exist, they restrict to choices of the lists of Thms. YM3_I:D, YM3_I:AJ and YM3_I:AD, and at each of them the standing conditions of the frame of [7, Def. B.14] hold, as do the locality for all pairs of regions and the gauge covariance at all decoupling parameters of the Gaussian structure among the construction results, the tower hypotheses of §5.2 and the inequality $\alpha_0+\alpha_1\le a^\sharp$ with the constant $a^\sharp$ in the radius conditions YM3_I:(867). At every choice $L>4\cdot10^{19}$, as a consequence of the condition $L\ge2\mathsf{C}_{\rm ex}$ among the construction results and of the lower bound $\mathsf{C}_{\rm ex}\ge2\mathsf{C}_{\rm cs}$ of the extended list, with $\mathsf{C}_{\rm cs}\ge24C_{\rm E}^2e^{24}$.

(a) *Lifted local terms.* For all K,m and $1\le j<k<K$ with $2L^{m+K-k}\ge C_LM$, that is, for which the lattice (ℓ,K), $\ell=2L^m$, is allowed at the height k (Section YM3_I:50.4), $C_L\ge8\mathsf{n}_{\rm e}L$ being the room and $\mathsf{n}_{\rm e}\ge1$ the halo width of Section YM3_I:50.4, so that $m+K-k\ge m'+2$, $\mathcal{D}_j$ and $\mathcal{D}_k$ are defined and a box of at most two cubes of the level k along each axis does not wrap around the torus, which has at least $8L$ such cubes along each axis, every $Z\in\mathcal{D}_j$ short at k, every $P'\in\mathcal{U}_k(X_Z)$ and every term E on $\mathcal{U}_j(Z)$,

$$|E(\rho_Z^{k\to j}P')-E(\mathbf{1})|\le C_Q\sup|E|\,(d_j(Z)+2)^4L^{-4(k-j)},$$

and the same with $(d_j(Z)+2)^2$ in the place of $(d_j(Z)+2)^4$ and with $72M^2\bar C_2^2(c_3^Q)^{-2}\le C_Q$ in the place of C_Q when $L^{-(k-j)}\le\lambda_Q$ and $(d_j(Z)+4)L^{-(k-j)}\le c_s$, with the constants λ_Q, $c_s=c_s^Q$, $\bar C_2$ and c_3^Q of item 8 of the list; C_Q is a function of constants placed before γ, it lays no condition on any constant, and it depends on no j,k,K,m,Z,P',E. The same holds on every larger base in $\mathcal{D}_k$ that holds the box of Z, for cores at any positive radii, and on the plain cores for its local terms of level j. The power 4 is attained: for every short Z and every plaquette with its bonds in the boundary of Z there are a term E with $\sup|E|\le\frac12$ and a triple P' with $E(\rho_Z^{k\to j}P')-E(\mathbf{1})\ge c_QL^{-4(k-j)}$, $c_Q>0$ a constant. For the abelian group $\mathrm{U}(1)^3$ the inequality fails.

(b) *The lift.* Let K,m and $1\le k<K$ be as in (a), $2L^{m+K-k}\ge C_LM$, and let $\kappa_{\rm lf}\ge4\log18432$ be a rate. For terms $g^{(j)}(Z,\cdot)$ on $\mathcal{U}_j(Z)$, $1\le j\le k$, $Z\in\mathcal{D}_j$, with $B_j=\sup_Z e^{\kappa_{\rm lf}d_j(Z)}\sup|g^{(j)}(Z,\cdot)|<\infty$, the lift

$$\mathrm{Lf}[g](X;P')=\sum_{j\le k}\sum_{Z\in\mathcal{D}_j:\,X_Z=X}[g^{(j)}(Z;\rho_Z^{k\to j}P')-g^{(j)}(Z;\mathbf{1})],\qquad X\in\mathcal{D}_k,\ P'\in\mathcal{U}_k(X),$$

($X_Z=Z^{\rm up}$, the union of the cubes of the level k that hold a cube of Z, for the Z that are not short, among them all $Z\in\mathcal{D}_k$, for which $Z^{\rm up}=Z$) is a term on $\mathcal{U}_k(X)$, linear in g, depending on g at the Z in X alone, and

$$|\mathrm{Lf}[g](X;P')|\le2C_E(140C_Q+1)\sum_{j\le k}L^{-(k-j)}B_j\,e^{-\kappa_{\rm lf}d_k(X)},\qquad C_E=4096,$$

a bound that is uniform in the level.

(c) *One step at all levels.* Let K,m with $L^m\ge MR(x_K)$ and let $x_0>\cdots>x_K\ge\gamma^{-2}$ be an admissible sequence of couplings in the frame of [7, Def. B.14]; by [7, eq. 215] the frame has $C_L\le2LR(\gamma^{-2})\le2LR(x_K)$, so that $2L^{m+K-k}\ge2LMR(x_K)\ge C_LM$ at every $k<K$, and (a) and (b) apply at every level. Let $\sigma,\tau\ge0$ with $\sigma+2\tau\le c_{\rm w}$, $c_{\rm w}=w_6>0$ the constant of item 8, placed before γ, and let ω be a complex plaquette function whose plane weights w_κ, $\kappa\in\{12,13,23\}$, obey $\sup|w_\kappa|\le\sigma(x_K-\bar B)$ and $M\mathrm{Lip}_\infty w_\kappa\le\tau x_K$, with no tie between the planes.

Put $v_n = -L^{K-n}\omega/x_n$, $g_n = x_n^{-1/2}$, $y_n = \log(x_n/2)$, $\mathsf{b}_n = A_1(\log x_n)^{\mathfrak{p}_0}$ the threshold of the level n (equation YM3_I:(634)), and, for a weight w, $A_n(w;U) = A^{\eta_n}(w,U) = L^n\sum_p w(p)a_p(U)$, the weighted action of [7, eq. 216]. For $1 \le n \le K$, $X \in \mathcal{D}_n$ and $\mathsf{a} = (\mathsf{a}_1, \mathsf{a}_2) \in (0,\alpha_0] \times (0,\alpha_2]$, $\alpha_2 = \alpha_2^{\rm pl}$ the third radius of item 8, the *core of the size* a, $\mathcal{U}_n(X;\mathsf{a})$, is the set of triples P in the same hereditary core of Def. YM3_I:49.1 at the radii (α_0,α_1) with $|\partial\mathbf{U}_P(p) - \mathbf{1}| < \mathsf{a}_1 L^{-2n}$ at the plaquettes formed on X and $\mathrm{hol}_n(P;X) < \mathsf{a}_2$; $\mathcal{U}_n(X) = \mathcal{U}_n(X;(\alpha_0,\alpha_2))$ is the core with the third radius, and $N_n(\mathsf{a}) = \sigma\mathsf{a}_2/\alpha_2 + 2\tau L^{-(K-n)}\mathsf{a}_1/\alpha_0$.

The following objects are formed by recursion on the step $k = 0, 1, \dots, K-1$, which integrates the fluctuation field of the level k at the coupling x_k over the fibre of a block field W of the level $k+1$, at the triple $Q = P_W$. *The classical action and the trees:* the classical action $\Psi^{(k)}(Q;\cdot)$ of the step is its Gaussian form, with the covariance $C_*(Q)$ of the step, together with its linear term and its higher potentials, the weight v_k and the earlier trees being entered as potentials, so that the action of the step carries no explicit weight; $c_*^{(k)}$ is its stationary point; the tree $\mathrm{ta}_k(Z';\omega,\cdot)$, $Z' \in \mathcal{D}_{k+1}$, is the value $\Psi^{(k)}(Q;c_*^{(k)})$, written as a sum of local terms and re-sorted into terms on the cores $\mathcal{U}_{k+1}(Z')$; and the sum of the trees is $\mathrm{Tr}_n(P) = \sum_{j=1}^n (x_{j-1}/x_n)\sum_{Z\in\mathcal{D}_j}\mathrm{ta}_{j-1}(Z;\omega,\rho_Z P)$, $\mathrm{Tr}_0 = 0$, so that $x_{k+1}\mathrm{Tr}_{k+1}(Q) = x_k[\mathrm{Tr}_k(\rho Q) + \Psi^{(k)}(Q;c_*^{(k)})]$.

The non-classical array: the non-classical term $\delta_{k+1}(Z';\omega,\cdot)$ consists of the potential V^J of the Jacobians of the chart (the localised exponent J_W of the Jacobians in the weight wt_W of the chart integral below, one of the potentials of the step) and of the potential V' of the arrays of the step, both taken at $c_*^{(k)}$, and of the ideal term $E_k^{\rm id}$ below at the centre $g_k^{-1}c_*^{(k)}$ less the same object at $\omega = 0$, all re-sorted in the same way; $\mathrm{Lf}_k = \mathrm{Lf}[g]$ is the lift of (b) at $g^{(j)} = \delta_j$; $S_n^c[\omega](U) = S[\mathfrak{f}_n](U) + \sum_{j\le n}\sum_{Z\in\mathcal{D}_j}[\delta_j(Z;\omega,p_Z^{(j)}(U)) - \delta_j(Z;\omega,\mathbf{1})]$, where $S[\mathfrak{f}_n]$ is the action of the bare arrays of the never-large term at the zero source ([7, Prop. B.15], at $w = 0$) and $p_Z^{(j)}(U)$ the plain pair of U on Z, the configuration and its current of the level j on $\mathrm{Bd}(Z)$; and $E_n[\omega] = E_{(n)} + \sum_{i\le n}\sum_{X\in\mathcal{D}_i}\delta_i(X;\omega,\mathbf{1})$, $E_{(n)}$ the number of that term.

The centres, the ideal term and the tiny terms: the true centres are $\nu_j^{(n)}(V) = g_j^{-1}c_*^{(j)}$, read at the triple ρP_V of the level $j+1$, $0 \le j < n$, and the centre of the step is $\mathsf{m} = \nu_k^{(k+1)}(W)$; for a system V of potentials $V^o(Y;\cdot)$, $Y \in \mathcal{D}_k$, and a centre n, $\Xi_{\mathsf{n}}[V](Q) = \int d\mathsf{G}_{\mathsf{n}}(B)\,\mathbf{1}\{B \in \mathrm{Rg}_{x_k}\}\exp\sum_{Y,o}V^o(Y;B-\mathsf{n})$, G_{n} being the Gaussian measure of the covariance $C_*(Q)$ and the mean n; V_{m} is the system of the three potentials of the step about the centre m (the potentials of the step, the potential of the arrays, and the remainder of the shift of the centre), without the carried tiny terms; $E_k^{\rm id}(X';Q,\mu)$, $X' \in \mathcal{D}_{k+1}$, is the cluster sum at the centre 0 of the system V_μ on X', so that $\Xi_0[V_\mu](Q) = \exp\sum_{X'}E_k^{\rm id}(X';Q|_{X'},\mu|_{X'})$; and $\mathrm{Rf}^{(k+1)}(X';Q,\nu)$ is the cluster sum at the centre ν_k of the system V_{ν_k} enlarged, for $k \ge 1$, by the carried term $\mathrm{Rf}^{(k)}$ as one further potential, less the cluster sum at the centre 0 of V_{ν_k} without it.

The classical factor of the step: $c_k(W) = n_k(W)\exp[\mathcal{E}_k(W) - x_k\Psi^{(k)}(Q;c_*^{(k)}) + \sum_{Y\in\mathcal{D}_k}(V^J + V')(Y;Q|_Y;c_*^{(k)}|_Y)]$, where $\mathcal{E}_k(W)$ is the exponent of $\Phi_k[\omega]$ below at $V_W(0)$ ($\Phi_0[\omega] = \rho_0$), V^J and V' are the potential of the Jacobians of the chart and the potential of the arrays of the step, as in δ_{k+1}, and $n_k(W) = \mathsf{pf}_k(x_k)\,\mathsf{Z}^{(k)}$ is the normalisation of the step, with the prefactor equation YM3_I:(907) and the Gaussian integral in Prop. YM3_I:50.2 at the triple of W; so $\mathrm{wt}_W(B)\Phi_k[\omega](V_W(B))$ is $c_k(W)$ times the density of G_{m} at B times $\exp\sum_{Y,o}V_{\mathsf{m}}^o(Y;B-\mathsf{m})$.

Then for $1 \le n \le K$ the never-large term of $\rho_0 = \exp[-A(x_0 - L^K\omega, \cdot)]$ at the level n has the density

$$f_n(V) = \Phi_n[\omega](V) \exp \sum_{X \in \mathcal{D}_n} \mathrm{Rf}^{(n)}(X; P_V, \nu^{(n)}(V)) \quad \text{on } \mathcal{V}_n^\chi, \qquad f_n = 0 \quad \text{off it},$$

$\mathcal{V}_n^\chi$ being the set of the configurations of the level n that pass all the small-field tests ([7, Def. B.14]), and

$$\Phi_n[\omega] = \exp\big[E_n[\omega] - x_n A_n(1 + v_n; U_n(V)) - x_n \mathrm{Tr}_n(P_V) + S_n^c[\omega](U_n(V))\big], \qquad v_n = -L^{K-n}\omega/x_n,$$

where Tr_n, the sum of the trees (the classical remainders) of the earlier steps, the non-classical array $S_n^c[\omega]$ and the number $E_n[\omega]$ are those above, and P_V is the frozen triple of the background $U_n(V)$ of [7, Prop. B.15] and Thm. YM3_I:E in the setting of Section YM3_I:52.7. Each step $0 \le k < K$, at every $W \in \mathcal{V}_{k+1}^\chi$ and with $f_0 = \rho_0$, is the identity

$$\int_{\mathrm{Rg}_{x_k}} \mathrm{wt}_W(B) f_k(V_W(B))\, \mathrm{d}B = c_k(W)\, \Xi_0[V_\mathsf{m}](Q) \exp \sum_{X' \in \mathcal{D}_{k+1}} \mathrm{Rf}^{(k+1)}(X'; Q, \nu^{(k+1)}(W)), \qquad c_k(W)\, \Xi_0[V_\mathsf{m}](Q) = \Phi_{k+1}[\omega](W),$$

whose left member is $f_{k+1}(W)$, the chart integral $\mathsf{I}[f_k](W)$ at $\mathsf{t} = x_k$ ([7, Def. B.2] and equation [7, eq. 191], [7, eq. 191]): Rg_{x_k} is the set of the scaled fields below the threshold $\mathfrak{b}(x_k)$ ([7, §B.2]), $V_W(B)$ the chart of the fibre over W ([7, Lem. B.3]), and $\mathrm{wt}_W(B) = \mathsf{pf}_k(x_k)\, z_{V_0}^{(k)}(\Gamma) \exp[-x_k G_W(B) + \mathsf{J}_W(B)]$, $V_0 = V_k^{\mathrm{bf}}(W)$, the weight of the chart integral; $Q = P_W$, $\mathsf{m} = \nu_k^{(k+1)}(W)$ is the centre of the step, and $c_k(W)$, Ξ_0 and V_m are those above; so each step is classical, times ideal, times the exponential of a tiny term.

The ideal term is $\exp \sum_{X'} E_k^{\mathrm{id}}(X'; \cdot)$ with $|E_k^{\mathrm{id}}| \le \frac{1}{4} e^{-\kappa'' d_{k+1}(X')}$, so it does not vanish. $\mathrm{Rf}^{(n)}(X; P, \nu)$ is holomorphic on $\mathcal{U}_n(X)$ times the polydisc of the centres, one 1-form ν_j for every earlier level $j < n$, with $|\mathrm{Re}\, \nu_j(b)| < \vartheta_{j,n} \theta_1' \mathsf{b}_j$ and $|\mathrm{Im}\, \nu_j(b)| < \vartheta_{j,n} \theta_1'' \mathsf{b}_j$, $\vartheta_{j,n} = \frac{1}{2}(1 + 2^{-(n-j-1)})$, gauge invariant when the centres are carried along, and $|\mathrm{Rf}^{(n)}| \le \bar{K}_{\mathrm{w}} e^{-c^{\mathrm{pl}} y_n^2} e^{-\kappa d_n(X)}$, $y_n = \log(x_n/2)$, with the constants $\bar{K}_{\mathrm{w}}$ and c^{pl} of item 7; the true centres obey $|\nu_j^{(n)}(V)(b)| \le \frac{\theta_*}{2} L^{-2(n-j-1)} \mathsf{b}_j$, $\mathsf{b}_j = A_1 (\log x_j)^{\mathfrak{p}_0}$ the threshold of the level j (equation YM3_I:(634)); the trees obey $e^{r_1 d_n(Z)} |\mathrm{ta}_{n-1}(Z; \omega, P)| \le K_{\mathrm{ta}} N_n(\mathsf{a})^2$ on the cores $\mathcal{U}_n(Z; \mathsf{a})$ of the size a, at every $\mathsf{a} \in (0, \alpha_0] \times (0, \alpha_2]$, with one K_{ta} and one rate r_1 at all levels; the non-classical terms obey $e^{r_1 d_n(Z)} |\delta_n(Z; \omega, P)| \le 129 \min\{1, (\sigma + 2\tau)/c_{\mathrm{w}}\}$ on $\mathcal{U}_n(Z)$; and the lift of the non-classical array, which is (b) at $g^{(j)} = \delta_j$, $B_j \le 129$ and the rate $\kappa_{\mathrm{lf}} = r_1$, obeys $e^{r_1 d_k(X)} |\mathrm{Lf}_k(X; \omega, P')| \le 129 K_{\mathrm{Lf}} L/(L-1)$, $K_{\mathrm{Lf}} = 2C_E(140 C_Q + 1)$, as $\sum_{j \le k} L^{-(k-j)} \le L/(L-1)$.

The rate r_1 is admissible in (b): $r_1 = \kappa'' - 8 \log 2$ with $\kappa'' = (1 + 4\beta_{\mathrm{s}})\kappa + 1 + \lambda_{\mathrm{s}}$ and $\lambda_{\mathrm{s}} \ge 1$ (Section YM3_I:82, where $\lambda_{\mathrm{s}} = \log 18432$ and the formula for κ'' is recalled), so that by item 3, $r_1 \ge E_q + 2 - 8 \log 2 > \mathsf{C}_{\mathrm{cs}} > 4 \log 18432$. Every number is a constant placed before γ or an absolute constant. Everything is holomorphic in ω on the open set of the plaquette functions that lie, for some $\sigma, \tau \ge 0$ with $\sigma + 2\tau \le c_{\mathrm{w}}$, in the class with strict inequalities $|v_n|_\infty < \sigma$ and $\mathrm{sw}_n(v_n) < 2\tau L^{-(K-n)}$ at every $0 \le n \le K$, $\mathrm{sw}_n(v)$ being the largest difference $|v(p_{\mu\nu}(x')) - v(p_{\mu\nu}(x))|$ over the parallel plaquettes with $x' - x$ along e_μ or e_ν and of length at most one spacing of the level n.

At $\omega = 0$ the trees, the δ_n and Tr_n vanish, the true centres are $\nu_j^{(n)}(V) = 0$, and $\mathrm{Rf}^{(n)}(X; P, 0) = 0$ for every $P \in \mathcal{U}_n(X)$, at the zero centres; hence $f_n = \Phi_n[0]$ is the never-large term. The background is $U_n(V)$ from [7, Prop. B.15], and the centres of the Gaussian integrals are free arguments; the cores with the third radius coincide with the plain cores at the levels $n \le (\frac{L}{2} + 29.39)/\log L$, so at the levels $n \le 13$ for every odd $L \ge 13$, and

are proper subsets of them from some level on; at $k = 0$, where the step is written with bond labels, its centre is at most $K_Z\sigma$ on the whole plain core, K_Z a constant placed before γ.

■□

Theorem R (Weighted backgrounds)**.** In the setting of Thm. Q, part (a) uses the regular-reference and geometric setting of Def. 5.4; its analytic response and Hessian estimates are proved in Prop. 5.5. Parts (c) and (d) use the complex scalar gauge-transfer estimates proved in Lem. 5.21. Evaluation of the algebraic identity in (e) on these moved backgrounds uses the same construction and its stated regularity and geometric conditions; the algebraic identity and its bounds for already given families do not require that construction.

(a) *Critical backgrounds.* Use the real regular reference, natural-pair and separated-region setting of Def. 5.4. For the plane and symmetric corner weights $\mathbf{h} = (h, v)$, Prop. 5.5 gives a unique critical background in the slice $RD^{\mathsf{T}}A = 0$, $\mathcal{Q}(A) = b$, among fields with $\sup_y \|A\|_y^{\beta_H} \le r_*$, where this norm includes the scaled first-gradient Hölder seminorm defined there. It is holomorphic in the row datum and the weights on $\sup|b| < \rho_*$, $\|\mathbf{h}\| < z_*$, real at real data, and obeys the localized response bounds eq. 5.iv–eq. 5.v. Its Hessian in the block datum satisfies eq. 5.vi on $Y_0 + iY_0$, with the longer-sequence hypotheses in that proposition. The unweighted comparison is at the same reference background. The Hölder norm of the weights cannot be replaced by their supremum in a uniform Hölder estimate for this response. There is a separate finite-lattice assertion at $U_0 = 1$, when all regions are the whole torus. For real plane weights $v_{\mu\nu} = \epsilon k_{\mu\nu}$, $|k| \le 1$, or for a single real mixed tensor term $\epsilon\varkappa\langle f_{\mu\nu}, f_{\mu\lambda}\rangle$, $|\varkappa| \le 1$, with $0 \le \epsilon < 1$, the critical backgrounds near the zero block datum form a real-analytic family and their block Hessian at that datum obeys $(1-\epsilon)Z_0 \le Z_\epsilon \le (1+\epsilon)Z_0$. This assertion uses each of these two perturbations separately.

A common constant $c \ne -1$ in the plane weights factors out: the weights $(h, v+c)$ have the same critical fields as $\mathbf{h}' = (h, v)/(1+c)$ and their multipliers and Hessians are multiplied by $1+c$. The smallness conditions are those on $\mathbf{h}'$. For real $c > -1$ the positive Hessian floor is multiplied by $1+c$ as well.

After lowering γ by the ceilings specified in the proof, the following holds on every history. At each step $t \ge 2$, the background and row datum of the run satisfy the hypotheses above, and for real weights with $\|\mathbf{h}\| \le z_N^+$ the block Hessian is at least $\frac{1}{2}\gamma^{\mathrm{pc}}\|B\|^2$ on real $B \in Y_0$. At the first step the real part of the second variation of the action itself is at least that amount for $B \in Y_0 + iY_0$ and complex weights with $\|h\|_\infty + \sup|v| \le z_N^1$. The latter assertion uses the two-region natural-pair hypothesis and the explicit background and block-field ceilings in the proof.

(b) *Walk terms.* Under the natural-pair and separated-region hypotheses of Prop. 5.37, there are numbers $r_\tau, r'_\tau > 0$ and a power M^τ of L, depending only on L, such that all four weighted systems are invertible for $|v|_\infty < r_\tau$, radii at most r'_τ, and block number at least M^τ. Their inverses satisfy the mean-square divergence-source estimate eq. 5.xliv, and their local walk terms satisfy eq. 5.xlv, at a common exponential rate. The Hessian changes by the local plaquette form of Lem. 5.36. Grouping a walk by its visited pieces gives zero for a disconnected set S and a bound with the factor $\epsilon_*^{\#S-1}$ for a connected set, after a further increase of M^τ for the prescribed ϵ_*, as proved after Prop. 5.37. The pieces and their adjacency are those specified there.

(c) *The moved background on a label.* On the label of a history with large-field regions, in every frame of Thms. YM3_I:AJ and YM3_I:AD (every such frame satisfies the standing conditions under which these labels are formed), after lowering γ by three further upper bounds of the shape of the list, $\sup_{x \ge \gamma^{-2}} \max\{w_{\mathrm{G}}, w_{\mathrm{lin}}(L)\}(\log x)^{-r} \le 1$ with $w_{\mathrm{G}} = 4c_U^{-1}(7\log L + \log 2)$, which gives $MR(\gamma^{-2}) \ge \max\{w_{\mathrm{G}}, w_{\mathrm{lin}}(L)\}$, $\sup_{x \ge \gamma^{-2}} C_m\varepsilon(x) \le \alpha_\star$ and $\sup_{x \ge \gamma^{-2}} 2K_5' C_m \varepsilon(x) \le b_2$, where C_m is fixed before γ and depends on L and the previously

chosen construction constants, including B_3, and is the constant with which a label supplies its background, regular at the radius $C_m\varepsilon_k$ with the envelope $e(x) = C_m\varepsilon_{j(x)}$, ε_j being the radius of regularity of the layer j; $\alpha_\star$ is a positive number of L with $\alpha_\star \le \min\{\alpha_{\rm lin}(L), e_0(L)\}$, the numbers of Lem. 5.19, so that the second ceiling gives the radius of regularity $\alpha \le \alpha_{\rm lin}(L)$ and the bound $\sup e \le e_0(L)$ on the envelope under which Lem. 5.19 applies; K_5' is a positive number formed with L through the constant $K_{\rm lin}$ of Lem. 5.19 and the constant K_5 below; and $b_2 > 0$ is the radius with which the substitution with three pairs of radii among the construction results (Prop. 5.27) carries the cores at half their radii into the cores at three quarters of them, and for anisotropies an of the plaquettes with $t(\mathrm{an}) < t_0$, $t_0 > 0$ a number of L with $t_0 \le t_{\rm lin}(L) \le \frac14$, $t_{\rm lin}(L)$ the number of Lem. 5.19, t being the supremum plus the scaled Hölder-$\frac12$ quotients: the moved background $W = e^{i\eta_k A}U$, $\eta_k = L^{-k}$ the spacing of the level, exists, is critical for $\sum_p(1+\mathrm{an}(p))a_p$ on the surface of the prescribed block averages up to a complex gauge, and is holomorphic in an; if $U = \tilde U$ at the bonds with an end point within the weighted distance R of a point x_1, then, with $e'(x) = e(x)\exp[-c_U(\mathsf{R} - d_B(x,x_1))_+]$,

$$l(b)\,|\eta_k A[U](b) - \eta_k A[\tilde U](b)| \le K_U\, t(\mathrm{an})\, e'(b_-) \qquad \text{at every bond } b,$$

b_- the initial point of b, with no gauge transformation between; the same bound holds, at every plaquette p, site x and row c of the block constraints, for the curvatures, $l(p)^2\eta_k^2|D_U A[U](p) - D_{\tilde U}A[\tilde U](p)| \le K_U t(\mathrm{an})e'(x_p)$, x_p the base point of p, for the divergences, $l(x)^2\eta_k^2|D_U^* A[U](x) - D_{\tilde U}^* A[\tilde U](x)| \le K_U t(\mathrm{an})e'(x)$, and for the multipliers ω of the pair, $|\omega[U](c) - \omega[\tilde U](c)| \le K_U t(\mathrm{an})e'(c_-)$, c_- the base point of c; if $\mathsf{R} \ge w_*$, then at every point x_* of a layer $n \ge 2$ with $d_B(x_*, x_1) \le \mathsf{R} - w_*$, in one gauge that is a member of gauge systems of both U and $\tilde U$, the Hölder-$\frac12$ quotients of $\eta_k^2(D_U A[U] - D_{\tilde U}A[\tilde U])$, over the plaquettes of one orientation in the cube of the bare sites within $\frac14 l_n$ bare spacings of x_*, and of $\eta_k(A[U] - A[\tilde U])$, over the bonds of one direction based within $\frac18 l_n - 2$ bare spacings of x_*, obey $(\frac14 l_n)^{1/2}[\cdot]_{1/2} \le K_U t(\mathrm{an})e'(x_*)l_n^{-2}$, resp. $l_n^{1/2}[\cdot]_{1/2} \le K_U t(\mathrm{an})e'(x_*)l_n^{-1}$; and the multiplier ω_0 of the plain background obeys $|\omega_0[U](c) - \omega_0[\tilde U](c)| \le K_O e'(c_-)$.

Here $l(b), l(p), l(x)$ are the scales of the layers, $l_n = L^n$, D_U is the covariant lattice exterior derivative and D_U^* its adjoint, and d_B is the weighted distance of Def. YM3_I:12.3; c_U, K_U, K_O depend on L alone, $c_U \le 1/(6L)$, $K_U = 7K_{\rm lin}^2(1+2L^3)^2 e^{c_U w_*}$ with the constant $K_{\rm lin}$ of Lem. 5.19 and $w_* = w + 21L$, $w = w(L)$ being the width of the cut-off used in the proof. The map $(\mathrm{an}, U) \mapsto A$ is jointly continuous, and the moved background, its triples and its currents are Borel functions of the configuration; every critical point on the fibre of norm $\le r_2(L)$ is carried to W by a unique small gauge transformation that induces the identity at every top of the sequence; the full covariant gradient obeys $l(x)^2|\eta_k^2(\nabla_\kappa A_\nu)(x)| \le K_5\, t(\mathrm{an})\, e(x)$; W lies in the cores at three quarters of their radii; the construction commutes with the translations by vectors of the coarsest lattice; and $\sum_p |(1+\mathrm{an}(p))a_p(W) - a_p(U)| \le K|\mathrm{an}|_\infty \sum_p a_p(U)$.

(d) *The centres on a label.* Let $0 < \theta_3 \le 1$ be fixed before γ. In the setting of (c), let γ be lowered further by upper bounds of the shape of the list formed with θ_3, A_0, A_1, $\mathfrak{p}_0$, r, $\varepsilon_{\rm v}$ and numbers of L, among them $C_\chi(1+2\theta_3)\delta(x) < \frac18 r_c$ for $x \ge \gamma^{-2}$, with two numbers C_χ, r_c of L; this keeps the box $|\mathrm{Re}\, B_1(b)| + |\mathrm{Im}\, B_1(b)| < (1+2\theta_3)\mathsf{b}_k$ of the complex block data of the step, of the unscaled radius $(1+2\theta_3)\delta(x_k)$, and with it the polydisc $|\mathsf{n}(b)| < \theta_3\mathsf{b}_k$ below, of the unscaled radius $\theta_3\delta(x_k)$, inside the ball, of a radius of L alone, in which the system of the centre has exactly one solution.

Let the envelope of (c) be taken with C'/L in the place of C_m, $C' = \max\{LC_m, 2LC_\chi(1+2\theta_3)A_1/(\rho_{\rm u}A_0)\}$, $\rho_{\rm u} = \rho_{\rm u}(L)$ being the radius of that ball; C' is the one number that is enlarged. Among the upper bounds on γ above are the ceilings $\sup_{x\ge\gamma^{-2}} C'\varepsilon(x) \le c(L)$, $c(L) > 0$ numbers of L, and the block-number and small-radius conditions of Prop. 5.37, under which Cor. 5.38 gives the global energy bound for the linear system; the nonlinear

field-energy bound is supplied by Lem. 5.20. Let the anisotropies obey moreover $t(\mathrm{an}) < t_\theta$, where $t_\theta = \min\{t_0, t_{\mathrm{u}}(L), c_{\mathrm{ce}}\theta_3 A_1/A_0\}$, $t_{\mathrm{u}}(L)$ being the number of L below which the moved backgrounds with a complex block datum are formed and $c_{\mathrm{ce}} > 0$ a number formed with L and C'; so $t_\theta \in (0, t_0]$ is formed with θ_3 and with constants placed before γ.

At every step $0 \le k < K$ and every regular base of the step (a regular background of the coarser level together with its block datum), the *centre* of the step is defined by two conditions: it is a complex stationary point n of the weighted action on the fibre of the block field, and it is in the block axial gauge. There is exactly one n with $|\mathsf{n}(b)| < \theta_3 \mathsf{b}_k$ at all bonds b of the step that obeys the two conditions; it is $\mathsf{n} = g_k^{-1} a''$, a'' the 1-form, in the axial gauge, of the finer block average of $W[\mathrm{an}; b_0]$ read about the finer block average of $W[0; b_0]$. Here both backgrounds are formed over the same regular reference background U_{b} on the longer sequence, with the same complex block datum b_0 obtained from the base of the step; $|\mathsf{n}(b)| \le C_\theta t(\mathrm{an})\mathsf{b}_k \le \frac{1}{2}\theta_3 \mathsf{b}_k$, C_θ depending on L, θ_3, A_1/A_0 and C_m. At $b_0 = 0$, the base being U_{b} with its own block averages, $\sum_b |\mathsf{n}(b)|^2 \le C^2 |\mathrm{an}|_\infty^2 A(\beta_t, U_{\mathrm{b}})$, C a number of L, the sum running over all the bonds of the step for $1 \le k < K$ and over its external bonds at $k = 0$, $A(\beta_t, U_{\mathrm{b}})$ being the action of the reference background with the couplings of the step; and n is holomorphic in an, real at real an and real b_0, and 0 when an is constant.

The linearised system of the centre (the second variation of the constrained weighted action with its multipliers, the averaging constraint and the block axial gauge) is invertible, in ℓ^2 and in a weighted maximum norm, with bounds depending on L alone. The centres $\mathsf{n}^{(i)}$ of the earlier steps $0 \le i < k$, formed by the same rule about the moved background of the label, vanish at $\mathrm{an} = 0$, are holomorphic in an and obey $|g_i \mathsf{n}^{(i)}(b)| \le C t(\mathrm{an})\varepsilon_i$, that is $|\mathsf{n}^{(i)}(b)| \le C t(\mathrm{an})(A_0/A_1)\mathsf{b}_i \le \frac{1}{2}\theta_3 \mathsf{b}_i$, with one number C of L for all levels, $\varepsilon_i = \varepsilon(x_i)$ the radius of regularity of the level i and $g_i = x_i^{-1/2}$; and their squares, summed over the stored bonds, which are the external bonds of these steps, obey $\sum_{i<k} \sum_b |\mathsf{n}^{(i)}(b)|^2 \le C^2 |\mathrm{an}|_\infty^2 A(\beta_k, U)$, $A(\beta_k, U)$ being the action of the background with the couplings of the level k.

(e) *The counterterms on a label.* On the label of a history with large-field regions the part of every coupling counterterm among the construction results that sees the anisotropy moves exactly into the local terms and the boundary terms (Lem. 5.22): the anisotropy stands in the main weight and the local terms, and in the moved background whenever (c) is applied, and in no counterterm; the counterterms, the values at the trivial triple and the trivial values of the sums are unchanged; at a fixed triple, with the original families and counterterms held fixed, the changes in the re-filed local and boundary entries are linear in an with explicit Lipschitz constants. The following size bounds require a margin. Let $0 < \theta < 1$; let the families $E^{(j)}$ before the re-filing obey $\|E^{(j)}\|_{j,\kappa+1} \le (1-\theta)E_0$ at every level and every source; let $\|\mathrm{an}\|$ be the norm of the anisotropies in the format of the families, with $|\mathrm{an}|_\infty \le \sigma_3 \|\mathrm{an}\|$, the anisotropies ranging over $\|\mathrm{an}\| < 1$, and, if $B_\sharp > 0$, let $\sigma_3 \le \theta E_0 e^{-7(\kappa+1)}/(\frac{3}{2} B_\sharp \alpha_0^2)$.

Then the re-filed families $\hat{E}^{(j)}$ obey $\|\hat{E}^{(j)}\|_{j,\kappa+1} \le E_0$ and $\|\hat{E}^{(j)} - E^{(j)}\|_{j,\kappa+1} \le \theta E_0 \|\mathrm{an}\|$; if $\|E^{(j)} - E_0^{(j)}\|_{j,\kappa+1} \le D$, $E_0^{(j)}$ the family at zero source, then $\|\hat{E}^{(j)} - E_0^{(j)}\|_{j,\kappa+1} \le D + \theta E_0 \|\mathrm{an}\|$, and for the re-filed families this bound stands in the place of the quadratic bound among the construction results; in a frame of Thms. YM3_I:AJ and YM3_I:AD the added parts of the boundary entries are at most $\theta M^3 E_0 e^{-7} \|\mathrm{an}\| e^{-\kappa d_j(X)}$; the trivial values are unchanged. ■□

Definition 5.4 (Weighted critical backgrounds)**.** Fix $0 < \beta_H < 1$. Use a separated sequence through level $1 \le k < K$, with bare spacing $\eta = L^{-k}$, the interior and outward determining rows, and no inward or edge rows. Retain the natural-pair, block-number, separation and torus-size hypotheses of the regular response construction in Props. YM3_I:31.1 and YM3_I:34.2. In particular the terminal

torus has at least $6R_1\mathfrak{M}$ sites per direction. The common block number and width are chosen large enough for the ordinary Hölder kernels and the positive fluctuation form used below: $\mathfrak{M} \geq M^{\mathrm{pc}}$, $R_1\mathfrak{M} \geq \mathfrak{r}^{\mathrm{pr}}$ and $\delta_* R_1\mathfrak{M} \geq 512 \log L$. Here M^{pc} is enlarged to make the cutoff defect in the proof at most $1/8$, and the regularity ceiling a^{pc} is decreased to give the local inverse comparison there.

Let U_0 be real, critical for the unweighted action on the fibre of its block averages. Write $\mathcal{Q}$ for the straightened row map and $a_{\mathcal{Q}}$ for its fixed analytic potential radius, chosen so that this map and its local Taylor bounds hold on $\sup_y \|A\|_y < 2a_{\mathcal{Q}}$. Write $K_A = \mathcal{Q}'(A)$, $K_0 = K_A|_{A=0}$, and

$$F(A) = \eta^{-1} \sum_p a_p(e^{i\eta A} U_0), \qquad F'(0) = K_0^{\mathsf{T}} \omega_0.$$

The fine pairing has weight η^3, and the dimensionless row pairing has weight ξ_c^{-1}. Suppose U_0 is regular at radius α: its plaquette defects are at most $\alpha\eta^2\xi_y^{-2}$ and $\xi_y^3|F'(0)| \leq \alpha$. Require the pair $(U_0, F'(0))$ to belong to the natural class of that sequence with radii (α, α, α) and real factor U_0. This retains the local real gauges in that class.

The weights are $\mathbf{h} = (h, v)$. The plane weight v is a function of inner plaquettes. At each bare site x, the symmetric matrix $h_{ww'}(x)$ is indexed by the 24 oriented corners $w = (\iota, \iota')$, $|\iota| \neq |\iota'|$, with $\iota, \iota' \in \{\pm 1, \pm 2, \pm 3\}$. If f_w is the field strength of the corresponding plaquette loop based at x, put

$$T_{\mathbf{h}}(A) = T_h(A) + T_v(A), \quad T_h(A) = \eta^3 \sum_{x,w,w'} h_{ww'}(x) \langle f_w, f_{w'} \rangle, \quad T_v(A) = \eta^{-1} \sum_p v(p) a_p.$$

All loops are contained in the ambient region. Set $|h(x)| = \max_w \sum_{w'} 2|h_{ww'}(x)|$. The norm $\|\mathbf{h}\|$ is the supremum over tops y of the sum of the local suprema of $|h|$ and $|v|$ and their C^{β_H} seminorms multiplied by $\xi_y^{\beta_H}$. The pairs in these seminorms have one base point in the cell of y and distance at most $\xi_y/4$; plane weights are compared in the same plane.

For potentials use

$$\|A\|_y = \max\{\xi_y |A|_y, \xi_y^2 |\nabla A|_y\}, \qquad \|A\|_y^{\beta_H} = \max\{\|A\|_y, \xi_y^{2+\beta_H} [\nabla A]_{\beta_H, y}\}.$$

The last seminorm uses the transports of U_0 along inner combs, over the same distances. With N_y denoting the appropriate potential, row or weight norm, write

$$\|u\|_{\theta, y_0} = \sup_y e^{\theta d_{\mathbf{B}}(y, y_0)} N_y(u).$$

For the Hessian assertion add the next region Ω_{k+1} and require the same natural-pair bounds on this longer sequence. Let Y_0 be the real top-layer fields about Ω_{k+1} whose next linearized averages vanish and which vanish on the block axial trees. Its norm and pairing are the top-row ones. ■

5.1.4. *The constrained background and its multiplier.* The weighted-background problem is finite dimensional at each lattice cutoff. Its uniform estimates, however, must not deteriorate with the number of regions or their layers. Fix a real regular reference U_0, critical for the unweighted action on the fibre of its block averages. Write a nearby field as $e^{i\eta A} U_0$. Let $\mathcal{Q}(A)$ be the straightened averaging map, put $K_0 = \mathcal{Q}'(0)$, and write

$$F(A) = \eta^{-1} \sum_p a_p(e^{i\eta A} U_0), \qquad F'(0) = K_0^{\mathsf{T}} \omega_0.$$

The multiplier ω_0 is part of the reference critical equation. It need not vanish when the reference is nontrivial. Thus the linearized constrained Hessian contains its correction W_0, defined by

$$\langle X, W_0 Y \rangle = \langle \omega_0, \mathcal{Q}''(0)[X, Y] \rangle.$$

Using $F''(0)$ alone would discard a term in the derivative of the constraint equation.

Let $\Delta = F''(0)$. The propagators invert the gauge-fixed operator built from $\Delta - W_0$, and their elimination gives maps H and $\widetilde{G}$ with

$$K_0 H = 1, \qquad K_0 \widetilde{G} = 0, \qquad \widetilde{G}(\Delta - W_0 + DRD^{\mathsf{T}}) = 1 - HK_0. \tag{5.ii}$$

Here $RD^{\mathsf{T}}A = 0$ is the chosen gauge slice. The exponential and Hölder bounds for these maps are inherited from the regular-reference inverse estimates at the specified region separation and block sizes. They provide a uniform inverse for the coupled equation, rather than an inverse for each local coordinate considered independently.

Write $C(A) = \mathcal{Q}(A) - K_0 A$ and $F'(A) = F'(0) + \Delta A + \mathcal{V}(A)$. For a weighted action term $T_{\mathbf{h}}$ and prescribed row datum b, the critical equation and constraint are equivalent to

$$\begin{aligned} A &= H(b - C(A)) + \widetilde{G}(C'(A)^{\mathsf{T}}\omega - F'(0) - \mathcal{V}(A) - W_0 A - T'_{\mathbf{h}}(A)), \\ \omega &= H^{\mathsf{T}}(F'(A) + T'_{\mathbf{h}}(A) + DRD^{\mathsf{T}}A - C'(A)^{\mathsf{T}}\omega). \end{aligned} \tag{5.iii}$$

The symbol b in this system is the block datum. The proof uses eq. 5.ii to recover both original equations from a fixed point. Gauge invariance removes the slice term at real data, and holomorphy carries that identity to the admitted complex parameter domain.

Most derivatives of the right side of eq. 5.iii contain the small potential, the reference regularity radius or the small weights. One field-to-multiplier derivative has only a fixed bound K_8. This is why the pair norm weights the multiplier component:

$$\|(A,\omega)\|_{\theta,y_0} = \max\{\|A\|^{\beta_H}_{\theta,y_0}, \Theta^{-1}\|\omega\|_{\theta,y_0}\}, \qquad \Theta \geq 4K_8.$$

Choosing the remaining radii in order gives contraction constant $1/2$ for $0 \leq \theta \leq \delta_*/4$. The displacement at $(0,\omega_0)$ is bounded by $C(\sup|b| + \alpha\|\mathbf{h}\|)$, so iteration remains in the chosen ball. This supplies existence, uniqueness in the stated Hölder norm, holomorphy and reality at real data.

Differentiating the fixed point in a single row gives $(1 - \partial_X \Phi^{\mathbf{h}}) d_b X e_{c_0} = (He_{c_0}, 0)$. The inverse has norm at most 2, and the right side has the exponential decay of H. Hence

$$\|d_b A e_{c_0}\|^{\beta_H}_y \leq 2B_* e^{-\delta_* d_{\mathbf{B}}(y,y_{c_0})/4}|e|.$$

The weight derivative is treated by the same inverse. Its source is $T'_{\mathbf{k}}(A)$, bounded by the product of the weight norm and the reference plus perturbed field strength. The first variation has a covariant-divergence term. Estimating it in the first-gradient Hölder norm uses the Hölder seminorm of the weights, through the product estimate for that divergence source. A supremum bound for the weights cannot supply this part of the response estimate.

5.1.5. *The Hessian bound.* The positive Hessian bound requires the longer region sequence. Its domain is $Y_0 + iY_0$, where real fields in Y_0 vanish on the block axial trees and have zero next linearized average. These conditions remove the gauge and retained block directions. The proof compares each local reduced inverse with the flat inverse in a real gauge. The resolvent estimate makes their difference $O(\alpha)$ with exponential decay. A partition of unity assembles these local inverses; derivatives of its cutoffs give a small commutator defect when the comparison blocks are sufficiently large. The resulting real unweighted form has a uniform positive floor. Holomorphic perturbation in the row datum and weights then gives exactly

$$\mathrm{Re}\langle \overline{B}, ZB\rangle \geq (\gamma^{\mathrm{pc}} - e_0\alpha - \kappa_0 s)\|B\|^2, \qquad B \in Y_0 + iY_0,$$

with $Z = d_b\omega$ and s as defined in proposition 5.5. The reference in this comparison is the unweighted Hessian over the same U_0.

Proposition 5.5 (Weighted critical response)**.** In Def. 5.4, there are positive radii a_*, r_*, ρ_*, z_* and constants B_*, Θ, C_T, depending only on L and β_H, with $a_* \leq \min\{a^{\mathrm{pc}}, a_{\mathcal{Q}}, L^{-2}/4\}$ and the following property. If $\alpha \leq a_*$, $\sup|b| < \rho_*$ and $\|\mathbf{h}\| < z_*$, there is exactly one critical background $e^{i\eta A}U_0$ with

$$RD^{\mathsf{T}}A = 0, \qquad \mathcal{Q}(A) = b, \qquad \sup_y \|A\|^{\beta_H}_y \leq r_*.$$

Its pair (A,ω) is holomorphic in $(b,\mathbf{h})$ and real at real data, and equals $(0,\omega_0)$ at $(0,0)$. It satisfies $\sup_y \|A\|_y^{\beta_H} < r_*$. For a row c_0 and a datum e_{c_0} supported there,

$$\|d_b A\, e_{c_0}\|_y^{\beta_H} \le 2B_* e^{-\delta_* d_{\mathbf{B}}(y,y_{c_0})/4}|e|, \qquad (5.iv)$$
$$|d_b\omega\, e_{c_0}|_y \le 2\Theta B_* e^{-\delta_* d_{\mathbf{B}}(y,y_{c_0})/4}|e|.$$

For every weight direction $\mathbf{k}$ and $0 \le \theta \le \delta_*/4$,

$$\max\{\|d_{\mathbf{h}} A\,\mathbf{k}\|_{\theta,y_0}^{\beta_H}, \Theta^{-1}\|d_{\mathbf{h}}\omega\,\mathbf{k}\|_{\theta,y_0}\} \le 2C_T(\alpha + \sup_y \|A\|_y^{\beta_H})\|\mathbf{k}\|_{\theta,y_0}. \qquad (5.v)$$

Here δ_* is the fixed propagator rate, with the response kernels taken at rate $3\delta_*/8$ before their weighted summation. With the longer-sequence hypotheses, let $Z = d_b\omega$ be the Hessian of the weighted critical value and put $s = \max\{\sup|b|/\rho_*, \|\mathbf{h}\|/z_*\}$. After decreasing the radii, there are $\gamma^{\mathrm{pc}}, e_0, \kappa_0 > 0$ such that

$$\mathrm{Re}\langle \bar{B}, ZB\rangle \ge (\gamma^{\mathrm{pc}} - e_0\alpha - \kappa_0 s)\|B\|^2, \qquad B \in Y_0 + iY_0. \qquad (5.vi)$$

The reference for the perturbation in s is the unweighted Hessian at $b=0$ over this same U_0. ■

Proof. We keep the exact constrained response at the reference. Its Hessian includes the multiplier correction. Put $\Delta = F''(0)$, and define $W(A,\omega)$ by $\langle X, W(A,\omega)Y\rangle = \langle \omega, \mathcal{Q}''(A)[X,Y]\rangle$. Thus $W_0 = W(0,\omega_0)$. Set

$$G_1 = (\Delta - W_0 + DRD^{\mathsf{T}} + K_0^{\mathsf{T}} a K_0)^{-1}, \quad H = G_1 K_0^{\mathsf{T}}(K_0 G_1 K_0^{\mathsf{T}})^{-1}, \quad \widetilde{G} = G_1 - HK_0G_1.$$

The physical and dimensionless row conversions are those in the proof of Prop. YM3_I:34.2; a is the positive row coefficient of that construction, including its boundary extension. This extension cancels on $N(K_0)$. Elimination of the auxiliary variables in Prop. YM3_I:31.1 gives

$$K_0H = 1, \quad K_0\widetilde{G} = 0, \quad \widetilde{G}(\Delta - W_0 + DRD^{\mathsf{T}}) = 1 - HK_0, \quad RD^{\mathsf{T}}H = 0.$$

The ordinary response kernels and their first differences decay exponentially. Their Hölder bounds follow as in Lem. 5.34, with the fixed exponents β_H and $(1+\beta_H)/2$ in that proof. The exponent gap in its ordinary-input argument is positive. For a divergence source of exponent β_H, use instead the same-exponent parametrix in the proof of Lem. 5.19, with that exponent in the Green-kernel cancellation estimate of Lem. YM3_I:19.4. The local multiplier term W_0 has size $C\alpha$ and is one of the lower-order terms in that parametrix. Thus all constants are uniform in the region sequence and its depth.

Write

$$C(A) = \mathcal{Q}(A) - K_0A, \qquad F'(A) = F'(0) + \Delta A + \mathcal{V}(A).$$

The critical equation with the slice term and its constraint is equivalent to the following two equations:

$$A = H(b - C(A)) + \widetilde{G}(C'(A)^{\mathsf{T}}\omega - F'(0) - \mathcal{V}(A) - W_0A - T'_{\mathbf{h}}(A)),$$
$$\omega = H^{\mathsf{T}}(F'(A) + T'_{\mathbf{h}}(A) + DRD^{\mathsf{T}}A - C'(A)^{\mathsf{T}}\omega).$$

Denote this map by $\Phi^{\mathbf{h}}$. The exact identities above recover both equations from its fixed point. Conversely a solution gives a fixed point by the same elimination. The slice term vanishes at a real fixed point by gauge invariance and the scalar projection identity; holomorphy extends that identity to the complex parameter domain.

Here is the uniform smallness estimate behind the contraction. The Taylor kernels of the row map are confined to the double block beneath a row. The local action kernels are confined to a plaquette star. In the scaled local norms their bounds are

$$\|C'(A)\| + \|C'(A)^{\mathsf{T}}\| \le Cr, \quad \|\mathcal{V}'(A)\| \le Cr, \quad \|W(A,\omega) - W_0\| \le C(|\omega-\omega_0| + r|\omega_0|)$$

on $\sup\|A\|_y \le r$, while $|\omega_0| \le C\alpha$. These are the local estimates used in Prop. YM3_I:33.1; their passage to exponentially weighted norms costs a fixed summation constant. The first component of $\partial_X \Phi^0$ is therefore small. In the second component, the only term without a small factor is $H^{\mathsf{T}}(D^{\mathsf{T}}D + DRD^{\mathsf{T}})\delta A$, which has a fixed bound $K_8\|\delta A\|$. Choose $\Theta \ge 4K_8$ and measure pairs by

$$\|X\|_{\theta,y_0} = \max\{\|A\|_{\theta,y_0}^{\beta_H}, \Theta^{-1}\|\omega\|_{\theta,y_0}\}.$$

First reducing a_*, r_* makes the two remaining derivative bounds at most $1/8$ each. The product inequality $[fg]_{\beta_H} \le |f|[g]_{\beta_H} + [f]_{\beta_H}|g|$ gives the same bounds for the Hölder terms. The added weighted action contributes at most $C\|\mathbf{h}\|$ to the derivative: its leading term is the covariant divergence of the weight times the field strength, and its other terms are the local curvature commutators. The divergence estimate just described uses the C^{β_H} norm of that product. Reduce z_* to absorb this contribution. We obtain

$$\|\partial_X \Phi^{\mathbf{h}}\|_{\theta,y_0} \le \tfrac{1}{2}, \qquad 0 \le \theta \le \delta_*/4. \tag{5.vii}$$

This contraction proves uniqueness in the stated norm, including its first-gradient Hölder term.

At $X_0 = (0, \omega_0)$ the displacement of the map is bounded by $C(\sup|b| + \alpha\|\mathbf{h}\|)$. Reducing ρ_*, z_* puts this displacement within half the chosen ball. Iteration gives the fixed point and its holomorphy, uniformly on smaller parameter balls. Conjugation gives reality. If a critical point in the slice has $\sup\|A\|_y^{\beta_H} \le r_*$, the row equation $K_A H = 1 + C'(A)H$ has an inverse bounded by its Neumann series. It gives the multiplier bound needed to place that point in the same pair ball. Thus it is the fixed point. The actual block field is obtained from the reduced datum by the accumulated averaging gauge, as in Cor. YM3_I:41.3.

Differentiate the fixed-point equation in a single row. It gives

$$(1 - \partial_X \Phi^{\mathbf{h}}) d_b X\, e_{c_0} = (H e_{c_0}, 0).$$

The right side has norm at most $B_*|e|$ in the exponential norm centred at y_{c_0}. The inverse on the left has norm at most 2, proving eq. 5.iv. In a weight direction the right side is formed from $T'_{\mathbf{k}}(A)$. This first variation is linear in the field strength and is bounded by $C_T(\alpha + \sup\|A\|_y^{\beta_H})\|\mathbf{k}\|_{\theta,y_0}$. The same inverse gives eq. 5.v.

We next record the unweighted lower bound used for the Hessian. The longer-sequence hypothesis is needed here. Let T^{ax} be the fluctuation form in nonpivot axial coordinates. In the local inversion proof of Thm. YM3_I:L, the flat axial form T^0 has the gap in equation YM3_I:(593). The finite support of the pivot-coordinate map and the exponential bound for the row inverse give an upper bound $T^0 \le a_{\mathrm{pc}} I$ by the row and column sum test. Hence

$$a_{\mathrm{pc}}^{-1} I \le (T^0)^{-1} \le \gamma_{\mathrm{c}}^{-1} I.$$

The natural class supplies a real local gauge on each comparison cube in which the potential and its first differences are small. In that gauge, the local system differs from the flat one by the local defect bounded in the proof of Thm. YM3_I:L. The resolvent expansion bounds the difference of the inverse blocks by $C\alpha e^{-\delta d(b,b')}$. Summing its rows and columns, and decreasing the regularity radius, gives

$$\|C_\kappa - (T^0)^{-1}\| \le (2a_{\mathrm{pc}})^{-1}.$$

Here C_κ is the block of the local inverse from the nonpivot-field source to the nonpivot field. Elimination of its other components identifies it with the inverse of the local reduced form. At real data it is real and symmetric. The last bound therefore gives

$$(2a_{\mathrm{pc}})^{-1} I \le C_\kappa \le p_{\mathrm{pc}} I, \qquad p_{\mathrm{pc}} = \gamma_{\mathrm{c}}^{-1} + (2a_{\mathrm{pc}})^{-1}.$$

Returning to the original local gauge is unitary and preserves these inequalities. With the partition from that same local inversion proof, put

$$P = \sum_\kappa h_\kappa C_\kappa h_\kappa, \qquad \sum_\kappa h_\kappa^2 = 1.$$

The local inverse bounds give $(2a_{\mathrm{pc}})^{-1}I \le P \le p_{\mathrm{pc}}I$. To compare PT^{ax} with the identity, lift a nonpivot field B to the auxiliary solution Φ in the reduced system. Its equation has only the last source component $T^{\mathrm{ax}}B$, and the ordinary inverse estimate gives $\|\Phi\|_\infty \le C_{\mathrm{pc}}\|B\|_\infty$. On the enlarged comparison cube the local and global equations agree. Multiply by a cutoff which equals one on the support of h_κ and apply the local inverse. The main term is $h_\kappa B$. The error is the inverse applied to the cutoff commutator. Every difference of the cutoff costs $\mathfrak{M}^{-1}$; the enlarged cubes have bounded overlap. Multiplying by h_κ and summing consequently gives

$$PT^{\mathrm{ax}} = 1 - E, \qquad \|E\|_\infty \le 2N_G B_{\mathrm{pc}} C'_{\mathrm{cut}} C_{\mathrm{cut}} C_{\mathrm{pc}} \mathfrak{M}^{-1} \le \tfrac{1}{8}.$$

Here N_G is the overlap bound of the enlarged comparison cubes, B_{pc} the local inverse bound, C_{cut} the bound for the scaled cutoff differences, and C'_{cut} the commutator bound per such difference. The constant C_{pc} is the lifting bound for Φ above. All depend only on L. The last inequality is imposed by the common block floor. Although PT^{ax} need not be symmetric, it is similar to the real symmetric matrix $P^{1/2}T^{\mathrm{ax}}P^{1/2}$. Its eigenvalues lie in $[7/8, 9/8]$. Consequently $T^{\mathrm{ax}} \ge 7/(8p_{\mathrm{pc}})$. For $B \in Y_0$, the next-average constraint recovers its pivot entries from the nonpivot entries: $B = \mathsf{C}\check{B}$. Here C solves the linear next-average equations for the pivot bonds in the axial gauge. Its entries are bounded by C_K, and each bond shares a coordinate block with at most $n_{\mathrm{pc}} = 36L^3$ positively oriented bonds. The finite block-coordinate estimate gives $\|B\|^2 \le 2(C_K n_{\mathrm{pc}})^2\|\check{B}\|^2$. Hence the unweighted fluctuation form has a positive floor γ^{pc} on Y_0. The difference between that Lagrangian form and the Hessian of the critical value is the second derivative of the row map paired with the reference current. Its local kernel bound gives a loss at most $e_0\alpha\|B\|^2$. Thus at $(b, \mathbf{h}) = (0, 0)$,

$$\langle B, Z(0,0;U_0)B\rangle \ge (\gamma^{\mathrm{pc}} - e_0\alpha)\|B\|^2.$$

The derivative of the multiplier is the block Hessian, because the critical equation cancels the field variation in the first derivative of the critical value. The analytic response and its localized derivative estimates above bound the rows and columns of this Hessian by a summable exponential kernel on the top layer. Cauchy's estimate on the common parameter balls, followed by that same row and column sum, gives

$$\|Z(b,\mathbf{h};U_0) - Z(0,0;U_0)\|_{Y_0^{\mathbb{C}} \to (Y_0^{\mathbb{C}})^*} \le \kappa_0 s.$$

Reducing the stated radii once leaves the larger analytic balls needed for Cauchy's estimate. Taking the real part proves eq. 5.vi. The unweighted comparison throughout is at the same reference U_0. □

5.1.6. *Construction estimates.* We use the following results among the construction results. Thm. YM3_I:D, and Thms. YM3_I:AJ and YM3_I:AD. The weighted progression and the never-large term with a source, with the statements on one term and on families: Section YM3_I:51.9, Prop. YM3_I:51.15 and Thm. YM3_I:V; and the sections on sources, equation [7, eq. 218] and [7, Prop. B.15]. The rarity of a first creation of a large field on one torus, Prop. 4.54, and the rarity of the large-field region uniformly in the volume, Prop. 4.52. The law of the field and of the history, Prop. YM3_I:44.1, with the measures in equation YM3_I:(647).

The count of polymers, [7, Lem. O.1], whose extension to any number of sources is Lem. 5.6. The frame, [7, Def. B.14]; its background and the history with no large-field region, [7, Prop. B.15]; and the local step with a source, Prop. 6.32. The hereditary cores, Def. YM3_I:49.1 and §5.2; the Hölder bound for real backgrounds, Thm. [7, Thm. BT]; and the ordinary Hölder response, Prop. 5.35. The boards of the lattices, [7, §B.1]; reflection positivity at the site and bond mirrors, which are the walls of the boards, Prop. YM3_I:91.4; the chessboard estimate, Prop. YM3_I:86.3 and, for the torus measures, [7, Cor. B.1]; the domination of periodic Gibbs states by the torus measures through their conditional laws, Prop. YM3_I:91.9; and the bound $0 \le p_{\ell'} - p_\ell \le G_g$ of the setting on the pressures formed with the plain partition functions of Section YM3_I:86, [7, Cor. B.1] with Prop. YM3_I:89.2 and [7, Lem. N.1].

The bound on connected functions of $\mathcal{O}$ on a torus at separated supports, $|S^{T}_{n;K,\ell,\beta}(f)| \le C_n(\mathsf{d})e^{ncV\ell^3}\prod_i \|f_i\|_\sharp$ for real Lipschitz f_i two of whose supports are $\ge \mathsf{d}$ apart, as proved in Lem. 5.1, and its per-ball form, Prop. YM3_I:89.4, for supports in sup-balls of radius R and $\ell \ge 16n(R+1)$. The two-point law on a torus of small side, Thm. YM3_I:AU with Prop. YM3_I:126.4, and the kernel bounds $|C^{\mathrm{free}}_K(p,q)| \le (1+3C_{\mathrm{lat}})a^3|u|^{-3}$ and $|\epsilon_p\epsilon_q C^{\mathrm{free}}_K(p,q) - a^3\mathbb{K}_3(u)_{\lambda(p)\lambda(q)}| \le C_{\mathrm{lat}}a^3(a^2|u|^{-5}+\ell^{-3})$, u the least lift of $x(p)-x(q)$ and $\epsilon_p = \pm 1$ the orientations, of equations YM3_I:(2245)–(2246).

The comparison of two lattices at equal label, Thm. YM3_I:AR; the laws of the block fields, Def. 2.7 and Cor. 2.8; and the limits $S^{(\ell,\mathsf{b})}_n$ of $S^{T}_{n;K,\ell,L^K\mathsf{b}}$ at a fixed label, with the rate $K^pL^{-K/10}$, Thm. YM3_I:AS, with the all-side extension in Cor. YM3_I:119.2. Reflection positivity of the periodic Gibbs states at all the families of mirrors, Prop. YM3_I:103.4, and the reconstruction data of the limits of Gibbs sequences, Thm. YM3_I:AP. The large-field part: the histories and their labels defined in Section YM3_I:44, with the finite exploration in equations YM3_I:(643)–(644); the separated layers, physical row spaces and unweighted constrained systems of Sections YM3_I:23 and YM3_I:27; their weighted mean-square extension in Prop. 5.37 and Cor. 5.38; and the layer-weighted distance of Lem. YM3_I:12.3. The same-exponent Hölder estimate for the moved background is proved in Lem. 5.19, with its normed spaces and envelope class.

The lattice calculus of [7, §J], [7, result J] and equations YM3_I:(2104)–(2105) and YM3_I:112, the integration by parts [7, Lem. J.1] of [7, Lem. J.1], and the rotational Ward identity at finite cutoff, Prop. YM3_I:112.1. For the step with plane couplings and the backgrounds: the lattices allowed at a height, with the room and the width of the halos, Section YM3_I:50.4; the scaled fields below the threshold, [7, §B.2], the chart of the fibre, [7, Lem. B.3], and the chart integral, [7, Def. B.2] and equation [7, eq. 191] with [7, eq. 191], its prefactor equation YM3_I:(907) and the Gaussian integral in Prop. YM3_I:50.2; the weighted action [7, eq. 216] and the density [7, eq. 217] at zero source; the common real and sourced choices in Prop. YM3_I:69.1 and Section YM3_I:82; and the substitution with three pairs of radii, Prop. 5.27.

The arbitrary-order short-distance comparison remains a sketch at ($\mathrm{Q_n}$) of §5.1.2. Its uniform interacting small-field remainder remains to be established. The free Gaussian contribution is computed by expanding $-\frac{3}{2}\log\det(1-\mathcal{S}W_\lambda)$: the mixed coefficients are the cyclic contractions of the covariance $\mathcal{S}$. The explicit first-creation weights give the large-field term in ω_n. These two calculations do not supply the remaining arbitrary-order interacting error. From the other sections we use: for $0 < g \le \gamma^{\mathrm{sd}}_W$, the bound on centred products of any number of tensor entries at separated supports in every periodic Gibbs state, uniformly in the cutoff, the side, the state and the position, [7, Thm. C](a), and the bound $\|\mathcal{T}_K(H) - c\|_{\mathrm{ch}} \le 2\mathsf{C}_T Y_i^{\mathsf{q}_T} a_i^{-11/2}\mathsf{b}^{-1}|H|_\infty$, $Y_i = 2[\mathsf{c}_T + \log\frac{1}{a_i}]$, on the chessboard seminorm of a centred tensor entry in a cell of side a_i, [7, eq. 4] of [7, Thm. B](a), with $\mathsf{C}_T, \mathsf{q}_T, \mathsf{c}_T$ the constants C_E, q_E, c_E of that bound ([7, §2]); and, for $0 < g \le \gamma^{\mathrm{sd}}_W$, the invariance $\sigma(RX) = \sigma(X)$, $R \in \mathrm{O}(3)$, of tensor limits at tuples with one non-scalar entry among scalar ones ([7, §5], [7, Thm. U], which gives [7, Thm. R](c) at these tuples).

5.1.7. *Auxiliary estimates.*

Lemma 5.6 (Several sources)**.** Let $0 < g \le \gamma$ with $g^{-2} \ge x_{\mathrm{J}}$, which holds for every $0 < g \le \gamma^{\mathrm{ref}}_W$; let $(K,\beta) \in \mathfrak{D}(g)$ with $K \ge 3$, $x_0 = \beta > x_1 > \cdots > x_K$ its run, $g^{-2} < x_K \le X_g$, and $\ell = 2L^m$ with $MR(x_K) \le L^m$, which holds on every side $\ell \in \mathfrak{L}(g)$; let $s \ge 1$ be a depth and $k = \max\{K-s, 2\}$ the working level, so that $2 \le k \le K-1$; and let (F_a, o_a, w_a), $a = 1,\dots,p$, be dominated local sources with the comparison weights w_a. A *dominated local source* (F,o,w) consists of a real function F of the configurations of the bare bonds that is a gauge invariant polynomial in the entries of the bond variables; a local representation o of F, that is, families of local terms at the heights $1 \le j \le k$ whose sums, less their values at the trivial configuration, are F, and a family at the height 0 that sums to the fluctuation variation $F(e^{iB'}\mathbf{U}) - F(\mathbf{U})$ over the background $\mathbf{U}$ of the level-1 block field; and a real weight $w \ge 0$ of the plaquettes with $|F| \le \mathcal{O}(w)$ on all configurations, with sizes of o at most

those of a source of the first size $|w|_\infty$, $m_0(o) \le 12\mathsf{c}^\flat M^3\alpha_0^2 L^K|w|_\infty$ and $m_j(o) \le 12M^3\alpha_0^2 L^{K-j}|w|_\infty$ for $1 \le j \le k$, $m_0(o)$ being the size of the family at the height 0, $m_j(o)$ twice the norm $\|\cdot\|_{j,\kappa}$ of the family at the height j and $\mathsf{c}^\flat = 81C_P^2\mathsf{C}_{\mathcal{A}}^2/K_\Lambda$, and such that the corners of the plaquettes of $\operatorname{supp} w$ carry o: its members vanish on the regions that hold none of these corners. Zero sources are omitted before the parameter radii are formed.

(a) Let $\Lambda_0 = \{\lambda \in \mathbb{C}^p : \sum_a |\lambda_a| m_{0,a} < N_\star^{[0]}\}$ and, for $1 \le j \le k$, $\Lambda_j = \{\lambda \in \Lambda_{j-1} : \sum_a |\lambda_a| n_{j,a} < N_\star^{[j]} - (\sum_{n\le j}\omega_n)(E_0 + \frac{3}{2}\alpha_0^2 B_\sharp)\}$, where $m_{j,a} = m_j(o_a)$, $n_{j,a} = m_{j,a} + \sum_{i=1}^{j}\omega_{j+1-i}d_{i,a}$, $d_{1,a} = \theta(x_0)m_{0,a}$, $d_{j+1,a} = \theta(x_j)n_{j,a}$, $\omega_n = (n+1)L^{-(n-1)}$, θ is the size of a source of (b), and $N_\star^{[j]} = w_\star\alpha_0/(4K_\Lambda(1+\beta_{\mathrm{s}})\delta(x_j))$ is the bound on the size of the bare array that the step of the level j carries (equations [7, eq. 196] and eq. 2.i). These are open convex neighbourhoods of 0, decreasing in j; their thresholds read no source and not p, so the sources share the room of each step, and a polydisc $\{|\lambda_a| < \rho_a\}$ lies in Λ_{t-1} when $\sum_a \rho_a m_{0,a} \le N_\star^{[0]}$ and $\sum_a \rho_a n_{j,a}$ is at most the right member of the condition of Λ_j for $1 \le j < t$; in particular $\Lambda_k = \bigcap_{0\le j\le k}\Lambda_j$ holds the polydisc of the radii $\rho_a = p^{-1}\min_{0\le j\le k} N_j/n_{j,a}$, N_j being the right member of the condition of Λ_j and $n_{0,a} = m_{0,a}$: these radii are p^{-1} times numbers that depend on the sizes of the sources alone.

For $1 \le t \le k+1$ and $\lambda \in \Lambda_{t-1}$, $\tau_{h_t}[\rho_0 e^{\sum_a \lambda_a F_a}] = e^{D_t^\lambda(U_t(\cdot))}\mathsf{m}_t$ on the small block fields of the level t, with D_t^λ holomorphic in λ and formed from the members $E_\lambda^{(i)}(X)$ of the weighted progression; $\partial_B E_\lambda^{(i)}(X) = 0$ unless X meets the carrier of every F_a, $a \in B$, and the mixed derivatives at $\lambda = 0$ of the level i carry the factor $\exp(-[L^{t-i}\mathsf{D}_t/M - 2]_+)$, D_t the least sup distance of the carriers in level-t units. The centred moments at the never-large history are $\Phi[\prod_{a\in B}(F_a - c_a)] = \sum_\pi \prod_{\beta\in\pi}\kappa_\beta$, π over the partitions of B into blocks β, with $\kappa_\beta = \partial_\beta D_t^\lambda|_{\lambda=0}$ for $\#\beta \ge 2$ and $\kappa_{\{a\}} = \partial_a D_t^\lambda|_{\lambda=0} - c_a$ at the one-element blocks.

Here $\rho_0 = e^{-\beta A}$ is the bare density, h_t the never-large history of the level t and $\tau_{h_t}[\cdot]$ its term, $\mathsf{m}_t = \tau_{h_t}[\rho_0]$ a finite positive measure on the block fields of the level t, U_t the background, c_a the constant of F_a at the level t, and $\Phi[Y]$ the bounded density of $\tau_{h_t}[Y\rho_0]$ with respect to m_t.

(b) Let $p \ge 2$, $0 < g \le \gamma_W^{\mathrm{ref}}$, $0 < \mathsf{d} < 1$ with $s = s(\mathsf{d}) \le K$ and $2\mathsf{T}_p(3)\theta(L^{s-2}g^{-2}) \le 1$, and let the weights $w_a \ne 0$ be carried by balls $B(x_a, \frac{7}{8}\mathsf{d})$ at mutual distances $\ge \frac{5}{4}\mathsf{d}$. Then $|\langle F_1;\dots;F_p\rangle_{K,\ell,\beta}| \le k_p^{\mathrm{loc}}(g,s;\ell)\,\mathsf{b}^{-p}\prod_a |w_a|_\infty$, with k_p^{loc} depending on g,p,ℓ,s alone. Here $\mathsf{T}_p(u) = \sum_{n\ge1}(n+1)L^{-(n-1)}[(\frac{4}{3})^{1/2}L^{n/2}(1+n)^{\mathfrak{p}_0}]^p e^{-4\beta_s\kappa[L^{n-1}u-2]_+}$ is the series of the mixed derivatives of the order p and $\theta(x) = K_{\mathbf{M}}(1+\beta_s)A_1(\log x)^{\mathfrak{p}_0}x^{-1/2}/\alpha_0$ the size of a source at the coupling x, with the constants $\beta_s, K_{\mathbf{M}}$.

(c) In (b), let moreover $2\max\{\tilde{\mathsf{T}}_p(3), (\frac{4}{3})^{1/2}L\,\mathsf{S}_{\mathrm{nat}}\}\,\theta(L^{s-2}g^{-2}) \le 1$, where $\tilde{\mathsf{T}}_p$ is the series T_p with the further factor $L^{n/4}$ in its n-th term and $\mathsf{S}_{\mathrm{nat}} = \sum_{n\ge1}(n+1)^{1+\mathfrak{p}_0}L^{-n/4}$; this condition implies that of (b), and $\tilde{s}(p)$ denotes the least integer $s \ge 3$ with it. If the sizes of the local representation of F_1 at the heights $1 \le j \le k$, and its values on the backgrounds of the levels $1 \le t \le k$, are smaller than those of a source of the weight w_1 by the factors $L^{-j/4}$, resp. $L^{-t/4}$, that is, $m_j(o_1) \le 12M^3\alpha_0^2L^{K-j}L^{-j/4}|w_1|_\infty$ and $|F_1(U_t(W))| \le L^{-t/4}\,2B_3^2\varepsilon(x_t)^2L^{K-4t}\sum_p w_1(p)$ at the backgrounds of the configurations $W \in \mathcal{V}_t^\chi$, then

$$|\langle F_1;\dots;F_p\rangle_{K,\ell,\beta}| \le L^{-(k-1)/4}\,\tilde{k}_p(g,s;\ell)\,\mathsf{b}^{-p}\prod_a |w_a|_\infty,$$

with a number $\tilde{k}_p(g,s;\ell) \ge k_p^{\mathrm{loc}}(g,s;\ell)$ that depends on g,p,ℓ,s alone.

(d) Let $c = \sum_i \kappa_i m_i$ be a block form of Lem. 5.9(b): each monomial m is a product of $n_2(m)$ factors $\langle \mathbf{w}, \mathbf{w}'\rangle$, $n_3(m)$ factors $\det(\mathbf{w},\mathbf{w}',\mathbf{w}'')$ and $n_a(m)$ factors $1 - \frac{1}{2}\operatorname{tr} V$, where V runs

over the holonomies of the lassos of the block of 27 sites centred at a site x, the holonomies of its plaquettes carried to x along paths of at most three steps in the block, and $\mathbf{w} = \mathbf{w}(V)$; the order of m is $2n_2(m) + 3n_3(m) + 2n_a(m)$, and $\|c\| = \sum_i |\kappa_i| 2^{n_a(m_i)}$ is the norm of c of that lemma. For a real function k of the sites, the source $\mathrm{Fm}^c(k) = a^{-1} \sum_x k(x) c_x(U)$ of Lem. 5.9(b) is a dominated local source with the comparison weight $\varpi \mathbf{1}_{\mathrm{Pl}(k)}$ for every $\varpi \geq 76 \|c\| |k|_\infty$, $\mathrm{Pl}(k)$ being the set of the plaquettes of the blocks of the sites with $k \neq 0$, and $|\mathrm{Fm}^c(k)| \leq \mathcal{O}(2\|c\||k|_\infty \mathbf{1}_{\mathrm{Pl}(k)})$ on all configurations; if all its monomials have an order ≥ 3, the hypothesis of (c) holds for it with the comparison weight $\Gamma_c^\circ \|c\| |k|_\infty \mathbf{1}_{\mathrm{Pl}(k)}$, $\Gamma_c^\circ = \max\{76,\ 34 \max\{1, 5.04\alpha_0\}^{\mathrm{Ord}(c)-2},\ 2\max\{1, 2B_3\}^{\mathrm{Ord}(c)-2}\}$, $\mathrm{Ord}(c)$ the largest order of its monomials, even with L^{-2j} and L^{-2t} in the places of $L^{-j/4}$ and $L^{-t/4}$.

(e) Let $\ell \in \mathfrak{L}(g)$, let (F_a, o_a, w_a), $a = 1, 2$, be two dominated local sources whose carriers are at the sup distance d', and let $2 \leq t \leq k$, k the working level, $s' = K - t$ (so that $1 \leq s' \leq K-2$, as $k \leq K-1$), with $L^{s'} \mathsf{d}' \geq 3M$, so that $\mathsf{D}_t \geq 3M$, and $\mathsf{h}_2(s') \leq 1$, where $\mathsf{h}_2(s') = \frac{4}{3} \mathsf{k}_2 L\, \theta(\frac{3}{4} L^{s'+1} g^{-2})\, \mathsf{T}_2^\circ(L^{s'} \mathsf{d}'/M)$ with the series $\mathsf{T}_2^\circ(u) = \sum_{n \geq 1} (n+1)^{2\mathfrak{p}_0+1} e^{-4\beta_s \kappa [L^{n-1} u - 2]_+}$ and the number $\mathsf{k}_2 = \max\{\frac{1}{2}(\mathsf{c}^\flat)^2, 5\}$. Then, with $y_a \leq 4\ell^3 L^{4(K-t)} |w_a|_\infty$: the real part of the density $D_{t,2}$ of F_2 at the never-large history differs from the real part of its constant c_2 by at most $C_1 y_2$, and the mixed part $D_{t,12}$ of the density of $F_1 F_2$ is at most $C_2 e^{3 - \mathsf{D}_t/M} y_1 y_2$. ■

Proof sketch. $MR(x_K) \leq L^m$ on the sides of $\mathfrak{L}(g)$:

$$L^{m_{\mathfrak{L}}(g)} \geq LM\, R(X_g + \bar{B})$$

by equation YM3_I:(1528), and $x_K \leq X_g + \bar{B}$. (a) is the derivative calculus for the never-large term with one source (Prop. 6.32 and equation [7, eq. 218]), carried out in p parameters. Each source enters each step as one more local term, and the p sources share the room of each step, the bound of the step on the size of what it carries: with $m_{0,a}$ the size of F_a at the height 0 and $n_{j,a} = m_{j,a} + \sum_{i=1}^{j} \omega_{j+1-i} d_{i,a}$ the size at the level j of F_a together with all that the steps below the level j have made of it, as in the statement, the parameter λ ranges at the level t, $1 \leq t \leq k+1$, over the set $\Lambda_{t-1} = \bigcap_{0 \leq j < t} \Lambda_j$, the sets Λ_j being those of the statement,

$$\Lambda_0 = \Big\{\lambda : \sum_a |\lambda_a|\, m_{0,a} < N_0\Big\}, \qquad \Lambda_j = \Big\{\lambda \in \Lambda_{j-1} : \sum_a |\lambda_a|\, n_{j,a} < N_j\Big\} \quad (1 \leq j \leq k),$$

with $N_0 = N_\star^{[0]}$ and $N_j = N_\star^{[j]} - (\sum_{n \leq j} \omega_n)(E_0 + \frac{3}{2}\alpha_0^2 B_\sharp)$ for $1 \leq j \leq k$, $\omega_n = (n+1) L^{-(n-1)}$ as in the statement: $N_\star^{[j]}$ is the bound on the size of what the step of the level j carries, and for $1 \leq j \leq k$ the array of the level j without the sources has a size at most $(\sum_{n \leq j} \omega_n)(E_0 + \frac{3}{2}\alpha_0^2 B_\sharp) \leq \frac{1}{2} N_\star^{[j]}$ (the weighted-lag array bound in the proof of [7, Prop. B.8], together with $8 K_\Lambda N_\star s_j \leq w_\star$ from item 9 before Thm. Q), to which the sources add at most $\sum_a |\lambda_a|\, n_{j,a}$; so $N_j \geq \frac{1}{2} N_\star^{[j]}$ is the room that the step of the level j leaves to the sources, a number that reads no p. The sets Λ_j are open and convex and decrease with j, and Λ_{t-1} holds the polydisc of the radii $\min_{0 \leq j < t} N_j/(p\, n_{j,a})$, $n_{0,a} = m_{0,a}$, since on it $\sum_a |\lambda_a|\, n_{j,a} < N_j$ for $0 \leq j < t$; so Λ_k, which lies in every Λ_{t-1}, holds the polydisc of the radii $\min_{0 \leq j \leq k} N_j/(p\, n_{j,a})$.

These radii are p^{-1} times numbers that depend on the sizes of the sources alone, while the thresholds N_j read no p. On Λ_{t-1} the members of the progression up to the level t are holomorphic in λ; the derivative in λ_a of a member vanishes unless its polymer meets the carrier of F_a, and the count of polymers in [7, Lem. O.1], applied at every level $1 \leq j \leq k+1$ to the polymers that meet a given set of sites, gives the decay in the distance of the carriers: a polymer of the level i that meets the carriers of all the F_a, $a \in B$, has a linear size at least $L^{t-i} \mathsf{D}_t/M - 2$, whence the factor $\exp(-[L^{t-i} \mathsf{D}_t/M - 2]_+)$ of the statement; the formula for the centred moments is the moment–cumulant formula for the quotient of the never-large term with the sources by the term

without them, in which the centring moves the first cumulant alone: $\kappa_{\{a\}} = \partial_a D_t^\lambda|_{\lambda=0} - c_a$, and $\kappa_\beta = \partial_\beta D_t^\lambda|_{\lambda=0}$ for $\#\beta \geq 2$. (b) The cumulant is a mixed derivative at 0 of the logarithm of the normalised partition function with the sources.

The histories with a first creation of a large field at a level t weigh at most w_t. Along the never-large history the mixed derivative families of the order p are bounded in the norm of the rate $\kappa^+ = (1+4\beta_s)\kappa$; read at the rate κ they keep the factor $e^{-4\beta_s\kappa[D_j/M-2]_+}$, D_j the least sup distance of the carriers in level-j units. On the way from the finer levels up to the working level k, a family of the level i enters the step of the level $j \geq i$ with the weight $(n+1)L^{-(n-1)}$ of the lag $n = j+1-i$, while the size of a source against the bound of the step is at the level $i-1$ at most $(\frac{4}{3})^{1/2}L^{n/2}(1+n)^{\mathfrak{p}_0}$ times that at the level j; so the series T_p of the statement sums the mixed derivatives of the order p over the levels finer than the working level,

$$\mathsf{T}_p(u) = \sum_{n\geq 1}(n+1)L^{-(n-1)}\Big[(\tfrac{4}{3})^{1/2}L^{n/2}(1+n)^{\mathfrak{p}_0}\Big]^p e^{-4\beta_s\kappa[L^{n-1}u-2]_+},$$

and the induction over the levels up to $k+1$ closes if $\theta(x_j)\,\mathsf{T}_p(D_j/M) \leq \frac{1}{2}$ for $1 \leq j \leq k$. Since $D_j/M \geq 3$, $\theta(x_j) \leq \theta(L^{s-2}g^{-2})$ and T_p does not increase, this holds under the condition $2\mathsf{T}_p(3)\theta(L^{s-2}g^{-2}) \leq 1$ of the statement. Cauchy's estimates in each parameter then give the factor $\mathsf{b}^{-p}\prod_a |w_a|_\infty$. (c) is a recursion of numbers. If the sizes of F_1 fall like $L^{-j/4}$, the recursion of (a) gives $n_{j,1} \leq \Gamma L^{-j/4}$ times the size at the level j of a source of the weight w_1, Γ a number formed with the constants: against the induction hypothesis at the level j, the lag n contributes at most $(\frac{4}{3})^{1/2}L\,\theta(x_j)(n+1)^{1+\mathfrak{p}_0}L^{-n/4}$, and the induction closes with the sum $\mathsf{S}_{\rm nat} = \sum_{n\geq 1}(n+1)^{1+\mathfrak{p}_0}L^{-n/4}$ of the statement.

In the induction of (b) the falling sizes of the first source compare the level $i-1$ with the level j with the further factor $L^{n/4}$, so that the series $\tilde{\mathsf{T}}_p$ of the statement, which is T_p with the further factor $L^{n/4}$ in its n-th term, takes the place of T_p. Both inductions close under the condition $2\max\{\tilde{\mathsf{T}}_p(3), (\frac{4}{3})^{1/2}L\,\mathsf{S}_{\rm nat}\}\,\theta(L^{s-2}g^{-2}) \leq 1$ of (c), which implies the one of (b) at the same s; then the bound of (b) holds with $L^{-(k-1)/4}\tilde{k}_p(g,s;\ell)$ in the place of $k_p^{\rm loc}(g,s;\ell)$, $\tilde{k}_p \geq k_p^{\rm loc}$ a number of the same kind formed with these series, for the sizes at the levels $1 \leq j \leq k$ and the values on the backgrounds of the levels $1 \leq t \leq k$.

So the recursion closes under the condition of (c); the least integer $s \geq 3$ at which it holds with $p = r$ is the number $\tilde{s}(r)$ of the statement, and we write $\tilde{k}_r$ for the number $\tilde{k}_r(g,s;\ell)$ of the bound of (c). (d) Let $c = \sum_i \kappa_i m_i$ be a block form of Lem. 5.9(b), $\|c\| = \sum_i |\kappa_i| 2^{n_a(m_i)}$ its norm as in the statement, with m_i written also for the order of the monomial m_i, and k a real function of the sites; the source is $\mathrm{Fm}^c(k) = a^{-1}\sum_x k(x)c_x(U)$. At a height $j \geq 1$ it is represented by the family $\eta_j^{-1}\sum_x L^{K-j}k(x)c_x$, $\eta_j = L^{-j}$, so that $\eta_j^{-1}L^{K-j} = a^{-1}$, the sum over the sites x whose block lies in a given region of the level j; such a region has at most 8 cubes and the linear size 0, and at most $8M^3\eta_j^{-3}$ sites have one region (each of the at most eight cubes contains $(M/\eta_j)^3$ bare sites).

On the cores of radii (b_0, b_1) in Def. YM3_I:49.1 the vector part of the holonomy of a lasso is at most $5.04\,b_0\eta_j^2$ in level-j units and its plaquette term at most $(5.04\,b_0\eta_j^2)^2$, so a monomial of the order m is at most $(5.04\,b_0)^m\eta_j^{2m}$, and the family has the size $8M^3L^{K-j}|k|_\infty\sum_i|\kappa_i|(5.04\,b_0)^{m_i}\eta_j^{2m_i-4}$. The sizes are taken at $(b_0,b_1) = (\alpha_0,\alpha_1)$: as $5.04\,\alpha_0L^{-2} \leq 1$, $(5.04\,\alpha_0)^{m_i}\eta_j^{2m_i-4} \leq 25.41\,\alpha_0^2$, and with the factor 2 of the size of a representation at a height $j \geq 1$ this is at most $34\|c\||k|_\infty$ in the unit $12M^3\alpha_0^2L^{K-j}$ of the size of a source of the weight 1.

At the height 0 the representation is the family in the fourth place $a^{-1}\sum_x k(x)[c_x(e^{iB'}U) - c_x(U)]$, B' the fluctuation field with its values at the pivot bonds, which sums to the fluctuation variation of the source; a lasso has at most ten factors, and telescoping in them gives the size $911\,\mathsf{c}^\flat M^3\alpha_0^2L^K\|c\||k|_\infty$, that is $\frac{911}{12}\|c\||k|_\infty \leq 76\|c\||k|_\infty$ in the unit $12\mathsf{c}^\flat M^3\alpha_0^2L^K$ of the height 0. So $76\|c\||k|_\infty$ bounds the sizes at all the heights, being at least $34\|c\||k|_\infty$ at the heights $j \geq 1$

and $\frac{911}{12}\|c\||k|_\infty$ at the height 0. On the real configurations one factor of a monomial is at most $a_p + a_{p'}$, p, p' plaquettes of its lassos: a factor $\langle w, w'\rangle$ or $\det(w, w', w'')$, since $|w|^2 \le 2a_p$ and $|w| \le 1$, or a plaquette term itself; every other factor is at most 1, or 2 if it is a plaquette term; so $|\mathrm{Fm}^c(k)| \le \mathcal{O}(2\|c\||k|_\infty \mathbf{1}_{\mathrm{Pl}(k)})$, $\mathrm{Pl}(k)$ the plaquettes of the blocks of the sites with $k \ne 0$, and the set $\{k \ne 0\}$ carries all the representations: $\mathrm{Fm}^c(k)$ is a dominated local source with the comparison weight $\varpi \mathbf{1}_{\mathrm{Pl}(k)}$ for every $\varpi \ge 76\|c\||k|_\infty$.

If all the orders are ≥ 3, the factor $(5.04\,\alpha_0\eta_j^2)^{m_i-2} \le \max\{1, 5.04\,\alpha_0\}^{\mathrm{Ord}(c)-2} L^{-2j}$, $\mathrm{Ord}(c)$ the largest order, gives the fall L^{-2j} of the sizes; on the backgrounds of the level t the plaquette terms are below $\bar a = 2B_3^2\varepsilon_t^2 L^{-4t}$, ε_t the radius of regularity of the level t, a factor of the order $q \in \{2, 3\}$ is at most $(2\bar a)^{q/2}$, and the values fall like L^{-2t}. With the comparison weight $\Gamma_c^\circ\|c\||k|_\infty \mathbf{1}_{\mathrm{Pl}(k)}$ of the statement, $\Gamma_c^\circ = \max\{76,\, 34\max\{1, 5.04\,\alpha_0\}^{\mathrm{Ord}(c)-2},\, 2\max\{1, 2B_3\}^{\mathrm{Ord}(c)-2}\}$, this is the hypothesis of (c), with L^{-2j} and L^{-2t}. (e) The density of a source at the never-large history of the level t is its first derivative in (a); its oscillation is bounded by the size of the source at that level, which is y_2 up to a constant, and the mixed part is the second mixed derivative, which carries the factor of the distance from (a), with $\mathsf{D}_t \ge L^{s'}\mathsf{d}' \ge 3M$.

The second mixed derivatives are summed over the levels finer than t as in (b) at $p = 2$, $\frac{4}{3}L\,\mathsf{T}_2^\circ$ being the series T_2, and the condition $\mathsf{h}_2(s') \le 1$ closes that induction up to the level t. □

Lemma 5.7 (Rough entries)**.** A petal at a site x is a pair $\pi = (\alpha, \alpha')$ of signed directions with $|\alpha| \ne |\alpha'|$, $W_x(\pi)$ the holonomy from x of the plaquette that it spans, and $q_x(\pi, \pi') = \langle \mathbf{w}(W_x(\pi)), \mathbf{w}(W_x(\pi'))\rangle$. For real $k = (k_{\pi\pi'}(x))$ put $\mathrm{Pf}(k) = a^{-1}\sum_x \sum_{\pi,\pi'} k_{\pi\pi'}(x) q_x(\pi, \pi')$ and $|k|_1 = \sup_x \sum_{\pi,\pi'} |k_{\pi\pi'}(x)|$. A *rough entry* is $F = \mathrm{Pf}(k) + \mathcal{O}(v)$ with a real weight v; its norm is $\|F\| = 13|k|_1 + |v|_\infty$, and it is carried by the barycentres of the plaquettes of $\operatorname{supp} v$ and of those with a corner at which $k \ne 0$. Let $0 < g \le \gamma_W^{\mathrm{ref}}$, $\ell' \in \mathfrak{L}(g)$, $r \ge 2$ and $0 < \mathsf{d} < 1$. There are $K_0'(r, \mathsf{d})$ and $k'(r, \mathsf{d}) \le C_{r,p}\mathsf{d}^{-pr}$ $(p > 7)$ such that

$$|\langle F_1; \dots; F_r\rangle_{K,\ell',\beta}| \le k'(r, \mathsf{d}) \prod_a \|F_a\|$$

for all $(K, \beta) \in \mathfrak{D}(g)$ with $K \ge K_0'(r, \mathsf{d})$ and all rough entries whose carriers are pairwise $\ge \mathsf{d}$ apart. Moreover $\Theta_\beta[\eta] = \mathsf{b}\,\mathrm{Pf}(k^\eta)$ with $|k^\eta|_1 \le 12\max_\rho \mathrm{Lip}_\infty \eta^\rho$, so $\|\mathrm{Pf}(k^\eta)\| \le 156\max_\rho \mathrm{Lip}_\infty \eta^\rho$; $-\beta\sum_p h_{\omega,\chi}(p) = \mathsf{b}\,\mathrm{Pf}(k^h)$ with $|k^h|_1 \le 12|\omega|\,\mathrm{Lip}(\chi) R_1(\chi) \le 36\ell|\omega|\,\mathrm{Lip}_\infty \chi$; $\mathfrak{B}_{\omega,\chi} = \mathsf{b}\,\mathrm{Pf}(k^h) + \mathcal{O}(v^{\mathfrak{B}})$ with $|v^{\mathfrak{B}}|_\infty \le (2\mathsf{b} + 3a)|\omega|\,\mathrm{Lip}(\chi) R_1'(\chi)$, where $R_1'(\chi) \le \frac{3}{2}\ell$ is the number of Thm. N(a), the largest $|x(p)|_1$, taken in the chart $(-\frac{\ell}{2}, \frac{\ell}{2})^3$, at the plaquettes where $\nabla_\omega^s \chi$ or v_χ does not vanish (at their barycentres $x(p)$, χ is not constant on the points $x(p) \pm a e_\rho$, $x(p) \pm \frac{a}{2} e_\rho$, $\rho = 1, 2, 3$); the entries $\mathsf{b}\,\mathrm{Pf}(k^\eta)$, $\mathsf{b}\,\mathrm{Pf}(k^h)$ and $\mathfrak{B}_{\omega,\chi}$ are carried within $2a$ of $\operatorname{supp}\eta$, resp. $\mathrm{Var}(\chi)$; and $T_{lr}(\varphi) = \mathsf{b}\,(\mathrm{Pf}(k^{lr}) + \mathcal{O}(v^{lr}))$ with $13|k^{lr}|_1 + |v^{lr}|_\infty \le C_T \sup|\varphi|$, C_T an absolute constant. ■

Proof sketch. A rough entry is a dominated local source in the sense of Lem. 5.6: a gauge invariant polynomial with a local representation at every height, with $|F| \le \mathcal{O}(w)$ for a weight w of size $\|F\|$ on its carrier. Put $\sum_a \lambda_a F_a$ into the weighted progression (Lem. 5.6(a)). Its families are holomorphic in λ near 0 with sizes that depend on the $\sup|w_a|$ alone, and they are local: $\partial_B E_\lambda^{(i)}(X) = 0$ unless X meets every carrier in B, so that $d_i(X) \ge D_i/M - 2$, D_i being the least sup distance of these carriers in level-i units, and a factor $e^{-\kappa^+[D_i/M-2]_+}$ appears. We split the mean over the level t of the first creation of a large field. Up to that level the history is the never-large one, and

$$\tau_{h_t}\big[\rho_0 e^{\sum_a \lambda_a F_a}\big] = e^{D_t^\lambda(U_t(\cdot))}\,\mathsf{m}_t, \qquad \Phi\Big[\prod_{a\in B}(F_a - c_a)\Big] = \sum_\pi \prod_{\beta\in\pi} \kappa_\beta, \quad \kappa_\beta = \partial_\beta D_t^\lambda|_{\lambda=0}\ (\#\beta \ge 2), \quad \kappa_{\{a\}} = \partial_a D_t^\lambda|_{\lambda=0} - c_a,$$

with the signs of Lem. 5.6(a); each κ_β with $\#\beta \ge 2$ is bounded by the mixed derivatives above. Behind the level t only a positive weight enters, of the mass $\mathsf{w}_t \le 2M^{-3}\ell'^3 e^{B(X_g)\ell'^3} L^{3(K-t)} e^{-c' p_0(g_t)^2}$,

which is smaller than every power of $L^{-(K-t)}$, while the centred insertions carried to the level $t-1$ are at most powers of L^{K-t}. The mixed derivatives of the orders $\le r$ converge at the depth $s(\mathsf{d}'')$, $\mathsf{d}''=\min\{\frac{4}{5}\mathsf{d},L^{-\bar s(r)}\}$, where $\bar s(r)$ is the least integer $s\ge 3$ with $2\mathsf{T}_r(3)\theta(L^{s-2}g^{-2})\le 1$, with the series T_r and the size θ of Lem. 5.6(b); at every $s\ge\bar s(r)$ the condition holds, since θ does not increase.

Apply [7, Prop. 2.10](a′), (b) to the dominated local sources just constructed. Its summability condition is the one defining $\bar s(r)$ above, and it permits the fixed side ℓ': the local cumulant constant may be chosen with $k_r(g,s;\ell')\le C_r(g,\ell')Y_s^{8r\mathfrak{p}_0}$. Cutting every weight into at most $(2\ell'/\mathsf{d}'')^3$ pieces and using $\mathsf{b}^{-r}\le g^{2r}$ therefore gives $k'(r,\mathsf{d})\le C_r(g,\ell')g^{2r}(2\ell'/\mathsf{d}'')^{3r}Y^{8r\mathfrak{p}_0}_{s(\mathsf{d}'')}$. The same proposition bounds $Y_{s(\mathsf{d}'')}\le 2\log(1/\mathsf{d}'')+C(g)$; since $\mathsf{d}''=\min\{4\mathsf{d}/5,L^{-\bar s(r)}\}$, this proves the asserted $C_{r,p}\mathsf{d}^{-pr}$ bound for every $p>7$. The same decomposition, at the least depth s with $L^s\mathsf{d}\ge 3M$ at which the mixed derivatives of the order q are summable, gives the second bound of Thm. M(a): $L^s\le\max\{1,LM\mathsf{u}_q/\mathsf{d}\}$ with $\mathsf{u}_q=O(q\log q)$, the sums over partitions are at most $C^q(q!)^{3/2+\mathfrak{p}_0}$, and the rarity of a first creation, at which all the q insertions are bounded by their suprema, costs $\exp[c_L(4q+4)^2]$. The formulas for k^η, k^h and k^{lr} are the definitions written in petals: for $b=\langle x,x+ae_\mu\rangle$ and $p=p(b,\alpha'')$, $\mathfrak{d}_{b,p,\rho}=-\frac12[q_x((\mu,\rho),(\mu,\alpha''))-q_x((\mu,-\rho),(\mu,\alpha''))]$, and the shell lies in $|x|_1\le\frac12+\frac32\ell$.

For the whole flux insertion, $\mathfrak{B}_{\omega,\chi}-\mathsf{b}\,\mathrm{Pf}(k^h)=\mathcal{O}(v^{\mathfrak{B}})$ with $v^{\mathfrak{B}}(p)=\mathsf{b}\,\nabla^s_\omega\chi(x(p))-\frac32 a\,v_\chi(p)$; as $\sum_\rho|(\omega y)_\rho|\le 2|\omega||y|_1$ and the difference quotients of χ in $\nabla^s_\omega\chi$ and in v_χ are at most $\mathrm{Lip}(\chi)$, $|v^{\mathfrak{B}}(p)|\le(2\mathsf{b}+3a)|\omega|\,\mathrm{Lip}(\chi)|x(p)|_1$, and $v^{\mathfrak{B}}(p)\ne 0$ only at the plaquettes where $\nabla^s_\omega\chi$ or v_χ does not vanish, at whose barycentres χ is not constant on the points $x(p)\pm ae_\rho$, $x(p)\pm\frac a2 e_\rho$: this gives the factor $R_1'(\chi)$ of Thm. N(a). □

Lemma 5.8 (Chessboard bound)**.** Let $0<g\le\gamma$, $\ell'\in\mathfrak{L}(g)$, $(K,\beta)\in\mathfrak{D}(g)$, and let a board of cubic cells of side l' dividing $\frac{\ell'}{2}$ be placed on $T_{\ell'}$, on every T_ℓ with $\ell\ge\ell'$ in $\mathfrak{L}(g)$ and on $a_K\mathbb{Z}^3$; $N'=(\ell'/\mathsf{l}')^3$. For a bounded Borel function F of the bonds inside one cell put $\|F\|_{\mathrm{ch}}=|\langle\prod_\kappa F^{[\kappa]}\rangle_{K,\ell',\beta}|^{1/N'}$, the product running over the mirror copies of F, conjugated on odd reflection parity, in the N' cells of $T_{\ell'}$. Then

$$|s(F_1\cdots F_q)|\le\prod_i e^{G_g\mathsf{l}'^3}\|F_i\|_{\mathrm{ch}}$$

for functions $F_1,\dots,F_q$ of distinct cells, s being a periodic Gibbs state at (K,β) or the measure of T_ℓ, $\ell\in\mathfrak{L}(g)$, $\ell\ge\ell'$. In particular, for real F_1,F_2 of two distinct cells and real c_1,c_2: $|s(F_1)-c_1|\le e^{G_g\mathsf{l}'^3}\|F_1-c_1\|_{\mathrm{ch}}$ and $|s(A^s[F_1]A^s[F_2])|\le 2e^{2G_g\mathsf{l}'^3}\|F_1-c_1\|_{\mathrm{ch}}\|F_2-c_2\|_{\mathrm{ch}}$. ■

Proof sketch. On T_ℓ, $\ell\ge\ell'$, reflection positivity at the walls gives the chessboard estimate $|\langle F_1\cdots F_q\rangle_{K,\ell,\beta}|\le\prod_i\langle\prod_\kappa F_i^{[\kappa]}\rangle^{1/N_\ell}_{K,\ell,\beta}$, $N_\ell=(\ell/\mathsf{l}')^3$. The product of the mirror copies over all the cells of T_ℓ is a periodisation of the product over the cells of $T_{\ell'}$, and the chessboard seminorms on the two sides compare by the inequality $\langle\prod_\kappa F^{[\kappa]}\rangle^{1/N_\ell}_{K,\ell,\beta}\le\langle\prod_\kappa F^{[\kappa]}\rangle^{1/N'}_{K,\ell',\beta}e^{\mathsf{l}'^3(p_{\ell'}-p_\ell)}$ ([7, Cor. B.1](e)). It is reflection positivity at the walls applied once more, to the periodised product: the pattern of the copies has the period $2\mathsf{l}'$, which divides ℓ', so that a slab of that length is a half slab and its mirror image, the transfer operator across it is positive, and $\mathrm{Tr}\,T^n\le(\mathrm{Tr}\,T)^n$ in each of the three directions; this bounds the unnormalised N_ℓ-th root by the unnormalised N'-th root, and the ratio $Z^{1/N'}_{K,\ell',\beta}/Z^{1/N_\ell}_{K,\ell,\beta}$ of the plain partition functions per cell is $e^{\mathsf{l}'^3(p_{\ell'}-p_\ell)}\le e^{G_g\mathsf{l}'^3}$, the pressures $p_\ell=\ell^{-3}\log Z_{K,\ell,\beta}$ being formed with the plain partition functions (Section YM3_I:86).

A periodic Gibbs state is dominated by the torus measures through its conditional laws (Prop. YM3_I:91.9), and this domination carries the bound to every periodic Gibbs state. The last two bounds are the cases $q=1$ and $q=2$, with $s(A^s[F_1]A^s[F_2])=s((F_1-c_1)(F_2-c_2))-(s(F_1)-c_1)(s(F_2)-c_2)$. □

Lemma 5.9 (Exact identities)**.**

(a) Let $\mathbf{X}$ be a polynomial field of directions that vanishes off a finite set Λ of bonds: at each positively oriented bond b a $\mathfrak{g}$-valued polynomial $\mathbf{X}(b)$ in the plaquette variables, 0 for $b \notin \Lambda$; let $v_{\mathbf{X}}F = \sum_b \sum_{\mathsf{a}} \langle \sigma_{\mathsf{a}}, \mathbf{X}(b)\rangle \partial_b^{\mathsf{a}} F$ and $\operatorname{div} v_{\mathbf{X}} = \sum_b \sum_{\mathsf{a}} \partial_b^{\mathsf{a}} \langle \sigma_{\mathsf{a}}, \mathbf{X}(b)\rangle$, $\sigma_1, \sigma_2, \sigma_3$ the Pauli matrices and ∂_b^{a} the derivative along $e^{i\epsilon\sigma_{\mathsf{a}}}U(b)$ at $\epsilon = 0$. Then $\nu((\beta v_{\mathbf{X}} A - \operatorname{div} v_{\mathbf{X}})Y) = 0$ for every Gibbs state or torus measure ν at (a_K, β) and every bounded Borel Y that does not depend on the bonds of Λ; so every joint cumulant under ν with the entry $\beta v_{\mathbf{X}} A - \operatorname{div} v_{\mathbf{X}}$ and further entries not depending on Λ vanishes. $N_\beta[\eta] = \Theta_\beta[\eta] + \mathfrak{K}^{\mathrm{v}}[\beta\eta] - \mathsf{b}\,\mathcal{O}(\operatorname{div}^s \eta) + \frac{3}{2} a\,\mathcal{O}(e^\eta)$ is such a null term.

With the contact terms, for all weights f_i and every vector field η, $\langle N_\beta[\eta]; \mathcal{O}(f_\cdot)\rangle + \sum_i \langle \mathrm{Ct}[f_i; \eta]; (\mathcal{O}(f_l))_{l \neq i}\rangle = 0$, where $\mathrm{Ct}[f; \eta] = \mathcal{O}(\operatorname{div}^s(f\eta)) - a^{-1}\Theta_f[\eta] - a^{-1}\mathfrak{K}^{\mathrm{v}}[f\eta]$.

(b) Identically in U, $\mathfrak{K}^{\mathrm{v}}[\eta] = -\sum_p \sum_\rho (\nabla_\rho^+ \eta^\rho)(x(p)) \Psi_\rho(p) + \mathfrak{K}^\sharp[\eta]$; for $\rho \in P(p)$, $\mathfrak{r}_\rho^\sharp(p) = 0$ and $-(a_p + a_{p+ae_\rho}) \le \Psi_\rho(p) \le 0$; $\mathfrak{r}^\sharp$ transforms as a vector under the coordinate mirrors of the lattice: $\mathfrak{r}_\rho^\sharp$ is odd under the mirror normal to ρ and even under the two mirrors that contain e_ρ, so that it is odd, for every ρ, under the inversion $y \mapsto 2z - y$ through a site z, the product of the three mirrors through z. Hence $\langle \mathfrak{K}^\sharp[\eta]\rangle_{K,\ell,\beta} = 0$, and by translation invariance also $\langle \mathfrak{K}^{\mathrm{v}}[\eta]\rangle_{K,\ell,\beta} = 0$; and if a field ζ has $\zeta^\rho(y + se_\rho) = \zeta^\rho(y)$ for all y, s and ρ, then $\langle \mathfrak{K}^\sharp[\zeta]; \mathcal{O}(f)\rangle_{K,\ell,\beta} = 0$ for every weight f.

A *block form* is a combination $c = \sum_i \kappa_i m_i$ of monomials in the lassos of the block $\mathrm{Bk}(x) = x + a\{-1, 0, 1\}^3$ of 27 sites centred at a site x: a lasso is a path of at most three steps from x inside the block to a corner y of a plaquette of the block, the holonomy of that plaquette from y, and the path back; with $w = \mathbf{w}(W)$ and $a_{\mathrm{lz}} = 1 - \frac{1}{2}\operatorname{tr} W$ the vector part and the plaquette term of the holonomy W of a lasso, a monomial is a product of n_2 factors $\langle w, w'\rangle$, n_3 factors $\det(w, w', w'')$ and n_a factors a_{lz}, and $2n_2 + 3n_3 + 2n_a$ is its order in the field strength. The same lassos are taken at every site, c_x is the form at the block of x, the order of c is the least order of its monomials, and its norm is the number $\|c\| = \sum_i |\kappa_i| 2^{n_a(m_i)}$ of Lem. 5.6(d).

For a real function k of the sites, $\mathrm{Fm}^c(k) = a^{-1}\sum_x k(x) c_x(U)$ is a dominated local source of size $\le 76\|c\||k|_\infty$, carried by $\{k \neq 0\}$, with $|\mathrm{Fm}^c(k)| \le \mathcal{O}(2\|c\||k|_\infty \mathbf{1}_{\mathrm{Pl}(k)})$, $\mathrm{Pl}(k)$ the plaquettes of the blocks of the sites with $k \neq 0$; and if all the monomials of c have order ≥ 3, as for the commutator terms, the defects of the linear Bianchi identity, a_p^2 and $a_p a_{p_+}$, its sizes fall like L^{-2j} with the level j and its values on the backgrounds like L^{-2t} with the level t, with the comparison weight $\Gamma_c^\circ \|c\||k|_\infty \mathbf{1}_{\mathrm{Pl}(k)}$ of Lem. 5.6(d). $\Psi_{\lambda(p)}(p)$ is a block form of order 2 and norm 1, and $\mathfrak{r}_\rho^\sharp(p)$, $\rho = \lambda(p)$, a block form of order 2 and norm ≤ 8, both at the block of the lowest corner of p; so for a real weight k the sums $a^{-1}\sum_p k(p)\Psi_{\lambda(p)}(p)$ and $a^{-1}\sum_p k(p)\mathfrak{r}_{\lambda(p)}^\sharp(p)$ over the plaquettes of one normal are sources $\mathrm{Fm}^c(\tilde{k})$, $\tilde{k}(x)$ the value of k at the plaquette of that normal with the lowest corner x, and they are dominated local sources of the sizes $\le 76|k|_\infty$ and $\le 608|k|_\infty$; as a function $|\mathfrak{r}_\rho^\sharp(p)| \le C\bar{a}^{3/2}$, $\bar{a}$ the largest a_q near p, and $\beta\mathfrak{K}^\sharp$ is treated by parity (Lems. 5.10, 5.11).

(c) For a vector field k, $\beta\sum_{p,\rho} \mathsf{k}^\rho(x(p))\Psi_\rho(p) = -\frac{1}{4}\mathfrak{N}[\mathsf{k}] + \sum_{v=1}^5 \mathfrak{J}_v[\mathsf{k}] - \frac{1}{4}c_{\mathsf{k}}$, where $\mathfrak{N}[\mathsf{k}] = \beta v_{\mathbf{X}^{\mathsf{k}}} A - \operatorname{div} v_{\mathbf{X}^{\mathsf{k}}}$ is the null term of (a) for the field of directions $\mathbf{X}^{\mathsf{k}}(b) = \sum_{\rho \neq \mu_b} \mathsf{k}^\rho(x_b) G_\rho(b)$, $c_{\mathsf{k}} = 6\sum_b \sum_{\rho \neq \mu_b} \mathsf{k}^\rho(x_b)$ is a number, and $\mathfrak{J}_1, \dots, \mathfrak{J}_5$ are sums of local densities of the orders $2, 2, 2, 3, 4$ with weights at most $3|\mathsf{k}|_\infty + \frac{3}{2}\mathsf{b}\mathrm{Lip}_\infty \mathsf{k}$, $\frac{1}{8}\mathsf{b}\mathrm{Lip}_\infty \mathsf{k}$, $\frac{1}{2}\mathsf{b}\mathrm{Lip}_\infty \mathsf{k}$, $\frac{1}{4}\beta|\mathsf{k}|_\infty$ and $\beta|\mathsf{k}|_\infty$, vanishing beyond the sup distance a of $\operatorname{supp}\mathsf{k}$. Consequently, with $\mathsf{k}_\eta^\rho = \nabla_\rho^+ \eta^\rho$, $\mathrm{St}_K[\eta] = N_\beta[\eta] - \frac{1}{4}\mathfrak{N}[\mathsf{k}_\eta] - \mathrm{Rm}_K[\eta] - \frac{1}{4}c_{\mathsf{k}_\eta}$, where $\mathrm{Rm}_K[\eta] = \beta\mathfrak{K}^\sharp[\eta] - \sum_v \mathfrak{J}_v[\mathsf{k}_\eta] + \frac{3}{2} a\,\mathcal{O}(e^\eta)$ is the remainder of the stress insertion.

(d) $\Delta_K[\eta] = F^0 + F^1$, rough entries carried within $2a$ of $\operatorname{supp}\eta$, with $\|F^0\| \le 286\mathsf{b}\sup|\nabla\eta|$ and $\|F^1\| \le 780\mathsf{b}\,a\sup|\nabla^2\eta|$; F^0 is null: its classical density, a quadratic form in the field strength with coefficients linear in $\nabla\eta$, vanishes identically.

(e) For a positively oriented bond b and directions λ,ρ: $\sum_{p\ni b} a^{-1}(x_b - x(p))_\lambda \mathfrak{d}_{b,p,\rho} = \langle \mathbf{u}_{b,\lambda}, \mathbf{u}_{b,\rho}\rangle$, so that $\sum_p \mathfrak{s}_{\lambda\rho}(p) = \sum_b \langle \mathbf{u}_{b,\lambda}, \mathbf{u}_{b,\rho}\rangle$ is a symmetric positive semidefinite matrix; for $\lambda \ne \mu_b$ and $p_\pm = p(b, \pm\lambda)$, $|\mathbf{u}_{b,\lambda}|^2 = (a_{p_+} + a_{p_-}) - \frac14(a_{p_+}^2 + a_{p_-}^2) + \Psi_\lambda(p_-)$ and $\Psi_\lambda(p_-) = -\frac14|G_\lambda(b)|^2 - \frac14(a_{p_+}^2 + a_{p_-}^2)$; for $\nu \in P(p)$, $\mathfrak{s}_{\nu\nu}(p) = \frac32 a_p + \frac14(a_{p+ae_\nu} + a_{p-ae_\nu}) - \frac12 a_p^2 + \frac12(\Psi_\nu(p) + \Psi_\nu(p-ae_\nu))$, and $\mathfrak{s}_{\lambda\rho}(p) = 0$ if $\lambda = \lambda(p)$. Consequently, identically in U,

$$T_{mm}(\varphi) - \mathsf{b}\check{D}_m(\varphi) = \beta \sum_{p:\, m\in P(p)} \varphi(x(p))\Big[\tfrac12\big(\Psi_m(p) + \Psi_m(p - ae_m)\big) - \tfrac12 a_p^2\Big] + \mathsf{b}\,\mathcal{O}(\tilde\varphi_m),$$

$\tilde\varphi_m(p) = \frac14[\varphi(x(p) + ae_m) - 2\varphi(x(p)) + \varphi(x(p) - ae_m)]\mathbf{1}\{m \in P(p)\}$, so $|\tilde\varphi_m| \le \frac12 a \operatorname{Lip}\varphi$; and $\sum_m T_{mm}(\varphi) - \mathsf{b}\,\mathcal{O}(\varphi) = \sum_m (T_{mm}(\varphi) - \mathsf{b}\check{D}_m(\varphi))$.

(f) *The null term against sources that meet its carrier.* For a complex vector field k and a complex function f on T_ℓ put $\mathsf{e}^{\mathsf{k}}(p) = \sum_{\rho\in P(p)}[\mathsf{k}^\rho(x(p) + \frac{a}{2}e_\rho) + \mathsf{k}^\rho(x(p) - \frac{a}{2}e_\rho)]$ and

$$\mathfrak{C}'[\mathsf{k}; f] = \sum_p \sum_{b\subset\partial p} \sum_{\rho\ne\mu_b} a^{-1}(f(x(p)) - f(x_b))\mathsf{k}^\rho(x_b)\,\langle G_\rho(b), \mathbf{w}(W_{b,p})\rangle, \qquad \mathfrak{R}[\mathsf{k}; f] = \mathsf{b}^{-1}\operatorname{div} v_{\mathbf{X}^{f\mathsf{k}}} + \mathfrak{C}'[\mathsf{k}; f],$$

with $\mathbf{X}^{\mathsf{k}}$ and $\mathfrak{N}[\mathsf{k}]$ as in (c). Then: (N1) $v_{\mathbf{X}^{\mathsf{k}}}\mathcal{O}(f) = \mathsf{b}^{-1}\mathfrak{N}[f\mathsf{k}] + \mathfrak{R}[\mathsf{k}; f]$, identically in U; (N2) $\int \mathfrak{N}[\mathsf{k}]\, e^{\mathcal{O}(w)}e^{-\beta A}\,\mathrm{d}U = \int (v_{\mathbf{X}^{\mathsf{k}}}\mathcal{O}(w))e^{\mathcal{O}(w)}e^{-\beta A}\,\mathrm{d}U$ for every complex function w on T_ℓ; (N3) for a finite set A and complex functions f_l, $l \in \mathsf{A}$,

$$\big\langle \mathfrak{N}[\mathsf{k}]; (\mathcal{O}(f_l))_{l\in\mathsf{A}}\big\rangle_{K,\ell,\beta} = \sum_{j=1}^{\#\mathsf{A}} \sum_{(q_1,\dots,q_j)} \mathsf{b}^{-(j-1)}\Big\langle \mathfrak{R}\big[f_{q_1}\cdots f_{q_{j-1}}\mathsf{k}; f_{q_j}\big]; (\mathcal{O}(f_l))_{l\in\mathsf{A}\setminus\{q_1,\dots,q_j\}}\Big\rangle_{K,\ell,\beta},$$

the inner sum over the sequences of j distinct elements of A (for $\mathsf{A} = \emptyset$ the left member is the mean, and it is 0); (N4) $\mathfrak{R}[\mathsf{k}; f] = \mathsf{b}^{-1}c_{f\mathsf{k}} - 3\mathsf{b}^{-1}\sum_p a_p\,\mathsf{e}^{f\mathsf{k}}(p) + \mathfrak{C}'[\mathsf{k}; f]$ is a number, the density a_p with weights $\le 12\,\mathsf{b}^{-1}\sup|f|\sup|\mathsf{k}|$ and the densities $\langle G_\rho(b), \mathbf{w}(W_{b,p})\rangle$ of the order 2 with weights $\le \frac12 \mathrm{Lip}_\infty f\,\sup|\mathsf{k}|$: no factor β enters. Its weight at a plaquette p vanishes unless a bond b of p has $\mathsf{k}(x_b) \ne 0$ and $f \ne 0$ at x_b or at $x(p)$, so that its densities are carried within the sup distance $2a$ of $\operatorname{supp}\mathsf{k}$ and of $\operatorname{supp} f$, and the number vanishes unless $\mathsf{k}(x_b)f(x_b) \ne 0$ at some bond.

Consequently, if the spread D of $\operatorname{supp}\mathsf{k}$ and the $\operatorname{supp} f_l$, $l \in \mathsf{A}$, is $> 4a$, the terms of (N3) with $j = \#\mathsf{A}$ vanish, the numbers drop from the other terms, and in each of them the entry is carried by a set whose spread with the supports of the remaining f_l is $\ge D - 2a$. This rests on the merging of sets: if $S_0, \dots, S_n$ have the spread D, $Q \subset \{1,\dots,n\}$, $2r < D$, and T lies within the sup distance r of S_0 and of every S_q, $q \in Q$, then $T = \emptyset$ if $Q = \{1,\dots,n\}$, and otherwise the spread of T and the S_l, $l \notin Q$, is $\ge D - r$.

■

Proof sketch. (a) The Haar measure is invariant; with $\partial_b^{\mathsf{a}}\langle \sigma_{\mathsf{a}}, \mathbf{w}(W_{b,p})\rangle = 1 - a_p$, an integration by parts in each bond variable against $e^{-\beta A}$ gives $\nu((\beta v_{\mathbf{X}}A - \operatorname{div} v_{\mathbf{X}})Y) = 0$ in finite volume, and in a Gibbs state through its conditional laws. With sources the same integration by parts against $e^{-A(\beta - a^{-1}w_z,\cdot)}$, $w_z = \sum_i z_i f_i$, followed by the mixed derivative in z at 0, gives the contact terms, since $v_\eta A(v,\cdot)$ is linear in the weight v. (b) For $q_\pm = p(b, \pm\rho)$ in one plane, $\langle \mathbf{w}(W_{b,q_-}), \mathbf{w}(W_{b,q_+})\rangle = -2\Psi_\rho(q_-) - a_{q_-} - a_{q_+}$; so $\mathfrak{r}_\rho(p) = \Psi_\rho(p) - \Psi_\rho(p_-)$ in the plane, and the split is a summation by parts.

Under the mirror r normal to ρ, $\mathfrak{d}_{\mathsf{r}b,\mathsf{r}p,\rho}(\mathsf{r}U) = -\mathfrak{d}_{b,p,\rho}(U)$ and $\Psi_\rho(\mathsf{r}p)(\mathsf{r}U) = \Psi_\rho(p_-)(U)$, and the measure of a torus is invariant under r. Under a coordinate mirror that contains e_ρ these quantities are carried to the same quantities at the image bond and plaquette: the holonomy $W_{b,p}$ is taken from the initial point of b along b, and where the mirror reverses the orientation of b both factors of $\mathfrak{d}_{b,p,\rho}$ change their sign together; so $\mathfrak{r}^\sharp_\rho$ is even under it, and the three coordinate mirrors through a site compose to the inversion through it.

For ζ: $\langle \mathfrak{K}^\sharp[\zeta]; \mathcal{O}(f)\rangle = a^{-1}\sum_{p'} f(p')\langle \mathfrak{K}^\sharp[\zeta]; a_{p'}\rangle$, and for each p' and ρ the mirror normal to ρ through $x(p')$ keeps $a_{p'}$, the measure and ζ^ρ, carries the plaquettes normal to ρ among themselves, and changes the sign of $\mathfrak{r}^\sharp_\rho$ on them. A block form has local representations at every height of size $76\|c\|\|k\|_\infty$ (the proof of Lem. 5.6(d), with the lassos from the centre of the block, which reach its 27 sites in at most three steps); the field strength is $O(L^{-2j})$ in level-j units on the backgrounds, which gives the fall for monomials of order ≥ 3, with the comparison weight $\Gamma_c^\circ\|c\|\|k\|_\infty \mathbf{1}_{\mathrm{Pl}(k)}$ of Lem. 5.6(d). (c) Take in (a) $\mathbf{X}(b) = \sum_\rho \mathsf{k}^\rho(x_b)[\mathbf{w}(W_{b,p(b,\rho)}) + \mathbf{w}(W_{b,p(b,-\rho)})]$; then $v_{\mathbf{X}} a_p = \sum_{b\subset\partial p}\langle \mathbf{X}(b), \mathbf{w}(W_{b,p})\rangle$, $\operatorname{div} v_{\mathbf{X}}$ is the number c_{k} minus $3\sum_p a_p$ times a weight formed from k, and the in-plane remainder is $-2a_p^2$, of order 4. (d) At a constant field strength, with $G = \nabla\eta$ and $T_{lr} = \operatorname{tr} B_l B_r$, the classical density of $\Delta_K[\eta]$ is $\mathsf{b}[\operatorname{tr}G \operatorname{tr}T - \operatorname{Tr} GT] - \frac{\mathsf{b}}{2}\operatorname{tr}G \operatorname{tr}T - \frac{\mathsf{b}}{2}\operatorname{Tr}(H^{\mathrm{cons}}T) = 0$, since $\operatorname{Tr}(H^{\mathrm{cons}}T) = \operatorname{tr}G \operatorname{tr}T - 2\operatorname{Tr} GT$; the Taylor remainder of η gives F^1. (e) $\sum_{p\ni b} a^{-1}(x_b - x(p))_\lambda \mathbf{w}(W_{b,p}) = \mathbf{u}_{b,\lambda}$ and the form is bilinear; with $w_\pm = \mathbf{w}(W_{b,p_\pm})$, $|w_\pm|^2 = a_{p_\pm}(2 - a_{p_\pm})$ and $\Psi_\lambda(p_-) = -\frac14|w_+ + w_-|^2 - \frac14(a_{p_-}^2 + a_{p_+}^2)$, and the parallelogram law $4|\mathbf{u}_{b,\lambda}|^2 = |w_+ - w_-|^2 = 2|w_+|^2 + 2|w_-|^2 - |w_+ + w_-|^2$ gives the first identity; the formula for $\mathfrak{s}_{\nu\nu}$ is the same computation at the two bonds of p of the direction other than ν, and no vector $x_b - x(p)$ has a component normal to p.

Inserting these in $T_{mm}(\varphi) = \beta\sum_p \varphi(x(p))[\mathfrak{s}_{mm}(p) - a_p]$ and $\mathsf{b}\check{D}_m(\varphi) = \beta\sum_p \varphi(x(p))a_p(1 - 2\cdot \mathbf{1}\{\lambda(p) = m\})$, the plaquettes normal to m cancel, and moving the shifts $a_{p\pm ae_m}$ onto φ gives $\tilde\varphi_m$. Finally $\sum_m \check{D}_m = 3\mathcal{O} - 2\mathcal{O} = \mathcal{O}$. (f) $\mathcal{O}(f) = a^{-1}\sum_p f(x(p))a_p$ and $v_{\mathbf{X}}a_p = \sum_{b\subset\partial p}\langle \mathbf{X}(b), \mathbf{w}(W_{b,p})\rangle$; with $f(x(p)) = f(x_b) + (f(x(p)) - f(x_b))$ and $f(x_b)\mathbf{X}^{\mathsf{k}}(b) = \mathbf{X}^{f\mathsf{k}}(b)$, the first part is $a^{-1}v_{\mathbf{X}^{f\mathsf{k}}}A = \mathsf{b}^{-1}\beta v_{\mathbf{X}^{f\mathsf{k}}}A = \mathsf{b}^{-1}(\mathfrak{N}[f\mathsf{k}] + \operatorname{div} v_{\mathbf{X}^{f\mathsf{k}}})$ and the second is $\mathfrak{C}'[\mathsf{k}; f]$: this is (N1). (N2) is the integration by parts of (a) along $\mathbf{X}^{\mathsf{k}}$ against $e^{\mathcal{O}(w)}e^{-\beta A}$.

At the source $w_z = \sum_{l\in\mathsf{A}} z_l f_l$, (N2) says that the mean of $\mathfrak{N}[\mathsf{k}] - \sum_q z_q v_{\mathbf{X}^{\mathsf{k}}}\mathcal{O}(f_q)$ under the measure with the source vanishes, and this function is affine in z; the mixed derivative in z at 0 gives $\langle \mathfrak{N}[\mathsf{k}]; (\mathcal{O}(f_l))_{l\in\mathsf{A}}\rangle = \sum_{q\in\mathsf{A}}\langle v_{\mathbf{X}^{\mathsf{k}}}\mathcal{O}(f_q); (\mathcal{O}(f_l))_{l\neq q}\rangle$, and inserting (N1), whose first part is a null term of the same kind with $f_q\mathsf{k}$ for k, an induction on $\#\mathsf{A}$ gives (N3). (N4): $\operatorname{div} v_{\mathbf{X}^{\mathsf{k}}} = c_{\mathsf{k}} - 3\sum_p a_p\, \mathsf{e}^{\mathsf{k}}(p)$, by $\partial_b^{\mathsf{a}}\langle \sigma_{\mathsf{a}}, \mathbf{w}(W_{b,p})\rangle = 1 - a_p$; $|\mathsf{e}^{f\mathsf{k}}| \leq 4\sup|f\mathsf{k}|$, $|x(p) - x_b|_\infty = \frac{a}{2}$, and the boxes of the densities lie within the sup distance $\frac{3}{2}a$ of $x(p)$.

The merging: if $Q = \{1, \dots, n\}$, points of $S_0, \dots, S_n$ within r of one point of T are pairwise at most $2r < D$ apart, which the spread forbids; otherwise, for $t \in T$, $y_l \in S_l$ ($l \notin Q$) and points $y_0 \in S_0$, $y_q \in S_q$ within r of t, some pair of the family $y_0, (y_q), (y_l)$ is at least D apart, it is no pair among $y_0, (y_q)$, and so two of the y_l, or one y_l and t, are at least $D - r$ apart. □

Lemma 5.10 (The chain with an entry; the cubic insertion)**.** Let $\gamma^{-2} \leq \mathsf{b}_- \leq \mathsf{b}_+$, $MR(\mathsf{b}_+) \leq L^m$, $\ell = 2L^m$, $0 < \tau \leq \frac14$, $0 < \phi < \frac{1}{42}$, $u = \lceil \phi K\rceil$, $t = K - u$ and $c_0' = \min\{\frac{1}{14} - \frac{\tau}{7}, \frac{1}{14} - 3\phi\}$. Let K_ϕ be the least integer $K \geq 2$ with $(1-\phi)K \geq 2$, $L^{\phi K}\gamma^{-2} \geq \mathsf{t}_\natural$ and $\phi K > 16\mathfrak{p}_0^2 c_{\log}/(\log L)^2$, $\mathsf{t}_\natural$ being the calibration of §2.3.1 and $c_{\log} = \log L/\log\gamma^{-2}$; it depends on the constants and on ϕ alone. For $K \geq K_\phi$ and $\mathsf{b}' \in [\mathsf{b}_-, \mathsf{b}_+]$ let $\rho_0 dU$ be the measure of T_ℓ at $\beta = L^K\mathsf{b}'$; the data (K, b', m) with the top level t form the setting (Th) (§2.3.1), and the small-field chain is formed along the couplings $x_j = L^{K-j}\mathsf{b}'$ up to the level t. Let $\mathcal{M}^k$ be the block map to the level k, Cut_k the field cut and $\mathrm{In}_k \subset \mathrm{Cut}_k$ the inner ball of the level k (the definitions preceding Lem. YM3_I:113.10), $\mathfrak{G}_k[x_k] \geq 0$ the gauge-fixing factor and $\chi^{\mathrm{sf}}_{x_k} \in \{0, 1\}$ the fluctuation test of the step from the level k ([7, §B.2]),

and

$$\mathsf{p}_k(U) = \mathbf{1}_{\mathrm{Cut}_k}(\mathcal{M}^k U) \prod_{j<k} [\mathbf{1}_{\mathrm{Cut}_j}\, \mathfrak{G}_j[x_j]\, \chi^{\mathrm{sf}}_{x_j}](\mathcal{M}^j U),$$

the weight of the small-field chain up to the level k: a bounded Borel function with values in $[0,\infty)$, with $\mathsf{p}_0 = \mathbf{1}_{\mathrm{Cut}_0}$ and $\mathsf{p}_{k+1} = (\mathbf{1}_{\mathrm{Cut}_{k+1}} \circ \mathcal{M}^{k+1})\, ((\mathfrak{G}_k[x_k]\chi^{\mathrm{sf}}_{x_k}) \circ \mathcal{M}^k)\, \mathsf{p}_k$; and $(\mathbf{1}_{\mathrm{In}_t} \circ \mathcal{M}^t)\, \mathsf{p}_t$ is the weight $\mathfrak{w}^{\mathrm{sf}}_t$ of the small-field chain of the definitions preceding Prop. YM3_I:115.2. Let $\nu_k = (\mathcal{M}^k)_*(\mathsf{p}_k \rho_0 dU)$, the measure of the level k of the chain among the construction results at the source 0 (Prop. YM3_I:114.2 and the proof of Prop. YM3_I:115.2), a finite positive measure with $\nu_{k+1} = \mathbf{1}_{\mathrm{Cut}_{k+1}} \mathbf{T}(\mathfrak{G}_k[x_k]\chi^{\mathrm{sf}}_{x_k} \nu_k)$, $\mathbf{T}$ the image under the block averaging; let $Q_k h$ be the density of $\mathbf{1}_{\mathrm{Cut}_{k+1}} \mathbf{T}(\mathfrak{G}_k[x_k]\chi^{\mathrm{sf}}_{x_k} h\, \nu_k)$ with respect to ν_{k+1}, the mean of h over the range Rg_{x_k} of the chart of the fibre of the step ([7, Lem. B.3]), and $Q_{t\leftarrow 0} = Q_{t-1} \cdots Q_0$, so that $(\mathcal{M}^t)_*(\mathsf{p}_t Y \rho_0 dU) = (Q_{t\leftarrow 0} Y)\, \nu_t$ for bounded Borel Y: $(Q_{t\leftarrow 0} Y) \circ \mathcal{M}^t$ is the conditional mean of Y given $\mathcal{M}^t U$ under $\mathsf{p}_t \rho_0 dU$.

An *entry of the type* $(m_J, c, \hat{c}, N)$ is a finite family $J = (\mathsf{Q}_\iota, j_\iota, \varkappa_\iota)_\iota$ of boxes $\mathsf{Q}_\iota = \{x + a\sum_\mu n_\mu e_\mu : 0 \le n_\mu \le r_\mu\}$, $(r_1, r_2, r_3) \in \{0,1,2\}^3$ with at least two $r_\mu \ge 1$, x the lowest site of the box, of densities j_ι on them and of complex weights $\varkappa_\iota$, at most N indices having one lowest site. A density on a box is a polynomial in the entries of the complex configurations $\mathbf{U}(b) \in \mathrm{SL}(2,\mathbb{C})$ at its bonds, $\mathbf{U}(\bar{b}) = \mathbf{U}(b)^{-1}$, invariant under the gauge transformations of the box; it has the height $\hat{c}$ if $|j| \le \hat{c}$ wherever $|\mathbf{U}(b)| \le 2$ at all the bonds, and the order m_J with the constant c if $|j| \le c\, \epsilon^{m_J}$ wherever, besides, $|\partial \mathbf{U}(q) - \mathbf{1}| \le \epsilon$ at all the plaquettes q of the box, $0 < \epsilon \le 1$.

Zero insertions and zero scalar sources are omitted before inverse norms and source radii are formed; the multilinear estimates are immediate when an insertion vanishes. We write $J(U) = \sum_\iota \varkappa_\iota j_\iota(U|_{\mathsf{Q}_\iota})$ and $|\varkappa|_\infty = \max_\iota |\varkappa_\iota|$; the carrier of J is the set of the sites of the boxes with $\varkappa_\iota \ne 0$.

(a) For $K \ge K_\phi$, an entry of the type $(m_J, c, \hat{c}, N)$ with $m_J \ge 2$, $|s| < r_J$ with $r_J^{-1} \le C|\varkappa|_\infty$, and $1 \le k \le t$:

$$(\mathcal{M}^k)_*(\mathsf{p}_k e^{sJ} \rho_0 dU) = \exp\Big[\sum_{i\le k} c_i(s)\Big] \mathbf{1}_{\mathrm{Cut}_k} e^{\Phi^s_k}\, \mathrm{d}V,$$

the numbers $c_i(s)$ and the action Φ^s_k being holomorphic in s and sums of local families, each of which depends on $\varkappa$ in its domain only. $Q_{t\leftarrow 0} J$ is a number plus $\partial_s \Phi^s_t|_{s=0}$ ν_t-almost everywhere on Cut_t, and its oscillation is at most $CK^{2\mathfrak{p}_0} L^{4u-K} |\varkappa|_\infty$ for $m_J = 2$ and $CK^{3\mathfrak{p}_0} L^{4u-\frac{3}{2}K} |\varkappa|_\infty$ for $m_J \ge 3$. Here C depends on the constants, ℓ, b_+, ϕ and the type of the entry.

(b) There are $K_2 \ge K_\phi$, C_2, p_2, depending on these data alone, such that for $K \ge K_2$ and every $\mathsf{b}' \in [\mathsf{b}_-, \mathsf{b}_+]$ the following holds. For every field Ξ there is a number a with $|Q_{t\leftarrow 0}(\beta \mathfrak{K}^\sharp[\Xi]) - \mathsf{a}| \le C_2 \mathsf{H}(\Xi) K^{p_2} L^{-c_0' K}$ ν_t-almost everywhere on Cut_t. The same holds at real sources: let $n \ge 1$, $f_1, \dots, f_n$ real Lipschitz functions, $\mathsf{M}_t \le CK^{2\mathfrak{p}_0} \ell^3 L^{3u}$ the inverse radius of the sources of the chain at the level t, and x real in the polydisc $\mathbb{D}_t$ of the radii $r_i = (8n\mathsf{M}_t \|f_i\|_\sharp)^{-1}$; then $w_x = \sum_i x_i f_i$ lies in the class of sources of the setting (Th), and with the measure $\rho_0 e^{\mathcal{O}(w_x)} dU$ in the place of $\rho_0 dU$, its chain measure $\nu^{w_x}_t$ and its conditional mean $Q^{w_x}_{t\leftarrow 0}$, and the entry with the moved weight $\mathfrak{K}^\sharp[(\beta - a^{-1} w_x)\Xi]$ in the place of $\beta \mathfrak{K}^\sharp[\Xi]$, the same bound holds with a number $\mathsf{a}(x)$, the numbers C_2, p_2, K_2 being free of x and of b'.

The number $\mathsf{a}(x)$ is the sum, over the levels $j \le t$, of the values at the configuration $\mathbf{1}$ of local families of the level j whose member at a polymer X reads w_x and Ξ on the plaquettes of X alone and vanishes if Ξ vanishes at the barycentres of these plaquettes. For free photons the cubic piece vanishes identically.

■

Proof sketch. (a) The entry is typed anew at each level. At a height $j \ge 1$ the sum of the $\varkappa_\iota j_\iota$ over the boxes whose cubes of the level j form a given polymer is a pointed family of the level j: at most

$8N(ML^j)^3$ boxes have one polymer, and on the cores each density is at most $c\,\alpha_0^{m_J}L^{-2m_Jj}$ by its order, so that the family has the size $8NM^3c\,\alpha_0^{m_J}|\varkappa|_\infty L^{-(2m_J-3)j}$; this is the power counting, of order $|\varkappa|_\infty L^{(3-2m_J)j}$ in level-j units. At the height 0 the family in the fourth place $\sum_\iota \varkappa_\iota[j_\iota(e^{iB'}\mathbf{U})-j_\iota(\mathbf{U})]$, B' the fluctuation field with its values at the pivot bonds, sums to $J(e^{iB'}\mathbf{U})-J(\mathbf{U})$ and has the size of a number times $N\hat{c}\,\delta(x_0)|\varkappa|_\infty$, by the height and Schwarz's lemma in the fluctuation field.

So for $|s|<r_J$, $r_J^{-1}\le C|\varkappa|_\infty$ with C reading N, c and $\hat{c}$, the entry enters each step as one more local term, inside the radii of the one-step theorem with sources (Prop. 6.32); the members of the families are holomorphic in s and read s only on the polymers that the entry meets. The identity of the measures follows by induction on k from the recursion of p_k: the images $\nu_k^s=(\mathcal{M}^k)_*(\mathsf{p}_k e^{sJ}\rho_0 dU)$ obey $\nu_{k+1}^s=\mathbf{1}_{\mathrm{Cut}_{k+1}}\mathbf{T}(\mathfrak{G}_k[x_k]\chi_{x_k}^{\mathrm{sf}}\nu_k^s)$, the small-field pushforward in [7, Lem. B.3], a chart integral over Rg_{x_k} that the one-step theorem evaluates; one function $e^{\Phi_k^s}$ is both the density reached at the level k and the integrand of the step from the level k.

The derivative of the exponent in s at 0 is the conditional mean up to the number $\dot{c}_t=\sum_i \partial_s c_i(0)$. On Cut_t the plaquette variables of the background are within $\mathsf{e}_t^{\mathrm{c}}L^{-2t}$ of $\mathbf{1}$, $\mathsf{e}_t^{\mathrm{c}}$ the radius of the field cut, so that the entry at the background is at most $N(\ell L^K)^3c\,(\mathsf{e}_t^{\mathrm{c}})^{m_J}L^{-2m_Jt}|\varkappa|_\infty$; with what the families of the earlier levels carry, $|\Phi_t^s-\Phi_t^{s'}|\le|s-s'|\,\mathsf{o}_t$ with o_t of the order of the stated oscillation, which is at most $2\mathsf{o}_t$. The threshold $K\ge K_\phi$ makes t a top level of the setting (Th): $(1-\phi)K\ge 2$ gives $1\le t\le K-1$, $L^{\phi K}\gamma^{-2}\ge \mathsf{t}_\natural$ gives $x_t=L^u\mathsf{b}'\ge\mathsf{t}_\natural$, and the third condition keeps the size of a source at the coupling x_k, $k\le t$, below the bound of the step; this size θ_k obeys $L\theta_k\le\Theta_K L^{-(K-k)/2}$, Θ_K a power of K. (b) Let $S=\beta\mathfrak{K}^\sharp[\Xi]=\sum_p\varkappa(p)\mathfrak{r}^\sharp_{\lambda(p)}(p)$, $\varkappa(p)=L^{2K}\mathsf{b}'\,\Xi^{\lambda(p)}(x(p))$, so that $\max|\varkappa|\le L^{2K}\mathsf{b}_+\mathsf{H}(\Xi)$; the density vanishes to the third order at $\mathbf{1}$ (Lem. 5.9(b)).

At a real source the weight is moved, $\varkappa(p)=L^{2K}(\mathsf{b}'-w_x(x(p)))\Xi^{\lambda(p)}(x(p))$, and the chain carries the source; we write the source 0, the other case being the same with the index w_x. Let $s_0=S$ and $s_j=S\circ U_j$ on Cut_j, the entry read at the background of the level j. For a pointed family f of a level $\le n$ let $\Sigma_n[\mathsf{f}](V)$, $V\in\mathrm{Cut}_n$, be the sum over the polymers of the members of f at the background $U_n(V)$, and $\mathrm{Sum}[\mathsf{f}](\mathbf{1})$ the same sum at the configuration $\mathbf{1}$, a number. Let $S_n\mathsf{f}$ be the first-order response of the step from the level n to a family f of a level $i\le n$, a family of the level $n+1$ with $\|S_n\mathsf{f}\|_{n+1}\le\theta_nL^{-(n-i)}\|\mathsf{f}\|_i$, so that $Q_n\Sigma_n[\mathsf{f}]=\Sigma_{n+1}[\mathsf{f}+S_n\mathsf{f}]$ on Cut_{n+1}; and let $D^{(j)}=D^{(j)}[w;\varkappa]$ be the family born at the level j from the entry at the source w and the weight $\varkappa$: $Q_{j-1}s_{j-1}-s_j=\Sigma_j[D^{(j)}]$ on Cut_j.

With $T_t=\mathrm{id}$ and $T_n\mathsf{g}=T_{n+1}(\mathsf{g}+S_n\mathsf{g})$, the carriage of a family from the level n to the level t with all that it feeds, $Q_{t-1}\cdots Q_n\Sigma_n[\mathsf{g}]=\Sigma_t[T_n\mathsf{g}]$; so, with $n_0=K-\lceil\frac{K}{7}\rceil$ and $Q_{t\leftarrow n_0}=Q_{t-1}\cdots Q_{n_0}$,

$$Q_{t\leftarrow 0}S=Q_{t\leftarrow n_0}s_{n_0}+\sum_{j\le n_0}\Sigma_t[T_jD^{(j)}]\qquad\text{on }\mathrm{Cut}_t.$$

The births are small by the parity in the fluctuation field: the part of the variation of the density that is odd in the fluctuation field has the size $g_{j-1}=x_{j-1}^{-1/2}$ and the Gaussian mean 0, so that it meets only the odd part of the interaction, of the size $g_{j-1}(\log x_{j-1})^{\mathfrak{p}_0}$, while the even part has the size g_{j-1}^2; so $\|D^{(j)}\|\le C(\log x_{j-1})^{\mathfrak{p}_0}g_{j-1}^2L^{-3(j-1)}\max|\varkappa|$ once $K-j$ exceeds a fixed construction threshold, and $L^j\|D^{(j)}\|\le CK^{\mathfrak{p}_0}L^{K-j}\mathsf{H}(\Xi)$. $T_jD^{(j)}$ is $D^{(j)}$ plus the sum of the chains $S_{k_m}\cdots S_{k_1}D^{(j)}$, $j\le k_1<\cdots<k_m<t$, of the families that it feeds; each chain is stopped at a level n of its own, chosen with the free fraction τ, at which the variation $\mathrm{Sum}[\mathsf{f}](U_n(V))-\mathrm{Sum}[\mathsf{f}](\mathbf{1})$ of a family f of the level i is at most $\beta_nL^i\|\mathsf{f}\|_i$, $\beta_n=CK^pL^{3(K-n)-K}$, the square of the radius of regularity of the level n entering. $Q_{t\leftarrow n}$ does not enlarge an oscillation, and each feeding costs $L\theta_k\le\Theta_KL^{-(K-k)/2}$, so that what is fed is forgotten.

The variation of $\mathrm{Sum}[D^{(j)}]$ over Cut_{n_0}, $j<n_0$, is where the parity of the cubic density enters. The step is covariant under the isometries of the lattice of its level, and $\mathfrak{r}^\sharp$ transforms as a vector (Lem. 5.9(b)); so under the inversion $r_z:y\mapsto 2z-y$ through a site z of the lattice of the

level j, $\mathrm{Sum}[D^{(j)}[w;\varkappa]](r_zU) = -\mathrm{Sum}[D^{(j)}[r_z^*w;r_z^*\varkappa]](U)$, $(r_z^*v)(p) = v(r_zp)$. With a partition of unity $\psi_z \geq 0$ over these sites, $\psi_z \circ r_z = \psi_z$, $\psi_z(p) = 0$ unless $|x(p) - z|_\infty \leq \frac{1}{2}L^{j-K}$, and $\Sigma_z[w;\varkappa] = \mathrm{Sum}[D^{(j)}[w;\psi_z\varkappa]]$, this gives the exact split

$$\Sigma_z[w;\varkappa](U) = \tfrac{1}{2}\big(\Sigma_z[w;\varkappa](U) - \Sigma_z[w;\varkappa](r_zU)\big) + \tfrac{1}{2}\Sigma_z[w;\varkappa - r_z^*\varkappa](U) + \tfrac{1}{2}\big(\Sigma_z[w;r_z^*\varkappa](U) - \Sigma_z[r_z^*w;r_z^*\varkappa](U)\big).$$

(I) The first part compares the background with its image under r_z. In the Hölder gauge (Thm. [7, Thm. BT] for real backgrounds) the curvature, a 2-form, is even under the inversion, so that the linear part of the potential is kept and the potential and its image differ by the odd part of the Hölder remainder, at most $R^2\vartheta^{2+b}e_3$ on a cube of the radius R about z, $\vartheta = L^{-(n_0-j)}$, $b = \frac{3}{4}$, e_3 of the order of the radius of regularity of the level n_0: a gain ϑ^b. (II) The second part carries the asymmetry of the weight, $|\varkappa - r_z^*\varkappa| \leq L^{2K}\mathrm{Lip}(\Xi)L^{j-K}$ where $\psi_z \neq 0$. (III) The third part is the asymmetry of the source in the step: it vanishes at the source 0, and at a real source it is of the second order at the trivial triple, of the size $\mathrm{Lip}(w_x)\,\delta(x_{j-1})$, $\delta(x_{j-1})$ of the order of $L^{-(K-j)/2}$ being the threshold of the level.

Summed over $j < n_0$, the parts (I), (II) and (III) are $O(L^{-3K/14})$, $O(L^{-4K/7})$ and $O(L^{-K/14})$; the stopped entry contributes $O(L^{-5K/14})$ and the family born at n_0 $O(L^{-3K/7})$; forgetting leaves $O(L^{-(\frac{1}{14}-\frac{\tau}{7})K})$ of what is fed, and against the factor L^{3u} of the levels above t it leaves $O(L^{-(\frac{1}{14}-3\phi)K})$; all this up to powers of K. The least of these exponents is c_0'; at the source 0 the last two meet at $\frac{1}{14}$ as $\tau, \phi \to 0$, and at a real source the part (III) meets them there. The number a, the sum of the values at the configuration $\mathbf{1}$ of the families born and fed up to their levels of stopping, depends on no V, and each of these families reads the source and the weight on its polymer alone and vanishes where the weight vanishes.

The numbers are uniform on $[\mathsf{b}_-, \mathsf{b}_+]$ because the frame of [7, Def. B.14] holds for the chain along $x_j = L^{K-j}\mathsf{b}'$ at every label of the interval and b' enters only through $x_i \in [\gamma^{-2}L^{K-i}, \mathsf{b}_+L^{K-i}]$ and $\max|\varkappa|$, each time in a monotone way; on the polydisc $\mathbb{D}_t$ the source stays in the class of the setting (Th) and the mass of the small-field chain stays away from 0, and at real x the measures of the chain are positive. □

Lemma 5.11 (The cubic remainder)**.** In the setting of Lem. 5.10, let $n \geq 1$, $\mathsf{d} > 0$ and $c < \frac{1}{14}$. There are C, p, free of K and of b', such that for all $K \geq 2$, $\mathsf{b}' \in [\mathsf{b}_-, \mathsf{b}_+]$, $\beta = L^K\mathsf{b}'$, all real Lipschitz f_i and all fields η, the spread of $\{\eta \neq 0\}, \mathrm{supp}\, f_1, \dots, \mathrm{supp}\, f_n$ being $\geq \mathsf{d}$,

$$|\mathfrak{U}^\sharp[\eta](f)| \leq CK^pL^{-cK}\mathsf{H}(\eta)\prod_i \|f_i\|_\sharp.$$

The spread condition holds if two of the sets are at sup distance $\geq \mathsf{d}$, and if $\eta = 0$ within the sup distance d of one support. By $x \mapsto L^zx$ the same holds at every label and on every side. ■

Proof sketch. Moving the weight of the entry with the sources produces the contact terms:

$$\mathfrak{U}^\sharp[\eta](f) = -\partial_s\partial_{z_1}\cdots\partial_{z_n}\log\Big\langle\exp\Big(s\,\mathfrak{K}^\sharp[(\beta - a^{-1}\textstyle\sum_i z_if_i)\eta] + \sum_i z_i\mathcal{O}(f_i)\Big)\Big\rangle\Big|_{s=z=0}.$$

After discarding the histories with large fields, Lem. 5.10(b), applied to the chain with the real sources x in its polydisc $\mathbb{D}_t$ and to the field η, makes the conditional mean of the entry with the moved weight equal, up to $\varepsilon = K^pL^{-c_0'K}$, to its number $\mathsf{a}(x)$; and $\mathsf{a}(x)$ is a sum $\sum_i c_i(x)$ of functions each of which does not depend on x_i, up to a remainder below every power of L^{-K}. Indeed, let $c_i(x)$ collect the members of the families of $\mathsf{a}(x)$ at the polymers on whose plaquettes f_i is the first of the sources that vanishes at all the barycentres; such a member reads the sources on its polymer alone, so it does not read x_i; the members at the other polymers, which meet $\{\eta \neq 0\}$ and all the $\mathrm{supp}\, f_i$, are left over; under the spread such a polymer of the level j has the linear size $\geq \mathsf{d}L^{K-j}/M - 3$, so that these members cost $e^{-\mathsf{d}L^{K-j}/M}$ against the sizes.

Such sums have no mixed derivative. At complex sources the mean is crudely at most $B = C\ell^3 L^{3K}\max|\varkappa|$, $\varkappa$ the weight of the entry. Interpolating between the two bounds, the mixed derivative is at most $2^n\varepsilon^{1-\sigma}B^\sigma(\frac{4n}{\pi\sigma})^n\prod_i r_i^{-1}$ with $r_i^{-1}\le CK^{2\mathfrak{p}_0}\ell^3L^{3u}\|f_i\|_\sharp$, and σ,ϕ,τ small give every $c<\frac{1}{14}$. The finitely many $K<K_2$ are paid by the crude bound of the cumulant by the suprema of its entries, which goes into C. For $n\ge 2$ the bound holds also where $\{\eta\neq 0\}$ covers the supports, as in the uses below, since the spread of the $n+1$ sets is at least that of the supports among themselves. □

Lemma 5.12 (Chessboard bound in one cell)**.** Let $0<g\le\gamma$, $\ell'=2\mathsf{l}\in\mathfrak{L}(g)$, $0<\phi\le\frac{1}{84}$, $u=\lceil\phi K\rceil$, $t=K-u$, $0<\delta\le\frac{1}{8}\mathsf{l}$ and $m\ge 1$. There are C,p,K_3, depending on g,ℓ',ϕ,m,δ, such that for $(K,\beta)\in\mathfrak{D}(g)$ with $K\ge K_3$, every board of side l with offset in $a_{K-t}\mathbb{Z}^3=L^{-u}\mathbb{Z}^3$, every cell, every d with $\mathsf{d}L^u\ge K^2$, and m slots $X_1,\dots,X_m$ supported in the cell at sup distance $\ge\delta$ from its walls and pairwise $\ge\mathsf{d}$ apart,

$$\Big\|\prod_a\big(X_a-\langle X_a\rangle_{K,\ell',\beta}\big)\Big\|_{\mathrm{ch}}\le C\prod_a\varpi(X_a)\,\mathsf{n}(X_a),$$

where a slot is $\mathcal{T}_K(H)$, with $\varpi=K^pL^{4u}$ and $\mathsf{n}=|H|_\infty$, or one of

$$\mathcal{T}_K(H^{\mathrm{cons}}[\eta]),\mathrm{St}_K[\eta],\Delta_K[\eta],\mathrm{Rm}_K[\eta],\mathfrak{K}^{\mathrm{v}}[\beta\eta],$$

with $\varpi=K^pL^{3u-K/28}$ and $\mathsf{n}=\|\eta\|_{C^2}$. For $m=1$ and $\phi=\frac{1}{400}$ we have $\|\mathrm{St}_K[\eta]-\langle\cdot\rangle\|_{\mathrm{ch}}\le CK^pL^{-K/28}\|\eta\|_{C^2}$ and

$$\|\Delta_K[\eta]-\langle\cdot\rangle\|_{\mathrm{ch}}+\|\mathcal{T}_K(H^{\mathrm{cons}}[\eta])-\langle\cdot\rangle\|_{\mathrm{ch}}\le CK^pL^{-K/36}\|\eta\|_{C^2};$$

for a plain tensor entry $\|\mathcal{T}_K(H')-\langle\cdot\rangle\|_{\mathrm{ch}}\le CK^{2\mathfrak{p}_0}L^{K/100}|H'|_\infty$. At $m=1$ and $\phi=\frac{1}{400}$, moreover, $\|\mathrm{Rm}_K[\eta]-\langle\cdot\rangle\|_{\mathrm{ch}}=\|\mathrm{St}_K[\eta]-\langle\cdot\rangle\|_{\mathrm{ch}}$, and the bound $CK^pL^{-K/28}\|\eta\|_{C^2}$ holds also for $\|F_\eta\|_{\mathrm{ch}}$, where

$$F_\eta=\mathfrak{K}^{\mathrm{v}}[\beta\eta]-\tfrac14\mathfrak{N}[\mathsf{k}_\eta]=\beta\mathfrak{K}^\sharp[\eta]-\sum_{v=1}^5\mathfrak{J}_v[\mathsf{k}_\eta]+\tfrac14 c_{\mathsf{k}_\eta},\qquad\langle F_\eta\rangle_{K,\ell',\beta}=0,$$

with $\mathsf{k}_\eta^\rho=\nabla_\rho^+\eta^\rho$ and the objects of Lem. 5.9(c); for every state s of Lem. 5.8, $s(\mathfrak{K}^{\mathrm{v}}[\beta\eta])=s(F_\eta)$, so that $|s(\mathfrak{K}^{\mathrm{v}}[\beta\eta])|\le e^{G_g\mathsf{l}^3}\|F_\eta\|_{\mathrm{ch}}$ by Lem. 5.8 at $q=1$, $c_1=0$. ■

Proof sketch. By Lem. 5.9(c), $\mathrm{St}_K[\eta]=N_\beta[\eta]-\frac14\mathfrak{N}[\mathsf{k}_\eta]-\mathrm{Rm}_K[\eta]-\frac14 c_{\mathsf{k}_\eta}$, so that the slots $\mathrm{St}_K[\eta]$ and $-\mathrm{Rm}_K[\eta]$ differ by null terms and a number; inside the mean of the $8m$ mirror copies on $T_{\ell'}$ the null terms drop, since no other copy depends on their bonds. A product of centred slots in one cell expands into joint cumulants over the partitions into blocks of at least two members. Apply [7, Prop. 3.15](a), with its common insertion radius r_1, size $\bar{\mathsf{d}}$ and polymer sum $\mathsf{t}_\times$. It bounds the higher mixed jets of the exponent as well as the separate conditional means and gives

$$|\langle J_0;\dots;J_r\rangle|\le\mathrm{op}_{r+1}\prod_a\omega_t(J_a)+9\cdot2^{r+1}(r+2)\bar{\mathsf{d}}\,\mathsf{t}_\times r_1^{-r}+4\,\mathrm{op}'_{r+1}\mathsf{R}_t\prod_a\sup|J_a|,\qquad\mathrm{op}_n=\sum_\pi(|\pi|-1)!,$$

π over the partitions of n points, with op'_n as in [7, Prop. 3.15]; ω_t is the oscillation of the conditional mean (Lem. 5.10) and R_t the weight of the histories with large fields. After allowing the carrier collars, take $\mathsf{d}_\times\ge c\min\{\mathsf{d},\delta\}$. The hypothesis $\mathsf{d}L^u\ge K^2$ gives $\mathsf{d}_\times L^u\ge cK^2$ for all sufficiently large K, uniformly in d. The prefactor estimates in the proof of [7, Prop. 3.24] bound the second term by a polynomial in K times $L^{AK}e^{-cK^2}$; thus the condition of [7, Prop. 3.15](a) holds and this term is below any prescribed L^{-BK}. The large-field term is controlled there separately. Six of the seven pieces of Rm_K are small by power counting, the seventh is the cubic one, of oscillation $K^pL^{-c_0'K}$. The principal part of Δ_K is null (Lem. 5.9(d)): on the backgrounds of the level $K-\lceil\frac{K}{14}\rceil$ its value is at most $CK^{2\mathfrak{p}_0}L^{\frac72\lceil K/14\rceil-\frac{K}{2}}\mathsf{b}\sup|\nabla\eta|$, what is born below that level is born at a small coupling and is forgotten, and a conditional mean is never larger than a supremum.

Each crude tensor entry costs L^{4u}. At $m=1$: by (a)–(c) of Lem. 5.9, $\mathfrak{K}^{\mathrm{v}}[\beta\eta]=\frac14\mathfrak{N}[\mathsf{k}_\eta]+\beta\mathfrak{K}^\sharp[\eta]-\sum_v\mathfrak{J}_v[\mathsf{k}_\eta]+\frac14 c_{\mathsf{k}_\eta}$, so that F_η consists of six of the seven pieces of $\mathrm{Rm}_K[\eta]$ and a number. The cubic piece has the oscillation $K^pL^{-K/28}\|\eta\|_{C^2}$ by Lem. 5.10(b) at $\tau=\frac14$, where $c_0'=\frac{1}{28}$ for $\phi\le\frac{1}{84}$; the five pieces $\mathfrak{J}_v$, of the orders $2,2,2$ with weights of the order of $\|\eta\|_{C^2}$ and of the orders $3,4$ with weights at most $\beta\|\eta\|_{C^2}$, have oscillations at most $K^{3\mathfrak{p}_0}L^{4u-K/2}\|\eta\|_{C^2}$ by Lem. 5.10(a); so no factor L^{3u} enters the bound of F_η; in the bounds of the slots of the kind of η it enters through Δ_K alone.

The means of $\mathfrak{K}^{\mathrm{v}}[\beta\eta]$ and of $\mathfrak{N}[\mathsf{k}_\eta]$ on $T_{\ell'}$ are 0 (Lem. 5.9(a), (b)), so is that of F_η, and $s(\mathfrak{N}[\mathsf{k}_\eta])=0$ in every state s by Lem. 5.9(a). □

Lemma 5.13 (From sup norms to an integrable density)**.** Let $H\ge0$ be self-adjoint, $T_t=e^{-tH}$, and for every $\eta>0$ let $\rho\mapsto S_\eta[\rho]$ be linear from $C_c((-\eta,\eta))$ to the bounded operators, with $\|S_\eta[\rho]\|\le A_J\sup|\rho|$ for $\operatorname{supp}\rho$ in a compact $J\subset(-\eta,\eta)$, and $S_\eta[\rho(\cdot-c)]=T_{\delta+c}S_{\eta-\delta}[\rho]T_{\delta-c}$ for $|c|\le\delta<\eta$. Then there is a unique family of bounded operators $B(s,s')$, $s,s'>0$, with $T_tB(s,s')T_{t'}=B(s+t,s'+t')$ and $S_\eta[\rho]=\int\rho(c)B(\eta+c,\eta-c)\,\mathrm{d}c$; for $0<\gamma'<\eta$ and $\operatorname{supp}\rho\subset[-(\eta-\gamma'),\eta-\gamma']$, $\|S_\eta[\rho]\|\le\|S_{\gamma'/2}[\mathrm{tent}_{\gamma'/8}]\|\,\|\rho\|_{L^1}$; and $B(s,s')^*=B(s',s)$ if $S_\eta[\rho]^*=S_\eta[\bar\rho(-\cdot)]$. The same holds in the space directions for commuting momenta P with $H\ge|P_k|$ in the direction k. ■

Proof sketch. For G holomorphic and bounded on the strip $\{|\mathrm{Re}\,z|<h'\}$, $h'>2h$, integrating $\Gamma(z)\sin^{-2}(\pi z/2h)$, $\Gamma'=G$, over the boundary of the strip gives

$$G(0)=\int\mu_2(\,\mathrm{d}s)\int\mathrm{tent}_{2h}(c)\,G(c+is)\,\mathrm{d}c,\qquad\mu_2=\mu_1*\mu_1,\quad\mu_1(\,\mathrm{d}s)=\tfrac{\pi}{4h}\mathrm{sech}^2(\tfrac{\pi s}{2h})\,\mathrm{d}s.$$

Thus an average in real time inverts a tent smearing in Euclidean time. Apply it to $z\mapsto\langle T_{\delta+\bar z}\psi_2,S[\rho]T_{\delta-z}\psi_1\rangle$ for vectors ψ_1,ψ_2, holomorphic by the covariance, and use $\rho_0*\rho_1=\rho_1*\rho_0$: the μ_2-average of $e^{-isH}S_{\gamma'/2}[\mathrm{tent}]e^{isH}$ is the operator $\mathbb{B}$ of the density, $S_\eta[\rho]=\int\rho(c)T_{\eta-\gamma'/2+c}\mathbb{B}T_{\eta-\gamma'/2-c}\,\mathrm{d}c$. In space, $G(0)=\int h_{d'}(x)\frac12[G(x+id')+G(x-id')]\,\mathrm{d}x$ with $h_{d'}=(2d')^{-1}\mathrm{sech}(\pi x/2d')$, with the imaginary translations that $H\ge|P|$ allows. □

Lemma 5.14 (Functions of the block field)**.** For a history h let $\mathfrak{w}_h\ge0$ be its weight in the law of the field and of the history and $\mathsf{t}(h)=\int\mathfrak{w}_h\rho_0\,\mathrm{d}U$ its mass, so that $\sum_h\mathsf{t}(h)=Z$ (equations YM3_I:(646)–(648) and Prop. YM3_I:44.1).

(i) In the setting of Lem. 5.10, let $1\le t\le K-1$ be a level up to which the chain is formed (a top level of the chain at the label b'), h_t the history without large-field regions of the level t, $\mathrm{In}_t=\{V:\varpi(V)<2\varepsilon(x_t)\}\subset\mathrm{Cut}_t$ the inner ball of the level t (the definitions preceding Lem. YM3_I:113.10; $\varpi(V)$ is the largest $|\partial V(p')-\mathbf{1}|$ over the plaquettes p' of the level t), and $\mathfrak{w}_t^{\mathrm{sf}}$ the weight of the level t of the small-field chain with its number R_t (the definitions preceding Prop. YM3_I:115.2, the definition preceding Prop. YM3_I:115.2, YM3_I:(2187)): $\mathfrak{w}_t^{\mathrm{sf}}(U)=\mathbf{1}_{\mathrm{In}_t}(\mathcal{M}^tU)\prod_{i<t}[\mathbf{1}_{\mathrm{Cut}_i}\mathfrak{G}_i[x_i]\chi_{x_i}^{\mathrm{sf}}](\mathcal{M}^iU)$, with the gauge-fixing factors $\mathfrak{G}_i[x_i]$ and the fluctuation tests $\chi_{x_i}^{\mathrm{sf}}$ among the construction results, so that $\mathfrak{w}_t^{\mathrm{sf}}=(\mathbf{1}_{\mathrm{In}_t}\circ\mathcal{M}^t)\,\mathsf{p}_t$ with $\mathsf{p}_t(U):=\mathbf{1}_{\mathrm{Cut}_t}(\mathcal{M}^tU)\prod_{i<t}[\mathbf{1}_{\mathrm{Cut}_i}\mathfrak{G}_i[x_i]\chi_{x_i}^{\mathrm{sf}}](\mathcal{M}^iU)$.

The measure ν_t of the level t of the chain is $(\mathcal{M}^t)_*(\mathsf{p}_t\rho_0\,\mathrm{d}U)$ (Prop. YM3_I:114.2, the measure recursion in Prop. YM3_I:114.2), and $Q_{t\leftarrow0}$ is the conditional mean given $\mathcal{M}^tU$ under $\mathsf{p}_t\rho_0\,\mathrm{d}U$; the tower of Lem. 5.10 is read with this p_t. By Prop. YM3_I:115.2, $\mathfrak{w}_t^{\mathrm{sf}}$ is a bounded Borel function with values in $[0,\infty)$, $(\mathcal{M}^t)_*(\mathfrak{w}_t^{\mathrm{sf}}\rho_0\,\mathrm{d}U)=\mathbf{1}_{\mathrm{In}_t}\nu_t$ YM3_I:(2188), $0\le\mathfrak{w}_{h_t}\le\mathfrak{w}_t^{\mathrm{sf}}$ at every point YM3_I:(2189), and $0\le Z-\int\mathfrak{w}_t^{\mathrm{sf}}\rho_0\,\mathrm{d}U\le Z-\mathsf{t}(h_t)\le\mathsf{R}_tZ$ YM3_I:(2190): R_t bounds the weight of the histories with a large field up to the level t.

For every bounded real Borel gauge invariant Y, real a_0 and $\epsilon\ge0$ with $|Q_{t\leftarrow0}Y-a_0|\le\epsilon\,\nu_t$-almost everywhere on In_t, and every bounded Borel function ψ of the block fields of the level t, $|\langle Y;\psi\circ\mathcal{M}^t\rangle|\le2[\epsilon+2(\sup|Y|+|a_0|)\mathsf{R}_t]\sup|\psi|$; if besides $\mathsf{R}_t\le\frac12$, then $|a_0|\le\sup|Y|+\epsilon$.

(ii) In the setting of Lem. 5.6 at $0 < g \le \gamma$ with $g^{-2} \ge x_{\mathrm{J}}$ (a first-passage datum $(K,\beta) \in \mathfrak{D}(g)$, a side $\ell \in \mathfrak{L}(g)$, a depth s and the working level k), let $2 \le T \le k$ be a level, h_T the history without large-field regions of the level T, $\mathsf{m}_T = \tau_{h_T}[\rho_0]$ (Lem. 5.6(a)), and $W_T < 1$ the weight of a first creation of a large field at a level $\le T$, so that $Z - \mathsf{t}(h_T) = W_T Z$. A dominated local source F_2 acts on a dominated local source F_1 at the sup distance d', with $L^{K-T}\mathsf{d}' \ge 3M$ and $\mathsf{h}_2(K-T) \le 1$ (the conditions of Lem. 5.6(e) at $t = T$), through a bounded function of the block field of the level T: let r_2 be the real part of the bounded density $D_{T,2}$ of $\tau_{h_T}[F_2\rho_0]$ with respect to m_T, c real, E the essential supremum of $|r_2 - c|$ for m_T, and $\psi = (r_2 - c)\mathbf{1}\{|r_2 - c| \le E\}$, a bounded real Borel function with $\sup|\psi| \le E$ and $\psi = r_2 - c$ m_T-almost everywhere; then

$$|\langle F_1; F_2\rangle - \langle F_1; \psi \circ \mathcal{M}^T\rangle| \le C_2 e^{3-\mathsf{D}_T/M} y_1 y_2 + 2\sup|F_1|(\sup|F_2| + \sup|\psi| + |c|)W_T,$$

with $y_a \le 4\ell^3 L^{4(K-T)}$ times the sizes of the sources and $\mathsf{D}_T \ge L^{K-T}\mathsf{d}'$ the least sup distance of the carriers in level-T units; at $c = \operatorname{Re} c_2$, c_2 the constant of F_2 at the level T, $E \le C_1 y_2$ and $|c| \le \sup|F_2| + C_1 y_2$. The display is additive in F_1.

■

Proof sketch. The law of the field and of the history, Prop. YM3_I:44.1 with the measures in equation YM3_I:(647), gives

$$\mathbf{T}^k(\Psi\rho_0\,\mathrm{d}U) = \sum_h \tau_h[\Psi\rho_0],$$
$$\int F\,\mathrm{d}\tau_h[\Psi\rho_0] = \int F(\mathcal{M}^k U)\,\mathfrak{w}_h(U)\Psi(U)\rho_0(U)\,\mathrm{d}U$$

for every bounded gauge invariant Ψ, when $\varepsilon_j \le \varepsilon_\flat = c_2^{\mathrm{SO}}$ for $j < k$; hence $Z\langle\Psi\cdot\psi\circ\mathcal{M}^k\rangle = \sum_h \int \psi\,\mathrm{d}\tau_h[\Psi\rho_0]$, and, as $\mathfrak{w}_h \ge 0$ and $\sum_h \mathsf{t}(h) = Z$,

$$\left|Z\langle\Psi\cdot\psi\circ\mathcal{M}^k\rangle - \int \psi\,\mathrm{d}\tau_{h_k}[\Psi\rho_0]\right| \le \sup|\Psi|\sup|\psi|\,(Z - \mathsf{t}(h_k)).$$

(i) At the level t replace $\mathfrak{w}_{h_t}$ by $\mathfrak{w}_t^{\mathrm{sf}}$: the difference $\mathfrak{w}_t^{\mathrm{sf}} - \mathfrak{w}_{h_t}$ is ≥ 0 at every point, of the mass $\int \mathfrak{w}_t^{\mathrm{sf}}\rho_0\,\mathrm{d}U - \mathsf{t}(h_t) \le Z - \mathsf{t}(h_t) \le \mathsf{R}_t Z$; with the histories other than h_t this gives $|Z\langle\Psi\cdot\psi\circ\mathcal{M}^t\rangle - \int \Psi\,\psi(\mathcal{M}^t U)\,\mathfrak{w}_t^{\mathrm{sf}}\rho_0\,\mathrm{d}U| \le 2\sup|\Psi|\sup|\psi|\mathsf{R}_t Z$, the factor 2 counting the two parts.

By the tower property, $\mathfrak{w}_t^{\mathrm{sf}} = (\mathbf{1}_{\mathrm{In}_t}\circ\mathcal{M}^t)\mathsf{p}_t$ and $\mathrm{In}_t \subset \mathrm{Cut}_t$, $\int \psi(\mathcal{M}^t U)\,(Y - a_0)\,\mathfrak{w}_t^{\mathrm{sf}}\rho_0\,\mathrm{d}U = \int_{\mathrm{In}_t} \psi\,(Q_{t\leftarrow 0}Y - a_0)\,\mathrm{d}\nu_t$, of a modulus at most $\epsilon\int_{\mathrm{In}_t}|\psi|\,\mathrm{d}\nu_t \le \epsilon\sup|\psi|\,Z$, since $\nu_t \ge 0$ and $\nu_t(\mathrm{In}_t) = \int \mathfrak{w}_t^{\mathrm{sf}}\rho_0\,\mathrm{d}U \le Z$. Applied at $\Psi = Y - a_0$, for ψ and for 1, with $\langle Y;\psi\circ\mathcal{M}^t\rangle = \langle(Y-a_0)\,\psi\circ\mathcal{M}^t\rangle - \langle Y - a_0\rangle\langle\psi\circ\mathcal{M}^t\rangle$, this is the bound of (i). If $\mathsf{R}_t \le \frac12$: $|Q_{t\leftarrow 0}Y| \le \sup|Y|$ ν_t-almost everywhere and $\nu_t(\mathrm{In}_t) \ge (1 - \mathsf{R}_t)Z > 0$, so In_t holds points at which $|a_0| \le |Q_{t\leftarrow 0}Y| + \epsilon \le \sup|Y| + \epsilon$. (ii) By the first display at the level T, where $Z - \mathsf{t}(h_T) = W_T Z$, the means $\langle F_1F_2\rangle$, $\langle F_1\,\psi\circ\mathcal{M}^T\rangle$, $\langle F_1\rangle$, $\langle F_2\rangle$ and $\langle\psi\circ\mathcal{M}^T\rangle$ are the integrals against $Z^{-1}\mathsf{m}_T$ of the bounded densities of these functions at h_T, up to the weight W_T times their suprema, and $Z^{-1}\mathsf{m}_T(1) = 1 - W_T$.

The densities of F_1, F_2 and F_1F_2 are $D_{T,1}$, $D_{T,2}$ and $D_{T,1}D_{T,2} + D_{T,12}$; the real part r_2 of $D_{T,2}$ is again a density of the real source F_2, two bounded densities of one function agree m_T-almost everywhere, and $r_2 = \psi + c$ there; so the density of F_1F_2 is $D_{T,12} + D_{T,1}(\psi + c)$ almost everywhere, and subtracting, $\langle F_1;F_2\rangle - \langle F_1;\psi\circ\mathcal{M}^T\rangle$ is $Z^{-1}\int D_{T,12}\,\mathrm{d}\mathsf{m}_T$, at most $C_2 e^{3-\mathsf{D}_T/M}y_1y_2$ by Lem. 5.6(e), plus two brackets of errors that together are at most $2\sup|F_1|(\sup|F_2| + \sup|\psi| + |c|)W_T$. At $c = \operatorname{Re} c_2$: $|r_2 - c| \le |D_{T,2} - c_2| \le C_1 y_2$ on the small block fields of the level T, which carry m_T (Lem. 5.6(e)), and $|r_2| \le \sup|F_2|$ there.

Every member but $D_{T,12}$ is formed with F_1 linearly, and $D_{T,12}$ is additive in F_1. □

Lemma 5.15 (One entry against separated sources)**.** Let $0 < g \le \gamma$, $\ell \in \mathfrak{L}(g)$, $n \ge 1$, $\mathsf{d} > 0$, $0 < c_* < \frac12$, and let $J = \sum_\iota \varkappa_\iota j_\iota$ be an entry of the order $m \ge 2$ (Lem. 5.10) of the type $(m, c, \hat{c}, N)$:

at complex configurations with $|\mathbf{U}(b)| \le 2$ at all the bonds of its box, $|j_\iota| \le c\, r^m$ where the plaquette variables of the box are within $r \le 1$ of $\mathbf{1}$, and $|j_\iota| \le \hat{c}$ everywhere; and at most N indices have boxes with one lowest site. There are C, K_1, depending on g, ℓ, n, d, c_* and the type, such that for $K \ge K_1$, every label $\mathsf{b}' \in [\gamma^{-2}, X_g + \bar{B}]$ (so at every first-passage datum $(K,\beta) \in \mathfrak{D}(g)$, and at $\beta = L^K \mathsf{b}'$ for every $\mathsf{b}' \in I(g)$) and all real Lipschitz f_q, if the spread of the $n+1$ sets formed by the carrier of J and the $\operatorname{supp} f_q$ is $\ge \mathsf{d}$, as it is when two of them are at sup distance $\ge \mathsf{d}$,

$$|\langle J; \mathcal{O}(f_1); \dots; \mathcal{O}(f_n)\rangle_{K,\ell,L^K\mathsf{b}'}| \le C|\varkappa|_\infty L^{-\mathsf{g}_m K} K^{(2n+1)\mathfrak{p}_0} L^{(\frac12 - c_*)K} \prod_q \|f_q\|_\sharp, \qquad \mathsf{g}_2 = 1,\ \mathsf{g}_m = \tfrac32\ (m \ge 3).$$

On all the sides at once: let besides $R > 0$ and $\ell' \in \mathfrak{L}(g)$ with $\ell' \ge 2R + 2\mathsf{d}$, and let the weights $\varkappa_\iota$ be real and the densities real on the G-valued configurations. There are C', K_1', depending on g, ℓ', n, d, R, c_* and the type, and on no side, such that for all $(K,\beta) \in \mathfrak{D}(g)$ with $K \ge K_1'$, every side $\ell \ge \ell'$ in $\mathfrak{L}(g)$, every entry of the type whose boxes lie in one sup ball of radius R, and all real Lipschitz f_q on $\mathbb{R}^3$ that vanish outside that ball, read on T_ℓ, with the carrier of the entry at sup distance $\ge \mathsf{d}$ from every $\operatorname{supp} f_q$ (the sources lie anywhere among themselves),

$$|\langle J; \mathcal{O}(f_1); \dots; \mathcal{O}(f_n)\rangle_{K,\ell,\beta}| \le C'|\varkappa|_\infty L^{-\mathsf{g}_m K} K^{1+\mathfrak{p}_0} L^{(\frac12 - c_*)K} \prod_q \|f_q\|_\sharp;$$

the same holds under every periodic Gibbs state at (K,β) and under every weak limit point of the torus laws as $\ell \to \infty$ in $\mathfrak{L}(g)$, and, with the further factor 2^{n+1}, for complex weights and complex Lipschitz f_q. ■

Proof sketch. Put $\phi = (\frac12 - c_*)/(3n+4)$, $u = \lceil \phi K \rceil$, $t = K - u$. Since $M\, R(X_g + \bar{B}) \le L^m$ and R increases, the frame of [7, Def. B.14] holds for the chain along $x_j = L^{K-j}\mathsf{b}'$ at every label of $[\gamma^{-2}, X_g + \bar{B}]$, and from a K_1 on t is a top level of it with $\mathsf{R}_t \le \frac12$ and $8a \le \mathsf{d}$.

The cut. Let $D \ge \mathsf{d}$ be the spread. For every site y_0 of the carrier there is a source q with $\operatorname{dist}_\infty(y_0, \operatorname{supp} f_q) \ge \frac12 D$: otherwise there are points $y_q \in \operatorname{supp} f_q$ with $|y_0 - y_q|_\infty < \frac12 D$, and $y_0, y_1, \dots, y_n$ are pairwise closer than D, against the definition of the spread. Give each index ι of J the least such q for the lowest site of its box; this cuts J into at most n entries J_q of the same type, the carrier of J_q being at sup distance $\ge \frac12 D - 2a \ge \frac14 \mathsf{d}$ from $\operatorname{supp} f_q$, while the other sources lie anywhere. The cumulant is linear in its first entry and symmetric in the others; in the cumulant of J_q the far source is f_q.

The chain with the entry and a complex source. The chain of Lem. 5.10(a) is formed with the entry sJ_q and with the source $w = \sum_q \zeta_q f_q$ at once, for $|s| < r_J$ and

$$\|w\|_\flat = \max\{\sup|w|, \frac{40}{\sqrt{3}} M \mathrm{Lip}_\infty w\} < 2\varrho$$

(the polydisc of the sources of Thms. YM3_I:AJ and YM3_I:AD): its families $E^{(i)}[w,s]$ are pointed families of the levels i, holomorphic in (w,s) and local, the member at a domain X reading w on the plaquettes of X alone, and s only if a box of the entry lies in X; with them $(\mathcal{M}^t)_*(\mathfrak{w}_t^{\mathrm{sf}} e^{sJ_q} \rho_0^w \,\mathrm{d}U) = \exp[\sum_{i \le t} c_i(w,s)] \mathbf{1}_{\mathrm{In}_t} e^{\Phi_t^{w,s}} dV$, $\rho_0^w = \rho_0 e^{\mathcal{O}(w)}$, and on Cut_t

$$|\Phi_t^{w,s} - \Phi_t^{w,s'}| \le |s - s'|\, \mathsf{o}_t, \qquad |\Phi_t^{w,0} - \Phi_t^{0,0}| \le N_t \mathsf{C}_t \|w\|_\flat,$$

wherever the sources lie, with $N_t = \ell^3 L^{3u}$ the number of the sites of the level t and $\mathsf{C}_t \le C K^{2\mathfrak{p}_0}$. The entry enters each step as one more local term, as in Lem. 5.10(a); bounding the radius $\mathsf{e}_t^{\mathrm{c}} \le 1$ of the field cut, raised to the order of the entry, by $\mathsf{e}_t^{\mathrm{c}}$ itself, the size of the entry at the level t and what the steps made of it give $\mathsf{o}_t \le C K^{\mathfrak{p}_0} L^{4u - \mathsf{g}_m K} |\varkappa|_\infty$.

The bound. Let $T(s,\zeta) = \int e^{sJ_q + \sum_{q'} \zeta_{q'} \mathcal{O}(f_{q'})}\, \mathfrak{w}_t^{\mathrm{sf}} \rho_0 \,\mathrm{d}U$. The cumulant under the positive measure $\mathfrak{w}_t^{\mathrm{sf}} \rho_0 \,\mathrm{d}U$ is $\partial_s \partial_{\zeta_1} \cdots \partial_{\zeta_n} \log T(0,0)$, and it differs from the cumulant under the measure of the torus

by at most $4\,\mathrm{op}_{n+1}\mathsf{R}_t \sup|J_q| \prod_{q'} 6\ell^3 L^{4K} \sup|f_{q'}|$ (Prop. YM3_I:115.2, its cumulant bound), which is smaller than every power of L^{-K}. By the identity of measures, $\partial_s \log T(0,\zeta) = \partial_s \mathsf{C}(0,\zeta) + \langle \partial_s \mathsf{D}\rangle_\zeta$, where $\mathsf{C}(s,\zeta) = \sum_{i\le t}(c_i(w,s) - c_i(w,0))$ collects the values of the families at the trivial triple, $\mathsf{D} = \Phi_t^{w,s} - \Phi_t^{w,0}$, and $\langle\cdot\rangle_\zeta$ is the mean under $e^{\Phi_t^{w,0} - \Phi_t^{0,0}}$ times the image of $\mathfrak{w}_t^{\mathrm{sf}}\rho_0\,\mathrm{d}U$, normalised.

On the polydisc $|\zeta_{q'}| < (2nN_t\mathsf{C}_t\|f_{q'}\|_\sharp)^{-1}$ the exponent $\Phi_t^{w,0} - \Phi_t^{0,0}$ is at most $\frac12$, so $|\langle G\rangle_\zeta| \le 3.1 \sup|G|$, and the second member is at most $3.1\,\mathsf{o}_t$. In the first, a member at a domain that does not hold both a box of J_q and a plaquette of the far source f_q does not depend on ζ_q (locality), and has no mixed derivative; the members at the domains X of the levels i that hold both have $d_i(X) \ge (\frac14\mathsf{d} - a)L^{K-i}/M - 2$, and with the count of the domains their sum is at most a power of L^K times $e^{-\mathsf{d}L^u/8M}$. Cauchy's estimates in the n sources on this polydisc cost $\prod_{q'}(nCK^{2\mathfrak{p}_0}L^{3u}\|f_{q'}\|_\sharp)$.

Together, $K^{(2n+1)\mathfrak{p}_0}L^{(3n+4)u - \mathfrak{g}_m K}$, and $(3n+4)u \le (\frac12 - c_*)K + 3n + 4$.

On all the sides. Let $\mathsf{l} = L^{-i}$ be the largest power of L with $\mathsf{l} \le \frac12\mathsf{d}$; then $\ell' \ge 2R + 4\mathsf{l}$. On $T_{\ell'}$ and on every T_ℓ, $\ell \ge \ell'$, a board of side l puts the boxes of the entry and the supports of the sources, which lie in one sup ball of radius R inside one period box of side ℓ', in different cells. The cumulant on T_ℓ is a mixed derivative at 0 of a quotient whose coefficients are means on T_ℓ of products of functions of different cells of that period box; the chessboard bound (Lem. 5.8) bounds each by a product of chessboard seminorms on $T_{\ell'}$, at the price $e^{\mathsf{l}^3(p_{\ell'} - p_\ell)} \le e^{G_g\mathsf{l}^3}$ for a cell; this ratio $Z_{K,\ell',\beta}^{1/N'}/Z_{K,\ell,\beta}^{1/N_\ell}$ of the plain partition functions per cell, N' and N_ℓ being the numbers of the cells of $T_{\ell'}$ and of T_ℓ, is the one number that depends on ℓ, and its bound is free of ℓ.

After a subtraction of Wick type in the functions of the cells, which removes the cumulants of the families inside one cell, the chessboard seminorm of the function of the cell of the entry, the mean on $T_{\ell'}$ of the product of its mirror copies, is by the moment–cumulant formula a sum over the partitions of the copies all of whose blocks join different cells; each such cumulant is small by the bound on one torus, read at a free depth u with $C_{\mathrm{u}}K^{1/(2\mathfrak{p}_0)} \le u \le \frac12 K$, $C_{\mathrm{u}} \ge 1$ a number formed with the data and the type: a factor $|\varkappa|_\infty K^{1+\mathfrak{p}_0}L^{4u - \mathfrak{g}_m K}$ for each copy of the entry and a factor of Cauchy's estimate for each copy of a source.

The number of the cells that the ball of radius R meets, at most $(2R/\mathsf{l} + 3)^3$, enters C', and at $u = \lceil\frac14(\frac12 - c_*)K\rceil$ the bound follows. Every periodic Gibbs state is dominated by the torus measures through its conditional laws (Prop. YM3_I:91.9), and the weak limit points of the torus laws inherit the bound; for complex data the real and imaginary parts are taken apart. □

Lemma 5.16 (The cubic remainder uniformly in the side)**.** Let $0 < g \le \gamma$, $\ell' \in \mathfrak{L}(g)$, $n \ge 2$, $\mathsf{d} > 0$, $c < \frac{1}{14}$, and let Q be a cube of side $\frac14\ell'$. There are C, p, K_0, depending on $g, \ell', n, \mathsf{d}, c$ and independent of the larger side ℓ, such that for all $\ell \ge \ell'$ in $\mathfrak{L}(g)$, all $(K,\beta) \in \mathfrak{D}(g)$ with $K \ge K_0$, all Lipschitz fields η and real Lipschitz f_q carried by Q with the supports of the f_q pairwise $\ge \mathsf{d}$ apart,

$$|\mathfrak{U}^\sharp[\eta](f)_{T_\ell}| + |\mathfrak{U}^\sharp[\eta](f)_{T_\ell} - \mathfrak{U}^\sharp[\eta](f)_{T_{\ell'}}| \le CK^p L^{-cK}\mathsf{H}(\eta)\prod_q \|f_q\|_\sharp;$$

the finitely many sides of $\mathfrak{L}(g)$ below ℓ' are covered by Lem. 5.11. ■

Proof sketch. Every label of $\mathfrak{D}(g)$ lies in $[\gamma^{-2}, X_g + \bar B]$, and $MR(X_g + \bar B) \le L^m$ on every side of $\mathfrak{L}(g)$; so the frame of [7, Def. B.14] holds for the chain along $x_j = L^{K-j}\mathsf{b}$ at the label of every first-passage datum, and Lem. 5.10(b) holds with one triple of numbers for all of them, also with real sources $\sum_i x_i f_i$ in its polydisc. The chain is formed with several entries and complex sources as well, as in the proof of Lem. 5.15 (Lem. 5.10(a), polarisation and locality in each entry).

The reduction. With $\mathcal{S}[\Xi] = a^{-2}\sum_p \Xi^{\lambda(p)}(x(p))\,\mathfrak{r}^\sharp_{\lambda(p)}(p)$ for a field Ξ we have $\beta\mathfrak{K}^\sharp[\eta] = \mathcal{S}[\mathsf{b}\eta]$ and $a^{-1}\mathfrak{K}^\sharp[f_i\eta] = \mathcal{S}[f_i\eta]$, so $\mathfrak{U}^\sharp[\eta](f) = -\langle\mathcal{S}[\mathsf{b}\eta]; \mathcal{O}(f_1); \dots; \mathcal{O}(f_n)\rangle + \sum_i\langle\mathcal{S}[f_i\eta]; (\mathcal{O}(f_l))_{l\ne i}\rangle$, with $\mathsf{H}(\mathsf{b}\eta) \le (X_g + \bar B)\mathsf{H}(\eta)$ and $\mathsf{H}(f_i\eta) \le 2\|f_i\|_\sharp\mathsf{H}(\eta)$. It suffices therefore to compare $G_\ell = \langle\mathcal{S}[\Xi]; \mathcal{O}(h_1); \dots; \mathcal{O}(h_{n'})\rangle_{K,\ell,\beta}$ with $G_{\ell'}$, $n' \in \{n-1, n\}$, $n' \ge 1$, for Lipschitz fields Ξ and real

Lipschitz h_q carried by Q whose supports are pairwise $\geq \mathsf{d}$ apart, nothing being asked of Ξ relative to them.

The fold with one factor for each copy. Place Q in the middle of one cell of the board of side $\mathsf{l} = \frac{1}{2}\ell'$; it lies at sup distance $\geq \frac{1}{8}\ell' \geq 1$ from the walls. On every T_ℓ, $\ell \geq \ell'$, $\mathcal{S}[\Xi]$ and the $\mathcal{O}(h_q)$ are the realisations of one model function each, and their mirror copies in the eight cells of $T_{\ell'}$ are again a cubic entry $\mathcal{S}[\Xi^\kappa]$, $\mathsf{H}(\Xi^\kappa) \leq \mathsf{H}(\Xi)$, and sources $\mathcal{O}(h_{(q,\kappa)})$ with $\|h_{(q,\kappa)}\|_\sharp = \|h_q\|_\sharp$; copies in different cells are ≥ 1 apart. The chessboard bound (Lem. 5.8) carries the cumulant on T_ℓ to the mean on $T_{\ell'}$ of the product of the eight mirror copies, at the price $e^{G_g \mathsf{l}^3}$ for each function, and by the moment–cumulant formula of the copies (an identity of formal power series, as in the proof of Thm. M(g)(ii)) the difference $G_\ell - G_{\ell'}$ reads the joint cumulants on $T_{\ell'}$ of the families B of copies that join different cells alone, the families inside one cell entering nowhere. If each of these is at most $\Gamma \prod_{(i,\kappa)\in B} \omega_i$, where $\Gamma \geq 1$, with one factor ω_0 for each copy of the entry and one factor ω_q for each copy of the source h_q, then

$$|G_\ell - G_{\ell'}| \leq \mathsf{c}_{n'}\, e^{(n'+1)G_g \mathsf{l}^3}\, \Gamma^{(n'+1)/2} \prod_{i=0}^{n'} \omega_i,$$

To see the exponent, use the within-cell normalization and logarithm identities of Lem. 4.26. In a fold of q entries, every remaining block joins cells and has at least two members; a partition of the $8q$ copies has at most $4q$ such blocks. Its bound is at most $\Gamma^{4q} \prod_i \omega_i^8$, and the eighth root gives $\Gamma^{q/2} \prod_i \omega_i$. The subsequent logarithm partitions preserve this exponent, since their subset sizes sum to $q = n' + 1$. This proves the display, including the stated pressure factor. Here $\mathsf{c}_{n'}$ is a pure number. With $u = \lceil \phi K \rceil$, $t = K - u$, the factors $\omega_0 = K^p L^{-c_1 K} \mathsf{H}(\Xi)$ and $\omega_q = K^{2\mathfrak{p}_0} L^{3u} \|h_q\|_\sharp$ serve, in three cases. (i) No copy of the entry: the family holds two copies of sources in different cells, ≥ 1 apart, and the bound eq. 5.i on $T_{\ell'}$ at separated supports applies. (ii) One copy $\mathcal{S}[\Xi^\kappa]$: moving the weight of the entry with the sources writes the cumulant as $-\mathfrak{U}^\sharp[\Xi^\kappa/\mathsf{b}](h)$ plus, when two or more sources are present, the cubic remainders $-\mathfrak{U}^\sharp[h_\alpha \Xi^\kappa/\mathsf{b}^2]((h_{\alpha'})_{\alpha'\neq\alpha})$ (the supports of the h's being disjoint), and with one source the contact member is a mean of $\mathfrak{K}^\sharp$, which vanishes; a source of another cell is ≥ 1 from the entry, and $h_\alpha \Xi^\kappa$ is carried by $\operatorname{supp} h_\alpha$, at sup distance $\geq \min\{\mathsf{d}, 1\}$ from the other supports: the spread condition holds, and Lem. 5.11 on $T_{\ell'}$ at the rate c_1 applies. (iii) Two to eight copies of the entry, in different cells, the sources anywhere: the cumulant is a mixed derivative of the logarithm of the generating function of the chain with these entries and the complex source $w_\zeta = \sum \zeta_\alpha h_\alpha$, at the level t. For case (iii), retain the higher mixed jets as in the proof of [7, Prop. 3.15](b). [7, Prop. 3.13](v) permits the fixed complex source w and the several insertion parameters $\mathbf{s}$ together. Write their small-field generating function as $T_w(\mathbf{s}) = T_w(0) e^{\mathsf{C}_w(\mathbf{s})} \langle e^{\mathsf{D}_{w,\mathbf{s}}} \rangle_w$, with the normalized block-field functional $\langle \cdot \rangle_w$. For q copies of the entry and I with $|I| \geq 2$, locality and the Cauchy estimate in that proof give $\max\{|\partial_I \mathsf{C}_w(0)|, \sup |\partial_I \mathsf{D}_{w,\mathbf{s}}|_{\mathbf{s}=0}\} \leq 2E_0 \mathsf{t}'_\times \prod_{a\in I} (q/r_a)$, where r_a are the insertion radii and $\mathsf{t}'_\times$ is the polymer sum of that proposition. Copies of the entry lie in distinct cells with a fixed positive margin. Their inverse radii are at most $CL^{2K}\mathsf{H}(\Xi^a)$, whereas $\mathsf{t}'_\times \leq CK^p L^{AK} e^{-cL^u}$. Hence the mixed-jet bound is below $\prod_{a\in I} \epsilon_a$ for large K, uniformly on a fixed fraction of the displayed, K-dependent source polydisc. Zero entries are omitted. At real sources the block-field functional is positive; at complex sources it has norm at most 3.1 on the smaller polydisc, by the exponent bound in the proof of Lem. 5.15. The moment and logarithm partition formulas in [7, Prop. 3.15](b) therefore apply in both cases, with finite constants adjusted in the complex case. At a real source x in the polydisc of the radii $(CK^{2\mathfrak{p}_0} L^{3u} \|h_\alpha\|_\sharp)^{-1}$, Lem. 5.10(b), applied to the field $\Xi^\kappa/(\mathsf{b} - w_x)$, makes the conditional mean of each copy given the block field of the level t a number up to $\epsilon_a = C\mathsf{H}(\Xi) K^{p_2} L^{-c'_0 K}$. Subtracting these constants from the first jets, and using the higher mixed-jet bound above, the same partition formulas bound the cumulant by a constant times $\prod_a \epsilon_a$. At a complex source, the norm bound for the normalized functional and the mixed-jet estimate instead give a constant times the product of the crude bounds $\hat{\mathsf{o}}_a = C\mathsf{H}(\Xi) K^{\mathfrak{p}_0} L^{4u+\frac{1}{2}K}$ of

the oscillations of the cubic entries, of the order 3 and of the weight a^{-2}. The estimate with two constants for a function holomorphic on a polydisc of the radii r_α, bounded by B there and by ϵ at its real points, $|\partial_{\zeta_1}\cdots\partial_{\zeta_{n''}}(0)| \le \epsilon^{1-\sigma}B^\sigma(\frac{4n''}{\pi\sigma})^{n''}\prod_\alpha r_\alpha^{-1}$, gives one factor $\epsilon_a^{1-\sigma}\mathsf{b}_a^\sigma \le C\mathsf{H}(\Xi)K^pL^{-c_1K}$ for each copy of the entry, $c_1 = (1-\sigma)c_0' - \sigma(\frac12 + 4\phi)$.

In all three cases the histories with a large field up to the level t cost on $T_{\ell'}$ less than every power of L^{-K}. With $\delta = \frac{1}{14} - c$, $\tau = \delta$, $\phi = \delta/(12(n+1))$ and $\sigma = \frac14\delta$ one has $c_1 - 3n\phi > c$, and the product $\omega_0\prod_q\omega_q$ gives the bound of the difference with the rate c and with numbers independent of the larger side ℓ; on $T_{\ell'}$ itself the bound is Lem. 5.11. □

Lemma 5.17 (The flow with an insertion)**.** Let $\ell = 2L^m$ be any side, $0 < \mathsf{b}_- \le \mathsf{b} \le \mathsf{b}_+$, $\beta = L^K\mathsf{b}$, $n \ge 1$, $\mathsf{d} > 0$ and f_i real Lipschitz; let $S_\varkappa = \sum_l \varkappa(l)\,\mathrm{dn}_l$ be an insertion of brick densities dn_l (gauge invariant polynomials on boxes of at most five sites along each axis, each holding a plaquette, vanishing at $\mathbf{1}$) of the order r, and let two of the $n+1$ sets formed by the carrier of $S_\varkappa$ and the supp f_i be at sup distance $\ge \mathsf{d}$, the others in any position. For γ small enough in terms of the other constants there are C, p, K_0, depending on $\ell, \mathsf{b}_\pm, n, \mathsf{d}$ and the type, such that for $K \ge K_0$:

(a) if $r \ge 3$ and $|\varkappa|_\infty \le \beta h$: $|\langle S_\varkappa; \mathcal{O}(f.)\rangle| \le CK^pL^{-\lambda_L K/4}h\prod_i \|f_i\|_\sharp$; no norm of $\varkappa$ but its supremum enters;

(b) if $r = 2$, $0 \le \sigma \le 1$, $|\varkappa|_\infty \le L^{\sigma K}h$ and $0 < \epsilon \le 1$: $|\langle S_\varkappa; \mathcal{O}(f.)\rangle| \le C_\epsilon K^pL^{-(1-\sigma-\epsilon)K}h\prod_i\|f_i\|_\sharp$; in particular $|\langle\mathcal{O}(w); \mathcal{O}(f.)\rangle| \le C_\epsilon a^{-\epsilon}\max|w|\prod_i\|f_i\|_\sharp$ for every real weight w, in contact with the f_i or not;

(c) if $r = 2$, the insertion is real with $|\varkappa|_\infty \le \beta h$, and its densities are of derivative kind, that is, $|G_\nu(b)|^2$ or $\Psi_{\lambda(p)}(p)$: $|\langle S_\varkappa; \mathcal{O}(f.)\rangle| \le CK^pL^{-c_nK}h\prod_i\|f_i\|_\sharp$, $c_n = \lambda_L/(4(6+3n))$; for every $0 < \theta_1 < 1$ and every real insertion of the order 2 with $|\varkappa|_\infty \le \beta h$, $\langle S_\varkappa - J_{t_1}; \mathcal{O}(f.)\rangle$ obeys the same bound with $L^{-\lambda_L\theta_1K/4}$, the numbers C, p, K_0 depending besides on θ_1, where $t_1 = K - \lceil\theta_1 K\rceil$ and J_{t_1} is the classical value of the insertion at the background of the level t_1, a function of the block field.

■

Proof sketch. Run the chain with the insertion, as in Lem. 5.10(a), down to a level t at which the large fields are rare. A transfer inequality bounds the cumulant by

$$2\delta_t(\varkappa)\prod_i(8nM_t\|f_i\|_\sharp) + \mathrm{Con}_t + 4\,\mathrm{op}_{n+1}\mathsf{R}_t\sup|S_\varkappa|\prod_i(6\ell^3L^{4K}\sup|f_i|),$$

with δ_t the oscillation of the conditional mean of the insertion given the block field of the level t, M_t the inverse radius of the sources at that level, and Con_t the mixed derivative of the constants of the chain, which is small by locality when a pair of sets is separated (when the separated pair is the insertion and one source, the constants of the sources that meet the insertion are handled by a slow cut-off at the scale d, at the price $(2 + 480M/\mathsf{d})^{n-1}$). By power counting an insertion of the order r has the size $|\varkappa|_\infty L^{(3-2r)i}$ at the level i, and what the steps make of it is forgotten at the rate $L^{-\lambda_L/2}$ per level, $(L/2)^{-x} = L^{-\lambda_L x}$; this gives (a) and (b).

For (c) two levels are used. Let $0 < \theta_1 < 1$, $\tau_1 = \lceil\theta_1 K\rceil$, $t_1 = K - \tau_1$, and $t = K - \tau$, $\tau = \lceil\theta K\rceil$, $\theta = \lambda_L\theta_1/(4(7+3n)) < \theta_1$. The insertion is put into the chain only up to the level t_1, as early members; there it equals its classical value J_{t_1} on the background of the level t_1 plus what the steps up to t_1 made of it, which in level-t_1 terms has the size $C|\varkappa|_\infty L^{-(t_1-1)}L^{-\lambda_L\tau_1/2}$ (what a step makes of an insertion of the order 2 is forgotten at the rate $L^{-\lambda_L/2}$ per level). From the level t_1 to the level t this part is forgotten by the factor $L^{-(t-t_1)}$ of the run, on the field cut of the level t; at the level t the histories with a large field are discarded, at a cost below every power of L^{-K}, and the volume $N_t = \ell^3L^{3\tau}$ and Cauchy's estimates in the n sources cost $L^{3(n+1)\tau}$ up to powers of K; the mixed derivatives of the constants of the chain in the sources are small by locality, two of the sets being separated.

The derivative in the insertion is taken as a difference quotient at a small step s, its second derivative being bounded crudely. Together the exponent is $(7+3n)\tau+1-\frac{1}{2}\lambda_L\tau_1 \le -\frac{1}{4}\lambda_L\theta_1 K+8+3n$: this is the second clause of (c), for every θ_1. For the first clause the classical part is bounded besides: J_{t_1} is a function of the block field of the level t_1, its cumulant with the sources is bounded by Cauchy's estimates in the sources at that level, at the price $L^{3n\tau_1}$, times its supremum, and for densities of derivative kind the supremum is at most $\beta\ell^3 L^{3K}\, C\varepsilon_{t_1}^2 (1+K\log L)^2 L^{-6t_1}$ by the estimate below; the exponent is $K+3K-6(K-\tau_1)+3n\tau_1 = -2K+(6+3n)\tau_1$, which at $\theta_1 = 1/(6+3n)$ is at most $-K+6+3n$.

With the second clause at this θ_1 the rate is $\min\{1, \frac{1}{4}\lambda_L\theta_1\} = c_n$. For a lattice field B on a box of radius R with $|B| \le A_0$ and $|\mathrm{div}^+ B|, |\mathrm{curl}^- B| \le \epsilon_1$, the div–curl estimate $|\partial_j^{\pm} B_i| \le C_*[A_0/R + \epsilon_1(1+\log R)]$ on the quarter box, applied with the Bianchi identity on a cube, gives $|G_\nu(b)|^2, |\Psi_{\lambda(p)}(p)| \le C\varepsilon_i^2\eta_i^6(1+i\log L)^2$ at the regular backgrounds of the level i, $\eta_i = L^{-i}$, which is the order that (c) needs. □

Proposition 5.18 (The mean plaquette)**.** Let $0 < g \le \gamma$, $\ell \in \mathfrak{L}(g)$, $n \ge 1$, and $E(\beta) = \langle a_p \rangle$ on a periodic lattice of N_s sites along each axis (it does not depend on p).

(a) For every $R \ge 3$, $N_s \ge R$ and β with $116R^3(\beta/2)^{-1/2} \le \frac{1}{5}$ and $\bar\eta(R,\beta) \le \frac{1}{4}$, $|\beta E(\beta) - 1| \le 2\bar\eta(R,\beta)^{1/2}$, with $\bar\eta(R,\beta) \le \frac{9\log(2\beta)+55}{R^3} + \frac{18\log(1+256R^3)+15}{R} + 1.21\cdot 10^9 R^9 \beta^{-1/2}$: at $R = \lceil \log\beta \rceil$ this tends to 0 uniformly in all volumes $N_s \ge \log\beta$.

(b) On the side ℓ, with $N_s = \ell L^K$: there are C, K_0, depending on g, ℓ, n, such that $\sup_{\mathsf{b}' \in I(g)} |\beta^n E^{(n-1)}(\beta) - (-1)^{n-1}(n-1)!| \le C K^{2n+3} L^{-K/20}$ for $\beta = L^K \mathsf{b}'$ and $K \ge K_0$.

(c) For all real Lipschitz $f_1, \dots, f_n$ on T_ℓ, with nothing asked of the supports, $|a_K^3 S^T_{n;K,\ell,L^K\mathsf{b}'}(f) - 3(n-1)!\, \mathsf{b}'^{-n} \int_{T_\ell} \prod_i f_i| \le C K^{2n+3} L^{-K/20} \prod_i \max\{\sup|f_i|, \frac{40}{\sqrt{3}} M \mathrm{Lip}_\infty f_i\}$.

■

Proof sketch. (a) Cut the lattice into blocks of R to $2R$ sites along each axis, and in each block take a comb tree. Bounding the partition function with plaquette weights $\beta_p \in [\beta_-, \beta_+]$ from above (dropping a few plaquettes) and from below (freezing a few bonds), both bounds become the same Gaussian integral, $\sum_B (2|B|-2)\log \mathsf{g}(\beta_B) - \mathsf{D}$ with $\mathsf{g}(\beta) = (2/\pi)^{1/2}\beta^{-3/2}$ and a number D of the geometry, up to an error $N\eta$ that is small per site; this uses weak coupling on blocks of $R \le \beta^{1/20}$ spacings alone. The convexity of $\log Z$ in β then pins down its first derivative. (b) For the higher derivatives analyticity in β is needed.

On a torus of a fixed physical side, Thms. YM3_I:AJ and YM3_I:AD write $Z[\zeta f] = \exp(E + \sum_{j,X} e_j(X;\zeta f))\widetilde{Z}(\zeta f)$ for real f with $\max\{\sup|f|, \frac{40}{\sqrt{3}} M\mathrm{Lip}_\infty f\} \le 1$, with $|e_j(X;w)| \le C_{\mathrm{S}} e^{-\kappa d_j(X)}$, $|\widetilde{Z}| \le e^{C_{\mathrm{S}}\ell^3}$ and $\widetilde{Z}(0) \ge e^{-C_g\ell^3}$; the factor $\widetilde{Z}$ may vanish. Put $N = (\ell L^K)^3$, $h_f(\zeta) = \widetilde{Z}(\zeta f)/\widetilde{Z}(0)$ and $D_f(\zeta) = (3N)^{-1}[\sum_{j,X} e_j(X;\zeta f) + \sum_p \log(1 - \zeta f(x(p))/\mathsf{b}')]$, the per-site exponent; by the count of polymers $|D_f| \le M_0$ on a disc $|\zeta| < \rho_1$, with M_0 and ρ_1 free of K. Then $h_f(0) = 1$, $h_f > 0$ on the real axis and $|h_f| \le e^\Gamma$ with $\Gamma = (C_{\mathrm{S}} + C_g)\ell^3$, a number free of K.

First, Cauchy's estimate on circles of radius $\frac{1}{2}\rho_1$ gives $|h_f - 1| \le \frac{1}{2}$ on $|\zeta| \le \frac{1}{4}\rho_1 e^{-\Gamma}$, so that $\log h_f$ is defined there and $|(\log h_f)^{(n)}(0)| \le n!\,(4e^\Gamma/\rho_1)^n$: this member does not grow with the cutoff, and it is of the order $1/N$ per site. Second, the Poisson–Jensen formula and Harnack's inequality, applied to the non-negative function $\Gamma - \log|h_f|$, give $\int_{-\rho_1/2}^{\rho_1/2} |\log|h_f(x)||\,\mathrm{d}x \le 9\rho_1\Gamma$; with this, the two-sided real-axis bound of (a), at the plaquette weights $\beta(1 - x f(x(p))/\mathsf{b}')$, controls the per-site exponent in L^1 on the segment: $\int_{-\rho_1/2}^{\rho_1/2} |D_f(x)|\,\mathrm{d}x \le \rho_1\delta$ with $\delta = \frac{1}{3}(2\bar\eta(R,\beta) + Ra_K + R^{-3}) + 3(C_{\mathrm{S}} + C_g)L^{-3K}$.

Third, A. Markov's inequality for the derivatives of the Taylor polynomial of D_f of the order $\mathsf{m} = \max\{n, \lceil \log_2(M_0/\delta)\rceil\}$, with Cauchy's estimate for the rest of the series, gives $|D_f^{(n)}(0)| \le 4\,(2/\rho_1)^n \mathsf{m}^{2n}(\mathsf{m}+1)^2\delta$, and at $R = \lfloor \beta^{1/20} \rfloor$ this is the rate. By Jensen's formula h_f has at most $\Gamma/\log 2$ zeros in $|\zeta| < \frac{1}{2}\rho_1$, a number that does not grow with the cutoff; the bound on

$(\log h_f)^{(n)}(0)$ above does not need it. At $f \equiv 1$, where $S_n^T(1,\dots,1) = a_K^{-n}(-1)^{n-1}3N\,E^{(n-1)}(\beta)$, the identity of (c) below gives (b). (c) is the same argument at a general f: $S_n^T(f,\dots,f) = (n-1)!\mathfrak{b}'^{-n}\sum_p f(x(p))^n + 3ND_f^{(n)}(0) + (\log h_f)^{(n)}(0)$, with the per-site exponent D_f and the factor h_f of Thms. YM3_I:AJ and YM3_I:AD; polarisation gives distinct entries. □

Use the separated regions and the actual averaging constraints of Prop. 5.37, with its block-number and small-triple hypotheses in addition to the width condition below. Require also the geometric and regularity hypotheses of Thm. YM3_I:I and Prop. YM3_I:29.1, with their block-number, torus-size and gap floors. The matrix-valued plaquette-divergence bound in that theorem follows from the regular curvature and current bounds by equation YM3_I:(579), with a factor 2 absorbed in the regularity ceiling. These are numerical floors and small-radius conditions on the data, not assumptions about the inverse estimate to be proved. Also retain the four local-system radii at most a_H, the block floor $\mathfrak{M} \geq M_H$, and the permitted decay rate of Lem. 5.34. Write $\xi = \eta_k$ and $\ell_n = \xi L^n$ for the physical layer scale. For clarity distinguish D_0, the covariant differential from sites to bonds, from D_1, the covariant differential from bonds to plaquettes. Let U be a real background, critical for the unweighted action on the kernel of the averaging constraint, regular at radius $\alpha \leq \alpha_{\mathrm{lin}}(L) \leq (16L^2)^{-1}$, with curvature bounded by the envelope $e(x) = \varepsilon_{j(x)}$ in the scaled norm below, and $\sup e \leq e_0(L)$. The layer radii satisfy $0 < \varepsilon_j \leq \varepsilon_{j+1} \leq L\varepsilon_j$. Require $R_1\mathfrak{M} \geq w_{\mathrm{lin}}(L)$.

At each bare site x_* of layer $n \geq 2$, fix a real adjoint gauge g_{x_*} on the cube of physical sup radius $\ell_n/4$ such that

$$|g_{x_*}(b_-)\,\mathrm{Ad}\,U(b)g_{x_*}(b_+)^{-1} - I| \leq L^2\alpha/L^n.$$

The gauge hypotheses are part of the setting. All Hölder quotients below compare components of the same orientation in these gauges. The cube for a potential is the concentric cube of radius $\ell_n/8 - 2\xi$; those for a curvature and a source two-form have radius $\ell_n/4$. Quotients are taken over distinct admissible base points in the indicated cube, with the usual physical Euclidean distance.

We use the following scaled spaces. For a field of inverse length degree r, put

$$|f|_{r,h} = \sup_i \frac{\ell(i)^r|f(i)|}{h(x_i)}, \qquad [f]_{r,1/2,h} = \sup_{n\geq 2,x_*} \frac{\ell_n^{r+1/2}}{h(x_*)}[g_{x_*}f]_{1/2;C(x_*)}.$$

For an index i, let $\ell(i)$ be the largest layer scale of the sites in the stencil defining that field value or first difference. For a row use its prescribed scale ξ_c. The envelope is read at the indicated base point x_i. This fixes the norm convention also for indices meeting a layer boundary. The adjacent-layer comparison of Lem. YM3_I:21.2 makes it equivalent, with constants depending only on L, to the base-point scale convention. Set $\|f\|_{r,1/2,h} = |f|_{r,h} + [f]_{r,1/2,h}$. For the first two layers only the supremum term is required; their ratios $\ell_n/\xi \leq L$ are uniformly bounded. The curvature hypothesis above means $\|F_U\|_{2,1/2,e} \leq 1$.

The row units are dimensionless. If Q_{ph} is the physical averaging derivative with row pairing of weight ξ_c^3, put

$$QA(c) = \xi_c Q_{\mathrm{ph}}A(c), \qquad \omega(c) = \xi_c^3\omega_{\mathrm{ph}}(c).$$

Use the row pairing of weight ξ_c^{-1} for Q^{T}. Then $Q^{\mathsf{T}}\omega = Q_{\mathrm{ph}}^{\mathsf{T}}\omega_{\mathrm{ph}}$, since both pairings with A are $\sum_c \frac{1}{2}\,\mathrm{tr}(Q_{\mathrm{ph}}A(c)\omega(c))$. This conversion changes units only; it retains the same row set and averaging germ. Row data b and multipliers have degree zero.

Let $h = e\theta$, where $\theta > 0$ and $|\log\theta(x) - \log\theta(x')| \leq c_U d(x,x')$. Choose c_U at most one eighth of each decay rate used below and at most $1/(6L)$. Thus, for fixed constants $C_h = C_h(L)$ and $0 < \nu \leq \delta_2/2$,

$$h(z)/h(y) \leq C_h e^{\nu d(y,z)}. \tag{5.viii}$$

Take the same condition on the local comparison cubes. Define

$$\|(A,\omega)\|_{\mathcal{X}_h} = \|A\|_{1,1/2,h} + \|D_1 A\|_{2,1/2,h} + |D_0^{\mathsf{T}} A|_{2,h} + |\omega|_{0,h},$$
$$\|(Y,f,b)\|_{\mathcal{F}_h} = \|Y\|_{2,1/2,h} + |f|_{3,h} + |b|_{0,h}.$$

The ordinary and divergence sources are read only through $f + D_1^{\mathsf{T}} Y$. For a complex anisotropy an, its size $t(\mathrm{an})$ is its supremum plus the scaled Hölder-1/2 seminorm on the same plaquette cubes.

Lemma 5.19 (Linear response at a variable anisotropy)**.** There are positive numbers $w_{\mathrm{lin}}, \alpha_{\mathrm{lin}}, e_0$, $t_{\mathrm{lin}}, K_{\mathrm{lin}}$, depending only on L, such that the following holds in the setting above. If $t(\mathrm{an}) \le t_{\mathrm{lin}} \le 1/4$, the system

$$QA = b, \qquad T_\tau A = f + D_1^{\mathsf{T}} Y + Q^{\mathsf{T}} \omega, \qquad \tau = 1 + \mathrm{an}, \tag{5.ix}$$

has a unique solution, linear in the data, with

$$\|(A,\omega)\|_{\mathcal{X}_h} \le K_{\mathrm{lin}} \|(Y,f,b)\|_{\mathcal{F}_h}. \tag{5.x}$$

Here $T_\tau = D_1^{\mathsf{T}} \tau c D_1 + V_0' + D_0 R D_0^{\mathsf{T}}$ is the second variation of the weighted action at the reference field, with the slice term. The coefficient c_p is determined by $\partial U(p) = c_p I + i\xi^2 F_U(p)$, and V_0' contains the curvature commutator terms of that second variation. The nonlinear block-constraint remainder and its multiplier are kept in the forcing when the lemma is applied to the nonlinear equation. The row and slice conventions are those of the setting. All constants are uniform in the volume, the number of layers and the permitted envelopes. For the mean-scheme rows of a label, rotated into the comb frame, the same estimate holds with $2K_{\mathrm{lin}}$. ■

Proof of the linear estimate. We first treat the unweighted operator. The block constraint must be retained in this argument: the inverse of the gauge-fixed operator without the constraint is not the required Green operator. Put

$$T = T_\tau\big|_{\mathrm{an}=0}, \qquad G = (T + Q^{\mathsf{T}} Q)^{-1}, \qquad M = QGQ^{\mathsf{T}},$$

and define

$$H = GQ^{\mathsf{T}} M^{-1}, \qquad \widetilde{G} = G - HQG. \tag{5.xi}$$

Thus $QH = 1$, $Q\widetilde{G} = 0$ and $\widetilde{G}T = 1 - HQ$. The rows are the interior and outward determining bonds; inward bonds are discarded, so there are no edge rows. On a row of level j, the derivative of the iterated comb average is

$$QA(c) = \xi \left(\mathbf{Q}(W_{j-1}) \cdots \mathbf{Q}(W_0) A\right)(c).$$

The factors read only the double block beneath that row. Separation keeps these blocks inside the domains on which the averages are defined. The row conversion above therefore identifies this Q with the physical averaging derivative in the regular constrained system, without changing the set of constraints.

The regular current bound supplies the matrix-divergence hypothesis used for that system. Indeed, at a full bond the quaternion decomposition gives

$$D_U^{\mathsf{T}} \partial U = aI + i\xi^2 J(U), \qquad |a| \le \xi^{-1} \max_{p \ni b} |\partial U(p) - I|^2.$$

Since the adjacent plaquette scale is at least the bond scale and the latter is at least ξ, the regularity bounds give $|D_U^{\mathsf{T}} \partial U| \le 2\alpha \xi^2 \ell(b)^{-3}$. This is equation YM3_I:(579); our choice $2\alpha_{\mathrm{lin}} \le a_Q$ permits the application below.

Here are the unweighted bounds that will be used. Write $|f|_{(r)} = \sup_i \ell(i)^r |f(i)|$ and $\|A\|_X = \max\{|A|_{(1)}, |\nabla A|_{(2)}\}$. For sufficiently small regularity radius, there are $K_1, K_7 \ge 1$, depending only on L, such that

$$\begin{gathered} \|\widetilde{G} f\|_X \le K_7 |f|_{(3)}, \qquad \|Hb\|_X \le K_1 \sup|b|, \qquad \sup|H^{\mathsf{T}} f| \le K_1 |f|_{(3)}, \\ \sup|(D_1 H)^{\mathsf{T}} Y| \le 3K_1 |Y|_{(2)}, \qquad \sup|(D_0^{\mathsf{T}} H)^{\mathsf{T}} \lambda| \le 3K_1 |\lambda|_{(2)}, \qquad |R\lambda|_{(2)} \le (1 + K_1^3) |\lambda|_{(2)}. \end{gathered} \tag{5.xii}$$

These estimates are unchanged by passing to the units of any layer. The averaging rows are dimensionless, their pairing has weight $\ell(c)^{-1}$, and their transposes are taken for that pairing. They are precisely the bounds of Prop. YM3_I:29.1, applied after the row identification just made. Its proof also records their exponential versions. For the transpose estimates, test against a datum supported at one row c. The row pairing contributes a factor $\ell(c)$, while the number of bonds or plaquettes in a cell of scale ℓ_y is at most $3(\ell_y/\xi)^3$. The resulting kernel sum has factor $\ell(c)/\ell_y$, which is absorbed by the exponential decay. The same calculation with the first differences gives the fourth and fifth estimates, including the zero-extension terms at the boundary. These are equations YM3_I:(459) and the following divergence estimate. The scalar estimate follows from $R = 1 - G'Q'^{\mathsf{T}}(Q'G'^2Q'^{\mathsf{T}})^{-1}Q'G'$. At a bond meeting the complement of the ambient region, the zero-extension difference has local scale ξ and is bounded by $|A|_{(1)}$. Thus these estimates also include boundary differences. For an empty ambient region all spaces are zero. The operators preserve real fields, so the estimates extend to complex fields by their real and imaginary parts, with constants enlarged by a fixed factor.

The same operators have exponential decay in the weighted distance. Their first-difference Hölder estimates follow from the ordinary kernel estimates in Lem. 5.34. Multiplying the input at z and output at y by the envelope ratio costs $h(z)/h(y)$. The permitted growth of this ratio consumes only part of the exponential decay. The remaining kernel is summable by Lem. YM3_I:21.2. This gives the envelope versions of eq. 5.xii, with constants independent of the torus and the number of layers.

For $q = f + D_1^{\mathsf{T}}Y$, elimination of the multiplier in $TA = q + Q^{\mathsf{T}}\omega$, $QA = b$, gives the exact formulas

$$A = \widetilde{G}q + Hb, \qquad \omega = H^{\mathsf{T}}(THb - q). \tag{5.xiii}$$

Indeed $(T + Q^{\mathsf{T}}Q)A = q + Q^{\mathsf{T}}(\omega + b)$, and applying QG determines $\omega + b$. The identity $M^{-1} = H^{\mathsf{T}}TH + 1$ gives the displayed formula for ω. The six bounds control the ordinary forcing, the lift, and all multiplier terms. The remaining issue is the first-slot divergence $D_1^{\mathsf{T}}Y$.

Cover each layer of index at least two by cubes of size comparable to its layer scale. In the local gauges, write $U = e^{i\xi a}$, with a and its first differences bounded in the units of that layer. A partition of unity reduces the divergence source to the flat equation on these cubes. Convolution with the lattice Laplace Green kernel gives a local potential; its first differences involve second differences of that kernel. These preserve the Hölder exponent by Lem. YM3_I:19.4. The cancellation in that lemma is essential: a bound on the absolute kernel alone would give a logarithmic loss.

Cut off these local potentials and carry them back to the original gauge. Their sum Φ satisfies

$$T\Phi = D_1^{\mathsf{T}}Y + J, \qquad |J|_{3,h} \le C\|Y\|_{2,1/2,h}.$$

Here J consists of cutoff commutators, the terms containing the local potential and curvature, the lower-order term $V_0'\Phi$, the term $-D_0(1-R)D_0^{\mathsf{T}}\Phi$, and the untreated sources in layers zero and one. The first differences of the local gauge potential bound the coefficient terms; the cutoff derivatives have the inverse powers of the local scale. In the first two layers the scale-to-spacing ratio is bounded by L, so a bare difference is bounded by the corresponding scaled supremum. These are ordinary sources of degree three. The parametrix identity is

$$\widetilde{G}D_1^{\mathsf{T}}Y = \Phi - HQ\Phi - \widetilde{G}J. \tag{5.xiv}$$

The row datum $Q\Phi$ is bounded in the dimensionless row norm: each average reads a double block on which the scale and envelope are comparable, and the factor $\ell(c)$ in the physical path average cancels the scale ℓ^{-1} of Φ. The remainder J has the degree-three norm displayed above. Thus both correction terms lie in the ordinary-input classes of Lem. 5.34. Its estimates are read at $s = 1$, with the same rows and their row pairing. This proves the required potential and curvature bounds. The first-difference Hölder bounds for the correction terms are the ordinary kernel bounds; those for Φ are the same-exponent Green-kernel bounds just used. Their supremum estimates also bound the divergence. To pass from the comparison cubes of radius $\ell/56$ to the cubes in the norm, cover

the latter by comparison cubes and use the local gauge transition bounds. On an overlap those transitions have differences bounded by $C(L)\alpha/\ell$; the Hölder product inequality and the supremum bound therefore control the change of gauge. The bounded overlap gives a constant depending only on L. No Hölder estimate is obtained here by applying an estimate with a positive loss of exponent at zero loss.

For completeness, the potential Hölder norm can also be read directly from its curvature and divergence. On a cube C_R in bare units, let the adjoint transports differ from the identity by at most v/R. If $6v \le \sigma_H$, then

$$\sup_{C_{R/2-2}} |E| + (R/2-2)^{1/2}[E]_{1/2;C_{R/2-2}} \le 2C_H\big((R-1)^{-3/2}\|E\|_2 + R|DE| + R|D_0^* E|\big).$$

To obtain it, apply Prop. YM3_I:18.1 to each component, whose flat Laplacian is the divergence of its flat curvature and divergence. Replacing the covariant quantities by the flat ones adds at most Cv times the supremum on the next larger cube. On radii increasing from $3R/4$ by $R/8, R/16, \dots$, choose v so that this factor is at most $1/12$. Iteration absorbs the supremum; the terminal error is at most $12^{8-\log_2 R}\|E\|_2$. A second application on cubes of radius $R/4$ gives the Hölder bound. For $R < 64$ the supremum is bounded by the finite-volume ℓ^2 norm. At $R = L^n/4$, the condition follows by reducing the regularity radius so that $\alpha \le \sigma_H/(8L^2)$.

We have proved the estimate at zero anisotropy with a constant $K^{(0)}$. At a nonzero anisotropy, the same solution solves the unweighted system with data

$$(Y - \operatorname{an} cD_1 A,\ f - V_0'[\operatorname{an}]A,\ b).$$

The Hölder product inequality bounds the first change by $Ct(\operatorname{an})\|D_1 A\|_{2,1/2,h}$; the second is bounded by $C|\operatorname{an}|_\infty \alpha |A|_{1,h}$. Choose $K^{(0)} C t_{\mathrm{lin}} \le 1/2$ and absorb these terms. The bound holds with $K_{\mathrm{lin}} = 2K^{(0)}$.

The real part of the form of T_τ on $N(Q)$ is coercive at these radii, as proved below. Hence a homogeneous solution has $A = 0$, by pairing its equation with $\bar{A}$, and then $Q^{\mathsf{T}}\omega = 0$ gives $\omega = 0$. The finite square system is therefore invertible. This proves existence and uniqueness as well as the estimate.

Finally, the mean-scheme rows of a label, rotated into the comb frame, have the form $Q + E$. Their comparison kernel is supported in the double block of the row and obeys

$$|EA|_{0,h} \le c_E \alpha |A|_{1,h}, \qquad |E^{\mathsf{T}}\omega|_{3,h} \le c_E \alpha |\omega|_{0,h}.$$

The system with these rows is the one just solved with data $(Y, f + E^{\mathsf{T}}\omega, b - EA)$. Taking $K_{\mathrm{lin}} c_E \alpha \le 1/2$ proves its bound, with $2K_{\mathrm{lin}}$, by a contraction. This is the row comparison needed when the estimate is applied to a label. □

Lemma 5.20 (The energy of the displacement)**.** In the setting of Lem. 5.19, decrease the regularity radius and the envelope ceiling by constants depending only on L. Put

$$\mathcal{E}(A) = \xi^3 \sum_p |D_1 A(p)|^2 + \xi^3 \sum_b \ell(b)^{-2}|A(b)|^2, \qquad |b|^2_{\mathrm{row}} = \sum_c \ell(c)^{-1}|b(c)|^2.$$

For the linear solution with data (Y, f, b),

$$\mathcal{E}(A) \le K^{(2)} \left(\xi^3 \sum_p |Y(p)|^2 + \xi^3 \sum_b \ell(b)^2 |f(b)|^2 + |b|^2_{\mathrm{row}} \right). \tag{5.xv}$$

For the small fixed point with zero block datum,

$$\mathcal{E}(A[\operatorname{an}; 0]) \le C|\operatorname{an}|^2_\infty \xi^3 \sum_p |F_U(p)|^2. \tag{5.xvi}$$

At a permitted complex block datum b_0, the difference $A' = A[\mathrm{an}; b_0] - A[0; b_0]$ instead satisfies

$$\mathcal{E}(A') \le C|\mathrm{an}|_\infty^2 \left(\xi^3 \sum_p |F_U(p)|^2 + |b_0|_{\mathrm{row}}^2 \right). \tag{5.xvii}$$

The constants are independent of the number of layers and the volume. The last two statements concern fixed points in the small ball of the linear estimate. ∎

Proof. We need an estimate on the field energy, including the forcing from the nonlinear block constraint. A contraction in a stronger energy norm of the field and multiplier is unnecessary.

At zero anisotropy the operator $T + Q^{\mathsf{T}}Q$ is real, symmetric and positive by Thm. YM3_I:I. Its restriction to $N(Q)$ is T, so the constrained Green operator is positive on that subspace. Apply the first estimate of eq. 5.xii to an eigenvector of the largest eigenvalue of $\ell^{-1}\widetilde{G}\ell^{-1}$. If that eigenvalue is λ, it gives

$$\lambda \sup_b \ell(b)^2 |g(b)| \le K_7 \sup_b \ell(b)^2 |g(b)|.$$

Thus the quadratic form of T on $N(Q)$ controls the weighted potential square sum with constant K_7^{-1}. Its expression as a curvature form, a slice term and V_0' gives the second lower bound. The plaquette coefficient satisfies $c_p \ge 1/2$, while the form of V_0' is bounded by $C_V\alpha$ times the weighted potential square sum. Since $|\mathrm{an}|_\infty \le 1/4$, reducing α so that $3C_V K_7 \alpha \le 1$ gives

$$\mathrm{Re}(\bar{A}, T_\tau A)_\xi \ge c^{(2)} \mathcal{E}(A) + \tfrac{1}{2}\|RD_0^* A\|_\xi^2, \qquad A \in N(Q), \tag{5.xviii}$$

where $c^{(2)} = \min\{1/8, (4K_7)^{-1}\}$. Pairing the equation with $\bar{A}$ removes the multiplier. Cauchy–Schwarz bounds the remaining pairing by $\mathcal{E}(A)^{1/2}$ times the square root of the first two data sums in eq. 5.xv.

For nonzero row data use the lift $A^\circ = Hb$. Then $A - A^\circ \in N(Q)$ and its equation has the additional forcing $-T_\tau A^\circ$. The bounds on H, its first differences and their transposes in eq. 5.xii, applied by the weighted Schur test, give

$$\mathcal{E}(A^\circ) + \|D_0^* A^\circ\|_\xi^2 \le C_H |b|_{\mathrm{row}}^2.$$

The row weight is $\ell(c)^{-1}$; using an unweighted row sum would not give a constant uniform in the layers. Apply eq. 5.xviii to $A - A^\circ$ and use this lift estimate. This proves eq. 5.xv.

At the nonlinear fixed point the data are

$$(-\mathrm{an}F_U - Y_3(A),\ -f_3(A) + (dC_2(U,A))^{\mathsf{T}}(\omega_0 + \omega),\ -C_2(U,A)).$$

Here Y_3, f_3 are the terms of order at least two in the displacement in the first variation, and C_2 is the nonlinear remainder of the block constraint. The last entry is not zero. The small-ball bound on A and the multiplier implies that the square-sum norm of the nonlinear data is at most $c_{\mathrm{ms}} \sup e\, \mathcal{E}(A)^{1/2}$. To see the row estimate in the same units, write $l(c) = \ell(c)/\xi$. The Taylor kernel of the block average and Cauchy–Schwarz give

$$|C_2(U,A)(c)| \le C l(c)^{-1} \sum_{b \text{ beneath } c} |\xi A(b)|^2.$$

There are at most $Cl(c)^3$ bonds beneath the row, their scales are comparable to $l(c)$, and each bond lies beneath at most a constant number of rows. Hence the row square sum is bounded by $C(\sup_b \ell(b)|A(b)|)^2 \mathcal{E}(A)$. The same bilinear kernel bounds the transposed derivative paired with $\omega_0 + \omega$, whose scaled supremum is at most $C \sup e$. The plaquette and bond remainders are treated by their local product bounds, using one small factor and one energy factor.

Choose the envelope ceiling so that $K^{(2)} c_{\mathrm{ms}}^2 (\sup e)^2 \le 1/16$. The nonlinear data can then be absorbed in eq. 5.xv. The remaining forcing is $-\mathrm{an}F_U$, which proves eq. 5.xvi.

At $\mathrm{an} = 0$ and datum b_0 the same argument gives $\mathcal{E}(A[0; b_0]) \le 4K^{(2)}|b_0|_{\mathrm{row}}^2$. Subtract the two fixed-point equations at the same b_0. The terms linear in an are bounded by its supremum times

the square root of $\xi^3 \sum |F_U|^2 + \mathcal{E}(A[0;b_0])$. The remaining differences have the factor $C \sup e$; the multiplier difference is bounded by testing against the same row lift. One further reduction of the envelope ceiling absorbs these differences. This proves eq. 5.xvii. □

Lemma 5.21 (Complex scalar gauge transfer)**.** Use the regular-reference setting of Lem. 5.19. After decreasing its envelope ceiling and the admitted radii by constants depending only on L, the following estimates hold. Consider a complex potential a in a sufficiently small ball for $\mathcal{X}_e$ and the small gauge $e^{i\lambda}$ relative to its real reference configuration. Define N and q by the exact identities

$$D^* a^{e^{i\lambda}} = D^* a - D^* D\lambda + N(a,\lambda),$$
$$T^{-1} i^{-1} \log \mathcal{R}(a; e^{i\lambda}) = Q'\lambda + q(a,\lambda).$$

Here $\mathcal{R}$ is the induced gauge at the retained block tops, and T is the orthogonal transport in its linearization; these are the maps in the proof of Lem. YM3_I:30.4. Their linear scalar system is precisely the compression $\mathbb{K}^{[2]}$ of Section YM3_I:23.2, in the covariant form of Section YM3_I:25:

$$\mathbb{K}^{[2]}(\lambda, w, \sigma) = (\Delta'_a \lambda + w,\ \Delta'_a w - Q'^* \sigma,\ Q'\lambda), \qquad \Delta'_a = \Delta_D + Q'^* \mathsf{a}' Q'.$$

Here Δ_D is the scalar Dirichlet Laplacian; the boundary identity is $RD^*D = R\Delta_D$. Let $\mathcal{S}(f,b)$ be the first component of the solution with data $(f + Q'^* \mathsf{a}' b, 0, b)$. Its first and third rows give $w = f - \Delta_D \lambda$, while its second row says $Rw = 0$. Consequently the two desired gauge conditions are exactly the fixed point $\lambda = \mathcal{S}(D^* a + N(a,\lambda), -q(a,\lambda))$. The mass term $Q'^* \mathsf{a}' b$ is necessary when $b \neq 0$. Put $\|\lambda\|_{\mathcal{Z}_h} := |\lambda|_{0,h} + \|(D\lambda, 0)\|_{\mathcal{X}_h}$. For $\lambda, \tilde{\lambda}$ in one common small ball for $\mathcal{Z}_e$, and every admitted envelope h,

$$\begin{aligned} \|\mathcal{S}(N(a,\lambda) - N(a,\tilde{\lambda}), -q(a,\lambda) + q(a,\tilde{\lambda}))\|_{\mathcal{Z}_h} &\le \tfrac{1}{2} \|\lambda - \tilde{\lambda}\|_{\mathcal{Z}_h}, \\ \|(a^{e^{i\lambda}} - a, 0)\|_{\mathcal{X}_h} &\le C \|\lambda\|_{\mathcal{Z}_h}. \end{aligned} \tag{5.xix}$$

The scalar fixed-point map preserves a sufficiently small common complex ball for $\mathcal{Z}_e$. The estimates also hold for differences about a given solution and for a cutoff repair with its commutator forcing, in the exponentially weighted envelopes. All constants are independent of the volume and number of layers. ■

Proof. The divergence remainder. The divergence remainder has a uniform complex estimate. Let a, λ and $\widetilde{\lambda}$ lie in one sufficiently small ball for the unlocalized norms $\mathcal{X}_e$, $\mathcal{Z}_e$, and let the sum of their norms be at most r. Then

$$|N(a,\lambda) - N(a,\widetilde{\lambda})|_{2,h} \le C(L) r \sup e \, \|\lambda - \widetilde{\lambda}\|_{\mathcal{Z}_h}. \tag{5.xx}$$

Here a localized norm measures the difference; the other factor in each product is bounded on the common unlocalized ball.

For completeness, put $k(l) = (e^{i \operatorname{ad} l} - 1)/(i \operatorname{ad} l)$, with its value at zero defined by the power series. At a site x, use the outgoing and incoming bond logarithms

$$\Psi_{\mathrm{out}}(l;u,d) = i^{-1} \log(e^{il} e^{iu} e^{-i(l+d)}), \qquad \Psi_{\mathrm{in}}(l;u,d) = i^{-1} \log(e^{i(l-d)} e^{iu} e^{-il}).$$

On a fixed complex ball these are holomorphic and have the form

$$\Psi(l;u,d) = \mathrm{Ad}(e^{il})u - k(l)d + \Psi^{(2)}(l;u,d), \qquad |\Psi^{(2)}| \le 14|d| \max\{|u|,|d|\}.$$

The signed bond sum therefore gives the exact identity

$$\begin{aligned} N(a,\lambda)(x) = {} & [\mathrm{Ad}(e^{i\lambda(x)}) - 1] D^* a(x) - [k(\lambda(x)) - 1] D^* D\lambda(x) \\ & + \xi^{-2} \sum_{b \ni x} \epsilon_{x,b} \Psi^{(2)}_{x,b}(\lambda(x); \xi a_{x,b}, \xi D\lambda_{x,b}). \end{aligned}$$

Incoming values are transported to x by the real reference configuration, and $\epsilon_{x,b}$ is the divergence sign. The holomorphic remainder vanishes for $d = 0$ and has no terms of degree less than two in

(u, d). On a smaller fixed ball, Cauchy's estimate bounds its differential by products containing one variation and at least one additional small factor, with a constant independent of the lattice. Subtract the formula at λ and $\widetilde{\lambda}$. Every term contains one difference, among $\lambda - \widetilde{\lambda}$, $D(\lambda - \widetilde{\lambda})$ and $D^*D(\lambda - \widetilde{\lambda})$, and a small factor bounded by Cre in its scaled degree. Adjacent scales and envelopes are comparable with constants depending only on L. Dividing by h and multiplying by the degree-two scale gives eq. 5.xx.

The induced top factors. The complex tower estimates of Prop. YM3_I:30.2 give the top remainder a common complex domain. To verify this without iterating estimates for complex group angles, fix a retained top y of level j and work on its tower. Here $\xi_j = \ell_j = \xi L^j$ is the layer scale, $B^j(y)$ is the block beneath y, and $\mathsf{P}^j(y)$ is its tower bond set, as in the cited proposition. Write $\alpha^{(j)}$ for the local bound on the reference plaquette angles at that scale. Put

$$a_y = \xi_j \sup_{\mathsf{P}^j(y)} |a|, \qquad d_y(\lambda) = \max\left\{ \sup_{B^j(y)} |\lambda|, \xi_j \sup_{\mathsf{P}^j(y)} |D\lambda| \right\}.$$

For $a_y + d_y(\lambda)$ below a fixed number of L, the complex bond logarithm for $a^{e^{i\lambda}}$ is defined and obeys

$$\xi_j \sup_{\mathsf{P}^j(y)} |a^{e^{i\lambda}}| \le C\{a_y + d_y(\lambda)\}.$$

Use the straightened averages and their accumulated factors $v_i[a]$, for both a and $a^{e^{i\lambda}}$. Their complex tower estimates give, with constants independent of j,

$$|v_j[a]^{\pm 1} - 1| \le C(L) a_y, \qquad |v_j[a^{e^{i\lambda}}]^{\pm 1} - 1| \le C(L)\{a_y + d_y(\lambda)\}.$$

In particular the expression

$$r_j(a, \lambda)(y) = v_j[a^{e^{i\lambda}}](y)^{-1} e^{i\lambda(y)} v_j[a](y)$$

and its small logarithm are holomorphic on a common complex ball independent of the depth. The factorization identity for the mean scheme identifies r_j with the induced gauge at real arguments. The identity continues to the complex ball. Apply the same bounded-factor formula at every intermediate level $i \le j$. Factorization identifies the straightened averages of $a^{e^{i\lambda}}$ with those of a transformed by r_i, so every intermediate factor and small logarithm used by the recursion is admitted on this ball.

At $\lambda = 0$ one has $r_j = 1$. The derivative at $(a, \lambda) = (0, 0)$ of $T^{-1} i^{-1} \log r_j$ in λ is $Q' + E'$, where E' is the comparison of the scalar mean and comb averages. Its local bound is

$$|E'\lambda(y)| \le \kappa_F \alpha^{(j)} \sup_{B^j(y)} |\lambda|.$$

The Taylor expansion of $q(a, \lambda) - E'\lambda$ contains only terms involving λ and of total degree at least two in (a, λ). Cauchy's estimate on a smaller common complex ball therefore gives

$$|q(a, \lambda)(y) - q(a, \widetilde{\lambda})(y)| \le C(L)\{\alpha^{(j)} + a_y + d_y(\lambda) + d_y(\widetilde{\lambda})\} d_y(\lambda - \widetilde{\lambda}).$$

For the weighted setting, the layer index is constant on $B^j(y)$ and is j at a retained top. Consequently $e = e(y)$ there and the internal bond scales are ξ_j. Its diameter in the weighted distance is bounded by a number depending only on L. The envelope condition therefore gives $h(x) \le C(L) h(y)$ for $x \in B^j(y)$. The local reference curvature parameter obeys $\alpha^{(j)} \le C(L) e(y)$: use the bound on F_U in the same pillar and the comparison between the small real plaquette angle and its vector part. For a, λ and $\widetilde{\lambda}$ in the same unlocalized radius-r ball as above, division by $h(y)$ now yields

$$|q(a, \lambda) - q(a, \widetilde{\lambda})|_{0,h} \le C(L)(1 + r) \sup e \, \|\lambda - \widetilde{\lambda}\|_{\mathcal{Z}_h}. \tag{5.xxi}$$

At level zero $q = 0$.

The scalar inverse and repaired field. The scalar inverse has the stronger norm required here. Its three equations are

$$\Delta'_a\lambda + w = f + Q'^*\mathsf{a}'b, \qquad \Delta'_a w - Q'^*\sigma = 0, \qquad Q'\lambda = b.$$

The scalar solution seminorms and forcing degrees are those of equations YM3_I:(327)–(328); the regional inverse is Thm. YM3_I:G, with its scalar compression. The solution seminorm retains $|\lambda|_{0;y}$, $|D\lambda|_{-1;y}$ and $|w|_{-2;y}$; its forcing seminorm retains $|f_2|_{-2;y}$, $|f_3|_{-4;y}$ and $|f_4|_{0;y}$. The regional inverse estimate is $|\mathbb{E}^{[2]}g|_y \le B\sup_z e^{-\delta d(y,z)}|g|_z$. Apply it with $g = (f + Q'^*\mathsf{a}'b, 0, b)$. At a bare site of the cell of a level-j top y, $Q'^*\mathsf{a}'b$ is the value $\varkappa_j\xi_j^{-2}b(y)$, transported by the real reference configuration, with $\varkappa_j \le 1$. The input therefore has the claimed degrees, uniformly in the cell size. The envelope inequality eq. 5.viii, with its exponent strictly below δ, bounds the weighted supremum in the regional inverse estimate. It gives

$$|\lambda|_{0,h} + |D\lambda|_{1,h} + |w|_{2,h} \le C_{\mathrm{sc}}(L)(|f|_{2,h} + |b|_{0,h}).$$

The first and third rows therefore imply

$$\Delta_D\lambda = f - w, \qquad |\Delta_D\lambda|_{2,h} \le C(L)(|f|_{2,h} + |b|_{0,h}).$$

At a site meeting the boundary of the first region, $\Delta_D\lambda - D^*D\lambda$ is the missing-neighbour mass $\xi^{-2}n\lambda$. Such sites have layer zero and layer scale ξ, so this term has the same bound.

To add the Hölder terms, put $B = D\lambda$. The exact covariant curl of this gradient is reference curvature acting on a transported value of λ, and its divergence has just been bounded. Apply the local curl–divergence estimate proved in the proof of Lem. 5.19 to B in the same comparison gauges. It gives the degree-one Hölder estimate for $D\lambda$, uniformly in the bare spacing. The pointwise gradient bound also makes λ locally Lipschitz after transport. The reference curvature is Hölder, so the exact formula for $D_1D\lambda$ gives its degree-two Hölder norm by the product rule. The first two layers require only supremum bounds. No additional regularity of the scalar forcing is used. We have proved

$$\|\mathcal{S}(f,b)\|_{\mathcal{Z}_h} \le K_{\mathrm{sc}}(L)(|f|_{2,h} + |b|_{0,h}). \tag{5.xxii}$$

The regional solution seminorm also retains $|Dw|_{3,h}$ and $|\sigma|_{4,h}$. For general three-row forcing $g = (g_1, g_2, g_3)$, the same inverse and the same local regularity calculation give

$$\|[\mathbb{E}^{[2]}g]_\lambda\|_{\mathcal{Z}_h} + |[\mathbb{E}^{[2]}g]_w|_{2,h} + |D[\mathbb{E}^{[2]}g]_w|_{3,h} + |[\mathbb{E}^{[2]}g]_\sigma|_{4,h} \le K_{\mathrm{sc}}(L)(|g_1|_{2,h} + |g_2|_{4,h} + |g_3|_{0,h}),$$

after enlarging K_{sc}. Indeed its first and third rows bound $\Delta_D\lambda = g_1 - w - Q'^*\mathsf{a}'g_3$; the rest of the Hölder argument is unchanged.

The field repaired by this scalar has the full stated norm. The bond logarithm is a holomorphic function of λ, ξa and $\xi D\lambda$, with its linear term $\xi(a - D\lambda)$. Its product and difference estimates on a fixed smaller complex ball give

$$\|a^{e^{i\lambda}} - a\|_{1,1/2,h} \le C(L)\|\lambda\|_{\mathcal{Z}_h}.$$

For its divergence use the exact formula for N above. For its curl use the exact plaquette identity

$$\partial(e^{i\xi a}U)^{e^{i\lambda}}(p) = e^{i\lambda(x_p)}\partial(e^{i\xi a}U)(p)e^{-i\lambda(x_p)},$$

before taking differences. The one-plaquette expansion in the complex potential has the form

$$F_{e^{i\xi a}U}(p) = F_U(p) + M_U(a)(p)D_1a(p) + R_U(a)(p),$$

where $M_U(a)$ and its inverse are bounded on the common ball. More explicitly, with the notation of the one-plaquette calculation, one can take $M_U(a) = c_p\mathsf{f}(-i\xi\,\mathrm{ad}\,a_1)^{-1}$, where $c_p = \frac{1}{2}\operatorname{tr}\partial U(p)$, $\mathsf{f}(z) = (e^z - 1)/z$, $\mathsf{f}(0) = 1$, and a_1 is the potential at the first positively oriented bond of the plaquette based at x_p; the terms left in R_U are linear in a with coefficient $O(\xi F_U)$, or of order at least two in the four potential values. The reference plaquette is small, so c_p stays away from zero.

Write $a' = a^{e^{i\lambda}}$ and $B = a' - a$. Subtracting the last identity and using plaquette conjugation gives

$$M_U(a')D_1B = (\mathrm{Ad}(e^{i\lambda(x_p)}) - 1)F_{e^{i\xi a}U} + (M_U(a) - M_U(a'))D_1a - R_U(a') + R_U(a).$$

Every term on the right contains λ or B. The quadratic holomorphic remainder has difference bound $C(|a| + |a'|)|B|$, while its curvature term costs $C\xi|F_U||B|$. The same estimates hold in the scaled Hölder norm by the product rule on the fixed comparison cubes. In these gauges a reference bond transport differs from the identity by $O_L(\alpha\xi/\ell)$; its scaled Hölder seminorm is bounded by $C(L)\alpha$, using the supremum bound at distances at least ξ. For the curvature norm on the larger cube, cover it by a fixed number of smaller comparison cubes. On overlaps the two real gauges have relative first difference $O_L(\alpha/\ell)$, because both transformed reference bond fields have the stated small bound. Product norms therefore pass between these gauges with a uniform constant. Pairs of points farther apart than a fixed fraction of ℓ are controlled by the supremum norm. The preceding bound for B and the reference curvature and D_1a bounds therefore give the degree-two supremum and Hölder estimates for D_1B. No second difference of λ is estimated by separately differencing a bond logarithm. Together with the divergence identity this proves

$$\|(a^{e^{i\lambda}} - a, 0)\|_{\mathcal{X}_h} \le C(L)\|\lambda\|_{\mathcal{Z}_h}. \tag{5.xxiii}$$

Subtracting the formulas at two scalars proves the corresponding localized difference bound on the same ball.

Combining eq. 5.xx, eq. 5.xxi and eq. 5.xxii, the nonlinear scalar map has Lipschitz constant at most $C(L)K_{\mathrm{sc}}(1 + r)\sup e$. Reduce the envelope ceiling so that this is at most $1/2$. On the unlocalized ball of radius $2K_{\mathrm{sc}}|D^*a|_{2,e}$ the same map is invariant, since $N(a, 0) = q(a, 0) = 0$; reduce the admitted radius of a so that this ball lies in the common complex ball above. Its fixed point exists and is holomorphic. For localization, retain the auxiliary variables of the scalar system. If a already lies in the slice, put $f = D^*a$. Then $Rf = 0$ and there is a top field σ_0 with $\Delta'_a f = Q'^*\sigma_0$. Thus $\Psi_0 = (0, f, \sigma_0)$ satisfies $\mathbb{K}^{[2]}\Psi_0 = (f, 0, 0)$. The retained inverse seminorm gives $|Df|_{3,h} + |\sigma_0|_{4,h} \le C(L)|f|_{2,h}$. Multiply the scalar and top slots by the same slow cutoff, read at their own base points. Subtract the resulting $\chi\Psi_0$ from the scalar system with first datum $D^*(\chi a)$. Its forcing is

$$([D^*, \chi]a, 0, 0) - [\mathbb{K}^{[2]}, \chi]\Psi_0.$$

It is supported in the enlarged collar where the cutoff varies: the Laplacian reads adjacent sites, and every scalar averaging row reads one tower block of bounded weighted diameter. If the cutoff has scaled first and second differences bounded by C/w and C/w^2, its commutator is bounded in degrees $(2, 4, 0)$ by $C(L)(w^{-1} + w^{-2})$ times the retained solution seminorm on that collar. In particular its second row uses both Df and σ_0. This is the local scalar system used in the cutoff comparison, with coefficients agreeing where the compared backgrounds agree. Use the general three-row inverse estimate above for this forcing, whose second row need not vanish. The localized inverse and the nonlinear estimates therefore localize the scalar correction. Subtracting a given small solution uses the difference estimates on the same complex ball. Together with eq. 5.xxiii, this proves the cutoff and difference assertions. □

Lemma 5.22 (The re-filing of the counterterms)**.** Let $1 \le j \le K$, $m' < m + K - j$, $b_0, b_1, \kappa' > 0$ and let c' be a complex function of the plaquettes. For a site z of the level j let $\square_z$ be the cube of π_j (of side M in level-j units) that holds z, and $X_z = \square_z^{\sim}$ the block of the $3 \times 3 \times 3$ cubes of π_j about it, the box of the site z: $X_z \in \mathcal{D}_j$, $z \in X_z$, $d_j(X_z) \le 7$ for the linear size d_j of YM3_I:72 (the definitions of $\mathcal{D}_j$ and d_j at its beginning) (Section YM3_I:72 (the definitions of $\mathcal{D}_j$ and d_j at its beginning)), every plaquette p with $h_z^{(j)}(p) \ne 0$ has its four corners in X_z and its four bonds in

$\mathrm{Bd}(X_z)$, and at most M^3 sites z' have $X_{z'} = X_z$, namely those of $\square_z$ (the tent domains and support assertions are those preceding equation YM3_I:(762) and in Lem. YM3_I:49.7).

Here $h_z^{(j)}$ is the tent of the level j at z, the product of the hat functions $\max\{1 - |u_\mu|, 0\}$ of the coordinates in level-j units, read at the barycentres (equation YM3_I:(760)): $0 \le h_z^{(j)} \le 1$, $h_z^{(j)}(y) \ne 0$ exactly when $|y - z|_\infty < 1$ in level-j units, $\sum_z h_z^{(j)} = 1$, and $\sum_p h_z^{(j)}(p) = 3\eta_j^{-3}$ (Prop. YM3_I:49.6 and the plaquette-count calculation in the proof of Lem. YM3_I:49.7). For a complex weight c and a complex bare configuration $\mathbf{U}$ put $A^\eta(c, \mathbf{U}) = \eta^{-1} \sum_p c(p) a_p(\mathbf{U})$ (equations YM3_I:(6)–(7)), and for a triple $P = (U, A', J)$ of a level q let $\mathbf{U}_P = e^{i\eta_q A'} U$ be its configuration. The tent family at the weight c',

$$A^t[c'](X, P, z) = A^{\eta_j}(c' h_z^{(j)}, \mathbf{U}_P) = \eta_j^{-1} \sum_p c'(p) h_z^{(j)}(p) a_p(\mathbf{U}_P) \ \text{ if } X = X_z, \qquad A^t[c'](X, P, z) = 0 \ \text{ for the other } X \in \mathcal{D}_j,$$

$z \in X$ a site of the level j, is, on the cores $\mathcal{U}_j^\circ(X; b_0, b_1)$, a pointed family of the level j at (b_0, b_1) (Defs. YM3_I:49.4 and YM3_I:49.5), complex linear in c' and 0 at the trivial triple; its size YM3_I:(759), the supremum of $e^{\kappa' d_j(X)} |A^t[c'](X, P, z)|$ over the $X \in \mathcal{D}_j$, the sites $z \in X$ and the triples P of the core, obeys $\|A^t[c']\|_{j,\kappa'} \le \frac{3}{2} |c'|_\infty b_0^2 e^{7\kappa'}$; at $c' = 1$ it is the tent family of equation YM3_I:(762); and $A^{\eta_j}(c', \mathbf{U}) = \sum_z A^{\eta_j}(c' h_z^{(j)}, \mathbf{U})$ for every complex $\mathbf{U}$. Consequently, let $1 \le n \le K$ with $m' < m + K - n$, and let $E^{(i)}$, $1 \le i \le n$, be pointed families of the levels i at (α_0, α_1) with complex counterterms e_i, let an be a complex weight with $|\mathrm{an}|_\infty = s$, and $\hat{E}^{(i)} = E^{(i)} - e_i A_i^t[\mathrm{an}]$, A_i^t the tent family of the level i; let P be a triple of one level $k \ge n$ on the whole torus whose restrictions $\rho_X P = \rho_X^{k \to i} P$ to the domains X of the levels $i \le n$ lie in the cores; since $\eta_i L^{-(k-i)} = \eta_k$, the configuration of $\rho_X P$ is $\mathbf{U}_P$ on X. Then, identically in P,

$$\sum_i \Big[\sum_z \sum_{X \ni z} (E^{(i)}(X, \rho_X P, z) - E^{(i)}(X, \mathbf{1}, z)) - e_i A^{\eta_i}(1 + \mathrm{an}, \mathbf{U}_P) \Big] = \sum_i \Big[\sum_z \sum_{X \ni z} (\hat{E}^{(i)}(X, \rho_X P, z) - \hat{E}^{(i)}(X, \mathbf{1}, z)) - e_i A^{\eta_i}(1, \mathbf{U}_P) \Big],$$

$\hat{E}^{(i)}(X, \mathbf{1}, z) = E^{(i)}(X, \mathbf{1}, z)$, and $\|\hat{E}^{(i)} - E^{(i)}\|_{i,\kappa'} \le \frac{3}{2} s |e_i| \alpha_0^2 e^{7\kappa'}$.

On the label $\mathcal{S}$ of a history $h \in \mathcal{H}_k$ with regular regions Λ_j, assume $m' < m + K - j$ at every level used. The same identity then holds in the following form. Let φ_j be the profile of Λ_j (equation YM3_I:(666) and the definition of φ_j following it): $0 \le \varphi_j \le 1$, $\varphi_j h_z^{(j)} = h_z^{(j)}$ at the sites z of $\Lambda_j^{\sim -1}$, which is Λ_j less one layer of its cells, and $\varphi_j(x(p)) \ne 0$ only if all the corners of p lie in Λ_j (the tent identity in Section YM3_I:44.4; the corner assertion also follows since a nonzero profile is more than one quarter of a cell width from Z_j, while every plaquette corner is less than one bare spacing from its center). The exponent on the label (equation YM3_I:(1476)) holds, besides its Wilson term and its vacuum constant, the regular sums in equation YM3_I:(1474), over the set $\mathsf{lx}_j^{\mathrm{rs}}(\mathcal{S})$ of the places (z, X) with $X \subset \Lambda_j$ and $z \in \Lambda_j^{\sim -1}$, with the counterterms $-e_j A^{\eta_j}(\varphi_j, \cdot)$, and the boundary sums in equation YM3_I:(1475), which hold the made boundary entries $\mathscr{B}(j, X, \mathrm{mk})$ at the filing places $X \in \mathsf{lx}_j^{\mathrm{mk}}(\mathcal{S})$ and the edge parts of equation YM3_I:(1388) of the families, the members with base points outside $\Lambda_j^{\sim -1}$, at the places $X \in \mathsf{lx}_j^{\mathrm{ed}}(\mathcal{S})$ (the index sets and domains preceding Prop. YM3_I:62.2); each member is filed at exactly one place (by the separate level, domain and origin indexing in Prop. YM3_I:62.2); the entries are boundary terms of their filing places, additive in the exponent, of the sizes $\le \mathsf{B}\, e^{-\kappa d_j(X)}$ of a well filed system (Prop. YM3_I:62.2, on its stated layered output spaces); with the anisotropy the counterterms are $-e_j A^{\eta_j}((1 + \mathrm{an})\varphi_j, \mathbf{U}_P)$.

For a site y of the level j outside $\Lambda_j^{\sim -1}$ put $\epsilon_y = 1$ if $X_y \subset \Lambda_j$ and $\epsilon_y = 0$ otherwise, and

$$r_{(j,y)} = -e_j \eta_j^{-1} \sum_p \mathrm{an}(p)\,[\varphi_j(x(p)) - \epsilon_y]\, h_y^{(j)}(p)\, a_p(\mathbf{U}_P).$$

Then the exponent formed with the families $E^{(j)}$, the entries $\mathscr{B}$ and the counterterms $-e_j A^{\eta_j}((1+\mathrm{an})\varphi_j, \mathbf{U}_P)$ equals, identically, the exponent formed with the families $\hat{E}^{(j)}$ at the regular places and in the edge parts, the entries $\mathscr{B}(j, X, \mathrm{mk}) + \sum\{r_{(j,y)} : y \notin \Lambda_j^{\sim -1},\ X_y = X\}$ and the counterterms $-e_j A^{\eta_j}(\varphi_j, \mathbf{U}_P)$. Here $r_{(j,y)} = 0$ unless the tent of y meets the set where $\varphi_j \neq \epsilon_y$, a collar of the boundary of Λ_j, and then $X_y \in \mathsf{lx}_j^{\mathrm{mk}}(\mathcal{S})$; $r_{(j,y)}$ is linear in an, reads $\mathbf{U}_P$ in X_y alone and is 0 at the trivial triple; and on the space of the filing place (j, X_y, mk), namely the layered output space at level j specified before Prop. YM3_I:62.2, where the plaquettes concerned have their plaquette variables within $r_0\eta_j^2$ of $\mathbf{1}$, $r_0 = \mathsf{r}_0((j, \mathrm{out}); j)$ being the prescribed output-layer curvature radius at level j (the same radius used for the edge-tent estimate in the proof of Prop. YM3_I:58.22), $|r_{(j,y)}| \leq \frac{3}{2} s |e_j| r_0^2$; in a frame of Thms. YM3_I:AJ and YM3_I:AD the radius of the level is at most $\alpha_0/(1+\beta_{\mathrm{s}}) < \alpha_0$. The values at the trivial triple are unchanged. ■

Proof. The room condition gives $2L^{m+K-j-m'} \geq 2L$ cubes of π_j in each coordinate direction. The three consecutive cubes in each projection of X_z therefore determine its middle cube. Thus $X_{z'} = X_z$ exactly for the M^3 level-j sites $z' \in \square_z$.

On the cores $|\partial \mathbf{U}_P(p) - \mathbf{1}| < b_0 \eta_j^2$ at the plaquettes with their bonds in $\mathrm{Bd}(X_z)$, and $a_p = \frac{1}{2}\det(\partial \mathbf{U}_P(p) - \mathbf{1})$ for a matrix of determinant 1, so $|a_p| \leq \frac{1}{2} b_0^2 \eta_j^4$; with $h_z^{(j)} \geq 0$ and $\sum_p h_z^{(j)}(p) = 3\eta_j^{-3}$ this gives $|A^t[c'](X_z, P, z)| \leq \eta_j^{-1} |c'|_\infty \cdot 3\eta_j^{-3} \cdot \frac{1}{2} b_0^2 \eta_j^4 = \frac{3}{2} |c'|_\infty b_0^2$, and $d_j(X_z) \leq 7$ gives the factor $e^{7\kappa'}$. Each member is a polynomial in the entries of the $U(b)^{\pm 1}$ and $e^{\pm i \eta_j A'(b)}$ at the bonds of $\mathrm{Bd}(X_z)$, entire in (A', J), a function of the pair, gauge invariant, and 0 at the trivial triple (Lem. YM3_I:49.7).

The tents form a partition of unity, so $A^{\eta_j}(c', \mathbf{U}) = \sum_z A^{\eta_j}(c' h_z^{(j)}, \mathbf{U})$; at $\mathbf{U} = \mathbf{U}_P$ the counterterm $e_i A^{\eta_i}(\mathrm{an}, \mathbf{U}_P)$ is the sum over the sites z of the members $e_i A_i^t[\mathrm{an}](X_z, \rho_{X_z} P, z)$, which vanish at the trivial triple, and moving each from the counterterm into the family of its box is the display, term by term: the two members differ by $\sum_i e_i \eta_i^{-1} \sum_p \mathrm{an}(p)[a_p(\mathbf{U}_P) - a_p(\mathbf{U}_P)] = 0$, the counterterm and the families being read at the one configuration $\mathbf{U}_P$. On a label, by the partition of unity and $\varphi_j h_z^{(j)} = h_z^{(j)}$ on $\Lambda_j^{\sim -1}$, for every rule $\epsilon_y \in \{0, 1\}$,

$$A^{\eta_j}(\mathrm{an}\,\varphi_j, \mathbf{U}) = \sum_{z \in \Lambda_j^{\sim -1}} A^{\eta_j}(\mathrm{an}\, h_z^{(j)}, \mathbf{U}) + \sum_{y \notin \Lambda_j^{\sim -1}} \epsilon_y A^{\eta_j}(\mathrm{an}\, h_y^{(j)}, \mathbf{U}) + \sum_{y \notin \Lambda_j^{\sim -1}} A^{\eta_j}(\mathrm{an}\,(\varphi_j - \epsilon_y) h_y^{(j)}, \mathbf{U}).$$

For $z \in \Lambda_j^{\sim -1}$ the place (z, X_z) is a regular place, and for $y \notin \Lambda_j^{\sim -1}$ the domain X_y is an edge place exactly when $X_y \subset \Lambda_j$; the exponent with $\hat{E}^{(j)}$ holds $-e_j A_j^t[\mathrm{an}]$ at the regular places and in the edge parts, which are the first sum and the members of the second with $\epsilon_y = 1$: this is why $\epsilon_y = 1$ at the edge places and $\epsilon_y = 0$ elsewhere, and $-e_j$ times the third sum is $\sum_y r_{(j,y)}$, the contribution to the made boundary entries. This identity also holds at $e_j = 0$. If $\mathrm{an}(p)[\varphi_j(x(p)) - \epsilon_y] h_y^{(j)}(p) \neq 0$, the corners of p lie in X_y, and in Λ_j (by the profile corner property just stated if $\varphi_j(x(p)) \neq 0$; and if $\varphi_j(x(p)) = 0$, then $\epsilon_y = 1$ and $X_y \subset \Lambda_j$); as $\Lambda_j \subset \Omega_j$, its bonds lie in the bonds of the label in X_y, and its base point, a corner, is a site of X_y and of Ω_j, so that X_y and Ω_j have a bare cube in common; and as $y \in X_y$ lies outside $\Lambda_j^{\sim -1}$, $X_y \in \mathsf{lx}_j^{\mathrm{mk}}(\mathcal{S})$: this index set consists precisely of domains sharing a bare cube with Ω_j and not contained in $\Lambda_j^{\sim -1}$, as defined before Prop. YM3_I:62.2.

The size: $|\varphi_j - \epsilon_y| \leq 1$, the mass $3\eta_j^{-3}$ of the tent and $|a_p| \leq \frac{1}{2} r_0^2 \eta_j^4$ on the space of the filing place give $\frac{3}{2} s |e_j| r_0^2$; in a frame of Thms. YM3_I:AJ and YM3_I:AD, $r_0 \leq \alpha_0/(1+\beta_{\mathrm{s}})$. The tent families

and φ_j, ϵ_y read no history beyond the label, and the values at the trivial triple are unchanged, the tent families vanishing there. □

5.1.8. *Proofs of the theorems.*

Proof of Thms. M–P, Q and R, and of the remarks. Step 1: Thm. M(a)–(c). (a) is Lem. 5.7 for the rough entries $F_a = \mathcal{O}(w_a)$, and its second bound is the second decomposition in the sketch of that lemma, applied to the weights $u_a \mathbf{1}\{\lambda(\cdot) = \lambda_a\}$; for a weight that vanishes the cumulant is 0. (b) Let $\ell' = 2\mathsf{l}$, $\mathsf{d} = \min\{\mathsf{d}_0, \frac{\mathsf{l}}{4n}, \frac{1}{2}\}$, and let K_1 be the least $K \geq K_0(r, \mathsf{d})$, $2 \leq r \leq 8n$, with $a_K \leq \frac{\mathsf{l}}{4n}$. For $K \geq K_1$ cut each f_i into at most $(\frac{R}{\rho} + 2)^3$ pieces in balls of radius $\rho = \frac{\mathsf{l}}{8n}$; each choice of pieces lies, for a suitable offset (a pigeonhole on the n centres), in cells of a board of side l at sup distance $\geq \frac{\mathsf{d}}{2}$ from the walls, so that the mirror copies stay $\geq \mathsf{d}$ apart, and Lem. 5.8 with (a) gives

$$\Big\| \prod_{a \leq r} (\mathcal{O}(w_a) - c_a) \Big\|_{\mathrm{ch}} \leq e^{G_g \mathsf{l}^3} \Big[\sum_{\pi} \prod_{I \in \pi} k(\#I, \mathsf{d}) \Big]^{1/8} \prod_a \sup |w_a|,$$

π over the partitions of $8r$ points into blocks of at least two; the moment–cumulant formula, $\langle F_1; \dots; F_n \rangle = \sum_\pi (-1)^{|\pi|-1} (|\pi| - 1)! \prod_{B \in \pi} \langle \prod_{a \in B} (F_a - c_a) \rangle$, gives a bound $T_n(\mathsf{d})$. For $K < K_1$ we use $|\mathcal{O}(f)| \leq m_K(R) \sup |f|$, $m_K(R) = 6 a_K^{-1} (\frac{2R}{a_K} + 1)^3$, increasing in K; so $C_n = \max\{(\frac{R}{\rho} + 2)^{3n} T_n(\mathsf{d}), m_{K_1}(R)^n \sum_\pi (\#\pi - 1)!\}$. On T_ℓ the same board is placed with $\ell \geq L\ell'$, and a periodic Gibbs state is dominated through its conditional laws (Lem. 5.8). The bound on $\nu' - \nu$ is the case $n = 1$ with the centring at the mean of the torus. (c) The bounds of (b) give equicontinuity in the sup norm, and a diagonal argument over a countable dense family of tuples gives a convergent subsequence; the lattice symmetries and the reflection positivity of the states pass to the limit, at the three coordinate mirrors in both cases and, for periodic Gibbs states, also at the six diagonal mirrors; the translations by all $v \in \mathbb{R}^3$ follow through $v_j \in a_{K_j} \mathbb{Z}^3$, $v_j \to v$, and the Lipschitz continuity of the limits; the growth in n follows by cutting into $\lceil \frac{2\mathsf{l}}{\mathsf{d}_0} \rceil^3$ cells and counting, with $B = 2e^{3+p} (\frac{8R}{\mathsf{l}} + 2)^3 \mathsf{d}_*^{-p} \max_{r \leq P} [e^{G_g \mathsf{l}^3} \mathrm{Bl}_{8r}^{1/8}]^{1/r}$, $\mathsf{d}_* = \min\{\mathsf{d}_0, \frac{\mathsf{l}}{4}, \frac{1}{2}\}$, $P = \lceil \frac{2\mathsf{l}}{\mathsf{d}_0} \rceil^3$; the sum over the planes is the scalar cumulant by linearity.

Step 2: Thm. M(d). Put $\sigma = \lfloor K/(1260n) \rfloor$. (i) Let $s = \lfloor \sigma/(8n+3) \rfloor$ be the coarse depth, $T = K - s$, and $\nu = \sigma - s \geq (8n+2)s$ the number of the levels between the fine depth σ and the coarse one. By forgetting, with the never-large term and the sources switched on at the level $K - \sigma$, each $\mathcal{O}(w_a)$ is exchanged for the weighted energy, plane by plane, of the background over the block field $\mathcal{M}^{K-\sigma} U$: what a source creates at the levels between $K - \sigma$ and T is forgotten by $L^{-\nu/2}$ over these ν levels. For the n sources, at mutual corner distances $\geq \mathsf{d}' = \frac{1}{2}\mathsf{d}$, the cumulants of the $\mathcal{O}(w_a)$ and of the energies differ by at most

$$\mathsf{K}_n \big[2L^{4n} \mathsf{V}_\ell(b) L^{-(b+1)s} + n\, C_{\mathrm{fg}} (\nu+1)^{2\mathsf{p}_0+2} L^{-\nu/2} + 2e^{3 - L^s \mathsf{d}'/M} \big] \prod_a y_a, \qquad y_a \leq 4\ell^3 L^{4s} \sup |w_a|,$$

for every real $b \geq 4n$, with numbers K_n, C_{fg} and $\mathsf{V}_\ell(b)$ that read no K: the sup norms of the sources are paid at the coarse depth, which brings the factor L^{4ns}, and the histories with a first creation of a large field up to the level T bring $\mathsf{V}_\ell(b) L^{-(b+1)s}$; the member $2e^{3 - L^s \mathsf{d}'/M}$ is the factor of the distance of the mixed parts at the level T (Lem. 5.6(a), (e)), read under the conditions $L^s \mathsf{d}' \geq 3M$ and $\mathsf{h}_2(s) \leq 1$ of Lem. 5.6(e) at $t = T$, which hold for $K \geq K_1$, since s grows with K and $\mathsf{h}_2(s) \to 0$ as $s \to \infty$. At $b = 4n+1$, $L^{4ns} L^{-(4n+1)s} = L^{-s}$; and $L^{4ns} L^{-\nu/2} \leq L^{-s}$ since $\nu \geq (8n+2)s$; so this error is at most $C\ell^{3n} K^{2\mathsf{p}_0+2} L^{-s} \prod_a \sup |w_a|$, the factor ℓ^{3n} of $\ell^{3(n+1)}$ in the statement coming from $\prod_a y_a$ (the further ℓ^3 comes from (iii)), and $s \geq \theta_n K - 2$. (ii) For one block field at the level t and the cutoffs K, $K+1$ at equal label the plane-wise energies of the two backgrounds, formed with the coefficient function c of the weights, differ per block site by $C t^{1/2} L^{-t/2} (\sup |\mathsf{c}| + L^{t-K} \mathrm{Lip}_\infty \mathsf{c})$.

For the sum of the three planes this is equation YM3_I:(2156). For one plane, retain the parallel-plaquette support in the proof of Prop. YM3_I:15.1. More precisely, for a set S of coarse

plaquettes and a finer real field W with $\varpi = \varpi_{\mathcal{N}(S)}(W) \le \varpi'_L$,

$$\Big(\sum_{p'\in S} a_{p'}(\mathcal{M}(W))\Big)^{1/2} \le L^{1/2}\Big(\sum_{p\in\mathcal{N}^{\mathsf{W}}(S)} a_p(W)\Big)^{1/2} + \sqrt{12}\,C_0^{\mathrm{SO}}\,\varpi\Big(\sum_{p\in\mathcal{N}(S)} a_p(W)\Big)^{1/2},$$

where $\mathcal{N}(S)$ consists of the fine plaquettes in the cones of S, and $\mathcal{N}^{\mathsf{W}}(S)$ consists of those parallel to a corresponding plaquette of S. To verify the estimate, put $f_W(p) = |\partial W(p) - 1|$. Lem. YM3_I:14.9 bounds $f_{\mathcal{M}(W)}(p')$ by $(\mathsf{W}f_W)(p') + C_0^{\mathrm{SO}}\varpi_{p'}^2$. The kernel in Lem. YM3_I:14.8 uses only parallel plaquettes, so its squared-norm bound is $L\|f_W\|^2_{\mathcal{N}^{\mathsf{W}}(S)}$. Equation YM3_I:(146) gives

$$\sum_{p'\in S} \varpi_{p'}^4 \le 12\varpi^2\|f_W\|^2_{\mathcal{N}(S)}.$$

The triangle inequality and $f_W(p)^2 = 2a_p(W)$ give the displayed energy bound. Applied to the plaquettes of one plane at which the coefficient exceeds a level λ, and integrated over λ, this bounds the plane-wise energy of the block average by that of the finer background up to the factor $1 + C_{\mathrm{en}}\varpi$ and up to the variation of the coefficient over a block; this is the one place where the Lipschitz constants enter, with the factor $L^{-\sigma}$. (iii) The laws of the block fields of depth at most $\lceil K/1260\rceil$ at the two cutoffs at equal label differ, on every bounded Borel gauge invariant function F, by at most $C_{\mathrm{bf}}\ell^3 K^{p'} L^{-K/10}\sup|F|$: this is Cor. 2.8(a), (2.xvii), with the number C_{bf} of Def. 2.7(5) and the exponent $p' = 2\mathfrak{p}_0 + \frac12$, the exponent defined before Thm. YM3_I:AR at $n = 0$.

With $\sigma = \lfloor K/(1260n)\rfloor$ and $s = \lfloor \sigma/(8n+3)\rfloor$ the three errors are at most $L^{-s} \le L^2 L^{-\theta_n K}$, $L^{4n\sigma}L^{-(K-\sigma)/2} \le L^{-K/3}$ and $L^{4n\sigma}L^{-K/10} \le L^{-K/11}$, which gives θ_n and the exponent $\max\{2\mathfrak{p}_0 + 2, p', 1\}$ of K; it equals $2\mathfrak{p}_0 + 2$, since $p' = 2\mathfrak{p}_0 + \frac12$ and $\mathfrak{p}_0 \ge 1$. The comparison of two lattices at equal label, Thm. YM3_I:AR in the setting (E) of §2.3.1, applies: it asks $\mathsf{b} \ge \gamma^{-2}$, which holds because $\mathsf{b} > g^{-2} + \bar B > \gamma^{-2}$, and $MR(\mathsf{b}) \le L^{m-1}$, which holds because $L^{m_{\mathfrak{L}}(g)} \ge LM\,R(L(g^{-2}+\bar B))$ by the definition of the sides (equation YM3_I:(1528)), $\mathsf{b} \le L(g^{-2} + \bar B)$ and R increases.

For $\mathsf{b} \ge 2\max\{\gamma^{-2}, x_{\mathrm{J}}\}$ and $g = (\mathsf{b}/2)^{-1/2}$ we have $g^{-2} = \frac{\mathsf{b}}{2} \ge x_{\mathrm{J}}$ and $\mathsf{b} \in I(g)$. On a compact interval of labels the numbers are monotone in b; the dilation is the change of units $x \mapsto L^z x$, which carries $\mu_{K,\ell,\beta}$ to $\mu_{K-z,L^z\ell,\beta}$ and multiplies the label by L^z. For continuous ϕ_a with supports at sup distances $\ge 2\mathsf{d}$, uniform approximation by Lipschitz functions with supports at distances $\ge \mathsf{d}$ and the bound of (b), uniform in K, make (G_K) a Cauchy sequence. With one plane field among action densities, (ii) is needed for the plane field alone, the action densities being compared at the two cutoffs by the comparison of two lattices at equal label (Thm. YM3_I:AR).

The plane field is a connected function at both cutoffs: with the tent $\theta_K(y) = \max\{0, 1 - (2/a_{K+1})\mathrm{dist}_\infty(y, \mathrm{Bc}_\lambda)\}$ about the barycentres Bc_λ of the plaquettes normal to λ at the cutoff $K+1$, $k_0 = k\,\theta_K$ is Lipschitz, $\mathrm{supp}\,k_0 \subset \mathrm{supp}\,k$, $\sup|k_0| \le \sup|k|$, and $k_0(x(p)) = k(x(p))\mathbf{1}\{\lambda(p) = \lambda\}$ at the plaquettes of both cutoffs, L being odd; so $\langle \mathcal{O}_\lambda(k); \mathcal{O}(f.)\rangle = S^T_{n+1}(f_1, \dots, f_n, k_0)$ at both cutoffs. The comparison statements on two cutoffs at equal label (Thm. YM3_I:AR with the lemmas on which it rests, Def. 2.7 and Cor. 2.8) hold, with the same proofs, at any two depths $u < s$ in the places of the depths there: their proofs read of the depths only conditions on the cutoff, which hold for $K \ge K_0$ when u and s are fixed fractions of K.

Take $u = \lceil K/(80(3n+8))\rceil$, at whose level $t_1 = K - u$ the sources are carried by the run, and $s = \lceil K/20\rceil$, at whose level $K - s$ the plane field is compared. Three members are left, each with the factor L^{3nu} of Cauchy's estimates in the n sources at the level t_1. What the plane field makes below the level $K - s$ is forgotten over the $s - u$ levels up to t_1, by $L^{-(s-u)}$, against the volume L^{3u} of the level t_1 and the size $L^{s/2}$ of its first variation at the level $K - s$: the exponent $(3n+4)u - \frac12 s \le -\frac{1}{80}K + 3n + 4$. The forgetting of the run between the two depths, against the size of the first variation at the level t_1, gives the exponent $(3n + \frac{15}{2})u - s \le -\frac{3}{80}K + 3n + 8$.

The classical comparison of the plane-wise energies of the two backgrounds at the level $K - s$, as in (ii), with the factor $K^{1/2}L^{-(K-s)/2}(\sup|k| + L^{-s}\mathrm{Lip}_\infty k)$ against the volume L^{3s}, bounds the

difference of the two first variations at the real points of the polydisc of the sources; at its complex points the first variations are bounded crudely, and the estimate with two constants for holomorphic functions, with the exponents $\frac{3}{4}$ and $\frac{1}{4}$, gives the exponent $3nu+\frac{23}{4}s-\frac{3}{8}K\le-\frac{6}{80}K+3n+6$. The histories with a first creation of a large field up to the level t_1, and the mixed derivatives of the constants of the run in the sources, which are small by locality, cost less than every power of L^{-K}.

The largest of the three exponents gives the rate $L^{-K/80}$, and the powers of K of Cauchy's estimates in the n sources give $K^{2\mathfrak{p}_0(n+2)+2}$. For the sharpness: L being odd, the barycentres of $a_K\mathbb{Z}^3$ are barycentres of $a_{K+1}\mathbb{Z}^3$ in the same plane, of density L^{-3} among them; so the indicator of the points of a ball that are barycentres at an even cutoff and at no smaller cutoff samples to a weight of density close to 1 in the ball at the even cutoffs and close to L^{-3} at the odd ones, and for suitable smooth spectators the cumulants along the even and along the odd cutoffs have different limits.

Step 3: Thm. M(e)–(g). (e) $\mathcal{O}_\lambda(f)=\mathcal{O}(f\chi_\lambda)$, χ_λ the indicator of the plaquettes normal to λ. Prop. YM3_I:126.4, the two-point law on a torus of small side against the lattice kernel, is stated for every test pair of $\mathfrak{T}_{T_\ell}(\mathsf{d};\vartheta)$ and every shape parameter $\vartheta\in(0,1)$, with a tolerance $\epsilon_g(s(\mathsf{d}))$ that depends neither on ϑ nor on a_K/d. A pair of non-negative Lipschitz functions, neither identically 0, carried by balls of radius d whose supports are between d and $6\mathsf{d}$ apart, is a test pair for every $\vartheta\in(0,1)$ below four positive numbers formed from their masses, their sup norms and their Lipschitz constants ([7, Lem. I.1]).

For non-negative weights w_1,w_2 carried by balls of radius $\mathsf{d}-a_K$ about x_0,y_0 with $3\mathsf{d}\le|x_0-y_0|\le 4\mathsf{d}$, the functions $\sum_p w_a(p)\,\mathrm{pk}_p$, pk_p the tent of height 1 and sup radius $\frac{1}{2}a_K$ at $x(p)$, are Lipschitz, take the values w_a at the barycentres, so that $\mathcal{O}(\sum_p w_a(p)\mathrm{pk}_p)=\mathcal{O}(w_a)$, and form such a pair; so Prop. YM3_I:126.4 applies to them with its tolerance; real weights are differences of two non-negative ones (four terms), complex weights have real and imaginary parts (sixteen terms). This applies to the weights $f\chi_\lambda$, $h\chi_{\lambda'}$, which depend on the plane, and the kernel bounds of equations YM3_I:(2245)–(2246), with the bounds that enter the template sums of Prop. YM3_I:121.2 and Lem. 6.34, compare the lattice kernel with $\mathbb{K}_3$ plane by plane at the Lipschitz pair (f,h).

The lower bound on $\mathcal{K}_{\lambda\lambda'}$ follows from the masses and the Lipschitz constants of test pairs. For $\mathcal{O}_1-\mathcal{O}_2$: with $\hat u=u/|u|$ and $\mathbb{K}_3(u)_{\lambda\mu}=(3\hat u_\lambda\hat u_\mu-\delta_{\lambda\mu})/(4\pi|u|^3)$,

$$[\mathbb{K}_3(u)_{11}]^2-2[\mathbb{K}_3(u)_{12}]^2+[\mathbb{K}_3(u)_{22}]^2=\frac{(3\hat u_1^2-1)^2-18\hat u_1^2\hat u_2^2+(3\hat u_2^2-1)^2}{16\pi^2|u|^6}$$
$$=\frac{9(\hat u_1^2-\hat u_2^2)^2-6(\hat u_1^2+\hat u_2^2)+2}{16\pi^2|u|^6},$$

which is at least $(32\pi^2|u|^6)^{-1}$ where $\hat u_1^2+\hat u_2^2\le\frac{1}{4}$, the bracket being then at least $2-\frac{3}{2}$; this holds at all $u\in\operatorname{supp}f-\operatorname{supp}h$ for the test pairs of the statement, for instance for those carried by balls of radius d whose centres are $4\mathsf{d}e_3$ apart. With $|u|\le 10\mathsf{d}$ there ([7, Lem. I.1](a)) and the masses $\ge\vartheta\mathsf{d}^3\|f\|_\infty$, $\vartheta\mathsf{d}^3\|h\|_\infty$ of the non-negative pair ([7, Lem. I.1]), the main term of the two-point function of $\mathcal{O}_1-\mathcal{O}_2$ is at least $\frac{3}{2}\mathsf{b}^{-2}(32\pi^2)^{-1}10^{-6}\vartheta^2\|f\|_\infty\|h\|_\infty$, and its error is at most 4 times the error of the display, one for each of the four two-point functions of the planes $1,2$; so it is positive for small d, whereas a plane field that is a multiple of $\mathcal{O}$ would make $\mathcal{O}_1-\mathcal{O}_2$ vanish.

The combination $\mathcal{O}-2\mathcal{O}_m$ has coefficients of modulus 1 in three planes, whence the factor 9 for two entries. Multilinearity gives one further factor b in both the main term and the error for each entry multiplied by b. (f) For free photons, expansion of $\log\det(1-\mathsf{b}^{-1}M_\lambda)^{-3/2}$, where $M_\lambda=\sum_a\lambda_a Q^{\mathrm{T}}\mathrm{diag}(w_a)Q$, gives $3\mathsf{b}^{-n}\mathcal{K}_K^{(n)}$ exactly. For the interacting weights the lattice comparison is ($\mathrm{Q_n}$). The kernel comparison and bound YM3_I:(2245)–(2246), applied to the factors of each ring and telescoped plane by plane, give the displayed C_n' error; multilinearity gives the factors for plane combinations and rescaled entries. (g)(i) Lem. 5.8 on T_{ℓ_g}, applied to the Wick monomials of

the tuple, whose eight mirror copies have as mean a mixture over histories, with jets that are nearly constant at the working level and with large fields that are rare on the small side.

A real multilinear form that is invariant under the lattice translations and bounded by a at every choice of signs of a packing of $N_c \sim \mathsf{d}^{-3(1-\sigma)}$ translates of the tuple in one cube of side $\sim \mathsf{d}^{\sigma}$ is bounded by a/N_c; this gives the factor $\mathsf{d}^{3(1-\sigma)}$. Each derivative in a source is paid by Cauchy's estimate on the polydisc of the height, which costs $(\frac{4}{3}\bar{\mathsf{b}}/\mathsf{d})^{[\frac{n}{2}-1]_+}$, $\bar{\mathsf{b}} = L(g^{-2}+\bar{B})$; the sizes at the working depth give a factor $\Lambda(\mathsf{d}) \le C(\log \frac{1}{\mathsf{d}})^{c}$; so $\mathsf{e}'_n = 2^{n-1}(\frac{42L}{M})^3(\frac{4}{3}\bar{\mathsf{b}})^{[\frac{n}{2}-1]_+}[c_n^{\circ}Q_0\Lambda(\mathsf{d})]^n\mathsf{d}^{3(1-\sigma)-[\frac{n}{2}-1]_+}$ with $c_n^{\circ} = n2^{3n}(1+n4^{n-1})$ and Q_0 a constant of the stability bounds on T_{ℓ_g}.

Every periodic Gibbs state is a state at g. (ii) For real weights, the mean of the product of the eight mirror copies of the Wick polynomials Ξ_B, centred at the cumulants of $T_{\ell'}$, is exactly the sum over the partitions of the copies all of whose blocks link different copies of the products of the cumulants of the blocks (an identity of formal power series). Hypotheses (Q_m), $2 \le m \le 8n$, apply to the real plaquette weights without an upper bound on their mutual distances. Together with the following Gaussian ring bound, they show that a linking cumulant is at most $\mathsf{e}_m + \mathsf{v}_m$, where $\mathsf{v}_m = \frac{3}{2}(m-1)!(81(1+3C_{\mathrm{lat}}))^m(\frac{4\mathsf{d}}{\ell'})^6$ bounds the linking lattice rings, two copies in different cells being $\ge \frac{1}{4}\ell'$ apart; with $\mathrm{Bl}_N(k) = \sum_{\pi}\prod_{I\in\pi} k_{\#I}$ over the partitions of N points into blocks of at least two, Lem. 5.8 gives the second display with $\eta^{\sharp} = 2^n e^{G_g|^3}\max_{m\le n}\mathrm{Bl}_{8m}(\mathsf{e}+\mathsf{v})^{1/8}$ and $\mathsf{e}^{\sharp}_n = \mathsf{e}_n + \eta^{\sharp}(1+\eta^{\sharp})^{n-1}$.

Step 4: Thm. M(h). Choose γ_σ so that every $g \le \gamma_\sigma$ satisfies (Γ), (Γ3) and $g \le \gamma_{\mathrm{R}}$, and decrease $\mathsf{d}_\sigma(g)$ below $\min\{2\mathsf{d}_4(g), \mathsf{d}_5(g), 1, e^{-2}, \ell_g/32\}$. Use the common real and sourced construction constants and domains. Set $E_2 = \mathsf{E}_2$ and $E_3 = \mathsf{E}_3$ of Def. 4.1(4).

(i) Apply Thm. I(C.4)(d) with $\mathsf{d} = \mathsf{d}_0/2$. Its state class, plane weights and support geometry are exactly those of (h)(i). Its bracket is $E_2(g,\mathsf{d}) - \frac{3}{2}C_{\mathrm{Rs}}(\mathsf{d}/\ell_g)^3$ and is at most $E_2(g,\mathsf{d})$. This proves the lattice-kernel bound for bounded Borel weights in every state. For Lipschitz weights, Lem. 4.22(c) adds $\frac{3}{2}C_{\mathrm{Rs}}[a_K/(\vartheta\rho) + (\mathsf{d}/\ell_g)^3]$; the finite-volume terms cancel, giving the asserted continuum-kernel bound. The definition of E_2 gives its stated small-distance rate.

(ii) The bound just proved and the free-kernel bound give uniform bounds in the sup norms on each pair of small separated balls. Lems. 4.30 and 4.23 give a common subsequence and the nine plane kernels. Translation invariance, multilinearity and $\mathcal{O} = \sum_\lambda \mathcal{O}_\lambda$ give the remaining claims exactly as in the proof of Thm. I(C.4).

(iii) Thm. G(A.1), followed by (A.4), supplies the finite-cutoff three-source bound and its transfer to every state, uniformly in the ratio of the near and far distances. Under the stated continuity assumption, (A.4)(ii) gives the claimed pointwise limit with $E_3 = \mathsf{E}_3$. Its definition includes the narrow error, the lattice-ring comparison and the state-transfer error and has the stated rate.

(iv) Apply Thm. I(C.4)(c) on the same subsequence. The null set is fixed before r is named. Its tolerance is $12\sup_{3\mathsf{d}/4\le t\le 17\mathsf{d}/16} E_3(g,t) + 6\epsilon_\sigma(g,\mathsf{d})$. The assertion for an arbitrary unit vector follows from (C.1)(iii).

(v) Take r_* below the thresholds of Thm. K, including r_α, $r_\alpha^{\mathcal{W}}$, $(\mathsf{d}/5LM)^2$, $\mathsf{d}/64$ and 1. Partition the far source into cells of side $r/8$, choosing a support point as the centre of an enclosing cube of side $r/4$ for each nonzero piece. Then every centre is at distance at most $6\mathsf{d}$ from every point of the near support, as required by that theorem. Use (E.1) under (Γ3) for the nonexplicit part, and use the three-term estimate in the proof of Lem. 4.40 before its final weakening. With $\rho_0 < Lr^{1/2}$ and $e^{-u} \le 2u^{-2}$, its bracket is bounded by

$$C(g,\mathsf{b})\,\mathsf{d}^{-9/2}\big[r^{\beta_H/72} + (a_K/r)^{1/18}\big].$$

The additional crossing cumulants in the exact decomposition in the proof of (E.1) are smaller. Summing the $O((\mathsf{d}/r)^3)$ pieces and integrating against ϕ_1 cancels the natural factor $\mathsf{d}^{-6}r^3$ with the two support-volume factors. The resulting tensor is $T_\lambda(\phi_1 * \chi^{r/8})$ exactly. Replacing it by $T_\lambda(\phi_1)$ costs at most $C(g,\mathsf{b},\vartheta_0)r\mathsf{d}^{-2}\|\phi_1\|_\infty\|\phi_2\|_\infty$ by Lem. 4.21; this is covered by the displayed

bound. Thus the quantitative estimate in (h)(v) follows with $C(g,\mathsf{b},\mathsf{d},\vartheta_0)=C(g,\mathsf{b},\vartheta_0)\mathsf{d}^{-9/2}$. The fixed-$r$ cutoff limit and then $r\to 0$ follow by the argument of Thm. K(E.3). The separate existence assertion for arbitrary disjoint supports retains the earlier invocation of (d).

For the extension of the narrow Lipschitz law to $g\le\gamma$, use Lem. 4.31 with its coupling threshold enlarged to $\max\{x_{\mathrm{N}},\gamma_\sigma^{-2}\}$. The two resulting reference couplings are below γ_σ; the lemma supplies the actual first-passage data, admissible sides and witness-depth inequalities. The dilation identities preserve the covariance normalization and the dimensionless geometric ratios.

Step 5: Thm. N(a), (b). By Lem. 5.7, $-\beta\sum_p h_{\omega,\chi}=\mathsf{b}\,\mathrm{Pf}(k^h)$, with

$$\|\mathrm{Pf}(k^h)\|\le 156|\omega|\,\mathrm{Lip}(\chi)R_1(\chi)\le 468\ell'|\omega|\,\mathrm{Lip}_\infty(\chi)$$

, and (i) is Lem. 5.7 at $r=n+1$, with $C_n^\flat(\rho)=468\ell' k_n^\flat(\rho)$, $k_n^\flat(\rho)=\max\{k'(n+1,\frac{\rho}{2}),\text{the crude bound below } K_0'\}$. Both clauses follow the one rule of the statement for a rough entry F: its norm $\|F\|$ times a number, and, free of the side and of the state, times the counts N_F and N_i of the cubes of the mesh that meet its carrier and the supports. For (ii) we put the rough entry alone in its cell of a board and the Lipschitz fields in other cells, in any mutual position.

For groups G_j of entries, $R_G=\partial_G[e^{\sum_{e\in G}z_eX_e}/\nu(e^{\sum_{e\in G}z_eX_e})]_{z=0}$ has $\nu(\prod_j R_{G_j})=\sum_\pi\prod_{I\in\pi}\kappa(I)$ over the partitions all of whose blocks meet two groups; and, for integers $n\ge 0$, $p\ge 1$ and numbers $s_1,\dots,s_n>0$, $\rho_*>0$, $Q\ge 1$, $\alpha_1,\dots,\alpha_p\ge 0$: if $A_1,\dots,A_p,B$ are holomorphic on the domain $\mathsf{D}=\{z\in\mathbb{C}^n:\sum_r|z_r|s_r<\rho_*\}$, with $B(0)=1$, $|A_j(z)|\le\alpha_jQ^{\#z}$ and $|B(z)|\le Q^{\#z}$ on D, $\#z$ being the number of the non-zero coordinates of z, then $A_1\cdots A_pB^{-p}$ is holomorphic near 0 and $|\partial_{[n]}(A_1\cdots A_pB^{-p})(0)|\le\binom{n+p-1}{n}n!\prod_j\alpha_j\prod_r(2eQs_r/\rho_*)$.

The chessboard bound of Lem. 5.8 and the bounds of Thms. YM3_I:AJ and YM3_I:AD on tori for the scalar fields, through their cumulant polynomial, with $e^{nc_V\ell'^3}$, conclude. On the larger sides the chessboard seminorm of a centred rough entry is at most $e^{G_g|^3}k_{N_0}(\mathsf{d})\|F\|$, $k_N(\mathsf{d})=[\sum_\pi\prod_{I\in\pi}k'(\#I,\mathsf{d})]^{1/N}$ over the partitions of N points into blocks of at least two, by its N_0 mirror copies on the reference side; a partition of unity by tents of the mesh $h(n,\rho)$ cuts the shell and the fields, which gives the counts N_χ,N_i; the domination of the periodic Gibbs states (Lem. 5.8) gives the states, and the change of units $x\mapsto L^zx$ gives every label and side.

The whole flux insertion is the rough entry $\mathsf{b}\,\mathrm{Pf}(k^h)+\mathcal{O}(v^{\mathfrak{B}})$, of the norm $13\mathsf{b}|k^h|_1+|v^{\mathfrak{B}}|_\infty\le(158\mathsf{b}+3a)|\omega|\,\mathrm{Lip}(\chi)R'(\chi)$, $R'(\chi)=\max\{R_1(\chi),R_1'(\chi)\}$ (Lem. 5.7), and the same rule bounds it. For $\Theta_\beta[\eta]$ we use $\Theta_\beta[\eta]=\mathsf{b}\,\mathrm{Pf}(k^\eta)$; for $\mathfrak{K}^{\mathrm{v}}[\beta\eta]$, Lem. 5.9(a) gives $\mathfrak{K}^{\mathrm{v}}[\beta\eta]=N_\beta[\eta]-\Theta_\beta[\eta]+\mathsf{b}\,\mathcal{O}(\mathrm{div}^s\eta)-\frac{3}{2}a\,\mathcal{O}(e^\eta)$, and the null term drops beside fields that do not depend on the bonds near $\mathrm{supp}\,\eta$. (b) The spin densities written in petals give $T_{lr}(\varphi)=\mathsf{b}(\mathrm{Pf}(k^{lr})+\mathcal{O}(v^{lr}))$; for two entries the two-source mixed derivatives of Lem. 5.6(a), (e) give the bound first at $0<g\le\gamma$ with $g^{-2}\ge x_{\mathrm{J}}$, the setting of that lemma, with the factor $(2\ell/\mathsf{d}'')^6k_2(g,s(\mathsf{d}''))g^4$.

Every $0<g\le\gamma$ is then reached by the change of units $x\mapsto L^zx$, which carries $\mu_{K,\ell,\beta}$ to $\mu_{K-z,L^z\ell,\beta}$ and multiplies the label by L^z: take z with $L^z\gamma^{-2}\ge 2\max\{\gamma^{-2},x_{\mathrm{J}}\}$. The labels of $\mathfrak{D}(g)$ lie in the compact interval $[\gamma^{-2},X_g+\bar{B}]$, whose image under the multiplication by L^z lies in $[2\max\{\gamma^{-2},x_{\mathrm{J}}\},\infty)$; every b' there lies inside $I(g')$ at $g'=(\mathsf{b}'/2)^{-1/2}\le\gamma$, where $g'^{-2}=\frac{1}{2}\mathsf{b}'\ge x_{\mathrm{J}}$, so finitely many intervals $I(g_i')$ cover the image, and at the cutoffs $K>z$ the bound at the couplings g_i' applies; the cutoffs $K\le z$ are bounded crudely, by the suprema of the entries.

The limits are formed as in Step 1. For the symmetry in (l,r) of the limits: for Lipschitz φ, by Lem. 5.9(e), $(T_{lr}-T_{rl})(\varphi)=\beta\sum_b\sum_{p\ni b}(\varphi(x(p))-\varphi(x_b))a^{-1}[(x_b-x(p))_l\mathfrak{d}_{b,p,r}-(x_b-x(p))_r\mathfrak{d}_{b,p,l}]$, which is b times a rough entry of norm $\le 15a\,\mathrm{Lip}\,\varphi$; so, by the two-entry bounds just proved, at every first-passage datum and every $0<g\le\gamma$, both $\langle(T_{lr}-T_{rl})(\varphi);\mathcal{O}(f)\rangle$ and $\langle(T_{lr}-T_{rl})(\varphi);T_{l'r'}(\varphi')\rangle$ are $O(a)$ at separated supports. The limits are continuous in the sup norms of the entries, by the same bounds, and the Lipschitz functions are dense in the continuous ones in the sup norm; so every limit point is symmetric in (l,r).

Step 6: Thm. N(c), (d). (c) By Lem. 5.9(a) the null term drops, and $a\,\mathcal{O}(e^{\eta})$ contributes $O(a)$ by Thm. M(a). By Lem. 5.9(b), $\mathfrak{K}^{\mathrm{v}}[\beta\eta] = -\mathsf{b}\sum_\rho P_\rho(\nabla^+_\rho \eta^\rho) + \beta\mathfrak{K}^\sharp[\eta]$, $P_\rho(\mathsf{k}) = a^{-1}\sum_p \mathsf{k}(x(p))\Psi_\rho(p)$: a bilinear half and a cubic half. In the bilinear half an exact identity in U, direction by direction and with one integration by parts, is used. For a real weight W of the plaquettes and a direction ρ, let (μ,ν,ρ) be the cyclic permutation of $(1,2,3)$ that ends with ρ; for a bare site y let $p(y;\alpha,\alpha')$ be the plaquette with the corners y, $y+ae_\alpha$, $y+ae_\alpha+ae_{\alpha'}$, $y+ae_{\alpha'}$, and $w_0(y) = W(p(y;\mu,\nu))$, the weight of the plaquette normal to ρ at y; let $g_\mu(y) = G_\rho(\langle y, y+ae_\mu\rangle)$ and $g_\nu(y) = G_\rho(\langle y, y+ae_\nu\rangle)$, the transported differences of the field strengths of the setting, and $E_\mu(y) = \sum_{p\ni b}\mathbf{w}(W_{b,p})$, the sum over the four plaquettes that hold the bond $b = \langle y, y+ae_\mu\rangle$, the lattice equation of motion at b, and $E_\nu(y)$ in the same way. Then

$$-4\sum_p W(p)\Psi_\rho(p) = \sum_y w_0(y)\big[\langle g_\mu, E_\mu\rangle + \langle g_\nu, E_\nu\rangle\big] + \mathrm{Bil} + \mathrm{Cub} + \mathrm{Q4},$$

where Bil comes from the differences of the weight W, Cub from the defect of the linear Bianchi identity and from commutators (factors $\mathrm{Ad}(W_{b,p}) - \mathbf{1}$ of plaquette holonomies), and Q4 from $a_p^2 + a_{p_+}^2$. For each plaquette $p = p(y;\mu,\alpha')$ through the bond $b = \langle y, y+ae_\mu\rangle$, the one-bond term $\beta\langle \mathbf{w}(W_{b,p}), E_\mu(y)\rangle - 3(1-a_p)$ has vanishing cumulants with every entry that does not depend on $U(b)$ (an integration by parts in $U(b)$, as in Lem. 5.9(a)), and $g_\mu(y)$ is the sum of the $\mathbf{w}(W_{b,p})$ over the two plaquettes through b in the directions $\pm\rho$; so β times the first sum is, up to a number, $-3a\,\mathcal{O}(v)$ inside cumulants whose other entries avoid the bonds concerned, with the weight $v = w_0(y) + w_0(y+ae_\rho)$ on the plaquettes $p(y;\rho,\mu)$ and $p(y;\rho,\nu)$ and $v = 0$ on the plaquettes normal to ρ; $|v| \le 2\sup|W|$ gives the factor $\frac{3}{2}$ below; $\beta\,\mathrm{Bil}$ is $a\mathsf{b}\,\mathrm{Lip}_\infty\mathsf{k}$ times a bounded petal form; Cub and Q4 are block forms all of whose monomials have an order ≥ 3, whose sizes fall like L^{-2j} with the level j and give $L^{-(K-s-1)/4}$ at the working depth s (Lem. 5.6(c), (d)).

So, for $r \ge 1$ spectators F_q of the sizes ω_q (block forms with weights, or $\mathcal{O}(v_q)$) whose carriers are at Euclidean distances $\ge \mathsf{d}$ from one another and from $\{\mathsf{k}\ne 0\}$, at all first-passage data with $36a \le \mathsf{d}$, $K \ge s(\mathsf{d}_1'')$ and $K \ge s(\mathsf{d}_1''')$, $\mathsf{d}_1 = \frac{3}{4}\mathsf{d}$,

$$|\mathsf{b}\langle P_\rho(\mathsf{k}\circ x); F_1;\dots;F_r\rangle| \le \Big\{a\big[\tfrac{3}{2}\sup|\mathsf{k}| + 190\,\mathsf{b}\,\mathrm{Lip}_\infty\mathsf{k}\big]e_2 + 50\,\mathsf{b}\,\Gamma L^{-(K-s-1)/4}e_3\sup|\mathsf{k}|\Big\}\prod_q \omega_q,$$

where $e_2 = e_2(r+1,\mathsf{d}_1)$ and $e_3 = e_3(r+1,\mathsf{d}_1)$, the cumulant having $r+1$ entries; for a number $r' \ge 2$ of entries and a distance d_1, $e_2(r',\mathsf{d}_1) = (3\ell'/\mathsf{d}_1'')^{3r'}k^{\mathrm{loc}}_{r'}(g, s(\mathsf{d}_1'');\ell')g^{2r'}$, $\mathsf{d}_1'' = \min\{\frac{4}{5}\mathsf{d}_1, L^{-\bar s(r')}\}$, with the depth $\bar s(r')$ of the proof of Lem. 5.7, the least integer $s \ge 3$ with the condition $2\mathsf{T}_{r'}(3)\theta(L^{s-2}g^{-2}) \le 1$ of Lem. 5.6(b), read with the series $\mathsf{T}_{r'}$ of the mixed derivatives of the order r' and the size θ stated there, and the number $k^{\mathrm{loc}}_{r'}$ of Lem. 5.6(b); $e_3(r',\mathsf{d}_1) = (3\ell'/\mathsf{d}_1''')^{3r'}\tilde k_{r'}(g, s(\mathsf{d}_1''');\ell')g^{2r'}$ is the like number at the second depth $\mathsf{d}_1''' = \min\{\frac{3}{4}\mathsf{d}_1, L^{-\tilde s(r')}\}$, $\tilde s(r')$ the least integer $s \ge 3$ with the condition of Lem. 5.6(c) at $p = r'$, and $\tilde k_{r'}$ the number of the bound of Lem. 5.6(c); $\Gamma = \max\{\Gamma^\circ_{\mathrm{Cub}}, \Gamma^\circ_{\mathrm{Q4}}\}$, the larger of the comparison weights Γ°_c of Lem. 5.6(d) of the two block forms Cub and Q4; and $s = s(\mathsf{d}_1''')$ at $r' = r+1$.

In (c) this is used with $r = n$ spectators $\mathcal{O}(f_i)$ and $\frac{1}{2}\mathsf{d}$ in the place of d, so that the numbers are formed at $(n+1, \frac{3}{8}\mathsf{d})$. These numbers depend on g, ℓ', r, d and on no K or β. The cubic half is Lem. 5.11, the spread being $\ge \mathsf{d}$; the norm $\|\cdot\|_\sharp$ and the rate $c < \frac{1}{14}$ come from it alone. (d) For the cubic half in sup norms we couple on one probability space the chain of the first-passage datum and the chain along $x_j = L^{K-j}\mathsf{b}$ of Lem. 5.10, which share only the block map, the Haar measure and the bare density, and use the law of total covariance given the block field of the level $t = K - \lceil\phi K\rceil$:

$$\langle X; F\rangle = \int (r_{XF} - r_X r_F)\,\mathrm{d}\mathsf{m}_t + \int (r_X - \mathsf{a})(r_F - f_0)\,\mathrm{d}\mathsf{m}_t + (\text{large fields}),$$

r_X, r_F, r_{XF} the densities at the never-large history. The first term is a mixed derivative of two separated sources, $\le C_2 e^{3-L^u\mathsf{d}'/M} y_X y_F$, $\mathsf{d}' = \frac{1}{2}\mathsf{d}$, $y_X = 4\ell'^3 L^{4u}\cdot 608\,\mathsf{b}L^K\sup|\eta|$, by Lem. 5.6(e) at

the level t, whose conditions $L^u\mathsf{d}' \ge 3M$ and $\mathsf{h}_2(u) \le 1$ hold for large K; the second is the oscillation of Lem. 5.10(b) times $3C_1 y_F$, $y_F = 4\ell'^3 L^{4u} \sup|w|$, and $4\phi < c_0'$; the rest is at most $\mathsf{V}_{\ell'}(\mathsf{b})L^{-11K}$. In words (Lem. 5.14): a spectator acts on the cubic insertion only through a bounded function of the block field, and the cubic insertion is uncorrelated with every function of the block field up to its oscillation, at the price $L^{4\phi K}$, which a small ϕ pays.

For n spectators the mixed derivatives of every order of the never-large term are bounded by Cauchy's estimates, locality and a count, $e^{-L^u\mathsf{d}/2M}$ paying every power of L^K. In the bilinear half in sup norms, $\mathsf{b}\, a^{-1}\sum_p w(p)\Psi_{\lambda(p)}(p)$, summed over the plaquettes of one normal, is a dominated local source of the weight $76\mathsf{b}\sup|w|$ (Lem. 5.9(b)), and a source cut at a level, with the Hölder bound [7, Thm. BT], gives $L^{-K/200}$. For two cubic insertions the same argument is used twice, and the factor L^{4u} does not enter; two orbital insertions and two stress insertions follow with the bilinear half and the null identity.

The numbers of Lem. 5.10(b) are uniform on $[\gamma^{-2}, X_g + \bar{B}]$, which holds every label of $\mathfrak{D}(g)$. The several-source statements used above hold clause by clause in their own ranges: Lem. 5.6(a), (e), which the clauses on the cubic insertion alone read, at $0 < g \le \gamma$ with $g^{-2} \ge x_{\mathrm{J}}$; Lem. 5.6(b), (c), which the bilinear half reads, at $0 < g \le \gamma_W^{\mathrm{ref}}$, and so does Thm. M(a), which bounds the term $a\,\mathcal{O}(e^{\eta})$. So at every first-passage datum at g on every side of $\mathfrak{L}(g)$, with numbers that depend on g and on no label, the clauses on the cubic insertion alone hold for $0 < g \le \gamma$ with $g^{-2} \ge x_{\mathrm{J}}$, and the other clauses for $0 < g \le \gamma_W^{\mathrm{ref}}$, where $g^{-2} \ge x_{\mathrm{J}}$ holds of itself; and $x \mapsto L^z x$ gives every fixed label and side.

Step 7: Thm. N(e)–(g). In the mean of the mirror copies $\mathrm{St}_K[\eta]$ may be replaced by $-\mathrm{Rm}_K[\eta]$ (Lem. 5.9(a) and the proof of Lem. 5.12). A partition of unity cuts η, η', H' into at most $C(R/\mathsf{d})^3$ pieces of sup diameter $\le \frac{1}{8}\mathsf{d}$, at the cost $C\mathsf{d}^{-2}$ in the C^2 norms. For two sets of sup diameter $\le \frac{1}{8}\delta_0$ that are δ_0 apart along one direction there is an offset in $h\mathbb{Z}^3$, $h \le \frac{1}{64}\delta_0$, of a board of side $\frac{1}{2}\ell_g$ for which they lie in different cells at sup distance $\ge \frac{1}{16}\delta_0$ from the walls; Lems. 5.8 and 5.12 give $|\nu(A^\nu[F_1]A^\nu[F_2])| \le 2e^{2G_g|^3}\|F_1 - c_1\|_{\mathrm{ch}}\|F_2 - c_2\|_{\mathrm{ch}}$.

Two stress insertions give $L^{-K/28}$ each, hence $L^{-K/14}$; for two conserved tensors $L^{-K/36}$ each gives $L^{-K/18} \le L^{-K/20}$; against a plain tensor entry, which costs $L^{4u} = L^{K/100}$, the factor $L^{-K/36}$ leaves $L^{-K/60}$; and $\mathsf{b}\,\mathcal{T}_K(H^{\mathrm{cons}}[\eta]) = \mathrm{St}_K[\eta] - \Delta_K[\eta]$. For the mean of the orbital insertion, cut into the pieces of the partition of unity, $\nu(\mathfrak{K}^{\mathrm{v}}[\beta\eta]) = \nu(F_\eta)$ with $F_\eta = \mathfrak{K}^{\mathrm{v}}[\beta\eta] - \frac{1}{4}\mathfrak{N}[\mathsf{k}_\eta]$, $\mathsf{k}_\eta^\rho = \nabla_\rho^+\eta^\rho$, since $\nu(\mathfrak{N}[\mathsf{k}_\eta]) = 0$ by Lem. 5.9(a); Lem. 5.12 at $m = 1$, $\phi = \frac{1}{400}$ gives $\|F_\eta\|_{\mathrm{ch}} \le CK^pL^{-K/28}\|\eta\|_{C^2}$, F_η having the mean 0 on $T_{\ell'}$, and Lem. 5.8 at $q = 1$, $c_1 = 0$ gives $|\nu(\mathfrak{K}^{\mathrm{v}}[\beta\eta])| \le e^{G_g|^3}CK^pL^{-K/28}\|\eta\|_{C^2}$. (f) n sets of sup diameter $\le \frac{1}{4n}$ lie at sup distance $> \frac{1}{8n}$ from all the walls of some board with offset on a grid of mesh $< \frac{1}{2n}$ (a pigeonhole); several slots may share a cell (Lem. 5.12); Lem. 5.8 holds with any number of cells; recentring from the torus mean to ν costs 2^n terms; and $4\phi(n-1) < \min\{c_0', \frac{1}{28} - 3\phi\}$ pays the factors L^{4u}.

At $n_1 = 1$, $c = \frac{1}{56}$, a fixed $\mathsf{d} > 0$ satisfies $\mathsf{d}L^u \ge K^2$ for $K \ge K_0(\mathsf{d})$. (g) $\sigma(H^{\mathrm{cons}}[\eta], H_2, \dots, H_n)$ is the limit of the left member of (f) at $n_1 = 1$, which tends to 0 at every fixed $\mathsf{d} > 0$, and disjoint compact supports have a positive sup distance. By reflection positivity of σ at the mirror θ, $\sigma(H^{\mathrm{cons}}[\theta\eta], H^{\mathrm{cons}}[\eta])$ is the square of the norm of the vector of $\mathcal{T}(H^{\mathrm{cons}}[\eta])$ in the reconstruction, which therefore vanishes. The equivalence along the sequence is Lem. 5.9(a) with the bound on Δ_K in (e), the labels lying in $[\gamma^{-2}, X_g + \bar{B}]$.

For $0 < g \le \gamma_W^{\mathrm{sd}}$ the cited bound on centred products of tensor entries and a diagonal argument give the convergent subsequences; the covariance and the reflection positivity pass to the limits from the periodic Gibbs states, and $\mathcal{T}_K(f\mathbf{1}) = \mathcal{O}(f)$ gives the trace part.

Step 8: Thm. O(a), (b). (a) We write $\mathfrak{U}_{\omega,\chi} = (\mathfrak{U}_{\omega,\chi} - \mathfrak{U}^\sharp[\chi\,\omega y]) + \mathfrak{U}^\sharp[\chi\,\omega y]$, where $\chi\,\omega y$ is a field with $\mathsf{H}(\chi\,\omega y) \le (2 + 3\ell)|\omega|\|\chi\|_{C^{1,1}}$; since χ covers the f_i, $f_i\chi = f_i$, and the cubic member of the bulk term is $\mathfrak{U}^\sharp[\chi\,\omega y](f)$. The first part has $2(n+1)$ members, a spin member and a

bilinear orbital member for each $\varphi \in \{\chi, f_1, \dots, f_n\}$. The spin members are $-\beta\langle \mathfrak{S}_\omega[\chi]; \mathcal{O}(f_\cdot)\rangle$, with $\beta\,\mathfrak{S}_\omega[\chi] = \mathsf{b}\, a^{-1}\mathfrak{S}_\omega[\chi]$, and $\langle a^{-1}\mathfrak{S}_\omega[f_i]; (\mathcal{O}(f_l))_{l\neq i}\rangle$. By Lem. 5.9(e), $\sum_{p\ni b} a^{-1}(x_b - x(p))_\lambda \mathfrak{d}_{b,p,\rho}$ is symmetric in (λ,ρ) and ω is antisymmetric, so that

$$a^{-1}\mathfrak{S}_\omega[\varphi] = \sum_b \sum_{p\ni b} \sum_\rho k^\varphi_{b,p,\rho}\, \mathfrak{d}_{b,p,\rho}, \qquad k^\varphi_{b,p,\rho} = a^{-1}(\varphi(x(p)) - \varphi(x_b)) \sum_\lambda \omega_{\rho\lambda}\, a^{-1}(x_b - x(p))_\lambda :$$

an entry of the order 2 with real weights $|k^\varphi| \le \frac14 |\omega| \mathrm{Lip}\,\varphi$, which vanish unless $\varphi(x(p)) \neq \varphi(x_b)$; $\beta\,\mathfrak{S}_\omega[\chi]$ is b times such an entry. By Lem. 5.9(b), and since $\nabla^+_\rho(\varphi\,(\omega y)_\rho) = (\omega y)_\rho \nabla^+_\rho \varphi$, $(\omega y)_\rho$ being free of y_ρ, the bilinear members are $\langle \mathfrak{J}^{\mathrm{bl}}[\mathsf{k}^\chi]; \mathcal{O}(f_\cdot)\rangle$ and $-\mathsf{b}^{-1}\langle \mathfrak{J}^{\mathrm{bl}}[\mathsf{k}^{f_i}]; (\mathcal{O}(f_l))_{l\neq i}\rangle$, where $\mathfrak{J}^{\mathrm{bl}}[\mathsf{k}] = \beta \sum_{p,\rho} \mathsf{k}^\rho(x(p))\Psi_\rho(p)$ and $\mathsf{k}^{\varphi,\rho}(y) = (\omega y)_\rho \nabla^+_\rho \varphi(y)$; these vector fields are Lipschitz, with norm $\le 5|\omega|(1+\ell)\|\varphi\|_{C^{1,1}}$, which is where the norms $\|f_i\|_{C^{1,1}}$ enter, and supported in the a-neighbourhood of $\mathrm{supp}\,\varphi$.

By Lem. 5.9(c), $\mathfrak{J}^{\mathrm{bl}}[\mathsf{k}] = -\frac14 \mathfrak{N}[\mathsf{k}] + \sum_{v=1}^5 \mathfrak{J}_v[\mathsf{k}] - \frac14 c_{\mathsf{k}}$: a null term, entries of the orders $2,2,2,3,4$ with weights of the order of the norm of k and, for the orders $3,4$, of β times it, and a number, which drops from the cumulants. In (i) let $D \ge \mathsf{d}$ be the spread of the supports. Two rules on spreads are used. A family of sets has a spread at least that of each of its subfamilies of two or more sets. And if $2r' < D$ and a set T lies within the sup distance r' of one of the sets and of each set of an index set Q, then T is empty when these are all the sets, and otherwise T and the remaining sets have a spread $\ge D - r'$: points of T and of the remaining sets that were closer would give points of all the sets that are pairwise closer than D.

With χ in the first entry, the carrier of the entry and the supports have a spread $\ge D$. With f_i in the first entry, the carrier lies within $\frac32 a$ of $\mathrm{supp}\, f_i$ and the cumulant is taken with the $\mathcal{O}(f_l)$, $l \neq i$: the spread is $\ge D - \frac32 a$. The null terms meet the sources: the covering fixes χ at the barycentres and at the bond midpoints of the plaquettes of the supports, while k^χ reads χ at the points $y + a e_\rho$ as well, so that $\mathrm{supp}\,\mathsf{k}^\chi$ may meet a support; and $\mathrm{supp}\,\mathsf{k}^{f_i}$ may meet $\mathrm{supp}\, f_l$. They are taken apart by the contact formula of Lem. 5.9(f): by (N3) the cumulant of $\mathfrak{N}[\mathsf{k}]$ with the sources of a set A is a sum of cumulants of the entries $\mathfrak{N}[f_{q_1}\cdots f_{q_{j-1}}\mathsf{k}; f_{q_j}]$ with the remaining sources, and by (N4) each such entry is a number, the density a_p and densities of the order 2, with no factor β, carried within the sup distance $2a$ of $\mathrm{supp}\,\mathsf{k}$ and of the supports that it absorbs.

The spread of $\mathrm{supp}\,\mathsf{k}^\chi$ and the supports is $\ge D$, and that of $\mathrm{supp}\,\mathsf{k}^{f_i}$ and the $\mathrm{supp}\, f_l$, $l \neq i$, is $\ge D - a$; once these exceed $4a$, the terms that absorb all the sources vanish, the numbers drop from the others, and by the merging of sets of Lem. 5.9(f) every remaining entry has with the remaining supports a spread $\ge D - 2a$, resp. $\ge D - 3a$. So in every term that is left the spread is $\ge \mathsf{d} - 3a$, which is $\ge \frac12 \mathsf{d}$ once $6a \le \mathsf{d}$, and Lem. 5.15 gives $L^{-K+(\frac12 - c_*)K} \le L^{-c_* K}$ for the orders 2 and $L^K \cdot L^{-\frac32 K + (\frac12 - c_*)K} = L^{-c_* K}$ for the orders $3,4$. The cubic part is Lem. 5.11, whose spread condition holds because the support of the field $\chi\,\omega y$ is added to the supports, whose spread is D.

In (ii), $n = 1$: the members with f_1 in the first entry are means, of the spin insertion and of the bilinear density $\mathfrak{K}^{\mathrm{v}} - \mathfrak{K}^\sharp$, and vanish by the mirror symmetries of the measure (Lem. 5.9(b)); in the members with χ the weights vanish unless χ varies across the plaquette or the bond, so, χ being 1 on the d-neighbourhood of $\mathrm{supp}\, f_1$, their carriers are at sup distance $\ge \mathsf{d} - 2a$ from $\mathrm{supp}\, f_1$, the null term meets no source, and Lem. 5.15 applies to the pair. The cubic member is $\mathfrak{U}^\sharp[\chi\,\omega y](f_1) = -\beta\langle \mathfrak{K}^\sharp[\chi\,\omega y]; \mathcal{O}(f_1)\rangle$, the contact member $a^{-1}\langle \mathfrak{K}^\sharp[f_1 \chi\,\omega y]\rangle$ being a mean of $\mathfrak{K}^\sharp$, which vanishes (Lem. 5.9(b)).

Here $\mathrm{supp}\, f_1$ lies in the set where the field does not vanish, and a periodic field is subtracted: let s be the ℓ-periodic continuous function with $\mathsf{s}(\tau) = \tau$ for $|\tau| \le \frac{\ell}{2} - \delta_0$ that is affine on $[\frac{\ell}{2} - \delta_0, \frac{\ell}{2} + \delta_0]$, and $\zeta^\rho(y) = \sum_\lambda \omega_{\rho\lambda} \mathsf{s}(y_\lambda)$. Then $\zeta = \omega y$ on the closure of the cube of the margin, ζ^ρ does not depend on y_ρ, $\sup|\zeta^\rho| \le |\omega|\ell$ and $\mathrm{Lip}_\infty \zeta^\rho \le |\omega|\ell/\delta_0$. As $\zeta^\rho(y + s e_\rho) = \zeta^\rho(y)$ for all y, s and ρ, the clause of Lem. 5.9(b) on such fields gives $\langle \mathfrak{K}^\sharp[\zeta]; \mathcal{O}(f_1)\rangle = 0$: for each direction ρ and each plaquette q, the

mirror normal to ρ through the barycentre of q keeps the measure, a_q and ζ^ρ, and changes the sign of $\mathfrak{K}^\sharp[\zeta^\rho e_\rho]$.

The field $\eta' = \chi\,\omega y - \zeta$ vanishes on the d-neighbourhood of $\operatorname{supp} f_1$, where $\chi = 1$ and $\zeta = \omega y$; so $f_1\eta' = 0$, $\mathfrak{U}^\sharp[\chi\,\omega y](f_1) = \mathfrak{U}^\sharp[\eta'](f_1)$, and Lem. 5.11 applies, η' vanishing within d of the support, with $\mathsf{H}(\eta') \le |\omega|[(\ell+2)\sup|\chi| + \ell\operatorname{Lip}_\infty\chi + \ell + \ell/\delta_0]$: here the margin δ_0 enters C. In (iii), $\chi - \chi'$ vanishes within d of all the supports, so the exact identity gives $\langle \mathfrak{B}_{\omega,\chi} - \mathfrak{B}_{\omega,\chi'}; \mathcal{O}(f_\cdot)\rangle = \beta\langle \mathfrak{S}_\omega[\chi-\chi'] + \mathfrak{K}^{\mathrm{v}}[(\chi-\chi')\omega y]; \mathcal{O}(f_\cdot)\rangle$, an insertion that vanishes near the sources, and no condition on the supports is needed.

The limit of the flux follows from the exact identity, $\nabla^s_\omega f = (\omega y)\cdot\nabla f + D_f$ with $\|D_f\|_\sharp \le C_0 M|\omega|(1+\ell)a\|f\|_{C^3}$ by Taylor's formula, the bound eq. 5.i on connected functions at separated supports, and the convergence of S_n^T at a fixed label with the rate $K^pL^{-K/10}$ (Cor. YM3_I:117.1). For the quarter turn: $e^{\frac{\pi}{2}\omega_{13}}$ is a signed permutation of the axes, an exact symmetry of the lattice measure on T_ℓ, so $S_n^T(f^{\pi/2}) = S_n^T(f)$; since $\frac{d}{dt}f^t = -(\omega y)\cdot\nabla f^t$, $\int_0^{\pi/2}\sum_i S_n^T(\dots,(\omega y)\cdot\nabla f_i^t,\dots)dt = 0$ at every K; with the exact identity and $\nabla^s_\omega f^t - (\omega y)\cdot\nabla f^t = O(a)$ this is the first display, and in the limit the second. (b) In $\mathfrak{U} - \mathfrak{U}^\sharp$ the entries sit where χ_R varies, in $\{\frac43 R \le |y| \le \frac53 R\}$ up to $2a$, or near one of the supports, which lie in $\{|y| < \frac{R}{2}\}$ at mutual distances $\ge \mathsf{d}_0$; all the boxes and supports lie in the sup ball of radius $2R$ about 0, and in each cumulant the carrier of the entry is at a sup distance $\ge \min\{\frac{5R}{6\sqrt3}, \frac{\mathsf{d}_0}{\sqrt3}\} - 2a \ge \mathsf{d}_{\mathrm{b}} := \frac12\min\{\frac{5R}{6\sqrt3}, \frac{\mathsf{d}_0}{\sqrt3}\}$ from every source of that cumulant, for K large, the cumulants with f_i in the first entry being taken with the other sources.

So Lem. 5.15 in its form on all the sides, at the radius $2R$, the distance d_{b} and on the sides $\ell \ge \ell'$ with $\ell' \ge 4R + 2\mathsf{d}_{\mathrm{b}}$, gives the bracket with a number free of the side, the finitely many smaller sides being covered by the form on one torus; the spin part comes with $L^{-(\frac12+c_*)K}$. The cubic member $\mathfrak{U}^\sharp[\chi_R\,\omega y](f)$ has its entry on top of the sources, so no wall parts them; it is Lem. 5.16, with Lem. 5.11 for the finitely many smaller sides. Weak limit points of the torus laws inherit the bounds. Along a sequence with converging connected functions, the exact identity gives $\mathfrak{U}^{(j)} + \langle \mathfrak{B}; \mathcal{O}(f_\cdot)\rangle^{(j)} = \sum_q S_n^T(f_1,\dots,\nabla^s_\omega f_q,\dots) \to -\frac{d}{dt}S_n(f(e^{-t\omega}\cdot))|_{t=0}$, and $\mathfrak{U}^{(j)} \to 0$.

For the volume-first limits Thm. YM3_I:AP gives invariance under O(3), so the flux tends to 0. With the limits one after the other: at a label $\mathsf{b}' \in I(g)$, on each torus T_ℓ, $\ell \in \mathfrak{L}(g)$, (a) gives the limit in K of the flux, $\sum_i S_n^{(\ell,\mathsf{b}')}(f_1,\dots,(\omega y)\cdot\nabla f_i,\dots,f_n)$, through every enclosing shell. Along the sides $\ell_k \to \infty$ the convergence $S_n^{(\ell_k,\mathsf{b}')} \to S_n^\circ$ is the hypothesis of the clause; the functions $(\omega y)\cdot\nabla f_i$ are Lipschitz, with the supports of the f_i, so the limit in k is $\sum_i S_n^\circ(f_1,\dots,(\omega y)\cdot\nabla f_i,\dots,f_n)$; and this is $-\frac{d}{dt}S_n^\circ(f(e^{-t\omega}\cdot))|_{t=0}$, as S_n° is multilinear and bounded per ball in the norms $\|\cdot\|_\sharp$, the per-ball bound of Prop. YM3_I:91.13 passing to the limit once $\ell \ge 16n(R+1)$ and the fixed supports lie in the central cube, and $t \mapsto f_i(e^{-t\omega}\cdot)$ is differentiable in these norms with the derivative $-(\omega y)\cdot\nabla f_i$ at $t = 0$.

The integers $K_k^* \ge k$ are chosen so that for $K \ge K_k^*$ and $n \le k$ the connected functions on T_{ℓ_k} at the label b' lie within $\frac1k$ of $S_n^{(\ell_k,\mathsf{b}')}$ at all tuples of norms $\|\cdot\|_\sharp \le 1$ with supports in the ball of radius $R_k = \min\{k, \frac18\ell_k\}$ about 0 at mutual distances $\ge \frac1k$, by the rate $K^pL^{-K/10}$ of Cor. YM3_I:117.1; every tuple of the clause lies, up to a factor, in these classes from some k on, so along every sequence $K_k \ge K_k^*$ the connected functions converge to S_n°. For given f_i and χ_k, raising K_k^* makes the bulk term on T_{ℓ_k} at most $\frac1k$ as well, by (a), and the flux converges along such sequences to the limit of the display.

For slowly growing tori, the threshold in the cutoff comparison must also be uniform in the side. Fix $n \ge 2$ and a positive separation d. Take the label interval

$$J = [\mathsf{b}_-, \mathsf{b}_+], \qquad \mathsf{b}_- = g^{-2} + B_\sharp - L^{-1}\bar B \ge g^{-2}, \qquad \mathsf{b}_+ = L(g^{-2} + \bar B).$$

It contains all first-passage labels by Cor. YM3_I:74.2. For large sides $\ell = 2L^m$, $MR(\mathsf{b}_+) \le L^m$, so the interval is admitted by Thm. YM3_I:AR. Its cutoff comparison uses

$$\phi_1 = \frac{1}{180(3n+4)}, \qquad u = \lceil \phi_1 K \rceil, \qquad s = \lceil K/9 \rceil.$$

All its smallness conditions are products of powers of ℓ and quantities decaying exponentially or faster in K, except for the volume factor in the discarded mass. The proof of that theorem writes the latter majorant as

$$\bar{\mathsf{R}} = 2M^{-3}\ell^3 L^{3u} \exp\Big[-c' A_0^2 (\log(L^u \mathsf{b}_-))^{2\mathfrak{p}_0} + C_3 \ell^3\Big], \qquad C_3 = \mathsf{C}^{\Sigma}_{\mathrm{maj}} + 2 + C_{\mathrm{low}}(\mathsf{b}_+).$$

Put

$$c_\ell(n,g) = \left[\frac{c' A_0^2 (\phi_1 \log L)^{2\mathfrak{p}_0}}{2C_3}\right]^{1/3}.$$

If $\ell \le c_\ell(n,g) K^{2\mathfrak{p}_0/3}$, the positive volume exponent uses at most half of the negative exponent, since $\mathsf{b}_- \ge \gamma^{-2} \ge 1$ and $\log(L^u \mathsf{b}_-) \ge \phi_1 K \log L$. As $2\mathfrak{p}_0 > 1$, the remaining bound decays faster than every exponential in K, uniformly over these sides. The other comparison conditions then hold beyond a threshold independent of the side; they continue to hold at every $K' \ge K$ with the side fixed. Thm. YM3_I:AR and Cor. YM3_I:117.1 give

$$\sum_{K' \ge K} |S^T_{n;K'+1,\ell} - S^T_{n;K',\ell}| \le C \ell^{3(n+1)} K^{\mathfrak{p}_n} L^{-K/10} \prod_i \|f_i\|_\sharp, \qquad \mathfrak{p}_n = 2\mathfrak{p}_0(n+1) + \tfrac{1}{2}.$$

Under the stated side bound this is at most $C' K^{2\mathfrak{p}_0(n+1)+\mathfrak{p}_n} L^{-K/10} \prod_i \|f_i\|_\sharp$, which tends to zero. Thus the finite-cutoff connected functions and their spacing limits on those tori have the same limit, and increasing the cutoffs changes the values by at most the same tail.

For completeness, the label bounds follow directly from the canonical flow, as in Prop. YM3_I:74.1 and Cor. YM3_I:74.2. Put $\mathcal{W}_K(g) = \{\beta : (K,\beta) \in \mathfrak{D}(g)\}$ and $X_g = L(g^{-2} + B_\sharp)$. The flow has increments of modulus at most $B_\sharp$ and satisfies

$$|x_j(\beta) - L^{-j}\beta| \le \bar{B}(1 - L^{-j}).$$

At its first passage, $x_K(\beta) \le X_g < x_{K-1}(\beta)$. Thus, for $\mathsf{b} = L^{-K}\beta$,

$$L\mathsf{b} > X_g - \bar{B}, \qquad \mathsf{b} \le X_g + \bar{B} - \bar{B}L^{-K} \le X_g + \bar{B}.$$

These imply the outer inclusion because $L^{-1}(X_g - \bar{B}) = g^{-2} + B_\sharp - L^{-1}\bar{B}$ and $X_g + \bar{B} = L(g^{-2} + \bar{B})$. To prove the inner inclusion, take $\mathsf{b} \in I(g) = (L^{-1}(X_g + \bar{B}), X_g - \bar{B}]$ and put $\beta = L^K \mathsf{b}$. The first passage exists at some $K' \ge 1$, since $\beta > X_g$. Its label $\mathsf{b}' = L^{K-K'}\mathsf{b}$ obeys the preceding two inequalities. If $K' \ge K+1$, then $L\mathsf{b}' \le \mathsf{b} \le X_g - \bar{B}$, a contradiction. If $K' \le K-1$, then $\mathsf{b}' \ge L\mathsf{b} > X_g + \bar{B}$, also a contradiction. Hence $K' = K$. Existence and uniqueness of the first passage also show that the sets $\mathcal{W}_K(g)$ are disjoint and have union (X_g, ∞).

Step 9: Thm. O(c)–(e). (c) By Lem. 5.9(e), $T_{mm}(\varphi) - \mathsf{b}\check{D}_m(\varphi)$ is β times orbital potentials in the plane, which are $-\frac{1}{4}|G_m(b)|^2$ up to densities of the order 4, plus β times densities of the order 4 with weights $\le \sup|\varphi|$, plus $\mathsf{b}\,\mathcal{O}(\tilde{\varphi}_m)$ with $|\tilde{\varphi}_m| \le \frac{1}{2} a \operatorname{Lip}\varphi$. Lem. 5.17(c) bounds the first by $CK^p L^{-c_n K} \sup|\varphi| \prod_i \|f_i\|_\sharp$, Lem. 5.17(a) the second by $CK^p L^{-\lambda_L K/4}$, and Lem. 5.17(b) the third by $C_\epsilon a^{1-\epsilon} \operatorname{Lip}\varphi$. The sum over m gives the trace. By Lem. 5.9(e), $\sum_{p \ni b} a^{-1}(x_b - x(p))_\lambda \mathfrak{d}_{b,p,\rho}$ is symmetric in (λ,ρ), so $(T_{lr} - T_{rl})(\varphi) = \beta \sum_b \sum_{p \ni b} (\varphi(x(p)) - \varphi(x_b)) a^{-1}[(x_b - x(p))_l \mathfrak{d}_{b,p,r} - (x_b - x(p))_r \mathfrak{d}_{b,p,l}]$, an insertion of the order 2 with weights $\le \beta a \operatorname{Lip}\varphi$, and Lem. 5.17(b) at $\sigma = 0$ gives $O(a^{1-\epsilon})\operatorname{Lip}\varphi$; at $\sigma = 1$ it gives $C_\epsilon a^{-\epsilon} \sup|\varphi|$ for T_{lr} itself.

By Lem. 5.9(b), $\mathfrak{K}^{\mathrm{v}}[\beta\eta] - \beta\mathfrak{K}^\sharp[\eta] = -\mathsf{b}\sum_\rho P_\rho(\nabla^+_\rho \eta^\rho)$, whose in-plane potentials are $-\frac{1}{4}|G|^2$ up to the order 4 and whose normal potentials are $\Psi_{\lambda(p)}(p)$: Lem. 5.17(a), (c). The limit at a fixed label follows from the clause of Thm. M(d) on one plane field among action densities, with the

convergence of the connected functions of $\mathcal{O}$ at a fixed label, since $\check{D}_m = \mathcal{O} - 2\mathcal{O}_m$. (d) One lattice step along e_r is $U(b) \mapsto W(\pm r, \mu_b)U(b)$ up to a gauge, and for every gauge invariant Φ

$$a\, v_{e_r}\Phi = \tfrac{1}{2}(\mathrm{Sh}_r^+ - \mathrm{Sh}_r^-)\Phi + \mathcal{P}_r\Phi - \mathrm{Sh}_r^-\mathcal{P}_r\Phi + \mathcal{R}_r\Phi, \qquad |\mathcal{P}_r\Phi| \le C\bar{a}, \quad |\mathcal{R}_r\Phi| \le C\bar{a}^{3/2},$$

$\mathrm{Sh}_r^\pm$ the shifts by $\pm a e_r$ and $\bar{a}$ the largest plaquette term near the support of Φ. With the null identity of Lem. 5.9(a), its contact form and (c), every member of the Ward identity for the field $\chi\xi$ but $\mathfrak{U}^\sharp[\chi\xi](f)$ is its stated limit up to $O(K^p L^{-c_n K})$, and $\mathfrak{U}^\sharp[\chi\xi](f)$ is bounded by Lem. 5.11, the spread being positive (the field $\chi\xi$ vanishes near the supports of the f_i with $i \notin I$, and near the boundary of the chart for $\xi = \omega y, y$). For $\xi = y$, $\partial_l y^r = \delta_{lr}$, so the contraction with T_{lr} contains the trace, which by (c) is $\mathsf{b}\,\mathcal{O}$; this gives the terms $|I|S_n - \mathsf{b}S_{n+1}(\chi, f)$. For $\xi = e_r$ and χ a function of y_3 alone, $\sum_l T_{lr}(\xi^r \partial_l \chi) = T_{3r}(\chi')$. (e) The equality is the exact identity for the constant field e_ρ (Lem. 5.9(a) with the contact terms; the stress functional of a constant field vanishes). Put $(\tau_\pm f)(y) = f(y \pm a e_\rho)$, $g_i = \nabla_\rho^s f_i$, $f_i^e = \frac{1}{2}(\tau_+ f_i + \tau_- f_i)$ and $h_i = (f_i^e - f_i)/a^2$. The measure is invariant under the lattice translations, so $S_n^T(\tau_+ f) = S_n^T(\tau_- f)$; with $\tau_\pm f_i = f_i^e \pm a g_i$ this gives $0 = 2\sum_{|J|\ \mathrm{odd}} a^{|J|} S_n^T(g_J, f_{J^c}^e)$, and with $f^e = f + a^2 h$,

$$\sum_i S_n^T(g_i, (f_l)_{l\neq i}) = -\sum_{|J|\ge 3\ \mathrm{odd}} a^{|J|-1} S_n^T(g_J, f_{J^c}^e) - \sum_i \sum_{J'\neq\emptyset} a^{2|J'|} S_n^T(g_i, h_{J'}, f_{\mathrm{rest}}),$$

at most $2^n + n2^n$ terms, each with a factor a^k, $k \ge 2$. The functions g_i and h_i are averages of translates of $\partial_\rho f_i$ and of $\frac{1}{2}\partial_\rho^2 f_i$, so their norms $\|\cdot\|_\sharp$ are bounded by those of these derivatives, and their supports lie in the a-neighbourhoods of the supp f_i; the bound eq. 5.i on connected functions at separated supports bounds each term. At $n = 2$, $\langle \mathcal{O}(\tau_\pm f_1); \mathcal{O}(f_2)\rangle = \langle \mathcal{O}(f_1); \mathcal{O}(\tau_\mp f_2)\rangle$ by the invariance, and the two members cancel exactly: the statement has content for $n \ge 3$ and expresses the invariance under lattice translations and the failure of the Leibniz rule for ∇_ρ^s. For the second display, the contact form of Lem. 5.9(a) at the field χe_ρ, where $\Theta_{f_i}[\chi e_\rho] = 0$ because $\chi(x_b) = \chi(x(p))$ at every plaquette p at which f_i does not vanish and at its four bonds b, and the vector weight $f_i \chi e_\rho$ is $f_i e_\rho$ for $i \in I$ and 0 otherwise, gives the left member as $-\sum_{i\in I} S_n^T(\dots, \nabla_\rho^s f_i, \dots)$; and $\nabla_\rho^s f - \partial_\rho f$ is an average of translates of $\partial_\rho^3 f$ with the total weight $\frac{a^2}{6}$.

Step 10: Remark 5.3. For a trilinear form that is invariant under the lattice translations along e_ρ, write the functions through their Fourier coefficients at $k_1 + k_2 + k_3 = 0$ and put $z_i = e^{iak_i}$; then ∇_ρ^s and ∇_ρ^+ act by $(z - z^{-1})/2a$ and $(z-1)/a$, and since $z_1 z_2 z_3 = 1$,

$$\prod_{i=1}^3 (z_i - 1) = \sum_i z_i - \sum_{i<j} z_i z_j = \sum_i (z_i - z_i^{-1}),$$

which is the displayed identity. Prop. 5.18(c) gives $a^3 S_3^T(g_1, g_2, g_3) \to 6\mathsf{b}'^{-3} \int g_1 g_2 g_3$ for Lipschitz functions, without a separation assumption. Thus the two sides can have order a^{-1} when the supports meet. In the rotational identity the flux through a shell separated from the supports stays bounded, by Thm. N(a), whereas the difference between the symmetric and continuous generators contains three derivatives.

We first give the free-photon calculation. Its local part requires a difference estimate for the curvature kernel. For equally oriented parallel plaquettes p, p', put $u = x(p) - x(q)$, using a least lift. If $|x(p) - x(p')| \le |u|/4$ and $|u| \ge 8a$, then

$$|C_K^{\mathrm{free}}(p, q) - C_K^{\mathrm{free}}(p', q)| \le C_\nabla a^3 |x(p) - x(p')|\, |u|^{-4}, \qquad C_\nabla = 256\sqrt{3} C_{\mathrm{h}}(1 + 3C_{\mathrm{lat}}). \tag{5.xxiv}$$

Here C_{h} is an absolute constant in the interior gradient estimate for a scalar discrete harmonic function; Thm. 6.3.8(a) of [5] gives this estimate, with the finitely many small radii absorbed into the constant. Indeed, $dd^* + d^*d$ acts on each plane component as the scalar lattice Laplacian and commutes with the projection onto exact 2-forms. Applied to the column $C_K^{\mathrm{free}}(\cdot, q)$, it gives $dd^*\delta_q$, which vanishes beyond the plaquettes sharing a bond with q. The column is therefore harmonic at

distance at least $2a$ from $x(q)$. Join $x(p)$ to $x(p')$ by a lattice path of length at most $\sqrt{3}|x(p)-x(p')|$. This path stays at distance at least $|u|/2$ from $x(q)$. On balls of radius $|u|/4$ about its points, equation YM3_I:(2246) bounds the column by $4^3(1+3C_{\mathrm{lat}})a^3|u|^{-3}$. The gradient estimate bounds each lattice step by $4^4C_{\mathrm{h}}(1+3C_{\mathrm{lat}})a^4|u|^{-4}$. Summing along the path proves eq. 5.xxiv.

Let the real functions f_i have compact supports inside the margin and bounded derivatives through order five, and let $\chi=1$ on a neighbourhood of these supports. Write θ_p for the three-colour Gaussian curvature of covariance $\beta^{-1}C_K^{\mathrm{free}}$. In this model

$$a_p=\tfrac12|\theta_p|^2,\qquad \mathbf{w}(W_{b,p})=\pm\theta_p,\qquad \Psi_\rho(p)=-\tfrac14|\theta_{p_+}-\theta_p|^2.$$

The sign records the plaquette orientation. The spin densities and the other insertions are quadratic. The linear Bianchi identity gives $\mathfrak{r}^\sharp=0$, hence $\mathfrak{U}^\sharp=0$, and Wick's formula gives the connected functions as rings. The exact Ward identity remains valid after removing the terms $\frac32 a\,\mathcal{O}(\cdot)$ coming from the curvature of $SU(2)$; against separated sources these terms are $O(a)$. The expression with the continuous generator is therefore, up to $O(a)$,

$$\mathfrak{U}_{\omega,\chi}(f)-\sum_i S_n^T(\dots,\nabla^s_\omega f_i-(\omega y)\cdot\nabla f_i,\dots). \tag{5.xxv}$$

The terms with χ in the first slot have their quadratic insertion separated from the supports. The ring formula makes them $O(a)$. The terms with f_i in that slot are local. For the bilinear part their kernel is

$$M_\rho(p,q)=[C_K^{\mathrm{free}}(p_+,q)-C_K^{\mathrm{free}}(p,q)]^2.$$

Equation eq. 5.xxiv bounds it by $Ca^8|u|^{-8}$ for $|u|\ge 8a$; at the remaining distances use the kernel bound and the projection bound on the diagonal. The spin part has one additional factor a and kernels $C_K^{\mathrm{free}}(q,q')C_K^{\mathrm{free}}(q',p)$, where the plaquettes q meet a bond of p.

Expand the test functions about the midpoint of $x(p),x(p_+)$ in the first case and about x_b in the second. In lattice units these kernels have uniformly bounded moments through orders four and two, respectively; the third-order tail in the second case is even and has degree -6. Together with

$$\nabla^s_\omega f-(\omega y)\cdot\nabla f=\frac{a^2}{6}\sum_\rho(\omega y)_\rho\partial_\rho^3 f+O(a^4),$$

the expansion writes eq. 5.xxv as local forms with factors a^{-3},a^{-2},a^{-1},a^0, plus an $O(a)$ remainder for $n=2$ and an $O(a\log(1/a))$ remainder for $n=3$. The infinite-lattice coefficients are independent of a by scaling. The torus correction changes each term by $O(a)$. At order three, configurations with two nearby entries and a third at distance $|u|$ use the fourth moment of a kernel of degree -7. Their contribution is bounded by a times the sum of $a^3|u|^{-3}$ over $a\le|u|\le1$, which gives the logarithm. Five derivatives suffice for these Taylor remainders and for the $O(a^4)$ error in replacing the lattice sum of $(\omega y)\cdot\nabla\prod_i f_i$ by its integral. The constants use the suprema of derivatives through order five.

All four local terms vanish. Inversion through the centre of the chart preserves the Gaussian measure and the expression, and removes the odd orders in this local expansion, namely the coefficients of a^{-2} and a^0. The coefficient of a^{-3} is a multiple of

$$\int(\omega y)\cdot\nabla\prod_i f_i=0,$$

since $\operatorname{div}(\omega y)=0$. For the coefficient of a^{-1}, use the other expression for the same quantity: the variation of S_n^T under continuous rotations minus the bounded shell flux. The potentially divergent terms are rotation derivatives of cubic-invariant local forms with at most two derivatives. A cubic-invariant symmetric tensor of rank two is a scalar multiple of the identity, so these forms are rotation invariant. At $n=2$ the terms can also be written as combinations of

$$\int(\omega y)_\rho\partial_\rho(\partial_\rho f_1\partial_\rho f_2)=0,\qquad \int\big[(\omega y\cdot\nabla f_1)\Delta f_2+(\omega y\cdot\nabla f_2)\Delta f_1\big]=0.$$

The first identity uses that $(\omega y)_\rho$ is independent of y_ρ. The first cubic-invariant local form that need not be rotation invariant has four derivatives. Thus only the stated remainders remain. Finally, $a^3 S_3^T \to 6\mathsf{b}^{-3}\int$ in the Gaussian model and

$$\sum_i \int (\omega y)_\rho \partial_\rho^3 f_i \prod_{l\neq i} f_l = 3\int (\omega y)_\rho \prod_i \partial_\rho f_i.$$

Multiplying the difference of the generators by a gives the asserted limit of $a\mathfrak{U}_{\omega,\chi}$.

At order one the neighbourhood condition has a concrete role. Take

$$S=[-1,1]\times[-2,2]\times[-1,1], \qquad \kappa(y)=3-y_1y_2, \qquad f_1=\kappa\ \mathrm{dist}(\cdot,\partial S)\ \text{on } S, \qquad f_1=0\ \text{off } S.$$

The faces of S are lattice planes at every cutoff, and f_1 is Lipschitz with a kink at its boundary. Let $\chi=\varsigma(\mathrm{dist}(\cdot,S))$, where $\varsigma\in C^{1,1}$ equals $1-\mu s^2/2$ for $0\le s\le s_0$, with $\mu>0$, and decreases to zero inside the margin. Then $\chi=1$ on S but on no neighbourhood of it. It covers f_1: the barycentre and bond midpoints of every contributing plaquette lie in S. Choose $\omega_{21}=-\omega_{12}=1$, with the other entries zero. The contact terms are means of the spin and vector insertions and vanish by reflection, as in Lem. 5.9(b). Hence

$$\mathfrak{U}_{\omega,\chi}(f_1)=-\beta\langle \mathfrak{S}_\omega[\chi]+\mathfrak{K}^{\mathrm{v}}[\chi\omega y];\mathcal{O}(f_1)\rangle.$$

Its bilinear part is

$$-\tfrac{3}{4}\mathsf{b}^{-1}\sum_{p,q,\rho}(\omega x(p))_\rho \nabla^+_\rho\chi(x(p))f_1(x(q))M_\rho(p,q), \qquad M_\rho\ge 0.$$

Away from the edges of a face with normal $n_F=\pm e_\nu$, only $\rho=\nu$ contributes. The factor $-(n_F)_\nu\nabla^+_\nu\chi$ is μa times a nonnegative function of the exterior lattice distance; f_1 is κa times the interior lattice distance. Summing the kernel over pairs across the plane and then over the a^{-2} plaquettes per unit area gives

$$\tfrac{3}{4}\mathsf{b}^{-1}\mu c_{\mathrm{lat}}\int_{\partial S}(\omega y)\cdot n_F\,\kappa\,\mathrm{d}S.$$

The positive constant c_{lat} is the pair sum per unit lattice face. Cubic symmetry makes it the same for every face; its convergence is bounded by $\int_0^\infty\int_0^\infty st(s+t+1)^{-6}\,\mathrm{d}s\,\mathrm{d}t$. Pairs a fixed distance apart contribute $O(a^2)$. The contribution near the edges vanishes with the width of the edge neighbourhood; the spin part is $O(a\log(1/a))$ by its $a^6|u|^{-6}$ kernel, and the torus correction is $O(a)$. The divergence theorem gives

$$\int_{\partial S}(\omega y)\cdot n_F\,\kappa\,\mathrm{d}S=\int_S(\omega y)\cdot\nabla\kappa=\int_S(y_2^2-y_1^2)=16.$$

Consequently $\mathfrak{U}_{\omega,\chi}(f_1)\to 12\mathsf{b}^{-1}\mu c_{\mathrm{lat}}>0$.

For the $SU(2)$ example, choose an even smooth function $\psi:\mathbb{R}\to[0,1]$, equal to one on $[-1,1]$ and zero off $[-2,2]$, and put $\mathsf{c}=100$ and

$$\mathit{\Pi}_{(r_1,r_2,r_3)}(y)=\psi((y_1-\mathsf{c})/r_1)\psi(y_2/r_2)\psi(y_3/r_3).$$

Set $g_3=\mathit{\Pi}_{(1,1,1)}$, $g_2=\mathit{\Pi}_{(3,3,3)}$, $g_1=\mathit{\Pi}_{(7,7,7)}$, and

$$G_i(y)=\int_{-\infty}^{y_2} g_i(y_1,t,y_3)\,\mathrm{d}t, \qquad f_i=G_i(\cdot+30e_2)+G_i-2G_i(\cdot-30e_2)\quad (i\le 3).$$

For $3<j\le n$ take $f_j=\mathit{\Pi}_{(15,45,15)}$ and take $\chi=\mathit{\Pi}_{(31,91,31)}$, with the same ω as above and $\delta_0=\mathsf{d}=1$. These fixed smooth data have compact supports with a common point. The cutoff is one on their closed unit neighbourhood and vanishes outside $\{|y|_\infty\le 182\}$, inside the margin of every admitted side, since its exponent m is at least seven.

Reflection in $y_1 = \mathsf{c}$ removes the rotation about the axis through $\mathsf{c}e_1$; the remaining vector field is $\mathsf{c}e_2$. Translation invariance therefore writes the left side of the exact identity as

$$\frac{\mathsf{c}}{2a} \sum_{|J|\geq 3} (|J|-2) a^{|J|} S_n^T\big((\nabla_2^+ f_i)_{i\in J}, (f_j)_{j\notin J}\big).$$

The bulk term is this expression minus the flux. The latter has a quadratic insertion, weights $O(\beta)$ and distance at least $1/2$ from the supports. Lem. 5.15, at $c_* = 1/4$, bounds it by $CK^{(2n+1)\mathfrak{p}_0} L^{K/4}$. If J contains an index above three, its derivative entry vanishes near the support of f_3, so the same bound applies. For $J = \{1,2,3\}$ the three derivatives are translates of $g_i^K = a^{-1}\int_0^a g_i(\cdot + te_2)\,\mathrm{d}t$ with coefficients $(1,1,-2)$. Replace the spectators by one, discard triples with different translates, and replace g_1^K, g_2^K by one. Each error has an entry separated by at least $1/2$ and is again covered by that lemma. The equal translates sum to $1+1-8 = -6$. With $I_K = (a^3/3)\sum_p g_3^K(x(p))$, translation invariance gives

$$S_n^T(g_3^K, 1, \dots, 1) = 3a^{-n-3} I_K (-1)^{n-1} E^{(n-1)}(\beta).$$

Thus, up to $CK^{(2n+1)\mathfrak{p}_0} L^{-3K/4}$,

$$a_K \mathfrak{U}_{\omega,\chi}(f) = 900 I_K (-1)^n \mathsf{b}'^{-n} \beta^n E^{(n-1)}(\beta).$$

The inner and outer boxes of g_3^K give $1 \leq I_K \leq 125$, and its Lipschitz approximation gives $I_K \to \int g_3$. Prop. 5.18(b) gives $(-1)^n \beta^n E^{(n-1)}(\beta) = -(n-1)! + O(K^{2n+3} L^{-K/20})$, uniformly in $\mathsf{b}' \in I(g)$. Finally $900/2 = 450$ and $900 \cdot 125 \cdot 2 \leq 2.3 \cdot 10^5$ give the asserted two-sided bounds.

Step 11: Thm. P. Let a board of side $a_i \leq \frac{\ell'}{2}$ be placed, $\{x_3 = c\}$ one of its wall planes, θ_c the mirror at it, $\tau_{a_i e_3}$ the translation by $a_i e_3$, and let $\tilde{w}$ be a real weight carried by the cell between $\{x_3 = c\}$ and its translate by a_i that vanishes within $\frac{1}{2}\mathsf{d}$ of the walls. The centred entry $D = \mathcal{O}_K(\tilde{w}) - c_0$, c_0 its mean on $T_{\ell'}$, has the chessboard seminorm $\|D\|_{\mathrm{ch}} \leq \mathsf{q}_i(\mathsf{d}) \sup|\tilde{w}|$ for $K \geq K^{\#}(i,\mathsf{d})$, with $\mathsf{q}_i = e^{G_g a_i^3} \mathrm{Bl}_{N_i'}(k(\cdot,\mathsf{d}))^{1/N_i'}$, $N_i' = (\ell' L^i)^3$ and $\mathrm{Bl}_N(k) = \sum_\pi \prod_{I\in\pi} k(\#I, \mathsf{d})$ over the partitions of N points into blocks of at least two: the mean on $T_{\ell'}$ of the product of its N_i' mirror copies is, by the moment–cumulant formula, the sum over these partitions of products of cumulants of copies that are $\geq \mathsf{d}$ apart, each bounded by Lem. 5.7 at the orders $\leq N_i'$ (Thm. M(a)), and the passage between the sides costs $e^{G_g a_i^3}$ (Lem. 5.8). A law ν of the lattice that is invariant under its symmetries, reflection positive at the wall planes of the boards, and dominated by the torus measures through its conditional laws obeys the reflection inequality

$$|\nu(\overline{\theta_c F'}\, D\, \tau_{a_i e_3} F)| \leq \|D\|_{\mathrm{ch}}\, \|F'\|\, \|F\|, \qquad \|F\| = \nu(\overline{\theta_c F} F)^{1/2},$$

for bounded local functions F, F' of the bonds in the half space $\{x_3 > c\}$: the form $(F,G) \mapsto \nu(\overline{\theta_c F} G)$ is positive, and the Cauchy–Schwarz inequality for it, repeated at the walls of the board on the dominating tori (the chessboard estimate, Prop. YM3_I:86.3, with the Cauchy–Schwarz inequalities of Prop. YM3_I:91.4), bounds the left member by the chessboard seminorm of D read along these tori. Every periodic Gibbs state has the three properties: it is invariant under the lattice symmetries, it is reflection positive at all the families of mirrors (Prop. YM3_I:103.4), and it is dominated by the torus measures (Prop. YM3_I:91.9). Recentring from c_0 to $\nu(\mathcal{O}_K(\tilde{w}))$ costs $\mathsf{q}_i(\mathsf{d}) \sup|\tilde{w}|\, \|F'\|\, \|F\|$ more, by Lem. 5.8 at $q = 1$ and $|\nu(\overline{\theta_c F'}\, \tau_{a_i e_3} F)| \leq \|F'\|\, \|F\|$; whence the factor 2:

$$|\nu(\overline{\theta_c F'}\, [\mathcal{O}_K(\tilde{w}) - \nu(\cdot)]\, \tau_{a_i e_3} F)| \leq 2\mathsf{q}_i(\mathsf{d}) \sup|\tilde{w}|\, \|F'\|\, \|F\| \qquad (K \geq K^{\#}(i,\mathsf{d}));$$

eight shifted boards cover all positions. Along subsequences the limits exist (Thm. M(c)), and in the reconstruction the bound is what Lem. 5.13 asks, and that lemma gives $B^{(m)}$, the density and the L^1 bound. The reconstruction of the limits of Gibbs sequences at $0 < g \leq \gamma$ has $H \geq 0$ (Thm. YM3_I: AP), the hypothesis of Lem. 5.13 in the direction x_3; and it has $H \geq |P_k|$, $k = 1,2$, for the commuting momenta P_k that generate the translations $U(\vec{x})$ in the directions orthogonal to e_3, the hypothesis of the last sentence of that lemma in the directions x_1, x_2: every periodic Gibbs state obeys the

five-point inequality ([7, eq. 241]) of Lem. 4.46 at the mirror normal to e_3 and the direction e_k, which passes to the limit and puts the spectral measure of every class on $\{\cosh(tE)+\cos(tq_k)\ge 2\}$ for a dense set of $t>0$, hence on $\{E\ge|q_k|\}$, as $\cosh(tE)+\cos(tq_k)=2+\frac12 t^2(E^2-q_k^2)+O(t^4)$.

The same holds for the reconstruction of a tensor limit. So Lem. 5.13 applies in the three directions, which gives the density in all three coordinates and the L^1 bound in them. For the bound on $\|B^{(m)}\|$: for $g\le\gamma_W^{\mathrm{sd}}$ the chessboard seminorm of a centred tensor entry in a cell of side a_i is at most $2\mathsf{C}_T Y_i^{\mathsf{q}_T} a_i^{-11/2}\mathsf{b}^{-1}|H|_\infty$, uniformly in the cutoff, the datum, the board, the side and the state ([7, §2], [7, eq. 4] of [7, Thm. B](a)); with Lem. 5.13 every tensor limit has the insertion bound with every $p>\frac{17}{2}$, and $B^{(m)}$ is the compression to the scalar subspace of the point operator at $e=\mathbf{1}-2E_{mm}$, $|e|=\sqrt{3}$; that every limit with converging labels arises from a tensor limit along a further subsequence is the extraction of Thm. N(g).

For the behaviour under turns: by [7, §5], every tensor limit at $0<g\le\gamma_W^{\mathrm{sd}}$ obeys $\sigma(RX)=\sigma(X)$, $R\in\mathrm{O}(3)$, at tuples with one non-scalar entry among scalar ones ([7, Thm. U], which gives [7, Thm. R](c) at these tuples); $R(\mathbf{1}-2E_{33})R^{\mathrm{T}}=\mathbf{1}-2nn^{\mathrm{T}}$ with $n=Re_3$, and the sum over $\pm\alpha$ cancels the off-diagonal part of nn^{T}, leaving $2\cos^2\alpha(\mathbf{1}-2E_{33})+2\sin^2\alpha(\mathbf{1}-2E_{ii})$; the entries of nn^{T} are combinations of 1, $\cos2\alpha$ and $\sin2\alpha$, which gives the form $c_0+c_1\cos2\alpha+c_2\sin2\alpha$ and the second derivative.

Step 12: Thm. Q. (a) Put $\lambda=L^{-(k-j)}$, $d=d_j(Z)$ and $S=\sup|E|$; the left member is at most $2S$, and the numbers a_3, C_2^Q, $\bar C_2=\max\{C_2^Q,1\}$, c_3^Q, $c_s=c_s^Q$ and λ_Q are those of the extended list. The trivial cases come first: if $\lambda>\lambda_Q$, then $2S\le 2\lambda_Q^{-4}\lambda^4 S$; if $(d+4)\lambda>c_s$, then $2S\le 2((d+4)\lambda/c_s)^4 S\le 32c_s^{-4}(d+2)^4\lambda^4 S$, as $d+4\le 2(d+2)$. Let now $\lambda\le\lambda_Q$ and $(d+4)\lambda\le c_s$. Then $\lambda\le 1/(4a_3)$, and on a cube that holds the box of Z, whose sites lie within the ℓ^1-distance $l=3(d+4)M$ (level-j units) of its least site, the definition of the cores gives a gauge in which the real part of the potential obeys $|A|\le\lambda$ and $|\nabla A|\le\lambda^2$ in level-j units; with the clause on the complex part this gives a complex potential A^0 of the sizes λ and λ^2, up to the complex gauge transformations $g_t=e^{itm}$, $0\le t\le 1$, with $|m|\le l\lambda a_3\le 3Ma_3c_s\le\frac14$, as $c_s\le 1/(12Ma_3)$, so of the size $e^{1/4}$.

The constant part, of the first order, is removed by g_1: a term is invariant under real gauge transformations and holomorphic, so it is invariant under every complex gauge transformation that is reached by a path inside its domain, and the path $t\mapsto g_t$ stays there, its triples lying in the cores and below the third radius. What is left, $\tilde A$, vanishes at the least site of the cube and has the gradient $|\nabla\tilde A|\le C_2^Q\lambda^2\le\bar C_2\lambda^2$, so that $|\tilde A|\le l\bar C_2\lambda^2=3(d+4)M\bar C_2\lambda^2$ on the cube; the disc $\zeta\mapsto(\mathbf{1},\zeta\tilde A,\zeta\tilde J)$ of the radius $R=c_3^Q/(l\bar C_2\lambda^2)=c_3^Q/(3(d+4)M\bar C_2\lambda^2)$, of the order $((d+2)\lambda^2)^{-1}$, stays in the domain, because the Hölder number depends only on the curl of the potential, up to squares, and scales with the power $\frac52>2$ under $\rho_Z^{k\to j}$; at the trivial field the first derivative of a gauge invariant term vanishes, since the adjoint representation of SU(2) has no invariant vector; and the Schwarz lemma gives $|E((\mathbf{1},\tilde A,\tilde J))-E(\mathbf{1})|\le 2SR^{-2}=2l^2\bar C_2^2(c_3^Q)^{-2}\lambda^4 S\le 72M^2\bar C_2^2(c_3^Q)^{-2}(d+2)^2\lambda^4 S$, since $l\le 6(d+2)M$ (if $R\le 1$ the bound $2S\le 2SR^{-2}$ holds already).

This is the clause with $(d+2)^2$, and with the trivial cases it gives

$$C_Q=\max\{2\lambda_Q^{-4},32c_s^{-4},72M^2\bar C_2^2(c_3^Q)^{-2}\}$$

and the power $(d+2)^4$. For $\mathrm{U}(1)^3$ the field strength scales like λ^2 and the linear term survives. The power 4 is attained by a term of the second order in the field strength at one plaquette, against a triple with a constant field strength of the largest admitted size. (b) A block of the level k holds $L^{3(k-j)}$ regions of the level j, each constant up to $L^{-4(k-j)}$ by (a); the count of regions among the construction results with the rate $\kappa_{\mathrm{lf}}\ge 4\log 18432$ of (b) gives the bound, uniformly in k. (c) The complex centre of each Gaussian integral is kept as a free argument, one 1-form for every earlier level, in a polydisc.

The first order of a plane weight gives $\sup|w|L^{-5/2}\mathrm{hol}$, hence the third radius, which is small at the true points and stable under the shift system among the construction results; the weight and the trees enter as potentials, so that the action of the step carries no explicit weight; a tree of the level j weighs L^{k-j+1} seen from the level k, there are $L^{3(k-j)}$ of them and each is smaller by $L^{-5(k-j)}$; the non-classical terms stay at their levels and are lifted afresh at every step by (b), at the rate r_1, which (b) admits since $r_1 > \mathsf{C}_{\mathrm{cs}} > 4\log 18432$. Each step is then the identity of the statement, classical times ideal times the exponential of a tiny term, with $c_k(W)\Xi_0[V_{\mathsf{m}}](Q) = \Phi_{k+1}[\omega](W)$; at $k = 0$ the step is written with bond labels.

On the plain cores the Hölder number is at most α_0 times a factor that grows like $L^{n/2}$ with the level, while $\alpha_2/\alpha_0 \geq 4.99 \cdot 10^6 e^{L/4}$; so the condition of the third radius follows from the plain core at the levels $n \leq (\frac{L}{2} + 29.39)/\log L$. At $k = 0$ the weight enters linearly through the bond labels, and the centre, the solution of the linear system of the step with a source of the size σ, is at most $K_Z\sigma$. The extended list, stated before Thm. Q, places each new constant after the constants on which it depends and before γ; its four new lower bounds are laid on the chosen constants C_{ex}, κ, $\bar{\kappa}$ and M, which are free upward, each by a function of earlier constants, its new conditions on γ are ceilings whose left members tend to 0, and it lays no condition on L itself; so it keeps the admissible shape, its choices exist, and they restrict to choices of the lists of Thms. YM3_I:D, YM3_I:AJ and YM3_I:AD.

At every choice $L > 4 \cdot 10^{19}$: the retained floor $L \geq L_\Pi = 2\mathsf{C}_{\mathrm{ex}}(1 + 4\beta_{\mathrm{s}})/(1 - 2\delta_{\mathrm{w}}) \geq 2\mathsf{C}_{\mathrm{ex}}$ (equation YM3_I:(904), with $\beta_{\mathrm{s}} > 0$ and $0 < \delta_{\mathrm{w}} < \frac{1}{2}$), the new lower bound $\mathsf{C}_{\mathrm{ex}} \geq 2\mathsf{C}_{\mathrm{cs}}$, and $\mathsf{C}_{\mathrm{cs}} \geq 24C_{\mathrm{E}}^2 e^{24}$ by the choice preceding equation YM3_I:(1332), $C_{\mathrm{E}} = 4096$, give $L \geq 4\mathsf{C}_{\mathrm{cs}} \geq 96C_{\mathrm{E}}^2 e^{24} > 4 \cdot 10^{19}$. The further conditions named in the statement are of three kinds. The tower hypotheses of §5.2 and the inequality $\alpha_0 + \alpha_1 \leq a^\sharp$, even $2(\alpha_0 + \alpha_1) \leq a^\sharp$, are closed inequalities between constants that hold at every choice of the list of Thm. YM3_I:D, and so at every choice of the extended list, which restricts to such a choice.

The standing conditions of the frame of [7, Def. B.14] follow, condition by condition, from members of the lists of Thms. YM3_I:D, YM3_I:AJ and YM3_I:AD, to which the choices of the extended list restrict. The locality for all pairs of regions and the gauge covariance at all decoupling parameters are properties of the Gaussian structure among the construction results as it is formed, at all admitted constants: its structures form a local system, and they are covariant under real gauge transformations at all decoupling parameters s of the closed polydisc (Thm. YM3_I:U(2) and the covariance of the operators in YM3_I:(765): restrictions commute with the unitary gauge actions, as do inversion and the functional calculus used to form the covariance and coupling).

Step 13: Thm. R. (a) Prop. 5.5 proves the uniform critical-background and Hessian assertions. We give the remaining comparisons and check the actual step data.

At $U_0 = 1$ with all regions the whole torus, the quadratic form for plane weights is

$$\langle B, Z_\epsilon B\rangle = \min_{Qd=B}\{(\partial d, \mathsf{a}\,\partial d) + \|RD^{\mathsf{T}}d\|^2\}, \qquad 1 - \epsilon \leq \mathsf{a} \leq 1 + \epsilon.$$

Each form before minimization lies between $1 - \epsilon$ and $1 + \epsilon$ times the unweighted form, including its nonnegative slice term. Taking the infimum gives the claimed comparison of block Hessians. For the single mixed tensor term, $2|\langle f_{\mu\nu}, f_{\mu\lambda}\rangle| \leq |f_{\mu\nu}|^2 + |f_{\mu\lambda}|^2$ gives the same form comparison in the stated normalization. Adding the square of the linearized block averages preserves it. Thus the linearization of the constrained system is invertible at zero. On each fixed finite lattice the analytic implicit-function theorem gives the stated family near zero.

The reason a uniform Hölder estimate needs the Hölder norm of the weight can be seen in this same quadratic problem. Fix $B \in Y_0$, $B \neq 0$, and write $h_\epsilon = d_b A \cdot B$ at $b = 0$. Take the torus side $N = 2L^{m+K-k}$ in level units, as in the construction; it has the even number $N/\eta_k = 2L^{m+K}$ of bare sites in each direction. Choose a plane $(\mu\nu)$ in which $F^0 = \partial h_0$ has largest component norm, and

put $v_{\mu\nu}(x) = \epsilon(-1)^{x_\mu/\eta_k}$, with all other plane weights zero. Then

$$Qh_\epsilon = B, \quad RD^{\mathsf{T}}h_\epsilon = 0, \quad \partial^{\mathsf{T}}\mathsf{a}\,\partial h_\epsilon = Q^{\mathsf{T}}Z_\epsilon B.$$

The supremum of the source on the right is at most $(1+\epsilon)C_{\mathrm{tr}}c_Z\|B\|$, with $c_Z \geq \|Z_0\|$ and the row-to-supremum norm C_{tr} of Q^{T} depending only on L. Since $\delta = h_\epsilon - h_0$ has $Q\delta = 0$, subtracting the two variational equations and testing by δ gives

$$\|\partial\delta\|^2 + \|RD^{\mathsf{T}}\delta\|^2 = -\epsilon((-1)^{x_\mu/\eta_k}(\partial h_\epsilon)_{\mu\nu}, (\partial\delta)_{\mu\nu}).$$

It follows that

$$\|(\partial h_\epsilon)_{\mu\nu}\| \geq (1+\epsilon)^{-1}\|F^0_{\mu\nu}\| \geq (1+\epsilon)^{-1}(\gamma^{\mathrm{pc}}/3)^{1/2}\|B\|.$$

On the torus of side N in level units, its supremum f_ϵ is therefore at least $(1+\epsilon)^{-1}(\gamma^{\mathrm{pc}}/(3N^3))^{1/2}\|B\|$. At two neighbouring sites in direction μ, use the bond equation at the site carrying the factor $1+\epsilon$. It expresses $2\epsilon(\partial h_\epsilon)_{\mu\nu}$ as η_k times the source minus two first differences of curvature components, with coefficients at most $1+\epsilon$ and 1. Consequently

$$2\epsilon f_\epsilon \leq (2+\epsilon)\,\mathrm{osc}(\partial h_\epsilon) + \eta_k(1+\epsilon)C_{\mathrm{tr}}c_Z\|B\|.$$

Here osc is the largest modulus of the change of one curvature component between neighbouring bare sites. For $\eta_k \leq c_\eta\epsilon N^{-3/2}$ this gives

$$[\partial h_\epsilon]_{\beta_H} \geq \frac{1}{6}\Big(\frac{\gamma^{\mathrm{pc}}}{3N^3}\Big)^{1/2}\epsilon\eta_k^{-\beta_H}\|B\|.$$

Here the seminorm includes neighbouring sites. At fixed N this diverges as $k \to \infty$, although the weight has supremum ϵ. Since the background vanishes at zero block datum, the example concerns its derivative there. Cauchy's estimate would bound that derivative if the backgrounds had a uniform Hölder bound on a common block-datum ball.

The constant-weight assertion follows from the exact identity

$$F + T_{(h,v+c)} = (1+c)(F + T_{(h,v)/(1+c)}).$$

Multiplying the critical equation multiplies its multiplier by $1+c$ and leaves its constraint and gauge unchanged. Uniqueness in Prop. 5.5 then gives the asserted identity of critical fields.

For the history application use the collared sequence before the step and its completion by the next large-field region. The background of the run is its real unweighted critical configuration at the prescribed averages. The block field A_k is read about its finer average in the axial gauge, and its row datum is $b = JA_k$, extended by zero to the remaining rows. The regular-background and chart estimates for these actual step fields, Lem. YM3_I:37.5, Prop. YM3_I:53.12, and the fibre-chart bounds in [7, §B.2], give

$$\alpha \leq 2C_{\mathfrak{P}}\mathfrak{M}B_3\varepsilon(x_t), \qquad \sup|b| \leq 2C_\chi \mathsf{c}_{\mathrm{fl}}\delta(x_k), \quad C_{\mathfrak{P}} = 504L^2B_1^{\mathrm{p}}.$$

For the natural-class assertion, apply Lem. YM3_I:37.5 with the separated witness $m = \mathfrak{M}$, increasing the common block floor if necessary. The layerwise regularity radius of the real background is at most $2B_3\varepsilon(x_t)$, so that lemma gives the displayed natural-class radius on the completed sequence. The chart converts the fluctuation field to the row datum with the displayed factor $2C_\chi$. Choose

$$a_N \leq \min\{a_*, a^{\mathrm{pc}}, \gamma^{\mathrm{pc}}/(4e_0)\}, \quad \rho_N^+ \leq \frac{\gamma^{\mathrm{pc}}\rho_*}{4\kappa_0}, \quad z_N^+ \leq \frac{\gamma^{\mathrm{pc}}z_*}{4\kappa_0},$$

also keeping $\rho_N^+ < \rho_*$ and $z_N^+ < z_*$. Impose the ceilings, for all $x \geq \gamma^{-2}$,

$$2C_{\mathfrak{P}}\mathfrak{M}B_3\varepsilon(x) \leq a_N, \qquad 2C_\chi \mathsf{c}_{\mathrm{fl}}\delta(x) \leq \rho_N^+.$$

Their left sides tend to zero with γ. Thus $e_0\alpha \leq \gamma^{\mathrm{pc}}/4$ and $\kappa_0 s \leq \gamma^{\mathrm{pc}}/4$ at every step $t \geq 2$, which gives the asserted half-floor.

At the first step $k = 0$, the lattice spacing is one, the block averages are the identity and the slice projection vanishes. There is no minimization at this step. At the real regular reference the

fluctuation floor differs from the action Hessian by the second derivative of the block constraint paired with the current. Its loss is at most $8c_{\mathrm{pc}}\alpha$, where $c_{\mathrm{pc}} = \sqrt{2}C_K n_{\mathrm{pc}}$ and $n_{\mathrm{pc}} = 36L^3$. For a complex block field $A^{(0)}$ with $r_A = \sup|A^{(0)}|$, differentiating the four link factors in each plaquette and summing their incident bonds gives the following three operator-norm bounds: the change in the plain action Hessian is at most $128 r_A e^{4r_A}$; the tensor term has norm at most $512 e^{8r_A}\|h\|_\infty$; and the plane term has norm at most $32 e^{4r_A}\sup|v|$. Therefore

$$\begin{aligned}\mathrm{Re}(F + T_{\mathbf{h}})''(JA^{(0)})[J\bar{B}, JB] \geq (\gamma^{\mathrm{pc}} - 8c_{\mathrm{pc}}\alpha - 128 r_A e^{4r_A} \\ - 512 e^{8r_A}\|h\|_\infty - 32 e^{4r_A}\sup|v|)\|B\|^2.\end{aligned}$$

This uses the real natural pair on the two-region sequence (Ω_0, Ω_1) and $\mathfrak{M} \geq M^{\mathrm{pc}}$, exactly as in the unweighted positivity argument above. Take

$$a_N^1 \leq \min\{a^{\mathrm{pc}}, \varpi_\chi, \gamma^{\mathrm{pc}}/(64c_{\mathrm{pc}})\}, \quad r_N^1 = \min\{1/4, \gamma^{\mathrm{pc}}/(1024e)\}, \quad z_N^1 = \gamma^{\mathrm{pc}}/(2048e^2).$$

The ceilings

$$2C_{\mathfrak{P}}\mathfrak{M}B_3\varepsilon(x) \leq a_N^1, \qquad 2C_\chi \mathsf{c}_{\mathrm{fl}}\delta(x) \leq \min\{r_N^1, r_\chi\}, \qquad x \geq \gamma^{-2},$$

put the first-step data in these ranges. The losses through α, through r_A, and through the weights are at most $\gamma^{\mathrm{pc}}/8$, $\gamma^{\mathrm{pc}}/8$, and $\gamma^{\mathrm{pc}}/4$ respectively. This leaves $\gamma^{\mathrm{pc}}/2$ for the stated complex data.

(b) Lem. 5.36 identifies the perturbation as $D^{(1)\mathsf{T}} v D^{(1)} + Z_{1,v}$. The complete mean-square argument is Prop. 5.37: its proof first controls the divergence-input system with the actual averaging rows, inserts $vD^{(1)}A$ as a source, and then localizes with a divergence-form cutoff. It proves both inverse existence and the termwise walk bounds without taking a difference of v.

The grouping by visited pieces after Prop. 5.37 gives the connected-set factor in (b). (c) The label supplies its background regular at the radius $C_m\varepsilon_k$ with the envelope $e(x) = C_m\varepsilon_{j(x)}$. Two of the three ceilings of the statement are the label's: $\sup_{x\geq\gamma^{-2}} C_m\varepsilon(x) \leq \alpha_\star$ bounds the radius of regularity and the envelope at every level by the number $\alpha_\star$ of L, and $\sup_{x\geq\gamma^{-2}} \max\{w_{\mathrm{G}}, w_{\mathrm{lin}}\}(\log x)^{-r} \leq 1$ gives $MR(\gamma^{-2}) \geq \max\{w_{\mathrm{G}}, w_{\mathrm{lin}}\}$, $w_{\mathrm{G}} = 4c_U^{-1}(7\log L + \log 2)$; under these, and with $t(\mathrm{an}) < t_0 \leq t_{\mathrm{lin}}(L) \leq \frac{1}{4}$, Lem. 5.19 and the locality below are applied; the third, the membership ceiling, is read below.

Use Lems. 5.19 and 5.21. The pair (A, ω) is the fixed point of the critical-point contraction built on eq. 5.x and the local Hölder bounds. To localize it, write $G' = (\Delta_D + Q'^*\mathsf{a}'Q')^{-1}$ in the scalar notation and retain $Q'Q'^* = 1$. Since $\ker R = \mathrm{ran}(G'Q'^*)$, the slice condition is equivalently the local equation $(1 - Q'^*Q')\Delta_D D^* A = 0$. Cut the far field off with a slow cutoff ch of width $w(L)$ and bounded scaled first and second differences. In the augmented scalar system the cutoff error is supported where ch varies. Its inverse is the actual compression $\mathbb{K}^{[2]}$ described in Lem. 5.21; the two nonlinear remainders are precisely N and q there. Estimate eq. 5.xix therefore makes the scalar correction a contraction in the exponentially weighted norm. It repairs the slice and the induced block gauge and retains exponential decay away from the cutoff. The repaired field nearly solves the critical-point system about U. Applying its contraction gives the stated locality estimate with c_U and $K_U = 7K_{\mathrm{lin}}^2(1 + 2L^3)^2 e^{c_U w_*}$, after the permitted enlargement of K_{lin}, where $w_* = w + 21L$.

Continuity: a uniform limit of iterates, with the error $\frac{2}{3}4^{-n}$. Uniqueness is in the normalized class: the gauge functions preserve the block averages and induce the identity at the tops. If $v_j[A']$ denotes the accumulated averaging gauge, this means

$$u(y) = v_j[A'^u](y)v_j[A'](y)^{-1} \quad \text{at every top } y \text{ of level } j.$$

The scalar transfer carries a sufficiently small critical point into the slice. Its multiplier is absorbed in the fixed-point equation using the same small-radius condition, so the resulting pair lies in the contraction ball and equals the constructed pair. Two normalized gauges with this property agree at every top. Their quotient preserves W, hence is covariantly constant along its bonds. It is therefore the identity along every pillar from a top, and those pillars cover the bare sites.

This proves the stated uniqueness. Energy: apply eq. 5.xvi to the fixed point, with its actual row forcing $-C_2(U,A)$. Since $|\xi^2 F_U(p)|^2 \le 2a_p(U)$, the local plaquette product estimate gives $\sum_p |\partial U(p)^{-1}\partial W(p) - 1|^2 \le C(L)|\mathrm{an}|_\infty^2 \sum_p a_p(U)$. Cauchy–Schwarz with the unperturbed plaquette defects gives the stated action comparison. Full covariant gradient: let x_* lie in a layer $n \ge 2$, of the scale l_n, and pass to the axial gauge in the cube of radius $\frac14 l_n$ about x_*, built from the transports of U along coordinate paths; the transports differ from $\mathbf{1}$ there by $O(e(x_*)l_n^{-1})$ in modulus and in Hölder-$\frac12$ quotient.

In this gauge the potential A solves a flat system for its curl and its divergence, $dA = \Phi$, $d^*A = \varpi$, where Φ is the curvature and ϖ the divergence of A, each up to products of A with the transports minus $\mathbf{1}$; by the curvature norm of Lem. 5.19 and the slice estimate for the divergence below, Φ and ϖ are bounded and Hölder-$\frac12$ in the scale of the layer, with the sizes $Kt(\mathrm{an})e(x_*)l_n^{-2}$ and, for the Hölder quotients, $Kt(\mathrm{an})e(x_*)l_n^{-5/2}$, K a constant of L alone. The divergence is even Lipschitz: on the slice $RD^*A = 0$, so $D^*A = G'Q'^*m$ for some m, and the range of $G'Q'^*$ has first differences bounded in the scale of the layer, $|\eta_k^3(DD^*A)(b)| \le Kt(\mathrm{an})e(x_b)l(b)^{-3}$.

After a cut-off at the radius $\frac18 l_n$ whose first and second differences are $O(l_n^{-1})$ and $O(l_n^{-2})$, the identity $\Delta = d^*d + dd^*$ writes each first difference of A as second differences of the lattice Green function of the Laplacian applied to the cut data; an interior estimate of Schauder type for these kernels against Hölder-$\frac12$ data, on a cube of radius $\sim l_n$, bounds all nine covariant first differences, $l(x)^2|\eta_k^2(\nabla_\kappa A_\nu)(x)| \le K_5\, t(\mathrm{an})\, e(x)$ with K_5 a function of L alone; in the layers $0, 1$ the bound $|\nabla_\kappa A_\nu(x)| \le |A_\nu(x+e_\kappa)| + |A_\nu(x)|$ suffices. Membership: the plain triple of U lies in the cores at half the radii (Cor. YM3_I:58.4, under its support, radius and smallness hypotheses); the moved triple is the plain triple with the displacement substituted, and in the units of each layer λ the displacement, its first differences, the change of the plaquette variables and the change of the current are at most $K_5'\, t(\mathrm{an})\, C_m\varepsilon_\lambda$, by Lem. 5.19 and the gradient estimate, where $C_m\varepsilon_{j(x)}$ is the envelope with which the label supplies its background, K_5' is a number formed with L through K_{lin} and K_5, and $t(\mathrm{an}) < t_0 \le t_{\mathrm{lin}}(L) \le \frac14$, Lem. 5.19 being applied at the anisotropy an.

These are small against the radii, as the substitution asks, under the third ceiling of the statement,

$$\sup_{x \ge \gamma^{-2}} 2K_5' C_m\, \varepsilon(x) \le b_2, \qquad \varepsilon(x) = A_0 x^{-1/2} (\log x)^{\mathfrak{p}_0} \text{ (equation YM3_I:(634))},$$

b_2 being the radius of Prop. 5.27 among the construction results, with which the substitution with three pairs of radii carries the cores at half their radii into the cores at three quarters of them: under this ceiling they are at most $\frac18 b_2$, and the ceiling is of the admissible shape, its left member tending to 0 as $\gamma \to 0$. Then this substitution puts the moved triple in the cores at three quarters of the radii, all the tower clauses included. Frames retain the following six admissions, supplied by the ordered choice of Prop. YM3_I:69.1 and its source continuation in Section YM3_I:82: (1) the geometric and background hypotheses used in that choice; (2) consistent data and a coupling sequence with flow bound $B_\sharp$; (3) at every level n, $x_n > \bar B$, $y_n > 2\mathfrak{p}_0$, $g_n \le \gamma$ and $2\varepsilon_n \le \bar\varepsilon$; (4) $\varepsilon_q \le \varepsilon_\flat$ at every level q; (5) at every level t,

$$C_{\mathrm{reg}}\varepsilon_t \le \varepsilon_{\mathrm{v}}, \qquad L^2 C_{\mathrm{reg}}\varepsilon_t \le c_2^{\mathrm{SO}}, \qquad 2L^2 C_{\mathrm{reg}}\varepsilon_t \le \bar\varepsilon;$$

and (6) $x_t \ge 2B_\sharp$ at every level. The layer and support conclusions used here are those of Cor. YM3_I:58.4, with these actual admissions; the pivot evaluation additionally retains the ceilings preceding Lem. YM3_I:66.1.

The real and sourced constructions use the same coupling sequence, tests and canonical real domains, by Lems. YM3_I:82.4 and YM3_I:82.5. Condition (1) is part of the geometric choice in Prop. YM3_I:69.1; condition (2) is the admissible-sequence and room hypothesis of Thm. YM3_I:D. The proof of Prop. YM3_I:69.1 imposes $x \ge 2B_\sharp$, $x > \bar B$ and $\log x > 2\mathfrak{p}_0$ uniformly for $x \ge \gamma^{-2}$. The additional ceilings in the proof of Lem. YM3_I:82.5 give $2\varepsilon_t \le \bar\varepsilon$, $\varepsilon_t \le \varepsilon_\flat$ and $L^2 C_{\mathrm{reg}}\varepsilon_t \le c_2^{\mathrm{SO}}$ at every level, as well as the remaining graded-field conditions there. These are (3)–(6). (d) The Hessian

of the Lagrangian at a critical background annihilates, against every field, a gauge direction whose gauge transformation preserves the block averages of every configuration. For a gauge direction whose linearized block average vanishes only at the background, it annihilates the fields in $N(Q)$; against other fields the remaining term lies in the range of Q^{T} and is absorbed by changing the multiplier. Thus Lem. 5.19, applied on the longer sequence in its own gauge, inverts the linearized system in the axial gauge. Its weighted maximum bound is that lemma. Its square-sum bound follows from Lem. 5.20; the multiplier is bounded by testing against a row lift $H\varsigma$ in

$$\langle \omega, \varsigma \rangle = \langle T_\tau A - f - D_1^{\mathsf{T}} Y, H\varsigma \rangle .$$

All constants depend only on L.

With Cauchy's inequalities for the remainder of the system this gives exactly one centre in the polydisc $|\mathsf{n}(b)| < \theta_3 \mathsf{b}_k$ of the statement, $\mathsf{b}_k = A_1 (\log x_k)^{\natural_0}$ the threshold of the level k, and it lies in the half polydisc, $|\mathsf{n}(b)| \le \frac{1}{2}\theta_3 \mathsf{b}_k$, once the following hold. The envelope of the step is that of (c) with C'/L in the place of C_m, where $C' = \max\{LC_m, 2LC_\chi(1+2\theta_3)A_1/(\rho_{\mathrm{u}} A_0)\}$ is the number of the statement and $\rho_{\mathrm{u}} = \rho_{\mathrm{u}}(L)$ the radius of the ball in which the centre is unique: the first member keeps the envelope at least that of (c), and the second puts every complex block datum of the box of the step, read in the chart of the fibre, inside that ball in the norm of the envelope; this is the one number that is enlarged, and it is read by the two conditions that follow.

The size of the anisotropies is below the number of the statement, $t_\theta = \min\{t_0, t_{\mathrm{u}}(L), c_{\mathrm{ce}}\theta_3 A_1/A_0\}$, where $t_{\mathrm{u}}(L)$ is the number below which the moved backgrounds with a complex block datum are formed and $c_{\mathrm{ce}} > 0$ is formed with L and C'; so $t_\theta \in (0, t_0]$ is formed with θ_3 and with constants placed before γ. And γ is below ceilings of the shape $\sup_{x \ge \gamma^{-2}} C'\varepsilon(x) \le c(L)$, of the admissible shape, among them the ceiling $C_\chi(1+2\theta_3)\delta(x) < \frac{1}{8} r_c$, $x \ge \gamma^{-2}$, of the statement, with the numbers C_χ, r_c of L: it keeps the box $|\mathrm{Re}\, B_1(b)| + |\mathrm{Im}\, B_1(b)| < (1+2\theta_3)\mathsf{b}_k$ of the complex block data of the step, of the unscaled radius $(1+2\theta_3)\delta(x_k)$, and with it the polydisc $|\mathsf{n}(b)| < \theta_3 \mathsf{b}_k$, of the unscaled radius $\theta_3\delta(x_k)$, inside the ball of the radius r_c, in the unscaled norm, in which the system of the centre has exactly one solution.

The centre is $\mathsf{n} = g_k^{-1} a''$, $g_k = x_k^{-1/2}$, where a' is the finer block average of $W[\mathrm{an}; b_0]$ read about that of $W[0; b_0]$, the two backgrounds having the same row datum, and a'' is this relative field turned into the block axial gauge; n is holomorphic in an, real at real an and real b_0, and 0 if an is one number. By the bound of (c) on the displacement in the units of the level, read with the envelope $(C'/L)\varepsilon_j$, and the passage to the block axial gauge, $|a''(b)| \le K'' t(\mathrm{an})(C'/L)\varepsilon_k$ with a number K'' of L; as $x_k^{1/2}\varepsilon_k = (A_0/A_1)\mathsf{b}_k$, this is $|\mathsf{n}(b)| \le C_\theta t(\mathrm{an})\mathsf{b}_k$ with $C_\theta = K''C'A_0/(LA_1)$, a number of L, θ_3, A_1/A_0 and C_m, and $C_\theta t(\mathrm{an}) \le \frac{1}{2}\theta_3$ at $t(\mathrm{an}) < t_\theta$ once $c_{\mathrm{ce}} \le L/(2K''C')$.

The square sum for $1 \le k < K$, over all the bonds of the step, follows in mean square, in three inequalities. The passage to the axial gauge: $|a''(c)|^2$ is at most a number of L times $|a'(c)|^2$ plus the sum of $|a'(c')|^2$ over the bonds c' inside the two blocks that c meets, a bond c' being counted at most $12L^3$ times. The averages in the mean square: the square sum of the block averages a' over a set of bonds is at most a number of L times the square sum of the displacement A' over the bonds beneath them, each in the units of its level. The displacement is the difference $A' = A[\mathrm{an}; b_0] - A[0; b_0]$ at the same complex row datum. Its energy is bounded by eq. 5.xvii. At $b_0 = 0$ this is at most $C|\mathrm{an}|_\infty^2 \xi^3 \sum_p |F_U(p)|^2$; in bare units it is the corresponding bound by $C|\mathrm{an}|_\infty^2 \sum_p |\tilde F_U(p)|^2$, where $\tilde F_U = \xi^2 F_U$ and $|\tilde F_U(p)|^2 \le 2a_p(U)$.

At $b_0 = 0$, since $|\mathsf{n}|^2 = x_k |a''|^2$ and, in a frame, $x_k L^k \le \frac{37}{11}\beta_t(y)$ at every point y, this gives $\sum_b |\mathsf{n}(b)|^2 \le C^2 |\mathrm{an}|_\infty^2 A(\beta_t, U)$, C a number of L; at $k = 0$ the same holds at the external bonds of the step. The centres $\mathsf{n}^{(i)}$ of the earlier steps $0 \le i < k$ are formed by the same rule about the moved background of the label, so that $g_i \mathsf{n}^{(i)}$, $g_i = x_i^{-1/2}$, is the block average at the level i of its displacement, turned into the block axial gauge; by the bound of (c) on the displacement in the units of each layer, each obeys $|g_i \mathsf{n}^{(i)}(b)| \le C t(\mathrm{an})\varepsilon_i$, that is $|\mathsf{n}^{(i)}(b)| \le C t(\mathrm{an})(A_0/A_1)\mathsf{b}_i$,

as $x_i^{1/2}\varepsilon_i = A_0(\log x_i)^{\mathfrak{p}_0} = (A_0/A_1)\mathsf{b}_i$, with one C of L for all the levels; this is at most $\frac{1}{2}\theta_3\mathsf{b}_i$ at $t(\mathrm{an}) < t_\theta$ once $c_{\mathrm{ce}} \le 1/(2C)$, and $c_{\mathrm{ce}} = \min\{L/(2K''C'), 1/(2C)\}$ is a number formed with L and C'.

The stored bonds, which are the external bonds of these steps, lie for the level i within $2L^i$ (bare units) of the layer i; the square sums of the block averages of all the levels over the bonds near their own layers are bounded together, with one number of L, by the square sum of the displacement weighted by the scales of the layers, because the layers are wide and the bonds near the own layers of different levels overlap boundedly; and the ℓ^2 bound of the linear system, as for the centre of the step, then gives $\sum_{i<k}\sum_b |\mathsf{n}^{(i)}(b)|^2 \le C^2|\mathrm{an}|_\infty^2 A(\beta_k, U)$, b running over the stored bonds, with one C of L, $A(\beta_k, U)$ being the action of the background with the couplings of the level k. (e) The room hypothesis of Lem. 5.22 holds at every level: $L^m \ge MR(x_K) \ge LM$ gives $m > m'$, hence $m' < m + K - j$ for $j \le K$. The identity is that lemma on a label, for any list of families with counterterms, with the explicit boundary terms of its clause on a label; the trivial values do not change because the tent families vanish at the trivial triple, and, at a fixed triple with the original families and counterterms held fixed, the linear changes in the re-filed local and boundary entries are those of the tent families.

The size bounds require the stated margin: by Lem. 5.22 with $\kappa' = \kappa + 1$ and $|e_j| \le B_\sharp$, $\|\hat{E}^{(j)} - E^{(j)}\|_{j,\kappa+1} \le \frac{3}{2}|\mathrm{an}|_\infty B_\sharp \alpha_0^2 e^{7(\kappa+1)} \le \frac{3}{2}\sigma_3\|\mathrm{an}\|B_\sharp\alpha_0^2 e^{7(\kappa+1)} \le \theta E_0\|\mathrm{an}\|$, and the triangle inequality, with the bound $(1-\theta)E_0$ of the families before the re-filing, gives $\|\hat{E}^{(j)}\|_{j,\kappa+1} \le E_0$ and the distance from the family at zero source; the boundary terms are at most $\frac{3}{2}M^3|\mathrm{an}|_\infty B_\sharp\alpha_0^2 e^{7\kappa}e^{-\kappa d_j(X)} \le \theta M^3 E_0 e^{-7}\|\mathrm{an}\|e^{-\kappa d_j(X)}$ in a frame of Thms. YM3_I:AJ and YM3_I:AD, where the radius of the level is at most α_0. □

5.1.9. *Scope of the conclusions.* Thms. M(d), N(c), (d) and O(a), (c), (d) live on one torus, with constants that depend on its side; so does Thm. O(e), whose numbers are free of the side when the closed a-neighbourhoods of the supports lie in balls of radius R and $\ell \ge 16n(R+1)$. On $\mathbb{R}^3$ the statements concern limit points and tensor limits, at separated or disjoint supports. Thm. O(d) concerns cumulants with one stress entry among action densities on one torus, at a fixed label. The kernels of Thm. M(h) and the operators of Thm. P are those of each limit, taken along the subsequence that gives it. Thms. Q and R concern one step, the never-large term, and the backgrounds, centres and counterterms of the construction.

Along growing tori and in the periodic Gibbs states the section asserts the following. The cumulants of the plane fields have limit points along every sequence of first-passage data at g with $K_j \to \infty$, in periodic Gibbs states and on tori T_{ℓ_j} with $\ell_j \to \infty$, whatever the relation between K_j and ℓ_j, with the bounds, the invariance and the reflection positivity of Thm. M(c). The bulk term of the rotational Ward identity vanishes with a rate and with numbers that read no side, so that a limit along tori is invariant under rotations exactly when the flux through a shell about the supports tends to 0; this flux tends to 0 for the volume-first scalar limits; and, with the limits taken one after the other, at a label $\mathsf{b} \in I(g)$ the limit in K of the flux on each torus T_ℓ is $\sum_i S_n^{(\ell,\mathsf{b})}(f_1, \dots, (\omega y)\cdot\nabla f_i, \dots, f_n)$ through every enclosing shell, and along sides $\ell_k \to \infty$ in $\mathfrak{L}(g)$ on which the limits $S_n^\circ = \lim_k S_n^{(\ell_k,\mathsf{b})}$ exist, the limit in k of these limits, through any shell that covers the supports, a fixed one or a far one, is $-\frac{d}{dt}S_n^\circ(f(e^{-t\omega}\cdot))|_{t=0}$, and there are integers K_k^* such that S_n° is the limit of the connected functions along $(K_k, \ell_k, L^{K_k}\mathsf{b})$ for every sequence $K_k \ge K_k^*$ (Thm. O(b)).

For each fixed order $n \ge 2$, along sides $\ell_j \to \infty$ with $\ell_j \le c_\ell(n,g)K_j^{2\mathfrak{p}_0/3}$, a limit along tori at first-passage labels is the limit taken spacing first, then volume, and raising the cutoffs at fixed sides and labels does not change it (Thm. O(b)). With numbers that read no side, in every periodic Gibbs state the stress insertion collapses onto its classical tensor and the conserved tensor decorrelates from any number of tensor entries at separated supports (Thm. N(e), (f)); every tensor limit is

conserved at disjoint supports (Thm. N(g)); and the translation identity of Thm. O(e), with its null term and its contact terms, holds per ball with numbers free of the side.

5.1.10. *The conjugation specific to two by two matrices.* The complex conjugation used for SU(2) preserves the matrix norm. This fact is needed when comparing the domains of a local term and its conjugate; it is not a general property of $X \mapsto (X^*)^{-1}$ in higher matrix dimension.

Lemma 5.23. Put $\epsilon = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$ and $\Theta(X) = \epsilon \overline{X} \epsilon^{-1}$. Then Θ is a conjugate-linear multiplicative involution on complex two by two matrices and

$$|\Theta(X)| = |X|, \qquad \Theta(X) = \overline{\operatorname{tr} X}\, I - X^*.$$

For $U \in \mathrm{SL}(2, \mathbb{C})$ and traceless A,

$$\Theta(U) = (U^*)^{-1}, \qquad \Theta(iA) = iA^*.$$

In particular $|(U^*)^{-1} - I| = |U - I|$. ■

Proof. Entrywise conjugation and conjugation by ϵ preserve products. Since ϵ is real and $\epsilon^2 = -I$, their composite squares to the identity. Entrywise conjugation preserves the operator norm, as does conjugation by the unitary matrix ϵ. Direct multiplication gives $\epsilon Y^{\mathsf{T}} \epsilon^{-1} = (\operatorname{tr} Y) I - Y$. Take $Y = X^*$ to obtain the displayed formula for $\Theta(X)$. For determinant-one Y, the same two by two identity gives $Y^{-1} = (\operatorname{tr} Y) I - Y$; apply it to $Y = U^*$. For traceless A, the formula gives $\Theta(iA) = -(iA)^* = iA^*$. Finally apply norm preservation to $U - I$. □

Lemma 5.24. Let j be a level with $m' < m + K - j$, let $D_j = L^{j-K}$, and put $W = MD_j$. Thus the cubes of level j have physical side W and the torus has side $\ell = 2L^m$. Let $\mathrm{pr} : \mathbb{R}^3 \to T_\ell$ be the covering map. A finite localization domain $\widehat{X}$ in the infinite level-j cube lattice is called near if it lies in a cuboid whose three side lengths are strictly less than $\ell/2$. A torus domain is non-wrapping if it is the image of a near set. Then the following hold.

(1) The image of every infinite localization domain belongs to $\mathcal{D}_j$. Every near lift of a non-wrapping domain belongs to the infinite domain family. Its cubes correspond bijectively to those of the torus domain, adjacency is preserved in both directions, and $d_j(\widehat{X}) = d_j(X)$.
(2) All near lifts of one domain are its translates by $\ell\mathbb{Z}^3$. They are pairwise disjoint; specifying one lift of a point of the domain specifies exactly one near lift containing that point.
(3) Every infinite domain $\widehat{X}$ lies in a cuboid with side lengths at most $W(d_j(\widehat{X}) + 2)$. A torus domain with $W(d_j(X) + 2) < \ell/2$ is non-wrapping. Consequently a wrapping domain, or an infinite domain which is not near, has size at least $\ell/(2W) - 2$.

■

Proof. The hypothesis gives $\ell/W = 2L^{m+K-j-m'} \geq 2L$. Projection carries cubes to cubes and shared faces to shared faces. It therefore carries a finite wall-connected union to a wall-connected union. On a near set projection is injective. The coordinate differences of two of its points have magnitude less than $\ell/2$; a different lift of such a difference differs by a period. Each cube of the image lifts to a whole cube: the difference between the inverse chart on that cube and any one of its cube lifts is continuous and period-valued, hence constant. Thus the inverse chart preserves the faces, edges and corners of its cubes, as well as their adjacency. Lifting a wall chain shows that a near lift is a localization domain. Connected edge sets meeting every cube are carried in both directions with their number of edges unchanged. Taking the minimum of that number proves the equality of the linear sizes.

To compare two near lifts, translate one by a period so that one pair of points over the same torus point agrees. For any other pair over the same point, their displacement is a period. Each coordinate of this displacement is the difference of two displacements from the matched points, each of magnitude less than $\ell/2$. It has magnitude less than ℓ, so is zero. The lifts therefore agree after

that translation. Distinct period translates are disjoint, since two points in a near cuboid cannot differ by a nonzero period. Translating a chosen lift to contain the prescribed point proves existence and uniqueness in the second assertion.

For an infinite domain choose a connected edge set with $d = d_j(\widehat{X})$ edges meeting every cube. If $d = 0$ this set is one corner. Otherwise enumerate its edges so that each edge after the first meets an earlier edge. Adding an edge of length W increases each coordinate span by at most W. The edge set is therefore contained in a cuboid with side lengths at most dW. Every cube of the domain meets it and lies within sup distance W of it. Enlarging the cuboid by W on both sides proves the bound $W(d+2)$.

The same construction lifts a small torus domain. Enumerate a minimal connected edge set as above, lift the first edge, and translate each later edge by a period so that it meets the chosen lift of an earlier edge. Their union projects onto the whole edge set and lies in a cuboid of side lengths at most dW. For every cube of the domain choose a lift containing a lifted point where that cube meets the edge set. The union of these cubes projects onto the domain and lies in the W-enlargement of the cuboid. If $W(d+2) < \ell/2$, this is a near lift. The last two lower bounds are the contrapositives of the cuboid bounds. □

The brackets of equation YM3_I:(806) are alternating finite differences. Their useful support property has an elementary proof which we record to make the localization of a shifted local term explicit.

Lemma 5.25. Let P be a finite set of cubes and let $\mathsf{N}_0 \subset \mathsf{P}$ be nonempty and wall-connected. Suppose that, at every vertex $\mathbf{1}_W$, $F(\mathbf{1}_W)$ depends only on the wall-components of W meeting N_0: deleting all other components does not change its value. For $\mathsf{N}_0 \subset W \subset \mathsf{P}$,

$$F(\mathbf{1}_W) - F(0) = \sum_{\mathcal{K} \subset W \setminus \mathsf{N}_0} F^*[\mathcal{K} \mid \mathsf{N}_0]. \tag{5.xxvi}$$

Every term whose union $\mathcal{K} \cup \mathsf{N}_0$ is not wall-connected is zero. ■

Proof. The resummation is equation YM3_I:(809): expand the alternating sums, and use $\sum_{A \subset B} (-1)^{\#A} = 0$ for nonempty B. For the second assertion choose a wall-component c of $\mathcal{K} \cup \mathsf{N}_0$ disjoint from N_0 and a cube $\Delta \in c$. For every $S \subset \mathcal{K} \setminus \{\Delta\}$, all components within c are unrooted, even after Δ is added. Thus $F(\mathbf{1}_{S \cup \mathsf{N}_0}) = F(\mathbf{1}_{S \cup \{\Delta\} \cup \mathsf{N}_0})$. The two terms have opposite signs in the bracket. Pairing them proves its vanishing. The same cancellation applies to $F - F(0)$. □

5.2. **Local analytic cores and substitution.** We use the simple domains and hereditary analytic cores of Def. YM3_I:49.1, with the geometric and regularity thresholds of Prop. YM3_I:51.9. These thresholds, together with the positivity of the local gauge constants, will be called the tower hypotheses below. In particular, the four geometric parameters are powers of L, $L \geq 13$ is odd, $R_1 \geq L$, $M_1 \geq R_1 \mathfrak{M}$, and $M \geq L^4 M_1$; the background and response thresholds are those fixed at the beginning of Section YM3_I:51.5. Write $a^\sharp \leq 1/60$ and $K_{\mathrm{tw}}^\sharp \geq 1$ for its tower radius and bound. Every use below keeps the domain and subtower restrictions in that proposition. A domain on the infinite lattice is finite; the same statement there is read in a containing lattice chart, with the chart compatibility established in Section YM3_I:52.5, equation YM3_I:(926).

The triple is $P = (U, A', \mathbf{J})$, where U is real and $\mathbf{p}(P) = (e^{i\eta_j A'} U, \mathbf{J})$ is its represented pair. Its independent bond coordinates and gauge action are those preceding Def. YM3_I:49.1. For a real bare configuration V, its pair at level j is $(V, J^{(j)}(V))$ restricted to the incident bonds of the domain. A local term is evaluated at this pair only when that pair belongs to its core. This convention does not assert that an arbitrary real configuration belongs to the core.

For a block field W, the background $U_j(W)$ denotes an element of the unique regular critical orbit over its determining datum, when that orbit exists (Defs. YM3_I:9.3 and YM3_I:9.5). On the full torus the reduced determining set consists exactly of the terminal bonds. Gauge invariant expressions

have the same value at every representative. The total of a pointed family is equation YM3_I:(935); it is defined only if the orbit exists and every pair read by the sum belongs to the respective core.

Lemma 5.26 (Open pair domains)**.** Under the tower hypotheses, let X be a localization domain and $b_0, b_1, c_0 > 0$, $b_0 + b_1 \le a^\sharp$. For fixed real U, the set

$$\mathcal{O}_U = \{(A', \mathbf{J}) : (U, A', \mathbf{J}) \in \mathcal{U}_j^\circ(X; b_0, b_1, c_0)\}$$

is open in the independent bond coordinates. The map $\Psi_U(A', \mathbf{J}) = (e^{i\eta_j A'}U, \mathbf{J})$ is biholomorphic from this set onto an open set of pairs. The union of these pair domains over U is open. A function on the core is analytic and depends only on the represented pair if and only if it descends to a holomorphic function on that union. ■

Proof. The conditions on the real factor and its real subtowers are fixed when U is fixed. If any fails, the slice is empty. Otherwise the simple-domain conditions on A' and $\mathbf{J}$ are finitely many strict inequalities between continuous functions. On this open set, Prop. YM3_I:51.9(f) constructs every required complex subtower holomorphically. There are finitely many subtowers, bonds and plaquettes in X. Their strict curvature and current inequalities therefore define an open subset. The holomorphic-continuation clause of the core is preserved on this subset by the same tower construction.

On the slice, $\eta_j|A'(b)| < \eta_j b_1 < 1/2 < \log 2$. Thus the logarithm at I gives the holomorphic inverse

$$(\mathbf{U}, \mathbf{J}) \longmapsto \big((i\eta_j)^{-1}\log(\mathbf{U}U^{-1}), \mathbf{J}\big)$$

of Ψ_U. The exponential and this logarithm identify open neighbourhoods in the bond-coordinate spaces, so the image of the open slice is open. Their union is open as well. Composition proves one direction of the final assertion. Conversely a function of the pair gives a single well-defined function on the union. On each open image it is the analytic slice function composed with the displayed inverse. These local holomorphic functions agree on intersections, and hence define the required holomorphic function. □

The chart at the trivial pair is $P[v] = (I, A, \mathbf{J})$, where $\|v\|_X = \max\{\sup|A|, \sup|\nabla A|, \sup|\mathbf{J}|\}$ and only differences defined on the incident bonds enter the maximum. Equation YM3_I:(933), proved in Section YM3_I:52.6, gives

$$r_{\mathrm{us}} = \min\{b_1, a^\sharp/4, b_0/(4K_{\mathrm{tw}}^\sharp)\}, \qquad P[\{\|v\|_X < r_{\mathrm{us}}\}] \subset \mathcal{U}_j^\circ(X; b_0, b_1).$$

In particular this is a genuine ball in the full core, not just in the simple domain.

Proposition 5.27 (Three pairs of radii)**.** Assume the tower hypotheses and let X be a level-$(j-1)$ domain, with lengths measured in level-j units. Let positive radii satisfy

$$\begin{gathered}\beta_0 < b_0 \le \gamma_0, \quad \beta_1 < b_1 \le \gamma_1, \quad \gamma_0 + \gamma_1 \le a^\sharp,\\ \beta_0 + \tfrac{9}{4}s \le \gamma_0, \quad \beta_1 + \tfrac{11}{10}s \le \gamma_1,\\ \beta_0 + \tfrac{9}{4}b_2 \le b_0, \quad \beta_1 + \tfrac{11}{10}b_2 \le b_1,\\ 2K_{\mathrm{tw}}^\sharp(\gamma_0 + \gamma_1^2)b_2 \le (b_0 - \beta_0)s.\end{gathered}$$

If $P = (U, A', \mathbf{J}) \in \mathcal{U}_j^\circ(X; \beta_0, \beta_1, c_0)$, $\mathbf{U} = e^{i\eta_j A'}U$, and $|\mathbf{A}|, |\nabla_{\mathbf{U}}\mathbf{A}| < b_2$, $|\mathbf{J} + \mathbf{K}| < c_0'$, then

$$P \dotplus (\mathbf{A}, \mathbf{K}) = \big(U, (i\eta_j)^{-1}\log(e^{i\eta_j \mathbf{A}}e^{i\eta_j A'}), \mathbf{J} + \mathbf{K}\big) \in \mathcal{U}_j^\circ(X; b_0, b_1, c_0').$$

Its pair is $(e^{i\eta_j \mathbf{A}}\mathbf{U}, \mathbf{J} + \mathbf{K})$. ■

Proof. The logarithm and the simple-domain bounds follow from the pointwise increment calculation in the proof of Prop. YM3_I:51.10: the potential and its first derivative increase by less than $11b_2/10$, and the plaquette bound by less than $9b_2/4$. The real factor does not change. Put $R = s/b_2$.

The last inequality and $K_{\mathrm{tw}}^{\sharp} \geq 1$ imply $R > 2$. For $|z| < R$ the same increment calculation, with $z\mathbf{A}$ in place of $\mathbf{A}$, places the substituted field in the simple domain at outer radii (γ_0, γ_1). Prop. YM3_I:51.9(f) therefore constructs all its subtowers holomorphically on that disc.

Consider a level-p subtower and set $\lambda_p = L^{-(j-p)}$. After its curvature and current are multiplied by L^{2p} and $L^{-2(n-p)}$, respectively, each is bounded on the disc by $\lambda_p K_{\mathrm{tw}}^{\sharp}(\gamma_0 + \gamma_1^2)$. At $z = 0$ its norm is strictly below $\lambda_p \beta_0$. The Schwarz estimate for the difference from its value at zero, as used in the same proof of Prop. YM3_I:51.10, bounds its increment at $z = 1$ by

$$\frac{2}{R} \lambda_p K_{\mathrm{tw}}^{\sharp}(\gamma_0 + \gamma_1^2) \leq \lambda_p (b_0 - \beta_0).$$

This proves the strict target inequality for each complex subtower. The real subtower inequalities are unchanged and their old radius β_0 is smaller than b_0. All clauses of the target core follow. □

For the passage from half the fixed radii to three quarters, take $\beta_i = \alpha_i/2$, $b_i = 3\alpha_i/4$, $\gamma_i = \alpha_i$, and $s = \min\{2\alpha_0/9, 5\alpha_1/11\}$. A positive b_2 satisfying

$$b_2 \leq \min\left\{ \alpha_0/9, 5\alpha_1/22, \frac{\alpha_0 s}{8K_{\mathrm{tw}}^{\sharp}(\alpha_0 + \alpha_1^2)} \right\}$$

satisfies all four displayed conditions. This choice depends only on the fixed radii and the tower constants.

5.2.1. *Normalizers for overlapping entries.* In the setting of [7, Lem. N.1], let $A_g = C_{\mathrm{S}} + C_g + C_{\mathrm{gap}} = G_g$. We use the following normalizer and cumulant constants:

$$\begin{aligned} \mathsf{Q}_1(L) &= \exp\{A_g L^3 + \Phi_1(2, 1; L)\}, \\ Q^{\natural}(n, R) &= \exp\{A_g h(n, R)^3 + \Phi_1(R, R+1; M)\}, \\ C_n^{\mathrm{per}}(d; R) &= (n/\varrho)^n \big[\log 2\,(2Q^{\natural}(n, R)^n + 2)^n + \mathsf{U}(R, d)\big]. \end{aligned} \tag{5.xxvii}$$

Here $h(n, R)$ is the least power of L at least $8n(R+1)$; Φ_1 and U are the functions in Section YM3_I:89, with source-energy constants (C_{S}, κ). These are equations YM3_I:(1844) and YM3_I:(1847) with the floor and gap constants of [7, Lem. N.1].

We will use one part of the normalizer proof for tuples whose supports may overlap. Fix a board of side l, and suppose each entry is supported in a ball of radius R whose centre is at distance at least $R + D$ from the walls, where $D > 0$. On the least admissible torus of side at least $2l$, form the cell normalizers and set $E(z) = \sum_c E_c(z)$. Each E_c reads only source coordinates assigned to cell c. The construction in the proof of Prop. YM3_I:91.12 gives, on the source polydisc of radius ϱ,

$$|e^{-E(z)}| \Pi_\ell(z) \leq \exp\{A_g l^3 + \Phi_1(R, D; l)\}^{\#B(z)} \qquad (\ell \in \mathfrak{L}(g),\ \ell \geq 2l). \tag{5.xxviii}$$

Indeed, the cell bound gives one factor $\exp\{A_g l^3 + \#B_c(z)\Phi_1(R, D; l)\}$ for each cell containing an active coordinate. The number of such cells is at most $\#B(z)$, and $\sum_c \#B_c(z) = \#B(z)$. Multiplying proves the inequality. Neither this count nor the holomorphy of E uses separation of the supports. If at least two cells are occupied, its full mixed derivative vanishes since every summand omits a coordinate. Separation is needed only for the additional bound on that derivative when all entries occupy one cell. The same normalizer works for periodic Gibbs states by Prop. YM3_I:91.11.

5.3. **Background estimates and the normalized step.** The background estimates used here are those for reduced determining sets. The region sequence may have an arbitrary first member. It is separated, its terminal lattice has at least $6R_1\mathfrak{M}$ sites in each direction, and the geometric, regular-reference and datum thresholds are those of Section YM3_I:51.5 and the localization construction in Section YM3_I:53.4. In particular we retain

$$0 < \varepsilon_{\mathrm{r}} \leq \bar{\varepsilon}, \qquad C_{\mathfrak{P}}\mathfrak{M} B_3 \varepsilon_{\mathrm{r}} \leq a_{\mathrm{R}}, \qquad \varrho = \min\{r_{\mathrm{B}}, \varrho_{\mathrm{R}}^0\}, \qquad B_3 \varepsilon_{\mathrm{r}} \leq L^{-2} a_{\mathsf{m}}.$$

Here a_{m} is the radius of Prop. YM3_I:30.2; the last inequality ensures its curvature hypothesis for the regular backgrounds, whose bare curvature is at most $B_3\varepsilon_{\mathrm{r}}\eta_n^2$. All depth thresholds chosen from this regularity radius use the decreased value. Thm. YM3_I:E gives a single regular critical orbit and its holomorphic axial representative. Lem. YM3_I:51.6 writes it as $(e^{i\eta_n\mathcal{H}(B)}U_0)^{u_B}$ on $\sup|B|<\varrho$, with $\mathcal{H}(0)=0$, $u_0=I$ and the stated Landau condition.

For the response constants in this paper we take $B_5^{\mathrm{e}}\geq\max\{8,B_5\}$ and $\delta_0^{\mathrm{e}}=\delta_{\mathrm{lc}}$ from equations YM3_I:(996) and YM3_I:(997). Thus, with $m_y(B)=\sum_{c>0}e^{-\delta_0^{\mathrm{e}}d(y,y_c)}|B(c)|$,

$$\xi_y|\mathcal{H}(B)(b)|\leq B_5^{\mathrm{e}}m_y(B),\quad \xi_y^2|\nabla_{U_0}\mathcal{H}(B)(x)|\leq B_5^{\mathrm{e}}m_y(B),\quad |u_B(y)-I|\leq 4\kappa_{\mathrm{av}}B_5^{\mathrm{e}}m_y(B).$$

The first two bounds also hold for differences at two real data, with $m_y(B-B')$ on the right. The derivative estimate YM3_I:(838) supplies the corresponding linear response and its second covariant difference, where all incident plaquettes are inner. These estimates hold on the specified datum ball; they are not assertions for arbitrary block configurations. Here κ_{av} denotes the joint contour constant called κ_6 in the localization construction. When a bound for the derivative of the gauge at a top is written with a single coefficient k_6, we take $k_6=4\kappa_{\mathrm{av}}$.

The localization estimate itself is Prop. YM3_I:53.5. Its hypotheses include the one- or two-component complement, the four-layer collars and their level assignments, the separation of different components, the joined sequence and its regular datum, and $\varepsilon_{\mathrm{r}}\leq\epsilon_{\mathrm{lc}}$. The definitions and all those conditions are equations YM3_I:(998)–YM3_I:(1000) and the paragraphs preceding the proposition. We keep its wide three-layer collars and its weighted distance to the tower rows.

Lemma 5.28 (Iterated plaquette bound)**.** Let p' be a plaquette at level $q\geq 1$. At level $j\leq q$, take the union of the four $(q-j)$-blocks below its corners and let ϖ_j be the largest plaquette curvature of $\mathcal{M}^jU$ within that union, in all three orientations. Set

$$\theta_L=\exp\frac{1}{2(L^2-1)},\qquad c_2^{\mathrm{SO}}=\min\{\varpi'_L/2,L^2/(4C_0^{\mathrm{SO}})\}.$$

The constants in the latter formula are those of Lem. YM3_I:14.9. If $\beta=L^{2q}\varpi_0\leq c_2^{\mathrm{SO}}$, then every average in this cone is defined and

$$\varpi_j\leq\theta_L L^{2(j-q)}\beta\quad(0\leq j\leq q),\qquad 1<\theta_L<1.003.$$

■

Proof. The cone at level $j+1$ is a union of blocks contained in the cone at level j. Consequently the one-step plaquette estimate in Lem. YM3_I:14.9 applies there whenever $\varpi_j\leq\varpi'_L$, and gives

$$\varpi_{j+1}\leq L^2\varpi_j(1+C_0^{\mathrm{SO}}L^{-2}\varpi_j).$$

Put $\theta_0=1$ and $\theta_j=\prod_{i<j}(1+\frac{1}{2}L^{2(i-q)})$. The geometric series and $1+x\leq e^x$ give $\theta_j\leq\theta_L<2$. If $\varpi_j\leq\theta_jL^{2(j-q)}\beta$ and $j<q$, then $\varpi_j\leq 2L^{2(j-q)}c_2^{\mathrm{SO}}\leq\varpi'_L$, so the small-curvature branch of the real average applies. Moreover $C_0^{\mathrm{SO}}L^{-2}\varpi_j\leq\frac{1}{2}L^{2(j-q)}$ by the definition of c_2^{SO}. The recurrence proves the next induction step. The case $j=0$ is the definition of β. Finally $L\geq 13$ gives $\theta_L\leq e^{1/336}\leq 336/335<1.003$. □

Lemma 5.29 (Averaging along a straight closed line)**.** Let U be a real SU(2)-valued configuration and $q\geq 1$. Suppose every bare plaquette has curvature at most ω_0 and $L^{2q}\omega_0\leq c_2^{\mathrm{SO}}$, with the constant of Lem. 5.28. Let Γ be a straight closed line of N bare bonds through a level-q site, where L^q divides N. Let Γ_q be the same line in the level-q lattice. Then

$$|\mathcal{M}^qU(\Gamma_q)-U(\Gamma)|\leq 13.3\,NL^q\omega_0.$$

The assertion also holds on a rectangular periodic lattice under these same local hypotheses. ■

Proof. We first keep the curvature parameter in the proof of Lem. YM3_I:4.4. Write $t = L^2\omega \le 1/200$ at one step. The contour diameter is at most $d = e^{9t/8} - 1$, and the rearrangement bound for the middle path is $e^{25t/8} - 1$ (Lem. YM3_I:4.1). Prop. YM3_I:4.3 writes the contour mean relative to any chosen contour as e^Y, with Y skew-Hermitian and $|Y| \le 2d$. Since $|e^Y - 1| \le |Y|$, the two contour replacements and the middle-path replacement, telescoped between unitary matrices, give

$$|W_x - 1| \le 4(e^{9t/8} - 1) + (e^{25t/8} - 1) < 8t.$$

Indeed $e^u - 1 \le u/(1-u)$ and $t \le 1/200$ give the coefficient $7200/1591 + 200/63 < 8$. Hence $|\log W_x| \le 8t/(1-8t) < 12t$. The average in YM3_I:(37) is the exponential of the mean of these skew-Hermitian logarithms times the straight transport. It differs from that transport by at most $12L^2\omega$.

At level j, Lem. 5.28 gives $\omega_j \le \theta_L L^{2j}\omega_0$. The small-curvature admission in that lemma also gives $L^2\omega_j \le L^2\varpi'_L < 1/200$ for $j < q$. The line has N/L^{j+1} bonds after the next average. A product of unitary matrices changes by at most the sum of the changes of its factors, so summing the one-step errors gives

$$\sum_{j=0}^{q-1} 12\frac{N}{L^{j+1}}L^2\omega_j \le 12\theta_L N\omega_0 \sum_{j=0}^{q-1} L^{j+1} \le \frac{12L\theta_L}{L-1}\,NL^q\omega_0 < 13.3\,NL^q\omega_0.$$

Here $L \ge 13$ and $\theta_L < 1.003$. All paths and exchanges are local to their double blocks. The proof is unchanged when the periods in the three directions differ. □

Lemma 5.30 (Derivatives and remainder of the plain average)**.** Take $a_Q = a_{\mathrm{plain}}$, $c_Q = c_{\mathrm{plain}}$ and C_Q from Prop. YM3_I:51.1, with the smallness choices made in its proof, and put $C'_Q = 4C_Q/c_Q$. On its tower over a double block, let U be real with $\varpi(U) \le a_Q L^{-2j}$. For a complex bond form a, write

$$Q_j(U,a)(c) = \frac{1}{i}\log[\mathcal{M}^j(e^{ia}U)(c)\mathcal{M}^j(U)(c)^{-1}], \qquad \tau_c = L^j \sup_{\mathsf{D}_c^j} |a| < c_Q.$$

The form is defined on the positive bare bonds inside the endpoint double-block tower; negative bonds are extended covariantly. Then

$$|\dot{Q}_j(U)a(c)| \le C_Q\tau_c, \qquad |Q_j(U,a)(c) - \dot{Q}_j(U)a(c)| \le C'_Q\tau_c^2. \tag{5.xxix}$$

At $U = 1$, $\dot{Q}_j(1) = \mathbf{Q}_j^{\mathrm{lin}}$, the actual iterated linear average of Lem. YM3_I:4.6. On a whole real periodic lattice satisfying the same curvature bound, the left-trivialized derivative obeys

$$\|D\mathcal{M}^j(U)w\| \le \sqrt{6}C_Q[w]_j, \qquad [w]_j^2 = \sum_y \left(L^j \max_{b_- \in B^j(y)} |w(b)|\right)^2, \tag{5.xxx}$$

where both sums defining the lattice ℓ^2 norms use positive bonds and counting measure. ■

Proof. Fix the row c. Prop. YM3_I:51.1 gives a holomorphic function $F(a) = Q_j(U,a)(c)$ on the ball $\tau_c < c_Q$, with $F(0) = 0$ and $|F(a)| \le C_Q\tau_c$. On every complex radial line the Cauchy bound gives $|F'(0)a| \le C_Q\tau_c$. If $\tau_c \le c_Q/2$, its Taylor remainder is bounded by

$$\sum_{n\ge 2} C_Q c_Q(\tau_c/c_Q)^n = \frac{C_Q\tau_c^2}{c_Q - \tau_c} \le \frac{2C_Q}{c_Q}\tau_c^2.$$

The coefficients can first be estimated on circles of radius less than c_Q and then passed to their supremum. If $c_Q/2 < \tau_c < c_Q$, the bounds for F and $F'(0)$ instead give $|F(a) - F'(0)a| \le 2C_Q\tau_c \le 4C_Q\tau_c^2/c_Q$. The derivative of the logarithm at 1 is the identity, so the linearization of the iterated average in Lem. YM3_I:4.6 identifies the derivative when $U = 1$.

For the final assertion put $m_y = L^j \max_{b_- \in B^j(y)} |w(b)|$. Every bond in the determining double block of c has its initial site in one of its two endpoint blocks. Differentiating the same row estimate at $a = 0$ therefore gives

$$|D\mathcal{M}^j(U)w(c)|^2 \le C_Q^2 \max\{m_{c_-}, m_{c_+}\}^2 \le C_Q^2(m_{c_-}^2 + m_{c_+}^2).$$

Each site is an endpoint of six positive bonds. Summing proves eq. 5.xxx. For $j = 0$ the derivative is the identity and the same bound follows by the same count. □

5.3.1. *The normalized fluctuation measure.* Consider a real canonical step of height t. Retain its actual array-size bounds, the full-core radii and numerical ceilings of Thm. YM3_I:W, and the whole-chart hypotheses of Prop. YM3_I:52.3. For the tame class $\mathsf{O}_{t+1} = \{\varpi(W) < 2\varepsilon_{t+1}\}$ these hypotheses follow under the coupling ceiling used in the chain; explicitly it is chosen so that

$$2\varepsilon_{t+1} \le \bar{\varepsilon}, \quad 2\varepsilon_{t+1} \le \varepsilon_{\mathcal{W}}\alpha_0, \quad \delta_t \le \varepsilon_{\mathcal{W}}, \quad 4\varepsilon_0 < \varepsilon_{\mathrm{v}}.$$

All geometric room and response requirements of that proposition are retained. At a tame W choose U_+ in its regular orbit so that $\mathcal{M}^t U_+$ is in the axial gauge, and put

$$P_W = (U_+, 0, J^{(t+1)}(U_+)), \qquad \mathsf{Rg}_{x_t} = \{B \text{ real on nonpivot bonds} : |B(b)| < \mathfrak{b}_t\}.$$

Thm. YM3_I:E gives regularity at radius $2B_3\varepsilon_{t+1}$, including the current bound in level-$(t+1)$ units. Prop. YM3_I:52.3 places every restriction of P_W in its full core and supplies the whole chart $V_W(B)$ and its lower backgrounds. The complete Gaussian structure has real positive covariance $C_W = (\mathsf{B}_W + \widehat{\mathsf{G}}_W)^{-1}$, as in Prop. YM3_I:52.2 and Thm. YM3_I:W.

Let S_t be the action of the canonical bare array, $S_0 = 0$, and set

$$F_t(V) = \exp[-x_t A^{\eta_t}(U_t(V)) + S_t(U_t(V))]$$

on its defined set. The chart only reads this set. Define

$$\mathsf{E}_W(B) = \sum_{Y \in \mathcal{D}_t} (\mathbb{V}^{\mathrm{J}} + \mathbb{V}^{\mathrm{G}} + \mathbb{V}^{\mathrm{A}} + \mathbb{V}'[S_t])(Y; P_W|_Y, B|_Y), \tag{5.xxxi}$$

$$\Xi_W = \int_{\mathsf{Rg}_{x_t}} e^{\mathsf{E}_W(B)}\, \mathrm{d}\mu_{C_W}(B), \qquad d\nu_W = \Xi_W^{-1} \mathbf{1}_{\mathsf{Rg}_{x_t}} e^{\mathsf{E}_W}\, \mathrm{d}\mu_{C_W}. \tag{5.xxxii}$$

Proposition 5.31. The integral Ξ_W converges absolutely and is nonzero. Thus ν_W is a complex Borel measure of finite variation and mass one. Its pushforward $\kappa_{t+1}(W, \cdot) = (V_W)_* \nu_W$ is supported on the fibre over W and, for bounded Borel G, satisfies

$$\int G\, \mathrm{d}\kappa_{t+1}(W, \cdot) = \frac{\mathsf{I}_t[GF_t](W)}{\mathsf{I}_t[F_t](W)}. \tag{5.xxxiii}$$

The two potentials $\mathbb{V}^{\mathrm{int}} = \mathbb{V}^{\mathrm{J}} + \mathbb{V}^{\mathrm{G}} + \mathbb{V}^{\mathrm{A}}$ and $\mathbb{V}'[S_t]$, read on real fields, have quadratic-growth norms bounded respectively by

$$\tfrac{7}{2} K_P \delta_t \le \tfrac{1}{2} w_\star, \qquad K_\Lambda s_t^{\mathrm{ab}} N_\star \le \tfrac{1}{4} w_\star, \qquad s_t^{\mathrm{ab}} = (1 + \beta_{\mathrm{s}})\delta_t/\alpha_0.$$

They are therefore each at most a constant times δ_t. On the actual all-small-field history measures and with their step factor h_{t+1}, the quotient has the disintegration identity

$$\int H(\mathcal{M}V) G(V) \mathsf{h}_{t+1}(V)\, \mathrm{d}m_t(V) = \int H(W) \left[\int G\, \mathrm{d}\kappa_{t+1}(W, \cdot) \right] \mathrm{d}m_{t+1}(W).$$

All statements use the prescribed tests and gauge factors of the canonical history. ■

Proof. The actual construction in Thm. YM3_I:W supplies the Gaussian structure and all four local potentials on the full core, with Def. YM3_I:52.1's bounds. Restricting the field polydisc to the real cutoff box gives the quadratic-growth norm. The triangle inequality and the calculation in the proof of Thm. YM3_I:V give the displayed two bounds (its interaction bound is in fact $7w_\star/16$). The fourth norm uses the admitted bare-array bound $N(\mathfrak{f}_t) \le N_\star$. The exact integral identity YM3_I:

(902) at $x = \mathsf{t} = x_t$, hence $\tau = 1$, is precisely Ξ_W: the additional quadratic potential is zero there. It proves absolute convergence and nonvanishing. The sum defining E_W is finite and continuous on the box. Division by Ξ_W gives finite variation and mass one.

To identify the quotient, the action and gauge remainders in the chart satisfy

$$e^{-x_t G_W(B)+\mathsf{J}_W(B)} F_t(V_W(B)) = e^{v_W} e^{-(B,C_W^{-1}B)/2} e^{\mathsf{E}_W(B)}, \qquad v_W = -x_t A^{\eta_t}(U_+) + S_t(U_+).$$

This is the sum of the definitions of the Jacobian, gauge and action remainders; their quadratic parts are $\widehat{\mathsf{G}}_W$ and B_W. The physical chart identity of [7, Lem. B.3] therefore gives

$$\mathsf{I}_t[GF_t](W) = k_W \int_{\mathsf{Rg}_{x_t}} G(V_W(B)) e^{\mathsf{E}_W(B)} \, \mathrm{d}\mu_{C_W}(B)$$

with $k_W \neq 0$ independent of G. Divide by its value at $G = 1$. The chart is continuous and $\mathcal{M}V_W(B) = W$, proving the Borel and fibre assertions. The averaged axial field and the complete operators are independent of the representative U_+; the action and the array are gauge invariant. Thus the measure is independent of that choice as well.

For disintegration, use the actual density and passing-test identity proved in Prop. 4.42 at zero source. On the passing input set, $dm_t = e^{\hat{E}_{(t)}} F_t \, \mathrm{d}V$; in the step factor the compatibility tests equal one on the whole small-field chart. The physical chart formula consequently gives

$$\int H(\mathcal{M}V) G(V) \mathsf{h}_{t+1}(V) \, \mathrm{d}m_t(V) = e^{E_{(t)}} \int \chi_{t+1}(W) H(W) \mathsf{I}_t[GF_t](W) \, \mathrm{d}W.$$

At $G = 1$ this is the density formula for m_{t+1}. Insert eq. 5.xxxiii to obtain the assertion. Null sets in the density identities do not affect these integrals. Bounded Borel insertions are allowed by the absolute chart formula; their measurability and integrability follow from that same formula. At $t = 0$ the bare density is $F_0 = \rho_0$ on the passing set, using $4\varepsilon_0 < \varepsilon_{\mathrm{v}}$, so the argument is unchanged. □

5.3.2. *Recovering the datum from the response.* The response estimates give an upper bound for the fine field in terms of its datum. For the comparison of metrics we also need the reverse bound. Only the constant frame is needed: the bare spacing is $\eta = L^{-j}$, the terminal spacing is one, all regions through level j are the whole torus and the next region is empty. Thus every constraint row has scale one and there are no interface rows. All norms in the following statement use counting measure on positive bonds. Retain the geometric and response hypotheses of §5.3.

Lemma 5.32. Let U be a regular real background in this constant frame, with $|\partial U(p) - 1| \le \alpha_0 \eta^2$ and $\alpha_0 \le L^{-2} a_{\mathsf{m}}$, where a_{m} is the radius in Prop. YM3_I:30.2. Let $\mathcal{Q}(A) = a_j(A)$ be its centered straightened row logarithm, and set $K_0 = D\mathcal{Q}(0)$. Then

$$\sup_c \sum_b |K_0(c,b)| \le 2, \qquad \sup_b \sum_c |K_0(c,b)| \le 2L^{-3j}, \qquad \|K_0\|_{2\to 2} \le 2L^{-3j/2}. \tag{5.xxxiv}$$

For the normalized response of equation YM3_I:(837), $\mathcal{Q}(\mathcal{H}_U(\zeta)) = \zeta$ near zero. Consequently, with $\mathsf{Z} = L^{-j} D\mathcal{H}_U(0)$,

$$\|\xi\| \le 2L^{-j/2} \|\mathsf{Z}\xi\| \le C_{\mathcal{Q}} L^{-j/2} \|\mathsf{Z}\xi\|, \qquad C_{\mathcal{Q}} := 34(L^3 + 2). \tag{5.xxxv}$$

At $j = 0$ the same conclusions hold with $K_0 = \mathsf{Z} = 1$. ■

Proof. The row estimate is the zero-field estimate for Q_j^{m} in Prop. YM3_I:30.2, with $\xi_j = 1$ and $\vartheta_c = 1$. Its level-j column estimate is equation YM3_I:(468), again with $\xi_j = 1$ and $\eta = L^{-j}$. These are bounds on sums of the norms of the actual derivative kernels. For clarity, the counting-measure Schur inequality here is

$$\sum_c \left| \sum_b K_0(c,b) A(b) \right|^2 \le 2 \sum_{c,b} |K_0(c,b)| \, |A(b)|^2 \le 4L^{-3j} \sum_b |A(b)|^2.$$

It gives the asserted norm bound.

The straightening recursion gives

$$\mathcal{M}^j(e^{i\eta A}U) = (e^{i\mathcal{Q}(A)}V)^{v_j[A]}, \qquad V = \mathcal{M}^j U.$$

This is the accumulated one-step identity used in Prop. YM3_I:30.2 and in the gauge-removed row formula before Lem. YM3_I:30.6. For the response, equation YM3_I:(837) says that its gauge factor u_ζ satisfies $u_\zeta(y)v_j[\mathcal{H}_U(\zeta)](y) = 1$ at every terminal site. Its actual average is $e^{i\zeta}V$. Gauge covariance and the displayed cancellation therefore give $e^{i\mathcal{Q}(\mathcal{H}_U(\zeta))}V = e^{i\zeta}V$. The two logarithms lie on the branch through zero, proving their equality. Differentiating at zero yields $K_0 D\mathcal{H}_U(0)\xi = \xi$, or $L^j K_0 \mathsf{Z}\xi = \xi$. Equation eq. 5.xxxv follows from eq. 5.xxxiv; the retained constant $34(L^3+2)$ is larger than two. At $j = 0$ no averaging is performed, so both maps are the identity. □

Lemma 5.33 (Continuity on the full core)**.** Let $j \geq 1$ and let X be an admitted finite domain. On the hereditary core of Def. YM3_I:49.1, take $0 < b_1 \leq \pi L^j$. Suppose F depends only on the represented pair and is holomorphic on every slice with fixed real factor. Then F is continuous on the full core, with its topology of matrix entries and bond-form coordinates. ■

Proof. Put $\xi = L^{-j}$ and let $P_\nu = (U_\nu, A_\nu, J_\nu) \to P_0$ in the core. Write $\mathbf{U}_\nu = e^{i\xi A_\nu}U_\nu$. We move the real factor to U_0, keeping this represented pair fixed. For each positive bond, $|i\xi A_0(b)| < \pi$. The differential of the exponential at this matrix is invertible: after a left translation it is the entire function $(1 - e^{-\mathrm{ad}Z})/(\mathrm{ad}Z)$, whose only possible zeros have nonzero eigenvalues in $2\pi i\mathbb{Z}$; here $\|\mathrm{ad}Z\| < 2\pi$. A local inverse Λ_b therefore exists at $e^{i\xi A_0(b)}$. For all large ν set

$$A'_\nu(b) = (i\xi)^{-1}\Lambda_b(\mathbf{U}_\nu(b)U_0(b)^{-1}), \qquad \widetilde{P}_\nu = (U_0, A'_\nu, J_\nu).$$

There are finitely many bonds, so these choices hold together, $A'_\nu \to A_0$, and $\widetilde{P}_\nu$ represents the pair $(\mathbf{U}_\nu, J_\nu)$ of P_ν.

We check that $\widetilde{P}_\nu$ remains in the core. Its real conditions hold at U_0, and its pair-only conditions hold because they hold for P_ν. The finitely many strict bounds on A'_ν and its covariant differences follow by continuity at fixed U_0. For each tower, its complex averaged datum is a continuous function of $\mathbf{U}_\nu$ on the admitted analytic branch. Its real datum is now the fixed datum of U_0. Their logarithmic quotient thus converges to the quotient for P_0. At P_0 that quotient is strictly inside the response ball; the same ball contains the quotients for all large ν. Holomorphy of the tower response gives convergence of its configuration, curvature and current. The strict bounds defining that tower consequently persist. There are only finitely many subdomains and tower tests on X, so all these assertions hold simultaneously. This proves $\widetilde{P}_\nu$ lies in the core for all large ν.

Both $\widetilde{P}_\nu$ and P_0 are now on the slice with real factor U_0. Slice holomorphy and pair dependence give

$$F(P_\nu) = F(\widetilde{P}_\nu) \longrightarrow F(P_0).$$

Since the core is a subspace of a finite-dimensional metric space, sequential continuity proves the assertion. □

5.3.3. *The half-Hölder estimate for the ordinary response.* The ordinary response to a block datum has a half-Hölder first covariant difference. This assertion concerns ordinary forcing in the constrained system. It does not give the same-exponent estimate for a variable anisotropy in Lem. 5.19. We retain the natural-pair geometry, actual averaging rows, local operators and decoupling parameters of Thm. YM3_I:M. All constants below depend only on L. The common regularity radius and block-number floor must meet the local inverse conditions used below.

For $0 < \beta < 1$ use the transported Hölder seminorm of Prop. YM3_I:34.1, with exponent β:

$$[f]_{\mathbf{U},\beta;y} = \sup_{i \neq i'} \left(\frac{\xi_y}{|i - i'|_1}\right)^\beta |\,\mathrm{Ad}(\mathbf{U}(\Gamma_{x,x'}))f(i') - f(i)|.$$

The indices have the same kind, their base points lie in the prescribed enlargement $\Delta^{3/2}(y)$, and the coordinate comb $\Gamma_{x,x'}$ stays in the local domain. At a boundary only admissible indices and combs are used. Write $|f|^{\beta}_{s;y} = \xi_y^{-s}(\sup|f| + [f]_{\mathbf{U},\beta;y})$ and $|\Psi|^{+\beta}_y = \max\{|\Psi|_y, |\nabla_{\mathbf{U}} A|^{\beta}_{-2;y}\}$ for the enhanced solution norm. The forcing norms and their divergence injection are those of eq. 5.xliii, with suprema instead of mean squares; only the first-slot divergence tensor receives the additional Hölder norm.

Lemma 5.34. There are a radius $a_H > 0$, a block floor M_H, a decay rate $\delta_H > 0$ and constants $\mathsf{B}_\sharp(0), \mathsf{B}_\sharp(1/2)$ such that the following holds. Take the local response setting above, with the four local-system radii at most a_H and block number $\mathfrak{M} \geq M_H$. Choose the common decay rate $\mathring{\delta}_0 \leq \delta_H$ in the decoupled construction. Then its normalized operators $H_s, K_s, \mathfrak{G}_s$ satisfy

$$\begin{aligned} \mathfrak{n}_{y,\zeta}(H_s X), \quad \mathfrak{n}_{y,\zeta}(K_s X) &\leq 6L^3 \mathsf{B}_\sharp(1/2) e^{8\varkappa} \sum_z e^{-\mu d(y,z)} |X|_z, \\ \mathfrak{n}_{y,\zeta}(\mathfrak{G}_s f) &\leq 6L^3 \mathsf{B}_\sharp(1/2) e^{8\varkappa} \sum_z e^{-\mu d(y,z)} |f|^\sharp_z, \qquad \mu = 3\mathring{\delta}_0/16, \end{aligned} \tag{5.xxxvi}$$

where $|f|^\sharp_z = \xi_z^3 \sup_{\mathsf{y}(b)=z} |f(b)|$ and

$$\mathfrak{n}_{y,\zeta}(A) = \frac{\xi_y^{5/2} [\zeta \nabla_{\mathbf{U}} A]_{1/2;\xi_y}}{\xi_y^{1/2} [\zeta]_{1/2;\xi_y} + |\zeta|}.$$

Here $\zeta \neq 0$ is a smooth scalar cutoff supported in the local comparison cube, whose intersection with the ambient region lies in W. The subscript $1/2;\xi_y$ restricts pairs to physical Manhattan distance at most ξ_y, and uses the same transports and domain restriction. In this display $[\cdot]_{1/2;\xi_y}$ is the physical quotient, with denominator $|x-x'|_1^{1/2}$; thus $\xi_y^{1/2}[\zeta]_{1/2;\xi_y}$ is dimensionless. The constants are uniform for all admitted local domains and $|s(\Delta)| \leq e^{\varkappa}$ and for arbitrary subsums of the localized expansion. ■

Proof. We give the extension of the two local estimates in the proof of Prop. YM3_I:34.1. Only the fixed exponents $1/2$ and $3/4$ are needed.

At the identity, eliminate the scalar and row components as in Section YM3_I:23.4. The potential is the vector Green operator applied to

$$f_1' + \nabla^{\mathsf{T}} X^{(p)} + \nabla^{\mathsf{T}} X_1 + Q^{\mathsf{T}} \omega.$$

The term X_1 is absent for ordinary forcing; $Q^{\mathsf{T}}\omega$ is absent in system [4]. To extend the cutoff proof there from $\beta = 1/2$ to a fixed $0 < \beta < 1$, take $\beta' = (1+\beta)/2$. The scalar elimination bounds $X^{(p)}$ in $C^{\beta'}$ with weight -2, and f_1' and $Q^{\mathsf{T}}\omega$ with weight -3. Apply the general Green-kernel Hölder bounds YM3_I:(225) and YM3_I:(227), with output exponent β and, for $X^{(p)}$, input exponent β'; their gap is $(1-\beta)/2 > 0$. The same cutoff calculation therefore bounds the first difference of the potential in C^β for ordinary forcing. We use this extension only at $\beta = 3/4$; the case $\beta = 1/2$ is equation YM3_I:349 itself. For the other estimate needed below, that same equation YM3_I:349 already bounds the $C^{1/2}$ first difference for a $C^{3/4}$ divergence source. Its exponent gap is $1/4$ and is not set to zero.

For completeness, the passage to a small regular field does not require absorption in the enhanced norm. In a local real gauge write $\mathbb{K}_0 - \mathbb{K} = \iota_0 V$. The already proved ordinary inverse controls $|\Psi|_y$, and the local coefficient estimate gives

$$|V\Psi|_y \leq C\epsilon \max_{d(y,z)\leq\rho} |\Psi|_z.$$

For $\Psi = \mathbb{E}\iota_0 g$ the identity $\Psi = \mathbb{E}_0\iota_0(g + V\Psi)$ puts only ordinary forcing into the flat enhanced estimate. For a divergence datum g^+ one uses $\Psi = \mathbb{E}_0\iota_0^+(g^+ + (V\Psi, 0))$ instead; the ordinary bound for that solution first uses the positive-exponent divergence estimate. The extra term has

no divergence component, so no Hölder norm of $V\Psi$ is required. Converting the injection and the transported norms between the flat and small gauges costs a fixed local factor. These are the same coefficient, shift and transport comparisons used to prove equation YM3_I:519. At layer zero, every comparison distance is a bounded number of bare spacings and the massive boundary inverse gives the estimates directly. This proves the three required local estimates, with one radius and floor chosen for the two fixed exponents.

The cutoff commutator has ordinary forcing. Consequently in each walk the enhanced estimate is needed only at its output factor; the remaining factors use the ordinary strong inverse. More explicitly, if T_κ is the local inverse with output cutoff and $R^\sharp_\kappa$ is its commutator forcing, the ordinary walk is

$$T_{\kappa_0} R^\sharp_{\kappa_1} \cdots R^\sharp_{\kappa_m}.$$

The output factor is bounded in $|\cdot|^{+\beta}$, and each of the other factors costs $q = C\mathfrak{M}^{-1}e^{-\delta D_0}$. The enlarged comparison cube and its two difference shifts lie in the common $[5/6]$ region, so the rows, transports and seminorms used by the local and global systems agree. This is the same identification used in Prop. 5.37. A finite number of pieces can follow a given piece. Increasing the common block floor until their number times q is at most $1/2$ gives the summed bound and the termwise walk bound.

For H_s and K_s, apply this estimate to the last-row forcing. For $\mathfrak{G}_s$ use the doubled inverse in the proof of Prop. YM3_I:34.1. Its coupling is $D(D^{\mathsf{T}}A' - w')$, a divergence source plus the explicitly retained layer-zero boundary correction. Apply the ordinary $3/4$ estimate to the second copy. The transported product and shift formulas in that proof bound the coupling tensor in $C^{3/4}$, with weight -2. Apply the divergence estimate with exponents $1/2, 1/4$ to the first copy. This bounds its first difference in $C^{1/2}$. The doubled cutoff commutator in equation YM3_I:526 again has ordinary forcing, so the same walk multiplication and summation apply. The local multiplier insertions see only the weak norm, already bounded by these output norms; their tree factors and summability are unchanged. This identifies the estimate for the actual projected operator, rather than for a different scalar Green function.

Finally insert the parameters. The carrier count in the decoupled construction gives

$$\#\mathsf{P}(\omega) \le 8 + 68D_\omega/\mathsf{w}, \qquad |s^\omega| e^{-aD_\omega} \le e^{8\varkappa} e^{-(a-68\varkappa/\mathsf{w})D_\omega}.$$

The standing lower bound on w leaves rate $\mathring{\delta}_0/4$. Fix $\mathsf{B}_\sharp(1/2)$ large enough to include the preceding local constants, the absolute tree masses and the bounded number of output-window cells. The normalized column bounds are therefore

$$[\zeta \nabla H_s e_c]_{1/2;\xi_y}, \quad [\zeta \nabla K_s e_c]_{1/2;\xi_y} \le \mathsf{B}_\sharp(1/2) e^{8\varkappa} (\xi_y^{1/2} [\zeta]_{1/2;\xi_y} + |\zeta|) \xi_y^{-3/2} \xi_z^{-1} e^{-\mathring{\delta}_0 d(y,z)/4} |e|,$$
$$[\zeta \nabla \mathfrak{G}_s f]_{1/2;\xi_y} \le \mathsf{B}_\sharp(1/2) e^{8\varkappa} (\xi_y^{1/2} [\zeta]_{1/2;\xi_y} + |\zeta|) \xi_y^{1/2} e^{-\mathring{\delta}_0 d(y,z)/4} \sup |f|,$$

when c is at z and f is supported over z. The ordinary product inequality $[\zeta v]_\beta \le |\zeta| [v]_\beta + [\zeta]_\beta |v|$ explains the cutoff factor. After multiplying by $\xi_y^{5/2}$, the scale ratios are ξ_y/ξ_z and $(\xi_y/\xi_z)^3$ respectively. The layer comparison absorbs them with a factor L^3 and a loss $\mathring{\delta}_0/16$ in the rate. There are at most six rows at each top. Summing the columns or source cells proves eq. 5.xxxvi. All steps use absolute bounds on individual terms, so arbitrary subsums obey the same estimate. □

Proposition 5.35. Retain the hypotheses of Lem. 5.34 and the datum radius of Thm. YM3_I:M. Choose $\mathsf{B}_\sharp(0)$ as the ordinary kernel constant in that theorem, $\mathsf{B}_\sharp(1/2) \ge \mathsf{B}_\sharp(0)$ as above, and put

$$\mathsf{b}_0 = 6L^3 \mathsf{B}_\sharp(0), \qquad \mathsf{b}_{\mathrm{D}} = \mathsf{b}_0 c_S(\mathring{\delta}_0/16), \qquad \widehat{\mathsf{b}} = \mathsf{b}_{\mathrm{D}} e^{8\varkappa}.$$

For its ordinary response $\mathbf{H}_s$ and any two admissible block data,

$$\sup_y w(y) \mathfrak{n}_{y,\zeta}(\mathbf{H}_s(B_1) - \mathbf{H}_s(B_2)) \le 3\widehat{\mathsf{b}} \frac{\mathsf{B}_\sharp(1/2)}{\mathsf{B}_\sharp(0)} |B_1 - B_2|_w. \tag{5.xxxvii}$$

Here w is any of the admissible exponential weights of Thm. YM3_I:M. In particular, taking $B_2 = 0$ and $w = 1$ gives the same bound for the response itself. ■

Proof. Write $A'_i = K_s B_i + \mathbf{A}_0(B_i)$. The response equations and constraint correction give the exact difference identity

$$\mathbf{H}_s(B_1) - \mathbf{H}_s(B_2) = K_s(B_1 - B_2) - \mathfrak{G}_s(W_s(A'_1) - W_s(A'_2)) - H_s(D_s(A'_1) - D_s(A'_2)).$$

The proof of Thm. YM3_I:J, with the radius in Thm. YM3_I:M, gives

$$|W_s(A'_1) - W_s(A'_2)|^\sharp_w \le |B_1 - B_2|_w, \qquad |D_s(A'_1) - D_s(A'_2)|_w \le \tfrac{1}{2}|B_1 - B_2|_w.$$

Indeed $\|A'_i\| \le 2\widehat{\mathsf{b}} \sup|B_i|$ and $\|A'_1 - A'_2\|_w \le 2\widehat{\mathsf{b}}|B_1 - B_2|_w$; the first bound uses $8C_W\widehat{\mathsf{b}}^4 \sup|B_i| \le 1$, and the second uses $32C_2\widehat{\mathsf{e}\mathsf{b}}\varepsilon_3 \le 1$, exactly the radius inequalities in equation YM3_I:494. Multiplication by w in eq. 5.xxxvi leaves decay rate $\mathring{\delta}_0/16$ for the top sum. Each operator is thus bounded by $\widehat{\mathsf{b}}\mathsf{B}_\sharp(1/2)/\mathsf{B}_\sharp(0)$ in the indicated seminorm. The triangle inequality gives the factor $1 + 1 + 1/2 \le 3$. Finally $\mathbf{H}_s(0) = 0$. □

5.3.4. *The local perturbation of the Hessian.* A plaquette weight changes the principal part of the Hessian. The remainder is a local curvature term. This elementary fact identifies the perturbation that a weighted inverse estimate must control.

Lemma 5.36. Let A be a finite ambient region of a three-dimensional lattice of spacing ξ, and let $\mathbf{U}$ be an $\mathrm{SL}(2,\mathbb{C})$ configuration on its inner bonds. Give one-forms the complex bilinear pairing $(A,B)_\xi = \xi^3 \sum_b \frac{1}{2}\operatorname{tr}(A(b)B(b))$, where the sum uses positive inner bonds. For a complex function w on the positive inner plaquettes, define

$$F_w(A) = \xi^{-1} \sum_p w(p)(1 - \tfrac{1}{2}\operatorname{tr}\partial(e^{i\xi A}\mathbf{U})(p)).$$

There is a unique symmetric Hessian Δ_w with $F''_w(0)[E,A] = (E, \Delta_w A)_\xi$. Its entries depend linearly on w and polynomially on the matrices $\mathbf{U}(b)^{\pm 1}$.

Write ∇_μ for the covariant forward difference, set $D^{(1)}A = \nabla_\mu A_\nu - \nabla_\nu A_\mu$ on a $\mu\nu$ plaquette, and use the ξ^3-weighted plaquette pairing to define its transpose. Then

$$\Delta_w = D^{(1)\mathsf{T}} w D^{(1)} + Z_{1,w}. \tag{5.xxxviii}$$

Suppose that $|\partial\mathbf{U}(p) - \mathbf{1}| \le 1$ and $|\operatorname{Ad}(\mathbf{U}(b'))^{\pm 1}| \le 2$ on every plaquette containing a given bond b and on its four bonds. Put

$$w_b = \max_{p \ni b} |w(p)|, \qquad f_b = \xi^{-2} \max_{p \ni b} |\partial\mathbf{U}(p) - \mathbf{1}|, \qquad |A|_{(b)} = \max_{p \ni b} \max_{b' \in p} |A(b')|.$$

With absolute constants, the kernel of Δ_w satisfies $|\Delta_w(b,b')| \le C w_b \xi^{-2}$, and

$$|(Z_{1,w}A)(b)| \le C w_b f_b |A|_{(b)}. \tag{5.xxxix}$$

Both kernels vanish unless b, b' belong to a common plaquette. The same assertions hold after restriction to a subregion, with fields extended by zero and with the ambient plaquettes retained. ■

Proof. We give the weighted version of the plaquette calculation in Lem. YM3_I:25.3. At $p = p_{\mu\nu}(x)$ put $Y_p = \partial\mathbf{U}(p)$, $y_0 = \frac{1}{2}\operatorname{tr} Y_p$, $\mathsf{U}_\mu = \operatorname{Ad}(\mathbf{U}_\mu)$ and

$$a_1 = A_\mu(x), \qquad a_2 = \mathsf{U}_\mu(x) A_\nu(x + \xi e_\mu),$$
$$a_3 = \operatorname{Ad}(Y_p)\mathsf{U}_\nu(x) A_\mu(x + \xi e_\nu), \quad a_4 = \operatorname{Ad}(Y_p) A_\nu(x).$$

For $\epsilon = (1,1,-1,-1)$ and $S_p(A) = \sum_i \epsilon_i a_i$, transporting the four factors to x gives

$$\partial(e^{it\xi A}\mathbf{U})(p) = e^{it\xi a_1} e^{it\xi a_2} e^{-it\xi a_3} e^{-it\xi a_4} Y_p.$$

The quadratic term of an ordered product is $\frac{1}{2}(\sum_i X_i)^2+\frac{1}{2}\sum_{i<j}[X_i,X_j]$. Since a traceless two-by-two matrix S has $S^2=\frac{1}{2}\operatorname{tr}(S^2)\mathbf{1}$, and $Y_p=y_0\mathbf{1}+i\xi^2F_{\mathbf{U}}(p)$, differentiation yields

$$(A,\Delta_wA)_\xi=\xi^3\sum_p w(p)\left\{y_0\langle\xi^{-1}S_p(A),\xi^{-1}S_p(A)\rangle+i\sum_{i<j}\epsilon_i\epsilon_j\langle[a_i,a_j],F_{\mathbf{U}}(p)\rangle\right\}. \tag{5.xl}$$

The pairing is nondegenerate; polarizing this finite quadratic form therefore defines exactly one symmetric operator. The formula also proves its asserted dependence on w and $\mathbf{U}$.

The exact identity

$$\xi^{-1}S_p(A)=D^{(1)}A(p)+\rho_p(A),\qquad \rho_p(A)=\xi^{-1}(I-\mathrm{Ad}(Y_p))\{\mathsf{U}_\nu(x)A_\mu(x+\xi e_\nu)+A_\nu(x)\}$$

shows that the quadratic form of $Z_{1,w}$ is the sum, multiplied by $\xi^3w(p)$, of

$$\begin{aligned}&(y_0-1)\langle D^{(1)}A,D^{(1)}A\rangle+2y_0\langle D^{(1)}A,\rho_p(A)\rangle+y_0\langle\rho_p(A),\rho_p(A)\rangle\\&\qquad+i\sum_{i<j}\epsilon_i\epsilon_j\langle[a_i,a_j],F_{\mathbf{U}}(p)\rangle.\end{aligned}$$

Here all values in one summand are at p. If $\kappa_p=|Y_p-\mathbf{1}|\le1$, the two-by-two determinant identity gives

$$|y_0-1|\le\kappa_p,\quad \xi^2|F_{\mathbf{U}}(p)|\le\kappa_p,\quad |I-\mathrm{Ad}(Y_p^{\pm1})|\le3\kappa_p.$$

Consequently $|D^{(1)}A(p)|\le C\xi^{-1}|A|_{(p)}$ and $|\rho_p(A)|\le C\xi^{-1}\kappa_p|A|_{(p)}$. Polarize against a one-form supported at b, of norm at most one. Each of the at most four contributing plaquettes then costs $C|w(p)|\xi^{-2}\kappa_p|A|_{(p)}$; the factor ξ^3 cancels against that of the one-bond pairing. The nondegenerate finite dimensional pairing gives eq. 5.xxxix. The same polarization of eq. 5.xl, with the second field supported on one bond, gives the kernel bound. Every summand involves only the four bonds of its plaquette. This proves the locality assertions, including restriction by zero extension. □

In particular the perturbation at $w=1+v$ is

$$\Delta_{1+v}-\Delta_1=D^{(1)\mathsf{T}}vD^{(1)}+Z_{1,v}.$$

The principal insertion contains the divergence source $vD^{(1)}A$. Multiplication by v costs at most $|v|_\infty$ in L^2. This calculation identifies the insertion; it does not assert the uniform inverse or the endpoint Hölder estimate for that insertion.

5.3.5. *Mean-square bounds for the weighted system.* The weight in Lem. 5.36 need not have bounded differences. We keep its principal contribution in divergence form. The relevant inverse estimate is therefore a mean-square estimate with a divergence source. We give the extension of the proof of Thm. YM3_I:G that is needed for this purpose.

We use precisely the separated regions, inner positive bonds, reduced coarse rows and four systems of Sections YM3_I:23 and YM3_I:27. In particular the row space consists of the positive level-n bonds with neither endpoint in Ω_{n+1}. The bilinear pairings have weights ξ^3 on fine sites and bonds, ξ_y^3 on block tops, and ξ_c^3 on a coarse row c. All transposes below use these pairings. The averaging row is the derivative of the physical constraint,

$$QA(c)=\left.\frac{d}{ds}\right|_{s=0}\xi_c^{-1}\mathcal{Q}_n(\mathbf{U};s\xi_cA)(c),$$

with the convention of equation YM3_I:370; Q' is the ordered scalar average. At $\mathbf{U}=\mathbf{1}$, these are the straight bond average and the arithmetic scalar average. At nontrivial fields we retain their actual first and second derivatives. In particular,

$$(A,\Delta^{(2)}[\mathrm{j}]B)_\xi=2(C^{(2)}(A,B),\mathrm{j})_{\mathrm{rows}}.$$

Thus the result below does not replace the physical constraint by its linearization at the identity.

For a fine-indexed field f define the local mean-square seminorm

$$\|f\|_{s;y} = \xi_y^{-s-3/2} \left(\xi^3 \sum_{\mathsf{y}(i)=y} |f(i)|^2 \right)^{1/2} . \tag{5.xli}$$

For a field on tops or rows, use $\|f\|_{s;y} = \xi_y^{-s} \max_{\mathsf{y}(i)=y} |f(i)|$. Values are zero outside the region. The system differences omit translated indices outside their inner-bond or site index set. The ambient Hessian retains boundary plaquettes of the zero-extended field; its additional boundary terms are treated explicitly in the proof below. For $\Psi = (A; v, w, \sigma; p, q, r; \omega)$ put

$$\begin{aligned} \|\Psi\|_y = \max\{&\|A\|_{-1;y}, \|\nabla A\|_{-2;y}, \|v\|_{0;y}, \|Dv\|_{-1;y}, \|w\|_{-2;y}, \|Dw\|_{-3;y}, \|\sigma\|_{-4;y}, \\ &\|p\|_{-2;y}, \|q\|_{0;y}, \|Dq\|_{-1;y}, \|r\|_{-4;y}, \|\omega\|_{-3;y}\}. \end{aligned} \tag{5.xlii}$$

Absent components are omitted for a compressed system. Write $g^+ = (f_1, \dots, f_8; X_5, X_1)$ and set

$$\begin{aligned} \|g^+\|_y = \max\{&\|f_1\|_{-3;y}, \|f_2\|_{-2;y}, \|f_3\|_{-4;y}, \|f_4\|_{0;y}, \|f_5\|_{-4;y}, \|f_6\|_{-2;y}, \|f_7\|_{0;y}, \\ &\|f_8\|_{-1;y}, \|X_5\|_{-3;y}, \|X_1\|_{-2;y}\}. \end{aligned} \tag{5.xliii}$$

Here X_1 has one fine bond index and one difference direction. The source map ι^+ replaces f_1 by $f_1 + \nabla^{\mathsf{T}} X_1$ and f_5 by $f_5 + D^{\mathsf{T}} X_5$. The original map ι is the case $X_1 = 0$.

Proposition 5.37. There are constants $a_2, r_2, B_2, \delta_2 > 0$ and a power M_2 of L, depending only on L, with the following properties. Take the full geometric hypotheses and the natural-pair class of Thm. YM3_I:G, with block number $\mathfrak{M} \geq M_2$ and all four radii $\alpha_0, \alpha_1, \gamma_0, \gamma'$ at most a_2. For any complex plaquette function v with $|v|_\infty < r_2$, replace Δ_Y by Δ_{1+v} in the first slot of each applicable system; retain the other coefficients and rows. The resulting four systems $\mathbb{K}_v^{[i]}$, $i = 1, 2, 4, 5$, are invertible. For $i = 4, 5$,

$$\|(\mathbb{K}_v^{[i]})^{-1} \iota^+ g^+\|_y \leq B_2 \sup_z e^{-\delta_2 d(y,z)} \|g^+\|_z. \tag{5.xliv}$$

The two scalar systems are unchanged. Their bounds are the corresponding compressed bounds. Constants are independent of the volume, the number of layers and the differences of v.

The inverse has a sum over walks of the pieces of Section YM3_I:27.1. If $\varpi = (\kappa_0, \dots, \kappa_m)$, its term $E_{v,\varpi} \iota^+$ vanishes unless consecutive enlarged pieces meet. It is supported in $\kappa_0^{[2/3]}$ at its output and depends only on the coefficients and weights in the union of the local regions of the visited pieces. With $y_0 = y$ it satisfies

$$\|E_{v,\varpi} \iota^+ g^+\|_y \leq B_2 q_2^m \sup \left\{ e^{-\delta_2 \sum_{a=0}^m d(y_a, y_{a+1})} \|g^+\|_{y_{m+1}} \right\}, \qquad q_2 = \frac{C_2}{\mathfrak{M}} e^{-\delta_2 D_0}. \tag{5.xlv}$$

The intermediate tops belong to $\kappa_{a-1}^{[5/6]} \cap \kappa_a^{[5/6]}$ and the final top to $\kappa_m^{[5/6]}$. The number D_0 is the separation between the input cutoff and the commutator support in the same piece construction. An empty supremum is zero. The rate can be fixed first below the unweighted local rate; increasing M_2 and decreasing a_2, r_2 then preserves that rate for all weights under consideration. ∎

Proof. We retain the proof's order: the flat system, a small field, and then localization to regular pieces. This also checks that the mean-square estimate uses the same rows as the nonlinear problem.

The flat inverse. Write

$$\begin{gathered} G' = (\Delta'_a)^{-1}, \quad C' = (Q' G'^2 Q'^{\mathsf{T}})^{-1}, \quad N = G' Q'^{\mathsf{T}} C', \quad P = N Q' G', \quad R = I - P, \\ G_0 = (\Delta_Y(\mathbf{1}) + D R D^{\mathsf{T}} + Q^{\mathsf{T}} \mathsf{m}^{-2} Q)^{-1}, \qquad Z = Q G_0 Q^{\mathsf{T}}. \end{gathered}$$

These are the scalar, vector and row operators in Section YM3_I:23.2. In the notation $T : s \Rightarrow s'$, the arrow means an estimate $\|Tf\|_{s';y} \leq C \sup_z e^{-\delta d(y,z)} \|f\|_{s;z}$. At a common sufficiently small rate,

their mean-square bounds are

$$\begin{aligned} &O : s \Rightarrow s+2, \qquad \nabla O, O\nabla^{\mathsf T} : s \Rightarrow s+1, \qquad \nabla O\nabla^{\mathsf T} : s \Rightarrow s, \\ &\nabla\nabla G_0 : s \Rightarrow s, \qquad C' : 0 \Rightarrow -4, \qquad Z^{-1} : s \Rightarrow s-2, \qquad Q, Q^{\mathsf T}, Q', Q'^{\mathsf T} : s \Rightarrow s. \end{aligned} \tag{5.xlvi}$$

Here $O = G', G_0$, and only weights $-4 \le s \le 2$ are used. For completeness, these bounds follow from the local L^2 estimates in the proof of Prop. YM3_I:21.3, the six bounds of Prop. YM3_I:21.1, and the top and row kernel estimates in Sections YM3_I:21 and YM3_I:22, as follows. If $N_y(f) = (\xi^3 \sum_{\mathsf y(i)=y} |f(i)|^2)^{1/2}$, the local bounds have the form

$$N_y(T\mathbf{1}_z f) \le C\xi_y^a e^{-\delta_0 d(y,z)} N_z(f), \qquad a = 2, 1, 0.$$

Normalization introduces $(\xi_z/\xi_y)^{s+3/2}$. The layer-gap estimate of Lem. YM3_I:21.2 bounds this factor by $L^6 e^{\delta_0 d(y,z)/4}$ after increasing the separation threshold. Summation over z spends another quarter of the decay; the remaining half gives the arrow. The same argument applies to the differentiated scalar energy estimate: pairing $\Delta'_a u = \nabla^{\mathsf T} X$ with an exponentially weighted u bounds its weighted gradient by $C\|X\|_2$, using the coercive conjugated form in that proof. Thus the mixed scalar arrow costs no difference of X.

The top and row seminorms are suprema, so their arrows are their counting-kernel bounds. For an average, Cauchy–Schwarz gives

$$|Qf(c)|^2 \le \Big(\sum_b |Q(c,b)|\Big) \sum_b |Q(c,b)|\,|f(b)|^2.$$

The straight average has row sum one and $|Q(c,b)| \le (\xi/\xi_c)^3$; Q' is the arithmetic mean. A double block meets only $C(L)$ cells at comparable scales. This proves their mean-square arrows. The weighted transposes have bounded row sums, and top or row input is already measured by its supremum; there are at most nine fine components per site. This proves the two transpose arrows with no mesh-dependent factor. These observations also justify compositions of the arrows: the supremum estimate composes by the triangle inequality for d.

The elimination formulas are unchanged by adding X_1. Set

$$p' = RG'f_5 + Pf_6 + Nf_7, \qquad p'' = RG'D^{\mathsf T}X_5, \qquad \varphi = Pf_2 + RG'f_3 - Nf_4,$$
$$f_0 = f_1 + D\varphi + Dp' + Dp'', \qquad f_* = f_0 + \nabla^{\mathsf T} X_1.$$

Direct substitution in the eight equations gives

$$\begin{aligned} &p = p' + p'', && q = G'(f_6 - p), \\ &r = C'\{Q'G'^2(f_5 + D^{\mathsf T}X_5) - Q'G'f_6 - f_7\}, && \\ &A = G_0 f_* && (i=4), \\ &\omega = Z^{-1}(f_8 - QG_0 f_*), && A = G_0(f_* + Q^{\mathsf T}\omega) \quad (i=5), \\ &w = PD^{\mathsf T}A + \varphi, && v = G'(f_2 + D^{\mathsf T}A - w), \\ &\sigma = C'\{Q'G'(f_2 + D^{\mathsf T}A) - Q'G'^2 f_3 - f_4\}. && \end{aligned} \tag{5.xlvii}$$

For example, p', p'', φ have weight -2 and their gradients have weight -3; for Dp'' use $DRG'D^{\mathsf T} = DG'D^{\mathsf T} - (DP)G'D^{\mathsf T}$. Hence f_0 has weight -3. The mixed arrow in eq. 5.xlvi gives A of weight -1 and ∇A of weight -2 even when X_1 is only in mean square. The row expression gives ω of weight -3.

To finish the remaining slots, write $D^{\mathsf T}A = T_1\nabla A + T_2 A$. The first matrix has at most three entries per row and one per column, each of size one. The second is supported on the layer of bare cells meeting the boundary, where $\xi_y = \xi$, and has at most three entries per row and two per column, each of size ξ^{-1}. Cauchy–Schwarz therefore bounds $D^{\mathsf T}A$ with weight -2. The last line of

eq. 5.xlvii now gives exactly the remaining entries of eq. 5.xlii. This proves the flat estimate. When $X_1 = 0$ the same calculation also bounds $\nabla\nabla A$ with weight -3.

Small backgrounds and the actual rows. The defect factorization of Section YM3_I:25.4 is $\mathbb{K}_0 - \mathbb{K} = \iota_0 V$. Its coefficients satisfy the same bound in the mean-square norms,

$$\|V\Psi\|_y \le C\epsilon \max_{d(y,z)\le\rho_0} \|\Psi\|_z, \qquad \rho_0 = C(L). \tag{5.xlviii}$$

Here is the kernel check needed to pass from the displayed pointwise defect to this assertion. The local difference and current terms have uniformly bounded row and column multiplicities, scale factors ξ_y^{-a}, and coefficients $O(\epsilon)$; their second-order part is expressed using one difference of A. Cauchy–Schwarz gives eq. 5.xlviii for these terms. For the averaging terms, the proof of Lem. YM3_I: 25.2 gives

$$\sum_b |\delta Q(c,b)| \le C\epsilon, \qquad |\delta Q(c,b)| \le C\epsilon(\xi/\xi_c)^3,$$

with double-block support; the same estimates hold for Q'. The preceding Cauchy–Schwarz calculation proves their mean-square bounds. Their transposes use the specified physical pairings and the bounded number of incident rows. Products such as $\delta(Q^{\mathsf{T}}\mathsf{m}^{-2}Q)$ are expanded with one changed factor, so satisfy the required weighted bound.

Finally $\Delta^{(2)}[\mathrm{j}]$ has a symmetric bilinear kernel supported in the same double blocks. The bound for $C^{(2)}$ in that lemma, polarized against a field supported at one bond, bounds its absolute row sum by $C\epsilon\xi_y^{-2}$. In Pauli coordinates the bilinear transpose has norm at most twice the original matrix norm; therefore the column sum obeys the same bound with a fixed extra factor. The weighted Schur inequality proves its mean-square bound. These are all the types of terms in V. In particular this argument uses the Hessian of the actual nonlinear row, not merely a coincident row label.

Choose ϵ so that the norm of $g^+ \mapsto (V\mathbb{E}_0\iota_0^+ g^+, 0)$ is at most $1/4$ at the chosen decay rate. Then

$$\mathbb{E}\iota_0^+ = \mathbb{E}_0\iota_0^+[I - (V\mathbb{E}_0\iota_0^+, 0)]^{-1}$$

converges in these norms. The difference $\iota^+ - \iota_0^+$ is a local source with coefficients $O(\epsilon)$, since it contains only the changed transports in the two divergences. Covariant and ordinary first differences differ by local zeroth-order terms of the same size. This proves the small-field estimate in the covariant seminorms. Gauge changes by SU(2) matrices preserve all the mean-square norms, so the conclusion also holds for a triple small in a local real gauge.

Inserting the weight. Only the first equation changes. By Lem. 5.36,

$$(\mathbb{K}_v - \mathbb{K})\Psi = e_1\{D^{(1)\mathsf{T}} v D^{(1)} A + Z_{1,v}A\} = \iota^+ V_v \Psi.$$

To read the last expression in the system's source space, put the antisymmetric components of $vD^{(1)}A$ whose indices belong to that space into X_1. Put $Z_{1,v}A$ and the omitted boundary contributions into f_1; the other entries are zero. Indeed, the ambient plaquette form retains plaquettes meeting the boundary, whereas the system's ∇ omits a translated bond outside its inner-bond index set. Their difference is supported on bonds a bounded number of bare steps from the boundary. Every such cell is in the bare layer, where $\xi_y = \xi$. Each difference has bounded row and column multiplicity and size at most $C|v|_\infty\xi^{-2}|A|$ on neighboring bonds. The weighted Schur estimate therefore gives it source weight -3 from the potential weight -1. This is the same boundary treatment used above for $D^{\mathsf{T}}A$. In the interior the identification with the transpose of $D^{(1)}A = \nabla_\mu A_\nu - \nabla_\nu A_\mu$ is exact. The curvature bound of the natural pair and the local plaquette support consequently give

$$\|V_v\Psi\|_y \le C|v|_\infty \max_{d(y,z)\le\rho_0} \|\Psi\|_z. \tag{5.xlix}$$

No derivative of v occurs. For a small-field local inverse, $\|V_v\mathbb{E}\iota^+\| \le 1/4$ follows by decreasing r_2. Consequently

$$\mathbb{E}_v\iota^+ = \mathbb{E}\iota^+[I + V_v\mathbb{E}\iota^+]^{-1}$$

converges at the same fixed local rate. It is a right inverse on all sources: ι^+ is surjective, since the plain source components alone span the equations. Finite dimensionality therefore gives invertibility. The scalar compressions do not contain the changed first equation. This proves the local weighted conclusion for all four systems.

Localization. Use the pieces, $\widetilde{h}_\kappa$, θ_κ and local sequences of Lem. YM3_I:27.5. Lem. YM3_I:27.6 identifies the global and local physical averaging germs, including $C^{(2)}$, on $\kappa^{[5/6]}$. The pair on a regular piece has a real gauge in which its size is at most $8L^2(\alpha_0+\alpha_1+\gamma_0+\gamma')$. Choose a_2 to place this below the small-field threshold above. A boundary piece has only the bare layer; its mean-square norms are finite-component Euclidean norms, uniformly equivalent to the local supremum norms. The boundary-piece inverse in the proof of Thm. YM3_I:G, followed by the same bounded insertion, gives its local estimate. Thus every local weighted inverse has a uniform mean-square bound at a rate $2\delta_2$.

The source cutoff must respect divergence form. Define $h\star g^+$ by multiplying all components by h and subtracting $[\nabla^{\mathsf T},h]X_1$ from the first component and $[D^{\mathsf T},h]X_5$ from the fifth. Then

$$\iota^+(h\star g^+)=h\,\iota^+g^+. \tag{5.l}$$

The additional commutators have coefficients bounded by the first scaled difference of h, so are bounded in mean square. For the system commutator write

$$[\mathbb{K}_v,h]=\iota^+C^+_{v,h}.$$

The unweighted terms have the same three kernel types checked in eq. 5.xlviii, with ϵ replaced by the scaled first or second difference of h. For example, $[Q,h](c,b)=Q(c,b)(h(b)-h(c))$; the quadratic-row term is treated by its symmetric Schur bounds. The weighted principal term is exactly

$$[D^{(1)\mathsf T}vD^{(1)},h]A=D^{(1)\mathsf T}\{v[D^{(1)},h]A\}+[D^{(1)\mathsf T},h]vD^{(1)}A.$$

Keep the admitted indices of its first term as a divergence source. Its omitted boundary indices are placed in the ordinary first slot, by the same bare-layer calculation as for V_v. Commuting their local kernel with h introduces the difference of h across at most a fixed number of bare steps. Thus both the interior and boundary terms cost a first scaled difference of h and $|v|_\infty$; no difference of v is taken. The local remainder $[Z_{1,v},h]$ has the same bound by its plaquette support. The piece cutoffs consequently give

$$\|C^+_{v,\widetilde{h}_\kappa}\Psi\|_y\le C\mathfrak{M}^{-1}\max_{d(y,z)\le 2\rho_0}\|\Psi\|_z.$$

The commutator is supported where the cutoff is nonconstant, at distance at least D_0 from the input cutoff. All these norms coincide under the local/global identification: the values, differences and one-step shifts that occur here lie in $\kappa^{[5/6]}$. This is the same margin used in Lem. YM3_I:27.6; raising M_2 to ensure $\mathfrak{M}\ge 32L^4$ also includes the two shifts in the differentiated norms.

Define maps on divergence-form sources by

$$T^+_\kappa=\widetilde{h}_\kappa\mathbb{E}_{v,\kappa}\iota^+(\theta_\kappa\star\cdot),\qquad R^+_\kappa=-C^+_{v,\widetilde{h}_\kappa}\,\mathbb{E}_{v,\kappa}\iota^+(\theta_\kappa\star\cdot).$$

The identity $\sum_\kappa\theta_\kappa=1$ gives

$$\mathbb{K}_v\sum_\kappa T^+_\kappa=\iota^+\left(I-\sum_\kappa R^+_\kappa\right).$$

Use half the local exponential decay for the input-to-commutator distance. The remaining half yields $\|T^+_\kappa\|\le C$ and $\|R^+_\kappa\|\le q_2$ with the supports in the statement. The local geometry bounds the number of possible next pieces by a number $N=N(L)$; at a fixed output top at most N first pieces contribute. Increase M_2 until $Nq_2\le 1/2$. Then

$$\mathbb{E}_v\iota^+=\sum_{m\ge 0}\sum_{\kappa_0,\dots,\kappa_m}T^+_{\kappa_0}R^+_{\kappa_1}\cdots R^+_{\kappa_m}.$$

Composition gives eq. 5.xlv; summing the geometric series $\sum_m N^{m+1}q_2^m$ gives eq. 5.xliv. The local construction also proves the stated dependence on the visited regions. This completes the proof. □

A convenient precise meaning of a connected walk term is now available. Make a graph whose vertices are the pieces and whose edges join pieces with meeting [5/6] enlargements. For a finite vertex set S, sum the walks whose set of visited pieces is exactly S. There is no such walk when S is disconnected. If $|S| = s$, every contributing walk has at least $s-1$ transitions. Given $0 < \epsilon_* < 1$, increase M_2 until $Nq_2 \le \min\{1/2, \epsilon_*\}$. At each fixed output top its sum is then bounded, in eq. 5.xliv, by

$$C \sum_{m \ge s-1} (Nq_2)^m \le 2C\epsilon_*^{s-1}.$$

Locality follows from the union of the corresponding local regions. The same grouping applies to a difference of two local inverses: $E_v - E_0 = -E_0(\mathbb{K}_v - \mathbb{K}_0)E_v$ and the displayed local vertex identify its support. This is the connected-set factor used for the weighted fluctuation form; its cubes are these pieces, with the indicated enlargements, rather than an arbitrary partition.

The proposition also permits an envelope h with $h(z)/h(y) \le C_h e^{\delta_2 d(y,z)/2}$. Dividing eq. 5.xliv by $h(y)$ gives the corresponding weighted mean-square estimate, with constant B_2C_h and at least half the decay rate. This quantitative ratio is what is meant here when an envelope varies slowly.

Corollary 5.38. Under the hypotheses of Prop. 5.37, let the only forcing be the first-slot divergence $D^{(1)\mathsf{T}}Y$, with homogeneous row data. The potential component of the solution satisfies

$$\xi^3 \sum_b \xi_{y(b)}^{-2} |A(b)|^2 + \xi^3 \sum_{b,\mu} |\nabla_\mu A(b)|^2 \le C\xi^3 \sum_p |Y(p)|^2. \tag{5.li}$$

Ambient boundary plaquettes are included on the right. The constant depends only on L. ■□

Proof. Decompose the divergence source by its cell positions and apply the local estimate to one cell at a time. The antisymmetric embedding and its bare-layer boundary correction have uniformly bounded multiplicities, as in the proof of eq. 5.xlix. Thus the block norm of the map from Y to $(A, \nabla A)$ is at most $Ce^{-\delta_2 d(y,z)}$, with weights -2 on Y and ∇A and -1 on A. The squares of these normalized cell norms acquire the common factor ξ_y^{-1} when expressed in the energy norm of eq. 5.li. On the resulting ordinary sequence ℓ^2 spaces the block bound is therefore

$$Ce^{-\delta_2 d(y,z)}(\xi_z/\xi_y)^{1/2} \le C' L^{1/2} e^{-\delta_2 d(y,z)/2},$$

after the same increase of the layer separation used in the flat estimate. The row and column sums of this nonnegative matrix are bounded by the cell-summation estimate of Lem. YM3_I:21.2. Cauchy–Schwarz, first with its row sum and then with its column sum, proves its boundedness on ℓ^2. This is eq. 5.li. □

6. The two-term short-distance law

The leading action-density covariance is the free kernel of degree -6. Its first correction has degree -5. A test pair of scale d and shape $\vartheta \in (0,1)$ consists of nonzero Lipschitz functions $f, h \ge 0$ with supports contained in closed balls of radius d. The distance between the supports lies in $[d, 6d]$, and the functions satisfy

$$\int f(x)\,\mathrm{d}x \ge \vartheta |f|_\infty d^3, \qquad \operatorname{Lip} f \le |f|_\infty/(\vartheta d),$$

and the same bounds for h. On a torus require $20d \le \ell$. Write this class as $\mathfrak{T}(d;\vartheta)$ and put

$$\mathcal{K}_q(f,h) = \iint f(x)h(y)|x-y|^{-q}\,\mathrm{d}x\,\mathrm{d}y, \qquad L_b^+ = c_2 b^{-2}(\mathcal{K}_6 + c_1 b^{-1}\mathcal{K}_5),$$

where

$$c_2 = \frac{9}{16\pi^2}, \qquad c_1 = \frac{8}{3\pi}.$$

On the least admissible torus the distance is read in the small Euclidean lift containing the test pair.

Sketch E (The first correction). At sufficiently small reference coupling, fixed label $b \in I(g)$ and sufficiently small d, the connected two-point function satisfies

$$|S_{2;K} - L_b^+|(f,h) \le \left[\bar{\varepsilon}(d) b^{-3} d + C_{\mathrm{lat}} b^{-2} \frac{a_K}{d}\right] |f|_\infty |h|_\infty, \qquad \bar{\varepsilon}(d) \longrightarrow 0.$$

The working depth of d must lie below K. After removal of the cutoff, the first error can be bounded by $Cb^{-3}d^{3/2}|f|_\infty|h|_\infty$. The law transfers, in the separately stated ranges, to other admitted tori, periodic states, volume-first limits and their components. ■□

Sketch of the argument. There are two jobs: identify the coefficient and control the interacting remainder. A finite-box perturbative calculation does only the first. To connect it with the full correlation, introduce two source parameters and differentiate the local normalizing energies twice. Sum these mixed derivatives over the renormalisation levels. An auxiliary complex field scale λ, multiplying every fluctuation field, makes this source-energy sum an even holomorphic function. The free contribution is explicit; the coefficient of λ^6 contains the first correction. Parity removes the half-integer orders in the equivalent Laplace expansion.

The coefficient is compared through a sequence of finite objects. First truncate the levels at a working scale, with a bound for the omitted tail. At zero field scale, the fluctuation laws are Gaussian laws restricted to their small-field boxes. Their precision matrices have a uniform positive lower bound. Gaussian tails therefore control the error made by removing the boxes. The coefficient is now a finite Gaussian expression with the coarsest background fixed at the identity. Compare these expressions in different boxes, then exhaust the lattice and compare the resulting lattice kernel with its homogeneous continuum part. Each comparison has its own geometry and error; the auxiliary box and cutoff are chosen so those errors are all small relative to $b^{-3}d$.

The last comparison is where the numerical value enters. The six two-loop graph classes are reduced to a finite residue functional. Their sum gives

$$\kappa^{\mathrm{lat}} = \frac{3}{2\pi^3} = c_2 c_1.$$

The finite-box coefficient is independent of the intermediate gauge parameter. Its infinite-lattice limit is obtained by taking the large-inverse-coupling expansion first at fixed box, then exhausting the lattice. This order does not establish a uniform remainder for growing boxes. The renormalisation comparisons above are needed precisely to return from that coefficient to the interacting measure.

Initially the expansion is proved in a cutoff window $L^K \le d^{-P}$. At fixed label, consecutive-cutoff comparison allows one to stop at a sufficiently fine $K_\sharp(d)$ inside that window and estimate every larger cutoff by it. Thus the window disappears from the fixed-label conclusion. The two errors in the result remain conceptually different: a_K/d is a lattice approximation error, whereas $\bar{\varepsilon}(d)b^{-3}d$ controls the correction beyond the two retained continuum kernels.

This is a two-term short-distance statement. It neither gives the arbitrary-order free law left open in the preceding sketch nor permits the cutoff and short-distance limits to be exchanged without their displayed scale restrictions.

6.1. **The two-term short-distance law on the torus.** For the action density, the first correction to the free two-point law is of relative order d/b. More precisely, the two-term approximation is

$$c_2 b^{-2}[\mathcal{K}_6 + c_1 b^{-1}\mathcal{K}_5](f,h), \qquad c_2 = \frac{9}{16\pi^2}, \quad c_1 = \frac{8}{3\pi},$$

The error is $[\bar{\varepsilon}(d)b^{-3}d + C_{\mathrm{lat}}b^{-2}a_K/d]\|f\|_\infty\|h\|_\infty$. Thm. S states the cutoff and label ranges, the improvement to a remainder of order $d^{3/2}$, and the averaged logarithmic correction. Cor. 6.4 passes the law to every periodic state and to the continuum limits and their components.

The proof first expresses the covariance as an average of source-energy jets over translated pairs. A complex field scale separates the free term, of order four, from the correction, of order six. A composed block-field chart identifies the latter with a constrained lattice Laplace coefficient. Comparison of two bare tori and removal of the flat block constraint then connect it to the infinite-lattice coefficient. The lattice calculation is proved in §7.1.

That calculation first takes the weak-coupling limit in a fixed box and then exhausts the lattice. Its box-dependent remainder is not a bound in the simultaneous continuum and large-volume limit. The renormalisation and finite-torus comparisons below provide the separate passage needed here.

6.1.1. *Parameters and test pairs.* We use the Wilson $G = \mathrm{SU}(2)$ model in three dimensions. Its Lie algebra consists of the hermitian traceless 2×2 matrices with $\langle X, Y\rangle = \frac{1}{2}\operatorname{tr} XY$, extended complex bilinearly. The trace is the ordinary matrix trace, and $|X|$ is the operator norm on $\mathbb{C}^2$. Near the identity we use $\log X = -\sum_{n\geq 1}(I-X)^n/n$, for $|X - I| < 1$. For $U \in \mathrm{SU}(2) \setminus \{-I\}$ the intrinsic branch is $\log U = iY$, where Y is Hermitian and traceless and $|Y| < \pi$; these branches agree on their common domain. Fix a mollifier, a source radius $2\varrho > 0$, and one common choice of the constants in Prop. YM3_I:69.1 and Section YM3_I:82. In particular $L \geq 13$ is odd and $a_j = L^{-j}$. We retain all the real-step conditions when imposing the sourced-step conditions: restricting to the real construction leaves every retained constant unchanged. The two radius conditions are

$$2\varrho \leq \sigma^{\mathrm{S}}(x - \bar{B}), \qquad 2\varrho x^{-1/2}(\log x)^{\mathfrak{p}_0} \leq \sigma^{\mathrm{S}}_{\mathrm{L}} \qquad (x \geq \gamma^{-2}).$$

They are imposed by Prop. YM3_I:82.2, after the radius-independent choices of Section YM3_I:82. Thus all uses of the real flow, the sourced progression and the sourced chain below refer to the same numerical assignment and the same canonical data.

The running couplings and the data at a reference coupling, the sides, the torus measures and the states, the weighted progression, the sourced chain on the least torus and its histories are therefore formed with this one choice, and every statement of this section is made at it; where Thms. YM3_I:AJ and YM3_I:AD enter, they are read at the radius ϱ and at the choice of the longer list. Among the constants are $B_\sharp > 0$, with $\bar{B} = B_\sharp L/(L-1)$, the powers M, C_L of L in the geometric setting of Section YM3_I:52.7, $A_0, A_1 > 0$ and $p_0 \in \mathbb{N}$. The real and source hypotheses are those imposed jointly in Section YM3_I:82. A number is *determined by the constants* if it depends on them alone.

For $\ell \in \mathfrak{L} = \{2L^m : m \in \mathbb{N}\}$ the torus $T_\ell = \mathbb{R}^3/\ell\mathbb{Z}^3$ carries the lattice of spacing a_K. The *lattice* (ℓ, K), a side with a cutoff, carries the lattices $\mathbb{T}^{(j)} = a_{K-j}\mathbb{Z}^3/\ell\mathbb{Z}^3$ of the levels $0 \leq j \leq K$, of spacing $D_j = a_{K-j}$; the level 0 is the bare lattice, which in bare units is $\Lambda_N = \mathbb{Z}^3/N\mathbb{Z}^3$, $N = \ell L^K$. The *height* i is the averaging step from the level i to the level $i+1$. Blocks, the bonds of the coarser lattices and the non-pivot bonds are those of Section YM3_I:4 and the pivot definition preceding Lem. YM3_I:40.1, contours those of Section YM3_I:4 (coordinate-order contours), and $\mathcal{M}$ is the block-averaging map (equations YM3_I:(37)–(39)).

For a plaquette p, $a_p(U) = 1 - \frac{1}{2}\operatorname{tr}\partial U(p)$ and $x(p)$ is its barycentre; $A = \sum_p a_p$, $\mu_{K,\ell,\beta}(dU) = Z^{-1}e^{-\beta A(U)}dU$, $\mathcal{O}(f) = a_K^{-1}\sum_p f(x(p))a_p$, for a complex function w on the bare plaquettes, a *source*, $\mathcal{O}(w) = a_K^{-1}\sum_p w(p)a_p$, so that $\mathcal{O}(f) = \mathcal{O}(f \circ x)$, and $S^T_{2;K,\ell,\beta}(f,h)$ is the covariance of $\mathcal{O}(f)$ and $\mathcal{O}(h)$ under $\mu_{K,\ell,\beta}$. The lattice isometries preserve $\mu_{K,\ell,\beta}$ and carry $\mathcal{O}(f)$ to $\mathcal{O}$ of the carried function ([7, Lem. G.1]: plaquette holonomies are conjugated or inverted, and the positive-bond variables are permuted or inverted under product Haar measure); so $S^T_{2;K,\ell,\beta}(\tau_v f, \tau_v h) = S^T_{2;K,\ell,\beta}(f,h)$ for every $v \in a_K\mathbb{Z}^3$ and all real Lipschitz f, h, whatever their supports, τ_v being the translation by v.

For a reference coupling $0 < g \leq \gamma$ (γ the last constant) put $X_g = L(g^{-2} + B_\sharp)$, $b_+ = L(g^{-2} + \bar{B})$ and $I(g) = (g^{-2} + \bar{B}, X_g - \bar{B}]$. The sides at g form $\mathfrak{L}(g) = \{2L^m : m \geq m_{\mathfrak{L}}(g)\}$ (equation YM3_I:(1528)), with least member ℓ_g; so $\ell_g L^s \in \mathfrak{L}(g)$ for every integer $s \geq 0$. A *datum* at g is a first-passage datum (K, β) at g (Cor. YM3_I:74.2): its running couplings $x_0 = \beta > x_1 > \cdots > x_K$ obey $x_{j+1} = L^{-1}x_j - e_{j+1}$ with $|e_{j+1}| \leq B_\sharp$, and K is the least index with $x_K \leq X_g$; its label is $b = a_K\beta$. The running couplings depend on (K, β) alone.

Every label at g lies in $[g^{-2}, b_+]$ YM3_I:(1531), and for $b \in I(g)$ the pair $(K, L^K b)$ is a datum at g for every $K \in \mathbb{N}$ (Cor. YM3_I:74.2). The *witness depth* is $s(d) = \min\{s \geq 0 : L^s d \geq C_W(s)\}$, $C_W(s) = C_{W,0} A_0^2 (\log[L^{s+1}(g^{-2} + \bar{B})])^{2p_0}$, $C_{W,0} = \max\{1, 4L^2 M/A_0^2\}$, as in equation YM3_I:(2321) and Thm. YM3_I:AV; it depends on g, d and the constants, and on no cutoff, label or side. As $C_W(s) \geq 4L^2 M$, for $d \leq L^{-1}$ and $K > s(d)$ one has $L^{s(d)} \geq 4L^3 M$, $s(d) \geq 2$, $K \geq 3$ and $a_K \leq L^{-1-s(d)} < d/13$; moreover $L^{s(d)} \geq 1/d$, so $C_W(s(d)) \geq (\log \frac{1}{d})^2$ and $d/a_K \geq L C_W(s(d)) \geq L(\log \frac{1}{d})^2$.

Finally $\gamma_W^{\mathrm{sd}} = \min\{\gamma_{W,0}^{\mathrm{sd}}, \gamma_{\mathrm{ev}}\} \leq \gamma_W^{\mathrm{ref}} \leq \gamma$, with the original thresholds of equations YM3_I:(2326)–(2327) and eq. 2.iii; $(\gamma_W^{\mathrm{sd}})^{-2} \geq 4\bar{B}$.

For $\vartheta \in (0,1)$, $d > 0$, and $\mathbb{E} = \mathbb{R}^3$ or $\mathbb{E} = T_\ell$ with $20d \leq \ell$, $\mathfrak{T}_{\mathbb{E}}(d;\vartheta)$ is the following class of test pairs: Lipschitz functions $f, h \geq 0$ on $\mathbb{E}$, neither 0, with supports in closed balls of radius d at a distance in $[d, 6d]$, $\int f \geq \vartheta \|f\|_\infty d^3$, $\operatorname{Lip} f \leq \|f\|_\infty/(\vartheta d)$, and the same for h. Then both supports lie in one ball of radius $5d$, and $d \leq |x-y| \leq 10d$ for $x \in \operatorname{supp} f$, $y \in \operatorname{supp} h$. The conditions involve distances alone. So the class is invariant under translations; and for $36d < \ell$ the periodisation of a pair of $\mathfrak{T}_{\mathbb{R}^3}(d;\vartheta)$ is a pair of $\mathfrak{T}_{T_\ell}(d;\vartheta)$ with the same sup norms and the same kernels $\mathcal{K}_q$ below, every pair of $\mathfrak{T}_{T_\ell}(d;\vartheta)$ being the periodisation of one such pair, its *lift*, since all the distances that occur are at most $10d < \ell/2$.

For two sides $\ell \leq \check{\ell}$, the *pair with the same lift* as a pair (f,h) on T_ℓ is the periodisation on $T_{\check{\ell}}$ of the lift of (f,h).

We put $\eta = \|f\|_\infty \|h\|_\infty$, $\mathcal{K}_q(f,h) = \iint f(x)h(y)|x-y|^{-q}\,\mathrm{d}x\,\mathrm{d}y$, $\varkappa = c_1 c_2 = \frac{3}{2\pi^3}$ and $L_b^+ = c_2 b^{-2}[\mathcal{K}_6 + c_1 b^{-1}\mathcal{K}_5]$. On $\mathbb{Z}^3$, $\mathsf{B}_1^{\mathrm{lat}}(p,q)$ is the coefficient of β^{-3} in the covariance of a_p and a_q (Def. 7.10), and $\mathsf{P}_1^{\mathrm{lat}}(f,h;a) = a\sum_{p,q} f(ax(p))h(ax(q))\mathsf{B}_1^{\mathrm{lat}}(p,q)$. There are a pure number C_{T} and a constant κ^{lat} with

$$|\mathsf{P}_1^{\mathrm{lat}}(f,h;a) - \kappa^{\mathrm{lat}}\mathcal{K}_5(f,h)| \leq C_{\mathrm{T}}\vartheta^{-1} a \lg(d/a)\,\eta \qquad ((f,h) \in \mathfrak{T}_{\mathbb{R}^3}(d;\vartheta),\ 0 < a \leq d),$$

$\lg t = 1 + |\log t|$ (Cor. 7.37), and $\kappa^{\mathrm{lat}} = \frac{3}{2\pi^3} = c_1 c_2$ (Thms. W, X). For real Lipschitz f, h on T_ℓ, $\ell \in \mathfrak{L}(g)$, with disjoint supports and $b \in \overline{I(g)}$, $S_2^{(\ell,b)}(f,h) = \lim_{K\to\infty} S^T_{2;K,\ell,L^K b}(f,h)$ is the limit of Thm. YM3_I:A. That theorem gives existence for every $b > 0$, continuity in b on $(0,\infty)$, and bilinearity on real Lipschitz functions supported in two fixed disjoint closed sets. A function on $\mathbb{R}^3$ with support in a ball of radius $< \ell/4$ is read on T_ℓ by periodisation, and $f_\rho := f(\cdot/\rho)$.

Lemma 6.1 (Kernel bounds for test pairs)**.** For $(f,h) \in \mathfrak{T}_{\mathbb{E}}(d;\vartheta)$ and $q > 0$,

$$10^{-q}\vartheta^2 d^{6-q}\|f\|_\infty\|h\|_\infty \leq \mathcal{K}_q(f,h) \leq (4\pi/3)^2 d^{6-q}\|f\|_\infty\|h\|_\infty.$$

Moreover $d \leq \mathcal{K}_5(f,h)/\mathcal{K}_6(f,h) \leq 10d$. ■

Proof. All cross-distances of the two supports lie between d and $10d$. Positivity gives

$$(10d)^{-q}\int f \int h \leq \mathcal{K}_q(f,h) \leq d^{-q}\int f \int h.$$

The defining mass bounds give the lower estimate, and the volumes of the two radius-d balls give the upper estimate. On a torus the balls have the same volumes because $20d \leq \ell$. Finally $|x-y|^{-5} = |x-y|\,|x-y|^{-6}$ on the product of the supports; integration gives the ratio bound. Its denominator is positive by the lower estimate. □

Definition 6.2. Let N_3 be the threefold convolution of $\mathbf{1}_{[-1/2,1/2)}$, $\Phi(x) = \prod_{\mu=1}^3 N_3(x_\mu)$, $\zeta = 1$ on $[0,7]$, $\zeta(t) = 1 - 3(t-7)^2 + 2(t-7)^3$ on $[7,8]$, $\zeta = 0$ on $[8,\infty)$, $P(y) = |y|^2 - \frac{3}{4}$ and $\Xi(y) = P(y)\zeta(|y|)$. For odd $q \geq 1$ let $J_q = \{-\frac{q-1}{2}, \dots, \frac{q-1}{2}\}$, $c_j^q = \prod_\mu q^{-2}\#\{\iota \in J_q^3 : \iota_1 + \iota_2 + \iota_3 = j_\mu\}$ for $j \in \mathbb{Z}^3$, and

$$\Upsilon^{(q)}(y) = q^{-1}\sum_{j\in\mathbb{Z}^3} c_j^q\,\Xi((y+j)/q) - \Xi(y).$$

With $e(u) = \int \Phi(x)\Xi(x+u)\,\mathrm{d}x$ let $\mathsf{k}_6 = \int_0^\infty (4\pi - t^{-2}\int_{S^2} e(t\omega)\,\mathrm{d}\omega)t^{-2}\,\mathrm{d}t$. ∎

6.1.2. *The scale of the remainder.* The two terms in the covariance have different homogeneities. For a test pair of scale d, the kernel $|x-y|^{-6}$ integrates to a quantity of order one, while $|x-y|^{-5}$ integrates to a quantity of order d. Thus the leading contribution is of size b^{-2} and its first correction is of size $b^{-3}d$. To identify that correction, an error merely tending to zero relative to the leading term is inadequate. The error must be $o(b^{-3}d)$ after the mesh error has been separated.

The nonnegativity and mass bounds in the test-pair definition give

$$10^{-q}\vartheta^2 d^{6-q}|f|_\infty|h|_\infty \le \mathcal{K}_q(f,h) \le (4\pi/3)^2 d^{6-q}|f|_\infty|h|_\infty. \tag{6.i}$$

These floors do more than normalize the functions. They show that the second term is detected by every pair in the class, and they permit division by $\mathcal{K}_5$ when recovering its coefficient. The lattice comparison retains the Lipschitz bound, which measures the error from replacing an integral by samples at plaquette barycentres. These sampling errors and the interacting remainder are kept as separate terms in theorem S.

The coefficient calculation later in the section is finite dimensional. Its remainder depends on the box. The present renormalization argument supplies the estimates needed to use that coefficient in an interacting continuum limit. This distinction determines the order of the comparison: begin with the actual covariance, express it through the sourced progression, and only then compare the coefficient of that progression with a lattice Laplace coefficient.

6.1.3. *The even expansion and its two retained orders.* Let n_j be the mixed derivative in the two sources of the local normalizing energy at level j, summed over its domains. Its sum is J_2. A packing of translated pairs gives

$$\left| S_2^T(f,h) - \frac{1}{\#\mathsf{Pk}} \sum_{v\in\mathsf{Pk}} \mathsf{J}_2(\tau_v f, \tau_v h) \right| \le C(g)\vartheta^{-1} d^2 b^{-2} |f|_\infty |h|_\infty.$$

The covariance is unchanged by the lattice translations. The packing therefore allows the contributions left by the conditioning procedure to be averaged while keeping the original quantity. In the units $b^{-3}d|f|_\infty|h|_\infty$, its cost is $C(g)\vartheta^{-1}bd$, which tends to zero at fixed admitted labels.

Multiply every fluctuation field by one complex parameter λ and move the quadratic coefficients of the action by

$$\xi_j(\lambda) = L^{-j}\beta + \lambda^2(x_j - L^{-j}\beta).$$

Write $\mathsf{J}_2(\lambda)$ for the deformed sum, so that $\mathsf{J}_2 = \mathsf{J}_2(1)$, and put $\mathsf{h}_r = [\lambda^r]\mathsf{J}_2(\lambda)$ for $r = 4, 6$. This deformation preserves the identities relating the steps. The simultaneous analyticity domains in proposition 6.6 justify using the same parameter at every height. The jets are even holomorphic functions of λ, and their constant and quadratic coefficients vanish. At the quadratic order the possible term is a covariance of linear colour forms at the identity; invariance under constant gauge rotations makes those linear forms zero. Cauchy's estimate then gives

$$|\mathsf{J}_2 - \mathsf{h}_4 - \mathsf{h}_6| \le C b^{-4} d^2 \big(\max\{1, \log(2b/d)\}\big)^{8p_0} |f|_\infty |h|_\infty. \tag{6.ii}$$

The orders four and six are the leading covariance and its first correction. The power d^2 in the remaining orders makes eq. 6.ii small on the scale required for identifying the second term.

The order-four tower is compared with the free curvature template. Its covariance is a sum of nested curvature projections, so the telescoping slice identity identifies the free kernel. The supplied estimate retains the error from the truncated Gaussian boxes, the coarsest background and the lattice-to-integral comparison. In particular the order-four error is bounded by $C(g)b^{-3}d^{3/2} + C_{\mathrm{lat}}b^{-2}a_K/d$, multiplied by the two suprema. The slowest part of the eventual continuum error comes from this comparison.

For the sixth order, let $\mathsf{h}_6^{(n)}$ be the sum of the sixth-order coefficients through level n, and stop at a level with spacing $D_n \geq d$. The tail estimate is

$$|\mathsf{h}_6 - \mathsf{h}_6^{(n)}| \leq Cb^{-3}d\big(\max\{1, \log(2b/d)\}\big)^{6p_0}(d/D_n)^{3/2}|f|_\infty|h|_\infty.$$

This is the freedom used to compare finite objects while keeping the discarded coarse levels below the second-term scale.

6.1.4. *The comparison of the sixth-order coefficient.* The composed block-field chart puts all earlier fluctuation fields on the fibre where the coarsest block field is the identity. At real field scale its exact density contains the Wilson action, the gauge-fixing terms and the chart Jacobian. The run equation gives the identity

$$\lambda^{-2}\xi_j(\lambda)L^j + \sum_{i\leq j} L^i e_i = \lambda^{-2}\beta. \tag{6.iii}$$

Thus the intermediate counterterms disappear from the total phase. The field changes between matched charts are triangular colour rotations. They preserve Lebesgue measure and the Euclidean balls used for the fluctuation cutoffs. Consequently the sixth-order coefficient agrees exactly with the second Laplace coefficient in that chart, apart from the separately bounded removal of the Gaussian boxes. Coordinate cubes in the colour components would not be preserved by this change; the shape of the boxes enters the argument here.

Set $\mathsf{H}^{(n)} = \beta^3 a_K^2 \mathsf{h}_6^{(n)}$ and let $\mathsf{T}[N, n]$ be the coefficient on a bare torus of N sites per direction, with the n-fold block constraint. The same $\mathsf{T}[N, n]$ can be reached from different running couplings, because eq. 6.iii leaves no dependence on that run. Let D be the original datum on a bare torus of side N_0, and let D' be the comparison datum on an enlarged bare torus of side N'. The truncation depths n_0, n_1 are chosen with the geometric and error inequalities of the detailed comparison below. In the case requiring the enlarged torus, the comparison is:

$$\begin{aligned}\mathsf{H}^{(K)}[D] &\longrightarrow \mathsf{H}^{(n_0)}[D] \longrightarrow \mathsf{T}[N_0, n_0] \longrightarrow \mathsf{T}[N', n_0] \longrightarrow \mathsf{H}^{(n_0)}[D'] \\ &\longrightarrow \mathsf{H}^{(n_1)}[D'] \longrightarrow \mathsf{T}[N', n_1] \longrightarrow \mathsf{T}^{\mathrm{lat}} \longrightarrow a_K^{-1}\kappa^{\mathrm{lat}}\mathcal{K}_5.\end{aligned} \tag{6.iv}$$

Each kind of comparison addresses a different obstruction. The first and fifth arrows discard coarse tails using the preceding estimate. Passing between H and T removes Gaussian boxes; their precisions have a uniform positive lower bound, so the error is an exponential in minus the squared cutoff radius, times the stated polynomial in the finite size. The change of bare torus uses locality of the chart and connected Wick pairings. A pairing reaching the remote part must contain a long line, whose exponential bound pays for the polynomial count. The lattice anchor removes the remaining block constraint, and the last arrow replaces the infinite-lattice kernel by its homogeneous part of degree -5. The numerical value of c_1 enters only at this last arrow.

The choices of K', n_0, n_1 in the proof make all these costs small in the common units $r|f|_\infty|h|_\infty$, where $r = d/a_K$. Their dependence is displayed because it is what permits the box-removal exponential to beat its finite-size polynomial. An unqualified passage from finite-box perturbation theory to the continuum covariance would omit this comparison.

6.1.5. *Removing the cutoff window and accumulating scales.* The initial expansion holds in a window $L^K \leq d^{-P}$. For a fixed label, proposition 6.19 supplies a cutoff $K_\sharp(d)$ inside that window and a bound for every finer cutoff by the quantity at $K_\sharp(d)$. Its error is $O(d^{178})$ with the stated shape factor. Apply the expansion at that one cutoff, then compare all larger cutoffs with it. The full-sequence limit follows from the already supplied cutoff comparison. This procedure removes the window without exchanging a short-distance limit with an uncontrolled finite-box expansion. The separate same-datum comparisons pass the result to other sides and states, and the conditioned-state limits pass it to the canonical components with their common label.

There is also a reason to retain the improved error, beyond identification of c_1. The spline identity in lemma 6.24 writes the logarithmic observable as a sum of differences at scales $\varrho_k = L^{-k} r_0$. After subtracting the two explicit kernels, the kth term is bounded by a constant times $\varrho_k^{1/2} b^{-3}$. These bounds are summable. They give the uniform limit of Δ_n and its geometric tail. A tolerance only tending to zero, without a summable rate, would not justify that accumulation over all scales.

6.1.6. *The quantitative two-term law.*

Remark 6.3 (Nonlinear coefficient comparison). In the setting of Prop. 6.18, let $\mathsf{B}_1^{\hat{E}}$ be the Gaussian member defined in its proof. The nonlinear block constraint and the top gauge measure contribute a remainder bounded by

$$|\mathsf{g}_2[\ell L^K, K](p,q) - \mathsf{B}_1^{\hat{E}}(p,q)| \le C\big[\ell^{13} K L^{-K/2} + \ell^{18} L^{-2K} + \ell^{19} K^2 L^{-5K}\big],$$

uniformly in the side and the cutoff. The proof of Prop. 6.18 expands this remainder into eleven cumulants and estimates them using the linear curvature response and connected Gaussian pairings.

Theorem S. There is $\gamma_{\mathrm{e}} \in (0, \frac{1}{2}]$, determined by the constants, with the following property. Let $0 < g \le \gamma_{\mathrm{e}}$, $\vartheta \in (0,1)$, $\ell = \ell_g$, $P \ge P_\sharp := 3601$, and put $C_{\mathrm{lat}} = \frac{3}{2} C_{\mathrm{tm}} \vartheta^{-1}$ and $C_{\mathrm{lat}}^\flat = \frac{9}{2} C_{\mathrm{tm}} \vartheta^{-1} + 1$, where C_{tm}, determined by the constants, is the constant of the template sums of test pairs (Prop. YM3_I:121.2 and Lem. 6.34). There are $d_{\mathrm{e}} > 0$, $c_{1/2} < \infty$, $c_{\mathrm{r}} < \infty$ and a non-decreasing $\bar\varepsilon : (0, d_{\mathrm{e}}] \to [0,\infty)$ with $\bar\varepsilon(d) \to 0$ as $d \to 0$, depending on g, ϑ, P and the constants, and on no cutoff, label, pair or position, such that for $0 < d \le d_{\mathrm{e}}$ and $(f,h) \in \mathfrak{T}_{T_\ell}(d;\vartheta)$ the following hold.

(a) At every datum (K,β) at g with $s(d) < K$ and $L^K \le d^{-P}$, with $b = a_K \beta$,

$$|S^T_{2;K,\ell,\beta} - L_b^+|(f,h) \le [\bar\varepsilon(d)\, b^{-3} d + C_{\mathrm{lat}}\, b^{-2} a_K/d]\eta. \tag{6.v}$$

(b) For every $b \in I(g)$, eq. 6.v holds at $(K, L^K b)$ for every $K > s(d)$, and $|S_2^{(\ell,b)} - L_b^+|(f,h) \le \min\{\bar\varepsilon(d), c_{1/2} d^{1/2}\}\, b^{-3} d\, \eta$.

(c) If moreover $g \le L^{-1/4} \min\{\gamma_{\mathrm{e}}, \gamma_W^{\mathrm{sd}}\}$, there are $d'_{\mathrm{e}} > 0$ and $\bar\varepsilon'$ with $\bar\varepsilon'(d) \to 0$, depending on g, ϑ, P, the radius ϱ and the choice of the longer list of which the constants are the restriction, and on no cutoff, label, pair or position, such that eq. 6.v, with $\bar\varepsilon'$ in the place of $\bar\varepsilon$, holds for $0 < d \le d'_{\mathrm{e}}$, $(f,h) \in \mathfrak{T}_{T_\ell}(d;\vartheta)$ and every datum (K,β) at g with $K > s(d)$.

(d) Let $k_0 \in \mathbb{Z}$ with $44 L^{-k_0} < \ell_g$, $r_0 = L^{-k_0}$, and for $n \ge 1$ let $r_1 = L^{-n} r_0$ and

$$\Delta_n(b) = r_1^{-1} S_2^{(\ell,b)}(\Phi_{r_1}, \Upsilon_{r_1}^{(L^n)}) - c_2 b^{-2} \mathsf{k}_6\,(r_1^{-1} - r_0^{-1}) - \frac{6}{\pi^2}\, b^{-3} \log \frac{r_0}{r_1}$$

(the pair has disjoint supports, so that the limit is defined at it). Then Δ_n converges uniformly on $I(g)$ to a bounded continuous function Δ_∞, and there are $C_{\mathrm{co}} < \infty$ and $n_{\mathrm{co}} \in \mathbb{N}$, depending on g, k_0 and the constants, with $\sup_{I(g)} |\Delta_n| \le C_{\mathrm{co}}$ for all $n \ge 1$ and $\sup_{I(g)} |\Delta_n - \Delta_\infty| \le C_{\mathrm{co}} L^{-n/2}$ for $n \ge n_{\mathrm{co}}$.

(e) For every $b \in I(g)$ and every $K > s(d)$,

$$|S^T_{2;K,\ell,L^K b} - L_b^+|(f,h) \le [c_{\mathrm{r}}\, b^{-3} d^{3/2} + C_{\mathrm{lat}}^\flat\, b^{-2} a_K/d]\eta; \tag{6.vi}$$

and eq. 6.vi, with β for $L^K b$, holds at every datum (K,β) at g with $s(d) < K \le K_\sharp(d)$ and $b = a_K \beta$, where $K_\sharp(d)$ is the cutoff of Prop. 6.19 at the side ℓ_g.

(f) Under the assumptions of (c) there are $d''_{\mathrm{e}} > 0$ and $c'_{\mathrm{r}} < \infty$, depending on g, ϑ, the radius ϱ and the choice of the longer list of which the constants are the restriction, and on no P, which enters them as $P_\sharp$, and no cutoff, label, pair or position, such that eq. 6.vi, with c'_{r} for c_{r} and β for $L^K b$, holds for $0 < d \le d''_{\mathrm{e}}$, $(f,h) \in \mathfrak{T}_{T_\ell}(d;\vartheta)$ and every datum (K,β) at g with $K > s(d)$, $b = a_K \beta$.

■□

As $|x-y|\le 10d$ on the supports, $\mathcal{K}_5(f,h)\ge (10d)^{-5}\int f\int h\ge 10^{-5}\vartheta^2 d\,\eta$ by the nonnegativity and integral floors in [7, Lem. I.1]. So clause (b) determines the number:

$$\Big|\frac{(S_2^{(\ell,b)}-c_2b^{-2}\mathcal{K}_6)(f,h)}{c_2b^{-3}\mathcal{K}_5(f,h)}-\frac{8}{3\pi}\Big|\le 10^5c_2^{-1}\vartheta^{-2}\min\{\bar\varepsilon(d),c_{1/2}d^{1/2}\}.$$

The continuum limit on T_{ℓ_g} at a label of $I(g)$ determines first the label, since

$$S_2^{(\ell,b)}(f,h)/(c_2\mathcal{K}_6(f,h))\to b^{-2}$$

along pairs of the class with $d\to 0$, and then the number $\frac{8}{3\pi}$. The number enters the proof at one place, the last link of the chain of Step 3 below, through $\kappa^{\mathrm{lat}}=c_1c_2$; it depends on no L, g, side or label.

Corollary 6.4 (Every state; every side; the limits on $\mathbb{R}^3$)**.** Let g, ϑ and P be as in clause (c) of Thm. S, with $d'_{\mathrm{e}},\bar\varepsilon'$ of (c) and $d''_{\mathrm{e}},c'_{\mathrm{r}}$ of (f); let $C_2^{\mathrm{T}}(g)$ and $\mathsf{d}_\star(g,\vartheta)$ be the numbers of Prop. 6.20, and put

$$d_\infty=\min\{d'_{\mathrm{e}},d''_{\mathrm{e}},\tfrac{1}{180},\mathsf{d}_\star(g,\vartheta),L^{-1}\},\qquad \bar\varepsilon_\infty(d)=\min\{\bar\varepsilon'(d),c'_{\mathrm{r}}d^{1/2}\}+C_2^{\mathrm{T}}(g)\vartheta^{-1}b_+d.$$

The numbers d_∞ and $\bar\varepsilon_\infty(d)$ depend on g, ϑ, P, the radius ϱ and the choice of the longer list of which the constants are the restriction, and on no cutoff, label, side, state, sequence, component, pair or position. Let $0<d\le d_\infty$ and $(\tilde f,\tilde h)\in\mathfrak{T}_{\mathbb{R}^3}(d;\vartheta)$, with $\eta=\|\tilde f\|_\infty\|\tilde h\|_\infty$.

(i) For every state $\mathfrak{s}$ at g at a datum (K,β) at g with $K>s(d)$ (Prop. 6.20), $b=a_K\beta$:

$$|S_2^T[\mathfrak{s}]-L_b^+|(\tilde f,\tilde h)\le\big[(\bar\varepsilon'(d)+C_2^{\mathrm{T}}(g)\vartheta^{-1}b_+d)\,b^{-3}d+C_{\mathrm{lat}}b^{-2}a_K/d\big]\eta,$$

and the same with $c'_{\mathrm{r}}d^{1/2}$ in the place of $\bar\varepsilon'(d)$ and $C^\flat_{\mathrm{lat}}$ in the place of C_{lat}. In particular this holds for the torus measure of every side of $\mathfrak{L}(g)$ at the periodisations of the pair, and for every periodic Gibbs state at (a_K,β).

(ii) For every side $\ell'\in\mathfrak{L}(g)$, every $b\in I(g)$ and every $(f,h)\in\mathfrak{T}_{T_{\ell'}}(d;\vartheta)$: $|S_2^{(\ell',b)}-L_b^+|(f,h)\le\bar\varepsilon_\infty(d)\,b^{-3}d\,\eta$.

(iii) Let $\mathfrak{s}_j$ be states at g at data (K_j,β_j) at g with $K_j\to\infty$ and labels $b_j=a_{K_j}\beta_j\to b$; the torus measures of sides $\ell_j\in\mathfrak{L}(g)$ that grow with j are among them. Then $b\in[g^{-2},b_+]$, and every limit point S of the sequence $S_2^T[\mathfrak{s}_j](\tilde f,\tilde h)$ satisfies $|S-L_b^+(\tilde f,\tilde h)|\le\bar\varepsilon_\infty(d)\,b^{-3}d\,\eta$. The labels are determined by the two-point functions: if $\mathfrak{s}_j$ are states at g at data (K_j,β_j) at g with $K_j\to\infty$, and $S_2^T[\mathfrak{s}_j](F)$ converges at every pair F of a set that holds, for arbitrarily small $d\in(0,d_\infty]$, a pair of $\mathfrak{T}_{\mathbb{R}^3}(d;\vartheta)$, then the labels $a_{K_j}\beta_j$ converge, to a number $b\in[g^{-2},b_+]$.

Consequently the labels of every convergent Gibbs sequence at g (a sequence (K_j,β_j,ν_j) of data at g with $K_j\to\infty$ and periodic Gibbs states ν_j at (a_{K_j},β_j)) converge, to a number b; and the connected two-point function S_2^σ of the action density of its limit σ, and that of every component of a canonical decomposition of σ, satisfy $|S_2^\sigma-L_b^+|(\tilde f,\tilde h)\le\bar\varepsilon_\infty(d)\,b^{-3}d\,\eta$ whenever $\tilde f$ and $\tilde h$ are of class C^1. Such a pair is a separated tuple, at which these functionals are defined, and $\mathfrak{T}_{\mathbb{R}^3}(d;\vartheta)$ holds pairs of class C^1 for every $d>0$ and $\vartheta\in(0,1)$.

Every volume-first limit at g has one label b, the same along all its defining sequences, and the same inequality holds for it and for its ergodic components ([7, §K.1] and Def. 13.4(2)) at the pairs of $\mathfrak{T}_{\mathbb{R}^3}(d;\vartheta)$ of class C^1. Every $b\in I(g)$ is the label of such limits.

(iv) In the cases (ii) and (iii), with S the two-point function in question at a pair at which (ii) or (iii) states the inequality, $|(S-c_2b^{-2}\mathcal{K}_6)/(c_2b^{-3}\mathcal{K}_5)-\frac{8}{3\pi}|\le 10^5c_2^{-1}\vartheta^{-2}\bar\varepsilon_\infty(d)$: one number $\frac{8}{3\pi}$ holds for all the limits on the tori of the sides at g, the limits of states at g and their components, and each of these limits determines its label and the number. In the case (i), at a state at a finite cutoff, the same quotient differs from $\frac{8}{3\pi}$ by at most

$10^5 c_2^{-1}\vartheta^{-2}[\bar\varepsilon'(d)+C_2^{\mathrm T}(g)\vartheta^{-1}b_+ d+C_{\mathrm{lat}}\,b\,a_K/d^2]$, and by at most the same with $c_{\mathrm r}' d^{1/2}$ in the place of $\bar\varepsilon'(d)$ and $C_{\mathrm{lat}}^{\flat}$ in the place of C_{lat}.

■□

6.1.7. *Renormalisation and coefficient inputs.* Thm. YM3_I:D stands behind all the statements on the renormalisation group that we use, and Thms. YM3_I:AJ and YM3_I:AD behind the existence of the limit on a torus, the rate of the cutoffs and the agreement of states, into which it enters through the bounds on the partition functions of the tori at the source 0. We use the following construction results: the flow of the coupling and its data (Section YM3_I:74, YM3_I:74, §6.2, Prop. 6.32, equation YM3_I:(1528), YM3_I:74.2, Section YM3_I:82, Prop. YM3_I:74.1, Cor. YM3_I:74.2, the ordered real choice in Prop. YM3_I:69.1 and the common source choice of Section YM3_I:82, the coupling-sequence quantities in equation YM3_I:(634) and the function R of equation YM3_I:(634)); the averaging chart and gauge factor of [7, §B.2], with the measure identity of [7, Lem. B.3] and the cube-form sum in equation YM3_I:(828), and the free response on tori (the construction and periodization in Section YM3_I:52.6, and the curvature bound YM3_I:(2234)); the localisation domains of a torus and of the infinite lattice, their near lifts and their linear sizes (Section YM3_I:72 (the definitions of $\mathcal D_j$ and d_j at its beginning), [7, Lem. O.1], Lem. 5.24); the sizes YM3_I:(759) of families (Def. YM3_I:49.5) and the two sizes $|v|_\infty$ and $\mathrm{sw}_k(v)$ of a plaquette weight in §6.2.1; the bracket localisation of a bare array and the octet property (Props. YM3_I:51.14 and YM3_I:51.16, with equation YM3_I:(877)); the one-step theorem (Thm. YM3_I:V); the naturality of one step under the charts between two lattices with one cutoff (Section YM3_I:52.5, in particular equation YM3_I:(928)); the regular progression on all the lattices of a level (Thm. YM3_I:Y); the weighted progression, its steps at a complex field scale and its chain classes ([7, Def. B.7] and equation [7, eq. 200], [7, Def. B.16], the definitions preceding [7, eq. 228], the definitions preceding equation YM3_I:(2276), Section YM3_I:124, eq. 2.vi, 6.26, the bracket-localization constants in (Wp4), [7, eq. 196], [7, eq. 196], [7, eq. 198], [7, eq. 199], [7, eq. 228], equation YM3_I:(2286), Prop. YM3_I:122.1, equation YM3_I:(2258), [7, Lem. B.10], equations YM3_I:(2286)–(2287), YM3_I:124.4, YM3_I:124.3, equation YM3_I:(2291), Props. YM3_I:122.2, YM3_I:123.1, [7, Prop. B.21], YM3_I:124.2, YM3_I:124.5, 2.4, 6.27, [7, Cor. B.9], Lem. YM3_I:126.2), with the restricted Gaussian measures of one step (Lem. 4.53); the level-wise sourced density and packed-source amplitudes (Prop. 4.42 and Lem. 4.44), the first-creation decomposition and its coefficient bounds (Prop. 4.43, Cor. 4.45 and Prop. 4.54), and the least-torus comparison proved in Prop. 6.20; the free template and the template sums (Section YM3_I:120, Section YM3_I:120, YM3_I:(2231), Lem. YM3_I:120.1, the test-pair bounds eq. 8.i, Prop. YM3_I:120.3, YM3_I:120.5 and YM3_I:120.6, YM3_I:121.2 and Lem. 6.34, Prop. YM3_I:120.3); Laplace's method and the lattice coefficient (Defs. 7.7, 7.10, Lems. 7.6, 7.9, 7.11, 7.24, 7.26, 7.27, 7.38, (7.xviii), (7.xix), (7.xxxi), Cor. 7.37, Thms. W, X); the rate of the cutoffs on a torus (the constants and thresholds in Thm. YM3_I:AR, YM3_I:(2206), the two threshold inequalities at the end of the proof of Thm. YM3_I:AR, Cor. YM3_I:117.1); the limit at every label on a torus (Thm. YM3_I:AS, Prop. YM3_I:118.3); the separated tuples and the test pairs of class C^1 ([7, §K.1], the convention in equation YM3_I:(2343), [7, Lem. I.1] and equation YM3_I:(2247)); the sequences of pairs of laws, their convergence and the connected functions of their limit functionals (Prop. YM3_I:92.1 and its moment–cumulant identity YM3_I:(1856)); the Gibbs sequences, their limits and the components of these (Def. 13.4(2), Thm. YM3_I:AP); and the common construction choice of Section YM3_I:82, with the torus limits of Thm. YM3_I:A and the growing-torus and volume-first limits of Prop. YM3_I:92.1.

6.1.8. *Source-energy jets and the field-scale expansion. Notation.* Throughout, $D=(K,\beta)$ is a datum at g with label b, $\ell=2L^m\in\mathfrak L(g)$, and f,h are real bounded functions on T_ℓ; $w_1=f\circ x$ and $w_2=h\circ x$ are the lattice sources on the bare plaquettes. We write $N=\ell L^K$, $\mathsf N=N+\beta$, $r=d/a_K$, $l=\max\{1,\log(2b/d)\}$, $\mathfrak b(x)=A_1(\log x)^{p_0}$, $\delta(x)=\mathfrak b(x)x^{-1/2}$ and $\langle z\rangle=\max\{1,|z|\}$. With $E_\mu^{(j)}$ the

local terms of the level j of the weighted progression ([7, eq. 200]) at the datum, with the weight $\mu_1 w_1 + \mu_2 w_2$ of the plaquettes, and $\mathcal{D}_j$ its domains of the level j, the second jet of the level j is

$$\mathsf{n}_j = \sum_{X\in\mathcal{D}_j} \sum_{z\in X\cap\mathbb{T}^{(j)}} \partial_{\mu_1}\partial_{\mu_2} E_\mu^{(j)}(X,\mathbf{1},z)\big|_{\mu=0}, \qquad \mathsf{J}_2 = \sum_{j=1}^{K} \mathsf{n}_j.$$

The same construction at a complex field scale λ (every fluctuation field multiplied by λ, the coefficient of the action at the height j moved to $\xi_j(\lambda) = L^{-j}x_0 + \lambda^2(x_j - L^{-j}x_0)$) gives the local terms $E_\mu^{(j),\lambda}$ of the level j at the field scale λ, with $E_\mu^{(j),1} = E_\mu^{(j)}$, and their second jets $\mathsf{n}_j^\lambda = \sum_{X\in\mathcal{D}_j}\sum_{z\in X\cap\mathbb{T}^{(j)}} \partial_{\mu_1}\partial_{\mu_2} E_\mu^{(j),\lambda}(X,\mathbf{1},z)|_{\mu=0} = \sum_{m\ge 0} \mathsf{n}_{j,[2m]}\lambda^{2m}$ (Prop. 6.6); $\mathsf{h}_{2m}^{(n)} = \sum_{j\le n}\mathsf{n}_{j,[2m]}$, $\mathsf{h}_{2m} = \mathsf{h}_{2m}^{(K)}$, and $\mathsf{H}^{(n)}[D] = \beta^3 a_K^2 \mathsf{h}_6^{(n)}$. As $\beta^{-3}a_K^{-2} = b^{-3}a_K$ and $a_K r = d$, a size $r\eta$ of H is a size $b^{-3}d\eta$ of h_6. Where the side matters we write $\mathsf{J}_2[\ell]$, $\mathsf{h}_6[\ell]$, $\mathsf{H}^{(n)}[D,\ell]$.

The free template (Section YM3_I:120) is the free abelian model of the curvature: cochains, plaquette averages and nested projections. At the height $0 \le i < K$, $\mathsf{A}_i^{\mathrm{cv}} = \mathsf{B}_i^{\mathrm{cv}} + \hat{\mathsf{G}}$ is its one-step quadratic form, a sum of two real symmetric matrices ≥ 0 on the non-pivot bonds of $\mathbb{T}^{(i)}$, and $\mathsf{A}_i^{\mathrm{cv}} \ge \gamma^{\mathrm{tr}}$ with a lower bound $\gamma^{\mathrm{tr}} > 0$ (Prop. YM3_I:120.3); T_i is its curvature map onto the bare plaquettes (Prop. YM3_I:120.3). Π_0^{tp} is the orthogonal projection onto the exact bare curvatures and Π_n^{tp}, $n \ge 1$, the projection onto those that the plaquette averages of the level n determine, so that $\Pi_0^{\mathrm{tp}} - \Pi_n^{\mathrm{tp}}$ is the orthogonal projection onto the curvatures whose plaquette averages of the level n vanish; $\mathcal{C}_{(j)} = \Pi_{j-1}^{\mathrm{tp}} - \Pi_j^{\mathrm{tp}}$ are the curvature slices, and $C_{K,K-n}^{\mathrm{fl}} = \Pi_0^{\mathrm{tp}} - \Pi_n^{\mathrm{tp}} = \sum_{1\le j\le n}\mathcal{C}_{(j)}$ is the fluctuation kernel, the conditional covariance of the free lattice curvature given its plaquette averages of the level n; on a lattice with t levels we write $\Pi_t^{\mathrm{fl}} = \Pi_0^{\mathrm{tp}} - \Pi_t^{\mathrm{tp}}$.

The numbers $\delta_{\mathrm{tp}} \in (0, \frac{1}{12}]$ and $C_{\mathrm{tp}} \ge 1$ of the decay of the slices (Props. YM3_I:120.5 and YM3_I:120.6), and the number $C_{\mathrm{tp}}^{\mathrm{row}}$ of the row sums of the curvature maps,

$$\sum_b |\mathsf{T}_i(p,b)| \le C_{\mathrm{tp}}^{\mathrm{row}} L^{-2i} \qquad \text{for every bare plaquette } p,$$

uniformly in the volume, at the height $i = 0$ and at every height at which the side ℓL^{K-i} of the level i, in its units, is at least N_{H}, so at every height if $\ell \ge N_{\mathsf{H}}$ (Prop. YM3_I:120.6), depend on L and on the least admissible side N_{H} fixed with C_{H} and δ_{H} in Section YM3_I:52.6 and the beginning of Section YM3_I:120.3; they, γ^{tr} and N_{H} are determined by the constants.

The Gaussian measures of the heights. Put $\tau_i = L^{-i}x_0/x_i$ and $\mathsf{M}_i = \tau_i \mathsf{B}_i^{\mathrm{cv}} + \hat{\mathsf{G}}$. ν_i° is the centred Gaussian measure with the precision $\mathsf{M}_i \otimes 1$ on the real $\mathfrak{g}$-valued 1-forms B_i on the non-pivot bonds of $\mathbb{T}^{(i)}$, and ν_i^{r} is ν_i° conditioned on the box $\{|B_i(b)| < \mathfrak{b}(x_i)\ \forall b\}$, $|\cdot|$ being the Euclidean norm of $\mathfrak{g} \cong \mathbb{R}^3$: the boxes are products of Euclidean balls. As $\tau_i \in [\frac{3}{4}, \frac{4}{3}]$ (Prop. 6.5) and both matrices are ≥ 0, $\mathsf{M}_i \ge \min\{\tau_i, 1\}\mathsf{A}_i^{\mathrm{cv}} \ge \frac{3}{4}\gamma^{\mathrm{tr}}$, so every variance in ν_i° is at most $\frac{4}{3\gamma^{\mathrm{tr}}}$.

The measures ν_i° are tied to the steps by Lem. 6.13(iii): at $\lambda = 0$, at the trivial configuration, the density of the step of the height i is proportional to $\exp[-\frac{1}{2}(B_i, (\mathsf{M}_i \otimes 1)B_i)]$ on the box of the height, the height 0 included, where $\tau_0 = 1$ and $\mathsf{M}_0 = \mathsf{A}_0^{\mathrm{cv}}$. For polynomials $X_{a,m}$ ($a = 1, 2$, $m = 0, 1, 2$), W_1, W_2 in these fields and a probability measure ν, with the joint cumulants $\langle\cdot;\dots;\cdot\rangle$ taken in ν,

$$\begin{aligned}\gamma_2[\nu] = {} & \langle X_{1,2}; X_{2,0}\rangle + \langle X_{1,0}; X_{2,2}\rangle + \langle X_{1,1}; X_{2,1}\rangle - \langle X_{1,1}; X_{2,0}; W_1\rangle \\ & - \langle X_{1,0}; X_{2,1}; W_1\rangle - \langle X_{1,0}; X_{2,0}; W_2\rangle + \tfrac{1}{2}\langle X_{1,0}; X_{2,0}; W_1^2\rangle; \end{aligned} \tag{6.vii}$$

$\gamma_2^{\mathrm{r}} = \gamma_2[\bigotimes_{0\le i<n}\nu_i^{\mathrm{r}}]$ and $\gamma_2^\circ = \gamma_2[\bigotimes_{0\le i<n}\nu_i^\circ]$, with the polynomials of Prop. 6.12. If ν is invariant under $B \mapsto -B$, the $X_{a,0}$ are even and W_1 is odd, then $\frac{1}{2}\langle X_{1,0}; X_{2,0}; W_1^2\rangle = \frac{1}{2}\langle X_{1,0}; X_{2,0}; W_1; W_1\rangle$: a cumulant with a product entry is $\langle X; Y; W^2\rangle = \langle X; Y; W; W\rangle + 2\langle X; W\rangle\langle Y; W\rangle + 2\langle W\rangle\langle X; Y; W\rangle$, and the last two terms vanish by parity.

Laplace's method (Lem. 7.11). Let ϕ, ψ, X, Y be real and smooth near $0 \in \mathbb{R}^m$, $\phi(0) = 0$, $\nabla\phi(0) = 0$, with a positive definite Hessian H at 0, and let X, Y vanish at 0 with their gradients. Let $\phi_j, \psi_j, X'_j, Y'_j$ be their homogeneous Taylor polynomials of order j at 0, and $W_j = \phi_{j+2} + \psi_j$, $X_{1,j} = X'_{j+2}$, $X_{2,j} = Y'_{j+2}$. Then the covariance of X and Y under the density proportional to $e^{-\beta\phi-\psi}$ on a small window about 0 is

$$\beta^{-2}\gamma_0^{\mathrm{L}} + \beta^{-3}\gamma_2^{\mathrm{L}} + O(\beta^{-4}) \qquad (\beta \to \infty), \tag{6.viii}$$

where $\gamma_0^{\mathrm{L}} = \langle X_{1,0}; X_{2,0}\rangle$ and γ_2^{L} is eq. 6.vii, both in the centred Gaussian measure of covariance H^{-1}. In the variable $\beta^{1/2}x$ the density is the Gaussian one times $e^{-\beta^{-1/2}W_1 - \beta^{-1}W_2 - \cdots}$ and the observables are $\beta^{-1}(X_{a,0} + \beta^{-1/2}X_{a,1} + \beta^{-1}X_{a,2} + \dots)$; the coefficient of β^{-3} of the covariance, expanded in joint cumulants, is eq. 6.vii, written with the parity just noted, and the half-integer orders vanish by parity. We write $\gamma^{\mathrm{L}}_{m'}[\phi, \psi, X, Y]$, $m' = 0, 2$. The numbers $\mathsf{g}_{m'}[N, t](p, q)$ of a flat top are set in Prop. 6.14; for functions w_1, w_2 on the plaquettes of Λ_N, $\mathsf{T}[N, t](w) = \sum_{p,q} w_1(p)w_2(q)\mathsf{g}_2[N, t](p, q)$, and for functions $\tilde{w}_1, \tilde{w}_2$ of finite support on the plaquettes of $\mathbb{Z}^3$, $\mathsf{T}^{\mathrm{lat}}(\tilde{w}) = \sum \tilde{w}_1(\tilde{p})\tilde{w}_2(\tilde{q})\mathsf{B}_1^{\mathrm{lat}}(\tilde{p}, \tilde{q})$.

If $\tilde{w}_1(\tilde{p}) = \tilde{f}(a_K x(\tilde{p}))$, $\tilde{w}_2(\tilde{p}) = \tilde{h}(a_K x(\tilde{p}))$, then $a_K\mathsf{T}^{\mathrm{lat}}(\tilde{w}) = \mathsf{P}_1^{\mathrm{lat}}(\tilde{f}, \tilde{h}; a_K)$.

Proposition 6.5 (The setting of the expansion)**.** There is $\gamma_{\mathrm{s}} \in (0, \min\{\gamma_W^{\mathrm{ref}}, \frac{1}{2}\}]$, determined by the constants, such that for $0 < g \le \gamma_{\mathrm{s}}$, every datum (K, β) at g with $K \ge 3$, every $\ell = 2L^m \in \mathfrak{L}(g)$, and real bounded f, h with supports in balls of radius d at a distance $\ge d$, $8a_K \le d$, at every position of (f, h):

(i) $2L^{m+K-j+1} \ge C_L M$ for $1 \le j \le K$, in particular $\ell L \ge C_L M$; and $\ell \ge N_{\mathsf{H}}$;
(ii) $x_0 > x_1 > \cdots > x_K > g^{-2}$, $x_{K-1} > Lg^{-2} \ge 4L$, and for $0 \le j < K$: $\frac{3}{4}b/D_j \le x_j \le 2b/D_j$ and $\frac{3}{4} \le \tau_j \le \frac{4}{3}$;
(iii) at every level $j < K$ the constants, the lattice of the step $j \to j+1$ and x_j are data under (Im) of [7, Def. B.2];
(iv) the data of the weighted progression ([7, Def. B.7]) are given with $k = K - 1$, the sources w_1, w_2 and the running couplings as thresholds; the setting (J) of [7, Def. B.16] holds at $k = K - 1$; $\log x_{K-1} \ge \max\{2p_0, 1 + \log L\}$; and $\delta(x_{K-1})$ lies below the threshold of the chain classes (equations YM3_I:(2285) and eq. 2.viii) and below the smallness that Prop. 6.6 asks of its discs.

No condition is put on the witness depth, on the signs or on the Lipschitz constants of f and h. ■

Proof sketch. (i) For a side at g the lattice of a datum is large at every level, by the flow of the coupling (Section YM3_I:74, YM3_I:74) and Lem. YM3_I:126.2; at $j = K$ this is $\ell L \ge C_L M$. The condition on N_{H} and its two companion numbers is a member of the collective condition, which holds at the fixed constants and under which the geometric hypotheses of Section YM3_I:52.7 hold; it is a condition on these three numbers, and no side occurs in it. The bound $\ell \ge N_{\mathsf{H}}$ is a property of the sides at g: for $\ell = 2L^m \in \mathfrak{L}(g)$ the flow of the coupling gives $LM\,R(x_K) \le L^m$ (Prop. YM3_I: 74.1; R is the function of equation YM3_I:(634)), so $M\,R(x_K) \le L^m$, and with this Lem. YM3_I: 126.2 gives $\ell \ge N_{\mathsf{H}}$, together with the frames of the datum on T_ℓ; the setting (J) at $k = K - 1$ follows in (iv).

(ii) The run equation gives $|x_j - L^{-j}x_0| \le B_\sharp \sum_{i \ge 0} L^{-i} = \bar{B}$, and $L^{-j}x_0 = b/D_j$. As K is the first passage, $x_j \ge x_{K-1} > X_g \ge Lg^{-2}$ for $j < K$, and $x_K = L^{-1}x_{K-1} - e_K > g^{-2}$; γ_{s} is so small that $X_g \ge 12\bar{B}$. Hence $|\tau_j - 1| \le \bar{B}/x_j \le \frac{1}{12}$, and $\frac{12}{13}b/D_j \le x_j \le b/D_j + \bar{B} \le 2b/D_j$. (iii), (iv) $\gamma_{\mathrm{s}} \le \gamma_W^{\mathrm{ref}}$ is the least of two thresholds determined by the constants. For $g \le \gamma_{\mathrm{s}}$, [7, Cor. B.9], [7, Prop. B.21] and Lem. YM3_I:126.2 give the data of the progression and the setting (J) at $k = K - 1$; in that setting, at every level, the constants, the lattice of the step and x_j are data under (Im), so that in particular the ratio A_0/A_1 lies above the number that (Im) asks.

As $x_{K-1} > L\gamma_{\rm s}^{-2}$ and δ decreases for large arguments, $\delta(x_{K-1}) \le \delta(L\gamma_{\rm s}^{-2})$, which lies below both thresholds for $\gamma_{\rm s}$ small, and $\log x_{K-1} \ge \log(L\gamma_{\rm s}^{-2}) \ge \max\{2p_0, 1+\log L\}$. The sources enter through their supports alone. □

Proposition 6.6 (One field scale)**.** Under the assumptions of Prop. 6.5 the progression is formed at one complex field scale for all its steps. There are $R_j > 1$ $(0 \le j < K)$ with $R_j^{-2} \le c_0(\log x_j)^{2p_0} D_j/b$, and $\rho \in (0, \frac12]$ with $(\rho R_j)^2 \ge 2$, such that, with $c_0, \rho, C_{\mathsf n}, C_{\rm E}, C_{{\rm E},m}, C_{\rm R}, C_{\rm J}$ determined by the constants (they depend on no cutoff, side, run, scale, position or pair), the following hold.

(i) $\lambda \mapsto \mathsf n_{j+1}^\lambda$ is holomorphic and even on $|\lambda| < R_j$, $\mathsf n_{j+1}^1 = \mathsf n_{j+1}$, its Taylor coefficients $\mathsf n_{j+1,[0]}$, $\mathsf n_{j+1,[2]}$ of the orders 0 and 2 vanish, and for $|\lambda| \le \rho R_j$

$$|\mathsf n_{j+1}^\lambda| \le C_{\mathsf n}|\lambda|^2 R_j^{-2} D_j^{-2} \min\{1, (d/D_j)^{5/2}\}\, \Pi_{j+1}\, \eta, \qquad \Pi_{j+1} = \max\{1, (d/D_{j+1})^3\} e^{-[d/(4MD_{j+1})-2]_+}.$$

(ii) $|\mathsf J_2 - \mathsf h_4 - \mathsf h_6| \le C_{\rm E} b^{-4} d^2 l^{8p_0}\eta$ and $|\mathsf h_{2m}| \le C_{{\rm E},m} b^{-m} d^{m-2} l^{2mp_0}\eta$ for $m = 2, 3$.

(iii) For $0 \le n \le K-1$ with $D_n \ge d$ and $m = 2, 3$:

$$\sum_{j=n}^{K-1} |\mathsf n_{j+1,[2m]}| \le C_{\rm R} b^{-m} l^{2mp_0} d^{m-2} (d/D_n)^{\frac52 - (m-2)} \eta.$$

In particular, for $n \le n' \le K$ with $L^n \ge r$: $|\mathsf H^{(n')}[D] - \mathsf H^{(n)}[D]| \le C_{\rm R} l^{6p_0} (r/L^n)^{3/2} r\,\eta$.

(iv) Let $\dot E_a^{(i+1),\lambda} = \partial_{\mu_a} E_\mu^{(i+1),\lambda}|_{\mu=0}$ $(a = 1, 2)$ and $\ddot E^{(i+1),\lambda} = \partial_{\mu_1}\partial_{\mu_2} E_\mu^{(i+1),\lambda}|_{\mu=0}$ be the first and the second jets of the level $i+1$ at the field scale λ, pointed families of the level $i+1$, and let $\psi_i = \min\{1, (d/D_i)^3\}$, $\chi_i = \min\{1, (d/D_i)^{5/2}\}$. At every λ at which they are formed, the member of $\dot E_a^{(i+1),\lambda}$ on a domain X vanishes unless X holds a plaquette of the support of w_a, and the member of $\ddot E^{(i+1),\lambda}$ on X vanishes unless X holds a plaquette of each of the two supports. For $0 \le i \le j < K$ and $|\lambda| \le \rho R_j$, their sizes YM3_I:(759) at the rate $\kappa^+ = (1 + 4\beta_{\rm s})\kappa$, at which the parts of the local terms that depend on the sources decay in the linear size of the domain (Prop. YM3_I:122.1), obey

$$\|\dot E_a^{(i+1),\lambda}\|_{i+1,\kappa^+} \le C_{\rm J}|\lambda|^2 R_i^{-2} D_i^{-1} \psi_i \|w_a\|_\infty, \qquad \|\ddot E^{(i+1),\lambda}\|_{i+1,\kappa^+} \le C_{\rm J}|\lambda|^2 R_i^{-2} D_i^{-2} \chi_i\, \eta.$$

If w_a vanishes identically, all the jets in the direction w_a vanish; so (i)–(iii) hold trivially if w_1 or w_2 vanishes, and so do the bounds of (iv) that involve a vanishing source. ■

Proof sketch. One step at a complex field scale is Prop. 6.27, applied with a parameter set (Def. 6.26) that holds the field scales of the earlier steps; the scale is put on the diagonal afterwards. A lower calibration $\mathsf t_\flat$, fixed with the constants, gives room for the size of the deformed bare array, the bound of the step being uniform in the parameter. The counterterms e_i are kept, and the coefficient of the action is moved along $\xi_j(\lambda)$, which stays in the admitted disc about x_j for $|\lambda| < R_j := \delta(\mathsf t_\flat)/(2\delta(x_j))$, because $|\xi_j(\lambda) - x_j| = |\lambda^2 - 1|\,|x_j - L^{-j}x_0| < 2R_j^2\bar B$. To make the last bound smaller than $\sigma_* x_j$, decrease $\gamma_{\rm s}$, if necessary, so that

$$\bar B \delta(\mathsf t_\flat)^2 < 2\sigma_* A_1^2 (\log x_j)^{2p_0}$$

for every coupling of the run. This is possible by the lower bound on x_j in Prop. 6.5; substitution of the formula for R_j gives $2R_j^2\bar B < \sigma_* x_j$. Thus the simultaneous substitution in the field scale and the quadratic coefficient stays within the product domain of Prop. 6.27.

By Prop. 6.5(ii), $R_j^{-2} = 4\delta(x_j)^2/\delta(\mathsf t_\flat)^2 \le \frac{16}{3} A_1^2 \delta(\mathsf t_\flat)^{-2} (\log x_j)^{2p_0} D_j/b$. Induction over the heights gives the holomorphy, the evenness, $\mathsf n_{j+1}^1 = \mathsf n_{j+1}$ and the locality in the weight of (iv): the member of the level $i+1$ on a domain X is formed from the members of the lower levels on the domains in X and from the weight on the plaquettes of X, at every field scale as at the field scale 1 (Prop. YM3_I: 122.1(c)). The derivatives in the weights obey two linear recursions over the levels, by the chain rule

and Cauchy's estimate in the direction of the bare array: one step at the field scale λ multiplies the size of the derivative in the direction of a bare array by at most $c\,|\lambda|^2 R_i^{-2}$ (clauses (cx3), (cx8) of Prop. 6.27, uniformly in the parameter), and the weight member of the level i in the direction w_a has the size at most $36M^3\alpha_0^2 D_i^{-1}\psi_i\|w_a\|_\infty$, its source meeting at most $35(d/a_K)^3$ plaquettes; so the first derivatives follow the profile $\psi_i = \min\{1,(d/D_i)^3\}$, the second the profile $\chi_i = \min\{1,(d/D_i)^{5/2}\}$.

They close on these profiles for $|\lambda| \le \rho R_j$ by the locality in the weight, the bare array of a source with support of diameter $\le 2d$ meeting at most $77\max\{1,(d/D_i)^3\}$ cubes of the level i, the cubes of side MD_i of the partition π_i of the torus whose wall-connected unions are the domains of the level i (the level-i localization partition of Section YM3_I:45); ρ is fixed by the two inequalities that close the induction, and the closed recursions are the two bounds of (iv). A domain that meets both supports holds two points at a distance at least $d/4$, so that its linear size in the units of its level is at least $[d/(4MD_{j+1}) - 2]_+$; with the count of such domains this gives the display of (i).

At the order 2 the earlier members are taken at the field scale 0, where they do not depend on the sources, and the coefficient is a covariance of two linear forms ℓ_Y of the field at the trivial configuration. A constant gauge rotation u fixes the trivial configuration, so $\ell_Y(\mathrm{Ad}(u)B) = \ell_Y(B)$, and $\int_G \mathrm{Ad}(u)Y\,\mathrm{d}u = 0$ for $Y \in \mathfrak{g}$ gives $\ell_Y = 0$.

(ii), (iii): an even holomorphic H with $|H| \le \mathsf{B}$ on $|\lambda| < R$ has $|H_{[2m]}| \le \mathsf{B}R^{-2m}$ and $|H(1) - \sum_{m\le p} H_{[2m]}| \le \mathsf{B}R^{-2p-2}/(1-R^{-2})$. Applied on $|\lambda| < \rho R_j$, where $(\rho R_j)^2 \ge 2$, with the bound of (i), each level gives at most $c\,b^{-m}(\log x_j)^{2mp_0} D_j^{m-2}\chi_j \Pi_{j+1}\eta$ at the order $2m$, and $m = 4$ for the remainder. The levels with $D_j \ge d$ have $\Pi_{j+1} = 1$, $\log x_j \le l$ and $D_j^{m-2}\chi_j = d^{m-2}(d/D_j)^{\frac{5}{2}-(m-2)}$: a geometric series over the levels. The other levels are summed against Π_{j+1}, which is summable against every power of the logarithm. For (iii) at $m = 3$: $\mathsf{h}_6^{(n')} - \mathsf{h}_6^{(n)} = \sum_{n\le j<n'} \mathsf{n}_{j+1,[6]}$, $D_n \ge d$ if and only if $L^n \ge r$, $d/D_n = r/L^n$ and $\beta^3 a_K^2 b^{-3} d = r$. □

Proposition 6.7 (The front on the least torus)**.** There are $\gamma_{\mathrm{C}} > 0$, determined by the constants, and for $0 < g \le \gamma_{\mathrm{C}}$ numbers $d_{\mathrm{A}}(g) > 0$ and $C_{\mathrm{A}}(g) < \infty$, depending on g and the constants and on no K, β, b, ϑ, d, pair or position, such that for every datum (K,β) at g, $\ell = \ell_g$, $0 < d \le d_{\mathrm{A}}(g)$ with $s(d) \le K$, $\vartheta \in (0,1)$ and $(f,h) \in \mathfrak{T}_{T_\ell}(d;\vartheta)$: $K \ge 3$ and $a_K \le d$; the set

$$\mathsf{Pk} = \{v \in \delta\mathbb{Z}^3 :\ |v_\mu| \le 0.575 - 9d\ (\mu = 1,2,3)\}, \qquad \delta = a_K\lceil 18d/a_K\rceil,$$

which lies in $a_K\mathbb{Z}^3$ and depends on a_K and d alone, consists of vectors of length < 1, has $\#\mathsf{Pk} \ge (30d)^{-3}$, and $(\tau_v f, \tau_v h) \in \mathfrak{T}_{T_\ell}(d;\vartheta)$ for $v \in \mathsf{Pk}$; and

$$\Big|S^T_{2;K,\ell,\beta}(f,h) - \frac{1}{\#\mathsf{Pk}}\sum_{v\in\mathsf{Pk}} \mathsf{J}_2(\tau_v f, \tau_v h)\Big| \le C_{\mathrm{A}}(g)\vartheta^{-1}d^2 b^{-2}\eta.$$

■

Proof. Decrease $\gamma_{\mathrm{C}} \le \min\{\gamma_{\mathrm{s}}, \gamma_{\mathrm{ev}}\}$ to satisfy the source and radius conditions of §4.2. These are conditions on the fixed constants and on the lower coupling threshold. Choose $d_{\mathrm{A}}(g)$ small enough for the packing and amplitude bounds below.

If $f \circ x = 0$, every lattice translate of this bare weight is zero. Both the covariance and its source-energy mixed derivative therefore vanish; the assertion follows. The same applies to $h \circ x$. Henceforth neither is zero. Choose x_1 with $\operatorname{supp} f \subset \mathbb{B}(x_1, d)$. Both supports lie in $\mathbb{B}(x_1, 9d)$. The balls $\mathbb{B}(x_1 + v, 9d)$, $v \in \mathsf{Pk}$, have disjoint interiors, since $\delta \ge 18d$, and lie in $\mathbb{B}(x_1, 1)$. Indeed, $|v| \le \sqrt{3}(0.575 - 9d)$ and $\sqrt{3}(0.575 - 9d) + 9d < 1$. The condition $s(d) \le K$ gives $L^K d \ge C_W(s(d)) \ge 4L^2 M$, so $a_K \le d$; for d small it also gives $K \ge 3$. Now $\delta \le 18d + a_K \le 19d$. Thus, for d small,

$$\#\mathsf{Pk} = \left(2\left\lfloor\frac{0.575 - 9d}{\delta}\right\rfloor + 1\right)^3 \ge (30d)^{-3}.$$

The translates preserve the support separation, amplitude and Lipschitz conditions defining the test class.

Use the norm eq. 4.viii and the holomorphic total source energy $\mathcal{E}$ of Prop. 4.42. Lem. 4.44 bounds a disjoint sum $w = (\sum_\iota \zeta_\iota u_\iota)\circ x$ by

$$\|w\|_{\mathrm{lw}} \le \max_\iota |\zeta_\iota| \mathsf{A}_\iota/\mathsf{A}_\star(d), \qquad \mathsf{A}_\star(d) = \min\left\{\sigma_{\mathrm{src}} b^\vee, \frac{\mu_\star}{K_{\mathrm{L}}^\vee}\frac{b^\vee}{(b^\wedge)^{1/2}}\left(\frac{\vartheta d}{4\sqrt{3}M}\right)^{1/2}\right\},$$

where $\|u_\iota\|_\infty \le \mathsf{A}_\iota$, $\operatorname{Lip} u_\iota \le \mathsf{A}_\iota/(\vartheta d)$, $b^\vee = g^{-2} - \bar{B}$ and $b^\wedge = b_+$. The total-energy identity in Prop. 4.42 shows that J_2 is the mixed derivative of $\mathcal{E}$ at zero. Therefore $S_2^T - \mathsf{J}_2$ is the mixed derivative of $\log(Z[\cdot]/Z[0]) - \mathcal{E}$.

For signs $\epsilon \in \{\pm 1\}^{\mathsf{Pk}}$ put

$$F^\epsilon = \sum_v \epsilon_v \tau_v f, \qquad H^\epsilon = \sum_v \epsilon_v \tau_v h, \qquad w_z = (z_1 F^\epsilon + z_2 H^\epsilon)\circ x, \qquad \Omega_\epsilon(z) = Z[w_z] e^{-\mathcal{E}(w_z)}/Z[0].$$

The nonzero sets of the individual bumps are pairwise disjoint. With $u_1 = \|f\|_\infty/\mathsf{A}_\star(d)$ and $u_2 = \|h\|_\infty/\mathsf{A}_\star(d)$, the amplitude bound gives $\|w_z\|_{\mathrm{lw}} \le \max_a |z_a| u_a$. Thus Ω_ϵ is holomorphic on $\mathbb{P} = \{|z_1|u_1 < 1,\ |z_2|u_2 < 1\}$ and $\Omega_\epsilon(0) = 1$.

Split the sourced partition by its first large-field creation, using eq. 4.xvi:

$$\Omega_\epsilon = \Omega_\epsilon^\emptyset + \sum_{t=1}^K \Omega_\epsilon^t, \qquad \Omega_\epsilon^\emptyset = e^{-\mathcal{E}(w_z)} m_K^{w_z}(\mathcal{V}_K)/Z[0], \qquad \Omega_\epsilon^t = e^{-\mathcal{E}(w_z)} \mathsf{Cr}_t(w_z)/Z[0].$$

Every source here is supported in the one ball $\mathbb{B}(x_1, 1)$. Prop. 4.43 therefore gives

$$|\Omega_\epsilon^\emptyset| \le e^{\mathsf{G}_K} \quad \text{on } \mathbb{P}, \qquad |\Omega_\epsilon^t| \le e^{\|w_z\|_{\mathrm{lw}}(\mathsf{G}_{t-1}+\mathsf{H}_{t-1})}\mathsf{w}_t.$$

Choose $\mathsf{C}_{\mathrm{cnt}} = C_{\mathrm{cnt}}(g) \ge 1$ in eq. 4.xv, and set $\mathsf{Cnt}_t = \mathsf{C}_{\mathrm{cnt}} L^{4(K-t+1)}$. On $\mathbb{P}_t = \{|z_a| u_a \mathsf{Cnt}_t < 1, a = 1, 2\}$ the second bound is at most $e\mathsf{w}_t$. For $\emptyset \ne B \subset \{1,2\}$ write $[z^B]\Omega = (\prod_{a\in B}\partial_{z_a})\Omega(0)$ and $u^B = \prod_{a\in B} u_a$. Cauchy's formula on these polydiscs gives

$$|[z^B]\Omega_\epsilon^\emptyset| \le e^{\mathsf{G}_K} u^B, \qquad |[z^B]\Omega_\epsilon^t| \le e\mathsf{w}_t \mathsf{Cnt}_t^{|B|} u^B \le e(\mathsf{C}_{\mathrm{cnt}} L^4)^2 L^{8(K-t)} \mathsf{w}_t u^B.$$

By eq. 4.xviii, the latter coefficients sum to at most $e(\mathsf{C}_{\mathrm{cnt}}L^4)^2 C_8(g) u^B$. Choose $\mathsf{Q}(g) \ge 1$ above this coefficient bound plus $e^{\sup \mathsf{G}_K}$. It is finite and independent of the datum, the positions and the signs. Thus

$$|[z^B]\Omega_\epsilon| \le \mathsf{Q}(g) u^B, \qquad |\partial_{12} \log \Omega_\epsilon(0)| \le (\mathsf{Q} + \mathsf{Q}^2) u_1 u_2 \le 2\mathsf{Q}^2 \mathsf{A}_\star(d)^{-2}\eta.$$

The logarithm is needed only near zero, where $\Omega_\epsilon(0) = 1$ ensures its existence.

By bilinearity the last mixed derivative is

$$\sum_{v,v'} \epsilon_v \epsilon_{v'} (S_2^T - \mathsf{J}_2)(\tau_v f, \tau_{v'} h).$$

Average over the independent signs. The terms $v \ne v'$ vanish and the diagonal terms remain. Translation invariance of S_2^T and the packing count then give

$$\left| S_2^T(f,h) - \frac{1}{\#\mathsf{Pk}} \sum_v \mathsf{J}_2(\tau_v f, \tau_v h) \right| \le (30d)^3\, 2\mathsf{Q}^2 \mathsf{A}_\star(d)^{-2}\eta.$$

For $d \le 4\sqrt{3}Mg^{-2}(K_{\mathrm{L}}^\vee/K_{\mathrm{Q}}^\vee)^2$ the second entry in the minimum defining $\mathsf{A}_\star$ is the smaller one. Substitution bounds the last expression by $C(g)\vartheta^{-1}d^2\eta$. Since the label satisfies $b \le b_+$, enlarging this fixed-g constant by b_+^2 gives the asserted $C_{\mathrm{A}}(g)\vartheta^{-1}d^2 b^{-2}\eta$. The finitely many ceilings on d determine $d_{\mathrm{A}}(g)$. □

Proposition 6.8 (The order four)**.** Let $0 < g \le \gamma_{\mathrm{s}}$, (K,β) a datum at g, $\ell \in \mathfrak{L}(g)$, $0 < d \le L^{-1}$, $s(d) < K$, $20d \le \ell$, $\vartheta \in (0,1)$ and $(f,h) \in \mathfrak{T}_{T_\ell}(d;\vartheta)$, at any position. Then $K \ge 3$, $8a_K \le d$, $d/a_K \ge L(\log \frac{1}{d})^2$, and

$$|\mathsf{h}_4 - c_2 b^{-2}\mathcal{K}_6|(f,h) \le \big[\bar\varepsilon_4(g,d)\, b^{-3} d + C_{\mathrm{lat}}\, b^{-2} a_K/d\big]\eta,$$

where, with $l' = \max\{1, \log \frac{2b'}{d}\}$,

$$\bar\varepsilon_4(g,d) = \sup_{g^{-2}<b'\le b_+} \frac{b'}{d}\Big\{3C_{\mathrm{tm}} d^3 + C_{\mathrm{Bx}}\big[d^{3/2} + e^{-\mathsf{c}_\chi(\log \frac{3b'}{4d^{1/2}})^2}\big] + C_{\mathrm{Bt}}\, l'^{\,4p_0} e^{-\mathsf{c}_\flat L(\log\frac{1}{d})^2}\Big\},$$

$C_{\mathrm{Bx}}, C_{\mathrm{Bt}}, \mathsf{c}_\chi > 0$ being determined by the constants and $\mathsf{c}_\flat = \min\{\frac{1}{8ML}, \frac{\delta_{\mathrm{tp}}}{2\sqrt{3}}\}$. The function $\bar\varepsilon_4(g,\cdot)$ depends on g and the constants alone and is the same for all sides and data at g; and $\bar\varepsilon_4(g,d) \le C_4(g) d^{1/2}$, with $C_4(g)$ depending on g and the constants. ∎

Proof sketch. The first three assertions are those of the setting on the witness depth. Let $2 \le n \le K$ with $D_n \ge d$.

(1)

The tower at a real field scale. Let $0 < \lambda \le 1$, $g_i = x_i^{-1/2}$, and let $c_{\mathfrak{h}}$ be the constant of the shift systems of the steps. The chain of the levels at the field scale λ is formed downwards from the flat top of the level n: $A_n^\lambda := 0$ and, for $i = n-1, \dots, 1$,

$$A_i^\lambda := A^{\mathrm{lw}}(v^{[i+1]}_{A^\lambda_{i+1}}, \lambda g_i B_i), \qquad \tilde U_i^\lambda := e^{iL^{-i}A_i^\lambda},$$

with the lowered 1-form A^{lw} of equation YM3_I:(2286) and the chart point $v_A^{[i+1]}$ of a chain class (equation eq. 2.vi). A *chart point* of a level k is a pair $v = (A, \mathbf{J})$ of 1-forms on the bare bonds; $P[v] = (\mathbf{1}, A, \mathbf{J})$ is its triple in the chart at the trivial pair (Section YM3_I:124), and $P[0]$ is the trivial triple; its norm $\|v\|$ is the largest of the numbers $|A(b)|$, $|\mathbf{J}(b)|$ over the bare bonds b and of the moduli of the difference quotients of A at the spacing L^{-k}; and $\mathsf{Cc}_k(r)$ is the chain class of the level k and the radius r. The A_i^λ are real 1-forms on the bare bonds, and $\tilde U_i^\lambda$, a configuration on the bare bonds, is the *background of the level* i, as in the proof of Prop. 6.12(i): up to a gauge transformation it is a background of the level i in the sense of the actual-background condition in eq. 2.vi; $\mathcal{O}(w_a)$ and the a_p are taken at it.

By Prop. YM3_I:124.2, equation YM3_I:(2288), read at the field λB_i, which lies in the box, as the box is a product of Euclidean balls, and by the bounds of equation YM3_I:(2287) and its domain verification at $\sup|\lambda g_i B_i| \le \lambda\delta(x_i)$, $A_i^\lambda \in \mathsf{Cc}_i(\lambda r_i)$, with $r_n := 0$ and $r_i := L^{-1}r_{i+1} + \frac{11}{10}c_{\mathfrak{h}}\delta(x_i) \le \frac{8}{5}c_{\mathfrak{h}}\delta(x_i)$; the conditions of that Proposition hold at every height, and $r_1 \le \frac{1}{2}r_\alpha$, r_α being the usable radius of the fixed radii (equations YM3_I:(933) and YM3_I:(2285)), as $\delta(x_{K-1})$ lies below the threshold of the chain classes (Prop. 6.5(iv)).

The same Proposition gives $v^{[i]}_{A_i^\lambda} = v^{\mathrm{lw}}(v^{[i+1]}_{A^\lambda_{i+1}}, \lambda g_i B_i)$, the lowered point of equation YM3_I:(2286); the lowered points are holomorphic (the holomorphy following equation YM3_I:(2286)) and obey the bounds of equation YM3_I:(2287) and its domain verification at complex points as well, so the same recursion, with the lowered points, continues the chain to the complex field scales $|\lambda| \le 1$, where the A_i^λ take complex values and the $\tilde U_i^\lambda$ lie in $\mathrm{SL}(2,\mathbb{C})$; one name, $\tilde U_i^\lambda$, serves for both. For $\mu \in \mathbb{C}^2$ near 0 put $w_\mu = \mu_1 w_1 + \mu_2 w_2$ and

$$D_i^{\lambda,\mu} = \mathcal{O}(w_\mu) + \sum_{i'\le i} \mathsf{Sum}\big[E_\mu^{(i'),\lambda} - E_0^{(i'),\lambda}\big],$$

$\mathsf{Sum}[\cdot]$ being the sum of a pointed family at a configuration or at a triple (the definitions preceding equation YM3_I:(2276)). Each step at a height $1 \le i < n$ obeys an identity of its own at the field

scale λ, whose case $\lambda = 1$ uses the quotient argument of Prop. 2.4: for $r \le r_{i+1}$, $A \in \mathsf{Cc}_{i+1}(r)$, $v = v_A^{[i+1]}$, $\tilde U = e^{iL^{-i-1}A}$ and $\tilde U^{\mathrm{lw}}(B) = e^{iL^{-i}A^{\mathrm{lw}}(v,\lambda g_i B)}$,

$$\exp D_{i+1}^{\lambda,\mu}(\tilde U) = \int \exp D_i^{\lambda,\mu}(\tilde U^{\mathrm{lw}}(B))\, \mathrm{d}\nu_i^\lambda[v](B).$$

Here $\nu_i^\lambda[v]$ is the complex measure of mass 1 on the box of the height i whose density is the integrand of the integral form of the step at the field scale λ, at the source 0 and at the triple $P[v]$ (clause (cx10) of Prop. 6.27), divided by its integral, which is not 0: the Gaussian density of the covariance $\tau_i(\lambda)^{-1}\mathbf{C}_*(P[v])$, with $\tau_i(\lambda) = \xi_i(\lambda)/x_i$, positive for real $|\lambda| \le 1$, and the whole covariance $\mathbf{C}_*$ of the step, times $\exp[-\frac{1-\tau_i(\lambda)}{2}\sum_\square(B, \hat{\mathsf{G}}_\square(P[v]|_\square)B)]$, with the gauge-fixing operators $\hat{\mathsf{G}}_\square$ of the fine cubes (equation YM3_I:(823)), times the exponential of the potentials of the step at the field scale λ. At complex λ this is the holomorphic continuation supplied by the same one-step theorem, with its nonvanishing normalizer; no positivity is asserted there.

The identity is proved as follows. The integral form of the step at the triple $P[v]$, at the source μ and at the source 0 (Prop. 6.27(cx10), at the data of Prop. 6.6), and the quotient of the two give

$$\exp \mathsf{Sum}\big[E_\mu^{(i+1),\lambda} - E_0^{(i+1),\lambda}\big](P[v]) = \int \exp\big[\mathbb{V}'[\mathfrak{g}]^{\mathrm{tot}}(P[v]; \lambda B)\big]\, \mathrm{d}\nu_i^\lambda[v](B),$$

because μ stands in the rule of the step alone and the rule is linear in the bare array, $\mathfrak{g}$ being the difference of the bare arrays of the height i at μ and at 0, whose members are the $E_\mu^{(i'),\lambda} - E_0^{(i'),\lambda}$, $i' \le i$, and the weight member of w_μ. Prop. YM3_I:124.2, equation YM3_I:(2289), which holds for every bare array and every field of the box, read at λB, in the plaquette form of equation YM3_I:(2290) and by linearity, gives $\mathbb{V}'[\mathfrak{g}]^{\mathrm{tot}}(P[v]; \lambda B) = D_i^{\lambda,\mu}(\tilde U^{\mathrm{lw}}(B)) - D_i^{\lambda,\mu}(\tilde U)$; and the proof of Prop. YM3_I: 124.2 gives $\mathsf{Sum}[\cdot](P[v]) = \mathsf{Sum}[\cdot](\tilde U)$ for the family of the level $i+1$. Multiplying by $\exp D_i^{\lambda,\mu}(\tilde U)$ gives the identity. Iterating it from the height $n-1$, where $A = 0$ and $\tilde U = \mathbf{1}$, down to the height 1 along the chain, and taking the mixed derivative in μ at 0 of the logarithms, gives

$$\sum_{j=2}^n \mathsf{n}_j^\lambda = \langle \mathcal{F}_1^\lambda(\tilde U_1^\lambda); \mathcal{F}_2^\lambda(\tilde U_1^\lambda)\rangle_{\mathbb{Q}_n^\lambda} + \int [\mathsf{F}_1^\lambda(\tilde U_1^\lambda) - \mathsf{F}_1^\lambda(\mathbf{1})]\, d\mathbb{Q}_n^\lambda,$$

where $\mathbb{Q}_n^\lambda(dB) = \prod_{1\le i<n} \nu_i^\lambda[v^{[i+1]}_{A_{i+1}^\lambda}](dB_i)$ is a complex measure of mass 1 on the product of the boxes, with a bounded continuous density, each factor being formed at the chart point that the chain gives above its height; $\mathcal{F}_a^\lambda = \mathcal{O}(w_a) + \mathsf{Sum}[\dot E_a^{(1),\lambda}]$ is $\mathcal{O}(w_a)$ plus the first jet of the level 1 in the direction w_a, and $\mathsf{F}_1^\lambda = \mathsf{Sum}[\ddot E^{(1),\lambda}]$ is the second jet of the level 1, with $\mathsf{F}_1^\lambda(\mathbf{1}) = \mathsf{n}_1^\lambda$ (the jets of Prop. 6.6(iv)).

(2)

The limit $\lambda \to 0$. The lattice being fixed, let $\lambda \downarrow 0$ in the identity of (1) divided by λ^4. Its left side tends to $\sum_{j=2}^n \mathsf{n}_{j,[4]}$, by Prop. 6.6(i).

The measures. The chart points of the chain have $\sup_B \|v^{[i+1]}_{A_{i+1}^\lambda}\| \le \lambda r_{i+1} \to 0$, and $\tau_i(\lambda) \to \tau_i(0) = \tau_i$. Two facts tie the step at the trivial configuration to the free template: the whole covariance of the step at the trivial triple is $(\mathsf{A}_i^{\mathrm{cv}})^{-1} \otimes 1$ (equation YM3_I:(2297); Prop. YM3_I:120.3); and the gauge-fixing operators of the fine cubes at the trivial triple add up to the gauge-fixing operator at the block field $\mathbf{1}$, which is $\hat{\mathsf{G}} \otimes 1$, as the fine cubes partition the sites of the level $i+1$ and the operator of a cube is the gauge-fixing form restricted to the blocks of that cube (equation YM3_I: (823); Prop. YM3_I:120.3).

The covariance and the gauge-fixing operators at the triple $P[v]$ tend to these, entry by entry, as $v \to 0$ (equation YM3_I:(2296); the kernels of the $\hat{\mathsf{G}}_\square$ are holomorphic in the chart point), and the potentials at the field scale λ are at most $c\lambda$ times a function bounded on the box (eq. 6.xiii of Def. 6.26). So the density of $\mathbb{Q}_n^\lambda$ tends, uniformly on the product of the boxes, to the product over

the heights of the densities proportional to

$$e^{-\frac12(B_i,(\tau_i\mathsf{A}_i^{\mathrm{cv}}+(1-\tau_i)\hat{\mathsf{G}})B_i)}=e^{-\frac12(B_i,\mathsf{M}_iB_i)};$$

as the boxes have finite volume and the limits of the normalisations are positive, $\mathbb{Q}_n^\lambda\to\mathbb{Q}_n^0:=\bigotimes_{1\le i<n}\nu_i^{\mathrm{r}}$ in total variation.

The plaquette terms. At every height the linear part of one step of the chain in the field B_i, read in the plaquette angle, is $\mathsf{sg}_i(\mathsf{T}_i\otimes 1)$ with a sign $\mathsf{sg}_i\in\{1,-1\}$ (Lem. YM3_I:124.3, which reads $\ell\ge N_{\mathsf{H}}$ of Prop. 6.5(i); its map $\mathsf{T}_i^{\mathrm{cv}}$ is T_i, the definitions preceding Lem. YM3_I:124.3). With the expansion of the plaquette terms and of the plaquette angles to the second order at one step (Lem. YM3_I:124.4), summed along the chain, this gives, uniformly on the boxes and for every bare plaquette p,

$$|a_p(\tilde U_1^\lambda)-\tfrac12\lambda^2|\Theta_p(B)|^2|\le c_\Theta\lambda^3,\qquad \Theta_p(B):=\sum_{1\le i<n}\mathsf{sg}_i\,x_i^{-1/2}(\mathsf{T}_iB_i)(p),$$

with c_Θ determined by the constants. Hence $\lambda^{-2}\mathcal{O}(w_a)(\tilde U_1^\lambda)\to q_a(B):=\frac12 a_K^{-1}\sum_p w_a(p)|\Theta_p(B)|^2$ uniformly on the boxes.

The level 1. The local terms of the level 1 at $\tilde U_1^\lambda$ take the values of their members at the restrictions of the triple $P[v]$, $v=v^{[1]}_{A_1^\lambda}$, as in the proof of Prop. YM3_I:124.2. For a pointed family f of the level 1, a domain X of the level 1, a site z of X and $\kappa'>0$, $d_1(X)$ being the linear size of X in the units of the level 1,

$$\big|\mathsf{f}(X,P[v]|_X,z)-\mathsf{f}(X,\mathbf{1},z)\big|\le 2\|\mathsf{f}\|_{1,\kappa'}e^{-\kappa' d_1(X)}\lambda^2(r_1/r_\alpha)^2:$$

the function $\zeta\mapsto\mathsf{f}(X,P[\zeta v]|_X,z)$ is holomorphic on $|\zeta|<r_\alpha/\|v\|$, where $r_\alpha/\|v\|\ge r_\alpha/(\lambda r_1)\ge 2$, and bounded there by $\|\mathsf{f}\|_{1,\kappa'}e^{-\kappa' d_1(X)}$; its derivative at 0 vanishes, because a constant gauge rotation u fixes the trivial configuration and acts on the chart point by $\mathrm{Ad}(u)$, the local terms are invariant, and $\int_G\mathrm{Ad}(u)Y\,\mathrm{d}u=0$ for $Y\in\mathfrak{g}$; Cauchy's estimate bounds the rest of the series. At $\mathsf{f}=\dot E_a^{(1),\lambda}$, whose size at the rate κ^+ is at most $C_{\mathrm{J}}\lambda^2R_0^{-2}a_K^{-1}\|w_a\|_\infty$ (Prop. 6.6(iv) at $i=0$, where $\psi_0=1$), this shows that the first jet of the level 1 changes along the chain by $O(\lambda^4)$, uniformly on the boxes; as a covariance does not change when constants are added, $\lambda^{-4}\langle\mathcal{F}_1^\lambda(\tilde U_1^\lambda);\mathcal{F}_2^\lambda(\tilde U_1^\lambda)\rangle_{\mathbb{Q}_n^\lambda}\to\langle q_1;q_2\rangle_{\mathbb{Q}_n^0}$.

At $\mathsf{f}=\ddot E^{(1),\lambda}$, whose size is at most $C_{\mathrm{J}}\lambda^2R_0^{-2}a_K^{-2}\eta$ and whose members vanish on the domains that do not meet both supports (Prop. 6.6(iv) at $i=0$), the count of the domains of the level 1 that meet both supports, weighted with $e^{-\kappa^+d_1(X)}$ ([7, Lem. B.10]), brings the factor Π_1, as in the proof of Prop. 6.6(i), and gives

$$|\mathsf{F}_1^\lambda(\tilde U_1^\lambda)-\mathsf{F}_1^\lambda(\mathbf{1})|\le\lambda^4\,\mathrm{Er}_\flat,\qquad \mathrm{Er}_\flat=c\,R_0^{-2}a_K^{-2}(r_1/r_\alpha)^2\,\Pi_1\,\eta.$$

So the limit $\mathsf{e}_\flat$ of $\lambda^{-4}\int[\mathsf{F}_1^\lambda(\tilde U_1^\lambda)-\mathsf{F}_1^\lambda(\mathbf{1})]\,\mathrm{d}\mathbb{Q}_n^\lambda$ exists, $|\mathsf{e}_\flat|\le\mathrm{Er}_\flat$, the total variation of $\mathbb{Q}_n^\lambda$ tending to 1, and

$$\sum_{j=2}^n\mathsf{n}_{j,[4]}=\langle q_1;q_2\rangle_{\mathbb{Q}_n^0}+\mathsf{e}_\flat.$$

The covariance $\langle q_1;q_2\rangle_{\mathbb{Q}_n^0}$ does not depend on the signs sg_i: every ν_i^{r} is invariant under $B_i\mapsto -B_i$, its box being symmetric, so that the law of $(\mathsf{sg}_iB_i)_i$ under $\mathbb{Q}_n^0$ is $\mathbb{Q}_n^0$. By Prop. 6.6(i) at $j=0$ and Cauchy's estimate on $|\lambda|=\rho R_0$, $|\mathsf{n}_{1,[4]}|\le C_{\mathsf{n}}\rho^{-2}R_0^{-4}a_K^{-2}\Pi_1\eta$. As $x_0=\beta=b/a_K$, $R_0^{-2}\le c_0(\log x_0)^{2p_0}a_K/b$, $r_1\le\frac85c_\flat\delta(x_1)$ and $\delta(x_1)^2\le\frac43A_1^2(\log x_0)^{2p_0}La_K/b$ (Prop. 6.5(ii)), both $|\mathsf{e}_\flat|$ and $|\mathsf{n}_{1,[4]}|$ are at most $c\,b^{-2}(\log\frac{b}{a_K})^{4p_0}\Pi_1\eta$.

With $u_0=d/a_K\ge 8$, $\log\frac{b}{a_K}\le l\,(1+\log u_0)$ and $\Pi_1\le\max\{1,u_0^3\}e^{2-u_0/(4ML)}$; as $(1+\log u)^{4p_0}\max\{1,u^3\}e^{2-u/(8ML)}$ is bounded on $u\ge 8$ and $\mathsf{c}_\flat\le\frac{1}{8ML}$, this is below $C\,b^{-2}l^{4p_0}e^{-\mathsf{c}_\flat d/a_K}\eta$.

(3)

The boxes out at the order four. The cumulants of the orders 2 and 4 of the components $B_i^\alpha(b)$ in ν_i^{r} and in ν_i° differ by at most $C_\chi e^{-\mathsf{c}_\chi(\log x_i)^{2p_0}}$, uniformly in the volume, with C_χ, c_χ determined by the constants (Lem. 4.53, read with the one quadratic system of the height and the threshold $\mathfrak{b}(x_i)$); the odd cumulants vanish in both measures by the symmetry $B \mapsto -B$, and the Gaussian ones are $\delta_{\alpha\beta}\mathsf{M}_i^{-1}(b,b')$ at the order 2 and 0 at the order 4.

(4)

Wick's rule and the slices. The B_i are independent in $\mathbb{Q}_n^0$ and each ν_i^{r} is even. With $F_i^\alpha(p) = \mathsf{sg}_i\, x_i^{-1/2}\sum_b \mathsf{T}_i(p,b)B_i^\alpha(b)$, so that $|\Theta_p|^2 = \sum_\alpha\big(\sum_i F_i^\alpha(p)\big)^2$ in an orthonormal basis (t_α) of $\mathfrak{g}$, $c_i^{\alpha\beta}(p,q) = \langle F_i^\alpha(p)F_i^\beta(q)\rangle_{\nu_i^{\mathrm{r}}}$ and $\kappa_i^{\alpha\beta}(p,q)$ the joint fourth cumulant of $F_i^\alpha(p), F_i^\alpha(p), F_i^\beta(q), F_i^\beta(q)$,

$$\langle q_1;q_2\rangle_{\mathbb{Q}_n^0} = \tfrac14 a_K^{-2}\sum_{p,q} w_1(p)w_2(q)\sum_{\alpha,\beta}\Big[2\Big(\sum_i c_i^{\alpha\beta}(p,q)\Big)^2 + \sum_i \kappa_i^{\alpha\beta}(p,q)\Big].$$

By the slice identity $\mathsf{T}_i\mathsf{M}_i^{-1}\mathsf{T}_i^{\mathsf{T}} = L^{-i}\tau_i^{-1}\mathcal{C}_{(i+1)}$ (Prop. YM3_I:120.3, valid at every $\tau_i \neq 0$) and

$$x_i^{-1}L^{-i}\tau_i^{-1} = x_0^{-1},$$

the running couplings drop out: in the Gaussian measures $c_i^{\alpha\beta} = \delta_{\alpha\beta}x_0^{-1}\mathcal{C}_{(i+1)}$. As $a_K x_0 = b$, the Gaussian value of $\langle q_1;q_2\rangle$ is $\frac32 b^{-2}\sum_{p,q} w_1(p)w_2(q)\Sigma(p,q)^2$ with

$$\Sigma = \sum_{1\le i<n}\mathcal{C}_{(i+1)} = C^{\mathrm{fl}}_{K,K-n} - \mathcal{C}_{(1)}.$$

The differences of (3) are carried to the F_i^α through the row sums of T_i, which are at most $C_{\mathrm{tp}}^{\mathrm{row}}L^{-2i}$ at every height, uniformly in the volume, as $\ell \ge N_{\mathsf{H}}$ (Prop. YM3_I:120.6; Prop. 6.5(i)). By multilinearity, with $\epsilon_i = C_\chi e^{-\mathsf{c}_\chi(\log x_i)^{2p_0}}$, $\mathsf{sg}_i^2 = 1$ and $x_i^{-1}L^{-4i} = \tau_i x_0^{-1}L^{-3i} \le \frac43 x_0^{-1}L^{-3i}$,

$$|c_i^{\alpha\beta}(p,q) - \delta_{\alpha\beta}x_0^{-1}\mathcal{C}_{(i+1)}(p,q)| \le \bar e_i := \tfrac43 (C_{\mathrm{tp}}^{\mathrm{row}})^2 x_0^{-1}L^{-3i}\epsilon_i, \qquad |\kappa_i^{\alpha\beta}(p,q)| \le \tfrac{16}{9}(C_{\mathrm{tp}}^{\mathrm{row}})^4 x_0^{-2}L^{-6i}\epsilon_i.$$

There are at most $35^2(d/a_K)^6$ pairs (p,q) with $w_1(p)w_2(q)\neq 0$, and for them $|x(p)-x(q)|_\infty \ge d/\sqrt{3}$; with $u_i = d/D_i$ one has $L^{-3i}(d/a_K)^3 = u_i^3$, and Prop. YM3_I:120.6 gives $(d/a_K)^3(|\Sigma|+|\mathcal{C}_{(1)}|)(p,q) \le C_{\mathrm{tp}}S_\Sigma$, $S_\Sigma = \sup_{u>0}\sum_{s\in\mathbb{Z}}(uL^s)^3 e^{-\delta_{\mathrm{tp}}uL^s/\sqrt{3}}$. Expanding the square, with $\mathsf{E}_k = \sum_{1\le i<n} u_i^k\epsilon_i$ for $k = 3, 6$,

$$\Big|\langle q_1;q_2\rangle_{\mathbb{Q}_n^0} - \tfrac32 b^{-2}\sum_{p,q} w_1(p)w_2(q)\Sigma(p,q)^2\Big| \le \tfrac{35^2}{4}\, b^{-2}\big[16C_{\mathrm{tp}}S_\Sigma(C_{\mathrm{tp}}^{\mathrm{row}})^2\mathsf{E}_3 + 32(C_{\mathrm{tp}}^{\mathrm{row}})^4\mathsf{E}_3^2 + 16(C_{\mathrm{tp}}^{\mathrm{row}})^4\mathsf{E}_6\big]\eta.$$

The numbers E_k are bounded in three ranges of levels. As $\log x_i \ge \log\frac{3b}{4D_i} \ge 0$ (Prop. 6.5(ii), and $\frac34 b \ge 1 \ge D_i$), $\log x_i \ge 1$ and $2p_0 \ge 2$, one has $\epsilon_i \le C_\chi e^{-\mathsf{c}_\chi(\log\frac{3b}{4D_i})^2}$. At the levels with $D_i < d$, $\log\frac{3b}{4D_i} = \Lambda_b + t_i$ with $\Lambda_b = \log\frac{3b}{4d} \ge \log\frac{3b}{4d^{1/2}}$ and $t_i = \log u_i > 0$, the t_i being spaced by $\log L$, and $(\Lambda_b + t)^2 \ge \Lambda_b^2 + t^2$, so that $u_i^k\epsilon_i \le C_\chi e^{-\mathsf{c}_\chi\Lambda_b^2}e^{6t_i - \mathsf{c}_\chi t_i^2}$ for $k \le 6$; so these levels give at most $C_\chi S_\chi e^{-\mathsf{c}_\chi\Lambda_b^2}$, with $S_\chi = \sup_{t_0\ge 0}\sum_{s\ge 0} e^{6(t_0+s\log L) - \mathsf{c}_\chi(t_0+s\log L)^2}$. At the levels with $d \le D_i \le d^{1/2}$, $\sum u_i^3 \le 2$ and $\epsilon_i \le C_\chi e^{-\mathsf{c}_\chi(\log\frac{3b}{4d^{1/2}})^2}$. At the levels with $D_i > d^{1/2}$, $\sum u_i^3 \le 2d^{3/2}$ and $\epsilon_i \le C_\chi$. So

$$\mathsf{E}_3,\ \mathsf{E}_6 \le C_\chi(S_\chi+2)\big[d^{3/2} + e^{-\mathsf{c}_\chi(\log\frac{3b}{4d^{1/2}})^2}\big] \le 2C_\chi(S_\chi+2),$$

and the bracket above gives the member $C_{\mathrm{Bx}}b^{-2}[d^{3/2} + e^{-\mathsf{c}_\chi(\log(3b/(4d^{1/2})))^2}]\eta$, with C_{Bx} formed with C_{tp}, S_Σ, $C_{\mathrm{tp}}^{\mathrm{row}}$, C_χ and S_χ, and so determined by the constants. The missing first slice is small: by Prop. YM3_I:120.6, $(d/a_K)^3|\mathcal{C}_{(1)}(p,q)| \le C_{\mathrm{tp}}u_0^3 e^{-\delta_{\mathrm{tp}}u_0/\sqrt{3}}$, $u_0 = d/a_K$, at $|x(p)-x(q)| \ge d/\sqrt{3}$; with the terms of the level 1 of (2) it gives the member $C_{\mathrm{Bt}}l^{4p_0}e^{-\mathsf{c}_\flat d/a_K}b^{-2}\eta$.

(5)

The free value. Lem. 6.34 (the template sums) gives $|\sum_{p,q} w_1 w_2 (C^{\mathrm{fl}}_{K,K-n})^2 - \frac{3}{8\pi^2}\mathcal{K}_6| \le C_{\mathrm{tm}}[\frac{a_K}{\vartheta d} + (\frac{d}{\ell})^3 + (\frac{d}{D_n})^3]\eta$, and $\frac{3}{2}\cdot\frac{3}{8\pi^2} = c_2$. Take $n = K$, $D_K = 1$: $(d/\ell)^3 + d^3 \le 2d^3$, the member $\frac{3}{2}C_{\mathrm{tm}}\vartheta^{-1}b^{-2}a_K/d$ is $C_{\mathrm{lat}}b^{-2}a_K/d$, and $b^{-2} = \frac{b}{d}\,b^{-3}d$; the supremum over the labels at g and $d/a_K \ge L(\log\frac{1}{d})^2$ give $\bar\varepsilon_4$. Finally $\log\frac{3b'}{4d^{1/2}} \ge \frac{1}{2}\log\frac{1}{d}$, as $\frac{3}{4}b' \ge 1$; so $\frac{b'}{d}e^{-\mathsf{c}_\chi(\cdot)^2} \le b_+ e^{\log\frac{1}{d} - \frac{1}{4}\mathsf{c}_\chi(\log\frac{1}{d})^2}$, $\frac{b'}{d}C_{\mathrm{Bx}}d^{3/2} \le C_{\mathrm{Bx}}b_+ d^{1/2}$, and the other members are smaller: $\bar\varepsilon_4(g,d) \le C_4(g)d^{1/2}$. □

Remark 6.9. With Prop. 6.6(ii) at $m = 3$, Prop. 6.8 gives, on every side $\ell \in \mathfrak{L}(g)$ and at every position,

$$|\mathsf{J}_2 - c_2 b^{-2}\mathcal{K}_6|(f,h) \le \big[(C_{\mathrm{E},3}l^{6p_0} + C_{\mathrm{E}}b^{-1}d\,l^{8p_0} + \bar\varepsilon_4(g,d))\,b^{-3}d + C_{\mathrm{lat}}b^{-2}a_K/d\big]\eta,$$

and on T_{ℓ_g}, under the assumptions of Prop. 6.7 besides ($g \le \gamma_{\mathrm{C}}$, $d \le d_{\mathrm{A}}(g)$), by that Proposition, the same for $S^T_{2;K,\ell_g,\beta}(f,h)$ with $C_{\mathrm{A}}(g)\vartheta^{-1}bd$ added to the coefficient of $b^{-3}d$. The member $\bar\varepsilon_4(g,d)\,b^{-3}d\,\eta$ holds the two finite-size terms of the free value, $\frac{3}{2}C_{\mathrm{tm}}b^{-2}[(d/\ell)^3 + (d/D_K)^3]\eta$, of the torus and of the constraint at the top level $D_K = 1$, which the proof of Prop. 6.8 bounds by $3C_{\mathrm{tm}}b^{-2}d^3\eta = 3C_{\mathrm{tm}}bd^2\cdot b^{-3}d\,\eta$: they are at most $3C_{\mathrm{tm}}\,b^{-3}d\,\eta$ for $d^2 \le 1/b$, and of the order bd^2 relative to the first correction beyond.

The other members of $\bar\varepsilon_4(g,d)\,b^{-3}d\,\eta$ are at most $C_4(g)\,d^{1/2}\,b^{-3}d\,\eta$ (Prop. 6.8), and $C_{\mathrm{E}}b^{-1}d\,l^{8p_0} \le C_{\mathrm{E}}c_{\mathfrak{p}}\,l^{6p_0}$ with $c_{\mathfrak{p}} = \sup_{0<t\le 1} t\max\{1,\log\frac{2}{t}\}^{2p_0}$. So, beside the lattice term $C_{\mathrm{lat}}b^{-2}a_K/d$ and the two finite-size terms, the deviation from the free law is at most $C\,l^{6p_0}\,b^{-3}d\,\eta$, with a C formed with g and the constants (and with ϑ on T_{ℓ_g}): of the order of the first correction, up to the factor l^{6p_0}; for $d^2 \le 1/b$ the two finite-size terms are of that order too. The coefficient of the first correction is fixed by the order six.

Lemma 6.10 (Gaussian measures conditioned on a convex set)**.** Let ν be a centred Gaussian measure on $\mathbb{R}^I$, I finite, with variances $\le \sigma_i^2$, and $Q \supset \prod_i(-c_i,c_i)$ open, bounded, convex and symmetric; put $t_i = c_i/\sigma_i$ and $\epsilon = \sum_i e^{-t_i^2/2}$. Let $F_1,\dots,F_r$ ($r = 2,3$) be complex polynomials of orders $\le k_j$ with $|F_j| \le S_j$ on Q, and $k = \sum_j k_j$. If $t_i^2 \ge \max\{4,2k\}$ for all i and $2^k\epsilon \le 1$, the joint cumulants of $F_1,\dots,F_r$ in $\nu(\cdot\,|\,Q)$ and in ν differ by at most $q_r(1+2^k)\epsilon\prod_j S_j$, with $q_2 = 5$, $q_3 = 37$. ■

Proof sketch. For $z \notin Q$ let $s = \max_i |z_i|/c_i \ge 1$. The segment from $-z/s$ to z/s lies in $\bar Q$, so $\tau \mapsto F_j(\tau z/s)$ is a polynomial of order $\le k_j$ bounded by S_j on $[-1,1]$, and the extremal property of Chebyshev's polynomials gives $|F_j(z)| \le S_j(2s)^{k_j}$. On Q^{c}, $s^k \le \sum_i(|z_i|/c_i)^k\mathbf{1}\{|z_i| \ge c_i\}$, and for a centred Gaussian variable of variance $\le \sigma_i^2$, $\mathbb{E}[(|z_i|/c_i)^k; |z_i| \ge c_i] \le e^{-t_i^2/2}$ once $t_i^2 \ge \max\{4,2k\}$; so

$$\nu(Q^{\mathrm{c}}) \le \epsilon, \qquad \int_{Q^{\mathrm{c}}} \prod_j |F_j|\,\mathrm{d}\nu \le 2^k\epsilon\prod_j S_j.$$

Let F be a product of some of the F_j, S the product of their S_j, and m, m^{r} its moments in ν and in $\nu(\cdot\,|\,Q)$. Then $m^{\mathrm{r}} - m = m^{\mathrm{r}}\nu(Q^{\mathrm{c}}) - \int_{Q^{\mathrm{c}}} F\,\mathrm{d}\nu$ and $|m^{\mathrm{r}}| \le S$, so $|m^{\mathrm{r}} - m| \le (1+2^k)\epsilon S$; and $|m| \le S + 2^k\epsilon S \le 2S$. A joint cumulant of order r is a sum, over the partitions into p blocks, of $(-1)^{p-1}(p-1)!$ times products of p moments; telescoping each product, with the factors bounded by S and by $2S$, bounds its change by $p\,2^{p-1}(1+2^k)\epsilon\prod_j S_j$. So $q_r = \sum_p \mathrm{S}(r,p)\,p!\,2^{p-1}$, S the Stirling numbers of the second kind: $q_2 = 1+4$, $q_3 = 1+12+24$. □

Lemma 6.11 (The coefficients of a logarithm)**.** Let G be holomorphic on $|\lambda| < 1$ with $G(0) = 0$ and $\mathrm{Re}\,G \le \mathsf{E}$. Then every Taylor coefficient $G_{[m]}$, $m \ge 1$, has $|G_{[m]}| \le 2\mathsf{E}$. Consequently, if F is holomorphic on $|\lambda| < 1$, $F(0) = 1$, $F \ne 0$, $|F| \le e^{\mathsf{E}}$, and $F = 1 + \lambda\mathsf{R}_1 + \lambda^2\mathsf{R}_2 + O(\lambda^3)$, then $W_1 = -\mathsf{R}_1$ and $W_2 = \frac{1}{2}\mathsf{R}_1^2 - \mathsf{R}_2$ have $|W_1|, |W_2| \le 2\mathsf{E}$. The condition $F \ne 0$ cannot be dropped. ■

Proof sketch. For $0 < \rho < 1$ and $m \ge 1$, $G_{[m]}\rho^m = -\pi^{-1}\int_0^{2\pi}(\mathsf{E} - \mathrm{Re}\,G(\rho e^{i\phi}))e^{-im\phi}\,\mathrm{d}\phi$; the factor $\mathsf{E} - \mathrm{Re}\,G$ is ≥ 0 and has the mean E over the circle, as $\mathrm{Re}\,G(0) = 0$; let $\rho \to 1$. For

F, the function $G = \log F$ is defined on the disc, as F has no zero there; $\operatorname{Re} G = \log|F| \le \mathsf{E}$ and $G = -\lambda W_1 - \lambda^2 W_2 + O(\lambda^3)$. For $F = 1 + \mathsf{c}\lambda$ with $\mathsf{c} \ge 3$, which has a zero in the disc, $|\mathsf{R}_1| = \mathsf{c} > 2\log(1+\mathsf{c}) = 2\mathsf{E}$. □

Proposition 6.12 (The order six at one level)**.** Under the assumptions of Prop. 6.5, let $2 \le n \le K$ and $\mathsf{Bx} = \prod_{0\le i<n}\{B_i : |B_i(b)| < \mathfrak{b}(x_i)\ \forall b\}$, a product of Euclidean balls in $\mathfrak{g}$, one for each non-pivot bond of the heights $0 \le i < n$.

(i) There are functions O_1^λ, O_2^λ, D^λ on Bx, continuous and bounded on $\{|\lambda| \le 1\} \times \mathsf{Bx}$ and holomorphic in $|\lambda| < 1$, with $\mathsf{O}_a^\lambda(B) = \mathsf{o}_a(\lambda B)$ for a function o_a holomorphic near 0, $\mathsf{O}_a^{-\lambda}(-B) = \mathsf{O}_a^\lambda(B)$, $\mathsf{D}^{-\lambda}(-B) = \mathsf{D}^\lambda(B)$, $\int_{\mathsf{Bx}} \mathsf{D}^\lambda dB = 1$, $\mathsf{D}^0 dB = \bigotimes_{0\le i<n} \nu_i^{\mathrm{r}}$, and

$$\sum_{j\le n} \mathsf{n}_j^\lambda = \langle \mathsf{O}_1^\lambda; \mathsf{O}_2^\lambda \rangle_{\mathsf{D}^\lambda dB} \qquad (|\lambda| < 1).$$

Here D^λ is the product over the heights $0 \le i < n$ of the normalised densities of the steps along the chain of the backgrounds, which reads (λ, B) through λB; for $0 \le i < n$, the height 0 included, the density of the step of the height i at $\lambda = 0$, at the trivial configuration, is proportional to $\exp[-\frac12(B_i, (\mathsf{M}_i \otimes 1)B_i)]$ on the box of the height, with the real symmetric $\mathsf{M}_i \ge \frac34\gamma^{\mathrm{tr}}$ of the measures ν_i° (Lem. 6.13(iii)); and for real $0 < \lambda \le 1$, $\log(\mathsf{D}^\lambda/\mathsf{D}^0)$ is the sum of a term that does not depend on B, of terms $\lambda^{-2}F(\lambda, \lambda B)$ and of terms $F(\lambda, \lambda B)$, with functions $F(\lambda, \zeta)$ holomorphic near $(0,0)$, those of the first kind with $F = O(|\zeta|^2)$.

Uniformly on Bx, $\mathsf{O}_a^\lambda = \lambda^2(X_{a,0} + \lambda X_{a,1} + \lambda^2 X_{a,2}) + O(\lambda^5)$ and $\mathsf{D}^\lambda = \mathsf{D}^0 \exp[-\lambda W_1 - \lambda^2 W_2 + O(\lambda^3)]$, where $X_{a,m}$ is homogeneous of order $m+2$, $X_{a,0} = q_a + q_a^{(0)}$ with the q_a of the proof of Prop. 6.8, formed with the signs sg_i of Lem. YM3_I:124.3 there, and a quadratic form $q_a^{(0)}$ that vanishes where $B_0 = 0$, W_1 is odd of order ≤ 3, and W_2 is, up to the constant that the term without B gives, even of order ≤ 4. These are polynomials on the whole space, with Taylor coefficients at the trivial configuration formed with L and the running couplings, so that the cumulants of eq. 6.vii are formed in $\bigotimes_{0\le i<n} \nu_i^\circ$ as well.

Hence $\mathsf{h}_4^{(n)} = \langle X_{1,0}; X_{2,0}\rangle$ and $\mathsf{h}_6^{(n)} = \gamma_2^{\mathrm{r}}$, the cumulants being taken in $\bigotimes_{0\le i<n} \nu_i^{\mathrm{r}}$.

(ii) If $\gamma^{\mathrm{tr}}\mathfrak{b}(x_{n-1})^2 \ge 80$ and $10240 N^3 e^{-\gamma^{\mathrm{tr}}\mathfrak{b}(x_{n-1})^2/8} \le 1$, then

$$\beta^3 a_K^2 |\gamma_2^{\mathrm{r}} - \gamma_2^\circ| \le \mathcal{U}(x_{n-1}, \mathsf{N}) \|w_1\|_\infty \|w_2\|_\infty, \qquad \mathcal{U}(x, \mathsf{N}) = C_{\mathrm{X}} \mathsf{N}^{36} \exp[-\tfrac18 \gamma^{\mathrm{tr}} A_1^2 (\log x)^{2p_0}],$$

with C_{X} determined by the constants.

■

Proof sketch. (i)

The chain of the backgrounds. For $|\lambda| \le 1$ and $B \in \mathsf{Bx}$ put $v_n^\lambda := 0$ and, downwards for $i = n-1, \dots, 1$, let v_i^λ be the lowered point (equation YM3_I:(2286)) that the step $i+1$ forms of the point v_{i+1}^λ of the level $i+1$ and of the fluctuation $\lambda x_i^{-1/2} B_i$. Its first member A_i^λ is a 1-form on the bare bonds, and $\tilde U_i^\lambda := \exp(iL^{-i}A_i^\lambda)$, a configuration on the bare bonds, is the *background of the level i*. The maps of the step are holomorphic on their domain, at real and at complex points (the holomorphy following equation YM3_I:(2286)); by their bounds (equation YM3_I:(2287) and its domain verification), as $\sup|\lambda x_i^{-1/2} B_i| \le |\lambda|\delta(x_i)$ on the box, $\|v_i^\lambda\| \le |\lambda| r_i$ at every height, with the radii $r_n = 0$, $r_i = L^{-1} r_{i+1} + \frac{11}{10} c_{\mathfrak{h}} \delta(x_i) \le \frac85 c_{\mathfrak{h}} \delta(x_i)$ of equation YM3_I:(2291), which keep the chain in the domain of the maps, as $\delta(x_{K-1})$ lies below the threshold of the chain classes (Prop. 6.5(iv)).

So the chain is formed for $|\lambda| \le 1$, it reads (λ, B) through λB, and for real $\lambda \in [0,1]$ it stays in the chain classes (Prop. YM3_I:124.2, equation YM3_I:(2288)). At complex λ the background has values in $\mathrm{SL}(2,\mathbb{C})$, and a_p is read there as $1 - \frac12 \operatorname{tr}$ of the product of the variables around p; on $\mathrm{SU}(2)$ this is a_p. The observable is $\mathsf{O}_a^\lambda(B) := a_K^{-1} \sum_p w_a(p)\, a_p(\tilde U_1^\lambda) + \mathsf{w}_a^\lambda(B_0)$: its first member is $\mathcal{O}(w_a)$ at the background of the level 1, which carries the fields of the heights $1 \le i < n$, and its

second, the source term of the height 0, is the sum over the domains of the level 0 of the action term of the source w_a at the point v_1^λ and the field $\lambda x_0^{-1/2} B_0$; the field of the height 0 enters O_a^λ through this term alone.

The density is $\mathsf{D}^\lambda(B) := \prod_{0\le i<n} \rho_i^\lambda / Z_i^\lambda$, with ρ_i^λ the density of the step of the height i at the field scale λ at the point v_{i+1}^λ, the exponential of the quadratic forms of the step in B_i and of its potentials, and Z_i^λ its integral over the box of the height.

The identity. At real $0 < \lambda \le 1$ each step at a height $1 \le i < n$ obeys the identity of one step at the field scale λ of step (1) of the proof of Prop. 6.8, whose case $\lambda = 1$ uses the quotient argument of Prop. 2.4, and at the height 0 the source enters linearly, through the action term of the sources. Iterating from the height $n-1$ down,

$$\exp D_n^{\lambda,\mu} = \int_{\mathsf{Bx}} \exp[\mu_1 \mathsf{O}_1^\lambda + \mu_2 \mathsf{O}_2^\lambda] \mathsf{D}^\lambda \, \mathrm{d}B,$$

$D_n^{\lambda,\mu}$ the exponent of the progression at the trivial configuration with the sources $\mu_1 w_1 + \mu_2 w_2$. Both sides are holomorphic in μ near 0, and holomorphic in $|\lambda| < 1$, as the maps of the chain are holomorphic at complex points and the normalisations Z_i^λ have no zero (Lem. 6.13(ii)); they agree on $(0,1]$. The mixed derivative in μ at 0 of the logarithms is the covariance.

At $\lambda = 0$. The chain is trivial and the potentials vanish. At the heights $1 \le i < n$ the density of the step is $\exp[-\frac12 (B_i, ((\tau_i \mathsf{A}_i^{\mathrm{cv}} + (1-\tau_i)\hat{\mathsf{G}}) \otimes 1) B_i)] = \exp[-\frac12 (B_i, (\mathsf{M}_i \otimes 1) B_i)]$, and at the height 0, where $\tau_0 = 1$, it is $\exp[-\frac12 (B_0, (\mathsf{A}_0^{\mathrm{cv}} \otimes 1) B_0)]$, $\mathsf{M}_0 = \mathsf{A}_0^{\mathrm{cv}}$; the sources of these two identities are given with Lem. 6.13(iii). So $\mathsf{D}^0 dB = \bigotimes_{0\le i<n} \nu_i^{\mathrm{r}}$.

The orders. The orders 0 and 1 of O_a^λ vanish, the order 1 by the constant gauge rotations, as in Prop. 6.6. As $\mathsf{O}_a^\lambda = \mathsf{o}_a(\lambda B)$ with o_a holomorphic near 0, the coefficient of λ^{m+2} is a homogeneous polynomial of order $m+2$. For real $0 < \lambda \le 1$ the integral identity of the step at a complex field scale (Prop. 6.27), read at the points of the chain, writes each Z_i^λ as $\det^{1/2}$ of the covariance of the step times the exponential of a sum of local terms of the step; in the product of the quotients the families of the progression drop out, by the sum rule of the chain classes (Prop. YM3_I:124.2, equation YM3_I:(2289)), which telescopes over the heights, and $\mathsf{D}^\lambda(B)$ is a number $c_n(\lambda)$, which does not depend on B, times the exponential of a sum of terms $\lambda^{-2} F(\lambda, \lambda B)$ with $F = O(|\zeta|^2)$, namely the quadratic forms of the steps, λ^{-2} times forms in λB_i whose coefficients read λB through the chain, and the potentials of the marks of the gauge fixing and of the action, $x_i \lambda^{-2}$ times functions of $\lambda x_i^{-1/2} B_i$, and of terms $F(\lambda, \lambda B)$, namely the other potentials, the logarithms of the determinants of the covariances along the chain, the localised Gaussian families of the steps and the counterterms at the backgrounds; in all of them $F(\lambda, \zeta)$ is holomorphic near $(0,0)$.

This is the form of $\log(\mathsf{D}^\lambda / \mathsf{D}^0)$ stated in (i), the term without B being $\log(c_n(\lambda)/c_n(0))$. With $F = \sum_{k,\alpha} f_{k\alpha} \lambda^k \zeta^\alpha$ the coefficient of λ^m of $\lambda^{-2} F(\lambda, \lambda B)$ is $\sum_{k+|\alpha|=m+2} f_{k\alpha} B^\alpha$, a polynomial of order $\le m+2$, and that of $F(\lambda, \lambda B)$ one of order $\le m$; this gives the orders of W_1 and W_2, and the identity between holomorphic functions of λ passes from $(0,1]$ to a disc about 0. Odd functions have the mean 0 in $\mathsf{D}^0 dB$, the mass 1 removes the mean of the order 2 of $\mathsf{D}^\lambda / \mathsf{D}^0$, and the coefficient of λ^2 in $\lambda^{-4} \langle \mathsf{O}_1^\lambda; \mathsf{O}_2^\lambda \rangle$ is eq. 6.vii; at λ^0 it is $\langle X_{1,0}; X_{2,0} \rangle$.

By Prop. 6.6(i), $\lambda^{-4} \sum_{j\le n} \mathsf{n}_j^\lambda = \mathsf{h}_4^{(n)} + \lambda^2 \mathsf{h}_6^{(n)} + O(\lambda^4)$.

(ii) By Lem. 6.13, on Bx the $X_{a,m}$ are at most $C_{\mathrm{S}} \mathsf{N}^8 \|w_a\|_\infty$ and W_1, W_2 at most $2 C_{\mathrm{S}} \mathsf{N}^7$; a cumulant of order ≥ 2 does not see the constant in W_2. By (iii) of that Lemma the variances of ν_i° are at most $\frac{4}{3\gamma^{\mathrm{tr}}}$; Bx holds the cube of the half-sides $\mathfrak{b}(x_{n-1})/\sqrt{3}$ in the components, as $x_i \ge x_{n-1}$ for $i < n$; so $t_i^2 \ge \frac14 \gamma^{\mathrm{tr}} \mathfrak{b}(x_{n-1})^2 \ge 20$ and $\epsilon \le 10 N^3 e^{-\gamma^{\mathrm{tr}} \mathfrak{b}(x_{n-1})^2/8}$. Lem. 6.10 at the seven cumulants, whose entries have the total order $k \le 10$, and $\beta^3 a_K^2 \le \mathsf{N}^3$ give (ii). □

Lemma 6.13 (The suprema on the boxes)**.** In the setting of Prop. 6.12(i) there is $C_{\mathrm{S}} \ge 1$, determined by the constants, such that for $|\lambda| < 1$ and $B \in \mathsf{Bx}$:

(i) $|\mathsf{O}_a^\lambda(B)| \le C_{\mathrm{S}}\mathsf{N}^8\|w_a\|_\infty$ for $a = 1, 2$;
(ii) $\mathsf{D}^\lambda(B) \ne 0$ and $|\mathsf{D}^\lambda(B)| \le e^{C_{\mathrm{S}}\mathsf{N}^7}|\mathsf{D}^0(B)|$;
(iii) for $0 \le i < n$, the height 0 included, the density of the step of the height i at $\lambda = 0$, at the trivial configuration, is proportional to $\exp[-\frac{1}{2}(B_i, (\mathsf{M}_i \otimes 1)B_i)]$ on the box of the height: its Gaussian part is ν_i°, the centred Gaussian measure with the real symmetric precision $\mathsf{M}_i \otimes 1 \ge \frac{3}{4}\gamma^{\mathrm{tr}}$, and its normalised restriction to the box is ν_i^{r}, a probability measure; at the height 0, $\tau_0 = 1$ and $\mathsf{M}_0 = \mathsf{A}_0^{\mathrm{cv}}$.

Consequently, on Bx, $|X_{a,m}| \le C_{\mathrm{S}}\mathsf{N}^8\|w_a\|_\infty$ for $m = 0, 1, 2$, and $|W_1|, |W_2| \le 2C_{\mathrm{S}}\mathsf{N}^7$. ■

Proof sketch. (i) The first member of O_a^λ is $a_K^{-1}\sum_p w_a(p)a_p(\tilde{U}_1^\lambda)$, at the background $\tilde{U}_1^\lambda$ of the level 1 of the chain of the proof of Prop. 6.12(i), a configuration on the bare bonds whose values stay near $\mathbf{1}$ in $\mathrm{SL}(2,\mathbb{C})$ by the radii of the chain, with at most $3N^3$ plaquettes and $a_K^{-1} \le N$; the second, the source term of the height 0, is a sum over at most $8N^3$ domains of members of the family of the source, each bounded on the box by the size bounds of the step at a complex field scale (Prop. 6.27) and by $\mathfrak{b}(\beta)^2 \le C\beta$.

(ii) $\mathsf{D}^\lambda = \prod_{0\le i<n}\rho_i^\lambda/Z_i^\lambda$ along the chain at the complex field scale λ, a product over the at most $K \le N$ heights. Each ρ_i^λ is the exponential of a finite sum, so $\rho_i^\lambda \ne 0$; its exponent is a sum over at most $8N^3$ places of quadratic forms in the field, whose coefficients are holomorphic along the chain, and of potentials, and on the box each of them is bounded by the size bounds of the step at a complex field scale (Prop. 6.27) and by $\mathfrak{b}(x_i)^2 \le C\beta$, the radii of the chain keeping the points of all the heights in the discs where these bounds hold.

By the integral identity of the step at a complex field scale, Z_i^λ is $\det^{1/2}$ of the covariance of the step times the exponential of a sum of local terms of the step, each bounded by the size bounds of the step; so $Z_i^\lambda \ne 0$, and $|Z_i^\lambda|$ is bounded below by a determinant bound for its Gaussian part. Hence $\mathsf{D}^\lambda \ne 0$, and, with at most CN^3 terms a height, each at most a number determined by the constants times $\mathfrak{b}(\beta)^2 \le C\beta \le C\mathsf{N}$, this gives the bound with N^7. (iii) For a height $1 \le i < n$, at $\lambda = 0$ the chain is trivial and the potentials vanish, and the density of the step at the trivial triple is $\exp[-\frac{\tau_i}{2}(B, \mathbf{C}_*^{-1}B) - \frac{1-\tau_i}{2}\sum_\square(B, \hat{\mathsf{G}}_\square B)]$.

Here the whole covariance of the step at the trivial triple is $\mathbf{C}_* = (\mathsf{A}_i^{\mathrm{cv}})^{-1} \otimes 1$ (equation YM3_I:(2297), from Prop. YM3_I:120.3 at the place of the height i), and the sum over the cubes of the local gauge-fixing forms is $\hat{\mathsf{G}}\otimes 1$ (equation YM3_I:(828) at the trivial block field, with Prop. YM3_I:120.3); so the density is $\exp[-\frac{1}{2}(B, ((\tau_i\mathsf{A}_i^{\mathrm{cv}} + (1-\tau_i)\hat{\mathsf{G}}) \otimes 1)B)] = \exp[-\frac{1}{2}(B, (\mathsf{M}_i \otimes 1)B)]$, as $\mathsf{A}_i^{\mathrm{cv}} = \mathsf{B}_i^{\mathrm{cv}} + \hat{\mathsf{G}}$. At the height 0, $\tau_0 = 1$, because $\xi_0 \equiv x_0$, and the gauge-fixing term drops out.

The provision of the step 1 is the canonical provision of the step at $(x_0; \mathfrak{a}_0^{\mathrm{abs}})$, the quadruple $(0, x_0, \mathfrak{a}_0^{\mathrm{abs}}, \mathfrak{w})$ being a place by the setting (J) of [7, Def. B.16] at $k = K - 1$ (Prop. 6.5(iv)); by Prop. YM3_I:120.3 and Prop. YM3_I:120.6 at $k = 0$, its whole covariance at the trivial triple is the scalar operator with the real matrix $(\mathsf{A}_0^{\mathrm{cv}})^{-1}$. So the density of the height 0 is $\exp[-\frac{1}{2}(B_0, (\mathsf{A}_0^{\mathrm{cv}} \otimes 1)B_0)]$, and $\mathsf{M}_0 = \mathsf{A}_0^{\mathrm{cv}}$. At every height $\mathsf{B}_i^{\mathrm{cv}}$ and $\hat{\mathsf{G}}$ are real symmetric matrices ≥ 0 (equation YM3_I:(2231)), $\mathsf{A}_i^{\mathrm{cv}} \ge \gamma^{\mathrm{tr}}$ (Prop. YM3_I:120.3) and $\frac{3}{4} \le \tau_i \le \frac{4}{3}$ (Prop. 6.5(ii)); so M_i is real and symmetric, $\mathsf{M}_i \ge \min\{\tau_i, 1\}\mathsf{A}_i^{\mathrm{cv}} \ge \frac{3}{4}\gamma^{\mathrm{tr}}$, the Gaussian part of the density is ν_i°, and ν_i^{r} is a probability measure.

The consequences: Cauchy's estimate on $|\lambda| < 1$ for $\lambda \mapsto \mathsf{o}_a(\lambda B)$, whose coefficient of λ^{m+2} is $X_{a,m}(B)$, and Lem. 6.11 for $\lambda \mapsto \mathsf{D}^\lambda(B)/\mathsf{D}^0(B)$. □

Proposition 6.14 (The composed chart; the lattice number of a flat top)**.** Let $N = \ell L^K$, $\ell = 2L^m$, and $1 \le t \le K$ with $2L^{m+K-t+1} \ge C_L M$; the levels are those of the lattice (ℓ, K), $\mathcal{V}_j$ is the space of the configurations of the level j, and Y the space of the $y = (\mathcal{A}_j)_{j<t}$, $\mathcal{A}_j$ a real $\mathfrak{g}$-valued 1-form on the non-pivot bonds of the level j (Section YM3_I:4 and the pivot definition preceding Lem. YM3_I:40.1).

(i) There are an open neighbourhood O_t of $\mathbf{1}$ in $\mathcal{V}_t$, $\rho_2 \in (0,1]$ and $C_{\mathrm{H}} > 0$ such that, with $\mathsf{Y}_{\rho_2} = \{y \in \mathsf{Y} : |\mathcal{A}_j| < \rho_2,\ j < t\}$: the chain $W_t = W$, $W_j = V_{W_{j+1}}[\mathcal{A}_j]$ of the charts of [7, §B.2] is defined on $\mathsf{O}_t \times \mathsf{Y}_{\rho_2}$, with $\mathcal{M}(W_j) = W_{j+1}$; $\Psi(W,y) = W_0$ is a real analytic diffeomorphism of $\mathsf{O}_t \times \mathsf{Y}_{\rho_2}$ onto an open set $\mathsf{U} \ni \mathbf{1}$; and there is a real continuous θ, real analytic near $(\mathbf{1},0)$, with $\int H \,\mathrm{d}U = C_{\mathrm{H}} \int_{\mathsf{O}_t} \mathrm{d}W \int_{\mathsf{Y}_{\rho_2}} \mathrm{d}y\, e^{-\theta} H \circ \Psi$ for every bounded Borel H that vanishes off U.

(ii) For $\kappa \in (0,\infty)^t$ let $F_\kappa = A + \sum_{j<t} \kappa_j G_j(\mathcal{M}^j \cdot)$, G_j the gauge-fixing function YM3_I:(54) at the level j, and $\phi_\kappa = F_\kappa \circ \Psi(\mathbf{1},\cdot)$. Then $\phi_\kappa \geq 0$, $\phi_\kappa(0) = 0$, $\nabla\phi_\kappa(0) = 0$, and the Hessian of ϕ_κ at 0 is block diagonal for the decomposition of Y by the levels and positive definite.

(iii) The numbers $\mathfrak{g}_{m'}[N,t](p,q) = \gamma^{\mathrm{L}}_{m'}[\phi_\kappa, \theta(\mathbf{1},\cdot), a_p \circ \Psi(\mathbf{1},\cdot), a_q \circ \Psi(\mathbf{1},\cdot)]$, $m' = 0, 2$, do not depend on κ, and they are the same in every adapted chart. An *adapted chart* of $(\mathcal{V}_0, dU, \mathrm{Lg}\,\mathcal{M}^t)$, Lg being the Lie algebra logarithm of the intrinsic logarithm on $G \setminus \{-1\}$, is $(\mathsf{E}', \Psi', \theta', C', \varrho')$ with: E' an open neighbourhood of $(0,0)$ in $\mathsf{W}_t \times \mathbb{R}^{\dim \mathsf{Y}}$, W_t the space of the $\mathfrak{g}$-valued 1-forms w on the bonds of the level t; Ψ' a C^∞ diffeomorphism of E' onto an open subset of the domain of $\mathrm{Lg}\,\mathcal{M}^t$, with $\Psi'(0,0) = \mathbf{1}$ and $\mathrm{Lg}\,\mathcal{M}^t\Psi'(w,y) = w$; θ' a real continuous function on E', with $\theta'(0,\cdot)$ of class C^∞; $C' > 0$; $\varrho' > 0$ a continuous function of w alone; and $\int H \,\mathrm{d}U = C' \int_{\mathsf{E}'} \varrho'(w)\, e^{-\theta'(w,y)} H(\Psi'(w,y))\,\mathrm{d}w\,\mathrm{d}y$ for every bounded Borel H that vanishes off $\Psi'(\mathsf{E}')$.

In it the numbers are $\gamma^{\mathrm{L}}_{m'}[F_\kappa \circ \Psi'(0,\cdot), \theta'(0,\cdot), a_p \circ \Psi'(0,\cdot), a_q \circ \Psi'(0,\cdot)]$. They depend on N, t, L, p, q alone.

(iv) One may take $\mathsf{O}_t = \{e^{iw} : |w| < \rho_2\}$ for every $\rho_2 \in (0, \rho_\star]$, with a $\rho_\star \in (0,1]$. Then W_j, $V_j^{\mathrm{bf}}(W_{j+1})$, $G_j(W_j)$, θ and $a_p \circ \Psi$ are real analytic in (w,y) on $\mathsf{O}_t \times \mathsf{Y}_{\rho_2}$, $W_j(\mathbf{1},0) = \mathbf{1}$, $0 < e^{-\theta} \leq \prod_{j<t} (27L^9)^{\#\Gamma_j}$, and $C_{\mathrm{H}} = \prod_{j<t} (2\pi^2)^{-\#\mathfrak{X}_j^+} L^{-3\#\Gamma_j}$, where $\mathfrak{X}_j^+$ are the positively oriented non-pivot bonds of the level j and Γ_j the positively oriented bonds of the level $j+1$. One may also take $\mathsf{O}_t = \{W \in \mathcal{V}_t : \max_p |\partial W(p) - \mathbf{1}| < \varepsilon_*\}$, p running over the plaquettes of the level t, and $\rho_2 = \rho_*$, with numbers $\varepsilon_*, \rho_* \in (0,1]$ determined by L and the constants and independent of t, K and m; (i)–(iii) hold for this choice as well.

■

Proof sketch. (i), (iv) One real number $\mathfrak{t}_* > 1$ serves at all levels: with it the constants, the lattice of each step and $\mathfrak{t}_*$ are data under (Im) of [7, Def. B.2]. For such data [7, Lem. B.3] gives, on an open set of block fields W and for $|\mathcal{A}|$ below the bound that belongs to $\mathfrak{t}_*$: the averaged configuration $V_j^{\mathrm{bf}}(W)$ is defined and $\mathcal{M}(V_j^{\mathrm{bf}}(W)) = W$; $\mathcal{M}(V_W[\mathcal{A}]) = W$; and $\mathrm{Lg}(V_W[\mathcal{A}](b) V_j^{\mathrm{bf}}(W)(b)^{-1}) = \mathcal{A}(b)$ on the non-pivot bonds. [7, Lem. B.3] gives the one-step integral formula, with the constant $(2\pi^2)^{-\#\mathfrak{X}_j^+} L^{-3\#\Gamma_j}$, a Jacobian factor in $(0, (27/8)^{\#\Gamma_j}]$ and a normalisation in $(0, (8L^9)^{\#\Gamma_j}]$.

Neighbourhoods of $\mathbf{1}$ chosen downwards, by continuity in $(W,\mathcal{A})$, keep the chain where these statements apply; by induction over the levels the integral formula holds for every Borel subset of the natural domain of the chain, with C_{H} the product of the constants. The map

$$U \mapsto \Big(\mathcal{M}^t U,\ \big(\mathrm{Lg}(\mathcal{M}^j U(b)\, V_j^{\mathrm{bf}}(\mathcal{M}^{j+1}U)(b)^{-1})\big)_{j<t} \Big)$$

is a real analytic left inverse of Ψ with injective differential, and $\dim \mathcal{V}_t + \dim \mathsf{Y} = \dim \mathcal{V}_0$; the inverse function theorem gives (i) after shrinking, and (iv) with $W = e^{iw}$, $dW = \varrho_{\mathrm{H}}(w)\,\mathrm{d}w$, $\varrho_{\mathrm{H}} > 0$ the continuous density of the Haar measure of $\mathcal{V}_t$ in the variable w, on one set that carries (i)–(iii). For the second choice in (iv) one follows, level by level, a regularity number $\mathfrak{r}_j(W_j)$, which measures the plaquette variables of the averaged configuration of the level j and is comparable with $\max_p |\partial W_j(p) - \mathbf{1}|$: one step gives $\mathfrak{r}_j(W_j) \leq L^{-2}\mathfrak{r}_{j+1}(W_{j+1}) + C_{\mathfrak{r}}|\mathcal{A}_j|$, with $C_{\mathfrak{r}}$ determined by L and the constants.

So a threshold $\mathfrak{r}_\flat$ for $\mathfrak{r}$, with ε_* a fixed fraction of it and $\rho_* \leq (1 - L^{-2})\mathfrak{r}_\flat / C_{\mathfrak{r}}$, keeps every level in the sets where the statements of one step apply; nothing in this sees t, K or m. (ii) $a_p \geq 0$ and

$G_j \geq 0$; at $\mathbf{1}$, $a_p(e^{iA}) = \frac{1}{2}|\mathrm{d}A(p)|^2 + O(|A|^3)$, G_j is half the sum of the squares of the linearised contour variables up to $O(|A|^3)$, and a fibre variable whose linearised curvature and contour variables vanish is 0. At the linear order the composed chart is the chart of the free template: the bare curvature of the field of the level j is T_j applied to it, with values in the slice $\mathcal{C}_{(j+1)}$ (Prop. YM3_I:120.3), the slices are mutually orthogonal, and the linearised contour variables of the level j are those of $\mathcal{A}_j$ alone; this makes the Hessian block diagonal.

(iii) By (i) and (iv), near $(0,0)$ the composed chart $(w,y) \mapsto \Psi(e^{iw},y)$, with C_{H}, ϱ_{H} and $\theta(e^{iw},y)$, is an adapted chart: it is a real analytic diffeomorphism onto an open set, $\Psi(\mathbf{1},0) = \mathbf{1}$ ($W_j(\mathbf{1},0) = \mathbf{1}$), $\operatorname{Lg}\mathcal{M}^t\Psi(e^{iw},y) = w$, and θ is real analytic. Let $(\mathsf{E}',\Psi',\theta',C',\varrho')$ be an adapted chart, and E_1 an open neighbourhood of $(0,0)$ that the composed chart maps into $\Psi'(\mathsf{E}')$. Then $\sigma := \Psi'^{-1}\circ\Psi$ is a C^∞ diffeomorphism of E_1 onto an open set of the form $\sigma(w,y) = (w,s(w,y))$, both charts carrying $\operatorname{Lg}\mathcal{M}^t$ to the first variable, and $\sigma(0,0) = (0,0)$, as $\Psi(\mathbf{1},0) = \mathbf{1} = \Psi'(0,0)$.

The two formulas for dU give

$$C_{\mathrm{H}}\,\varrho_{\mathrm{H}}(w)\,e^{-\theta(e^{iw},y)} = C'\varrho'(w)\,e^{-\theta'(\sigma(w,y))}\,|\det D_y s(w,y)|$$

almost everywhere on E_1, and, both sides being continuous, at every point of E_1; in particular on the fibre $w=0$, a null set, where the numbers are read. So $y \mapsto s(0,y)$, a diffeomorphism near 0 that fixes 0, carries the phase $F_\kappa\circ\Psi(\mathbf{1},\cdot)$, the observables and the density $e^{-\theta(\mathbf{1},y)}dy$ into $F_\kappa\circ\Psi'(0,\cdot)$, the observables and $e^{-\theta'(0,y')}dy'$, up to the constant factor $C'\varrho'(0)/(C_{\mathrm{H}}\varrho_{\mathrm{H}}(0))$; the Taylor polynomials are taken at the minimum 0 in both charts. The two coefficients of Laplace's method in either chart are those of the expansion of one quantity, the covariance of a_p and a_q under $e^{-\beta F_\kappa}$ on the fibre $\mathcal{M}^tU = \mathbf{1}$ near $\mathbf{1}$, and so they are the same in both.

The gauge transformations equal to $\mathbf{1}$ on the top lattice preserve the fibre, A and the a_p, and the normalised gauge-fixing factor (Def. YM3_I:6.1 and equations YM3_I:(51)–(52)) integrates to one over their orbits at every strength (Lem. YM3_I:5.3 and equation YM3_I:(55)); integrating over the orbits, the finest sites first, the covariance does not depend on κ up to errors $e^{-c\beta}$, and neither do its coefficients. Near $(0,0)$ the composed chart is itself an adapted chart. □

Proposition 6.15 (The identification of the order six)**.** Under the assumptions of Prop. 6.12, $\beta^3 a_K^2\gamma_2^\circ = \mathsf{T}[N,n](w)$, the plaquettes being read in bare units on Λ_N; $\mathsf{T}[N,n](w)$ depends on N, n, L and the lattice sources alone. The identity has no error term and holds at every size of the boxes. ■

Proof sketch. By Prop. 6.5(i), Prop. 6.14 applies at $t = n$ on the lattice of the datum. For real $0 < \lambda \leq 1$ the exact formula of the steps (Prop. YM3_I:123.1), read at the fields λB, and the sum rules of the chain classes (Prop. YM3_I:124.2, equation YM3_I:(2289)) add the exponents of the steps of Prop. 6.12(i) up to

$$\mathsf{D}^\lambda(B) = c_n(\lambda)\exp\Big[-\lambda^{-2}\Big(x_0 A(W_0) + \sum_{j<n} x_j G_j(W_j)\Big) - \theta\Big],$$

where $c_n(\lambda)$ does not depend on B and (W_j) is the composed chart over the flat top $\mathbf{1}$ of the level n at the fields $\mathcal{A}_i = \lambda x_i^{-1/2}\check{B}_i$, $\check{B}_i = \mathrm{Ad}(u_{i+1})B_i$, with $u_{i+1} \in G$ a function of the fields of the heights $> i$ alone: the chain of Prop. 6.12 runs through configurations in another gauge than the composed chart. The observables become $a_K^{-1}\sum_p w_a(p)a_p(W_0)$ at the same fields. No τ_i and no counterterm e_i is left, because

$$\lambda^{-2}\xi_j(\lambda)L^j + \sum_{i\leq j} L^i e_i = \lambda^{-2}x_0$$

by the run equation; this is what ξ_j is chosen for. So the density is the Wilson weight at $\beta\lambda^{-2}$ with the gauge-fixing strengths $\kappa_j = x_j/\beta$ on the fibre $\mathcal{M}^nU = \mathbf{1}$. The triangular change $B \mapsto \check{B}$ preserves the volume and the boxes, which are Euclidean balls $|B_i(b)| < \mathfrak{b}(x_i)$ and are preserved by

the rotations $\mathrm{Ad}(u)$; its linear part at 0 is a constant rotation of the colours, so it preserves the Gaussian part $\bigotimes_{i<n}\nu_i^\circ$; and the two numbers of Laplace's method are kept by a diffeomorphism that carries phase, observables and density into each other.

So γ_2° is the second coefficient of Laplace's method eq. 6.viii at the parameter $\beta\lambda^{-2}$, and $\beta^3 a_K^2\gamma_2^\circ = \sum_{p,q} w_1(p)w_2(q)\gamma_2^{\mathrm{L}}$ in the chart of Prop. 6.14, which does not depend on κ. The dilation $x \mapsto L^{K-n}x$ carries the lattice (ℓ, K) with its flat top at the level n to the lattice $(\ell L^{K-n}, n)$, with the same bare torus, plaquettes to plaquettes with their barycentres; so the number is $\mathsf{g}_2[N,n](p,q)$. The argument uses that the boxes are Euclidean balls; it would not apply to boxes that are cubes in the components. □

Lemma 6.16 (An implicit function, Stokes's formula in a box, Wick's rule)**.**

(i) Let V be a finite-dimensional complex normed space, $0 < \varsigma \le \frac{1}{120}$, and F holomorphic from $\{\|y\| < \varsigma\} \times \{\lambda \in \mathfrak{g}^{\mathbb{C}} : |\lambda| < \varsigma\}$ to $\mathfrak{g}^{\mathbb{C}}$ with $|F| \le \frac{1}{40}$, $F(0,0) = 0$ and $D_\lambda F(0,0) = 1$. Then for $\|y\| < \varsigma^3$ there is exactly one $h(y)$ with $|h(y)| \le \varsigma^2$ and $F(y, h(y)) = 0$; h is holomorphic, $|h(y)| \le \|y\|/(15\varsigma)$, and h is $\mathfrak{g}$-valued at the real points if F is.

(ii) Let B be the injective image in a bare torus of a box of $\mathbb{Z}^3$. For every closed bare path γ in B there is an integer function k_γ on the plaquettes with all corners in B, $|\mathsf{k}_\gamma| \le |\gamma|$ (the number of bonds of γ), with $a(\gamma) = \sum_p \mathsf{k}_\gamma(p)\,(\mathrm{d}a)(p)$ for every 1-cochain a.

(iii) Let (ζ_ι) be a centred Gaussian family with covariance C, and $P_1, \dots, P_r$ homogeneous polynomials in it of orders $n_1, \dots, n_r$, $n = \sum n_a$, $\|P\|_1$ the sum of the moduli of the coefficients. For monomials the joint cumulant $\langle P_1; \dots; P_r\rangle$ is the sum, over the pairings of the factors that are connected (the graph on $\{1, \dots, r\}$ with an edge for every pair that joins two of the P_a is connected), of the products of the covariances, and by multilinearity this extends to polynomials; the cumulant vanishes if n is odd. With $\mathsf{c} \ge \sup|C|$ on the supports, $|\langle P_1; \dots; P_r\rangle| \le (n-1)!! \prod_a \|P_a\|_1 \mathsf{c}^{n/2}$; if every connected pairing has a pair with $|C| \le \varepsilon$, the bound holds with $\mathsf{c}^{n/2-1}\varepsilon$ for $\mathsf{c}^{n/2}$; and for two covariances C, C' bounded by c the two cumulants differ by at most $(n-1)!!\frac{n}{2}\prod_a \|P_a\|_1 \mathsf{c}^{n/2-1}\sup|C - C'|$.

■

Proof sketch. (i) Put $g(y,\lambda) = \lambda - F(y,\lambda)$, so $|g| \le \frac{1}{40} + \varsigma \le \frac{1}{30}$, $g(0,0) = 0$, $D_\lambda g(0,0) = 0$. Cauchy's estimate and Schwarz's lemma give $|D_\lambda g(y,\lambda)| \le \frac{2}{15}\varsigma^{-2}\max\{\|y\|, |\lambda|\}$ for $\max\{\|y\|, |\lambda|\} \le \frac{1}{2}\varsigma$, and $|g(y,0)| \le \|y\|/(30\varsigma)$. So for $\|y\| < \varsigma^3$ the map $\lambda \mapsto g(y,\lambda)$ sends $\{|\lambda| \le \varsigma^2\}$ into itself and contracts by $\frac{2}{15}$; its fixed point is $h(y)$, the uniform limit of holomorphic iterates, and $|h| \le |g(y,0)| + \frac{2}{15}|h|$. (ii) With the paths P_z from a corner o of the box to z that run first in the direction 1, then 2, then 3, the quantity $a(P_z) + a(e) - a(P_{z+e_\mu})$ of a bond $e = \langle z, z+e_\mu\rangle$ is the sum of $\pm(\mathrm{d}a)$ over a strip of plaquettes of the box; along a closed path the terms $a(P_\cdot)$ telescope.

(iii) By Wick's theorem the moments are sums over all pairings; a pairing determines the partition of the factors into the components of its graph and is connected on each block, which identifies the sums over connected pairings with the joint cumulants. The bounds count the $(n-1)!!$ pairings and use $|\prod_{i\le m} c_i - \prod_{i\le m} c_i'| \le m\,\mathsf{c}^{m-1}\max|c_i - c_i'|$. □

Proposition 6.17 (Two bare tori under one flat top)**.** Let $N = 2L^{\mathsf{m}}$, $N' = L^s N$ $(s \ge 1)$, $1 \le t < \mathsf{m}$, $\pi : \Lambda_{N'} \to \Lambda_N$ the reduction, and

$$\nu = N/L^t \ge \nu_{\mathrm{W}} = \max\Big\{192,\ \frac{48(3\log L + \log 2)}{\delta_{\mathrm{tp}} L(L-1)},\ \frac{C_L M}{L}\Big\}.$$

For plaquettes p', q' of $\Lambda_{N'}$ whose base points lie within $N/8$ of each other (the base point of the plaquette $p_{\mu\nu}(x)$ being x, the distance being the sup distance of the bare torus, in bare units) and $m' = 0, 2$,

$$|\mathsf{g}_{m'}[N,t](\pi p', \pi q') - \mathsf{g}_{m'}[N',t](p',q')| \le C'_{\mathrm{W}} N'^{237} e^{-\delta_{\mathrm{tp}} L\nu/48},$$

with C'_{W} determined by L and C_{tp}. Hence, for real w_1, w_2 on the plaquettes of Λ_N with supports in balls of radius $r \geq 8$, all base points within $N/8$ of each other, and $\breve{w}_a$ equal to $w_a \circ \pi$ on one lift of a ball of radius $N/4$ that holds the supports and 0 elsewhere,

$$|\mathsf{T}[N,t](w) - \mathsf{T}[N',t](\breve{w})| \leq 900 r^6 C'_{\mathrm{W}} N'^{237} e^{-\delta_{\mathrm{tp}} L\nu/48} \|w_1\|_\infty \|w_2\|_\infty.$$

■

Proof sketch. Signs. Let $S \in \{N, N'\}$, and read the objects of the lattice $(S/L^t, t)$ in bare units on Λ_S; distances are those of the bare torus, $|\cdot|_\infty$ its sup distance, and $x(p)$ is the barycentre of a plaquette p, its base point being x if $p = p_{\mu\nu}(x)$. For a bond c of the top lattice $\mathbb{T}^{(t)}$, $B^t(c_\pm)$ are the blocks of order t of its end points, its double block is their union, and its corridor $\mathrm{cor}(c)$ is the set of the bare bonds b with $b_- \in B^t(c_-)$, $b_+ \in B^t(c_+)$; $\mathsf{1}_c$ is the bare 1-cochain that is 1 on $\mathrm{cor}(c)$, -1 on the reversed bonds and 0 elsewhere. As $\nu \geq 8$, every bare bond lies in one top block or in exactly one corridor, and no bond of the corridor of c' has both end points in the double block of c unless $c' \in \{c, \bar{c}\}$; the bonds of the double block of c are those with both end points in it.

For a bare site $x \notin \mathbb{T}^{(t)}$, $j(x) = \max\{j : x \in \mathbb{T}^{(j)}\} < t$, its parent $\hat{x} \in \mathbb{T}^{(j(x)+1)}$ is the centre of the block of order 1 that holds x, and $\mathrm{anc}(z) = \{z, \hat{z}, \hat{\hat{z}}, \dots\} \setminus \mathbb{T}^{(t)}$; the sums over x below run over the bare sites off $\mathbb{T}^{(t)}$. Q_k is the linear part of the k-fold block average, and $\xi_A(x) = (\mathsf{Q}_{j(x)}A)(\hat{x}, x)$ is the linearised contour variable of the site x. A $\mathfrak{g}$-valued 1-form y has the coordinates $y(\iota)$, $\iota = (b, \alpha)$, $b > 0$ a bare bond and α a colour; $\iota + \gamma$ is the translate of ι by γ, and $\mathrm{dist}(\iota, \iota') = |b_- - b'_-|_\infty$.

By Prop. 6.14(iii) the numbers do not depend on κ; we take $\kappa \equiv 1$.

A chart made for locality. Q_k is a mean over bare paths: for every bond c of $\mathbb{T}^{(k)}$ there is a probability measure on the bare paths from c_- to c_+ in the double block of c, with at most $(4L)^k$ bonds, whose mean of $A(\gamma)$ is $(\mathsf{Q}_k A)(c)$; and $\mathsf{Q}_t \mathsf{1}_c$ is 1 at c and 0 at the other top bonds. $\mathcal{M}^t(U)(c)$ depends on U on the double block of the top bond c alone. So, for y in $Y_S := \ker \mathsf{Q}_t$ near 0, the constraint $\mathcal{M}^t(e^{i\hat{y}}) = \mathbf{1}$ for $\hat{y} := y + \sum_{c>0} h_c(y)\mathsf{1}_c$ is solved one top bond at a time, with one element $h_c(y) \in \mathfrak{g}$ (three unknowns) for each: Lem. 6.16(i) at $\varsigma = \varsigma_t := (75L)^{-t}/(60L)$, applied to $(y, \lambda) \mapsto \mathrm{Lg}\, \mathcal{M}^t(e^{i(y+\lambda \mathsf{1}_c)})(c)$, which is holomorphic and bounded by $\frac{1}{40}$ on the polydisc of the Lemma, vanishes at $(0,0)$ and has the derivative 1 in λ there, gives exactly one holomorphic h_c with $|h_c(y)| \leq \|y\|_\infty/(15\varsigma_t)$ for $\|y\|_\infty < \varsigma_t^3$, $\mathfrak{g}$-valued at real y; it reads y on the double block of c alone, $\hat{y} = y$ on the top blocks, and $\hat{y} = y + O(y^2)$ on Y_S.

The map $A \mapsto (\mathrm{Lg}\, \mathcal{M}^t(e^{iA}),\ A - \sum_{c>0}(\mathsf{Q}_t A)(c)\mathsf{1}_c)$ is a real analytic diffeomorphism near 0, as its differential at 0 has the inverse $(w, y) \mapsto y + \sum_c w(c)\mathsf{1}_c$; its inverse, followed by $A \mapsto e^{iA}$, is an adapted chart Ψ' in the sense of Prop. 6.14(iii), with $\varrho' = 1$, a real analytic θ' and $\Psi'(0, y) = e^{i\hat{y}}$ for $y \in Y_S$; so both numbers may be computed in it. In it the action, the gauge terms, the Haar density (the exponential-coordinate Haar formula following equation YM3_I:(911)) and the Jacobian, the product over the top bonds c of the determinants of the derivatives of $\mathrm{Lg}\, \mathcal{M}^t(e^{i\cdot})(c)$ in the direction $\mathsf{1}_c$, are sums of terms, each holomorphic and bounded by 1 on the polydisc $\|y\|_\infty < \varsigma_t^3$ and depending on y on at most $12L^{3t}$ bonds with end points in a set of diameter $\leq 2L^t$; for a plaquette p these are the top blocks that hold a corner of p, within L^t of its base point.

The reduction π carries the lattices, blocks, contours, corridors and the block average of $\Lambda_{N'}$ to those of Λ_N on every set of diameter $\leq N/2$, so these are the same functions on both tori after the renaming of the variables by π.

The lines. The gauge terms and the Haar terms of the chart read y on the bonds, and not its curvature alone; so the lines of Wick's rule in this chart are the entries of the covariance C_S of the chart variable $y \in Y_S$ under the centred Gaussian measure with the density proportional to $e^{-\frac{1}{2}\mathsf{H}_S(y,y)}$, $\frac{1}{2}\mathsf{H}_S(y,y)$ being the second Taylor polynomial at 0 of the action with the gauge terms,

$$\mathsf{H}_S(A, A') = (\mathrm{d}A, \mathrm{d}A') + \sum_x \langle \xi_A(x), \xi_{A'}(x) \rangle.$$

Let $\mathcal{X}_S$ be the space of the $\mathfrak{g}$-valued functions on the bare sites that vanish on $\mathbb{T}^{(t)}$. Then $\mathrm{d}\mathcal{X}_S \subset Y_S$, as $\mathsf{Q}_t \mathrm{d}\chi = \mathrm{d}(\chi|_{\mathbb{T}^{(t)}})$, and $\chi \mapsto \xi_{\mathrm{d}\chi} = (\chi(x) - \chi(\hat{x}))_x$ is a bijection of $\mathcal{X}_S$ onto the functions on the sites off $\mathbb{T}^{(t)}$, with the inverse $\chi(x) = \sum_{x'' \in \mathrm{anc}(x)} \xi(x'')$. So Y_S is the direct sum of its axial part $Y_S^{\mathrm{ax}} = \{A \in Y_S : \xi_A = 0\}$ and of $\mathrm{d}\mathcal{X}_S$, $y = y^{\mathrm{ax}} + \mathrm{d}\chi_y$, and

$$\mathsf{H}_S(y,y) = \|\mathrm{d}y^{\mathrm{ax}}\|^2 + \sum_x |\xi_y(x)|^2;$$

d maps Y_S^{ax} one to one onto the curvatures whose plaquette averages of the order t vanish, and H_S is positive definite on Y_S. In the Gaussian measure y^{ax} and the $\xi(x)$ are independent: the $\xi(x)$, one for each site off the top lattice, are independent with the variance 1 in each colour, $(\mathrm{d}\chi)(b) = \sum_x \sigma(b,x)\xi(x)$ with $\sigma(b,x) = \mathbf{1}[x \in \mathrm{anc}(b_+)] - \mathbf{1}[x \in \mathrm{anc}(b_-)]$, and $\mathrm{d}y^{\mathrm{ax}}$ has the covariance $\Pi_t^{\mathrm{fl}} = \Pi_0^{\mathrm{tp}} - \Pi_t^{\mathrm{tp}}$ in each colour. On Y_S^{ax} the means of y over two families of bare paths vanish: over the paths from the centre of the top block of a site x to x, inside that block and with at most $(4L)^t$ bonds, whose mean of $y(\gamma)$ is $\sum_{x'' \in \mathrm{anc}(x)} \xi_y(x'') = 0$ (the contour averages of the levels composed from the top down), and over the paths of Q_t along a top bond, as $y \in \ker \mathsf{Q}_t$.

So $y(b)$ is the mean of the values of y on closed bare paths of at most $4(4L)^t$ bonds in the box of b, the top block that holds b, or the double block of c if $b \in \mathrm{cor}(c)$, and by Lem. 6.16(ii), the box being the injective image of a box of $\mathbb{Z}^3$, $y(b) = \sum_p \tau(b,p)\,(\mathrm{d}y)(p)$ with real Stokes coefficients, $|\tau(b,p)| \le 4(4L)^t$, $\tau(b,p) = 0$ unless all the corners of p lie in the box of b. Hence $C_S((b,\alpha),(b',\alpha')) = \delta_{\alpha\alpha'}[C_S^{\mathrm{ax}} + C_S^{\mathrm{g}}](b,b')$,

$$C_S^{\mathrm{ax}}(b,b') = \sum_{p,p'} \tau(b,p)\tau(b',p')\,\Pi_t^{\mathrm{fl}}(p,p'), \qquad C_S^{\mathrm{g}}(b,b') = \sum_x \sigma(b,x)\sigma(b',x),$$

and $C_S^{\mathrm{g}}(b,b') = 0$ unless an end point of b and one of b' lie in one top block. A box holds at most $6L^{3t}$ plaquettes, $|\Pi_t^{\mathrm{fl}}| \le 1$ (the orthogonal projections in YM3_I:(2226)), and $\sigma(b,\cdot) \ne 0$ at no more than $2t$ sites; so $\sup|C_S| \le \mathsf{c}_\tau^2 + 2t$, $\mathsf{c}_\tau = 24L^{3t}(4L)^t$. By Prop. YM3_I:120.6 on the lattice $(S/L^t, t)$ at $s = 0$, uniformly in the volume,

$$|\Pi_t^{\mathrm{fl}}(p,q)| \le 2L^3 C_{\mathrm{tp}} L^{-3t} e^{-\delta_{\mathrm{tp}} L|x(p)-x(q)|_\infty/L^t} \qquad \text{if } \delta_{\mathrm{tp}} L(L-1)|x(p)-x(q)|_\infty \ge (3\log L + \log 2)L^t,$$

the barycentres being taken in bare units. The barycentre of a plaquette in the box of b lies within $2L^t$ of b_-; so if $\mathrm{dist}(\iota,\iota') \ge R \ge 8L^t$ and $\delta_{\mathrm{tp}} L(L-1)(R-4L^t) \ge (3\log L + \log 2)L^t$, then $C_S^{\mathrm{g}}(b,b') = 0$ and

$$|C_S(\iota,\iota')| \le \mathsf{c}_\tau^2\, 2L^{3-3t} C_{\mathrm{tp}} \exp\bigl[-\delta_{\mathrm{tp}} L(R-4L^t)/L^t\bigr].$$

At $R = N/24$ these conditions hold, and the exponent is at least $\delta_{\mathrm{tp}} L\nu/48$: $R = \frac{\nu}{24}L^t \ge 8L^t$ and $R - 4L^t \ge \frac{\nu}{48}L^t$ as $\nu \ge 192$, and $\delta_{\mathrm{tp}} L(L-1)\frac{\nu}{48} \ge 3\log L + \log 2$ by the second member of ν_{W}. Finally the covariance on Λ_N is the periodisation of that on $\Lambda_{N'}$, exactly: with $\Gamma = N\mathbb{Z}^3/N'\mathbb{Z}^3$, $|\Gamma| = (N'/N)^3$,

$$C_N(\pi\iota,\pi\iota') = \sum_{\gamma\in\Gamma} C_{N'}(\iota,\iota'+\gamma).$$

For π^* commutes with d, the Q_k and ξ, and π is $|\Gamma|$ to one on the plaquettes and on the sites, so that $\mathsf{H}_{N'}(\pi^*A,\pi^*A') = |\Gamma|\,\mathsf{H}_N(A,A')$ and $\pi^*Y_N = Y_{N'} \cap \pi^*\mathcal{E}_N$, $\mathcal{E}_N$ the 1-forms on Λ_N; $u = \sum_\gamma C_{N'}(\cdot,\iota'+\gamma)$ is the one solution in $Y_{N'}$ of $\mathsf{H}_{N'}(u,v) = \sum_\gamma v(\iota'+\gamma)$ for all $v \in Y_{N'}$, a problem invariant under Γ, so $u = \pi^*u_0$ with $u_0 \in Y_N$, and the test functions $v = \pi^*v_0$ give $\mathsf{H}_N(u_0,v_0) = v_0(\pi\iota')$ for all $v_0 \in Y_N$, that is $u_0 = C_N(\cdot,\pi\iota')$.

Near and far. By Laplace's method in the chart, each of the seven cumulants of $\mathsf{g}_{m'}$ is, by multilinearity, a sum over at most $64N'^6$ tuples of places of cumulants of homogeneous Taylor polynomials of the terms, of total order ≤ 10, in the Gaussian family with the covariance C_S; Lem. 6.16(iii) writes each as a sum over connected pairings of products of lines. A place is near if the set on which its term reads y lies within $N/4$ of the base point of p', and far otherwise; π maps the near places of $\Lambda_{N'}$ one to one onto those of Λ_N, and p', q' are near, as $L^t \le N/192$. A tuple of

near places gives the same polynomials on both tori, renamed by π, so that its cumulant on Λ_N is that of the polynomials of $\Lambda_{N'}$ in the covariance $C_N(\pi\iota, \pi\iota')$; its indices lie within $N/2$ of each other, so that by the periodisation the two covariances differ by $\sum_{\gamma\neq 0} C_{N'}(\iota, \iota'+\gamma)$, a sum of fewer than $|\Gamma|$ lines of length $\geq N/2$.

If a tuple has a far place, every connected pairing of it contains a line of length $\geq N/24$, on either torus: a path without repetitions in the graph of the pairing leads from the place p' to the far place in at most three edges, the polynomials of one place depend on y on a set of diameter $\leq 2L^t \leq \frac{2}{192}N$, and so the lengths of the at most three lines of the path add up to more than $\frac{N}{4} - 7L^t > \frac{N}{8}$. By the bound of the lines at $R = N/24$, such a line is at most $\varepsilon_C = 2L^{3-3t} C_{\mathrm{tp}} \mathsf{c}_\tau^2 e^{-\delta_{\mathrm{tp}} L\nu/48}$, and $\sup|C_S| \leq 2\mathsf{c}_\tau^2$; the bounds of Lem. 6.16(iii) then bound the far tuples on each torus by a multiple of ε_C and the difference at a near tuple by a multiple of $|\Gamma|\varepsilon_C$, with Cauchy's estimate on the polydisc for the coefficients of the polynomials, at most $(108L^{3t}\varsigma_t^{-3})^k$ in the sum of their moduli at the order k.

The powers of $L^t \leq N'$, the factor $|\Gamma| \leq N'^3$ and the number of the tuples are collected in N'^{237}, with C'_{W} determined by L and C_{tp}. A ball of radius $r \geq 8$ holds at most $30r^3$ barycentres of plaquettes, which gives the second display. □

Proposition 6.18 (The anchor)**.** Let $\ell = 2L^m \geq N_{\mathsf{H}}$ with $\ell L \geq C_L M$, and $K \geq 1$. For all plaquettes p, q of $\Lambda_{\ell L^K}$ and all plaquettes $\tilde{p}, \tilde{q}$ of $\mathbb{Z}^3$ over them whose base points are at sup-distance $\leq \frac{1}{8}\ell L^K$,

$$|\mathsf{g}_2[\ell L^K, K](p,q) - \mathsf{B}_1^{\mathrm{lat}}(\tilde{p},\tilde{q})| \leq C_\star[\ell^{29/2} K L^{-K/2} + \ell^{18} L^{-2K} + \ell^{19} K^2 L^{-5K}] \leq 3C_\star \ell^{19} K L^{-K/2},$$

with $C_\star$ determined by the constants and independent of the side; if $\ell^3 \leq L^K$, the left member is at most $3C_\star \ell^{29/2} K L^{-K/2}$. ■

Proof. Write $\mathsf{N} = \ell L^K$. Distances are those of the bare torus Λ_{N}, in bare units, $|\cdot|$ its Euclidean distance and $|\cdot|_\infty$ its sup distance, $\langle z\rangle = \max\{1, |z|\}$; x_p is the base point of the plaquette p, $\langle b - b'\rangle := \langle b_- - b'_-\rangle$, $\langle x_p - b\rangle := \langle x_p - b_-\rangle$; and o is a fixed site of the top lattice $\mathbb{T}^{(K)}$.

The covariant gauge below the flat top. By Prop. 6.14(iii), $\mathsf{g}_2[\mathsf{N}, K](p,q)$ is the coefficient of β^{-3} in the covariance of a_p and a_q in the local fibre $\mathcal{M}^K U = \mathbf{1}$. Use the group of bare gauge transformations fixed at o. Its restriction to the top sites is onto the top gauge group fixed at o. For each top transformation, a bare lift preserves the action and insertions and carries the conditional Haar density of the original fibre to that of the transformed fibre. Integrating the transformed fibre against the top Haar measure thus preserves the coefficient. The top field in the enlarged fibre is the pure gauge with logarithm

$$w(\mu)(c) = \mathrm{Lg}(e^{i\mu(c_-)} e^{-i\mu(c_+)}), \qquad \mu(o) = 0.$$

Along the based gauge orbit the coordinate $g \mapsto \mathrm{d}^* \mathrm{Lg}\, U^g$ has invertible derivative at the trivial configuration, from based functions to mean-zero functions. For the parameter $g = e^{it\varepsilon}$ the derivative is $-\mathrm{d}^*\mathrm{d}$. Pulling a normalized Gaussian back by this coordinate gives the covariant gauge of Def. 7.7 at parameter 1 and the local Faddeev–Popov determinant as in Lem. 7.6. At fixed N, the discarded tails of the coordinate Gaussian are exponentially small in β and do not affect the Laplace coefficient.

To compare the based determinant with the mean-zero one, write $J_{\mathsf{a}}\varepsilon = \frac{\mathrm{d}}{\mathrm{d}t}\mathrm{d}^* \mathrm{Lg}((e^{i\mathsf{a}})^{e^{it\varepsilon}})\Big|_{t=0}$ and let J'_{a} be its restriction to mean-zero functions. For constant κ, $J_{\mathsf{a}}\kappa = i[\kappa, \mathrm{d}^*\mathsf{a}]$. The map $\varepsilon \mapsto \varepsilon - \varepsilon(o)$ identifies the mean-zero and based domains, with constant Jacobian. The determinant ratio in these coordinates is $\det_{\mathfrak{g}}(1 - \tau_{\mathsf{a}})$, where $\tau_{\mathsf{a}}\kappa = (J'^{-1}_{\mathsf{a}} i[\kappa, \mathrm{d}^*\mathsf{a}])(o)$. The linear term in its logarithm has zero trace because the adjoint action is traceless. Each quadratic monomial has a $\mathrm{d}^*\mathsf{a}$ slot. In the Gaussian below, $\mathrm{Cov}(\mathrm{d}\mathsf{a}, \mathrm{d}^*\mathsf{a}) = 0$, so the connected cumulant of this quadratic correction with the two quadratic plaquette insertions vanishes. The mean-zero determinant supplies the usual vertices with periodic Green functions.

The nonlinear constraint. In the linearization of one block step, the difference between the ordinary and straight averages is $-\mathrm{d}\varphi_\omega$, where φ_ω is the average of the linearized contour variable within the block. Set $v_\omega(z) = e^{i\varphi_\omega(z)}$ and use the one-step map

$$m^s(\omega)(c) = \mathrm{Lg}(v_\omega(c_-)^{-1}\mathcal{M}(e^{i\omega})(c)v_\omega(c_+)).$$

It has linear part Q_1^s. Its quadratic and cubic Taylor coefficients are bounded on each double block by constants depending only on the one-step map. This follows from analyticity at the trivial configuration and the fixed number of bonds in the double block. By gauge covariance, the iterate $s^{(K)}$ of m^s satisfies $e^{is^{(K)}(\mathsf{a})(c)} = u_{\mathsf{a}}(c_-)^{-1}\mathcal{M}^K(e^{i\mathsf{a}})(c)u_{\mathsf{a}}(c_+)$, where u_{a} is the product of the successive frame changes, restricted to the top sites. Replace $e^{i\mu(z)}$ by $u_{\mathsf{a}}(z)^{-1}e^{i\mu(z)}u_{\mathsf{a}}(o)$. At fixed a this preserves the based top Haar measure. The constraint delta function relative to bond Haar measure is likewise invariant under the change of frames. We continue to write μ for the new coordinate. The constraint is now $s^{(K)}(\mathsf{a}) - w(\mu) = 0$. Its derivative is $Q_K^s\mathsf{a} + \mathrm{d}\mu$, where $\mathrm{d}Q_K^s = Q_K^{\mathrm{pl}}\mathrm{d}$ by Lem. YM3_I:120.1.

Let $R\omega = (\mathsf{a}_\omega, \mu_\omega)$ be the harmonic right inverse, defined by minimizing $\|\mathrm{d}\mathsf{a}_\omega\|^2 + \|\mathrm{d}^*\mathsf{a}_\omega\|^2$ among the pairs with constraint value ω and $\mu_\omega(o) = 0$. This form is positive definite on the kernel $\hat{E}$ of the based linear constraint. A zero-energy member has constant a, and summing the constraint in each coordinate direction makes that constant zero. Then $\mathrm{d}\mu = 0$ and $\mu(o) = 0$. Surjectivity of the linear constraint also follows from the composed chart in Prop. 6.14(i), since the straight frame changes add a coboundary at the linear level. Let $e \in \hat{E}$ and let q_2, q_3 be the polarized homogeneous terms of degrees two and three of $s^{(K)} - w$. In the splitting $\hat{E} \oplus R(\text{top 1-forms})$ the graph of the nonlinear constraint is

$$e - \zeta - \zeta_3 + O(|e|^4), \qquad \zeta = Rq_2(e,e), \qquad \zeta_3 = R\big(q_3(e,e,e) - 2q_2(e,\zeta)\big).$$

Indeed, substitution cancels the degree-two and degree-three terms. The transverse derivative in these coordinates is $1 + N_1 + N_2 + O(|e|^3)$, where

$$N_1 = 2q_2(e, R\,\cdot), \qquad N_2 = 3q_3(e,e,R\,\cdot) - 2q_2(\zeta, R\,\cdot).$$

The delta function contributes the inverse determinant, so the quadratic addition to the negative log density is $\operatorname{tr} N_2 - \frac{1}{2}\operatorname{tr} N_1^2$. The trace of N_1 vanishes because each colour coefficient of the equivariant bilinear map q_2 is a multiple of the cross product and R is diagonal in the colours. The top Haar factor and the inverse Haar density of the delta function give the additional quadratic form $J_{\mathrm{top}} = (\|\mu\|_2^2 - \|\mathrm{d}\mu\|_2^2)/3$, using the Haar density proportional to $(\sin|\mu|/|\mu|)^2$ in exponential coordinates.

Write X_i, Y_i for the terms of degree $i+2$ of the two plaquette observables, for $i = 0, 1, 2$, and H for the Hessian of the Gaussian phase on the ambient linear space. In eq. 6.vii, the cubic action S_3 is unchanged, by H-orthogonality of e to the range of R. The corrected X_1 and X_2 are $X_1 - DX_0[\zeta]$ and $X_2 - DX_1[\zeta] - DX_0[\zeta_3] + X_0(\zeta)$. The same replacements hold for Y. The degree-four action gains $-DS_3[\zeta] + \frac{1}{2}(\zeta, H\zeta)$. The degree-two negative log density gains $\operatorname{tr} N_2 - \frac{1}{2}\operatorname{tr} N_1^2 + J_{\mathrm{top}}$. Consequently $\mathsf{g}_2 - \mathsf{B}_1^{\hat{E}} = \sum_{i=1}^{11} t_i$, where the cumulants below are taken in the Gaussian on $\hat{E}$, and derivatives are evaluated at e:

$$\begin{aligned}
t_1 &= -\langle DX_0[\zeta]; Y_1\rangle - \langle X_1; DY_0[\zeta]\rangle, & t_2 &= \langle DX_0[\zeta]; DY_0[\zeta]\rangle,\\
t_3 &= -\langle DX_1[\zeta]; Y_0\rangle - \langle X_0; DY_1[\zeta]\rangle, & t_4 &= -\langle DX_0[\zeta_3]; Y_0\rangle - \langle X_0; DY_0[\zeta_3]\rangle,\\
t_5 &= \langle X_0(\zeta); Y_0\rangle + \langle X_0; Y_0(\zeta)\rangle, & &\\
t_6 &= \langle DX_0[\zeta]; Y_0; S_3\rangle + \langle X_0; DY_0[\zeta]; S_3\rangle, & t_7 &= \langle X_0; Y_0; DS_3[\zeta]\rangle,\\
t_8 &= -\tfrac{1}{2}\langle X_0; Y_0; (\zeta, H\zeta)\rangle, & t_9 &= -\langle X_0; Y_0; \operatorname{tr} N_2\rangle,\\
t_{10} &= \tfrac{1}{2}\langle X_0; Y_0; \operatorname{tr} N_1^2\rangle, & t_{11} &= -\langle X_0; Y_0; J_{\mathrm{top}}\rangle.
\end{aligned}$$

The term $\mathsf{B}_1^{\hat E}$ is eq. 7.xviii at gauge parameter 1, with the Gaussian covariance on $\hat E$ and the ghost kernels formed from the periodic Green function.

The lines. Let $\mathsf{C}^{\mathrm{tor}}$ be the inverse of $\mathrm{d}^*\mathrm{d}+\mathrm{d}\mathrm{d}^*$ off the constant 1-forms and 0 on them, and $\mathsf{S}_0 := \mathsf{C}^{\mathrm{tor}}\mathrm{d}^*$ on the exact curvatures: S_0F is the primitive of F in the range of d^*, and $\mathrm{d}\mathsf{S}_0 = 1$. The a-members of $\hat E$ are the $\mathsf{a} = \mathsf{S}_0F + \mathrm{d}\lambda$ with F a curvature whose plaquette averages of the level K vanish; in the Gaussian measure F and λ are independent, with the covariances $\Pi_0^{\mathrm{tp}} - \Pi_K^{\mathrm{tp}}$ and $(\mathsf{G}^{\mathrm{tor}})^2$. So the covariance of a is $C_{\hat E} = \mathsf{C}^{\mathrm{tor}} - \mathsf{S}_0\Pi_K^{\mathrm{tp}}\mathsf{S}_0^{\mathsf{T}}$, with $\mathrm{d}C_{\hat E}\,\mathrm{d}^* = \Pi := \Pi_0^{\mathrm{tp}} - \Pi_K^{\mathrm{tp}}$; and the response is $\mathsf{a}_\omega = \mathsf{S}_0\mathsf{H}_K^{\mathrm{cv}}\omega$ plus a constant 1-form, with $\mathrm{d}\mathsf{a}_\omega = \mathsf{H}_K^{\mathrm{cv}}\omega$, the curvature response of the level K, and $\mathrm{d}^*\mathsf{a}_\omega = 0$.

The kernels $\mathsf{C}^{\mathrm{tor}}$, $\mathrm{d}\mathsf{C}^{\mathrm{tor}}$ and $\Pi_0^{\mathrm{tp}} = \mathrm{d}\mathsf{C}^{\mathrm{tor}}\mathrm{d}^*$ are Fourier sums $\mathsf{N}^{-3}\sum_{k\neq 0} e^{ik\cdot z}\sigma(k)$ over the non-zero modes of the torus, z the difference of the base points, with the symbols σ of the lines of $\mathbb{Z}^3$ (Lem. 7.24) at the gauge parameter 1, of the classes $\mathcal{S}^{-2}$, $\mathcal{S}^{-1}$, $\mathcal{S}^0$; by Lem. 7.38(a) they are at most $c_{\mathrm{T}}\langle z\rangle^{-1}$, $c_{\mathrm{T}}\langle z\rangle^{-2}$, $c_{\mathrm{T}}\langle z\rangle^{-3}$, with one number c_{T} for all sides. With $|\Pi_K^{\mathrm{tp}}(p,q)| \le CL^{-3K} = C\ell^3\mathsf{N}^{-3}$ uniformly in the volume (Prop. YM3_I:120.5) and $\sum_z\langle z\rangle^{-2} \le 14\mathsf{N}$ over the torus, the corrections $\mathsf{S}_0\Pi_K^{\mathrm{tp}}\mathsf{S}_0^{\mathsf{T}}$, $\Pi_K^{\mathrm{tp}}\mathsf{S}_0^{\mathsf{T}}$ and Π_K^{tp} are at most $c\,\ell^3\mathsf{N}^{-1}$, $c\,\ell^3\mathsf{N}^{-2}$ and $c\,\ell^3\mathsf{N}^{-3}$, every distance of the torus is below N, and $|\Pi| \le 1$, Π being the kernel of an orthogonal projection (the orthogonal projections in YM3_I:(2226)).

So, with $\mathsf{c}_0 = c\,\ell^3$, the lines obey

$$|C_{\hat E}(b,b')| \le \mathsf{c}_0\langle b-b'\rangle^{-1}, \qquad |\mathrm{d}C_{\hat E}(p,b)| \le \mathsf{c}_0\langle x_p - b\rangle^{-2}, \qquad |\Pi(p,q)| \le \min\{1, \mathsf{c}_0\langle x_p - x_q\rangle^{-3}\},$$

and the kernels g, D formed with $\mathsf{G}^{\mathrm{tor}}$ obey the majorants of (7.xxxi) in the same way. The response has $|\mathsf{a}_\omega| \le C\ell L^{-K}$ and $|\mathrm{d}\mathsf{a}_\omega| \le CL^{-2K}$ in the sup norm of ω: the latter from the row sums $\le CL^{-2K}$ of the response kernel at every admissible side (equation YM3_I:(2234), whose row sum follows by summing its exponential kernel bound), the former from these and the row sums $\le c\,\mathsf{N}$ of S_0.

The Gaussian member against $\mathbb{Z}^3$. Let $\mathcal{L}_{\mathsf{N}}$ be a Fourier sum over the non-zero modes of the torus of side N with a symbol of the class $\mathcal{S}^s$, $s > -3$, and $\mathcal{L}_\infty$ the kernel of $\mathbb{Z}^3$ with the same symbol. Then $\sum_{v\in\{0,1\}^3}\mathcal{L}_{2\mathsf{N}}(z+\mathsf{N}v) = \mathcal{L}_{\mathsf{N}}(z)$ exactly, as $\sum_v e^{ik\cdot\mathsf{N}v}$ is 8 on the modes of the smaller torus and 0 on the others; for $|z|_\infty \le \frac34\mathsf{N}$ the seven images with $v\neq 0$ are at most $C\mathsf{N}^{-3-s}$ each, by Lem. 7.38(a), and telescoping over the sides $2^i\mathsf{N}$, with Lem. 7.38(b), gives $|\mathcal{L}_{\mathsf{N}}(z) - \mathcal{L}_\infty(z)| \le C\mathsf{N}^{-3-s}$.

With the three corrections from Π_K^{tp}, each line difference in the range $|z|_\infty \le 3\mathsf{N}/4$ is bounded by $C\mathsf{N}^{-1/2}$ times its majorant with the decay exponent lowered by $1/2$. We use the graph list of Lems. 7.26 and 7.27 to sum these majorants with one weakened line on either the torus or $\mathbb{Z}^3$. For the one-position graphs, the two decay exponents at the summed vertex have sum at least $7/2$ after weakening, and Hölder's inequality applies. In T_1, bound the line between the two summed vertices by a constant, leaving two such one-position sums. For the two-position chains of T_{I}, the unweakened decay powers of the three convolution factors are $(2,4,2)$, $(2,3,3)$, $(3,2,3)$, or $(3,3,2)$, up to reversal. If one is lowered by $1/2$, the sum of the powers capped at 3 is still strictly greater than 6. Choose three reciprocal Young exponents summing to 2, each at most 1 and strictly below its power divided by 3. The three norms are uniform on the finite torus and on $\mathbb{Z}^3$. These arguments bound each graph sum by $C\mathsf{c}_0^5$.

The terms with all positions in the box of sup radius $\mathsf{N}/4$ about x_p correspond on the two lattices and differ by at most $C\mathsf{N}^{-1/2}\mathsf{c}_0^5$ by telescoping the line differences. A term with a position outside this box has a long edge along a path from that vertex to an external vertex. There are at most two edges in a shortest such path, and both external vertices are within $\mathsf{N}/8$ of x_p. Thus an edge has length at least $\mathsf{N}/16$. Extracting $\mathsf{N}^{-1/2}$ from that edge gives the same bound on the tail. Therefore $|\mathsf{B}_1^{\hat E} - \mathsf{B}_1^{\mathrm{lat}}| \le C\ell^{15}(\ell L^K)^{-1/2}$.

Estimates for the constraint vertices. Set $T = L^K$, so that $\mathsf{N} = \ell T$. For a component B of a bare curl or a bare bond, denote the covariance direction $\mathrm{Cov}(e,B)$ by h_F or h_A, respectively. Use $\|\cdot\|_r$ for the maximum of the componentwise L^r norms. For $u = L^j$ write $e_j = Q_j^s\mathsf{a}$ and use the

corresponding notation for the averaged potential component of any input direction. The absolute values of the averaged curl and bond covariance directions are bounded at a point x of the mesh u by

$$P_{F,u}(x) = \frac{C\mathsf{c}_0 u}{(u + \operatorname{dist}(x, x_p))^2}, \qquad P_{A,u}(x) = \frac{C\mathsf{c}_0 u}{u + \operatorname{dist}(x, b_-)}.$$

These estimates follow by summing the bare bounds over a box of diameter $O(u)$ when the source meets an enlargement of that box. Otherwise the bare bound varies by a bounded factor on the box. Each straight average has entries $O(u^{-2})$ and row sum u. Since $C_{\hat{E}} \le \mathsf{C}^{\mathrm{tor}}$ as quadratic forms, the free kernel bound gives $\sup_x \|e_j(x)\|_{L^r} \le C_r u^{1/2}$. In fact its variance is bounded by $Cu^{-4} \cdot u^3 \cdot u^2 = Cu$ by summing the free covariance on the box. The response at mesh u has bound $C\ell u/T$ times the norm of its input.

For an average from mesh u to mesh v, the entries and column sums are $O((u/v)^2)$ and the row sums are $O(v/u)$. The column bound follows from bounded overlap of supports of diameter $O(v)$ at mesh v. For the iterate from mesh Lu to v the same bounds apply up to constants depending on L. The quadratic term of the composed map is the sum of the one-step quadratic vertices evaluated on linearly averaged inputs and carried to the top by these averages. The cubic term has in addition the quadratic vertices with one input replaced by the accumulated quadratic term at that mesh.

In a box of diameter $O(T)$ on the mesh u, the sums of products of the two profiles, divided by c_0^2, are bounded by Cu^{-2} for FF, $Cu^{-1}(1+\log(T/u))$ for FA, and CT/u for AA, uniformly in the two centers. For FF use Cauchy–Schwarz and the summability of $(1 + \operatorname{dist}/u)^{-4}$. For FA use Hölder's inequality with exponents $3/2$ and 3, for which the two powers both have logarithmic sums. For AA use Cauchy–Schwarz and the truncated sums of $(1 + \operatorname{dist}/u)^{-2}$.

At the top, the quadratic term of w is proportional to $[\mu(c_-), \mathrm{d}\mu(c)]$. Every monomial in its cubic term, written in $\mu(c_-), \mathrm{d}\mu(c)$, contains a difference slot because w vanishes when the endpoints agree. The top linear constraint determines $\mathrm{d}\mu$ and an undifferenced slot costs at most $C\ell$ times the bound on $\mathrm{d}\mu$ by summing along a path from o. For $e, h_F, h_A, R\omega$, respectively, the bounds on the difference are $C_r T^{1/2}$, $C\mathsf{c}_0/T$, $C\mathsf{c}_0$, and $C\ell\|\omega\|_\infty$. For the random arguments we use maximum componentwise L^r norms, with constants allowed to depend on the fixed moment order. All applications of the response to a random input use its absolute row sums in this norm.

The resulting bounds for q_2 are as follows, with unit inputs to R in the last three applicable entries:

inputs	bound	inputs	bound
e, e	$(K+\ell)T$	e, R	$\ell^2 T^{1/2}$
h_F, e	$\ell\mathsf{c}_0 T^{-1/2}$	h_F, h_F	$\ell\mathsf{c}_0^2 K T^{-2}$
h_F, h_A	$\ell\mathsf{c}_0^2 T^{-1}$	h_A, h_A	$\ell\mathsf{c}_0^2$
h_F, R	$\ell^2\mathsf{c}_0 T^{-1}$	R, R	ℓ^3

For the entries with two deterministic profiles use the three product sums and the outer weight $O((u/T)^2)$. For a single curl profile use $\sum P_{F,u} \le C\mathsf{c}_0 T/u^2$ on the outer support. In particular, $q_2(h_F, e)$ costs $C_r\mathsf{c}_0 u^{1/2}/T$ at each mesh before summation. The other entries use the outer row sum $O(T/u)$ together with the moment and response bounds. The top terms use the difference estimates above.

We also need

$$\|q_3(e,e,e)\|_r \le C_r(K+\ell^2)T^{3/2}, \qquad \|q_3(h_F, h_F, R\omega)\|_\infty \le C\mathsf{c}_0^2\ell^4 K T^{-2}\|\omega\|_\infty.$$

For the first bound the quadratic iterate at mesh $u = L^j$ costs $C_r j u$. Its product with the third field, multiplied by the outer row sum $O(T/u)$, is bounded by $C_r j T u^{1/2}$. Sum over j and include the direct cubic term and the top term $O_r(\ell^2 T^{3/2})$.

For the second bound, the inner two-curl-line iterate at mesh u has spatial ℓ^1 norm at most $C\mathsf{c}_0^2 j/u^2$ by the column bound. The inner curl-line/response iterate at a point of that mesh is bounded by $C(\ell u/T)P_{F,u}$ for a unit response input. Indeed, for $v \le u$ the sum of the finer curl

profile over a ball of radius $O(u)$ centered a distance d from the source is $O(\mathsf{c}_0 u^3/[v^2(u+d)^2])$. Multiply by the averaging weight $O((v/u)^2)$ and by $\ell v/T$, then sum geometrically in v. The two possible nestings are thus bounded at mesh u by $C\mathsf{c}_0^2\ell(1+j)u/T^3$, using the outer weight $O((u/T)^2)$ and the two-curl product sum. The direct cubic term obeys the same bound. Summing gives $C\mathsf{c}_0^2\ell K/T^2$. Even the crude estimate of the top term using three undifferenced slots gives $C\mathsf{c}_0^2\ell^4/T^2$.

The connected cumulants. Throughout the following estimates, bounded local moments follow from $C_{\hat{E}} \le \mathsf{C}^{\mathrm{tor}}$ and $|\Pi| \le 1$. In particular, $\|\mathrm{d}\zeta\|_r \le C_r(K+\ell)/T$. Hölder's inequality gives $|t_1| \le C\ell K/T$ and $|t_2| + |t_5| \le C\ell^2K^2/T^2$. The cubic bound and the estimate for $q_2(e, R\omega)$ give $\|\mathrm{d}\zeta_3\|_r \le C_r\ell^3K/T^{1/2}$ and hence $|t_4| \le C\ell^3K/T^{1/2}$.

For t_3, integration by parts against a quadratic curl observable places two derivatives in covariance directions of the same colour. Both hitting the local cubic slots give zero by the alternating colour tensor. Both hitting q_2 give zero for the same reason. The split is bounded by the h_F, e estimate followed by the potential response $O(\ell/T)$, giving $|t_3| \le C\ell^5/T^{3/2}$.

For t_6 and t_7, write E for the curl–bond covariance. The norms of the covariance rows satisfy

$$\|E\|_1 \le C\mathsf{c}_0\mathsf{N}, \qquad \|E\|_2 \le C\mathsf{c}_0, \qquad \|\Pi\|_1 \le C\mathsf{c}_0(1+\log\mathsf{N}), \qquad \|\Pi\|_2 \le 1.$$

The first three follow by summing the powers of distance above. The last follows from the projection identity. Each local vertex of S_3 has one curl and two bond legs with three distinct colours by Lem. 7.26(d).

In t_6, internal pairs in either cubic entry vanish by colour, and an internal pair in the quadratic entry disconnects the graph. The two cubic entries have two edges between them and one edge each to the quadratic. If the explicit curl in $DX_0[\zeta]$ meets the quadratic entry, q_2 has AA or FA inputs from the action vertex. The remaining row to sum is Π or E, respectively. Including $\mathrm{d}R$, these terms cost at most $C\ell^{11}K/T^2$. If the explicit curl meets the action vertex, q_2 has FF or FA inputs. The two remaining action rows are E, E or E, Π, respectively. Cauchy–Schwarz gives $C(\ell^{13}K/T^4 + \ell^{10}/T^3)$. The term with the plaquettes exchanged obeys the same bounds.

In t_7, the derivative of S_3 replaces either its curl by $\mathrm{d}Rq_2$ or a bond by Rq_2. If there is a direct p–q edge, the other external legs must split between q_2 and the local slots, since otherwise an internal pair has equal colours in an alternating tensor. For a curl replacement use the FA bound and sum one E row. For a bond replacement use the FF bound and sum an E row, or use the FA bound and sum a Π row. The total cost is at most $C\ell^{12}K/T^2$. Without a p–q edge all four external legs hit the action derivative. The FF bound applies to q_2 and Cauchy–Schwarz sums the other two rows. The costs for the two replacements are bounded by $C(\ell^{13}K/T^4 + \ell^{11}K/T^3)$.

In t_8, $\mathrm{d}^*R = 0$ so the energy insertion is a sum of squares of $\mathrm{d}Rq_2$. With a p–q edge the other two external legs must split across the quadratic factors, since a self-pair within q_2 vanishes. The bound is $C\mathsf{N}^3(T^{-2}\ell\mathsf{c}_0T^{-1/2})^2$. Without a p–q edge there are four external attachments and the bound is $C\mathsf{N}^3(T^{-2}\ell\mathsf{c}_0^2KT^{-2})^2$. Together these give $C(\ell^{11}/T^2 + \ell^{17}K^2/T^5)$.

Finally, in t_9, t_{10}, t_{11} the connected graph on the three quadratic entries is a triangle. It evaluates the polarized quadratic insertion on two curl covariance directions, with a factor $\Pi(p,q)$ of modulus at most 1. For N_2, the cubic bound and the R, R bound applied to $q_2(Rq_2(h_F, h_F), R\,\cdot)$ give row norm $C\mathsf{c}_0^2\ell^4K/T^2$. The trace costs $O(\ell^3)$, so $|t_9| \le C\ell^{13}K/T^2$. For N_1^2, multiply the two h_F, R row norms and use the same trace bound to get $|t_{10}| \le C\ell^{13}/T^2$. The top Haar insertion sums $O(\ell^3)$ products of path values bounded by $C\ell\mathsf{c}_0/T$, giving $|t_{11}| \le C\ell^{11}/T^2$.

Summing the eleven bounds gives

$$\sum_{i=1}^{11} |t_i| \le C\big[\ell^{13}KT^{-1/2} + \ell^{18}T^{-2} + \ell^{19}K^2T^{-5}\big].$$

Combining this with the Gaussian comparison proves the stated estimate. If $\ell^3 \le T$, both $\ell^{18}T^{-2}$ and $\ell^{19}K^2T^{-5}$ are bounded by $\ell^{29/2}KT^{-1/2}$. □

Proposition 6.19 (One lattice is enough)**.** Let $0 < g \le \gamma$ and $\ell \in \mathfrak{L}(g)$. There are integers $K_0, K_\flat$ and numbers $C_\diamond < \infty$, $d_{\mathrm{K}} > 0$, depending on g, ℓ and the constants, such that, with $K_\sharp(d) = \max\{K_0, K_\flat, \lceil 3600 \log_L(\sqrt{3}M/d)\rceil\}$: for $0 < d \le 1$, $20d \le \ell$, $\vartheta \in (0,1)$, $(f,h) \in \mathfrak{T}_{T_\ell}(d;\vartheta)$, $b \in [\gamma^{-2}, L^2(g^{-2}+\bar{B})]$ and $K' \ge K \ge K_\sharp(d)$,

$$|S^T_{2;K',\ell,L^{K'}b} - S^T_{2;K,\ell,L^K b}|(f,h) \le C_\diamond \vartheta^{-2} d^{178}\eta,$$

also with $S_2^{(\ell,b)}$ for the first member; and for $0 < d \le d_{\mathrm{K}}$, $s(d) < K_\sharp(d)$ and $(\sqrt{3}M/d)^{3600} \le L^{K_\sharp(d)} \le d^{-3601}$. ■

Proof sketch. Cor. YM3_I:117.1, equation YM3_I:(2207) gives, for $K' \ge K \ge K_1(\mathsf{d}_\infty)$ and the labels of the interval $[\gamma^{-2}, L^2(g^{-2}+\bar{B})]$,

$$|S^T_{2;K'} - S^T_{2;K}|(f,h) \le \mathsf{C}'\ell^9 K^{\mathsf{p}_2} L^{-K/10}\|f\|_\sharp\|h\|_\sharp, \qquad \|f\|_\sharp = \max\{\|f\|_\infty, 40M \operatorname{Lip} f\} \le 40M\|f\|_\infty/(\vartheta d),$$

with C' and $\mathsf{p}_2 = 6p_0 + \frac{1}{2}$ the constants of that statement; the limit $K' \to \infty$ exists along the full sequence. Here $\mathsf{d}_\infty \ge d/\sqrt{3}$ is the sup distance of the supports, and $K_1(\mathsf{d}_\infty)$ (the constants and thresholds in Thm. YM3_I:AR) is the least integer $\ge K_0$ from which on, with $u = \lceil K/1800\rceil$, the two threshold inequalities at the end of the proof of Thm. YM3_I:AR hold: $\mathsf{d}_\infty L^u \ge 6M$, and $c_9\ell^3 L^{3u} e^{-\mathsf{d}_\infty L^u/M} \le \frac{1}{2}\mathsf{C}\ell^9 K^{\mathsf{p}_2} L^{-K/10}$, with the numbers c_9, C of the two threshold inequalities at the end of the proof of Thm. YM3_I:AR at $n = 2$: C is the constant YM3_I:(2206), and $c_9\ell^3 L^{3u}$ is the factor of $e^{-\mathsf{d}_\infty L^u/M}$ in the left member of the second condition.

Let $K \ge K_\sharp(d)$ and $y = K/1800$. The second condition follows, in logarithms, from $\mathsf{d}_\infty L^u/M \ge c_{10} + (3u + \frac{K}{10})\log L$, $c_{10} = \max\{0, \log(2c_9/\mathsf{C})\}$, and this from the two inequalities $L^{y/2} \ge c_{10} + 3\log L + 200y\log L$, which defines $K_\flat = \lceil 1800 y_0\rceil$ through the least y_0 from which on it holds, and $L^{y/2} \ge \sqrt{3}M/d$, which is $K \ge 3600\log_L(\sqrt{3}M/d)$; the first condition follows from the same product. So $K_1(\mathsf{d}_\infty) \le K_\sharp(d)$, and $K^{\mathsf{p}_2}L^{-K/10} \le cL^{-K/20} \le c(d/\sqrt{3}M)^{180}$, which gives the factor $C_\diamond\vartheta^{-2}d^{178}$. By the minimality of $s(d)$, $L^{s(d)}d < LC_{W,0}A_0^2(\log L^{s(d)} + \log b_+)^{2p_0}$, so that $L^{s(d)} \le d^{-2}$ for small d; and $(\sqrt{3}M/d)^{3600} \le L^{K_\sharp(d)} < L(\sqrt{3}M/d)^{3600} \le d^{-3601}$ for small d. □

Proposition 6.20 (The agreement of states)**.** Let $0 < g \le \gamma_W^{\mathrm{sd}}$, (K,β) a datum at g with $K \ge 3$ and label b, $\vartheta \in (0,1)$, and $0 < d \le \frac{1}{180}$ with $a_K \le 9d$. A *state at g at (K,β)* is the periodically extended torus measure $\tilde{\mu}_{K,\ell',\beta}$ of a side $\ell' \in \mathfrak{L}(g)$, or a periodic Gibbs state at (a_K,β); $S_2^T[\tilde{\mu}_{K,\ell',\beta}](\tilde{f},\tilde{h})$ is $S^T_{2;K,\ell',\beta}$ at the periodisations of $\tilde{f},\tilde{h}$ on $T_{\ell'}$. Put $\mathsf{d}_\star(g,\vartheta) = 4\sqrt{3}M\vartheta^{-1}(K_{\mathrm{L}}^\vee/K_{\mathrm{Q}}^\vee)^2 g^{-2}$, with the pair $K_{\mathrm{L}}^\vee, K_{\mathrm{Q}}^\vee$ of equation YM3_I:(2166). There are $\mathsf{Q}_2^{\mathrm{T}}(g) \ge 1$ and $C_2^{\mathrm{T}}(g) < \infty$, depending on g, on the fixed radius ϱ and the common sourced choice of Section YM3_I:82 specified at the beginning of this section, and on no K, β, side, state, d, ϑ, pair or position, with the following property.

Let $(\tilde{f},\tilde{h})$ be real Lipschitz functions on $\mathbb{R}^3$, neither 0, each vanishing off a closed ball of radius d, with disjoint sets $\{\tilde{f} \ne 0\}$, $\{\tilde{h} \ne 0\}$, both vanishing off one ball of radius $9d$, and with $\operatorname{Lip}\tilde{f} \le \|\tilde{f}\|_\infty/(\vartheta d)$, $\operatorname{Lip}\tilde{h} \le \|\tilde{h}\|_\infty/(\vartheta d)$; no sign and no lower bound of the integrals is asked. Then for every state $\mathfrak{s}$ at g at (K,β)

$$|S_2^T[\mathfrak{s}] - S_2^T[\tilde{\mu}_{K,\ell_g,\beta}]|(\tilde{f},\tilde{h}) \le (30d)^3 \cdot 2\,\mathsf{Q}_2^{\mathrm{T}}(g)^2\,\mathsf{A}_\star(d)^{-2}\eta,$$

with a number $\mathsf{A}_\star(d)$ of the order $(b\vartheta d)^{1/2}$ formed with the datum; and for $d \le \mathsf{d}_\star(g,\vartheta)$, in particular for the pairs of $\mathfrak{T}_{\mathbb{R}^3}(d;\vartheta)$,

$$|S_2^T[\mathfrak{s}] - S_2^T[\tilde{\mu}_{K,\ell_g,\beta}]|(\tilde{f},\tilde{h}) \le C_2^{\mathrm{T}}(g)\vartheta^{-1}d^2 b^{-2}\eta. \tag{6.ix}$$

Any two states at g at (K,β) differ at such pairs by at most twice this. ■

Proof sketch. Let $\ell' = \ell_g = 2L^m$, $\mu' = \tilde{\mu}_{K,\ell',\beta}$ and $\mathsf{l} = \ell'/2 = L^m$. Let x_1 be the centre of the ball of radius $9d$ off which $\tilde{f}$ and $\tilde{h}$ vanish, and let $o \in a_K\mathbb{Z}^3$ be a site with $|o_\mu - (x_1)_\mu| \le \frac{1}{2}a_K$ for

$\mu = 1, 2, 3$. Consider the board of the least torus: the eight cells $\square_{\mathbf{k}} = o + \mathsf{l}\mathbf{k} + [-\frac{\mathsf{l}}{2}, \frac{\mathsf{l}}{2}]^3$, $\mathbf{k} \in \{0, 1\}^3$, with the reflections $x_\mu \mapsto 2o_\mu + \mathsf{l} - x_\mu$ in the planes $x_\mu = o_\mu \pm \frac{\mathsf{l}}{2}$, which carry one cell into another. As $o \in a_K \mathbb{Z}^3$ and l is a multiple of a_K, these reflections are maps of the lattice, and they carry plaquettes to plaquettes.

The source families used below are those of Prop. 4.42, at the common sourced choice fixed in this section. Its numerical source conditions are the ones retained in §4.2. The radius condition for Prop. 4.43 holds at the present coupling: by eq. 2.iii, $g \le \gamma_W^{\mathrm{sd}} \le \gamma_{\mathrm{ev}}$, and every running $x_t \ge g^{-2}$ has $\varepsilon(x_t) \le c_A \alpha_0/(10 M B_3)$. Also $\mathsf{l} \ge L^2 M$ by the least-side definition, so $L^{m+K-t} \ge L^2 M \ge 10M + 5$. Both entries in the minimum defining $d_*(t) + 2$ in eq. 4.xiii are at least 5, and $d_*(t) \ge 3$. The calibrated rarity hypotheses are those of the same real history, as in Prop. 4.54. Thus Cor. 4.45 applies without a further restriction on this datum.

As $1 + \frac{1}{2} a_K < \frac{1}{2}\mathsf{l}$, the ball $\mathbb{B}(x_1, 1)$ lies in the interior of $\square_0$; the signed sums of (3) below vanish off it. Its images under the reflections are eight balls of radius 1, one in each cell, with the centres $o_\mu + \mathsf{l} k_\mu + (-1)^{k_\mu}(x_1 - o)_\mu$ ($\mu = 1, 2, 3$); any two of these centres differ, in a coordinate μ in which their cells differ, by $\pm(\mathsf{l} - 2(x_1 - o)_\mu)$ modulo $2\mathsf{l}$, so that they lie at a sup distance at least $\mathsf{l} - 2|(x_1 - o)_\mu| \ge \mathsf{l} - a_K$ on $T_{\ell'}$.

(1)

One cell against the least torus. For a bounded complex Borel function F of the inner bonds of one cell, the reflected copies, conjugated on odd reflection parity, satisfy

$$\|F\|_{\mathrm{ch}}^8 = \mu'\Big(\prod_{\mathbf{k} \in \{0,1\}^3} F^{[\mathbf{k}]} \Big) \ge 0.$$

For $\ell \in \mathfrak{L}(g)$, [7, Cor. B.1] gives

$$|\tilde{\mu}_{K,\ell,\beta}(F)| \le \|F\|_{\mathrm{ch},\ell} \le \|F\|_{\mathrm{ch}} \exp[\mathsf{l}^3 (p_{\ell'} - p_\ell)].$$

[7, Lem. N.1] bounds $p_{\ell'} - p_\ell$ by $C_{\mathrm{gap}} + C_{\mathrm{S}} + C_g \le 2C_{\mathrm{S}} + C_g$. Thus one may take $\mathsf{Q}_0 = \exp[\mathsf{l}^3(2C_{\mathrm{S}} + C_g)] \ge 1$. For a periodic Gibbs state, Prop. YM3_I:91.9 gives

$$|\mathfrak{s}(F)| \le \liminf_{r \to \infty} \|F\|_{\mathrm{ch}, \ell_g L^r} \le \mathsf{Q}_0 \|F\|_{\mathrm{ch}}.$$

The constants are those of [7, Lem. N.1], with $C_{\mathrm{S}} \ge C_{\mathrm{gap}}$ chosen there. In particular Q_0 is independent of the cutoff and the larger side.

This is the one place where Thms. YM3_I:AJ and YM3_I:AD enter, and the place through which $\mathsf{Q}_2^{\mathrm{T}}(g)$ and $C_2^{\mathrm{T}}(g)$ depend on the radius and on the choice of the longer list.

(2)

Only linking cumulants are left. With $X_a = \mathcal{O}(F_a)$, $\mathcal{P}_a = X_a - \mu'(X_a)$ and $\mathcal{P}_{12} = \mathcal{P}_1 \mathcal{P}_2 - S_2^T[\mu'](F_1, F_2)$ one has $S_2^T[\mathfrak{s}](F_1, F_2) - S_2^T[\mu'](F_1, F_2) = \mathfrak{s}(\mathcal{P}_{12}) - \mathfrak{s}(\mathcal{P}_1)\mathfrak{s}(\mathcal{P}_2)$. Expanding $\mu'(\prod_{\mathbf{k}} \mathcal{P}_{12}^{[\mathbf{k}]})$ in joint cumulants of the sixteen copies $X_{a,\mathbf{k}}$, the blocks that stay in one cell cancel exactly, because $\mathcal{P}_{12}$ is centred by the covariance of μ' itself; what is left is a sum, over the at most Bl_{16} partitions into linking blocks (blocks that meet two cells), of products of linking cumulants, Bl_n being the Bell numbers.

So if every linking cumulant of the copies is at most $C_{\mathrm{LK}} \prod u_a$, then $|S_2^T[\mathfrak{s}] - S_2^T[\mu']|(F_1, F_2) \le [\mathsf{Q}_0 \mathsf{Bl}_{16}^{1/8} + \mathsf{Q}_0^2 \mathsf{Bl}_8^{1/4}] C_{\mathrm{LK}} u_1 u_2 \le 2\mathsf{Q}_1^2 u_1 u_2$, $\mathsf{Q}_1 = \mathsf{Q}_0 \mathsf{Bl}_{16}^{1/8} C_{\mathrm{LK}}$.

(3)

The packing. Apply (2) to the signed sums $F_a^\epsilon = \sum_v \epsilon_v \tau_v f_a$, $f_1 = \tilde{f}$, $f_2 = \tilde{h}$, over a packing of translates $\mathbb{B}(x_1 + v, 9d)$, $v \in a_K \mathbb{Z}^3$, in the ball $\mathbb{B}(x_1, 1)$, with at least $(30d)^{-3}$ members; the mean over the signs and the invariance of every state under the lattice translations give $|S_2^T[\mathfrak{s}] - S_2^T[\mu']|(\tilde{f}, \tilde{h}) \le (30d)^3 \cdot 2\mathsf{Q}_1^2 u_1 u_2$.

(4)

The linking cumulants under μ'. Their generating function is $Z[W_z]/Z$ on $T_{\ell'}$, with the source $W_z = \sum_{\mathbf{k},a} z_{\mathbf{k},a} F^{\epsilon}_{a,\mathbf{k}} \circ x$ read on the torus, $F^{\epsilon}_{a,\mathbf{k}}$ the copy in the cell $\mathbf{k}$. By the property of the level-wise class recalled in the proof of Prop. 6.7, with $b^{\pm}_K = x_K \pm \bar{B}$ in the place of $b^{\vee}$, $b^{\wedge}$, the source lies in the class with norm at most $\max |z_{\mathbf{k},a}| u_a$, $u_a = \|f_a\|_\infty / \mathsf{A}_\star(d)$. Write $\log(Z[W_z]/Z) = \log \Omega_8(z) + [\mathcal{E}(W_z) - \sum_{\mathbf{k}} \mathcal{E}(w^{(\mathbf{k})})] + \sum_{\mathbf{k}} \mathcal{E}(w^{(\mathbf{k})})$, $\Omega_8 = Z[W_z] e^{-\mathcal{E}(W_z)}/Z$, $w^{(\mathbf{k})}$ the part of W_z in the cell $\mathbf{k}$.

A linking derivative kills the last sum. Apply Cor. 4.45 to the eight reflected balls and their sixteen variables. Its hypotheses on the bumps hold because reflections preserve their amplitude, Lipschitz bounds and disjoint nonzero sets. The same amplitude calculation in Lem. 4.44, now with $b^-_K = x_K - \bar{B}$ and $b^+_K = x_K + \bar{B}$, applies: these bound $a_{K-j} x_j$ below and above. Thus the square-free derivatives of Ω_8 at zero are at most $\mathsf{Q}_{\mathrm{D}} \prod u_a$, with $\mathsf{Q}_{\mathrm{D}} = Q_{8,16}(g)$.

We estimate the bracket in the generating function by using the local energies of Prop. 4.42:

$$e_j(X; w) = \sum_{z \in X \cap \mathbb{T}^{(j)}} \Delta F_j[w](X, \mathbf{1}, z).$$

They depend only on the source plaquettes in X and vanish at zero. The counting calculation in [7, eq. 213], applied to eq. 4.ix, gives

$$|e_j(X; w)| \le 16 M^3 E_{\mathrm{s}} \|w\|_{\mathrm{lw}} e^{-\kappa d_j(X)}.$$

Here E_{s} is the fixed family ceiling of §4.2. If X meets source plaquettes in at most one reflected ball, its contribution to $\mathcal{E}(W_z) - \sum_{\mathbf{k}} \mathcal{E}(w^{(\mathbf{k})})$ is zero: locality identifies the one possible nonzero term, and all other terms vanish. Only domains meeting two balls remain.

The centres of the reflected balls are at sup distance at least $\mathsf{l} - a_K$. Their source barycentres are therefore at distance at least $\mathsf{l} - a_K - 2$, and a corner is within $a_K/2$ of its barycentre. Thus a domain that contributes contains two points at distance at least $\mathsf{l} - 3$, since $K \ge 3$ and $L \ge 13$ imply $2a_K \le 1$. Lem. 5.24 gives

$$d_j(X) \ge \frac{\mathsf{l} - 3}{M D_j} - 2 \ge 12 L^{K-j} - 2, \qquad D_j = L^{j-K}.$$

For a wrapped domain the bound follows instead from $M(d_j(X) + 2) \ge \ell'/(2D_j) = \mathsf{l}/D_j$, which is stronger. In the last inequality we used $\mathsf{l} \ge L^2 M$ and $M \ge 5$. The least-side formula YM3_I:(1528) gives this side bound: its maximum includes $L^{\lceil \log_L C_{\mathrm{s}} \rceil} \ge L$. In particular the displayed lower bound is positive also at $j = K$.

Put $r = \max_{\mathbf{k},a} |z_{\mathbf{k},a}| u_a < 1$. The full source and each cell source have level-wise norm at most r, so the contribution of one domain is at most $9 \cdot 16 M^3 E_{\mathrm{s}} r e^{-\kappa d_j(X)}$. A ball of radius 1 meets source plaquettes whose corners lie in a cube of side $2 + a_K \le 3$. There are at most $(3L^{K-j}/M + 2)^3 \le 27 L^{3(K-j)}$ level-j cubes containing these corners. The eight balls therefore require at most $216 L^{3(K-j)}$ such cubes. Apply [7, Lem. B.10] with $t = j$, $q = 0$ and $D = 12L^{K-j} - 2$. Its rate condition holds because $\delta_{\mathrm{w}} \kappa > \kappa_{\mathcal{D}} + 2$ and $0 < \delta_{\mathrm{w}} < 1/2$ in equation YM3_I:(877). The extra factor $\#(X \cap \mathbb{T}^{(j)})$ in that lemma is at least one. Summing first in X and then in j gives

$$\left| \mathcal{E}(W_z) - \sum_{\mathbf{k}} \mathcal{E}(w^{(\mathbf{k})}) \right| \le \mathsf{C}_\Delta r, \qquad \mathsf{C}_\Delta = 248832 M^6 E_{\mathrm{s}} C_{\mathrm{E},1} \sum_{n \ge 0} L^{3n} e^{2 - 12 L^n}.$$

The series converges, since the ratio of successive terms is $L^3 e^{-12(L-1)L^n} \to 0$. Cauchy's formula on the source polydisc bounds each nonempty square-free derivative of this bracket by $\mathsf{C}_\Delta \prod u_a$.

So $C_{\mathrm{LK}} = \mathsf{cu}_{16} \mathsf{Q}_{\mathrm{D}}^{16} + \mathsf{C}_\Delta$, with a combinatorial number cu_{16}, and $\mathsf{Q}^{\mathrm{T}}_2(g) := \mathsf{Q}_1$ gives the first display.

(5) For $d \le \mathsf{d}_\star(g, \vartheta)$ the second member of the minimum that defines $\mathsf{A}_\star(d)$ is the smaller one, so that $\mathsf{A}_\star(d)^{-2} = (K^{\vee}_{\mathrm{L}}/\mu_\star)^2 (b^+_K/(b^-_K)^2)\, 4\sqrt{3} M/(\vartheta d)$, and $b^+_K/(b^-_K)^2 \le b^{-2}(X_g + \bar{B})^3/(g^{-2} - \bar{B})^2$. With $30^3 \cdot 2 = 54000$ this is eq. 6.ix with $C^{\mathrm{T}}_2(g) = 54000 [\mathsf{Q}^{\mathrm{T}}_2(g) (K^{\vee}_{\mathrm{L}}/\mu_\star)(X_g + \bar{B})^{3/2} (g^{-2} - \bar{B})^{-1} (4\sqrt{3} M)^{1/2}]^2$.

The last assertion is the triangle inequality through μ'. Of this proposition the proof of clause (c) uses eq. 6.ix alone, for one torus measure against the least torus. □

Proposition 6.21 (One datum on two sides)**.** Let $0 < g \le \gamma_{\mathrm{s}}$, (K,β) a datum at g with $K \ge 3$ and label b, $8a_K \le d \le L^{-2}$, $\vartheta \in (0,1)$, $s \ge 0$, $(f,h) \in \mathfrak{T}_{T_{\ell_g}}(d;\vartheta)$, and $(\breve f, \breve h)$ the pair on $T_{\ell_g L^s}$ with the same lift; the datum, with its one run, is taken on both tori. Put $\dagger = \log(10LMb/d)$.

(i) For every level $1 \le j \le K$, $\mathsf{n}_j[\ell_g](f,h) - \mathsf{n}_j[\ell_g L^s](\breve f,\breve h) = \mathsf{R}_j[\ell_g] - \mathsf{R}_j[\ell_g L^s]$, where $\mathsf{R}_j[\ell]$ is the part of the sum that defines n_j on T_ℓ over the domains X that meet both supports and are not the image of a domain of the infinite lattice lying in a cuboid of sides $< \frac12 \ell_g$; each such domain has linear size $\ge \ell_g/(2MD_j) - 2$.

(ii) On every side of $\mathfrak{L}(g)$, at every level j with $D_j > Ld/8$,

$$|\mathsf{n}_j(f,h)| \le C'_{\mathsf{n}}\, \dagger^{3p_0} \Big(\frac{d}{D_j}\Big)^3 [b^{-3/2} D_j^{-1/2} + b^{-1} d^3 D_j^{-4}]\,\eta.$$

(iii) $|\mathsf{J}_2[\ell_g](f,h) - \mathsf{J}_2[\ell_g L^s](\breve f,\breve h)| \le C_{\mathrm{V}}\, \dagger^{3p_0}\, b^{-1} d^3\, \eta$.

C'_{n} and C_{V} are determined by the constants; the bounds depend on no s, K, ϑ or pair beyond η, and no bound on L^K and no witness depth is asked. ■

Proof sketch. If a source vanishes, all the terms vanish. (i) A domain of the level j of the infinite lattice that lies in a cuboid of sides $< \frac12 \ell_g$ and meets the lifted supports projects onto exactly one domain of each of the two lattices (ℓ_g, K) and $(\ell_g L^s, K)$, with their levels; these do not wrap, and it is a near lift of each. Every domain of either lattice that meets both supports and is not of this kind either wraps the torus or lies in no such cuboid, and in both cases its linear size is $\ge \ell_g/(2MD_j) - 2$. These are the following facts on the localisation domains (Lem. 5.24). The projection of a domain of the infinite lattice is a domain of the lattice (ℓ, K); a domain of the lattice (ℓ, K) does not wrap if and only if it is the projection of a domain of the infinite lattice that lies in a cuboid of lengths less than half the side; such domains are its near lifts, and they are translates of one another by periods and pairwise disjoint; the projection keeps the linear size $d_j(X)$, the number of edges of a least edge tree of X in the units of the level j (Section YM3_I:72 (the definitions of $\mathcal{D}_j$ and d_j at its beginning)); every domain of the infinite lattice lies in a cuboid of lengths $\le MD_j(d_j(X)+2)$; and on T_ℓ a domain with $M(d_j(X)+2) < \ell/(2D_j)$ does not wrap.

A cuboid of lengths $< \frac12 \ell_g$ has lengths less than half the side of either lattice. On a near lift the two pairs of lattice sources agree: the lifted supports lie in one ball of radius $6d$, and their translates by non-zero periods are more than $\frac12 \ell_g$ away. On a domain that does not wrap the local terms of the weighted progression are then the same number on the two tori. This is proved by induction over the levels: the regular progression without sources is one family for all the lattices (ℓ, K) of the level (Thm. YM3_I:Y, including its common-companion assertion); one step of the progression is natural under the chart of the domain from one lattice to the other, the two lattices having one K: bare arrays of the two lattices that correspond under the chart (the common-input condition at the beginning of Section YM3_I:52.5) give local terms of the next level that are defined together and are equal (equation YM3_I:(928), including its separately equal three summands); and the bare arrays of the two lattices correspond under this chart, their members of the lower levels by the induction over the levels and their weight members because the sources agree on the plaquettes of the domain.

So the terms of these domains cancel in the difference, which is (i).

(ii) The first jets of the coarse levels gain the factor $(d/D_j)^3$. Here $\dot E_a^{(j)}$ and $\ddot E^{(j)}$ are the derivatives $\partial_{\mu_a} E_\mu^{(j)}$ and $\partial_{\mu_1}\partial_{\mu_2} E_\mu^{(j)}$ at $\mu = 0$ of the local terms of the weighted progression, and $\|\cdot\|$ is the size YM3_I:(759) of a family of the level j at the rate $\kappa^+ = (1+4\beta_{\mathrm{s}})\kappa$, the rate at which the part of the local terms that depends on the sources decays in the linear size of the domain (Prop. YM3_I:122.1);

κ is the decay rate and $\beta_{\mathrm{s}} \in (0, \frac{1}{5}]$ the enlargement fraction among the constants ([7, Def. B.7]), and $\|\cdot\|_{j,\kappa} \le \|\cdot\|$ is the size at the rate κ.

The members of $\dot{E}_a^{(i)}$ vanish on the domains that hold no plaquette of $\operatorname{supp} w_a$, and those of $\ddot{E}^{(i)}$ on the domains that do not hold a plaquette of each support (Prop. YM3_I:122.1(c)). The cubes of the level i are the cubes of side MD_i of the partition π_i of the torus whose wall-connected unions are the domains of the level i (the level-i localization partition of Section YM3_I:45); for a set $\mathcal{S}$ of plaquettes, $\mathsf{N}_i(\mathcal{S})$ is the number of the cubes of the level i in which a corner of a plaquette of $\mathcal{S}$ lies.

The level j_1. Let r be the least sup distance between a corner of a plaquette of $\operatorname{supp} w_1$ and a corner of a plaquette of $\operatorname{supp} w_2$. The barycentres of the plaquettes of $\operatorname{supp} w_1$ and of $\operatorname{supp} w_2$ lie in $\operatorname{supp} f$ and $\operatorname{supp} h$, which lie in balls of radius d with centres at most $8d$ apart and are at a Euclidean distance at least d, and a corner lies at the sup distance $\frac{1}{2}a_K \le \frac{1}{16}d$ from the barycentre; so $\frac{2}{5}d \le \mathsf{r} < 11d$. Let j_1 be the least level $j \ge 1$ with $\mathsf{r} < 3MD_j$. As $3MD_{K-1} = 3M/L > 11d$, it exists and $j_1 \le K-1$; and $\mathsf{r} \ge 3MD_j$ at every level $1 \le j < j_1$.

As a consequence of this definition, D_{j_1} lies in the interval $(\frac{2d}{15M}, \frac{Ld}{8}]$: $D_{j_1} > \mathsf{r}/3M \ge \frac{2d}{15M}$; and $D_{j_1} \le \frac{Ld}{8}$, since $D_{j_1} = LD_{j_1-1} \le L\mathsf{r}/3M < \frac{11Ld}{3M} \le \frac{Ld}{8}$ if $j_1 \ge 2$, and $D_{j_1} = La_K \le \frac{Ld}{8}$ if $j_1 = 1$. The level j_1 is formed on each of the two tori; the bounds below are the same on both. All the corners of the plaquettes of $\operatorname{supp} w_a$ lie in a cube of side $\frac{17}{8}d$, so that $\mathsf{N}_i(\operatorname{supp} w_a) \le (\frac{17d}{8MD_i} + 2)^3$; as $d/MD_{j_1} < \frac{15}{2}$, this is less than 18^3 for $i \ge j_1$, and less than $18^3 L^{3(j_1-i)}$ for $i \le j_1$.

So the bare array of the first derivative in the direction w_a sits at few places at the levels from j_1 on. For such an array the size of the family that one step forms is bounded with the weights $L^{-4(j-i)}$ times the number of the places, where the general size has $L^{3(j-i)}$ places times $L^{-4(j-i)}$: if the counterterms of a bare array are 0, and its members $G^{(i)}$, $i \le j$, and its weight member $H^{(j)}$ vanish on the domains that hold no plaquette of a set $\mathcal{S}$ of plaquettes, then the family that the rule of the step $j+1$ forms of it has a size at most

$$c_\sharp \Big[\sum_{i \le j} L^{-4(j-i)}\, \mathsf{N}_i(\mathcal{S})\, \|G^{(i)}\|_{i,\kappa} + \mathsf{N}_j(\mathcal{S})\, \|H^{(j)}\|_{j,\kappa} \Big],$$

$H^{(j)}$ being the weight member and $c_\sharp$ a number determined by the constants. The proof is that of the bound of the bracket localisation of a bare array (the proof of Prop. YM3_I:51.16, especially the rooted-domain count and equation YM3_I:(881)), with the count of the places changed: as the counterterms vanish, the members enter the rule as they stand; a member of the level i enters the rule of the height j either on a domain whose linear size is at least $\max\{2L^{j-i-1} - 2, 0\}$, which costs $e^{-\max\{2L^{j-i-1}-2,0\}} \le L^{4-4(j-i)}$, or through its octet function, which lies within $\mathsf{C}_{\mathrm{oc}} \|G^{(i)}\|_{i,\kappa} (d_i(X)+2)^2 L^{4-4(j-i)}$ of a constant (Prop. YM3_I:51.14, at fixed radii), where C_{oc} is the larger of 1 and its constant C_{oc}; the weight member enters at the level j alone; and the sum over the domains that hold a plaquette of $\mathcal{S}$ is bounded by [7, Lem. B.10], which gives the factor $\mathsf{N}_i(\mathcal{S})$ in the place of the $L^{3(j-i)}$ places of the general bound.

Two bare arrays are of this kind. The bare array of the first derivative in the direction w_a at the height j has the members $\dot{E}_a^{(i)}$, $i \le j$, the counterterms 0, since the counterterms do not depend on the sources (Prop. YM3_I:122.1(e)), and as weight member that of the source $L^{K-j} w_a$, whose members sit at the domains $X(p)$ of the plaquettes $p \in \operatorname{supp} w_a$, each of which holds its plaquette; it is of this kind with $\mathcal{S} = \operatorname{supp} w_a$. The bare array of the second derivative has the members $\ddot{E}^{(i)}$, $i \le j$, the counterterms 0 and the weight member 0; it is of this kind with $\mathcal{S} = \operatorname{supp} w_1$.

The step depends on its bare array through the size of this family: by the Lipschitz bound and the holomorphy of the step in the family that the rule forms (Thm. YM3_I:V, including equation YM3_I:(899)), the first derivative of the step at a bare array of the progression in the direction of an array is at most θ_j times the size of the family formed of that array, and the mixed second derivative in the directions of two arrays at most $2\theta_j / N_\star^{[j]}$ times the product of the two sizes,

by Cauchy's estimate; here $\theta_j = \theta^{[j]} = K_{\mathbf{M}}(1+\beta_{\mathrm{s}})\delta(x_j)/\alpha_0$ and $N_\star^{[j]}$ are the numbers of [7, eq. 196], $K_{\mathbf{M}} \geq 1$ is the step constant and α_0 the first fixed radius in [7, Def. B.7].

Every running coupling of the datum lies above the threshold x_{J} of the definitions preceding [7, eq. 228] (Lem. YM3_I:126.2, as in the proof of Prop. 6.5), and the conditions [7, eq. 228] that define x_{J} hold at every $x \geq x_{\mathrm{J}}$, as δ decreases there; so $C_0^\flat \mathsf{S}_{\mathrm{A}}\theta_j \leq 1$ and $\frac{4}{3}C_1^\flat L\mathcal{T}(3)\theta_j \leq 1$ at every height $j < K$, with the numbers $C_0^\flat$, $C_1^\flat$ of [7, §B.5], S_{A} of [7, eq. 198] and the function $\mathcal{T}(u) = \sum_{n\geq 1}(n+1)^{2p_0+1}e^{-4\beta_{\mathrm{s}}\kappa[L^{n-1}u-2]_+}$, written T in [7, eq. 199], all determined by the constants.

The first jets. Fix a; let $\dot{\mathsf{B}}_j = \sum_{i\leq j} L^{-4(j-i)}\mathsf{N}_i(\operatorname{supp} w_a)\|\dot{E}_a^{(i)}\|$ be the weighted sum of the sizes of the first jets of the levels up to j, and v_j the size of the weight member of the level j. For $j_1 \leq j \leq K-1$ the bound just stated, with $\mathsf{N}_j(\operatorname{supp} w_a) < 18^3$, gives the family formed of the bare array of the first derivative at the height j a size at most $c_\sharp[\dot{\mathsf{B}}_j + 18^3\mathsf{v}_j]$; the first derivative of the step gives $\|\dot{E}_a^{(j+1)}\|$ at most θ_j times this size; and $\dot{\mathsf{B}}_{j+1} = L^{-4}\dot{\mathsf{B}}_j + \mathsf{N}_{j+1}(\operatorname{supp} w_a)\|\dot{E}_a^{(j+1)}\|$.

The power $d^3D_j^{-4}$ comes from two inputs. The first is the inhomogeneous term: the weight member of the level j is the action term $\eta_j^{-1}\sum_p L^{K-j}w_a(p)\,a_p$, $\eta_j = L^{-j}$ being the bare spacing in the units of the level j; as $\operatorname{supp} w_a$ has at most $3(\frac{17}{8})^3(d/a_K)^3 < 29(d/a_K)^3$ plaquettes, as $|a_p| < \frac{1}{2}\alpha_0^2\eta_j^4$ at the fields of the radii (α_0, α_1), and as the domains $X(p)$ have the linear size 0, its size is $\mathsf{v}_j \leq \frac{29}{2}\alpha_0^2 d^3 D_j^{-4}|w_a|_\infty$. The second is the start: by the general bound of the first jets, $\|\dot{E}_a^{(i)}\| \leq C_0^\flat\mu_{i-1,a}$ with $\mu_{i,a} = 12M^3\alpha_0^2\theta_i|w_a|_\infty/D_i$ (Prop. YM3_I:122.1, equation YM3_I:(2258), [7, eq. 198]), by $\mu_{i-1,a} \leq (\frac{4}{3})^{1/2}L^{n/2}(1+n)^{p_0}\mu_{j_1-1,a}$ for $n = j_1 - i \geq 0$ (the running-coupling comparison following equation YM3_I:(2258)) and by $\mathsf{N}_i(\operatorname{supp} w_a) < 18^3L^{3n}$, the powers of L leave the convergent series $\sum_n(1+n)^{p_0}L^{-n/2}$, and $\dot{\mathsf{B}}_{j_1} \leq C\mu_{j_1-1,a} \leq C'\alpha_0^2|w_a|_\infty/D_{j_1} \leq C'(\frac{L}{8})^3\alpha_0^2|w_a|_\infty d^3D_{j_1}^{-4}$, as $\theta_{j_1-1} \leq (C_0^\flat\mathsf{S}_{\mathrm{A}})^{-1}$ and $D_{j_1} \leq \frac{Ld}{8}$.

With $\hat{\mathsf{v}} = \frac{29}{2}\alpha_0^2 d^3 D_{j_1}^{-4}|w_a|_\infty$, so that $L^{4(j-j_1)}\mathsf{v}_j \leq \hat{\mathsf{v}}$ for $j \geq j_1$, the three relations give

$$L^{4(j+1-j_1)}\dot{\mathsf{B}}_{j+1} + 18^3\hat{\mathsf{v}} \leq (1 + L^4\cdot 18^3 c_\sharp\theta_j)\,(L^{4(j-j_1)}\dot{\mathsf{B}}_j + 18^3\hat{\mathsf{v}}) \qquad (j_1 \leq j \leq K-2).$$

So it is the quantity $L^{4(j-j_1)}\dot{\mathsf{B}}_j + 18^3\hat{\mathsf{v}}$, and not the weighted sum itself, that grows from the level j_1 on by at most the factor $\prod_i(1 + L^4\cdot 18^3 c_\sharp\theta_i) \leq \exp[L^4\cdot 18^3 c_\sharp\sum_i\theta_i]$; and $\sum_{i<K}\theta_i \leq \mathsf{S}_{\mathrm{B}}\theta_{K-1} \leq \mathsf{S}_{\mathrm{B}}/(C_0^\flat\mathsf{S}_{\mathrm{A}})$ (the running-coupling comparison following equation YM3_I:(2258), with the number S_{B} of [7, eq. 198]). Hence $\dot{\mathsf{B}}_j + 18^3\mathsf{v}_j \leq L^{-4(j-j_1)}(L^{4(j-j_1)}\dot{\mathsf{B}}_j + 18^3\hat{\mathsf{v}}) \leq Cd^3D_{j_1}^{-4}L^{-4(j-j_1)}\alpha_0^2|w_a|_\infty = Cd^3D_j^{-4}\alpha_0^2|w_a|_\infty$; the family formed of the bare array of the first derivative at the height j has a size at most $C_{\mathrm{F}}\alpha_0^2|w_a|_\infty d^3D_j^{-4}$, and the recursion of the first jets closes as $\|\dot{E}_a^{(j+1)}\| \leq \theta_j C_{\mathrm{F}}\alpha_0^2|w_a|_\infty d^3D_j^{-4}$ for $j_1 \leq j \leq K-1$, with a number C_{F} determined by the constants.

The second jets. At the levels up to j_1 the general bound holds: $\|\ddot{E}^{(i)}\| \leq C_1^\flat\varpi_{i-1}$ for $1 \leq i \leq j_1$, where $\varpi_i = c_\nu\delta(x_i)^2|w_1|_\infty|w_2|_\infty D_i^{-2}$ is the number written ν_i in [7, eq. 199] and c_ν is determined by the constants. This is the induction of Prop. YM3_I:122.2, carried up to the level j_1: its condition $\frac{4}{3}C_1^\flat L\theta_j\mathcal{T}(\mathsf{r}/MD_j) \leq 1$ holds at the levels $1 \leq j < j_1$, as $\mathsf{r}/MD_j \geq 3$ there and $\mathcal{T}$ decreases (the monotonicity in [7, eq. 199]). Let $\mathsf{B}_j = \sum_{i\leq j} L^{-4(j-i)}\mathsf{N}_i(\operatorname{supp} w_1)\|\ddot{E}^{(i)}\|_{i,\kappa}$ collect the second jets of the levels up to j, at the rate κ.

By the two derivatives of the step, at the arrays of the first and of the second derivative, the second jets obey $\|\ddot{E}^{(j+1)}\| \leq \nu_j^\sharp + \theta_j c_\sharp\mathsf{B}_j$, where $\nu_j^\sharp \leq C\delta(x_j)^2 d^6 D_j^{-8}|w_1|_\infty|w_2|_\infty$ for $j \geq j_1$ comes from the product of the two first jets; and $\mathsf{B}_{j+1} = L^{-4}\mathsf{B}_j + \mathsf{N}_{j+1}(\operatorname{supp} w_1)\|\ddot{E}^{(j+1)}\|_{j+1,\kappa}$. The recursion has the same form as that of the first jets. The start: at a level $i < j_1$ a domain that holds plaquettes of both supports has a linear size at least $\mathsf{r}/MD_i - 2$, so that $\|\ddot{E}^{(i)}\|_{i,\kappa} \leq e^{-4\beta_{\mathrm{s}}\kappa[\mathsf{r}/MD_i-2]_+}\|\ddot{E}^{(i)}\|$ (equation YM3_I:(2259)), with $\mathsf{r}/MD_i \geq 3L^{j_1-1-i}$; with $\varpi_{i-1} \leq \frac{4}{3}L^n(1+n)^{2p_0}\varpi_{j_1-1}$ and $\mathsf{N}_i(\operatorname{supp} w_1) < 18^3L^{3n}$, $n = j_1 - i$, the powers of L cancel and the series $\sum_{n\geq 1}(1+$

$n)^{2p_0}e^{-4\beta_s\kappa[3L^{n-1}-2]_+}$ converges, so that $\mathsf{B}_{j_1} \le C\varpi_{j_1-1} \le C'(\log x_{j_1-1})^{2p_0}|w_1|_\infty|w_2|_\infty/(bD_{j_1})$, by $x_{j_1-1} \ge \frac34 b/D_{j_1-1}$.

The quantity $L^{4(j-j_1)}\mathsf{B}_j$ grows from one level to the next by at most the factor $1+L^4\cdot 18^3 c_\sharp\theta_j$, beside the inhomogeneous terms $L^{4(j+1-j_1)}18^3\nu_j^\sharp$, whose sum over $j \ge j_1$ is at most

$$C(\log x_{j_1-1})^{2p_0}b^{-1}|w_1|_\infty|w_2|_\infty d^6 D_{j_1}^{-7} \le C'(\log x_{j_1-1})^{2p_0}|w_1|_\infty|w_2|_\infty/(bD_{j_1}),$$

as $\delta(x_j)^2 \le \frac43 A_1^2(\log x_j)^{2p_0}D_j/b$, $\sum_{j\ge j_1} D_j^{-3} \le D_{j_1}^{-3}/(1-L^{-3})$ and $d/D_{j_1} < \frac{15}{2}M$. So, as for the first jets,

$$\mathsf{B}_j \le C(\log x_{j_1-1})^{2p_0}b^{-1}|w_1|_\infty|w_2|_\infty d^3 D_j^{-4}$$

for $j \ge j_1$, by $D_{j_1} \le \frac{Ld}{8}$; and $\|\ddot E^{(j+1)}\| \le \nu_j^\sharp + \theta_j c_\sharp \mathsf{B}_j$, with $\theta_j \le C(\log x_j)^{p_0}(D_j/b)^{1/2}$, gives

$$\|\ddot E^{(j)}\| \le C\dagger^{3p_0}[b^{-3/2}d^3D_j^{-7/2} + b^{-1}d^6D_j^{-7}]|w_1|_\infty|w_2|_\infty$$

for $j > j_1$, as $\log x_j \le \log x_{j_1-1} < \dagger$ for $j \ge j_1 - 1$, by $x_{j_1-1} \le \frac{12}{11}b/D_{j_1-1}$ (proof of Prop. 6.5(ii)) and $D_{j_1-1} = D_{j_1}/L > \frac{2d}{15LM}$.

A level with $D_j > \frac{Ld}{8} \ge D_{j_1}$ lies above j_1, and the count of the domains that meet a support ([7, Lem. B.10], at most 18^3 cubes of the level j holding a corner of a plaquette of $\operatorname{supp} w_1$) gives (ii).

(iii) By (i) the difference is at most $\sum_j(|\mathsf{R}_j[\ell_g]| + |\mathsf{R}_j[\ell_g L^s]|)$. On each torus the local second jets decay like $e^{-\kappa^+ d_j(X)}$ in the linear size $d_j(X)$, with the rate $\kappa^+ = (1+4\beta_s)\kappa$ of (ii) (Prop. YM3_I: 122.1); and the count of the domains of linear size at least R that meet a support, $\sum_X \#(X\cap\mathbb{T}^{(j)})\,e^{-\kappa^+ d_j(X)} \le c\,\nu\,e^{-R}$ over the domains X with $d_j(X) \ge R$ that meet the ν cubes of the level j in which a corner of a plaquette of the support lies ([7, Lem. B.10], as κ^+ exceeds the number $\kappa_{\mathcal D}+2$ that it asks), taken at $R = A/D_j - 2$, gives $|\mathsf{R}_j| \le C\|\ddot E^{(j)}\|e^{2-A/D_j}$, $A = \ell_g/(2M) \ge L$.

For the levels above j_1, (ii) and $\sum_{n\ge0} L^{7n}e^{-AL^n} \le \frac85 e^{-A}$ give $C\dagger^{3p_0}b^{-1}d^3\eta$. For the levels $j \le j_1$ the second jets grow at most like $L^n(1+n)^{2p_0}$, $n = j_1 - j$, by their general bound $C_1^\flat\varpi_{j-1} \le \frac43 C_1^\flat L^n(1+n)^{2p_0}\varpi_{j_1-1}$ up to the level j_1 in the proof of (ii), against $e^{-A/D_j} \le e^{-8/d}e^{-8L^2(L^n-1)}$ and $e^{-8/d} \le \frac{3}{512}d^4$; this gives again at most $C\dagger^{2p_0}b^{-1}d^3\eta$. Double for the two tori. □

Lemma 6.22 (Two sides at the order six)**.** Let $0 < g \le \gamma_s$, (K,β) a datum at g, $\ell \le \breve\ell$ in $\mathfrak L(g)$, $d \le L^{-1}$, $s(d) < K$, $20d \le \ell$, $(f,h) \in \mathfrak T_{T_\ell}(d;\vartheta)$, and $(\breve f,\breve h)$ the pair on $T_{\breve\ell}$ with the same lift. Then

$$|\mathsf{h}_6[\ell](f,h) - \mathsf{h}_6[\breve\ell](\breve f,\breve h)| \le |\mathsf{J}_2[\ell](f,h) - \mathsf{J}_2[\breve\ell](\breve f,\breve h)| + \Big[(2C_{\mathrm E}b^{-1}d\,l^{8p_0} + 2\bar\varepsilon_4(g,d))b^{-3}d + 3C_{\mathrm{tm}}\vartheta^{-1}b^{-2}\frac{a_K}{d}\Big]\eta.$$

■

Proof sketch. On each side, $\mathsf{h}_6 = \mathsf{J}_2 - \mathsf{h}_4 - [\mathsf{J}_2 - \mathsf{h}_4 - \mathsf{h}_6]$. The bracket is at most $C_{\mathrm E}b^{-1}dl^{8p_0}\cdot b^{-3}d\,\eta$ on each side (Prop. 6.6(ii)), and h_4 is $c_2b^{-2}\mathcal K_6$ up to $[\bar\varepsilon_4 b^{-3}d + C_{\mathrm{lat}}b^{-2}a_K/d]\eta$ on each side (Prop. 6.8, with one $\bar\varepsilon_4$ for all sides and data at g); $\mathcal K_6$ and η are the same for the two pairs. Subtract. □

Proposition 6.23 (Two lattices with one bare torus)**.** Let $0 < g \le \gamma_s$; $D_0 = (K_0,\beta_0)$ a datum at g with label b; $d \le L^{-1}$, $s(d) < K_0$; $(f,h) \in \mathfrak T_{T_{\ell_g}}(d;\vartheta)$, $81d \le \ell_g$, $r = dL^{K_0}$; $b_* \in I(g)$; and integers $K' \ge K_0$ and $2 \le n_0 \le n_1$ with $n_0 \le K_0$, $n_1 \le K'$, $L^{n_0} \ge r$. Put $\breve\ell = \ell_g L^{K'-K_0}$, $N' = \ell_g L^{K'} = \breve\ell L^{K_0}$, $D' = (K', L^{K'}b_*)$, $l' = \max\{1, \log(2b_*L^{K'}/r)\}$, $\mathsf N' = N' + \max\{\beta_0, L^{K'}b_*\}$, and let $(\breve f,\breve h)$ be the pair on $T_{\breve\ell}$ with the same lift. Let the two conditions of Prop. 6.12(ii) hold at the three places $(D_0,\breve\ell;n_0)$, $(D',\ell_g;n_0)$ and $(D',\ell_g;n_1)$, and let $x_\flat$ denote the running coupling x_{n-1} of the place; and let $\sigma \ge 0$ be a number with $|\mathsf{J}_2[\ell_g](f,h) - \mathsf{J}_2[\breve\ell](\breve f,\breve h)| \le \sigma b^{-3}d\eta$ at D_0. Then, with $\tilde w$ the lattice sources of

the lift on $\mathbb{Z}^3$,

$$|\mathsf{H}^{(K_0)}[D_0,\ell_g]-\mathsf{T}^{\mathrm{lat}}(\tilde w)|\le\Big\{[\sigma+2C_{\mathrm E}b^{-1}dl^{8p_0}+2\bar\varepsilon_4(g,d)+C_{\mathrm R}(l^{6p_0}+l'^{6p_0})(r/L^{n_0})^{3/2}]r \\ +\frac{3C_{\mathrm{tm}}b}{\vartheta d}+\sum_{\text{three places}}\mathcal{U}(x_\flat,\mathsf{N}')+2700C_\star(\ell_gL^{K'-n_1})^{19}n_1L^{-n_1/2}r^6\Big\}\eta.$$

■

Proof sketch. If $\tilde w_1$ or $\tilde w_2$ vanishes, both numbers are 0. Consider three places: (1) the datum D_0 on T_{ℓ_g} with (f,h); (2) the datum D_0 on $T_{\breve\ell}$ with $(\breve f,\breve h)$, where $\breve\ell\in\mathfrak{L}(g)$; (3) the datum D' on T_{ℓ_g} with $f'=\breve f\circ\mathsf{D}^{-1}$, $h'=\breve h\circ\mathsf{D}^{-1}$, $\mathsf{D}:T_{\breve\ell}\to T_{\ell_g}$ the dilation $x\mapsto L^{-(K'-K_0)}x$. D' is a datum at g, as $b_*\in I(g)$; f',h' are real and bounded, with the norms of f,h, with supports in balls of radius $d'=rL^{-K'}$ at a distance in $[d',6d']$, and $8a_{K'}\le d'$ as $r\ge8$; so Prop. 6.5 applies at the three places, as it asks no depth.

D carries the bare lattice of $(\breve\ell,K_0)$ onto that of (ℓ_g,K'), plaquettes to plaquettes and barycentres to barycentres: in bare units the sources of (2) and of (3) are one pair of functions on the plaquettes of $\Lambda_{N'}$, the periodisation of $\tilde w$, and r is the same at the three places. The chain has seven links. 1. $\mathsf{H}^{(K_0)}[D_0,\ell_g]\to\mathsf{H}^{(K_0)}[D_0,\breve\ell]$: Lem. 6.22 with σ, multiplied by $\beta_0^3a_{K_0}^2=b^3/a_{K_0}$. 2. $\to\mathsf{H}^{(n_0)}[D_0,\breve\ell]$: Prop. 6.6(iii) at (2). 3. $\to\mathsf{T}[N',n_0]$: Prop. 6.12(i), (ii), with Lem. 6.13, and Prop. 6.15 at (2), as for the link X of Step 3 below. 4.

$\to\mathsf{H}^{(n_0)}[D',\ell_g]$: the same at (3); the number $\mathsf{T}[N',n_0]$ is one, as it depends on N', n_0, L and the lattice sources alone, and the two runs that meet in it leave no trace, by Prop. 6.14(iii). 5. $\to\mathsf{H}^{(n_1)}[D',\ell_g]$: Prop. 6.6(iii) at (3), where l is l' and $d'/D'_{n_0}=r/L^{n_0}$, $D'_{n_0}=a_{K'-n_0}$ being the spacing of the level n_0 of the lattice (ℓ_g,K'). 6. $\to\mathsf{T}[N',n_1]$: Prop. 6.12(i), (ii), with Lem. 6.13, and Prop. 6.15 at (3). 7. $\to\mathsf{T}^{\mathrm{lat}}(\tilde w)$: Prop. 6.18 on the lattice with $\ell'=N'/L^{n_1}=\ell_gL^{K'-n_1}$ top sites and n_1 levels, where $\ell'\ge\ell_g\ge N_{\mathsf H}$, $\ell'L\ge C_LM$, and all base points lie within $10r+1$ of each other, $10r+1\le\frac18N'$ as $81d\le\ell_g$ and $r\ge8$; at most $900r^6$ pairs.

In link 1, $b^3a_{K_0}^{-1}\cdot b^{-3}d=r$ and $b^3a_{K_0}^{-1}\cdot3C_{\mathrm{tm}}\vartheta^{-1}b^{-2}a_{K_0}/d=3C_{\mathrm{tm}}b/(\vartheta d)$. Add the errors of the seven links. □

Lemma 6.24 (The splines of Def. 6.2)**.** Let $q,q'\ge1$ be odd.

(i) $N_3(t)=\frac34-t^2$ for $|t|\le\frac12$, $N_3(t)=\frac12(\frac32-|t|)^2$ for $\frac12\le|t|\le\frac32$, and $N_3(t)=0$ beyond; Φ is of class C^1, $0\le\Phi\le(\frac34)^3$ and $\{\Phi\ne0\}=(-\frac32,\frac32)^3$, so $|x|<2.6$ on $\operatorname{supp}\Phi$; $\int\Phi=1$, $\int\Phi x_\mu=0$ and $\int\Phi|x|^2=\frac34$.

(ii) $\Phi(x/q)=\sum_j c_j^q\Phi(x-j)$, with $c_j^q\ge0$ and $c_j^q=0$ unless $|j_\mu|\le\frac32(q-1)$ for all μ; $\sum_j c_j^q=q^3$, $\sum_j c_j^qj=0$, $\sum_j c_j^q|j|^2=\frac34q^3(q^2-1)$; hence $q^{-1}\sum_j c_j^qP(\frac{y+j}{q})=P(y)$ for all y.

(iii) For $q\ge3$, $\Upsilon^{(q)}$ is of class C^1 with support in $\{7\le|y|\le11q\}$, and the distance of the supports of Φ and $\Upsilon^{(q)}$ exceeds 4.4; $0\le\Upsilon^{(q)}\le64q^2$, and $\Upsilon^{(q)}=63.25$ on $|y|=8$.

(iv) $e(u)=|u|^2$ for $|u|\le4.4$, $e(u)=0$ for $|u|\ge10.6$, $0\le e(u)\le|u|^2$ otherwise, and $\int\Phi(x)\Upsilon^{(q)}(x+u)\,\mathrm dx=q^2e(u/q)-e(u)$.

(v) $\mathsf k_6\in[\frac{4\pi}{10.6},\frac{4\pi}{4.4}]$, and for $q\ge3$, on $\mathbb{R}^3$: $\mathcal{K}_5(\Phi,\Upsilon^{(q)})=4\pi\log q$ and $\mathcal{K}_6(\Phi,\Upsilon^{(q)})=(1-q^{-1})\mathsf k_6$.

(vi) $\Upsilon^{(qq')}(y)=\Upsilon^{(q)}(y)+q^{-1}\sum_j c_j^q\Upsilon^{(q')}(\frac{y+j}{q})$.

■

Proof sketch. (i) N_3 is the density of the sum of three independent variables uniform on $[-\frac12,\frac12)$: integral 1, mean 0, variance $\frac14$. (ii) The intervals $[i-\frac12,i+\frac12)$, $i\in J_q$, tile $[-\frac q2,\frac q2)$, so $\mathbf1_{[-1/2,1/2)}(\cdot/q)=\sum_{i\in J_q}\mathbf1_{[-1/2,1/2)}(\cdot-i)$, and the threefold convolution gives $q^2N_3(x/q)=\sum_{i_1,i_2,i_3\in J_q}N_3(x-i_1-i_2-i_3)$; the moments of the c_j^q are those of the sum of three variables uniform on J_q, and the identity for P

follows by expanding the square. (iii) If $|y| \le 4.4q + 2.6$ and $c_j^q \neq 0$, then $|y + j| \le 7q$, so all the $\Xi(\frac{y+j}{q})$ are $P(\frac{y+j}{q})$, and by (ii) $\Upsilon^{(q)}(y) = P(y)(1 - \zeta(|y|))$, which is 0 for $|y| \le 7$ and ≥ 0 beyond.

If $|y| \ge 4.4q + 2.6$, then $\Xi(y) = 0$ and $|y + j|/q > 1.8$, where $\Xi \ge 0$. If $|y| > 10.6q - 2.6$ all the terms vanish. The upper bound follows from $\Xi \le 64 - \frac{3}{4}$. (iv) For $|u| \le 4.4$, $|x + u| < 7$ on $\operatorname{supp}\Phi$, and (i) gives $e(u) = \frac{3}{4} + |u|^2 - \frac{3}{4}$; the last formula is (ii) after the change of variables $x \mapsto x - j$. (v) With $\varepsilon(t) = t^{-2}\int_{S^2} e(t\omega)\,\mathrm{d}\omega$, which is 4π on $(0, 4.4]$ and 0 on $[10.6, \infty)$, polar coordinates and (iv) give $\mathcal{K}_p(\Phi, \Upsilon^{(q)}) = \int_0^\infty t^{4-p}[\varepsilon(t/q) - \varepsilon(t)]\,\mathrm{d}t$. At $p = 5$ the integral over $t \ge \delta$ is $\int_{\delta/q}^{\delta} \varepsilon(s)\frac{ds}{s} = 4\pi \log q$ for small δ; at $p = 6$ the substitution $t = qs$ gives $(1 - q^{-1})\mathsf{k}_6$.

(vi) $(i, i') \mapsto i + qi'$ is a bijection of $J_q \times J_{q'}$ onto $J_{qq'}$, so $c_J^{qq'} = \sum_{j+qj'=J} c_j^q c_{j'}^{q'}$. □

6.1.9. *Proof of Thm. S. Step 1: the front and the expansion.* We take $\gamma_{\mathrm{e}} = \min\{\gamma_{\mathrm{s}}, \gamma_{\mathrm{C}}\}$. The number d_{e} is the least of L^{-13}, $\frac{1}{200}$, $d_{\mathrm{A}}(g)$, d_{K} of Prop. 6.19 and of the thresholds of Steps 2, 4 and 8: the smallness of d needed for $2 \le n_0 < n_1$ and $12r + 2 \le N_0/8$, for the two conditions of Prop. 6.12(ii) at the places of the chain, and for the three inequalities on d stated at the head of Step 8. Let $D = (K, \beta)$ be a datum at g with $s(d) < K$ and $L^K \le d^{-P}$, and $(f, h) \in \mathfrak{T}_{T_\ell}(d; \vartheta)$; then $K \ge 3$, $8a_K \le d$ and $r \ge L^{s(d)}d \ge C_W(s(d)) \ge (\log \frac{1}{d})^2 \ge 8$. If w_1 or w_2 vanishes, all the jets vanish and only the last link of Step 3 is used. Props. 6.7, 6.6(ii) and 6.8, the last two at every translate $(\tau_v f, \tau_v h)$, $v \in \mathsf{Pk}$, give, as L_b^+ depends on distances alone,

$$|S_2^T - L_b^+|(f, h) \le \Big[\sum_{i=1}^{3} E_i\, b^{-3}d + C_{\mathrm{lat}} b^{-2}\frac{a_K}{d}\Big]\eta + b^{-3}a_K \max_{v\in\mathsf{Pk}} |\mathsf{H}^{(K)}[D] - \varkappa\mathcal{K}_5/a_K|(\tau_v f, \tau_v h), \quad (6.\mathrm{x})$$

with $E_1 = C_{\mathrm{A}}(g)\vartheta^{-1}bd$, $E_2 = C_{\mathrm{E}}b^{-1}dl^{8p_0}$ and $E_3 = \bar{\varepsilon}_4(g, d)$; here $\mathsf{h}_6 = b^{-3}a_K\mathsf{H}^{(K)}[D]$, and $b^{-3}a_K \cdot r\eta = b^{-3}d\eta$. It remains to bound $|\mathsf{H}^{(K)}[D] - \varkappa\mathcal{K}_5/a_K|$ in multiples of $r\eta$ at every position; we write (f, h) for the translate.

Step 2: the levels and a second datum. Put

$$K' = \max\{K, \lceil 24 \log_L r\rceil\}, \quad N_0 = \ell L^K, \quad N' = \ell L^{K'}, \quad n_1 = K' - \lceil K'/78\rceil,$$

so that $r^{24} \le L^{K'} \le \max\{d^{-P}, Lr^{24}\} \le Ld^{-24P}$. If $K' = K$ (Case A) let $D' = D$. If $K' > K$ (Case B), then $r^{24} > L^K = r/d$, that is $r > d^{-1/23}$; fix $b_* \in I(g)$ once, let $D' = (K', L^{K'}b_*)$, a datum at g (Cor. YM3_I:74.2), and let n_0 be the level with $D_{n_0} = a_{K-n_0} \in [d^{1/2}, Ld^{1/2})$. Then $L^{n_0} \in [rd^{-1/2}, Lrd^{-1/2})$; $2 \le n_0 \le K$ for $d \le L^{-2}$, as $L^{n_0} \ge 8L$ and $D_{n_0} < 1$; and $n_0 < n_1$ for small d, as $L^{n_0} < Lrd^{-1/2} < Lr^{12.5}$ and $L^{n_1} \ge L^{77K'/78-1} \ge r^{23.6}/L$. In bare units the base points of the plaquettes of the two supports lie within $12r + 2$ of each other, and $12r + 2 \le N_0/8 = \ell r/(8d)$ for small d.

The lifts $\breve{w}_a$ of Prop. 6.17 on $\Lambda_{N'}$ are the lattice sources, at D', of real bounded functions on T_ℓ with supports in balls of radius $d' = ra_{K'}$ at a distance $\ge d'$, $8a_{K'} \le d'$; Prop. 6.5 applies to them at D', as it puts no condition on the depth, the signs or the Lipschitz constants. The unit $r\eta$ is the same at D and at D', since $d'L^{K'} = r$.

Step 3: the chain. In Case B,

$$\begin{aligned}\mathsf{H}^{(K)}[D] &\xrightarrow{\mathrm{R}} \mathsf{H}^{(n_0)}[D] \xrightarrow{\mathrm{X}} \mathsf{T}[N_0, n_0](w) \xrightarrow{\mathrm{W}} \mathsf{T}[N', n_0](\breve{w}) \xrightarrow{\mathrm{X}} \mathsf{H}^{(n_0)}[D'] \\ &\xrightarrow{\mathrm{R}} \mathsf{H}^{(n_1)}[D'] \xrightarrow{\mathrm{X}} \mathsf{T}[N', n_1](\breve{w}) \xrightarrow{\mathrm{A}} a_K^{-1}\mathsf{P}_1^{\mathrm{lat}}(\tilde{f}, \tilde{h}; a_K) \xrightarrow{\mathrm{T}} \varkappa\mathcal{K}_5(f, h)/a_K. \end{aligned} \quad (6.\mathrm{xi})$$

Here R is Prop. 6.6(iii). X is Prop. 6.12(i), (ii), with Lem. 6.13, and Prop. 6.15: $\mathsf{H}^{(n)}[D] = \beta^3 a_K^2 \mathsf{h}_6^{(n)} = \beta^3 a_K^2 \gamma_2^{\mathrm{r}}$ by Prop. 6.12(i), the cumulants being taken in $\mathsf{D}^0 dB = \bigotimes_{0\le i<n} \nu_i^{\mathrm{r}}$; $\beta^3 a_K^2 \gamma_2^\circ = \mathsf{T}[N, n](w)$ by Prop. 6.15; and $\beta^3 a_K^2 |\gamma_2^{\mathrm{r}} - \gamma_2^\circ| \le \mathcal{U}(x_{n-1}, \mathsf{N})\eta$ by Prop. 6.12(ii) under the two conditions there, its proof reading the bounds of Lem. 6.13 on the boxes and the precisions $\mathsf{M}_i \ge \frac{3}{4}\gamma^{\mathrm{tr}}$ of the Gaussian measures of the heights.

So $|\mathsf{H}^{(n)}[D] - \mathsf{T}[N,n](w)| \le \mathcal{U}(x_{n-1},\mathsf{N})\eta$ under the two conditions; the two middle members $\mathsf{T}[N',n_0](\breve{w})$ are one number, as it depends on no datum. W is Prop. 6.17 at $t=n_0$, $\nu = \ell/D_{n_0} > \ell d^{-1/2}/L \ge \nu_{\rm W}$, with the base points within $12r+2 \le N_0/8$. A is Prop. 6.18 on the lattice with $\ell' = N'/L^{n_1} = \ell L^{\lceil K'/78\rceil}$ top sites and n_1 levels: $\ell' \ge \ell \ge N_{\mathsf{H}}$ and $\ell' L \ge C_L M$ by Prop. 6.5(i), the base points lie within $12r+2 \le \frac{1}{8}N' - 1$, and the sum runs over at most $900r^6$ pairs; with $\frac{19}{78} + \frac{1}{156} - \frac{1}{2} = -\frac{1}{4}$ and $r^5 \le L^{5K'/24}$ the error is E_{10} below, and $a_K \mathsf{T}^{\rm lat}(\tilde{w}) = \mathsf{P}_1^{\rm lat}(\tilde{f},\tilde{h};a_K)$ for the lift $(\tilde{f},\tilde{h}) \in \mathfrak{T}_{\mathbb{R}^3}(d;\vartheta)$.

T is Cor. 7.37 for the lift: $|\mathsf{P}_1^{\rm lat}(\tilde{f},\tilde{h};a) - \kappa^{\rm lat}\mathcal{K}_5(f,h)| \le C_{\rm T}\vartheta^{-1} a \lg(d/a)\eta$ for $0 < a \le d$, with $\kappa^{\rm lat} = \varkappa = c_1c_2$ (Thms. W, X); only here enters $c_1 = \kappa^{\rm lat}/c_2$, which depends on no L, g or label. The errors of the links, in multiples of $r\eta$, are

$$E_4 = C_{\rm R} l^{6p_0} d^{3/4}, \quad E_5 = \mathcal{U}(x_{n_0-1},\mathsf{N}_0)/r, \quad E_6 = 900 r^5 C'_{\rm W} N'^{237} e^{-\delta_{\rm tp}\ell d^{-1/2}/48},$$
$$E_7 = \mathcal{U}(x'_{n_0-1},\mathsf{N}')/r, \quad E_8 = C_{\rm R} l'^{6p_0} d^{3/4}, \quad E_9 = \mathcal{U}(x'_{n_1-1},\mathsf{N}')/r,$$
$$E_{10} = 2700 C_\star \ell^{19} L^{20} K' L^{-K'/24}, \quad E_{11} = C_{\rm T}\vartheta^{-1}\lg(r)/r, \tag{6.xii}$$

where x, x' are the running couplings of D, D'; $\mathsf{N}_0 = N_0 + \beta \le (\ell + b_+)d^{-P}$ and $\mathsf{N}' = N' + L^{K'}b_* \le (\ell+b_+)Lr^{24}$; by Prop. 6.5(ii), $x_{n_0-1} \ge \frac{3}{4}bL/D_{n_0} \ge \frac{3}{4}d^{-1/2}$, $x'_{n_0-1} \ge \frac{3}{4}r^{23}d^{1/2} \ge \frac{3}{4}\max\{r^{11.5}, d^{-1/2}\}$ and $x'_{n_1-1} \ge L^{K'/78}$; $l' \le \log(2b_+) + K'\log L$; and $(r/L^{n_0})^{3/2} \le d^{3/4}$. In Case A the chain is $\mathsf{H}^{(K)}[D] \to \mathsf{H}^{(n_1)}[D]$ by R, with the error $C_{\rm R} l^{6p_0} L^{\frac{3}{2}(1-0.94K)}$, as $r/L^{n_1} \le L^{1-0.94K}$ and $l \le \log(2b_+) + K\log L$, followed by the last three links with $D' = D$.

Step 4: clause (a). At each place of X the two conditions of Prop. 6.12(ii) hold once $\frac{1}{8}\gamma^{\rm tr}A_1^2(\log x_\flat)^{2p_0} \ge 10 + \log(10240\mathsf{N}^3)$, $x_\flat$ being the lower bound of the running coupling at that place in Step 3; and $\log\mathsf{N} \le C_1 + C_2\log x_\flat$ with $C_1 = \log((\ell+b_+)L) + P$ and $C_2 = 2P$ at the first place ($\log\frac{1}{d} \le 2\log x_\flat + 0.6$), $C_2 = 3$ at the second ($\log x_\flat \ge 11\log r$ for $r \ge 2$), $C_2 = 78$ at the third ($\log x_\flat \ge \frac{K'}{78}\log L$). As $2p_0 \ge 2$,

$$\mathcal{U}(x_\flat,\mathsf{N}) \le C_{\rm X}\exp\bigl[36C_1 + 36C_2\log x_\flat - \tfrac{1}{8}\gamma^{\rm tr}A_1^2(\log x_\flat)^{2p_0}\bigr]$$

tends to 0 faster than every power of $x_\flat^{-1}$, and $x_\flat \to \infty$ uniformly as $d \to 0$: $x_\flat \ge \frac{3}{4}d^{-1/2}$ at the first two places and $x_\flat \ge L^{s(d)/78}$ at the third. Further $E_6 \le 900 C'_{\rm W}(\ell L)^{237} d^{-5P-5688P} e^{-\delta_{\rm tp}\ell d^{-1/2}/48}$; $E_4, E_8 \le C_{\rm R}[\log(2b_+L) + 24P\log\frac{1}{d}]^{6p_0} d^{3/4}$; E_{10} and the error of Case A tend to 0 as $K' \ge K > s(d) \to \infty$; and $E_{11} \le C_{\rm T}\vartheta^{-1}\sup_{\rho \ge (\log 1/d)^2}\lg(\rho)/\rho$. As $b \in [g^{-2}, b_+]$, each E_i is bounded by a function of (g,ϑ,P,d) alone that tends to 0 with d. With $\bar{\varepsilon}_{\rm a}(d)$ the supremum over $d'' \le d$ of the sum of these bounds, eq. 6.x gives (a).

Step 5: clause (b) and the rate. Let $b \in I(g)$; every $(K, L^K b)$ is a datum at g, and $I(g) \subset [\gamma^{-2}, L^2(g^{-2} + \bar{B})]$. For $s(d) < K \le K_\sharp(d)$, (a) applies, as $L^K \le d^{-3601} \le d^{-P}$ (Prop. 6.19). For $K > K_\sharp = K_\sharp(d)$, and for the limit, Prop. 6.19 and (a) at $K_\sharp$ give, with $a_{K_\sharp} \le (d/\sqrt{3}M)^{3600}$,

$$|S^T_{2;K,\ell,L^K b} - L^+_b|(f,h) \le [\bar{\varepsilon}_{\rm a}(d) + C_\diamond\vartheta^{-2}b_+^3 d^{177} + C_{\rm lat}b_+ d^{3598}]b^{-3}d\,\eta,$$

so that $\bar{\varepsilon}(d) = \bar{\varepsilon}_{\rm a}(d) + \sup_{d''\le d}[C_\diamond\vartheta^{-2}b_+^3 d''^{177} + C_{\rm lat}b_+ d''^{3598}]$ serves in (a) and (b). For the limit we apply (a) at the single cutoff $K_\sharp$, where $r \ge d^{-3599}$ and Case B holds: E_5, E_6, E_7, E_9 are below every power of d; $E_{10}, E_{11} \le Cd^{3598}\log\frac{1}{d}$; $E_1 + E_2 \le Cd(\log\frac{1}{d})^{8p_0}$; $E_4 + E_8 \le Cd^{3/4}(\log\frac{1}{d})^{6p_0}$; and $E_3 \le C_4(g)d^{1/2}$ (Prop. 6.8). So the error of the limit is at most $c_{1/2}d^{1/2}b^{-3}d\,\eta$; its slowest member is that of the order four.

Step 6: clause (c). Let $g_{\rm a} = L^{1/4}g$ and $g_{\rm b} = L^{-1/4}g$. For $g'' \in \{g, g_{\rm a}, g_{\rm b}\}$, $g'' \le \min\{\gamma_{\rm e}, \gamma_W^{\rm sd}\}$, and $\mathsf{x} := g^{-2} \ge L^{1/2}(\gamma_W^{\rm sd})^{-2} \ge 4L^{1/2}\bar{B} \ge 5\bar{B}$. All these reference couplings use the single assignment fixed at the beginning of the section. The real constants are its unchanged restriction; hence the canonical flow and every sourced construction are evaluated at the same values. This is the

common-choice convention of Section YM3_I:82, with its radius conditions imposed uniformly for $x \geq \gamma^{-2}$.

So there is one flow of the coupling: at every coupling the running couplings of a datum, which depend on (K,β) alone, the data, the histories of the sourced chain and the torus measures are the same objects in the real restriction of the common assignment and in its sourced construction, as fixed in Section YM3_I:82; and $C_{W,0}$ and P are fixed before the couplings. Hence (a) and (b), which hold at this choice, hold at the three couplings g'', and these clauses and Prop. 6.20 speak of the same objects. The labels at g lie in $[\mathsf{x}, L(\mathsf{x}+\bar B)]$, and $I(g)$ leaves out $[\mathsf{x}, \mathsf{x}+\bar B]$ and $(L(\mathsf{x}+B_\sharp)-\bar B, L(\mathsf{x}+\bar B)]$.

As $\mathsf{x} \geq 5\bar B$ and $L \geq 13$,

$$L^{-1/2}\mathsf{x}+\bar B < \mathsf{x}, \quad \mathsf{x}+\bar B \leq L^{1/2}\mathsf{x}+LB_\sharp-\bar B, \quad L^{1/2}\mathsf{x}+\bar B \leq L(\mathsf{x}+B_\sharp)-\bar B, \quad L(\mathsf{x}+\bar B) \leq L(L^{1/2}\mathsf{x}+B_\sharp)-\bar B,$$

which place the first interval in $I(g_{\mathrm{a}}) = (L^{-1/2}\mathsf{x}+\bar B, L^{1/2}\mathsf{x}+LB_\sharp-\bar B]$ and the second in $I(g_{\mathrm{b}}) = (L^{1/2}\mathsf{x}+\bar B, L^{3/2}\mathsf{x}+LB_\sharp-\bar B]$. Let d'_{e} be the least, over the three couplings g'', of $\frac{1}{180}$, L^{-2}, $d_{\mathrm{e}}(g'')$, $\mathsf{d}_\star(g'',\vartheta)$ and $(LC_w(g''))^{-2}$, where $C_w(g'') = \sup_{s\geq 0} C_W(s)L^{-s/2}$ is formed at g''; for $d \leq (LC_w(g''))^{-2}$ the witness depth at g'' has $L^{s(d)} \leq d^{-3}$, since $L^{s-1}d < C_W(s-1) \leq C_w L^{(s-1)/2}$ for $s = s(d) \geq 1$. Let (K,β) be a datum at g with $K > s(d)$. If $L^K \leq d^{-P}$, (a) applies; if $b \in I(g)$, (b) applies.

Otherwise $b \in I(g_\bullet)$ with $g_\bullet \in \{g_{\mathrm{a}}, g_{\mathrm{b}}\}$, and $(K,\beta) = (K, L^K b)$ is a datum at $g_\bullet$ (Cor. YM3_I:74.2), with $K \geq 3$, $a_K \leq d$ and $K > s(d)$ at $g_\bullet$, since $L^K > d^{-P} \geq d^{-3}$. Clause (b) at $g_\bullet$ gives the law on $T_{\ell_{g_\bullet}}$ for the pair with the same lift as (f,h), which has the same η, $\mathcal{K}_5$ and $\mathcal{K}_6$. Of the sides ℓ_g, $\ell_{g_\bullet}$, both of the form $2L^m$, let $g_\circ \in \{g, g_\bullet\}$ be the coupling of the smaller; the larger lies in $\mathfrak{L}(g_\circ)$, so both torus measures are states at $g_\circ$ at the datum (K,β), one of them on the least torus, and eq. 6.ix at $g_\circ$ bounds their difference at the lift by

$$C_2^{\mathrm{T}}(g_\circ)\vartheta^{-1}d^2 b^{-2}\eta \leq C_2^{\mathrm{T}}(g_\circ)\vartheta^{-1}b_+ d \cdot b^{-3} d\,\eta.$$

So (c) holds with $\bar\varepsilon'(d) = \max_{g''} \bar\varepsilon_{g''}(d) + \max_{g''} C_2^{\mathrm{T}}(g'')\vartheta^{-1}b_+ d$, $\bar\varepsilon_{g''}$ being the function of (a) and (b) at g''.

Step 7: clause (d). (1) By Lem. 6.24(ii), $\Phi(x/q) = \sum_j c_j^q \Phi(x-j)$; so every bilinear form S invariant under the translations by $\rho\mathbb{Z}^3$ satisfies

$$(q\rho)^{-1}\mathsf{S}(\Phi_{q\rho}, \Xi_{q\rho}) - \rho^{-1}\mathsf{S}(\Phi_\rho, \Xi_\rho) = \rho^{-1}\mathsf{S}(\Phi_\rho, \Upsilon_\rho^{(q)}).$$

At a cutoff $K \geq k_0 + n$ the covariance $S^T_{2;K,\ell,L^K b}$ is bilinear and invariant under the translations by $a_K\mathbb{Z}^3 \supset \varrho_k\mathbb{Z}^3$, $\varrho_k = L^{-k}r_0$, whatever the supports (Setting and notation). Taken at $\rho = \varrho_k$, $q = L$, summed over $1 \leq k \leq n$, and compared with the identity at $\rho = r_1$, $q = L^n$, it gives

$$r_1^{-1} S(\Phi_{r_1}, \Upsilon_{r_1}^{(L^n)}) = \sum_{k=1}^n \varrho_k^{-1} S(\Phi_{\varrho_k}, \Upsilon_{\varrho_k}^{(L)});$$

these pairs have disjoint supports (Lem. 6.24(iii)), which lie in balls of radius $11r_0 < \ell/4$, and $K \to \infty$ is taken member by member, by the existence and the bilinearity of the limit. (2) By Lem. 6.24(v), $\mathcal{K}_5(\Phi, \Upsilon^{(q)}) = 4\pi \log q$ and $\mathcal{K}_6(\Phi, \Upsilon^{(q)}) = (1-q^{-1})\mathsf{k}_6$; with $\mathcal{K}_q(f_\rho, h_\rho) = \rho^{6-q}\mathcal{K}_q(f,h)$ and $4\pi c_1 c_2 = \frac{6}{\pi^2}$, the subtracted members of Δ_n are $\sum_{k\leq n} \varrho_k^{-1} L_b^+(\Phi_{\varrho_k}, \Upsilon_{\varrho_k}^{(L)})$, so that

$$\Delta_n = \sum_{k=1}^n \delta_k, \qquad \delta_k(b) = \varrho_k^{-1}\big[S_2^{(\ell,b)} - L_b^+\big](\Phi_{\varrho_k}, \Upsilon_{\varrho_k}^{(L)}).$$

(3) The tents $\chi_z(x) = \prod_\mu \max\{0, 1-|2x_\mu - z_\mu|\}$, $z \in \mathbb{Z}^3$, form a partition of unity and split Φ and $\Upsilon^{(L)}$, which is ≥ 0 (Lem. 6.24(iii)), into finitely many nonnegative Lipschitz pieces. Each pair

of pieces $(\chi_z\Phi, \chi_{z'}\Upsilon^{(L)})$, neither 0, lies in $\mathfrak{T}_{\mathbb{R}^3}(d_{zz'};\vartheta_0)$ with $d_{zz'} = \max\{1, \frac{1}{6}\operatorname{dist}\} \in [1, 2L]$, the distance of the two supports lying in $(4.4, 11L + 2.6)$, and with $\vartheta_0 \in (0, \frac{1}{2}]$ depending on L alone; scaled by ϱ_k and read on T_ℓ, it lies in $\mathfrak{T}_{T_\ell}(\varrho_k d_{zz'};\vartheta_0)$, since all the points of its two supports lie within $22L\varrho_k \le 22r_0 < \ell/2$ of each other and $20\varrho_k d_{zz'} \le 40r_0 < \ell$. Clause (b) at $(g, \vartheta_0, P_\sharp)$, with its d^0_{e} and $c^0_{1/2}$, gives for $2L\varrho_k \le d^0_{\mathrm{e}}$

$$|\delta_k(b)| \le C_\Psi c^0_{1/2}(2L\varrho_k)^{1/2} b^{-3}, \qquad C_\Psi = \sum_{z,z'} d_{zz'} \|\chi_z\Phi\|_\infty \|\chi_{z'}\Upsilon^{(L)}\|_\infty,$$

C_Ψ depending on L alone. (4) For the finitely many k with $2L\varrho_k > d^0_{\mathrm{e}}$, δ_k is continuous on the compact set $\overline{I(g)} \subset (0,\infty)$, by the continuity of the limit in b, so it is bounded there. (5) As $b^{-3} < g^6$ on $I(g)$, $\sum_k \sup_{I(g)} |\delta_k| < \infty$; so $\Delta_n \to \Delta_\infty = \sum_k \delta_k$ uniformly on $I(g)$, Δ_∞ is bounded and continuous, $\sup_{I(g)} |\Delta_n| \le \sum_k \sup_{I(g)} |\delta_k|$, and for n beyond the coarse scales

$$\sup_{I(g)} |\Delta_n - \Delta_\infty| \le C_\Psi g^6 c^0_{1/2}(2Lr_0)^{1/2} \frac{L^{-(n+1)/2}}{1 - L^{-1/2}}.$$

Of the rate in (b) the argument uses only that the tolerances at the scales $2L\varrho_k$ are summable over k; a tolerance of the order $1/\log(1/d)$ would not give it.

Step 8: clause (e). Here the levels are cut at the scale $d^{1/5}$, and the bare torus is changed through a change of the side at one datum. Let $\Lambda = \log\frac{1}{d}$, $l_+ = \log(2b_+) + \Lambda$, $l'_+ = \log(2b_+L) + 23(P_\sharp - 1)\Lambda$, $A_\ell = 2700 C_\star \ell_g^{19} L^{39/2}$,

$$\mathsf{N}_{\max}(d) = (\ell_g + b_+) L d^{-24(P_\sharp - 1)}, \qquad E_{\mathrm{X}}(d) = C_{\mathrm{X}} \mathsf{N}_{\max}(d)^{36} \exp\big[-\tfrac{1}{8}\gamma^{\mathrm{tr}} A_1^2 (\Lambda/78)^{2p_0}\big],$$

and recall that the number $d_{\mathrm{e}} \le \frac{1}{200}$ of Step 1 is so small that for $0 < d \le d_{\mathrm{e}}$: $\gamma^{\mathrm{tr}} A_1^2 (\Lambda/78)^{2p_0} \ge 80$, $10240\, \mathsf{N}_{\max}(d)^3 \exp[-\frac{1}{8}\gamma^{\mathrm{tr}} A_1^2 (\Lambda/78)^{2p_0}] \le 1$, and

$$\epsilon_{\mathrm{r}}(d) := g^2 d\big[A_\ell (1 + 24(P_\sharp - 1)\Lambda/\log L) + C_{\mathrm{T}} \vartheta^{-1}(1 + (P_\sharp - 1)\Lambda)\big] \le 1.$$

Let $0 < d \le d_{\mathrm{e}}$, $(f, h) \in \mathfrak{T}_{T_\ell}(d;\vartheta)$, and let (K, β) be a datum at g with $s(d) < K \le K_\sharp(d)$; for the first assertion, $\beta = L^K b$ with $b \in I(g)$. Write $K_0 = K$, $D_0 = (K_0, \beta)$ and $r = dL^{K_0}$; then $L^{K_0} \le d^{-P_\sharp}$ (Prop. 6.19), $K_0 \ge 3$, $8a_{K_0} \le d$ and $r \ge L\Lambda^2$ (Prop. 6.8). As in Step 1,

$$\Big|S_2^T - c_2 b^{-2}\mathcal{K}_6 - \frac{1}{\#\mathsf{Pk}} \sum_v \mathsf{h}_6(\tau_v f, \tau_v h)\Big| \le \Big[(C_{\mathrm{A}}\vartheta^{-1} b d + C_{\mathrm{E}} b^{-1} d l^{8p_0} + \bar\varepsilon_4)\, b^{-3} d + C_{\mathrm{lat}} b^{-2} \frac{a_{K_0}}{d}\Big]\eta,$$

and, at every position, by the last link T,

$$|\mathsf{h}_6 - \varkappa b^{-3}\mathcal{K}_5| \le b^{-3} a_{K_0} |\mathsf{H}^{(K_0)}[D_0, \ell_g] - \mathsf{T}^{\mathrm{lat}}(\tilde w)| + b^{-3} C_{\mathrm{T}} \vartheta^{-1} a_{K_0} \lg(r)\, \eta.$$

Let n_0 be the level with $D_{n_0} \in [d^{1/5}, Ld^{1/5})$; then $4 \le n_0 \le K_0 - 1$ and $r/L^{n_0} \le d^{4/5}$.

Case A, $r^{24} \le L^{K_0}$. With $e = \lceil K_0/78 \rceil$ and $n_1 = K_0 - e$: $2 \le n_1 \le K_0 - 1$, $(r/L^{n_1})^{3/2} \le d^{6/5}$, $L^e \ge d^{-1/78}$ and $r^6 \le L^{K_0/4}$. Props. 6.6(iii) from K_0 to n_1, 6.12(i), (ii) and 6.15 at n_1, and 6.18 on the lattice with $\ell_g L^e$ top sites and n_1 levels give $|\mathsf{H}^{(K_0)} - \mathsf{T}^{\mathrm{lat}}| \le \{C_{\mathrm{R}} l^{6p_0} d^{6/5} r + E_{\mathrm{X}}(d) + A_\ell K_0\}\eta$, since $19e - \frac{1}{2}n_1 < \frac{39}{2} - \frac{K_0}{4}$ gives $(\ell_g L^e)^{19} n_1 L^{-n_1/2} r^6 \le \ell_g^{19} L^{39/2} K_0$.

Case B, $r^{24} > L^{K_0}$. With $K' = \lceil 24 \log_L r \rceil$, $e = \lceil K'/78 \rceil$ and $n_1 = K' - e$: $K' > K_0$, $r^{23} > 1/d$, $r^{24} \le L^{K'} < Lr^{24} \le Ld^{-24(P_\sharp - 1)}$, $n_0 < n_1 \le K' - 1$, $L^e \ge d^{-1/78}$ and $K' \le 1 + 24(P_\sharp - 1)\Lambda/\log L$. Prop. 6.23, with $b_* = b$ if $b \in I(g)$ and a fixed $b_* \in I(g)$ otherwise, and with $\sigma = C_{\mathrm{V}} \dagger^{3p_0} b^2 d^2$ from Prop. 6.21(iii) at D_0 and $s = K' - K_0$, gives

$$|\mathsf{H}^{(K_0)} - \mathsf{T}^{\mathrm{lat}}| \le \Big\{\big[\sigma + 2C_{\mathrm{E}} b^{-1} d l^{8p_0} + 2\bar\varepsilon_4 + C_{\mathrm{R}}(l^{6p_0} + l'^{6p_0}) d^{6/5}\big] r + \frac{3C_{\mathrm{tm}} b}{\vartheta d} + 3E_{\mathrm{X}}(d) + A_\ell K'\Big\}\eta,$$

with $l' \le \log(2b_+ L r^{23}) \le l'_+$.

The places of Prop. 6.12(ii). At every one of them $\mathsf{N} \le \mathsf{N}_{\max}(d)$, and the running coupling $x_\flat = x_{n-1}$ of the place exceeds $d^{-1/78}$: $x_\flat \ge \frac{3}{4} b L^{K_0 - n_0 + 1} > d^{-1/5}$ at the level n_0, on either side, and

$x_\flat \ge \frac34 b_* L^{e+1} \ge L^e$ at the level n_1, as $\frac34 b \ge 1$. So the two conditions hold by the choice of d_e, and every $\mathcal{U}$ is at most $E_\mathrm{X}(d)$, which is smaller than every power of d, as $2p_0 \ge 2$.

Collecting. Multiply by $b^{-3}a_{K_0}$: $b^{-3}a_{K_0}r = b^{-3}d$; $b^{-3}a_{K_0}\cdot 3E_\mathrm{X} \le \frac38 E_\mathrm{X} b^{-3}d$, as $r \ge 8$; and $b^{-3}a_{K_0}\cdot 3C_\mathrm{tm}b/(\vartheta d) = 3C_\mathrm{tm}\vartheta^{-1}b^{-2}a_{K_0}/d$. The two members that grow with r go into the lattice term: as $\lg r \le 1 + (P_\sharp - 1)\Lambda$ and $b \ge g^{-2}$,

$$b^{-3}a_{K_0}\big[A_\ell K' + C_\mathrm{T}\vartheta^{-1}\lg r\big] = \frac{d}{b}\big[A_\ell K' + C_\mathrm{T}\vartheta^{-1}\lg r\big]\, b^{-2}\frac{a_{K_0}}{d} \le \epsilon_\mathrm{r}(d)\, b^{-2}\frac{a_{K_0}}{d} \le b^{-2}\frac{a_{K_0}}{d},$$

with K_0 for K' in Case A. So for $s(d) < K \le K_\sharp(d)$

$$|S_2^T - L_b^+|(f,h) \le \Big[\bar\varepsilon_\flat(d)\, b^{-3}d + (\tfrac32+3)C_\mathrm{tm}\vartheta^{-1}b^{-2}\frac{a_K}{d} + b^{-2}\frac{a_K}{d}\Big]\eta = \Big[\bar\varepsilon_\flat(d)\, b^{-3}d + C^\flat_\mathrm{lat}b^{-2}\frac{a_K}{d}\Big]\eta,$$

where $C_\mathrm{E}b^{-1}dl^{8p_0}$ and $\bar\varepsilon_4$ enter once through the front and twice through the first link of Prop. 6.23, and

$$\bar\varepsilon_\flat(d) = C_\mathrm{A}\vartheta^{-1}b_+ d + 3C_\mathrm{E}g^2 d\, l_+^{8p_0} + 3\bar\varepsilon_4(g,d) + C_\mathrm{V}[\log(10LMb_+/d)]^{3p_0}b_+^2 d^2 + C_\mathrm{R}\big[l_+^{6p_0} + l_+'^{6p_0}\big]d^{6/5} + \tfrac38 E_\mathrm{X}(d).$$

This is the second assertion of (e). For $b \in I(g)$ and $K > K_\sharp(d)$, and for the limit, Prop. 6.19 at $K_\sharp(d)$ adds $\epsilon_\diamond(d)b^{-3}d\,\eta$, $\epsilon_\diamond(d) = C_\diamond\vartheta^{-2}b_+^3 d^{177} + C^\flat_\mathrm{lat}b_+ d^{3598}$, and the lattice term may be dropped, as in Step 5. Finally every member is at most a multiple of $d^{1/2}$: $d\, l_+^{8p_0} \le Cd^{1/2}$; $\bar\varepsilon_4(g,d) \le C_4(g)d^{1/2}$; the member of the two sides is $O(d^2\Lambda^{3p_0})$; $d^{6/5}l_+'^{6p_0} \le Cd$; E_X and $\epsilon_\diamond$ are smaller. So $\sup_{d''\le d}[\bar\varepsilon_\flat(d'') + \epsilon_\diamond(d'')] \le c_\mathrm{r}d^{1/2}$, with c_r depending on g, ϑ and the constants; this is (e). The slowest member is again that of the order four.

Step 9: clause (f). Let g, g_a and g_b be as in Step 6, and let d''_e be the least, over the three couplings g'', of the number $d_\mathrm{e}(g'')$ of Step 1 at $P = P_\sharp$, $\frac{1}{180}$, L^{-2}, $\mathsf{d}_\star(g'',\vartheta)$ and $(LC_w(g''))^{-2}$. Let (K,β) be a datum at g with $K > s(d)$ and label b. If $K \le K_\sharp(d)$, the second assertion of (e) at g applies. If $K > K_\sharp(d)$ and $b \in I(g)$, the first assertion of (e) at g applies. Otherwise $b \in I(g_\bullet)$ with $g_\bullet \in \{g_\mathrm{a}, g_\mathrm{b}\}$ (Step 6), (K,β) is a datum at $g_\bullet$, and $K > s(d)$ at $g_\bullet$, since $L^K > L^{K_\sharp(d)} \ge (\sqrt3 M/d)^{3600} > d^{-3}$; the first assertion of (e) at $g_\bullet$ gives the law with the rate on $T_{\ell_{g_\bullet}}$ at the pair with the same lift, and eq. 6.ix, as in Step 6, bounds the difference of the two torus measures at the lift by $C_2^\mathrm{T}(g_\circ)\vartheta^{-1}b_+ d\cdot b^{-3}d\,\eta$.

So (f) holds with $c'_\mathrm{r} = \max_{g''} c_\mathrm{r}(g'') + \max_{g''} C_2^\mathrm{T}(g'')\vartheta^{-1}b_+(d''_\mathrm{e})^{1/2}$.

6.1.10. *Proof of Cor. 6.4.* (i) Let $\mathfrak{s}$ be a state at g at a datum (K,β) with $K > s(d)$; as $d \le L^{-1}$, $K \ge 3$ and $a_K < d/13 < 9d$. By eq. 6.ix,

$$|S_2^T[\mathfrak{s}] - S_2^T[\tilde\mu_{K,\ell_g,\beta}]|(\tilde f,\tilde h) \le C_2^\mathrm{T}(g)\vartheta^{-1}d^2b^{-2}\eta \le C_2^\mathrm{T}(g)\vartheta^{-1}b_+ d\cdot b^{-3}d\,\eta,$$

and $S_2^T[\tilde\mu_{K,\ell_g,\beta}](\tilde f,\tilde h) = S^T_{2;K,\ell_g,\beta}(f,h)$ at the periodisation $(f,h) \in \mathfrak{T}_{T_{\ell_g}}(d;\vartheta)$, to which clauses (c) and (f) of Thm. S apply; $L_b^+(f,h) = L_b^+(\tilde f,\tilde h)$. The torus measure of a side of $\mathfrak{L}(g)$, periodically extended, and every periodic Gibbs state at (a_K,β) are states at g at the datum. (ii) For $b \in I(g)$ every $(K, L^K b)$ is a datum at g. Apply (i) to the torus measure of the side ℓ' at the lift of (f,h), and let $K \to \infty$: the lattice term tends to 0 and $S^T_{2;K,\ell',L^Kb}(f,h) \to S_2^{(\ell',b)}(f,h)$. (iii) The labels at g lie in $[g^{-2}, b_+]$, and so does their limit. The witness depth depends on no cutoff, so $K_j > s(d)$ for large j; apply (i) at (K_j,β_j) and let $j\to\infty$: $a_{K_j}\to 0$ and $L^+_{b_j} \to L^+_b$.

The labels. Let the states $\mathfrak{s}_j$ and the set of pairs be as in the second assertion of (iii), and let b, b' be two limit points of the labels, along two subsequences. Along each of them the first assertion applies, and at a pair $F = (\tilde f,\tilde h)$ of the set that lies in $\mathfrak{T}_{\mathbb{R}^3}(d;\vartheta)$, $d \le d_\infty$, the limit $S(F)$ of $S_2^T[\mathfrak{s}_j](F)$ is a limit point along both; so $|L_b^+(F) - L_{b'}^+(F)| \le \bar\varepsilon_\infty(d)(b^{-3} + b'^{-3})d\,\eta$. On the class, $\mathcal{K}_6(F) \ge (10d)^{-6}\int\tilde f\int\tilde h \ge 10^{-6}\vartheta^2\eta$, and $\mathcal{K}_5(F) \le d^{-5}\int\tilde f\int\tilde h \le (\frac{4\pi}{3})^2 d\,\eta$, as $|x-y| \ge d$ on the

supports and each support lies in a ball of radius d. Hence

$$10^{-6}\vartheta^2 c_2\,|b^{-2}-b'^{-2}| \le \Big[\bar\varepsilon_\infty(d)(b^{-3}+b'^{-3}) + (\tfrac{4\pi}{3})^2 c_1 c_2\,|b^{-3}-b'^{-3}|\Big]d,$$

and $d\to 0$ along the scales of the set gives $b=b'$. The labels lie in the compact interval $[g^{-2},b_+]$ and have one limit point, so they converge.

Limits and components. Let (K_j,β_j,ν_j) be a convergent Gibbs sequence at g and σ its limit (Def. 13.4(2)). Then $S_2^\sigma(X)=\lim_j S_2^T[\nu_j](X)$ at every separated tuple $X\in\mathfrak{S}_2(\mathbb{R}^3)$ (Thm. YM3_I:AP). A pair of $\mathfrak{T}_{\mathbb{R}^3}(d;\vartheta)$ whose entries are of class C^1 is a separated tuple: its entries lie in $C^1_c(\mathbb{R}^3)$ and have disjoint supports ([7, §K.1] and the test-pair convention preceding equation YM3_I:(2343)); and for every $d>0$ and $\vartheta\in(0,1)$ the class $\mathfrak{T}_{\mathbb{R}^3}(d;\vartheta)$ holds a pair of C^∞ functions ([7, Lem. I.1](c)).

So the second assertion of (iii), with the states ν_j and the C^1 pairs in the class, shows that the labels converge, to a number b. For a component σ_ξ of a canonical decomposition of σ, Thms. YM3_I:AL and YM3_I:AP show that σ_ξ is a limit in the frame of the sequence along all j: $\sigma_\xi(X)=\lim_j \rho_j(X^{\nu_j}_{K_j})$ for every $X\in\mathfrak{S}(\mathbb{R}^3)$, $X^{\nu_j}_{K_j}$ being the product of the centred observables $\mathcal{O}(f_q)-\nu_j(\mathcal{O}(f_q))$ of the entries of X, with the approximants $\rho_j(F)=\nu_j(\mathbf{1}_{E_j}F)/\nu_j(E_j)$, where E_j is a Borel set invariant under the translations by $a_{K_j}\mathbb{Z}^3$ with $\nu_j(E_j)>0$; and ρ_j is a periodic Gibbs state at (a_{K_j},β_j) (Thm. YM3_I:AP).

The convergence on every separated tuple also gives convergence of all connected functions. Indeed, put $\sigma_j(X)=\rho_j(X^{\nu_j}_{K_j})$. The inverse moment–cumulant identity in the proof of Prop. YM3_I:92.1 gives, for $n\ge 2$,

$$S_n^T[\rho_j](X)=\sum_{\pi\in\mathrm{Part}(n)}(-1)^{|\pi|-1}(|\pi|-1)!\prod_{B\in\pi}\sigma_j(X_B).$$

Centering changes only the first cumulant. Every subtuple X_B remains separated, so each factor tends to $\sigma_\xi(X_B)$. Passing to the limit in this finite sum proves $S_n^{\sigma_\xi}(X)=\lim_j S_n^T[\rho_j](X)$ for every $n\ge 2$ and $X\in\mathfrak{S}_n(\mathbb{R}^3)$. For σ itself $\rho_j=\nu_j$. In both cases $S_2^{\sigma'}$, $\sigma'\in\{\sigma,\sigma_\xi\}$, is at a C^1 pair in the class the limit of $S_2^T[\rho_j]$ along states at g at the data (K_j,β_j), whose labels converge to b, and the first assertion of (iii) gives the law.

Volume-first limits. A defining sequence of states of a volume-first limit S at g is a Gibbs sequence at g; it converges, and its limit is the centred functional σ_S of S (Prop. YM3_I:92.1 and its moment–cumulant identity); the ergodic components of S are the components of the canonical decompositions of σ_S ([7, §K.1] and Def. 13.4(2)). So the labels along every defining sequence converge, and the law holds for S and its ergodic components at the pairs of class C^1. Two defining sequences give one label: the one function $S_2=S_2^{\sigma_S}$ obeys the law with both limits at the C^1 pairs in the class at every small d, and the inequality displayed above gives their equality.

This is the unique label of Thm. YM3_I:AW. For $b\in I(g)$ the pairs $(K,L^K b)$, $K\in\mathbb{N}$, are data at g (Cor. YM3_I:74.2), and Prop. YM3_I:92.1, applied to torus-limit states at these data, gives a volume-first limit at g with a defining sequence of states (K_j,β_j,ν_j) with $a_{K_j}\beta_j=b$ for every j; its label is b.

(iv) $\mathcal{K}_5\ge 10^{-5}\vartheta^2 d\,\eta$ on the class; divide by $c_2 b^{-3}\mathcal{K}_5$. In the case (i) the lattice term $C_{\mathrm{lat}}b^{-2}(a_K/d)\,\eta$, divided by $c_2b^{-3}\cdot 10^{-5}\vartheta^2 d\,\eta$, gives the member $10^5 c_2^{-1}\vartheta^{-2}C_{\mathrm{lat}}\,b\,a_K/d^2$, which is small only if a_K is small against d^2/b. In the cases (ii) and (iii), $|S-c_2b^{-2}\mathcal{K}_6|\le[(\frac{4\pi}{3})^2c_1c_2+\bar\varepsilon_\infty(d)]\,b^{-3}d\,\eta$ and $\mathcal{K}_6\ge 10^{-6}\vartheta^2\eta$, so that $S/(c_2\mathcal{K}_6)\to b^{-2}$ as $d\to 0$ along the pairs at which the law holds: each of these limits determines its label, and then the quotient determines the number $\frac{8}{3\pi}$. A law of this form with a tolerance that tends to 0 with d thus fixes both its label and its coefficient.

Remark 6.25. At a finite cutoff the tolerance of Thm. S is $\bar\varepsilon(d)$, or $c_{\mathrm{r}}d^{1/2}$, together with the lattice term $C_{\mathrm{lat}}b^{-2}a_K/d$, which is larger than the correction $c_1c_2b^{-3}\mathcal{K}_5$, of the order $b^{-3}d$, unless a_K is small against d^2/b; the number c_1 is read off at such cutoffs or in the continuum limit. Clause (d) is

weaker than clause (b): a mean over scales does not see an oscillation in the scale. The arguments use one finite lattice or one torus at a time, at scales below the least side; the agreement of states uses besides one uniformity in the volume, which comes from reflection positivity and from the bounds of Thms. YM3_I:AJ and YM3_I:AD on the partition functions of the tori at the source 0, and no property of the correlations at large distances.

6.2. **A complex scale for one fluctuation step.** The coefficient arguments in §§6.1 and 8.1 require a disc of fluctuation scales whose radius increases as the running coupling increases. Analyticity near the real step alone does not give this disc. We obtain it by constructing the potentials at a fixed lower real coupling, where the unscaled field has a larger admissible range, and then evaluating the same functions at $\lambda \mathtt{t}^{-1/2} B$.

We use the local input and the localized expansion of Def. YM3_I:52.1 and Thm. YM3_I:V. In particular $k+1$ is a single admitted step, $0 \le k < K$, $M = L^{m'}$ satisfies $m' < m + K - k - 1$, and $P = (U, A', \mathbf{J})$ lies in the output core $\mathcal{U}^\circ(X)$. On a fixed real-U slice its variables $(A', \mathbf{J})$ lie in the open set $\mathcal{O}_U(X)$. The Gaussian structure, its restriction consistency, the gauge-fixing operators and their bounds are those in that definition. We retain the constants and numerical assumptions of Thm. YM3_I:V. Write

$$\delta(t) = A_1 t^{-1/2} (\log t)^{\mathfrak{p}_0}, \qquad \mathfrak{b}(t) = t^{1/2}\delta(t), \qquad s_{\mathfrak{a}}(t) = \delta(t)/a_{0,k+1},$$

and let $\mathbb{D}^{\mathrm{f}}_t(Y)$ be the complex field polydisc $\sup_b |B(b)| < C_{\mathcal{A}} \mathfrak{b}(t)$ on the nonpivot bonds in Y. The potential norm uses the weight

$$Q_Y(B) = 1 + M^{-3} \sum_{b \in \mathfrak{X}^{\circ +} \langle Y \rangle} |B(b)|^2.$$

All parameter sets below are either points or open subsets of a finite-dimensional complex vector space; a point factor is omitted from a product domain.

Definition 6.26 (Scaled input)**.** Fix $\mathtt{t} > 1$ in the admitted range and a radius $R > 0$. Suppose that four potentials $V^{J,\lambda}, V^{G,\lambda}, V^{A,\lambda}, V'^{,\lambda}_\theta$ are given for $|\lambda| < R$, $\theta \in \Theta$, on every local core and field polydisc. They have the pair dependence and gauge covariance of Def. YM3_I:52.1, and are jointly holomorphic on

$$D(0,R) \times \Theta \times \mathcal{O}_U(Y) \times \mathbb{D}^{\mathrm{f}}_{\mathtt{t}}(Y).$$

The first three are independent of θ. The J and fourth potentials vanish at $B = 0$; the G and A potentials vanish through degree two in B. Suppose also that

$$\begin{aligned} |V^{o,\lambda}(Y;P,B)| &\le |\lambda| k_{\mathrm{int}} e^{-\lambda_V d_k(Y)} Q_Y(B) \quad (o = J, G, A), \\ |V'^{,\lambda}_\theta(Y;P,B)| &\le |\lambda| k_{\mathrm{iv}} e^{-\lambda_V d_k(Y)} Q_Y(B), \\ R k_{\mathrm{int}} \le w_*/8, &\qquad R k_{\mathrm{iv}} \le w_*/4, \end{aligned} \tag{6.xiii}$$

and $V^{o,-\lambda}(Y;P,-B) = V^{o,\lambda}(Y;P,B)$ for all four potentials. The Gaussian structure and gauge-fixing operators are fixed as λ, θ vary. The coefficient x is an independent parameter in $D(\mathtt{t}, \sigma_* \mathtt{t})$, and $\tau = x/\mathtt{t}$. ■

Use exactly the localized interpolation and the three bond terms in the proof of Thm. YM3_I:V: $\mathbf{G}_\tau$ is the Gaussian term, E^χ the cutoff term, and $E^{\bullet,\lambda}_\theta$ the interpolated interaction term. For the last term the interpolation is

$$V^{\mathrm{int}}_v(B) = V^{J,\lambda}(vB) + v^{-2}[V^{G,\lambda} + \tau V^{A,\lambda}](vB), \qquad V'_v(B) = V'^{,\lambda}_\theta(vB).$$

The value at $v = 0$ is its removable continuation. Distribute bond terms to sites with the same nonnegative tents h_z:

$$\mathbf{G}_\tau(X,P,z) = \sum_b h_z(b) \mathbf{G}_\tau(X,b;P),$$

$$F_\theta^\lambda(X,P,z)=\sum_b h_z(b)(E^\chi+E_\theta^{\bullet,\lambda})(X,b;P),\qquad F^\chi(X,P,z)=\sum_b h_z(b)E^\chi(X,b;P).$$

Thus $F^\lambda-F^\chi$ is the interaction-dependent part. It is this difference that vanishes at $\lambda=0$.

Proposition 6.27 (The step on the full complex disc)**.** For the data of Def. 6.26, put $M_\bullet = K_\Pi R(32k_{\rm int}+9k_{\rm iv}) \le (25/4)K_\Pi w_*$ and $d=d_{k+1}(X)$. The following assertions hold on the full product domain

$$D(\mathsf{t},\sigma_*\mathsf{t})\times D(0,R)\times\Theta\times\mathcal{O}_U(X). \tag{6.xiv}$$

(cx1) All three bond terms are defined and vanish unless $b^\sharp\subset X$. Their tent sums are pointed local families.

(cx2) For fixed x,P, $\mathbf{G}_\tau$ and E^χ are independent of λ,θ and of all four potentials. They have the Gaussian and cutoff bounds of Thm. YM3_I:V.

(cx3) The interaction term obeys

$$|E_\theta^{\bullet,\lambda}(X,b;P)|\le K_\Pi|\lambda|(32k_{\rm int}+9k_{\rm iv})e^{-\lambda_{\rm out}d}.$$

For $F^\lambda-F^\chi$ the same bound holds with the factor $3L^3$. In particular the respective bounds on the disc are $M_\bullet e^{-\lambda_{\rm out}d}$ and $3L^3M_\bullet e^{-\lambda_{\rm out}d}$.

(cx4) $E_\theta^{\bullet,0}=0$ and $F_\theta^0=F^\chi$.

(cx5) All terms are jointly holomorphic on eq. 6.xiv; the terms in (cx2) are holomorphic in x and the slice variables alone.

(cx6) Each individual bond term $E_\theta^{\bullet,\lambda}$ and each site term F_θ^λ is even in λ.

(cx7) The terms depend only on the local pair and are gauge invariant, with the same restriction and locality properties as the original step. No sum over domains is needed for these properties.

(cx8) The Taylor series starts at order two:

$$F_\theta^\lambda=F^\chi+\sum_{m\ge1}\lambda^{2m}F_{[2m],\theta},\qquad |F_{[2m],\theta}(X,P,z)|\le 3L^3M_\bullet R^{-2m}e^{-\lambda_{\rm out}d}.$$

The coefficients are jointly holomorphic in all remaining parameters and inherit locality and gauge invariance. The corresponding bond bounds omit $3L^3$.

(cx9) If $R>1$, then

$$|F^1-F^\chi|\le 3L^3M_\bullet R^{-2}e^{-\lambda_{\rm out}d},\qquad |F^1-F^\chi-F_{[2]}|\le\frac{3L^3M_\bullet R^{-4}}{1-R^{-2}}e^{-\lambda_{\rm out}d}.$$

(cx10) The local sum rules and the whole-torus integral identity of Thm. YM3_I:V hold for these potentials. More explicitly, put $V^{{\rm int},\lambda}=V^{J,\lambda}+V^{G,\lambda}+\tau V^{A,\lambda}$ and $V^{\rm q}=-(1-\tau)\sum_\square(B,\widehat{G}_\square B)/2$. Then

$$\begin{aligned}\Xi_\theta^\lambda(x;P_*)&:=\int d\mu_{\tau^{-1}C_*(P_*)}(B)\prod_{b\in\mathfrak{X}^+}\mathbf{1}_{\{|B(b)|<\mathfrak{b}(\mathsf{t})\}}\exp\sum_Y(V^{\rm q}+V^{{\rm int},\lambda}+V_\theta'^{,\lambda})(Y;P_*|_Y,B|_Y)\\&=\exp\sum_{X,z}(\mathbf{G}_\tau+F_\theta^\lambda)(X,P_*|_X,z)\neq0.\end{aligned}$$

The inverse covariance has positive real part, the integral converges absolutely, and its integrand is defined as zero where the cutoff product vanishes. The Gaussian measure and integral are complex when their data are complex.

(cx11) If $R>1$ and the four potentials at $\lambda=1$ are the admitted potentials of the original step, the terms at $\lambda=1$ are exactly its terms, with the same localization.

■

Proof. We check the quantitative bounds in the construction rather than increasing the constant K_P in its global smallness conditions. The Gaussian scaling calculation in Thm. YM3_I:V depends only on x, the fixed structure and the numerical constants. Thus on its stated x disc the scaled structure $(\tau^{-1}C, \tau^{1/2}\Gamma)$ is admitted, the whole covariance has inverse with positive real part, and $\|V^{\mathrm{q}}\| \le w_*/2$.

Put $\nu_{\mathrm{int}} = |\lambda| k_{\mathrm{int}}$ and $\nu_{\mathrm{iv}} = |\lambda| k_{\mathrm{iv}}$. The interpolation estimate, Prop. YM3_I:48.4, uses only the potential norms and their indicated orders of vanishing. It gives

$$\|V_v^{\mathrm{int}}\| \le \tfrac{7}{2}\nu_{\mathrm{int}} \le 7w_*/16, \quad \|V_v'\| \le \nu_{\mathrm{iv}} \le w_*/4, \quad \|\dot V_{v,b}^{\mathrm{int}}\| \le 32\nu_{\mathrm{int}}, \quad \|\dot V_{v,b}'\| \le 9\nu_{\mathrm{iv}}.$$

These are precisely the margins used in the proof of Thm. YM3_I:V. Its insertion expansion therefore defines $E_\theta^{\bullet,\lambda}$ and gives (cx3), with its support. The first two terms use no interaction potentials, proving (cx2). The tent mass at one site is at most $3L^3$, giving the site bounds. At $\lambda = 0$ all four potentials vanish by eq. 6.xiii; hence so do the interpolation and its insertion, proving (cx4).

The same insertion majorant is uniform on the full product domain. Each refined integrand is holomorphic in its parameters on the slice and fixed field polydisc. The common majorant permits parameter integration and locally uniform summation, so the terms are jointly holomorphic. The cutoff terms use their fixed-density formula, whose differentiated masses have the same summable majorant, exactly as in that theorem. This proves (cx5).

For parity apply $B \mapsto -B$ to each refined Gaussian integral. Its covariance, every quadratic gauge term and every cutoff are unchanged. The hypothesis on the four potentials gives $V_v^{o,-\lambda}(-B) = V_v^{o,\lambda}(B)$ for the localized interpolations, including their bond velocities: multiplication by the interpolation field commutes with -1, and v^{-2} is unchanged. In a refined derivative the signs from differentiating the potentials are matched by those from the corresponding Gaussian derivatives; equivalently both are conjugated by the orthogonal map $-I$ on the bond-coordinate space. The insertion expansion is covariant under this map. Thus equality holds for each domain and each marked bond before summation. It proves (cx6). Pair dependence, gauge covariance and restriction commute with the same construction, proving (cx7).

For fixed remaining parameters, $E_\theta^{\bullet,\lambda}$ is even, holomorphic, zero at zero and bounded by $M_\bullet e^{-\lambda_{\mathrm{out}} d}$. Cauchy's formula on $|\lambda| = r < R$ bounds its degree-$2m$ coefficient by $M_\bullet r^{-2m} e^{-\lambda_{\mathrm{out}} d}$; let $r \uparrow R$. The odd coefficients vanish. Coefficient extraction by this contour integral preserves holomorphy in the remaining parameters and all the established invariances. Finite tent summation gives (cx8). Schwarz's lemma applied after the order-two zero gives the first bound in (cx9); summing the coefficient bounds for $m \ge 2$ gives its second bound.

For (cx10), integrate the insertion along v: the sum of its bond velocities is the derivative of the interpolated potential. Thus its sum is the difference of the full cluster sum and the one with only V^{q}. The cutoff term supplies the difference between the finite-threshold and infinite-threshold sums for V^{q}, and the Gaussian term supplies the latter sum. The tents have total mass one when summed over their sites. Restriction consistency identifies the domain terms with those computed on the whole torus. The complete Gaussian expansion therefore gives exactly the displayed exponential identity. Its common majorant proves absolute convergence; its exponential form proves nonvanishing. Finally the formulas defining every term are unchanged at $\lambda = 1$, which proves (cx11). □

We now construct this input on a large disc. The required one-variable estimate is included to distinguish the gain from the mere vanishing of a potential at the origin.

Lemma 6.28 (Rescaling a holomorphic potential)**.** Let $E_r = \{B : \sup|B| < r\}$, $0 < r_\flat \le r$, and $|\mu| \le r_\flat/r$. Suppose f is holomorphic on $E_{r_\flat}$. If $f(0) = 0$ and $|f(B')| \le aQ_Y(B')$, then

$$|f(\mu B)| \le a|\mu|(r/r_\flat)Q_Y(B), \qquad B \in E_r.$$

If the bound is a instead, omit Q_Y. If f vanishes through degree two, then the stronger bound is

$$|f(\mu B)| \le a|\mu|^3(r/r_\flat)(1 + M^3/r_\flat^2)(Q_Y(B) - 1).$$

■

Proof. For the first assertion apply Schwarz's lemma to $z \mapsto f(z(r_\flat/r)B)$ on the unit disc. Its bound is $aQ_Y(B)$, since $Q_Y(cB) = 1 + |c|^2(Q_Y(B) - 1)$. For the third-order assertion let $\beta = \sup|B| > 0$; the case $B = 0$ is immediate. For $|\mu| \le s < r_\flat/\beta$, the function $z \mapsto f(zsB)$ vanishes to order three and is bounded on the unit disc by $a[1 + s^2(Q_Y(B) - 1)]$. Schwarz's lemma of order three gives

$$|f(\mu B)| \le a|\mu|^3[s^{-3} + s^{-1}(Q_Y(B) - 1)].$$

Let $s \uparrow r_\flat/\beta$ and use $\beta^2 \le M^3(Q_Y(B) - 1)$ and $\beta < r$. This proves the displayed estimate. □

Definition 6.29 (A lower calibration)**.** Retain one lattice, one system of radii, its open slices and one Gaussian structure and gauge-fixing family. Let $1 < \mathfrak{t}_\flat \le \mathfrak{t}$ and $\delta(\mathfrak{t}) \le \delta(\mathfrak{t}_\flat)$. All numerical and radius conditions of Thm. YM3_I:V are required at $\mathfrak{t}_\flat$, with $\varepsilon(\mathfrak{t}_\flat), \delta(\mathfrak{t}_\flat) \le 1$, $\mathfrak{b}(\mathfrak{t}_\flat) \ge \mathfrak{b}_*$ and $s_\mathfrak{a}(\mathfrak{t}_\flat) \le s^*$. Suppose four functions of the unscaled field $\mathcal{A}$ are defined for $\sup|\mathcal{A}| < C_\mathcal{A}\delta(\mathfrak{t}_\flat)$. For $s \in \{\mathfrak{t}_\flat, \mathfrak{t}\}$ set

$$V^J[s](B) = v^J(s^{-1/2}B), \quad V^o[s](B) = -s\, v^o(s^{-1/2}B)\ (o = G, A), \quad V'_\theta[s](B) = v'_\theta(s^{-1/2}B).$$

The dependence on Y, P is suppressed. At $s = \mathfrak{t}_\flat$ require the full local input of Def. YM3_I:52.1, jointly in $\theta \in \Theta$, with fourth-family size at most $n_\flat \le N_*$. No bound at a larger unscaled field is assumed. ■

Proposition 6.30 (The disc supplied by a lower calibration)**.** For Def. 6.29, put

$$R = \frac{\delta(\mathfrak{t}_\flat)}{2\delta(\mathfrak{t})}, \qquad k_{\mathrm{int}} = 2K_P\delta(\mathfrak{t}), \qquad k_{\mathrm{iv}} = K_\Lambda s_\mathfrak{a}(\mathfrak{t})n_\flat.$$

On $|\lambda| < 2R$, define

$$\begin{aligned} V^{J,\lambda}(B) &= v^J(\lambda\mathfrak{t}^{-1/2}B), \\ V'^{,\lambda}_\theta(B) &= v'_\theta(\lambda\mathfrak{t}^{-1/2}B), \\ V^{o,\lambda}(B) &= -\mathfrak{t}\lambda^{-2}v^o(\lambda\mathfrak{t}^{-1/2}B) \quad (o = G, A). \end{aligned} \tag{6.xv}$$

with the last two values at zero defined as zero. Restricted to $|\lambda| < R$, these satisfy Def. 6.26. In particular Prop. 6.27 holds on its full product domain, with

$$M_\bullet = K_\Pi[32K_P\delta(\mathfrak{t}_\flat) + \tfrac{9}{2}K_\Lambda s_\mathfrak{a}(\mathfrak{t}_\flat)n_\flat] \le \tfrac{41}{8}K_\Pi w_*.$$

If $R > 1$ and the input at $\mathfrak{t}$ is the original admitted input, the value at $\lambda = 1$ is its original step. The order-two gain and fourth-order remainder carry respectively the factors $R^{-2} = 4\delta(\mathfrak{t})^2/\delta(\mathfrak{t}_\flat)^2$ and $R^{-4} = 16\delta(\mathfrak{t})^4/\delta(\mathfrak{t}_\flat)^4$. ■

Proof. Let $r = C_\mathcal{A}\mathfrak{b}(\mathfrak{t})$, $r_\flat = C_\mathcal{A}\mathfrak{b}(\mathfrak{t}_\flat)$ and $\mu = \lambda(\mathfrak{t}_\flat/\mathfrak{t})^{1/2}$. Then $r_\flat \le r$ and $|\mu|r/r_\flat = |\lambda|/(2R)$. Thus μB lies in the lower field polydisc for $|\lambda| < 2R$, and eq. 6.xv equals

$$V^J[\mathfrak{t}_\flat](\mu B), \qquad V'_\theta[\mathfrak{t}_\flat](\mu B), \qquad \mu^{-2}V^o[\mathfrak{t}_\flat](\mu B).$$

The last functions extend holomorphically through $\lambda = 0$: their numerators vanish through order two, so division by λ^2 has a removable zero, also jointly in the other parameters by Cauchy's coefficient formula.

Apply Lem. 6.28 at the lower place. For J the bound is $|\lambda|K_P\delta(\mathfrak{t})Q_Y(B)e^{-\lambda_V d_k(Y)}$. For G, A, division by μ^2 leaves the bound

$$|\lambda|K_P\delta(\mathfrak{t})(1 + M^3/r_\flat^2)(Q_Y(B) - 1)e^{-\lambda_V d_k(Y)}.$$

The threshold floor gives $\mathfrak{b}(\mathfrak{t}_\flat)^2 > (LM)^3/2$, and $C_\mathcal{A} \ge 3$, so $1 + M^3/r_\flat^2 < 11/9 < 2$. The constant-bound version of the lemma gives for the fourth potential $|\lambda|K_\Lambda s_\mathfrak{a}(\mathfrak{t})n_\flat e^{-\lambda_V d_k(Y)}$. Moreover

$$Rk_{\mathrm{int}} = K_P\delta(\mathfrak{t}_\flat) \le w_*/8, \qquad Rk_{\mathrm{iv}} = \tfrac{1}{2}K_\Lambda s_\mathfrak{a}(\mathfrak{t}_\flat)n_\flat \le w_*/8.$$

These prove all the size requirements. Pair dependence and gauge covariance are preserved by scalar multiplication of the field. The products $(-\lambda)(-B)$ and λB agree, as do $(-\lambda)^{-2}$ and λ^{-2}, proving parity. The orders of vanishing in B are unchanged. At $\lambda = 1$ the defining formulas are precisely those at t. The coefficient and remainder claims now follow from Prop. 6.27. □

Proposition 6.31 (The actual construction at two calibrations)**.** Fix the constants, the tower and shift systems, and one lag-weight case (ω, K) of the actual step Thm. YM3_I:W. Suppose its geometric, core, shift and octet hypotheses hold at a lower real calibration $\mathsf{t}_\flat > 1$, with the same radii $\mathfrak{a}$, and choose positive lag weights satisfying

$$\varrho_j^2 L^{-(k-j)} \le c_\omega \omega_{k+1-j} \quad (1 \le j \le k), \qquad \mathsf{K} = K_\Lambda c_\omega.$$

Retain the rate bounds and all numerical hypotheses of Thm. YM3_I:V at $\mathsf{t}_\flat$, with K_Λ replaced by this K. Choose $\check{s} > 0$ such that

$$\check{s} \le s^*, \qquad 8K_P\check{s} \le w_*, \qquad 4\mathsf{K}N_*\check{s} \le w_*, \qquad s_{\mathfrak{a}}(\mathsf{t}_\flat) \le \check{s}, \quad \mathfrak{b}(\mathsf{t}_\flat) \ge \mathfrak{b}_*.$$

Let $\mathsf{t} \ge \mathsf{t}_\flat$ with $\delta(\mathsf{t}) \le \delta(\mathsf{t}_\flat)$. For a bare array $f_\circ$ on these same cores, assume $N_{\mathfrak{a},\omega}(f_\circ) \le N_*$; at height zero the array is empty. Then the canonical Gaussian structure and the actual four potentials give Def. 6.29, with K_Λ replaced by K and $n_\flat \le N_{\mathfrak{a},\omega}(f_\circ)$. The step is consequently defined on

$$x \in D(\mathsf{t}, \sigma_*\mathsf{t}), \qquad |\lambda| < \delta(\mathsf{t}_\flat)/(2\delta(\mathsf{t})).$$

If the latter radius exceeds one, its terms at one coincide with the actual canonical step terms. This statement concerns $\mathbf{G}_\tau, F^\lambda$; the separate local normalizer is unchanged and is not included in F^λ.

The same conclusion holds jointly for a holomorphic family of arrays with this uniform size bound. If an array has a reserve $N_{\mathfrak{a},\omega}(f_\circ) + 2q \le N_*$ with $q > 0$, its first and mixed second array derivatives, in directions h_1, h_2, satisfy

$$\begin{aligned} |D_f F^\lambda[h_1](X,P,z)| &\le Cq^{-1}\|h_1\| \frac{(|\lambda|/R)^2}{1-(|\lambda|/R)^2} e^{-\lambda_{\mathrm{out}} d}, \\ |D_f^2 F^\lambda[h_1,h_2](X,P,z)| &\le Cq^{-2}\|h_1\|\|h_2\| \frac{(|\lambda|/R)^2}{1-(|\lambda|/R)^2} e^{-\lambda_{\mathrm{out}} d}, \end{aligned} \tag{6.xvi}$$

where $R = \delta(\mathsf{t}_\flat)/(2\delta(\mathsf{t}))$, $\|h\| = N_{\mathfrak{a},\omega}(h)$, and C depends only on the fixed constants and $\check{s}$. The derivatives retain the local dependence on the array and weights. ■

Proof. We verify both the common objects and their domains. The radius and octet conditions of the actual step involve only the chosen radii, geometric constants and shift systems. Their coupling-dependent inequalities involve $\delta(t)$, $s_{\mathfrak{a}}(t)$ and the lower bound on $\mathfrak{b}(t)$. The first two decrease between the two specified calibrations and the third increases. Thus every such hypothesis valid at $\mathsf{t}_\flat$ is valid at t. The lag ratios and their weight inequalities do not involve t.

The actual covariance, its square root, the local backgrounds and the gauge-fixing matrices are built from the pair, the lattice, the fixed shifts and the radii. Changing t with these data fixed changes none of them. In the Jacobian formula the field occurs as $\mathcal{A} = t^{-1/2}B$. In the gauge and action potentials the same field occurs inside their cubic remainders, with the sole additional factor $-t$. In the fourth-potential formula the substitutions, finite vertex differences and octet evaluations are fixed; the field again enters only as $\mathcal{A}$. These facts follow directly from the actual formulas in the proofs of Thm. YM3_I:W and Prop. YM3_I:51.16. They supply the four unscaled functions in Def. 6.29 on $\sup|\mathcal{A}| < C_{\mathcal{A}}\delta(\mathsf{t}_\flat)$: this is exactly the image of the lower field polydisc under $B \mapsto \mathsf{t}_\flat^{-1/2}B$.

At that lower calibration Thm. YM3_I:W gives the three canonical interaction bounds and the Gaussian input. Prop. YM3_I:51.16 bounds the fourth potential by

$$\mathsf{K}s_{\mathfrak{a}}(\mathsf{t}_\flat)N_{\mathfrak{a},\omega}(f_\circ)e^{-\lambda_V d_k(Y)}.$$

It is linear in the array, holomorphic, and zero at zero field. Thus its lower size is at most $N_{\mathfrak{a},\omega}(f_\circ)$. All premises of Def. 6.29 are verified, with K as its fourth-potential constant. Prop. 6.30 proves the asserted domain and the identification at one. It also gives the uniform bound

$$M_\bullet \le K_\Pi(32K_P + \tfrac{9}{2}\mathsf{K}N_*)\check{s}.$$

Uniform holomorphic array parameters are permitted throughout the localized construction, so the parameter statement follows.

For the derivative estimates, vary the array by $z_1h_1+z_2h_2$ on the two discs $|z_i| < q/\|h_i\|$, omitting a zero direction. The reserve keeps their sum within the admitted ball. The term F^χ is independent of the array. For each even coefficient of $F^\lambda - F^\chi$, (cx8) and Cauchy's formula in the z_i give respectively the additional factors $q^{-1}\|h_1\|$ and $q^{-2}\|h_1\|\|h_2\|$. Sum the coefficient bounds for $m \ge 1$ to obtain eq. 6.xvi. The finite local substitutions use only input domains contained in the output domain, and taking these derivatives does not change that dependence. □

When the step is subsequently evaluated at an even holomorphic $x = \xi(\lambda)$, this substitution is legitimate only while $|\xi(\lambda) - \mathsf{t}| < \sigma_*\mathsf{t}$. It preserves holomorphy and evenness. Although the Gaussian and cutoff terms may then vary with λ through ξ, they remain independent of the bare array; the array-derivative argument is unchanged.

6.2.1. *Source families and the local step.* An insertion changes the potentials integrated in a step. It does not change its Gaussian structure. We specify the norm in which this change is small and the parameter dependence needed to differentiate the resulting local terms.

Use one admitted step of Def. YM3_I:52.1 and Thm. YM3_I:V, including their lattice, core, Gaussian and numerical hypotheses. At height k write P for a background triple and let B range over the complex nonpivot-field polydisc $\sup_b |B(b)| < C_{\mathcal{A}}\mathfrak{b}(\mathsf{t})$. For a family V define

$$\mathsf{n}^{\mathrm{f}}(V) = \sup_{X,P} \sup_{Y^\sharp \subset X,\, B} \frac{e^{\lambda_V d_k(Y)}|V(Y;P|_Y,B)|}{1+q_Y(B)}, \qquad q_Y(B) = M^{-3} \sum_{b \in \mathfrak{X}^{\circ+}\langle Y\rangle} |B(b)|^2. \tag{6.xvii}$$

Keeping X, P fixed defines $\mathsf{n}^{\mathrm{f}}_{X,P}$; keeping only X fixed defines $\mathsf{n}^{\mathrm{f}}_X$. A family in the fourth place is holomorphic on the open real-U slices and the field polydiscs, depends on the background pair, is gauge invariant, vanishes at $B = 0$, and has finite norm eq. 6.xvii. Holomorphic families of such inputs are required to be jointly holomorphic in their parameters and these variables, with a uniform norm bound.

The three ordinary interactions $V^{\mathrm{J}}, V^{\mathrm{G}}, V^{\mathrm{A}}$ are also families in the fourth place. Indeed their analyticity, pair dependence and gauge invariance are part of the local input; $V^{\mathrm{J}}(0) = 0$, and $V^{\mathrm{G}}, V^{\mathrm{A}}$ vanish through degree two. Their pointwise bounds give

$$\mathsf{n}^{\mathrm{f}}_X(V^\natural) \le \mathsf{n}^{\mathrm{f}}(V^\natural) \le K_P\delta(\mathsf{t}), \qquad \natural = \mathrm{J}, \mathrm{G}, \mathrm{A}. \tag{6.xviii}$$

The same bound holds for their even and odd parts, since $B \mapsto -B$ preserves the polydisc and q_Y.

For the canonical absolute-radius step, the source potentials are defined on the larger unscaled-field sets

$$\mathfrak{D}^{\mathrm{src}}(Y) = \{(P,\mathcal{A}) : P \in \mathcal{U}^\circ(Y), \quad \sup_b |\mathcal{A}(b)| < \varepsilon_{\mathrm{src}}\}, \qquad \varepsilon_{\mathrm{src}} = \begin{cases} \min\{\varepsilon_{\mathfrak{h}}, c_{\mathrm{os}}\alpha_0/(2c_{\mathfrak{h}})\}, & k \ge 1, \\ r_P^\circ, & k = 0. \end{cases} \tag{6.xix}$$

Retain $C_{\mathcal{A}}\delta(\mathsf{t}) \le \varepsilon_{\mathrm{src}}/2$ and $0 < \lambda_V \le \bar{\kappa} - 2$, as in Lem. YM3_I:114.1. Thus $\mathcal{A} = \mathsf{t}^{-1/2}B$ lies strictly inside this source domain. Its radius is fixed by the cores and does not shrink with t.

Let $\mathbb{W}^{\mathrm{src}}[v]$ be the direct weighted-action family of that lemma, with the reference linear term subtracted bond by bond. Choose $K_{\mathrm{Q}}^{\vee}, K_{\mathrm{L}}^{\vee}$ satisfying equation YM3_I:(2166), and put

$$K^{\mathrm{d}} = \tfrac{351}{2} L^3 K_{\Pi}, \qquad \mu_\star = \min\{w_\star/32, E_0/(16K_{\mathbf{M}}^{\mathrm{f}})\}, \qquad K^{\mathrm{d}} \le K_{\mathbf{M}}^{\mathrm{f}},$$
$$\sigma_{\mathrm{src}} = \mu_\star/K_{\mathrm{Q}}^{\vee}, \qquad o_{\mathrm{src}}(\mathsf{t}) = \mathsf{t}^{-1/2}\mu_\star/K_{\mathrm{L}}^{\vee}, \tag{6.xx}$$
$$\mathsf{Wgt}_k^{\mathrm{src}}(\mathsf{t}) = \{v : |v|_\infty < \sigma_{\mathrm{src}}, \quad \mathrm{sw}_k(v) < o_{\mathrm{src}}(\mathsf{t})\}.$$

Here E_0 denotes the size E_0^{s} in the recursion preceding Prop. YM3_I:114.2, and $K_{\mathbf{M}}^{\mathrm{f}}$ is its constant $3L^3 \max\{K_E e^{4\lambda_1}, 32K_\Pi K_P, 117K_\Pi/2\}$. The oscillation compares plaquettes whose barycentres are at sup distance at most M in level-k units. Write

$$\mathsf{sz}_X(v) = K_{\mathrm{Q}}^{\vee}(1 + 2C_{\mathcal{A}}\delta/\varepsilon_{\mathrm{src}})|v|_X + K_{\mathrm{L}}^{\vee}\mathsf{t}^{1/2}\,\mathrm{sw}_k(v).$$

Proposition 6.32 (Changing the fourth family)**.** Retain the preceding admitted step. At the real field scale $x = \mathsf{t}$, let the three internal families have total norm at most $3w_\star/8$. The pointed integration map $\mathbf{F}$ is holomorphic in a holomorphic fourth-family input on its admitted norm ball and is local in that input. It preserves pair dependence, gauge invariance, conjugation and the lattice covariances of the data. If V_0, V_1 have fourth-family norms at most $w_\star/4$ and $\mathsf{n}^{\mathrm{f}}(V_1 - V_0) \le w_\star/2$, then

$$|\mathbf{F}[V_1] - \mathbf{F}[V_0]|(X, P, z) \le K^{\mathrm{d}}\mathsf{n}_X^{\mathrm{f}}(V_1 - V_0)e^{-\lambda_{\mathrm{out}}d_{k+1}(X)}.$$

In particular let $\mathfrak{f}$ be a canonical old array satisfying $8K_\Lambda s_{\mathfrak{a}} N_{\mathfrak{a}}(\mathfrak{f}) \le w_\star$, with the empty array at $k = 0$. For $v \in \mathsf{Wgt}_k^{\mathrm{src}}$, the fourth family $V'[\mathfrak{f}] - \mathbb{W}^{\mathrm{src}}[v]$ has norm less than $7w_\star/32$. Its output $E_v = \mathbf{N} + \mathbf{F}[V'[\mathfrak{f}] - \mathbb{W}^{\mathrm{src}}[v]]$ is therefore defined on the full output core, jointly holomorphic in the weight and any holomorphic old-array parameters, and

$$|E_v - E_{v=0}|(X, P, z) \le K^{\mathrm{d}}\mathsf{sz}_X(v)e^{-\lambda_{\mathrm{out}}d_{k+1}(X)}.$$

The maps obey the exact local and whole-torus sum rules of Thm. YM3_I:V. On an actual real fibre admitted by [7, Lem. B.3], the source term sums to the weighted-action difference of equation YM3_I:(2168). ■

Proof. At a fixed background the four inputs are families on the field polydiscs. Lem. YM3_I:77.1 constructs their bond insertion terms, and Prop. YM3_I:77.2 compares two fourth families. In its second estimate use $n_{\mathrm{int}} \le 3w_\star/8$, $n_{\mathrm{iv}} \le w_\star/4$ and $n_W \le w_\star/2$. Its coefficient is at most

$$9(3L^3 - 3)K_\Pi\{1 + 8(3/8 + 1/4)\} < 162L^3K_\Pi < K^{\mathrm{d}}.$$

The same proof at fixed X gives the local norm. Holomorphy follows from the parameter-dependent insertion theorem with its common summable majorant. Restriction identifies every Gaussian matrix, potential and insertion on a smaller domain, and hence identifies the localized terms. The covariance assertions follow from the corresponding changes of variables in those same terms.

Prop. YM3_I:51.16 bounds the old fourth family by $K_\Lambda s_{\mathfrak{a}} N_{\mathfrak{a}}(\mathfrak{f}) \le w_\star/8$. Lem. YM3_I:114.1 gives $\mathsf{n}_X^{\mathrm{f}}(\mathbb{W}^{\mathrm{src}}[v]) \le \mathsf{sz}_X(v)$. The half-radius condition and eq. 6.xx imply

$$\mathsf{sz}_{T_\ell}(v) < 2\mu_\star + \mu_\star \le 3w_\star/32.$$

Their sum is strictly below $7w_\star/32 < w_\star/4$. The old-potential map preserves holomorphic parameters and the source family is linear in v, so the preceding parameter result applies. Here $E_{v=0}$ is the output with the zero weight and the same old array. The source-independent normalizer cancels in $E_v - E_{v=0}$. Finally the local sum rule is precisely the pointed insertion identity. The whole-torus identity follows by restriction and summation of the tents. For the source summation the reference linear terms cancel bond by bond by criticality of the real background; the remaining terms are its weighted-action difference, as proved in Lem. YM3_I:114.1. □

Proposition 6.33 (Holomorphic parameters and parity)**.** Use the same local Gaussian and numerical hypotheses, allowing $x \in D(\mathtt{t}, \sigma_\star \mathtt{t})$ and $\tau = x/\mathtt{t}$. Let four families depend jointly holomorphically on $(\mu, \zeta) \in D(0,\rho) \times \mathcal{O}$, where $\mathcal{O}$ is an open finite-dimensional parameter set or a point. The Gaussian structure, τ and the gauge-fixing operators do not depend on μ. Suppose each internal family has norm at most $\bar\nu_{\mathrm{int}} \le w_\star/8$, the fourth has norm at most $\bar\nu' \le w_\star/4$, and all four retain the stated vanishing orders at $B = 0$. Put

$$\bar B = K_\Pi(32\bar\nu_{\mathrm{int}} + 9\bar\nu'), \qquad \Phi(\mu,\zeta) = \sum_b h_z(b) E^\bullet(X, b; \mu, \zeta), \qquad d = d_{k+1}(X).$$

Then the Gaussian and cutoff terms do not depend on μ, Φ is holomorphic, and $|\Phi| \le 3L^3 \bar B e^{-\lambda_{\mathrm{out}} d}$. If each of the four families satisfies $V^\natural_{-\mu,\zeta}(Y; -B) = V^\natural_{\mu,\zeta}(Y; B)$, then $\Phi(-\mu,\zeta) = \Phi(\mu,\zeta)$. Its odd Taylor coefficients vanish. With $\Phi_{(n)} = (n!)^{-1}\partial_\mu^n \Phi(0,\zeta)$ one has

$$|\Phi_{(n)}| \le 3L^3 \bar B \rho^{-n} e^{-\lambda_{\mathrm{out}} d}, \qquad \left| \Phi - \sum_{n \le N} \Phi_{(n)} \mu^n \right| \le 3L^3 \bar B \frac{(|\mu|/\rho)^{N+1}}{1 - |\mu|/\rho} e^{-\lambda_{\mathrm{out}} d}.$$

For $\rho > 1$, parity improves the remainder after degree two at $\mu = 1$ to $3L^3 \bar B \rho^{-4}(1-\rho^{-2})^{-1} e^{-\lambda_{\mathrm{out}} d}$. ■

Proof. Use the interpolation displayed in §6.2. The bond insertions from the three internal families have total norm at most $32\bar\nu_{\mathrm{int}}$, and that from the fourth has norm at most $9\bar\nu'$. The insertion theorem, Thm. YM3_I:S, bounds their interpolated integral by $\bar B e^{-\lambda_{\mathrm{out}} d}$. The nonnegative tents have total at most $3L^3$, giving the asserted site bound. The common majorant also permits integration of the holomorphic parameter dependence. The Gaussian and cutoff terms use none of the four families.

For parity take the orthogonal map $O_b = -I$ at every nonpivot bond. It preserves the field polydiscs and cutoffs. The covariance, coupling and gauge-fixing operators are unchanged, since $(-I)T(-I) = T$ for each of these linear operators. Their quadratic potentials and marked quadratic insertions are unchanged as well. The interpolation commutes with $B \mapsto -B$, by Prop. YM3_I:48.4. Thus the families and bond insertions at μ are carried to those at $-\mu$. The orthogonal covariance of the insertion and cutoff expansions, Thms. YM3_I:S and YM3_I:T, identifies their individual localized terms. Integration of these identical terms and then summation with the fixed tents proves that Φ is even. This argument compares the localized terms, not only their total partition function.

Cauchy's formula on circles of radius increasing to ρ gives the coefficient bounds. Summing their geometric majorant gives the remainder estimate. Evenness removes the odd coefficients; at $\mu = 1$ the remaining tail is bounded by the geometric series $\rho^{-4} + \rho^{-6} + \cdots$. □

Lemma 6.34. The free-kernel comparison of Prop. YM3_I:121.2 also holds for $k = C^{\mathrm{fl}}_{K,s}$ when $a_s \ge d$, with the additional error $e = (d/a_s)^3$. Its constant may be enlarged by a fixed number depending only on the constants of the flat curvature kernels. In particular, its statement for $k = C^{\mathrm{fl}}_{K,K-n}$ has $e = (d/a_{K-n})^3$ whenever $a_{K-n} \ge d$. ■

Proof. The orthogonal decomposition YM3_I:(2227) gives $C^{\mathrm{fl}}_{K,s} = C^{\mathrm{free}}_K - C^{\mathrm{bg}}_{K,s}$. The all-side background estimate of Prop. YM3_I:120.5 is $|C^{\mathrm{bg}}_{K,s}(p,q)| \le C^{\mathrm{bg}}_{\mathrm{tp}}(a_K/a_s)^3$. On the separated supports put $t = d/a_s \le 1$. Thus the two kernels differ by at most $C^{\mathrm{bg}}_{\mathrm{tp}} t^3 a_K^3 d^{-3}$, whereas YM3_I:(2246) bounds the free kernel by $(1 + 3C_{\mathrm{lat}})a_K^3 d^{-3}$. The squared-kernel estimate in the proof of Prop. YM3_I:121.2, obtained by applying $|u^2 - v^2| \le |u - v|(2|v| + |u - v|)$ to its plaquette sum, consequently gives

$$|\mathcal{K}[C^{\mathrm{fl}}_{K,s}] - \mathcal{K}[C^{\mathrm{free}}_K]| \le 324 C^{\mathrm{bg}}_{\mathrm{tp}}(2 + 6C_{\mathrm{lat}} + C^{\mathrm{bg}}_{\mathrm{tp}}) t^3 \|f\|_\infty \|h\|_\infty.$$

Adding its free-kernel conclusion proves the assertion with the stated fixed enlargement, under the same $8a_K \le d$ and $20d \le \ell$ hypotheses. □

When explicit kernel constants are needed, we use

$$C_\star^{\mathrm{fl}} = C_{\mathrm{tp}}^{\mathrm{fl}}(1/\sqrt{3}), \qquad C_\flat^{\mathrm{tm}} = \left(\frac{3\sqrt{3}}{e\delta_{\mathrm{tp}}L}\right)^3. \tag{6.xxi}$$

The first constant is the convergent slice sum of Prop. YM3_I:120.6; the second bounds $\sup_{t\geq 0} t^3 e^{-\delta_{\mathrm{tp}}Lt/\sqrt{3}}$, whose maximum is at $t = 3\sqrt{3}/(\delta_{\mathrm{tp}}L)$. The comparison constant C_{tm} is chosen to cover both Prop. YM3_I:121.2 and Lem. 6.34.

7. The lattice two-loop coefficient

The first correction in the action-density two-point function can be computed in a fixed finite box before studying the interacting continuum remainder. Let $a_p = 1 - \frac{1}{2}\operatorname{tr} U(\partial p)$, and let $\mathsf{B}_0^{[n]}(p,q)$ be the leading Gaussian coefficient in a box Λ_n. The next coefficient is obtained from

$$\mathsf{B}_1^{\mathrm{lat}}(p,q) = \lim_{n\to\infty}\lim_{\beta\to\infty} \beta^3 \left[\langle a_{p+o_n}; a_{q+o_n}\rangle_{\Lambda_n,\beta} - \beta^{-2}\mathsf{B}_0^{[n]}(p+o_n, q+o_n)\right],$$

where the translations o_n put the fixed plaquettes in the interior of the boxes. The semicolon denotes a connected expectation. The order of these two limits is part of the definition.

Sketch F (The coefficient of degree -5). For plaquettes with normals i, j, the large-distance coefficient has the form

$$\mathsf{B}_1^{\mathrm{lat}}(p,q) = \mathsf{U}_{ij}(x(q)-x(p)) + O(r^{-6}\log r), \qquad r = |x(q)-x(p)| \longrightarrow \infty,$$

where U_{ij} is homogeneous of degree -5. Its scalar sum and the relative short-distance coefficient are

$$\sum_{i,j}\mathsf{U}_{ij}(u) = \kappa^{\mathrm{lat}}|u|^{-5}, \qquad \kappa^{\mathrm{lat}} = \frac{3}{2\pi^3}, \qquad c_1 = \frac{\kappa^{\mathrm{lat}}}{c_2} = \frac{8}{3\pi}, \quad c_2 = \frac{9}{16\pi^2}.$$

■□

Sketch of the argument. Expand the Wilson density and the plaquette insertions in a fixed box, retaining all terms that contribute to the coefficient of β^{-3}. A gauge may be used to evaluate the Gaussian integrals; the connected Wilson expectation defining the coefficient is gauge independent. The source organizes the answer into finitely many contraction kernels. The propagators converge as the box grows, so these kernels define $\mathsf{B}_1^{\mathrm{lat}}$.

The important cancellation occurs before estimating their sum. Terms localized at one of the two plaquettes have symbols whose value and first derivatives vanish at zero momentum. Consequently they contribute only at order r^{-6} or smaller. The remaining convolutions have homogeneous terms of degree -5; subtracting those terms leaves the stated logarithmic remainder. Thus the coefficient comes from a finite homogeneous calculation, rather than from a termwise estimate of all lattice sums.

The six scalar homogeneous contributions reduce to a finite residue calculation. Their normalized values sum to $-2/3$, giving

$$\kappa^{\mathrm{lat}} = -\frac{3}{32\pi^3}\,24\left(-\frac{2}{3}\right) = \frac{3}{2\pi^3}.$$

The contraction kernels and their residues are calculated below. The interacting error estimate is developed separately in §6.

For two nonnegative test functions supported in balls of radius d at distance comparable to d, the lattice-to-continuum comparison has relative error $O((a_K/d)(1+|\log(a_K/d)|))$, with the test shapes fixed. This identifies the same c_1 after smearing. It does not interchange the fixed-box weak-coupling expansion with a continuum limit of the interacting measure. That interchange requires the renormalization estimates described in Sketch E.

7.1. **The lattice two-loop coefficient.** This section concerns the truncated two-point function $\langle a_p; a_q\rangle_{n,\beta}$ of the actions of two plaquettes p, q under the Wilson measure of $SU(2)$ lattice gauge theory in a box $\Lambda_n \subset \mathbb{Z}^3$ with Dirichlet boundary data, at large β. We expand it up to order β^{-3}, pass to the infinite lattice in the coefficients of the expansion, determine the behaviour of the coefficient $\mathsf{B}_1^{\mathrm{lat}}(p,q)$ of order β^{-3} at large separation of p and q, and compute the constant κ^{lat} that governs this behaviour. The argument has four stages.

First, in the box Λ_n the change of variables of Faddeev and Popov and Laplace's method give a constant C, depending on n, τ_{gf}, p and q, such that

$$\big|\langle a_p; a_q\rangle_{n,\beta} - \beta^{-2}\,\mathsf{B}_0^{[n]}(p,q) - \beta^{-3}\,\mathsf{B}_1^{[n]}(p,q;\tau_{\mathrm{gf}})\big| \le C\beta^{-4} \qquad (\beta \ge 1),$$

and neither coefficient depends on the gauge parameter $\tau_{\mathrm{gf}} > 0$ (Prop. 7.12).

Second, the coefficients $\mathsf{B}_0^{\mathrm{lat}}$ and $\mathsf{B}_1^{\mathrm{lat}}$ of the lattice $\mathbb{Z}^3$ are defined by the same cumulants of a Gaussian family as those of a box (Def. 7.10). Here $\mathsf{B}_0^{\mathrm{lat}}(p,q) = \frac{\mathsf{n}}{2}\,\Pi^{\mathrm{lat}}(p,q)^2$ by eq. 7.xvii, while $\mathsf{B}_1^{\mathrm{lat}}(p,q)$ is an absolutely convergent sum of lattice graphs with two loops (Lem. 7.27). By the method of images, the coefficients of the plaquettes $p + o_n$, $q + o_n$, translated to the centre o_n of the box, converge to $\mathsf{B}_j^{\mathrm{lat}}(p,q)$ as $n \to \infty$ (Prop. 7.39). Consequently

$$\mathsf{B}_1^{\mathrm{lat}}(p,q) = \lim_{n\to\infty}\ \lim_{\beta\to\infty}\ \beta^3\big[\langle a_{p+o_n}; a_{q+o_n}\rangle_{n,\beta} - \beta^{-2}\,\mathsf{B}_0^{[n]}(p+o_n, q+o_n)\big],$$

the two limits being taken one after the other, first in β and then in n (Thm. X(c)).

Third, a calculus of lattice kernels, combined with identities at zero momentum for the self-energy and for the vertex function, yields functions $\mathsf{U}_{ij} \in C^1(\mathbb{R}^3 \setminus \{0\})$, $i, j \in \{1,2,3\}$, homogeneous of degree -5, and pure constants C, r_0 such that

$$|\mathsf{B}_1^{\mathrm{lat}}(p,q) - \mathsf{U}_{\lambda(p)\lambda(q)}(u)| \le C\,r^{-6}\lg r \quad (r \ge r_0), \qquad \sum_{i,j} \mathsf{U}_{ij}(u) = \kappa^{\mathrm{lat}}|u|^{-5},$$

where $u = x(q) - x(p)$ joins the barycentres of the plaquettes, $r = |u|$, $\lambda(p)$ and $\lambda(q)$ are their dual directions, and the sum is taken over the nine pairs of dual directions (Thm. V). From this we obtain a rate for the convergence of the smeared coefficient $\mathsf{P}_1^{\mathrm{lat}}(f,h;a)$ to $\kappa^{\mathrm{lat}}\mathcal{K}_5(f,h)$ on test pairs (f,h) (Cor. 7.37).

Fourth, we compute κ^{lat}. Six two-loop integrands of a continuum model carry the terms of degree -5 of the lattice classes of graphs; the constant κ^{lat} is tied to their coefficients of $\mathsf{p}^2 \log|\mathsf{p}|$, and these are evaluated by means of finite parts, an expansion by regions and a functional $\mathcal{R}$. The outcome is $\kappa^{\mathrm{lat}} = \frac{3}{2\pi^3}$ (Thm. W), and hence $c_1 := \kappa^{\mathrm{lat}}/\mathsf{c}_2 = \frac{8}{3\pi}$ (Thm. X). The relation of this coefficient calculation to the interacting continuum two-point function is stated separately in §6.1.

The proof below includes the gauge-fixed finite-box expansion, convergence of the lattice graph sums, comparison with the six continuum integrands, and the residue calculation giving the coefficient. All finite-box remainder constants retain their stated volume dependence. The final scope statement records the order of the limits.

7.1.1. *Notation and hypotheses.* We collect the conventions, the standing data and the facts on $SU(2)$ that are used throughout the section, and then summarise its notation.

Definition 7.1 (Standing setting)**.** The following conventions hold throughout this section.

(1) *The group.* We take $G = SU(2)$ and $N = 2$, and we put $\mathsf{n} := N^2 - 1 = 3$. We write tr for the trace of 2×2 matrices, not divided by 2, and for a complex 2×2 matrix W we denote by $|W|$ its operator norm. The group G is a compact subset of $\mathbb{C}^{2\times 2}$ for the distance of this norm. Its normalised Haar measure du is the Borel probability measure on G that is invariant under left translations; it is unique, and it is also invariant under right translations and under $g \mapsto g^{-1}$. We denote by $\mathfrak{g}$ the real vector space of Hermitian traceless 2×2 matrices, with the inner product $\langle X, Y\rangle := \frac{1}{2}\operatorname{tr} XY$ and the norm $|X| := \langle X, X\rangle^{1/2}$, which

on $\mathfrak{g}$ coincides with the operator norm; dY is the Lebesgue measure of the Euclidean space $(\mathfrak{g}, \langle\cdot,\cdot\rangle)$.

The Pauli matrices

$$\mathsf{e}_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \mathsf{e}_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \qquad \mathsf{e}_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

form an orthonormal basis of $\mathfrak{g}$, and $\mathsf{e}_a\mathsf{e}_b = \delta_{ab}\mathbf{1} + i\sum_c \epsilon_{abc}\mathsf{e}_c$, where ϵ_{abc} is totally antisymmetric in a, b, c with $\epsilon_{123} = 1$: it is the sign of the permutation (a, b, c) of $(1, 2, 3)$, and it vanishes when two of its indices agree. Writing X^a for the coordinates of $X \in \mathfrak{g}$ in this basis, we define the cross product $X \times Y := \sum_{a,b,c} \epsilon_{abc} X^a Y^b \mathsf{e}_c$. For $X, Y \in \mathfrak{g}$ we then have

$$XY = \langle X, Y\rangle\mathbf{1} + i\, X \times Y, \qquad [X, Y] = 2i\, X \times Y, \qquad e^{iX} = \cos|X|\,\mathbf{1} + i\,\frac{\sin|X|}{|X|}\,X,$$
$$\tfrac{1}{2}\operatorname{tr} e^{iX} = \cos|X|;$$

the third identity follows from $X^2 = |X|^2\mathbf{1}$, its right side being $\mathbf{1}$ at $X = 0$. Colour indices $a, a', c, c', \ldots$, as well as the index b in ϵ_{abc}, take their values in $\{1, 2, 3\}$.

(2) *Facts on G; smooth maps.* Put $\sigma_{\mathrm{H}} = \frac{1}{2\pi^2}$; the entire function $\sigma_1(w) = \big(\sum_{n\geq 0}(-w)^n/(2n+1)!\big)^2$, which satisfies $\sigma_1(|Y|^2) = (\sin|Y|/|Y|)^2 \in (0, 1]$ for $|Y| < \pi$; and, for $Y \in \mathfrak{g}$, the linear map $\mathsf{dE}(Y)H = \int_0^1 e^{isY} H e^{-isY}\,\mathrm{d}s$ of $\mathfrak{g}$. For $\varrho > 0$ we write $\mathbb{B}_\varrho = \{Y \in \mathfrak{g} : |Y| < \varrho\}$. The following formulas will be used in the finite-box change of variables. Their proofs follow this definition.

- For $X_1, \ldots, X_n \in i\mathfrak{g}$ one has $|e^{X_1}\cdots e^{X_n} - \mathbf{1}| \leq e^{|X_1|+\cdots+|X_n|} - 1$, all norms being operator norms as in (1). The map $Y \mapsto e^{iY}$ is a homeomorphism of $\mathbb{B}_\pi$ onto $G \setminus \{-\mathbf{1}\}$, whose inverse is denoted by $\frac{1}{i}\log$.
- For $Y, H \in \mathfrak{g}$, $\frac{d}{ds}e^{i(Y+sH)}\big|_{s=0} = i\big(\mathsf{dE}(Y)H\big)e^{iY}$; more generally, $\frac{d}{ds}e^{iZ(s)}\big|_{s=0} = i\big(\mathsf{dE}(Z(0))Z'(0)\big)e^{iZ(0)}$ for every curve $s \mapsto Z(s)$ in $\mathfrak{g}$ that is differentiable at $s = 0$. Moreover $\det \mathsf{dE}(Y) = \sigma_1(|Y|^2)$.
- For every bounded Borel function F on G, $\int_G F\,\mathrm{d}u = \sigma_{\mathrm{H}} \int_{\mathbb{B}_\pi} F(e^{iY})\,\sigma_1(|Y|^2)\,\mathrm{d}Y$. Further, $\int_{\mathfrak{g}} e^{-\frac{1}{2}|Y|^2}\,\mathrm{d}Y = (2\pi)^{3/2}$.

Smooth means C^∞. A map f between subsets of finite-dimensional real vector spaces is smooth if every point of its domain has an open neighbourhood O in the ambient space and a C^∞ map $\tilde{f}$ on O such that $f = \tilde{f}$ on the intersection of O with the domain of f; it is a diffeomorphism if it is bijective and both f and f^{-1} are smooth. Composites of smooth maps are smooth, and for maps between open sets these are the usual notions. The group G and its powers G^m lie in powers of $\mathbb{C}^{2\times 2}$; the product of G and the inversion $g \mapsto g^{-1} = g^*$ are smooth, since they are restrictions of real polynomial maps. Finally, $Y \mapsto e^{iY}$ is a diffeomorphism of $\mathbb{B}_\pi$ onto $G \setminus \{-\mathbf{1}\}$; the proof is given after the present definition.

(3) *Standard facts.* The following standard facts are used without further reference.

- *Measure and integration:* the existence of Haar measure on G; the theorems of Fubini and Tonelli and the monotone and dominated convergence theorems, for integrals and for sums; the Borel measurability of partial integrals; the change of variables formula for a diffeomorphism between open subsets of $\mathbb{R}^m$, in particular for translations, reflections, rotations, dilations and polar coordinates in $\mathbb{R}^3$ and in $\mathbb{R}^6$; the Cauchy–Schwarz inequality; the continuity of translations on $L^2(\mathbb{T})$; Gauss' divergence theorem on a ball and on a cube less a ball. An integer matrix of determinant ± 1 acting on $(\mathbb{R}^3)^{\mathsf{m}}$ induces a measure-preserving bijection of $\mathbb{T}^{\mathsf{m}}$.

  For a function that is continuous on an interval $[a, b]$ or on a compact cube, the Riemann sums over equal subintervals or equal subcubes converge to the integral as the mesh tends to 0. Integration by parts holds in one variable; on $\mathbb{R}^3$, when one of the two

factors has a compact support or lies in the Schwartz space $\mathcal{S}(\mathbb{R}^3)$ of rapidly decreasing smooth functions; and over a period, for smooth periodic functions, where no boundary terms arise.

- *Calculus and topology:* the Leibniz rule, the chain rule, the mean value theorem, the intermediate value theorem and the fundamental theorem of calculus; Taylor's formula with integral remainder, in one and in several variables; the inverse function theorem for C^∞ maps; the Heine–Borel theorem; the density of $\mathbb{Q}^3$ in $\mathbb{R}^3$. A function with continuous partial derivatives is of class C^1. Differentiation under the integral sign with respect to a real or a complex parameter is allowed when the integrand and its derivative have integrable majorants that are locally uniform in the parameter, and a series of C^j functions may be differentiated termwise up to the order j when the suprema of the derivatives of orders $\le j$ are summable. A continuous function on a compact set is bounded and uniformly continuous and attains its minimum, and a finite product of compact metric spaces and a closed subset of a compact metric space are compact.
- *Series:* the power series of exp, cos and sin, the series of $\log(1+w)$ and the binomial series of $(1+w)^{-t}$ for $|w|<1$, and the Cauchy product; the inequalities $\frac{2}{\pi}t \le \sin t \le t$ for $t \in [0, \frac{\pi}{2}]$. A function holomorphic on a disc about 0 is the sum of its Taylor series on that disc, and such a series may be evaluated at a square matrix of norm smaller than its radius of convergence, sums and products of series then giving sums and products of the matrices. For $\|\mathsf{A}\|<1$ we have the Neumann series $(\mathbf{1}+\mathsf{A})^{-1} = \sum_{n\ge0}(-\mathsf{A})^n$ and the expansion $\log\det(\mathbf{1}+\mathsf{A}) = \sum_{n\ge1}\frac{(-1)^{n-1}}{n}\operatorname{tr}\mathsf{A}^n$.
  Formal power series in finitely many variables form a ring, in which division by a series with constant term 1 is possible, and exp, applied to series without constant term, and log, applied to series with constant term 1, are mutually inverse. Real analytic functions on open subsets of $\mathbb{R}^m$, with real, complex or matrix values, are given locally by absolutely convergent power series, are smooth, are locally the sums of their homogeneous Taylor polynomials, and are stable under sums, products, composites and quotients with non-vanishing denominator; examples are the polynomials, the restrictions of holomorphic functions to real arguments, log on $(0,\infty)$, and a power series as above evaluated at a matrix with real analytic entries and norm smaller than the radius of convergence.
- *Linear algebra:* trace, determinant, adjoint, orthonormal bases and the polarisation of a quadratic form. A positive definite operator is invertible; the determinant is an affine function of a rank-one summand; a polynomial vanishing on a non-empty open set is zero; and Lagrange's identity $\sum_{\mu<\nu}|v_\mu w_\nu - v_\nu w_\mu|^2 = |v|^2|w|^2 - |v^*w|^2$ holds for $v, w \in \mathbb{C}^3$. The Cayley map $\mathsf{A} \mapsto (\mathbf{1}-\mathsf{A})(\mathbf{1}+\mathsf{A})^{-1}$ is continuous from the real antisymmetric 3×3 matrices onto the rotations without eigenvalue -1, and these rotations are dense in $SO(3)$.
- *Fourier analysis:* Plancherel's theorem on $\mathbb{R}^3$ and for the map $L^2(\mathbb{T}, d^\circ k) \to \ell^2(\mathbb{Z}^3)$; the uniqueness of the Fourier coefficients of an integrable function on a torus; the Poisson summation formula for functions in $C_c^\infty(\mathbb{R}^D)$, the smooth functions on $\mathbb{R}^D$ with a compact support; and the finite inversion formula

$$f(z) = (2n)^{-3} \sum_{k \in \frac{\pi}{n}\mathbb{Z}^3 \cap (-\pi,\pi]^3} e^{ik\cdot z} \sum_{z'} e^{-ik\cdot z'} f(z')$$

  for functions f on $(\mathbb{Z}/2n)^3$, where z' ranges over $(\mathbb{Z}/2n)^3$. The inverse Fourier transform $\mathcal{F}^{-1}$ of (5) maps $\mathcal{S}(\mathbb{R}^3)$ into itself, and $\mathcal{F}^{-1}(x_j\phi) = -i\partial_{p_j}\mathcal{F}^{-1}\phi$ when ϕ is written as a function of x and $\mathcal{F}^{-1}\phi$ as a function of p; further, $\widehat{\partial_j w} = ip_j\hat{w}$ for every continuously

differentiable function w with a compact support, and $\int_{\mathbb{R}^3} e^{-tp^2+ip\cdot x}\,\mathrm{d}p = (\pi/t)^{3/2}e^{-x^2/4t}$ for $t>0$.

- *Probability:* for every positive semidefinite covariance C there is a centred Gaussian family (ϕ_i) with covariance C; its law is determined by C, and $\langle e^{\sum_i s_i\phi_i}\rangle = e^{\frac12\sum_{i,j}s_is_j\mathsf{C}(i,j)}$ for real s_i, finitely many of them non-zero. If H is a real symmetric positive definite $m\times m$ matrix, then under the probability measure on $\mathbb{R}^m$ with a density proportional to $e^{-\frac12(z,Hz)}$ the coordinates of $\mathbb{R}^m$ form a centred Gaussian family with the covariance H^{-1}.
- *Complex analysis:* the identity theorem on connected open sets, and Euler's Gamma function, which we write Γ_{E}: it is given by $\Gamma_{\mathrm{E}}(z)=\int_0^\infty t^{z-1}e^{-t}\,\mathrm{d}t$ and is holomorphic for $\operatorname{Re}z>0$, it satisfies $\Gamma_{\mathrm{E}}(z+1)=z\,\Gamma_{\mathrm{E}}(z)$ and $\Gamma_{\mathrm{E}}(\frac12)=\sqrt{\pi}$, and $1/\Gamma_{\mathrm{E}}$ extends to an entire function.

(4) *Three functions.* For $x\in\mathbb{R}^D$ we put $\langle x\rangle := \max\{1,|x|\}$. For $t>0$ we put $\lg t := 1+|\log t|$; then $\lg t\ge 1$, $\lg(1/t)=\lg t$, and $\lg(st)\le \lg s\,\lg t$ for $s,t>0$. Finally $\log_+ t := \max\{0,\log t\}$, $\log_+\infty:=\infty$, and $t/0:=\infty$ for $t>0$.

(5) *Fourier conventions.* We put $\mathbb{T}:=\mathbb{R}^3/2\pi\mathbb{Z}^3$ and identify functions on $\mathbb{T}$ with the functions on $\mathbb{R}^3$ that are periodic with the periods $2\pi e_\mu$, $\mu=1,2,3$; integrals over $\mathbb{T}$ are integrals over $[-\pi,\pi]^3$. We write $|k|_{\mathbb{T}} := \operatorname{dist}(k,2\pi\mathbb{Z}^3)$, which equals $|k|$ on $[-\pi,\pi]^3$, and $d^\circ k := (2\pi)^{-3}dk$. We put $\mathsf{k}_\mu(k):=e^{ik_\mu}-1$ and $\mathsf{k}(k):=(\mathsf{k}_\mu(k))_\mu\in\mathbb{C}^3$. Then

$$|\mathsf{k}(k)|^2=\sum_\mu(2-2\cos k_\mu),\qquad \frac{4}{\pi^2}|k|^2\le|\mathsf{k}(k)|^2\le|k|^2\quad\text{on }[-\pi,\pi]^3,$$

the inequalities following from $2-2\cos k_\mu=4\sin^2\frac{k_\mu}{2}$ and the bounds on $\sin t$ in (3). The *kernel with the symbol* $\sigma\in L^1(\mathbb{T})$ is the function $K(x):=\int_{\mathbb{T}}e^{ik\cdot x}\sigma(k)\,d^\circ k$ of $x\in\mathbb{Z}^3$. A bound on a symbol or on an integrand that is said to hold "on $[-\pi,\pi]^3$" or "for $|\kappa|\le c_0$" holds almost everywhere if the function is an L^1-class, and at every point $\neq 0$ if the function is given by a formula. For a function ϕ on $\mathbb{R}^D$ that lies in the Schwartz space $\mathcal{S}(\mathbb{R}^D)$, or is continuous with a compact support, we put $\hat\phi(\kappa):=\int e^{-i\kappa\cdot\zeta}\phi(\zeta)\,\mathrm{d}\zeta$, and on $\mathbb{R}^3$ we put $(\mathcal{F}^{-1}\phi)(x):=\int e^{ip\cdot x}\phi(p)\,d^\circ p$. On $\mathbb{Z}^D$ we use the differences $(\nabla_j f)(\zeta):=f(\zeta+e_j)-f(\zeta)$ and, for a multi-index γ, $\nabla^\gamma:=\prod_j\nabla_j^{\gamma_j}$.

(6) *Data and constants.* The data of a statement are the real numbers and integers quantified in it, and each of them may take every value in its range. The ranges are $\tau_{\mathrm{gf}}>0$ for the gauge parameter (abbreviated τ where this is said), $\beta>0$, $a>0$, $\mathsf{b}>0$, $\mathsf{d}>0$, $\vartheta\in(0,1)$, $\Lambda>1$, $L>1$ for the real number L of Lem. 7.53, and $n\ge 2$ for the side of the box Λ_n. The *tools* are the eight statements marked (K), (S), (P), (L), (D), (E), (V) and (T), namely Lems. 7.17–7.22, Prop. 7.23 and Lem. 7.38; the ranges of their data are those stated in their hypotheses.

A relation between data holds only where a statement imposes it; thus $a\le\mathsf{d}$ is assumed in Cor. 7.37 and in Thm. X(b). For each constant whose existence a statement asserts, such as C, C_γ, c_0, r_0, ϱ_1, ϱ_2, ϱ_3, the statement specifies the quantities on which the constant depends. A *pure constant* is a constant that depends on nothing. In proofs, $C, C', c, \dots$ denote positive constants whose values may change from one occurrence to the next and which depend only on the quantities on which the statement in question allows its constants to depend.

(7) *Length scale.* We work on the lattice $\mathbb{Z}^3$ of spacing 1. For a function f on $\mathbb{R}^3$, $\operatorname{supp}f$ is the closure of $\{f\neq 0\}$, $\|f\|_\infty=\sup|f|$, $\operatorname{Lip}f=\sup_{x\neq y}|f(x)-f(y)|/|x-y|$, $\int f$ is the Lebesgue integral, and $\mathbb{B}(x,\rho)=\{y:|y-x|\le\rho\}$.

(8) *Brackets, indices and notations with several meanings.* On complex forms, the bracket $(\cdot,\cdot)$ of Def. 7.2(2) is Hermitian and antilinear in its first argument: it is the sum of the

sesquilinear extension of $\langle\cdot,\cdot\rangle$, evaluated at the values of the two forms. On $\mathfrak{g}\otimes\mathbb{C}$, on the other hand, $\langle X,Y\rangle$ denotes the complex bilinear extension. For two sites x,x', $\langle x,x'\rangle$ is a bond. The brackets $\langle\cdot\rangle_{n,\beta}$, $\langle\cdot\rangle_{\mathcal{W},\beta}$ and $\langle\cdot\rangle_0$ (Def. 7.3, Lem. 7.11 and Def. 7.8) are means; the same brackets with entries separated by semicolons are cumulants; $\langle x\rangle$ is the function of (4).

When a,a',c,c' occur as upper indices of forms or of legs, as indices of e, δ, ϵ or ∂, or as the second entry of a pair (b,a), they are colour indices; b,b' denote bonds. Forms and site functions are $\mathfrak{g}$-valued unless they are said to be $\mathbb{R}$-valued or complex. Finally, D denotes a dimension, equal to 3 or to 6.

Several notations have more than one meaning in this section, and the following rules tell the meanings apart. Without index, Γ ranges over $\{1,\dots,6\}$ and numbers the integrands F_Γ of Def. 7.32; Γ_q is the kernel of Def. 7.13(5), and Γ_{E} is Euler's function of (3). With the index n, Λ_n is the box; in Lem. 7.20 and wherever that lemma is applied, Λ is the function $\Lambda(\kappa)$ defined there, also when its argument is not written; otherwise Λ is a real number >1. The momenta are $\kappa,\kappa_1,\kappa_2,\dots$, the κ_J are the cumulants of Def. 7.8(2), κ^{lat} and $\kappa^{\mathrm{lat}}_\Gamma$ are numbers, and in the statement of Lem. 7.36 κ alone is a constant.

Before Def. 7.16, σ_1 is the entire function of (2); from that Definition on, $\sigma,\sigma_1,\sigma_2,\dots$ denote symbols, and an s or s_i that accompanies a symbol or a line is its degree, whereas $(s_1,s_2,s_3,s_4)=(+,+,-,-)$ are the signs of the four bonds of a plaquette (Lem. 7.5). Further, n is the number 3 of (1), except in Lem. 7.26(a), (b) and in its proof, where it is a matrix $(n_{vv'})$ of integers. As an index, β is a direction or a multi-index; in Prop. 7.46 and in the proof of Prop. 7.50 it is a real coefficient; otherwise it is the coupling $\beta>0$.

As a datum, $\vartheta\in(0,1)$, whereas the argument ϑ of the function j of Lem. 7.14 is a real variable. As a datum, the integer $n\geq 2$ is the side of the box Λ_n; other integers n, such as the degree $n=s+3$ of a line, the index of the class $\mathfrak{K}(n)$, the exponent $n\in\{4,5\}$ of the sums s_n and the index $n\in\mathbb{Z}$ of the spaces $\mathcal{L}_n$, are introduced where they occur. Besides the kernel q of Lem. 7.22 and the kernels derived from it, q is a vector of $\mathbb{R}^3$ where it is the argument of ρ or of a symbol, and $\mathsf{q}(t)$ is the polynomial of Lem. 7.48.

In Def. 7.47(2), Lem. 7.48 and Lem. 7.49, $\mathsf{a},\mathsf{b},\mathsf{c}$ are integers, while the datum $\mathsf{b}>0$ of (6) is a real number; in $\mathsf{d}_{b,a}$ (Lem. 7.28(I)) the d belongs to the name of a polynomial; and τ is a variable of integration in the proofs of Lems. 7.49 and 7.53. With an argument, $\mathcal{S}(\mathbb{R}^D)$ is the Schwartz space; with an upper index, $\mathcal{S}^s$ is a class of symbols (Def. 7.16(2)).

(9) *Further definitions.* The lattice and its forms are defined in Def. 7.2, the finite-box measure in Def. 7.3, and Gaussian cumulants in Def. 7.8. The test pairs and kernels are defined in §6.1; their bounds are Lem. 6.1.

■

Proof of the group formulas in (2). The product bound follows by writing $e^{X_j}=1+Z_j$, using $|Z_j|\leq e^{|X_j|}-1$, and expanding the product: the sum of the norms of its nonconstant terms is at most $\prod_j(1+|Z_j|)-1\leq e^{\sum_j|X_j|}-1$. Every element of $SU(2)$ has the unique form $a_0\mathbf{1}+i\sum_a a_a\mathsf{e}_a$ with real coefficients and $a_0^2+\sum_a a_a^2=1$. The exponential formula in (1) therefore gives a bijection from $\mathbb{B}_\pi$ to $G\setminus\{-\mathbf{1}\}$. Its smooth inverse is verified below.

For matrices A,B, differentiate $s\mapsto e^{s(A+rB)}e^{(1-s)A}$ and integrate to obtain

$$e^{A+rB}-e^A=r\int_0^1 e^{s(A+rB)}Be^{(1-s)A}\,\mathrm{d}s.$$

Dividing by r and passing to zero proves the stated differential formula. If $\vartheta=|Y|>0$, conjugation by e^{isY} fixes the line spanned by Y and acts on its orthogonal plane as a rotation through $2s\vartheta$. Indeed, for $H\perp Y$ the Pauli identities give $HY=-YH$ and $e^{isY}He^{-isY}=\cos(2s\vartheta)H+\sin(2s\vartheta)iYH/\vartheta$.

Consequently the determinant of its average is

$$\det \mathsf{dE}(Y) = \left(\frac{\sin(2\vartheta)}{2\vartheta}\right)^2 + \left(\frac{1-\cos(2\vartheta)}{2\vartheta}\right)^2 = \left(\frac{\sin\vartheta}{\vartheta}\right)^2.$$

The value at zero follows by continuity.

For completeness, this determinant also identifies the Haar density. Let μ be the pushforward of $\sigma_1(|Y|^2)dY$ on $\mathbb{B}_\pi$ under the exponential. Right translation by $u \in G$ induces $\tau_u(Y) = i^{-1}\log(e^{iY}u)$ wherever its argument is not $-\mathbf{1}$. It is a diffeomorphism after removing at most one point from each chart, with inverse $\tau_{u^{-1}}$. Differentiating $e^{i\tau_u(Y)} = e^{iY}u$ gives

$$\mathsf{dE}(\tau_u(Y))D\tau_u(Y) = \mathsf{dE}(Y), \qquad \sigma_1(|\tau_u(Y)|^2)|\det D\tau_u(Y)| = \sigma_1(|Y|^2).$$

The change of variables formula, with those null points omitted, shows that μ is right invariant. Its mass is $4\pi\int_0^\pi \sin^2\vartheta\,\mathrm{d}\vartheta = 2\pi^2$. Uniqueness of normalized Haar measure proves the integration formula. The Gaussian integral is the product of the three one-dimensional Gaussian integrals in the orthonormal Pauli basis. □

Smoothness of the inverse exponential in (2). Let $\mathrm{cs}(w) := \sum_{n\geq 0}(-w)^n/(2n)!$ and $\mathrm{sn}(w) := \sum_{n\geq 0}(-w)^n/(2n+1)!$. These series define entire functions which are real on $\mathbb{R}$, and comparing them term by term with the series of cos and sin, and differentiating term by term, we obtain

$$\mathrm{cs}(\varphi^2) = \cos\varphi, \qquad \mathrm{sn}(\varphi^2) = \frac{\sin\varphi}{\varphi}, \qquad \sigma_1 = \mathrm{sn}^2, \qquad \mathrm{cs}' = -\tfrac{1}{2}\,\mathrm{sn},$$

for real φ, with $\varphi \neq 0$ in the second identity; moreover $\mathrm{sn}(0) = 1$. By (1), $e^{iY} = \mathrm{cs}(|Y|^2)\mathbf{1} + i\,\mathrm{sn}(|Y|^2)\,Y$ for $Y \in \mathfrak{g}$, so the map $Y \mapsto e^{iY}$ is smooth on $\mathfrak{g}$. On the interval $(-\infty, \pi^2)$ the function sn is positive: for $w < 0$ all terms of its series are positive, and for $0 \leq w < \pi^2$ it equals $\sin\sqrt{w}/\sqrt{w} > 0$, with the value 1 at $w = 0$. Hence cs is strictly decreasing on $(-\infty, \pi^2)$ and its derivative does not vanish there. Moreover $\mathrm{cs}(w) \to +\infty$ as $w \to -\infty$, and $\mathrm{cs}(\pi^2) = \cos\pi = -1$. By the intermediate value theorem cs maps $(-\infty, \pi^2)$ onto $(-1, \infty)$, and by the inverse function theorem the inverse map $\mathrm{cs}^{-1} : (-1, \infty) \to (-\infty, \pi^2)$ is smooth.

Now let $W = e^{iY}$ with $Y \in \mathbb{B}_\pi$. Since $\operatorname{tr} Y = 0$, $\operatorname{tr}\mathbf{1} = 2$ and $Y^* = Y$, the formula above gives $\frac{1}{2}\operatorname{tr} W = \mathrm{cs}(|Y|^2)$ and $\frac{1}{2i}(W - W^*) = \mathrm{sn}(|Y|^2)\,Y$. As $|Y|^2 \in [0, \pi^2)$, the first identity yields $|Y|^2 = \mathrm{cs}^{-1}(\frac{1}{2}\operatorname{Re}\operatorname{tr} W)$, and the second then yields

$$\tfrac{1}{i}\log W = \Big[\mathrm{sn}\big(\mathrm{cs}^{-1}(\tfrac{1}{2}\operatorname{Re}\operatorname{tr} W)\big)\Big]^{-1} \tfrac{1}{2i}(W - W^*) \qquad (W \in G \setminus \{-\mathbf{1}\}).$$

The right side is defined and smooth on the open set $\{W \in \mathbb{C}^{2\times 2} : \frac{1}{2}\operatorname{Re}\operatorname{tr} W > -1\}$: there cs^{-1} is smooth and takes its values in $(-\infty, \pi^2)$, where sn is positive. This open set contains $G \setminus \{-\mathbf{1}\}$, because $\frac{1}{2}\operatorname{tr} e^{iY} = \cos|Y| > -1$ for $Y \in \mathbb{B}_\pi$. Hence the inverse map $\frac{1}{i}\log$ is smooth, and the bijection $Y \mapsto e^{iY}$ of $\mathbb{B}_\pi$ onto $G \setminus \{-\mathbf{1}\}$ given by (2) is a diffeomorphism. □

For reference we summarise the notation introduced in the rest of the section, together with the places where it is defined.

Lattice and boxes (Def. 7.2). We write $(x, \mu) := \langle x, x + e_\mu\rangle$ for the positively oriented bonds and, for a 1-form A, $A_\mu(x) := A((x, \mu))$, with components $A^a_\mu(x)$. Bonds b and plaquettes p have the base points b_- and x_p; the plaquette $p = p_{\mu\nu}(x)$ has the barycentre $x(p) = x + \frac{1}{2}(e_\mu + e_\nu)$ and the dual direction $\lambda(p) \in \{1, 2, 3\} \setminus \{\mu, \nu\}$; $p_{[i]}(x)$ is the plaquette with base point x and dual direction i; and $b + y$, $p + y$ are translates. The differences are

$$(\mathrm{d}\lambda)(b) := \lambda(b_+) - \lambda(b_-), \qquad (\mathrm{d}A)(p) := A(b_1) + A(b_2) - A(b_3) - A(b_4), \qquad (\mathrm{d}^*A)(x) := \sum_\mu \big[A_\mu(x - e_\mu) - A_\mu(x)\big],$$

and $(\cdot,\cdot)$, $\|\cdot\|$ are the sums of $\langle\cdot,\cdot\rangle$ over sites, over positively oriented bonds or over plaquettes; further $\bar\lambda(b) := \frac12(\lambda(b_-)+\lambda(b_+))$ and $(A\times\bar\lambda)(b) := A(b)\times\bar\lambda(b)$. The box is $\Lambda_n := \{0,\dots,n\}^3$, with $\Lambda_n^\circ := \{1,\dots,n-1\}^3$ and $\partial\Lambda_n := \Lambda_n\setminus\Lambda_n^\circ$. Attached to it are the set Bd_n of *free bonds*, which are the positively oriented bonds with both end points in Λ_n and at least one of them in the interior Λ_n°, the plaquettes Pl_n, the configurations $\mathcal{U}_n \cong G^{\mathsf{Bd}_n}$, the forms Form_n, the site functions X_n, the group $\mathcal{G}_n := \{u:\Lambda_n\to G,\ u=\mathbf{1} \text{ on } \partial\Lambda_n\}$, and the $\mathbb{R}$-valued analogues $\mathsf{Form}_n^{\mathbb{R}}$, $\mathsf{X}_n^{\mathbb{R}}$ (Def. 7.2(3)).

Finally $|A|_\infty := \max_b|A(b)|$, $|\lambda|_\infty := \max_x|\lambda(x)|$, and the centre of the box is $o_n := (\lfloor n/2\rfloor,\lfloor n/2\rfloor,\lfloor n/2\rfloor)$.

Measure (Def. 7.3). The Wilson measure of the box is

$$d\mu_{n,\beta}(U) := Z_{n,\beta}^{-1}\exp\Big[-\beta\sum_{p\in\mathsf{Pl}_n}a_p(U)\Big]\prod_{b\in\mathsf{Bd}_n}dU(b),$$

with the mean $\langle\cdot\rangle_{n,\beta}$ and the truncated two-point function $\langle X;Y\rangle_{n,\beta} = \langle XY\rangle_{n,\beta} - \langle X\rangle_{n,\beta}\langle Y\rangle_{n,\beta}$.

Chart (Def. 7.4, Lem. 7.5). The coefficients $\mathsf{a}_p^{(m)}(A)$ are given by $a_p(e^{itA}) = \sum_{m\ge0} t^m\,\mathsf{a}_p^{(m)}(A)$. We put $A^\lambda(b) := \frac1i\log(e^{i\lambda(b_-)}e^{iA(b)}e^{-i\lambda(b_+)})$; the gauge function is $A\mapsto \mathrm{d}^*A$, and the Faddeev–Popov operator is $\mathsf{M}(A)\lambda := \frac{d}{ds}\mathrm{d}^*(A^{s\lambda})\big|_{s=0}$, with $\mathsf{M}(tA) = \sum_{j\ge0}t^j\mathsf{M}^{(j)}(A)$. The signs of the four bonds of a plaquette are $(s_1,s_2,s_3,s_4) := (+,+,-,-)$. With $Z := \frac1i\log\partial(e^{itA})(p)$ we put $\mathfrak{z}_j := \frac{1}{(j+1)!}\partial_t^{j+1}Z\big|_{t=0}$, and $\mathfrak{z}_j(p)$ denotes its value at the values of A on the bonds of p. Finally $\mathsf{a}_Y := -2\,Y\times\cdot$, and $\mathsf{r}(Y)$ is the remainder of fourth order in eq. 7.vi.

Slice (Lem. 7.6). We write $S(U) := \sum_{p\in\mathsf{Pl}_n}a_p(U)$, $\mathsf{T}_n := \{\langle x-e_1,x\rangle : x\in\Lambda_n^\circ\}$, $\mathcal{U}_n^{\mathsf{T}} := \{V\in\mathcal{U}_n : V=\mathbf{1} \text{ on } \mathsf{T}_n\}$ and $\mathsf{Form}_n^{\mathsf{T}} := \{A\in\mathsf{Form}_n : A=0 \text{ on } \mathsf{T}_n\}$, and j for the map $(u,V)\mapsto V^u$. The lemma provides the radii $\varrho_1,\varrho_2,\varrho_3$, the set $\mathfrak{D}$, the map $\Xi:(\lambda,A)\mapsto A^\lambda$ and the set $\mathcal{W} := \Xi(\mathfrak{D})$.

Action, propagators and vertices (Def. 7.7). The upper index lat refers to $\mathbb{Z}^3$ and the upper index $[n]$ to Λ_n. We put $\mathsf{S}_2(A) := \frac12\|\mathrm{d}A\|^2 + \frac{1}{2\tau_{\mathrm{gf}}}\|\mathrm{d}^*A\|^2$, $\mathsf{C}_{\tau_{\mathrm{gf}}} = (\mathrm{d}^*\mathrm{d}+\tau_{\mathrm{gf}}^{-1}\mathrm{d}\mathrm{d}^*)^{-1}$ and $\mathsf{G} = (\mathrm{d}^*\mathrm{d})^{-1}$, and

$$\mathsf{E}_p(b) := \sum_i s_i\,\mathsf{C}_{\tau_{\mathrm{gf}}}(b_i(p),b),\qquad \Pi(p,q) := \sum_j s_j\,\mathsf{E}_p(b_j(q)),\qquad \mathsf{g}(b) := \mathsf{G}(b_+,b_+)+\mathsf{G}(b_-,b_-)-2\mathsf{G}(b_+,b_-),$$

$$\mathsf{D}(b;b') := \tfrac12\sum_{x\in\{b'_-,b'_+\}}\big[\mathsf{G}(x,b_+)-\mathsf{G}(x,b_-)\big].$$

The vertices are $\mathsf{S}_3 := \sum_p\mathsf{a}_p^{(3)}$, the quartic Wilson vertex $\mathsf{S}_4 := \sum_p\mathsf{a}_p^{(4)}$, and

$$\mathsf{V}_2 := \mathsf{S}_4 + \frac13\sum_b|A(b)|^2 + \frac23\sum_b\mathsf{g}(b)|A(b)|^2 - 4\sum_{b,b'}\mathsf{D}(b;b')\mathsf{D}(b';b)\langle A(b),A(b')\rangle,$$

whose last three terms are the *Haar term*, which is the quadratic term contributed by the Haar density in the exponential chart, the ghost tadpole and the ghost loop.

Gaussian calculus (Def. 7.8, Lem. 7.26). The mean $\langle\cdot\rangle_0$ belongs to the centred Gaussian family with $\langle A^a(b)A^{a'}(b')\rangle_0 = \delta_{aa'}\mathsf{C}^\Omega_{\tau_{\mathrm{gf}}}(b,b')$, and $\langle X_1;\dots;X_r\rangle_0$ is the joint cumulant of polynomials $X_1,\dots,X_r$ in this family. A *leg* is a linear form that occurs as a factor of a monomial of a vertex: a *plain leg* is a variable $A^a(b)$, and a *curl leg* is $(\mathrm{d}A)^a(p) = \sum_i s_iA^a(b_i(p))$. The *standard writing* of $\mathsf{a}_p^{(2)}$, $\mathsf{a}_p^{(3)}$ and $\mathsf{a}_p^{(4)}$ is a fixed way of writing them as sums of monomials in plain and curl legs, chosen so that the counting of powers takes the curl legs into account.

On $\mathbb{Z}^3$, S_3 and V_2 are *formal sums of local terms*: the local terms of S_3 are the $\mathsf{a}_p^{(3)}$, and those of V_2 are the $\mathsf{a}_p^{(4)}$, the terms of the sums over bonds b and the terms of the sum over pairs (b,b'); the cumulants of formal sums are defined term by term under a condition of absolute convergence (Def. 7.8(3)). Further $\mathsf{F}[\![X_1,\dots,X_m]\!] := \partial_{s_1}\cdots\partial_{s_m}\mathsf{F}(\sum_i s_iX_i)$; for $\mathsf{i}=(b,a)$, δ_{i} is the form equal to e_a at b and to 0 at the other positively oriented bonds; $\mathsf{C}(\mathsf{i},\mathsf{i}') := \delta_{aa'}\mathsf{C}(b,b')$ and $\mathsf{E}_p^a := \mathsf{E}_p(\cdot)\mathsf{e}_a$; the

values $\mathsf{val}(\mathsf{n})$ and the *classes* $\mathsf{T}_\mathrm{I}, \mathsf{T}_1, \mathsf{T}_4, \mathsf{T}_{4'}, \mathsf{T}_5, \mathsf{T}_6$, which are the topologies of the connected graphs of B_1, are those of Lem. 7.26; and $\mathsf{t}_P(X,Y,Z) := -\langle \mathrm{d}X(P), \sum_{i<j} s_i s_j\, Y(b_i) \times Z(b_j)\rangle$.

Coefficients (Def. 7.10). We put $\mathsf{B}_0^\Omega(p,q) := \langle \mathsf{a}_p^{(2)}; \mathsf{a}_q^{(2)}\rangle_0$; $\mathsf{B}_1^\Omega(p,q;\tau_\mathrm{gf})$ is the combination of seven cumulants in eq. 7.xviii; $\mathsf{B}_j^{[n]}$ and $\mathsf{B}_j^\mathrm{lat}$ are these coefficients for $\Omega = \Lambda_n$ and for $\Omega = \mathbb{Z}^3$; and

$$\mathsf{B}_1^\mathrm{lat}(p,q) := \mathsf{B}_1^\mathrm{lat}(p,q;1), \qquad \mathsf{P}_1^\mathrm{lat}(f,h;a) := a\sum_{p,q} f(a\,x(p))\, h(a\,x(q))\, \mathsf{B}_1^\mathrm{lat}(p,q).$$

Laplace's method (Lem. 7.11). Here $\langle\cdot\rangle_{\mathcal{W},\beta}$ is the mean for the density proportional to $e^{-\beta\Phi-\Theta}$ on $\mathcal{W}$; we put $W_j := \Phi_{j+2} + \Theta_j$, $X_j := X'_{j+2}$ and $Y_j := Y'_{j+2}$, and γ_0, γ_2 are the two coefficients defined in the lemma.

Zero momentum (Def. 7.13, Lems. 7.14 and 7.15). We write $\partial_{b,a}$ for the derivative with respect to $A^a(b)$. The matrices $\mathsf{N}(B)$ and $\mathsf{h}_q(B)$ are the Hessians at $A = B$ of $\sum_p a_p(e^{iA}) + \frac{1}{2\tau_\mathrm{gf}}\sum_x |\mathrm{d}^* A(x)|^2$ and of $a_q(e^{iA})$, respectively. The constant abelian field is $Y_\gamma((x,\mu)) := \gamma_\mu \mathsf{e}_3$. We put $\mathsf{e}_\pm := 2^{-1/2}(\mathsf{e}_1 \pm i\mathsf{e}_2)$ and $\mathsf{e}_0 := \mathsf{e}_3$, call the index $e \in \{+1,-1,0\}$ of e_e its *charge*, which is the eigenvalue of $\frac{1}{2}\operatorname{ad}\mathsf{e}_3$ on e_e, and set $\upsilon := 2e\gamma$.

The symbols at Y_γ are $\hat{\mathsf{N}}_e$, $\hat{\mathsf{h}}_{q,e}$ and $\hat{\mathsf{M}}_e$; an additional index j, as in $\hat{\mathsf{N}}_{e,j}$, $\hat{\mathsf{h}}_{q,e,j}$ and $\hat{\mathsf{M}}_{e,j}$, marks the part that is homogeneous of degree j in γ; and $\hat{\mathsf{C}} := \hat{\mathsf{N}}_{e,0}^{-1}$. The integrands $\mathfrak{s}^{(i)}$ and the numbers $c_{\mu\nu}^{(i)}(\tau_\mathrm{gf})$, $i \le 5$, describe the self-energy, and $Z_0 := \int_\mathbb{T} d^\circ k/|\mathsf{k}(k)|^2$; the integrands $\mathfrak{v}_q^{(j)}$ and the numbers $\mathsf{g}_{q;\mu\nu}^{(j)}(\tau_\mathrm{gf})$, $j \le 4$, describe the vertex function; and Σ_tot, Γ_q are the kernels of Def. 7.13(5).

Moreover $j(\vartheta) := (e^{i\vartheta}-1)/i\vartheta$ for real $\vartheta \neq 0$ and $j(0) := 1$, $J := \mathrm{diag}(j(\upsilon_\mu))$, $P(k) := |\mathsf{k}(k)|^2\mathbf{1} - \mathsf{k}(k)\mathsf{k}(k)^*$, and, for $q = p_{\mu\nu}(x)$, $D_q(k)\alpha = \mathsf{k}_\mu(k)\alpha_\nu - \mathsf{k}_\nu(k)\alpha_\mu$ and $\varpi_q(k) := (|\mathsf{k}_\mu(k)|^2 + |\mathsf{k}_\nu(k)|^2)/|\mathsf{k}(k)|^2$. Finally $R_c : x \mapsto 2c - x$ is the point reflection at c, and $(x,\mu)^R = (2c - x - e_\mu, \mu)$.

Kernel calculus (Def. 7.16 and the tools). The cut-offs are $\theta_0(t) := e^{-1/t}$ for $t > 0$, $\theta_0(t) := 0$ for $t \le 0$,

$$\theta(s) := \frac{\theta_0(1-s)}{\theta_0(1-s) + \theta_0(s-\frac{1}{2})}, \qquad \chi(\kappa) := \theta(|\kappa|/2).$$

The symbol classes are the $\mathcal{S}^s$. A *line symbol* of degree s, an integer ≥ -2, is a symbol in $\mathcal{S}^s$ with a smooth leading part σ^h that is homogeneous of degree s and of parity $(-1)^s$, the remainder $\sigma - \chi\sigma^h$ lying in $\mathcal{S}^{s+1}$; its kernel is an *elementary line* of degree $n = s+3$, which decays like $|x|^{-n}$. The *line class* $\mathfrak{K}(n)$ is defined by (K1)–(K3): its elements are kernels that have a homogeneous leading symbol and a homogeneous leading kernel, up to one order and a logarithm, and an element K has its *leading data* (σ^h, K^h) and its *constants*.

A *slowly varying function* is a lattice function whose differences of orders up to ten each gain a factor R^{-1}, as measured by $\|g\|_{(R)} := \max_{|\gamma|\le 10} R^{|\gamma|} \sup_\zeta |\nabla^\gamma g(\zeta)|$; with it we use $\check{g}(\kappa) := \sum_\zeta g(\zeta) e^{-i\kappa\cdot\zeta}$ and $\|G\|_{C^{10}}$. The *majorants* of degree $s > -D$ are the Borel functions on $\mathbb{R}^D$ with values in $[0,\infty]$ that are homogeneous of degree s and integrable over the ball $\{|\kappa| \le 1\}$; they form the class $\mathcal{H}_s(D)$. The kernel $\mathcal{F}_{\neq 0}^{-1}\sigma^h$ of a homogeneous symbol σ^h is defined in Lem. 7.17(b).

For the one-loop functions of Lem. 7.20, which are built from $\mathsf{m} \in \{2,3\}$ line symbols $\sigma_1, \dots, \sigma_\mathsf{m}$ of the degrees $s_1, \dots, s_\mathsf{m}$, from signs $\epsilon_i \in \{\pm 1\}$ and from linear maps $\lambda_i : \mathbb{R}^D \to \mathbb{R}^3$ with $D = 3(\mathsf{m}-1)$, we use

$$\mathsf{I}(\kappa) := \int_\mathbb{T} \prod_i \sigma_i(\epsilon_i l + \lambda_i(\kappa))\, d^\circ l, \qquad l_i(\kappa) := -\epsilon_i\lambda_i(\kappa), \qquad \omega := 3 + \sum_i s_i, \qquad \Lambda(\kappa) := 1 + \sum_{i<j} \log_+ \frac{|\kappa|}{|l_i(\kappa) - l_j(\kappa)|},$$

together with *the polynomial* Q and *the homogeneous part* I^h of I, which are its Taylor part and its non-analytic homogeneous part at small κ. In Lem. 7.22, a *chain* is a convolution of lines, $(X * Y)(u) :=$

$\sum_z X(z)Y(u-z)$, and a *bubble* is a product $K_\mathrm{a}K_\mathrm{b}$ of two lines, that is, a one-loop two-point graph; with the polynomial Q of the symbol of the bubble we put $\tilde{\mathsf{Q}}(k) := \mathsf{Q}(0) + \sum_j \partial_j \mathsf{Q}(0) \sin k_j$, and q is the kernel with the symbol $\tilde{\mathsf{Q}}$. The *kite* of Prop. 7.23 is the two-loop two-point graph with five elementary lines $K_1, \dots, K_5$ of the degrees $n_1, \dots, n_5$,

$$\mathsf{T}(u) := \sum_{z,w} K_1(z)K_2(w)K_3(w-z)K_4(u-z)K_5(u-w), \qquad \Delta := \sum_i n_i - 6,$$

and with it we use the functions $\hat{\mathsf{T}}_\mathsf{p}, \hat{\mathsf{T}}_\mathsf{q}$, the numbers $\omega_\mathsf{p}, \omega_\mathsf{q}$, the polynomials $\mathsf{Q}_\bullet$ with the corresponding $\tilde{\mathsf{Q}}_\bullet$ and kernels $\mathsf{q}_\bullet$, the function U, and the *short* and the *long lines*, which are the lines of the kite cut off to $|y| \lesssim \varepsilon r$ and to $|y| \gtrsim \varepsilon r$, respectively. On discrete tori (Lem. 7.38) we use $\mathbb{T}_n := \frac{\pi}{n}\mathbb{Z}^3 \cap (-\pi, \pi]^3$, $|z|_n$, $\langle z \rangle_n$ and $K^{(n)}(z) := (2n)^{-3} \sum_{k \in \mathbb{T}_n \setminus \{0\}} e^{ik\cdot z} \sigma(k)$.

Graphs on $\mathbb{Z}^3$. For $p_{[\iota]} = p_{\alpha\beta}$ we put $\mathsf{c}_{\iota,\nu}(k) := \mathsf{k}_\alpha(k)\delta_{\nu\beta} - \mathsf{k}_\beta(k)\delta_{\nu\alpha}$, and $\hat{\mathsf{C}}_{\mu\nu}(k)$ denotes the bracket in the Fourier representation of $\mathsf{C}^{\mathrm{lat}}_{\tau_\mathrm{gf}}$ in Def. 7.7(1) (Lem. 7.24). For $p = p_{\mu\nu}(x)$, $\epsilon_p := -1$ if $(\mu, \nu) = (1, 3)$ and $\epsilon_p := +1$ otherwise, and $p^\flat := (x, \lambda(p))$ (Lem. 7.25). Lem. 7.28 uses $u' := x_q - x_p$, the index sets $\mathfrak{G}_\bullet(\iota, \iota')$, the coefficients c_G of the *elementary terms* G, each of them the contribution of one monomial and one pairing to one of the classes $\mathsf{T}_\bullet$, the lines $K^{\rightarrow}_{\iota\mu}(y) := \mathsf{E}_{p_{[\iota](0)}}((y, \mu))$ and $K^{\leftarrow}_{\nu\iota'}(y) := \mathsf{E}_{p_{[\iota']}(y)}((0, \nu))$, the kernel $\Sigma_{\mu\nu}(y) := \Sigma_{\mathrm{tot}}((x, \mu), (x + y, \nu))$, the signs $\mathsf{s}_P(b)$, the decomposition $\partial_{b,a}\mathsf{S}_3 = \mathsf{d}_{b,a} + \mathsf{u}_{b,a}$, the kernel $\mathsf{q}^0_{\mu\nu}$ of finite support, and the difference operators $\mathsf{f}_G(\nabla)$ of degree d_G.

Lems. 7.29 and 7.35 use $\mathsf{q}_X, \mathsf{q}_Y$, $X' := X - \mathsf{q}_X$, $Y' := Y - \mathsf{q}_Y$ and the functions U_G. Lem. 7.31 uses the kernel q_G of the bubble $K_\mathrm{a}K_\mathrm{b}$ of each term G of $\Sigma_{\mu\nu}$, and it splits $\Sigma_{\mu\nu} = \mathsf{q}^\Sigma_{\mu\nu} + \mathsf{W}_{\mu\nu}$ into a kernel $\mathsf{q}^\Sigma_{\mu\nu}$ of finite support and a remainder $\mathsf{W}_{\mu\nu} \in \mathfrak{K}(4)$,

$$\mathsf{q}^\Sigma_{\mu\nu} := \mathsf{q}^0_{\mu\nu} + \sum_G c_G\, \mathsf{f}_G(\nabla)\mathsf{q}_G, \qquad \mathsf{W}_{\mu\nu} := \sum_G c_G\, \mathsf{f}_G(\nabla)[K_\mathrm{a}K_\mathrm{b} - \mathsf{q}_G];$$

it also introduces a symmetric kernel γ_q of finite range and the kernels $\gamma_{q;\mu\nu}$ derived from it. The *local parts* are kernels of finite support that carry the Taylor polynomials, of degree at most 1, of one-loop subgraphs, such as q, $\mathsf{q}_\bullet$, $\mathsf{q}^\Sigma_{\mu\nu}$ and γ_q; by Lem. 7.31, the symbols of $\mathsf{q}^\Sigma_{\mu\nu}$ and of $\gamma_{q;\mu\nu}$ vanish to second order at zero momentum. In a second sense of the words, distinct from the local terms of the formal sums S_3 and V_2, the proofs of Lem. 7.35 and Thm. V call local terms those terms of the expansions of the elementary terms that contain one of the kernels q_X, q_Y, q_p, q_q, together with the two halves of $\mathsf{T}_{4'}$; they are grouped into the local terms at q and those at p.

Two integrals over a loop momentum l agree *integrand by integrand* if their integrands coincide before integration, up to substitutions $l \to \pm l + \lambda(\kappa)$ with λ linear.

Model (Def. 7.32, Lems. 7.34, 7.35 and 7.36, Thm. V). The tensors are

$$V_2(k_1, k_2)_{\mu\nu} := -(k_1 \cdot k_2)\delta_{\mu\nu} + k_{1\nu}k_{2\mu}, \qquad V_3(k_1, k_2, k_3)_{\alpha\beta\gamma} := \delta_{\alpha\gamma}(k_1 - k_3)_\beta + \delta_{\alpha\beta}(k_2 - k_1)_\gamma + \delta_{\beta\gamma}(k_3 - k_2)_\alpha,$$

and the variables are $\mathsf{x}_0 := \mathsf{p}^2$, $\mathsf{x}_1 := k^2$, $\mathsf{x}'_1 := (k - \mathsf{p})^2$, $\mathsf{x}_2 := l^2$, $\mathsf{x}'_2 := (l - \mathsf{p})^2$, $\mathsf{x}_3 := (k - l)^2$. The six integrands $F_1, \dots, F_6$ are those of eq. 7.xxxv, each with its *lines*, which are the vectors whose squares occur in its denominator, counted with multiplicity. With $\rho_0 := \frac{\pi}{16}$ and $\rho(\mathsf{q}) := \theta(|\mathsf{q}|/2\rho_0)$, R_Γ is the product of ρ over the lines of Γ, and

$$\Psi_\Gamma(\mathsf{p}) := \iint R_\Gamma F_\Gamma \, d^\circ k \, d^\circ l, \qquad \mathsf{T}^{\mathrm{mod}}_\Gamma(u) := 24 \int_{\mathbb{R}^3} e^{i\mathsf{p}\cdot u} \Psi_\Gamma(\mathsf{p}) \, d^\circ \mathsf{p}.$$

The *model* consists of the continuum two-loop integrals Ψ_Γ, with the smooth ultraviolet cut-off ρ on every line, and of the functions $\mathsf{T}^{\mathrm{mod}}_\Gamma$. Lem. 7.34 introduces the *leading integrand* F_G of an elementary term G, which is the product of the leading parts of the symbols of its lines, the sets $\mathfrak{G}_2, \mathfrak{G}_3$ and the symbols $\sigma_\Pi, \sigma_\rightarrow, \sigma_\leftarrow$; Lem. 7.35 the functions U_Γ, I_Γ and $\Pi^h := \mathcal{F}^{-1}_{\neq 0}\sigma^h_\Pi$; Lem. 7.36

the numbers $c_{4,\Gamma}$ and $\kappa^{\mathrm{lat}}_\Gamma$, given by $\mathsf{I}_\Gamma = c_{4,\Gamma}|u|^{-4}$ and $\mathsf{U}_\Gamma(u) = \kappa^{\mathrm{lat}}_\Gamma|u|^{-5}$; and Thm. V the functions U_{ij} and the constant $\kappa^{\mathrm{lat}} := \sum_{\Gamma=1}^{6} \kappa^{\mathrm{lat}}_\Gamma$.

Value. Lem. 7.41 introduces $c(s) := \dfrac{2^s\,\Gamma_{\mathrm{E}}(\frac{3+s}{2})}{\pi^{3/2}\,\Gamma_{\mathrm{E}}(-\frac{s}{2})}$, and Lem. 7.42 the sums $\mathsf{s}_n(\mathsf{p}) := \sum_{u\in\mathbb{Z}^3\setminus\{0\}} |u|^{-n} e^{-i\mathsf{p}\cdot u}$ and the functions $\mathsf{e}_4, \mathsf{e}_5$. Lem. 7.43 uses the numbers $\mathsf{a}_0, \mathsf{a}_1, \mathsf{a}_2$, the function o, and $R^\Lambda_\Gamma(k,l,\mathsf{p}) := R_\Gamma(k/\Lambda, l/\Lambda, \mathsf{p}/\Lambda)$. Def. 7.44 introduces $\mathbb{D} := \{\mathrm{Re}\,\mathsf{z} > -\frac12\} \setminus \{\frac12, \frac32, \dots\}$, the homogeneous factors of degree d, the class $\mathfrak{P}$, the finite part $\mathrm{f.p.}\int \mathsf{g}\,\mathrm{d}k$, that is, the value at $\mathsf{z}=0$ of the continuation of $\int |k|^{2\mathsf{z}}\mathsf{g}\,\mathrm{d}k$, the cut-off $\eta_c(k) := \theta(4|k|/c)$, the class $\mathfrak{P}_2(c)$ defined by (Φ0)–(Φ3), the finite part $\mathrm{f.p.}\iint$, the class $\mathcal{Y}(c)$ defined by (Y1), (Y2), the scaled cut-offs Υ^Λ, and

$$\overline{G_2}(k,l) := \tfrac16\,\Delta_{\mathsf{p}} G(k,l,\mathsf{p})\big|_{\mathsf{p}=0}, \qquad \Xi_\Gamma[\Upsilon](\mathsf{p}) := \iint \Upsilon F_\Gamma\, d^\circ k\, d^\circ l.$$

The homogeneous factors and the classes $\mathfrak{P}$, $\mathfrak{P}_2(c)$ and $\mathcal{Y}(c)$ describe the domain of the finite part in one and in two variables and the admissible cut-offs, which vanish near the origin. Lem. 7.45(d) treats a *soft line*, that is, a factor $S(k,\mathsf{p})$ of degree -2 that is singular at $k=0$ and at $k=\mathsf{p}$, and it uses $[SJ]_i(k;\mathsf{p}) := \sum_{|\alpha|=i} \frac{\mathsf{p}^\alpha}{\alpha!}\partial^\alpha_{\mathsf{p}}(SJ)(k,0)$ and the function h_S; Prop. 7.46 defines $\mathfrak{f}_\Gamma[\Upsilon] := (2\pi)^{-6}\,\mathrm{f.p.}\iint \overline{(\Upsilon F_\Gamma)_2}\,\mathrm{d}k\,\mathrm{d}l$.

Def. 7.47 introduces the spaces $\mathcal{L}_n$, the coefficient r_{111} of $(\mathsf{x}_1\mathsf{x}_2\mathsf{x}_3)^{-1}$, the space $\mathfrak{F}$ and the functional $\mathcal{R}[F] := r_{111}(\overline{F_2})$, which takes the coefficient of $(\mathsf{x}_1\mathsf{x}_2\mathsf{x}_3)^{-1}$ in the angular mean of the Taylor coefficient of order p^2 of F; Lem. 7.48 uses $\mathsf{q}(t) := \frac13 t(2t-1)$; Prop. 7.50, the *bridge*, expresses $\kappa^{\mathrm{lat}}_\Gamma$ through $\mathcal{R}[F_\Gamma]$ and defines $d_{2,\Gamma} := \mathcal{R}[F_\Gamma]/16\pi^2$; Def. 7.52 defines $c_1 := \kappa^{\mathrm{lat}}/\mathsf{c}_2$; and Lem. 7.53 uses the *shell*, that is, the annulus $\mathsf{Ann}_r := \{u\in\mathbb{R}^3 : r \le |u| < Lr\}$ with a real number $L>1$.

The lattice conventions below specify the oriented cells, gauge transformations and differences. A subscript D on a bond or plaquette set denotes restriction to the domain D.

The joint cumulants are given by the partition formulas in Def. 7.8. The Gibbs kernel and the dilation of the smeared observable are written explicitly where they enter the finite-box argument.

The continuum test pairs $\mathfrak{T}_{\mathbb{R}^3}(\mathsf{d};\vartheta)$ and the kernels $\mathcal{K}_q(f,h)$ are those of §6.1. The leading constant is $\mathsf{c}_2 = 9/(16\pi^2)$. The curl projection and its Fourier kernel are specified below.

7.1.2. *Lattice, boxes, chart.*

Definition 7.2 (Lattice, forms, boxes with Dirichlet data)**.** (1)

The lattice. We work on the lattice $\mathbb{Z}^3$ of spacing 1. A bond $b = \langle b_-, b_+\rangle$ is an ordered pair of neighbouring sites, $\bar b := \langle b_+, b_-\rangle$ is its reverse, and the positively oriented bonds are

$$(x,\mu) := \langle x, x+e_\mu\rangle \qquad (x\in\mathbb{Z}^3,\ \mu\in\{1,2,3\});$$

the base point of (x,μ) is $x = b_-$. For $x\in\mathbb{Z}^3$ and $1\le\mu<\nu\le 3$ the plaquette $p = p_{\mu\nu}(x)$ has the base point $x_p := x$, the four corners x, $x+e_\mu$, $x+e_\mu+e_\nu$, $x+e_\nu$, and the four bonds

$$b_1 = (x,\mu), \qquad b_2 = (x+e_\mu,\nu), \qquad b_3 = (x+e_\nu,\mu), \qquad b_4 = (x,\nu).$$

A plaquette is determined by the triple (x,μ,ν).

Let $\mathcal{B}$ be a set of bonds which contains the reverse of each of its elements. A G-valued configuration on $\mathcal{B}$ is a map U on $\mathcal{B}$ with $U(\bar b) = U(b)^{-1}$; the configuration with the constant value $\mathbf{1}$ is again denoted by $\mathbf{1}$. If the bonds of a plaquette p belong to $\mathcal{B}$, we set

$$\partial U(p) := U(b_1)\,U(b_2)\,U(b_3)^{-1}\,U(b_4)^{-1},$$

and a G-valued function u on a set of sites containing the end points of the bonds of $\mathcal{B}$ acts on U by the gauge transformation $U^u(b) := u(b_-)\,U(b)\,u(b_+)^{-1}$. A $\mathfrak{g}$-valued 1-form is a map A on bonds with $A(\bar b) = -A(b)$, and e^{iA} denotes the configuration $b\mapsto e^{iA(b)}$. The configurations on the set of all bonds form the space $\mathcal{V}(\mathbb{L}_1)$; a configuration on a set B of positively oriented bonds, extended to

the reverses by $U(\bar b) = U(b)^{-1}$, is an element of G^B. We write $A_\mu(x) := A((x,\mu))$, and $A^a_\mu(x)$ for the components of $A_\mu(x)$ in the basis $\mathsf{e}_1, \mathsf{e}_2, \mathsf{e}_3$.

The barycentre of $p = p_{\mu\nu}(x)$ is $x(p) = x + \frac12(e_\mu + e_\nu)$, and its *dual direction* is the index $\lambda(p) \in \{1,2,3\} \setminus \{\mu,\nu\}$. For $x \in \mathbb{Z}^3$ and $i \in \{1,2,3\}$ we denote by $p_{[i]}(x)$ the plaquette with base point x and dual direction i. The plaquettes correspond to the pairs (z,λ), the barycentre of (z,λ) being $z + \frac12(e_1+e_2+e_3-e_\lambda)$: the map $(z,\lambda) \mapsto p_{\mu\nu}(z)$, where $\{\mu,\nu\} = \{1,2,3\} \setminus \{\lambda\}$, is a bijection which preserves base point, barycentre and dual direction, and it sends (x,i) to $p_{[i]}(x)$. For $y \in \mathbb{Z}^3$ the translates $b+y$ and $p+y$ are the images of b and p under the translation τ_y.

For $U \in \mathcal{V}(\mathbb{L}_1)$ the plaquette action is $a_p(U) = 1 - \frac12 \operatorname{tr} \partial U(p)$. It depends on the values of U at the four bonds of p only, and we use the same formula for every configuration U on a set of bonds which is stable under reversal and contains the bonds of p. For every such U, and every G-valued u on a set of sites containing the end points of the bonds on which U is given,

$$a_p(U) \in [0,2], \qquad a_p(U^u) = a_p(U).$$

For the first relation we recall that every $g \in G$ can be written as e^{iY} with $Y \in \mathfrak{g}$, $|Y| \le \pi$: for $g \ne -\mathbf{1}$ this is the exponential chart in Def. 7.1(2), and $-\mathbf{1} = e^{i\pi \mathsf{e}_3}$. Since $\frac12 \operatorname{tr} e^{iY} = \cos|Y|$, the choice $e^{iY} = \partial U(p)$ gives $a_p(U) = 1 - \cos|Y| \in [0,2]$. For the second relation let x be the base point of p; in the product

$$\begin{aligned}\partial U^u(p) = {} & u(x)\,U(b_1)\,u(x+e_\mu)^{-1} \cdot u(x+e_\mu)\,U(b_2)\,u(x+e_\mu+e_\nu)^{-1} \\ & \cdot u(x+e_\mu+e_\nu)\,U(b_3)^{-1}\,u(x+e_\nu)^{-1} \cdot u(x+e_\nu)\,U(b_4)^{-1}\,u(x)^{-1}\end{aligned}$$

the inner factors cancel, so that $\partial U^u(p) = u(x)\,\partial U(p)\,u(x)^{-1}$, and the trace is invariant under conjugation.

(2)

Differences. For a site function λ, $\mathfrak{g}$-valued or complex, and a 1-form A we set

$$(\mathrm{d}\lambda)(b) := \lambda(b_+) - \lambda(b_-), \qquad (\mathrm{d}A)(p) := A(b_1) + A(b_2) - A(b_3) - A(b_4),$$
$$(\mathrm{d}^*A)(x) := \sum_\mu [A_\mu(x-e_\mu) - A_\mu(x)].$$

The relation $\mathrm{dd}\lambda = 0$ holds: at $p = p_{\mu\nu}(x)$,

$$\begin{aligned}(\mathrm{dd}\lambda)(p) = {} & [\lambda(x+e_\mu) - \lambda(x)] + [\lambda(x+e_\mu+e_\nu) - \lambda(x+e_\mu)] \\ & - [\lambda(x+e_\mu+e_\nu) - \lambda(x+e_\nu)] - [\lambda(x+e_\nu) - \lambda(x)] = 0.\end{aligned}$$

On $\mathfrak{g}$-valued λ and A the three operators act componentwise in the basis e_a. By $(\cdot,\cdot)$ and $\|\cdot\|$ we denote the sums of $\langle\cdot,\cdot\rangle$ over the sites, over the positively oriented bonds or over the plaquettes, according to the arguments; on complex forms $(\cdot,\cdot)$ is the Hermitian sum, antilinear in its first argument. If λ and A have finite support, a shift of the summation variable gives

$$\begin{aligned}(\mathrm{d}\lambda, A) &= \sum_{x,\mu} \big\langle \lambda(x+e_\mu) - \lambda(x), A_\mu(x) \big\rangle \\ &= \sum_x \Big\langle \lambda(x), \sum_\mu [A_\mu(x-e_\mu) - A_\mu(x)] \Big\rangle = (\lambda, \mathrm{d}^*A).\end{aligned}$$

On plaquette functions, d^* denotes the adjoint of d on 1-forms, which is again a difference operator. To write it down, we put $F_{\mu\nu}(x) := F(p_{\mu\nu}(x))$ for a plaquette function F and $\mu < \nu$, and $F_{\nu\mu}(x) := -F_{\mu\nu}(x)$, $F_{\mu\mu}(x) := 0$. Since $(\mathrm{d}A)(p_{\mu\nu}(x)) = [A_\nu(x+e_\mu) - A_\nu(x)] - [A_\mu(x+e_\nu) - A_\mu(x)]$, the antisymmetry of $F_{\mu\nu}$ in (μ,ν) and a shift of the summation variable give, for A and F of finite

support,

$$(\mathrm{d}A, F) = \sum_x \sum_{\mu,\nu} \langle A_\nu(x+e_\mu) - A_\nu(x), F_{\mu\nu}(x)\rangle$$
$$= \sum_{x,\rho} \Big\langle A_\rho(x), \sum_\nu [F_{\nu\rho}(x-e_\nu) - F_{\nu\rho}(x)]\Big\rangle,$$

so that

$$(\mathrm{d}^* F)((x,\rho)) = -\sum_\nu [F_{\rho\nu}(x-e_\nu) - F_{\rho\nu}(x)];$$

by this formula we apply d^* to plaquette functions of arbitrary support. Finally we set $\bar\lambda(b) := \frac{1}{2}(\lambda(b_-) + \lambda(b_+))$ and $(A \times \bar\lambda)(b) := A(b) \times \bar\lambda(b)$.

(3)

Boxes. Let $n \geq 2$ be an integer, and let

$$\Lambda_n := \{0, \dots, n\}^3, \qquad \Lambda_n^\circ := \{1, \dots, n-1\}^3, \qquad \partial\Lambda_n := \Lambda_n \setminus \Lambda_n^\circ. \tag{7.i}$$

A positively oriented bond is *free* if both its end points lie in Λ_n and at least one of them lies in Λ_n°; Bd_n denotes the set of free bonds, and Pl_n the set of plaquettes all of whose corners lie in Λ_n. A bond with both end points in Λ_n which is neither free nor the reverse of a free bond has both end points in $\partial\Lambda_n$.

- $\mathcal{U}_n$ is the set of configurations on the bonds with both end points in Λ_n which take the value $\mathbf{1}$ at every bond that is neither free nor the reverse of a free bond; restriction to Bd_n identifies $\mathcal{U}_n$ with G^{Bd_n}.
- Form_n is the space of 1-forms on the same bonds which vanish at every bond that is neither free nor the reverse of a free bond. X_n is the space of $\mathfrak{g}$-valued functions on Λ_n which vanish on $\partial\Lambda_n$, and
$$\mathcal{G}_n := \{u : \Lambda_n \to G : \ u = \mathbf{1} \text{ on } \partial\Lambda_n\}.$$
- For $A \in \mathsf{Form}_n$, $\mathrm{d}A$ is considered on Pl_n, and d^*A on Λ_n°, where it involves only bonds with both end points in Λ_n; we extend d^*A by 0 to $\partial\Lambda_n$, so that $\mathrm{d}^*A \in \mathsf{X}_n$. For $\lambda \in \mathsf{X}_n$ we have $\mathrm{d}\lambda \in \mathsf{Form}_n$, because λ vanishes at both end points of a bond in Λ_n which is not free. The relation $(\mathrm{d}\lambda, A) = (\lambda, \mathrm{d}^*A)$ for $\lambda \in \mathsf{X}_n$, $A \in \mathsf{Form}_n$ follows from (2), applied to the extensions of λ and A by zero. For $u \in \mathcal{G}_n$ and $U \in \mathcal{U}_n$ we have $U^u \in \mathcal{U}_n$, since $U^u(b) = u(b_-)u(b_+)^{-1} = \mathbf{1}$ at a bond in Λ_n which is not free.
- $\mathsf{Form}_n^{\mathbb{R}}$ and $\mathsf{X}_n^{\mathbb{R}}$ denote the real-valued analogues of Form_n and X_n; an element of Form_n, resp. of X_n, has three components in $\mathsf{Form}_n^{\mathbb{R}}$, resp. in $\mathsf{X}_n^{\mathbb{R}}$. We write $|A|_\infty := \max_b |A(b)|$ and $|\lambda|_\infty := \max_x |\lambda(x)|$, and $o_n := (\lfloor n/2 \rfloor, \lfloor n/2 \rfloor, \lfloor n/2 \rfloor)$ is the centre of the box.
- Geometrically, the plaquette $p_{\mu\nu}(x)$ is the square $\{x + se_\mu + te_\nu : \ s,t \in [0,1]\}$ with the four corners listed in (1), and the set of a bond b is $[b_-, b_+] := \{(1-s)b_- + sb_+ : \ s \in [0,1]\}$. Write $\mathsf{Pl}(D)$ and $\mathsf{Bd}(D)$ for the plaquettes and positively oriented bonds contained in a set D. Then $\mathsf{Pl}_n = \mathsf{Pl}([0,n]^3)$, and Bd_n is the set of the bonds of $\mathsf{Bd}([0,n]^3)$ with an end point in Λ_n°. Indeed, the square of a plaquette is the convex hull of its four corners, and $[b_-, b_+]$ is the convex hull of b_- and b_+; since the cube $[0,n]^3$ is convex, it contains the square, resp. the set $[b_-, b_+]$, if and only if it contains these sites, and the sites of $\mathbb{Z}^3$ in $[0,n]^3$ are the points of Λ_n.

■

Definition 7.3 (Wilson measure of a box; truncated two-point function)**.** Let $n \geq 2$ and $\beta > 0$. The *Wilson measure* of the box Λ_n is the probability measure

$$d\mu_{n,\beta}(U) := Z_{n,\beta}^{-1} \exp\Big[-\beta \sum_{p \in \mathsf{Pl}_n} a_p(U)\Big] \prod_{b \in \mathsf{Bd}_n} dU(b) \tag{7.ii}$$

on $\mathcal{U}_n \cong G^{\mathsf{Bd}_n}$, where $dU(b)$ is the normalised Haar measure in the variable $U(b)$ and $Z_{n,\beta} > 0$ is the integral of the exponential with respect to the product measure, so that $\mu_{n,\beta}$ has total mass 1; its mean is denoted by $\langle\cdot\rangle_{n,\beta}$. For bounded Borel functions X, Y on $\mathcal{U}_n$ define their joint cumulant by

$$\langle X;Y\rangle_{n,\beta} = \partial_s\partial_t \log\langle e^{sX+tY}\rangle_{n,\beta}\Big|_{s=t=0}.$$

The exponential mean is holomorphic and equals 1 at the origin, so its logarithm is defined near the origin. Differentiating the logarithm gives

$$\langle X;Y\rangle_{n,\beta} = \langle XY\rangle_{n,\beta} - \langle X\rangle_{n,\beta}\langle Y\rangle_{n,\beta}.$$

The measure $\mu_{n,\beta}$ is the Gibbs kernel of the set of free bonds, evaluated at the configuration $\mathbf{1}$, in the following sense. For a finite set B of positively oriented bonds define its Gibbs kernel by

$$(\gamma_B F)(U) = \frac{\int F(V \vee U_{B^c}) e^{H_B(V\vee U_{B^c})}\,\mathrm{d}V_B}{\int e^{H_B(V\vee U_{B^c})}\,\mathrm{d}V_B}, \qquad H_B = -\beta\sum_{p\in\mathsf{Pl}[B]} a_p,$$

where $V \vee U_{B^c}$ equals V on B and U off B, $\mathsf{Pl}[B]$ consists of the plaquettes containing a bond of B, and dV_B is product Haar measure on G^B. Take $B = \mathsf{Bd}_n$, and for $V \in G^B$ let $V \vee \mathbf{1}_{B^c}$ denote the configuration which equals V on B and $\mathbf{1}$ at all other positively oriented bonds. Then, for every bounded Borel function F on $\mathcal{V}(\mathbb{L}_1)$,

$$(\gamma_B F)(\mathbf{1}) = \int F(V \vee \mathbf{1}_{B^c})\,\mathrm{d}\mu_{n,\beta}(V).$$

Indeed, by definition the left side is the quotient of $\int F e^{H_B}\,\mathrm{d}V_B$ by $\int e^{H_B}\,\mathrm{d}V_B$, both integrands being evaluated at $V \vee \mathbf{1}_{B^c}$. Here dV_B is the product, over the bonds $b \in B$, of the normalised Haar measures in the variables $V(b)$, and $H_B = -\beta\sum_{p\in\mathsf{Pl}[B]} a_p$, the sum running over the set $\mathsf{Pl}[B]$ of the plaquettes which contain a bond of B. Every plaquette of $\mathsf{Pl}[B]$, that is, every plaquette containing a free bond (x,μ), lies in Pl_n: one end point of (x,μ) lies in Λ_n°, so both end points have the coordinates $x_\nu \in \{1,\dots,n-1\}$ for $\nu \neq \mu$, and the corners of such a plaquette are the end points of (x,μ) and their translates by one of the vectors $\pm e_\nu$, $\nu \neq \mu$; all of them lie in Λ_n.

Conversely, the bonds of a plaquette of $\mathsf{Pl}_n \setminus \mathsf{Pl}[B]$, which contains no free bond, have both end points in Λ_n and are not free, so every configuration of $\mathcal{U}_n$ equals $\mathbf{1}$ on them, and $a_p = 0$ on $\mathcal{U}_n$. Hence $H_B(V \vee \mathbf{1}_{B^c}) = -\beta\sum_{p\in\mathsf{Pl}_n} a_p(V)$, and the quotient is the integral of $F(V \vee \mathbf{1}_{B^c})$ with respect to $\mu_{n,\beta}$. ∎

Definition 7.4 (Exponential chart, gauge function, Faddeev–Popov operator)**.** Let Ω be $\mathbb{Z}^3$ or a box Λ_n; in the second case forms are taken in Form_n, site functions in X_n and plaquettes in Pl_n.

(1) For a plaquette p, a 1-form A and $m \geq 0$, the number $\mathsf{a}_p^{(m)}(A)$ is the m-th Taylor coefficient of the plaquette action along the curve $t \mapsto e^{itA}$:

$$a_p(e^{itA}) = \sum_{m\geq 0} t^m\,\mathsf{a}_p^{(m)}(A) \qquad (t \in \mathbb{R}); \tag{7.iii}$$

the series converges for every real t by Lem. 7.5(a).

(2) If a 1-form A and a site function λ satisfy $|A(b)| + |\lambda(b_-)| + |\lambda(b_+)| < 1$ for every bond b, we set

$$A^\lambda(b) := \tfrac{1}{i}\log(e^{i\lambda(b_-)}e^{iA(b)}e^{-i\lambda(b_+)}), \qquad \text{that is,} \qquad e^{iA^\lambda} = (e^{iA})^u, \quad u = e^{i\lambda}.$$

This is well defined: by the product estimate in Def. 7.1(2),

$$|e^{i\lambda(b_-)}e^{iA(b)}e^{-i\lambda(b_+)} - \mathbf{1}| \leq e^{|\lambda(b_-)|+|A(b)|+|\lambda(b_+)|} - 1 < e - 1 < 2 = |-\mathbf{1}-\mathbf{1}|,$$

so the argument of the logarithm lies in $G \setminus \{-\mathbf{1}\}$, where $\frac{1}{i}\log$, the inverse of $Y \mapsto e^{iY}$ on $\mathbb{B}_\pi$, is defined (the exponential chart in Def. 7.1(2)).

(3) The *gauge function* is the map $A \mapsto \mathrm{d}^* A$. For a 1-form A with $|A(b)| < 1$ at every bond b, the Faddeev–Popov operator $\mathsf{M}(A)$ acts on site functions λ by

$$\mathsf{M}(A)\lambda := \frac{d}{ds}\,\mathrm{d}^*(A^{s\lambda})\Big|_{s=0}, \qquad \mathsf{M}(tA) = \sum_{j\ge 0} t^j\,\mathsf{M}^{(j)}(A); \tag{7.iv}$$

the second formula is the expansion at $t = 0$ of the real-analytic function $t \mapsto \mathsf{M}(tA)$ (Lem. 7.5(d)), and it defines the operators $\mathsf{M}^{(j)}(A)$; since $\mathsf{M}(t(cA)) = \mathsf{M}((tc)A)$, the operator $\mathsf{M}^{(j)}(A)$ is homogeneous of degree j in A. The derivative is taken site by site: $\mathrm{d}^*(A^{s\lambda})(x)$ involves the six bonds with end point x, and for each of them $A^{s\lambda}(b)$ is defined and smooth in s for $|s|$ small. In particular the definition applies to every site function λ, bounded or not. ■

Lemma 7.5 (Coefficients of the chart)**.** Let p be a plaquette with bonds $b_1, \dots, b_4$, let A be a 1-form, $A_i := A(b_i)$, and $(s_1, s_2, s_3, s_4) := (+, +, -, -)$, so that $\mathrm{d}A(p) = \sum_i s_i A_i$.

(a) The series eq. 7.iii converges for every $t \in \mathbb{R}$. Each $\mathsf{a}_p^{(m)}$ is a homogeneous polynomial of degree m in the twelve components of $A_1, \dots, A_4$, with real coefficients which do not depend on p. Moreover $\mathsf{a}_p^{(0)} = \mathsf{a}_p^{(1)} = 0$, and $\mathsf{a}_p^{(m)}(gAg^{-1}) = \mathsf{a}_p^{(m)}(A)$ for every $g \in G$, where gAg^{-1} denotes the form $b \mapsto gA(b)g^{-1}$.

(b) On the set $\{(t, A_1, \dots, A_4) \in \mathbb{R} \times \mathfrak{g}^4 :\ |t| \sum_i |A_i| < \log 3\}$ the function $Z := \frac{1}{i} \log \partial(e^{itA})(p) \in \mathbb{B}_\pi$ is defined and smooth. For $j = 0, 1, 2$ the maps $\mathfrak{z}_j := \frac{1}{(j+1)!}\partial_t^{j+1} Z|_{t=0}$ are homogeneous polynomial maps $\mathfrak{g}^4 \to \mathfrak{g}$ of degree $j+1$, which do not depend on p; $\mathfrak{z}_0 = \mathrm{d}A(p)$, and

$$\begin{gathered} \mathsf{a}_p^{(2)} = \tfrac{1}{2}|\mathrm{d}A(p)|^2, \qquad \mathsf{a}_p^{(3)} = \langle \mathrm{d}A(p), \mathfrak{z}_1 \rangle, \\ \mathsf{a}_p^{(4)} = \tfrac{1}{2}|\mathfrak{z}_1|^2 + \langle \mathrm{d}A(p), \mathfrak{z}_2 \rangle - \tfrac{1}{24}|\mathrm{d}A(p)|^4, \qquad \mathfrak{z}_1 = -\sum_{i<j} s_i s_j\, A_i \times A_j. \end{gathered} \tag{7.v}$$

We write $\mathfrak{z}_j(p)$ for the value of $\mathfrak{z}_j$ at $(A(b_1(p)), \dots, A(b_4(p)))$.

(c) The function $Y \mapsto -\log \sigma_1(|Y|^2)$ is real analytic and even on $\mathbb{B}_\pi$, and its Taylor polynomial of degree 3 at 0 is $\frac{1}{3}|Y|^2$.

(d) For $Y \in \mathfrak{g}$ let $\mathsf{a}_Y := -2\,Y \times \cdot$ on $\mathfrak{g}$. Then $e^{isY} X e^{-isY} = e^{s\mathsf{a}_Y} X$ for $s \in \mathbb{R}$ and $X \in \mathfrak{g}$. If $|A(b)| < 1$ and λ is a site function, then

$$\begin{aligned} \frac{d}{ds} A^{s\lambda}(b)\Big|_{s=0} &= \mathsf{dE}(A(b))^{-1}\big(\lambda(b_-) - e^{\mathsf{a}_{A(b)}}\lambda(b_+)\big) \\ &= -\mathrm{d}\lambda(b) + 2A(b) \times \bar\lambda(b) - \tfrac{1}{3}A(b) \times \big(A(b) \times \mathrm{d}\lambda(b)\big) + \mathsf{r}(A(b))\,\mathrm{d}\lambda(b), \end{aligned} \tag{7.vi}$$

where $\mathsf{r} : \mathbb{B}_\pi \to \operatorname{End}\mathfrak{g}$ is real analytic and vanishes at 0 to fourth order. Consequently, if $|A(b)| < 1$ at every bond, $\mathsf{M}(tA)$ is real analytic in t, and

$$\begin{gathered} \mathsf{M}^{(0)} = -\mathrm{d}^*\mathrm{d}, \qquad \mathsf{M}^{(1)}(A)\lambda = 2\,\mathrm{d}^*(A \times \bar\lambda), \\ \mathsf{M}^{(2)}(A)\lambda = -\tfrac{1}{3}\,\mathrm{d}^*(A \times (A \times \mathrm{d}\lambda)), \qquad \mathsf{M}^{(3)}(A) = 0. \end{gathered}$$

■

Proof. Step 1: claim (a). Since $U(b_3)^{-1} = e^{-itA_3}$ and $U(b_4)^{-1} = e^{-itA_4}$ for $U = e^{itA}$, we have $\partial(e^{itA})(p) = \prod_{i=1}^4 e^{its_iA_i}$, a product of four exponential series in t which converge for every t; their Cauchy product converges for every t, and hence so does eq. 7.iii. For $m \ge 1$ the coefficient of t^m is

$$\mathsf{a}_p^{(m)}(A) = -\frac{1}{2} \sum_{m_1+\dots+m_4=m} \operatorname{tr}\Big[\prod_{i=1}^4 \frac{(is_iA_i)^{m_i}}{m_i!}\Big],$$

that is, $-\frac{1}{2}\operatorname{tr}$ of a sum of products of m factors is_iA_i. It is therefore a homogeneous polynomial of degree m in the components of $A_1, \dots, A_4$, given by the same formula for every plaquette. It takes real values on real arguments, since a_p is real; hence, if we write it as $\sum_\alpha c_\alpha y^\alpha$ in the twelve real

components y of $A_1,\dots,A_4$, the polynomial $\sum_\alpha(\operatorname{Im} c_\alpha)\,y^\alpha$ vanishes on $\mathbb{R}^{12}$, so that $\operatorname{Im} c_\alpha = 0$ and the coefficients are real. Replacing every A_i by gA_ig^{-1} conjugates each product of the factors is_iA_i by g, which does not change the trace. The relations $\mathsf{a}_p^{(0)} = \mathsf{a}_p^{(1)} = 0$ follow from (b), see Step 5.

Step 2: existence and smoothness of Z. Let $X_i := its_iA_i$ and $\mathsf{x} := \partial(e^{itA})(p) = e^{X_1}e^{X_2}e^{X_3}e^{X_4}$. Since $|X_i| = |t|\,|A_i|$, the product estimate in Def. 7.1(2) gives on the set of (b)

$$|\mathsf{x}-\mathbf{1}| \le e^{|t|\sum_i|A_i|} - 1 < e^{\log 3} - 1 = 2 = |-\mathbf{1}-\mathbf{1}|;$$

hence $\mathsf{x} \ne -\mathbf{1}$, and $Z = \frac{1}{i}\log\mathsf{x} \in \mathbb{B}_\pi$ is defined. The matrix x depends smoothly on $(t, A_1,\dots,A_4)$, and $W \mapsto \frac{1}{i}\log W$ is smooth on $G\setminus\{-\mathbf{1}\}$, the inverse of the diffeomorphism $Y \mapsto e^{iY}$ of $\mathbb{B}_\pi$ onto $G\setminus\{-\mathbf{1}\}$ (Def. 7.1); thus Z is smooth. At $t=0$ we have $\mathsf{x}=\mathbf{1}$ and $Z=0$. The maps $\mathfrak{z}_j$ are t-derivatives of Z at $t=0$, hence smooth maps $\mathfrak{g}^4 \to \mathfrak{g}$, given by the same formula for all plaquettes.

Step 3: homogeneity. For $c>0$ the substitution $(t,A)\mapsto(t/c, cA)$ preserves the domain of Z and leaves x unchanged. Hence $Z(t/c,cA) = Z(t,A)$, that is, $Z(s,cA) = Z(cs,A)$, and differentiating $j+1$ times at $s=0$ we find $\mathfrak{z}_j(cA) = c^{j+1}\mathfrak{z}_j(A)$. It remains to use the following fact:

a smooth function f on a finite-dimensional real vector space with $f(cA) = c^m f(A)$ for all $c>0$ is a homogeneous polynomial of degree m. To see it, let f_i denote the homogeneous Taylor polynomials of f at 0; Taylor's formula at the points cA gives

$$c^m f(A) = f(cA) = \sum_{i\le m} c^i f_i(A) + O(c^{m+1}) \qquad (c\to 0),$$

and comparing the powers of c as $c\to 0$ we obtain $f_i(A)=0$ for $i<m$ and $f(A) = f_m(A)$. Applied to the components of $\mathfrak{z}_j$ in a basis of $\mathfrak{g}$, this shows that $\mathfrak{z}_j$ is a homogeneous polynomial map of degree $j+1$.

Step 4: $\mathfrak{z}_0$ and $\mathfrak{z}_1$. At fixed A, Taylor's formula in t gives $Z = t(\mathfrak{z}_0 + t\mathfrak{z}_1 + t^2\mathfrak{z}_2) + O(t^4)$. We compare the two expansions of $e^{iZ} = \mathsf{x}$,

$$e^{iZ} = \mathbf{1} + iZ - \tfrac12 Z^2 + O(t^3), \qquad \mathsf{x} = \mathbf{1} + \sum_i X_i + \tfrac12\sum_i X_i^2 + \sum_{i<j} X_iX_j + O(t^3).$$

The terms of order t give $it\mathfrak{z}_0 = \sum_i X_i$, hence $\mathfrak{z}_0 = \sum_i s_iA_i = \mathrm{d}A(p)$. Since $-\frac12 Z^2 = \frac12(it\mathfrak{z}_0)^2 + O(t^3) = \frac12\big(\sum_i X_i\big)^2 + O(t^3)$, the terms of order t^2 give

$$\begin{aligned} it^2\mathfrak{z}_1 &= \tfrac12\sum_i X_i^2 + \sum_{i<j} X_iX_j - \tfrac12\Big(\sum_i X_i\Big)^2 = \tfrac12\sum_{i<j}[X_i,X_j] \\ &= -\tfrac12 t^2\sum_{i<j} s_is_j[A_i,A_j] = -it^2\sum_{i<j} s_is_j\, A_i\times A_j, \end{aligned}$$

where $[X,Y] = 2i\,X\times Y$ was used in the last step. This is the formula for $\mathfrak{z}_1$ in eq. 7.v.

Step 5: the coefficients $\mathsf{a}_p^{(m)}$, $m\le 4$. Since $\frac12\operatorname{tr} e^{iZ} = \cos|Z|$, we have $a_p(e^{itA}) = 1-\cos|Z| = \frac12|Z|^2 - \frac{1}{24}|Z|^4 + O(t^6)$. By Step 4,

$$|Z|^2 = t^2\Big[|\mathfrak{z}_0|^2 + 2t\langle\mathfrak{z}_0,\mathfrak{z}_1\rangle + t^2\big(|\mathfrak{z}_1|^2 + 2\langle\mathfrak{z}_0,\mathfrak{z}_2\rangle\big) + O(t^3)\Big],$$

and therefore

$$a_p(e^{itA}) = \tfrac12 t^2|\mathfrak{z}_0|^2 + t^3\langle\mathfrak{z}_0,\mathfrak{z}_1\rangle + t^4\Big(\tfrac12|\mathfrak{z}_1|^2 + \langle\mathfrak{z}_0,\mathfrak{z}_2\rangle - \tfrac{1}{24}|\mathfrak{z}_0|^4\Big) + O(t^5).$$

Comparing with the Taylor series eq. 7.iii we obtain $\mathsf{a}_p^{(0)} = \mathsf{a}_p^{(1)} = 0$ and eq. 7.v.

Step 6: claim (c). From its power series, $\sigma_1(w) = \big(1 - \frac{w}{6} + O(w^2)\big)^2 = 1 - \frac{w}{3} + O(w^2)$, and $\sigma_1(|Y|^2) = (\sin|Y|/|Y|)^2 > 0$ on $\mathbb{B}_\pi$. Thus $Y\mapsto -\log\sigma_1(|Y|^2)$ is the composite of the real analytic maps $Y\mapsto|Y|^2$, σ_1 at real arguments, and log on $(0,\infty)$, hence real analytic. It depends on $|Y|^2$ only and is therefore even, and $-\log(1-\frac13|Y|^2 + O(|Y|^4)) = \frac13|Y|^2 + O(|Y|^4)$ shows that its Taylor polynomial of degree 3 at 0 is $\frac13|Y|^2$.

Step 7: conjugation. By $[Y,X] = 2iY \times X$ we have $i[Y,X] = -2Y \times X = \mathsf{a}_Y X$. The function $x(s) := e^{isY} X e^{-isY}$ takes values in $\mathfrak{g}$ and satisfies $x'(s) = i[Y, x(s)] = \mathsf{a}_Y x(s)$, so that $\frac{d}{ds}(e^{-s\mathsf{a}_Y} x(s)) = 0$; together with $x(0) = X$ this gives $x(s) = e^{s\mathsf{a}_Y} X$. Consequently

$$\mathsf{dE}(Y) = \int_0^1 e^{s\mathsf{a}_Y}\, \mathrm{d}s = \mathsf{f}(\mathsf{a}_Y), \qquad \mathsf{f}(w) := \frac{e^w - 1}{w} = \sum_{n\geq 0} \frac{w^n}{(n+1)!}.$$

Step 8: first equality in eq. 7.vi. Let $Y := A(b)$ and $\delta := \frac{d}{ds} A^{s\lambda}(b)|_{s=0}$. For $|s|$ small, $A^{s\lambda}(b)$ is defined, depends smoothly on s, equals Y at $s = 0$, and satisfies $e^{iA^{s\lambda}(b)} = e^{is\lambda(b_-)} e^{iY} e^{-is\lambda(b_+)}$. We differentiate this identity at $s = 0$, using the differential formula in Def. 7.1(2) on the left and Step 7 on the right:

$$i\big(\mathsf{dE}(Y)\delta\big) e^{iY} = i\lambda(b_-) e^{iY} - i e^{iY} \lambda(b_+) = i\big(\lambda(b_-) - e^{\mathsf{a}_Y} \lambda(b_+)\big) e^{iY}.$$

By the determinant formula in Def. 7.1(2), $\det \mathsf{dE}(Y) = \sigma_1(|Y|^2) > 0$, so $\mathsf{dE}(Y)$ is invertible, and the first equality follows.

Step 9: expansion. The double cross product formula $X \times (Y \times Z) = \langle X, Z\rangle Y - \langle X, Y\rangle Z$ gives $\mathsf{a}_Y Y = 0$ and $\mathsf{a}_Y^2 = -4|Y|^2$ on the orthogonal complement of Y; so a_Y has the eigenvalues 0 and $\pm 2i|Y|$, and Lagrange's identity $|Y \times X|^2 = |Y|^2|X|^2 - \langle Y, X\rangle^2$ gives $\|\mathsf{a}_Y\| \leq 2|Y| < 2\pi$ on $\mathbb{B}_\pi$. The function $1/\mathsf{f}(w) = w/(e^w - 1)$ is holomorphic for $|w| < 2\pi$, and since $w/(e^w - 1) + w/2$ is even,

$$\frac{1}{\mathsf{f}(w)} = 1 - \frac{w}{2} + \frac{w^2}{12} + w^4 \mathsf{f}_4(w), \qquad \frac{we^w}{e^w - 1} = w + \frac{w}{e^w - 1} \qquad (|w| < 2\pi),$$

with f_4 holomorphic for $|w| < 2\pi$. Evaluating these power series at a_Y, Steps 7 and 8 give

$$\begin{aligned} \delta &= \tfrac{1}{\mathsf{f}}(\mathsf{a}_Y)\lambda(b_-) - \big(\mathsf{a}_Y + \tfrac{1}{\mathsf{f}}(\mathsf{a}_Y)\big)\lambda(b_+) = -\tfrac{1}{\mathsf{f}}(\mathsf{a}_Y)\, \mathrm{d}\lambda(b) - \mathsf{a}_Y \lambda(b_+) \\ &= -\mathrm{d}\lambda(b) - \mathsf{a}_Y\big(\lambda(b_+) - \tfrac{1}{2}\mathrm{d}\lambda(b)\big) - \tfrac{1}{12}\mathsf{a}_Y^2\, \mathrm{d}\lambda(b) - \mathsf{a}_Y^4 \mathsf{f}_4(\mathsf{a}_Y)\, \mathrm{d}\lambda(b). \end{aligned}$$

Now $\lambda(b_+) - \frac{1}{2}\mathrm{d}\lambda(b) = \bar\lambda(b)$, $-\mathsf{a}_Y \bar\lambda(b) = 2Y \times \bar\lambda(b)$ and $\mathsf{a}_Y^2 = 4\, Y \times (Y \times \cdot)$; with $\mathsf{r}(Y) := -\mathsf{a}_Y^4 \mathsf{f}_4(\mathsf{a}_Y)$ this is the second equality in eq. 7.vi. The map r is real analytic on $\mathbb{B}_\pi$, being the sum of a power series at an operator which depends linearly on Y and has norm $< 2\pi$, and it vanishes at 0 to fourth order because of the factor a_Y^4. Since d^* involves at each site finitely many bonds, $\mathsf{M}(A)\lambda = \mathrm{d}^*\delta$, where δ is the form given at every bond by eq. 7.vi. If A is replaced by tA, the first three terms of eq. 7.vi are of degree 0, 1, 2 in t, and the last one is real analytic in t and of order t^4. This proves that $t \mapsto \mathsf{M}(tA)$ is real analytic, and it gives the formulas for $\mathsf{M}^{(0)}, \dots, \mathsf{M}^{(3)}$. □

Lemma 7.6 (Free gauge action, absence of zero modes, a slice, and the Faddeev–Popov change of variables in a box)**.** Let $n \geq 2$. On $\mathcal{U}_n$ let $S(U) := \sum_{p \in \mathsf{Pl}_n} a_p(U)$, and let

$$\mathsf{T}_n := \{\langle x - e_1, x\rangle :\ x \in \Lambda_n^\circ\} \subset \mathsf{Bd}_n,$$

$$\mathcal{U}_n^{\mathsf{T}} := \{V \in \mathcal{U}_n :\ V = \mathbf{1} \text{ on } \mathsf{T}_n\}, \qquad \mathsf{Form}_n^{\mathsf{T}} := \{A \in \mathsf{Form}_n :\ A = 0 \text{ on } \mathsf{T}_n\}.$$

(a) A positively oriented bond (x, μ) with both end points in Λ_n is free if and only if $x_\nu \in \{1, \dots, n-1\}$ for both $\nu \neq \mu$; otherwise it lies in a face $\{x_\nu = 0\}$ or $\{x_\nu = n\}$ of the box, for some $\nu \neq \mu$.

(b) The map $\mathsf{j} : \mathcal{G}_n \times \mathcal{U}_n^{\mathsf{T}} \to \mathcal{U}_n$, $(u, V) \mapsto V^u$, is a diffeomorphism, and for every bounded Borel function F on $\mathcal{U}_n$

$$\int F(U) \prod_{b \in \mathsf{Bd}_n} \mathrm{d}U(b) = \int \prod_{x \in \Lambda_n^\circ} \mathrm{d}u(x) \int \prod_{b \in \mathsf{Bd}_n \setminus \mathsf{T}_n} \mathrm{d}V(b)\ F(V^u). \tag{7.vii}$$

The group $\mathcal{G}_n$ acts freely on $\mathcal{U}_n$. One has $S(U) = 0$ if and only if $U = \mathbf{1}^u$ for some $u \in \mathcal{G}_n$; the only zero of S in $\mathcal{U}_n^{\mathsf{T}}$ is $\mathbf{1}$.

(c) If $A \in \mathsf{Form}_n$ satisfies $\mathrm{d}A = 0$ on Pl_n and $\mathrm{d}^*A = 0$, then $A = 0$. Hence, for every $\tau_{\mathrm{gf}} > 0$, the quadratic form $\frac{1}{2}\|\mathrm{d}A\|^2 + \frac{1}{2\tau_{\mathrm{gf}}}\|\mathrm{d}^*A\|^2$ is positive definite on Form_n, and $\mathrm{d}^*\mathrm{d}$ is positive definite on X_n.

(d) There are numbers $0 < \varrho_2, \varrho_3 \le \varrho_1 \le \frac{1}{20}$, depending on n, with the following properties, where

$$\mathfrak{D} := \{(\lambda, A) \in \mathsf{X}_n \times \mathsf{Form}_n^{\mathsf{T}} :\ |\lambda|_\infty < \varrho_1,\ |A|_\infty < \varrho_3\}.$$

(i) The map $\Xi : (\lambda, A) \mapsto A^\lambda$ extends to a diffeomorphism of a neighbourhood of $\overline{\mathfrak{D}}$ onto an open subset of $\{A' \in \mathsf{Form}_n :\ |A'|_\infty < 1\}$.

(ii) $\det \mathsf{M}(A') \neq 0$ for every $A' \in \overline{\mathcal{W}}$, where $\mathcal{W} := \Xi(\mathfrak{D})$.

(iii) If $A \in \mathsf{Form}_n^{\mathsf{T}}$ and $|A|_\infty \le \varrho_3$, then $\lambda \mapsto \mathrm{d}^*(A^\lambda)$ is injective on $\{\lambda \in \mathsf{X}_n :\ |\lambda|_\infty \le \varrho_1\}$, and the image of $\{\lambda \in \mathsf{X}_n :\ |\lambda|_\infty < \varrho_1\}$ contains $\{c \in \mathsf{X}_n :\ |c|_\infty < \varrho_2\}$.

(iv) For every bounded Borel function Φ on $\mathcal{W}$ and every bounded Borel function $w \ge 0$ on X_n,

$$\begin{aligned}\int_{\mathcal{W}} \Phi(A')\, w(\mathrm{d}^*A')\, |\det \mathsf{M}(A')| \prod_{b \in \mathsf{Bd}_n} \sigma_1(|A'(b)|^2)\, \mathrm{d}A' \\ = \int_{|A|_\infty < \varrho_3} \prod_{b \in \mathsf{Bd}_n \setminus \mathsf{T}_n} \sigma_1(|A(b)|^2)\, \mathrm{d}A \int_{\{\mathrm{d}^*(A^\lambda):\, |\lambda|_\infty < \varrho_1\}} w(c)\, \Phi\big(A^{\lambda(c)}\big)\, \mathrm{d}c,\end{aligned} \tag{7.viii}$$

where $\lambda(c)$ is the solution of $\mathrm{d}^*(A^\lambda) = c$ with $|\lambda|_\infty < \varrho_1$, unique by (iii), and dA', dA, dc are the Lebesgue measures of the Euclidean spaces Form_n, $\mathsf{Form}_n^{\mathsf{T}}$, X_n, with the scalar product $(\cdot,\cdot)$ of Def. 7.2(2). Thus $dA' = \prod_{b \in \mathsf{Bd}_n} dA'(b)$, where $dA'(b)$ is the Lebesgue measure dY of $(\mathfrak{g}, \langle\cdot,\cdot\rangle)$ in the variable $A'(b)$; likewise dA is the product of these measures over the bonds of $\mathsf{Bd}_n \setminus \mathsf{T}_n$, and dc the product over the sites of Λ_n°.

■

Proof. Step 1: claim (a). Let (x, μ) be a positively oriented bond with both end points in Λ_n; then $x_\mu \in \{0, \dots, n-1\}$, and the end points x, $x + e_\mu$ have the same coordinates x_ν, $\nu \neq \mu$. If both these coordinates lie in $\{1, \dots, n-1\}$, then, as $n \ge 2$, one of the numbers x_μ, $x_\mu + 1$ lies in $\{1, \dots, n-1\}$, and the corresponding end point lies in Λ_n°: the bond is free. If $x_\nu \in \{0, n\}$ for some $\nu \neq \mu$, then both end points have this coordinate, neither lies in Λ_n°, and the bond lies in the face $\{x_\nu = 0\}$ or $\{x_\nu = n\}$.

Step 2: j *is bijective and smooth.* Let $U \in \mathcal{U}_n$. For $u' \in \mathcal{G}_n$ the condition $U^{u'} = \mathbf{1}$ on T_n is equivalent to

$$u'(x) = u'(x - e_1)\, U(\langle x - e_1, x\rangle) \qquad (x \in \Lambda_n^\circ).$$

Since $u' = \mathbf{1}$ at the sites with first coordinate 0, which lie in $\partial\Lambda_n$, this recursion in $x_1 = 1, \dots, n-1$ determines u' uniquely: $u'(x)$ is the ordered product of the values of U at the bonds $\langle y - e_1, y\rangle$ on the line from $(0, x_2, x_3)$ to x. Let $V := U^{u'} \in \mathcal{U}_n^{\mathsf{T}}$ and $u := u'^{-1} \in \mathcal{G}_n$; since $(U^v)^w = U^{wv}$, we have $V^u = U$. Conversely, if $U = V^u$ with $(u, V) \in \mathcal{G}_n \times \mathcal{U}_n^{\mathsf{T}}$, then $U^{u^{-1}} = V$ equals $\mathbf{1}$ on T_n, so $u^{-1} = u'$ and $V = U^{u'}$. Hence j is bijective. Both j and j^{-1} are composed of products and inverses of the values of u, V, U, and are therefore smooth (Def. 7.1).

Step 3: the action is free. If $U^u = U$ for some $u \in \mathcal{G}_n$, then $u(b_+) = U(b)^{-1} u(b_-) U(b)$ at every bond $b \in \mathsf{T}_n$, i.e. $u(x) = U(\langle x - e_1, x\rangle)^{-1} u(x - e_1)\, U(\langle x - e_1, x\rangle)$ for $x \in \Lambda_n^\circ$; the recursion of Step 2, started from $u = \mathbf{1}$ at $x_1 = 0$, gives $u = \mathbf{1}$.

Step 4: the identity eq. 7.vii. We fix $u \in \mathcal{G}_n$ on the right side of eq. 7.vii and substitute $U(b) := u(b_-) V(b) u(b_+)^{-1}$ for $b \in \mathsf{Bd}_n \setminus \mathsf{T}_n$. By Fubini's theorem and the invariance of the Haar measure under the maps $g \mapsto hgh'^{-1}$, $h, h' \in G$ (left and right invariance of Haar measure), the inner integral becomes $\int \prod_{b \in \mathsf{Bd}_n \setminus \mathsf{T}_n} \mathrm{d}U(b)\, F(U)$, where U takes the values $U(\langle x - e_1, x\rangle) = u(x - e_1) u(x)^{-1}$ on T_n. In the integral over u we now replace, on each line parallel to e_1 and for $x_1 = n-1, n-2, \dots, 1$ in this order, the variable $u(x)$ by the variable $U(\langle x - e_1, x\rangle) = u(x - e_1) u(x)^{-1}$, at fixed $u(x - e_1)$.

When this is done at x, the variable $u(x)$ enters no other value of U on T_n (the value at $\langle x, x+e_1\rangle$, if $x+e_1 \in \Lambda_n^\circ$, has already been replaced), and the map $u(x) \mapsto u(x-e_1)u(x)^{-1}$, an inversion followed by a left translation, preserves the Haar measure (left and right invariance of Haar measure). After the last substitution we arrive at the left side of eq. 7.vii.

Step 5: zeros of S. If $\partial U(p) = e^{iY}$ with $|Y| \le \pi$, then $a_p(U) = 1 - \cos|Y|$; hence $a_p(U) = 0$ if and only if $\partial U(p) = \mathbf{1}$, and since $a_p \ge 0$, $S(U) = 0$ if and only if $\partial U(p) = \mathbf{1}$ for every $p \in \mathsf{Pl}_n$. By the gauge invariance of a_p (Def. 7.2(1)), $S(\mathbf{1}^u) = S(\mathbf{1}) = 0$ for $u \in \mathcal{G}_n$. Now let $V \in \mathcal{U}_n^{\mathsf{T}}$ with $S(V) = 0$. We show that V equals $\mathbf{1}$ at every positively oriented bond with both end points in Λ_n, direction by direction.

- Bonds $(y,1)$ with $y_1 \le n-2$: if $y+e_1 \in \Lambda_n^\circ$, then $(y,1) \in \mathsf{T}_n$; otherwise one of y_2, y_3 lies in $\{0,n\}$, and $(y,1)$ is not free by (a). In both cases $V((y,1)) = \mathbf{1}$.
- Bonds (y,ν), $\nu \in \{2,3\}$: for $y_1 \le n-2$ the plaquette $p_{1\nu}(y)$ lies in Pl_n, and $\partial V(p_{1\nu}(y)) = \mathbf{1}$ together with $V((y,1)) = V((y+e_\nu,1)) = \mathbf{1}$ gives $V((y+e_1,\nu)) = V((y,\nu))$. The bonds (y,ν) with $y_1 = 0$ are not free, so $V((y,\nu)) = \mathbf{1}$ for $y_1 = 0$, and by induction for $y_1 \le n-1$; the bonds with $y_1 = n$ are not free.
- Bonds $(y,1)$ with $y_1 = n-1$: for $y_2 \le n-1$ the plaquette $p_{12}(y)$ gives $V((y+e_2,1)) = V((y,1))$, because $V = \mathbf{1}$ at the bonds in direction 2. The bonds with $y_2 = 0$ are not free, and induction in y_2 gives $V((y,1)) = \mathbf{1}$ for $y_2 \le n-1$; the bonds with $y_2 = n$ are not free.

Hence $V = \mathbf{1}$. Finally let $U \in \mathcal{U}_n$ with $S(U) = 0$. By (b), $U = V^u$ with $(u,V) \in \mathcal{G}_n \times \mathcal{U}_n^{\mathsf{T}}$, and $S(V) = S(U^{u^{-1}}) = S(U) = 0$; thus $V = \mathbf{1}$ and $U = \mathbf{1}^u$.

Step 6: claim (c). Let $A \in \mathsf{Form}_n$ with $\mathrm{d}A = 0$ on Pl_n and $\mathrm{d}^*A = 0$. We define $\lambda \in \mathsf{X}_n$ by $\lambda = 0$ on $\partial\Lambda_n$ and $\lambda(x) = \lambda(x-e_1) + A(\langle x-e_1, x\rangle)$ for $x \in \Lambda_n^\circ$, recursively in x_1. Then $A' := A - \mathrm{d}\lambda \in \mathsf{Form}_n$ vanishes on T_n, and $\mathrm{d}A' = \mathrm{d}A - \mathrm{d}\mathrm{d}\lambda = 0$ on Pl_n. The walk of Step 5, in additive form, shows $A' = 0$: A' vanishes at the bonds $(y,1)$ with $y_1 \le n-2$, which lie in T_n or are not free; for $\nu \in \{2,3\}$ and $y_1 \le n-2$,

$$0 = \mathrm{d}A'(p_{1\nu}(y)) = A'((y+e_1,\nu)) - A'((y,\nu)),$$

so $A' = 0$ at the bonds in the directions 2 and 3; and for $y_1 = n-1$,

$$0 = \mathrm{d}A'(p_{12}(y)) = A'((y,1)) - A'((y+e_2,1)),$$

so $A' = 0$ at the remaining bonds. Hence $A = \mathrm{d}\lambda$, and $0 = (\lambda, \mathrm{d}^*A) = (\lambda, \mathrm{d}^*\mathrm{d}\lambda) = \|\mathrm{d}\lambda\|^2$, so that $A = \mathrm{d}\lambda = 0$. Consequently, for $\tau_{\mathrm{gf}} > 0$ the quadratic form $\frac{1}{2}\|\mathrm{d}A\|^2 + \frac{1}{2\tau_{\mathrm{gf}}}\|\mathrm{d}^*A\|^2$ is non-negative and vanishes only at $A = 0$; it is therefore positive definite. Similarly, $(\lambda, \mathrm{d}^*\mathrm{d}\lambda) = \|\mathrm{d}\lambda\|^2 \ge 0$ for $\lambda \in \mathsf{X}_n$, and $\mathrm{d}\lambda = 0$ implies $\lambda(x) = \lambda(x-e_1)$ for $x \in \Lambda_n^\circ$, hence $\lambda = 0$, since $\lambda = 0$ at $x_1 = 0$; so $\mathrm{d}^*\mathrm{d}$ is positive definite on X_n.

Step 7: size. If $|\lambda|_\infty < \frac{1}{3}$ and $|A|_\infty < \frac{1}{3}$, then $|A(b)| + |\lambda(b_-)| + |\lambda(b_+)| < 1$ at every bond, and A^λ is defined. If $|\lambda|_\infty \le \frac{1}{10}$ and $|A|_\infty \le \frac{1}{10}$, then

$$|A^\lambda(b)| \le \tfrac{\pi}{2}\,|e^{iA^\lambda(b)} - \mathbf{1}| \le \tfrac{\pi}{2}(e^{3/10} - 1) < 1.$$

The second inequality is the product estimate in Def. 7.1(2). For the first one, let $Y \in \mathbb{B}_\pi$; then $(e^{iY}-\mathbf{1})^*(e^{iY}-\mathbf{1}) = (2-2\cos|Y|)\mathbf{1}$, so that $|e^{iY}-\mathbf{1}| = 2\sin\frac{|Y|}{2} \ge \frac{2}{\pi}|Y|$, by the inequality $\sin t \ge \frac{2}{\pi}t$ on $[0, \frac{\pi}{2}]$.

Step 8: Ξ in the exponential charts. The maps $\lambda \mapsto e^{i\lambda}$, $A \mapsto e^{iA}$, $A' \mapsto e^{iA'}$, taken site by site and bond by bond, are diffeomorphisms of the sets $\{|\lambda|_\infty < \pi\} \subset \mathsf{X}_n$, $\{|A|_\infty < \pi\} \subset \mathsf{Form}_n^{\mathsf{T}}$, $\{|A'|_\infty < \pi\} \subset \mathsf{Form}_n$ onto relatively open subsets of $\mathcal{G}_n$, $\mathcal{U}_n^{\mathsf{T}}$, $\mathcal{U}_n$ (Def. 7.1). On $\{|\lambda|_\infty < \frac{1}{3},\ |A|_\infty < \frac{1}{3}\}$ we have $e^{iA^\lambda} = \mathsf{j}(e^{i\lambda}, e^{iA})$ and $|A^\lambda|_\infty < \pi$; thus, in these charts, Ξ is the map j, and (b) shows that Ξ is a diffeomorphism of $\{|\lambda|_\infty < \frac{1}{3},\ |A|_\infty < \frac{1}{3}\}$ onto an open subset of Form_n, with $\Xi(0,0) = 0$. We note that $\dim \mathsf{X}_n + \dim \mathsf{Form}_n^{\mathsf{T}} = \dim \mathsf{Form}_n$, since T_n has as many elements as Λ_n°.

Let $\mathfrak{D}' \subset \{|\lambda|_\infty < \frac{1}{3},\ |A|_\infty < \frac{1}{3}\}$ be open, and let Φ' be a bounded Borel function on $\Xi(\mathfrak{D}')$. We apply eq. 7.vii to the function F on $\mathcal{U}_n$ which equals $\Phi'(A')$ at $U = e^{iA'}$, $A' \in \Xi(\mathfrak{D}')$, and 0 elsewhere, and we express every Haar integral by the Haar formula in Def. 7.1(2) at $u_0 = \mathbf{1}$, bond by bond and site by site. Each of these integrals carries the measure dY of $(\mathfrak{g}, \langle\cdot,\cdot\rangle)$ in the variable of its bond or site; since $(\cdot,\cdot)$ is the sum of $\langle\cdot,\cdot\rangle$ over the bonds, resp. the sites, the products of these measures are the Lebesgue measures dA', dA, $d\lambda$ of the Euclidean spaces Form_n, $\mathsf{Form}_n^{\mathsf{T}}$, X_n, as in (iv).

On the left side this produces the factor $\sigma_{\mathrm{H}}^{\#\mathsf{Bd}_n}$ and an integral over $\Xi(\mathfrak{D}')$. On the right side, the injectivity of j and of the exponential charts shows that, for (λ, A) in the domain of the charts, $(e^{iA})^{e^{i\lambda}} = e^{iA'}$ with $A' \in \Xi(\mathfrak{D}')$ holds if and only if $(\lambda, A) \in \mathfrak{D}'$, and then $A' = A^\lambda$; this produces the factor $\sigma_{\mathrm{H}}^{\#\Lambda_n^\circ + \#(\mathsf{Bd}_n \setminus \mathsf{T}_n)} = \sigma_{\mathrm{H}}^{\#\mathsf{Bd}_n}$ and an integral over $\mathfrak{D}'$. The powers of σ_{H} cancel, and

$$\int_{\Xi(\mathfrak{D}')} \Phi'(A') \prod_{b\in\mathsf{Bd}_n} \sigma_1(|A'(b)|^2)\,\mathrm{d}A' \\ = \int_{\mathfrak{D}'} \Phi'(A^\lambda) \prod_{x\in\Lambda_n^\circ} \sigma_1(|\lambda(x)|^2) \prod_{b\in\mathsf{Bd}_n\setminus\mathsf{T}_n} \sigma_1(|A(b)|^2)\,\mathrm{d}\lambda\,\mathrm{d}A.$$

Step 9: the derivative of the gauge function in λ. Let $\mathsf{c}(\lambda, A) := \mathrm{d}^*(A^\lambda)$ for $\lambda \in \mathsf{X}_n$ and $A \in \mathsf{Form}_n^{\mathsf{T}}$ with $|\lambda|_\infty < \frac{1}{10}$, $|A|_\infty < \frac{1}{10}$. For $H \in \mathsf{X}_n$ and $|s|$ small let $\mu(s) \in \mathsf{X}_n$ be defined, site by site and with small values, by $e^{i\mu(s)} := e^{i(\lambda+sH)}e^{-i\lambda}$. Then $\mu(0) = 0$, and μ is smooth in s. Differentiating $e^{i\mu(s)} = e^{i(\lambda+sH)}e^{-i\lambda}$ at $s = 0$ with the differential formula in Def. 7.1(2) on both sides, and using $\mathsf{dE}(0) = \mathbf{1}$, we find at every site

$$i\mu'(0) = i\big(\mathsf{dE}(\lambda)H\big)e^{i\lambda}e^{-i\lambda}, \qquad \text{that is,} \qquad \mu'(0) = \vartheta_\lambda H, \quad \vartheta_\lambda := \bigoplus_x \mathsf{dE}(\lambda(x)).$$

Since $e^{i(\lambda+sH)} = e^{i\mu(s)}e^{i\lambda}$, we have $(e^{iA})^{e^{i(\lambda+sH)}} = \big((e^{iA})^{e^{i\lambda}}\big)^{e^{i\mu(s)}}$, that is, $A^{\lambda+sH} = (A^\lambda)^{\mu(s)}$ for $|s|$ small, both sides lying in $\mathbb{B}_\pi$ at every bond and having the same exponential. By Step 7, $|A^\lambda|_\infty < 1$, and the definition eq. 7.iv of M with the chain rule gives

$$\partial_\lambda \mathsf{c}(\lambda, A)H = \frac{d}{ds}\,\mathrm{d}^*\big((A^\lambda)^{\mu(s)}\big)\Big|_{s=0} = \mathsf{M}(A^\lambda)\,\vartheta_\lambda H.$$

By the determinant formula in Def. 7.1(2), $\det\vartheta_\lambda = \prod_x \sigma_1(|\lambda(x)|^2) > 0$. At $(\lambda, A) = (0,0)$ we have $\vartheta_0 = \mathbf{1}$ and $\partial_\lambda\mathsf{c}(0,0) = \mathsf{M}(0) = \mathsf{M}^{(0)} = -\mathrm{d}^*\mathrm{d}$, which is invertible by (c).

Step 10: choice of the radii; claims (i)–(iii). The map $\Theta_* : (\lambda, A) \mapsto (\mathsf{c}(\lambda, A), A)$, defined on the set of the pairs $(\lambda, A) \in \mathsf{X}_n \times \mathsf{Form}_n^{\mathsf{T}}$ with $|\lambda|_\infty < \frac{1}{10}$, $|A|_\infty < \frac{1}{10}$ and taking values in $\mathsf{X}_n \times \mathsf{Form}_n^{\mathsf{T}}$, is smooth, and its differential at $(0,0)$ is block triangular with the invertible diagonal blocks $-\mathrm{d}^*\mathrm{d}$ and $\mathbf{1}$. By the inverse function theorem there is an open neighbourhood $\mathcal{N}$ of $(0,0)$ on which Θ_* is a diffeomorphism onto an open set; on $\mathcal{N}$ the differential of Θ_* is invertible, and therefore so is $\partial_\lambda\mathsf{c}$. We first choose $\varrho_1 \le \frac{1}{20}$ with $\{|\lambda|_\infty \le 2\varrho_1,\ |A|_\infty \le 2\varrho_1\} \subset \mathcal{N}$, and then $\varrho_2, \varrho_3 \le \varrho_1$ with

$$\{c \in \mathsf{X}_n :\ |c|_\infty < \varrho_2\} \times \{A \in \mathsf{Form}_n^{\mathsf{T}} :\ |A|_\infty \le \varrho_3\} \subset \Theta_*(\{|\lambda|_\infty < \varrho_1,\ |A|_\infty < 2\varrho_1\});$$

this is possible because the right side is an open set containing $\Theta_*(0,0) = (0,0)$. These numbers depend on n only.

For (iii), let $A \in \mathsf{Form}_n^{\mathsf{T}}$ with $|A|_\infty \le \varrho_3$. Since Θ_* is injective on $\mathcal{N}$ and keeps the second component, $\lambda \mapsto \mathsf{c}(\lambda, A)$ is injective on $\{|\lambda|_\infty \le \varrho_1\}$. If $|c|_\infty < \varrho_2$, then $(c, A) = \Theta_*(\lambda, A'')$ with $|\lambda|_\infty < \varrho_1$, and necessarily $A'' = A$; thus $c = \mathrm{d}^*(A^\lambda)$ with $|\lambda|_\infty < \varrho_1$. For (i): since $\varrho_1, \varrho_3 \le \frac{1}{20}$, the closure $\overline{\mathfrak{D}} = \{|\lambda|_\infty \le \varrho_1,\ |A|_\infty \le \varrho_3\}$ lies in the open set $\{|\lambda|_\infty < \frac{1}{10},\ |A|_\infty < \frac{1}{10}\}$, on which Ξ is a diffeomorphism onto an open set (Step 8) with values in $\{|A'|_\infty < 1\}$ (Step 7).

For (ii): $\overline{\mathfrak{D}}$ is compact, so $\Xi(\overline{\mathfrak{D}})$ is compact; it contains $\mathcal{W}$ and is contained in $\overline{\mathcal{W}}$, hence equals $\overline{\mathcal{W}}$. For $(\lambda, A) \in \overline{\mathfrak{D}} \subset \mathcal{N}$ the operator $\partial_\lambda \mathsf{c}(\lambda, A) = \mathsf{M}(A^\lambda)\vartheta_\lambda$ is invertible and $\det \vartheta_\lambda \neq 0$; hence $\det \mathsf{M}(A^\lambda) \neq 0$.

Step 11: claim (iv). By eq. 7.vi, the entries of $\mathsf{M}(A')$ are continuous functions of A' for $|A'|_\infty < 1$; hence $|\det \mathsf{M}|$ is bounded on the compact set $\overline{\mathcal{W}}$. We apply the identity of Step 8 with $\mathfrak{D}' := \mathfrak{D}$ and the bounded Borel function $\Phi' := \Phi \cdot (w \circ \mathrm{d}^*) \cdot |\det \mathsf{M}|$ on $\mathcal{W}$. Its left side is the left side of eq. 7.viii. On its right side we integrate first over λ, at fixed A with $|A|_\infty < \varrho_3$ (Fubini's theorem). By (iii) and Step 9, $\lambda \mapsto c = \mathsf{c}(\lambda, A)$ is an injective smooth map on $\{|\lambda|_\infty < \varrho_1\}$ with the invertible differential $\mathsf{M}(A^\lambda)\vartheta_\lambda$; hence it is a diffeomorphism onto its open image $\{\mathrm{d}^*(A^\lambda) : |\lambda|_\infty < \varrho_1\}$, and

$$dc = |\det \mathsf{M}(A^\lambda)| \prod_{x \in \Lambda_n^\circ} \sigma_1(|\lambda(x)|^2)\, \mathrm{d}\lambda.$$

Since $\Phi'(A^\lambda) = \Phi(A^\lambda)\, w(\mathsf{c}(\lambda, A))\, |\det \mathsf{M}(A^\lambda)|$, the substitution $c = \mathsf{c}(\lambda, A)$ turns the inner integral into $\int w(c)\, \Phi(A^{\lambda(c)})\, \mathrm{d}c$ over this image, and eq. 7.viii follows. □

7.1.3. *A coefficient defined before the continuum limit.* The quantity computed here is a coefficient of the Wilson covariance in a fixed Dirichlet box. Gauge fixing provides coordinates for its Gaussian evaluation. It does not change the observable whose asymptotic coefficient is being defined. For each fixed box, the construction gives

$$\langle a_p; a_q \rangle_{n,\beta} = \beta^{-2} \mathsf{B}_0^{[n]}(p, q) + \beta^{-3} \mathsf{B}_1^{[n]}(p, q) + O_n(\beta^{-4}).$$

The constant in this remainder retains its box dependence. The infinite-lattice coefficient is obtained by exhausting the boxes after extracting the coefficient. The two limits are therefore iterated. The relation of this quantity to an interacting continuum covariance is supplied by the preceding section's renormalization comparisons.

The finite-box calculation starts with a gauge slice on which the quadratic action has no zero modes. The Faddeev–Popov change of variables supplies its determinant, and exponential coordinates supply the Haar density. Both contribute to the coefficient. Write $t = \beta^{-1/2}$. If X_j, Y_j are the homogeneous insertion terms of degree $j + 2$, and W_j is the sum of the phase term of degree $j + 2$ and the density term of degree j, the second connected Laplace coefficient is

$$\begin{aligned} \gamma_2 = {} & \langle X_1; Y_1 \rangle_0 + \langle X_0; Y_2 \rangle_0 + \langle X_2; Y_0 \rangle_0 - \langle X_1; Y_0; W_1 \rangle_0 \\ & - \langle X_0; Y_1; W_1 \rangle_0 - \langle X_0; Y_0; W_2 \rangle_0 + \tfrac{1}{2} \langle X_0; Y_0; W_1; W_1 \rangle_0. \end{aligned} \tag{7.ix}$$

Parity makes the intervening odd coefficients zero. This formula is useful before any graph notation: it states exactly which Taylor terms of the action, density and two insertions must be retained. Omitting the density terms changes the answer.

For Gaussian polynomials, a joint cumulant is the sum of Wick pairings whose graph connects all its polynomial vertices. The partition formula for cumulants removes the disconnected pairings. Applying that rule to eq. 7.ix gives the finite graph classes and their numerical multiplicities. The source retains the counting by incidence matrices, so each factor can be checked without relying on a picture or an implicit convention for equivalent contractions.

7.1.4. *Cancellations before long-distance estimates.* The plain, mixed and curl lines have different decay. In Feynman gauge their leading orders are respectively $|u|^{-1}$, $|u|^{-2}$ and $|u|^{-3}$. Products and convolutions of these lines generate local contributions as well as the desired homogeneous kernel. Estimating every local contribution before adding the relevant classes would retain terms that cancel. The zero-momentum identities are what remove them.

For the self-energy, the five contributions comprise the Wilson vertices, the ghost terms and the Haar term. Evaluating the Hessian at a constant abelian background expresses their sum as a

second derivative of a periodic logarithm. The integral of that derivative vanishes: opposite faces of a period cell cancel, and the flux through a sphere of radius ε is $O(\varepsilon)$. Put

$$Z_0 = \int_{[-\pi,\pi]^3} \left(\sum_{\mu=1}^{3} 4\sin^2(k_\mu/2) \right)^{-1} \frac{\mathrm{d}k}{(2\pi)^3}.$$

At gauge parameter one, the five diagonal coefficients are

$$-20Z_0 + \tfrac{8}{3}, \quad 16Z_0 - \tfrac{28}{9}, \quad 4Z_0 - \tfrac{2}{3}, \quad \tfrac{4}{9}, \quad \tfrac{2}{3}.$$

Their sum is zero. The plaquette vertex has a corresponding four-term identity. Point reflections supply the required vanishing first derivatives. Together these facts say that the symbols of the combined local insertion terms vanish at zero momentum to the required order.

The kernel lemmas then use these moment cancellations. A local insertion with vanishing low moments gains differences of the external lines instead of contributing its unweighted mass. In the two-point coefficient this places the local pieces at order $|u|^{-6}$ or smaller. The surviving chain, bubble and kite terms have homogeneous pieces of degree -5. Their subtractions and borderline convolution sums give the remainder $O(|u|^{-6}(1+\log|u|))$. The proof retains the individual kernel estimates because absolute convergence, existence of a homogeneous term and the error after subtraction are distinct claims.

The six continuum integrands reproduce precisely these homogeneous pieces. Their rotationally invariant scalar sum determines κ^{lat}; the individual plane kernels remain the functions U_{ij} of theorem V. The method-of-images estimates separately show that the finite-box coefficients converge to the lattice sums. They complete the definition of the lattice coefficient, rather than supplying a joint bound for growing boxes and couplings.

7.1.5. *Why a residue determines the number.* For a kernel of degree -5 in dimension three, the relevant Fourier term is $\mathsf{p}^2 \log|\mathsf{p}|$. The normalization is fixed by lemma 7.41. With the Fourier convention used there, analytic continuation of the transform of $|p|^s$ gives

$$c(s) = \frac{2^s \Gamma_{\mathrm{E}}((3+s)/2)}{\pi^{3/2}\Gamma_{\mathrm{E}}(-s/2)}, \qquad c(2) = 0, \qquad c'(2) = \frac{3}{2\pi}.$$

Differentiation at $s = 2$ converts $p^2 \log|p|$ into $(3/2\pi)|x|^{-5}$ away from the origin. Stating this constant before the graph evaluation fixes the Fourier normalization and the eventual sign.

For the six integrands $F_\Gamma(k, l, \mathsf{p})$, introduce

$$A = k^2, \quad A' = (k-\mathsf{p})^2, \quad B = l^2, \quad B' = (l-\mathsf{p})^2, \quad C = (k-l)^2, \quad P = \mathsf{p}^2.$$

All contracted numerators become polynomials in these quantities. Take the quadratic term in p, average its angular dependence as in definition 7.47, and express the result as Laurent monomials in A, B, C. The functional $\mathcal{R}$ extracts the coefficient of $(ABC)^{-1}$. Lemma 7.48 provides an explicit rule on each monomial, so the remaining six evaluations are finite algebraic calculations with recorded symmetry factors.

This coefficient is singled out by the finite-part integral under a dilation of the cutoff. For a smooth radial cutoff ϕ equal to one near the origin, the contributing integral is

$$\iint \frac{\phi((k,l)/\Lambda) - \phi(k,l)}{k^2 l^2 (k-l)^2} \, \mathrm{d}k \, \mathrm{d}l = 4\pi^4 \log\Lambda.$$

In six-dimensional polar coordinates, the radial factor is the integral of the cutoff difference against $\mathrm{d}s/s$, hence $\log\Lambda$. The angular integral is $4\pi^4$, computed in lemma 7.49. The other monomials allowed by the graph numerators have zero finite-part dilation difference. After the two Fourier measures this leaves $\mathcal{R}[F_\Gamma]/(16\pi^2)$ as the coefficient of $\log\Lambda$.

The expansion by regions justifies using that finite part: it separates the small-momentum regions, subtracts their specified singular terms and bounds the remainder. Comparing the cutoff-dilation expansion with the Fourier expansion then gives the bridge

$$\kappa_\Gamma^{\mathrm{lat}} = -\frac{3}{32\pi^3}\, 24\, \mathcal{R}[F_\Gamma].$$

The six values are

$$-\frac{4}{3}, \quad -\frac{13}{12}, \quad -\frac{1}{12}, \quad 0, \quad \frac{13}{6}, \quad -\frac{1}{3}.$$

Their sum is $-2/3$, and therefore $\kappa^{\mathrm{lat}} = 3/(2\pi^3)$ and $c_1 = \kappa^{\mathrm{lat}}/c_2 = 8/(3\pi)$. The intermediate cancellations and residue calculation explain the number; the final arithmetic alone does not constitute its derivation.

For separated test pairs, the kernel remainder and the lattice sampling estimate give an error of order $a(1 + |\log(d/a)|)$ in the smeared coefficient. The lower bound for $\mathcal{K}_5$ then gives the relative error in theorem X. This conclusion concerns separated supports. It supplies no coincident-plaquette term and does not extend the coefficient calculation to constant or overlapping sources.

7.1.6. *The coefficient.*

Definition 7.7 (Quadratic form, covariances, vertices)**.** Let Ω be either $\mathbb{Z}^3$, indicated by the upper index lat, or a box Λ_n, $n \geq 2$, indicated by the upper index $[n]$, and let $\tau_{\mathrm{gf}} > 0$; the index is omitted when Ω is fixed.

(1) We set

$$\begin{aligned} \mathsf{S}_2(A) &:= \tfrac{1}{2}\|\mathrm{d}A\|^2 + \tfrac{1}{2\tau_{\mathrm{gf}}}\|\mathrm{d}^*A\|^2, \\ \mathsf{C}_{\tau_{\mathrm{gf}}} &= \left(\mathrm{d}^*\mathrm{d} + \tau_{\mathrm{gf}}^{-1}\mathrm{d}\mathrm{d}^*\right)^{-1}, \qquad \mathsf{G} = (\mathrm{d}^*\mathrm{d})^{-1} \text{ on site functions.} \end{aligned} \tag{7.x}$$

In the box, $\mathrm{d}^*\mathrm{d} + \tau_{\mathrm{gf}}^{-1}\mathrm{d}\mathrm{d}^*$ is positive definite on $\mathsf{Form}_n^{\mathbb{R}}$ and $\mathrm{d}^*\mathrm{d}$ is positive definite on $\mathsf{X}_n^{\mathbb{R}}$, by Lem. 7.6(c) applied to forms and functions with a single non-vanishing component. $\mathsf{C}_{\tau_{\mathrm{gf}}}^{[n]}$ and $\mathsf{G}^{[n]}$ are the inverses of these operators, with the matrices $\mathsf{C}_{\tau_{\mathrm{gf}}}^{[n]}(b,b')$, $b,b' \in \mathsf{Bd}_n$, and $\mathsf{G}^{[n]}(x,y)$, $x,y \in \Lambda_n^\circ$; we set $\mathsf{C}_{\tau_{\mathrm{gf}}}^{[n]}(b,b') := 0$ if b or b' is not a free bond, and $\mathsf{G}^{[n]}(x,y) := 0$ if x or y does not lie in Λ_n°. On $\mathbb{Z}^3$ we define, with $\mathsf{k} = \mathsf{k}(k)$,

$$\begin{aligned} \mathsf{C}_{\tau_{\mathrm{gf}}}^{\mathrm{lat}}((x,\mu),(y,\nu)) &:= \int_{\mathbb{T}} e^{ik\cdot(x-y)}\Big[\frac{\delta_{\mu\nu}}{|\mathsf{k}|^2} + (\tau_{\mathrm{gf}}-1)\frac{\mathsf{k}_\mu\overline{\mathsf{k}_\nu}}{|\mathsf{k}|^4}\Big]d^\circ k, \\ \mathsf{G}^{\mathrm{lat}}(x,y) &:= \int_{\mathbb{T}} \frac{e^{ik\cdot(x-y)}}{|\mathsf{k}|^2}\, d^\circ k. \end{aligned}$$

The matrix in brackets is Hermitian, with the eigenvalues $|\mathsf{k}|^{-2}$ (twice) and $\tau_{\mathrm{gf}}|\mathsf{k}|^{-2}$; as $|\mathsf{k}|^2 \geq \frac{4}{\pi^2}|k|^2$ on $[-\pi,\pi]^3$, the integrands are bounded by $\frac{\pi^2}{4}\max\{1,\tau_{\mathrm{gf}}\}|k|^{-2}$, which is integrable. The bracket is positive, and it is replaced by its complex conjugate under $k \to -k$, because $\mathsf{k}_\mu(-k) = \overline{\mathsf{k}_\mu(k)}$. Hence the kernels are real and symmetric. They are also positive semidefinite: for finitely supported complex functions f on the positively oriented bonds, resp. on the sites,

$$\begin{aligned} \sum_{b,b'} \overline{f(b)}\, \mathsf{C}_{\tau_{\mathrm{gf}}}^{\mathrm{lat}}(b,b')f(b') &= \int_{\mathbb{T}} v(k)^*[\cdots]\, v(k)\, d^\circ k \geq 0, \\ \sum_{x,y} \overline{f(x)}\, \mathsf{G}^{\mathrm{lat}}(x,y)f(y) &= \int_{\mathbb{T}} \Big|\sum_y e^{-ik\cdot y} f(y)\Big|^2 |\mathsf{k}|^{-2}\, d^\circ k \geq 0, \end{aligned}$$

where $v_\nu(k) := \sum_y e^{-ik\cdot y} f((y,\nu))$ and $[\cdots]$ is the bracket.

The definition on $\mathbb{Z}^3$ is explained by plane waves. Let $A_\mu(x) = e^{ik\cdot x}\alpha_\mu$ with $\alpha \in \mathbb{C}^3$, let $\lambda(x) = e^{ik\cdot x}c$ with $c \in \mathbb{C}$, and let F be the plaquette function with $F_{\rho\nu}(x) = e^{ik\cdot x} f_{\rho\nu}$ for all ρ,ν,

where $f_{\nu\rho} = -f_{\rho\nu}$; we write $\mathsf{k}^*\alpha := \sum_\mu \overline{\mathsf{k}_\mu}\,\alpha_\mu$. The formulas for d and d^* in Def. 7.2(2), together with $e^{ik_\mu} - 1 = \mathsf{k}_\mu$ and $e^{-ik_\mu} - 1 = \overline{\mathsf{k}_\mu}$, give

$$\mathrm{d}A(p_{\mu\nu}(x)) = e^{ik\cdot x}(\mathsf{k}_\mu\alpha_\nu - \mathsf{k}_\nu\alpha_\mu), \qquad \mathrm{d}^*A(x) = e^{ik\cdot x}\,\mathsf{k}^*\alpha,$$
$$(\mathrm{d}\lambda)((x,\rho)) = e^{ik\cdot x}\,\mathsf{k}_\rho\,c, \qquad (\mathrm{d}^*F)((x,\rho)) = -e^{ik\cdot x}\sum_\nu \overline{\mathsf{k}_\nu}\,f_{\rho\nu}.$$

For $F = \mathrm{d}A$ we have $f_{\rho\nu} = \mathsf{k}_\rho\alpha_\nu - \mathsf{k}_\nu\alpha_\rho$, and for $\lambda = \mathrm{d}^*A$ we have $c = \mathsf{k}^*\alpha$; therefore

$$(\mathrm{d}^*\mathrm{d}A)((x,\rho)) = e^{ik\cdot x}(|\mathsf{k}|^2\alpha_\rho - \mathsf{k}_\rho\,\mathsf{k}^*\alpha), \qquad (\mathrm{d}\mathrm{d}^*A)((x,\rho)) = e^{ik\cdot x}\,\mathsf{k}_\rho\,\mathsf{k}^*\alpha.$$

Thus $\mathrm{d}^*\mathrm{d} + \tau_{\mathrm{gf}}^{-1}\mathrm{d}\mathrm{d}^*$ has the symbol $|\mathsf{k}|^2\mathbf{1} - (1-\tau_{\mathrm{gf}}^{-1})\mathsf{k}\mathsf{k}^*$: it maps the plane wave with the amplitude α to the plane wave with the amplitude $(|\mathsf{k}|^2\mathbf{1} - (1-\tau_{\mathrm{gf}}^{-1})\mathsf{k}\mathsf{k}^*)\alpha$. The inverse of this matrix is the bracket: as $\mathsf{k}^*\mathsf{k} = |\mathsf{k}|^2$,

$$(|\mathsf{k}|^2\mathbf{1} - (1-\tau_{\mathrm{gf}}^{-1})\mathsf{k}\mathsf{k}^*)\Big[\frac{\mathbf{1}}{|\mathsf{k}|^2} + (\tau_{\mathrm{gf}}-1)\frac{\mathsf{k}\mathsf{k}^*}{|\mathsf{k}|^4}\Big]$$
$$= \mathbf{1} + \big[(\tau_{\mathrm{gf}}-1) - (1-\tau_{\mathrm{gf}}^{-1}) - (1-\tau_{\mathrm{gf}}^{-1})(\tau_{\mathrm{gf}}-1)\big]\frac{\mathsf{k}\mathsf{k}^*}{|\mathsf{k}|^2} = \mathbf{1},$$

since $(1-\tau_{\mathrm{gf}}^{-1})(1+(\tau_{\mathrm{gf}}-1)) = \tau_{\mathrm{gf}} - 1$.

(2)

Derived kernels. With $b_i(p)$ and s_i as in Lem. 7.5 we set

$$\mathsf{E}_p(b) := \sum_i s_i\,\mathsf{C}_{\tau_{\mathrm{gf}}}(b_i(p), b), \qquad \Pi(p,q) := \sum_j s_j\,\mathsf{E}_p(b_j(q)),$$
$$\mathsf{g}(b) := \mathsf{G}(b_+,b_+) + \mathsf{G}(b_-,b_-) - 2\mathsf{G}(b_+,b_-), \qquad \mathsf{D}(b;b') := \tfrac{1}{2}\sum_{x\in\{b'_-,b'_+\}}[\mathsf{G}(x,b_+) - \mathsf{G}(x,b_-)].$$

The kernels E_p and Π do not depend on τ_{gf}. On $\mathbb{Z}^3$, for $p = p_{\mu\nu}(x)$ and a bond (y,ρ), the four terms of E_p combine to

$$\sum_i s_i\,\mathsf{C}^{\mathrm{lat}}_{\tau_{\mathrm{gf}}}(b_i(p),(y,\rho)) = \int_{\mathbb{T}} e^{ik\cdot(x-y)}(\mathsf{k}_\mu\,[\cdots]_{\nu\rho} - \mathsf{k}_\nu\,[\cdots]_{\mu\rho})\,d^\circ k,$$

where $[\cdots]_{\alpha\rho}$ denotes the entry (α,ρ) of the bracket in (1); the τ_{gf}-dependent term of the bracket contributes $(\tau_{\mathrm{gf}}-1)(\mathsf{k}_\mu\mathsf{k}_\nu - \mathsf{k}_\nu\mathsf{k}_\mu)\overline{\mathsf{k}_\rho}|\mathsf{k}|^{-4} = 0$. In the box,

$$\mathsf{C}_{\tau_{\mathrm{gf}}} = \mathsf{C}_1 + (\tau_{\mathrm{gf}}-1)\,\mathrm{d}\,\mathsf{G}^2\mathrm{d}^*, \qquad \mathrm{d}\mathrm{d} = 0;$$

since $\mathsf{E}_p = \mathsf{C}_{\tau_{\mathrm{gf}}}\mathrm{d}^*\delta_p$ by the symmetry of $\mathsf{C}_{\tau_{\mathrm{gf}}}$, where δ_p is the plaquette function with the value 1 at p and 0 elsewhere, and $\mathrm{d}^*\mathrm{d}^* = (\mathrm{d}\mathrm{d})^* = 0$, the second term does not contribute to E_p or to Π. To prove the identity for $\mathsf{C}_{\tau_{\mathrm{gf}}}$, let $\tau := \tau_{\mathrm{gf}}$ and $\mathsf{L}_\tau := \mathrm{d}^*\mathrm{d} + \tau^{-1}\mathrm{d}\mathrm{d}^*$. From $\mathrm{d}\mathrm{d} = 0$ we get $\mathsf{L}_\tau\mathrm{d} = \tau^{-1}\mathrm{d}(\mathrm{d}^*\mathrm{d})$, hence $\mathsf{L}_\tau\,\mathrm{d}\mathsf{G}^2\mathrm{d}^* = \tau^{-1}\mathrm{d}\mathsf{G}\mathrm{d}^*$.

Moreover $\mathrm{d}^*\mathsf{L}_1 = \mathrm{d}^*\mathrm{d}\,\mathrm{d}^*$ gives $\mathrm{d}^*\mathsf{C}_1 = \mathsf{G}\mathrm{d}^*$, and therefore $\mathsf{L}_\tau\mathsf{C}_1 = \mathbf{1} + (\tau^{-1}-1)\mathrm{d}\mathrm{d}^*\mathsf{C}_1 = \mathbf{1} + (\tau^{-1}-1)\mathrm{d}\mathsf{G}\mathrm{d}^*$. Adding the first identity multiplied by $\tau - 1$ to the second we obtain $\mathsf{L}_\tau[\mathsf{C}_1 + (\tau-1)\mathrm{d}\mathsf{G}^2\mathrm{d}^*] = \mathbf{1}$.

On $\mathbb{Z}^3$, for $b = (x,\mu)$, the reality and symmetry of $\mathsf{G}^{\mathrm{lat}}$ give

$$\mathsf{g}(b) = \int_{\mathbb{T}} \frac{2 - e^{ik_\mu} - e^{-ik_\mu}}{|\mathsf{k}|^2}\,d^\circ k = \int_{\mathbb{T}} \frac{|\mathsf{k}_\mu|^2}{|\mathsf{k}|^2}\,d^\circ k = \frac{1}{3},$$

since $|\mathsf{k}|^2$ is invariant under permutations of the coordinates of k, so that the three integrals $\int |\mathsf{k}_\mu|^2|\mathsf{k}|^{-2}d^\circ k$ are equal, with sum 1.

(3)

Vertices. With sums over the plaquettes and the positively oriented bonds of Ω, that is, over Pl_n and Bd_n in the box, we set

$$\begin{aligned}\mathsf{S}_3 &:= \sum_p \mathsf{a}_p^{(3)},\\ \mathsf{V}_2 &:= \sum_p \mathsf{a}_p^{(4)} + \frac{1}{3}\sum_b |A(b)|^2 + \frac{2}{3}\sum_b \mathsf{g}(b)|A(b)|^2 - 4\sum_{b,b'} \mathsf{D}(b;b')\mathsf{D}(b';b)\langle A(b), A(b')\rangle.\end{aligned} \tag{7.xi}$$

The four terms of V_2 are called the *quartic Wilson vertex* $\mathsf{S}_4 := \sum_p \mathsf{a}_p^{(4)}$, the *Haar term*, the *ghost tadpole* and the *ghost loop*. On $\mathbb{Z}^3$, S_3 and V_2 are formal sums of local terms in the sense of Def. 7.8(3). In the box,

$$\begin{aligned}\tfrac{2}{3}\sum_b \mathsf{g}(b)|A(b)|^2 &= -\operatorname{tr}\big[(\mathsf{M}^{(0)})^{-1}\mathsf{M}^{(2)}(A)\big],\\ -4\sum_{b,b'} \mathsf{D}(b;b')\mathsf{D}(b';b)\langle A(b), A(b')\rangle &= \tfrac{1}{2}\operatorname{tr}\big[\big((\mathsf{M}^{(0)})^{-1}\mathsf{M}^{(1)}(A)\big)^2\big],\end{aligned} \tag{7.xii}$$

the traces being taken on X_n, and $\operatorname{tr}\big[(\mathsf{M}^{(0)})^{-1}\mathsf{M}^{(1)}(A)\big] = 0$.

To prove these identities we use $(\mathsf{M}^{(0)})^{-1} = -\mathsf{G}$ (Lem. 7.5(d)). For a family $T = (T_b)$ of endomorphisms $T_b \in \operatorname{End}\mathfrak{g}$, acting on forms bond by bond, the cyclicity of the trace gives

$$\operatorname{tr}\big[\mathsf{G}\,\mathrm{d}^* T \mathrm{d}\big] = \sum_b \operatorname{tr} T_b \cdot (\mathrm{d}\mathsf{G}\mathrm{d}^*)(b,b) = \sum_b \mathsf{g}(b)\operatorname{tr} T_b,$$

because the diagonal element of the kernel of $\mathrm{d}\mathsf{G}\mathrm{d}^*$ at b is $\mathsf{G}(b_+,b_+) - \mathsf{G}(b_+,b_-) - \mathsf{G}(b_-,b_+) + \mathsf{G}(b_-,b_-) = \mathsf{g}(b)$. The formula $X\times(Y\times Z) = \langle X,Z\rangle Y - \langle X,Y\rangle Z$ gives $\operatorname{tr}[X\times(Y\times\cdot)] = \langle X,Y\rangle - 3\langle X,Y\rangle = -2\langle X,Y\rangle$. By Lem. 7.5(d), $\mathsf{M}^{(2)}(A) = \mathrm{d}^*T\mathrm{d}$ with $T_b = -\frac{1}{3}A(b)\times(A(b)\times\cdot)$, so $\operatorname{tr} T_b = \frac{2}{3}|A(b)|^2$, and the first identity follows. Next let $\mathsf{R} := \mathsf{G}\,\mathrm{d}^*(A\times\bar{\cdot})$, so that $(\mathsf{M}^{(0)})^{-1}\mathsf{M}^{(1)}(A) = -2\mathsf{R}$. Its kernel is

$$\mathsf{R}(x,y) = \sum_b \big[\mathsf{G}(x,b_+) - \mathsf{G}(x,b_-)\big]\,\tfrac{1}{2}\big[\delta_{y,b_-} + \delta_{y,b_+}\big]\,A(b)\times\cdot \;\in \operatorname{End}\mathfrak{g}.$$

Since $\operatorname{tr}[X\times\cdot] = 0$, we get $\operatorname{tr}\mathsf{R} = 0$, which is the last assertion. Summing first over x and y in $\operatorname{tr}\mathsf{R}^2 = \sum_{x,y}\operatorname{tr}[\mathsf{R}(x,y)\mathsf{R}(y,x)]$ we find

$$\operatorname{tr}\mathsf{R}^2 = \sum_{b,b'} \mathsf{D}(b;b')\mathsf{D}(b';b)\operatorname{tr}\big[A(b)\times(A(b')\times\cdot)\big] = -2\sum_{b,b'} \mathsf{D}(b;b')\mathsf{D}(b';b)\langle A(b),A(b')\rangle,$$

and $\frac{1}{2}\operatorname{tr}[(-2\mathsf{R})^2] = 2\operatorname{tr}\mathsf{R}^2$ gives the second identity. ∎

Definition 7.8 (Gaussian family; cumulants of polynomials; formal sums)**.** Let Ω and τ_{gf} be as in Def. 7.7.

(1) The variables $A^a(b)$, where b ranges over the positively oriented bonds of Ω (over the free bonds if $\Omega = \Lambda_n$) and $a \in \{1,2,3\}$, form a centred Gaussian family with

$$\langle A^a(b)A^{a'}(b')\rangle_0 = \delta_{aa'}\mathsf{C}^{\Omega}_{\tau_{\mathrm{gf}}}(b,b');$$

such a family exists because this covariance is positive semidefinite (Def. 7.7(1)), and its mean is denoted by $\langle\cdot\rangle_0$. In the box we put $A^a(b) := 0$ at the bonds which are not free, in accordance with the definition of Form_n. A *leg* is a variable $A^a(b)$ (*plain leg*) or a variable $(\mathrm{d}A)^a(p) = \sum_i s_i A^a(b_i(p))$ (*curl leg*). By the definitions of E_p and Π,

$$\begin{aligned}\langle A^a(b)A^{a'}(b')\rangle_0 &= \delta_{aa'}\mathsf{C}(b,b'), \qquad \langle(\mathrm{d}A)^a(p)A^{a'}(b)\rangle_0 = \delta_{aa'}\mathsf{E}_p(b),\\ \langle(\mathrm{d}A)^a(p)(\mathrm{d}A)^{a'}(q)\rangle_0 &= \delta_{aa'}\Pi(p,q).\end{aligned} \tag{7.xiii}$$

(2) Let $X_1, \dots, X_r$ be polynomials in the variables of the family, and $m_J := \langle \prod_{i\in J} X_i\rangle_0$ for $\emptyset \neq J \subset \{1, \dots, r\}$. Their *joint cumulants* $\langle X_i; i \in J\rangle_0 := \kappa_J$ are defined by the partition formula

$$\kappa_J = \sum_{\pi\in \mathrm{Part}(J)} (-1)^{|\pi|-1}(|\pi|-1)! \prod_{I\in\pi} m_I. \tag{7.xiv}$$

Here $\mathrm{Part}(J)$ denotes the partitions π of J into nonempty blocks I, and $|\pi|$ is the number of blocks. Equivalently, (κ_J) is the unique family satisfying

$$m_J = \sum_{\pi\in \mathrm{Part}(J)} \prod_{I\in\pi} \kappa_I. \tag{7.xv}$$

The equivalence is proved below. The cumulants are symmetric and multilinear in the X_i: each index belongs to exactly one block of each partition, every moment is linear in each factor, and the formula involves J only as a set. Formula eq. 7.xiv requires only the indicated moments; no exponential moment of a polynomial in the Gaussian family is assumed.

When exponential moments exist, this definition agrees with differentiation of their logarithm. For bounded complex variables, the exponential mean is holomorphic near zero and can be differentiated under the integral. For real variables $Y_i = X_i$, $i \in J = \{1, \dots, n\}$, assume instead that $\nu(e^{t\sum_i |Y_i|}) < \infty$ for some $t > 0$, where $\nu(\cdot) = \langle\cdot\rangle_0$. Then $h(z) := \nu(e^{\sum_i z_i Y_i})$ is holomorphic on $\{|\operatorname{Re} z_i| < t\}$ and $h(0) = 1$. Write $\partial_I = \prod_{i\in I} \partial_{z_i}$. For $|\operatorname{Re} z_i| \le t/2$ and $I \subset J$,

$$\prod_{i\in I} |Y_i|\, |e^{\sum_i z_i Y_i}| \le \Big(\frac{2}{t}\Big)^{|I|} e^{t\sum_i |Y_i|},$$

by $s \le e^s$ at $s = t|Y_i|/2$. This integrable majorant gives $\partial_I h(0) = m_I$. The mixed derivative formula for the logarithm, proved below, is

$$\partial_J \log h(0) = \sum_{\pi\in \mathrm{Part}(J)} (-1)^{|\pi|-1}(|\pi|-1)! \prod_{I\in\pi} \partial_I h(0). \tag{7.xvi}$$

It identifies this derivative with κ_J.

(3) A polynomial in the Gaussian family can be written in several ways as a sum of monomials in legs, since a curl leg is a signed sum of four plain legs. The *standard writing* of $\mathsf{a}_p^{(m)}$, $m = 2, 3, 4$, is obtained from eq. 7.v by multiplying out in components, writing each displayed factor $\mathrm{d}A(p)$ in curl legs $(\mathrm{d}A)^a(p)$ and the polynomials $\mathfrak{z}_1$, $\mathfrak{z}_2$ in plain legs $A^a(b_i(p))$. Thus $\mathsf{a}_p^{(2)} = \frac{1}{2}\sum_a \big((\mathrm{d}A)^a(p)\big)^2$; every monomial of $\mathsf{a}_p^{(3)}$ consists of one curl leg and two plain legs; every monomial of $\mathsf{a}_p^{(4)}$ consists of four plain legs (those coming from $\frac{1}{2}|\mathfrak{z}_1|^2$), or of one curl leg and three plain legs, or of four curl legs. The other local terms of V_2 are written in plain legs.

Formal sums. Let $X_i = \sum_{z\in \mathsf{Z}_i} X_{i,z}$, $i = 1, \dots, r$, be formal sums of polynomials $X_{i,z}$ over countable index sets Z_i. We set

$$\langle X_1; \dots; X_r\rangle_0 := \sum_{z=(z_1,\dots,z_r)} \langle X_{1,z_1}; \dots; X_{r,z_r}\rangle_0,$$

provided the following absolute convergence holds: when every entry is in its standard writing and each cumulant of monomials is expanded as the sum over pairings of Lem. 7.9, the sum of the moduli of all the resulting terms is finite. Under this proviso $\sum_z |\langle X_{1,z_1}; \dots; X_{r,z_r}\rangle_0| < \infty$, because at fixed z the cumulant is a number, the same in every writing, and it is bounded by the sum of the moduli of its terms; hence the value of the sum does not depend on the writing or on the order of summation. The *local terms* are the $\mathsf{a}_p^{(3)}$ for S_3; for V_2 they are the $\mathsf{a}_p^{(4)}$, the terms of the two sums over b, and the terms of the sum over pairs (b, b'). ∎

Proof of the cumulant identities in Def. 7.8. Fix a finite nonempty index set J_0, and write $z^I = \prod_{i\in I} z_i$ for $I \subset J_0$. For a holomorphic function h of finitely many variables and a one-variable holomorphic function ϕ near $h(0)$, repeated differentiation gives

$$\partial_J(\phi\circ h) = \sum_{\pi\in\mathrm{Part}(J)} \phi^{(|\pi|)}(h)\prod_{I\in\pi}\partial_I h.$$

Indeed, when a new variable is differentiated, it either differentiates $\phi^{(|\pi|)}(h)$ and forms a new singleton block, or differentiates one factor $\partial_I h$ and joins that block. Every partition of the enlarged set arises exactly once. This proves the formula by induction from the one-variable chain rule. At $h(0)=1$, the derivatives $(\log)^{(k)}(1) = (-1)^{k-1}(k-1)!$ give eq. 7.xvi. With $\phi=\exp$ and $h(0)=0$, the same formula gives

$$\partial_J e^h(0) = \sum_{\pi\in\mathrm{Part}(J)}\prod_{I\in\pi}\partial_I h(0).$$

For arbitrary moments m_I, put $H(z) = 1+\sum_{\emptyset\neq I\subset J_0} m_I z^I$ and $\varphi = \log H$ near zero. The logarithm formula gives $\partial_I\varphi(0)=\kappa_I$ as defined by eq. 7.xiv. Since $e^\varphi = H$, the exponential formula proves eq. 7.xv. Conversely, suppose eq. 7.xv holds and put $\varphi(z)=\sum_{\emptyset\neq I\subset J_0}\kappa_I z^I$. The exponential formula gives $\partial_I e^\varphi(0) = m_I$, and applying the logarithm formula to e^φ recovers eq. 7.xiv. This also proves uniqueness. □

Lemma 7.9 (Cumulants of monomials: connected pairings)**.** Let $(\phi_i)_{i\in I}$ be a centred Gaussian family with covariance $\mathsf{C}(i,j)$ and mean $\langle\cdot\rangle_0$, the cumulants of polynomials being those of Def. 7.8(2), and let $\mathsf{m}_v = \prod_{e\in E_v}\phi_{i(e)}$, $v=1,\dots,r$, where the sets E_v of legs are finite and pairwise disjoint and $i(e)\in I$. Then

$$\langle \mathsf{m}_1;\dots;\mathsf{m}_r\rangle_0 = \sum_{\mathsf{P}}\prod_{\{e,e'\}\in\mathsf{P}}\mathsf{C}(i(e),i(e')),$$

where the sum ranges over the pairings P of $E_1\cup\cdots\cup E_r$ whose graph on $\{1,\dots,r\}$ is connected; this graph has an edge vv' for each pair $\{e,e'\}\in\mathsf{P}$ with $e\in E_v$, $e'\in E_{v'}$, $v\neq v'$. ■

Proof. Step 1: moments. Let E be a finite set of legs, with $i(e)\in I$ for $e\in E$, and let $s=(s_e)_{e\in E}$ be real variables. The variables $\phi_{i(e)}$, $e\in E$, form a centred Gaussian family with covariance $\mathsf{C}(i(e),i(e'))$, so that

$$\Big\langle \exp\sum_{e\in E} s_e\phi_{i(e)}\Big\rangle_0 = \exp\Big(\tfrac12\sum_{e,e'\in E} s_e s_{e'}\,\mathsf{C}(i(e),i(e'))\Big).$$

Both sides are real analytic in s, and the left side may be differentiated under the mean. We compare the coefficients of the monomial $\prod_{e\in E}s_e$. On the left it is $\langle\prod_{e\in E}\phi_{i(e)}\rangle_0$. On the right, this monomial occurs only in the term $\frac{1}{m!}\big(\frac12\sum_{e,e'}s_e s_{e'}\mathsf{C}(i(e),i(e'))\big)^m$ with $2m=\#E$; when the power is expanded, each pairing of E arises from $m!\,2^m$ choices, namely the order of its m pairs and the order within each pair. Hence

$$\Big\langle\prod_{e\in E}\phi_{i(e)}\Big\rangle_0 = \sum_{\mathsf{P}}\prod_{\{e,e'\}\in\mathsf{P}}\mathsf{C}(i(e),i(e')),$$

the sum ranging over all pairings P of E; it vanishes if $\#E$ is odd.

Step 2: connected pairings. For $\emptyset\neq J\subset\{1,\dots,r\}$ let $\tilde\kappa_J$ be the sum over the pairings of $\bigcup_{v\in J}E_v$ whose graph on J is connected. A pairing of $\bigcup_{v\in J}E_v$ is the same as a partition π of J, namely into the vertex sets of the connected components of its graph, together with a pairing of $\bigcup_{v\in I}E_v$ with connected graph on I for each block $I\in\pi$; the product of the covariances factorises accordingly. Step 1, applied to $E=\bigcup_{v\in J}E_v$, therefore gives

$$m_J := \Big\langle\prod_{v\in J}\mathsf{m}_v\Big\rangle_0 = \sum_{\pi\in\mathrm{Part}(J)}\prod_{I\in\pi}\tilde\kappa_I,$$

which is eq. 7.xv. By Def. 7.8(2), $\tilde\kappa_J$ is the joint cumulant κ_J of the m_v, $v \in J$; for $J = \{1,\dots,r\}$ this is the claim. □

Definition 7.10 (The lattice coefficients B_0, B_1, $\mathsf{P}_1^{\mathrm{lat}}$)**.** Let Ω and τ_{gf} be as in Def. 7.7, let p, q be plaquettes of Ω, and let $\langle\cdot\rangle_0$ refer to the Gaussian family of Ω and τ_{gf} (Def. 7.8). We set

$$\mathsf{B}_0^{\Omega}(p,q) := \langle \mathsf{a}_p^{(2)}; \mathsf{a}_q^{(2)} \rangle_0 = \tfrac{\mathsf{n}}{2}\, \Pi^{\Omega}(p,q)^2. \tag{7.xvii}$$

The second equality follows from Lem. 7.9 and eq. 7.xiii, applied to the legs, which form a centred Gaussian family as linear combinations of the variables $A^a(b)$. Indeed, $\mathsf{a}_p^{(2)} = \frac12 \sum_a ((\mathrm{d}A)^a(p))^2$ and multilinearity give $\mathsf{B}_0^{\Omega}(p,q) = \frac14 \sum_{a,a'} \langle ((\mathrm{d}A)^a(p))^2; ((\mathrm{d}A)^{a'}(q))^2 \rangle_0$; in each of these cumulants exactly two pairings are connected, namely those which pair each leg at p with a leg at q, and each of them contributes $\delta_{aa'}\Pi^{\Omega}(p,q)^2$. Hence $\mathsf{B}_0^{\Omega}(p,q) = \frac14 \cdot 2 \cdot 3\, \Pi^{\Omega}(p,q)^2 = \frac{\mathsf{n}}{2}\Pi^{\Omega}(p,q)^2$. Next we set

$$\begin{aligned} \mathsf{B}_1^{\Omega}(p,q;\tau_{\mathrm{gf}}) := {} & \langle \mathsf{a}_p^{(3)}; \mathsf{a}_q^{(3)} \rangle_0 + \langle \mathsf{a}_p^{(2)}; \mathsf{a}_q^{(4)} \rangle_0 + \langle \mathsf{a}_p^{(4)}; \mathsf{a}_q^{(2)} \rangle_0 \\ & - \langle \mathsf{a}_p^{(3)}; \mathsf{a}_q^{(2)}; \mathsf{S}_3 \rangle_0 - \langle \mathsf{a}_p^{(2)}; \mathsf{a}_q^{(3)}; \mathsf{S}_3 \rangle_0 \\ & - \langle \mathsf{a}_p^{(2)}; \mathsf{a}_q^{(2)}; \mathsf{V}_2 \rangle_0 + \tfrac12 \langle \mathsf{a}_p^{(2)}; \mathsf{a}_q^{(2)}; \mathsf{S}_3; \mathsf{S}_3 \rangle_0 . \end{aligned} \tag{7.xviii}$$

We write $\mathsf{B}_j^{[n]}$ for $\Omega = \Lambda_n$ and $\mathsf{B}_j^{\mathrm{lat}}$ for $\Omega = \mathbb{Z}^3$, $j = 0, 1$. On $\mathbb{Z}^3$ the four cumulants containing S_3 or V_2 are formal sums in the sense of Def. 7.8(3); the proviso of absolute convergence is satisfied for every $\tau_{\mathrm{gf}} > 0$ by Lem. 7.27. Finally, for bounded Borel functions f, h on $\mathbb{R}^3$ with compact supports and $a > 0$,

$$\mathsf{B}_1^{\mathrm{lat}}(p,q) := \mathsf{B}_1^{\mathrm{lat}}(p,q;1), \qquad \mathsf{P}_1^{\mathrm{lat}}(f,h;a) := a \sum_{p,q} f(a\,x(p))\, h(a\,x(q))\, \mathsf{B}_1^{\mathrm{lat}}(p,q); \tag{7.xix}$$

the sum over pairs of plaquettes of $\mathbb{Z}^3$ is finite, because $f(a\,x(p)) \neq 0$ only for the finitely many plaquettes with $a\,x(p)$ in the compact support of f, and likewise for h.

The normalisation in eq. 7.xix follows from dilation. On the spacing-a lattice define

$$\mathcal{O}_a(f)(V) = a^{-1} \sum_P f(x(P)) a_P(V),$$

with the sum over its plaquettes. The dilation sends a unit bond b to ab and defines $(\mathsf{dil}_a U)(ab) = U(b)$. It therefore preserves the ordered plaquette product and sends its barycentre $x(p)$ to $ax(p)$. Consequently

$$\mathcal{O}_a(f)(\mathsf{dil}_a U) = a^{-1} \sum_p f(a\,x(p))\, a_p(U)$$

for every configuration $U \in \mathcal{V}(\mathbb{L}_1)$, the sum running over the plaquettes of $\mathbb{L}_1 = \mathbb{Z}^3$. If $\beta = \mathsf{b}/a$ with $\mathsf{b} > 0$, then $a^{-2}\beta^{-3} = a\,\mathsf{b}^{-3}$, and eq. 7.xix gives

$$a^{-2} \sum_{p,q} f(a\,x(p))\, h(a\,x(q))\, \beta^{-3}\, \mathsf{B}_1^{\mathrm{lat}}(p,q) = \mathsf{b}^{-3}\, \mathsf{P}_1^{\mathrm{lat}}(f,h;a).$$

■

7.1.7. *The asymptotic expansion in a box.*

Lemma 7.11 (Laplace's method for a truncated two-point function)**.** Let $m \geq 1$ and $\rho > 0$, let $\mathcal{W} \subset \mathbb{R}^m$ be a bounded Borel set containing $\{|y| < \rho\}$, and let Φ, Θ, X, Y be real C^∞ functions on an open set containing $\overline{\mathcal{W}}$ with the following properties: $\Phi(0) = 0$, $\nabla\Phi(0) = 0$, the Hessian H of Φ at 0 is positive definite, and $\inf\{\Phi(y) : \ y \in \mathcal{W},\ |y| \geq \rho'\} > 0$ for every $\rho' \in (0, \rho]$; X and Y vanish at 0 together with their gradients. For $\beta \geq 1$ let $\langle\cdot\rangle_{\mathcal{W},\beta}$ be the mean with respect to the probability density on $\mathcal{W}$, relative to the Lebesgue measure, proportional to $e^{-\beta\Phi-\Theta}$,

and $\langle X;Y\rangle_{\mathcal{W},\beta} := \langle XY\rangle_{\mathcal{W},\beta} - \langle X\rangle_{\mathcal{W},\beta}\langle Y\rangle_{\mathcal{W},\beta}$. Let Φ_j, Θ_j, X'_j, Y'_j be the homogeneous Taylor polynomials of degree j at 0 of Φ, Θ, X, Y, and

$$W_j := \Phi_{j+2} + \Theta_j \quad (j \geq 1), \qquad X_j := X'_{j+2}, \quad Y_j := Y'_{j+2} \quad (j \geq 0).$$

Let $\langle\cdot\rangle_0$ be the centred Gaussian measure on $\mathbb{R}^m$ with covariance H^{-1}, the cumulants of polynomials being taken as in Def. 7.8(2). Then there is $C < \infty$, depending on the data $m, \rho, \mathcal{W}, \Phi, \Theta, X, Y$, such that for all $\beta \geq 1$

$$\left|\langle X;Y\rangle_{\mathcal{W},\beta} - \beta^{-2}\gamma_0 - \beta^{-3}\gamma_2\right| \leq C\beta^{-4},$$

where $\gamma_0 := \langle X_0;Y_0\rangle_0$ and

$$\begin{aligned}\gamma_2 := {} & \langle X_1;Y_1\rangle_0 + \langle X_0;Y_2\rangle_0 + \langle X_2;Y_0\rangle_0 - \langle X_1;Y_0;W_1\rangle_0 - \langle X_0;Y_1;W_1\rangle_0 \\ & - \langle X_0;Y_0;W_2\rangle_0 + \tfrac{1}{2}\langle X_0;Y_0;W_1;W_1\rangle_0.\end{aligned}$$

■

Proof. Step 1: localisation and scaling. By Taylor's formula at 0 and the positivity of H, there is $\rho_0 \in (0,\rho]$ with $\Phi(y) \geq \frac{1}{4}(y,Hy)$ for $|y| \leq \rho_0$. Let $t := \beta^{-1/2} \in (0,1]$, and for bounded Borel functions f on $\mathcal{W}$ let

$$\mathsf{N}(f) := \int_{\mathcal{W}} f\, e^{-\beta\Phi-\Theta}\,\mathrm{d}y.$$

The contribution of $\{y \in \mathcal{W} : |y| \geq \rho_0\}$ to $\mathsf{N}(f)$ is bounded by $C\sup|f|\,e^{-c_*\beta}$, where $c_* := \inf\{\Phi(y) : y \in \mathcal{W},\ |y| \geq \rho_0\} > 0$. On $\{|y| < \rho_0\} \subset \mathcal{W}$ we substitute $y = tz$. Taylor's formula with integral remainder provides smooth functions Φ_α ($|\alpha| = 3$) and X_α, Y_α ($|\alpha| = 2$) on $\{|y| \leq \rho_0\}$ such that

$$\Phi(y) = \tfrac{1}{2}(y,Hy) + \sum_{|\alpha|=3} y^\alpha \Phi_\alpha(y),$$

$$X(y) = \sum_{|\alpha|=2} y^\alpha X_\alpha(y), \qquad Y(y) = \sum_{|\alpha|=2} y^\alpha Y_\alpha(y).$$

We set

$$\varphi(t,z) := t^{-2}\Phi(tz) - \tfrac{1}{2}(z,Hz) + \Theta(tz) - \Theta(0) = t\sum_\alpha z^\alpha \Phi_\alpha(tz) + \Theta(tz) - \Theta(0),$$

$$\mathsf{x}(t,z) := t^{-2}X(tz) = \sum_\alpha z^\alpha X_\alpha(tz), \qquad \mathsf{y}(t,z) := t^{-2}Y(tz) = \sum_\alpha z^\alpha Y_\alpha(tz).$$

These functions are smooth on $\{(t,z) : |tz| \leq \rho_0\}$, the value $t = 0$ included, and they satisfy $|\partial_t^j\varphi|, |\partial_t^j\mathsf{x}|, |\partial_t^j\mathsf{y}| \leq C_j\langle z\rangle^{j+3}$ there. Since the Taylor coefficients at $t = 0$ of $t \mapsto F(tz)$ are the homogeneous Taylor polynomials $F_j(z)$, their formal Taylor series in t are $\sum_{j\geq 1} t^j W_j(z)$, $\sum_{j\geq 0} t^j X_j(z)$ and $\sum_{j\geq 0} t^j Y_j(z)$. Moreover

$$e^{-\frac{1}{2}(z,Hz)-\varphi(t,z)} = e^{-t^{-2}\Phi(tz)-\Theta(tz)+\Theta(0)} \leq Ce^{-\frac{1}{4}(z,Hz)} \qquad (|tz| \leq \rho_0).$$

Step 2: expansion of the integrals. For $\mathsf{f} \in \{1,\mathsf{x},\mathsf{y},\mathsf{xy}\}$ let $\mathsf{F} := \mathsf{f}e^{-\varphi}$. By Step 1, $|\partial_t^j\mathsf{F}| \leq C_j\langle z\rangle^{4j+6}e^{-\varphi}$ on $\{|tz| \leq \rho_0\}$; if $|tz| \leq \rho_0$, the points (s,z) with $0 \leq s \leq t$ belong to this set as well; and the coefficients $\mathsf{F}_j(z) := \frac{1}{j!}\partial_t^j\mathsf{F}(0,z)$ are polynomials in z, since $\varphi(0,z) = 0$ and the t-derivatives of φ, x, y at $t = 0$ are polynomials. Taylor's formula in t with a remainder of order 4, together with the bound of Step 1 at the points (s,z), gives

$$\left|\mathsf{F}(t,z) - \sum_{j<4} t^j\mathsf{F}_j(z)\right| e^{-\frac{1}{2}(z,Hz)} \leq Ct^4\langle z\rangle^{22}e^{-\frac{1}{4}(z,Hz)} \qquad (|tz| \leq \rho_0).$$

Let $\mathsf{Z} := t^m e^{-\Theta(0)}\int e^{-\frac{1}{2}(z,Hz)}\,\mathrm{d}z$, and let $\nu \in \{0,1,1,2\}$ be the number of factors X, Y in $f \in \{1,X,Y,XY\}$, with the corresponding $\mathsf{f} \in \{1,\mathsf{x},\mathsf{y},\mathsf{xy}\}$. We substitute $y = tz$ in $\mathsf{N}(f)$, use $X(tz) = t^2\mathsf{x}(t,z)$, $Y(tz) = t^2\mathsf{y}(t,z)$ and $e^{-\beta\Phi(tz)-\Theta(tz)} = e^{-\Theta(0)}e^{-\frac{1}{2}(z,Hz)-\varphi(t,z)}$, integrate the last bound over

$|z| < \rho_0/t$, and add the Gaussian tails $\int_{|z|\geq\rho_0/t} |\mathsf{F}_j| e^{-\frac12(z,Hz)}\,\mathrm{d}z = O(e^{-c/t^2})$. Since the Gaussian measure with covariance H^{-1} has a density proportional to $e^{-\frac12(z,Hz)}$, we obtain

$$\mathsf{N}(f) = \mathsf{Z}\,t^{2\nu}\Big[\sum_{j<4} t^j \langle \mathsf{F}_j\rangle_0 + O(t^4)\Big] \qquad (f = 1, X, Y, XY),$$

where the contribution of order e^{-c_*/t^2} of $\{|y| \geq \rho_0\}$ is also absorbed in $O(t^4)$, because $\mathsf{Z}\,t^{2\nu+4} \geq c\,t^{m+8}$. For $f = 1$ we have $\mathsf{F}_0 = 1$; hence there is $t_0 \in (0,1]$ with $\mathsf{N}(1) \geq \frac12 \mathsf{Z}$ for $t \leq t_0$.

In the range $t_0 < t \leq 1$, that is $1 \leq \beta < t_0^{-2}$, the claim holds with $C = t_0^{-8}(2\sup|X|\sup|Y| + |\gamma_0| + |\gamma_2|)$: the functions X and Y are continuous, hence bounded on the compact set $\overline{\mathcal{W}}$ (Heine–Borel), so that $|\langle X;Y\rangle_{\mathcal{W},\beta}| \leq 2\sup|X|\sup|Y|$, and $\beta^{-4} > t_0^8$.

In the range $t \leq t_0$ we have

$$\langle X;Y\rangle_{\mathcal{W},\beta} = \frac{\mathsf{N}(XY)}{\mathsf{N}(1)} - \frac{\mathsf{N}(X)\,\mathsf{N}(Y)}{\mathsf{N}(1)^2} = t^4\Big[\sum_{j<4} t^j\gamma_j + O(t^4)\Big],$$

where $\sum_j t^j\gamma_j$ is the formal power series

$$\mathsf{e} := \frac{\langle \mathsf{x}\mathsf{y}e^{-\varphi}\rangle_0}{\langle e^{-\varphi}\rangle_0} - \frac{\langle \mathsf{x}e^{-\varphi}\rangle_0\,\langle \mathsf{y}e^{-\varphi}\rangle_0}{\langle e^{-\varphi}\rangle_0^2} \in \mathbb{R}[[t]],$$

in which $\mathsf{x} = \sum_j t^j X_j$, $\mathsf{y} = \sum_j t^j Y_j$ and $\varphi = \sum_{j\geq1} t^j W_j$ are formal series and $\langle\cdot\rangle_0$ acts coefficientwise. Indeed, the formal series $\sum_j t^j\langle \mathsf{F}_j\rangle_0$ are the series $\langle \mathsf{f}e^{-\varphi}\rangle_0$; the series $\langle e^{-\varphi}\rangle_0$ has the constant term 1, so that the quotients are defined in $\mathbb{R}[[t]]$; and, the denominators being at least $\frac12$, the quotients of the truncated expansions differ from the truncations of the formal quotients by $O(t^4)$. Since $t^4 = \beta^{-2}$, $t^6 = \beta^{-3}$ and $t^8 = \beta^{-4}$, it remains to identify γ_0 and γ_2 and to show $\gamma_1 = \gamma_3 = 0$.

Step 3: the coefficients. Let α_1, α_2 be formal variables, and let $Z := \alpha_1\mathsf{x} + \alpha_2\mathsf{y} - \varphi$, a formal series in (α_1, α_2, t) without constant term whose coefficients are polynomials in z. By eq. 7.xv and multilinearity,

$$\langle Z^r\rangle_0 = \sum_{\pi\in\mathrm{Part}(\{1,\dots,r\})} \prod_{I\in\pi} \varkappa_{\#I}, \qquad \varkappa_j := \langle Z;\dots;Z\rangle_0 \quad (j \text{ entries}).$$

The number of partitions of $\{1,\dots,r\}$ with m_j blocks of size j ($\sum_j jm_j = r$) is $r!/\prod_j (j!)^{m_j} m_j!$; hence

$$\langle e^Z\rangle_0 = \sum_{r\geq0}\frac{1}{r!}\langle Z^r\rangle_0 = \sum_{(m_j)}\prod_j \frac{1}{m_j!}\Big(\frac{\varkappa_j}{j!}\Big)^{m_j} = \exp\sum_{j\geq1}\frac{1}{j!}\varkappa_j.$$

We compare the coefficients of $\alpha_1\alpha_2$ in $\log\langle e^Z\rangle_0$. Since $\langle e^Z\rangle_0 = \langle e^{-\varphi}\rangle_0 + \alpha_1\langle \mathsf{x}e^{-\varphi}\rangle_0 + \alpha_2\langle \mathsf{y}e^{-\varphi}\rangle_0 + \alpha_1\alpha_2\langle \mathsf{x}\mathsf{y}e^{-\varphi}\rangle_0$ up to terms containing α_1^2 or α_2^2, this coefficient is e. On the other hand, by multilinearity and symmetry of the cumulants, the coefficient of $\alpha_1\alpha_2$ in $\varkappa_j$ is $j(j-1)\langle \mathsf{x};\mathsf{y};-\varphi;\dots;-\varphi\rangle_0$, with $j-2$ entries $-\varphi$. Therefore

$$\mathsf{e} = \sum_{j\geq2}\frac{j(j-1)}{j!}\langle \mathsf{x};\mathsf{y};-\varphi;\dots;-\varphi\rangle_0 = \sum_{r\geq0}\frac{(-1)^r}{r!}\langle \mathsf{x};\mathsf{y};\underbrace{\varphi;\dots;\varphi}_{r}\rangle_0.$$

Inserting the series for x, y, φ, the coefficient of t^0 is $\langle X_0;Y_0\rangle_0 = \gamma_0$. In the coefficient of t^2 the term $r=0$ contributes $\langle X_1;Y_1\rangle_0 + \langle X_0;Y_2\rangle_0 + \langle X_2;Y_0\rangle_0$, the term $r=1$ contributes $-\langle X_1;Y_0;W_1\rangle_0 - \langle X_0;Y_1;W_1\rangle_0 - \langle X_0;Y_0;W_2\rangle_0$, and the term $r=2$ contributes $\frac12\langle X_0;Y_0;W_1;W_1\rangle_0$; the sum is γ_2. The coefficient of an odd power t^k is a sum of cumulants of polynomials X_{j_1}, Y_{j_2}, $W_{i_1},\dots,W_{i_r}$ with $j_1+j_2+i_1+\cdots+i_r = k$. Each of X_j, Y_j, Φ_{j+2}, Θ_j is homogeneous of a degree with the parity of j; hence, in each term of eq. 7.xiv for such a cumulant, at least one moment is the mean of a polynomial which is odd under $z\mapsto -z$, and it vanishes, because the Gaussian measure is invariant under $z\mapsto -z$. Thus $\gamma_1 = \gamma_3 = 0$. □

Proposition 7.12 (The asymptotic expansion in a box)**.** Let $n \geq 2$ be an integer, $\tau_{\mathrm{gf}} > 0$, and $p, q \in \mathsf{Pl}_n$. There is $C < \infty$, depending on n, τ_{gf}, p, q, such that for all $\beta \geq 1$

$$\big|\langle a_p; a_q\rangle_{n,\beta} - \beta^{-2}\,\mathsf{B}_0^{[n]}(p,q) - \beta^{-3}\,\mathsf{B}_1^{[n]}(p,q;\tau_{\mathrm{gf}})\big| \leq C\beta^{-4}. \tag{7.xx}$$

In particular, $\mathsf{B}_1^{[n]}(p,q;\tau_{\mathrm{gf}})$ does not depend on τ_{gf}. ∎

Proof. We write $\tau := \tau_{\mathrm{gf}}$, and we use S, T_n, ϱ_1, ϱ_2, ϱ_3, $\mathfrak{D}$, Ξ, $\mathcal{W}$ of Lem. 7.6. Throughout, F denotes one of the functions 1, a_p, a_q, a_pa_q on $\mathcal{U}_n$; each of them satisfies $|F| \leq 4$ and, like S, is invariant under $\mathcal{G}_n$ (Def. 7.2(1)). Let $d_0 := \dim \mathsf{X}_n$ and $d_1 := \dim \mathsf{Form}_n^{\mathsf{T}}$. The constants $C, c > 0$ below may depend on n, τ, p, q.

Step 1: axial gauge. By eq. 7.vii and the $\mathcal{G}_n$-invariance of $Fe^{-\beta S}$, and then by the Haar formula in Def. 7.1(2) at $u_0 = \mathbf{1}$, bond by bond,

$$\begin{aligned}\int Fe^{-\beta S}\prod_{b\in\mathsf{Bd}_n}\mathrm{d}U(b) &= \int F(V)\,e^{-\beta S(V)}\prod_{b\in\mathsf{Bd}_n\setminus\mathsf{T}_n}\mathrm{d}V(b)\\ &= \sigma_{\mathrm{H}}^{d_1/3}\int_{\{A\in\mathsf{Form}_n^{\mathsf{T}}:\,|A(b)|<\pi\}}(Fe^{-\beta S})(e^{iA})\prod_{b\in\mathsf{Bd}_n\setminus\mathsf{T}_n}\sigma_1(|A(b)|^2)\,\mathrm{d}A,\end{aligned}$$

as $\mathsf{Bd}_n \setminus \mathsf{T}_n$ has $d_1/3$ elements. Let $\mathsf{I}(F)$ denote the last integral restricted to $|A|_\infty < \varrho_3$. The space $\mathcal{U}_n^{\mathsf{T}}$ is compact, being a finite power of the compact group G, and the image of $\{|A|_\infty < \varrho_3\}$ under $A \mapsto e^{iA}$ is relatively open in it (Def. 7.1); hence its complement K in $\mathcal{U}_n^{\mathsf{T}}$ is compact, and by the injectivity of the exponential on $\mathbb{B}_\pi$, $e^{iA} \in K$ for the remaining A in the integral. The continuous function S has no zero on K, because the only zero of S in $\mathcal{U}_n^{\mathsf{T}}$ is $\mathbf{1} = e^{i0}$ (Lem. 7.6(b)); so its minimum c_* on K is positive, and the integral over the remaining A is bounded by $Ce^{-c_*\beta}$.

By Lem. 7.5(a),(b), $A \mapsto S(e^{iA})$ is smooth and vanishes at 0 together with its gradient; hence $S(e^{iA}) \leq C|A|_\infty^2$ for $|A|_\infty \leq \varrho_3$, and since $\sigma_1(|A(b)|^2) \geq c$ there,

$$\mathsf{I}(1) \geq c\int_{|A|_\infty<\min\{\varrho_3,\beta^{-1/2}\}} e^{-C\beta|A|_\infty^2}\,\mathrm{d}A \geq c\,\beta^{-d_1/2}.$$

As $|\mathsf{I}(F)| \leq 4\,\mathsf{I}(1)$, it follows that $\langle F\rangle_{n,\beta} = \mathsf{I}(F)/\mathsf{I}(1) + O(e^{-c\beta})$.

Step 2: the gauge fixing term. We apply eq. 7.viii with the bounded Borel functions $\Phi(A') := (Fe^{-\beta S})(e^{iA'})$ and $w(c) := e^{-\beta\|c\|^2/2\tau}$. Since $e^{iA^\lambda} = (e^{iA})^{e^{i\lambda}}$ and $Fe^{-\beta S}$ is $\mathcal{G}_n$-invariant, $\Phi(A^\lambda) = \Phi(A)$; thus the inner integral on the right side of eq. 7.viii is $\Phi(A)$ times the integral of w over $\{\mathrm{d}^*(A^\lambda) : |\lambda|_\infty < \varrho_1\}$, a set which contains $\{|c|_\infty < \varrho_2\}$ by Lem. 7.6(d)(iii). Since $\|c\|^2 = \sum_{x\in\Lambda_n^\circ}|c(x)|^2$ and dc is the product of the measures dY at the $d_0/3$ sites of Λ_n° (Lem. 7.6(d)(iv)), the substitution $c \to (\tau/\beta)^{1/2}c$ and the Gaussian integral in Def. 7.1(2), site by site, give

$$\int_{\mathsf{X}_n} w\,\mathrm{d}c = \Big(\frac{\tau}{\beta}\Big)^{d_0/2}\prod_{x\in\Lambda_n^\circ}\int_{\mathfrak{g}} e^{-\frac12|Y|^2}\,\mathrm{d}Y = \Big(\frac{\tau}{\beta}\Big)^{d_0/2}(2\pi)^{d_0/2} = \Big(\frac{2\pi\tau}{\beta}\Big)^{d_0/2},$$

and the integral of w over $\{|c|_\infty \geq \varrho_2\}$ is at most $Ce^{-c\beta}(2\pi\tau/\beta)^{d_0/2}$. Hence the inner integral equals $\Phi(A)(2\pi\tau/\beta)^{d_0/2}(1-\epsilon(A))$ with $0 \leq \epsilon(A) \leq Ce^{-c\beta}$. The left side of eq. 7.viii is

$$\mathsf{J}(F) := \int_{\mathcal{W}} F(e^{iA'})\,e^{-\beta\Phi_\tau(A')-\Theta(A')}\,\mathrm{d}A',$$

$$\Phi_\tau(A') := S(e^{iA'}) + \tfrac{1}{2\tau}\|\mathrm{d}^*A'\|^2, \qquad \Theta(A') := -\sum_{b\in\mathsf{Bd}_n}\log\sigma_1(|A'(b)|^2) - \log|\det\mathsf{M}(A')|;$$

here Θ is defined on $\overline{\mathcal{W}}$, since $\sigma_1 > 0$ on $\mathbb{B}_\pi$ and $\det\mathsf{M} \neq 0$ on $\overline{\mathcal{W}}$ (Lem. 7.6(d)(ii)). Comparing with the definition of $\mathsf{I}(F)$ we obtain

$$\big|\mathsf{J}(F) - (2\pi\tau/\beta)^{d_0/2}\,\mathsf{I}(F)\big| \leq Ce^{-c\beta}(2\pi\tau/\beta)^{d_0/2}\,\mathsf{I}(1).$$

Together with Step 1, and since $\mathsf{J}(F)/\mathsf{J}(1)$ and $\mathsf{I}(F)/\mathsf{I}(1)$ are bounded by 4, this gives $\langle F\rangle_{n,\beta} = \mathsf{J}(F)/\mathsf{J}(1) + O(e^{-c\beta})$. Consequently, up to $O(e^{-c\beta})$, $\langle a_p; a_q\rangle_{n,\beta}$ is the truncated two-point function $\langle X; Y\rangle_{\mathcal{W},\beta}$ of the functions $X(A') := a_p(e^{iA'})$ and $Y(A') := a_q(e^{iA'})$ for the probability density on $\mathcal{W}$ proportional to $e^{-\beta\Phi_\tau - \Theta}$.

Step 3: the hypotheses of Lem. 7.11. We identify $\mathbb{R}^m$ with the Euclidean space Form_n, so that $m = \dim \mathsf{Form}_n$, by means of an orthonormal basis for the scalar product $(\cdot,\cdot)$ of Def. 7.2(2). Then the Lebesgue measure of $\mathbb{R}^m$ becomes the measure dA' of Lem. 7.6(d)(iv), the norm $|y|$ becomes $|A'| := \|A'\|$, and (y, Hy) becomes (A', HA'). We take the set $\mathcal{W}$ and the functions Φ_τ, Θ, X, Y of Step 2. The set $\mathcal{W}$ is open and bounded and contains $0 = \Xi(0,0)$, hence a ball $\{|y| < \rho\}$; its closure is compact (Heine–Borel), contained in $\{|A'|_\infty < 1\}$, and $\det \mathsf{M} \neq 0$ on it (Lem. 7.6(d)).

The four functions are smooth on an open set containing $\overline{\mathcal{W}}$: the maps $A' \mapsto a_P(e^{iA'})$ are smooth for every plaquette P (Def. 7.1); $Y \mapsto \log\sigma_1(|Y|^2)$ is smooth on $\mathbb{B}_\pi$ (Lem. 7.5(c)); and by eq. 7.vi the entries of $\mathsf{M}(A')$ are smooth on $\{|A'|_\infty < 1\}$, being polynomials in the components of the $A'(b)$ plus entries of the $\mathsf{r}(A'(b))$; hence $\det\mathsf{M}$ is continuous, it does not vanish on an open set containing $\overline{\mathcal{W}}$, and $\log|\det\mathsf{M}|$ is smooth there. By Lem. 7.5(a),(b), X, Y and Φ_τ vanish at 0 together with their gradients, and the homogeneous Taylor polynomials of $A' \mapsto a_P(e^{iA'})$ at 0 are the $\mathsf{a}_P^{(j)}(A')$, $j \geq 0$, by eq. 7.iii and the homogeneity of the $\mathsf{a}_P^{(j)}$.

In particular the Taylor polynomial of degree 2 of Φ_τ is S_2 (by eq. 7.v and eq. 7.x), which is positive definite by Lem. 7.6(c). Since $\frac12(A', HA') = \mathsf{S}_2(A') = \frac12(A', (\mathrm{d}^*\mathrm{d} + \tau^{-1}\mathrm{d}\mathrm{d}^*)A')$, the Hessian H acts in each colour as $\mathrm{d}^*\mathrm{d} + \tau^{-1}\mathrm{d}\mathrm{d}^*$, so that $H^{-1} = \mathsf{C}_\tau^{[n]}$ in each colour, and the Gaussian measure of Lem. 7.11 is the Gaussian family of Λ_n and τ (Def. 7.8(1)). Finally let $\rho' \in (0, \rho]$. The function $\Phi_\tau \circ \Xi$ is continuous on the compact set $\overline{\mathfrak{D}}$, and by the gauge invariance of S,

$$\Phi_\tau(\Xi(\lambda, A)) = S(e^{iA}) + \tfrac{1}{2\tau}\|\mathrm{d}^*(A^\lambda)\|^2.$$

If this vanishes, then $S(e^{iA}) = 0$, hence $e^{iA} = \mathbf{1}$ and $A = 0$ (Lem. 7.6(b)), and $\mathrm{d}^*(0^\lambda) = 0 = \mathrm{d}^*(0^0)$, hence $\lambda = 0$ by the injectivity in Lem. 7.6(d)(iii). Therefore $\Phi_\tau \circ \Xi$ has a positive minimum on the compact set $\overline{\mathfrak{D}} \setminus \Xi^{-1}(\{|A'| < \rho'\})$, which does not contain $(0,0)$, and $\inf\{\Phi_\tau(A') : \ A' \in \mathcal{W},\ |A'| \geq \rho'\} > 0$.

Step 4: the Taylor polynomials. By Step 3, $X_j = \mathsf{a}_p^{(j+2)}$, $Y_j = \mathsf{a}_q^{(j+2)}$, $\Phi_{\tau,3} = \mathsf{S}_3$ and $\Phi_{\tau,4} = \sum_P \mathsf{a}_P^{(4)}$, the sums being over Pl_n. By Lem. 7.5(c), as $t \to 0$,

$$-\sum_b \log\sigma_1(t^2|A(b)|^2) = \tfrac13 t^2 \sum_b |A(b)|^2 + O(t^4).$$

By Lem. 7.5(d), $\mathsf{M}(tA) = \mathsf{M}^{(0)} + t\mathsf{M}^{(1)}(A) + t^2\mathsf{M}^{(2)}(A) + O(t^4)$, where $\mathsf{M}^{(0)} = -\mathrm{d}^*\mathrm{d}$ is invertible. With $\mathsf{A} := (\mathsf{M}^{(0)})^{-1}(\mathsf{M}(tA) - \mathsf{M}^{(0)}) = O(t)$ we have $\det(\mathbf{1} + \mathsf{A}) > 0$ for small t, and the series $\log\det(\mathbf{1} + \mathsf{A}) = \sum_{k\geq1} \frac{(-1)^{k-1}}{k} \operatorname{tr}\mathsf{A}^k$ gives

$$\begin{aligned}\log|\det\mathsf{M}(tA)| - \log|\det\mathsf{M}^{(0)}| &= t\operatorname{tr}[(\mathsf{M}^{(0)})^{-1}\mathsf{M}^{(1)}(A)] \\ &\quad + t^2\Big\{\operatorname{tr}[(\mathsf{M}^{(0)})^{-1}\mathsf{M}^{(2)}(A)] - \tfrac12\operatorname{tr}\big[\big((\mathsf{M}^{(0)})^{-1}\mathsf{M}^{(1)}(A)\big)^2\big]\Big\} + O(t^3).\end{aligned}$$

By eq. 7.xii we conclude that $\Theta_1 = 0$ and

$$\Theta_2 = \tfrac13\sum_b |A(b)|^2 + \tfrac23\sum_b \mathsf{g}(b)|A(b)|^2 - 4\sum_{b,b'} \mathsf{D}(b;b')\mathsf{D}(b';b)\langle A(b), A(b')\rangle,$$

so that $W_1 = \mathsf{S}_3$ and $W_2 = \mathsf{V}_2$, by eq. 7.xi. Lem. 7.11 now gives eq. 7.xx, with $\gamma_0 = \langle \mathsf{a}_p^{(2)}; \mathsf{a}_q^{(2)}\rangle_0 = \mathsf{B}_0^{[n]}(p,q)$ by eq. 7.xvii and $\gamma_2 = \mathsf{B}_1^{[n]}(p,q;\tau)$ by eq. 7.xviii; the error $O(e^{-c\beta})$ of Step 2 is bounded by $C\beta^{-4}$.

Step 5: independence of the gauge parameter. Neither $\langle a_p; a_q\rangle_{n,\beta}$ nor $\mathsf{B}_0^{[n]}(p,q) = \frac{\mathfrak{n}}{2}\Pi^{[n]}(p,q)^2$ depends on τ (Def. 7.7(2)). Multiplying eq. 7.xx by β^3 we obtain

$$\beta^3\big[\langle a_p; a_q\rangle_{n,\beta} - \beta^{-2}\mathsf{B}_0^{[n]}(p,q)\big] = \mathsf{B}_1^{[n]}(p,q;\tau) + O(\beta^{-1}),$$

and letting $\beta\to\infty$ we see that $\mathsf{B}_1^{[n]}(p,q;\tau)$ does not depend on τ. □

7.1.8. *Zero momentum.* In this subsection we compute the values at zero momentum of the self-energy and of the vertex function of a plaquette action. Both are expressed through Hessians of the action at constant abelian fields, whose symbols can be computed explicitly.

Definition 7.13 (Hessians; self-energy and vertex function; their values at zero momentum)**.** We work on $\mathbb{Z}^3$ and fix a gauge parameter $\tau_{\rm gf} > 0$. For a positively oriented bond b and a colour index a we write $\partial_{b,a}$ for the derivative with respect to the variable $A^a(b)$. On a formal sum $\mathsf{F} = \sum_z \mathsf{F}_z$ of local terms (Def. 7.8(3)) the derivative acts term by term, and the terms on which it vanishes are omitted. The derivatives $\partial_{b,a}\mathsf{S}_3$, $\partial_{b,a}\partial_{b',a'}\mathsf{S}_4$ and $\partial_{b,a}\partial_{b',a'}\mathsf{V}_2$ which occur below are then finite sums: a bond (x,μ) belongs to the two plaquettes on its two sides in each of the other two coordinate directions, hence to four plaquettes; of a sum over bonds b'' only the term with $b''=b$ survives; and of the sum over pairs of bonds in V_2 only the terms of the pairs (b,b') and (b',b) survive.

(1) Hessians. Let B be a 1-form on $\mathbb{Z}^3$ with $\sup_b|B(b)|<1$, and let q be a plaquette. We define real symmetric matrices $\mathsf{N}(B)$ and $\mathsf{h}_q(B)$, indexed by the pairs (b,a) of a positively oriented bond b and a colour index a, by

$$\mathsf{N}(B)_{(b,a),(b',a')} := \partial_{b,a}\partial_{b',a'}\Big[\sum_p a_p(e^{iA}) + \frac{1}{2\tau_{\rm gf}}\sum_x |\mathrm{d}^*A(x)|^2\Big]_{A=B}, \qquad \mathsf{h}_q(B)_{(b,a),(b',a')} :$$
$$= \partial_{b,a}\partial_{b',a'}\, a_q(e^{iA})\big|_{A=B}.$$

The infinite sums over all plaquettes p and over all sites x are understood as follows: under $\partial_{b,a}\partial_{b',a'}$ only the finitely many terms which depend on $A(b)$ contribute, namely those of the plaquettes containing b and those of the two end points of b. Consequently an entry of either matrix vanishes unless b and b' lie in a common plaquette or have a common end point. For fixed (b,a) only finitely many entries $\mathsf{N}(B)_{(b,a),(b',a')}$ and $\mathsf{h}_q(B)_{(b,a),(b',a')}$ are different from 0, and both matrices act on bounded complex 1-forms A by $(\mathsf{N}(B)A)^a(b) := \sum_{(b',a')}\mathsf{N}(B)_{(b,a),(b',a')}A^{a'}(b')$, and likewise for $\mathsf{h}_q(B)$.

(2) Constant abelian fields. For $\gamma\in\mathbb{R}^3$ with $|\gamma|<1$ let Y_γ be the constant 1-form $Y_\gamma((x,\mu)) := \gamma_\mu \mathsf{e}_3$. We put $\mathsf{e}_\pm := 2^{-1/2}(\mathsf{e}_1 \pm i\mathsf{e}_2)$ and $\mathsf{e}_0 := \mathsf{e}_3$. These three vectors form a Hermitian-orthonormal basis of $\mathfrak{g}\otimes\mathbb{C}$, and

$$[\mathsf{e}_3, \mathsf{e}_e] = 2e\,\mathsf{e}_e \qquad (e\in\{+1,-1,0\});$$

we call e the *charge*, and we write $\upsilon := 2e\gamma \in \mathbb{R}^3$. For $k\in\mathbb{R}^3$ we define two 3×3 matrices $\hat{\mathsf{N}}_e(k;\gamma)$, $\hat{\mathsf{h}}_{q,e}(k;\gamma)$ and a number $\hat{\mathsf{M}}_e(k;\gamma)$ by

$$\mathsf{N}(Y_\gamma)(e^{ik\cdot x}\alpha_\mu\mathsf{e}_e) = e^{ik\cdot x}(\hat{\mathsf{N}}_e\alpha)_\mu\,\mathsf{e}_e, \qquad (e^{ik\cdot x}\alpha'\mathsf{e}_e,\, \mathsf{h}_q(Y_\gamma)\,e^{ik\cdot x}\alpha\,\mathsf{e}_e) = \alpha'^*\hat{\mathsf{h}}_{q,e}\alpha,$$
$$\mathsf{M}(Y_\gamma)(e^{ik\cdot x}\mathsf{e}_e) = \hat{\mathsf{M}}_e\,e^{ik\cdot x}\mathsf{e}_e$$

for all $\alpha,\alpha'\in\mathbb{C}^3$. Here $e^{ik\cdot x}\alpha_\mu\mathsf{e}_e$ stands for the complex 1-form $(x,\mu)\mapsto e^{ik\cdot x}\alpha_\mu\mathsf{e}_e$ and $e^{ik\cdot x}\mathsf{e}_e$ for the complex site function $x\mapsto e^{ik\cdot x}\mathsf{e}_e$; the bracket $(\cdot,\cdot)$ is Hermitian, and the sum which defines it is finite, since $\mathsf{h}_q(Y_\gamma)$ has non-zero entries only at bonds of q; the operator $\mathsf{M}(Y_\gamma)$ of Def. 7.4(3) is extended complex linearly. That the left sides have these forms is shown in Lem. 7.14, which also gives explicit formulas; by these formulas, for fixed k, the three objects are real analytic functions of γ in $\{|\gamma|<1\}$.

A second index j denotes the homogeneous polynomial of degree j in γ in the Taylor expansion at $\gamma=0$, so that $\hat{\mathsf{N}}_e = \sum_j \hat{\mathsf{N}}_{e,j}$, $\hat{\mathsf{h}}_{q,e} = \sum_j \hat{\mathsf{h}}_{q,e,j}$ and $\hat{\mathsf{M}}_e = \sum_j \hat{\mathsf{M}}_{e,j}$. For $k\notin 2\pi\mathbb{Z}^3$ we put $\hat{\mathsf{C}} := \hat{\mathsf{N}}_{e,0}^{-1}$.

By Lem. 7.14 at $\gamma = 0$ we have $\hat{\mathsf{N}}_{e,0}(k) = |\mathsf{k}|^2\mathbf{1} - (1-\tau_{\mathrm{gf}}^{-1})\mathsf{k}\mathsf{k}^*$ with $\mathsf{k} = \mathsf{k}(k)$, so that $\hat{\mathsf{C}}$ does not depend on e and is the matrix with the entries

$$\hat{\mathsf{C}}_{\mu\nu}(k) = \frac{\delta_{\mu\nu}}{|\mathsf{k}|^2} + (\tau_{\mathrm{gf}} - 1)\frac{\mathsf{k}_\mu \overline{\mathsf{k}_\nu}}{|\mathsf{k}|^4}$$

which appears in the formula for $\mathsf{C}^{\mathrm{lat}}_{\tau_{\mathrm{gf}}}$ in Def. 7.7(1); moreover $\hat{\mathsf{M}}_{e,0} = -|\mathsf{k}|^2$.

(3) Five numbers of the self-energy. For $k \in [-\pi,\pi]^3 \setminus \{0\}$ and $\gamma \in \mathbb{R}^3$ we put, with all functions taken at $(k;\gamma)$,

$$\begin{aligned}
\mathfrak{s}^{(1)} &:= -\tfrac{1}{4}\sum_e \mathrm{tr}\big[(\hat{\mathsf{C}}\hat{\mathsf{N}}_{e,1})^2\big] && \text{(gluon loop)},\\
\mathfrak{s}^{(2)} &:= \tfrac{1}{2}\sum_e \mathrm{tr}\big[\hat{\mathsf{C}}\hat{\mathsf{N}}_{e,2}\big] && \text{(gluon tadpole)},\\
\mathfrak{s}^{(3)} &:= \tfrac{1}{2}\sum_e \mathrm{Re}\big[(\hat{\mathsf{M}}_{e,1}/\hat{\mathsf{M}}_{e,0})^2\big] && \text{(ghost loop)},\\
\mathfrak{s}^{(4)} &:= -\sum_e \mathrm{Re}\big[\hat{\mathsf{M}}_{e,2}/\hat{\mathsf{M}}_{e,0}\big] && \text{(ghost tadpole)},
\end{aligned}$$

and $\mathfrak{s}^{(5)}(k;\gamma) := \frac{1}{3}|\gamma|^2$ (Haar term). For fixed k each $\mathfrak{s}^{(i)}(k;\cdot)$ is a homogeneous polynomial of degree 2 in γ, bounded by $C|\gamma|^2|k|^{-2}$ (for $i \le 4$ by Thm. T(i), for $i = 5$ because $|k|$ is bounded on the cube); hence it is integrable over $\mathbb{T}$, and its integral is a quadratic form in γ. For $i = 1,\dots,5$ we let $c^{(i)}_{\mu\nu}(\tau_{\mathrm{gf}})$, $1 \le \mu,\nu \le 3$, be the entries of the symmetric matrix with

$$\frac{1}{2}\sum_{\mu\nu} c^{(i)}_{\mu\nu}(\tau_{\mathrm{gf}})\,\gamma_\mu\gamma_\nu = \int_{\mathbb{T}} \mathfrak{s}^{(i)}(k;\gamma)\,d^\circ k \qquad (\gamma \in \mathbb{R}^3).$$

We also need the number

$$Z_0 := \int_{\mathbb{T}} \frac{d^\circ k}{|\mathsf{k}(k)|^2} = \int_{[-\pi,\pi]^3} \frac{dk}{(2\pi)^3}\Big(\sum_\mu 4\sin^2\frac{k_\mu}{2}\Big)^{-1}. \tag{7.xxi}$$

(4) Four numbers of the vertex function of a_q. For a plaquette q, $k \in [-\pi,\pi]^3 \setminus \{0\}$ and $\gamma \in \mathbb{R}^3$ we put, again with all functions taken at $(k;\gamma)$,

$$\begin{aligned}
\mathfrak{v}^{(1)}_q &:= \tfrac{1}{2}\sum_e \mathrm{tr}\big[\hat{\mathsf{h}}_{q,e,2}\hat{\mathsf{C}}\big], & \mathfrak{v}^{(2)}_q &:= -\tfrac{1}{2}\sum_e \mathrm{tr}\big[\hat{\mathsf{h}}_{q,e,1}\hat{\mathsf{C}}\hat{\mathsf{N}}_{e,1}\hat{\mathsf{C}}\big],\\
\mathfrak{v}^{(3)}_q &:= -\tfrac{1}{2}\sum_e \mathrm{tr}\big[\hat{\mathsf{h}}_{q,e,0}\hat{\mathsf{C}}\hat{\mathsf{N}}_{e,2}\hat{\mathsf{C}}\big], & \mathfrak{v}^{(4)}_q &:= \tfrac{1}{2}\sum_e \mathrm{tr}\big[\hat{\mathsf{h}}_{q,e,0}\hat{\mathsf{C}}\hat{\mathsf{N}}_{e,1}\hat{\mathsf{C}}\hat{\mathsf{N}}_{e,1}\hat{\mathsf{C}}\big].
\end{aligned}$$

These are homogeneous polynomials of degree 2 in γ, bounded by $C|\gamma|^2|k|^{-2}$ (Thm. U). For $j = 1,\dots,4$ we let $\mathsf{g}^{(j)}_{q;\mu\nu}(\tau_{\mathrm{gf}})$ be the entries of the symmetric matrix with

$$\frac{1}{2}\sum_{\mu\nu} \mathsf{g}^{(j)}_{q;\mu\nu}(\tau_{\mathrm{gf}})\,\gamma_\mu\gamma_\nu = \int_{\mathbb{T}} \mathfrak{v}^{(j)}_q(k;\gamma)\,d^\circ k \qquad (\gamma \in \mathbb{R}^3).$$

(5) Kernels. Let $\langle\cdot\rangle_0$ refer to the Gaussian family of $\mathbb{Z}^3$ and τ_{gf} (Def. 7.8(1)), and write $\partial = \partial_{b,a}$, $\partial' = \partial_{b',a'}$. We define

$$\begin{aligned}
\Sigma_{\mathrm{tot}}(b,a;b',a') &:= \langle \partial\partial' \mathsf{V}_2\rangle_0 - \langle \partial\mathsf{S}_3; \partial'\mathsf{S}_3\rangle_0,\\
\Gamma_q(b,a;b',a') &:= \langle \partial\partial' \mathsf{a}^{(4)}_q\rangle_0 - \langle \partial\mathsf{a}^{(3)}_q; \partial'\mathsf{S}_3\rangle_0 - \langle \partial'\mathsf{a}^{(3)}_q; \partial\mathsf{S}_3\rangle_0 - \langle \mathsf{a}^{(2)}_q; \partial\partial'\mathsf{S}_4\rangle_0 + \langle \mathsf{a}^{(2)}_q; \partial\mathsf{S}_3; \partial'\mathsf{S}_3\rangle_0.
\end{aligned}$$

These are means and cumulants (Def. 7.8(2)) of polynomials in the Gaussian family, and both kernels are symmetric under the exchange $(b,a) \leftrightarrow (b',a')$.

We determine the colour structure of the two kernels. For $g \in G$ the map $\mathrm{Ad}\,g : X \mapsto gXg^{-1}$ is orthogonal on $(\mathfrak{g}, \langle\cdot,\cdot\rangle)$; let $(\mathrm{Ad}\,g)_{aa'}$ be its matrix in the basis $\mathsf{e}_1, \mathsf{e}_2, \mathsf{e}_3$. The components $(gA(b)g^{-1})^a = \sum_{a'}(\mathrm{Ad}\,g)_{aa'}A^{a'}(b)$ of the $\mathfrak{g}$-valued family $(gA(b)g^{-1})_b$ are real linear combinations

of the variables of the Gaussian family, so that this family is again centred Gaussian. By Def. 7.8(1) and the orthogonality of $\operatorname{Ad} g$, its covariance is

$$\big\langle (gA(b)g^{-1})^a\,(gA(b')g^{-1})^c\big\rangle_0 = \sum_{a'} (\operatorname{Ad} g)_{aa'}(\operatorname{Ad} g)_{ca'}\,\mathsf{C}^{\mathrm{lat}}_{\tau_{\mathrm{gf}}}(b,b') = \delta_{ac}\,\mathsf{C}^{\mathrm{lat}}_{\tau_{\mathrm{gf}}}(b,b'),$$

which is the covariance of $(A(b))_b$; as a centred Gaussian family is determined in law by its covariance, $(gA(b)g^{-1})_b$ has the law of $(A(b))_b$. By Lem. 7.5(a) the polynomials $\mathsf{a}_p^{(m)}$ are invariant under $A \mapsto gAg^{-1}$, and so are S_3, S_4 and, term by term, V_2, whose remaining terms involve A only through $|A(b)|^2$ and $\langle A(b), A(b')\rangle$. By the chain rule, each of the two kernels, regarded at fixed b, b' as a matrix K in (a, a'), satisfies $K = (\operatorname{Ad} g)^{\mathsf{T}} K \operatorname{Ad} g$, that is, it commutes with every $\operatorname{Ad} g$. For $g = e^{isY}$, $Y \in \mathfrak{g}$, we have $\operatorname{Ad} g = e^{s\mathsf{a}_Y}$ (Lem. 7.5(d)), and differentiation at $s = 0$ shows that K commutes with $\mathsf{e}_c \times \cdot$ for $c = 1, 2, 3$.

Such a matrix is a multiple of $\mathbf{1}$: it preserves $\ker(\mathsf{e}_c \times \cdot) = \mathbb{R}\mathsf{e}_c$, so it is diagonal in the basis $\mathsf{e}_1, \mathsf{e}_2, \mathsf{e}_3$, with entries m_1, m_2, m_3 say; applying it to $\mathsf{e}_1 \times \mathsf{e}_2 = \mathsf{e}_3$ and to $\mathsf{e}_3 \times \mathsf{e}_1 = \mathsf{e}_2$ gives $m_3 = m_2$ and $m_2 = m_1$. Hence

$$\Sigma_{\mathrm{tot}}(b, a; b', a') = \delta_{aa'}\,\Sigma_{\mathrm{tot}}(b, b'), \qquad \Gamma_q(b, a; b', a') = \delta_{aa'}\,\Gamma_q(b, b'),$$

and these identities define the kernels $\Sigma_{\mathrm{tot}}(b, b')$ and $\Gamma_q(b, b')$. Finally, the kernels are covariant under the translations of $\mathbb{Z}^3$. Indeed, $\mathsf{C}^{\mathrm{lat}}_{\tau_{\mathrm{gf}}}((x,\mu),(y,\nu))$ and $\mathsf{G}^{\mathrm{lat}}(x, y)$ depend on the sites x, y only through $x - y$ (Def. 7.7(1)), so that the law of the Gaussian family is invariant under translations; the coefficients of the polynomials $\mathsf{a}_p^{(m)}$ do not depend on p (Lem. 7.5(a)); and $\mathsf{g}(b) = \frac{1}{3}$ as well as $\mathsf{D}(b; b')$, which are built from $\mathsf{G}^{\mathrm{lat}}$ (Def. 7.7(2)), do not change when b and b' are translated together. Hence the formal sums S_3, S_4, V_2 are invariant under translations as well, and $\Sigma_{\mathrm{tot}}(b+y, b'+y) = \Sigma_{\mathrm{tot}}(b, b')$ and $\Gamma_{q+y}(b + y, b' + y) = \Gamma_q(b, b')$ for all $y \in \mathbb{Z}^3$. ∎

The symbols of Def. 7.13(2) can be computed explicitly.

Lemma 7.14 (The Hessian at a constant abelian field)**.** Let $\gamma \in \mathbb{R}^3$ with $|\gamma| < 1$, let $e \in \{\pm 1, 0\}$, $\upsilon = 2e\gamma$ and $k \in \mathbb{R}^3$. Let $j(\vartheta) := (e^{i\vartheta} - 1)/i\vartheta$ for real $\vartheta \neq 0$ and $j(0) := 1$, and put

$$J := \operatorname{diag}\big(j(\upsilon_1), j(\upsilon_2), j(\upsilon_3)\big), \qquad P(k) := |\mathsf{k}(k)|^2\mathbf{1} - \mathsf{k}(k)\mathsf{k}(k)^*;$$

the matrix J is invertible, since $|\upsilon_\mu| < 2$ and $j(\vartheta) \neq 0$ for $|\vartheta| < 2\pi$. For a plaquette $q = p_{\mu\nu}(x)$ let $D_q(k)$ be the 1×3 matrix with $D_q(k)\alpha = \mathsf{k}_\mu(k)\alpha_\nu - \mathsf{k}_\nu(k)\alpha_\mu$, $\alpha \in \mathbb{C}^3$. Then the left sides of the three identities in Def. 7.13(2) have the forms stated there, with

$$\hat{\mathsf{N}}_e(k;\gamma) = J^*P(k+\upsilon)J + \tau_{\mathrm{gf}}^{-1}\mathsf{k}(k)\mathsf{k}(k)^*, \qquad \hat{\mathsf{h}}_{q,e}(k;\gamma) = J^*D_q(k+\upsilon)^*D_q(k+\upsilon)J,$$
$$\hat{\mathsf{M}}_e(k;\gamma) = -\mathsf{k}(k)^*J^{-1}\mathsf{k}(k+\upsilon).$$

Moreover the following hold.

(a) We have

$$\det \hat{\mathsf{N}}_e(k;\gamma) = \tau_{\mathrm{gf}}^{-1}\,|\det J|^2\,|\mathsf{k}(k+\upsilon)|^2\,|\hat{\mathsf{M}}_e(k;\gamma)|^2, \qquad \prod_e |\det J| = \prod_\mu \sigma_1(\gamma_\mu^2), \tag{7.xxii}$$

where in the second identity J is taken at $\upsilon = 2e\gamma$ for each $e \in \{+1, -1, 0\}$.

(b) If $\mathsf{k}(k+\upsilon) \neq 0$ and $\hat{\mathsf{M}}_e(k;\gamma) \neq 0$, then $\hat{\mathsf{N}}_e(k;\gamma)$ is invertible, and for $q = p_{\mu\nu}(x)$

$$\operatorname{tr}\big[\hat{\mathsf{h}}_{q,e}(k;\gamma)\hat{\mathsf{N}}_e(k;\gamma)^{-1}\big] = \varpi_q(k+\upsilon), \qquad \varpi_q(k) := \frac{|\mathsf{k}_\mu(k)|^2 + |\mathsf{k}_\nu(k)|^2}{|\mathsf{k}(k)|^2}. \tag{7.xxiii}$$

(c) With $\mathsf{k} = \mathsf{k}(k)$,

$$\hat{\mathsf{M}}_{e,1} = -\sum_\mu \upsilon_\mu \sin k_\mu, \qquad \hat{\mathsf{M}}_{e,2} = \frac{1}{12}\sum_\mu \upsilon_\mu^2|\mathsf{k}_\mu|^2, \qquad \operatorname{tr}\hat{\mathsf{N}}_{e,2} = \sum_\mu \upsilon_\mu^2\Big[2\cos k_\mu - \frac{1}{12}\big(|\mathsf{k}|^2 - |\mathsf{k}_\mu|^2\big)\Big].$$

∎

Proof. Step 1: second variation. Let a be a 1-form of finite support, and let $u_\mu := e^{i\gamma_\mu \mathsf{e}_3}$; these three matrices commute with each other. For a bond b and small real s the product $e^{i(Y_\gamma(b)+sa(b))}e^{-iY_\gamma(b)}$ lies close to $\mathbf{1}$, in particular in $G\setminus\{-\mathbf{1}\}$. By Def. 7.1(2) the map $Y\mapsto e^{iY}$ is a diffeomorphism of $\mathbb{B}_\pi$ onto $G\setminus\{-\mathbf{1}\}$, with the smooth inverse $\frac{1}{i}\log$; we may therefore write the product as $e^{i\eta_s(b)}$, where $\eta_s(b) := \frac{1}{i}\log(e^{i(Y_\gamma(b)+sa(b))}e^{-iY_\gamma(b)}) \in \mathfrak{g}$ is a smooth function of s with $\eta_0(b)=0$. The differential formula in Def. 7.1(2), applied with $Y=Y_\gamma(b)$ and $H=a(b)$, shows that the derivative of the product at $s=0$ equals $i\,\mathsf{dE}(Y_\gamma(b))a(b)$; hence

$$\eta_s(b) = s\,\mathsf{dE}(Y_\gamma(b))\,a(b) + O(s^2).$$

Let $p=p_{\mu\nu}(x)$ have the bonds $b_1,\dots,b_4$, and put $\eta_i := \eta_s(b_i)$. Since $Y_\gamma(b_1)=Y_\gamma(b_3)=\gamma_\mu\mathsf{e}_3$ and $Y_\gamma(b_2)=Y_\gamma(b_4)=\gamma_\nu\mathsf{e}_3$, since u_μ and u_ν commute, and since $ue^{iX}u^{-1}=e^{iuXu^{-1}}$, we find

$$\partial\big(e^{i(Y_\gamma+sa)}\big)(p) = e^{i\eta_1}u_\mu\,e^{i\eta_2}u_\nu\,u_\mu^{-1}e^{-i\eta_3}\,u_\nu^{-1}e^{-i\eta_4} = e^{i\eta_1}\,e^{iu_\mu\eta_2u_\mu^{-1}}\,e^{-iu_\nu\eta_3u_\nu^{-1}}\,e^{-i\eta_4} = e^{iZ},$$

where $Z\in\mathfrak{g}$ is the image of this product under $\frac{1}{i}\log$; for small s the product is close to $\mathbf{1}$, and by the diffeomorphism property recalled above Z is a smooth function of s, with $Z=0$ at $s=0$. To first order

$$Z = \eta_1 + u_\mu\eta_2u_\mu^{-1} - u_\nu\eta_3u_\nu^{-1} - \eta_4 + O(s^2).$$

Since $a_p(U) = 1-\frac{1}{2}\operatorname{tr}\partial U(p)$ and $\frac{1}{2}\operatorname{tr}e^{iZ}=\cos|Z|$, we have $a_p(e^{i(Y_\gamma+sa)}) = 1-\cos|Z| = \frac{1}{2}|Z|^2 + O(|Z|^4)$. We put $(\mathsf{J}_\gamma a)(b) := \mathsf{dE}(Y_\gamma(b))a(b)$ and, for a 1-form a', $(\mathsf{D}_\gamma a')(p) := a'(b_1) + u_\mu a'(b_2)u_\mu^{-1} - u_\nu a'(b_3)u_\nu^{-1} - a'(b_4)$; by the expansion of $\eta_s(b)$, $Z = s\,(\mathsf{D}_\gamma\mathsf{J}_\gamma a)(p) + O(s^2)$, and therefore

$$a_p(e^{i(Y_\gamma+sa)}) = \tfrac{1}{2}s^2\big|(\mathsf{D}_\gamma\mathsf{J}_\gamma a)(p)\big|^2 + O(s^3).$$

The constant form Y_γ satisfies $\mathrm{d}^*Y_\gamma=0$, so that $\frac{1}{2\tau_{\mathrm{gf}}}\sum_x|\mathrm{d}^*(Y_\gamma+sa)(x)|^2 = \frac{s^2}{2\tau_{\mathrm{gf}}}\|\mathrm{d}^*a\|^2$. Taking second derivatives in s at $s=0$ we obtain the quadratic forms of the two Hessians:

$$(a,\mathsf{h}_p(Y_\gamma)a) = \big|(\mathsf{D}_\gamma\mathsf{J}_\gamma a)(p)\big|^2, \qquad (a,\mathsf{N}(Y_\gamma)a) = \sum_p\big|(\mathsf{D}_\gamma\mathsf{J}_\gamma a)(p)\big|^2 + \tau_{\mathrm{gf}}^{-1}\|\mathrm{d}^*a\|^2,$$

the sum over p being finite. A real symmetric matrix of the kind considered in Def. 7.13(1) is determined by its quadratic form on the forms of finite support; by polarisation, therefore,

$$\mathsf{h}_p(Y_\gamma) = (\delta_p\mathsf{D}_\gamma\mathsf{J}_\gamma)^*(\delta_p\mathsf{D}_\gamma\mathsf{J}_\gamma), \qquad \mathsf{N}(Y_\gamma) = (\mathsf{D}_\gamma\mathsf{J}_\gamma)^*\mathsf{D}_\gamma\mathsf{J}_\gamma + \tau_{\mathrm{gf}}^{-1}\mathrm{dd}^*,$$

where δ_p is the evaluation at p and the adjoints are taken with respect to the brackets $(\cdot,\cdot)$.

Step 2: symbols. We extend all operators complex linearly. From $[\mathsf{e}_3,\mathsf{e}_e] = 2e\,\mathsf{e}_e$ we get $u_\mu\mathsf{e}_eu_\mu^{-1} = e^{2ie\gamma_\mu}\mathsf{e}_e = e^{iv_\mu}\mathsf{e}_e$, and hence

$$\mathsf{dE}(\gamma_\mu\mathsf{e}_3)\,\mathsf{e}_e = \int_0^1 e^{isv_\mu}\,\mathrm{d}s\;\mathsf{e}_e = j(v_\mu)\,\mathsf{e}_e.$$

So J_γ maps the form $e^{ik\cdot x}\alpha\,\mathsf{e}_e$ to $e^{ik\cdot x}J\alpha\,\mathsf{e}_e$, and for $\alpha'\in\mathbb{C}^3$

$$\mathsf{D}_\gamma(e^{ik\cdot x}\alpha'\mathsf{e}_e)(p_{\mu\nu}(x)) = e^{ik\cdot x}\big[\alpha'_\mu + e^{i(v_\mu+k_\mu)}\alpha'_\nu - e^{i(v_\nu+k_\nu)}\alpha'_\mu - \alpha'_\nu\big]\mathsf{e}_e = e^{ik\cdot x}D_{p_{\mu\nu}(x)}(k+v)\,\alpha'\,\mathsf{e}_e.$$

Thus J_γ, D_γ and dd^* commute with the translations of $\mathbb{Z}^3$ and map a plane wave of momentum k and charge e to a plane wave of the same momentum and charge, the coefficient vector being multiplied by J, by the 1×3 matrices $D_{p_{\mu\nu}(x)}(k+v)$ $(\mu<\nu)$, and by $\mathsf{k}(k)\mathsf{k}(k)^*$, respectively; the last symbol follows from the plane-wave formulas of Def. 7.7(1). The adjoint of such an operator acts on plane waves through the adjoint matrix. By Lagrange's identity,

$$\sum_{\mu<\nu}|v_\mu\alpha_\nu - v_\nu\alpha_\mu|^2 = |v|^2|\alpha|^2 - |v^*\alpha|^2 = \alpha^*\big(|v|^2\mathbf{1} - vv^*\big)\alpha \qquad (v,\alpha\in\mathbb{C}^3),$$

so that $\sum_{\mu<\nu}D_{p_{\mu\nu}(x)}(k')^*D_{p_{\mu\nu}(x)}(k') = P(k')$ for $k'\in\mathbb{R}^3$. Together with Step 1 this gives the formula for $\hat{\mathsf{N}}_e$; for $\hat{\mathsf{h}}_{q,e}$ only the plaquette q enters, and the factors $e^{ik\cdot x}$ have modulus 1.

For $\hat{\mathsf{M}}_e$ we use the middle expression in eq. 7.vi, which is real linear in λ and is extended complex linearly, together with $e^{\mathsf{a}_Y}\mathsf{e}_e = u_\mu \mathsf{e}_e u_\mu^{-1}$ for $Y = \gamma_\mu \mathsf{e}_3$ (Lem. 7.5(d)). At $A = Y_\gamma$, $\lambda = e^{ik\cdot x}\mathsf{e}_e$ and the bond (x,μ) it gives

$$\frac{d}{ds}A^{s\lambda}((x,\mu))\Big|_{s=0} = \mathsf{dE}(\gamma_\mu \mathsf{e}_3)^{-1}\big(e^{ik\cdot x}\mathsf{e}_e - e^{iv_\mu}e^{ik\cdot(x+e_\mu)}\mathsf{e}_e\big) = j(v_\mu)^{-1}\big(1-e^{i(v_\mu+k_\mu)}\big)e^{ik\cdot x}\mathsf{e}_e = \\ - j(v_\mu)^{-1}\mathsf{k}_\mu(k+v)\,e^{ik\cdot x}\mathsf{e}_e.$$

Since $\mathsf{M}(A)\lambda$ is d^* of this derivative and d^* multiplies plane waves by $\mathsf{k}(k)^*$ (Def. 7.7(1)), we obtain $\hat{\mathsf{M}}_e = -\mathsf{k}(k)^* J^{-1}\mathsf{k}(k+v)$. Here we have used that $j(\vartheta)\neq 0$ for $|\vartheta|<2\pi$.

Step 3: proof of (a). Let $v := \mathsf{k}(k+v)$ and $\tilde{u} := (J^*)^{-1}\mathsf{k}(k)$. Then

$$\hat{\mathsf{N}}_e = J^*\big(P(k+v) + \tau_{\mathrm{gf}}^{-1}\tilde{u}\tilde{u}^*\big)J, \qquad v^*\tilde{u} = (J^{-1}v)^*\mathsf{k}(k) = -\overline{\hat{\mathsf{M}}_e},$$

and $P(k+v) = |v|^2\mathbf{1} - vv^*$. For $v,w\in\mathbb{C}^3$ we have $\det\big(|v|^2\mathbf{1} - vv^* + ww^*\big) = |v|^2|v^*w|^2$: for $v=0$ both sides vanish; for $v\neq 0$ we use an orthonormal basis whose first vector is $v/|v|$; in it the matrix is $|v|^2\,\mathrm{diag}(0,1,1) + ww^*$, the determinant is an affine function of the rank-one summand, and its value is $0 + |v|^4|w_1|^2$ with $w_1 = v^*w/|v|$. With $w = \tau_{\mathrm{gf}}^{-1/2}\tilde{u}$ this gives $\det\hat{\mathsf{N}}_e = |\det J|^2\,|v|^2\,\tau_{\mathrm{gf}}^{-1}|v^*\tilde{u}|^2$, the first identity in eq. 7.xxii. For the second, $|j(\vartheta)|^2 = (2-2\cos\vartheta)/\vartheta^2 = \big(\sin\frac{\vartheta}{2}/\frac{\vartheta}{2}\big)^2$; since $\det J = \prod_\mu j(2e\gamma_\mu)$, the product over the three charges contributes, for each μ, the factor $|j(2\gamma_\mu)|^2 = (\sin\gamma_\mu/\gamma_\mu)^2 = \sigma_1(\gamma_\mu^2)$.

Step 4: proof of (b). By (a), $\hat{\mathsf{N}}_e$ is invertible. By the cyclicity of the trace,

$$\mathrm{tr}\big[\hat{\mathsf{h}}_{q,e}\hat{\mathsf{N}}_e^{-1}\big] = D_q\big(P + \tau_{\mathrm{gf}}^{-1}\tilde{u}\tilde{u}^*\big)^{-1}D_q^*,$$

with D_q and P taken at $k+v$. Now $D_q v = v_\mu v_\nu - v_\nu v_\mu = 0$, hence $v^*D_q^* = 0$ and $PD_q^* = |v|^2 D_q^*$; moreover $Pv = 0$, and $\tilde{u}^*v \neq 0$ because $v^*\tilde{u} = -\overline{\hat{\mathsf{M}}_e} \neq 0$. The vector

$$\mathsf{X} := |v|^{-2}D_q^* - \frac{\tilde{u}^*D_q^*}{|v|^2\,\tilde{u}^*v}\,v$$

satisfies $P\mathsf{X} = D_q^*$ and $\tilde{u}^*\mathsf{X} = 0$, hence $(P+\tau_{\mathrm{gf}}^{-1}\tilde{u}\tilde{u}^*)\mathsf{X} = D_q^*$. Therefore the trace equals $D_q\mathsf{X} = |v|^{-2}D_qD_q^* = |v|^{-2}(|v_\mu|^2 + |v_\nu|^2) = \varpi_q(k+v)$.

Step 5: proof of (c). We expand in powers of ϑ and of v; the dots stand for terms of degree at least 3:

$$j(\vartheta) = 1 + \frac{i\vartheta}{2} - \frac{\vartheta^2}{6} + \dots, \qquad j(\vartheta)^{-1} = 1 - \frac{i\vartheta}{2} - \frac{\vartheta^2}{12} + \dots, \qquad |j(\vartheta)|^2 = 1 - \frac{\vartheta^2}{12} + \dots,$$
$$\mathsf{k}_\mu(k+v) = \mathsf{k}_\mu + e^{ik_\mu}\big(iv_\mu - \tfrac{1}{2}v_\mu^2 + \dots\big), \qquad |\mathsf{k}_\mu(k+v)|^2 = 2 - 2\cos(k_\mu+v_\mu) = |\mathsf{k}_\mu|^2 \\ + 2v_\mu\sin k_\mu + v_\mu^2\cos k_\mu + \dots.$$

In $\hat{\mathsf{M}}_e = -\sum_\mu \overline{\mathsf{k}_\mu}\,j(v_\mu)^{-1}\mathsf{k}_\mu(k+v)$ the terms of degree 1 are

$$-i\sum_\mu v_\mu\overline{\mathsf{k}_\mu}\big[e^{ik_\mu} - \tfrac{1}{2}\mathsf{k}_\mu\big] = -i\sum_\mu v_\mu\overline{\mathsf{k}_\mu}\,\tfrac{1}{2}(e^{ik_\mu}+1) = -\sum_\mu v_\mu\sin k_\mu,$$

because $\overline{\mathsf{k}_\mu}(e^{ik_\mu}+1) = -2i\sin k_\mu$, and the terms of degree 2 are

$$-\sum_\mu v_\mu^2\,\overline{\mathsf{k}_\mu}\big[-\tfrac{1}{12}\mathsf{k}_\mu + \tfrac{1}{2}e^{ik_\mu} - \tfrac{1}{2}e^{ik_\mu}\big] = \frac{1}{12}\sum_\mu v_\mu^2|\mathsf{k}_\mu|^2.$$

The term $\tau_{\mathrm{gf}}^{-1}\mathsf{k}\mathsf{k}^*$ of $\hat{\mathsf{N}}_e$ does not depend on γ. Hence $\mathrm{tr}\,\hat{\mathsf{N}}_{e,2}$ is the homogeneous polynomial of degree 2 in the expansion of

$$\mathrm{tr}\big[J^*P(k+v)J\big] = \sum_\mu |j(v_\mu)|^2\sum_{\nu\neq\mu}|\mathsf{k}_\nu(k+v)|^2,$$

namely

$$\sum_\mu \Big[-\frac{1}{12} v_\mu^2 \sum_{\nu\neq\mu} |\mathsf{k}_\nu|^2 + \sum_{\nu\neq\mu} v_\nu^2 \cos k_\nu \Big] = \sum_\mu v_\mu^2 \Big[2\cos k_\mu - \frac{1}{12}\big(|\mathsf{k}|^2 - |\mathsf{k}_\mu|^2\big)\Big].$$

This is the asserted formula for $\operatorname{tr} \hat{\mathsf{N}}_{e,2}$, and (c) is proved. □

With these formulas we evaluate the self-energy at zero momentum.

Theorem T (The five contributions to the self-energy at zero momentum)**.** Let $\tau_{\mathrm{gf}} > 0$.

(i) For $i = 1, \dots, 4$, all $k \in [-\pi,\pi]^3 \setminus \{0\}$ and all $\gamma \in \mathbb{R}^3$ we have $|\mathfrak{s}^{(i)}(k;\gamma)| \le C|\gamma|^2 |k|^{-2}$, with $C = C(\tau_{\mathrm{gf}})$.

(ii) $\sum_{i=1}^5 c_{\mu\nu}^{(i)}(\tau_{\mathrm{gf}}) = 0$ for all μ, ν.

(iii) At $\tau_{\mathrm{gf}} = 1$,

$$(c_{\mu\nu}^{(1)}, \dots, c_{\mu\nu}^{(5)})(1) = \delta_{\mu\nu} \cdot \Big(-20 Z_0 + \tfrac{8}{3},\ \ 16 Z_0 - \tfrac{28}{9},\ \ 4 Z_0 - \tfrac{2}{3},\ \ \tfrac{4}{9},\ \ \tfrac{2}{3} \Big). \tag{7.xxiv}$$

■□

Proof. Step 1: bounds. By the formulas of Lem. 7.14, the entries of $J^{\pm 1}$ and of $\mathsf{k}(k+v)$ are real analytic functions of (k,γ). On $[-\pi,\pi]^3$ we have $P(k) = O(|k|^2)$ and $\partial_k P(k) = O(|k|)$. The terms of degree 1 in γ of $J^* P(k+v) J$ are terms containing the factor $P(k)$ and the term $\sum_\mu v_\mu \partial_\mu P(k)$; hence $|\hat{\mathsf{N}}_{e,1}| \le C|\gamma||k|$, while $|\hat{\mathsf{N}}_{e,2}| \le C|\gamma|^2$. Every term of $\hat{\mathsf{M}}_e$ contains the factor $\mathsf{k}(k)^*$, so that $|\hat{\mathsf{M}}_{e,j}| \le C|\gamma|^j |k|$. Finally $|\hat{\mathsf{C}}| \le C|k|^{-2}$ by the formula for $\hat{\mathsf{C}}$ in Def. 7.13(2), and $|\hat{\mathsf{M}}_{e,0}|^{-1} = |\mathsf{k}(k)|^{-2} \le C|k|^{-2}$, as $|\mathsf{k}(k)|^2 \ge \frac{4}{\pi^2}|k|^2$.

Inserting these bounds into the definitions of $\mathfrak{s}^{(1)}, \dots, \mathfrak{s}^{(4)}$ we obtain $|\hat{\mathsf{C}}\hat{\mathsf{N}}_{e,1}|^2 \le C|\gamma|^2|k|^{-2}$, $|\hat{\mathsf{C}}\hat{\mathsf{N}}_{e,2}| \le C|\gamma|^2|k|^{-2}$, $|\hat{\mathsf{M}}_{e,1}/\hat{\mathsf{M}}_{e,0}|^2 \le C|\gamma|^2|k|^{-2}$ and $|\hat{\mathsf{M}}_{e,2}/\hat{\mathsf{M}}_{e,0}| \le C|\gamma|^2|k|^{-1} \le C'|\gamma|^2|k|^{-2}$, which proves (i).

Step 2: identity of the integrands. Fix $k \in [-\pi,\pi]^3 \setminus \{0\}$, and let γ be so close to 0 that $\mathsf{k}(k+v) \neq 0$ and $\hat{\mathsf{M}}_e(k;\gamma) \neq 0$ for all e. We divide the first identity in eq. 7.xxii by its value at $\gamma = 0$, take one half of its logarithm, sum over e and use the second identity in eq. 7.xxii:

$$\sum_e \Big\{ \tfrac{1}{2} \log\det\big[\hat{\mathsf{C}}\hat{\mathsf{N}}_e(k;\gamma)\big] - \log\big|\hat{\mathsf{M}}_e(k;\gamma)/\hat{\mathsf{M}}_{e,0}\big| \Big\} - \sum_\mu \log \sigma_1(\gamma_\mu^2) = \tfrac{1}{2} \sum_e \log \frac{|\mathsf{k}(k+2e\gamma)|^2}{|\mathsf{k}(k)|^2}.$$

Both sides are real analytic in γ near 0, and we compare their terms of degree 2. On the left, the expansion $\log\det(\mathbf{1}+\mathsf{A}) = \operatorname{tr}\mathsf{A} - \frac{1}{2}\operatorname{tr}\mathsf{A}^2 + \dots$ with $\mathsf{A} = \hat{\mathsf{C}}(\hat{\mathsf{N}}_{e,1} + \hat{\mathsf{N}}_{e,2} + \dots)$ yields $\mathfrak{s}^{(2)} + \mathfrak{s}^{(1)}$; the expansion $\log|1+z| = \operatorname{Re}(z - \frac{1}{2}z^2 + \dots)$ with $z = (\hat{\mathsf{M}}_{e,1} + \hat{\mathsf{M}}_{e,2} + \dots)/\hat{\mathsf{M}}_{e,0}$ yields, with the sign in front, $\mathfrak{s}^{(4)} + \mathfrak{s}^{(3)}$; and by Lem. 7.5(c), applied to $Y = \gamma_\mu \mathsf{e}_3$, the term $-\sum_\mu \log\sigma_1(\gamma_\mu^2)$ yields $\frac{1}{3}|\gamma|^2 = \mathfrak{s}^{(5)}$. On the right, with $\mathsf{l}(k) := \log|\mathsf{k}(k)|^2$, the terms of degree 2 are $\frac{1}{2}\sum_{e=\pm1}\frac{1}{2}\sum_{\mu\nu} 4\gamma_\mu\gamma_\nu\,\partial_\mu\partial_\nu \mathsf{l}(k)$. Hence

$$\sum_{i=1}^5 \mathfrak{s}^{(i)}(k;\gamma) = 2\sum_{\mu\nu} \gamma_\mu\gamma_\nu\, \partial_\mu\partial_\nu \mathsf{l}(k) \qquad (k \neq 0),$$

an identity between homogeneous polynomials of degree 2 in γ, valid for all $\gamma \in \mathbb{R}^3$.

Step 3: flux argument. The function l is smooth on $\mathbb{R}^3 \setminus 2\pi\mathbb{Z}^3$ and periodic, and on the cube $|\partial\mathsf{l}| \le C|k|^{-1}$ and $|\partial\partial\mathsf{l}| \le C|k|^{-2}$. For $0 < \varepsilon < 1$ we apply Gauss' divergence theorem on the cube $[-\pi,\pi]^3$ less the ball $\{|k| \le \varepsilon\}$; the contributions of opposite faces of the cube cancel by periodicity, and

$$\int_{[-\pi,\pi]^3 \cap \{|k|>\varepsilon\}} \partial_\mu\partial_\nu \mathsf{l}\, \mathrm{d}k = -\int_{|k|=\varepsilon} \frac{k_\mu}{\varepsilon}\, \partial_\nu \mathsf{l}\, \mathrm{d}S = O(\varepsilon).$$

Letting $\varepsilon \to 0$ we get $\int_{\mathbb{T}} \partial_\mu\partial_\nu \mathsf{l}\, d^\circ k = 0$. By Step 2 and the definition of the numbers $c_{\mu\nu}^{(i)}$, $\frac{1}{2}\sum_{\mu\nu}(\sum_i c_{\mu\nu}^{(i)})\gamma_\mu\gamma_\nu = 0$ for all γ; as the matrices are symmetric, (ii) follows.

Step 4: elementary integrals. We write $\mathsf{k} = \mathsf{k}(k)$. By the symmetry under permutations of the coordinates, $\int_{\mathbb{T}} |\mathsf{k}_\mu|^2 |\mathsf{k}|^{-2} d^\circ k = \frac{1}{3}$. As $\cos k_\mu = 1 - \frac{1}{2}|\mathsf{k}_\mu|^2$, we get $\int_{\mathbb{T}} \cos k_\mu |\mathsf{k}|^{-2} d^\circ k = Z_0 - \frac{1}{6}$. Since $\partial_\mu |\mathsf{k}|^2 = 2 \sin k_\mu$,

$$\partial_\mu (\sin k_\mu |\mathsf{k}|^{-2}) = \cos k_\mu |\mathsf{k}|^{-2} - 2 \sin^2 k_\mu |\mathsf{k}|^{-4};$$

the function $\sin k_\mu |\mathsf{k}|^{-2}$ is periodic and $O(|k|^{-1})$, so the flux argument of Step 3 shows that the integral of the left side over $\mathbb{T}$ vanishes, and $\int_{\mathbb{T}} \sin^2 k_\mu |\mathsf{k}|^{-4} d^\circ k = \frac{1}{2}(Z_0 - \frac{1}{6})$. Finally $\int_{\mathbb{T}} \sin k_\mu \sin k_\nu |\mathsf{k}|^{-4} d^\circ k = 0$ for $\mu \neq \nu$, the integrand being odd under $k_\mu \to -k_\mu$.

Step 5: values. We use Lem. 7.14(c) with $\upsilon = 2e\gamma$, and $\hat{\mathsf{M}}_{e,0} = -|\mathsf{k}|^2$. For $e = 0$ we have $\upsilon = 0$, so that $\hat{\mathsf{N}}_{0,j} = 0$ and $\hat{\mathsf{M}}_{0,j} = 0$ for $j \geq 1$; the charge 0 contributes nothing, and for $e = \pm 1$ we have $\upsilon_\mu^2 = 4\gamma_\mu^2$. We write δ for the matrix $(\delta_{\mu\nu})$.

- $\mathfrak{s}^{(5)} = \frac{1}{3}|\gamma|^2$, whose integral is $\frac{1}{3}|\gamma|^2$; hence $c^{(5)} = \frac{2}{3}\delta$.
- $\mathfrak{s}^{(4)} = \sum_{e=\pm 1} \frac{4}{12} \sum_\mu \gamma_\mu^2 |\mathsf{k}_\mu|^2 |\mathsf{k}|^{-2}$, whose integral is $\frac{2}{9}|\gamma|^2$ by Step 4; hence $c^{(4)} = \frac{4}{9}\delta$.
- $\mathfrak{s}^{(3)} = 4\big(\sum_\mu \gamma_\mu \sin k_\mu\big)^2 |\mathsf{k}|^{-4}$, whose integral is $4|\gamma|^2 \cdot \frac{1}{2}(Z_0 - \frac{1}{6})$; hence $c^{(3)} = (4Z_0 - \frac{2}{3})\delta$.
- At $\tau_{\mathrm{gf}} = 1$ we have $\hat{\mathsf{C}} = |\mathsf{k}|^{-2}\mathbf{1}$ and

$$\mathfrak{s}^{(2)} = \tfrac{1}{2} \sum_{e=\pm 1} |\mathsf{k}|^{-2} \operatorname{tr} \hat{\mathsf{N}}_{e,2} = 4 \sum_\mu \gamma_\mu^2 \Big[2 \cos k_\mu - \frac{1}{12}\big(|\mathsf{k}|^2 - |\mathsf{k}_\mu|^2\big) \Big] |\mathsf{k}|^{-2},$$

  whose integral is $4|\gamma|^2 [2Z_0 - \frac{1}{3} - \frac{1}{12} \cdot \frac{2}{3}] = |\gamma|^2 (8Z_0 - \frac{14}{9})$; hence $c^{(2)} = (16Z_0 - \frac{28}{9})\delta$.
- By (ii), $c^{(1)} = -\sum_{i \geq 2} c^{(i)} = -(20Z_0 - \frac{28}{9} - \frac{2}{3} + \frac{4}{9} + \frac{2}{3})\delta = (-20Z_0 + \frac{8}{3})\delta$.

This proves (iii). □

The same method applies to the vertex function of a plaquette action.

Theorem U (The vertex function at zero momenta)**.** Let $\tau_{\mathrm{gf}} > 0$ and let q be a plaquette. Then $|\mathfrak{v}_q^{(j)}(k;\gamma)| \leq C|\gamma|^2 |k|^{-2}$ for $j = 1, \dots, 4$, all $k \in [-\pi, \pi]^3 \setminus \{0\}$ and all $\gamma \in \mathbb{R}^3$, where C depends on τ_{gf} only; and

$$\sum_{j=1}^4 \mathsf{g}_{q;\mu\nu}^{(j)}(\tau_{\mathrm{gf}}) = 0 \qquad \text{for all } \mu, \nu.$$

■□

Proof. Step 1: bound. The 1×3 matrix $D_q(k)$ is smooth in k, satisfies $D_q(k) = O(|k|)$, and depends on q only through the two directions of q. In the expansion of $\hat{\mathsf{h}}_{q,e} = J^* D_q(k + \upsilon)^* D_q(k + \upsilon) J$ in powers of γ, each term of degree $j \leq 1$ retains at least $2 - j$ factors $D_q(k)$ or $D_q(k)^*$; hence $|\hat{\mathsf{h}}_{q,e,j}| \leq C|\gamma|^j |k|^{2-j}$ for $j = 0, 1, 2$, with C independent of q. As in Step 1 of the proof of Thm. T, Lem. 7.14 gives $|\hat{\mathsf{N}}_{e,1}| \leq C|\gamma||k|$ and $|\hat{\mathsf{N}}_{e,2}| \leq C|\gamma|^2$, and Def. 7.7(1) gives $|\hat{\mathsf{C}}| \leq C|k|^{-2}$. Inserted into the definitions of $\mathfrak{v}_q^{(1)}, \dots, \mathfrak{v}_q^{(4)}$, these bounds give the asserted bound.

Step 2: identity of the integrands. Fix $k \in [-\pi, \pi]^3 \setminus \{0\}$. For small γ we write $\hat{\mathsf{N}}_e = \hat{\mathsf{N}}_{e,0}(\mathbf{1} + \hat{\mathsf{C}}(\hat{\mathsf{N}}_e - \hat{\mathsf{N}}_{e,0}))$, and the Neumann series gives

$$\hat{\mathsf{N}}_e^{-1} = \hat{\mathsf{C}} - \hat{\mathsf{C}}\hat{\mathsf{N}}_{e,1}\hat{\mathsf{C}} - \hat{\mathsf{C}}\hat{\mathsf{N}}_{e,2}\hat{\mathsf{C}} + \hat{\mathsf{C}}\hat{\mathsf{N}}_{e,1}\hat{\mathsf{C}}\hat{\mathsf{N}}_{e,1}\hat{\mathsf{C}} + O(|\gamma|^3).$$

Hence the terms of degree 2 in γ of $\frac{1}{2}\sum_e \operatorname{tr}[\hat{\mathsf{h}}_{q,e}\hat{\mathsf{N}}_e^{-1}]$ are $\sum_{j=1}^4 \mathfrak{v}_q^{(j)}(k;\gamma)$. On the other hand, for small γ the hypotheses of Lem. 7.14(b) hold, and by eq. 7.xxiii the same expression equals $\frac{1}{2}\sum_e \varpi_q(k + 2e\gamma)$, whose terms of degree 2 are $2\sum_{\mu\nu} \gamma_\mu \gamma_\nu \partial_\mu \partial_\nu \varpi_q(k)$. Therefore

$$\sum_{j=1}^4 \mathfrak{v}_q^{(j)}(k;\gamma) = 2 \sum_{\mu\nu} \gamma_\mu \gamma_\nu \, \partial_\mu \partial_\nu \varpi_q(k) \qquad (k \neq 0, \ \gamma \in \mathbb{R}^3).$$

Step 3: flux argument. The function ϖ_q is smooth on $\mathbb{R}^3 \setminus 2\pi\mathbb{Z}^3$ and periodic, and on the cube $|\partial\varpi_q| \le C|k|^{-1}$ and $|\partial\partial\varpi_q| \le C|k|^{-2}$, since it is a quotient of smooth functions vanishing to second order at 0 whose denominator is at least $\frac{4}{\pi^2}|k|^2$. As in Step 3 of the proof of Thm. T,

$$\int_{[-\pi,\pi]^3\cap\{|k|>\varepsilon\}} \partial_\mu\partial_\nu\varpi_q \,\mathrm{d}k = -\int_{|k|=\varepsilon} \frac{k_\mu}{\varepsilon}\,\partial_\nu\varpi_q \,\mathrm{d}S = O(\varepsilon),$$

hence $\int_{\mathbb{T}} \partial_\mu\partial_\nu\varpi_q \, d^\circ k = 0$. By Step 2, $\frac{1}{2}\sum_{\mu\nu}(\sum_j \mathsf{g}^{(j)}_{q;\mu\nu})\gamma_\mu\gamma_\nu = 0$ for all γ, and the symmetry of the matrices gives the claim. □

We next record the behaviour of these objects under point reflections.

Lemma 7.15 (Point reflections)**.** Let $c \in \frac{1}{2}\mathbb{Z}^3$, and let $R_c : x \mapsto 2c - x$ on $\mathbb{Z}^3$ (note that $2c \in \mathbb{Z}^3$). We let R_c act on bonds by $R_c\langle x, x'\rangle := \langle R_c x, R_c x'\rangle$, on plaquettes through their corners (the corners of $R_c p$ are the images of the corners of p), and on configurations U, 1-forms A and site functions λ by

$$(R_cU)(b) := U(R_cb), \qquad (R_cA)(b) := A(R_cb), \qquad (R_c\lambda)(x) := \lambda(R_cx);$$

thus $(R_cA)_\mu(x) = -A_\mu(2c - x - e_\mu)$. Since $2c \in \mathbb{Z}^3$, R_c preserves the lattice. The following hold.

(a) For all plaquettes p, configurations U, 1-forms A on $\mathbb{Z}^3$ and $m \ge 0$: $a_p(R_cU) = a_{R_cp}(U)$ and $\mathsf{a}^{(m)}_p(R_cA) = \mathsf{a}^{(m)}_{R_cp}(A)$. Moreover $\mathrm{d}^*R_c = R_c\mathrm{d}^*$ on 1-forms, and $\mathrm{d}R_c = R_c\mathrm{d}$ on site functions.

(b) For every $\tau_{\mathrm{gf}} > 0$ the family $\big((R_cA)^a(b)\big)_{b,a}$ has the law of the Gaussian family of $\mathbb{Z}^3$ and τ_{gf}. Moreover $\mathsf{D}(R_cb; R_cb') = \mathsf{D}(b; b')$ for all bonds b, b', where D is taken with oriented arguments, that is, it is given by the formula of Def. 7.7(2) for arbitrary bonds; in particular $\mathsf{D}(\bar{b}; b') = -\mathsf{D}(b; b')$ and $\mathsf{D}(b; \bar{b}') = \mathsf{D}(b; b')$.

(c) For a positively oriented bond b let b^R be the positively oriented one of the bonds R_cb and $\overline{R_cb}$, that is, $(x,\mu)^R = (2c - x - e_\mu, \mu)$. Then for all positively oriented bonds b, b' and all plaquettes q

$$\Sigma_{\mathrm{tot}}(b^R, b'^R) = \Sigma_{\mathrm{tot}}(b, b'), \qquad \Gamma_{R_cq}(b^R, b'^R) = \Gamma_q(b, b').$$

■

Proof. The actions in the statement are defined by the same involution on sites; in particular R_c^2 is the identity on sites, bonds, plaquettes and configurations.

Step 1: proof of (a). Let $p = p_{\mu\nu}(x)$. The loop $\partial p = (b_1, b_2, \bar{b}_3, \bar{b}_4)$ passes through x, $x + e_\mu$, $x + e_\mu + e_\nu$, $x + e_\nu$. The map R_c carries it to the loop through $y + e_\mu + e_\nu$, $y + e_\nu$, y, $y + e_\mu$, where $y := 2c - x - e_\mu - e_\nu$ is the base point of $R_cp = p_{\mu\nu}(y)$; this image loop is $\partial(R_cp)$, traversed in the same sense but started at the opposite corner. Since $\partial(R_cU)(p)$ is the product of the values of U along the image loop, it is conjugate in G to $\partial U(R_cp)$, and the two matrices have the same trace; hence $a_p(R_cU) = a_{R_cp}(U)$. As $R_c(e^{itA}) = e^{itR_cA}$, comparison of the Taylor coefficients in eq. 7.iii gives $\mathsf{a}^{(m)}_p(R_cA) = \mathsf{a}^{(m)}_{R_cp}(A)$.

Next, $(\mathrm{d}^*A)(x) = -\sum_{b:\, b_- = x} A(b)$, the sum over the six bonds starting at x, and R_c maps the bonds starting at x bijectively onto those starting at R_cx; therefore $\mathrm{d}^*R_cA = R_c\mathrm{d}^*A$. Finally $(\mathrm{d}R_c\lambda)(b) = \lambda(R_cb_+) - \lambda(R_cb_-) = (\mathrm{d}\lambda)(R_cb)$.

Step 2: proof of (b). The variables $(R_cA)_\mu(x) = -A_\mu(2c - x - e_\mu)$ form a centred Gaussian family, with the same colour structure as A. By Def. 7.7(1) the covariance of $(R_cA)_\mu(x)$ and $(R_cA)_\nu(y)$, for equal colours, is

$$\mathsf{C}\big((2c - x - e_\mu, \mu), (2c - y - e_\nu, \nu)\big) = \int_{\mathbb{T}} e^{-ik\cdot(x-y)}\, e^{-i(k_\mu - k_\nu)}\, \hat{\mathsf{C}}_{\mu\nu}(k)\, d^\circ k.$$

Since $e^{-ik_\mu}\mathsf{k}_\mu(k) = -\mathsf{k}_\mu(-k)$, and correspondingly $e^{ik_\nu}\overline{\mathsf{k}_\nu(k)} = -\overline{\mathsf{k}_\nu(-k)}$, we have $e^{-i(k_\mu - k_\nu)}\hat{\mathsf{C}}_{\mu\nu}(k) = \hat{\mathsf{C}}_{\mu\nu}(-k)$, also for $\mu = \nu$. The substitution $k \to -k$ shows that the covariance equals $\mathsf{C}((x,\mu),(y,\nu))$,

and a centred Gaussian family is determined in law by its covariance. Moreover $\mathsf{G}(R_c x, R_c y) = \mathsf{G}(x,y)$, again by the substitution $k \to -k$ in the formula for $\mathsf{G}^{\mathrm{lat}}$; with $(R_c b)_\pm = R_c(b_\pm)$ the formula for D in Def. 7.7(2) gives $\mathsf{D}(R_c b; R_c b') = \mathsf{D}(b; b')$. The same formula shows that reversing the first argument changes the sign of D, while reversing the second does not change it.

Step 3: proof of (c). By (a) and (b), S_3 and V_2 are invariant under $A \to R_c A$, term by term: the term of a plaquette p is carried to that of $R_c p$, and the terms of a bond b and of a pair (b, b') in V_2 to those of b^R and of (b^R, b'^R); here $\mathsf{g}(b) = \frac{1}{3}$ is constant on $\mathbb{Z}^3$, $|A(R_c b)| = |A(b^R)|$, and the signs coming from reversed arguments of D cancel in pairs. Likewise $\mathsf{a}_q^{(m)}(A) = \mathsf{a}_{R_c q}^{(m)}(R_c A)$. Since $(R_c A)(b^R) = A(\bar{b}) = -A(b)$, the chain rule gives

$$\partial_{b,a}(\mathsf{F} \circ R_c) = -(\partial_{b^R,a}\mathsf{F}) \circ R_c$$

for every polynomial F. Each term of the kernels in Def. 7.13(5) contains exactly two derivatives, so that the signs cancel, and by (b) the means and cumulants do not change under the substitution $A \to R_c A$. Hence $\Sigma_{\mathrm{tot}}(b, a; b', a') = \Sigma_{\mathrm{tot}}(b^R, a; b'^R, a')$ and $\Gamma_q(b, a; b', a') = \Gamma_{R_c q}(b^R, a; b'^R, a')$, which is (c). □

7.1.9. *Tools on kernels.* We now develop a calculus of kernels on $\mathbb{Z}^3$ given by symbols on $\mathbb{T}$; it serves to determine the behaviour of lattice graphs at large distances. Its basic objects are the lines of Def. 7.16(4): kernels whose symbols near 0 and whose values at large distances are given, up to one order and a logarithm, by homogeneous functions. The lemmas after the definition relate these two homogeneous functions to each other; they compare lattice sums with integrals, and a kernel tested against a slowly varying function with its leading part; they expand one-loop integrals of line symbols at small external momenta; and they show that, under conditions on the degrees, convolutions of lines and, up to local terms, products of two lines are again lines.

Definition 7.16 (Symbols, lines, slowly varying functions, majorants)**.** *(1) Two cut-offs.* Let $\theta_0(t) = e^{-1/t}$ for $t > 0$ and $\theta_0(t) = 0$ for $t \le 0$. This function is smooth. To check the only issue, at zero, induction gives $\theta_0^{(n)}(t) = P_n(1/t)e^{-1/t}$ for $t > 0$, where $P_0 = 1$ and $P_{n+1}(u) = u^2(P_n(u) - P_n'(u))$. Every polynomial times e^{-u} tends to zero as $u \to \infty$, as does $uP_n(u)e^{-u}$. Thus each derivative extends continuously by zero and has derivative zero at the origin. For $s \ge 0$ put

$$\theta(s) := \frac{\theta_0(1-s)}{\theta_0(1-s) + \theta_0(s - \frac{1}{2})};$$

the denominator is positive, since one of the numbers $1 - s$, $s - \frac{1}{2}$ is positive. Hence $\theta \in C^\infty([0,\infty))$, $0 \le \theta \le 1$, $\theta = 1$ on $[0, \frac{1}{2}]$ and $\theta = 0$ on $[1, \infty)$. For $\kappa \in \mathbb{R}^D$, $D \in \{3, 6\}$, let $\chi(\kappa) := \theta(|\kappa|/2)$. Then χ is smooth, $\chi = 1$ for $|\kappa| \le 1$ and $\chi = 0$ for $|\kappa| \ge 2$; since $2 < \pi$, we also regard χ as a function on $\mathbb{T}^{D/3}$. The functions θ and χ are fixed by formulas, so constants which depend on θ and χ alone are pure constants.

(2) Symbol classes. For real s let

$$\mathcal{S}^s := \Big\{\sigma \in C^\infty(\mathbb{T} \setminus \{0\}) : \sup_{k \in [-\pi,\pi]^3 \setminus \{0\}} |k|^{|\alpha| - s} |\partial^\alpha \sigma(k)| < \infty \text{ for every multi-index } \alpha\Big\}.$$

By Leibniz' rule $\mathcal{S}^s \mathcal{S}^{s'} \subset \mathcal{S}^{s+s'}$, and $\mathcal{S}^{s+1} \subset \mathcal{S}^s$ because $|k|$ is bounded on the cube.

(3) Line symbols. Let $s \ge -2$ be an integer. A *line symbol of degree* s is a $\sigma \in \mathcal{S}^s$ for which there is a *leading part* $\sigma^h \in C^\infty(\mathbb{R}^3 \setminus \{0\})$ with

$$\sigma^h(tk) = t^s \sigma^h(k) \quad (t > 0), \qquad \sigma^h(-k) = (-1)^s \sigma^h(k), \qquad \sigma - \chi\sigma^h \in \mathcal{S}^{s+1};$$

here $\chi\sigma^h$ is regarded as a function on $\mathbb{T} \setminus \{0\}$, which is possible since χ vanishes near the boundary of the cube. The kernel K with the symbol σ is then called an *elementary line of degree* $n = s + 3$. Sums of line symbols of the same degree are line symbols, and their leading parts add; the product

of two line symbols whose degrees have a sum ≥ -2 is a line symbol, and the leading parts multiply. For products this follows from

$$\sigma_1\sigma_2 - \chi\sigma_1^h\sigma_2^h = (\sigma_1 - \chi\sigma_1^h)\sigma_2 + \chi\sigma_1^h(\sigma_2 - \chi\sigma_2^h) + (\chi^2 - \chi)\sigma_1^h\sigma_2^h,$$

whose last term vanishes for $|k| \leq 1$ and for $|k| \geq 2$. We list some examples.

- $e^{ik\cdot y}$, $y \in \mathbb{Z}^3$: degree 0, leading part 1.
- k_μ and $\overline{\mathsf{k}_\mu}$: degree 1, leading parts ik_μ and $-ik_\mu$.
- $|\mathsf{k}|^{-2}$: degree -2, leading part $|k|^{-2}$. Indeed, on the cube less 0 we write $|\mathsf{k}|^2 = |k|^2(1+\mathsf{w})$ with $\mathsf{w} := |\mathsf{k}|^2/|k|^2 - 1$; the numerator of w, $|\mathsf{k}|^2 - |k|^2$, vanishes to fourth order at 0, so that w is smooth with the bounds of $\mathcal{S}^2$, and $1 + \mathsf{w} \geq \frac{4}{\pi^2}$; then
$$|\mathsf{k}|^{-2} - \chi|k|^{-2} = \chi|k|^{-2}((1+\mathsf{w})^{-1} - 1) + (1-\chi)|\mathsf{k}|^{-2} \in \mathcal{S}^0.$$
- The periodic continuation of $\rho\sigma^h$, where σ^h is as above and $\rho \in C_c^\infty(\mathbb{R}^3)$ satisfies $\operatorname{supp}\rho \subset \{|k| < \pi\}$ and $\rho = 1$ near 0: its leading part is σ^h.

(4) Lines. For an integer $n \geq 2$ let $\mathfrak{K}(n)$ be the set of functions $K : \mathbb{Z}^3 \to \mathbb{C}$ with the following properties.

(K1) K is the kernel with a symbol $\sigma \in L^1(\mathbb{T})$.

(K2) There are a continuous function $\sigma^h : \mathbb{R}^3 \setminus \{0\} \to \mathbb{C}$, homogeneous of degree $n-3$, and a number c with $|\sigma(k) - \chi(k)\sigma^h(k)| \leq c\,|k|^{n-2}\lg|k|$ on $[-\pi,\pi]^3$.

(K3) There are a function $K^h \in C^\infty(\mathbb{R}^3 \setminus \{0\})$, homogeneous of degree $-n$, and numbers c_γ with $|\nabla^\gamma(K - K^h)(x)| \leq c_\gamma|x|^{-n-1-|\gamma|}\lg|x|$ for all multi-indices γ and all $x \in \mathbb{Z}^3$ with $|x| \geq |\gamma| + 1$.

The function K determines the pair (σ^h, K^h), which we call the *leading data* of K. Indeed, σ is determined almost everywhere by K (uniqueness of Fourier coefficients). If σ^h and σ'^h both satisfy (K2), then $|\sigma^h - \sigma'^h|(k) \leq C|k|^{n-2}\lg|k|$ for $0 < |k| \leq 1$, first almost everywhere and then everywhere by continuity; at the points tk with $t \to 0$ this gives $t^{n-3}|\sigma^h(k) - \sigma'^h(k)| \leq Ct^{n-2}|k|^{n-2}\lg(t|k|)$, whence $\sigma^h = \sigma'^h$. If K^h and K'^h both satisfy (K3), we take $\gamma = 0$ and $x = j\mathsf{m}$ with $\mathsf{m} \in \mathbb{Z}^3 \setminus \{0\}$ and $j \to \infty$: from $j^{-n}|K^h(\mathsf{m}) - K'^h(\mathsf{m})| \leq Cj^{-n-1}\lg(j|\mathsf{m}|)$ we get $K^h = K'^h$ on $\mathbb{Z}^3 \setminus \{0\}$, hence on $\mathbb{Q}^3 \setminus \{0\}$ by homogeneity, and everywhere by density and continuity.

The *constants of* K are c, the numbers c_γ, $\|\sigma\|_{L^1}$, and the suprema of $|\sigma^h|$ and of the derivatives of K^h on S^2.

(5) Slowly varying functions. For $D \in \{3, 6\}$, $R \geq 1$ and $g : \mathbb{Z}^D \to \mathbb{C}$ we put

$$\|g\|_{(R)} := \max_{|\gamma|\leq 10} R^{|\gamma|} \sup_\zeta |\nabla^\gamma g(\zeta)|,$$

and if g has finite support, $\check{g}(\kappa) := \sum_\zeta g(\zeta)e^{-i\kappa\cdot\zeta}$. For $G \in C_c^\infty(\mathbb{R}^D)$ we put $\|G\|_{C^{10}} := \max_{|\gamma|\leq 10}\sup|\partial^\gamma G|$.

(6) Majorants. For real $s > -D$ let $\mathcal{H}_s(D)$ be the set of Borel functions $\mathsf{H} : \mathbb{R}^D \to [0,\infty]$ which are homogeneous of degree s and integrable over $\{|\kappa| \leq 1\}$. If $s < 9 - D$ and $\mathsf{H} \in \mathcal{H}_s(D)$, then $\int_{\mathbb{R}^D} \mathsf{H}(\kappa)\lg|\kappa|\,\langle\kappa\rangle^{-10}d\kappa < \infty$, as one sees in polar coordinates. ■

Lemma 7.17 ((K) From a symbol to its kernel)**.** (a) Let $s > -3$ be real, $\sigma \in \mathcal{S}^s$, and let K be the kernel with the symbol σ. Then for every multi-index γ and all $x \in \mathbb{Z}^3$

$$|\nabla^\gamma K(x)| \leq C_\gamma\langle x\rangle^{-3-s-|\gamma|},$$

where C_γ depends on finitely many of the suprema in Def. 7.16(2).

(b) Let $s > -3$ be real and let $\sigma^h \in C^\infty(\mathbb{R}^3 \setminus \{0\})$ be homogeneous of degree s. There is exactly one continuous function K^h on $\mathbb{R}^3 \setminus \{0\}$, homogeneous of degree $-3-s$, such that for every $\rho \in C_c^\infty(\mathbb{R}^3)$ with $\rho = 1$ near 0 and every N

$$\sup_{|x|\geq 1} |x|^N \Big| \int_{\mathbb{R}^3} e^{ik\cdot x} \rho(k) \sigma^h(k)\, d^\circ k - K^h(x) \Big| < \infty.$$

The function K^h is smooth. If $\sigma^h(-k) = \pm\sigma^h(k)$, then $K^h(-x) = \pm K^h(x)$; replacing σ^h by $(ik)^\gamma \sigma^h$ replaces K^h by $\partial^\gamma K^h$; and replacing σ^h by $\sigma^h \circ O^{-1}$, O a rotation, replaces K^h by $K^h \circ O^{-1}$. We write $K^h =: \mathcal{F}_{\neq 0}^{-1} \sigma^h$.

(c) Let K be an elementary line of degree n with the symbol σ, and $K^h := \mathcal{F}_{\neq 0}^{-1}\sigma^h$. Then

$$|\nabla^\gamma (K - K^h)(x)| \leq C_\gamma |x|^{-n-1-|\gamma|} \qquad (|x| \geq |\gamma| + 1),$$

with $C_\gamma = C_\gamma(\gamma, \sigma)$. In particular, if $n \geq 2$, then $K \in \mathfrak{K}(n)$ with the leading data $(\sigma^h, \mathcal{F}_{\neq 0}^{-1}\sigma^h)$.

■

Proof. Step 1: proof of (a). The kernel $\nabla^\gamma K$ has the symbol $\mathsf{k}^\gamma \sigma := \prod_j \mathsf{k}_j^{\gamma_j} \sigma$, which lies in $\mathcal{S}^{s+|\gamma|}$ since $\mathsf{k}_j \in \mathcal{S}^1$; as $s + |\gamma| > -3$, it suffices to treat $\gamma = 0$. Always $|K(x)| \leq \|\sigma\|_{L^1}$, which settles $x = 0$. Let $|x| \geq 1$, and split $\sigma = \sigma' + \sigma''$ with $\sigma' := \chi(|x|k)\sigma$, where $\chi(|x|k)$ vanishes for $|k| \geq 2/|x|$ and is regarded as a function on $\mathbb{T}$. Then

$$\int_{\mathbb{T}} |\sigma'|\, d^\circ k \leq C \int_{|k|\leq 2/|x|} |k|^s \, \mathrm{d}k \leq C|x|^{-3-s}.$$

The function σ'' is smooth and periodic, vanishes for $|k| \leq 1/|x|$, and satisfies $|\partial^\alpha \sigma''(k)| \leq C|k|^{s-|\alpha|}$: a derivative falling on $\chi(|x|k)$ produces a factor $|x|$, and $|x| \leq 2/|k|$ on the support of the derivatives of $\chi(|x|k)$. We choose j with $|x_j| \geq |x|/\sqrt{3}$ and an integer $N > s+3$. After N integrations by parts in k_j, which produce no boundary terms by periodicity,

$$\Big| \int_{\mathbb{T}} e^{ik\cdot x} \sigma''(k)\, d^\circ k \Big| \leq |x_j|^{-N} \int_{\mathbb{T}} |\partial_j^N \sigma''|\, d^\circ k \leq C|x|^{-N} \int_{|k|\geq 1/|x|} |k|^{s-N}\, \mathrm{d}k \leq C|x|^{-3-s}.$$

These bounds involve only the finitely many suprema in Def. 7.16(2) with $|\alpha| \leq N$.

Step 2: uniqueness in (b). If K^h and K'^h both have the property in (b), then, with one and the same ρ, their difference is homogeneous of degree $-3-s$ and $O(|x|^{-N})$ on $\{|x| \geq 1\}$ for every N; taking $N > 3 + s$ and letting $x = t\omega$, $|\omega| = 1$, $t \to \infty$, we see that it vanishes.

Step 3: existence in (b). Let ρ be as in (b), $x \neq 0$, and let N be an integer with $N > \frac{1}{2}(s+3)$. We put

$$K_\rho^h(x) := \int e^{ik\cdot x} \rho \sigma^h\, d^\circ k + \mathsf{E}_\rho(x), \qquad \mathsf{E}_\rho(x) := |x|^{-2N} \int e^{ik\cdot x} (-\Delta_k)^N \big[(1-\rho)\sigma^h\big]\, d^\circ k,$$

with integrals over $\mathbb{R}^3$. The function $(1-\rho)\sigma^h$ is smooth, and its derivatives of order j are $O(|k|^{s-j})$; as $2N > s+3$, the second integral converges absolutely. Since $|x|^2 e^{ik\cdot x} = -\Delta_k e^{ik\cdot x}$, two integrations by parts over balls (Gauss' theorem) lead from N to $N+1$; the boundary terms on the sphere of radius r are $O((1+|x|)\mathsf{r}^2 \cdot \mathsf{r}^{s-2N})$, which tends to 0 as $\mathsf{r} \to \infty$ because $2N > s+3$. Hence E_ρ does not depend on N, so that $|\mathsf{E}_\rho(x)| \leq C_N |x|^{-2N}$ for every N. For the same reason E_ρ is smooth on $\mathbb{R}^3 \setminus \{0\}$: given an order j, we take N with $2N > s + 3 + j$; since $(-\Delta_k)^N[(1-\rho)\sigma^h] = O(|k|^{s-2N})$ and each derivative of $e^{ik\cdot x}$ in x brings a factor of modulus at most $|k|$, the derivatives of order $\leq j$ in x may be taken under the integral sign, with an integrable majorant independent of x, and $|x|^{-2N}$ is smooth on $\mathbb{R}^3 \setminus \{0\}$.

If ρ' is a second function as in (b), then $(\rho - \rho')\sigma^h \in C_c^\infty(\mathbb{R}^3 \setminus \{0\})$, and integration by parts gives $\int e^{ik\cdot x}(-\Delta_k)^N[(\rho-\rho')\sigma^h]\, d^\circ k = |x|^{2N} \int e^{ik\cdot x}(\rho - \rho')\sigma^h\, d^\circ k$; therefore $K_\rho^h = K_{\rho'}^h =: K^h$. This function is smooth on $\mathbb{R}^3 \setminus \{0\}$, and it has the property in (b), since $\int e^{ik\cdot x} \rho\sigma^h\, d^\circ k - K^h(x) = -\mathsf{E}_\rho(x)$.

The substitution $k \to k/t$, $t > 0$, gives $K^h_\rho(tx) = t^{-3-s} K^h_{\rho(\cdot/t)}(x)$, so that K^h is homogeneous of degree $-3-s$. The parity follows from the substitution $k \to -k$ with an even ρ, the behaviour under rotations from $k \to Ok$ with a radial ρ.

Finally, $\mathsf{E}_\rho(x) = \lim_{t\to\infty} \int e^{ik\cdot x}(1-\rho)\sigma^h \rho(k/t)\, d^\circ k$, together with all derivatives in x and locally uniformly in x, as one sees by integrating by parts as above; so we may differentiate under the integrals, and this gives the rule for $(ik)^\gamma \sigma^h$, a smooth function homogeneous of degree $s + |\gamma| > -3$.

Step 4: proof of (c). Let $s := n - 3$. By (b) with $\rho = \chi$ and by (a) applied to $\sigma - \chi\sigma^h \in \mathcal{S}^{s+1}$,

$$K(x) - K^h(x) = \int_{\mathbb{T}} e^{ik\cdot x}(\sigma - \chi\sigma^h)\, d^\circ k + \Big[\int_{\mathbb{R}^3} e^{ik\cdot x}\chi\sigma^h\, d^\circ k - K^h(x)\Big] = O(|x|^{-4-s}) \qquad (x \in \mathbb{Z}^3 \setminus \{0\}).$$

The same argument applies to the line symbol $\mathsf{k}^\gamma\sigma$ of degree $s + |\gamma|$: its kernel is $\nabla^\gamma K$, its leading part is $(ik)^\gamma\sigma^h$, and $\mathcal{F}^{-1}_{\neq 0}[(ik)^\gamma\sigma^h] = \partial^\gamma K^h$ by (b); hence $|\nabla^\gamma K - \partial^\gamma K^h| \le C|x|^{-4-s-|\gamma|}$. By Taylor's formula and the homogeneity of the derivatives of K^h, $|\nabla^\gamma K^h(x) - \partial^\gamma K^h(x)| \le C|x|^{-4-s-|\gamma|}$ for $|x| \ge |\gamma| + 1$, because the points at which K^h and its derivatives are evaluated have modulus at least $|x|/(|\gamma|+1)$. As $-4-s = -n-1$, this is the asserted bound. For $n \ge 2$, (K1) holds as $\sigma \in \mathcal{S}^{n-3} \subset L^1(\mathbb{T})$, (K2) holds as $|\sigma - \chi\sigma^h| \le C|k|^{n-2}$ on the cube, and (K3) holds since $\lg \ge 1$. □

Lemma 7.18 ((S) Sums)**.** Let $D \in \{3, 6\}$ and $R \ge 1$.

(a) For real $a > D$, $\sum_{\zeta\in\mathbb{Z}^D}(R + |\zeta|)^{-a} \le C_a R^{D-a}$. If $y, \zeta \in \mathbb{R}^D$ and $|y| \le R$, then $R + |\zeta - y| \ge \frac{1}{2}(R + |\zeta|)$.

(b) If $F \in C^1(\mathbb{R}^D)$ and $\sum_\zeta \sup_{\zeta+[0,1]^D}(|F| + |\nabla F|) < \infty$, then

$$\Big|\sum_{\zeta\in\mathbb{Z}^D} F(\zeta) - \int_{\mathbb{R}^D} F\Big| \le \sqrt{D}\sum_\zeta \sup_{\zeta+[0,1]^D} |\nabla F|.$$

(c) For real $a, a' \ge 0$ with $a + a' > 3$ and for $x \in \mathbb{Z}^3$,

$$\sum_{z\in\mathbb{Z}^3} \langle z\rangle^{-a}\langle x - z\rangle^{-a'} \le C\langle x\rangle^{-\min\{a,a',a+a'-3\}} \lg\langle x\rangle, \qquad C = C(a, a').$$

■

Proof. (a) For $\zeta' \in \zeta + [0,1]^D$ we have $R + |\zeta'| \le R + |\zeta| + \sqrt{D} \le (1 + \sqrt{D})(R + |\zeta|)$, as $R \ge 1$; so the factor $R + |\zeta|$ changes at most by the factor $1 + \sqrt{D}$ over a cube of side 1, and the sum is at most

$$(1 + \sqrt{D})^a \int_{\mathbb{R}^D} (R + |\zeta|)^{-a}\, \mathrm{d}\zeta = (1 + \sqrt{D})^a R^{D-a} \int_{\mathbb{R}^D} (1 + |\zeta|)^{-a}\, \mathrm{d}\zeta,$$

where the last integral is finite since $a > D$. If $|y| \le R$, then $R + |\zeta - y| \ge R$ and $R + |\zeta - y| \ge R + |\zeta| - |y| \ge |\zeta|$, so that $R + |\zeta - y| \ge \max\{R, |\zeta|\} \ge \frac{1}{2}(R + |\zeta|)$.

(b) The cube $\zeta + [0,1]^D$ has volume 1 and diameter $\sqrt{D}$; by the mean value theorem $|F(\zeta) - \int_{\zeta+[0,1]^D} F| \le \sqrt{D}\sup_{\zeta+[0,1]^D} |\nabla F|$. Summing over ζ, with absolutely convergent series, we obtain the claim.

(c) Let first $|x| < 2$. Then $|z| \le |x - z| + |x| < |x - z| + 2 \le 3\langle x - z\rangle$, so that $\langle x - z\rangle \ge \frac{1}{3}\langle z\rangle$ for all z, and the sum is at most $3^{a'}\sum_z \langle z\rangle^{-a-a'}$, which is finite as $a + a' > 3$. On the other hand, the exponent $m := \min\{a, a', a + a' - 3\}$ is not negative, and $1 \le \langle x\rangle < 2$, $\lg\langle x\rangle \ge 1$; so the right side of the claim is at least $2^{-m}C$, and the claim holds for $|x| < 2$ once $C \ge 2^m 3^{a'}\sum_z \langle z\rangle^{-a-a'}$. Let now $|x| \ge 2$. On $\{|z| \le |x|/2\}$ we have $\langle x - z\rangle \ge |x|/2$ and $\sum_{|z|\le|x|/2}\langle z\rangle^{-a} \le C\max\{1, |x|^{3-a}\}\lg|x|$; the sum over this set is therefore at most $C|x|^{-a'}\max\{1, |x|^{3-a}\}\lg|x| \le C|x|^{-\min\{a',a+a'-3\}}\lg|x|$.

The sum over $\{|x - z| \le |x|/2\}$ is treated in the same way, with a and a' exchanged. On the remaining set we have $|z| \ge |x|/2$ and $|x - z| \ge |x|/2$, hence $|z| \le |x| + |x - z| \le 3|x - z|$ and $\langle x - z\rangle \ge \frac{1}{3}\langle z\rangle$; the sum over it is at most $3^{a'}\sum_{|z|\ge|x|/2}\langle z\rangle^{-a-a'} \le C|x|^{3-a-a'}$. □

Lemma 7.19 ((P) A kernel against a slowly varying function)**.** Let $D \in \{3,6\}$, $R \ge 1$, and let $g : \mathbb{Z}^D \to \mathbb{C}$ vanish outside $\{|\zeta| \le R\}$. We write $d^\circ\kappa := (2\pi)^{-D} d\kappa$.

(a) $|\check g(\kappa)| \le C\, R^D \langle R\kappa\rangle^{-10} \|g\|_{(R)}$ for $\kappa \in [-\pi,\pi]^D$, with a pure constant C.

(b) Let $\hat{\mathsf T} \in L^1(\mathbb{T}^{D/3})$, $0 < c_0 \le 1$, $-D < s < 9-D$, $\mathsf H \in \mathcal{H}_s(D)$ and $\iota \in \{0,1\}$, and assume $|\hat{\mathsf T}(\kappa)| \le \mathsf H(\kappa)(\lg|\kappa|)^\iota$ for $|\kappa| \le c_0$. Then

$$\Big|\int_{\mathbb{T}^{D/3}} \hat{\mathsf T}(\kappa)\, \check g(-\kappa)\, d^\circ\kappa\Big| \le C\, R^{-s} (\lg R)^\iota \|g\|_{(R)}, \qquad C = C(\mathsf H, c_0, \|\hat{\mathsf T}\|_{L^1}).$$

(c) Let, besides the hypotheses of (b), $s < 8-D$. Let $\hat{\mathsf T}^h$ be a Borel function on $\mathbb{R}^D$, homogeneous of degree s, with $|\hat{\mathsf T}^h| \le \mathsf H$, and let $|\hat{\mathsf T} - \hat{\mathsf T}^h| \le \mathsf H' \lg|\kappa|$ on $\{|\kappa| \le c_0\}$ for some $\mathsf H' \in \mathcal{H}_{s+1}(D)$. Let $a \in \mathbb{R}$, $A \ge 0$ and $G \in C_c^\infty(\mathbb{R}^D)$ with $\operatorname{supp} G \subset \{|\zeta| \le 1\}$, and assume $\|g - R^{-a} G(\cdot/R)\|_{(R)} \le A R^{-a-1} \lg R$. Then

$$\Big|\int_{\mathbb{T}^{D/3}} \hat{\mathsf T}(\kappa)\, \check g(-\kappa)\, d^\circ\kappa - R^{-a-s} \int_{\mathbb{R}^D} \hat{\mathsf T}^h(\kappa)\, \hat G(-\kappa)\, d^\circ\kappa\Big| \le C(A + \|G\|_{C^{10}}) R^{-a-s-1} \lg R$$

and $\big|\int_{\mathbb{R}^D} \hat{\mathsf T}^h(\kappa) \hat G(-\kappa)\, d^\circ\kappa\big| \le C\|G\|_{C^{10}}$, where $C = C(\mathsf H, \mathsf H', c_0, \|\hat{\mathsf T}\|_{L^1})$ only, independent of g, G, R, a and A.

■

Proof. Step 1: proof of (a). For $j \in \{1,\dots,D\}$ we have $\sum_\zeta g(\zeta + e_j) e^{-i\kappa\cdot\zeta} = e^{i\kappa_j} \check g(\kappa)$; hence the function $\nabla_j^N g$, which again has finite support, satisfies

$$\sum_\zeta (\nabla_j^N g)(\zeta)\, e^{-i\kappa\cdot\zeta} = (e^{i\kappa_j} - 1)^N \check g(\kappa).$$

For $N \le 10$ the function $\nabla_j^N g$ vanishes outside $\{|\zeta| \le 11R\}$, a set of at most CR^D points, and $|\nabla_j^N g| \le R^{-N}\|g\|_{(R)}$. For $N = 0$ this gives $|\check g| \le CR^D \|g\|_{(R)}$. Choosing j with $|\kappa_j| \ge |\kappa|/\sqrt{D}$ and using $|e^{i\kappa_j} - 1| \ge \frac{2}{\pi}|\kappa_j|$ for $|\kappa_j| \le \pi$, we obtain from $N = 10$ that $|\check g(\kappa)| \le C|\kappa|^{-10} R^{D-10} \|g\|_{(R)}$. The two bounds together give (a).

Step 2: proof of (b). We split the integral at $|\kappa| = c_0$. For $|\kappa| \ge c_0$, (a) gives $|\check g(-\kappa)| \le CR^{D-10} c_0^{-10} \|g\|_{(R)}$; since $D - 10 < -s$ and $R \ge 1$, the integral over $\{|\kappa| \ge c_0\}$ is at most $C\|\hat{\mathsf T}\|_{L^1} c_0^{-10} R^{-s} \|g\|_{(R)}$. For $|\kappa| \le c_0$ we use the hypothesis and (a), and substitute $\kappa = \kappa'/R$; as $\mathsf H(\kappa'/R) = R^{-s}\mathsf H(\kappa')$ and $\lg(|\kappa'|/R) \le \lg|\kappa'| \lg R$,

$$\int_{|\kappa|\le c_0} \mathsf H(\kappa)(\lg|\kappa|)^\iota R^D \langle R\kappa\rangle^{-10}\, \mathrm{d}\kappa \le R^{-s}(\lg R)^\iota \int_{\mathbb{R}^D} \mathsf H(\kappa')(\lg|\kappa'|)^\iota \langle\kappa'\rangle^{-10}\, \mathrm{d}\kappa',$$

and the last integral is finite by Def. 7.16(6), because $s < 9 - D$.

Step 3: proof of (c). Let $g^h := R^{-a} G(\cdot/R)$ on $\mathbb{Z}^D$. It vanishes outside $\{|\zeta| \le R\}$, and $\|g^h\|_{(R)} \le R^{-a}\|G\|_{C^{10}}$, since a difference of order γ of $G(\cdot/R)$ is an integral of a derivative of order $|\gamma|$ of $G(\cdot/R)$, which is bounded by $R^{-|\gamma|}\|G\|_{C^{10}}$. On $\{|\kappa| \le c_0\}$ we have $|\hat{\mathsf T}| \le \mathsf H + \mathsf H' \lg|\kappa| \le \mathsf H + \mathsf H'/|\kappa|$, because $t \lg t \le 1$ for $0 < t \le 1$; and $\mathsf H + \mathsf H'/|\kappa| \in \mathcal{H}_s(D)$, as one sees in polar coordinates. The function $h := g - g^h$ vanishes outside $\{|\zeta| \le R\}$, and assertion (b), applied to h in place of g, with this majorant and with $\iota = 0$, gives

$$\Big|\int \hat{\mathsf T}(\kappa)\, \check h(-\kappa)\, d^\circ\kappa\Big| \le CR^{-s}\|h\|_{(R)} \le CA\, R^{-s-a-1} \lg R.$$

Next, the function $\hat{\mathsf T} - \chi(\cdot/c_0)\hat{\mathsf T}^h$ is integrable on the cube and bounded by $\mathsf H' \lg|\kappa|$ on $\{|\kappa| \le c_0\}$, where $\chi(\kappa/c_0) = 1$. Assertion (b), applied with $s+1$ in place of s (admissible, since $s + 1 < 9 - D$), with the majorant $\mathsf H'$, with $\iota = 1$ and with g^h in place of g, gives

$$\Big|\int (\hat{\mathsf T} - \chi(\cdot/c_0)\hat{\mathsf T}^h)(\kappa)\, \check g^h(-\kappa)\, d^\circ\kappa\Big| \le CR^{-s-1} \lg R\, \|g^h\|_{(R)} \le CR^{-s-1-a} \lg R\, \|G\|_{C^{10}}.$$

It remains to treat $\int \chi(\kappa/c_0)\hat{\mathsf{T}}^h(\kappa)\check{g}^h(-\kappa)\,d^\circ\kappa$. By the Poisson summation formula,

$$\check{g}^h(\kappa) = R^{D-a}\sum_{\mathsf{m}\in\mathbb{Z}^D}\hat{G}(R(\kappa+2\pi\mathsf{m})), \qquad \text{and} \qquad |\hat{G}(\xi)| \le C\langle\xi\rangle^{-10}\|G\|_{C^{10}}.$$

For $|\kappa|\le 2c_0\le 2$ and $\mathsf{m}\neq 0$ we have $|\kappa+2\pi\mathsf{m}|\ge\pi|\mathsf{m}|$; hence the terms with $\mathsf{m}\neq 0$ contribute at most $CR^{D-a-10}\|G\|_{C^{10}}\le CR^{-a-s-1}\|G\|_{C^{10}}$, as $D-10<-s-1$. The term with $\mathsf{m}=0$ gives, after the substitution $\kappa=\kappa'/R$,

$$R^{-a-s}\int_{\mathbb{R}^D}\chi(\kappa'/c_0R)\,\hat{\mathsf{T}}^h(\kappa')\,\hat{G}(-\kappa')\,d^\circ\kappa'.$$

Since $\int_{|\kappa'|\ge c_0R}\mathsf{H}(\kappa')\langle\kappa'\rangle^{-10}d\kappa'\le CR^{s+D-10}\le CR^{-1}$, replacing $\chi(\kappa'/c_0R)$ by 1 changes this expression by at most $CR^{-a-s-1}\|G\|_{C^{10}}$. The bound $|\hat{\mathsf{T}}^h(\kappa')\hat{G}(-\kappa')|\le C\mathsf{H}(\kappa')\langle\kappa'\rangle^{-10}\|G\|_{C^{10}}$ also gives the last assertion of (c). Collecting the estimates we obtain (c). □

Lemma 7.20 ((L) One-loop functions)**.** Let $\mathsf{m}\in\{2,3\}$ and $D:=3(\mathsf{m}-1)$. Let $\sigma_1,\dots,\sigma_{\mathsf{m}}$ be line symbols of degrees $s_1,\dots,s_{\mathsf{m}}$, let $\epsilon_i\in\{\pm1\}$, let $\lambda_i:\mathbb{R}^D\to\mathbb{R}^3$ be linear, and put $l_i(\kappa):=-\epsilon_i\lambda_i(\kappa)$. We assume:

- there is a number $c_\lambda\ge 1$ with $\max_i|l_i(\kappa)|\le c_\lambda|\kappa|\le c_\lambda^2\max_{i<j}|l_i(\kappa)-l_j(\kappa)|$ for all $\kappa\in\mathbb{R}^D$;
- $l_i\neq l_j$ as maps, for $i\neq j$;
- the linear map $(l,\kappa)\mapsto(\epsilon_i l+\lambda_i(\kappa))_{i=1,\dots,\mathsf{m}}$ of $(\mathbb{R}^3)^{\mathsf{m}}=\mathbb{R}^3\times\mathbb{R}^D$ has an integer matrix of determinant ±1;
- if $\mathsf{m}=3$, then $s_i+s_j\ge -3$ for $i\neq j$;
- $\omega:=3+\sum_i s_i\ge -1$.

We put

$$\mathsf{I}(\kappa):=\int_{\mathbb{T}}\prod_{i=1}^{\mathsf{m}}\sigma_i(\epsilon_i l+\lambda_i(\kappa))\,d^\circ l, \qquad \Lambda(\kappa):=1+\sum_{i<j}\log_+\frac{|\kappa|}{|l_i(\kappa)-l_j(\kappa)|} \quad (\kappa\neq 0).$$

The function Λ is homogeneous of degree 0 and even, and $\Lambda(\kappa)=\infty$ exactly where two of the points $l_i(\kappa)$ agree.

(0) The integral defining $\mathsf{I}(\kappa)$ converges absolutely for almost every κ, and $\|\mathsf{I}\|_{L^1(\mathbb{T}^{D/3})}\le\prod_i\|\sigma_i\|_{L^1(\mathbb{T})}$. It converges absolutely for every κ with $0<c_\lambda|\kappa|\le\frac12$ at which the points $l_i(\kappa)$ are pairwise different.

Moreover, there are numbers $c_0\in(0,1]$ and C, depending on the σ_i, ϵ_i, λ_i and c_λ, a polynomial Q in κ of degree $\le\omega$, with $\mathsf{Q}=0$ if $\omega=-1$, and a Borel function I^h on $\mathbb{R}^D$, vanishing at $\kappa=0$ and wherever two of the points $l_i(\kappa)$ agree, with the following properties.

(a) I^h is homogeneous of degree ω, $\mathsf{I}^h(-\kappa)=(-1)^{\omega+1}\mathsf{I}^h(\kappa)$, and $|\mathsf{I}^h(\kappa)|\le C|\kappa|^\omega\Lambda(\kappa)$. It depends only on the σ_i^h, ϵ_i and λ_i, and multilinearly on the σ_i^h. If O is a rotation acting on each of the $\mathsf{m}-1$ arguments in $\mathbb{R}^3$ of κ, and the λ_i commute with this action, then replacing every σ_i^h by $\sigma_i^h\circ O^{-1}$ replaces I^h by $\mathsf{I}^h\circ O^{-1}$. If $\mathsf{m}=2$, then I^h is continuous on $\mathbb{R}^D\setminus\{0\}$.

(b) $|\mathsf{I}(\kappa)-\mathsf{Q}(\kappa)-\mathsf{I}^h(\kappa)|\le C|\kappa|^{\omega+1}\lg|\kappa|\,\Lambda(\kappa)$ for $0<|\kappa|\le c_0$ whenever the points $l_i(\kappa)$ are pairwise different.

(c) If $\omega\ge 1$, then $\prod_i\sigma_i(\epsilon_i l)$ is integrable over $\mathbb{T}$, and $\mathsf{Q}(0)=\int_{\mathbb{T}}\prod_i\sigma_i(\epsilon_i l)\,d^\circ l$.

(d) (Uniqueness.) Let $\mathsf{K}^\sharp\subset\mathbb{R}^D\setminus\{0\}$ be open and non-empty, with $t\mathsf{K}^\sharp=\mathsf{K}^\sharp$ for every real $t\neq 0$, such that the l_i are pairwise different at every point of $\mathsf{K}^\sharp$. Let $\mathsf{Q}^\sharp$ be a polynomial of degree $\le\omega$, with $\mathsf{Q}^\sharp=0$ if $\omega=-1$, and let $\mathsf{I}^\sharp$ be a function on $\mathsf{K}^\sharp$, homogeneous of degree ω, with $\mathsf{I}^\sharp(-\kappa)=(-1)^{\omega+1}\mathsf{I}^\sharp(\kappa)$. If $t^{-\omega}|\mathsf{I}(t\kappa)-\mathsf{Q}^\sharp(t\kappa)-\mathsf{I}^\sharp(t\kappa)|\to 0$ as $t\downarrow 0$ for every $\kappa\in\mathsf{K}^\sharp$, then $\mathsf{Q}^\sharp=\mathsf{Q}$, and $\mathsf{I}^\sharp=\mathsf{I}^h$ on $\mathsf{K}^\sharp$.

Thus Q and I^h are determined by I near 0, and they are independent of c_λ; we call them *the polynomial* and *the homogeneous part* of I. If $\tilde{\mathsf{I}}$ is a function of the same kind, with data

$\tilde\sigma_i$, $\tilde\epsilon_i$, $\tilde\lambda_i$ and the same m and ω, and if $\tilde{\mathsf{I}}(\kappa) = \mathsf{I}(-\kappa)$ for all small $\kappa \neq 0$ at which both families of points, the $l_i(\kappa)$ and the $\tilde l_i(\kappa)$, are pairwise different, then the polynomial of $\tilde{\mathsf{I}}$ is the polynomial $\kappa \mapsto \mathsf{Q}(-\kappa)$, and the homogeneous part of $\tilde{\mathsf{I}}$ equals $\mathsf{I}^h(-\kappa)$ at every $\kappa \neq 0$ at which both families of points are pairwise different.

(e) $|\kappa|^s \Lambda(\kappa) \in \mathcal{H}_s(D)$ for every real $s > -D$.

■

Proof. Step 0: two bounds. Let $t_1, \dots, t_{\mathsf{m}} > -3$ be real, with $t_i + t_j \geq -3$ for $i \neq j$ if $\mathsf{m} = 3$; let $\kappa \neq 0$ and $\varepsilon := c_\lambda|\kappa|$. We first show

$$\int_{|l|\leq 2\varepsilon} \prod_i |l - l_i(\kappa)|^{t_i} \,\mathrm{d}l \leq C\varepsilon^{3+\sum_i t_i} \Lambda(\kappa). \tag{7.xxv}$$

By the scaling $l \to \varepsilon l$, $l_i(\kappa) \to \varepsilon l_i(\kappa)$ it suffices to treat $\varepsilon = 1$; then $|l_i| \leq 1$ and $\max_{i<j} |l_i - l_j| \geq c_\lambda^{-2} =: 2c$, and, the scaled distances being $|l_i - l_j| = |l_i(\kappa) - l_j(\kappa)|/(c_\lambda|\kappa|)$, we have $1 + \log \frac{1}{|l_i - l_j|} \leq (1 + \log c_\lambda)\Lambda(\kappa)$ for $i \neq j$. If $\mathsf{m} = 2$, the two singularities lie at distance at least $2c$ from each other, each of them is integrable, and the integral is at most C. Let $\mathsf{m} = 3$, and number the points so that $d := |l_1 - l_2|$ is the least of the three distances. If $d \geq c/2$, the three singularities are apart from each other, and again the integral is at most C.

If $d < c/2$, then $|l_3 - l_1|, |l_3 - l_2| \geq c$, since one of them is at least $2c$ and they differ at most by d. On $\{|l - l_3| \leq c/2\}$ the integrand is at most $C|l - l_3|^{t_3}$; off this set it is at most $C|l - l_1|^{t_1}|l - l_2|^{t_2}$. The integral of the latter over $\{|l - l_1| \leq 2d\}$ is at most $Cd^{3+t_1+t_2} \leq C$, as $t_1 + t_2 \geq -3$ and $d < 1$; over $\{2d \leq |l - l_1| \leq 3\}$, where $\frac12|l - l_1| \leq |l - l_2| \leq \frac32|l - l_1|$, it is at most $C\int_{2d}^3 \varrho^{2+t_1+t_2}\,\mathrm{d}\varrho \leq C(1 + \log\frac1d) \leq C\Lambda(\kappa)$. This proves eq. 7.xxv.

We also prove (e) here. The function $|\kappa|^s\Lambda(\kappa)$ is non-negative, Borel and homogeneous of degree s, and, as $s > -D$, it is integrable over $\{|\kappa| \leq 1\}$ if Λ is integrable over the sphere $\{|\kappa| = 1\}$. For $i < j$ the linear map $l_i - l_j$ is not 0, so there are a vector $v \in \mathbb{R}^D$ with $|v| = 1$ and a number $c' > 0$ with $|l_i(\kappa) - l_j(\kappa)| \geq c'|v \cdot \kappa|$; and $\log_+(1/(c'|v \cdot \kappa|))$ is integrable over the sphere.

Next let $\tau_i \in \mathcal{S}^{t_i}$, $\mathsf{A}(l, \kappa) := \prod_i \tau_i(\epsilon_i l + \lambda_i(\kappa))$, and $0 < \varepsilon \leq \frac12$. For $l \in [-\pi, \pi]^3$ with $|l| \geq 2\varepsilon$ and for $|\kappa'| \leq |\kappa|$ we have $\epsilon_i l + \lambda_i(\kappa') = \epsilon_i(l - l_i(\kappa'))$ and $|l_i(\kappa')| \leq \varepsilon \leq \frac12|l|$, so that $\frac14|l| \leq |\epsilon_i l + \lambda_i(\kappa')|_{\mathbb{T}} \leq \frac32|l|$. With the chain and Leibniz rules this gives

$$|\partial_\kappa^\alpha \mathsf{A}(l, \kappa')| \leq C_\alpha |l|^{\sum_i t_i - |\alpha|} \qquad (|\kappa'| \leq |\kappa|). \tag{7.xxvi}$$

The same bound holds on $\mathbb{R}^3$, for all l with $|l| \geq 2\varepsilon$, if the τ_i are smooth on $\mathbb{R}^3 \setminus \{0\}$ and homogeneous of degree t_i.

We now prove (0). By Fubini's theorem and the substitution $(l, \kappa) \to (\epsilon_i l + \lambda_i(\kappa))_i$, which induces a measure-preserving bijection of $\mathbb{T}^{\mathsf{m}}$ since its matrix is an integer matrix of determinant ±1, we get $\int_{\mathbb{T}^{D/3}} \int_{\mathbb{T}} \prod_i |\sigma_i(\epsilon_i l + \lambda_i(\kappa))|\, d^\circ l\, d^\circ\kappa = \prod_i \|\sigma_i\|_{L^1(\mathbb{T})}$; this gives the first two assertions. For $0 < c_\lambda|\kappa| \leq \frac12$ with pairwise different $l_i(\kappa)$ we have $\Lambda(\kappa) < \infty$; the integral over $\{|l| \leq 2\varepsilon\}$ converges by eq. 7.xxv with $t_i = s_i$, since $|\sigma_i(\epsilon_i l + \lambda_i(\kappa))| \leq C|l - l_i(\kappa)|^{s_i}$ there, and the integral over $\{|l| \geq 2\varepsilon\}$ by eq. 7.xxvi with $\alpha = 0$.

Step 1: crude expansion. Let $\tau_i \in \mathcal{S}^{t_i}$ with integers $t_i \geq -2$ as in Step 0, $\omega' := 3 + \sum_i t_i \geq 0$, A as in Step 0, and $\mathsf{J}(\kappa) := \int_{\mathbb{T}} \mathsf{A}(l, \kappa)\, d^\circ l$. We claim: there is a polynomial Q' of degree $< \omega'$ with

$$|\mathsf{J}(\kappa) - \mathsf{Q}'(\kappa)| \leq C|\kappa|^{\omega'} \lg|\kappa|\, \Lambda(\kappa) \qquad (0 < c_\lambda|\kappa| \leq \tfrac12),$$

and $\mathsf{Q}'(0) = \int_{\mathbb{T}} \mathsf{A}(l, 0)\, d^\circ l$ if $\omega' \geq 1$. Indeed, the integral over $\{|l| \leq 2\varepsilon\}$ is $O(\varepsilon^{\omega'}\Lambda(\kappa))$ by eq. 7.xxv. For $|l| \geq 2\varepsilon$ we expand by Taylor's formula in κ at 0,

$$\mathsf{A}(l, \kappa) = \sum_{|\alpha|<\omega'} \kappa^\alpha \mathsf{A}_\alpha(l) + \mathsf{A}_{\mathrm{rem}}(l, \kappa), \qquad |\mathsf{A}_{\mathrm{rem}}(l, \kappa)| \leq C|\kappa|^{\omega'}|l|^{-3}$$

by eq. 7.xxvi, so that $\int_{|l|\ge 2\varepsilon} |\mathsf{A}_{\rm rem}|\, d^\circ l = O(|\kappa|^{\omega'} \lg|\kappa|)$. Moreover $|\mathsf{A}_\alpha(l)| \le C|l|^{\omega'-3-|\alpha|}$, which is integrable over $\mathbb{T}$ for $|\alpha| < \omega'$, and $|\kappa|^{|\alpha|}\int_{|l|\le 2\varepsilon}|\mathsf{A}_\alpha|\, d^\circ l \le C|\kappa|^{\omega'}$. Hence the claim holds with $\mathsf{Q}'(\kappa) := \sum_{|\alpha|<\omega'} \kappa^\alpha \int_{\mathbb{T}} \mathsf{A}_\alpha\, d^\circ l$.

Step 2: leading parts. Let $\mathsf{A}^h(l,\kappa) := \prod_i \sigma_i^h(\epsilon_i l + \lambda_i(\kappa))$ on $\mathbb{R}^3\times\mathbb{R}^D$. It is homogeneous of degree $\omega-3$, and $\mathsf{A}^h(-l,-\kappa) = (-1)^{\omega+1}\mathsf{A}^h(l,\kappa)$. For $|l|\ge 2\varepsilon$ we expand by Taylor's formula in κ at 0,

$$\mathsf{A}^h(l,\kappa) = \sum_{j\le\omega} \mathsf{A}^h_j(l;\kappa) + \mathsf{A}^h_{\rm rem}(l;\kappa);$$

here A^h_j is defined for $l\ne 0$, homogeneous of degree j in κ and of degree $\omega-3-j$ in l, with $\mathsf{A}^h_j(-l;\kappa) = (-1)^{\omega+1+j}\mathsf{A}^h_j(l;\kappa)$, and $|\mathsf{A}^h_{\rm rem}(l;\kappa)| \le C|\kappa|^{\omega+1}|l|^{-4}$ by the bound eq. 7.xxvi for homogeneous factors. With $\varepsilon = c_\lambda|\kappa|$ we define, where $\kappa\ne 0$ and the points $l_i(\kappa)$ are pairwise different,

$$\mathsf{I}^h(\kappa) := \int_{|l|\le 2\varepsilon} \mathsf{A}^h\, d^\circ l + \int_{|l|>2\varepsilon} \mathsf{A}^h_{\rm rem}\, d^\circ l - \sum_{j<\omega}\int_{|l|\le 2\varepsilon} \mathsf{A}^h_j\, d^\circ l,$$

and $\mathsf{I}^h := 0$ elsewhere, that is, at 0 and on a finite union of linear subspaces. The substitution $l\to tl$ gives the homogeneity, $l\to -l$ the parity, and $l\to Ol$ the behaviour under rotations. Each of the three terms is $O(|\kappa|^\omega\Lambda(\kappa))$, by eq. 7.xxv and eq. 7.xxvi. The function I^h is Borel, as an absolutely convergent integral of a Borel function of (l,κ). At fixed c_λ it is built from the σ_i^h, ϵ_i and λ_i alone, and it is multilinear in the σ_i^h.

Step 2′: continuity for $\mathsf{m}=2$. Here $l_1(\kappa)\ne l_2(\kappa)$ for all $\kappa\ne 0$, and by homogeneity it suffices to prove continuity on $\{|\kappa|=1\}$, where $\varepsilon = c_\lambda$ is fixed. The third term in the definition of I^h is a polynomial in κ: $\mathsf{A}^h_j(l;\kappa) = \sum_{|\alpha|=j}\kappa^\alpha \mathsf{A}^h_\alpha(l)$ with $|\mathsf{A}^h_\alpha(l)| \le C|l|^{\omega-3-j}$, a function integrable near 0 for $j<\omega$, so that the coefficients $\int_{|l|\le 2\varepsilon}\mathsf{A}^h_\alpha\, d^\circ l$ are finite. In the second term the integrand is continuous in κ for every l with $|l|>2\varepsilon$ and has the integrable majorant $C|l|^{-4}$; dominated convergence applies.

In the first term, $\mathsf{A}^h = \sigma_1^h(\epsilon_1(l-l_1(\kappa)))\,\sigma_2^h(\epsilon_2(l-l_2(\kappa)))$, and by the first hypothesis of the lemma $|l_1(\kappa)-l_2(\kappa)| \ge c_\lambda^{-1} =: 4\delta$. We fix κ_0 with $|\kappa_0|=1$, put $\psi_1(l) := \theta(|l-l_1(\kappa_0)|/2\delta)$ and $\psi_2 := 1-\psi_1$, and consider κ with $|\kappa|=1$ so close to κ_0 that $|l_i(\kappa)-l_i(\kappa_0)| \le \frac12\delta$ for $i=1,2$. As $\theta=1$ on $[0,\frac12]$ and $\theta=0$ on $[1,\infty)$, we have $\psi_1=1$ where $|l-l_1(\kappa_0)|\le\delta$ and $\psi_1=0$ where $|l-l_1(\kappa_0)|\ge 2\delta$. Hence, where $\psi_1\ne 0$,

$$|l-l_2(\kappa)| \ge |l_1(\kappa_0)-l_2(\kappa_0)| - |l-l_1(\kappa_0)| - |l_2(\kappa_0)-l_2(\kappa)| \ge 4\delta - 2\delta - \tfrac12\delta = \tfrac32\delta,$$

and where $\psi_2\ne 0$, that is, where $|l-l_1(\kappa_0)|>\delta$, we have $|l-l_1(\kappa)|\ge \delta-\frac12\delta = \frac12\delta$. We split the first term by the factors ψ_1 and ψ_2. In the integral with the factor ψ_1 we substitute $l\to l+l_1(\kappa)$; since $|l_1(\kappa)|\le c_\lambda|\kappa| = \varepsilon$, it becomes the integral over $\{|l+l_1(\kappa)|\le 2\varepsilon\}\subset\{|l|\le 3\varepsilon\}$ of

$$\psi_1(l+l_1(\kappa))\,\sigma_1^h(\epsilon_1 l)\,\sigma_2^h(\epsilon_2(l+l_1(\kappa)-l_2(\kappa))).$$

The domain of integration moves with κ; we extend the integrand by 0 off this domain and integrate over $\mathbb{R}^3$. Where $\psi_1(l+l_1(\kappa))\ne 0$, the last factor is evaluated at points of modulus in $[\frac32\delta, 4\varepsilon]$, a compact set not containing 0, on which σ_2^h is bounded and continuous. The singular factor $\sigma_1^h(\epsilon_1 l)$ does not depend on κ, and $|\sigma_1^h(\epsilon_1 l)| \le C|l|^{s_1}$ by homogeneity. Hence the extended integrand has the majorant $C|l|^{s_1}$, which does not depend on κ and is integrable over $\{|l|\le 3\varepsilon\}$ as $s_1\ge -2$; and as $\kappa\to\kappa_0$ it converges at every $l\ne 0$ with $|l+l_1(\kappa_0)|\ne 2\varepsilon$, that is, outside a null set. So dominated convergence applies. The integral with the factor ψ_2 is treated in the same way, with the substitution $l\to l+l_2(\kappa)$ and the majorant $C|l|^{s_2}$.

Step 2″: comparison. Let $\mathsf{X}(l,\kappa) := \prod_i \chi(\epsilon_i l + \lambda_i(\kappa))$, let $\chi_*(l) := \chi(l)^{\mathsf{m}}$, which is radial, and let $\mathsf{I}^{(0)}(\kappa) := \int_{\mathbb{R}^3}\mathsf{X}\mathsf{A}^h\, d^\circ l$ for $\varepsilon\le\frac14$. The function $\mathsf{X}-\chi_*$ is smooth and vanishes for $|l|\le\frac12$ and for $|l|\ge 3$; therefore $\int(\mathsf{X}-\chi_*)\mathsf{A}^h\, d^\circ l$ is smooth in κ and vanishes at $\kappa=0$, so it equals a polynomial of

degree $\le \omega$ without constant term up to $O(|\kappa|^{\omega+1})$. Moreover

$$\int \chi_* \mathsf{A}^h \, d^\circ l = \int_{|l|\le 2\varepsilon} \mathsf{A}^h \, d^\circ l + \int_{|l|>2\varepsilon} \mathsf{A}^h_{\mathrm{rem}} \, d^\circ l - \int (1-\chi_*) \mathsf{A}^h_{\mathrm{rem}} \, d^\circ l + \sum_{j\le\omega} \int_{|l|>2\varepsilon} \chi_* \mathsf{A}^h_j \, d^\circ l.$$

The third term is $O(|\kappa|^{\omega+1})$. For $j<\omega$ we have $\int_{|l|>2\varepsilon} \chi_* \mathsf{A}^h_j = \int \chi_* \mathsf{A}^h_j - \int_{|l|\le 2\varepsilon} \mathsf{A}^h_j$, and $\int \chi_* \mathsf{A}^h_j \, d^\circ l$ is a homogeneous polynomial of degree j in κ. For $j=\omega$ the function $\mathsf{A}^h_\omega(\cdot;\kappa)$ is odd, while χ_* and the domain $\{|l|>2\varepsilon\}$ are even, so that this term vanishes. Consequently

$$\mathsf{I}^{(0)} = \mathsf{Q}^{(0)} + \mathsf{I}^h + O(|\kappa|^{\omega+1}),$$

with a polynomial $\mathsf{Q}^{(0)}$ of degree $\le\omega$; if $\omega\ge 1$, then $\mathsf{Q}^{(0)}(0) = \int \chi_* \mathsf{A}^h_0 \, d^\circ l = \int \mathsf{X}(l,0)\mathsf{A}^h(l,0)\, d^\circ l$.

Step 3: proof of (b) and (c). We write $\sigma_i = \chi\sigma_i^h + \sigma_i^\flat$ with $\sigma_i^\flat \in \mathcal{S}^{s_i+1}$ and multiply out the product in the definition of I. For $\varepsilon \le \frac14$ the term in which all factors are leading is $\mathsf{I}^{(0)}$. Every other term is a function J as in Step 1, with $t_i \in \{s_i, s_i+1\}$ and $\omega' \ge \omega+1 \ge 0$, since $\chi\sigma_i^h \in \mathcal{S}^{s_i}$; the terms of degree $>\omega$ of its polynomial Q' are $O(|\kappa|^{\omega+1})$. This proves (b), with Q the sum of $\mathsf{Q}^{(0)}$ and of the polynomials Q' cut at degree ω, and with $c_0 := \frac{1}{4c_\lambda}$, which lies in $(0,\frac14]$ as $c_\lambda \ge 1$; for $0<|\kappa|\le c_0$ we have $\varepsilon \le \frac14$, as Step 2″ requires, and $c_\lambda|\kappa| \le \frac12$, as Step 1 requires.

If $\omega \ge 1$, then $|\prod_i \sigma_i(\epsilon_i l)| \le C|l|^{\omega-3}$ is integrable, and adding the constant terms we obtain $\mathsf{Q}(0) = \int_{\mathbb{T}} \prod_i (\chi\sigma_i^h + \sigma_i^\flat)(\epsilon_i l)\, d^\circ l = \int_{\mathbb{T}} \prod_i \sigma_i(\epsilon_i l)\, d^\circ l$, which is (c).

Step 4: uniqueness. Let $\kappa \in \mathsf{K}^\sharp$. By (b), and since $\Lambda(t\kappa) = \Lambda(\kappa) < \infty$,

$$t^{-\omega}\big|\mathsf{I}(t\kappa) - \mathsf{Q}(t\kappa) - \mathsf{I}^h(t\kappa)\big| \le Ct|\kappa|^{\omega+1} \lg(t|\kappa|)\, \Lambda(\kappa) \to 0 \qquad (t\downarrow 0).$$

Subtracting the hypothesis of (d) and writing Q_j, $\mathsf{Q}_j^\sharp$ for the homogeneous terms of degree j, we get

$$\sum_{j<\omega} t^{j-\omega}(\mathsf{Q}_j - \mathsf{Q}_j^\sharp)(\kappa) + (\mathsf{Q}_\omega - \mathsf{Q}_\omega^\sharp)(\kappa) + (\mathsf{I}^h - \mathsf{I}^\sharp)(\kappa) \to 0.$$

Beginning with the least j, we conclude $(\mathsf{Q}_j - \mathsf{Q}_j^\sharp)(\kappa) = 0$ for $j<\omega$, and then $(\mathsf{Q}_\omega - \mathsf{Q}_\omega^\sharp)(\kappa) + (\mathsf{I}^h - \mathsf{I}^\sharp)(\kappa) = 0$. The same holds at $-\kappa \in \mathsf{K}^\sharp$, where the two terms acquire the factors $(-1)^\omega$ and $(-1)^{\omega+1}$; hence also $(\mathsf{Q}_\omega - \mathsf{Q}_\omega^\sharp)(\kappa) - (\mathsf{I}^h - \mathsf{I}^\sharp)(\kappa) = 0$, and both terms vanish. (If $\omega=-1$, only the term $\mathsf{I}^h - \mathsf{I}^\sharp$ is present.) Polynomials which vanish on a non-empty open set vanish identically; so $\mathsf{Q}^\sharp = \mathsf{Q}$, and $\mathsf{I}^\sharp = \mathsf{I}^h$ on $\mathsf{K}^\sharp$.

The set of all $\kappa \ne 0$ at which the $l_i(\kappa)$ are pairwise different is such a set $\mathsf{K}^\sharp$, being the complement of finitely many proper linear subspaces, and $\mathsf{I}^h = 0$ outside it; therefore the pair $(\mathsf{Q}, \mathsf{I}^h)$ with (a) and (b) is unique and independent of c_λ. As, by Step 2, at fixed c_λ the function I^h is built from the σ_i^h, ϵ_i and λ_i alone, it depends only on these data.

For the last assertion of (d) let $\tilde{\mathsf{Q}}$ and $\tilde{\mathsf{I}}^h$ be the polynomial and the homogeneous part of $\tilde{\mathsf{I}}$, and let $\mathsf{K}^\sharp$ be the set of all $\kappa \ne 0$ at which the $l_i(\kappa)$ are pairwise different and so are the $\tilde{l}_i(\kappa)$. It is again the complement of finitely many proper linear subspaces, hence open, non-empty and invariant under $\kappa \to t\kappa$ for real $t \ne 0$. We put $\mathsf{Q}^\sharp(\kappa) := \tilde{\mathsf{Q}}(-\kappa)$ and $\mathsf{I}^\sharp(\kappa) := \tilde{\mathsf{I}}^h(-\kappa)$ on $\mathsf{K}^\sharp$; they have the degree, the homogeneity and the parity required in (d). For $\kappa \in \mathsf{K}^\sharp$ and small $t>0$ the hypothesis gives $\mathsf{I}(t\kappa) = \tilde{\mathsf{I}}(-t\kappa)$, and (b) for the data of $\tilde{\mathsf{I}}$, at the point $-t\kappa$, gives

$$t^{-\omega}\big|\mathsf{I}(t\kappa) - \mathsf{Q}^\sharp(t\kappa) - \mathsf{I}^\sharp(t\kappa)\big| = t^{-\omega}\big|\tilde{\mathsf{I}}(-t\kappa) - \tilde{\mathsf{Q}}(-t\kappa) - \tilde{\mathsf{I}}^h(-t\kappa)\big| \to 0 \qquad (t\downarrow 0).$$

By the first part of (d), $\mathsf{Q}^\sharp = \mathsf{Q}$ and $\mathsf{I}^\sharp = \mathsf{I}^h$ on $\mathsf{K}^\sharp$. Thus $\tilde{\mathsf{Q}}(\kappa) = \mathsf{Q}(-\kappa)$ for all $\kappa \in \mathbb{R}^D$, as an identity of polynomials, and, $\mathsf{K}^\sharp$ being symmetric, $\tilde{\mathsf{I}}^h(\kappa) = \mathsf{I}^h(-\kappa)$ for all $\kappa \in \mathsf{K}^\sharp$. □

Lemma 7.21 ((D) Differences of an asymptotic behaviour)**.** Let $n>0$ be real and $f : \mathbb{Z}^3 \to \mathbb{C}$. Assume that for every multi-index γ there are a continuous function H_γ on $\mathbb{R}^3 \setminus \{0\}$, homogeneous of degree $-n-|\gamma|$, and numbers c_γ, r_γ with

$$|\nabla^\gamma f(x) - \mathsf{H}_\gamma(x)| \le c_\gamma |x|^{-n-|\gamma|-1} \lg|x| \qquad (x \in \mathbb{Z}^3\setminus\{0\},\ |x| \ge r_\gamma).$$

Then $\mathsf{H}_0 \in C^\infty(\mathbb{R}^3 \setminus \{0\})$, $\partial^\gamma \mathsf{H}_0 = \mathsf{H}_\gamma$ for every γ, and

$$|\nabla^\gamma (f - \mathsf{H}_0)(x)| \le c'_\gamma |x|^{-n-|\gamma|-1} \lg|x| \qquad (|x| \ge \max\{r_\gamma, |\gamma|+1\}),$$

with $c'_\gamma = c'_\gamma(n, \gamma, c_\gamma, \mathsf{H}_0)$. ■

Proof. Let $y \in \mathbb{R}^3 \setminus \{0\}$, $\mu \in \{1,2,3\}$, and let $t > 0$ be so small that $y + se_\mu \ne 0$ for $0 \le s \le t$. For $R > 0$ let $x_R := (\lfloor R y_\nu \rfloor)_\nu \in \mathbb{Z}^3$ and $N_R := \lfloor Rt \rfloor$. Then

$$\nabla^\gamma f(x_R + N_R e_\mu) - \nabla^\gamma f(x_R) = \sum_{j < N_R} \nabla^{\gamma + e_\mu} f(x_R + j e_\mu).$$

We multiply by $R^{n+|\gamma|}$ and let $R \to \infty$. By the hypothesis, the homogeneity and the continuity of H_γ, the left side tends to $\mathsf{H}_\gamma(y + te_\mu) - \mathsf{H}_\gamma(y)$. The points $x_R + je_\mu$, $j < N_R$, have modulus at least cR for some $c > 0$; by the hypothesis for $\gamma + e_\mu$ and the homogeneity of $\mathsf{H}_{\gamma+e_\mu}$, the right side equals

$$\frac{1}{R} \sum_{j < N_R} \mathsf{H}_{\gamma+e_\mu}\big((x_R + je_\mu)/R\big) + O(R^{-1} \lg R),$$

and this tends to $\int_0^t \mathsf{H}_{\gamma+e_\mu}(y + se_\mu)\,\mathrm{d}s$: it is a Riemann sum, up to the shift $x_R/R - y = O(R^{-1})$ and the uniform continuity of $\mathsf{H}_{\gamma+e_\mu}$ on a compact neighbourhood of $\{y + se_\mu : 0 \le s \le t\}$. Hence $\mathsf{H}_\gamma(y + te_\mu) - \mathsf{H}_\gamma(y) = \int_0^t \mathsf{H}_{\gamma+e_\mu}(y + se_\mu)\,\mathrm{d}s$ for small $t > 0$, and, applied at the point $y - te_\mu$, the corresponding identity holds for small $t < 0$. Therefore $\partial_\mu \mathsf{H}_\gamma$ exists and equals the continuous function $\mathsf{H}_{\gamma+e_\mu}$. By induction on $|\gamma|$, all partial derivatives of H_0 exist, are continuous, and $\partial^\gamma \mathsf{H}_0 = \mathsf{H}_\gamma$; thus $\mathsf{H}_0 \in C^\infty(\mathbb{R}^3 \setminus \{0\})$.

Finally, Taylor's formula and the homogeneity of the derivatives of order $|\gamma| + 1$ of H_0 give $|\nabla^\gamma \mathsf{H}_0(x) - \partial^\gamma \mathsf{H}_0(x)| \le C|x|^{-n-|\gamma|-1}$ for $|x| \ge |\gamma| + 1$, with C depending on n, γ and H_0; together with the hypothesis this proves the bound for $\nabla^\gamma(f - \mathsf{H}_0)$. □

Lemma 7.22 ((E) Chains (convolutions of lines), bubbles)**.** For $X, Y : \mathbb{Z}^3 \to \mathbb{C}$ we write $(X * Y)(u) := \sum_z X(z) Y(u - z)$, and $r := |u|$.

(a) (Bound.) Let X, Y be kernels with symbols $\hat X, \hat Y \in L^1(\mathbb{T})$; let $a_X, a_Y \in (-3, 3)$ and $n_X, n_Y \ge 0$ with $n_X + n_Y > 3$. Assume $|\hat X(k)| \le c|k|^{a_X} \lg|k|$ on $[-\pi, \pi]^3$ and $|\nabla^\gamma X(x)| \le c\langle x\rangle^{-n_X - |\gamma|}$ for $|\gamma| \le 10$ and all x, and the same for Y with a_Y, n_Y. Then the sum defining $(X * Y)(u)$ converges absolutely, and for $r \ge 1$

$$|(X * Y)(u)| \le C\big(r^{-a_X - n_Y} + r^{-a_Y - n_X} + r^{3 - n_X - n_Y}\big) \lg r, \qquad C = C(c, a_X, a_Y, n_X, n_Y).$$

(b) (Class.) Let $X \in \mathfrak{K}(n_X)$ and $Y \in \mathfrak{K}(n_Y)$ with $n_X, n_Y \le 4$ and $n_X + n_Y \ge 5$; we write $\hat X$ and $\hat Y$ for their symbols, as in (K1), and $(\hat X^h, X^h)$ and $(\hat Y^h, Y^h)$ for their leading data in the sense of Def. 7.16(4). Then $X * Y \in \mathfrak{K}(n_X + n_Y - 3)$, with the symbol $\hat X \hat Y$. Its leading data consist of $\hat X^h \hat Y^h$ and of a function which depends only on the leading data of X and Y, bilinearly. The constants of $X * Y$ depend only on those of X and Y. The leading data are rotation covariant: if the leading data of X and of Y are composed with O^{-1}, O a rotation, then so are those of $X * Y$.

(c) (Bubbles.) Let K_a, K_b be elementary lines of degrees $n_\mathrm{a}, n_\mathrm{b} \in \{1,2,3\}$, with symbols σ_a, σ_b; let $X := K_\mathrm{a} K_\mathrm{b}$ (pointwise product) and $n := n_\mathrm{a} + n_\mathrm{b}$. Then X is the kernel with the symbol

$$\mathsf{I}(k) = \int_{\mathbb{T}} \sigma_\mathrm{a}(k - l)\, \sigma_\mathrm{b}(l)\, d^\circ l,$$

a function as in Lem. 7.20 with $\mathsf{m} = 2$ and $\omega = n - 3$. With its polynomial Q and its homogeneous part I^h we put

$$\tilde{\mathsf{Q}}(k) := \mathsf{Q}(0) + \sum_j \partial_j \mathsf{Q}(0) \sin k_j,$$

and we let q be the kernel with the symbol $\tilde{\mathsf{Q}}$; it vanishes outside $\{0, \pm e_1, \pm e_2, \pm e_3\}$. Then $|\nabla^\gamma X(x)| \le C_\gamma \langle x\rangle^{-n-|\gamma|}$ for all x and γ, and $|\mathsf{I}(k) - \tilde{\mathsf{Q}}(k)| \le C|k|^{\min\{n-3,2\}}$ on $[-\pi,\pi]^3$, where C_γ and C depend on γ, σ_{a}, σ_{b}. If $n \le 4$, then $X - \mathsf{q} \in \mathfrak{K}(n)$, with the leading data $(\mathsf{I}^h, K^h_{\mathrm{a}} K^h_{\mathrm{b}})$, where $K^h_{\mathrm{a}} = \mathcal{F}^{-1}_{\neq 0}\sigma^h_{\mathrm{a}}$ and $K^h_{\mathrm{b}} = \mathcal{F}^{-1}_{\neq 0}\sigma^h_{\mathrm{b}}$.

(d) (Uniqueness.) Let $K \in \mathfrak{K}(n)$ with $n \le 6$ and leading data (σ^h, K^h). If $\sigma^h = 0$, then $K^h = 0$. Hence, within $\mathfrak{K}(n)$, K^h is determined by σ^h, and depends linearly on it.

■

Proof. Step 0: splitting. Let $r \ge 8$; for $1 \le r < 8$ the bound in (a) follows from Lem. 7.18(c). For $z \in \mathbb{R}^3$ we put $\theta_1(z) := \theta(4|z|/r)$ and $\theta_2(z) := \theta(4|u-z|/r)$. These are smooth functions on $\mathbb{R}^3$, as $\theta \in C^\infty([0,\infty))$ is constant on $[0, \frac12]$; they vanish for $|z| \ge r/4$, resp. for $|u-z| \ge r/4$, so that their supports are disjoint, as $|u| = r$. We write $X * Y = \mathsf{T}_1 + \mathsf{T}_2 + \mathsf{T}_0$, where the three sums carry the factors θ_1, θ_2 and $1 - \theta_1 - \theta_2$. The function $g_1 := \theta_1\, Y(u - \cdot)$ on $\mathbb{Z}^3$ vanishes outside $\{|z| \le r/4\}$, a set on which $|u - z| \ge \frac34 r$. Since g_1 has finite support and $X(z) = \int_{\mathbb{T}} e^{ik\cdot z}\hat{X}(k)\, d^\circ k$, we may interchange the sum and the integral:

$$\mathsf{T}_1 = \sum_z X(z) g_1(z) = \int_{\mathbb{T}} \hat{X}(k) \sum_z g_1(z)\, e^{ik\cdot z}\, d^\circ k = \int_{\mathbb{T}} \hat{X}(k)\, \check{g}_1(-k)\, d^\circ k.$$

After the substitution $z \to u - z$ the same argument gives $\mathsf{T}_2 = \int_{\mathbb{T}} \hat{Y}(k)\, \check{g}_2(-k)\, d^\circ k$ with $g_2 := \theta(4|\cdot|/r)\, X(u - \cdot)$.

Step 1: proof of (a). The sum converges absolutely by Lem. 7.18(c) with $a = n_X$, $a' = n_Y$. By the discrete Leibniz rule $\nabla_j(fg) = (\nabla_j f)\, g(\cdot + e_j) + f\, \nabla_j g$, a difference of order ≤ 10 of g_1 is a sum of products of differences of θ_1, bounded by $|\nabla^\beta \theta_1| \le C r^{-|\beta|}$, with differences of Y of order ≤ 10 taken at points $u - z'$ with $|z'| \le r/4 + 10$. At these points $\langle u - z'\rangle \ge r/40$: for $r \le 40$ because $\langle u - z'\rangle \ge 1$, and for $r > 40$ because $|u - z'| \ge \frac34 r - 10 \ge r/40$. With the bounds on the differences of Y this gives $\|g_1\|_{(r)} \le C r^{-n_Y}$; in the same way $\|g_2\|_{(r)} \le C r^{-n_X}$.

Lem. 7.19(b), applied with $D = 3$, $R = r$, $g = g_1$, $\hat{\mathsf{T}} = \hat{X}$, the majorant $\mathsf{H} = c|k|^{a_X} \in \mathcal{H}_{a_X}(3)$ (admissible since $-3 < a_X < 3 < 6$), $\iota = 1$ and $c_0 = 1$, gives $|\mathsf{T}_1| \le C r^{-a_X - n_Y} \lg r$; likewise $|\mathsf{T}_2| \le C r^{-a_Y - n_X} \lg r$. On the support of $1 - \theta_1 - \theta_2$ we have $|z| \ge r/8$ and $|u - z| \ge r/8$, hence $\langle z\rangle \ge \frac19(r + |z|)$ and $\langle u - z\rangle \ge \frac19(r + |u - z|)$; together with $r + |u - z| \ge \frac12(r + |z|)$ (Lem. 7.18(a)) this gives

$$|X(z)Y(u-z)| \le C(r + |z|)^{-n_X}(r + |u - z|)^{-n_Y} \le C(r + |z|)^{-n_X - n_Y},$$

and Lem. 7.18(a) with the exponent $n_X + n_Y > 3$ gives $|\mathsf{T}_0| \le C r^{3 - n_X - n_Y}$.

Step 2: proof of (b), conditions (K1) and (K2). As $n_X, n_Y \ge 2$, conditions (K3) and the homogeneity of X^h, Y^h give $|X(x)| \le C\langle x\rangle^{-n_X}$ and $|Y(x)| \le C\langle x\rangle^{-n_Y}$, so that $X, Y \in \ell^2(\mathbb{Z}^3)$; by the uniqueness of Fourier coefficients $\hat{X}, \hat{Y} \in L^2(\mathbb{T})$, and Plancherel's theorem gives $(X * Y)(u) = \int_{\mathbb{T}} e^{ik\cdot u}\hat{X}\hat{Y}\, d^\circ k$, which is (K1) with the symbol $\hat{X}\hat{Y} \in L^1(\mathbb{T})$. For (K2) let $s_X := n_X - 3$ and $s_Y := n_Y - 3$, both ≥ -1; then

$$\hat{X}\hat{Y} - \chi\hat{X}^h\hat{Y}^h = (\hat{X} - \chi\hat{X}^h)\hat{Y} + \chi\hat{X}^h(\hat{Y} - \chi\hat{Y}^h) + (\chi^2 - \chi)\hat{X}^h\hat{Y}^h = O(|k|^{s_X + s_Y + 1} \lg|k|),$$

and $\hat{X}^h\hat{Y}^h$ is continuous and homogeneous of degree $s_X + s_Y = (n_X + n_Y - 3) - 3$.

Step 3: proof of (b), condition (K3) for $\gamma = 0$. Let $n := n_X + n_Y - 3$, and let γ be a multi-index; in this step $\gamma = 0$, in Step 4 it is arbitrary. We take $r \ge 200 + 8|\gamma|$. In T_1 the line Y enters through g_1, and in T_2 the line X through g_2; in the estimates below these lines are evaluated, together with their differences of the orders $\le |\gamma| + 10$ (the order γ of Step 4 and up to ten further differences from the norm $\|\cdot\|_{(r)}$), at points of modulus at least $r/2 \ge |\gamma| + 11$. In T_0 both lines, with their differences of the orders $\le |\gamma|$, are evaluated at points of modulus at least $r/8 \ge |\gamma| + 1$. Thus all these points lie in the range $|x| \ge |\gamma'| + 1$ in which (K3) for X and Y bounds the differences of

order γ'. Let $\hat u := u/r$ and

$$G_1(\zeta) := \theta(4|\zeta|)\, Y^h(\hat u - \zeta), \qquad G_2(\zeta) := \theta(4|\zeta|)\, X^h(\hat u - \zeta);$$

these functions lie in $C^\infty_c(\mathbb{R}^3)$, with support in $\{|\zeta| \le \frac14\}$, and $r^{-n_Y} G_1(z/r) = \theta_1(z) Y^h(u-z)$ by homogeneity. By (K3) for Y, $\|g_1 - r^{-n_Y} G_1(\cdot/r)\|_{(r)} \le C r^{-n_Y-1} \lg r$. We use Lem. 7.19(c) with $D=3$, $R=r$, $g=g_1$, $\hat{\mathsf T} = \hat X$, $\hat{\mathsf T}^h = \hat X^h$, $s = s_X \le 1$, $\iota = 1$, $c_0 = 1$, $a = n_Y$, $G = G_1$, the constant of the preceding bound as A, and the majorants $\mathsf H = 2c|k|^{s_X}$, $\mathsf H' = c|k|^{s_X+1}$, where c bounds both $|\hat X^h|$ on the unit sphere and the number in (K2) for X. The hypotheses of (b) and (c) of that lemma are satisfied: on $\{|k| \le 1\}$ we have $\chi = 1$, so that (K2) for X gives $|\hat X - \hat X^h| \le \mathsf H' \lg|k|$ there, and hence $|\hat X| \le \mathsf H \lg|k|$, while $|\hat X^h| \le \mathsf H$ everywhere. We obtain

$$\mathsf T_1 = r^{-s_X - n_Y} \int \hat X^h(k)\, \hat G_1(-k)\, d^\circ k + O(r^{-s_X-n_Y-1} \lg r),$$

and in the same way $\mathsf T_2 = r^{-s_Y-n_X} \int \hat Y^h(k)\, \hat G_2(-k)\, d^\circ k + O(r^{-s_Y-n_X-1}\lg r)$; note $s_X + n_Y = s_Y + n_X = n$. In $\mathsf T_0$ we replace X, Y by X^h, Y^h; by (K3) the error is at most

$$C \sum_z (r+|z|)^{-n_X-n_Y-1} \lg(r+|z|) \le C r^{2-n_X-n_Y} \lg r,$$

since $\lg(r+|z|) \le \lg r \lg(1+|z|/r) \le \lg r\, (r+|z|)/r$, and by Lem. 7.18(a) with the exponent $n_X + n_Y > 3$. Next we replace the sum by the integral of $\mathsf F(z) := (1-\theta_1-\theta_2)(z) X^h(z) Y^h(u-z)$, $z \in \mathbb{R}^3$. The factor $1-\theta_1-\theta_2$ vanishes for $|z| \le r/8$ and for $|u-z| \le r/8$, that is, near the points 0 and u at which X^h and $Y^h(u-\cdot)$ are singular; so $\mathsf F$, extended by 0 at these two points, is smooth on $\mathbb{R}^3$, in particular $\mathsf F \in C^1(\mathbb{R}^3)$. Since moreover $|\mathsf F(z)| \le C(r+|z|)^{-n_X-n_Y}$ and $|\nabla \mathsf F(z)| \le C(r+|z|)^{-n_X-n_Y-1}$, Lem. 7.18(b) applies, and together with (a) of that lemma it bounds the error by $Cr^{2-n_X-n_Y}$. The substitution $z = r\zeta$ gives

$$\int_{\mathbb{R}^3} \mathsf F = r^{3-n_X-n_Y} \int_{\mathbb{R}^3} (1 - \theta(4|\zeta|) - \theta(4|\hat u - \zeta|)) X^h(\zeta)\, Y^h(\hat u - \zeta)\, \mathrm{d}\zeta.$$

Altogether $(X * Y)(u) = \mathsf H_0(u) + O(r^{-n-1} \lg r)$, where

$$\mathsf H_0(u) := r^{-n}\Big[\int \hat X^h(k)\, \hat G_1(-k)\, d^\circ k + \int \hat Y^h(k)\, \hat G_2(-k)\, d^\circ k + \int (1 - \theta(4|\zeta|) - \theta(4|\hat u - \zeta|)) X^h(\zeta)\, Y^h(\hat u - \zeta)\, \mathrm{d}\zeta\Big].$$

The function $\mathsf H_0$ is homogeneous of degree $-n$, the bracket depending on $\hat u$ only. It is continuous: $\hat u \mapsto G_i$ is continuous with respect to $\|\cdot\|_{C^10}$, so that the last bound of Lem. 7.19(c) applies to the first two integrals; and the third integral depends continuously on $\hat u$ by dominated convergence, since on the support of its integrand $|\zeta| \ge \frac18$ and $|\hat u - \zeta| \ge \frac18$, so that the integrand is bounded by $C\langle\zeta\rangle^{-n_X-n_Y}$ uniformly in $\hat u$, and $n_X + n_Y > 3$. It is bilinear in the leading data of X and Y, and it is rotation covariant, by the substitutions $k \to Ok$, $\zeta \to O\zeta$.

Step 4: proof of (b), condition (K3) for all γ. Let γ be a multi-index and $r \ge 200 + 8|\gamma|$. We have $\nabla^\gamma (X*Y)(u) = \sum_z X(z) (\nabla^\gamma Y)(u-z)$, which we split by the factors θ_1, θ_2 and $1-\theta_1-\theta_2$ as before.

In the first sum the function $g_{1,\gamma} := \theta_1 (\nabla^\gamma Y)(u-\cdot)$ takes the place of g_1; it vanishes outside $\{|z| \le r/4\}$. The function $G_{1,\gamma}(\zeta) := \theta(4|\zeta|)(\partial^\gamma Y^h)(\hat u - \zeta)$ lies in $C^\infty_c(\mathbb{R}^3)$, with support in $\{|\zeta| \le \frac14\}$, and $r^{-n_Y-|\gamma|} G_{1,\gamma}(z/r) = \theta_1(z)(\partial^\gamma Y^h)(u-z)$ by homogeneity. By (K3) for Y, and since $|(\nabla^\gamma - \partial^\gamma) Y^h(x)| \le C|x|^{-n_Y-|\gamma|-1}$ at the points concerned, the function $g_{1,\gamma}$ differs from $r^{-n_Y-|\gamma|} G_{1,\gamma}(\cdot/r)$ by at most $Cr^{-n_Y-|\gamma|-1}\lg r$ in $\|\cdot\|_{(r)}$.

As in Step 0 the first sum equals $\int_{\mathbb{T}} \hat X(k)\, \check g_{1,\gamma}(-k)\, d^\circ k$, and Lem. 7.19(c), with the data of Step 3 except that now $g = g_{1,\gamma}$, $a = n_Y + |\gamma|$, $G = G_{1,\gamma}$ and A is the constant of the bound just stated, shows that it equals $r^{-n-|\gamma|} \int \hat X^h(k)\, \hat G_{1,\gamma}(-k)\, d^\circ k + O(r^{-n-|\gamma|-1}\lg r)$; here $s_X + n_Y = n$.

In the second sum we sum by parts $|\gamma|$ times, by the rule $\sum_z f(z)(\nabla_j Y)(u-z) = \sum_z (\nabla_j f)(z) Y(u-z)$ for f of finite support, and obtain $\sum_{z'} Y(z') g_{2,\gamma}(z')$ with $g_{2,\gamma}(z') := \nabla^\gamma(\theta_2 X)(u-z')$. This function vanishes outside $\{|z'| \le r/4 + |\gamma|\}$, a subset of $\{|z'| \le r\}$. The function

$$G_{2,\gamma}(\zeta) := \partial_y^\gamma \big[\theta(4|\hat u - y|) X^h(y)\big]_{y=\hat u-\zeta}$$

lies in $C_c^\infty(\mathbb{R}^3)$, with support in $\{|\zeta| \le \frac14\}$, and by (K3) for X, the discrete Leibniz rule and Taylor's formula, as for the first sum, $g_{2,\gamma}$ differs from $r^{-n_X-|\gamma|} G_{2,\gamma}(\cdot/r)$ by at most $Cr^{-n_X-|\gamma|-1}\lg r$ in $\|\cdot\|_{(r)}$. The second sum equals $\int_{\mathbb{T}} \hat Y(k)\, \check g_{2,\gamma}(-k)\, d^\circ k$, and Lem. 7.19(c), now taken with $D=3$, $R=r$, $g = g_{2,\gamma}$, $\hat{\mathsf{T}} = \hat Y$, $\hat{\mathsf{T}}^h = \hat Y^h$, $s = s_Y$, $\iota = 1$, $c_0 = 1$, $\mathsf{H} = 2c|k|^{s_Y}$, $\mathsf{H}' = c|k|^{s_Y+1}$ (with c chosen for Y as it was chosen for X in Step 3), $a = n_X + |\gamma|$ and $G = G_{2,\gamma}$, yields the value $r^{-n-|\gamma|}\int \hat Y^h(k)\, \hat G_{2,\gamma}(-k)\, d^\circ k + O(r^{-n-|\gamma|-1}\lg r)$, as $s_Y + n_X = n$.

In the third sum we replace X and $\nabla^\gamma Y$ by X^h and $\partial^\gamma Y^h$, and then the sum by the integral over $\mathbb{R}^3$, exactly as in Step 3; the errors are $O(r^{-n-|\gamma|-1}\lg r)$, and the substitution $z = r\zeta$ produces the factor $r^{-n-|\gamma|}$. Altogether $\nabla^\gamma(X*Y)(u) = \mathsf{H}_\gamma(u) + O(r^{-n-|\gamma|-1}\lg r)$, where

$$\mathsf{H}_\gamma(u) := r^{-n-|\gamma|}\Big[\int \hat X^h(k)\, \hat G_{1,\gamma}(-k)\, d^\circ k + \int \hat Y^h(k)\, \hat G_{2,\gamma}(-k)\, d^\circ k + \int (1-\theta(4|\zeta|) - \theta(4|\hat u - \zeta|)) X^h(\zeta)\, (\partial^\gamma Y^h)(\hat u - \zeta)\, \mathrm{d}\zeta\Big].$$

For the reasons given for H_0 in Step 3, the function H_γ is homogeneous of degree $-n-|\gamma|$ and continuous on $\mathbb{R}^3\setminus\{0\}$: the bracket depends on $\hat u$ only; $\hat u \mapsto G_{1,\gamma}$ and $\hat u \mapsto G_{2,\gamma}$ are continuous with respect to $\|\cdot\|_{C^{10}}$, so that the last bound of Lem. 7.19(c) applies; and the third integrand is bounded by $C\langle\zeta\rangle^{-n_X-n_Y}$ uniformly in $\hat u$, so that dominated convergence applies. Lem. 7.21, applied to $f = X*Y$, with the present $n>0$, the functions H_γ and $r_\gamma = 200 + 8|\gamma|$, gives (K3) with $K^h = \mathsf{H}_0$ for $|u| \ge 200 + 8|\gamma|$.

For the finitely many other u with $|u| \ge |\gamma| + 1$, the left side of (K3) is a finite number, bounded through the constants of X and Y, as H_0 is evaluated only at points different from 0; so (K3) holds with a larger c_γ, which still depends only on the constants of X and Y. This proves (b).

Step 5: proof of (c). We have $X(x) = \iint e^{i(k'+l)\cdot x}\sigma_{\mathrm{a}}(k')\sigma_{\mathrm{b}}(l)\, d^\circ k'\, d^\circ l$, and the substitution $k = k'+l$ shows that X is the kernel with the symbol I. The data of Lem. 7.20 are $\lambda_1(k) = k$, $\epsilon_1 = -1$, $\lambda_2 = 0$, $\epsilon_2 = 1$, so that $l_1(k) = k$, $l_2 = 0$; the hypotheses hold with $c_\lambda = 1$, the map $(l,k) \mapsto (k-l, l)$ having an integer matrix of determinant -1, and $\omega = 3 + (n_{\mathrm{a}} - 3) + (n_{\mathrm{b}} - 3) = n - 3 \ge -1$; moreover $\Lambda = 1$. The bound on $\nabla^\gamma X$ follows from Lem. 7.17(a) and the discrete Leibniz rule. For $|k| \ge c_0$ the function I is bounded, as an integral with two integrable singularities at distance at least c_0, and so is $\tilde{\mathsf{Q}}$. For $0 < |k| \le c_0$, Lem. 7.20(b) gives

$$\mathsf{I} - \tilde{\mathsf{Q}} = (\mathsf{Q} - \tilde{\mathsf{Q}}) + \mathsf{I}^h + O(|k|^{\omega+1}\lg|k|),$$

where $|\mathsf{I}^h(k)| \le C|k|^\omega$, and $\mathsf{Q} - \tilde{\mathsf{Q}} = O(|k|^2)$, even $O(|k|^3)$ if $\deg \mathsf{Q} \le 1$. This gives the bound on $\mathsf{I} - \tilde{\mathsf{Q}}$, and, for $n \le 4$ (where $\omega \le 1$), conditions (K1) and (K2) for $X - \mathsf{q}$ with $\sigma^h = \mathsf{I}^h$, which is continuous on $\mathbb{R}^3\setminus\{0\}$ by Lem. 7.20(a). For (K3): at points where no difference involves a point of $\operatorname{supp}\mathsf{q}$,

$$X - K_{\mathrm{a}}^h K_{\mathrm{b}}^h = (K_{\mathrm{a}} - K_{\mathrm{a}}^h)K_{\mathrm{b}} + K_{\mathrm{a}}^h(K_{\mathrm{b}} - K_{\mathrm{b}}^h),$$

and Lem. 7.17(a),(c) gives the required bounds; at the finitely many x with $|x| \ge |\gamma| + 1$ at which a difference of order γ involves a point of $\operatorname{supp}\mathsf{q}$, the left side of (K3) is a finite number, which we absorb into c_γ.

Step 6: proof of (d). Let $G \in C_c^\infty(\mathbb{R}^3)$ with support in $\{\frac12 < |\zeta| < 1\}$, let $R \ge 2$ and $g := G(\cdot/R)$ on $\mathbb{Z}^3$. By (K2) with $\sigma^h = 0$ we have $|\sigma(k)| \le c|k|^{n-2}\lg|k|$ on the cube, and Lem. 7.19(b), with

$D=3$, $s=n-2<6$, $\mathsf{H}=c|k|^{n-2}$, $\iota=1$ and $c_0=1$, gives, since $\|g\|_{(R)}\le\|G\|_{C^{10}}$,

$$\sum_x K(x)g(x)=\int_{\mathbb{T}}\sigma(k)\,\check g(-k)\,d^\circ k=O(R^{2-n}\lg R).$$

On the other hand, g vanishes outside $\{R/2<|x|<R\}$, a set of at most CR^3 points; by (K3) and by Lem. 7.18(b), applied to the function $K^hG(\cdot/R)$, whose gradient is bounded by CR^{-n-1},

$$\sum_x K(x)g(x)=\sum_x K^h(x)g(x)+O(R^{2-n}\lg R)=R^{3-n}\int_{\mathbb{R}^3}K^hG\,\mathrm{d}\zeta+O(R^{2-n}\lg R).$$

Letting $R\to\infty$ we obtain $\int K^hG\,\mathrm{d}\zeta=0$ for all such G; as K^h is continuous and homogeneous, $K^h=0$. □

Proposition 7.23 ((V) The kite)**.** Let $K_1,\dots,K_5$ be elementary lines with symbols $\sigma_1,\dots,\sigma_5$ and degrees $n_1,\dots,n_5$, where

$$n_1,n_2,n_4,n_5\in\{2,3\},\quad n_3\in\{1,2,3\},\quad n_1+n_2+n_3\le 7,\quad n_3+n_4+n_5\le 7,$$

and put $\Delta:=\sum_{i=1}^5 n_i-6$. For $u\in\mathbb{Z}^3$ let $r:=|u|$ and

$$\mathsf{T}(u):=\sum_{z,w\in\mathbb{Z}^3}K_1(z)\,K_2(w)\,K_3(w-z)\,K_4(u-z)\,K_5(u-w).$$

(i) There is a constant C, depending on $n_1,\dots,n_5$ only, such that for every $u\in\mathbb{Z}^3$

$$\sum_{z,w\in\mathbb{Z}^3}\langle z\rangle^{-n_1}\langle w\rangle^{-n_2}\langle w-z\rangle^{-n_3}\langle u-z\rangle^{-n_4}\langle u-w\rangle^{-n_5}\le C.$$

In particular the series defining $\mathsf{T}(u)$ converges absolutely.

(ii) For $\kappa=(\kappa_1,\kappa_2)\in\mathbb{R}^3\times\mathbb{R}^3$ put

$$\hat{\mathsf{T}}_{\mathsf{p}}(\kappa_1,\kappa_2):=\int_{\mathbb{T}}\sigma_1(\kappa_1+l)\,\sigma_2(\kappa_2-l)\,\sigma_3(l)\,d^\circ l,$$
$$\hat{\mathsf{T}}_{\mathsf{q}}(\kappa_1,\kappa_2):=\int_{\mathbb{T}}\sigma_4(\kappa_1-l)\,\sigma_5(\kappa_2+l)\,\sigma_3(l)\,d^\circ l.$$

Both are functions I of the kind treated in Lem. 7.20, with $\mathsf{m}=3$ and with data satisfying the hypotheses of that lemma; their numbers ω are

$$\omega_{\mathsf{p}}:=n_1+n_2+n_3-6\qquad\text{and}\qquad\omega_{\mathsf{q}}:=n_3+n_4+n_5-6,$$

respectively, and both lie in $\{-1,0,1\}$. For every function g of finite support on $\mathbb{Z}^6$

$$\begin{aligned}\sum_{z,w}K_1(z)\,K_2(w)\,K_3(w-z)\,g(z,w)&=\int_{\mathbb{T}^2}\hat{\mathsf{T}}_{\mathsf{p}}(\kappa)\,\check g(-\kappa)\,d^\circ\kappa,\\ \sum_{z,w}K_4(z)\,K_5(w)\,K_3(z-w)\,g(z,w)&=\int_{\mathbb{T}^2}\hat{\mathsf{T}}_{\mathsf{q}}(\kappa)\,\check g(-\kappa)\,d^\circ\kappa.\end{aligned}\tag{7.xxvii}$$

(iii) Let Q_{p} and Q_{q} be the polynomials of $\hat{\mathsf{T}}_{\mathsf{p}}$ and of $\hat{\mathsf{T}}_{\mathsf{q}}$ in the sense of Lem. 7.20(d), let

$$\tilde{\mathsf{Q}}_\bullet(\kappa):=\mathsf{Q}_\bullet(0)+\sum_{j=1}^6\partial_j\mathsf{Q}_\bullet(0)\,\sin\kappa_j\qquad(\bullet\in\{\mathsf{p},\mathsf{q}\}),$$

and let $\mathsf{q}_{\mathsf{p}},\mathsf{q}_{\mathsf{q}}$ be the kernels on $\mathbb{Z}^6$ with the symbols $\tilde{\mathsf{Q}}_{\mathsf{p}},\tilde{\mathsf{Q}}_{\mathsf{q}}$, that is, $\mathsf{q}_\bullet(\zeta)=\int_{\mathbb{T}^2}e^{i\kappa\cdot\zeta}\,\tilde{\mathsf{Q}}_\bullet(\kappa)\,d^\circ\kappa$; they vanish off $\{0,\pm e_1,\dots,\pm e_6\}$. There are constants r_0 and C, depending on $\sigma_1,\dots,\sigma_5$

only, and a function $\mathsf{U} \in C^1(\mathbb{R}^3 \setminus \{0\})$, homogeneous of degree $-\Delta$, such that for every u with $r \geq r_0$

$$\Big|\mathsf{T}(u) - \sum_{z,w} \mathsf{q}_{\mathsf{p}}(z,w)\, K_4(u-z)\, K_5(u-w) \\ - \sum_{z,w} \mathsf{q}_{\mathsf{q}}(z,w)\, K_1(u-z)\, K_2(u-w) - \mathsf{U}(u)\Big| \leq C\, r^{-\Delta-1} \lg r.$$

(iv) The function U depends only on the leading parts $\sigma_1^h, \dots, \sigma_5^h$, and it is linear in each of them. If every σ_i^h is replaced by $\sigma_i^h \circ O^{-1}$, with one and the same rotation O, then U is replaced by $\mathsf{U} \circ O^{-1}$.

■

Proof. Step 1: convergence. The symbol σ_i of K_i is a line symbol of degree $s_i := n_i - 3 \geq -2 > -3$. Lem. 7.17(a), applied with $s = s_i$ and $\gamma = 0$, therefore gives $|K_i(y)| \leq C\langle y\rangle^{-n_i}$ for all $y \in \mathbb{Z}^3$, and the absolute convergence of the series for $\mathsf{T}(u)$ follows from the bound in (i). To prove that bound we estimate $\langle w - z\rangle^{-n_3} \leq 1$; what remains factorises into

$$\Big(\sum_z \langle z\rangle^{-n_1} \langle u - z\rangle^{-n_4}\Big)\Big(\sum_w \langle w\rangle^{-n_2} \langle u - w\rangle^{-n_5}\Big).$$

Lem. 7.18(c) applies to the first factor with $(a, a', x) = (n_1, n_4, u)$, since $a, a' \geq 2$ and $a + a' \geq 4 > 3$; it bounds that factor by $C\langle u\rangle^{-\min\{n_1, n_4, n_1+n_4-3\}} \lg\langle u\rangle \leq C\langle u\rangle^{-1} \lg\langle u\rangle \leq C$, because $t^{-1} \lg t = t^{-1}(1 + \log t) \leq 1$ for $t \geq 1$. The second factor is bounded in the same way, with (n_2, n_5) in place of (n_1, n_4).

Step 2: the hypotheses of Lem. 7.20. For $\hat{\mathsf{T}}_{\mathsf{p}}$ we take $\mathsf{m} = 3$, $D = 6$, the line symbols $\sigma_1, \sigma_2, \sigma_3$ of degrees s_1, s_2, s_3, the signs $(\epsilon_1, \epsilon_2, \epsilon_3) = (1, -1, 1)$ and the linear maps $\lambda_1(\kappa) = \kappa_1$, $\lambda_2(\kappa) = \kappa_2$, $\lambda_3(\kappa) = 0$. The integrand of $\hat{\mathsf{T}}_{\mathsf{p}}$ is then $\prod_i \sigma_i(\epsilon_i l + \lambda_i(\kappa))$, and $(l_1, l_2, l_3)(\kappa) = (-\kappa_1, \kappa_2, 0)$. Up to signs, the pairwise differences of the $l_i(\kappa)$ are $\kappa_1 + \kappa_2$, κ_1 and κ_2. Hence $\max_i |l_i(\kappa)| \leq |\kappa| \leq 2|\kappa|$ and $|\kappa| \leq \sqrt{2}\max\{|\kappa_1|, |\kappa_2|\} \leq 2\max_{i<j} |l_i(\kappa) - l_j(\kappa)|$, so that the condition on the l_i holds with $c_\lambda = 2$; the three maps l_i are pairwise different.

The map $(l, \kappa_1, \kappa_2) \mapsto (\kappa_1 + l, \kappa_2 - l, l)$ of $(\mathbb{R}^3)^3$ acts on the three components in $\mathbb{R}^3$ by the integer matrix

$$\begin{pmatrix} 1 & 1 & 0 \\ -1 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix},$$

whose determinant is 1. Since $n_1, n_2 \geq 2$ and $n_3 \geq 1$, we have $s_1, s_2 \geq -1$ and $s_3 \geq -2$, hence $s_i + s_j \geq -3$ for $i \neq j$. Finally $\omega = 3 + s_1 + s_2 + s_3 = n_1 + n_2 + n_3 - 6 = \omega_{\mathsf{p}}$, and this number lies between $2 + 2 + 1 - 6 = -1$ and $7 - 6 = 1$. For $\hat{\mathsf{T}}_{\mathsf{q}}$ the line symbols are $\sigma_4, \sigma_5, \sigma_3$, the signs are $(-1, 1, 1)$ and the maps are $\kappa \mapsto \kappa_1$, $\kappa \mapsto \kappa_2$, $\kappa \mapsto 0$, so that $(l_1, l_2, l_3)(\kappa) = (\kappa_1, -\kappa_2, 0)$; the matrix of $(l, \kappa_1, \kappa_2) \mapsto (\kappa_1 - l, \kappa_2 + l, l)$ is $\left(\begin{smallmatrix} -1 & 1 & 0 \\ 1 & 0 & 1 \\ 1 & 0 & 0 \end{smallmatrix}\right)$, again of determinant 1, and the remaining conditions follow as before, now with $n_3 + n_4 + n_5 \leq 7$.

Step 3: proof of eq. 7.xxvii. We insert $K_1(z) = \int_{\mathbb{T}} e^{ik_1 \cdot z} \sigma_1(k_1)\, d^\circ k_1$, $K_2(w) = \int_{\mathbb{T}} e^{ik_2 \cdot w} \sigma_2(k_2)\, d^\circ k_2$ and $K_3(w - z) = \int_{\mathbb{T}} e^{il \cdot (w-z)} \sigma_3(l)\, d^\circ l$. The sum over (z, w) is finite and the three symbols are integrable, so Fubini's theorem gives

$$\sum_{z,w} K_1(z)\, K_2(w)\, K_3(w - z)\, g(z,w) \\ = \iiint \sigma_1(k_1)\, \sigma_2(k_2)\, \sigma_3(l)\, \check{g}(-(k_1 - l), -(k_2 + l))\, d^\circ k_1\, d^\circ k_2\, d^\circ l,$$

since $\sum_{z,w} g(z,w)\, e^{i(k_1 - l)\cdot z + i(k_2 + l)\cdot w} = \check{g}(-(k_1 - l), -(k_2 + l))$. For fixed l we substitute $k_1 = \kappa_1 + l$ and $k_2 = \kappa_2 - l$, which are translations of $\mathbb{T}$, and we then carry out the integration over l first; this

is the first identity. The function $y \mapsto K_3(-y)$ is the kernel with the symbol $l \mapsto \sigma_3(-l)$, because $K_3(-y) = \int e^{-il\cdot y}\sigma_3(l)\,d^\circ l = \int e^{il\cdot y}\sigma_3(-l)\,d^\circ l$. The derivation of the first identity used only the integrability of the three symbols. Applied to K_4, K_5 and $K_3(-\cdot)$ in place of K_1, K_2 and K_3, it represents $\sum_{z,w} K_4(z)K_5(w)K_3(z-w)g(z,w)$ by the function $\kappa \mapsto \int_{\mathbb{T}} \sigma_4(\kappa_1+l)\,\sigma_5(\kappa_2-l)\,\sigma_3(-l)\,d^\circ l$, which becomes $\hat{\mathsf{T}}_{\mathsf{q}}(\kappa)$ after the substitution $l \to -l$.

Step 4: short and long lines. Let $\varepsilon := \frac{1}{8}$ and, for $r > 0$, $\theta^r(y) := \theta(|y|/\varepsilon r)$ and $\bar\theta^r := 1 - \theta^r$, as functions on $\mathbb{R}^3$ and, by restriction, on $\mathbb{Z}^3$. By Def. 7.16(1), $\theta^r(y) = 1$ for $|y| \le \frac{1}{2}\varepsilon r$ and $\theta^r(y) = 0$ for $|y| \ge \varepsilon r$. For the arguments of the five lines we write

$$(y_1, y_2, y_3, y_4, y_5) := (z,\ w,\ w - z,\ u - z,\ u - w).$$

For $\mathsf{S} \subset \{1, \dots, 5\}$ let $\mathsf{T}_{\mathsf{S}}(u)$ be the series defining $\mathsf{T}(u)$, with the additional factor

$$\prod_{i\in\mathsf{S}} \theta^r(y_i) \prod_{i\notin\mathsf{S}} \bar\theta^r(y_i)$$

in each term; the lines with an index in S are called short, the others long. These series converge absolutely by (i), the additional factors taking values in $[0, 1]$, and multiplying out $\prod_{i=1}^5 (\theta^r(y_i) + \bar\theta^r(y_i)) = 1$ we obtain $\mathsf{T} = \sum_{\mathsf{S}} \mathsf{T}_{\mathsf{S}}$.

The arguments satisfy

$$u = y_1 + y_4 = y_2 + y_5 = y_1 + y_3 + y_5 = y_2 - y_3 + y_4.$$

If S contains one of the sets $\{1, 4\}$, $\{2, 5\}$, $\{1, 3, 5\}$, $\{2, 3, 4\}$, then a non-vanishing term of $\mathsf{T}_{\mathsf{S}}(u)$ requires $|y_i| < \varepsilon r$ for the indices i of that set, hence $r = |u| < 3\varepsilon r < r$; thus $\mathsf{T}_{\mathsf{S}} = 0$. There are three possibilities for $\mathsf{S} \cap \{1, 4\}$, three for $\mathsf{S} \cap \{2, 5\}$ and two for $\mathsf{S} \cap \{3\}$; of the resulting $3 \cdot 3 \cdot 2 = 18$ sets, two, namely $\{1, 3, 5\}$ and $\{2, 3, 4\}$, contain one of the forbidden triples, and 16 sets remain.

The substitutions $(z, w) \to (w, z)$ and $(z, w) \to (u - z, u - w)$ are bijections of $\mathbb{Z}^6$; they carry $(y_1, \dots, y_5)$ into $(y_2, y_1, -y_3, y_5, y_4)$ and into $(y_4, y_5, -y_3, y_1, y_2)$, respectively. As θ^r is even, they show that T_{S}, computed for the lines $(K_1, K_2, K_3, K_4, K_5)$, equals $\mathsf{T}_{\mathsf{S}'}$ computed for $(K_2, K_1, K_3(-\cdot), K_5, K_4)$, where S' is the image of S under the permutation (12)(45), and that it equals $\mathsf{T}_{\mathsf{S}''}$ computed for $(K_4, K_5, K_3(-\cdot), K_1, K_2)$, where S'' is the image of S under (14)(25). The new families satisfy the hypotheses of the proposition: $K_3(-\cdot)$ is an elementary line of degree n_3, its symbol $\sigma_3(-k)$ being a line symbol with leading part $\sigma_3^h(-k)$ (the cut-off χ is even), and the conditions on the degrees are invariant under both permutations.

Under the group generated by (12)(45) and (14)(25) the sixteen sets fall into seven classes: the set $\emptyset$ alone; the sets $\{1\}, \{2\}, \{4\}, \{5\}$; the set $\{3\}$ alone; the sets $\{1, 2\}, \{4, 5\}$; the sets $\{1, 5\}, \{2, 4\}$; the sets $\{1, 3\}, \{2, 3\}, \{3, 4\}, \{3, 5\}$; and the sets $\{1, 2, 3\}, \{3, 4, 5\}$. We choose the representatives

$$\emptyset, \quad \{1\}, \quad \{3\}, \quad \{1, 2\}, \quad \{1, 5\}, \quad \{1, 3\}, \quad \{1, 2, 3\}.$$

It therefore suffices to expand T_{S} for these seven sets, for an arbitrary family of lines obeying the hypotheses; for each of the remaining nine sets we use the expansion of its representative, computed for the transformed family.

Step 5: long factors. Let $L_i := \bar\theta^r K_i$ on $\mathbb{Z}^3$, let $K_i^h := \mathcal{F}_{\neq 0}^{-1}\sigma_i^h$, which by Lem. 7.17(b) is smooth on $\mathbb{R}^3 \setminus \{0\}$ and homogeneous of degree $-3 - s_i = -n_i$, and let $L_i^h := \bar\theta^r K_i^h$, extended by 0 at the origin. Since $\bar\theta^r$ vanishes on $\{|y| \le \frac{1}{2}\varepsilon r\}$, the function L_i^h is smooth on $\mathbb{R}^3$, and by homogeneity $L_i^h(y) = r^{-n_i}\,\bar\theta^1(y/r)\,K_i^h(y/r)$. We claim that there are r_0 and C such that for $r \ge r_0$, $i = 1, \dots, 5$ and $|\gamma| \le 10$

$$\begin{aligned} |\nabla^\gamma L_i(y)| + |\partial^\gamma L_i^h(y)| &\le C\,(r + |y|)^{-n_i - |\gamma|}, \\ |\nabla^\gamma (L_i - L_i^h)(y)| &\le C\,(r + |y|)^{-n_i - 1 - |\gamma|}; \end{aligned} \tag{7.xxviii}$$

here $y \in \mathbb{Z}^3$, except in the term $\partial^\gamma L_i^h(y)$, where $y \in \mathbb{R}^3$. Indeed, Lem. 7.17(a) gives $|\nabla^\gamma K_i(x)| \le C\langle x\rangle^{-n_i-|\gamma|}$, and Lem. 7.17(c) gives $|\nabla^\gamma (K_i - K_i^h)(x)| \le C|x|^{-n_i-1-|\gamma|}$ for $|x| \ge |\gamma| + 1$. The

derivatives of order γ of K_i^h are homogeneous of degree $-n_i-|\gamma|$, hence bounded by $C|x|^{-n_i-|\gamma|}$; the derivatives of order $j\ge 1$ of $\bar\theta^r$ are bounded by Cr^{-j} and vanish outside $\{\frac12\varepsilon r\le|y|\le\varepsilon r\}$. The discrete Leibniz rule $\nabla_j(fg)=(\nabla_j f)\,g(\cdot+e_j)+f\,\nabla_j g$ distributes ∇^γ over the two factors, the arguments being shifted by at most $|\gamma|\le 10$.

Only points y' with $|y'|\ge\frac12\varepsilon r$ contribute, and at such points, for $r\ge r_0$ with r_0 large enough, we have $|y'|\ge|\gamma|+1$ and $r+|y|\le C|y'|$ whenever $|y-y'|\le 10$. These facts together give eq. 7.xxviii.

Step 6: the scheme for $\mathsf S\neq\emptyset$. For each of the six non-empty representatives we write

$$\mathsf T_{\mathsf S}(u)=\sum_{\zeta\in\mathbb Z^D}\mathsf T^{\mathsf S}(\zeta)\,g_{\mathsf S}(\zeta),$$

where the kernel $\mathsf T^{\mathsf S}$ on $\mathbb Z^D$ collects the short lines, and we let s be the degree of the leading part $\hat{\mathsf T}^{\mathsf S,h}$ of the symbol of $\mathsf T^{\mathsf S}$. The data are as follows.

- $\mathsf S=\{1\}$, $D=3$, $\zeta=z$: $\mathsf T^{\mathsf S}(z)=K_1(z)$, with symbol $\sigma_1(\kappa)$ and $s=n_1-3$; and $g_{\mathsf S}(z)=\theta^r(z)\,L_4(u-z)\sum_w L_2(w)\,L_3(w-z)\,L_5(u-w)$.
- $\mathsf S=\{3\}$, $D=3$, $\zeta=y$: $\mathsf T^{\mathsf S}(y)=K_3(y)$, with symbol $\sigma_3(\kappa)$ and $s=n_3-3$; and $g_{\mathsf S}(y)=\theta^r(y)\sum_z L_1(z)\,L_2(z+y)\,L_4(u-z)\,L_5(u-z-y)$.
- $\mathsf S=\{1,2\}$, $D=6$, $\zeta=(z,w)$: $\mathsf T^{\mathsf S}(z,w)=K_1(z)K_2(w)$, with symbol $\sigma_1(\kappa_1)\sigma_2(\kappa_2)$ and $s=n_1+n_2-6$; and $g_{\mathsf S}(z,w)=\theta^r(z)\,\theta^r(w)\,L_3(w-z)\,L_4(u-z)\,L_5(u-w)$.
- $\mathsf S=\{1,5\}$, $D=6$, $\zeta=(z,w')$: $\mathsf T^{\mathsf S}(z,w')=K_1(z)K_5(w')$, with symbol $\sigma_1(\kappa_1)\sigma_5(\kappa_2)$ and $s=n_1+n_5-6$; and $g_{\mathsf S}(z,w')=\theta^r(z)\,\theta^r(w')\,L_2(u-w')\,L_3(u-w'-z)\,L_4(u-z)$.
- $\mathsf S=\{1,3\}$, $D=6$, $\zeta=(z,y)$: $\mathsf T^{\mathsf S}(z,y)=K_1(z)K_3(y)$, with symbol $\sigma_1(\kappa_1)\sigma_3(\kappa_2)$ and $s=n_1+n_3-6$; and $g_{\mathsf S}(z,y)=\theta^r(z)\,\theta^r(y)\,L_2(z+y)\,L_4(u-z)\,L_5(u-z-y)$.
- $\mathsf S=\{1,2,3\}$, $D=6$, $\zeta=(z,w)$: $\mathsf T^{\mathsf S}(z,w)=K_1(z)K_2(w)K_3(w-z)$, with symbol $\hat{\mathsf T}_{\mathsf p}(\kappa)$ (by eq. 7.xxvii, applied to $g=\delta_\zeta$) and $s=\omega_{\mathsf p}$; and $g_{\mathsf S}(z,w)=\theta^r(z)\,\theta^r(w)\,\theta^r(w-z)\,L_4(u-z)\,L_5(u-w)$.

In the cases $\{1,5\}$ and $\{1,3\}$ we have substituted $w=u-w'$, resp. $w=z+y$, in the sum over w; in the cases $\{1\}$ and $\{3\}$ the inner sums converge absolutely by Step 1. Each $g_{\mathsf S}$ vanishes off $\{|\zeta|\le r\}$, since the short factors confine every component of ζ to $\{|\cdot|<\varepsilon r\}$ and $\sqrt2\,\varepsilon<1$.

Let $g^h_{\mathsf S}$ be the function on $\mathbb R^D$ obtained from $g_{\mathsf S}$ by replacing every L_i by L_i^h and, in the cases $\{1\}$ and $\{3\}$, the inner sum by the integral over $\mathbb R^3$. Since $\theta^r(y)=\theta^1(y/r)$ and $L_i^h(y)=r^{-n_i}(\bar\theta^1K_i^h)(y/r)$, the substitution of r times the variable in the inner integrals gives

$$g^h_{\mathsf S}(\zeta)=r^{-a}\,G_{\mathsf S}(\zeta/r),\qquad\text{where}\quad a:=\begin{cases}\sum_{i\notin\mathsf S}n_i & \text{if } D=6,\\ \sum_{i\notin\mathsf S}n_i-3 & \text{if } \mathsf S=\{1\},\{3\},\end{cases}$$

so that $a+s=\Delta$ in all six cases. Here $G_{\mathsf S}$ is the function $g^h_{\mathsf S}$ computed with $r=1$ and with $\hat u:=u/r$ in place of u; it belongs to $C^\infty_c(\mathbb R^D)$ and has its support in $\{|\zeta|\le 1\}$. We claim

$$\|g_{\mathsf S}-g^h_{\mathsf S}\|_{(r)}\le C\,r^{-a-1}\qquad(r\ge r_0),\tag{7.xxix}$$

where $g^h_{\mathsf S}$ is restricted to $\mathbb Z^D$ and $\|\cdot\|_{(r)}$ is the norm of Def. 7.16(5).

In the cases $\{1,2\}$, $\{1,5\}$, $\{1,3\}$, $\{1,2,3\}$ the functions $g_{\mathsf S}$ and $g^h_{\mathsf S}$ are products of the same short factors with the long factors L_i, resp. L_i^h, $i\notin\mathsf S$, evaluated at linear functions of ζ. We write the difference as a sum of terms in each of which one factor L_i has been replaced by $L_i-L_i^h$, and we distribute the differences ∇^γ, $|\gamma|\le 10$, by the Leibniz rule. By eq. 7.xxviii a difference falling on a long factor gains a factor $C(r+|y_i|)^{-1}\le Cr^{-1}$, a difference falling on a short factor gains Cr^{-1}, and the factor $L_i-L_i^h$ gains $(r+|y_i|)^{-1}\le r^{-1}$ in comparison with L_i; moreover $\prod_{i\notin\mathsf S}(r+|y_i|)^{-n_i}\le r^{-a}$. This gives $r^{|\gamma|}|\nabla^\gamma(g_{\mathsf S}-g^h_{\mathsf S})|\le Cr^{-a-1}$.

In the case $\mathsf{S}=\{1\}$ let

$$\mathsf{w}(z) := \sum_w L_2(w)\, L_3(w-z)\, L_5(u-w),$$
$$\mathsf{w}^h(z) := \int_{\mathbb{R}^3} L_2^h(w)\, L_3^h(w-z)\, L_5^h(u-w)\,\mathrm{d}w.$$

For $|z|\le r$ the differences in z fall on $L_3(w-z)$ only. By eq. 7.xxviii, by $r+|u-w|\ge\frac12(r+|w|)$ (second assertion of Lem. 7.18(a), as $|u|=r$) and by the first assertion of Lem. 7.18(a) with $D=3$, $R=r$ and exponent $n_2+n_5\ge 4>3$, we obtain

$$|\nabla^\gamma \mathsf{w}(z)| \le Cr^{-n_3-|\gamma|}\sum_w (r+|w|)^{-n_2-n_5} \le Cr^{3-n_2-n_3-n_5-|\gamma|}.$$

Each replacement of an L_i by L_i^h in w costs a factor Cr^{-1}, by the second inequality of eq. 7.xxviii. The replacement of the sum over w by the integral costs a factor Cr^{-1} as well: we apply Lem. 7.18(b), with $D=3$, to $F(w):=L_2^h(w)\,(\nabla_z^\gamma L_3^h)(w-z)\,L_5^h(u-w)$, whose gradient is bounded by $Cr^{-n_3-|\gamma|-1}(r+|w|)^{-n_2-n_5}$ on each cube $w+[0,1]^3$. Since $g_{\{1\}}(z)=\theta^r(z)L_4(u-z)\mathsf{w}(z)$, since $|\nabla^\gamma L_4(u-\cdot)|\le Cr^{-n_4-|\gamma|}$ on $\{|z|\le r\}$ by eq. 7.xxviii, and since the differences of order γ of θ^r are bounded by $Cr^{-|\gamma|}$, the Leibniz rule gives eq. 7.xxix with $a=n_2+n_3+n_4+n_5-3$.

In the case $\mathsf{S}=\{3\}$ let $\mathsf{w}(y):=\sum_z L_1(z)\,L_2(z+y)\,L_4(u-z)\,L_5(u-z-y)$, and let w^h be the integral over $\mathbb{R}^3$ of the corresponding product of the L_i^h. For $|y|\le r$ the differences in y fall on L_2 and on L_5 (and, in $g_{\{3\}}$, on the factor $\theta^r(y)$), each difference gaining a factor Cr^{-1}. By Lem. 7.18(a) we have $r+|z+y|\ge\frac12(r+|z|)$, $r+|u-z|\ge\frac12(r+|z|)$, and, with $R=2r\ge|u-y|$, also $r+|u-y-z|\ge\frac14(r+|z|)$; the first assertion of that lemma, with exponent $n_1+n_2+n_4+n_5\ge 8$, gives $\sum_z(r+|z|)^{-n_1-n_2-n_4-n_5}\le Cr^{3-n_1-n_2-n_4-n_5}$. Hence $|\nabla^\gamma\mathsf{w}(y)|\le Cr^{3-n_1-n_2-n_4-n_5-|\gamma|}$, and, as in the case $\{1\}$, the replacements $L_i\to L_i^h$ and the replacement of the sum by the integral (Lem. 7.18(b)) cost a factor Cr^{-1} each. This proves eq. 7.xxix in all six cases.

Step 7: symbols, and the pairing with slowly varying functions. We verify the hypotheses of Lem. 7.19(c) for $\hat{\mathsf{T}}$, the symbol of T^{S}, and $\hat{\mathsf{T}}^h:=\hat{\mathsf{T}}^{\mathsf{S},h}$. In the cases $\{1\}$ and $\{3\}$ ($D=3$, $i=1$, resp. $i=3$, $s=s_i$) we take $c_0=1$. On $\{|\kappa|\le 1\}$ we have $\chi=1$, so $\sigma_i-\chi\sigma_i^h\in\mathcal{S}^{s+1}$ gives $|\sigma_i-\sigma_i^h|\le C|\kappa|^{s+1}$ there; the majorants are $\mathsf{H}:=C|\kappa|^s\in\mathcal{H}_s(3)$ for σ_i^h and $\mathsf{H}':=C|\kappa|^{s+1}\in\mathcal{H}_{s+1}(3)$, which belong to these classes since $s>-3$; the required bound $|\sigma_i-\sigma_i^h|\le\mathsf{H}'\lg|\kappa|$ holds on $\{|\kappa|\le1\}$, as $\lg|\kappa|\ge1$. In the cases $\{1,2\}$, $\{1,5\}$, $\{1,3\}$ ($D=6$) the symbol is $\sigma_i(\kappa_1)\sigma_j(\kappa_2)$ with $(i,j)=(1,2),(1,5),(1,3)$, its leading part is $\sigma_i^h(\kappa_1)\sigma_j^h(\kappa_2)$, and $s=s_i+s_j$ with $s_i,s_j>-3$. Then $|\sigma_i^h(\kappa_1)\sigma_j^h(\kappa_2)|\le C|\kappa_1|^{s_i}|\kappa_2|^{s_j}$, a function in $\mathcal{H}_{s_i+s_j}(6)$, and on $\{|\kappa|\le1\}$

$$\big|\sigma_i(\kappa_1)\sigma_j(\kappa_2)-\sigma_i^h(\kappa_1)\sigma_j^h(\kappa_2)\big| \le C\big(|\kappa_1|^{s_i+1}|\kappa_2|^{s_j}+|\kappa_1|^{s_i}|\kappa_2|^{s_j+1}\big),$$

a function in $\mathcal{H}_{s_i+s_j+1}(6)$; here we wrote the difference as $(\sigma_i-\sigma_i^h)\sigma_j+\sigma_i^h(\sigma_j-\sigma_j^h)$ and used $|\sigma_j(\kappa_2)|\le C|\kappa_2|^{s_j}$. In the case $\{1,2,3\}$ we use Lem. 7.20, whose hypotheses hold by Step 2, with the number c_0 given there. By part (b) of that lemma, $|\hat{\mathsf{T}}_{\mathsf{p}}-\mathsf{Q}_{\mathsf{p}}-\mathsf{I}_{\mathsf{p}}^h|\le C|\kappa|^{\omega_{\mathsf{p}}+1}\lg|\kappa|\,\Lambda(\kappa)$ for almost every κ with $|\kappa|\le c_0$, where $\mathsf{I}_{\mathsf{p}}^h$ is the homogeneous part of $\hat{\mathsf{T}}_{\mathsf{p}}$. The polynomial Q_{p} has degree at most $\omega_{\mathsf{p}}\le1$, so $\mathsf{Q}_{\mathsf{p}}-\tilde{\mathsf{Q}}_{\mathsf{p}}=\sum_j\partial_j\mathsf{Q}_{\mathsf{p}}(0)(\kappa_j-\sin\kappa_j)=O(|\kappa|^3)$, and $|\kappa|^3\le|\kappa|^{\omega_{\mathsf{p}}+1}\lg|\kappa|\,\Lambda(\kappa)$ for $|\kappa|\le1$.

We therefore take $\hat{\mathsf{T}}:=\hat{\mathsf{T}}_{\mathsf{p}}-\tilde{\mathsf{Q}}_{\mathsf{p}}$, which is integrable on $\mathbb{T}^2$: Lem. 7.20 bounds the L^1 norm of $\hat{\mathsf{T}}_{\mathsf{p}}$ by the product of the L^1 norms of $\sigma_1,\sigma_2,\sigma_3$, and $\tilde{\mathsf{Q}}_{\mathsf{p}}$ is a trigonometric polynomial. We further take $\hat{\mathsf{T}}^h:=\mathsf{I}_{\mathsf{p}}^h$ and the majorants $\mathsf{H}:=C|\kappa|^{\omega_{\mathsf{p}}}\Lambda$ (Lem. 7.20(a)) and $\mathsf{H}':=C|\kappa|^{\omega_{\mathsf{p}}+1}\Lambda$. They belong to $\mathcal{H}_{\omega_{\mathsf{p}}}(6)$ and to $\mathcal{H}_{\omega_{\mathsf{p}}+1}(6)$, respectively, since the same lemma asserts that $|\kappa|^s\Lambda(\kappa)$ belongs to $\mathcal{H}_s(6)$ for every real $s>-6$, and since $\omega_{\mathsf{p}}\ge-1$. In all cases $-D<s<8-D$, as Lem. 7.19(c) requires: for $D=3$ we have $s\in\{-2,-1,0\}\subset(-3,2)$, and for $D=6$ we have $s\in\{-3,\dots,1\}\subset(-6,2)$.

Inserting the Fourier integrals of the short lines, we have $\mathsf{T}_\mathsf{S}(u) = \int_{\mathbb{T}^{D/3}} \hat{\mathsf{T}}^\mathsf{S}(\kappa)\, \check{g}_\mathsf{S}(-\kappa)\, d^\circ\kappa$, where $\hat{\mathsf{T}}^\mathsf{S}$ is the symbol of T^S: for the product kernels this follows from Fubini's theorem as in Step 3, and for $\{1,2,3\}$ it is eq. 7.xxvii. We apply Lem. 7.19(c) with $R = r$, $g = g_\mathsf{S}$, which vanishes off $\{|\zeta| \le r\}$, $G = G_\mathsf{S}$, the number a of Step 6, and $A = C$, which is admissible by eq. 7.xxix because $\lg r \ge 1$. The norm $\|G_\mathsf{S}\|_{C^{10}}$ is bounded uniformly in $\hat{u} \in S^2$, by the smoothness in $(\zeta, \hat{u})$ established in Step 9 and the compactness of S^2. In the first five cases, $\{1\}$, $\{3\}$, $\{1,2\}$, $\{1,5\}$, $\{1,3\}$, we obtain

$$\mathsf{T}_\mathsf{S}(u) = \mathsf{U}_\mathsf{S}(u) + O(r^{-\Delta-1} \lg r), \qquad \mathsf{U}_\mathsf{S}(u) := r^{-\Delta} \int_{\mathbb{R}^D} \hat{\mathsf{T}}^{\mathsf{S},h}(\kappa)\, \hat{G}_\mathsf{S}(-\kappa)\, d^\circ\kappa,$$

because $a + s = \Delta$. In the case $\{1,2,3\}$, where $\hat{\mathsf{T}}^{\mathsf{S},h} := \mathsf{I}^h_\mathsf{p}$, the lemma is applied to $\hat{\mathsf{T}}_\mathsf{p} - \tilde{\mathsf{Q}}_\mathsf{p}$, and the remaining term is $\int_{\mathbb{T}^2} \tilde{\mathsf{Q}}_\mathsf{p}(\kappa) \check{g}_\mathsf{S}(-\kappa)\, d^\circ\kappa = \sum_{z,w} \mathsf{q}_\mathsf{p}(z,w)\, g_\mathsf{S}(z,w)$; thus

$$\mathsf{T}_{\{1,2,3\}}(u) = \sum_{z,w} \mathsf{q}_\mathsf{p}(z,w)\, g_{\{1,2,3\}}(z,w) + \mathsf{U}_{\{1,2,3\}}(u) + O(r^{-\Delta-1} \lg r).$$

On the support of q_p, that is, for $(z,w) \in \{0, \pm e_1, \dots, \pm e_6\}$, we have $|z|, |w|, |w-z| \le 1 \le \frac{1}{2}\varepsilon r$ and $|u-z|, |u-w| \ge r - 1 \ge \varepsilon r$ as soon as $r \ge r_0 \ge 16$; hence $g_{\{1,2,3\}}(z,w) = K_4(u-z)K_5(u-w)$ there, which produces the first local term in (iii). The image of $\{1,2,3\}$ under $(z,w) \to (u-z, u-w)$ is $\{3,4,5\}$, and the transformed family $(K_4, K_5, K_3(-\cdot), K_1, K_2)$ has, in place of $\hat{\mathsf{T}}_\mathsf{p}$, the function $\int_\mathbb{T} \sigma_4(\kappa_1 + l)\sigma_5(\kappa_2 - l)\sigma_3(-l)\, d^\circ l = \hat{\mathsf{T}}_\mathsf{q}(\kappa)$; its local kernel is therefore q_q, and $\mathsf{T}_{\{3,4,5\}}$ produces the second local term in (iii).

Step 8: the case $\mathsf{S} = \emptyset$. Here $\mathsf{T}_\emptyset(u) = \sum_{z,w} \prod_{i=1}^5 L_i(y_i)$. By eq. 7.xxviii with $\gamma = 0$, by $(r + |y_3|)^{-n_3} \le r^{-n_3}$ and by Lem. 7.18(a) (as in Step 6, with $r + |u-z| \ge \frac{1}{2}(r + |z|)$ and $r + |u-w| \ge \frac{1}{2}(r + |w|)$),

$$\sum_{z,w} \prod_{i=1}^5 (r + |y_i|)^{-n_i} \le C\, r^{-n_3}\, r^{3-n_1-n_4}\, r^{3-n_2-n_5} = C\, r^{-\Delta}.$$

Replacing each L_i by L_i^h costs a factor Cr^{-1}, by the second inequality of eq. 7.xxviii; replacing the sum over $(z,w) \in \mathbb{Z}^6$ by the integral over $\mathbb{R}^6$ costs a factor Cr^{-1} by Lem. 7.18(b) with $D = 6$, applied to $F(z,w) := \prod_i L_i^h(y_i)$, whose gradient is bounded by $Cr^{-1} \prod_i (r + |y_i|)^{-n_i}$ on each cube of side 1. Hence $\mathsf{T}_\emptyset(u) = \mathsf{U}_\emptyset(u) + O(r^{-\Delta-1})$ with

$$\mathsf{U}_\emptyset(u) := \iint_{\mathbb{R}^3 \times \mathbb{R}^3} \prod_{i=1}^5 L_i^h(y_i)\, \mathrm{d}z\, \mathrm{d}w,$$

which, after the substitution $(z,w) = r(z',w')$, is seen to be homogeneous of degree $6 - \sum_i n_i = -\Delta$.

Step 9: the function U. Let U be the sum of the sixteen functions U_S, where for a set S which is not a representative, U_S is the function of its representative computed for the transformed family (Step 4). By Steps 4, 7 and 8, the inequality in (iii) holds for $r \ge r_0$. Every U_S is homogeneous of degree $-\Delta$, and so is U. We let a vector $v \in \mathbb{R}^3$ take the place of $\hat{u}$ in the definition of G_S, and we write $G_\mathsf{S}(\zeta; v)$. In the factors $\bar{\theta}^1 K_i^h$ the argument of K_i^h has modulus at least $\frac{1}{2}\varepsilon$ wherever $\bar{\theta}^1$ does not vanish, so these factors are smooth on $\mathbb{R}^3$; the inner integrals in the cases $\{1\}$ and $\{3\}$ may be differentiated under the integral sign, with integrable majorants that are uniform for (ζ, v) in compact sets.

Hence $(\zeta, v) \mapsto G_\mathsf{S}(\zeta; v)$ is smooth, with support in $\{|\zeta| \le 1\}$ for every v, and $v \mapsto G_\mathsf{S}(\cdot\,; v)$ is of class C^1 with respect to the norm $\|\cdot\|_{C^{10}}$. The last bound of Lem. 7.19(c), $|\int \hat{\mathsf{T}}^h \hat{G}(-\kappa)\, d^\circ\kappa| \le C\|G\|_{C^{10}}$, which is linear in G, then shows that $\Phi_\mathsf{S}(v) := \int \hat{\mathsf{T}}^{\mathsf{S},h}(\kappa)\, \hat{G}_\mathsf{S}(-\kappa; v)\, d^\circ\kappa$ is of class C^1 in v. Since $\mathsf{U}_\mathsf{S}(u) = |u|^{-\Delta}\, \Phi_\mathsf{S}(u/|u|)$, the function U_S is of class C^1 on $\mathbb{R}^3 \setminus \{0\}$.

For $\mathsf{U}_\emptyset$ we differentiate under the integral sign.

The functions U_S are built from the leading parts σ_i^h of the short lines, from the functions $K_i^h = \mathcal{F}_{\neq 0}^{-1}\sigma_i^h$ of the long lines, which depend linearly on σ_i^h (Lem. 7.17(b)), from the homogeneous parts I_p^h and I_q^h, which depend only on the leading parts and multilinearly on them (Lem. 7.20(a)), and from radial cut-offs; every line enters each U_S exactly once. Hence U depends only on $\sigma_1^h, \dots, \sigma_5^h$ and is linear in each of them. If every σ_i^h is replaced by $\sigma_i^h \circ O^{-1}$, then K_i^h is replaced by $K_i^h \circ O^{-1}$ (Lem. 7.17(b)), and I_p^h, I_q^h are replaced by their compositions with O^{-1} acting on both arguments, since the linear maps λ_i of Step 2 commute with this action (Lem. 7.20(a)).

The substitutions $\zeta \to O\zeta$ in the integrals defining G_S and $\mathsf{U}_\emptyset$, and $\kappa \to O\kappa$ in the integral defining U_S, together with the invariance of the radial cut-offs, show that U is replaced by $\mathsf{U} \circ O^{-1}$. This proves (iii) and (iv). □

7.1.10. *Classes of pairings; convergence.*

Lemma 7.24 (The lines of the lattice)**.** Let $\tau_{\mathrm{gf}} > 0$; we work on $\mathbb{Z}^3$. For $\iota \in \{1,2,3\}$ write $p_{[\iota]} = p_{\alpha\beta}$ with $\alpha < \beta$, and put

$$\mathsf{c}_{\iota,\nu}(k) := \mathsf{k}_\alpha(k)\,\delta_{\nu\beta} - \mathsf{k}_\beta(k)\,\delta_{\nu\alpha} \qquad (\nu \in \{1,2,3\},\ k \in \mathbb{R}^3),$$

so that for $v \in \mathbb{C}^3$ the form $(x,\nu) \mapsto e^{ik\cdot x}v_\nu$ satisfies $\mathrm{d}(e^{ik\cdot x}v_\nu)(p_{[\iota]}(x)) = e^{ik\cdot x}\sum_\nu \mathsf{c}_{\iota,\nu}(k)\,v_\nu$. Let $\hat{\mathsf{C}}_{\mu\nu}(k)$ be the bracket in the formula for $\mathsf{C}^{\mathrm{lat}}_{\tau_{\mathrm{gf}}}$ in Def. 7.7(1),

$$\hat{\mathsf{C}}_{\mu\nu}(k) = \frac{\delta_{\mu\nu}}{|\mathsf{k}|^2} + (\tau_{\mathrm{gf}} - 1)\,\frac{\mathsf{k}_\mu \overline{\mathsf{k}_\nu}}{|\mathsf{k}|^4}.$$

Then for every $x \in \mathbb{Z}^3$, all directions μ,ν and all ι,ι', with $p_{[\iota']} = p_{\alpha'\beta'}$, $\alpha' < \beta'$, the following functions of $y \in \mathbb{Z}^3$ are elementary lines, with the symbols, the degrees n and the leading parts of the symbols listed:

- $y \mapsto \mathsf{C}\big((x,\mu),(x+y,\nu)\big)$: symbol $\hat{\mathsf{C}}_{\nu\mu}(k)$; $n=1$; leading part $\delta_{\mu\nu}|k|^{-2} + (\tau_{\mathrm{gf}}-1)\,k_\mu k_\nu |k|^{-4}$;
- $y \mapsto \mathsf{E}_{p_{[\iota]}(x)}\big((x+y,\nu)\big)$: symbol $\sum_\mu \overline{\mathsf{c}_{\iota,\mu}(k)}\,\hat{\mathsf{C}}_{\nu\mu}(k) = \overline{\mathsf{c}_{\iota,\nu}(k)}\,|\mathsf{k}|^{-2}$; $n=2$; leading part $-i(k_\alpha\delta_{\nu\beta} - k_\beta\delta_{\nu\alpha})|k|^{-2}$;
- $y \mapsto \Pi(p_{[\iota]}(x), p_{[\iota']}(x+y))$: symbol $\sum_\nu \overline{\mathsf{c}_{\iota,\nu}(k)}\,\mathsf{c}_{\iota',\nu}(k)\,|\mathsf{k}|^{-2}$; $n=3$; leading part $\sum_\nu (k_\alpha\delta_{\nu\beta} - k_\beta\delta_{\nu\alpha})(k_{\alpha'}\delta_{\nu\beta'} - k_{\beta'}\delta_{\nu\alpha'})|k|^{-2}$;
- $y \mapsto \mathsf{D}\big((x,\mu);(x+y,\nu)\big)$: symbol $\frac{1}{2}(1+e^{ik_\nu})\,\overline{\mathsf{k}_\mu(k)}\,|\mathsf{k}|^{-2}$; $n=2$; leading part $-ik_\mu|k|^{-2}$.

If $K(y)$ is an elementary line with symbol $\sigma(k)$ and $y_0 \in \mathbb{Z}^3$, then $K(-y)$ and $K(y+y_0)$ are elementary lines of the same degree, with the symbols $\sigma(-k)$ and $e^{ik\cdot y_0}\sigma(k)$. Consequently, the contraction eq. 7.xiii of a leg at a bond or a plaquette with base point $x+\xi$ with a leg at a bond or a plaquette with base point $x+y+\xi'$ equals $\delta_{aa'}$ times an elementary line in y of degree $1 + \#\{\text{curl legs among the two}\}$. ■

Proof. The matrix $\hat{\mathsf{C}}(k)$ is Hermitian, and $\hat{\mathsf{C}}(-k) = \overline{\hat{\mathsf{C}}(k)}$ because $\mathsf{k}_\mu(-k) = \overline{\mathsf{k}_\mu(k)}$. By Def. 7.7(1) and the substitution $k \to -k$,

$$\begin{aligned}\mathsf{C}\big((x,\mu),(x+y,\nu)\big) &= \int_{\mathbb{T}} e^{-ik\cdot y}\,\hat{\mathsf{C}}_{\mu\nu}(k)\,d^\circ k \\ &= \int_{\mathbb{T}} e^{ik\cdot y}\,\overline{\hat{\mathsf{C}}_{\mu\nu}(k)}\,d^\circ k = \int_{\mathbb{T}} e^{ik\cdot y}\,\hat{\mathsf{C}}_{\nu\mu}(k)\,d^\circ k.\end{aligned}$$

The plaquette $p = p_{[\iota]}(x) = p_{\alpha\beta}(x)$ has the bonds $(x,\alpha),(x+e_\alpha,\beta),(x+e_\beta,\alpha),(x,\beta)$, with the signs $+,+,-,-$. By the formula just proved, $\mathsf{C}\big((x+\xi,\mu'),(x+y,\nu)\big) = \int e^{ik\cdot(y-\xi)}\hat{\mathsf{C}}_{\nu\mu'}(k)\,d^\circ k$, and

therefore

$$\begin{aligned}\mathsf{E}_p((x+y,\nu)) &= \sum_i s_i\,\mathsf{C}(b_i(p),(x+y,\nu))\\ &= \int_{\mathbb{T}} e^{ik\cdot y}\Big[\hat{\mathsf{C}}_{\nu\alpha}(k)(1-e^{-ik_\beta})+\hat{\mathsf{C}}_{\nu\beta}(k)(e^{-ik_\alpha}-1)\Big]\,d^\circ k.\end{aligned}$$

As $\overline{\mathsf{k}_\mu(k)} = e^{-ik_\mu}-1$, the bracket equals $\sum_\mu \overline{\mathsf{c}_{\iota,\mu}(k)}\,\hat{\mathsf{C}}_{\nu\mu}(k)$. In this sum the part of $\hat{\mathsf{C}}_{\nu\mu}$ containing $\mathsf{k}_\nu\overline{\mathsf{k}_\mu}$ drops out, since $\sum_\mu \overline{\mathsf{c}_{\iota,\mu}}\,\mathsf{k}_\nu\overline{\mathsf{k}_\mu} = \mathsf{k}_\nu\,\overline{\sum_\mu \mathsf{c}_{\iota,\mu}\mathsf{k}_\mu}$ and $\sum_\mu \mathsf{c}_{\iota,\mu}\mathsf{k}_\mu = \mathsf{k}_\alpha\mathsf{k}_\beta - \mathsf{k}_\beta\mathsf{k}_\alpha = 0$; what remains is $\overline{\mathsf{c}_{\iota,\nu}}\,|\mathsf{k}|^{-2}$. The kernel Π arises from E_p by the operation d in the second argument (Def. 7.7(2)).

With $q = p_{[\iota']}(x+y) = p_{\alpha'\beta'}(x+y)$, the four bonds $(x+y,\alpha')$, $(x+y+e_{\alpha'},\beta')$, $(x+y+e_{\beta'},\alpha')$, $(x+y,\beta')$ of q contribute, with the signs $+,+,-,-$, the symbols $\overline{\mathsf{c}_{\iota,\alpha'}}|\mathsf{k}|^{-2}$, $e^{ik_{\alpha'}}\overline{\mathsf{c}_{\iota,\beta'}}|\mathsf{k}|^{-2}$, $e^{ik_{\beta'}}\overline{\mathsf{c}_{\iota,\alpha'}}|\mathsf{k}|^{-2}$, $\overline{\mathsf{c}_{\iota,\beta'}}|\mathsf{k}|^{-2}$, and their sum is

$$(\mathsf{k}_{\alpha'}\,\overline{\mathsf{c}_{\iota,\beta'}} - \mathsf{k}_{\beta'}\,\overline{\mathsf{c}_{\iota,\alpha'}})|\mathsf{k}|^{-2} = \sum_\nu \overline{\mathsf{c}_{\iota,\nu}}\,\mathsf{c}_{\iota',\nu}\,|\mathsf{k}|^{-2};$$

that is, d in the second argument multiplies the symbol by $\mathsf{c}_{\iota',\nu}(k)$ and sums over ν. Finally, with $\mathsf{G}(x',x'') = \int e^{ik\cdot(x'-x'')}|\mathsf{k}|^{-2}\,d^\circ k$,

$$\begin{aligned}\mathsf{D}((x,\mu);(x+y,\nu)) &= \tfrac12 \sum_{x'\in\{x+y,\,x+y+e_\nu\}} [\mathsf{G}(x',x+e_\mu)-\mathsf{G}(x',x)]\\ &= \int_{\mathbb{T}} e^{ik\cdot y}\,\tfrac12(1+e^{ik_\nu})(e^{-ik_\mu}-1)\,|\mathsf{k}|^{-2}\,d^\circ k.\end{aligned}$$

All four symbols are sums of products of the examples of Def. 7.16(3), namely $e^{ik\cdot y}$ (degree 0, leading part 1), k_μ and $\overline{\mathsf{k}_\mu}$ (degree 1, leading parts ik_μ and $-ik_\mu$) and $|\mathsf{k}|^{-2}$ (degree -2, leading part $|k|^{-2}$), and no product of degree below -2 occurs: the term $\mathsf{k}_\nu\overline{\mathsf{k}_\mu}|\mathsf{k}|^{-4}$ is the product $(\mathsf{k}_\nu|\mathsf{k}|^{-2})(\overline{\mathsf{k}_\mu}|\mathsf{k}|^{-2})$ of two line symbols of degree -1. By the rules for sums and products in Def. 7.16(3), the four symbols are line symbols of the degrees $-2,-1,0,-1$, so the kernels are elementary lines of the degrees $1,2,3,2$; their leading parts arise by replacing k_μ, $\overline{\mathsf{k}_\mu}$, $|\mathsf{k}|^{-2}$, e^{ik_ν} by ik_μ, $-ik_\mu$, $|k|^{-2}$, 1, which gives the leading parts listed.

If K has the symbol σ, then $K(-y) = \int e^{-ik\cdot y}\sigma(k)\,d^\circ k = \int e^{ik\cdot y}\sigma(-k)\,d^\circ k$ and $K(y+y_0) = \int e^{ik\cdot y}e^{ik\cdot y_0}\sigma(k)\,d^\circ k$. The function $\sigma(-k)$ is a line symbol of the same degree, with leading part $\sigma^h(-k)$, as χ is even; and $e^{ik\cdot y_0}\sigma(k)$ is the product of σ with a line symbol of degree 0 and leading part 1. For the consequence we recall from eq. 7.xiii that the contraction of two plain legs is $\delta_{aa'}\mathsf{C}$, that of a curl leg with a plain leg is $\delta_{aa'}\mathsf{E}$, and that of two curl legs is $\delta_{aa'}\Pi$. These kernels are invariant under simultaneous translations of their arguments, by their Fourier representations. Hence, by the list above and by the two rules just proved (which take care of the order of the legs and of the shifts ξ,ξ'), each contraction is an elementary line in y of degree 1, 2, 3, respectively. □

Lemma 7.25 (Π^{lat} is the curvature projection Π_∞ up to signs)**.** For a plaquette $p = p_{\mu\nu}(x)$, $\mu<\nu$, of $\mathbb{Z}^3$, let $\epsilon_p = -1$ for $(\mu,\nu) = (1,3)$ and $\epsilon_p = 1$ otherwise. This converts the positive orientation to the cyclic orientation indexed by the dual direction. Write $p^\flat = (x,\lambda(p))$ in these cyclic coordinates, and let Π_∞ be the orthogonal projection onto the closed range of curl in ℓ^2 of cyclic plaquette cochains. Its Fourier formula is equation YM3_I:(2242). Then for all plaquettes p,q of $\mathbb{Z}^3$

$$\Pi^{\mathrm{lat}}(p,q) = \epsilon_p\,\epsilon_q\,\Pi_\infty(p^\flat,q^\flat), \qquad \text{and hence} \qquad \mathsf{B}_0^{\mathrm{lat}}(p,q) = \tfrac{\mathsf{n}}{2}\,\Pi_\infty(p^\flat,q^\flat)^2.$$

■

Proof. Step 1. Write $p = p_{[\iota]}(x)$ and $q = p_{[\iota']}(x+y)$, and let $\epsilon_\iota := -1$ for $\iota = 2$ and $\epsilon_\iota := 1$ for $\iota\in\{1,3\}$. Since $p_{[2]} = p_{13}$, we have $\epsilon_p = \epsilon_\iota$ and $\epsilon_q = \epsilon_{\iota'}$. By the entry for Π in Lem. 7.24,

$$\Pi^{\mathrm{lat}}(p,q) = \int_{\mathbb{T}} e^{ik\cdot y}\sum_\nu \overline{\mathsf{c}_{\iota,\nu}(k)}\,\mathsf{c}_{\iota',\nu}(k)\,|\mathsf{k}|^{-2}\,d^\circ k.$$

Step 2. We show that

$$\sum_\nu \overline{\mathsf{c}_{\iota,\nu}}\,\mathsf{c}_{\iota',\nu} = \epsilon_\iota\,\epsilon_{\iota'}\big(\delta_{\iota\iota'}\,|\mathsf{k}|^2 - \mathsf{k}_\iota\,\overline{\mathsf{k}_{\iota'}}\big).$$

For $\iota = \iota'$, with $p_{[\iota]} = p_{\alpha\beta}$, the left side is $|\mathsf{k}_\alpha|^2 + |\mathsf{k}_\beta|^2 = |\mathsf{k}|^2 - |\mathsf{k}_\iota|^2$, and $\epsilon_\iota^2 = 1$. Let $\iota \neq \iota'$, and let ι'' be the third direction. As $\mathsf{c}_{\iota,\nu} = 0$ for $\nu = \iota$ and $\mathsf{c}_{\iota',\nu} = 0$ for $\nu = \iota'$, only $\nu = \iota''$ contributes. By definition $\mathsf{c}_{\iota,\iota''} = \pm\mathsf{k}_{\iota'}$, with the sign $+$ if and only if $\iota' < \iota''$, and $\mathsf{c}_{\iota',\iota''} = \pm\mathsf{k}_\iota$, with the sign $+$ if and only if $\iota < \iota''$. So the left side is $\pm\overline{\mathsf{k}_{\iota'}}\,\mathsf{k}_\iota$, with the sign $-$ if and only if ι'' lies between ι and ι', that is, if and only if $\iota'' = 2$. On the right side the factor of $\mathsf{k}_\iota\overline{\mathsf{k}_{\iota'}}$ is $-\epsilon_\iota\epsilon_{\iota'}$, which equals -1 if and only if neither ι nor ι' is 2, that is, if and only if $\iota'' = 2$.

Step 3. By Fubini's theorem, $\int_{\mathbb{T}} e^{ik\cdot y}\,d^\circ k = \prod_\mu \frac{1}{2\pi}\int_{-\pi}^{\pi} e^{ity_\mu}\,\mathrm{d}t = \delta_{y,0}$, because $\int_{-\pi}^{\pi} e^{itm}\,\mathrm{d}t = 0$ for integers $m \neq 0$. The substitution $k \to -k$, together with $\mathsf{k}_\mu(-k) = \overline{\mathsf{k}_\mu(k)}$, gives

$$\int_{\mathbb{T}} e^{ik\cdot y}\,\frac{\mathsf{k}_\iota\,\overline{\mathsf{k}_{\iota'}}}{|\mathsf{k}|^2}\,d^\circ k = \int_{[-\pi,\pi]^3} e^{-ik\cdot y}\,\frac{\overline{\mathsf{k}_\iota(k)}\,\mathsf{k}_{\iota'}(k)}{|\mathsf{k}(k)|^2}\,\frac{dk}{(2\pi)^3} = T_{\iota\iota'}(-y),$$

where $T_{\lambda\lambda'}(y) = \int_{[-\pi,\pi]^3} e^{ik\cdot y}\,\overline{\mathsf{q}_\lambda}\,\mathsf{q}_{\lambda'}\,|\mathsf{q}|^{-2}\,\frac{dk}{(2\pi)^3}$ and $\mathsf{q}_\mu(k) := \mathsf{k}_\mu(k) = e^{ik_\mu} - 1$. Together with Steps 1 and 2 we obtain

$$\Pi^{\mathrm{lat}}(p,q) = \epsilon_\iota\,\epsilon_{\iota'}\,[\delta_{\iota\iota'}\,\delta_{y,0} - T_{\iota\iota'}(-y)].$$

The Fourier formula in equation YM3_I:(2242) states that $\Pi_\infty((z,\lambda),(z',\lambda')) = \delta_{pq} - T_{\lambda\lambda'}(z-z')$ for the plaquettes $p = (z,\lambda)$, $q = (z',\lambda')$ of $\mathbb{Z}^3$. We apply it to $p^\flat = (x,\iota)$ and $q^\flat = (x+y,\iota')$, for which $z - z' = -y$ and $\delta_{p^\flat q^\flat} = \delta_{\iota\iota'}\delta_{y,0}$; this is the first formula. The second follows from eq. 7.xvii, as $\epsilon_p^2 = \epsilon_q^2 = 1$. □

Lemma 7.26 (The list of the pairings)**.** Let Ω and τ_{gf} be as in Def. 7.7, and let $\langle\cdot\rangle_0$ refer to the Gaussian family of Ω and τ_{gf}. For a homogeneous polynomial F of degree m in the variables $A^a(b)$ and for $\mathfrak{g}\otimes\mathbb{C}$-valued forms $X_1,\dots,X_m$ we put

$$\mathsf{F}[\![X_1,\dots,X_m]\!] := \partial_{s_1}\cdots\partial_{s_m}\,\mathsf{F}\Big(\sum_{i=1}^m s_i X_i\Big);$$

this expression is symmetric and multilinear in $X_1,\dots,X_m$, and $\mathsf{F}[\![A,\dots,A]\!] = m!\,\mathsf{F}(A)$. For an index $\mathsf{i} = (b,a)$, where b is a positively oriented bond of Ω (a free bond if $\Omega = \Lambda_n$) and $a \in \{1,2,3\}$, let δ_{i} be the form which equals e_a at b and 0 at all other positively oriented bonds. We put $\mathsf{C}(\mathsf{i},\mathsf{i}') := \delta_{aa'}\,\mathsf{C}(b,b')$ for $\mathsf{i}' = (b',a')$, and $\mathsf{E}^a_p := \mathsf{E}_p(\cdot)\,\mathsf{e}_a$.

(a) (Symmetry factors.) Let $\mathsf{F}_1,\dots,\mathsf{F}_r$ be homogeneous polynomials of degrees $m_1,\dots,m_r$. Then $\langle\mathsf{F}_1;\dots;\mathsf{F}_r\rangle_0 = \sum_{\mathsf{n}}\mathsf{val}(\mathsf{n})$. The sum extends over the symmetric matrices $\mathsf{n} = (n_{vv'})_{v,v'=1}^r$ of non-negative integers such that $2n_{vv} + \sum_{v'\neq v} n_{vv'} = m_v$ for every v and such that the graph on $\{1,\dots,r\}$ with $n_{vv'}$ edges between v and $v' \neq v$ is connected; and

$$\mathsf{val}(\mathsf{n}) = \Big[\prod_{v<v'} n_{vv'}!\prod_v 2^{n_{vv}}\,n_{vv}!\Big]^{-1}\sum\prod_{v=1}^r \mathsf{F}_v[\![\delta_{\mathsf{i}(v,1)},\dots,\delta_{\mathsf{i}(v,m_v)}]\!] \times \prod_{\text{lines }\{e,e'\}}\mathsf{C}(\mathsf{i}(e),\mathsf{i}(e')).$$

Here, for every such n, we fix one pairing of the slots $(v,1),\dots,(v,m_v)$, $v = 1,\dots,r$, with exactly $n_{vv'}$ pairs between the slots of v and those of v', and n_{vv} pairs among the slots of v; its pairs $\{e,e'\}$ are called lines, and the sum extends over all indices $\mathsf{i}(e)$ attached to the slots e. In (a), in (b) and in the proof of the lemma the symbol n stands for such a matrix; in the formula for T_{I} and in eq. 7.xxx below it is the number $\mathsf{n} = 3$ of the setting.

(b) (Colour.) For every plaquette P, every real kernel c on pairs of bonds, symmetric or not, and every form X,

$$\sum_{\mathsf{i},\mathsf{i}'} \mathsf{a}_P^{(3)}[\![\delta_{\mathsf{i}}, \delta_{\mathsf{i}'}, X]\!]\, \mathsf{c}(b,b')\, \delta_{aa'} = 0 \qquad (\mathsf{i} = (b,a),\ \mathsf{i}' = (b',a')).$$

Consequently $\mathsf{val}(\mathsf{n}) = 0$ if $n_{vv} \geq 1$ for some v with $\mathsf{F}_v = \mathsf{a}_P^{(3)}$, and also if $n_{vv'} = 2$ for some v, v' with $\mathsf{F}_v = \mathsf{a}_p^{(2)}$ and $\mathsf{F}_{v'} = \mathsf{a}_P^{(3)}$.

(c) (Classes.) Let Σ_{tot} and Γ_q be defined by the formulas of Def. 7.13(5), with the Gaussian family and the vertices of Ω. Then

$$\mathsf{B}_1^{\Omega}(p,q;\tau_{\mathrm{gf}}) = \mathsf{T}_{\mathrm{I}} + \mathsf{T}_1 + \mathsf{T}_4 + \mathsf{T}_{4'} + \mathsf{T}_5 + \mathsf{T}_6,$$

where, with P, P' denoting plaquettes and $\mathsf{i}, \mathsf{i}', \mathsf{j}, \mathsf{j}', \dots$ indices of Ω,

$$\mathsf{T}_{\mathrm{I}} := -\mathsf{n}\,\Pi(p,q) \sum_{b,b'} \mathsf{E}_p(b)\, \Sigma_{\mathrm{tot}}(b,b')\, \mathsf{E}_q(b'),$$

$$\mathsf{T}_1 := \tfrac{1}{2} \sum_{P,P'} \sum_{a,a'} \sum_{\mathsf{i},\mathsf{j}} \mathsf{a}_P^{(3)}[\![\mathsf{E}_p^a, \mathsf{E}_q^{a'}, \delta_{\mathsf{i}}]\!]\, \mathsf{C}(\mathsf{i},\mathsf{j})\, \mathsf{a}_{P'}^{(3)}[\![\mathsf{E}_p^a, \mathsf{E}_q^{a'}, \delta_{\mathsf{j}}]\!],$$

$$\mathsf{T}_4 := -\tfrac{1}{4} \sum_{P} \sum_{a,a'} \mathsf{a}_P^{(4)}[\![\mathsf{E}_p^a, \mathsf{E}_p^a, \mathsf{E}_q^{a'}, \mathsf{E}_q^{a'}]\!],$$

$$\mathsf{T}_{4'} := \tfrac{1}{4} \sum_{a} \sum_{\mathsf{i},\mathsf{j}} \mathsf{C}(\mathsf{i},\mathsf{j}) \{ \mathsf{a}_q^{(4)}[\![\mathsf{E}_p^a, \mathsf{E}_p^a, \delta_{\mathsf{i}}, \delta_{\mathsf{j}}]\!] + \mathsf{a}_p^{(4)}[\![\mathsf{E}_q^a, \mathsf{E}_q^a, \delta_{\mathsf{i}}, \delta_{\mathsf{j}}]\!] \},$$

$$\mathsf{T}_5 := -\tfrac{1}{2} \sum_{P} \sum_{a} \sum_{\mathsf{i},\mathsf{i}',\mathsf{j},\mathsf{j}'} \mathsf{C}(\mathsf{i},\mathsf{j})\, \mathsf{C}(\mathsf{i}',\mathsf{j}') \{ \mathsf{a}_q^{(3)}[\![\mathsf{E}_p^a, \delta_{\mathsf{i}}, \delta_{\mathsf{i}'}]\!]\, \mathsf{a}_P^{(3)}[\![\mathsf{E}_p^a, \delta_{\mathsf{j}}, \delta_{\mathsf{j}'}]\!] + \mathsf{a}_p^{(3)}[\![\mathsf{E}_q^a, \delta_{\mathsf{i}}, \delta_{\mathsf{i}'}]\!]\, \mathsf{a}_P^{(3)}[\![\mathsf{E}_q^a, \delta_{\mathsf{j}}, \delta_{\mathsf{j}'}]\!] \},$$

$$\mathsf{T}_6 := \tfrac{1}{6} \sum \mathsf{a}_p^{(3)}[\![\delta_{\mathsf{i}_1}, \delta_{\mathsf{i}_2}, \delta_{\mathsf{i}_3}]\!]\, \mathsf{a}_q^{(3)}[\![\delta_{\mathsf{j}_1}, \delta_{\mathsf{j}_2}, \delta_{\mathsf{j}_3}]\!] \prod_{\varsigma=1}^{3} \mathsf{C}(\mathsf{i}_\varsigma, \mathsf{j}_\varsigma),$$

the last sum extending over all indices $\mathsf{i}_1, \mathsf{i}_2, \mathsf{i}_3, \mathsf{j}_1, \mathsf{j}_2, \mathsf{j}_3$. By the first half of $\mathsf{T}_{4'}$, resp. of T_5, we mean the term with the first summand in braces, by the second half the term with the second summand. On $\Omega = \mathbb{Z}^3$ the formulas hold provided the sums converge absolutely, which is shown in Lem. 7.27. Moreover, if the summands of Γ_q are taken in the order of Def. 7.13(5), the second and the third being grouped together, then

$$\tfrac{\mathsf{n}}{2} \sum_{b,b'} \Gamma_q(b,b')\, \mathsf{E}_p(b)\, \mathsf{E}_p(b') = [\text{first half of } \mathsf{T}_{4'}] + [\text{first half of } \mathsf{T}_5] + \mathsf{T}_4 + \mathsf{T}_1 \tag{7.xxx}$$

term by term, the four groups of summands of Γ_q producing the four terms on the right in this order; the same holds with p and q exchanged and with the second halves of $\mathsf{T}_{4'}$ and T_5.

(d) (Vertices.) For forms X, Y, Z we have $\mathsf{a}_P^{(3)}[\![X,Y,Z]\!] = \sum \mathsf{t}_P(X',Y',Z')$, the sum extending over the six orderings (X',Y',Z') of (X,Y,Z), where

$$\mathsf{t}_P(X,Y,Z) := -\Big\langle \mathrm{d}X(P), \sum_{i<j} s_i s_j\, Y(b_i) \times Z(b_j) \Big\rangle,$$

with $b_i = b_i(P)$ and with the complex bilinear extensions of $\langle\cdot,\cdot\rangle$ and of $\times$. The standard writing of $\mathsf{a}_P^{(4)}$ consists of monomials of three kinds: with four plain legs (these come from $\frac{1}{2}|\mathfrak{z}_1|^2$), with one curl leg and three plain legs, and with four curl legs.

■

Proof. Step 1: proof of (a). Writing $A_{\mathsf{i}} := A^a(b)$ for $\mathsf{i} = (b,a)$, we have $A = \sum_{\mathsf{i}} A_{\mathsf{i}}\,\delta_{\mathsf{i}}$, and multilinearity together with $\mathsf{F}_v(A) = \frac{1}{m_v!}\mathsf{F}_v[\![A,\dots,A]\!]$ gives the finite sum

$$\mathsf{F}_v = \frac{1}{m_v!}\sum_{\mathsf{i}_1,\dots,\mathsf{i}_{m_v}} \mathsf{F}_v[\![\delta_{\mathsf{i}_1},\dots,\delta_{\mathsf{i}_{m_v}}]\!]\,A_{\mathsf{i}_1}\cdots A_{\mathsf{i}_{m_v}}.$$

We use the multilinearity of cumulants and Lem. 7.9, applied to the monomials $A_{\mathsf{i}(v,1)}\cdots A_{\mathsf{i}(v,m_v)}$, with the slots $(v,1),\dots,(v,m_v)$ as legs and with the covariance $\langle A_{\mathsf{i}}A_{\mathsf{i}'}\rangle_0 = \mathsf{C}(\mathsf{i},\mathsf{i}')$ of Def. 7.8(1). We find that $\langle \mathsf{F}_1;\dots;\mathsf{F}_r\rangle_0$ is $\prod_v \frac{1}{m_v!}$ times the sum, over all indices and over all pairings of the slots with connected graph on $\{1,\dots,r\}$, of $\prod_v \mathsf{F}_v[\![\cdots]\!]\prod_{\text{pairs}}\mathsf{C}$. As the forms $\mathsf{F}_v[\![\cdot]\!]$ are symmetric, the contribution of a pairing, summed over the indices, depends only on its matrix $(n_{vv'})$; and the graph of a pairing is connected if and only if that of its matrix is.

It remains to count the pairings with a given matrix n. At each vertex v we decide which slots are joined to which $v' \neq v$ and which are paired among themselves: there are $m_v!/[\prod_{v'\neq v} n_{vv'}!\,(2n_{vv})!]$ possibilities. For each pair $v < v'$ we match the chosen slots of v with those of v': $n_{vv'}!$ possibilities. At each v we pair the $2n_{vv}$ slots reserved for this purpose: $(2n_{vv})!/(2^{n_{vv}}n_{vv}!)$ possibilities. Since $\prod_v\prod_{v'\neq v} n_{vv'}! = \prod_{v<v'}(n_{vv'}!)^2$, the number of pairings is

$$\prod_v m_v!\Big/\Big[\prod_{v<v'} n_{vv'}!\prod_v 2^{n_{vv}}\,n_{vv}!\Big],$$

and multiplication by $\prod_v \frac{1}{m_v!}$ gives the factor in $\mathsf{val}(\mathsf{n})$.

Step 2: proof of (d) and (b). By eq. 7.v, $\mathsf{a}_P^{(3)}(A) = \langle \mathrm{d}A(P), \mathfrak{z}_1(P)\rangle$ with $\mathfrak{z}_1 = -\sum_{i<j} s_i s_j\, A(b_i)\times A(b_j)$; thus $\mathsf{a}_P^{(3)}(A) = \mathsf{t}_P(A,A,A)$ with the trilinear form t_P, and

$$\partial_{s_1}\partial_{s_2}\partial_{s_3}\,\mathsf{t}_P\big(\textstyle\sum s_iX_i, \sum s_iX_i, \sum s_iX_i\big)$$

is the sum of t_P over the six orderings of X_1,X_2,X_3. The three kinds of monomials of $\mathsf{a}_P^{(4)}$ are those of the three terms $\frac12|\mathfrak{z}_1|^2$, $\langle \mathrm{d}A(P),\mathfrak{z}_2\rangle$, $-\frac{1}{24}|\mathrm{d}A(P)|^4$ of eq. 7.v, written as in Def. 7.8(3).

For (b) we insert (d): $\mathsf{a}_P^{(3)}[\![\delta_{(b,a)},\delta_{(b',a)},X]\!]$ is a sum of terms, each the product of a real number which depends on b, b' and on the ordering, but not on a, with one of $\langle \mathsf{e}_a, \mathsf{e}_a\times X'\rangle$, $\langle \mathsf{e}_a, X'\times \mathsf{e}_a\rangle$, $\langle X', \mathsf{e}_a\times \mathsf{e}_a\rangle$, where X' is a value of X or of $\mathrm{d}X$. Summed over a, each of these three expressions vanishes, since $Y\times X'$ is orthogonal to Y and $Y\times Y = 0$; multiplying by $\mathsf{c}(b,b')$ and summing over b,b' we obtain (b). A line between two slots of a vertex $\mathsf{a}_P^{(3)}$ produces exactly the sum in (b) with $\mathsf{c} = \mathsf{C}$.

By the rule for $\mathsf{a}_p^{(2)}$ in Step 3 below, two lines from $\mathsf{a}_p^{(2)} = \frac12\sum_a((\mathrm{d}A)^a(p))^2$ to the same vertex $\mathsf{a}_P^{(3)}$ produce the sum in (b) with $\mathsf{c}(b,b') = \mathsf{E}_p(b)\,\mathsf{E}_p(b')$. In both cases $\mathsf{val}(\mathsf{n}) = 0$.

Step 3: rules for (c). From $\mathsf{a}_p^{(2)} = \frac12\sum_a((\mathrm{d}A)^a(p))^2$ we obtain $\mathsf{a}_p^{(2)}[\![\delta_{\mathsf{i}},\delta_{\mathsf{i}'}]\!] = \delta_{aa'}\,\mathsf{s}_p(b)\,\mathsf{s}_p(b')$, where $\mathsf{s}_p(b_i(p)) = s_i$ and $\mathsf{s}_p(b) = 0$ for the bonds b which do not belong to p. Since $\sum_b \mathsf{s}_p(b)\,\mathsf{C}(b,b'') = \mathsf{E}_p(b'')$, a line from a slot of $\mathsf{a}_p^{(2)}$ to a slot of another vertex inserts E_p^a into that slot, the colour a being summed together with the other slot of $\mathsf{a}_p^{(2)}$. A line from $\mathsf{a}_p^{(2)}$ to $\mathsf{a}_q^{(2)}$ produces $\Pi(p,q)$ and identifies the colours of the two remaining slots. A vertex $\mathsf{a}^{(2)}$ carrying a line between its own two slots is not connected to the other vertices.

Step 4: enumeration for (c). We go through the seven cumulants of eq. 7.xviii and list the matrices n with connected graph whose value does not vanish by (b) or by Step 3; in particular there are no lines between two slots of a vertex $\mathsf{a}^{(2)}$ or $\mathsf{a}^{(3)}$. We index these matrices by names of the vertices: the vertices $\mathsf{a}_p^{(m)}$ and $\mathsf{a}_q^{(m)}$ are called p and q, the vertices $\mathsf{a}_P^{(m)}$ and $\mathsf{a}_{P'}^{(3)}$ coming from S_3 or from V_2 are called P and P', and a quadratic local term F of V_2, taken as a vertex, is called F. Thus n_{pq} is the number of lines between the slots of p and those of q, n_{qq} the number of lines joining two slots of q, and similarly for the other entries.

- $\langle \mathsf{a}_p^{(3)}; \mathsf{a}_q^{(3)} \rangle_0$: here $n_{pq} = 3$, with the factor $\frac{1}{3!} = \frac{1}{6}$; this is T_6.
- $\langle \mathsf{a}_p^{(2)}; \mathsf{a}_q^{(4)} \rangle_0$: here $n_{pq} = 2$ and $n_{qq} = 1$, with the factor $\frac{1}{2\cdot 2}$; this is the first half of $\mathsf{T}_{4'}$. In the same way $\langle \mathsf{a}_p^{(4)}; \mathsf{a}_q^{(2)} \rangle_0$ gives the second half.
- $\langle \mathsf{a}_p^{(2)}; \mathsf{a}_q^{(3)}; \mathsf{a}_P^{(3)} \rangle_0$, from the term $-\langle \mathsf{a}_p^{(2)}; \mathsf{a}_q^{(3)}; \mathsf{S}_3 \rangle_0$: the conditions $n_{pq}+n_{pP} = 2$, $n_{pq}+n_{qP} = 3$, $n_{pP} + n_{qP} = 3$ force $n_{pq} = n_{pP} = 1$ and $n_{qP} = 2$, with the factor $\frac{1}{2}$; with the sign of eq. 7.xviii and the sum over P this is the first half of T_5. In the same way $-\langle \mathsf{a}_p^{(3)}; \mathsf{a}_q^{(2)}; \mathsf{S}_3 \rangle_0$ gives the second half.
- $\langle \mathsf{a}_p^{(2)}; \mathsf{a}_q^{(2)}; \mathsf{F} \rangle_0$ with a quadratic local term F of V_2: here $n_{pq} = n_{p\mathsf{F}} = n_{q\mathsf{F}} = 1$, with the factor 1, and the value is $\Pi(p,q) \sum_a \mathsf{F}[\![\mathsf{E}_p^a, \mathsf{E}_q^a]\!] = \Pi(p,q) \sum_a \sum_{b,b'} \mathsf{E}_p(b)\, \mathsf{E}_q(b')\, \partial_{b,a}\partial_{b',a}\mathsf{F}$.
- $\langle \mathsf{a}_p^{(2)}; \mathsf{a}_q^{(2)}; \mathsf{a}_P^{(4)} \rangle_0$: either $n_{pq} = n_{pP} = n_{qP} = n_{PP} = 1$, with the factor $\frac{1}{2}$ and the value
$$\Pi(p,q) \sum_a \tfrac{1}{2} \sum_{\mathsf{i},\mathsf{j}} \mathsf{C}(\mathsf{i},\mathsf{j})\, \mathsf{a}_P^{(4)}[\![\mathsf{E}_p^a, \mathsf{E}_q^a, \delta_\mathsf{i}, \delta_\mathsf{j}]\!]$$
$$= \Pi(p,q) \sum_a \sum_{b,b'} \mathsf{E}_p(b)\, \mathsf{E}_q(b')\, \langle \partial_{b,a}\partial_{b',a} \mathsf{a}_P^{(4)} \rangle_0$$
(by (a) for the quadratic polynomial $\partial_{b,a}\partial_{b',a}\mathsf{a}_P^{(4)}$, whose polarisation is $\mathsf{a}_P^{(4)}[\![\delta_{(b,a)}, \delta_{(b',a)}, \cdot, \cdot]\!]$); or $n_{pP} = n_{qP} = 2$, with the factor $\frac{1}{4}$, which with the sign of eq. 7.xviii and the sum over P gives T_4. The matrix with $n_{pq} = 2$ has a disconnected graph.
- $\langle \mathsf{a}_p^{(2)}; \mathsf{a}_q^{(2)}; \mathsf{a}_P^{(3)}; \mathsf{a}_{P'}^{(3)} \rangle_0$: subtracting the conditions at p and q from those at P and P' we get $n_{PP'} = 1 + n_{pq}$. For $n_{pq} = 2$ the graph is disconnected. For $n_{pq} = 1$ we get $n_{PP'} = 2$ and either $n_{pP} = n_{qP'} = 1$ or $n_{pP'} = n_{qP} = 1$, each with the factor $\frac{1}{2}$; summed over P, P' and multiplied by the $\frac{1}{2}$ of eq. 7.xviii, they give
$$\Pi(p,q) \sum_a \tfrac{1}{2} \sum \mathsf{a}_P^{(3)}[\![\mathsf{E}_p^a, \delta_\mathsf{i}, \delta_{\mathsf{i}'}]\!]\, \mathsf{a}_{P'}^{(3)}[\![\mathsf{E}_q^a, \delta_\mathsf{j}, \delta_{\mathsf{j}'}]\!]\, \mathsf{C}(\mathsf{i},\mathsf{j})\, \mathsf{C}(\mathsf{i}',\mathsf{j}')$$
$$= \Pi(p,q) \sum_a \sum_{b,b'} \mathsf{E}_p(b)\, \mathsf{E}_q(b')\, \langle \partial_{b,a}\mathsf{S}_3; \partial_{b',a}\mathsf{S}_3 \rangle_0,$$
where the sum on the left extends over P, P' and all indices, and the equality is (a) for two quadratic polynomials. For $n_{pq} = 0$ we get $n_{PP'} = 1$ and $n_{pP} + n_{qP} = 2 = n_{pP'} + n_{qP'}$; the possibilities $(n_{pP}, n_{qP}) = (2,0)$ and $(0,2)$ vanish by (b), and there remains $n_{pP} = n_{qP} = n_{pP'} = n_{qP'} = 1$, with the factor 1; with the $\frac{1}{2}$ of eq. 7.xviii this is T_1.
- We collect the terms with $n_{pq} = 1$ found above, with the signs of eq. 7.xviii. By the definition of Σ_{tot} in Def. 7.13(5) they add up to $-\Pi(p,q) \sum_a \sum_{b,b'} \mathsf{E}_p(b)\, \mathsf{E}_q(b')\, \Sigma_{\mathrm{tot}}(b,a;b',a)$. The argument on the colour structure given there applies in the box as well and shows $\Sigma_{\mathrm{tot}}(b,a;b',a) = \Sigma_{\mathrm{tot}}(b,b')$ for every colour a; summing over the three colours we obtain T_I.

Step 5: proof of eq. 7.xxx. Let $\partial = \partial_{b,a}$, $\partial' = \partial_{b',a'}$, $\delta = \delta_{(b,a)}$, $\delta' = \delta_{(b',a')}$. By (a),
$$\langle \partial\partial' \mathsf{a}_q^{(4)} \rangle_0 = \tfrac{1}{2} \sum_{\mathsf{i},\mathsf{j}} \mathsf{C}(\mathsf{i},\mathsf{j})\, \mathsf{a}_q^{(4)}[\![\delta, \delta', \delta_\mathsf{i}, \delta_\mathsf{j}]\!],$$
$$\langle \partial \mathsf{a}_q^{(3)}; \partial' \mathsf{S}_3 \rangle_0 = \tfrac{1}{2} \sum_P \sum \mathsf{C}(\mathsf{i},\mathsf{j})\, \mathsf{C}(\mathsf{i}',\mathsf{j}')\, \mathsf{a}_q^{(3)}[\![\delta, \delta_\mathsf{i}, \delta_{\mathsf{i}'}]\!]\, \mathsf{a}_P^{(3)}[\![\delta', \delta_\mathsf{j}, \delta_{\mathsf{j}'}]\!],$$
$$\langle \mathsf{a}_q^{(2)}; \partial\partial' \mathsf{S}_4 \rangle_0 = \tfrac{1}{2} \sum_P \sum_{a''} \mathsf{a}_P^{(4)}[\![\delta, \delta', \mathsf{E}_q^{a''}, \mathsf{E}_q^{a''}]\!],$$
$$\langle \mathsf{a}_q^{(2)}; \partial \mathsf{S}_3; \partial' \mathsf{S}_3 \rangle_0 = \sum_{P,P'} \sum_{a''} \sum_{\mathsf{i},\mathsf{j}} \mathsf{C}(\mathsf{i},\mathsf{j})\, \mathsf{a}_P^{(3)}[\![\delta, \mathsf{E}_q^{a''}, \delta_\mathsf{i}]\!]$$
$$\times\, \mathsf{a}_{P'}^{(3)}[\![\delta', \mathsf{E}_q^{a''}, \delta_\mathsf{j}]\!];$$

in the last line the three polynomials are quadratic, the only connected matrix has one line between any two of them, and its factor is 1. We contract these expressions, and the one obtained from the second by exchanging (b,a) and (b',a'), with $\frac12\sum_a \mathsf{E}_p^a\otimes\mathsf{E}_p^a$; that is, we set $a'=a$, multiply by $\frac12\,\mathsf{E}_p(b)\,\mathsf{E}_p(b')$ and sum over b,b',a. Since $\Gamma_q(b,a;b',a)=\Gamma_q(b,b')$, this turns Γ_q into the left side of eq. 7.xxx. By multilinearity, and with the signs of Def. 7.13(5), the first summand of Γ_q produces the first half of $\mathsf{T}_{4'}$; the second and the third, whose contributions coincide after the symmetric contraction, together produce the first half of T_5; the fourth produces T_4; and the fifth produces T_1. Exchanging p and q gives the statement for the second halves. □

Lemma 7.27 (Convergence of the graph sums on $\mathbb{Z}^3$)**.** Let $\tau_{\mathrm{gf}}>0$.

(i) On $\mathbb{Z}^3$, for all plaquettes p,q, the sums over the positions of the local terms of S_3 and V_2 in eq. 7.xviii, taken monomial by monomial in the standard writing and pairing by pairing, converge absolutely. Consequently $\mathsf{B}_1^{\mathrm{lat}}(p,q;\tau_{\mathrm{gf}})$ is defined, the formulas of Lem. 7.26(c) hold on $\mathbb{Z}^3$, and $\sup_{p,q}|\mathsf{B}_1^{\mathrm{lat}}(p,q;\tau_{\mathrm{gf}})|<\infty$.

(ii) (Uniform form.) Let $\mathsf{c}_0\ge 1$, let $\Omega=\mathbb{Z}^3$ or $\Omega=\Lambda_n$, and substitute for $\mathsf{C},\mathsf{E},\Pi,\mathsf{D},\mathsf{g}$ arbitrary real kernels which satisfy, with the base points b_- of the bonds and x_p of the plaquettes,

$$|\mathsf{C}(b,b')|\le\frac{\mathsf{c}_0}{\langle b_- - b'_-\rangle},\qquad |\mathsf{E}_p(b)|\le\frac{\mathsf{c}_0}{\langle x_p-b_-\rangle^2},$$
$$|\mathsf{D}(b;b')|\le\frac{\mathsf{c}_0}{\langle b_- - b'_-\rangle^2},\qquad |\Pi(p,q)|\le\frac{\mathsf{c}_0}{\langle x_p-x_q\rangle^3},\qquad |\mathsf{g}(b)|\le\mathsf{c}_0. \tag{7.xxxi}$$

Then the sum of the absolute values of all terms of the seven cumulants of eq. 7.xviii, over monomials, pairings and positions, is at most $C\,\mathsf{c}_0^5$, with a pure constant C.

■

Proof. Step 1. By Lem. 7.24, the kernels C, E, Π, D of $\mathbb{Z}^3$ are, as functions of the difference of the base points, elementary lines of the degrees $1,2,3,2$, with finitely many different symbols. Lem. 7.17(a), applied with $s=-2,-1,0,-1$ and $\gamma=0$, bounds them by constants times $\langle\cdot\rangle^{-1}$, $\langle\cdot\rangle^{-2}$, $\langle\cdot\rangle^{-3}$, $\langle\cdot\rangle^{-2}$, and $\mathsf{g}=\frac13$ on $\mathbb{Z}^3$ by Def. 7.7(2). Hence these kernels satisfy eq. 7.xxxi with some $\mathsf{c}_0=\mathsf{c}_0(\tau_{\mathrm{gf}})$, and (i) follows from (ii): the bound $C\mathsf{c}_0^5$, which does not depend on p,q, is the absolute convergence required in Def. 7.8(3), so that $\mathsf{B}_1^{\mathrm{lat}}$ is defined and those rearrangements made in the proof of Lem. 7.26(c) are allowed whose terms are bounded by the majorants of Step 2 below, such as regroupings of whole terms of positions and regroupings which split one curl leg or, as in Step 5 of that proof and for $n_{pq}=n_{pP}=n_{qP}=n_{PP}=1$ in its Step 4, curl legs at P or P' joined to p or q, whose lines then have the degree 2 which these majorants have there.

For the kernels of $\mathbb{Z}^3$ we obtain the equality of the display for $n_{pq}=1$ in Step 4 of that proof, whose right side splits the curl legs at P and at P' joined to p and to q, as in $\Pi(p,P)=\sum_b \mathsf{s}_P(b)\,\mathsf{E}_p(b)$, by regrouping one vertex at a time, with u,z,w as in Step 2. At P we write $\mathsf{a}_P^{(3)}[\![\mathsf{E}_p^a,\cdot,\cdot]\!]=\sum_b \mathsf{E}_p(b)\,\mathsf{a}_P^{(3)}[\![\delta_{(b,a)},\cdot,\cdot]\!]$, split the quadratic polynomial $\partial_{b,a}\mathsf{a}_P^{(3)}$, whose polarisation is the last factor, into its two terms as in eq. 7.xxxii, which follows from eq. 7.v by the Leibniz rule, and keep the vertex at P' in the standard writing; the resulting terms, indexed by (P,b,P'), the monomials and the pairings, are bounded by $C\langle z\rangle^{-2}\langle w-z\rangle^{-n_3}\langle u-w\rangle^{-n_5}$, with a constant C which depends only on τ_{gf}, as the kernels of $\mathbb{Z}^3$ satisfy eq. 7.xxxi with $\mathsf{c}_0=\mathsf{c}_0(\tau_{\mathrm{gf}})$, and with the four pairs (n_3,n_5) of Step 2, for which the computation there holds for every $n_1\ge 2$; so they may be grouped by (b,P'), and the sum over $P\ni b$ turns the first entry into $\partial_{b,a}\mathsf{S}_3$.

Next, with z,w the positions of b,P' relative to the base point of p and with $C=C(\tau_{\mathrm{gf}})$, we have $|\langle\partial_{b,a}\mathsf{S}_3;\partial_{b',a}\mathsf{a}_{P'}^{(3)}\rangle_0|\le C\langle w-z\rangle^{-3}$: by eq. 7.xxxii, a monomial of $\mathsf{u}_{b,a}$ has a curl leg and a plain leg, so that the two lines joining it to a monomial of $\partial_{b',a}\mathsf{a}_{P'}^{(3)}$ have degrees of sum at least 3, and eq. 7.xxxi applies; and $\mathsf{d}_{b,a}$ contributes $\sum_{P\ni b}\mathsf{s}_P(b)\,\mathsf{F}(P)$ with $\mathsf{F}(P):=\langle\langle\mathsf{e}_a,\mathfrak{z}_1(P)\rangle;\partial_{b',a}\mathsf{a}_{P'}^{(3)}\rangle_0$, by

Lem. 7.26(a) and Lem. 7.24 a finite sum of products $K_{\mathrm{a}}K_{\mathrm{b}}$ of elementary lines in the base point of P with $n_{\mathrm{a}}+n_{\mathrm{b}}\geq 2$; since the four plaquettes containing b form two pairs, each of one dual direction, with base points differing by one of the vectors e_1,e_2,e_3 and with opposite signs, this is a sum of two first differences in the base point of P, which Lem. 7.22(c) with $|\gamma|=1$ bounds by $C\langle w-z\rangle^{-3}$.

At P', by Lem. 7.26(a) for two quadratic polynomials, the term of (b,P') so obtained is the sum over b' of $\mathsf{E}_p(b)\,\mathsf{E}_q(b')\,\langle\partial_{b,a}\mathsf{S}_3;\partial_{b',a}\mathsf{a}^{(3)}_{P'}\rangle_0$; these terms, indexed by (b,P',b'), are bounded by $C\langle z\rangle^{-2}\langle w-z\rangle^{-3}\langle u-w\rangle^{-2}$, the pair $(3,2)$ of Step 2; so they may be grouped by (b,b'), and the sum over $P'\ni b'$ gives the right side of that display. Thus this equality holds on $\mathbb{Z}^3$, and the sum over b,b' in the formula for T_{I} converges absolutely: for the summand $-\langle\partial_{b,a}\mathsf{S}_3;\partial_{b',a}\mathsf{S}_3\rangle_0$ of Σ_{tot} by these majorants, and for the summands of $\langle\partial_{b,a}\partial_{b',a}\mathsf{V}_2\rangle_0$, which by eq. 7.xi and eq. 7.xxxi are bounded and of finite range, or bounded by $8\mathsf{c}_0^2\langle b_--b'_-\rangle^{-4}$, by the sums of Step 2.

Step 2. We prove (ii). We write every vertex in the standard writing and count a curl leg as one leg (Lem. 7.26(d)); at every position there are finitely many monomials, with coefficients independent of the position. We follow the enumeration in the proof of Lem. 7.26(c); at fixed positions all sums there are finite. By eq. 7.xxxi a term is bounded, up to a bounded factor, by $\prod_\ell \mathsf{c}_0\langle y_\ell\rangle^{-n_\ell}$, where ℓ ranges over its lines, y_ℓ is the difference of the positions of the two vertices joined by ℓ, and $n_\ell=1+\#(\text{curl legs at }\ell)$; the bounded factor accounts for shifts of length at most 3 between the base points of the legs and the positions of the vertices, since $\langle y\rangle\leq 4\langle y+v\rangle$ for $|v|\leq 3$.

Every term has at most five lines, $\mathsf{c}_0\geq 1$, and the numbers of monomials and pairings are pure constants; it therefore suffices to bound the sums over the positions of the products $\prod_\ell\langle y_\ell\rangle^{-n_\ell}$ by pure constants. Let u be the base point of q minus that of p, and let z,w be the positions of P,P' (or of the bonds of the local terms of V_2) relative to the base point of p.

- T_1: the lines starting at p or ending at q carry a curl leg of p or of q, so their degrees are at least 2, and the line between P and P' has degree at least 1. As $\langle\cdot\rangle^{-n}$ decreases in n, the sum is at most the sum in Prop. 7.23(i) with $(n_1,\dots,n_5)=(2,2,1,2,2)$, which satisfies the hypotheses of that proposition.
- The terms of T_{I} with two cubic vertices, or with the ghost loop, have sums of the form
$$\langle u\rangle^{-3}\sum_{z,w}\langle z\rangle^{-n_1}\langle w-z\rangle^{-n_3}\langle u-w\rangle^{-n_5},$$
$$n_1,n_5\in\{2,3\},\qquad n_3\geq 2,\qquad n_1+n_3+n_5=8;$$
the factor $\langle u\rangle^{-3}$ comes from $\Pi(p,q)$, and $\langle w-z\rangle^{-n_3}$ from the two lines between the cubic vertices, resp. from $\mathsf{D}(b;b')\mathsf{D}(b';b)$. The possible pairs are $(n_3,n_5)=(2,3),(3,2),(3,3),(4,2)$, and for each of them Lem. 7.18(c) gives $\sum_w\langle w-z\rangle^{-n_3}\langle u-w\rangle^{-n_5}\leq C\langle u-z\rangle^{-2}\lg\langle u-z\rangle$. Next, $\sum_z\langle z\rangle^{-2}\langle u-z\rangle^{-2}\lg\langle u-z\rangle\leq C$: by $\lg t\leq 2t^{1/2}$ for $t\geq 1$ and by Lem. 7.18(c) with $(a,a')=(2,\frac{3}{2})$ this sum is at most $C\langle u\rangle^{-1/2}\lg\langle u\rangle\leq C$. Finally $\langle u\rangle^{-3}\leq 1$.
- The other terms of T_{I} (with a line between two slots of the quartic vertex at P, with the Haar term, with the ghost tadpole) have sums bounded by $\sum_z\langle z\rangle^{-2}\langle u-z\rangle^{-2}\leq C$, by Lem. 7.18(c).
- T_4 leads to $\sum_z\langle z\rangle^{-4}\langle u-z\rangle^{-4}$, and T_5 to $\sum_z\langle z\rangle^{-n_X}\langle u-z\rangle^{-n_Y}$ with $n_X,n_Y\geq 2$, the line between p and q being bounded by c_0; both sums are bounded by Lem. 7.18(c). The sums of $\mathsf{T}_{4'}$ and T_6 are finite, as these classes have no vertex besides p and q.
- The graphs which vanish by colour in Lem. 7.26(b) vanish pairing by pairing, so they contribute no terms. Indeed $\langle\mathrm{d}A,A_i\times A_j\rangle=\sum\epsilon_{abc}(\mathrm{d}A)^aA_i^bA_j^c$, so every monomial of $\mathsf{a}^{(3)}_P$ has three legs of three different colours, while $((\mathrm{d}A)^a(p))^2$ has two legs of one colour; since the covariance is diagonal in the colours, a line between two legs of one monomial of $\mathsf{a}^{(3)}_P$, or two lines from $((\mathrm{d}A)^a(p))^2$ to one monomial of $\mathsf{a}^{(3)}_P$, give the factor 0.

In the box the positions range over subsets of $\mathbb{Z}^3$, and the same bounds apply. □

7.1.11. *Elementary terms, their expansions, the local parts.*

Lemma 7.28 (Elementary terms and their degrees)**.** Let $\tau_{\mathrm{gf}} > 0$, let us work on $\mathbb{Z}^3$, let $p = p_{[\iota]}(x_p)$ and $q = p_{[\iota']}(x_q)$ be plaquettes, and put $u' := x_q - x_p$. We write every vertex occurring in the formulas of Lem. 7.26(c) in the standard writing (Def. 7.8(3), Lem. 7.26(d)), we carry out the sums over the colours, and we write every contraction as an elementary line by Lem. 7.24. Then there are finite index sets $\mathfrak{G}_\bullet = \mathfrak{G}_\bullet(\iota, \iota')$, real numbers c_G and elementary lines $K^G_\bullet$, all independent of x_p and x_q, such that the following representations hold; every G is called an elementary term. We omit the index G on the lines, and $*$ denotes the convolution of Lem. 7.22.

(1) $\mathsf{T}_1 = \sum_{G\in\mathfrak{G}_1} c_G\, \mathsf{T}^G(u')$, where every T^G is a kite as in Prop. 7.23, with degrees satisfying the hypotheses of that proposition and with $\Delta = 5$. Here K_1 and K_2 are contractions of $(\mathrm{d}A)(p)$ with legs at P and at P', K_4 and K_5 are contractions of legs at P and at P' with $(\mathrm{d}A)(q)$, and K_3 is the contraction of a leg at P with a leg at P'.

(4) $\mathsf{T}_4 = \sum_{G\in\mathfrak{G}_4} c_G\, (X * Y)(u')$, where $X = K_1K_2$ and $Y = K_3K_4$ with $n_i \in \{2,3\}$, and $(n_X, n_Y) := (n_1+n_2, n_3+n_4) \in \{(4,4),(5,4),(4,5),(6,6)\}$, with $(n_X, n_Y) = (4,4)$ for the monomials of $\frac{1}{2}|\mathfrak{z}1|^2$.

(4′) The first half of $\mathsf{T}_{4'}$ equals $\sum_{G\in\mathfrak{G}_{4'}} c_G\, K_1(u')\, K_2(u')$, where $n_i \in \{2,3\}$ and K_1, K_2 are contractions of $(\mathrm{d}A)(p)$ with legs at q.

(5) The first half of T_5 equals $\sum_{G\in\mathfrak{G}_5} c_G\, K_0(u')\, (X * Y)(u')$ with $Y = K_\mathrm{a}K_\mathrm{b}$. Here K_0 and X are contractions of $(\mathrm{d}A)(p)$ with a leg at q, resp. at P, and $n_0, n_X \in \{2,3\}$; $K_\mathrm{a}, K_\mathrm{b}$ are contractions of legs at P with legs at q, and $n_\mathrm{a}, n_\mathrm{b} \in \{1,2,3\}$; finally $n_0 + n_X + n_\mathrm{a} + n_\mathrm{b} = 8$, so that $n_Y := n_\mathrm{a} + n_\mathrm{b} \in \{2,3,4\}$.

(6) $\mathsf{T}_6 = \sum_{G\in\mathfrak{G}_6} c_G\, K_1(u')\, K_2(u')\, K_3(u')$, with $n_1 + n_2 + n_3 = 5$.

(I) $\mathsf{T}_\mathrm{I} = -\mathsf{n}\,\Pi(p,q) \sum_{\mu,\nu} (K^{\rightarrow}_{\iota\mu} * \Sigma_{\mu\nu} * K^{\leftarrow}_{\nu\iota'})(u')$, where $K^{\rightarrow}_{\iota\mu}(y) := \mathsf{E}_{p_{[\iota](0)}}((y,\mu))$ and $K^{\leftarrow}_{\nu\iota'}(y) := \mathsf{E}_{p_{[\iota']}(y)}((0,\nu))$ are elementary lines of degree 2, and $\Sigma_{\mu\nu}(y) := \Sigma_{\mathrm{tot}}((x,\mu),(x+y,\nu))$, which does not depend on x. With $\mathsf{s}_P(b) := s_i$ if $b = b_i(P)$ and $\mathsf{s}_P(b) := 0$ otherwise,

$$\partial_{b,a}\mathsf{S}_3 = \mathsf{d}_{b,a} + \mathsf{u}_{b,a},$$
$$\mathsf{d}_{b,a} := \sum_{P\ni b} \mathsf{s}_P(b)\, \langle \mathsf{e}_a, \mathfrak{z}1(P)\rangle, \qquad \mathsf{u}_{b,a} := \sum_{P\ni b} \langle \mathrm{d}A(P), \partial_{b,a}\mathfrak{z}1(P)\rangle; \tag{7.xxxii}$$

the monomials of $\mathsf{d}_{b,a}$ have two plain legs, those of $\mathsf{u}_{b,a}$ a curl leg and a plain leg. Moreover

$$\Sigma_{\mu\nu} = \mathsf{q}^0_{\mu\nu} + \sum_{G\in\mathfrak{G}_\mathrm{I}(\mu,\nu)} c_G\, \mathsf{f}_G(\nabla)[K_\mathrm{a}K_\mathrm{b}], \tag{7.xxxiii}$$

where $\mathsf{q}^0_{\mu\nu}$ has finite support (it comes from the line between two slots of the quartic Wilson vertex, from the Haar term and from the ghost tadpole); $\mathsf{f}_G(k)$ is the product of two trigonometric polynomials, each a line symbol of degree 0 or 1, of total degree $d_G \in \{0,1,2\}$; $\mathsf{f}_G(\nabla)$ is the finite difference operator which multiplies symbols by f_G; $n_\mathrm{a}, n_\mathrm{b} \in \{1,2,3\}$; and $n_\mathrm{a} + n_\mathrm{b} + d_G = 4$. The terms G are the ghost loop $-8\,\mathsf{D}(b;b')\,\mathsf{D}(b';b)$, with $d_G = 0$, and the terms of $-\langle\partial\mathsf{S}_3;\partial'\mathsf{S}_3\rangle_0$ multiplied out by eq. 7.xxxii, with d_G the number of entries d.

The second halves of $\mathsf{T}_{4'}$ and T_5 have the same forms, with p and q exchanged and u' replaced by $-u'$. ■

Proof. We use two counting rules: every monomial of $\mathsf{a}^{(3)}$ has exactly one curl leg and $\mathsf{a}^{(2)}$ has two (Def. 7.8(3)); and the degree of a line is $1 + \#$(its curl legs) (Lem. 7.24). By the consequence stated in Lem. 7.24, every contraction is an elementary line in the difference of the base points of the two vertices which it joins, its symbol depending only on the two legs; the shifts between the base points of the legs and those of the vertices are absorbed into the lines.

(1) A kite has five lines. There are four curl legs at p and q and one at each of P and P', hence $\sum_i n_i = 5 + 6 = 11$ and $\Delta = 5$. For $i \in \{1,2,4,5\}$ one end of the line i is a curl leg of p or of q, so

that $n_i \geq 2$, and $n_i \leq 3$. Besides the two curl legs of p, the lines $1,2,3$ carry at most two further curl legs, one of P and one of P'; hence $n_1+n_2+n_3 \leq 2+2+1+2 = 7$, and the same count applies to the lines $3,4,5$. With the base points x_p+z and x_p+w of P and P', the sum over P, P' becomes the sum over z, w and over the dual directions of P, P', and the five lines become functions of z, w, $w-z$, $u'-z$, $u'-w$.

(4) Each of the four lines carries a curl leg at p or at q; it has degree 2, or degree 3 if its leg at P is a curl leg. The three kinds of monomials of $\mathsf{a}_P^{(4)}$ (Lem. 7.26(d)) give $(n_X, n_Y) = (4,4)$; $(5,4)$ or $(4,5)$; and $(6,6)$. With the base point x_p+z of P, the two lines from p make up $X(z)$, the two lines to q make up $Y(u'-z)$, and the sum over z is $(X*Y)(u')$.

(4′) There are two lines from p to q, each with its curl leg at p; the line between two slots of the vertex at q is a number, independent of the positions, and it is absorbed into c_G.

(5) There are four lines, and there are $2+1+1$ curl legs, at p, q and P; hence $\sum n = 4+4 = 8$. With the base point x_p+z of P, the line from p to P is $X(z)$, the two lines from P to q make up $Y(u'-z)$, and the line from p to q is $K_0(u')$.

(6) There are three lines, and there are $1+1$ curl legs, at p and q; hence $\sum n = 3+2 = 5$.

(I) In the formula for T_I we write $b = (x,\mu)$ and $b' = (x',\nu)$ with $x = x_p+z$, $x' = x_p+w$. By translation invariance $\mathsf{E}_p(b) = K^{\rightarrow}_{\iota\mu}(z)$ and $\mathsf{E}_q(b') = K^{\leftarrow}_{\nu\iota'}(u'-w)$, and by the translation covariance in Def. 7.13(5) $\Sigma_\mathrm{tot}(b,b') = \Sigma_{\mu\nu}(w-z)$; this gives the formula for T_I. The functions $K^{\rightarrow}_{\iota\mu}$ and $K^{\leftarrow}_{\nu\iota'}$ are elementary lines of degree 2 by Lem. 7.24, the second because $K^{\leftarrow}_{\nu\iota'}(y) = \mathsf{E}_{p_{[\iota']}(0)}((-y,\nu))$. By Def. 7.13(5) and eq. 7.xi, with $\partial = \partial_{b,a}$ and $\partial' = \partial_{b',a}$,

$$\Sigma_\mathrm{tot}(b,b') = \Big[\langle \partial\partial' \mathsf{S}_4\rangle_0 + (\tfrac{2}{3} + \tfrac{4}{3}\mathsf{g}(b))\delta_{bb'}\Big] \\ - 4\big[\mathsf{D}(b;b')\mathsf{D}(b';b) + \mathsf{D}(b';b)\mathsf{D}(b;b')\big] - \langle \partial\mathsf{S}_3; \partial'\mathsf{S}_3\rangle_0.$$

The bracket has finite range and is translation invariant; it gives $\mathsf{q}^0_{\mu\nu}$. The middle term is the ghost loop $-8\,\mathsf{D}(b;b')\,\mathsf{D}(b';b)$, a product of two elementary lines of degree 2 in y (Lem. 7.24), with $\mathsf{f}_G = 1$. From eq. 7.v and the Leibniz rule we have

$$\partial_{b,a}\langle \mathrm{d}A(P), \mathfrak{z}_1(P)\rangle = \mathsf{s}_P(b)\,\langle \mathsf{e}_a, \mathfrak{z}_1(P)\rangle + \langle \mathrm{d}A(P), \partial_{b,a}\mathfrak{z}_1(P)\rangle,$$

which gives eq. 7.xxxii. For a function F on the plaquettes of one dual direction, the map $b \mapsto \sum_{P\ni b} \mathsf{s}_P(b)\,\mathsf{F}(P)$ is a difference operator whose symbol is $\pm\overline{\mathsf{k}_\alpha}$ up to a factor $e^{ik\cdot y_0}$, a line symbol of degree 1; this operator yields the factor of f_G which belongs to an entry d. An entry u we split into its monomials and treat each of them separately. By the consequence stated in Lem. 7.24, the contraction of a leg with base point $x+\xi$ with a leg with base point $x+y+\xi'$ is an elementary line in y in which the shifts ξ and ξ' are already contained; hence the factor of f_G which belongs to an entry u is 1, a line symbol of degree 0 with leading part 1, just as for the ghost loop.

By Lem. 7.26(a), the cumulant of two quadratic monomials is a sum of two products of two lines. Hence every term of $-\langle \partial\mathsf{S}_3; \partial'\mathsf{S}_3\rangle_0$, multiplied out by eq. 7.xxxii, has the form $c_G\,\mathsf{f}_G(\nabla)[K_\mathrm{a}K_\mathrm{b}]$, with d_G the number of entries d and with $n_\mathrm{a}+n_\mathrm{b} = 2+\#(\text{curl legs on the two lines})$. As d has no curl leg and u has one, $n_\mathrm{a}+n_\mathrm{b} = 2+(2-d_G)$. The second halves of $\mathsf{T}_{4'}$ and T_5 are treated in the same way, the roles of p and q being exchanged. □

Lemma 7.29 (Expansions of the elementary terms, with errors)**.** Let the setting be that of Lem. 7.28, let $r' := |u'|$ be large, and let $K^h := \mathcal{F}^{-1}_{\neq 0}\sigma^h$ for every line K with symbol σ. For the bubbles X and Y let q_X and q_Y be the kernels q of Lem. 7.22(c), and put $X' := X - \mathsf{q}_X$, $Y' := Y - \mathsf{q}_Y$. Then the elementary terms of Lem. 7.28 have the following expansions.

(6) $K_1K_2K_3(u') = K_1^hK_2^hK_3^h(u') + O(r'^{-6})$.

(5) $K_0(u')\,(X{*}Y)(u') = K_0(u')\,(\mathsf{q}_Y{*}X)(u') + K_0^h(u')\,(X{*}Y')^h(u') + O(r'^{-6}\lg r')$, where $(X{*}Y')^h$ is the second component of the leading data of $X*Y' \in \mathfrak{K}(n_X+n_Y-3)$.

(4) $(X*Y)(u') = (\mathsf{q}_X * Y)(u') + (\mathsf{q}_Y * X)(u') + \mathsf{U}_G(u') + O(r'^{-6}\lg r')$, where $\mathsf{U}_G := (X'*Y')^h$ if $(n_X, n_Y) = (4,4)$ and $\mathsf{U}_G := 0$ otherwise.
(1) The expansion of Prop. 7.23(iii) holds, with the error $O(r'^{-6}\lg r')$.
(∗) In all four cases the function of degree -5 in the expansion (that is, $K_1^h K_2^h K_3^h$, $K_0^h (X*Y')^h$, U_G, resp. the function U of Prop. 7.23) is determined by the leading parts σ^h of the lines alone; it is multilinear in them, it belongs to $C^1(\mathbb{R}^3 \setminus \{0\})$, and it is rotation covariant as in Prop. 7.23(iv).

■

Proof. (6) By Lem. 7.17(a),(c), $K_i(u') = O(r'^{-n_i})$, $K_i^h(u') = O(r'^{-n_i})$ and $K_i(u') = K_i^h(u') + O(r'^{-n_i-1})$. As $n_1+n_2+n_3 = 5$, the product $K_1K_2K_3(u')$ differs from $K_1^hK_2^hK_3^h(u')$ by $O(r'^{-6})$, and $K_1^hK_2^hK_3^h$ is homogeneous of degree -5.

(5) The line X is an elementary line of degree $n_X \in \{2,3\}$, so $X \in \mathfrak{K}(n_X)$ by Lem. 7.17(c). The bubble $Y = K_\mathrm{a}K_\mathrm{b}$ has $n_\mathrm{a}, n_\mathrm{b} \in \{1,2,3\}$ and $n_Y \le 4$, so $Y' \in \mathfrak{K}(n_Y)$ by Lem. 7.22(c). We write $X*Y = \mathsf{q}_Y*X + X*Y'$. Lem. 7.22(b) applies to X and Y', as $n_X, n_Y \le 4$ and $n_X+n_Y = 8-n_0 \ge 5$; it gives $X*Y' \in \mathfrak{K}(n_X+n_Y-3)$, and (K3) with $\gamma = 0$ gives $X*Y' = (X*Y')^h + O(r'^{-(n_X+n_Y-3)-1}\lg r')$. Moreover $K_0 = K_0^h + O(r'^{-n_0-1})$ by Lem. 7.17(c), and K_0, K_0^h, $X*Y'$, $(X*Y')^h$ are bounded by the corresponding powers of r'. As $n_0 + (n_X+n_Y-3) = 8-3 = 5$, the claim follows.

(4) Multiplying out $X*Y = (\mathsf{q}_X + X') * (\mathsf{q}_Y + Y')$ we obtain

$$X*Y = \mathsf{q}_X * Y + \mathsf{q}_Y * X - \mathsf{q}_X * \mathsf{q}_Y + X'*Y',$$

and $\mathsf{q}_X * \mathsf{q}_Y$ vanishes off $\{|u'| \le 2\}$, because q_X and q_Y vanish off $\{0, \pm e_j\}$. If $(n_X, n_Y) = (4,4)$, then $X', Y' \in \mathfrak{K}(4)$ by Lem. 7.22(c), and Lem. 7.22(b) gives $X'*Y' \in \mathfrak{K}(5)$ and $X'*Y' = (X'*Y')^h + O(r'^{-6}\lg r')$. Otherwise we apply Lem. 7.22(a) to X' and Y', with $a_X = \min\{n_X - 3, 2\}$ and $a_Y = \min\{n_Y - 3, 2\}$. Its hypotheses hold by Lem. 7.22(c), which bounds the differences of a bubble of degree n by $C_\gamma \langle x\rangle^{-n-|\gamma|}$ (the kernels X', Y' differ from X, Y at finitely many points only) and the symbol of X', resp. Y', by $|\mathsf{I} - \tilde{\mathsf{Q}}| \le C|k|^{\min\{n-3,2\}}$. The three exponents in the bound of Lem. 7.22(a) are $-2-4$, $-1-5$, $3-9$ for $(n_X, n_Y) = (5,4)$, the same by symmetry for $(4,5)$, and at most -8 for $(6,6)$; so $X'*Y' = O(r'^{-6}\lg r')$ in these cases.

(1) This is Prop. 7.23(iii) with $\Delta = 5$ (Lem. 7.28(1)), so that $r^{-\Delta-1}\lg r = r'^{-6}\lg r'$.

(∗) In the case (6) this follows from Lem. 7.17(b). In the cases (5) and (4), Lem. 7.22(b) shows that the second component of the leading data of a convolution depends only on the leading data of the two factors, bilinearly and rotation covariantly. By Lem. 7.22(c) the leading data of Y', and of X' in the case (4), are $(\mathsf{I}^h, K_\mathrm{a}^h K_\mathrm{b}^h)$, where I^h depends only on $\sigma_\mathrm{a}^h, \sigma_\mathrm{b}^h$, bilinearly and rotation covariantly, by Lem. 7.20(a); the leading data of the lines X and K_0 are (σ^h, K^h) with $K^h = \mathcal{F}_{\neq 0}^{-1}\sigma^h$, by Lem. 7.17(b),(c). The functions K^h and the second components of leading data are smooth off the origin. In the case (1) the assertions are Prop. 7.23(iii),(iv). □

Lemma 7.30 (A local insertion with vanishing moments)**.**

(a) Let $\gamma_{\mu\nu}$, $1 \le \mu, \nu \le 3$, be functions of finite support on $\mathbb{Z}^3 \times \mathbb{Z}^3$ whose symbols

$$\hat\gamma_{\mu\nu}(\kappa_1, \kappa_2) := \sum_{y,y'} \gamma_{\mu\nu}(y,y')\, e^{-i(\kappa_1\cdot y + \kappa_2 \cdot y')}$$

vanish at $\kappa = 0$ together with their gradients, and let K_1, K_2, K_3 be elementary lines of degree 2. Then

$$\Big|\sum_{\mu,\nu}\sum_{y,y'} \gamma_{\mu\nu}(y,y')\, K_\mu(u+y)\, K_\nu(u+y')\Big| \le C\,|u|^{-6} \qquad (u \in \mathbb{Z}^3 \setminus \{0\}),$$

where C depends on the $\gamma_{\mu\nu}$ and on the symbols of the K_μ.

(b) Let q be a function of finite support on $\mathbb{Z}^3$ with $\hat{\mathsf{q}}(k) := \sum_y \mathsf{q}(y)\, e^{-ik\cdot y} = O(|k|^2)$ at $k = 0$, and let K_1, K_2 be elementary lines of degree 2. Then $|(K_1 * \mathsf{q} * K_2)(u)| \le C\langle u\rangle^{-3}$ for all $u \in \mathbb{Z}^3$, where C depends on q and on the symbols of K_1, K_2.

■

Proof. (a) As the $\gamma_{\mu\nu}$ have finite support, only points y, y' in a fixed ball occur; let $|u|$ be large. By Lem. 7.17(a) with $s = -1$, $K_\mu(u) = O(|u|^{-2})$, $\nabla K_\mu(u) = O(|u|^{-3})$, and the second differences of K_μ near u are $O(|u|^{-4})$. We write $K_\mu(u+y) - K_\mu(u)$ as a sum of first differences along a nearest-neighbour path from u to $u+y$ and compare each of them with $\nabla_j K_\mu(u)$; this gives

$$K_\mu(u+y) = K_\mu(u) + \sum_j y_j\, \nabla_j K_\mu(u) + O(|u|^{-4}).$$

We insert this expansion for $K_\mu(u+y)$ and for $K_\nu(u+y')$. The terms of the orders $|u|^{-4}$ and $|u|^{-5}$ then carry the coefficients

$$\sum_{y,y'} \gamma_{\mu\nu}(y,y') = \hat\gamma_{\mu\nu}(0), \qquad \sum_{y,y'} \gamma_{\mu\nu}(y,y')\, y_j, \qquad \sum_{y,y'} \gamma_{\mu\nu}(y,y')\, y'_j,$$

and the last two are i times the derivatives of $\hat\gamma_{\mu\nu}$ with respect to $\kappa_{1,j}$, resp. $\kappa_{2,j}$, at 0. All three vanish by hypothesis, and the remaining terms are $O(|u|^{-6})$. For $u \neq 0$ in a bounded set the left side is bounded and $|u|^{-6}$ is bounded below, so the bound holds for all $u \neq 0$ after enlarging C.

(b) The trigonometric polynomial $\hat{\mathsf{q}}$ is smooth and vanishes at 0 together with its gradient, hence $\hat{\mathsf{q}} \in \mathcal{S}^2$. The kernel $\mathsf{q} * K_2$, a finite combination of translates of K_2, has the symbol $\hat{\mathsf{q}}\sigma_2 \in \mathcal{S}^1$, where $\sigma_2 \in \mathcal{S}^{-1}$ is the symbol of K_2; by Lem. 7.17(a) it is $O(\langle x\rangle^{-4})$ and lies in $\ell^2(\mathbb{Z}^3)$, as does $K_1 = O(\langle x\rangle^{-2})$. By Plancherel's theorem $K_1 * (\mathsf{q} * K_2)$ is the kernel with the symbol $\sigma_1\hat{\mathsf{q}}\sigma_2 \in \mathcal{S}^0$, and Lem. 7.17(a) with $s = 0$ gives the bound $C\langle u\rangle^{-3}$. □

Lemma 7.31 (The local parts vanish to the second order)**.** Let $\tau_{\mathrm{gf}} > 0$; we work on $\mathbb{Z}^3$.

(a) (Self-energy.) In eq. 7.xxxiii let q_G be the kernel q of the bubble $K_{\mathrm{a}}K_{\mathrm{b}}$ in the sense of Lem. 7.22(c), and put

$$\mathsf{q}^{\Sigma}_{\mu\nu} := \mathsf{q}^{0}_{\mu\nu} + \sum_G c_G\, \mathsf{f}_G(\nabla)\mathsf{q}_G, \qquad \mathsf{W}_{\mu\nu} := \sum_G c_G\, \mathsf{f}_G(\nabla)[K_{\mathrm{a}}K_{\mathrm{b}} - \mathsf{q}_G],$$

so that $\Sigma_{\mu\nu} = \mathsf{q}^{\Sigma}_{\mu\nu} + \mathsf{W}_{\mu\nu}$. Then $\mathsf{q}^{\Sigma}_{\mu\nu}$ has finite support. Each $\mathsf{f}_G(\nabla)[K_{\mathrm{a}}K_{\mathrm{b}} - \mathsf{q}_G]$ belongs to $\mathfrak{K}(4)$, and the first component of its leading data is $\mathsf{f}^h_G\, \mathsf{I}^h_G$, where f^h_G is the leading part of f_G and I^h_G is the homogeneous part of the symbol I_G of the bubble (Lem. 7.22(c), Lem. 7.20(d)). Hence $\mathsf{W}_{\mu\nu} \in \mathfrak{K}(4)$, and its leading data are functions of the leading parts σ^h and of the f^h_G occurring in eq. 7.xxxiii alone. Moreover the symbol of $\mathsf{q}^{\Sigma}_{\mu\nu}$ satisfies $\hat{\mathsf{q}}^{\Sigma}_{\mu\nu}(k) = O(|k|^2)$ at $k = 0$.

(b) (Vertex.) Let $q = p_{[\iota']}(x_q)$. Let $\gamma_q(b,b')$ be the symmetric kernel of finite range defined below; then, for every plaquette p with $|u'|$ large, $\frac{\mathsf{n}}{2}\sum_{b,b'} \gamma_q(b,b')\, \mathsf{E}_p(b)\, \mathsf{E}_p(b')$ is the sum of the following terms of Lem. 7.29: the first half of $\mathsf{T}_{4'}$; the terms $c_G\, K_0(u')\, (\mathsf{q}_Y * X)(u')$ of the first half of T_5; the terms $c_G\, (\mathsf{q}_Y * X)(u')$ of T_4; and the terms $c_G \sum_{z,w} \mathsf{q}_{\mathsf{q}}(z,w)\, K_1(u'-z)\, K_2(u'-w)$ of T_1. The kernel γ_q is defined as follows. Each of K_0, X, K_1, K_2, at the points where it is evaluated in these terms, is either a value $\mathsf{E}_p(b)$ with b a bond near q, or a value $\Pi(p,P) = \sum_i s_i\, \mathsf{E}_p(b_i(P))$ with P a plaquette near q.

Substituting this term by term in the four sums we obtain a quadratic expression in the values $\mathsf{E}_p(b)$, and $\frac{\mathsf{n}}{2}\gamma_q$ is its coefficient kernel, symmetrised. This kernel does not depend on p. Indeed, by the rule for $\mathsf{a}^{(2)}_p$ in the proof of Lem. 7.26(c), the plaquette p, with its base point and its dual direction, enters the four sums only through the forms E^a_p inserted into the two slots joined to p, that is, only through the values of K_0, X, K_1, K_2 just described. An elementary term G of these sums is given by the monomials at the vertices at q, P, P'

and by the choice of their legs which are contracted with p, with q and with one another, so that the numbers c_G do not involve p; and the kernels q_Y, q_q and the line between two slots of the vertex at q are built from lines among q, P, P' alone.

We define γ_q through this way of writing, since the numbers $\sum_{b,b'} \gamma_q(b,b')\mathsf{E}_p(b)\mathsf{E}_p(b')$ alone would not fix a kernel. Let $\gamma_{q;\mu\nu}(y,y') := \gamma_q((x_q+y,\mu),(x_q+y',\nu))$. Then the symbols $\hat\gamma_{q;\mu\nu}$, defined as in Lem. 7.30(a), vanish at $\kappa=0$ together with their gradients.

■

Proof. Step 1: a trace formula. Let $\mathsf{m}^1,\dots,\mathsf{m}^\mathsf{r}$, $\mathsf{r}\le 3$, be real symmetric matrices indexed by $\mathsf{i}=(x,\mu,a)$, that is, by a bond (x,μ) and a colour a, each with finitely many non-zero entries. For $y\in\mathbb{Z}^3$ let m_y be the translate of m by y, $(\mathsf{m}_y)_{(x,\mu,a),(x',\mu',a')} := \mathsf{m}_{(x-y,\mu,a),(x'-y,\mu',a')}$, and let $\hat{\mathsf{m}}(l',l)$ be the 9×9 matrix, indexed by (μ,a) and (μ',a'), with the entries

$$\sum_{x,x'} e^{-il'\cdot x}\,\mathsf{m}_{(x,\mu,a),(x',\mu',a')}\,e^{il\cdot x'}.$$

Then $\hat{\mathsf{m}}_y(l',l)=e^{-i(l'-l)\cdot y}\,\hat{\mathsf{m}}(l',l)$, and $\hat{\mathsf{m}}(l,l)$ is the symbol of the translation invariant matrix $\sum_y \mathsf{m}_y$, in the sense that $\sum_y \mathsf{m}_y$ maps $e^{il\cdot x}\alpha$ to $e^{il\cdot x}\,\hat{\mathsf{m}}(l,l)\,\alpha$. We insert $\mathsf{C}(\mathsf{i},\mathsf{i}')=\delta_{aa'}\int_{\mathbb{T}} e^{il\cdot(x-x')}\,\hat{\mathsf{C}}_{\mu\mu'}(l)\,d^\circ l$ and let $\hat{\mathsf{C}}(l)$ act on the nine-dimensional space as $\hat{\mathsf{C}}(l)$ in the directions and as the identity in the colours. For $\mathsf{r}=3$ we obtain

$$\begin{aligned}&\operatorname{Tr}\big[\mathsf{m}^1\mathsf{C}\,\mathsf{m}^2_{y_2}\mathsf{C}\,\mathsf{m}^3_{y_3}\mathsf{C}\big]\\&\quad=\iint e^{i(\kappa_2\cdot y_2+\kappa_3\cdot y_3)}\Big\{\int \operatorname{tr}\big[\hat{\mathsf{m}}^1(l_3,l)\,\hat{\mathsf{C}}(l)\,\hat{\mathsf{m}}^2(l,l_2)\,\hat{\mathsf{C}}(l_2)\\&\qquad\hat{\mathsf{m}}^3(l_2,l_3)\,\hat{\mathsf{C}}(l_3)\big]\,d^\circ l\Big\}\,d^\circ\kappa_2\,d^\circ\kappa_3,\end{aligned}\tag{7.xxxiv}$$

with $l_2:=l+\kappa_2$, $l_3:=l+\kappa_2+\kappa_3$, all integrals being over $\mathbb{T}$. Indeed, the sums over the indices are finite and the integrals over the variables l,l_2,l_3 of the three covariances converge absolutely; Fubini's theorem allows us to carry out the sums first and then to pass to the variables $\kappa_2=l_2-l$, $\kappa_3=l_3-l_2$. For $\mathsf{r}=2$ one omits m^3, the last factor $\hat{\mathsf{C}}$ and κ_3; for $\mathsf{r}=1$ one has $\operatorname{Tr}[\mathsf{m}^1\mathsf{C}]=\int\operatorname{tr}[\hat{\mathsf{m}}^1(l,l)\,\hat{\mathsf{C}}(l)]\,d^\circ l$. For the quadratic forms $\mathsf{F}_j:=\frac12(A,\mathsf{m}^jA)$, Lem. 7.26(a) gives

$$\langle\mathsf{F}_1\rangle_0=\tfrac12\operatorname{Tr}[\mathsf{m}^1\mathsf{C}],\qquad\langle\mathsf{F}_1;\mathsf{F}_2\rangle_0=\tfrac12\operatorname{Tr}[\mathsf{m}^1\mathsf{C}\,\mathsf{m}^2\mathsf{C}],$$
$$\langle\mathsf{F}_1;\mathsf{F}_2;\mathsf{F}_3\rangle_0=\operatorname{Tr}[\mathsf{m}^1\mathsf{C}\,\mathsf{m}^2\mathsf{C}\,\mathsf{m}^3\mathsf{C}].$$

The formula eq. 7.xxxiv also identifies integrands. For a leg ℓ of colour a we define $\mathsf{v}_\ell(k)\in\mathbb{C}^3$ by $\ell(e^{ik\cdot x}\alpha\,\mathsf{e}_a)=\mathsf{v}_\ell(k)\cdot\alpha$, where $e^{ik\cdot x}\alpha\,\mathsf{e}_a$ denotes the form $(x,\mu)\mapsto e^{ik\cdot x}\alpha_\mu\mathsf{e}_a$; thus $\mathsf{v}_\ell(k)=e^{ik\cdot\xi}e_\nu$ for the plain leg at the bond (ξ,ν), and $\mathsf{v}_\ell(k)=e^{ik\cdot\xi}\big(\mathsf{c}_{\iota,\nu}(k)\big)_{\nu=1,2,3}$ for the curl leg at $p_{[\iota]}(\xi)$. The contraction of ℓ with the translate by y of a leg ℓ' of the same colour is $\int e^{-ik\cdot y}\,\mathsf{v}_\ell(k)^\mathsf{T}\,\hat{\mathsf{C}}(k)\,\mathsf{v}_{\ell'}(-k)\,d^\circ k$, and the Hessian m of the product $\ell_1\ell_2$ of two legs has

$$\hat{\mathsf{m}}(l',l)=\mathsf{v}_{\ell_1}(-l')\,\mathsf{v}_{\ell_2}(l)^\mathsf{T}+\mathsf{v}_{\ell_2}(-l')\,\mathsf{v}_{\ell_1}(l)^\mathsf{T},$$

the colour indices being those of the legs. The matrices m^j used below are sums of Hessians of monomials in legs. Multiplying out, we find that for all (l,κ) the integrand in braces in eq. 7.xxxiv is a sum of products of the symbols of r lines evaluated at $\pm l,\pm l_2,\pm l_3$, and that these products are, term by term, the integrands of the one-loop functions (Lem. 7.20, Lem. 7.22(c), Prop. 7.23) of the elementary terms of Lem. 7.28, up to substitutions $l\to\pm l+\lambda(\kappa)$ with λ linear. At $\kappa=0$ the integrand in braces is the sum of the integrands of these one-loop functions at $\kappa=0$, up to the substitution $l\to -l$ in single terms. We express this by saying that the two sides agree integrand by integrand.

Step 2: the Hessians at a constant abelian field. By eq. 7.iii with $t=1$, $a_p(e^{iA})=\sum_m \mathsf{a}_p^{(m)}(A)$. Let $\mathsf{N}_j(B)$ and $\mathsf{h}_q^{(j)}(B)$ be the parts of degree j in B of the Hessians $\mathsf{N}(B)$ and $\mathsf{h}_q(B)$ of Def. 7.13(1).

Since S_3 is cubic and S_4 quartic, $\mathsf{N}_1(B)$ is the Hessian of the quadratic polynomial $\sum_{b,a} B^a(b)\,\partial_{b,a}\mathsf{S}_3$, and $\mathsf{N}_2(B)$ is that of $\frac12\sum B^a(b)\,B^{a'}(b')\,\partial_{b,a}\partial_{b',a'}\mathsf{S}_4$; the matrix $\mathsf{h}_q^{(j)}(B)$ is obtained in the same way from $\mathsf{a}_q^{(2+j)}$.

Let H_b be the Hessian of $\partial_{b,3}\mathsf{S}_3$, and let $\mathsf{H}_b^{\mathsf{u}}$ and $\mathsf{H}_b^{\mathsf{d}}$ be those of $\mathsf{u}_{b,3}$ and $\mathsf{d}_{b,3}$ in eq. 7.xxxii; these matrices are translation covariant, $\mathsf{H}_{b+y}=(\mathsf{H}_b)_y$. For the constant field Y_γ of Def. 7.13(2), with $\mu(b)$ the direction of b, we have

$$\sum_b \gamma_{\mu(b)}\,\mathsf{d}_{b,3}=\sum_P\langle \mathrm{d}Y_\gamma(P),\mathfrak{z}_1(P)\rangle=0$$

(as a quadratic form, every entry of its Hessian being a finite sum), because $\sum_b\gamma_{\mu(b)}\,\mathsf{s}_P(b)\,\mathsf{e}_3=\mathrm{d}Y_\gamma(P)$ and $\mathrm{d}Y_\gamma=0$. Hence $\mathsf{N}_1(Y_\gamma)=\sum_b\gamma_{\mu(b)}\,\mathsf{H}_b^{\mathsf{u}}=\sum_\mu\gamma_\mu\sum_y(\mathsf{H}^{\mathsf{u}}_{(0,\mu)})_y$, and by Step 1 and Def. 7.13(2) its symbol is

$$\sum_\mu\gamma_\mu\,\hat{\mathsf{H}}^{\mathsf{u}}_{(0,\mu)}(l,l)=\bigoplus_e\hat{\mathsf{N}}_{e,1}(l;\gamma)$$

in the basis e_e of the colours.

Step 3: the structure in (a). Let $K\in\mathfrak{K}(n)$ have the symbol σ and the leading data (σ^h,K^h), and let $y_0\in\mathbb{Z}^3$. The kernels with the symbols $e^{ik\cdot y_0}\sigma$, $\mathsf{k}_\mu\sigma$ and $\overline{\mathsf{k}_\mu}\sigma$ lie in $\mathfrak{K}(n)$, $\mathfrak{K}(n+1)$ and $\mathfrak{K}(n+1)$, respectively, with the leading data (σ^h,K^h), $(ik_\mu\sigma^h,\partial_\mu K^h)$ and $(-ik_\mu\sigma^h,-\partial_\mu K^h)$. Indeed, the symbol $\mathsf{k}_\mu\sigma$ belongs to the kernel $\nabla_\mu K$. For (K2) we write

$$\mathsf{k}_\mu\sigma-\chi\,ik_\mu\sigma^h=\mathsf{k}_\mu(\sigma-\chi\sigma^h)+(\mathsf{k}_\mu-ik_\mu)\,\chi\sigma^h=O(|k|^{n-1}\lg|k|),$$

and for (K3), by Taylor's formula and the homogeneity of the derivatives of K^h,

$$\nabla^\gamma(\nabla_\mu K-\partial_\mu K^h)=\nabla^{\gamma+e_\mu}(K-K^h)+\nabla^\gamma(\nabla_\mu-\partial_\mu)K^h=O(|x|^{-n-2-|\gamma|}\lg|x|);$$

at the finitely many points with $|\gamma|+1\le|x|<|\gamma|+2$ the left side is a finite number, which is absorbed into the constant. Next, $\overline{\mathsf{k}_\mu}=-e^{-ik_\mu}\mathsf{k}_\mu$. For the shift, $(e^{ik\cdot y_0}-1)\,\chi\sigma^h=O(|k|^{n-2})$ and $\nabla^\gamma[K^h(\cdot+y_0)-K^h]=O(|x|^{-n-1-|\gamma|})$. By Lem. 7.22(c), $K_{\mathrm a}K_{\mathrm b}-\mathsf{q}_G\in\mathfrak{K}(n_{\mathrm a}+n_{\mathrm b})$, with the leading data $(\mathsf{I}_G^h,K_{\mathrm a}^hK_{\mathrm b}^h)$; that lemma applies, since $n_{\mathrm a},n_{\mathrm b}\in\{1,2,3\}$ and $n_{\mathrm a}+n_{\mathrm b}=4-d_G\le4$. A trigonometric polynomial f is a finite sum of terms $c\,e^{ik\cdot y_0}$.

As a line symbol of degree 0 it has the leading part $\mathsf{f}(0)$. As a line symbol of degree 1 it has $\mathsf{f}(0)=0$, its leading part is its linear part at 0, and f is a finite sum of terms $c\,e^{ik\cdot y_0}\mathsf{k}_\mu$: indeed f is a linear combination of the functions $e^{ik\cdot y}-1$, and each of these is a telescoping sum, along a nearest-neighbour path from 0 to y, of terms $\pm e^{ik\cdot y_0}\mathsf{k}_\mu$ (with $\overline{\mathsf{k}_\mu}=-e^{-ik_\mu}\mathsf{k}_\mu$ for the steps in the negative direction). Since $\mathfrak{K}(n)$ is a linear space and the leading data depend linearly on the kernel, multiplication of the symbol by f maps $\mathfrak{K}(n)$ into $\mathfrak{K}(n)$, resp. $\mathfrak{K}(n+1)$, and multiplies the first component of the leading data by f^h.

Applying this once for each of the two factors of f_G, we obtain $\mathsf{f}_G(\nabla)[K_{\mathrm a}K_{\mathrm b}-\mathsf{q}_G]\in\mathfrak{K}(n_{\mathrm a}+n_{\mathrm b}+d_G)=\mathfrak{K}(4)$, with $\mathsf{f}_G^h\,\mathsf{I}_G^h$ as the first component of its leading data. Hence $\mathsf{W}_{\mu\nu}\in\mathfrak{K}(4)$. The first component of its leading data, $\sum_G c_G\,\mathsf{f}_G^h\,\mathsf{I}_G^h$, depends only on the leading parts σ^h (Lem. 7.20(a)) and on the f_G^h; by Lem. 7.22(d) the second component is determined by the first, linearly. Finally $\mathsf{q}_{\mu\nu}^\Sigma$ has finite support, since $\mathsf{q}_{\mu\nu}^0$ and the q_G have and $\mathsf{f}_G(\nabla)$ is a finite difference operator.

Step 4: the value of $\hat{\mathsf{q}}_{\mu\nu}^\Sigma$ at 0. The symbol of q_G is $\tilde{\mathsf{Q}}_G$, whose value at 0 is $\mathsf{Q}_G(0)$, and $\mathsf{f}_G(0)=0$ if $d_G\ge1$. Hence

$$\hat{\mathsf{q}}_{\mu\nu}^\Sigma(0)=\hat{\mathsf{q}}_{\mu\nu}^0(0)+\sum_{G:\,d_G=0}c_G\,\mathsf{f}_G(0)\,\mathsf{Q}_G(0).$$

For $d_G=0$ we have $n_{\mathrm a}+n_{\mathrm b}=4$, so that the bubble symbol has $\omega=1$, and Lem. 7.20(c), with the data of the bubble given in the proof of Lem. 7.22(c), yields $\mathsf{Q}_G(0)=\int_{\mathbb T}\sigma_{\mathrm a}(-l)\,\sigma_{\mathrm b}(l)\,d^\circ l$. The terms with $d_G=0$ are the ghost loop and the terms of $-\langle\mathsf{u}_{b,3};\mathsf{u}_{b',3}\rangle_0=-\frac12\,\mathrm{Tr}[\mathsf{H}_b^{\mathsf{u}}\mathsf{C}\,\mathsf{H}_{b'}^{\mathsf{u}}\mathsf{C}]$ (Step 1). We compute $\frac12\sum_{\mu,\nu}\gamma_\mu\gamma_\nu$ times each contribution to $\hat{\mathsf{q}}_{\mu\nu}^\Sigma(0)$.

- Gluon loop. By eq. 7.xxxiv, $-\frac12\operatorname{Tr}[\mathsf{H}^{\mathsf{u}}_{(0,\mu)}\mathsf{C}\,\mathsf{H}^{\mathsf{u}}_{(y,\nu)}\mathsf{C}]$ is the kernel in $y=b'_--b_-$ with the symbol $\kappa_2\mapsto-\frac12\int\operatorname{tr}[\hat{\mathsf{H}}^{\mathsf{u}}_{(0,\mu)}(l_2,l)\,\hat{\mathsf{C}}(l)\,\hat{\mathsf{H}}^{\mathsf{u}}_{(0,\nu)}(l,l_2)\,\hat{\mathsf{C}}(l_2)]\,d^\circ l$, and this symbol is $\sum c_G\,\mathsf{f}_G\,\mathsf{I}_G$, summed over the corresponding G, integrand by integrand. At zero momentum the integrand is $-\frac12\operatorname{tr}[\hat{\mathsf{H}}^{\mathsf{u}}_{(0,\mu)}(l,l)\,\hat{\mathsf{C}}(l)\,\hat{\mathsf{H}}^{\mathsf{u}}_{(0,\nu)}(l,l)\,\hat{\mathsf{C}}(l)]$, and by Step 2
$$\tfrac12\sum_{\mu,\nu}\gamma_\mu\gamma_\nu\Big(-\tfrac12\int\operatorname{tr}[\hat{\mathsf{H}}^{\mathsf{u}}_{(0,\mu)}(l,l)\,\hat{\mathsf{C}}\,\hat{\mathsf{H}}^{\mathsf{u}}_{(0,\nu)}(l,l)\,\hat{\mathsf{C}}]\,d^\circ l\Big)$$
$$=-\tfrac14\sum_e\int\operatorname{tr}[(\hat{\mathsf{C}}\hat{\mathsf{N}}_{e,1})^2]\,d^\circ l=\int_{\mathbb{T}}\mathfrak{s}^{(1)}(l;\gamma)\,d^\circ l.$$
- Gluon tadpole. Let $\mathsf{H}_{bb'}$ be the Hessian of $\partial_{b,3}\partial_{b',3}\mathsf{S}_4$, and let $\mathsf{m}:=\frac12\sum_{\mu,\nu}\sum_y\gamma_\mu\gamma_\nu\,\mathsf{H}_{(0,\mu),(y,\nu)}$, a finite sum. By Step 2 and translation covariance $\mathsf{N}_2(Y_\gamma)=\sum_x\mathsf{m}_x$, hence $\hat{\mathsf{m}}(l,l)=\bigoplus_e\hat{\mathsf{N}}_{e,2}(l;\gamma)$, and Step 1 with $\mathsf{r}=1$ gives
$$\tfrac12\sum_{\mu,\nu}\gamma_\mu\gamma_\nu\sum_y\langle\partial_{(0,\mu),3}\partial_{(y,\nu),3}\mathsf{S}_4\rangle_0=\tfrac12\operatorname{Tr}[\mathsf{m}\,\mathsf{C}]$$
$$=\tfrac12\int\operatorname{tr}[\hat{\mathsf{m}}(l,l)\,\hat{\mathsf{C}}(l)]\,d^\circ l=\int_{\mathbb{T}}\mathfrak{s}^{(2)}(l;\gamma)\,d^\circ l.$$
- Haar term: $\frac12\cdot\frac23|\gamma|^2=\frac13|\gamma|^2=\int\mathfrak{s}^{(5)}$. Ghost tadpole: $\frac12\cdot\frac43\cdot\frac13|\gamma|^2=\frac29|\gamma|^2=\int\mathfrak{s}^{(4)}$, since by Lem. 7.14(c) $\int\mathfrak{s}^{(4)}=\sum_{e=\pm1}\frac{4}{12}\sum_\mu\gamma_\mu^2\int|\mathsf{k}_\mu|^2|\mathsf{k}|^{-2}\,d^\circ k$, and $\int|\mathsf{k}_\mu|^2|\mathsf{k}|^{-2}\,d^\circ k=\frac13$.
- Ghost loop. Here $\mathfrak{s}^{(3)}=4\big(\sum_\mu\gamma_\mu\sin k_\mu\big)^2|\mathsf{k}|^{-4}$. By Lem. 7.24 the symbols of $y\mapsto\mathsf{D}((x,\mu);(x+y,\nu))$ and of $y\mapsto\mathsf{D}((x+y,\nu);(x,\mu))$ are $\frac12(1+e^{ik_\nu})\,\overline{\mathsf{k}_\mu}\,|\mathsf{k}|^{-2}$ and its image under $\mu\leftrightarrow\nu$, $k\to-k$. Since $(1+e^{ik_\nu})\,\overline{\mathsf{k}_\nu}=-2i\sin k_\nu$, the ghost loop G, for which $c_G=-8$ and $\mathsf{f}_G=1$, has $\mathsf{Q}_G(0)=\int\frac14(-2i\sin l_\mu)(-2i\sin l_\nu)\,|\mathsf{k}(l)|^{-4}\,d^\circ l$, and $\frac12\sum_{\mu,\nu}\gamma_\mu\gamma_\nu\,(-8)\,\mathsf{Q}_G(0)=\int\mathfrak{s}^{(3)}$.

Summing up, we find $\frac12\sum_{\mu,\nu}\gamma_\mu\gamma_\nu\,\hat{\mathsf{q}}^\Sigma_{\mu\nu}(0)=\sum_{i=1}^5\int\mathfrak{s}^{(i)}=\frac12\sum_{\mu,\nu}(\sum_i c^{(i)}_{\mu\nu})\gamma_\mu\gamma_\nu$ for small γ (Def. 7.13(3)); hence $\hat{\mathsf{q}}^\Sigma_{\mu\nu}(0)+\hat{\mathsf{q}}^\Sigma_{\nu\mu}(0)=2\sum_i c^{(i)}_{\mu\nu}$, and this number vanishes, because by Thm. T the five matrices $c^{(i)}$ add up to zero, $\sum_{i=1}^5 c^{(i)}_{\mu\nu}=0$ for all μ,ν. It remains to show that $\hat{\mathsf{q}}^\Sigma_{\mu\nu}(0)=\hat{\mathsf{q}}^\Sigma_{\nu\mu}(0)$. Let $\hat\Sigma_{\mu\nu}:=\hat{\mathsf{q}}^\Sigma_{\mu\nu}+\hat{\mathsf{W}}_{\mu\nu}$ be the symbol of $\Sigma_{\mu\nu}$.

Since $\Sigma_{\rm tot}$ is symmetric and translation covariant, $\Sigma_{\mu\nu}(y)=\Sigma_{\nu\mu}(-y)$, and the uniqueness of Fourier coefficients gives $\hat\Sigma_{\mu\nu}(k)=\hat\Sigma_{\nu\mu}(-k)$ for almost every k. As $\hat{\mathsf{q}}^\Sigma$ is continuous and $\hat{\mathsf{W}}(k)\to0$ for $k\to0$, by (K2) for $\mathsf{W}\in\mathfrak{K}(4)$, letting $k\to0$ gives the symmetry, and therefore $\hat{\mathsf{q}}^\Sigma_{\mu\nu}(0)=0$.

Step 5: the gradient of $\hat{\mathsf{q}}^\Sigma_{\mu\nu}$ at 0. The function $\hat\Sigma_{\mu\nu}$ is integrable on $\mathbb{T}$, with the Fourier coefficients $\Sigma_{\mu\nu}(y)$. Lem. 7.15(c) with $c=0$, for which $(x,\mu)^R=(-x-e_\mu,\mu)$, gives
$$\Sigma_{\mu\nu}(y)=\Sigma_{\rm tot}((-e_\mu,\mu),(-y-e_\nu,\nu))=\Sigma_{\mu\nu}(-y+e_\mu-e_\nu),$$
hence $\hat\Sigma_{\mu\nu}(k)=e^{-ik\cdot(e_\mu-e_\nu)}\,\hat\Sigma_{\mu\nu}(-k)$ for almost every k, by the uniqueness of Fourier coefficients. With $\varphi(k):=\frac12(k_\mu-k_\nu)$ the function $\tilde\Sigma(k):=e^{i\varphi(k)}\hat\Sigma_{\mu\nu}(k)$ is therefore even. By (K2) for $\mathsf{W}_{\mu\nu}\in\mathfrak{K}(4)$, $\hat{\mathsf{W}}_{\mu\nu}=\hat{\mathsf{W}}^h+O(|k|^2\lg|k|)$ near 0, where $\hat{\mathsf{W}}^h=\sum_G c_G\,\mathsf{f}^h_G\,\mathsf{I}^h_G$ is homogeneous of degree 1 and even: by Lem. 7.20(a) the parity of $\mathsf{f}^h_G\mathsf{I}^h_G$ is $(-1)^{d_G}(-1)^{\omega_G+1}$, where $\omega_G=n_{\rm a}+n_{\rm b}-3$, and $\omega_G+d_G=1$.

Consequently the odd part of the smooth function $e^{i\varphi}\hat{\mathsf{q}}^\Sigma_{\mu\nu}=\tilde\Sigma-e^{i\varphi}\hat{\mathsf{W}}_{\mu\nu}$ equals the odd part of $-e^{i\varphi}\hat{\mathsf{W}}^h+O(|k|^2\lg|k|)$, which is $O(|k|^2\lg|k|)$, because $\hat{\mathsf{W}}^h$ is even and $(e^{i\varphi}-1)\hat{\mathsf{W}}^h=O(|k|^2)$. So the gradient of $e^{i\varphi}\hat{\mathsf{q}}^\Sigma_{\mu\nu}$ at 0 vanishes; as $\hat{\mathsf{q}}^\Sigma_{\mu\nu}(0)=0$ by Step 4, the gradient of $\hat{\mathsf{q}}^\Sigma_{\mu\nu}$ at 0 vanishes as well. Since $\hat{\mathsf{q}}^\Sigma_{\mu\nu}$ is a trigonometric polynomial, this proves $\hat{\mathsf{q}}^\Sigma_{\mu\nu}(k)=O(|k|^2)$ and completes the proof of (a).

Step 6: set-up for (b). Let $\Gamma_{q;\mu\nu}(y,y') := \Gamma_q((x_q+y,\mu),(x_q+y',\nu))$. In the rest of the proof c_G denotes $\frac{2}{\mathsf{n}}$ times the number c_G of Lem. 7.28; the reason is that the sums with the numbers c_G of that lemma equal $\frac{\mathsf{n}}{2}\Gamma_q$, by eq. 7.xxx, resp. $\frac{\mathsf{n}}{2}\gamma_q$, by the definition of γ_q in (b). By the first, second, third and fourth summand we mean the four groups of summands of Γ_q as in eq. 7.xxx. In each summand the derivatives $\partial_{b,a}$ and $\partial_{b',a'}$ of Def. 7.13(5) act on two legs of the vertices at q, P, P'; we call these two legs the outer legs.

In the terms on the right of eq. 7.xxx the outer legs are the legs which are contracted with the two curl legs $(\mathrm{d}A)(p)$, namely the legs at which the lines K_1, K_2 of Lem. 7.28(4′), (4) and (1), resp. the lines K_0, X of Lem. 7.28(5), end. An outer leg is either plain or a curl leg; in the second case we speak of an outer curl leg. The symbol of a kernel on $\mathbb{Z}^6$, such as $\Gamma_{q;\mu\nu}$, is defined as in Lem. 7.30(a), so that κ_1 is the variable dual to y and κ_2 the variable dual to y'; accordingly we say that the outer leg on which $\partial_{b,a}$ acts, $b=(x_q+y,\mu)$, carries the momentum κ_1, and that the outer leg on which $\partial_{b',a'}$ acts, $b'=(x_q+y',\nu)$, carries the momentum κ_2.

By eq. 7.xxx and Lem. 7.28, $\Gamma_{q;\mu\nu}$ is a finite sum of kernels on $\mathbb{Z}^6$ with the symbols

$$c_G\,\mathsf{f}_{G,1}(\kappa_1)\,\mathsf{f}_{G,2}(\kappa_2)\,\mathsf{J}_G(\kappa),$$

symmetrised under $(\mu,\kappa_1)\leftrightarrow(\nu,\kappa_2)$. Here the $\mathsf{f}_{G,i}$ are trigonometric polynomials, line symbols of degree 0 if the corresponding outer leg is plain and of degree 1 if it is a curl leg. Further, $\mathsf{J}_G=1$ for the first summand; J_G is the bubble symbol $\mathsf{I}_G(\pm\kappa_2)$ of Y for the second summand, the outer leg at q carrying the momentum κ_1; $\mathsf{J}_G=\mathsf{I}_G(\pm(\kappa_1+\kappa_2))$ for the third summand, both outer legs sitting at P; and $\mathsf{J}_G=\hat{\mathsf{T}}_{\mathsf{q}}(\pm\kappa)$ for the fourth summand, the Fourier representation being eq. 7.xxvii.

By the definition of γ_q in (b), the symbol $\hat\gamma_{q;\mu\nu}$ is the same sum with J_G replaced by $\tilde{\mathsf{Q}}_G$, where $\tilde{\mathsf{Q}}_G:=1$ for the first summand, which is entirely local (the line between two slots of the vertex at q is a number, a factor of c_G), and otherwise $\tilde{\mathsf{Q}}_G$ is the function $\tilde{\mathsf{Q}}$ of Lem. 7.22(c), resp. $\tilde{\mathsf{Q}}_{\mathsf{q}}$ of Prop. 7.23, taken at the argument of J_G. Indeed, each of the local terms listed in (b) arises from the corresponding full term of Lem. 7.28 by replacing a single kernel by its local kernel, while the lines which end at the outer legs are left as they are.

In the first half of T_5 the full term is $K_0(u')\sum_z Y(z)\,X(u'-z)$, and the local term has q_Y in place of Y, in accordance with $X*Y=\mathsf{q}_Y*X+X*Y'$; in T_4 the full term is $\sum_z Y(z)\,(K_1K_2)(u'-z)$, and again Y is replaced by q_Y. In T_1 the substitution $(z,w)\to(u'-z,u'-w)$ turns the full term into

$$\sum_{z,w}\big[K_4(z)\,K_5(w)\,K_3(z-w)\big]\,K_1(u'-z)\,K_2(u'-w),$$

where the bracket, a kernel in (z,w), has the symbol $\hat{\mathsf{T}}_{\mathsf{q}}$ by the second identity of eq. 7.xxvii; the local term has $\mathsf{q}_{\mathsf{q}}(z,w)$, with the symbol $\tilde{\mathsf{Q}}_{\mathsf{q}}$, in place of the bracket. The first half of $\mathsf{T}_{4'}$ is entirely local and is its own local term. In each case the symbol J_G of the replaced kernel turns into $\tilde{\mathsf{Q}}_G$, while the number c_G, the factors $\mathsf{f}_{G,1}$, $\mathsf{f}_{G,2}$ and the argument of J_G remain the same. Let $\hat\Gamma:=\hat\Gamma_{q;\mu\nu}$ be the sum of the symbols of $\Gamma_{q;\mu\nu}$; in $\hat\Gamma-\hat\gamma_{q;\mu\nu}$ the first summand cancels.

Step 7: counting degrees. Let $d_{G,i}\in\{0,1\}$ be the degree of $\mathsf{f}_{G,i}$ and $d_G:=d_{G,1}+d_{G,2}$, the number of outer curl legs; for G in the second, third and fourth summand let ω_G be the number ω of J_G. We use again that the degree of a line is $1+\#$(its curl legs) (Lem. 7.24), and that $\mathsf{a}^{(2)}$ has two curl legs and every monomial of $\mathsf{a}^{(3)}$ one (Def. 7.8(3)). In the second summand the vertices at q and P have two curl legs, of which $2-d_G$ lie on the two bubble lines, so that $\omega_G=2+(2-d_G)-3=1-d_G$. In the fourth summand the vertices at q, P, P' have four curl legs, of which $4-d_G$ lie on the three lines of $\hat{\mathsf{T}}_{\mathsf{q}}$, so that $\omega_G=3+(4-d_G)-6=1-d_G$.

In the third summand the bubble lines carry the two curl legs at q: for the monomials of $\frac12|\mathfrak{z}_1|^2$ we get $(d_G,\omega_G)=(0,1)$; for the monomials of $\mathsf{a}_P^{(4)}$ with one curl leg we get $(1,1)$ or $(0,2)$, according as this leg is outer or not; for those with four curl legs we get $(2,3)$. Thus $d_G+\omega_G=1$, except for these last three kinds, for which $d_G+\min\{\omega_G,2\}\geq 2$ and $\omega_G\geq 1$.

Step 8: expansion of $\hat\Gamma - \hat\gamma_{q;\mu\nu}$. We show that for almost every κ with $|\kappa| \le c$, c small,

$$\hat\Gamma - \hat\gamma_{q;\mu\nu} = \hat\Gamma^h + O(|\kappa|^2 \lg|\kappa|\, \Lambda(\kappa)),$$
$$\hat\Gamma^h := \sum_{G:\, d_G+\omega_G=1} c_G\, \mathsf{f}^h_{G,1}\, \mathsf{f}^h_{G,2}\, \mathsf{J}^h_G, \qquad |\hat\Gamma^h| \le C|\kappa|\, \Lambda(\kappa).$$

Here J^h_G is the homogeneous part of J_G (Lem. 7.20(d)), taken at the argument of J_G, and $\Lambda \ge 1$ is the function Λ of Lem. 7.20 for $\hat{\mathsf{T}}_{\mathsf{q}}$. The function $\hat\Gamma^h$ is homogeneous of degree 1 and even, each of its terms having the parity $(-1)^{d_G}(-1)^{\omega_G+1} = 1$. Let κ' denote the argument of J_G; then $|\kappa'| \le \sqrt{2}|\kappa|$. For $d_G + \omega_G \ge 2$, Lem. 7.22(c) gives

$$|\mathsf{f}_{G,1}\mathsf{f}_{G,2}\,(\mathsf{J}_G - \tilde{\mathsf{Q}}_G)| \le C|\kappa|^{d_G}\, |\kappa'|^{\min\{\omega_G,2\}} \le C|\kappa|^2.$$

For the other G we have $\omega_G \le 1$, hence $\deg \mathsf{Q}_G \le 1$ and $\mathsf{Q}_G - \tilde{\mathsf{Q}}_G = O(|\kappa'|^3)$, and Lem. 7.20(b) gives

$$\mathsf{J}_G - \tilde{\mathsf{Q}}_G = \mathsf{J}^h_G + O(|\kappa'|^{\omega_G+1} \lg|\kappa'|\, \Lambda_G(\kappa')) + O(|\kappa'|^3),$$

where Λ_G is bounded for the bubbles and equals Λ for $\hat{\mathsf{T}}_{\mathsf{q}}$. Moreover $\mathsf{f}_{G,i}(\kappa_i) = \mathsf{f}^h_{G,i}(\kappa_i) + O(|\kappa_i|^{d_{G,i}+1})$, and the functions $t\lg t$ and $t^2\lg t$ are increasing on $(0,1]$, their derivatives being $-\log t$ and $t(1-2\log t)$. We go through the summands.

- Third summand, $\omega_G = 1$, $d_G = 0$: $|\mathsf{J}^h_G| \le C|\kappa'| \le C|\kappa|$, and the errors are $O(|\kappa|^2\lg|\kappa|)$.
- Fourth summand: $\kappa' = \pm\kappa$, $|\mathsf{J}^h_G| \le C|\kappa|^{\omega_G}\Lambda$ and $|\mathsf{f}_{G,1}\mathsf{f}_{G,2}| \le C|\kappa|^{d_G}$; the leading term is $O(|\kappa|\Lambda)$, and the errors are $O(|\kappa|^2\lg|\kappa|\,\Lambda)$.
- Second summand: $\kappa' = \pm\kappa_2$, and $\omega_G = -1$ occurs. We take the factor of the outer leg at P, which carries the momentum κ_2, together with J_G. As $d_{G,2} + \omega_G = 1 - d_{G,1} \ge 0$, we have $|\mathsf{f}^h_{G,2}(\kappa_2)\, \mathsf{J}^h_G| \le C|\kappa_2|^{1-d_{G,1}}$ and $|\mathsf{f}^h_{G,1}\mathsf{f}^h_{G,2}\, \mathsf{J}^h_G| \le C|\kappa_1|^{d_{G,1}}|\kappa_2|^{1-d_{G,1}} \le C|\kappa|$. Replacing $\mathsf{f}_{G,i}$ by its leading part costs a factor $|\kappa_i|$, and

$$|\mathsf{f}_{G,1}\mathsf{f}_{G,2}|\, |\kappa_2|^{\omega_G+1} \lg|\kappa_2| \le C|\kappa_1|^{d_{G,1}}\, |\kappa_2|^{2-d_{G,1}} \lg|\kappa_2| \le C|\kappa|^2\lg|\kappa|.$$

The symmetrised terms are treated in the same way, with the indices 1 and 2 exchanged.

Step 9: the value of $\hat\gamma_{q;\mu\nu}$ *at* 0. We have $\hat\gamma_{q;\mu\nu}(0) = \sum c_G\, \mathsf{f}_{G,1}(0)\, \mathsf{f}_{G,2}(0)\, \tilde{\mathsf{Q}}_G(0)$, the sum extending over the G with two plain outer legs, as an outer curl leg gives $\mathsf{f}_{G,i}(0) = 0$. For such G in the first summand, $\tilde{\mathsf{Q}}_G = 1$, and the line between two slots of the vertex at q, contained in c_G, is the integral of its symbol (the case $\mathsf{r} = 1$ of Step 1). For such G in the other summands, $d_G = 0$, hence $\omega_G \ge 1$ by Step 7, and $\tilde{\mathsf{Q}}_G(0) = \mathsf{Q}_G(0)$ is the integral of the integrand at $\kappa = 0$ (Lem. 7.20(c)). By Step 1 the four summands of $\Gamma_q(b,3;b',3)$ have the trace forms

$$\tfrac12 \operatorname{Tr}[\mathsf{m}_{bb'}\mathsf{C}], \qquad -\tfrac12 \operatorname{Tr}[\mathsf{m}_b\mathsf{C}\,\mathsf{H}_{b'}\mathsf{C}] - (b \leftrightarrow b'),$$
$$-\tfrac12 \operatorname{Tr}[\mathsf{h}^{(0)}_q\mathsf{C}\,\mathsf{H}_{bb'}\mathsf{C}], \qquad \operatorname{Tr}[\mathsf{h}^{(0)}_q\mathsf{C}\,\mathsf{H}_b\mathsf{C}\,\mathsf{H}_{b'}\mathsf{C}],$$

where $\mathsf{m}_{bb'}$, m_b and $\mathsf{H}_{bb'}$ are the Hessians of $\partial\partial'\mathsf{a}^{(4)}_q$, $\partial\mathsf{a}^{(3)}_q$ and $\partial\partial'\mathsf{S}_4$ ($\partial = \partial_{b,3}$, $\partial' = \partial_{b',3}$), H_b is the Hessian of $\partial_{b,3}\mathsf{S}_3$ (Step 2), and $\mathsf{h}^{(0)}_q$ is the Hessian of $\mathsf{a}^{(2)}_q$. By Step 2,

$$\sum_{b,b'} \tfrac12 \gamma_{\mu(b)}\gamma_{\mu(b')}\, \mathsf{m}_{bb'} = \mathsf{h}^{(2)}_q(Y_\gamma), \qquad \sum_b \gamma_{\mu(b)}\, \mathsf{m}_b = \mathsf{h}^{(1)}_q(Y_\gamma),$$
$$\sum_\mu \gamma_\mu\, \hat{\mathsf{H}}_{(0,\mu)}(l,l) = \bigoplus_e \hat{\mathsf{N}}_{e,1}(l;\gamma),$$
$$\sum_{\mu,\nu}\sum_y \tfrac12 \gamma_\mu\gamma_\nu\, \hat{\mathsf{H}}_{(0,\mu),(y,\nu)}(l,l) = \bigoplus_e \hat{\mathsf{N}}_{e,2}(l;\gamma),$$

the sums in the first line being finite; and $\hat{\mathsf{h}}(l,l)$ is the matrix of the form $(e^{il\cdot x}\alpha'\mathsf{e}_e, \mathsf{h}\, e^{il\cdot x}\alpha\, \mathsf{e}_e)$, which for $\mathsf{h} = \mathsf{h}^{(j)}_q(Y_\gamma)$ is $\bigoplus_e \hat{\mathsf{h}}_{q,e,j}(l;\gamma)$ by Def. 7.13(2). The parts of the Hessians which come from an outer curl leg drop out here, since $\mathrm{d}Y_\gamma = 0$; they belong to the G with $\mathsf{f}_{G,1}(0)\,\mathsf{f}_{G,2}(0) = 0$. We

use eq. 7.xxxiv at $\kappa = 0$, integrand by integrand, and write $\hat{\mathsf{h}}_j := \bigoplus_e \hat{\mathsf{h}}_{q,e,j}$ and $\hat{\mathsf{N}}_j := \bigoplus_e \hat{\mathsf{N}}_{e,j}$. The four summands contribute to $\frac{1}{2}\sum_{\mu,\nu}\gamma_\mu\gamma_\nu\,\hat\gamma_{q;\mu\nu}(0)$ the amounts

$$\tfrac{1}{2}\int \mathrm{tr}[\hat{\mathsf{h}}_2\hat{\mathsf{C}}], \qquad \tfrac{1}{2}\cdot 2\cdot(-\tfrac{1}{2})\int \mathrm{tr}[\hat{\mathsf{h}}_1\hat{\mathsf{C}}\hat{\mathsf{N}}_1\hat{\mathsf{C}}],$$
$$-\tfrac{1}{2}\int \mathrm{tr}[\hat{\mathsf{h}}_0\hat{\mathsf{C}}\hat{\mathsf{N}}_2\hat{\mathsf{C}}], \qquad \tfrac{1}{2}\int \mathrm{tr}[\hat{\mathsf{h}}_0\hat{\mathsf{C}}\hat{\mathsf{N}}_1\hat{\mathsf{C}}\hat{\mathsf{N}}_1\hat{\mathsf{C}}]$$

(integrals over $\mathbb{T}$ with respect to $d^\circ l$), and these are $\int \mathfrak{v}_q^{(j)}(l;\gamma)\,d^\circ l$, $j = 1,\dots,4$, by Def. 7.13(4). Hence, by Def. 7.13(4) and Thm. U,

$$\tfrac{1}{2}\sum_{\mu,\nu}\gamma_\mu\gamma_\nu\,\hat\gamma_{q;\mu\nu}(0) = \tfrac{1}{2}\sum_{\mu,\nu}\gamma_\mu\gamma_\nu\sum_{j=1}^{4}\mathsf{g}^{(j)}_{q;\mu\nu} = 0$$

for all small γ. Since γ_q is a symmetric kernel, $\hat\gamma_{q;\mu\nu}(0)$ is symmetric in μ,ν, and therefore $\hat\gamma_{q;\mu\nu}(0) = 0$.

Step 10: the gradient of $\hat\gamma_{q;\mu\nu}$ at 0. Lem. 7.15(c) with $c = x(q)$, for which $R_c q = q$, gives

$$\Gamma_{q;\mu\nu}(y,y') = \Gamma_{q;\mu\nu}(\mathsf{o} - e_\mu - y,\ \mathsf{o} - e_\nu - y'), \qquad \mathsf{o} := 2(x(q) - x_q),$$

and the uniqueness of Fourier coefficients gives $\hat\Gamma(\kappa) = e^{-2i\varphi(\kappa)}\,\hat\Gamma(-\kappa)$ for almost every κ, with the real linear form $\varphi(\kappa) := \frac{1}{2}[\kappa_1\cdot(\mathsf{o}-e_\mu) + \kappa_2\cdot(\mathsf{o}-e_\nu)]$; hence $e^{i\varphi}\hat\Gamma$ is even almost everywhere. Since $\hat\Gamma^h$ is even, the odd part of $e^{i\varphi}\hat\Gamma^h$ is $O(|\kappa|^2\Lambda)$. By Step 8 the odd part of the smooth function $e^{i\varphi}\hat\gamma_{q;\mu\nu}$ is therefore $O(|\kappa|^2\lg|\kappa|\,\Lambda)$ almost everywhere near 0. The function Λ is bounded on an open cone. Indeed, for $\hat{\mathsf{T}}_{\mathsf{q}}$ we have $(l_1,l_2,l_3)(\kappa) = (\kappa_1,-\kappa_2,0)$ (Step 2 of the proof of Prop. 7.23), so that the differences $l_i(\kappa) - l_j(\kappa)$, $i<j$, are $\kappa_1+\kappa_2$, κ_1 and $-\kappa_2$. Let

$$\mathsf{K} := \{\kappa\in\mathbb{R}^6 :\ \min\{|\kappa_1|,|\kappa_2|,|\kappa_1+\kappa_2|\} > \tfrac{1}{2}|\kappa|\}.$$

This set is open, it is invariant under $\kappa\to t\kappa$ for real $t\neq 0$, and it is not empty, since it contains (v,v) for every $v\in\mathbb{R}^3\setminus\{0\}$: there $|\kappa| = \sqrt{2}\,|v|$, while $|\kappa_1| = |\kappa_2| = |v|$ and $|\kappa_1+\kappa_2| = 2|v|$. On K each of the three terms $\log_+(|\kappa|/|l_i(\kappa)-l_j(\kappa)|)$ in the definition of Λ is at most $\log 2$, so that $\Lambda\le 1+3\log 2$ there. On K the odd part is therefore $O(|\kappa|^2\lg|\kappa|)$ almost everywhere near 0. The odd part of the smooth function $e^{i\varphi}\hat\gamma_{q;\mu\nu}$ differs from its linear part $\kappa\mapsto\kappa\cdot\nabla(e^{i\varphi}\hat\gamma_{q;\mu\nu})(0)$ by $O(|\kappa|^3)$.

Hence $|\kappa\cdot\nabla(e^{i\varphi}\hat\gamma_{q;\mu\nu})(0)|\le C|\kappa|^2\lg|\kappa|$ for almost every, and by continuity for every, $\kappa\in\mathsf{K}$ with $|\kappa|\le c$. Replacing κ by $t\kappa$ and letting $t\downarrow 0$, we see that this linear function vanishes on K, a non-empty open set, and hence everywhere: the gradient of $e^{i\varphi}\hat\gamma_{q;\mu\nu}$ at 0 vanishes. As $\hat\gamma_{q;\mu\nu}(0) = 0$ by Step 9, the gradient of $\hat\gamma_{q;\mu\nu}$ at 0 vanishes as well. This proves (b). □

7.1.12. *The six terms, the continuum model, the kernel of degree -5.* In this subsection we introduce six continuum two-loop integrands and a model built from them, with a smooth ultraviolet cut-off on every line. We show that the model has the same parts of degree -5 as the lattice classes of Lem. 7.28, that these parts are multiples of $|u|^{-5}$, and we deduce the large-distance asymptotics of $\mathsf{B}_1^{\mathrm{lat}}$ and its consequence for smeared plaquette observables.

Definition 7.32 (The six two-loop integrands; the model)**.** *(1) Tensors and variables.* For $k_1,k_2,k_3\in\mathbb{R}^3$ we put

$$V_2(k_1,k_2)_{\mu\nu} := -(k_1\cdot k_2)\,\delta_{\mu\nu} + k_{1\nu}k_{2\mu},$$
$$V_3(k_1,k_2,k_3)_{\alpha\beta\gamma} := \delta_{\alpha\gamma}(k_1-k_3)_\beta + \delta_{\alpha\beta}(k_2-k_1)_\gamma + \delta_{\beta\gamma}(k_3-k_2)_\alpha.$$

The tensor V_2 is even and the tensor V_3 is odd under the simultaneous change of sign $k_i\to -k_i$ of all arguments, and $|V_2(k_1,k_2)_{\mu\nu}|\le 2|k_1||k_2|$ for all μ,ν. For $k,l,\mathsf{p}\in\mathbb{R}^3$ we write

$$\mathsf{x}_0 := \mathsf{p}^2, \quad \mathsf{x}_1 := k^2, \quad \mathsf{x}_1' := (k-\mathsf{p})^2, \quad \mathsf{x}_2 := l^2, \quad \mathsf{x}_2' := (l-\mathsf{p})^2, \quad \mathsf{x}_3 := (k-l)^2.$$

(2) The six integrands. With summation over repeated tensor indices we define

$$\begin{aligned}
F_1 &:= -\tfrac{1}{2}\, V_2(k,\mathsf{p}-k)_{\mu\nu}\, V_2(-l,l-\mathsf{p})_{\rho\sigma} \\
&\qquad \times V_3(-k,l,k-l)_{\mu\rho\alpha}\, V_3(k-\mathsf{p},\mathsf{p}-l,l-k)_{\nu\sigma\alpha}\,/\,\mathsf{x}_1\mathsf{x}_1'\mathsf{x}_2\mathsf{x}_2'\mathsf{x}_3, \\
F_2 &:= -\tfrac{1}{2}\, V_2(k,\mathsf{p}-k)_{\mu\nu}\, V_2(-k,k-\mathsf{p})_{\rho\nu} \\
&\qquad \times V_3(-k,l,k-l)_{\mu\beta\gamma}\, V_3(k,-l,l-k)_{\rho\beta\gamma}\,/\,\mathsf{x}_1^2\mathsf{x}_1'\mathsf{x}_2\mathsf{x}_3, \\
F_3 &:= -\, V_2(k,\mathsf{p}-k)_{\mu\nu}\, V_2(-k,k-\mathsf{p})_{\rho\nu}\, l_\rho(l-k)_\mu\,/\,\mathsf{x}_1^2\mathsf{x}_1'\mathsf{x}_2\mathsf{x}_3, \\
F_4 &:= -\tfrac{1}{2}\big[V_2(k,\mathsf{p}-k)_{\mu\mu} V_2(l,\mathsf{p}-l)_{\nu\nu} \\
&\qquad -\tfrac{1}{2} V_2(k,\mathsf{p}-k)_{\mu\nu}\big(V_2(l,\mathsf{p}-l)_{\mu\nu}+V_2(l,\mathsf{p}-l)_{\nu\mu}\big)\big]\,/\,\mathsf{x}_1\mathsf{x}_1'\mathsf{x}_2\mathsf{x}_2', \\
F_5 &:= V_3(\mathsf{p}-k,l,k-l)_{\alpha\beta\gamma}\, V_3(k,-l,l-k)_{\sigma\beta\gamma}\, V_2(k-\mathsf{p},-k)_{\alpha\sigma}\,/\,\mathsf{x}_1'\mathsf{x}_1\mathsf{x}_2\mathsf{x}_3, \\
F_6 &:= -\tfrac{1}{6}\, V_3(l,k-l,\mathsf{p}-k)_{\alpha\beta\gamma}\, V_3(-l,l-k,k-\mathsf{p})_{\alpha\beta\gamma}\,/\,\mathsf{x}_2\mathsf{x}_3\mathsf{x}_1'.
\end{aligned} \tag{7.xxxv}$$

We call F_1 the kite, F_2 the gluon loop on a line, F_3 the ghost loop on a line, F_4 the quartic vertex, F_5 the term of $\mathsf{a}^{(3)}$ with S_3, and F_6 the term of $\mathsf{a}^{(3)}$ with $\mathsf{a}^{(3)}$. Each F_Γ, $\Gamma\in\{1,\dots,6\}$, is a rational function of (k,l,p), homogeneous of degree -4, even under $(k,l,\mathsf{p})\to-(k,l,\mathsf{p})$, and invariant under simultaneous rotations of k, l and p. The *lines of* Γ are the vectors whose squares occur in the denominator of F_Γ, counted with multiplicity: for $\Gamma=1$ the vectors $k,\mathsf{p}-k,l,\mathsf{p}-l,k-l$; for $\Gamma=2,3$ the vector k twice and the vectors $\mathsf{p}-k,l,k-l$; for $\Gamma=4$ the vectors $k,\mathsf{p}-k,l,\mathsf{p}-l$; for $\Gamma=5$ the vectors $k,\mathsf{p}-k,l,k-l$; for $\Gamma=6$ the vectors $\mathsf{p}-k,l,k-l$.

(3) The model. Let $\rho_0:=\frac{\pi}{16}$, and let $\rho(\mathsf{q}):=\theta(|\mathsf{q}|/2\rho_0)$ for $\mathsf{q}\in\mathbb{R}^3$, with the function θ of Def. 7.16(1); thus $\rho(\mathsf{q})=1$ for $|\mathsf{q}|\le\rho_0$ and $\rho(\mathsf{q})=0$ for $|\mathsf{q}|\ge2\rho_0$. Let $R_\Gamma(k,l,\mathsf{p})$ be the product of the values of ρ at the lines of Γ. We define

$$\Psi_\Gamma(\mathsf{p}):=\iint R_\Gamma F_\Gamma\, d^\circ k\, d^\circ l \quad (\mathsf{p}\in\mathbb{R}^3\setminus\{0\}), \qquad \mathsf{T}_\Gamma^{\mathrm{mod}}(u):=24\int_{\mathbb{R}^3} e^{i\mathsf{p}\cdot u}\,\Psi_\Gamma(\mathsf{p})\, d^\circ\mathsf{p} \quad (u\in\mathbb{Z}^3),$$

the first integral extending over $\mathbb{R}^3\times\mathbb{R}^3$.

These integrals converge absolutely. Indeed, every entry of $V_3(k_1,k_2,k_3)$ is bounded by $C\max_i|k_i|$; together with the bound on the entries of V_2 and with the denominators this gives, on every bounded set of points (k,l,p) and with a constant C depending on that set,

$$\begin{gathered}
|F_1|\le\frac{C}{|k|\,|\mathsf{p}-k|\,|l|\,|\mathsf{p}-l|\,|k-l|^2}, \qquad |F_2|,\ |F_3|\le\frac{C(|k|+|l|)}{|k|^2|l|^2|k-l|^2}, \\
|F_4|\le\frac{C}{|k|\,|\mathsf{p}-k|\,|l|\,|\mathsf{p}-l|}, \qquad |F_5|\le\frac{C}{|k|\,|\mathsf{p}-k|\,|l|^2|k-l|^2}, \qquad |F_6|\le\frac{C}{|\mathsf{p}-k|^2|l|^2|k-l|^2}.
\end{gathered} \tag{7.xxxvi}$$

We integrate these majorants first over l and then over k. For v,w in a bounded subset of $\mathbb{R}^3$, and with integrals over a bounded set of l, we have the elementary bounds

$$\int|l-v|^{-2}|l-w|^{-2}\,\mathrm{d}l\le C|v-w|^{-1}, \qquad \int|l-v|^{-1}|l-w|^{-2}\,\mathrm{d}l\le C\lg|v-w|,$$

with constants depending on these sets; moreover $|l|^{-1}|\mathsf{p}-l|^{-1}\le|\mathsf{p}|^{-1}\big(|l|^{-1}+|\mathsf{p}-l|^{-1}\big)$, since $|\mathsf{p}|\le|l|+|\mathsf{p}-l|$, and $|k|+|l|\le2|k|+|k-l|$. For $\Gamma=1$ the majorant is at most $C|\mathsf{p}|^{-2}\big(|k|^{-1}+|\mathsf{p}-k|^{-1}\big)\big(|l|^{-1}+|\mathsf{p}-l|^{-1}\big)|k-l|^{-2}$; its integral over l is at most $C|\mathsf{p}|^{-2}\big(|k|^{-1}+|\mathsf{p}-k|^{-1}\big)\big(\lg|k|+\lg|k-\mathsf{p}|\big)$, which is integrable over bounded sets of k, with a bound uniform for p in a bounded set after the factor $|\mathsf{p}|^{-2}$ is removed. For $\Gamma=4$ the same argument applies without the factor $|k-l|^{-2}$.

For $\Gamma=2,3$ the majorant is at most $2C|k|^{-1}|l|^{-2}|k-l|^{-2}+C|k|^{-2}|l|^{-2}|k-l|^{-1}$, whose integral over l is at most $C|k|^{-2}\lg|k|$, a function integrable over bounded sets of k. For $\Gamma=5$ the integral over l is at most $C|k|^{-2}|\mathsf{p}-k|^{-1}$, and the subsequent integral over k is at most $C\lg|\mathsf{p}|$. For $\Gamma=6$ the integral over l is at most $C|k|^{-1}|\mathsf{p}-k|^{-2}$, and again the integral over k is at most $C\lg|\mathsf{p}|$. Hence,

for p in a bounded set, the integral of each majorant over a bounded set of (k,l) is at most $C|\mathsf{p}|^{-2}$, since 1 and $\lg|\mathsf{p}|$ are bounded by $C|\mathsf{p}|^{-2}$ there.

Where $R_\Gamma \neq 0$, every line of Γ has modulus less than $2\rho_0$. Hence $|l| < 2\rho_0$ and $|k| < 4\rho_0$ there (by the lines l and k, and for $\Gamma = 6$ by the lines l and $k-l$), and $|\mathsf{p}| \le |\mathsf{p}-k| + |k| < 6\rho_0$, by the line $\mathsf{p}-k$, which occurs for every Γ. Consequently the integral defining $\Psi_\Gamma(\mathsf{p})$ converges absolutely, $|\Psi_\Gamma(\mathsf{p})| \le C|\mathsf{p}|^{-2}$, and Ψ_Γ is integrable, radial (by the rotation invariance of F_Γ and of R_Γ, and the substitution $(k,l) \to (Ok, Ol)$), and equal to 0 for $|\mathsf{p}| \ge 6\rho_0 = \frac{3\pi}{8}$; in particular $\mathsf{T}_\Gamma^{\mathrm{mod}}(u)$ is well defined. ∎

In order to compare the model with the elementary terms of Lem. 7.28, we write these terms, too, as integrals over momenta.

Lemma 7.33 (Momentum-space forms)**.** (a) Let $K_1, \dots, K_5$ be elementary lines as in Prop. 7.23, with symbols $\sigma_1, \dots, \sigma_5$, and let T be their kite. Then, for every $u \in \mathbb{Z}^3$,

$$\mathsf{T}(u) = \int_{\mathbb{T}} e^{i\mathsf{p}\cdot u} \Big[\iint_{\mathbb{T}^2} \sigma_1(k)\, \sigma_2(\mathsf{p}-k)\, \sigma_3(k-l)\, \sigma_4(l)\, \sigma_5(\mathsf{p}-l)\, d^\circ k\, d^\circ l \Big] d^\circ \mathsf{p},$$

and the triple integral converges absolutely.

(b) If two kernels have symbols $\sigma_{\mathrm{a}}, \sigma_{\mathrm{b}} \in L^1(\mathbb{T})$, then their product is the kernel whose symbol is the convolution $k \mapsto \int_{\mathbb{T}} \sigma_{\mathrm{a}}(k-l)\, \sigma_{\mathrm{b}}(l)\, d^\circ l$ of the two symbols. If $X, Y \in \ell^2(\mathbb{Z}^3)$ are kernels with symbols $\hat{X}, \hat{Y} \in L^1(\mathbb{T})$, then $X * Y$ (Lem. 7.22) is the kernel with the symbol $\hat{X}\hat{Y}$. ∎

Proof. We begin with (b). Let $K_{\mathrm{a}}, K_{\mathrm{b}}$ be the two kernels. The function $(k,l) \mapsto \sigma_{\mathrm{a}}(k)\sigma_{\mathrm{b}}(l)$ is integrable over $\mathbb{T}^2$; by Fubini's theorem and the substitution $k \to k-l$ on $\mathbb{T}$,

$$K_{\mathrm{a}}(x)\, K_{\mathrm{b}}(x) = \iint e^{i(k+l)\cdot x}\, \sigma_{\mathrm{a}}(k)\, \sigma_{\mathrm{b}}(l)\, d^\circ k\, d^\circ l = \int_{\mathbb{T}} e^{ik\cdot x} \Big[\int_{\mathbb{T}} \sigma_{\mathrm{a}}(k-l)\, \sigma_{\mathrm{b}}(l)\, d^\circ l \Big] d^\circ k,$$

and the bracket is an integrable function of k. For the second assertion we note that $X \in \ell^2(\mathbb{Z}^3)$ is, by Plancherel's theorem, the sequence of Fourier coefficients of a function in $L^2(\mathbb{T})$, which coincides with $\hat{X}$ by the uniqueness of Fourier coefficients; thus $\hat{X} \in L^2(\mathbb{T})$, and likewise $\hat{Y} \in L^2(\mathbb{T})$. The kernel $z \mapsto Y(u-z)$ has the symbol $k \mapsto e^{-ik\cdot u}\hat{Y}(-k)$, and Plancherel's theorem in the form $\sum_z a(z)b(z) = \int_{\mathbb{T}} \hat{a}(-k)\, \hat{b}(k)\, d^\circ k$, for square summable kernels a, b with symbols $\hat{a}, \hat{b}$, gives $(X*Y)(u) = \int_{\mathbb{T}} e^{ik\cdot u} \hat{X}(k)\hat{Y}(k)\, d^\circ k$.

We turn to (a).

Step 1. For $i \neq 3$ the degree n_i is at least 2, so that $|\sigma_i(\mathsf{q})| \le C|\mathsf{q}|^{-1}$ on $[-\pi,\pi]^3$ (Def. 7.16(2),(3)); hence $\sigma_i \in L^2(\mathbb{T})$ for $i \neq 3$, while $\sigma_3 \in L^1(\mathbb{T})$. The functions

$$\mathsf{A}(v) := \int_{\mathbb{T}} \sigma_1(v+\mathsf{q})\, \sigma_4(\mathsf{q})\, e^{iu\cdot\mathsf{q}}\, d^\circ \mathsf{q}, \qquad \mathsf{B}(v) := \int_{\mathbb{T}} \sigma_2(v+\mathsf{q})\, \sigma_5(\mathsf{q})\, e^{iu\cdot\mathsf{q}}\, d^\circ \mathsf{q}$$

are bounded, by the Cauchy–Schwarz inequality, and continuous, since translations act continuously on $L^2(\mathbb{T})$. Consequently

$$\Phi(\eta_1, \eta_2) := \int_{\mathbb{T}} \sigma_3(\mathsf{q})\, \mathsf{A}(\eta_1+\mathsf{q})\, \mathsf{B}(\eta_2-\mathsf{q})\, d^\circ \mathsf{q}$$

is bounded and continuous on $\mathbb{T}^2$, by dominated convergence, and the integral over $(\mathsf{q}_3, \mathsf{q}_4, \mathsf{q}_5) \in \mathbb{T}^3$ which results from inserting the definitions of A and B (with q_3 in place of q and $\mathsf{q}_4, \mathsf{q}_5$ the variables of A, B) converges absolutely.

Step 2. Let $\phi \in C_c^\infty(\mathbb{R}^3)$ with support in $\{|x| \le 1\}$ and $\phi(0) = 1$, and let $\phi_R := \phi(\cdot/R)$. By Prop. 7.23, the double sum defining $\mathsf{T}(u)$ converges absolutely; by dominated convergence, $\mathsf{T}(u) = \lim_{R\to\infty} \mathsf{T}_R(u)$, where $\mathsf{T}_R(u)$ denotes the same sum with the additional factor $\phi_R(z)\phi_R(w)$. For fixed R we insert $K_i(y) = \int_{\mathbb{T}} e^{i\mathsf{q}_i\cdot y} \sigma_i(\mathsf{q}_i)\, d^\circ \mathsf{q}_i$. The sums over z and w are then finite, and by Fubini's theorem they produce the factor

$$\check{\phi}_R(-(\mathsf{q}_1 - \mathsf{q}_3 - \mathsf{q}_4))\, \check{\phi}_R(-(\mathsf{q}_2 + \mathsf{q}_3 - \mathsf{q}_5))\, e^{iu\cdot(\mathsf{q}_4+\mathsf{q}_5)}.$$

We pass to the variables $\eta_1 := \mathsf{q}_1 - \mathsf{q}_3 - \mathsf{q}_4$, $\eta_2 := \mathsf{q}_2 + \mathsf{q}_3 - \mathsf{q}_5$, q_3, q_4, q_5; this is a linear change of variables with integer matrix of determinant ± 1, hence a measure preserving bijection of $\mathbb{T}^5$, and it gives

$$\mathsf{T}_R(u) = \iint_{\mathbb{T}^2} \check{\phi}_R(-\eta_1)\, \check{\phi}_R(-\eta_2)\, \Phi(\eta_1, \eta_2)\, d^\circ\eta_1\, d^\circ\eta_2.$$

Here $\int_{\mathbb{T}} \check{\phi}_R(-\eta)\, d^\circ\eta = \phi_R(0) = 1$, and $|\check{\phi}_R(\eta)| \le CR^3\langle R\eta\rangle^{-10}$ by Lem. 7.19(a), applied with $D = 3$ to the restriction of ϕ_R to $\mathbb{Z}^3$; this function vanishes off $\{|\zeta| \le R\}$, and its norm $\|\phi_R\|_{(R)}$ is at most $\|\phi\|_{C^{10}}$, a difference being an integral of a derivative. Thus the functions $\eta \mapsto \check{\phi}_R(-\eta)$ have integral 1, bounded L^1 norms, and integrals over $\{|\eta| \ge \delta\}$ tending to 0 as $R \to \infty$, for every $\delta > 0$. As Φ is bounded and continuous at $(0,0)$, $\mathsf{T}_R(u) \to \Phi(0,0)$.

Step 3. In the triple integral $\Phi(0,0)$ we change variables to $l = \mathsf{q}_4$, $k = \mathsf{q}_3 + \mathsf{q}_4$, $\mathsf{p} = \mathsf{q}_4 + \mathsf{q}_5$, again a measure preserving bijection of $\mathbb{T}^3$. Then $\mathsf{q}_3 = k - l$, $\mathsf{q}_5 = \mathsf{p} - l$ and $\mathsf{q}_5 - \mathsf{q}_3 = \mathsf{p} - k$, so that the integrand becomes $\sigma_1(k)\,\sigma_2(\mathsf{p}-k)\,\sigma_3(k-l)\,\sigma_4(l)\,\sigma_5(\mathsf{p}-l)\,e^{i\mathsf{p}\cdot u}$. The triple integral converges absolutely by Step 1, and Fubini's theorem yields the formula of (a). □

The next lemma computes the weights with which the leading parts of the elementary terms add up to the six integrands of Def. 7.32.

Lemma 7.34 (The weights of the six terms)**.** Let $\tau_{\mathrm{gf}} = 1$, and let $p = p_{[\iota]}(x_p)$, $q = p_{[\iota']}(x_q)$. To an elementary term G of Lem. 7.28, with its lines named as there, we attach its *leading integrand* $\mathsf{F}_G(k, l, \mathsf{p})$, where σ^h denotes the leading part of the symbol of the line in question:

$$\begin{aligned}
&G \in \mathfrak{G}_1 : && \sigma_1^h(k)\,\sigma_2^h(\mathsf{p}-k)\,\sigma_3^h(k-l)\,\sigma_4^h(l)\,\sigma_5^h(\mathsf{p}-l);\\
&G \in \mathfrak{G}_4,\ (n_X, n_Y) = (4,4) : && \sigma_1^h(k)\,\sigma_2^h(\mathsf{p}-k)\,\sigma_3^h(l)\,\sigma_4^h(\mathsf{p}-l);\\
&G \in \mathfrak{G}_5 : && \sigma_0^h(\mathsf{p}-k)\,\sigma_X^h(k)\,\sigma_{\mathrm{a}}^h(k-l)\,\sigma_{\mathrm{b}}^h(l);\\
&G \in \mathfrak{G}_6 : && \sigma_1^h(\mathsf{p}-k)\,\sigma_2^h(k-l)\,\sigma_3^h(l);\\
&G \in \mathfrak{G}_{\mathrm{I}}(\mu,\nu) : && \sigma_\Pi^h(\mathsf{p}-k)\,\sigma^h_{\rightarrow,\iota\mu}(k)\,\mathsf{f}_G^h(k)\,\sigma_{\mathrm{a}}^h(k-l)\,\sigma_{\mathrm{b}}^h(l)\,\sigma^h_{\leftarrow,\nu\iota'}(k);
\end{aligned}$$

here σ_Π, $\sigma_\rightarrow$ and $\sigma_\leftarrow$ are the symbols of $y \mapsto \Pi(p_{[\iota]}(0), p_{[\iota']}(y))$, of $K^{\rightarrow}_{\iota\mu}$ and of $K^{\leftarrow}_{\nu\iota'}$. Let $\mathfrak{G}_2 = \mathfrak{G}_2(\iota,\iota')$, resp. $\mathfrak{G}_3 = \mathfrak{G}_3(\iota,\iota')$, be the set of the triples (μ,ν,G) such that $G \in \mathfrak{G}_{\mathrm{I}}(\mu,\nu)$ is a term of $-\langle \partial\mathsf{S}_3; \partial'\mathsf{S}_3\rangle_0$, resp. of the ghost loop, in eq. 7.xxxiii; such a triple carries the coefficient $-\mathsf{n}\,c_G$, which takes the place of c_G in the sums below. Then, with sums over $\iota,\iota' \in \{1,2,3\}$ and over $G \in \mathfrak{G}_\Gamma(\iota,\iota')$, where for $\Gamma = 4$ only the terms with $(n_X, n_Y) = (4,4)$ are taken,

$$\sum_{\iota,\iota'}\sum_{G} c_G\,\mathsf{F}_G = 24\,F_\Gamma \quad (\Gamma = 1,2,3,4,6), \qquad \sum_{\iota,\iota'}\sum_{G\in\mathfrak{G}_5} c_G\,\mathsf{F}_G = 12\,F_5.$$

■

Proof. Step 1 (Feynman rules). Let $K(y) = \int e^{i\mathsf{q}\cdot y}\sigma(\mathsf{q})\, d^\circ\mathsf{q}$ be a line, where y is the base point of its second leg minus that of its first leg; we say that the line carries the momentum q from the first vertex to the second. The momentum entering a vertex through this line is then q at the second vertex and $-\mathsf{q}$ at the first. Leading parts of line symbols are additive and multiplicative (Def. 7.16(3)); hence the sum over the monomials of a vertex may be carried out on the leading parts. By Lem. 7.24 at $\tau_{\mathrm{gf}} = 1$, every line contributes a factor $\delta_{\mu\nu}|\mathsf{q}|^{-2}$, where at a plain leg the index of the line is the direction of the bond of that leg; a curl leg at a plaquette $p_{\alpha\beta}$ contributes the factor $i(\mathsf{q}'_\alpha\delta_{\nu\beta} - \mathsf{q}'_\beta\delta_{\nu\alpha})$, where q' is the momentum entering the vertex; and shifts of base points contribute the factor 1, the leading part of $e^{ik\cdot y_0}$.

Step 2 (the vertex $\mathsf{a}^{(2)}$). Two curl legs at $p_{[\iota]}$, with entering momenta k_1, k_2 and line indices μ, ν, give after summation over ι, that is, over $\alpha < \beta$,

$$i^2 \sum_{\alpha<\beta} (k_{1\alpha}\delta_{\mu\beta} - k_{1\beta}\delta_{\mu\alpha})(k_{2\alpha}\delta_{\nu\beta} - k_{2\beta}\delta_{\nu\alpha}) \\ = -\big[(k_1 \cdot k_2)\,\delta_{\mu\nu} - k_{1\nu}k_{2\mu}\big] = V_2(k_1, k_2)_{\mu\nu}.$$

Step 3 (the vertex $\mathsf{a}^{(3)}$). By Lem. 7.26(d), $\mathsf{a}_P^{(3)}[\![X, Y, Z]\!]$ is the sum of t_P over the six orders of X, Y, Z. Let $P = p_{\mu\nu}$. To leading order, $(Y(b_1), \dots, Y(b_4))$ is replaced by $(Y_\mu, Y_\nu, Y_\mu, Y_\nu)$; hence

$$\sum_{i<j} s_i s_j\, Y(b_i) \times Z(b_j) \to Y_\mu \times Z_\nu - Y_\nu \times Z_\mu - Y_\mu \times Z_\mu - Y_\nu \times Z_\nu,$$

and, by the antisymmetry of the cross product,

$$\mathsf{t}_P(X, Y, Z) + \mathsf{t}_P(X, Z, Y) \to -2\langle (\mathrm{d}X)_{\mu\nu},\, Y_\mu \times Z_\nu - Y_\nu \times Z_\mu\rangle.$$

Both factors are antisymmetric in $\mu\nu$. Summing over $\mu < \nu$ and inserting $(\mathrm{d}X)_{\mu\nu} \to i(k_{1\mu}X_\nu - k_{1\nu}X_\mu)$, we obtain

$$-2i \sum_{\mu,\nu} k_{1\mu} \langle X_\nu,\, Y_\mu \times Z_\nu - Y_\nu \times Z_\mu\rangle.$$

For legs of colours a, b, c, with indices α, β, γ and entering momenta k_1, k_2, k_3, this equals $-2i\,\epsilon_{abc}[\delta_{\alpha\gamma}k_{1\beta} - \delta_{\alpha\beta}k_{1\gamma}]$. We add the terms in which the second, resp. the third, leg is the curl leg; they arise by cyclic permutation, under which ϵ_{abc} is invariant. The leading part of $\sum_\iota \mathsf{a}^{(3)}_{p_{[\iota]}}[\![\cdot,\cdot,\cdot]\!]$ is therefore

$$-2i\,\epsilon_{abc}\, V_3(k_1, k_2, k_3)_{\alpha\beta\gamma}.$$

Step 4 ($\Gamma = 1$). By Lem. 7.33(a) the five lines $1, \dots, 5$ of the kite carry k (from p to P), $\mathsf{p} - k$ (from p to P'), $k - l$ (from P to P'), l (from P to q) and $\mathsf{p} - l$ (from P' to q). The entering momenta are $(-k, k - \mathsf{p})$ at p; $(k, -l, l - k)$ at P, for the legs towards p, q, P'; $(\mathsf{p} - k, l - \mathsf{p}, k - l)$ at P'; and $(l, \mathsf{p} - l)$ at q. At both cubic vertices the colours are (a, a', c), and $\sum_{a,a',c} \epsilon^2_{aa'c} = 6$. With the factor $\frac{1}{2}$ of T_1 in Lem. 7.26(c) we obtain

$$\tfrac{1}{2}(-2i)^2 \cdot 6 \cdot V_2(-k, k - \mathsf{p})_{\mu\nu}\, V_2(l, \mathsf{p} - l)_{\rho\sigma} \\ \times V_3(k, -l, l - k)_{\mu\rho\alpha}\, V_3(\mathsf{p} - k, l - \mathsf{p}, k - l)_{\nu\sigma\alpha} \,/\, \mathsf{x}_1\mathsf{x}'_1\mathsf{x}_2\mathsf{x}'_2\mathsf{x}_3,$$

and $\frac{1}{2}(-4) \cdot 6 = 24 \cdot (-\frac{1}{2})$. Since V_2 is even, and the product of the two factors V_3 is even under the change of sign of all their arguments, this is $24F_1$.

Step 5 ($\Gamma = 6$). The factor is $\frac{1}{6}(-2i)^2 \cdot 6 = 24 \cdot (-\frac{1}{6})$. The entering momenta are $-(\mathsf{p} - k)$, $-(k - l)$, $-l$ at p and the opposite ones at q. The tensor V_3 is invariant under the cyclic permutations of its arguments together with its indices, and it changes sign under a transposition, like ϵ_{abc}; this gives $24F_6$.

Step 6 ($\Gamma = 5$). The line K_0 carries $\mathsf{p} - k$ from p to q, the line X carries k from p to P, and K_a, K_b carry $k - l$ and l from P to q. The factor is $-\frac{1}{2}(-2i)^2 \cdot 6 = 12$, with the factor $-\frac{1}{2}$ of T_5, and the tensors are

$$V_2(k - \mathsf{p}, -k)_{\alpha\sigma}\, V_3(\mathsf{p} - k, l, k - l)_{\alpha\beta\gamma}\, V_3(k, -l, l - k)_{\sigma\beta\gamma};$$

with the factors $|\mathsf{q}|^{-2}$ of the four lines this is $12F_5$.

Step 7 ($\Gamma = 2$). By Lem. 7.26(a), applied to two quadratic polynomials,

$$-\langle \partial_{b,3}\mathsf{S}_3; \partial_{b',3}\mathsf{S}_3\rangle_0 = -\tfrac{1}{2} \sum \mathsf{a}^{(3)}_P[\![\delta_{b,3}, \delta_\mathsf{i}, \delta_{\mathsf{i}'}]\!]\, \mathsf{a}^{(3)}_{P'}[\![\delta_{b',3}, \delta_\mathsf{j}, \delta_{\mathsf{j}'}]\!]\, \mathsf{C}(\mathsf{i}, \mathsf{j})\, \mathsf{C}(\mathsf{i}', \mathsf{j}'),$$

the sum extending over P, P' and over the indices $\mathsf{i}, \mathsf{i}', \mathsf{j}, \mathsf{j}'$. Here $\sum_{c,c'} \epsilon^2_{3cc'} = 2$, and together with the factor $-\mathsf{n}$ of T_I the total factor is $-\mathsf{n} \cdot (-\frac{1}{2})(-2i)^2 \cdot 2 = -12 = 24 \cdot (-\frac{1}{2})$. The line Π carries $\mathsf{p} - k$ from p to q, and the chain carries k. The curl legs give $V_2(-k, k - \mathsf{p})_{\mu\nu}$ at p and $V_2(k, \mathsf{p} - k)_{\rho\nu}$ at q, where ν is the index of the line Π, and the cubic vertices give $V_3(k, -l, l - k)_{\mu\beta\gamma}\, V_3(-k, l, k - l)_{\rho\beta\gamma}$.

With the factors of the five lines, whose squares give the denominator $\mathsf{x}_1^2\mathsf{x}_1'\mathsf{x}_2\mathsf{x}_3$, and with the parities of V_2 and V_3, this is $24F_2$.

Step 8 ($\Gamma=3$). The ghost loop enters Σ_{tot} through the kernel $-8\,\mathsf{D}(b;b')\,\mathsf{D}(b';b)$. By Lem. 7.24, the leading parts of the symbols of $y\mapsto \mathsf{D}((x,\mu);(x+y,\rho))$ and of $y\mapsto \mathsf{D}((x+y,\rho);(x,\mu))$ are $-i\mathsf{q}_\mu|\mathsf{q}|^{-2}$ and $+i\mathsf{q}_\rho|\mathsf{q}|^{-2}$. With $\mathsf{q}=k-l$, resp. $\mathsf{q}=l$, and with the factor $-\mathsf{n}$ of T_{I} we obtain

$$-\mathsf{n}\,(-8)\,(k-l)_\mu\,l_\rho\,/\,\mathsf{x}_3\mathsf{x}_2 = 24\cdot(-1)\,l_\rho(l-k)_\mu\,/\,\mathsf{x}_2\mathsf{x}_3;$$

together with the factors at p and q and on the lines Π, $K^{\rightarrow}$, $K^{\leftarrow}$, as in Step 7, this is $24F_3$.

Step 9 ($\Gamma=4$). By Lem. 7.28(4), the terms $G\in\mathfrak{G}_4$ with $(n_X,n_Y)=(4,4)$ are exactly those which arise from the monomials of $\frac12|\mathfrak{z}_1|^2$ in the standard writing of $\mathsf{a}_P^{(4)}$ (Lem. 7.26(d)); here, by eq. 7.v in Lem. 7.5(b), with $A_i:=A(b_i(P))$,

$$\mathsf{a}_P^{(4)}=\tfrac12|\mathfrak{z}_1|^2+\langle \mathrm{d}A(P),\mathfrak{z}_2\rangle-\tfrac{1}{24}|\mathrm{d}A(P)|^4,\qquad \mathfrak{z}_1=-\sum_{i<j}s_is_j\,A_i\times A_j.$$

Thus the vertex of the class $\Gamma=4$ is $\frac12|\mathfrak{z}_1|^2$. At $P=p_{\alpha\beta}$ the replacement of Step 3, $(A_1,\dots,A_4)\to(A_\alpha,A_\beta,A_\alpha,A_\beta)$, gives to leading order

$$\mathfrak{z}_1\to-(A_\alpha\times A_\beta-A_\beta\times A_\alpha-A_\alpha\times A_\alpha-A_\beta\times A_\beta)=-2A_\alpha\times A_\beta;$$

since $\frac12|2A_\alpha\times A_\beta|^2=2|A_\alpha\times A_\beta|^2$ and the terms with $\alpha=\beta$ vanish, the sum over ι, that is, over $\alpha<\beta$, gives $\sum_\iota\frac12|\mathfrak{z}_1|^2\to\mathsf{Q}(A):=\sum_{\alpha,\beta}|A_\alpha\times A_\beta|^2$, where $|Z|^2:=\langle Z,Z\rangle$ with the complex bilinear extension of $\langle\cdot,\cdot\rangle$. For $X=\xi\mathsf{e}_a$ and $Y=\eta\mathsf{e}_{a'}$ with $\xi,\eta\in\mathbb{C}^3$, that is, $X_\alpha=\xi_\alpha\mathsf{e}_a$ and $Y_\alpha=\eta_\alpha\mathsf{e}_{a'}$, we have, with the bilinear dot product on $\mathbb{C}^3$,

$$\mathsf{Q}(sX+tY)=s^2t^2(1-\delta_{aa'})\sum_{\alpha,\beta}(\xi_\alpha\eta_\beta-\xi_\beta\eta_\alpha)^2,$$

$$\sum_{\alpha,\beta}(\xi_\alpha\eta_\beta-\xi_\beta\eta_\alpha)^2=2\big[(\xi\cdot\xi)(\eta\cdot\eta)-(\xi\cdot\eta)^2\big].$$

Polarising, with $X_i=\xi_i\mathsf{e}_a$ and $Y_j=\eta_j\mathsf{e}_{a'}$ $(i,j\in\{1,2\})$, we find

$$\mathsf{Q}[\![X_1,X_2,Y_1,Y_2]\!]=4(1-\delta_{aa'})\cdot2\big[(\xi_1\cdot\xi_2)(\eta_1\cdot\eta_2)-\tfrac12(\xi_1\cdot\eta_1)(\xi_2\cdot\eta_2)-\tfrac12(\xi_1\cdot\eta_2)(\xi_2\cdot\eta_1)\big].$$

Since $\sum_{a,a'}(1-\delta_{aa'})=6$, the factor is $-\frac14\cdot4\cdot2\cdot6=-12=24\cdot(-\frac12)$, with the factor $-\frac14$ of T_4. The pairs of legs contracted with p and with q give $\xi_{1\mu}\xi_{2\nu}\to V_2(k,\mathsf{p}-k)_{\mu\nu}/\mathsf{x}_1\mathsf{x}_1'$ and $\eta_{1\mu}\eta_{2\nu}\to V_2(l,\mathsf{p}-l)_{\mu\nu}/\mathsf{x}_2\mathsf{x}_2'$, and the result is $24F_4$. □

We now show that the model $\mathsf{T}_\Gamma^{\mathrm{mod}}$ of Def. 7.32 has, up to a part of degree -4, the same part of degree -5 as the corresponding lattice terms. For a kernel K in a class $\mathfrak{K}(n)$ we write K^h, or $(K)^h$ when K is given by a longer expression, for the second component of its leading data (Def. 7.16(4)), that is, for the function K^h of condition (K3), and we call it the *leading kernel* of K; it is smooth off 0 and homogeneous of degree $-n$.

Lemma 7.35 (The model has the degree -5 parts of the lattice classes)**.** Let $\tau_{\mathrm{gf}}=1$. For $G\in\mathfrak{G}_1\cup\mathfrak{G}_4\cup\mathfrak{G}_5\cup\mathfrak{G}_6$ let U_G be the function of degree -5 attached to G in Lem. 7.29. For $(\mu,\nu,G)\in\mathfrak{G}_2\cup\mathfrak{G}_3$ let

$$\mathsf{U}_G:=\Pi^h\cdot\big(K^{\rightarrow}_{\iota\mu}*\mathsf{f}_G(\nabla)[K_{\mathrm{a}}K_{\mathrm{b}}-\mathsf{q}_G]*K^{\leftarrow}_{\nu\iota'}\big)^h,\qquad \Pi^h:=\mathcal{F}^{-1}_{\neq0}\sigma^h_\Pi,$$

where the chain lies in $\mathfrak{K}(2)$ by Lem. 7.31(a) and two applications of Lem. 7.22(b). Let

$$\mathsf{U}_\Gamma:=\sum_{\iota,\iota'}\sum_{G\in\mathfrak{G}_\Gamma}c_G\,\mathsf{U}_G\quad(\Gamma\neq5),$$

$$\mathsf{U}_5(u):=\sum_{\iota,\iota'}\sum_{G\in\mathfrak{G}_5}c_G\big[\mathsf{U}_G(u)+\mathsf{U}_G(-u)\big],$$

with the coefficients $-\mathsf{n}\,c_G$ of Lem. 7.34 for $\Gamma = 2, 3$. Then for each $\Gamma = 1, \dots, 6$ there are a continuous function I_Γ on $\mathbb{R}^3 \setminus \{0\}$, homogeneous of degree -4, and a pure constant C such that

$$|\mathsf{T}_\Gamma^{\mathrm{mod}}(u) - \mathsf{I}_\Gamma(u) - \mathsf{U}_\Gamma(u)| \le C|u|^{-6} \lg|u| \qquad (u \in \mathbb{Z}^3 \setminus \{0\}).$$

■

Proof. Step 0 (model lines). For a line whose symbol has the leading part σ^h, of degree s, let σ^{mod} be the periodic continuation of $\rho\sigma^h$, with ρ of Def. 7.32(3). As $\rho \in C_c^\infty(\mathbb{R}^3)$, $\operatorname{supp}\rho \subset \{|k| \le 2\rho_0\} \subset \{|k| < \pi\}$ and $\rho = 1$ near 0, the function σ^{mod} is a line symbol with the same leading part (Def. 7.16(3)), and its parity is exact: $\sigma^{\mathrm{mod}}(-k) = (-1)^s\sigma^{\mathrm{mod}}(k)$. Its kernel $K^{\mathrm{mod}}(x) = \int_{\mathbb{R}^3} e^{ik\cdot x}\rho\,\sigma^h\, d^\circ k$ satisfies $K^{\mathrm{mod}} = K^h + O(|x|^{-N})$ for every N, with $K^h = \mathcal{F}_{\neq 0}^{-1}\sigma^h$, by Lem. 7.17(b). For $(\mu, \nu, G) \in \mathfrak{G}_2 \cup \mathfrak{G}_3$ the elementary term is, by Lem. 7.28(I), the value at $u = u'$ of the function

$$u \mapsto K_\Pi(u)\,(K_{\iota\mu}^{\rightarrow} * \mathsf{f}_G(\nabla)[K_{\mathrm{a}}K_{\mathrm{b}}] * K_{\nu\iota'}^{\leftarrow})(u), \qquad K_\Pi(y) := \Pi(p_{[\iota]}(0), p_{[\iota']}(y)) \quad (y \in \mathbb{Z}^3);$$

here K_Π is the elementary line of degree 3 of Lem. 7.24, with the symbol σ_Π of Lem. 7.34, and $\Pi(p, q) = K_\Pi(u')$ by translation invariance. For an elementary term G let $G^{\mathrm{mod}}(u)$ be the same expression with every line K replaced by K^{mod}, in particular K_Π by K_Π^{mod}; for G of type (I) we moreover drop the operator $\mathsf{f}_G(\nabla)$ and replace $K^{\rightarrow}$ by the elementary line K_1' whose symbol is the periodic continuation of $\rho\,\sigma_{\rightarrow}^h\mathsf{f}_G^h$; it has degree $2 + d_G$.

Step 1 (the model in momentum space). We show that $\mathsf{T}_\Gamma^{\mathrm{mod}} = \sum c_G\, G^{\mathrm{mod}}$, the sum extending over ι, ι' and G as in Lem. 7.34, where for $\Gamma = 5$ the term $G^{\mathrm{mod}}(u)$ is replaced by $G^{\mathrm{mod}}(u) + G^{\mathrm{mod}}(-u)$. By Lem. 7.33,

$$G^{\mathrm{mod}}(u) = \int_{\mathbb{T}} e^{i\mathsf{p}\cdot u}\Big[\iint_{\mathbb{T}^2} \prod \sigma^{\mathrm{mod}}\, d^\circ k\, d^\circ l\Big] d^\circ \mathsf{p},$$

where the product extends over the lines of G and the arguments of the symbols are those in F_G. For the kites this is assertion (a) of that lemma, the model lines having the degrees of the lattice lines. For the other types we apply assertion (b) to the products and convolutions of which G^{mod} is built, one at a time, and we verify its hypotheses. A model line of degree n is bounded by $C\langle x\rangle^{-n}$ (Lem. 7.17(a)), and its symbol is integrable. A kernel in $\ell^2(\mathbb{Z}^3)$ with an integrable symbol has its symbol in $L^2(\mathbb{T})$, as in the proof of Lem. 7.33(b); hence the symbol of the convolution of two such kernels, a product of two functions in $L^2(\mathbb{T})$, is integrable, and so is the symbol of the product of two kernels with integrable symbols, a convolution of two integrable functions.

- For $G \in \mathfrak{G}_6$, G^{mod} is a product of three model lines, and the rule for products is used twice.
- For $G \in \mathfrak{G}_4$ with $(n_X, n_Y) = (4, 4)$, $G^{\mathrm{mod}} = X * Y$, where X and Y are products of two model lines of degree 2. Their symbols are given by the rule for products, and $|X(x)|, |Y(x)| \le C\langle x\rangle^{-4}$, so that $X, Y \in \ell^2(\mathbb{Z}^3)$ and the rule for $X * Y$ applies.
- For $G \in \mathfrak{G}_5$, $G^{\mathrm{mod}}(u) = K_0(u)\,(X * Y)(u)$ with model lines K_0, X and a product Y of two model lines, where $n_X \in \{2, 3\}$ and $n_Y \in \{2, 3, 4\}$. Thus $|X(x)| \le C\langle x\rangle^{-n_X}$ and $|Y(x)| \le C\langle x\rangle^{-n_Y}$ with exponents at least 2, so that $X, Y \in \ell^2(\mathbb{Z}^3)$; the kernel $X * Y$ has the symbol $\hat{X}\hat{Y}$, and the product with K_0 is treated by the rule for products.
- For G of type (I), $G^{\mathrm{mod}} = K_\Pi^{\mathrm{mod}} \cdot (K_1' * X * K^{\leftarrow,\mathrm{mod}})$ with $X := K_{\mathrm{a}}^{\mathrm{mod}}K_{\mathrm{b}}^{\mathrm{mod}}$; the degrees of K_1', X and $K^{\leftarrow,\mathrm{mod}}$ are $2 + d_G$, $4 - d_G$ and 2, all at least 2. Hence $K_1' * X$ has the symbol $\hat{K}_1'\hat{X}$, and Lem. 7.18(c), with $(a, a') = (2 + d_G, 4 - d_G)$ and $a + a' - 3 = 3$, gives

  $$|(K_1' * X)(x)| \le C\langle x\rangle^{-\min\{2+d_G,\, 4-d_G,\, 3\}} \lg\langle x\rangle;$$

  the exponent is at least 2, so that $K_1' * X \in \ell^2(\mathbb{Z}^3)$, the logarithm included. The rule for convolutions applies once more to $(K_1' * X) * K^{\leftarrow,\mathrm{mod}}$, and the product with K_Π^{mod} is treated by the rule for products.

For $k, l, \mathsf{p} \in [-\pi, \pi]^3$ the periodic continuations may be replaced by $\rho\sigma^h$. We use the following observation: if $v \in [-\pi, \pi]^3$, $|w| < 4\rho_0$ and $|v - w|_{\mathbb{T}} < 2\rho_0$, then $|v - w| < 2\rho_0$, because the distance

of $v-w$ from any point $2\pi\mathsf{m}$ with $\mathsf{m}\in\mathbb{Z}^3\setminus\{0\}$ is at least $\pi-4\rho_0>2\rho_0$. Let $k,l,\mathsf{p}\in[-\pi,\pi]^3$, and assume first that every line q of Γ satisfies $|\mathsf{q}|_{\mathbb{T}}<2\rho_0$; this is the case wherever all the factors σ^{mod} are non-zero. Then $|l|=|l|_{\mathbb{T}}<2\rho_0$, since $l\in[-\pi,\pi]^3$. For $\Gamma\neq 6$ we have likewise $|k|<2\rho_0$; then $|k-l|<4\rho_0<\pi$, so that $|k-l|=|k-l|_{\mathbb{T}}<2\rho_0$ wherever $k-l$ is a line ($\Gamma=1,2,3,5$), and the observation with $v=\mathsf{p}$ and $w=k$, resp. $w=l$, gives $|\mathsf{p}-k|<2\rho_0$, resp. $|\mathsf{p}-l|<2\rho_0$, wherever these vectors are lines.

For $\Gamma=6$ the observation with $(v,w)=(k,l)$ gives $|k-l|<2\rho_0$, hence $|k|<4\rho_0$, and with $(v,w)=(\mathsf{p},k)$ it gives $|\mathsf{p}-k|<2\rho_0$. Thus every line satisfies $|\mathsf{q}|<2\rho_0$, and $\sigma^{\mathrm{mod}}(\mathsf{q})=\rho(\mathsf{q})\sigma^h(\mathsf{q})$. If, on the other hand, some line satisfies $|\mathsf{q}|_{\mathbb{T}}\geq 2\rho_0$, then $\sigma^{\mathrm{mod}}(\mathsf{q})=0$ and also $\rho(\mathsf{q})=0$, since $|\mathsf{q}|\geq|\mathsf{q}|_{\mathbb{T}}$. In either case the product of the factors σ^{mod} equals $R_\Gamma\mathsf{F}_G$ at (k,l,p), the lines of G being the lines of Γ (for type (I) the factor $\rho\,\sigma^h_{\rightarrow}\mathsf{f}^h_G$ of K'_1 provides the factors $\sigma^h_{\rightarrow}(k)\,\mathsf{f}^h_G(k)$ of F_G).

Since R_Γ vanishes unless $k,l,\mathsf{p}\in[-\pi,\pi]^3$ (Def. 7.32(3)), we obtain

$$G^{\mathrm{mod}}(u)=\int_{\mathbb{R}^3}e^{i\mathsf{p}\cdot u}\Big[\iint R_\Gamma\,\mathsf{F}_G\,d^\circ k\,d^\circ l\Big]d^\circ\mathsf{p}.$$

By Lem. 7.34, $\sum c_G\iint R_\Gamma\mathsf{F}_G\,d^\circ k\,d^\circ l=24\Psi_\Gamma$ for $\Gamma\neq 5$, whence $\sum c_G G^{\mathrm{mod}}=\mathsf{T}^{\mathrm{mod}}_\Gamma$. For $\Gamma=5$ the term $G^{\mathrm{mod}}(-u)$ carries the factor $e^{-i\mathsf{p}\cdot u}$, and $\sum c_G\iint R_5\mathsf{F}_G\,d^\circ k\,d^\circ l=12\Psi_5$ is even; hence $\sum c_G[G^{\mathrm{mod}}(u)+G^{\mathrm{mod}}(-u)]=12\int(e^{i\mathsf{p}\cdot u}+e^{-i\mathsf{p}\cdot u})\Psi_5\,d^\circ\mathsf{p}=\mathsf{T}^{\mathrm{mod}}_5(u)$.

Step 2 (expansions). Lem. 7.29 applies to G^{mod}, the model lines being elementary lines of the same degrees as the lattice lines; the functions of degree -5 which it provides are determined by the leading parts of the symbols of the lines alone (Lem. 7.29), and therefore coincide with the U_G. For G of type (I) let $X:=K^{\mathrm{mod}}_{\mathrm{a}}K^{\mathrm{mod}}_{\mathrm{b}}$, of degree $n_X=4-d_G$, and write $X=\mathsf{q}_X+X'$ with $X'\in\mathfrak{K}(n_X)$ (Lem. 7.22(c)). Since K'_1 is a line of degree $2+d_G\leq 4$, two applications of Lem. 7.22(b) give $K'_1*X'*K^{\leftarrow,\mathrm{mod}}\in\mathfrak{K}(2)$, and the first component of its leading data is $\sigma^h_{\rightarrow}\mathsf{f}^h_G\mathsf{I}^h\sigma^h_{\leftarrow}$, where the homogeneous part I^h of the bubble depends only on σ^h_{a}, σ^h_{b} (Lem. 7.20(a)).

The lattice chain $K^{\rightarrow}_{\iota\mu}*\mathsf{f}_G(\nabla)[K_{\mathrm{a}}K_{\mathrm{b}}-\mathsf{q}_G]*K^{\leftarrow}_{\nu\iota'}$ has the same first component of its leading data, by Lem. 7.31(a) and by Lem. 7.22(b), used for each of its two convolutions. Within $\mathfrak{K}(2)$ the leading kernel is determined by the first component of the leading data (Lem. 7.22(d)); hence the two chains have the same leading kernel. Now $G^{\mathrm{mod}}=K^{\mathrm{mod}}_\Pi\cdot(K'_1*\mathsf{q}_X*K^{\leftarrow,\mathrm{mod}})+K^{\mathrm{mod}}_\Pi\cdot(K'_1*X'*K^{\leftarrow,\mathrm{mod}})$, and the first term is local. In the second term, $K^{\mathrm{mod}}_\Pi=\Pi^h+O(|u|^{-N})$ with $\Pi^h=O(|u|^{-3})$, and $K'_1*X'*K^{\leftarrow,\mathrm{mod}}$ differs from its leading kernel, which is $O(|u|^{-2})$, by $O(|u|^{-3}\lg|u|)$, by condition (K3).

We obtain in all cases

$$G^{\mathrm{mod}}=[\text{local terms}]+\mathsf{U}_G+O(|u|^{-6}\lg|u|).$$

Step 3 (local terms). In this step all lines are model lines, and we omit the superscript mod. Because of the exact parity of the model symbols, every one-loop function I built from model lines satisfies $\mathsf{I}(-\kappa)=(-1)^{\omega+1}\mathsf{I}(\kappa)$, as one sees by the substitution $l\to-l$; its homogeneous part I^h has the same parity (Lem. 7.20(a)). By Lem. 7.20(b) the polynomial $\mathsf{Q}(\kappa)-(-1)^{\omega+1}\mathsf{Q}(-\kappa)$, of degree at most ω, is therefore $O(|\kappa|^{\omega+1}\lg|\kappa|\,\Lambda(\kappa))$, hence equal to 0. Consequently $\mathsf{Q}=0$ if $\omega\leq 0$, and $\mathsf{Q}=\mathsf{Q}(0)$ if $\omega=1$; in both cases $\mathsf{q}=\mathsf{Q}(0)\,\delta_0$.

The local terms are thus: $\mathsf{Q}_{\mathsf{p}}(0)K_4K_5(u)+\mathsf{Q}_{\mathsf{q}}(0)K_1K_2(u)$ for the kites, where $n_4+n_5=4$ if $\omega_{\mathsf{p}}=1$ and $n_1+n_2=4$ if $\omega_{\mathsf{q}}=1$; $\mathsf{Q}_Y(0)K_0X(u)$, with $n_0+n_X=4$, for $\mathfrak{G}_5$; $\mathsf{Q}_X(0)Y(u)+\mathsf{Q}_Y(0)X(u)$, with $n_X=n_Y=4$, for $\mathfrak{G}_4$; and, for type (I) with $d_G=0$, $\mathsf{Q}_X(0)K_\Pi(u)\,(K'_1*K^{\leftarrow})(u)$, where $K'_1*K^{\leftarrow}$ is, by Lem. 7.33(b), the elementary line of degree 1 whose symbol is the periodic continuation of $\rho^2\sigma^h_{\rightarrow}\mathsf{f}^h_G\sigma^h_{\leftarrow}$ (a line symbol by Def. 7.16(3), with ρ^2 in place of ρ), $\mathsf{f}^h_G=\mathsf{f}_G(0)$ being a number.

By Step 0, and by Lem. 7.17(b) with ρ^2 in place of ρ for the line $K'_1*K^{\leftarrow}$, each of these local terms is, up to $O(|u|^{-N})$, its factor $\mathsf{Q}_\bullet(0)$ times the product of the functions $K^h=\mathcal{F}^{-1}_{\neq 0}\sigma^h$ of its lines, a function homogeneous of degree -4 and smooth off 0. We let I_Γ be the sum of these functions, each taken with the coefficient with which its term G^{mod} enters $\mathsf{T}^{\mathrm{mod}}_\Gamma$ in Step 1. For $\Gamma=5$, where

$\mathsf{T}_5^{\mathrm{mod}}(u) = \sum c_G\big[G^{\mathrm{mod}}(u) + G^{\mathrm{mod}}(-u)\big]$, the expansion of Step 2 is used at u and at $-u$; accordingly, every such function h of a term $G \in \mathfrak{G}_5$ enters I_5 together with its reflection $u \mapsto \mathsf{h}(-u)$, just as U_5 contains $\mathsf{U}_G(u)$ and $\mathsf{U}_G(-u)$.

Adding the expansions of Step 2 with these coefficients, we obtain the claimed bound for large $|u|$; for small $|u|$ all terms are bounded. All data being fixed, the constant C is pure. □

The functions I_Γ and U_Γ are in fact multiples of $|u|^{-4}$ and $|u|^{-5}$, since $\mathsf{T}_\Gamma^{\mathrm{mod}}$ is radial.

Lemma 7.36 (Rotation invariance of the homogeneous parts; definition of $\kappa_\Gamma^{\mathrm{lat}}$)**.** Let $\mathsf{t}\colon [0,\infty) \to \mathbb{C}$, let I, U be continuous functions on $\mathbb{R}^3 \setminus \{0\}$, homogeneous of degrees -4 and -5 respectively, and assume that for some C

$$|\mathsf{t}(|u|) - \mathsf{I}(u) - \mathsf{U}(u)| \le C|u|^{-6} \lg|u| \qquad (u \in \mathbb{Z}^3 \setminus \{0\}).$$

Then $\mathsf{I}(u) = c_4|u|^{-4}$ and $\mathsf{U}(u) = \kappa|u|^{-5}$ with constants c_4, κ. In particular the functions of Lem. 7.35 satisfy $\mathsf{I}_\Gamma(u) = c_{4,\Gamma}|u|^{-4}$ and

$$\mathsf{U}_\Gamma(u) = \kappa_\Gamma^{\mathrm{lat}}\,|u|^{-5} \qquad (\Gamma = 1,\dots,6);$$

this defines the numbers $c_{4,\Gamma}$ and $\kappa_\Gamma^{\mathrm{lat}}$. ■

Proof. Let O be a rotation with rational entries, N a common denominator of these entries, and $\mathsf{m} \in \mathbb{Z}^3 \setminus \{0\}$. For integers $j \ge 1$ the points $j\mathsf{N}\mathsf{m}$ and $j\mathsf{N}O\mathsf{m}$ lie in $\mathbb{Z}^3$ and have the same norm jr_1, $r_1 := \mathsf{N}|\mathsf{m}|$. Let $\hat{\mathsf{m}} := \mathsf{m}/|\mathsf{m}|$. Subtracting the hypothesis at these two points, at which t takes the same value, we obtain by homogeneity

$$\Big|(jr_1)^{-4}\big[\mathsf{I}(\hat{\mathsf{m}}) - \mathsf{I}(O\hat{\mathsf{m}})\big] + (jr_1)^{-5}\big[\mathsf{U}(\hat{\mathsf{m}}) - \mathsf{U}(O\hat{\mathsf{m}})\big]\Big| \le 2C(jr_1)^{-6}\lg(jr_1).$$

Multiplying by $(jr_1)^4$ and letting $j \to \infty$ we obtain $\mathsf{I}(\hat{\mathsf{m}}) = \mathsf{I}(O\hat{\mathsf{m}})$; then, multiplying the remaining difference by $(jr_1)^5$, we obtain $\mathsf{U}(\hat{\mathsf{m}}) = \mathsf{U}(O\hat{\mathsf{m}})$. The directions $\mathsf{m}/|\mathsf{m}|$ are dense in S^2, since $\mathbb{Q}^3$ is dense in $\mathbb{R}^3$. The rational rotations are dense in $SO(3)$: the Cayley map $\mathsf{A} \mapsto (\mathbf{1}-\mathsf{A})(\mathbf{1}+\mathsf{A})^{-1}$ is continuous from the real antisymmetric 3×3 matrices onto the rotations without eigenvalue -1, which are dense in $SO(3)$; the antisymmetric matrices with rational entries are dense among the real ones, and by the formula they are mapped to rotations with rational entries, since the inverse of an invertible matrix with rational entries has rational entries.

By continuity, I and U are invariant under all rotations; as $SO(3)$ acts transitively on S^2, they are constant on S^2, and homogeneity gives the claim. For the last assertion we note that $\mathsf{T}_\Gamma^{\mathrm{mod}}$ is the restriction to $\mathbb{Z}^3$ of a radial function, since Ψ_Γ is radial (Def. 7.32(3)); thus $\mathsf{T}_\Gamma^{\mathrm{mod}}(u) = \mathsf{t}(|u|)$ for a function t, and Lem. 7.35 provides the hypothesis. □

We can now describe $\mathsf{B}_1^{\mathrm{lat}}(p,q)$ at large distances of the two plaquettes. The local parts of the elementary terms drop out by the zero-momentum identities of Lem. 7.31, and the parts of degree -5 add up, by Lems. 7.35 and 7.36, to a multiple of $|u|^{-5}$.

Theorem V (The kernel of degree -5: large-distance asymptotics of $\mathsf{B}_1^{\mathrm{lat}}$)**.** There are functions $\mathsf{U}_{ij} \in C^1(\mathbb{R}^3 \setminus \{0\})$, $i,j \in \{1,2,3\}$, homogeneous of degree -5, and pure constants C, r_0 such that for all plaquettes p, q of $\mathbb{Z}^3$, with $u := x(q) - x(p)$ and $r := |u| \ge r_0$,

$$\begin{aligned} &|\mathsf{B}_1^{\mathrm{lat}}(p,q) - \mathsf{U}_{\lambda(p)\lambda(q)}(u)| \le C\,r^{-6}\lg r, \\ &\sum_{i,j} \mathsf{U}_{ij}(u) = \kappa^{\mathrm{lat}}|u|^{-5}, \qquad \kappa^{\mathrm{lat}} := \sum_{\Gamma=1}^{6} \kappa_\Gamma^{\mathrm{lat}}, \end{aligned} \tag{7.xxxvii}$$

with the numbers $\kappa_\Gamma^{\mathrm{lat}}$ of Lem. 7.36. Moreover $\mathsf{B}_1^{\mathrm{lat}}$ is bounded. ■□

Proof. We work at $\tau_{\mathrm{gf}} = 1$; recall that $\mathsf{B}_1^{\mathrm{lat}}(p,q) = \mathsf{B}_1^{\mathrm{lat}}(p,q;1)$ by eq. 7.xix. Boundedness is contained in Lem. 7.27, which gives $\sup_{p,q} |\mathsf{B}_1^{\mathrm{lat}}(p,q;\tau_{\mathrm{gf}})| < \infty$ for every $\tau_{\mathrm{gf}} > 0$. Let $p = p_{[\iota]}(x_p)$, $q = p_{[\iota']}(x_q)$ and $u' := x_q - x_p$; then $|u - u'| \le \sqrt{2}$, and we take r large. By Lem. 7.26(c) and Lem. 7.28, $\mathsf{B}_1^{\mathrm{lat}}(p,q)$ is the sum of T_{I} and of finitely many elementary terms, to which Lem. 7.29 applies.

Step 1 (local terms at q). These are the first half of $\mathsf{T}_{4'}$ and the terms with q_Y in the first half of T_5, with q_Y in T_4 and with q_{q} in T_1, in the expansions of Lem. 7.29. By Lem. 7.31(b), and since $\mathsf{E}_p((x_q + y, \mu)) = K^{\rightarrow}_{\iota\mu}(u' + y)$ by translation invariance, their sum equals

$$\tfrac{\mathsf{n}}{2} \sum_{b,b'} \gamma_q(b,b')\, \mathsf{E}_p(b)\, \mathsf{E}_p(b') = \tfrac{\mathsf{n}}{2} \sum_{\mu,\nu} \sum_{y,y'} \gamma_{q;\mu\nu}(y,y')\, K^{\rightarrow}_{\iota\mu}(u' + y)\, K^{\rightarrow}_{\iota\nu}(u' + y').$$

The kernels $\gamma_{q;\mu\nu}$ have finite support, and their symbols vanish at $\kappa = 0$ together with their gradients (Lem. 7.31(b)); the $K^{\rightarrow}_{\iota\mu}$ are elementary lines of degree 2. Lem. 7.30(a), applied with $\gamma_{\mu\nu} := \gamma_{q;\mu\nu}$ and $K_\mu := K^{\rightarrow}_{\iota\mu}$, shows that the sum is $O(|u'|^{-6}) = O(r^{-6})$. By the translation invariance in Lem. 7.28, the kernels $\gamma_{q;\mu\nu}$ depend on q only through ι', so that only finitely many constants occur.

Step 2 (local terms at p). These are the second half of $\mathsf{T}_{4'}$ and the terms with q_Y in the second half of T_5, with q_X in T_4 and with q_{p} in T_1. We show that they are the local terms at the second plaquette of the ordered pair (q,p); by Step 1, applied to this pair, their sum is then $\frac{\mathsf{n}}{2} \sum_{b,b'} \gamma_p(b,b')\, \mathsf{E}_q(b)\, \mathsf{E}_q(b') = O(r^{-6})$. For the second halves of $\mathsf{T}_{4'}$ and T_5 this holds because they are the first halves for (q,p) (Lem. 7.28). The formulas for T_1 and T_4 in Lem. 7.26(c) are symmetric under $p \leftrightarrow q$.

An elementary term G of T_1 or T_4 is given by the monomials at P, P' and by the choice of the legs contracted with p, with q, and with one another; taken for the ordered pair (q,p), it becomes a term $\tilde{G}$ with $c_{\tilde{G}} = c_G$, and $G \mapsto \tilde{G}$ is a bijection $\mathfrak{G}_\bullet(\iota,\iota') \to \mathfrak{G}_\bullet(\iota',\iota)$. In $\tilde{G}$ the base points are counted from x_q, u' is replaced by $-u'$, and the contractions, which are symmetric in their two legs, are taken from the other end: for the kite

$$\tilde{K}_1 = K_4(-\cdot), \quad \tilde{K}_2 = K_5(-\cdot), \quad \tilde{K}_3 = K_3, \quad \tilde{K}_4 = K_1(-\cdot), \quad \tilde{K}_5 = K_2(-\cdot),$$

and for T_4 we have $\tilde{X} = Y(-\cdot)$, $\tilde{Y} = X(-\cdot)$. Since $K(-\cdot)$ has the symbol $\sigma(-k)$, the function $\hat{\mathsf{T}}_{\mathsf{q}}$ of $\tilde{G}$ is

$$\int_{\mathbb{T}} \sigma_1(-\kappa_1 + l)\, \sigma_2(-\kappa_2 - l)\, \sigma_3(l)\, d^\circ l = \hat{\mathsf{T}}_{\mathsf{p}}(-\kappa),$$

and the symbol of $\tilde{Y}$ is $\mathsf{I}_X(-k)$, after the substitution $l \to -l$. By the last assertion of Lem. 7.20(d), whose hypothesis holds here for all κ, the corresponding polynomials are $\mathsf{Q}_{\mathsf{p}}(-\kappa)$ and $\mathsf{Q}_X(-k)$. The passage from a polynomial Q to $\tilde{\mathsf{Q}}(\kappa) = \mathsf{Q}(0) + \sum_j \partial_j \mathsf{Q}(0) \sin \kappa_j$ in Prop. 7.23 and Lem. 7.22(c) commutes with the change of sign of the argument, the sine being odd. Hence the kernel q_{q} of $\tilde{G}$ is $(z,w) \mapsto \mathsf{q}_{\mathsf{p}}(-z,-w)$, and $\mathsf{q}_{\tilde{Y}} = \mathsf{q}_X(-\cdot)$, so that

$$\sum_{z,w} \mathsf{q}_{\mathsf{p}}(-z,-w)\, \tilde{K}_1(-u' - z)\, \tilde{K}_2(-u' - w) = \sum_{z,w} \mathsf{q}_{\mathsf{p}}(z,w)\, K_4(u' - z)\, K_5(u' - w),$$

$$(\mathsf{q}_{\tilde{Y}} * \tilde{X})(-u') = (\mathsf{q}_X * Y)(u').$$

These are the local terms at p in T_1 and in T_4.

Step 3 (the class T_{I}). By Lem. 7.31(a),

$$K^{\rightarrow} * \Sigma_{\mu\nu} * K^{\leftarrow} = K^{\rightarrow} * \mathsf{q}^{\Sigma}_{\mu\nu} * K^{\leftarrow} + K^{\rightarrow} * \mathsf{W}_{\mu\nu} * K^{\leftarrow}.$$

The first term is $O(r^{-3})$ by Lem. 7.30(b), since $\hat{\mathsf{q}}^{\Sigma}_{\mu\nu}(k) = O(|k|^2)$. The second lies in $\mathfrak{K}(2)$, by Lem. 7.22(b) applied to each of its two convolutions, and it equals its leading kernel, of degree -2, up to $O(r^{-3} \lg r)$; by the linearity of the leading data, this leading kernel is the sum over G, with the coefficients c_G, of the leading kernels of the chains in the definition of U_G in Lem. 7.35. Moreover

$\Pi(p,q) = \Pi^h(u') + O(r^{-4})$, with Π^h homogeneous of degree -3 (Lem. 7.24, Lem. 7.17(c)). Therefore

$$\mathsf{T}_{\mathrm{I}} = \sum_{(\mu,\nu,G)\in\mathfrak{G}_2\cup\mathfrak{G}_3} (-\mathsf{n}\, c_G)\, \mathsf{U}_G(u') + O(r^{-6}\lg r).$$

Step 4 (assembly). Let

$$\mathsf{U}_{\iota\iota'}(u) := \sum_{\Gamma\neq 5} \sum_{G\in\mathfrak{G}_\Gamma(\iota,\iota')} c_G\, \mathsf{U}_G(u) + \sum_{G\in\mathfrak{G}_5(\iota,\iota')} c_G\, \mathsf{U}_G(u) + \sum_{G\in\mathfrak{G}_5(\iota',\iota)} c_G\, \mathsf{U}_G(-u),$$

with the coefficients $-\mathsf{n}\, c_G$ for $\Gamma = 2,3$; the last sum comes from the second half of T_5, which is the first half for (q,p) with u' replaced by $-u'$. The function $\mathsf{U}_{\iota\iota'}$ is homogeneous of degree -5 and of class C^1 off 0, by Lem. 7.29 and since the leading kernels of the classes $\mathfrak{K}(n)$ are smooth off 0. By Steps 1–3 and Lem. 7.29, $\mathsf{B}_1^{\mathrm{lat}}(p,q) = \mathsf{U}_{\iota\iota'}(u') + O(r^{-6}\lg r)$. The gradient of $\mathsf{U}_{\iota\iota'}$ is homogeneous of degree -6, and all points $u' + t(u-u')$, $0\le t\le 1$, have modulus at least $r/2$; hence $|\mathsf{U}_{\iota\iota'}(u') - \mathsf{U}_{\iota\iota'}(u)| \le Cr^{-6}$.

With $\mathsf{U}_{ij} := \mathsf{U}_{\iota\iota'}$ for $(i,j) = (\iota,\iota') = (\lambda(p),\lambda(q))$ we obtain the bound in eq. 7.xxxvii; as $\tau_{\mathrm{gf}} = 1$, the finite sets $\mathfrak{G}_\bullet(\iota,\iota')$, the coefficients c_G and the lines are fixed once and for all, the constants C, r_0 are pure. Finally, exchanging ι and ι' in the last sum and using Lems. 7.35 and 7.36,

$$\sum_{\iota,\iota'} \mathsf{U}_{\iota\iota'} = \sum_{\Gamma=1}^{6} \mathsf{U}_\Gamma = \sum_{\Gamma=1}^{6} \kappa_\Gamma^{\mathrm{lat}}\, |u|^{-5}.$$

□

For the smeared observables of eq. 7.xix the theorem gives the following rate of convergence on test pairs. Here $\mathfrak{T}_{\mathbb{R}^3}(\mathsf{d};\vartheta)$ is the set of test pairs at the scale d with the shape parameter ϑ defined in §6.1, with $\mathbb{E} = \mathbb{R}^3$.

Corollary 7.37 (Rate of convergence of $\mathsf{P}_1^{\mathrm{lat}}$ to $\kappa^{\mathrm{lat}}\mathcal{K}_5$)**.** There is a pure constant C such that the following holds for all $\mathsf{d} > 0$, $\vartheta \in (0,1)$, $(f,h) \in \mathfrak{T}_{\mathbb{R}^3}(\mathsf{d};\vartheta)$ and $0 < a \le \mathsf{d}$. For the kernel of §6.1, at $q = 5$,

$$\mathcal{K}_5(f,h) = \iint f(x)\, h(y)\, |x-y|^{-5}\, \mathrm{d}x\, \mathrm{d}y,$$

the integrand being 0 where $f(x)h(y) = 0$. Then

$$|\mathsf{P}_1^{\mathrm{lat}}(f,h;a) - \kappa^{\mathrm{lat}}\, \mathcal{K}_5(f,h)| \le C\, \vartheta^{-1}\, a\, \lg(\mathsf{d}/a)\, \|f\|_\infty \|h\|_\infty. \tag{7.xxxviii}$$

■□

We recall three properties of a test pair $(f,h) \in \mathfrak{T}_{\mathbb{R}^3}(\mathsf{d};\vartheta)$: the supports of f and h lie in closed balls $\mathbb{B}(x_f,\mathsf{d})$ and $\mathbb{B}(x_h,\mathsf{d})$ of radius d; $\mathsf{d} \le |v-w| \le 10\,\mathsf{d}$ for $v \in \operatorname{supp} f$ and $w \in \operatorname{supp} h$ (the support separation in §6.1); and $\operatorname{Lip} f \le \|f\|_\infty/\vartheta\mathsf{d}$, $\operatorname{Lip} h \le \|h\|_\infty/\vartheta\mathsf{d}$ (by the definition of that test class).

Proof. Let $\varrho := \mathsf{d}/a \ge 1$, $\mathsf{A} := \|f\|_\infty\|h\|_\infty$ and $\varrho_0 := \max\{r_0, 8\}$, with r_0 of Thm. V. A ball of radius $\varrho + 2$ contains at most $C\varrho^3$ points of $\frac{1}{2}\mathbb{Z}^3$. The barycentres $x(p)$ lie in $\frac{1}{2}\mathbb{Z}^3$, distinct plaquettes have distinct barycentres, and $f(ax(p)) \neq 0$ implies $x(p) \in \mathbb{B}(x_f/a, \varrho)$; likewise for h.

Step 1 ($\varrho \le \varrho_0$). Then the sum defining $\mathsf{P}_1^{\mathrm{lat}}(f,h;a)$ has at most C non-zero terms, and since $\mathsf{B}_1^{\mathrm{lat}}$ is bounded (Thm. V), $|\mathsf{P}_1^{\mathrm{lat}}(f,h;a)| \le Ca\mathsf{A}$. Moreover $|x-y| \ge \mathsf{d}$ where $f(x)h(y) \neq 0$, whence $|\mathcal{K}_5(f,h)| \le (\frac{4\pi}{3}\mathsf{d}^3)^2\mathsf{d}^{-5}\mathsf{A} \le C\varrho_0 a\mathsf{A}$. As $\vartheta^{-1} > 1$ and $\lg(\mathsf{d}/a) \ge 1$, eq. 7.xxxviii holds in this case.

Step 2 ($\varrho > \varrho_0$, first error). If $f(ax(p))h(ax(q)) \neq 0$, then $r = |x(q) - x(p)| \in [\varrho, 10\varrho]$ by the separation of the supports; in particular $r \ge r_0$. There are at most $C\varrho^6$ such pairs, and $r^{-6}\lg r \le$

$\varrho^{-6}\lg(10\varrho)$ for these r. By eq. 7.xxxvii, the replacement of $\mathsf{B}_1^{\mathrm{lat}}(p,q)$ by $\mathsf{U}_{\lambda(p)\lambda(q)}(x(q)-x(p))$ in $\mathsf{P}_1^{\mathrm{lat}}(f,h;a)$ costs at most

$$a\cdot C\varrho^6\cdot C\varrho^{-6}\lg(10\varrho)\,\mathsf{A}\le Ca\lg\varrho\,\mathsf{A}.$$

Step 3 (second error). We write $p=p_{[i]}(x)$, $q=p_{[j]}(y)$, so that $x(p)=x+\mathsf{o}_i$ with $\mathsf{o}_i:=x(p_{[i]}(0))$, $|\mathsf{o}_i|=2^{-1/2}$; all terms below are set equal to 0 where the product of the values of f and h vanishes. After Step 2 we have to compare

$$\mathfrak{R}_1:=a\sum_{i,j}\sum_{x,y\in\mathbb{Z}^3} f(a(x+\mathsf{o}_i))\,h(a(y+\mathsf{o}_j))\,\mathsf{U}_{ij}(y-x+\mathsf{o}_j-\mathsf{o}_i)$$

with

$$\mathfrak{R}_2:=a\sum_{x,y} f(ax)\,h(ay)\sum_{i,j}\mathsf{U}_{ij}(y-x)=a\,\kappa^{\mathrm{lat}}\sum_{x,y}\frac{f(ax)\,h(ay)}{|y-x|^5},$$

the last equality by eq. 7.xxxvii. We replace f, h and U_{ij} one after the other. In every non-zero term, the arguments of U_{ij} which occur, and the points between them, have modulus at least $\varrho-2\ge\varrho/2$, and x,y lie in balls of radius $\varrho+1$. By homogeneity, $|\mathsf{U}_{ij}|\le C\varrho^{-5}$ and $|\nabla\mathsf{U}_{ij}|\le C\varrho^{-6}$ at these points, and the Lipschitz bound gives $|f(a(x+\mathsf{o}_i))-f(ax)|\le a\|f\|_\infty/\vartheta\mathsf{d}$, and likewise for h. Therefore

$$|\mathfrak{R}_1-\mathfrak{R}_2|\le a\,C\varrho^6\Big[\frac{a}{\vartheta\mathsf{d}}\,\varrho^{-5}+\varrho^{-6}\Big]\mathsf{A}\le Ca\vartheta^{-1}\mathsf{A}.$$

Step 4 (third error). Let $\mathsf{F}(v,w):=f(v)h(w)|v-w|^{-5}$, equal to 0 where $f(v)h(w)=0$. Then $\mathfrak{R}_2=\kappa^{\mathrm{lat}}\sum_{x,y}a^6\mathsf{F}(ax,ay)$ and $\mathcal{K}_5(f,h)=\iint\mathsf{F}$. Let $(v,w)\in(ax,ay)+[0,a)^6$. Unless F vanishes at both points (v,w) and (ax,ay), we have $|v-w|\ge\mathsf{d}-2\sqrt{3}a\ge\mathsf{d}/2$ at both points and at all points $(1-t)(ax,ay)+t(v,w)$, $0\le t\le1$ (recall $\varrho>8$), and the Lipschitz bounds for f,h together with the bound $5|z|^{-6}$ for the gradient of $|z|^{-5}$ give

$$|\mathsf{F}(v,w)-\mathsf{F}(ax,ay)|\le\sqrt{3}a\Big[\frac{2}{\vartheta\mathsf{d}}\Big(\frac{\mathsf{d}}{2}\Big)^{-5}+10\Big(\frac{\mathsf{d}}{2}\Big)^{-6}\Big]\mathsf{A}.$$

At most $C\varrho^6$ of the cubes $(ax,ay)+[0,a)^6$, of volume a^6, contain points where $\mathsf{F}\ne0$. Hence

$$|\mathfrak{R}_2-\kappa^{\mathrm{lat}}\mathcal{K}_5(f,h)|\le C\varrho^6a^6\cdot a\vartheta^{-1}\mathsf{d}^{-6}\mathsf{A}=Ca\vartheta^{-1}\mathsf{A}.$$

Adding the bounds of Steps 2–4 and using $\vartheta^{-1}>1$, $\lg\varrho\ge1$, we obtain eq. 7.xxxviii. □

7.1.13. *The boxes exhaust the lattice.* We now show that the coefficients $\mathsf{B}_j^{[n]}$ of the boxes converge to the coefficients $\mathsf{B}_j^{\mathrm{lat}}$ of the lattice $\mathbb{Z}^3$; together with Prop. 7.12, this shows that $\mathsf{B}_1^{\mathrm{lat}}$ is independent of the gauge parameter. The kernels of a box are expressed by the method of images through kernels on a discrete torus, which we first bound uniformly in the size of the torus.

Lemma 7.38 (Kernels on discrete tori, uniformly)**.** For an integer $n\ge2$ let $\mathbb{T}_n:=\frac{\pi}{n}\mathbb{Z}^3\cap(-\pi,\pi]^3$; for $z\in\mathbb{Z}^3$ let $|z|_n$ be the least norm of the points of $z+2n\mathbb{Z}^3$, and $\langle z\rangle_n:=\max\{1,|z|_n\}$. For real $s>-3$ and $\sigma\in\mathcal{S}^s$ let

$$K^{(n)}(z):=(2n)^{-3}\sum_{k\in\mathbb{T}_n\setminus\{0\}}e^{ik\cdot z}\,\sigma(k)\qquad(z\in\mathbb{Z}^3).$$

(a) $|K^{(n)}(z)|\le C\langle z\rangle_n^{-3-s}$ for all $n\ge2$ and $z\in\mathbb{Z}^3$, where C depends on s and on finitely many of the suprema in Def. 7.16(2), but not on n and z.

(b) For every $z\in\mathbb{Z}^3$, $K^{(n)}(z)\to K(z)=\int_{\mathbb{T}}e^{ik\cdot z}\sigma(k)\,d^\circ k$ as $n\to\infty$. ■

Proof. Let $\mathsf{h}:=\pi/n$, so that $(2n)^{-3}=(2\pi)^{-3}\mathsf{h}^3$.

Step 0. If $0 \neq k \in \mathsf{h}\mathbb{Z}^3$ and k' lies in the cube of side h centred at k, then $(1-\frac{\sqrt{3}}{2})|k| \le |k'| \le (1+\frac{\sqrt{3}}{2})|k|$, as $|k'-k| \le \frac{\sqrt{3}}{2}\mathsf{h} \le \frac{\sqrt{3}}{2}|k|$. Comparing every term with the integral over its cube, we obtain

$$(2n)^{-3} \sum_{k\in\mathsf{h}\mathbb{Z}^3,\ 0<|k|\le\varrho} |k|^t \le C\varrho^{3+t} \quad (t>-3),$$
$$(2n)^{-3} \sum_{k\in\mathsf{h}\mathbb{Z}^3,\ |k|\ge\varrho} |k|^t \le C\varrho^{3+t} \quad (t<-3),$$

with C depending on t only.

Step 1 (proof of (b)). Let $0<\varepsilon<1$, and split $\sigma = \chi(k/\varepsilon)\sigma + (1-\chi(k/\varepsilon))\sigma$ on $[-\pi,\pi]^3$. The first summand is bounded by $C|k|^s$ and vanishes for $|k| \ge 2\varepsilon$; by Step 0 its contributions to $K^{(n)}(z)$ and to $K(z)$ are $O(\varepsilon^{3+s})$, uniformly in n. The second summand, set equal to 0 at $k=0$, is continuous on $[-\pi,\pi]^3$; its contribution to $K^{(n)}(z)$ is a Riemann sum over the $(2n)^3$ subcubes of side h, which converges to its contribution to $K(z)$. Letting first $n\to\infty$ and then $\varepsilon\to 0$ we obtain (b).

Step 2 (proof of (a)). Since $K^{(n)}$ is $2n$-periodic in each coordinate, we may take the representative of z with $|z| = |z|_n$; then $|z_j| \le n$ for all j. For $z=0$ the bound follows from Step 0. Let $|z| \ge 1$, and write $\sigma = \sigma' + \sigma''$ with $\sigma' := \chi(|z|k)\sigma$ on $[-\pi,\pi]^3$. As σ' is bounded by $C|k|^s$ and vanishes for $|k| \ge 2/|z|$, Step 0 shows that the contribution of σ' is at most $C|z|^{-3-s}$. The function σ'' is smooth and periodic, it vanishes where $|k|_{\mathbb{T}} \le 1/|z|$, and $|\partial^\alpha \sigma''(k)| \le C|k|_{\mathbb{T}}^{s-|\alpha|}$, because every derivative falling on $\chi(|z|k)$ produces a factor $|z| \le 2/|k|$ on its support. Since $\mathbb{T}_n$ represents $\mathsf{h}\mathbb{Z}^3/2\pi\mathbb{Z}^3$, we have $\sum_k e^{ik\cdot z}\sigma''(k-\mathsf{h}e_j) = e^{i\mathsf{h}z_j}\sum_k e^{ik\cdot z}\sigma''(k)$; with $(\delta_j \mathsf{F})(k) := \mathsf{F}(k) - \mathsf{F}(k-\mathsf{h}e_j)$ it follows that

$$(1-e^{i\mathsf{h}z_j})^N \sum_{k\in\mathbb{T}_n} e^{ik\cdot z}\,\sigma''(k) = \sum_{k\in\mathbb{T}_n} e^{ik\cdot z}\,\delta_j^N\sigma''(k)$$

for every integer $N \ge 1$; the term $k=0$ vanishes, as $\sigma''(0)=0$. Here $|1-e^{i\mathsf{h}z_j}| = 2|\sin\frac{\pi z_j}{2n}| \ge 2|z_j|/n$, since $\sin t \ge \frac{2}{\pi}t$ on $[0,\frac{\pi}{2}]$ and $|z_j| \le n$; moreover $|\delta_j^N\sigma''(k)| \le \mathsf{h}^N \sup_{0\le t\le N\mathsf{h}} |\partial_j^N \sigma''(k-te_j)|$. We fix an integer $N > s+3$; as $|z| \le \sqrt{3}n$, we have $N\mathsf{h} \le C/|z|$. For $|k| \ge 2N\mathsf{h}$ the points $k-te_j$, $0\le t\le N\mathsf{h}$, satisfy $|k-te_j|_{\mathbb{T}} \in [\frac{1}{2}|k|, \frac{3}{2}|k|]$; they may leave $[-\pi,\pi]^3$, but their distance from $2\pi\mathbb{Z}^3$ equals $|k|$ at $t=0$ and changes by at most $t \le \frac{1}{2}|k|$. Hence for these k the supremum is at most $C|k|^{s-N}$, and it vanishes unless $|k| \ge \frac{2}{3|z|}$. For $|k| < 2N\mathsf{h}$ the supremum is at most $C|z|^{N-s}$, as σ'' vanishes where $|k|_{\mathbb{T}} \le 1/|z|$ and $s-N<0$; there are at most C such points k, and $(2n)^{-3} \le C|z|^{-3}$. Together with Step 0 we find

$$(2n)^{-3} \sum_{k\in\mathbb{T}_n} |\delta_j^N\sigma''(k)| \le C\mathsf{h}^N|z|^{N-s-3}.$$

Choosing j with $|z_j| \ge |z|/\sqrt{3}$, we conclude that the contribution of σ'' is at most

$$(n/2|z_j|)^N(\pi/n)^N\, C|z|^{N-s-3} \le C|z|^{-3-s}.$$

This proves (a). □

Proposition 7.39 (The boxes exhaust the lattice)**.** Let $\tau_{\mathrm{gf}} > 0$, let p, q be plaquettes of $\mathbb{Z}^3$ and $j \in \{0,1\}$, and write $\mathsf{B}_0^\Omega(p,q;\tau_{\mathrm{gf}}) := \mathsf{B}_0^\Omega(p,q)$. The numbers $\mathsf{B}_j^{[n]}(p+o_n, q+o_n;\tau_{\mathrm{gf}})$ are defined for all $n \ge 2$ with $p+o_n, q+o_n \in \mathsf{Pl}_n$, that is, for all but finitely many n, and

$$\lim_{n\to\infty} \mathsf{B}_j^{[n]}(p+o_n, q+o_n;\tau_{\mathrm{gf}}) = \mathsf{B}_j^{\mathrm{lat}}(p,q;\tau_{\mathrm{gf}}). \tag{7.xxxix}$$

■

Proof. We use the method of images. Fix n and write $\tau := \tau_{\mathrm{gf}}$.

Step 1 (reflections). We regard forms and functions on the torus $(\mathbb{Z}/2n)^3$ as $2ne_\nu$-periodic forms and functions on $\mathbb{Z}^3$, and we write $(\cdot,\cdot)_{\mathsf{t}}$ for the sums over one period. Let ϱ_ν be the reflection

$z_\nu \to -z_\nu$, acting on bonds by $\varrho_\nu\langle z,z'\rangle = \langle \varrho_\nu z, \varrho_\nu z'\rangle$, and put $(\varrho_\nu^* \tilde A)(b) := \tilde A(\varrho_\nu b)$ and $\varrho_\nu^*\lambda := \lambda\circ\varrho_\nu$; these maps commute with d and d^*. Let F^{as} and X^{as} be the spaces of real periodic forms $\tilde A$ and functions λ with $\varrho_\nu^*\tilde A = -\tilde A$ and $\varrho_\nu^*\lambda = -\lambda$ for all ν; in components, $\tilde A_\mu(\varrho_\nu z) = -\tilde A_\mu(z)$ for $\nu\neq\mu$, and $\tilde A_\mu(\varrho_\mu z - e_\mu) = \tilde A_\mu(z)$.

We claim that restriction to the box is a bijection of F^{as} onto $\mathsf{Form}_n^{\mathbb{R}}$ and of X^{as} onto $\mathsf{X}_n^{\mathbb{R}}$. If $z_\nu\in\{0,n\}$, then $\varrho_\nu z \equiv z$ modulo $2n$, hence $\tilde A_\mu(z)=0$ for $\nu\neq\mu$, and $\lambda(z)=0$; by Lem. 7.6(a), $\tilde A$ vanishes on the non-free bonds of the box, and λ vanishes on $\partial\Lambda_n$. For $\varepsilon\in\{0,1\}^3$ let $R_\varepsilon := \prod_\nu \varrho_\nu^{\varepsilon_\nu}$, $|\varepsilon| := \sum_\nu \varepsilon_\nu$, and $R_\varepsilon(z,\mu) := (R_\varepsilon z - \varepsilon_\mu e_\mu, \mu)$, the positively oriented one among the image bond and its reverse.

Every positively oriented bond of the torus has the form $R_\varepsilon(x,\mu)$ with $x_\mu\in\{0,\dots,n-1\}$ and $x_\nu\in\{0,\dots,n\}$ for $\nu\neq\mu$; here (x,μ) is unique, and ε is unique if (x,μ) is free. Hence every $A\in\mathsf{Form}_n^{\mathbb{R}}$ has the unique continuation $\tilde A(R_\varepsilon b) = (-1)^{|\varepsilon|+\varepsilon_\mu}A(b)$, $b=(x,\mu)$, and every $\lambda\in\mathsf{X}_n^{\mathbb{R}}$ the unique continuation $\tilde\lambda(R_\varepsilon x) = (-1)^{|\varepsilon|}\lambda(x)$.

Step 2 (operators). The eight boxes $R_\varepsilon[0,n]^3$ cover the torus, and a plaquette lying in two of them lies in a hyperplane $\{z_\nu=0\}$ or $\{z_\nu=n\}$, on whose bonds $\tilde A=0$. The functions $|\mathrm{d}\tilde A|$ and $|\mathrm{d}^*\tilde A|$ are invariant under the ϱ_ν, and $\mathrm{d}^*\tilde A\in\mathsf{X}^{\mathrm{as}}$ vanishes on these hyperplanes. Hence

$$(\tilde A,\tilde A')_{\mathsf{t}} = 8(A,A'),\qquad \|\mathrm{d}\tilde A\|_{\mathsf{t}}^2 = 8\|\mathrm{d}A\|^2,\qquad \|\mathrm{d}^*\tilde A\|_{\mathsf{t}}^2 = 8\|\mathrm{d}^*A\|^2.$$

The operator $\mathsf{L}_\tau := \mathrm{d}^*\mathrm{d} + \tau^{-1}\mathrm{d}\mathrm{d}^*$ on the torus commutes with the ϱ_ν^* and preserves F^{as}. By polarisation, $(\tilde A',\mathsf{L}_\tau\tilde A)_{\mathsf{t}} = 8(A',\mathsf{L}_\tau^{[n]}A)$ for all $A,A'\in\mathsf{Form}_n^{\mathbb{R}}$, where $\mathsf{L}_\tau^{[n]}$ is the operator on $\mathsf{Form}_n^{\mathbb{R}}$ of the quadratic form in Def. 7.7(1); hence $\mathsf{L}_\tau\tilde A$ is the continuation of $\mathsf{L}_\tau^{[n]}A$. The forms in F^{as} have mean 0, and on periodic forms of mean 0 the inverse of L_τ is, by the finite Fourier inversion formula, the convolution with

$$\mathsf{C}_{\mu\nu}^{(n)}(z) := (2n)^{-3}\sum_{k\in\mathbb{T}_n\setminus\{0\}} e^{ik\cdot z}\,\hat{\mathsf{C}}_{\mu\nu}(k).$$

The continuation of $\delta_{b'}$, for a free bond $b'=(x',\mu')$, is $\sum_\varepsilon(-1)^{|\varepsilon|+\varepsilon_{\mu'}}\delta_{R_\varepsilon b'}$. For functions we have in the same way $(\tilde\lambda,\tilde\lambda')_{\mathsf{t}} = 8(\lambda,\lambda')$ and $\|\mathrm{d}\tilde\lambda\|_{\mathsf{t}}^2 = 8\|\mathrm{d}\lambda\|^2$, since $\mathrm{d}\tilde\lambda\in\mathsf{F}^{\mathrm{as}}$ is the continuation of $\mathrm{d}\lambda$; the elements of X^{as} have mean 0, and the continuation of $\delta_{x'}$, $x'\in\Lambda_n^\circ$, is $\sum_\varepsilon(-1)^{|\varepsilon|}\delta_{R_\varepsilon x'}$. By polarisation, as for forms, $\mathrm{d}^*\mathrm{d}\tilde\lambda$ is the continuation of $\mathrm{d}^*\mathrm{d}\lambda$, and on periodic functions of mean 0 the inverse of $\mathrm{d}^*\mathrm{d}$ is the convolution with $\mathsf{G}^{(n)}(z) := (2n)^{-3}\sum_{k\in\mathbb{T}_n\setminus\{0\}} e^{ik\cdot z}|\mathsf{k}(k)|^{-2}$.

We apply this to $B := (\mathsf{L}_\tau^{[n]})^{-1}\delta_{b'}$, whose value at a free bond b is $\mathsf{C}_\tau^{[n]}(b,b')$; its continuation $\tilde B$ has mean 0 and satisfies $\mathsf{L}_\tau\tilde B = \tilde\delta_{b'}$, the continuation of $\delta_{b'}$, so that $\tilde B$ is the convolution of $\mathsf{C}^{(n)}$ with $\tilde\delta_{b'}$. In the same way we treat $(\mathrm{d}^*\mathrm{d})^{-1}\delta_{x'}$ on $\mathsf{X}_n^{\mathbb{R}}$, whose value at $x\in\Lambda_n^\circ$ is $\mathsf{G}^{[n]}(x,x')$. Consequently, for free bonds $(x,\mu),(x',\mu')$ and for $x,x'\in\Lambda_n^\circ$,

$$\begin{aligned}\mathsf{C}_\tau^{[n]}((x,\mu),(x',\mu')) &= \sum_\varepsilon (-1)^{|\varepsilon|+\varepsilon_{\mu'}}\,\mathsf{C}_{\mu\mu'}^{(n)}(x - R_\varepsilon x' + \varepsilon_{\mu'}e_{\mu'}),\\ \mathsf{G}^{[n]}(x,x') &= \sum_\varepsilon (-1)^{|\varepsilon|}\,\mathsf{G}^{(n)}(x - R_\varepsilon x').\end{aligned}\tag{7.xl}$$

In the first argument the right sides are values of elements of F^{as}, resp. X^{as}, and therefore vanish at non-free bonds, resp. on $\partial\Lambda_n$; in the second argument the same follows by symmetry. Since the box kernels vanish there by definition, eq. 7.xl holds for all bonds and sites of Λ_n.

Step 3 (uniform bounds). The kernels $\mathsf{E}^{[n]},\Pi^{[n]},\mathsf{D}^{[n]},\mathsf{g}^{[n]}$ arise from eq. 7.xl by the differences and means of Def. 7.7(2). A difference in the first argument acts on $\mathsf{C}^{(n)}$ and $\mathsf{G}^{(n)}$. In the second argument, R_ε maps the four bonds of q onto those of $R_\varepsilon q$, and $\sum_j s_j(-1)^{\varepsilon_{\mu_j}}\delta_{R_\varepsilon b_j(q)} = \pm\sum_j s_j\delta_{b_j(R_\varepsilon q)}$, with μ_j the direction of $b_j(q)$, since d commutes with the ϱ_ν^*; the same holds for the end points of a bond. Hence each of these kernels is a sum of eight terms $\pm K^{(n)}(x - R_\varepsilon x' + y_0)$ with $|y_0|\le 3$, where x and x' are the base points of its two arguments, as in eq. 7.xxxi (the base point b_- of a bond b, the base

point x_p of a plaquette p; for $\mathsf{g}^{[n]}(b)$ both are b_-), and where $K^{(n)}$ is built as in Lem. 7.38 with the symbol of the corresponding line of Lem. 7.24, an element of $\mathcal{S}^{n_\ell-3}$, n_ℓ being the degree of the line.

For $x, x' \in \Lambda_n$ we have $|x - R_\varepsilon x'|_n \geq |x - x'|$: if $\varepsilon_\nu = 0$, then $|x_\nu - x'_\nu| \leq n$, and if $\varepsilon_\nu = 1$, then $\min\{x_\nu + x'_\nu, 2n - x_\nu - x'_\nu\} \geq |x_\nu - x'_\nu|$. By Lem. 7.38(a), the kernels of all boxes therefore satisfy the bounds eq. 7.xxxi with one constant $\mathsf{c}_0 = \mathsf{c}_0(\tau_{\mathrm{gf}})$.

Step 4 (pointwise limits). The arguments are translates by o_n of fixed bonds, sites and plaquettes. For $\varepsilon = 0$ the term $K^{(n)}(x - x' + y_0)$ has a fixed argument and converges to the corresponding kernel of $\mathbb{Z}^3$, by Lem. 7.38(b) and Lem. 7.24. If $\varepsilon_\nu = 1$, the ν-th coordinate of the argument is $\tilde{x}_\nu + \tilde{x}'_\nu + 2\lfloor n/2 \rfloor + O(1)$, where $\tilde{x} = x - o_n$ and $\tilde{x}' = x' - o_n$ are fixed; its distance from $2n\mathbb{Z}$ is therefore at least $n - C$, and the term tends to 0 by Lem. 7.38(a).

Step 5 (conclusion). For $j = 0$, $\mathsf{B}_0^{[n]} = \frac{\mathsf{n}}{2}(\Pi^{[n]})^2$ by eq. 7.xvii, which converges by Step 4 to $\frac{\mathsf{n}}{2}(\Pi^{\mathrm{lat}})^2 = \mathsf{B}_0^{\mathrm{lat}}$. For $j = 1$, the formulas of Lem. 7.26(c) in the box, which are finite sums, express $\mathsf{B}_1^{[n]}(p + o_n, q + o_n; \tau_{\mathrm{gf}})$ as a sum, over monomials, pairings and positions, of products of box kernels. We write the positions in the box as translates $P + o_n$, $P' + o_n$, $b + o_n, \dots$ of plaquettes P, P' and bonds $b, \dots$ of $\mathbb{Z}^3$, and we index the terms by the monomial, the pairing and the positions $P, P', b, \dots$ in $\mathbb{Z}^3$; this set of indices does not depend on n, and a term whose positions do not all lie in the box is set equal to 0.

For fixed $P, P', b, \dots$ all positions lie in the box for every large n, and by Step 4 the term converges, as $n \to \infty$, to the term of the formulas of Lem. 7.26(c) on $\mathbb{Z}^3$ with the same monomial, the same pairing and the positions $P, P', b, \dots$. By Step 3, the box kernels satisfy eq. 7.xxxi with one constant c_0; these bounds depend only on differences of base points, which the translation by o_n does not change, and the uniform form of Lem. 7.27, valid for any kernels with these bounds, therefore dominates the moduli of the terms, uniformly in n, by a family of majorants over the same set of indices whose sum is at most $C\mathsf{c}_0^5$.

By dominated convergence, $\mathsf{B}_1^{[n]}(p + o_n, q + o_n; \tau_{\mathrm{gf}})$ tends to the sum of the terms on $\mathbb{Z}^3$, which converges absolutely and equals $\mathsf{B}_1^{\mathrm{lat}}(p, q; \tau_{\mathrm{gf}})$ by Lem. 7.27. This is eq. 7.xxxix. □

Corollary 7.40 (Independence of the gauge parameter)**.** For every $\tau_{\mathrm{gf}} > 0$ and all plaquettes p, q of $\mathbb{Z}^3$,

$$\mathsf{B}_1^{\mathrm{lat}}(p, q; \tau_{\mathrm{gf}}) = \mathsf{B}_1^{\mathrm{lat}}(p, q).$$

■□

Proof. By Prop. 7.12, $\mathsf{B}_1^{[n]}(p+o_n, q+o_n; \tau_{\mathrm{gf}}) = \mathsf{B}_1^{[n]}(p+o_n, q+o_n; 1)$ for every n with $p+o_n, q+o_n \in \mathsf{Pl}_n$. Letting $n \to \infty$, Prop. 7.39 gives $\mathsf{B}_1^{\mathrm{lat}}(p, q; \tau_{\mathrm{gf}}) = \mathsf{B}_1^{\mathrm{lat}}(p, q; 1)$, and the right side is $\mathsf{B}_1^{\mathrm{lat}}(p, q)$ by eq. 7.xix. □

7.1.14. *The value: Fourier constants, coefficient of* $\mathsf{p}^2 \log|\mathsf{p}|$. To compute κ^{lat} we relate each $\kappa_\Gamma^{\mathrm{lat}}$ to the coefficient of $\mathsf{p}^2 \log|\mathsf{p}|$ in the expansion of Ψ_Γ at small p. This requires two Fourier constants and the corresponding expansions of two lattice sums. In this subsection $(\mathcal{F}^{-1}\phi)(p) = \int e^{ip\cdot x}\phi(x)\, d^\circ x$ for a function ϕ of $x \in \mathbb{R}^3$.

Lemma 7.41 (Two Fourier constants)**.** Let $\phi \in \mathcal{S}(\mathbb{R}^3)$ vanish near 0.

(a) For $s \in \mathbb{C}$ with $\operatorname{Re} s > -3$,

$$\int_{\mathbb{R}^3} |p|^s\, (\mathcal{F}^{-1}\phi)(p)\, \mathrm{d}p = c(s) \int_{\mathbb{R}^3} |x|^{-3-s}\, \phi(x)\, \mathrm{d}x,$$
$$c(s) := \frac{2^s\, \Gamma_{\mathrm{E}}(\frac{3+s}{2})}{\pi^{3/2}\, \Gamma_{\mathrm{E}}(-\frac{s}{2})}, \tag{7.xli}$$

where $1/\Gamma_{\mathrm{E}}$ is entire. Moreover $c(s) = 0$ for $s = 0, 2, 4, \dots$, and $c(1) = -\pi^{-2}$.

(b) We have

$$\int_{\mathbb{R}^3} p^2 \log|p|\,(\mathcal{F}^{-1}\phi)(p)\,\mathrm{d}p = c'(2)\int_{\mathbb{R}^3}|x|^{-5}\,\phi(x)\,\mathrm{d}x, \qquad c'(2)=\frac{3}{2\pi}. \tag{7.xlii}$$

■

Proof. Step 1. Let $\psi := \mathcal{F}^{-1}\phi \in \mathcal{S}(\mathbb{R}^3)$, and assume first $-3 < \operatorname{Re} s < 0$. The substitution $t \to t/p^2$ in Euler's integral gives $\int_0^\infty t^{-s/2-1}e^{-tp^2}\,\mathrm{d}t = \Gamma_{\mathrm{E}}(-\frac{s}{2})\,|p|^s$ for $p \neq 0$. Since

$$\iint t^{-\operatorname{Re} s/2-1}e^{-tp^2}\,|\psi(p)|\,\mathrm{d}t\,\mathrm{d}p = \Gamma_{\mathrm{E}}(-\tfrac{1}{2}\operatorname{Re} s)\int |p|^{\operatorname{Re} s}|\psi(p)|\,\mathrm{d}p < \infty,$$

Fubini's theorem yields

$$\int |p|^s\psi(p)\,\mathrm{d}p = \Gamma_{\mathrm{E}}(-\tfrac{s}{2})^{-1}\int_0^\infty t^{-s/2-1}\Big[\int e^{-tp^2}\psi(p)\,\mathrm{d}p\Big]\,\mathrm{d}t.$$

By Fubini's theorem and the Gaussian integral $\int_{\mathbb{R}^3} e^{-tp^2+ip\cdot x}\,\mathrm{d}p = (\pi/t)^{3/2}e^{-x^2/4t}$,

$$\int e^{-tp^2}\psi(p)\,\mathrm{d}p = \int \phi(x)\Big[\int e^{-tp^2+ip\cdot x}\,d^\circ p\Big]\,\mathrm{d}x = \int (4\pi t)^{-3/2}e^{-x^2/4t}\,\phi(x)\,\mathrm{d}x.$$

For $x \neq 0$ the substitution $t \to x^2/4t$ gives

$$\int_0^\infty t^{-(s+3)/2-1}e^{-x^2/4t}\,\mathrm{d}t = \Gamma_{\mathrm{E}}\big(\tfrac{s+3}{2}\big)\Big(\frac{x^2}{4}\Big)^{-(s+3)/2},$$

as $\operatorname{Re} s + 3 > 0$. The same identity with $\operatorname{Re} s$ in place of s,

$$\int_0^\infty t^{-(\operatorname{Re} s+3)/2-1}e^{-x^2/4t}\,\mathrm{d}t = \Gamma_{\mathrm{E}}\big(\tfrac{\operatorname{Re} s+3}{2}\big)\Big(\frac{x^2}{4}\Big)^{-(\operatorname{Re} s+3)/2},$$

together with $\int |x|^{-\operatorname{Re} s-3}|\phi(x)|\,\mathrm{d}x < \infty$, which holds as ϕ vanishes near 0 and lies in $\mathcal{S}(\mathbb{R}^3)$, shows that the modulus $(4\pi)^{-3/2}t^{-(\operatorname{Re} s+3)/2-1}e^{-x^2/4t}|\phi(x)|$ of the integrand $t^{-s/2-1}(4\pi t)^{-3/2}e^{-x^2/4t}\phi(x)$ is integrable over $(0,\infty)\times\mathbb{R}^3$. Hence Fubini's theorem applies once more, and

$$\int |p|^s\psi(p)\,\mathrm{d}p = \Gamma_{\mathrm{E}}(-\tfrac{s}{2})^{-1}(4\pi)^{-3/2}\,2^{s+3}\,\Gamma_{\mathrm{E}}\big(\tfrac{s+3}{2}\big)\int |x|^{-s-3}\phi(x)\,\mathrm{d}x,$$

with $(4\pi)^{-3/2}2^{s+3} = 2^s\pi^{-3/2}$. This is eq. 7.xli for $-3 < \operatorname{Re} s < 0$.

Step 2. The left side of eq. 7.xli is holomorphic in s on $\{\operatorname{Re} s > -3\}$, since $|p|^{\operatorname{Re} s}|\psi|$ provides integrable majorants locally uniformly in s; the function $s \mapsto \int |x|^{-3-s}\phi(x)\,\mathrm{d}x$ is entire, since ϕ vanishes near 0 and lies in $\mathcal{S}(\mathbb{R}^3)$; and c is holomorphic on $\{\operatorname{Re} s > -3\}$, since Γ_{E} is holomorphic on the right half plane and $1/\Gamma_{\mathrm{E}}$ is entire. Both sides agree for $-3 < \operatorname{Re} s < 0$, hence on the connected set $\{\operatorname{Re} s > -3\}$ by the identity theorem. The entire functions $1/\Gamma_{\mathrm{E}}(z)$ and $z/\Gamma_{\mathrm{E}}(z+1)$ agree for $\operatorname{Re} z > 0$, hence for all z; therefore $1/\Gamma_{\mathrm{E}}$ vanishes at 0, and then successively at $-1, -2, \ldots$, which gives $c(s) = 0$ for $s = 0, 2, 4, \ldots$.

Moreover $1/\Gamma_{\mathrm{E}}(-\frac{1}{2}) = -\frac{1}{2}/\Gamma_{\mathrm{E}}(\frac{1}{2}) = -\frac{1}{2\sqrt{\pi}}$, whence $c(1) = \frac{2\,\Gamma_{\mathrm{E}}(2)}{\pi^{3/2}}\cdot(-\frac{1}{2\sqrt{\pi}}) = -\pi^{-2}$.

Step 3. For (b) we differentiate eq. 7.xli with respect to s at $s = 2$. On the left we obtain $\int p^2\log|p|\,\psi(p)\,\mathrm{d}p$, by differentiation under the integral; on the right $c'(2)\int|x|^{-5}\phi - c(2)\int|x|^{-5}\log|x|\,\phi$, where $c(2) = 0$. From $1/\Gamma_{\mathrm{E}}(z) = z(z+1)/\Gamma_{\mathrm{E}}(z+2)$ the derivative of $1/\Gamma_{\mathrm{E}}$ at $z = -1$ is $(2z+1)/\Gamma_{\mathrm{E}}(z+2)|_{z=-1} = -1$; with $z = -\frac{s}{2}$ this gives $\partial_s\big[1/\Gamma_{\mathrm{E}}(-\frac{s}{2})\big]_{s=2} = \frac{1}{2}$. As $1/\Gamma_{\mathrm{E}}(-1) = 0$, only this derivative contributes, and $c'(2) = 2^2\,\Gamma_{\mathrm{E}}(\frac{5}{2})\,\pi^{-3/2}\cdot\frac{1}{2}$ with $\Gamma_{\mathrm{E}}(\frac{5}{2}) = \frac{3}{2}\cdot\frac{1}{2}\sqrt{\pi} = \frac{3}{4}\sqrt{\pi}$; thus $c'(2) = \frac{3}{2\pi}$. □

Lemma 7.42 (Two Fourier series: Epstein-type sums near $\mathsf{p} = 0$)**.** For $n \in \{4, 5\}$ and $\mathsf{p} \in \mathbb{R}^3$ let $\mathsf{s}_n(\mathsf{p}) := \sum_{u\in\mathbb{Z}^3\setminus\{0\}}|u|^{-n}e^{-i\mathsf{p}\cdot u}$. There are functions $\mathsf{e}_4, \mathsf{e}_5 \in C^\infty(\{|\mathsf{p}| < 1\})$, invariant under the 48 signed permutations of the coordinates, such that

$$\mathsf{s}_4(\mathsf{p}) = -\pi^2|\mathsf{p}| + \mathsf{e}_4(\mathsf{p}), \qquad \mathsf{s}_5(\mathsf{p}) = \tfrac{2\pi}{3}\,\mathsf{p}^2\log|\mathsf{p}| + \mathsf{e}_5(\mathsf{p}) \qquad (0 < |\mathsf{p}| < 1).$$

■

Proof. We treat the two cases $(\varpi, n, \mathsf{c}) = (|p|, 4, -\pi^{-2})$ and $(\varpi, n, \mathsf{c}) = (p^2 \log|p|, 5, \frac{3}{2\pi})$ together. The function ϖ is continuous, radial and smooth off 0, and its derivatives of order j are $O(|p|^{2-j} \lg|p|)$ at infinity. For an integer $\mathsf{N} \geq 3$ let

$$\mathsf{K}(x) := \int e^{ip\cdot x} \chi(p)\, \varpi(p)\, d^\circ p, \qquad \mathsf{E}(x) := |x|^{-2\mathsf{N}} \int e^{ip\cdot x} (-\Delta_p)^{\mathsf{N}} [(1-\chi)\varpi](p)\, d^\circ p \quad (x \neq 0).$$

Step 1. The integrand of E vanishes for $|p| \leq 1$ and is $O(|p|^{2-2\mathsf{N}} \lg|p|)$, with $2\mathsf{N} - 2 > 3$; hence the integral converges absolutely. The value of $\mathsf{E}(x)$ does not depend on N: we write $|x|^2 e^{ip\cdot x} = -\Delta_p e^{ip\cdot x}$ and integrate by parts twice over balls (Gauss' theorem); the boundary terms on the sphere of radius r are $O((1+|x|)\mathsf{r}^2 \cdot \mathsf{r}^{2-2\mathsf{N}} \lg \mathsf{r})$, which tends to 0 as $\mathsf{r} \to \infty$ because $\mathsf{N} \geq 3$; hence N and $\mathsf{N}+1$ give the same value. Consequently E is continuous off 0 and radial, and $|\mathsf{E}(x)| \leq C_{\mathsf{N}} |x|^{-2\mathsf{N}}$ for every $\mathsf{N} \geq 3$.

Step 2. We show $\mathsf{K} + \mathsf{E} = \mathsf{c}\,|x|^{-n}$ on $\mathbb{R}^3 \backslash \{0\}$. Let $\phi \in C_c^\infty(\mathbb{R}^3 \backslash \{0\})$. By Fubini's theorem, $\int \mathsf{K}\phi\, dx = \int \chi\varpi\, \mathcal{F}^{-1}\phi\, dp$. Let $\phi_{\mathsf{N}} := |x|^{-2\mathsf{N}}\phi \in C_c^\infty(\mathbb{R}^3 \setminus \{0\})$. By Fubini's theorem and $2\mathsf{N}$ integrations by parts, which are permitted as $\mathcal{F}^{-1}\phi_{\mathsf{N}} \in \mathcal{S}(\mathbb{R}^3)$, and by $(-\Delta_p)^{\mathsf{N}} \mathcal{F}^{-1}\phi_{\mathsf{N}} = \mathcal{F}^{-1}(|x|^{2\mathsf{N}}\phi_{\mathsf{N}}) = \mathcal{F}^{-1}\phi$,

$$\int \mathsf{E}\phi\, dx = \int (-\Delta_p)^{\mathsf{N}} [(1-\chi)\varpi]\, \mathcal{F}^{-1}\phi_{\mathsf{N}}\, dp = \int (1-\chi)\varpi\, \mathcal{F}^{-1}\phi\, dp.$$

The sum is $\int \varpi\, \mathcal{F}^{-1}\phi\, dp = \mathsf{c} \int |x|^{-n}\phi\, dx$, by Lem. 7.41(a) at $s = 1$ in the first case and by Lem. 7.41(b) in the second. As $\mathsf{K} + \mathsf{E} - \mathsf{c}|x|^{-n}$ is continuous on $\mathbb{R}^3 \setminus \{0\}$, it vanishes there.

Step 3. The periodic continuation of $\chi\varpi$, whose support lies in $\{|p| \leq 2\}$ and thus in the interior of $[-\pi, \pi]^3$, is continuous, and its Fourier coefficients are the numbers $\mathsf{K}(u)$, $u \in \mathbb{Z}^3$. By Steps 1 and 2, $\sum_u |\mathsf{K}(u)| < \infty$; hence the continuous function $\sum_u \mathsf{K}(u) e^{-i\mathsf{p}\cdot u}$ has the same Fourier coefficients and, both functions being continuous, coincides with the periodic continuation of $\chi\varpi$. For $|\mathsf{p}| < 1$ we have $\chi(\mathsf{p}) = 1$, and therefore

$$\varpi(\mathsf{p}) = \mathsf{K}(0) + \mathsf{c}\,\mathsf{s}_n(\mathsf{p}) - \sum_{u \neq 0} \mathsf{E}(u)\, e^{-i\mathsf{p}\cdot u}.$$

We put $\mathsf{e}_n(\mathsf{p}) := \mathsf{c}^{-1}\big[\sum_{u\neq 0} \mathsf{E}(u) e^{-i\mathsf{p}\cdot u} - \mathsf{K}(0)\big]$. This function is smooth, by the rapid decay of E, and invariant under the signed permutations of the coordinates, since E is radial. Finally $\mathsf{c}^{-1} = -\pi^2$, resp. $\mathsf{c}^{-1} = \frac{2\pi}{3}$, which gives the claim. □

Lemma 7.43 (Coefficient of $\mathsf{p}^2 \log|\mathsf{p}|$ in Ψ_Γ; scaling of the cut-off)**.** Let $\Gamma \in \{1, \dots, 6\}$.

(a) There are numbers $\mathsf{a}_0, \mathsf{a}_1, \mathsf{a}_2$ and a function o on $\{|\mathsf{p}| < 1\}$ with $\mathsf{o}(\mathsf{p})/\mathsf{p}^2 \to 0$ as $\mathsf{p} \to 0$, depending on Γ only, such that for almost every p with $|\mathsf{p}| < 1$

$$\Psi_\Gamma(\mathsf{p}) = \mathsf{a}_0 + \mathsf{a}_1|\mathsf{p}| + \mathsf{a}_2\mathsf{p}^2 + \tfrac{\pi}{36}\,\kappa_\Gamma^{\mathrm{lat}}\, \mathsf{p}^2 \log|\mathsf{p}| + \mathsf{o}(\mathsf{p}).$$

(b) For real $\Lambda > 1$ let $R_\Gamma^\Lambda(k, l, \mathsf{p}) := R_\Gamma(k/\Lambda, l/\Lambda, \mathsf{p}/\Lambda)$. Then $\iint R_\Gamma^\Lambda F_\Gamma\, d^\circ k\, d^\circ l = \Lambda^2 \Psi_\Gamma(\mathsf{p}/\Lambda)$ for every $\mathsf{p} \neq 0$, and for almost every p with $|\mathsf{p}| < 1$

$$\iint (R_\Gamma^\Lambda - R_\Gamma) F_\Gamma\, d^\circ k\, d^\circ l = (\Lambda^2 - 1)\mathsf{a}_0 + (\Lambda - 1)\mathsf{a}_1|\mathsf{p}| - \tfrac{\pi}{36}\,\kappa_\Gamma^{\mathrm{lat}} \log\Lambda\, \mathsf{p}^2 + \Lambda^2 \mathsf{o}(\mathsf{p}/\Lambda) - \mathsf{o}(\mathsf{p}).$$

■

Proof. Step 1. By Lems. 7.35 and 7.36,

$$\mathsf{T}_\Gamma^{\mathrm{mod}}(u) = c_{4,\Gamma}|u|^{-4} + \kappa_\Gamma^{\mathrm{lat}}|u|^{-5} + \mathsf{r}(u),$$
$$|\mathsf{r}(u)| \leq C|u|^{-6} \lg|u| \qquad (u \in \mathbb{Z}^3 \setminus \{0\});$$

we put $\mathsf{r}(0) := \mathsf{T}_\Gamma^{\mathrm{mod}}(0)$. Hence $\sum_u |\mathsf{T}_\Gamma^{\mathrm{mod}}(u)| < \infty$. The function Ψ_Γ is integrable and vanishes for $|\mathsf{p}| \geq \frac{3\pi}{8}$ (Def. 7.32(3)), that is, outside a ball contained in $[-\pi, \pi]^3$. Therefore the periodic function which equals $24\Psi_\Gamma$ on $[-\pi, \pi]^3$ has the Fourier coefficients $24 \int_{[-\pi,\pi]^3} e^{i\mathsf{p}\cdot u} \Psi_\Gamma(\mathsf{p})\, d^\circ \mathsf{p} = \mathsf{T}_\Gamma^{\mathrm{mod}}(u)$, which

are also the Fourier coefficients of the continuous function $\sum_u \mathsf{T}^{\mathrm{mod}}_\Gamma(u)e^{-i\mathsf{p}\cdot u}$. By the uniqueness of Fourier coefficients the two periodic functions agree almost everywhere; as the first one coincides with $24\Psi_\Gamma$ on $[-\pi,\pi]^3$, we obtain, for almost every $\mathsf{p}\in[-\pi,\pi]^3$,

$$24\,\Psi_\Gamma(\mathsf{p}) = c_{4,\Gamma}\,\mathsf{s}_4(\mathsf{p}) + \kappa^{\mathrm{lat}}_\Gamma\,\mathsf{s}_5(\mathsf{p}) + \hat{\mathsf{r}}(\mathsf{p}),$$
$$\hat{\mathsf{r}}(\mathsf{p}) := \sum_u \mathsf{r}(u)\,e^{-i\mathsf{p}\cdot u}.$$

Step 2. Since $\sum_u |u|^2|\mathsf{r}(u)|<\infty$, the function $\hat{\mathsf{r}}$ is of class C^2. As $\mathsf{T}^{\mathrm{mod}}_\Gamma$ is the restriction to $\mathbb{Z}^3$ of a radial function, r and $\hat{\mathsf{r}}$ are invariant under the signed permutations of the coordinates. A function f of class C^2 near 0 with this invariance has vanishing gradient at 0 (invariance under $-\mathbf{1}$), a Hessian at 0 with vanishing off-diagonal entries (change of sign of one coordinate) and equal diagonal entries (permutations); hence $\mathsf{f}(\mathsf{p}) = \mathsf{f}(0) + \mathsf{a}\,\mathsf{p}^2 + o(\mathsf{p}^2)$. We apply this to $\hat{\mathsf{r}}$, e_4 and e_5 and insert Lem. 7.42; this gives (a), the coefficient of $\mathsf{p}^2\log|\mathsf{p}|$ being $\frac{1}{24}\cdot\frac{2\pi}{3}\,\kappa^{\mathrm{lat}}_\Gamma = \frac{\pi}{36}\,\kappa^{\mathrm{lat}}_\Gamma$, and o being $\frac{1}{24}$ times the sum of the remainders $o(\mathsf{p}^2)$ of $c_{4,\Gamma}\mathsf{e}_4$, $\kappa^{\mathrm{lat}}_\Gamma\mathsf{e}_5$ and $\hat{\mathsf{r}}$.

Step 3. For (b) we substitute $k\to\Lambda k$, $l\to\Lambda l$ and use $F_\Gamma(\Lambda k,\Lambda l,\Lambda\mathsf{q}) = \Lambda^{-4}F_\Gamma(k,l,\mathsf{q})$ with $\mathsf{q}=\mathsf{p}/\Lambda$; this produces the factor $\Lambda^{6-4}=\Lambda^2$ and gives the first assertion. For the second we insert (a) at p/Λ and at p, both of modulus less than 1. The terms with a_2 cancel, the terms with a_0, a_1 give $(\Lambda^2-1)\mathsf{a}_0 + (\Lambda-1)\mathsf{a}_1|\mathsf{p}|$, and

$$\Lambda^2\,\tfrac{\pi}{36}\,\kappa^{\mathrm{lat}}_\Gamma\Lambda^{-2}\mathsf{p}^2(\log|\mathsf{p}| - \log\Lambda) - \tfrac{\pi}{36}\,\kappa^{\mathrm{lat}}_\Gamma\,\mathsf{p}^2\log|\mathsf{p}| = -\tfrac{\pi}{36}\,\kappa^{\mathrm{lat}}_\Gamma\log\Lambda\;\mathsf{p}^2.$$

□

7.1.15. *The value: finite parts and regions.* The coefficient $\frac{\pi}{36}\kappa^{\mathrm{lat}}_\Gamma$ of Lem. 7.43 will be computed by an expansion by regions of two-loop integrals with cut-offs vanishing near the origin. We first define the finite parts which occur in this expansion, by analytic regularisation, and the classes of cut-offs.

Definition 7.44 (Finite parts (analytic regularisation); cut-offs vanishing near the origin)**.** Let $\mathbb{D} := \{\mathsf{z}\in\mathbb{C} : \mathrm{Re}\,\mathsf{z} > -\frac12\}\setminus\{\frac12,\frac32,\frac52,\dots\}$; this set is open and connected, and it contains 0.

(1) One variable. A *homogeneous factor of degree* $d\in\mathbb{Z}$ is a function $\mathsf{h}\in C^\infty(\mathbb{R}^3\setminus\{0\})$ with $\mathsf{h}(tk) = t^d\mathsf{h}(k)$ for $t>0$ and $\mathsf{h}(-k) = (-1)^d\mathsf{h}(k)$. Let $\mathfrak{P}$ be the set of functions on $\mathbb{R}^3\setminus\{0\}$ of the form $\mathsf{g} = \sum_\iota \mathsf{h}_\iota\mathsf{g}_\iota$, a finite sum, with homogeneous factors h_ι of degrees d_ι and with $\mathsf{g}_\iota\in C^\infty_c(\mathbb{R}^3)$. For such g the integral $\int|k|^{2\mathsf{z}}\mathsf{g}(k)\,\mathrm{d}k$ converges absolutely and is holomorphic in z for $2\,\mathrm{Re}\,\mathsf{z} + \min_\iota d_\iota > -3$; it has a holomorphic continuation to $\mathbb{D}$ (Lem. 7.45(a)), which is unique. The *finite part* $\mathrm{f.p.}\int\mathsf{g}\,\mathrm{d}k$ is the value of this continuation at $\mathsf{z}=0$; it depends on g only, and it is linear in g.

(2) Two variables. Let $c>0$ and $\eta_c(k) := \theta(4|k|/c)$, so that $\eta_c(k)=1$ for $|k|\le\frac{c}{8}$ and $\eta_c(k)=0$ for $|k|\ge\frac{c}{4}$. Let $\mathfrak{P}_2(c)$ be the set of Borel functions Φ on $\{(k,l) : k\neq0,\ l\neq0,\ k\neq l\}$ for which there is a constant C_Φ with the following properties.

- (Φ0) $\Phi(k,l)=0$ if $|k|+|l|<c$ or $|k|+|l|>C_\Phi$.
- (Φ1) For $0<|k|<\frac{c}{3}$, $\Phi(k,l) = \sum_\iota \mathsf{h}_\iota(k)\Phi_\iota(k,l)$, a finite sum with homogeneous factors h_ι and with functions $\Phi_\iota\in C^\infty(\{|k|<\frac{c}{3}\}\times\mathbb{R}^3)$ which vanish where $|k|+|l|<c$ or $|k|+|l|>C_\Phi$.
- (Φ2) For $0<|l|<\frac{c}{3}$, $\Phi(k,l) = \sum_\iota \mathsf{h}'_\iota(l)\Phi'_\iota(k,l)$, with the same properties after the exchange of k and l.
- (Φ3) $|\Phi(k,l)|\le C_\Phi|k-l|^{-2}$ for $|k|,|l|\ge\frac{c}{8}$.

For $\Phi\in\mathfrak{P}_2(c)$ the function

$$k\mapsto\eta_c(k)\int\Phi(k,l)\,\mathrm{d}l = \sum_\iota \mathsf{h}_\iota(k)\,\eta_c(k)\int\Phi_\iota(k,l)\,\mathrm{d}l$$

lies in $\mathfrak{P}$, since η_c vanishes for $|k| \geq \frac{c}{4}$ and the Φ_ι are smooth and vanish for $|l| > C_\Phi$; likewise $l \mapsto \eta_c(l) \int \Phi(k,l)\,\mathrm{d}k$ lies in $\mathfrak{P}$. We define

$$\begin{aligned}\mathrm{f.p.}\iint \Phi\,\mathrm{d}k\,\mathrm{d}l := {} & \iint (1-\eta_c(k)-\eta_c(l))\Phi\,\mathrm{d}k\,\mathrm{d}l \\ & + \mathrm{f.p.}\int \Big[\eta_c(k)\int \Phi(k,l)\,\mathrm{d}l\Big]\,\mathrm{d}k + \mathrm{f.p.}\int \Big[\eta_c(l)\int \Phi(k,l)\,\mathrm{d}k\Big]\,\mathrm{d}l.\end{aligned}$$

On the support of Φ, $|k| < \frac{c}{8}$ implies $|l| > \frac{7}{8}c$, hence $\eta_c(k) = 1$ and $\eta_c(l) = 0$; the same holds with k and l exchanged. Therefore the first integrand vanishes unless $|k|, |l| \geq \frac{c}{8}$, and the first integral converges absolutely, by (Φ0) and (Φ3): its integrand is bounded by $C_\Phi |k-l|^{-2}$, which is integrable over bounded sets of (k,l), and it vanishes for $|k|+|l| > C_\Phi$. This finite part is linear on $\mathfrak{P}_2(c)$ and symmetric under the exchange of k and l.

(3) Cut-offs. Let $\mathcal{Y}(c)$ be the set of functions $\Upsilon \in C^\infty(\mathbb{R}^3 \times \mathbb{R}^3 \times \mathbb{R}^3)$ for which there is a constant C_Υ with the following properties.

(Y1) $\Upsilon(k,l,\mathsf{p}) = 0$ if $|\mathsf{p}| < c$ and ($|k|+|l| < c$ or $|k|+|l| > C_\Upsilon$).

(Y2) $\Upsilon(-k,-l,-\mathsf{p}) = \Upsilon(k,l,\mathsf{p}) = \Upsilon(Ok, Ol, O\mathsf{p})$ for every rotation O.

Then $\mathcal{Y}(c') \subset \mathcal{Y}(c)$ for $c < c'$. For $\Lambda > 1$ and a function Υ of (k,l,p) or of (k,l), let Υ^Λ denote Υ evaluated at the arguments divided by Λ; if $\Upsilon \in \mathcal{Y}(c)$, then $\Upsilon^\Lambda \in \mathcal{Y}(\Lambda c) \subset \mathcal{Y}(c)$. For a function $G(k,l,\mathsf{p})$ which is of class C^2 in p near 0 we put

$$\overline{G_2}(k,l) := \tfrac{1}{6}\,\Delta_{\mathsf{p}} G(k,l,\mathsf{p})\big|_{\mathsf{p}=0};$$

this is the mean over the directions of p of the term of second order in the Taylor expansion of G in p, divided by p^2, since the mean of $\mathsf{p}_i\mathsf{p}_j$ over $\{|\mathsf{p}| = t\}$ is $\frac{1}{3}t^2\delta_{ij}$. Finally, for $\Upsilon \in \mathcal{Y}(c)$ and $0 < |\mathsf{p}| < c$ we put

$$\Xi_\Gamma[\Upsilon](\mathsf{p}) := \iint \Upsilon F_\Gamma\, d^\circ k\, d^\circ l;$$

this integral converges absolutely by eq. 7.xxxvi and the estimates following it in Def. 7.32(3), since by (Y1) the integrand vanishes for $|k|+|l| > C_\Upsilon$. ∎

Lemma 7.45 (Properties of finite parts)**.**

(a) (Continuation with a parameter.) Let h be a homogeneous factor of degree d, and let $\mathsf{g}\colon \mathbb{R}^3 \times \mathbb{C} \to \mathbb{C}$ have the following properties: for every $\mathsf{z} \in \mathbb{C}$ the function $\mathsf{g}(\cdot;\mathsf{z})$ is smooth, and its support lies in a compact set which does not depend on z; for every multi-index α the derivative $\partial_k^\alpha \mathsf{g}$ is continuous on $\mathbb{R}^3 \times \mathbb{C}$ and, for each fixed k, an entire function of z. Then the function

$$\mathsf{z} \longmapsto \int |k|^{2\mathsf{z}}\, \mathsf{h}(k)\, \mathsf{g}(k;\mathsf{z})\,\mathrm{d}k,$$

which is defined for $2\,\mathrm{Re}\,\mathsf{z} + d > -3$, has a holomorphic continuation to $\mathbb{D}$, and the value of this continuation at $\mathsf{z} = 0$ is $\mathrm{f.p.}\int \mathsf{h}\,\mathsf{g}(\cdot;0)\,\mathrm{d}k$.

(b) If $\mathsf{g} \in \mathfrak{P}$ is integrable, then $\mathrm{f.p.}\int \mathsf{g}\,\mathrm{d}k = \int \mathsf{g}\,\mathrm{d}k$.

(c) Let h be a homogeneous factor of degree $d \leq -4$, and let $\eta(k) = \theta_1(|k|)$, where $\theta_1 \in C^\infty([0,\infty))$ has compact support and is equal to 1 near 0. Then

$$\mathrm{f.p.}\int \eta\,\mathsf{h}\,\mathrm{d}k = -\int (1-\eta)\,\mathsf{h}\,\mathrm{d}k.$$

(d) (A soft line.) Let $\mu_0 > 0$ and $\eta(k) := \theta(|k|/2\mu_0)$. Let

$$S(k,\mathsf{p}) = \mathsf{P}(k,\mathsf{p})\,|k|^{-2\mathsf{a}}\,|k-\mathsf{p}|^{-2\mathsf{a}'},$$

where $\mathsf{a}, \mathsf{a}' \geq 0$ are integers and P is a polynomial, homogeneous of degree $2\mathsf{a}+2\mathsf{a}'-2$, and assume that $|S(k,\mathsf{p})| \leq C_S\big(|k|^{-2} + |k-\mathsf{p}|^{-2}\big)$. Let $J \in C^\infty\big(\{|k| < 3\mu_0\} \times \{|\mathsf{p}| < \mu_0\}\big)$ satisfy

$J(-k,-\mathsf{p}) = J(k,\mathsf{p})$, and for $k \neq 0$ and $i \in \{0,1,2\}$ put

$$[SJ]_i(k;\mathsf{p}) := \sum_{|\alpha|=i} \frac{\mathsf{p}^\alpha}{\alpha!}\, \partial_{\mathsf{p}}^\alpha (SJ)(k,0).$$

Then $\eta\,[SJ]_i(\cdot;\mathsf{p}) \in \mathfrak{P}$. Moreover, there are a Borel function h_S on $\mathbb{R}^3$, which depends on S only, is bounded on the unit sphere and satisfies $\mathsf{h}_S(t\mathsf{p}) = t\,\mathsf{h}_S(\mathsf{p})$ for $t>0$, and a constant $C = C(\mu_0, S, J)$, such that for $0 < |\mathsf{p}| \le \frac{1}{2}\mu_0$

$$\Big| \int \eta SJ \,\mathrm{d}k - \mathrm{f.p.} \int \eta\,[SJ]_0 \,\mathrm{d}k - \mathrm{f.p.} \int \eta\,[SJ]_2(k;\mathsf{p})\,\mathrm{d}k - J(0,0)\,\mathsf{h}_S(\mathsf{p}) \Big| \le C\,|\mathsf{p}|^3 \lg|\mathsf{p}|.$$

(e) (Two variables.) Let $\Phi \in \mathfrak{P}_2(c)$ and $(\epsilon,\epsilon') \in \{(1,1),(1,0),(0,1)\}$. If $\epsilon = 0$, assume that all degrees of the homogeneous factors in (Φ1) are ≥ -2; if $\epsilon' = 0$, assume that all degrees of the homogeneous factors in (Φ2) are ≥ -2. Then the integral

$$\iint |k|^{2\epsilon\mathsf{z}}\, |l|^{2\epsilon'\mathsf{z}}\, \Phi \,\mathrm{d}k\,\mathrm{d}l$$

converges absolutely for large $\mathrm{Re}\,\mathsf{z}$, and it has a holomorphic continuation to $\mathbb{D}$ whose value at $\mathsf{z}=0$ is $\mathrm{f.p.}\iint \Phi\,\mathrm{d}k\,\mathrm{d}l$. If $0 < c' < c$, then $\mathfrak{P}_2(c) \subset \mathfrak{P}_2(c')$, and every $\Phi \in \mathfrak{P}_2(c)$ has the same finite part whether it is regarded as an element of $\mathfrak{P}_2(c)$ or of $\mathfrak{P}_2(c')$. If $\Phi \in \mathfrak{P}_2(c)$ is integrable, then $\mathrm{f.p.}\iint \Phi\,\mathrm{d}k\,\mathrm{d}l = \iint \Phi\,\mathrm{d}k\,\mathrm{d}l$.

■

Proof. Proof of (a). Put $m := \max\{0, -d-3\}$; then $d+m+1 \ge -2$. For the proof of (a) let $\eta(k) := \theta(|k|)$, with the function θ of Def. 7.16(1), so that $\eta = 1$ for $|k| \le \frac{1}{2}$ and $\eta = 0$ for $|k| \ge 1$, and let

$$T\mathsf{g}(k;\mathsf{z}) := \sum_{|\alpha|\le m} \frac{k^\alpha}{\alpha!}\, \partial_k^\alpha \mathsf{g}(0;\mathsf{z})$$

be the Taylor polynomial of degree m of $\mathsf{g}(\cdot;\mathsf{z})$ at $k=0$. For $2\,\mathrm{Re}\,\mathsf{z} + d > -3$ we decompose

$$\int |k|^{2\mathsf{z}} \mathsf{h}\,\mathsf{g}\,\mathrm{d}k = \int |k|^{2\mathsf{z}} \mathsf{h}\,[\mathsf{g} - \eta\, T\mathsf{g}]\,\mathrm{d}k + \sum_{|\alpha|\le m} \frac{\partial_k^\alpha \mathsf{g}(0;\mathsf{z})}{\alpha!} \int |k|^{2\mathsf{z}}\,\eta(k)\,\mathsf{h}(k)\,k^\alpha\,\mathrm{d}k.$$

The integrand of the first integral on the right vanishes outside a compact set which does not depend on z. For $|k| \le \frac{1}{2}$ we have $\eta = 1$, and Taylor's formula, together with the continuity of the derivatives of order $m+1$ of g on $\mathbb{R}^3 \times \mathbb{C}$, bounds the modulus of the integrand by $C|k|^{2\,\mathrm{Re}\,\mathsf{z}+d+m+1}$, with a constant C that stays bounded as long as z ranges over a compact set. Since $d+m+1 \ge -2$, the exponent is larger than -3 as soon as $\mathrm{Re}\,\mathsf{z} > -\frac{1}{2}$. For each $k \neq 0$ the integrand is an entire function of z; differentiating under the integral sign, we see that the first integral is holomorphic on $\{\mathrm{Re}\,\mathsf{z} > -\frac{1}{2}\}$.

In the integrals of the sum we pass to polar coordinates and integrate by parts once. With

$$\mathsf{m}_\alpha := \int_{S^2} \mathsf{h}(\omega)\,\omega^\alpha\,\mathrm{d}\omega$$

we obtain, for $2\,\mathrm{Re}\,\mathsf{z} + d + |\alpha| > -3$,

$$\int |k|^{2\mathsf{z}}\,\eta\,\mathsf{h}\,k^\alpha\,\mathrm{d}k = \mathsf{m}_\alpha \int_0^\infty \theta(s)\,s^{2\mathsf{z}+d+|\alpha|+2}\,\mathrm{d}s = -\frac{\mathsf{m}_\alpha}{2\mathsf{z}+d+|\alpha|+3} \int_0^\infty \theta'(s)\,s^{2\mathsf{z}+d+|\alpha|+3}\,\mathrm{d}s.$$

As θ' is supported in $[\frac{1}{2},1]$, the last integral is an entire function of z. If $d+|\alpha|$ is odd, then $\omega \mapsto \mathsf{h}(\omega)\omega^\alpha$ is odd and $\mathsf{m}_\alpha = 0$. If $d+|\alpha|$ is even, the denominator vanishes only at $\mathsf{z} = -\frac{1}{2}(d+|\alpha|+3)$, which is half an odd integer and hence not a point of $\mathbb{D}$. Consequently the right side of the decomposition is holomorphic on $\mathbb{D}$; it is the asserted continuation, which is unique because $\mathbb{D}$ is connected.

The value of this continuation at $\mathsf{z}=0$ involves g only through $\mathsf{g}(\cdot;0)$: it contains the function $\mathsf{g}(\cdot;0)-\eta\,T\mathsf{g}(\cdot;0)$ and the numbers $\partial_k^\alpha \mathsf{g}(0;0)$. It is therefore equal to the value at 0 of the same expression built with the function $\mathsf{g}(\cdot;0)$, which does not depend on z. By the argument just given, applied to this function, that expression continues $\mathsf{z}\mapsto \int |k|^{2\mathsf{z}}\mathsf{h}\,\mathsf{g}(\cdot;0)\,\mathrm{d}k$ to $\mathbb{D}$, and by Def. 7.44(1) its value at 0 is $\mathrm{f.p.}\int \mathsf{h}\,\mathsf{g}(\cdot;0)\,\mathrm{d}k$.

Proof of (b). Let $\mathsf{g}\in\mathfrak{P}$ be integrable. Like every element of $\mathfrak{P}$, the function g vanishes outside a bounded set; hence for every $\mathsf{N}>0$ there is C_N with $|k|^{2\,\mathrm{Re}\,\mathsf{z}}|\mathsf{g}(k)|\le C_\mathsf{N}|\mathsf{g}(k)|$ whenever $0\le \mathrm{Re}\,\mathsf{z}\le \mathsf{N}$. By differentiation under the integral sign and by dominated convergence, $\int |k|^{2\mathsf{z}}\mathsf{g}\,\mathrm{d}k$ is holomorphic on $\{\mathrm{Re}\,\mathsf{z}>0\}$ and continuous on $\{\mathrm{Re}\,\mathsf{z}\ge 0\}$. The open set $\mathbb{D}\cap\{\mathrm{Re}\,\mathsf{z}>0\}$ is connected, and on it this function coincides with the continuation of Def. 7.44(1), because the two functions agree for large $\mathrm{Re}\,\mathsf{z}$ (identity theorem). Letting $\mathsf{z}\downarrow 0$ along the real axis, we obtain $\mathrm{f.p.}\int \mathsf{g}\,\mathrm{d}k=\int \mathsf{g}\,\mathrm{d}k$.

Proof of (c). Let $\mathsf{m}_0:=\int_{S^2}\mathsf{h}\,\mathrm{d}\omega$. For $2\,\mathrm{Re}\,\mathsf{z}+d>-3$, polar coordinates and one integration by parts give

$$\int |k|^{2\mathsf{z}}\,\eta\,\mathsf{h}\,\mathrm{d}k=\mathsf{m}_0\int_0^\infty \theta_1(s)\,s^{2\mathsf{z}+d+2}\,\mathrm{d}s=-\frac{\mathsf{m}_0}{2\mathsf{z}+d+3}\int_0^\infty \theta_1'(s)\,s^{2\mathsf{z}+d+3}\,\mathrm{d}s.$$

The support of θ_1' is a compact subset of $(0,\infty)$, so the last integral is an entire function of z. As in the proof of (a), either d is odd and $\mathsf{m}_0=0$, or the zero of the denominator is half an odd integer and lies outside $\mathbb{D}$. Hence the right side is the continuation of the left side to $\mathbb{D}$, and its value at $\mathsf{z}=0$ is

$$\mathrm{f.p.}\int \eta\,\mathsf{h}\,\mathrm{d}k=-\frac{\mathsf{m}_0}{d+3}\int_0^\infty \theta_1'(s)\,s^{d+3}\,\mathrm{d}s.$$

On the other hand, $1-\theta_1$ vanishes near 0 and $d+3<0$; an integration by parts without boundary terms therefore yields

$$\int (1-\eta)\,\mathsf{h}\,\mathrm{d}k=\mathsf{m}_0\int_0^\infty (1-\theta_1(s))\,s^{d+2}\,\mathrm{d}s=\frac{\mathsf{m}_0}{d+3}\int_0^\infty \theta_1'(s)\,s^{d+3}\,\mathrm{d}s.$$

Comparing the last two displays, we obtain (c).

Proof of (d)*: preparation.* The function S is smooth on $\{k\neq 0,\ k\neq \mathsf{p}\}$, homogeneous of degree -2 in (k,p) jointly, and even, since the polynomial P is homogeneous of even degree. The set $\{(k,\mathsf{q}):|k|=1,\ |\mathsf{q}|\le\tfrac12\}$ is compact, and S is smooth in a neighbourhood of it; together with homogeneity this gives, for every multi-index α,

$$|\partial_\mathsf{p}^\alpha S(k,\mathsf{q})|\le C_\alpha\,|k|^{-2-|\alpha|}\qquad (|\mathsf{q}|\le\tfrac12|k|).$$

For every multi-index β the function $\partial_\mathsf{p}^\beta S(\cdot,0)$ is a homogeneous factor of degree $-2-|\beta|$: it is smooth off 0, homogeneous of this degree, and of parity $(-1)^{|\beta|}$ because S is even. By the Leibniz rule, $\eta\,[SJ]_i(\cdot;\mathsf{p})$ is a finite sum of products of such factors with the functions $\eta\,\partial_\mathsf{p}^\gamma J(\cdot,0)$; these belong to $C_c^\infty(\mathbb{R}^3)$, since η vanishes for $|k|\ge 2\mu_0$. Hence $\eta\,[SJ]_i(\cdot;\mathsf{p})\in\mathfrak{P}$.

Since J is even, the terms of odd degree in its Taylor expansion at $(0,0)$ vanish, and its Taylor polynomial of degree 3 is $J(0,0)+J_2$ with a quadratic form J_2 in (k,p). Thus $J=J(0,0)+J_2+J_4$, and Taylor's formula with integral remainder, applied to J and to its derivatives, gives

$$|\partial_\mathsf{p}^\alpha J_4(k,\mathsf{p})|\le C\,(|k|+|\mathsf{p}|)^{4-|\alpha|}\qquad (|\alpha|\le 3)\quad\text{on }\{|k|\le 2\mu_0,\ |\mathsf{p}|\le\tfrac12\mu_0\}.$$

Each of the four expressions inside the modulus in the asserted estimate, namely $\int \eta SJ\,\mathrm{d}k$, $\mathrm{f.p.}\int \eta\,[SJ]_0\,\mathrm{d}k$, $\mathrm{f.p.}\int \eta\,[SJ]_2(k;\mathsf{p})\,\mathrm{d}k$ and $J(0,0)\,\mathsf{h}_S(\mathsf{p})$, depends linearly on J, because $J\mapsto[SJ]_i$ and the finite part are linear. We insert $J=J(0,0)+J_2+J_4$. The quadratic form J_2 and the function J_4 are even and vanish at $(0,0)$, so that for them the last of the four expressions is 0. It therefore suffices to consider, in place of SJ, the three functions

$$N:=S,\qquad N:=SJ_2,\qquad N:=SJ_4,$$

and to show that the difference $\int \eta N\,\mathrm{d}k-\mathrm{f.p.}\int \eta\,[N]_0\,\mathrm{d}k-\mathrm{f.p.}\int \eta\,[N]_2\,\mathrm{d}k$ is equal to $\mathsf{h}_S(\mathsf{p})+O(|\mathsf{p}|^3)$ for $N=S$, the case which enters with the factor $J(0,0)$, and that it is $O(|\mathsf{p}|^3\lg|\mathsf{p}|)$ for $N=SJ_2$

and for $N = SJ_4$. For each of the three functions we write $N_i := [N]_i$ $(i = 0, 1, 2)$, and for $|k| \geq 2|\mathsf{p}|$ we put $N_{\mathrm{r}} := N - N_0 - N_1 - N_2$; moreover, $\mathsf{B} := \{|k| \leq 2|\mathsf{p}|\}$. Since $|\mathsf{p}| \leq \frac{1}{2}\mu_0$, we have $\eta = 1$ on B. In all three cases $N(-k, -\mathsf{p}) = N(k, \mathsf{p})$, as S, J_2 and J_4 are even; differentiating with respect to p at $\mathsf{p} = 0$, we see that N_1 is odd in k while N_0 and N_2 are even.

The case $N = S$. By the bound on the derivatives of S, $|N_i| \leq C|\mathsf{p}|^i|k|^{-2-i}$. On the complement B^c of B, Taylor's formula in p, applied at the points $t\mathsf{p}$ with $0 \leq t \leq 1$, all of which satisfy $|t\mathsf{p}| \leq \frac{1}{2}|k|$, gives $|N_{\mathrm{r}}| \leq C|\mathsf{p}|^3|k|^{-5}$. We write

$$\int \eta N \,\mathrm{d}k = \int_{\mathsf{B}} N \,\mathrm{d}k + \int_{\mathsf{B}^c} \eta\,(N_0 + N_1 + N_2 + N_{\mathrm{r}})\,\mathrm{d}k$$

and consider the four terms of the second integral one after the other. First, $\int_{\mathsf{B}^c} \eta N_0 = \int \eta N_0 - \int_{\mathsf{B}} N_0$. Second, $\int_{\mathsf{B}^c} \eta N_1 = 0$, because N_1 is odd and integrable over $\mathsf{B}^c \cap \operatorname{supp}\eta$, while η and B^c are invariant under $k \to -k$. Third, $N_2(\cdot\,; \mathsf{p})$ is a homogeneous factor of degree -4; assertion (c), applied with $\theta_1(s) := \theta(s/2\mu_0)$, and the vanishing of $1 - \eta$ on B give

$$\int_{\mathsf{B}^c} \eta N_2 = \int_{\mathsf{B}^c} N_2 - \int (1 - \eta) N_2 = \int_{\mathsf{B}^c} N_2 + \mathrm{f.\,p.} \int \eta N_2.$$

Fourth, $\int_{\mathsf{B}^c} \eta N_{\mathrm{r}} = \int_{\mathsf{B}^c} N_{\mathrm{r}} - \int (1 - \eta) N_{\mathrm{r}}$, and

$$\Big| \int (1 - \eta) N_{\mathrm{r}} \Big| \leq C|\mathsf{p}|^3 \int_{|k| \geq \mu_0} |k|^{-5} \,\mathrm{d}k \leq C|\mathsf{p}|^3.$$

Collecting the terms, we arrive at

$$\int \eta S \,\mathrm{d}k = \int \eta N_0 \,\mathrm{d}k + \mathrm{f.\,p.} \int \eta N_2 \,\mathrm{d}k + \mathsf{h}_S(\mathsf{p}) + O(|\mathsf{p}|^3), \qquad \mathsf{h}_S(\mathsf{p}) := \int_{|k| \leq 2|\mathsf{p}|} (S - N_0) \,\mathrm{d}k + \int_{|k| > 2|\mathsf{p}|} (N_2 + N_{\mathrm{r}}) \,\mathrm{d}k.$$

Both integrals in the definition of h_S converge absolutely, because $|S| + |N_0| \leq C(|k|^{-2} + |k - \mathsf{p}|^{-2})$, and $|N_2 + N_{\mathrm{r}}| \leq C|\mathsf{p}|^2|k|^{-4}$ on B^c. The substitution $k \to tk$, together with $N_i(tk; t\mathsf{p}) = t^{-2} N_i(k; \mathsf{p})$, yields $\mathsf{h}_S(t\mathsf{p}) = t\,\mathsf{h}_S(\mathsf{p})$ for $t > 0$; the same bounds, taken at $|\mathsf{p}| = 1$, show that h_S is bounded on the unit sphere $\{|\mathsf{p}| = 1\}$, so that $|\mathsf{h}_S(\mathsf{p})| \leq C|\mathsf{p}|$; and h_S is a Borel function by Fubini's theorem. By its definition h_S depends on S only. Finally, $\eta N_0 \in \mathfrak{P}$ is integrable, so that $\int \eta N_0 \,\mathrm{d}k$ is its finite part by (b).

The case $N = SJ_2$. Now N is homogeneous of degree 0 in (k, p), and as above $|\partial_{\mathsf{p}}^{\alpha} N(k, \mathsf{q})| \leq C|k|^{-|\alpha|}$ for $|\mathsf{q}| \leq \frac{1}{2}|k|$. Hence $|N_i| \leq C|\mathsf{p}|^i|k|^{-i}$, which is integrable near $k = 0$ for $i \leq 2$; further $|N_{\mathrm{r}}| \leq C|\mathsf{p}|^3|k|^{-3}$ on B^c, and $|N| \leq C|\mathsf{p}|^2(|k|^{-2} + |k - \mathsf{p}|^{-2})$ on B. Therefore

$$\int \eta N \,\mathrm{d}k = \sum_{i \leq 2} \int \eta N_i \,\mathrm{d}k + \Big[\int_{\mathsf{B}} (N - N_0 - N_1 - N_2) \,\mathrm{d}k + \int_{\mathsf{B}^c} \eta N_{\mathrm{r}} \,\mathrm{d}k \Big],$$

and the bracket is bounded in modulus by

$$C|\mathsf{p}|^3 + C|\mathsf{p}|^3 \int_{2|\mathsf{p}| \leq |k| \leq 2\mu_0} |k|^{-3} \,\mathrm{d}k \leq C|\mathsf{p}|^3 \lg|\mathsf{p}|.$$

The term with $i = 1$ vanishes because N_1 is odd, and the terms with $i = 0$ and $i = 2$ are finite parts by (b).

The case $N = SJ_4$. By the Leibniz rule and the bounds on the derivatives of S and of J_4,

$$|\partial_{\mathsf{p}}^{\alpha} N(k, \mathsf{q})| \leq C|k|^{2-|\alpha|} \qquad (|\alpha| \leq 3,\ |\mathsf{q}| \leq \tfrac{1}{2}|k|,\ |\mathsf{q}| \leq \tfrac{1}{2}\mu_0,\ |k| \leq 2\mu_0).$$

Hence $|N_i| \leq C|\mathsf{p}|^i|k|^{2-i}$, $|N_{\mathrm{r}}| \leq C|\mathsf{p}|^3|k|^{-1}$ on $\mathsf{B}^c \cap \operatorname{supp}\eta$, and $|N| \leq C|\mathsf{p}|^4(|k|^{-2} + |k - \mathsf{p}|^{-2})$ on B. Arguing as in the previous case, we find $\int \eta N \,\mathrm{d}k = \int \eta N_0 \,\mathrm{d}k + \int \eta N_2 \,\mathrm{d}k + O(|\mathsf{p}|^3)$, and the two integrals on the right are finite parts by (b).

Adding the three cases, the first one multiplied by $J(0,0)$, and using $|\mathsf{p}|^3 \le |\mathsf{p}|^3 \lg|\mathsf{p}|$, we obtain the estimate of (d), with a constant depending on μ_0, S and J.

Proof of (e). We split $1 = \big(1-\eta_c(k)-\eta_c(l)\big)+\eta_c(k)+\eta_c(l)$ and treat the three resulting integrals separately; each of them converges absolutely for large $\operatorname{Re}\mathsf{z}$, as the following discussion shows.

On the support of Φ, $|k| < \frac{c}{8}$ implies $|l| > \frac{7}{8}c$ by (Φ0), hence $\eta_c(k) = 1$ and $\eta_c(l) = 0$; the same holds with k and l exchanged. Therefore the first integrand, which contains the factor $1-\eta_c(k)-\eta_c(l)$, vanishes unless $|k|, |l| \ge \frac{c}{8}$, and by (Φ0) it vanishes unless $|k|+|l| \le C_\Phi$. On this bounded set the factor $|k|^{2\epsilon\mathsf{z}}|l|^{2\epsilon'\mathsf{z}}$ is an entire function of z, bounded for z in compact sets, while $|\Phi| \le C_\Phi |k-l|^{-2}$ by (Φ3). Hence the first integral is an entire function of z, whose value at $\mathsf{z}=0$ is $\iint \big(1-\eta_c(k)-\eta_c(l)\big)\Phi \,\mathrm{d}k\,\mathrm{d}l$.

In the second integral, $\eta_c(k) \ne 0$ forces $|k| \le \frac{c}{4} < \frac{c}{3}$, so that ($\Phi$1) applies, and Fubini's theorem turns this integral into

$$\sum_\iota \int |k|^{2\epsilon\mathsf{z}}\, \mathsf{h}_\iota(k)\, \mathsf{g}_\iota(k;\mathsf{z})\,\mathrm{d}k, \qquad \mathsf{g}_\iota(k;\mathsf{z}) := \eta_c(k) \int |l|^{2\epsilon'\mathsf{z}}\, \Phi_\iota(k,l)\,\mathrm{d}l.$$

Since Φ_ι vanishes where $|k|+|l| < c$ or $|k|+|l| > C_\Phi$, the integrand defining g_ι vanishes unless $\frac{3}{4}c \le |l| \le C_\Phi$; consequently g_ι has the properties required in (a). If $\epsilon = 1$, assertion (a) provides a holomorphic continuation to $\mathbb{D}$ whose value at $\mathsf{z}=0$ is $\mathrm{f.p.}\int \mathsf{h}_\iota\, \mathsf{g}_\iota(\cdot;0)\,\mathrm{d}k$. If $\epsilon = 0$, all degrees of the h_ι are ≥ -2; the k-integral then converges absolutely for every z and defines an entire function, whose value at 0 is again the finite part, by (b). Summing over ι, we obtain at $\mathsf{z}=0$ the number $\mathrm{f.p.}\int \big[\eta_c(k)\int \Phi(k,l)\,\mathrm{d}l\big]\,\mathrm{d}k$.

The hypotheses of (e) are symmetric under the exchange of k with l, of ϵ with ϵ' and of (Φ1) with (Φ2): the set $\{(1,1),(1,0),(0,1)\}$ is invariant under $(\epsilon,\epsilon') \mapsto (\epsilon',\epsilon)$, conditions ($\Phi$0) and ($\Phi$3) do not change when k and l are interchanged, and the assumptions on the degrees in (Φ1) and (Φ2) are interchanged. Under this exchange the third integral, which contains the factor $\eta_c(l)$, becomes the second one, and the argument just given applies to it. In detail: $\eta_c(l) \ne 0$ forces $|l| \le \frac{c}{4} < \frac{c}{3}$, so that ($\Phi$2) applies, and Fubini's theorem turns the third integral into

$$\sum_\iota \int |l|^{2\epsilon'\mathsf{z}}\, \mathsf{h}'_\iota(l)\, \mathsf{g}'_\iota(l;\mathsf{z})\,\mathrm{d}l, \qquad \mathsf{g}'_\iota(l;\mathsf{z}) := \eta_c(l) \int |k|^{2\epsilon\mathsf{z}}\, \Phi'_\iota(k,l)\,\mathrm{d}k.$$

Since Φ'_ι vanishes where $|k|+|l| < c$ or $|k|+|l| > C_\Phi$, the integrand defining g'_ι vanishes unless $\frac{3}{4}c \le |k| \le C_\Phi$, and g'_ι has the properties required in (a). If $\epsilon' = 1$, assertion (a) provides a holomorphic continuation to $\mathbb{D}$ whose value at $\mathsf{z}=0$ is $\mathrm{f.p.}\int \mathsf{h}'_\iota\, \mathsf{g}'_\iota(\cdot;0)\,\mathrm{d}l$. If $\epsilon' = 0$, all degrees of the h'_ι are ≥ -2; the l-integral then converges absolutely for every z and defines an entire function, whose value at 0 is the finite part, by (b). Summing over ι, we obtain at $\mathsf{z}=0$ the number $\mathrm{f.p.}\int \big[\eta_c(l)\int \Phi(k,l)\,\mathrm{d}k\big]\,\mathrm{d}l$. The sum of the three values at $\mathsf{z}=0$ is the right side of the definition of $\mathrm{f.p.}\iint \Phi\,\mathrm{d}k\,\mathrm{d}l$ in Def. 7.44(2).

Change of the parameter c. Let $0 < c' < c$ and $\Phi \in \mathfrak{P}_2(c)$. Conditions ($\Phi$0), ($\Phi$1) and ($\Phi$2) at c' follow from the same conditions at c by restriction. For (Φ3) at c', let $|k|, |l| \ge \frac{c'}{8}$ and $\Phi(k,l) \ne 0$. If $|k|, |l| \ge \frac{c}{8}$, (Φ3) at c applies. Both $|k|$ and $|l|$ smaller than $\frac{c}{8}$ is impossible by (Φ0). If $\frac{c'}{8} \le |k| < \frac{c}{8}$, we use $\Phi = \sum_\iota \mathsf{h}_\iota \Phi_\iota$ from (Φ1): the h_ι are bounded on $\{\frac{c'}{8} \le |k| \le \frac{c}{8}\}$, the Φ_ι are bounded there as well (they are continuous on $\{|k| \le \frac{c}{8}\} \times \mathbb{R}^3$ and vanish for $|l| > C_\Phi$), and $|k-l| \le |k|+|l| \le C_\Phi$; hence $|\Phi| \le C \le C\, C_\Phi^2 |k-l|^{-2}$.

The case $\frac{c'}{8} \le |l| < \frac{c}{8}$ is symmetric, with (Φ2). Thus $\mathfrak{P}_2(c) \subset \mathfrak{P}_2(c')$. The integral with $(\epsilon,\epsilon') = (1,1)$ does not involve c, so the value at $\mathsf{z}=0$ of its continuation, which is the finite part with respect to c and also the finite part with respect to c', is the same.

Finally, let Φ be integrable. By Fubini's theorem the functions $k \mapsto \eta_c(k)\int \Phi(k,l)\,\mathrm{d}l$ and $l \mapsto \eta_c(l)\int \Phi(k,l)\,\mathrm{d}k$ are integrable elements of $\mathfrak{P}$, so by (b) their finite parts are their integrals, and the definition of $\mathrm{f.p.}\iint \Phi$ reduces to $\iint \Phi\,\mathrm{d}k\,\mathrm{d}l$. □

Proposition 7.46 (Expansion by regions of $\Xi_\Gamma[\Upsilon]$ at small p)**.** Let $\Gamma \in \{1,\dots,6\}$, $c>0$ and $\Upsilon \in \mathcal{Y}(c)$. Then $\overline{(\Upsilon F_\Gamma)_2} \in \mathfrak{P}_2(c)$, and there are numbers α, β and C, depending only on Γ, c and Υ, such that

$$|\Xi_\Gamma[\Upsilon](\mathsf{p}) - \alpha - \beta\,|\mathsf{p}| - \mathfrak{f}_\Gamma[\Upsilon]\,\mathsf{p}^2| \le C\,|\mathsf{p}|^3 \lg|\mathsf{p}| \qquad (0<|\mathsf{p}|\le \tfrac{c}{16}),$$

where

$$\mathfrak{f}_\Gamma[\Upsilon] := (2\pi)^{-6}\,\mathrm{f.\,p.}\iint \overline{(\Upsilon F_\Gamma)_2}\,\mathrm{d}k\,\mathrm{d}l.$$

The map $\Upsilon \mapsto \mathfrak{f}_\Gamma[\Upsilon]$ is linear on $\mathcal{Y}(c)$, and $\mathfrak{f}_\Gamma[\Upsilon^\Lambda] = \mathfrak{f}_\Gamma[\Upsilon]$ for every $\Lambda>1$. ■

Proof. Step 1: two factorisations of F_Γ. We write the integrand F_Γ of eq. 7.xxxv in two ways,

$$F_\Gamma = \sum S\cdot H = \sum S'\cdot H',$$

where the sums extend over the tensor indices carried by S, resp. by S', the factor S depends on (k,p) and S' on (l,p) as in the following table, and H, resp. H', denotes the product of all the other factors of F_Γ:

Γ	$S(k,\mathsf{p})$	$S'(l,\mathsf{p})$
1	$V_2(k,\mathsf{p}-k)_{\mu\nu}/\mathsf{x}_1\mathsf{x}'_1$	$V_2(-l,l-\mathsf{p})_{\rho\sigma}/\mathsf{x}_2\mathsf{x}'_2$
2, 3	$V_2(k,\mathsf{p}-k)_{\mu\nu}V_2(-k,k-\mathsf{p})_{\rho\nu}/\mathsf{x}_1^2\mathsf{x}'_1$	$1/\mathsf{x}_2$
4	$V_2(k,\mathsf{p}-k)_{\mu\nu}/\mathsf{x}_1\mathsf{x}'_1$	$V_2(l,\mathsf{p}-l)_{\mu\nu}/\mathsf{x}_2\mathsf{x}'_2$
5	$V_2(k-\mathsf{p},-k)_{\alpha\sigma}/\mathsf{x}_1\mathsf{x}'_1$	$1/\mathsf{x}_2$
6	$1/\mathsf{x}'_1$	$1/\mathsf{x}_2$

For $\Gamma=4$ this means

$$H_{\mu\nu} = -\tfrac12[\delta_{\mu\nu}V_2(l,\mathsf{p}-l)_{\lambda\lambda} - \tfrac12 V_2(l,\mathsf{p}-l)_{\mu\nu} - \tfrac12 V_2(l,\mathsf{p}-l)_{\nu\mu}]/\mathsf{x}_2\mathsf{x}'_2,$$

and H' has the same form, with $V_2(k,\mathsf{p}-k)$ and $\mathsf{x}_1\mathsf{x}'_1$ in place of $V_2(l,\mathsf{p}-l)$ and $\mathsf{x}_2\mathsf{x}'_2$. Every component of S has the form required in Lem. 7.45(d): it is a homogeneous polynomial divided by $|k|^{2\mathsf{a}}|k-\mathsf{p}|^{2\mathsf{a}'}$, with $(\mathsf{a},\mathsf{a}')=(1,1)$ for $\Gamma\in\{1,4,5\}$, $(2,1)$ for $\Gamma\in\{2,3\}$ and $(0,1)$ for $\Gamma=6$, and since $|V_2(k_1,k_2)|\le 2|k_1||k_2|$ its modulus is at most $2/|k||k-\mathsf{p}| \le |k|^{-2}+|k-\mathsf{p}|^{-2}$ for $\Gamma\in\{1,4,5\}$, at most $4|k|^{-2}$ for $\Gamma\in\{2,3\}$, and equal to $|k-\mathsf{p}|^{-2}$ for $\Gamma=6$.

The same holds for the components of S', as functions of l. The factor S contains all factors $\mathsf{x}_1,\mathsf{x}'_1$ of the denominator of F_Γ, and S' all factors $\mathsf{x}_2,\mathsf{x}'_2$. Hence H is a polynomial divided by powers of $\mathsf{x}_2,\mathsf{x}'_2,\mathsf{x}_3$, and H' a polynomial divided by powers of $\mathsf{x}_1,\mathsf{x}'_1,\mathsf{x}_3$; in both, x_3 occurs at most to the first power. Moreover, H and H' are even under the reflection of all their arguments: they are quotients of homogeneous polynomials of even degree, because besides factors V_2 each of them contains two factors V_3, or the factor $l_\rho(l-k)_\mu$, or neither.

Step 2: where the lines are long. Put $\mu_0 := \frac{c}{8}$ and $\eta := \eta_c = \theta(|\cdot|/2\mu_0)$. Let $|\mathsf{p}|<\mu_0$. By (Y1), $\Upsilon(k,l,\mathsf{p})\neq 0$ implies $|k|+|l|\ge 8\mu_0$. If, in addition, $|k|<3\mu_0$, then

$$|l|>5\mu_0, \qquad |l-\mathsf{p}|>4\mu_0, \qquad |k-l|>2\mu_0;$$

these are the lengths whose squares $\mathsf{x}_2,\mathsf{x}'_2,\mathsf{x}_3$ occur in the denominator of H. Consequently the product ΥH, extended by 0 where $\Upsilon=0$, is smooth on $\{|k|<3\mu_0\}\times\mathbb{R}^3\times\{|\mathsf{p}|<\mu_0\}$: this set is covered by the open set $\{|k|+|l|<8\mu_0,\ |\mathsf{p}|<c\}$, on which Υ vanishes identically, and by the open set $\{|k|<3\mu_0,\ |l|>5\mu_0,\ |\mathsf{p}|<\mu_0\}$, on which H is smooth. Symmetrically, if $\Upsilon(k,l,\mathsf{p})\neq 0$ and $|l|<3\mu_0$, then $|k|\ge 8\mu_0-|l|$, and

$$|k|>5\mu_0, \qquad |k-\mathsf{p}|>4\mu_0, \qquad |k-l|>2\mu_0;$$

these are the lengths whose squares $\mathsf{x}_1,\mathsf{x}'_1,\mathsf{x}_3$ occur in the denominator of H'. The set $\mathbb{R}^3\times\{|l|<3\mu_0\}\times\{|\mathsf{p}|<\mu_0\}$ is covered by the open set $\{|k|+|l|<8\mu_0,\ |\mathsf{p}|<c\}$, on which Υ vanishes identically, and by the open set $\{|k|>5\mu_0,\ |l|<3\mu_0,\ |\mathsf{p}|<\mu_0\}$, on which H' is smooth; hence $\Upsilon H'$, extended by 0 where $\Upsilon=0$, is smooth on it. Finally, if $|k|,|l|\ge\mu_0$ and $|\mathsf{p}|\le\frac12\mu_0$, then $\mathsf{x}_1,\mathsf{x}'_1,\mathsf{x}_2,\mathsf{x}'_2\ge\frac14\mu_0^2$, and therefore $F_\Gamma = F^\sharp/\mathsf{x}_3$ with a function $F^\sharp$ which is smooth on this set.

Step 3: $\Phi := \overline{(\Upsilon F_\Gamma)_2}$ *belongs to* $\mathfrak{P}_2(c)$. Condition (Φ0) follows from (Y1): where $|k|+|l|<c$ or $|k|+|l|>C_\Upsilon$, the function ΥF_Γ vanishes for all $|\mathsf{p}|<c$, and so does Φ. For (Φ1) let $0<|k|<\frac{c}{3}$; then $|k|<3\mu_0$, and

$$\Phi = \frac{1}{6}\sum_j \sum \Big[\partial_{\mathsf{p}_j}^2 S\cdot \Upsilon H + 2\,\partial_{\mathsf{p}_j} S\cdot \partial_{\mathsf{p}_j}(\Upsilon H) + S\cdot \partial_{\mathsf{p}_j}^2(\Upsilon H)\Big]_{\mathsf{p}=0}$$

by the Leibniz rule. Here $\partial_{\mathsf{p}}^\beta S(\cdot,0)$ is a homogeneous factor of degree $-2-|\beta|$, and $\partial_{\mathsf{p}}^\beta(\Upsilon H)(k,l,0)$ is smooth by Step 2. The latter function vanishes wherever $\Upsilon(\cdot,\cdot,\mathsf{p})=0$ for all small p, that is, by (Y1), where $|k|+|l|<c$ or $|k|+|l|>C_\Upsilon$; this is the support condition of (Φ1), with any $C_\Phi \ge C_\Upsilon$. Condition (Φ2) is obtained from the factorisation $F_\Gamma = \sum S'\cdot H'$ by the same argument with the roles of k and l exchanged: for $0<|l|<\frac{c}{3}$ the Leibniz rule writes Φ as the corresponding sum of products of the functions $\partial_{\mathsf{p}}^\beta S'(\cdot,0)$, which are homogeneous factors of degree $-2-|\beta|$ in l, with the functions $\partial_{\mathsf{p}}^\beta(\Upsilon H')(k,l,0)$, which are smooth on $\mathbb{R}^3\times\{|l|<3\mu_0\}$ by Step 2 and vanish where $|k|+|l|<c$ or $|k|+|l|>C_\Upsilon$.

For (Φ3) let $|k|,|l|\ge\frac{c}{8}=\mu_0$; by Step 2,

$$\Phi = \tfrac{1}{6}\,\Delta_{\mathsf{p}}(\Upsilon F^\sharp)\big|_{\mathsf{p}=0}/\mathsf{x}_3,$$

and the numerator is bounded, being smooth with compact support in (k,l).

Step 4: three regions. For $0<|\mathsf{p}|\le\frac12\mu_0$ we split $\Xi_\Gamma[\Upsilon]=\Xi_{\mathrm{H}}+\Xi_{\mathrm{A}}+\Xi_{\mathrm{B}}$, inserting into the integral the factors $1-\eta(k)-\eta(l)$, $\eta(k)$ and $\eta(l)$, respectively. If $\Upsilon\ne0$ and $|k|<\mu_0$, then $|l|>7\mu_0$, hence $\eta(k)=1$ and $\eta(l)=0$; thus on the support of Υ the first factor vanishes unless $|k|,|l|\ge\mu_0$, and the same holds with k and l exchanged. Each of the three terms is invariant under rotations of p, by (Y2), by the rotation invariance of F_Γ, and because η is radial.

Step 5: the hard region. The integrand of Ξ_{H} is $(1-\eta(k)-\eta(l))\,\Upsilon F^\sharp/\mathsf{x}_3$; it is supported in $\{\mu_0\le|k|,|l|\le C_\Upsilon\}$, and all its derivatives with respect to p are bounded by $C|k-l|^{-2}$. Hence Ξ_{H} is smooth on $\{|\mathsf{p}|<\frac12\mu_0\}$; it is even, by (Y2) and the evenness of F_Γ, and rotation invariant. At $\mathsf{p}=0$ its gradient and its third derivatives therefore vanish, and its Hessian is a multiple of the identity, so that

$$\Xi_{\mathrm{H}}(\mathsf{p}) = \Xi_{\mathrm{H}}(0) + \tfrac16\,\Delta\Xi_{\mathrm{H}}(0)\,\mathsf{p}^2 + O(|\mathsf{p}|^4), \qquad \tfrac16\,\Delta\Xi_{\mathrm{H}}(0) = (2\pi)^{-6}\iint (1-\eta(k)-\eta(l))\,\Phi\,\mathrm{d}k\,\mathrm{d}l.$$

Step 6: k soft. Let $J(k,\mathsf{p}) := \int \Upsilon H\, d^\circ l$, one function for each value of the tensor indices of S. By Step 2, and since ΥH vanishes for $|l|>C_\Upsilon$, the function J is smooth on $\{|k|<3\mu_0\}\times\{|\mathsf{p}|<\mu_0\}$, and it is even, as the substitution $l\to -l$ shows. By Fubini's theorem $\Xi_{\mathrm{A}} = \sum\int \eta S J\, d^\circ k$. We apply Lem. 7.45(d), with the present μ_0, to each component of S and the corresponding component of J; for $0<|\mathsf{p}|\le\frac12\mu_0$ this gives

$$(2\pi)^3\,\Xi_{\mathrm{A}}(\mathsf{p}) = \sum\Big\{\text{f. p.}\int \eta\,[SJ]_0\,\mathrm{d}k + \sum_{i,j}\mathsf{p}_i\mathsf{p}_j\,\tfrac12\,\text{f. p.}\int \eta\,\partial_{\mathsf{p}_i}\partial_{\mathsf{p}_j}(SJ)(k,0)\,\mathrm{d}k + J(0,0)\,\mathsf{h}_S(\mathsf{p})\Big\} + O(|\mathsf{p}|^3\lg|\mathsf{p}|).$$

We take the mean of both sides over the sphere $|\mathsf{p}|=t$. The left side is constant there, by Step 4. The mean of h_S is t times its mean over the unit sphere, by homogeneity, and the mean of $\mathsf{p}_i\mathsf{p}_j$ is $\frac13 t^2\delta_{ij}$. For $k\ne0$, differentiation under the integral sign gives $\sum\frac16\Delta_{\mathsf{p}}(SJ)(k,0) = \int\Phi(k,l)\,d^\circ l$. Since the finite part is linear, we conclude that

$$\Xi_{\mathrm{A}}(\mathsf{p}) = \alpha_{\mathrm{A}} + \beta_{\mathrm{A}}|\mathsf{p}| + (2\pi)^{-6}\,\mathsf{p}^2\,\text{f. p.}\int\Big[\eta(k)\int\Phi(k,l)\,\mathrm{d}l\Big]\,\mathrm{d}k + O(|\mathsf{p}|^3\lg|\mathsf{p}|)$$

with numbers α_{A}, β_{A}.

Step 7: l soft, and the sum. For Ξ_{B} we exchange the roles of k and l in Step 6. We use the factorisation $F_\Gamma=\sum S'\cdot H'$ of Step 1 and put $J'(l,\mathsf{p}) := \int\Upsilon H'\,d^\circ k$, one function for each value of

the tensor indices of S'. By Step 2, and since $\Upsilon H'$ vanishes for $|k| > C_\Upsilon$, the function J' is smooth on $\{|l| < 3\mu_0\} \times \{|\mathsf{p}| < \mu_0\}$, and it is even, as the substitution $k \to -k$ shows; by Fubini's theorem $\Xi_{\mathrm{B}} = \sum \int \eta S' J' \, d^\circ l$. We apply Lem. 7.45(d), with l in place of k, to each component of S' and the corresponding component of J', and we take the mean over the sphere $|\mathsf{p}| = t$ as in Step 6, now with $\sum \frac{1}{6}\Delta_{\mathsf{p}}(S'J')(l,0) = \int \Phi(k,l)\, d^\circ k$ for $l \neq 0$. This gives

$$\Xi_{\mathrm{B}}(\mathsf{p}) = \alpha_{\mathrm{B}} + \beta_{\mathrm{B}}|\mathsf{p}| + (2\pi)^{-6}\,\mathsf{p}^2\,\mathrm{f.p.}\int \Big[\eta(l)\int \Phi(k,l)\,\mathrm{d}k\Big]\,\mathrm{d}l + O(|\mathsf{p}|^3 \lg|\mathsf{p}|) \qquad (0 < |\mathsf{p}| \le \tfrac{1}{2}\mu_0)$$

with numbers α_{B}, β_{B}. Since $\eta = \eta_c$ and $\Phi \in \mathfrak{P}_2(c)$ by Step 3, Def. 7.44(2) shows that the sum of the three coefficients of p^2 obtained in Steps 5–7 is $\mathfrak{f}_\Gamma[\Upsilon]$. Adding the three expansions, and noting that $\frac{1}{2}\mu_0 = \frac{c}{16}$, we obtain the asserted estimate, with $\alpha := \Xi_{\mathrm{H}}(0) + \alpha_{\mathrm{A}} + \alpha_{\mathrm{B}}$ and $\beta := \beta_{\mathrm{A}} + \beta_{\mathrm{B}}$.

Step 8: linearity and scaling. The map $\Upsilon \mapsto \overline{(\Upsilon F_\Gamma)_2}$ and the finite part are linear; hence so is $\mathfrak{f}_\Gamma$. Let $\Lambda > 1$. The substitution $k \to \Lambda k$, $l \to \Lambda l$, together with the homogeneity of degree -4 of F_Γ, gives $\Xi_\Gamma[\Upsilon^\Lambda](\mathsf{p}) = \Lambda^2\,\Xi_\Gamma[\Upsilon](\mathsf{p}/\Lambda)$, so that by the estimate already proved

$$\Xi_\Gamma[\Upsilon^\Lambda](\mathsf{p}) = \Lambda^2\alpha + \Lambda\beta\,|\mathsf{p}| + \mathfrak{f}_\Gamma[\Upsilon]\,\mathsf{p}^2 + O(|\mathsf{p}|^3 \lg|\mathsf{p}|).$$

On the other hand $\Upsilon^\Lambda \in \mathcal{Y}(c)$, and the estimate applied to Υ^Λ gives $\Xi_\Gamma[\Upsilon^\Lambda](\mathsf{p}) = \alpha' + \beta'|\mathsf{p}| + \mathfrak{f}_\Gamma[\Upsilon^\Lambda]\,\mathsf{p}^2 + O(|\mathsf{p}|^3\lg|\mathsf{p}|)$ with numbers α', β'. If $\mathsf{a} + \mathsf{b}t + \mathsf{c}t^2 = o(t^2)$ as $t \downarrow 0$, then $\mathsf{a} = \mathsf{b} = \mathsf{c} = 0$; comparing the two expansions, we obtain in particular $\mathfrak{f}_\Gamma[\Upsilon^\Lambda] = \mathfrak{f}_\Gamma[\Upsilon]$. □

7.1.16. *The value: the functional $\mathcal{R}$, the residue, the bridge.* By Prop. 7.46, the coefficient of p^2 in the expansion of $\Xi_\Gamma[\Upsilon]$ at small p is the number $\mathfrak{f}_\Gamma[\Upsilon]$, which is given by a finite part. We now compute it for the differences $R_\Gamma^\Lambda - R_\Gamma$ of the cut-offs of Lem. 7.43(b). The result is $d_{2,\Gamma}\log\Lambda$ with $d_{2,\Gamma} = \mathcal{R}[F_\Gamma]/16\pi^2$, where $\mathcal{R}$ is a linear functional defined below; the comparison with Lem. 7.43 then expresses $\kappa_\Gamma^{\mathrm{lat}}$ through $\mathcal{R}[F_\Gamma]$.

Definition 7.47 (The functional $\mathcal{R}$)**.** We use the variables $\mathsf{x}_0, \mathsf{x}_1, \mathsf{x}_1', \mathsf{x}_2, \mathsf{x}_2', \mathsf{x}_3$ of Def. 7.32(1).

(1) The scalar products of k, l, p are linear in these variables:

$$k\cdot l = \tfrac{1}{2}(\mathsf{x}_1 + \mathsf{x}_2 - \mathsf{x}_3), \qquad k\cdot\mathsf{p} = \tfrac{1}{2}(\mathsf{x}_1 - \mathsf{x}_1' + \mathsf{x}_0), \qquad l\cdot\mathsf{p} = \tfrac{1}{2}(\mathsf{x}_2 - \mathsf{x}_2' + \mathsf{x}_0).$$

(2) For $n \in \mathbb{Z}$ let $\mathcal{L}_n$ be the real linear span of the functions $\mathsf{x}_1^{-\mathsf{a}}\mathsf{x}_2^{-\mathsf{b}}\mathsf{x}_3^{-\mathsf{c}}$, where $\mathsf{a},\mathsf{b},\mathsf{c} \in \mathbb{Z}$ and $\mathsf{a}+\mathsf{b}+\mathsf{c} = -n$, on the set $\{(k,l) : k \neq 0,\ l \neq 0,\ k \neq l\}$. These functions are linearly independent. Indeed, the image of the map $(k,l) \mapsto (\mathsf{x}_1,\mathsf{x}_2,\mathsf{x}_3)$ contains a non-empty open set: for $k = (t_1,t_2,0)$ and $l = (0,t_3,0)$ the map becomes

$$(t_1,t_2,t_3) \longmapsto (t_1^2 + t_2^2,\ t_3^2,\ t_1^2 + (t_2 - t_3)^2),$$

whose differential at $(1,0,1)$ has determinant 8, so that the inverse function theorem applies. A vanishing finite linear combination of the functions, multiplied by a sufficiently high power of $\mathsf{x}_1\mathsf{x}_2\mathsf{x}_3$, becomes a polynomial in $(\mathsf{x}_1,\mathsf{x}_2,\mathsf{x}_3)$ which vanishes on a non-empty open set, and such a polynomial is zero. We denote by $r_{111}\colon \mathcal{L}_{-3} \to \mathbb{R}$ the linear map which assigns to an element of $\mathcal{L}_{-3}$ its coefficient of $(\mathsf{x}_1\mathsf{x}_2\mathsf{x}_3)^{-1}$.

(3) Let $\mathfrak{F}$ be the real linear span of the monomials

$$\mathsf{x}_0^n\,\mathsf{x}_1^{-a_1}\mathsf{x}_1'^{-a_2}\mathsf{x}_2^{-a_3}\mathsf{x}_2'^{-a_4}\mathsf{x}_3^{-a_5}, \qquad \text{integers } n \ge 0,\ a_1,\dots,a_5, \quad \textstyle\sum_i a_i = 2+n;$$

these monomials are homogeneous of degree -4. The numerators of the integrands in eq. 7.xxxv are complete contractions of tensors built from k, l, p and Kronecker symbols; hence they are polynomials in the scalar products of k, l, p, and by (1) polynomials in the six variables. Since moreover x_3 occurs in the denominators of eq. 7.xxxv at most to the first power, and not at all in F_4, each F_Γ belongs to $\mathfrak{F}$: it is a linear combination of monomials with $a_5 \le 1$, and F_4 is a linear combination of monomials with $a_5 \le 0$.

(4) For $F \in \mathfrak{F}$ we put

$$\mathcal{R}[F] := r_{111}(\overline{F_2}), \tag{7.xliii}$$

with $\overline{F_2}$ as in Def. 7.44(3); by Lem. 7.48 below, $\overline{F_2} \in \mathcal{L}_{-3}$.

■

Lemma 7.48 (Values of $\mathcal{R}$ on monomials)**.** For every $F \in \mathfrak{F}$ the function $\overline{F_2}$ belongs to $\mathcal{L}_{-3}$; hence $\mathcal{R}$ is well defined on $\mathfrak{F}$, and it is linear. If F is a linear combination of monomials with $a_5 \le 1$, then $\overline{F_2}$ is a linear combination of functions $\mathsf{x}_1^{-\mathsf{a}}\mathsf{x}_2^{-\mathsf{b}}\mathsf{x}_3^{-\mathsf{c}}$ with $\mathsf{c} \le 1$. For integers $n \ge 0$ and $a_1, \dots, a_5$ with $\sum_i a_i = 2 + n$, put

$$(\alpha_1, \alpha_2, \alpha_3) := (a_1 + a_2,\ a_3 + a_4,\ a_5), \qquad \mathsf{q}(t) := \tfrac{1}{3}t(2t-1).$$

Then

$$\mathcal{R}\big[\mathsf{x}_0^n\, \mathsf{x}_1^{-a_1}\mathsf{x}_1'^{-a_2}\mathsf{x}_2^{-a_3}\mathsf{x}_2'^{-a_4}\mathsf{x}_3^{-a_5}\big] = \begin{cases} 1 & n = 1,\ (\alpha_1,\alpha_2,\alpha_3) = (1,1,1), \\ \mathsf{q}(a_2) + \frac{2}{3}a_2a_4 & n = 0,\ (\alpha_1,\alpha_2,\alpha_3) = (0,1,1), \\ \mathsf{q}(a_4) + \frac{2}{3}a_2a_4 & n = 0,\ (\alpha_1,\alpha_2,\alpha_3) = (1,0,1), \\ -\frac{2}{3}a_2a_4 & n = 0,\ (\alpha_1,\alpha_2,\alpha_3) = (0,0,2), \\ 0 & \text{otherwise.} \end{cases} \tag{7.xliv}$$

■

Proof. The map $F \mapsto \overline{F_2}$ is linear; it therefore suffices to consider a monomial F as on the left of eq. 7.xliv. We write $x := k \cdot \mathsf{p}$ and $y := l \cdot \mathsf{p}$; then $\mathsf{x}_1' = \mathsf{x}_1 - 2x + \mathsf{x}_0$ and $\mathsf{x}_2' = \mathsf{x}_2 - 2y + \mathsf{x}_0$. For an integer t, the binomial series $(1+w)^{-t} = 1 - tw + \frac{1}{2}t(t+1)w^2 + O(w^3)$, taken at $w = (\mathsf{x}_0 - 2x)/\mathsf{x}_1$, gives for fixed (k,l) and small p

$$\mathsf{x}_1'^{-t} = \mathsf{x}_1^{-t}\Big[1 + 2t\,\frac{x}{\mathsf{x}_1} - t\,\frac{\mathsf{x}_0}{\mathsf{x}_1} + 2t(t+1)\,\frac{x^2}{\mathsf{x}_1^2} + O(|\mathsf{p}|^3)\Big],$$

and $\mathsf{x}_2'^{-t}$ has the same expansion, with y and x_2 in place of x and x_1. We use these expansions at $t = a_2$ and at $t = a_4$. On the terms of second order the operator $\frac{1}{6}\Delta_{\mathsf{p}}$ acts by

$$\tfrac{1}{6}\Delta_{\mathsf{p}}\mathsf{x}_0 = 1, \qquad \tfrac{1}{6}\Delta_{\mathsf{p}}x^2 = \tfrac{1}{3}\mathsf{x}_1, \qquad \tfrac{1}{6}\Delta_{\mathsf{p}}y^2 = \tfrac{1}{3}\mathsf{x}_2, \qquad \tfrac{1}{6}\Delta_{\mathsf{p}}(xy) = \tfrac{1}{3}\,k\cdot l = \tfrac{1}{6}(\mathsf{x}_1 + \mathsf{x}_2 - \mathsf{x}_3),$$

the last equality by Def. 7.47(1).

If $n \ge 2$, then $F = O(|\mathsf{p}|^4)$ and $\overline{F_2} = 0$. If $n = 1$, the term of second order in p of F is $\mathsf{x}_0\,\mathsf{x}_1^{-\alpha_1}\mathsf{x}_2^{-\alpha_2}\mathsf{x}_3^{-\alpha_3}$, and $\overline{F_2} = \mathsf{x}_1^{-\alpha_1}\mathsf{x}_2^{-\alpha_2}\mathsf{x}_3^{-\alpha_3}$. If $n = 0$, the terms of second order are

$$\mathsf{x}_1^{-\alpha_1}\mathsf{x}_2^{-\alpha_2}\mathsf{x}_3^{-\alpha_3}\Big\{\Big[-a_2\frac{\mathsf{x}_0}{\mathsf{x}_1} + 2a_2(a_2+1)\frac{x^2}{\mathsf{x}_1^2}\Big] + \Big[-a_4\frac{\mathsf{x}_0}{\mathsf{x}_2} + 2a_4(a_4+1)\frac{y^2}{\mathsf{x}_2^2}\Big] + 4a_2a_4\,\frac{xy}{\mathsf{x}_1\mathsf{x}_2}\Big\},$$

and since $-t + \frac{2}{3}t(t+1) = \mathsf{q}(t)$, the rules above give

$$\overline{F_2} = \mathsf{x}_1^{-\alpha_1}\mathsf{x}_2^{-\alpha_2}\mathsf{x}_3^{-\alpha_3}\Big\{\mathsf{q}(a_2)\,\mathsf{x}_1^{-1} + \mathsf{q}(a_4)\,\mathsf{x}_2^{-1} + \tfrac{2}{3}a_2a_4\big[\mathsf{x}_2^{-1} + \mathsf{x}_1^{-1} - \mathsf{x}_3\mathsf{x}_1^{-1}\mathsf{x}_2^{-1}\big]\Big\}.$$

In each case $\overline{F_2} \in \mathcal{L}_{-3}$, because $\alpha_1 + \alpha_2 + \alpha_3 = 2 + n$; and the exponent c of x_3 in the terms obtained is a_5 or $a_5 - 1$, which proves the assertion on $\mathsf{c} \le 1$. Finally, the monomial $(\mathsf{x}_1\mathsf{x}_2\mathsf{x}_3)^{-1}$ occurs for $n = 1$ exactly when $(\alpha_1,\alpha_2,\alpha_3) = (1,1,1)$, with coefficient 1. For $n = 0$ it arises from the terms containing x_1^{-1} exactly when $(\alpha_1,\alpha_2,\alpha_3) = (0,1,1)$, from the terms containing x_2^{-1} exactly when $(\alpha_1,\alpha_2,\alpha_3) = (1,0,1)$, and from the last term exactly when $(\alpha_1,\alpha_2,\alpha_3) = (0,0,2)$. Collecting the coefficients, we obtain eq. 7.xliv. □

Lemma 7.49 (The residue)**.** Let $\mathsf{M} = \mathsf{x}_1^{-\mathsf{a}}\mathsf{x}_2^{-\mathsf{b}}\mathsf{x}_3^{-\mathsf{c}}$ with integers $\mathsf{a},\mathsf{b},\mathsf{c}$ such that $\mathsf{a}+\mathsf{b}+\mathsf{c}=3$ and $\mathsf{c}\le 1$. Let $\phi\in C_c^\infty(\mathbb{R}^3\times\mathbb{R}^3)$ be equal to 1 near 0, let $\Lambda>1$, and let ϕ^Λ be as in Def. 7.44(3). Then $(\phi^\Lambda-\phi)\mathsf{M}\in\mathfrak{P}_2(c)$ for all sufficiently small $c>0$, and

$$\text{f. p.}\iint(\phi^\Lambda-\phi)\,\mathsf{M}\,\mathrm{d}k\,\mathrm{d}l=\begin{cases}4\pi^4\log\Lambda & (\mathsf{a},\mathsf{b},\mathsf{c})=(1,1,1),\\ 0 & \text{otherwise;}\end{cases}$$

$$\oint_{S^5}\frac{\mathrm{d}\omega}{\mathsf{x}_1\mathsf{x}_2\mathsf{x}_3}=4\pi^4, \tag{7.xlv}$$

$$\frac{4\pi^4}{(2\pi)^6}=\frac{1}{16\pi^2},$$

where $d\omega$ denotes the surface measure of the sphere $S^5\subset\mathbb{R}^3\times\mathbb{R}^3$. ■

Proof. Step 0: the class. Let $\Upsilon_0\in C_c^\infty(\mathbb{R}^6)$ vanish for $|k|+|l|<c$. We show that $\Upsilon_0\mathsf{M}\in\mathfrak{P}_2(c)$. Condition ($\Phi$0) holds, since Υ_0 vanishes for $|k|+|l|<c$ and has compact support. If $|k|<\frac{c}{3}$ and $\Upsilon_0(k,l)\neq 0$, then $|l|\ge c-|k|>\frac{2}{3}c$ and $|k-l|\ge|l|-|k|>\frac{c}{3}$. Hence $\Upsilon_0\,\mathsf{x}_2^{-\mathsf{b}}\mathsf{x}_3^{-\mathsf{c}}$, extended by 0 where Υ_0 vanishes, is smooth on $\{|k|<\frac{c}{3}\}\times\mathbb{R}^3$ (it vanishes on the open set $\{|k|+|l|<c\}$, and it is smooth on the open set $\{|k|<\frac{c}{3},\ |l|>\frac{2}{3}c\}$), and it vanishes where $|k|+|l|<c$ or where (k,l) lies outside the support of Υ_0.

Thus (Φ1) holds with the single homogeneous factor $|k|^{-2\mathsf{a}}$, of degree $-2\mathsf{a}$. Symmetrically, if $|l|<\frac{c}{3}$ and $\Upsilon_0(k,l)\neq 0$, then $|k|\ge c-|l|>\frac{2}{3}c$ and $|k-l|\ge|k|-|l|>\frac{c}{3}$. Hence $\Upsilon_0\,\mathsf{x}_1^{-\mathsf{a}}\mathsf{x}_3^{-\mathsf{c}}$, extended by 0 where Υ_0 vanishes, is smooth on $\mathbb{R}^3\times\{|l|<\frac{c}{3}\}$ (it vanishes on the open set $\{|k|+|l|<c\}$, and it is smooth on the open set $\{|k|>\frac{2}{3}c,\ |l|<\frac{c}{3}\}$), and it vanishes where $|k|+|l|<c$ or where (k,l) lies outside the support of Υ_0. Thus (Φ2) holds with the single homogeneous factor $|l|^{-2\mathsf{b}}$, of degree $-2\mathsf{b}$.

Condition (Φ3) holds because $\mathsf{c}\le 1$ and Υ_0 has compact support. We apply this to $\Upsilon_0=\phi^\Lambda-\phi$, which vanishes near 0, since ϕ and ϕ^Λ are both equal to 1 there.

Step 1: independence of the cut-off. Let Υ_0 be as in Step 0. For large $\mathrm{Re}\,\mathsf{z}$ the substitution $(k,l)\to(\Lambda k,\Lambda l)$ and the homogeneity of degree -6 of M give

$$\iint(|k||l|)^{2\mathsf{z}}\,\Upsilon_0^\Lambda\,\mathsf{M}\,\mathrm{d}k\,\mathrm{d}l=\Lambda^{4\mathsf{z}+6-6}\iint(|k||l|)^{2\mathsf{z}}\,\Upsilon_0\,\mathsf{M}\,\mathrm{d}k\,\mathrm{d}l.$$

Since Υ_0^Λ also vanishes for $|k|+|l|<c$, Lem. 7.45(e), with $(\epsilon,\epsilon')=(1,1)$, continues both sides holomorphically to $\mathbb{D}$, and at $\mathsf{z}=0$ we obtain $\text{f. p.}\iint\Upsilon_0^\Lambda\mathsf{M}=\text{f. p.}\iint\Upsilon_0\mathsf{M}$. If $\tilde\phi$ is a second function with the properties of ϕ, we take $\Upsilon_0:=\phi-\tilde\phi$; then

$$(\phi^\Lambda-\phi)-(\tilde\phi^\Lambda-\tilde\phi)=\Upsilon_0^\Lambda-\Upsilon_0,$$

and the finite part of $(\phi^\Lambda-\phi)\mathsf{M}$ does not depend on the choice of ϕ. We may therefore choose ϕ conveniently in each case. In the following steps we use the function $\varphi(s):=\theta(s)$ of one variable $s\ge 0$; it is equal to 1 on $[0,\frac{1}{2}]$ and to 0 on $[1,\infty)$.

Step 2: the case $\mathsf{c}\le 0$. We choose $\phi(k,l):=\varphi(|k|)\varphi(|l|)$. Since $\mathsf{x}_3^{-\mathsf{c}}=(k^2-2k\cdot l+l^2)^{|\mathsf{c}|}$ is a polynomial, M is a finite sum

$$\mathsf{M}=\sum c_{\alpha\beta}\,|k|^{-2\mathsf{a}_\alpha}k^\alpha\cdot|l|^{-2\mathsf{b}_\beta}l^\beta$$

with numbers $c_{\alpha\beta}$, multi-indices α,β and integers $\mathsf{a}_\alpha,\mathsf{b}_\beta$. For large $\mathrm{Re}\,\mathsf{z}$ put

$$\mathsf{Z}(\mathsf{z}):=\iint(|k||l|)^{2\mathsf{z}}\,\phi\,\mathsf{M}\,\mathrm{d}k\,\mathrm{d}l=\sum c_{\alpha\beta}\int|k|^{2\mathsf{z}-2\mathsf{a}_\alpha}k^\alpha\varphi(|k|)\,\mathrm{d}k\int|l|^{2\mathsf{z}-2\mathsf{b}_\beta}l^\beta\varphi(|l|)\,\mathrm{d}l.$$

As M is homogeneous of degree -6, we have $\iint(|k||l|)^{2\mathsf{z}}\phi^\Lambda\mathsf{M}\,\mathrm{d}k\,\mathrm{d}l=\Lambda^{4\mathsf{z}}\mathsf{Z}(\mathsf{z})$. The functions $|k|^{-2\mathsf{a}_\alpha}k^\alpha$ and $|l|^{-2\mathsf{b}_\beta}l^\beta$ are homogeneous factors, so by Lem. 7.45(a) each factor in the expression

for Z continues holomorphically to $\mathbb{D}$. Hence the continuation of $\iint(|k||l|)^{2\mathsf{z}}(\phi^\Lambda-\phi)\mathsf{M}\,\mathrm{d}k\,\mathrm{d}l$ is $(\Lambda^{4\mathsf{z}}-1)\mathsf{Z}(\mathsf{z})$, which vanishes at $\mathsf{z}=0$; by Lem. 7.45(e) this value is the finite part.

Step 3: the case $\mathsf{c}=1$, $(\mathsf{a},\mathsf{b})\neq(1,1)$. Now $\mathsf{a}+\mathsf{b}=2$. The finite part is symmetric under $k\leftrightarrow l$ (Def. 7.44(2)), and ϕ may be chosen freely; we may therefore assume $\mathsf{b}\le 0$ and $\mathsf{a}\ge 2$. The degree of the homogeneous factor in (Φ2) is then $-2\mathsf{b}\ge 0$, so that Lem. 7.45(e) applies with $(\epsilon,\epsilon')=(1,0)$. We choose $\phi(k,l):=\varphi(|k|)\varphi(|l-k|)$ and write $\mathsf{m}:=l-k$. Then

$$\mathsf{M}=|k|^{-2\mathsf{a}}\,|k+\mathsf{m}|^{2|\mathsf{b}|}\,|\mathsf{m}|^{-2}=\sum c_{\alpha\beta}\,|k|^{-2\mathsf{a}}k^\alpha\cdot\mathsf{m}^\beta|\mathsf{m}|^{-2},$$

and for large $\operatorname{Re}\mathsf{z}$

$$\mathsf{Z}(\mathsf{z}):=\iint|k|^{2\mathsf{z}}\,\phi\,\mathsf{M}\,\mathrm{d}k\,\mathrm{d}l=\sum c_{\alpha\beta}\int|k|^{2\mathsf{z}-2\mathsf{a}}k^\alpha\varphi(|k|)\,\mathrm{d}k\int\mathsf{m}^\beta|\mathsf{m}|^{-2}\varphi(|\mathsf{m}|)\,\mathrm{d}\mathsf{m},$$

while $\iint|k|^{2\mathsf{z}}\phi^\Lambda\mathsf{M}\,\mathrm{d}k\,\mathrm{d}l=\Lambda^{2\mathsf{z}}\mathsf{Z}(\mathsf{z})$. The integrals over m converge absolutely, and the integrals over k continue to $\mathbb{D}$ by Lem. 7.45(a). The continuation $(\Lambda^{2\mathsf{z}}-1)\mathsf{Z}(\mathsf{z})$ vanishes at $\mathsf{z}=0$.

Step 4: the case $(\mathsf{a},\mathsf{b},\mathsf{c})=(1,1,1)$. We choose $\phi(k,l):=\varphi((k^2+l^2)^{1/2})$. All degrees of the homogeneous factors in (Φ1) and (Φ2) are now -2; hence $(\phi^\Lambda-\phi)\mathsf{M}$ is integrable, and by Lem. 7.45(e) its finite part is its integral. In polar coordinates of $\mathbb{R}^6$,

$$\iint(\phi^\Lambda-\phi)\,\mathsf{M}\,\mathrm{d}k\,\mathrm{d}l=\oint_{S^5}\mathsf{M}\,\mathrm{d}\omega\cdot\int_0^\infty[\varphi(s/\Lambda)-\varphi(s)]\,\frac{\mathrm{d}s}{s}.$$

The integrand on the right vanishes for $s\le\frac12$, and for $0<\varepsilon\le\frac12$

$$\int_\varepsilon^\infty[\varphi(s/\Lambda)-\varphi(s)]\frac{ds}{s}=\int_{\varepsilon/\Lambda}^{\varepsilon}\varphi(s)\,\frac{ds}{s}=\log\Lambda.$$

Together with the value of the integral over S^5, computed in Step 5, this gives $4\pi^4\log\Lambda$.

Step 5: the integral over the sphere. We parametrise S^5 by $k=\cos\psi\,\omega_1$, $l=\sin\psi\,\omega_2$ with $\psi\in(0,\frac{\pi}{2})$ and $\omega_1,\omega_2\in S^2$. In $\mathbb{R}^6$ we have $dk\,\mathrm{d}l=\mathsf{r}_1^2d\mathsf{r}_1\,\mathrm{d}\omega_1\,\mathsf{r}_2^2d\mathsf{r}_2\,\mathrm{d}\omega_2$, with $\mathsf{r}_1=s\cos\psi$, $\mathsf{r}_2=s\sin\psi$ and $d\mathsf{r}_1\,\mathrm{d}\mathsf{r}_2=s\,\mathrm{d}s\,\mathrm{d}\psi$; hence

$$d\omega=\cos^2\psi\,\sin^2\psi\,\mathrm{d}\psi\,\mathrm{d}\omega_1\,\mathrm{d}\omega_2.$$

On S^5 we have $\mathsf{x}_1\mathsf{x}_2=\cos^2\psi\sin^2\psi$ and $\mathsf{x}_3=1-\sin2\psi\;\omega_1\cdot\omega_2$. For Borel functions $g\ge0$ on $[-1,1]$ we have $\iint_{S^2\times S^2}g(\omega_1\cdot\omega_2)\,\mathrm{d}\omega_1\,\mathrm{d}\omega_2=4\pi\cdot2\pi\int_{-1}^1g(\tau)\,\mathrm{d}\tau$, and for $0\le z<1$

$$\int_{-1}^1\frac{d\tau}{1-z\tau}=\sum_{j\ge0}z^j\int_{-1}^1\tau^j\,\mathrm{d}\tau=2\sum_{n\ge0}\frac{z^{2n}}{2n+1}.$$

Applying this with $z=\sin2\psi$ for $\psi\neq\frac{\pi}{4}$, and interchanging sum and integral, which is allowed since all terms are non-negative, we find

$$\oint_{S^5}\frac{d\omega}{\mathsf{x}_1\mathsf{x}_2\mathsf{x}_3}=8\pi^2\int_0^{\pi/2}2\sum_{n\ge0}\frac{\sin^{2n}2\psi}{2n+1}\,\mathrm{d}\psi=16\pi^2\sum_{n\ge0}\frac{\mathsf{w}_n}{2n+1},\qquad\mathsf{w}_n:=\int_0^{\pi/2}\sin^{2n}2\psi\,\mathrm{d}\psi$$
$$=\int_0^{\pi/2}\sin^{2n}\psi\,\mathrm{d}\psi.$$

An integration by parts gives $2n\,\mathsf{w}_n=(2n-1)\,\mathsf{w}_{n-1}$, and $\mathsf{w}_0=\frac{\pi}{2}$. Hence $\mathsf{w}_n=\frac{\pi}{2}\,\mathsf{b}_n$ with

$$\mathsf{b}_n:=\prod_{j=1}^n\frac{2j-1}{2j},$$

the coefficients of the binomial series $(1-t^2)^{-1/2}=\sum_n\mathsf{b}_nt^{2n}$, $|t|<1$. Integrating this series over $[0,x]$ and letting $x\uparrow1$ (monotone convergence), we obtain

$$\sum_n\frac{\mathsf{b}_n}{2n+1}=\int_0^1(1-t^2)^{-1/2}\,\mathrm{d}t=\frac{\pi}{2},$$

where we have substituted $t = \sin\psi$, $0 \le \psi < \frac{\pi}{2}$, so that $(1-t^2)^{-1/2}dt = d\psi$. Therefore the integral over S^5 equals $16\pi^2 \cdot \frac{\pi}{2} \cdot \frac{\pi}{2} = 4\pi^4$, and $4\pi^4/(2\pi)^6 = 1/16\pi^2$. □

Proposition 7.50 (The bridge: $\kappa_\Gamma^{\mathrm{lat}}$ from $\mathcal{R}[F_\Gamma]$)**.** Let $\Gamma \in \{1,\dots,6\}$ and $\Lambda > 1$; let $\mathfrak{f}_\Gamma$ be as in Prop. 7.46, R_Γ^Λ as in Lem. 7.43(b), and $\kappa_\Gamma^{\mathrm{lat}}$ as in Lem. 7.36. Then

$$\mathfrak{f}_\Gamma[R_\Gamma^\Lambda - R_\Gamma] = d_{2,\Gamma}\log\Lambda, \quad d_{2,\Gamma} := \frac{\mathcal{R}[F_\Gamma]}{16\pi^2}; \qquad \kappa_\Gamma^{\mathrm{lat}} = -\frac{3}{2\pi}\cdot 24\cdot d_{2,\Gamma} = -\frac{3}{32\pi^3}\cdot 24\cdot\mathcal{R}[F_\Gamma]. \tag{7.xlvi}$$

■

Proof. Step 1: cut-offs. The function R_Γ is smooth, even and rotation invariant. If $|k|+|l| < \frac12\rho_0$ and $|\mathsf{p}| < \frac12\rho_0$, then each of the vectors k, l, $k-l$, $\mathsf{p}-k$, $\mathsf{p}-l$, among which are the lines of Γ, has length less than ρ_0, and therefore $R_\Gamma = 1 = R_\Gamma^\Lambda$ there. If $R_\Gamma \neq 0$, then $|l| < 2\rho_0$ and $|k| < 4\rho_0$: for $\Gamma \neq 6$ because l and k are lines of Γ, and for $\Gamma = 6$ because l and $k-l$ are. Put $\phi(k,l) := \theta((k^2+l^2)^{1/2})$; then $\phi = 1$ for $|k|+|l| \le \frac12$. Consequently the functions

$$\Theta := R_\Gamma^\Lambda - R_\Gamma, \qquad \Upsilon_0 := R_\Gamma - \phi, \qquad \phi^\Lambda - \phi,$$

regarded as functions of (k,l,p), belong to $\mathcal{Y}(c)$ with $c := \frac12\rho_0$. Indeed, they are smooth and satisfy (Y2); for $|\mathsf{p}| < c$ and $|k|+|l| < c$ they vanish, because R_Γ, R_Γ^Λ, ϕ and ϕ^Λ are all equal to 1 there ($c < \frac12$); and they vanish when $|k|+|l|$ is large, because R_Γ, R_Γ^Λ, ϕ and ϕ^Λ do. We note that, by Lem. 7.45(e), the finite part, and with it $\mathfrak{f}_\Gamma$, does not depend on the parameter c with respect to which an element of $\mathcal{Y}(c)$ is considered.

Step 2: $\mathfrak{f}_\Gamma[\Theta] = -\frac{\pi}{36}\kappa_\Gamma^{\mathrm{lat}}\log\Lambda$. The function $\Xi_\Gamma[\Theta](\mathsf{p})$ is the left side of the identity in Lem. 7.43(b). We thus have two expansions of it: that of Lem. 7.43(b), valid for almost every p with $|\mathsf{p}| < 1$, and that of Prop. 7.46, valid for every p with $0 < |\mathsf{p}| \le \frac{c}{16}$. Since the remainder $\Lambda^2\mathsf{o}(\mathsf{p}/\Lambda) - \mathsf{o}(\mathsf{p})$ of the former is $o(\mathsf{p}^2)$, they give

$$(\Lambda^2-1)\,\mathsf{a}_0 + (\Lambda-1)\,\mathsf{a}_1|\mathsf{p}| - \tfrac{\pi}{36}\,\kappa_\Gamma^{\mathrm{lat}}\log\Lambda\;\mathsf{p}^2 + o(\mathsf{p}^2) = \alpha + \beta\,|\mathsf{p}| + \mathfrak{f}_\Gamma[\Theta]\,\mathsf{p}^2 + o(\mathsf{p}^2).$$

Along a sequence $\mathsf{p} \to 0$ of points at which both expansions hold, we compare first the constant terms, then the coefficients of $|\mathsf{p}|$, and then those of p^2.

Step 3: $\mathfrak{f}_\Gamma[\Theta] = \mathfrak{f}_\Gamma[\phi^\Lambda - \phi]$. We have $\Theta = (\phi^\Lambda - \phi) + (\Upsilon_0^\Lambda - \Upsilon_0)$. By the linearity of $\mathfrak{f}_\Gamma$ and the identity $\mathfrak{f}_\Gamma[\Upsilon_0^\Lambda] = \mathfrak{f}_\Gamma[\Upsilon_0]$ of Prop. 7.46, the second difference contributes nothing.

Step 4: the value. Since ϕ does not depend on p, we have $\overline{((\phi^\Lambda-\phi)F_\Gamma)_2} = (\phi^\Lambda-\phi)\,\overline{(F_\Gamma)_2}$. By Def. 7.47(3) and Lem. 7.48, $\overline{(F_\Gamma)_2} = \sum_{\mathsf{M}} r_{\mathsf{M}}\mathsf{M}$ is a finite linear combination of monomials M as in Lem. 7.49, and the coefficient of $\mathsf{M} = (\mathsf{x}_1\mathsf{x}_2\mathsf{x}_3)^{-1}$ is $r_{\mathsf{M}} = \mathcal{R}[F_\Gamma]$. The finite part is linear and independent of the parameter c (Lem. 7.45(e)); hence eq. 7.xlv yields

$$\mathfrak{f}_\Gamma[\phi^\Lambda - \phi] = (2\pi)^{-6}\cdot 4\pi^4\log\Lambda\cdot\mathcal{R}[F_\Gamma] = d_{2,\Gamma}\log\Lambda.$$

Step 5: conclusion. By Steps 2–4, $\mathfrak{f}_\Gamma[R_\Gamma^\Lambda - R_\Gamma] = \mathfrak{f}_\Gamma[\Theta] = d_{2,\Gamma}\log\Lambda$ and $-\frac{\pi}{36}\kappa_\Gamma^{\mathrm{lat}} = d_{2,\Gamma}$. Hence

$$\kappa_\Gamma^{\mathrm{lat}} = -\frac{36}{\pi}\,d_{2,\Gamma} = -\frac{3}{2\pi}\cdot 24\cdot d_{2,\Gamma},$$

and $\frac{3}{2\pi}\cdot\frac{1}{16\pi^2} = \frac{3}{32\pi^3}$ gives the last expression in eq. 7.xlvi. □

Proposition 7.51 (The six numbers $\mathcal{R}[F_\Gamma]$)**.** For the integrands $F_1,\dots,F_6$ of eq. 7.xxxv,

$$\mathcal{R}[F_1] = -\tfrac43, \quad \mathcal{R}[F_2] = -\tfrac{13}{12}, \quad \mathcal{R}[F_3] = -\tfrac{1}{12}, \quad \mathcal{R}[F_4] = 0, \quad \mathcal{R}[F_5] = +\tfrac{13}{6}, \quad \mathcal{R}[F_6] = -\tfrac13; \qquad \sum_{\Gamma=1}^{6}\mathcal{R}[F_\Gamma] = -\tfrac23. \tag{7.xlvii}$$

■

Proof. Step 0: notation. Throughout this proof we use the abbreviations

$$A,\ A',\ B,\ B',\ C,\ P := \mathsf{x}_1,\ \mathsf{x}'_1,\ \mathsf{x}_2,\ \mathsf{x}'_2,\ \mathsf{x}_3,\ \mathsf{x}_0,$$

and we put $k' := k - \mathsf{p}$, $l' := l - \mathsf{p}$ and $q := k - l = k' - l'$, so that $A' = k'^2$, $B' = l'^2$ and $C = q^2$. The relations $P = (k-k')^2 = (l-l')^2$, $C = (k-l)^2 = (k'-l')^2$, $k\cdot\mathsf{p} = A - k\cdot k'$ and $l\cdot\mathsf{p} = B - l\cdot l'$ express the remaining scalar products through these variables:

$$\begin{aligned} a &:= k\cdot k' = \tfrac12(A + A' - P),\\ b &:= l\cdot l' = \tfrac12(B + B' - P),\\ e &:= k\cdot l = \tfrac12(A + B - C),\\ e' &:= k'\cdot l' = \tfrac12(A' + B' - C),\\ f &:= k\cdot l' = e - k\cdot\mathsf{p} = \tfrac12(A' + B - C - P),\\ f' &:= k'\cdot l = e - l\cdot\mathsf{p} = \tfrac12(A + B' - C - P). \end{aligned}$$

We shall use two symmetries. The exchange of primes, $(k,l) \leftrightarrow (k',l')$, interchanges $A \leftrightarrow A'$, $B \leftrightarrow B'$, $e \leftrightarrow e'$, $f \leftrightarrow f'$ and fixes a and b. The exchange $(k,k') \leftrightarrow (l,l')$ interchanges $(A,A') \leftrightarrow (B,B')$, $a \leftrightarrow b$, $f \leftrightarrow f'$ and fixes e and e'. Finally, $\mathsf{q}(1) = \frac13$, $\mathsf{q}(0) = 0$ and $\mathsf{q}(-1) = 1$.

Step 1: two identities for V_3. Let

$$V := V_3(k_1,k_2,k_3)_{\alpha\beta\gamma} = \delta_{\alpha\gamma}\mathsf{a}_\beta + \delta_{\alpha\beta}\mathsf{b}_\gamma + \delta_{\beta\gamma}\mathsf{c}_\alpha, \qquad \mathsf{a} := k_1 - k_3, \quad \mathsf{b} := k_2 - k_1, \quad \mathsf{c} := k_3 - k_2.$$

(i) Since $\mathsf{a} + \mathsf{b} + \mathsf{c} = 0$,

$$V_{\alpha\beta\gamma}V_{\alpha\beta\gamma} = 3(\mathsf{a}^2+\mathsf{b}^2+\mathsf{c}^2)+2(\mathsf{a}\cdot\mathsf{b}+\mathsf{a}\cdot\mathsf{c}+\mathsf{b}\cdot\mathsf{c}) = 2(\mathsf{a}^2+\mathsf{b}^2+\mathsf{c}^2), \qquad \mathsf{a}^2+\mathsf{b}^2+\mathsf{c}^2 = 3\sum_i k_i^2 - \Big(\sum_i k_i\Big)^2.$$

(ii) If $k_1 + k_2 + k_3 = 0$, then

$$k_{1\alpha}V_{\alpha\beta\gamma} = (\mathsf{P}_2 - \mathsf{P}_3)_{\beta\gamma}, \qquad \mathsf{P}_i := k_i^2\,\delta_{\beta\gamma} - k_{i\beta}k_{i\gamma}.$$

Indeed, $k_{1\alpha}V_{\alpha\beta\gamma} = k_{1\gamma}(k_1-k_3)_\beta + k_{1\beta}(k_2-k_1)_\gamma + \delta_{\beta\gamma}\,k_1\cdot(k_3-k_2)$; here $k_{1\gamma}(k_1-k_3)_\beta + k_{1\beta}(k_2-k_1)_\gamma = k_{1\beta}k_{2\gamma} - k_{1\gamma}k_{3\beta}$, which equals $k_{3\beta}k_{3\gamma} - k_{2\beta}k_{2\gamma}$ because $k_1 = -k_2 - k_3$, and for the same reason $k_1\cdot(k_3 - k_2) = k_2^2 - k_3^2$.

Step 2: $\Gamma = 4$. By Def. 7.47(3), F_4 is a linear combination of monomials with $a_5 \le 0$, whereas every non-zero case of eq. 7.xliv has $\alpha_3 = a_5 \ge 1$. Hence $\mathcal{R}[F_4] = 0$.

Step 3: $\Gamma = 6$. Since V_3 is odd, $V_3(-l, l-k, k-\mathsf{p}) = -V_3(l, k-l, \mathsf{p}-k)$, and therefore

$$F_6 = +\tfrac16\, V_{\alpha\beta\gamma}V_{\alpha\beta\gamma}/A'BC, \qquad V := V_3(l, q, -k'),$$

where the three arguments of V_3 add up to $l+q-k' = \mathsf{p}$. By Step 1(i), $V_{\alpha\beta\gamma}V_{\alpha\beta\gamma} = 2\big[3(B+C+A')-P\big]$, hence

$$F_6 = \frac{1}{A'C} + \frac{1}{A'B} + \frac{1}{BC} - \frac13\,\frac{P}{A'BC}.$$

We evaluate $\mathcal{R}$ by eq. 7.xliv. The monomial $1/A'C$ has $n = 0$, $(\alpha_1,\alpha_2,\alpha_3) = (1,0,1)$ and $a_2 = 1$, $a_4 = 0$, so its value is $\mathsf{q}(0) = 0$; the monomial $1/A'B$ has $(\alpha_1,\alpha_2,\alpha_3) = (1,1,0)$ and value 0; the monomial $1/BC$ has $n = 0$, $(\alpha_1,\alpha_2,\alpha_3) = (0,1,1)$ and $a_2 = 0$, so its value is $\mathsf{q}(0) = 0$; and $P/A'BC$ has $n = 1$ and $(\alpha_1,\alpha_2,\alpha_3) = (1,1,1)$, so its value is 1. Therefore $\mathcal{R}[F_6] = -\frac13$.

Step 4: $\Gamma = 2, 3$, a common factor. Put

$$T_{\mu\nu} := V_2(k, \mathsf{p}-k)_{\mu\nu} = V_2(k, -k')_{\mu\nu} = a\,\delta_{\mu\nu} - k_\nu k'_\mu;$$

as V_2 is even, also $V_2(-k, k-\mathsf{p})_{\rho\nu} = T_{\rho\nu}$. Let

$$H_{\mu\rho} := T_{\mu\nu}T_{\rho\nu}/A' = \big[a^2\delta_{\mu\rho} - a\,(k_\rho k'_\mu + k_\mu k'_\rho) + A\,k'_\mu k'_\rho\big]/A',$$

and $V := V_3(-k, l, q)_{\mu\beta\gamma}$, so that $V_3(k, -l, -q) = -V$. In this notation

$$F_2 = \tfrac12\, H_{\mu\rho}V_{\mu\beta\gamma}V_{\rho\beta\gamma}/A^2BC, \qquad F_3 = -H_{\mu\rho}\,l_\rho(l-k)_\mu/A^2BC,$$

and p enters these expressions through H only. We compute $\overline{(H_{\mu\rho})_2}$. We insert $k' = k - \mathsf{p}$, $a = A - x$ and $A' = A - 2x + P$, with $x := k \cdot \mathsf{p}$, into H and expand: the term of second order in p is a sum of products of $\mathsf{p}_i \mathsf{p}_j$ with tensors built from δ and k, the coefficients being powers of A. Applying $\frac{1}{6}\Delta_{\mathsf{p}}$, we obtain

$$\overline{(H_{\mu\rho})_2} = \mathsf{a}_1\, \delta_{\mu\rho} + \mathsf{a}_2\, k_\mu k_\rho / A$$

with numbers $\mathsf{a}_1, \mathsf{a}_2$, since H is homogeneous of degree 2 in (k, p). Next, $k_\mu T_{\mu\nu} = a\, k_\nu - k_\nu (k \cdot k') = 0$ for all p; hence $k_\mu \overline{(H_{\mu\rho})_2} = 0$, which gives $\mathsf{a}_2 = -\mathsf{a}_1$. To determine a_1 we take the trace. From $T_{\mu\nu} T_{\mu\nu} = 3a^2 - 2a^2 + AA'$ we get $H_{\mu\mu} = a^2/A' + A$, and with $a = A - x$, $A' = A - 2x + P$,

$$\frac{a^2}{A'} = \Big(A - 2x + \frac{x^2}{A}\Big)\Big(1 + \frac{2x}{A} - \frac{P}{A} + \frac{4x^2}{A^2} + O(|\mathsf{p}|^3)\Big),$$

whose term of second order is $-P + 4x^2/A - 4x^2/A + x^2/A$; its image under $\frac{1}{6}\Delta_{\mathsf{p}}$ is $-1 + \frac{1}{3}$. Since the trace of $\overline{(H_{\mu\rho})_2}$ is $3\mathsf{a}_1 + \mathsf{a}_2 = 2\mathsf{a}_1$, we obtain $2\mathsf{a}_1 = -\frac{2}{3}$, that is,

$$\overline{(H_{\mu\rho})_2} = -\tfrac{1}{3}\,(\delta_{\mu\rho} - k_\mu k_\rho / A).$$

Step 5: $\Gamma = 2$. By Step 4,

$$\overline{(F_2)_2} = -\tfrac{1}{6}\,\mathsf{W}/A^2BC, \qquad \mathsf{W} := V_{\mu\beta\gamma} V_{\mu\beta\gamma} - |k_\mu V_{\mu\beta\gamma}|^2 / A.$$

The function $A\mathsf{W}$ is a polynomial in A, B, C, homogeneous of degree 2, and $\overline{(F_2)_2} = -\frac{1}{6} A\mathsf{W}/A^3BC$. A monomial $A^i B^j C^m$ of $A\mathsf{W}$ thus contributes a multiple of $A^{i-3}B^{j-1}C^{m-1}$, and only A^2 yields $(ABC)^{-1}$; we may therefore set $B = C = 0$. By Step 1(i), applied to the arguments $-k, l, q$, whose sum is 0, $V_{\mu\beta\gamma}V_{\mu\beta\gamma} = 6(A + B + C)$, which becomes $6A$. By Step 1(ii), applied with $(k_1, k_2, k_3) = (-k, l, q)$, we have $|k_\mu V_{\mu\beta\gamma}|^2 = \operatorname{tr}(\mathsf{P}_l - \mathsf{P}_q)^2$ with $\mathsf{P}_l = B\delta - ll^{\mathsf{T}}$ and $\mathsf{P}_q = C\delta - qq^{\mathsf{T}}$. Here $\operatorname{tr}\mathsf{P}_l^2 = 2B^2$, $\operatorname{tr}\mathsf{P}_q^2 = 2C^2$, $\operatorname{tr}\mathsf{P}_l\mathsf{P}_q = BC + (l \cdot q)^2$ and $l \cdot q = e - B = \frac{1}{2}(A - B - C)$; thus $|k_\mu V_{\mu\beta\gamma}|^2 = 2B^2 + 2C^2 - 2BC - 2(l \cdot q)^2$ is a polynomial in A, B, C, which becomes $-2(\frac{1}{2}A)^2 = -\frac{1}{2}A^2$ at $B = C = 0$.

Hence $A\mathsf{W}$ becomes $6A^2 + \frac{1}{2}A^2$, and

$$\mathcal{R}[F_2] = -\tfrac{1}{6} \cdot \tfrac{13}{2} = -\tfrac{13}{12}.$$

Step 6: $\Gamma = 3$. By Step 4,

$$\overline{(F_3)_2} = +\tfrac{1}{3}\,\mathsf{W}'/A^2BC, \qquad \mathsf{W}' := (\delta_{\mu\rho} - k_\mu k_\rho/A)\, l_\rho (l - k)_\mu = l \cdot (l - k) - \frac{(k \cdot l)\,(k \cdot (l - k))}{A}$$
$$= B - e - \frac{e(e - A)}{A}.$$

Again $A\mathsf{W}'$ is a polynomial in A, B, C, homogeneous of degree 2, and as in Step 5 only its coefficient of A^2 matters. At $B = C = 0$ we have $e = \frac{1}{2}A$, and $A\mathsf{W}'$ becomes $-\frac{1}{2}A^2 + \frac{1}{4}A^2$. Hence

$$\mathcal{R}[F_3] = \tfrac{1}{3} \cdot (-\tfrac{1}{4}) = -\tfrac{1}{12}.$$

Step 7: values for the denominator $AA'BC$. For a monomial $\mathsf{m} = P^n A^i A'^{i'} B^j B'^{j'} C^m$ of degree 2 we put $\mathsf{L}_5(\mathsf{m}) := \mathcal{R}[\mathsf{m}/AA'BC]$. In the notation of Lem. 7.48, the monomial $\mathsf{m}/AA'BC$ has the exponents $(a_1, \dots, a_5) = (1-i, 1-i', 1-j, -j', 1-m)$, and $(\alpha_1, \alpha_2, \alpha_3) = (2-i-i',\ 1-j-j',\ 1-m)$. By eq. 7.xliv, $\mathsf{L}_5(\mathsf{m}) \neq 0$ requires $m = 0$, and the values are as follows.

- $n = 1$ (then $i + i' = 1$, $j = j' = 0$): $\mathsf{L}_5(PA) = \mathsf{L}_5(PA') = 1$.
- $n = 0$, $(\alpha_1, \alpha_2, \alpha_3) = (0, 1, 1)$ (then $i + i' = 2$, $a_4 = 0$, value $\mathsf{q}(1 - i')$): $\mathsf{L}_5(A^2) = \frac{1}{3}$, $\mathsf{L}_5(AA') = 0$, $\mathsf{L}_5(A'^2) = 1$.
- $n = 0$, $(\alpha_1, \alpha_2, \alpha_3) = (1, 0, 1)$ (then $i + i' = 1$, $j + j' = 1$, value $\mathsf{q}(-j') - \frac{2}{3}(1 - i')j'$): $\mathsf{L}_5(AB) = \mathsf{L}_5(A'B) = 0$, $\mathsf{L}_5(AB') = 1 - \frac{2}{3} = \frac{1}{3}$, $\mathsf{L}_5(A'B') = 1$.

All other monomials of degree 2 have the value 0; in particular, every monomial containing B or C has the value 0.

Step 8: $\Gamma = 5$. Put

$$\tilde{T}_{\alpha\sigma} := V_2(k - \mathsf{p}, -k)_{\alpha\sigma} = V_2(k', -k)_{\alpha\sigma} = a\,\delta_{\alpha\sigma} - k_\alpha k'_\sigma.$$

Then $k'_\alpha \tilde{T}_{\alpha\sigma} = 0 = \tilde{T}_{\alpha\sigma} k_\sigma$; we write $\tilde{T}(X,Y) := X_\alpha \tilde{T}_{\alpha\sigma} Y_\sigma = a\,X\cdot Y - (k\cdot X)(k'\cdot Y)$, and we note $\tilde{T}_{\alpha\alpha} = 2a$. Further,

$$\begin{aligned} V_3(\mathsf{p}-k, l, k-l) &= V_3(-k', l, q) \\ &= \delta_{\alpha\gamma} u_{1\beta} + \delta_{\alpha\beta} v_{1\gamma} + \delta_{\beta\gamma} w_{1\alpha}, \\ u_1 = -k' - q, &\qquad v_1 = l + k', \qquad w_1 = q - l, \\ V_3(k, -l, l-k) &= \delta_{\sigma\gamma} u_{2\beta} + \delta_{\sigma\beta} v_{2\gamma} + \delta_{\beta\gamma} w_{2\sigma}, \\ u_2 = k + q, &\qquad v_2 = -(k+l), \qquad w_2 = l - q. \end{aligned}$$

Hence $F_5 = N_5/AA'BC$, where, after multiplying out the nine products and summing over β, γ,

$$\begin{aligned} N_5 = 2a\,(u_1\cdot u_2 + v_1\cdot v_2) + \tilde{T}(v_2,u_1) + \tilde{T}(u_1,w_2) + \tilde{T}(u_2,v_1) + \tilde{T}(v_1,w_2) + \tilde{T}(w_1,u_2) \\ + \tilde{T}(w_1,v_2) + 3\,\tilde{T}(w_1,w_2). \end{aligned}$$

The first argument of $\tilde{T}$ matters only modulo multiples of k', the second only modulo multiples of k. Using $q = k - l$ we find, in the first argument,

$$u_1 \equiv l - k, \qquad v_1 \equiv l, \qquad w_1 = k - 2l, \qquad u_2 = 2k - l, \qquad v_2 = -(k+l),$$

and in the second argument

$$u_1 \equiv l - k', \qquad v_1 = l + k', \qquad u_2 \equiv -l, \qquad v_2 \equiv -l, \qquad w_2 \equiv 2l.$$

Therefore

$$\begin{gathered} \tilde{T}(u_1,w_2) = -2\tilde{T}(k,l) + 2\tilde{T}(l,l), \\ \tilde{T}(v_1,w_2) = 2\tilde{T}(l,l), \\ \tilde{T}(w_1,u_2) = \tilde{T}(w_1,v_2) = -\tilde{T}(k,l) + 2\tilde{T}(l,l), \\ 3\,\tilde{T}(w_1,w_2) = 6\tilde{T}(k,l) - 12\tilde{T}(l,l), \\ \tilde{T}(v_2,u_1) + \tilde{T}(u_2,v_1) = -\tilde{T}(k+l,l-k') + \tilde{T}(2k-l,l+k') = \tilde{T}(k,l) - 2\tilde{T}(l,l) + 3\tilde{T}(k,k'), \end{gathered}$$

and the sum of these seven terms is $3\tilde{T}(k,l) - 6\tilde{T}(l,l) + 3\tilde{T}(k,k')$. We regard N_5 as a polynomial in A, A', B, B', C, P; by the last assertion of Step 7 we may set $B = C = 0$ in it. Then $e = \frac{1}{2}A$,

$f' = \frac12(A + B' - P)$, $k\cdot q = A - e = \frac12 A$ and $k'\cdot q = a - f'$, and with the values of Step 7 we find:

$$\begin{aligned}
\tilde T(k,l) = ae - Af' = \tfrac14 A\,(-A + A' - 2B' + P) :\\
\mathsf{L}_5 = \tfrac14(-\tfrac13 + 0 - \tfrac23 + 1) = 0;\\
\tilde T(l,l) = aB - ef' = -\tfrac14 A\,(A + B' - P) :\\
\mathsf{L}_5 = -\tfrac14(\tfrac13 + \tfrac13 - 1) = \tfrac{1}{12};\\
\tilde T(k,k') = a^2 - AA' = \tfrac14(A + A' - P)^2 - AA' :\\
\mathsf{L}_5 = \tfrac14(\tfrac13 + 1 - 2 - 2) = -\tfrac23;\\
u_1\cdot u_2 + v_1\cdot v_2\\
= -(a + k'\cdot q + k\cdot q + C) - (e + B + a + f')\\
= -(3a + A);\\
a\,(3a + A) = \tfrac34(A + A' - P)^2 + \tfrac12 A\,(A + A' - P) :\\
\mathsf{L}_5 = \tfrac34(-\tfrac83) + \tfrac12(\tfrac13 - 1) = -\tfrac73.
\end{aligned}$$

In the third line the four numbers in the bracket come from the monomials A^2, A'^2, $-2PA$, $-2PA'$ of $(A + A' - P)^2$, the others having the value 0; thus $\mathsf{L}_5((A + A' - P)^2) = \frac13 + 1 - 2 - 2 = -\frac83$, the value used in the last line. Consequently

$$\mathcal{R}[F_5] = -2(-\tfrac73) + 3\cdot 0 - 6\cdot\tfrac{1}{12} + 3(-\tfrac23) = \tfrac{14}{3} - \tfrac52 = \tfrac{13}{6}.$$

Step 9: values for the denominator $AA'BB'C$. For a monomial $\mathsf{m} = P^n A^i A'^{i'} B^j B'^{j'} C^m$ of degree 3 we put $\mathsf{L}(\mathsf{m}) := \mathcal{R}[\mathsf{m}/AA'BB'C]$. Now $(a_1,\dots,a_5) = (1-i, 1-i', 1-j, 1-j', 1-m)$ and $(\alpha_1,\alpha_2,\alpha_3) = (2-i-i',\ 2-j-j',\ 1-m)$, and again $\mathsf{L}(\mathsf{m}) \neq 0$ requires $m = 0$. Let $X \in \{A, A'\}$ and $Y \in \{B, B'\}$. The values are as follows.

- $n = 1$: $\mathsf{L}(PXY) = 1$.
- $n = 0$, $(\alpha_1,\alpha_2,\alpha_3) = (0,1,1)$ (then $i + i' = 2$, $j + j' = 1$, value $\mathsf{q}(1-i') + \frac23(1-i')(1-j')$): $\mathsf{L}(A^2B) = \frac13 + \frac23 = 1$, $\mathsf{L}(A^2B') = \frac13$, $\mathsf{L}(A'^2B) = 1 - \frac23 = \frac13$, $\mathsf{L}(A'^2B') = 1$, $\mathsf{L}(AA'Y) = 0$.
- $n = 0$, $(\alpha_1,\alpha_2,\alpha_3) = (1,0,1)$ (then $i + i' = 1$, $j + j' = 2$, value $\mathsf{q}(1-j') + \frac23(1-i')(1-j')$): $\mathsf{L}(AB^2) = 1$, $\mathsf{L}(A'B^2) = \frac13$, $\mathsf{L}(AB'^2) = \frac13$, $\mathsf{L}(A'B'^2) = 1$, $\mathsf{L}(XBB') = 0$.

All other monomials of degree 3 have the value 0: those containing C or P^2, and those of the forms PX_1X_2, PY_1Y_2, $X_1X_2X_3$, $Y_1Y_2Y_3$ with $X_i \in \{A, A'\}$, $Y_i \in \{B, B'\}$. These values are invariant under the exchange of primes and under $(A, A') \leftrightarrow (B, B')$.

Step 10: $\Gamma = 1$, reduction. As in Step 4, $T_{\mu\nu} = V_2(k, \mathsf{p} - k)_{\mu\nu} = a\,\delta_{\mu\nu} - k_\nu k'_\mu$, and $k_\mu T_{\mu\nu} = 0 = T_{\mu\nu}k'_\nu$. Similarly,

$$S_{\rho\sigma} := V_2(-l, l - \mathsf{p})_{\rho\sigma} = V_2(-l, l')_{\rho\sigma} = b\,\delta_{\rho\sigma} - l_\sigma l'_\rho, \qquad l_\rho S_{\rho\sigma} = 0 = S_{\rho\sigma} l'_\sigma.$$

The tensor $V_3(-k, l, q)_{\mu\rho\alpha} = \delta_{\mu\alpha}(l - 2k)_\rho + \delta_{\mu\rho}(k + l)_\alpha + \delta_{\rho\alpha}(k - 2l)_\mu$ is contracted with $T_{\mu\nu}S_{\rho\sigma}$, which annihilates the terms with l_ρ and with k_μ; in this contraction it may therefore be replaced by

$$\mathsf{W}_{\mu\rho\alpha} := -2k_\rho\,\delta_{\mu\alpha} + (k + l)_\alpha\,\delta_{\mu\rho} - 2l_\mu\,\delta_{\rho\alpha}.$$

Likewise, $V_3(k - \mathsf{p}, \mathsf{p} - l, l - k)_{\nu\sigma\alpha} = V_3(k', -l', l' - k')_{\nu\sigma\alpha} = \delta_{\nu\alpha}(2k' - l')_\sigma - \delta_{\nu\sigma}(k' + l')_\alpha + \delta_{\sigma\alpha}(2l' - k')_\nu$, contracted with $T_{\mu\nu}S_{\rho\sigma}$, may be replaced by $-\mathsf{W}'_{\nu\sigma\alpha}$, where W' denotes W with (k', l') in place of (k, l). Hence

$$F_1 = +\tfrac12\, N'/AA'BB'C, \qquad N' := T_{\mu\nu}S_{\rho\sigma}\,\mathsf{W}_{\mu\rho\alpha}\mathsf{W}'_{\nu\sigma\alpha}.$$

We use the bilinear forms

$$T(X,Y) := X_\mu T_{\mu\nu} Y_\nu = a\,X\cdot Y - (k'\cdot X)(k\cdot Y), \qquad S(X,Y) := X_\rho S_{\rho\sigma} Y_\sigma = b\,X\cdot Y - (l'\cdot X)(l\cdot Y),$$

the vectors $\mathsf{s}_\sigma := k_\rho S_{\rho\sigma} = (bk - fl)_\sigma$ and $\mathsf{t}_\rho := S_{\rho\sigma} k'_\sigma = (bk' - f'l')_\rho$, and the identities $T_{\mu\mu} = 2a$, $S_{\rho\rho} = 2b$, $T_{\mu\nu}S_{\mu\nu} = 3ab - ab - ab + ee' = ab + ee'$. Let N'_{ij} be the contribution to N' of the i-th term of W and the j-th term of W'. Then

$$\begin{aligned} N'_{11} &= 4T_{\mu\mu}S(k,k') = 8a(ab - ff'),\\ N'_{33} &= 4T(l,l')S_{\rho\rho} = 8b(ab - ff'),\\ N'_{22} &= (k+l)\cdot(k'+l')\,(ab + ee'),\\ N'_{13} &= 4T(\mathsf{s},l'),\\ N'_{31} &= 4T(l,\mathsf{t}),\\ N'_{12} &= -2T(k'+l',\mathsf{s}),\\ N'_{21} &= -2T(\mathsf{t},k+l),\\ N'_{23} &= -2\,T_{\mu\nu}l'_\nu\,S_{\mu\sigma}(k+l)_\sigma,\\ N'_{32} &= -2\,l_\mu T_{\mu\nu}\,(k'+l')_\rho S_{\rho\nu}. \end{aligned}$$

Under the exchange of primes, $T_{\mu\nu}$ goes over into $T_{\nu\mu}$, so that $T(X,Y)$ becomes the value of T at the images of Y and X, in this order, and s goes over into t; hence N'_{13} goes over into N'_{31}, and N'_{12} into N'_{21}. Under the exchange $(k,k') \leftrightarrow (l,l')$, the tensors $T_{\mu\nu}$ and $S_{\mu\nu}$ are interchanged; hence $N'_{12} = -2(k'+l')_\mu T_{\mu\nu}k_\rho S_{\rho\nu}$ goes over into N'_{32}, $N'_{21} = -2S_{\mu\sigma}k'_\sigma T_{\mu\nu}(k+l)_\nu$ into N'_{23}, and N'_{11} into N'_{33}. By the invariance of the values in Step 9,

$$\mathsf{L}(N'_{33}) = \mathsf{L}(N'_{11}), \qquad \mathsf{L}(N'_{31}) = \mathsf{L}(N'_{13}), \qquad \mathsf{L}(N'_{21}) = \mathsf{L}(N'_{32}) = \mathsf{L}(N'_{23}) = \mathsf{L}(N'_{12}).$$

Step 11: $\Gamma = 1$*, values.* By Step 9 we may set $C = 0$. Then

$$4(ab - ff') = (A + A' - P)(B + B' - P) - (A' + B - P)(A + B' - P) = AB' + A'B - AA' - BB' =: \mathsf{D}.$$

- $N'_{11} = (A + A' - P)\,\mathsf{D}$. Of its monomials only A^2B', A'^2B, $-PAB'$ and $-PA'B$ have non-zero values, and $\mathsf{L}(N'_{11}) = \frac{1}{3} + \frac{1}{3} - 1 - 1 = -\frac{4}{3}$.
- N'_{13}: since $\mathsf{s}\cdot l' = bf - fb = 0$ and $k'\cdot\mathsf{s} = ab - ff'$, we have $N'_{13} = -4f(ab - ff') = -\frac{1}{2}(A' + B - P)\,\mathsf{D}$. In $(A' + B - P)\,\mathsf{D}$ the monomials with non-zero values are A'^2B, $A'B^2$, $-PAB'$ and $-PA'B$; they give $\frac{1}{3} + \frac{1}{3} - 2 = -\frac{4}{3}$, and therefore $\mathsf{L}(N'_{13}) = \frac{2}{3}$.
- N'_{22}: here $(k+l)\cdot(k'+l') = a + f + f' + b = A + A' + B + B' - 2P$, and

$$4(ab + ee') = (A + A' - P)(B + B' - P) + (A + B)(A' + B') = \mathsf{E} + AA' + BB' - P(A + A' + B + B') + P^2, \qquad \mathsf{E} := AB + 2AB' + 2A'B + A'B'.$$

  We have $\mathsf{L}\big[(A+A'+B+B')\,\mathsf{E}\big] = (1+\frac{2}{3})+(\frac{2}{3}+1)+(1+\frac{2}{3})+(\frac{2}{3}+1) = \frac{20}{3}$, the four contributions coming from $A(AB+2AB')$, $A'(2A'B+A'B')$, $B(AB+2A'B)$ and $B'(2AB'+A'B')$. Further, $\mathsf{L}\big[(A + A' + B + B')(AA' + BB')\big] = 0$, $\mathsf{L}\big[-P(A + A' + B + B')^2\big] = -2\cdot 4 = -8$ and $\mathsf{L}[-2P\,\mathsf{E}] = -2\cdot 6 = -12$, while the remaining monomials, which contain P^2 or have one of the forms PX_1X_2, PY_1Y_2, have the value 0. Hence $\mathsf{L}(N'_{22}) = \frac{1}{4}\big(\frac{20}{3} - 20\big) = -\frac{10}{3}$.
- $N'_{12} = -2\big[a\,(k'+l')\cdot\mathsf{s} - \big(k'\cdot(k'+l')\big)(k\cdot\mathsf{s})\big]$. Since $(k'+l')\cdot\mathsf{s} = ab - ff'$, the first term is $a(ab - ff') = \frac{1}{8}(A + A' - P)\,\mathsf{D}$, with the value $\frac{1}{8}\cdot(-\frac{4}{3}) = -\frac{1}{6}$. Next, $k'\cdot(k'+l') = A' + e' = \frac{1}{2}(3A' + B')$ and

$$k\cdot\mathsf{s} = bA - fe = \tfrac{1}{4}\big[2A(B+B'-P) - (A'+B-P)(A+B)\big] = \tfrac{1}{4}\big[AB+2AB'-AA'-A'B-B^2-PA+PB\big].$$

  In $(3A'+B')\big[AB+2AB'-AA'-A'B-B^2-PA+PB\big]$ the monomials with non-zero values are $-3A'^2B$, $-3A'B^2$, $+3PA'B$, $+2AB'^2$ and $-PAB'$; they give $-1-1+3+\frac{2}{3}-1 = \frac{2}{3}$. The second term, $\frac{1}{8}(3A' + B')\big[\cdots\big]$, therefore has the value $\frac{1}{8}\cdot\frac{2}{3} = \frac{1}{12}$, and $\mathsf{L}(N'_{12}) = -2\big(-\frac{1}{6} - \frac{1}{12}\big) = \frac{1}{2}$.
- Summing, with the identities at the end of Step 10,

$$\mathsf{L}(N') = 2\big(-\tfrac{4}{3}\big) + 2\cdot\tfrac{2}{3} - \tfrac{10}{3} + 4\cdot\tfrac{1}{2} = -\tfrac{8}{3}, \qquad \text{hence} \qquad \mathcal{R}[F_1] = \tfrac{1}{2}\big(-\tfrac{8}{3}\big) = -\tfrac{4}{3}.$$

Step 12: the sum. Adding the six values obtained in Steps 2, 3, 5, 6, 8 and 11, we find

$$\sum_{\Gamma=1}^{6}\mathcal{R}[F_\Gamma]=-\tfrac{4}{3}-\tfrac{13}{12}-\tfrac{1}{12}+0+\tfrac{13}{6}-\tfrac{1}{3}=\frac{-16-13-1+26-4}{12}=-\tfrac{2}{3},$$

which completes the proof of eq. 7.xlvii. □

7.1.17. *The value: κ^{lat} and c_1.* Adding the six contributions of Prop. 7.50 and Prop. 7.51, we now obtain the value of the constant κ^{lat} of Thm. V, and with it the number c_1.

Theorem W (The value of κ^{lat})**.** The number κ^{lat} of Thm. V is

$$\kappa^{\mathrm{lat}}=\frac{3}{2\pi^3}.$$

■□

Proof. In eq. 7.xxxvii, κ^{lat} is the sum of the numbers $\kappa_\Gamma^{\mathrm{lat}}$ of Lem. 7.36. Expressing each of them by eq. 7.xlvi and inserting the sum from eq. 7.xlvii, we obtain

$$\kappa^{\mathrm{lat}}=\sum_{\Gamma=1}^{6}\kappa_\Gamma^{\mathrm{lat}}=-\frac{3}{32\pi^3}\cdot 24\cdot\sum_{\Gamma=1}^{6}\mathcal{R}[F_\Gamma]=-\frac{3}{32\pi^3}\cdot 24\cdot\Big(-\frac{2}{3}\Big)=\frac{48}{32\pi^3}=\frac{3}{2\pi^3},$$

as asserted. □

Definition 7.52 (The number c_1)**.** We put

$$c_1:=\frac{\kappa^{\mathrm{lat}}}{\mathsf{c}_2},$$

where κ^{lat} is the number of Thm. V and $\mathsf{c}_2=\frac{3\dim\mathfrak{g}}{16\pi^2}=\frac{9}{16\pi^2}$ is the leading constant fixed in §6.1; here $\dim\mathfrak{g}=3=\mathsf{n}$.

This fixes the coefficient used in §6.1; it does not by itself give a cutoff-uniform remainder for the interacting measure. ■

Theorem X (The number c_1)**.**

(a) $c_1=\dfrac{8}{3\pi}$, and $\mathsf{c}_2c_1=\dfrac{3}{2\pi^3}$.

(b) There is a pure constant C with the following property. Let $\mathsf{d}>0$, $\vartheta\in(0,1)$, $(f,h)\in\mathfrak{T}_{\mathbb{R}^3}(\mathsf{d};\vartheta)$ and $0<a\le\mathsf{d}$, the class of test pairs and the functional $\mathcal{K}_5$ being those of Cor. 7.37. Then

$$\Big|\frac{\mathsf{P}_1^{\mathrm{lat}}(f,h;a)}{\mathsf{c}_2\,\mathcal{K}_5(f,h)}-c_1\Big|\le C\,\vartheta^{-3}\,\frac{a}{\mathsf{d}}\,\lg\frac{\mathsf{d}}{a}.\tag{7.xlviii}$$

Consequently, $\lim_{a\to0}\mathsf{P}_1^{\mathrm{lat}}(f,h;a)/\mathsf{c}_2\mathcal{K}_5(f,h)=c_1$ for every such pair.

(c) For all plaquettes p,q of $\mathbb{Z}^3$, with n ranging over the integers for which $p+o_n,q+o_n\in\mathsf{Pl}_n$,

$$\mathsf{B}_1^{\mathrm{lat}}(p,q)=\lim_{n\to\infty}\lim_{\beta\to\infty}\ \beta^3\Big[\langle a_{p+o_n};a_{q+o_n}\rangle_{n,\beta}-\beta^{-2}\,\mathsf{B}_0^{[n]}(p+o_n,q+o_n)\Big].$$

The right side is gauge independent: no gauge fixing enters it.

■□

Proof. (a) By Thm. W,

$$c_1=\frac{3}{2\pi^3}\cdot\frac{16\pi^2}{9}=\frac{48}{18\pi}=\frac{8}{3\pi},$$

and $\mathsf{c}_2c_1=\kappa^{\mathrm{lat}}=\frac{3}{2\pi^3}$.

(b) Lem. 6.1 on $\mathbb{E}=\mathbb{R}^3$, with $q=5$, gives

$$\mathcal{K}_5(f,h)\ge 10^{-5}\,\vartheta^2\,\mathsf{d}\,\|f\|_\infty\|h\|_\infty,$$

and the right side is positive, since neither f nor h vanishes identically. We divide eq. 7.xxxviii by $\mathsf{c}_2\mathcal{K}_5(f,h)$ and use $\kappa^{\mathrm{lat}} = \mathsf{c}_2 c_1$. The left side of eq. 7.xlviii is then at most

$$C\,\vartheta^{-1}\,a\,\lg(\mathsf{d}/a)\cdot 10^5\,\mathsf{c}_2^{-1}\,\vartheta^{-2}\,\mathsf{d}^{-1},$$

which is the asserted bound, with a pure constant. Since $\frac{a}{\mathsf{d}}\lg\frac{\mathsf{d}}{a}\to 0$ as $a\to 0$, the limit follows.

(c) We apply eq. 7.xx at $\tau_{\mathrm{gf}}=1$ to the plaquettes $p+o_n, q+o_n\in\mathsf{Pl}_n$: for $\beta\geq 1$ the expression $\beta^3[\cdots]$ differs from $\mathsf{B}_1^{[n]}(p+o_n,q+o_n;1)$ by at most $C\beta^{-1}$, with a constant C depending on n, p and q. This gives the inner limit. By eq. 7.xxxix with $j=1$ and $\tau_{\mathrm{gf}}=1$, the outer limit equals $\mathsf{B}_1^{\mathrm{lat}}(p,q;1)$, which is $\mathsf{B}_1^{\mathrm{lat}}(p,q)$ by eq. 7.xix. The mean $\langle\cdot\rangle_{n,\beta}$ is taken in the Wilson measure of the box, and $\mathsf{B}_0^{[n]}=\frac{\mathsf{n}}{2}(\Pi^{\Lambda_n})^2$ by eq. 7.xvii, where Π^{Λ_n} does not depend on τ_{gf} (Def. 7.7(2)); thus no gauge fixing enters the right side. □

7.1.18. *What one shell adds; scope.* We close the section with two items. The first, Lem. 7.53, concerns a kernel $\mathsf{U}(u)=\mathsf{c}\,|u|^{-5}\varphi(u)$ with $\varphi(Lu)=\varphi(u)$ for a real number $L>1$, for instance $\kappa^{\mathrm{lat}}|u|^{-5}$, the case $\mathsf{c}=\kappa^{\mathrm{lat}}$, $\varphi\equiv 1$; it computes the part of the quadratic form $\frac12\iint w(x)\,w(y)\,\mathsf{U}(x-y)\,\mathrm{d}x\,\mathrm{d}y$ that comes from the shell $r\leq|x-y|<Lr$, which covers the scales from r to Lr. The second, Remark 7.54, delimits the results of the section.

Lemma 7.53 (What one shell adds)**.** Let $L>1$ and $\mathsf{c}\geq 0$ be real numbers, and let $\varphi\colon\mathbb{R}^3\setminus\{0\}\to\mathbb{R}$ be bounded, Borel and even, with $\varphi(Lu)=\varphi(u)$. Put $\mathsf{U}(u):=\mathsf{c}\,|u|^{-5}\varphi(u)$ and, for $r>0$,

$$\mathsf{Ann}_r:=\{u\in\mathbb{R}^3: r\leq|u|<Lr\}.$$

(a) For every $r>0$,

$$\int_{\mathsf{Ann}_r}\mathsf{U}(u)\,|u|^2\,\mathrm{d}u=\mathsf{c}\int_{S^2}\int_1^L\varphi(t\omega)\,\frac{dt}{t}\,\mathrm{d}\omega;\tag{7.xlix}$$

for $\varphi\equiv 1$ this equals $4\pi\,\mathsf{c}\log L$.

(b) Assume moreover that $\int_{\mathsf{Ann}_r}\mathsf{U}(u)\,u_iu_j\,\mathrm{d}u$ is a multiple of δ_{ij}; this is the case if φ is invariant under the signed permutations of the coordinates. Then for every real $w\in C_c^2(\mathbb{R}^3)$

$$\begin{aligned}\tfrac12\iint_{x-y\in\mathsf{Ann}_r} w(x)\,w(y)\,\mathsf{U}(x-y)\,\mathrm{d}x\,\mathrm{d}y=\\ \tfrac12\Big(\int_{\mathsf{Ann}_r}\mathsf{U}\Big)\int w^2\\ -\tfrac{1}{12}\Big(\int_{\mathsf{Ann}_r}\mathsf{U}(u)\,|u|^2\,\mathrm{d}u\Big)\int|\nabla w|^2+\mathrm{Rem},\\ |\mathrm{Rem}|\leq\tfrac{\pi}{24}\,\mathsf{c}\,\|\varphi\|_\infty\,(Lr)^2\,\|\Delta w\|_2^2.\end{aligned}\tag{7.l}$$

(c) If $\mathsf{c}=\mathsf{c}_2c_1\mathsf{b}^{-3}$ with $\mathsf{b}>0$, and $\varphi\equiv 1$, then the middle term on the right of eq. 7.l equals

$$-\frac{1}{2\pi^2}\,\mathsf{b}^{-1}\log L\int|\nabla(w/\mathsf{b})|^2.$$

■

Proof. (a) In polar coordinates the left side of eq. 7.xlix equals

$$\mathsf{c}\int_{S^2}\int_r^{Lr}t^{-5}\varphi(t\omega)\,t^2\,t^2\,\mathrm{d}t\,\mathrm{d}\omega=\mathsf{c}\int_{S^2}\int_r^{Lr}\varphi(t\omega)\,\frac{dt}{t}\,\mathrm{d}\omega.$$

For fixed ω the function $\tau\mapsto\varphi(e^\tau\omega)$ is bounded, Borel and periodic with period $\log L$; its integral over an interval of length $\log L$ is therefore the same for all such intervals, and

$$\int_r^{Lr}\varphi(t\omega)\,\frac{dt}{t}=\int_{\log r}^{\log r+\log L}\varphi(e^\tau\omega)\,\mathrm{d}\tau=\int_1^L\varphi(t\omega)\,\frac{dt}{t}.$$

For $\varphi\equiv 1$ the right side of eq. 7.xlix is $\mathsf{c}\cdot 4\pi\cdot\log L$.

(b) If φ is invariant under the signed permutations of the coordinates, then so are Ann_r and U. Changing the sign of u_i shows that the off-diagonal entries of the matrix $\big(\int_{\mathsf{Ann}_r} \mathsf{U}\, u_i u_j\big)_{ij}$ vanish, and permuting the coordinates shows that its diagonal entries are equal.

We turn to the expansion. The substitution $x = y+u$ (Fubini's theorem and translation invariance) turns the left side of eq. 7.l into

$$\tfrac{1}{2}\int_{\mathsf{Ann}_r} \mathsf{U}(u)\,\mathsf{a}(u)\,\mathrm{d}u, \qquad \mathsf{a}(u) := \int w(y+u)\,w(y)\,\mathrm{d}y.$$

Since $w(\cdot+u)$ has the transform $e^{ip\cdot u}\hat{w}(p)$, Plancherel's theorem gives

$$\mathsf{a}(u) = \int |\hat{w}(p)|^2\, e^{ip\cdot u}\, d^\circ p = \int |\hat{w}(p)|^2 \cos(p\cdot u)\, d^\circ p,$$

the second equality because a is real. For real ϑ', Taylor's formula with integral remainder and $|\cos^{(4)}| \le 1$ give $|\cos\vartheta' - 1 + \tfrac{1}{2}\vartheta'^2| \le \tfrac{1}{24}\vartheta'^4$. The functions w, $\partial_i w$ and Δw belong to L^2, with the transforms $\hat{w}$, $ip_i\hat{w}$ and $-|p|^2\hat{w}$, by the rule for the transform of a derivative, applied to w and to $\partial_i w \in C^1_c(\mathbb{R}^3)$. Plancherel's theorem therefore yields

$$\int |\hat{w}|^2\, d^\circ p = \int w^2, \qquad \int |\hat{w}|^2\, p_i p_j\, d^\circ p = \int \partial_i w\, \partial_j w, \qquad \int |\hat{w}|^2\, |p|^4\, d^\circ p = \|\Delta w\|_2^2,$$

and, with $\vartheta' = p\cdot u$ and $|p\cdot u| \le |p||u|$,

$$\Big|\mathsf{a}(u) - \int w^2 + \tfrac{1}{2}\sum_{i,j} u_i u_j \int \partial_i w\, \partial_j w\Big| \le \tfrac{1}{24}\,|u|^4\,\|\Delta w\|_2^2.$$

By the hypothesis of (b), taking the trace, $\int_{\mathsf{Ann}_r} \mathsf{U}\, u_i u_j = \tfrac{1}{3}\delta_{ij}\int_{\mathsf{Ann}_r} \mathsf{U}\,|u|^2$. Inserting the expansion of a, we obtain the right side of eq. 7.l: its first term is $\tfrac{1}{2}(\int_{\mathsf{Ann}_r}\mathsf{U})\int w^2$, its second term has the coefficient $\tfrac{1}{2}\cdot\tfrac{1}{2}\cdot\tfrac{1}{3} = \tfrac{1}{12}$, and the remainder satisfies

$$|\mathrm{Rem}| \le \tfrac{1}{2}\cdot\tfrac{1}{24}\,\|\Delta w\|_2^2 \int_{\mathsf{Ann}_r} |\mathsf{U}|\,|u|^4\,\mathrm{d}u \le \tfrac{1}{48}\,\|\Delta w\|_2^2\,\mathsf{c}\,\|\varphi\|_\infty\, 4\pi \int_r^{Lr} t\,\mathrm{d}t \le \tfrac{1}{48}\,\|\Delta w\|_2^2\,\mathsf{c}\,\|\varphi\|_\infty\, 4\pi\,\frac{(Lr)^2}{2},$$

which is the bound in eq. 7.l.

(c) By (a) and Thm. X(a), the middle term equals

$$-\tfrac{1}{12}\cdot 4\pi\cdot\frac{3}{2\pi^3}\,\mathsf{b}^{-3}\log L\int|\nabla w|^2 = -\frac{1}{2\pi^2}\,\mathsf{b}^{-3}\log L\int|\nabla w|^2,$$

and $\mathsf{b}^{-3}\int|\nabla w|^2 = \mathsf{b}^{-1}\int|\nabla(w/\mathsf{b})|^2$. □

Remark 7.54 (Scope of the results). The results of this section are delimited as follows.

(i) We make no statement about the cumulants of the smeared plaquette observables $\mathcal{O}_a(f)$ in Wilson measures on tori. In particular, it is not asserted that $\mathsf{B}_1^{\mathrm{lat}}$ is an expansion coefficient of these cumulants when $\beta = \mathsf{b}/a$ and the side of the torus, measured in lattice spacings, grow jointly as $a \to 0$.

(ii) The constant C of eq. 7.xx is not controlled as a function of n. Accordingly, the limits in Thm. X(c) are iterated limits, to be taken in the order written there: first $\beta\to\infty$, then $n\to\infty$.

(iii) Cor. 7.37 concerns test pairs, whose supports are at distance at least d from each other; it gives no information on the terms with $p = q$ of the double sum eq. 7.xix, on constant functions f, h, or on functions f, h with intersecting supports.

(iv) The group is $SU(2)$ throughout; nothing is asserted for $N > 2$.

(v) Of the term of degree -6 in the large-distance behaviour of $\mathsf{B}_1^{\mathrm{lat}}$, we prove only the bound eq. 7.xxxvii.

8. Natural scaling and failure of dilation covariance

The interacting comparison. The natural two-term laws and failure of dilation covariance use the nonlinear comparison in Prop. 6.18, through Lem. 8.8(iii) and Prop. 8.10. The physical label b has the dimension of a length. The natural small parameter in the preceding law is consequently d/b. Writing the error in this variable makes the range of the law independent of the reference coupling used to describe a fixed physical sequence.

For a test pair $\Psi = (f, h)$ put $|\Psi| = |f|_\infty |h|_\infty$ and write $\Psi_r = (f(\cdot/r), h(\cdot/r))$. The normalization of these weights is unchanged: they are not multiplied by r^{-3}. Hence

$$\mathcal{K}_q(\Psi_r) = r^{6-q}\mathcal{K}_q(\Psi).$$

In particular the leading $\mathcal{K}_6$ term is constant in r, while the next term is linear.

Sketch G (The natural-scale law). For every volume-first limit in the stated short-distance coupling range and every canonical component, the two-point function S satisfies

$$|\mathsf{S}(\Psi) - L_b^+(\Psi)| \le \varepsilon^{\mathrm{nat}}(d/b) b^{-3} d|\Psi|, \qquad \varepsilon^{\mathrm{nat}}(\delta) \le C\delta^{1/2},$$

for the fixed-shape pairs at $0 < d \le \delta_c b$. The constants depend on the admitted construction and the shape, not on the component. On a fixed torus there is the corresponding law at every positive label, in its stated range. None of these two-point functions is dilation covariant. ■□

Sketch of the argument. The transfer to natural scales uses the exact dilation of the lattices and Wilson measures. Dilating a sequence moves its label and its observation scale by the same factor. One may therefore move to an admitted reference window where the short-distance estimate is available, apply it there, and carry the result back. The ratio d/b is unchanged. The comparison with the least admissible torus then transfers the law to the periodic-state limit and its components. The error constants must be uniform in the component for this last assertion.

For a fixed nonzero positive test pair at scale d_1, put

$$A = c_2 b^{-2}\mathcal{K}_6(\Psi), \qquad B = \frac{3}{2\pi^3} b^{-3}\mathcal{K}_5(\Psi) > 0.$$

The law gives

$$\mathsf{S}(\Psi_r) = A + Br + O(r^{3/2}).$$

This exhibits two distinct homogeneities with nonzero coefficients. Dilation covariance, in the sense $\mathsf{S}(\Phi_r) = r^{\varpi}\mathsf{S}(\Phi)$ for all separated pairs and all $r > 0$, is incompatible with them. The nonzero constant limit fixes the possible exponent, and the nonzero linear term excludes that exponent.

A three-scale combination isolates the correction without subtracting A separately. For $t > 1$, define

$$\mathcal{N}_t(\Psi; r) = \mathsf{S}(\Psi_{t^2 r}) - (1 + t^6)\mathsf{S}(\Psi_{tr}) + t^6\mathsf{S}(\Psi_r).$$

Its weights annul constants and terms proportional to r^6. For the linear correction they give $-t(t-1)(t^5-1)Br$. At $t = 2$ this reads

$$\mathsf{S}(\Psi_{4r}) - 65\mathsf{S}(\Psi_{2r}) + 64\mathsf{S}(\Psi_r) = -62Br + O(r^{3/2}).$$

It is negative at all sufficiently small r. The estimate retains a quantitative tolerance, so this is a conclusion about the correlation function, not merely a formal comparison of degrees. The same test works in every canonical component because the coefficient and the tolerance survive the component transfer.

Failure of dilation covariance does not by itself establish interaction. A quadratic observable of a Gaussian field can have the same two-term law; the construction gives such an example. Its three-point behaviour distinguishes it from the torus limits under discussion. A claim of non-Gaussianity must therefore use that additional information, rather than the second two-point coefficient alone.

8.1. **Natural scaling and failure of dilation covariance.** We now express the two-term law in the natural parameter d/b. The resulting tolerance is uniform over the admitted reference couplings. The law holds at every positive label on a fixed torus and, on $\mathbb{R}^3$, in every volume-first limit at the stated coupling ceiling and in each of its canonical components. The nonzero second term excludes dilation covariance of the two-point function.

We give two passages to this conclusion: a direct transfer of the law between labels and an alternative using blocking factors with incommensurable logarithms. We also retain the leading law for signed test functions, with its sharp linear rate. These conclusions do not by themselves establish interaction. Prop. 8.18 constructs a quadratic Gaussian model with the same two-term law; its three-point behavior distinguishes it from the torus limits considered here.

8.1.1. *Parameters and test pairs.* We retain the construction setting of §6.1, and write b and d for the label b and scale d. A *choice* $\mathfrak{c}$ consists of a mollifier, a radius $\varrho > 0$, the common real and sourced numerical assignment of Prop. YM3_I:69.1 and Section YM3_I:82 at source radius 2ϱ, and the canonical auxiliary data for these constructions. Every constant in the real restriction keeps its value. The choice fixes an odd $L \geq 13$, $B_\sharp \geq 0$, $\bar{B} = B_\sharp L/(L-1)$, M, and $0 < \gamma \leq 1/2$. The short-distance ceilings γ_W of Thm. YM3_I:AV and γ_W^{sd} of the choices in equations YM3_I:(2326)–(2327) are both at most γ. The ceiling $\gamma_{\mathrm{e}} \in (0, \gamma]$, depending only on the constants, is fixed at the beginning of the subsection on the least torus, and $\Gamma := \min\{\gamma_{\mathrm{e}}, \gamma_W^{\mathrm{sd}}\}$.

We write $a_K = L^{-K}$, $X_g = L(g^{-2} + B_\sharp)$, $I(g) = (g^{-2} + \bar{B}, X_g - \bar{B}]$ for the interval of labels at g (Section YM3_I:74) and $I^\circ(g)$ for its interior, $\mathfrak{L}(g) = \{2L^m : m \geq m_{\mathfrak{L}}(g)\}$ for the set of the sides at g (equation YM3_I:(1528)) and ℓ_g for its least member, and $T_\ell = \mathbb{R}^3/\ell\mathbb{Z}^3$. A triple (g, b, ℓ) is *admissible* if $0 < g \leq \gamma_W^{\mathrm{sd}}$, $b \in I^\circ(g)$ and $\ell \in \mathfrak{L}(g)$. Since $LB_\sharp = (L-1)\bar{B}$, we have $I^\circ(g) = (g^{-2} + \bar{B},\, Lg^{-2} + (L-2)\bar{B})$, an interval of length $(L-1)g^{-2} + (L-3)\bar{B} > 0$.

We use the lattices, first-passage data, labels and torus cumulants of §6.1, with the same fixed choice of construction data. The continuum torus cumulants $S_n^{(\ell,b)}$ exist for every $b > 0$ on Lipschitz tuples with no common point, by Thm. YM3_I:A. A volume-first limit S at g is defined by a sequence (K_j, β_j, ν_j) as in [7, §K.1]; for $g \leq \gamma_W$ it has one label $\mathsf{b}(S) = \lim_j a_{K_j}\beta_j$. $(\mathsf{X}, \lambda, (\sigma_\xi))$ denotes a canonical decomposition of its functional σ_S, with the probability measure λ on X; S_2 and S_2^ξ are the two-point functions of σ_S and σ_ξ, and $\mathsf{N} \subset \mathsf{X}$ is the set of the ξ for which S_2^ξ is not dilation covariant.

The class $\mathfrak{T}_{\mathbb{E}}(d;\vartheta)$ is the test-pair class of §6.1. $\mathfrak{T}^1(d;\vartheta)$ is the set of its pairs on $\mathbb{R}^3$ of class C^1. Over $\mathbb{R}^3$ the pairs of real C^1 functions with compact disjoint supports are called *separated*; the pairs of $\mathfrak{T}^1(d;\vartheta)$ are separated. On a torus we admit real Lipschitz entries with disjoint supports. Every $\mathfrak{T}^1(d;\vartheta)$, $d > 0$, $\vartheta \in (0,1)$, contains a pair ([7, Lem. I.1](c)); for $\vartheta \leq 3\sqrt{3}/8$ one may take $f(u) = (1 - |u-x|^2/d^2)_+^2$, $h(u) = (1 - |u-y|^2/d^2)_+^2$ with $3d \leq |x-y| \leq 8d$. For a pair we write $|\Psi| = \|f\|_\infty \|h\|_\infty$, $\Psi_r = (f(\cdot/r), h(\cdot/r))$ for $r > 0$, $\Psi^{\langle\ell\rangle}$ for the periodisation, and

$$\mathcal{K}_q(\Psi) = \iint f(x)h(y)\,|x-y|^{-q}\,\mathrm{d}x\,\mathrm{d}y,$$

on T_ℓ with the Euclidean distance of T_ℓ; further $T^{(\ell,b)}(\Psi) = S_2^{(\ell,b)}(\Psi^{\langle\ell\rangle})$, $c_2 = \frac{9}{16\pi^2}$, $c_1 = \frac{8}{3\pi}$, $\varkappa = c_1c_2 = \frac{3}{2\pi^3}$ and $L_b^+ = c_2 b^{-2}[\mathcal{K}_6 + c_1 b^{-1}\mathcal{K}_5]$. By [7, Lem. I.1] and equation YM3_I:(2247) and [7, Lem. I.1](b): $\Psi_r \in \mathfrak{T}^1(rd;\vartheta)$ for $\Psi \in \mathfrak{T}^1(d;\vartheta)$, with the same ϑ, $|\Psi_r| = |\Psi|$ and $\mathcal{K}_q(\Psi_r) = r^{6-q}\mathcal{K}_q(\Psi)$; translations keep the classes, the suprema and the $\mathcal{K}_q$; for $d < \ell/20$ the periodisation maps $\mathfrak{T}_{\mathbb{R}^3}(d;\vartheta)$ into $\mathfrak{T}_{T_\ell}(d;\vartheta)$, keeps the suprema, and $\mathcal{K}_q(\Psi^{\langle\ell\rangle}) = \mathcal{K}_q(\Psi)$ for every real $q > 0$. As $d \leq |x-y| \leq 10d$ on the supports of a pair of the class, $(10d)^{-q}\int f \int h \leq \mathcal{K}_q(\Psi) \leq d^{-q}|\Psi|(\frac{4\pi}{3}d^3)^2$, that is,

$$10^{-5}\vartheta^2 d\,|\Psi| \leq \mathcal{K}_5(\Psi) \leq \left(\tfrac{4\pi}{3}\right)^2 d\,|\Psi|, \qquad 10^{-6}\vartheta^2|\Psi| \leq \mathcal{K}_6(\Psi) \leq \left(\tfrac{4\pi}{3}\right)^2|\Psi|, \qquad \varkappa\left(\tfrac{4\pi}{3}\right)^2 = \tfrac{8}{3\pi}; \quad (8.\mathrm{i})$$

note $10^{-5}\varkappa = 4.837\ldots\cdot 10^{-7}$.

A functional σ on separated tuples over $\mathbb{R}^3$ *obeys the law* (L) *with* $(b, c, \varepsilon, d_7, \vartheta)$, where ε is non-decreasing on $(0, d_7]$ and $\varepsilon(d) \to 0$ as $d \to 0$, if

$$|S_2^\sigma(\Psi) - c_2 b^{-2}[\mathcal{K}_6 + cb^{-1}\mathcal{K}_5](\Psi)| \le \varepsilon(d)\, b^{-3} d\, |\Psi| \qquad (0 < d \le d_7,\ \Psi \in \mathfrak{T}^1(d;\vartheta)).$$

S_2^σ is *dilation covariant* if there is a real ϖ with $S_2^\sigma(\Phi_r) = r^\varpi S_2^\sigma(\Phi)$ for all $r > 0$ and all separated pairs Φ. For a function S on pairs, $t > 1$ and $r > 0$ we call

$$\mathcal{N}_t^{\mathsf{S}}(\Psi; r) = \mathsf{S}(\Psi_{t^2 r}) - (1 + t^6)\, \mathsf{S}(\Psi_{tr}) + t^6\, \mathsf{S}(\Psi_r)$$

the t-scale number; at $t = 2$ it is the *three-scale number* $\mathsf{S}(\Psi_{4r}) - 65\, \mathsf{S}(\Psi_{2r}) + 64\, \mathsf{S}(\Psi_r)$. The weights annul constants and multiples of r^6, and give the function $r \mapsto r$ the factor $t^2 - (1+t^6)t + t^6 = -t(t-1)(t^5 - 1)$, which is -62 at $t = 2$. For a triple $\tau = (g, b, \ell)$ and a pair Ψ we put $E_\tau = 2C_2^{\mathrm{T}}(g)\vartheta^{-1}b^2$, with the number C_2^{T} of Prop. 8.3, and $Y_\tau(\delta) = b^2\, T^{(\ell,b)}(\Psi_{\delta b})$.

8.1.2. *Changing units and keeping the label.* The natural short-distance variable is d/b. The label b has the dimension of length, and under a lattice dilation the mesh, side and label change together. At powers of the blocking factor, the exact identity is

$$T^{(\ell,b)}(\Psi) = L^{2s} T^{(L^s \ell, L^s b)}(\Psi_{L^s}), \qquad \Psi_r = (f(\cdot/r), h(\cdot/r)). \tag{8.ii}$$

It follows at finite cutoff by transporting the same Wilson measure to the dilated lattice, keeping the bare inverse coupling fixed, and using the factor L^s in each smeared observable. The full-sequence cutoff limits then give eq. 8.ii. This identity compares theories at different labels. Dilation covariance of one two-point function would instead keep its label fixed and require one power of r for every separated pair.

The two explicit kernels transform compatibly with eq. 8.ii: $\mathcal{K}_6(\Psi_r) = \mathcal{K}_6(\Psi)$ and $\mathcal{K}_5(\Psi_r) = r\mathcal{K}_5(\Psi)$. Their common natural parameter is consequently d/b. The purpose of theorem Y is to make the tolerance uniform in that parameter across the admitted labels and canonical components. A bound $C(g)d^{1/2}$ on the least torus does not yet have this uniformity.

The supplied proof obtains it by putting labels into one multiplicative period. Choose Λ once from the fixed construction constants. Every $b > 0$ has a unique dilate $b' = L^z b$ in $[\Lambda, L\Lambda)$. Two fixed reference couplings have admissible label intervals covering that period. The least-torus law is therefore needed at only these two couplings. The state comparison of lemma 8.11 bounds the difference between a continuum component and the corresponding torus function by a constant times $(d/b)^2$ after multiplication by b^2. At the dilated label this comparison constant is bounded throughout the fixed period.

The torus tolerance contributes a constant times $(d/b)^{1/2}$ in the normalized remainder, and the state comparison contributes a constant times d/b. Taking their nondecreasing majorant gives $\varepsilon^{\mathrm{nat}}(d/b) \le C(d/b)^{1/2}$. For components, the comparison is justified by their conditioned periodic-Gibbs approximants. Centring affects only the first cumulant; the covariance is the limit of those approximants. Thus the same transfer applies to each component with the same recovered label. No averaging over components is used to obtain a pointwise assertion about an individual component.

8.1.3. *The nonzero linear term.* Fix a test pair Ψ of scale d_1 in the positive class. The natural-scale law gives

$$\mathsf{S}(\Psi_r) = A + Br + O(r^{3/2}), \qquad A = c_2 b^{-2}\mathcal{K}_6(\Psi) > 0, \qquad B = \kappa^{\mathrm{lat}} b^{-3}\mathcal{K}_5(\Psi) > 0, \tag{8.iii}$$

for the volume-first two-point function and for each canonical component. Positivity of A and B follows from the mass floors of the test class, not merely from the signs of their displayed numerical coefficients. The error is uniform in the small-radius range specified by the natural variable.

If S were dilation covariant with exponent ϖ, its value at Ψ_r would be $r^\varpi \mathsf{S}(\Psi)$. The positive finite limit A as $r \to 0$ forces $\varpi = 0$. Covariance would then make the value constant in r, whereas dividing eq. 8.iii minus A by r gives the positive limit B. This is the direct contradiction in lemma 8.12.

The scale difference used in the source gives a quantitative version. For $t > 1$, the combination

$$\mathcal{N}_t^{\mathsf{S}}(\Psi; r) = \mathsf{S}(\Psi_{t^2 r}) - (1 + t^6)\mathsf{S}(\Psi_{tr}) + t^6 \mathsf{S}(\Psi_r)$$

annuls constants and terms proportional to r^6. Its value on the linear term is $-t(t-1)(t^5-1)Br$. At $t = 2$ this is $-62Br$, and the sum of the three remainder bounds has coefficient 198. Hence the three-scale number is negative for sufficiently small r, with an explicit threshold given by the normalized tolerance. The test records a property of the two-point function at fixed label, rather than the exact change-of-units identity between labels.

The alternative proof in lemma 8.15 and proposition 8.17 transfers this obstruction from a single admitted triple. Choose a second blocking factor whose logarithm is incommensurable with the first. Sequences of powers of the two factors approximate any positive ratio of scales. On each pair of corresponding lattices, transport the same bare periodic Gibbs state. Its two rescaled covariances agree exactly after multiplication by the squares of their labels. Comparing each with its torus limit leaves an error $O((d/b)^2)$. Thus a hypothetical quadratic-order agreement with the free law in one limit would transfer to the triple where the nonzero linear correction is already known. It cannot absorb that correction. This second route retains the original comparison estimates and continuity in the varying label and test pair; density of scale ratios alone would not justify passage to the limit.

8.1.4. *Signed pairs and what the two-term law detects.* The positivity conditions on the test class are used to identify the coefficient and its sign. A leading short-distance estimate for arbitrary signed separated Lipschitz functions follows by a finite decomposition into signed nonnegative pieces. A common test shape serves the finitely many pairs of pieces. Apply the two-term estimate to each pair and use bilinearity. This gives the $O(r)$ error in corollary 8.1 uniformly in position and on compact label intervals. A nonzero positive reflected pair has $\mathcal{K}_5 > 0$, so its linear correction shows that this rate is attained. The label derivative identity is retained separately: inserting the constant function differentiates the coupling, and integration over a compact interval of labels gives the stated bound involving $S_3(1, f_r, h_r)$.

The failure of dilation covariance is not by itself a test of interaction. Proposition 8.18 supplies a concrete comparison model. On T_ℓ, take three independent Gaussian fields with covariance symbol

$$\widehat{G}_b(k) = (b|k|^2 + \alpha|k| + \upsilon)^{-1}, \qquad \upsilon = \upsilon_0/\ell > 0.$$

Its positive spectral representation is

$$(bu^2 + \alpha u + \upsilon)^{-1} = \int_0^\infty \frac{\rho_{\mathrm{sp}}(m)}{u^2 + m^2}\,\mathrm{d}m, \qquad \rho_{\mathrm{sp}}(m) = \frac{2\alpha}{\pi}\frac{m^2}{(\upsilon - bm^2)^2 + \alpha^2 m^2}.$$

Reflection positivity follows by integrating the reflection-positive massive covariances against this positive density. The model's observable is a quadratic functional of mollified fields, combining $-\frac{1}{2}{:}|\nabla\varphi|^2{:}$ with $-{:}\varphi\Delta\varphi{:}$ as specified there. Its separated cumulants have limits independent of the mollifier, although its variance at a nonzero single test function diverges. Consequently it is important to retain the separated-cumulant formulation and not claim an L^2 limiting observable.

Let $\mathcal{C}$ be its covariance kernel and $\mathcal{H} = \Delta\mathcal{C}$ off the origin. Wick's rule gives the two-point kernel

$$\frac{3}{2}\left[\mathrm{tr}(\nabla\nabla^{\mathsf{T}}\mathcal{C})^2 + 4\nabla\mathcal{C}\cdot\nabla\mathcal{H} + 2\mathcal{H}^2 + 2\mathcal{C}\Delta\mathcal{H}\right].$$

The supplied expansion of $\mathcal{C}$ makes its first two singular terms

$$c_2 b^{-2}\left[|x|^{-6} + \frac{4\alpha}{3\pi}b^{-1}|x|^{-5}\right].$$

At $\alpha = 2$ the relative coefficient is exactly $8/(3\pi)$. After smearing, the remaining error is $O(d^2(1 + \log(\ell/d)))$, which is smaller than the linear correction. This model therefore reproduces the same two-term law.

The source distinguishes it using a three-point property of the torus limits. Their cutoff estimates imply convergence to zero for tuples with uniformly bounded supremum norms and Lipschitz constants, fixed positive spread, and a product of supremum norms tending to zero. Here the spread of supports $S_1, \dots, S_n$ is $\inf_{x_a \in S_a} \max_{a,a'} |x_a - x_{a'}|_\infty$. For $f_{\mathbf{q}} = |\mathbf{q}|^{-1} \psi \cos(\mathbf{q} \cdot x)$ with ψ separated from a nonzero $h \geq 0$, these conditions hold. The first two supports may coincide; the third stays separated from them. The quadratic Gaussian model instead has the limit

$$\mathcal{S}_3(f_{\mathbf{q}}, f_{\mathbf{q}}, h) \longrightarrow -\frac{3}{2b} \iint \psi(x)^2 h(y) \big[|\nabla \mathcal{C}|^2 + 2\mathcal{C} \Delta \mathcal{C} \big](x - y) \, \mathrm{d}x \, \mathrm{d}y < 0.$$

The spectral representation makes the bracket positive off the origin. This is the supplied test that excludes that comparison model. It also explains why the two-point coefficient, despite detecting a scale, cannot alone identify the full field theory.

8.1.5. *The law at natural scales.*

Theorem Y (The components)**.** Let $\mathfrak{c}$ be a choice and $\vartheta \in (0,1)$. There are $\delta_{\mathrm{c}} > 0$ and a non-decreasing function $\varepsilon^{\mathrm{nat}}$ on $(0, \delta_{\mathrm{c}}]$ with $\varepsilon^{\mathrm{nat}}(\delta) \leq C \delta^{1/2}$, where δ_{c}, $\varepsilon^{\mathrm{nat}}$ and C depend only on $\mathfrak{c}$ and ϑ, such that for every $0 < g \leq \gamma_W$, every volume-first limit S at g with $b = \mathsf{b}(S)$, every canonical decomposition $(\mathsf{X}, \lambda, (\sigma_\xi))$ of σ_S, and for $\mathsf{S} = S_2$ and $\mathsf{S} = S_2^\xi$, $\xi \in \mathsf{X}$, the following holds.

(1) $|\mathsf{S}(\Psi) - L_b^+(\Psi)| \leq \varepsilon^{\mathrm{nat}}(d/b)\, b^{-3} d\, |\Psi|$ for $0 < d \leq \delta_{\mathrm{c}} b$ and $\Psi \in \mathfrak{T}^1(d; \vartheta)$. So σ_S and every σ_ξ obey (L) with $(b, \frac{8}{3\pi}, \varepsilon^S, d^S, \vartheta)$, where $\varepsilon^S(d) = \varepsilon^{\mathrm{nat}}(g^2 d)$ and $d^S = \delta_{\mathrm{c}} g^{-2}$ depend only on $\mathfrak{c}$, g and ϑ. They obey (L) also with a tolerance formed at any dilated label: if z is an integer with $L^z b \geq 2\Gamma^{-2}$ and $g' = (2/(L^z b))^{1/2}$, then (L) holds with $(b, \frac{8}{3\pi}, \varepsilon', d', \vartheta)$, where $\varepsilon'(d) = \varepsilon^L_{g',\vartheta}(L^z d) + 2C_2^{\mathrm{T}}(g') \vartheta^{-1} L^{2z} b\, d$ and $d' = L^{-z} \min\{d^L, d_\flat\}(g', \vartheta)$, with the objects of Thm. Z (a) and Prop. 8.3.

(2) Let $d_1 > 0$, $\Psi \in \mathfrak{T}^1(d_1; \vartheta)$ and $B_S = \varkappa b^{-3} \mathcal{K}_5(\Psi)$, so that $B_S \geq 10^{-5} \varkappa \vartheta^2 b^{-3} d_1 |\Psi| > 0$; let $t > 1$. Then

$$|\mathsf{S}(\Psi_r) - c_2 b^{-2} \mathcal{K}_6(\Psi) - B_S r| \leq \varepsilon^{\mathrm{nat}}(r d_1/b)\, b^{-3} d_1 |\Psi|\, r \qquad (r d_1 \leq \delta_{\mathrm{c}} b),$$

$$|\mathcal{N}_t^{\mathsf{S}}(\Psi; r) + t(t-1)(t^5 - 1) B_S r| \leq (t^2 + t + t^6 + t^7)\, \varepsilon^{\mathrm{nat}}(t^2 r d_1/b)\, b^{-3} d_1 |\Psi|\, r \qquad (t^2 r d_1 \leq \delta_{\mathrm{c}} b).$$

So $\mathsf{S}(\Psi_r) = c_2 b^{-2} \mathcal{K}_6(\Psi) + B_S r + O(r^{3/2})$; $|r^{-1}[\mathsf{S}(\Psi_{tr}) - \mathsf{S}(\Psi_r)] - (t-1) B_S| \leq (t+1)\, \varepsilon^{\mathrm{nat}}(t r d_1/b)\, b^{-3} d_1 |\Psi|$, so that $r^{-1}[\mathsf{S}(\Psi_{tr}) - \mathsf{S}(\Psi_r)] \to (t-1) B_S$; $r^{-1} \mathcal{N}_t^{\mathsf{S}}(\Psi; r) \to -t(t-1)(t^5 - 1) B_S$, and so do the means $n^{-1} \sum_{j<n} (t^{-j} r_0)^{-1} \mathcal{N}_t^{\mathsf{S}}(\Psi; t^{-j} r_0)$ as $n \to \infty$, for every r_0 with $t^2 r_0 d_1 \leq \delta_{\mathrm{c}} b$; and $\mathcal{N}_t^{\mathsf{S}}(\Psi; r) < 0$ as soon as $(t^2 + t + t^6 + t^7)\, \varepsilon^{\mathrm{nat}}(t^2 r d_1/b) < t(t-1)(t^5 - 1) \cdot 10^{-5} \varkappa \vartheta^2$. At $t = 2$: the three-scale number is $-62 B_S r + O(r^{3/2})$, within $198\, \varepsilon^{\mathrm{nat}}(4 r d_1/b)\, b^{-3} d_1 |\Psi|\, r$ of $-62 B_S r$, and negative as soon as $198\, \varepsilon^{\mathrm{nat}}(4 r d_1/b) < 62 \cdot 10^{-5} \varkappa \vartheta^2$, that is, for all small r.

(3) Neither S_2 nor any S_2^ξ is dilation covariant: $\mathsf{N} = \mathsf{X}$ and $\lambda(\mathsf{N}) = 1$.

■□

Theorem Z (The tori)**.** Let $\mathfrak{c}$ be a choice and $\vartheta \in (0,1)$; let d_{e}, $\bar{\varepsilon} = \bar{\varepsilon}_{g,\vartheta}$, $\bar{\varepsilon}_\infty = \bar{\varepsilon}_{\infty;g,\vartheta}$ and C_{cut} be the objects of Prop. 8.10 and the depth $s(d)$ that of the subsection on the least torus, and $d_\flat$, C_2^{T} those of Prop. 8.3.

(a) *The least torus.* For $0 < g \leq \gamma_{\mathrm{e}}$ there are $d^L(g, \vartheta) > 0$ and a non-decreasing function $\varepsilon^L_{g,\vartheta}$ on $(0, d^L(g, \vartheta)]$ with $\varepsilon^L_{g,\vartheta}(d) \leq C(g, \vartheta) d^{1/2}$, all depending only on $\mathfrak{c}$, g and ϑ, such that

$$|T^{(\ell_g, b)}(\Psi) - L_b^+(\Psi)| \leq \varepsilon^L_{g,\vartheta}(d)\, b^{-3} d\, |\Psi| \qquad (b \in I(g),\ 0 < d \leq d^L(g, \vartheta),\ \Psi \in \mathfrak{T}_{\mathbb{R}^3}(d; \vartheta)).$$

(b) *Every torus of the family; every state at a finite cutoff.* Let $0 < g \leq \Gamma$, $b \in I(g)$, $\ell \in \mathfrak{L}(g)$, $0 < d \leq \min\{d_{\mathrm{e}}, \ell_g/20, d_\flat\}(g, \vartheta)$ and $\Psi \in \mathfrak{T}_{\mathbb{R}^3}(d; \vartheta)$. Then

$$|T^{(\ell, b)}(\Psi) - L_b^+(\Psi)| \leq \big[\bar{\varepsilon}_\infty(d) + 2 C_2^{\mathrm{T}}(g) \vartheta^{-1} b d \big]\, b^{-3} d\, |\Psi|,$$

and the bracket is at most $C(g,\vartheta)(1+b)d^{1/2}$. Moreover, for every cutoff $K > s(d)$ and every state $\mathfrak{s}$ at the datum $(K, L^K b)$ in the sense of Prop. 8.3,

$$|S_2^T[\mathfrak{s}](\Psi) - L_b^+(\Psi)| \le \Big[(\bar\varepsilon(d) + C_2^{\mathrm{T}}(g)\vartheta^{-1}bd)\, b^{-3}d + C_{\mathrm{cut}}\, b^{-2}\frac{a_K}{d}\Big]|\Psi|,$$

and the same holds with $C_{\mathrm{r}}d^{1/2}$ in the place of $\bar\varepsilon(d)$ and C'_{cut} in the place of C_{cut}, with the numbers of Prop. 8.10 (c).

(c) *Every side, every label.* Let $\ell = 2L^m$, $m \ge 1$, $b > 0$, $s \ge 0$ an integer and $0 < g'' \le \Gamma$ a coupling for which $(g'', L^s b, L^s \ell)$ is admissible; this holds for all large s with $g'' = (2/(L^s b))^{1/2}$. Then for $0 < d \le L^{-s}\min\{d_{\mathrm{e}}, \ell_{g''}/20, d_{\mathrm{b}}\}(g'', \vartheta)$ and $\Psi \in \mathfrak{T}^1(d;\vartheta)$

$$|T^{(\ell,b)}(\Psi) - L_b^+(\Psi)| \le \big[\bar\varepsilon_\infty(L^s d) + 2C_2^{\mathrm{T}}(g'')\vartheta^{-1}L^{2s}bd\big]\, b^{-3}d\,|\Psi|,$$

with $\bar\varepsilon_\infty$ formed at (g'', ϑ); the bracket is at most $Cd^{1/2}$, with C depending on $\mathfrak{c}$, g'', s, b and ϑ.

(d) *Exact dilation by powers of L.* $T^{(\ell,b)}(\Psi) = L^{2s}\, T^{(L^s\ell, L^s b)}(\Psi_{L^s})$ for $\ell = 2L^m$, $b > 0$, integers $s \ge 0$ and pairs Ψ of real Lipschitz functions with compact supports whose periodisations to T_ℓ have disjoint supports. In particular, for $0 < g \le \gamma_{\mathrm{e}}$, $b \in I(g)$ and every integer $z \ge 0$, the law of (a) holds on the side $L^z\ell_g$ at the label $L^z b$ with no comparison of states:

$$|T^{(L^z\ell_g, L^z b)}(\Psi) - L^+_{L^z b}(\Psi)| \le \varepsilon^L_{g,\vartheta}(L^{-z}d)\,(L^z b)^{-3}d\,|\Psi| \qquad (0 < d \le L^z d^L(g,\vartheta),\ \Psi \in \mathfrak{T}_{\mathbb{R}^3}(d;\vartheta)).$$

(e) *One coefficient.* If at one side $2L^m$ and one label the inequality of (c) holds with a number c and with a number c' in the place of $\frac{8}{3\pi}$, each with a tolerance that tends to 0, then $c = c'$.

(f) *Compact sets of labels.* At a fixed side, one tolerance, one range of d and one constant C serve in (c) for all the labels of a compact subset of $(0,\infty)$.

Let now $0 < g \le \gamma_{\mathrm{e}}$, $b \in I(g)$, $T = T^{(\ell_g, b)}$, $\varepsilon^L = \varepsilon^L_{g,\vartheta}$, $d^L = d^L(g,\vartheta)$, $d_1 > 0$, $\Psi \in \mathfrak{T}^1(d_1;\vartheta)$, $\mathsf{A} = c_2 b^{-2}\mathcal{K}_6(\Psi)$, $\mathsf{B} = \varkappa b^{-3}\mathcal{K}_5(\Psi)$ and $\mathsf{u} = b^{-3}d_1|\Psi|$. Then $10^{-5}\varkappa\vartheta^2\mathsf{u} \le \mathsf{B} \le \frac{8}{3\pi}\mathsf{u}$, and for $0 < rd_1 \le d^L$

$$|T(\Psi_r) - \mathsf{A} - \mathsf{B}r| \le \varepsilon^L(rd_1)\,\mathsf{u}\,r \le C(g,\vartheta)\, d_1^{1/2}\,\mathsf{u}\,r^{3/2}. \tag{8.iv}$$

(i) For $t > 1$ and $0 < trd_1 \le d^L$: $|r^{-1}[T(\Psi_{tr}) - T(\Psi_r)] - (t-1)\mathsf{B}| \le (t+1)\,\varepsilon^L(trd_1)\,\mathsf{u} \to 0$; for $0 < t^2 rd_1 \le d^L$: $|r^{-1}\mathcal{N}_t^T(\Psi;r) + t(t-1)(t^5-1)\mathsf{B}| \le (t^2+t+t^6+t^7)\,\varepsilon^L(t^2rd_1)\,\mathsf{u} \to 0$. At $t = 2$: $|r^{-1}[T(\Psi_{2r}) - T(\Psi_r)] - \mathsf{B}| \le 3\varepsilon^L(2rd_1)\mathsf{u}$.

(ii) $r^{-2}|T(\Psi_r) - \mathsf{A}| \ge r^{-1}\mathsf{u}\,[10^{-5}\varkappa\vartheta^2 - \varepsilon^L(rd_1)] \to \infty$ as $r \to 0$.

(iii) For all $d_n \to 0$ and all $\Psi_n \in \mathfrak{T}^1(d_n;\vartheta)$: $|T(\Psi_n) - c_2 b^{-2}\mathcal{K}_6(\Psi_n)|/(d_n^2|\Psi_n|) \ge b^{-3}[10^{-5}\varkappa\vartheta^2 - \varepsilon^L(d_n)]/d_n \to \infty$.

(iv) Let $g \le \Gamma$ and $b \in I^\circ(g)$. For every $0 < C < \infty$, every $d \le d^L$ with $\varepsilon^L(d) \le \frac12\cdot 10^{-5}\varkappa\vartheta^2$ and $d/b < \frac12\cdot 10^{-5}\varkappa\vartheta^3/C$, and every $\Psi \in \mathfrak{T}^1(d;\vartheta)$: $b^2|T(\Psi) - c_2 b^{-2}\mathcal{K}_6(\Psi)| > C\vartheta^{-1}(d/b)^2|\Psi|$.

■□

Corollary 8.1 (The leading term with the rate r, for signed functions)**.** Let $\mathfrak{c}$ be a choice, $\ell = 2L^m$ with $m \ge 1$, $b > 0$, $p \in T_\ell$ with a lift $\tilde p$, let f_0, h_0 be real Lipschitz functions on $\mathbb{R}^3$ with compact disjoint supports, and put $f_r = (f_0((\cdot - \tilde p)/r))^{\langle\ell\rangle}$ and $h_r = (h_0((\cdot - \tilde p)/r))^{\langle\ell\rangle}$ for $r > 0$.

(a) There are $C, r_* > 0$, independent of p and the same for all the labels of a compact subset of $(0,\infty)$, with

$$|S_2^{(\ell,b)}(f_r, h_r) - c_2 b^{-2}\mathcal{K}_6(f_0, h_0)| \le Cr \qquad (0 < r \le r_*).$$

In particular, for $\theta_\mu x = x - 2x_\mu e_\mu$ and every real Lipschitz function ψ_0 with compact support in $\{x_\mu > 0\}$, the pair $(f_0, h_0) = (\psi_0\circ\theta_\mu, \psi_0)$ gives $S_2^{(\ell,b)}(f_r, h_r) = c_2 b^{-2}\mathcal{K}_6(\psi_0\circ\theta_\mu, \psi_0) + O(r)$.

(b) The rate r is attained: if $\psi_0 \geq 0$ is of class C^1, not 0, with support in the ball $\mathbb{B}(e_\mu, \frac{1}{2})$, then

$$S_2^{(\ell,b)}(f_r, h_r) - c_2 b^{-2} \mathcal{K}_6(\psi_0 \circ \theta_\mu, \psi_0) = \varkappa\, b^{-3} \mathcal{K}_5(\psi_0 \circ \theta_\mu, \psi_0)\, r + O(r^{3/2}), \qquad \mathcal{K}_5(\psi_0 \circ \theta_\mu, \psi_0) > 0;$$

so no bound $o(r)$ holds in (a).

(c) For $0 < b_1 < b_2$, with the constant C of (a) for the labels of $[b_1, b_2]$ and $0 < r \leq r_*$,

$$\Big| \int_{b_1}^{b_2} [b^2 S_3^{(\ell,b)}(1, f_r, h_r) - 2b\, S_2^{(\ell,b)}(f_r, h_r)]\, \mathrm{d}b \Big| \leq (b_1^2 + b_2^2)\, C\, r,$$

where 1 is the constant function on T_ℓ.

■□

8.1.6. *Construction inputs.* The proof rests on Thms. YM3_I:D, YM3_I:AJ and YM3_I:AD among the construction results, with its conditions on the constants, through the statements cited below. Among them are Section YM3_I:74, [7, result G.2], YM3_I:(1528) and YM3_I:82, the real-sequence conditions and smallness floor YM3_I:(972) preceding Thm. YM3_I:Y, the bounds YM3_I:(1531) on labels, Cor. YM3_I:74.2 and [7, Lem. G.4] on couplings and labels, which we use in the form of Lem. 8.2 below, the exact dilation by powers of L YM3_I:(2215), Thm. YM3_I:AS on the limits at a fixed label (their existence, the rate of removal of the cutoff, the bounds YM3_I:(2214) uniform in the cutoff and in the labels of a compact set, the continuity in the label and the convergence along converging labels), Prop. YM3_I:118.3 on the derivatives in the label, the uniform cutoff estimate of Thm. YM3_I:AR on the removal of the cutoff, the weighted progression and its source energies ([7, Def. B.7] and [7, Cor. B.9]), its setting (J) for the canonical objects ([7, Def. B.16] and equation [7, eq. 228] and [7, Prop. B.21]), the places with a lower place of a step at a complex field scale (Def. 6.29 and Prop. 6.31), the sourced renormalisation-group chain and its source energies ([7, Prop. B.12] and equations [7, eq. 212]–[7, eq. 214]), the histories, the sum over the level of the first creation of a large-field region and the rarity of large fields (Section YM3_I:123, equation eq. 4.xxxi and Prop. 4.54), the reflection positivity of the limits on the tori (proved in Prop. 8.18(c) from Thm. YM3_I:A and finite-cutoff positivity, for every side $2L^m$, every $b > 0$ and C^1 entries at every coordinate mirror) and that of the Wilson measures on the tori at the site and bond mirrors (Prop. YM3_I:91.4), which, with the existence of the limits along the full sequence of the cutoffs, gives it for real Lipschitz entries at the mirrors $\{x_\mu = c\}$, $c \in \bigcup_{K \in \mathbb{N}} \frac{1}{2} L^{-K} \mathbb{Z}$, of the lattices of the cutoffs, the block averages ([7, Lem. B.3]), the template sum of the free lattice curvature (Prop. YM3_I:121.2 and Lem. 6.34), Laplace's method (Lem. 7.11) and the coefficients of lattice perturbation theory (Def. 7.10, Cor. 7.37, Thms. W and X), the chessboard estimate ([7, Cor. B.1]), the representation of the components by conditioned periodic Gibbs states (Thm. YM3_I:AP), and, for $g \leq \gamma_W$, the leading law of Thm. YM3_I:AW for σ_S and, using the conditioned approximants in Thm. YM3_I:AP, for its components: for $0 < d \leq \mathsf{d}_0(g, \vartheta)$ and the C^1 pairs of $\mathfrak{T}_{\mathbb{R}^3}(d; \vartheta)$, $\mathsf{S}(\Psi) = c_2 b^{-2} \mathcal{K}_6(\Psi)(1 + \eta)$ with $|\eta| \leq \eta_{g,\vartheta}(d) \leq \frac{1}{3}$ and $\eta_{g,\vartheta}(d) \to 0$ as $d \to 0$.

Two further statements are proved below from these: the agreement of all states at one datum at two points, which uses no decay of correlations (Prop. 8.3), and the next-order law on the least torus (Prop. 8.10).

Lemma 8.2 (Couplings and labels)**.**

(a) $\gamma^{-2} > \bar{B}$. For every real $b' > 2\bar{B}$, in particular for every $b' \geq 2\gamma^{-2}$, the coupling $g' = (2/b')^{1/2}$ satisfies $b' \in I^\circ(g')$; if $b' \geq 2\gamma^{-2}$, then $g' \leq \gamma$.

(b) Let $0 < g' \leq \gamma$, and let (K_j, β_j) be pairs of integers $K_j \geq 1$ and reals $\beta_j > 0$ with $a_{K_j} \beta_j \to b' \in I^\circ(g')$. Then (K_j, β_j) is a first-passage datum at g' for all large j.

■

Proof sketch. (a) Cor. YM3_I:74.2 and [7, Lem. G.4] at the label $2\gamma^{-2}$, where the coupling is γ, puts $2\gamma^{-2}$ in $I(\gamma) = (\gamma^{-2} + \bar{B}, X_\gamma - \bar{B}]$; so $\gamma^{-2} > \bar{B}$. With $g'^{-2} = b'/2$ the formula for I° in the

setting gives $I^\circ(g') = (\frac12 b' + \bar B,\ \frac12 Lb' + (L-2)\bar B)$. Its upper end exceeds b', as $L \ge 3$, and its lower end is less than b' if and only if $b' > 2\bar B$, which holds for $b' \ge 2\gamma^{-2}$; and then $g' \le (2/(2\gamma^{-2}))^{1/2} = \gamma$. (b) As $I^\circ(g')$ is open, $a_{K_j}\beta_j \in I(g')$ for all large j, and Cor. YM3_I:74.2 at the label $a_{K_j}\beta_j$ says that $(K_j, L^{K_j}a_{K_j}\beta_j) = (K_j, \beta_j)$ is a first-passage datum at g'. □

8.1.7. *Agreement of states at two points.*

Proposition 8.3 (Agreement of two states at two points)**.** Let $\mathfrak c$ be a choice, $0 < g' \le \gamma_W^{\rm sd}$, (K,β) a first-passage datum at g' with $K \ge 3$ and label $b = a_K\beta$, $\vartheta \in (0,1)$, $0 < d \le d_\flat(g',\vartheta)$ and $a_K \le 9d$. A *periodic Gibbs state* of the Wilson action at β on $a_K\mathbb Z^3$ is a Gibbs state at (a_K,β) that is invariant under the translations of a subgroup of finite index of $a_K\mathbb Z^3$ (Section YM3_I:91); these states form the set $\mathcal G^{\rm per}_{K,\beta}$. For every K and every $\beta > 0$ this set is the set $\mathcal G^{\rm inv}_{K,\beta}$ of the periodic Gibbs states that are invariant under all the translations of $a_K\mathbb Z^3$ (Prop. YM3_I:103.4; [7, §K.1]), and it contains the states of the tori, the weak limit points of the periodic extensions of the Wilson measures on the tori T_{2L^m}, $m \to \infty$ (Prop. YM3_I:91.2; [7, §K.1]).

A *state at the datum* is the periodic extension to $a_K\mathbb Z^3$ of the Wilson measure on T_ℓ at (K,β), for some $\ell \in \mathfrak L(g')$, or a periodic Gibbs state of the Wilson action at β on $a_K\mathbb Z^3$: a state at g' at (K,β) in the sense of [7, §K.1]. Every state at the datum is invariant under the translations of $a_K\mathbb Z^3$, and so are its cumulants: $S_n^T[\mathfrak s](\tau_v f_1,\dots,\tau_v f_n) = S_n^T[\mathfrak s](f_1,\dots,f_n)$ for all $v \in a_K\mathbb Z^3$ ([7, §K.1]). Let μ' be the periodic extension of the Wilson measure on the least torus $T_{\ell_{g'}}$ at (K,β).

(a) Every state $\mathfrak s$ at the datum satisfies $|S_2^T[\mathfrak s](\Psi) - S_2^T[\mu'](\Psi)| \le C_2^{\rm T}(g')\vartheta^{-1}d^2b^{-2}|\Psi|$ for all $\Psi \in \mathfrak T_{\mathbb R^3}(d;\vartheta)$; hence any two states $\mathfrak s, \mathfrak s'$ at the datum have

$$|S_2^T[\mathfrak s](\Psi) - S_2^T[\mathfrak s'](\Psi)| \le 2C_2^{\rm T}(g')\,\vartheta^{-1}d^2(a_K\beta)^{-2}|\Psi|.$$

(b) Here $d_\flat(g',\vartheta) = \min\{\frac{1}{180}, d_\star(g',\vartheta)\}$ with $d_\star(g',\vartheta) = c_{\rm a}\vartheta^{-1}g'^{-2}$, and

$$C_2^{\rm T}(g') = C_{\rm a}\big[Q_0(g')\,Q_1(g')\big]^2\,\frac{(X_{g'}+\bar B)^3}{(g'^{-2}-\bar B)^2}, \qquad Q_0(g') = \exp\big[\tfrac18\ell_{g'}^3(2C_{\rm S}+C_{g'})\big],$$

where $C_{\rm S}$ and $C_{g'}$ are the numbers of the bounds of Thms. YM3_I:AJ and YM3_I:AD on the partition functions of the tori, $Q_1(g') \ge 1$ is the bound on the linking cumulants of step (4) of the proof, and $c_{\rm a}, C_{\rm a} > 0$ depend only on the constants. So $d_\star > 0$ and $C_2^{\rm T}(g') < \infty$ depend only on the radius, the constants and g' (and on ϑ, for $d_\star$), and on no K, β, side, state, pair or position.

■

Proof sketch. Let $\ell' = \ell_{g'}$, let x_1 be a centre of a ball of radius d that holds $\operatorname{supp} f$, and consider on $a_K\mathbb Z^3$ the board of cubic cells of side $\ell'/2$ one of which, $\square_0$, is centred within $\frac12 a_K$ of x_1; on $T_{\ell'}$ there are eight cells, the mirror images of $\square_0$ under lattice reflections.

(1)

One cell against the least torus. For a bounded function F of the configuration that depends only on the bonds in the interior of $\square_0$ put $\|F\|_{\rm ch}^8 = \mu'(\prod_{\mathbf k\in\{0,1\}^3} F^{[\mathbf k]}) \ge 0$, the $F^{[\mathbf k]}$ being the mirror copies of F in the eight cells, conjugated when the number of reflections is odd. This is the chessboard seminorm $\|F\|_{{\rm ch},\ell'}$ of the side ℓ'. Then $|\mathfrak s(F)| \le Q_0\|F\|_{\rm ch}$ for every state at the datum. For the torus of a side $\ell \ge \ell'$ of $\mathfrak L(g')$ the chessboard estimate at the mirrors of the cells gives $|\mathfrak s(F)| \le \|F\|_{{\rm ch},\ell}$ (clause (c) of [7, Cor. B.1]), and the comparison of two sides gives $\|F\|_{{\rm ch},\ell} \le \|F\|_{{\rm ch},\ell'}\,e^{(\ell'/2)^3(p_{\ell'}-p_\ell)}$ with the pressures $p_\ell = \ell^{-3}\log Z_{K,\ell,\beta}$ (clause (e) there); Thms. YM3_I:AJ and YM3_I:AD at the source 0 bound $p_{\ell'} - p_\ell$ by $2C_{\rm S} + C_{g'}$, so that $e^{(\ell'/2)^3(p_{\ell'}-p_\ell)} \le Q_0$.

A periodic Gibbs state $\mathfrak s$ is dominated through tori of growing sides: by the domination of the means of a periodic Gibbs state by the chessboard seminorms of the tori (Prop. YM3_I:

91.9, for the board above, the one cell $\square_0$ and a sequence of sides $\ell_k \in \mathfrak{L}(g')$ with $\ell_k \to \infty$), $|\mathfrak{s}(F)| \le \liminf_k \|F\|_{\mathrm{ch},\ell_k}$, and every $\|F\|_{\mathrm{ch},\ell_k}$ is at most $Q_0\|F\|_{\mathrm{ch}}$ by the comparison of two sides.

(2)

Normalised moments; only linking cumulants remain. For bounded real F_1, F_2 localised in $\square_0$ let $X_a = \mathcal{O}_K(F_a)$, $M'(z) = \mu'(e^{z_1X_1+z_2X_2})$, $M_{\mathfrak{s}}(z) = \mathfrak{s}(e^{z_1X_1+z_2X_2})$, and $c_I = \partial_I(1/M')(0)$, $I \subset \{1,2\}$, which are real. For $\emptyset \ne B \subset \{1,2\}$ the function $P_B = \sum_{I\subset B} c_{B\setminus I} \prod_{a\in I} X_a$ has $\mathfrak{s}(P_B) = \partial_B(M_{\mathfrak{s}}/M')(0)$. Its mirror copies are the same polynomials in the mirror images $X_{a,\mathbf{k}} = \mathcal{O}_K(F_a \circ \hat{\mathsf{r}}_{\mathbf{k}}^{-1})$, and as μ' is invariant under the lattice isometries,

$$\|P_B\|_{\mathrm{ch}}^8 = \mu'\Big(\prod_{\mathbf{k}} P_B^{[\mathbf{k}]}\Big) = \sum_{\pi} \prod_{\gamma\in\pi} \langle X_{a,\mathbf{k}};\, (\mathbf{k},a)\in\gamma\rangle_{\mu'},$$

the sum running over the partitions π of $\{0,1\}^3 \times B$ into *linking* blocks, blocks that hold images in two different cells: all cumulants within one cell cancel. So if every linking cumulant is at most $Q_1 \prod_{(\mathbf{k},a)\in\gamma} u_a$, then $|\partial_B(M_{\mathfrak{s}}/M')(0)| \le Q_0 \mathsf{Bl}_{16}^{1/8} Q_1^{|B|/2} \prod_{a\in B} u_a$, Bl_N the number of partitions of a set of N elements, and the two cumulants $\partial_1\partial_2 \log M_{\mathfrak{s}}(0)$, $\partial_1\partial_2 \log M'(0)$ differ by $|\partial_1\partial_2 \log(M_{\mathfrak{s}}/M')(0)| \le 2(Q_0\mathsf{Bl}_{16}^{1/8}Q_1)^2 u_1u_2$.

(3)

Packing and signs. Let $\iota = a_K\lceil 18d/a_K\rceil$ and $\mathsf{Pk} = \{v \in \iota\mathbb{Z}^3 : |v_\mu| \le 0.575 - 9d\}$. The balls $\mathbb{B}(x_1+v, 9d)$, $v \in \mathsf{Pk}$, are disjoint and lie in $\mathbb{B}(x_1,1) \subset \square_0$, both supports of Ψ lie in $\mathbb{B}(x_1,9d)$, and $\#\mathsf{Pk} \ge (0.3/9d)^3 = (30d)^{-3}$. For signs $\epsilon = (\epsilon_v)$ apply (2) to $F_1 = \sum_v \epsilon_v\tau_v f$, $F_2 = \sum_v \epsilon_v\tau_v h$. The difference $\mathsf{D} = S_2^T[\mathfrak{s}] - S_2^T[\mu']$ is bilinear, its mean over the signs at (F_1,F_2) is $\sum_v \mathsf{D}(\tau_v f, \tau_v h)$, and both cumulants are invariant under the translations of $a_K\mathbb{Z}^3$ ([7, §K.1]); so $|\mathsf{D}(f,h)| \le (30d)^3 \sup_\epsilon |\mathsf{D}(F_1,F_2)|$.

(4)

The linking cumulants on the least torus. For a complex source w on the plaquettes of $T_{\ell'}$ let $Z[w] = \int \rho_0\, e^{\mathcal{O}(w)}\, \mathrm{d}U$ be the partition function of $T_{\ell'}$ with the source w, ρ_0 the Wilson density at (K,β) and $\mathcal{O}(w)(U) = L^K \sum_p w(p)\, a_p(U)$, and $Z = Z[0]$. The generating function of the sixteen mirrored, packed and signed observables is $Z[W_z]/Z$, with the source $W_z = \sum_{\mathbf{k},a} z_{\mathbf{k},a}(F_a \circ \hat{\mathsf{r}}_{\mathbf{k}}^{-1})^{\langle\ell'\rangle} \circ x$; it vanishes off the plaquettes whose barycentres lie in the eight mirror images of $\mathbb{B}(x_1,1)$, closed balls of radius 1 whose centres are at pairwise sup distances $\ge \frac12\ell' - a_K$. Write

$$\log\frac{Z[W]}{Z} = \log\frac{Z[W]e^{-\mathcal{E}(W)}}{Z} + \Big[\mathcal{E}(W) - \sum_{\mathbf{k}} \mathcal{E}(w^{(\mathbf{k})})\Big] + \sum_{\mathbf{k}} \mathcal{E}(w^{(\mathbf{k})}),$$

$\mathcal{E}$ the source energy of $T_{\ell'}$, the sum over the levels and localisation domains of the source energies [7, eq. 212] of the sourced renormalisation-group chain [7, eq. 205], and $w^{(\mathbf{k})}$ the part of W in the cell $\mathbf{k}$.

(4a) *The setting on $T_{\ell'}$.* Let $\ell' = 2L^m$, $m = m_{\mathfrak{L}}(g')$. By YM3_I:(1528), $L^m = LM\max\{R(L(g'^{-2}+\bar B)), L^{\lceil \log_L C_{\mathfrak{s}}\rceil}\}$, and the values of the cube-size function R are powers L^{u} with $\mathsf{u} \ge 1$; so the cells have the side $\mathsf{l} := \frac12\ell' \ge L^2M$. As $g' \le \gamma_W^{\mathrm{sd}}$, the least side is small over the run: $MR(x_K) \le L^m$ (Lem. YM3_I:126.3). As $K \ge 3$, Lem. YM3_I:126.2, whose head asks $K \ge 3$, gives $x_j \ge x_{\mathrm{J}}$ for $0 \le j \le K$ and makes the cuts of the run, on the lattice (ℓ', K), frames of the construction; so the setting (J) holds with the last height $K-1$ ([7, Prop. B.21]), and the canonical sourced chain of $T_{\ell'}$ is formed, its conditions holding by [7, Cor. B.13], at the sources of a class of small sources, an open neighbourhood of 0.

By the running estimate YM3_I:(1530) at the index K and the endpoint bounds YM3_I:(1529), $b_- \le a_{K-j}x_j \le b_+$ for $0 \le j \le K$, with $b_\pm = x_K \pm \bar B$; and $g'^{-2} - \bar B < b_-$, $b_+ \le X_{g'} + \bar B$. The places, the classes of the sum rule, the image sets and the chart integrals of the steps, the backgrounds and the minimal orbits that the setting (J) names ([7, Lem. B.17] and [7, Prop. B.18]) are the objects that the sourced chain names ([7, Prop. B.15] and its one-step proof): both use the same definitions

(among them [7, Def. B.2] and equation [7, eq. 191] of the chart integral and [7, Def. B.2] of the image set), at the same lattice (ℓ', K), the same height j and the same coupling $\mathsf{t} = x_j$.

(4b)

The level-wise norm; the families and the energies. For a source w let $v_j^w = -L^{K-j} w/x_j$ be its weight at the level j. The *level-wise norm* $\|w\|_{\mathrm{lw}}$ is the largest, over $0 \le j < K$, of $|v_j^w|_\infty/\sigma_{\mathrm{a}}$ and of $\mathrm{sw}_j(v_j^w)/(\mu_{\mathrm{a}} x_j^{-1/2})$, where $\mathrm{sw}_j(v)$ is the largest difference of the values of v at two plaquettes whose barycentres are at a sup distance $\le M a_{K-j}$, and $\sigma_{\mathrm{a}} = \sigma_{\mathrm{src}}$, $\mu_{\mathrm{a}} x_j^{-1/2} = o_{\mathrm{src}}(x_j)$ are the radii of the class of the weights of eq. 6.xx, σ_{a} and μ_{a} depending only on the constants.

It is a norm; its open unit ball $\mathsf{Src}^{\mathrm{lw}}$ is convex, consists of the sources whose weights lie at every level in the class of the weights at which Prop. 6.32 is stated, and contains the class of small sources of (4a); and $|w|_\infty \le \sigma_{\mathrm{a}} b_+ \|w\|_{\mathrm{lw}}$, as $|v_j^w|_\infty = |w|_\infty/(a_{K-j} x_j)$. [7, Prop. B.12] runs the induction on $\mathsf{Src}^{\mathrm{lw}}$; consequently the sourced families of all levels are formed there, holomorphic in the source, bounded by E_0, a constant of the canonical list, in the norm of the localised terms at the rate $\kappa + 1$, and the family on a domain reads the source on the plaquettes of that domain alone.

So the source energies $e_j(X; w)$ of the domains X of the level j are holomorphic on $\mathsf{Src}^{\mathrm{lw}}$, vanish at $w = 0$ and read w on the plaquettes of X alone (equation [7, eq. 212] and its proof), and, by [7, eq. 213] at the bound $2E_0$ of the differences of the families and by Schwarz's lemma for $\zeta \mapsto e_j(X; \zeta w/\|w\|_{\mathrm{lw}})$ on the unit disc,

$$|e_j(X; w)| \le 16 M^3 E_0 \, \|w\|_{\mathrm{lw}} \, e^{-\kappa d_j(X)},$$

$d_j(X)$ the linear size of X in the units of the level j. Put $\mathcal{E}_t(w) = \sum_{j \le t} \sum_{X \in \mathcal{D}_j} e_j(X; w)$, $\mathcal{D}_j$ the localisation domains of the level j; so $\mathcal{E} = \mathcal{E}_K$, and these functions are holomorphic on $\mathsf{Src}^{\mathrm{lw}}$ and vanish at 0.

(4c)

The term without large fields, with a source. For $0 \le t \le K$ let h_t be the history without large-field regions up to the level t and $\tau_{h_t}[\cdot]$ its term (Section YM3_I:123), a complex Borel measure on the configurations $\mathcal{V}_t$ of the level t, and $\mathsf{m}_t = \tau_{h_t}[\rho_0]$: a positive measure of mass at most Z (equation eq. 4.xxxi), equal to $\rho_0 \, \mathrm{d}U$ at $t = 0$ and carried, for $t \ge 1$, by the open set $\mathcal{V}_t^\chi$ of the configurations that pass the density tests of the level (Prop. YM3_I:123.1). For $w \in \mathsf{Src}^{\mathrm{lw}}$ and $0 \le t \le K$,

$$\tau_{h_t}\big[\rho_0 e^{\mathcal{O}(w)}\big] = e^{\mathcal{E}_t(w)} \, e^{\mathsf{R}_t(w,\cdot)} \, \mathsf{m}_t,$$

an identity of complex Borel measures on $\mathcal{V}_t$, with $\mathsf{R}_0(w, \cdot) = \mathcal{O}(w)$ and, for $t \ge 1$ and $W \in \mathcal{V}_t^\chi$,

$$\mathsf{R}_t(w, W) = \mathcal{O}(w)(U) + S[\mathfrak{f}_t[w]](U) - S[\mathfrak{f}_t[0]](U),$$

U an element of the minimal orbit over W and $S[\mathfrak{f}_t[w]]$ the action of the sourced bare array of the level t; this value does not depend on the element of the orbit, the domain of definition of the action being a condition on the configuration alone, and $\mathsf{R}_t(w, \cdot)$ is a bounded Borel function on $\mathcal{V}_t^\chi$. For the small sources of (4a) and $1 \le t \le K$, the density of $\tau_{h_t}[\rho_0 e^{\mathcal{O}(w)}]$ is

$$\exp\Big[\mathsf{C}_t(w) - A^{\eta_t}(x_t^w, U_t(W)) + S[\mathfrak{f}_t[w]](U_t(W))\Big] \quad \text{on } \mathcal{V}_t^\chi, \text{ and 0 off it,} \qquad \mathsf{C}_t(w) = \sum_{1 \le j \le t} \mathsf{c}_j + \mathcal{E}_t(w),$$

with numbers c_j formed with the source 0 alone, the weight $x_t^w = x_t - L^{K-t} w$ of the weighted action A^{η_t}, $\eta_t = L^{-t}$, and the background $U_t(W)$ over W. This follows by induction over t. At $t = 0$ the term is $\rho_0 e^{\mathcal{O}(w)} dU$, and $\rho_0 e^{\mathcal{O}(w)} = e^{-A^{\eta_0}(x_0^w, \cdot)}$ (equation [7, eq. 216]). From t to $t+1$: the term of the history is carried by one step, $\tau_{h_{t+1}}[\cdot] = \mathbf{T}(\mathsf{g}_{t+1} \tau_{h_t}[\cdot])$ with the factor $\mathsf{g}_{t+1} \ge 0$ of the step, and a measure whose density on $\mathcal{V}_t^\chi$ is e^{e} times a bounded Borel function F is carried to the density $e^{\mathsf{e}} \, \mathsf{I}[F]$ on $\mathcal{V}_{t+1}^\chi$, and 0 off it, $\mathsf{I}[F]$ being the chart integral of the step ([7, Prop. B.18]); here, for $t \ge 1$, $\mathsf{e} = \mathsf{C}_t(w)$ and F is the function F_t^w in the proof of [7, Prop. B.15] at $k = t$, which on $\mathcal{V}_t^\chi$ is $\exp[-A^{\eta_t}(x_t^w, U_t(V)) + S[\mathfrak{f}_t[w]](U_t(V))]$ (the definition preceding [7, eq. 218]), and for $t = 0$, $\mathsf{e} = 0$

and $F = \rho_0 e^{\mathcal{O}(w)}$, whose chart integral is that of F_0^w on the class of the sum rule (the height-zero case in that proof); $\mathcal{V}_{t+1}^{\chi}$ lies in the class of the sum rule of the step and in the set on which the chart integral is formed ([7, Lem. B.17]); and there the one-step identity of the sourced chain [7, eq. 218] gives

$$\mathsf{I}[F_t^w](W) = \exp\Big[\mathsf{c}_{t+1} + \sum_{X\in\mathcal{D}_{t+1}} e_{t+1}(X;w) - A^{\eta_{t+1}}(x_{t+1}^w, U_{t+1}(W)) + S[\mathsf{f}_{t+1}[w]](U_{t+1}(W))\Big].$$

The weighted action is linear in the weight, $A^{\eta_t}(x_t^w, U) = A^{\eta_t}(x_t, U) - \mathcal{O}(w)(U)$; so the quotient of the density by the same density at $w = 0$, which is the density of m_t, is $e^{\mathcal{E}_t(w)+\mathsf{R}_t(w,\cdot)}$ on $\mathcal{V}_t^{\chi}$: the identity holds for the small sources. For $w \in \mathsf{Src}^{\mathrm{lw}}$, $w \neq 0$, and a bounded Borel function ψ on $\mathcal{V}_t$, the function $\zeta \mapsto \int \psi \,\mathrm{d}\tau_{h_t}[\rho_0 e^{\zeta\mathcal{O}(w)}]$ is entire, and $\zeta \mapsto e^{\mathcal{E}_t(\zeta w)} \int \psi\, e^{\mathsf{R}_t(\zeta w,\cdot)}\,\mathrm{d}\mathsf{m}_t$ is holomorphic on $\{|\zeta|\,\|w\|_{\mathrm{lw}} < 1\}$, by (4b); they agree for small ζ, hence at $\zeta = 1$.

(4d)

The remainders. Let $w \in \mathsf{Src}^{\mathrm{lw}}$ vanish off the plaquettes whose barycentres lie in N_{b} closed balls of radius 1. Then

$$|\mathsf{R}_K(w,\cdot)| \le \|w\|_{\mathrm{lw}} N_{\mathrm{b}}\, \mathsf{g}_\emptyset \ \text{ on } \mathcal{V}_K^{\chi}, \qquad |\mathsf{R}_{n-1}(w,\cdot)| + |\mathcal{E}(w) - \mathcal{E}_{n-1}(w)| \le \|w\|_{\mathrm{lw}} N_{\mathrm{b}}\, C_{\mathrm{cnt}}\, L^{4(K-n+1)} \ \text{ on } \mathcal{V}_{n-1}^{\chi} \quad (1 \le n \le K),$$

on all of $\mathcal{V}_0$ for $n = 1$, with numbers $\mathsf{g}_\emptyset$ and $C_{\mathrm{cnt}} \ge 1$ that depend only on g' and the constants. Indeed, let $W \in \mathcal{V}_s^{\chi}$. The background over W has $0 \le a_p(U_s(W)) < 2B_3^2\varepsilon_s^2 L^{-4s}$ at every bare plaquette [7, eq. 221], $\varepsilon_s = \varepsilon(x_s) \le \varepsilon(g'^{-2})$ being the threshold of the small fields at the coupling x_s and B_3 a constant of the canonical list; at most $81N_{\mathrm{b}}L^{3K}$ plaquettes have their barycentres in the balls; and $|w|_\infty \le \sigma_{\mathrm{a}} b_+ \|w\|_{\mathrm{lw}}$. So $|\mathcal{O}(w)(U_s(W))| \le 162 N_{\mathrm{b}} B_3^2 \varepsilon_s^2 \sigma_{\mathrm{a}} b_+ L^{4(K-s)} \|w\|_{\mathrm{lw}}$, and at $s = 0$, where $0 \le a_p \le 2$, $|\mathcal{O}(w)| \le 162 N_{\mathrm{b}} \sigma_{\mathrm{a}} b_+ L^{4K} \|w\|_{\mathrm{lw}}$.

The difference of the two actions is the sum, over the levels $j \le s$, the domains X of the level j that meet the support of w and their points z, of the differences between the values at U and at $\mathbf{1}$ of the local terms $(E^{(j)}[w] - E^{(j)}[0])(X,\cdot,z)$, each of a norm at most $2E_0\|w\|_{\mathrm{lw}} e^{-(\kappa+1)d_j(X)}$ by (4b). Every element of the minimal orbit over W is regular of the level s at the radius $2B_3\varepsilon_s$ ([7, Lem. B.17]); at such a configuration a local term of the level j moves from its value at $\mathbf{1}$ by at most its norm times $C(\varepsilon_s/\alpha_0)^2 L^{-4(s-j)}(1 + d_j(X))^2$ on the domains that are small against the radius, and by at most twice its norm on the other ones, whose linear sizes are large (equation YM3_I: (2263) and its proof), α_0 being the fixed radius of the localised terms; the bound in [7, Lem. B.10], with at most $27N_{\mathrm{b}}L^{3(K-s)}$ cubes of the level s holding a corner of a plaquette of the support, sums these differences to at most $CN_{\mathrm{b}}L^{3(K-s)}\|w\|_{\mathrm{lw}}$.

The energies of the levels $> s$ are bounded in the same way, by (4b) and (q3), by $CN_{\mathrm{b}}L^{3(K-s)}\|w\|_{\mathrm{lw}}$. With $b_+ \le X_{g'} + \bar{B}$ this gives the two bounds.

(4e) *The coefficients of $Z[w]e^{-\mathcal{E}(w)}/Z$.* By eq. 4.xxxi at level K, applied to the bounded gauge invariant function $e^{\mathcal{O}(w)}$, with the canonical history measures of Section YM3_I:123,

$$Z[w] = Z_\emptyset[w] + \sum_{n=1}^{K} Z_n[w], \qquad Z_\emptyset[w] = \tau_{h_K}[\rho_0 e^{\mathcal{O}(w)}](\mathcal{V}_K), \qquad Z_n[w] = \int \mathsf{g}_n^{\mathrm{w}}\,\mathrm{d}\tau_{h_{n-1}}[\rho_0 e^{\mathcal{O}(w)}],$$

where $\mathsf{g}_n^{\mathrm{w}} \ge 0$ is the factor of the step n at which a large-field region is first created, and $\mathsf{w}_n = Z^{-1}\int \mathsf{g}_n^{\mathrm{w}}\,\mathrm{d}\mathsf{m}_{n-1}$ is the weight of that event; each term is an entire function of w. By (4c),

$$Z_\emptyset[w]e^{-\mathcal{E}(w)} = \int e^{\mathsf{R}_K(w,\cdot)}\,\mathrm{d}\mathsf{m}_K, \qquad Z_n[w]e^{-\mathcal{E}(w)} = e^{-(\mathcal{E}-\mathcal{E}_{n-1})(w)} \int \mathsf{g}_n^{\mathrm{w}}\, e^{\mathsf{R}_{n-1}(w,\cdot)}\,\mathrm{d}\mathsf{m}_{n-1},$$

so that, by (4d) and as m_K has a mass at most Z,

$$\big|Z_\emptyset[w]e^{-\mathcal{E}(w)}\big| \le e^{\|w\|_{\mathrm{lw}} N_{\mathrm{b}} \mathsf{g}_\emptyset}\, Z, \qquad \big|Z_n[w]e^{-\mathcal{E}(w)}\big| \le \exp\big[\|w\|_{\mathrm{lw}} N_{\mathrm{b}} C_{\mathrm{cnt}} L^{4(K-n+1)}\big]\, \mathsf{w}_n\, Z.$$

At $w = W_z$ take $N_{\mathrm{b}} = 8$ and the polydisc $\mathbb{P} = \{|z_{\mathbf{k},a}| < r_a\}$, $r_a = A_\star(d)/\|f_a\|_\infty$, on which $\|W_z\|_{\mathrm{lw}} \le \max_{\mathbf{k},a} |z_{\mathbf{k},a}|/r_a < 1$ by (4g). The function $\Phi_{\mathrm{D}}(z) = Z[W_z]e^{-\mathcal{E}(W_z)}/Z$ is holomorphic on $\mathbb{P}$, $Z[w]$ being entire, and $\Phi_{\mathrm{D}}(0) = 1$; its term without large fields is at most $e^{8\mathsf{g}_\emptyset}$ on the whole of $\mathbb{P}$, and its term of the level n at most $e\,\mathsf{w}_n$ on the polydisc smaller by the factor $8C_{\mathrm{cnt}}L^{4(K-n+1)}$. Cauchy's estimate on each polydisc gives, for every set J of at most sixteen of the variables,

$$|\partial_J \Phi_{\mathrm{D}}(0)| \le \Big[e^{8\mathsf{g}_\emptyset} + e\,(8C_{\mathrm{cnt}}L^4)^{16} \sum_{n=1}^{K} L^{64(K-n)}\mathsf{w}_n\Big] \prod_{i\in J} r_i^{-1}.$$

By eq. 4.xxxi, $\mathsf{w}_n \le \tilde{\nu}_K\{Z_n \ne \emptyset\}$. Equation eq. 4.xxxiv, at $q = 64$ and $t = K$, therefore bounds $\sum_n L^{64(K-n)}\mathsf{w}_n$ by

$$4M^{-3}(\ell')^3 e^{\mathsf{B}(X_{g'})(\ell')^3}\mathsf{C}_{\mathrm{ag}}(65)e^{-c'p_0(g_K)^2/2}.$$

The assumptions hold by the canonical choice of the fixed history thresholds in Prop. 4.54; the ceiling $g' \le \gamma_W^{\mathrm{sd}}$ preserves them. Since ℓ' depends only on g' and the constants, the square bracket is bounded by a number $Q_{\mathrm{D}}(g') \ge 1$ with the same dependence.

(4f)

The linking cumulants. The last sum of the decomposition is a sum of functions each of which reads the variables of one cell; it is annulled by every linking derivative. In the bracket only the domains X that meet the supports of two parts $w^{(\mathbf{k})}$ contribute, the source energies being local and vanishing at 0. Such a domain of the level j holds corners of plaquettes whose barycentres lie in two different balls, and two such corners are at a sup distance $\ge \mathsf{l}-3$, while two points of a domain of the level j are at a sup distance at most $M(d_j(X)+2)a_{K-j}$; so $d_j(X) \ge (\mathsf{l}-3)L^{K-j}/M - 2 \ge 166L^{K-j}$, as $\mathsf{l} \ge L^2M$ and $L \ge 13$. The bound of (4b) at the nine sources W, $w^{(\mathbf{k})}$ and the bound in [7, Lem. B.10] give

$$\Big|\mathcal{E}(W) - \sum_{\mathbf{k}} \mathcal{E}(w^{(\mathbf{k})})\Big| \le C_\Delta \max_{\mathbf{k}}\{\|W\|_{\mathrm{lw}}, \|w^{(\mathbf{k})}\|_{\mathrm{lw}}\},$$

this exponential decay of the source energies in the size of their domains summing over the levels to a number C_Δ depending only on the constants. On $\mathbb{P}$ the bracket is holomorphic and at most C_Δ, so that Cauchy's estimate bounds its squarefree coefficients by $C_\Delta \prod_i r_i^{-1}$. A derivative ∂_J at 0 of the logarithm of a function equal to 1 at 0 is the sum, over the partitions π of J, of $(-1)^{|\pi|-1}(|\pi|-1)!$ times the product over the blocks of its derivatives; so the squarefree coefficients of $\log\Phi_{\mathrm{D}}$ are at most $\mathsf{cu}_{|J|}Q_{\mathrm{D}}^{|J|}\prod_{i\in J} r_i^{-1}$, with $\mathsf{cu}_N = \sum_\pi (|\pi|-1)!$ over the partitions of N elements.

(4g)

The packed bumps. For a source made of bumps with pairwise disjoint sets of non-vanishing, of height at most A and Lipschitz constant at most $A/(\vartheta d)$, the supremum of the weight of the level j is at most A/b_-, and its oscillation at most $(A/b_-)\min\{2, 2\sqrt{3}Ma_{K-j}/(\vartheta d)\}$; as $a^{-1/2}\min\{2, ca\} \le (2c)^{1/2}$ for all $a > 0$, its level-wise norm is at most $A/A_\star(d)$ at every level, with

$$A_\star(d) = \min\Big\{\sigma_{\mathrm{a}}b_-,\ \mu_{\mathrm{a}}\frac{b_-}{b_+^{1/2}}\Big(\frac{\vartheta d}{4\sqrt{3}M}\Big)^{1/2}\Big\} \ge c\,(g'^{-2} - \bar{B})(X_{g'} + \bar{B})^{-3/2}(\vartheta d)^{1/2}\,b \qquad (d \le d_\star(g',\vartheta)),$$

$d_\star$ being the scale below which the second member of the minimum is the smaller one. The bumps of W_z are the mirror images of the $z_{\mathbf{k},a}\epsilon_v\tau_v f_a$; their sets of non-vanishing are pairwise disjoint, their heights are at most $A = \max_{\mathbf{k},a}|z_{\mathbf{k},a}|\,\|f_a\|_\infty$ and their Lipschitz constants at most $A/(\vartheta d)$, as $\mathrm{Lip} f_a \le \|f_a\|_\infty/(\vartheta d)$; so $\|W_z\|_{\mathrm{lw}} \le \max_{\mathbf{k},a}|z_{\mathbf{k},a}|/r_a$. So, by (4f), every linking cumulant is at most $(\mathsf{cu}_{16}Q_{\mathrm{D}}^{16} + C_\Delta)\prod_{(\mathbf{k},a)\in\gamma} r_a^{-1}$: the linking cumulants obey the bound of (2) with $u_a = \|f_a\|_\infty/A_\star(d)$ and $Q_1 = \mathsf{cu}_{16}Q_{\mathrm{D}}^{16} + C_\Delta$.

(5)

Conclusion. By (1)–(4),

$$|S_2^T[\mathfrak{s}](\Psi) - S_2^T[\mu'](\Psi)| \le (30d)^3\,2(Q_0\mathsf{BI}_{16}^{1/8}Q_1)^2 A_\star(d)^{-2}|\Psi| \le C_2^{\mathrm{T}}(g')\vartheta^{-1}d^2b^{-2}|\Psi|$$

for $d \le d_\flat$, which is (a) with the constants of (b); two states differ by at most twice this. □

Remark 8.4. The proof uses only the following properties of the pair: real Lipschitz functions, neither 0, each vanishing off a ball of radius d, whose sets of non-vanishing are disjoint and lie in one ball of radius $9d$, with $\mathrm{Lip} f_a \le \|f_a\|_\infty/(\vartheta d)$. So Prop. 8.3 holds also for such pairs of either sign. With n entries in the place of two, every pair of which lies in $\mathfrak{T}_{\mathbb{R}^3}(d;\vartheta)$, the same argument gives a difference of at most $2C_n^{\mathrm{T}}(g')\vartheta^{-n/2}d^{3-n/2}b^{-n}\prod_a\|f_a\|_\infty$, which is small against the natural size $b^{-n}\prod_a\|f_a\|_\infty$ for $n \le 5$; we use only $n = 2$.

8.1.8. *The coefficient on the least torus.* Here the constants are an admissible assignment of the canonical list. The statements used in Lems. 8.5–8.8 are made for the data (Wp1)–(Wp6) of a weighted progression ([7, Def. B.7]), in its setting (J) ([7, Def. B.16]) and, where a step is continued to a complex field scale, at a place with a lower place (Def. 6.29); we call this the setting of the weighted progression (the statements used in Lem. 8.9, the constants and thresholds in Thm. YM3_I:AR, Thm. YM3_I:AR, equation YM3_I:(2203), Cor. YM3_I:117.1, Lem. 2.5 and Cor. YM3_I:118.4, are made in the setting (Th1), (Th2), whose datum is the choice of the constants).

For the canonical objects at every admissible assignment of the canonical list it holds: the data are given by [7, Cor. B.9]; the setting (J) holds by [7, Prop. B.21] as soon as the couplings are at least the threshold x_{J} of the definitions preceding [7, eq. 228], a number depending only on the constants, and at a first-passage datum at g this is the case whenever $g^{-2} \ge x_{\mathrm{J}}$, all the couplings of the run exceeding g^{-2} (Lem. 8.5); and the places with a lower place are those of Prop. 6.31. Besides the constants the setting involves two auxiliary real numbers: the lower place $\mathsf{t}_\flat \ge \gamma^{-2}$, and a room $\mathsf{q} > 0$, such that a bound on the sizes of the bare arrays formed with the constants alone, increased by $2\mathsf{q}$, does not exceed the size ceiling at the lower place.

For the canonical objects $\mathsf{q} = 1$, and $\mathsf{t}_\flat$ is a number depending only on the constants. No further condition on the constants enters. $\gamma_{\mathrm{e}} \in (0,\gamma]$ is the largest number under which the setting holds and all the conditions on g of these statements hold (each has the form $g \le \gamma'$ or $\log g^{-2} \ge Y$ with numbers γ', Y depending only on the constants), together with $g^{-2} \ge 4\bar{B}$; for $0 < g \le \gamma_{\mathrm{e}}$, $d_{\mathrm{e}}(g,\vartheta) \in (0, L^{-13}]$ is the largest number below which their conditions on d and those of the proof of Prop. 8.10 hold (the arithmetic of the levels there uses $d \le L^{-13}$, and the removal of the cutoff there uses $d \le d_{\mathrm{K}}$, the number of Lem. 8.9 at $\ell = \ell_g$).

The depth of the scale d at g is the witness depth $s(d) = \min\{s \ge 0 : L^s d \ge C_W(s)\}$, $C_W(s) = \mathsf{c}_W(\log[L^{s+1}(g^{-2}+\bar{B})])^{2p_0}$, with a number $\mathsf{c}_W \ge 1$ depending only on the constants (equation YM3_I:(2321)); so $C_W(s(d)) \le L^{s(d)}d < LC_W(s(d))$, $C_W(s(d)) \ge (\log\frac{1}{d})^2$, $s(d) \to \infty$ as $d \to 0$, and $K > s(d)$ entails $d/a_K \ge L(\log\frac{1}{d})^2$, hence $K \ge 3$ and $8a_K \le d$. We write $D_j = a_{K-j}$ (the spacing of the level j), $\mathsf{l} = \max\{1, \log(2b/d)\}$, $\mathfrak{b}(x) = A_1(\log x)^{p_0}$ for the field cut-off, A_1 and the integer $p_0 \ge 1$ being constants of the canonical list, $\|u\|_\sharp = \max\{\|u\|_\infty, 40M\,\mathrm{Lip}\,u\}$ and $\mathbb{Z}^3_N = \mathbb{Z}^3/N\mathbb{Z}^3$; C denotes numbers that depend only on the constants, and on g and ϑ where this is shown.

Lemma 8.5 (The run)**.** Let $0 < g \le \gamma_{\mathrm{e}}$ and (K,β) a first-passage datum at g with $K \ge 3$ and label b. Then $|x_j - L^{-(j-i)}x_i| \le \bar{B}$ for $0 \le i \le j \le K$, $g^{-2} < x_K \le X_g$, and $\frac{3}{4}b/D_j \le x_j \le 2b/D_j$ for $0 \le j \le K$. On every $\ell \in \mathfrak{L}(g)$ the setting of the weighted progression holds with the sources $f\circ x$, $h\circ x$ for all real bounded f, h on T_ℓ supported in closed balls of radius d at a distance $\ge d$, with $8a_K \le d$, neither $f\circ x$ nor $h\circ x$ vanishing identically; no other condition on d enters. If $f\circ x$ or $h\circ x$ vanishes identically, then so does $(\tau_v f)\circ x$, respectively $(\tau_v h)\circ x$, for every $v \in a_K\mathbb{Z}^3$, and $S^T_{2;K,\ell,\beta}(f,h)$, the jets $\mathsf{n}_j(f,h)$ and $\mathsf{n}^{\mathsf{z}}_j(f,h)$ of Lem. 8.7 and every sum $\sum_{p,q}(f\circ x)(p)\,(h\circ x)(q)\,\mathsf{k}(p,q)$ against a kernel k on the plaquettes vanish.

So, for $b_* \in I(g)$, $K' \ge 3$ and real functions w_1, w_2 on the plaquettes of $\mathbb{Z}^3_{\ell L^{K'}}$, neither identically 0, supported in balls of radius $r \ge 8$ at a distance $\ge r$, $w_a = f'_a \circ x$ for such a pair (f'_1, f'_2) at the scale $ra_{K'}$ and the first-passage datum $(K', L^{K'}b_*)$. ■

Proof sketch. The first two assertions are the running and endpoint estimates YM3_I:(1529)–(1530). As $L^{-j}x_0 = b/D_j$ and $x_j \geq x_K > g^{-2} \geq 4\bar{B}$: $b/D_j \leq x_j + \bar{B} \leq \frac{4}{3}x_j$, and $x_j \leq b/D_j + \bar{B} \leq 2b/D_j$ because $b/D_j \geq b > g^{-2} \geq \bar{B}$. The data of the weighted progression are those of the canonical objects ([7, Cor. B.9]) at the calibrations $x_0, \dots, x_{K-1}$ of the run, with the lower place $\mathfrak{t}_\flat$ and the room $\mathfrak{q} = 1$; the lattice (ℓ, K), with the run, is admissible over every cut of the run, for every $\ell \in \mathfrak{L}(g)$ (Prop. YM3_I:74.1); and the setting (J) holds, as $x_j \geq x_K > g^{-2} \geq x_{\mathrm{J}}$ ([7, Prop. B.21]).

The other conditions of the setting are lower bounds on x_{K-1} and on $\log x_K$ by numbers formed with the constants and with $\mathfrak{t}_\flat$; they hold as $x_{K-1} > X_g \geq Lg^{-2}$ and $x_K > g^{-2}$ once g is small, and they enter γ_{e}. Of the sources the setting reads only that they are real and bounded, supported in balls of radius d at a distance $\geq d$, with $8a_K \leq d$, and not identically 0; a translation by $v \in a_K\mathbb{Z}^3$ permutes the barycentres of the plaquettes. If $f \circ x \equiv 0$, then $\mathcal{O}_K(f) = 0$, the source $\mu_1 f \circ x + \mu_2 h \circ x$ does not read μ_1, so that the source energies of the weighted progression do not read it at any field scale, and a sum of the values $(f \circ x)(p)$ against anything vanishes.

For the last sentence let f'_a take the values of w_a at the barycentres of the plaquettes of spacing $a_{K'}$ and vanish elsewhere: they are real and bounded, supported in balls of radius $ra_{K'}$ at a distance $\geq ra_{K'}$, with $8a_{K'} \leq ra_{K'}$, and no condition on the depth of the scale $ra_{K'}$ enters; $(K', L^{K'}b_*)$ is a first-passage datum since $b_* \in I(g)$. □

Lemma 8.6 (The chart of the iterated block averages)**.** Let $\ell = 2L^m$, $1 \leq n \leq K$ and $2L^{m+K-n+1} \geq C_L M$, C_L a constant of the canonical list. For a block field W of the level n and fluctuation fields $y = (\mathcal{A}_j)_{j<n}$, $\mathcal{A}_j$ an $\mathfrak{su}(2)$-valued 1-form on the non-pivot bonds of the level j, let $W_n = W$, $W_j = V_{W_{j+1}}[\mathcal{A}_j]$ ([7, §B.2]), $\Upsilon(W, y) = W_0$, and $\theta_{\mathrm{ch}} = -\sum_{j<n}[\mathsf{J}_{V_j}[\mathcal{A}_j] + \log z^{(j)}_{V_j}]$, with the Jacobian term J and the normalisation $z^{(j)}$ of [7, Lem. B.3], $V_j = V_j^{\mathrm{bf}}(W_{j+1})$ being the averaged background over the block field W_{j+1} ([7, §B.2]).

There are an open neighbourhood O_n of $\mathbf{1}$ among the block fields of the level n and a number $\rho_2 \in (0, 1]$, both defined by bounds that depend only on the constants and not on n, K or m (they are written out in the proof), such that, with $\Omega = \{|\mathcal{A}_j| < \rho_2,\ j < n\}$ and

$$C_{\mathrm{H}} = \prod_{j<n} (2\pi^2)^{-\#\mathfrak{X}_j^+} L^{-3\#\mathfrak{B}_{j+1}^+},$$

$\mathfrak{X}_j^+$ the positively oriented non-pivot bonds of the level j and $\mathfrak{B}_{j+1}^+$ the positively oriented bonds of the level $j+1$:

(i) on $\mathsf{O}_n \times \Omega$ the chain and θ_{ch} are formed, W_{j+1} is the block average of W_j, and θ_{ch} is real analytic;

(ii) Υ is a real analytic diffeomorphism of $\mathsf{O}_n \times \Omega$ onto an open set $\mathcal{U} \ni \mathbf{1}$;

(iii) $\int H \,\mathrm{d}U = C_{\mathrm{H}} \int_{\mathsf{O}_n} \mathrm{d}W \int_\Omega \mathrm{d}y\, e^{-\theta_{\mathrm{ch}}} H(\Upsilon(W, y))$ for every bounded Borel function H that vanishes off $\mathcal{U}$, and the same holds with every open subset of $\mathsf{O}_n \times \Omega$ and its image.

■

Proof sketch. The numbers and the good sets. For a configuration V of a level let $\varpi(V) = \max_p |\partial V(p) - \mathbf{1}|$, the maximum over the plaquettes of that level, and $\varpi_c(V)$ the same maximum over the plaquettes of the double block of a bond c of the next level; $\mathcal{M}$ is the block average and Lg the $\mathfrak{su}(2)$-valued logarithm. With the numbers ϖ_χ, $\varpi_1 = 2\varpi_\chi$, r_χ, C_χ of the chart of the fibre ([7, §B.2]), which depend only on L, the small-curvature radius c_2^{SO} of Prop. YM3_I:14.1 and the response constant K_{B} of [7, Lem. P.5], and the constants B_3, $\bar{\varepsilon}$, ε_{v}, r_{B} of the canonical list, put

$$\mu_\flat = \min\Big\{c_2^{\mathrm{SO}},\ \tfrac{1}{2}\bar{\varepsilon},\ \tfrac{1}{4}L^2\varpi_\chi,\ \tfrac{1}{8}\varepsilon_{\mathrm{v}}\Big\}, \qquad \rho_2 = \min\Big\{\frac{r_\chi}{C_\chi},\ \frac{r_{\mathrm{B}}}{4C_\chi},\ \frac{(1-L^{-2})\,\mu_\flat}{4K_{\mathrm{B}}C_\chi},\ \frac{\varepsilon_{\mathrm{v}}}{64C_\chi}\Big\}, \qquad \mathsf{O}_n = \Big\{W :\ \varpi(W) < \frac{\mu_\flat}{B_3}\Big\};$$

they read neither n nor K nor m, and $\rho_2 \le r_\chi/C_\chi \le 1$. For $1 \le i \le n$ let Gd_i be the set of the block fields W of the level i over which the background exists and has a regularity number $\mu_i(W) < \mu_\flat$ ([7, Def. B.2]; the number is the same at all the elements of the minimal orbit, by unitary gauge invariance), and $\mathsf{Gd}_0 = \{V : \varpi(V) < \frac14 \varepsilon_{\mathrm{v}}\}$.

(1)

The averaged background over a good block field. Gd_i is open and $\{\varpi < \mu_\flat/B_3\} \subset \mathsf{Gd}_i \subset \{\varpi < 2\mu_\flat\}$ for $1 \le i \le n$: over a block field with $\varpi < \mu_\flat/B_3$ the background exists and has a regularity number $< \mu_\flat$ (Thm. YM3_I:E in the setting of Section YM3_I:52.7), a regularity number $< \mu_\flat$ gives $\varpi < 2\mu_\flat \le \bar\varepsilon$ (Lem. 5.28), and the representative of the background is a real analytic function of the block field (Thm. YM3_I:E and [7, Lem. P.5]); so $\mathbf{1} \in \mathsf{O}_n \subset \mathsf{Gd}_n$. Let $0 \le j < n$ and $W \in \mathsf{Gd}_{j+1}$. Then $V_j(W) = V_j^{\mathrm{bf}}(W)$ is defined and $\mathcal{M} V_j(W) = W$: it is the average $\mathcal{M}^j U$ of an element U of the minimal orbit over W, carried to the axial gauge by a gauge transformation (the definition of $p(W)$ in [7, Def. B.2]), which conjugates the plaquette variables; so Lem. 5.28 gives $\varpi(V_j(W)) \le 2L^{-2}\mu_{j+1}(W) < \frac12 \varpi_\chi$.

The map $W \mapsto V_j(W)$ is real analytic on Gd_{j+1}: $\mathcal{M}^j U^\circ(W)$, with the representative $U^\circ(W)$ of the background, is real analytic by Thm. YM3_I:E and Lem. 5.28, on the good-set and actual-average domains just established, and the gauge transformation to the axial gauge is $v(x) = (\mathcal{M}^j U^\circ(W))(y,x)$, x in the block of y, an averaged contour variable, a real analytic function of the configuration where it is formed. And $V_j(\mathbf{1}) = \mathbf{1}$: $\mathbf{1}$ is flat, lies in the minimal orbit over $\mathbf{1}$, and every path product and its average are $\mathbf{1}$; uniqueness of the minimal orbit in Thm. YM3_I:E identifies its axial representative with $\mathbf{1}$.

(2)

One step. Let O be an open subset of Gd_{j+1}, $0 < \rho \le r_\chi/C_\chi$, $G(\rho) = \{u \in SU(2) : |\operatorname{Lg} u| < \rho\}$, and

$$\mathcal{N} = \{V : \ \mathcal{M}V \in \mathsf{O}, \ \ V(\mathsf{e})\, V_j(\mathcal{M}V)(\mathsf{e})^{-1} \in G(\rho) \text{ at every non-pivot bond } \mathsf{e}\}.$$

At $W' \in \mathsf{O}$ the configuration $V_j(W')$, with $\varpi_c(V_j(W')) \le \varpi_\chi$, is a reference configuration of the chart of the fibre, and Prop. YM3_I:53.2, with empty exterior, gives for $|\mathcal{A}| < \rho$: $V_j(W')$ is real and admissible, with invertible pivot coefficients; the chart is formed at $\mathcal{A}$, $V_{W'}[\mathcal{A}] = e^{iB'} V_j(W')$ with a real B', $\sup|B'| \le 2C_\chi|\mathcal{A}|$; $\mathcal{M}V_{W'}[\mathcal{A}] = W'$; $\operatorname{Lg}(V_{W'}[\mathcal{A}](\mathsf{e}) V_j(W')(\mathsf{e})^{-1}) = \mathcal{A}(\mathsf{e})$ at the non-pivot bonds; $\varpi_c(V_{W'}[\mathcal{A}]) < \varpi_1$; and $\mathsf{J}_{V_j}[\mathcal{A}]$, $z_{V_j}^{(j)}$ are formed, real and continuous (the continuity statement and the local bounds in its proof).

By the converse in that proposition, $\Phi_j(W', \mathcal{A}) = V_{W'}[\mathcal{A}]$ is a bijection of $\mathsf{O} \times \{|\mathcal{A}| < \rho\}$ onto the open set $\mathcal{N}$, with the inverse $\Xi_j(V) = (\mathcal{M}V, (\operatorname{Lg}(V(\mathsf{e}) V_j(\mathcal{M}V)(\mathsf{e})^{-1}))_{\mathsf{e}})$. The map Ξ_j is real analytic on $\mathcal{N}$: $\mathcal{M}$ is real analytic at configurations whose plaquettes in the double blocks are small, V_j is by (1), and Lg is the logarithmic series where $|u - \mathbf{1}| < 1$. Its differential is injective: if the derivatives of $\mathcal{M}V$ and of the second member of $\Xi_j(V)$ vanish along a tangent vector $\dot a$, then $\dot a = 0$ at the non-pivot bonds, and then the derivative of $(\mathcal{M}V)(c)$ at a bond c of the level $j+1$ is $\mathsf{n}_c\, \dot a(b_0(c))$, with the invertible kernel n_c at the pivot bond $b_0(c)$ (the pivot differential preceding equation YM3_I:(911)), so that $\dot a = 0$.

The dimensions agree, the positively oriented bonds of the level j being the non-pivot ones and one pivot bond for each positively oriented bond of the level $j+1$. By the inverse function theorem Ξ_j is a local real analytic diffeomorphism; being injective, it has the real analytic inverse Φ_j, a real analytic diffeomorphism of $\mathsf{O} \times \{|\mathcal{A}| < \rho\}$ onto $\mathcal{N}$. The integral identity of the chart of the fibre (equation YM3_I:(986), with $e^{\mathsf{z}_c(V)} = L^{-3} z_V^{(j)}(c)$ because $\mathsf{n}_c = L\mathsf{m}_c$ acts on a three-dimensional pivot space) gives, for every Borel function $H \ge 0$ on the configurations of the level j,

$$\int_{\mathcal{N}} H \,\mathrm{d}V = c_j \int_{\mathsf{O}} dW'\, z_{V_j}^{(j)} \int_{|\mathcal{A}|<\rho} e^{\mathsf{J}_{V_j}[\mathcal{A}]}\, H(V_{W'}[\mathcal{A}])\,\mathrm{d}\mathcal{A}, \qquad V_j = V_j(W'), \quad c_j = (2\pi^2)^{-\#\mathfrak{X}_j^+} L^{-3\#\mathfrak{B}_{j+1}^+},$$

and the exponent has the closed form

$$\mathsf{J}_{V_j}[\mathcal{A}] + \log z_{V_j}^{(j)} = \sum_{\mathsf{e}\in\mathfrak{X}_j^+} \log\sigma_1(|\mathcal{A}(\mathsf{e})|^2) + \sum_c \mathsf{z}_c(V_{W'}[\mathcal{A}]) + 3\,\#\mathfrak{B}_{j+1}^+ \log L, \qquad \sigma_1(s^2) = \Big(\frac{\sin s}{s}\Big)^2,$$

where $\mathsf{z}_c = -\log\det \mathsf{n}_c$ is a real analytic function of the configuration on $\{\varpi_c < \varpi_1\}$.

(3)

The step keeps the good sets, with constants alone. Let $W' \in \mathsf{Gd}_{j+1}$, $|\mathcal{A}| < \rho_2$ and $W'' = V_{W'}[\mathcal{A}]$. For $j \geq 1$: $V_j(W') = \mathcal{M}^j U_+$ with an element U_+ of the minimal orbit over W', which lies in the minimal orbit over $V_j(W')$ ([7, Lem. B.4](2)), so that $\mu_j(V_j(W')) = L^{-2}\mu_{j+1}(W')$ (the level scaling in [7, Def. B.2]); and $W'' = e^{iB'}V_j(W')$ with $\sup|B'| \leq 2C_\chi|\mathcal{A}| < \frac{1}{2}r_{\mathrm{B}}$ moves the representative of the background so that its regularity number moves by at most $2K_{\mathrm{B}}\sup|B'|$ ([7, Lem. P.5]): the background over W'' exists and

$$\mu_j(W'') \leq L^{-2}\mu_{j+1}(W') + 4K_{\mathrm{B}}C_\chi|\mathcal{A}| < L^{-2}\mu_\flat + (1-L^{-2})\mu_\flat = \mu_\flat.$$

For $j = 0$: every bond moves by $|e^{iB'(\mathsf{e})} - \mathbf{1}| \leq |B'(\mathsf{e})|$, and a plaquette variable by at most the sum over its four bonds, so that $\varpi(W'') \leq \varpi(V_0(W')) + 8C_\chi|\mathcal{A}| \leq 2L^{-2}\mu_1(W') + 8C_\chi|\mathcal{A}| < \frac{1}{4}\varepsilon_{\mathrm{v}}$. So the step maps $\mathsf{Gd}_{j+1} \times \{|\mathcal{A}| < \rho_2\}$ into Gd_j.

(4)

Composition. Put $\mathcal{U}_n = \mathsf{O}_n$ and, downwards, $\mathcal{U}_j = \Phi_j(\mathcal{U}_{j+1} \times \{|\mathcal{A}_j| < \rho_2\})$. By (2) at $\mathsf{O} = \mathcal{U}_{j+1}$, which lies in Gd_{j+1} by (3), every $\mathcal{U}_j$ is open and lies in Gd_j, and $\Phi_j \times \mathrm{id}$ is a real analytic diffeomorphism. Υ is their composite: a real analytic diffeomorphism of $\mathsf{O}_n \times \Omega$ onto the open set $\mathcal{U} = \mathcal{U}_0$, injective on all of $\mathsf{O}_n \times \Omega$, with the inverse $U \mapsto (\mathcal{M}^n U, (\mathrm{Lg}(\mathcal{M}^j U(\mathsf{e})\, V_j(\mathcal{M}^{j+1}U)(\mathsf{e})^{-1}))_{\mathsf{e},j})$, and $\Upsilon(\mathbf{1}, 0) = \mathbf{1}$ by (1). Along the chain the backgrounds exist (at level 0 the background is $U_0(W_0) = W_0$, and $|J(W_0)| \leq 4\varpi(W_0) < \varepsilon_{\mathrm{v}}$, as in the height-zero part of the proof of Prop. YM3_I: 52.3), $\mathcal{M}W_j = W_{j+1}$ and θ_{ch} is formed; this is (i) and (ii), and by the closed form of (2)

$$\theta_{\mathrm{ch}} = -\sum_{j<n}\Big[\sum_{\mathsf{e}\in\mathfrak{X}_j^+} \log\sigma_1(|\mathcal{A}_j(\mathsf{e})|^2) + \sum_c \mathsf{z}_c(W_j)\Big] - 3\log L \sum_{j<n} \#\mathfrak{B}_{j+1}^+,$$

a real analytic function on $\mathsf{O}_n \times \Omega$. The identities of the steps, applied downwards with Tonelli's theorem, give (iii) for Borel $H \geq 0$ that vanish off $\mathcal{U}$, with $C_{\mathrm{H}} = \prod_j c_j$, and for bounded complex Borel H by linearity. As Υ is injective on all of $\mathsf{O}_n \times \Omega$, a function that vanishes off the image of an open subset E gives an integrand that vanishes off E: this is (iii) for E. The density is fixed by (iii): if a continuous θ' satisfied (iii) as well, the function H equal to the sign of $e^{-\theta'} - e^{-\theta_{\mathrm{ch}}}$, carried to $\mathcal{U}$ by Υ, would give $\int |e^{-\theta'} - e^{-\theta_{\mathrm{ch}}}|\,\mathrm{d}W\,\mathrm{d}y = 0$, and a continuous function with this property vanishes. □

Lemma 8.7 (The jets of the weighted progression)**.** Let $0 < g \leq \gamma_{\mathrm{e}}$, (K, β) and b as in Lem. 8.5, $\ell \in \mathfrak{L}(g)$, and f, h as there, with $d < 1$; $\Psi = (f, h)$. $\mathsf{n}_j(f, h)$ is the mixed second derivative at 0 in (μ_1, μ_2) of the source energy of the level j of the weighted progression with the source $\mu_1 f \circ x + \mu_2 h \circ x$, summed over the localisation domains and points of the level j; $\mathsf{J}_2 = \sum_{j\leq K} \mathsf{n}_j$. When all fluctuation fields are multiplied by a complex field scale z and each x_i is replaced by $L^{-i}\beta + \mathsf{z}^2(x_i - L^{-i}\beta)$, n_{j+1} becomes an even function $\mathsf{n}_{j+1}^{\mathsf{z}} = \sum_m \mathsf{n}_{j+1,[2m]}\mathsf{z}^{2m}$, holomorphic for $|\mathsf{z}| < R_j$ with $R_j^{-2} \leq C(\log x_j)^{2p_0} D_j/b$, and $\mathsf{n}_{j+1}^1 = \mathsf{n}_{j+1}$; put $\mathsf{h}_{2m}^{(n)} = \sum_{j<n} \mathsf{n}_{j+1,[2m]}$ and $\mathsf{h}_{2m} = \mathsf{h}_{2m}^{(K)}$. Then:

(i) $\mathsf{n}_{j,[0]} = \mathsf{n}_{j,[2]} = 0$, and $|\mathsf{J}_2 - \mathsf{h}_4 - \mathsf{h}_6| \leq Cb^{-4}d^2|^{8p_0}|\Psi|$;

(ii) for $0 \leq n \leq K-1$ with $D_n \geq d$: $|\mathsf{h}_6 - \mathsf{h}_6^{(n)}| \leq Cb^{-3}|^{6p_0} d\,(d/D_n)^{3/2}|\Psi|$;

(iii) for $(f, h) \in \mathfrak{T}_{T_\ell}(d; \vartheta)$, $20d \leq \ell$, $d \leq L^{-1}$ and $K > s(d)$: $|\mathsf{h}_4 - c_2 b^{-2}\mathcal{K}_6(f, h)| \leq [\bar{\varepsilon}_4(g, d)\, b^{-3}d + \frac{3}{2}C_{\mathrm{tm}}\vartheta^{-1}b^{-2}a_K/d]|\Psi|$, with $\bar{\varepsilon}_4(g, d) \leq C(g)d^{1/2}$ and the constant C_{tm} of Prop. YM3_I:121.2 and Lem. 6.34;

(iv) for $\ell = \ell_g$, $(f,h) \in \mathfrak{T}_{T_\ell}(d;\vartheta)$, $d \le d_{\mathrm{e}}(g,\vartheta)$ and $s(d) \le K$: there is a set $\mathsf{Pk} = \mathsf{Pk}(a_K, d) \subset a_K\mathbb{Z}^3$, depending on a_K and d alone, with $|v| < 1$ on it and $\#\mathsf{Pk} \ge (30d)^{-3}$, such that

$$\Big| S^T_{2;K,\ell,\beta}(f,h) - \frac{1}{\#\mathsf{Pk}} \sum_{v\in\mathsf{Pk}} \mathsf{J}_2(\tau_v f, \tau_v h) \Big| \le C_{\mathrm{A}}(g)\, \vartheta^{-1} d^2 b^{-2} |\Psi|,$$

τ_v the translation by v and $C_{\mathrm{A}}(g)$ a number depending only on g and the constants.

■

Proof sketch. (i), (ii). By Lem. 8.5, if $f \circ x$ or $h \circ x$ vanishes identically, all the jets vanish; let neither vanish.

The steps at one field scale. When the fluctuation fields of all the steps are multiplied by one complex field scale z and the coefficients of the action are moved as in the statement, the step of the level $j+1$ is a step at the place x_j with the lower place $\mathsf{t}_\flat$ (Def. 6.29; Prop. 6.31 for the canonical objects), the deformed activities of all the earlier levels standing in its bare array: the room q bounds their size uniformly in z, so that one induction over the levels forms all the steps at once. So the activities of the level $j+1$ are pointed families, holomorphic and even in z on $|\mathsf{z}| < R_j$, $R_j = \delta(\mathsf{t}_\flat)/(2\delta(x_j))$ with $\delta(\mathsf{t}) = \mathsf{t}^{-1/2}\mathfrak{b}(\mathsf{t})$, equal at $\mathsf{z} = 1$ to the activities of the weighted progression, and local in the weights; and $R_j^{-2} = 4\delta(x_j)^2/\delta(\mathsf{t}_\flat)^2 \le C(\log x_j)^{2p_0} D_j/b$, as $x_j \ge \frac{3}{4}b/D_j$ (Lem. 8.5).

One step at the field scale z, as a map of its bare array (the activities of the earlier levels and the weight of the level), has first and second derivatives at most $C\, q(|\mathsf{z}|/R_j)$ times the sizes of the increments, $q(t) = t^2/(1-t^2)$, by Cauchy's estimate, its order 0 in z reading no bare array (Prop. 6.27 with Prop. 6.30); the chain rule gives the derivatives in the two parameters μ_1, μ_2 of the activities of the level $j+1$ from those of the earlier levels and from the weight.

The derivatives in the sources. The weight of the source a at the level i is $L^{K-i}w_a$, $w_1 = f \circ x$, $w_2 = h \circ x$, of size $\|f_a\|_\infty/D_i$ at a plaquette, $f_1 = f$, $f_2 = h$. The weight member of the bare array of the level i is L^i times its sum against the a_p over the plaquettes of a domain, a_p being at most of the order L^{-4i} on the analyticity core; a domain of one cube holds at most of the order L^{3i} plaquettes, and at most $35(d/a_K)^3$ plaquettes carry w_a; as $L^{-3i}(d/a_K)^3 = (d/D_i)^3$, the size of this member is at most $C(\|f_a\|_\infty/D_i)\psi_i$, $\psi_i = \min\{1, (d/D_i)^3\}$. An activity of an earlier level $i-n$ that vanishes off the domains meeting a set of sites enters the bare array of the level i with a size at most $C[\varpi_n + (n+2)L^{-4n}\min\{L^{3n}, N_{\mathrm{cb}}\}]$ times its norm, $\varpi_n = L^{3n}e^{-\max\{2L^{n-1}-2,0\}}$ and N_{cb} the number of the cubes of the level $i-n$ that hold such a site; for the supports of the sources $N_{\mathrm{cb}} \le 77\max\{1, (d/D_{i-n})^3\}$.

Along the run $\delta(x_{i-n}) \le C(1+n)^{p_0}L^{-n/2}\delta(x_i)$, so $R_{i-n}^{-2} \le C(1+n)^{2p_0}L^{-n}R_i^{-2}$. By induction over $i \le j$ one obtains, for $|\mathsf{z}| \le \eta_0 R_j$, $\eta_0 \in (0, \frac{1}{2}]$ a number depending only on the constants: the first derivative in μ_a of the activity of the level $i+1$ has a norm at most $C|\mathsf{z}|^2R_i^{-2}(\|f_a\|_\infty/D_i)\psi_i$, and the mixed second derivative a norm at most $C|\mathsf{z}|^2R_i^{-2}(\|f\|_\infty/D_i)(\|h\|_\infty/D_i)\chi_i$, $\chi_i = \min\{1, (d/D_i)^{5/2}\}$; here $\psi_i^2 \le \chi_i$ bounds the product of two first derivatives, and the second derivatives of the earlier levels, carried up with the sizes above, sum to at most $C\chi_i$, the levels below the scale d through the decay of the activities across the distance of the two supports, those above it through the factor $L^{-n/2}$ that is left when the factor L^{-4n} of the size is set against the growth L^n of the radii and weights and $L^{5n/2}$ of the profile.

The jet. $\mathsf{n}^{\mathsf{z}}_{j+1}$ is the sum, over the domains of the level $j+1$ and their points, of the mixed second derivative at the trivial configuration. Only the domains that meet both supports contribute, the activities being local in the weights; the cubes of the level $j+1$ that hold a corner of a plaquette of a support number at most $77\max\{1, (d/D_{j+1})^3\}$; two corners of plaquettes of the two supports are at a sup distance $\ge d/4$, so that such a domain has a linear size $\ge [d/(4MD_{j+1}) - 2]_+$ in the units of the level $j+1$, and the decay of the localised terms in the size of their domains, summed by the

bound in [7, Lem. B.10], gives the factor $e^{-[d/(4MD_{j+1})-2]_+}$. Hence, on $|\mathsf{z}| \le \eta_0 R_j$,

$$|\mathsf{n}^{\mathsf{z}}_{j+1}| \le C|\mathsf{z}|^2 R_j^{-2} D_j^{-2} \chi_j \, \Xi_{j+1} |\Psi|, \qquad \chi_j = \min\{1, (d/D_j)^{5/2}\}, \qquad \Xi_{j+1} = \max\{1, (d/D_{j+1})^3\}\, e^{-[d/(4MD_{j+1})-2]_+};$$

χ_j records that a source of width d is seen at the spacing D_j through its mean, and Ξ_{j+1} counts the cubes of the level $j+1$ that meet the supports and the decay of the activities across the distance of the two supports; the bound is local, and it reads no K, side, position or pair.

The orders. At $\mathsf{z} = 0$ the steps add nothing that reads the sources, so the order 0 in z vanishes. The order z^2 comes from the step of the level $j+1$ alone, the earlier levels being taken at the field scale 0, where they read no source; it is the covariance, in the measure of the step, of the parts linear in the fluctuation field of the two source potentials at the trivial configuration. These linear forms are invariant under the adjoint action of the constant gauge transformations, which leave the trivial configuration and the gauge invariant terms unchanged, and the mean of $\mathrm{Ad}(u)$ over $u \in SU(2)$ is 0 on $\mathfrak{su}(2)$; so they vanish, and the order 2 vanishes. Cauchy's estimate on $|\mathsf{z}| = \eta_0 R_j$ gives

$$|\mathsf{n}_{j+1,[2m]}| \le C\eta_0^{-2m} R_j^{-2m} \, D_j^{-2} \chi_j \, \Xi_{j+1} |\Psi| \le C b^{-m} (\log x_j)^{2mp_0} D_j^{m-2} \chi_j \, \Xi_{j+1} |\Psi|,$$

and the same with $m = 4$ for the sum of the orders ≥ 8 at $\mathsf{z} = 1$, as $\eta_0^2 R_j^2 \ge 2$ for $g \le \gamma_{\mathrm{e}}$ (geometric summation of the even coefficient bounds in Prop. 6.27(cx8)).

The sums over the levels. For $D_j \ge d$: $\Xi_{j+1} = 1$, $D_j^{m-2}\chi_j = d^{m-2}(d/D_j)^{9/2-m}$ is geometric in j with the ratio $L^{-(9/2-m)} \le L^{-1/2}$ for $m \le 4$, and $\log x_j \le \log(2b/D_j) \le \mathsf{l}$. For $D_j < d$: $D_j^{m-2}\chi_j \le d^{m-2}$, $\log x_j \le \mathsf{l}\,(1 + (r+1)\log L)$ at the r-th such level counted from above, and $\Xi_{j+1} \le \sup_{u \in [L^{r-1}, L^r]} \max\{1, u^3\} e^{-[u/4M-2]_+}$, which is summable against every power of r. Summing over all the levels gives (i), with $|\mathsf{h}_{2m}| \le C b^{-m} d^{m-2} \mathsf{l}^{2mp_0} |\Psi|$ for $m = 2, 3$; summing over the levels $j \ge n$ alone, where $D_j \ge D_n \ge d$, gives $\sum_{j \ge n} |\mathsf{n}_{j+1,[6]}| \le C b^{-3} \mathsf{l}^{6p_0} d\,(d/D_n)^{3/2} |\Psi|$, which is (ii).

(iii). If $f \circ x$ or $h \circ x$ vanishes identically, $\mathsf{h}_4 = 0$ and the template sum below vanishes, so that its bound alone gives the assertion; let neither vanish.

At a real field scale. The coefficients $\xi_i(\mathsf{z}) = L^{-i}\beta + \mathsf{z}^2(x_i - L^{-i}\beta)$ of the action are chosen so that, the counterterms $e_{i'}$ of the progression being kept, $\mathsf{z}^{-2}\xi_i(\mathsf{z}) L^i + \sum_{1 \le i' \le i} L^{i'} e_{i'} = \mathsf{z}^{-2}\beta$, by the run equation $L^i x_i + \sum_{1 \le i' \le i} L^{i'} e_{i'} = \beta$; and the sum rules of the steps are stated for every field of the box. So at a real field scale $0 < \mathsf{z} \le 1$ the chain of the weighted progression is the chain of the chart integrals of the Wilson weight at β/z^2, read at the fields $\mathsf{z}\zeta$, $\zeta = (\zeta_i)_i$ the fluctuation fields of the levels in their boxes $|\zeta_i| < \mathfrak{b}(x_i)$, and $\sum_{2 \le j \le K} \mathsf{n}^{\mathsf{z}}_j$ is, up to a term of the first level, the covariance of the first source derivatives of the exponent of the level 1, taken at the background that the chain builds at the level 1 from these fields, in a complex measure of mass 1 on the product of the boxes, the product over the levels $1 \le i < K$ of normalised measures on the box of the level i, each of which depends on the fields of the higher levels.

As $\mathsf{z} \to 0$ at a fixed lattice, the backgrounds tend to the trivial configuration uniformly on the boxes, all the potentials vanish, and the coefficient of the action of the level i tends to $L^{-i}x_0$, so that this measure tends in total variation to the product $\mathbb{Q}$ of the measures ν_i^{r} below; dividing by z^4,

$$\sum_{2 \le j \le K} \mathsf{n}_{j,[4]} = \langle q_1; q_2 \rangle_{\mathbb{Q}} + \mathsf{e}_1, \qquad q_a = \tfrac{1}{2} a_K^{-1} \sum_p w_a(p)\, |\Theta_p|^2,$$

Θ_p the linear curvature of the plaquette p in the fields ζ_i, $\mathbb{Q} = \bigotimes_{1 \le i < K} \nu_i^{\mathrm{r}}$, ν_i^{r} the centred Gaussian measure of the quadratic part of the step of the level i at the coefficient $L^{-i}x_0$ of the action, restricted to the box of the level i and normalised, and e_1 a term of the first level. With $\mathsf{n}_{1,[4]}$ and the slice of the level 1 below, e_1 costs at most $C\, b^{-2} \mathsf{l}^{4p_0} e^{-c_\flat d/a_K} |\Psi|$, $c_\flat = \min\{1/(8ML), \delta_{\mathrm{tp}}/(2\sqrt{3})\}$, δ_{tp} the rate of Props. YM3_I:120.5 and YM3_I:120.6, as the supports are d/a_K bare spacings apart.

The Gaussian values; the running couplings drop out. Under the Gaussian measure of the level i at the coefficient $L^{-i}x_0$ the curvature of the fluctuation field of the level i has the covariance

$x_0^{-1}\mathcal{C}_{(i+1)}$, $\mathcal{C}_{(i+1)}$ the slice of the level $i+1$ of the free lattice curvature (Prop. YM3_I:120.3): the coupling x_i enters the scaled field and the coefficient of the action in inverse proportions and drops out. The slices $\mathcal{C}_{(i+1)}$, $1 \le i < K$, add up to $C^{\mathrm{fl}}_{K,0} - \mathcal{C}_{(1)}$, where $C^{\mathrm{fl}}_{K,0} = \sum_{1\le j\le K} \mathcal{C}_{(j)}$ is the fluctuation kernel of depth 0 of Section YM3_I:120 (Props. YM3_I:120.5 and YM3_I:120.6), the kernel of the orthogonal projection of the plaquette 2-cochains, in each colour, onto the curvatures of the fields whose K-fold linear block average vanishes, that is, onto $\mathsf{Fl}_K = \mathrm{d}(\ker \mathbf{Q}^s_K)$, $\mathbf{Q}^s_K$ the straight average of Section YM3_I:120; it has no dimension and $|C^{\mathrm{fl}}_{K,0}| \le 1$; the slice $\mathcal{C}_{(1)}$ of the level 1 goes into e_1.

The variables of different levels are independent and every ν^{r}_i is even; by Wick's rule, with three colours, the Gaussian value of $\langle q_1; q_2\rangle$ is, up to that slice,

$$\tfrac14 a_K^{-2}\cdot 2\cdot 3\, x_0^{-2} \sum_{p,q} w_1(p)\, w_2(q)\, C^{\mathrm{fl}}_{K,0}(p,q)^2 = \tfrac32\, b^{-2}\, \mathcal{K}[C^{\mathrm{fl}}_{K,0}](f,h), \qquad b = a_K x_0,$$

with the template sum $\mathcal{K}[\mathsf{k}](f,h) = \sum_{p,q} f(x(p))\, h(x(q))\, \mathsf{k}(p,q)^2$; no further power of a_K enters.

The boxes. Every joint cumulant of order 2 or 4 of the components of the field of the level i under ν^{r}_i differs from the Gaussian one by at most $C_{\mathrm{bx}} e^{-\frac12 \gamma_2 \mathfrak{b}(x_i)^2}$, uniformly in the volume and in the cutoff (Lem. 4.53, the components entering as insertions), with numbers C_{bx}, γ_2 depending only on the constants; the odd cumulants vanish. The map from the fields of the level i to the curvature has row sums at most CL^{-2i}, and at most $35^2(d/a_K)^6$ pairs of plaquettes have $w_1(p)w_2(q) \ne 0$; with $L^{-3i}(d/a_K)^3 = (d/D_i)^3$ the boxes change $\langle q_1; q_2\rangle$ by at most $Cb^{-2}[\mathsf{E}^{(3)} + (\mathsf{E}^{(3)})^2 + \mathsf{E}^{(6)}]|\Psi|$, $\mathsf{E}^{(q)} = \sum_{1\le i<K} u_i^q \epsilon_i$, $u_i = d/D_i$, $\epsilon_i = e^{-\frac12\gamma_2\mathfrak{b}(x_i)^2} \le \phi_{\mathrm{bx}}(\frac34 b/D_i)$, where $\phi_{\mathrm{bx}}(s) = e^{-c_{\mathrm{bx}}(\log s)^2}$, $c_{\mathrm{bx}} = \frac12\gamma_2 A_1^2$, since $\mathfrak{b}(x_i)^2 \ge A_1^2(\log x_i)^2$ and $x_i \ge \frac34 b/D_i$ (Lem. 8.5).

For $s, s' \ge 1$, ϕ_{bx} decreases, $\phi_{\mathrm{bx}}(ss') \le \phi_{\mathrm{bx}}(s)\phi_{\mathrm{bx}}(s')$ and $s^q\phi_{\mathrm{bx}}(s) \le e^{q^2/(4c_{\mathrm{bx}})}$. Cut the levels at $D_i \approx d^{1/2}$: the levels with $D_i < d$ give at most $C\phi_{\mathrm{bx}}(\frac34 b)\phi_{\mathrm{bx}}(d^{-1/2})$, their u_i being spaced by the factor L; those with $d \le D_i \le d^{1/2}$ have $u_i \le 1$, $\sum u_i^3 \le 2$ and $\epsilon_i \le \phi_{\mathrm{bx}}(\frac34 b)\phi_{\mathrm{bx}}(d^{-1/2})$; those with $D_i > d^{1/2}$ have $\sum u_i^3 \le 2d^{3/2}$ and $\epsilon_i \le \phi_{\mathrm{bx}}(\frac34 b)$. As $\phi_{\mathrm{bx}}(d^{-1/2}) \le Cd^{3/2}$ and $\phi_{\mathrm{bx}}(\frac34 b) \le C/b$, $\mathsf{E}^{(3)}, \mathsf{E}^{(6)} \le Cb^{-1}d^{3/2}$, and also $\le C$; so the boxes cost at most $Cb^{-3}d^{3/2}|\Psi|$, with C depending only on the constants.

The bound is uniform in K and in the side through the locality of the cumulant bounds; no count of the restrictions enters.

The template sum. Lem. 6.34 at the depth 0, where $a_0 = 1 \ge d$:

$$\Big|\mathcal{K}[C^{\mathrm{fl}}_{K,0}](f,h) - \frac{3}{8\pi^2}\mathcal{K}_6(f,h)\Big| \le C_{\mathrm{tm}}\Big[\frac{a_K}{\vartheta d} + \Big(\frac{d}{\ell}\Big)^3 + d^3\Big]|\Psi|,$$

and $\frac32 \cdot \frac{3}{8\pi^2} = c_2$. Collecting,

$$|\mathsf{h}_4 - c_2 b^{-2}\mathcal{K}_6(f,h)| \le b^{-2}\Big\{\tfrac32 C_{\mathrm{tm}}\Big[\frac{a_K}{\vartheta d} + \Big(\frac{d}{\ell}\Big)^3 + d^3\Big] + C\,\frac{d}{b}\,d^{1/2} + C\,\mathsf{I}^{4p_0} e^{-c_\flat d/a_K}\Big\}|\Psi|.$$

As $\ell \ge 1$, $d/a_K \ge L(\log\frac1d)^2$ for $K > s(d)$, and $g^{-2} < b \le L(g^{-2} + \bar B)$, the members other than $\frac32 C_{\mathrm{tm}} b^{-2} a_K/(\vartheta d)$ are at most $\bar\varepsilon_4(g,d)\, b^{-3} d\, |\Psi|$ with

$$\bar\varepsilon_4(g,d) = \sup_{g^{-2}<b\le L(g^{-2}+\bar B)} \Big\{3C_{\mathrm{tm}}\, b\, d^2 + C\, d^{1/2} + C\,\frac{b}{d}\,\mathsf{I}^{4p_0} e^{-c_\flat L(\log\frac1d)^2}\Big\} \le C(g)\, d^{1/2},$$

$e^{-c(\log\frac1d)^2}$ being below every power of d.

(iv). Let $Z[w]$ be the generating function of T_ℓ, the partition function with the source w, and $\mathcal{E}(w)$ its source energy, the sum over the levels and localisation domains of the source energies [7, eq. 212] of the sourced chain on T_ℓ [7, eq. 205]. On the least torus the setting and the identity of (4a)–(4c) in the proof of Prop. 8.3 hold, the conditions on g of the statements read there entering γ_{e}: the term without large-field regions in the sum over the level of the first creation of a large-field region (equation eq. 4.xxxi, with the histories of Section YM3_I:123) is $e^{\mathcal{E}(w)} \int e^{\mathsf{R}_K(w,\cdot)}\, \mathsf{dm}_K$ for the

sources of level-wise norm < 1, with $|\mathsf{R}_K(w,\cdot)| \le \|w\|_{\mathrm{lw}}\mathsf{g}_\emptyset$ for a source carried by one ball of radius 1, and with a measure m_K of mass at most $Z[0]$ (steps (4c), (4d) there).

The weighted progression ([7, Def. B.7]) and the sourced chain are made level by level by the same renormalisation step, with the same couplings, the same auxiliary data and the same lattice, and the source enters both only through the weights $x_j - L^{K-j}w$ of the plaquettes. At the trivial configuration every localised term is taken at the trivial pair, and $a_p(\mathbf{1}) = 0$, so that there the relative exponent of the weighted progression after the level K is the sum over the levels and localisation domains of its source energies; following the two constructions level by level, this sum is $\mathcal{E}(w)$ for all sources near 0, with no condition on the depth.

So the mixed second derivative of $\mathcal{E}$ at $(\tau_v f, \tau_v h)$ is $\mathsf{J}_2(\tau_v f, \tau_v h)$, and $\log(Z[w]/Z[0]) - \mathcal{E}(w)$ is holomorphic near 0 with the mixed second derivative $\mathsf{R} = S_2^T - \mathsf{J}_2$. Take $\iota = a_K\lceil 18d/a_K\rceil$ and $\mathsf{Pk} = \{v \in \iota\mathbb{Z}^3 : |v_\mu| \le 0.575 - 9d\}$; the balls of radius $9d$ about the x_1+v, x_1 a centre of a ball of radius d that holds $\mathrm{supp} f$, are disjoint and lie in $\mathbb{B}(x_1, 1)$. For signs ϵ_v the source $z_1\sum_v \epsilon_v\tau_v f + z_2\sum_v \epsilon_v\tau_v h$ has level-wise norm < 1 for $|z_1|\|f\|_\infty, |z_2|\|h\|_\infty < A_\star$, with $A_\star \ge c(g)(\vartheta d)^{1/2}b$, and on this polydisc the squarefree coefficients of $Z[w]e^{-\mathcal{E}(w)}/Z[0]$ in (z_1, z_2) are at most $Q_{\mathrm{D}}(g)\prod_a \|f_a\|_\infty/A_\star$, with the number $Q_{\mathrm{D}}(g)$ of (4e) in the proof of Prop. 8.3 formed with one ball and two variables: split at the level of the first creation of a large field, bound the part in which no field is large, and apply Cauchy's estimate on a polydisc for each level, as in (4e) there.

So $|\mathsf{R}(\sum_v \epsilon_v\tau_v f, \sum_v \epsilon_v\tau_v h)| \le 2Q_{\mathrm{D}}^2 A_\star^{-2}|\Psi|$; its mean over the signs is $\sum_v \mathsf{R}(\tau_v f, \tau_v h)$; S_2^T is invariant under the translations of $a_K\mathbb{Z}^3$; divide by $\#\mathsf{Pk} \ge (30d)^{-3}$. □

For $N = 2L^{\mathsf{M}}$, $1 \le n < \mathsf{M}$ with $2L^{\mathsf{M}-n+1} \ge C_L M$, and plaquettes p, q of $\mathbb{Z}^3_N$, let $\mathsf{g}_2[N,n](p,q)$ be the coefficient of β^{-3} in the expansion by Laplace's method (Lem. 7.11), as $\beta \to \infty$, of the covariance of a_p and a_q on the fibre $\{\mathcal{M}^n U = \mathbf{1}\}$ of $\mathbb{Z}^3_N$ under $\exp[-\beta(A + \sum_{j<n}\mathsf{k}_j G_j \circ \mathcal{M}^j)]$, written in the chart $\Upsilon(\mathbf{1},\cdot)$ of Lem. 8.6; here $A = \sum_p a_p$ is the Wilson action, G_j the gauge-fixing function YM3_I: (54) of the level j, and $\mathsf{k} \in (0,\infty)^n$ are strengths. This is the number $\mathsf{g}_2[N,n]$ of Prop. 6.14(iii): both definitions take the coefficient of the covariance on the same fibre, and that proposition proves independence from the adapted chart and from the gauge-fixing strengths. In particular, it depends only on the bare torus, the number of blocking levels, L and the plaquettes, with no running coupling or cutoff remaining. $\mathsf{B}_1^{\mathrm{lat}}$ and $\mathsf{P}_1^{\mathrm{lat}}(f,h;a) = a\sum_{\tilde p,\tilde q} f(ax(\tilde p))h(ax(\tilde q))\mathsf{B}_1^{\mathrm{lat}}(\tilde p,\tilde q)$, sums over the plaquettes of $\mathbb{Z}^3$, are the coefficients of Def. 7.10; Cor. 7.37 and Thms. W, X (a) give, for $(f,h) \in \mathfrak{T}_{\mathbb{R}^3}(d;\vartheta)$, $0 < a \le d$ and $\lg(\cdot) = 1 + |\log(\cdot)|$,

$$|\mathsf{P}_1^{\mathrm{lat}}(f,h;a) - \varkappa\mathcal{K}_5(f,h)| \le C_\varkappa\vartheta^{-1}a\lg(d/a)\,\|f\|_\infty\|h\|_\infty, \qquad \varkappa = \tfrac{3}{2\pi^3} = c_1c_2, \tag{8.v}$$

with an absolute number $C_\varkappa$.

Lemma 8.8 (Lattice perturbation theory)**.**

(i) In the setting of Lem. 8.7 let $N = \ell L^K$, $w_1 = f\circ x$ and $w_2 = h\circ x$ on $\mathbb{Z}^3_N$, $2 \le n \le K$,

$$\mathsf{H}^{(n)} = \beta^3 a_K^2 \mathsf{h}_6^{(n)}$$

and

$$\mathsf{T}[N,n](w) = \sum_{p,q} w_1(p)w_2(q)\mathsf{g}_2[N,n](p,q).$$

If $\gamma^{\mathrm{tr}}\mathfrak{b}(x_{n-1})^2 \ge 80$ and

$$10240\,N^3 e^{-\gamma^{\mathrm{tr}}\mathfrak{b}(x_{n-1})^2/8} \le 1,$$

$\gamma^{\mathrm{tr}} > 0$ a number depending only on the constants, then

$$|\mathsf{H}^{(n)} - \mathsf{T}[N,n](w)| \le C(N+\beta)^{36}e^{-\gamma^{\mathrm{tr}}\mathfrak{b}(x_{n-1})^2/8}\,\|w_1\|_\infty\|w_2\|_\infty.$$

(ii) Let $N = 2L^{\mathsf{M}}$, $N' = L^s N$ with $s \geq 1$, $1 \leq n < \mathsf{M}$ with $N/L^n \geq \max\{192,\, 48(3\log L + \log 2)/(\delta_{\mathrm{tp}} L(L-1)),\, C_L M/L\}$, $\delta_{\mathrm{tp}} > 0$ the rate of Props. YM3_I:120.5 and YM3_I:120.6, and π the reduction $\mathbb{Z}^3_{N'} \to \mathbb{Z}^3_N$. For plaquettes p', q' of $\mathbb{Z}^3_{N'}$ whose base points have a distance $\leq N/8$: $|\mathsf{g}_2[N,n](\pi p', \pi q') - \mathsf{g}_2[N',n](p',q')| \leq C N'^{237} e^{-\delta_{\mathrm{tp}} N/(48L^{n-1})}$.

(iii) Let $\ell' = 2L^{m'} \geq N_{\mathsf{H}}$, the side threshold fixed in Section YM3_I:52.6 and retained for the kernel bounds in Section YM3_I:120.3, $\ell' L \geq C_L M$ and $K \geq 1$. For plaquettes p, q of $\mathbb{Z}^3_{\ell' L^K}$ and plaquettes $\tilde p, \tilde q$ of $\mathbb{Z}^3$ over them with $|x_{\tilde p} - x_{\tilde q}|_\infty \leq \ell' L^K/8$: $|\mathsf{g}_2[\ell' L^K, K](p,q) - \mathsf{B}_1^{\mathrm{lat}}(\tilde p, \tilde q)| \leq 3 C_\star \ell'^{19} K L^{-K/2}$, with a number $C_\star$ independent of the side.

■

Proof sketch. (i) In the chart of Lem. 8.6 at the trivial block field of the level n, an exact identity, made one level at a time along matched chains of fields, shows that at a real field scale z the sum of the jets of the levels $\leq n$ is the covariance of two observables in the probability measure $\propto \exp[-\mathsf{z}^{-2}\mathcal{Q}(\mathsf{z}g\zeta) - \mathcal{Q}'(\mathsf{z}g\zeta)]d\zeta$ on the product of the boxes, $g\zeta = (x_i^{-1/2}\zeta_i)_i$. Here $\mathcal{Q} \geq 0$ has a positive definite Hessian, block diagonal over the levels; $\mathcal{Q}$, $\mathcal{Q}'$ and the observables are real analytic near 0 and contain no z, no counterterm and no box; and, up to colour rotations at each level, which keep the boxes and the Lebesgue measure, they are β times the action with the gauge fixing of the strengths $\mathsf{k}_j = x_j/\beta$, the function θ_{ch}, and $\mathcal{O}(w_a) \circ \Upsilon$.

The chart is read there on the product of the boxes $|\mathcal{A}_i| < x_i^{-1/2}\mathfrak{b}(x_i)$ of the fields $\mathcal{A}_i = x_i^{-1/2}\zeta_i$, and these boxes lie in Ω once $\delta(x_i) = x_i^{-1/2}\mathfrak{b}(x_i) < \rho_2$; as ρ_2 depends only on the constants (it is written out in the proof of Lem. 8.6), and as δ decreases on $[e^{2p_0}, \infty)$ while all the couplings of the run exceed the inverse square of the coupling g of the datum (Lem. 8.5), this is a condition on the coupling g alone, and it enters γ_{e}. So $\mathsf{h}_6^{(n)}$ is the second Laplace coefficient computed with the Gaussian measures restricted to the boxes.

Without the boxes, $\beta^3 a_K^2$ times this coefficient is $\mathsf{T}[N,n](w)$, by the intrinsic fibre-coefficient identity of Prop. 6.14(iii) and because the second Laplace coefficient of $c\,\mathcal{Q}$ scales like c^{-3}. For these matched chains, Prop. 6.12(ii) bounds the change on restricting the Gaussian cumulants to the boxes: the total polynomial order is at most 10, there are at most N^3 places, and its Gaussian precision and normalisation bounds give the stated $(N+\beta)^{36}$ times exponential error. (ii) Apply Prop. 6.17 with $t = n$ and $m' = 2$. Its lower bound on N/L^n is the one in this statement, and $\delta_{\mathrm{tp}} L(N/L^n)/48 = \delta_{\mathrm{tp}} N/(48L^{n-1})$. The coefficients are the same intrinsic fibre coefficients by Prop. 6.14(iii), so its estimate gives the assertion.

(iii) Apply the coarser bound of Prop. 6.18 at side ℓ'. The side threshold, room condition and separation of the lifted plaquettes are those of that proposition. □

Lemma 8.9 (Removal of the cutoff at a fixed label)**.** Let $0 < g \leq \gamma_{\mathrm{e}}$, $\ell \in \mathfrak{L}(g)$ and $\vartheta \in (0,1)$. Let $\mathsf{J}_g = [\gamma^{-2}, L(X_g + \bar B)]$, let $\mathsf{C}' = \mathsf{C}'_2(\mathsf{J}_g)$ be the number of the definition in Cor. YM3_I:117.1 at $n = 2$ on the side ℓ, $\mathsf{p}_2 = 6p_0 + \frac12$ the exponent defined before Thm. YM3_I:AR at $n = 2$, $c_\diamond = \sup_{k \geq 1} k^{\mathsf{p}_2} L^{-k/20} < \infty$, and

$$\epsilon_\diamond(d) = (X_g + \bar B)^3\, \mathsf{C}'\, \ell^9\, c_\diamond \Big(\frac{40M}{\vartheta}\Big)^2 d^{87}.$$

There are $d_{\mathrm{K}} \in (0, \frac{1}{200}]$ and integers $K_\sharp(d) > s(d)$ with $d^{-1800} < L^{K_\sharp(d)} \leq d^{-1801}$ for $0 < d \leq d_{\mathrm{K}}$, such that the following holds at every label $b \in I(g)$, at which every pair $(K, L^K b)$, $K \in \mathbb{N}$, is a first-passage datum at g, for $0 < d \leq d_{\mathrm{K}}$ and $(f,h) \in \mathfrak{T}_{T_\ell}(d;\vartheta)$.

(a) For every $K \geq K_\sharp(d)$

$$|S^T_{2;K,\ell,L^K b}(f,h) - S^T_{2;K_\sharp(d),\ell,L^{K_\sharp(d)}b}(f,h)| \leq \epsilon_\diamond(d)\, b^{-3} d\, \|f\|_\infty \|h\|_\infty,$$

and the same with $S_2^{(\ell,b)}(f,h)$ in the place of the first term.

(b) If, for numbers $\epsilon_{\mathrm{h}}, C_{\mathrm{h}} \ge 0$, the cutoff $K_\sharp(d)$ has $|S^T_{2;K_\sharp(d),\ell,L^{K_\sharp(d)}b} - L^+_b|(f,h) \le [\epsilon_{\mathrm{h}}\, b^{-3}d + C_{\mathrm{h}}\, b^{-2} a_{K_\sharp(d)}/d]\, \|f\|_\infty \|h\|_\infty$, then for every $K \ge K_\sharp(d)$

$$|S^T_{2;K,\ell,L^K b} - L^+_b|(f,h) \le \big[\epsilon_{\mathrm{h}} + \epsilon_\diamond(d) + C_{\mathrm{h}}(X_g + \bar B)\, d^{1798}\big]\, b^{-3} d\, \|f\|_\infty \|h\|_\infty,$$

with no term in a_K, and the same with $S^{(\ell,b)}_2$ in the place of $S^T_{2;K,\ell,L^K b}$.

■

Proof sketch. The rate. the constants and thresholds in Thm. YM3_I:AR and Thm. YM3_I:AR, equation YM3_I:(2203) are made in the setting (Th1), (Th2), whose datum is the choice of the constants and whose numbers are formed from it. The interval J_g contains $I(g)$, and it is an interval of labels under (Tor) on the side ℓ (Lem. 2.5). Let $\mathsf{C} = \mathsf{C}_2(\mathsf{J}_g)$, $K_0 = K_0(\ell, 2, \mathsf{J}_g)$ and $K_1(\cdot) = K_1(\ell, 2, \mathsf{J}_g, \cdot)$ be the other numbers of the constants and thresholds in Thm. YM3_I:AR at $n = 2$; so $\mathsf{C}' = \mathsf{C}\sum_{j\ge 0}(1+j)^{\mathsf{p}_2} L^{-j/10}$. For real Lipschitz functions f, h on T_ℓ, neither of them 0, whose supports have the spread $\mathsf{d}_\infty(f,h) \ge d' > 0$ in the sup distance, the uniform cutoff estimate of Thm. YM3_I:AR gives, at every label $b \in \mathsf{J}_g$ and every integer $K \ge K_1(d')$, the inequality YM3_I:(2203) at $n = 2$,

$$|S^T_{2;K+1,\ell,L^{K+1}b}(f,h) - S^T_{2;K,\ell,L^K b}(f,h)| \le \mathsf{C}\, \ell^9 K^{\mathsf{p}_2} L^{-K/10}\, \|f\|_\sharp \|h\|_\sharp,$$

and its sum over the cutoffs, with $(K+j)^{\mathsf{p}_2} \le K^{\mathsf{p}_2}(1+j)^{\mathsf{p}_2}$, is the inequality YM3_I:(2207) of Cor. YM3_I:117.1:

$$|S^T_{2;K',\ell,L^{K'}b}(f,h) - S^T_{2;K,\ell,L^K b}(f,h)| \le \mathsf{C}'\, \ell^9 K^{\mathsf{p}_2} L^{-K/10}\, \|f\|_\sharp \|h\|_\sharp \qquad (K' \ge K \ge K_1(d')).$$

The supports of a pair $(f,h) \in \mathfrak{T}_{T_\ell}(d;\vartheta)$ are at the Euclidean distance $\ge d$, so at the sup distance $\ge d/\sqrt{3}$, both distances of T_ℓ being minima over the lifts and $|v|_\infty \ge |v|/\sqrt{3}$ on $\mathbb{R}^3$; so $\mathsf{d}_\infty(f,h) \ge d/\sqrt{3}$ (the definition preceding Lem. YM3_I:114.6), and $d' = d/\sqrt{3}$ serves; put $K_\diamond(d) = K_1(d/\sqrt{3})$. Moreover $\|f\|_\sharp \le 40M\|f\|_\infty/(\vartheta d)$, as $\operatorname{Lip} f \le \|f\|_\infty/(\vartheta d)$ and $40M \ge \vartheta d$, and the same for h.

The threshold. By the two threshold inequalities at the end of the proof of Thm. YM3_I:AR at $n = 2$, $K_1(d')$ is the least integer $k \ge K_0$ such that every integer $K \ge k$ has

$$d' L^u \ge 6M \qquad \text{and} \qquad A_\diamond\, N_{[u]}\, e^{-d' L^u/M} \le \tfrac{1}{2}\mathsf{C}\, \ell^9 K^{\mathsf{p}_2} L^{-K/10},$$

with the depth $u = \lceil K/1800 \rceil$ of the depth choice preceding Thm. YM3_I:AR at $n = 2$, $N_{[u]} = \ell^3 L^{3u}$, and $A_\diamond = 8e^3 E_0 C_{\mathrm{E},0} \varrho^{-2}$ the factor in front of $N_{[u]}$ in the two threshold inequalities at the end of the proof of Thm. YM3_I:AR, ϱ being the radius of the class of sources in §2.3.1 that stands in the formulas of the constants and thresholds in Thm. YM3_I:AR, not the radius of the choice. The first condition at $K = K_\diamond(d)$, with $u < K/1800 + 1$, gives $L^{K_\diamond(d)} > (6\sqrt{3}M/(Ld))^{1800}$; as $6\sqrt{3}M > L$ (the cube size is $M = L^{m'}$ with $m' \ge 5$), $a_{K_\diamond(d)} < d^{1800}$. Conversely let $\Lambda_\diamond(d) = 4000 + 366\log(\sqrt{3}M/d)$, and let K be an integer whose number $v = L^u$ has $v \ge (\sqrt{3}M/d)\Lambda_\diamond(d)$.

The first condition holds at $d' = d/\sqrt{3}$, as $\Lambda_\diamond(d) \ge 6$. As $K \le 1800u$, $L^{-K/10} \ge v^{-180}$; with $K^{\mathsf{p}_2} \ge 1$ and $\ell \ge 1$ the second condition follows from $A_\diamond v^{183} e^{-d'v/M} \le \frac{1}{2}\mathsf{C}$, that is, from $\varphi_\diamond(v) = d'v/M - 183\log v \ge \log(2A_\diamond/\mathsf{C})$, and $2A_\diamond < \mathsf{C}$: at $n = 2$, as $\log(L \cdot L(X_g + \bar B)) \ge 1$ and $\bar{\mathsf{C}}_{\mathsf{M}} \ge \varrho^{-1}$ in the majorant definitions in the proof of Thm. YM3_I:AR, the number C of YM3_I:(2206) is at least $4\,(16\varrho^{-1})^2\, 2^3 (640/\pi)^2 (1 + 8E_0 C_{\mathrm{E},0}) > 2A_\diamond$. So $\varphi_\diamond(v) \ge 0$ suffices. The function $\varphi_\diamond$ increases for $v \ge 183M/d'$, and at $v = (M/d')\Lambda_\diamond(d) = (\sqrt{3}M/d)\Lambda_\diamond(d)$, since $183\log\lambda < \frac{1}{2}\lambda$ for $\lambda \ge 4000$, $\varphi_\diamond > \frac{1}{2}\Lambda_\diamond(d) - 183\log(\sqrt{3}M/d) = 2000$; so $\varphi_\diamond(v) > 0$ for all these v.

Every $K \ge 1800 \log_L(\sqrt{3}M\Lambda_\diamond(d)/d)$ has such a v, and $L^{\lceil x \rceil} < L \cdot L^x$. So

$$\Big(\frac{6\sqrt{3}M}{Ld}\Big)^{1800} < L^{K_\diamond(d)} \le \max\Big\{L^{K_0},\ L\Big(\frac{\sqrt{3}M\,\Lambda_\diamond(d)}{d}\Big)^{1800}\Big\}.$$

(a). For $K \ge K_\diamond(d)$: $K^{\mathsf{p}_2} L^{-K/10} \le c_\diamond L^{-K/20} \le c_\diamond\, a_{K_\diamond(d)}^{1/20} < c_\diamond d^{90}$. So for $b \in I(g) \subset \mathsf{J}_g$, $K' \ge K \ge K_\diamond(d)$ and $(f,h) \in \mathfrak{T}_{T_\ell}(d;\vartheta)$

$$|S^T_{2;K',\ell,L^{K'}b}(f,h) - S^T_{2;K,\ell,L^K b}(f,h)| \le \mathsf{C}'\ell^9 c_\diamond \Big(\frac{40M}{\vartheta}\Big)^2 d^{88}\,\|f\|_\infty\|h\|_\infty \le \epsilon_\diamond(d)\, b^{-3}d\,\|f\|_\infty\|h\|_\infty,$$

as $d^{88} = b^{-3}d \cdot b^3 d^{87}$ and $b \le X_g + \bar B$ on $I(g)$. Put $K_\sharp(d) = \max\{K_\diamond(d), s(d)+1\}$; then $L^{K_\sharp(d)} \ge L^{K_\diamond(d)} > d^{-1800}$. From $L^{s(d)}d < LC_W(s(d)) = L\mathsf{c}_W(\log L^{s(d)} + \log(X_g+\bar B))^{2p_0}$ one gets $L^{s(d)} \le d^{-2}$, so $L^{s(d)+1} \le d^{-3}$, for small d; and, K_0 being fixed with g and ℓ, $L^{K_0} \le L(\sqrt{3}M\Lambda_\diamond(d)/d)^{1800} \le d^{-1801}$ for small d. Let $d_{\mathrm{K}} \in (0, \frac{1}{200}]$ be a number below which these hold; then $L^{K_\sharp(d)} \le d^{-1801}$ for $0 < d \le d_{\mathrm{K}}$. The last display, with $K \ge K_\sharp(d)$ in the place of K' and $K_\sharp(d)$ in the place of K, is (a) at the finite cutoffs. At the labels of $I(g)$ the limit $S_2^{(\ell,b)}(f,h)$ exists along the full sequence of the cutoffs (Cor. YM3_I:118.4), and a bound that holds at all large K with one right member holds for the limit.

(b). For $b \in I(g)$ the pairs $(K, L^K b)$, $K \in \mathbb{N}$, are the first-passage data at g with the label b (Cor. YM3_I:118.4). The main term $L_b^+(f,h)$ reads the label and the pair, not the cutoff. By the triangle inequality, (a) and the inequality assumed at $K_\sharp(d)$, the left member at $K \ge K_\sharp(d)$, and at the limit, is at most $[\epsilon_{\mathrm{h}} b^{-3}d + \epsilon_\diamond(d) b^{-3}d + C_{\mathrm{h}} b^{-2} a_{K_\sharp(d)}/d]\|f\|_\infty\|h\|_\infty$, and $b^{-2}a_{K_\sharp(d)}/d = b^{-3}d \cdot b\, a_{K_\sharp(d)} d^{-2} < b^{-3}d \cdot (X_g + \bar B)\, d^{1798}$, as $a_{K_\sharp(d)} \le a_{K_\diamond(d)} < d^{1800}$. □

Proposition 8.10 (The law on the least torus)**.** Let $0 < g \le \gamma_{\mathrm{e}}$, $\vartheta \in (0,1)$, $\ell = \ell_g$ and $C_{\mathrm{cut}} = \frac{3}{2} C_{\mathrm{tm}} \vartheta^{-1}$. There is a non-decreasing function $\bar\varepsilon = \bar\varepsilon_{g,\vartheta}$ on $(0, d_{\mathrm{e}}(g,\vartheta)]$, $\bar\varepsilon(d) \to 0$, depending on g, ϑ and the constants and on no K, β, label, pair or position, such that for $0 < d \le d_{\mathrm{e}}(g,\vartheta)$ and $\Psi = (f,h) \in \mathfrak{T}_{T_\ell}(d;\vartheta)$:

(a) at every first-passage datum (K,β) at g with $s(d) < K$ and $L^K \le d^{-1801}$, whatever its label $b = a_K\beta$,
$$|S^T_{2;K,\ell,\beta} - L_b^+|(f,h) \le [\bar\varepsilon(d)\, b^{-3}d + C_{\mathrm{cut}}\, b^{-2} a_K/d]|\Psi|;$$

(b) for $b \in I(g)$ the same holds at $(K, L^K b)$ for every $K > s(d)$, and $|S_2^{(\ell,b)} - L_b^+|(f,h) \le \bar\varepsilon_\infty(d)\, b^{-3}d\,|\Psi|$ with a non-decreasing $\bar\varepsilon_\infty = \bar\varepsilon_{\infty;g,\vartheta} \le \bar\varepsilon$ and $\bar\varepsilon_\infty(d) \le C(g,\vartheta) d^{1/2}$;

(c) with $C'_{\mathrm{cut}} = \frac{9}{2} C_{\mathrm{tm}}\vartheta^{-1} + 1$ in the place of C_{cut}, the tolerance has the rate $d^{1/2}$ at every cutoff: there is $C_{\mathrm{r}}(g,\vartheta) < \infty$ with $|S^T_{2;K,\ell,L^K b} - L_b^+|(f,h) \le [C_{\mathrm{r}}\, b^{-3}d^{3/2} + C'_{\mathrm{cut}}\, b^{-2}a_K/d]|\Psi|$ for all $b \in I(g)$ and $K > s(d)$, and $|S_2^{(\ell,b)} - L_b^+|(f,h) \le C_{\mathrm{r}}\, b^{-3}d^{3/2}|\Psi|$.

■

Proof sketch. Before the chain. If $f \circ x$ or $h \circ x$ vanishes identically, then $S^T_{2;K,\ell,\beta}(f,h)$, all the jets and every lattice sum of the sources vanish (Lem. 8.5), and Lem. 8.7 (iii) holds by the template sum alone; in the chain below every arrow but the last then reads $0 = 0$, and the last is eq. 8.v as it stands, with $\mathsf{P}_1^{\mathrm{lat}} = 0$. Let $\mathsf{r} = d/a_K$, so that $\mathsf{r} \ge L(\log\frac{1}{d})^2$. Translations keep the class and the $\mathcal{K}_q$, so Lem. 8.7 (iv), (i), (iii) at the pairs $(\tau_v f, \tau_v h)$ give

$$|S^T_{2;K,\ell,\beta} - L_b^+|(f,h) \le \Big[C(g,\vartheta)(d^{1/2} + d(\log\tfrac{1}{d})^{8p_0})b^{-3}d + C_{\mathrm{cut}}\, b^{-2}\frac{a_K}{d}\Big]|\Psi| + \max_{v\in\mathsf{Pk}}|\mathsf{h}_6 - \varkappa b^{-3}\mathcal{K}_5|(\tau_v f, \tau_v h),$$

the packing mean costing $C_{\mathrm{A}}(g)\vartheta^{-1}bd$ and the orders ≥ 8 costing $Cb^{-1}d\,\mathsf{I}^{8p_0}$ in the unit $b^{-3}d|\Psi|$. As $\mathsf{h}_6 = b^{-3}a_K\mathsf{H}^{(K)}$ and $b^{-3}a_K\mathsf{r} = b^{-3}d$, it remains to show $|\mathsf{H}^{(K)} - \varkappa\mathcal{K}_5/a_K| \le \eta(d)\,\mathsf{r}\,|\Psi|$ with $\eta(d) \to 0$ uniformly; all the costs below are in the unit $\mathsf{r}|\Psi|$.

The levels. Let $K' = \max\{K, \lceil 24\log_L \mathsf{r}\rceil\}$, so that $L^{K'} \ge \mathsf{r}^{24}$ and $K' \le C\log\frac{1}{d}$; let $n_1 = K' - \lceil K'/78\rceil$, $N_0 = \ell L^K$ and $N' = \ell L^{K'}$. If $K' > K$, let n_0 be the level with $D_{n_0} \in [d^{1/2}, Ld^{1/2})$, $b_* \in I(g)$ fixed, $D' = (K', L^{K'}b_*)$, a first-passage datum at g, and $\breve w_a$ the sources lifted to one copy

in $\mathbb{Z}^3_{N'}$ of a ball of radius $N_0/4$ that holds the supports; by Lem. 8.5 the lifted sources are a pair at the datum D', with no condition on its depth. The chain is

$$\mathsf{H}^{(K)} \to \mathsf{H}^{(n_0)} \to \mathsf{T}[N_0, n_0] \to \mathsf{T}[N', n_0] \to \mathsf{H}^{(n_0)}[D'] \to \mathsf{H}^{(n_1)}[D'] \to \mathsf{T}[N', n_1] \to \mathsf{T}^{\mathrm{lat}} \to \varkappa\mathcal{K}_5/a_K,$$

with $\mathsf{T}^{\mathrm{lat}} = \sum \tilde{w}_1(\tilde{p})\tilde{w}_2(\tilde{q})\mathsf{B}^{\mathrm{lat}}_1(\tilde{p},\tilde{q})$ for the sources of the lifts $\tilde{f},\tilde{h}$ to $\mathbb{R}^3$. By Lem. 8.7 (ii) at $n = n_0$, the first arrow costs at most $C\mathsf{l}^{6p_0}(d/D_{n_0})^{3/2} \le C\mathsf{l}^{6p_0}d^{3/4}$. By Lem. 8.8 (i), with $x_{n_0-1} \ge \frac{3}{4}b/D_{n_0-1} \ge \frac{3}{4}d^{-1/2}$ (Lem. 8.5) and $N_0 + \beta \le (\ell + X_g + \bar{B})d^{-1801}$, as $b \le L(g^{-2} + \bar{B}) = X_g + \bar{B}$, the second costs at most $e^{-c(\log\frac{1}{d})^{2p_0}}$ times a power of $1/d$. By Lem. 8.8 (ii), applied pair by pair to the at most $900\mathsf{r}^6$ pairs of plaquettes of the supports, with $N_0/L^{n_0} = \ell/D_{n_0} \ge \ell d^{-1/2}/L$, the third costs at most $900\mathsf{r}^5CN'^{237}e^{-\delta_{\mathrm{tp}}\ell d^{-1/2}/48}$.

By Lem. 8.8 (i) at (D', n_0), the number $\mathsf{T}[N', n_0]$ being free of β, and with $x'_{n_0-1} \ge \frac{3}{4}\mathsf{r}^{11}$, the fourth costs less than every power of d. By Lem. 8.7 (ii) at D' the fifth costs at most $C\mathsf{l}'^{6p_0}d^{3/4}$, $\mathsf{l}' \le \log(2X_g + 2\bar{B}) + K'\log L$. By Lem. 8.8 (i) at (D', n_1), with $x'_{n_1-1} \ge L^{K'/78}$, the sixth costs less than every power of d. By Lem. 8.8 (iii) on the torus of the side $N'/L^{n_1} = \ell L^{\lceil K'/78\rceil}$ with n_1 levels, over at most $900\mathsf{r}^6$ pairs, the seventh costs at most $C\ell^{19}K'\mathsf{r}^5L^{-K'/4} \le CK'/\mathsf{r}$, since $\mathsf{r}^{24} \le L^{K'}$.

By eq. 8.v at $a = a_K$, as $a_K\mathsf{T}^{\mathrm{lat}} = \mathsf{P}^{\mathrm{lat}}_1(\tilde{f},\tilde{h};a_K)$, the eighth costs at most $C_\varkappa\vartheta^{-1}\lg(\mathsf{r})/\mathsf{r}$. If $K' = K$, that is $\mathsf{r}^{24} \le L^K$, there is no second datum and no second torus: the chain is $\mathsf{H}^{(K)} \to \mathsf{H}^{(n_1)} \to \mathsf{T}[N_0, n_1] \to \mathsf{T}^{\mathrm{lat}} \to \varkappa\mathcal{K}_5/a_K$, its first arrow by Lem. 8.7 (ii) at $n = n_1$, with the cost $C\mathsf{l}^{6p_0}(\mathsf{r}/L^{n_1})^{3/2}$, $\mathsf{r}/L^{n_1} \le L^{1-0.94K}$ and $\mathsf{l} \le \log(2X_g + 2\bar{B}) + K\log L$, and the other three as the last three above with $D' = D$ and $N' = N_0$.

The conditions and the limits. At the three places where Lem. 8.8 (i) is used its two conditions hold for $d \le d_{\mathrm{e}}$, because $\frac{1}{8}\gamma^{\mathrm{tr}}\mathfrak{b}(X)^2 = \frac{1}{8}\gamma^{\mathrm{tr}}A_1^2(\log X)^{2p_0}$ with $2p_0 \ge 2$ outgrows $10 + \log(10240N^3)$, the logarithm of N being at most a constant plus a fixed multiple of $\log X$ at the lower bounds X of the three places. Every cost tends to 0 with d, uniformly ($\mathsf{r} \to \infty$, $K' \le C\log\frac{1}{d}$), and depends only on g, ϑ and the constants; $\bar{\varepsilon}(d)$ is the supremum over $d'' \le d$ of their sum. This is (a).

Removal of the cutoff. For $b \in I(g)$ the pairs $(K, L^K b)$ are first-passage data at g, and for $s(d) < K \le K_\sharp(d)$, (a) applies, as $L^K \le d^{-1801}$. At $K_\sharp(d)$ the inequality of (a) is the one that Lem. 8.9 (b) asks there, with $\epsilon_{\mathrm{h}} = \bar{\varepsilon}(d)$ and $C_{\mathrm{h}} = C_{\mathrm{cut}}$; so for $K \ge K_\sharp(d)$ and for the limit the inequality holds with the tolerance $\bar{\varepsilon}(d) + \epsilon_\diamond(d) + C_{\mathrm{cut}}(X_g + \bar{B})d^{1798}$ and without the second member, and $\bar{\varepsilon} + \sup_{d'' \le d}[\epsilon_\diamond(d'') + C_{\mathrm{cut}}(X_g + \bar{B})d''^{1798}]$ serves in (a) and (b). At $K_\sharp$ one has $\mathsf{r} \ge d^{-3}$, so the second, third, fourth and sixth costs are below every power of d, K'/r and $\lg(\mathsf{r})/\mathsf{r}$ are at most Cd^2, the packing mean and the orders ≥ 8 give $Cd(\log\frac{1}{d})^{8p_0}$, the order six $C\mathsf{l}^{6p_0}d^{3/4}$, and the slowest member is the order four, $\bar{\varepsilon}_4(g,d) \le C(g)d^{1/2}$: this is the rate of $\bar{\varepsilon}_\infty$.

The rate at every cutoff. Run the same chain with the cut at $D_{n_0} \in [d^{1/5}, Ld^{1/5})$; the arithmetic of the levels goes through as before for $d \le L^{-13}$. The first and fifth arrows now cost at most $C\mathsf{l}^{6p_0}d^{6/5}$ and $C\mathsf{l}'^{6p_0}d^{6/5}$, the third $e^{-c\ell d^{-1/5}}$ times a power of $1/d$, and the second, fourth and sixth stay below every power of d. The seventh and eighth members, which near the smallest r decay only like a power of $\log\frac{1}{d}$, contribute at most $C(K' + \lg\mathsf{r})\mathsf{r}^{-1}\,b^{-3}d|\Psi| = C(K' + \lg\mathsf{r})(d/b)\,b^{-2}(a_K/d)|\Psi| \le \eta(d)\,b^{-2}(a_K/d)|\Psi|$, with $\eta(d) = Cg^2d\log\frac{1}{d} \le 1$ for $d \le d_{\mathrm{e}}$; they go into the second member, whose constant becomes at most $C_{\mathrm{cut}} + 1 \le C'_{\mathrm{cut}}$.

All the remaining members are $O(d(\log\frac{1}{d})^{8p_0})$ but the order four, which is at most $C(g)d^{1/2}$. Beyond $K_\sharp(d)$, Lem. 8.9 (b), with $C_{\mathrm{h}} = C'_{\mathrm{cut}}$, removes the second member at the price $\epsilon_\diamond(d) + C'_{\mathrm{cut}}(X_g + \bar{B})d^{1798}$, at most a constant times $d^{1/2}$. This is (c). □

The coefficient $\frac{8}{3\pi}$ comes from $\varkappa = c_1c_2$ in eq. 8.v alone; in particular it does not depend on L, g or the label.

8.1.9. *Transfer to continuum components.*

Lemma 8.11 (From a limit at a fixed label to the components)**.** Let $\mathfrak{c}$ be a choice, $0 < g \le \gamma_W$, S a volume-first limit at g, $b = \mathsf{b}(S)$, S the function S_2 or a function S_2^ξ of a canonical decomposition of σ_S, $\vartheta \in (0,1)$, $z \in \mathbb{Z}$, and $\tau' = (g', b', \ell')$ an admissible triple with $b' = L^z b$. For $L^z d \le d_\flat(g', \vartheta)$ and $\Psi \in \mathfrak{T}^1(d;\vartheta)$,

$$|b^2 \mathsf{S}(\Psi) - b'^2 T^{(\ell',b')}(\Psi_{L^z})| \le E_{\tau'}\,(d/b)^2 |\Psi|.$$

■

Proof sketch. Let (K_j, β_j, ν_j) define S; then $a_{K_j}\beta_j \to b$. S is the limit of $S_2^T[\rho_j]$ for periodic Gibbs states ρ_j at (K_j, β_j), the class of Prop. 8.3, which is $\mathcal{G}^{\mathrm{per}}_{K_j,\beta_j} = \mathcal{G}^{\mathrm{inv}}_{K_j,\beta_j}$ and holds the states of the tori: $\rho_j = \nu_j$ for S_2 ([7, §K.1]), and for S_2^ξ the conditioned states of Thms. YM3_I:AL and YM3_I:AP. By Lem. 8.13 (a) with $s = L^z$ the dilates $\rho_j^{(z)}$ are such states at $(K_j - z, \beta_j)$, and $S_2^T[\rho_j](\Phi) = L^{2z} S_2^T[\rho_j^{(z)}](\Phi_{L^z})$. As $a_{K_j - z}\beta_j = L^z a_{K_j}\beta_j \to b' \in I^\circ(g')$, the pairs $(K_j - z, \beta_j)$ are first-passage data at g' for large j (Lem. 8.2 (b)). Prop. 8.3 there, the $\rho_j^{(z)}$ being states at the datum, at the scale $L^z d \le \frac{1}{180} < \ell'/20$ and with $a_{K_j - z} \le 9 L^z d$ for large j, gives

$$|S_2^T[\rho_j^{(z)}](\Psi_{L^z}) - S^T_{2;K_j - z, \ell', \beta_j}((\Psi_{L^z})^{\langle \ell' \rangle})| \le 2 C_2^{\mathrm{T}}(g') \vartheta^{-1} (L^z d)^2 (a_{K_j - z}\beta_j)^{-2} |\Psi|.$$

Let $j \to \infty$, using the convergence along converging labels (Thm. YM3_I:AS), and multiply by $b'^2 = L^{2z} b^2$; then $2 C_2^{\mathrm{T}}(g') \vartheta^{-1} L^{2z} d^2 = E_{\tau'}(d/b)^2$. □

Lemma 8.12 (A second term excludes covariance)**.** Let a functional σ on separated tuples over $\mathbb{R}^3$ obey (L) with $(b, c, \varepsilon, d_7, \vartheta)$, $c \ne 0$; let $d_1 > 0$, $\Psi \in \mathfrak{T}^1(d_1;\vartheta)$, $\mathsf{B} = c\, c_2 b^{-3} \mathcal{K}_5(\Psi)$, $\mathsf{u} = b^{-3} d_1 |\Psi|$ and $t > 1$. Then

$$|S_2^\sigma(\Psi_r) - c_2 b^{-2} \mathcal{K}_6(\Psi) - \mathsf{B} r| \le \varepsilon(r d_1)\, \mathsf{u}\, r \quad (r d_1 \le d_7), \qquad |\mathcal{N}_t^{S_2^\sigma}(\Psi; r) + t(t-1)(t^5 - 1)\mathsf{B} r| \le (t^2 + t + t^6 + t^7)\, \varepsilon(t^2 r d_1)\, \mathsf{u}\, r \quad (t^2 r d_1 \le d_7);$$

$\mathcal{N}_t^{S_2^\sigma}(\Psi; r) \ne 0$, with the sign of $-c$, as soon as $(t^2 + t + t^6 + t^7)\varepsilon(t^2 r d_1) < t(t-1)(t^5 - 1) \cdot 10^{-5} |c| c_2 \vartheta^2$; and S_2^σ is not dilation covariant. ■

Proof sketch. (L) at $\Psi_r \in \mathfrak{T}^1(r d_1; \vartheta)$, with $|\Psi_r| = |\Psi|$, $\mathcal{K}_6(\Psi_r) = \mathcal{K}_6(\Psi)$ and $\mathcal{K}_5(\Psi_r) = r \mathcal{K}_5(\Psi)$, gives the first bound. In $\mathcal{N}_t$ the weights $1, -(1 + t^6), t^6$ at $t^2 r, t r, r$ annul the constant $c_2 b^{-2} \mathcal{K}_6(\Psi)$, give $\mathsf{B} r$ the factor $t^2 - (1 + t^6) t + t^6 = -t(t-1)(t^5 - 1)$, and give the remainders a sum at most $(t^2 + t + t^6 + t^7)\varepsilon(t^2 r d_1)\mathsf{u} r$, ε being non-decreasing; at $t = 2$ these are -62 and 198. By eq. 8.i, $|\mathsf{B}| \ge 10^{-5} |c| c_2 \vartheta^2 \mathsf{u}$, which gives the sign. If $S_2^\sigma(\Phi_r) = r^\varpi S_2^\sigma(\Phi)$ for all $r > 0$ and all separated Φ, take $\Phi = \Psi$: by the first bound the left side tends to $c_2 b^{-2} \mathcal{K}_6(\Psi) \in (0, \infty)$ as $r \to 0$, so $S_2^\sigma(\Psi) \ne 0$ and r^ϖ has a finite non-zero limit, whence $\varpi = 0$; then $S_2^\sigma(\Psi_r) = S_2^\sigma(\Psi) = c_2 b^{-2} \mathcal{K}_6(\Psi)$ for all r, and the first bound gives $|\mathsf{B}| \le \varepsilon(r d_1)\mathsf{u} \to 0$, against $|\mathsf{B}| \ge 10^{-5} |c| c_2 \vartheta^2 \mathsf{u} > 0$. □

8.1.10. *An alternative transfer using two blocking factors.*

Lemma 8.13 (Three facts on the lattices)**.**

(a) For $a, s, \beta > 0$ the map $x \mapsto s x$ carries the Gibbs states of the Wilson action at β on $a\mathbb{Z}^3$, and the periodic ones, onto those on $sa\mathbb{Z}^3$. For the image $D_s\nu$ of ν, $S_n^T[\nu](f_1, \dots, f_n) = s^n S_n^T[D_s\nu](f_1(\cdot/s), \dots, f_n(\cdot/s))$; so $(sa\beta)^2 S_2^T[D_s\nu](\Psi_{sr}) = (a\beta)^2 S_2^T[\nu](\Psi_r)$.

(b) For $\ell = 2L^m$, a compact set $J' \subset (0, \infty)$ and $d_* > 0$ there is C with $|S^T_{2;K,\ell,L^K b}(u, w)| \le C \|u\|_\sharp \|w\|_\sharp$ for all K, all $b \in J'$ and all real Lipschitz u, w on T_ℓ at a sup-distance $\ge d_*$. Hence, if $K_j \to \infty$, $a_{K_j}\beta_j \to b > 0$ and $r_j \to r > 0$, then $S^T_{2;K_j,\ell,\beta_j}((\Psi_{r_j})^{\langle \ell \rangle}) \to T^{(\ell,b)}(\Psi_r)$ for every separated pair Ψ whose dilate Ψ_r has both supports in one ball of radius $< \ell/16$; for $\Psi \in \mathfrak{T}^1(d_1;\vartheta)$ it suffices that $80 r d_1 < \ell$.

(c) If $\log L_1 / \log L_2 \notin \mathbb{Q}$ and $\sigma_0 > 0$, there are integers $K_j, K_j' \to \infty$ with $L_1^{K_j} L_2^{-K_j'} \to \sigma_0$.

■

Proof sketch. (a) The action and the Haar measure depend on the lattice as a graph, and $a^{-1}\sum_p f(x(p))a_p = s\,(sa)^{-1}\sum_p f(sx(p)/s)\,a_p$, the plaquette p of $a\mathbb{Z}^3$ going to the plaquette sp of $sa\mathbb{Z}^3$; cumulants are multilinear. (b) The bound is Thm. YM3_I:AS and its uniform bound YM3_I:(2214). For C^1 entries with compact supports $\|f(\cdot/r_j) - f(\cdot/r)\|_\sharp \to 0$, and the dilated supports stay at a distance bounded below; so, by bilinearity, the pair may be fixed at Ψ_r, and the convergence along converging labels, Thm. YM3_I:AS, applies. A pair of $\mathfrak{T}^1(d_1;\vartheta)$ dilated by r lies in two balls of radius rd_1 whose centres are at most $8rd_1$ apart, so in one ball of radius $5rd_1$. (c) The multiples $K\log L_1$ are dense modulo $\log L_2$ (Kronecker's theorem). □

Lemma 8.14 (A second choice)**.** Let $\varrho > 0$, values of the constants of $\mathcal{T}^{\mathrm{S}}[2\varrho]$ with the blocking factor L, and an odd integer $Q \geq 1$ be given. There is j_0 such that for every $j \geq j_0$ there is an admissible assignment of $\mathcal{T}^{\mathrm{S}}[2\varrho]$ in which the constants before L keep their values and the blocking factor is $\tilde{L} = (LQ)^j + 2$. $\tilde{L}$ is odd and prime to L and to Q; $\log L/\log\tilde{L} \notin \mathbb{Q}$, and $\log Q/\log\tilde{L} \notin \mathbb{Q}$ if $Q > 1$. With the radius and the mollifier of a choice $\mathfrak{c}$ it is a *second choice* $\tilde{\mathfrak{c}}$ of $\mathfrak{c}$; at $Q = L$, $\tilde{L} = L^{2j} + 2$. ■

Proof sketch. By the ordered choices in the proofs of Prop. YM3_I:69.1 and Lem. YM3_I:82.1, for an admissible assignment σ_0 of the constants before L there is an odd integer $L_\uparrow(\sigma_0)$ such that (σ_0, L') is admissible for every odd $L' \geq L_\uparrow(\sigma_0)$: the conditions attached to the blocking factor are oddness and floors formed with the earlier constants; and every admissible assignment up to a place of the order of the constants is the truncation of a full admissible assignment. For large j, $\tilde{L} \geq L_\uparrow$. $\tilde{L}$ is odd; a prime that divides $\tilde{L}$ and L (or Q) divides 2, which is impossible. If $\log L/\log\tilde{L} = p/q$ with integers $p, q \geq 1$, then $L^q = \tilde{L}^p$ with coprime $L, \tilde{L} > 1$, which is impossible; the same for $Q > 1$. □

Lemma 8.15 (Two blocking factors)**.** Let $\mathfrak{c}_1, \mathfrak{c}_2$ be choices with $\log L_1/\log L_2 \notin \mathbb{Q}$, $\tau_i = (g_i, b_i, \ell_i)$ an admissible triple of $\mathfrak{c}_i$ (for $i = 1$, $b_1 \in I(g_1)$ suffices), $\vartheta \in (0,1)$, $d_1 > 0$, $\Psi \in \mathfrak{T}^1(d_1;\vartheta)$, and $\delta > 0$ with $\delta b_1 d_1 \leq d_\flat^{(1)}(g_1,\vartheta)$ and $\delta b_2 d_1 < d_\flat^{(2)}(g_2,\vartheta)$, the thresholds of Prop. 8.3 at $\mathfrak{c}_1$ and $\mathfrak{c}_2$. With T, E and Y formed at the respective choice,

$$|Y_{\tau_1}(\delta) - Y_{\tau_2}(\delta)| \leq (E_{\tau_1} + E_{\tau_2})\, d_1^2 |\Psi|\, \delta^2.$$

■

Proof sketch. Lem. 8.13 (c) gives $K_j, K_j' \to \infty$ with $s_j = L_2^{K_j'} L_1^{-K_j} \to b_1/b_2$. Put $\beta_j = L_1^{K_j} b_1$: (K_j, β_j) is a first-passage datum of $\mathfrak{c}_1$ at g_1 with the label b_1, and $b_{2,j} = L_2^{-K_j'}\beta_j = b_1/s_j \to b_2 \in I^\circ(g_2)$, so that (K_j', β_j) is a first-passage datum of $\mathfrak{c}_2$ at g_2 for large j (Lem. 8.2 (b) at $\mathfrak{c}_2$). Both data have the bare coupling β_j on the lattice $\mathbb{Z}^3$ in lattice units. Let ρ_j be a periodic Gibbs state at β_j there, and $\rho_j^{(1)}$, $\rho_j^{(2)}$ its images on $L_1^{-K_j}\mathbb{Z}^3$ and $L_2^{-K_j'}\mathbb{Z}^3$; then $\rho_j^{(1)} = D_{s_j}\rho_j^{(2)}$, and by Lem. 8.13 (a)

$$b_1^2 S_2^T[\rho_j^{(1)}](\Psi_{\delta b_1}) = b_{2,j}^2 S_2^T[\rho_j^{(2)}](\Psi_{\delta b_{2,j}}).$$

Prop. 8.3 at $\mathfrak{c}_1$ and at $\mathfrak{c}_2$ compares $\rho_j^{(1)}$ and $\rho_j^{(2)}$ with the Wilson measures on T_{ℓ_1} and T_{ℓ_2}; after multiplication by b_1^2 and $b_{2,j}^2$ the errors are $E_{\tau_1} d_1^2|\Psi|\delta^2$ and, in the limit, $E_{\tau_2} d_1^2|\Psi|\delta^2$. As $j \to \infty$ the torus terms tend to $Y_{\tau_1}(\delta)$, by the limit at a fixed label along the full sequence, and to $Y_{\tau_2}(\delta)$, by Lem. 8.13 (b), since $80\,\delta b_2 d_1 < 1 < \ell_2$. □

Corollary 8.16 (Two labels; all real dilations)**.** Let $\mathfrak{c}$ be a choice, $\tilde{\mathfrak{c}}$ a second choice of $\mathfrak{c}$ (Lem. 8.14 with $Q = L$) with an admissible triple $\tau_2 = (g_2, b_2, \ell_2)$ and the threshold $\tilde{d}_\flat$, and $\vartheta \in (0,1)$.

(i) For admissible triples $\tau=(g,b,\ell)$ and $\tau'=(g',b',\ell')$ of $\mathfrak{c}$, $d_1>0$, $\Psi\in\mathfrak{T}^1(d_1;\vartheta)$ and $\delta>0$ with $\delta b d_1\le d_\flat(g,\vartheta)$, $\delta b' d_1\le d_\flat(g',\vartheta)$ and $\delta b_2 d_1<\tilde d_\flat(g_2,\vartheta)$,

$$|Y_\tau(\delta)-Y_{\tau'}(\delta)|\le (E_\tau+E_{\tau'}+2E_{\tau_2})\,d_1^2|\Psi|\,\delta^2.$$

Equivalently, with $s=b'/b$, for $\Phi\in\mathfrak{T}^1(d;\vartheta)$ with $d\le d_\flat(g,\vartheta)$, $sd\le d_\flat(g',\vartheta)$ and $(b_2/b)d<\tilde d_\flat(g_2,\vartheta)$: $|b'^2T^{(\ell',b')}(\Phi_s)-b^2T^{(\ell,b)}(\Phi)|\le (E_\tau+E_{\tau'}+2E_{\tau_2})(d/b)^2|\Phi|$. So, up to errors $O((d/b)^2)$, the limits on the tori at the labels b and b' are carried into each other by the real dilation by b'/b; for b'/b a power of L and $\ell'/\ell=b'/b$ this holds exactly, by Thm. Z (d).

(ii) If at one admissible triple $\tau_*=(g_*,b_*,\ell_*)$ of $\mathfrak{c}$ a function $T_*=T^{(\ell_*,b_*)}$ satisfies $|T_*(\Phi)-c_2b_*^{-2}[\mathcal{K}_6+cb_*^{-1}\mathcal{K}_5](\Phi)|\le\varepsilon(d)b_*^{-3}d|\Phi|$ for $0<d\le d_7$ and $\Phi\in\mathfrak{T}^1(d;\vartheta)$, with a real c and a non-decreasing $\varepsilon\to0$, then at every admissible triple $\tau=(g,b,\ell)$ of $\mathfrak{c}$ the same inequality holds with the same c, with b in the place of b_* and with the tolerance $\varepsilon(b_*d/b)+(E_\tau+E_{\tau_*}+2E_{\tau_2})b^{-1}d$, for d below the least of $(b/b_*)d_7$, $(b/b_*)d_\flat(g_*,\vartheta)$, $d_\flat(g,\vartheta)$ and $\frac12(b/b_2)\tilde d_\flat(g_2,\vartheta)$.

In particular the law of Thm. Z (a) at one admissible triple $(g^\circ,b^\circ,\ell_{g^\circ})$ with $g^\circ\le\Gamma$ gives the law with the same $\frac{8}{3\pi}$ at every admissible triple of $\mathfrak{c}$, on every side of $\mathfrak{L}(g)$, with the tolerance $\varepsilon^L_{g^\circ,\vartheta}(b^\circ d/b)+Cb^{-1}d$.

(iii) Let S and S' be volume-first limits at couplings g and g' with $0<g,g'\le\gamma_W$, with the labels $b=\mathsf{b}(S)$ and $b'=\mathsf{b}(S')$; let S be S_2 or a function S_2^ξ of a canonical decomposition of σ_S, and S' be S_2' or a function $S_2'^\eta$ of a canonical decomposition of $\sigma_{S'}$; let $z,z'\in\mathbb{Z}$, and let $\tau=(g_\tau,L^zb,\ell_\tau)$ and $\tau'=(g_{\tau'},L^{z'}b',\ell_{\tau'})$ be admissible triples of $\mathfrak{c}$. For $d_1>0$, $\Psi\in\mathfrak{T}^1(d_1;\vartheta)$ and $\delta>0$ with $\delta L^zbd_1\le d_\flat(g_\tau,\vartheta)$, $\delta L^{z'}b'd_1\le d_\flat(g_{\tau'},\vartheta)$ and $\delta b_2d_1<\tilde d_\flat(g_2,\vartheta)$,

$$|b^2\mathsf{S}(\Psi_{\delta b})-b'^2\mathsf{S}'(\Psi_{\delta b'})|\le(2E_\tau+2E_{\tau'}+2E_{\tau_2})\,d_1^2|\Psi|\,\delta^2.$$

Equivalently, with $s=b'/b$, for $\Phi\in\mathfrak{T}^1(d;\vartheta)$ with $L^zd\le d_\flat(g_\tau,\vartheta)$, $L^{z'}sd\le d_\flat(g_{\tau'},\vartheta)$ and $(b_2/b)d<\tilde d_\flat(g_2,\vartheta)$: $|b'^2\mathsf{S}'(\Phi_s)-b^2\mathsf{S}(\Phi)|\le(2E_\tau+2E_{\tau'}+2E_{\tau_2})(d/b)^2|\Phi|$. So, up to errors $O((d/b)^2)$, the two-point functions of the volume-first limits and of all their components at the couplings $g\le\gamma_W$ are carried into each other by the real dilation by the quotient of their labels. The law of Thm. Y (1) at the two labels gives the same comparison with the error $2\varepsilon^{\mathrm{nat}}(d/b)(d/b)|\Phi|$, which is only $O((d/b)^{3/2})|\Phi|$.

■□

Proof sketch. (i) Lem. 8.15 for the pair $(\mathfrak{c},\tilde{\mathfrak{c}})$, whose blocking factors have incommensurable logarithms, at (τ,τ_2) and at (τ',τ_2), and the triangle inequality. For the second form put $\Phi=\Psi_{\delta b}$ and $d=\delta bd_1$; then $\Psi_{\delta b'}=\Phi_{b'/b}$ and $d_1^2\delta^2=(d/b)^2$. (ii) Let $\Phi\in\mathfrak{T}^1(d;\vartheta)$ and $\Phi_*=\Phi_{b_*/b}\in\mathfrak{T}^1(b_*d/b;\vartheta)$. By (i), $b^2T^{(\ell,b)}(\Phi)$ is within $(E_\tau+E_{\tau_*}+2E_{\tau_2})(d/b)^2|\Phi|$ of $b_*^2T_*(\Phi_*)$, and by the assumption of (ii) $b_*^2T_*(\Phi_*)$ is within $\varepsilon(b_*d/b)(d/b)|\Phi|$ of $c_2[\mathcal{K}_6(\Phi_*)+cb_*^{-1}\mathcal{K}_5(\Phi_*)]=c_2[\mathcal{K}_6(\Phi)+cb^{-1}\mathcal{K}_5(\Phi)]$; divide by b^2.

(iii) Lem. 8.11, for S with z and τ at the pair $\Psi_{\delta b}\in\mathfrak{T}^1(\delta bd_1;\vartheta)$, and for S' with z' and τ' at the pair $\Psi_{\delta b'}$, gives $|b^2\mathsf{S}(\Psi_{\delta b})-Y_\tau(\delta)|\le E_\tau d_1^2|\Psi|\delta^2$ and $|b'^2\mathsf{S}'(\Psi_{\delta b'})-Y_{\tau'}(\delta)|\le E_{\tau'}d_1^2|\Psi|\delta^2$, as $(\Psi_{\delta b})_{L^z}=\Psi_{\delta L^zb}$ and $(d/b)^2=d_1^2\delta^2$ at $d=\delta bd_1$; (i) at (τ,τ') and the triangle inequality give the bound, and the second form follows as in (i). By Thm. Y (1) at Φ and at $\Phi_s\in\mathfrak{T}^1(sd;\vartheta)$, $b^2\mathsf{S}(\Phi)$ and $b'^2\mathsf{S}'(\Phi_s)$ lie within $\varepsilon^{\mathrm{nat}}(d/b)(d/b)|\Phi|$ of the one number $c_2[\mathcal{K}_6(\Phi)+\frac{8}{3\pi}b^{-1}\mathcal{K}_5(\Phi)]$, as $sd/b'=d/b$ and $b'^{-1}\mathcal{K}_5(\Phi_s)=b^{-1}\mathcal{K}_5(\Phi)$. □

Proposition 8.17 (Non-covariance from the law at one triple)**.** Let $\mathfrak{c}_*$ and $\mathfrak{c}^\circ$ be choices, equal or not, $\vartheta\in(0,1)$, and $\tau^\circ=(g^\circ,b^\circ,\ell^\circ)$ an admissible triple of $\mathfrak{c}^\circ$ such that, with $T^\circ=T^{(\ell^\circ,b^\circ)}$ formed at $\mathfrak{c}^\circ$, there are $d_n\to0$ and $\Psi_n\in\mathfrak{T}^1(d_n;\vartheta)$ with $|T^\circ(\Psi_n)-c_2b^{\circ-2}\mathcal{K}_6(\Psi_n)|/(d_n^2|\Psi_n|)\to\infty$. Then for every $0<g\le\gamma_W$ of $\mathfrak{c}_*$, every volume-first limit S at g and every canonical decomposition $(\mathsf{X},\lambda,(\sigma_\xi))$ of σ_S:

(i) neither S_2 nor any S_2^ξ is dilation covariant: $\mathsf{N}=\mathsf{X}$ and $\lambda(\mathsf{N})=1$;

(ii) there is no $d_* > 0$ such that the three-scale number $S_2(\Psi_{4r}) - 65S_2(\Psi_{2r}) + 64S_2(\Psi_r)$ vanishes for all d, r with $4rd \le d_*$ and all $\Psi \in \mathfrak{T}^1(d;\vartheta)$.

■

Proof sketch. Let $b = \mathsf{b}(S)$ and $\mathsf{S} = S_2$ or S_2^{ξ}. Both assertions follow once a bound

$$|\mathsf{S}(\Phi) - c_2 b^{-2}\mathcal{K}_6(\Phi)| \le C\, d^2 |\Phi| \qquad (0 < d \le d_0,\ \Phi \in \mathfrak{T}^1(d;\vartheta)) \tag{8.vi}$$

is excluded for every C and $d_0 > 0$. If S is dilation covariant with the exponent ϖ, the leading laws YM3_I:(2353) and YM3_I:(2356) gives $\mathsf{S}(\Phi_r) \to c_2 b^{-2}\mathcal{K}_6(\Phi) \in (0,\infty)$ as $r \to 0$, so $\varpi = 0$ and $\mathsf{S} = c_2 b^{-2}\mathcal{K}_6$ on the pairs of all the classes $\mathfrak{T}^1(d;\vartheta)$: eq. 8.vi with $C = 0$. If the three-scale number vanishes as in (ii), we may take $d_* \le 4\mathsf{d}_0(g,\vartheta)$; for $\Phi \in \mathfrak{T}^1(d';\vartheta)$ with $d_*/8 < d' \le d_*/4$ the numbers $u_k = S_2(\Phi_{2^{-k}})$ satisfy $u_k - 65u_{k+1} + 64u_{k+2} = 0$, so $u_k = \alpha + \alpha' 64^{-k}$; by the leading law $\alpha = \lim_k u_k = c_2 b^{-2}\mathcal{K}_6(\Phi)$ and $|\alpha'| = |u_0 - \alpha| \le \frac{1}{3} c_2 b^{-2}\mathcal{K}_6(\Phi) \le C|\Phi|$.

Every pair of $\mathfrak{T}^1(d;\vartheta)$ with $d \le d_*/4$ is $\Phi_{2^{-k}}$ for such a Φ and some $k \ge 0$, so $|S_2(\Phi_{2^{-k}}) - c_2 b^{-2}\mathcal{K}_6(\Phi_{2^{-k}})| \le C|\Phi|\, 2^{-6k} \le C(8/d_*)^6 d^6 |\Phi|$: eq. 8.vi with d^6 in the place of d^2.

Suppose now eq. 8.vi. Take z with $b' = L^z b \ge 2(\gamma_W^{\mathrm{sd}})^{-2}$ and $g' = (2/b')^{1/2}$; then $b' \in I^\circ(g')$ by Lem. 8.2 (a), as $b' \ge 2(\gamma_W^{\mathrm{sd}})^{-2} \ge 2\gamma^{-2}$, and $\tau' = (g', b', \ell_{g'})$ is an admissible triple of $\mathfrak{c}_*$. For $\Psi \in \mathfrak{T}^1(d_1;\vartheta)$, eq. 8.vi at $\Phi = \Psi_{\delta b}$ and Lem. 8.11 give

$$|Y_{\tau'}(\delta) - c_2\mathcal{K}_6(\Psi)| \le (Cb^4 + E_{\tau'})\, d_1^2 |\Psi|\, \delta^2$$

for small δ. Let $\tilde{\mathfrak{c}}$ be a second choice of $\mathfrak{c}_*$ with Q the blocking factor of $\mathfrak{c}^\circ$ (Lem. 8.14), and τ_2 an admissible triple of $\tilde{\mathfrak{c}}$. Lem. 8.15, for the pair $(\mathfrak{c}_*, \tilde{\mathfrak{c}})$ at (τ', τ_2) and for the pair $(\mathfrak{c}^\circ, \tilde{\mathfrak{c}})$ at (τ°, τ_2), gives for small δ

$$|Y_{\tau^\circ}(\delta) - c_2\mathcal{K}_6(\Psi)| \le C' d_1^2 |\Psi|\, \delta^2, \qquad C' = Cb^4 + 2E_{\tau'} + 2E_{\tau_2} + E_{\tau^\circ}.$$

At $d_1 = 1$, $\delta = d_n/b^\circ$ and $\Psi = (\Psi_n)_{1/d_n}$ this reads $|T^\circ(\Psi_n) - c_2 b^{\circ -2}\mathcal{K}_6(\Psi_n)| \le C' b^{\circ -4} d_n^2 |\Psi_n|$, against the assumption on T°. As λ is a probability measure, $\lambda(\mathsf{N}) = \lambda(\mathsf{X}) = 1$. □

8.1.11. *A Gaussian model satisfying the same two-term law.*

Proposition 8.18 (A quadratic functional of a Gaussian field)**.** Let $\ell > 0$, $\alpha > 0$, $\upsilon_0 > 0$, $\upsilon = \upsilon_0/\ell$ and $b > 0$. Let $\varphi_1, \varphi_2, \varphi_3$ be independent centred Gaussian fields on T_ℓ with the covariance operator $\hat{G}_b(D)$, $\hat{G}_b(k) = (b|k|^2 + \alpha|k| + \upsilon)^{-1}$ for $k \in \frac{2\pi}{\ell}\mathbb{Z}^3$, and let $\mathcal{C}$ be the kernel of $\hat{G}_b(D)$. A *mollifier* is a real radial function $\jmath \in C_c^\infty(\mathbb{B}(0,1))$ with $\int \jmath = 1$. For a mollifier $\jmath$ and $0 < \varepsilon_{\mathrm{m}} < \frac{\ell}{2}$ put $\jmath_{\varepsilon_{\mathrm{m}}} = \varepsilon_{\mathrm{m}}^{-3}\jmath(\cdot/\varepsilon_{\mathrm{m}})$ and $\varphi_{a,\varepsilon_{\mathrm{m}}} = \varphi_a * \jmath_{\varepsilon_{\mathrm{m}}}$, and for a real Lipschitz function f on T_ℓ let

$$\mathcal{O}_{\varepsilon_{\mathrm{m}}}(f) = \sum_a \int f\, [-\tfrac{1}{2}{:}|\nabla\varphi_{a,\varepsilon_{\mathrm{m}}}|^2{:} - {:}\varphi_{a,\varepsilon_{\mathrm{m}}}\Delta\varphi_{a,\varepsilon_{\mathrm{m}}}{:}]$$

be the observable of the mollified field. For $f \ne 0$ the variance of $\mathcal{O}_{\varepsilon_{\mathrm{m}}}(f)$ tends to ∞ as $\varepsilon_{\mathrm{m}} \to 0$, so that there is no L^2 limit observable. For $n \ge 2$ and real Lipschitz functions $f_1, \dots, f_n$ on T_ℓ whose supports have no common point we put

$$\mathcal{S}_n^{(\ell,b)}(f_1, \dots, f_n) := \lim_{\varepsilon_{\mathrm{m}} \to 0} \langle \mathcal{O}_{\varepsilon_{\mathrm{m}}}(f_1); \dots; \mathcal{O}_{\varepsilon_{\mathrm{m}}}(f_n) \rangle,$$

the limit of the joint cumulants of the observables of the mollified field; d_∞ is the spread of such a tuple, as in Thm. YM3_I:AS.

(a) $(bu^2 + \alpha u + \upsilon)^{-1} = \int_0^\infty \rho_{\mathrm{sp}}(m)\,(u^2 + m^2)^{-1} dm$ for $u \ge 0$, with $\rho_{\mathrm{sp}}(m) = \frac{2\alpha}{\pi}\, m^2 \big((\upsilon - bm^2)^2 + \alpha^2 m^2\big)^{-1} > 0$ and $\int_0^\infty \rho_{\mathrm{sp}} = \frac{1}{b}$.

(b) The limit exists, depends on no mollifier, and is symmetric and multilinear, as long as the tuples keep supports without a common point; at tuples with pairwise disjoint supports it is the Wick integral of the kernels of the vertices and of $\mathcal{C}$. For every $n \ge 2$, every compact subset of $(0,\infty)$ and every $d_0 > 0$ there is c with $|\mathcal{S}_n^{(\ell,b)}(f_1, \dots, f_n)| \le c \prod_i \|f_i\|_\sharp$ for all labels

b of that subset and all such tuples with $\mathsf{d}_\infty \ge d_0$. At pairs (f,h) of real Lipschitz functions on T_ℓ with disjoint supports, $\mathcal{S}_2^{(\ell,b)}(f,h) = \iint f(x)h(y)\,s_2(x-y)\,\mathrm{d}x\,\mathrm{d}y$ with a kernel s_2 smooth off 0 and

$$s_2(x) = c_2 b^{-2}\Big[|x|^{-6} + \frac{4\alpha}{3\pi}\,b^{-1}|x|^{-5}\Big] + O\big(|x|^{-4}(1+\log^+(\ell/|x|))\big) \qquad (x\to 0).$$

So at every label b, for $(f,h)\in\mathfrak{T}_{T_\ell}(d;\vartheta)$ and small d,

$$|\mathcal{S}_2^{(\ell,b)}(f,h) - c_2 b^{-2}[\mathcal{K}_6 + \frac{4\alpha}{3\pi}b^{-1}\mathcal{K}_5](f,h)| \le C(\ell,\alpha,v_0,b)\,d^2(1+\log(\ell/d))\,\|f\|_\infty\|h\|_\infty$$

: the two-term law with the coefficient $\frac{4\alpha}{3\pi}$, which is $\frac{8}{3\pi}$ at $\alpha=2$, and with a remainder $O(d^2\log\frac{1}{d})$.

(c) Let $\{x_\mu = s\}$ be a coordinate mirror of T_ℓ, s any real number, θ the reflection in it and $\Sigma_+ = \{s < x_\mu < s+\frac{\ell}{2}\}$ one of its open sides. Then $(w\circ\theta, \mathcal{C}w)\ge 0$ for every real function w carried by Σ_+; hence the Gaussian measure is reflection positive. For a tuple $\mathcal{F} = (f_i)_{i\in I}$, I a finite set, let $\mathfrak{m}(\mathcal{F}) = \sum_\pi \prod_{J\in\pi}\mathcal{S}_{|J|}^{(\ell,b)}((f_i)_{i\in J})$, the sum running over the partitions π of I into blocks of at least two elements, and $\mathfrak{m}(\varnothing)=1$; let $\theta\mathcal{F} = (f_i\circ\theta)_{i\in I}$, and let $\theta\mathcal{F}\cdot\mathcal{F}'$ be the tuple of the entries of $\theta\mathcal{F}$ and of $\mathcal{F}'$. Then

$$\sum_{i,i'} \bar z_i z_{i'}\,\mathfrak{m}(\theta\mathcal{F}_i\cdot\mathcal{F}_{i'}) \ge 0$$

for every finite family of complex numbers z_i and of tuples $\mathcal{F}_i$, the empty tuple among them, of real Lipschitz functions with pairwise disjoint supports in compact subsets of Σ_+; the same holds for entries that overlap on Σ_+, as long as every cumulant that occurs stands at a tuple whose supports have no common point. The limits on the tori have the same property at the mirrors of the lattices of the cutoffs: for $\ell = 2L^m$, every $b>0$, $\mu\in\{1,2,3\}$ and every $c\in\bigcup_{K\in\mathbb{N}}\frac12 L^{-K}\mathbb{Z}$, with θ the reflection in $\{x_\mu = c\}$ and $\Sigma_+ = \{c<x_\mu<c+\frac{\ell}{2}\}$, the inequality above holds with the $S_n^{(\ell,b)}$ in the place of the $\mathcal{S}_n^{(\ell,b)}$, for every finite family of complex numbers z_i and of tuples $\mathcal{F}_i$, the empty tuple among them, of real Lipschitz functions with pairwise disjoint supports in compact subsets of Σ_+.

For entries of class C^1 the same inequality holds at every mirror $\{x_\mu = c\}$, c any real number (the moving-mirror limit argument in the proof below, using Thm. YM3_I:A). As the model is reflection positive at every coordinate mirror, it is so at all the mirrors at which the limits are. On the tuples whose supports have no common point, the family $(\mathcal{S}_n^{(\ell,b)})_n$ is invariant under the isometries of T_ℓ, and

$$\mathcal{S}_n^{(r\ell,rb)}(f_1(\cdot/r),\dots,f_n(\cdot/r)) = r^{-n}\mathcal{S}_n^{(\ell,b)}(f_1,\dots,f_n) \qquad (r>0),$$

the number v_0 being kept.

(d) For every such tuple $\underline{f} = (f_1,\dots,f_n)$ the map $b\mapsto\mathcal{S}_n^{(\ell,b)}(\underline{f})$ extends holomorphically to the half plane $\mathrm{Re}\,b>0$, and

$$\partial_b\mathcal{S}_n^{(\ell,b)}(\underline{f}) = -\mathcal{S}_{n+1}^{(\ell,b)}(1,\underline{f}), \qquad (-\partial_b)^{\mathsf{j}}\mathcal{S}_n^{(\ell,b)}(\underline{f}) = \mathcal{S}_{n+\mathsf{j}}^{(\ell,b)}(1,\dots,1,\underline{f}) \quad (\mathsf{j}\ge 0),$$

with j entries equal to the constant function 1 on T_ℓ: the rule that Prop. YM3_I:118.3, equation YM3_I:(2217) gives for the limits on the tori. Consequently, for $0<b_1<b_2$ and every pair (f,h) of real Lipschitz functions on T_ℓ with disjoint supports,

$$\int_{b_1}^{b_2}[b^2\mathcal{S}_3^{(\ell,b)}(1,f,h) - 2b\,\mathcal{S}_2^{(\ell,b)}(f,h)]\,\mathrm{d}b = b_1^2\mathcal{S}_2^{(\ell,b_1)}(f,h) - b_2^2\mathcal{S}_2^{(\ell,b_2)}(f,h),$$

the identity on which Cor. 8.1 (c) rests.

(e) Let $\ell \in \{2L, 2L^2, 2L^3, \dots\}$, $b > 0$ and $n \geq 2$. The limits on the tori have the following property: if $\underline{f}^{(j)}$, $j \in \mathbb{N}$, are tuples of n real Lipschitz functions on T_ℓ, none of them 0, whose supports have no common point, with spreads bounded below by a positive number, norms $\|f_i^{(j)}\|_\sharp$ bounded in j, and $\prod_i \|f_i^{(j)}\|_\infty \to 0$, then $S_n^{(\ell,b)}(\underline{f}^{(j)}) \to 0$. The model fails this at $n = 3$ and at every label: if $\psi \in C^\infty(T_\ell)$ is real and $h \geq 0$ is Lipschitz, neither of them 0, with $\operatorname{supp}\psi \cap \operatorname{supp} h = \varnothing$, and $f_{\mathbf{q}} = |\mathbf{q}|^{-1}\psi\cos(\mathbf{q}\cdot x)$ for $\mathbf{q} \in \frac{2\pi}{\ell}\mathbb{Z}^3 \setminus \{0\}$, then the triples $(f_{\mathbf{q}}, f_{\mathbf{q}}, h)$ satisfy these conditions as $|\mathbf{q}| \to \infty$, and

$$\mathcal{S}_3^{(\ell,b)}(f_{\mathbf{q}}, f_{\mathbf{q}}, h) \to -\frac{3}{2b}\iint \psi(x)^2 h(y)\,\mathsf{p}(x-y)\,\mathrm{d}x\,\mathrm{d}y < 0 \qquad (|\mathbf{q}| \to \infty),$$

with $\mathsf{p} = |\nabla\mathcal{C}|^2 + 2\,\mathcal{C}\Delta\mathcal{C}$, which is positive off 0.

■

Proof sketch. (a) With $P(\zeta) = b\zeta^2 + \alpha\zeta + \upsilon$, whose roots have negative real parts, $\rho_{\mathrm{sp}}(m) = \frac{1}{\pi i}\,m\,[P(-im)^{-1} - P(im)^{-1}]$, and closing the contour of $\int_{\mathbb{R}} m\,\mathrm{d}m/((m^2+u^2)P(im))$ in the lower half plane, where the only pole is $m = -iu$, gives $1/P(u)$; letting $u \to \infty$ in $u^2/P(u)$ gives the mass $\frac{1}{b}$. (b)

The limit and the bound. For smooth u, v the vertex $\mathsf{V}_f(u,v) = \int f\,\nabla u\cdot\nabla v + \int \nabla f\cdot\nabla(uv)$ equals $\int f[-\nabla u\cdot\nabla v - u\Delta v - v\Delta u]$, so the observable of the mollified field is, in each colour, the Wick ordered half of V_f at that field. The operator V_f with $(u, V_f v) = \mathsf{V}_f(u,v)$ maps H^1 to H^{-1} with a norm at most $\|f\|_\infty + \operatorname{Lip} f \leq c\|f\|_\sharp$, and $\hat{G}_b(D)$ maps H^{-1} to H^1, uniformly for b in a compact subset of $\operatorname{Re} b > 0$. The joint cumulant of half squares of a Gaussian vector is half the sum, over the cyclic orders of the entries, of the traces of the products of the forms against the covariance; so, the three colours giving the factor 3, the joint cumulant of the $\mathcal{O}_{\varepsilon_{\mathrm{m}}}(f_i)$ is $\frac{3}{2}$ times the sum, over the $(n-1)!$ cyclic orders of $f_1, \dots, f_n$, of the traces of the products of the operators $V_{f_i}|\hat{\jmath}(\varepsilon_{\mathrm{m}}D)|^2\hat{G}_b(D)$ in these orders, $\hat{\jmath}$ the Fourier transform of $\jmath$.

If the supports have no common point, split every propagator into a part of range less than d_∞/n and a smooth remainder. The product of the short parts vanishes: the vertices are local, and a closed chain of n short propagators through the n supports would give points of all the supports at mutual distances less than the spread. Every other product contains a smooth propagator, a trace class operator, among bounded ones. So the products are trace class, with trace norms at most $c\prod_i \|f_i\|_\sharp$ for b in a compact subset of $\operatorname{Re} b > 0$ and $\mathsf{d}_\infty \geq d_0$, and their traces are continuous under the convergence of the propagators as $\varepsilon_{\mathrm{m}} \to 0$.

This gives the limit, its independence of the mollifier, its symmetry and multilinearity, and the bound; at pairwise disjoint supports the traces are the integrals of $\prod_i f_i(x_i)$ against the products of the kernels off the diagonal, the Wick integrals.

The variance. At $n = 2$ and $f_1 = f_2 = f$ the same trace formula is the variance of $\mathcal{O}_{\varepsilon_{\mathrm{m}}}(f)$, and it is a sum of terms ≥ 0. In the orthonormal basis $e_k = \ell^{-3/2}e^{ik\cdot x}$, $k \in \frac{2\pi}{\ell}\mathbb{Z}^3$, the matrix of V_f has the entries $\int \bar{e}_p V_f e_q = \ell^{-3}v(p,q)\hat{f}(p-q)$, with $v(p,q) = |p|^2 + |q|^2 - p\cdot q = p\cdot q + |p-q|^2$ and $\hat{f}(k) = \int f(x)e^{-ik\cdot x}\,\mathrm{d}x$: the first member of V_f gives $p\cdot q$, and the second, after one integration by parts, $|p-q|^2$. As f is real, the entries at (p,q) and at (q,p) are conjugate; as $\hat{\jmath}$ is real, the mollified field has the covariance $\hat{\jmath}(\varepsilon_{\mathrm{m}}D)^2\hat{G}_b(D)$, and

$$\operatorname{Var}\mathcal{O}_{\varepsilon_{\mathrm{m}}}(f) = \frac{3}{2}\,\ell^{-6}\sum_{p,q\in\frac{2\pi}{\ell}\mathbb{Z}^3} \hat{\jmath}(\varepsilon_{\mathrm{m}}p)^2\,\hat{\jmath}(\varepsilon_{\mathrm{m}}q)^2\,\hat{G}_b(p)\,\hat{G}_b(q)\,v(p,q)^2\,|\hat{f}(p-q)|^2,$$

every term of which is ≥ 0. Let $f \neq 0$, and let k_0 have $\hat{f}(k_0) \neq 0$. Without the mollifier, the terms with $p = q + k_0$ tend to $b^{-2}|\hat{f}(k_0)|^2 > 0$ as $|q| \to \infty$, since $v(q+k_0, q) = |q|^2 + q\cdot k_0 + |k_0|^2$ and $|q|^2\hat{G}_b(q) \to b^{-1}$; so their series diverges. As $\hat{\jmath}(\varepsilon_{\mathrm{m}}k) \to \hat{\jmath}(0) = \int \jmath = 1$ at every k when $\varepsilon_{\mathrm{m}} \to 0$, Fatou's lemma for sums of non-negative terms gives $\operatorname{Var}\mathcal{O}_{\varepsilon_{\mathrm{m}}}(f) \to \infty$.

The kernel. By (a) the covariance kernel is $\mathcal{C} = \int_0^\infty \rho_{\mathrm{sp}}(m)\mathcal{C}_m \,\mathrm{d}m$,

$$\mathcal{C}_m(x) = \sum_{j\in\mathbb{Z}^3} e^{-m|x+\ell j|}/(4\pi|x+\ell j|) > 0,$$

and $\mathcal{H} = \Delta\mathcal{C} = \int \rho_{\mathrm{sp}}(m)m^2\mathcal{C}_m \,\mathrm{d}m$ off 0. Since $m^2\rho_{\mathrm{sp}}(m) = \frac{2\alpha}{\pi b^2} + O((1+m^2)^{-1})$, the Laplace transform $\hat\rho_{\mathrm{sp}}$ of ρ_{sp} has $\hat\rho_{\mathrm{sp}}(0) = \frac{1}{b}$ and $\hat\rho''_{\mathrm{sp}}(|x|) = \frac{2\alpha}{\pi b^2}|x|^{-1} + O(1)$, so that, with $\omega_0 = \frac{1}{4\pi b}$ and $\omega_1 = \frac{\alpha}{2\pi^2 b^2}$,

$$\begin{aligned}
\mathcal{C} &= \frac{\omega_0}{|x|} + O(\lg(\ell/|x|)),\\
\nabla\mathcal{C} &= -\omega_0\frac{\hat x}{|x|^2} + O\Big(\frac{\lg(\ell/|x|)}{|x|}\Big),\\
\nabla\nabla^{\mathsf T}\mathcal{C} &= \omega_0\frac{3\hat x\hat x^{\mathsf T}-1}{|x|^3} + \omega_1\frac{1-2\hat x\hat x^{\mathsf T}}{|x|^2} + O(|x|^{-1}),\\
\mathcal{H} &= \frac{\omega_1}{|x|^2} + O(|x|^{-1}),\\
\nabla\mathcal{H} &= -\frac{2\omega_1\hat x}{|x|^3} + O(|x|^{-2}),\\
\Delta\mathcal{H} &= \frac{2\omega_1}{|x|^4} + O(|x|^{-2}),
\end{aligned}$$

with $\hat x = x/|x|$ and $\lg(\ell/|x|) = 1 + \log(\ell/|x|)$, as $|x| < \ell$ on T_ℓ, the images under $\ell\mathbb{Z}^3$ being smooth near 0. The observable is half the quadratic form of the vertex in each colour, so Wick's rule, with three colours, gives

$$s_2 = \frac{3}{2}[\mathrm{tr}(\nabla\nabla^{\mathsf T}\mathcal{C})^2 + 4\nabla\mathcal{C}\cdot\nabla\mathcal{H} + 2\mathcal{H}^2 + 2\mathcal{C}\Delta\mathcal{H}]$$

off 0. The leading terms give $\frac{3}{2}\cdot 6\,\omega_0^2|x|^{-6} = c_2 b^{-2}|x|^{-6}$; the terms of the order $|x|^{-5}$ are $-8\omega_0\omega_1$, $+8\omega_0\omega_1$ and $+4\omega_0\omega_1$, in all

$$\frac{3}{2}\cdot 4\,\omega_0\omega_1|x|^{-5} = \frac{3\alpha}{4\pi^3}b^{-3}|x|^{-5} = c_2\frac{4\alpha}{3\pi}b^{-3}|x|^{-5};$$

the rest is $O(\lg(\ell/|x|)\,|x|^{-4})$. Integrating the remainder against a pair of the class, on whose supports $d \le |x-y| \le 10d$, gives the bound.

(c) Write $x = (x_\mu, x_\perp)$, and let w be carried by Σ_+. A Fourier series in $x_\perp$, with the frequency $k_\perp$ and $m_\perp = (|k_\perp|^2 + m^2)^{1/2}$, writes $(w\circ\theta, \mathcal{C}_m w)$ as a sum over $k_\perp$ of double integrals, in the distances $\mathsf{v},\mathsf{v}' \in (0, \frac{\ell}{2})$ of two points of Σ_+ to the mirror, against the kernel $\cosh m_\perp(\frac{\ell}{2} - \mathsf{v} - \mathsf{v}')/(2m_\perp \sinh\frac{m_\perp\ell}{2})$, and $\cosh m_\perp(\frac{\ell}{2} - \mathsf{v} - \mathsf{v}') = \cosh m_\perp(\frac{\ell}{4} - \mathsf{v})\cosh m_\perp(\frac{\ell}{4} - \mathsf{v}') + \sinh m_\perp(\frac{\ell}{4} - \mathsf{v})\sinh m_\perp(\frac{\ell}{4} - \mathsf{v}')$ makes each of them a sum of two squares; integrating against $\rho_{\mathrm{sp}} \ge 0$ gives $(w\circ\theta, \mathcal{C}w) \ge 0$, and with it the reflection positivity of the Gaussian measure, as for every centred Gaussian measure with a reflection positive covariance.

For ε_{m} less than the distance of the supports from the mirror, the observables of the mollified field of functions carried by compact subsets of Σ_+ are functions of the field on Σ_+, and, the mollifier being radial, the reflection carries $\mathcal{O}_{\varepsilon_{\mathrm{m}}}(f)$ to $\mathcal{O}_{\varepsilon_{\mathrm{m}}}(f\circ\theta)$. The moments of centred variables are the sums, over the partitions without blocks of one element, of the products of their cumulants; so $\sum \bar z_i z_{i'}\mathfrak{m}_{\varepsilon_{\mathrm{m}}}(\theta\mathcal{F}_i\cdot\mathcal{F}_{i'})$, with $\mathfrak{m}_{\varepsilon_{\mathrm{m}}}$ formed with the cumulants of the mollified field, is the expectation of $\overline{\Theta F}\,F$ for $F = \sum_{i'} z_{i'}\prod_{f\in\mathcal{F}_{i'}}\mathcal{O}_{\varepsilon_{\mathrm{m}}}(f)$, Θ the reflection of the field, and it is ≥ 0.

As $\varepsilon_{\mathrm{m}} \to 0$, each cumulant that occurs stands at a tuple whose supports have no common point and converges by (b). For the limits on the tori, let $2c \in a_{K_0}\mathbb{Z}$. For every $K \ge K_0$ the plane $\{x_\mu = c\}$ is a site or a bond mirror of the lattice of the spacing a_K, at which the Wilson measure on T_ℓ at $\beta = L^K b$ is reflection positive (Prop. YM3_I:91.4). For large K the function

$F_K = \sum_{i'} z_{i'} \prod_{f\in\mathcal{F}_{i'}} (\mathcal{O}_K(f) - \langle \mathcal{O}_K(f)\rangle)$ reads only bonds in Σ_+, the measure is invariant under the reflection, and $\langle \overline{\Theta F_K}\, F_K\rangle \geq 0$ is the inequality with the cumulants $S^T_{n;K,\ell,L^K b}$ in the place of the $\mathcal{S}^{(\ell,b)}_n$.

At the tuples that occur, whose supports are pairwise disjoint, the cumulants converge along the full sequence of cutoffs by Thm. YM3_I:A; hence the finite positive matrix converges to the asserted matrix. For a real mirror position c, choose $c_j \in \bigcup_K \frac{1}{2}L^{-K}\mathbb{Z}$ with $c_j \to c$. The fixed compact positive-side supports remain in the positive half-torus for all large j, and their reflected supports remain in disjoint compact neighbourhoods. For C^1 entries the translated reflections converge in the sharp norm. The multilinear bound in Thm. YM3_I:AS, with the all-side dilation of Cor. YM3_I:119.2, therefore passes each cumulant, and hence each centred moment, to the limit. Positivity is closed under this finite matrix limit. This is the moving-mirror argument of Prop. YM3_I:92.4, here on the two half-tori. It proves the assertion at every real c, for every $b > 0$ and every side $2L^m$ allowed by Thm. YM3_I:A.

The invariance is that of $\hat{G}_b$ and of the mollifiers, which are radial; the dilation identity follows from $\hat{G}_{rb}(k/r)$ on $T_{r\ell}$ being $r\hat{G}_b(k)$ on T_ℓ, with $v = v_0/\ell$: the field on $T_{r\ell}$ has the law of $x \mapsto r^{-1}\varphi(x/r)$, the mollifier at the scale ε_{m} goes to that at the scale $r\varepsilon_{\mathrm{m}}$, and each observable, with its two derivatives, scales by r^{-1}; the identity passes to the limits.

(d) The trace formula of (b) defines $\mathcal{S}^{(\ell,b)}_n$ for $\operatorname{Re} b > 0$, and its terms are holomorphic in b, locally uniformly, as $\hat{G}_b$ is; so $\mathcal{S}^{(\ell,b)}_n(\underline{f})$ is holomorphic there. The vertex of the constant function 1 is $V_1 = -\Delta$, and $\partial_b \hat{G}_b = -|k|^2\hat{G}_b^2$, that is, $\partial_b \hat{G}_b(D) = -\hat{G}_b(D)V_1\hat{G}_b(D)$: the derivative of a propagator inserts the vertex of 1 between its ends. Summing over the n propagators of each product and over the cyclic orders of $f_1,\dots,f_n$ gives the cyclic orders of $(1,f_1,\dots,f_n)$; hence $\partial_b\mathcal{S}^{(\ell,b)}_n(\underline{f}) = -\mathcal{S}^{(\ell,b)}_{n+1}(1,\underline{f})$, and the higher derivatives follow by induction, the tuples with entries 1 having supports without a common point.

At the entry 1 the observable of the mollified field is $\frac{1}{2}\sum_a \int :|\nabla\varphi_{a,\varepsilon_{\mathrm{m}}}|^2:$, since $\int \varphi\Delta\varphi = -\int|\nabla\varphi|^2$ on the torus. The integral identity is the integral over $[b_1,b_2]$ of $\partial_b[-b^2\mathcal{S}^{(\ell,b)}_2(f,h)] = b^2\mathcal{S}^{(\ell,b)}_3(1,f,h) - 2b\,\mathcal{S}^{(\ell,b)}_2(f,h)$.

(e) For the limits, let $d_0 > 0$ bound the spreads from below. By Thm. YM3_I:AS, at the interval $\{b\}$ and with d_0 in the place of d, for all large K the term of index K of the sequence of the cutoffs differs from $S^{(\ell,b)}_n(\underline{f}^{(j)})$ by at most $\prod_i \|f_i^{(j)}\|_\sharp$ times a number that tends to 0 as $K\to\infty$ and does not depend on j; and, as $|\mathcal{O}_K(f)| \leq 6\ell^3L^{4K}\|f\|_\infty$, the term of index K is at most a number depending only on n times $\prod_i(6\ell^3L^{4K}\|f_i^{(j)}\|_\infty)$, as in the first member of YM3_I:(2214). Let $j\to\infty$ and then $K\to\infty$.

For the model, $\|f_{\mathbf{q}}\|_\infty \leq |\mathbf{q}|^{-1}\|\psi\|_\infty$, $\operatorname{Lip} f_{\mathbf{q}}$ is bounded in $\mathbf{q}$, and the spread of $(f_{\mathbf{q}}, f_{\mathbf{q}}, h)$ is at least the distance of $\operatorname{supp}\psi$ and $\operatorname{supp} h$ in the norm $|\cdot|_\infty$; so the conditions hold. By (b), $\mathcal{S}^{(\ell,b)}_3(f_{\mathbf{q}}, f_{\mathbf{q}}, h)$ is three times the trace of $V_{f_{\mathbf{q}}}\hat{G}_b(D)V_{f_{\mathbf{q}}}\hat{G}_b(D)V_h\hat{G}_b(D)$, the two cyclic orders giving equal traces. Now $V_{f_{\mathbf{q}}}u = -f_{\mathbf{q}}\Delta u - \nabla\cdot(u\nabla f_{\mathbf{q}})$; the first member tends to 0, and the second, of the order 1 in H^{-1}, carries the frequencies $\pm\mathbf{q} + O(1)$, at which $\hat{G}_b(k)kk^{\mathsf{T}} \to b^{-1}\hat{\mathbf{q}}\hat{\mathbf{q}}^{\mathsf{T}}$, $\hat{\mathbf{q}} = \mathbf{q}/|\mathbf{q}|$.

So $V_{f_{\mathbf{q}}}\hat{G}_b(D)V_{f_{\mathbf{q}}}$ tends to the multiplication by $(2b)^{-1}\psi^2$, the mean of $b^{-1}\psi^2\sin^2(\mathbf{q}\cdot x)$; as the supports of ψ and h are disjoint, $\hat{G}_b(D)V_h\hat{G}_b(D)$ has a smooth kernel near $\operatorname{supp}\psi$, and the traces tend to $\frac{3}{2b}\operatorname{Tr}[\psi^2\hat{G}_b(D)V_h\hat{G}_b(D)] = -\frac{3}{2b}\iint \psi(x)^2h(y)\,\mathsf{p}(x-y)\,dx\,dy$, since $\mathsf{V}_h(u,u) = -\int h(|\nabla u|^2 + 2u\Delta u)$ at $u = \mathcal{C}(x-\cdot)$. Finally $\mathsf{p} = |\nabla\mathcal{C}|^2 + 2\mathcal{C}\mathcal{H} > 0$ off 0, as $\mathcal{C} > 0$ and $\mathcal{H} > 0$ there, by the representation of $\mathcal{C}$ above. □

8.1.12. *Proof of Thm. Z. Step 1: the choice.* At $\pi^+(\varsigma^{\mathrm{S}})$ every constant keeps its value (the unchanged restriction of Section YM3_I:82); so L, $I(g)$, $\mathfrak{L}(g)$, ℓ_g, the first-passage data and the functions $S^T_{2;K,\ell,\beta}$ formed there are those of $\mathfrak{c}$, and Prop. 8.10 speaks of $\mathfrak{c}$.

Step 2: (a). Put $d^L = \min\{d_{\mathrm{e}}, \ell_g/21\}$. For $d \le d^L$ and $\Psi \in \mathfrak{T}_{\mathbb{R}^3}(d;\vartheta)$ the periodisation $\Psi^{\langle \ell_g\rangle}$ lies in $\mathfrak{T}_{T_{\ell_g}}(d;\vartheta)$ and has the suprema and the kernels $\mathcal{K}_q$ of Ψ, and Prop. 8.10 (b) gives

$$|T^{(\ell_g,b)}(\Psi) - L_b^+(\Psi)| \le \bar\varepsilon_\infty(d)\, b^{-3} d\, |\Psi| \qquad (b \in I(g)).$$

Let $\varepsilon^L_{g,\vartheta}(d)$ be the supremum, over $0 < d'' \le d$, $b \in I(g)$ and $\Psi \in \mathfrak{T}_{\mathbb{R}^3}(d'';\vartheta)$, of $|T^{(\ell_g,b)}(\Psi) - L_b^+(\Psi)|/(b^{-3}d''|\Psi|)$: the least non-decreasing tolerance; it is at most $\bar\varepsilon_\infty \le C(g,\vartheta)d^{1/2}$.

Step 3: the consequences (i)–(iv). (a) at $\Psi_r \in \mathfrak{T}^1(rd_1;\vartheta)$, with $\mathcal{K}_6(\Psi_r) = \mathcal{K}_6(\Psi)$, $\mathcal{K}_5(\Psi_r) = r\mathcal{K}_5(\Psi)$ and $|\Psi_r| = |\Psi|$, gives eq. 8.iv: a term of the order r against an error $O(r^{3/2})$ and the threshold r^2 of a dilation covariant function; the bounds on B are eq. 8.i. Write $T(\Psi_r) = \mathsf{A} + \mathsf{B}r + \rho(r)$ with $|\rho(r)| \le \varepsilon^L(rd_1)\mathsf{u}r$; then $T(\Psi_{tr}) - T(\Psi_r) = (t-1)\mathsf{B}r + \rho(tr) - \rho(r)$ with $|\rho(tr) - \rho(r)| \le (t+1)\varepsilon^L(trd_1)\mathsf{u}r$, and $\mathcal{N}_t^T$ is treated as in Lem. 8.12: (i). For $\Phi \in \mathfrak{T}^1(d;\vartheta)$ and $d \le d^L$, by eq. 8.i,

$$|T(\Phi) - c_2 b^{-2}\mathcal{K}_6(\Phi)| \ge \varkappa b^{-3}\mathcal{K}_5(\Phi) - \varepsilon^L(d)\, b^{-3} d|\Phi| \ge [10^{-5}\varkappa\vartheta^2 - \varepsilon^L(d)]\, b^{-3} d\, |\Phi|;$$

at $\Phi = \Psi_r$, $d = rd_1$, divided by r^2, this is (ii), and at $\Phi = \Psi_n$, divided by $d_n^2|\Psi_n|$, it is (iii). Under $\varepsilon^L(d) \le \frac12 \cdot 10^{-5}\varkappa\vartheta^2$ the same line gives $b^2|T(\Psi) - c_2 b^{-2}\mathcal{K}_6(\Psi)| \ge \frac12 \cdot 10^{-5}\varkappa\vartheta^2 b^{-1} d|\Psi|$, which exceeds $C\vartheta^{-1}(d/b)^2|\Psi|$ if and only if $d/b < \frac12 \cdot 10^{-5}\varkappa\vartheta^3/C$; such d exist, as $\varepsilon^L \to 0$: (iv).

Step 4: (b). As $d_\flat \le \frac{1}{180} < \ell_g/21$, Step 2 applies. At the data $(K, L^K b)$ with $a_K \le 9d$ the periodic extensions of the Wilson measures on T_ℓ and T_{ℓ_g} are two states at the datum; Prop. 8.3 and $K \to \infty$, along the full sequence, give

$$|T^{(\ell,b)}(\Psi) - T^{(\ell_g,b)}(\Psi)| \le 2C_2^{\mathrm{T}}(g)\vartheta^{-1}d^2b^{-2}|\Psi| = 2C_2^{\mathrm{T}}(g)\vartheta^{-1}bd \cdot b^{-3}d\,|\Psi|,$$

and Prop. 8.10 (b) on T_{ℓ_g} gives the first assertion; $\bar\varepsilon_\infty(d) \le C(g,\vartheta)d^{1/2}$ and $d < 1$ bound the bracket. At a cutoff $K > s(d)$ one has $K \ge 3$ and $8a_K \le d$, so $(K, L^K b)$ is a first-passage datum at g at which Prop. 8.3 (a) compares every state with the periodic extension μ' of the Wilson measure on T_{ℓ_g}; and $S_2^T[\mu'](\Psi) = S^T_{2;K,\ell_g,L^K b}(\Psi^{\langle\ell_g\rangle})$, to which Prop. 8.10 (b), respectively (c), applies.

Step 5: (d). This is YM3_I:(2215) at $n = 2$: the dilation by L^s carries T_ℓ with the spacing a_K onto $T_{L^s\ell}$ with the spacing a_{K-s}, keeps $\beta = L^K b = L^{K-s}(L^s b)$, and $\mathcal{O}_K(f) = L^s\mathcal{O}_{K-s}(f(\cdot/L^s))$; so $S^T_{2;K,\ell,L^K b}(\Psi^{\langle\ell\rangle}) = L^{2s}S^T_{2;K-s,L^s\ell,L^{K-s}L^s b}((\Psi_{L^s})^{\langle L^s\ell\rangle})$, and we let $K \to \infty$. For the second sentence let $\Psi \in \mathfrak{T}_{\mathbb{R}^3}(d;\vartheta)$ with $d \le L^z d^L$; then $\Psi_{L^{-z}} \in \mathfrak{T}_{\mathbb{R}^3}(L^{-z}d;\vartheta)$, $T^{(L^z\ell_g, L^z b)}(\Psi) = L^{-2z}T^{(\ell_g,b)}(\Psi_{L^{-z}})$, $L^{-2z}L_b^+(\Psi_{L^{-z}}) = L^+_{L^z b}(\Psi)$, and $L^{-2z}b^{-3}L^{-z}d = (L^z b)^{-3}d$; apply (a) at $\Psi_{L^{-z}}$.

Step 6: (c). For large s the coupling $g'' = (2/(L^s b))^{1/2}$ is $\le \Gamma$, $L^s b \in I^\circ(g'')$ by Lem. 8.2 (a), as $L^s b \ge 2\Gamma^{-2} \ge 2\gamma^{-2}$ for large s, and $L^s\ell \in \mathfrak{L}(g'')$ (equation YM3_I:(1528)), as $m_{\mathfrak{L}}(g'')$ grows only like a multiple of $\log\log g''^{-2}$ while $m + s$ grows like s. Apply (b) at $(g'', L^s b, L^s\ell)$ to $\Psi_{L^s} \in \mathfrak{T}^1(L^s d;\vartheta)$, and (d), with

$$\mathcal{K}_6(\Psi_{L^s}) = \mathcal{K}_6(\Psi), \qquad (L^s b)^{-1}\mathcal{K}_5(\Psi_{L^s}) = b^{-1}\mathcal{K}_5(\Psi), \qquad L^{2s}(L^s b)^{-3}L^s d = b^{-3}d;$$

the bracket of (b) at $(g'', L^s b, L^s d)$ is $\bar\varepsilon_\infty(L^s d) + 2C_2^{\mathrm{T}}(g'')\vartheta^{-1}L^{2s}bd \le C(g'',\vartheta)(1 + L^s b)(L^s d)^{1/2}$.

Step 7: (e), (f). Two inequalities as in (e), at a pair $\Psi \in \mathfrak{T}^1(d;\vartheta)$, give $|c - c'|\, c_2 b^{-3}\mathcal{K}_5(\Psi) \le (\varepsilon + \varepsilon')(d)\, b^{-3} d|\Psi|$, and $\mathcal{K}_5(\Psi) \ge 10^{-5}\vartheta^2 d|\Psi|$; so $|c - c'|\, c_2\, 10^{-5}\vartheta^2 \le (\varepsilon + \varepsilon')(d) \to 0$. For a compact set J of labels one s serves for all $b \in J$, and the compact set $L^s J$ is covered by finitely many open intervals $I^\circ(g_i)$, $g_i \le \Gamma$, $L^s\ell \in \mathfrak{L}(g_i)$, since each point b'' of $L^s J$ lies in $I^\circ((2/b'')^{1/2})$ (Lem. 8.2 (a)); the least of the ranges and the largest of the tolerances and constants of the g_i serve for all $b \in J$. □

8.1.13. *Proof of Thm. Y. Step 1: a period of labels.* Let $\gamma_o = \min\{\gamma_{\mathrm{e}}, \gamma_W, \gamma_W^{\mathrm{sd}}\}$, $y = \gamma_o^{-2}$, $\Lambda = 2y$, $\hat g_1 = \gamma_o$ and $\hat g_2 = \gamma_o/\sqrt{2}$. As $y \ge \gamma_{\mathrm{e}}^{-2} \ge 4\bar B$ and $y \ge 4$, we have $y > \bar B$, so that

$$I^\circ(\hat g_1) = (y + \bar B,\, Ly + (L-2)\bar B) \supset [2y, Ly), \qquad I^\circ(\hat g_2) = (2y + \bar B,\, 2Ly + (L-2)\bar B) \supset [3y, 2Ly),$$

and, as $L \geq 3$, $[\Lambda, L\Lambda) \subset I^\circ(\hat g_1) \cup I^\circ(\hat g_2)$. Put $\delta_{\mathrm c} = (L\Lambda)^{-1} \min_i \min\{d^L, d_\flat\}(\hat g_i, \vartheta)$ and, for $0 < \delta \leq \delta_{\mathrm c}$,

$$\varepsilon^{\mathrm{nat}}(\delta) = \max_{i=1,2} \varepsilon^L_{\hat g_i,\vartheta}(L\Lambda\delta) + 2 \max_{i=1,2} C_2^{\mathrm T}(\hat g_i)\, \vartheta^{-1} L^2\Lambda^2\delta :$$

non-decreasing, depending only on $\mathfrak c$ and ϑ, and $\varepsilon^{\mathrm{nat}}(\delta) \leq C\delta^{1/2}$ by Thm. Z (a).

Step 2: a dilated label. For S with the label b there is exactly one $z \in \mathbb Z$ with $b' = L^z b \in [\Lambda, L\Lambda)$; take i with $b' \in I^\circ(\hat g_i)$. As $\hat g_i \leq \gamma_W^{\mathrm{sd}}$, the triple $\tau' = (\hat g_i, b', \ell_{\hat g_i})$ is admissible, and $E_{\tau'} = 2C_2^{\mathrm T}(\hat g_i)\vartheta^{-1}b'^2 \leq 2C_2^{\mathrm T}(\hat g_i)\vartheta^{-1}L^2\Lambda^2$.

Step 3: (1). Let $d \leq \delta_{\mathrm c} b$, $\Psi \in \mathfrak T^1(d;\vartheta)$ and $d' = L^z d = b'd/b \leq L\Lambda\delta_{\mathrm c}$. Thm. Z (a) at $(\hat g_i, b')$, where $\hat g_i \leq \gamma_{\mathrm e}$ and $b' \in I(\hat g_i)$, for the pair $\Psi_{L^z} \in \mathfrak T^1(d';\vartheta)$, with $\mathcal K_6(\Psi_{L^z}) = \mathcal K_6(\Psi)$, $b'^{-1}\mathcal K_5(\Psi_{L^z}) = b^{-1}\mathcal K_5(\Psi)$ and $b'^{-1}d' = b^{-1}d$, together with Lem. 8.11, gives

$$|b^2\mathsf S(\Psi) - c_2[\mathcal K_6 + \tfrac{8}{3\pi}b^{-1}\mathcal K_5](\Psi)| \leq \Big[\varepsilon^L_{\hat g_i,\vartheta}\Big(L\Lambda\frac{d}{b}\Big) + 2C_2^{\mathrm T}(\hat g_i)\vartheta^{-1}L^2\Lambda^2\frac{d}{b}\Big]\frac{d}{b}\,|\Psi| \leq \varepsilon^{\mathrm{nat}}(d/b)\, b^{-1}d\,|\Psi|;$$

divide by b^2. The labels of first-passage data at g exceed g^{-2} by YM3_I:(1531), so $b \geq g^{-2}$, $d/b \leq g^2 d$, and $d \leq d^S$ implies $d \leq \delta_{\mathrm c} b$: the law with ε^S and d^S. For the form at a dilated label, let $L^z b \geq 2\Gamma^{-2}$ and $g' = (2/(L^z b))^{1/2}$; then $g' \leq \Gamma$ and $L^z b \in I^\circ(g')$ by Lem. 8.2 (a), as $2\Gamma^{-2} \geq 2\gamma^{-2}$, so $\tau' = (g', L^z b, \ell_{g'})$ is admissible; the same computation, with Thm. Z (a) at $(g', L^z b)$ and Lem. 8.11, gives the tolerance $\varepsilon^L_{g',\vartheta}(L^z d) + E_{\tau'} b^{-1} d$ for $L^z d \leq \min\{d^L, d_\flat\}(g', \vartheta)$, and $E_{\tau'} b^{-1} = 2C_2^{\mathrm T}(g')\vartheta^{-1}L^{2z}b$.

Step 4: (2). Lem. 8.12 with $c = \frac{8}{3\pi}$, $\varepsilon(d) = \varepsilon^{\mathrm{nat}}(d/b)$ and $d_7 = \delta_{\mathrm c} b$; there $\mathsf B = cc_2 b^{-3}\mathcal K_5(\Psi) = B_S$, and the lower bound on B_S is eq. 8.i. The first bound at tr and at r gives the bound on $r^{-1}[\mathsf S(\Psi_{tr}) - \mathsf S(\Psi_r)]$; $\varepsilon^{\mathrm{nat}}(\delta) \leq C\delta^{1/2}$ turns the remainders $o(r)$ into $O(r^{3/2})$; the means converge because their terms do; and at $t = 2$ the numbers $t(t-1)(t^5-1)$ and $t^2 + t + t^6 + t^7$ are 62 and 198.

Step 5: (3). By Lem. 8.12, S_2 and every S_2^ξ fail to be dilation covariant; so $\mathsf N = \mathsf X$, and $\lambda(\mathsf N) = 1$ as λ is a probability measure.

Step 6: (3) from the law at a single triple of any choice. Let $b^\circ \in I^\circ(\Gamma)$. The triple $(\Gamma, b^\circ, \ell_\Gamma)$ is admissible, and clause (iii) of Thm. Z holds there, as $\Gamma \leq \gamma_{\mathrm e}$. Prop. 8.17 with $\mathfrak c_* = \mathfrak c^\circ = \mathfrak c$ gives (3) again, and moreover the failure of the three-scale number of S_2 to vanish at pairs and r as small as is wished. This second proof uses the law at one triple and one ϑ only, and no law at the labels of the limits; the triple may belong to any choice. □

Remark 8.19 (The constants in the second proof of Thm. Y (3)). The departure of clause (iv) of Thm. Z dominates explicitly the errors of the comparisons by which the proof of Prop. 8.17 passes from a limit to the least torus, when the label of the limit is first dilated into one period $[\Lambda_W, L\Lambda_W)$. Let $g \leq \Gamma$ and $b \in I^\circ(g)$; let $\tilde{\mathfrak c}$ be a second choice of $\mathfrak c$ (Lem. 8.14 with $Q = L$), with its number $\tilde C_2^{\mathrm T}$ and an admissible triple $\tau_2 = (g_2, b_2, \ell_2)$; put $\Lambda_W = 2(\gamma_W^{\mathrm{sd}})^{-2}$, $g_1' = \gamma_W^{\mathrm{sd}}$ and $g_2' = \gamma_W^{\mathrm{sd}}/\sqrt{2}$. Since $(\gamma_W^{\mathrm{sd}})^{-2} \geq \gamma^{-2} > \bar B$ (Lem. 8.2 (a)), the computation of Step 1 above gives $[\Lambda_W, L\Lambda_W) \subset I^\circ(g_1') \cup I^\circ(g_2')$; every $b' > 0$ has exactly one dilate $L^z b'$ in $[\Lambda_W, L\Lambda_W)$. The number

$$c_{\max} = \max_{i=1,2} \sup\{2C_2^{\mathrm T}(g_i')\, b'^2 :\ b' \in [\Lambda_W, L\Lambda_W) \cap I^\circ(g_i')\} \leq 2L^2\Lambda_W^2 \max_i C_2^{\mathrm T}(g_i')$$

is finite, and so is $\mathsf C_{\mathrm{pair}} = 2C_2^{\mathrm T}(g)b^2 + 4\tilde C_2^{\mathrm T}(g_2)b_2^2 + 2c_{\max}$. For every admissible triple $\tau' = (g_i', b', \ell')$ with $b' \in [\Lambda_W, L\Lambda_W) \cap I^\circ(g_i')$,

$$\vartheta^{-1}\mathsf C_{\mathrm{pair}} \geq 2E_{\tau'} + 2E_{\tau_2} + E_{(g,b,\ell_g)},$$

the sum of the constants of the three comparisons by which the proof of Prop. 8.17, with τ' of this kind and $\tau^\circ = (g, b, \ell_g)$, passes from a limit at the label b', through the second choice, to $T^{(\ell_g, b)}$. Clause (iv) of Thm. Z, at $C = \mathsf C_{\mathrm{pair}}$, says that for small d/b the departure $b^2|T^{(\ell_g,b)}(\Psi) - c_2 b^{-2}\mathcal K_6(\Psi)|$ exceeds $(2E_{\tau'} + 2E_{\tau_2} + E_{(g,b,\ell_g)})(d/b)^2|\Psi|$ at every pair $\Psi \in \mathfrak T^1(d;\vartheta)$.

8.1.14. *Proof of Cor. 8.1. Step 1: pieces.* Let $\delta_0 = \mathrm{dist}(\mathrm{supp} f_0, \mathrm{supp} h_0) > 0$ and $\rho_0 = \delta_0/4$, and let $\phi_i, \chi_k \geq 0$, finitely many, be Lipschitz functions supported in closed balls of radius ρ_0 that lie within $\delta_0/4$ of $\mathrm{supp} f_0$, respectively of $\mathrm{supp} h_0$, with $\sum_i \phi_i = 1$ on $\mathrm{supp} f_0$ and $\sum_k \chi_k = 1$ on $\mathrm{supp} h_0$. With $M_0 = \|f_0\|_\infty + 1$, $f_0 = \sum_i [(f_0 + M_0)\phi_i - M_0\phi_i]$ is a signed sum of non-negative Lipschitz pieces, and likewise h_0; we drop the pieces that vanish. A piece F of f_0 and a piece H of h_0 lie in balls of radius ρ_0 at the distance $\delta' = \mathrm{dist}(\mathrm{supp} F, \mathrm{supp} H) \geq \delta_0/2 = 2\rho_0$; so $(F, H) \in \mathfrak{T}_{\mathbb{R}^3}(\delta'/2; \vartheta_0)$ for one $\vartheta_0 \in (0,1)$ that serves for all the finitely many pairs of pieces.

Step 2: one pair of pieces. Let s and g'' be as in Step 6 of the proof of Thm. Z. Clauses (b) at $(g'', L^s b, L^s \ell)$ and (d), which hold on all of $\mathfrak{T}_{\mathbb{R}^3}$, applied to the translate by $\tilde{p}$ of $(F, H)_r \in \mathfrak{T}_{\mathbb{R}^3}(r\delta'/2; \vartheta_0)$, with $\mathcal{K}_6((F,H)_r) = \mathcal{K}_6(F,H)$, $\mathcal{K}_5((F,H)_r) = r\,\mathcal{K}_5(F,H)$ and eq. 8.i, give for small r, with a tolerance ε independent of the position,

$$\big|T^{(\ell,b)}((F,H)_r(\cdot - \tilde{p})) - c_2 b^{-2}\mathcal{K}_6(F,H)\big| \leq \Big[\frac{8}{3\pi} + \varepsilon(r\delta'/2)\Big]\, b^{-3} r\, \frac{\delta'}{2}\, \|F\|_\infty \|H\|_\infty .$$

Step 3: (a). $S_2^{(\ell,b)}$ is bilinear on the functions supported in two disjoint closed sets, and so is $\mathcal{K}_6$; summing over the pairs of pieces, with their signs, gives $|S_2^{(\ell,b)}(f_r, h_r) - c_2 b^{-2}\mathcal{K}_6(f_0, h_0)| \leq Cr$, with C independent of p and, by the covering argument of Step 7 of the proof of Thm. Z, the same for the labels of a compact set. For $(\psi_0 \circ \theta_\mu, \psi_0)$ the supports are compact subsets of $\{x_\mu < 0\}$ and $\{x_\mu > 0\}$, so at a positive distance.

Step 4: (b). The supports of $\psi_0 \circ \theta_\mu$ and ψ_0 lie in $\mathbb{B}(-e_\mu, \frac{1}{2})$ and $\mathbb{B}(e_\mu, \frac{1}{2})$, at a distance in $[1,3]$; so $(\psi_0 \circ \theta_\mu, \psi_0) \in \mathfrak{T}^1(\frac{1}{2}; \vartheta_1)$ for some $\vartheta_1 \in (0,1)$, and its translate by $\tilde{p}$ dilated by r lies in $\mathfrak{T}^1(\frac{r}{2}; \vartheta_1)$. Clause (c) of Thm. Z, whose bracket is at most $C(r/2)^{1/2}$, gives the expansion, and $\mathcal{K}_5(\psi_0 \circ \theta_\mu, \psi_0) > 0$ by eq. 8.i.

Step 5: (c). At a finite cutoff $\beta = L^K b$, so that $\partial_b = L^K \partial_\beta$, and ∂_β of a cumulant of observables under $e^{-\beta A}$ is minus the cumulant with the action $A = \sum_p a_p = a_K \mathcal{O}_K(1)$ added; hence $\partial_b S^T_{2;K,\ell,L^K b}(f,h) = -S^T_{3;K,\ell,L^K b}(1,f,h)$, which is the finite-cutoff differentiation identity in the proof of Prop. YM3_I:118.3 for the first derivative. By Prop. YM3_I:118.3, equation YM3_I:(2217), applied to the pair (f_r, h_r), whose supports are disjoint, $b \mapsto S_2^{(\ell,b)}(f_r, h_r)$ is infinitely differentiable on $(0,\infty)$ and $\partial_b S_2^{(\ell,b)}(f_r, h_r) = -S_3^{(\ell,b)}(1, f_r, h_r)$, the derivatives at the finite cutoffs converging to it uniformly on $[b_1, b_2]$; so the integrand is $-\partial_b[b^2 S_2^{(\ell,b)}(f_r, h_r)]$, and the integral equals

$$b_1^2[S_2^{(\ell,b_1)}(f_r,h_r) - c_2 b_1^{-2}\mathcal{K}_6(f_0,h_0)] - b_2^2[S_2^{(\ell,b_2)}(f_r,h_r) - c_2 b_2^{-2}\mathcal{K}_6(f_0,h_0)],$$

each bracket being at most Cr by (a). □

8.1.15. *Scope of the conclusions.* The theorems concern short distances at a fixed coupling; on $\mathbb{R}^3$ they hold for $0 < g \leq \gamma_W$. Not being dilation covariant is not being interacting: three free photon fields together with an independent generalised free field of two-point kernel $\varkappa b^{-3}|x-y|^{-5}$, whose spectral density $\varkappa b^{-3}\frac{2\pi}{3} m^3$ is positive, obey (L) with $c = \frac{8}{3\pi}$ and remainder 0 and have $\mathsf{N} = \mathsf{X}$, and the Gaussian model of Prop. 8.18 has the same two-term law on a torus. The numbers γ_{e}, d_{e}, d^L, δ_{c} and the tolerances are given by existence.

9. Compact gauge groups

Let $G \subset U(N)$ be compact, connected and simply connected, with simple Lie algebra $\mathfrak{g}$. Use the representation to normalize $\langle X, Y\rangle = N^{-1}\,\mathrm{tr}(XY)$ on the Hermitian form of $\mathfrak{g}$, and choose an orthonormal basis $(t_a)_{a=1}^n$, where $n = \dim \mathfrak{g}$. Put $\Omega = \sum_a t_a^2$. The two Casimir numbers in the calculation are

$$C_A = \mathrm{tr}_{\mathfrak{g}^{\mathbb{C}}}(\mathrm{ad}\, H)^2, \qquad C_\rho = N^{-1}\,\mathrm{tr}(\Omega H^2), \qquad \langle H, H\rangle = 1.$$

For a simple algebra these numbers do not depend on H.

Sketch H (Extension to a simple compact group). The proposed extension of the $SU(2)$ construction has the same lattice geometry, with constants depending on G and its representation. Its leading action-density two-point coefficient and relative next perturbative coefficient are

$$c_2(G) = \frac{3n}{16\pi^2}, \qquad c_1(G) = \frac{\varphi_G}{3\pi n}, \qquad \varphi_G = nC_A.$$

The coefficient calculation is local. The full extension still requires the uniform construction at nontrivial regular backgrounds described below. ■□

Sketch of the argument. At the identity configuration, a linear operator of the construction has the form $K \otimes 1_{\mathfrak{g}^{\mathbb{C}}}$, where K depends only on the lattice. The quadratic Gaussian theory therefore consists of n identical colour components. This gives the factor n in $c_2(G)$. At one loop, summing the colour indices reduces the commutator terms to C_A and the representation terms to C_ρ. Let Ψ_k and Ψ_k^G denote the one-loop functionals for $SU(2)$ and G, and write $b[\Psi]$ for the quadratic coefficient. Colour contraction gives

$$b[\Psi_k^G] = \frac{C_A}{8} b[\Psi_k] + \left(\frac{C_\rho}{3} - \frac{C_A}{8}\right)(L^{-k} - L^{-4k}).$$

This also identifies the increments of the Gaussian normalizer. For $SU(3)$ in its defining representation, $(n, C_A, C_\rho) = (8, 18, 8)$, and hence $c_2 = 3/(2\pi^2)$ and $c_1 = 6/\pi$.

The nonlinear construction needs more than this colour count. The logarithm must lie in $\mathfrak{g}^{\mathbb{C}}$, the Haar density must be retained in the local chart, and projection and transpose bounds must be read in their correct norms. Complex domains must control both $|X - 1|$ and $|X^{-1} - 1|$. These changes give local formulas; they do not establish uniform inverses at every regular background.

The nonlinear step is to construct these inverses, critical orbits, localized responses and Gaussian families with bounds uniform in depth and volume, with radii satisfying all the estimates. These uniform constructions remain to be supplied. The sourced induction also requires uniform complex-weight estimates. Passage to continuum limits requires the transferred state bounds and cutoff comparisons. For the finite-box perturbative coefficient, expand at fixed box before increasing its size. For separately acting products of $SU(2)$ factors, the Wilson measures factor exactly, giving a special case without this general-group transfer problem.

9.1. **Compact gauge groups.** For a general compact gauge group, the lattice geometry of the construction is unchanged, but the colour algebra and the matrix chart are not. At the trivial field every linear operator is a scalar lattice operator tensored with the colour space. This observation determines the leading two-point coefficient. At one loop the dependence on the representation is carried by two Casimir numbers. Away from the trivial field, the construction also requires uniform estimates for propagators, critical configurations and localised Gaussian integrals.

We first state the local calculations and a conditional extension of the multiscale construction. The hypotheses for that extension are given explicitly below. They separate the local group calculations from the estimates at general regular backgrounds. The finite-box calculation of the next perturbative coefficient is independent of this extension. For products acting on separate blocks, the measures factor, and the corresponding conclusions follow directly from the $SU(2)$ construction. For quotients, the preliminary-gauge and quantitative local assertions of Prop. 9.33(iii),(iv) retain their separately stated analytic hypotheses.

9.1.1. *Notation and hypotheses.*

Definition 9.1 (The pair)**.** Let $N \geq 2$ and let $G \subset U(N)$ be nontrivial, compact and connected; the *pair* is G with its representation on $\mathbb{C}^N$. Let $\mathfrak{g} = \{Y = Y^* : e^{itY} \in G \text{ for all } t \in \mathbb{R}\}$, $n = \dim \mathfrak{g}$, $\mathfrak{g}^c = \mathfrak{g} + i\mathfrak{g}$, and G^c the group generated by $\exp \mathfrak{g}^c$; $\langle X, Y\rangle = \frac{1}{N}\operatorname{tr} XY$ with the norm $\|X\| = (\frac{1}{N}\operatorname{tr} X^*X)^{1/2}$, and $|\cdot|$ the operator norm; P the projection of $\mathbb{C}^{N\times N}$ onto $\mathfrak{g}^c$

along $\{M : \operatorname{tr} MZ = 0 \text{ for all } Z \in \mathfrak{g}^c\}$; $(t_a)_{a\le n}$ an orthonormal basis of $\mathfrak{g}$ and $\Omega = \sum_a t_a^2$; $\kappa_G = \max_{0\neq Y\in\mathfrak{g}} |Y|/\|Y\|$ and $\tau_G = \sup_{\mathbf{1}\neq V\in G} |V-\mathbf{1}|/\|V-\mathbf{1}\|$;

$G^\pi = \exp(i\{Y \in \mathfrak{g} : |Y| < \pi\})$; r_G the largest r with $\{u \in G : |u-\mathbf{1}| < r\} \subset G^\pi$, and $\rho_G = 2\arcsin\frac{r_G}{2} \in (0,\pi]$, the largest number such that every $u \in G$ whose eigen-angles are below ρ_G in modulus lies in G^π; $z_G = \min\{1, \operatorname{dist}(\mathbf{1}, Z(G)\setminus\{\mathbf{1}\})\}$, and $z_G = 1$ if $Z(G) = \{\mathbf{1}\}$.

With log the power series on $|X-\mathbf{1}| < 1$, let $D^c = \{X \in G^c : |X-\mathbf{1}| < 1,\ \log X \in \mathfrak{g}^c\}$ and $D = D^c \cap G$; further $X^c = (X^*)^{-1}$, $m(X) = \max\{|X-\mathbf{1}|, |X^{-1}-\mathbf{1}|\}$, $c(X) = \frac12(X+X^{-1})$ and $H_X = \frac{1}{2i}(X-X^{-1})$; $|P|$ is the norm of P for $|\cdot|$, and $\varsigma_G = \sup\{|PH_V|/|V-\mathbf{1}| : \mathbf{1} \neq V \in G\}$, $\varsigma_G^D = \sup\{|PH_V|/|V-\mathbf{1}| : \mathbf{1} \neq V \in D\}$, $\kappa' = \kappa'(G) = 4\sup\{|(1-P)H_V|/|V-\mathbf{1}|^3 : \mathbf{1} \neq V \in D\}$, which Lem. 9.12(ii) bounds, and $j_G = \max\{1+\kappa' z_G^2, 3\varsigma_G^D\}$ are numbers of the pair, $(1-P)H_V$ being of third order in $V-\mathbf{1}$.

The action density is $a_p = a(\partial U(p))$ with $a(X) = 1 - \frac{1}{2N}\operatorname{tr}(X+X^{-1})$, which is $1 - \frac1N \operatorname{Re}\operatorname{tr} X$ on G; $F(p) = \xi^{-2} P H_{\partial U(p)}$ and $J = D^{\mathrm T}F$, with ∂U, D and ξ of Def. YM3_I:9.1; the *Casimir density* is $k_p = n - \frac1N \operatorname{Re}\operatorname{tr}(\Omega\,\partial U(p))$.

For simple $\mathfrak{g}$: $C_A = \operatorname{tr}(\operatorname{ad} H)^2$, the trace taken on $\mathfrak{g}^c$, and $C_\rho = \frac1N \operatorname{tr}(\Omega H^2)$ for $H \in \mathfrak{g}$ with $\|H\| = 1$, numbers that do not depend on H (Lem. 9.15); $\varphi_G = \sum_a \operatorname{tr}(\operatorname{ad} t_a)^2 = nC_A$; $\gamma_G = \frac12 C_\rho - \frac{3}{16}C_A$; $c_2(G) = \frac{3n}{16\pi^2}$ and $c_1(G) = \frac{\varphi_G}{3\pi n}$.

For semisimple $\mathfrak{g} = \bigoplus_{i\le r}\mathfrak{g}_i$ with simple ideals $\mathfrak{g}_i$: $n_i = \dim\mathfrak{g}_i$, P_i the orthogonal projection onto $\mathfrak{g}_i$, $\kappa_i = \max_{0\neq Y\in\mathfrak{g}_i}|Y|/\|Y\|$, and $C_A^{(i)}$, θ_i are formed as C_A, C_ρ with $H \in \mathfrak{g}_i$. ■

If $\mathfrak{g}$ is semisimple, then $G \subset SU(N)$: det restricted to G is a character, and its differential tr vanishes on $[\mathfrak{g},\mathfrak{g}] = \mathfrak{g}$. For every G, $|P| \le N^{1/2}$: $\mathfrak{g}^c$ being closed under adjoints, P is the orthogonal projection for $\operatorname{tr} X^*Y$, so that $|PZ| \le (\operatorname{tr} Z^*Z)^{1/2} \le N^{1/2}|Z|$. For $SU(N)$: $n = N^2-1$, $\kappa_G^2 = \tau_G^2 = N-1$, $r_G = 2\sin\frac{\pi}{N}$, $\rho_G = \frac{2\pi}{N}$, $z_G = \min\{1, r_G\}$, which is 1 exactly for $N \le 6$, $\Omega = n\mathbf{1}$, $|P| \le 2$, $\varsigma_G \le 2 - \frac{2}{N}$ (Remark 9.6), $\kappa' \le 2 - \pi/\sqrt3$ (Lem. 9.12(ii)), $\varsigma_G^D = 1$ and $j_G = 3$ (on D, where $|V-\mathbf{1}| < 1$, $|PH_V| \le |H_V| + |(1-P)H_V| \le |V-\mathbf{1}| - \frac18|V-\mathbf{1}|^3 + \frac14\kappa'|V-\mathbf{1}|^3 < |V-\mathbf{1}|$, and the quotient tends to 1 at $\mathbf{1}$), $(C_A, C_\rho) = (2N^2, N^2-1)$, $\gamma_G = \frac{N^2-4}{8}$ and $c_1(G) = \frac{2N^2}{3\pi}$; at $SU(2)$ this is $(C_A, C_\rho) = (8,3)$, $\gamma_G = 0$, $c_1 = \frac{8}{3\pi}$, and at $SU(3)$ it is $(n, C_A, C_\rho, \gamma_G) = (8, 18, 8, \frac58)$, $r_G = \sqrt3$, $z_G = 1$, $\kappa_G^2 = 2$. The blocking factor L, the cutoff K, the levels, $\eta = L^{-k}$ and $a_K = L^{-K}$ are those of Section YM3_I:3; $B_\sharp$ bounds the increments $L^{-1}x_{k-1} - x_k$ of the run, $\bar B = B_\sharp \frac{L}{L-1}$, $\mathsf{G}_\sharp$ bounds the coefficient, and $x_k \ge \gamma^{-2}$, as in Thm. YM3_I:D; the reference coupling and first-passage notation are those of Prop. YM3_I:74.1: $0 < g \le \gamma$, $X_g = L(g^{-2} + B_\sharp)$ and $I(g) = (g^{-2} + \bar B, X_g - \bar B]$, with the admitted sides determined by the geometric room condition there. The short-distance ceiling $\gamma_W \le \gamma$ and the scale and cutoff restrictions are those of Thm. YM3_I:AV, transferred under Def. 9.5 and the construction hypotheses below. Set $\mathcal{K}_q(f,h) = \iint f(x)h(y)|x-y|^{-q}\,\mathrm dx\,\mathrm dy$.

The averaging radius of the $SU(2)$ construction is $\omega_M = \frac{1}{200L^2}$. A *number of L and the pair* is a number that depends on L and on G with its representation, and on nothing else.

Definition 9.2 (Regular classes at the local scale)**.** For a separated sequence $\Omega_0 \supset \cdots \supset \Omega_k$, put $\eta = L^{-k}$. The local index of a bare bond or plaquette $\mathfrak{e}$ meeting Ω_0 is the largest $j \le k$ for which one of its corners lies in Ω_j, and $\mathsf{L}(\mathfrak{e}) = L^{j-k}$. A plaquette is inner if all its corners lie in Ω_0; a bond is full if its four incident plaquettes are inner. These are the scales of Def. YM3_I:9.3.

Both $\mathcal{U}'(r) = \mathfrak{U}'_k(\mathbf{B}, r)$ and $\mathcal{U}(r) = \mathfrak{U}_k(\mathbf{B}, r)$ consist of real G-valued configurations on the inner bare bonds with

$$|\partial U(p) - \mathbf{1}| < r\eta^2 \mathsf{L}(p)^{-2} \qquad (p \text{ inner}).$$

The additional condition, at every full bond b, is respectively

$$|D_U^{\mathrm T}\partial U(b)| < r\eta^2\mathsf{L}(b)^{-3}, \qquad |J(U)(b)| < r\mathsf{L}(b)^{-3}.$$

Here the divergence and current use bare spacing η, and J is the projected current of Def. 9.1. The definitions themselves impose no inclusion between the two classes. ■

Lemma 9.3 (Transpose bound and depth zero)**.** For a complex linear map T of $\mathfrak{g}^c$, let T^{T} be its transpose for $\langle X, Y\rangle = N^{-1}\mathrm{tr}(XY)$. With operator norms induced by $|\cdot|$,

$$|T^{\mathrm{T}}| \le N|T|.$$

At depth zero the determining set consists of all inner bare bonds, and its constraint fibre over a real datum V is $\{V\}$. If each inner plaquette of V has curvature less than ε_1, then

$$|J(V)(b)| < 4\varsigma_G\varepsilon_1 \quad (b \text{ full}), \qquad V \in \mathcal{U}(4\varsigma_G\varepsilon_1).$$

In particular, if $B_3 \ge 4\varsigma_G$ and $\varepsilon_0 \ge B_3\varepsilon_1$, this singleton is the set of critical configurations in the regularity class of radius ε_0. ■

Proof. In the real orthonormal basis (t_a), the trace pairing is the ordinary bilinear coordinate pairing and $\|X\|^2 = \sum_a |x_a|^2$. Singular values give $\|X\| \le |X| \le \sqrt{N}\|X\|$. The Hilbert operator norm is unchanged by transposition, since the transpose is the complex conjugate of the adjoint. Applying the two norm comparisons, first to the output and then to the input, gives

$$|T^{\mathrm{T}}X| \le \sqrt{N}\|T^{\mathrm{T}}\|_{\mathrm{op}}\|X\| = \sqrt{N}\|T\|_{\mathrm{op}}\|X\| \le N|T|\,|X|.$$

At depth zero there are no inward bonds and the averaging map is the identity. Thus the fibre is a singleton and every curve in it is constant. In bare units the current at a full bond is a sum of four transported terms $\pm PH_{\partial V(p)}$. Each has norm less than $\varsigma_G\varepsilon_1$; unitary transport preserves this norm. This proves the current bound. Since G has positive dimension, the ratio defining ς_G tends to one along e^{itY} for any nonzero $Y \in \mathfrak{g}$; hence $\varsigma_G \ge 1$, and the curvature bound also has the claimed radius. The final assertion follows by increasing that radius to ε_0. □

Lemma 9.4 (The averaging row constant)**.** Suppose that the linearized average at a positive-level row c has the counting kernel

$$(\hat{Q}A)(c) = \sum_{b\in\mathsf{P}_c} k(c,b)\xi_c A(b), \qquad |k(c,b)| \le C_{\mathrm{av}}(\xi/\xi_c)^3,$$

where ξ is the bare spacing and $C_{\mathrm{av}} \ge 1$. Use the actual double blocks and neighbourhoods of the averaging map: each P_c has at most $6(\xi_c/\xi)^3$ positive bonds; each bare bond lies in at most eighteen such blocks; and the scale ratios used in either row or column sum are at most L. Then the two linear averaging bounds in equation YM3_I:(1045) hold with

$$C_Q = 18NL^3C_{\mathrm{av}}.$$

At level zero the average is the identity. The choice $C_{\mathrm{av}} = 3$ therefore gives $C_Q = 54NL^3$, including $108L^3$ at $SU(2)$. For a general pair this numerical choice requires the displayed kernel estimate with $C_{\mathrm{av}} = 3$ on the admitted averaging domain; it is a quantitative averaging input in the transferred construction. ■

Proof. For a row over y, the row sum is bounded by $6LC_{\mathrm{av}}\max_{N_L(y)}|A|$, with the dimensionless field norms of equation YM3_I:(1045). The transpose uses the bare weight ξ^3 and the row weight ξ_c^{-1}. The latter cancels the factor ξ_c in the counting kernel, giving

$$(\hat{Q}^{\mathrm{T}}\varphi)(b) = \xi^{-3}\sum_{c:b\in\mathsf{P}_c} k(c,b)^{\mathrm{T}}\varphi(c).$$

If b is over y, Lem. 9.3 and the eighteen-block count give

$$\xi_y^3|\hat{Q}^{\mathrm{T}}\varphi(b)| \le \sum_{c:b\in\mathsf{P}_c} NC_{\mathrm{av}}(\xi_y/\xi_c)^3|\varphi(c)| \le 18NL^3C_{\mathrm{av}}\max_{N_L(y)}|\varphi|.$$

The same constant dominates the row bound and the identity rows. This is the row/column count following YM3_I:(1045), with the transpose factor N retained. □

Definition 9.5 (Transfer to G)**.** The transfer to G of a definition or a statement of the $SU(2)$ construction is made by kind.

(1) What stands for one object is replaced alike at all places: $N\times N$ matrices, $\frac{1}{N}\mathrm{tr}$, G, G^c, $\mathfrak{g}$, $\mathfrak{g}^c$ for 2×2 matrices, $\frac{1}{2}\mathrm{tr}$, $SU(2)$, $SL(2,\mathbb{C})$ and their algebras; colour indices run to n; a_p, F and J are those of Def. 9.1, and $a(X)$ stands for every trace of a group-valued argument at complex fields, the gauge-fixing term included; $P[\frac{1}{2}\{c(X),\theta\}]$ stands where $SU(2)$ has $\frac{1}{2}(\mathrm{tr}\,X)\theta$; $|\cdot|$ stands in suprema, radii, series, domains and tests, $\|\cdot\|$ and $\langle\cdot,\cdot\rangle$ in Hilbert sums, pairings and quadratic forms; the cut-off of the bond shift (the cutoff preceding Lem. YM3_I:53.3), $\vartheta_{\mathrm{co}}(|Y|/\delta_k)$, is $\vartheta_{\mathrm{co}}(\frac{4}{3}+\lambda(\nu(Y)/\delta_k-\frac{4}{3}))$, with $\nu(Y)=(\frac{1}{N}\mathrm{tr}\,Y^{2p})^{1/2p}$, p an integer with $w=N^{1/2p}<\frac{5}{4}$, and $\lambda=\frac{w}{5-4w}$, and $\lambda\|\vartheta'_{\mathrm{co}}\|_\infty$ stands for $\|\vartheta'_{\mathrm{co}}\|_\infty$ in every condition and bound; as $\nu\le|\cdot|\le w\nu$, it is smooth and invariant, 0 on $|Y|\le\frac{4}{3}\delta_k$ and 1 on $|Y|\ge\frac{5}{3}\delta_k$ (Lem. 9.14(ii)); and "a constant depending only on L and the pair" stands for "a number of L".

(2) A numeral that counts colours, compares $|\cdot|$ with $\|\cdot\|$, comes from a transpose or comes from the projection P takes its value for G: n, $\frac{3n}{2}$, $3n$ for 3, $\frac{9}{2}$, 9; the mean over n colours replaces the mean over three colours in the quadratic coefficient of a local term; N replaces the 2 of $|T^{\mathrm{T}}|\le 2|T|$ (Lem. 9.3); $\sqrt{N}$ for $\sqrt{2}$; a factor $N^{1/2}$ where an inverse bounded in ℓ^2 is read in $\sup|\cdot|$; $4\varsigma_G$ for the 4 at depth 0 (Lem. 9.3), which is at most $8(1-\frac{1}{N})$ for $SU(N)$; $3|P|$ for the 3 of the bound $3\alpha_0$ of the current at an inner bond that is not full (Prop. YM3_I:30.1), which is at most 6 for $SU(N)$, and, under $\alpha_0\le z_G$, j_G for the 3 of that lemma, which is 3 for $SU(N)$ (Def. 9.1, Remark 9.6, Lem. 9.12(ii)); $6\hat{\varsigma}_G$ for the 12 of the bound $\xi^3|J_W(U)(b)|\le 12a$ of the $SU(2)$ construction on the current J_W of a region W at a bond b, at G^c-valued fields whose plaquette variables at b lie within a of $\mathbf{1}$ and whose bond variables there have condition number $|u||u^{-1}|\le 2$, $\hat{\varsigma}_G=\sup|PH_g|/|g-\mathbf{1}|$ over the plaquette variables $g\ne\mathbf{1}$ of such fields, which is at most 2 at $SU(2)$ and at most $\frac{17}{2}|P|$ for semisimple $\mathfrak{g}$ (Remark 9.6); the factor $|P|$ in the kernel and Weitzenböck bounds of the Hessian in $|\cdot|$ (Lem. 9.19); for the averaging row constant of Lem. 9.4, $C_Q=18NL^3C_{\mathrm{av}}$, which is $54NL^3$ when the normalized kernel bound $C_{\mathrm{av}}=3$ is supplied, and $108L^3$ at $SU(2)$; the constants formed from $6LC$ and $12LC$ move with this value; $z<(\frac{5}{4})^nL^{3n}$ for the bound $z<2L^9$ of Lem. YM3_I:65.3; and an inequality that absorbs a colour factor, such as $\frac{3}{4}\le 1$ in the determinant counting argument of Lem. 9.25, must hold anew with the new numbers.

(3) The following statements of the $SU(2)$ construction are false for G as worded, and their transfers are the forms named, each with the statement of this section that bears it. The inclusion $\mathcal{U}'(r)\subset\mathcal{U}(r)$ of the two regular classes $\mathcal{U}'(r)=\mathfrak{U}'_k(\mathbf{B},r)$ and $\mathcal{U}(r)=\mathfrak{U}_k(\mathbf{B},r)$ of Def. 9.2 (the $SU(2)$ inclusions are Lem. YM3_I:9.4 and equation YM3_I:(465)), the first bounding $D_U^{\mathrm{T}}\partial U$ and the second the current, becomes $\mathcal{U}'(r)\subset\mathcal{U}(\kappa_G r)$, and for $r\le z_G$ the sharper $\mathcal{U}'(r)\subset\mathcal{U}((1+\kappa' r^2)r)$, the form that Lem. 9.12(ii) states for $SU(N)$ (Remark 9.6). The second inclusion of the two classes in the $SU(2)$ construction, $\mathcal{U}(r)\subset\mathcal{U}'(r+r^2)$, becomes $\mathcal{U}(r)\subset\mathcal{U}'(r+r^2+\kappa' r^3)$ for every G and $r\le z_G$, which at $SU(2)$, where $\kappa'=0$ and $z_G=1$, is the inclusion of the $SU(2)$ construction for $r\le 1$: the plaquette variables of a field of $\mathcal{U}(r)$ or of $\mathcal{U}'(r)$ lie within z_G of $\mathbf{1}$, so in D (Lem. 9.9(iii)), where $|(1-P)H_V|\le\frac{1}{4}\kappa'|V-\mathbf{1}|^3$ (Def. 9.1), and the two bounds $\eta|(D^{\mathrm{T}}\partial U)(b)|\le\rho^2+\eta^3|J(b)|+\kappa'\rho^3$ and $\eta^3|J(b)|\le\eta|(D^{\mathrm{T}}\partial U)(b)|+\kappa'\rho^3$ at a bond b whose plaquette variables lie in D within $\rho\le z_G$ of $\mathbf{1}$, which Lem. 9.12(ii) states for every G, give the two inclusions at each full bond, as in the proof of that lemma for $SU(N)$.

The $SU(2)$ null-set exception $G\setminus G^\pi=\{-\mathbf{1}\}$ becomes the indicator of G^π in the integrand (Lem. 9.14(i)); and where the $SU(2)$ construction uses $|\log j|<1$ on the ball of radius $\frac{3}{8}$, the bound $|\log j_G(Y)|\le 0.173\,n|Y|^2$ stands, its constant entering a maximum (Lem. 9.14(i)).

"$|X-\mathbf{1}|<1$" becomes "$X\in D^c$" wherever a series logarithm is to lie in $\mathfrak{g}^c$ (Lem. YM3_I:5.1; Lem. 9.9(iii)). The mean of a family of diameter below $\frac{1}{40}$ (Prop. YM3_I:4.3) takes the proviso that the quotients lie in D^c, which holds if the diameter is below z_G (Lem. 9.9(iv)). The second assertion of Lem. YM3_I:14.2 takes the proviso that every eigen-angle of each W_x is below ρ_G in modulus, or that W_x is a product of exponentials $e^{i\gamma_j}$, $\gamma_j\in\mathfrak{g}^c$, $\sum_j|\gamma_j|<\ln 2$ (Lem. 9.11(ii)). "$u\neq-\mathbf{1}$" becomes "$u\in G^\pi$" (Lems. 9.9(ii), 9.11(ii)). $|X-\mathbf{1}|$ becomes $m(X)$ in conditions on complex fields that are to be kept by $X\mapsto X^c$ and wherever $|X^{-1}-\mathbf{1}|=|X-\mathbf{1}|=|X^*-\mathbf{1}|$ was used; "the conjugate point lies in the domain if and only if the point does" (Lem. 5.23) and "the function z_c of the elimination factor is formed at $(U^*)^{-1}$ if and only if at U" take the two-sided guard $m(X)$; and the symmetric part Q^s of a complex domain is replaced by its real-connected part Q^r, or moved into one, the conditions of Lem. 9.21(iii) on the radii standing then among the conditions on the constants of §9.1.1(K) (Lems. 9.20(ii), 9.21(iii), 9.14(iv)).

The averaging radius ω_M (equation YM3_I:(30)) becomes $\omega^G=\min\{\omega_M,z_G\}$, or the floor $200L^2z_G\geq 1$ is laid once (Lem. 9.11(i)).

The radius $\frac{1}{2}$ that is to keep $\det T$ off $(-\infty,0]$ (the three-dimensional determinant argument in the proof of Prop. YM3_I:50.2) becomes $r_n=\sin\frac{\pi}{2n}$, with the numbers of Lem. 9.14(iv) in the chains in the proof of Prop. YM3_I:50.2 and in the elimination-factor estimates of Lem. 9.14(iii), (iv). The full constant C_{mob} in the local determinant bound takes the factor $\max\{1,n2^{-11}\}$, as shown in Lem. 9.25; its trace-dependent part is evaluated with n colours.

The comb gauge (Lem. YM3_I:14.4) takes the guard $\varpi<z_G$ (Lem. 9.12(iii)), and Lem. 9.10 the guard $\varpi_y(W)<z_G$, under which its contour quotients lie in D^c and their means in G (Lem. 9.9(iii), (iv)). The quantitative cube-gauge input (Lem. 9.12(iii)) takes constants c, B of the pair (Lem. 9.12(iii)). The orthogonal field maps in the covariance argument following Prop. YM3_I:47.4 are required here to keep $|\cdot|$, as $\mathrm{Ad}(G)$ does (Prop. 9.24). The membership in $\mathcal{U}'(2s)$, $s=2\alpha_0+3\alpha_2$, of the curvature/current conversion in Lem. 9.12(ii), with both of its quantitative hypotheses, takes the guard $\alpha_0+3\alpha_2<z_G$, under which the plaquette variables there, within $\alpha_0+3\alpha_2$ of $\mathbf{1}$, lie in D (Lem. 9.9(iii)); the bound of Lem. 9.12(ii) at a full bond b, with $\rho=(\alpha_0+3\alpha_2)\eta^2\mathsf{L}(b)^{-2}$ and the bound $s\mathsf{L}(b)^{-3}$ of the current there, then gives for every G the membership in $\mathcal{U}'(s+s^2+\kappa's^3)$, which lies in $\mathcal{U}'(2s)$ as soon as $s(1+\kappa's)\leq 1$, so at $SU(N)$ for $s\leq\frac{1}{3}$, κ' being at most $2-\pi/\sqrt{3}$ there.

The bound $|\mathrm{Bl}(\mathsf{g},\mathsf{f})|\leq\|\mathsf{g}\|_2\|\mathsf{f}\|_2$ that the $SU(2)$ construction draws by polarisation from $|\mathrm{Qf}(\mathsf{a})|\leq\|\mathsf{a}\|_2^2$, Qf a quadratic form on $\mathfrak{g}^c$-valued functions a on bonds, Bl its polar form, $\|\mathsf{a}\|_2^2=\sum_b|\mathsf{a}(b)|^2$, at real g, f, takes the factor κ_G^2, and keeps the factor 1 where $|X+iY|^2\leq|X|^2+|Y|^2$ for all $X,Y\in\mathfrak{g}$ (Remark 9.6, Lem. 9.9(i)).

For $G\subset SU(N)$ the energy of block averages takes $\frac{1}{4\tau_G^2}$ and the threshold $e_*=e_L/(2\tau_G^2)$, or $N-1$ in the place of τ_G^2, and the lower bounds of the action on large-field regions take the factor τ_G^{-2} (Lem. 9.13).

The assembly of the bond maps (Lem. YM3_I:53.3) takes r_G for 2, with the condition $\mathsf{c}_L c_4\leq 2Lr_G$ (Prop. 9.22(iv)).

The regularity radius ε_{v} of the critical configurations (Def. YM3_I:9.5) takes the ceilings $\varepsilon_{\mathrm{v}}\leq a_G$, with a fixed admitted positive orbit radius a_G from §9.1.6, $\varepsilon_{\mathrm{v}}\leq\varepsilon_G$, ε_G the number of the uniqueness clause of §9.1.1(V), and $\varepsilon_{\mathrm{v}}\leq L^2r_G$ (Prop. 9.26(f)).

The π of the ceiling $\alpha_1<\pi L^j$, $1\leq j\leq k$, under which the local terms are continuous on the analyticity cores (Lem. 5.33), which a proof of the chain from level k to level $k+1$ quotes, becomes ρ_G, the bound on the eigen-angles under which an element of G lies in G^π (Def. 9.1); the ceiling reads $\alpha_1<\rho_GL^j$ and is met by the row $\alpha_1<\rho_GL$ of the table (Prop. 9.24).

These rules define the transferred statement. They do not, by themselves, prove it. When no correction in (3) applies, the convention uses (1) and (2) alone. ■

Remark 9.6 (Where the literal transfer fails). A rule that replaced the signs alike and kept every numeral would not transfer the proofs, as the following instances show. Each names the statement of the $SU(2)$ construction that it concerns, the least N at which the literal transfer fails for $SU(N)$, or the least n, and the statement of this section that proves the form that holds.

Numerals. The 2 of $|T^{\mathrm{T}}| \le 2|T|$ (Lem. 9.3) is N, sharply at $SU(N)$, $\kappa^2 = N-1$ being attained by general kernels (Prop. 9.26(g)). The 4 of Lem. 9.3 at $SU(2)$, without its general-group factor ς_G, fails from $N = 6$: let Ω_0 hold a bare bond b with its four plaquettes, and let the datum be $X_N = \mathrm{diag}(e^{ia}, e^{-ia}, \dots, e^{-ia}) \in SU(N)$, $a = \frac{2\pi}{N-2}$, at b and $\mathbf{1}$ at the other bonds; with $\varpi = |X_N - \mathbf{1}| = 2\sin\frac{\pi}{N-2}$ its plaquette variables are within ϖ of $\mathbf{1}$, while its current at b is $4PH_{X_N}$ up to sign, of norm $4r_N\varpi$, $r_N = (2-\frac{2}{N})\cos\frac{\pi}{N-2} \ge 1.178$ for $N \ge 6$; so for $\varepsilon_1 \in (\varpi, r_N\varpi]$ the datum is ε_1-regular and does not lie in $\mathfrak{U}(\mathbf{B}; 4\varepsilon_1)$. The current at b is a sum of four transported terms $\pm PH_{\partial V(p)}$, so the lemma holds for every G with $4\varsigma_G$ for 4. For $SU(N)$ this is at most $8(1-\frac{1}{N})$: for $V \in SU(N)$ with eigen-angles θ_j the eigenvalues of PH_V are $\sin\theta_j - \frac{1}{N}\sum_l \sin\theta_l$, so that $|PH_V| \le (2-\frac{2}{N})\max_j|\sin\theta_j| \le (2-\frac{2}{N})|V-\mathbf{1}|$ (Prop. 9.26(g)). The value $8(1-\frac{1}{N})$ is that of $SU(N)$. For the spin representation of $Spin(2m+1)$ on $\mathbb{C}^{2^m}$, simple, simply connected and with $z_G = 1$, let $(h_a)_{a\le m}$ be the orthonormal basis of a Cartan subalgebra that acts on a basis of weight vectors by the diagonal matrices of signs; P commutes with the adjoint action of the maximal torus, so it maps the diagonal matrices to the Cartan subalgebra, and at $V = \exp(i\sum_a x_a h_a)$

$$PH_V = \sum_a \sin x_a \prod_{b\ne a} \cos x_b \; h_a, \qquad |PH_V| = \sum_a |\sin x_a| \prod_{b\ne a} |\cos x_b|,$$

every sign vector being a weight. At $x_a = x$, $\tan^2 x = \frac{1}{m-1}$, this is $\sqrt{m}(1-\frac{1}{m})^{(m-1)/2}$, which exceeds 4 from $m = 43$; as $|V-\mathbf{1}| \le 2$, $\varsigma_G > 2$ there. The inequality $\frac{3}{4} \le 1$ by which the proof of Lem. 9.25 absorbs a colour factor reads $\frac{n}{4} \le 1$, false from $n = 5$, and $(\frac{5}{4})^3 < 2$ behind the bound $2L^9$ of Lem. YM3_I:65.3 fails from $n = 4$ (Prop. 9.26(d)). The 12 of the bound $\xi^3|J_W(U)(b)| \le 12a$ on the current of a region at G^c-valued fields whose bond variables have condition number at most 2 (Def. 9.5(2)) fails for every $N \ge 5$: on the thirteen bonds of the four plaquettes at b a field with values in $SL(N,\mathbb{R})$ of condition number at most 2, products of plane rotations and of diagonal matrices $\mathrm{diag}(d_1, d_2, d_3, d_3, d_3, 1, \dots, 1)$ with rational entries, and $\mathbf{1}$ at the other bonds, has its four plaquette variables at b within $a = \frac{103}{100}$ of $\mathbf{1}$ and $\xi^3|J_W(U)(b)| > 12.98 > 12a$. The current at b is ξ^{-3} times a sum over the two directions ν of the plaquettes at b, each term being $\pm PH_g$ at the plaquette variable g of one plaquette of direction ν and $\mp u^{-1}PH_{g'}u$ at that of the other, u a bond variable, with $|u^{-1}Xu| \le |u||u^{-1}||X| \le 2|X|$; so the bound holds for every semisimple $\mathfrak{g}$ with $2(1+2)\hat{\varsigma}_G = 6\hat{\varsigma}_G$ for 12. At $SU(2)$, $g - g^{-1} = 2(g-\mathbf{1}) - \mathrm{tr}(g-\mathbf{1})\mathbf{1}$ gives $|H_g| \le 2|g-\mathbf{1}|$, so that $\hat{\varsigma}_G \le 2$ and $6\hat{\varsigma}_G \le 12$; for semisimple $\mathfrak{g}$, $G^c \subset SL(N,\mathbb{C})$ (Lem. 9.9(iii)), so a bond variable u of condition number at most 2 has $|u| \le 2$ and $|u^{-1}| \le 2$, a plaquette variable g has $|g^{-1}| \le 16$, and $|PH_g| \le \frac{1}{2}|P|(|g-\mathbf{1}| + |g^{-1}||g-\mathbf{1}|) \le \frac{17}{2}|P||g-\mathbf{1}|$.

One matrix and the logarithm. The identity $X + X^{-1} = (\mathrm{tr}\,X)\mathbf{1}$ of YM3_I:(70), which (1) transfers to $X + X^{-1} = \frac{2}{N}(\mathrm{tr}\,X)\mathbf{1}$, holds for every $X \in G$ if and only if $X + X^{-1}$ is scalar for every $X \in G$, that is, if and only if G is conjugate to $\{u \otimes \mathbf{1}_k : u \in SU(2)\}$ (for $n \ge 2$; Lem. 9.9(i)); with it fail at $SU(3)$ the identity $a_p = \frac{1}{2}|\partial U(p) - \mathbf{1}|^2$, the membership $H_X \in \mathfrak{g}^c$, $|Y| = \|Y\|$ on $\mathfrak{g}$ and $|X^{-1} - \mathbf{1}| = |X - \mathbf{1}|$ on G^c (at $X = \mathrm{diag}(\sqrt{2}, \sqrt{2}, \frac{1}{2})$); P, the two norms, $m(X)$ and G^π take their places (Lems. 9.9(i), 9.20(ii)). "$|X-\mathbf{1}| < 1$ gives $\log X \in \mathfrak{g}^c$" (Lem. YM3_I:5.1) fails for $N \ge 7$, at $X = e^{2\pi i/N}\mathbf{1}$; "$X \in D^c$" takes its place (Lem. 9.9(iii)). "A family in G of diameter less than $\frac{1}{40}$ has its mean in G" (Prop. YM3_I:4.3) fails for $N \ge 252$: the mean of $\mathbf{1}$ and $e^{2\pi i/N}\mathbf{1}$ is $e^{\pi i/N}\mathbf{1}$, of determinant -1; the quotients in D^c take its place (Lem. 9.9(iv)).

The averaging radius ω_M of equation YM3_I:(30) fails for $N \geq 212372$ at $L = 13$; ω^G, or the floor $200L^2 z_G \geq 1$, takes its place (Lem. 9.11(i)). "$|\mathbf{1} - T| \leq \frac{1}{2}$ keeps $\det T$ off $(-\infty, 0]$" (the three-dimensional determinant argument in the proof of Prop. YM3_I:50.2) fails for $n \geq 6$; the radius r_n takes its place (Lem. 9.14(iv)).

Fields. The inclusion $\mathcal{U}'(r) \subset \mathcal{U}(r)$ of the two regular classes (Lem. YM3_I:9.4) fails for $N \geq 3$ at every small r (Lem. 9.12(ii)). For every G it holds at the radius $\kappa_G r$: the current is $J(U) = \eta^{-2} PT$ with $T = \frac{1}{2i}(S - S^*)$ and $S = D_U^{\mathrm{T}} \partial U$, the transports commuting with the adjoint, and P is the orthogonal projection onto $\mathfrak{g}$ on hermitian matrices, so that $|PT| \leq \kappa_G \|PT\| \leq \kappa_G |T| \leq \kappa_G |S|$; and for $r \leq z_G$ it holds at the radius $(1 + \kappa' r^2) r$, by the second bound at a bond of Lem. 9.12(ii), the plaquette variables lying in D (Def. 9.5(3)). The bound $3\alpha_0$ of the current at an inner bond that is not full (Prop. YM3_I:30.1) fails for $N \geq 8$ (Lem. 9.12(ii)). It holds for every G with $3|P|\alpha_0$, as $|PH_V| \leq |P||H_V| \leq |P||V - \mathbf{1}|$ at unitary V, which is at most $6\alpha_0$ at $SU(N)$; and under $\alpha_0 \leq z_G$, where the plaquette variables lie in D, the bound of that lemma holds with $j_G \alpha_0$: at a full bond by the inclusion at the radius $(1 + \kappa' \alpha_0^2)\alpha_0$, and at an inner bond b that is not full, where $\mathsf{L}(b)^3$ times the current is a sum of at most three transported terms $\pm PH_{\partial U(p)}$, with $3\varsigma_G^D \alpha_0$. The 3 fails under $\alpha_0 \leq z_G$ at the pair of $SU(2)$ on $\mathbb{C}^4 \oplus (\mathbb{C}^2)^{100}$, spin $\frac{3}{2}$ and a hundred copies of spin $\frac{1}{2}$, which is simply connected, with simple Lie algebra and $z_G = 1$: with J_3 the image of $\frac{1}{2}\sigma_3$, $\operatorname{tr} J_3^2 = 55$, and $w = e^{itJ_3}$ one has $PH_w = c(t) J_3$, $c(t) = \operatorname{tr}(J_3 \sin t J_3)/\operatorname{tr} J_3^2 = \frac{1}{55}(3 \sin \frac{3t}{2} + 101 \sin \frac{t}{2})$, so that $|PH_w| = \frac{3}{2} c(t) = \frac{3}{2} t - 0.1034\, t^3 + O(t^5)$ exceeds $|w - \mathbf{1}| = 2 \sin \frac{3t}{4} = \frac{3}{2} t - 0.1406\, t^3 + O(t^5)$ for small $t > 0$; with Ω_0 not the whole torus, the field U_0 equal to w at an inner bond b in the outermost layer of Ω_0 along a face, three of whose four plaquettes are inner, and to $\mathbf{1}$ at the other bonds lies in $\mathfrak{U}'_k(\mathbf{B}, \alpha_0)$ for $\alpha_0 > |w - \mathbf{1}|$, and $\mathsf{L}(b)^3 |J_{\Omega_0}(U_0)(b)| = 3|PH_w|$, above $3\alpha_0$ for $\alpha_0 < |PH_w|$.

The comb gauge of a box of Lem. YM3_I:14.4 needs $\varpi < z_G$ for $N \geq 7$, and the form of that clause with some gauge of the box in the place of the comb, that $2s\varpi < 2$ alone gives a gauge in which $W^h = e^{ia^0}$ with a^0 in $\mathfrak{g}$ and $\sup |a^0| \leq \pi s \varpi$, fails for $N \geq 14$; the gauge on a cube of Thm. YM3_I:F has the quantitative general-group counterpart assumed in Lem. 9.12(iii). That $G \setminus G^\pi$ is a null set, as it is for $SU(2)$, fails for $N \geq 3$; the indicator of G^π takes its place (Lem. 9.14(i)). The cut-off $\vartheta_{\mathrm{co}}(|Y|/\delta_k)$ of the cutoff preceding Lem. YM3_I:53.3 is not differentiable for rank at least 2; the cut-off $\vartheta_{\mathrm{co}}(\frac{4}{3} + \lambda(\nu(Y)/\delta_k - \frac{4}{3}))$ of Def. 9.5(1) takes its place, with $\lambda \|\vartheta'_{\mathrm{co}}\|_\infty$ for $\|\vartheta'_{\mathrm{co}}\|_\infty$; at $SU(3)$, $p = 3$ and $\lambda = 6.1194$ (Lem. 9.14(ii)). The orthogonal colour maps in the covariance argument following Prop. YM3_I:47.4 keep the ball of $|\cdot|$ for $SU(2)$, and general rotations of $\mathfrak{g}$ do not keep it for rank at least 2; the rotations are to keep $|\cdot|$, as $\mathrm{Ad}(G)$ does, and covariance under $\mathrm{Ad}(G)$ is what the proofs of the induction use (Prop. 9.24).

The bound $|\mathrm{Bl}(\mathsf{g}, \mathsf{f})| \leq \|\mathsf{g}\|_2 \|\mathsf{f}\|_2$ that the $SU(2)$ construction draws by polarisation from $|\mathrm{Qf}(\mathsf{a})| \leq \|\mathsf{a}\|_2^2$ (Def. 9.5(3)) fails at $SU(3)$, of rank 2: on two bonds b_1, b_2 let x, y, z be the diagonal entries of $\mathsf{a}(b_1)$, $\zeta_\pm = \frac{1}{2}(\mathsf{a}(b_2)_{11} - \mathsf{a}(b_2)_{22} \pm i\mathsf{a}(b_2)_{12} \pm i\mathsf{a}(b_2)_{21})$, $\psi_1 = \frac{1}{2}(e^{-i\pi/4}\zeta_+ + e^{i\pi/4}\zeta_-)$, $\psi_2 = \frac{1}{2i}(e^{i\pi/4}\zeta_- - e^{-i\pi/4}\zeta_+)$ and $\mathrm{Qf}(\mathsf{a}) = \frac{1}{2}(x - z)\psi_1 + \frac{3}{4}\psi_1^2 + \frac{3\sqrt{6}}{4} y \psi_2 - \frac{1}{8}\psi_2^2$; then $|\mathrm{Qf}(\mathsf{a})| \leq |\mathsf{a}(b_1)|^2 + |\mathsf{a}(b_2)|^2$ for all $\mathfrak{sl}(3, \mathbb{C})$-valued a, while the real $\mathsf{g} = (\frac{3}{5}\mathrm{diag}(1, -1, 0), \frac{4}{5}\sigma_3 \oplus 0)$ and $\mathsf{f} = (\frac{3}{5}\mathrm{diag}(0, 1, -1), \frac{4}{5}\sigma_1 \oplus 0)$ at (b_1, b_2), σ_1 and σ_3 the Pauli matrices, have $\|\mathsf{g}\|_2 = \|\mathsf{f}\|_2 = 1$ and $\mathrm{Bl}(\mathsf{g}, \mathsf{f}) = \frac{1}{25}(7 + 3\sqrt{2} + 9\sqrt{3}) > 1.07$. For every G the bound holds with κ_G^2: as $\|\cdot\| \leq |\cdot| \leq \kappa_G \|\cdot\|$ on $\mathfrak{g}$ (Lem. 9.9(i)) and $\|\cdot\|$ obeys the parallelogram law, $4|\mathrm{Bl}(\mathsf{g}, \mathsf{f})| \leq \|\mathsf{g} + \mathsf{f}\|_2^2 + \|\mathsf{g} - \mathsf{f}\|_2^2 \leq 2\kappa_G^2(\|\mathsf{g}\|_2^2 + \|\mathsf{f}\|_2^2)$, and $\mathsf{g} \mapsto t\mathsf{g}$, $\mathsf{f} \mapsto t^{-1}\mathsf{f}$ gives $|\mathrm{Bl}(\mathsf{g}, \mathsf{f})| \leq \kappa_G^2 \|\mathsf{g}\|_2 \|\mathsf{f}\|_2$, which is the bound of the $SU(2)$ construction at $SU(2)$; and where $|X + iY|^2 \leq |X|^2 + |Y|^2$ for all $X, Y \in \mathfrak{g}$ it holds with 1, by $4i\,\mathrm{Bl}(\mathsf{g}, \mathsf{f}) = \mathrm{Qf}(\mathsf{g} + i\mathsf{f}) - \mathrm{Qf}(\mathsf{g} - i\mathsf{f})$ and $|X - iY| = |X + iY|$ for hermitian X, Y.

Complex fields. The bare trace of the gauge-fixing term (the definition of Ψ^{G} in Section YM3_I:51, p. 393) is not real at real fields for $N \geq 3$; $a(X)$ takes its place (Prop. 9.23). "The conjugate point lies in the domain if and only if the point does", which for $SU(2)$ rests on the isometry Θ of Lem. 5.23, fails for $N \geq 3$; the two-sided guard $m(X)$ takes its place (Lem. 9.20(ii), (iv)).

For $SU(N)$ with $N \le 6$ the forms with D^c coincide with the statements of the $SU(2)$ construction, and the proviso on means of Lem. 9.9(iv) is void for $N \le 251$.

What remains in the general-group construction. Let G be simply connected with $\mathfrak{g}$ simple, $L \ge 13$ odd and $200L^2 z_G \ge 1$; constants are constants depending only on L and the pair, uniform in the volume. The following discussion identifies the analytic work still needed to extend the complete multiscale construction. The local matrix identities do not establish these estimates.

Propagators and covariances (P).. One must establish the uniform linear estimates at regular backgrounds required below. Their $SU(2)$ forms are Thms. YM3_I:G, YM3_I:I and YM3_I:L, and Prop. YM3_I:31.1: the constrained inverse and its local expansion, the real regional inverse, the fluctuation covariance, and the current multiplier. For G these are assumptions on the natural field classes and on every regional geometry used by the construction; the cited reduced-row theorems alone do not assert their extension to other determining sets. The quantitative local gauges in Lem. 9.12(iii), and the local current, difference, commutator, action-remainder and layer estimates used to construct these inverses, are also required wherever used. The required inverse bounds include: for a separated sequence of regions with M above a floor and fields of the natural class with small parameters, the operator K of the random-walk expansion is invertible, $\mathsf{K}^{-1} = \sum_w E_w$ over finite sequences w of pieces, local and holomorphic, with

$$|E_w \iota g|_y \le 8 B_4 \, q_M^{|w|} \, \sup e^{-\delta_2 \sum_i d(y_i, y_{i+1})} |g|, \qquad q_M = C_q M^{-1} e^{-\delta_2 M/(8L)},$$

the rate δ_2 being that of the $SU(2)$ construction and the floor on M a constant depending only on L and the pair; and for G-valued U in the class $\mathfrak{U}'_k(\mathbf{B}, \alpha)$ of Def. 9.2, α small, $\Delta_U + Q'^{*} \mathsf{a} Q'$ is positive definite, with an inverse G' that obeys $|G' f|_y \le B_{\mathrm{ps}} \xi_y^2 e^{-\delta_{\mathrm{ps}} d(y,y')} \sup |f|$, and the like for its covariant derivative, for $(Q' G'^2 Q'^{*})^{-1}$ and for the gauge-fixed operator.

Critical configurations off the trivial field (V).. One must construct a regular critical orbit over every admitted regular datum, prove uniqueness of that orbit at one fixed positive regularity radius and improvement to a constant times the datum radius, and establish holomorphic response with exponential decay uniformly in depth and volume. The $SU(2)$ reduced-data models are Thm. YM3_I:E, Prop. YM3_I:37.1, Thms. YM3_I:K and YM3_I:M, and Lem. YM3_I:51.6. The precise existence requirement is stated in §9.1.8. Restrictions, joined data and localised data must remain in the admitted classes; their critical backgrounds must satisfy the comparison in Prop. 9.22(iii). These estimates remain to be established for G, including the uniform near-trivial background, tower and analyticity-core bounds of Prop. 9.22(ii), with the response constant $C_\partial = 12\kappa_6 + 4L$, on the objects of a step at the fields over a regular base (the global and the localised background of the step and the minimal orbits of its three data), and on the towers off the trivial field. Among them, the uniqueness of the regular critical orbit near the trivial field and its normalized representatives as stated in Prop. 9.17, for the full regional geometry of §9.1.6 and every reversal-invariant set $\mathfrak{F}$ of removed inward bonds. The radii and bounds there are uniform in depth, sub-domain, cutoff and volume.

Conjugation and domain preservation (C).. With $a(X)$ for every trace of a group-valued argument at complex fields, the gauge-fixing term included, and with the domains written with $m(X)$ and with D^c at the logarithms of the averages, every map between sets of complex fields that the induction uses commutes with $p \mapsto p^c$ on the real-connected parts of its domains (Lem. 9.21) and maps them into the real-connected parts of its targets. The covariance assertion follows from Prop. 9.23(i) for these maps wherever they are formed; the source and target domain memberships required to apply it remain among (V), (II), and (K).

The nonlinear step (II).. One must verify that the transfers to G of the statements of the second part of the $SU(2)$ construction that the induction cites hold, in the forms of Prop. 9.23(ii), every proposition that the proof of Thm. YM3_I:D quotes among them, and the objects of the definitions that these statements read, formed for G, have the properties that the $SU(2)$ construction proves

for them; so do the statements of the second part that use the averaging map off the small fields, the pivot bond or the measure of $G \setminus G^{\pi}$, the uniform box-gauge bound of Lem. 9.12(iv), and the large-field chain of Lem. 9.13(ii); and at every citation in the proofs named in Prop. 9.24 the assumptions of the cited statement, in its transferred form, hold at the data of the citation. Among them: the pivot bond (Prop. YM3_I:14.11) holds for G in all its clauses with the constants of equation YM3_I:(128), no numeral changed; and the localised Gaussian family (Prop. YM3_I:50.2) is a pointed family of its level on the analyticity cores of G, holomorphic on balls of one radius, with the contraction bound of its expansion.

Compatible constants (K).. One must show that every condition on the constants read for G, the places where z_G enters included, passes its test in Prop. 9.26(b): it holds on a large set, and at each two-sided constant the set of the earlier choices at which its window meets all its ends together is large. The conditions read for G are every condition of the $SU(2)$ construction transferred to G, every condition of Prop. 9.26(c)–(g), the conditions $\varepsilon_{\mathrm{v}} \le \varepsilon_G$ and $c_{11} \le r_G/(2C_{\mathrm{pg}})$ of the uniqueness clause of (V), c_{11} the radius of §9.1.6 and C_{pg} the admitted preliminary-gauge constant of §9.1.6, the ceiling $\mathsf{c}_L c_4 \le 2Lr_G$ of Prop. 9.22(iv), the row $\alpha_1 < \rho_G L$ of Prop. 9.24, and, at the memberships reached from memberships of restrictions had apart, the conditions of Lem. 9.21(iii): the moved first radius $b_0' = b_0 + 2e^{4\xi b_1} b_1^2$, for the regular pairs $\alpha_0 + 2\vartheta^2 e^{4\vartheta\alpha_1}\alpha_1^2$ with $\vartheta \ge 1$ the largest ratio of the spacings around a plaquette, at most the first radius B_0 at which the statement read there is had, and for the analyticity cores besides $b_1 \le B_1$, $K(b_0' + b_1^2) \le B_0$ and $b_0' + b_1 \le a^\sharp$, with $K \ge 16L^2$ and $a^\sharp$ the constant and the radius of that lemma, which are positive ceilings on the second radius where $b_0 < B_0$, $Kb_0 < B_0$ and $b_0 < a^\sharp$. Each number of G that stands in such a condition has a place before γ in the order of the table and is a function of the constants before that place and of the pair, which reads no radius.

The near-trivial uniqueness clause of (V) is reduced in Prop. 9.17 to the transferred uniform slice and contraction estimates; their uniform nonlinear bounds remain to be established there. The orbit and energy hypothesis of §9.1.6, with a fixed admitted positive radius a_G, is also required wherever its ceiling is used in (K). For quotients, the sector statements are those of Prop. 9.33.

Source and continuum transfer. For the sourced induction in (b), one must transfer the local response and extraction estimates, their holomorphic bounds for complex weights, and the source-energy and normalized partition-function bounds uniformly in the cutoff and volume. The admissibility conditions of the unsourced induction also apply to the sourced induction. For the continuum conclusions in (d) and (e), one must also transfer the comparison of states, the cutoff and volume limit estimates, and the short-distance bounds under the hypotheses used at each passage. These inputs are part of the construction programme in Sketch AA.

9.1.2. *What changes with the gauge group.* The representation of the group fixes the normalization of the action. For $G \subset U(N)$ we use Hermitian Lie-algebra elements with scalar product $\langle X, Y\rangle = N^{-1}\operatorname{tr}(XY)$ and action

$$a(U) = 1 - \frac{1}{2N}\operatorname{tr}(U + U^{-1}).$$

On G this is the real Wilson action. Its displayed form also specifies its holomorphic continuation. The pair consisting of G and its representation is therefore part of the data. A numerical Casimir computed with another scalar product cannot be inserted without changing the coupling normalization.

Let $n = \dim \mathfrak{g}$ and choose an orthonormal basis (t_a). At the trivial background the differential operators act independently on the n colour coordinates: a scalar lattice operator K becomes $K \otimes 1_{\mathfrak{g}}$. This explains the factor n in the leading action-density coefficient. The lattice projection, the Gaussian covariance and their decay estimates at this reference are the same spatial operators

as before. The two-point Gaussian contraction therefore gives

$$c_2(G) = \frac{3n}{16\pi^2}.$$

This calculation identifies a local coefficient; it does not construct the interacting measure at a nontrivial background.

For simple $\mathfrak{g}$, invariance reduces the next colour contractions to two scalars. Put $\Omega = \sum_a t_a^2$ and, for $\|H\| = 1$, define

$$C_A = \operatorname{tr}_{\mathfrak{g}^c}(\operatorname{ad} H)^2, \qquad C_\rho = N^{-1}\operatorname{tr}(\Omega H^2).$$

These numbers are independent of H in the simple case. The source's one-loop coefficient identity is

$$b[\Psi_k^G] = \frac{C_A}{8} b[\Psi_k] + \left(\frac{C_\rho}{3} - \frac{C_A}{8}\right)(L^{-k} - L^{-4k}).$$

Here $b[\Psi_k]$ is the quadratic colour coefficient in the SU(2) calculation, and $b[\Psi_k^G]$ is its counterpart for the pair. The second term is explicit in k. Thus the convergence and bounds of the original coefficient sequence transfer through an algebraic identity, rather than through a new interacting construction. The detailed contraction below also gives

$$\varphi_G = \sum_a \operatorname{tr}(\operatorname{ad} t_a)^2 = nC_A, \qquad c_1(G) = \frac{\varphi_G}{3\pi n}.$$

At complex fields the matrix changes are consequential. For SU(2), special conjugation identities simplify the trace and the domains of logarithms. They fail in higher matrix dimension. The holomorphic action above replaces a bare trace where the latter ceases to be real on real fields, and the control of

$$m(X) = \max\{|X - 1|, |X^{-1} - 1|\}$$

keeps the domain compatible with $X \mapsto (X^*)^{-1}$. The projection onto $\mathfrak{g}^c$, the chart radius, and the distance to nontrivial central elements also enter the estimates. These changes are part of defining the maps on the correct domains, before one can attempt the induction.

The general-group sketch must then address regular backgrounds. The first task is to invert the covariant operators with uniform local decay. The second is to construct a critical orbit over each admitted datum, control its response, and compare restrictions and localized backgrounds. The nonlinear fluctuation integral must stay in those domains at every step, and its Gaussian family must satisfy the required contraction bounds. Finally, the constants must meet their upper and lower constraints simultaneously. The colour calculation at the identity gives the local coefficients. The detailed reductions below specify the separate slice, contraction and orbit estimates needed for the nonlinear construction.

There is an exact special case that avoids this transfer problem. For separately acting SU(2) factors with their stated positive weights, the action splits and the Wilson measure factors. A cumulant of the sum of the factor observables is the sum of their cumulants with the corresponding weights. This is a product construction using already supplied factors. A quotient changes the global topology and sectors, while a representation tying several factors together changes the analytic operators. Those cases need the separate local and global analysis retained below; they do not follow merely by adding colour dimensions.

9.1.3. *Construction and local coefficients.*

Detailed sketch AA (Construction programme and local coefficients). Let $G \subset U(N)$ be compact and connected with $\mathfrak{g}$ simple, and let $L \geq 13$ be odd with $200L^2 z_G \geq 1$ (for $SU(N)$ at $L = 13$ this is $N \leq 212371$). Every constant below is a positive number of L and the pair, uniform in the volume. The construction clauses in (a), the second half of (b), the existence assertion of (d), and the second half of (e) describe the proposed extension. Their regular-background and nonlinear multiscale estimates remain to be established as described in §9.1.1. The fixed-lattice identities,

one-loop coefficients, finite-box expansion and assertions about an already convergent sequence have separate supplied arguments.

(a) For simply connected G, the construction aims to establish that Thm. YM3_I:D holds for G at every choice of the constants of the $SU(2)$ construction that meets the conditions on the constants read for G (§9.1.1(K)): its conditions transferred to G and the conditions of Prop. 9.26, among them $\varepsilon_{\mathrm{v}} \le a_G$, $\varepsilon_{\mathrm{v}} \le \varepsilon_G$ and $\varepsilon_{\mathrm{v}} \le L^2 r_G$ on the regularity radius ε_{v} of the critical configurations (Def. YM3_I:9.5), a_G a fixed positive radius admitted by the hypothesis of §9.1.6 and ε_G the number of the uniqueness clause of §9.1.1(V), and the ceiling $\mathsf{c}_L c_4 \le 2Lr_G$ of Prop. 9.22(iv); such choices exist, in the order of the $SU(2)$ construction with γ last (§9.1.1(K) and Prop. 9.26(b)). The run is the one sequence $x_0 = \beta$, $x_k = L^{-1}x_{k-1} - b_k$, b_k the mean of the n quadratic colour coefficients of the local terms of second order of step k: it is real, does not depend on the torus, and $|b_k| \le \mathsf{G}_\sharp \le B_\sharp$.

(b) Let $A(v,V) = \sum_p v(p)a_p(V)$ for a complex weight v of the plaquettes, and $\mathcal{O}(w) = a_K^{-1}A(w,\cdot)$ for a source w. For every compact $G \neq \{\mathbf{1}\}$ in $U(N)$: the chessboard norm of a source is $\|e^{\mathcal{O}(w)}\|_{\mathrm{ch}}^{\mathsf{N}} = Z_\beta[w^{\mathrm{ch}}]/Z_\beta \ge 0$ (equation YM3_I:(1759), transferred to G); a source w with $\operatorname{Re} w \le (1-c_{\mathrm{pr}})a_K\beta$, $c_{\mathrm{pr}} \in (0,1]$, leaves every plaquette at least the fraction c_{pr} of its price, $|e^{-A(\beta - a_K^{-1}w, V)}| \le e^{-c_{\mathrm{pr}}\beta\sum_p a_p(V)}$, as $a_p \ge 0$ on G; and on a torus with $|\Lambda|$ sites the Wilson measure with one coupling $s \ge 1$ at all plaquettes, which at $s = c_{\mathrm{pr}}\beta$ dominates the weight with the source by the price, has

$$\langle a_p\rangle_s \le \frac{C}{s} + \frac{n\log s}{|\Lambda|\, s}$$

at every plaquette, C a number of the pair, while for plaquette couplings $v(p)$ with $s \le v(p) \le C's$, $C' \ge 1$, the mean $\frac{1}{3|\Lambda|}\sum_p \langle a_p\rangle_v$ over the plaquettes of the torus is at most $(C + n\log C')/s + n\log s/(|\Lambda|s)$, and at couplings invariant under the lattice symmetries every plaquette has this mean; for $c_{\mathrm{pr}}\beta \ge 1$ the couplings $\beta - a_K^{-1}\operatorname{Re} w(p)$ of a source with $0 \le \operatorname{Re} w \le (1-c_{\mathrm{pr}})a_K\beta$ are of this kind, with $s = c_{\mathrm{pr}}\beta$ and $C' = 1/c_{\mathrm{pr}}$. On the tori of the continuum theorem at its first-passage data, which have at least $2L^K$ sites a side and couplings at most $L^{K+1}(g^{-2} + \bar{B})$, $n|\Lambda|^{-1}\log s$ is below a number of g, the pair and the constants. For simply connected G, the intended sourced-density and stability conclusions are those of Thms. YM3_I:AJ and YM3_I:AD with the run of (a): the effective densities with a source have at every level the structure that the $SU(2)$ construction proves, and the stability holds through the whole induction with complex plaquette weights, uniformly in the volume; writing $\widetilde{Z}(w) = e^{-E_{(K)} - \mathcal{E}_K(w)} Z_{K,\ell,\beta}[w]$, with $\mathcal{E}_K$ defined by [7, eq. 212], gives

$$\int e^{-A(\beta - a_K^{-1}w, V)}\, \mathrm{d}V = \exp\Big(E_{(K)} + \sum_{j=1}^{K}\sum_{X\in\mathcal{D}_j} e_j(X;w)\Big)\, \widetilde{Z}(w),$$

$E_{(K)}$ the vacuum constant at the source 0, $e_j(X;w)$ the source energy of level j on the domain X of level j [7, eq. 212] and $\widetilde{Z}(w)$ the normalised partition function with the source, with the bounds of Thm. YM3_I:AD on them. With it the connected functions of the action density obey the bound per ball of Prop. YM3_I:89.4: for every reference coupling $0 < g \le \gamma$, every order $\mathsf{m} \ge 2$, every $R > 0$ and every $\delta > 0$, at every first-passage datum (K,β) at g, on every side ℓ admitted at g with $\ell \ge 16\mathsf{m}(R+1)$ and for real Lipschitz functions $f_1, \dots, f_{\mathsf{m}}$ on the torus, each supported in a ball of radius R, the supports pairwise at a distance at least δ, $|\langle \mathcal{O}(f_1); \dots; \mathcal{O}(f_{\mathsf{m}})\rangle|$ is at most the constant $C_{\mathsf{m}}^{\infty}(\delta;R)$ of that proposition times the product of the norms of the f_i in that proposition, $C_{\mathsf{m}}^{\infty}(\delta;R)$ being formed from g, m, δ, R and the constants of Thm. YM3_I:AD for G alone, free of K, β and ℓ: the connected functions are derivatives at 0 of the logarithm of the normalised partition function with

sources, and Cauchy's inequality on the polydiscs of sources on which the stability holds gives the bound, as in Prop. YM3_I:89.4.

(c) For $k \geq 1$ and on every torus, the one-loop functional Ψ_k^G of Thm. YM3_I:O for G is tempered (Section YM3_I:43.1), with the coefficient

$$b[\Psi_k^G] = \frac{C_A}{8}\, b[\Psi_k] + \Big(\frac{C_\rho}{3} - \frac{C_A}{8}\Big)(L^{-k} - L^{-4k}),$$

Ψ_k that of $SU(2)$; Thms. YM3_I:O and AC hold for G with constants of L, C_A and C_ρ; and near the trivial block field the quadratic coefficient of the logarithm of the Gaussian normaliser of the step to level l is the one-loop increment (Thm. YM3_I:O)

$$b_l^0(G) = \frac{C_A}{8}\, b_l^0 + \Big(\frac{C_\rho}{3} - \frac{C_A}{8}\Big)(L^3 - 1)L^{-4l}.$$

(d) Let the run have real increments that depend on no torus and are bounded by $B_\sharp$, as in (a). Let $0 < g \leq \gamma$, $X_g = L(g^{-2} + B_\sharp)$, $K(\beta, g)$ the least k with $x_k \leq X_g$, and let $K_j \to \infty$, let $l_j \to \infty$ be sides admitted at g, and $K(\beta_j, g) = K_j$. For $\mathsf{m} \geq 2$ and real C^1 functions f_i on $\mathbb{R}^3$ of compact pairwise disjoint supports let $S_{\mathsf{m}}^{(j)}(f_1, \dots, f_{\mathsf{m}}) = \langle \mathcal{O}(f_1); \dots; \mathcal{O}(f_{\mathsf{m}})\rangle$ on the torus of side l_j at (K_j, β_j), $\mathcal{O}(f) = a_{K_j}^{-1} \sum_p f(x(p)) a_p$, f periodised. Then along a subsequence $a_{K_j}\beta_j$ converges in $[g^{-2} + B_\sharp - \bar{B}/L,\ L(g^{-2} + \bar{B})]$. For simply connected G, after completion of the construction tasks, with the run of (a), the $S_{\mathsf{m}}^{(j)}$ obey the bound per ball of (b) uniformly in j, which holds at every first-passage datum and on every side admitted at g and at least $16\mathsf{m}(R+1)$ with one constant for given m, R and δ, each of the compact pairwise disjoint supports of the f_i lying in a ball of some radius R and the supports lying at mutual distances at least some $\delta > 0$; so along a further subsequence every $S_{\mathsf{m}}^{(j)}$ converges. For simply connected G, the following are the transferred conclusions sought from the uniform construction in §9.1.1, with the same parameter restrictions. On a fixed torus the fixed-label and moving-label cutoff limits have the properties of Thm. YM3_I:A. Growing-torus and periodic-Gibbs sequences have the subsequential compactness, multilinear bounds and inherited coordinate reflection positivity of Props. YM3_I:92.1, YM3_I:92.3 and YM3_I:92.4. For periodic-Gibbs limits the canonical component and reconstruction conclusions are those of Thms. YM3_I:AL and YM3_I:AP: each component is a conditioned periodic-Gibbs limit, is Euclidean invariant and reflection positive, clusters, and has a unique vacuum; the undecomposed limit has the same reconstruction, with zero-energy space $L^2(\mathsf{X}, \lambda)$. For a general growing-torus limit, full rotation invariance retains the Ward criterion in (g). The short-distance conclusions retain the restrictions in (e) and use $c_2(G)$ in place of c_2. For every compact connected G and along every sequence of tori on which all $S_{\mathsf{m}}^{(j)}$ converge, each limit S_{m} is m-linear, real and symmetric, obeys $|S_{\mathsf{m}}(f_1, \dots, f_{\mathsf{m}})| \leq C(A_1, \dots, A_{\mathsf{m}}) \prod_i \|f_i\|_{C^1}$ for supports in fixed disjoint compact sets A_i, with C depending on the sequence, is invariant under the translations of $\mathbb{R}^3$ and the 48 signed permutations of the axes, and is reflection positive at every coordinate mirror. On one torus of side $l = 2L^m$ with $K_j \to \infty$ the same holds with the translations of the torus, the signed permutations and the pairs of mirrors $\{y_\mu = s\} \cup \{y_\mu = s + \frac{l}{2}\}$.

(e) The Gaussian measure defined by the quadratic action at $\mathbf{1}$ is the product of n identical one-colour measures. Its leading two-point coefficient is

$$c_2(G) = \frac{3n}{16\pi^2}.$$

If G is simply connected and satisfies all the construction hypotheses, the finite-cutoff estimate of Thm. YM3_I:AV and its continuum version, Thm. YM3_I:AW, transfer to its

tori and their limits:

$$\langle \mathcal{O}(f); \mathcal{O}(h) \rangle = c_2(G) \mathsf{b}^{-2} \mathcal{K}_6(f,h)(1+\varepsilon'), \qquad |\varepsilon'| \le \bar{\varepsilon}_g(\mathsf{d}),$$

At finite cutoff this means $0 < g \le \gamma_W$, $(K,\beta) \in \mathfrak{d}(g)$, $0 < \vartheta < 1$, $0 < \mathsf{d} \le d_0(g,\vartheta)$, $K > s(\mathsf{d})$, $\ell \in \mathfrak{L}(g)$, and a test pair $(f,h) \in \mathfrak{T}_{T_\ell}(\mathsf{d};\vartheta)$; then $\mathsf{b} = a_K \beta$ and the tolerance is the transferred $\bar{\varepsilon}_{g,\vartheta}(\mathsf{d}) \le 1/3$ of Thm. YM3_I:AV. In a continuum limit the pairs are C^1, the cutoffs tend to infinity, and b is the limiting label of Thm. YM3_I:AW. For block sums of $SU(2)$, the corresponding statement is Thm. AB(iii).

(f) For every compact connected G, the fixed-box expansion is that of Lem. 9.28. Its coefficient of β^{-3} has an infinite-box limit; after summing over orientations, its leading part at separation r is $\varphi_G \mathsf{r}^{-5}/(16\pi^3)$. Relative to the coefficient in (e), its ratio is

$$c_1(G) = \frac{\varphi_G}{3\pi n}.$$

The expansion in β^{-1} is taken at fixed box. A two-term torus law for G would also require a uniform comparison with the interacting coefficient at varying cutoff.

(g) At every cutoff, on every torus and for every $\beta > 0$, the finite-cutoff rotational Ward identity used in §5.1 holds for G with $\frac{1}{2}k_p$ in the place of $\frac{3}{2}a_p$; and $k_p = n a_p$ if and only if $\Omega = n\mathbf{1}$, as for an irreducible representation or a sum of copies of one. Let (K_j, l_j, β_j) be a sequence of tori with $l_j \to \infty$ along which all connected functions converge, S_{m} their limits as in (d), and let $(\omega, \mathsf{m}, \mathcal{R}, f)$ be an admitted quadruple (Section YM3_I:112, before equation YM3_I:(2115)): $\omega \in \mathfrak{so}(3)$, $\mathsf{m} \ge 2$, $\mathcal{R} > 0$, and $f = (f_1, \dots, f_{\mathsf{m}})$ smooth functions with compact pairwise disjoint supports in $\{|y| < \frac{1}{2}\mathcal{R}\}$. For the j with $2\mathcal{R} + 2a_{K_j} < \frac{1}{2}l_j$ the bulk term and the flux term of the identity at the cut-off radius $\mathcal{R}$ are defined, and their sum has a limit, which equals

$$\sum_i S_{\mathsf{m}}(f_1, \dots, L_\omega f_i, \dots, f_{\mathsf{m}}) = -\frac{d}{dt} S_{\mathsf{m}}(e^{t\omega} f)\Big|_{t=0},$$

L_ω the rotation field of ω, $t \mapsto S_{\mathsf{m}}(e^{t\omega} f)$ being differentiable, and which does not depend on $\mathcal{R}$. The limit of the connected functions is invariant under $O(3)$ at all C^1 tuples of (d) if and only if this limit of the sum vanishes for one $\omega_0 \neq 0$ and every admitted quadruple $(\omega_0, \mathsf{m}, \mathcal{R}, f)$, and if and only if it vanishes at every admitted quadruple: the conjugates of ω_0 under the signed permutations span $\mathfrak{so}(3)$, and the invariance passes from smooth tuples to C^1 tuples by mollifying, with the bound of (d). It is the limit of the flux term alone wherever the bulk term tends to 0. If the limit is invariant, the rotations and reflections about the third axis act on the Hilbert space of the reflection positive form of the limit by a strongly continuous unitary representation U of $O(2)$ that fixes the vacuum vector, and $\mathsf{U}(R)[\tau_{te_3} F] = [\tau_{te_3}((R \oplus 1)F)]$ for $t \ge 0$ on the classes of the functions F supported in $\{y_3 > 0\}$, τ_{te_3} being the translation by te_3.

(h) Let instead $\mathfrak{g} = \bigoplus_{i \le r} \mathfrak{g}_i$ be semisimple, $r \ge 2$.

(h1) The mean of the n quadratic colour coefficients of a local term of second order is $\mathsf{cf} = \sum_i \frac{n_i}{n} \mathsf{cf}_i$, where cf_i is its coefficient on the i-th ideal; moreover, $|\mathsf{cf}_i| \le \kappa_i^2 C_{\mathrm{am}} r_0^{-2} \|E\| = (\kappa_i/\kappa_G)^2 \mathsf{G}_\sharp \|E\|/E_0$ for the terms E of Thm. YM3_I:D, C_{am} the constant of the coefficient bound YM3_I:(940), $r_0 = a^{\mathrm{us}}(\alpha_0, \alpha_1)$ the radius of the $SU(2)$ construction and $\mathsf{G}_\sharp = \kappa_G^2 C_{\mathrm{am}} r_0^{-2} E_0$ (Prop. 9.24), C_{am} and r_0 reading constants of the $SU(2)$ construction that come before them in its order. If $|e_k^{(i)}| \le \Gamma$, the sequences $x_0^{(i)} = \beta$, $x_k^{(i)} = L^{-1} x_{k-1}^{(i)} - e_k^{(i)}$ and their mean $x_k = \sum_i \frac{n_i}{n} x_k^{(i)}$ satisfy $x_k = L^{-1} x_{k-1} - \sum_i \frac{n_i}{n} e_k^{(i)}$ and $|x_k^{(i)} - x_k| \le 2(1 - \frac{n_i}{n}) \Gamma \frac{L}{L-1}(1 - L^{-k})$.

(h2) The one-loop theorem holds in block form at every semisimple pair, the ideals being tied by the representation or not: on $\mathfrak{g}_i$, $b^{(i)}[\Psi_k^G] = \frac{C_A^{(i)}}{8} b[\Psi_k] + (\theta_i - \frac{3}{8}C_A^{(i)})t_k$, $t_k = \frac{1}{3}(L^{-k} - L^{-4k})$, $b^{(i)}[\Psi_k^G]$ being the coefficient on $\mathfrak{g}_i$ of the one-loop functional Ψ_k^G of the pair and Ψ_k that of $SU(2)$, with one set of sequences for all ideals; here $\theta_i = \frac{1}{N}\mathrm{tr}(\Omega H^2)$ for a unit $H \in \mathfrak{g}_i$, Ω being summed over a basis of all of $\mathfrak{g}$, so that $\theta_i = \sum_j n_j \tau_{ij}$ with numbers τ_{ij} of the pair (Prop. 9.18(iv)). Equivalently, $b^{(i)}[\Psi_k^G]$ is the coefficient of the simple pair of $\mathfrak{g}_i$ plus $(\sum_{j\neq i} n_j\tau_{ij})$ times the coefficient t_k of one scalar form Υ_k of the lattice, which is tempered. For a unit $H \in \mathfrak{g}_i$ the numbers $\mathrm{tr}(\mathrm{ad}\, H)^2$ and $\frac{1}{N}\mathrm{tr}(\Omega H^2)$ that Def. 9.1 forms for simple $\mathfrak{g}$ are $C_A^{(i)}$ and θ_i; they depend on the ideal wherever these pairs are not all equal, as at $SU(2)\times SU(3)$ on $\mathbb{C}^2 \oplus \mathbb{C}^3$, where $C_A^{(i)}$ is 20 and 30 (Lem. 9.15(i)), and the coefficient is then given ideal by ideal, as above. The r coefficients are bounded ideal by ideal, $|b^{(i)}[\Psi_k^G]| \le \frac{C_A^{(i)}}{8}\sup_k |b[\Psi_k]| + |\frac{\theta_i}{3} - \frac{C_A^{(i)}}{8}|L^{-1}$, and so by the largest of these r bounds (Prop. 9.18(iv)).

(h3) For the tied action, the missing transfer task is to establish open slices for the backgrounds in the setting preceding Def. YM3_I:52.1, the Gaussian structure of Thm. YM3_I:U and normaliser of Prop. YM3_I:50.2 with their stated properties, localisation as in Prop. YM3_I:51.16, and one step as in Thm. YM3_I:V, with the counterterm numbers free. Thm. YM3_I:X must also be verified at the simple pairs (G_i, Λ) on $\mathbb{C}^N$, with the fixed weight $\Lambda = \frac{1}{N}\mathbf{1}$ of the action of G, which lies in the commutant (Prop. 9.24). This is the transfer task (T). Once this transfer is established, the intended one-run conclusion uses the mean $x_k = L^{-1}x_{k-1} - \sum_i \frac{n_i}{n} e_k^{(i)}$, $e_k^{(i)}$ the coefficient on $\mathfrak{g}_i$ of the local terms of second order of step k: the sequences $x_k^{(i)}$ of (h1) stay within $2(1-\frac{n_i}{n})\mathsf{G}_\sharp \frac{L}{L-1}(1-L^{-k})$ of x_k, by (h1) with $\Gamma = \mathsf{G}_\sharp$, as $|e_k^{(i)}| \le (\kappa_i/\kappa_G)^2 \mathsf{G}_\sharp \le \mathsf{G}_\sharp$, and all clauses of the $SU(2)$ construction on the increments hold, the increment theorem, the regular progression, the sequence and the bounds on polydiscs among them. At the least tied pair, the additional propagator assertions require the regular-domain and relative-smallness hypotheses stated in Prop. 9.30(v); under them the kinematic and interaction parts of the first step, the Hessian lemma and the propagators at every level hold (Prop. 9.30(v)).

(h4) If $G = G_1 \times \cdots \times G_r$ acts on $\mathbb{C}^{N_1} \oplus \cdots \oplus \mathbb{C}^{N_r}$ by blocks, the Wilson measure at β is the product of those of the G_i at $\beta N_i/N$, cumulants add with the weights $(N_i/N)^{\mathsf{m}}$, and for simply connected G_i with simple Lie algebras and $200L^2 z_{G_i} \ge 1$, each satisfying the construction hypotheses, clauses (a)–(g) hold factor by factor at these couplings, with precisely the qualifications stated there.

■□

Remark 9.7 (Leading law and perturbative coefficients). *The constants.* The Hessian of a at $\mathbf{1}$ is $\langle\cdot,\cdot\rangle$, so to leading order the field consists of n independent copies of the free field with this form, colour by colour, each contributing $\frac{3}{16\pi^2}$ as at $SU(2)$, where $3\cdot\frac{3}{16\pi^2} = \frac{9}{16\pi^2}$; this is $c_2(G) = \frac{3n}{16\pi^2}$, the constant of the law of leading order. For $SU(2)^r$ the product $c_2(G)\,\mathsf{b}^{-2}$ is the sum $\sum_i \lambda_i^2 \frac{9}{16\pi^2}(\lambda_i\mathsf{b})^{-2} = r\frac{9}{16\pi^2}\mathsf{b}^{-2}$ that Step 3 of the proof of Thm. AB forms from the laws of the factors. The next term is governed by the coefficient of β^{-3} of $\langle a_p; a_q\rangle$; by Lem. 9.28 its leading part, summed over orientations, is $\frac{\varphi_G}{16\pi^3}\mathsf{r}^{-5}$ at distance r, and $c_1(G) = \frac{\varphi_G}{16\pi^3}/c_2(G) = \frac{\varphi_G}{3\pi n}$ is the ratio of the next perturbative coefficient to the leading coefficient. At $SU(2)$, $\varphi = 24$, and this is $\frac{3}{2\pi^3}/\frac{9}{16\pi^2} = \frac{8}{3\pi}$, the coefficient c_1 of Thm. X.

The law of leading order. Let G be simply connected with $\mathfrak{g}$ simple, $L \ge 13$ odd and $200L^2 z_G \ge 1$, and assume the construction hypotheses. On the tori of the continuum theorem and in their limits along tori, under the finite-cutoff restrictions of Thm. YM3_I:AV and the continuum restrictions of

Thm. YM3_I:AW, as stated in Thm. AA(e),

$$\langle \mathcal{O}(f); \mathcal{O}(h) \rangle = c_2(G)\, \mathsf{b}^{-2} \mathcal{K}_6(f,h)\,(1+\varepsilon'), \qquad |\varepsilon'| \le \bar{\varepsilon}_g(\mathsf{d}),$$

$\mathsf{b} = a_K \beta$ the label and $\bar{\varepsilon}_g$ the number of that law, which tends to 0 with d; this is the law of leading order of Thm. AA(e). For $SU(2)^r$ the law holds with one label (Thm. AB(iii)). For simply connected G with $\mathfrak{g}$ simple its proof is the proof of the law of Thms. YM3_I:AV and YM3_I:AW, with the colour sum running to n. That proof rests on Thms. YM3_I:D, YM3_I:AJ and YM3_I:AD and on the count of colours: Thm. AA(a) gives Thm. YM3_I:D for G along the one run; Thms. YM3_I:AJ and YM3_I:AD for G with that run (Thm. AA(b)) gives the stability with sources; the Gaussian measure that the quadratic part of the action at the trivial field defines is the product of n copies of that of one colour of $SU(2)$, with the same normalisation (Thm. AA(e)); and the remainders are bounded by the same terms of the effective action as for $SU(2)$. The passage to limits along tori is that of Thm. AA(d). The law concerns the tori and their limits along tori.

The fixed-box expansion in Lem. 9.28 has an error constant depending on the box. Its large-box limit is taken after the expansion at fixed box. Thus the calculation of $c_1(G)$ in this lemma is an iterated perturbative limit.

Corollary 9.8 ($SU(3)$)**.** Let $G = SU(3)$ and $L \ge 13$ odd. Then $z_G = 1$, so that no floor on L is needed and the forms with D^c are the statements of the $SU(2)$ construction; no number of the averaging layer depends on the group; $r_G = \sqrt{3}$, $\kappa_G^2 = \tau_G^2 = 2$, $n = 8$; the least admissible p of the cut-off is 3, with $\lambda = 6.1194$; and the extra conditions of Lem. 9.11(v) and Prop. 9.22(iv) are void. The local coefficients in Thm. AA have the following values. The construction conclusions require the separate tasks of §9.1.1:

$$b[\Psi_k^G] = \frac{9}{4}\, b[\Psi_k] + \frac{5}{12}(L^{-k} - L^{-4k}), \qquad c_2(G) = \frac{3}{2\pi^2}, \qquad c_1(G) = \frac{6}{\pi}.$$

Besides, $\mathfrak{su}(3)$ has a basis, orthonormal for $\frac{1}{2}\mathrm{tr}\, XY$, whose eight members have the eigenvalues $1, -1, 0$, as the Pauli matrices, with the eigenvalues $1, -1$, are such a basis of $\mathfrak{su}(2)$. ■□

Sketch of proof. The numbers are those of $SU(N)$ at $N = 3$ (Def. 9.1); $\frac{C_A}{8} = \frac{9}{4}$, $\frac{C_\rho}{3} - \frac{C_A}{8} = \frac{5}{12}$, $\frac{3\cdot 8}{16\pi^2} = \frac{3}{2\pi^2}$, $\frac{C_A}{3\pi} = \frac{6}{\pi}$. The cut-off asks $3^{1/2p} < \frac{5}{4}$, first met at $p = 3$, and $\lambda = w/(5-4w)$ at $w = 3^{1/6}$. With $r_G = \sqrt{3}$ the condition $\varpi_L < r_G$ and the condition $\mathsf{c}_L c_4 \le 2 L r_G$ are met by the ceilings of the $SU(2)$ construction. For the basis: the rotation R by $\frac{2\pi}{5}$ about e_3 acts on the real symmetric traceless matrices, an orthogonal sum $V_0 \oplus V_1 \oplus V_2$ of the multiples of $\mathrm{diag}(-1,-1,2)$, of the span of $e_1 e_3^{\mathrm{T}} + e_3 e_1^{\mathrm{T}}$ and $e_2 e_3^{\mathrm{T}} + e_3 e_2^{\mathrm{T}}$, and of the span of $\mathrm{diag}(1,-1,0)$ and $e_1 e_2^{\mathrm{T}} + e_2 e_1^{\mathrm{T}}$, as the identity, the rotation by $\frac{2\pi}{5}$ and the rotation by $\frac{4\pi}{5}$. So for X of unit norm with squared components (c_0, c_1, c_2), $\frac{1}{2}\mathrm{tr}(R^j X R^{-j} R^k X R^{-k}) = c_0 + c_1 \cos\frac{2\pi(j-k)}{5} + c_2 \cos\frac{4\pi(j-k)}{5}$, which vanishes for $j \not\equiv k$ exactly when $(c_0, c_1, c_2) = (\frac{1}{5}, \frac{2}{5}, \frac{2}{5})$. For $X = ab^{\mathrm{T}} + ba^{\mathrm{T}}$ with orthonormal $a, b \in \mathbb{R}^3$, which has the eigenvalues $1, -1, 0$, one finds $c_0 = 3a_3^2 b_3^2$ and $c_2 = (1-a_3^2)(1-b_3^2)$; the two conditions say that a_3^2, b_3^2 are the roots $\frac{5 \pm \sqrt{10}}{15}$ of $15x^2 - 10x + 1 = 0$, whose sum $\frac{2}{3}$ leaves $\frac{1}{3}$ for the square of the third component of $a \times b$, so such a, b exist. The five $R^k X R^{-k}$ and the three imaginary antisymmetric Gell-Mann matrices, which are orthogonal to all real symmetric matrices, have the eigenvalues $1, -1, 0$ and are pairwise orthogonal. □

Theorem AB (Sums of copies of $SU(2)$)**.** Let $G = SU(2)^r$ act on $\mathbb{C}^2 \oplus \cdots \oplus \mathbb{C}^2$ by blocks, $a_p = \sum_i \lambda_i (1 - \frac{1}{2}\mathrm{tr}\, \partial U^{(i)}(p))$ with $\lambda_i > 0$, $\sum_i \lambda_i = 1$ (a block repeated m_i times in $\mathbb{C}^N$ gives $\lambda_i = 2m_i/N$), one coupling β, $n = 3r$, $\lambda_\pm$ the largest and the least λ_i, and let the constants of the $SU(2)$ construction, $\mathsf{G}_\sharp$ among them with its value for $SU(2)$ (Prop. 9.24), obey besides $\lambda_- B_\sharp \ge \lambda_+ \mathsf{G}_\sharp$.

(i) On every finite lattice and for $\beta > 0$ the Wilson measure is the product of those of $SU(2)$ at $\lambda_i \beta$, and $\langle \mathcal{O}(f_1); \dots; \mathcal{O}(f_{\mathsf{m}}) \rangle_\beta = \sum_i \lambda_i^{\mathsf{m}} \langle \mathcal{O}(f_1); \dots; \mathcal{O}(f_{\mathsf{m}}) \rangle_{\lambda_i \beta}^{SU(2)}$.

(ii) For $\lambda_-\beta \geq \gamma^{-2}$ the mean run $x_0 = \beta$, $x_k = L^{-1}x_{k-1} - \frac{1}{r}\sum_i \lambda_i^{-1} c_k^{(i)}$, with $c_k^{(i)}$ the coefficient of step k of $SU(2)$ along the couplings $\lambda_i x_0, \dots, \lambda_i x_{k-1}$, is defined and real while $\lambda_- x_{k-1} \geq \gamma^{-2}$. Along it Thm. YM3_I:D holds in every factor, the step of the factor i at level j being the step of $SU(2)$ at $\lambda_i x_j$ with its own thresholds and histories; and $x_0^{(i)} = \beta$, $x_k^{(i)} = L^{-1}x_{k-1}^{(i)} - \lambda_i^{-1} c_k^{(i)}$ have the mean x_k and $|x_k^{(i)} - x_k| \leq 2\lambda_-^{-1} \mathsf{G}_\sharp \frac{L}{L-1}$.

(iii) Let $\lambda_- g^{-2} \geq \gamma_W^{-2} + \bar{B}$, K the least k with $x_k \leq L(g^{-2} + \frac{\mathsf{G}_\sharp}{r}\sum_i \lambda_i^{-1})$, $\mathsf{b} = a_K \beta$, and let the hypotheses of the law of two points of Thm. YM3_I:AV on the scale d, the cutoff and the test pair (f, h) hold separately at the $2r$ reference couplings $g_{i,1}$, $g_{i,2}$ with $g_{i,1}^{-2} = \lambda_i g^{-2} - \bar{B}$, $g_{i,2}^{-2} = L\lambda_i g^{-2} - 3\bar{B}$, and let the tori have sides admitted at each of the $2r$ couplings $g_{i,s}$, that is, sides $2L^m$ with m at least the largest of the $2r$ least exponents admitted at them, an unbounded set. Then

$$\langle \mathcal{O}(f); \mathcal{O}(h) \rangle = \frac{3n}{16\pi^2}\, \mathsf{b}^{-2} \mathcal{K}_6(f,h)\,(1+\varepsilon'), \qquad |\varepsilon'| \leq \max_{i,s} \bar{\varepsilon}_{g_{i,s}}(\mathsf{d}) + \tfrac{r-1}{12}\varepsilon^\times(\mathsf{d}),$$

uniformly in K and in β in its window, on these tori, in their limits along such tori, and in every Gibbs state of $e^{-\beta\sum_p a_p}$ whose marginals on the factors are periodic; $\bar{\varepsilon}_g$ is the number of that law, and $\varepsilon^\times(\mathsf{d})$, a number of g, the λ_i, the parameter of the test pairs and the constants, is at most a constant times d and vanishes on the tori and in product states. ■□

9.1.4. *Inputs from the $SU(2)$ construction.* We start from Thms. YM3_I:D, YM3_I:AJ and YM3_I:AD for $SU(2)$ and from their proofs: the inductive step, Prop. YM3_I:69.3 and Thm. YM3_I:AC; the flow bound YM3_I:(1468) and the intrinsic coupling construction in Section YM3_I:52.10; the large-field chain, Prop. YM3_I:44.4, Prop. YM3_I:44.5 and Lem. YM3_I:65.2 and the local-deposit argument in the proof of Prop. YM3_I:65.4; and the verification of the conditions on the constants, the ordered choice and its numerical verification in Prop. YM3_I:69.1. From the first part we use the mean and the block average (equation YM3_I:(25) and YM3_I:(37)–(39), Prop. YM3_I:4.3), the propagators in a background (the precise input (P) of §9.1.1), the critical configurations and their localisation (input (V) there and Prop. 9.22(iii)), the bond map (Lem. YM3_I:53.3), the elimination factor (Lem. 9.14(iii)), the one-loop Thm. YM3_I:O, with the determinant representation and telescoping argument in Section YM3_I:43.3 and Lem. 9.40, and its limit, Thm. AC; and the continuum theorems with their proofs. Of G we use the closed subgroup theorem, the spectral theorem, Schur's lemma, the Haar measure, the Peter–Weyl theorem, and for groups other than $SU(N)$ maximal tori and roots.

9.1.5. *Matrix charts, averages and gauges.*

Lemma 9.9 (Norms, the domain D^c and means)**.**

(i) $\|Y\| \leq |Y| \leq \kappa_G \|Y\|$ on $\mathfrak{g}$, with $\kappa_G^2 \leq N-1$ for semisimple $\mathfrak{g}$; $a(V) = \frac{1}{2}\|V - \mathbf{1}\|^2$ on $U(N)$; on $SU(N)$, $\frac{1}{2(N-1)}|V-\mathbf{1}|^2 \leq a(V) \leq \frac{1}{2}|V-\mathbf{1}|^2$ and $|V-\mathbf{1}| \leq (N-1)^{1/2}\|V-\mathbf{1}\|$, all three sharp. So $a(V) \geq \frac{1}{2}\tau_G^{-2}|V-\mathbf{1}|^2$ on G, with $\tau_G^2 \leq N-1$ for $G \subset SU(N)$. For $n \geq 2$, $X + X^{-1}$ is scalar for every $X \in G$ if and only if G is conjugate to $\{u \otimes \mathbf{1}_k : u \in SU(2)\}$.

(ii) $Y \mapsto e^{iY}$ maps $\{Y \in \mathfrak{g} : |Y| < \pi\}$ bijectively onto G^π, which is the component of $\mathbf{1}$ in $\{u \in G : -1 \notin \operatorname{spec} u\}$. If $A_j \in \mathfrak{g}$ and $\sum_j |A_j| < \pi$, then $e^{iA_1}\cdots e^{iA_m} = e^{iA}$ with $A \in \mathfrak{g}$, $|A| \leq \sum_j |A_j|$. For hermitian Y, Y' with $|Y|, |Y'| \leq t < \pi$: $|Y - Y'| \leq \frac{t}{\sin t}|e^{iY} - e^{iY'}|$. For a matrix X with $s = |X - \mathbf{1}| < 1$: $|X^{-1} - \mathbf{1}| \leq \frac{s}{1-s}$; and $|X^{-1}| \leq |\operatorname{Ad} X|$ for $X \in SL(N,\mathbb{C})$, $\operatorname{Ad} X$ being read on all $N \times N$ matrices, where $|\operatorname{Ad} X| = |X|\,|X^{-1}|$; with the norm of $\operatorname{Ad} X$ on $\mathfrak{g}^c$ alone in its place the inequality fails for some pairs, as the proof of Lem. 9.19 shows.

(iii) For compact connected $G \subset U(N)$ the statements $\mathfrak{g}^c = \{Z : \operatorname{tr} Z = 0\}$, $G^c = \{X : \det X = 1\}$ and $G = SU(N)$ are equivalent; for semisimple $\mathfrak{g}$, $\operatorname{tr} Z = 0$ on $\mathfrak{g}^c$ and $\det X = 1$ on G^c, and either inclusion the other way holds only at $SU(N)$. For every compact connected G:

if $Z_1, \dots, Z_q \in \mathfrak{g}^c$ and $\sum_j |Z_j| < \ln 2$, then $|e^{Z_1} \cdots e^{Z_q} - \mathbf{1}| < 1$ and $\log(e^{Z_1} \cdots e^{Z_q}) \in \mathfrak{g}^c$, and it lies in $i\mathfrak{g}$ if the Z_j do. Let $\mathfrak{g}$ be semisimple. Then $Z(G^c) = Z(G)$ is a finite set of unitary matrices; for $X \in G^c$ with $|X - \mathbf{1}| < 1$, $z(X) = \exp((1-P)\log X)$ lies in $Z(G)$, $|z(X) - \mathbf{1}| \le |X - \mathbf{1}|$, $\log X = \log z(X) + P \log X$, and z is locally constant. So $D^c = \{z(X) = \mathbf{1}\}$ is the connected component of $\mathbf{1}$ in $\{X \in G^c : |X - \mathbf{1}| < 1\}$ and the trace on G^c of an open set of matrices; it contains $\{X \in G^c : |X - \mathbf{1}| < z_G\}$, and no larger ball if $z_G < 1$; it is kept by the powers $X^t = e^{t \log X}$, $0 \le t \le 1$, by continuous paths in the unit ball that begin in it, by products of elements X_j of D^c with $\prod_j (1 + |X_j - \mathbf{1}|) < 2$, by the products $e^{Z_1} \cdots e^{Z_q}$ with $Z_j \in \mathfrak{g}^c$, $\sum_j |Z_j| < \ln 2$, by conjugation with G, by inverses in the unit ball and by adjoints; $\{X \in G^c : |\log X| < 2 \arcsin \frac{z_G}{2}\} \subset D^c$; and $D = \exp(i\{Y \in \mathfrak{g} : |Y| < \frac{\pi}{3}\}) = G^\pi \cap \{|X - \mathbf{1}| < 1\}$. Let $0 < c_j \le 1$. Within the domain cut out in stages by $|X_j - \mathbf{1}| < c_j$ for continuous G^c-valued maps, retain these radius conditions and add $X_j \in D^c$ at each stage. The resulting domain is open and closed in the original domain, and equals it if $z_G = 1$.

(iv) Let $V_1, \dots, V_q$ be invertible, $d = \max_{i,j} |V_i V_j^{-1} - \mathbf{1}| \le \frac{1}{40}$. Then

$$\Phi(Y) = Y + \frac{1}{q} \sum_i \log(V_i V_1^{-1} e^{-Y})$$

is a contraction with factor $\frac{2}{5}$ of $\{|Y| \le \frac{1}{8}\}$; its fixed point has $|Y_*| \le 2d$, and $\mathrm{av} = e^{Y_*} V_1$ satisfies $\sum_i \log(V_i \mathrm{av}^{-1}) = 0$ and $|V_i \mathrm{av}^{-1} - \mathbf{1}| \le 4d$, is holomorphic in (V_i), and is the only unitary mean of a unitary family. Let $\mathfrak{g}$ be semisimple, let the V_i lie in G^c, and let the quotients $V_i V_1^{-1}$ lie in D^c, which holds if $d < z_G$. Then $Y_* \in \mathfrak{g}^c$, $\mathrm{av} \in G^c$ and $V_i \mathrm{av}^{-1} \in D^c$ for all i; and $\mathrm{av} \in G$ for a family in G. Further,

$$\mathrm{av}(W \mathbf{V} W') = W \,\mathrm{av}(\mathbf{V})\, W'$$

for $W, W' \in G$. For semisimple $\mathfrak{g}$, if $z_G \le \frac{1}{40}$ and $z \in Z(G)$ has $|z - \mathbf{1}| = z_G$, the family $(\mathbf{1}, z)$ has no mean in G.

■

Sketch of proof. (i) For V with the eigen-angles θ_j, $|V - \mathbf{1}| = \max_j 2|\sin \frac{\theta_j}{2}|$ and $a(V) = \frac{2}{N} \sum_j \sin^2 \frac{\theta_j}{2} = \frac{1}{2}\|V - \mathbf{1}\|^2$. On $SU(N)$, $\sum_j \theta_j \in 2\pi\mathbb{Z}$, so $|\sin \frac{\theta_1}{2}| \le \sum_{j \ge 2} |\sin \frac{\theta_j}{2}|$, and Cauchy's inequality gives $\sin^2 \frac{\theta_1}{2} \le (N-1) \sum_{j \ge 2} \sin^2 \frac{\theta_j}{2}$, which is $|V - \mathbf{1}|^2 \le (N-1)\|V - \mathbf{1}\|^2$; the same argument with the eigenvalues of a traceless hermitian Y gives $\kappa_G^2 \le N - 1$. The constants are approached along $\exp(it\,\mathrm{diag}(N-1, -1, \dots, -1))$, $t \to 0$, and $\frac{1}{2}$ is attained at $e^{2\pi i/N}\mathbf{1}$. If $X + X^{-1}$ is scalar on G, then $2\cos(tY)$ is scalar for $Y \in \mathfrak{g}$ and all t, so $Y^2 = \|Y\|^2 \mathbf{1}$ and orthogonal elements of $\mathfrak{g}$ anticommute; for orthonormal $X, Y \in \mathfrak{g}$ the element $Z = iXY = \frac{i}{2}[X, Y]$ lies in $\mathfrak{g}$, and an element W orthogonal to X, Y, Z would anticommute with X and Y, hence commute with Z, and anticommute with Z, which is absurd as $Z^2 = W^2 = \mathbf{1}$; so $\mathfrak{g}$ is spanned by X, Y, Z, these generate a quaternion algebra, and $\mathbb{C}^N$ is a multiple of its two-dimensional representation.

(ii) The path $s \mapsto e^{isA_1} \cdots e^{isA_m}$ has speed at most $\sum_j |A_j|$, so its eigen-angles stay below π, and its logarithm, continuous along it and in $\mathfrak{g}$ near $\mathbf{1}$ by the closed subgroup theorem, stays in $\mathfrak{g}$; the Lipschitz bound is the spectral calculus of $Y \mapsto e^{iY}$; the bound on X^{-1} is the series $\sum_j (\mathbf{1} - X)^j$; and for $X \in SL(N, \mathbb{C})$ with singular values $\sigma_1 \ge \cdots \ge \sigma_N$, $\prod \sigma_j = 1$, one has $|X^{-1}| = \sigma_N^{-1} \le \sigma_1 \sigma_N^{-1} = |X|\,|X^{-1}| = |\mathrm{Ad}\, X|$, the norm of $\mathrm{Ad}\, X$ on all $N \times N$ matrices being attained at vw^* for unit vectors v, w with $|Xv| = |X|$ and $|(X^{-1})^* w| = |X^{-1}|$.

(iii) $G^c = G \exp(\mathfrak{g}) = \{ue^H : u \in G,\ H \in \mathfrak{g}\}$, e^H being positive (a polar decomposition got by integrating e^{tZ} inside the set $\{ue^H\}$), and $\det e^Z = e^{\mathrm{tr}\, Z}$; for semisimple $\mathfrak{g}$ every element of $\mathfrak{g}^c$ is a sum of brackets, so its trace on every invariant subspace vanishes. The rule for products holds for every compact connected G: along $X_t = e^{tZ_1} \cdots e^{tZ_q}$, $0 \le t \le 1$, $|X_t - \mathbf{1}| \le e^{t \sum_j |Z_j|} - 1 < 1$,

and $X_t'X_t^{-1} = \sum_j \mathrm{Ad}(e^{tZ_1}\cdots e^{tZ_{j-1}})Z_j$ lies in $\mathfrak{g}^c$, as $\mathrm{Ad}(G^c)$ keeps $\mathfrak{g}^c$; $W_t = \log X_t$ has $X_t'X_t^{-1} = \Xi(\mathrm{ad}\, W_t)W_t'$ with $\Xi(w) = (e^w-1)/w$, and $\Xi(\mathrm{ad}\, W_t)$ is invertible, the eigenvalues of $\mathrm{ad}\, W_t$ being differences of logarithms of eigenvalues of X_t, with imaginary parts in $(-\pi,\pi)$; so W_t solves $W' = \Xi(\mathrm{ad}\, W)^{-1}(X'X^{-1})$ from $W_0 = 0$. For a real Lie algebra $\mathfrak{h}$ of matrices that holds W, $\Xi(\mathrm{ad}\, W)$, a power series in $\mathrm{ad}\, W$ with real coefficients, maps $\mathfrak{h}$ into itself and, being invertible, onto itself; so the equation keeps $\mathfrak{h} = \mathfrak{g}^c$, and $\mathfrak{h} = i\mathfrak{g}$ if the Z_j lie there, and W_t stays in $\mathfrak{h}$. With $W = \log X$ the eigenvalues of $\mathrm{ad}\, W$ have imaginary parts in $(-\pi,\pi)$, so $\mathrm{ad}\, W$ is a polynomial in $e^{\mathrm{ad}\, W} = \mathrm{Ad}(X)$ and keeps $\mathfrak{g}^c$; hence $(1-P)W$ commutes with $\mathfrak{g}^c$, it is a scalar on each isotypic component, the mean there of the eigenvalues of W, and $z(X) = Xe^{-PW} \in G^c$ is central. A mean c of numbers $\log\lambda_j$ with $|\lambda_j - 1| < r \le 1$ has $|e^c - 1| < r$, the set $\{|e^{x+iy}-1| < r,\ |y| < \frac{\pi}{2}\}$ being convex; this gives $|z(X) - \mathbf{1}| \le |X - \mathbf{1}|$, and z, continuous with values in the finite set $Z(G)$, is locally constant. The products are joined to $\mathbf{1}$ inside the unit ball by $t \mapsto e^{tZ_1}\cdots e^{tZ_q}$ and $t \mapsto X_1^t\cdots X_q^t$; the inner ball comes from $|\log X| \ge \theta(z(X))$, $\theta(V)$ the largest modulus of the argument of an eigenvalue of V.

(iv) $|\log(\mathbf{1}+E)| \le -\log(1-|E|)$ and the bound on the derivative of log for $|E| \le \frac{1}{10}$ give the contraction; if the quotients lie in D^c, so does $V_iV_1^{-1}e^{-Y}$ for $Y \in \mathfrak{g}^c$, $|Y| \le \frac{1}{8}$, by the rule for products of (iii), so Φ keeps $\mathfrak{g}^c$, $Y_* \in \mathfrak{g}^c$ and $V_i\mathrm{av}^{-1} = V_iV_1^{-1}e^{-Y_*} \in D^c$; and $\mathrm{av} = e^{Y_*}V_1 \in G^c$. For unitary V_i, V' with $\sum_i \log(V_iV'^{-1}) = 0$ the spectral theorem gives that every angle of $V_jV'^{-1}$ is at most $\frac{2\pi}{3\sqrt{3}}d$, whence uniqueness. At $(\mathbf{1}, z)$, $\mathrm{av} = e^{\frac{1}{2}\log z}$, which is not in G^c (for $SU(N)$ its determinant is -1). □

Lemma 9.10 (A bond in the contour gauge)**.** Let W be a real G-valued field on the bonds of a block $B(y)$, and let $\varpi_y(W)$ be the largest plaquette distance from $\mathbf{1}$ in that block. Suppose

$$\tfrac{3}{4}(L-1)^2\varpi_y(W) < \omega_{\mathrm{c}}, \qquad \varpi_y(W) < z_G,$$

where $\omega_{\mathrm{c}} := 1/40$, the contour-mean radius admitted by Prop. YM3_I:4.3. Then the contour means $W(y,x)$ exist in G, and every bond b inside the block satisfies

$$|W(y,b_-)W(b)W(y,b_+)^{-1} - \mathbf{1}| \le 2\kappa_1^{\mathrm{SO}}\varpi_y(W).$$

Here $\kappa_1^{\mathrm{SO}} = \pi(L-1)^2/(2\sqrt{3})$. ■

Proof. Two coordinate-order paths from y to x differ by at most $\frac{3}{4}(L-1)^2$ plaquettes. Telescoping their transports gives the displayed diameter bound for the contour family. Its quotients are products of conjugated plaquette variables. The guard $\varpi_y(W) < z_G$ and Lem. 9.9(iii) put their logarithms in $\mathfrak{g}$; Lem. 9.9(iv) therefore gives the mean in G. Apply the inverse-exponential inequality of Lem. 9.9(ii) to the intrinsic logarithms of these quotients, then the zero-sum-logarithm argument of Lem. YM3_I:14.5. This gives

$$|W(\Gamma)W(y,x)^{-1} - \mathbf{1}| \le \kappa_1^{\mathrm{SO}}\varpi_y(W)$$

for every coordinate-order path Γ from y to x. Reverse b if necessary so that its step moves away from y in its coordinate direction. Choose that direction last in the coordinate order. The resulting path Γ' from y to b_+ ends with b, and deleting this last bond gives a coordinate-order path Γ to b_-. Thus

$$W(y,b_-)W(b)W(y,b_+)^{-1} = [W(\Gamma)W(y,b_-)^{-1}]^{-1}[W(\Gamma')W(y,b_+)^{-1}].$$

The two factors lie in G and have distance at most $\kappa_1^{\mathrm{SO}}\varpi_y(W)$ from $\mathbf{1}$. The triangle inequality and unitary invariance prove the claim. □

Lemma 9.11 (Block averages)**.** Let $L \ge 13$ be odd and let c be a coarse bond.

(i) Let U be G-valued with $|\partial U(p) - \mathbf{1}| < \min\{\omega_M, r_G\}$ at the plaquettes of the double block of c. Then the contour quotients lie in D within $\frac{1}{160}$ of $\mathbf{1}$, each contour family (Section YM3_I:4 (coordinate-order contours)) has exactly one mean in G, within $\frac{1}{40}$, the W_x of equations YM3_I:(37)–(39) lie in D within $\frac{1}{14}$, and $\mathcal{M}(U)(c) = \exp(L^{-3}\sum_x \log W_x)\, U[c] \in$

G is real-analytic in U. This holds for every compact connected G; with ω_M in the place of $\min\{\omega_M, r_G\}$ it fails for every G with $r_G < \omega_M$. Let now the centre of $\mathfrak{g}$ be 0. Then $Z(G)$ is finite and $\min\{1, r_G\} = z_G$, so that the statement holds with ω_M in the place of $\min\{\omega_M, r_G\}$ if and only if $\omega_M \le z_G$. At G^c-valued fields the same formula is holomorphic in the matrix entries where the contour quotients and the W_x lie in D^c; without this condition it leaves G^c for every such G with $z_G < 1$.

(ii) The averaging map of equations YM3_I:(37)–(39) transferred to G, its three cases read with "$u \in G^\pi$" and $\operatorname{Log} e^{iY} = iY$ for $Y \in \mathfrak{g}$, $|Y| < \pi$, is G-valued and Borel on all fields, depends on U on the double block of c alone, is compatible with the reversal of bonds, and $\mathcal{M}(U^h)(c) = h(c_-)\mathcal{M}(U)(c)h(c_+)^{-1}$ with the same case; so $\mathcal{M}^q(U^h) = (\mathcal{M}^q U)^h$. As a map it needs no condition on z_G. For $SU(N)$ with $\omega^G = \min\{\omega_M, 2\sin\frac{\pi}{N}\}$ the averaging map and the lemmas of the $SU(2)$ construction on it, Lem. YM3_I:14.2 among them, hold with their constants but for one clause: the second assertion of Lem. YM3_I:14.2, that $L^{-3}\sum_x \log W_x \in i\mathfrak{g}$ and $\mathcal{M}(W)(c) \in G$ for every G-valued W of its domain, holds for every G with $\rho_G \ge \frac{\pi}{3}$, that is $r_G \ge 1$, and fails where $\rho_G < \frac{\pi}{3}$, that is $r_G < 1$, so for $SU(N)$ exactly when $N \ge 7$. It holds for every G with the proviso that every eigen-angle of each W_x is below ρ_G in modulus, or that W_x is a product of exponentials $e^{i\gamma_j}$, $\gamma_j \in \mathfrak{g}^c$, with $\sum_j |\gamma_j| < \ln 2$; the proviso is met under the averaging radius.

(iii) If $|\partial U(p) - \mathbf{1}| \le \alpha_0 L^{-2k}$ at all bare plaquettes and $\alpha_0 \le c(G) = \min\{3.9\cdot 10^{-9} L^{-2}, \frac{1}{4} r_G\}$, then for $0 \le q \le k$ the plaquette variables of $\mathcal{M}^q U$ are within $2\alpha_0 L^{2(q-k)}$ of $\mathbf{1}$, $\mathcal{M}^k U(c) U(\Gamma_c)^{-1}$ is a product of exponentials e^{iA_j}, $A_j \in \mathfrak{g}$, of total size at most $12\alpha_0$, so that it lies in G^π, and $|\mathcal{M}^k U(c) - U(\Gamma_c)| \le 12\alpha_0$, Γ_c the straight bare path from c_- to c_+. The same holds for every G-valued averaging map that is the formula of (i) on small fields, and in a local form, for fields given beneath the double block of c alone.

(iv) Let $G = \tilde{G}/\mathsf{Z}$ with $\tilde{G}$ simply connected. On a torus a field as in (iii) has a sector $\zeta(U) \in \mathsf{Z}^3$, the class of the Z-valued cocycle of the central parts of the plaquette variables of a lift of U to $\tilde{G}$, and $\mathcal{M}^k U$ lies in the sector of U. More generally, a coarse field V with $|V(c) - U(\Gamma_c)| \le \epsilon$ over a fine field with plaquettes within κ of $\mathbf{1}$ lies in the sector of U as soon as $\mathcal{L}^2\kappa + 4\epsilon < r_G$, $\mathcal{L}$ the ratio of the spacings; in angles the threshold is π, and it is sharp for $SO(3)$. Fields whose block averages are all $\mathbf{1}$ lie in the trivial sector, and the block averages alone fix the sector.

(v) Let $1 \le j \le K$, let Y be the two end points of a bond of level j and $\mathsf{Y}_i = B^{j-i}(\mathsf{Y})$, $0 \le i \le j$, the sites of level i in their blocks of order $j - i$; let U_0 be G-valued on the bare bonds with both end points in Y_0, with $|\partial U_0(p) - \mathbf{1}| \le \alpha L^{-2j}$ at the bare plaquettes with all corners in Y_0, where $0 < \alpha \le c_2^{\mathrm{g}}$, and let A_0 be $\mathfrak{g}$-valued on these bonds with $L^j \sup|A_0| < c_4$; here $c_2^{\mathrm{g}} \le \frac{1}{4}\min\{\omega_M, r_G\}$ and the radius $c_4 \le \frac{1}{4} r_L$ of the averages over a box, r_L the radius of the one step below, are numbers of L and r_G, so small that $\sum_l C_{\mathrm{P}} \varpi_l \le \ln 2$ and $\sum_l C\lambda_l \le \ln 2$ in the proof. Then, uniformly in the number j of steps, for $0 \le i \le j$ the averages $\mathcal{M}^i U_0$ and $\mathcal{M}^i(e^{iA_0} U_0)$ are formed on the bonds with both end points in Y_i, by the formula of (i) at every step, the plaquette variables of $\mathcal{M}^i U_0$ lie within $2\alpha L^{-2(j-i)}$ of $\mathbf{1}$, and there are $\mathfrak{g}$-valued A_i and $f_i : \mathsf{Y}_i \to G$, $f_0 = \mathbf{1}$, with

$$\mathcal{M}^i(e^{iA_0} U_0)(c) = f_i(c_-)\, e^{iA_i(c)}\, (\mathcal{M}^i U_0)(c)\, f_i(c_+)^{-1}, \qquad |A_i| \le 2L^i \sup|A_0|,$$

$f_{i+1}(y) = f_i(y) e^{i\sigma_i(y)}$ with $\sigma_i(y) \in \mathfrak{g}$, $|\sigma_i(y)| \le \kappa_6 \sup|A_i|$, and $f_i \in G^\pi$ with $|\frac{1}{i}\log f_i| \le 6L^i \sup|A_0|$, so that the bound $|f_i - \mathbf{1}| \le C_v L^i \sup|A_0|$ of the averages over a box holds with $C_v = 6$. One step is governed by the numbers $r_L = \frac{1}{1600 L^2}$, $\varpi_L = \frac{1}{2}\min\{\omega_M, r_G\}$ and $\kappa_6 = 2L$: if W_0 is G-valued on the double block of a bond c with plaquette variables within ϖ_L of $\mathbf{1}$, and a is $\mathfrak{g}$-valued there with $\sup|a| < r_L$, then the contour variables of $W = e^{ia} W_0$ and of W_0 are formed, $|W(y,x) W_0(y,x)^{-1} - \mathbf{1}| < 1$, and its logarithm lies

in $i\mathfrak{g}$ with $|\frac{1}{i}\log(W(y,x)W_0(y,x)^{-1})| \le \kappa_6 \sup|a|$. The condition $\varpi_L < r_G$ cannot be spared. At complex arguments there is a number $a_c > 0$ of L and the pair such that, at every real background U_0 beneath a bond c of level q whose bare plaquette variables lie within $a_c L^{-2q}$ of $\mathbf{1}$, and for $\mathfrak{g}^c$-valued a with $L^q \sup|a| < a_c$, $\mathcal{M}^q(e^{ia}U_0)(c) = e^{iQ}\,\mathcal{M}^q U_0(c)$ with $Q \in \mathfrak{g}^c$ holomorphic in a, in $\mathfrak{g}$ for real a, and $|Q| \le 167\, L^q \sup|a|$, uniformly in q; and $|\frac{1}{i}\log \mathcal{M}^q(e^{ia})| \le 9.2\, L^q \sup|a|$, uniformly in q and N. At the trivial background, for $\xi = L^{-k}$ and a $\mathfrak{g}^c$-valued form f beneath the double block of c with $\sup|f| < 4\cdot 10^{-7}$, the averages $\mathcal{M}^j(e^{i\xi f})$, $0 \le j \le k$, are e^{ia_j} with $\mathfrak{g}^c$-valued a_j, holomorphic in f, real for real f, and $\sup|a_j| \le 8L^{j-k}\sup|f|$. For $SU(3)$ no number changes.

■

Sketch of proof. (i) A contour quotient is a product of at most $\frac{9}{8}L^2$ conjugated plaquette variables, those of the area between two shortest paths in a block, so it lies within $(1+\omega_M)^{9L^2/8} - 1 \le e^{9/1600} - 1 \le \frac{1}{160}$ of $\mathbf{1}$. A plaquette variable within r_G of $\mathbf{1}$ is e^{iF} with $F \in \mathfrak{g}$, $|F| \le \frac{\pi}{2}|\partial U(p) - \mathbf{1}|$, and so is each of its conjugates by G; so the quotient is e^{iA} with $A \in \mathfrak{g}$, $|A| \le \frac{9\pi}{16}L^2\omega_M < \frac{1}{100}$, and lies in D (Lem. 9.9(ii)). Lem. 9.9(iv) gives exactly one unitary mean, within $4d \le \frac{1}{40}$, and its map Φ keeps $i\mathfrak{g}$, each $V_i V_1^{-1} e^{-Y}$ with $Y \in i\mathfrak{g}$, $|Y| \le \frac{1}{8}$, being a product of two exponentials of elements of $i\mathfrak{g}$ of total size below $\ln 2$ (Lem. 9.9(iii)); so the fixed point lies in $i\mathfrak{g}$ and the mean in G. $P_x = U(\Gamma)U([x,x'])U(\Gamma')^{-1}U[c]^{-1}$ takes $\frac{25}{8}L^2$ plaquettes and lies within $e^{1/64} - 1 \le \frac{1}{60}$; then $|W_x - \mathbf{1}| \le \frac{6541}{96000} < \frac{1}{14}$, and W_x, the product of P_x with conjugates of the quotients of two contour variables by their means, is again e^{iB} with $B \in \mathfrak{g}$ small, so it lies in D. Holomorphy holds on the open sets of matrices cut out by the conditions with D^c, on which av and log are holomorphic. If $r_G < \omega_M$, one bond set to an element $u_* \in G \setminus G^\pi$ with $|u_* - \mathbf{1}| = r_G$ gives a contour family without a mean in G, as in (v). Let the centre of $\mathfrak{g}$ be 0. A central $z \ne \mathbf{1}$ lies outside G^π, since $z = e^{iY}$ with $Y \in \mathfrak{g}$, $|Y| < \pi$, would give $e^{i\,\mathrm{ad}\,Y} = 1$ with the eigenvalues of $\mathrm{ad}\,Y$ below 2π in modulus, so $\mathrm{ad}\,Y = 0$ and $Y = 0$; and an element $u \in G \setminus G^\pi$ with $|u - \mathbf{1}| < 1$ has the central factor $z(u) \ne \mathbf{1}$, with $|z(u) - \mathbf{1}| \le |u - \mathbf{1}|$ (Lem. 9.9(iii)); so $\min\{1, r_G\} = z_G$. If $z_G < \omega_M$, one bond set to a central z with $|z - \mathbf{1}| = z_G$ gives the contour family $(z, \mathbf{1})$, which has no mean in G (Lem. 9.9(iv)).

(ii) The three cases are read with G^π, on which Log inverts the chart of Lem. 9.9(ii); each is covariant, as $\mathrm{av}(h[V]h') = h\,\mathrm{av}([V])h'$ and $\mathrm{Log}(huh^{-1}) = h(\mathrm{Log}\,u)h^{-1}$; the sets on which the cases apply are open, closed and Borel, and the map is continuous on each. The domain of Lem. YM3_I: 14.2 asks $|W_x - \mathbf{1}| < 1$, so the eigen-angles of W_x are below $\frac{\pi}{3}$ in modulus; if $\rho_G \ge \frac{\pi}{3}$, then $\log W_x \in i\mathfrak{g}$ by the definition of ρ_G and Lem. 9.9(ii), and under the proviso the same holds by Lem. 9.9(ii), (iii). If $\rho_G < \frac{\pi}{3}$, there is $V \in G$ whose eigen-angles are at most ρ_G in modulus and whose hermitian logarithm Y does not lie in $\mathfrak{g}$, for $SU(N)$ the element $e^{2\pi i/N}\mathbf{1}$; with V on one bond of the corridor of c other than the pivot bond and $\mathbf{1}$ at the others, $L^{-3}\sum_x \log W_x = iL^{-2}Y \notin i\mathfrak{g}$.

(iii) We never take the logarithm of an element that is merely near $\mathbf{1}$: every object of a step is written as a product of exponentials e^{iA_j}, $A_j \in \mathfrak{g}$ small, and $\sum_j |A_j|$ is added up (Lem. 9.9(ii)). One step gives $\mathcal{M}(W)(c) = e^{iY}W[c]$ with $Y \in \mathfrak{g}$, $|Y| \le \kappa_3 \max_p |\partial W(p) - \mathbf{1}|$, $\kappa_3 < 6L^2$; the plaquette variables of an average are within $L^2\varpi + C_0\varpi^2$ of $\mathbf{1}$ if those beneath are within ϖ, C_0 a number of L; and the corrections sum over the levels as geometric series, with the factor $\frac{2\kappa_3}{L(L-1)} < 11.99$; the number $c(G)$ reads of the pair only r_G.

(iv) If $\sum_j |X_j| < \pi$ in the norm of $\mathbb{C}^N$, then $\prod_j e^{i\tilde{X}_j}$ is the canonical lift of $\prod_j e^{iX_j}$; by (iii) the plaquette variables of every average are such products of small exponentials of those beneath, so the central parts of the plaquette variables of the lift multiply under blocking. A field with trivial block averages is a gauge transform of e^{ia}, $4\sup|a| < \pi$ (Prop. 9.33), whose lift $e^{i\tilde{a}}$ has central parts $\mathbf{1}$. Four bonds turned by $\frac{\pi}{4} + \delta$ about one axis around one plaquette of $SO(3)$ give a field near the trivial one in a twisted sector.

(v) One step, relative to the background. On the double block of c a tree gauge, the comb of Lem. YM3_I:14.4, writes $W_0 = e^{iA_0}$ and $W = e^{ia}W_0 = e^{iA}$ with $\mathfrak{g}$-valued A_0, A of size at most $2\pi L\varpi + \sup|a|$ and $\sup|A - A_0| \le 1.01\sup|a|$ (Lem. 9.9(ii)). In this gauge every contour variable, every W_x and the average are exponentials e^{iX} of elements $X \in \mathfrak{g}$ formed from A by means and by logarithms of products of small exponentials, and the Lipschitz bound of the remainder of Campbell and Hausdorff, $|F(X) - F(X')| \le 4S\sum_j |X_j - X'_j|$ for $F(X) = \log(e^{X_1}\cdots e^{X_r}) - \sum_j X_j$ and $S = \sum_j \max\{|X_j|, |X'_j|\} \le \frac{1}{4}$, compares each of them with its linear part, a straight average with real weights. Let $\xi(y,x) = \frac{1}{i}\log(W(y,x)W_0(y,x)^{-1})$, so that $|\xi| \le \kappa_6 \sup|a|$, and let $\sigma(y) = L^{-3}\sum_{x\in B(y)}\xi(y,x)$. Then $\mathcal{M}(W)(c) = e^{i\sigma(c_-)}e^{ia'(c)}\mathcal{M}(W_0)(c)e^{-i\sigma(c_+)}$ with $a'(c) \in \mathfrak{g}$: in the linear part the means of $A - A_0$ over the contours from $c_\pm$ cancel against $\sigma(c_\pm)$, and what is left is $L^{-3}\sum_{x\in B(c_-)}(A - A_0)([x,x'])$, a sum over the L bonds of each $[x,x']$, whence the exact factor L; the rest is bounded by the Lipschitz bound, so that $|a'(c)| \le L\sup|a|\,(1 + C\lambda)$ with λ a number of L times $\varpi + \sup|a|$. In the same way the plaquette variables of the average lie within $L^2\varpi(1 + C_{\mathrm{P}}\varpi)$ of $\mathbf{1}$ if those beneath lie within ϖ, C_{P} a number of L. Over the levels the plaquette bounds $\varpi_i = 2\alpha L^{-2(j-i)}$ and the numbers λ_i, a number of L times $\varpi_i + 2c_4L^{i-j}$, decrease geometrically in $L^{-(j-i)}$, with $\sum_{l<j}\varpi_l \le \frac{2c_2^{\mathrm{g}}}{L^2-1}$ and $\sum_{l<j} 2c_4L^{l-j} \le \frac{2c_4}{L-1}$. At level i the one step applies, as $\varpi_i \le 2c_2^{\mathrm{g}} \le \varpi_L$ and $\sup|A_i| \le 2L^i\sup|A_0| < 2c_4L^{i-j} \le \frac{1}{2}r_L$; and the two conditions on c_2^{g} and c_4, $\sum_l C_{\mathrm{P}}\varpi_l \le \ln 2$ and $\sum_l C\lambda_l \le \ln 2$, keep the products $\prod_{l<i}(1 + C_{\mathrm{P}}\varpi_l)$ and $\prod_{l<i}(1 + C\lambda_l)$ at most 2, as $1 + x \le e^x$; this closes the induction on i and gives $|A_i| \le 2L^i\sup|A_0|$ and the plaquettes within $2\alpha L^{-2(j-i)}$. Further $f_{i+1} = f_ie^{i\sigma_i}$ with $\sum_{l<i}|\sigma_l| \le \kappa_6\sum_{l<i}2L^l\sup|A_0| \le 6L^i\sup|A_0| < \pi$, so $f_i \in G^\pi$ (Lem. 9.9(ii)), and $|f_i - \mathbf{1}| \le |\frac{1}{i}\log f_i|$, as $|e^{i\theta} - 1| \le |\theta|$ for real θ. At complex arguments the same recursion runs with $\mathfrak{g}^c$-valued exponents: the bound of Campbell and Hausdorff holds for complex matrices, the products $e^{Z_1}\cdots e^{Z_q}$ with $Z_k \in \mathfrak{g}^c$, $\sum_k |Z_k| < \ln 2$, have their logarithms in $\mathfrak{g}^c$ (Lem. 9.9(iii)), and the averages, composites of exponentials, products, means and logarithms, are holomorphic in a; this gives $Q \in \mathfrak{g}^c$ and the bounds with 167, 9.2 and 8, uniformly in q. If $\varpi_L \ge r_G$, one bond set to $u_* \in G \setminus G^\pi$ with $|u_* - \mathbf{1}| = r_G$ gives two contours without a mean in G. □

Lemma 9.12 (Regular fields and gauges)**.**

(i) For every G the gradient of $\xi^{-1}\sum_p a_p$ for the pairing of Def. YM3_I:9.1 is $J = D^{\mathrm{T}}F$, at G- and at G^c-valued fields.

(ii) Let κ' be the number of the pair of Def. 9.1. Then $4|(1 - P)H_V| \le \kappa'|V - \mathbf{1}|^3$ for $V \in D$ with $|V - \mathbf{1}| \le z_G$; $|(1 - P)H_V| \le (\frac{\pi}{3} - \frac{\sqrt{3}}{2})|1 - P|\,|V - \mathbf{1}|^3$ for every $V \in D$, $|1 - P|$ the norm of $1 - P$ for $|\cdot|$, so that $\kappa' \le 4(\frac{\pi}{3} - \frac{\sqrt{3}}{2})|1 - P|$; at $SU(2)$, $\kappa' = 0$; and at $SU(N)$, for every N, $4|(1 - P)H_V| \le (2 - \pi/\sqrt{3})|V - \mathbf{1}|^3$ for every $V \in D$, so that $\kappa' \le 2 - \pi/\sqrt{3}$, $|(1 - P)H_V| \le \frac{1}{5}|V - \mathbf{1}|^3$ and $|PH_V| \le (1 + \frac{1}{5}|V - \mathbf{1}|^2)|V - \mathbf{1}|$ on D. For every G, if the plaquette variables at the four plaquettes of a bond b lie in D within $\rho \le z_G$ of $\mathbf{1}$, then

$$\eta|(D^{\mathrm{T}}\partial U)(b)| \le \rho^2 + \eta^3|J(b)| + \kappa'\rho^3, \qquad \eta^3|J(b)| \le \eta|(D^{\mathrm{T}}\partial U)(b)| + \kappa'\rho^3.$$

For every G and every r, $\mathcal{U}'(r) \subset \mathcal{U}(\kappa_G r)$, the classes being those of Def. 9.2; for every G and $r \le z_G$, $\mathcal{U}(r) \subset \mathcal{U}'(r + r^2 + \kappa' r^3)$; Moreover, let $\alpha_0, \alpha_2 > 0$ and let E be a real field whose plaquette variables obey $|\partial E(p) - \mathbf{1}| \le (\alpha_0 + 3\alpha_2)\eta^2\mathsf{L}(p)^{-2}$ and whose current at every full bond obeys $|J(E)(b)| \le s\mathsf{L}(b)^{-3}$, where $s = 2\alpha_0 + 3\alpha_2$ and $\alpha_0 + 3\alpha_2 < z_G$. Then $E \in \mathcal{U}'(s + s^2 + \kappa' s^3)$, and hence $E \in \mathcal{U}'(2s)$ if $s(1 + \kappa' s) \le 1$. For $SU(2)$ the two hypotheses are supplied by the action expansion under equations YM3_I:(466)–YM3_I:(467). Their general-group counterparts are quantitative inputs in (P), wherever this conversion is applied to a solved field. The bound $\mathsf{L}(b)^3|J_{\Omega_0}(U_0)(b)| \le 3\alpha_0$ of Prop. YM3_I:30.1 at an inner bond that is not full holds for every G with $3|P|\alpha_0$ in the place of $3\alpha_0$. Let now $G = SU(N)$ and $r \le z_G$. Then $\mathcal{U}'(r) \subset \mathcal{U}((1 + \kappa' r^2)r)$; at $SU(2)$, where $\kappa_G = 1$, $\kappa' = 0$ and $z_G = 1$, the inclusions are the two inclusions of the $SU(2)$ construction. Under the same curvature and

current hypotheses, the membership in $\mathcal{U}'(2s)$ holds for $s \le \frac{1}{3}$; and the bound at an inner bond that is not full holds with $6\alpha_0$, with $3\alpha_0$ exactly when $N \le 7$, and for every N with $3\alpha_0$ if $\alpha_0 \le 2\sin\frac{\pi}{N}$, so if $\alpha_0 \le z_G$. If $|\partial U(p) - \mathbf{1}| \le \rho \le 1$, $\rho < z_G$, at the four plaquettes of a bond b, then $J(U)(b) \in \mathfrak{g}$ and

$$|(D^{\mathrm{T}}\partial U)(b) - i\eta^2 J(U)(b)| \le \eta^{-1}\rho^2\big(1 + \tfrac{3}{5}\rho^2\big)^{1/2} \le \tfrac{13}{10}\eta^{-1}\rho^2.$$

The inclusion $\mathcal{U}'(r) \subset \mathcal{U}(r)$ is false for $N \ge 3$ at every small r, and the guard $\rho < z_G$ cannot be dropped.

(iii) The following quantitative cube-gauge assertion is an additional local analytic input when it is used for a general closed $G \subset U(N)$ of positive dimension. It requires constants $c, B > 0$ depending only on the pair: if Q is a cube of l^3 blocks at level k, the concentric cube $\tilde{Q}$ of side $l + 4$ is shorter than half the torus, $l\alpha \le c$, and $|\partial U(p) - \mathbf{1}|$ and the covariant divergence of ∂U are below $\alpha\eta^2$ on $\tilde{Q}$, then on $\tilde{Q}$ the field is gauge equivalent to $e^{i\eta A}$ with $|A|, |\nabla A|, |d^*dA|, |\Delta A| < Bl\alpha$ on Q. For $SU(2)$ all four bounds hold. The first three are Thm. YM3_I:F; the Laplacian bound follows from the same construction, as proved below, after enlarging B.

The comb gauge of a box of side s (Lem. YM3_I:14.4) gives $|W^h(b) - \mathbf{1}| \le \min\{2s, |x - y|_1\}\varpi$, and $W^h = e^{ia^0}$ with $\sup|a^0| \le \pi s\varpi$ if besides $2s\varpi < 2$ also $\varpi < z_G$; at $SU(N)$ this guard bites from $N = 7$ on. Without the guard, the extension with some gauge in place of the comb is proved by this argument for $N \le 6$. To extend the argument to $7 \le N \le 13$, we must prove the exceptional finite-cube estimates described below. The unguarded clause is false for $N \ge 14$. The quantitative regular-cube assertion for $\mathrm{SU}(N)$ uses the separate analytic input just stated; the comb estimate alone does not supply its derivative bounds.

(iv) For $SU(2)$ the following box-gauge estimate is Prop. 15.8. For other G with centre-free Lie algebra it is an additional analytic input, required in (II): there is C such that for every s and $0 < \kappa \le C^{-2}$ every G-valued field on $\{0, \dots, s\}^3$ with $|\partial U(p) - \mathbf{1}| \le \kappa$ has a gauge transform within $C\sqrt{\kappa}$ of $\mathbf{1}$ at all bonds. The general-group assertion requires a uniform quantitative filling estimate. For a closed G that is not connected this holds if and only if it holds for the identity component, with $C(G) = \max\{C(G_0), (2/r_G)^{1/2}\}$. It fails for every G whose Lie algebra has a centre, and on periodic lattices for $SO(3)$.

■

Sketch of proof. (i) For $\theta \in \mathfrak{g}^c$ and every $X \in G^c$, near $\mathbf{1}$ or not, $\frac{d}{dt}a(e^{it\theta}X)|_{t=0} = \frac{1}{N}\mathrm{tr}(\theta H_X) = \frac{1}{N}\mathrm{tr}(\theta P H_X) = \langle\theta, PH_X\rangle$, since $(1-P)M$ is trace-orthogonal to $\mathfrak{g}^c$ by the definition of P (Def. 9.1). A change $e^{it\theta}$ of one bond changes the plaquette variables of its four plaquettes by conjugates of $e^{\pm it\theta}$ under transports, and $P\,\mathrm{Ad}(g) = \mathrm{Ad}(g)P$ for $g \in G^c$, as $\mathrm{Ad}(g)$ keeps $\mathfrak{g}^c$ and the trace form; summing over the four plaquettes, with the factors ξ of Def. YM3_I:9.1, gives $J = D^{\mathrm{T}}F$ with $F = \xi^{-2}PH_{\partial U}$.

(ii) For $V = e^{iY} \in D$, $Y \in \mathfrak{g}$, $t = |Y| < \frac{\pi}{3}$: $H_V = \sin Y$ and $(1-P)Y = 0$, so $(1-P)H_V = (1-P)(\sin Y - Y)$, of norm at most $|1-P|(t - \sin t) \le (\frac{\pi}{3} - \frac{\sqrt{3}}{2})|1-P|\,|V-\mathbf{1}|^3$, as $|V-\mathbf{1}| = 2\sin\frac{t}{2}$ and $(t-\sin t)/(2\sin\frac{t}{2})^3$ increases on $(0, \frac{\pi}{3}]$. For $V \in G$ with $0 < |V-\mathbf{1}| < z_G$, $V \in D$ (Lem. 9.9(iii)), so that κ' is at most four times the supremum of $|(1-P)H_V|/|V-\mathbf{1}|^3$ over $D \setminus \{\mathbf{1}\}$; and at $V = e^{iY} \in D$ with $|V-\mathbf{1}| = z_G$ the bound $4|(1-P)H_V| \le \kappa'|V-\mathbf{1}|^3$ is the limit of those at e^{isY}, $s \uparrow 1$, which lie in G within z_G of $\mathbf{1}$. At $SU(2)$, H_V lies in $\mathfrak{g}^c$. At $SU(N)$, $|1-P| = 1$ and $(1-P)H_V = \frac{1}{N}\mathrm{Im}\,\mathrm{tr}\,V \cdot \mathbf{1} = \frac{1}{N}\sum_j(\sin\theta_j - \theta_j)\mathbf{1}$, the eigen-angles θ_j of V being at most $t < \frac{\pi}{3}$ in modulus and summing to 0. With $f(x) = x - \sin x$, which is odd, convex on $[0,t]$ and concave on $[-t,0]$, the sum is extreme with a share $p \le \frac{1}{2}$ of the angles at t and the rest equal, at $-\frac{pt}{1-p}$: $\frac{1}{N}|\sum_j f(\theta_j)| \le \max_{0\le p\le 1/2}\psi(p)$, $\psi(p) = pf(t) - (1-p)f(\frac{pt}{1-p})$. The share $p = \frac{1}{3}$, which is extreme to third order in t, gives $\psi(\frac{1}{3}) = \frac{2}{3}\sin\frac{t}{2}\,(1-\cos\frac{t}{2})$, that is $4\psi(\frac{1}{3}) = \frac{|V-\mathbf{1}|^3}{3(1+\cos(t/2))}$, at most $0.1787\,|V-\mathbf{1}|^3$.

The supremum of $4\psi(p)/|V-\mathbf{1}|^3$ over p and $t \le \frac{\pi}{3}$ is below 0.179, so that $4|(1-P)H_V| \le 0.179\,|V-\mathbf{1}|^3$, below $(2-\pi/\sqrt{3})|V-\mathbf{1}|^3$. The bound with $2-\pi/\sqrt{3}$ comes from the tangent of ψ at $p=\frac{1}{3}$: ψ is concave, $\psi''(p) = -t^2(1-p)^{-3}\sin\frac{pt}{1-p}$; with $s=\frac{t}{2}$, $\psi'(\frac{1}{3}) = 3s\cos s - \sin s - \sin 2s \le 0$, as $2\sin s + \tan s \ge 3s$; so $\psi \le \psi(\frac{1}{3}) - \frac{1}{3}\psi'(\frac{1}{3}) = \sin s - s\cos s$ on $[0,\frac{1}{2}]$, and $4(\sin s - s\cos s)/(2\sin s)^3$ increases in s, its derivative having the sign of $x(2+\cos x) - 3\sin x \ge 0$ at $x=2s$, to $2-\pi/\sqrt{3}$ at $s=\frac{\pi}{6}$. At a bond b, $S = \eta(D^{\mathrm{T}}\partial U)(b)$ is the sum of the four plaquette variables transported to b, two with the sign $+$ and two with the sign $-$, and $S = R + iT$ with R, T the like signed sums of the transported $c(\partial U(p))$ and $H_{\partial U(p)}$, both hermitian at real fields, and $PT = \eta^3 J(U)(b)$, P commuting with the transports. The four constants $\mathbf{1}$ cancel and each $c(\partial U(p)) - \mathbf{1}$ lies between $-\frac{1}{2}\rho^2$ and 0, so $|R| \le \rho^2$; and $|(1-P)T| \le \kappa'\rho^3$, the four plaquette variables lying in D within $\rho \le z_G$ of $\mathbf{1}$. Hence $|S| \le |R| + |PT| + |(1-P)T| \le \rho^2 + \eta^3|J(b)| + \kappa'\rho^3$, and $\eta^3|J(b)| = |PT| \le |T| + |(1-P)T| \le |S| + \kappa'\rho^3$. At G-valued fields, for every G, T is hermitian and P commutes with taking adjoints, $\mathfrak{g}^c$ being closed under them, so $PT \in \mathfrak{g}$ and $|PT| \le \kappa_G\|PT\| \le \kappa_G\|T\| \le \kappa_G|T| \le \kappa_G|S|$, P being orthogonal for $\langle\cdot,\cdot\rangle$ (Def. 9.1, Lem. 9.9(i)); at a full bond of a field of $\mathcal{U}'(r)$ this gives $|J(U)(b)| < \kappa_G r\,\mathsf{L}(b)^{-3}$, and as $\kappa_G \ge 1$, $\mathcal{U}'(r) \subset \mathcal{U}(\kappa_G r)$. For every G and $r \le z_G$ the plaquette variables of the two classes lie within z_G of $\mathbf{1}$, so in D (Lem. 9.9(iii)); at a full bond b they are within $\rho = r\eta^2\mathsf{L}(b)^{-2}$ of $\mathbf{1}$ at the four plaquettes p of b, as $\mathsf{L}(p) \ge \mathsf{L}(b)$, each p containing the end points of b (Def. 9.2); with $\eta \le \mathsf{L}(b)$ the first bound gives $|(D^{\mathrm{T}}\partial U)(b)| < (r + r^2\frac{\eta}{\mathsf{L}(b)} + \kappa' r^3\frac{\eta^3}{\mathsf{L}(b)^3})\eta^2\mathsf{L}(b)^{-3} \le (r + r^2 + \kappa' r^3)\eta^2\mathsf{L}(b)^{-3}$ on $\mathcal{U}(r)$, which is the second inclusion, and the second bound gives $|J(U)(b)| < (1+\kappa' r^2) r\,\mathsf{L}(b)^{-3}$ on $\mathcal{U}'(r)$, which is the inclusion $\mathcal{U}'(r) \subset \mathcal{U}((1+\kappa' r^2)r)$. Under the stated curvature and current hypotheses the plaquette variables at a full bond b lie within $\rho = (\alpha_0 + 3\alpha_2)\eta^2\mathsf{L}(b)^{-2}$ of $\mathbf{1}$, so in D, and the current there is at most $s\,\mathsf{L}(b)^{-3}$; the first bound gives $|(D^{\mathrm{T}}\partial U)(b)| \le (s + s^2 + \kappa' s^3)\eta^2\mathsf{L}(b)^{-3}$, as $\alpha_0 + 3\alpha_2 \le s$ and $\eta \le \mathsf{L}(b)$. As $\eta^3 J = PT$ and $|PT| \le |P|\,|T|$, the bound $3\alpha_0$ at an inner bond that is not full holds for every G with $3|P|\alpha_0$, its proof bounding T as at $SU(2)$, where $PT = T$; at $SU(N)$, $|P| \le 2$. At $SU(N)$, $(1-P)T$ is a real multiple of $\mathbf{1}$, so $|R + i(1-P)T|^2 \le |R|^2 + |(1-P)T|^2 \le \rho^4(1+\kappa'^2\rho^2)$, and $(D^{\mathrm{T}}\partial U)(b) - i\eta^2 J(U)(b) = \eta^{-1}(R + i(1-P)T)$ gives the display, as $\kappa'^2 \le \frac{3}{5}$. One excited bond with all other variables $\mathbf{1}$ refutes $\mathcal{U}'(r) \subset \mathcal{U}(r)$; the central plaquette variables $z\mathbf{1}, \mathbf{1}, \mathbf{1}, z\mathbf{1}$, $z = e^{2\pi i/N}$, have $J = 0$ and the left member $2\eta^{-1}z_G$. So each later proof that uses the bound of the regular class goes through with $\kappa_G\alpha_0$, and at $SU(N)$ under $\alpha_0 \le z_G$ with $(1+\kappa')\alpha_0$, each threshold on α_0 read as a ceiling being divided by κ_G, or by $1+\kappa'$.

(iii) For $SU(2)$, use the gauge constructed in the proof of Thm. YM3_I:F. We record the ordinary Laplacian bound as well. In its box notation, put $\epsilon = s\alpha$, $s = l + 4 \le 5l$, and let $d(x)$ be the depth of a site in the box. The smoothed section defines a quaternion-valued one-form ω and a smooth map Θ with

$$A = \eta^{-1}\Theta(\mathbf{1} + \eta\omega), \qquad |\omega| \le 20\epsilon, \qquad |\nabla^\eta\omega| \le C_M\epsilon \quad (d \ge 1).$$

Its equation and the interior estimate in that proof give

$$|\Delta^\eta\omega_\mu(x)| \le 55\epsilon + 120\epsilon \max_{x'\sim x} \eta^{-1}|\omega_\mu(x') - \omega_\mu(x)| \le (55 + C_M)\epsilon \quad (d(x) \ge 1 + \eta),$$

provided $120\epsilon \le 1$. Choose c_Θ to bound the first two derivatives of Θ on $|q - \mathbf{1}| \le 1/2$. Taylor's formula at each of the six neighbours, with the constant terms cancelling and the linear terms combining into the Laplacian of ω, yields

$$|\Delta^\eta A_\mu(x)| \le c_\Theta|\Delta^\eta\omega_\mu(x)| + 6c_\Theta\eta \max_{x'\sim x}\big(\eta^{-1}|\omega_\mu(x') - \omega_\mu(x)|\big)^2 \le c_\Theta(55 + C_M)\epsilon + 6c_\Theta\eta C_M^2\epsilon^2.$$

All arguments of Θ lie in this ball by $20\eta\epsilon \le 1/2$. Since $\eta \le 1/13$ and ϵ has a fixed upper bound, the last expression is at most $C\epsilon$. Bonds with an endpoint in Q have base depth at least $2 - \eta \ge 3/2 > 1 + \eta$; hence the bound is at most $5Cl\alpha$ there. Enlarging B proves the fourth estimate without a further hypothesis for $SU(2)$.

For general G, the quantitative assertion also requires the corresponding derivative bounds for the retracted smoothed section; it remains the local input stated in (iii). For the elementary comb estimate: under $\varpi < z_G$ the bond variables of the comb are products of at most $2s$ conjugated plaquette variables, with logarithms in $\mathfrak{g}$ of size at most $\pi s\varpi$ (Lem. 9.9(ii)). To extend this clause without the guard to SU(N), $N \le 13$, it remains to prove the exceptional cases $s = 1$, $7 \le N \le 13$, and $s = 2$, $N = 13$: these are the only possibilities with $z_G \le \varpi$ and $2s\varpi < 2$, since $2\sin\frac{\pi}{13} < \frac{1}{2} < 2\sin\frac{\pi}{12}$ and $\frac{1}{3} < 2\sin\frac{\pi}{18}$. We must therefore prove the following simultaneous gauge bounds: for $s = 1$, $7 \le N \le 13$, $\varpi < 1$, the six face plaquette bounds on $\{0,1\}^3$ give a simultaneous gauge $U^v = e^{iA}$ with $A(b) \in \mathfrak{su}(N)$ and $|A(b)| \le 6\arcsin(\varpi/2) \le \pi\varpi$ at every bond; and for $s = 2$, $N = 13$, $\varpi \in [2\sin(\pi/13), \frac{1}{2})$, such a gauge on $\{0,1,2\}^3$ has $|A(b)| \le 6\theta \le 2\pi\varpi$ at all 54 bonds, $\theta = 2\arcsin(\varpi/2)$. The construction must achieve these intrinsic-logarithm bounds simultaneously at every bond. At $N = 13$, $\varpi = z_G$, the unit-cube bound is $6\pi/13$, so the necessary one-plaquette lower bound does not by itself prove the simultaneous assertion. For $N \ge 14$ one plaquette variable $e^{2\pi i/N}\mathbf{1}$, at the distance $2\sin\frac{\pi}{N} < 1$, refutes the clause at $s = 1$: $e^{iA_1}\cdots e^{iA_4} = e^{\pm 2\pi i/N}\mathbf{1}$ with $A_j \in \mathfrak{su}(N)$ forces $\sum_j |A_j| \ge 2\pi\frac{N-1}{N}$, so $\max_j |A_j| \ge \frac{\pi}{2}\frac{N-1}{N} > 2\pi\sin\frac{\pi}{N}$.

(iv) The $SU(2)$ proof uses a quantitative filling of boundary maps in the quaternionic sphere. For general simply connected G, avoidance of the singular set $\{u : \dim\ker(\mathrm{Ad}\,u - 1) \ge \mathrm{rk}\,G + 2\}$ alone does not give a logarithm or extension with a uniform Lipschitz bound. Such a quantitative filling bound, together with its lifting to quotients, is needed to deduce the asserted estimate; it is retained as part of the analytic input in (II). For $G = U(1)$ the box estimate fails at constant field strength: let U have the plaquette angle κ at every plaquette of one plane direction and 0 at the others, as $U(\langle x, x + e_2\rangle) = e^{i\kappa x_1}$ and $\mathbf{1}$ at the other bonds does. Its plaquette variables are within κ of $\mathbf{1}$. In a gauge with all bonds within $C\sqrt{\kappa}$ of $\mathbf{1}$, $\kappa \le \frac{1}{4}C^{-2}$, the angles $a(b)$ of the bonds obey $|a(b)| \le \frac{\pi}{2}C\sqrt{\kappa}$, so $da = \kappa$ exactly at those plaquettes, and summing over a square of side l in that plane gives $\kappa l^2 \le 4l \cdot \frac{\pi}{2}C\sqrt{\kappa}$, which is false for $l > 2\pi C\kappa^{-1/2}$; so no C serves for all s, and the same field strength in a central direction gives the failure whenever $\mathfrak{g}$ has a centre. With the lemma fail the vanishing of invariant linear forms (the constant-gauge argument following equation YM3_I:(943)) and further statements, among them a step of the proof of Prop. YM3_I:42.1, so the theorems ask $\mathfrak{g}$ semisimple. A flat $SO(3)$-field in a twisted sector of a torus is no gauge transform of $\mathbf{1}$. □

Lemma 9.13 (Energy of block averages and large fields)**.**

(i) Let $G \subset SU(N)$ be compact, $L \ge 13$ odd, $\mathcal{M}$ a G-valued averaging map that equals the formula of Lem. 9.11(i) where $|\partial U(p) - \mathbf{1}| \le \varpi'(L, r_G) \le \frac{1}{2}r_G$, and $e_* = e_L/(2\tau_G^2)$, where $e_L > 0$ is a constant depending only on L and the pair with $(2e_L/L)^{1/2} \le \varpi'(L, r_G)$ and $(N-1)^{1/2}C_{\mathrm{en}}(L)(2e_L)^{1/2} \le \frac{1}{2}(L^{1/2} - 1)\ln 2$, $C_{\mathrm{en}}(L)$ being the number of L in the cost of one step of the proof. For all K, tori, $1 \le t \le K$, sets S of plaquettes of level t and fields U, with $\mathcal{N}(S)$ the bare plaquettes beneath S, of all three pairs of directions (the cones defined at the beginning of Section YM3_I:15),

$$L^t \sum_{\mathcal{N}(S)} a_p(U) \ge \min\Big\{\frac{1}{4\tau_G^2}\sum_{p'\in S}|\partial\mathcal{M}^t U(p') - \mathbf{1}|^2,\ e_*\Big\};$$

as $\tau_G^2 \le N - 1$, the same holds with $N - 1$ in the place of τ_G^2. No number above $\frac{1}{2\kappa_G^2}$ serves for $\frac{1}{4\tau_G^2}$, and $\frac{1}{4}$ fails where $\kappa_G^2 > 2$, so at $SU(N)$ for $N \ge 4$. At $SU(3)$, $\frac{1}{4}$ serves for every set S in which no site is a corner of two plaquettes of S of different pairs of directions, with a threshold $e(L) > 0$ in the place of e_*; and $\frac{1}{4(1+\delta)}$ serves for every S and every $\delta > 0$, with a threshold of L and δ.

(ii) Let $\mathfrak{g}$ be semisimple, so that $G \subset SU(N)$ and (i) applies, $L \ge 13$ odd, and let the averaging radius $\min\{\omega_M, r_G\}$ stand for ω_M. Assume the transferred witness and local-term estimates in (II). Then the large-field chain holds for G with the following prices. Its first link is

Prop. YM3_I:44.4 and Prop. YM3_I:44.5: a failed plaquette test or a failed approximate fluctuation test of a test cube (equations YM3_I:(636)–(637); conditions in $|\cdot|$ on plaquette variables and on averaged contour variables) forces a plaquette variable or an averaged contour variable farther than the numbers of the $SU(2)$ construction from $\mathbf{1}$, or an averaged contour variable that is not formed. The fluctuation-test implication is used with $V_t = \mathcal{M}(V_{t-1})$ on the inner level-t bonds of the tested region, as required by Prop. YM3_I:44.5; the successive block fields satisfy this local averaging identity. Its second link is Lem. YM3_I:65.2, on the witnesses of the cells created at one step, with "$\notin G^\pi$" for "$= -\mathbf{1}$"; The local-deposit argument in the proof of Prop. YM3_I:65.4 uses its alternatives for a failed plaquette or fluctuation test, a failed density test, and a large or undefined stored bond. Its third link is the local-deposit argument in the proof of Prop. YM3_I:65.4, the lower bounds of the amount of action near a created cell, which hold with the expressions of the $SU(2)$ construction, taken in constants of the pair, times τ_G^{-2}. Here an energy pair (c, e) means that the action beneath one level-t plaquette p' obeys

$$L^t \sum_{p \in \mathcal{N}(p')} a_p(U) \geq \min\{c|\partial \mathcal{M}^t U(p') - \mathbf{1}|^2, e\}.$$

Taking $S = \{p'\}$ in (i) gives the pair $(\frac{1}{4\tau_G^2}, e_*)$; and the deposit constant

$$c_\star = 5^{-3} \min\{\kappa_{\mathrm{W}}, \kappa_{\mathrm{P}}, \kappa_{\mathrm{Q}}, \kappa_{\mathrm{F}}\}$$

of equation YM3_I:(1429), where

$$\kappa_{\mathrm{W}} = \tfrac{1}{2}, \qquad \kappa_{\mathrm{P}} = \tfrac{1}{2} c_{\mathrm{P}}, \qquad \kappa_{\mathrm{Q}} = \tfrac{1}{2} c_{\mathrm{Q}} (A_1/A_0)^2, \qquad \kappa_{\mathrm{F}} = \tfrac{1}{4} \gamma_{\mathrm{kp}} \varkappa_{\mathrm{w}}^2 (A_1/A_0)^2,$$

formed from the four prices per created cell, so that each created cell of level t gives the small factor $e^{-c_\star p_0(g_t)^2}$, is at least τ_G^{-2} times its expression in the $SU(2)$ construction, taken in constants of the pair.

■

Sketch of proof. (i) Sizes are followed in $|\cdot|$ and energies in $\|\cdot\|$, where $a = \frac{1}{2}\|V - \mathbf{1}\|^2$ exactly, and at one plaquette $|V - \mathbf{1}|^2 \leq 2\tau_G^2 a(V)$ (Def. 9.1, Lem. 9.9(i)). Let E_j be the energy $\sum a_p(\mathcal{M}^j U)$ over the plaquettes of level j beneath S, of all three pairs of directions, and ϖ_j the largest $|\partial \mathcal{M}^j U(p) - \mathbf{1}|$ among them. Where $\varpi_j \leq \varpi'(L, r_G)$ the averages are given by the formula of Lem. 9.11(i), and one step costs, between square roots of energies, the factor $L^{1/2}(1 + (N-1)^{1/2} C_{\mathrm{en}}(L) \varpi_j)$, the exact $L^{1/2}$ carrying the comparison of two cutoffs. Let $L^t E_0 < e_*$, the other case being plain. By induction on j, $E_j \leq 2L^j E_0$: if this holds up to $j < t$, then $\varpi_i^2 \leq 2\tau_G^2 E_i \leq 4\tau_G^2 L^i E_0 < 2e_L L^{-(t-i)}$ for $i \leq j$, so that $\varpi_i \leq (2e_L/L)^{1/2} \leq \varpi'(L, r_G)$, and the product of the factors up to j is at most $\exp((N-1)^{1/2} C_{\mathrm{en}}(L)(2e_L)^{1/2} \sum_{m \geq 1} L^{-m/2}) \leq \sqrt{2}$, which gives $E_{j+1} \leq 2L^{j+1} E_0$. At $j = t$, $\sum_{p' \in S} |\partial \mathcal{M}^t U(p') - \mathbf{1}|^2 \leq 2\tau_G^2 E_t \leq 4\tau_G^2 L^t E_0$, which is the inequality. The constant $\frac{1}{2\kappa_G^2}$ is approached by abelian fields $e^{i\theta Y}$ of small constant field strength in one plane along a direction $Y \in \mathfrak{g}$ with $|Y| = \kappa_G \|Y\|$: the averages of such a field are exact, L^t times the bare energy beneath a coarse plaquette, carried by L^{3t} parallel bare plaquettes of angle φ, is $\frac{1}{2} L^{4t} \varphi^2 \|Y\|^2$ to leading order, and the coarse plaquette variable is within $L^{2t}\varphi|Y|$ of $\mathbf{1}$. At $SU(3)$ let S have one pair of directions, and let $E_\parallel$ and $E_\perp$ be the sums of $a_p(U)$ over the bare plaquettes beneath S parallel and transverse to that pair. Below a threshold of L, $\sum_{p' \in S} a_{p'}(\mathcal{M}^t U) \leq L^t(E_\parallel + w_\perp E_\perp)$ with a weight $w_\perp \leq C(L)\big(L^t \sum_{\mathcal{N}(S)} a_p(U)\big)^{1/2} \leq \frac{1}{2}$. For $W = e^{i\epsilon A}$ with constant field strengths $F_{\mu\nu} = dA$ in $\mathfrak{g}$, the defect of one step at one coarse plaquette of the pair 12 contains the cubic term $-\epsilon^3 \frac{L^2-1}{12} L^4 \langle F_{12}, i[F_{13}, F_{23}] \rangle$, which has either sign; it stands at every coarse plaquette, coming from the variance $\frac{L^2-1}{12}$ of the transverse coordinate of the bare plaquettes beneath, and it is paid by a small part of the energy of the transverse plaquettes beneath, whose weights stay below

1 over the t steps, the errors summing as a geometric series in $L^{-1/2}$. If no site is a corner of two plaquettes of S of different pairs of directions, the cones of members of S of different pairs are disjoint, so the transverse energy spent for one pair is claimed by no plaquette of S, and $\sum_{p'\in S} a_{p'}(\mathcal{M}^t U) \le L^t \sum_{\mathcal{N}(S)} a_p(U)$. For general S the plaquettes beneath members of two pairs add at most $2C(L)\big(L^t \sum_{\mathcal{N}(S)} a_p(U)\big)^{1/2}$ times $L^t \sum_{\mathcal{N}(S)} a_p(U)$, which below a threshold of L and δ is at most δ times it. With $a(V) \ge \frac14 |V-\mathbf{1}|^2$ on $SU(3)$ this gives the two forms.

(ii) A test is a condition in $|\cdot|$ on plaquette variables and on averaged contour variables. A test passed at all plaquettes of a double block puts the field under the averaging radius $\min\{\omega_M, r_G\}$, where every contour family has exactly one mean, within $4d$ (Lems. 9.9(iv), 9.11(i)); by contraposition a failed test forces a plaquette variable or an averaged contour variable farther than the numbers of the $SU(2)$ construction from $\mathbf{1}$, or an averaged contour variable that is not formed: this is the first link, with the numerals of the $SU(2)$ construction, the straightening of Lem. 9.11(iii) holding for every closed G. The geometric and running-coupling statements used in the local-deposit argument of Prop. YM3_I:65.4, besides Lem. YM3_I:65.2 involve cubes, regions and runs of couplings alone, read no group and hold for G as written. The witness and local-deposit estimates of Lem. YM3_I:65.2 and the local-deposit argument in the proof of Prop. YM3_I:65.4 are the second-part inputs assumed in (II); their inputs ask that the terms of each level be bounded in norm by E_0 and the increments of the run by $B_\sharp$, separately, and the bounds of a step add these sizes; and a family of local terms splits off from the effective action of a large-field region, with a contribution bounded by a small factor times its norm times a factor that decays exponentially in the size of the region.

The group enters the prices alone, where the local-deposit argument in the proof of Prop. YM3_I:65.4 turns the witnesses of Lem. YM3_I:65.2 into lower bounds of the action: at one plaquette through $a(V) \ge \frac12 \tau_G^{-2} |V-\mathbf{1}|^2$ (Lem. 9.9(i)), and at averaged variables through (i), whose pair $(\frac{1}{4\tau_G^2}, \frac{e_L}{2\tau_G^2})$ takes the place of the pair $(\frac14, \frac{e_L}{2})$ of the $SU(2)$ construction. So every price of the local-deposit argument in the proof of Prop. YM3_I:65.4 is the expression of the $SU(2)$ construction, in constants of the pair, times τ_G^{-2}, each of the four prices per created cell is at least τ_G^{-2} times its expression, and so is $c_\star$. For simply connected G with $\mathfrak{g}$ simple this chain is among the statements of §9.1.1(II). □

Lemma 9.14 (The Haar measure in the chart, the cut-off and determinants)**.**

(i) For $u_0 \in G$ and Borel $F \ge 0$, $\int_{G^\pi u_0} F \,\mathrm{d}u = \mathrm{vol}(G)^{-1} \int_{|Y|<\pi} F(e^{iY}u_0)\, j_G(Y)\,\mathrm{d}Y$, $j_G = \prod_\alpha (\frac{\sin(\alpha/2)}{\alpha/2})^2$ over the positive eigenvalues α of $\mathrm{ad}\, Y$; $|\log j_G(Y)| \le (n - \mathrm{rk}\, G)(-\log \frac{\sin|Y|}{|Y|})$ on $|Y| < \pi$, and $|\log j_G(Y)| \le 0.173\, n|Y|^2$ on $|Y| \le 1$; the quadratic part of $\log j_G$ is $-\frac{1}{24}\sum_i C_A^{(i)} \|P_i Y\|^2$. The identities of the $SU(2)$ construction that integrate in the chart hold for G with the indicator of $e^{iA}e^{iH} \in G^\pi$ in the integrand, $A, H \in \mathfrak{g}$ being the two exponents of the product that such an identity integrates in the chart. $G \setminus G^\pi$ is a null set if and only if $r_G = 2$; otherwise it has inner points.

(ii) Let ϑ_{co} be the profile of the cut-off of the bond shift (the cutoff preceding Lem. YM3_I:53.3), smooth, 0 on $(-\infty, \frac43]$ and 1 on $[\frac53, \infty)$, with the absolute constant $\|\vartheta'_{\mathrm{co}}\|_\infty \ge 3$, and $\delta \in (0,1]$ its scale. If $w = N^{1/2p} < \frac54$ and $\lambda = \frac{w}{5-4w}$, then, with $\vartheta_G(s) := \vartheta_{\mathrm{co}}(\frac43 + \lambda(s - \frac43))$, the function $\vartheta_G(\nu(Y)/\delta)$ is smooth and invariant, 0 on $|Y| \le \frac43\delta$ and 1 on $|Y| \ge \frac53\delta$, and $\|\vartheta'_G\|_\infty = \lambda \|\vartheta'_{\mathrm{co}}\|_\infty \ge 3$; the cut-off $\vartheta_{\mathrm{co}}(|Y|/\delta)$ itself is not differentiable for rank at least 2.

(iii) If $T_j = L^3 \mathrm{Ad}(g_j)\, \mathsf{m}\, \mathrm{Ad}(g'_j)^{-1}$, $g_j, g'_j \in G$, $|T_j - 1| \le r < 1$ $(j = 1,2)$, then m is invertible, $|\mathrm{tr} \log T_j| \le -n \log(1-r)$ and $\mathrm{tr} \log T_1 = \mathrm{tr} \log T_2$. For every compact connected G the elimination factor $\mathsf{m}_c(V)$ of Lem. 9.14(iii) is formed, as a complex linear map of $\mathfrak{g}^c$ into itself, at every V of the domain of the complex average at c: the G^c-valued V on P_c whose contour families have diameter below $\frac{1}{40}$ and whose contour quotients and variables W_x lie

in D^c; at G-valued V it maps $\mathfrak{g}$ into $\mathfrak{g}$. For $\mathfrak{g}^c$-valued a on P_c with $\sup|a| < \frac{3}{4}r_*$,

$$|\mathsf{m}_c(e^{ia}) - L^{-3}| \le L^{-1}C_S \sup|a|,$$

where $0 < r_* \le \frac{1}{600L}$, $r_1 = \frac{r_*}{40}$, $m_1 = \max\{\frac{1}{r_1}, \frac{4}{r_1^2}\}$ and $C_S \ge \frac{8m_1}{3r_*}$: numbers of L, free of k, N and G.

(iv) Let $n \ge 2$. If T acts on an n-dimensional space and $|1-T| \le r_n := \sin\frac{\pi}{2n}$, then $\det T \notin (-\infty, 0]$, $\operatorname{Log}\det T = \operatorname{tr}\log T$ and $|\operatorname{Log}\det T| \le -n\log(1-r_n) =: \rho_n$, with $\frac{\pi}{2} < \rho_n < \frac{5}{2}$; and $|\operatorname{Log}\det(1+E)| \le c_n|E| \le \rho_n$ for $|E| \le r_n$, where $c_n := \rho_n/r_n = -\frac{n}{r_n}\log(1-r_n)$, which is below 5 for $n \le 3$ and above 5 for $n \ge 4$.

The radius $\frac{1}{2}$ keeps $\det T$ off $(-\infty, 0]$ if and only if $n \le 5$. The following two forms of the elimination estimate use this radius; their three-dimensional counterparts are the proof of Prop. YM3_I:50.2 and the argument preceding equation YM3_I:(1393). Let $K := LC_SC''_{\mathrm{ia}}$, with C_S, r_* as in (iii) and $0 < c_\sharp \le 4\cdot 10^{-7}$ and $C''_{\mathrm{ia}} \ge 8$, using the last, trivial-background estimate of Lem. 9.11(v), read at the trivial background. In the chain of the localised Gaussian family, whose radius a_{N} is taken with $0 < a_{\mathsf{N}} \le \min\{\frac{1}{2}, \frac{Lc_\sharp}{8}, \frac{Lr_*}{8C''_{\mathrm{ia}}}, \frac{r_n}{4K}\}$, the normalised factor T stays within r_n of 1 and the function z_c of the elimination factor obeys $|z_c - 2n\log L| \le \rho_n$, for every G. The corresponding linear estimate is $|z_c - 2n\log L| \le C_z a$ on fields of size a for every $C_z \ge \max\{1, 4c_nK\}$ and every a with

$$0 < a \le \min\Big\{\frac{1}{2},\ \frac{Lc_\sharp}{8},\ \frac{3Lr_*}{32C''_{\mathrm{ia}}},\ \frac{r_n}{4K},\ \frac{L\varrho_G^{\log}}{4}\Big\},$$

$\varrho_G^{\log} \in (0, \log 2]$ being the radius of the logarithm on G^c, the largest $\varrho \le \log 2$ such that $X \in G^c$, $|X - \mathbf{1}| < 1$ and $|\log X| < \varrho$ give $\log X \in \mathfrak{g}^c$; $\varrho_G^{\log}$ is positive, $G^c = G\exp(\mathfrak{g})$ being a closed subgroup of $GL(N,\mathbb{C})$ with the Lie algebra $\mathfrak{g}^c$, it is at least $\min\{\log 2, 2\arcsin\frac{z_G}{2}\}$ for semisimple $\mathfrak{g}$ (Lem. 9.9(iii)), and at $SU(N)$ and at $U(N)$ the last member of the minimum is implied by the others. So all these constants are explicit in L and n at $SU(N)$ and $U(N)$, and for the other G the radius of the second chain reads the pair through $\varrho_G^{\log}$.

■

Sketch of proof. (i) For $SU(2)$, write $U = \cos r\,\mathbf{1} + i\sin r\,\omega$, where $0 \le r \le \pi$ and $\omega \in \mathfrak{su}(2)$ has norm one. This identifies $SU(2)$ with the unit sphere in $\mathbb{R}\mathbf{1} + i\mathfrak{su}(2)$. Left and right multiplication preserve its Euclidean norm, so its normalized surface measure is Haar measure. In these coordinates it is $(2\pi^2)^{-1}\sin^2 r\,\mathrm{d}r\,\mathrm{d}\omega$, whereas $dY = r^2 dr\,\mathrm{d}\omega$ for $Y = r\omega$. Therefore

$$\int_{SU(2)} F(U)\,\mathrm{d}U = \frac{1}{2\pi^2}\int_{|Y|<\pi} F(e^{iY})\left(\frac{\sin|Y|}{|Y|}\right)^2 dY.$$

The omitted point $-\mathbf{1}$ has surface measure zero. For general G, the differential of $\exp(iY)$, after translation back to the identity, is $i\int_0^1 e^{-it\,\mathrm{ad}Y}\,\mathrm{d}t$. The paired eigenvalues $\alpha, -\alpha$ of $\mathrm{ad}Y$ contribute $(\sin(\alpha/2)/(\alpha/2))^2$ to its real Jacobian; a zero eigenvalue contributes one. The principal logarithm is unique on $|Y| < \pi$, so change of variables on $G^\pi u_0$ gives the formula in (i), with the normalization $\mathrm{vol}(G)^{-1}$. The range of the chart is the one in Lem. 9.9(ii); the eigenvalues of $\mathrm{ad}\,Y$ are at most $2|Y|$ in modulus and at least $\mathrm{rk}\,G$ of them vanish, and on $|Y| \le 1$ the bound $0.173\,n|Y|^2$ follows because $x^{-2}(-\log\frac{\sin x}{x})$ increases and is 0.1726 at $x = 1$. For the nonnegative function F in (i), the same formula with $\int_G$ on the left holds whenever F vanishes Haar-almost everywhere off $G^\pi u_0$. Conversely, if the integral over $G^\pi u_0$ is finite, equality requires this vanishing. The $SU(2)$ construction bounds the one-bond functions ls_b defined before equation YM3_I:(1381), the logarithms $-\log\sigma_1$ of the density at the arguments $g_k\mathsf{a}(b)$, by 1 on their domain $\{g_k|\mathsf{a}(b)| < 1\}$ (the Haar-term calculation in the proof of Prop. YM3_I:62.1), and evaluates them with room at arguments of modulus below $\frac{3}{8}$ (the same calculation: $g_k|\mathsf{a}+\mathsf{e}| < 1/8 + 2\rho_{\mathrm{bd}} \le 3/8$). For G the logarithm of j_G is formed and

holomorphic on $\{Y \in \mathfrak{g}^c : |Y| < \pi\}$, so the domain stays, and with $\lambda(x) = -\log \frac{\sin x}{x}$ its modulus is at most $(n - \operatorname{rk} G)\lambda(1)$, a number of the pair, on $\{Y \in \mathfrak{g}^c : |Y| < 1\}$. On $\mathfrak{g}$ at $SU(N)$, λ being convex and increasing, the supremum of $|\log j_G|$ on a ball $|Y| \le r$ is taken with the eigenvalues of Y at $\pm r$ (and one at 0 for odd N): it is at most $\frac{N^2}{2}\lambda(r)$, with equality for even N, so about $0.0863N^2$ at $r = 1$ and $0.0118N^2$ at $r = \frac{3}{8}$, and at $r = \frac{3}{8}$ it is below 1 exactly when $N \le 9$. The bound enters Prop. YM3_I:53.15, the transfer of these one-bond functions, linearly, as the size of the function transferred in equation YM3_I:(1039); so a constant of the pair takes the place of 1 there, the constants that use this transfer bound absorb it in a maximum, and neither the domain nor the radius $\frac{3}{8}$ moves.

(ii) ν is smooth off 0, invariant, and $\nu \le |\cdot| \le w\nu$; so $\nu(Y)/\delta \le \frac{4}{3}$ on $|Y| \le \frac{4}{3}\delta$, and on $|Y| \ge \frac{5}{3}\delta$ the argument of ϑ_{co} is at least $\frac{4}{3} + \lambda(\frac{5}{3w} - \frac{4}{3}) = \frac{5}{3}$.

(iii) T_2 is conjugate to $T_1 W^{-1}$ with $W = \mathrm{Ad}(h)$; $\operatorname{tr}\log X - \operatorname{tr}\log(XW^{-1})$ lies in $2\pi i\mathbb{Z}$, is constant on the convex set $\{|X-1| \le r, |X-W| \le r\}$, and vanishes at $\frac{1}{2}(1+W)$, the eigen-angles of $\mathrm{Ad}(h)$ being symmetric. The elimination factor is read off the relative average $B \mapsto \frac{1}{i}\log[\mathcal{M}(e^{iB}V)(c)\mathcal{M}(V)(c)^{-1}]$: $L\,\mathsf{m}_c(V)X$ is the derivative at $s = 0$ of its value at the field equal to sX on the pivot bond and to 0 on the other bonds of P_c, so that $L\,\mathsf{m}_c(V)$ is the coefficient of the pivot bond in its derivative at $B = 0$. For V in the domain and B small, $e^{iB}V$ stays in it, its quotients and its W_x staying in D^c by the rule for products of Lem. 9.9(iii), so the relative average is formed by the series, holomorphic and $\mathfrak{g}^c$-valued, $\mathfrak{g}$-valued at G-valued V and real B, and $\mathsf{m}_c(V)$ maps $\mathfrak{g}^c$ into itself. At $V = e^{ia}$ with $\sup|a| < \frac{3}{4}r_*$ and $\sup|B| < r_1$ the relative average is at most 0.55 and holomorphic in (a, B); so, by Cauchy's estimate in B, $|L\,\mathsf{m}_c(e^{ia})| \le \frac{1}{r_1}$, and $L\,\mathsf{m}_c(e^{ia})$ is holomorphic in a. At $V = \mathbf{1}$ the relative average is $\frac{1}{i}\log\mathcal{M}(e^{iB})(c)$, whose derivative at $B = 0$ is the linear average with real weights, which gives the pivot bond the weight L^{-2}; so $L\,\mathsf{m}_c(\mathbf{1}) = L^{-2}$, that is $\mathsf{m}_c(\mathbf{1}) = L^{-3}$. Finally $h(a) = L\,\mathsf{m}_c(e^{ia}) - L^{-2}$ is holomorphic on $\sup|a| < \frac{3}{4}r_*$, vanishes at 0 and has $|h| \le \frac{1}{r_1} + L^{-2} \le 2m_1$, so Schwarz's lemma gives $|h(a)| \le \frac{8m_1}{3r_*}\sup|a|$; as $\mathsf{m}_c(e^{ia}) - L^{-3} = L^{-1}h(a)$, this is the bound for every $C_S \ge \frac{8m_1}{3r_*}$.

(iv) Each eigenvalue of T lies within r_n of 1 and so has an argument of modulus at most $\arcsin r_n = \frac{\pi}{2n}$; the arguments add up to at most $\frac{\pi}{2}$; at $\cos\frac{\pi}{n}\,e^{i\pi/n}1$ the radius $\frac{1}{2}$ fails for $n \ge 6$. As $-\log(1-x)$ is convex on $[0, r_n]$, $|\mathrm{Log}\det(1+E)| \le -n\log(1-|E|) \le \frac{|E|}{r_n}\rho_n = c_n|E|$; and $c_2 < 3.48$, $c_3 = 6\log 2 < 4.16$, $c_n > 5$ for $n \ge 4$. In the chains the radii are bounded from above only and the bounds from below only. What the chains need at G is $4Ka \le r_n$, which keeps T within r_n of 1 through $|1 - L^3\mathsf{m}_c| \le L^2 C_S C''_{\mathrm{ia}}\sup|\varphi|$ at potentials with $\sup|\varphi| < \frac{4a}{L}$, and $C_z \ge 4c_nK$; for $n \ge 4$ the radii of the $SU(2)$ construction, which give $4Ka \le \frac{1}{2}$, fall by the factor $2r_n$, and its $C_z \ge 20K$ rises by the factor $\frac{c_n}{5}$, which gives these two inequalities, the two of the $SU(2)$ construction holding besides as $r_n < \frac{1}{2}$ and $c_n > 5$; for $n = 2, 3$ the constants of the $SU(2)$ construction serve, as $r_n \ge \frac{1}{2}$ and $c_n < 5$. The member $\frac{L\varrho_G^{\log}}{4}$ of the second radius keeps $\frac{4a}{L} \le \varrho_G^{\log}$, so that the logarithms of that chain lie in $\mathfrak{g}^c$. The one input "z_c is formed at $(U^*)^{-1}$ if and only if at U" fails from $SU(3)$ on (at $\mathrm{diag}(1.9, 1.9, \frac{1}{3.61})$); the chain needs only conjugate values where both are formed (Lem. 9.20), and with the two-sided guard $m(X)$ the clause holds for every G. □

9.1.6. *Scalar kernels and the one-loop calculation.*

Lemma 9.15 (Colour at the trivial field)**.**

(i) Let $\mathfrak{g}$ be semisimple. Every $\mathrm{Ad}(G)$-invariant complex bilinear form on $\mathfrak{g}^c$ is $q = \sum_i q_i\langle P_i\cdot, P_i\cdot\rangle$, $q_i = \frac{1}{n_i}\sum_a q(t_a^{(i)}, t_a^{(i)})$ over an orthonormal basis of $\mathfrak{g}_i$; every invariant form is a multiple of $\langle\cdot,\cdot\rangle$ if and only if $\mathfrak{g}$ is simple, and then C_A and C_ρ do not depend on H; invariant linear forms vanish, and a vector fixed by $\mathrm{Ad}(G)$ is 0. For semisimple $\mathfrak{g}$ and a unit H, $C_A(H) = \operatorname{tr}(\operatorname{ad} H)^2 = \sum_i C_A^{(i)}\|P_iH\|^2$, which does not depend on H if and only if all $C_A^{(i)}$ agree: for

$SU(2)\times SU(3)$ on $\mathbb{C}^2\oplus\mathbb{C}^3$ the $C_A^{(i)}$ are 20 and 30, for $SU(2)\times SU(2)$ on $\mathbb{C}^2\oplus\mathbb{C}^2$ both are 16. For simple $\mathfrak{g}$ and $z>0$ the Hessian at $U=\mathbf{1}$ of $\ln\det(D^{\mathrm{T}}D+z)-\frac{1}{2}\ln\det(A''+DD^{\mathrm{T}}+z)$, A'' the Hessian of the action, depends on the pair only through C_A and C_ρ, exactly, on every finite lattice.

(ii) At $U=\mathbf{1}$ every linear operator of a step (averaging operators, Laplacians, gauge-fixing projections, propagators, the fluctuation operator) is $K\otimes 1_{\mathfrak{g}^c}$, K a real matrix formed from the lattice alone; so the linear average is $Q_k\otimes 1$, the Hessian is $d^*d\otimes 1$, the projection R is $R_{\mathrm{real}}\otimes 1$ and the propagator is $\mathsf{M}^{-1}\otimes 1$, $\mathsf{M}=d^*d+dR_{\mathrm{real}}d^*+Q_k^*Q_k$. For such operators: $K\mapsto K\otimes 1_V$ respects sums, products, inverses and adjoints; for pairwise disjoint sets P_j that cover the support of $K(x',\cdot)$ and numbers $w_j\geq 0$, $|(K_Vf)(x')|\leq\sum_j w_j\sup_{P_j}|f|$ holds for one normed space V if and only if $\sum_{x\in P_j}|K(x',x)|\leq w_j$, so for one V if and only if for all; the best constants of ℓ^2 bounds and of lower bounds for inner product norms do not depend on V; and $|l_0(f)|\leq\Phi(|l_1(f)|,\dots,|l_m(f)|)$, with real linear forms l_j and non-decreasing Φ, holds for functions with values in one normed space if and only if it holds for real functions. So the bounds of the $SU(2)$ construction at the trivial field hold for G with their constants, each in the norms to which its kind passes: the bounds in suprema, by row sums and by non-decreasing functions of real linear forms in $|\cdot|$ and in $\|\cdot\|$ alike; the ℓ^2 bounds, the quadratic forms and the lower bounds, the lower bound of the fluctuation operator among them, with its number of L alone, in the inner-product norm $\|\cdot\|$, the sign $|X|^2$ of the $SU(2)$ construction for $X\in\mathfrak{g}$ being read as $\langle X,X\rangle$. A bound of a sum of squares of $|\cdot|$ retains its constant where it follows from Schur's test on $|K|$ or from convexity. Otherwise the comparison $\|X\|\leq|X|\leq N^{1/2}\|X\|$ costs at most $N^{1/2}$ in an unsquared ℓ^2 norm estimate and at most N in its squared form.

■

Sketch of proof. (i) is Schur's lemma for $\mathrm{Ad}(G)$ on each simple ideal; the ideals are pairwise orthogonal for every invariant form, and invariant linear forms vanish because $[\mathfrak{g},\mathfrak{g}]=\mathfrak{g}$. For $\mathfrak{su}(m)$, $\mathrm{tr}(\mathrm{ad}\,H)^2=2m\,\mathrm{tr}\,H^2$, and $\|H\|=1$ in $\frac{1}{5}\mathrm{tr}$ means $\mathrm{tr}\,H^2=5$; this gives 20 and 30. The statement on the Hessian is the computation of Prop. 9.18, made on one finite lattice.

(ii) The averages of $\mathbf{1}$ are $\mathbf{1}$; by the chain rule the first-order part of the average at $\mathbf{1}$ is the linear average, the straight average plus the gradient of the block mean of the contour sums, and that of the straightened averages is the straight average, the contour terms cancelling; all have real weights formed from the lattice alone; the second variation of the action at $\mathbf{1}$ is $\sum_p\|dA(p)\|^2$, colour by colour; and a holomorphic map that is $\mathfrak{g}$-valued at real points is $\mathfrak{g}^c$-valued. The transfer of bounds is linear algebra, and a norming functional reduces a normed space to $\mathbb{R}$; for the row sums, f equal to the sign of $K(x',\cdot)$ times a unit vector on one P_j gives the converse. In an orthonormal basis (t_a) of $\mathfrak{g}$, $(F,(X\otimes 1)F)=\sum_a\langle F^a,XF^a\rangle$ for real F, so quadratic forms and their lower bounds pass between inner-product norms; $|\cdot|$ on $\mathfrak{g}$ is no inner-product norm for rank at least 2, and read with the operator norm a clause of this kind can fail: at $SU(3)$ the floor of the $SU(2)$ construction for a datum $X\delta_b$, $X=\mathrm{diag}(2,-1,-1)$, gives $4\|X\|^2=8$, against $4|X|^2=16$. Where a sharp linear bound for G is wanted, it is read off a nonlinear bound of the $SU(2)$ construction along $f=tg\sigma_3$, $t\to 0$, with Cauchy's inequality for the rest. □

The uniqueness argument needs a uniform gauge slice and a contraction estimate in that slice. We state the required input explicitly, including the extra rows at the interfaces. The result remains conditional for a general group. The reduced $SU(2)$ problem is treated in Props. YM3_I:30.9 and YM3_I:37.3; those statements do not assert the additional interface-row estimates below.
The slice and nonlinear estimates still required. We specify the regional geometry and row maps for the near-trivial uniqueness reduction. The uniform slice and nonlinear contraction estimates described here remain to be proved for the general pair. Fix the pair, an odd $L\geq 13$ with

$200L^2 z_G \geq 1$, and the separated regional geometry, with

$$\mathfrak{M} \geq M_{\mathrm{ps}}, \qquad R_1 \geq L, \qquad R_1\mathfrak{M} \geq w_{\mathrm{S}}, \qquad \#\mathbb{T}^{(k)} \text{ in each direction} \geq 6LM_{\mathrm{ps}}.$$

The separation condition is that of $\mathfrak{r}_1$ for the determining set $\mathbf{B}$. Permit every reversal-invariant set $\mathfrak{F}$ of removed inward bonds. Lengths are level-k lengths: $\eta = L^{-k}$ and $\xi_j = L^j\eta$. All constants sought in these estimates must depend only on L and the pair, uniformly in this geometry, $\mathfrak{F}$, depth, cutoff and volume.

The reference is $U_0 = \mathbf{1}$. Let $a_j[A]$ and $v_j[A]$ be the centered averaging logarithms and the accumulated straightening factors, so that, on a bond b of level j,

$$\mathcal{M}^j(e^{i\eta A})(b) = v_j[A](b_-)\,\mathsf{x}_A(b)\,v_j[A](b_+)^{-1}, \qquad \mathsf{x}_A(b) = e^{ia_j[A](b)}. \tag{9.i}$$

The value at a reversed bond is its inverse. At the outer endpoint z of an outward bond the factors satisfy $v_j[A](z) = v_{j-1}[A](z)\,\mathsf{v}(a_{j-1}[A])(z)$. The last factor depends only on interior bonds in the preceding block; write $\kappa_6 \geq 1$ for its uniform small-field bound.

There is one row for each positively oriented interior or outward bond. At a seam node x, order the retained inward bonds ending there as $b_1(x), \dots, b_m(x)$ and add the rows $(b_i(x), b_1(x))$, $2 \leq i \leq m$. Their scale is ξ_j if the bonds have level j. Let $\mathfrak{W}$ be the space of $\mathfrak{g}$-valued row functions, with norm $\sup|\cdot|$ and inner product

$$(w, w')_1 = \sum_{\iota} \xi_\iota^{-1}\langle w(\iota), w'(\iota)\rangle.$$

Here $\langle\cdot,\cdot\rangle$ is the inner product of the pair; it is not replaced by the operator norm. Fine-field adjoints use the η^3-weighted inner product. The nonlinear row map is

$$\begin{aligned} \mathcal{Q}^\natural(A)(c) &= a_j[A](c) && (c \text{ interior or outward}),\\ \mathcal{Q}^\natural(A)(b_i, b_1) &= \frac{1}{i}\log\{\mathsf{x}_A(b_i)\mathsf{x}_A(b_1)^{-1}\} && (\text{seam rows}). \end{aligned} \tag{9.ii}$$

For $\sup|B| \leq c_{11}/2$, the products defining the datum in eq. 9.viii lie on the intrinsic branch and obey eq. 9.ix. All logarithms below are on the intrinsic branch through $\mathbf{1}$. The construction also needs these logarithms and maps to exist and be analytic on neighbourhoods of the closed balls used below.

The uniform slice. The required estimate would provide $c_{11} > 0$, $\mathsf{B}_1 \geq 1$ and $C_\sharp \geq 1 + \kappa_6$. A regular real configuration U with regularity parameter α_0 and datum e^{iB}, where $\alpha_1 = \sup|e^{iB} - \mathbf{1}|$ and $\alpha_0 + \alpha_1 \leq c_{11}$, has a chart and gauge satisfying

$$U^g = e^{i\eta A}, \qquad g = v_j[A] \text{ on } \Lambda_j, \qquad \mathsf{R}D^*A = 0, \qquad \|A\|_{X,\mathbf{B}} \leq \tfrac{1}{4}\mathsf{B}_1(\alpha_0 + C_\sharp\alpha_1). \tag{9.iii}$$

At every retained inward bond b ending at a seam node x,

$$\mathsf{x}_A(b)\,v_j[A](x)^{-1}g(x) = e^{iB_j(b)}. \tag{9.iv}$$

The preliminary gauge used to obtain this chart bounds the discrepancy from $\mathbf{1}$ at every level by $C_{\mathrm{pg}}(\alpha_0 + \alpha_1)$; its radius is included in c_{11}, and $c_{11} \leq r_G/(2C_{\mathrm{pg}})$. The regular class here bounds curvature by $\alpha_0\eta^2\mathsf{L}(p)^{-2}$ and the current on full bonds by $\alpha_0\mathsf{L}(b)^{-3}$.

The nonlinear contraction. The required estimates would provide positive regularity ceilings c_8, c_9, a chart radius $r_\circ$ and constants $K_1 \geq 4$, $K_7 \geq 1$. Set $K_5 = 2(K_7 + K_1)$ and $K_8 = K_1(16 + 9K_1^3)$. For regularity parameters at most $\min\{c_8, c_9\}$, the following operators and estimates hold on $\|A\|_{X,\mathbf{B}} \leq r_\circ$. Let $F(A)$ be the real action in the chart, $F'(0) = 0$, and write

$$\Delta_{\mathsf{R}} = F''(0) + D\mathsf{R}D^*, \qquad F'(A) + D\mathsf{R}D^*A = \Delta_{\mathsf{R}}A + \mathcal{V}(A).$$

There are real linear maps $Q, H, \widetilde{G}$ and an isometric row map T preserving both the row supremum norm and its inner product, with

$$QH = 1, \quad Q\widetilde{G} = 0, \quad \widetilde{G}Q^* = 0, \quad \widetilde{G}\Delta_{\mathsf{R}} = 1 - HQ, \quad \ker\widetilde{G} = \operatorname{ran}Q^*. \tag{9.v}$$

Define

$$\mathcal{C}(A) = T^{-1}\mathcal{Q}^\natural(A) - QA, \qquad K_A = D\mathcal{Q}^\natural(A), \qquad S_A = 1 + H^*\mathcal{C}'(A)^*.$$

The map S_A is invertible, with $\|S_A^{-1}\|_{\infty\to\infty} \le 8/7$. The differential of the actual constraint, after removing the gauge factors, is $K_A^\flat = \Xi'_A K_A$, with Ξ'_A invertible. In particular a configuration $e^{i\eta A}$ is critical for its actual averaging constraint precisely when $F'(A)$ annihilates $\ker K_A$. For $B^\natural \in \mathfrak{W}$, put

$$\begin{aligned} w(A) &= S_A^{-1}H^*(F'(A) + D\mathsf{R}D^*A), \\ N(A) &= -\mathcal{V}(A) + \mathcal{C}'(A)^* w(A), \\ \Psi_{B^\natural}(A) &= \widetilde{G}N(A) + H\{T^{-1}B^\natural - \mathcal{C}(A)\}. \end{aligned} \tag{9.vi}$$

Whenever $\beta = \sup|B^\natural|$, $K_5\beta \le r \le r_\circ$ and $\|A\|, \|A_1\|, \|A_2\| \le r$, the uniform bounds are

$$\|\Psi_{B^\natural}(A)\| \le \tfrac{1}{2}K_5\beta + \tfrac{1}{2}r, \qquad \|\Psi_{B^\natural}(A_1) - \Psi_{B^\natural}(A_2)\| \le \tfrac{1}{2}\|A_1 - A_2\|. \tag{9.vii}$$

The norms in this display are $\|\cdot\|_{X,\mathbf{B}}$. The radii are also chosen so that the chart and row transformations just described remain on their intrinsic logarithm branches.

The uniform slice and nonlinear estimates remain to be verified. Colour-scalar linear operators at the trivial reference do not by themselves establish either task.

We describe explicitly which row datum corresponds to e^{iB}. At an interior bond put $B^\natural(c) = B_j(c)$. For an outward bond with outer endpoint z, put $v_z[B] = \mathsf{v}(B_{j-1})(z)$ and

$$\begin{aligned} B^\natural(c) &= \frac{1}{i}\log(e^{iB_j(c)}v_z[B]) && (z = c_+), \\ B^\natural(c) &= \frac{1}{i}\log(v_z[B]^{-1}e^{iB_j(c)}) && (z = c_-), \\ B^\natural(b_i, b_1) &= \frac{1}{i}\log(e^{iB_j(b_i)}e^{-iB_j(b_1)}) && \text{(seam rows)}. \end{aligned} \tag{9.viii}$$

The uniform small-field bounds included in the hypotheses give

$$\sup|B^\natural| \le (1 + \kappa_6)\sup|B| \le C_\sharp \sup|B|. \tag{9.ix}$$

Indeed the outward products have logarithm norm at most $|B_j(c)| + \kappa_6 \sup|B|$; the seam products have norm at most $2\sup|B|$, and $\kappa_6 \ge 1$. These products are taken within the specified branch, as are the one-step factors defining $v_z[B]$.

Lemma 9.16. The following reduction describes the consequence of establishing the slice estimates in §9.1.6: the row derivative K_A is onto for $\|A\|_{X,\mathbf{B}} \le r_\circ$. For $K_5\sup|B^\natural| \le r_\circ$ the system

$$F'(A) + D\mathsf{R}D^*A = K_A^*\omega, \qquad \mathcal{Q}^\natural(A) = B^\natural \tag{9.x}$$

has exactly one solution with $\|A\|_{X,\mathbf{B}} \le r_\circ$. The multiplier is unique without any further bound on its size. ∎

Proof. Write $\widehat{K}_A = T^{-1}K_A = Q + \mathcal{C}'(A)$ and $\widehat{\omega} = T^{-1}\omega$. Since $H^*Q^* = 1$, $H^*\widehat{K}_A^* = S_A$. The invertibility of S_A implies the injectivity of K_A^*, and hence the surjectivity of K_A in these finite-dimensional real inner-product spaces.

Set $\mathcal{G}(A) = F'(A) + D\mathsf{R}D^*A$. The first equation in eq. 9.x implies $S_A\widehat{\omega} = H^*\mathcal{G}(A)$, so $\widehat{\omega} = w(A)$. With this value of the multiplier the two equations become

$$\Delta_{\mathsf{R}}A - N(A) = Q^*w(A), \qquad QA = T^{-1}B^\natural - \mathcal{C}(A).$$

Apply $\widetilde{G}$ and use eq. 9.v: these equations imply $A = \Psi_{B^\natural}(A)$. Conversely, a fixed point has the second equation after applying Q. Applying $\widetilde{G}$ shows that $\Delta_{\mathsf{R}}A - N(A)$ belongs to $\operatorname{ran}Q^*$. Its coefficient is $w(A)$, because applying H^* gives $H^*\mathcal{G}(A) - H^*\mathcal{C}'(A)^*w(A) = w(A)$. Thus the fixed-point equation is equivalent to the system.

By eq. 9.vii, $\Psi_{B^\natural}$ maps the closed radius-$r_\circ$ ball into itself and contracts distances by $1/2$. This ball is complete, so it has one fixed point. The formula $\omega = Tw(A)$ gives the only multiplier. The same argument on the radius-$K_5\beta$ ball, or directly the fixed-point bound when $\beta = 0$, gives $\|A\|_{X,\mathbf{B}} \le K_5\beta$. □

Proposition 9.17 (Uniqueness from uniform slice estimates)**.** Let G be compact and connected and let $L \ge 13$ be odd with $200L^2 z_G \ge 1$. The supplied reduction shows what would follow once the uniform slice and nonlinear estimates in §9.1.6 are established: there would be $\varepsilon_G, \varrho_G > 0$, depending only on L and the pair, with the following property in every geometry described there. For any $\mathfrak{g}$-valued datum B on the positively oriented bonds of $\mathbf{B}_{\mathfrak{F}}$ with $\sup|B| < \varrho_G$, any two real critical configurations in $C(\mathbf{B}_{\mathfrak{F}}, e^{iB})$ satisfying

$$|\partial U(p) - \mathbf{1}| < \varepsilon_G \eta^2 \mathsf{L}(p)^{-2}, \qquad |J(U)(b)| < \varepsilon_G \mathsf{L}(b)^{-3}$$

at inner plaquettes and full bonds, respectively, belong to the same residual gauge orbit. Thus $U' = U^u$, where $u = \mathbf{1}$ at every node of $\mathbf{B}_{\mathfrak{F}}$. Each configuration has a representative $U^g = e^{i\eta A}$ with $g = v_j[A]$ on the tops, $\mathsf{R}D^*A = 0$ and

$$\|A\|_{X,\mathbf{B}} \le \tfrac{1}{4}\mathsf{B}_1(\varepsilon_G + C_\sharp \sup|B|).$$

The two radii are uniform in depth, sub-domain, cutoff and volume. ■

Proof. Take the constants of §9.1.6 and put

$$\varepsilon_G = \min\{c_{11}/2, c_8, c_9, r_\circ/(K_5 L\mathsf{B}_1)\}, \qquad \varrho_G = \min\{c_{11}/2, r_\circ/(K_5 L\mathsf{B}_1 C_\sharp)\}.$$

For two configurations U_a, U_b as in the statement set $\alpha_0 = \varepsilon_G$ and $\alpha_1 = \sup|e^{iB} - \mathbf{1}| \le \sup|B| < \varrho_G$. Hypothesis (S) gives A_*, g_*, $* = a, b$, with

$$U_*^{g_*} = e^{i\eta A_*}, \qquad \mathsf{R}D^*A_* = 0, \qquad \|A_*\|_{X,\mathbf{B}} \le \tfrac{1}{4}\mathsf{B}_1(\alpha_0 + C_\sharp\alpha_1) \le \frac{r_\circ}{2K_5 L}.$$

This gives the asserted representatives.

Both have the row datum eq. 9.viii. To check this, at an interior bond the two endpoint gauges in eq. 9.i cancel, giving $e^{ia_j[A_*](c)} = e^{iB_j(c)}$ and hence $a_j[A_*](c) = B_j(c)$. At an outward bond the uncancelled factor is $\mathsf{v}(a_{j-1}[A_*])(z)$; it equals $v_z[B]$, since it reads only the interior bonds of the preceding block. Moving this factor to the other side gives the first two formulas in eq. 9.viii. Finally, eq. 9.iv at b_i and b_1 gives

$$\mathsf{x}_{A_*}(b_i)\mathsf{x}_{A_*}(b_1)^{-1} = e^{iB_j(b_i)}e^{-iB_j(b_1)}.$$

Taking the stipulated intrinsic logarithms proves $\mathcal{Q}^\natural(A_a) = B^\natural = \mathcal{Q}^\natural(A_b)$. By eq. 9.ix, $K_5 \sup|B^\natural| \le K_5 C_\sharp \sup|B| < r_\circ/(L\mathsf{B}_1) \le r_\circ$.

Criticality is preserved by gauge transformations. The criticality clause of (N), together with the surjectivity in Lem. 9.16, gives a multiplier ω_* with $F'(A_*) = K^*_{A_*}\omega_*$. Since $\mathsf{R}D^*A_* = 0$, both pairs solve eq. 9.x for the same datum. Lem. 9.16 therefore gives $A_a = A_b$. Consequently $U_b = U_a^{g_b^{-1}g_a}$. The two gauges agree at every top by eq. 9.iii. At a seam node, one retained inward bond in eq. 9.iv determines $g_*(x)$ from A_* and B, so they agree there as well. Every node of $\mathbf{B}_{\mathfrak{F}}$ is a top or such a seam node. Thus $g_b^{-1}g_a$ is a residual gauge transformation. □

The orbit and energy estimates still required. Let $G \subset U(N)$ be compact and connected and $L \ge 13$ odd. Let $\mathcal{M}$ be a G-valued averaging map that, at every coarse bond c, is the formula of Lem. 9.11(i) at every field whose plaquettes on the double block of c are within $\omega_G = \min\{\omega_M, r_G\}$ of $\mathbf{1}$, the radius below which Lem. 9.11(i) forms that formula in G for every compact connected G, also where $\mathfrak{g}$ has a centre and $z_G = 0$; where the centre of $\mathfrak{g}$ is 0, ω_G is the radius $\omega^G = \min\{\omega_M, z_G\}$ of Def. 9.5(3). The radial reduction below reads $\mathcal{M}$ at such fields alone. Let $\Lambda_0, \dots, \Lambda_t$ be the lattices of the tower, Λ_0 the bare one and Λ_t the block lattice. In this definition and its reductions, in exception to Def. 9.5(1), for functions f on the bonds or on the plaquettes of one lattice $\|f\|^2 = \sum|f(\cdot)|^2$,

$\|f\|_1 = \sum |f(\cdot)|$ and $|f|_\infty = \max |f(\cdot)|$, the values being measured in the operator norm; the energy norm of a $\mathfrak{g}$-valued 1-form δ on Λ_0 is $\|\delta\|_E^2 = \|d\delta\|^2 + L^{-2t}\|\delta\|^2$, in bare units; and the two matrix norms are compared only through $\|X\| \le |X| \le N^{1/2}\|X\|$. The orbit-and-energy task is to obtain $a_G, \varepsilon_* \in (0,1]$, constants depending only on L and the pair that read ω_G, such that for every $t \ge 1$, on every torus whose lattice Λ_t has at least 12 sites a side, all regions being the whole torus:
Flatness of a small critical field. The target is that a configuration U with $\mathcal{M}^t U = \mathbf{1}$ that is critical for the action on $\{\mathcal{M}^t U = \mathbf{1}\}$ and has $|\partial U(p) - \mathbf{1}| < \epsilon L^{-2t}$ at all bare plaquettes, $\epsilon \le a_G$, is flat;
Flat fields over the trivial datum. Independently of that critical-orbit estimate, flat fields U with $\mathcal{M}^t U = \mathbf{1}$ are the $\mathbf{1}^u$ with $u = \mathbf{1}$ on Λ_t; once the preceding flatness target is established, therefore, the small critical configurations over $\mathbf{1}$ with these plaquettes are exactly this orbit;
Nearby critical orbits. One must also show that for each t, each torus and each $\epsilon' \le a_G$ there is a neighbourhood of $\mathbf{1}$, depending on them, at each point V of which two critical configurations of $\{\mathcal{M}^t U = V\}$ with $|\partial U(p) - \mathbf{1}| < \epsilon' L^{-2t}$ lie in one orbit of the gauge functions equal to $\mathbf{1}$ on Λ_t, which depends holomorphically on V;
The energy comparison. The required estimate is that if U is critical for the action on $\{\mathcal{M}^t U' = V\}$, $V = \mathcal{M}^t U$, and $|\partial U(p) - \mathbf{1}| \le \varepsilon_* L^{-2t}$, then

$$\sum_p |\partial U(p) - \mathbf{1}|^2 \le 4N\varepsilon_* L^{-t} \sum_{p'} \Big| \frac{1}{i} \log \partial V(p') \Big|,$$

the sums running over the plaquettes of Λ_0 and of Λ_t; in particular over a flat block field the critical configurations with these plaquettes are flat, and over a block field near $\mathbf{1}$ the regular critical configurations form one gauge orbit.

The required estimates must make a_G and ε_* to be independent of t and the torus. If all the uniform analytic estimates in the reduction below hold, its radial argument permits $a_G = \min\{\varepsilon_5/2, [4N(68C_sC_\alpha + 132C_R)]^{-1}\}$ with the numbers ε_5, C_s, C_α, C_R of those estimates. Write $a_*(G)$ for the supremum of the admitted orbit radii, taking it to be 0 if no positive radius is admitted. The construction uses a fixed admitted positive a_G, not an assumed maximiser. Positivity of this radius for general G, and a lower bound uniform in N for $Na_*(SU(N))$, are not established by the matrix estimates alone.

Flat fields and conditional reductions. The flatness reduction requires a uniform gauge g, equal to $\mathbf{1}$ on Λ_t, in which the potentials of U and of all its averages are small, $(\mathcal{M}^i U)^g = e^{i\alpha_i}$ with $|\alpha_i|_\infty \le C_s \epsilon L^{-(t-i)}$ and $\|\alpha_i\| \le C_\alpha L^{t-i/2}\mathcal{E}^{1/2}$, $\mathcal{E} = \sum_p |\partial U(p) - \mathbf{1}|^2$. Let $F(a')$ be the action at $e^{ia'}$, $a = \alpha_0$, and T the derivative at a of the t-fold average read in the chart. The required first-variation estimate is $dF(a) \ge (\frac{5}{6N} - 68C_sC_\alpha\epsilon)\mathcal{E}$, from $\langle \theta, H_{e^{i\theta}} \rangle \ge \frac{5}{6}\|e^{i\theta} - \mathbf{1}\|^2$ for $|\theta| \le 1$. Along the ray $s \mapsto e^{isa}$ an abelian field would stay in the constraint set; for G it leaves it at second order, and we correct it. It also requires the derivative to have the form $T\delta = d\lambda_\delta + K\delta$ with $K = (Q' + E_{t-1}) \cdots (Q' + E_0)$, $\|Q'\rho\| \le L^{-1/2}\|\rho\|$, $\|E_i\rho\| \le C(|\alpha_i|_\infty + |\alpha_{i+1}|_\infty)\|\rho\|$; the interpolation H_0 of the straight average, composed with the inverse of its average, has $Q'^t H_0 = 1$ and $\|H_0\psi\|_E \le 33L^{-t/2}\|\psi\|$. The number ε_5 is the last of a falling chain $\varepsilon_1 \ge \varepsilon_2 \ge \cdots \ge \varepsilon_5$ in $(0,1]$; existence of these positive thresholds uniformly in depth and torus is part of the analytic input: below ε_1 all the averages $\mathcal{M}^i U$ are formed by the formula, their plaquettes staying within $2\epsilon L^{-2(t-i)}$ of $\mathbf{1}$, below ω_G, and $\|\partial \mathcal{M}^i U - \mathbf{1}\| \le 2L^{i/2}\mathcal{E}^{1/2}$; below ε_2 the gauge g exists with the two bounds above; below ε_3 the first variation obeys the bound above; below ε_4 the derivative T has the structure above, and the radial direction the form below; and below ε_5, besides, $\|(K - Q'^t)H_0\| \le \frac{1}{2}$. For $\epsilon \le \varepsilon_5$, $KH_0 = 1 + \Delta$ with $\|\Delta\| \le \frac{1}{2}$, so K has the right inverse $H = H_0(1+\Delta)^{-1}$ with $\|H\psi\|_E \le 66L^{-t/2}\|\psi\|$, and T is onto; and $Ta = d\lambda_* + R$ with $\|R\| \le C_R \epsilon L^{t/2}\mathcal{E}^{1/2}$. The direction $a - \delta_{\lambda_*}a - HR + \delta_{\lambda'}a$, δ_λ the infinitesimal gauge transformations and $\lambda' = \lambda_{HR}$, lies in $\ker T$, and

dF vanishes on gauge directions; so the critical equation gives

$$\Big(\frac{5}{6N} - 68C_sC_\alpha\epsilon\Big)\mathcal{E} \le dF(a) = dF(HR) \le 2\mathcal{E}^{1/2}\|HR\|_E \le 132\,\mathcal{E}^{1/2}L^{-t/2}\|R\| \le 132\,C_R\epsilon\,\mathcal{E},$$

and $\mathcal{E} = 0$ for $\epsilon \le a_G$. The same argument proves flatness with $2a_G$ in place of a_G. Indeed, $2a_G \le \varepsilon_5$ and

$$(68C_sC_\alpha + 132C_R)2a_G \le \frac{1}{2N} < \frac{5}{6N}.$$

This proves the radial reduction under the displayed uniform estimates. For the independent flat-field assertion, a flat field has trivial transports around closed paths in boxes, and its averages are the transports along straight paths. This gives the flat-field assertion without the uniform analytic input: lifting to $\mathbb{Z}^3$ and using the comb gauge on growing cubes gives $U = \mathbf{1}^u$, and the averages $\mathbf{1}$ force $u = \mathbf{1}$ on Λ_t. For nearby critical orbits, this margin also excludes other critical branches near the trivial datum, once the gauge-fixed constrained Hessian at the flat orbit is invertible. Fix the torus, t and $0 < \epsilon' \le a_G$. The implicit function theorem gives a critical orbit near the flat orbit for all data V near $\mathbf{1}$, with uniqueness in a fixed neighbourhood of that orbit. Suppose that critical configurations U_m over $V_m \to \mathbf{1}$, with $|\partial U_m - \mathbf{1}| < \epsilon' L^{-2t}$, escape this neighbourhood. Compactness of the finite product of copies of G gives a subsequence $U_m \to U_\infty$. The averaging map is smooth near U_∞: the bounds through ε_5 keep every average in its small-field formula. At U_∞ the differential is onto, by the right inverse in the radial reduction. It remains onto on a neighbourhood of U_∞. The tangent kernels therefore vary continuously along the subsequence, and constrained criticality passes to U_∞. This limit lies over $\mathbf{1}$ and has curvature at most $a_G L^{-2t} < 2a_G L^{-2t}$, so it is flat. By the flat-field assertion, a residual gauge transformation carries it to $\mathbf{1}$. Applying this fixed transformation to U_m places the subsequence in the neighbourhood where the implicit function theorem gives uniqueness, a contradiction. Thus the nearby-orbit conclusion follows from the radial estimates and this local invertibility; it requires no separate exclusion of remote branches. The neighbourhood obtained in this way may depend on the torus, t and ϵ'. The terminal-field energy bound is an additional analytic input. Its descending construction would require the following estimates. Let $V_j = \mathcal{M}^j U$, $\phi_V(p) = |\partial V(p) - \mathbf{1}|$, $\Phi_V(p) = \frac{1}{i}\log \partial V(p)$, $D_j = \max_p \phi_{V_j}(p)$, and D_V the derivative of the plaquette variables along a tangent vector in the left chart. The required hierarchical axial-gauge descent solves, one step down, $\mathcal{M}'(V_j)X_j = X_{j+1}$ with $\|X_j\| \le L(1+CD_j)\|X_{j+1}\| + C\|\phi_{V_j}\|$, starting from $X_t = 0$, and the defect $E_j = D_{V_j}X_j - \Phi_{V_j}$ obeys $\|E_j\|_1 \le L\|E_{j+1}\|_1 + C\|\phi_{V_j}\|(\|\phi_{V_j}\| + \|X_{j+1}\|)$; X_0 is tangent to the constraint set, so $0 = \sum_p \langle \Phi_U(p) + E_0(p), \sin \Phi_U(p)\rangle$, and $\langle \Phi, \sin\Phi\rangle \ge \frac{1}{2N}|\Phi|^2$ does not by itself provide the uniform aggregate error bound needed for (iv); that bound is part of the terminal-field input. At $SU(N)$ a central element $e^{2\pi i/N}\mathbf{1}$ on one bond off the block axes gives a nonflat configuration with zero current and plaquettes at distance $2\sin(\pi/N)$. If the chosen extension of $\mathcal{M}$ averages this field to $\mathbf{1}$, it obstructs an orbit radius above $2L^2\sin(\pi/N)$ at the first level. Agreement with the small-field averaging formula alone does not impose this behaviour on the extension. □

Proposition 9.18 (One loop)**.** Let $L \ge 13$ be odd and $k \ge 1$.

(i) Let $\mathfrak{g}$ be simple. On every torus Ψ_k^G is tempered: real-analytic near $\mathbf{1}$, zero at $\mathbf{1}$, invariant under gauge transformations and lattice isometries, with the symbol $\delta_{aa'}\hat{\pi}^G_{\mu\nu}$, $\hat{\pi}^G = \frac{C_A}{8}\hat{\pi}^{\Psi_k} + \gamma_G\hat{\pi}^{\mathfrak{B}_k}$, holomorphic and bounded on a strip; here $\mathfrak{B}_k = (\mathfrak{B}_{k,s})_s$ is the scalar functional of the tadpole, formed for $SU(2)$ on the tower of height k and side s (Section YM3_I:42.1),

$$\mathfrak{B}_{k,s}(W) = \sum_p w_{k,s}(p)\,a_p(c_k(W)) \qquad (W \text{ near } \mathbf{1} \text{ on } \Lambda_k),$$

c_k the critical branch (Thm. YM3_I:N) and $w_{k,s}(p) \in [0,1]$ the diagonal of the orthogonal projection of the plaquette functions of Λ_0 onto $d(\ker \mathsf{Q}^{[k]})$, $\mathsf{Q}^{[k]}$ the k-fold linear average, which reads L, k, s and no group; $\mathfrak{B}_0 = 0$, and with the weights 1 in the place of $w_{k,s}$

it would be $\mathcal{S}_k = A \circ c_k$. The functional $\mathfrak{B}_k$ is tempered (Section YM3_I:43.1), with $b[\mathfrak{B}_k] = j_k := \frac{2}{3}(L^{-k} - L^{-4k})$; the coefficient of Ψ_k^G is that of Thm. AA(c), and the increments $b[\Psi_l^G] - L^{-1}b[\Psi_{l-1}^G]$ are the coefficients of tempered functionals and telescope as in Section YM3_I:43.3. Thm. YM3_I:O holds for G with the remainder constant $\frac{C_A}{8}C_{\rm rem}(L) + \frac{2}{3}L^{-4}|\gamma_G|\delta_0^2$, $C_{\rm rem}(L)$ and δ_0 those of $SU(2)$, the least that the argument gives, and

$$\sup_k |b[\Psi_k^G]| \le \frac{C_A}{8}\sup_k |b[\Psi_k]| + \Big|\frac{C_\rho}{3} - \frac{C_A}{8}\Big|L^{-1};$$

so the condition $B_\sharp \ge (1+L^{-1})c_2 + 1$ of the $SU(2)$ construction, c_2 that of Thm. YM3_I:O, and Lem. 9.40 stand with this number, which depends on L, C_A, C_ρ alone. For $SU(N)$ the two factors are $\frac{N^2}{4}$ and $\frac{N^2-4}{12}$.

(ii) Let $\mathfrak{g}$ be simple. Thm. AC holds for G in every clause: the numbers $b_l^0(G)$ converge geometrically,

$$\Big|b_l^0(G) - \frac{C_A}{8}b_\infty^0\Big| \le \frac{C_A}{8}C_{\rm geo}L^{-l/6} + \frac{2}{3}|\gamma_G|L^{3-4l},$$

the functionals of that theorem for G are $\frac{C_A}{8}$ times those of $SU(2)$ plus γ_G times one scalar functional bounded by $5C_h^2L^{-k}$, $C_{\rm re}(G) \le \frac{C_A}{8}C_{\rm re} + 10|\gamma_G|C_h^2$, and $C_{\rm inc}(G) \le 17(2C_{\rm rem}(G)C_{\rm re}(G))^{1/2}\delta_0^{-2}$ with $C_{\rm rem}(G) = \frac{C_A}{8}C_{\rm rem} + 5|\gamma_G|C_h^2/L$ in this clause, a larger number than the remainder constant of (i).

(iii) Let G be compact and connected, $l \ge 1$ and, in this clause, $k = l-1 \ge 0$. On the tower of height l and side s, formed for G, let $p_j(W)$ be the critical point near $\mathbf{1}$ of $\mathcal{S}_{j-1} + \mathrm{GF}_j$ on the fibre $\{\mathcal{M}(V) = W\}$, $c_l = p_1 \circ \cdots \circ p_l$ the critical branch, with $\mathcal{M}^l c_l(W) = W$ and $c_l(\mathbf{1}) = \mathbf{1}$, and $\mathcal{S}_l = A \circ c_l$ (Thm. YM3_I:N); let $\mathsf{Z}_l(W)$ be the normaliser of the first part formed for G, the Gaussian integral of the Hessian of $\mathcal{S}_{l-1} + \mathrm{GF}_l$ on the fibre at $p_l(W)$ (Thm. YM3_I:N), whose logarithm has the quadratic coefficient $b_l^0(G)$ for simple $\mathfrak{g}$ (Thm. YM3_I:O, Thm. AA(c)); and let $\mathsf{Z}^{(k)}[\mathbf{P}] = \prod_c |\det \mathsf{m}_c(V)|^{-1}\int \mathrm{d}B\, e^{-\frac{1}{2}(B,\mathsf{A}^\sharp(\mathbf{P})B)_1}$, $V = \mathcal{M}^k U$, be the Gaussian integral of the step of the second part at a triple $\mathbf{P} = (U, 0, J')$ (Prop. YM3_I:50.2, with $\dim \mathfrak{g}$ real coordinates per positive nonpivot bond), with the fluctuation operator $\mathsf{A}^\sharp$ of the step, whose value at the plain triples over W is the Gaussian normaliser $Z^{(k)}(W) = Z_k[P_W]$ preceding equation YM3_I:(957). Near $W = \mathbf{1}$ it is formed at the plain triple $\mathbf{P}_W = (c_l(W), 0, L^3 J(c_l(W)))$ of the branch, is positive, is the same along the gauge orbit of this specified branch, and identification with every regular critical configuration over W requires the orbit hypothesis of §9.1.6, or (V) in its regular domain. At the specified branch one has

$$\mathsf{Z}^{(k)}[\mathbf{P}_W] = L^{nn_l}L^{-kd_l/2}\mathsf{Z}_l(W), \qquad n_l = 3s^3, \quad d_l = 2n(L^3-1)s^3;$$

so the quadratic coefficient of $\log Z^{(l-1)}$ is $b_l^0(G)$ for simple $\mathfrak{g}$, and for semisimple $\mathfrak{g}$ it is, on each ideal $\mathfrak{g}_i$, $b_l^{0,(i)} = \frac{C_A^{(i)}}{8}b_l^0 + (\theta_i - \frac{3}{8}C_A^{(i)})\frac{1}{3}(L^3-1)L^{-4l}$. The localised Gaussian family (Prop. YM3_I:50.2), formed for G, has its members formed and holomorphic near the trivial triple, and there the same Hessian as $\log Z^{(k)}$; its coefficient there is one number for all data: for simple $\mathfrak{g}$

$$\frac{C_A}{8}b_{k+1}^0 + \gamma_G(j_{k+1} - L^{-1}j_k) = b_{k+1}^0(G),$$

with $j_k = \frac{2}{3}(L^{-k} - L^{-4k})$ and b_{k+1}^0 the increment of $SU(2)$, and for semisimple $\mathfrak{g}$, on $\mathfrak{g}_i$, $\frac{C_A^{(i)}}{8}b_{k+1}^0 + \gamma^{(i)}(j_{k+1} - L^{-1}j_k) = b_{k+1}^{0,(i)}$, $\gamma^{(i)} = \frac{1}{2}\theta_i - \frac{3}{16}C_A^{(i)}$. That this family is a pointed family of its level on the analyticity cores of G, holomorphic on balls of one radius, with the

contraction bound of its expansion, is part of (II) of §9.1.1 (Prop. 9.23(iii)), and therefore is asserted here under hypothesis (II) for simply connected G with simple $\mathfrak{g}$ and $200L^2 z_G \geq 1$.

(iv) Let $\mathfrak{g} = \bigoplus_i \mathfrak{g}_i$ be semisimple, Ω_i the sum of t_a^2 over an orthonormal basis of $\mathfrak{g}_i$, $\tau_{ij} = \frac{1}{N}\mathrm{tr}(t^2 s^2)$ for unit $t \in \mathfrak{g}_i$, $s \in \mathfrak{g}_j$, $j \neq i$, and $\tau_{ii} = \frac{1}{n_i N}\mathrm{tr}(\Omega_i t^2)$ for unit $t \in \mathfrak{g}_i$, numbers of the pair; so $\theta_i = \frac{1}{N}\mathrm{tr}(\Omega H^2) = \sum_j n_j \tau_{ij}$ for a unit $H \in \mathfrak{g}_i$, and $n_i \tau_{ii}$ is the number C_ρ of the simple pair of $\mathfrak{g}_i$, the subgroup with Lie algebra $\mathfrak{g}_i$ on $\mathbb{C}^N$. With one set of sequences and functions for all ideals and all groups, the one-loop coefficients on $\mathfrak{g}_i$ are $\frac{C_A^{(i)}}{8}$ times those of $SU(2)$ plus $(\theta_i - \frac{3}{8}C_A^{(i)})$ times one scalar sequence: at every semisimple pair, the ideals being tied by the representation or not,

$$b^{(i)}[\Psi_k^G] = \frac{C_A^{(i)}}{8} b[\Psi_k] + \Big(\theta_i - \frac{3}{8}C_A^{(i)}\Big)t_k, \qquad t_k = \tfrac{1}{3}(L^{-k} - L^{-4k}),$$

$b^{(i)}[\Psi_k^G]$ being the coefficient on $\mathfrak{g}_i$ of the one-loop functional Ψ_k^G of the pair and Ψ_k that of $SU(2)$. The same holds with weights, for an action $\sum_m w_m$ times the Wilson actions in representations ρ_m, $w_m > 0$, with $C_A^{(i)}(w)$, $\theta_i(w)$ formed with the weighted form $\langle X, Y\rangle_w = \sum_m w_m N_m^{-1} \mathrm{tr}\,\rho_m(X)\rho_m(Y)$ in the place of $\langle\cdot,\cdot\rangle$, a rational weight being a multiplicity; the limits of the coefficients, $\frac{C_A^{(i)}(w)}{8} b_\infty^0$, read the weights through this normalisation alone, and the limit as a form on $\mathfrak{g}$, $\frac{b_\infty^0}{8}\mathrm{tr}(\mathrm{ad}\,X)^2$, does not read them. Equivalently, for a unit vector t of $\mathfrak{g}_i$, with $\Pi^F = F''(\mathbf{1})$ the polarisation of a functional F (Section YM3_I:43.1),

$$\Pi^{\Psi_k^G}[tf, tf'] = \Pi^{\Psi_k^{(i)}}[tf, tf'] + \Big(\sum_{j\neq i} n_j \tau_{ij}\Big)\Upsilon_k[f, f'], \qquad 0 \leq \Upsilon_k[f,f] \leq \tfrac{1}{2}\mathsf{h}_k(f,f),$$

$\Psi_k^{(i)}$ that of the simple pair of $\mathfrak{g}_i$, $\mathsf{h}_k = \mathcal{S}_k''(\mathbf{1})$ the Hessian at $\mathbf{1}$, for one colour, of $\mathcal{S}_k = A \circ c_k$ (Thm. YM3_I:N), and $\Upsilon_k[f,f'] = \frac{1}{2}\sum_p \mathsf{v}_k(p)(d\mathsf{c}_k f)(p)(d\mathsf{c}_k f')(p)$ one scalar form of the lattice that reads no group, $\mathsf{v}_k(p) \in [0,1]$ the diagonal of the orthogonal projection onto $d(\ker \mathsf{Q}^{[k]})$, a variance of the scalar Gaussian theory on $\ker \mathsf{Q}^{[k]}$, and c_k the linear part of the chain at $\mathbf{1}$. The form $\Upsilon_k = \frac{1}{2}\Pi^{\mathfrak{B}_k}$ is the polarisation of the tempered functional $\frac{1}{2}\mathfrak{B}_k$ of (i), with $b[\Upsilon_k] = t_k \in [0, \frac{1}{3}L^{-k}]$; so $b^{(i)}[\Psi_k^G] = b[\Psi_k^{(i)}] + (\sum_{j\neq i} n_j\tau_{ij})\, t_k$. For block sums $\tau_{ij} = 0$ for $j \neq i$. The form of the coefficients on $\mathfrak{g}$ is $\sum_i b^{(i)}[\Psi_k^G]\langle P_i\cdot, P_i\cdot\rangle$, with r numbers, which are bounded ideal by ideal,

$$|b^{(i)}[\Psi_k^G]| \leq \frac{C_A^{(i)}}{8}\sup_k |b[\Psi_k]| + \Big|\frac{\theta_i}{3} - \frac{C_A^{(i)}}{8}\Big| L^{-1},$$

and so by the largest of these r bounds. For a unit $H \in \mathfrak{g}_i$ the numbers $\mathrm{tr}(\mathrm{ad}\,H)^2$ and $\frac{1}{N}\mathrm{tr}(\Omega H^2)$ that Def. 9.1 forms for simple $\mathfrak{g}$ are $C_A^{(i)}$ and θ_i. The pairs $(C_A^{(i)}, \theta_i)$ may differ between the ideals, as at $SU(2)\times SU(3)$ on $\mathbb{C}^2 \oplus \mathbb{C}^3$, where $C_A^{(i)}$ is 20 and 30 (Lem. 9.15(i)), the coefficient being then given ideal by ideal, as above, or agree, as $(16, 6)$ on both ideals of $SU(2)\times SU(2)$ on $\mathbb{C}^2 \oplus \mathbb{C}^2$ and $(8,6)$ on both on $\mathbb{C}^2 \otimes \mathbb{C}^2$; and $b^{(1)}[\Psi_k^G] - b^{(2)}[\Psi_k^G] = \frac{1}{8}(C_A^{(1)} - C_A^{(2)})(b[\Psi_k] - 3t_k) + (\theta_1 - \theta_2)t_k$.

■

Sketch of proof. (i) By invariance the background may be taken, to second order, along one unit $H \in \mathfrak{g}$, and the fluctuations split into the eigenspaces of $\mathrm{ad}\,H$. Every object of a step but the action is built from products, logarithms and means of group elements, which commute with homomorphisms of Lie algebras; with a parameter whose cube vanishes this shows that the sector of eigenvalue c of $\mathrm{ad}\,H$ in the field θH is the charged sector of $SU(2)$ in the field $\frac{c}{2}\theta t_3$, $\mathrm{ad}\,t_3$ having the eigenvalues $0, \pm 2$. Summing over the sectors, the group enters through $\sum c^2 = C_A$. The action adds one

term: $\cos(fH) = \mathbf{1} - \frac{1}{2}f^2H^2 + O(f^4)$ gives the tadpole $-\frac{1}{2}d^{\mathrm{T}}(d\theta)^2 d \otimes M_H$, $M_H Y = \frac{1}{2}P\{H^2, Y\}$, $\operatorname{tr} M_H = C_\rho$, whose propagator is the orthogonal projection onto $d(\ker Q^{[k]})$, $Q^{[k]}$ the k-fold linear average; its trace is $2s^3(\eta^{-3} - 1)$, two thirds of the plaquettes less the block lattice. Eliminating the group-free second differential,

$$\Xi^{G\prime\prime}(\mathbf{1})(\theta H, \theta H) = \frac{C_A}{8}\,\Xi^{\prime\prime}(\mathbf{1})(\theta t_3, \theta t_3) + \frac{1}{2}\Big(C_\rho - \frac{3}{8}C_A\Big)\sum_p w_k(p)(d\theta)(p)^2,$$

Ξ the logarithm of the normaliser and w_k the diagonal of the projection; the count of the trace gives the formula for $b[\Psi_k^G]$. The functional $\mathfrak{B}_k$, formed here, is tempered by a step of its own. It is real-analytic near $\mathbf{1}$ and 0 at $\mathbf{1}$; it is gauge invariant, as $c_k(W^v) = c_k(W)^{\tilde{v}}$ for v near $\mathbf{1}$ or constant and a_p is a class function; and it is invariant under the isometries of the torus that map Λ_k onto itself, which keep $d(\ker \mathsf{Q}^{[k]})$ and so $w_{k,s}$. On every torus the orthogonal projection onto $d(\ker \mathsf{Q}^{[k]})$ is the matrix of the periodic operator $\mathcal{P} = d(\mathsf{X}^{-1} - \mathsf{X}^{-1}Q^{\mathrm{T}}C^{-1}Q\mathsf{X}^{-1})d^{\mathrm{T}}$, $\mathsf{X} = d^{\mathrm{T}}d + d\varrho d^{\mathrm{T}} + Q^{\mathrm{T}}Q$, $\varrho = 1 - R^{\mathrm{T}}R$, $C = Q\mathsf{X}^{-1}Q^{\mathrm{T}}$, formed with the operators Q, R of the response of the $SU(2)$ construction (Section YM3_I:43.2), which is real, acts trivially on the colour and has an exponentially decaying kernel: its quadratic form at a plaquette function g is the maximum of $2(dB, g) - |dB|^2 - |\varrho d^{\mathrm{T}}B|^2$ over $B \in \ker Q$, and a gauge direction $d\lambda$, $\lambda \in \ker R$, removes the last term, which leaves $|\mathsf{P}_{\mathcal{C}}g|^2$, $\mathsf{P}_{\mathcal{C}}$ the projection onto $\mathcal{C} = d(\ker Q) = d(\ker \mathsf{Q}^{[k]})$. So $w_{k,s}(p) - w_k(p) = \sum_{m\neq 0}\mathcal{P}(p, p + sm)$ tends to 0 exponentially in s, w_k the diagonal at the infinite side, and the symbol $\hat{\pi}_{\mu\nu}^{\mathfrak{B}_k}(q) = \eta^4 \sum_{p \bmod \mathbb{Z}^3} w_k(p)\,(\hat{f}(q)e_\mu)(p)\,(\hat{f}(-q)e_\nu)(p)$, $\hat{f} = \hat{d}\hat{h}$ with the symbol $\hat{h}$ of the response (Section YM3_I:43.2), is holomorphic and bounded on the strip of width $2\delta_0$. The trace $\sum_p w_{k,s}(p) = \dim d(\ker \mathsf{Q}^{[k]}) = 2s^3(\eta^{-3} - 1)$, a third on each orientation, gives $b[\mathfrak{B}_k] = \eta^4 \cdot \frac{2}{3}(\eta^{-3} - 1) = j_k$. With this the theorems of the $SU(2)$ construction on tempered functionals apply to $\mathfrak{B}_k$: the coefficient (Section YM3_I:43.1); the composition with one step of the branch (Section YM3_I:43.3), $b[\mathfrak{B}_{l-1} \circ p_l] = L^{-1}b[\mathfrak{B}_{l-1}]$, hence the increments and their telescoping sum (Section YM3_I:43.3); and the remainder, the term of $\mathfrak{B}_k$ adding at most $\frac{2}{3}|\gamma_G|L^{-4}\delta_0^2$ to the constant. Temperedness, the telescoping sum, the remainder and the uniform bound of Ψ_k^G follow from the theorems for $SU(2)$ applied to the two terms, with $0 \le L^{-k} - L^{-4k} \le L^{-1}$. The gauge-fixing term is a trace too; it enters at its zero, the branch being in the axial gauge, with the Hessian $|\mathsf{L}_U B|^2$, L_U being built from products, logarithms and means, so it brings no second tadpole.

A second proof walks the chain of the $SU(2)$ construction for G: at the trivial configuration all operators are scalar times $1_{\mathfrak{g}^c}$ (Lem. 9.15(ii)), so the numbers δ_0, C_0, C_h pass from three colours to n.

(ii) Along one colour direction every block of the limit theorem is a scalar kernel tensor a polynomial in $\operatorname{ad} H$, but for the tadpole, and a $\log\det$ sees $\operatorname{tr}(\operatorname{ad} H)^2$ and $\operatorname{tr} M_H$ alone; the two scalar contributions in the proof of Thm. AC have the same lattice kernels. Their estimates add, with factors $C_A/8$ and $|\gamma_G|$, giving the bounds in (ii).

(iii) For the Gaussian identity, work at the specified local critical branch. Identification of all regular critical configurations with this branch requires §9.1.6, or (V) on its regular domain. The difference $\Theta = A^\eta - L^k\mathcal{S}_k \circ \mathcal{M}^k$ of the action $A^\eta = \eta^{-1}A = L^kA$, written in the units of level k, and the effective action of the averaged field is gauge invariant, vanishes on the branch and is stationary there. Its second-order expansion identifies the quadratic form at this branch with the Hessian of $L^k\mathcal{S}_k$ plus the gauge-fixing term on the fibre $\{\mathcal{M} = W\}$, a Schur complement. At $W = \mathbf{1}$ the pivot coefficients are $L^{-3}\,\mathrm{id}$ and the fluctuation operator plus $Q^{\mathrm{T}}Q$ is positive definite, so everything is formed near $\mathbf{1}$; the Gaussian integral at the point of the branch gives the powers of L, with nn_l for $3n_l$ and $d_l = 2n(L^3 - 1)s^3$; $\operatorname{Log}\det$ is taken at the radius r_n (Lem. 9.14(iv)); and the value is the same on the orbit, the mean being covariant under all of $G \times G$ and the pivot determinants being those of their restrictions to $\mathfrak{g}$. The step of the $SU(2)$ construction that $|1 - T| \le \frac{1}{2}$ keeps $\det T$ off

the negative axis (the three-dimensional determinant argument in the proof of Prop. YM3_I:50.2) fails from $n = 6$ and is replaced by the radius r_n.

The localised Gaussian family is formed for G near the trivial triple from the operators $\mathsf{A}^\sharp_Y$ of the regions Y, of whose tr log its members are the differences over regions; at complex fields near $\mathbf{1}$ these operators are holomorphic germs, built from means, averages, logarithms, the action and its derivatives, and so are the members. The coefficient of the family reads of it the terms of second order at the trivial triple alone, in the directions $f\,t_a$, f scalar; these read the lattice, L and the pair and none of the data, so the coefficient is one number for all data. Along an abelian direction θH every operator of which $\mathsf{A}^\sharp_Y$ is made is $o_0 \otimes 1 + o_1 \otimes \mathrm{ad}(\theta H) + o_2 \otimes (\mathrm{ad}(\theta H))^2 + o_\tau \otimes M_{\theta H} + O(\theta^3)$ with matrices o that read no group, as in the proof of (i); so the term of second order for G is $\frac{C_A}{8}$ times that of $SU(2)$ plus γ_G times that of one gauge invariant family of $SU(2)$, formed with the Wilson action with the numerical weights of the tadpole on the plaquettes, to which the statements of the second part at $SU(2)$ apply as they stand. The same splitting holds for $\log Z^{(k)}$ by (i) and the identity above; and at $SU(2)$ the coefficient of the localised family is that of $\log Z^{(k)}$ (equation YM3_I:(790) identifies their totals up to a constant, and Thm. YM3_I:X identifies the coefficient of such totals), for both families. This gives the Hessian of the family at the trivial triple and its coefficient, $\frac{C_A}{8} b^0_{k+1} + \gamma_G(j_{k+1} - L^{-1} j_k)$, the scalar family having the coefficient $j_{k+1} - L^{-1} j_k = b[\mathfrak{B}_{k+1}] - L^{-1} b[\mathfrak{B}_k]$ by (i); this is the germ of second order of the family at the trivial triple, which the proof of Prop. 9.23(iii) computes in the same way, by the sectors of (i); on an ideal $C_A^{(i)}$, $\gamma^{(i)}$ take the places of C_A, γ_G, by the splitting of (iv).

(iv) A basis split along the ideals. For weights the form $\langle \cdot, \cdot \rangle_w$, the Hessian of the weighted action at $\mathbf{1}$, takes the place of $\langle \cdot, \cdot \rangle$, and $C_A^{(i)}$, θ_i, $\gamma^{(i)} = \frac{1}{2}\theta_i - \frac{3}{16} C_A^{(i)}$ become $C_A^{(i)}(w)$, $\theta_i(w)$, $\gamma^{(i)}(w)$; a rational weight is a multiplicity of a representation, which brings the weighted action to an unweighted one in a larger representation, at a coupling multiplied by a number, and both members of the display are analytic in the weights, the rationals being dense: so it holds at all positive weights. The term with $\gamma^{(i)}(w)$ drops in the limit, $t_k \to 0$, and $C_A^{(i)}(w) \langle P_i X, P_i X \rangle_w = \mathrm{tr}(\mathrm{ad}\, P_i X)^2$, whose sum over i is $\mathrm{tr}(\mathrm{ad}\, X)^2$. For G_i-valued W near $\mathbf{1}$ the chain of the pair is that of the simple pair of $\mathfrak{g}_i$; the linear averages and the second variation split over the orthogonal sum $\bigoplus_l \mathfrak{g}_l \otimes \mathbb{R}^{\mathsf{b}_0}$, $\mathbb{R}^{\mathsf{b}_0}$ the real functions on the bonds of Λ_0, the ideals $j \neq i$ seeing the plaquettes of U only through scalar weights; so the normaliser factorises into that of the simple pair times n_j-th powers of scalar Gaussian integrals, whose second derivative is the display. These scalar integrals are those of the tadpole of (i), whose propagator is the projection onto $d(\ker \mathsf{Q}^{[k]})$ with the diagonal $\mathsf{v}_k = w_k$; so $\Upsilon_k = \frac{1}{2} \Pi^{\mathfrak{B}_k}$, which is real, symmetric, vanishes on closed forms, is invariant under the isometries and is tempered by (i), with the coefficient $\frac{1}{2} j_k = t_k$, $j_k = \frac{2}{3}(L^{-k} - L^{-4k})$ being that of $\mathfrak{B}_k$ by (i), (iii) and Thm. AA(c); $0 \leq \mathsf{v}_k \leq 1$ gives $0 \leq \Upsilon_k \leq \frac{1}{2}\mathsf{h}_k$, with $b[\mathsf{h}_k] = L^{-k}$, the composite $A \circ p_1 \circ \cdots \circ p_k$ having the coefficient L^{-k}. Adding the coefficient $\frac{C_A^{(i)}}{8} b[\Psi_k] + (n_i \tau_{ii} - \frac{3}{8} C_A^{(i)}) t_k$ of the simple pair and $(\sum_{j \neq i} n_j \tau_{ij}) t_k$ gives the formula with θ_i. The bound ideal by ideal is that of (i) with $C_A^{(i)}$, θ_i for C_A, C_ρ, as $0 \leq t_k \leq \frac{1}{3} L^{-1}$. At $SU(2) \times SU(2)$ a unit H of an ideal has a factor $h \in \mathfrak{su}(2)$ and $C_A^{(i)} = \mathrm{tr}(\mathrm{ad}\, h)^2 = 4\,\mathrm{tr}\, h^2$, which is 16 on $\mathbb{C}^2 \oplus \mathbb{C}^2$, where $\mathrm{tr}\, h^2 = 4$, and 8 on $\mathbb{C}^2 \otimes \mathbb{C}^2$, where $\mathrm{tr}\, h^2 = 2$; in both $\Omega = 6 \cdot \mathbf{1}$, so $\theta_i = 6$ on both ideals. □

9.1.7. *Hessians and conjugation at complex fields.*

Lemma 9.19 (The Hessian at complex fields)**.** Let U be $SL(N, \mathbb{C})$-valued, p a plaquette with its bonds $b_1, \ldots, b_4$ numbered as in the $SU(2)$ construction, $U_i = U(b_i)$ and $Y_p = \partial U(p) = U_1 U_2 U_3^{-1} U_4^{-1}$; for an $\mathfrak{sl}(N, \mathbb{C})$-valued form A let $a_1, \ldots, a_4$ be its values on the bonds of p transported to the base point along the plaquette in its order,

$$a_1 = A(b_1), \quad a_2 = \mathrm{Ad}(U_1) A(b_2), \quad a_3 = \mathrm{Ad}(U_1 U_2 U_3^{-1}) A(b_3), \quad a_4 = \mathrm{Ad}(Y_p) A(b_4),$$

so that $\partial(e^{it\xi A}U)(p) = e^{it\xi a_1}e^{it\xi a_2}e^{-it\xi a_3}e^{-it\xi a_4}Y_p$; with the signs $\epsilon = (1,1,-1,-1)$ let $S_p = \sum_i \epsilon_i a_i$. Here and below $|\operatorname{Ad} g| = |g|\,|g^{-1}|$ is the norm of $X \mapsto gXg^{-1}$ on all $N \times N$ matrices. With no condition on U, the Hessian of the action of $SU(N)$ is

$$(A, \Delta(U)A) = \xi^3 \sum_p \Big\{ \tfrac{1}{N}\operatorname{tr}\big(\xi^{-1}S_p\, c(Y_p)\, \xi^{-1}S_p\big) + i \sum_{i<j} \epsilon_i\epsilon_j \langle [a_i,a_j], F(p)\rangle \Big\}.$$

Let $f^\circ(b) = \xi^{-2}\max_{p\ni b}|Y_p - \mathbf{1}|$, and $f(b)$ the same maximum over the plaquettes with a corner within sup distance ξ of b_-. If $|\operatorname{Ad}(U(b))^{\pm1}| \le 2$ at all bonds and $|Y_p - \mathbf{1}| \le \frac12$, the kernel $\Delta(U)(b,b')$ vanishes unless b and b' lie in a common plaquette, $\xi^2|\Delta(U)(b,b')| \le 259$, and on forms on a region R interior to the ambient one

$$\Delta_{\mathsf{R}}(U) + DD^{\mathrm{T}} = \Delta^\circ + Z(U), \qquad |(Z(U)A)(b)| \le 1346 f^\circ(b)\max|A| \le 3456 f(b)\max|A|, \quad Z(\mathbf{1}) = 0,$$

Δ_{R} the compression of $\Delta(U)$ to the forms on R and Δ° the covariant Dirichlet Laplacian, $(\Delta^\circ\phi)(x) = \xi^{-2}[6\phi(x) - \sum_\nu(\operatorname{Ad}_\nu(x)\phi(x+\xi e_\nu) + \operatorname{Ad}_\nu(x-\xi e_\nu)^{-1}\phi(x-\xi e_\nu))]$, $\operatorname{Ad}_\nu(x) = \operatorname{Ad}(U(x, x+\xi e_\nu))$. Under the guard $|Y_p - \mathbf{1}| \le 1$ of the $SU(2)$ construction the kernel is at most $180\xi^{-2}$ and $|(Z(U)A)(b)| \le 868 f(b)\max|A|$; so the numerals 1536 and 3456 of the $SU(2)$ construction serve in all its clauses. If U is $SU(N)$-valued with $|Y_p - \mathbf{1}| \le \sqrt{2}$, then $-12\sup f^\circ\,(A,A) \le (A,\Delta(U)A) \le (16\xi^{-2} + 12\sup f^\circ)(A,A)$ for $\mathfrak{su}(N)$-valued A. The constants are absolute, the same for every $N \ge 2$. Let $\mathfrak{g}$ be semisimple, so that $G \subset SU(N)$ and $G^c \subset SL(N,\mathbb{C})$, and let U be G^c-valued. Then the Hessian of the action of G is this form restricted to $\mathfrak{g}^c$-valued forms, and its operator and its Weitzenböck term are those of $SU(N)$ followed by P, the orthogonal projection onto $\mathfrak{g}^c$ for $\operatorname{tr} X^*Y$: the bounds on the quadratic form and the bounds in the normalised trace norm stay, and the kernel and Weitzenböck bounds in $|\cdot|$ hold with the factor $|P| \le N^{1/2}$, the norm of P for $|\cdot|$, the condition $|\operatorname{Ad}(U(b))^{\pm1}| \le 2$ being read on all $N \times N$ matrices. Read with Ad on $\mathfrak{g}^c$ alone, the clauses under $|Y_p - \mathbf{1}| \le \frac12$ stand, with $|Y_p^{-1}| \le 2$ by the Neumann series, and under the guard $|Y_p - \mathbf{1}| \le 1$ the inverse obeys $|Y_p^{-1}| \le |Y_p|^{N-1} \le 2^{N-1}$, a number of the pair, in the place of 16. ∎

Sketch of proof. The identity is the second derivative of $a(e^{it\xi a_1}\cdots)$ written with $c(Y_p)$ and H_{Y_p}; in operator form the bilinear maps N_{ij}, such as $X \mapsto \frac12(XY_p + Y_p^{-1}X)$, have norm at most $\frac12(|Y_p| + |Y_p^{-1}|)$ and differ from the identity by at most $\frac12(|Y_p - \mathbf{1}| + |Y_p^{-1} - \mathbf{1}|)$.

The inverse of a plaquette variable is the one new quantity, and $|X^{-1}| \le |\operatorname{Ad} X|$ on $SL(N,\mathbb{C})$, Ad acting on all $N \times N$ matrices (Lem. 9.9(ii)), bounds it with no N: $|\operatorname{Ad}(U(b))^{\pm1}| \le 2$ at four bonds gives $|Y_p^{-1}| \le |\operatorname{Ad} Y_p| \le 16$; on $\mathfrak{g}^c$ alone the inequality $|X^{-1}| \le |\operatorname{Ad} X|$ fails, as at the representation of spin 2 of $SU(2)$ at $X = e^{(\log 2)J_3}$, where $|X^{-1}| = 4$ and the norm of $\operatorname{Ad} X$ on $\mathfrak{g}^c$ is 2.

Before the projection P the count of the kernel is the same in $|\cdot|$ and in $\|\cdot\|$: $|N_{ij}| \le \frac12(\frac32 + 2) = \frac74$, the transports give $|R_i^{-1}||R_i| = 1$ at the first position of a bond in a plaquette and at most $(3\cdot2)^2 = 36$ at the third, $R_3 = \operatorname{Ad}(Y_pU_4)$, and the four plaquettes of a bond hold it at most at the positions $1,3,1,3$, as for a bond of the first direction: so $\xi^2|\Delta(U)(b,b')| \le \frac74 \cdot 74 = 129.5$ before P; P has the norm 1 for $\|\cdot\|$ and $2 - \frac{2}{N} < 2$ for $|\cdot|$, whence 259 in $|\cdot|$, with no passage between the two norms; the formula of Weitzenböck, with the covariant Laplacian, which is the operator of U itself and not the free one, gives 1346, and $f^\circ \le f$.

Under the guard $|Y_p - \mathbf{1}| \le 1$ the finer count, in which $\operatorname{Ad}(Y_p)$ drops out of every block, gives 180 and 868.

With the eigen-angles of a unitary Y_p the bounds on the form follow as in the $SU(2)$ construction. For semisimple $\mathfrak{g}$ the action of G is the restriction of that of $SU(N)$ to G^c-valued fields, $\operatorname{Ad}(U(b))$, DD^{T} and the covariant Laplacian keep $\mathfrak{g}^c$-valued forms, and $\langle [a_i,a_j], F(p)\rangle = \langle [a_i,a_j], PF(p)\rangle$, $\langle A, PB\rangle = \langle A, B\rangle$ for $A \in \mathfrak{g}^c$; so the form is the restriction, and the operator and Z are P times

those of $SU(N)$. As P is orthogonal for $\operatorname{tr} X^*Y$, the form bounds and the bounds in the trace norm stay; in $|\cdot|$, $|PX| \le (\operatorname{tr}(PX)^*PX)^{1/2} \le N^{1/2}|X|$. □

Lemma 9.20 (Conjugation)**.**

(i) $X \mapsto X^c = (X^*)^{-1}$ is an antiholomorphic involutive automorphism of $GL(N,\mathbb{C})$ whose fixed points are the unitary matrices; it keeps G^c; $X^c - \mathbf{1} = (X^{-1} - \mathbf{1})^*$; $\partial(U^c)(p) = \partial U(p)^c$, $H_{X^c} = H_X^*$, and F, J and covariant differences are covariant likewise; $a(X^c) = \overline{a(X)}$.

(ii) For a set Q of points, $Q^s = Q \cap Q^c$ is the largest subset kept by the conjugation; it holds the real points, commutes with intersections and is open with Q; if $Q = \{|X_\alpha(p) - \mathbf{1}| < r_\alpha\}$ with covariant X_α of group type, then $Q^s = \{m(X_\alpha) < r_\alpha\}$. $m(X^{-1}) = m(X^*) = m(X^c) = m(gXg^{-1}) = m(X)$ for unitary g, and $x \le m(X) \le \frac{x}{1-x}$ for $x = |X - \mathbf{1}| < 1$. If $|Y - \mathbf{1}| \le \varphi$ on a set kept by the conjugation, Y covariant and φ symmetric, then $m(Y) \le \varphi$. If $m(X) < 1$, then $\log X^{-1} = -\log X$ and $\frac{1}{i}\log X^c = (\frac{1}{i}\log X)^*$; and $\operatorname{av}(V^c) = \operatorname{av}(V)^c$.

(iii) The chain of objects of a step from the fields to the fluctuation operator commutes with the conjugation wherever both sides are formed, $A^\sharp(p^c) = \theta A^\sharp(p)\theta^{-1}$ with $\theta Z = Z^*$, on the domain of the $SU(2)$ construction written with $m(X) < 1$ and with $X \in D^c$ at the logarithms of the averages, the latter condition being void for $SU(N)$ with $N \le 6$. On this domain the symmetric action $a(X)$ suffices while the proviso on means of Lem. 9.9(iv), the contour quotients in D^c, is void, that is for $z_G > \frac{1}{40}$, at $SU(N)$ exactly for $N \le 251$; from $N = 252$ on the quotients in D^c are added to the domain. This applies in particular to the intrinsic cube response of Prop. YM3_I:54.2, with the uniqueness ball and radii of Prop. YM3_I:54.1; to the unparameterized response obtained there by setting $s = \mathbf{1}$ and $\varkappa = 0$; and to the potential and axial gauge (A, h) of Thm. YM3_I:E, constructed in Section YM3_I:38. Each is taken on its stated domain. For another group the same covariance assertion applies wherever the corresponding construction and uniqueness bounds are supplied under (V); it does not supply those bounds.

(iv) For $SU(N)$, $N \ge 3$, the moduli are not kept: with $H = \operatorname{diag}(1, \dots, 1, -(N-1))$ and $s > 0$, e^{-sH} is the conjugate of e^{sH} and $|e^{sH} - \mathbf{1}| = \max\{e^s - 1, 1 - e^{-(N-1)s}\} < e^{(N-1)s} - 1 = |e^{-sH} - \mathbf{1}|$, at $SU(3)$ with $H = \operatorname{diag}(1, 1, -2)$; so neither the analyticity cores of the $SU(2)$ construction nor the moduli of the quantities of its towers are kept by $X \mapsto X^c$ unless the conditions are written with $m(X)$.

■

Sketch of proof. (i) is direct, G^c being closed under adjoints and inverses; all of it says that $X \mapsto (X, X^{-1})$ carries the conjugation to the involution $\Theta(E, F) = (F^*, E^*)$ of the ring $M_N \times M_N^{\mathrm{op}}$, of which m, a and H_X are a norm, a trace and the odd part. (ii) The adjoint keeps the operator norm, so $m(X^c) = \max\{|X^{-1} - \mathbf{1}|, |X - \mathbf{1}|\}$; the bound by $\frac{x}{1-x}$ is Lem. 9.9(ii); for a covariant Y the bound at the point and at the conjugate point gives both members of m; the logarithms follow from the series, and the mean from its uniqueness (Lem. 9.9(iv)). (iii) Products, inverses, means, logarithms, the action a and its derivatives commute with the stated conjugation by (i), (ii). For the regional response retain the two equations YM3_I:(1051), with the regional substitutions YM3_I:(1052). In particular the identity in $S - I$ remains untruncated. The current is extended by zero, and the nonlinear maps and every surviving walk term use only the pair in the region. The proof of Prop. YM3_I:54.2 identifies the regional first-difference norm on the response carrier with that for any continuation. Thus conjugation carries both equations and their uniqueness ball to the equations and ball at the conjugate data:

$$\max\{\|T\|, |\lambda - H^{\mathfrak{k}}(s)^{\mathrm{T}}\mathbf{J}^0|_\infty\} \le C_{\mathrm{D,s}}\varepsilon_{\mathrm{D,s}}e^{-8\varkappa}.$$

The parameter and datum restrictions remain $|s| \le e^{\varkappa}$ and $|B'|_\infty, \gamma_0 \le \varepsilon_{\mathrm{D,s}}e^{-16\varkappa}$. Here the field and multiplier are conjugated by $Z \mapsto Z^*$, which preserves $|Z|$ and intertwines the covariant differences;

the group-valued domain restrictions use m. Uniqueness gives the claimed response identity. The same argument at $s = \mathbf{1}$ uses the unparameterized bounds, with $\varkappa = 0$.

For the canonical pair, the inverse datum twist YM3_I:(580) commutes with conjugation. Its constrained response is the one of Thm. YM3_I:K, and the inverse formula for its multiplier in Section YM3_I:38 consists of covariant operations. Finally the pillar recursion YM3_I:(584) constructs h from the response by products, inverses and means. These operations give $A(B^*) = A(B)^*$ and $h(B^*) = h(B)^c$ at the real reference of that construction. Every radius used here is the radius of the cited construction, restricted further by this lemma's two-sided guards. For G, the same calculation applies to the supplied counterparts under (V). (iv) is a computation with the eigenvalues. □

Lemma 9.21 (Reflection on the real-connected part)**.** Let Q be a set of points $p = (\varrho, v)$, ϱ real parameters and v complex fields and numbers; let $p \mapsto p^c$ act by $X \mapsto X^c$ on group-valued members, by $Z \mapsto Z^*$ on $\mathfrak{g}^c$-valued ones and by complex conjugation on numbers; p is real if $p^c = p$. Let $Q^s = \{p \in Q : p^c \in Q\}$, and let Q^r, the *real-connected part*, be the set of $p \in Q^s$ whose connected component in the slice $(Q^s)_\varrho$ holds a real point.

(i) Q^r is kept by the conjugation, holds the real points of Q, is monotone in Q, has open slices if Q has, and is the largest subset of Q^s every slice component of which holds a real point; polydiscs and products pass; $Q^r = Q^s$ if the slices of Q^s are star-shaped about real points, or if the slices of Q are convex and have no group-valued member, the conjugation being then an antilinear involution; the convex clause asks for this last condition, as at $G = U(2)$, where the segment from $\mathrm{diag}(2,3)$ to $\mathrm{diag}(\frac{1}{2}, \frac{1}{3})$ has conjugate ends and keeps the distance 0.08 from $U(2)$, its open 0.05-neighbourhood Q in $|\cdot|$, which is convex, lies in $GL(2,\mathbb{C})$ and holds no unitary matrix, has $Q^s \neq \emptyset$ and $Q^r = \emptyset$; and $(\bigcap_i Q_i)^r \subset \bigcap_i Q_i^r$, possibly strictly.

(ii) Let the slices of Q_1^r be open and let Φ be holomorphic in v there, with real members constant on slice components. Then $\Phi(p^c) = \Phi(p)^c$ on Q_1^r if and only if $\Phi(p)$ is real for every real $p \in Q_1$; and then $\Phi(Q_1^r) \subset Q_2$ implies $\Phi(Q_1^r) \subset Q_2^r$. Composites of such maps and of continuous covariant maps are covariant; a composite of maps built from products, inverses, means and logarithms and of continuations on convex subsets of a vector space kept by the conjugation (slices without group-valued member) is covariant on all of Q^s, and so are the towers wherever they exist at both points. If the slices of Q are open, the function equal to i on $Q^s \setminus Q^r$ and to 0 on Q^r is holomorphic, real at real points, and not covariant at any point of $Q^s \setminus Q^r$.

(iii) Let $\mathcal{Q}(b_0, b_1, c_0)$ be the set of the $SU(2)$ construction of the triples $(U, A', \mathbf{J})$, U real with $|\partial U(p) - \mathbf{1}| < b_0\xi^2$, A' with $|A'|$ and its covariant differences below b_1, $|\partial(e^{i\xi A'}U)(p) - \mathbf{1}| < b_0\xi^2$ at the plaquettes and $|\mathbf{J}| < c_0$; its symmetric part has $m(\partial(e^{i\xi A'}U)(p))$ in the place of the one-sided distance. Then

$$\mathcal{Q}(b_0, b_1, c_0)^s \subset \mathcal{Q}\big(b_0 + 2e^{4\xi b_1} b_1^2, b_1, c_0\big)^r,$$

the ray $(U, tA', t\mathbf{J})$, $0 \le t \le 1$, joining each point to the real point $(U, 0, 0)$ inside the larger set. For the classes of regular pairs the same holds with $\alpha_0 + 2\vartheta^2 e^{4\vartheta\alpha_1}\alpha_1^2$ in the place of α_0, $\vartheta \ge 1$ the largest ratio of the spacings around a plaquette, and the like holds for the natural classes. For the analyticity cores, $\mathcal{U}^\circ(b_0, b_1, c_0)^s \subset \mathcal{U}^\circ(B_0, B_1, c_0)^r$ whenever $b_0' = b_0 + 2e^{4\xi b_1} b_1^2 \le B_0$, $b_1 \le B_1$, $K(b_0' + b_1^2) \le B_0$ and $b_0' + b_1 \le a^\sharp$, where $K \ge 16L^2$ and $a^\sharp$ are the constant and the radius of the inclusion of the simple sets in the cores, in its symmetric form, assumed for G through Prop. 9.22(ii) and the uniqueness clause of §9.1.1(V). With Q^r in the place of Q^s no number moves but at a membership reached from memberships of restrictions had apart: such a membership is reached in Q^s, and the inclusions above carry it into a real-connected part, at the radii named there. The symmetric parts of the domains of the $SU(2)$ construction are cut out by conditions $m(X) < r$ and

commute with intersections (Lem. 9.20(ii)); the real-connected parts pass to intersections as the inclusion $(\bigcap_i Q_i)^r \subset \bigcap_i Q_i^r$ of (i).

■

Sketch of proof. (i) is set topology. In the instance at $U(2)$ the distance in $|\cdot|$ of a positive diagonal matrix from $U(2)$ is the largest distance of its entries from 1, and from the singular matrices its least entry, singular values moving by at most the norm of a difference; on the segment, $\mathrm{diag}(2-\frac{3}{2}t, 3-\frac{8}{3}t)$ with $0 \le t \le 1$, the first is least at $t = \frac{18}{25}$, where it is 0.08, and the second is at least $\frac{1}{3}$. (ii) $p \mapsto \Phi(p^c)^c$ is holomorphic in v, as a composite of two antiholomorphic maps, and agrees with Φ at the real points, which form in each slice a totally real submanifold of maximal dimension; by the identity theorem the two agree on every component that holds a real point. No formula of Φ is used, so sums of group elements, complex constants and maps defined as the one solution in a ball are covered. Conversely covariance at a real point says that its image is real. (iii) Along the ray tA' a product of four exponentials differs from its chord by

$$|X_t - (1-t)X_0 - tX_1| \le 8\, t(1-t)\, \xi^2 d^2 e^{4\xi b_1},$$

where d is the maximum operator norm of the four transported parameter increments A' in this plaquette product, so $d < b_1$; the same estimate holds for X_t^{-1}. Thus $m(X_t) \le (1-t)m(X_0) + tm(X_1) + 8t(1-t)\xi^2 d^2 e^{4\xi b_1}$, and with $X_0 = \partial U(p)$, $m(X_0) = |\partial U(p) - \mathbf{1}| < b_0\xi^2$, $m(X_1) < b_0\xi^2$ at a point of the symmetric part and $d < b_1$ this gives $m(X_t) < b_0\xi^2 + 2e^{4\xi b_1} b_1^2 \xi^2$, as $8t(1-t) \le 2$, the bounds on tA' and $t\mathbf{J}$ being smaller than those on A', $\mathbf{J}$; so the ray from $(U, A', \mathbf{J})$ to the real point $(U, 0, 0)$ stays in the larger set. For the regular pairs the ray is the same, the factor ϑ entering through the spacings of the plaquettes. For the cores, $\mathcal{U}^\circ(b_0, b_1, c_0)^s \subset \mathcal{Q}(b_0, b_1, c_0)^s$, and the ray lies in $\mathcal{Q}(b_0', b_1, c_0)^s$, which lies in $\mathcal{U}^\circ(B_0, B_1, c_0)^s$ by the inclusion of the simple sets under the conditions named, and it ends at a real point. □

Proposition 9.22 (Fields off the trivial field)**.** Let $\mathfrak{g}$ be simple, $L \ge 13$ odd, $200L^2 z_G \ge 1$; constants are constants depending only on L and the pair, uniform in the volume. In (i), the completed propagator construction assumes (P). The uniform critical-background, tower and core assertions in (ii), as well as (iii)–(v), assume (V). The explicit Hessian identities and bounds, the averaging formulas on their guarded domains, and holomorphic germs at each fixed finite tower are local statements. A fixed finite-dimensional germ does not assert a radius or a nonlinear majorant uniform in depth and volume.

(i) Let $G = SU(N)$, $N \ge 2$ arbitrary. The Hessian is given by Lem. 9.19, the current by Lem. 9.12(i), and the trivial-field operators by Lem. 9.15. Their gauge covariance follows from their transport formulas. The remaining quantitative inputs—the local action remainders, difference and commutator estimates in the natural norms, their variable-layer forms, the local gauges, and the order-zero inverse at rate $\frac{1}{4}\log(37/36)$—are included in (P). For $SU(2)$ the local nonlinear estimates are supplied by Props. YM3_I:30.1 and YM3_I:33.1; the completed inverse is Thm. YM3_I:G. The following Wilson-Hessian bounds are proved directly below. In the Wilson part $\Delta_U^W = D^{\mathrm{T}}D + \Delta'(U)$ the operator $\Delta'(U)$ is local, symmetric and real on real forms, with row sums at most $18\alpha \mathsf{L}_{\mathbf{B}}(b)^{-2}$ in $|\cdot|$ and $20\alpha \mathsf{L}_{\mathbf{B}}(b)^{-2}$ in $\|\cdot\|$ for U in the class $\mathfrak{U}_k(\mathbf{B}, \alpha)$ of Def. 9.2 with $\alpha \le \frac{3}{8}$, and, for U in that class at every α, $(A, \Delta_U^W A) \ge \eta^3 \sum \|DA\|^2 - 32\alpha\eta^3 \sum_b \mathsf{L}_{\mathbf{B}}(b)^{-2} \|A(b)\|^2$, with no N, $\mathsf{L}_{\mathbf{B}}$ being the local block side of the determining set $\mathbf{B}$ (Def. YM3_I:9.3): $\mathsf{L}_{\mathbf{B}}(\mathfrak{e}) = L^{j-k}$, j the largest index for which the bond or plaquette $\mathfrak{e}$ has an end point or a corner in Ω_j, so that $\eta^2 \mathsf{L}_{\mathbf{B}}(\mathfrak{e})^{-2} = L^{-2j}$; and $|\tau^* - U(\Gamma)| \le \kappa_7 \alpha_0$ for the transports of the chains, with the transport-comparison constant $\kappa_7(L)$ in the proof of Lem. YM3_I:30.3. The same path and mean comparisons apply on the guarded domains of Lem. 9.11. The completed propagator and covariance estimates are hypothesis (P) of §9.1.1; bounds of them in $|\cdot|$ give the same bounds in the normalised trace norm, with the factors $(N-1)^{1/2}$ and $N^{1/2}$ between the two (Lem. 9.9(i)). For the

other simple G of the proposition Lem. 9.19 gives the Hessian among these inputs, with the factor $|P|$ in its kernel and Weitzenböck bounds.

(ii) For every G of the proposition, averages exist, with logarithms in $\mathfrak{g}^c$, at every regular real background. Over every real e^{iB} with $\sup|B|$ small there is a regular critical configuration with a holomorphic continuation $W(B)$ on a polydisc whose radius depends on no scale, and $|\partial W - \mathbf{1}| \le C \sup|B|\, \eta^2 \mathsf{L}_{\mathbf{B}}^{-2}$, $|J(W)| \le C \sup|B|\, \mathsf{L}_{\mathbf{B}}^{-3}$ at the plaquettes and the bonds, $\mathbf{B}$ the determining set and $\mathsf{L}_{\mathbf{B}}$ its local block side, as in (i); the flatness L^{-2q} of towers of depth q, with plaquettes within cL^{-2q} of $\mathbf{1}$ at every truncated depth, is included in this hypothesis. The linearised tower kernel is independently the same real scalar kernel for all G; control of the nonlinear remainder at this scale is required in (V). Under the uniform estimates of Prop. 9.17, small fields lie in the analyticity cores of G uniformly in the scale, the sub-domain, the cutoff and the volume.

(iii) The localisation input in (V) is the following counterpart of Prop. YM3_I:53.5. Use exactly the separated joined sequence, the one or two disjoint wide collars, the regular original and joined data, and the torus-size hypothesis of Section YM3_I:53.4. For arbitrary elements U, U' of the two minimal orbits, require gauges w, w' and a real odd field a such that $(U|_\Omega)^w = e^{ia}(U')^{w'}$, and, outside the wide collars,

$$|a(b)| \le \mathsf{C}_{\mathrm{lc}} \varepsilon_{\mathrm{r}} L^{-j(b)} e^{-\delta_{\mathrm{lc}} D(b)/2}, \qquad w'(y) = 1, \quad |w(y) - 1| \le \mathsf{C}_{\mathrm{lc}} \varepsilon_{\mathrm{r}} e^{-\delta_{\mathrm{lc}} D(y)/2}$$

at the nodes y. Here $j(b)$ is the largest level containing an endpoint, and D is the weighted distance to the endpoints of the joined tower bonds, with $D = +\infty$ when there are none. The constants are independent of depth and volume. Retain all five entries of the smallness ceiling in equation YM3_I:(1000), evaluated at the admitted constants of the pair, as well as its geometric floors and response hypotheses. This is the comparison used in (iv); no localisation estimate for a different joining geometry is inferred.

(iv) Let the localisation of (iii) hold for G, and let the objects of a step be formed for G at the fields over a regular base, which is part of (V): the global and the localised background of the step are defined there; the minimal orbits of the three data of the step exist and have elements whose k-fold averages are in the axial gauge; their elements U' have $|\partial U'(p) - \mathbf{1}| < 2B_3 \varepsilon_t L^{-2j}$ at the inner bare plaquettes p, j the largest index for which p has a corner in Ω'_j; and the data lie in the regular sets of radius $2\varepsilon_t$. For simply connected G they follow under (V), by (iii). The zone-bond conditions use the following explicit ceilings. The background constants are those admitted in (V), and c_{Q}, c_{s} and c_{fl} have the formulas in equation YM3_I:(1026), evaluated at the constants of the pair. Put

$$\begin{aligned}
\epsilon_{\mathrm{Q}} &= \min\{\varepsilon_{\flat}, \bar{\varepsilon}/2, c_2^{\mathrm{SO}}/(2B_3), \omega_{\mathrm{p}}/(2 + 4B_3 L^{-2}), 1/520\},\\
\epsilon_{\mathrm{fl}} &= \min\{\epsilon_{\mathrm{Q}}, (1 + \mathsf{c}_{\mathrm{s}})^{-1}\}, \qquad \mathsf{c}_{\mathrm{H}} = \pi \kappa_{\mathsf{c}} \sqrt{2L} \max\{2\bar{C}_{\mathrm{reg}}, 4B_3\},\\
\epsilon_{\mathrm{H}} &= \min\{\epsilon_{\mathrm{fl}}, c_2^{\mathrm{SO}}/(2B_3), \pi/[\pi(1 + \mathsf{c}_{\mathrm{s}}) + \mathsf{c}_{\mathrm{H}}]\},\\
\epsilon_{\mathrm{cu}} &= \min\{\epsilon_{\mathrm{lc}}/2, c_2^{\mathrm{g}}/(2B_3), c_4/(4\mathsf{C}_{\mathrm{lc}})\}, \qquad \mathsf{c}_L = 4 + 2C_v + 4\kappa_6,\\
\epsilon_{\mathrm{xp}} &= \min\{\epsilon_{\mathrm{H}}, \epsilon_{\mathrm{cu}}, \epsilon_{\mathrm{lc}}/2, c_2^{\mathrm{g}}/(2L^2 B_3), c_4/(4L\mathsf{C}_{\mathrm{lc}}), (2\mathsf{c}_L \mathsf{C}_{\mathrm{lc}})^{-1}, (2 + \mathsf{c}_{\mathrm{s}})^{-1}\}.
\end{aligned}$$

Here $\bar{C}_{\mathrm{reg}}$ is the coupling-independent regularity bound, $\varepsilon_{\flat}$ is the background radius, κ_{c} is the contour-comparison constant, and $c_2^{\mathrm{g}}, c_4, C_v$ are the curvature, perturbation and gauge-factor constants of the centered averaging estimate in Lem. 9.11(v). Assume $\varepsilon_t \le \epsilon_{\mathrm{xp}}$; $A_1/A_0 \le c_{\mathrm{Q}}$; $\gamma \le \gamma_{\mathrm{xp}}^G$, where $\gamma_{\mathrm{xp}}^G = e^{-y_{\mathrm{xp}}^G/2}$ and y_{xp}^G is formed by equation YM3_I:(1016) with $\|\vartheta'_G\|_\infty = \lambda \|\vartheta'_{\mathrm{co}}\|_\infty$ of Lem. 9.14(ii) in the place of $\|\vartheta'_{\mathrm{co}}\|_\infty$ in Υ_1, the one formula of the chain that changes; and $C_\chi \mathsf{c}_{\mathrm{fl}} \delta_k \le r_\chi \le 1 \le C_\chi$, with the chart constants r_χ, C_χ of [7, §B.2], whose transferred bounds are among the construction hypotheses. Here ε_t and δ_k are the thresholds of the levels t and $k = t - 1$ of the step, g_k the running coupling and

$\check R_t$ the cube size (equation YM3_I:(634), YM3_I:(634)), $\mathfrak{b}_k = \delta_k/g_k$ is the radius of the scaled fluctuation field (equation YM3_I:(634)), c_{fl} is the constant of the fluctuation range (equation YM3_I:(1026)), C_{lc}, δ_{lc} are those of the localisation, and $\mathsf{c}_L = 4 + 2C_v + 4\kappa_6$ as above, with the radius c_4 and the constants C_v, κ_6 of the averages over a box and of the contour variables (Lem. 9.11(v), with the one-step contour estimate of Lem. YM3_I:14.7), which for G are constants depending only on L and the pair with $|f_i - \mathbf{1}| \le C_v L^i \sup|A_0|$ whenever $L^j \sup|A_0| < c_4$, in the notation of Lem. 9.11(v). Put $\varsigma = 2\mathsf{C}_{\mathrm{lc}}\varepsilon_t \exp(-\frac14 \delta_{\mathrm{lc}} M \check R_t)$; the fifth member of ϵ_{xp}, $4L\mathsf{C}_{\mathrm{lc}}\varepsilon_t \le c_4$, gives $\varsigma < \frac{c_4}{2L}$. Let finally $\mathsf{c}_L c_4 \le 2Lr_G$, which is void at $SU(3)$. Then at each bond b of the cut-off zone the quotient E_b of the global and the localised background has $|E_b - \mathbf{1}| \le \mathsf{c}_L\varsigma < r_G$; $H_b = \frac1i \log E_b \in \mathfrak{g}$, the shift of YM3_I:(989), has $|H_b| \le \frac{\pi}{2}\mathsf{c}_L\varsigma$ and $8\|\vartheta'_G\|_\infty |H_b| = 8\lambda\|\vartheta'_{\mathrm{co}}\|_\infty |H_b| \le \delta_k$; with $\theta(W) = \vartheta_G(\nu(Y(W))/\delta_k)$ where -1 is no eigenvalue of W, $Y(W)$ the principal hermitian logarithm, and $\theta(W) = 1$ elsewhere, the bond map $W \mapsto We^{-i\theta(W)H_b}e^{iH_b}$ of Lem. YM3_I:53.3 is a diffeomorphism of G onto itself, its inverse at W being the fixed point of the contraction $W' \mapsto We^{-iH_b}e^{i\theta(W')H_b}$ of factor at most $\frac{\pi}{16}$, with Jacobian in $[\frac78, \frac98]$; and as $g_k\mathsf{c}_{\mathrm{fl}}\mathfrak{b}_k + |H_b| = \mathsf{c}_{\mathrm{fl}}\delta_k + |H_b| \le 1 + \frac{1}{24} < \pi$, the substitution $W \mapsto We^{-i\theta(W)H_b}$, $W = e^{ig_kY(b)}$, is a homeomorphism between open sets each holding $\{g_k|Y(b)| + |H_b| < \pi\}$, outside of which the integrands vanish.

(v) The domains of analyticity are defined for G, are gauge stable, and are the same on all large tori and on $\mathbb{R}^3$; the totals do not depend on the chosen background; and at the trivial real factor the continuations and the bounds of the towers are those of (ii). The continuations and the towers off the trivial field are those of (V), which are assumed in (V) for the simply connected construction.

■

Sketch of proof. (i) The local identities follow from the current and Hessian formulas. The quantitative inverse and local-remainder bounds specified in (P) are assumptions here. The following calculation proves the asserted Wilson-Hessian bounds without using those assumptions.

The group enters them through the Hessian; through the commutators, $|[X,Y]| \le 2|X||Y|$; through sums, products and Neumann series of decaying kernels in $|\cdot|$, which are sums over the lattice and keep the constants of the $SU(2)$ construction for every N; through the transpose of a kernel, which takes N for the 2 of $|T^{\mathrm T}| \le 2|T|$, and the bounds of Schur's kind, whose numeral 3 becomes n (Def. 9.5(2), Prop. 9.26(g)); through the table at the trivial field (Lem. 9.15); and through the small-field theorem in the gauge of the class (Lem. 9.12(iii)).

The Wilson part $\Delta'(U) = \Delta(U) - D^{\mathrm T}D$ is read off the identity of Lem. 9.19, with $\xi = \eta$: at a unitary Y_p its terms are $\xi^{-2}\frac1N \mathrm{tr}(S_p(c(Y_p) - \mathbf{1})S_p)$ and $i\sum_{i<j}\epsilon_i\epsilon_j\langle[a_i,a_j],F(p)\rangle$, where $|c(Y_p) - \mathbf{1}| = \frac12 x_p^2$ with $x_p = |Y_p - \mathbf{1}|$, and $\langle[a_i,a_j],F(p)\rangle = \xi^{-2}\langle[a_i,a_j],H_{Y_p}\rangle$ with $|H_{Y_p}| \le x_p$, as $\langle A, PB\rangle = \langle A, B\rangle$ for $A \in \mathfrak{g}^c$. A bond lies in four plaquettes and a plaquette has four bonds. So, with x the largest x_p at the plaquettes of b, which is $\xi^2 f^\circ(b)$, the row sums of $\Delta'(U)$ at b are at most $(12+8x)f^\circ(b)$ in $\|\cdot\|$ and $(12+16x)f^\circ(b)$ in $|\cdot|$, where $|P| \le 2$ at $SU(N)$; in the class $\mathfrak{U}_k(\mathbf{B},\alpha)$ one has $f^\circ(b) \le \alpha \mathsf{L}_{\mathbf B}(b)^{-2}$ and $x \le \alpha$, and for $\alpha \le \frac38$ this gives $18\alpha\mathsf{L}_{\mathbf B}(b)^{-2}$ in $|\cdot|$ and $15\alpha\mathsf{L}_{\mathbf B}(b)^{-2} \le 20\alpha\mathsf{L}_{\mathbf B}(b)^{-2}$ in $\|\cdot\|$. As $\Delta(U)$ and $D^{\mathrm T}D$ are Hessians of real functions at G-valued U, $\Delta'(U)$ is real on real forms. With $\|S_p\|^2 \le 4\sum_i \|a_i\|^2$ the same count gives, bond by bond, $(A, \Delta'(U)A) \ge -\eta^3 \sum_b (12+8x)f^\circ(b)\|A(b)\|^2$, x taken at b, and $12 + 8x \le 28 \le 32$ at every unitary Y_p, where $x \le 2$: this is the lower bound. The transport-comparison constant $\kappa_7(L)$ in the proof of Lem. YM3_I:30.3 reads no N (Prop. 9.26(g)).

(ii) The averages: Lems. 9.9(iii) and 9.11 with the rule D^c.

At a fixed finite tower the implicit function theorem gives a local critical germ over e^{iB}. The uniform radius and nonlinear majorants asserted in (ii) are part of (V), not consequences of this finite-dimensional theorem. The linearisation is formed from the operators of Lem. 9.15(ii), with Lem. 9.19; a holomorphic map that is $\mathfrak{g}$-valued at real points is $\mathfrak{g}^c$-valued. On a tower of depth q, truncated at a level $q' \le q$, the towers ask at this configuration the flatness L^{-2q}, which a bound

of first order in $\sup|B|$ gives only up to the factor $L^{q-q'}$. The linear term has a kernel $\hat{D}$ that is real, scalar and the same for all G: for a real scalar 1-form ψ, $|\hat{D}[\psi]| \le K_{\mathrm{lin}} \sup_p \mathsf{L}_-(p)^2 |d\psi(p)|$, where $\mathsf{L}_-(p)$ is the least of the local scales $L^{\lambda(x)-k}$ at the corners x of p, $\lambda(x)$ being the level of the sequence of regions at x and $\eta = L^{-k}$, and the constant K_{lin} of the $SU(2)$ construction is read off its nonlinear bound of the towers along ψ times a fixed unit of $\mathfrak{su}(2)$, as the factor tends to 0 (Lem. 9.15(ii)); Cauchy's estimate controls higher orders only on an admitted polydisc with a uniform majorant. The uniform nonlinear tower-flatness estimate itself remains part of (V). To lie in the analyticity core a small field asks, on every sub-domain, for both tower rules and at every depth, that its averages exist with logarithms in $\mathfrak{g}^c$ and that over each real datum near $\mathbf{1}$ the regular critical configurations form one gauge orbit whose representative continues holomorphically with the tower bounds. These uniform assertions are the near-trivial input in (V); the slice and contraction reduction of Prop. 9.17 gives the one-orbit conclusion when its uniform hypotheses hold. The local matrix and colour-scalar linear estimates do not establish those hypotheses by themselves.

(iii) The $SU(2)$ statement, including the choice of one common triple for both the bond and node estimates, is proved in Section YM3_I:53.4. Its general-group counterpart with the same domains is the explicit localisation input of (V). The logarithms used to align the collar data are formed from products of small exponentials under Lem. 9.9(ii).

(iv) At a bond b of the cut-off zone E_b is a product of seven factors, and their chords add: the localisation of (iii) bounds the relative field of the two backgrounds beneath b by $L^{-k}\varsigma$ and gives two factors at the ends within ς of $\mathbf{1}$; Lem. 9.11(v), as L^k times that field is below $\varsigma < c_4$, gives the averaged exponent, at most 2ς, the two factors f_k at the ends, within $C_v\varsigma$ of $\mathbf{1}$, and the two exponents of the contour variables at the ends, at most $2\kappa_6\varsigma$ each; so $|E_b - \mathbf{1}| \le (4+2C_v+4\kappa_6)\varsigma = \mathsf{c}_L\varsigma$. The fifth member of ϵ_{xp} gives $\varsigma < \frac{c_4}{2L}$, so $\mathsf{c}_L c_4 \le 2Lr_G$ gives $|E_b - \mathbf{1}| < r_G$, $E_b \in G^\pi$ and $|H_b| \le \frac{\pi}{2}|E_b - \mathbf{1}|$; the bound $8\|\vartheta'_G\|_\infty |H_b| \le \delta_k$ comes from $A_1/A_0 \le c_{\mathrm{Q}}$ and $\gamma \le \gamma^G_{\mathrm{xp}}$, and $\mathsf{c}_{\mathrm{fl}}\delta_k \le 1$ from $C_\chi \mathsf{c}_{\mathrm{fl}}\delta_k \le r_\chi \le 1$; the assembly is that of the $SU(2)$ construction with r_G for 2.

A further argument gives $E_b \in G^\pi$ from the block constraints. On the double block of the coarse bond c of b let V and V' be the k-fold averages of the global and of the localised background. They have the same contour variables $X_\pm$ at the two end blocks, both being in the axial gauge at the sites of the region of level $k+1$ and both being given by the datum at the others, as an element of a minimal orbit satisfies its block constraints and the joined datum is the datum at the common bonds; and they have the same average X_0 at c. So $V(b) = X_-^{-1} e^{i\beta} X_0 X_+$ and $V'(b) = X_-^{-1} e^{i\beta'} X_0 X_+$ with $\beta, \beta' \in \mathfrak{g}$, $|\beta| + |\beta'| < \pi$, and $E_b = V(b)V'(b)^{-1} = X_-^{-1} e^{i\beta} e^{-i\beta'} X_-$, which lies in G^π by Lem. 9.9(ii), G^π being kept by conjugation with G. The statement that $G \setminus G^\pi$ is a null set (the $SU(2)$ case of Lem. 9.14(i) and Prop. YM3_I:53.4) is false for $SU(N)$, $N \ge 3$, at every $H_b \neq 0$, and it is not needed: the substitution enters only integrands that vanish outside the chart (the chart-supported substitution in the proof of Prop. YM3_I:53.4).

(v) The proofs of the $SU(2)$ construction stand; "trace 0" becomes $\mathrm{Ad}(g)\mathfrak{g}^c = \mathfrak{g}^c$, and $F = \xi^{-2} PH$ with $P\,\mathrm{Ad}(g) = \mathrm{Ad}(g)P$; a holomorphic map that is $\mathfrak{g}$-valued at real points is $\mathfrak{g}^c$-valued; and $G^c = G\exp(\mathfrak{g})$ is closed. □

9.1.8. *Dependence on regular backgrounds.* The construction hypotheses have different roles. The propagator estimates control inverses at general regular fields. The variational estimates then construct and localise the constrained critical orbit. The second-part hypotheses provide the holomorphic Gaussian families and the one-step estimates on their full domains. Finally, the constant conditions must be compatible in the prescribed order. The following reductions identify the group dependence in these requirements.

Propagators. Prop. 9.22(i) proves the local Hessian and current identities and separates them from the quantitative inputs in (P). For other simple G the Hessian bounds include the factor $|P|$. Changing the colour norm, taking a transpose, and summing components may introduce the factors κ_G, $N^{1/2}$ and N in Def. 9.5(2). The constants of an entire inverse or tree expansion must

be chosen from its admitted quantitative bounds; these norm comparisons alone do not certify a particular numerical choice. Hölder and ℓ^2 seminorms of $\mathfrak{g}^c$-valued functions compare with those of real functions by the factors $N^{1/2}$ and $(2n)^{1/2}$, since $\|\cdot\| \le |\cdot| \le N^{1/2}\|\cdot\|$ and such a function has $2n$ real components; the estimate required in (P) keeps the rate δ_2, and the floor on M moves with N through the natural bound B_{nat}, of the order $N^{1/2}$. The covariance numbers, thresholds, rates and cell sizes are formed from those of the expansion.

Variational estimates. The variational estimates in (V) require the background estimates of (P) with the changes of Def. 9.5, among which these concern the variational part: a proof that uses the current bound of the regular class runs with the class $\mathcal{U}(r)$, into which $\mathcal{U}'(r)$ falls at the radius $\kappa_G r$, and for $r \le z_G$ at $(1+\kappa' r^2)r$, thresholds on α_0 read as ceilings being divided by κ_G, or by $1+\kappa'$ (Remark 9.6, Lem. 9.12(ii)); the bound $3\alpha_0$ of Prop. YM3_I:30.1 becomes $3|P|\alpha_0$, at most $6\alpha_0$ at $SU(N)$, and $j_G\alpha_0$ under $\alpha_0 \le z_G$, which is $3\alpha_0$ at $SU(N)$; the second inclusion becomes $\mathcal{U}(r) \subset \mathcal{U}'(r + r^2 + \kappa' r^3)$ for every G and $r \le z_G$, and the membership in $\mathcal{U}'(2s)$, $s = 2\alpha_0 + 3\alpha_2$, of the curvature/current conversion in Lem. 9.12(ii), with both of its quantitative hypotheses, holds for every G under $\alpha_0 + 3\alpha_2 < z_G$ and $s(1+\kappa' s) \le 1$, so for $SU(N)$ under $\alpha_0 + 3\alpha_2 < z_G$ and $s \le \frac{1}{3}$, both by the bound at a bond of Lem. 9.12(ii), for the membership at each full bond b with $\rho = (\alpha_0 + 3\alpha_2)\eta^2 \mathsf{L}(b)^{-2}$ (Def. 9.5(3)); Lem. 9.10, whose averaged contour variables are means of contour families (Prop. YM3_I:4.3), takes the guard $\varpi_y(W) < z_G$, under which the contour quotients, products of at most $\frac{9}{8}L^2$ conjugated plaquette variables within $\varpi_y(W)$ of $\mathbf{1}$, lie in D^c by the rule for products, so that the means lie in G (Lem. 9.9(iii), (iv)), and the comb gauge of Lem. YM3_I:14.4 takes the guard $\varpi < z_G$ (Lem. 9.12(iii)); and wherever the towers use $|X^c - \mathbf{1}| = |X - \mathbf{1}|$, an equality of moduli at the plaquettes alone, they take the form with $m(X)$, with $|\mathrm{Ad}(g) - 1| \le m(g)(2 + m(g))$, and a logarithm is guarded by $X \in D^c$.

The existence assertion required in (V) has the following uniform content: there must be a power $\mathfrak{r}(G)$ of L, a number $B_3(G) \ge 1$ and, for every power $\rho \ge \mathfrak{r}(G)$ of L, a number $a_1(\rho, G) > 0$, numbers of L, ρ and the pair with $\kappa_G B_3(G) a_1(\rho, G) \le a_0(G)$, $a_0(G) > 0$ the fixed uniqueness radius required in (V), such that in every setting of the $SU(2)$ construction at the separation ρ, for every reversal-invariant set $\mathfrak{F}$ of inward bonds and over every real datum V on $\mathbf{B}_{\mathfrak{F}}$ whose plaquette variables, and on a sequence of regions whose holonomies at the tested plaquettes, lie within $\varepsilon_1 \le a_1(\rho, G)$ of $\mathbf{1}$, the set $C(\mathbf{B}_{\mathfrak{F}}, V)$ holds a critical configuration in $\mathcal{U}'(B_3(G)\varepsilon_1)$. Together with uniqueness of the regular critical orbit this assumption gives one nonempty regular critical orbit, contained in $\mathcal{U}'(B_3(G)\varepsilon_1)$; near the trivial datum it is the conditional assertion of Prop. 9.22(ii). Its route is the corresponding $SU(2)$ proof, the existence of a critical configuration in the class and the improvement of the regularity radius (the $SU(2)$ model is Prop. YM3_I:37.1) being run for G with the bounds that the variational part takes from the earlier sections at a regular reference configuration $U_0 \ne \mathbf{1}$: those of the linear operators at U_0, which rest on the propagators of (P) at regular backgrounds, and those of the straightened averages, the action and the gauge transformations in the chart, the rows and the gauge functional at U_0.

Conjugation. Prop. 9.23(i) gives (C) for the maps listed there and for their composites, finite sums and continuations on convex subsets of a vector space kept by the conjugation (slices without group-valued member, Lem. 9.21(i), (ii)). The maps of the induction are of these kinds: each is a chain of factors, each factor being built from products, inverses, means, logarithms in D^c, the action a and its derivatives, which commute with the conjugation by Lem. 9.20(i), (ii), or one of the three implicit systems of Lem. 9.20(iii), or a tower, or a continuation on a convex subset of a vector space kept by the conjugation, or a map holomorphic on the open slices of the real-connected part of its domain that takes real fields to real values, every trace of a group-valued argument at complex fields being $a(X)$ (Def. 9.5(1)); and the domains of the $SU(2)$ construction pass into real-connected parts by Lem. 9.21(iii). A map that is holomorphic on the open slices of the real-connected part of its domain, its real members being constant on slice components, is of them as soon as it takes real

fields to real values, and it then maps the real-connected part of its domain into that of its target as soon as it maps it into the target (Lem. 9.21(ii)).

The pivot bond and Gaussian coefficient. For the pivot bond, the unknown element $x \in G$ at the pivot bond enters the quotient $R = W[c]G_1[c]^{-1}$ through a homeomorphism of G onto itself; in each of the three cases of the averaging map a solution has the eigen-angles of R below $\frac{1}{272L^2}$ in modulus, so that it lies among the fields with plaquette variables within ω_M of $\mathbf{1}$ on the double block, where the average is the formula of Lem. 9.11(i) and the other two cases have no solution; every element that occurs is a product of conjugated plaquette variables within $\omega_M \le z_G \le r_G$ of $\mathbf{1}$, so that its logarithm lies in $i\mathfrak{g}$ (Lem. 9.9(ii), (iii)); and in the chart the equation reads $r = L^2(t - \bar{s} - E(r))$ for $r = \log R$, with t and $\bar{s}$ averages over the block of logarithms that do not read x and with E holomorphic, the right member being a contraction of $\{r \in i\mathfrak{g} : |r| \le \frac{1}{50}\}$ into itself with constant below $\frac{1}{121L^2}$; its fixed point gives the one solution and the bound at the pivot bond, with the numerals of the $SU(2)$ construction. The large-field chain is Lem. 9.13(ii). The localised Gaussian family is formed from the fluctuation operators of (P) by the formulas of the $SU(2)$ construction, and at the trivial triple its Hessian is that of the normaliser (Prop. 9.18(iii)).

Choice of constants. Prop. 9.26 gives the shape of the table and the criterion: one test for each condition and one joint test for each two-sided constant. A floor on a constant free upward and a positive ceiling on a constant free downward are admissible. A condition whose last constant is determined must be checked from its formula. The conclusion that all the conditions pass these tests is hypothesis (K).

The conditions of Lem. 9.21(iii) at the memberships reached from memberships of restrictions had apart are positive ceilings on the second radius where the first radius lies below B_0, and for the cores $Kb_0 < B_0$ and $b_0 < a^\sharp$; the ceiling $\mathsf{c}_L c_4 \le 2Lr_G$ is a positive ceiling on c_4, and the row $\alpha_1 < \rho_G L$ one on α_1 (Props. 9.22(iv), 9.24).

The comparison of the two norms, the regularity window and the central-element guards are specified in Prop. 9.26(c), (e), (f). The numbers of the pair must occur before γ in the order of choice; they may depend only on the constants preceding them and on the pair.

9.1.9. *Multiscale induction and the choice of constants.*

Proposition 9.23 (Conjugation and the inputs of the induction)**.** The conjugation assertions apply to the maps wherever they are formed. The global second-part assertions and the pointed-family assertion in (iii) assume (II) of §9.1.1. Let $\mathfrak{g}$ be simple, $L \ge 13$ odd, $200L^2 z_G \ge 1$; constants are constants depending only on L and the pair, uniform in the volume.

(i) Let $a(X)$ stand for every trace of a group-valued argument at complex fields, the gauge-fixing term included, and let the domains be written with $m(X)$ and with D^c at the logarithms of the averages. Then plaquettes, path products, gauge transforms, covariant differences, F, J, means, averages and their iterates, the chain to the fluctuation operator and the three implicit systems of Lem. 9.20(iii), the towers, and the composites and finite sums of these maps and their continuations on convex subsets of a vector space kept by the conjugation (slices without group-valued member) commute with $p \mapsto p^c$ on the real-connected parts of their domains and map them into the real-connected parts of their targets. A map holomorphic on the open slices of Q_1^r, with real members constant on slice components, commutes with $p \mapsto p^c$ there if and only if it takes the real points of Q_1 to real points, and then it maps Q_1^r into Q_2^r as soon as it maps Q_1^r into Q_2 (Lem. 9.21(ii)).

(ii) In the second-part statements assumed in (II), the clauses that are false for G as worded are required in the following corrected forms, among them these: $m(X)$ where $|X^{-1} - \mathbf{1}| = |X - \mathbf{1}| = |X^* - \mathbf{1}|$ was used, which is false at $SU(3)$; the radius r_n where $|1 - T| \le \frac{1}{2}$ was to keep $\det T$ off $(-\infty, 0]$ (the three-dimensional determinant argument in the proof of Prop. YM3_I:50.2), which is false from $n = 6$; the factor $\max\{1, n2^{-11}\}$ in the bound eq. 9.xii (Prop. 9.26(d)); the bound $z < (\frac{5}{4})^n L^{3n}$ in the place of $z < 2L^9$ in Lem. YM3_I:65.3;

orthogonal colour maps that keep $|\cdot|$ in the covariance argument following Prop. YM3_I:47.4; the indicator of G^{π} where $G \setminus G^{\pi}$ is used as a null set, and integrands that vanish outside the chart in the substitution of Prop. YM3_I:53.4; $a(X)$ for the bare trace of the gauge-fixing term of the definition of Ψ^{G} in Section YM3_I:51, p. 393; $E^{\mathrm{fam}} \geq e_n$ in the bound for the local Gaussian constants, as derived in Prop. 9.26(d); and the two-sided guard $m(X)$ in the elimination-factor estimate of Lem. 9.14(iv). Of the clauses in which a proof of the second part works with one colour, those at the trivial field and in the averaging layer hold for G with the constants of the $SU(2)$ construction; those that concern the background near trivial block data, and the radius $r_0 = a^{\mathrm{us}}(\alpha_0, \alpha_1)$ in $\mathsf{G}_\sharp$ (Prop. 9.24), hold as soon as on every truncated tower the background exists, is one orbit, is holomorphic on a polydisc of uniform radius and has there the bounds of Prop. 9.22(ii), whose uniform conclusions here are assumed in (V), including uniqueness of the critical orbit; and one more ceiling, $\varepsilon_{\mathrm{v}} \leq a_G$, enters on the regularity radius ε_{v} of the critical configurations (Def. YM3_I:9.5), with a fixed admitted positive radius a_G from §9.1.6 (Prop. 9.26(f)).

(iii) Near $W = \mathbf{1}$ the localised Gaussian family of the second part (Prop. YM3_I:50.2) has at the trivial triple the Hessian of the Gaussian normaliser of Prop. 9.18(iii); its quadratic coefficient there is one number for all data, $b_l^0(G)$. Under (II), for simply connected G the family is a pointed family of its level on the analyticity cores of G, holomorphic on balls of one radius, with the contraction bound of its expansion (§9.1.1(II)).

■

Sketch of proof. (i) A map is covariant on the real-connected part if it is a chain of factors each of which is either built from products, inverses, means, logarithms in D^c, the action and its derivatives, and so covariant by Lem. 9.20, or holomorphic on open slices and real at real fields, and so covariant by Lem. 9.21(ii), or a continuation on a convex subset of a vector space kept by the conjugation (a slice without group-valued member); a second way is to check that each of its formulas is real and holds at the doubled pair $X \mapsto \mathrm{diag}(X, (X^{\mathrm{T}})^{-1})$, for which $|\mathrm{diag}(X, (X^{\mathrm{T}})^{-1}) - \mathbf{1}| = m(X)$ and the action, the norm and the form restrict to those of G. The maps listed in (i) are of these kinds: plaquettes, path products, gauge transforms, covariant differences, F, J, means, averages and their iterates and the chain to the fluctuation operator are built from products, inverses, means, logarithms in D^c, the action and its derivatives (Lem. 9.20(i)–(iii)); the three implicit systems are covariant by Lem. 9.20(iii); the towers and the continuations on convex subsets of a vector space kept by the conjugation by Lem. 9.21(ii); and composites and finite sums of covariant maps are covariant. Lem. 9.21(iii) moves the domains of the $SU(2)$ construction into real-connected parts at the cost named there. Two statements of the $SU(2)$ construction take another form for G: the gauge-fixing term of the definition of Ψ^{G} in Section YM3_I:51, p. 393, set with a bare trace at complex fields, whose imaginary part $\frac{t}{N}\mathrm{tr}\sin(tH)$ at $SU(3)$ does not vanish at a real field, takes $a(X)$, which keeps its quadratic part and its bounds; and the statement that the conjugate point lies in the domain if and only if the point does, which for $SU(2)$ rests on the isometry Θ of Lem. 5.23 and is false beyond $SU(2)$, takes the two-sided guard $m(X)$.

(ii) The forms that hold are those of Lems. 9.20(ii), 9.14(iv) and Prop. 9.26(d) for the first three, and those of Def. 9.5(2), (3), Remark 9.6, (i) above, Props. 9.22(iv), 9.26(d) and Lem. 9.14(iv) for the others. At the trivial pair every operator is scalar times $1_{\mathfrak{g}^c}$ (Lem. 9.15(ii)), with the lower bound of the fluctuation operator and only 3 replaced by n, and the averaging layer is Lem. 9.11; near trivial block data the clauses need the existence, one orbit, holomorphy on a polydisc of uniform radius and the bounds of the background on every truncated tower, of which Prop. 9.22(ii) gives these conclusions under (V). The ceiling uses a fixed admitted orbit radius from §9.1.6; the central-field obstruction described there applies only when the chosen averaging extension sends that field to the trivial block datum. It does not establish existence or a sharp value of a positive orbit radius for every averaging extension.

(iii) At the trivial triple the germ of second order of the localised family is computed as that of the normaliser in Prop. 9.18(i): by invariance the background is taken, to second order, along one unit $H \in \mathfrak{g}$; the fluctuations split into the eigenspaces of $\operatorname{ad} H$, each of which is a charged sector of $SU(2)$; and the action adds the one tadpole, with $\operatorname{tr} M_H = C_\rho$. So the coefficient of the family there is $\frac{C_A}{8}$ times that of the localised family of $SU(2)$, which is one number for all data and equals the coefficient of $\log Z^{(k)}$ (equation YM3_I:(790) identifies their totals up to a constant, and Thm. YM3_I:X identifies the coefficient of such totals), the increment b^0_{k+1}, plus $\gamma_G(j_{k+1} - L^{-1} j_k) = (\frac{C_\rho}{3} - \frac{C_A}{8})(L^3 - 1)L^{-4(k+1)}$, $j_k = \frac{2}{3}(L^{-k} - L^{-4k})$; this is $b^0_{k+1}(G)$, the coefficient of the normaliser (Thm. AA(c), Prop. 9.18(iii)), and it reads no datum. □

Proposition 9.24 (Induction for the proposed construction)**.** For a simple Lie algebra, the construction remains to be completed through the tasks (P)–(K) of §9.1.1. For a semisimple Lie algebra, the transferred one-step statements must likewise be established and their applicability checked at the data of each citation below. For tied ideals, this also requires property (T) of Thm. AA(h3). The proposition records the induction that would then give the construction conclusions. The coefficient estimates below apply whenever the indicated terms and their analyticity bounds are given. Let $\mathfrak{g}$ be semisimple. Each of the following proofs of the $SU(2)$ construction derives its transferred statement from the transferred statements that it cites, the hypotheses of each citation, those that the forms (3) of Def. 9.5 add to a cited statement among them, being met at its data, as in §9.1.1(II): the regular progression in Thm. YM3_I:Y; the actual one-step estimates of Thm. YM3_I:W, their quadratic coefficient in Thm. YM3_I:X, and the canonical coupling construction in Section YM3_I:52.10, with $\mathsf{G}_\sharp$ as its coefficient bound; the historical step in Prop. YM3_I:69.3, its iteration in Thm. YM3_I:AC and Cor. YM3_I:69.4; and the history bounds of Section YM3_I:70, which give Thm. YM3_I:D. In the historical step the proof explicitly checks the coordinate changes and tests, the separate on-shell identities for the old exponent and retained factors, the conditional Gaussian and cluster identity, the normalizer, the kept and rejected forms, the strip sum, and the canonical identification of the new exponents. The positive change-of-variables theorem used there is Thm. YM3_I:AB; the boundary and retained-factor bounds are those of Section YM3_I:62. Their transferred hypotheses, including their domains and measurability assertions, remain required here. Of the group these proofs use only that $\mathrm{Ad}(G)$ keeps $\langle \cdot, \cdot \rangle$ and $|\cdot|$, that invariant linear forms vanish, that $|t_a| \le \kappa_G$, and that $a(wuw^{-1}) = a(u)$. The part of second order of the effective terms enters them as one real number, the mean coefficient cf, the mean of its n quadratic colour coefficients: it is defined, bounded, additive, real and holomorphic in the parameters, never a multiple of the Hessian of the action, with $|\mathsf{cf}| \le \mathsf{G}_\sharp = \kappa_G^2 C_{\mathrm{am}} r_0^{-2} E_0 \le B_\sharp$, where C_{am} is the constant of the coefficient bound YM3_I:(940) and $r_0 = a^{\mathrm{us}}(\alpha_0, \alpha_1)$ the radius of the $SU(2)$ construction; C_{am} and r_0 read constants of the $SU(2)$ construction that come before them in its order, $\mathsf{G}_\sharp$ reads besides the pair, through κ_G, and E_0, and none of them reads γ, $B_\sharp$, a level, the cutoff, β or a field. On an ideal $\mathfrak{g}_i$, $|\mathsf{cf}_i| \le \kappa_i^2 C_{\mathrm{am}} r_0^{-2} E_0 = (\kappa_i/\kappa_G)^2 \mathsf{G}_\sharp$. The numbers $\frac{9}{2}$ and 3 of Thm. YM3_I:D become $\frac{3n}{2}$ and n; the π of the ceiling $\alpha_1 < \pi L^j$, $1 \le j \le k$, under which the local terms are continuous on the analyticity cores, which a proof of the chain from level k to level $k+1$ quotes, becomes ρ_G of Def. 9.1, the bound on the eigen-angles under which an element of G lies in G^π, the domain of the chart, so that the ceiling is $\alpha_1 < \rho_G L$, one more row of the table of Prop. 9.26(a), a positive ceiling on α_1, which holds of itself for $\rho_G > \frac{1}{13}$, as $\alpha_1 \le 1$ and $L \ge 13$; as $\rho_G \le \pi$, this row gives $\alpha_1 < \pi L^j$, under which the exponential of $\mathfrak{g}^c$ is a local diffeomorphism at every Z_0 with $|Z_0| < \pi$, the eigenvalues of $\operatorname{ad} Z_0$ being at most $2|Z_0| < 2\pi$ in modulus for matrices of every size; the $\frac{3}{2}$ in the size of an array stays if the analyticity condition is read with $m(\partial U)$, since $|a(u)| \le \frac{1}{2}|u - \mathbf{1}||u^{-1} - \mathbf{1}|$; and the transferred statements of these proofs differ from those of the $SU(2)$ construction only in these numbers and in "of L and the pair" for "of L", the sign $\mathsf{G}_\sharp$ keeping its places with its value for G. The large-field chain of the $SU(2)$ construction, Prop. YM3_I:44.4, Prop. YM3_I:44.5 and Lem. YM3_I:65.2 and the local-deposit argument in the

proof of Prop. YM3_I:65.4, is transferred by Lem. 9.13(ii), its statements taking the prices of the $SU(2)$ construction times τ_G^{-2} and the averaging radius $\min\{\omega_M, r_G\}$. The same holds for an action $a(u) = 1 - \frac{1}{2}\mathrm{tr}\,\Lambda(u + u^{-1})$ with a weight $\Lambda > 0$ in the commutant of G, $\mathrm{tr}\,\Lambda = 1$, the form $\mathrm{tr}(\Lambda XY)$ and $\kappa_\Lambda = \max |Y|/\mathrm{tr}(\Lambda Y^2)^{1/2}$ in the places of $\langle\cdot,\cdot\rangle$ and κ_G. ■

Sketch of proof. The construction hypotheses provide the cited one-step estimates and their applicability on the transferred domains. There are five additional guards to retain. Logarithms of group variables are taken in D^c; a logarithm in the real group is taken in G^π; complex domains use $m(X)$; determinant logarithms use the radius r_n; and local terms on the analyticity cores satisfy $\alpha_1 < \rho_G L^j$. Lems. 9.9, 9.14, 9.20 and Prop. 9.22(iv) supply the first four at their stated data. The row $\alpha_1 < \rho_G L$ supplies the last because $L^j \geq L$. Compatibility of these rows is part of (K).

The colour algebra in the induction consists of invariance of the trace form and of the operator norm, vanishing of invariant linear forms, $|t_a| \leq \kappa_G$, and conjugation invariance of a. The comparison of the two volume exponents in Thm. YM3_I:D becomes $n \leq 3n/2$. With the one-step bounds and domain conditions assumed, the induction proceeds on the same histories.

Terms of first order in the fluctuation field are bounded by Schwarz's lemma: if h is holomorphic on $|\lambda| < \rho$ with $\rho > 1$, $h(0) = 0$ and $|h| \leq c$, then $|h(1)| \leq c/\rho$, nothing being asked of $h'(0)$. The damping $L^{-(k-j)}$ of old levels at $U = \mathbf{1}$ comes from the vanishing of invariant linear forms. In the nine proofs of the chain from level k to level $k+1$ the inputs ask $\|E^{(j)}\| \leq E_0$ and $|b_j| \leq B_\sharp$ separately, and the bounds of a step add these sizes. The increments enter only as the real numbers $e_j = b_j$ with $|b_j| \leq B_\sharp$, no relation between them and the entries of a term or their derivatives is used, and the one stationarity used is that of the one action. The bounds of the coefficient: by Cauchy's estimate on the ball of radius r_0 the part q of second order of a term E obeys $|q(X,X)| \leq C_{\mathrm{am}} r_0^{-2} \|E\| |X|^2$, as in the $SU(2)$ construction, where the orthonormal basis has $|t_a| = 1$; with $|t_a| \leq \kappa_G$, and $|t_a| \leq \kappa_i$ on $\mathfrak{g}_i$, this gives $|\mathsf{cf}_i| \leq \kappa_i^2 C_{\mathrm{am}} r_0^{-2} E_0$ and $|\mathsf{cf}| \leq \sum_i \frac{n_i}{n} \kappa_i^2 C_{\mathrm{am}} r_0^{-2} E_0 \leq \mathsf{G}_\sharp$. The factor κ_i^2 against C_{am} cannot be dropped: for $SU(2)^r$ on blocks, $r \geq 2$, one has $\kappa_i^2 = r$, and a term that reads one factor has on its ideal r times its coefficient for $SU(2)$. □

The colour factor in the determinant estimate is most easily checked before it is absorbed into the constant. The following calculation records the precise contraction and closed-chain bounds it requires. Their availability for the transferred operators remains part of the construction hypotheses.

Lemma 9.25 (The colour factor in a local determinant bound)**.** Let $n \geq 1$, $c > 0$, $0 < q < 1$, $Q > 1$ and $\delta_1 > 0$. Put

$$\delta = \frac{\delta_1}{128} \frac{\log(1/q)}{\log Q + \log(1/q)},$$
$$C_{\mathrm{a}} = n \left(|\log c| + \log \frac{1}{1-q} \right),$$
$$C_{\mathrm{d}} = \frac{Q}{Q-1} + \frac{1}{1-q}, \qquad C_{\mathrm{mob}} = 2^7 C_{\mathrm{a}} + C_{\mathrm{d}}.$$

Assume $\delta M \geq 9 \log 2$. Let X be a finite set of coarse cubes, $d = d_t(X) \geq 0$, and let $\emptyset \neq S \subset X$ satisfy $d_t(S) = 0$ and $\#X \leq 8(d+1)$. For each $S \subset Y \subset X$ let T_Y be an operator on bond fields with n colour coordinates, with $\|T_Y\| \leq q$, and with a common bond b in its domain. Define

$$f_b(Y) = n \log c - \sum_{\nu \geq 1} \frac{1}{\nu} \mathrm{tr}\, T_Y^\nu(b,b), \qquad E(X,b) = \sum_{S \subset Y \subset X} (-1)^{\#X - \#Y} f_b(Y).$$

The trace is over the colour coordinates. When $d > 0$, suppose the closed-chain sums

$$\Sigma_\nu = \sum_{S \subset Y \subset X} (-1)^{\#X - \#Y} \mathrm{tr}\, T_Y^\nu(b,b)$$

satisfy, with $\mathcal{N} = 2^{\#X-\#S}$,

$$|\Sigma_\nu| \le n\mathcal{N}q^\nu, \qquad |\Sigma_\nu| \le n\mathcal{N}Q^\nu e^{-\delta_1 M(d+1)/64}. \tag{9.xi}$$

Then

$$|E(X,b)| \le 2^{-d}\max\{1, n2^{-11}\}C_{\mathrm{mob}}e^{-\delta M d}. \tag{9.xii}$$

■

Proof. The contraction bound makes each series absolutely convergent and gives $|f_b(Y)| \le C_{\mathrm{a}}$. There are at most $2^{8(d+1)-1}$ sets Y. If $d = 0$, their sum is bounded by $2^7 C_{\mathrm{a}} \le C_{\mathrm{mob}}$. For $d > 0$, $X \ne S$, so the alternating sum of the constant $n \log c$ vanishes. Thus $E(X,b) = -\sum_{\nu\ge1}\Sigma_\nu/\nu$.

Put

$$\theta = \frac{\delta_1}{64(\log Q + \log(1/q))}, \qquad \nu_* = \lfloor \theta M(d+1) \rfloor.$$

Use the second majorant up to ν_* and the first afterwards. Since $\theta\log(1/q) = 2\delta$ and $\theta\log Q - \delta_1/64 = -2\delta$, the two geometric sums give

$$|E(X,b)| \le n\mathcal{N}\left(\frac{Q}{Q-1} + \frac{1}{1-q}\right)e^{-2\delta M(d+1)}.$$

One exponential absorbs the region count:

$$|E(X,b)| \le \frac{n}{4}\,2^{-d}C_{\mathrm{d}}e^{-\delta M(d+1)} \le n2^{-11}\,2^{-d}C_{\mathrm{d}}e^{-\delta M d}.$$

Here the second inequality retains the further factor $e^{-\delta M} \le 2^{-9}$. Together with the case $d = 0$ and $C_{\mathrm{d}} \le C_{\mathrm{mob}}$, this proves the claim. □

Proposition 9.26 (The constants)**.** Let $L \ge 13$ be odd and $200L^2 z_G \ge 1$. The constants are chosen in the order of the $SU(2)$ construction, γ last, each chosen constant ranging, given the earlier ones, over the range that the $SU(2)$ construction declares for it; each is free upward, free downward, two-sided (its admissible values, given the earlier ones, form a window) or determined by a formula from earlier ones. The table of the constants of the $SU(2)$ construction carries besides a radius $\rho > 0$, which is not chosen and stands in the rows on γ alone. A set of choices of all the constants is *large* if it is large at the empty choice, where, from the last constant down to the first, a set is large at a choice of the constants before the i-th one if the values of the i-th constant at which it is large at the longer choice hold, for some t and some $c > 0$, the part $[t,\infty)$ of its range when this constant is free upward, the interval $(0,c]$ when it is free downward, all its admissible values when it is two-sided, and its value when it is determined; at a choice of all the constants a set is large if it holds that choice. A set that reads γ or ρ is large if so is the set of the choices of the constants before γ at which, for every $\rho > 0$, some interval $(0,c]$ consists of admissible values of γ at which it holds.

(a) For $SU(N)$, every $N \ge 2$, the table of the constants of the $SU(2)$ construction, formed with the numbers of G as data (any positive values of these numbers in the ranges that the $SU(2)$ construction states for them serve for its shape), with the changes of Def. 9.5 and with those conditions of (c)–(g) that are a floor on a constant free upward or a positive ceiling on a constant free downward laid as rows, has admissible shape, the shape under which, at every admissible choice of the earlier constants, the admissible values of a constant free upward hold a half-line of its range, those of a constant free downward an interval $(0,c]$, and the window of a two-sided constant is not empty; γ is last, and choices exist at every value of the radius ρ, the admissible choices of the constants before γ being the same for all ρ; a further condition read for G is laid on it as a row when it is large. For a semisimple pair with a weight in the commutant, the table of the second part of the $SU(2)$ construction, its chosen constants, their order, kinds and ranges kept, is, as the table of the $SU(2)$ construction is, of admissible shape and without cap, with γ last, and choices exist, given that the numbers

of the first part of the pair exist with the ranges that the $SU(2)$ construction states for them. In both, a floor on a constant free upward and a positive ceiling on a constant free downward never obstruct; L is free upward, with floors alone, and a floor on L may be laid once at the root, every other constant keeping the same admissible set. This is the shape of the table; that every condition read for G is large, so that all of them are rows of a table of admissible shape and hold at each of its choices, is (K).

(b) Large sets form a filter: the intersection of two sets of choices is large if and only if both are. Finitely many conditions are each large if and only if there is a table obtained from that of (a) by floors on free upward constants, positive ceilings on free downward constants, and further constants determined by formulas from earlier ones, at every choice of which all the conditions hold. These additions retain every original two-sided window whole; one floor or ceiling on each free constant suffices. Separately, further ends on a two-sided constant are jointly feasible at an earlier choice exactly when its original window meets all those ends together. Requiring a large set of such earlier choices is a compatibility test for the restricted windows, not largeness of the ends as subsets of the original full table. Hypothesis (K) requires both the largeness conditions and the joint-window compatibility tests used by the construction.

(c) For $SU(N)$ the two-sided conditions that compare $|\cdot|$ with $\|\cdot\|$ hold together for every $N \geq 2$ at the ratio $\frac{1}{N-1} = \kappa_G^{-2}$ of the squares of the two norms, with the kernel bounds of (P) read in the normalised trace norm, which the bounds in $|\cdot|$ give with the factors $(N-1)^{1/2}$ and $N^{1/2}$ between the two norms (Prop. 9.22(i)). The two-sided condition on the regularity radius ε_{v}, which comes from the inclusions $\mathcal{U}'(r) \subset \mathcal{U}((1+\kappa' r^2)r)$ and $\mathcal{U}(r) \subset \mathcal{U}'(r + r^2 + \kappa' r^3)$, $r \leq z_G$, of Lem. 9.12(ii), is not empty while $\frac{1}{400L^2} < 2\sin\frac{\pi}{N}$, a constant that precedes ε_{v} in the order being multiplied by $1+\kappa'$, with the numbers of the critical configuration of (V), and one floor on the separation constant R_1 makes its ends compatible; for the two-sided conditions that a central element near $\mathbf{1}$ enters when $z_G < 1$ see (e).

(d) The chains that use Log det (the proof of Prop. YM3_I:50.2 and the elimination-factor estimates of Lem. 9.14(iii), (iv)) hold at the radius r_n, with $4Ka \leq r_n$ and $C_z \geq 4c_n K$, $c_n = \rho_n / r_n$, the pair $4Ka \leq \frac{1}{2}$, $C_z \geq 20K$ of the $SU(2)$ construction holding besides, and with the further member of the radius of the second chain that Lem. 9.14(iv) names, which keeps its logarithms in $\mathfrak{g}^c$; their constants are constants depending only on L and the pair, explicit in L and n at $SU(N)$, that obey the conditions in which they enter: there the radii are bounded from above only and the bounds from below only, so the ordered choice closes again provided every subsequent use of the normalizer and elimination constants requires only these upper bounds on radii and lower bounds on estimate constants. This compatibility with the later inequalities remains part of (K). Write E^{fam} for the local Gaussian bound constant, the counterpart of E_b in Prop. YM3_I:50.2; here E_b already denotes the quotient of Prop. 9.22(iv). Require

$$E^{\mathrm{fam}} \geq e_n := \frac{2(2n \log L + \rho_n)}{L^3 + 1}.$$

Indeed, Lem. 9.14(iv) then gives

$$|z_c| \leq 2n \log L + \rho_n \leq \tfrac{1}{2}(L^3+1)E^{\mathrm{fam}},$$

which is the bound needed for the local Gaussian constant at an elimination term. The chosen constants must satisfy this guard; it is not inferred merely from a dimension-independent lower bound on E^{fam}. The inequality $\frac{3}{4} \leq 1$ by which the proof of Lem. 9.25 absorbs a colour factor reads $\frac{n}{4} \leq 1$, which fails from $n = 5$; a factor 2^{-9} of the same estimate, which that proof does not use, restores it for $n \leq 2048$, and for every n the precise bound is eq. 9.xii. Here C_{mob} is the full constant, including the zero-size term, with its trace contribution evaluated

in dimension n. $(\frac{5}{4})^3 < 2$ becomes $(\frac{5}{4})^n$, which fails from $n = 4$, and the bound $z < 2L^9$ of Lem. YM3_I:65.3 becomes $z < (\frac{5}{4})^n L^{3n}$. The devices by which the $SU(2)$ construction fixes the branch of a logarithm or of a root or power of a determinant ask something of the operator and not of the value of its determinant, and serve in every dimension, but for those that read the value: the principal logarithm of the elimination factor (Lem. 9.14(iii)); the power series of the logarithm about 1 in the Jacobian of the exponential map (Lem. 9.14(i)); the power series of the logarithm about 1 at the Jacobian $1 - \mathsf{jc}$ of the bond shift with its cut-off (the cutoff preceding Lem. YM3_I:53.3), jc being the derivative of the cut-off in the direction of the shift, so that $1 - \mathsf{jc}$ is the determinant of a perturbation of rank one of a map of determinant 1, whose value is computed outright and in which no count of the dimension enters; and the power series of the logarithm about 1 at a quotient $\det(\mathsf{T}(0)^{-1}\mathsf{T}(\zeta))$ of the determinants of a holomorphic family $\mathsf{T}(\zeta)$ of operators on $\mathfrak{g}^c$-valued fields, formed near $\zeta = 0$, where the quotient is near 1 by continuity. Among the devices that ask something of the operator is the number $2n \log L - \operatorname{tr} \log T$ of the elimination, $T = L^3 \mathrm{Ad}(g)\, \mathsf{m}\, \mathrm{Ad}(g')^{-1}$ with $g, g' \in G$ as in Lem. 9.14(iii): it asks $|T - 1| < 1$ at some g, g', and it does not depend on their choice among those with $|T - 1| < 1$, for every n (Lem. 9.14(iii)). Where a count of the dimension, $n = 3$ against π or 2π or $n - \operatorname{rk} G = 2$ against 1, fixes a branch, as in the determinant argument in Prop. YM3_I:50.2, the elimination estimates of Lem. 9.14(iii), (iv), the auxiliary-frame independence proved in Lem. 9.14(iii) and the bound for the Gaussian constants just proved, the radius r_n and the numbers of Lem. 9.14(iv) take its place, or, where it compares $\operatorname{tr} \log T$ at two choices of g, g', the equality of Lem. 9.14(iii); and where the sign $(-1)^3 = (-1)^{\dim \mathfrak{g}}$ stands twice in one identity, in the adjoint determinant calculation, it cancels for every n. That the branch is fixed in one of these ways at every place where the $SU(2)$ construction forms a logarithm or a root or power of a determinant over $\mathfrak{g}^c$ is part of (II) and (K).

(e) If $z_G < 1$, the conditions that fall to z_G include a ceiling on γ, a floor on L and the radii that the $SU(2)$ construction defines as minima with the averaging radius. For the rows that fall to z_G the floor on L is $L^2 \geq \frac{1}{200 z_G}$ together with the row $\alpha_0 \leq z_G^2/(4C^2)$, C of Lem. 9.12(iv), or $L > \frac{1}{z_G}$; the floor $L > 2 z_G^{-2}$, which implies $L > \frac{1}{z_G}$ and $200 L^2 z_G \geq 1$, carries with a ceiling on γ all the relations in N of the table. The numbers $\frac{1}{200}$, $\frac{1}{2}$, $\frac{1}{3}$, the radius 1 and the like do not fall, for there "$X \in D^c$" is supplied by a product or a path; the mean and the block average keep their numerals under $\omega_M \leq z_G$, which the floor $200 L^2 z_G \geq 1$ gives; so do the charts, the gluing of gauges and the gauges at fields whose plaquette variables lie in D under the averaging radius, while the gauges that Def. 9.5(3) names take there the forms named: the comb gauge of Lem. YM3_I:14.4 takes the guard $\varpi < z_G$ and Lem. 9.10 the guard $\varpi_y(W) < z_G$, which bite at $SU(N)$ from $N = 7$, and the quantitative cube-gauge input of Lem. 9.12(iii) takes constants c, B of the pair; the box clause of Lem. YM3_I:14.4 read with some gauge in place of the comb is retained under $\varpi < z_G$; for SU(N), $N \leq 13$, its unguarded extension requires the exceptional finite-cube gauge construction described in Lem. 9.12(iii), and it fails for $N \geq 14$; an element known by a chord $|V - V'|$ alone takes z_G in its radius; and the statements that place a logarithm in $\mathfrak{g}^c$ by "$|X - \mathbf{1}| < 1$" are false for $z_G < 1$ as worded and hold with "$X \in D^c$". Among the conditions that the centre enters, a condition can bound N only through an inequality with no free constant, $b < 2 \sin \frac{\pi}{N}$, where a proof concludes from "X is within b of $\mathbf{1}$" that $\log X$ is traceless; read with D^c, such a conclusion asks instead for a product or a path, while a ball of radius z_G in the place of D^c would let the number $\frac{1}{14}$ of Lem. 9.11(i) cap N at 87. The other inequalities with no free constant that bound n or N take the forms named at their places: $\frac{n}{4} \leq 1$, $(\frac{5}{4})^n < 2$ and $|\mathbf{1} - T| \leq \frac{1}{2}$ in (d), $|\log j| < 1$ on the ball of radius $\frac{3}{8}$ in Lem. 9.14(i), and the ratio of the two norms in (c). A field whose bond variables lie within r of $\mathbf{1}$, $(1 + r)^4 < 2$, is ζe^{iA} with

$\zeta(b) = z(U(b)) \in Z(G)$ and $\mathfrak{g}$-valued A, and $\partial\zeta(p) = z(\partial U(p))$; so ζ is flat exactly when every plaquette variable lies in D^c, and a flat ζ is a central gauge transform of $\mathbf{1}$ on a box and is fixed by its three holonomies up to such a transform on a torus. A gauge function h on a connected set with $|h(x)-\mathbf{1}| < 1$ and $h(x)h(y)^{-1} \in D^c$ at neighbouring sites x, y, as when h carries a field with bond variables in D to another such field, is a central constant times $e^{i\psi}$, ψ being $\mathfrak{g}$-valued. For the actual intrinsic-logarithm sectors and radii of the construction, we must prove the connectedness assertions needed in **(K)**: the sets cut out by the block, double-block and tower plaquette tests are connected to $\mathbf{1}$ inside the corresponding sector and test set; and a field passing the box test t is joined to $\mathbf{1}$ inside that sector with the test enlarged to $t + 4.49t^2$. The paths must stay in the actual intrinsic-logarithm sectors; a chordal bound $t < z_G$ alone does not give the required paths. We must also verify $X \in D^c$ at every use of a logarithm and show that the inherited-radius operations preserve these domains.

(f) For every G, also at $z_G = 1$, the conditions include $\varepsilon_{\mathrm{v}} \le L^2 r_G$ and $\varepsilon_{\mathrm{v}} \le a_G$, where a positive orbit radius a_G admitted by §9.1.6 is fixed as part of the analytic input. No unconditional positivity or two-sided $SU(N)$ estimate for the supremum $a_*(G)$ is used. At $SU(N)$, $N \ge 3$, the central-field obstruction at $\varepsilon_{\mathrm{v}} > 2L^2 \sin(\pi/N)$ applies if the chosen averaging extension sends the displayed central field to $\mathbf{1}$: the nonflat zero-current field of §9.1.6 is then admitted at the first level. The conditions also include $\varepsilon_{\mathrm{v}} \le \varepsilon_G$ and $c_{11} \le r_G/(2C_{\mathrm{pg}})$ of the uniqueness clause of §9.1.1(V), c_{11} the radius of §9.1.6 and C_{pg} the admitted preliminary-gauge constant of §9.1.6; the ceiling $\mathsf{c}_L c_4 \le 2Lr_G$ of Prop. 9.22(iv), a positive ceiling on c_4, void at $SU(3)$; the row $\alpha_1 < \rho_G L$ of Prop. 9.24, a positive ceiling on α_1; and, at the memberships reached from memberships of restrictions had apart, the conditions of Lem. 9.21(iii): the moved first radius $b_0' = b_0 + 2e^{4\xi b_1} b_1^2$, for regular pairs $\alpha_0 + 2\vartheta^2 e^{4\vartheta\alpha_1}\alpha_1^2$, at most the first radius B_0 at which the statement read there is had, and for the analyticity cores besides $b_1 \le B_1$, $K(b_0' + b_1^2) \le B_0$ and $b_0' + b_1 \le a^\sharp$, with $K \ge 16L^2$ and $a^\sharp$ of that lemma; where $b_0 < B_0$, and for the cores $Kb_0 < B_0$ and $b_0 < a^\sharp$, these are positive ceilings on the second radius.

(g) The constants in the matrix product, logarithm and contour comparisons of Lems. 9.9(ii), 9.11(iii) and 9.19 have the stated dependence on the pair. The $SU(2)$ contour coefficients are those of Lems. YM3_I:14.5 and YM3_I:14.6; the accumulated transport coefficient is constructed in the proof of Lem. YM3_I:30.3. Exact numerical choices for a general-group Hessian or nonlinear-remainder bound must be supplied as part of (P). In particular, this paragraph does not deduce the values $12, 1536, 3456$, the action radius and bound $1/128, 808$, or the gauge radius and bound $1/64, 93$ from matrix norm comparison alone. Of the constants of the first part of the $SU(2)$ construction these move with N: the averaging radius; the 2 of $|T^{\mathrm{T}}| \le 2|T|$ (Lem. 9.3), which becomes N, sharply; a numeral 3 that counts colours in a bound of Schur's kind of the opening lemmas of the propagator construction (Prop. 9.22(i)), which becomes n (Def. 9.5(2)); the unmodified $SU(2)$ coefficient 4 in Lem. 9.3 at depth 0, which becomes $4\varsigma_G$, at most $8(1 - \frac{1}{N})$ for $SU(N)$; the averaging constant $C_Q = 18NL^3 C_{\mathrm{av}}$ of Lem. 9.4, with its admitted kernel bound. When $C_{\mathrm{av}} = 3$ is supplied, its value is $54NL^3$, compared with $108L^3$ at $SU(2)$; the constants formed from $6LC$ and $12LC$ then move by $N/2$ and their reciprocal thresholds by $2/N$; and one constant, which takes the factor κ_G, for $SU(N)$ and radii $r \le z_G$ the factor $1 + \kappa'$, the thresholds on α_0 being divided by it. Two bounds of the $SU(2)$ construction take new numbers besides (Def. 9.5(2), (3), Remark 9.6): the 12 of the bound $\xi^3 |J_W(U)(b)| \le 12a$ on the current of a region at complex fields whose bond variables have condition number at most 2 becomes $6\hat{\varsigma}_G$, the 12 failing at $SU(N)$ from $N = 5$; and the factor 1 of the bound $|\mathrm{Bl}(\mathsf{g}, \mathsf{f})| \le \|\mathsf{g}\|_2 \|\mathsf{f}\|_2$ drawn by polarisation becomes κ_G^2, and stays 1 where $|X + iY|^2 \le |X|^2 + |Y|^2$ on $\mathfrak{g}$. Further numbers move in Def. 9.5(2) and Lems. 9.12(ii), (iii), 9.13(i) and 9.14(i), (ii). The constants of the propagators and the covariances, those of the critical configurations and those of the sections, which rest on both,

are set by the statements of Prop. 9.22 and of (P), (V); and all relations among the constants follow from the form of the displays. A condition of the table need not stay true when a threshold that stands in a conclusion is lowered: the floor $\delta_{2,\mathrm{b}} R_1 \mathfrak{M} \geq 64(3\log L + \log 2)$ in the required separation conditions of (K) fails when the rate $\delta_{2,\mathrm{b}}$ is lowered with R_1 and $\mathfrak{M}$ fixed, already at $SU(2)$; the ordered choice of (a) uses no such monotonicity.

■

Sketch of proof. (a), (b) The table of (a) follows the ordered choice of Prop. YM3_I:69.1, with the numbers of G as data and the compatibility required in §9.1.1(K): the constants of $SU(2)$ are kept, the value for G stands behind each function symbol, and new letters enter only where a formula needs them; a floor on a constant free upward or a positive ceiling on one free downward keeps the shape admissible and leaves the admissible sets of the other constants as they are, and a condition whose last constant is determined is read off its formula. The recursion that defines large sets commutes with intersections, so the intersection of two sets is large exactly when both are; a large set holds, at each choice of the earlier constants at which it is large, a half-line, an interval $(0, c]$ or the whole window of the next constant, so one floor or ceiling on each free constant, laid in the order of the choice, makes every admissible choice lie in it, and a set in which every choice of such a table lies is large, provided the original two-sided windows have been retained whole. Further ends on a two-sided constant instead restrict its window. Their joint feasibility at an earlier choice means exactly that the original window meets their intersection; this does not make the ends large in the original table. The radius stands in rows on γ alone, so the tables at two radii differ in rows on γ alone and have the same admissible choices before γ; a set that reads γ or ρ and is large is met by laying its set of earlier choices as rows before γ, which read no ρ, and then at each ρ a ceiling on γ. (c) On $\mathfrak{su}(N)$, $|\cdot| \leq (N-1)^{1/2}\|\cdot\|$ (Lem. 9.9(i)), that is $\|\cdot\|^2 \geq \frac{1}{N-1}|\cdot|^2$, and the conditions that compare the norms hold together at this ratio of the squares. The two-sided condition on ε_{v} is that of the inclusions of Lem. 9.12(ii), valid for $r \leq z_G$: its window, which uses the two inclusions, is not empty with the numbers of the critical configuration of (V), and its ends are made compatible by one floor on R_1. (d) Lem. 9.14(iv): the chains need $4Ka \leq r_n$, which keeps the normalised factor within r_n of 1, and $C_z \geq 4c_nK$; for $n \geq 4$ the radii of the $SU(2)$ construction fall by the factor $2r_n$ and its constant C_z rises by the factor $\frac{c_n}{5}$, which gives both, the pair $4Ka \leq \frac{1}{2}$, $C_z \geq 20K$ of the $SU(2)$ construction holding besides as $r_n < \frac{1}{2}$ and $c_n > 5$; for $n = 2, 3$ the constants of the $SU(2)$ construction serve, as $r_n \geq \frac{1}{2}$ and $c_n < 5$; and the further member of the radius of the second chain keeps the logarithms of that chain in $\mathfrak{g}^c$. The other devices, a trace series, an integral along a segment, a Gaussian integral, positivity along a path, uniqueness by continuity and the number $2n\log L - \operatorname{tr}\log T$ of the elimination, ask their conditions of the operator or of each of its eigenvalues by itself, and so read no dimension. The last asks $|T-1| < 1$ at one choice of g, g'; at two choices T_1, T_2 with $|T_j - 1| \leq r < 1$, T_2 is conjugate to $T_1 W^{-1}$, $W = \mathrm{Ad}(h)$ with $h \in G$, and $\operatorname{tr}\log X - \operatorname{tr}\log(XW^{-1})$ lies in $2\pi i\mathbb{Z}$, is constant on the convex set $\{|X-1| \leq r,\ |X-W| \leq r\}$ and vanishes at $\frac{1}{2}(1+W)$, the eigen-angles of W being symmetric, so that the number is the same at both, for every n (Lem. 9.14(iii)). Of the devices that read the value, the series at the Jacobian of the bond shift and the series at the quotient of determinants read it without a count: $\det(A + u\otimes v) = \det A\,(1 + v(A^{-1}u))$ makes the Jacobian of the bond shift, a map of determinant 1 plus the rank-one term that the cut-off brings, the scalar $1 - \mathsf{jc}$, and the bound $8\|\vartheta'_G\|_\infty |H_b| \leq \delta_k$ on the shift keeps $|\mathsf{jc}|$ below 1, the Jacobian lying in $[\frac{7}{8}, \frac{9}{8}]$ (Prop. 9.22(iv)); and $\det(\mathsf{T}(0)^{-1}\mathsf{T}(\zeta))$ is 1 at $\zeta = 0$, so within 1 of 1 near $\zeta = 0$. (e) The central factor $z(X)$ is a local homomorphism, invariant under conjugation in G^c, and $|\log X| \geq \theta(z(X))$; so the inequality by which a product is shown to lie near $\mathbf{1}$ shows its membership in D^c, and $\{|\log X| < 2\arcsin\frac{z_G}{2}\} \subset D^c$ exactly (Lem. 9.9(iii)); a radius known only through a chord meets the central elements at distance z_G. Around a plaquette whose bond variables lie within r of $\mathbf{1}$, $(1+r)^4 < 2$, the partial products stay in the unit ball, so $z(\partial U(p)) = \partial\zeta(p)$; and a gauge function with $h(x)h(y)^{-1} \in D^c$ at neighbours has $z(h(x)) = z(h(y))$

there, so $z \circ h$ is constant on a connected set. (f) Use the admitted orbit radius of §9.1.6; for an averaging extension that sends its central field to $\mathbf{1}$: at $SU(N)$ the field with $e^{2\pi i/N}\mathbf{1}$ on one bond off the block axes is critical, without current, averages to $\mathbf{1}$ and is no pure gauge, and its plaquettes are at distance $2\sin\frac{\pi}{N}$ from $\mathbf{1}$, so it is admitted at the first level as soon as $2\sin\frac{\pi}{N} < \varepsilon_{\mathrm{v}} L^{-2}$; the further conditions of (f) are those that §9.1.1(V), Props. 9.22(iv), 9.24 and Lem. 9.21(iii) set. (g) Each constant is given by the statement that sets it, in which N enters only as named; $|T^{\mathrm{T}}| \le N|T|$ is sharp, $\kappa^2 = N-1$ being attained by general kernels; $4\varsigma_G$, $6\hat{\varsigma}_G$ and κ_G^2 are those of Remark 9.6, where $8(1-\frac{1}{N})$ is the bound at $SU(N)$ and $4\varsigma_G$ exceeds 8 at the spin representation of $Spin(2m+1)$ from $m = 43$. The displayed separation floor bounds from below a product in which $\delta_{2,\mathrm{b}}$ is a factor, so it is lost when $\delta_{2,\mathrm{b}}$ alone is lowered; in the ordered choice each constant is chosen after those it depends on. □

Proposition 9.27 (Reflection positivity and Gibbs states)**.** Let $G \neq \{\mathbf{1}\}$ be a closed subgroup of $U(N)$ and $\beta \ge 0$, and consider the Wilson measure of a periodic lattice.

(i) With $\hat{g} = \mathrm{diag}(g, \bar{g})$,

$$e^{-\beta a(gg'^{-1})} = e^{-\beta} \sum_{\mathsf{n}\ge 0} \frac{(\beta/2N)^{\mathsf{n}}}{\mathsf{n}!} \sum_{I,J} \hat{g}_{IJ}\, \overline{\hat{g}'_{IJ}},$$

absolutely and uniformly convergent, with positive coefficients, where $I = (i_1, \dots, i_{\mathsf{n}})$ and $J = (j_1, \dots, j_{\mathsf{n}})$ run over $\{1, \dots, 2N\}^{\mathsf{n}}$ and $\hat{g}_{IJ} = \prod_k \hat{g}_{i_k j_k}$, the empty product being 1; so the measure is reflection positive at every mirror $v \cdot x = c$, $v = e_\mu$ with $2c \in \mathbb{Z}$ or $v = e_\mu \pm e_\nu$ with $c \in \mathbb{Z}$, whose period lattice, with periods at least 3, splits along v, for every bounded Borel function of the bonds on one side, gauge invariant or not; and for $\beta > 0$ every $\lambda_\pi(\beta) = d_\pi^{-1} \int e^{-\beta a} \chi_\pi$ obeys $\lambda_\pi(\beta) \ge e^{-\beta} \frac{(\beta/2N)^{\mathsf{n}(\pi)}}{\mathsf{n}(\pi)!\, d_\pi} > 0$, $\mathsf{n}(\pi)$ being an n for which π occurs in the n-th tensor power of $\mathbb{C}^N \oplus \overline{\mathbb{C}^N}$. At a mirror through the midpoints of bonds reflection positivity holds if and only if the crossing operator K is positive; K is the average over gauge transformations times the convolution with the weight on every bond, with the eigenvalues $\prod_b c_{\pi_b}$, $c_\pi = d_\pi^{-1} \int k\, \chi_\pi$ for a real even class weight k in the place of $e^{-\beta a}$, so that $c_\pi = \lambda_\pi(\beta)$ at $k = e^{-\beta a}$; two site mirrors ask nothing of the weight, a bond mirror asks these products to be non-negative, and a diagonal pair asks every $c_\pi \ge 0$.

(ii) The two chessboard estimates of the $SU(2)$ construction hold, with mirror copies conjugated at odd reflection parity in the slab products and in Fold:

$$|\mu(\prod_{j \le 2M} f_j^{[j]})| \le \prod_j \mu(\prod_i f_j^{[i]})^{1/2M}$$

for slabs and

$$|\langle \prod_\kappa F_\kappa \rangle_\beta| \le \prod_\kappa \langle \mathrm{Fold}\, F_\kappa \rangle_\beta^{1/\mathsf{N}}$$

for cubic cells, and so do the statements of the $SU(2)$ construction that follow them: for sources $\|e^{\mathcal{O}(w)}\|_{\mathrm{ch}}^{\mathsf{N}} = Z_\beta[w^{\mathrm{ch}}]/Z_\beta \ge 0$ (equation YM3_I:(1759)); subadditivity,

$$Z_{\underline{\mathsf{n}}}[\mathrm{Fold}\, F] \le Z_{\underline{\mathsf{n}}'}[\mathrm{Fold}\, F]^{\prod_\mu \mathsf{n}_\mu/\mathsf{n}'_\mu}$$

for boards with even numbers $\mathsf{n}'_\mu \le \mathsf{n}_\mu$ of cells along the axes and one function F of a cell on both (Prop. YM3_I:86.5); and the volume (Cor. YM3_I:86.6): on tori of sides $\ell \ge \ell' \ge 2\ell_{\mathrm{c}}$, ℓ_{c} the side of the cells, with $p_\ell = \ell^{-3} \log Z_\beta$ on the torus of side ℓ, $p_\ell \le p_{\ell'} \le 0$ and $\|F\|_{\mathrm{ch},\ell} \le \|F\|_{\mathrm{ch},\ell'} e^{\ell_{\mathrm{c}}^3 (p_{\ell'} - p_\ell)}$. The comparison of the Gibbs kernel of a box with the torus measure (Prop. YM3_I:91.8) holds for every such G, as it reads of the group only $0 \le a_p \le 2$. The Wilson measure on a torus and on the periodic configurations of the infinite lattice are one measure. Given $\mathsf{K} \ge 0$ in the three directions, reflection positivity and the

two chessboard estimates hold for every real bounded even class function in the place of βa, and so do the statements that follow them, with $p_\ell \le p_{\ell'}$, and $p_{\ell'} \le 0$ for a non-negative function.

(iii) For representations ρ_i whose sum is the inclusion and all $\beta_i > 0$, every periodic Gibbs state of $\prod_p \exp[-\sum_i \beta_i(1 - \frac{1}{N_i}\mathrm{Re}\,\mathrm{tr}\,\rho_i(\partial U(p)))]$ is invariant under the reflections in bond mirrors and under the lattice isometries, and is reflection positive at the nine families of mirrors; for one block this holds also at $\beta = 0$; and $\nu(\mathrm{Im}\,\chi_\pi(\partial U(p))) = 0$ at every plaquette. The evenness $k(g^{-1}) = k(g)$ of the weight cannot be dropped.

(iv) Let G be compact, connected and semisimple, $w(V) = P[\frac{1}{2i}(V - V^*)]$, and W the holonomy of a plaquette $p \ni b$ based at b. Then $\sum_a \frac{d}{ds}\big|_{s=0}\langle t_a, w(e^{ist_a}W)\rangle = n - k_p$. With the isotypic decomposition $\mathbb{C}^N = \bigoplus_s W_s \otimes \mathbb{C}^{k_s}$, $d_s = \dim W_s$, Ω acting on W_s by ω_s and $w_s = 1 - \mathrm{Re}\,\chi_s/d_s$: $a_p = \sum_s \frac{d_s k_s}{N} w_s(\partial U(p))$ and $k_p = \sum_s \omega_s \frac{d_s k_s}{N} w_s(\partial U(p))$; so k_p is a multiple of a_p if and only if the ω_s of the non-trivial constituents agree, and then $k_p = \frac{nN}{N - k_0} a_p$, k_0 the multiplicity of the trivial constituent, on which $w_s = 0$; under this condition, $k_p = n a_p$ if no constituent is trivial, as for an irreducible representation and for sums of copies of one, and $k_p = \frac{9}{2} a_p$ at $\{\mathrm{diag}(u, 1) : u \in SU(2)\} \subset U(3)$. For $SU(2)$ on $\mathbb{C}^2 \oplus \mathbb{C}^3$, with $a_p = 1 - \frac{1}{5}\mathrm{Re}\,\mathrm{tr}\,\partial U(p)$, no multiple of a_p serves for k_p. With the weight $\Lambda = \mathrm{diag}(0.26, 0.26, 0.16, 0.16, 0.16)$ in the commutant, at first order of strong coupling, the integrated identity holds with the Casimir density, both sides being 0.206, and fails with $\frac{n}{2} a_p$ in the place of $\frac{1}{2} k_p$, the side that carries it becoming 0.2796 against 0.206. The same holds for an action with a weight Λ in the commutant: the bonds move by the vector part w, the action is differentiated by the gradient part, which is $P[N\Lambda(V - V^*)/2i]$ for the pairing $\langle \cdot, \cdot \rangle$, w itself at $\Lambda = \frac{1}{N}\mathbf{1}$, and $P[\Lambda(V - V^*)/2i]$ for the pairing $\mathrm{tr}\,XY$, and the flux carries the same Casimir density.

■

Sketch of proof. (i) We expand $e^{\frac{\beta}{2N}\mathrm{tr}(\hat{g}\hat{g}'^*)}$; every term is $f(g)\overline{f(g')}$ with a positive coefficient, which gives reflection positivity, and one term bounds λ_π; the square root of the weight for the convolution, $k(gh^{-1}) = \int_G q(gx^{-1})q(xh^{-1})\,\mathrm{d}x$ with q a real continuous even class function whose coefficients are the $\lambda_\pi^{1/2}$, makes each plaquette cut by a mirror a square, $\partial U(p) = g_p(U)g_p(\theta U)^{-1}$; at a bond mirror the partition function is a trace of powers of $\mathsf{K}^{1/2}\mathsf{S}\mathsf{K}^{1/2}$, K holding the crossing plaquettes alone and S, the slab operator, the plaquettes inside a layer.

(ii) The partition function with insertions is the trace of a product in which the crossing operator alternates with the slab operators, which carry the plaquettes inside a layer and the insertions; the eigenvalues of the crossing operator are products of the $\lambda_\pi \ge 0$, so $\mathsf{K} \ge 0$, and Hölder's inequality for traces, $|\mathrm{Tr}(A_{2M} \cdots A_1)| \le \prod_j \|A_j\|_{2M}$, gives the estimates; where the proofs of the chessboard estimates of the $SU(2)$ construction (Prop. YM3_I:86.3) use $\mathrm{tr}\,X^{-1} = \mathrm{tr}\,X$, false at $i \in U(1)$, they need only $a(X^{-1}) = a(X) = a(hXh^{-1})$, that is, that flipping a plaquette keeps its action. Reflections and translations keep action, observable and measure; folded integrals on two tori are traces of powers of one positive operator, so that $\mathrm{Tr}\,B^{\mathsf{p}} \le (\mathrm{Tr}\,B^{\mathsf{p}'})^{\mathsf{p}/\mathsf{p}'}$ for $B \ge 0$ and integers $\mathsf{p} \ge \mathsf{p}' \ge 1$ gives subadditivity and the comparison of the volumes for every weight with $\mathsf{K} \ge 0$, and $Z_\beta \le 1$, that is $p_{\ell'} \le 0$, where the weight is at most 1; the proof of Prop. YM3_I:91.8 reads of the group only $0 \le a_p \le 2$, which holds on every $G \subset U(N)$; and the cell estimate also follows from the slab estimate in three directions, with no trace formula.

(iii) $\log(Z_\Gamma[t\mathbb{A}]/Z_\Gamma[0]) \ge t|\mathbb{A}|\nu(A)$ minus a surface term; the trace formula with insertions has modulus at most $\prod \|X\|$; at $X = e^{\pm\varepsilon\mathsf{D}}$, D real antisymmetric, $\nu(A_{\mathsf{D}}) = 0$, so neighbouring layers are exchangeable, and two layers fix the chain of layers because the kernel is injective on measures, all λ_π being positive. For the nine families, reflection positivity on adapted tori, Schwarz's inequality with a longest run and a separation argument identify every periodic Gibbs state with a reflection positive one.

(iv) The derivative of $\langle t_a, w(e^{ist_a}W)\rangle$ at 0 is $\frac{1}{N}\mathrm{Re}\,\mathrm{tr}(t_a^2 W)$; summing gives $\frac{1}{N}\mathrm{Re}\,\mathrm{tr}(\Omega W) = n - k_p$. The decomposition follows from Schur's lemma. At the trivial constituent $w_s = 0$; a constituent s and its conjugate s^* have $w_s = w_{s^*}$ and $\omega_s = \omega_{s^*}$; and the w_s of the non-trivial constituents, one for each pair $\{s, s^*\}$, are linearly independent functions on G, as 1 and the real parts of their characters are, by the orthogonality of characters. So $k_p = ca_p$ if and only if $\omega_s = c$ at every non-trivial constituent, and $\frac{1}{N}\mathrm{tr}\,\Omega = \sum_s \omega_s \frac{d_s k_s}{N} = n$ gives $c = \frac{nN}{N-k_0}$. For the action $1 - \frac{1}{2}\mathrm{tr}\,\Lambda(u+u^{-1})$ with Λ in the commutant, $\frac{d}{ds}\big|_{s=0}(1 - \frac{1}{2}\mathrm{tr}\,\Lambda(e^{isX}W + W^{-1}e^{-isX})) = \mathrm{tr}(X\Lambda H_W) = \langle X, P[N\Lambda H_W]\rangle$ for $X \in \mathfrak{g}$, as Λ commutes with W. □

Lemma 9.28 (The coefficient of β^{-3} in a box)**.** Let $G \subset U(N)$ be compact and connected, $n \geq 1$, semisimple or not, and consider the box $\{0,\dots,s\}^3$ with trivial Dirichlet data, $\beta \geq 1$. Then $|\langle a_p; a_q\rangle - \beta^{-2}B_0^{[s]}(p,q) - \beta^{-3}B_1^{[s]}(p,q)| \leq C\beta^{-4}$, with C depending on the box, as in Prop. 7.12 for $SU(2)$, and

$$B_1^{[s]}(G) = \frac{\varphi_G}{24}\,B_1^{[s]}(SU(2)) + c_G\,Y^{[s]}, \qquad c_G = \frac{1}{N}\mathrm{tr}\,\Omega^2 - \frac{3}{8}\varphi_G,$$

$\varphi_G = \sum_a \mathrm{tr}(\mathrm{ad}\,t_a)^2$ and $Y^{[s]}$ the coefficient for $U(1)$, $0 \leq Y^{[s]}(p,q) \leq C\langle x_q - x_p\rangle^{-6}$. As $s \to \infty$ the coefficient converges, the points being recentred at the centre of the box, and summed over orientations its part at distance r is $\frac{\varphi_G}{16\pi^3}\mathsf{r}^{-5}$ up to $C_G\mathsf{r}^{-6}(1 + |\log \mathsf{r}|)$. ■

Sketch of proof. Laplace's method about the gauge orbit of **1** in the box gives the expansion. The Wick pairings that decay like the fifth inverse power fall into six classes, and in each the structure constants enter through $\sum_{abc} f_{abc}^2 = \varphi_G$: the numeral 24 of $SU(2)$ becomes φ_G in five classes and 12 becomes $\frac{1}{2}\varphi_G$ in one. The representation enters through Ω only in terms that decay like the sixth power. The limit and the leading part are those of $SU(2)$ (Thm. V), whose constant $\frac{3}{2\pi^3}$ is multiplied by $\frac{\varphi_G}{24}$. □

Lemma 9.29 (Coefficients on the simple ideals)**.** Let $\mathfrak{g} = \bigoplus_{i\leq r}\mathfrak{g}_i$ be semisimple. The mean of the n quadratic colour coefficients of a local term of second order is $\sum_i \frac{n_i}{n}\mathsf{cf}_i$, cf_i its coefficient on $\mathfrak{g}_i$, and $|\mathsf{cf}_i| \leq \kappa_i^2 C_{\mathrm{am}} r_0^{-2} E_0 = (\kappa_i/\kappa_G)^2 \mathsf{G}_\sharp$ for the terms of Thm. YM3_I:D, with C_{am}, r_0 and $\mathsf{G}_\sharp$ of Prop. 9.24. If $|e_k^{(i)}| \leq \Gamma$, the sequences $x_0^{(i)} = \beta$, $x_k^{(i)} = L^{-1}x_{k-1}^{(i)} - e_k^{(i)}$ and $x_k = \sum_i \frac{n_i}{n}x_k^{(i)}$ satisfy $x_k = L^{-1}x_{k-1} - \sum_i \frac{n_i}{n}e_k^{(i)}$ and

$$|x_k^{(i)} - x_k| \leq 2\Big(1 - \frac{n_i}{n}\Big)\Gamma\frac{L}{L-1}(1 - L^{-k}).$$

■

Sketch of proof. The first claim is the splitting of an orthonormal basis along the ideals; the bound is that of the $SU(2)$ construction with $|t_a| \leq \kappa_i$ on $\mathfrak{g}_i$ (Prop. 9.24). Further $x_k^{(i)} - L^{-k}\beta = -\sum_{s\leq k} L^{-(k-s)}e_s^{(i)}$, so $|x_k^{(i)} - x_k^{(m)}| \leq 2\Gamma\frac{L}{L-1}(1-L^{-k})$, and $x_k^{(i)} - x_k = \sum_{m\neq i}\frac{n_m}{n}(x_k^{(i)} - x_k^{(m)})$. □

Proposition 9.30 (Semisimple Lie algebras and tied ideals)**.** Let $\mathfrak{g} = \bigoplus_{i\leq r}\mathfrak{g}_i$ be semisimple, $r \geq 2$, and G_i the subgroup with Lie algebra $\mathfrak{g}_i$.

(i) The averages $E_j(M) = \int_{G_j} hMh^{-1}\,\mathrm{d}h$ and their products over $j \notin I$ are conditional expectations; so the thresholds of the $SU(2)$ construction pass down to the ideals with the constant 1, and back up with the factor r.

(ii) For $u_i = e^{iZ_i} \in G_i$, $a(u_1\cdots u_r) = \sum_i a(u_i) - t(u)$ with the tie $t = \frac{1}{4}\sum_{i<j}\tau_{ij}\langle Z_i, Z_i\rangle\langle Z_j, Z_j\rangle + O(Z^6)$, τ_{ij} of Prop. 9.18(iv); the tie vanishes exactly for block sums. In the Hessian of the action at $u = u_i v$, $v = \prod_{j\neq i} u_j$, the block of the ideal $\mathfrak{g}_i$ is $\frac{1}{N}\mathrm{Re}\,\mathrm{tr}(Sc(u)T)$, $S, T \in \mathfrak{g}_i$, with $c(u) = c(u_i)c(v) - H_{u_i}H_v$: it is the Hessian of the simple pair of $\mathfrak{g}_i$ with the weight $c(v)/N$, which is hermitian, lies in the commutant of G_i, is positive near **1** and reads the field v of the other factors, less the term $\frac{1}{N}\mathrm{Re}\,\mathrm{tr}(H_v S H_{u_i} T)$, at most $|S||T||u_i - \mathbf{1}||v - \mathbf{1}|$. This term

vanishes when on each constituent $W_s \otimes M_s$ of $\mathbb{C}^N$, W_s irreducible under G_i and M_s its space of multiplicity under the other factors, one of W_s and M_s is self-conjugate, so at every ideal $SU(2)$ and at $SU(2) \times SU(3)$ on $\mathbb{C}^2 \otimes \mathbb{C}^3$; at $SU(3) \times SU(3)$ on $\mathbb{C}^3 \otimes \mathbb{C}^3$ it is not zero in general. The gradient on $\mathfrak{g}_i$ carries the like term $P_i[c(u_i)H_v]$ beside $P_i[c(v)H_{u_i}]$. Between two ideals stands one mixed block $\frac{1}{N}\mathrm{Re}\,\mathrm{tr}(SS'(c(u)-\mathbf{1}))$, $S \in \mathfrak{g}_i$, $S' \in \mathfrak{g}_j$, $i \neq j$, at most $|S||S'||u_i - \mathbf{1}||u_j - \mathbf{1}|$, which vanishes at $u = \mathbf{1}$ and need not vanish at a background (at $SU(2) \times SU(2)$ on $\mathbb{C}^2 \otimes \mathbb{C}^2$, with $S = \sigma_3 \otimes \mathbf{1}$, $S' = \mathbf{1} \otimes \sigma_3$ and $u = e^{it\sigma_3} \otimes e^{it\sigma_3}$, it is $-\sin^2 t$). The block averages on small fields and the exponential map are products over the factors, the density $j_G = \prod_i j_{G_i} \circ P_i$ is a product, and the quadratic part of the gauge-fixing term is a sum over the ideals; the gauge-fixing term $a(X)$ itself is $\sum_i a(X_i) - t(X)$, with the tie of fourth order; and the domain $\{|Y| < \pi\}$ of the chart, G^π with its indicator and the cut-off ν are compared with those of the factors through (i), with the constant 1 down to the ideals and the factor r back up.

(iii) Let $F_i = P_i F$ be the part of the field strength on $\mathfrak{g}_i$, and $Q_i = \sum_p \frac{1}{2}\langle P_i H_{\partial U(p)}, P_i H_{\partial U(p)}\rangle$, the sum running over the bare plaquettes. By the transferred statements of the step on its lists, its potentials and their edge parts (Prop. 9.24), the leftover $\sum_i(\mathsf{cf}_i - \mathsf{cf})\frac{1}{2}\langle F_i, F_i\rangle$ of a level splits off at the level of the potentials, with size at most $\frac{7}{4}\alpha_0^2 \max_i |\mathsf{cf}_i - \mathsf{cf}| \leq \frac{7}{2}(1 - \min_i \frac{n_i}{n})\alpha_0^2 \max_i |\mathsf{cf}_i|$, its edge parts $8M^3$ times this; its terms of first order are of third order in the field strength and are bounded by Cauchy's estimate whether or not they vanish; in dimension three the part of second order of an earlier level is bounded and is not taken out; the leftover of level j is $L^j \sum_i (e_j^{(i)} - b_j)Q_i$, $e_j^{(i)}$ the coefficient on $\mathfrak{g}_i$ and b_j their mean; and the leftovers of all levels add up to $-L^k \sum_i (x_k^{(i)} - x_k)Q_i$, which is bounded uniformly in k, as $L^k Q_i$ is of the order α_0^2 on a block of level k and $|x_k^{(i)} - x_k|$ is bounded by Lem. 9.29.

(iv) If the action of G has property (T) of Thm. AA(h3) at a choice of the constants, then at that choice one run, the mean, suffices: the r couplings $x_k^{(i)}$ stay within $2(1 - \frac{n_i}{n})\mathsf{G}_\sharp \frac{L}{L-1}$ of it (Lem. 9.29 with $\Gamma = \mathsf{G}_\sharp$, as $|e_k^{(i)}| \leq (\kappa_i/\kappa_G)^2 \mathsf{G}_\sharp \leq \mathsf{G}_\sharp$, $\kappa_i \leq \kappa_G$ since $\mathfrak{g}_i \subset \mathfrak{g}$), and all clauses on the increments hold, the increment theorem, the regular progression, the sequence and the bounds on polydiscs among them; a format with one exponent, one integrand, one total and a carried term is the format of one run.

(v) At the least tied pair, $SU(2) \times SU(2)$ acting on $\mathbb{C}^4 \oplus \mathbb{C}^2$ by $(u_1, u_2) \mapsto (u_1 \otimes u_2) \oplus u_2$ with every weight, that is, with the action $\lambda_1(1 - \frac{1}{4}\mathrm{Re}\,\mathrm{tr}(u_1 \otimes u_2)) + (1 - \lambda_1)(1 - \frac{1}{2}\mathrm{Re}\,\mathrm{tr}\,u_2)$ for every $\lambda_1 \in (0,1)$, one has $1 - a = (1 - a^{(2)})(1 - \lambda_1 a^{(1)})$, $a^{(i)} = 1 - \frac{1}{2}\mathrm{tr}\,u_i$; so the gradient and the Hessian of the action are the weighted ones of the $SU(2)$ construction, with the plaquette weights $\lambda_1(1 - a_p^{(2)})$ on the first factor and $1 - \lambda_1 a_p^{(1)}$ on the second, plus the block $-\lambda_1(\nabla a^{(1)} \otimes \nabla a^{(2)} + \nabla a^{(2)} \otimes \nabla a^{(1)})$, which annihilates infinitesimal gauge directions. The fluctuation operator of the first step is $A^\sharp(p_1) \oplus A^\sharp(p_2) + C^{\mathrm{T}}TC$, where p_1, p_2 are the points of the two factors, each with its plaquette weights, $A^\sharp$ is the fluctuation operator of the $SU(2)$ construction, C is the linear map from the fluctuation fields to the forms on the bonds through which the $SU(2)$ construction forms $A^\sharp$ from the Hessian of the action, and T, the Hessian of the action less the direct sum of the Hessians of the two factors, is symmetric, of the range of one plaquette, vanishes at $\mathbf{1}$ and at fields with values in one factor, and obeys $|(TA)(b)| \leq 36300 \max_{p \ni b} |\partial U(p) - \mathbf{1}|^2 \, |A|_{(b)}$, $|A|_{(b)}$ the largest $|A|$ on the bonds of the plaquettes that hold b. To obtain the propagator conclusion one still has to establish at each level a regular domain on which the diagonal weighted operator $A_0 = A^\sharp(p_1) \oplus A^\sharp(p_2)$ has the required inverse and decay bounds, and $\|A_0^{-1} C^{\mathrm{T}} TC\| < 1$ in the corresponding forcing and decay norm, uniformly in the level. That relative bound would justify the Neumann inverse. The kinematic, interaction and Hessian conclusions

require the corresponding regular-domain estimates; the algebraic identities alone do not establish them or property (T).

■

Sketch of proof. (i) E_j is the average of the conjugation over a compact group, so it is a positive unital projection onto the commutant of G_j, a conditional expectation, and does not increase $|\cdot|$, also at complex arguments; the E_j commute, and their product over $j \notin I$ is the average over the product of these G_j, a conditional expectation onto its commutant. On $\mathfrak{g}^c$, E_j is 0 on $\mathfrak{g}_j^c$, a vector of the simple ideal $\mathfrak{g}_j$ fixed by $\mathrm{Ad}(G_j)$ being 0 (Lem. 9.15(i)), and the identity on the other ideals, which commute with G_j; so $\prod_{j\neq i} E_j = P_i$ there, whence $|P_i X| \le |X|$ and $|X| \le \sum_i |P_i X| \le r \max_i |P_i X|$.

(ii) For the commuting Z_i, $1 - \frac{1}{N}\mathrm{Re}\,\mathrm{tr}(u_1 \cdots u_r) = 1 - \frac{1}{N}\mathrm{tr}\cos(\sum_i Z_i)$. Its terms of second order are those of the $a(u_i)$, as $\mathrm{tr}\, Z_i Z_j = 0$ for $i \neq j$; of its terms of fourth order those in which some Z_j stands to the first power vanish, the trace of an element of a simple ideal being 0 on every invariant subspace, and the cross terms that remain are $-\frac{1}{4N}\sum_{i<j}\mathrm{tr}(Z_i^2 Z_j^2)$, where $\frac{1}{N}\mathrm{tr}(Z_i^2 Z_j^2) = \tau_{ij}\langle Z_i, Z_i\rangle\langle Z_j, Z_j\rangle$, an invariant quadratic form on a simple ideal being a multiple of $\langle\cdot,\cdot\rangle$. They vanish exactly when $Z_i^2 Z_j^2 = 0$ for all $i \neq j$, a product of two commuting positive matrices with trace 0 being 0, that is, when the G_i act on separate blocks. At $u = u_i v$, $v = \prod_{j\neq i} u_j$, the elements u_i and v commute, so that $c(u_i v) = c(u_i)c(v) - H_{u_i}H_v$ and $H_{u_i v} = H_{u_i}c(v) + c(u_i)H_v$, and $c(v)$ and H_v commute with G_i and with $\mathfrak{g}_i$. So the block $\frac{1}{N}\mathrm{Re}\,\mathrm{tr}(Sc(u)T)$ of $\mathfrak{g}_i$, the second derivative along $S, T \in \mathfrak{g}_i$, is $\frac{1}{N}\mathrm{Re}\,\mathrm{tr}(c(v)\,Sc(u_i)T)$, the Hessian of the simple pair of $\mathfrak{g}_i$ with the weight $c(v)/N$, less $\frac{1}{N}\mathrm{Re}\,\mathrm{tr}(H_v S H_{u_i} T)$, whose modulus is at most $|S||T||H_{u_i}||H_v| \le |S||T||u_i - \mathbf{1}||v - \mathbf{1}|$, as $|H_w| \le |w - \mathbf{1}|$ for unitary w; and the gradient on $\mathfrak{g}_i$ is $P_i H_u = P_i[c(v)H_{u_i}] + P_i[c(u_i)H_v]$. On a constituent $W_s \otimes M_s$ the elements S, T and H_{u_i} act on W_s and H_v on M_s, with $\mathrm{tr}_{M_s} H_v = \mathrm{Im}\,\chi_{M_s}(v)$, and $\mathrm{Re}\,\mathrm{tr}(SHT) = \mathrm{tr}(\frac{1}{2}\{S,T\}H)$ for hermitian S, H, T; so the odd term is $\frac{1}{N}\sum_s \mathrm{tr}_{W_s}(\frac{1}{2}\{S,T\}H_{u_i})\,\mathrm{Im}\,\chi_{M_s}(v)$. An antiunitary J that commutes with the group on a self-conjugate side has $JSJ^{-1} = -S$ for the elements S of its Lie algebra and $JH_wJ^{-1} = -H_w$ for its elements w, and $\mathrm{tr}(JXJ^{-1}) = \overline{\mathrm{tr}\, X}$; so the factor of that side, a real number, equals its negative and vanishes. Every representation of $SU(2)$ is self-conjugate, and at $SU(2) \times SU(3)$ on $\mathbb{C}^2 \otimes \mathbb{C}^3$ the side of the other factor of the ideal of $SU(3)$ is the $\mathbb{C}^2$ of $SU(2)$; at $SU(3) \times SU(3)$ on $\mathbb{C}^3 \otimes \mathbb{C}^3$, with $S = T = D$, $Z_i = \epsilon D$ and $v = e^{i\epsilon' D}$, $D = \mathrm{diag}(1, 1, -2)$, the two factors are $\epsilon\,\mathrm{tr}\, D^3 + O(\epsilon^3)$ and $-\frac{\epsilon'^3}{6}\mathrm{tr}\, D^3 + O(\epsilon'^5)$, with $\mathrm{tr}\, D^3 = -6$. Between two ideals the second derivative along $S \in \mathfrak{g}_i$, $S' \in \mathfrak{g}_j$ is $\frac{1}{N}\mathrm{Re}\,\mathrm{tr}(SS'u) = \frac{1}{N}\mathrm{Re}\,\mathrm{tr}(SS'(c(u) - \mathbf{1}))$, as $\mathrm{tr}\, SS' = 0$; on a constituent of the factor i, the factor j and the rest, $\mathrm{tr}(SS'u)$ is $\mathrm{tr}(S(u_i - \mathbf{1}))\,\mathrm{tr}(S'(u_j - \mathbf{1}))$ times the trace of the rest, S and S' having trace 0 there, which gives the bound; and the example is a computation. The block averages on small fields and the exponential map are products over the factors, the mean of a family of products of commuting factors being the product of their means and the logarithm of a product of small commuting factors the sum of their logarithms; $j_G(Y) = \prod_i j_{G_i}(P_i Y)$, the eigenvalues of $\mathrm{ad}\, Y$ being those of the $\mathrm{ad}\, P_i Y$ on the $\mathfrak{g}_i$; the gauge-fixing term is a trace $a(X)$, to which the expansion above applies at complex fields, so that its quadratic part is a sum over the ideals; and the chart, G^π with its indicator and ν are compared with those of the factors by the bounds of (i).

(iii) The leftover is a sum over the ideals of the difference of their coefficient and the mean times their part $\frac{1}{2}\langle F_i, F_i\rangle$ of the Yang–Mills term. By the transferred statements of the step on its lists, its potentials and their edge parts (Prop. 9.24) it splits off as a named term of the potentials of each level, as the $SU(2)$ construction splits off a term of the potentials, with the size $\frac{7}{4}\alpha_0^2 \max_i |\mathsf{cf}_i - \mathsf{cf}|$ and its edge parts $8M^3$ times this; and $\mathsf{cf}_i - \mathsf{cf} = \sum_m \frac{n_m}{n}(\mathsf{cf}_i - \mathsf{cf}_m)$ gives $|\mathsf{cf}_i - \mathsf{cf}| \le 2(1 - \frac{n_i}{n}) \max_m |\mathsf{cf}_m|$, whence the second bound. Its terms of first order in the fluctuation field are of third order in the field strength, and Cauchy's estimate bounds them, as Schwarz's lemma bounds such terms in the proof of Prop. 9.24, whether or not they vanish. In dimension three the part of second order of an earlier level is bounded, as in the $SU(2)$ construction, and stays in the potentials. At level j the

leftover is $L^j\sum_i(e_j^{(i)}-b_j)Q_i$; as $x_k^{(i)}-x_k=-\sum_{j\le k}L^{-(k-j)}(e_j^{(i)}-b_j)$, the leftovers of the levels up to k add up to $-L^k\sum_i(x_k^{(i)}-x_k)Q_i$, which is bounded uniformly in k, L^kQ_i being of the order α_0^2 on a block of level k and $|x_k^{(i)}-x_k|$ bounded by Lem. 9.29.

(iv) The proof of the $SU(2)$ construction meets the colour through one theorem on a whole class of families of terms, Thm. YM3_I:X, so a term of the tied action may be restricted to an ideal, and the leftover of (iii) is carried as a term; with property (T) the remaining steps are those of the $SU(2)$ construction, and Lem. 9.29 with $\Gamma=\mathsf{G}_\sharp$ gives the distance of the couplings, as $|e_k^{(i)}|=|\mathsf{cf}_i|\le(\kappa_i/\kappa_G)^2\mathsf{G}_\sharp\le\mathsf{G}_\sharp$ (Prop. 9.24), $\kappa_i\le\kappa_G$ since $\mathfrak{g}_i\subset\mathfrak{g}$. A format with one exponent, one integrand and one total, the leftover of (iii) being the carried term, is the format of the $SU(2)$ construction for one run.

(v) At this pair $1-a=(1-a^{(2)})(1-\lambda_1a^{(1)})$, $a^{(1)}$ and $a^{(2)}$ being the plaquette functions of the two factors, so that $1-a$ is a product of two factors each of which reads one group; hence the gradient and the Hessian of the action are the weighted ones of the $SU(2)$ construction, one weight on each ideal, $\lambda_1(1-a_p^{(2)})$ and $1-\lambda_1a_p^{(1)}$, plus the block $-\lambda_1(\nabla a^{(1)}\otimes\nabla a^{(2)}+\nabla a^{(2)}\otimes\nabla a^{(1)})$, which annihilates infinitesimal gauge directions, the $a_p^{(i)}$ being gauge invariant. The fluctuation operator of the first step is the direct sum $A^\sharp(p_1)\oplus A^\sharp(p_2)$ plus $C^{\mathrm{T}}TC$, T being this block: it is symmetric, of the range of one plaquette and of second order in the plaquettes, each gradient being of first order in its plaquette variable, and it vanishes at $\mathbf{1}$ and at fields with values in one factor, where the gradient of the factor whose field is $\mathbf{1}$ vanishes. If the regular-domain inverse hypotheses of (v) hold, $(A_0+C^{\mathrm{T}}TC)^{-1}=\sum_{j\ge0}(-A_0^{-1}C^{\mathrm{T}}TC)^jA_0^{-1}$ in the stated norm. A pointwise second-order bound on T does not establish this relative-smallness condition after the maps C, scale changes and gauge fixing, nor does it supply the all-level regular-domain estimates. These remain analytic inputs for the additional assertions of (v). □

Lemma 9.31 (Block sums)**.** Let $G=G_1\times\cdots\times G_r$ act on $\mathbb{C}^{N_1}\oplus\cdots\oplus\mathbb{C}^{N_r}$ by blocks, $G_i\neq\{\mathbf{1}\}$ closed.

(i) $a(u)=\sum_i\frac{N_i}{N}a_i(u_i)$ and the Haar measure is a product; so on every finite lattice the Wilson measure at β is the product of those of the G_i at $\beta N_i/N$, the fields of the factors are independent, and $\langle\mathcal{O}(f_1);\dots;\mathcal{O}(f_\mathsf{m})\rangle_\beta=\sum_i(\frac{N_i}{N})^\mathsf{m}\langle\mathcal{O}^{(i)}(f_1);\dots;\mathcal{O}^{(i)}(f_\mathsf{m})\rangle_{\beta N_i/N}$; with a weight $\Lambda=\bigoplus\Lambda_i$, $\frac{N_i}{N}$ is replaced by $\operatorname{tr}\Lambda_i$.

(ii) Let ν be a Gibbs state of the product weight. Then $\nu(\prod_i F_i(U^{(i)}))=\nu(\prod_i\bar F_i(U^{(i)}))$ for bounded Borel functions F_i, $\bar F_i$ any representative of the conditional expectation of F_i given the tail field $\mathcal{T}$ in the marginal ν_i: the factors are independent given the tails. For a Gibbs state ν_1 of one factor and $\mathsf{m}\ge1$ there is exactly one probability on m copies whose value at $F_1\otimes\cdots\otimes F_\mathsf{m}$ is $\nu_1(\prod_a\mathbb{E}_{\nu_1}[F_a|\mathcal{T}])$; it is a Gibbs state of the product weight of m copies at equal couplings, all its marginals are ν_1, and it is periodic if ν_1 is. Let the marginals ν_i of ν be periodic for the factors i of a set I', and for $i\in I'$ let $F_\mathbf{k}^{(i)}\in\mathcal{A}(\square_\mathbf{k}^{(i)})$, $\mathbf{k}$ in a finite set $\mathcal{K}_0^{(i)}$, be functions of the cells of a board of $\mathbb{L}_K$, each factor with its own board and its own sequence of sides $\ell_j^{(i)}\to\infty$. Then the chessboard domination of the $SU(2)$ construction (Prop. YM3_I:91.9) holds for their product:

$$\Big|\nu\Big(\prod_{i\in I'}\prod_\mathbf{k}F_\mathbf{k}^{(i)}(U^{(i)})\Big)\Big|\le\prod_{i\in I'}\liminf_{j\to\infty}\prod_\mathbf{k}\|F_\mathbf{k}^{(i)}\|^{(i)}_{\mathrm{ch},\ell_j^{(i)}},$$

$\|\cdot\|^{(i)}_{\mathrm{ch},\ell}$ being the chessboard seminorm of the Wilson measure of G_i at its coupling.

(iii) Let $G=SU(2)^r$ in $\mathbb{C}^2\oplus\cdots\oplus\mathbb{C}^2$, $0<g_i\le\gamma$, $(K;(\beta_i))$ a first-passage datum at (g_i) with the labels $\mathsf{b}_i=a_K\beta_i$, ν a Gibbs state whose marginals ν_i, ν_j are periodic, $i\neq j$, $\vartheta\in(0,1)$,

$a_K \le \mathsf{d} \le \frac{1}{20}$, and (f,h) a test pair of scale d and parameter ϑ as defined in §6.1. Then

$$|\langle \mathcal{O}^{(i)}(f); \mathcal{O}^{(j)}(h)\rangle_\nu| \le \frac{1}{4}\varepsilon_{ij}^{\times}(\mathsf{d})\,\frac{3}{16\pi^2}(\mathsf{b}_i\mathsf{b}_j)^{-1}\mathcal{K}_6(f,h),$$

with $\varepsilon_{ij}^{\times}(\mathsf{d})$ at most a number of g_i, g_j, ϑ and the constants times d, reading none of K, β_i, ν, f, h. In every Gibbs state of the product weight whose marginals are periodic in all the factors $i_1,\dots,i_{\mathsf{m}}$ of a mixed cumulant $\langle \mathcal{O}^{(i_1)}(f_1);\dots;\mathcal{O}^{(i_{\mathsf{m}})}(f_{\mathsf{m}})\rangle$, $\mathsf{m}\ge 2$, the i_q not all equal, this cumulant obeys the per-ball and factorial growth estimates obtained by the normalizer argument in the proof of Prop. YM3_I:91.13, with constants depending on the factor ceilings and the radii of the balls containing the supports. No separation of the supports is required: the mixed derivative of every single-factor normalizing exponent is zero, as checked below. ■

Sketch of proof. (i) is the product structure; cumulants of sums of independent families add, and $\mathcal{O} = \sum_i \frac{N_i}{N}\mathcal{O}^{(i)}$ gives the powers.

(ii) The Gibbs kernels of the product weight are products of those of the factors, and the marginal ν_i is a Gibbs state of the factor i. For finite sets of bonds $B_1 \subset B_2 \subset \cdots$ that exhaust the lattice, $\nu(\prod_i F_i) = \nu(\prod_i \gamma^{(i)}_{B_l} F_i)$, $\gamma^{(i)}_{B_l}$ the Gibbs kernel of the factor i; $\gamma^{(i)}_{B_l}F_i$ is the conditional expectation of F_i in ν_i given the bonds off B_l, these σ-fields decrease to $\mathcal{T}$, and the downward martingale theorem gives the convergence to $\bar F_i$ off a ν_i-null set; dominated convergence gives the first statement. The space of configurations being compact and metrisable, ν_1 has a regular conditional probability π given $\mathcal{T}$, and the m-copy state is $\int \nu_1(\,\mathrm{d}\omega)\,\pi(\omega,\cdot)^{\otimes\mathsf{m}}$: its values at products fix it; it obeys the Gibbs equations because $\mathcal{T}$ lies in the σ-field of the bonds off every finite set; and a translation that keeps ν_1 keeps $\mathcal{T}$ and so keeps it. The domination is the proof of Prop. YM3_I:91.9 run with the factors, first for a periodic Gibbs state of the product weight and functions that are products over the factors: the comparison of the Gibbs kernel of a box with the torus measure (Prop. YM3_I:91.8) holds factor by factor, by Fubini, the exponents adding, and reads of a group only $0 \le a_p \le 2$; the master inequality (the proof of Prop. YM3_I:91.9) holds with the diagonal translations, the geometry of boards (the proof of Prop. YM3_I:91.9) reading no group; and after the expansion of the product of the translates each term is a product over the factors of functions of one factor alone, whose mean under the product of the torus measures is the product of the means, each bounded by the chessboard estimate of its factor (Prop. 9.27(ii)). Periodic marginals suffice: with $\bar F^{(i)}$ the conditional expectation given $\mathcal{T}$ of the product $F^{(i)}$ of the functions of the factor i, the first statement and Hölder's inequality give $|\nu(\prod_i F^{(i)})| \le \prod_{i\in I'} \nu_i(|\bar F^{(i)}|^{\mathsf{m}})^{1/\mathsf{m}}$, m the least even integer at least $\#I'$; and $\nu_i(|\bar F^{(i)}|^{\mathsf{m}})$ is the value of the m-copy state of ν_i, a periodic Gibbs state of the product weight of m copies, at the product of $\frac{\mathsf{m}}{2}$ copies of $F^{(i)}$ and $\frac{\mathsf{m}}{2}$ copies of its conjugate, which the domination for the product weight bounds, the conjugate having the seminorm of the function.

(iii) By (ii) the mixed covariance is the covariance of the conditional expectations given the tails, so that $|\langle \mathcal{O}^{(i)}(f); \mathcal{O}^{(j)}(h)\rangle_\nu|^2 \le \mathsf{V}_{\nu_i}(f)\,\mathsf{V}_{\nu_j}(h)$, $\mathsf{V}_{\nu_i}(f)$ the variance of $\mathbb{E}_{\nu_i}[\mathcal{O}(f)|\mathcal{T}]$ in ν_i. Let $\mathsf{D}(f_1,f_2)$ be the covariance of $\mathbb{E}_{\nu_i}[\mathcal{O}(f_1)|\mathcal{T}]$ and $\mathbb{E}_{\nu_i}[\mathcal{O}(f_2)|\mathcal{T}]$ in ν_i, a real bilinear form; it is the mixed covariance of the two copies in the two-copy state of ν_i. *The unit scale.* Let F_1, F_2 vanish off $\mathbb{B}(0,1)$. The systems of the $SU(2)$ construction give at the first-passage datum (K,β_i) the normalisers (Nr1) of the one-entry tuples on the board of side L (equation YM3_I:(1844), Prop. YM3_I:91.12 (1)); they pass to the two-copy state, on the product of the polydiscs of the two copies, by the domination of (ii), as in the transfer principle (Prop. YM3_I:91.11); at this mixed word the derivative of the sum of the two normalising exponents vanishes, and Cauchy's estimate gives $|\mathsf{D}(F_1,F_2)| \le C'_V(g_i)\|F_1\|_\sharp\|F_2\|_\sharp$, with $C'_V(g)$ a number of g and the constants and $\|F\|_\sharp = \max\{\sup|F|, 40M\,\mathrm{Lip}\,F\}$. *The factor of the radius.* ν_i is invariant under all translations of $a_K\mathbb{Z}^3$ (Prop. 9.27(iii)), so $\mathsf{D}(\tau_v f_1,\tau_v f_2) = \mathsf{D}(f_1,f_2)$ for $v \in a_K\mathbb{Z}^3$. Let f_1, f_2 vanish off a ball $\mathbb{B}(x_0,\mathsf{r})$, $a_K \le \mathsf{r} \le \frac{1}{20}$. There is a packing of at least $(0.3/\mathsf{r})^3$ translates of this ball in $\mathbb{B}(0,1)$ by vectors of $a_K\mathbb{Z}^3$ (the packing argument in the

proof of Prop. YM3_I:127.1); the signed sums $\sum_v \epsilon_v \tau_v f_l$ over the packing, $\epsilon_v = \pm 1$, have the norms $\|f_l\|_\sharp$, and the mean over the signs of D at these sums is the sum over the packing of the $\mathsf{D}(\tau_v f_1, \tau_v f_2)$, the cross terms cancelling (the packing argument in the proof of Prop. YM3_I:127.1). So $|\mathsf{D}(f_1, f_2)| \le C'_V(g_i)(\mathsf{r}/0.3)^3 \|f_1\|_\sharp \|f_2\|_\sharp$, and $\mathsf{V}_{\nu_i}(f) \le C'_V(g_i)(\mathsf{r}/0.3)^3 \|f\|_\sharp^2$. *The test pairs.* The two functions of a test pair of scale d and parameter ϑ vanish off balls of radius d, so $\mathsf{r} = \mathsf{d}$; as $\vartheta < 1$, $\mathsf{d} \le \frac{1}{20}$ and $M \ge 1$, $\|f\|_\sharp \le \frac{40M}{\vartheta \mathsf{d}} \|f\|_\infty$, and the same for h; $\|f\|_\infty \|h\|_\infty \le 10^6 \vartheta^{-2} \mathcal{K}_6(f, h)$ and $\mathsf{b}_i \le L(g_i^{-2} + \bar{B})$; this gives the bound, with $\varepsilon_{ij}^\times(\mathsf{d})$ a number of g_i, g_j, ϑ and the constants times d. *The higher cumulants.* For each factor of the word, the graded normaliser that the systems give to the sub-tuple of its entries (equation eq. 5.xxviii), with no separation of the supports, passes to the state by the domination of (ii), on the product of the polydiscs of the factors; at a mixed word the derivative of the sum of the exponents vanishes, each exponent reading the entries of its own factor alone, so that Cauchy's estimate bounds the cumulant itself, and the per-ball bound and the growth bound follow from the placement and tent-decomposition calculations in the proof of Prop. YM3_I:91.13. The separation-dependent contribution there is the mixed derivative of the normalizing exponent; here that contribution vanishes because every exponent depends on the variables of only one factor. The remaining Cauchy and counting estimates require no separation. □

Lemma 9.32 (Change of units and one-step normalization)**.**

(i) Remove the last level of a regional sequence, and denote its length convention and determining set by $\eta^- = L\eta$ and $\mathbf{B}^-$. For a real form A^- put $A_0 = LA^-$. Then

$$\eta A_0 = \eta^- A^-, \qquad \|A^-\|_{X,\mathbf{B}^-} \le \|A_0\|_{X,\mathbf{B}} \le L^2 \|A^-\|_{X,\mathbf{B}^-}.$$

This holds for every closed matrix group and every background.

(ii) Let $k \ge 1$ and suppose all regions through level k are the whole torus. Let U_0, U' be regular backgrounds with parameter α_0, and let (A_0, g_0) be a real chart pair, $U'^{g_0} = e^{i\eta A_0} U_0$, normalized on the preceding top: $g_0 = v_{k-1}[A_0]$ on the level-k sites. Assume the gauge-correction estimate of Lem. YM3_I:30.4 and the slice estimate of Lem. YM3_I:30.10 hold at these backgrounds, with their admitted radii and constants, and assume the one-step factor bound

$$\sup_{\Lambda_k} |\mathsf{v}_{k-1}[A_0] - \mathbf{1}| \le 2\kappa_6 L^{-1} \|A_0\|_{X,\mathbf{B}}.$$

Here the bounds may be the transferred bounds when a transfer is explicitly supplied. In particular this statement does not assert an unconditional transfer to an arbitrary group. If α_0 and $\|A_0\|_{X,\mathbf{B}}$ are below the thresholds specified in the proof, and

$$\sup_c |\mathcal{M}^k U'(c) \mathcal{M}^k U_0(c)^{-1} - \mathbf{1}| \le \gamma,$$

there is a new chart pair (A, g) with

$$\mathsf{R} D^* A = 0, \qquad g = v_k[A] \text{ on } \Lambda_k, \qquad \sup_c |a_k[A](c)| \le \tfrac{\pi}{2}\gamma,$$

and

$$|A|_{(-1)} \le k_1 \|A_0\|_{X,\mathbf{B}}, \qquad \|A\|_{X,\mathbf{B}} + |D^* D A|_{(-3)} \le C_{\mathrm{sl}}(\alpha_0 + \gamma).$$

The constants and the thresholds depend only on those of the stated inputs and on L.

■

Proof. (i) If $j(b)$ is the level of a bare bond, its weighted field values in the two conventions are

$$L^{j(b)-k} L |A^-(b)|, \qquad L^{\min\{j(b),k-1\}-(k-1)} |A^-(b)|.$$

They agree unless $j(b) = k$, when their ratio is L. A covariant difference gains two factors of L, one from the form and one from the inverse bare spacing. Its weighted values therefore agree below level k and have ratio L^2 at level k. The background transports are identical. Taking the maxima

gives both inequalities. In particular, on the constant sequence this computation is independent of the sector.

(ii) Write $\alpha_3 = \|A_0\|_{X,\mathbf{B}}$ and prescribe $r = \mathsf{v}_{k-1}[A_0]$ on the top sites. Let a_{gc} be the radius admitted by gauge correction and a_{sl} the field radius admitted by the slice estimate, decreased to stay in the averaging chart. Set

$$k_1 = 1 + 2C_{\mathrm{gc}}K_3(3 + 2\kappa_6 L^{-1}).$$

Require α_0 below both input regularity ceilings and require

$$\alpha_3 \le a_{\mathrm{gc}}, \qquad 2\kappa_6 L^{-1}\alpha_3 \le a_{\mathrm{gc}}, \qquad k_1\alpha_3 \le a_{\mathrm{sl}}.$$

Gauge correction gives $w = e^{i\lambda}$ with $\mathsf{R}D^*A_0^w = 0$ and induced top factor r. Since $|D^*A_0|_{(-2)} \le 3\alpha_3$, its bounds give

$$\|\lambda\|_Y \le K_3(3 + 2\kappa_6 L^{-1})\alpha_3, \qquad |A_0^w|_{(-1)} \le k_1\alpha_3.$$

Put $A = A_0^w$ and $g = wg_0$. The initial residual factor at a level-k site is $v_k[A_0]^{-1}g_0 = \mathsf{v}_{k-1}[A_0]^{-1}$. Under the correction it is multiplied on the left by the prescribed r, so it becomes $\mathbf{1}$. Thus $g = v_k[A]$ on the top.

Every retained row on the constant sequence is an interior row at level k. The normalized factorization of the actual average gives

$$e^{ia_k[A](c)}\mathcal{M}^k U_0(c) = \mathcal{M}^k U'(c).$$

For a unitary matrix on the intrinsic branch, $|\log V| \le (\pi/2)|V - \mathbf{1}|$, by the scalar spectral formula; hence $\sup|a_k[A]| \le \pi\gamma/2$. Gauge invariance preserves regularity of $e^{i\eta A}U_0 = U'^g$. The slice estimate now applies with row bound $\pi\gamma/2$, proving the two final bounds after increasing C_{sl}. For the $SU(2)$ constants of Lem. YM3_I:30.10, one may take $C_{\mathrm{sl}} = 20\pi K_1$. This proof uses only the stated local bounds and group identities, so the same calculation applies whenever those bounds have been supplied in another sector. □

Proposition 9.33 (Quotients)**.** Let $G = \tilde{G}/\mathsf{Z}$ with $\tilde{G}$ simply connected and Z finite and central, and $L \ge 13$ odd. For $g = e^{iY} \in G^\pi$, $Y \in \mathfrak{g}$, $|Y| < \pi$, the *canonical lift* $s(g) \in \tilde{G}$ is $e^{i\tilde{Y}}$, $\tilde{Y}$ the element of the Lie algebra of $\tilde{G}$ that the covering carries to Y; for a lift $\tilde{U}$ to $\tilde{G}$ of a field U whose plaquette variables lie in G^π, the *central part* of $\tilde{U}$ at a plaquette p is $z_{\tilde{U}}(p) = \partial\tilde{U}(p)\, s(\partial U(p))^{-1} \in \mathsf{Z}$.

(i) The sector of Lem. 9.11(iv) is defined, gauge invariant and locally constant on the fields with small plaquettes, and kept by blocking. Let a separated sequence of regions have the whole torus at the bottom. Write the finite sequence as $\Omega_0, \dots, \Omega_k$, with $\Omega_{-1} = \Omega_0$ and $\Omega_{k+1} = \varnothing$. In bare units its members Ω_q are unions of cubes of side $\mathfrak{M}L^q$, with $\mathfrak{M}$ and R_1 powers of L, $R_1 \ge L$, $R_1\mathfrak{M} \ge L^2$, and $\mathrm{dist}_\infty(\Omega_q^c, \Omega_{q+1}) > R_1\mathfrak{M}L^q$. Let U be a field with $|\partial U(p) - \mathbf{1}| < \varepsilon L^{-2\mathsf{j}(p)}$ at every bare plaquette, $\mathsf{j}(p)$ the level of the sequence at p, where $L^2\varepsilon \le c(G)$ of Lem. 9.11(iii), and let V be its block datum. For a coordinate plane P through a site of the top lattice, the *plane number* of U at P is the product of the central parts $z_{\tilde{U}}(p)$ over the bare plaquettes p of P, for a lift $\tilde{U}$ of U; the *cells* S of P are defined as follows. Write $Q_r(p)$ for the square of side L^r in P containing the bare plaquette p, with corners on the level-r lattice. Its cell is $Q_q(p)$ for the largest $q \le k$ at which a corner lies in Ω_q. These cells tile P. A side of a level-q cell having an end in Ω_q is one determining bond of level q; a side with both ends outside Ω_q consists of L determining bonds of level $q-1$. Call these bonds the elements of the cell. Thus each cell has at most $4L$ elements, and $\partial_S V$ is their ordered product, with the signs of a walk around the boundary of S. Then every $\partial_S V$ lies in G^π; for any lifts $\hat{V}$ of the values of V, $z_{\hat{V}}(S) := \partial_S\hat{V}\, s(\partial_S V)^{-1}$ lies in Z; and the plane number of U at P is $\prod_S z_{\hat{V}}(S)$ over the cells S of P. So the plane numbers depend on V alone, and two such fields over one datum lie in one sector; over $V = e^{iB}$ with $4L\sup|B| < \pi$ the sector is trivial.

(ii) Let $G \subset U(N)$ be closed and let the averaging map agree with the formula of Lem. 9.11(i) on small fields. Let the sequence have the whole torus at the bottom and satisfy the following stronger separation conditions in bare units: Ω_q is a union of cubes of side L^{q+1}, and $\mathrm{dist}_\infty(\Omega_q^c, \Omega_{q+1}) > L^{q+2}$. Empty and full members are allowed. Then a field U with $|\partial U(p) - \mathbf{1}| < \varepsilon L^{-2\mathsf{j}(p)}$ at every bare plaquette, $\mathsf{j}(p)$ the level of the sequence at p, and block datum $\mathbf{1}$, $\varepsilon \le \varepsilon_2 = \min\{\varepsilon_2(L), \frac{1}{106} r_G\}$, has a gauge transform, by a gauge function equal to $\mathbf{1}$ at the tops and within $C\varepsilon$ of $\mathbf{1}$ at the nodes, of the form e^{ia} with $|a(b)| \le C\varepsilon L^{-\mathsf{j}(b)}$ and $\sum_b L^{-2\mathsf{j}(b)} |a(b)|^2 \le C \sum_p |\partial U(p) - \mathbf{1}|^2$, C a number of L; and a flat field with block datum $\mathbf{1}$ is a pure gauge. So these statements on the whole torus hold as worded; their hypothesis, block datum $\mathbf{1}$, selects the trivial sector.

(iii) Suppose that all regions are the whole torus. For every closed $G \subset U(N)$, the preliminary gauge, the change of level (Lem. 9.32(i)) and the section of a datum (Lem. YM3_I:36.3) hold in every sector. More precisely, two configurations in $\mathcal{U}'(\alpha_0)$ whose terminal data differ by at most α_1 admit a gauge equal to $\mathbf{1}$ at the top sites that identifies their averaged contour variables and obeys

$$|\mathcal{M}^n(U^{\hat g})(b) \mathcal{M}^n U_0(b)^{-1} - \mathbf{1}| \le C_{\mathrm{pg}}(\alpha_0 + \alpha_1)$$

at every level. Here C_{pg} depends only on L, and the ceiling on α_0 is $c(G)$ of Lem. 9.11(iii).

(iv) Let $G = SO(3) \subset U(3)$, all regions being the whole torus, and let U_0 be a background of the regular class $\mathfrak{U}'_k(\mathbf{B}, \alpha_0)$ of the $SU(2)$ construction, in any sector. There are two numbers of L, a ceiling on α_0 and a floor $6LM$ on the number of sites of the top lattice along each axis, $M \ge M_{\mathrm{ps}}$ a power of L, in the place of the floor $6LM_{\mathrm{ps}}$ of the $SU(2)$ construction (Thm. YM3_I:I), such that under them the operator $\Delta^{\mathrm{W}}_{U_0} + D\mathsf{R}D^* + \mathsf{Q}^*\mathsf{Q}$ of the $SU(2)$ construction is formed on the whole torus and is positive definite, and its inverse G_0, the operator $\mathsf{H}_0 = \mathsf{G}_0\mathsf{Q}^*(\mathsf{Q}\mathsf{G}_0\mathsf{Q}^*)^{-1}$ and $\mathsf{G}_0 - \mathsf{H}_0\mathsf{Q}\mathsf{G}_0$ obey the bounds of the $SU(2)$ construction with numbers of L in the places of its constants.

The clauses of the $SU(2)$ construction that read the background on pieces of bounded size hold for $SO(3)$ in every sector. A bound of such a clause that is of degree d in sizes of forms and of degree e in chords of values in the group, chords at most α, $\alpha \in (0, 1]$ the parameter of the regular class $\mathfrak{U}'_k(\mathbf{B}, \alpha)$ of the background, keeps its number C of the $SU(2)$ construction where $d + e \ge 2$, where $d = 1$ and $e = 0$ at real forms, and, for a linear map whose kernel is real, in bounds on the norms of its kernel and on their sums along rows, and takes $(1 + 0.04\alpha)C$ where $d = 0$ and $e = 1$, as the constant of the gauge on a cube in the improvement of the regularity radius does, at most $\sqrt{2}C$ in the other bounds with $d = 1$ and $e = 0$ at complex forms, and $2C$ where $d = e = 0$, while a bound on the norm from suprema to suprema of an operator on complex forms takes a factor at most $\sqrt{2}$.

The constant $C_{4,\mathrm{ec}} = 808$ of the $SU(2)$ normalization in the bounds of the action of Prop. YM3_I:30.1 gives way to an absolute number, which serves at real and at complex forms and at every scale of that clause, a scale being a number $\xi \ge \eta$ at which the plaquette variables at the bond meet the condition of the clause. The comparison of the two regular classes at complex fields holds with $r + 5r^2$ in the place of $r + 4r^2$ at the fields $e^{i\eta A} U_0$ with $\eta \sup |A| \le \frac{1}{8}$.

For an averaging map that is covariant at every field and the formula of Lem. 9.11(i) only below a smaller radius, the same holds with three ceilings, each the least of the number of the $SU(2)$ construction and a fixed multiple of that radius ϖ: on α_0, the ceiling $L^{-2} c_2^g$ of the global chart with $\frac{1}{4} L^{-2}\varpi$; on the size $|A|_{(-1)}$ of the potential A of a chart pair (A, g) of the global chart, the ceiling $L^{-1}c_4$, c_4 the number of the local chart estimates, with $\frac{1}{32} L^{-1}\varpi$; and on the radius of the one-step chart maps, the maps $a \mapsto \xi_\star(W_0, a)$, $\frac{1}{i}\log \mathsf{v}_\star(W_0, a)$ and $\mathcal{Q}^1_\star(W_0, a)$ of the one-step chart, the chart radius r_L with $\frac{1}{8}\varpi$. The backgrounds of

these maps then lie within $\frac{1}{2}\varpi$ of $\mathbf{1}$, and the fields of a step within ϖ, in the way in which Lem. 9.11(v) keeps them within $\min\{\omega_M, r_G\}$.

With these ceilings, the gauge correction (Lem. YM3_I:30.4) holds in every sector with constants K_3 and C_{gc} that are numbers of L; and one step (Lem. 9.32(ii)) and one step of the ascent (Prop. YM3_I:37.2) hold in every sector, with constants of L in the places of the numerals of the $SU(2)$ construction, the improvement of the regularity radius (Prop. YM3_I:37.1) holding with free constants. In particular, over every datum whose plaquette variables lie within ε' of $\mathbf{1}$ and which lies within ε' of the averages of a configuration of the regular class one level down, ε' below a number of L, there is a critical configuration of the regular class one level up. The sector does enter objects on the whole torus: at a flat $SO(3)$-field the covariant difference on $\mathfrak{g}$-valued site functions has a non-zero kernel in the trivial sector and none in a twisted one.

■

Sketch of proof. (i) The squares $Q_r(p)$ are nested, so the largest-level rule assigns the same cell to every bare plaquette of that cell. A level-q cell has a corner in Ω_q and no corner in Ω_{q+1}. The block structure then puts no level-q site of the cell in Ω_{q+1}. Separation places the cell and the double blocks of its boundary elements in Ω_{q-1}: their distance from that corner is at most $\frac{3}{2}L^q < L^{q+1}$. On a side with no end in Ω_q, the level-$(q-1)$ sites are outside Ω_q, so the L subdivisions are indeed determining bonds.

An element of level r has an end in Ω_r and neither end in Ω_{r+1}. The cell across it has level at least r and at most $r+1$: a cell of level $q' \geq r+2$ would put an endpoint within $L^{q'}$ of $\Omega_{q'}$, hence in $\Omega_{q'-1} \subset \Omega_{r+1}$, a contradiction. At level r the element is a whole side; at level $r+1$ it is one of the L subdivisions of a side. Consequently each element belongs to exactly two cell boundaries, with opposite orientations.

If $\sum_j |X_j| < \pi$ in the norm of $\mathbb{C}^N$, $X_j \in \mathfrak{g}$, then $\prod_j e^{i\tilde{X}_j}$ is the canonical lift of $\prod_j e^{iX_j}$, and a product of small exponentials that lies in Z is $\mathbf{1}$. By Lem. 9.11(iii) in its local form, under $L^2\varepsilon \leq c(G)$, the value of the datum at a bond is the transport of U along the straight bare path of the bond times a product of exponentials of elements of $\mathfrak{g}$ of small total size; the paths of the sides of a cell S compose its boundary, so that $\partial_S V$ is the transport of U around S, a product of conjugated plaquette variables of U in S, times such a product, of total size below π. So $\partial_S V \in G^\pi$, and, the values of V being lifted along the lifted paths, $\partial_S \hat{V} = \prod_{p \subset S} z_{\tilde{U}}(p) \cdot s(\partial_S V)$. Another choice of the lifts of the values of V changes the $z_{\hat{V}}(S)$ by central factors that cancel in the product over the cells of P, each bond of the datum in P bounding two cells with opposite orientations; and the product over the cells of the central parts of their plaquettes is the plane number. The plane numbers, with the cube numbers, which are $\mathbf{1}$ for small plaquettes, fix the sector. Over $V = e^{iB}$ the product around a cell has at most $4L$ factors $e^{\pm iB(e)}$, and their lifts $e^{\pm i\tilde{B}(e)}$ give $z_{\hat{V}}(S) = \mathbf{1}$.

(ii) We construct the gauge by descending through the layers. The interpolation used in the descent is needed both for the pointwise bound and for the energy estimate; setting every contour variable equal to the identity would not give the latter estimate.

Write n for the last level, put $\Omega_{-1} = T_\ell$ and $\Omega_{n+1} = \emptyset$, and set

$$j(x) = \max\{q : x \in \Omega_q\}, \qquad j(e) = \max_{x \text{ a corner of } e} j(x).$$

Here e may be a bond or a plaquette of any level. Let $\Lambda_q = \mathbb{T}^{(q)} \cap (\Omega_q \setminus \Omega_{q+1})$; these are the tops. The datum prescribes the averages on the level-q bonds with an endpoint in Λ_q, with any of the inward bonds omitted as permitted by the given bond set. In particular it prescribes all such bonds with no endpoint in Ω_{q+1}. A node is an endpoint of a retained bond. All bond and plaquette sums below use positive orientations.

We first record the geometric consequences of the stated separation. The blocks $B^q(y)$, $y \in \Lambda_q$, partition the bare sites, and $j = q$ throughout each such block. Indeed, membership in Ω_q, a union

of cubes of side L^{q+1}, is constant on each level-q block; assign a site to the block of the largest level whose region contains it. Furthermore,

$$|x - x'|_\infty \le L^{j(x)+1} \quad \Longrightarrow \quad j(x') \ge j(x) - 1. \tag{9.xiii}$$

For $j(x) \ge 1$ this follows from the distance between $\Omega_{j(x)}$ and $\Omega^c_{j(x)-1}$; for $j(x) = 0$ it is automatic. A node which is not a top is the inner endpoint of a retained inward bond whose other endpoint is a top.

For level i, let $\mathcal{B}_i$ and $\mathcal{P}_i$ be the bonds and plaquettes with all corners in Ω_{i-1}, and put

$$V_i = \mathcal{M}^i U|_{\mathcal{B}_i}, \qquad u_i(p) = |\partial V_i(p) - \mathbf{1}|.$$

The bare cone $Y_i(p)$ of a plaquette is the union of the level-i blocks of its four corners. The double block beneath a bond of $\mathcal{B}_{i+1}$ lies in Ω_i, so each averaging step below is local to the indicated sets.

The two estimates for an averaging step. We use constants depending only on L. If v is the largest plaquette distance on the four blocks of a coarse plaquette P, then, for sufficiently small v,

$$u_{i+1}(P) \le (\mathsf{W}_i u_i)(P) + C_0 v^2, \qquad (\mathsf{W}_i f)(P) = L^{-3} \sum_{x \in B(P_-)} \sum_{p \in S_x(P)} f(p). \tag{9.xiv}$$

Here P_- is the base point of P, and $S_x(P)$ is the $L \times L$ square of fine plaquettes parallel to P with base point x. The kernel is nonnegative and has row sum L^2 and column sum L^{-1}; consequently $\|\mathsf{W}_i\|_{2\to 2} \le L^{1/2}$.

For completeness, the matrix calculation behind this estimate is the one-step calculation of Lem. YM3_I:14.9. Put the field in the comb gauge on the four blocks. Each resulting bond is a product of at most C_L conjugated plaquette variables, so its logarithm has size at most $C_L v$. Differentiating the mean formula at the identity gives

$$(Qa)(c) = L^{-3} \sum_{x \in B(c_-)} [\bar a(c_-, x) + a([x, x']) - \bar a(c_+, x')], \qquad x' = x + c_+ - c_-,$$

where $\bar a(y, x)$ is the arithmetic mean of the line sums along the coordinate-order contours. On going around P, the contour terms at its four corners cancel. The remaining sum is $L^{-3} \sum_x \sum_{p \in S_x(P)} da(p)$. The product-exponential estimate and the quadratic Taylor remainder of the mean give $C_L v^2$ for the discarded terms. This proves eq. 9.xiv. All estimates are matrix norm estimates. Each logarithm is intrinsic to G: the comb products and the contour quotients are products of small exponentials in $i\mathfrak{g}$, and Lems. 9.9(ii), (iv) and 9.11(i) apply. Thus the argument uses no $SU(2)$ identity. Counting L^3 translated squares, each with L^2 plaquettes, gives the row sum; a fixed fine plaquette has total multiplicity L^2 in these squares as the coarse plaquette varies, giving the column sum L^{-1}.

These facts imply the two multiscale bounds

$$u_i(p) \le 2\varepsilon L^{2(i-k)} \qquad \text{if } Y_i(p) \subset \Omega_k, \quad k \ge \max\{i-1, 0\}, \tag{9.xv}$$

$$\sum_{p \in \mathcal{P}} u_i(p)^2 \le 2L^i \sum_{p_0 \subset \bigcup_{p \in \mathcal{P}} Y_i(p)} u_0(p_0)^2 \qquad (\mathcal{P} \subset \mathcal{P}_i). \tag{9.xvi}$$

The bare plaquettes on the right are counted once. To see the first bound, take maxima P_r on the nested cones beneath p. Equation eq. 9.xiv gives $P_{r+1} \le L^2 P_r + C_0 P_r^2$. Write $P_r = \varepsilon L^{2(r-k)} p_r$. Starting with $p_0 \le 1$, the induction $p_r \le 2$ closes because

$$p_r \le \exp\left(2C_0 \varepsilon L^{-2} \sum_{t<r} L^{2(t-k)}\right) < 2$$

when ε is below a number of L. For the second bound, restrict at each level to the cones of $\mathcal{P}$. A fine plaquette belongs to the four-block set of at most twelve coarse plaquettes. Hence the square root of the plaquette energy increases at one step by at most

$$L^{1/2}(1 + 4C_0 L^{-1/2} v_r), \qquad v_r \le 2\varepsilon L^{2(r-i+1)} \quad (r < i).$$

Since $\sum_{r<i} v_r < 4\varepsilon$, the product of the parentheses is at most $\sqrt{2}$, after decreasing the same ceiling on ε. This proves eq. 9.xvi.

Interpolation and its error. Measure a fine spacing as one, so the coarse lattice is $L\mathbb{Z}^3$. For $b = \langle x, x + e_\mu\rangle$, write $x = LY + r$, $0 \le r_\nu < L$, and put

$$(I\alpha)(b) = L^{-1} \sum_{s_\mu=0,\ s_\nu\in\{0,1\}} \prod_{\nu\neq\mu} (1 - r_\nu/L)^{1-s_\nu} (r_\nu/L)^{s_\nu} \alpha\big(\langle L(Y+s), L(Y+s) + Le_\mu\rangle\big).$$

This is periodic and independent of the chosen coarse origin. Its weights are nonnegative and sum to one. In particular

$$|I\alpha| \le I|\alpha| \le L^{-1} \sup|\alpha|, \qquad \|I\alpha\|_2 \le L^{1/2}\|\alpha\|_2.$$

For the second inequality apply convexity of the square: the total weight of the fine bonds reading any fixed coarse bond is L^3, which cancels two powers of the prefactor L^{-1}. The line sum of $I\alpha$ along a straight coarse bond is $\alpha(c)$. Directly taking transverse differences also gives $dI\alpha = L^{-2}$ times a convex combination of the two adjacent parallel values of $d\alpha$.

Suppose $s = \sup|\alpha| \le r_*/2$ on the coarse bonds within one coarse spacing of a double block, and v bounds the plaquette distances of $e^{i\alpha}$ there. Choose $r_* = r_*(L) > 0$ small enough that the logarithm of the mean at the identity is holomorphic and bounded by one on the complex ball of radius r_*, and $r_* \le 1/(200L)$. Such a choice follows from the last, trivial-background estimate of Lem. 9.11(v). For $\beta = I\alpha$, the preceding identities and the elementary product estimates give

$$\begin{aligned} \sup|\beta| &\le L^{-1}s, \\ \varpi_c(e^{i\beta}) &\le L^{-2}(v + 51s^2), \\ |\mathcal{M}(e^{i\beta})(c)e^{-i\alpha(c)} - \mathbf{1}| &\le v + (35 + 2r_*^{-2})s^2. \end{aligned} \tag{9.xvii}$$

Here is the error calculation. On a coarse plaquette, $|d\alpha| \le v + 35s^2$ by expanding its four exponentials. The curl identity for I and the same expansion on a fine plaquette add at most $16L^{-2}s^2$, giving the second line. The logarithm of the mean differs from $Q\beta$ by at most $2r_*^{-2}s^2$, by Cauchy's estimate. Moving each of the L straight steps past a coordinate-order contour uses at most $L(L-1)$ plaquettes. Thus $|Q\beta(c) - \alpha(c)| \le v + 35s^2$, proving the last line.

We also use the following rigidity estimate on a double block. If W, W_0 have identical averaged contour variables on its two blocks, then

$$|W(b)W_0(b)^{-1} - \mathbf{1}| \le |\mathcal{M}(W)(c)\mathcal{M}(W_0)(c)^{-1} - \mathbf{1}| + C_L \max\{\varpi_c(W), \varpi_c(W_0)\}.$$

Inside either block, choose a coordinate-order path whose last step is the bond. The two quotients of its transports by the endpoint contour means have size at most $C_L\varpi$, by the path comparison in Lem. 9.11(i); applying this to both fields gives the bound. For a bond between the blocks, insert the two endpoint contour means; move its straight crossing to the central coarse path and then replace that path by its mean. Each replacement costs C_L times the largest plaquette distance, and the endpoint means of the two fields cancel. This is precisely the double-block reconstruction calculation of Lem. YM3_I:44.3. It proves the displayed estimate with a constant of L for the present G as well. In addition, the quotients in this calculation are products of conjugated small plaquette exponentials and the mean corrections. Their intrinsic logarithms therefore lie in $\mathfrak{g}$ when the total exponent size is less than π, by Lem. 9.9(ii). For a closed group which is not connected these products lie in its identity component; the unrestricted transports themselves need not do so.

The descending gauge. Set $g = \mathbf{1}$ at the tops. The other bare sites form the disjoint sets

$$N_i = (\mathbb{T}^{(i)} \setminus \mathbb{T}^{(i+1)}) \cap \Omega_{i+1}, \qquad i = n-1, \dots, 0.$$

When g is known on $\mathbb{T}^{(i+1)}$, set

$$W_{i+1} = V_{i+1}^{\cdot g}, \quad a_{i+1} = \frac{1}{\mathrm{i}} \log W_{i+1}, \quad \beta_i = Ia_{i+1}, \quad W_{0,i} = e^{i\beta_i},$$

and, for $x \in B(y) \setminus \{y\}$, $y \in \mathbb{T}^{(i+1)} \cap \Omega_{i+1}$, define

$$g(x) = W_{0,i}(y,x)^{-1} g(y) V_i(y,x).$$

The expressions $W(y,x)$ are averaged contour variables. This choice makes $W_i = V_i^g$ and $W_{0,i}$ have the same contour variables on each such block; gauge covariance gives $\mathcal{M}(W_i) = W_{i+1}$.

A bond of $\mathcal{B}_i$ with no endpoint in Ω_{i+1} has both endpoints at tops, and $W_i = \mathbf{1}$ there. If one endpoint lies in Ω_i, this follows from the trivial datum. Otherwise its double block consists of bonds of the level-$i-1$ datum, all equal to $\mathbf{1}$, so its average is also $\mathbf{1}$. In particular $W_n = \mathbf{1}$ and $a_n = 0$. This observation also treats the exterior block of a double block crossing the boundary of Ω_{i+1}: both its contour variables and the interpolated reference are trivial.

Here are the estimates that justify every step of the construction. For a bond b of level i with an endpoint in Ω_{i+1}, choose a coarse double block containing it, with one centre $y \in \Omega_{i+1}$, and put $k = j(y) - 1 \geq i$. Enlarge the bare double block by $2L^{i+1}$. Equation eq. 9.xiii shows that $k \leq j \leq k+2$ throughout this enlargement. Let $v_i(b)$ be the maximum of u_i on the double block and of u_{i+1} within one coarse spacing of it, and let $s_{i+1}(b)$ be the maximum of $|a_{i+1}|$ on the coarse bonds in that neighbourhood. Inductively assume

$$|a_q(c)| \leq C_A \varepsilon L^{q-j(c)} \quad (q > i).$$

Equation eq. 9.xv gives

$$v_i(b) \leq 2\varepsilon L^{2(i+1-k)}, \qquad s_{i+1}(b) \leq C_A \varepsilon L^{i+1-k}, \qquad s_{i+1}(b)^2 \leq \varepsilon L^{2(i+1-k)}$$

if $C_A^2 \varepsilon \leq 1$. Hence the reference plaquettes are at distance at most 53ε from $\mathbf{1}$, by eq. 9.xvii. Rigidity and that equation give

$$W_i(b) = e^{iz_i(b)} e^{i\beta_i(b)}, \qquad |z_i(b)| \leq C_z(v_i(b) + s_{i+1}(b)^2). \tag{9.xviii}$$

The constant $C_z = C_z(L) \geq 1$ may be enlarged to control the total size of the small exponentials just described. Thus the logarithm defining z_i is in $\mathfrak{g}$, independently of the size of r_G. On bonds with no endpoint in Ω_{i+1} put $z_i = v_i = s_{i+1} = 0$.

Choose

$$N_2 = 3(8L)^3, \qquad C_A \geq \max\{4C_z L^8, 375 C_z^2 N_2 L\},$$

and choose $\varepsilon_2(L) > 0$ below the ceilings used in eq. 9.xv–eq. 9.xvi and such that

$$C_A^2 \varepsilon_2(L) \leq 1, \qquad C_A L \varepsilon_2(L) \leq r_*/2, \qquad 53\varepsilon_2(L) \leq \omega_M/2, \qquad 2C_z C_A N_2^{1/2} L^{3/2} \varepsilon_2(L) \leq \log 2.$$

We may decrease it further by constants depending only on L to keep the contour means in their stated domains. The additional ceiling $\varepsilon \leq r_G/106$ gives $53\varepsilon \leq r_G/2$. In particular every interpolated reference has the small plaquettes required by Lem. 9.11(i). The actual intermediate fields do as well: $C_A \geq 4L^8$ implies $\varepsilon_2(L) \leq L^{-16}/16 < 3.9 \cdot 10^{-9} L^{-2}$, so $\varepsilon \leq c(G)$. The local version of Lem. 9.11(iii) on each required double block gives intermediate plaquette distances at most 2ε. Thus each chosen averaging map agrees with the small-field formula throughout this construction.

The small exponential product in eq. 9.xviii and Lem. 9.9(ii) give $|a_i| \leq I|a_{i+1}| + |z_i|$. Iterating, using positivity of I and $a_n = 0$, gives

$$|a_i| \leq \sum_{r \geq i} I^{r-i} |z_r|.$$

For a fixed bond b of level i, put $J = j(b)$. A nonzero term at level r can influence b only if $r \leq J$; its endpoints lie within $2L^r$ of those of b, and its layer index is at least $J-1$, by eq. 9.xiii. The local bound above implies $|z_r(b')| \leq 3C_z L^6 \varepsilon L^{2r-2j(b')}$. Since each interpolation costs L^{-1} in supremum norm,

$$|a_i(b)| \leq 3C_z L^8 \varepsilon L^{i-2J} \sum_{r=i}^{J} L^r \leq 4C_z L^8 \varepsilon L^{i-J} \leq C_A \varepsilon L^{i-J}.$$

This closes the induction and justifies all logarithms in it. At $i = 0$ it proves the claimed pointwise bound. At a node which is not a top, the retained inward bond of level q joining it to a top has $V_q = \mathbf{1}$. Its transformed value is $g(z)$ or $g(z)^{-1}$, and the same bound gives $|g(z) - \mathbf{1}| \le C_A\varepsilon$.

The energy estimate. Keep the term quadratic in the coarse potential in eq. 9.xviii. With all functions extended by zero off their indicated bond sets, it gives

$$|a_r| \le T_r|a_{r+1}| + \zeta_r, \qquad \zeta_r = C_z v_r, \qquad T_r = I + B_r,$$

where, on an active bond b,

$$(B_r f)(b) = C_z C_A \varepsilon L^{r+3-j(b)} \max_{\mathcal{N}_b} f.$$

The set $\mathcal{N}_b$ is the coarse neighbourhood used in the definition of $s_{r+1}(b)$; on an inactive bond $B_r f = 0$. The maps T_r are monotone and subadditive on nonnegative functions. Split ζ_r according to its layer: $\zeta_r^J = \mathbf{1}_{\{j=J\}}\zeta_r$, which is zero unless $r < J$, and put

$$f^J = \sum_{r<J} T_0 \cdots T_{r-1}\zeta_r^J. \qquad \text{Then} \qquad |a| \le \sum_J f^J.$$

Each step spreads support by at most $3L^{r+1}$. Therefore all the descendants of ζ_r^J are within $4L^{J-1}$ of a site of layer J, and their layer indices lie in $[J-1, J+1]$, by eq. 9.xiii.

A coarse bond belongs to $\mathcal{N}_b$ for at most N_2 fine bonds: their base points lie in a cube of side $6L^{r+1}$, with three bond directions at each site. Thus, on such a layer-supported descendant,

$$\|T_t f\|_2 \le (L^{1/2} + C_z C_A N_2^{1/2} \varepsilon L^{t+4-J})\|f\|_2.$$

Since $t < r < J$, the sum of $L^{t+7/2-J}$ over $t < r$ is less than $2L^{3/2}$. The chosen smallness ceiling therefore gives

$$\|T_0 \cdots T_{r-1}\zeta_r^J\|_2 \le 2L^{r/2}\|\zeta_r^J\|_2.$$

Let

$$E^J = \sum_{p:\, |j(p)-J|\le 2} u_0(p)^2.$$

Every plaquette entering $v_r(b)$ for $j(b) = J$ has its bare cone in these layers. The same multiplicity bound N_2, now for plaquettes, and eq. 9.xvi give

$$\|\zeta_r^J\|_2^2 \le 4C_z^2 N_2 L^{r+1} E^J, \qquad \sum_J E^J \le 5\sum_p u_0(p)^2.$$

Consequently

$$\|f^J\|_2 \le 4C_z N_2^{1/2} L^{1/2} (E^J)^{1/2} \sum_{r<J} L^r \le 5C_z N_2^{1/2} L^{J-1/2} (E^J)^{1/2}.$$

At a bare bond at most three f^J are nonzero, those with $|J - j(b)| \le 1$. Cauchy–Schwarz now yields

$$\begin{aligned}\sum_b L^{-2j(b)}|a(b)|^2 &\le 3\sum_J L^{-2(J-1)}\|f^J\|_2^2 \\ &\le 75 C_z^2 N_2 L \sum_J E^J \le 375 C_z^2 N_2 L \sum_p |\partial U(p) - \mathbf{1}|^2.\end{aligned}$$

This is the asserted energy bound, with $C = C_A$.

Finally suppose the field is flat. In each pillar let $t(x)$ be transport from its top to x. Flatness makes this independent of the path inside the pillar. On a determining bond the straight transport equals its block average, hence is $\mathbf{1}$. For a bare bond within one pillar, its transport and the two pillar paths form a contractible loop. For a bond joining pillars of equal level, close the loop by their determining bond. For adjacent pillars of levels q and $q+1$, first pass through the level-$(q+1)$ double block, using its outer determining bond and the level-q determining bonds in the outer block. All closing transports are $\mathbf{1}$; the separation and block nesting place the loop inside that double

block. These three cases give $t(b_-)U(b)t(b_+)^{-1} = \mathbf{1}$ on every bare bond. The same closing path at a retained inward bond gives $t = \mathbf{1}$ at its inner node, as at every top. Thus $U = \mathbf{1}^{t^{-1}}$, with the residual gauge equal to $\mathbf{1}$ at all nodes. In the quotient case the constructed potential also satisfies $4\sup|a| \le 4C_A\varepsilon \le 4/C_A < \pi$; lifting its small exponentials therefore has trivial central plaquette parts. No sector condition was assumed in the construction; the trivial datum enforces it.

(iii) We give the preliminary gauge argument, including its estimate. Write $W_n = \mathcal{M}^n U$ and $W_{0,n} = \mathcal{M}^n U_0$. The local straightening estimate of Lem. 9.11(iii) keeps every mean used below in its small-field domain. Its contour means lie in G and are gauge covariant, by Lems. 9.9(iv) and 9.11(i). Set $g_k = \mathbf{1}$ on the top lattice and descend through the blocks. If $x \in B(y)$ at level n, set

$$g_n(x) = W_{0,n}(y,x)^{-1} g_{n+1}(y) W_n(y,x),$$

where $W_n(y,x)$ is the averaged contour variable from y to x. It equals $\mathbf{1}$ at $x = y$, so g_n extends g_{n+1}. Thus $\hat{g} = g_0$ is a gauge, equal to $\mathbf{1}$ at the top sites. Gauge covariance gives

$$(\mathcal{M}^n U^{\hat{g}})(y,x) = W_{0,n}(y,x).$$

These identities also determine $\hat{g}$ uniquely, by downward induction.

For the quantitative assertion, put

$$\delta_n(b) = |\mathcal{M}^n(U^{\hat{g}})(b) W_{0,n}(b)^{-1} - \mathbf{1}|.$$

At the terminal level $\delta_k \le \alpha_1$. Within a single block, the equality of the contour variables and the contour straightening bound give $\delta_n \le C_L\alpha_0$. Across two adjacent blocks, with coarse bond c, the same comparison gives

$$\delta_n(b) \le \delta_{n+1}(c) + C_L\alpha_0 L^{2(n-k)}.$$

Indeed, a path product changes by at most the sum of the changes of its factors, since the other factors are unitary. Replacing the paths by their means costs at most a constant of L times the largest plaquette discrepancy in the two blocks; the two end-point contour discrepancies vanish by the identities above. The straightening estimate bounds this plaquette discrepancy by $2\alpha_0 L^{2(n-k)}$. Iterating until the two end points enter one block or reach the top and summing $\sum_{j\ge1} L^{-2j} = (L^2-1)^{-1}$ proves the asserted bound. This argument uses only multiplication, inversion, unitarity and the mean estimates, so applies to every closed G. Change of level is a rescaling; the section of a datum is constructed from its group values by the same contour identities. Neither uses a lift or a choice of sector.

(iv) *What the operators read.* The operators of the $SU(2)$ construction read the background through $\mathrm{Ad}(U_0(b))$ at the bonds and through the plaquette functions, on pieces of bounded size. Unrolled to the covering lattice the field is periodic; on a box, which is simply connected, every sector is trivial, and the field has a lift $\tilde{U}$ to $SU(2)$ with $\partial\tilde{U}(p) = s(\partial U_0(p))$, which lies in the regular class of $SU(2)$ at the parameter $\frac{\alpha}{2}(1+0.04\alpha)$, $\alpha \le 1$ the parameter of U_0: at the plaquettes by the ratio $\frac{1}{2\cos(\theta/4)}$ of the chords of $s(g)$ and of g, θ the angle of g, which is at most $\frac{1}{2}(1+0.04\alpha)$ for chords at most α; and at the bonds, as the scalar part of the divergence of $\partial\tilde{U}$ at a bond b is at most $0.3\alpha^2\eta^3$ and the norm of its part $i\eta^2 J(\tilde{U})(b)$ at most half that of the divergence of ∂U_0 at b plus $0.3\alpha^3\eta^5$, which keeps the divergence of $\partial\tilde{U}$ below $\frac{\alpha}{2}(1+0.002\alpha^2)\eta^2$ at $\eta \le \frac{1}{13}$. At the lift the operators of $SO(3)$ are those of $SU(2)$ conjugated by the differential of the covering, which doubles the operator norm on $\mathfrak{su}(2)$, the pairings changing by the one factor $\frac{8}{3}$; and the action of the pair is $a_3 = a^\flat(1 - \frac{3}{16}a^\flat)$, with $a^\flat = \frac{8}{3}\, a_2 \circ s$ and a_2 the action of $SU(2)$.

The operator on the whole torus. The boxes are taken as the first members of separated sequences of larger tori, at growing radii; there the propagator statements of the $SU(2)$ construction for $SU(2)$ (Thm. YM3_I:I and its local inverse construction), read at the lift, give the operators with the Hessian $\Delta^{\mathrm{W},\flat}$ of $\sum_p a^\flat$, positive and with inverses that decay exponentially, uniformly in the box.

Their limits are kernels on the covering lattice, invariant under the periods of the torus, and their periodisations are, on the whole torus and in every sector, the operators with $\Delta^{\mathrm{W},\flat}_{U_0}$: positive definite, with inverses that decay exponentially. Since the distance d_T on the torus is the least distance d of two lifts, the sum over the periods keeps with its constant a bound on the sums along rows weighted by $e^{\delta d}$, and it carries a pointwise bound $Be^{-\delta d}$ of a kernel, d in the spacing of the top lattice, to $9Be^{-\delta d_T}$ once $LM\delta \geq 2$, which the floor $6LM$ gives for M large. This periodisation factor must be at least 8 on a torus of even side, where the point antipodal to a site has eight images at one distance from that site.

The Hessian of a_3 differs from $\Delta^{\mathrm{W},\flat}_{U_0}$ by a local symmetric operator E, $|(\mathsf{E}A)(b)| \leq 96\,\alpha_0^2\eta^2 \max|A|$, the maximum over the bonds of the four plaquettes at b, of the relative order $|\partial U_0(p) - \mathbf{1}|^2$; $\mathsf{G}_0 = (1 - \mathsf{G}_0^\flat\mathsf{E})^{-1}\mathsf{G}_0^\flat$, $\mathsf{G}_0^\flat$ the inverse with $\Delta^{\mathrm{W},\flat}_{U_0}$, by a Neumann series in norms weighted by the decay, for α_0 below the ceiling. Positivity holds along $\Delta^{\mathrm{W},\flat}_{U_0} - t\mathsf{E}$, $0 \leq t \leq 1$, on which the operator stays invertible.

The bounds take numbers of L in the places of the constants of the $SU(2)$ construction. With $q < 1$ the norm of $\mathsf{G}_0^\flat\mathsf{E}$ for the sums along rows weighted by the decay, kept below 1 by the ceiling on α_0, a bound B of these sums for $\mathsf{G}_0^\flat$ becomes $B/(1-q)$ for G_0, and its pointwise bound $9Be^{-\delta d_T}$, with the factor 9 of the sum over the periods, becomes $9B(1-q)^{-1}e^{-\delta d_T}$ at the same rate, and the bounds of H_0 and of $\mathsf{G}_0 - \mathsf{H}_0\mathsf{Q}\mathsf{G}_0$, formed from those of G_0, take numbers of L in the same way.

The local clauses. The same lifts, on boxes shorter than half the torus: the clause of the $SU(2)$ construction for $SU(2)$ is read at the lift and carried back by the covering. Its differential doubles $|\cdot|$ on $\mathfrak{su}(2)$ and multiplies it on $\mathfrak{su}(2)^c = \mathfrak{sl}(2,\mathbb{C})$ by a factor between $\sqrt{2}$ and 2, as $|z\cdot\sigma|^2 = |z|^2 + 2|x\times y|$ and $|2z\cdot T| = 2|z|$ for $z = x + iy \in \mathbb{C}^3$, σ the Pauli matrices and T the generators of $\mathfrak{so}(3)$ to which it takes $\frac{1}{2}\sigma$. The chords at most α of values in the group are halved at the lift up to the factor $1 + 0.04\alpha$, as above. The way back multiplies the size of a form by at most 2 and a chord by $2\cos(\theta/4) \leq 2$, keeps the norms of operators on real forms, and keeps on complex forms the norms of a real kernel and their sums along rows, a real 3×3 matrix having one norm on $\mathbb{R}^3$ and on $\mathbb{C}^3$.

Thus a bound $C\prod_i \sigma_i^{d_i}$ of a form or a chord, of degree d in sizes of forms and e in such chords, comes back with $2^{1-d-e}(1+0.04\alpha)^e C$ at real forms and with $2^{1-d/2-e}(1+0.04\alpha)^e C$ at complex ones: with its constant C at $d + e \geq 2$, and at $d = 1$, $e = 0$ for real forms; with $(1 + 0.04\alpha)C$ at $d = 0$, $e = 1$, the gauge on a cube among these, of degree one in α and none in forms, whose bound $Bl\alpha$ comes back as $2Bl\cdot\frac{\alpha}{2}(1+0.04\alpha)$; with $\sqrt{2}C$ at $d = 1$, $e = 0$ for complex forms, and with C there for a linear map whose kernel is real, in bounds on the norms of its kernel or on their sums along rows; and with $2C$ at $d = e = 0$. A bound on the norm from suprema to suprema of an operator on complex forms comes back with a factor at most $\sqrt{2}$, and a clause of $SU(2)$ on the forms below r serves for the complex forms below $\sqrt{2}r$ and the real ones below $2r$. The bounds of the action are taken anew for a_3, whence the absolute number in the place of $C_{4,\mathrm{ec}}$ and the form with $r + 5r^2$. The three ceilings keep the fields of a step below the radius under which the averaging map is the formula.

The three lemmas. The gauge correction, one step and the ascent are the proofs of the $SU(2)$ construction read at such backgrounds with the operator and these clauses, and so are the statements that these proofs cite and those that these cite in turn. Among them the estimate in the slice (Lem. YM3_I:30.10) and the solution of the system (Prop. YM3_I:30.9) are statements on the whole torus, whose proofs read the background only through the estimates on G_0 and H_0 and through clauses on pieces of bounded size, so that they too hold in every sector. The improvement of the regularity radius is the proof of Prop. YM3_I:37.1 read so with all its numerals free.

The constants are fixed in this order: K_3, C_{gc} and the constant of the gauge on a cube, at most 1.04 times its value in the $SU(2)$ construction, as numbers of L; then the numbers that the $SU(2)$ construction forms from them by its formulas, those of Prop. YM3_I:37.1 among them; and the ceilings on α_0 and on ε' last, so that the inequalities by which the improvement and the ascent

bring the radius back into the regular class, hold with the constants as moved, each being a ceiling on a constant chosen after the numbers that it reads; the rate δ of the decay stays that of the $SU(2)$ construction, and the floor $6LM$ is laid after it.

Flat fields. At a flat field the equation $D_U\lambda = 0$ says $\lambda(b_-) = \mathrm{Ad}(U(b))\lambda(b_+)$ at all bonds, and a twist leaves no invariant vector. □

9.1.10. *Proof of Thm. AA.* For the construction clauses, the argument reduces to the tasks (P)–(K). The following steps describe the construction once those estimates are established. The local calculations have the separate scope specified in the statement.

Step 1: the objects of a step. The following definitions of the first part of the $SU(2)$ construction, transferred to G by Def. 9.5, define objects with the properties that the $SU(2)$ construction proves: the logarithm on D^c, the mean, the block average on small fields and on all fields and its iterates (Lems. 9.9, 9.11); the gradient (Lem. 9.12(i)); the regular classes, with the inclusions $\mathfrak{U}'_k(\mathbf{B}, r) \subset \mathfrak{U}_k(\mathbf{B}, \kappa_G r)$ for every G (Remark 9.6) and $\mathfrak{U}_k(\mathbf{B}, r) \subset \mathfrak{U}'_k(\mathbf{B}, r + r^2 + \kappa' r^3)$ for every G and $r \le z_G$ (Def. 9.5(3), by the bound at a bond of Lem. 9.12(ii)), and for $SU(N)$ and $r \le z_G$ the sharper $\mathfrak{U}'_k(\mathbf{B}, r) \subset \mathfrak{U}_k(\mathbf{B}, (1 + \kappa' r^2) r)$ (Lem. 9.12(ii)); the gauges (Lem. 9.12(iii), (iv)); the chart with the indicator of G^π, the Haar density and the smooth cut-off (Lem. 9.14(i), (ii)); the elimination factor of Lem. 9.14(iii), a complex linear map of $\mathfrak{g}^c$ into itself at every V of the domain of the complex average at c, the G^c-valued V on P_c whose contour families have diameter below $\frac{1}{40}$ and whose contour quotients and variables W_x lie in D^c, with its branch of $\mathrm{tr}\log$ and $|\mathsf{m}_c(e^{ia}) - L^{-3}| \le L^{-1} C_S \sup|a|$ for $\mathfrak{g}^c$-valued a with $\sup|a| < \frac{3}{4} r_*$ (Lem. 9.14(iii)); Log det at the radius r_n (Lem. 9.14(iv)); and the domains of analyticity with their structural facts (Prop. 9.22(v)). The propagators, the covariances, the critical configurations and the towers off the trivial field are the objects of (P) and (V) of §9.1.1. Every logarithm is taken of an element of D^c, the averages being those of Lem. 9.11(i) with $W_x \in D$; integrals over $G \setminus G^\pi$, which has inner points wherever $r_G < 2$, so for $SU(N)$ with $N \ge 3$ (Lem. 9.14(i)), carry the indicator of G^π; and the uses of the averaging map off the small fields, of the pivot bond and of the measure of $G \setminus G^\pi$ in the second part are those of (II) in §9.1.1. Among them the pivot bond (Prop. YM3_I:14.11) holds for G in all its clauses with all its constants: the quotients that it reads are products of conjugated plaquette variables within $\omega_M \le z_G \le r_G$ of $\mathbf{1}$, so in G^π, the averaged contour variables have one mean each (Lem. 9.9(iv)), and the equation for the variable at the pivot bond is solved in the chart of G^π by a contraction, the nonlinear part being bounded by Cauchy's estimate.

Step 2: the trivial field; clause (c). By Lem. 9.15 every linear operator of a step at $\mathbf{1}$ is $K \otimes 1_{\mathfrak{g}^c}$, K a real matrix formed from the lattice alone, and every bound of the $SU(2)$ construction there holds for G with its constant, each in the norms to which its kind passes (Lem. 9.15(ii)). The bounds in suprema, by row sums and by non-decreasing functions of real linear forms hold in $|\cdot|$ and in $\|\cdot\|$ alike:

$$|l_0(f)| \le \Phi(|l_1(f)|, \dots, |l_m(f)|) \text{ for all } f : \Lambda \to \mathfrak{g}^c \iff \text{the same for all } f : \Lambda \to \mathbb{R}.$$

The ℓ^2 bounds, the quadratic forms and the lower bounds, the lower bound of the fluctuation operator among them, hold in $\|\cdot\|$, the sign $|X|^2$ of the $SU(2)$ construction being read as $\langle X, X\rangle$ for $X \in \mathfrak{g}$; sums of squares in $|\cdot|$ retain their constants under Schur's test on $|K|$ or convexity. Otherwise the comparison $\|X\| \le |X| \le N^{1/2}\|X\|$ costs $N^{1/2}$ for an unsquared ℓ^2 estimate and N for its squared form. Identification of all small critical configurations with the near-trivial branch requires the orbit input of §9.1.6; the local coefficient calculation in Prop. 9.18(i)–(iii) gives (c).

Step 3: fields off the trivial field. Lem. 9.19 gives the Hessian at $SL(N, \mathbb{C})$-valued fields with the absolute numerals 12, 1536, 3456 of the $SU(2)$ construction, the inverse of a plaquette variable being bounded by $|X^{-1}| \le |\mathrm{Ad}\, X|$, Ad acting on all $N \times N$ matrices (Lem. 9.9(ii)). For another G, that is, for semisimple $\mathfrak{g}$, so that $G \subset SU(N)$, and G^c-valued U, the Hessian of the action of G is the restriction of this form to $\mathfrak{g}^c$-valued forms; its two-sided bound and its bounds in the

normalised trace norm stay, and the kernel and Weitzenböck bounds in $|\cdot|$ carry the norm $|P| \le N^{1/2}$ of the projection of the pair, the condition $|\mathrm{Ad}(U(b))^{\pm 1}| \le 2$ being read on all $N \times N$ matrices. Prop. 9.22(i) gives the inputs of the propagator construction for $SU(N)$, and, for the other simple G, the Hessian among them with the factor $|P|$; (ii) the averages and the critical configuration near the trivial field; (iii) the localisation, assuming (V); and (iv) the bond maps, with $|E_b - \mathbf{1}| < r_G$ at every bond of the cut-off zone under the ceiling $\mathsf{c}_L c_4 \le 2Lr_G$ of Prop. 9.22(iv). This is a positive ceiling on the constant c_4, which is free downward, so that it never obstructs the ordered choice (Prop. 9.26(a)); it stands among the conditions on the constants read for G under which clause (a) holds. With (P) and (V) of §9.1.1 these are the propagators, the covariances, the minimisers over several scales with their towers, the localisation and the bond maps of the $SU(2)$ construction, transferred to G.

Step 4: conjugation. Each symmetric domain Q^s of the $SU(2)$ construction is read as its real-connected part Q^r (Lem. 9.21): no number moves, but at the memberships reached from memberships of restrictions had apart. The analyticity of the maps is had on the open domains cut out by conditions $m(X) < r$ and by $X \in D^c$ at the logarithms of the averages, and the clauses on the conjugation are stated on the real-connected parts; Q^r is the part of Q^s whose slice components hold real points, and it passes to intersections as the inclusion $(\bigcap_i Q_i)^r \subset \bigcap_i Q_i^r$. A map of the induction is a composite of maps built from products, inverses, means, logarithms in D^c, the action and its derivatives, which commute with $p \mapsto p^c$ by Lem. 9.20, of maps holomorphic on open slices and real at real points, and of continuations on convex subsets of a vector space kept by the conjugation (slices without group-valued member); by Lem. 9.21(ii) and Prop. 9.23(i) such a map commutes with $p \mapsto p^c$ on Q_1^r and maps Q_1^r into Q_2^r as soon as it maps Q_1^r into Q_2; that every map of the induction is of this kind is (C) of §9.1.1. A membership that is reached from memberships of restrictions had apart is reached in Q^s, and there Lem. 9.21(iii) carries it into a real-connected part: $\mathcal{Q}(b_0, b_1, c_0)^s \subset \mathcal{Q}(b_0 + 2e^{4\xi b_1} b_1^2, b_1, c_0)^r$, the like for regular pairs and the natural classes, and for the analyticity cores $\mathcal{U}^\circ(b_0, b_1, c_0)^s \subset \mathcal{U}^\circ(B_0, B_1, c_0)^r$ under the four conditions of Lem. 9.21(iii), $b_0' = b_0 + 2e^{4\xi b_1} b_1^2 \le B_0$, $b_1 \le B_1$, $K(b_0' + b_1^2) \le B_0$ and $b_0' + b_1 \le a^\sharp$, where $K \ge 16L^2$ and $a^\sharp$ are the constant and the radius of the inclusion of the simple sets in the cores. The moved first radius, which is to be at most the first radius B_0 at which the statement read at such a membership is had, and these conditions stand among the conditions on the constants read for G (Prop. 9.26): where the first radius of the membership lies below B_0, and for the cores $Kb_0 < B_0$ and $b_0 < a^\sharp$, they are positive ceilings on the second radius, which is free downward.

Step 5: large fields; sectors. Lem. 9.13 gives the energy of block averages and the large-field chain, the prices being those of the $SU(2)$ construction times τ_G^{-2}, by $a(V) = \frac{1}{2}\|V - \mathbf{1}\|^2 \ge \frac{1}{2}\tau_G^{-2}|V - \mathbf{1}|^2$. Lem. 9.13(i) holds with the pair $(\frac{1}{4\tau_G^2}, e_*)$, $e_* = e_L/(2\tau_G^2)$: at one plaquette its proof reads the pair through $|V - \mathbf{1}|^2 \le 2\tau_G^2 a(V)$ alone; and $\tau_G^2 \le N - 1$. The chain has three links, those of Lem. 9.13(ii). The first link is Prop. YM3_I:44.4 and Prop. YM3_I:44.5, on the failed tests. A failed small-field test forces a plaquette variable or an averaged contour variable farther than the constants of the $SU(2)$ construction from $\mathbf{1}$: by contraposition, a test passed at all plaquettes of a double block puts the field under the averaging radius $\min\{\omega_M, r_G\}$, where the contour families have one mean within $4d$ (Lems. 9.9(iv), 9.11(i)), the straightening of Lem. 9.11(iii) holding for every closed G. The second link is Lem. YM3_I:65.2, on the witnesses of the cells created at one step, with "$\notin G^\pi$" for "$= -\mathbf{1}$". The third link is the local-deposit argument in the proof of Prop. YM3_I:65.4, the lower bounds of the amount of action near a created cell, which cites clauses of the second. Beside these, the proof of the third link cites statements on cubes, regions and runs of couplings alone, which read no group and hold for G as written. The second and the third link hold for G by Lem. 9.13(ii), by which Prop. 9.24 transfers the chain and which (II) of §9.1.1 includes: their proofs derive them, by the device of the proof of Prop. 9.24, from the transferred statements that they cite, their inputs asking $\|E^{(j)}\| \le E_0$ and $|b_j| \le B_\sharp$ separately; and the prices follow from these three links and Lem. 9.13(i):

the group enters them where the third link turns the witnesses of the second into lower bounds of the action, at one plaquette through $a(V) \geq \frac{1}{2}\tau_G^{-2}|V-\mathbf{1}|^2$ and at averaged variables through the pair $(\frac{1}{4\tau_G^2}, e_*)$ of Lem. 9.13(i), so that the clauses of the third link hold with the prices of the $SU(2)$ construction divided by τ_G^2. As G is simply connected, the fields on a torus with small plaquettes form one sector, and the statements on the whole torus that the induction uses hold as for $SU(2)$; $SU(N)$ is simply connected, and for quotients Prop. 9.33 gives what holds in each sector.

Step 6: clause (a). Let G be simply connected. Assume the five construction hypotheses of §9.1.1. By Steps 1–5, Prop. 9.23 and (II), the statements cited in the proofs of the induction hold for G, with $m(X)$ where $|X^{-1}-\mathbf{1}| = |X-\mathbf{1}| = |X^*-\mathbf{1}|$ was used, with the radius r_n where $|1-T| \leq \frac{1}{2}$ was to keep $\det T$ off $(-\infty, 0]$, with $\max\{1, n2^{-11}\}$ in the bound eq. 9.xii, with the ceilings $\varepsilon_{\mathrm{v}} \leq a_G$, $\varepsilon_{\mathrm{v}} \leq \varepsilon_G$ and $\varepsilon_{\mathrm{v}} \leq L^2 r_G$ on the regularity radius (Prop. 9.23(ii), Prop. 9.26(f), §9.1.1(V)), the member $r_G/(2C_{\mathrm{pg}})$ of the radius c_{11} of §9.1.6, the ceiling $\mathsf{c}_L c_4 \leq 2Lr_G$ of Step 3 and the conditions of Step 4 on the radii at the memberships reached from memberships of restrictions had apart among the conditions on the constants (§9.1.1(K)), and with the colour clauses near trivial block data from Prop. 9.22(ii) and (V). The ceiling $b_1 \leq \pi L^j$ under which the local terms are continuous on the analyticity cores (Lem. 5.33), which a proof of the chain from level k to level $k+1$ quotes at the fixed radii as $\alpha_1 < \pi L^j$, $1 \leq j \leq k$, takes ρ_G of Def. 9.1 in the place of π (Prop. 9.24): $\alpha_1 < \rho_G L^j$, which the row $\alpha_1 < \rho_G L$ of the table gives, as $L^j \geq L$. This row is a positive ceiling on α_1 and holds of itself for $\rho_G > \frac{1}{13}$, as $\alpha_1 \leq 1$ and $L \geq 13$, so for $SU(N)$ with $N \leq 81$, where $\rho_G = \frac{2\pi}{N}$. As $\rho_G \leq \pi$, it gives $\alpha_1 < \pi L^j$, under which the exponential of $\mathfrak{g}^c$ is a local diffeomorphism at the Z_0 with $|Z_0| < \pi$ that the continuity on the cores reads, for matrices of every size, the eigenvalues of $\operatorname{ad} Z_0$ being at most $2|Z_0| < 2\pi$ in modulus. The maps of the induction commute with the conjugation by (C). By Prop. 9.24, the hypotheses of each citation, those that the forms (3) of Def. 9.5 add among them, being met at its data by (II), the proofs of the induction then give Thm. YM3_I:D for G, the mean coefficient being the one number of second order that they read, so that, with b_k the mean of the n quadratic colour coefficients of step k, the run is

$$x_k = L^{-1}x_{k-1} - b_k, \qquad |b_k| \leq \mathsf{G}_\sharp \leq B_\sharp,$$

real and free of the torus as for $SU(2)$. By (K) and Prop. 9.26(b) the conditions on the constants read for G, those of Prop. 9.26(c)–(g) included, hold together on a large set, so choices exist, in the order of the $SU(2)$ construction with γ last.

Step 7: clause (b). The chessboard norm of a source is Prop. 9.27(ii), for every compact $G \neq \{\mathbf{1}\}$. With $A(v,V) = \sum_p v(p)a_p(V)$ for a complex weight v of the plaquettes and $\mathcal{O}(w) = a_K^{-1}A(w,\cdot)$ for a source w, the weight with the source is $e^{-A(\beta-a_K^{-1}w,V)}$, of modulus $e^{-\sum_p(\beta-a_K^{-1}\operatorname{Re} w(p))a_p(V)}$, with $a_p \geq 0$ on G; so $\operatorname{Re} w \leq (1-c_{\mathrm{pr}})a_K\beta$, $c_{\mathrm{pr}} \in (0,1]$, gives $|e^{-A(\beta-a_K^{-1}w,V)}| \leq e^{-c_{\mathrm{pr}}\beta\sum_p a_p(V)}$: every plaquette keeps at least the fraction c_{pr} of its price, and the Wilson measure at the one coupling $s = c_{\mathrm{pr}}\beta$ dominates the weight with the source.

The bound on the mean plaquette reads of the group its dimension n and that a vanishes on G at $\mathbf{1}$ alone, through $\frac{1}{2}\tau_G^{-2}|V-\mathbf{1}|^2 \leq a(V) \leq \frac{1}{2}|V-\mathbf{1}|^2$ (Lem. 9.9(i)) and through the Haar measure of balls, $cr^n \leq \mathrm{Haar}\{|V-\mathbf{1}| < r\} \leq C''r^n$, the first for $r \leq 1$, c and C'' numbers of the pair, which the chart of Lem. 9.14(i) gives, its density being at most 1 and bounded below on $|Y| \leq \frac{\pi}{3}$. With the first of the two bounds on a they give $\int_G e^{-\frac{s}{2}a(W)}\,\mathrm{d}W \leq C_1 s^{-n/2}$ for $s > 0$, C_1 a number of the pair; in the chart, with $a(e^{iY}) \geq \frac{2}{\pi^2}\|Y\|^2$ for $|Y| < \pi$ and $a \geq \frac{1}{2}\tau_G^{-2}r_G^2$ off G^π, $\int_G e^{-ya(u)}\,\mathrm{d}u$ lies between two numbers of the pair times $y^{-n/2}$, the lower bound for $y \geq 1$ and the upper one for $y > 0$. Let the couplings of the plaquettes of a torus with $|\Lambda|$ sites satisfy $s \leq v(p) \leq C's$, $s \geq 1$, $C' \geq 1$, let $Z(v) = \int e^{-A(v,V)}\,\mathrm{d}V$, and let $\langle\cdot\rangle_v$ be the Wilson measure at the couplings v. As $\log Z(tv)$ is convex in t with the derivative $-\sum_p v(p)\langle a_p\rangle_{tv}$, this weighted sum does not increase with t, its derivative being minus a variance, and $\frac{s}{2}\sum_p\langle a_p\rangle_v \leq \frac{1}{2}\sum_p v(p)\langle a_p\rangle_v \leq \log Z(\frac{v}{2}) - \log Z(v)$.

From below: in the gauge in which a maximal tree carries $\mathbf{1}$ the $2|\Lambda|+1$ bonds off the tree are independent Haar variables, and restricting them to within $(C's)^{-1/2}$ of $\mathbf{1}$ gives $a_p \le \frac{1}{2}(4(C's)^{-1/2})^2$ by the second bound on a, so $v(p)a_p \le 8$ at each of the $3|\Lambda|$ plaquettes, and $Z(v) \ge (c(C's)^{-n/2})^{2|\Lambda|+1} e^{-24|\Lambda|}$.

From above: of all the bonds, $2|\Lambda|-2$ can be ordered and matched to distinct plaquettes so that each lies in its own plaquette and in none of the plaquettes of the earlier ones; dropping the other plaquettes, as $a_p \ge 0$, and integrating these bonds one by one, the last first, by the invariance of the Haar measure, gives $Z(\frac{v}{2}) \le (C_1 s^{-n/2})^{2|\Lambda|-2}$. So the mean $\frac{1}{3|\Lambda|}\sum_p \langle a_p\rangle_v$ is at most $(C + n\log C')/s + n\log s/(|\Lambda|s)$, C a number of the pair. At couplings invariant under the lattice symmetries every plaquette has this mean, so at the one coupling s, where $C'=1$, $\langle a_p\rangle_s \le C/s + n\log s/(|\Lambda|s)$ at every plaquette. The term $n\log s/(|\Lambda|s)$ comes from the three holonomies of the torus, $2|\Lambda|+1$ bonds off the tree against $2|\Lambda|-2$ matched bonds; on the tori of the continuum theorem, with at least $2L^{K+1}$ sites a side, at couplings at most $L^{K+1}(g^{-2}+\bar B)$, that of its first-passage data, $n|\Lambda|^{-1}\log s$ is below a number of g and the constants. At equal couplings equipartition gives the mean plaquette $\frac{n}{3s}(1+o(1))$ on large tori as $s\to\infty$; so the number C in the bound reads the pair, and it is at least $\frac{n}{3}$.

Let G be simply connected and assume the construction hypotheses. The sourced-density and stability assertions of Thms. YM3_I:AJ and YM3_I:AD hold for G with the run of (a), by the corresponding $SU(2)$ proof run for G: a source enters as a weight of the plaquette function a_p, which for simple $\mathfrak{g}$ is the same class function as in the action, so the steps of the induction of (a) run for the weighted action, whose gradient and Hessian are the weighted ones, through complex plaquette weights near β and uniformly in the volume, the price of a plaquette keeping the prices of the large fields; the effective densities with a source have at every level the structure that the $SU(2)$ construction proves, and writing $\tilde Z(w) = e^{-E_{(K)}-\mathcal{E}_K(w)} Z_{K,\ell,\beta}[w]$, with $\mathcal{E}_K$ defined by [7, eq. 212], gives the factorisation of the partition function with a source that (b) displays, with the bounds of Thm. YM3_I:AD on its factors. With it the bound per ball of (b) is the passage of the $SU(2)$ construction from the stability to Prop. YM3_I:89.4, run for G, at every reference coupling $0<g\le\gamma$, at every first-passage datum (K,β) at g and on every side ℓ admitted at g with $\ell \ge 16\mathsf{m}(R+1)$, for supports in balls of radius R pairwise at a distance at least δ: the tuple of test functions is placed on a board by a lattice translation (Lem. YM3_I:89.3); the stability gives on each cell a normaliser without zeros, with its bound per cell (Prop. YM3_I:89.1, Prop. YM3_I:89.1, Prop. YM3_I:89.2); the expectation with sources is dominated by the chessboard norms of the occupied cells, through subadditivity and the comparison of two volumes (equation YM3_I:(1759), Prop. YM3_I:86.5, Cor. YM3_I:86.6, which hold for G by Prop. 9.27(ii)), the price of a plaquette and the bound on the mean plaquette entering as in the $SU(2)$ construction; and the graded Cauchy inequality on the polydisc of sources (Lems. YM3_I:87.1 and YM3_I:87.2) gives the bound per ball, uniformly in the torus, with the constant $C^\infty_{\mathsf{m}}(\delta;R)$ of Prop. YM3_I:89.4, which these sections form from g, m, δ, R and the constants of Thm. YM3_I:AD for G alone, free of K, β and ℓ. These sections of the $SU(2)$ construction read the group through the statements named alone.

Step 8: clause (d). The increments of the run are real, depend on no torus and are bounded by $B_\sharp$, by (a); so $|x_j - L^{-(j-i)}x_i| \le \bar B(1-L^{-(j-i)})$, for $\beta > X_g$ there is exactly one first passage K, and

$$g^{-2} < x_K \le X_g, \qquad \frac{\beta}{L(g^{-2}+\bar B)} \le L^K < \beta g^2, \qquad g^{-2} + B_\sharp - \frac{\bar B}{L} < a_K\beta \le L(g^{-2}+\bar B);$$

the window $I(g)$ has the length $(L-1)g^{-2} + B_\sharp\frac{L(L-3)}{L-1} > 0$, and $L^K I(g)$ consists of first-passage couplings; so $a_{K_j}\beta_j$ has a convergent subsequence in the closed interval. For simply connected G satisfying the construction hypotheses, by (b) the bound per ball holds uniformly in j, at the first-passage data (K_j,β_j) and on the sides l_j admitted at g, from the j with $l_j \ge 16\mathsf{m}(R+1)$ on:

$|S^{(j)}_{\mathsf{m}}(f)| \le C^{\infty}_{\mathsf{m}}(\delta;R)\prod_i \|f_i\|$, in the norms of Lipschitz functions of Prop. YM3_I:89.4, each at most a constant times $\|f_i\|_{C^1}$, for supports in balls of radius R pairwise at a distance at least δ, $C^{\infty}_{\mathsf{m}}(\delta;R)$ being formed from g, m, δ, R and the constants of Thm. YM3_I:AD for G alone; on one torus of side l the bound $C_{\mathsf{m}}(\delta)e^{\mathsf{m}c_V l^3}\prod_i \|f_i\|_{C^1}$ holds (Prop. YM3_I:88.3); with it, a diagonal subsequence over a countable dense set of tuples and m-linearity give convergence. The properties of the limits need convergence alone and hold for every compact connected G: the bound by uniform boundedness for m-linear forms on Banach spaces, which is why C depends on the sequence; the translations and the signed permutations from the lattice symmetries, which keep the measure, the action and the observables; reflection positivity from Prop. 9.27(i); and on one torus its translations and the pairs of parallel mirrors at the distance $\frac{l}{2}$. For the additional conclusions assume throughout that G is simply connected and that (P), (V), (C), (II) and (K) hold. The transferred construction in (a), with the sourced estimates in (b), supplies the inputs of the fixed-torus comparison in Thm. YM3_I:A; its proof uses these estimates at the unchanged label and admitted sides. The bounds in (b) give the compactness and distributional limits of Props. YM3_I:92.1–YM3_I:92.4 by the same diagonal extraction and passage of finite positive matrices used above. For Gibbs limits, Prop. 9.27(iii) supplies periodic Gibbs states, their lattice symmetries and all the required reflection families. Conditioning on a translation-invariant event preserves the Gibbs specification by the conditional-expectation argument of Prop. YM3_I:106.1, which uses no special identity of the group. Together with the transferred cumulant and one-point bounds, these are exactly the hypotheses used for the component construction in Thm. YM3_I:AL and its application in Thm. YM3_I:AP. The reflection-Hilbert-space and reconstruction arguments concern the resulting scalar functionals and use no further identity of $SU(2)$. Thus they give the Gibbs conclusions stated in (d). For growing-torus limits this argument supplies coordinate reflection positivity; their full rotational invariance is the criterion proved in Step 10. The colour coefficient in the short-distance law is computed in Step 9.

Step 9: the leading law and the perturbative ratio. The quadratic part of the action at $\mathbf{1}$ is $\frac{1}{2}\sum_p \|dA(p)\|^2$, colour by colour (Lem. 9.15(ii)). In an orthonormal basis of $\mathfrak{g}$, its Gaussian measure is therefore the product of n copies of the one-colour Gaussian measure for $SU(2)$, with the same normalisation. Each colour contributes $3/(16\pi^2)$, and hence

$$c_2(G) = \frac{3n}{16\pi^2}.$$

Under the construction hypotheses, the estimates in (a) and (b) give the effective-action remainders used in the finite-cutoff leading law of Thm. YM3_I:AV. Its colour sum now runs to n, and the label and continuum passage use Thm. YM3_I:AW. The passage to limits is the one in (d). For $SU(2)^r$ the product argument is given in Thm. AB(iii).

For (f), Lem. 9.28 gives the coefficient of β^{-3} first at fixed box and then at infinite volume. Its leading part at large separation is $\varphi_G \mathsf{r}^{-5}/(16\pi^3)$, after summing over orientations. Dividing this coefficient by the leading one gives

$$c_1(G) = \frac{\varphi_G}{16\pi^3}\Big/\frac{3n}{16\pi^2} = \frac{\varphi_G}{3\pi n}.$$

For $SU(2)$, $\varphi_G = 24$, so that $c_1 = 8/(3\pi)$.

Step 10: clause (g). Prop. 9.27(iv) replaces the formula of the $SU(2)$ construction for each generator. The rest of its proof of the Ward identity at finite cutoff (Prop. YM3_I:112.1) is an integration by parts on G at every cutoff, $\int v_{\mathcal{X}}\Phi\,\mathrm{d}V = -\int \Phi \operatorname{div} v_{\mathcal{X}}\,\mathrm{d}V$ for a field of directions $\mathcal{X}$, and it is unchanged but for the divergence of the rotation field, which carries $\frac{1}{2}k_p$ where the $SU(2)$ construction has $\frac{3}{2}a_p$. For $\omega \in \mathfrak{so}(3)$ let $(L_\omega f)(y) = \sum_\rho (\omega y)_\rho\, \partial_\rho f(y)$, let $e^{t\omega}f$ be the tuple of the functions $f_i(e^{-t\omega}\,\cdot\,)$, and let $\nabla^{\mathrm{s}}_\omega$ be the symmetric lattice difference along ωy, $(\nabla^{\mathrm{s}}_\omega\phi)(y) = \sum_\rho (\omega y)_\rho (2a)^{-1}[\phi(y+ae_\rho)-\phi(y-ae_\rho)]$. A quadruple $(\omega, \mathsf{m}, \mathcal{R}, f)$ is admitted (Section YM3_I:112,

before equation YM3_I:(2115)) if $\omega \in \mathfrak{so}(3)$, $\mathsf{m} \geq 2$, $\mathcal{R} > 0$ and the f_i are smooth, with compact pairwise disjoint supports in $\{|y| < \frac{1}{2}\mathcal{R}\}$. Let (K_j, l_j, β_j) be a sequence of tori as in (d), $K_j \to \infty$, with $l_j \to \infty$. For the j with $2\mathcal{R} + 2a_{K_j} < \frac{1}{2}l_j$ the ball $\{|y| < \frac{5}{3}\mathcal{R} + a_{K_j}\}$, which carries the cut-off of radius $\mathcal{R}$ and its shell, has a diameter below l_j and lies in the torus without wrapping, so the bulk term and the flux term of the identity at the cut-off radius $\mathcal{R}$ are defined; $j_1(\mathcal{R}, f)$ is the least index from which on $2\mathcal{R} + 2a_{K_j} < \frac{1}{2}l_j$, $2\sqrt{3}\,a_{K_j} \leq \frac{1}{2}\mathcal{R}$ and $a_{K_j} \leq \frac{1}{4}\min\{1, \mathrm{sep}(f)\}$, $\mathrm{sep}(f)$ the least distance of two of the supports. For $j \geq j_1(\mathcal{R}, f)$ the identity at the cutoff K_j, with the radial cut-off $\chi_{\mathcal{R}}(y) = \chi_0(|y|/\mathcal{R})$ (choose χ_0 smooth, equal to 1 on $[0, 4/3]$ and zero on $[5/3, \infty)$, as permitted in Section YM3_I:112), which equals 1 on $\{|y| \leq \frac{4}{3}\mathcal{R}\}$ and so covers the f_i, reads

$$\sum_i S^{(j)}_{\mathsf{m}}(f_1, \dots, \nabla^{\mathrm{s}}_\omega f_i, \dots, f_{\mathsf{m}}) = \mathfrak{U}^{(j)}_{\omega,\mathcal{R}}(f) + \mathrm{Fl}^{(j)}_{\omega,\mathcal{R}}(f),$$

the bulk term $\mathfrak{U}^{(j)}_{\omega,\mathcal{R}}(f)$ collecting the spin and orbital insertions, and the flux $\mathrm{Fl}^{(j)}_{\omega,\mathcal{R}}(f)$ being the connected function of the $\mathcal{O}(f_i)$ with the shell flux, a sum over the plaquettes p with $\frac{4}{3}\mathcal{R} - a < |x(p)| < \frac{5}{3}\mathcal{R} + a$ that carries $\frac{1}{2}k_p$ (Prop. YM3_I:112.1, under YM3_I:(2116)). In the limit, along such a sequence on which all connected functions converge,

$$\lim_j [\mathfrak{U}^{(j)}_{\omega,\mathcal{R}}(f) + \mathrm{Fl}^{(j)}_{\omega,\mathcal{R}}(f)] = \sum_i S_{\mathsf{m}}(f_1, \dots, L_\omega f_i, \dots, f_{\mathsf{m}}) = -\frac{d}{dt} S_{\mathsf{m}}(e^{t\omega} f)\Big|_{t=0},$$

independently of $\mathcal{R}$ among the radii that admit f: the bound of (d) (Step 8), on the spaces of C^1 functions with supports in fixed disjoint compact neighbourhoods of the supports, with $\|\nabla^{\mathrm{s}}_\omega f_i - L_\omega f_i\|_\sharp \leq c\, a^2_{K_j}$ and $\|t^{-1}(e^{t\omega} f_i - f_i) + L_\omega f_i\|_\sharp \leq c|t|$, c a number of ω and f_i, gives the limit and the differentiability of $t \mapsto S_{\mathsf{m}}(e^{t\omega} f)$ (Prop. YM3_I:112.3). The conjugates of one $\omega_0 \neq 0$ under the signed permutations span $\mathfrak{so}(3)$, and the one-parameter group of ω_0 with the cubic group generates $O(3)$ (Lem. YM3_I:112.4 and the proof of Thm. YM3_I:AQ), which gives the equivalence at the admitted quadruples (Thm. YM3_I:AQ); the invariance under $O(3)$ passes to the C^1 tuples of (d) by mollifying, with the bound of (d): the mollified entries and their gradients converge uniformly, with supports in fixed disjoint compact sets, on whose C^1 functions that bound is continuous. The limit of $\mathrm{Fl}^{(j)}_{\omega,\mathcal{R}}(f)$ alone is this limit exactly when $\mathfrak{U}^{(j)}_{\omega,\mathcal{R}}(f) \to 0$. When the limit is invariant, the rotations about the third axis and the reflections in planes through it keep the half-space $\{y_3 > 0\}$, commute with the reflection in $\{y_3 = 0\}$ and keep the reflection positive form, so $\mathsf{U}(R)[F] = [(R \oplus 1)F]$, on the classes of the functions F supported in $\{y_3 > 0\}$, is well defined modulo the null space, unitary and fixes the vacuum vector; as $R \oplus 1$ commutes with the translation by te_3, $\mathsf{U}(R)[\tau_{te_3} F] = [\tau_{te_3}((R \oplus 1)F)]$ for $t \geq 0$. Strong continuity comes from the continuity of the S_{m}.

Step 11: clause (h). (h1) is Lem. 9.29 with the bound of Prop. 9.24; (h2) is Prop. 9.18(iv); (h3) is Prop. 9.30(iv), with Lem. 9.29 at $\Gamma = \mathsf{G}_\sharp$, as $|e^{(i)}_k| \leq (\kappa_i/\kappa_G)^2 \mathsf{G}_\sharp \leq \mathsf{G}_\sharp$ (Prop. 9.24), which gives

$$|x^{(i)}_k - x_k| \leq 2\Big(1 - \frac{n_i}{n}\Big)\mathsf{G}_\sharp \frac{L}{L-1}(1 - L^{-k}),$$

and so the same with $\kappa^2_G \mathsf{G}_\sharp$ in the place of $\mathsf{G}_\sharp$, as $\kappa_G \geq 1$; and (h4) is Lem. 9.31(i) with clauses (a)–(g) for each factor, in the forms stated there. □

9.1.11. *Proof of Thm. AB. Step 1: clause (i).* $a(u) = \sum_i \lambda_i a_i(u_i)$ and the Haar measure is a product, so the fields of the factors are independent (Lem. 9.31(i)); cumulants of sums of independent families add, and $\mathcal{O} = \sum_i \lambda_i \mathcal{O}^{(i)}$ gives the factor λ_i^{m}.

Step 2: clause (ii). Prop. YM3_I:52.4 proves covariance of the pointed totals for the actual arrays admitted by Thm. YM3_I:W. Its proof uses the earlier array through the sum in equation YM3_I:(963): earlier pointed totals, constants, and multiples of the plaquette action. The constants and the action are invariant, independently of the counterterm coefficients. Thus the covariance induction

applies to these arrays whenever the earlier pointed totals are covariant. So symmetry passes from level to level along every good sequence, $|L^{-1}x_{j-1} - x_j| \le B_\sharp$ and $x_j \ge \gamma^{-2}$, and along it the coefficient of $SU(2)$ is defined, real and at most $\mathsf{G}_\sharp$, and Thm. YM3_I:D holds whatever its quadratic leftovers. The sequences $\lambda_i x_j$ are good: their increments are $\lambda_i \frac{1}{r}\sum_m \lambda_m^{-1} c_j^{(m)}$, at most $\lambda_+\lambda_-^{-1}\mathsf{G}_\sharp \le B_\sharp$, and $\lambda_i x_j \ge \lambda_- x_j \ge \gamma^{-2}$. Then

$$|x_k^{(i)} - x_k| \le \mathsf{G}_\sharp\Big(\lambda_i^{-1} + \frac{1}{r}\sum_m \lambda_m^{-1}\Big)\frac{L}{L-1}(1-L^{-k}) \le 2\lambda_-^{-1}\mathsf{G}_\sharp\frac{L}{L-1}.$$

Step 3: clause (iii). The first-passage arithmetic of Prop. YM3_I:74.1 and Cor. YM3_I:74.2 applies to the mean run, whose increments are real and bounded by $B_\sharp$. The windows $I(g_{i,1}) = (\lambda_i g^{-2}, L\lambda_i g^{-2} - 2\bar{B}]$ and $I(g_{i,2}) = (L\lambda_i g^{-2} - 2\bar{B}, L^2\lambda_i g^{-2} - 2(L+1)\bar{B}]$ are adjacent, and $\lambda_i a_K \beta$ lies in one of them, $I(g_{i,s(i)})$, so $(K, \lambda_i\beta)$ is a first-passage datum of the factor i at the reference coupling $g_{i,s(i)}$, $s(i)$ being read off $(K, \lambda_i\beta)$. The law of the continuum theorem for the factor i is cited at $g_{i,s(i)}$, with the hypotheses on the scale, the cutoff and the test pair that (iii) lays at the $2r$ couplings, and on tori whose side is admitted at $g_{i,s(i)}$, which the hypothesis that the sides are admitted at all $2r$ couplings $g_{i,s}$ provides; these laws and (i) give

$$\sum_i \lambda_i^2 \frac{9}{16\pi^2}(\lambda_i \mathsf{b})^{-2}\mathcal{K}_6 = \frac{3n}{16\pi^2}\,\mathsf{b}^{-2}\mathcal{K}_6.$$

In a Gibbs state with periodic marginals the factors are independent given the tails, the diagonal entries are the laws of the factors in their periodic marginals, and the mixed entries are bounded by Lem. 9.31(iii), whose range $a_K \le \mathsf{d} \le \frac{1}{20}$ holds, as the hypotheses of the law on the scale and the cutoff give $a_K < \mathsf{d} \le \frac{1}{200}$; as $\sum_{i\ne j}\lambda_i\lambda_j\frac{1}{4}\varepsilon^\times\frac{3}{16\pi^2}(\lambda_i\lambda_j\mathsf{b}^2)^{-1} = \frac{r-1}{12}\varepsilon^\times\cdot\frac{9r}{16\pi^2}\mathsf{b}^{-2}$, this gives the error term $\frac{r-1}{12}\varepsilon^\times(\mathsf{d})$, the divergent parts never mixing the factors. □

9.1.12. *Scope of the conclusions.* The full construction for a simply connected simple group is conditional on §9.1.1. Its two principal outputs are the source stability in Thm. AA(b) and the resulting bound per ball. The latter yields subsequential limits. In contrast, the algebraic coefficients and the finite-box perturbative expansion require no global construction hypothesis. The error constant in that expansion depends on the box, so its stated limit is iterated.

For any compact connected group, convergence of the connected functions along a sequence of tori gives the translation symmetry, cubic symmetry and reflection positivity stated in Thm. AA(d). Its rotational Ward identity gives the criterion in (g): full $O(3)$ invariance is equivalent to the vanishing of the indicated bulk-plus-flux limit. No vanishing assertion is included in that criterion.

Products acting on separate blocks are governed by exact factorisation. For tied simple ideals, the one-loop calculation is block diagonal at the trivial field, while the nonlinear action couples the ideals. The assertion that the mean run suffices has property (T) as its explicit hypothesis. Quotients require the sector distinctions in Prop. 9.33. The regular-field construction here assumes a semisimple Lie algebra: a nonzero centre obstructs the box gauge in Lem. 9.12(iv), and invariant linear forms then need not vanish.

9.2. **Geometric convergence of the Gaussian one-loop coefficient.** The bounded Gaussian coefficient has a limit as the number of averaging steps increases. We prove this by comparing two successive meshes in units in which the terminal lattice has spacing one. Throughout this section the group is SU(2), $L \ge 13$ is odd, and the averages, Gaussian normalisers and coefficient functional are those of Sections YM3_I:42–YM3_I:43. In particular, Thm. YM3_I:O gives

$$c_k = \mathsf{b}[\Psi_k] = \sum_{l=1}^{k} L^{-(k-l)} b_l^0, \qquad c_0 = 0, \qquad c_2^{\mathrm{loop}} := \sup_{k\ge 0}|c_k| < \infty.$$

We write c_k here for the numbers denoted by c_k elsewhere.

Theorem AC. There are real numbers c_∞ and $b^0_\infty = (1 - L^{-1})c_\infty$, depending only on L, and a finite constant $C_{\mathrm{geo}} = C_{\mathrm{geo}}(L)$ such that

$$|c_k - c_\infty| \le \tfrac{1}{2} C_{\mathrm{geo}} L^{-k/6} \quad (k \ge 0), \qquad |b^0_l - b^0_\infty| \le C_{\mathrm{geo}} L^{-l/6} \quad (l \ge 1).$$

Moreover

$$|b^0_{l+1} - b^0_l| \le (1 - L^{-1/6}) C_{\mathrm{geo}} L^{-l/6} \quad (l \ge 1).$$

■□

The local determinant analysis already gives a uniformly holomorphic remainder. The additional issue is convergence of that remainder on changing Fourier spaces. We first remove the explicit massive term, then compare the remaining finite expression. The comparison will be made at real momentum; a separate finite-difference argument will recover the derivative defining the coefficient.

9.2.1. *The explicit massive term.* Put $\eta = L^{-k}$ and $\psi(q) = \sum_\mu (2 - 2\cos q_\mu)$, with normalised integration over $[-\pi, \pi]^3$. The massive determinant calculation in Prop. YM3_I:43.3 gives fixed real trigonometric polynomials g_1, g_2 such that

$$\widetilde{\beta}(z) = I_2[g_2](z) + I_1[g_1](z), \qquad I_n[g](z) = \int (\psi(q) + z)^{-n} g(q)\, \mathrm{d}q.$$

More explicitly, $g_2 = 4 \operatorname{tr} \sigma_1^2 - \psi_{11}\psi_{22}/3$ and $g_1 = -8 \operatorname{tr} \sigma_2 + (\psi^{(1)} + \psi^{(2)})/2$ in that calculation. Define

$$E(\eta) = \eta[\widetilde{\beta}(\eta^2) - \eta^2 \widetilde{\beta}'(\eta^2) + \tfrac{1}{2}\eta^4 \widetilde{\beta}''(\eta^2)].$$

With $\delta = \delta_{\mathrm{loop}} \le 1/64$, let

$$F_k(p) = \widehat{\pi}^0_{12}(p) - \mathcal{L}^\sharp_k(p), \qquad r_k = c_k - E(L^{-k}).$$

The determinant representation in Section YM3_I:43.3, Prop. YM3_I:43.2 and the massive calculation give

$$r_k = -\partial_1 \partial_2 F_k(0), \qquad |F_k(p)| \le C_{\mathrm{rem}} \quad (|p_\mu| < \delta), \tag{9.xix}$$

with C_{rem} independent of k.

Lemma 9.34. There is an absolute real number E_∞ such that $C_E := \sup_{0<\eta\le 1} \eta^{-1} |E(\eta) - E_\infty| < \infty$. ■

Proof. Differentiation under the integral gives $I_n' = -nI_{n+1}$. Writing $z = \eta^2$, we obtain

$$E(\eta) = z^{1/2}(I_2 + 2zI_3 + 3z^2 I_4)[g_2](z) + z^{1/2}(I_1 + zI_2 + z^2 I_3)[g_1](z).$$

For a fixed trigonometric polynomial g, put $i_n = (2\pi)^{-3} \int_{\mathbb{R}^3} (1 + |y|^2)^{-n}\, \mathrm{d}y$. We claim

$$|I_1[g](z)| \le C_g, \qquad |z^{n-3/2} I_n[g](z) - g(0) i_n| \le C_g \sqrt{z} \quad (n = 2, 3, 4; 0 < z \le 1). \tag{9.xx}$$

Indeed, on the integration cube,

$$a|q|^2 \le \psi(q) \le |q|^2, \qquad 0 \le |q|^2 - \psi(q) \le |q|^4/12, \qquad a = 4/\pi^2.$$

The first bound follows by integrating $C|q|^{-2}$ over a fixed ball. For the others replace g by its even part, whose difference from $g(0)$ is $O(|q|^2)$. The error is bounded by $C \int_{|q| \le \sqrt{3}\pi} |q|^2 (a|q|^2 + z)^{-n}\, \mathrm{d}q$. Replacing ψ by $|q|^2$ costs at most $C \int_{|q| \le \sqrt{3}\pi} |q|^4 (a|q|^2 + z)^{-n-1}\, \mathrm{d}q$. For either integral, multiplication by $z^{n-3/2}$ gives $O(\sqrt{z})$ when $n = 2$, by the integrable radial majorant $C|q|^{-2}$, and $O(z)$ when $n = 3, 4$, by $q = \sqrt{z}\, y$. Extending the cube to $\mathbb{R}^3$ costs $O(z^{n-3/2})$, since its complement has $|q| \ge \pi$. The same substitution then gives i_n exactly. This proves eq. 9.xx.

It follows that $E_\infty = (i_2 + 2i_3 + 3i_4) g_2(0)$ and $E(\eta) - E_\infty = O(\eta)$. The contribution of g_1 is $O(\sqrt{z})$ by the same estimates. Reality follows from that of the determinant coefficient. Since E is defined by $\widetilde{\beta}$, its limit and C_E do not depend on a choice of the polynomial representation. □

9.2.2. *A finite expression for the remainder.* We use the physical pairings of weight ℓ_j^3 on $\ell_j\mathbb{Z}^3$, $\ell_j = L^{j-k}$, and the scaled exterior derivative. The operators $D_U, Q_U, R_U, K_U, \Delta_U, \Xi_U, Y_U$ are those defined before Section YM3_I:43.2. Thus $Q_U = \eta\mathring{\mathcal{Q}}_U^{[k]}$ and $R_U = \mathfrak{a}_U^{[k]}$. Unsubscripted d, K, Δ, Q, R denote their free values, and $G = (K+1)^{-1}$, $g = (\Delta+1)^{-1}$. The response is $h = \Xi^{-1}Q^{\mathsf{T}}(Q\Xi^{-1}Q^{\mathsf{T}})^{-1}$. Along

$$\zeta_\pm = \eta e^{\pm ip\cdot x}\alpha_\pm, \qquad \alpha_\pm = \widehat{h}(\pm p)e_{\mu_\pm}t_3,$$

bold letters denote the jets with components $00, 10, 01, 11$. The two nonzero field directions and their curls are bounded uniformly in k on the fixed strip, by Section YM3_I:43.2. In this subsection either pair of directions is allowed; we later take $(\mu_+, \mu_-) = (1, 2)$.

For completeness, the real domains for these jets are $K^{00} = K = [-\pi-\delta, \pi+\delta]^3$, $K^{11} = [-\pi, \pi]^3$ and $K^{10} = K \cap (K-p)$, $K^{01} = K \cap (K+p)$. Their norm is the sum of the four supremum operator norms. Products are the jet products: if $a, b \in \{0, 1\}$, then

$$(\mathbf{T}^1 \cdots \mathbf{T}^s)_{ab}(\varkappa) = \sum_{\sum a_i = a,\, \sum b_i = b} \prod_{i=1}^{s} T^i_{a_i b_i}\left(\varkappa + \sum_{j>i}(a_j - b_j)p\right). \tag{9.xxi}$$

Every argument lies in its stated domain. Consequently the norm is submultiplicative, and a product containing three factors with zero 00 component vanishes. Transposes use the bilinear pairing; their arguments are $-\varkappa$, $-\varkappa - p$, $-\varkappa + p$, $-\varkappa$, respectively, and preserve the norm. Put

$$\mathsf{t}(\mathbf{T}) = \int \operatorname{tr} T_{11}(\varkappa)\,\mathrm{d}\varkappa,$$

and, when C_{00} is invertible,

$$\ell(\mathbf{C}) = \int \operatorname{tr}[C_{00}^{-1}C_{11} - C_{00}(\varkappa)^{-1}C_{10}(\varkappa - p)C_{00}(\varkappa - p)^{-1}C_{01}(\varkappa)]\,\mathrm{d}\varkappa.$$

The integrals are normalised over $[-\pi, \pi]^3$.

The local Hessian formula of Section YM3_I:43.4 reads $H_U = (d + E_U)^{\mathsf{T}} z_U (d + E_U) + f_U$. Write $\mathbf{e} = \mathbf{D} - d$, and define

$$\kappa = G^{1/2}(\mathbf{K} - K)G^{1/2}, \quad x = \mathbf{Q}G^{1/2}, \quad y = G^{1/2}\mathbf{D}\mathbf{R}^{\mathsf{T}}, \qquad \kappa_g = g^{1/2}(\mathbf{\Delta} - \Delta)g^{1/2}, \quad x_g = \mathbf{R}g^{1/2}.$$

All of these are jets. For either datum $(\mathsf{G}, \kappa, x, y) = (G, \kappa, x, y)$ or $(g, \kappa_g, x_g, 0)$, put

$$\omega = x^{\mathsf{T}}x - yy^{\mathsf{T}}, \quad B_t = (1 + \kappa + t\omega)^{-1}, \quad B_\tau^\flat = (1 + \kappa + \omega - (1-\tau)\mathsf{G})^{-1}.$$

The strip estimate, Prop. YM3_I:43.1, and the bounded jets in Section YM3_I:43.4 bound all these jets and inverses by one constant $A = A(L)$. Indeed the four free 00 inverses are precisely the sandwiched inverses in that proposition; the inverse jet follows by multiplying out its $10, 01, 11$ components, each involving at most three such inverses. The 00 inverse of $xB_0^\flat x^{\mathsf{T}}$ is bounded by a constant ι, also uniformly in k. These statements hold for $0 \le t, \tau \le 1$, continuously in those parameters.

For these data define the following explicit expression:

$$\begin{aligned}
\Phi[\mathsf{G}, \kappa, x, y] = {} & \int_0^1 [\mathsf{t}(xB_t x^{\mathsf{T}}) - \mathsf{t}(y^{\mathsf{T}}B_t y)]\,\mathrm{d}t + \ell(xB_0^\flat x^{\mathsf{T}}) \\
& - \int_0^1 (1-\tau)^2 \mathsf{t}(\mathsf{G}B_0\mathsf{G}B_0\mathsf{G}B_\tau^\flat)\,\mathrm{d}\tau \\
& + \int_0^1 [\mathsf{t}(xB_\tau^\flat\mathsf{G}B_0 x^{\mathsf{T}}) - \mathsf{t}(y^{\mathsf{T}}B_\tau^\flat\mathsf{G}B_0 y)]\,\mathrm{d}\tau \\
& + \int_0^1 (1-\tau)[\mathsf{t}(xB_\tau^\flat\mathsf{G}B_0\mathsf{G}B_0 x^{\mathsf{T}}) - \mathsf{t}(y^{\mathsf{T}}B_\tau^\flat\mathsf{G}B_0\mathsf{G}B_0 y)]\,\mathrm{d}\tau.
\end{aligned} \tag{9.xxii}$$

Lemma 9.35. For real $|p_\mu| \le \delta$,

$$\widehat{\pi}^{0}_{\mu_+\mu_-}(p) - \mathcal{L}^{\sharp}_{k}(p) = -\tfrac{1}{2}\Phi[G,\kappa,x,y] + \Phi[g,\kappa_g,x_g,0].$$

■

Proof. Set $V = \mathbf{Q}^{\mathsf{T}}\mathbf{Q} - \mathbf{D}\mathbf{R}^{\mathsf{T}}\mathbf{R}\mathbf{D}^{\mathsf{T}}$. The Schur factorisation of the vector saddle matrix gives $\ell(\mathbf{\Xi}) + \ell(\mathbf{Q}\mathbf{\Xi}^{-1}\mathbf{Q}^{\mathsf{T}})$; the scalar version gives $\ell(\mathbf{Y}) + \ell(\mathbf{R}\mathbf{Y}^{-1}\mathbf{R}^{\mathsf{T}})$. The triangular factors have determinant one and hence zero polarisation. The coarse factors are invertible by the strip estimate, so these identities take place inside the same jet algebra.

For either pair $(K_\circ, V_\circ) = (\mathbf{K}, V)$ or $(\mathbf{\Delta}, \mathbf{R}^{\mathsf{T}}\mathbf{R})$, let $G_U = (K_\circ + 1)^{-1}$ and $P_\tau = (K_\circ + V_\circ + \tau)^{-1}$. Integration of the logarithmic derivative along $tV_\circ$ and then along τ gives

$$\ell(K_\circ + V_\circ) = \ell(K_\circ + 1) + \int_0^1 \mathsf{t}((K_\circ + 1 + tV_\circ)^{-1}V_\circ)\,\mathrm{d}t - \int_0^1 \mathsf{t}(P_\tau)\,\mathrm{d}\tau.$$

Twice using the resolvent identity yields exactly

$$P_\tau = G_U + (1-\tau)G_U^2 + (1-\tau)^2 G_U^2 P_\tau - G_U V_\circ P_\tau - (1-\tau)G_U^2 V_\circ P_\tau.$$

The first two terms give $[1 - \partial_z + \tfrac{1}{2}\partial_z^2]\ell(K_\circ + z)|_{z=1}$. Substitute $G_U = \mathsf{G}^{1/2}B_0\mathsf{G}^{1/2}$, $P_\tau = \mathsf{G}^{1/2}B_\tau^{\flat}\mathsf{G}^{1/2}$ and $\mathsf{G}^{1/2}V_\circ\mathsf{G}^{1/2} = \omega$ in the remaining terms. Cyclicity of t gives the four integrals in eq. 9.xxii, and the coarse Schur factor gives its last logarithmic term. Finally take minus one half of the vector identity plus the scalar identity. This also explains why the only fine-space trace left has three propagator factors. □

9.2.3. *Comparing the two Fourier spaces.* We compare height k with height $k+1$; primes indicate the latter. Their bare spacings are η and $\eta' = \eta/L$, and all later lattices agree: $\Lambda'_{j+1} = \Lambda_j$. Let $N = (\eta^{-1} - 1)/2$ and $\mathfrak{m} = \{m \in \mathbb{Z}^3 : |m|_\infty \le N\}$, with $\mathfrak{m}'$ defined at η'. Fourier coefficients identify each fine site, bond or plaquette fibre isometrically with $\ell^2(\mathfrak{m};\mathsf{V})$, where V is its colour and component space. The bilinear pairing pairs m with $-m$. Let J extend coefficients by zero from $\mathfrak{m}$ to $\mathfrak{m}'$, and let $J = 1$ on the common coarse fibres. Then $J^{\mathsf{T}} = J^*$ and $J^*J = 1$. For $1 \le \Lambda \le N$ let P_Λ, P'_Λ retain the modes $|m|_\infty \le \Lambda$. These projections commute with J in the corresponding sense and are their own transposes.

For two jets between corresponding fibres define $\mathrm{df}(T', T) = \|T'J - JT\|$ in the real jet norm. The product rule eq. 9.xxi gives

$$\begin{aligned}
\mathrm{df}(T'S', TS) &\le \|T'\|\,\mathrm{df}(S', S) + \mathrm{df}(T', T)\|S\|,\\
\mathrm{df}(T'^{\mathsf{T}}, T^{\mathsf{T}}) &\le \mathrm{df}(T', T) + \|T'(1 - JJ^*)\|,\\
\mathrm{df}((1+S')^{-1}, (1+S)^{-1}) &\le \|(1+S')^{-1}\|\,\mathrm{df}(S', S)\|(1+S)^{-1}\|.
\end{aligned} \tag{9.xxiii}$$

The first and third follow by inserting J between the factors. For the second, insert $JJ^* + (1 - JJ^*)$ before transposing. This extra tail will be retained throughout the comparison.

Here are the elementary multiplier estimates we need. For $h = \eta, \eta'$ and $q = \varkappa + 2\pi m$, write

$$d^h_\mu(q) = h^{-1}(e^{ihq_\mu} - 1), \quad D_h(q) = \sum_\mu |d^h_\mu(q)|^2, \quad s_h(t) = h \sum_{|j| \le (h^{-1}-1)/2} e^{ihjt},$$

$$v_h(q) = \prod_\nu s_h(q_\nu), \qquad u_{h,\mu}(q) = e^{i(1-h)q_\mu/2}s_h(q_\mu)v_h(q), \qquad g_h(q) = (1 + D_h(q))^{-1/2}.$$

Define $s_1(t) = L^{-1}\sum_{|j| \le (L-1)/2} e^{i\eta' jt}$, $v_1 = \prod s_1$ and $u_{1,\mu} = e^{i(\eta - \eta')q_\mu/2}s_1(q_\mu)v_1$. Division with centred remainder gives

$$s_{\eta'} = s_\eta s_1, \qquad v_{\eta'} = v_\eta v_1, \qquad u_{\eta',\mu} = u_{\eta,\mu}u_{1,\mu}. \tag{9.xxiv}$$

The free finest-step averages therefore act by

$$(Q^{\mathrm{f}}\varphi')_{\mu,m}=\sum_{m'\equiv m}u_{1,\mu}(q_{m'})\varphi'_{\mu,m'},\qquad (R^{\mathrm{f}}\varphi')_{m}=\sum_{m'\equiv m}v_1(q_{m'})\varphi'_{m'},\tag{9.xxv}$$

where congruence is modulo $\eta^{-1}\mathbb{Z}^3$. They have norm at most one. The transpose of the second map sends φ_m to $v_1(q_{m'})\varphi_m$ at every such alias. These formulas follow by averaging each plane wave over the centred block and, for Q^{f}, over its straight path.

Put $w(n)=\max(1,2|n|-1)^{-1}$, $w^{(3)}(m)=\prod_\nu w(m_\nu)$ and $w_3(m)=w^{(3)}(m)/\max(1,|m|_\infty)$. For $\varkappa\in K$ and the respective mode sets,

$$\begin{gathered}D_h(q)\ge\tfrac{9}{25}|q|^2,\quad |u_{h,\mu}(q)|,|v_h(q)|\le 2w^{(3)}(m),\quad |q|\le 11(1+|m|_\infty),\\ |g_\eta-g_{\eta'}|\le 6\eta,\quad |g_\eta^2-g_{\eta'}^2|\le 3\eta^2,\quad |d_\mu^{\eta'}g_{\eta'}-d_\mu^{\eta}g_{\eta}|\le 8\eta|q|,\\ \sum_{|m|_\infty\ge\Lambda}w_3(m)\le 252\Lambda^{-1/2},\quad \sum_m w_3(m)\le 253,\quad \eta(3+\log\eta^{-1})^3\le 49\eta^{1/2}.\end{gathered}\tag{9.xxvi}$$

For the middle line the modes are common modes. To verify the bounds, use $h^{-1}|\sin(ht/2)|\ge 3|t|/10$ for $|t|\le\pi/h+1$, and $h^{-1}\sin(ht/2)s_h(t)=\sin(t/2)$. Taylor's formula gives $|d^\eta-d^{\eta'}|\le\eta|q|^2$ and $|D_\eta-D_{\eta'}|\le\eta^2|q|^4/12$; the root estimates follow by subtracting the two positive square roots. Also

$$|1-v_1|\le\eta^2|q|^2/8,\quad |1-u_{1,\mu}|\le\eta|q_\mu|/2+\eta^2|q|^2/4,$$

and $1-|u_{1,\mu}|^2,1-v_1^2\le\eta^2|q|^2/2$. For the sums, $\sum_n w(n)^2\le 4$ and its tail from n_0 is at most $4/n_0$. On the shell $|m|_\infty=n$,

$$\sum w_3(m)\le 6n^{-2}(3+\log 2n)^2\le 84n^{-3/2}.$$

Summing the shells proves the displayed tail. The final bound follows by maximising $e^{-u/2}(3+u)^3$ at $u=3$.

9.2.4. *The backgrounds at the two heights.* The comparison must include the response fields themselves. Write their components as $\alpha_{\pm,\mu}=a_\mu^{\pm}t_3$ and $f_{\pm,\mu\nu}=f_{\mu\nu}^{\pm}t_3$, where $d(e^{\pm ipx}\alpha_\pm)=e^{\pm ipx}f_\pm$. For any one of these scalar functions write $\widehat{a}_m$ for its Fourier coefficient. Constants below depend only on L.

Lemma 9.36. For common modes,

$$|\widehat{a}_m|,|\widehat{a}'_m|\le Cw_3(m),\qquad |\widehat{a}'_m-\widehat{a}_m|\le C\eta w^{(3)}(m).$$

The first bound also holds on all primed modes. Consequently

$$\sum|\widehat{a}_m|\le C,\quad \sum_{|m|_\infty\ge\Lambda}|\widehat{a}_m|\le C\Lambda^{-1/2},\quad \sum_{m\in\mathfrak{m}}|\widehat{a}'_m-\widehat{a}_m|+\sum_{m\in\mathfrak{m}'\setminus\mathfrak{m}}|\widehat{a}'_m|\le C\eta^{1/2}.\tag{9.xxvii}$$

Moreover $Q^{\mathrm{f}}(e^{\pm ipx}\alpha'_\pm)$ differs from $e^{\pm ipx}\alpha_\pm$ in supremum norm by $C\eta^{1/2}$, and $e^{\pm ipx}a'^{\pm}_\mu(x)$ has oscillation at most $C\eta^{1/2}$ between sites at distance at most η. ∎

Proof. It suffices to use the plus sign. For $b=\widehat{h}(p)\varepsilon$, $n=\widehat{\nu}(p)\varepsilon$ and $\gamma=\widehat{R}(p)\widehat{d^{\mathsf{T}}}(p)b$, Section YM3_I: 43.2 gives

$$D_\eta(q_m)b_{\mu,m}=u_{\eta,\mu}(-q_m)n_\mu+d_\mu^\eta(q_m)v_\eta(q_m)\gamma,\qquad \|b\|,|n|,|\gamma|\le C_h|\varepsilon|.\tag{9.xxviii}$$

For $m\neq 0$, $|q_m|\ge\pi|m|_\infty$. The multiplier bounds give $|b_m|\le Cw_3(m)$. Taking a curl cancels the two γ terms and gives

$$f_{\mu\nu,m}=\frac{d_\mu(q_m)u_\nu(-q_m)n_\nu-d_\nu(q_m)u_\mu(-q_m)n_\mu}{D_\eta(q_m)},$$

which obeys the same bound. The zero modes are bounded by the response norm.

To compare zero modes without dividing by $|p|$, use $Qh = 1$, $d^{\mathsf{T}}\nu = 0$ and $\sum_m v(q_m)^2 = 1$. They imply

$$\gamma = v(p)^{-1}\sum_\nu d_\nu(-p)b_{\nu,0}, \qquad n_\mu = u_\mu(-p)^{-1}[D_\eta(p)b_{\mu,0} - d_\mu(p)\sum_\nu d_\nu(-p)b_{\nu,0}].$$

Here $v(p) \geq (2/\pi)^3$ and $|u_\mu(p)| \geq (2/\pi)^4$. Set $A_\mu = \sum_{m\neq 0}|u_\mu(q_m)|^2/D_\eta(q_m)$ and $A_0 = \sum_{m\neq 0} v(q_m)^2/D_\eta(q_m)$. Write $d_\mu = d^\eta_\mu$ and $d^{\mathrm{c}}_\mu(p) = e^{ip_\mu} - 1$. The constraint becomes $Hb_0 = \varepsilon$, where

$$H_{\mu\nu} = \delta_{\mu\nu}[u_\mu(p) + A_\mu D_\eta(p)/u_\mu(-p)] + [d^{\mathrm{c}}_\mu(p)A_0/v(p) - A_\mu d_\mu(p)/u_\mu(-p)]d_\nu(-p).$$

This holds for every ε, so H is invertible and $\|H^{-1}\| \leq C_h$ by the response bound. Factorisation eq. 9.xxiv gives $|A'_\mu - A_\mu| + |A'_0 - A_0| \leq C\eta^2$: on common nonzero modes the reciprocal-denominator and averaging defects are $O(\eta^2)$ times $|u|^2$ or v^2, whose sums are at most one; on added modes $|q_m|^{-2} \leq C\eta^2$. All other entries in H change by $O(\eta)$, with denominators bounded away from zero. Thus $H' - H = O(\eta)$ and $b'_0 - b_0, n' - n, \gamma' - \gamma = O(\eta)$.

Subtract eq. 9.xxviii and the curl formula at a common nonzero mode, one factor at a time. The defects of u, v, d, D^{-1} are bounded respectively by $C\eta|q|w^{(3)}$, $C\eta^2|q|^2w^{(3)}$, $\eta|q|^2$ and $C\eta^2$. Since $|q| \geq 1$ and $\eta|q| \leq 6$, each resulting term is $O(\eta w^{(3)}(m))$. This proves the coefficient comparison. Summation uses eq. 9.xxvi; in particular $\eta\sum_{|m|_\infty\leq N} w^{(3)}(m) \leq \eta(3 + \log\eta^{-1})^3$. The added-mode tail starts at $N + 1 \geq (2\eta)^{-1}$.

Finally use the alias formula eq. 9.xxv. The principal alias contributes the coefficient difference and $(u_1 - 1)\widehat{a}'_m$, bounded by $C\eta(1 + |m|_\infty)w_3(m)$; other aliases are controlled by the added-mode tail. For the oscillation, multiply the coefficients by $\min(2, 19\eta(1 + |m|_\infty))$ and split at N. The same two sums give $C\eta^{1/2}$. □

9.2.5. *The actual averaging maps.* We now compare the field-dependent averages. This requires their actual local definitions: equality of the free multipliers alone would not identify their derivatives. Let ρ_*, C_* be the uniform field radius and bounds from Section YM3_I:43.4, and C_h the response bound from Section YM3_I:43.2. Enlarge C_* to be at least 16, so it also bounds $\ell_j^{-1}|\mathcal{M}^j(e^{i\eta a}) - 1|_\infty/|a|_\infty$. Decrease ρ_* if necessary so that $\rho_* \leq 1$, and put

$$r_1 = \frac{\rho_*}{5C_*}, \qquad r_2 = \frac{\rho_*}{2C_*L}, \qquad r_0 = \frac{r_1}{2C_h}.$$

For $|s_\pm| < r_0$ the fields $a(s) = s_+e^{ipx}\alpha_+ + s_-e^{-ipx}\alpha_-$ and $a'(s)$ have supremum norm less than r_1.

A dilation taking one averaging step to another takes centred blocks, coordinate contours and their orders to the corresponding blocks and contours. It therefore preserves their products and the equations defining the local mean. The holomorphic mean is the branch through the identity, so uniqueness on its defining polydisc preserves this branch as well. Differentiating these identities preserves the corrected differential average and the scalar transport average. In the operator norms the factor λ^3 in the physical pairing occurs on both sides and cancels. Thus the estimates for a step with terminal spacing one apply to every step. Splitting the product after the first step is consequently the exact identity

$$\mathbf{Q}' = \mathbf{Q}^\sharp\mathbf{Q}^{\mathrm{f}}, \qquad \mathbf{R}' = \mathbf{R}^\sharp\mathbf{R}^{\mathrm{f}}. \tag{9.xxix}$$

Here the sharp maps are the remaining height-k averages evaluated at $\mathcal{M}(e^{i\eta' a'(s)})$.

We record explicitly how field bounds give bounds on these jets. If a local operator function $T(s)$ is holomorphic on $|s_\pm| < r_0$ and $\|T(s) - T(0)\| \leq M$, pairing with two finitely supported unit vectors and applying Cauchy's formula bounds its $10, 01, 11$ derivatives by $M/r_0, M/r_0, M/r_0^2$. For real p, the first derivative kernels become periodic after multiplication at the output by $e^{\mp ipx}$, an isometry; the mixed derivative kernel is already periodic. The Fourier symbol norm is bounded by the corresponding operator norm. This proves the same bounds in the real jet norm. Applying this

argument to $T(s) - \widetilde{T}(s)$ also proves the difference version, including the 00 component when it is nonzero.

For a single step and an absorbed field A, the uniform bound and Schwarz's lemma on $|A| < \rho_*/L$ give

$$\|L^{-1}(\mathring{\mathcal{Q}}_{e^{iA}} - \mathring{\mathsf{Q}})\| + \|\mathfrak{a}_{e^{iA}} - \mathfrak{a}\| \le 4C_* L\rho_*^{-1}|A|_\infty.$$

Indeed each difference vanishes at zero and is bounded by $2C_*$ on that ball; apply the scalar Schwarz lemma after pairing. For the intermediate logarithm $A_{j-1} = i^{-1}\log \mathcal{M}^{j-1}(e^{i\eta a})$, the uniform mean bound gives $|A_{j-1}|_\infty \le 2C_*\ell_{j-1}|a|_\infty$. The preceding estimate and the jet Cauchy estimate therefore give

$$\mathbf{A}_j = \mathfrak{a} + \mathbf{a}_j, \quad (\mathbf{a}_j)_{00} = 0, \quad \|\mathbf{a}_j\| \le c_a\ell_{j-1}, \qquad \|\mathbf{Q}^{\mathrm{f}} - Q^{\mathrm{f}}\| + \|\mathbf{R}^{\mathrm{f}} - R^{\mathrm{f}}\| \le C\eta, \tag{9.xxx}$$

where $\mathbf{A}_j$ is the jet of the jth scalar step and $c_a = (8C_*^2L/\rho_*)r_1(2r_0^{-1} + r_0^{-2})$. All these constants depend only on L.

The scalar average also suppresses high Fourier modes:

$$\|\mathbf{R}(1 - P_\Lambda)\|, \|(1 - P_\Lambda)\mathbf{R}^{\mathsf{T}}\| \le 28(1 + c_a)^2\Lambda^{-1/2}. \tag{9.xxxi}$$

To prove this, the product $a^{[n]}$ of the first n free steps has multiplier

$$v^{(n)}(q) = \prod_\nu L^{-n} \sum_{|j|\le (L^n-1)/2} e^{i\eta j q_\nu}$$

. Its output modes have aliases $m = \bar{m} + \ell_n^{-1}e$. The sine identity for this finite sum gives $|v^{(n)}(q_m)|^2 \le 2w^{(3)}(e)^2$. If $|m|_\infty > \Lambda$ and $\ell_n\Lambda \ge 1$, some $|e_\nu| > \ell_n\Lambda - 1/2$; thus the sum of the squares in each alias class is at most $768/(\ell_n\Lambda)$. Cauchy–Schwarz gives $\|a^{[n]}(1-P_\Lambda)\| \le 28(\ell_n\Lambda)^{-1/2}$. When $\ell_n\Lambda < 1$ this follows instead from the contraction bound. Expand $\mathbf{R} = \prod_{j=k}^1(\mathfrak{a} + \mathbf{a}_j)$. There are only the free term, terms with one insertion, and terms with two insertions: three zero-00 jets have zero product. Applying the free tail bound to the rightmost free segment gives

$$\|\mathbf{R}(1 - P_\Lambda)\| \le 28\Lambda^{-1/2}\left(1 + c_a\sum_j \ell_{j-1}^{1/2} + c_a^2\sum_{i<j}\ell_{j-1}\ell_{i-1}^{1/2}\right).$$

Since $\sum_j \ell_{j-1}^{1/2} \le (\sqrt{L} - 1)^{-1} \le 1$, this proves the first bound; transposition proves the second.

It remains to compare the sharp maps with the unprimed maps. For the finest step define

$$\mathcal{N}(A) = i^{-1}\log \mathcal{M}(e^{iA}) - \mathsf{Q}A.$$

On $|A|_\infty \le r_2$, the mean differs from the identity by at most $\rho_*/2$, and hence the logarithm is defined. The logarithm and the linear average bound $\mathcal{N}$ by 3. Its value and first derivative vanish at zero because $\mathcal{M}'(1) = \mathsf{Q}$. Schwarz's lemma for a double zero gives $|\mathcal{N}(A)|_\infty \le 3r_2^{-2}|A|_\infty^2$. The relation between the ordinary and corrected linear averages is $\mathsf{Q} = \mathring{\mathsf{Q}} - d_0\mathsf{n}$, with d_0 the unscaled coarse difference. Therefore, for $A = \eta' a'(s)$,

$$a^\sharp = Q^{\mathrm{f}}a'(s) - \eta^{-1}d_0\mathsf{n}(\eta' a'(s)) + \eta^{-1}\mathcal{N}(\eta' a'(s)). \tag{9.xxxii}$$

The operator n annihilates constant forms; its paths have at most $3L/2$ bonds within one block. Subtract a constant value on each such block and use the oscillation bound in Lem. 9.36. The middle term in eq. 9.xxxii is bounded by $C\eta^{1/2}$; the last is bounded by $C\eta$. The first differs from $a(s)$ by $C\eta^{1/2}$. Thus $|a^\sharp - a(s)|_\infty \le C\eta^{1/2}$ on the fixed bidisc. Both fields remain within ρ_*, and $|a(s)| < \rho_*/2$.

The uniformly bounded full averages are Lipschitz between such fields, with constant $4C_*/\rho_*$. To see this, restrict to the complex line from the first field to the second. If their distance is at most

$\rho_*/2$, Schwarz's lemma on the disc of radius $\rho_*/(2|a_1-a_0|)$ gives the bound; for a larger distance the bound $2C_*$ suffices. Applying the difference Cauchy estimate on the bidisc now proves

$$\|\mathbf{Q}^\sharp-\mathbf{Q}\|+\|\mathbf{R}^\sharp-\mathbf{R}\|\le C\eta^{1/2}. \tag{9.xxxiii}$$

9.2.6. *The local multiplication terms.* We next compare the curvature and covariant-derivative jets. The comparison uses their actual first derivatives, including the lattice shifts. For a periodic scalar function a, let

$$M_h[a;\varpi;\chi_1,\chi_2]_{m,\widetilde{m}}=\widehat{a}_{m-\widetilde{m}}e^{ih(\varpi+2\pi(m-\widetilde{m}))\cdot\chi_1}e^{ihq_{\widetilde{m}}\cdot\chi_2},$$

where the coefficient index is read modulo $h^{-1}\mathbb{Z}^3$, $\varpi=\pm p$ and $\chi_1,\chi_2\in\{-1,0,1\}^3$. In position space this is multiplication by $e^{i\varpi\cdot(x+h\chi_1)}a(x+h\chi_1)$ followed by the shift $x\mapsto x+h\chi_2$. Schur's inequality gives $\|M_h\|\le\sum_m|\widehat{a}_m|$. If $3\Lambda\le N$, it also gives

$$\begin{aligned}\|(1-P_{2\Lambda})M_\eta P_\Lambda\|&\le\sum_{|m|_\infty\ge\Lambda}|\widehat{a}_m|,\\ \|P'_{2\Lambda}(M_{\eta'}J-JM_\eta)P_\Lambda\|&\le\sum_{|m|_\infty\le3\Lambda}|\widehat{a}'_m-\widehat{a}_m|+90\eta\Lambda\sum_m|\widehat{a}_m|.\end{aligned} \tag{9.xxxiv}$$

For the first inequality an output above 2Λ and an input below Λ have coefficient index at least Λ, even after periodic reduction, since $N+1-\Lambda\ge\Lambda$. For the second the coefficient indices lie below 3Λ, so no reduction is needed. Subtract the two phase factors and use $|e^{iu}-e^{iv}|\le|u-v|$; their difference is at most $90\eta\Lambda$. The row and column sums prove the claim.

Here are the particular multiplication terms in the Hessian. Write $\theta=\operatorname{ad}t_3$, so $\|\theta\|=2$ and $\theta^{\mathsf{T}}=-\theta$. On the commuting background $U=\exp(ihat_3)$,

$$(e_U\lambda)(b)=h^{-1}(e^{iha(b)\theta}-1)\lambda(b_+).$$

Thus its first derivative is $ia_\mu\theta$ followed by the shift he_μ. Its transpose has shifts $(-e_\mu,-e_\mu)$ and coefficient $-i\theta$. The mixed derivative is $-ha^+_\mu a^-_\mu\theta^2$ followed by that shift, hence $O(h)$.

On a positively oriented $\mu\nu$ plaquette enumerate the bonds as bottom, right, top, left, with signs $(1,1,-1,-1)$ and origins $o_1=o_4=0$, $o_2=e_\mu$, $o_3=e_\nu$. For their field values a_i, put $\vartheta_1=0$, $\vartheta_2=a_1$, $\vartheta_3=a_1+a_2-a_3$, $\vartheta_4=h\,\mathrm{d}a$. Then

$$E_Ub=h^{-1}\sum_i\varsigma_i(e^{ih\vartheta_i\theta}-1)b_i,\qquad z_U=\cos(h^2da).$$

The first derivative of E in this plane is

$$i\theta\{a_\mu(x)b_\nu(x+he_\mu)-[a_\mu(x)+a_\nu(x+he_\mu)-a_\mu(x+he_\nu)]b_\mu(x+he_\nu)-h\,\mathrm{d}a(p)b_\nu(x)\}.$$

This is a list of five multiplication terms per plane. Its mixed derivative is $-h\sum_i\varsigma_i\vartheta_i^+\vartheta_i^-\theta^2b_i$, which is $O(h)$. The first derivatives of z vanish and $z_{11}=-h^4f_+f_-$.

Finally the remaining curvature term has quadratic form

$$(b,f_Ub)=-ih^3\sum_p h^{-2}\sin(h^2da(p))\sum_{i<j}\varsigma_i\varsigma_j\langle b_i,\theta e^{ih(\vartheta_j-\vartheta_i)\theta}b_j\rangle.$$

Its first derivative is the symmetrisation of the matrix with entries $-i\varsigma_i\varsigma_j da\,\theta$ from b_j to b_i. Its multiplication terms have offsets $(-o_i,o_j-o_i)$, together with the transposed offsets $(-o_j,o_i-o_j)$, and coefficients of norm at most one. There are 36 terms over the three planes. The mixed derivative comes from differentiating the exponential once and da once; it is a sum of $h(da)(\vartheta_j-\vartheta_i)\theta^2$ and its polarisation, so is $O(h)$. The sine factor has no quadratic term.

Consequently the first jets of e,e^{T},E,f are finite matched lists of the operators in eq. 9.xxxiv, with coefficients $a^\pm_\mu$, $f^\pm_{\mu\nu}$ or $hf^\pm_{\mu\nu}$. The lists and shifts are identical at both heights. Lem. 9.36 controls their coefficient sums and differences. Their mixed jets are $O(\eta)$ and $O(\eta')$; this last observation is what permits the comparison without introducing additional second-order background coefficients.

9.2.7. *Defects of the determinant data.* Let $1 \le \Lambda$ and $3\Lambda \le N$, and put $\epsilon_1 = \Lambda^{-1/2} + \eta\Lambda + \eta^{1/2}$. All constants in the following comparison are independent of k, p, Λ in their stated ranges.

Lemma 9.37. For each of the vector and scalar data in eq. 9.xxii,

$$\mathrm{df}(\kappa', \kappa) + \mathrm{df}(x', x) + \|x'(1 - JJ^*)\| + \mathrm{df}(y', y) \le C\epsilon_1.$$

■

Proof. The free roots $G^{1/2}, g^{1/2}$ have diagonal entries $g_\eta(q_m)$. Denote this diagonal matrix by g_*, and the matrix on the same modes with entries $g_{\eta'}(q_m)$ by $g_\flat$. The elementary bounds imply

$$\|(1 - P_\Lambda)g_*\|, \|(1 - P_\Lambda)g_\flat\| \le 2/\Lambda, \quad \|G'^{1/2}(1 - JJ^*)\| \le 4\eta, \quad \|g_\flat - g_*\| \le 6\eta; \tag{9.xxxv}$$

the corresponding scalar statements hold as well. Indeed a nonzero mode has $|q_m| \ge \pi|m|_\infty/2$, and an added mode has $|m|_\infty \ge N + 1 \ge (2\eta)^{-1}$.

Consider the free blocks $D_1 = dG^{1/2}$, $D_0 = d^\mathsf{T}G^{1/2}$, $D_g = dg^{1/2}$ and their transposes. They are diagonal in the Fourier modes, with matrices of at most 3×3 entries 0 or $\pm d^\eta_\mu(\pm q)g_\eta(q)$ and the same positions and signs at both heights. Consequently

$$\|D\|, \|D'\| \le 3, \quad D'J - JD = J\delta_D, \quad \|\delta_D\| \le 6, \quad \|\delta_D P_\Lambda\| \le 528\eta\Lambda. \tag{9.xxxvi}$$

The last estimate follows from $|d^{\eta'}_\mu g_{\eta'} - d^\eta_\mu g_\eta| \le 8\eta|q|$ and $|q| \le 22\Lambda$ on these modes.

For a multiplication jet $M = E, e^\mathsf{T}, f$ or e, let $X = MG^{1/2}$ or $Mg^{1/2}$ with the appropriate spaces. We claim

$$\|X'(1 - JJ^*)\| \le C\eta, \quad \|(1 - P_{2\Lambda})X\| \le C\epsilon_1, \quad \mathrm{df}(X', X) \le C\epsilon_1. \tag{9.xxxvii}$$

The first bound is the added-mode root bound times the bounded multiplication jet. For the second split the input at P_Λ: the high input costs C/Λ, the low input followed by the high output costs $C\Lambda^{-1/2}$ by eq. 9.xxxiv and the coefficient tail, and the mixed jet costs $C\eta$. For the third use

$$X'J - JX = (M'J - JM)g_\flat + JM(g_\flat - g_*).$$

The second term is $O(\eta)$. Split the first at P_Λ on the input and $P'_{2\Lambda}$ on the output. The high input costs C/Λ; the low input and high output are bounded by the preceding tail argument at the two heights; on the remaining modes eq. 9.xxxiv costs $C(\eta^{1/2} + \eta\Lambda)$. This proves eq. 9.xxxvii.

Now put $X_1 = EG^{1/2}$, $X_0 = e^\mathsf{T}G^{1/2}$, $X_f = fG^{1/2}$ and $X_g = eg^{1/2}$. The Hessian formula gives

$$\begin{aligned}\kappa &= D_1^\mathsf{T}(z-1)D_1 + D_1^\mathsf{T}zX_1 + X_1^\mathsf{T}zD_1 + X_1^\mathsf{T}zX_1 + G^{1/2}X_f \\ &\quad + D_0^\mathsf{T}X_0 + X_0^\mathsf{T}D_0 + X_0^\mathsf{T}X_0, \\ \kappa_g &= D_g^\mathsf{T}X_g + X_g^\mathsf{T}D_g + X_g^\mathsf{T}X_g.\end{aligned}$$

The terms containing $z - 1$ are $O(\eta^4)$ at either height, since its only nonzero component is 11. For $D^\mathsf{T}X$, split X at $P_{2\Lambda}$ and use eq. 9.xxxvi on the low modes and the output tail in eq. 9.xxxvii on the other modes. This gives defect $C\epsilon_1$. For $X^\mathsf{T}D$ apply the transpose rule to $D^\mathsf{T}X$, retaining its added-mode tail, bounded by $3\|X'(1 - JJ^*)\|$. Products $X^\mathsf{T}X$ and $G^{1/2}X_f$ follow from the product and transpose rules. These observations prove the claims for κ and κ_g.

For x the exact split eq. 9.xxix gives

$$\begin{aligned}x'J - x &= (\mathbf{Q}^\sharp - \mathbf{Q})\mathbf{Q}^\mathrm{f}G'^{1/2}J \\ &\quad + \mathbf{Q}(\mathbf{Q}^\mathrm{f} - Q^\mathrm{f})G'^{1/2}J + \mathbf{Q}(Q^\mathrm{f}G'^{1/2}J - G^{1/2}).\end{aligned}$$

The first two terms cost $C\eta^{1/2}$ and $C\eta$. The last difference is diagonal, with entries $u_{1,\mu}g_{\eta'} - g_\eta$, of modulus at most 11η: use the bounds for $u_1 - 1$, $g_{\eta'}$ and $g_{\eta'} - g_\eta$. The scalar version uses $v_1 g_{\eta'} - g_\eta$, bounded by 8η. The added-mode tails of x' and x'_g are bounded by 4η times the uniform averaging bound. This proves the assertions concerning x.

For y, set $X' = G'^{1/2}\mathbf{D}'\mathbf{R}^{\mathrm{f}\mathsf{T}}$ and $X = G^{1/2}\mathbf{D}$. Then

$$\begin{aligned} y' - Jy &= X'(\mathbf{R}^{\sharp} - \mathbf{R})^{\mathsf{T}} + (X' - JX)P_{\Lambda}\mathbf{R}^{\mathsf{T}} \\ &\quad + (X' - JX)(1 - P_{\Lambda})\mathbf{R}^{\mathsf{T}}. \end{aligned}$$

The first term costs $C\eta^{1/2}$; the last costs $C\Lambda^{-1/2}$ by eq. 9.xxxi. To estimate the middle term, expand $X' - JX$ as

$$\begin{aligned} &[G'^{1/2}d'R^{\mathrm{f}\mathsf{T}} - JG^{1/2}d] \\ &\quad + G'^{1/2}\mathbf{D}'(\mathbf{R}^{\mathrm{f}} - R^{\mathrm{f}})^{\mathsf{T}} \\ &\quad + G'^{1/2}e'(R^{\mathrm{f}\mathsf{T}} - J) + [G'^{1/2}e'J - JG^{1/2}e]. \end{aligned}$$

The second line is $O(\eta)$ by eq. 9.xxx. For the free aliases at $|m|_{\infty} \le \Lambda$, the principal coefficient satisfies $|v_1(q_m) - 1| \le 11\eta\Lambda$. For any other alias, some component of $q_{m'}$ has modulus at least $\pi\eta^{-1}$; the sine quotient defining s_1 then gives $|v_1(q_{m'})| \le 6\eta\Lambda$. There are at most L^3 aliases. Hence

$$\|(R^{\mathrm{f}\mathsf{T}} - J)P_{\Lambda}\| \le 11L^{3/2}\eta\Lambda.$$

The same calculation, with the normalized derivative, bounds the first line by $187\sqrt{3}L^{3/2}\eta\Lambda$; on the principal alias use $|d^{\eta'}g_{\eta'} - d^{\eta}g_{\eta}| \le 8\eta|q|$, and on the others use $|d^{\eta'}g_{\eta'}| \le 1$. The third line's first term is thus $O(\eta\Lambda)$. For its last term, split the output at $P'_{2\Lambda}$. On the low output, the multiplication comparison and the root difference give $C(\eta^{1/2} + \eta\Lambda + \eta)$; on the high output the left root gives C/Λ. This proves $\mathrm{df}(y', y) \le C\epsilon_1$ and completes the proof. □

9.2.8. *Stability of the finite expression.* The coarse traces in eq. 9.xxii have dimension at most nine. Its one remaining fine trace is controlled instead by the three propagators in it. Write $\|\cdot\|_3$ for the Schatten norm of exponent three on a Fourier fibre.

Lemma 9.38. Suppose two data for eq. 9.xxii satisfy the following bounds, with $A, C_3, \iota \ge 1$ and $\epsilon > 0$. At every real $\varkappa \in K$ the diagonal propagators satisfy

$$\|\mathsf{G}\|_3, \|\mathsf{G}'\|_3 \le C_3, \qquad \|\mathsf{G}'(1 - JJ^*)\|_3, \|(\mathsf{G}'J - J\mathsf{G})J^*\|_3 \le \epsilon.$$

The jets $\kappa, x, y, B_t, B^{\flat}_{\tau}$ have norms at most A at both heights, for $t, \tau \in [0,1]$, and the 00 inverses of $xB^{\flat}_0x^{\mathsf{T}}$ have norms at most ι. Finally

$$\mathrm{df}(\kappa', \kappa), \quad \mathrm{df}(x', x), \quad \|x'(1 - JJ^*)\|, \quad \mathrm{df}(y', y) \le \epsilon.$$

Then, with $a = \max(A, C_3)$,

$$|\Phi[\mathsf{G}', \kappa', x', y'] - \Phi[\mathsf{G}, \kappa, x, y]| \le 2600\iota^3a^{11}\epsilon.$$

■

Proof. The propagator operator norms are at most C_3, and their defect is at most ϵ. The transpose defects of x, y are at most $2\epsilon, \epsilon$, respectively; for y its input is coarse and there is no added input space. The product and inverse rules therefore give

$$\mathrm{df}(\omega', \omega) \le 5A\epsilon, \quad \mathrm{df}(B'_t, B_t) \le 6A^3\epsilon, \quad \mathrm{df}(B^{\flat\prime}_{\tau}, B^{\flat}_{\tau}) \le 7A^3\epsilon.$$

A product of n displayed factors has defect at most $7na^{n+2}\epsilon$.

For an endomorphism of a coarse fibre, the trace of its 11 component is bounded by nine times its jet norm. The six coarse-trace products in eq. 9.xxii have respectively $3, 3, 5, 5, 7, 7$ factors. Remembering that $\int_0^1 (1 - \tau)\,\mathrm{d}\tau = 1/2$, their total difference is at most

$$9(42a^5 + 70a^7 + 49a^9)\epsilon \le 1449a^9\epsilon.$$

For the logarithmic term put $C = xB^{\flat}_0x^{\mathsf{T}}$. Then $\|C\|, \|C'\| \le a^3$ and $e := \|C' - C\| \le 21a^5\epsilon$. The inverse identity bounds the difference of C_{00}^{-1} by $\iota^2 e$. Subtracting the two products in the definition of ℓ one factor at a time gives at most $6\iota^3a^6e$ before the trace, hence $1134\iota^3a^{11}\epsilon$ after integration.

It remains to compare $W = \mathsf{G}B_0\mathsf{G}B_0\mathsf{G}B_\tau^\flat$. Insert the projection onto the common modes in the trace:

$$\operatorname{tr} W'_{11} - \operatorname{tr} W_{11} = \operatorname{tr}[W'_{11}(1 - JJ^*)] + \operatorname{tr}[(W'J - JW)_{11}J^*].$$

In the first term cycle the projection next to a propagator. Hölder's inequality for the three Schatten-three factors bounds it by $\epsilon C_3^2 A^3 \le a^5\epsilon$. In the second term telescope the six factors. Insert $J^*J = 1$ between common-space factors so that each term is a product on the primed space. The three propagator-difference terms are bounded by $3a^5\epsilon$ in total. The three inverse-jet differences, bounded by $6A^3\epsilon, 6A^3\epsilon, 7A^3\epsilon$, contribute at most $19a^8\epsilon$. In each case the other three propagator factors, including a difference when present, are estimated in Schatten norm; the jet product norm sums the finitely many 11 terms. The remaining integral has weight $(1-\tau)^2$, whose integral is $1/3$. Its contribution is therefore at most $8a^8\epsilon$. Adding the three bounds and using $a, \iota \ge 1$ proves the lemma. □

Proposition 9.39. There is a finite $C_{\mathrm{re}} = C_{\mathrm{re}}(L)$ such that

$$|F_{k+1}(p) - F_k(p)| \le C_{\mathrm{re}} L^{-k/3} \quad (k \ge 1,\ p \in \mathbb{R}^3,\ |p_\mu| \le \delta).$$

■

Proof. Choose $\Lambda = \lfloor \eta^{-2/3}/4 \rfloor$. For $\eta^{-1} \ge 13$ this is at least one, is at least $\eta^{-2/3}/8$, and satisfies $3\Lambda \le N$. For the last assertion the function $(y-1)/2 - 3y^{2/3}/4$ is positive at $y = 13$ and increasing thereafter. Thus

$$\epsilon_1 \le (\sqrt{8} + 1/4 + 1)\eta^{1/3} \le 5\eta^{1/3}.$$

Lem. 9.37 supplies the operator defects in Lem. 9.38 with $\epsilon = C\eta^{1/3}$. The uniform inverse and jet bounds were established before eq. 9.xxii.

We verify the required Schatten bounds explicitly. Both propagators have eigenvalues $g_h(q_m)^2$, each repeated at most nine times. Since $g_h \le \min(1, 2/|m|_\infty)$ away from zero and the shell $|m|_\infty = n$ has $24n^2 + 2 \le 26n^2$ points,

$$9\sum_m g_h(q_m)^6 \le 9\left(1 + 64 \cdot 26 \sum_{n\ge 1} n^{-4}\right).$$

This gives a uniform C_3. The sum over added modes starts beyond $N \ge 6\eta^{-1}/13$, and its tail is $O(\eta^3)$. On common modes eq. 9.xxvi gives $|g_{\eta'}^2 - g_\eta^2| \le 3\eta^2$; there are η^{-3} such modes, so their cubed Schatten difference is at most $9\eta^{-3}(3\eta^2)^3 = 243\eta^3$. Both relevant Schatten norms are therefore $O(\eta)$, and hence at most $C\eta^{1/3}$.

Apply Lem. 9.38 to the vector and scalar data and use Lem. 9.35. This proves the stated bound. In particular the choice

$$C_{\mathrm{re}} := \sup_{k\ge 1,\ |p_\mu|\le\delta,\ p\in\mathbb{R}^3} L^{k/3}|F_{k+1}(p) - F_k(p)|$$

is finite and depends only on L. □

9.2.9. *Recovering the coefficient and summing the increments.* The real-momentum bound does not by itself bound derivatives. We use the uniform complex bound from eq. 9.xix as follows. If f is holomorphic on $|p_\mu| < \delta$, bounded by M there and by ϵ on the real cube $|p_\mu| \le \delta/4$, then

$$|\partial_1\partial_2 f(0)| \le 17\sqrt{\epsilon M}\,\delta^{-2}. \tag{9.xxxviii}$$

Indeed expand $f(s,t,0) = \sum c_{ij}s^i t^j$, with $|c_{ij}| \le M\delta^{-i-j}$. The centred mixed difference at the four points $(\pm h, \pm h, 0)$ divided by $4h^2$ retains exactly the odd powers of s, t. For $h \le \delta/4$ the terms other than c_{11} total at most $3Mh^2\delta^{-4}$. Therefore

$$|c_{11}| \le \epsilon h^{-2} + 3Mh^2\delta^{-4}.$$

When $0 < \epsilon < M$, choose $h^2 = \delta^2\sqrt{\epsilon/M}/16$. When $\epsilon \geq M$, the ordinary Cauchy bound suffices; when $\epsilon = 0$, let h tend to zero. These cases prove eq. 9.xxxviii.

Proof of Thm. AC. Apply eq. 9.xxxviii to $F_{k+1} - F_k$, with $M = 2C_{\mathrm{rem}}$ and $\epsilon = C_{\mathrm{re}}L^{-k/3}$. By eq. 9.xix,

$$C_{\mathrm{inc}} := \sup_{k\geq 1} L^{k/6}|r_{k+1} - r_k| \leq 17(2C_{\mathrm{rem}}C_{\mathrm{re}})^{1/2}\delta^{-2} < \infty.$$

Define

$$C_\circ = \max\{C_{\mathrm{inc}} + 2C_E, c_2^{\mathrm{loop}}\}, \qquad C_{\mathrm{geo}} = \frac{2C_\circ}{1 - L^{-1/6}}. \tag{9.xxxix}$$

These are functions of L alone: each is defined from the Gaussian coefficients, their fixed massive term and their fixed determinant remainder. Lem. 9.34 gives

$$|c_{k+1} - c_k| \leq C_\circ L^{-k/6} \qquad (k \geq 1).$$

The same bound holds at $k = 0$ because $c_0 = 0$ and $|c_1| \leq c_2^{\mathrm{loop}}$. For $n > k \geq 0$ it follows that

$$|c_n - c_k| \leq \sum_{j\geq k} C_\circ L^{-j/6} = \tfrac{1}{2}C_{\mathrm{geo}}L^{-k/6}.$$

Thus c_k converges to a real c_∞, with the asserted error, and $|c_\infty| \leq c_2^{\mathrm{loop}}$. Also $r_k \to c_\infty - E_\infty$.

The defining identity for c_k gives $b_l^0 = c_l - L^{-1}c_{l-1}$ for $l \geq 1$. Therefore

$$|b_{l+1}^0 - b_l^0| \leq C_\circ(1 + L^{-5/6})L^{-l/6} \leq (1 - L^{-1/6})C_{\mathrm{geo}}L^{-l/6}.$$

This includes $l = 1$ by the already verified $k = 0$ bound. The limit is $(1 - L^{-1})c_\infty$; summing this last increment estimate from l to infinity gives $|b_l^0 - b_\infty^0| \leq C_{\mathrm{geo}}L^{-l/6}$, as claimed. □

9.2.10. *The bound used in the coupling flow.* The coupling recursion needs a bound on the increments, not only on their accumulated coefficients. The following elementary form also applies to the coefficients of a general group once their stated bound has been established.

Lemma 9.40. Let $L \geq 13$, let $c_0 = 0$ and $|c_k| \leq c_2^{\mathrm{loop}}$ for all $k \geq 0$, and put $b_l^0 = c_l - L^{-1}c_{l-1}$. If $B_\sharp \geq (1 + L^{-1})c_2^{\mathrm{loop}} + 1$ and $\bar{B} = LB_\sharp/(L - 1)$, then

$$B_\sharp \geq 1, \qquad |b_l^0| \leq B_\sharp - 1 \quad (l \geq 1), \qquad B_\sharp < \bar{B} \leq \tfrac{13}{12}B_\sharp.$$

■

Proof. The triangle inequality gives $|b_l^0| \leq (1 + L^{-1})c_2^{\mathrm{loop}} \leq B_\sharp - 1$. Since $c_2^{\mathrm{loop}} \geq 0$, we have $B_\sharp \geq 1$. Finally $1 < L/(L - 1) \leq 13/12$, which proves the bounds on $\bar{B}$. □

10. Renormalized Wilson loops

Fix a physical torus T_ℓ and a label $b > 0$, with $a_K = L^{-K}$ and $\beta = L^K b$. Let C be a self-avoiding closed polygon with finitely many axis-parallel pieces, corners in a fixed coarse lattice $a_{i_0}\mathbb{Z}^3/\ell\mathbb{Z}^3$, and contained in a ball of radius less than $\ell/4$. Write $|C|$ for its physical perimeter and $W_C = \frac{1}{2}\operatorname{tr}U(C)$. The same notation applies to a finite family of polygons at positive mutual distances, with the product of their traces and the sum of their perimeters.

Sketch I (A perimeter renormalization). If $\kappa = 3|C|/(4\pi b) < 1/2$, then

$$a_K^{-\kappa}\langle W_C\rangle_K \longrightarrow \mathcal{W}(C)$$

along the full cutoff sequence on the fixed torus. For coordinate rectangles the limit is strictly positive. This is a statement about expectations; it does not assert convergence of the renormalized sharp loop as a random variable. ■□

Sketch of the argument. Keep the line insertion when integrating one fluctuation scale. The leading factor is the expectation of the corresponding Gaussian line, and can be written

$$\lambda_n(C) = \exp\left[-\frac{3}{2b} a_K(X_C, \mathcal{C}_{(n)} X_C)\right].$$

Here $d^{\mathrm{T}} X_C$ is the loop current and $\mathcal{C}_{(n)}$ is the curvature covariance removed at the nth block step. The orthogonal covariance decomposition makes these factors multiply. Their logarithmic self-energy divergence has coefficient $\kappa \log(1/a_K)$; the remaining Coulomb energy has a finite part. Thus one power $a_K^{-\kappa}$ removes the divergence.

The remaining insertion is compared with the holonomy of the critical background. An ordered-product estimate controls the change of a noncommuting line when its individual bonds change. The fluctuation step gives an error with a summable power a_d^{ϑ} for every $\vartheta < 1/2$. After renormalization its sum has exponent $\vartheta - \kappa$, explaining the perimeter restriction.

To take the cutoff limit of the background loop, first fix a block depth d. The compatible block laws converge in total variation. A thread is a family of coarse fields compatible with the block averaging maps. The large-field tails are summable over d, so almost every thread is eventually in the regular domain. Over a regular block field at this fixed depth, the critical backgrounds on successively finer cutoff lattices give a smoothed loop in the limit $K \to \infty$. The line-integration estimate then compares the sharp loop expectation with a fixed-depth smoothed expectation. Take $K \to \infty$ first and $d \to \infty$ afterwards. Reflection positivity and bounds for small rectangles give the positive lower bound for rectangles.

Smoothed loops at different depths use different critical refinements of the thread, and are different random functions. The argument gives limits of means, not an almost sure limit as the depth increases. Also, the unrenormalized sharp trace has a semicircle limit law, whereas its mean tends to zero. Dividing by that mean does not produce a tight family. Mixed functions with local fields can be treated by keeping the field sources in the line integration, but require the stronger perimeter and support-separation conditions of that argument.

10.1. **Wilson loops on a fixed torus.** We consider three limits of lattice Wilson loops. Background-smoothed loops converge along almost every block-field thread. Sharp polygonal loops have renormalised expectation limits in the stated perimeter range. The unrenormalised trace has a semicircle limit, and division by its vanishing mean produces divergent variance.

These are different limiting statements. The sharp-loop theorem concerns expectations on a fixed torus; its stronger mixed-function conclusions have their own perimeter and separation restrictions. The proof retains the ordered-product estimate, the inserted-line renormalisation step and the summability restriction on the normalised errors.

10.1.1. *Notation and hypotheses. Constants.* Use a common real numerical choice as in Prop. YM3_I:69.1. We use the odd blocking factor $L \geq 13$; $M = L^{m'}$ with $m' \geq 5$; $r \in \mathbb{N}$; $\gamma \in (0, \frac{1}{2}]$ with $\log \gamma^{-2} \geq \log 4 > 1$; $A_0 > 0$, $\mathfrak{p}_0 \in \mathbb{N}$ and the threshold $\varepsilon(t) = A_0 t^{-1/2} (\log t)^{\mathfrak{p}_0}$, $t > 1$, of YM3_I:(634), which decreases for $\log t > 2\mathfrak{p}_0$; $B_3 \geq 1$; $c_\rho = \max\{2B_3, C_H A_1/((1 - L^{-3/2})A_0)\}$ with the constants C_H, A_1 of this formula (§2.3.1), so that $2 \leq 2B_3 \leq c_\rho < \infty$; $c' \in (0, \frac{1}{250})$, $\mathsf{C}^{\Sigma}_{\mathrm{maj}} \geq 0$ and $c_{\mathrm{low}} \geq 24$.

The number $t_\natural \geq \gamma^{-2}$ is the least real number from which on the three smallness conditions (cut1)–(cut3) of §2.3.1 on $\varepsilon(t)$ hold; they hold at every $t \geq t_\natural$, and there

$$2c_\rho \varepsilon(t) \leq \varepsilon_{\mathcal{W}} \alpha_0 \leq \bar{\varepsilon} \leq 1$$

, $\bar{\varepsilon}$ the radius of Thm. YM3_I:E in the setting of Section YM3_I:52.7 and $\varepsilon_{\mathcal{W}}$, α_0 the numbers of (cut1), formed from the constants alone (equation YM3_I:(2134)). $\Theta_\star \geq t_\natural$ is the least real number from which on the five conditions of the comparison of two consecutive cutoffs (Prop. YM3_I:113.9)

hold; both exist (the threshold choices in Sections YM3_I:113 and YM3_I:115, with [7, Cor. B.13]), and $2c_\rho\varepsilon(t) < 1/200L^2$ for $t \geq \Theta_\star$.

The lowered choice of the constants (Def. 4.47: A_1 multiplied by $\lambda'_\natural$, $B_\sharp$ not smaller, A_0 and the constants fixed before A_1 kept) gives the number $c'_\star$, which is $c_\star$ of eq. 4.xx at that choice, and the numbers $\mathsf{Y}_1^\natural, \dots, \mathsf{Y}_4^\natural$ of eq. 4.xxi formed from the two choices; we put $c^\natural = c'_\star/(4 \cdot 49^3) > 0$ and $Y^\natural = \max_{i \leq 4} \mathsf{Y}_i^\natural$. All these numbers depend on the constants alone.

The one-torus setting: a label $\mathsf{b} \geq \gamma^{-2}$ and a side $\ell = 2L^m$, $m \in \mathbb{N}$, with $L^m \geq M(\log \mathsf{b})^r$, equivalently $M\,Q(\mathsf{b}) \leq L^m$ (Lem. 10.1).

Fields and laws. $T_\ell = (\mathbb{R}/\ell\mathbb{Z})^3$, $a_d = L^{-d}$ and $\Lambda_d = a_d\mathbb{Z}^3/\ell\mathbb{Z}^3$ (Section YM3_I:3). $\mathcal{V}_d$ is the set of the fields on the bonds of Λ_d with values in $G = \mathrm{SU}(2)$, $V(\bar e) = V(e)^{-1}$, with the gauge transformations $V^v(e) = v(e_-)V(e)v(e_+)^{-1}$ (Section YM3_I:3); $V(\Gamma)$ is the ordered product along a path Γ, $\partial V(p)$ the ordered product along the boundary of a plaquette p, and $\varpi(V) = \max_p |\partial V(p) - \mathbf{1}|$, in the operator norm . $\mathcal{M} : \mathcal{V}_{d+1} \to \mathcal{V}_d$ is the block averaging map (equations YM3_I:(37)–(39)); it is Borel and gauge covariant, and one map serves all cutoffs.

The Wilson measure $\langle\cdot\rangle_K$ on $\mathcal{V}_K$ has the density $Z^{-1}e^{-\beta\sum_p a_p}$ with respect to the Haar measure dU, where $a_p = 1 - \frac12 \operatorname{tr} \partial U(p)$ and $\beta = L^K\mathsf{b}$. μ_d^K is the law of $\mathcal{M}^{K-d}U$ under $\langle\cdot\rangle_K$, and $\mu_d = \lim_K \mu_d^K$ in total variation along the full sequence of cutoffs (Cor. 2.8). $\mathcal{X} = \prod_{d\geq 0}\mathcal{V}_d$ carries the product topology and the coordinates π_d; $\mathsf{Thr} = \{V_\bullet : V_d = \mathcal{M}(V_{d+1})\ \forall d\}$ is the set of the threads; $\mathcal{F}_d = \sigma(\pi_0, \dots, \pi_d)$; $(\theta_v V_\bullet)_d = V_d^{v|\Lambda_d}$ for $v : \bigcup_d \Lambda_d \to G$.

Finally μ denotes the Borel probability measure on $\mathcal{X}$ whose image under $V_\bullet \mapsto (V_0, \dots, V_D)$ is, for every D, the image of μ_D under $V \mapsto (\mathcal{M}^D V, \dots, \mathcal{M}V, V)$; it exists and is unique (Prop. 10.2(a)).

Depths. We write $x_{[d]} = L^d\mathsf{b}$ and $\varepsilon_{[d]} = \varepsilon(x_{[d]})$, with brackets, so as to tell these depth quantities from the run quantities x_t, ε_t of the frame below. In $\mathcal{V}_d$ let $I_d = \{\varpi < 2\varepsilon_{[d]}\}$ and $J_d = \{\varpi < c_\rho\varepsilon_{[d]}/B_3\}$; $I_d \subset J_d$ since $c_\rho \geq 2B_3$. Put

$$R_d = \frac{2\ell^3 L^{3d}}{M^3}\, e^{-c' A_0^2 (\log x_{[d]})^{2\mathfrak{p}_0} + (\mathsf{C}^\Sigma_{\mathrm{maj}} + 2 + c_{\mathrm{low}} + \frac92 \log \mathsf{b})\ell^3}, \qquad R_d^\natural = \ell^3 L^{3d} e^{-c^\natural A_0^2 (\log x_{[d]})^{2\mathfrak{p}_0}}.$$

A depth d is admitted if $x_{[d]} \geq \Theta_\star$. $d_\natural$ and d_Θ are the least integers $d \geq 1$ with $x_{[d]} \geq t_\natural$, resp. $x_{[d]} \geq \Theta_\star$, and $s_\natural$ is the least integer $s \geq 0$ with $\log x_{[s]} \geq Y^\natural$; they depend on b and the constants, not on ℓ, and $d_\Theta \geq d_\natural$. $\tau_0(V_\bullet)$ is the least $D \geq d_\natural$ with $V_d \in I_d$ for all $d \geq D$ (∞ if there is none); the starting depth is $\tau = \max\{\tau_0, d_\Theta\}$, and $\Omega = \mathsf{Thr} \cap \{\tau < \infty\}$.

Loops and their smoothing. A lattice loop C of $\Lambda_{d'}$ with N bonds has the length $|C| = Na_{d'}$; for $j \geq d'$, $C^{\langle j\rangle}$ arises by replacing $j - d'$ times every bond by the straight path of L bonds of the next lattice, and $W_C(V) = \frac12 \operatorname{tr} V(C^{\langle j\rangle})$ for $V \in \mathcal{V}_j$. For $W \in \mathcal{V}_d$ and $K \geq d$ the backgrounds over W form the set $\mathcal{C}_K(W)$ of the $U \in \mathcal{V}_K$ with $\mathcal{M}^{K-d}U = W$ that are critical for $\sum_p a_p$ on this constraint set and regular at the radius ε_{v} (Defs. YM3_I:9.3 and YM3_I:9.5); $\mathcal{G}_K^d$ is the group of the gauge transformations of Λ_K that equal $\mathbf{1}$ on Λ_d, and $w_C^K(W) = W_C(U)$ for $U \in \mathcal{C}_K(W)$.

For $X \in \mathcal{V}_s$ let $H_s^K(X) = \mathcal{M}^{K-s-1}U_+ \in \mathcal{V}_{s+1}$, U_+ an element of $\mathcal{C}_K(X)$ for which the average $\mathcal{M}^{K-s-1}U_+$ is in the axial gauge on Λ_s ([7, §B.2]); $H_s^\infty = \lim_K H_s^K$ and $H_{d,n}^\bullet = H_{d+n-1}^\bullet \circ \cdots \circ H_d^\bullet$. The smoothed loop is $W_C^{[d]}(W) = \lim_n W_C(H_{d,n}^\infty(W))$; $F_C^{[d]}$ is $W_C^{[d]}$ on J_d and 0 off J_d; $Y_{d,C} = F_C^{[d]} \circ \pi_d$ and $Y_{d,C}^K = (\mathbf{1}_{J_d} w_C^K) \circ \pi_d$. A curve is a continuous $\hat\gamma : [0,1] \to \mathbb{R}^3$ of finite length $|\gamma|$ with $\hat\gamma(1) - \hat\gamma(0) \in \ell\mathbb{Z}^3$; a shadow $\Gamma^{(j)}$ of it on Λ_j cuts it into at most $|\gamma|L^j + 2$ pieces of sup-diameter $\leq a_j$, takes sites of Λ_j within $\frac12 a_j$ of the cut points and joins them by shortest paths of Λ_j; $W_\gamma^{[d]}(W) = \lim_n W_{\Gamma^{(d+n)}}(H_{d,n}^\infty(W))$, $w_\gamma^K(W) = W_{\Gamma^{(K)}}(U)$ for $U \in \mathcal{C}_K(W)$, and F_γ is $W_\gamma^{[d]}$ on J_d and 0 off it.

Polygons. A polygon is a closed self-avoiding lattice path with finitely many axis-parallel pieces (maximal straight parts) and corners in some Λ_{i_0}, lying in a ball of radius $< \ell/4$. A family $\mathsf{C} = (C_1, \dots, C_{\#\mathsf{C}})$ consists of polygons at pairwise positive distances; $|\mathsf{C}| = \sum_i |C_i|$ and

$W_{\mathsf C}=\prod_i W_{C_i}$. For a polygon or a family, $n_{\mathsf C}$ is the number of pieces, $s_{\mathsf C}$ the least length of a piece, $\delta_{\mathsf C}$ the least distance of two different parallel pieces of one polygon, $\mathfrak s_{\mathsf C}=n_{\mathsf C}^2+|\mathsf C|^2/\delta_{\mathsf C}^2$, $\mathfrak d_{\mathsf C}=\min\{s_{\mathsf C},\delta_{\mathsf C}\}$, $c_{\mathsf C}\in(0,1]$ a chord-arc constant ($|x-y|\ge c_{\mathsf C}\operatorname{arc}(x,y)$ for x,y on one C_i, arc the shorter arc), $\delta'_{\mathsf C}$ the least sup-distance of two different polygons (∞ for one polygon), and $\rho_{\mathsf C}>0$ the least $|x-y|_\infty$ over the pairs of points that lie on two different polygons, or on one polygon with both arcs between them of length $\ge\frac12 s_{\mathsf C}$.

We put $k=\frac{3}{4\pi}$, $\kappa=k|\mathsf C|/\mathsf b$ and $\mathcal W_K(\mathsf C)=a_K^{-\kappa}\langle W_{\mathsf C}\rangle_K$; for a square $n_{\mathsf C}=4$ and one may take $c_{\mathsf C}=\frac12$, numbers that dilations do not move.

Numbers of the free field. By this we mean numbers formed from the Gaussian field alone, in which no measure of the interacting theory enters. For loops C, D with corners in Λ_K, $\mathcal I_a(C,D)=a\sum_\mu\sum_{y,y'}j^C_\mu(y)\,g_N(\frac{y-y'}{a})\,j^D_\mu(y')$ is the lattice Coulomb energy at the spacing $a=a_K$, $N=\ell/a$, j^C the current of C and g_N the Green function of the lattice Laplacian on $\mathbb Z^3/N\mathbb Z^3$ without its zero mode. G_ℓ is the Green function of $-\Delta$ on T_ℓ without its zero mode, $\mathcal I_\ell(C,D)=\oint_C\oint_D G_\ell(x-y)\,\mathrm dx\cdot dy$ for loops at positive distance, $\mathcal J_\ell(C)$ the limit as $u\to0$ of the part $|x-y|>u$ of $\oint_C\oint_C G_\ell(x-y)\,\mathrm dx\cdot dy$ less $\frac{|C|}{2\pi}\log\frac1u$, and $\mathcal J(C)$ the same with $\frac{1}{4\pi|x|}$ in the place of G_ℓ; $\mathrm{Lg}(s)=(1+\log(e+s))^{3/2}$. With Euler's constant γ_E, $S(z)=\frac{2\sin(z/2)}{z}$, $p=q+2\pi n$, $\mathsf N=\sum_{n\in\mathbb Z^2}\frac{S(p_2)S(p_3)}{p_2^2+p_3^2}$ and $\mathsf D$ the same with S^2 in the place of S:

$$c_{\mathrm{lat}}=\frac{5\log2+2\gamma_E}{4\pi}=0.36766\ldots,\quad c_r=\frac{1}{4\pi^2}\int_{[-\pi,\pi]^2}\Big[\frac{\mathsf N^2}{\mathsf D}-\frac{1}{4\sin^2\frac{q_2}{2}+4\sin^2\frac{q_3}{2}}\Big]dq_2\,\mathrm dq_3=$$
$$-0.03149\ldots,\quad c_Q=c_{\mathrm{lat}}+c_r=0.33617\ldots$$

Action density; mirrors. $\mathcal O(f)=a_K^{-1}\sum_p f(x(p))a_p$ for real Lipschitz f, $x(p)$ the centre of p, and $\mathcal W_K(\mathsf C;f_1,\ldots,f_n)=a_K^{-\kappa}\langle W_{\mathsf C}\prod_q(\mathcal O(f_q)-\langle\mathcal O(f_q)\rangle_K)\rangle_K$. $S^{\mathsf T}_{m,K}(f_1,\ldots,f_m)$ is the joint cumulant of $\mathcal O(f_1),\ldots,\mathcal O(f_m)$ under $\langle\cdot\rangle_K$, the truncated m-point function of the action density; for pairwise disjoint supports its limit $S_m(f_1,\ldots,f_m)$ exists along the full sequence YM3_I:(2212). A coordinate mirror is $\theta x=x-2(x_\mu-c)e_\mu$ with $c\in\frac12 a_{i_0}\mathbb Z$; it fixes the planes $x_\mu=c$ and $x_\mu=c+\frac{\ell}{2}$, it is a site mirror at the cutoff K if $c\in a_K\mathbb Z$ and a bond mirror otherwise, and $S^+_\theta=\{c\le x_\mu\le c+\frac{\ell}{2}\}$. It acts on functions of the field by $(\theta F)(U)=F(\theta U)$ and on test functions by $(\theta f)(x)=f(\theta x)$, so that $\theta W_C=W_{\theta C}$ and $\theta\mathcal O(f)=\mathcal O(\theta f)$.

Three free photon fields. X_C is the exact bare 2-cochain with $d^{\mathsf T}X_C=j_C$; in lattice units $a_K(X_C,X_D)=\mathcal I_{a_K}(C,D)$. The three free photon fields at the label $\mathsf b$ are the curvatures $F^a=\beta^{-1/2}\Pi\xi^a$, $a=1,2,3$, ξ^a independent standard Gaussian variables on the bare plaquettes and Π the orthogonal projection onto the exact 2-cochains; $\phi^a_C=(F^a,X_C)$ is the flux of F^a through C, so that $\mathbb E\,\phi^a_C\phi^{a'}_D=\delta_{aa'}\mathcal I_{a_K}(C,D)/\mathsf b$; their loop is $W^{\mathrm{fr}}_C=\prod_{a=1}^3\cos\phi^a_C$, their action density $\mathcal O^{\mathrm{fr}}(f)=a_K^{-1}\sum_p f(x(p))\,\frac12\sum_a(F^a(p))^2$, and their block curvatures $F^{(d)}$ are the curvatures of the block averages at the depth d, linear images of F.

The frame of the renormalisation group (Thm. AE and Thm. AD(5)). A reference coupling $g\le\gamma$ with $g^{-2}\ge4\bar B$, $\log g^{-2}\ge\max\{6\mathfrak p_0,A_0^{-1/\mathfrak p_0},(1+\log LM)^{1/2}\}$ and $g(\log g^{-2})^{\mathfrak p_0}\le c_{\mathrm{th}}$, $c_{\mathrm{th}}>0$ formed from the constants (true for small g); the flow bound $B_\sharp$ of the run and $\bar B=\frac{L}{L-1}B_\sharp$; $\mathsf I(g)=(g^{-2}+\bar B,L(g^{-2}+B_\sharp)-\bar B]$ (Section YM3_I:74); a side $\ell\in\mathfrak L(g)$, the set of the sides at g (equation YM3_I:(1528)); a compact $\mathsf J\subset\mathsf I(g)$, $\mathsf b\in\mathsf J$, $\mathsf b_-=\min\mathsf J$, $\mathsf b_+=\max\mathsf J$.

The run is $x_0=\beta$, $x_t=x_{t-1}/L-u_t$ with $|u_t|\le B_\sharp$, so that $|x_t-L^{K-t}\mathsf b|\le\bar B$ and $\frac23 x_{[d]}\le x_{K-d}\le x_{[d]}+\bar B$; $g_t=x_t^{-1/2}$, $\varepsilon_t=\varepsilon(x_t)$, $\mathfrak b_t=A_1(\log x_t)^{\mathfrak p_0}$ and $\delta_t=g_t\mathfrak b_t$. The level t is the lattice Λ_{K-t}, with the tame set $\mathsf O_t=\{\varpi<2\varepsilon_t\}$. Thm. YM3_I:D, the inductive description of the effective densities, with the actual-history conclusion of Cor. YM3_I:69.4, provides the following objects. Finite positive Borel measures m_t on $\mathcal V_{K-t}$, carried by $\mathsf O_t$ for $t\ge1$, with $m_0=e^{-\beta\sum_p a_p}dU$; bounded Borel functions $\mathsf h_t,\mathsf h_t^{\mathrm w}\ge0$ on $\mathcal V_{K-t+1}$, the weights of the step t without and with a new large

field, with $\int G\,\mathrm{d}m_t = \int (G\circ\mathcal{M})\,\mathsf{h}_t\,\mathrm{d}m_{t-1}$; for every bounded Borel F on $\mathcal{V}_K$ quotients $\Phi^{(t)}[F]$ with $\Phi^{(0)}[F]=F$, $|\Phi^{(t)}[F]|\le \sup|F|$ and $\int G\,\Phi^{(t)}[F]\,\mathrm{d}m_t = \int (G\circ\mathcal{M})\,\Phi^{(t-1)}[F]\,\mathsf{h}_t\,\mathrm{d}m_{t-1}$ for bounded Borel G, out of which functions of coarser block fields pull, $\Phi^{(t)}[F\cdot(G\circ\mathcal{M}^{t'})] = \Phi^{(t)}[F]\cdot(G\circ\mathcal{M}^{t'-t})$ for $t\le t'$; and, for gauge invariant F and every last level $0\le \bar t\le K$,

$$Z\langle F\rangle_K = \int \Phi^{(\bar t)}[F]\,\mathrm{d}m_{\bar t} + \sum_{t\le \bar t}\int \Phi^{(t-1)}[F]\,\mathsf{h}_t^{\mathrm{w}}\,\mathrm{d}m_{t-1}, \qquad \mathsf{w}_t := \frac{1}{Z}\int \mathsf{h}_t^{\mathrm{w}}\,\mathrm{d}m_{t-1} \le \mathsf{V}_\ell C_{\mathrm{r}}(p) L^{-p(K-t)} \quad (p\ge 0), \tag{10.i}$$

the last inequality being the *rarity bound*; it gives $\sum_{t'\le t} L^{p(K-t')}\mathsf{w}_{t'} \le 2\mathsf{V}_\ell C_{\mathrm{r}}(p+1)L^{-(K-t)}$. Here $C_{\mathrm{r}}(p)$ is formed from the constants and p, and V_ℓ, free of K, is at most $2M^{-3}\ell^3 e^{B\ell^3}$ on every side of $\mathfrak{L}(g)$, with B formed from the constants and g, and equals 1 on the least sides of $\mathfrak{L}(g)$ when g is small. We take for $\mathfrak{su}(2)$ the real space of the Hermitian traceless 2×2 matrices with the scalar product $\langle X,Y\rangle = \frac12 \operatorname{tr} XY$, whose norm $|X|$ is the operator norm on it; t_1,t_2,t_3 is an orthonormal basis, with $t_a t_b + t_b t_a = 2\delta_{ab}\mathbf{1}$ and $\sum_a t_a^2 = 3\cdot\mathbf{1}$, the components of X are $X^a = \langle t_a, X\rangle$, a complex 2×2 matrix is written $x = x^0\mathbf{1} + \sum_a x^a\, i t_a$, and $\mathrm{R}(h)\in SO(3)$, $\mathrm{R}(h)_{ab} = \langle t_a, h t_b h^{-1}\rangle$, is the rotation of $h\in \mathrm{SU}(2)$.

For $W\in \mathsf{O}_{t+1}$ there are a chart $B\mapsto V_W(B)$, from the real $\mathfrak{su}(2)$-valued 1-forms B on a set of bonds of Λ_{K-t} (the fluctuation field of the step) with $|B|<\mathfrak{b}_t$ at every bond into $\mathcal{V}_{K-t}$, with $\mathcal{M}V_W(B) = W$ and $V_W(B)(b) = e^{ig_t B(b)}V_W(0)(b)$ at the fluctuation bonds, and the kernel ν_W of one step, a complex measure of mass 1 on $\{|B|<\mathfrak{b}_t\}$, with the *disintegration*

$$\int H(\mathcal{M}V)\,G(V)\,\mathsf{h}_{t+1}(V)\,m_t(\mathrm{d}V) = \int H(W)\Big[\int G(V_W(B))\,\nu_W(\mathrm{d}B)\Big] m_{t+1}(\mathrm{d}W)$$

for bounded Borel G, H; so $\Phi^{(t+1)}[F](W) = \int \Phi^{(t)}[F](V_W(B))\,\nu_W(\mathrm{d}B)$ for m_{t+1}-almost every W. On $\{|B|<\mathfrak{b}_t\}$ the kernel is $\nu_W = \Xi_W^{-1} e^{\mathsf{E}_W}\mu_{C_W}$: the Gaussian measure μ_{C_W} of the covariance C_W of the step at the background triple of W, which is formed from a background U over W with $\mathcal{M}^t U$ in the axial gauge and from the current of U (the triple defined before equation YM3_I:(827) and §5.3.1), times e^{E_W}, E_W the sum of two local potentials, the interaction and the part of the bare array of couplings, each with a norm with quadratic growth at most $c\,\delta_t$, c formed from the constants, and Ξ_W the number that makes the mass 1 (eq. 5.xxxii and Prop. 5.31). $W_{\mathsf{C}}^{(t)}(V) = W_{\mathsf{C}}(U)$ for a background U over $V\in\mathcal{V}_{K-t}$, and U_+ denotes a background over $W\in\mathsf{O}_{t+1}$. Π_n is the orthogonal projection of the bare 2-cochains onto the linearised curvatures of the backgrounds of n block steps (Π_0: onto the exact ones), $\mathcal{C}_{(n)} = \Pi_{n-1} - \Pi_n$, and

$$\lambda_n = \prod_i \exp\Big[-\frac{3}{2\mathfrak{b}}\,a_K(X_{C_i}, \mathcal{C}_{(n)}X_{C_i})\Big] \in (0,1], \qquad \Xi_t = \prod_{s\le t}\lambda_s, \qquad v_{K,d}(C) = a_K(X_C, \Pi_{K-d}X_C).$$

10.1.2. *The perimeter factor and the summable error.* Fix the physical torus and a permitted polygon C. At each fluctuation step the loop insertion has a leading Gaussian factor. If X_C is the exact plaquette cochain whose boundary is the loop current and $\mathcal{C}_{(n)}$ is the covariance of that step, this factor is

$$\lambda_n(C) = \exp\left[-\frac{3}{2b}a_K(X_C, \mathcal{C}_{(n)}X_C)\right].$$

The covariance pairing is the line's self-energy at that scale. Its singular part accumulates along the straight segments. Multiplying the one-step factors gives the perimeter power a_K^κ, with $\kappa = 3|C|/(4\pi b)$. Corners and interactions between separated segments enter the finite comparison terms and the errors, rather than changing this leading power.

The existence argument must control the remainder after dividing by that vanishing factor. Choose $\vartheta\in(\kappa,1/2)$. The inserted-line estimate gives a one-step error bounded by a constant times

$L^{-\vartheta j}$ at depth j. The accumulated normalization can cost $L^{\kappa j}$. Thus the error in the normalized expectation is controlled by

$$C \sum_{j \geq d} L^{-(\vartheta - \kappa) j},$$

where d is the retained coarse depth. This sum tends to zero as d increases. The strict restriction $\kappa < 1/2$ supplies room to choose ϑ; it is used at the step that sums the normalized errors. The source then compares successive cutoffs with these summable errors. A pointwise Gaussian approximation without this normalized summability would not establish the claimed renormalized limit.

The background loop provides the coarse comparison. The laws of the block fields are compatible under averaging, so they define a law on sequences of compatible fields. The large-field probability at depth d is bounded by R_d, with $\sum_d R_d < \infty$. The event that a sequence is regular at every depth above D has complement of probability at most $\sum_{d \geq D} R_d$. Almost every sequence is therefore eventually regular. One may then choose the minimal background over its field at depth D and study its loop as the finer cutoff is removed. The convergence at this fixed depth is uniform on the admitted regular set.

Two different limits occur here. At fixed D, the background loop becomes a bounded function of the block field V_D. Different choices of D give different functions of the sequence. The full limiting thread law permits their expectations and joint expectations to be compared. The estimates for these means do not state that the functions themselves converge almost surely as D increases. The sharp-loop calculation, meanwhile, removes its perimeter factor and produces a limit of expectations. It does not provide a random holonomy obtained by dividing a sharp trace by its vanishing mean.

Localization of the background uses gauge invariance in a concrete way. If the loop lies a positive distance inside a region, compare its background with the one constructed using only data in that region. In suitable gauges, each bond discrepancy is exponentially small in the distance to the boundary measured in coarse lattice units. For unitary matrices,

$$\left| \operatorname{tr} \prod_i A_i - \operatorname{tr} \prod_i B_i \right| \leq 2 \sum_i |A_i - B_i|.$$

Summing over the $|C|/a_K$ fine bonds cancels the fine spacing in the background estimate. The remaining error depends on the coarse depth and decays exponentially with the boundary distance. This is why the comparison survives removal of the fine cutoff while retaining a fixed loop and region.

The free-field loop, the background loop, and the sharp Wilson trace consequently play different roles in the calculation. The free field computes the self-energy; the background carries the coarse dependence; the inserted-line estimate controls their comparison with the interacting expectation. The detailed formulas below retain these roles and the stronger restrictions needed when local fields are inserted as well.

10.1.3. *Main results.*

Theorem AD (Smoothed loops along almost every thread)**.** Assume the one-torus setting.

(1) *The starting depth.* τ is a Borel function on $\mathcal{X}$ with values in $\{d_\Theta, d_\Theta + 1, \dots\} \cup \{\infty\}$; for every integer $D \geq d_\Theta$, $\{\tau \leq D\} = \bigcap_{d \geq D} \pi_d^{-1}(I_d)$, a countable intersection of open sets; Ω is Borel and $\mu(\Omega) = 1$. For $D \geq d_\Theta$: $\mu(\tau > D) \leq \sum_{d \geq D} R_d$, which is at most $\frac{L}{L-1} R_D$ if $c' A_0^2 \log x_{[D]} \geq 2$; both bounds exceed 1 as long as $c' A_0^2 (\log x_{[D]})^{2\mathfrak{p}_0} \leq 26 \ell^3$. For $D \geq \max\{d_\Theta, s_\natural\}$: $\mu(\tau > D) \leq \sum_{d \geq D} R_d^\natural$, which is at most $\frac{L}{L-1} R_D^\natural$ if $c^\natural A_0^2 \log x_{[D]} \geq 2$; and $R_D^\natural \leq \eta \in (0, 1]$ as soon as

$$D \log L \geq \max\Big\{ \frac{6}{c^\natural A_0^2}, \Big[\frac{2(3 \log \ell + \log \frac{1}{\eta})}{c^\natural A_0^2} \Big]^{1/2} \Big\}.$$

So τ is at most of the order $(\log \ell)^{1/2}$ on events of probability at least $1 - \frac{L}{L-1}\eta$.

(2) *Loops from the starting depth on.* If $K > d$ and $\varpi(W) < \bar\varepsilon$, $\bar\varepsilon$ the radius of Thm. YM3_I:E in the setting of Section YM3_I:52.7, then $\mathcal{C}_K(W)$ is one non-empty orbit of $\mathcal{G}_K^d$; and $\{\varpi < 2c_\rho \varepsilon_{[d]}\} \subset \{\varpi < \bar\varepsilon\}$ if $x_{[d]} \geq t_\natural$. Hence for $d \geq \tau_0(V_\bullet)$ and $K \geq d+1$ the backgrounds over V_d form one non-empty orbit, and $w_C^K(V_d)$ is one number for every closed path C of $\Lambda_{d'}$, $d' \leq K$. If $d \geq \tau(V_\bullet)$, then d is admitted, $V_d \in I_d \subset J_d$, and for every lattice loop C of $\Lambda_{d'}$, $d' \geq d$,

$$\lim_{K\to\infty} w_C^K(V_d) = W_C^{[d]}(V_d) = \lim_{n\to\infty} W_C(H_{d,n}^\infty(V_d)), \qquad |w_C^K(V_d) - W_C^{[d]}(V_d)| \leq |C| L^d \mathsf{c}_{\mathrm{f}} L^{-(K-d)/3} \quad (K \geq K_2),$$

with the numbers c_{f}, K_2 of Prop. 10.4. For a curve the same holds with w_γ^K, $W_\gamma^{[d]}$ and the bound $(|\gamma| L^d + 2)\mathsf{c}_{\mathrm{g}} L^{-(K-d)/3}$, and $W_\gamma^{[d]}$ does not depend on the shadows. At these V_d the smoothed loop, as a function on J_d, is continuous and gauge invariant with values in $[-1,1]$, and $0 \leq 1 - W_C^{[d]}(V_d) \leq \frac{1}{2}(2B_3 \varpi(V_d) L^{2d} s_1 s_2)^2$ if C bounds a coordinate rectangle with sides s_1, s_2. Ω depends on none of d, C, γ, K. The statements at finite cutoffs hold at every depth $d \geq \tau_0$, the limits as $K \to \infty$ at every depth $d \geq \tau$. No step of (2) and (3) uses the thread property $V_d = \mathcal{M}(V_{d+1})$: they hold at every point of $\{\tau < \infty\} \subset \mathcal{X}$.

(3) *Random variables.* Let d be admitted and C a lattice loop of $\Lambda_{d'}$, $d' \geq d$. Then $Y_{d,C}$ is Borel on $\mathcal{X}$, $[-1,1]$-valued, $\sigma(\pi_d)$-measurable, θ_v-invariant, and continuous on the open set $\pi_d^{-1}(J_d)$, where it is $W_C^{[d]}(V_d)$; for $K \geq \max\{2, d+1, d'\}$, $Y_{d,C}^K$ is Borel and $\sup_{\mathcal{X}} |Y_{d,C}^K - Y_{d,C}| \to 0$; $\{\tau \leq d\}$ is Borel, θ_v-invariant and contained in $\pi_d^{-1}(I_d)$, and $Y_{d,C} = \lim_K w_C^K(V_d)$ on it. With $Y_{d,C} := Y_{d,C^{\langle d\rangle}}$ for $d > d'$, $(Y_{d,C})_{d \geq d_\Theta}$ is an $(\mathcal{F}_d)$-adapted sequence with values in $[-1,1]$ on $(\mathcal{X}, \mu)$ whose members of index $\geq \tau$ are, on Ω, the smoothed loops along C. The same holds for rectifiable closed curves, with one shadow fixed on each Λ_j, with $F_\gamma \circ \pi_d$ and w_γ^K, $K \geq \max\{2, d+1\}$.

(4) *Means.* For admitted d,

$$\int Y_{d,C}\, \mathrm{d}\mu = \int F_C^{[d]}\, \mathrm{d}\mu_d = \lim_K \langle F_C^{[d]}(\mathcal{M}^{K-d}U)\rangle_K = \lim_K \langle (\mathbf{1}_{J_d} w_C^K)(\mathcal{M}^{K-d}U)\rangle_K,$$

with the rates of Prop. 10.4(d); the same holds for curves. For $d \geq d_\Theta$,

$$\Big| \int Y_{d,C}\, \mathrm{d}\mu - \int \mathbf{1}\{\tau \leq d\} Y_{d,C}\, \mathrm{d}\mu \Big| \leq \sum_{d' \geq d} R_{d'},$$

which is at most $\frac{L}{L-1} R_d$ if $c' A_0^2 \log x_{[d]} \geq 2$; if G_d is Borel on $\mathcal{V}_d$ with $|G_d| \leq 1$ and $G_d = W_C^{[d]}$ on J_d, then $G_d(V_d) = Y_{d,C}$ for all $d \geq \tau$ and $\int |G_d(V_d) - Y_{d,C}|\, \mathrm{d}\mu \leq R_d$. For admitted depths $d_1, \dots, d_n$ and loops C_i of depths $\geq d_i$, $\langle \prod_i F_{C_i}^{[d_i]}(\mathcal{M}^{K-d_i}U)\rangle_K \to \int \prod_i Y_{d_i,C_i}\, \mathrm{d}\mu$.

(5) *Locality.* In the frame, let d be admitted and at least the depth of the corners of C, $M_1 a_d \leq \frac{\delta_1}{8}$, $K > d$ and $2\varepsilon_{K-d} \leq \epsilon_{\mathrm{lc}}$; let Ω' be a slab of the depth d with faces in $M_1 a_d(\mathbb{Z} + \frac{1}{2})$, or the intersection of two such slabs, with $\mathrm{dist}(C, T_\ell \setminus \Omega') \geq \frac{\delta_1}{2}$, and let $Y_{K,d}(V) = W_C(U')$, U' the localised background formed from the bonds of V in Ω' (M_1, δ_{lc}, C_{lc}, ϵ_{lc} are the constants of the localisation of backgrounds, Prop. YM3_I:53.5 and Section YM3_I:53.4). Then for every $V \in \mathsf{O}_{K-d}$

$$|w_C^K(V) - Y_{K,d}(V)| \leq 2\mathsf{C}_{\mathrm{lc}} \hat\varepsilon_d |C| L^d e^{-\delta_{\mathrm{lc}}\delta_1 L^d/8}, \qquad \hat\varepsilon_d = A_0 (\tfrac{2}{3} x_{[d]})^{-1/2} \log^{\mathfrak{p}_0}(x_{[d]} + \bar B),$$

a bound free of K that falls faster than every power of a_d; so two fields of J_d that lie in O_{K-d} for all large K and agree on the bonds of Ω' have smoothed loops within twice this bound.

(6) *Every label.* Let $\mathsf{b} > 0$, $\ell = 2L^m$ with $m \in \mathbb{N}$, and let $s \geq 0$ be an integer with $L^s\mathsf{b} \geq \gamma^{-2}$ and $M\,Q(L^s\mathsf{b}) \leq L^{m+s}$ (Lem. 10.1). Then (1)–(4) hold at (ℓ, b) for the depths $d \geq s$, the objects of the depth d being those of $(L^s\ell, L^s\mathsf{b})$ at the depth $d-s$, carried back by $x \mapsto L^{-s}x$, with $R_{d-s}[L^s\ell, L^s\mathsf{b}]$ in the place of R_d and with the depths $d_\natural$, d_Θ, $s_\natural$ of $(L^s\ell, L^s\mathsf{b})$ increased by s; the backgrounds so obtained are backgrounds of the lattices of side ℓ. Indeed, $x \mapsto L^{-s}x$ maps Λ_{d-s} of side $L^s\ell$ onto Λ_d of side ℓ, one graph with $2L^{m+d}$ sites in each direction, and the cutoff $K-s$ onto the cutoff K, with the same number $K-d$ of block steps above the depth d; it carries the Wilson measure at $(K-s, L^s\ell, L^s\mathsf{b})$ onto the one at (K, ℓ, b), as $\beta = L^{K-s}(L^s\mathsf{b})$.

The block averaging map with its averaging radius $1/200L^2$ (equations YM3_I:(37)–(39)); the constraint sets, the action $\sum_p a_p$, criticality and the regularity at the radius ε_{v}, measured in the units of the level, with the bare spacing $L^{-(K-d)}$ (Defs. YM3_I:9.3 and YM3_I:9.5); the residual gauge groups and the minimal orbit (§5.2); the axial gauge of the averaged backgrounds ([7, §B.2]); the radius $\bar{\varepsilon}$ and the lattices of Thm. YM3_I:E in the setting of Section YM3_I:52.7, whose lattice hypotheses are imposed through the number of the sites in each direction at the levels used; and the plaquette size ϖ: all are formed from these graphs, from $K-d$ and from the constants, and hold neither the side, nor the label, nor the spacing.

And $x_{[d]} = L^d\mathsf{b} = L^{d-s}(L^s\mathsf{b})$, so that $\varepsilon_{[d]}$, I_d and J_d at (ℓ, b) are the sets of $(L^s\ell, L^s\mathsf{b})$ at the depth $d-s$. So $\mathcal{C}_K(W)$, H_d^K, w_C^K and $W_C^{[d]}$ at (ℓ, b) are the carried objects.

■□

Theorem AE (The renormalised sharp loop)**.** Work in the frame; C is a family of polygons and $\kappa = k|\mathsf{C}|/\mathsf{b}$.

(1) *Ceiling.* If $\kappa < \frac{1}{2}$, that is $|\mathsf{C}| < \frac{2\pi}{3}\mathsf{b} = 2.094\ldots\mathsf{b}$, then $\sup_{K>s_0} |\mathcal{W}_K(\mathsf{C})| < \infty$; precisely, for $\vartheta \in (\kappa, \frac{1}{2})$ and $K > d \geq s_0$,

$$\Big|\mathcal{W}_K(\mathsf{C}) - a_K^{-\kappa}\Xi_{K-d}\frac{1}{Z}\int W_{\mathsf{C}}^{(K-d)}\,\mathrm{d}m_{K-d}\Big| \leq \mathsf{E}'_d := \epsilon_1 L^{-(\vartheta-\kappa)(d+1)} + (c_\Xi L^\kappa + \epsilon_1)\,2\mathsf{V}_\ell C_{\mathrm{r}}(\kappa+1)L^{-d}, \tag{10.ii}$$

where $\epsilon_1 = c_\Xi C_\eta/(1-L^{\kappa-\vartheta})$, C_η and s_0 are the numbers of Prop. 10.6(a) at the exponent ϑ, and c_Ξ is the number of Prop. 10.7(c). So $\sup_{K>s_0}|\mathcal{W}_K(\mathsf{C})| \leq c_\Xi L^{\kappa s_0} + \mathsf{E}'_{s_0}$, and $|\Phi^{(t)}[W_{\mathsf{C}}]| \leq (c_\Xi + \epsilon_1)L^{-\kappa t}\ m_t$-almost everywhere for $t \leq K - s_0$. On tori with $\ell \leq 1.48\,\mathsf{b}$ every coordinate rectangle has $\kappa < \frac{1}{2}$. If $\kappa \geq \frac{1}{2}$, then $|\langle W_{\mathsf{C}}\rangle_K| \leq c_\vartheta a_K^\vartheta$ for every $\vartheta < \frac{1}{2}$.

(2) *The limit.* If $\kappa < \frac{1}{2}$, then $\mathcal{W}(\mathsf{C}) = \lim_K \mathcal{W}_K(\mathsf{C})$ exists along the full sequence of cutoffs. Let $F_{\mathsf{C}}^{[d]} = \prod_i W_{C_i}^{[d]}$ on J_d and 0 off J_d, and

$$\mathsf{I}_d = \lim_K a_K^{-\kappa}\Xi_{K-d} = a_d^{-\kappa}\exp\Big[-\frac{3}{2\mathsf{b}}\big((c_{\mathrm{lat}} - c_Q)|\mathsf{C}| + O_{\mathsf{C},\ell}(a_d(1+\log^+(1/a_d)))\big)\Big].$$

Then for all $d \geq \max\{s_0, d_0\}$, d_0 the depth of Prop. 10.4(e),

$$\Big|\mathcal{W}(\mathsf{C}) - \mathsf{I}_d\int F_{\mathsf{C}}^{[d]}\,\mathrm{d}\mu_d\Big| \leq \mathsf{E}_d := \mathsf{E}'_d + 2c_\Xi\mathsf{V}_\ell C_{\mathrm{r}}(\kappa+1)L^{-d} + 2c_\Xi L^{\kappa d}R_d \to 0,$$

and so $|a_d^{-\kappa}\int F_{\mathsf{C}}^{[d]}\,\mathrm{d}\mu_d - \mathcal{W}(\mathsf{C})\,e^{\frac{3}{2\mathsf{b}}(c_{\mathrm{lat}}-c_Q)|\mathsf{C}|}| \leq c\,(\mathsf{E}_d + a_d(1+\log^+(1/a_d))/\mathsf{b})$, whether or not $\mathcal{W}(\mathsf{C})$ vanishes; here $c_{\mathrm{lat}} - c_Q = 0.03149\ldots$ and $\frac{3}{2}(c_{\mathrm{lat}} - c_Q) = 0.047235\ldots$. One power of the spacing renormalises; no further divergent counterterm is needed. The ball on which $F_{\mathsf{C}}^{[d]}$ is formed may be taken at the label b or at the label $\mathsf{b} - \bar{B}$: the two means differ by at most R_d. If $k|\mathsf{C}|/\mathsf{b}_- < \frac{1}{2}$, then $\mathcal{W}_K \to \mathcal{W}$ uniformly on J, $\mathcal{W}$ is continuous on J, and $\mathcal{W}_K$ formed at labels $\mathsf{b}_K \in \mathsf{J}$ with $\mathsf{b}_K \to \mathsf{b}$, b_K in the exponent too, tends to $\mathcal{W}$ at b.

(3) *Every torus, every label.* For every integer m, $\ell = 2L^m$, every $\mathsf{b} > 0$ and every family with $\kappa < \frac{1}{2}$ the limit $\mathcal{W}^{(\ell,\mathsf{b})}(\mathsf{C})$ exists along the full sequence, and $\mathcal{W}^{(L^z\ell,L^z\mathsf{b})}(L^z\mathsf{C}) = L^{-z\kappa}\mathcal{W}^{(\ell,\mathsf{b})}(\mathsf{C})$ for every integer z.

(4) *Positivity; perimeter law.* Wherever they exist, the limits have the properties of limits listed in Prop. 10.9. For every coordinate rectangle C with $\kappa < \frac{1}{2}$ there are $0 < c \le c_2$ with $c \le \mathcal{W}_K(C) \le c_2$ for all large K, uniformly in $\mathsf{b} \in \mathsf{J}$ if $k|C|/\mathsf{b}_- < \frac{1}{2}$; so $\mathcal{W}(C) > 0$. For rectangles $C_i = n_1 a_i \times n_2 a_i$ with corners in Λ_i, n_1, n_2 fixed, $d_i = \lceil (\frac{4}{3} - \frac{u}{2})i \rceil$ with $u \in (0, \frac{1}{3})$, and a fixed $\vartheta \in (0, \frac{1}{2})$, let $s_0(i) = \max\{i, s_0'\}$ and $C_\eta(i) = C(\mathsf{J})L^{\vartheta i}$ be the numbers of the one-step bound linear in $|C_i|/\mathsf{b}$ (Prop. 10.6(b)) for C_i, $C(\mathsf{J})$ depending on the constants, ϑ, J, n_1 and n_2, and let i be so large that $(n_1^2 + n_2^2)a_i^2 < \ell^2/4$ (so that C_i is a polygon), $\kappa(C_i) \le k|C_i|/\mathsf{b}_- < \vartheta$, $s_0(i) \le d_i$ and $\theta_{d_i} \le 1$, with the number θ_{d_i} below, which tends to 0 as $i \to \infty$ (so that the factor $1 - \frac{1}{2}\theta_{d_i}^2$ is at least $\frac{1}{2}$). Then

$$e^{-\frac{3}{2\mathsf{b}_-}\eta_i}(1 - \tfrac{1}{2}\theta_{d_i}^2)(1 - 2\mathsf{V}_\ell C_{\mathrm{r}}(1)L^{-d_i}) - \mathsf{E}'_{d_i} \le \mathcal{W}_K(C_i) \le c_\Xi L^{\kappa d_i} + \mathsf{E}'_{d_i} \qquad (K > d_i,\ \mathsf{b} \in \mathsf{J}),$$

where E'_{d_i} is the number of eq. 10.ii formed with $C_\eta(i)$ and $s_0(i)$ in the place of the numbers of Prop. 10.6(a), $\eta_i = \sup_{K \ge i} |c_{\mathrm{lat}}|C_i| + \mathcal{J}_\ell(C_i) + E_{a_K}(C_i)|$ with E_a the remainder of the lattice Coulomb energy (Prop. 10.7(a)), and $\theta_{d_i} = 2B_3 \sup_{K > d_i, \mathsf{b} \in \mathsf{J}} \varepsilon_{K-d_i} L^{2d_i} n_1 n_2 a_i^2$ (Prop. 10.8(d)); both ends tend to 1: $\sup\{|\mathcal{W}_K(C_i) - 1| : \mathsf{b} \in \mathsf{J},\ K > d_i\} \to 0$ as $i \to \infty$. For the squares Q_i of side a_i with corners in Λ_i there are i_2 and n_Q with $\mathcal{W}_K(Q_i) \ge \frac{1}{8}$ for all $\mathsf{b} \in \mathsf{J}$, $i \ge i_2$ and $K > i + n_Q$. On sides with $\mathsf{V}_\ell > 1$ these small rectangles must be so small that $\mathsf{V}_\ell L^{-i}$ is small. If $s \times T$ and $s \times 2T$ are rectangles with $\kappa < \frac{1}{2}$, then $\liminf_K \mathcal{W}_K(s \times nT) \ge \mathcal{W}(s \times 2T)[\mathcal{W}(s \times 2T)/\mathcal{W}(s \times T)]^{n-2}$ for every $n \ge 2$ with $nT < \ell/2$.

(5) *Block fields; a static source.* Let $\kappa < \frac{1}{2}$ and H be a bounded Borel gauge invariant function on $\mathcal{V}_{s_1}$. For $K > d' \ge \max\{s_0, s_1\}$,

$$\Big| a_K^{-\kappa} \langle W_{\mathsf{C}} H(\mathcal{M}^{K-s_1} U) \rangle_K - a_K^{-\kappa} \Xi_{K-d'} \frac{1}{Z} \int W_{\mathsf{C}}^{(K-d')} \, (H \circ \mathcal{M}^{d'-s_1}) \, \mathrm{d}m_{K-d'} \Big| \le \sup |H| \, \mathsf{E}'_{d'};$$

so $\lim_K a_K^{-\kappa} \langle W_{\mathsf{C}} H(\mathcal{M}^{K-s_1} U) \rangle_K$ exists along the full sequence and equals $\lim_d \mathsf{l}_d \int F_{\mathsf{C}}^{[d]} \, (H \circ \mathcal{M}^{d-s_1}) \, \mathrm{d}\mu_d$. For a rectangle C the quotients $\langle W_C H(\mathcal{M}^{K-s_1} U) \rangle_K / \langle W_C \rangle_K$ converge; their limit $\omega(H)$ is a linear functional on the bounded gauge invariant Borel functions of all block fields, real on the real ones, normalised, $\omega(1) = 1$, consistent between the depths, $\omega(H \circ \mathcal{M}) = \omega(H)$, and bounded at each depth: $|\omega(H)| \le \sup |H| \, (c_\Xi L^{\kappa d'} + \mathsf{E}'_{d'})/\mathcal{W}(C)$ with $d' = \max\{s_0, s_1\}$, $\mathcal{W}(C) > 0$ by (4); $c^{-1} a_d^\kappa \le \int Y_{d,C} \, \mathrm{d}\mu \le c \, a_d^\kappa$ for large d, and $\int H(V_{s_1}) Y_{d,C} \, \mathrm{d}\mu / \int Y_{d,C} \, \mathrm{d}\mu$ tends to $\omega(H)$ as $d \to \infty$. For H bounded Borel but not gauge invariant all of this holds with the gauge average $\bar{H}(V) = \int H(V^v) \, \mathrm{d}v$ (Haar measure on the gauge transformations of Λ_{s_1}) in the place of H, the Wilson measure, μ and $Y_{d,C}$ being gauge invariant.

(6) *Mixed functions.* Let a coordinate mirror θ part C, at distance $\ge \delta_1 > 0$ from its planes, from the supports of real Lipschitz $f_1, \dots, f_n$, at distance $\ge \delta > 0$ from the planes, and let $2\kappa < \frac{1}{2}$. Then $Q_K(z) = a_K^{-\kappa} \langle W_{\mathsf{C}} e^{\mathcal{O}(w_z)} \rangle_K / \langle e^{\mathcal{O}(w_z)} \rangle_K$, $w_z = \sum_q z_q f_q$, obeys $|Q_K(z)| \le \sqrt{6} \, e^{\Delta(\delta)/2} \mathcal{W}_K(\mathsf{C}, \theta\mathsf{C})^{1/2}$ for $K > s_0$ on a polydisc free of K, and converges locally uniformly there to a holomorphic Q; the supports need not be disjoint. If they are pairwise disjoint, then $\lim_K \mathcal{W}_K(\mathsf{C}; f_1, \dots, f_n)$ exists along the full sequence and equals $\sum_{A \subset \{1,\dots,n\}} \partial_A Q(0) \sum_\pi \prod_{\beta \in \pi} S_{|\beta|}(f_\beta)$, π running over the partitions of the complement of A into blocks of at least two elements, $f_\beta = (f_q)_{q \in \beta}$.

The same holds for two orthogonal coordinate mirrors θ_1, θ_2 if $4\kappa < \frac{1}{2}$, C lies on one side of both at distance $\ge \delta_1$ from their planes, and the f_q fall into two groups, the first with supports on the other side of θ_1, the second with supports on the side of C of θ_1 and on

the other side of θ_2, the two groups at sup-distance $\ge 2\delta$ and all supports at distance $\ge \delta$ from the planes. For $n = 1$ it holds, if $2\kappa < \frac12$ and C has extent $< \frac{\ell}{2}$ in each coordinate direction, for every real Lipschitz f whose support misses the least closed coordinate box that holds C. Where no such mirror parts them, but N equal coordinate boxes between equally spaced walls in $\frac12 a_i\mathbb{Z}$, for an index i, hold each polygon in one box, the number of the boxes being even in every direction that the walls cut, so that both fixed planes c and $c + \frac{\ell}{2}$ of every wall mirror are walls (on $\ell = 2L^m$, $2L^j$ or $4L^j$ slabs in each cut direction), each f_q is supported in one box, the supports lie off the walls and no box meets both a polygon and a support, one has $\langle W_{\mathsf{C}^{\mathrm{f}}_\iota}\rangle_K \ge 0$ and $|Q_K(z)| \le M_S \prod_\iota \mathcal{W}_K(\mathsf{C}^{\mathrm{f}}_\iota)^{1/N}$ for $\sum_q |z_q| \|f_q\|_\sharp < \rho_D/N$, a polydisc free of K, with $\|f\|_\sharp$, ρ_D and M_S of Prop. 10.10, $\mathsf{C}^{\mathrm{f}}_\iota$ being the family of the images under the wall reflections of the polygons C_ι of the box ι, which has exactly one image in every box, with $\kappa(\mathsf{C}^{\mathrm{f}}_\iota) = N\kappa(\mathsf{C}_\iota)$; so, if $N\kappa(\mathsf{C}_\iota) < \frac12$ for every box ι, $\sup_K |\partial_A Q_K(0)| < \infty$ for every $A \subset \{1,\dots,n\}$, and this holds whatever the supports off the walls, a function whose support misses the walls being the sum of pieces each supported in one box; and $\sup_K |\mathcal{W}_K(\mathsf{C}; f_1,\dots,f_n)| < \infty$ when the supports are pairwise disjoint, the truncated functions of the action density being bounded in K there YM3_I:(2212); for $n = 1$, $\mathcal{W}_K(\mathsf{C}; f_1) = \partial_1 Q_K(0)$, with no further condition.

The limits obey the law of dilations $\mathcal{W}^{(L^z\ell, L^z\mathsf{b})}(L^z\mathsf{C}; f^{(z)}) = L^{-z\kappa - zn}\mathcal{W}^{(\ell,\mathsf{b})}(\mathsf{C}; f)$, $f_q^{(z)} = f_q(L^{-z}\cdot)$, and the positive form of Prop. 10.9(b) with mixed entries: let θ' be the coordinate mirror $\theta' x = x - 2(x_\mu - \hat c)e_\mu$ with a place $\hat c \in \frac12 a_{i_0}\mathbb{Z}$, let families $G_1,\dots,G_s$ lie in $\{\hat c < x_\mu < \hat c + \frac{\ell}{4}\}$ and tuples $f_1,\dots,f_s$ of real Lipschitz functions, each tuple with pairwise disjoint supports, have their supports in $\{\hat c + \frac{\ell}{4} < x_\mu < \hat c + \frac{\ell}{2}\}$, all at positive distance from the planes $x_\mu \in \hat c + \frac{\ell}{4}\mathbb{Z}$, and let $2[\kappa(G_i) + \kappa(G_j)] < \frac12$ for all i, j; then each entry $\mathcal{W}(\theta' G_i, G_j; \theta' f_i, f_j)$ is a limit of the kind above, the coordinate mirror with the place $\hat c + \frac{\ell}{4}$, which lies in $\frac12 a_{i_0}\mathbb{Z}$ as $\frac{\ell}{4} = \frac12 L^m$, parting the family $(\theta' G_i, G_j)$ from the supports of $\theta' f_i$ and f_j, and $\sum_{i,j} \bar c_i c_j \mathcal{W}(\theta' G_i, G_j; \theta' f_i, f_j) \ge 0$ for all complex $c_1,\dots,c_s$.

(7) *The action density beside a loop.* Let C be a polygon with $\psi = \langle W_C\rangle_K > 0$ (every coordinate rectangle has it), f bounded and real, and θ a site mirror with C in one open half torus and every plaquette p with $f(x(p)) \ne 0$ with its bonds in the other closed half S. Write $\langle X; Y\rangle_K = \langle XY\rangle_K - \langle X\rangle_K\langle Y\rangle_K$. At every cutoff, for every bounded real Borel function H_0 of the bonds of S and $H = \mathcal{O}(f) - H_0$: $\langle W_C; W_{\theta C}\rangle_K \ge 0$, $\langle \theta H; H\rangle_K \ge 0$, and

$$\Big|\frac{\langle W_C; \mathcal{O}(f)\rangle_K}{\psi} - \frac{\langle W_C; H_0\rangle_K}{\psi}\Big| \le \Big[\frac{\langle W_C W_{\theta C}\rangle_K}{\psi^2} - 1\Big]^{1/2} \langle \theta H; H\rangle_K^{1/2};$$

at $H_0 = 0$ the left side is $|\langle W_C \mathcal{O}(f)\rangle_K/\psi - \langle \mathcal{O}(f)\rangle_K|$. Let moreover, in the frame, $g \le \gamma_W$, the ceiling of equation YM3_I:(2346); let $\vartheta_{\mathrm{sh}} \in (0,1)$ be a shape parameter, $\mathsf{d} \in (0, \mathsf{d}_0]$ with the threshold $\mathsf{d}_0 = \mathsf{d}_0(g, \vartheta_{\mathrm{sh}}) \in (0, \frac{1}{200}]$ of equation YM3_I:(2350), and $K - s(\mathsf{d}) \ge 1$, $s(\mathsf{d})$ the witness depth at g of equation YM3_I:(2321); and let $(\theta f, f)$ be a test pair of the class $\mathfrak{T}_{T_\ell}(\mathsf{d}; \vartheta_{\mathrm{sh}})$ of [7, Lem. I.1], on a torus with $20\mathsf{d} \le \ell$: f is non-negative, Lipschitz and not $\equiv 0$, the supports of θf and f lie in closed balls of radius d and are at a distance in $[\mathsf{d}, 6\mathsf{d}]$, $\int f \ge \vartheta_{\mathrm{sh}} \|f\|_\infty \mathsf{d}^3$ and $\mathrm{Lip}\, f \le \|f\|_\infty/\vartheta_{\mathrm{sh}}\mathsf{d}$.

As $(K, L^K\mathsf{b})$ is a first-passage datum at g with the label $\mathsf{b} \in \mathsf{l}(g)$ (Cor. YM3_I:74.2), on the side $\ell \in \mathfrak{L}(g)$ of the frame, the law of the truncated two-point function of the action density, Thm. YM3_I:AV, gives $\langle \theta\mathcal{O}(f); \mathcal{O}(f)\rangle_K = \mathsf{c}_2\mathsf{b}^{-2} \iint f(\theta x) f(y) |x-y|^{-6}\, \mathrm{d}x\, \mathrm{d}y\, (1+\epsilon')$, with $\mathsf{c}_2 = \frac{9}{16\pi^2}$ the constant of YM3_I:C, the double integral the free kernel YM3_I:(2240) of the test pair, and $|\epsilon'| \le \epsilon_{\mathrm{tp}} := \bar\varepsilon_{g,\vartheta_{\mathrm{sh}}}(\mathsf{d})$, the tolerance of equation YM3_I:(2349), which is

at most $\frac{1}{3}$ in that theorem; with the bound above and $\mathsf{c}_2^{1/2} = \frac{3}{4\pi}$ this gives

$$\Big|\frac{\langle W_C\mathcal{O}(f)\rangle_K}{\psi} - \langle\mathcal{O}(f)\rangle_K\Big| \le \frac{3}{4\pi\mathsf{b}}\Big[(1+\epsilon_{\rm tp})\iint\frac{|f(\theta x)f(y)|}{|x-y|^6}\,\mathrm{d}x\,\mathrm{d}y\Big]^{1/2}\Big[\frac{\langle W_C W_{\theta C}\rangle_K}{\psi^2}-1\Big]^{1/2};$$

and, as the free kernel of a test pair is at most $(\frac{4\pi}{3})^2\|f\|_\infty^2$ (equation YM3_I:(2247); two balls of radius d at a distance $\ge\mathsf{d}$), the same two-point estimate gives on this whole class $\langle\theta\mathcal{O}(f);\mathcal{O}(f)\rangle_K \le \frac{4}{3}(\frac{4\pi}{3})^2\mathsf{c}_2\mathsf{b}^{-2}\|f\|_\infty^2 = \frac{4}{3}\mathsf{b}^{-2}\|f\|_\infty^2$, so that, with no condition beyond this class,

$$\Big|\frac{\langle W_C\mathcal{O}(f)\rangle_K}{\psi} - \langle\mathcal{O}(f)\rangle_K\Big| \le \frac{2}{\sqrt{3}}\,\mathsf{b}^{-1}\|f\|_\infty\Big[\frac{\langle W_C W_{\theta C}\rangle_K}{\psi^2}-1\Big]^{1/2}.$$

Besides, at every cutoff,

$$\frac{\langle W_C\mathcal{O}(f)\rangle_K}{\psi} - \langle\mathcal{O}(f)\rangle_K = -\frac{1}{a_K}\sum_p f(x(p))\langle W_{\partial p}\rangle_K\Big[\frac{\langle W_C W_{\partial p}\rangle_K}{\psi\langle W_{\partial p}\rangle_K}-1\Big],$$

$$\frac{d}{d\mathsf{b}}\log\langle W_C\rangle_K = -\Big[\frac{\langle W_C\mathcal{O}(1)\rangle_K}{\psi} - \langle\mathcal{O}(1)\rangle_K\Big].$$

For a rectangle with $2\kappa < \frac{1}{2}$ at positive distance from the planes of θ and Lipschitz f with support at positive distance from them, $\langle W_C;\mathcal{O}(f)\rangle_K/\psi = \mathcal{W}_K(C;f)/\mathcal{W}_K(C)$, $\langle W_C W_{\theta C}\rangle_K/\psi^2 = \mathcal{W}_K(C,\theta C)/\mathcal{W}_K(C)^2$ and $\langle\theta\mathcal{O}(f);\mathcal{O}(f)\rangle_K = S^{\mathsf{T}}_{2,K}(\theta f, f)$ converge as $K\to\infty$.

(8) *Short distances.* Let $0<\vartheta<\frac{1}{2}$ and $p\ge 1$. (a) For rectangles $C_i = n_1a_i\times n_2a_i$ with corners in Λ_i, n_1, n_2 fixed, $\varrho_i = \max\{n_1,n_2\}a_i$, $\mathsf{n} = \max\{L,n_1,n_2\}$, all large i and all $K>i$ with $\Delta_i\le\frac{1}{2}$,

$$\Big|\log\mathcal{W}_K(C_i) + \frac{3}{2\mathsf{b}}\big[c_{\rm lat}|C_i| + \mathcal{J}_\ell(C_i) + E_{a_K}(C_i)\big]\Big| \le \kappa\log L + c\frac{|C_i|}{\mathsf{b}} + 2\Delta_i,$$

$$\Delta_i = 4B_3^2A_0^2\mathsf{n}^3\frac{\varrho_i}{\mathsf{b}}\Big(1+\log\frac{\mathsf{n}\mathsf{b}}{\varrho_i}\Big)^{2\mathsf{p}_0} + C_1\frac{|C_i|}{\mathsf{b}}\Big(\frac{\varrho_i}{\mathsf{b}}\Big)^{\vartheta/2} + C_3\mathsf{V}_\ell C_{\rm r}(p)\varrho_i^{p-1/2},$$

E_a the remainder of Prop. 10.7(a) and c, C_1, C_3 depending on the constants, ϑ, n_1, n_2 and J: an error of the order of the main term, which gives the two-sided bound of (4) at these rectangles. (b) For rectangles C with sides in $[c_0\varrho, 10\varrho]$ and corners in Λ_i, i the integer with $a_i\le\frac{1}{2}c_0\varrho < La_i$, put $a_* = \varrho(\mathsf{b}/\varrho)^{\vartheta/(3+\vartheta)}$, a length coarser than the loop. At every level t with $a_{K-t}\le a_*$ the mean over the block field of the defect of one step, $\bar{\mathsf{D}}_{t+1}$ of Prop. 10.8(b) for C, has modulus at most $C_*\frac{\varrho}{\mathsf{b}}(\frac{a_{K-t}}{\mathsf{b}})^\vartheta$, C_* depending on the constants, c_0, ϑ and J alone: at the levels finer than the corners, $K-t>s_0=\max\{i,s_0'\}$, by Prop. 10.6(b), and at the levels with $a_i\le a_{K-t}\le a_*$, about $\frac{\vartheta}{3+\vartheta}\log_L\frac{\mathsf{b}}{\varrho}$ of them, where the loop passes a number of the cubes of the level that is bounded in terms of L and c_0, by the argument of one step with this bounded number of segments. Hence, as $\varrho\to 0$, with a constant depending on the constants, c_0, ϑ, J and ℓ,

$$\log\mathcal{W}(C) = -\frac{3}{2\mathsf{b}}\big[\mathcal{J}(C) + c_{\rm lat}|C|\big] + O\Big(\Big(\frac{\varrho}{\mathsf{b}}\Big)^{1+\frac{3\vartheta}{3+\vartheta}}\Big(1+\log\frac{\mathsf{b}}{\varrho}\Big)^{2\mathsf{p}_0} + \frac{\varrho^4}{\mathsf{b}\,\ell^3}\Big);$$

the exponent $1+\frac{3\vartheta}{3+\vartheta}$ reaches $\frac{10}{7}$ as $\vartheta\to\frac{1}{2}$, the length a_* balancing the step errors, $\frac{\varrho}{\mathsf{b}}(\frac{a_*}{\mathsf{b}})^\vartheta$, against the last level, $\frac{\varrho^4}{\mathsf{b}a_*^3}$, and the remainder, which holds $\log(\mathsf{b}/\varrho)$ and not $\log\varrho$, is invariant under dilations.

■□

Theorem AF (Without the renormalisation group)**.** Let C be a coordinate rectangle on a torus of side ℓ, $\mathsf{b}>0$, a the spacing, $\beta=\mathsf{b}/a\ge e^3$, $\Lambda=\max\{1,|C|/\mathsf{b}\}$, $\rho=\lfloor(\frac{\beta}{\Lambda\log^3\beta})^{1/4}\rfloor\ge 2$, the sides of C at least $(4\rho+1)a$, and the points within $(\rho+1)a$ of C in a cube of side $\le\frac{\ell}{2}$. Then

$$|\langle W_C\rangle|\le 6e^{3.1\mathsf{b}/\ell}\,1.58^{|C|/\mathsf{b}}\Big[\Lambda\frac{a}{\mathsf{b}}\log^3\frac{\mathsf{b}}{a}\Big]^{k|C|/4\mathsf{b}}.$$

With a dyadic $\rho=2^S$ instead, sides at least $(4\rho+1)a$ and $k|C|/\mathsf{b}<3$, the ceiling of the tube with the profiles of the scales $2,4,\dots,2^S$ reads

$$|\langle W_C\rangle|\le e^{\delta_0+2.2144k|C|/\mathsf{b}}\rho^{-k|C|/\mathsf{b}}\Big[1+\frac{72\,\bar a\,\rho^3\log^2\rho}{2^{3-k|C|/\mathsf{b}}-1}\Big]+35\,\bar a,\qquad \delta_0=\frac{16k\rho\log\rho+k^2(|C|/\mathsf{b})\log^2\rho}{\beta},$$

with the plaquette mean $\bar a=\langle a_p\rangle\le\beta^{-1}\log\mathfrak{M}_\nu$ (Lem. 10.12(f)), the condition $k|C|/\mathsf{b}<3$ keeping the denominator positive; at $\rho\in(\frac12\beta^{1/3},\beta^{1/3}]$, where $\bar a\rho^3\log^2\rho$ is at most a multiple of $\log^3\beta$, and $\beta\ge\beta_2$, a threshold depending on $k|C|/\mathsf{b}$ alone (the term $192\frac{\mathsf{b}}{\ell}$ of $\log\mathfrak{M}_\nu$ is at most $384\mathsf{b}/|C|$, as $|C|<2\ell$): $|\langle W_C\rangle|\le C(\frac{k|C|}{\mathsf{b}})(\frac{a}{\mathsf{b}})^{|C|/4\pi\mathsf{b}}\log^3\frac{\mathsf{b}}{a}$, as $\rho^{-k|C|/\mathsf{b}}\le 2^{k|C|/\mathsf{b}}(\frac{a}{\mathsf{b}})^{k|C|/3\mathsf{b}}$ and $\frac{k}{3}=\frac{1}{4\pi}$. The first bound holds in the same form, with a constant depending also on the number of pieces, for polygons whose pieces have at least $4\rho+1$ steps and whose pieces without a common corner are at sup-distance at least $(2\rho+1)a$, for families of such polygons in which also any two pieces of different polygons are at sup-distance at least $(2\rho+1)a$, also with a further factor G, $|G|\le 1$, a Borel function of the bonds farther than ρa from the polygons; and for the characters χ_j of spin j, with $\kappa_j=\frac43 j(j+1)\le 20\log\beta$ and $A=6e^{3.1\mathsf{b}/\ell}$:

$$|\langle\chi_j(U(C))\rangle|\le(2j+1)\big(A\,1.58^{|C|/\mathsf{b}}\big)^{\kappa_j}e^{\kappa_j^2/100}\Big[\Lambda\frac{a}{\mathsf{b}}\log^3\frac{\mathsf{b}}{a}\Big]^{\kappa_j k|C|/4\mathsf{b}}.$$

These bounds hold with 3 in the place of $6e^{3.1\mathsf{b}/\ell}$ for every periodic Gibbs state of the Wilson action at β on $\mathbb{Z}^3$ that is invariant under the isometries of $\mathbb{Z}^3$ and reflection positive in the lattice planes. So, as $a\to 0$ with b in a compact set, uniformly in $\ell\ge\ell_0$, for polygons: $\langle h(W_C)\rangle\to\sigma(h)=\frac{2}{\pi}\int_{-1}^1 h(w)\sqrt{1-w^2}\,dw$ for continuous h, $\langle W_C^2\rangle\to\frac14$, and for polygons C_i at pairwise positive distances, continuous h_i and Borel G, $|G|\le 1$, depending only on the bonds at distance $\ge\delta>0$ from them, $\langle\prod_i h_i(W_{C_i})\,G\rangle-\prod_i\sigma(h_i)\langle G\rangle\to 0$.

For a coordinate rectangle, whose mean $\langle W_C\rangle$ is positive (Prop. 10.9(c)), the normalised sharp loop has the variance $\mathrm{Var}(W_C/\langle W_C\rangle)=(1+\langle\chi_1(U(C))\rangle)/(4\langle W_C\rangle^2)-1$, which tends to infinity as $a\to 0$, at least like a positive constant times $[\Lambda\frac{a}{\mathsf{b}}\log^3\frac{\mathsf{b}}{a}]^{-k|C|/2\mathsf{b}}$; and no multiple c_aW_C of the sharp loop has both a limit law and a mean $c_a\langle W_C\rangle$ that stays away from 0. In the other direction, $\langle W_C\rangle\ge\langle W_{\partial p}\rangle^{A_C/a^2}\ge\exp[-\frac32\frac{A_C}{a\mathsf{b}}(192\frac{\mathsf{b}}{\ell}+56.6+12\log\beta)]$ for a rectangle of area A_C and sides $<\ell/2$, the second inequality when $192\frac{\mathsf{b}}{\ell}+56.6+12\log\beta\le\frac{\beta}{2}$; for the Gibbs states above and rectangles with even numbers of steps, $\langle W_C\rangle\ge\exp[-\frac{A_C}{a\mathsf{b}}(43+9\log\frac{\mathsf{b}}{a})]$ when $28.3+6\log\beta\le\frac{\beta}{2}$.

Without the renormalisation group the floor falls like the exponential of the area in units of $a\mathsf{b}$. ■□

10.1.4. *Construction estimates.* Thm. YM3_I:D with Cor. YM3_I:69.4 on the actual admissible histories enters through the objects of the frame and through several of the following statements, cited by their keys, which we use: the limit laws of the block fields, with one block map for all cutoffs, with their rate and with the two depths of their proof (Cor. 2.8, (2.xvii), §2.3.1), the numbers $\mathsf{C}_{\rm bf}$, $K_{\rm bf}$ of the rate being formed at an interval of labels under the condition (Tor) (Defs. 2.3.1 and 2.7(5)); the bound on the history without large-field regions (Prop. YM3_I:115.2, YM3_I:(2187)); the rarity of large-field regions (Prop. 4.54); the bound at one place at the fixed and at the lowered constants (Lem. 4.51, Def. 4.47); the three smallness conditions that define $t_\natural$ and the facts on them (§2.3.1, §2.3.1, §2.3.1); the comparison of two consecutive cutoffs (Prop. YM3_I:113.9, YM3_I:(2154)) with the competitors, the lower bound on their excess, the chart of the fibre at the

pivot bonds and the numbers of the comparison (Lem. YM3_I:113.8 and Prop. YM3_I:113.5, the chart bounds in equation YM3_I:(2138), the global hypotheses preceding equation YM3_I:(2132) and the constants in and before equation YM3_I:(2133)), the averaged backgrounds (Prop. YM3_I:52.3, the axial-normalization argument in the proof of Prop. YM3_I:52.3), and the response of the background to a change of its datum with the averages of a moved field (Thm. YM3_I:E and Lem. YM3_I:51.6, with the thresholds of §5.3, Lem. 9.11(v)), which give the comparison of the averaged backgrounds of two consecutive cutoffs over one block field and along the tower, and the Lipschitz bound of the tower in the block field (Lem. 10.3); the existence of the background over a small field as one orbit (Thm. YM3_I:E in the setting of Section YM3_I:52.7, §5.2); the localisation of backgrounds (Prop. YM3_I:53.5, Section YM3_I:53.4, including equations YM3_I:(998)–YM3_I:(1000)), with the localised background of a region (equation YM3_I:(77)) and the arguments that give its hypotheses for a slab of M_1-cubes (the geometric verification in Step 5 of the proof of Lem. 10.12, using Prop. YM3_I:44.2); the covariance of the block averages and of the backgrounds under the lattice isometries ([7, §G.1], [7, Lem. G.1]); for one step, the triples, the analyticity cores, the chart at the trivial pair and the ball about the trivial triple in the core (§5.2, Def. YM3_I:49.1, and equation YM3_I:(933)), the background triple and its regularity (the triple defined before equation YM3_I:(827) and §5.3.1), the kernel of one step (eq. 5.xxxii and Prop. 5.31), the provision of the step on the whole core (Def. YM3_I:52.1, Thm. YM3_I:W), the shift of Prop. YM3_I:51.5 with their brackets (equations YM3_I:(806)–YM3_I:(809) and Lem. 5.25), the chart and the pivot values at $t = 0$ (equation YM3_I:(804) and the bounds and covariance proved with it), the potential on a cube of any side (Thm. YM3_I:F), the cluster expansion of the step with up to four potential families (Thm. YM3_I:R, Cor. YM3_I:47.3 and Prop. YM3_I:47.4, with the following covariance argument), applied to the actual Gaussian structure and the two potentials of Prop. 5.31, the constrained variational equation for critical curves at the trivial field (the proof of Lem. YM3_I:124.1, through its gauge normalization), and its curvature identification (equation YM3_I:(2230)), the curvature response (the curvature-response construction preceding Lem. YM3_I:120.2, the curvature-response calculation preceding Lem. YM3_I:120.2), the slice of the free Coulomb energy at the trivial triple (Prop. YM3_I:120.3, Prop. YM3_I:120.3) and the linearised shift system (Lem. YM3_I:124.3, whose proof, with the two changes named in the proof of Prop. 10.6(a), runs for the objects of the frame on every side of $\mathfrak{L}(g)$); and the limits of the truncated functions of the action density YM3_I:(2212).

Thms. YM3_I:AJ and YM3_I:AD, the bound for a history with a source on a polydisc uniformly in K, is used in this section only in Prop. 10.10, hence in Thm. AE(6) and in the convergence of the three quantities of Thm. AE(7); the explicit ceiling of Thm. AE(7) uses besides the law of the truncated two-point function of the action density, Thm. YM3_I:AV, with its test pairs ([7, Lem. I.1]), the free kernel YM3_I:(2240) and its bound on a test pair (equation YM3_I:(2247)), the constant c_2 YM3_I:C, its ceiling γ_W equation YM3_I:(2346), its threshold d_0 equation YM3_I:(2350), its witness depth $s(\mathsf{d})$ in equation YM3_I:(2321) and its tolerance equation YM3_I:(2349), and the first-passage data (Cor. YM3_I:74.2).

The facts on the constants (Prop. YM3_I:69.1, the calibration in §2.3.1, and [7, Cor. B.13]) make fixed constants satisfy the conditions of the setting. Besides these we use standard facts: reflection positivity of the Wilson measure on the torus at site and bond mirrors, the chessboard estimate, Kolmogorov's extension theorem, Vitali's theorem, Schwarz's lemma and Cauchy's estimates. In Thm. AD(1)–(5) and in Props. 10.2 and 10.4(a)–(d) one torus and one label are fixed, and the limit is $K \to \infty$, with the limit $n \to \infty$ of the tower inside it; in the frame the numbers of the rate of Prop. 10.4(d) serve all the labels of J.

Thm. AE takes besides the limit $d \to \infty$ of means after $K \to \infty$ ((2), (5)), lets the label move in a compact set J and along labels $\mathsf{b}_K \to \mathsf{b}$ ((2)), lets rectangles shrink, $i \to \infty$ and $\varrho \to 0$ ((4), (8)), and moves the torus and the label together by dilations ((3), as Thm. AD(6) does). Thm. AF takes

$a \to 0$ uniformly in $\ell \geq \ell_0$ with b in a compact set, and holds also for periodic Gibbs states on $\mathbb{Z}^3$. In Thms. AD and AE the side ℓ of the torus is fixed, or moves together with the label, by dilations.

10.1.5. *Auxiliary estimates.*

Lemma 10.1. Let $L \geq 2$, $m' \in \mathbb{Z}_{\geq 0}$, $r, m \in \mathbb{N}$, $M = L^{m'}$, b real with $\log \mathsf{b} > 1$, and $Q(\mathsf{b}) = \min\{L^j : j \in \mathbb{Z}_{\geq 0},\ L^j \geq (\log \mathsf{b})^r\}$. Then $M\,Q(\mathsf{b}) \leq L^m$ if and only if $L^m \geq M(\log \mathsf{b})^r$. The hypothesis $\log \mathsf{b} > 1$ is needed: at $L = 3$, $m' = 2$, $m = 1$, $r = 2$, $\mathsf{b} = 1.2$ the second condition holds and the first fails. In the setting it holds since $\gamma \leq \frac{1}{2}$. ■

Sketch of proof. $Q \geq (\log \mathsf{b})^r$ gives $M(\log \mathsf{b})^r \leq MQ \leq L^m$. Conversely $L^{m-m'} \geq (\log \mathsf{b})^r > 1$, so $j = m - m'$ is a positive integer, L^j lies in the set whose least element is Q, and $Q \leq L^{m-m'}$. In the example $(\log 1.2)^2 = 0.033\ldots$, so $Q = 1$ and $MQ = 9 > 3 = L^m \geq 9 \cdot 0.033$. □

Proposition 10.2 (Laws on the threads; small plaquettes)**.** In the one-torus setting: (a) there is exactly one Borel probability measure μ on $\mathcal{X}$ whose image under $V_\bullet \mapsto (V_0, \ldots, V_D)$ is, for every D, the image of μ_D under $V \mapsto (\mathcal{M}^D V, \ldots, \mathcal{M}V, V)$; $\mu(\mathsf{Thr}) = 1$, the image of μ under π_d is μ_d, μ is θ_v-invariant, and the law of $(\mathcal{M}^K U, \ldots, \mathcal{M}^{K-D}U)$ under $\langle\cdot\rangle_K$ tends in total variation to the image of μ under $(\pi_0, \ldots, \pi_D)$; (b) each I_d is open and gauge invariant; for $d \geq d_\natural$ and $K > d$, $\mu_d^K(\mathcal{V}_d \setminus I_d) \leq R_d$ and $\mu_d(\mathcal{V}_d \setminus I_d) \leq R_d$; for $D \geq d_\natural$, $\mu(\tau_0 > D) \leq \sum_{d \geq D} R_d$, which is at most $\frac{L}{L-1} R_D$ if $c' A_0^2 \log x_{[D]} \geq 2$; and R_d tends to 0 faster than every L^{-pd}; (c) for $d \geq s_\natural$, $K > d$ and every set S of sites of Λ_d, the μ_d^K- and the μ_d-probability that some plaquette with a corner in S has $|\partial V(p) - \mathbf{1}| \geq 2\varepsilon_{[d]}$ are at most $\#S\, e^{-c^\natural A_0^2 (\log x_{[d]})^{2\mathfrak{p}_0}}$; (d) with the constants $C_5, \mathsf{C}_{\mathrm{maj}} \geq 0$ of Thm. YM3_I:D, $s_d = \frac{1}{4}\min\{\bar{\varepsilon}, B_3^{-1}\varepsilon_{[d]}\}$,

$$\mathsf{p}^{[d]} = \exp\big[-(c_{\mathrm{low}} + \tfrac{9}{2}\log x_{[d]} + \mathsf{C}_{\mathrm{maj}} + 2)(\ell L^d)^3\big], \qquad \mathsf{q}^{[d]} = \exp\Big[-\Big(C_5 + \mathsf{C}_{\mathrm{maj}} + 2 + 3\log\frac{7}{s_d^3} + 24 B_3^2 s_d^2 x_{[d]}\Big)(\ell L^d)^3\Big],$$

one has at every admitted depth d and for $K \geq \max\{2, d+1\}$: $\mu_d^K(J_d) \geq \mu_d^K(I_d) \geq \mathsf{q}^{[d]} > 0$, and $\mu_d^K(I_d) \geq \max\{1 - R_d, \mathsf{p}^{[d]}, \mathsf{q}^{[d]}\}$ if $d \geq 1$; the same holds for μ_d. The floors hold no K and fall like $e^{-c(\ell L^d)^3}$. ■

Sketch of proof. (a) By one block map, $\mu_{d+1} \circ \mathcal{M}^{-1} = \mu_d$ (Cor. 2.8(c)), so the images of the μ_D form a projective family of Borel probabilities on the finite products of the compact metric spaces $\mathcal{V}_d$, which has exactly one extension to $\mathcal{X}$ (Kolmogorov's theorem). Each marginal is carried by the Borel set $\{V_d = \mathcal{M}V_{d+1},\ d < D\}$, whence $\mu(\mathsf{Thr}) = 1$; gauge invariance of the μ_d and covariance of $\mathcal{M}$ give the invariance; the joint law at the cutoff K is the image of the single law μ_D^K under $V \mapsto (\mathcal{M}^D V, \ldots, V)$, and $\mu_D^K \to \mu_D$ in total variation. (b) ϖ is continuous and $\varpi(V^v) = \varpi(V)$.

Here Thm. YM3_I:D is applied along the line of couplings $x_{[d']} = L^{d'}\mathsf{b}$, $0 \leq d' \leq K$ (the run with all $u_t = 0$, at the label b), with the inner ball $I_{d'} = \{\varpi < 2\varepsilon_{[d']}\}$ at the depth d': the partition function is a sum over the histories of large-field regions, and the small-field chain of this line (Prop. YM3_I:115.2), which at the depths $d' > d$ cuts the fields at the radii $c_\rho \varepsilon_{[d']}$, measured on their backgrounds (the definitions preceding Lem. YM3_I:113.10), is a bounded Borel weight $\mathfrak{w} \geq 0$ on the bare fields with three properties: the image of $\mathfrak{w}\, e^{-\beta\sum_p a_p} dU$ under $\mathcal{M}^{K-d}$ is carried by I_d (clause (a) there); $\int F(1-\mathfrak{w}) e^{-\beta\sum_p a_p}\, \mathrm{d}U \geq 0$ for every gauge invariant bounded Borel $F \geq 0$ (clause (b), which places $\mathfrak{w}$ between the history without large-field regions and the full density); and $\int (1-\mathfrak{w}) e^{-\beta\sum_p a_p}\, \mathrm{d}U \leq R_d Z$, R_d being YM3_I:(2187) at the depth d and the label b (clause (c)).

As I_d is gauge invariant, $\mu_d^K(\mathcal{V}_d \setminus I_d) = Z^{-1}\int \mathbf{1}_{\mathcal{V}_d \setminus I_d}(\mathcal{M}^{K-d}U)(1-\mathfrak{w}) e^{-\beta\sum_p a_p}\, \mathrm{d}U \leq R_d$, uniformly in K. This chain of the line is another object than the measures m_t of the frame, which follow the

run, with $|x_{K-d} - x_{[d]}| \le \bar{B}$ and the thresholds $2\varepsilon(x_t)$: the tame set O_{K-d} that carries m_{K-d} is placed by Prop. 10.4(e) in the larger ball J'_d. The bound for μ_d^K passes to μ_d in total variation. For $D \ge d_\natural$, $\{\tau_0 > D\} = \bigcup_{d\ge D} \pi_d^{-1}(\mathcal{V}_d \setminus I_d)$, and the image of μ under π_d is μ_d by (a); countable subadditivity gives $\mu(\tau_0 > D) \le \sum_{d\ge D} R_d$.

As $(y+\log L)^{2\mathfrak{p}_0} - y^{2\mathfrak{p}_0} \ge 2y \log L$ for $y \ge 1$, and $\log x_{[d]} \ge \log \mathsf{b} > 1$, $R_{d+1}/R_d \le L^{3-2c'A_0^2 \log x_{[d]}}$, which is at most L^{-1} for all $d \ge D$ if $c'A_0^2 \log x_{[D]} \ge 2$; then $\sum_{d\ge D} R_d \le \frac{L}{L-1} R_D$. R_d falls faster than every L^{-pd} because $(\log x_{[d]})^{2\mathfrak{p}_0} \ge (d \log L)^2$. (c) One witness function of a large plaquette at one place has a mean at most $e^{-c^\natural A_0^2 (\log x_{[d]})^{2\mathfrak{p}_0}}$, free of the volume: Lem. 4.51, which applies Thm. YM3_I:D at the fixed constants and at the lowered choice and frees the bound of the volume by the chessboard estimate for the reflection positive Wilson measure.

Sum over the corners in S; the bound passes to μ_d in total variation. (d) For $\mathsf{p}^{[d]}$: Thm. YM3_I:D, along the line of couplings of (b), bounds, at the level of Λ_d, the mass of the history without large-field regions below by $e^E e^{-(c_{\mathrm{low}} + \frac{9}{2}\log x_{[d]})(\ell L^d)^3}$ and the partition function above by $e^E e^{(\mathsf{C}_{\mathrm{maj}}+2)(\ell L^d)^3}$, with one number E, and that history is carried by fields whose block field of the depth d lies in the inner ball I_d of the line. For $\mathsf{q}^{[d]}$: Thm. YM3_I:D along the line bounds the density of the block field of the depth d below, as a measure, by $e^{E - C_5(\ell L^d)^3}$ times the small-field indicator times the Boltzmann factor of the background over V; on the ball of the fields with $|V(b) - \mathbf{1}| < s_d$ at every bond, whose plaquettes are $< 4s_d \le \varepsilon_{[d]}$ so that it lies in I_d, the integral of this density is at least $\exp[-(3\log \frac{7}{s_d^3} + 24 B_3^2 s_d^2 x_{[d]})(\ell L^d)^3]$; with the upper bound on the partition function this gives $\mathsf{q}^{[d]}$.

The floors hold no K; let $K \to \infty$. □

Lemma 10.3 (Two consecutive cutoffs over one block field; the Lipschitz bound of the tower)**.** In the one-torus setting put $\varepsilon_{\mathrm{r}}(s) = 2c_\rho \varepsilon_{[s]}$; for two fields V, V' on one lattice put $\mathrm{dist}(V,V') = \max_b \theta(V(b)V'(b)^{-1})$, $\theta(X) \in [0,\pi]$ the angle of $X \in G$, $\cos\theta(X) = \frac{1}{2}\operatorname{tr} X$; for a real 1-form B on the bonds of a lattice (its values $B(b)$ hermitian and traceless, $B(\bar{b}) = -B(b)$) put $\sup|B| = \max_b |B(b)|$ and $(e^{iB}X')(b) = e^{iB(b)}X'(b)$ at the positively oriented bonds b. With the chart constant $C_{\mathrm{db}} = 8L^3$ and the constant $\gamma_{\mathrm{r}} \in (0,1]$ of the lower bound on the excess of a competitor (the global hypotheses preceding equation YM3_I:(2132), Prop. YM3_I:113.5), the constant K_u of the comparison of the background actions at two consecutive cutoffs (equation YM3_I:(2155), YM3_I:(2154)), the numbers $\varepsilon_{\mathrm{cp}}$, $\varrho_{\mathrm{no}} = \min\{r_{\mathrm{B}}, \varrho_{\mathrm{R}}^0\}$ and S_{e} of the definitions preceding and in equation YM3_I:(2133), the radius $c_4 \in (0,1]$ and the positive curvature ceiling $c_2' = c_2^{\mathrm{g}}$ of Lem. 9.11(v), decreasing c_4 to 1 if necessary, and the constant $B_5^{\mathrm{e}} \ge 8$ of the response of the background to a change of its datum (§5.3), we put

$$\mathsf{c}_{\mathrm{A}} = 16\, C_{\mathrm{db}}\, c_\rho\, (\mathsf{K}_u/\gamma_{\mathrm{r}})^{1/2}, \qquad a_s(K) = \mathsf{c}_{\mathrm{A}}\, \varepsilon_{[s]}\, (\ell L^s)^{3/2} L^{-(K-s+1)/2}, \qquad r_s(K) = \sum_{K'\ge K} a_s(K') = \frac{a_s(K)}{1-L^{-1/2}} = r_s(s) L^{-(K-s)/2},$$

$$\varepsilon_{\mathrm{L}} = \min\{\varepsilon_{\mathrm{cp}}, c_2'/B_3\}, \qquad \beta_{\mathrm{L}} = \min\Big\{1,\ \tfrac{1}{2}\varrho_{\mathrm{no}},\ \frac{c_4}{2B_5^{\mathrm{e}}\mathsf{S}_{\mathrm{e}}}\Big\}, \qquad \mathsf{c}_9 = \frac{2L}{L^{1/2}-1}\, B_5^{\mathrm{e}}\mathsf{S}_{\mathrm{e}};$$

c_{A}, ε_{L}, β_{L} and c_9 depend on the constants alone, $a_s(\cdot)$ decreases, and $\varepsilon_{\mathrm{r}}(s) \le \varepsilon_{\mathrm{L}}$ at every admitted depth s, by the conditions of the comparison of two consecutive cutoffs that define $\Theta_\star$. (a) *One depth.* Let s be admitted, $K \ge \max\{2, s+1\}$ and $X \in \mathcal{V}_s$ with $\varpi(X) < \varepsilon_{\mathrm{r}}(s)$. Then $H_s^K(X)$ and $H_s^{K+1}(X)$ are defined; each is in the axial gauge on Λ_s, has the average X and has plaquettes at most $2B_3 L^{-2}\varpi(X)$; and

$$\mathrm{dist}\big(H_s^{K+1}(X), H_s^K(X)\big) \le a_s(K).$$

Hence $H_s^\infty(X) = \lim_K H_s^K(X)$ exists, $\mathrm{dist}(H_s^\infty(X), H_s^K(X)) \le r_s(K)$ uniformly on $\{\varpi < \varepsilon_{\mathrm{r}}(s)\}$, H_s^∞ is continuous there, $\mathcal{M}H_s^\infty(X) = X$, $\varpi(H_s^\infty(X)) \le 2B_3 L^{-2}\varpi(X)$, and $H_s^\infty(X^u) = H_s^\infty(X)^{u\circ\pi}$

for every gauge transformation u of Λ_s, π mapping each site of Λ_{s+1} to the site of Λ_s whose block holds it. (b) *Along the tower.* Let d be admitted, $K \geq \max\{2, d+1\}$, $W \in J_d$, and $Y_j = H_{d,j}^{K+1}(W)$, $0 \leq j \leq K+1-d$, the tower over W at the cutoff $K+1$. Then for $0 \leq j < K-d$

$$\operatorname{dist}\bigl(Y_{j+1}, H_{d+j}^{K}(Y_j)\bigr) \leq L^{-j/2} a_d(K),$$

whereas (a) at the depth $d+j$ gives the number $a_{d+j}(K) = (\varepsilon_{[d+j]}/\varepsilon_{[d]}) L^{2j} a_d(K)$, up to $L^{2j} a_d(K)$. (c) *The Lipschitz bound in the block field.* Let $K \geq 2$, $d' < K$, $X' \in \mathcal{V}_{d'}$ with $\varpi(X') < \varepsilon_{\mathrm{L}}$, B a real 1-form on the bonds of $\Lambda_{d'}$ with $\sup|B| \leq \beta_{\mathrm{L}}$, and $X = e^{iB}X'$. Then $\mathcal{C}_K(X')$ and $\mathcal{C}_K(X)$ are non-empty, each one orbit of $\mathcal{G}_K^{d'}$, and for $U' \in \mathcal{C}_K(X')$, $U \in \mathcal{C}_K(X)$, every depth d'' with $d' \leq d'' \leq K$ and every closed path Γ of $\Lambda_{d''}$ with N bonds

$$\Bigl|\tfrac{1}{2}\operatorname{tr}(\mathcal{M}^{K-d''}U)(\Gamma) - \tfrac{1}{2}\operatorname{tr}(\mathcal{M}^{K-d''}U')(\Gamma)\Bigr| \leq 2B_5^{\mathrm{e}}\mathsf{S}_{\mathrm{e}}\, N\, L^{-(d''-d')} \sup|B|;$$

the constant holds neither K, nor d', nor the number $d''-d'$ of the depths between. (d) *Two consecutive cutoffs: the loops at the depth $d+n$.* Let d be admitted, $n \geq 1$, $K \geq \max\{2, d+n+1\}$ with $a_d(K) \leq \beta_{\mathrm{L}}$, $W \in J_d$, Γ a closed path of Λ_{d+n} with N bonds and $W_\Gamma(V) = \frac{1}{2}\operatorname{tr} V(\Gamma)$. Then

$$|W_\Gamma(H_{d,n}^{K+1}W) - W_\Gamma(H_{d,n}^{K}W)| \leq \mathsf{c}_9\, N\, L^{-n/2} a_d(K), \qquad |W_\Gamma(H_{d,n}^{\infty}W) - W_\Gamma(H_{d,n}^{K}W)| \leq \mathsf{c}_9\, N\, L^{-n/2} r_d(K),$$

$H_{d,n}^{\infty}W$ being the limit of the towers $H_{d,n}^{K}W$ (Prop. 10.4(a)). ■

Sketch of proof. (a) Let $t = K-s$ and $\tilde{U}$ a background over X at the cutoff $K+1$ (it exists by Thm. YM3_I:E in the setting of Section YM3_I:52.7, as $\varepsilon_{\mathrm{r}}(s) \leq \bar{\varepsilon}$). Its average $U' = \mathcal{M}\tilde{U}$, a field of the bare lattice of the cutoff K with $\mathcal{M}^t U' = X$, is a competitor there (Lem. YM3_I:113.8): its action, in the units of the level t, exceeds the action of the backgrounds over X at the cutoff K by at most $2\mathsf{K}_u \varepsilon_{\mathrm{r}}(s)^2 L^{-t}(\ell L^s)^3$, the comparison of the background actions at the cutoffs K and $K+1$ (equation YM3_I:(2155)) and the energy of the background (equation YM3_I:(2149)) bounding the two parts of this excess.

The excess is at least $\frac{1}{8}\gamma_{\mathrm{r}} \sum_{k<t} L^{t-k}\|\mathcal{A}_k\|^2$ (Prop. YM3_I:113.5), where $\|\cdot\|$ is the ℓ^2 norm over the bonds and $\mathcal{A}_k$ the real 1-form by which the average $\mathcal{M}^k U'$, in the gauge of its averaged contour variables, differs from the averaged background over $\mathcal{M}^{k+1}U'$; the term $k = t-1$ alone gives $\|\mathcal{A}_{t-1}\| \leq 4(\mathsf{K}_u/\gamma_{\mathrm{r}})^{1/2}\varepsilon_{\mathrm{r}}(s)(\ell L^s)^{3/2} L^{-(t+1)/2}$. In that gauge $\mathcal{M}^{t-1}U'$ is the averaged background over X at the cutoff $K+1$, that is $H_s^{K+1}(X)$ (the axial-normalization argument in the proof of Prop. YM3_I:52.3), and the chart of the fibre over X writes it as $e^{iB'}H_s^K(X)$ with $\sup|B'| \leq 2C_{\mathrm{db}} \sup|\mathcal{A}_{t-1}|$ (equation YM3_I:(2138)); with $\varepsilon_{\mathrm{r}}(s) = 2c_\rho \varepsilon_{[s]}$ this is $a_s(K)$.

The axial gauge, the average and the plaquettes are equation YM3_I:(953). Summed over the cutoffs, a geometric series of ratio $L^{-1/2}$, the bound makes $(H_s^K(X))_K$ a Cauchy sequence in a compact space, uniformly in X; each H_s^K is continuous (Prop. YM3_I:52.3) and covariant, a bare gauge transformation equal to $u \circ \pi$ on Λ_{s+1} carrying the elements of the minimal orbit over X whose averages are in the axial gauge to those over X^u; $\mathcal{M}$ is continuous on the fields with plaquettes $< 1/200L^2$, which holds the fields $H_s^K(X)$; and the limit inherits all this. (b) The same argument with $\tilde{U} = Y_{t+1}$, $t = K-d$, the bare field at the end of the tower of the cutoff $K+1$: it is a background over W and over every Y_j, with $\mathcal{M}^{t+1-j}\tilde{U} = Y_j$ (the minimal orbit over $H_{d+j}^{K+1}(Y_j)$ lies in the one over Y_j, [7, Lem. B.4](2)).

The competitor $U' = \mathcal{M}\tilde{U} = Y_t$ has the averages $\mathcal{M}^k U' = Y_{t-k}$, and its excess is bounded once, at the depth d, by $2\mathsf{K}_u \varepsilon_{\mathrm{r}}(d)^2 L^{-t}(\ell L^d)^3$. At $k = t-1-j$ the term of the lower bound has the weight $L^{t-k} = L^{j+1}$, so $\|\mathcal{A}_k\| \leq 4(\mathsf{K}_u/\gamma_{\mathrm{r}})^{1/2}\varepsilon_{\mathrm{r}}(d)(\ell L^d)^{3/2} L^{-(t+1)/2} L^{-j/2}$; at the point Y_j the chart of the fibre and the axial-normalization argument in the proof of Prop. YM3_I:52.3 give $Y_{j+1} = e^{iB'}H_{d+j}^K(Y_j)$ with $\sup|B'| \leq 2C_{\mathrm{db}} \sup|\mathcal{A}_k| \leq L^{-j/2} a_d(K)$. The single bound on the excess at the depth d pays for all the depths of the tower together. (c) The response of the background

to a change of its datum (Thm. YM3_I:E and Lem. YM3_I:51.6, with the thresholds of §5.3, for the $n = K - d'$ averaging steps on the whole torus, the datum X' and the radius ε_{L}): $\mathcal{C}_K(X')$ and $\mathcal{C}_K(e^{iB}X')$ are single orbits of $\mathcal{G}_K^{d'}$, and a background over X is, up to a bare gauge transformation, $e^{iL^{-n}\mathcal{H}(B)}U^\circ$ with $U^\circ \in \mathcal{C}_K(X')$ and $|\mathcal{H}(B)(b)| \le \sigma := B_5^{\mathrm{e}}\mathsf{S}_{\mathrm{e}} \sup|B|$ at every bare bond b, the exponentially decaying weights of the response summing to at most S_{e}.

The averages of every order $i \le n$ of a field so moved are, up to gauge, $e^{ia_i}\mathcal{M}^i U^\circ$ with $|a_i| \le 2L^{-(n-i)}\sigma$ (Lem. 9.11(v)). Along a closed path a gauge transformation conjugates the holonomy; the product bound $|\prod_i A_i - \prod_i A_i'| \le \sum_i |A_i A_i'^{-1} - \mathbf{1}|$ on G, $|e^{iY} - \mathbf{1}| \le |Y|$ and $|\operatorname{tr} Y| \le 2|Y|$ give the bound at $i = K - d''$. (d) A telescope over the depths $d + j$, $0 \le j < n$, along the tower $Y_j = H_{d,j}^{K+1}(W)$ of the finer cutoff. For a field Y of a depth between d and $d+n$ and $K' \in \{K, K+1\}$ let $\mathsf{u}^{K'}(Y)$ be the loop along Γ of the average down to Λ_{d+n} of a background over Y at the cutoff K': one number on the orbit, with $\mathsf{u}^{K'}(Y) = \mathsf{u}^{K'}(H_s^{K'}(Y))$ at the depth s of Y ([7, Lem. B.4](2)) and $\mathsf{u}^{K'}(Y) = W_\Gamma(Y)$ at the depth $d + n$.

With $\Delta_j = \mathsf{u}^{K+1}(Y_j) - \mathsf{u}^K(Y_j)$ one has $\Delta_n = 0$, Δ_0 the left side of the first bound, and $\Delta_j - \Delta_{j+1} = \mathsf{u}^K(Y_{j+1}) - \mathsf{u}^K(H_{d+j}^K(Y_j))$. By (b), $H_{d+j}^K(Y_j) = e^{iB}Y_{j+1}$ with $\sup|B| \le L^{-j/2}a_d(K) \le \beta_{\mathrm{L}} < \pi$, and $\varpi(Y_{j+1}) \le 2B_3\varpi(W)L^{-2(j+1)} < \varepsilon_{\mathrm{r}}(d+j+1) \le \varepsilon_{\mathrm{L}}$; so (c) at $d' = d+j+1$ and $d'' = d+n$, $n-j-1$ depths apart, gives $|\Delta_j - \Delta_{j+1}| \le 2B_5^{\mathrm{e}}\mathsf{S}_{\mathrm{e}}NL^{-(n-j-1)}L^{-j/2}a_d(K)$. The roles of the two fields are so chosen because (c) asks a bound on the plaquettes of X' alone, which the tower of the finer cutoff has. The sum over j is a geometric series of the ratio $L^{1/2}$, $\sum_{j<n} L^{j/2} \le L^{n/2}/(L^{1/2} - 1)$, which gives $\mathsf{c}_9 N L^{-n/2}a_d(K)$.

The conditions persist for $K' \ge K$, as a_d decreases, and the sum over $K' \ge K$ with $\sum_{K' \ge K} a_d(K') = r_d(K)$ gives the second bound. □

Proposition 10.4 (Smoothed loops over a fixed field)**.** In the one-torus setting let d be admitted, $W \in J_d$ and C a lattice loop of $\Lambda_{d'}$, $d' \ge d$. Put $\varepsilon_{\mathrm{r}} = 2c_\rho\varepsilon_{[d]}$; let $\mathsf{c}_{\mathrm{A}} > 0$ be the constant with which the comparison of two consecutive cutoffs (Prop. YM3_I:113.9) bounds the distance of the averaged backgrounds of two consecutive cutoffs over one block field (Lem. 10.3(a)), and $\mathsf{c}_9, \beta_{\mathrm{L}} > 0$ the constant and the radius of the Lipschitz bound of the tower in the block field (Lem. 10.3(c), (d)), the numbers formed in Lem. 10.3; all three depend on the constants alone. Put $a_s(K) = \mathsf{c}_{\mathrm{A}}\varepsilon_{[s]}(\ell L^s)^{3/2}L^{-(K-s+1)/2}$ and $r_d(K) = \sum_{K' \ge K} a_d(K') = r_d(d)L^{-(K-d)/2}$; let K_2 be the least $K \ge 2$ with $\lfloor \frac{K-d}{3} \rfloor \ge \max\{1, d' - d\}$ and $a_d(K) \le \beta_{\mathrm{L}}$; and put $\mathsf{c}_{\mathrm{f}} = \frac{28L\varepsilon_{\mathrm{r}}}{1-L^{-1}} + \mathsf{c}_9 r_d(d)$ and $\mathsf{c}_{\mathrm{g}} = \frac{1544L^2\varepsilon_{\mathrm{r}}}{1-L^{-1}} + 6\mathsf{c}_9 r_d(d)$, numbers that depend on d, ℓ, b and the constants (K_2 also on $d' - d$). (a) $V_n = H_{d,n}^\infty(W)$ satisfies $\mathcal{M}V_{n+1} = V_n$, $\varpi(V_n) \le 2B_3\varpi(W)L^{-2n}$ and, for $n \ge d' - d$, $|W_C(V_n) - W_C(V_{n+1})| \le 28B_3|C|L^d\varpi(W)L^{-n}$. So $W_C^{[d]}$ exists uniformly on J_d, with $|W_C^{[d]} - W_C(V_n)| \le \frac{14}{1-L^{-1}}|C|L^d\varepsilon_{\mathrm{r}}L^{-n}$ there; it is continuous and gauge invariant on J_d with values in $[-1, 1]$; $F_C^{[d]}$ is Borel; and $0 \le 1 - W_C^{[d]}(W) \le \frac{1}{2}(2B_3\varpi(W)L^{2d}s_1s_2)^2$ for a coordinate rectangle with sides s_1, s_2. (b) For $K \ge \max\{2, d+1, d'\}$, $\mathcal{C}_K(W)$ is one non-empty orbit of $\mathcal{G}_K^d$; w_C^K is continuous and gauge invariant on J_d; $\sup_{J_d}|w_C^K - W_C^{[d]}| \to 0$, and this supremum is at most $|C|L^d\mathsf{c}_{\mathrm{f}}L^{-(K-d)/3}$ for $K \ge K_2$. (c) For curves and any shadows, $|W_{\Gamma^{(d+n)}}(V_n) - W_{\Gamma^{(d+n+1)}}(V_{n+1})| \le 772(|\gamma|L^{d+n+1} + 2)L^{-2n}\varepsilon_{\mathrm{r}}$; so $W_\gamma^{[d]}$ exists uniformly on J_d, does not depend on the shadows, equals $W_C^{[d]}$ along a lattice loop C, and is continuous and gauge invariant, and F_γ is Borel. $w_\gamma^K(W)$ is one number on $\mathcal{C}_K(W)$; two shadows on the bare lattice change it by at most $1496(|\gamma|L^K + 2)L^{-2(K-d)}\varepsilon_{\mathrm{r}}$; and $\sup_{J_d}|w_\gamma^K - W_\gamma^{[d]}| \le (|\gamma|L^d + 2)\mathsf{c}_{\mathrm{g}}L^{-(K-d)/3}$ for $K \ge K_2$, taken at $d' = d$. (d) For $K \ge K_{\mathrm{bf}}$ with

$\lceil K/1260\rceil \ge d$,

$$\Big|\langle F_C^{[d]}(\mathcal{M}^{K-d}U)\rangle_K - \int F_C^{[d]}\,\mathrm{d}\mu_d\Big| \le \mathfrak{r}_K := \mathsf{C}_{\mathrm{bf}} S_L \ell^3 K^{2\mathfrak{p}_0+1/2} L^{-K/10}, \qquad S_L = \sum_{j\ge 0}(1+j)^{2\mathfrak{p}_0+1/2} L^{-j/10},$$

a bound free of C and d, where C_{bf} and K_{bf} are the numbers of Def. 2.7(5), formed at an interval of labels under the condition (Tor) on the side ℓ (§2.3.1) that holds $\mathfrak{b}$, and depend on the label only through that interval (Cor. 2.8(a)); in the frame $[\mathfrak{b}_-,\mathfrak{b}_+]$ is such an interval, and it gives one pair of numbers for all $\mathfrak{b}\in\mathsf{J}$, with the range $K\ge K_{\mathrm{bf}}$, $\lceil K/1260\rceil\ge d$; for $K\ge K_2$ besides, $|\langle(\mathbf{1}_{J_d}w_C^K)(\mathcal{M}^{K-d}U)\rangle_K - \int F_C^{[d]}\,\mathrm{d}\mu_d| \le |C|L^d\mathsf{c}_{\mathrm{f}}L^{-(K-d)/3}+\mathfrak{r}_K$. The same holds for curves, with F_γ, w_γ^K and $(|\gamma|L^d+2)\mathsf{c}_{\mathrm{g}}$. (e) In the frame put $\mathfrak{b}'=\mathfrak{b}-\bar B$ and $J'_d=\{\varpi<c_\rho\varepsilon(L^d\mathfrak{b}')/B_3\}$, and let $d\ge d_0$, the least integer $d_0\ge 1$ with $L^{d_0}\gamma^{-2}\ge\Theta_\star$ and $\log(L^{d_0}\gamma^{-2})>2\mathfrak{p}_0$. Then the one-torus setting holds at the labels $\mathfrak{b}$ and $\mathfrak{b}'$ and d is admitted at both; $I_d\subset J_d\subset J'_d$ and $\mathsf{O}_{K-d}\subset J'_d$ for every $K>d$, so that m_{K-d} gives no mass to $\mathcal{V}_d\setminus J'_d$, and $\mu_d^K(\mathcal{V}_d\setminus J'_d)\le R_d$; over every $V\in J'_d$ and every $K>d$ the backgrounds form one non-empty orbit, w_C^K is Borel, and (a), (b) hold on J'_d, also for families: $\sup_{J'_d}|w_{\mathsf{C}}^K-\prod_i W_{C_i}^{[d]}|\to 0$. ■

Sketch of proof. (a) Over W the backgrounds of the finer cutoffs are built one block step at a time, $H_{d,n}^K(W)$, and all of them are, up to gauge, block averages $\mathcal{M}^{K-d-n}U$ of one bare background U, whose plaquettes are $< B_3\varpi(W)(a_K/a_d)^2$ (Thm. YM3_I:E in the setting of Section YM3_I:52.7); so the factor $2B_3$ is paid once, not at every step: $\varpi(\mathcal{M}^{K-d-n}U)\le 2B_3\varpi(W)L^{-2n}<\varepsilon_{\mathrm{r}}L^{-2n}\le L^{-3n/2}\,2c_\rho\varepsilon_{[d+n]}$. Induction on n, uniform continuity on the compact set $\{\varpi\le L^{-3n/2}2c_\rho\varepsilon_{[d+n]}\}$ and Lem. 10.3(a) give $H_{d,n}^K\to H_{d,n}^\infty$ uniformly on J_d; the bound passes to V_n, which lies in $\{\varpi<2c_\rho\varepsilon_{[d+n]}\}$, so that $H_{d+n}^\infty(V_n)$ is defined.

The ball $\{\varpi<2c_\rho\varepsilon_{[d]}\}$, larger than J_d by the factor $2B_3\ge 2$, is the domain of the single step H_d^∞; the whole tower needs room for the factor $2B_3$ that the backgrounds pay, which is why the proposition asks $W\in J_d$, and why the smaller ball $I_d\subset J_d$, and not the larger one, enters the starting depth. One refinement costs $|W_\Gamma(\mathcal{M}V)-W_\Gamma(V)|\le 14L^2N\varpi(V)$ for a loop of N bonds if $\varpi(V)<1/200L^2$ (in a gauge fixed along the blocks the average of a nearly flat field moves each bond by $O(L^2\varpi(V))$), which holds as $2c_\rho\varepsilon_{[d]}<1/200L^2$; with $N=|C|L^{d+n}$ the number of bonds grows like L^n while the plaquettes fall like L^{-2n}: a geometric series, whose tail with $\varpi(W)<\varepsilon_{\mathrm{r}}/2B_3$ is the stated bound.

For a rectangle, $a_p=\frac12|\partial U(p)-\mathbf{1}|^2$ and the holonomy of ∂R is an ordered product of conjugated plaquette variables, so $1-W_{\partial R}\le(\sum_{p\subset R}\sqrt{a_p})^2$ (Prop. 10.9(c)), and the bare plaquettes of the background are $<2B_3\varpi(W)(a_K/a_d)^2$. (b) The tower ends in the non-empty set $\mathcal{C}_K(W)$, one orbit by Thm. YM3_I:E in the setting of Section YM3_I:52.7 and §5.2. The comparison of two consecutive cutoffs (Prop. YM3_I:113.9) gives $\mathrm{dist}(H_s^{K+1}(X),H_s^K(X))\le a_s(K)$ for $\varpi(X)<2c_\rho\varepsilon_{[s]}$ (Lem. 10.3(a)). At a depth $d+n$ we split into three terms. $|w_C^K(W)-W_C(\mathcal{M}^{K-d-n}U)|$ and $|W_C^{[d]}(W)-W_C(V_n)|$ are at most $\frac{14}{1-L^{-1}}|C|L^d\varepsilon_{\mathrm{r}}L^{-n}$, uniformly in K, by the series of (a); and $|W_C(H_{d,n}^KW)-W_C(V_n)|\le|C|L^{d+n}\sup\mathrm{dist}(H_{d,n}^K,H_{d,n}^\infty)\to 0$ at fixed n. Choose n, then K. For the rate we use the Lipschitz bound of the tower in the block field (Lem. 10.3(d)): for $n\ge1$, $K\ge\max\{2,d+n+1\}$ with $a_d(K)\le\beta_{\mathrm{L}}$, $W\in J_d$ and every closed path Γ of Λ_{d+n} with N bonds, $|W_\Gamma(H_{d,n}^{K+1}W)-W_\Gamma(H_{d,n}^KW)|\le\mathsf{c}_9NL^{-n/2}a_d(K)$.

It is a telescope over the depths $d+j$, $0\le j<n$, along the tower $Y_j=H_{d,j}^{K+1}(W)$ of the finer cutoff: at the depth $d+j$ the one-depth smoothings H_{d+j}^{K+1} and H_{d+j}^K of the two cutoffs, applied to Y_j, give fields of the depth $d+j+1$ at distance at most $L^{-j/2}a_d(K)$ (Lem. 10.3(b): the excess of the action of the averaged background of the finer cutoff is bounded once, at the depth d, and

the term of the depth $d+j$ in the lower bound on the excess carries the weight L^{j+1}; the bound $a_{d+j}(K)$ of one depth, applied at Y_j, is up to L^{2j} times larger), and the loop along Γ of the average down to Λ_{d+n} of the background over such a field, $n-j-1$ depths below it, moves by at most $2B_5^{\mathrm{e}}\mathsf{S}_{\mathrm{e}}NL^{-(n-j-1)}$ times that distance, a constant free of the cutoff and of the number of the depths between (Lem. 10.3(c)); the sum over j is a geometric series of the ratio $L^{1/2}$, which gives c_9.

Summed over $K'\ge K$, with $\sum_{K'\ge K}a_d(K')=r_d(K)$, at $\Gamma=C^{\langle d+n\rangle}$ it bounds the middle term by $\mathsf{c}_9|C|L^dL^{n/2}r_d(K)$; at $n=\lfloor\frac{K-d}{3}\rfloor$ both L^{-n} and $L^{n/2}L^{-(K-d)/2}$ are at most $L\cdot L^{-(K-d)/3}$, which gives c_{f}. Continuity and gauge invariance of w_C^K on J_d come with the tower. (c) Two shadows of consecutive lattices, evaluated at V_{n+1}, differ by a band of $O(|\gamma|L^{d+n+1}+2)$ plaquettes of size $O(L^{-2n}\varepsilon_{\mathrm{r}})$; two shadows on the bare lattice, evaluated at a background over W, differ by a band of $O(|\gamma|L^K+2)$ bare plaquettes of size $O(L^{-2(K-d)}\varepsilon_{\mathrm{r}})$; then argue as in (a), (b). (d) By Cor. 2.8 and (2.xvii), two consecutive cutoffs differ on a gauge invariant bounded Borel function F of the block field of a depth at most $\lceil K/1260\rceil$ ($1260=180(3N+4)$ at $N=1$ in §2.3.1) by at most $\mathsf{C}_{\mathrm{bf}}\ell^3K^{2\mathfrak{p}_0+1/2}L^{-K/10}\sup|F|$; the comparison there is made through the small-field chains of the two cutoffs at that depth, and its numbers, formed at an interval of labels under (Tor), serve every label of the interval (Cor. 2.8(a)). In the frame $[\mathsf{b}_-,\mathsf{b}_+]$ is such an interval on the side ℓ: $\mathsf{b}_->g^{-2}\ge\gamma^{-2}$, and $MQ(\mathsf{b}_+)\le L^m$, with Q of Lem. 10.1, holds on the sides of $\mathfrak{L}(g)$ at every label of $\mathsf{I}(g)$, as in (e). Sum over $K'\ge K$ with $F=F_C^{[d]}$; the second mean adds the rate of (b). (e) $\gamma^{-2}\le g^{-2}<\mathsf{b}'<\mathsf{b}<L(g^{-2}+\bar B)$, so the sides at g satisfy the condition on the side at both labels, and $L^d\mathsf{b}'\ge L^d\gamma^{-2}\ge\Theta_\star$; the constraint sets, the action and the regularity radius hold no label, so (a), (b) apply at the label b' on its ball J'_d. $x_{K-d}\ge L^d\mathsf{b}-\bar B\ge L^d\mathsf{b}'$, ε decreases for $\log t>2\mathfrak{p}_0$ and $c_\rho\ge 2B_3$; so $2\varepsilon(x_{K-d})\le 2\varepsilon(L^d\mathsf{b}')\le c_\rho\varepsilon(L^d\mathsf{b}')/B_3$, that is $\mathsf{O}_{K-d}\subset J'_d$, and $J_d\subset J'_d$ likewise. $\mu_d^K(\mathcal{V}_d\setminus J'_d)\le\mu_d^K(\mathcal{V}_d\setminus I_d)\le R_d$ by Prop. 10.2(b). For families $|\prod_i x_i-\prod_i y_i|\le\sum_i|x_i-y_i|$ on $[-1,1]$. □

The new estimate is one renormalisation step with lines inserted. Its soft part rests on three facts about ordered products of matrices under a measure.

Lemma 10.5 (Ordered products under a measure)**.** (a) Let $\mathfrak{A}$ be a unital normed algebra, I a finite totally ordered set, and for $j\in I$ let At_j be a finite set of atoms α, each with an element $\tau_\alpha\in\mathfrak{A}$, $|\tau_\alpha|\le 1$. A block is a non-empty set of atoms of pairwise different indices, a packing π a set of blocks with pairwise disjoint sets of indices, and $\tau^{\rightarrow}(\pi)$ the product of the τ_α, α in the blocks of π, in the order of I. For complex weights w on the blocks put $Z[w]=\sum_\pi\prod_{\beta\in\pi}w(\beta)\,\tau^{\rightarrow}(\pi)$ and $|w|_1=\sum_\beta|w(\beta)|$. Then $|Z[w]|\le e^{|w|_1}$ and $|Z[w_1+w_2]-Z[w_1]|\le|w_2|_1e^{|w_1|_1+|w_2|_1}$. If bounded functions f_α on a space with a complex measure ν of mass 1 and finite total variation have $\int e^{\sum_\alpha z_\alpha f_\alpha}\,\mathrm{d}\nu=e^{F(z)}$ with F holomorphic near 0 and $F(0)=0$, then

$$\int\prod_{j\in I}^{\rightarrow}\Big(\mathbf{1}+\sum_{\alpha\in\mathsf{At}_j}f_\alpha\tau_\alpha\Big)\,\mathrm{d}\nu=Z[\kappa],\qquad\kappa(\beta)=\Big(\prod_{\alpha\in\beta}\partial_{z_\alpha}\Big)F(0),$$

the joint cumulants of the f_α, $\alpha\in\beta$. *Generators.* Here and below $\mathfrak{su}(2)$ is the real space of the Hermitian traceless 2×2 matrices, as the chart $e^{ig_tB(b)}V_W(0)(b)$ of the frame asks, with the scalar product $\langle X,Y\rangle=\frac12\operatorname{tr}XY$, and t_1,t_2,t_3 is an orthonormal basis of it: $t_at_b+t_bt_a=2\delta_{ab}\mathbf{1}$, $t_a^2=\mathbf{1}$, $|t_a|=1$ in the operator norm and $\sum_a t_a^2=3\cdot\mathbf{1}$ (the Pauli matrices up to a rotation). The components of $X\in\mathfrak{su}(2)$ are $X^a=\langle t_a,X\rangle$, so that $X=\sum_aX^at_a$ and $|X|=(\sum_a(X^a)^2)^{1/2}$; this is used for the variables y_j, the fluctuation field B, the test forms φ and the rotations Υ below, and $(\varphi,B)=\sum_b\sum_a\varphi^a(b)B^a(b)$. A complex 2×2 matrix is written $x=x^0\mathbf{1}+\sum_a x^a\,it_a$, with $x^0=\frac12\operatorname{tr}x$, $x^a=-\frac{i}{2}\operatorname{tr}(xt_a)$ and $|x^0|,|x^a|\le|x|$; as $(\sum_a x^at_a)^2=\sum_a(x^a)^2\mathbf{1}$, $\det x=(x^0)^2+\sum_a(x^a)^2$. For $h\in\mathrm{SU}(2)$, $\mathrm{R}(h)\in SO(3)$ is the rotation with $ht_bh^{-1}=\sum_a\mathrm{R}(h)_{ab}t_a$, $\mathrm{R}(h)_{ab}=\langle t_a,ht_bh^{-1}\rangle$. (b) Let $\mathfrak{A}=M_2(\mathbb{C})^{\otimes r}$ with the operator norm and $I=I_1<\dots<I_r$ a union of intervals; let c_j ($j\in I$) and

p_e ($e=\{j<j'\}$ in a set of pairs, each inside one I_i) be complex numbers, and $w_{\rm g}$ the weight that is c_j at one block $\{\alpha^0_j\}$ with $\tau=\mathbf{1}$, p_e at the three blocks $\{\alpha^a_j,\alpha^a_{j'}\}$, $a=1,2,3$, with $\tau=it_a$ in the i-th factor, and 0 elsewhere. Two pairs cross if their end points interlace. With $\mathfrak{T}=\sum_j|c_j|+3\sum_e|p_e|$, $\mathfrak{X}=\sum 9|p_e||p_{e'}|$ over the crossing pairs of pairs, and $\mathfrak{O}=\sum|\hat w(\beta)||\hat w(\beta')|$ over the ordered pairs (equal ones included) of overlapping sets β,β' among the $\{j\}$ and the e, $\hat w=c_j$ resp. $-3p_e$:

$$\Big|Z[w_{\rm g}]-\exp\Big(\sum_j c_j-3\sum_e p_e\Big)\mathbf{1}\Big|\le(2\mathfrak{X}+\tfrac12\mathfrak{O})\,e^{\mathfrak{T}}.$$

(c) Let $J=J_1\sqcup\cdots\sqcup J_r$ be finite cyclically ordered sets (the lines), $g_j\in\mathrm{SU}(2)$, and $w(y)=\prod_i\frac12\operatorname{tr}\prod_{j\in J_i}(e^{i\sum_a y^a_j t_a}g_j)$ for $y=(y_j)\in(\mathbb{R}^3)^J$; every derivative of w of every order has modulus at most 1. Let ϱ be a complex Borel measure of mass 1 on $\{|y_j|\le\bar y\ \forall j\}$ whose logarithmic Laplace transform $\ell(z)=\log\int e^{(z,y)}d\varrho$ is holomorphic on a polydisc of radius $r_1>1$, with the cumulants $\kappa_\gamma=\partial^\gamma\ell(0)$, and let $s_{jj'}=s_{j'j}$ be real numbers, for the pairs $\{j,j'\}$ of one line and for $j=j'$, with $S=\sum_j s_{jj}+2\sum_{\{j,j'\}}s_{jj'}\ge0$. For a pair of one line let $j\to j'$ be the arc from j forward to j', $G_{j\to j'}=g_jg_{j+1}\cdots g_{j'-1}$, $\mathrm{R}(h)\in SO(3)$ the rotation with $ht_bh^{-1}=\sum_a\mathrm{R}(h)_{ab}t_a$, $\kappa^{(2)}_{jj'}=(\kappa_{e_{ja}+e_{j'b}})_{a,b}$, $d_{jj'}=\kappa^{(2)}_{jj'}-s_{jj'}\mathrm{R}(G_{j\to j'})$, $d_{jj}=\kappa^{(2)}_{jj}-s_{jj}\mathbf{1}$, and $|m|_1=\sum_{a,b}|m^{ab}|$. Then

$$\Big|\int w\,\mathrm{d}\varrho-e^{-\frac32 S}w(0)\Big|\le|\varrho|\,(E_1+E_2+E_3),\qquad E_1=\sum_\alpha|\kappa_{e_\alpha}|,\quad E_2=\sup_{|y_j|\le\bar y}\mathsf{e}(y),\quad E_3=\sum_{|\gamma|\ge3}\frac{|\gamma|}{\gamma!}|\kappa_\gamma|,$$

$|\varrho|$ the total variation, with

$$\mathsf{e}(y)=\sum_{\{j,j'\}}\Big[|d_{jj'}|_1+6|s_{jj'}|\sum_{i\in(j\to j')}|y_i|+|\kappa^{(2)}_{jj'}|_1(|y_j|+|y_{j'}|+|y_j||y_{j'}|)\Big]+\sum_j\big[\tfrac12|d_{jj}|_1+|\kappa^{(2)}_{jj}|_1|y_j|\big]+\sum_{\text{two lines}}|\kappa^{(2)}_{jj'}|_1,$$

the first sum over the pairs of one line, i over the indices strictly between j and j' on the arc. If $\ell=\sum_X\ell_X$, $|\ell_X|\le A_X$ on the polydisc, ℓ_X depending on at most N_X of the variables, and $r_1\ge2$, then $E_3\le3\sum_X A_X(N_X/r_1)^3e^{2(N_X+3)/r_1}$. ∎

Sketch of proof. (a) $|\tau^{\to}(\pi)|\le1$, and a packing with n blocks arises from $n!$ of the n-tuples of blocks, so $\sum_\pi\prod|w(\beta)|\le e^{|w|_1}$; expanding $\prod(w_1+w_2)(\beta)$ and isolating one block of w_2 gives the second bound. For the identity, multiply out: the left side is $\sum_{S\subset I}\sum_{(\alpha_j)_{j\in S}}\int\prod_{j\in S}f_{\alpha_j}\,\mathrm{d}\nu\prod^{\to}\tau_{\alpha_j}$; the moment is $\prod\partial_{z_{\alpha_j}}e^F$ at 0, and by induction from $\partial e^F=(\partial F)e^F$ the derivatives of e^F in pairwise different variables are, at 0, sums over the partitions of the variables of products of derivatives of F; a choice of S, of the atoms and of a partition is a packing. (b) A packing in the support of $w_{\rm g}$ is a set of singletons, a set m of disjoint pairs and a colouring of m.

If no two pairs of m cross, the sum over the colours is $(-3)^{\#\mathsf{m}}\mathbf{1}$: take a shortest pair; between its two factors τ_a stand only factors $\mathbf{1}$, and $\sum_a\tau_a\tau_a=-\sum_a t_a^2=-3$; induct. The packings with a crossing contribute at most $2\mathfrak{X}e^{\mathfrak{T}}$; and $\exp\sum\hat w-Z_{\rm nc}$, $Z_{\rm nc}$ the sum over the packings with the values $(-3)^{\#\mathsf{m}}$, is a sum over tuples of sets with an overlap, at most $\frac12\mathfrak{O}e^{\mathfrak{T}}$. (c) Let $f(\tau)=\int w(\tau y)\,\mathrm{d}\varrho$. The cumulant formula $\int y_\alpha G\,\mathrm{d}\varrho=\sum_\beta\frac{1}{\beta!}\kappa_{\beta+e_\alpha}\int\partial^\beta G\,\mathrm{d}\varrho$, for entire G with bounded derivatives at 0 (from $\partial_\alpha Z=Z\,\partial_\alpha\ell$ for $Z=e^\ell$, coefficient by coefficient, then by Taylor polynomials and dominated convergence), gives $f'(\tau)=\sum_{\alpha,\beta}\frac{\tau^{|\beta|}}{\beta!}\kappa_{\beta+e_\alpha}\int(\partial^\beta\partial_\alpha w)(\tau y)\,\mathrm{d}\varrho$.

The terms $|\beta|=0$ are at most $|\varrho|E_1$, those with $|\beta|\ge2$ at most $|\varrho|E_3$. For $|\beta|=1$ the matrix identity $\sum c^{ab}_{jj'}\partial_{y^a_j}\partial_{y^b_{j'}}w=-3Sw+\mathcal{R}$, $|\mathcal{R}|\le2\mathsf{e}(y)$, at $c=\kappa^{(2)}$ holds: by cyclicity the

trace of the line of a pair is $\frac12\operatorname{tr}(u_jHu_{j'}H')$ with $u_j=e^{i\sum_a y_j^a t_a}$; moving $e^{iy\cdot t}$ across t_a costs $|y|$; $\sum_{a,b}\mathrm{R}(G)_{ab}t_aHt_b=\sum_b(Gt_bG^{-1})Ht_b=3H+\rho$ with $|\rho|\le 6|H-G|\le 6\sum_{i\in(j\to j')}|y_i|$ (a telescoping product of unitaries; $\sum_b t_b^2=3$); on the diagonal $\partial_a\partial_b u=-\delta_{ab}u$ up to $2|y|$. So $f'=-3\tau Sf+\mathcal{E}$ with $|\mathcal{E}|\le|\varrho|(E_1+2\tau E_2+E_3)$, and $f(1)=e^{-\frac32 S}f(0)+\int_0^1 e^{-\frac32 S(1-\tau^2)}\mathcal{E}\,\mathrm{d}\tau$, $S\ge0$.

The bound on E_3: Cauchy's estimates for each ℓ_X and $\sum_m m\binom{m+N-1}{m}x^m=Nx(1-x)^{-N-1}$, $x=1/r_1\le\frac12$. □

Proposition 10.6 (One step with lines inserted)**.** In the frame let C be a family with corners in Λ_{i_0}. (a) There are $C_\eta<\infty$ and an integer $s_0\ge i_0$, depending on the constants, g, J and on C only through $\#\mathsf{C}$, $|\mathsf{C}|$, s_C, ρ_C and i_0, not on K, $\mathsf{b}\in\mathsf{J}$, ℓ, t, W, such that for $\mathsf{b}\in\mathsf{J}$, $K>s_0$, $0\le t<K-s_0$ and every $W\in\mathsf{O}_{t+1}$

$$\Big|\int W_\mathsf{C}^{(t)}(V_W(B))\,\nu_W(\,\mathrm{d}B)-\lambda_{t+1}W_\mathsf{C}^{(t+1)}(W)\Big|\le C_\eta L^{-(K-t)/4};$$

more precisely, with $s=K-t$ (in (a) and its proof), $\Lambda_\mathsf{C}=\frac{3|\mathsf{C}|}{2M\mathsf{b}_-}$, $\mathsf{g}_s=(\frac{3}{2\mathsf{b}_-})^{1/2}L^{-s/2}$ and $\mathsf{y}_s=\log(2\mathsf{b}_+)+s\log L$, the left side is at most

$$C'(1+\#\mathsf{C}+\Lambda_\mathsf{C})e^{C''\Lambda_\mathsf{C}}\big[\mathsf{y}_s^{2\mathfrak{p}_0}(\mathsf{g}_s+L^{-s/2})+e^{-\rho_\mathsf{C}L^s/LM}\big]+\tfrac32\,\mathsf{c}\,C_X\mathsf{b}_-^{-2}|\mathsf{C}|L^{-s},\qquad \mathsf{c}=\tfrac54\bar B,$$

with C', C'', C_X depending on the constants alone; so for every $\vartheta<\frac12$ the same holds with $L^{-\vartheta(K-t)}$ and another C_η. For one polygon the same bound holds for the holonomy matrix, $|\int\mathrm{Hol}_C(e^{i\eta_t\mathbf{A}(B)}U_+)\,\nu_W(\,\mathrm{d}B)-\lambda_{t+1}\mathrm{Hol}_C(U_+)|\le C_\eta L^{-(K-t)/4}$ in the operator norm, from a fixed base point, $e^{i\eta_t\mathbf{A}(B)}U_+$ being the configuration of the shift formula of step 1 of the proof (at $t=0$, where $\eta_0=1$, the chart $e^{iB'}U_+$ itself). No positivity of ν_W is used. (b) If $\nu_W\ge0$, which is the case for m_{t+1}-almost every W, and $K-t>s_0$, the left side is at most

$$C_\sharp\frac{|\mathsf{C}|}{\mathsf{b}}\Big[(1+\mathfrak{b}_t)^2(\varepsilon_{t+1}+\delta_t+L^{-(K-t)})+e^{-\mathsf{D}_t}+\frac{\bar B}{\mathsf{b}}L^{-(K-t)}\Big],\qquad \mathsf{D}_t=\frac{\delta'_\mathsf{C}L^{K-t}}{LM}-2,$$

where $C_\sharp$ depends only on the constants, n_C and c_C, and s_0 is the least integer $\ge i_0$ such that every $x\ge\frac23\mathsf{b}_-L^{s_0-1}$ satisfies explicit smallness conditions formed from the constants and n_C, so that $s_0=\max\{i_0,s_0'\}$ with s_0' formed from the constants, b_- and n_C alone; with the Euclidean distance in the place of the sup-distance in δ'_C, D_t would carry a further factor $1/\sqrt3$. At every $W\in\mathsf{O}_{t+1}$ a bound of the same form holds with the total variation of ν_W as a further factor. The right side is at most $C_\eta L^{-\vartheta(K-t)}$ for every $\vartheta<\frac12$, with C_η depending on the constants, ϑ, J, $|\mathsf{C}|$, n_C, c_C, δ'_C, and on g through J alone. For the squares Q_i of side a_i with corners in Λ_i one has $s_0=\max\{i,s_0'\}=i+O(1)$, so $s_0=i$ for $i\ge s_0'$, and the bound $C(\mathsf{J})\frac{a_i}{\mathsf{b}}L^{-\vartheta(K-t)}\le C(\mathsf{J})L^{-\vartheta(K-t-i)}$, which depends on the scale of the step only through a_{K-t}/a_i. (c) (The drift along a small square.) Let C be a square of side a_s in a coordinate plane with corners in Λ_s, let $\sigma_1,\dots,\sigma_N$ be its $N=4L^{K-t-s}$ bonds of Λ_{K-t}, and let $\Upsilon_i(B)=\frac1i\log[\mathrm{hol}_{\sigma_i}(e^{i\eta_t\mathbf{A}(B)}U_+)\,\mathrm{hol}_{\sigma_i}(U_+)^{-1}]\in\mathfrak{su}(2)$ be the rotation of σ_i in the shift formula of step 1 of the proof of (a) (at $t=0$ the σ_i are bare bonds, $\Upsilon_i=B'(\sigma_i)$ at a positively oriented σ_i and $\Upsilon_i=-\operatorname{Ad}(U_+(\sigma_i))B'(\bar\sigma_i)$ at a reversed one). For every $\alpha<\frac12$ there are $c_1(\alpha)$ and s_A, depending on the constants, J, α and a lower bound of ℓ, not on g, K, t, s, b, W, such that for $s\ge s_A$, $K>s$, $0\le t<K-s$ and every $W\in\mathsf{O}_{t+1}$

$$\sum_{i,a}\Big|\int\Upsilon_i^a\,\mathrm{d}\nu_W\Big|\le c_1(\alpha)L^{-s}L^{-\alpha(K-t)}.$$

These are the rotations of the shift formula; the loop, being a trace, does not see the difference to the rotations of the background over $V_W(B)$ in any other gauge. (d) (The mean of the fluctuation field.) For every $W\in\mathsf{O}_{t+1}$ and every real 1-form φ on the fluctuation bonds, $|\int(\varphi,B)\,\nu_W(\,\mathrm{d}B)|\le C_4\delta_t(\varepsilon_{t+1}+a_{K-t}/\ell)\sum_b\max_a|\varphi^a(b)|$, C_4 depending on the constants: the mean of the fluctuation field is of the order of the coupling with its logarithm, δ_t, times the bound ε_{t+1} of the curvature of

a tame W, the term a_{K-t}/ℓ coming from the domains that do not fit into the torus; on tame W it is so of the order of the coupling squared, up to logarithms. The order g_t^2 times the curvature, up to a logarithm, is that of the mean of one rotation Υ_i of (c), which holds the further factor g_t of g_tB. (e) (Levels at least as coarse as the corners.) Let $0<\vartheta<\frac12$, $c_0>0$ and $\varrho>0$, let C be a rectangle with sides in $[c_0\varrho,10\varrho]$ and corners in Λ_i, i the integer with $a_i\le\frac12 c_0\varrho<La_i$, and put $a_*=\varrho(\mathsf{b}/\varrho)^{\vartheta/(3+\vartheta)}$, a length coarser than the loop for small ϱ/b. There is C_*, depending on the constants, c_0, ϑ and J alone, such that at every cutoff $K\ge i$, at every level $0\le t<K$ with $a_i\le a_{K-t}\le a_*$ and at every $W\in\mathsf{O}_{t+1}$ with $\nu_W\ge0$, hence for m_{t+1}-almost every W,

$$\Big|\int W_C^{(t)}(V_W(B))\,\nu_W(\,\mathrm{d}B)-\lambda_{t+1}W_C^{(t+1)}(W)\Big|\le C_*\frac{\varrho}{\mathsf{b}}\Big(\frac{a_{K-t}}{\mathsf{b}}\Big)^{\vartheta};$$

so the mean of the defect over the block field, $\bar{\mathsf{D}}_{t+1}$ of Prop. 10.8(b) for C, has modulus at most $C_*\frac{\varrho}{\mathsf{b}}(\frac{a_{K-t}}{\mathsf{b}})^{\vartheta}$, as $Z^{-1}m_{t+1}$ has mass at most 1 (Prop. 10.8(c)). At the levels finer than the corners, $K-t>s_0=\max\{i,s_0'\}$, (b) gives the same bound with another C_* of the same kind: there $|C|\le40\varrho$, $n_C=4$, the chord-arc constant of C is bounded below in terms of c_0, and the bracket of (b), whose distance term vanishes for one polygon ($\delta_C'=\infty$), is at most a multiple of $(a_{K-t}/\mathsf{b})^{\vartheta}$ formed from the constants, ϑ and J. The levels $i<K-t\le s_0'$ occur only when $i<s_0'$, and then $\varrho>2a_{s_0'}/c_0$ and $a_{K-t}\ge a_{s_0'}$, so that there the bound 2 of the defect at $\nu_W\ge0$ is at most $C_*\frac{\varrho}{\mathsf{b}}(\frac{a_{K-t}}{\mathsf{b}})^{\vartheta}$ for C_* of the same kind. So the bound of one step holds at every level with $a_{K-t}\le a_*$; this is the bound of one step in Thm. AE(8)(b). ∎

Sketch of proof. (a) *0. The objects of the step.* A *triple* on a domain X is $P=(U,A',\mathbf{J})$, U an SU(2)-valued configuration and A', $\mathbf{J}$ complex 1-forms on the bare bonds of X; it enters through its pair $\mathbf{p}(P)=(e^{i\eta_{t+1}A'}U,\mathbf{J})$, and a gauge transformation u acts by $P^u=(U^u,\mathrm{Ad}(u)A',\mathrm{Ad}(u)\mathbf{J})$ (§5.2). The *background triple* of $W\in\mathsf{O}_{t+1}$ is $P_W=(U_+,0,J^{(t+1)}(U_+))$, U_+ a background over W whose average $\mathcal{M}^tU_+$ is in the axial gauge and $J^{(t+1)}(U_+)$ its current (the background triple defined before equation YM3_I:(827), §5.3.1); by Thm. YM3_I:E in the setting of Section YM3_I:52.7 at the radius $2\varepsilon_{t+1}$, $|\partial U_+(p)-\mathbf{1}|$ and $|(D^{\mathsf{T}}_{U_+}\partial U_+)(b)|$ are below $\alpha\eta_{t+1}^2$ at all bare plaquettes and bonds, $\alpha:=2B_3\varepsilon_{t+1}$, and $|J^{(t+1)}(U_+)|<\alpha$ (§5.3.1 at the level $t+1$).

The *trivial triple* is $(\mathbf{1},0,0)$; the constant gauge transformations fix it. For a pair $v=(A,\mathbf{J})$ of 1-forms on X, $P[v]=(\mathbf{1},A,\mathbf{J})$, and $\|v\|_X$ is the largest of $|A|$, $|\nabla A|$, $|\mathbf{J}|$ on the bonds of X (§5.2). The *analyticity core* $\mathcal{U}^\circ(X)$ is a set of complex triples on X given by bounds whose radii are formed from the constants alone (Def. YM3_I:49.1); it is carried onto itself by $P\mapsto P^u$, kept by restriction to smaller domains, holds $P_W|_X$ for tame W (Prop. 5.31), and holds the ball of the $P[v]$, $\|v\|_X<r_\alpha$, about the trivial triple, r_α formed from the constants (equation YM3_I:(933)).

Three cited facts enter. (i) *The kernel.* On $\{|B|<\mathfrak{b}_t\}$, $\nu_W=\Xi_W^{-1}e^{\mathsf{E}_W}\mu_{C_W}$: the Gaussian measure μ_{C_W} of the covariance C_W of the step at P_W, times the exponential of two local potentials, the interaction and the part of the bare array of couplings, each with a norm with quadratic growth at most $c\,\delta_t$ and below the admitted norm $w_\star$ of the cluster expansion, normalised to the mass 1 (eq. 5.xxxii and Prop. 5.31). (ii) *The provision on the core.* At every triple P of $\mathcal{U}^\circ(X)$, complex ones included, the Gaussian structure and both potentials of the step are given, holomorphic in $(A',\mathbf{J})$ at a fixed U, functions of the pair, gauge covariant, local (their restrictions to a smaller domain are the objects formed there) and with the same bounds as at P_W (Def. YM3_I:52.1 and Thm. YM3_I:W); so the cluster expansion for data with up to four potentials (Thm. YM3_I:R for the logarithm and bound; Cor. YM3_I:47.3 for an added potential and the comparison with the Gaussian part; Prop. YM3_I:47.4 for holomorphy and locality, and the covariance argument following it) bounds every cluster term on the whole core, and not only at P_W. (iii) *A potential on a cube.* If $|\partial U-\mathbf{1}|$ and $|D^{\mathsf{T}}_U\partial U|$ are below $\alpha\eta^2$ near a cube Q of integer side l in level units, with $\mathsf{l}\alpha\le c_*$ and $\mathsf{l}+4$ less than half the side of the torus, then $U^{u^{-1}}=e^{i\eta A}$ on Q for a gauge transformation u and a real 1-form A with $|A|,|\nabla A|<B_*\mathsf{l}\alpha$, B_* and c_* formed from the constants and free of l

(Thm. YM3_I:F). *1. The exact formula.* For $t \geq 1$ the shift of Prop. YM3_I:51.5, taken at the background triple and the 1-form $g_t B$, gives a 1-form $\mathbf{A}(B)$ on the bare bonds such that $e^{i\eta_t \mathbf{A}(B)} U_+$ is a gauge transform of a background over $V_W(B)$ (equation YM3_I:(827)); the loop being gauge invariant, $W_{\mathsf{C}}^{(t)}(V_W(B)) = W_{\mathsf{C}}(e^{i\eta_t \mathbf{A}(B)} U_+)$ exactly, $\eta_t = a_K / a_{K-t}$. $\mathbf{A}$ is holomorphic in B and in the triple, at most $\mathsf{c}_{\mathfrak{h}} \sup |g_t B|$, local through the decoupling parameters, gauge covariant and real at real data (the shift properties proved in Prop. YM3_I:51.5). At $t = 0$ there is no shift system and no decoupling parameter: $\eta_0 = 1$, $W_{\mathsf{C}}^{(0)} = W_{\mathsf{C}}$, and the chart itself gives $V_W(B) = e^{iB'} U_+$, $U_+ = V_W(0)$, with $B' = g_0 B$ on the fluctuation bonds and, at the pivot bond $b_0(c)$ of each block bond c, $B'(b_0(c)) = \beta_c(P_W, g_0 B)$, a pivot value that reads the double block of c alone, vanishes at $B = 0$, is at most $2 C_P \sup |g_0 B|$, is holomorphic in the triple and in B, covariant at its first end point and $\mathfrak{su}(2)$-valued at real data (equation YM3_I:(804) and the bounds and covariance proved with it); we put $\mathbf{A}(B) := B'$.

In both cases $U' := e^{i\eta_t \mathbf{A}(B)} U_+$ is the configuration of the shift formula. *2. Segments.* The trace does not see where a closed path begins: each C_i is cut at one site, its base point, and run from there. For $s \geq i_0 + m' + 1$, which s_0 ensures, the corners are centres of cubes of side M of the level t, and the cubes that C passes are all different; we cut C into $N = |\mathsf{C}|/(M a_{K-t})$ segments, one for each such cube, the segment j being the steps of the path that start in the j-th cube. The segments of C_i form an interval I_i of indices, in the order of the path from the base point, and $I = I_1 < \cdots < I_r$. In the frames R_x carried along the loop by U_+, R_x the product of the $U_+(b')$ over the steps of C_i from the base point to x, the holonomy of U' from the base point is $\prod^{\rightarrow}_{j \in I_i} (\mathbf{1} + k_j) \cdot \mathrm{Hol}_{C_i}(U_+)$, where $\mathbf{1} + k_j$ is the ordered product over the steps b of the segment j of $R_b e^{i\eta_t \sigma_b \mathbf{A}(b^+)} R_b^{-1}$, b^+ the positively oriented bond of the step b, $\sigma_b = \pm 1$ its orientation and R_b the frame at the first end point of b^+ (a step against the orientation gives $U'(b) = U_+(b) e^{-i\eta_t \mathbf{A}(b^+)}$); $\det(\mathbf{1} + k_j) = 1$.

For $t \geq 1$ the brackets in the decoupling parameters of the shift system (equations YM3_I:(806)–YM3_I:(809) and Lem. 5.25) make each k_j a sum of local terms over the domains of the level t that hold the cube of j, each with a factor $e^{-c\,d}$ in its size d, $c > 1$ formed from the constants; at $t = 0$, k_j is one local term, on the cube of the segment with its six neighbours. The linear part of k_j in B, which has no component along $\mathbf{1}$ (its exponents have trace 0), and its remainder have, in the norm with quadratic growth of the cluster expansion, the sizes $\mathsf{n}_\ell \leq c\, g_t$ and $\mathsf{n}_{\mathrm{r}} \leq c\, g_t^2$, c formed from the constants, and not $c\,\delta_t$, which is what keeps every sum below of the order $|\mathsf{C}|/\mathsf{b}$; a sum of them over any set of segments has the same norms up to the factors 3 and 4.

In components, $k_j = \sum_a \ell_j^a\, it_a + \sum_{\mu=0}^3 r_j^\mu \tau_\mu$, $\tau_0 = \mathbf{1}$, $\tau_a = it_a$, with the linear parts ℓ_j^a and the remainders r_j^μ; so the atoms of j are three linear ones, with $\tau = it_a$, and four remainder ones, with $\tau = \mathbf{1}$ and $\tau = it_a$, and $k_j^0 = r_j^0$, $k_j^a = \ell_j^a + r_j^a$. *3. Cumulants.* The linear parts and the remainders are entered as two further potentials into the cluster expansion of the step (Thm. YM3_I:R, with four potential families, and Prop. YM3_I:47.4 for holomorphy and locality; the two families already in ν_W are those of Prop. 5.31), the linear parts as potentials given at every real B, the remainders as potentials given below the threshold $\mathfrak{b}_t$: $\log \int \exp(\sum_\alpha z_\alpha f_\alpha)\, \mathrm{d}\nu_W = \sum_X F_X(z)$, the sum over the domains X of the level $t+1$, with F_X holomorphic, $F_X(0) = 0$, $|F_X| \leq c\, e^{-\lambda d(X)}$, c and $\lambda > 2$ constants of the cluster expansion ($\lambda - 1$ exceeding the growth constant of the number of domains) and $d(X)$ the linear size of X, and F_X depending on the variables of the segments in X alone.

The polydisc is the one on which the inserted potentials stay below the admitted norm: its radii are $\rho_\ell = w_\star / (3\mathsf{n}_\ell)$, of the order $1/g_t$, in the variables of the linear atoms and $\rho_{\mathrm{r}} = w_\star / (4\mathsf{n}_{\mathrm{r}})$, of the order $1/g_t^2$, in those of the remainder atoms, the inverses of the sizes of step 2; they do not depend on N. Cauchy's estimates give all joint cumulants $\kappa(\beta) = \partial_\beta \sum_X F_X(0)$, with tree decay: a block β with n_ℓ linear and n_{r} remainder atoms has $|\kappa(\beta)| \leq \sum_X c\, \rho_\ell^{-n_\ell} \rho_{\mathrm{r}}^{-n_{\mathrm{r}}} e^{-\lambda d(X)}$, of the order $g_t^{n_\ell + 2n_{\mathrm{r}}}$, the sum over the X that hold the segments of β.

We write $E(X)$ for the X-term of this expansion with one or two insertions, that is a first or a second derivative of F_X at $z=0$, regarded as a function of the triple at which the step is formed. *4. The ordered product.* By Lem. 10.5(a), with $\mathfrak{A}=M_2(\mathbb{C})^{\otimes r}$, one tensor factor for each polygon, the integral of the ordered product $\bigotimes_i \prod^{\rightarrow}_{j\in I_i}(\mathbf{1}+k_j)$ is $Z[\kappa]$; the functional $T\mapsto 2^{-r}\operatorname{tr}(T\cdot\bigotimes_i \operatorname{Hol}_{C_i}(U_+))$ has norm at most 1, sends $\mathbf{1}$ to $W_{\mathsf C}^{(t+1)}(W)$ and the integral of the ordered product to $\int W_{\mathsf C}^{(t)}(V_W(B))\,\nu_W(\mathrm dB)$, so that the left side of (a) is at most $|Z[\kappa]-\lambda_{t+1}\mathbf{1}|$.

For all $j\le j'$ in I let $\delta_{cc'}p_{jj'}$ be the second cumulant of the linear parts ℓ_j^c, $\ell_{j'}^{c'}$ formed at the trivial triple with the frames $\mathbf{1}$, under the measure of the step there; it is a multiple of $\delta_{cc'}$ by step 5, it depends neither on W nor on the colour, and $p_{j'j}:=p_{jj'}$. Let $c_j:=\kappa(\{(j;\mathrm r,0)\})=\int k_j^0\,\mathrm d\nu_W$, the first cumulant of the central remainder atom. A pair $j<j'$ of one polygon is *near* if $j'-j\le N_\sharp:=\lfloor s_{\mathsf C}/(2Ma_{K-t})\rfloor$; the arc from the segment j to the end of the segment j' then has length at most $\frac12 s_{\mathsf C}+Ma_{K-t}\le s_{\mathsf C}$ (s_0 ensures $N_\sharp\ge1$), so that it holds at most one corner, and it does not pass the base point.

It is *straddling* if it is not near and $N_i-(j'-j)\le N_\sharp$, $N_i=\#I_i$ (near across the base point), and *far* otherwise; pairs on two polygons are far. A far pair is at sup-distance at least $\rho_{\mathsf C}/a_{K-t}$ in level units, so that the domains that hold it have $d(X)\ge\rho_{\mathsf C}/(LMa_{K-t})-2$. We write $\kappa=w_{\mathrm g}+w_{\mathrm b}$, $w_{\mathrm g}$ the weight of Lem. 10.5(b) with these c_j and with $p_e=p_{jj'}$ at the near pairs $e=\{j<j'\}$, and $w_{\mathrm b}=\kappa-w_{\mathrm g}$; $w_{\mathrm b}$ holds the coloured singletons, the colour defects $\kappa(\{(j;\ell,a),(j';\ell,a')\})-\delta_{aa'}p_{jj'}$ of the near pairs, the other blocks with a remainder atom (all but the central singletons), the blocks of three or more atoms, and the far and the straddling pairs.

With $\mathsf g_s\ge g_t$, $N\mathsf g_s^2=\Lambda_{\mathsf C}$, and the bounds of steps 3 and 5, $|w_{\mathrm b}|_1\le c(1+\Lambda_{\mathsf C})\mathsf y_s^{2\mathfrak p_0}(\mathsf g_s+L^{-s/2})+c\Lambda_{\mathsf C}e^{2-\rho_{\mathsf C}L^s/LM}+c\,\#\mathsf C\,\mathsf g_s^2$, the far pairs giving the term with $\rho_{\mathsf C}$ and the straddling ones, of the order g_t^2 for each polygon, the term with $\#\mathsf C$. Further $\mathfrak T\le c\Lambda_{\mathsf C}$, and $\mathfrak X,\mathfrak O\le c\Lambda_{\mathsf C}\mathsf g_s^2$: a near pair that crosses a near pair $\{j_1<j_2\}$ has an end point among the j_2-j_1 indices between them, and $\sum_{j'}|p_{jj'}|$ and $\sum_{j'}|j'-j||p_{jj'}|$, the latter over the near pairs, are at most $c\,g_t^2$.

Lem. 10.5(a), (b) give $Z[\kappa]=\exp(\sum_j c_j-3\sum_e p_e)\mathbf{1}$ up to these errors. *5. Symmetry.* What is not small by size, the first cumulants of the colour components and the colour defects of the near pairs and of the second cumulants $\langle\ell_j^a;\ell_j^{a'}\rangle$ of one segment, is small by symmetry. By fact (ii) each cluster term $E(X)$ is defined at every triple of $\mathcal U^\circ(X)$, holomorphic there in the sense of the slices, and bounded there as at P_W. Let X be small, $d(X)\le\frac12 s\log L$, and let Q_X be the cube of side $\mathsf l_X=2M(d(X)+2)+1$ (level-$(t+1)$ units) about a cube of X; it holds X and the arc of every near pair whose segments lie in X.

For $s\ge s_0$ the conditions of fact (iii) hold on Q_X for every small X ($\mathsf l_X\alpha\le cMs\log L\,\varepsilon_{t+1}$ is small and $\mathsf l_X+4$ is far below the side of the torus), and fact (iii) at the α of step 0 gives u and a real A with $U_+^u=e^{i\eta_{t+1}A}$ on Q_X and $|A|,|\nabla A|<B_*\mathsf l_X\alpha$; with $v_X:=(A,\operatorname{Ad}(u)J^{(t+1)}(U_+))$ on the bonds of X, $\|v_X\|_X<B'_*\mathsf l_X\alpha$, $B'_*=\max\{B_*,1\}$, and $(P_W|_X)^u$ and $P[v_X]$ have one pair. So the triple lies, up to the gauge transformation u, on the disc $\zeta\mapsto P[\zeta v_X]$, $|\zeta|<\rho_X$, about the trivial triple, inside the core by equation YM3_I:(933); its radius $\rho_X=\min\{r_\flat,1/(3\mathsf l_X)\}/(B'_*\mathsf l_X\alpha)$, $r_\flat$ formed from the constants, has $1/\rho_X\le cM^2\varepsilon_{t+1}(d(X)+2)^2$, c formed from the constants, keeps the frames along the arcs in Q_X within a factor e of $\mathbf{1}$, and is at least 2 for large s.

The locality, the covariance and the dependence on the pair alone of the cluster terms (Prop. YM3_I:47.4 and the covariance argument following it, applied with fact (ii)) turn $E(X)$ at P_W, formed on the torus, into $E(X)$ at $P[v_X]$, formed on X, the frames being changed by one constant rotation, so that the colours are turned by one $O_X\in SO(3)$. At $\zeta=0$ the constant gauge rotations fix the trivial triple: a colour vector that they turn, the first derivative of a term in one coloured atom, vanishes there, and a bilinear form in the colours that commutes with all of them is a multiple of δ_{ab}, $\mathbb C^3$ being irreducible under $SO(3)$. Schwarz' lemma for $\zeta\mapsto E(X)[P[\zeta v_X]]-E(X)[P[0]]$ on the disc gives the factor $1/\rho_X\le cM^2\varepsilon_{t+1}(d(X)+2)^2$, paid by the decay $e^{-\lambda d(X)}$; the domains with $d(X)>\frac12 s\log L$ cost $e^{-\frac12 s\log L}=L^{-s/2}$.

The size that this factor multiplies is the size of step 3, except for the first cumulant of a linear part, whose size is smaller: the Gaussian part of a cluster term, the same term formed with the linear insertions alone under the Gaussian measure of the step without threshold, is even under $B \mapsto -B$ (apply the covariance argument following Prop. YM3_I:47.4 to the reflection $B \mapsto -B$), so that its derivative in one linear atom vanishes at $z = 0$; the first cumulant of a linear part is therefore the derivative of the difference of the term and its Gaussian part, which is at most $c\,\delta_t\, e^{-\lambda d(X)}$ (Cor. YM3_I:47.3, equation YM3_I:(724)): it has the amplitude $c\,\delta_t \mathsf{n}_\ell$, that of the non-Gaussian potentials.

With $\alpha \le c\,\mathsf{g}_s \mathsf{y}_s^{\mathfrak{p}_0}$, $\delta_t \le A_1 \mathsf{g}_s \mathsf{y}_s^{\mathfrak{p}_0}$ and the sums over the N segments, these are the bounds of step 4. *6. Singletons.* $\det(\mathbf{1} + k_j) = 1$ and $\det x = (x^0)^2 + \sum_a (x^a)^2$ give $(1 + k_j^0)^2 + \sum_a (k_j^a)^2 = 1$, that is $k_j^0 = -\frac{1}{2}\sum_a (k_j^a)^2 - \frac{1}{2}(k_j^0)^2$; integrated with $\int f^2 \,\mathrm{d}\nu_W = \langle f; f\rangle + (\int f \,\mathrm{d}\nu_W)^2$, $\langle f; f\rangle$ the second cumulant, and with $\langle \ell_j^a; \ell_j^a\rangle = p_{jj}$ plus a diagonal defect, this gives $c_j = -\frac{3}{2} p_{jj}$ up to errors of the size of step 4 (the diagonal defects, the mixed second cumulants with a remainder, the squares of first cumulants, c_j^2).

As $\sum_{j,j' \in I_i} p_{jj'} = \sum_{j \in I_i} p_{jj} + 2\sum_{j<j'} p_{jj'}$,

$$\sum_j c_j - 3\sum_e p_e = -\frac{3}{2}\sum_i \sum_{j,j' \in I_i} p_{jj'} + \sum_j (c_j + \tfrac{3}{2} p_{jj}) + 3 \sum_{j<j' \text{ in one } I_i, \text{ not near}} p_{jj'},$$

and the last two sums are of the size of the errors of step 4. *7. The flat value.* At the trivial triple with the frames $\mathbf{1}$ the linear part of a segment is linear in B, and the sum over the segments of C_i of its colour component a is $g_t(j_{C_i}, \eta_t(\dot{\mathbf{A}}B)^a) = g_t(X_{C_i}, \eta_t\,\mathrm{d}(\dot{\mathbf{A}}B)^a)$, $\dot{\mathbf{A}}$ the derivative at 0 of $\mathcal{A} \mapsto \mathbf{A}$ formed at the trivial triple (at $t = 0$ the derivative $\mathsf{C}_{\mathbf{1}} = \mathsf{C}_{\mathbf{1},\mathbb{R}} \otimes 1$ of the chart), j_C the current of C and $\mathrm{d}^{\mathsf{T}} X_C = j_C$. The linear part of the shift system at the trivial triple produces, up to one sign, the curvature response of the step: on every side of $\mathfrak{L}(g)$ and for $1 \le t < K$ there is $\mathsf{sg}_t \in \{1, -1\}$ with $\eta_t\,\mathrm{d}(\dot{\mathbf{A}}\mathcal{A}) = \mathsf{sg}_t(\mathsf{T}_t^{\mathrm{cv}} \otimes 1)\mathcal{A}$ for every 1-form $\mathcal{A}$ on the fluctuation bonds, $\mathsf{T}_t^{\mathrm{cv}} = \mathsf{H}_t^{\mathrm{cv}} \mathsf{C}_{\mathbf{1},\mathbb{R}}$, $\mathsf{H}_t^{\mathrm{cv}}$ the curvature response of the level t (the curvature-response construction preceding Lem. YM3_I:120.2).

The proof follows that of Lem. YM3_I:124.3, which is stated for data with two real sources and on large sides only; two changes make it run for the objects of the frame on every side of $\mathfrak{L}(g)$. First, the block field $\mathbf{1}$ lies in O_{t+1}, so that it has the properties that the kernel lemma of fact (i) (Prop. 5.31) gives to every tame block field and on which the hypotheses of (F7) rest; the configuration $\mathbf{1}$ is a background over it, its average $\mathcal{M}^t\mathbf{1} = \mathbf{1}$ is in the axial gauge and its current vanishes, so that its background triple is the trivial triple; and the value of the shift on the curve below is taken from (F7) at the block field $\mathbf{1}$.

Second, we use only the constrained variational equation in the proof of Lem. YM3_I:124.1 and the curvature identity YM3_I:(2230). These calculations hold on every side; the large-side identification with the full Landau inverse is not needed. Constant gauge rotations fix the trivial triple, so covariance makes the shift derivative commute with every $\mathrm{Ad}(g_0)$, $g_0 \in \mathrm{SU}(2)$. It is therefore one real kernel times the identity on colours. For real $\mathcal{A}$, (F7) makes $U_\tau = e^{i\eta_t \mathbf{A}(\tau\mathcal{A})}$ a curve of configurations critical over their own averages. Write $\dot{A} = \dot{\mathbf{A}}\mathcal{A}$ and $\mathcal{M}^t U_\tau = \mathbf{1} + i\tau\dot{V} + o(\tau)$. The cited variational calculation gives a real fine-site function θ and a multiplier ω such that

$$h = \dot{A} + \nabla\theta, \qquad D_{\mathbf{1}}^{\mathsf{T}} D_{\mathbf{1}} h = Q^{\mathsf{T}}\omega, \qquad Qh = \dot{V}.$$

Indeed, differentiation of the action on the moving constraint kernel makes its flat Hessian pairing with $\dot{A}$ vanish on the kernel of the linearized average. Factoring through that average gives the multiplier; its value on gradients is zero, and the gauge normalization restores the straight average $Qh = \dot{V}$. Equation YM3_I:(2230) now gives

$$\eta_t\,\mathrm{d}\dot{A} = \eta_t^2 D_{\mathbf{1}} h = (\mathsf{H}_t^{\mathrm{cv}} \otimes 1)\dot{V}.$$

The chart over $\mathbf{1}$ identifies $\dot V$, up to a constant rotation $\mathrm{Ad}(g_*)$, with $\mathsf{C}_{\mathbf{1}}\mathcal{A}$ minus a gradient. The curvature response annihilates this gradient. On a single colour the remaining rotation gives a sign; linearity in $\mathcal{A}$ makes it the same sign for all $\mathcal{A}$, as in the last paragraph of the proof of Lem. YM3_I:124.3.

At $t=0$, $\mathsf{H}_0^{\mathrm{cv}}=\mathrm{d}$ (the curvature-response calculation preceding Lem. YM3_I:120.2) and the sign is $+$. At the trivial triple the covariance of the step is $\mathbf{C}_*(P_{\mathbf{1}})=(\mathsf{A}_t^{\mathrm{cv}})^{-1}\otimes 1$, and $\mathsf{T}_t^{\mathrm{cv}}(\mathsf{A}_t^{\mathrm{cv}})^{-1}(\mathsf{T}_t^{\mathrm{cv}})^{\mathsf{T}}=\eta_t\,\mathcal{C}_{(t+1)}$ (Prop. YM3_I:120.3, Prop. YM3_I:120.3); so the Gaussian variance of the sum of the linear parts along C_i is $g_t^2\eta_t(X_{C_i},\mathcal{C}_{(t+1)}X_{C_i})$ in each colour, the slice $(X_{C_i},\mathcal{C}_{(t+1)}X_{C_i})$ of the free Coulomb energy, and the sign drops out. The true pair sum differs from the Gaussian one, as in step 5, by the difference of the terms and their Gaussian parts: $\sum_{j,j'\in I_i}p_{jj'}=g_t^2\eta_t(X_{C_i},\mathcal{C}_{(t+1)}X_{C_i})$ up to $cN_i\delta_t g_t^2$, N_i the number of segments of C_i.

With step 6, $\sum_j c_j-3\sum_e p_e=-\frac{3}{2}S$ up to the errors of step 4, where

$$S:=g_t^2\eta_t\sum_i(X_{C_i},\mathcal{C}_{(t+1)}X_{C_i})\ge 0,$$

$\mathcal{C}_{(t+1)}$ being an orthogonal projection: three colours and $\frac{1}{2}\,\mathrm{tr}$ give $\frac{3}{2}$. Between different polygons all pairs are far, and

$$|a_K(X_{C_i},\mathcal{C}_{(t+1)}X_{C_{i'}})|\le C_X|\mathsf{C}|e^{2-\rho_{\mathsf{C}}/LMa_{K-t}}$$

for $i\ne i'$; and $0\le a_K(X_C,\mathcal{C}_{(t+1)}X_C)\le C_X|C|$, C_X formed from the constants. *8. The label.* $g_t^2\eta_t=a_K/(a_{K-t}x_t)$ and $|1/(a_{K-t}x_t)-1/\mathsf{b}|\le \mathsf{c}\,\mathsf{b}^{-2}L^{-(K-t)}$, $\mathsf{c}=\frac{5}{4}\bar B$, turn $e^{-\frac{3}{2}S}$ into λ_{t+1} up to $\frac{3}{2}\mathsf{c}C_X\mathsf{b}_-^{-2}|\mathsf{C}|L^{-s}$, both exponents being ≤ 0.

Steps 4 to 8 together give the bound, with C', C'' formed from the constants; $\sup_s \mathsf{y}_s^{2\mathfrak{p}_0}L^{-s/4}<\infty$ and $\sup_s L^{s/4}e^{-\rho_{\mathsf{C}}L^s/LM}<\infty$ give C_η. The matrix form is step 4 without the trace, for one polygon from its base point.

(b) The variables are now the rotations $\Upsilon_\sigma(B)=\frac{1}{i}\log[(e^{i\eta_t\mathbf{A}(B)}U_+)(\sigma)U_+(\sigma)^{-1}]$ of the $|\mathsf{C}|/a_{K-t}$ bonds σ of Λ_{K-t} along C, oriented as C runs (at $t=0$ the bare bonds of C, with $\Upsilon_\sigma=B'(\sigma)$ at a positively oriented σ and $-\mathrm{Ad}(U_+(\sigma))B'(\bar\sigma)$ at a reversed one), $|\Upsilon_\sigma|\le c\delta_t$, so that $W_{\mathsf{C}}^{(t)}(V_W(B))=w(\Upsilon(B))$, $W_{\mathsf{C}}^{(t+1)}(W)=w(0)$, with w as in Lem. 10.5(c), $g_\sigma=U_+(\sigma)$. By the brackets each Υ_σ is a sum of local terms, gauge covariant at σ_- ($\Upsilon_\sigma\mapsto\mathrm{Ad}(u(\sigma_-))\Upsilon_\sigma$); their linear parts and remainders enter the cluster expansion as in (a), with $\log\nu_W(e^{(z,\Upsilon)})=\sum_X\ell_X(z)$ on a polydisc of radius $r_1\ge c/(n_{\mathsf{C}}g_t)$, one variable $z_\sigma\in\mathbb{C}^3$ for the whole of Υ_σ.

Lem. 10.5(c), with ϱ the image of $\nu_W\ge 0$ under $B\mapsto\Upsilon(B)$, and with $s_{\sigma\sigma'}\delta_{aa'}$ the covariance of the linear parts of Υ_σ^a, $\Upsilon_{\sigma'}^{a'}$ formed at the trivial triple under the Gaussian measure of the step there (real, symmetric, free of W; a multiple of $\delta_{aa'}$ by the constant rotations), gives the bound with $E_1\le C_1\mathsf{l}g_t^2(\mathfrak{b}_t+\mathfrak{b}_t^2)\epsilon_t^\circ$, $E_2\le C_2\mathsf{l}g_t^2[\epsilon_t^\circ+(1+\mathfrak{b}_t)^2\delta_t+e^{-\mathsf{D}_t}]$ and $E_3\le C_3\mathsf{l}g_t^3$, $\mathsf{l}=|\mathsf{C}|/a_{K-t}$, where $\epsilon_t^\circ\le c(\varepsilon_{t+1}+L^{-(K-t)})$ is the price of the symmetry argument: the first cumulants and the defects of the second cumulants from $s_{\sigma\sigma'}\mathrm{R}(U_+(\sigma_-\to\sigma'_-))$ are small by the symmetry of (a), step 5, the transport running along the shorter arc, whose length the chord-arc constant bounds; the pairs on two polygons have $d(X)\ge\mathsf{D}_t$. $\sum_{\sigma,\sigma'\in C_i}s_{\sigma\sigma'}=g_t^2\eta_t(X_{C_i},\mathcal{C}_{(t+1)}X_{C_i})\ge 0$ as in (a), step 7, and $\mathsf{l}g_t^2\le\frac{3}{2}|\mathsf{C}|/\mathsf{b}$; the label costs $\frac{9}{4}C_2\bar B|\mathsf{C}|\mathsf{b}^{-2}L^{-(K-t)}$.

For a complex ν_W of mass 1 Lem. 10.5(c) holds with the total variation as a factor. Positivity: by the disintegration with m_t, m_{t+1}, $\mathsf{h}_{t+1}\ge 0$, $\int H(W)[\int G(V_W(B))\nu_W(\mathrm{d}B)]m_{t+1}(\mathrm{d}W)\ge 0$ for all bounded Borel $H,G\ge 0$; testing a countable set of continuous $G\ge 0$ dense among them, the image of ν_W under $B\mapsto V_W(B)$ is a positive measure of mass 1 for m_{t+1}-almost every W; as $V_W(B)(b)=e^{ig_tB(b)}V_W(0)(b)$ at the fluctuation bonds and $|g_tB(b)|<\delta_t<\log 2$, the chart is injective with a Borel inverse on $\{|B|<\mathfrak{b}_t\}$, so ν_W itself is ≥ 0 for almost every W. The exponent $\frac{1}{2}$ is the order of the error: five of its members are of the order $\frac{|\mathsf{C}|}{\mathsf{b}}x_t^{-1/2}$ up to logarithms, two of them of first order in the curvature of W.

For Q_i: $n_{Q_i} = 4$ and $c_{Q_i} = \frac{1}{2}$ are free of i, there is no distance term, and $|Q_i|/\mathsf{b} = 4a_i/\mathsf{b}$.

(c), (d) A covariant insertion based at a site x_0 is a family of local terms of the step that turns under gauge transformations as a vector at x_0. For such an insertion each term $D(X;P)$ of the first cumulant, formed at the triple P on a domain X (the derivative at 0 of the cluster term of X with the insertion multiplied by z), is holomorphic in P on the analyticity core and bounded there by $\mathsf{A}(\mathbb{W})e^{-\lambda d(X)}$ (fact (ii) in step 0 of the proof of (a)), is a function of the pair, turns as a vector under constant rotations, and so vanishes at the trivial triple; a tame background on X is gauge equivalent to a potential of norm $\le c\,M(d(X)+2)\varepsilon_{t+1}$, c formed from the constants (fact (iii) on the cube of side $2M(d(X)+2)+1$ about X), as long as this cube fits into the torus and $c\,M(d(X)+2)\varepsilon_{t+1}$ stays below a number formed from the constants, the other domains costing a_{K-t}/ℓ, resp. ε_{t+1}, with the factor $d(X)+2$; Schwarz' lemma gives $\sum_a |\int \mathbb{W}_a^{\mathrm{tot}}\,\mathrm{d}\nu_W| \le c\,\mathsf{A}(\mathbb{W})\varsigma_t$, $\varsigma_t = c\varepsilon_{t+1} + 6LMa_{K-t}/\ell$, where $\mathbb{W}_a^{\mathrm{tot}}$ is the sum of the local terms of the insertion and $\mathsf{A}(\mathbb{W})$ its amplitude in the cluster expansion.

For the fluctuation field, an odd insertion linear in B, the amplitude is that of the non-Gaussian potentials, $\lesssim \delta_t$, because the Gaussian part is even: this is (d). For the rotations Υ_i the linear part is an odd insertion of norm $\le cg_t$ and the remainder an insertion of norm $\le cg_t^2$, so each bond costs $cg_t^2(\log x_t)^{\mathfrak{p}_0}\varsigma_t$; and $Ng_t^2 \le c\,a_s/\mathsf{b}$ for the $N = 4L^{K-t-s}$ bonds, which gives (c). At $t=0$ the rotations are those of the chart, step 1 of the proof of (a), and the same argument runs.

(e) As $a_{K-t} \ge a_i > c_0\varrho/2L$ and $|C| \le 40\varrho$, $|C|/a_{K-t} < 80L/c_0$, and C lies in a cube of side less than $20L/c_0$ in the units of the level. If ϱ/b is at least a number $\eta_0 > 0$ formed from the constants, c_0, ϑ and J, the bound 2 of the defect at $\nu_W \ge 0$ is at most $C_*\frac{\varrho}{\mathsf{b}}(\frac{c_0\varrho}{2L\mathsf{b}})^{\vartheta} \le C_*\frac{\varrho}{\mathsf{b}}(\frac{a_{K-t}}{\mathsf{b}})^{\vartheta}$ for $C_* \ge 2(2L/c_0)^{\vartheta}\eta_0^{-1-\vartheta}$; so let $\varrho/\mathsf{b} < \eta_0$, and with it $a_{K-t}/\mathsf{b} \le a_*/\mathsf{b} = (\varrho/\mathsf{b})^{3/(3+\vartheta)}$, g_t, δ_t and ε_{t+1} be small, and $K-t$ large.

The proof of (b) runs with the pieces of C in the cubes of the level in the place of the bonds of the level. *The exact formula.* Step 1 of the proof of (a) holds for every bare loop, the corners entering nowhere: $W_C^{(t)}(V_W(B)) = W_C(U')$, $U' = e^{i\eta_t\mathbf{A}(B)}U_+$, at every level. *Pieces.* From a base point, C is cut into the maximal runs $C_1,\dots,C_n$ of consecutive steps that start in one of the half-open cubes of side a_{K-t} about the sites of Λ_{K-t}, the cubes of the level that C passes; each of the four sides of C gives at most its length over a_{K-t} plus 2 of them, so that $n < 80L/c_0 + 8$, a number formed from L and c_0. With $g_j = U_+(C_j)$, the holonomy of U_+ along the piece, and $\Upsilon_j(B) = \frac{1}{i}\log[U'(C_j)\,U_+(C_j)^{-1}] \in \mathfrak{su}(2)$, one has $|\Upsilon_j| \le c\,\delta_t|C_j|/a_{K-t}$, c formed from the constants (the shift is at most $\mathsf{c}_{\mathfrak{h}}\sup|g_tB|$, at $t=0$ at most $2C_P\sup|g_0B|$, at each of the $|C_j|/a_K$ bare steps, and $\eta_t|C_j|/a_K = |C_j|/a_{K-t}$), and $W_C^{(t)}(V_W(B)) = w(\Upsilon(B))$, $W_C^{(t+1)}(W) = w(0)$, with w as in Lem. 10.5(c) for one line.

Insertions. By the brackets each Υ_j is a sum of local terms of the level, gauge covariant at the first end point of C_j, with a linear part in B of norm at most $c\,g_t|C_j|/a_{K-t}$ and a remainder of norm at most $c\,g_t^2\frac{|C_j|}{a_{K-t}}(1+\frac{|C_j|}{a_{K-t}})$ in the norm of the cluster expansion; each piece is one insertion, of norm of the order g_t times its length over a_{K-t}. Entered into the cluster expansion of the step as in step 3 of the proof of (a), they give $\log\nu_W(e^{(z,\Upsilon)}) = \sum_X \ell_X(z)$, one variable $z_j \in \mathbb{C}^3$ for the whole of Υ_j, on a polydisc of radius $r_1 \ge c\,a_{K-t}/(g_t|C|) \ge c\,c_0/(Lg_t) \ge 2$, and Cauchy's estimates give a cumulant of order m the factor $(c\,g_t|C|/a_{K-t})^m$ and the decay of its domains.

Symmetry. Let $s_{jj'}\delta_{aa'}$ be the covariance of the linear parts of Υ_j^a, $\Upsilon_{j'}^{a'}$ formed at the trivial triple under the Gaussian measure of the step there (real, symmetric, free of W; a multiple of $\delta_{aa'}$ by the constant rotations). At the trivial triple the first cumulants, colour vectors, vanish, and the pair cumulants are multiples of $\delta_{aa'}$, which differ from $s_{jj'}\delta_{aa'}$ by parts of the relative size $c\,\delta_t$, from the non-Gaussian potentials, and $c\,g_t$, from the remainders (step 5 of the proof of (a)). For a domain X that holds pieces of C, with $d(X) \le \frac{1}{2}(K-t)\log L$, fact (iii) of step 0 of the proof of (a), on a cube of side at most $2M(d(X)+2)+1+40L/c_0$ that holds X and C, where its conditions hold for $\varrho/\mathsf{b} < \eta_0$, puts the background triple, up to a gauge transformation, on a disc about the trivial triple

whose radius ρ_X has $1/\rho_X \le c\,\varepsilon_{t+1}(d(X)+2)^2$, c formed from the constants, L and c_0, the frames along C staying within a factor e of $\mathbf{1}$; Schwarz' lemma on this disc bounds the terms of X of the first cumulants, and of the pair cumulants less their values at the trivial triple turned by the frames, by $1/\rho_X$ times their size, paid by the decay $e^{-\lambda d(X)}$, and the domains with $d(X) > \frac12(K-t)\log L$ cost $L^{-(K-t)/2}$.

So the first cumulants and the defects $d_{jj'}$ of the pair cumulants from $s_{jj'}\mathrm{R}(G_{j\to j'})$ carry, relative to their size, the factor $c\,(\varepsilon_{t+1} + L^{-(K-t)/2} + \delta_t + g_t)$, and the first cumulant of a linear part carries besides the amplitude $c\,\delta_t$ of the non-Gaussian potentials, its Gaussian part being even. *The pair term.* The argument of step 7 of the proof of (a) holds for every bare loop: at the trivial triple with the frames $\mathbf{1}$ the sum of the linear parts of the Υ_j is $g_t\eta_t(j_C, \dot{\mathbf{A}}B)$, so that $S := \sum_{j,j'} s_{jj'} = g_t^2\eta_t(X_C, \mathcal{C}_{(t+1)}X_C) \ge 0$, the slice of the free Coulomb energy at the step, at the depths coarser than the corners as at the finer ones.

By Prop. 10.7(c), (d), which hold at every depth, $0 \le a_K(X_C, \mathcal{C}_{(t+1)}X_C) = v_{K,K-t}(C) - v_{K,K-t-1}(C) \le C^{\mathrm{bg}}A_C^2/a_{K-t}^3 \le c\,|C|$, with the area $A_C \le 100\varrho^2$ of C, $|C| \ge 4c_0\varrho$ and c formed from L and c_0; so step 8 of the proof of (a) turns $e^{-\frac32 S}$ into λ_{t+1} up to $c\,|C|\mathsf{b}^{-2}a_{K-t}$. *The bound.* Lem. 10.5(c), its measure being the image of $\nu_W \ge 0$ under $B \mapsto \Upsilon(B)$, a probability measure on $\{|y_j| \le c\,\delta_t|C|/a_{K-t}\}$, bounds the defect by $E_1 + E_2 + E_3$ and this term of the label, the sum over two lines being empty. With $g_t^2 \le \frac32 a_{K-t}/\mathsf{b}$, $\delta_t = g_t\mathfrak{b}_t$, ε_{t+1} and $L^{-(K-t)/2}$ of the order $(a_{K-t}/\mathsf{b})^{1/2}$ up to logarithms, and the sizes above, each of E_1, E_2, E_3 is at most $\frac{|C|}{\mathsf{b}}(\frac{a_{K-t}}{\mathsf{b}})^{1/2}y_t^{\mathfrak{p}}$, $y_t = \log x_t$, $\mathfrak{p}$ formed from $\mathfrak{p}_0$, times a polynomial in $|C|/a_{K-t}$ with coefficients formed from the constants and J, and the term of the label is at most $c\frac{|C|}{\mathsf{b}}\frac{a_{K-t}}{\mathsf{b}}$.

As $|C|/a_{K-t} < 80L/c_0$, $|C| \le 40\varrho$ and $(\frac{a_{K-t}}{\mathsf{b}})^{1/2}y_t^{\mathfrak{p}} \le c\,(\frac{a_{K-t}}{\mathsf{b}})^{\vartheta}$ for $\vartheta < \frac12$, c formed from the constants, ϑ and J, the defect is at most $C_*\frac{\varrho}{\mathsf{b}}(\frac{a_{K-t}}{\mathsf{b}})^{\vartheta}$. The mean over the block field follows, $|\bar{\mathsf{D}}_{t+1}|$ being at most the bound of the defect at m_{t+1}-almost every W times $Z^{-1}m_{t+1}(\mathcal{V}_{K-t-1}) \le 1$. The levels finer than the corners are treated in the statement of (e) itself. □

Proposition 10.7 (Numbers of the free field)**.** Let $L \ge 3$ be odd, $\ell = 2L^m$, and C a polygon with corners in Λ_i; no measure and no SU(2) enter. (a) For $K \ge i$ and $a = a_K$: $\mathcal{I}_a(C,C) = \frac{|C|}{2\pi}\log\frac1a + c_{\mathrm{lat}}|C| + \mathcal{J}_\ell(C) + E_a(C)$ with $|E_a(C)| \le c[n_C a(1+\log\frac{|C|}{a}) + a^2|C|^2\mathrm{Lg}(\ell/a)(\delta_C^{-3} + \ell^{-3})]$, c absolute; pieces in perpendicular directions do not interact in the sum; $|\mathcal{J}_\ell(C) - \mathcal{J}(C)| \le c|C|^2\mathrm{diam}(C)^2\ell^{-3}$; for a rectangle $s \times t$,

$$\mathcal{J}(C) = \frac1\pi\Big[s\log s + t\log t - s\,\mathrm{asinh}\,\frac{s}{t} - t\,\mathrm{asinh}\,\frac{t}{s} + 2\sqrt{s^2+t^2} - 2(s+t)\Big].$$

For polygons C, D at distance $\delta > 0$: $|\mathcal{I}_a(C,D) - \mathcal{I}_\ell(C,D)| \le ca^2|C||D|\,\mathrm{Lg}(\ell/a)(\delta^{-3} + \ell^{-3})$. (b) For $0 \le d \le K$ and every closed path C' of Λ_K, $v_{K,d}(C')$ is the largest value of $2\mathcal{I}_{a_K}(j_{C'}, Q^{\mathsf{T}}\iota) - \mathcal{I}_{a_K}(Q^{\mathsf{T}}\iota, Q^{\mathsf{T}}\iota)$ over the closed currents ι of Λ_d without zero mode, Q the linearisation of $\mathcal{M}^{K-d}$ at the trivial field (the mean over a block of the straight bare paths): the energy of the best smeared current. For the polygon C and $i \le d \le K$,

$$v_{K,d}(C) = \frac{|C|}{2\pi}\log\frac{1}{a_d} + (c_{\mathrm{lat}} + c_{r,K-d})|C| + \mathcal{J}_\ell(C) + \mathrm{Rem}_{K,d}(C),$$

$|\mathrm{Rem}_{K,d}(C)| \le |E_{a_d}(C)| + c[n_C^2 a_d + n_C a_d(1+\log\frac{|C|}{a_d}) + a_d|C|^2/\delta_C^2]$, a bound free of K. Here the numbers $c_{r,j}$, $j \ge 0$ an integer, are formed as c_r in the setting, with $S_{(j)}(z) = \frac{\sin(z/2)}{L^j\sin(z/2L^j)}$ in the place of $S(z)$, $E_{(j)}(p_\nu) = 4L^{2j}\sin^2\frac{p_\nu}{2L^j}$ in the place of p_ν^2 in the denominators, and the sums over the $n \in \mathbb{Z}^2$ with $|n_2|, |n_3| \le \frac{L^j-1}{2}$ in the places of the sums over $\mathbb{Z}^2$, in N and in D; at $j = K-d$, $L^j = a_d/a_K$ is the ratio of the two spacings. The number $c_{r,j}$ depends on L^j alone, $c_{r,0} = 0$ (then $S_{(0)} = 1$ and only $n = 0$ enters), $|c_{r,j}| \le c_3$, and $c_{r,j} \to c_r$ as $j \to \infty$. The limit $v_d = \lim_K v_{K,d}$

exists and obeys the same remainder bound:

$$\Big|v_d - \frac{|C|}{2\pi}\log\frac{1}{a_d} - c_Q|C| - \mathcal{J}_\ell(C)\Big| \le |E_{a_d}(C)| + c\Big[n_C^2 a_d + n_C a_d\Big(1+\log\frac{|C|}{a_d}\Big) + \frac{a_d|C|^2}{\delta_C^2}\Big].$$

For a fixed polygon the right side is $O_{C,\ell}(a_d(1+\log^+(1/a_d)))$ as $d\to\infty$. The constants c and c_3 are free of the side; c_Q is a number of the plane, free of L, ℓ and C. (c) $\prod_{n\le K-d}\lambda_n = \exp[-\frac{3}{2\mathsf{b}}(\mathcal{I}_{a_K}(C,C) - v_{K,d}(C))]$ for all $0\le d\le K$. Hence for $i\le d\le K$

$$a_K^{-\kappa}\prod_{n\le K-d}\lambda_n = L^{\kappa d}\exp\Big[\frac{3}{2\mathsf{b}}\big(c_{r,K-d}|C| + \mathrm{Rem}_{K,d}(C) - E_{a_K}(C)\big)\Big],$$

and one step, $n=K-d$ with $d\ge i$, gives $\lambda_n = L^{-\kappa}\exp[-\frac{3}{2\mathsf{b}}((c_{r,n-1}-c_{r,n})|C| + \mathrm{Rem}_{K,d+1} - \mathrm{Rem}_{K,d})]$: each step takes away $\kappa\log L$ and a multiple of the length; the steps sum to $\sum_{n\ge1}(c_{r,n-1}-c_{r,n}) = -c_r = c_{\mathrm{lat}} - c_Q$, every term depending on L and the sum not. For families, $c_\Xi^{-1}L^{\kappa d}\le a_K^{-\kappa}\Xi_{K-d}\le c_\Xi L^{\kappa d}$ for $i_0\le d\le K$, with c_Ξ free of d and K and uniform for b in a compact set, and $\mathsf{l}_d = \lim_K a_K^{-\kappa}\Xi_{K-d} = a_d^{-\kappa}\exp[-\frac{3}{2\mathsf{b}}((c_{\mathrm{lat}}-c_Q)|\mathsf{C}| + O_{\mathsf{C},\ell}(a_d(1+\log^+(1/a_d))))]$, so $a_d^\kappa \mathsf{l}_d \to e^{-0.047235\ldots|\mathsf{C}|/\mathsf{b}}$. (d) For a rectangle C of area A with corners in Λ_K, in any position, and $0\le d\le K$: $0\le v_{K,d}(C)\le C^{\mathrm{bg}}A^2/a_d^3$, C^{bg} absolute. ■

Sketch of proof. (a) Fourier analysis of g_N: a component of the current couples only to the same component, so perpendicular pieces do not meet; the sum of g_N along an axis gives the logarithm and the constant c_{lat}, a computation with the lattice Green function on a line; the torus gives $\mathcal{J}_\ell$; distant pieces are compared with the continuum with an error $O(a^2)$, the corners with an error $O(a)$ each. The formula for the rectangle is the double integral of $\frac{1}{4\pi|x-y|}$ over its four sides, the parallel pieces interacting with the sign -1 and the perpendicular ones not at all. (b) The range of Π_{K-d} is spanned by the curvatures of the bare fields $\Delta^{-1}Q^{\mathsf{T}}\iota$, ι a closed current of Λ_d (the Lagrange multiplier of the linearised background problem); as $(X,\Pi X)$ is the largest value of $2(X,Y)-(Y,Y)$ over the range and $(X_{C'},\mathrm{d}A) = (j_{C'},A)$, $v_{K,d}$ is the stated maximum, at every depth and for every closed bare path.

Since the corners of the polygon lie in Λ_d for $d\ge i$, the Fourier transform of the sharp loop at the momenta $(q+2\pi n)/a_d$ is the transform $\hat\jmath(q)$ of its chain on Λ_d times the form factor of one bond; the quadratic form is diagonal in q and in the direction of the current, and the maximum is solved mode by mode. Without the constraint that ι be closed one gets $a_d N^{-3}\sum_{q\ne0}\sum_\mu|\hat\jmath_\mu(q)|^2\mathsf{m}_\mu^{(K-d)}(q)$, $N=\ell/a_d$, where $\mathsf{m}_\mu^{(j)} = (\mathsf{N}_\mu^{(j)})^2/\mathsf{D}_\mu^{(j)}$ is the propagator that the block averages see: with $p=q+2\pi n$, $s_{(j)}(p)=\prod_\nu S_{(j)}(p_\nu)$ and $\Delta_{(j)}(p)=\sum_\nu E_{(j)}(p_\nu)$, $\mathsf{N}_\mu^{(j)}$ sums, over the aliases (n in the L^{3j} classes $|n_\nu|\le\frac{L^j-1}{2}$ at a finite cutoff, $j=K-d$, and in $\mathbb{Z}^3$ at $K=\infty$, where $S_{(\infty)}=S$ and $E_{(\infty)}(z)=z^2$), the bare propagator $1/\Delta_{(j)}(p)$ times the form factors $S_{(j)}(p_\mu)$ of the sharp bond and $S_{(j)}(p_\mu)s_{(j)}(p)$ of the smeared bond, and $\mathsf{D}_\mu^{(j)}$ the same with the smeared bond twice; at $q_\mu=0$ only $n_\mu=0$ survives, and they reduce to the sums N, D of $c_{r,j}$ (of c_r at $K=\infty$).

The constraint costs a non-negative term at most $cn_C^2a_d$: with $\partial_\mu(q)=e^{iq_\mu}-1$ and $\langle z,z'\rangle = \sum_\mu\bar z_\mu z'_\mu$, $\langle\partial(q),\hat\jmath(q)\rangle = 0$ for a closed C (the sum over the steps telescopes), $|\bar\partial_\mu\hat\jmath_\mu|$ is at most twice the number of pieces in the direction μ, $\mathsf{N}_\mu^{(j)}/\mathsf{D}_\mu^{(j)}$ differs from $1/s_{(j)}(q)$, the inverse form factor of the block, which does not depend on μ, by $O(|q|^2)$, and $\sum_\mu|\partial_\mu|^2/\mathsf{D}_\mu^{(j)}\ge c|q|^4$. With $\Delta_1(q)=\sum_\nu 4\sin^2\frac{q_\nu}{2}$, the symbol $\mathsf{m}_\mu^{(j)} - 1/\Delta_1$ is bounded with derivatives at most $c|q|^{-|\alpha|}$, c free of j; its kernel falls like $|x|^{-2}$; along one straight piece of n bonds its sum gives $c_{r,j}$ per unit length (the sum along an axis is the integral over the plane $q_\mu=0$, whence the two-dimensional formula) up to $O(a_d(1+\log n))$, and parallel pieces at distance $\ge\delta_C$ interact through it by $O(a_d|C|^2/\delta_C^2)$.

With (a) at the spacing a_d this gives the law at every finite cutoff, with a remainder free of K; $c_{r,j}\to c_r$ by dominated convergence, and $|c_{r,j}|\le c_3$ by the bound on the symbol. (c)

$\sum_{n\le K-d}\mathcal{C}_{(n)} = \Pi_0 - \Pi_{K-d}$ at every d, and $a_K(X_C, \Pi_0 X_C) = \mathcal{I}_{a_K}(C,C)$; insert (a) and (b): $\frac{3}{2\mathsf{b}}\frac{|\bar C|}{2\pi}\log\frac{a_d}{a_K} = \kappa\log\frac{a_d}{a_K}$; one step is $a_K(X_C, \mathcal{C}_{(K-d)}X_C) = v_{K,d+1} - v_{K,d}$.

For families the factors multiply, and one may take $c_\Xi = \prod_i \exp[\frac{3}{2\mathsf{b}}(c_3|C_i| + 2\mathsf{r}_i)]$, r_i the supremum over $d \ge i_0$ of the bound on $|\mathrm{Rem}_{K,d}(C_i)|$ in (b), which also bounds $|E_{a_K}(C_i)|$ for $K \ge d$. (d) $v_{K,d} \ge 0$, Π_{K-d} being an orthogonal projection. It vanishes on the 2-cochains Y with $\mathrm{d}^\mathsf{T} Y = 0$, as its range consists of exact cochains; so $v_{K,d}(C) = a_K(X, \Pi_{K-d}X)$ for every 2-cochain X with $\mathrm{d}^\mathsf{T}X = j_C$, in particular for X the indicator of the bare plaquettes of the rectangle, with $\|X\|_1 = \sum_p |X(p)| = A/a_K^2$. The maximum of (b) holds at every depth $0 \le d \le K$ for this bare current, wherever the corners lie (corners in Λ_d are needed only to factor $\hat\jmath$ in the proof of (b)).

Drop the constraint that ι be closed: the maximum can only grow. With $\mathfrak{q} = a_d/a_K = L^{K-d}$, the bare momenta $\varkappa = (q + 2\pi n)/\mathfrak{q}$, $q \in (2\pi/N)\mathbb{Z}^3$, $|n_\nu| \le \frac{\mathfrak{q}-1}{2}$, and $p = \mathfrak{q}\varkappa$, the transform of $Q^\mathsf{T}\iota$ at $\varkappa$ is $\hat\iota_\mu(q)\varphi_\mu(\varkappa)$, $|\varphi_\mu(\varkappa)| = \mathfrak{q}|S_{(K-d)}(p_\mu)s_{(K-d)}(p)|$ the form factor of the smeared bond, and $\Delta_1(\varkappa) = \mathfrak{q}^{-2}\Delta_{(K-d)}(p)$; the maximum over all ι, taken mode by mode, gives

$$v_{K,d}(C) \le a_K(N\mathfrak{q})^{-3}\sum_{q\ne0}\sum_\mu \frac{|\mathcal{B}_\mu(q)|^2}{\mathfrak{q}^4\mathsf{D}_\mu^{(K-d)}(q)}, \qquad \mathcal{B}_\mu(q) = \sum_n \frac{\hat\jmath_\mu(\varkappa)\,\overline{\varphi_\mu(\varkappa)}}{\Delta_1(\varkappa)},$$

$\hat\jmath$ the transform of the bare current j_C. Three bounds hold uniformly in $\mathfrak{q}$: $|\hat\jmath_\mu(\varkappa)| \le \Delta_1(\varkappa)^{1/2}\|X\|_1$, as $j_C = \mathrm{d}^\mathsf{T}X$ and $\sum_\nu |e^{i\varkappa_\nu} - 1|^2 = \Delta_1(\varkappa)$; $\sum_n |S_{(K-d)}(p_\mu)s_{(K-d)}(p)|\Delta_{(K-d)}(p)^{-1/2} \le c/|q|$, the term $n = 0$ being at most $(0.3|q|)^{-1}$ and, for $n \ne 0$, each one-dimensional factor at most $51/\max\{1, |n_\nu|\}$ and $\Delta_{(K-d)}(p) \ge 0.2|n|^2$, so that the terms $n \ne 0$ add up to at most $c\sum_n \prod_\nu \max\{1, |n_\nu|\}^{-4/3}$; and $\mathsf{D}_\mu^{(K-d)}(q) \ge 0.3^8|q|^{-2}$ ($S_{(j)} \ge 0.3$ on $[-\frac32\pi, \frac32\pi]$ and $E_{(j)}(z) \le z^2$). So $|\mathcal{B}_\mu(q)| \le c\,\mathfrak{q}^2\|X\|_1/|q|$, each of the N^3 modes contributes at most $c\|X\|_1^2$, and $v_{K,d}(C) \le c\,a_K\mathfrak{q}^{-3}\|X\|_1^2 = c\,A^2/a_d^3$, as $a_K\mathfrak{q} = a_d$: C^{bg} is absolute, for every odd $L \ge 3$, every $0 \le d \le K$ and every position. Equivalently, the kernel of Π_{K-d} on the bare plaquettes is at most $c\,(a_K/a_d)^3$ in modulus: at one plaquette this is the bound on the diagonal, and Schwarz' inequality gives the rest. □

Proposition 10.8 (The tower)**.** In the frame let $F = W_\mathsf{C}$ and let $0 \le \bar t \le K$ be the last level of the tower. (a) Suppose $|\int W_\mathsf{C}^{(t)}(V_W(B))\,\nu_W(\,\mathrm{d}B) - \lambda_{t+1}W_\mathsf{C}^{(t+1)}(W)| \le \eta_{t+1}$ for m_{t+1}-almost every W and $0 \le t < \bar t$, and put $\mathsf{e}_t = \sum_{s\le t}\Xi_{s-1}\eta_s$. Then $|\Phi^{(t)}[F] - \Xi_t W_\mathsf{C}^{(t)}| \le \mathsf{e}_t$ m_t-almost everywhere for $t \le \bar t$, and

$$\Big|\langle F\rangle_K - \Xi_{\bar t}\frac{1}{Z}\int W_\mathsf{C}^{(\bar t)}\,\mathrm{d}m_{\bar t}\Big| \le \mathsf{e}_{\bar t} + \sum_{t\le\bar t}(\Xi_{t-1} + \mathsf{e}_{t-1})\mathsf{w}_t;$$

the same holds with a further factor $H(\mathcal{M}^{K-s_1}U)$ in $\langle F\rangle_K$ and $H\circ\mathcal{M}^{K-\bar t-s_1}$ in the integral, for H bounded Borel gauge invariant on $\mathcal{V}_{s_1}$ and $K - \bar t \ge s_1$, with $\sup|H|$ multiplying the right side. No positivity of ν_W is used. (b) Without any hypothesis, let $\mathsf{A}_t = Z^{-1}\int W_\mathsf{C}^{(t)}\,\mathrm{d}m_t$ and $\bar{\mathsf{D}}_t = Z^{-1}\int[\int W_\mathsf{C}^{(t-1)}(V_W(B))\,\nu_W(\,\mathrm{d}B) - \lambda_t W_\mathsf{C}^{(t)}(W)]\,m_t(\,\mathrm{d}W)$, the mean of the defect of the step over the block field. Then $\mathsf{A}_{t-1} = \lambda_t\mathsf{A}_t + \bar{\mathsf{D}}_t + \omega_t$ with $|\omega_t| \le \mathsf{w}_t$, and so

$$\langle W_\mathsf{C}\rangle_K = \Xi_{\bar t}\mathsf{A}_{\bar t} + \sum_{t\le\bar t}\Xi_{t-1}(\bar{\mathsf{D}}_t + \omega_t)\,:$$

as an identity between numbers, only the means of the defects over the block field enter. (c) On gauge invariant Borel sets $\mu_{K-\bar t}^K - Z^{-1}m_{\bar t}$ is a positive measure of mass $\sum_{t\le\bar t}\mathsf{w}_t$, and $1 - 2\mathsf{V}_\ell C_\mathrm{r}(1)L^{-(K-\bar t)} \le Z^{-1}m_{\bar t}(\mathcal{V}_{K-\bar t}) \le 1$. (d) For a coordinate rectangle C with sides s_1, s_2: $1 - \frac12\theta_d^2 \le W_C^{(K-d)} \le 1$ on O_{K-d}, with $\theta_d = 2B_3\varepsilon_{K-d}L^{2d}s_1s_2$. ■

Sketch of proof. (a) For m_{t+1}-almost every W,

$$\Phi^{(t+1)}[F](W) - \Xi_{t+1} W_{\mathsf{C}}^{(t+1)}(W) = \int [\Phi^{(t)}[F] - \Xi_t W_{\mathsf{C}}^{(t)}](V_W(B))\, \nu_W(\,\mathrm{d}B) + \Xi_t \Big[\int W_{\mathsf{C}}^{(t)}(V_W(B))\, \nu_W(\,\mathrm{d}B) - \lambda_{t+1} W_{\mathsf{C}}^{(t+1)}(W) \Big].$$

The second member is at most $\Xi_t \eta_{t+1}$ almost everywhere by hypothesis. The first, $\tilde\Psi(W) := \int \Psi_t(V_W(B))\, \nu_W(\,\mathrm{d}B)$ with $\Psi_t := \Phi^{(t)}[F] - \Xi_t W_{\mathsf{C}}^{(t)}$, is at most e_t in modulus for m_{t+1}-almost every W whenever $|\Psi_t| \le \mathsf{e}_t$ m_t-almost everywhere; this duality step reads only $m_t, m_{t+1}, \mathsf{h}_{t+1} \ge 0$. The disintegration at $G = \Psi_t$, a bounded Borel function, and the transport identity $\int G'\,\mathrm{d}m_{t+1} = \int (G' \circ \mathcal{M})\, \mathsf{h}_{t+1}\,\mathrm{d}m_t$ at $G' = |H|$ give, for every bounded Borel H,

$$\Big| \int H \tilde\Psi\,\mathrm{d}m_{t+1} \Big| = \Big| \int (H \circ \mathcal{M})\, \Psi_t\, \mathsf{h}_{t+1}\,\mathrm{d}m_t \Big| \le \mathsf{e}_t \int |H \circ \mathcal{M}|\, \mathsf{h}_{t+1}\,\mathrm{d}m_t = \mathsf{e}_t \int |H|\,\mathrm{d}m_{t+1};$$

at $H = \overline{\operatorname{sgn}\tilde\Psi}\, \mathbf{1}\{|\tilde\Psi| > \mathsf{e}_t + \epsilon\}$ this says $(\mathsf{e}_t + \epsilon)\, m_{t+1}(|\tilde\Psi| > \mathsf{e}_t + \epsilon) \le \mathsf{e}_t\, m_{t+1}(|\tilde\Psi| > \mathsf{e}_t + \epsilon)$, so the set has measure 0, for every $\epsilon > 0$. No law of $V_W(B)$, no null set of m_t under ν_W and no positivity of ν_W enter. Induction on t, with $\mathsf{e}_{t+1} = \mathsf{e}_t + \Xi_t \eta_{t+1}$; then eq. 10.i with $|W_{\mathsf{C}}^{(t)}| \le 1$ and $|\Phi^{(t-1)}[F]| \le \Xi_{t-1} + \mathsf{e}_{t-1}$. With H: $\Phi^{(t)}[W_{\mathsf{C}}\, H \circ \mathcal{M}^{K-s_1}] = \Phi^{(t)}[W_{\mathsf{C}}]\, H \circ \mathcal{M}^{K-t-s_1}$ for $K - t \ge s_1$. (b) For gauge invariant bounded Borel G on $\mathcal{V}_{K-t+1}$, eq. 10.i at $F = G \circ \mathcal{M}^{t-1}$ for $\bar t = t-1$ and $\bar t = t$, with $\Phi^{(t-1)}[G \circ \mathcal{M}^{t-1}] = G$ (functions pull out and $\Phi^{(t-1)}[1] = 1$), gives $\int G\,\mathrm{d}m_{t-1} = \int G\,(\mathsf{h}_t + \mathsf{h}_t^{\mathrm{w}})\,\mathrm{d}m_{t-1}$.

At $G = W_{\mathsf{C}}^{(t-1)}$ the disintegration with $H = 1$ turns $Z^{-1} \int W_{\mathsf{C}}^{(t-1)} \mathsf{h}_t\,\mathrm{d}m_{t-1}$ into $\lambda_t \mathsf{A}_t + \bar{\mathsf{D}}_t$, and $\omega_t = Z^{-1} \int W_{\mathsf{C}}^{(t-1)} \mathsf{h}_t^{\mathrm{w}}\,\mathrm{d}m_{t-1}$; $\mathsf{A}_0 = \langle W_{\mathsf{C}} \rangle_K$. (c) eq. 10.i at $F = \mathbf{1}_E \circ \mathcal{M}^{\bar t}$, E gauge invariant: $Z \mu_{K-\bar t}^K(E) - m_{\bar t}(E) = \sum_{t \le \bar t} \int (\mathbf{1}_E \circ \mathcal{M}^{\bar t - t+1}) \mathsf{h}_t^{\mathrm{w}}\,\mathrm{d}m_{t-1} \ge 0$, of total mass $Z \sum_{t \le \bar t} \mathsf{w}_t$; the rarity bound at $p = 0$ in its summed form. (d) Over a tame field the background has bare plaquettes $< 2B_3 \varepsilon_{K-d} (a_K/a_d)^2$ (Thm. YM3_I:E in the setting of Section YM3_I:52.7), and $1 - W_{\partial R} \le (\sum_{p \subset R} \sqrt{a_p})^2$ over the $s_1 s_2 / a_K^2$ plaquettes of the rectangle (Prop. 10.9(c)). □

Proposition 10.9 (Identities and inequalities at every cutoff)**.** For every $\beta > 0$, every torus and every cutoff: (a) $\mathcal{W}_{K-z}^{(L^z \ell, L^z \mathsf{b})}(L^z \mathsf{C}; f^{(z)}) = L^{-z\kappa - zn} \mathcal{W}_K^{(\ell, \mathsf{b})}(\mathsf{C}; f_1, \dots, f_n)$, $f_q^{(z)} = f_q(L^{-z}\cdot)$; (b) if families $G_1, \dots, G_s$ lie in the interior of S_θ^+ for a coordinate mirror θ and the supports of the tuples f_i lie in S_θ^+ (and do not meet the fixed planes if θ is a bond mirror), then $\sum_{i,j} \bar c_i c_j \mathcal{W}_K(\theta G_i, G_j; \theta f_i, f_j) \ge 0$ for all complex c_i; so $\mathcal{W}_K(G_1, \theta G_2)^2 \le \mathcal{W}_K(G_1, \theta G_1)\, \mathcal{W}_K(G_2, \theta G_2)$ for real entries; if θ cuts a loop C_{12} into two halves and C_{11}, C_{22} are the loops made of each half and its mirror image, then $\mathcal{W}_K(C_{12})^2 \le \mathcal{W}_K(C_{11})\, \mathcal{W}_K(C_{22})$; (c) the plaquette mean $\langle W_{\partial p} \rangle_K$ is > 0; for bare rectangles $p \times q$ with $p, q < \ell/2a_K$, $w(p,q) = \langle W_{p \times q} \rangle_K \ge w(1,1)^{pq} > 0$, and $\log w$ is convex in p and in q; and $0 \le 1 - W_{\partial R} \le (\sum_{p \subset R} \sqrt{a_p})^2$ for every field and every rectangle R. For the rectangles $s \times T$ of one coordinate plane, s and T lengths in $\bigcup_i a_i \mathbb{Z}$, put $\mathcal{W}(s \times T) := \lim_K \mathcal{W}_K(s \times T) = \lim_K a_K^{-\kappa} \langle W_{s \times T} \rangle_K$, $\kappa = \kappa(s \times T)$, wherever the limit exists. A *range* of the limits is a set of such rectangles on which they exist and are finite, closed downwards (with $s \times T$ it holds every $s' \times T'$ with $s' \le s$, $T' \le T$) and open upwards (with $s \times T$ it holds a rectangle with both sides larger); in the frame, and by dilation on every torus and at every label, the polygons $s \times T$ with $\kappa < \frac12$ form one (Thm. AE(1)–(3)). Every limit inherits (a), (b), $\mathcal{W} \ge 0$ and the midpoint inequality $\mathcal{W}(s \times \frac{T' + T''}{2})^2 \le \mathcal{W}(s \times T')\, \mathcal{W}(s \times T'')$, and the same in s, whenever the three rectangles lie in a range, the powers of a_K matching because $2|s \times \frac{T'+T''}{2}| = |s \times T'| + |s \times T''|$; a limit that is > 0 at one rectangle of a range is > 0 at all rectangles of that range. Strict positivity of a limit is not inherited; it comes from the floor of Thm. AE(4). ■

Sketch of proof. (a) The Wilson measure at (K, ℓ, b) is the one at $(K - z, L^z \ell, L^z \mathsf{b})$ relabelled, as $\beta = L^K \mathsf{b} = L^{K-z}(L^z \mathsf{b})$: lengths are multiplied by L^z, κ is unchanged, $a_K^{-\kappa}$ gains $L^{-z\kappa}$ and each $\mathcal{O}(f)$ the factor L^{-z}. (b) Reflection positivity of the Wilson measure at site and bond mirrors,

for functions of the bonds of one closed (resp. open) half torus, and Schwarz' inequality: each $a_K^{-\kappa(G_i)} W_{G_i} \prod_q (\mathcal{O}(f_{iq}) - \langle \mathcal{O}(f_{iq})\rangle_K)$ lives on one side of θ, and the powers of a_K split because lengths add. For the cut loop, $W_{C_{12}} = \frac{1}{2}\operatorname{tr}(U(\gamma_1)U(\gamma_2))$ with the two halves γ_1, γ_2, and reflection positivity for matrix-valued functions with Schwarz' inequality gives the bound; $|C_{11}| + |C_{22}| = 2|C_{12}|$. (c) $s \mapsto \log \int e^{s \sum_p \frac{1}{2}\operatorname{tr} \partial U(p)}\, \mathrm{d}U$ has the derivative 0 at $s = 0$ (the Haar mean of a non-trivial character vanishes) and is strictly convex, so its derivative at $s = \beta$, the number of plaquettes times the plaquette mean, is > 0.

Reflection positivity in the two directions of the plane of the rectangle gives $w(p,q)^2 \le w(p,q-1)\, w(p,q+1)$ and the same in p; with $w(p,0) = 1$ the ratios $w(p,q)/w(p,q-1)$ increase, so $w(p,q) \ge w(p,1)^q \ge w(1,1)^{pq}$. For $g \in \mathrm{SU}(2)$, $1 - \frac{1}{2}\operatorname{tr} g = \frac{1}{2}|g - \mathbf{1}|^2$; the holonomy of ∂R is an ordered product of conjugated plaquette variables, and $|\prod_i g_i - \mathbf{1}| \le \sum_i |g_i - \mathbf{1}|$. Limits keep ≥ 0, the positive forms and the inequalities; the midpoint inequality of $\mathcal{W}_K$ is the convexity of $\log w$ in each side, multiplied by $a_K^{-2\kappa(s\times\frac{T'+T''}{2})} = a_K^{-\kappa(s\times T')-\kappa(s\times T'')}$, at every cutoff at which the three rectangles have their corners in Λ_K.

Spreading in a range R, the limits being finite there (Thm. AE(1)): if $\mathcal{W}(s \times T_0) > 0$ and $s \times (T_0 \pm \epsilon) \in \mathsf{R}$, the product $\mathcal{W}(s \times (T_0 - \epsilon))\, \mathcal{W}(s \times (T_0 + \epsilon)) \ge \mathcal{W}(s \times T_0)^2 > 0$ of two finite numbers ≥ 0 makes both positive; going down from the lowest positive point with an upper point of R fixed, the distance to it doubles at each step, so that every T below T_0 is reached, and upwards in the same way with a lower point; then the same in s; and two rectangles of R are joined through the one with the smaller sides, which lies in R. The values of ϵ are taken in $\bigcup_i a_i\mathbb{Z}$. □

Proposition 10.10 (A source off the loop)**.** In the frame let $f_1, \dots, f_n$ be real Lipschitz, $\|f\|_\sharp = \max\{\sup|f|, 40M \operatorname{Lip} f\}$, and z in the polydisc $\sum_q |z_q| \|f_q\|_\sharp < \rho_D$, where $\rho_D = 2\varrho_S/(2e^{(C_S+C_g)\ell^3}+2)$ is formed from the radius ϱ_S and the constants C_S, C_g of Thms. YM3_I:AJ and YM3_I:AD; ρ_D depends on neither K nor $\mathsf{b} \in \mathsf{I}(g)$. Write $w_z = \sum_q z_q f_q$. The radius estimates below are stated for a family with $\max_q \|f_q\|_\sharp > 0$. If every f_q vanishes, the source is zero, its derivatives of positive order vanish, and the assertions reduce directly to the zero-source statements. (a) $\langle e^{\mathcal{O}(w_z)}\rangle_K = e^{\mathcal{E}(w_z)} h_K(z)$ with $\frac{1}{2} \le |h_K(z)| \le \frac{3}{2}$, whatever the supports, $\mathcal{E}(w)$ a sum of local energies of the source; the energies of sources $v_i = \sum_q z_q^{(i)} f_q^{(i)}$, $i = 1, \dots, s$, $f_q^{(i)}$ real Lipschitz, with $\sum_i \sum_q |z_q^{(i)}| \|f_q^{(i)}\|_\sharp < \rho_D$ and supports at pairwise sup-distance $\ge 2\delta$, add up to an error: $|\mathcal{E}(\sum_i v_i) - \sum_i \mathcal{E}(v_i)| \le \Delta_s(\delta) = c(s+1)(\ell/2\delta)^3$, c formed from the constants of Thms. YM3_I:AJ and YM3_I:AD, free of K; so, with $\mathsf{e}(v) = |\langle e^{\mathcal{O}(v)}\rangle_K|$, $\frac{1}{2}(\frac{2}{3})^s e^{-\Delta_s(\delta)} \prod_i \mathsf{e}(v_i) \le \mathsf{e}(\sum_i v_i) \le \frac{3}{2}\, 2^s e^{\Delta_s(\delta)} \prod_i \mathsf{e}(v_i)$; we write $\Delta(\delta) = \Delta_2(\delta)$. In particular $\langle e^{\mathcal{O}(w_z)}\rangle_K \ne 0$ on the polydisc, for all K. (b) The objects of the frame exist with the weight $e^{\mathcal{O}(w)}$. For complex bounded w and bounded Borel F on $\mathcal{V}_K$ let $m_t^w[F] = \Phi^{(t)}[F e^{\mathcal{O}(w)}]\, m_t$, a complex measure on $\mathcal{V}_{K-t}$, and $m_t^w = m_t^w[1]$. Then $\int G \,\mathrm{d}m_t^w[F] = \int (G \circ \mathcal{M})\, \mathsf{h}_t \,\mathrm{d}m_{t-1}^w[F]$ for bounded Borel G, and, for gauge invariant F and G and $0 \le \bar{t} \le K$, the sum over the level of the first creation of a large field

$$Z\big\langle F\, G(\mathcal{M}^{\bar{t}} U)\, e^{\mathcal{O}(w)}\big\rangle_K = \int G \,\mathrm{d}m_{\bar{t}}^w[F] + \sum_{t \le \bar{t}} \int (G \circ \mathcal{M}^{\bar{t}-t+1})\, \mathsf{h}_t^{\mathrm{w}} \,\mathrm{d}m_{t-1}^w[F].$$

At real w only, $\Phi^{(t)}[e^{\mathcal{O}(w)}] > 0$ m_t-almost everywhere, and the quotients

$$\Phi^{(t),w}[F] = \Phi^{(t)}[F e^{\mathcal{O}(w)}]/\Phi^{(t)}[e^{\mathcal{O}(w)}],$$

with $|\Phi^{(t),w}[F]| \le \sup|F|$, and the kernel

$$\nu_W^w(dB) = \Phi^{(t)}[e^{\mathcal{O}(w)}](V_W(B))\, \nu_W(dB)/\Phi^{(t+1)}[e^{\mathcal{O}(w)}](W),$$

a probability measure for m_{t+1}-almost every W, obey eq. 10.i and the disintegration for (m^w, ν^w), with $Z\langle e^{\mathcal{O}(w)}\rangle_K$ in the place of Z; m_t^w is then a positive measure with the null sets of m_t, and on gauge invariant sets $\mu_{K-\bar t}^{K,w} - m_{\bar t}^w/(Z\langle e^{\mathcal{O}(w)}\rangle_K)$ is a positive measure of mass $\sum_{t\le\bar t}\mathsf{w}_t^w$, $\mathsf{w}_t^w = (Z\langle e^{\mathcal{O}(w)}\rangle_K)^{-1}\int \mathsf{h}_t^{\mathrm{w}}\,\mathrm{d}m_{t-1}^w$, $\mu_d^{K,w}$ being the law of $\mathcal{M}^{K-d}U$ under $e^{\mathcal{O}(w)}\langle\cdot\rangle_K/\langle e^{\mathcal{O}(w)}\rangle_K$.

The rarity bound holds with a source, at complex z for the integral of $\mathsf{h}_t^{\mathrm{w}}$:

$$|Z\langle e^{\mathcal{O}(w)}\rangle_K|^{-1}\,|\int \mathsf{h}_t^{\mathrm{w}}\,\mathrm{d}m_{t-1}^w| \le \mathsf{V}_S C_{\mathrm{r}}^S(p) L^{-p(K-t)},$$

with V_S, $C_{\mathrm{r}}^S(p)$ formed from the constants of Thms. YM3_I:D, YM3_I:AJ and YM3_I:AD, p and ℓ; at real z this bounds w_t^w, and $\sum_{t\le\bar t} L^{\kappa(K-t)}\mathsf{w}_t^w \le 2\mathsf{V}_S C_{\mathrm{r}}^S(\kappa+1)L^{-(K-\bar t)}$. (c) For bounded Borel gauge invariant F on $\mathcal{V}_d$,

$$\mu_d^{K,w_z}(F) = \langle F(\mathcal{M}^{K-d}U)e^{\mathcal{O}(w_z)}\rangle_K/\langle e^{\mathcal{O}(w_z)}\rangle_K$$

obeys $|\mu_d^{K,w_z}(F)| \le B_d \sup|F|$, with B_d formed from the constants of Thms. YM3_I:D, YM3_I:AJ and YM3_I:AD, d, b and ℓ, and free of K. The limit $\mu_d^{w_z}(F) = \lim_K \mu_d^{K,w_z}(F)$ exists for z in the polydisc $|z_q| < \rho_* := \rho_D/(n\max_q \|f_q\|_\sharp)$, which lies in the polydisc of the head and is free of K, uniformly on its half $|z_q| \le \frac12\rho_*$ and in $\sup|F| \le 1$; it is holomorphic in z, a probability measure at real z, and $\mu_{d+1}^w(\mathcal{M}^{-1}E) = \mu_d^w(E)$. No condition on the supports is asked; so the derivatives at $z=0$, the mixed functions of $F(\mathcal{M}^{K-d}U)$ and of the $\mathcal{O}(f_q)$, converge too. (d) If a coordinate mirror θ parts C, at distance $\ge\delta_1$ from its planes, from the supports, at distance $\ge\delta$, and z lies in half the polydisc, then for every bounded function X of the bonds on the side of C

$$|\langle X e^{\mathcal{O}(w_z)}\rangle_K| \le M_0\|X\|_\theta\,|\langle e^{\mathcal{O}(w_z)}\rangle_K|, \qquad \|X\|_\theta^2 = \langle X\,\overline{\theta X}\rangle_K \ge 0, \qquad M_0 = \sqrt{6}\,e^{\Delta(\delta)/2};$$

so $|Q_K(z)| \le M_0\,\mathcal{W}_K(\mathsf{C},\theta\mathsf{C})^{1/2}$. For two orthogonal mirrors, the sources in two groups as in Thm. AE(6), and z in a quarter of the polydisc, $\sum_q |z_q|\|f_q\|_\sharp < \rho_D/4$ (for one mirror, half the polydisc is $\sum_q |z_q|\|f_q\|_\sharp < \rho_D/2$),

$$|\langle X e^{\mathcal{O}(w_z)}\rangle_K|^4 \le M_0'^4\langle X\,\overline{\theta_1 X}\,\overline{\theta_2 X}\,\theta_1\theta_2 X\rangle_K\,|\langle e^{\mathcal{O}(w_z)}\rangle_K|^4$$

for X on the side of C of both mirrors, $M_0'^4 = 24\cdot 36\cdot(\frac92)^4 e^{\Delta_4(\delta)+6\Delta_2(\delta)}$, the mean on the right holding no source and being ≥ 0. Where equally spaced coordinate walls part the polygons from the supports as in Thm. AE(6), with an even number of boxes in each direction that the walls cut, so that both fixed planes c and $c+\frac{\ell}{2}$ of every wall mirror are walls (N boxes in all; on $\ell = 2L^m$, with walls in $\frac12 a_i\mathbb{Z}$, the number of slabs in a cut direction divides $4L^{m+i}$, and it is taken even, $2L^j$ or $4L^j$), and with each f_q supported in one box: $\langle W_{\mathsf{C}_\iota^{\mathrm{f}}}\rangle_K \ge 0$, $\kappa(\mathsf{C}_\iota^{\mathrm{f}}) = N\kappa(\mathsf{C}_\iota)$, and

$$|Q_K(z)| \le M_S\prod_\iota \mathcal{W}_K(\mathsf{C}_\iota^{\mathrm{f}})^{1/N} \quad \text{for } \sum_q |z_q|\|f_q\|_\sharp < \rho_D/N, \qquad M_S = 2(\tfrac32)^{n_{\mathrm{b}}}e^{\Delta_{n_{\mathrm{b}}}(\delta)}\big[\tfrac32\,2^N e^{\Delta_N(\delta)}\big]^{n_{\mathrm{b}}/N},$$

n_{b} the number of boxes that meet the supports and δ the distance of the supports to the walls; a function whose support misses the walls is the sum of such pieces, one for each box. (e) $\mathcal{W}_K(\mathsf{C};f_1,\dots,f_n) = \sum_A \partial_A Q_K(0)\,\partial_{A^c}E_K(0)$, $E_K(z) = \langle\exp\sum_q z_q(\mathcal{O}(f_q) - \langle\mathcal{O}(f_q)\rangle_K)\rangle_K$, and $\partial_{A^c}E_K(0) = \sum_\pi\prod_{\beta\in\pi} S_{|\beta|,K}^{\mathsf{T}}(f_\beta)$, π over the partitions of A^c into blocks of at least two elements: only the derivatives of Q_K in pairwise different variables enter. (f) Let $0<\vartheta'\le 1$, $\bar t(K) = K - \lceil\vartheta' K\rceil$, $\eta = (24\ell^3\sum_q\sup|f_q|)^{-1}$, and let $q_{K,\bar t}(z) = a_K^{-\kappa}\int\Phi^{(\bar t)}[W_{\mathsf{C}}e^{\mathcal{O}(w_z)}]\,\mathrm{d}m_{\bar t}/\int\Phi^{(\bar t)}[e^{\mathcal{O}(w_z)}]\,\mathrm{d}m_{\bar t}$ be the same quotient formed with the measure $m_{\bar t}$ without large fields. For every multi-index α, with $p = (4|\alpha|+\kappa+1)/\vartheta'$, and all K with $2\mathsf{V}_\ell C_{\mathrm{r}}(p)L^{-p\vartheta' K} \le 0.16$: $|\partial^\alpha Q_K(0) - \partial^\alpha q_{K,\bar t(K)}(0)| \le 14\,\alpha!\,\eta^{-|\alpha|}\mathsf{V}_\ell C_{\mathrm{r}}(p)L^{-K}$: the Taylor coefficients of Q_K at 0 are those of the chain without large fields. ∎

Sketch of proof. (a) Thms. YM3_I:AJ and YM3_I:AD for the Wilson measure with the source w_z writes $\langle e^{\mathcal{O}(w_z)}\rangle_K$ as $e^{\mathcal{E}(w_z)}h$, $\mathcal{E}(w)=\sum_{j\le K}\sum_X e_j(X;w)$ a sum of local terms $e_j(X;w)$ over the levels j and the domains X of the level j, which read w near X alone and vanish at $w=0$, with $|e_j(X;w)|\le C_S e^{-\kappa_S d_j(X)}$, κ_S the decay number of Thms. YM3_I:AJ and YM3_I:AD and $d_j(X)$ the size of X in cubes of side M of the level j; and h holomorphic with $h(0)=1$ and $|h|\le e^{(C_S+C_g)\ell^3}$. With $\|z\|=\sum_q|z_q|\|f_q\|_\sharp$, Schwarz' lemma for $\zeta\mapsto h(w_{\zeta z})-1$ on $|\zeta|<2\varrho_S/\|z\|$ gives $|h-1|\le\frac12$ on the polydisc.

In $\mathcal{E}(\sum_i v_i)-\sum_i\mathcal{E}(v_i)$ a local term that sees one support alone cancels; one that sees two supports at sup-distance $\ge 2\delta$ belongs to a domain X with $d_j(X)\ge 2\delta L^{K-j}/M-4$ and is at most $(s+1)C_S e^{-\kappa_S d_j(X)}$; the sum over such domains and levels is at most $c(s+1)(\ell/2\delta)^3$. The bounds on $\mathsf{e}(\sum_i v_i)$ follow, as $\frac12 e^{\operatorname{Re}\mathcal{E}(v)}\le\mathsf{e}(v)\le\frac32 e^{\operatorname{Re}\mathcal{E}(v)}$ for every source v of the polydisc. (b) A history is linear in the bare density, so $\Phi^{(t)}[Fe^{\mathcal{O}(w)}]$ is the density of the history without large fields with the source, at complex w as well. The transport identity is that of the frame at $Fe^{\mathcal{O}(w)}$, and the sum over the first creation is eq. 10.i at the gauge invariant function $F\,(G\circ\mathcal{M}^{\bar t})\,e^{\mathcal{O}(w)}$, the factor $G\circ\mathcal{M}^{\bar t}$ pulling out of $\Phi^{(\bar t)}$ and of $\Phi^{(t-1)}$ as G and as $G\circ\mathcal{M}^{\bar t-t+1}$.

The quotients and the kernel are formed at real w: there $e^{\mathcal{O}(w)}\ge e^{-\sup|\mathcal{O}(w)|}>0$ gives, at every level t, $\Phi^{(t)}[e^{\mathcal{O}(w)}]\ge e^{-\sup|\mathcal{O}(w)|}$ off one m_t-null set $\mathcal{N}_t$, so m_t^w has the null sets of m_t. The kernel ν_W^w is ν_W times the density $B\mapsto\Phi^{(t)}[e^{\mathcal{O}(w)}](V_W(B))/\Phi^{(t+1)}[e^{\mathcal{O}(w)}](W)$, and it is a probability measure for m_{t+1}-almost every W: $\nu_W\ge0$ for m_{t+1}-almost every W (Prop. 10.6(b)); the denominator is $\ge e^{-\sup|\mathcal{O}(w)|}$ for $W\notin\mathcal{N}_{t+1}$; the numerator is $\ge e^{-\sup|\mathcal{O}(w)|}$ off $V_W^{-1}\mathcal{N}_t$, and $\nu_W(V_W^{-1}\mathcal{N}_t)=0$ for m_{t+1}-almost every W, the one set $\mathcal{N}_t$ being fixed first and the exceptional set of W depending on it: the disintegration at $G=\mathbf{1}_{\mathcal{N}_t}$ gives $\int H(W)\,\nu_W(V_W^{-1}\mathcal{N}_t)\,m_{t+1}(\,\mathrm{d}W)=\int(H\circ\mathcal{M})\,\mathbf{1}_{\mathcal{N}_t}\,\mathsf{h}_{t+1}\,\mathrm{d}m_t=0$ for every bounded Borel H; and the mass is 1 by the disintegration at $F=e^{\mathcal{O}(w)}$, $\Phi^{(t+1)}[e^{\mathcal{O}(w)}](W)=\int\Phi^{(t)}[e^{\mathcal{O}(w)}](V_W(B))\,\nu_W(\,\mathrm{d}B)$ for m_{t+1}-almost every W.

The disintegration for (m^w,ν^w) is that of the frame at $G\,\Phi^{(t)}[e^{\mathcal{O}(w)}]$. The wild mass is a sum over the histories of the level K in which a large field is created at the step t; Thms. YM3_I:AJ and YM3_I:AD bounds every history with a source as Thm. YM3_I:D bounds it without one, with the constants of Thms. YM3_I:AJ and YM3_I:AD, and (a) bounds the denominator below; the rarity bound then follows as in the square-completion estimate eq. 4.xxxv and the geometric-series calculation in the proof of Prop. 4.54, with the constants of Thms. YM3_I:AJ and YM3_I:AD. (c) Thms. YM3_I:AJ and YM3_I:AD at one depth bounds $|\langle F(\mathcal{M}^{K-d}U)e^{\mathcal{O}(w_z)}\rangle_K|$ by $\sup|F|$ times a number free of K times $|\langle e^{\mathcal{O}(w_z)}\rangle_K|$: this is B_d.

For the limit let $\sup|F|\le1$ and $G_K(z)=\mu_d^{K,w_z}(F)$, holomorphic and bounded by B_d on the polydisc $|z_q|<\rho_*$. We use an interpolation inequality: if G is holomorphic on the polydisc of radius ρ in n variables, $|G|\le B$ there and $|G|\le\sigma$ on the real cube $[-r,r]^n$, $0<r\le\rho/2$, then its Taylor coefficients at 0 obey $|c_\alpha(G)|\le C_{n,j}r^{-|\alpha|}(\sigma+2B(r/\rho)^{j+1})$ for $|\alpha|\le j$ (the Taylor polynomial of order j is within $2B(r/\rho)^{j+1}$ of G on the cube, by Cauchy's estimates on the complex lines through 0, and on the polynomials of order at most j in n variables the largest coefficient and the supremum on a cube are two norms).

It suffices that every coefficient $c_\alpha(G_K)$ converges, uniformly in F: Cauchy's estimate $|c_\alpha(G_K)|\le B_d\rho_*^{-|\alpha|}$ makes the tails of the Taylor series small on $|z_q|\le\frac12\rho_*$, uniformly in K and F. For $\alpha=0$ this is Cor. 2.8. Fix $j=|\alpha|\ge1$. The proof of the limit laws of the block fields compares two consecutive cutoffs through the small-field chains at the depth $u=\lceil\phi_1K\rceil$, $\phi_1=1/(180(3N'+4))$, N' the number of the test functions (§2.3.1); we prolong the tuple f by copies of f_1 to $N'=\max\{n,j^2\}$ functions, the new variables being 0, so that $u\ge d$ for large K. Let G_K^{sf} be the quotient of F formed with the weight of the history without large-field regions down to the depth u, and $D_K=G_K-G_K^{\mathrm{sf}}$; then $c_\alpha(G_{K+1})-c_\alpha(G_K)=c_\alpha(D_{K+1})-c_\alpha(D_K)+c_\alpha(G_{K+1}^{\mathrm{sf}}-G_K^{\mathrm{sf}})$, and the interpolation inequality is used twice, on two polydiscs. (i) On the polydisc of radius $\rho'=(24n\ell^3L^{4(K+1)}\max_q\sup|f_q|)^{-1}$,

on which $|\mathcal{O}(w_z)| \le \frac14$ at the cutoffs K and $K+1$ (there are $3(\ell/a_K)^3$ plaquettes and $0 \le a_p \le 2$), $|G_K|, |G_K^{\mathrm{sf}}| < 1.8$ and $|D_K| \le 3.6$; at its real points the measure without large-field regions is below the full one with a mass defect of the order of the rarity at the depth u, so that $|D_K| \le C\exp[3K\log L - cK^2]$, below every power of L^K; with $r = L^{-(4j+1)K}\rho'$ the coefficients of D_K are at most $C(L^{(4j^2+5j+3)K}e^{-cK^2} + L^{-K})$, summable in K. (ii) On the polydisc on which the proof of the limit laws bounds the small-field chains, of radii ρ'' with $1/\rho'' \le cK^{2\mathfrak{p}_0}L^{3u}$, so that $\rho''^{-j} \le (cK^{2\mathfrak{p}_0})^j L^{3j+K/(180j)}$ by $3\phi_1 j \le 1/(180j)$, which is where $N' \ge j^2$ enters, $|G_K^{\mathrm{sf}}| \le 1.8$; at its real points that proof goes through with the source, the contact terms of the source standing in a factor that drops out of the quotient, and gives $|G_{K+1}^{\mathrm{sf}} - G_K^{\mathrm{sf}}| \le cK^{2\mathfrak{p}_0+1/2}L^{-K/10}$; with $r = L^{-K/(90j)}\rho''$ the coefficients are at most a power of K times $L^{K(-\frac{1}{10}+\frac{1}{90}+\frac{1}{180j})} + L^{K(-\frac{1}{90j}+\frac{1}{180j})}$, summable in K, both exponents being negative.

So every $c_\alpha(G_K)$ converges with a summable rate, uniformly in $\sup|F| \le 1$, and G_K converges uniformly on $|z_q| \le \frac12\rho_*$; the limit is holomorphic, a complex measure of mass 1 in F (a limit, uniform in $\sup|F| \le 1$, of such measures), a probability measure at real z, where the $\mu_d^{K,w}$ are, and $\mathcal{M}$-compatible, as $\mu_{d+1}^{K,w}(\mathcal{M}^{-1}E) = \mu_d^{K,w}(E)$ at every K. (d) Reflection positivity at θ and Schwarz' inequality give $|\langle XH\rangle_K|^2 \le \|X\|_\theta^2\langle\overline{\theta H}H\rangle_K$ for H on the other side; at $H = e^{\mathcal{O}(w)}$, $\overline{\theta H}H = e^{\mathcal{O}(w+w^\theta)}$ with $w^\theta = \overline{\theta w} = \sum \bar z_q \theta f_q$.

The mean, not the energy, is invariant: $\langle e^{\mathcal{O}(w^\theta)}\rangle_K = \overline{\langle e^{\mathcal{O}(w)}\rangle_K}$, as θ is a symmetry of the Wilson measure and the a_p are real. By (a) for w, w^θ and $w + w^\theta$, which lie in the polydisc of the head as z lies in its half and whose supports lie 2δ apart, $\langle e^{\mathcal{O}(w+w^\theta)}\rangle_K \le \frac32 e^{\Delta}e^{\mathrm{Re}\,\mathcal{E}(w)+\mathrm{Re}\,\mathcal{E}(w^\theta)}$, and $e^{\mathrm{Re}\,\mathcal{E}(v)} \le 2|\langle e^{\mathcal{O}(v)}\rangle_K|$ at $v = w$ and $v = w^\theta$; so $\langle e^{\mathcal{O}(w+w^\theta)}\rangle_K \le 6e^{\Delta}|\langle e^{\mathcal{O}(w)}\rangle_K|^2$. At $X = W_{\mathsf{C}}$, $\|W_{\mathsf{C}}\|_\theta^2 = \langle W_{\mathsf{C}}W_{\theta\mathsf{C}}\rangle_K = a_K^{2\kappa}\mathcal{W}_K(\mathsf{C},\theta\mathsf{C})$.

Two mirrors: Schwarz at θ_1 with $Xe^{\mathcal{O}(w_2)}$ on the side of C, then at θ_2; the sources that arise hold four images of the second group and two of the first, pairwise 2δ apart, and together they stay in the polydisc of (a), $\sum_q |z_q|\|f_q\|_\sharp < \rho_D$ with each z_q counted as often as its images, as soon as z lies in a quarter of it; then (a) bounds the source factors. Walls: in a cut direction with an even number n of slabs the reflections in the walls map walls to walls, both fixed planes of each being walls, and generate the $n/2$ translations by even multiples of the width and $n/2$ reflections, a group of n elements that acts simply transitively on the slabs; so each box has exactly one image in every box, an image has a parity (the number of reflections modulo 2), and the folds below are defined. (With an odd number of slabs the translation by one width lies in the group, and so does the reflection of a box in its own mid-plane, which is no wall; the evenness excludes this.) The chessboard estimate $|\langle\prod_\iota F_\iota\rangle_K| \le \prod_\iota \langle \mathrm{Fold}\, F_\iota\rangle_K^{1/N}$ for functions F_ι of the bonds of the boxes, $\mathrm{Fold}\, F_\iota$ the product of the N images of F_ι under the group (conjugated at the odd ones), each $\langle \mathrm{Fold}\, F_\iota\rangle_K$ being ≥ 0, is applied with $F_\iota = W_{\mathsf{C}_\iota}$ for a box with polygons, $F_\iota = e^{\mathcal{O}(w_\iota)}$ for a box with a support, w_ι the part of w in it, and $F_\iota = 1$ otherwise. $\mathrm{Fold}\, W_{\mathsf{C}_\iota} = W_{\mathsf{C}_\iota^{\mathrm{f}}}$, with $\kappa(\mathsf{C}_\iota^{\mathrm{f}}) = N\kappa(\mathsf{C}_\iota)$, so that the powers of a_K split. $\mathrm{Fold}\, e^{\mathcal{O}(w_\iota)} = e^{\mathcal{O}(\sum_g w_\iota^g)}$, the N images w_ι^g lying in different boxes, pairwise 2δ apart, each with $\mathsf{e}(w_\iota^g) = \mathsf{e}(w_\iota)$ and together in the polydisc of (a) for $\sum_q |z_q|\|f_q\|_\sharp < \rho_D/N$; by (a) at $s = N$, $\langle \mathrm{Fold}\, e^{\mathcal{O}(w_\iota)}\rangle_K^{1/N} \le [\frac32\, 2^N e^{\Delta_N(\delta)}]^{1/N}\mathsf{e}(w_\iota)$.

In the denominator, (a) at $s = n_{\mathrm{b}}$ gives $|\langle e^{\mathcal{O}(w)}\rangle_K| \ge \frac12(\frac23)^{n_{\mathrm{b}}}e^{-\Delta_{n_{\mathrm{b}}}(\delta)}\prod_\iota \mathsf{e}(w_\iota)$; the factors $\mathsf{e}(w_\iota)$ cancel and M_S remains. (e) Leibniz' rule for $\partial_{\{1,\dots,n\}}[Q_K(z)E_K(z)]$ at 0; the derivatives of E_K at 0 in pairwise different variables are sums over the partitions of products of joint cumulants, the blocks of one element vanishing because the $\mathcal{O}(f_q)$ are centred. (f) eq. 10.i at $F = W_{\mathsf{C}}e^{\mathcal{O}(w_z)}$ and at $F = e^{\mathcal{O}(w_z)}$, on the polydisc $|z_q| \le \eta a_K^4$, where $|\mathcal{O}(w_z)| \le \frac14$ (there are $3(\ell/a_K)^3$ plaquettes and $0 \le a_p \le 2$), together with the rarity bound at p, bounds the difference of the two quotients

by $c\mathsf{V}_\ell C_{\mathrm{r}}(p) a_K^{-\kappa} L^{-p(K-\bar t)}$; Cauchy's estimate on this polydisc costs $\alpha!(\eta a_K^4)^{-|\alpha|}$, and $p(K-\bar t) \geq (4|\alpha|+\kappa+1)K$. □

Proposition 10.11 (Three free photon fields)**.** (a) Uniformly in K, $\mathbb{P}(\max_p |F^{(d)}(p)| \geq 2\varepsilon_{[d]}) \leq 18\ell^3 L^{3d} e^{-\frac{2}{3}A_0^2(\log x_{[d]})^{2\mathfrak{p}_0}}$, and the probability that some plaquette with a corner in a set S of sites of Λ_d has $|F^{(d)}(p)| \geq 2\varepsilon_{[d]}$ is at most $72\,\#S\, e^{-\frac{2}{3}A_0^2(\log x_{[d]})^{2\mathfrak{p}_0}}$. These are bounds at one depth, uniform in K: the first has the form of $R_d^\natural$, with 18 and $\frac{2}{3}$ in the place of 1 and $c^\natural$, the second the local form of Prop. 10.2(c), with 72 and $\frac{2}{3}$ in the place of 1 and $c^\natural$; summed over the depths $d \geq D$ (a union bound) they bound the probability that some depth $d \geq D$ has a plaquette with $|F^{(d)}(p)| \geq 2\varepsilon_{[d]}$. There the smoothing is linear and needs no ball. (b) For a family $\mathsf{C} = (C_1, \dots, C_r)$, at every length,

$$a_K^{-\kappa}\, \mathbb{E} \prod_i W_{C_i}^{\mathrm{fr}} \to \prod_i e^{-\frac{3}{2\mathsf{b}}[c_{\mathrm{lat}}|C_i| + \mathcal{J}_\ell(C_i)]} \Big[2^{-r} \sum_{\epsilon \in \{\pm 1\}^r} e^{-\frac{1}{\mathsf{b}} \sum_{i<j} \epsilon_i \epsilon_j \mathcal{I}_\ell(C_i, C_j)} \Big]^3 > 0;$$

at $r = 2$ the bracket is $\cosh(\mathcal{I}_\ell(C_1, C_2)/\mathsf{b})$. (c) For one loop the step is exact: the conditional mean of $W_C^{\mathrm{fr},(t)}$, the loop of the conditional mean of the field given the block field of the level t, given the block field of the level $t+1$ is $\lambda_{t+1} W_C^{\mathrm{fr},(t+1)}$; for a family the defect is at most $(e^{3x}-1)\lambda_{t+1}$, $x = \frac{1}{\mathsf{b}} \sum_{i<j} |a_K(X_{C_i}, \mathcal{C}_{(t+1)} X_{C_j})|$, which for polygons at fixed distances falls faster than every power of a_{K-t}; so for families the factor without cross terms is the right one. The exponent κ, the perimeter constant $c_{\mathrm{lat}}|C| + \mathcal{J}_\ell(C)$ and the factor λ_{t+1} of one step are those of three free photon fields. (d) At every cutoff K, for every polygon C with corners in Λ_K and every bounded real f,

$$\frac{\mathbb{E}[W_C^{\mathrm{fr}} \mathcal{O}^{\mathrm{fr}}(f)]}{\mathbb{E}\, W_C^{\mathrm{fr}}} - \mathbb{E}\, \mathcal{O}^{\mathrm{fr}}(f) = -\frac{3}{2\mathsf{b}^2}\, a_K^{-1} \sum_p f(x(p))\, \mathcal{I}_{a_K}(C, \partial p)^2.$$

If C has its corners in a fixed Λ_i and f is Lipschitz with support at positive distance from C, then as $K \to \infty$

$$-\frac{3}{2\mathsf{b}^2}\, a_K^{-1} \sum_p f(x(p))\, \mathcal{I}_{a_K}(C, \partial p)^2 \;\longrightarrow\; -\frac{3}{2\mathsf{b}^2} \int_{T_\ell} f(x)\, |B_{C,\ell}(x)|^2 \, \mathrm{d}x, \qquad B_{C,\ell}(x) = \mathrm{curl}_x \oint_C G_\ell(x-y)\, \mathrm{d}y,$$

the field of C on the torus; in the ball of radius $< \ell/4$ that holds the polygon, $|B_{C,\ell}(x) - B_C(x)| \leq c\,|C|\,\mathrm{diam}(C)\,\ell^{-3}$, c absolute, with $B_C(x) = \oint_C \frac{dy \times (x-y)}{4\pi|x-y|^3}$ the Biot–Savart field of C in $\mathbb{R}^3$. For a bounded f that is not Lipschitz the sums over the plaquette centres need not tend to the integral: an f that vanishes at the plaquette centres of all cutoffs, a countable set, gives 0 at every cutoff. ■

Sketch of proof. (a) $F^{(d)}$ is a linear image of F; each component at a plaquette of Λ_d is a centred Gaussian variable of variance at most $1/x_{[d]}$ (the smeared self-energy of the boundary of one plaquette of Λ_d, divided by β). $|F^{(d)}(p)| \geq 2\varepsilon$ forces a component $\geq 2\varepsilon/\sqrt{3}$, of probability at most $2e^{-\frac{2}{3}\varepsilon^2 x_{[d]}}$, and $\frac{2}{3}\varepsilon_{[d]}^2 x_{[d]} = \frac{2}{3}A_0^2(\log x_{[d]})^{2\mathfrak{p}_0}$; a union bound over the $3(\ell L^d)^3$ plaquettes and the three components gives 18, over the 12 plaquettes at a site 72. (b) $\mathbb{E}\prod_i \cos\phi^a_{C_i} = 2^{-r}\sum_\epsilon \mathbb{E}\, e^{i\sum_i \epsilon_i \phi^a_{C_i}} = 2^{-r}\sum_\epsilon \exp[-\frac{1}{2\mathsf{b}}\sum_{i,j}\epsilon_i\epsilon_j \mathcal{I}_{a_K}(C_i, C_j)]$; three independent colours; Prop. 10.7(a) for the diagonal terms and for the distant pairs. (c) Given the block field of the level $t+1$, the curvature is its conditional mean, the curvature $\Pi_{t+1}F$ of the linear background, plus an independent Gaussian fluctuation; between the levels t and $t+1$ the fluctuation of the flux through C has the variance $\frac{a_K}{\mathsf{b}}(X_C, \mathcal{C}_{(t+1)} X_C)$ per colour, and $\mathbb{E}[\cos(\phi+\xi) \mid \phi] = e^{-\frac{1}{2}\mathrm{Var}\,\xi}\cos\phi$; three colours give λ_{t+1}.

For a family the cross covariances $\frac{a_K}{\mathsf{b}}(X_{C_i}, \mathcal{C}_{(t+1)} X_{C_j})$ enter as factors $e^{\mp \frac{a_K}{\mathsf{b}}(X_{C_i}, \mathcal{C}_{(t+1)} X_{C_j})}$, at most e^x for each colour. (d) For jointly Gaussian ϕ, F: $\mathbb{E}[\cos\phi\, F^2]/\mathbb{E}\cos\phi - \mathbb{E}F^2 = -\mathrm{Cov}(\phi, F)^2$; $\mathrm{Cov}(\phi^a_C, F^a(p)) = \beta^{-1}(X_C, X_{\partial p}) = \mathcal{I}_{a_K}(C, \partial p)/\mathsf{b}$; $\frac{1}{2}$ and three colours give $\frac{3}{2}$. The identity holds

for every bounded f, as nothing in it is a limit. For the limit let the corners of C lie in a fixed Λ_i: away from C, uniformly on the support of f, $\mathcal{I}_{a_K}(C,\partial p) = a_K^2(B_{C,\ell}(x(p))\cdot n_p + O(a_K \mathrm{Lg}(\ell/a_K)))$, the mutual energy of C with a small loop being the flux through it of the field of C formed with the Green function of the torus: Prop. 10.7(a) for the two polygons C and ∂p at positive distance compares the lattice energy with $\mathcal{I}_\ell(C,\partial p) = \oint_{\partial p}\oint_C G_\ell(x-y)\,\mathrm{d}y\cdot dx$ up to $O(a_K^3\mathrm{Lg}(\ell/a_K))$, and Stokes' formula makes $\mathcal{I}_\ell(C,\partial p)$ the flux of $B_{C,\ell}$ through p; summing $a_K^3 f(x(p))(B_{C,\ell}(x(p))\cdot n_p)^2$ over the three orientations of the plaquettes gives a Riemann sum of $f|B_{C,\ell}|^2$, which converges for Lipschitz f.

Finally $G_\ell(x) = \frac{1}{4\pi|x|} + H_\ell(x)$ with $H_\ell(x) = \ell^{-1}H_1(x/\ell)$ smooth on $|x|<\ell$; as $\oint_C dy = 0$,

$$\operatorname{curl}\oint_C H_\ell(x-y)\,\mathrm{d}y = \oint_C [\nabla H_\ell(x-y) - \nabla H_\ell(x-y_0)]\times dy,$$

of modulus at most

$$|C|\operatorname{diam}(C)\sup_{|u|\le \ell/2}|\nabla^2 H_\ell(u)| = |C|\operatorname{diam}(C)\,\ell^{-3}\sup_{|u|\le 1/2}|\nabla^2 H_1(u)|.$$

□

The proof of Thm. AF uses the following tools; we state them in lattice units, with spacing 1 and $\beta = \mathsf{b}/a$.

Lemma 10.12 (Tools of the tube)**.** Let ν be the Wilson measure at β on a torus $(\mathbb{Z}/N\mathbb{Z})^3$, N even, or a periodic Gibbs state as in Thm. AF. For a step $c = \langle x_-, x_+\rangle$ of a closed path let $c(y)$, $y\in\mathbb{Z}^2$, be its translates in the lattice plane through x_- orthogonal to c; for an integer $\rho\ge 2$ let $T_\rho(c) = \{c(y) : |y|<\rho\}$ be the slab and $K_\rho(c) = \{x_- + y + se : |y|_\infty \le \rho,\ 0\le s\le 1\}$ the box of c, e its direction; let $\phi_\rho(y) = \min\{1, \log^+(\rho/|y|)/\log\rho\}$, $\phi_\rho(0)=1$, so that $\phi_\rho = 0$ at $|y|\ge\rho$, $E_\rho = (2\pi\log\rho + 7.63)/\log^2\rho$, and $c_j = j(j+1)$. (a) (Admissible steps.) A set I of steps of closed paths $C_1,\dots,C_r$ is admissible at ρ if no other step of the family runs through a bond of $T_\rho(c_i)$, $i\in I$, and no bond of $T_\rho(c_{i'})$, $i'\ne i$, lies in $K_\rho(c_i)$. For closed paths without self-intersection with s straight pieces in all, each of at least $4\rho+1$ steps, whose pieces without a common corner are at sup-distance at least $2\rho+1$, the steps $2\rho\le j\le l'-1-2\rho$ of each piece of l' steps form an admissible set of $n-4\rho s$ steps, n the number of steps; for a rectangle, $n-16\rho$. (b) (The profile.) $\sum_{\langle x,y\rangle}(\phi_\rho(x)-\phi_\rho(y))^2 \le E_\rho$, the sum over the bonds of $\mathbb{Z}^2$, and $(\log\rho - 1.2144)/2\pi \le 1/E_\rho \le \log\rho/2\pi$. (c) (An information inequality, three colours at once.) Let $\varrho>0$ be a C^2 density on $\mathrm{SU}(2)^Y$ with $0\in Y$, $\phi: Y\to[0,\infty)$ with $\phi_0=1$, frames $P_y\in\mathrm{SU}(2)$ with $P_0=\mathbf{1}$, and for a unit vector $\mathbf{n}=(\mathbf{n}_1,\mathbf{n}_2,\mathbf{n}_3)\in\mathbb{R}^3$ the moves $v_y\mapsto \exp[is\phi_y P_y^{-1}(\frac12\sum_a \mathbf{n}_a\sigma_a)P_y]v_y$, $\sigma_1,\sigma_2,\sigma_3$ the Pauli matrices, with the derivative $D_{\mathbf{n}}$ at $s=0$; let $\bar I = \sup_{|\mathbf{n}|=1}\mathbb{E}[-D_{\mathbf{n}}^2\log\varrho]$. Then $\|\mathbb{E}\,D^j(v_0)\|^2 \le \bar I/(\bar I + c_j)$ in the operator norm for every spin $j>0$, where D^j is the representation of spin j. The spin-zero representation has norm 1. (d) (The second variation.) For these moves on the bonds $c(y)$ of a slab, with any frames, a bond $e'=\langle x,y\rangle$ of the plane, $p(e')$ the plaquette spanned by e' and c, and $H_{e'} = P_y U(e')^{-1}P_x^{-1}$: $D_{\mathbf{n}}^2 a_{p(e')} \le \frac14(\phi_x-\phi_y)^2 + \phi_x\phi_y(3a(H_{e'}) + a_{p(e')})$, $a(g) = 1-\frac12\operatorname{tr} g$. (e) (Ceilings multiply over cells.) Let $B=\bigsqcup_l B_l$ be a set of bonds made of cells B_l, no plaquette having bonds in two cells, and $C=\gamma_1\delta_1\cdots\gamma_s\delta_s$ a closed path with γ_l through no bond of $B\setminus B_l$ and δ_l through no bond of B. If Q_l is the conditional mean under ν of $D^j(U(\gamma_l))$ given the bonds off B_l, then $\nu(\chi_j(U(C))\,G) = \nu(\mathrm{tr}[\prod_l Q_l D^j(U(\delta_l))]\,G)$ for every bounded Borel G of the bonds off B, and $|\nu(\chi_j(U(C))\,G)| \le (2j+1)\sup|G|\,\nu(\prod_l\|Q_l\|)$. (f) (One plaquette.) With $\mathfrak{M}_\nu = \sup_p \nu(e^{\beta a_p})$: $\nu(a_p)\le \beta^{-1}\log\mathfrak{M}_\nu$ and $\nu(a_p\ge s)\le \mathfrak{M}_\nu e^{-\beta s}$; for $\beta\ge1$, $\log\mathfrak{M}_\nu \le 192\frac{\mathsf{b}}{\ell} + 4(14.15 + 3\log\beta)$ on the torus of side ℓ, and $\log\mathfrak{M}_\nu \le 28.3 + 6\log\beta$ for the Gibbs states. (g) (The ceiling of a tube.) Let $\beta>0$, $\rho\ge2$, C_l closed paths with spins j_l, I admissible at ρ, $c_i = c_{j_l}$ for a step of C_l, G a bounded Borel function of the bonds off $B=\bigcup_{i\in I}T_\rho(c_i)$, $\mathcal{G} = \sum_{i\in I}\frac{2c_i}{\beta E_\rho}(1-\frac{2c_i}{\beta E_\rho})$ and $\omega_\rho = 40\rho^4\sum_{i\in I}c_i/(\beta E_\rho)^2$. If $\omega_\rho\le1$, then $|\nu(G\prod_l\chi_{j_l}(U(C_l)))| \le \sup|G|\prod_l(2j_l+1)\,e^{-\mathcal{G}}\mathfrak{M}_\nu^{\omega_\rho}$; and

$2c_j/E_\rho \ge \frac{4}{3}c_j k(\log\rho - 1.2144)$: the number per scale is k. (h) (The ceiling with dyadic profiles.) Let $\rho = 2^S \ge 2$ and C a coordinate rectangle of n steps with sides of at least $4\rho+1$ steps and, on the torus, with the points within $\rho+1$ of C in a cube of side at most $\frac{N}{2}$; let $W_C = \frac{1}{2}\operatorname{tr} U(C)$, $p = kn/\beta < 3$, $\delta_0 = (16k\rho\log\rho + pk\log^2\rho)/\beta$ and $\bar a = \nu(a_p)$, the plaquette mean. Then, with the profiles ϕ_r of the scales $r = 2, 4, \dots, 2^S$,

$$|\nu(W_C)| \le e^{\delta_0 + 2.2144p}\rho^{-p}\Big[1 + \frac{72\,\bar a\rho^3\log^2\rho}{2^{3-p}-1}\Big] + 35\,\bar a;$$

the plaquette mean enters linearly, and $p < 3$ keeps the denominator positive. ■

Sketch of proof. (a) A step at distance $\ge 2\rho$ from the ends of its piece has a slab that meets no other step, and the slabs of two steps of pieces without a common corner at sup-distance $\ge 2\rho+1$ do not reach each other's boxes. (b) The sum is a lattice form of $\int|\nabla\phi_\rho|^2 = 2\pi\int_1^\rho (r\log\rho)^{-2} r\,\mathrm{d}r = 2\pi/\log\rho$, with explicit corrections at the centre and at the rim. (c) The moves are one-parameter groups of maps of $\mathrm{SU}(2)^Y$ that keep the Haar measure; under the move of v_0, $D^j(v_0)$ is multiplied by $e^{isJ_{\mathbf n}}$, $J_{\mathbf n}$ the generator of spin j, the image of $\frac{1}{2}\sum_a \mathbf{n}_a\sigma_a$, and $\sum_{\mathbf n} J_{\mathbf n}^2 = c_j$ over an orthonormal basis of directions.

With $M = \mathbb{E}\,D^j(v_0)$ and a unit vector u, integration by parts gives $iJ_{\mathbf n}Mu = -\mathbb{E}[(D^j(v_0) - M)u\,D_{\mathbf n}\log\varrho]$. Put $s_a = D_{e_a}\log\varrho$ and $s = (s_1, s_2, s_3)$. Since the supremum defining $\bar I$ runs over every unit direction, integration by parts gives $\mathbb{E}(ss^{\mathsf T}) \le \bar I\mathbf{1}$. The vector form of Schwarz's inequality therefore gives $\sum_a |\mathbb{E}[Xs_a]|^2 \le \bar I\mathbb{E}|X|^2$ for $X = (D^j(v_0) - M)u$. Using $\sum_a |J_{e_a}Mu|^2 = c_j|Mu|^2$ and $\mathbb{E}|X|^2 = 1 - |Mu|^2$ yields $c_j|Mu|^2 \le \bar I(1 - |Mu|^2)$. (d) Expansion of $a_{p(e')}$ in s: the abelian part gives $\frac{1}{4}(\phi_x - \phi_y)^2$, and the commutator terms are bounded by the defect $a(H_{e'})$ of the frames and by $a_{p(e')}$. (e) The cells interact with the rest only through plaquettes with bonds in one cell and off B, so conditioning cell by cell replaces $D^j(U(\gamma_l))$ by Q_l, the δ_l and G being functions of the bonds off B; $|\operatorname{tr}\prod A_i| \le (2j+1)\prod\|A_i\|$. (f) Jensen's and Chebyshev's inequalities; the chessboard estimate bounds $\nu(e^{\beta a_p})$ by a power of a ratio of partition functions, and $\int\exp[-\beta\sum_{p\in Q} a_p]\,\mathrm{d}U \ge (\mathsf{K}\beta^3)^{-|Q|}$, $\mathsf{K} = 1.39\cdot 10^6$, for a box Q of sites ($\log\mathsf{K} < 14.15$); the torus adds $192\mathsf{b}/\ell$. (g) In each slab of I we use comb frames: in the plane through x_- orthogonal to c_i a coordinate axis through x_- is the base, the lines of the plane orthogonal to it are the teeth, and P_y is the transport from x_- along the base and then along the tooth to $x_- + y$, a product over bonds of the plane, which lie off B.

Then $H_{e'} = \mathbf{1}$ on the base and on the teeth, and at a rung e', a bond parallel to the base at the height $t_{e'} \ne 0$, $H_{e'}$ is conjugate to the holonomy of the boundary of the strip $S_{e'}$ of the $|t_{e'}|$ plaquettes between the rung and the base, so that $a(H_{e'}) \le |t_{e'}|\sum_{q\in S_{e'}} a_q$ (Prop. 10.9(c) and Schwarz' inequality). Write $\nu_B(F)$ for the conditional mean of F under ν given the bonds off B. As no plaquette has bonds in two cells, given the bonds off B the cells are independent, and the conditional law of the bonds of the slab of c_i given the bonds off it is their conditional law given the bonds off B. The bonds of the slab enter the action only through the plaquettes $p(e')$, e' a bond of the plane, so that $-D_{\mathbf n}^2$ of the logarithm of this conditional density is $\beta\sum_{e'} D_{\mathbf n}^2 a_{p(e')}$, with ϕ taken as 0 off the slab, whose bonds alone move; by (b), (d) its conditional mean is $\bar I \le \frac{\beta}{4}(E_\rho + X_i)$, with

$$X_i = 4\sum_{e'} \phi_x\phi_y\big[3a(H_{e'}) + \nu_B(a_{p(e')})\big],$$

a function of the bonds off B. It is at most $\sum_q w_q^{(i)}\,\nu_B(a_q)$, a sum over plaquettes with weights $w_q^{(i)} \ge 0$: $4\phi_x\phi_y$ at the plaquette $p(e')$ of every bond e' of the disc, and $12|t_{e'}|\phi_x\phi_y$ at each plaquette of the strip of every rung e', where $\nu_B(a_q) = a_q$, the bonds of the strip lying in the box $K_\rho(c_i)$ and so off B. The total weight $\sum_q w_q^{(i)}$ is at most $20\rho^4$: the strips give about $12\int_{|y|<\rho} y_2^2\,\mathrm{d}y = 3\pi\rho^4$, the plaquettes $p(e')$ a multiple of ρ^2. (c) gives $\|Q_i\|^2 \le (1 + 4c_i/(\beta(E_\rho + X_i)))^{-1}$, so, by $-\frac{1}{2}\log(1+z) \le$

$-\frac{z}{2}+\frac{z^2}{4}$ and $\frac{1}{E+X}\ge\frac{1}{E}-\frac{X}{E^2}$, $\log\|Q_i\|\le-\frac{2c_i}{\beta E_\rho}(1-\frac{2c_i}{\beta E_\rho})+\frac{2c_iX_i}{\beta E_\rho^2}$; (e) multiplies these over the disjoint cells.

Put $w_q=\sum_i 2c_i w_q^{(i)}/(\beta E_\rho^2)$. Jensen's inequality for the conditional mean ν_B gives $\exp\sum_i\frac{2c_iX_i}{\beta E_\rho^2}\le\nu_B(\prod_q e^{w_qa_q})$; as $\sum_q w_q/\beta\le 40\rho^4\sum_i c_i/(\beta E_\rho)^2=\omega_\rho\le 1$, Hölder's inequality with the exponents β/w_q and (f), with $\mathfrak{M}_\nu\ge 1$, give $\nu(\prod_q e^{w_qa_q})\le\prod_q\nu(e^{\beta a_q})^{w_q/\beta}\le\mathfrak{M}_\nu^{\omega_\rho}$. The last inequality is (b) with $\frac{4}{3}k=\frac{1}{\pi}$. (h) Keep the cells $T_\rho(c_i)$ of the $n-16\rho$ steps of the admissible set of (a) (spin $\frac12$, $c_{1/2}=\frac34$). On each cell and for every dyadic $2\le r\le\rho$ the moves with the profile ϕ_r, which vanishes at $|y|\ge r$, rotate bonds of the cell alone, and (b)–(e), as in the proof of (g), give at every field $\prod_i\|Q_i\|\le\min_r e^{-\mathcal{G}_r+Y_r}$, with $\mathcal{G}_r$ the number $\mathcal{G}$ of (g) formed with E_r, $Y_r=\sum_i 2c_{1/2}X_i^{(r)}/(\beta E_r^2)$ and $X_i^{(r)}$ the X_i of the proof of (g) formed with ϕ_r; by (b), $e^{-\mathcal{G}_r}\le e^{\delta_0+1.2144p}r^{-p}$, δ_0 paying for the 16ρ omitted steps and for the second-order term of $\mathcal{G}_r$, so that $e^{-\mathcal{G}_r+Y_r}\le e^{\delta_0+2.2144p}r^{-p}$ where $Y_r\le p$.

The sharper estimate of the non-abelian member of the second variation bounds the mean of Y_r in the place of an exponential moment: in comb frames $a(H_{e'})$ at a rung e' of height $t_{e'}$ is $1-W$ of the boundary of the strip $S_{e'}$ of $1\times|t_{e'}|$ plaquettes, whose mean is $1-w(1,|t_{e'}|)\le 1-w(1,1)^{|t_{e'}|}\le|t_{e'}|\,\bar a$ by Prop. 10.9(c) (for the Gibbs states the strips of even height by the reflections in the planes of sites, those of odd height by $\sqrt{a(gh)}\le\sqrt{a(g)}+\sqrt{a(h)}$), so that $\nu(X_i^{(r)})\le c\,r^3\bar a$ and $\nu(Y_r)\le c\,p\,r^3\log^2 r\,\bar a$: the plaquette mean enters linearly, and this bound of a mean takes the place of the condition $\omega_\rho\le 1$ and of the weight $40\rho^4$ of (g).

Let r_* be the largest dyadic $r\le\rho$ with $Y_r\le p$; for $r<\rho$ the event $\{r_*=r\}$ lies in $\{Y_{2r}>p\}$, of probability at most $\nu(Y_{2r})/p$ by Markov's inequality, and on the event that $Y_r>p$ at every r the bound 1 serves. The sum over the dyadic $r<\rho$ of $r^{-p}(2r)^3\log^2(2r)$ is geometric and led by its last term exactly when $p<3$, which gives the display. □

10.1.6. *Proof of Thm. AD. Step 1: (1).* For $D\ge d_\Theta\ge d_\natural$ (as $\Theta_\star\ge t_\natural$), $\{\tau\le D\}=\{\tau_0\le D\}=\bigcap_{d\ge D}\pi_d^{-1}(I_d)$: if $\tau_0\le D$, then $V_d\in I_d$ for all $d\ge\tau_0$, so for all $d\ge D$; if $V_d\in I_d$ for all $d\ge D$, then D is one of the integers of which τ_0 is the least. Each member is open (I_d is open, π_d continuous), $\{\tau\le D\}=\emptyset$ for $D<d_\Theta$, and $\{\tau=\infty\}=\{\tau_0=\infty\}$; a function into a countable set whose sublevel sets are Borel is Borel. As $\pi_d(\mu)=\mu_d$, Prop. 10.2(b) and countable subadditivity give the tail by the sum, and the same clause of Prop. 10.2(b) the bound $\frac{L}{L-1}R_D$ when $c'A_0^2\log x_{[D]}\ge 2$, as then $R_{d+1}\le L^{-1}R_d$ for all $d\ge D$; $\sum_d R_d<\infty$ and $\mu(\mathsf{Thr})=1$ give $\mu(\Omega)=1$, and for this the bound $\mu_d(\mathcal{V}_d\setminus I_d)\le R_d$ with $R_d\to 0$ faster than every power of L^{-d} suffices.

If $c'A_0^2(\log x_{[D]})^{2\mathfrak{p}_0}\le 26\ell^3$, the exponent in R_D is at least $(\mathsf{C}_{\mathrm{maj}}^{\Sigma}+2+c_{\mathrm{low}}-26)\ell^3\ge 0$ by $c_{\mathrm{low}}\ge 24$, and $2\ell^3L^{3D}M^{-3}\ge 16$ as $\ell\ge 2M$; so $R_D>1$, and the two bounds, $\sum_{d\ge D}R_d$ and $\frac{L}{L-1}R_D$, both at least R_D, exceed 1. Next, $\{\tau>D\}=\bigcup_{d\ge D}\pi_d^{-1}(\mathcal{V}_d\setminus I_d)$, and $\mathcal{V}_d\setminus I_d$ is the event of Prop. 10.2(c) for the set of all ℓ^3L^{3d} sites, which gives the bound by $\sum_{d\ge D}R_d^\natural$. As $(y+\log L)^{2\mathfrak{p}_0}-y^{2\mathfrak{p}_0}\ge 2y\log L$ for $y\ge 1$, $R_{d+1}^\natural/R_d^\natural\le L^{3-2c^\natural A_0^2\log x_{[d]}}$, which is at most L^{-1} for $d\ge D$ if $c^\natural A_0^2\log x_{[D]}\ge 2$.

With $y=D\log L\le\log x_{[D]}$, $\log R_D^\natural\le 3\log\ell+3y-c^\natural A_0^2y^2$; the first member of the condition gives $3y\le\frac12 c^\natural A_0^2y^2$ and $c^\natural A_0^2\log x_{[D]}\ge 6$, the second $\frac12 c^\natural A_0^2y^2\ge 3\log\ell+\log\frac{1}{\eta}$; so $R_D^\natural\le\eta$.

Step 2: (2). Thm. YM3_I:E in the setting of Section YM3_I:52.7 and §5.2 give the orbit for $\varpi(W)<\bar\varepsilon$; the inclusion $\{\varpi<2c_\rho\varepsilon_{[d]}\}\subset\{\varpi<\bar\varepsilon\}$ for $x_{[d]}\ge t_\natural$ is the condition (cut1) at $t=x_{[d]}$, which gives $2c_\rho\varepsilon(t)\le\varepsilon_{\mathcal{W}}\alpha_0$, with $\varepsilon_{\mathcal{W}}\alpha_0\le\bar\varepsilon$ (equation YM3_I:(2134)). For $d\ge\tau_0\ge d_\natural$, $x_{[d]}\ge t_\natural$ and $V_d\in I_d\subset\{\varpi<2c_\rho\varepsilon_{[d]}\}$. If $d\ge\tau(V_\bullet)$, then $d\ge d_\Theta$ is admitted and $d\ge\tau_0$, so $V_d\in I_d\subset J_d$. Each clause of Prop. 10.4 asks of W only that it lie in J_d at an admitted depth, not that it be a block average or a member of a thread; we substitute $W:=V_d$, and Prop. 10.4(a)–(c) give the limits, the rate, the curves, the continuity, the gauge invariance and the bound for rectangles. The thread property enters no step, and Ω is defined without d, C, γ, K.

Step 3: (3). J_d is open and $W_C^{[d]}$ is continuous and gauge invariant on it; the inverse image of a Borel set under $F_C^{[d]}$ is a Borel subset of J_d, joined with the closed set $\mathcal{V}_d \setminus J_d$ if it contains 0; π_d is continuous; so $Y_{d,C}$ is Borel and $\sigma(\pi_d)$-measurable, $\sigma(\pi_d) \subset \mathcal{F}_d$, and its restriction to the Borel set Thr is a Borel function on the threads. $\sup_{\mathcal{X}} |Y_{d,C}^K - Y_{d,C}| \le \sup_{J_d} |w_C^K - W_C^{[d]}| \to 0$ by Prop. 10.4(b). θ_v is a bijection, with inverse $\theta_{v^{-1}}$, that maps every $\pi_j^{-1}(I_j)$ onto itself, and $Y_{d,C}(\theta_v V_\bullet) = F_C^{[d]}(V_d^{v|\Lambda_d}) = Y_{d,C}(V_\bullet)$. $(C^{\langle d\rangle})^{\langle j\rangle} = C^{\langle j\rangle}$, so $W_{C^{\langle d\rangle}} = W_C$ on $\mathcal{V}_j$, and every $d \ge d_\Theta$ is admitted. For curves Prop. 10.4(c) replaces (a), (b); the shadows are fixed lattice by lattice because w_γ^K depends on the shadow on the bare lattice, and a curve has no refinement.

Step 4: (4). $\pi_d(\mu) = \mu_d$ gives the first equality, Prop. 10.4(b), (d) the others and the rates. $|Y_{d,C}| \le 1$ and Step 1 give $|\int Y_{d,C}\,\mathrm{d}\mu - \int \mathbf{1}\{\tau \le d\} Y_{d,C}\,\mathrm{d}\mu| \le \mu(\tau > d) \le \sum_{d' \ge d} R_{d'}$, at most $\frac{L}{L-1} R_d$ if $c' A_0^2 \log x_{[d]} \ge 2$ (Step 1). $G_d - F_C^{[d]}$ vanishes on J_d and has modulus at most 1 off it, and $\mu_d(\mathcal{V}_d \setminus J_d) \le \mu_d(\mathcal{V}_d \setminus I_d) \le R_d$; for $d \ge \tau$, $V_d \in J_d$ and so $G_d(V_d) = Y_{d,C}$. For the joint means, with $D = \max_i d_i$: the law of $(V_0, \dots, V_D)$ is the image of μ_D under $V \mapsto (\mathcal{M}^D V, \dots, V)$, $\prod_i F_{C_i}^{[d_i]} \circ \mathcal{M}^{D-d_i}$ is one bounded Borel function on $\mathcal{V}_D$, $\mathcal{M}^{K-d_i} = \mathcal{M}^{D-d_i}\mathcal{M}^{K-D}$, and $\mu_D^K \to \mu_D$ in total variation; no measure on an infinite product is needed here. The means depend on the single laws μ_d; what μ adds to them are the events that involve infinitely many depths, $\{\tau \le D\}$ and Ω.

Step 5: (5). We use the localisation of backgrounds (Prop. YM3_I:53.5) at $n = K - d$, for the constant sequence $\Omega_0 = \cdots = \Omega_{K-d} = T_\ell$, whose datum is V on the bonds of Λ_d and whose minimal orbit is the set of the backgrounds over V; for the region Ω', a region for the partition into the M_1-cubes of the depth d, with the one part $\mathsf{Y}_1 = T_\ell \setminus \Omega'$ of its complement and the index $l_1 = K - d$; at $\varepsilon_{\mathrm{r}} = 2\varepsilon_{K-d} \le \epsilon_{\mathrm{lc}}$; and at $V \in \mathsf{O}_{K-d}$. We verify its hypotheses for this region of M_1-cubes, using the joined sequence defined in equation YM3_I:(999): the joined sequence is the $\mathfrak{r}_2$-tower of Ω', whose minimal orbit over the datum induced by V defines the localised background U' (equation YM3_I:(77)); it is separated; its tower bonds are the bonds of its members of index $< K - d$, with both end points in the collar $\mathsf{Y}_1^{\sim 2} \setminus \mathsf{Y}_1$, which lies in the wide collar $\mathsf{Y}_1^{\sim 3} \setminus \mathsf{Y}_1$ (Section YM3_I:53.4), the points of Ω' within three layers of M_1-cubes of the depth d of $T_\ell \setminus \Omega'$; the core $\Omega'^{\sim -2}$ lies in its last member, so that $j(b) = K - d$ at the bonds of the core; and both data are regular at the radius $2\varepsilon_{K-d}$, as $V \in \mathsf{O}_{K-d}$ (Prop. YM3_I:44.2).

So, after gauge transformations, the background over V and U' differ by factors $e^{i\mathsf{a}(b)}$ at the bare bonds b with no end point in the wide collar, with $|\mathsf{a}(b)| \le \mathsf{C}_{\mathrm{lc}}\, 2\varepsilon_{K-d} L^{d-K} e^{-\delta_{\mathrm{lc}} D(b)/2}$, $D(b)$ the distance of b from the tower bonds in the weighted distance of the tower (Section YM3_I:53.4). The bare bonds of C are such bonds of the core, as $\mathrm{dist}(C, T_\ell \setminus \Omega') \ge \frac{\delta_1}{2} \ge 4M_1 a_d$. The tower-bond endpoints lie within the two-layer collar: a row below the last level has both endpoints outside the next tower member, which contains the core $\Omega'^{\sim -2}$. Moreover every local spacing in the joined sequence is at most a_d. For any connected path Γ joining two points, the layer pieces cover Γ, so its weighted length is at least a_d^{-1} length(Γ), and hence at least a_d^{-1} times the periodic sup-distance between those points. Taking the infimum over paths proves the same bound for weighted distance; the sup-distance to the complement of a slab, or of the intersection of two slabs, is the distance; so on C, $D(b) \ge \mathrm{dist}(C, T_\ell \setminus \Omega')/a_d - 2M_1 \ge \frac{\delta_1}{2} L^d - 2M_1 \ge \frac{\delta_1}{4} L^d$, as $M_1 a_d \le \frac{\delta_1}{8}$.

The loop is gauge invariant, $W_C = \frac{1}{2}\,\mathrm{tr}$, and $|\,\mathrm{tr}\prod_i A_i - \mathrm{tr}\prod_i B_i| \le 2\sum_i |A_i - B_i|$ for unitary matrices; summing $|e^{i\mathsf{a}(b)} - \mathbf{1}| \le |\mathsf{a}(b)|$ over the $|C|/a_K$ bare bonds of C gives $|w_C^K(V) - Y_{K,d}(V)| \le \frac{|C|}{a_K}\, 2\mathsf{C}_{\mathrm{lc}} \varepsilon_{K-d} L^{d-K} e^{-\delta_{\mathrm{lc}}\delta_1 L^d/8} = 2\mathsf{C}_{\mathrm{lc}} \varepsilon_{K-d} |C| L^d e^{-\delta_{\mathrm{lc}}\delta_1 L^d/8}$, and $\varepsilon_{K-d} \le \hat{\varepsilon}_d$ because $\frac{2}{3} x_{[d]} \le x_{K-d} \le x_{[d]} + \bar{B}$. Two fields that agree on the bonds of Ω' have the same $Y_{K,d}$; let $K \to \infty$ with Prop. 10.4(b), (e).

Step 6: (6). The Wilson measure at (K, ℓ, b) is the one at $(K - s, L^s\ell, L^s\mathsf{b})$ carried by $x \mapsto L^s x$, since $\beta = L^K \mathsf{b} = L^{K-s}(L^s\mathsf{b})$, and Λ_d goes to Λ_{d-s}; so the limit laws, the measure on the threads from the depth s on and Prop. 10.2 hold at every $\mathsf{b} > 0$ with $R_{d-s}[L^s\ell, L^s\mathsf{b}]$. At $(L^s\ell, L^s\mathsf{b})$ the

one-torus setting holds (Lem. 10.1), and (1)–(4) hold there at the depth $d-s$, with the depths $d_\natural$, d_Θ, $s_\natural$ of $(L^s\ell, L^s\mathsf{b})$; increased by s, these depths take the places of $d_\natural$, d_Θ, $s_\natural$ in (1)–(4) at (ℓ,b), the depth $d-s$ of $(L^s\ell, L^s\mathsf{b})$ being the depth d of (ℓ,b).

The dilation maps Λ_j of T_ℓ onto Λ_{j-s} of $T_{L^s\ell}$, one graph with the same $\ell L^j = 2L^{m+j}$ sites in each direction, bond onto bond, block onto block and plaquette onto plaquette, and the cutoff K onto the cutoff $K-s$, with the same number $K-d$ of block steps above the depth d; it leaves β, $x_{[d]} = L^d\mathsf{b} = L^{d-s}(L^s\mathsf{b})$, $\varepsilon_{[d]}$, the balls I_d, J_d and the length in units of the lattice, $|C|L^d$, unchanged. Every object in the definition of the smoothed loop is formed from these graphs, from $K-d$ and from the constants, in the units of the lattices, and holds neither the side, nor the label, nor the spacing. Item by item: the block averaging map with its averaging radius $1/200L^2$ (equations YM3_I:(37)–(39)); the constraint sets $\{\mathcal{M}^{K-d}U = W\}$, the action $\sum_p a_p$, criticality for it on them and the regularity at the radius $\varepsilon_{\rm v}$, a constant, all measured in the units of the level, with the bare spacing $L^{-(K-d)}$ (Defs. YM3_I:9.3 and YM3_I:9.5); the residual gauge groups $\mathcal{G}_K^d$ and the minimal orbit (§5.2); the axial gauge of the averaged backgrounds in the definition of H_s^K ([7, §B.2]); the radius $\bar\varepsilon$ and the lattices of Thm. YM3_I:E in the setting of Section YM3_I:52.7, whose lattice hypotheses are imposed through the number of the sites in each direction at the levels used, so that the backgrounds over a field with $\varpi < \bar\varepsilon$ form one non-empty orbit at (ℓ,b) and at $(L^s\ell, L^s\mathsf{b})$ alike; and the plaquette size ϖ, a maximum over the plaquettes of one graph .

So the transported backgrounds over V_d are the backgrounds over V_d on the lattices of side ℓ, the transported $\mathcal{C}_K(W)$, H_d^K, $H_{d,n}^\infty$, w_C^K, smoothed loops, balls and variables are those of (ℓ,b) at the depth d, and (1)–(4) at $(L^s\ell, L^s\mathsf{b})$ and the depth $d-s$ are (1)–(4) at (ℓ,b) and the depth d.

10.1.7. *Proof of Thm. AE. Step 1: the ceiling.* Fix $\vartheta \in (\kappa, \frac12)$. Prop. 10.6(a) gives the hypothesis of Prop. 10.8(a) with $\eta_s = C_\eta L^{-\vartheta(K-s+1)}$ up to the last level $\bar t = K-d$, $d \ge s_0$. By Prop. 10.7(c), $a_K^{-\kappa}\Xi_{s-1} \le c_\Xi L^{\kappa(K-s+1)}$, the depth $K-s+1 \ge d$ being at least i_0; so

$$a_K^{-\kappa}\mathsf{e}_t \le c_\Xi C_\eta \sum_{j\ge K-t+1} L^{-(\vartheta-\kappa)j} = \epsilon_1 L^{-(\vartheta-\kappa)(K-t+1)}\,:$$

the additive errors of the steps, weighed by $\Xi_t \approx L^{-\kappa t}$, sum after division by a_K^κ exactly when $\vartheta > \kappa$; what ends at $\kappa = \vartheta$ is the additive error, not the tower. The rarity bound at $p = \kappa+1$ gives $\sum_{t\le\bar t} L^{\kappa(K-t)}\mathsf{w}_t \le 2\mathsf{V}_\ell C_{\rm r}(\kappa+1)L^{-d}$, and $a_K^{-\kappa}(\Xi_{t-1}+\mathsf{e}_{t-1}) \le (c_\Xi L^\kappa + \epsilon_1)L^{\kappa(K-t)}$; Prop. 10.8(a) gives eq. 10.ii. At $d = s_0$, $|W_{\mathsf{C}}^{(\bar t)}| \le 1$ and $Z^{-1}m_{\bar t}(\mathcal{V}_{K-\bar t}) \le 1$ give the supremum, and $|\Phi^{(t)}[W_{\mathsf{C}}]| \le \Xi_t + \mathsf{e}_t \le (c_\Xi+\epsilon_1)L^{-\kappa t}$. A rectangle in a ball of radius $< \ell/4$ has $|C| < \sqrt{2}\,\ell$, which is $< \frac{2\pi}{3}\mathsf{b}$ if $\ell \le 1.48\,\mathsf{b}$. If $\kappa \ge \frac12 > \vartheta$, the same tower with $\Xi_{s-1} \le c\,L^{-\kappa(s-1)}$ gives $\mathsf{e}_{\bar t} \le c\,a_K^\vartheta$ and $\Xi_{\bar t} \le c\,a_K^\kappa$, and the rarity terms are smaller.

Step 2: the limit. Let $d \ge \max\{s_0, d_0\}$. By Prop. 10.4(e), m_{K-d} is carried by $\mathsf{O}_{K-d} \subset J'_d$, where $W_{\mathsf{C}}^{(K-d)} = w_{\mathsf{C}}^K \to F' := \prod_i W_{C_i}^{[d]}$ uniformly. By Prop. 10.8(c), $Z^{-1}m_{K-d}$ and μ_d^K differ on gauge invariant sets by a positive measure of mass $\sum_{t\le K-d}\mathsf{w}_t \le 2\mathsf{V}_\ell C_{\rm r}(\kappa+1)L^{-(\kappa+1)d}$; $\mu_d^K \to \mu_d$ in total variation; $a_K^{-\kappa}\Xi_{K-d} \to \mathsf{I}_d \le c_\Xi L^{\kappa d}$; and $\mathbf{1}_{J'_d}F'$ differs from $F_{\mathsf{C}}^{[d]}$ only off J_d, a set of μ_d-measure at most R_d, which is also the difference of the two means. With eq. 10.ii, $\limsup_K \mathcal{W}_K - \liminf_K \mathcal{W}_K \le 2\mathsf{E}_d \to 0$, so the limit exists, and $|\mathcal{W}(\mathsf{C}) - \mathsf{I}_d \int F_{\mathsf{C}}^{[d]}\,\mathrm{d}\mu_d| \le \mathsf{E}_d$.

The value of I_d (Prop. 10.7(c)) gives the asymptotics of $a_d^{-\kappa}\int F_{\mathsf{C}}^{[d]}\,\mathrm{d}\mu_d$, its remainder giving $c\,a_d(1+\log^+(1/a_d))/\mathsf{b}$ for a fixed polygon, which tends to zero. Uniformity on J: the numbers of Prop. 10.6 depend on J alone; c_Ξ is uniform on J; $a_K^{-\kappa}\prod_n \lambda_n \to \mathsf{I}_d$ locally uniformly in b; the interval $[\mathsf{b}_-, \mathsf{b}_+]$ is an interval of labels under (Tor) on the side ℓ (§2.3.1: $\mathsf{b}_- \ge \gamma^{-2}$, and $MQ(\mathsf{b}_+) \le L^m$, with Q of Lem. 10.1, which is the function R of that definition, holds on the sides of $\mathfrak{L}(g)$ at every label of $\mathsf{I}(g)$, as in the proof of Prop. 10.4(e)), so the numbers $\mathsf{C}_{\rm bf}$, $K_{\rm bf}$ of Def. 2.7(5) formed at this interval depend on the label only through it, and Cor. 2.8(a) bounds

the difference of the laws of the block field of the depth d at two consecutive cutoffs K', $K'+1$, on the gauge invariant sets, by $\mathsf{C}_{\rm bf}\ell^3 K'^{2\mathfrak{p}_0+1/2}L^{-K'/10}$ at every $\mathsf{b}\in\mathsf{J}$ and every $K'\ge K_{\rm bf}$ with $\lceil K'/1260\rceil \ge d$; summed over $K'\ge K$, with Cor. 2.8(b), (c) for the limit and for all Borel sets, $\sup_E|\mu_d^K(E)-\mu_d(E)|\le \sum_{K'\ge K}\mathsf{C}_{\rm bf}\ell^3K'^{2\mathfrak{p}_0+1/2}L^{-K'/10}$ for all $\mathsf{b}\in\mathsf{J}$ and all $K\ge K_{\rm bf}$ with $\lceil K/1260\rceil\ge d$; R_d and the rate of Prop. 10.4(b) are explicit and continuous in b; the balls J'_d may move with b.

So Steps 1 and 2 hold uniformly on J, $\mathcal{W}_K\to\mathcal{W}$ uniformly, each $\mathcal{W}_K$ is continuous in b, and the same bounds hold at labels $\mathsf{b}_K\in\mathsf{J}$.

Step 3: every torus and label. Given $m\in\mathbb{Z}$ and $\mathsf{b}>0$, for large s the number $g_s=(L^s\mathsf{b}-\bar B-1)^{-1/2}$ is a reference coupling below the thresholds of the frame, $L^s\mathsf{b}\in\mathsf{I}(g_s)$, and $2L^{m+s}\in\mathfrak{L}(g_s)$; the exponent of Prop. 10.6 is the same for all g, κ is invariant under dilations, and Prop. 10.9(a) gives $\mathcal{W}_K^{(\ell,\mathsf{b})}(\mathsf{C})=L^{s\kappa}\mathcal{W}_{K-s}^{(L^s\ell,L^s\mathsf{b})}(L^s\mathsf{C})$; so the limit exists and obeys the law of dilations.

Step 4: the floor. For $C_i=n_1a_i\times n_2a_i$, Prop. 10.6(b) applies with $n_{C_i}=4$ and c_{C_i} fixed, both invariant under dilations, with $s_0(i)=\max\{i,s'_0\}=i+O(1)$ and no distance term; its bound is at most $C(\mathsf{J})\frac{a_i}{\mathsf{b}}L^{-\vartheta(K-t)}\le C(\mathsf{J})L^{-\vartheta(K-t-i)}$, an error that depends on the scale of the step only through a_{K-t}/a_i; it holds for m_{t+1}-almost every W, which is all that Prop. 10.8(a) asks, so Step 1 runs with $C_\eta(i)=C(\mathsf{J})L^{\vartheta i}$ and $s_0(i)$ in the place of the numbers of Prop. 10.6(a). Let i be so large that $(n_1^2+n_2^2)a_i^2<\ell^2/4$, so that C_i lies in a ball of radius $<\ell/4$ and is a polygon, that $\kappa(C_i)\le k|C_i|/\mathsf{b}_-<\vartheta$ for all $\mathsf{b}\in\mathsf{J}$, that $s_0(i)\le d_i$, which holds for large i as $d_i\ge\frac76 i$, and that $\theta_{d_i}\le1$, which holds for large i as $\theta_{d_i}\to0$ (below).

We run Prop. 10.8 down to the depth $d_i=\lceil(\frac43-\frac{u}{2})i\rceil$. There

$$\theta_{d_i}=2B_3\sup_{K>d_i,\mathsf{J}}\varepsilon_{K-d_i}L^{2d_i}n_1n_2a_i^2\le c\,L^{\frac32 d_i-2i}(1+d_i)^{\mathfrak{p}_0}\to0,$$

as $\frac32 d_i\le(2-\frac34 u)i+\frac32$ and $\varepsilon_{K-d}\le A_0(2a_d/\mathsf{b}_-)^{1/2}(\log(\mathsf{b}_-/2a_d))^{\mathfrak{p}_0}$ for large d. The numbers of the free field of C_i tend to 0: $\eta_i\to0$ by Prop. 10.7(a), so $e^{-\frac{3}{2\mathsf{b}_-}\eta_i}\le a_K^{-\kappa}\Xi_{K-d_i}$ (as $v_{K,d}\ge0$), $c_\Xi\to1$ and $L^{\kappa(d_i+1)}\to1$ ($\kappa d_i\le c\,a_i i$); and $\mathsf{E}'_{d_i}\to0$, as $\epsilon_1\le 2C(\mathsf{J})L^{\vartheta i}/(1-L^{-\vartheta/2})$ and $d_i\ge(\frac43-\frac{u}{2})i$.

Prop. 10.8(a), (c), (d) give the display of (4): by (d) and $\theta_{d_i}\le1$, $W_{C_i}^{(K-d_i)}\ge1-\frac12\theta_{d_i}^2\ge\frac12$ on O_{K-d_i}, which carries m_{K-d_i}, so

$$Z^{-1}\int W_{C_i}^{(K-d_i)}\,{\rm d}m_{K-d_i}\ge(1-\frac12\theta_{d_i}^2)Z^{-1}m_{K-d_i}(\mathcal{V}_{d_i})\ge(1-\frac12\theta_{d_i}^2)(1-2\mathsf{V}_\ell C_{\rm r}(1)L^{-d_i})$$

by (c), whatever the sign of the last factor, and this integral, being ≥0, is multiplied by $a_K^{-\kappa}\Xi_{K-d_i}\ge e^{-\frac{3}{2\mathsf{b}_-}\eta_i}$; both ends are $1+o(1)$ uniformly in $K>d_i$ and $\mathsf{b}\in\mathsf{J}$. For the squares Q_i we use the mean form, Prop. 10.8(b), at one i: by Prop. 10.6(b) the mean defects are at most $C(\mathsf{J})L^{-\vartheta(K-t-i)}$.

With $d=i+n_Q$, choose n_Q and then i so large that $\kappa(Q_i)\le\frac{\vartheta}{2}$, the free factor $a_K^{-\kappa}\Xi_{K-d}$ lies within a factor $\sqrt2$ of $L^{\kappa d}$, $L^{\kappa(d+1)}\le\sqrt2$, the summed step errors $2C(\mathsf{J})L^{-\vartheta(n_Q+1)}/(1-L^{-\vartheta/2})$ are at most $\frac{1}{32}$, $2B_3L^{2n_Q}\sup_{K>d}\varepsilon_{K-d}\le1$ (so that the loop at the last level is at least $\frac12$ on tame fields), the mass of the measure without large fields is at least $\frac78$, and the rarity terms are at most $\frac{1}{32}$; then $\mathcal{W}_K(Q_i)\ge\frac{1}{\sqrt2}\cdot\frac12\cdot\frac78-\frac{1}{16}>\frac18$. For other rectangles we spread with Prop. 10.9(b): $\rho_K(s,T)=\mathcal{W}_K(s\times T)$ has $\rho_K(s,\frac{T'+T''}{2})^2\le\rho_K(s,T')\rho_K(s,T'')$, the powers of a_K matching because lengths add; so $T\mapsto\log\rho_K(s,T)$ is convex, $\rho_K(s,nT)\ge\rho_K(s,2T)[\rho_K(s,2T)/\rho_K(s,T)]^{n-2}$, and a finite lim sup (Step 1) on a set of rectangles closed downwards together with a positive lim inf at one of them gives a positive lim inf at all rectangles of the range, going down by $\log\rho(T)\ge2\log\rho(T')-\log\rho(2T'-T)$ and up by convexity, in each side.

Step 5: block fields; the static source. $H\circ\mathcal{M}^{K-s_1}$ pulls out of the quotients, $\Phi^{(t)}[W_{\mathsf{C}}\,H\circ\mathcal{M}^{K-s_1}]=\Phi^{(t)}[W_{\mathsf{C}}]\,H\circ\mathcal{M}^{K-t-s_1}$ for $K-t\ge s_1$; so Steps 1 and 2 run with the factor H, $\sup|H|$ multiplying the errors (Prop. 10.8(a)), and $Y_{d,C}$-means are those of $F_C^{[d]}$. For a rectangle we divide by $\mathcal{W}(C)>0$

(Step 4): the limit quotients $\omega(H) = \lim_K \langle W_C H(\mathcal{M}^{K-s_1}U)\rangle_K / \langle W_C\rangle_K$ form a real linear functional with $\omega(1) = 1$ on the bounded gauge invariant Borel functions of all block fields, consistent between depths, $\omega(H\circ\mathcal{M}) = \omega(H)$, because $H\circ\mathcal{M}^{d'-s_1}$ is the same function seen at a finer depth; at the depth s_1 it is bounded, $|\omega(H)| \le \sup|H|\,(c_\Xi L^{\kappa d'} + \mathsf{E}'_{d'})/\mathcal{W}(C)$ with $d' = \max\{s_0, s_1\}$, by the display of (5) with $|W_C^{(K-d')}| \le 1$, $Z^{-1}m_{K-d'}(\mathcal{V}_{d'}) \le 1$ and $a_K^{-\kappa}\Xi_{K-d'} \le c_\Xi L^{\kappa d'}$.

On the threads $H(V_{s_1}) = H(\mathcal{M}^{d-s_1}V_d)$ μ-almost surely, so $\int H(V_{s_1})Y_{d,C}\,\mathrm{d}\mu = \int F_C^{[d]}\,(H\circ\mathcal{M}^{d-s_1})\,\mathrm{d}\mu_d$, and Step 2, with $\mathsf{l}_d \asymp a_d^{-\kappa}$, gives the limits in d and the two-sided bound for $\int Y_{d,C}\,\mathrm{d}\mu$. For H not gauge invariant: for a gauge transformation v of Λ_K, $\mathcal{M}^{K-s_1}(U^v) = (\mathcal{M}^{K-s_1}U)^{v|\Lambda_{s_1}}$; averaging v over the Haar measure and using the invariance of the Wilson measure, of W_C, of μ and of $Y_{d,C}$ replaces H by $\bar H$.

Step 6: mixed functions. By Prop. 10.10(d) at $X = W_\mathsf{C}$, $|Q_K(z)| \le M_0\mathcal{W}_K(\mathsf{C},\theta\mathsf{C})^{1/2}$ on half the polydisc; $(\mathsf{C},\theta\mathsf{C})$ is a family (its polygons lie $2\delta_1$ apart) with $\kappa(\mathsf{C},\theta\mathsf{C}) = 2\kappa < \frac12$, so Step 1 bounds the right side uniformly in $K > s_0$. For the limit at real z we take the source off the loop by Prop. 10.10(d) applied to the difference of the loop and a function of the block field on the side of C; no step with a source is used. Let $\bar t = K - d$. As $c \in \frac12 a_{i_0}\mathbb{Z}$, θ is a lattice isometry of Iso_{i_0} ([7, §G.1]); let $d_1 := i_0 + \log_L M_1$, so that from the depth d_1 on θ maps Λ_d and the partition into the M_1-cubes of the depth d, cubes of side $M_1 a_d$, onto themselves ([7, Lem. G.1]).

Let d be at least $\max\{s_0(\mathsf{C}), s_0(\mathsf{C},\theta\mathsf{C}), d_0, d_1\}$, $s_0(\cdot)$ the number of Prop. 10.6(a) at the family named, and so large that $M_1 a_d \le \frac{\delta_1}{8}$ and $2\hat\varepsilon_d \le \epsilon_{\mathrm{lc}}$, so that $2\varepsilon_{K-d} \le \epsilon_{\mathrm{lc}}$ for every $K > d$ and Thm. AD(5) applies at d; let $K > d$. Then θ commutes with the block averages, $\mathcal{M}^j(\theta U) = \theta\,\mathcal{M}^j U$ for $j \le \bar t$, and carries the constraint sets, criticality, regularity and the backgrounds over V to those over θV ([7, Lem. G.1]); ϖ is θ-invariant, so $\mathsf{O}_{\bar t}$ is; and the lattice Green function is θ-invariant and the linearised block averages are θ-covariant ([7, Lem. G.1]), so the projections Π_n commute with θ and the lattice Coulomb energies are θ-invariant.

Hence $W_\mathsf{C}^{(\bar t)}(\theta^{-1}V) = W_{\theta\mathsf{C}}^{(\bar t)}(V)$ on $\mathsf{O}_{\bar t}$ and $\Xi_{\bar t}(\theta\mathsf{C}) = \Xi_{\bar t}(\mathsf{C})$. Let $\mathcal{Y}_{K,d}(V)$ be the loop of C at the background localised to a slab Ω' on the side of C as in Thm. AD(5); then $Y = \Xi_{\bar t}(\mathsf{C})\mathcal{Y}_{K,d}(\mathcal{M}^{\bar t}U)$ is a function of the bonds on the side of C, and $\sup_{\mathsf{O}_{\bar t}}|W_\mathsf{C}^{(\bar t)} - \mathcal{Y}_{K,d}| \le \epsilon_d^{\mathrm{loc}}$, the bound of Thm. AD(5). Now $\|W_\mathsf{C} - Y\|_\theta^2 = \langle W_\mathsf{C} W_{\theta\mathsf{C}}\rangle_K - 2\langle W_\mathsf{C}\,\theta Y\rangle_K + \langle Y\,\theta Y\rangle_K$, and three towers without source evaluate the three means: Prop. 10.8(a) for the family $(\mathsf{C},\theta\mathsf{C})$, whose loop at the background is $W_\mathsf{C}^{(\bar t)}W_{\theta\mathsf{C}}^{(\bar t)}$ and whose factor is $\Xi_{\bar t}^2$; Prop. 10.8(a) for C with the gauge invariant factor $\theta\mathcal{Y}_{K,d}$ of the block field; and Prop. 10.8(c) for the function $\mathcal{Y}_{K,d}\,\theta\mathcal{Y}_{K,d}$ of the block field.

The main terms add up to $\Xi_{\bar t}^2 Z^{-1}\int[W_\mathsf{C}^{(\bar t)}(W_{\theta\mathsf{C}}^{(\bar t)} - \theta\mathcal{Y}_{K,d}) - (W_\mathsf{C}^{(\bar t)} - \mathcal{Y}_{K,d})\,\theta\mathcal{Y}_{K,d}]\,\mathrm{d}m_{\bar t}$, of modulus at most $2\Xi_{\bar t}^2\epsilon_d^{\mathrm{loc}}$. With $a_K^{-\kappa}\Xi_{\bar t} \le c_\Xi L^{\kappa d}$ and the rarity bound at $p = 2\kappa+1$,

$$a_K^{-2\kappa}\|W_\mathsf{C} - Y\|_\theta^2 \le \sigma_d := c_\Xi^2 L^{2\kappa d}\big[2\epsilon_d^{\mathrm{loc}} + 2\mathsf{V}_\ell C_{\mathrm{r}}(2\kappa+1)L^{-(2\kappa+1)d}\big] + \mathsf{E}'_d(\mathsf{C},\theta\mathsf{C}) + 2c_\Xi L^{\kappa d}\mathsf{E}'_d(\mathsf{C}),$$

free of K, and $\sigma_d \to 0$ because $2\kappa < \vartheta < \frac12$ and $\epsilon_d^{\mathrm{loc}}$ falls faster than every power of a_d. At real z, Prop. 10.10(d) at $X = W_\mathsf{C} - Y$ gives $|Q_K(z) - a_K^{-\kappa}\Xi_{\bar t}\,\mu_d^{K,w_z}(\mathcal{Y}_{K,d})| \le M_0\sigma_d^{1/2}$. By Prop. 10.10(b) the probability $\mu_d^{K,w}$ gives at most $\sum_{t\le\bar t}\mathsf{w}_t^w \le 2\mathsf{V}_S C_{\mathrm{r}}^S(\kappa+1)L^{-(\kappa+1)d}$ to the complement of $\mathsf{O}_{\bar t}\cap J'_d$, since the sourced measure without large fields has the null sets of $m_{\bar t}$ and is carried by $\mathsf{O}_{\bar t}\subset J'_d$; on $\mathsf{O}_{\bar t}\cap J'_d$, $|\mathcal{Y}_{K,d} - F'_d| \le \epsilon_d^{\mathrm{loc}} + \sup_{J'_d}|w_\mathsf{C}^K - F'_d|$ with $F'_d = \mathbf{1}_{J'_d}\prod_i W_{C_i}^{[d]}$ (Prop. 10.4(e)). With $a_K^{-\kappa}\Xi_{\bar t} \to \mathsf{l}_d$ and $\mu_d^{K,w} \to \mu_d^w$ (Prop. 10.10(c)),

$$\limsup_K |Q_K(z) - \mathsf{l}_d\,\mu_d^{w_z}(F'_d)| \le M_0\sigma_d^{1/2} + c_\Xi L^{\kappa d}\epsilon_d^{\mathrm{loc}} + 4c_\Xi\mathsf{V}_S C_{\mathrm{r}}^S(\kappa+1)L^{-d} \to 0,$$

so $\lim_K Q_K(z) = \lim_d \mathsf{l}_d\,\mu_d^{w_z}(F'_d)$ exists at every real z of half the polydisc. A bounded family of holomorphic functions that converges on a real cube converges locally uniformly to a holomorphic limit (Vitali's theorem); so $Q_K \to Q$, the derivatives converge, and Prop. 10.10(e) with the limits

S_m of YM3_I:(2212), for pairwise disjoint supports, gives the limit of $\mathcal{W}_K(\mathsf{C};f)$. For two orthogonal mirrors the same argument runs with the four images of C and Prop. 10.10(d) twice, on the quarter $\sum_q |z_q|\|f_q\|_\sharp < \rho_D/4$ of the polydisc, the towers at the subfamilies of $(\mathsf{C},\theta_1\mathsf{C},\theta_2\mathsf{C},\theta_1\theta_2\mathsf{C})$, of κ-value up to $4\kappa<\frac12$, and a region Ω' that is the intersection of two slabs; the limit then exists at the real z with $|z_q|<\rho_D/(4n\max_q\|f_q\|_\sharp)$, a real cube in that quarter, and Vitali's theorem applies there.

For $n=1$, $\mathcal{W}_K(\mathsf{C};f)$ is linear in f; as each coordinate extent of C is $<\frac{\ell}{2}$, every point outside the least closed coordinate box that holds C lies, with a neighbourhood, beyond a coordinate mirror that misses the box, with $c\in\frac12 a_i\mathbb{Z}$ for some i; a Lipschitz partition of unity subordinate to finitely many such mirrors cuts f into pieces, each of which the case of one mirror covers. With walls, the number of boxes is even in every direction that the walls cut and the walls lie in $\frac12 a_i\mathbb{Z}$, so that both fixed planes c and $c+\frac{\ell}{2}$ of every wall mirror are walls and each box has exactly one image in every box; and each f_q is supported in one box, a function whose support misses the walls being the sum of its pieces in the boxes, with the derivatives in its variable the sums of those in the variables of its pieces.

Then Prop. 10.10(d) on the polydisc $\sum_q|z_q|\|f_q\|_\sharp<\rho_D/N$, Step 1 for each family $\mathsf{C}^{\mathrm{f}}_\iota$, of κ-value $N\kappa(\mathsf{C}_\iota)<\frac12$, and Cauchy's estimates bound the derivatives $\partial_A Q_K(0)$ uniformly in K, for all supports off the walls; by Prop. 10.10(e) this bounds $\mathcal{W}_K(\mathsf{C};f_1,\dots,f_n)$ uniformly in K for $n=1$, where $\mathcal{W}_K(\mathsf{C};f_1)=\partial_1 Q_K(0)$, and for pairwise disjoint supports, where the cumulants $S^{\mathsf{T}}_{|\beta|,K}(f_\beta)$ of the blocks of at least two elements converge YM3_I:(2212) and so are bounded in K. The law of dilations and the positive form with mixed entries hold at every cutoff (Prop. 10.9(a), (b)) and pass to the limits; for the latter the plane $x_\mu=\hat c+\frac{\ell}{4}$, $\hat c\in\frac12 a_{i_0}\mathbb{Z}$ the place of the mirror θ' of (6), a fixed plane of the coordinate mirror with the place $\hat c+\frac{\ell}{4}$, which lies in $\frac12 a_{i_0}\mathbb{Z}$ as $\frac{\ell}{4}=\frac12 L^m$, parts polygons and supports, so that every entry converges.

Step 7: beside the loop. Let $X=W_C-\psi$, a function of the bonds on the side of C, and $Y=H-\langle H\rangle_K$, a real function of the bonds of S. Reflection positivity at the site mirror and Schwarz' inequality (as in Prop. 10.9(b)) give $|\langle XY\rangle_K|^2\le\langle X\,\theta X\rangle_K\langle\theta Y\,Y\rangle_K$, with $\langle X\,\theta X\rangle_K=\langle W_C W_{\theta C}\rangle_K-\psi^2\ge0$ by the θ-invariance of the measure, $\langle\theta Y\,Y\rangle_K=\langle\theta H;H\rangle_K\ge0$, and $\langle XY\rangle_K=\langle W_C;\mathcal{O}(f)\rangle_K-\langle W_C;H_0\rangle_K$; divide by ψ. The explicit ceiling uses the law of the truncated two-point function of the action density, Thm. YM3_I:AV.

Let, as in (7), $g\le\gamma_W$ in the frame, γ_W the ceiling of equation YM3_I:(2346); let $\vartheta_{\mathrm{sh}}\in(0,1)$ be a shape parameter, and $\mathsf{d}\in(0,\mathsf{d}_0(g,\vartheta_{\mathrm{sh}})]$ with $K-s(\mathsf{d})\ge1$, $\mathsf{d}_0(g,\vartheta_{\mathrm{sh}})$ the threshold of equation YM3_I:(2350) and $s(\mathsf{d})$ the witness depth of equation YM3_I:(2321) at g. As $\mathsf{b}\in\mathsf{I}(g)$, in the frame $(K,L^K\mathsf{b})$ is a first-passage datum at g with the label b (Cor. YM3_I:74.2), and the side ℓ of the frame lies in $\mathfrak{L}(g)$. Then for every test pair $(h_1,h_2)\in\mathfrak{T}_{T_\ell}(\mathsf{d};\vartheta_{\mathrm{sh}})$ on a torus T_ℓ with $20\mathsf{d}\le\ell$ ([7, Lem. I.1]: non-negative Lipschitz functions, neither 0, with supports in closed balls of radius d at a distance between d and $6\mathsf{d}$, $\int h_j\ge\vartheta_{\mathrm{sh}}\|h_j\|_\infty\mathsf{d}^3$ and $\mathrm{Lip}\,h_j\le\|h_j\|_\infty/(\vartheta_{\mathrm{sh}}\mathsf{d})$)

$$S^{\mathsf{T}}_{2,K}(h_1,h_2)=a_K^{-2}\sum_{p,q}h_1(x(p))h_2(x(q))\langle W_{\partial p};W_{\partial q}\rangle_K=\frac{9}{16\pi^2}\mathsf{b}^{-2}\iint h_1(x)h_2(y)|x-y|^{-6}\,\mathrm{d}x\,\mathrm{d}y\,(1+\epsilon'),\qquad|\epsilon'|\le\epsilon_{\mathrm{tp}},$$

with $\frac{9}{16\pi^2}$ the constant c_2 of YM3_I:C, the double integral the free kernel YM3_I:(2240), and $\epsilon_{\mathrm{tp}}:=\bar\varepsilon_{g,\vartheta_{\mathrm{sh}}}(\mathsf{d})$, the tolerance equation YM3_I:(2349), one function for all these data, non-decreasing, tending to 0 with d and at most $\frac13$ on $(0,\mathsf{d}_0(g,\vartheta_{\mathrm{sh}})]$ (Thm. YM3_I:AV); so $|\epsilon'|\le\frac13$ and the bound $(\frac{4\pi}{3})^2\|h_1\|_\infty\|h_2\|_\infty$ of the free kernel of a test pair, two balls of radius d at a distance $\ge\mathsf{d}$ (equation YM3_I:(2247)), give $S^{\mathsf{T}}_{2,K}(h_1,h_2)\le\frac43(\frac{4\pi}{3})^2\frac{9}{16\pi^2}\mathsf{b}^{-2}\|h_1\|_\infty\|h_2\|_\infty$, the bound of Thm. YM3_I:AV.

As $\mathcal{O}(f)=a_K^{-1}\sum_p f(x(p))(1-W_{\partial p})$, $\langle\theta\mathcal{O}(f);\mathcal{O}(f)\rangle_K=a_K^{-2}\sum_{p,q}f(\theta x(p))f(x(q))\langle W_{\partial p};W_{\partial q}\rangle_K=S^{\mathsf{T}}_{2,K}(\theta f,f)$, and $\sqrt{9/16\pi^2}=\frac{3}{4\pi}$: for a test pair $(\theta f,f)$ the inequality at $H_0=0$ gives the bound of

(7) with the factor $1+\epsilon_{\rm tp}$. On the whole class of the law, $\frac{4}{3}(\frac{4\pi}{3})^2\frac{9}{16\pi^2}=\frac{4}{3}$ gives $\langle\theta\mathcal{O}(f);\mathcal{O}(f)\rangle_K\le\frac{4}{3}\mathsf{b}^{-2}\|f\|_\infty^2$, and the inequality at $H_0=0$ gives the ceiling $\frac{2}{\sqrt{3}}\mathsf{b}^{-1}\|f\|_\infty[\langle W_C W_{\theta C}\rangle_K/\psi^2-1]^{1/2}$ with no further condition.

The identity follows from the same expression of $\mathcal{O}(f)$:

$$\langle W_C\mathcal{O}(f)\rangle_K/\psi-\langle\mathcal{O}(f)\rangle_K=a_K^{-1}\sum_p f(x(p))[\langle W_{\partial p}\rangle_K-\langle W_C W_{\partial p}\rangle_K/\psi],$$

and $\langle W_{\partial p}\rangle_K>0$ (Prop. 10.9(c)). The derivative is $\frac{d}{d\mathsf{b}}=L^K\frac{d}{d\beta}$ applied to $\log\langle W_C\rangle_K$. For a rectangle with $2\kappa<\frac12$ the three quantities converge by (6) at $n=1$ with (4), by (2) at the family $(C,\theta C)$, and by YM3_I:(2212), the supports of θf and f being disjoint.

Step 8: short distances. (a) We run the tower in its mean form, Prop. 10.8(b), down to the depth i of the corners, that is to the level $t_i=K-i$; when $\max\{n_1,n_2\}<L$ this is the level t_ϱ with $a_{K-t_\varrho}\le\varrho_i<La_{K-t_\varrho}$. The number s_0 of Prop. 10.6(b) for C_i is $\max\{i,s_0'\}$, s_0' formed from the constants, b_- and $n_{C_i}=4$; so $s_0=i$ for all large i, and every step that the tower uses, at a level $t<t_i$, has $K-t>i=s_0$. There Prop. 10.6(b) bounds the steps with a constant free of the size of the loop; weighed by the free factors, whose ratios between the levels are bounded by Prop. 10.7(c), they contribute $C_1\frac{|C_i|}{\mathsf{b}}(\frac{\varrho_i}{\mathsf{b}})^{\vartheta/2}$, and the rarity terms contribute $C_3\mathsf{V}_\ell C_{\rm r}(p)\varrho_i^{p-1/2}$.

At the depth i a tame background has $1-W_{C_i}^{(t_i)}\le\frac12\theta^2=2B_3^2n_1^2n_2^2\varepsilon_{K-i}^2\le4B_3^2A_0^2\mathsf{n}^3\frac{\varrho_i}{\mathsf{b}}(1+\log\frac{\mathsf{nb}}{\varrho_i})^{2\mathfrak{p}_0}$ (Prop. 10.8(d), with $x_{K-i}\ge\frac23\mathsf{b}/a_i$, $n_1^2n_2^2a_i\le\mathsf{n}^3\varrho_i$ and $\mathsf{b}/a_i\le\mathsf{nb}/\varrho_i$). The free factor down to that level is $a_K^{-\kappa}\Xi_{t_i}=\exp[-\frac{3}{2\mathsf{b}}(c_{\rm lat}|C_i|+\mathcal{J}_\ell(C_i)+E_{a_K}(C_i)-v_{K,i}(C_i))]$ (Prop. 10.7(a), (c)), and $0\le\frac{3}{2\mathsf{b}}v_{K,i}(C_i)\le\kappa\log^+\frac{\varrho_i}{a_i}+c\frac{|C_i|}{\mathsf{b}}\le\kappa\log L+c\frac{|C_i|}{\mathsf{b}}$, c depending on n_1, n_2: at the depth of the corners the logarithm of v (Prop. 10.7(b)) and that of $\mathcal{J}_\ell$ cancel up to $\frac{|C_i|}{2\pi}\log\max\{n_1,n_2\}$ and a multiple of $|C_i|$, and $\kappa\log\max\{n_1,n_2\}\le\kappa\log L+k\frac{|C_i|}{\mathsf{b}}\log\frac{\mathsf{n}}{L}$; $|\log(1-z)|\le2|z|$ for $|z|\le\frac12$. (b) Let $a_*:=\varrho(\mathsf{b}/\varrho)^{\vartheta/(3+\vartheta)}$ and let d_* be the least depth with $a_{d_*}\le a_*$, so that $a_*/L<a_{d_*}\le a_*$; for small ϱ, $a_{d_*}>a_i$, and the depth d_* is coarser than the corners.

At all the levels with $a_{K-t}\le a_*$ the defect of one step is at most $C_*\frac{\varrho}{\mathsf{b}}(\frac{a_{K-t}}{\mathsf{b}})^\vartheta$ at m_{t+1}-almost every W, hence in the mean, $|\bar{\mathsf{D}}_{t+1}|\le C_*\frac{\varrho}{\mathsf{b}}(\frac{a_{K-t}}{\mathsf{b}})^\vartheta$ with $\bar{\mathsf{D}}_{t+1}$ of Prop. 10.8(b), C_* depending on the constants, c_0, ϑ and J alone: this is the bound of one step of (8)(b), shown at the end of this step. We stop the tower, in its mean form, Prop. 10.8(b), at the depth d_*. Prop. 10.7(c), valid at every depth, and (d) give the factor $e^{-\frac{3}{2\mathsf{b}}(\mathcal{I}_{a_K}(C,C)-v_{K,d_*})}$ with $0\le v_{K,d_*}(C)\le C^{\rm bg}A_C^2/a_{d_*}^3$, A_C the area of C; on tame fields $0\le1-W_C^{(K-d_*)}\le4B_3^2A_0^2\frac{A_C^2}{\mathsf{b}a_{d_*}^3}(1+\log\frac{\mathsf{b}}{a_{d_*}})^{2\mathfrak{p}_0}$ (Prop. 10.8(d)).

As $A_C\le100\varrho^2$ and $a_*/L<a_{d_*}\le a_*$, these are $O(\varrho^4/a_*^3)$ and $O(\frac{\varrho^4}{\mathsf{b}a_*^3}(1+\log\frac{\mathsf{b}}{a_*})^{2\mathfrak{p}_0})$, with constants that hold 10^4L^3 (and $\log L$ in the logarithm); the step errors add up to $O(\frac{\varrho}{\mathsf{b}}(\frac{a_*}{\mathsf{b}})^\vartheta)$; the terms of the large fields, $\sum_{t\le K-d_*}\Xi_{t-1}\omega_t$ with $|\omega_t|\le\mathsf{w}_t$, are by the rarity bound at $p=2$ at most $c\,\mathsf{V}_\ell C_{\rm r}(2)\,a_{d_*}^2\le c\,\mathsf{V}_\ell C_{\rm r}(2)\,a_*^2$ times the free factor, and $a_*^2=\mathsf{b}^2(\frac{\varrho}{\mathsf{b}})^{2-2\alpha}$ with $\alpha=\frac{\vartheta}{3+\vartheta}<\frac17$, below the remainder as $2-2\alpha>1+3\alpha$; and $\frac{\varrho}{\mathsf{b}}(\frac{a_*}{\mathsf{b}})^\vartheta=\frac{\varrho^4}{a_*^3\mathsf{b}}=(\frac{\varrho}{\mathsf{b}})^{1+\frac{3\vartheta}{3+\vartheta}}$: the choice of a_* balances the two.

Prop. 10.7(a) and $|\mathcal{J}_\ell-\mathcal{J}|\le c\varrho^4\ell^{-3}$ finish; $\log(\mathsf{b}/\varrho)$, and not $\log\varrho$, is the logarithm that dilations leave invariant. The bound of one step at the levels with $a_{K-t}\le a_*$. At the levels finer than the lattice of the corners, $K-t>s_0=\max\{i,s_0'\}$, which is i for small ϱ, it is Prop. 10.6(b): its constant depends on the constants, $n_C=4$ and the chord-arc constant of C, which is bounded below in terms of c_0, its factor $|C|/\mathsf{b}$ is at most $40\varrho/\mathsf{b}$, and its bracket, without a distance term for one polygon, is at most $c\,(a_{K-t}/\mathsf{b})^\vartheta$ with c depending on the constants, ϑ and J.

At the remaining levels, $a_i\le a_{K-t}\le a_*$, at least as coarse as the lattice of the corners, $a_{K-t}\ge a_i>c_0\varrho/2L$ and $|C|\le40\varrho$, so that the loop passes a number of cubes of the level that is bounded in terms of L and c_0, and $|C|/a_{K-t}<80L/c_0$. There the proof of Prop. 10.6(b) runs

with the segments C_j of C, one for each cube of the level that the loop passes, in the place of the bonds of the level, down to one segment where the cells are larger than the loop. The shift of Prop. YM3_I:51.5 gives $W_C^{(t)}(V_W(B)) = W_C(e^{i\eta_t \mathbf{A}(B)} U_+)$ exactly for every bare loop (step 1 of the proof of Prop. 10.6(a)); so $W_C^{(t)}(V_W(B)) = w(\Upsilon(B))$ and $W_C^{(t+1)}(W) = w(0)$, with w as in Lem. 10.5(c), $g_j = U_+(C_j)$ the holonomy of U_+ along the segment, and the rotations $\Upsilon_j(B) = \frac{1}{i}\log[(e^{i\eta_t \mathbf{A}(B)} U_+)(C_j)\, U_+(C_j)^{-1}]$ of the segments.

Each segment is one insertion into the four-family Gaussian cluster expansion of Thm. YM3_I:R, applied to the two potentials in ν_W as in Step 3 of the proof of Prop. 10.6(a), with a linear part of norm of the order $g_t|C_j|/a_{K-t}$ and a remainder of the order $(g_t|C_j|/a_{K-t})^2$, so that $\log \nu_W(e^{(z,\Upsilon)})$ is a sum of local terms on a polydisc of radius of the order $a_{K-t}/(g_t|C|)$. The first cumulants, and the defects of the pair cumulants from their values $s_{jj'}\mathrm{R}(G_{j\to j'})$ at the trivial triple, are colour vectors and colour defects that vanish at the flat background, and Schwarz' lemma in the background gives them the factor ε_{t+1}, as in step 5 of the proof of Prop. 10.6(a); the pair values at the trivial triple sum to the slice $g_t^2 \eta_t(X_C, \mathcal{C}_{(t+1)} X_C)$ of the free Coulomb energy, as in its step 7, at depths coarser than the corners as well (Prop. 10.7(c) holds at every depth); and $\nu_W \geq 0$ for m_{t+1}-almost every W.

Lem. 10.5(c) then bounds the defect of the step by $E_1 + E_2 + E_3$ and the term of the label, at most $c\,|C|\mathsf{b}^{-2} a_{K-t}$; each of E_1, E_2, E_3 is at most $\frac{|C|}{\mathsf{b}}(\frac{a_{K-t}}{\mathsf{b}})^{1/2} y_t^{\mathfrak{p}}$, $y_t = \log x_t$, $\mathfrak{p}$ formed from $\mathfrak{p}_0$, times a polynomial in $|C|/a_{K-t}$ with coefficients formed from the constants, since a cumulant of order m carries the factor $(g_t|C|/a_{K-t})^m$, the first cumulants besides the amplitude $\delta_t = g_t \mathfrak{b}_t$ of the non-Gaussian potentials, the Gaussian part being even, the variables are at most $c\,\delta_t|C|/a_{K-t}$, g_t^2 is of the order a_{K-t}/b, and the first cumulants and the defects of the pair cumulants carry the factor ε_{t+1}, of the order $(a_{K-t}/\mathsf{b})^{1/2} y_t^{\mathfrak{p}_0}$.

As $|C|/a_{K-t} < 80L/c_0$, $|C| \leq 40\varrho$ and $(\frac{a}{\mathsf{b}})^{1/2} y^{\mathfrak{p}} \leq c\,(\frac{a}{\mathsf{b}})^{\vartheta}$ for $\vartheta < \frac{1}{2}$, the bound is at most $C_* \frac{\varrho}{\mathsf{b}}(\frac{a_{K-t}}{\mathsf{b}})^{\vartheta}$. No level coarser than a_* is used; at the levels with $a_{K-t}^3 \geq \varrho^2 \mathsf{b}$, coarser than a_*, the step bound holds by itself, since there $1 - \lambda_{t+1}$ and $1 - W_C^{(t)}$ on tame fields are at most of the order $\varrho^4/(\mathsf{b} a_{K-t}^3)$ up to logarithms, below $C_* \frac{\varrho}{\mathsf{b}}(\frac{a_{K-t}}{\mathsf{b}})^{\vartheta}$.

10.1.8. *Proof of Thm. AF.* We work in lattice units, $\beta = \mathsf{b}/a$, $n = |C|/a$ steps. By Lem. 10.12(a) the steps of C at distance $\geq 2\rho$ from the corners form an admissible set I of $n - 16\rho$ steps (of $n - 4\rho s$ steps for polygons with s pieces), whose slabs lie in disjoint cells. Lem. 10.12(g) at spin $\frac{1}{2}$, $c_i = \frac{3}{4}$, gives $|\langle W_C \rangle| \leq e^{-\mathcal{G}} \mathfrak{M}_\nu^{\omega_\rho}$: across each admissible step the bonds of the slab that cross the disc of radius ρ about it are rotated together with the profile ϕ_ρ, whose Dirichlet energy is about $2\pi/\log\rho$, at an action cost $\approx 2\pi\beta/\log\rho$; the information inequality in three colours gives the factor $e^{-k\log\rho/\beta}$ per slab, up to the second-order term of $\mathcal{G}$ and the constant 1.2144, and the bounds multiply exactly over the disjoint cells; with n slabs, $e^{-kn\log\rho/\beta} = \rho^{-k|C|/\mathsf{b}}$.

The second variation (Lem. 10.12(d)) limits ρ: its second member, summed over the disc, asks $\omega_\rho = 40\rho^4 \sum_i c_i/(\beta E_\rho)^2 \leq 1$, $E_\rho \sim 2\pi/\log\rho$; with $\rho = \lfloor (\frac{\beta}{\Lambda \log^3 \beta})^{1/4} \rfloor$ one has $\omega_\rho \leq c/\log\beta$, so that $\mathfrak{M}_\nu^{\omega_\rho}$, with Lem. 10.12(f), the corrections of $\mathcal{G}$ and the 16ρ omitted steps give the prefactor $6e^{3.1\mathsf{b}/\ell} 1.58^{|C|/\mathsf{b}}$, and $\rho^{-1} \leq 2(\Lambda \log^3\beta/\beta)^{1/4}$ gives the power $[\Lambda \frac{a}{\mathsf{b}} \log^3 \frac{\mathsf{b}}{a}]^{k|C|/4\mathsf{b}}$, the factor $2^{k|C|/\mathsf{b}}$ going into the prefactor. For the second bound let $\rho = 2^S \in (\frac{1}{2}\beta^{1/3}, \beta^{1/3}]$ and $p = k|C|/\mathsf{b} = kn/\beta < 3$, and keep the cells $T_\rho(c_i)$ of the $n - 16\rho$ admissible steps.

On each cell, for every dyadic $2 \leq r \leq \rho$, the moves with the profile ϕ_r, taken as 0 at $|y| \geq r$, rotate bonds of the cell alone, and Lem. 10.12(b)–(e), as in the proof of (g), give at every field $\prod_i \|Q_i\| \leq \min_r e^{-\mathcal{G}_r + Y_r}$, with $\mathcal{G}_r = \sum_i \frac{2c_i}{\beta E_r}(1 - \frac{2c_i}{\beta E_r})$, $Y_r = \sum_i 2c_i X_i^{(r)}/(\beta E_r^2)$ and $X_i^{(r)}$ the X_i of that proof formed with ϕ_r; by (b), $e^{-\mathcal{G}_r} \leq e^{\delta_0 + 1.2144 p} r^{-p}$, where $\delta_0 = (16k\rho\log\rho + pk\log^2\rho)/\beta$ pays for the 16ρ omitted steps and the second-order term of $\mathcal{G}_r$. The sharper estimate bounds the mean of Y_r instead of an exponential moment: in comb frames $a(H_{e'})$ at a rung e' of height $t_{e'}$ is $1 - W$ of the boundary of the strip $S_{e'}$ of $1 \times |t_{e'}|$ plaquettes, whose mean is $1 - w(1, |t_{e'}|) \leq$

$1-w(1,1)^{|t_{e'}|}\le|t_{e'}|\,\nu(a_p)$ by Prop. 10.9(c), in the place of the bound $|t_{e'}|\sum_{q\in S_{e'}}a_q$ at every field; so $\nu(X_i^{(r)})\le c\,r^3\nu(a_p)$, r^3 in the place of r^4, and $\nu(Y_r)\le c\,p\,r^3\log^2 r\,\bar a$ with $\bar a=\nu(a_p)\le\beta^{-1}\log\mathfrak{M}_\nu$ (Lem. 10.12(f)): the plaquette mean enters linearly, and this bound of a mean takes the place of the condition $\omega_\rho\le 1$ and of the weight $40\rho^4$.

Let r_* be the largest dyadic $r\le\rho$ with $Y_r\le p$; on $\{r_*=r\}$, $r<\rho$, $Y_{2r}>p$, an event of probability at most $\nu(Y_{2r})/p$ by Markov's inequality, and on the event that $Y_r>p$ at every r the bound 1 serves. So, as Lem. 10.12(h) states,

$$|\langle W_C\rangle|\le e^{\delta_0+2.2144p}\rho^{-p}\Big[1+\frac{72\,\bar a\rho^3\log^2\rho}{2^{3-p}-1}\Big]+35\,\bar a,$$

the sum over the dyadic $r<\rho$ of $r^{-p}(2r)^3\log^2(2r)$ being geometric and led by its last term exactly when $p<3$. With $\rho^3\le\beta$, so that $\bar a\rho^3\le\log\mathfrak{M}_\nu$, with $\rho^{-p}\le 2^p\beta^{-p/3}$, $\bar a\le\beta^{-p/3}\log\mathfrak{M}_\nu$ and $\log\mathfrak{M}_\nu\le 192\frac{\mathsf{b}}{\ell}+4(14.15+3\log\beta)$, where $192\frac{\mathsf{b}}{\ell}\le\frac{384k}{p}$ as $|C|<2\ell$, this is at most $C(\frac{k|C|}{\mathsf{b}})(\frac{a}{\mathsf{b}})^{|C|/4\pi\mathsf{b}}\log^3\frac{\mathsf{b}}{a}$ for $\beta\ge\beta_2$, the threshold, depending on p alone, from which on $n\ge 16\rho+4$ and $\delta_0\le 1$: $\frac{p}{3}=\frac{|C|}{4\pi\mathsf{b}}$, and the third logarithm is $\log^2\rho$ times $\log\mathfrak{M}_\nu$.

For the Gibbs states the strips of even height are treated by the reflections in the planes of sites, those of odd height by $\sqrt{a(gh)}\le\sqrt{a(g)}+\sqrt{a(h)}$. For the spin j, $c_j=j(j+1)$ replaces $\frac{3}{4}$, that is $k\mapsto\frac{4}{3}c_jk=\frac{j(j+1)}{\pi}$, and $\kappa_j\le 20\log\beta$ keeps ω_ρ admissible. For a family of such polygons whose pieces, those of different polygons included, are at sup-distance at least $(2\rho+1)a$ when they have no common corner, Lem. 10.12(a) applies to the whole family and gives one admissible set of $n-4\rho s$ steps, n and s the numbers of all the steps and of all the pieces, and Lem. 10.12(e), (g) bound $\langle\prod_i W_{C_i}G\rangle$ in the same way, a factor G of the bonds farther than ρa from the polygons being a function of the bonds off B; a fixed family at positive distances has this property as soon as $(2\rho+1)a$ is below these distances, that is for all small a, which is what the limits as $a\to 0$ use.

The proof uses of ν only the conditional laws of finite sets of bonds, reflection positivity and invariance (through Lem. 10.12(f), and for the second bound through Prop. 10.9(c) for the strips); for the Gibbs states the bound of (f) gives 3 in the place of $6e^{3.1\mathsf{b}/\ell}$. So all characters of the holonomy of non-zero spin have means tending to 0: its class tends to the Haar law of SU(2), whose image under $\frac{1}{2}$tr has the density $\frac{2}{\pi}\sqrt{1-w^2}$ (the χ_j are the Chebyshev polynomials of the second kind in W_C, orthogonal for σ, and polynomials are dense among the continuous h); and $\langle W_C^2\rangle=\frac{1}{4}(1+\langle\chi_1(U(C))\rangle)\to\frac{1}{4}$, from $W_C^2=\frac{1}{4}\chi_{1/2}(U(C))^2=\frac{1}{4}(1+\chi_1(U(C)))$.

For a rectangle $\langle W_C\rangle>0$ (Prop. 10.9(c)), so $\mathrm{Var}(W_C/\langle W_C\rangle)=\langle W_C^2\rangle/\langle W_C\rangle^2-1=(1+\langle\chi_1(U(C))\rangle)/(4\langle W_C\rangle^2)-1$; as $\langle\chi_1(U(C))\rangle\to 0$ and the first bound gives

$$\langle W_C\rangle^{-2}\ge(6e^{3.1\mathsf{b}/\ell}1.58^{|C|/\mathsf{b}})^{-2}[\Lambda\frac{a}{\mathsf{b}}\log^3\frac{\mathsf{b}}{a}]^{-k|C|/2\mathsf{b}}$$

, the variance tends to infinity at least at this rate. If c_aW_C had a limit law while $|c_a\langle W_C\rangle|\ge c>0$, then $|c_a|\ge c/|\langle W_C\rangle|\to\infty$; as the law of W_C tends to the semicircle law, which gives no mass to 0, $\langle\mathbf{1}\{|c_aW_C|>R\}\rangle\to 1$ for every R, and the laws of c_aW_C are not tight: a contradiction.

The rotations touch only bonds within $\rho a\to 0$ of the polygons, act on several polygons at once and leave G unchanged, so $\langle\prod_i\chi_{j_i}(U(C_i))\,G\rangle\to 0$ unless all $j_i=0$, which is the asymptotic independence. The floor: $\langle W_C\rangle\ge\langle W_{\partial p}\rangle^{A_C/a^2}$ by Prop. 10.9(c), and $\langle W_{\partial p}\rangle=1-\langle a_p\rangle\ge 1-\beta^{-1}\log\mathfrak{M}_\nu\ge e^{-\frac{3}{2}\beta^{-1}\log\mathfrak{M}_\nu}$ when $\beta^{-1}\log\mathfrak{M}_\nu\le\frac{1}{2}$ (Lem. 10.12(f)), with $\frac{A_C}{a^2}\beta^{-1}=\frac{A_C}{a\mathsf{b}}$; for the Gibbs states the same argument, with reflections in planes of sites and even numbers of steps, and $\frac{3}{2}(28.3+6\log\beta)\le 43+9\log\beta$.

10.1.9. *Scope of the conclusions.* At a fixed depth the variable $Y_{d,C}$ of Thm. AD is the smoothed loop on $\{V_d\in J_d\}$ and 0 off it, and Prop. 10.2(d) bounds the probability of $\{V_d\in J_d\}$ below; the starting depth is random and by its form no stopping time, since $\{\tau\le D\}$ involves all the

coordinates of index $\geq D$; the loop of the depth d uses $H^{\infty}_{d,n}(V_d)$ and not the finer members of the thread, so that $W_C^{[d]}(V_d)$ and $W_C^{[d+1]}(V_{d+1})$ are two different functions on the threads, and the limit in the depth is a limit of means and of quotients of means (Thm. AE(2), (5)); for the sharp loop itself no multiple has both a limit law and a mean that stays away from 0, by the non-tightness consequence of the semicircle law (Thm. AF).

The bounds of the convergence in n and in K hold the factor $L^d\varepsilon_{[d]} = A_0\mathsf{b}^{-1/2}L^{d/2}(\log x_{[d]})^{\mathfrak{p}_0}$, which grows with d. On the side ℓ depend the tails of τ, which hold ℓ^3 or $e^{C\ell^3}$; the floors $\mathsf{p}^{[d]}$, $\mathsf{q}^{[d]}$; the speed of $K \to \infty$, as $a_s(K)$ holds $(\ell L^s)^{3/2}$, $\mathfrak{r}_K$ holds ℓ^3 and K_{bf} depends on ℓ; and V_ℓ; the event $\{V_d \in J_d\}$ asks all the $3(\ell L^d)^3$ plaquettes. Free of ℓ are the bounds in n on the tower, on curves and on small rectangles (Prop. 10.4(a), the first bound of (c)), the locality bound of Thm. AD(5), the numbers c_{lat}, c_Q, c and c_3 (Prop. 10.7), and, for $\ell \geq \ell_0$, all the ceilings of Thm. AF.

Thm. AE concerns $|\mathsf{C}| < \frac{2\pi}{3}\mathsf{b}$ and is a perimeter law, two-sided for rectangles. Three free photon fields at the label b have the same features: their block curvatures obey at each depth, uniformly in K, bounds of the same form as those of Prop. 10.2(c) and of $R_d^{\natural}$ (Prop. 10.11(a)), and their sharp loops have the same exponent κ, the same perimeter constant $c_{\mathrm{lat}}|C| + \mathcal{J}_\ell(C)$ and the same factor λ_{t+1} of one step as the loops of SU(2) in Thm. AE and Prop. 10.6 (Prop. 10.11(b), (c)). Thm. AE(7) is an upper bound: for three free photon fields the response of the action density is $-\frac{3}{2\mathsf{b}^2}\int f|B_{C,\ell}|^2\,\mathrm{d}x$, with the field $B_{C,\ell}$ of C on the torus, within $c|C|\operatorname{diam}(C)\ell^{-3}$ of its Biot–Savart field B_C in the ball that holds the polygon (Prop. 10.11(d)).

The quotients of Thm. AE(5) formed for three free photon fields, $\lim_K \mathbb{E}[W_C^{\mathrm{fr}}\,H]/\mathbb{E}\,W_C^{\mathrm{fr}}$ for H a bounded function of a block field, take negative values at functions $H \geq 0$: by Prop. 10.11(c) the conditional mean of W_C^{fr} given the block field of the level t is $\Xi_t W_C^{\mathrm{fr},(t)}$, and $W_C^{\mathrm{fr},(t)}$ is a product of three cosines of independent centred Gaussian variables of the variance $v_{K,K-t}(C)/\mathsf{b}$, which grows like $\frac{|C|}{2\pi\mathsf{b}}\log\frac{1}{a_{K-t}}$ (Prop. 10.7(b)); so $W_C^{\mathrm{fr},(t)}$ is negative with positive probability, and the quotient at the indicator of $\{W_C^{\mathrm{fr},(t)} < 0\}$ is negative.

Alternatively, the levels $\varrho < a_{K-t} \leq a_*$ of the proof of Thm. AE(8)(b), whose cells are larger than the loop, can be traded for a law of one point at the depth of the size of the loop: $1 - W_C^{(t)}$ is, on tame fields, a quadratic form in the curvature, and the law follows from a bound on $|Z^{-1}\int(1 - W_C^{(t)})\,\mathrm{d}m_t - \frac{3}{2\mathsf{b}}v_{K,K-t}(C)|$ at that level, the mean of one bounded function of the block field against its free value.

11. Centre flux

For $SU(2)$, reverse the plaquette trace on a column winding around a periodic box. A twist class has three signs, one for each coordinate two-torus. On choosing a time direction, write the class as (k, m), where $k \in \mathbb{Z}_2^2$ is electric and $m \in \mathbb{Z}_2$ is magnetic. Let T_m be the transfer operator on a spatial section with magnetic twist m, and let C^k be the product of its two centre-flip operators.

Sketch J (Positive electric-flux sectors). The twisted partition functions have the decomposition

$$Z[(k,m)] = \operatorname{Tr}(T_m^N C^k), \qquad Z_{e,m} = \frac{1}{4}\sum_k (-1)^{e\cdot k} Z[(k,m)] = \operatorname{Tr}(P_e T_m^N) \geq 0,$$

where N is the number of time layers and $P_e = \frac{1}{4}\sum_k (-1)^{e\cdot k} C^k$. Changing the time direction relates these sectors by a finite Fourier transform. Reflection-positive mixtures are obtained when their electric Fourier weights are nonnegative in each time direction. ■□

Sketch of the argument. Split the plaquette action into its spatial and crossing parts. The crossing kernel is positive, and multiplication by the square root of the spatial Boltzmann weight gives $T_m \geq 0$. The centre flips are commuting unitary involutions preserving this operator. Their four

character projections are the P_e. Closing the time direction with the chosen electric twist inserts C^k in the trace, and Fourier inversion gives the displayed formulas.

The same box can be traced in a second coordinate direction. Comparing the two formulas exchanges one electric index with the magnetic index while retaining the common-plane flux. This crossing relation supplies inequalities between the sector traces. The strictly positive transfer kernel has a simple largest eigenvalue in the zero-electric-flux sector. Ratios of eigenvalues then give the electric and magnetic free energies over the chosen spatial section.

For a mixture with weights $w(k,m)$, define its electric-sector coefficients by

$$\widehat{w}(e,m) = \sum_k (-1)^{e\cdot k} w(k,m).$$

Nonnegativity of these coefficients makes the reflected trace positive in that time direction. Requiring it for each direction gives the stated class of mixtures. A single twisted class does not automatically satisfy all these positivity conditions.

The continuum sector ratios require a further comparison of the twisted and untwisted small-field integrations. The source leaves full-Hessian control on doubled domains and a quantitative twisted-window estimate as additional work for its strongest uniform asymptotics. These are steps to establish, not consequences of the scalar Gaussian quadratic estimate. A gap found over one fixed section also does not establish the infinite-volume mass gap.

Finally, the direction of the confinement asymptotic matters. In the finite-size area-law regime the twisted quotient Z^-/Z approaches one. Its defect $1 - Z^-/Z$, or the associated odd-flux weight, is the small quantity. The trace decomposition makes this distinction explicit before any limiting argument is attempted.

11.1. **Centre flux and twisted partition functions.** Centre twists admit a transfer-operator description in terms of electric-flux sectors. We establish the crossing and Fourier relations, the positivity properties of admissible mixtures and the continuum limits that follow from the construction estimates.

A single twisted class need not be reflection positive in every direction. The flux asymptotics remain sketches at the Hessian, window and transport steps specified below. We also distinguish a gap over a fixed small section from a gap of the infinite-volume continuum theory. In the conditional area-law regime the quotient Z^-/Z tends to 1; the small quantity is its defect.

The continuum centre-flux argument requires two estimates: **(Hess)** for the full fluctuation Hamiltonian on a doubled domain, and **(Flux-window)** for the probability of a twisted window. They are described below but are not consequences of the Gaussian quadratic estimate alone.

11.1.1. *Notation and hypotheses. The lattice.* Throughout, $G = \mathrm{SU}(2)$, the directions are $\mu \in \{1,2,3\}$, and tr is the trace of 2×2 matrices, not halved. A box is a triple $N = (N_1, N_2, N_3)$ of integers ≥ 3 with its periodic lattice; $|N| = N_1N_2N_3$, a cube has equal sides, and $N^{\times 3} = (N,N,N)$. $\partial U(p)$ is the plaquette variable of a plaquette p, $t_p = \operatorname{tr}\partial U(p)$, and dU is the product of the Haar probability measures on the bonds. A sign field γ on the bonds acts by $(\gamma U)(b) = \gamma(b)U(b)$, and $d\gamma(p)$ is the product of γ over the four bonds of p. For a sign field η on the plaquettes, $\beta \geq 0$ and a bounded Borel function F of the field,

$$A_\eta(U) = \sum_p (1 - \tfrac{1}{2}\eta(p)\operatorname{tr}\partial U(p)), \qquad Z_\beta[\eta;F] = \int F\, e^{-\beta A_\eta(U)}\,\mathrm{d}U, \tag{11.i}$$

and $A = A_\eta$ at $\eta \equiv 1$.

Classes. A class is $n = (n_{12}, n_{13}, n_{23}) \in \mathbb{Z}_2^3$; a plane $\mu\nu$ with $n_{\mu\nu} = 1$ is a twisted plane of n, and $n_{\mu\nu}$ also names the class whose one twisted plane is $\mu\nu$. A column of the plane $\mu\nu$ is the set of the plaquettes of that plane over one point of it; η^n is -1 exactly on one column of each twisted plane; $Z[n;F] = Z_\beta[\eta^n;F]$, $Z[n] = Z[n;1]$, $r[n] = Z[n]/Z[0]$, $F^n = -\log r[n]$, $\mu_n = Z[n;\cdot\,]/Z[n]$ and $\langle\cdot\rangle = \mu_0$. On a cube, r_i is $r[n]$ for a class with i twisted planes; it depends on i alone. A function of

the field is centre-blind if it is invariant under $U \mapsto \gamma U$ for every sign field γ on the bonds. A sign field s on the plaquettes is closed if its product over the faces of each unit cube is 1; its product over a lattice plane parallel to $\mu\nu$ is then $(-1)^{n_{\mu\nu}}$ for one class $n(s)$, and $[n] \ni \eta^n$ is the set of the closed s with $n(s) = n$. A bounded function $G(U, s)$ of a field and a closed sign field, Borel in U, is covariant if $G(\gamma U, s\,\mathrm{d}\gamma) = G(U, s)$ for all sign fields γ on the bonds; for it $Z_\beta[s; G(\cdot, s)]$ is the same number $\mathsf{M}_n[G]$ for all $s \in [n]$, and $\mu_n(G) = \mathsf{M}_n[G]/Z[n]$.

Transfer operators. With μ as the time and $\nu < \rho$ the other two directions we write $n = (k, m)_\mu$, with the electric part $k = (n_{\mu\nu}, n_{\mu\rho})$ and the magnetic part $m = n_{\nu\rho}$. The section is the periodic lattice $N_\nu \times N_\rho$, and a layer is the field on one section. On L^2 of the fields of the section, $T_m = D_m \mathsf{K} D_m$, where K is the crossing operator defined in Section YM3_I:85 by integrating the time-like bonds between two layers and $D_m^2 = e^{-\beta A_\Sigma^m}$ is the Boltzmann weight of the section Σ, with the plaquette whose corner is a fixed site ξ_* of the section reversed if $m = 1$; a section with $m = 1$ is called twisted. For a site ξ of the section and $j \in \{\nu, \rho\}$, (ξ, j) is the bond from ξ to $\xi + e_j$, and the plaquette with the corner ξ has the bonds (ξ, ν), $(\xi + e_\nu, \rho)$, $(\xi + e_\rho, \nu)$, (ξ, ρ). The flip C_j reverses the bonds (ξ, j) with $\xi_j = (\xi_*)_j$, the bonds of direction j that leave the line of the section through ξ_* normal to j; $C^k = C_\nu^{k_1} C_\rho^{k_2}$ is composition with a map c^k of the fields; $P_e = \frac{1}{4}\sum_k (-1)^{e\cdot k} C^k$ for $e \in \mathbb{Z}_2^2$; and $Z_{e,m} = \mathrm{Tr}(P_e T_m^{N_\mu})$, written $Z_{e,m}^{[\mu]}$ where the time is to be named.

Numbers of a box. $F_0 = \log Z[0]$, $P = P(\beta) = \inf F_0/|N|$ over all boxes, the excess $\delta = F_0 - |N|P$, and $\delta^n = \log Z[n] - |N|P = \log r[n] + \delta$. Over a section and with the height M, $\mathsf{G}(M) = \log(\mathrm{Tr}\, T_0^M / \|T_0\|^M)$ and $\mathsf{d}(M) = 2F_0(M) - F_0(2M)$, the section being fixed. For integers $l \geq 3$, $D_q(l) = q^3 F_0(l^{\times 3}) - F_0((ql)^{\times 3})$, $D_2(l) = 8F_0(l, l, l) - F_0(2l, 2l, 2l)$ and $\mathsf{e}(l) = 2F_0(l, 2l, 2l) - F_0(2l, 2l, 2l)$; over a section (l_1, l_2), $\varepsilon_{\mathrm{v}}(l_1, l_2) = \log(\|T_0\|/\|T_1\|)$, and $\varepsilon_{\mathrm{v}}(l) = \varepsilon_{\mathrm{v}}(l, l)$. For a loop C, $W_C = \frac{1}{2}\mathrm{tr}\, U(C)$; $W_{R\times T}$ is W_C for the boundary of a rectangle $R \times T$ of a coordinate plane.

Gibbs states. For a box N, μ_N is the measure $\langle\cdot\rangle = \mu_0$ of the periodic box N (the index of μ_N is a box, that of μ_n a class); where it stands beside a measure on the fields of the infinite lattice, it is extended periodically. A Gibbs state at β on the infinite lattice $a\mathbb{Z}^3$, $a > 0$, is, as in Section YM3_I:91, a Borel probability measure ν on its fields with $\nu(\gamma_B F) = \nu(F)$ for every finite set B of bonds and every continuous function F of finitely many bonds, γ_B the Gibbs kernel of B with the weight $\exp[-\beta\sum_p (1 - \frac{1}{2}\mathrm{tr}\,\partial U(p))]$, the sum over the plaquettes p that hold a bond of B (equation YM3_I:(1818)); it is periodic if it is invariant under the translations of a subgroup of finite index of $a\mathbb{Z}^3$ (Section YM3_I:91), and every periodic Gibbs state is invariant under all isometries of the lattice (Prop. YM3_I:103.4). In Thm. AG(10), (11) the lattice is $\mathbb{Z}^3$.

Mixtures and boards. A weight $w\colon \mathbb{Z}_2^3 \to [0, \infty)$, $w \neq 0$, is admissible if

$$\hat{w}(e, m) = \sum_k (-1)^{e\cdot k} w((k, m)_\mu) \geq 0$$

for all μ, e, m; then $Z_w = \sum_n w(n) Z[n; \cdot]$, $|Z_w| = Z_w[1]$ and $\mu_w = Z_w/|Z_w|$. A board is a box with $N_\mu = \mathsf{c}\,\mathsf{n}_\mu$, $\mathsf{c} \geq 3$ odd and $\mathsf{n}_\mu \geq 2$ even, cut into $\mathsf{N} = \prod_\mu \mathsf{n}_\mu$ cells by mirrors through bonds, the columns placed where mirrors cross. For a bounded complex Borel function F of one cell, $\mathrm{Fold}\, F$ is the product over the cells of the mirror copies of F, the copy in a cell being complex conjugated if the cell is odd relative to the given one, that is, reached from it by an odd number of reflections in the mirrors (equation YM3_I:(1749), YM3_I:(1749), YM3_I:(1749)); for real F the copies are the mirror images, and for complex F the conjugation makes $Z[0; \mathrm{Fold}\, F] \geq 0$ (with the plain mirror images the constant $e^{i\pi/8}$ on a board of 8 cells would have the fold -1).

$$\|F\|_{\mathrm{ch}}^{\mathsf{N}} = Z[0; \mathrm{Fold}\, F]/Z[0]$$

is the periodic chessboard seminorm (Prop. YM3_I:86.3), and $\|F\|_w^{\mathsf{N}} = Z_w[\mathrm{Fold}\, F]/|Z_w|$.

Cutoffs. L, an odd integer with $L \geq 13$, and $a_K = L^{-K}$ are the blocking factor and the spacings; the label is $b > 0$ and $\beta = L^K b$. A box s of the cutoff K has sides $s_\mu \in a_K\mathbb{N}$, $s_\mu \geq 3a_K$; an index K on a number above denotes that number at $N_\mu = s_\mu/a_K$ and $\beta = L^K b$, as in $Z_K^z(s; b) = Z_K[z](s; b)$,

$r_K[z](s;b)$ and δ_K^0; with $p_\infty(K,b) = a_K^{-3}P(L^K b)$ we put $\delta_K^n(s;b) = \log Z_K[n](s;b) - |s|\,p_\infty(K,b)$. A length is L-adic if it lies in $\mathbb{D} = \bigcup_K a_K\mathbb{N}$. For $K, m \in \mathbb{N}$ the cube $\ell = 2L^m$ is the lattice $\Lambda_{K,\ell}$ of the torus T_ℓ (Section YM3_I:3); for other m it is reached by the dilation $Z_K[n](s;b) = Z_{K-j}[n](L^j s; L^j b)$. $Z_{K,\ell}$ is its periodic partition function, $\Delta_K[\ell, b] = \log Z_{K+1,\ell} - \log Z_{K,\ell}$, and on cubes $t = \ell/b$.

A box s is mixed for the class z if all $s_\mu \in L^{\mathbb{Z}} \cup 2L^{\mathbb{Z}}$ and each twisted plane of z has a side in $L^{\mathbb{Z}}$. We write $\rho^z(N,\beta)$ for $r[z]$ on the box N at β, and, for a fixed class z, $\phi_j(t) = \rho^z((2L^j)^{\times 3}, 2L^j/t)$. The constants γ, $M = L^{m'}$, $r_\star$ and $\mathfrak{p}_0$ are those of Prop. YM3_I:69.1, with $r_\star = r$ and $2\mathfrak{p}_0 > 3r_\star + 2$. Put $R(b) = \min\{L^a : a \in \mathbb{N}_0,\ L^a \geq (\log b)^{r_\star}\}$, $\ell_\flat(b) = 2MR(b)$ and $\ell_\sharp(b) = L\ell_\flat(b)$, and set

$$m_0(b) = 1 + m' + \max\{\log_L R(b + \bar{B}), \lceil \log_L C_{\mathrm{s}} \rceil\}, \qquad b \geq \gamma^{-2}.$$

Here C_{s} is the torus-floor constant, chosen at least C_L as in the paragraph following Prop. YM3_I: 74.5. With σ and $\mathfrak{b}$ of Thm. AG(3) at $\beta = L^K b$, $\sigma_K(l) = a_K^{-2}\sigma(l/a_K)$ and $\mathfrak{b}_K(t,\lambda;l)$ is $\mathfrak{b}$ on the box $(t,\lambda,l)/a_K$.

Levels and block fields. Over a section $l = (l_1, l_2)$ and with a time μ, the levels are $E_i^{K,m,e} = a_K^{-1}\log(\|T_0\|/\lambda_i)$, $\lambda_1 \geq \lambda_2 \geq \cdots > 0$ the eigenvalues of $P_e T_m$, each repeated according to its multiplicity, and $E_i^{K,m,e} = \infty$ beyond them. $\mathcal{M}$ is the averaging map; on the cube ℓ at the cutoff K the block field of depth $d \leq K$ is $\mathcal{M}^{K-d}U$, a field on the block lattice of spacing a_d, and μ_d^K is its law under the periodic measure. A block line is a line of a block lattice.

11.1.2. *Positivity belongs to the electric sectors.* Fix a time direction. A magnetic twist m determines a transfer space and a positive transfer operator T_m. The two independent centre flips commute with T_m and generate a representation $k \mapsto C^k$ of $\mathbb{Z}_2^2$. A temporal twist inserts C^k in the transfer trace. Thus the twisted partition function has the exact finite-cutoff expression

$$Z[(k,m)] = \mathrm{Tr}(T_m^N C^k).$$

The integer N is the number of time steps. Although T_m is positive, the insertion C^k may have both signs. This formula does not imply that a fixed twisted class has all the reflection positivity properties of the untwisted measure.

The character projections isolate the positive quantities. For $e \in \mathbb{Z}_2^2$, put

$$P_e = \frac{1}{4}\sum_{k \in \mathbb{Z}_2^2} (-1)^{e\cdot k} C^k.$$

Character orthogonality makes the P_e orthogonal projections summing to the identity. They commute with T_m, so

$$Z_{e,m} = \mathrm{Tr}(P_e T_m^N) = \frac{1}{4}\sum_k (-1)^{e\cdot k} Z[(k,m)] \geq 0.$$

The source's electric-flux decomposition is therefore a decomposition of one positive transfer operator into invariant subspaces. The positivity is visible before any limit is taken. Changing which coordinate is time gives a second transfer-trace description and the crossing identities between the finite Fourier transforms. Those identities are retained below with their finite-box parity and covering conventions.

The same distinction governs mixtures of classes. In a chosen time direction, nonnegative electric Fourier weights ensure reflection positivity. To obtain reflection positivity in every coordinate direction by this argument, impose the same condition in each description. Positivity in one description alone does not establish the other conditions.

The continuum flux sketch asks for more than these algebraic identities. Its window comparison concerns the actual interacting integral in boxes with different twists. The Hamiltonian differentiated there includes the interaction, the chart Jacobian and the boundary regularization. A lower bound for the Gaussian quadratic form at its reference field does not control that full Hessian. Likewise, a

change of sign in two plane means specifies a candidate window but does not estimate its probability or its transfer between cutoffs. The sketch below specifies these probability and transfer estimates. The finite-cutoff Fourier and transfer arguments do not depend on their completion.

There are also two different large-distance spectral questions. At a fixed small transverse section, the largest transfer eigenvalue and the remaining thermal weights define sector free energies and a transfer gap. An infinite-volume mass gap requires control as the transverse section grows. The finite-size confinement criterion in §17 supplies a separate route to that uniform control; the existence of one fixed-section gap does not supply it.

Finally, the twist ratio in an area-law regime approaches one. The decaying quantity is its defect. Keeping $1 - Z^-/Z$ visible in the finite Fourier formulas prevents the electric-flux suppression from being misread as decay of Z^-/Z itself. The quantitative tube estimates later in the confinement argument concern this small flux weight.

11.1.3. *Principal results.* The untwisted partition function supplies the reference for the centre-flux ratios. We record the stability estimate at a fixed physical label, including the room needed to compare two tori.

Lemma 11.1 (Stability at a fixed label)**.** Use the common construction constants, with $C_{\mathrm{s}} \ge C_L$ as in the torus-floor choice following Prop. YM3_I:74.5, and let $b \ge \gamma^{-2}$, $K \ge 1$ and $m_i \ge m_0(b)$ for $i = 1, 2$. Put $\ell_i = 2L^{m_i}$ and $\theta_j = L^{K-j}b$. The vacuum constants $E_i = E_{(K)}[((\ell_i, K), \theta_\bullet)]$ of the real progression are real, and

$$\begin{aligned} e^{-[c_{\mathrm{low}} + \frac{9}{2}\log b]\ell_i^3} \le e^{-E_i} Z_{K,\ell_i,L^K b} \le e^{(C_{\mathrm{maj}}+2)\ell_i^3}, \\ \left| E_1\ell_1^{-3} - E_2\ell_2^{-3} \right| \le 2C_{\mathrm{tt}}C'_{\mathrm{tt}}E_0 e^{-\kappa \min\{\ell_1,\ell_2\}/(5M)}. \end{aligned} \tag{11.ii}$$

Here $C_{\mathrm{tt}} = 4096\sum_{q\ge 0} e^{-q}$ and $C'_{\mathrm{tt}} = 2$; the rate and size constants are those of Prop. YM3_I: 74.5. In particular, the first logarithmic error is bounded by $(C_{\mathrm{lab}} + \frac{9}{2}\log b)\ell_i^3$, where $C_{\mathrm{lab}} = \max\{c_{\mathrm{low}}, C_{\mathrm{maj}} + 2\}$. ■

Proof. The increments of the sequence are $L^{-1}\theta_{j-1} - \theta_j = 0$, and $\theta_j \ge b \ge \gamma^{-2}$. Thus it is an admissible prescribed sequence in equation YM3_I:(1524), with last radius $R(b)$. The definition of m_0 gives

$$L^{m_i} \ge LMR(b + \bar{B}) \ge LMR(b), \qquad \ell_i = 2L^{m_i} \ge 2LMC_{\mathrm{s}} \ge C_L M.$$

Both tori are therefore admitted, including the stronger room required for their comparison. Apply Prop. YM3_I:74.4 at $k = K$, with $u_0 = L^K b$ and $u_K = b$, and Prop. YM3_I:74.5 to this same sequence on the two tori. These give the two bounds and the reality of the vacuum constants. The vacuum terms are those of the canonical progression of the prescribed sequence; its bare counterterms are its zero increments. Finally $b \ge \gamma^{-2} \ge 4$ implies $\log b > 0$, so the definition of C_{lab} gives the symmetric logarithmic bound. □

Theorem AG (One lattice)**.** Let N be a box, $\beta \ge 0$, and μ a direction, taken as the time, $\nu < \rho$ the other two.

(1) *(Transfer operator; order; momentum; crossing.)* $T_m \ge 0$ is Hilbert–Schmidt with a continuous symmetric kernel > 0; C_ν, C_ρ are commuting unitary involutions that commute with T_m. For every class $Z[(k,m)_\mu] = \mathrm{Tr}(T_m^{N_\mu} C^k)$, and $Z_{e,m} = \frac{1}{4}\sum_k (-1)^{e\cdot k} Z[(k,m)_\mu] = \sum_i \lambda_i(P_e T_m)^{N_\mu} \ge 0$ with eigenvalues $\lambda_i(P_e T_m)$ that do not depend on N_μ; $Z_{e,m} < Z_{00,m}$ for $e \ne 0$, and $Z_{e,m} > 0$ if $\beta > 0$. $\|T_m\|$ is a simple eigenvalue with an eigenfunction > 0 in the range of P_{00}, and $\|T_1\| \le \|T_0\|$. Moreover

$$e^{-2\beta(n_{23}N_1 + n_{13}N_2 + n_{12}N_3)} Z[0] \le Z[n] \le Z[(0,m)_\mu] \le Z[0],$$

the last two inequalities strict if $\beta > 0$ and the classes differ; so $r[n] \le r[n']$ for every one-plane class n' whose plane is twisted in n. Over a twisted section, with the reversed

plaquette at the corner ξ_* and the lines of C_ν, C_ρ through ξ_*, as in the setting, let $\mathcal{T}_\nu$ be the composition with the map $V \mapsto \mathsf{S}_\nu(\gamma_\nu V)$ of the fields, where $(\mathsf{S}_\nu V)_{(\xi,j)} = V_{(\xi+e_\nu,j)}$ is the step by one site in the direction ν and γ_ν reverses the one bond $(\xi_* + e_\nu, \rho)$, which the reversed plaquette shares with its neighbour in the direction ν; and let $\mathcal{T}_\rho$ be formed likewise, with ν and ρ exchanged and the bond $(\xi_* + e_\rho, \nu)$.

On all of L^2, $\mathcal{T}_\nu$ and $\mathcal{T}_\rho$ are unitaries that commute with T_1 and map the gauge invariant functions onto themselves, and $\mathcal{T}_\nu^{N_\nu} = C_\rho$, $\mathcal{T}_\rho^{N_\rho} = C_\nu$; so $\mathcal{T}_\nu$ commutes with C_ρ, and $\mathcal{T}_\rho$ with C_ν. The products $\mathcal{T}_\nu\mathcal{T}_\rho$ and $\mathcal{T}_\rho\mathcal{T}_\nu$ differ by the reversal of the four bonds at the site $\xi_* + e_\nu + e_\rho$, and $\mathcal{T}_\nu C_\nu$ and $C_\nu\mathcal{T}_\nu$, as $\mathcal{T}_\rho C_\rho$ and $C_\rho\mathcal{T}_\rho$, by the gauge transformation that is -1 at the sites of one line of the section and 1 elsewhere: gauge transformations with the values ± 1, which act as the identity on the gauge invariant functions.

On the gauge invariant functions, off which T_1 vanishes ($T_1 = \Pi T_1 \Pi$, Π the gauge average) and which for $\beta > 0$ form the closure of the range of T_1, the five operators T_1, $\mathcal{T}_\nu$, $\mathcal{T}_\rho$, C_ν, C_ρ commute pairwise. So the range of P_e, $e = (e_\nu, e_\rho)$, in the gauge invariant functions, on which $C_\nu = (-1)^{e_\nu}$ and $C_\rho = (-1)^{e_\rho}$, has an orthonormal basis of joint eigenvectors of T_1, $\mathcal{T}_\nu$ and $\mathcal{T}_\rho$, and the joint eigenvalues $(e^{i\theta_\nu}, e^{i\theta_\rho})$ of $(\mathcal{T}_\nu, \mathcal{T}_\rho)$ there have $\theta_\nu \in \frac{2\pi}{N_\nu}(\mathbb{Z} + \frac{e_\rho}{2})$ and $\theta_\rho \in \frac{2\pi}{N_\rho}(\mathbb{Z} + \frac{e_\nu}{2})$; and $Z[(k,1)_\mu] = \mathrm{Tr}(T_1^{N_\mu}\mathcal{T}_\nu^{k_2 N_\nu}\mathcal{T}_\rho^{k_1 N_\rho})$.

For the times 1 and 2, with $e = (e_1, e_2)$ dual to (n_{12}, n_{13}) and $m = n_{23}$ for the time 1, and (e_1, e_2') dual to (n_{12}, n_{23}) and $m' = n_{13}$ for the time 2, the crossing relation

$$Z^{[1]}_{e_1 e_2, m} = \tfrac{1}{2} \sum_{m', e_2' \in \{0,1\}} (-1)^{e_2 m' + e_2' m}\, Z^{[2]}_{e_1 e_2', m'}$$

holds, and likewise for every pair of times: the flux of the common plane is kept, and the pairs (e_2, m) and (m', e_2') are exchanged by the Fourier transform of $\mathbb{Z}_2^2$. Hence, for every time, $Z_{e_1 e_2, m} \le \min\{Z_{e_1 0,0}, Z_{0 e_2, 0}\}$, $\|T_m P_{e_1 e_2}\| \le \min\{\|T_0 P_{e_1 0}\|, \|T_0 P_{0 e_2}\|\}$ and $\|\frac{1}{2}(1 - C_\nu)T_0\| = \|T_0 P_{10}\|$: the least level of odd electric flux along ν has no flux along ρ. For $\beta > 0$ and a fixed section, $\varepsilon_{\mathrm{mag}} = \log(\|T_0\|/\|T_1\|) = -\lim_{N_\mu \to \infty} N_\mu^{-1} \log r[n_{\nu\rho}] \in [0, \infty)$ and $\varepsilon_{\mathrm{el}} = \log(\|T_0\|/\|T_0 P_{10}\|) = -\lim_{N_\mu \to \infty} N_\mu^{-1} \log \frac{Z[0] - Z[n_{\mu\nu}]}{Z[0] + Z[n_{\mu\nu}]} \in (0, \infty)$.

(2) *(Two lengths; the excess.)* Each of $\frac{1}{2}(Z[0] \pm Z[n_{12}])$ is a sum of N_1-th powers of numbers ≥ 0 that do not depend on N_1, and a sum of N_2-th powers of numbers ≥ 0 that do not depend on N_2. $F_0/|N|$ does not increase in any side and tends to $P \ge -6\beta$ as all sides tend to infinity. Over every section and for every time, $0 \le \mathsf{G} \le \delta$, G does not increase in the height, and $\mathsf{G}(M) \le \mathsf{d}(M) = 2\delta(M) - \delta(2M) = 2\mathsf{G}(M) - \mathsf{G}(2M) \le 2\mathsf{G}(M)$. For every integer $q \ge 2$, $qF_0(N) - F_0(N^{[\mu]}) = q\delta(N) - \delta(N^{[\mu]}) \ge 0$, $N^{[\mu]}$ the box with N_μ multiplied by q. And $\delta(N')/|N'| \le \min\{1, \sum_\mu N_\mu/N'_\mu\}\, \delta(N)/|N|$ if $N'_\mu \ge N_\mu$ for all μ.

(3) *(Electric flux tube.)* Let $\beta > 0$ and $l \ge 3$. Over the sections of sides λ and l let $E_0 = -\log\|T_0\|$ and $E_{\mathrm{e}} = -\log\|\frac{1}{2}(1 - C')T_0\| = -\log\|T_0 P_{10}\|$, C' the flip of the bonds parallel to the side λ, and $\varepsilon = E_{\mathrm{e}} - E_0$; then $0 \le E_0 \le E_{\mathrm{e}} < \infty$. For $f \in \{E_0, E_{\mathrm{e}}\}$ and $\lambda, \lambda' \ge 3$: $2f(\lambda) \ge f(\lambda - j) + f(\lambda + j)$ for $\lambda - j \ge 3$, $f(\lambda + \lambda') \ge f(\lambda) + f(\lambda')$, $f(\lambda)/\lambda$ does not decrease and tends to a real number s_f, $s_f \le [f(2\lambda) - f(\lambda)]/\lambda$, and $s_f\lambda - f(\lambda) \ge 0$ is convex and does not increase. With $s_0 = s_{E_0}$ and $R(\lambda) = s_0\lambda - E_0(\lambda)$, the tension $\sigma(l) = s_{E_{\mathrm{e}}} - s_0 = \lim \varepsilon(\lambda)/\lambda \in [0, \infty)$ obeys

$$\frac{\varepsilon(\lambda) - R(\lambda)}{\lambda} \le \sigma(l) \le \frac{\varepsilon(2\lambda) - \varepsilon(\lambda) + E_0(2\lambda) - 2E_0(\lambda)}{\lambda} \qquad (\lambda \ge 3).$$

On the boxes (t, λ, l) let $r = r[n_{12}]$, $q = \frac{1}{2}(Z[0] - Z[n_{12}])$, $p = \frac{1}{2}(Z[0] + Z[n_{12}])$, $\Phi = F_0 + t\lambda s_0$ and $\mathfrak{b} = \frac{1}{2}(1 - r)e^{\Phi} = q\, e^{t\lambda s_0}$. Then $-s_0 = \inf_{t', \lambda' \ge 3} F_0(t', \lambda', l)/t'\lambda' \ge lP$, $0 \le \Phi \le \delta$, $\Phi/t\lambda$

does not increase in t nor in λ, $p \ge e^{-t\lambda s_0}$ and so $\mathfrak{b} \le e^{\Phi} - 1$; and for $t \ge t_0 \ge 3$, $\lambda \ge \lambda_0 \ge 3$

$$\frac{1-r}{1+r} = \frac{q}{p} \le \mathfrak{b}(t,\lambda) \le \mathfrak{b}(t_0,\lambda_0)^{t\lambda/t_0\lambda_0} \le \big(e^{\Phi(t_0,\lambda_0)} - 1\big)^{t\lambda/t_0\lambda_0},$$

so that $\Phi(t_0,\lambda_0) < \log 2$, a number of periodic boxes, is enough for an area law, and $r(t,\lambda,l) \ge (1-x)/(1+x)$ with $x = \mathfrak{b}(t_0,\lambda_0)^{t\lambda/t_0\lambda_0}$. Further $-(t\lambda)^{-1}\log\mathfrak{b}(t,\lambda)$ does not decrease in t nor in λ, and

$$\sigma(l) = \sup_{t,\lambda} \frac{\log(1/\mathfrak{b}(t,\lambda))}{t\lambda} = \lim_{t,\lambda\to\infty} \frac{\log(1/\mathfrak{b}(t,\lambda))}{t\lambda};$$

so $\sigma(l) > 0$ if and only if $\mathfrak{b} < 1$ on one box. For $t > t_0$, $\lambda > \lambda_0$: $-\log\mathfrak{b}(t,\lambda) \ge (t-t_0)[(\lambda - \lambda_0)\sigma(l) - R(\lambda_0)] - (\lambda/\lambda_0)\Phi(t_0,\lambda_0)$. Finally $\varrho = q/Z[0]$ has $\varrho(2t) \le \varrho(t)^2 e^{2F_0(t) - F_0(2t)}$, and the same in λ.

(4) *(Ceilings of the tension; Casimir terms; slabs.)* Let $\beta > 0$ and $l \ge 3$, with the signs of (3). (a) For integer heights $3 \le h_0 < h_1$ and every length $\lambda \ge 3$,

$$(h_1 - h_0)\,\varepsilon(\lambda) \le \log(e^{\mathsf{G}(h_0;\lambda,l)} - 1) + \log\frac{2}{1 - r(h_1,\lambda,l)},$$

$\mathsf{G}(h_0;\lambda,l)$ being G at the height h_0 over the section (λ,l); and for $3 \le \lambda_0 < \lambda_1$, with $\kappa' = \log(2/(1 - r(h_1,\lambda_1,l)))$,

$$\sigma(l) \le \frac{1}{h_1 - h_0}\Big\{\frac{\Phi(h_0,\lambda_0)}{\lambda_0} + \frac{\kappa' + \Phi(h_1,\lambda_0)}{\lambda_1 - \lambda_0}\Big\} + \frac{\Phi(h_1,\lambda_0)}{\lambda_0 h_1},$$

and the same with $\delta(\cdot,\cdot,l)$ in the place of Φ. (b) Over a transverse side $b' \ge 3$, for all integers $a \ge 3$, $k' \ge 2$: $R(a) \le \mathsf{G}(a;b',k'a)/((k'-1)a)$, R formed over the sections (a,b'); and $\delta(M,M,M) \le \mathsf{G}(M;M,M) + \mathsf{G}(M;M,2M) + \frac{1}{2}\mathsf{G}(M;2M,2M) + \frac{1}{4}\mathsf{G}(2M;2M,4M)$ for every $M \ge 3$. (c) For all integers $t,\lambda_1,l \ge 3$,

$$\sigma(l) \ge \frac{1}{\lambda_1}\Big\{\frac{1}{t}\Big[\log\frac{2}{1 - r(t,\lambda_1,l)} - \log\frac{Z[0](t,\lambda_1,l)^2}{Z[0](2t,\lambda_1,l)}\Big] - \frac{1}{\lambda_1}\log\frac{Z[0](\lambda_1,2\lambda_1,l)^2}{Z[0](2\lambda_1,2\lambda_1,l)}\Big\}:$$

a positive tension is one inequality between five partition functions. (d) On the slab $\{0,\dots,A'\} \times \{0,\dots,B'\} \times \mathbb{Z}_{N'}$, free in the first two directions and periodic in the third, $N' \ge 3$, with the Wilson action on all its plaquettes, $|\langle \operatorname{tr} U(C)\rangle| \le 2(\tanh\beta N')^{RT}$ for the boundary C of every rectangle of $R \times T$ plaquettes in a plane normal to the third direction, at every $\beta \ge 0$. (e) On a box (t,λ,l) let the $t\lambda$ plaquettes of the periodic $t \times \lambda$ lattice of the twisted plane be parted into sets D, each connected through common edges; call a bond inner to D if it is parallel to the twisted plane and lies over an edge both of whose plaquettes lie in D, or normal to it and lies over a site all four of whose plaquettes lie in D. Then $q/p = \int \prod_D \rho_D \,\mathrm{d}\mu$, where μ is a probability measure on the configurations of the bonds that are inner to no D, and $\rho_D = (Z_D^+ - Z_D^-)/(Z_D^+ + Z_D^-) \in (-1,1)$, $Z_D^\pm$ the integrals over the bonds inner to D of the weight of the plaquettes parallel to the twisted plane over D and of the plaquettes normal to it over the inner edges of D, without and with one reversed column over D; Z_D^- does not depend on the place of that column.

(5) *(How floors and ceilings travel.)* The time μ and the section are fixed, the boxes differ in N_μ alone, and all numbers of layers are ≥ 3. (a) For $k \ne 0$ and $m \in \{0,1\}$,

$$Z[(k,m)_\mu](N_\mu)^2 \ge 2Z[(0,m)_\mu](2N_\mu) - Z[(0,m)_\mu](N_\mu)^2;$$

so with $\mathcal{R} = Z[0](N_\mu)^2/Z[0](2N_\mu)$, $r[(k,0)_\mu]^2 \ge 2/\mathcal{R} - 1$, and $1/\mathcal{R} \le \frac{1}{4}\sum_k r[(k,0)_\mu]^2$. With $Z' = Z[(0,m)_\mu]$ and $q_0 = Z_{00,m}/Z' \ge \|T_m\|^{N_\mu}/\operatorname{Tr} T_m^{N_\mu}$: $Z[(k,m)_\mu]/Z' \ge 2q_0 - 1$ for every k; $\sum_{k\ne0} Z[(k,m)_\mu]/Z' = 4q_0 - 1$, so that $\sum_{k\ne0} r[(k,0)_\mu] \ge 4e^{-\mathsf{G}(N_\mu)} - 1 \ge 4e^{-\delta} - 1$; and

$$Z[((1,1),m)_\mu] \ge Z[((1,0),m)_\mu] + Z[((0,1),m)_\mu] - Z'.$$

(b) *A ceiling moves inside a twisted plane.* Let $n = (k,0)_\mu$, $k \neq 0$, and $r(N_\mu) = r[n]$. For $N_\mu - j \geq 3$,

$$(1 - r(N_\mu - j))(1 - r(N_\mu + j)) \geq (1 - r(N_\mu))^2 e^{2\delta(N_\mu) - \delta(N_\mu - j) - \delta(N_\mu + j)}.$$

(c) *A floor moves along its own time.* For $\emptyset \neq S \subset \mathbb{Z}_2^2$, $P_S = \sum_{e \in S} P_e$, $f(N') = \mathrm{Tr}(P_S T_m^{N'})/\mathrm{Tr}\, T_0^{N'}$, $\lambda = \|P_S T_m\|$ and $\Theta = e^{\mathsf{G}}$:

$$(\lambda/\|T_0\|)^{N'}/\Theta(N') \leq f(N') \leq (\lambda/\|T_0\|)^{N' - N''}\Theta(N'')$$

for $3 \leq N'' < N'$; so for $3 \leq M_0 < M_1$ and $M_2 \geq 3$,

$$f(M_2) \geq \Theta(M_2)^{-1}[f(M_1)/\Theta(M_0)]^{M_2/(M_1 - M_0)}.$$

In particular, if $r[n_{\nu\rho}](M_1) \geq \varrho > 0$, then $\varepsilon_{\mathrm{v}} \leq [\mathsf{G}(M_0) + \log(1/\varrho)]/(M_1 - M_0)$ and $r[n_{\nu\rho}](M) \geq e^{-\mathsf{G}(M)}(\varrho e^{-\mathsf{G}(M_0)})^{M/(M_1 - M_0)}$ for every $M \geq 3$. (d) *Floors move inside a twisted plane.* With n and r as in (b): for $3 \leq N_1 < N_\mu < N_2$ and $\vartheta = (N_2 - N_\mu)/(N_2 - N_1)$,

$$1 - r(N_\mu) \leq (1 - r(N_1))^{\vartheta}(1 - r(N_2))^{1-\vartheta} e^{\vartheta\delta(N_1) + (1-\vartheta)\delta(N_2) - \delta(N_\mu)};$$

for $N_\mu \geq N_0 \geq 3$,

$$\frac{1}{2}(1 - r(N_\mu)) \leq [\frac{1}{2}(1 - r(N_0))e^{\mathsf{G}(N_0)}]^{N_\mu/N_0};$$

and for every class z one of whose twisted planes holds μ, all integers $n' \geq 1$, $m_1, m_2 \geq 2$, $m = m_1 + m_2$, and every odd $k' \geq 1$,

$$F^z(k'n' + m) \leq k' F^z(n' + m) + \tfrac{k'-1}{2}[\delta(2m_1) + \delta(2m_2)] + \delta(k'n' + m) - k'\delta(n' + m),$$

the argument being the number of layers in the time μ. (e) *Cubes.* For odd $L' \geq 3$, $N' = 2L'^j$ with $j \geq 2$, every class $z \neq 0$ and every β, on the cubes,

$$F^z(L'N') \leq (2L' + 1)^3 F^z(N') + C_{L'}\,\delta(N'/L'), \qquad C_{L'} = 34L'^5 + 53L'^4 + 27L'^3 + 4L'^2,$$

$\delta(N'/L')$ the excess of the cube of side N'/L'.

(6) *(The overlap; odd functions.)* Let $N_\mu = 2\tau$, $k \neq 0$, Ψ the kernel of T_m^τ, $\mu_2 = \Psi^2 dV\, \mathrm{d}W/Z[(0,m)_\mu]$ the law of two layers half a period apart in the class $(0,m)_\mu$, $\langle\cdot\rangle_2$ the mean under μ_2, $g(V,W) = (V, c^k W)$, $r' = Z[(k,m)_\mu]/Z[(0,m)_\mu]$, $\mathsf{m} = (\Psi^2 - \Psi^2 \circ g)/(\Psi^2 + \Psi^2 \circ g)$, p_1 the law of one layer and $E_c = \{\Psi \circ g \geq c\Psi\}$. Then $r' = \langle \Psi \circ g/\Psi\rangle_2 = \int \sqrt{d\mu_2\, d(g_*\mu_2)} = \langle(1 - \mathsf{m}^2)^{1/2}\rangle_2$; $1 - \mathrm{TV} \leq r' \leq (1 - \mathrm{TV}^2)^{1/2}$, TV the total variation distance of μ_2 and $g_*\mu_2$; $c\,\mu_2(E_c) \leq r' \leq c + \mu_2(E_c)^{1/2}$ for every $c > 0$; $r' \leq 2\mu_2(E)^{1/2}$ for every Borel E with $gE = E^c$; $1 - r' \leq \sup\langle\Phi\rangle_2^2/\langle\Phi^2\rangle_2 = \langle\mathsf{m}^2\rangle_2 \leq 1 - r'^2$, the supremum over the bounded Borel Φ with $\Phi \circ g = -\Phi$ and $\langle\Phi^2\rangle_2 > 0$; $r' \geq 1 - 2\,\mathrm{TV}(\mu_2, p_1 \otimes p_1)$; if $\mu_2 \geq c_0\, p_1 \otimes p_1$ on a Borel set $\mathcal{G}$, then $r' \geq c_0\,(p_1 \otimes p_1)(\mathcal{G} \cap g\mathcal{G})$; $-\log r' \leq \frac{1}{2}\mathrm{KL}(\mu_2 | g_*\mu_2) = \langle \log(\Psi/\Psi \circ g)\rangle_2 < \infty$; and for every bimeasurable bijection Φ of the fields that keeps dU and the layer 0 and maps the layer τ by c^k, with μ' the law of the class $(0,m)_\mu$ and $A' = A_{\eta^{(0,m)_\mu}}$,

$$r' \geq \int \sqrt{\mathrm{d}\mu'\, d(\Phi_*\mu')} \geq \exp[-\tfrac{\beta}{2}\mu'(A' \circ \Phi^{-1} - A')], \qquad \tfrac{1}{2}\mathrm{KL}(\mu_2 | g_*\mu_2) \leq \tfrac{\beta}{2}\mu'(A' \circ \Phi^{-1} - A').$$

Odd functions: let $n = (k,0)_\mu$ have one twisted plane, $N_\mu = 2k'\vartheta$ with integers $\vartheta \geq 2$, $k' \geq 2$, let D, D' be bounded complex Borel functions of the bonds inside one slab of ϑ layers, D' odd under c^k applied to all layers of the slab, and $\mathsf{t}_j D'$ the translate of D' by $j\vartheta$ layers, $1 \leq j \leq 2k' - 1$. Then

$$|\langle D\, \mathsf{t}_j D'\rangle| \leq \sup|D|\, \sup|D'| \sum_{\varsigma = \pm 1} \Big(\frac{1 + \varsigma r[n]}{2}\Big)^{\frac{2k' - j - 1}{2k'}} \Big(\frac{1 - \varsigma r[n]}{2}\Big)^{\frac{j-1}{2k'}},$$

and at $j = k'$, $|\langle D\,\mathsf{t}_{k'} D'\rangle| \le 2 \sup|D| \sup|D'| \,[\frac{1}{4}(1 - r[n]^2)]^{\frac{1}{2} - \frac{1}{2k'}}$.

(7) *(Reflection positivity.)* Let $N_\mu \ge 4$. On centre-blind functions $Z[n;\cdot\,]$ does not depend on the places of the columns, and for $\beta > 0$ it is reflection positive at the mirrors normal to μ if and only if μ lies in no twisted plane of n; when μ lies in a twisted plane, the form is negative at a centre-blind gauge invariant function of one slab, a product of even class functions of the plaquettes through μ in one crossing. For centre-blind F of a slab and each (e,m), $\sum_k (-1)^{e\cdot k} Z[(k,m)_\mu; \overline{\theta F} F] \ge 0$. For each of the eight sets $\mathfrak{s}$ of coordinate planes, $\sum\{Z[n;\cdot\,] : n_{\mu\nu} = 0 \text{ for } \mu\nu \notin \mathfrak{s}\}$ is reflection positive on centre-blind functions at the mirrors normal to every direction λ with $N_\lambda \ge 4$; of the single classes, $n = 0$ alone is so in all three directions; and $Z[0] \le \sum_n Z[n] \le 8 Z[0]$.

If N_μ is even, $2 \le \vartheta \le N_\mu/2$, F is a bounded Borel function of a slab of ϑ layers, and the columns of the planes through μ lie in a crossing outside the slab, then $|Z[(k,m)_\mu; F]|^2 \le Z[(0,m)_\mu]\, Z[(0,m)_\mu; \overline{\theta F} F]$, θ the reflection that leaves $\frac{1}{2}N_\mu - \vartheta + 1$ crossings on each side, and $Z[(k,m)_\mu; \mathbf{1}_E] \le Z[(0,m)_\mu]\, \mu_{(0,m)_\mu}(E)^{1/2}$ for every event E of the slab: what is rare without electric flux is rare with it.

(8) *(Chessboard bounds.)* On a board, for bounded complex Borel functions F_κ of the cells and every class, $|Z[n; \prod_\kappa F_\kappa]| \le Z[0] \prod_\kappa \|F_\kappa\|_{\mathrm{ch}}$, and the same with η^n replaced by any sign field carried by the plaquettes in the mirrors. For admissible w: Z_w is reflection positive at the mirrors of the three directions; $w(0) Z[0] \le |Z_w| = Z[0] \sum_n w(n) r[n] \le 8 w(0) Z[0]$; $|Z_w[\prod_\kappa F_\kappa]| \le |Z_w| \prod_\kappa \|F_\kappa\|_w$; $\|F\|_w^{\mathsf{N}} \le \frac{\sum_n w(n)}{\sum_n w(n) r[n]} \|F\|_{\mathrm{ch}}^{\mathsf{N}} \le 8 \|F\|_{\mathrm{ch}}^{\mathsf{N}}$, and for $F \ge 0$, $\|F\|_w^{\mathsf{N}} \ge \frac{w(0)}{\sum_n w(n) r[n]} \|F\|_{\mathrm{ch}}^{\mathsf{N}} \ge \frac{1}{8} \|F\|_{\mathrm{ch}}^{\mathsf{N}}$; so $|\mu_w(\prod_\kappa F_\kappa)| \le 8 \prod_\kappa \|F_\kappa\|_{\mathrm{ch}}$.

All this holds for covariant functions $G_\kappa(U,s)$ of the cells, with $F_\kappa = G_\kappa(\cdot, 1)$, every $s \in [n]$ and the columns anywhere. The constants are attained: the functions of the cells that cancel the weight of rods of plaquettes through the centres of the cells give equality in the first bound and in the bound of $\|\cdot\|_w$ at $w \equiv 1$; and with the fold taken in a class $n \ne 0$ itself the estimate fails for $\beta > 0$: there are positive functions of the cells with $Z[n; \prod_\kappa F_\kappa] = Z[0]$ and $\prod_\kappa Z[n; \mathrm{Fold}\, F_\kappa]^{1/\mathsf{N}} = Z[n] < Z[0]$. In a class: for real $q \ge 1$, $\mu_n(\prod_\kappa |G_\kappa|) \le r[n]^{-1/q} \prod_\kappa \||F_\kappa|^q\|_{\mathrm{ch}}^{1/q}$; for a covariant event E of one cell, $\mu_n(E) \le r[n]^{-1} \|\mathbf{1}_E\|_{\mathrm{ch}}$; and the law of the inner bonds of one cell under μ_n has a density h_n against their law under $\langle\cdot\rangle$ with $\int h_n \log h_n \le \mathsf{N} \log(1/r[n])$, so that the two laws differ in total variation by at most $(\frac{\mathsf{N}}{2} \log \frac{1}{r[n]})^{1/2}$.

Without a board: let $N_\mu = 2h$, $h \ge 2$, and let G be covariant and depend only on the bonds and signs of the h layers on one side of a mirror normal to μ, $F = G(\cdot, 1)$. If μ lies in every twisted plane of n, then $r[n] |\mu_n(G)| \le \langle \overline{\theta F} F\rangle^{1/2} \le \langle |F|^2\rangle^{1/2}$, the first inequality attained, and

$$|\mu_n(G) - \langle F\rangle| \le (r[n]^{-2} - 1)^{1/2} (\langle |F|^2\rangle - |\langle F\rangle|^2)^{1/2},$$

which is at most $\frac{1}{2}(r[n]^{-2} - 1)^{1/2}$ for events; the condition on μ cannot be dropped, since an event of a half torus may be determined by the flux through a plane inside it. If besides $N_\nu = 2h'$, $h' \ge 2$, and G depends only on a quarter of the torus cut by mirrors normal to μ and to ν, then $r[n] |\mu_n(G)| \le \langle |F|^4\rangle^{1/4}$ for every n, and for every $w \ge 0$, $w \ne 0$, and every covariant event E of the quarter, $\mathsf{M}_w[E]/|Z_w| \le \frac{\sum_n w(n)}{\sum_n w(n) r[n]} \langle \mathbf{1}_E\rangle^{1/4}$, where $\mathsf{M}_w = \sum_n w(n) \mathsf{M}_n$. Pairs: if $N_\mu = 2h$, $h \ge 2$, $w \ge 0$ has $\hat{w} \ge 0$ at the time μ, and $\mathbb{M}_w[G] = \sum_s w(n(s)) Z_\beta[s; G(\cdot, s)]$ over the closed s, then $\mathbb{M}_w[G\, \overline{\theta G}] \ge 0$ for every bounded $G(U,s)$, Borel in U and not supposed covariant, that depends only on the bonds and plaquette signs of the h layers on one side of a plane midway between two layers normal to μ, θ the reflection in it.

The infinite lattice: let $\beta > 0$, $K, m \in \mathbb{N}$, $\ell = 2L^m$, let pr be the projection of $a_K\mathbb{Z}^3$ onto $\Lambda_{K,\ell}$, on sites, bonds and plaquettes, and $\mathrm{per}_\ell U = U \circ \mathrm{pr}$, on the bonds, the periodic extension. For a class n and $s \in [n]$ the sign field $s \circ \mathrm{pr}$ on the plaquettes of $a_K\mathbb{Z}^3$ is closed; closed sign fields on the plaquettes of $a_K\mathbb{Z}^3$ being exact, $s \circ \mathrm{pr} = d\chi$ for a sign field χ on the bonds of $a_K\mathbb{Z}^3$, unique up to a factor $d\epsilon$, ϵ a sign field on the sites and $d\epsilon$ at a bond the product of ϵ at its two ends; and the plaquette variables of the twisted extension $\chi\,\mathrm{per}_\ell U$ are $\partial(\chi\,\mathrm{per}_\ell U)(p) = s_{\mathrm{pr}\,p}\,\partial U(\mathrm{pr}\,p)$.

The law $\tilde\mu_n$ of the class n on $a_K\mathbb{Z}^3$ is the image of $e^{-\beta A_s} dU/Z[n]$ under $U \mapsto \chi\,\mathrm{per}_\ell U$; on the functions invariant under $U \mapsto (d\epsilon)U$ for all ϵ it depends neither on $s \in [n]$ nor on χ (a change $s \mapsto s\,d\gamma$ is undone by $U \mapsto \gamma U$, with $\chi(\gamma \circ \mathrm{pr})$ for χ, and χ is fixed up to $d\epsilon$), and $\tilde\mu_0$ is the periodic measure extended periodically. For admissible w let $\tilde\mu_w = \sum_n w(n) r[n]\,\tilde\mu_n / \sum_n w(n) r[n]$; for a board of $a_K\mathbb{Z}^3$ with cells of side $\mathsf{l} \le \ell/2$ and a function F of a cell let $\|F\|_{w,\ell} = \|F \circ \mathrm{per}_\ell\|_w$, on the board of $\Lambda_{K,\ell}$ under it, and let $\|F\|_{\mathrm{ch},\ell}$, the seminorm of [7, Cor. B.1], be the same number at $w = \delta_0$.

Then on the local functions invariant under all $U \mapsto (d\epsilon)U$ the law $\tilde\mu_w$ satisfies [7, Cor. B.1](a)–(d) with $\|\cdot\|_{w,\ell}$ for $\|\cdot\|_{\mathrm{ch},\ell}$ and Z_w for the periodic partition function, and clause (e) of it with the pressure $p^w_\ell = \ell^{-3}\log(|Z_w|/\underline{w})$ for p_ℓ, where $|Z_w|$ is formed on $\Lambda_{K,\ell}$ and $\underline{w}$ is the least positive value of $\hat w(e,m)$ over all μ, e, m ($\underline{w} = 4$ at $w \equiv 1$, $\underline{w} = 1$ at $w = \delta_0$): for $\ell \ge \ell' \ge 2\mathsf{l}$, $p^w_\ell \le p^w_{\ell'}$ and $\|F\|_{w,\ell} \le \|F\|_{w,\ell'}\, e^{\mathsf{l}^3(p^w_{\ell'} - p^w_\ell)}$; at $w \equiv 1$ the same holds with $\ell^{-3}\log|Z_w|$ for p^w_ℓ. (The division by $\underline{w}$ makes $e^{\mathsf{l}^3 p^w_\ell}$, the root of order $(\ell/\mathsf{l})^3$ of $|Z_w|/\underline{w}$, non-increasing in ℓ: with the time μ, a fold integral of Z_w over n_μ cells in the direction μ is $\sum_{e,m} \hat w(e,m)\,\mathrm{Tr}\,B_{e,m}^{\mathsf{n}_\mu/2}$ with positive operators $B_{e,m}$ that do not depend on n_μ; for $q = \mathsf{n}_\mu/\mathsf{n}'_\mu \ge 1$, $\mathrm{Tr}\,B^{q\mathsf{n}'_\mu/2} \le (\mathrm{Tr}\,B^{\mathsf{n}'_\mu/2})^q$, $\omega \le \omega^q$ for $\omega = \hat w(e,m)/\underline{w} \in \{0\} \cup [1,\infty)$, and $\sum_i x_i^q \le (\sum_i x_i)^q$ for $x_i \ge 0$; the three directions are taken in turn.

At $w = t\delta_0$ the undivided pressure is $p_\ell + \ell^{-3}\log t$, and for small t it is larger at ℓ than at $\ell' < \ell$.) Further $r[n]\,|\tilde\mu_n(\prod_\kappa F_\kappa)| \le \prod_\kappa \|F_\kappa\|_{\mathrm{ch},\ell}$ for such functions F_κ of the cells of one period box. For all local functions F_κ of the cells, invariant or not, with the plain extension and the columns of every class on the edges of the board of $\Lambda_{K,\ell}$, so that the functional μ_w names the board, $|\mu_w(\prod_\kappa F_\kappa \circ \mathrm{per}_\ell)| \le \prod_\kappa \|F_\kappa\|_{w,\ell}$.

(9) *(The density of the mixture.)* Let $\beta > 0$, $\mathsf{w}_{\mathrm{mix}} = \sum_s e^{-\beta A_s}$ over the closed sign fields s, and $\hat{\mathsf{w}}_{\mathrm{mix}} = 2^{2-2|N|}\mathsf{w}_{\mathrm{mix}}$, $|N|$ the number of sites. For every class and every bounded centre-blind Borel F, $\int F \sum_{s\in[n]} e^{-\beta A_s}\,\mathrm{d}U = 2^{2|N|-2} Z[n;F]$, whatever the places of the columns; so $\hat{\mathsf{w}}_{\mathrm{mix}}$ is the density of $\sum_n Z[n;\cdot\,]$ on the centre-blind functions. It is the restriction of an entire function of the matrix entries, centre-blind, gauge invariant and invariant under the isometries of the lattice, and $\hat{\mathsf{w}}_{\mathrm{mix}} \le 8$. With $\bar A = \sum_p (1 - \frac12|t_p|)$, $z_p = e^{-\beta|t_p|}$ and $\mathsf{M}(U)$ the set of the unit cubes on whose faces the signs of the t_p have the product -1, $\mathsf{w}_{\mathrm{mix}} = e^{-\beta\bar A}\sum_\Gamma \prod_{p\in\Gamma} z_p$, the sum over the sets Γ of plaquettes whose boundary, the set of the cubes with an odd number of faces in Γ, is $\mathsf{M}(U)$. $\log \mathsf{w}_{\mathrm{mix}}$ is no sum of functions of the four bonds of single plaquettes.

If the angles $\theta_p = \arccos\frac12|t_p|$ of the faces of a unit cube add up to less than π, no t_p there vanishes and the product of their signs is 1, and π cannot be replaced by a larger number; so on $\mathfrak{S} = \{\theta_p < \frac{\pi}{6} \text{ for all } p\}$ the signs of the t_p form a closed sign field $s(U)$, and $\mathfrak{S}$ is the disjoint union of the eight open, non-empty, centre-blind, gauge invariant sets $\mathfrak{S}_n = \{U \in \mathfrak{S} : n(s(U)) = n\}$. Let now all sides be ≥ 4. On $\mathfrak{S}$, for $\beta \ge 4$,

$$\log\bigl(e^{\beta A_{s(U)}}\mathsf{w}_{\mathrm{mix}}(U)\bigr) = \sum_X E(X), \qquad |E(X)| \le \frac{|X|}{198}\prod_{p\in X}(400 z_p),$$

the sum over the connected sets X of plaquettes that are unions of vortices, where in this item two plaquettes are neighbours if they are faces of one unit cube, that is, dual bonds with a common dual site, and a set is connected if it is connected for this relation (so the consecutive plaquettes of a straight column, opposite faces of one cube, are neighbours; the sets D of (4)(e) are connected through common edges instead), a vortex being a non-empty connected set of plaquettes that every unit cube meets in an even number of faces (it has at least 4 plaquettes, and at least $\min_\mu N_\mu$ if the sign field that is -1 on it is not exact), with $E(X) = \log(1 + \prod_{p\in X} z_p)$ for a vortex X that holds no smaller one: thin vortices are local terms. For $\vartheta \le \frac{\pi}{6}$ with $\zeta = 100e^{-2\beta\cos\vartheta} \le \frac12$ and every centre-blind $F \ge 0$ vanishing off $\mathfrak{S}_n \cap \{\theta_p < \vartheta\ \forall p\}$, $(1-\epsilon)\sum_{n'} Z[n';F] \le Z[n;F] \le \sum_{n'} Z[n';F]$ with $\epsilon = \frac{3|N|}{50}\zeta^{\min_\mu N_\mu}$.

(10) *(Area laws from one inequality.)* Let $\beta > 0$ and $D_2(l_0) \le \frac{1}{20}$ at an integer $l_0 \ge 3$. Then $D_2(l_0) > 0$; put $m_* = l_0^{-1}\log\frac{1}{18D_2(l_0)}$, let l_1 be the least $2^i l_0$ with $m_* l_1 \ge 3$, and $\sigma_* = m_*/l_1$. Then $\sigma_* = m_*/l_0$ if $m_* l_0 \ge 3$ and $\sigma_* > \frac16 m_*^2$ otherwise; $D_2(l) \le \frac{1}{18}e^{-m_* l}$ for $l \in \{2^i l_0\}$; $l\,\varepsilon_{\rm v}(l) \le e^{-\sigma_* l^2}$ for $l \in \{2^j l_1\}$; for every integer $h \ge 4l_1$ and all $t, \lambda \ge (l_1 h)^{1/2}$, on the box (t,λ,h) whose first two sides span the twisted plane,

$$-\log[\tfrac12(1-r)] \ge \sigma_* t\lambda - \frac{4t\lambda}{l_1 h}\Big(\frac12\log\frac{4h}{l_1} - 0.66\Big),$$

in particular, on every cube of side $N \ge 4l_1$, $-\log[\frac12(1-r_1)] \ge \sigma_* N^2 - \frac{4N}{l_1}(\frac12\log\frac{4N}{l_1} - 0.66)$; over the section (N,N), $N \ge 4l_1$, the norm $\Lambda_-(N)$ of T_0 on the functions odd under one of the two flips has $\log(\|T_0\|/\Lambda_-(N)) \ge \sigma_* N - \frac{2}{l_1}\log\frac{4N}{l_1}$; and on the even cubes of side $N \ge \max\{2R, T+1, 4l_1\}$, $|\mu_{N^{\times 3}}(W_{R\times T})| \le \exp[-\sigma_* RT + \frac{T}{N}\log 8 + \frac{2RT}{l_1 N}\log\frac{4N}{l_1}]$. There is exactly one periodic Gibbs state ν, the measures of the periodic cubes converge to it, and $|\nu(W_C)| \le e^{-\sigma_*|A(C)|}$ for every simple closed path C of a coordinate plane that each line parallel to one fixed axis of the plane, through midpoints of bonds, meets at most twice, $|A(C)|$ the number of plaquettes enclosed; such a C is made of T rows of plaquettes along the axis, and if they lie in a segment of length w' along it, $|\mu_{N^{\times 3}}(W_C)| \le 8^{T/N}e^{1.13D_2(N/2)}[\frac12(1 - r_1(N))]^{|A(C)|/N^2}$ on the cubes $N \in \{2^j l_0\}$ with $N \ge \max\{4l_1, 2l_0, 2T, 2w'\}$. If $\mathsf{e}(l) \le 0.069$ at an integer $l \ge 3$, then $D_2(8l) \le 0.04736$, so that all of this holds at $l_0 = 8l$; and on every cube of side $N \ge 2l$, $\delta(N^{\times 3}) \le 0.13736\,\theta^{N/2l}$ and $1 - r[n] \le 0.4121\,\theta^{N/2l}$ for every $n \ne 0$, $\theta = 12.9\,\mathsf{e}(l) \le 0.8901$.

(11) *(Rectangular loops.)* Let $\beta > 0$, ν a periodic Gibbs state, $\mathsf{m}(R,T) = \nu(W_{R\times T})$ and $\mathsf{w} = \mathsf{m}(1,1)$. Then m depends on R,T alone, $\mathsf{m}(R,T) = \mathsf{m}(T,R)$, $\mathsf{m}(R,T)^2 \le \mathsf{m}(R,T-1)\,\mathsf{m}(R,T+1)$ with $\mathsf{m}(R,0) = 1$, $\mathsf{w} \ge \frac{\beta}{4(\beta+1)}$, $\mathsf{m}(R,T) \ge \mathsf{w}^{RT} > 0$, and $\mathsf{m}(R,T) \le (8\beta/(8\beta+3))^{2(R+T)-4}$ for $R,T \ge 2$. Hence $V(R) = \lim_T -T^{-1}\log\mathsf{m}(R,T) = \inf_T(-T^{-1}\log\mathsf{m}(R,T))$ exists, is concave and non-decreasing on $\{0,1,2,\dots\}$ with $V(0) = 0$; $\sigma_{\rm W} = \lim V(R)/R = \inf_R V(R)/R$; $\mathsf{m}(R,T) \le e^{-TV(R)} \le e^{-\sigma_{\rm W} RT}$; $V(1) \ge \log(1 + \frac{3}{8\beta})$ and $V(R) \ge 2\log(1+\frac{3}{8\beta})$ for $R \ge 2$, while $V(1) < 2\log(1+\frac{3}{8\beta})$ for $\beta < \frac{1}{24}$; and $0 \le \sigma_{\rm W} \le V(1) \le -\log\mathsf{w} \le \log\frac{4(\beta+1)}{\beta}$.

(12) *(Purity floor.)* Let $N_3 = 2h$, $n = (k,0)_3$ with $k \ne 0$, and $F \in [0,1]$ a Borel function of the fields on the closed half slab of the layers $0,\dots,h$ that is invariant under c^k applied to all its layers; F may be such a function of the block fields of the half slab. On the box $(N_1,N_2,2ih)$ lay F into each of the $2i$ half slabs, mirrored at the odd places; let $\mathcal{Z}_i$ be the periodic integral of the product, $w_F = \mathcal{Z}_1/Z[0]$. If $\mathcal{Z}_1 > 0$, put $\mathfrak{q}_i = \mathcal{Z}_i/\mathcal{Z}_1^i$. Then, for $i \ge 2$,

$$r[n] \ge w_F\,\frac{\mathfrak{q}_i - 2^{1-i}}{1-2^{1-i}}, \qquad r[n] \ge w_F\big((2\mathfrak{q}_2 - 1)_+\big)^{1/2}, \qquad r[n] \ge w_F\Big(\frac{2\|\mathsf{P}\|}{\operatorname{Tr}\mathsf{P}} - 1\Big),$$

where $\mathsf{P} = \mathsf{Y}^*\mathsf{Y}$, Y the operator of F on the half slab, so that $\mathcal{Z}_i = \operatorname{Tr}\mathsf{P}^i$. If $\mathcal{Z}_1 = 0$, positivity gives $\mathsf{P} = 0$ and $\mathcal{Z}_i = 0$ for every i; no purity quotient is asserted in this case.

(13) *(Small cubes, exactly.)* On every cube $N^{\times 3}$, $N \geq 3$, with $r_1 \geq r_2, r_3$ its flux quotients and every time: $|Z_{e,0}/Z[0] - \frac{1}{4}| \leq \frac{1}{4}(2r_1 + r_2) \leq \frac{3}{4}r_1$ for all e; $\delta(N^{\times 3}) \geq \mathsf{G}(N) \geq \log \frac{4}{1+2r_1+r_2} \geq \log \frac{4}{1+3r_1}$; $D_q(N) \geq q^2(q-1)\,\mathsf{G}(N)$ for $q \geq 2$; $Z[0](N^{\times 3})^2/Z[0](2N, N, N) \geq 4/(1+2r_1^2+r_2^2) \geq 4/(1+3r_1^2)$; if N is even, $N \geq 6$ and $r_1 \leq \frac{1}{3}$, then $\mathsf{e}(N/2) \geq \log 2$; and for $\beta > 0$ the bracket of (3) on the cube has $\mathfrak{b}(N, N) \geq \frac{1}{2}(1 - r_1)e^{\mathsf{G}(N)} \geq 2(1 - r_1)/(1 + 3r_1)$. So on a cube with $r_1 \leq \frac{1}{5}$: $\delta \geq \log \frac{5}{2}$, $\mathfrak{b} \geq 1$, and, when N is even with $N \geq 6$, $\mathsf{e}(N/2) \leq 0.069$ fails.

(14) *(A floor and the bulk by construction.)* Let $\lambda(\beta) = \frac{9}{2}\log\beta + 31.4$. (a) Let $\mu\nu$ be any coordinate plane and ρ the third direction. If $\beta \geq 1$ and $h' \geq 1$ is an integer with $2h' \leq \min\{N_\mu, N_\nu\}$, then

$$\log r[n_{\mu\nu}] \geq -N_\rho[32\beta/(h' - \tfrac{1}{2})^2 + 2432\, h'\lambda(\beta)];$$

if $\min\{N_\mu, N_\nu\} \geq 4 + 2(\beta/\lambda(\beta))^{1/3}$, then $\log r[n_{\mu\nu}] \geq -N_\rho[2464\,\beta^{1/3}\lambda(\beta)^{2/3} + 4864\,\lambda(\beta)]$; and if $\min_\lambda N_\lambda \geq 4h' + 2$, then for every $n \neq 0$, $\log r[n] \geq -[32\beta/(h' - \frac{1}{2})^2 + 2432\, h'\lambda(\beta)] \sum N_\rho$, the sum over the twisted planes of n, ρ normal to the plane. (b) With c_∞ of Prop. 11.14 and $P^{\mathrm{G}}(\beta) = \log \frac{2}{\pi\beta^3} - \frac{3}{2}c_\infty$: $P(\beta) - P^{\mathrm{G}}(\beta) \leq 2.2 \cdot 10^5 (\log\beta)\,\beta^{-1/6}$ for $\beta \geq e^{148}$.

■□

The mixed-box comparison still to be proved. The fixed-cutoff transfer identities do not complete the continuum centre-flux argument. One still needs the following estimates for the actual interacting integrals.

The full interacting Hessian (Hess). For every mixed box, every embedding in a periodic covering torus used in Lem. 11.9, every level of its small-field chain, and every admitted real background, remove the factors independent of the fluctuation field. The remaining real fluctuation integral has the form $\int_{D_r} e^{-H(B)}\,\mathrm{d}B$, where $D_r = \{B : |B_y| < r \text{ for every fluctuation site } y\}$. Here H includes the interaction, Jacobian, and all other field-dependent factors of this integral, and is invariant under the covering group. Its restriction to the fixed fields, with the action divided by the order of the group and with Lebesgue measure at orbit representatives, is the fluctuation integral of the mixed box. There are no additional boundary factors or tests in either integral; alternatively, any such factors have a common invariant convex regularization for which the following bounds hold uniformly and which converges to these integrals.

Let $c_- > 0$, $c_+ > 0$, $\mu > 0$ and $g_0 \geq 0$. The function H extends to a real smooth function on D_{2r}, and, on that whole set,

$$\nabla^2 H \geq c_-, \qquad \sup_y \sum_{y'} \sup_{D_{2r}} |\nabla_y \nabla_{y'} H| e^{\mu d(y,y')} \leq c_+, \qquad \sup_y |\nabla_y H(0)| \leq g_0.$$

The constants can be chosen uniformly over the cutoffs, backgrounds, embeddings and labels in a compact subset of $(0, \infty)$, with the dependence on the level, radius and box controlled so that the right side of Prop. 11.10 is at most

$$C|s|L^{3(K-i)} \exp[-cs_{\min}L^{K-i}]$$

at level i. The same realization and bounds hold for the intermediate small-field integrals and boundary regularizations used in the comparison of two consecutive cutoffs.

The twisted window (Flux-window). For the flip-invariant window F defined in Step 11 of the proof of Thm. AH, one must establish, for a fixed sufficiently large m_0, absolute constants $C_2 > 0$, $\epsilon_F \geq 0$, and $t_0 > 0$ such that, on every cube $\ell = 2L^m$, at every cutoff, for $t = \ell/b \leq t_0$ and the one-plane class $n = n_{13}$,

$$Z_K[0; F] \leq C_2 t Z_K[0], \qquad Z_K[n; 1 - F] \leq \epsilon_F t Z_K[0].$$

The window is read at the block layer next to a mirror at the cutoffs $K \geq K_\sharp$, and at the bare layer specified in that step below $K_\sharp$; the flux columns are placed away from these layers. The same

estimates hold for its reflected window. All constants are independent of K and of the representative obtained by the dilations used in that step.

The remaining analytic task is to control the full interacting Hessian and establish the window probabilities. The Gaussian lower bound (Prop. YM3_I:55.2, equation YM3_I:(1099)) and the flip of the two plane means do not supply these estimates. They enter the mixed-box and spectral-limit argument below; the fixed-cutoff transfer identities and confinement implication from the stated box inequality are independent of them.

Detailed sketch AH (Uniformity in the cutoff and the proposed limits). Let the constants be fixed as in Prop. YM3_I:69.1, and $b > 0$. The mixed-box floor in (2)(a) and two-cutoff comparison in (4) remain targets until the full interacting Hessian estimate described above is proved. The same Hessian estimate is required for the proposed consequences in (2)(b,c), the continuum and full-sequence spectral assertions of (3), (5)(b,c), the continuum-function and zero-free conclusions of (6), the full-sequence limits in (7), the uniform-cutoff and limiting assertions of (8), and (9). In (10), the large-N conclusion of (a) and the normalization by positive limiting class masses in (d) still require that step unless the stated positivity has been established separately.

The ceiling in (5)(a) and its proposed consequence (5)(d) still require the twisted-window estimate described above; the same applies to the strict sector-survival and $v_z < 1$ conclusions of (3) and their uses. For the quantitative windows and ultraviolet transport in (10), one must establish the estimates specified there. Fixed-cutoff identities, reflection positivity and estimates whose hypotheses do not invoke these comparisons are unaffected.

(1) *(Window.)* For every box s of the cutoff K and $n \neq 0$, $e^{-2bL^{2K}(n_{23}s_1+n_{13}s_2+n_{12}s_3)} \leq r_K[n](s;b) < 1$, and $r_K[n] \leq r_K[n']$ for every one-plane class n' contained in n; $\delta^0_K \geq 0$, $\delta^n_K = \log r_K[n] + \delta^0_K \leq \delta^0_K$, $P(L^K b) \geq -6L^K b$, and the identities and inequalities of Thm. AG(2) hold for δ^0_K in physical units. For every compact set $\mathsf{J} \subset (0,\infty)$ of labels and every $s_{\min} > 0$ there are j and C, depending on J, $s_{\min}$ and the constants, with $\delta^0_K(s;b) \leq 2C|s|$ for all $b \in \mathsf{J}$ and $K > j$ on every box whose sides are $\geq s_{\min}$, and $\operatorname{Tr} T_m^{t/a_K}/\|T_0\|^{t/a_K} \leq e^{2Ctl_1l_2}$ over the sections (l_1,l_2) for $t, l_1, l_2 \geq s_{\min}$, $m = 0, 1$, $b \in \mathsf{J}$ and $K > j$: with $\mathsf{J} \subset [b_-, b_+]$ one may take j least with $L^j b_- \geq \gamma^{-2}$ and $L^j s_{\min} \geq 2L^{m_0(L^j b_+)}$, and $C = L^{3j} C_*(L^j b_+)$, where $C_*(b)$ is the constant of the comparison $|\log Z_K[0](s;b) - \varepsilon_K|s|| \leq C_*(b)|s|$, with numbers $\varepsilon_K(b)$, on all boxes with sides $\geq 2L^{m_0(b)}$ at $b \geq \gamma^{-2}$, which Lem. 11.1 and the monotone density of F_0 give; $C_*(b)$ and $m_0(b)$ do not decrease in b.

On the cubes, $0 \leq \delta^0_K((2L^m)^{\times 3}; b) \leq (2L^m)^3[c_R + \frac{9}{2}\log b]$ for $b \geq \gamma^{-2}$, $m \geq m_0(b)$ and $K \geq 1$, with c_R formed from the constants of Lem. 11.1, and $\delta^0_K(\ell'^{\times 3}; b) \leq C_f \ell'^3 \log b$ at $\ell' \in \{\ell_\flat(b)/L, \ell_\flat(b)\}$ for $b \geq b_\sharp$ of (4) and $K \geq 4$, C_f formed from the constants; and the supremum of δ^0 over all lattices and couplings with N_μ/β in a compact subset of $(0,\infty)$ is finite. On every box with L-adic sides $\sup_K \delta^n_K < \infty$, and $\inf_K r_K[n] > 0$ if and only if $\inf_K \delta^n_K > -\infty$.

(2) *(Floor.)* (a) If s is mixed for z, there is $C(s,b) < \infty$, depending on s, b and the constants and bounded for b in compact sets, with $\log Z^z_K(s;b) - |s|\, p_\infty(K,b) \geq -C(s,b)$ at every K at which s is a box of the cutoff K; hence $\inf_K r_K[z](s;b) > 0$, uniformly for b in compact sets. (For b in a compact set J, (4) for z and for the class 0, for which s is mixed too, gives $|\log r_{K+1}[z] - \log r_K[z]| \leq 2CL^{-K/10}$ for $K \geq K_0$, with the C, K_0 of (4) at J; the first sentence of (1) bounds $r_K[z]$ from below, uniformly on J, at the finitely many $K < K_0$; and $\delta^0_K \geq 0$.) (b) For a class with one twisted plane the same holds on every box with L-adic sides, uniformly for b in compact sets, and $\inf\{\rho^z(N,\beta) : N_\mu \geq 3,\ N_\mu/\beta \in I\ \forall \mu\} > 0$ for every compact interval $I \subset (0,\infty)$. (c) For every class the same holds on every box with sides in $2L^{\mathbb{Z}}$, the cubes $\ell = 2L^m$ among them, uniformly for b in compact sets; $c(I) = \inf\{\rho^z(N^{\times 3},\beta) : N \geq 3,\ N/\beta \in I\} > 0$ for every compact interval $I \subset (0,\infty)$, so

that $\sup_{j\ge1}\sup_{t\in I}(-\log\phi_j(t))\le\log(1/c(I))$; and the infimum of ρ^z over the lattices with N/β in a compact subset of $\{v_z>0\}$ of (3) is positive.

On these boxes the weights $r_K[n]/\sum_{n'}r_K[n']$ of the mixture converge, to numbers in $[v_n/8,\frac12]$ for $n\neq0$.

(3) *(Limit.)* For every class there is a continuous $v_z\colon(0,\infty)^3\to[0,1]$ with $\rho^z(N_j,\beta_j)\to v_z(\tau)$ along every sequence with $\beta_j\to\infty$ and $N_j/\beta_j\to\tau$; in particular $r_K[z](s;b)\to V_z(s;b)=v_z(s/b)$ along the full sequence of the cutoffs on every box with L-adic sides, uniformly for b in compact sets. In each side, at all real values of the two others, v_z is real-analytic on $(0,\infty)$, and $|V_z(s;b)-V_z(s';b)|\le\frac{4}{s_-}e^{\Gamma}|s-s'|_1$ for $s,s'\in[s_-,s_+]^3$, where Γ is the supremum of δ^0 over the lattices with $N_\mu/\beta\in[\tau_-/2,2\tau_+]$ for all μ, $[\tau_-,\tau_+]$ an interval that holds $[s_-/b,s_+/b]$ in its interior; Γ is finite by (1). $v_z>0$ on the boxes of (2), $\{v_z>0\}$ is open, $v_z>0$ everywhere for one twisted plane, and $v_z<1$ on every box with L-adic sides.

For each time and all e, m, the sector weight $\frac14\sum_k(-1)^{e\cdot k}v_{(k,m)_\mu}\ge0$ is >0 on all of $(0,\infty)^3$ or $=0$ on all of it, and so is v_z for a class with one twisted plane; so a positive lower limit in the cutoff on one box gives one on all boxes, uniform for the sides divided by the label in a compact set; the limits obey the order of Thm. AG(1), $\frac14\sum_k(-1)^{e\cdot k}v_{(k,m)_\mu}\ge0$, and $v_{((1,1),m)_\mu}\ge v_{((1,0),m)_\mu}+v_{((0,1),m)_\mu}-v_{(0,m)_\mu}$. For each time, each section $l\in\mathbb{D}^2$ and all m,e,i, $E_i^{K,m,e}\to E_i^{m,e}(l;b)\in[0,\infty]$ along the full sequence; the limits extend to real sections, are continuous in b, and $E_i^{m,e}(\lambda' l;\lambda' b)=E_i^{m,e}(l;b)/\lambda'$ for real $\lambda'>0$.

Let $H_{l,m}$ be the diagonal operator whose eigenvalues are the finite limits, the eigenvectors labelled by (e,i), and $C^k=(-1)^{e\cdot k}$ on the label e. Then $H_{l,m}\ge0$ has finite multiplicities and no finite point of accumulation; $\operatorname{Tr}e^{-tH_{l,1}}\le\operatorname{Tr}e^{-tH_{l,0}}<\infty$ for $t>0$, and for every L-adic height h_0 the number of levels $\le\Lambda$ is at most $\exp[h_0\Lambda+2Ch_0l_1l_2]$, C that of (1) at $\mathsf{J}=\{b\}$ and $s_{\min}=\min\{h_0,l_1,l_2\}$; 0 is an eigenvalue of $H_{l,0}$ with label $e=00$; the space of $H_{l,1}$ and each sector e of $H_{l,0}$ are not $\{0\}$; with $H_{l,m}$ at the label b,

$$v_{(k,m)_\mu}((t,l)/b)=\operatorname{Tr}(e^{-tH_{l,m}}C^k)/\operatorname{Tr}e^{-tH_{l,0}},$$

and, for every time μ, the sector weights $w^{[\mu]}_{e,m}=\frac14\sum_k(-1)^{e\cdot k}v_{(k,m)_\mu}$, functions on $(0,\infty)^3$, are by this formula $w^{[\mu]}_{e,m}(s/b)=\operatorname{Tr}(P_ee^{-s_\mu H_{l,m}})/\operatorname{Tr}e^{-s_\mu H_{l,0}}$ on the boxes s with L-adic sides, l the section of s normal to μ and P_e the projection on the label e, each trace divided by the periodic trace of its own time and section; at every point of $(0,\infty)^3$ they obey the crossing relation of Thm. AG(1) with $w^{[\mu]}_{e,m}$ in the place of $Z^{[\mu]}_{e,m}$, $w^{[1]}_{e_1e_2,m}=\frac12\sum_{m',e_2'}(-1)^{e_2m'+e_2'm}w^{[2]}_{e_1e_2',m'}$, and likewise for every pair of times, since on each lattice the relation of Thm. AG(1) divided by $Z[0]$ is this relation between the numbers $\frac14\sum_k(-1)^{e\cdot k}r[(k,m)_\mu]$; $E_{\mathrm{mag}}(l)=\inf\operatorname{spec}H_{l,1}=\lim_K a_K^{-1}\log(\|T_0\|/\|T_1\|)$ is finite, and $E_{\mathrm{el}}(l)=\min\{E_1^{0,e}:e_1=1\}=-\lim_{t\to\infty}t^{-1}\log\frac{1-v}{1+v}$, $v=v_{((1,0),0)_\mu}((t,l)/b)$, is finite.

For each time, L-adic section l, direction ν of it and integer $q\ge2$, $a_K^{-1}[q\log\|T_0(l)\|-\log\|T_0(l^{[\nu]})\|]$ converges to a finite number ≥0, $l^{[\nu]}$ the section with l_ν multiplied by q. δ^0_K converges on every box with L-adic sides. On the cubes $\ell=2L^m$, $D(t)=\lim_K[L^3\log Z_{K,\ell}-\log Z_{K,L\ell}]$ exists, depends on $t=\ell/b$ alone, is continuous, and $D(t)-(L^3-1)\log(1/t)$ has a finite limit as $t\to0$; $\delta(t)=\lim_K\delta^0_K(\ell^{\times3};b)=\sum_{r\ge0}L^{-3(r+1)}D(L^rt)$, locally uniformly, and $\lim_K\delta^n_K=\log v_n(t,t,t)+\delta(t)$ is finite for every n. The limits do not depend on the choice of the constants: they are the same for the blocking factor of every other admissible choice.

By (6) and Thm. YM3_I:AD, $t\mapsto v_z(t,t,t)$ and D are real-analytic on $(0,\infty)$.

(4) *(Two consecutive cutoffs.)* If s is mixed for z, $\ell \in 2L^{\mathbb{Z}}$ and $\mathsf{J} \subset (0,\infty)$ is compact, there are C, K_0 with

$$\Big|\log Z^z_{K+1}(s;b) - \log Z^z_K(s;b) - \frac{|s|}{\ell^3}\Delta_K[\ell,b]\Big| \le CL^{-K/10} \qquad (b \in \mathsf{J},\ K \ge K_0);$$

so for two classes z, z' for which s is mixed, $|\log(Z^z_K/Z^{z'}_K)(s;b) - \log(V_z/V_{z'})(s;b)| \le C'L^{-K/10}$, the limits lying in $(0,\infty)$. On boxes with sides in $2L^{\mathbb{Z}}$, in every class, the left member tends to 0. Uniformly in the label: there are numbers $b_\sharp, c_1, C_V, C_G, j_0$, formed from the constants, such that with $\varpi = 3 + 2\mathfrak{p}_0/r_\star$ and $c_2 = 20\varpi$, for $b \ge b_\sharp$, every s mixed for z with sides in $L^{\mathbb{N}} \cup 2L^{\mathbb{N}}$, all $\ge \ell_\flat(b)$, $S = \max_\mu s_\mu$, and every $K \ge c_2 \log_L S$ with $((\lceil K/1260\rceil - j_0)\log L + \log b)^{2\mathfrak{p}_0} \ge C_V|s|\log b$, the left member at $\ell = \ell_\flat(b)$ is at most $c_1 S^\varpi L^{-K/10}$; the last condition is void where $|s| \le (\log b)^{2\mathfrak{p}_0-1}/C_V$, so that for every $0 \le q' < 2\mathfrak{p}_0 - 3r_\star - 1$, all large b and $s_\mu = \frac12 \ell_\sharp(b) L^{i_\mu}$ with $L^{i_1+i_2+i_3} \le (\log b)^{q'}$ only $K \ge c_2 \log_L S$ is asked. And $0 \le \frac{|s|}{\ell_\flat^3}\log Z_{K,\ell_\flat} - \log Z^z_K(s;b) \le C_G|s|\log b$ for $b \ge b_\sharp$, these boxes and $K \ge j_0 + 2$.

(5) *(Small cubes.)* (a) There are an absolute constant $C_\square$ and $t_0 > 0$ depending on the constants with $r_K[n] \le C_\square t$ on every cube $\ell = 2L^m$ with $t = \ell/b \le t_0$, at every K, for every $n \ne 0$; so $-\log r_K[n] \ge \log(1/t) - \log C_\square$, and no floor is uniform as $t \to 0$. (b) As $t \to 0$, $v_z(t,t,t)/t \to \mathscr{G}_z$, the number eq. 11.iii, whose numerical evaluation gives 0.10868, 0.08251, 0.07574 for one, two, three twisted planes; so $C^{-1}t \le v_z(t,t,t) \le Ct$ for $t \le t_0$; and for $z, z' \ne 0$, with ϑ and θ^a of Prop. 11.14,

$$\lim_{t\to 0}\frac{v_z(t,t,t)}{v_{z'}(t,t,t)} = \exp\Big\{\frac12\int_0^\infty \sum_a \Big[\prod_\mu \vartheta(u,\theta^a_\mu(z)) - \prod_\mu \vartheta(u,\theta^a_\mu(z'))\Big]\frac{du}{u}\Big\},$$

which for two and three twisted planes against one is 0.75926 and 0.69695. (c) On the odd cubes $s = \frac12\ell_\sharp(b)$, $b \ge b_\sharp$, at every cutoff, $|\log r_K[n](s^{\times 3};b) - \log(t\,C^n(N)/N)| \le \omega_1(t)$, $N = s/a_K$, $t = s/b$, $C^n(N) = \lim_{\beta\to\infty}\beta\rho^n(N^{\times 3},\beta)$, with $\omega_1(t) \to 0$ as $t \to 0$. (d) There is $t_1 > 0$, formed from the constants, such that on every cube $2L^m$ with $t \le t_1$, at every cutoff, $r_1 \le \frac15$; there, by Thm. AG(13), $\delta^0_K \ge \log\frac52$, $\mathfrak{b}_K \ge 1$ on the cube and $\mathsf{e} \le 0.069$ fails at the half side; and $\liminf_{t\to0}\inf_K \delta^0_K \ge \log 4$, the four electric sectors being equally filled in the limit.

(6) *(The label.)* Let a reference coupling $g \le \gamma$, a side $\ell \in \mathfrak{L}(g)$ (equation YM3_I:(1528)) and a label $b_0 \in I(g)$ (Cor. YM3_I:74.2) be given, with the radius and the constants of Thm. YM3_I:AD. There are $R_0, R_0' > 0$ and $\mathsf{M}_\mathrm{S} < \infty$, formed from these data and the constants and not depending on K, such that for every K the periodic partition function of the cube ℓ has no zero at the complex labels $b_0 - c$, $|c| < R_0$, and $c \mapsto r_K[n](\ell; b_0 - c)$ is holomorphic in $|c| < R_0$ with $|r_K[n]| \le 27\mathsf{M}_\mathrm{S}$ on $|c| \le R_0/8$ for one or two twisted planes, and holomorphic in $|c| < R_0'$ with $|r_K[n]| \le 12600\mathsf{M}_\mathrm{S}$ on $|c| \le R_0'/15$ for three; so $\sup_K |\partial_b^\nu r_K[n](\ell;b_0)| \le 27\mathsf{M}_\mathrm{S}\,\nu!\,(8/R_0)^\nu$ for one or two planes, $\le 12600\mathsf{M}_\mathrm{S}\,\nu!\,(15/R_0')^\nu$ for three.

For every $t' > 0$ there are j', $\varrho > 0$ and M such that for $j \ge j'$ the function ϕ_j extends holomorphically to $\{|t - t'| < \varrho\}$ with $|\phi_j| \le \mathsf{M}$ there; hence $\phi_j \to v_n(t,t,t)$ with all derivatives, locally uniformly on a complex neighbourhood of $(0,\infty)$. There is an open $\Omega \subset \mathbb{C}$ containing $(0,\infty)$ such that for every compact $Q \subset \Omega$ there is j_Q with $Z[n] \ne 0$ on the cube of $2L^j$ sites at $\beta = 2L^j/t$ for $t \in Q$, $j \ge j_Q$. With $m_0 = \inf_K r_K[n](\ell;b_0) > 0$, $\int \exp[-L^K(b_0 - c)A_{\eta^n}]\,\mathrm{d}U \ne 0$ for all complex $|c| < R_1$ and all K, where $R_1 = \min\{R_0/16, m_0R_0/864\mathsf{M}_\mathrm{S}\}$ for one or two twisted planes and $R_1 = \min\{R_0'/30, m_0R_0'/756000\mathsf{M}_\mathrm{S}\}$ for three.

For labels in $[b_1,b_2] \subset I(g)$, with $R_* = R_0(b_1)$, $R_*' = R_0'(b_1)$, m the infimum of $r_K[n](\ell;b)$ over K and $b \in [b_1,b_2]$, $\mathcal{O} = a_K^{-1}A$ under μ_0 and $\mathcal{O}^n = a_K^{-1}A_{\eta^n}$ under μ_n: at every K, for one

or two twisted planes, the means differ by at most $V_1 = 216\mathsf{M}_\mathrm{S}/(R_*\mathsf{m})$ and the variances by at most $3456\mathsf{M}_\mathrm{S}/(R_*^2\mathsf{m})+V_1^2$; for three, by $V_1 = 189000\mathsf{M}_\mathrm{S}/(R'_*\mathsf{m})$ and $5670000\mathsf{M}_\mathrm{S}/(R'^2_*\mathsf{m})+V_1^2$.

(7) *(Block pairs.)* Extend the averaging map $\mathcal{M}$ to pairs (U,s), s closed, by $\mathcal{M}_s(U)(c) = \gamma[c]\,\mathcal{M}(\gamma U)(c)$, where γ is a sign field on the bonds with $d\gamma = s$ on the two blocks at the ends of the block bond c and $\gamma[c]$ is its product over the L bonds of the straight path of c; the block sign field is the product of s over the L^2 plaquettes of a block plaquette, and $s^{(d)}$ is its $(K-d)$-th iterate. Let $w \geq 0$, $w \not\equiv 0$, $\ell = 2L^m$ with $m \geq 1$, $K \geq 2$ and $0 \leq d \leq K$. Under $\sum_n w(n) \sum_{s\in[n]} e^{-\beta A_s} dU$ the law of the block pairs $(\mathcal{M}_s^{K-d}U, s^{(d)})$ of depth d is reflection positive at the mirrors midway between neighbouring layers of the block lattice of depth d normal to each direction in which $\hat{w} \geq 0$; for admissible w in all three directions; and so is every weak limit point as $K \to \infty$ of the laws of the towers $((\mathcal{M}_s^{K-d}U, s^{(d)}))_{1\leq d\leq K}$. At every cutoff the block sign field $s^{(d)}$ of a closed s is closed and $n(s^{(d)}) = n(s)$, its product over a plane of the block lattice being the product of s over the plane of the lattice that holds it: the class is a function of the block pairs. Let $b \geq \gamma^{-2}$ and $MR(b) \leq L^m$, let $d_\natural$ be the least $d \in \mathbb{N}$ with $L^d b \geq \mathsf{t}_\natural$, and for $d \geq d_\natural$ let

$$\mathsf{R}(d) = 2M^{-3}(\ell L^d)^3 \exp\bigl[-c' A_0^2 (\log L^d b)^{2\mathfrak{p}_0}\bigr] \exp\bigl[(C + \tfrac{9}{2}\log b)\,\ell^3\bigr] \leq C(\ell,b) L^{3d} e^{-cd^2},$$

where $\mathsf{t}_\natural$, c', A_0, C, c are formed from the constants and $C(\ell,b)$ from ℓ, b and the constants; $\mathsf{R}(d)$ does not depend on K. For integers $d \geq d_\natural$ and $K \geq d+1$ let $\hat{E}_{K,d}$ be the event of the pairs (U,s) that for some depth $d' \in [d,K]$ and some block plaquette P of depth d' the matrix $s_P^{(d')}\,\partial(\mathcal{M}_s^{K-d'}U)(P)$ is farther than $1/400L^2$ from $\mathbf{1}$: a union of covariant local events, one for each block plaquette, which at $s=1$ is the event that some plaquette of the block field $\mathcal{M}^{K-d'}U$ of some depth $d' \in [d,K]$ is farther than $1/400L^2$ from $\mathbf{1}$. Its periodic probability is at most $\mathsf{R}(d)$; off it, at each depth $d' \in [d,K]$, the signs of the traces of the block plaquettes of the block field $\mathcal{M}_s^{K-d'}U$ form the block sign field $s^{(d')}$, whose class is $n(s)$, so that off $\hat{E}_{K,d}$ the class is a function of each of these block fields alone; and $r_K[n]\,\mu_n(\hat{E}_{K,d}) \leq 16\mathsf{R}(d)^{1/4}$, while under the normalised mixture of weight w with $w(0) > 0$ the probability of $\hat{E}_{K,d}$ is at most $\frac{\sum_n w(n)}{w(0)} 16\mathsf{R}(d)^{1/4}$.

Other pairs (ℓ,b) are brought into this range by the dilation by a power L^j with $L^j b \geq \gamma^{-2}$ and $MR(L^j b) \leq L^{m+j}$; then the same holds at the depths $d \geq j + d_\natural(L^j b)$, a threshold that depends on ℓ and b. Under these conditions, in every weak limit point, with $r_n = \lim_K r_K[n]$ and $\mathbb{P}^n$ the limit law of the towers in the class n: the class is, up to null sets, a function of the tower of block fields, and the classes with $r_n > 0$ are carried by disjoint sets of towers; each class is locally dominated by the periodic limit law, that is, for $d \geq 1$ and every covariant Borel event B of the block pairs of depth d that depends only on a quarter of the torus cut by block mirrors normal to two directions, $r_n \mathbb{P}^n(B) \leq \mu_d(B_1)^{1/4}$, where $B_1 = \{V : (V,1) \in B\}$ and μ_d is the periodic limit law of the block field of depth d, and $r_n\mathbb{P}^n(B) \leq \mu_d(B_1)^{1/2}$ if B depends only on a half of the torus whose normal lies in every twisted plane of n; and in every class with $r_n > 0$ the towers are consistent under $\mathcal{M}_s$: with (V_d, s_d) the pair of depth d of a tower, almost surely $(V_d, s_d) = (\mathcal{M}_{s_{d+1}} V_{d+1},$ the block sign field of $s_{d+1})$ for all $d \geq 1$.

(8) *(Tension of a slab.)* For $\ell = 2L^j$ and $K+j \geq 2$, with r_1 on the cube ℓ and $\kappa_K = \log(2/(1-r_1))$,

$$\Bigl(1-\frac{1}{L}\Bigr)\ell\, E_{\mathrm{el},K}(\ell,\ell) \leq L^2 \delta_K^0((\ell/L)^{\times 3}; b) + \kappa_K, \qquad \sigma_K(\ell)\,\ell^2 \leq \Bigl(\frac{L}{L-1}\Bigr)^2 \Bigl[L^3\,\delta_K^0((\ell/L)^{\times 3}; b) + \kappa_K\Bigr],$$

$E_{\mathrm{el},K} = a_K^{-1}\varepsilon_{\mathrm{el}}$ over the section (ℓ,ℓ), and $(L/(L-1))^2 \leq 1.1737$ for $L \geq 13$. Both right members are bounded in K; so $\sup_K \sigma_K(\ell) < \infty$; by Thm. AG(13) the lower limits in K of $\delta_K^0(\ell^{\times 3}; b)$, of G over the section of the cube and of every $D_q(\ell/a_K)$ are positive; and in every limit the least level of the sector $e = 10$ of $H_{(\ell,\ell),0}$ is at most $\frac{L}{(L-1)\ell}[L^2 \sup_K \delta_K^0((\ell/L)^{\times 3}; b) +$

$\log\frac{2}{1-v_1(t,t,t)}]$, $t=\ell/b$. For $l\in\mathbb{D}$, $\sigma_K(l)\ge -(t\lambda)^{-1}\log\mathfrak{b}_K(t,\lambda;l)$ at every L-adic (t,λ); for $t>t_0$, $\lambda>\lambda_0$ in $\mathbb{D}$ and $s'=2L^i\le\min\{t_0,\lambda_0,l\}$,

$$-\log\mathfrak{b}_K(t,\lambda;l)\ge(t-t_0)\Big[(\lambda-\lambda_0)\sigma_K(l)-\frac{2\lambda_0 l}{s'^3}\bar\delta(s')\Big]-\frac{t_0\lambda l}{s'^3}\bar\delta(s'),\qquad \bar\delta(s')=\sup_K\delta^0_K(s'^{\times 3};b),$$

at all K at which the boxes and the cube are formed; so $\liminf_K\sigma_K(l)>0$ if and only if $\mathfrak{b}_K(t,\lambda;l)\le c<1$ at one L-adic (t,λ) for all large K.

(9) *(Small twisted section.)* There is $c_0>0$, depending on the constants, such that for $l\in L^{\mathbb{Z}}$ with $t=l/b\le c_0$, over the section (l,l) with the time 1 at every cutoff with $l/a_K\in L^{\mathbb{N}}$, the following hold. In this item $\lambda_0\ge\lambda_1\ge\lambda_2\ge\dots$ are all the eigenvalues of T_1, the four sectors together, each repeated according to its multiplicity, and the levels of T_1 are the numbers $a_K^{-1}\log(\lambda_0/\lambda_i)$, $i\ge0$; as $\lambda_0=\|T_1\|$, they are the levels $E_j^{K,1,e}$ of the setting, the four sectors together, each diminished by $a_K^{-1}\log(\|T_0\|/\|T_1\|)$, and by (3) they converge along the full sequence to the eigenvalues of $H_{(l,l),1}$, each diminished by $E_{\mathrm{mag}}(l,l)=\inf\operatorname{spec}H_{(l,l),1}$, which is the least eigenvalue of the sector 00. (a) The top eigenvalue λ_0 of T_1 is simple, and the gap $a_K^{-1}\log(\lambda_0/\lambda_1)$, λ_1 the next eigenvalue, lies between $1.2142/l$ and $\pi L/((L-1)l)\le 3.4034/l$; for $t\le c_0'$, $c_0'\le c_0$ formed from the constants, it lies between $3.0117/l$ and $3.19/l$ at every such cutoff, and at all large cutoffs every level of T_1 but the least, in all four sectors, is at least $1.2519/l$, with $\sum e^{-lE}\le e^{0.2515}-1<0.2860$ over these levels E. (b) With $E(e)$ the least eigenvalue of $H_{(l,l),1}$ in the sector e minus its least eigenvalue $E_{\mathrm{mag}}(l,l)$, $lE(e)\to\pi|e|$ as $t\to0$: $\pi,\pi,\sqrt{2}\pi$ for $e=10,01,11$; and l times the second eigenvalue of $H_{(l,l),1}$ in the sector 00, counted with multiplicity, minus $E_{\mathrm{mag}}(l,l)$ tends to 2π as $t\to0$: in the normalisation of this item, l times the limit in the cutoff of the least level of T_1 in the sector 00 above the vacuum tends to 2π. (c) Let $t\le c_0'$, and let the cutoffs be those with $l/a_K=L^J$, $J\ge2$.

A plaquette function is a bounded Borel function of the numbers $1-\frac12\eta(p)\operatorname{tr}\partial U(p)$ of finitely many plaquettes p, η a closed sign field of the class; it is a covariant function. Let $\omega_K(G)=\lim_{h\to\infty}\mu_{n_{23}}(G)$ on the boxes $(h,l/a_K,l/a_K)$, for plaquette functions G of finitely many layers: the state of the twisted cylinder $C_l=\mathbb{R}\times(\mathbb{R}/l\mathbb{Z})^2$ at the cutoff K, invariant under the translations of its lattice and under the reflection θ_K in the plane at the time $a_K/2$, midway between two layers. For $f\in\operatorname{Lip}_c(C_l)$ let $\mathcal{O}_K(f)=a_K^{-1}\sum_p f(x(p))(1-\frac12\eta(p)\operatorname{tr}\partial U(p))$, $x(p)$ the barycentre of p, $e_K(f)=a_K^{-1}\sum_p f(x(p))\bar a_{K,p}$, $\bar a_{K,p}$ the mean of $1-\frac12\operatorname{tr}\partial U(p')$ in the periodic measure of the cube $2l/L$ at the cutoff K for a plaquette p' of the orientation of p, $\mathcal{O}_K^-(f)=\mathcal{O}_K(f)-e_K(f)$, and $\|f\|_\sharp=\max\{\sup|f|,40M\operatorname{Lip}f\}$.

For $n'\ge1$ and $d,R>0$ there are $C_{n'}(d,R)$ and $K_{n'}$, formed from n', d, R, l, t and the constants, with $|\omega_K(\prod_{q\le n'}\mathcal{O}_K^-(f_q))|\le C_{n'}(d,R)\prod_q\|f_q\|_\sharp$ for $K\ge K_{n'}$ whenever the supports of $f_1,\dots,f_{n'}$ have mutual distances $\ge d$ and each lies in a ball of radius R. Every sequence of cutoffs tending to infinity has a subsequence along which the levels of T_1 converge, to $E_0\le E_1\le\dots$, the eigenvalues of $H_{(l,l),1}$ diminished by $E_{\mathrm{mag}}(l,l)$, and $S_{n'}(f_1,\dots,f_{n'})=\lim\omega_K(\prod_q\mathcal{O}_K^-(f_q))$ exists for all n' and all tuples of functions of class C^1 with compact and disjoint supports; each $S_{n'}$ is multilinear, symmetric, invariant under the translations of C_l and the reflection of the time, bounded as above, and reflection positive.

For every such limit, the Hilbert space $\mathcal{H}$ formed from the tuples supported at positive times, with the vacuum Ω, carries a contraction semigroup e^{-sH} with $H\ge0$ of pure point spectrum contained in $\{E_i\}$; $E_0=0$ and $3.0117/l\le E_1\le 3.19/l$; $\ker H=\mathbb{C}\Omega$ and $\operatorname{spec}H\subset\{0\}\cup[3.0/l,\infty)$; for tuples F,G supported at positive times and $s\ge0$,

$$|S(\theta F,\tau_s G)-S(F)S(G)|\le e^{-3.0\,s/l}\left(S(\theta F,F)-S(F)^2\right)^{1/2}\left(S(\theta G,G)-S(G)^2\right)^{1/2},$$

θ the reflection of the time and τ_s the translation by s; with the law of (4) used also on the boxes (L^3l,l,l), 3.0 may be replaced by 6.1 in the last two assertions, the action density

being blind to the electric sectors other than 00. Finally $\mathcal{H} \neq \mathbb{C}\Omega$: with $l = L^m$, let $j(t)$ be the least integer for which the reference coupling $g = (L^{j(t)}/2t)^{-1/2}$ is at most the ceiling γ_W of the reference couplings of Thm. YM3_I:AV and $2L^{j(t)-1}$ is a side at g (equation YM3_I:(1528)), an integer formed from t and the constants; there is a number $d_* > 0$, formed from t and the constants, such that the bump $f^\circ(y) = (1 - |y - (2d^\circ, \xi)|^2/d^{\circ 2})_+^2$ of the radius $d^\circ = L^{m-j(t)}d_*$, ξ a point of the section, has $S_2(\theta f^\circ, f^\circ) - S_1(f^\circ)^2 \geq \frac{1}{8}\,10^{-6}c_{\mathrm{sd}}b^{-2} > 0$, b the label and $c_{\mathrm{sd}} = 9/16\pi^2$; so d° is l times a number formed from t and the constants.

(10) *(Even cubes.)* Let $n \neq 0$, $q_K(b) = r_K[n](\ell_\sharp(b)^{\times 3}; b)$, $K_1(b) = \lfloor \frac{1}{68}\log_L b \rfloor$ and $C^n(N) = \lim_{\beta\to\infty} \beta\rho^n(N^{\times 3}, \beta)$. There is $t_{\mathrm{w}} > 0$ such that for $t \leq t_{\mathrm{w}}$ there is a least integer $\mu(t)$ with $b'(t) = 2L^{\mu(t)}/t \geq b_\sharp$ and $\ell_\sharp(b'(t)) = 2L^{\mu(t)}$; then $\log\frac{1}{t} \leq \log b'(t) \leq 2\log\frac{1}{t}$ and $\phi_j(t) = q_{j-\mu(t)}(b'(t))$. (a) $|\log\rho^n(N^{\times 3}, N/t) - \log(tC^n(N)/N)| \leq \omega(t) \to 0$ $(t \to 0)$ for $N \in 2L^{\mathbb{N}}$ with $N \geq N(t)$, and for $N \in 2L^{\mathbb{N}}$ with $N \leq \ell_\sharp(b'(t))\, b'(t)^{1/68}$. (b) To complete the ultraviolet comparison one needs the estimate denoted **(R)**: for some $\kappa > 0$, $|\log q_{K+1} - \log q_K| \leq c_1\ell_\sharp^{\varpi}L^{-\kappa K}$ for $b \geq b_\sharp$, $K \geq c_2\log_L \ell_\sharp$, as (4) gives on mixed boxes.

That estimate would imply $\sup_{b'\geq b}\sum_{K\geq K_1(b')}|\log q_{K+1} - \log q_K|(b') \to 0$ as $b \to \infty$; the bound of (a) holds for all $N \in 2L^{\mathbb{N}}$ at $t \leq t_0$, so $ct \leq r_K[n] \leq Ct$ on the small cubes $2L^m$ at every cutoff; and $\sum_j |\phi_{j+1} - \phi_j| < \infty$ on $(0, t_{\mathrm{w}}]$. For the lower bound $ct \leq r_K[n]$ and for the bounded variation the one-sided form $\log q_{K+1} \geq \log q_K - c_1\ell_\sharp^{\varpi}L^{-\kappa K}$ of (R) is enough. (c) The window estimates in §11.1.5 and their transport between cutoffs remain to be established. The transport estimate sought, denoted **(W)**, is: for one $\Lambda < \lambda_\flat$ and some $c, \eta, t_* > 0$, $\varsigma_K \geq e^{-\Lambda}\varsigma_{K_\sharp} - e^{-c(\log b)^{1+\eta}}$ for $K \geq K_\sharp$, $t \leq t_*$ (Lem. 11.11 and §11.1.5, with F_t or F_t').

Then $r_1(N, N/t) \geq c't$ for small t and all $N \in 2L^{\mathbb{N}}$. $\Lambda \leq 0.55L - 1.1$ is admissible for F_t, and $\Lambda = \frac{L}{2}$, a loss of $\frac{1}{2}$ for each cube of the box $(\ell, \ell, L\ell)$, for F_t', as $L \geq 7$. (W) may be replaced by the same inequality for F_t' on the box $(\ell, \ell, i\ell)$ at one $2 \leq i \leq L$, with L replaced by i in the definition of ς_K and $\Lambda < \log(0.42 \cdot 1.8712^i)$; or by **(W′)**: for some $\gamma' > 0$, $\|\mathsf{P}_K\| \geq \frac{1}{2}e^{\gamma'}\operatorname{Tr}\mathsf{P}_K$ for $K \geq K_\sharp$ and $t \leq t_*$, P_K the operator P of Thm. AG(12) for F_t on the cube ℓ at the cutoff K; under (W′), $r_1(N, N/t) \geq \frac{1}{2}\gamma' c_\flat t$ for small t and all $N \in 2L^{\mathbb{N}}$.

With the window F_t'' of §11.1.5 each of these gives the same for r_2. (d) On a cube $2L^m$ with $b \geq \gamma^{-2}$ and $MR(b) \leq L^m$ the weak limit points of the laws of (7) are, in every class n with $\lim_K r_K[n] > 0$, carried by towers consistent under $\mathcal{M}_s$, by (7). Full convergence of the block-pair laws still requires the summable total-variation estimate denoted **(M)**: $|\mu_{\mathrm{mix}}^{K+1}(H \circ \mathcal{M}_s^{K+1-d}) - \mu_{\mathrm{mix}}^{K}(H \circ \mathcal{M}_s^{K-d})| \leq o_K \sup|H|$, $\sum_K o_K < \infty$, for $\mu_{\mathrm{mix}} = \mathbb{M}_w/\mathbb{M}_w[1]$ at $w \equiv 1$ and all bounded Borel H of the block pairs of depth d with $H(\gamma V, s\,\mathrm{d}\gamma) = H(V, s)$. This estimate would give convergence in total variation of the block-pair laws under μ_n in every class.

■□

Theorem AI (Area laws uniformly in the cutoff, under one inequality)**.** The final small-side exclusion requires **(Hess)** and **(Flux-window)**; the fixed-cutoff confinement implications use only their stated box hypothesis. Let the constants be fixed as in Prop. YM3_I:69.1 and $b > 0$, and assume the hypothesis

(H) there are $s_0 > 0$ and K_0 such that for every $K \geq K_0$ the number $l_K = s_0/2a_K$ is an integer ≥ 3 and $\mathsf{e}(l_K) \leq 0.069$ at $\beta = L^K b$.

At each cutoff $K \geq K_0$, (H) is one inequality between the partition functions of two periodic boxes, $\mathsf{e}(l_K) = 2F_0(l_K, 2l_K, 2l_K) - F_0(2l_K, 2l_K, 2l_K) \leq 0.069$; the theorem is conditional on this hypothesis. Then for every $K \geq K_0$, at $\beta = L^K b$: Thm. AG(10) applies at $l_0 = 8l_K$, with $l_0 a_K = 4s_0$, $D_2(l_0) \leq 0.047355$, $m_* l_0 \geq 0.1597$, $4s_0 \leq l_1 a_K \leq 150.4\, s_0$ and $\sigma_* a_K^{-2} \geq 2.65\cdot 10^{-4} s_0^{-2}$; on every cube of physical

side $\ell \geq 602\, s_0$, $-\log[\frac{1}{2}(1-r_1)] \geq 2.65 \cdot 10^{-4} \ell^2/s_0^2 - (\ell/2s_0)\log(\ell/s_0)$; the periodic Gibbs state ν_K of the lattice $a_K \mathbb{Z}^3$ (Section YM3_I:91) is unique and $|\nu_K(W_{R\times T})| \leq \exp[-2.65 \cdot 10^{-4}(Ra_K)(Ta_K)/s_0^2]$ for every rectangle; on every cube of physical side $\ell \geq s_0$, $1 - r_K[n] \leq 0.4121 \cdot 0.8901^{\ell/s_0}$ for every $n \neq 0$; and $\sigma_K(\ell) \geq 1.9\, \ell^{-2}$ for every $\ell \in \mathbb{D}$ with $\ell \geq s_0$ and $\ell/a_K \in \mathbb{N}$. All these bounds are uniform in $K \geq K_0$. Under **(Hess)** and **(Flux-window)**, with t_1 of Thm. AH(5), (H) fails if $s_0 \leq t_1 b$: it can hold only at physical sides of the order of the label or more. ■□

11.1.4. *Construction estimates.* We use Thm. YM3_I:D, the inductive description of the effective densities, with its constants and their conditions (Prop. YM3_I:69.1 and Cor. YM3_I:69.4), and the lattices and tori T_ℓ (Section YM3_I:3); through Thm. YM3_I:D we use the small-field chain on T_ℓ with its small-field tests and its control of large fields (Prop. 4.54), the comparison of two consecutive cutoffs on T_ℓ (Section YM3_I:116), reflection positivity on tori, the statements on blocks and on the averaging map $\mathcal{M}$ (its covariance under lattice isometries and sign changes), and the convergence in total variation of the laws of the block fields on T_ℓ (Cor. 2.8).

We use the stability bound on periodic cubes (Lem. 11.1): for $b \geq \gamma^{-2}$ and $m_\iota \geq m_0(b)$, $\log Z_{K,\ell_\iota}$ at $\ell_\iota = 2L^{m_\iota}$ lies within a constant times ℓ_ι^3 of a vacuum energy E_ι, and $|E_1/\ell_1^3 - E_2/\ell_2^3|$ is bounded, no constant depending on K or m_ι. We use Thm. YM3_I:AD, the local source-energy and stability theorem, with the windows of labels (Cor. YM3_I:74.2), the sides (equation YM3_I:(1528)), the extents ([7, eq. 321]) and the bound on the number of localization domains ([7, Lem. O.1]); and, of the constants, that the blocking factor L is an odd integer with $L \geq 13$, and that the ordered choice in the proof of Prop. YM3_I:69.1, with the further floors of Section YM3_I:82, permits every sufficiently large odd L with the preceding constants fixed, so that both L and $L+2$ serve.

For Thm. AH(9)(c) we use besides, with the radius and the constants of Thms. YM3_I:AJ and YM3_I:AD, the bounds on the joint cumulants of the action density on periodic tori, uniform in the cutoff ([7, Lem. N.2]), and the short-distance law of its two-point function (Thm. YM3_I:AV), whose leading term is $c_{\mathrm{sd}} b^{-2}$ times the free kernel, $c_{\mathrm{sd}} = 9/16\pi^2$ (Thm. YM3_I:C and equation YM3_I:(2240)), with the witness depth (equation YM3_I:(2321)); and, for its last assertion, the normalisers at the first-passage data (Prop. YM3_I:91.12 (1)), the transfer principle for one law and for two laws under one normaliser (Prop. YM3_I:91.11), the first facts on states ([7, §K.1]), the device of sign averaging over a packing with its count (the packing and sign-averaging argument in the proof of Prop. YM3_I:127.1), and the first-passage data at the labels of the windows (Cor. YM3_I:74.2 and [7, Lem. G.4]).

For Prop. 11.15 we use besides the lemma on rearranged paths (Lem. YM3_I:4.1). On one lattice we use the exact statements among the construction results: the periodic lattice and its means, the crossing operator, slabs and slab operators, boards and the periodic chessboard estimate (Prop. YM3_I:86.3), the chessboard theorem on the infinite lattice ([7, Cor. B.1]), and the statements on periodic Gibbs states (Prop. YM3_I:91.2 and Prop. YM3_I:91.9). A Gibbs state at β on the infinite lattice $a\mathbb{Z}^3$ is, as in Section YM3_I:91, a Borel probability measure ν on its fields with $\nu(\gamma_B F) = \nu(F)$ for every finite set B of bonds and every continuous function F of finitely many bonds, γ_B the conditional kernel of B with the weight $\exp[-\beta \sum_p (1 - \frac{1}{2}\mathrm{tr}\, \partial U(p))]$, the sum over the plaquettes p that hold a bond of B (equation YM3_I:(1818)); it is periodic if it is invariant under the translations of a subgroup of finite index of $a\mathbb{Z}^3$ (Section YM3_I:91), and every periodic Gibbs state is invariant under all isometries of the lattice (Prop. YM3_I:103.4).

For a box N, μ_N is the measure $\langle\cdot\rangle = \mu_0$ of the periodic box N; where it stands beside a measure on the fields of the infinite lattice, it is extended periodically. Thm. AI rests, besides, on its hypothesis (H): at each cutoff $K \geq K_0$, the one inequality $\mathsf{e}(l_K) \leq 0.069$ between the partition functions of the two periodic boxes $(l_K, 2l_K, 2l_K)$ and $(2l_K, 2l_K, 2l_K)$.

11.1.5. *Estimates and proofs.*

Lemma 11.2 (Power sums; lists of levels)**.** Let $T \ge 0$ be compact, Q a projection commuting with T, $z(N) = \operatorname{Tr}(QT^N) < \infty$ for integers $N \ge N_*$, and $\lambda = \|QT\|$. Then $z(N)^2 \le z(N-j)z(N+j)$ for $N - j \ge N_*$; $z(N)^{1/N}$ does not increase and tends to λ; $z(N+N') \le \lambda^{N'} z(N) \le z(N)z(N')$; and, if $\lambda > 0$, $z(N)/\lambda^N \ge 1$ does not increase and is at most $(z(N)^q/z(qN))^{1/(q-1)}$ for every integer $q \ge 2$. (a) If (E_i), (E'_i) are non-decreasing sequences in $[0,\infty]$ with $f(t) = \sum_i e^{-tE_i}$ and $f'(t) = \sum_i e^{-tE'_i}$ finite for all $t > 0$, and $f(t_r) = f'(t_r)$ along one sequence $t_r \to \infty$, then $E_i = E'_i$ for all i. (b) If non-decreasing lists $(E_i^K)_i$ in $[0,\infty]$ have $\sum_i e^{-t_0 E_i^K} \le B(t_0)$ for all large K and every t_0 of a set with infimum 0, then $E_i^K \ge t_0^{-1} \log(i/B(t_0))$; every infinite set of indices K holds an infinite subset along which all E_i^K converge in $[0,\infty]$; and along it $\sum_i e^{-tE_i^K} \to \sum_i e^{-tE_i} < \infty$ for every $t > 0$, locally uniformly on $(0,\infty)$, the limit list having finitely many members in every bounded set. ∎

Proof sketch. If $QT = 0$, all the unnormalised power sums vanish. Otherwise $\lambda > 0$. With $z(N) = \sum_i \lambda_i^N$, λ_i the eigenvalues of QT, the first assertions are the Cauchy–Schwarz inequality, the monotonicity of ℓ^p norms in p, and $\sum_i \lambda_i^{N+N'} \le \lambda^{N'} \sum_i \lambda_i^N$, $\lambda^{N'} \le z(N')$; with $\mu_i = \lambda_i/\lambda$, $x(N) = \sum_i \mu_i^N \ge 1$ does not increase and $x(qN) \le x(N)$, so $x(N)^q/x(qN) \ge x(N)^{q-1}$, the powers of λ cancelling in $z(N)^q/z(qN)$. For (a), with $f_i(s) = \sum_{p \ge i} e^{-sE_p}$, $e^{-sE_i} \le f_i(s) \le e^{-(s-s_1)E_i} f_i(s_1)$ for $s \ge s_1$, so $E_i = -\lim_r t_r^{-1} \log f_i(t_r)$; by induction f_i and f'_i agree along (t_r), and so do E_i and E'_i. Without the finiteness of f at all $t > 0$ the conclusion fails: $E_i = 1$ and $E'_i = 2$ for all i. For (b), $i\, e^{-t_0 E_i^K} \le \sum_{p \le i} e^{-t_0 E_p^K} \le B(t_0)$; $[0,\infty]^{\mathbb{N}}$ is compact and metrisable; and for $t \ge t_1 > t_0$ the terms are bounded by $\min\{1, (B(t_0)/i)^{t_1/t_0}\}$, summable in i, so dominated convergence applies. □

Lemma 11.3 (The path inequality)**.** Let X be a continuous, strictly positive, symmetric kernel on a compact metric space with a Borel probability measure, u, v continuous and strictly positive, and $a \ge 1$ odd. Then $\langle u, X^a v\rangle \ge \langle u, Xv\rangle^a (\|u\|_2 \|v\|_2)^{1-a}$. In particular, if $t^{(h)}$ is the kernel of T_m^h, g a flip c^k of the fields of a section, $\Delta(h) = \int t^{(h)}(V, gV)\,\mathrm{d}V$ and $\mathsf{Z}(h) = \int t^{(h)}(V,V)\,\mathrm{d}V$, then for integers $n' \ge 1$, $m_1, m_2 \ge 1$, $m = m_1 + m_2$, and odd k',

$$t^{(k'n'+m)}(V, gW) \ge \frac{t^{(n'+m)}(V, gW)^{k'}}{[t^{(2m_1)}(V,V)\, t^{(2m_2)}(W,W)]^{(k'-1)/2}}, \qquad \Delta(k'n'+m) \ge \frac{\Delta(n'+m)^{k'}}{[\mathsf{Z}(2m_1)\mathsf{Z}(2m_2)]^{(k'-1)/2}}.$$

∎

Proof sketch. Let $\|u\|_2 = \|v\|_2 = 1$, $c = \langle u, Xv\rangle$, and $\nu(x,y) = u(x)X(x,y)v(y)/c$, a probability density with the marginals $\nu_1 = u\,Xv/c$ and $\nu_2 = v\,Xu/c$ (symmetry). On $(v_0, \dots, v_a)$ the chain $\nu(v_0, v_1) \prod_{i=1}^{a-1} q_i$, with $q_i = \nu(v_{i+1}, v_i)/\nu_2(v_i)$ for odd i and $\nu(v_i, v_{i+1})/\nu_1(v_i)$ for even i, has the i-th marginal ν_1 for even i and ν_2 for odd i; as a is odd, v_a carries an odd index. Writing $\langle u, X^a v\rangle$ as the mean of a quotient under this chain, Jensen's inequality and Gibbs' inequality give $\log\langle u, X^a v\rangle \ge a \log c + \frac{a-1}{2}[\mathrm{KL}(\nu_1|u^2) + \mathrm{KL}(\nu_2|v^2)] \ge a \log c$. For the second part, $X(V,W) = t^{(n')}(V, gW)$ is symmetric, since g commutes with T_m and $g^2 = 1$, and $X^a(V,W) = t^{(an')}(V, g^a W)$; with $u = t^{(m_1)}(\cdot, V)$, $v = t^{(m_2)}(\cdot, W)$ and odd a, $\langle u, X^a v\rangle = t^{(an'+m)}(V, gW)$ and $\|u\|_2^2 = t^{(2m_1)}(V,V)$; apply the first part at $a = k'$ and $a = 1$, put $W = V$, and integrate with Hölder's inequality (exponents k', $k'/(k'-1)$) and the Cauchy–Schwarz inequality. □

Lemma 11.4 (Doubling)**.** Let $\beta \ge 0$, and for a box N and a direction μ let $\mathsf{d}^{(\mu)}(N) = 2F_0(N) - F_0(N^{(\mu)})$, $N^{(\mu)}$ the box with N_μ doubled. Then $\mathsf{d}^{(\mu)} \ge 0$, and $\mathsf{d}^{(\mu)}(N^{(\mu)}) \le \psi(\mathsf{d}^{(\mu)}(N))$ whenever $\mathsf{d}^{(\mu)}(N) < \log 2$, with $\psi(x) = 2\log(1 + \varrho_x^2)$, $\varrho_x = (e^x - 1)/(1 + \sqrt{1 - (e^x - 1)^2})$. Further $D_2(N) \ge 4\mathsf{G}(N) > 0$ for $\beta > 0$, G over the section (N, N) at the height N, and $\mathsf{e}(l) \le D_2(l)$; if $\mathsf{e}(l) \le \frac{1}{20}$, then $D_2(2l) \le 17.72\,\mathsf{e}(l)^2$; if $\mathsf{e}(l) \le 0.069$, then $\mathsf{e}(2l) \le 12.9\,\mathsf{e}(l)^2$, $D_2(2l) \le 20\,\mathsf{e}(l)^2$, and $\delta(M, M, M) \le 0.01717\,\theta^{M/l}$ for $M = 2^j l$, $j \ge 1$, $\theta = 12.9\,\mathsf{e}(l)$. On every cube of side N and for every $n \ne 0$, $1 - r[n] \le 3(1 - e^{-\mathsf{G}(N)})$, G over the section of the cube. ∎

Proof sketch. With the time μ, $Z[0] = \|T_0\|^{N_\mu}(1+\mathsf{r})$, $\mathsf{r} = \sum_{i\ge1}(\lambda_i/\lambda_0)^{N_\mu}$, and the doubled side has $\mathsf{r}' \le \mathsf{r}^2$; so $e^{\mathsf{d}^{(\mu)}} = (1+\mathsf{r})^2/(1+\mathsf{r}') \ge 1 + 2\mathsf{r}/(1+\mathsf{r}^2)$, which gives $\mathsf{r} \le \varrho_x$, and the defect at the doubled side is at most $2\log(1+\mathsf{r}^2)$. By Thm. AG(2), $D_2(N)$ is a sum of three defects of doubling, each at least $\mathsf{d} \ge \mathsf{G}$ of its section, the first of them $4\mathsf{d}^{(1)}(N,N,N)$. By the symmetry of the axes $\mathsf{e}(l) = \mathsf{d}^{(1)}(l,2l,2l) = \mathsf{d}^{(2)}(2l,l,2l) = \mathsf{d}^{(3)}(2l,2l,l)$; doubling once in each direction, the bulk cancelling in sums of defects, gives $\mathsf{a} = \mathsf{d}^{(2)}(2l,2l,2l) \le \psi(\mathsf{e})$, $\mathsf{b} = \mathsf{d}^{(3)}(2l,4l,2l) \le \psi(2\mathsf{e}+\mathsf{a})$, $\mathsf{d}^{(1)}(l,4l,4l) \le 4\mathsf{e}+2\mathsf{a}+\mathsf{b}$, $\mathsf{e}(2l) \le \psi(4\mathsf{e}+2\mathsf{a}+\mathsf{b})$, and

$$D_2(2l) = 4\mathsf{a}+2\mathsf{b}+\mathsf{e}(2l), \qquad \delta(M,M,M) = \sum_{s\ge0} 8^{-s-1} D_2(2^s M);$$

with $\kappa(y_0) = \varrho_{y_0}/y_0$, as ϱ_y/y increases and $\log(1+u) \le u$, $\psi(y) \le 2\kappa(y_0)^2 y^2$ on $[0,y_0]$, $y_0 < \log 2$, which gives the constants of the lemma. On a cube, $\sum_k r[(k,0)_\mu] = 4Z_{00,0}/Z[0] \ge 4e^{-\mathsf{G}}$; with $x_i = 1-r_i$ this is $2x_1 + x_2 \le 4(1-e^{-\mathsf{G}})$, and the order of Thm. AG(1) with $Z_{11,0}, Z_{11,1} \ge 0$ gives $x_1 \le x_2 \le 2x_1$ and $x_3 \le 2x_2 - x_1$. □

Lemma 11.5 (The fourth power; loops)**.** Let $\beta > 0$. On a box (t,t',h) whose first two sides span the plane of the class n put $\zeta = \frac12(1-r[n])$, and over the section (t,t'), with the third direction as the time, $\mathsf{r}_h = \operatorname{Tr} T_0^h/\|T_0\|^h - 1$ and $\mathsf{d}(h;t,t') = 2F_0(t,t',h) - F_0(t,t',2h)$. For integers $t,t',h \ge 3$ and $u \ge t$, $u' \ge t'$:

$$\zeta(u,u';h) \le \big[\zeta(t,t';h)\, e^{\delta(t,t',h)}\big]^{uu'/tt'} e^{-\delta(u,u',h)}, \qquad 2\zeta \le h\varepsilon_{\mathrm v}(t,t') + \mathsf{r}_h, \qquad h\varepsilon_{\mathrm v}(t,t') \le \mathsf{d}(h;t,t') - \log(1-2\zeta(t,t';2h)),$$

and $\zeta(t,t';h) \le 1 - e^{-\mathsf{G}(t;t',h)} \le \delta(t,t',h)$. If $\bar z = [(h\varepsilon_{\mathrm v}(t,t') + \frac12 \mathsf{r}_{2h}) e^{\delta(t,t',2h)}]^4 < \frac12$, then

$$h\,\varepsilon_{\mathrm v}(2t,2t') \le \mathsf{d}(h;2t,2t') - \log(1-2\bar z).$$

If t' is even and C is a rectangle parallel to that plane with sides $R \le t-1$ along t and $T \le t'-1$ along t', then $|\langle W_C\rangle| \le (4\langle \mathsf{PP}'\rangle)^{T/t'}$ and $0 \le \langle \mathsf{PP}'\rangle \le (1-\zeta)^{1-\varrho}\zeta^{\varrho} + \zeta^{1-\varrho}(1-\zeta)^{\varrho} \le 2\zeta^{\min\{\varrho,1-\varrho\}}$, $\varrho = R/t$, P,P' the half traces of the two straight closed lines through the sides of length T; the second chain holds for all $t' \ge 3$. ■

Proof sketch. Along the third side $Z[0] = \|T_0\|^h(1+\mathsf{r}_h)$ and $Z[n] = \operatorname{Tr} T_1^h \ge \|T_1\|^h$, so $2\zeta \le 1 - e^{-h\varepsilon_{\mathrm v}}/(1+\mathsf{r}_h) \le h\varepsilon_{\mathrm v} + \mathsf{r}_h$. By Thm. AG(2), $q = \frac12(Z[0]-Z[n])$ is a power sum in each length of the plane, so $q(u,u') \le q(t,t')^{uu'/tt'}$ with the volumes taken out by δ: this is the first display, and doubling both short sides raises ζ to the fourth power. Back along the third side, $\|T_1\|^h \ge \operatorname{Tr} T_1^{2h}/\operatorname{Tr} T_1^h \ge (1-2\zeta(2h))Z[0](2h)/Z[0](h)$ and $\|T_0\|^h \le Z[0](h)$ give the third inequality; with the time along t, $1-\zeta \ge Z_{00,0}/Z[0] \ge e^{-\mathsf{G}}$. The fourth power follows by chaining these. For the loop, $Z\langle W_C\rangle = \operatorname{Tr}(T_0^{t'-T}\Gamma^* T_{\mathrm{or}}^T \Gamma)$ with the transfer operator T_{or} with two static charges, acting on functions with values in 2×2 matrices; Hölder's inequality for the double series over the two eigenbases gives the first bound, and $\langle \mathsf{PP}'\rangle$, with the time along t, is a sum of the weights $1-\zeta$ and ζ of the two flux sectors at the distances R and $t-R$. □

Lemma 11.6 (Lipschitz bounds in the sides)**.** Let $\beta > 0$, N a box, λ a direction with $N_\lambda \ge 5$, z a class, $N^\flat$ the box N with $\lceil N_\lambda/2\rceil$ in place of N_λ and $N^\natural$ the box N with $N_\lambda + 1 + \lfloor N_\lambda/2\rfloor$ in place of N_λ. Then

$$|\rho^z(N+e_\lambda,\beta) - \rho^z(N,\beta)| \le \frac{4}{N_\lambda} e^{\delta(N^\flat)}, \qquad |\delta(N+e_\lambda) - \delta(N)| \le \frac{3}{N_\lambda}\max\{\delta(N^\flat),\delta(N^\natural)\}.$$

■

Proof sketch. With λ as the time, $\rho^z = \sum_e (-1)^{e\cdot k}\mathcal{W}(e;m)/\sum_e \mathcal{W}(e;0)$, $\mathcal{W}(e;m) = \sum_i \vartheta_i^{N_\lambda}$ with the eigenvalues $\vartheta_i \in [0,1]$ of $P_e T_m/\|T_0\|$, and the denominator is ≥ 1. One more layer changes each

term by $\vartheta^N(1-\vartheta) \le \vartheta^{\lceil N/2\rceil}/(\lfloor N/2\rfloor+1)$, and $\sum \vartheta_i^{\lceil N/2\rceil}$ over all sectors is at most $e^{\mathsf{G}} \le e^{\delta(N^\flat)}$ for each m, by $\operatorname{Tr} T_1^M \le \operatorname{Tr} T_0^M$. For δ, $n \mapsto \log\sum_i a_i^n$ is convex and $\delta \ge 0$: the increment at N_λ lies between the mean increments over the heights $\lceil N_\lambda/2\rceil \dots N_\lambda$ and $N_\lambda+1 \dots N_\lambda+1+\lfloor N_\lambda/2\rfloor$, and $\lfloor N_\lambda/2\rfloor \ge N_\lambda/3$. □

Lemma 11.7 (Periodic Gibbs states)**.** Let $\beta > 0$. (a) For every periodic Gibbs state ν, every bounded Borel function F of the bonds in a cube of side q and every cube of side $l \in 2q\mathbb{N}$, $|\nu(F) - \mu_{l^{\times 3}}(F)| \le \|F\|_\infty (2\delta(l^{\times 3}))^{1/2}$. (b) Every weak limit point of the measures of the periodic cubes, extended periodically, is a periodic Gibbs state. (c) So if $\delta(l^{\times 3}) \to 0$ as $l \to \infty$, there is exactly one periodic Gibbs state and the measures of the periodic cubes converge to it; this is so if $D_2(l_0) \le \frac{1}{20}$ at one $l_0 \ge 3$ or $\mathsf{e}(l) \le 0.069$ at one $l \ge 3$, and then $|\nu(F) - \mu_{l^{\times 3}}(F)| \le 0.126\|F\|_\infty e^{-m_* l/2}$ for $l \in \{2^i l_0\} \cap 2q\mathbb{N}$ in the first case. ■

Proof sketch. (a) Let A be a real continuous function of the bonds in the cube of side q. For $n \in 2q\mathbb{N}$ let $\mathbb{A}_n$ be the sum of the mirror images of A over the $(n/q)^3$ cells of side q of the cube n, an array periodic under $2q$, and $F_t(n) = \log Z[0; e^{t\mathbb{A}_n}](n^{\times 3})$. By reflection positivity at the mirrors between the cells, F_t is a sum of powers in each side in $2q\mathbb{N}$, so $F_t(n)/n^3$ does not increase (Lem. 11.2) and tends to its infimum $\pi(t)$; $\pi(0) = P$. The convex form of the domination of periodic Gibbs states by the chessboard seminorms (Prop. YM3_I:91.9; its proof, the Gibbs equations of ν on a large box and the chessboard estimate on the periodic cube, holds for cells of every side) gives the tangent property $\pi(t) - \pi(0) \ge t\nu(A)/q^3$.

On the cube l, with $N_c = (l/q)^3$ cells, the N_c mirror images of A have one mean $\mu_{l^{\times 3}}(A)$ under the measure $\mu_{l^{\times 3}}$ of the periodic cube, and $|\mathbb{A}_l| \le N_c\|A\|_\infty$; Hoeffding's lemma gives $F_t(l) \le F_0(l) + tN_c\mu_{l^{\times 3}}(A) + \frac{1}{2}t^2N_c^2\|A\|_\infty^2$. As $l^3\pi(t) \le F_t(l)$ and $F_0(l) = l^3P + \delta(l^{\times 3})$, with $s = tN_c > 0$ this is $\nu(A) - \mu_{l^{\times 3}}(A) \le \delta/s + s\|A\|_\infty^2/2$, whose infimum over s is $\|A\|_\infty(2\delta)^{1/2}$; the same holds for $-A$, and bounded Borel functions follow by the regularity of the two measures. (b) is Prop. YM3_I:91.2. (c) Two periodic Gibbs states agree on F up to $2\|F\|_\infty(2\delta(l^{\times 3}))^{1/2}$ for all large l; with (b) the periodic cubes converge.

Under $D_2(l_0) \le \frac{1}{20}$, Lem. 11.4 and Thm. AG(2) give $\delta(N^{\times 3}) \le 0.14108\,(N/l)^3 D_2(l) \le 8 \cdot 0.14108 \cdot \frac{1}{18}e^{-m_* N/2}$ with $l \le N < 2l$ in $\{2^i l_0\}$; under $\mathsf{e}(l) \le 0.069$, $\delta(N^{\times 3}) \le 0.13736\,\theta^{N/2l}$. At $l \in \{2^i l_0\}$, $(2 \cdot 0.14108 \cdot \frac{1}{18}e^{-m_* l})^{1/2} \le 0.126\,e^{-m_* l/2}$. □

Lemma 11.8 (Chessboard bounds for every number of cells; the cylinder over a twisted section)**.** Let $\beta \ge 0$, $\lambda' \ge 3$ odd and $\mathsf{q} \ge 2$ even, and let Q be the periodic cube of side $\mathsf{q}\lambda'$, a board with q^3 cells of side λ', with the fold and the chessboard seminorm (equation YM3_I:(1749), Prop. YM3_I:86.3): for a bounded complex Borel function ϕ of one cell, $\operatorname{Fold}\phi$ is the product over the cells of Q of the mirror copies of ϕ, the copy in a cell being complex conjugated if the cell is odd relative to the given one, $Z_Q[0; \operatorname{Fold}\phi] \ge 0$, and $\|\phi\|_{\mathrm{ch};Q} = (Z_Q[0; \operatorname{Fold}\phi]/Z_Q[0])^{1/\mathsf{q}^3}$; for real ϕ these are the fold and the seminorm of the setting. A plaquette function of a cell is a bounded complex Borel function ϕ of the numbers $1 - \frac{1}{2}\operatorname{tr}\partial U(p)$ of the plaquettes p of the cell; in a class it is taken with $1 - \frac{1}{2}\eta(p)\operatorname{tr}\partial U(p)$ in the place of these numbers, and $\phi^{(u)}$ is its copy in the cell u.

(a) On every box with $k_\mu\lambda'$ sites in the direction μ, $k_\mu \ge \mathsf{q}$ integers, odd ones admitted, cut into cells of side λ', and in every class n,

$$\Big|Z\Big[n; \prod_u \phi_u^{(u)}\Big]\Big| \le \prod_u Z_Q[0; \operatorname{Fold}\phi_u]^{1/\mathsf{q}^3}; \qquad \text{in particular} \quad Z[n] \le Z_Q[0]^{|N|/|Q|}.$$

(b) Let $\beta > 0$ and $N = k\lambda'$ with an integer $k \ge \mathsf{q}$; let $\omega = \lim_{h\to\infty}\mu_{n_{23}}$ on the boxes (h, N, N), the state of the twisted cylinder $\mathbb{Z} \times (\mathbb{Z}/N)^2$ on the plaquette functions of finitely many layers, $\lambda_0 > \lambda_1 \ge \lambda_2 \ge \dots$ the eigenvalues of T_1 over the twisted section (N, N) with the time 1, and $\Xi = Z_Q[0]^{k^2/\mathsf{q}^3}\lambda_0^{-\lambda'}$. For plaquette functions ϕ_u of the cells $u \in \mathbb{Z} \times (\mathbb{Z}/k)^2$, all

but finitely many equal to 1,

$$\Big|\omega\Big(\prod_u \phi_u^{(u)}\Big)\Big| \le \Xi^{J'} \prod_u \|\phi_u\|_{\mathrm{ch};Q},$$

J' the number of the slabs of cells in time that carry a function other than 1. Moreover $1 \le \Xi \le \exp[\Phi'/(k_0 - \mathsf{q})]$ for every integer $k_0 > \mathsf{q}$, where $\Phi' = \frac{k_0 k^2}{\mathsf{q}^3} \log Z_Q[0] - \log Z_{(k_0\lambda',N,N)}[n_{23}]$, and $\sum_i (\lambda_i/\lambda_0)^{j\lambda'} \le \Xi^j$ for every integer $j \ge \mathsf{q}$. With $N' = \mathsf{q}\lambda'$ the side of Q, $k^2/\mathsf{q}^3 = N^2\lambda'/N'^3$, so that $\log \Xi = \lambda'[(N^2/N'^3) \log Z_Q[0] - \log \lambda_0]$: at a fixed partner Q and a fixed section (N, N), $\log \Xi$ is linear in the number λ' of the sites of a side of a cell.

■

Proof sketch. (a) Without a twist, cut the box in a direction μ into its $k_\mu \ge \mathsf{q}$ slabs of cells. By the trace formula the integral is the trace of the product of the slab operators; Hölder's inequality for traces with q factors, the operator norm being at most the norm of order q, bounds its modulus by the product of the norms of order q of the slab operators. The adjoint of a slab operator Y is the operator of the mirror copy of the slab carrying the complex conjugate function; so the q-th power of the norm of order q, $\mathrm{Tr}\,(Y^*Y)^{\mathsf{q}/2}$, is the integral, over the box with q slabs in the direction μ, of the mirror copies of the function of the slab, complex conjugated at every other place.

The three directions in turn give the bound, the copies composing to the folds on Q, each of which is thus an integral ≥ 0. In a class n place the columns inside cells; on a reversed plaquette $e^{-\beta(1+\frac{1}{2}\mathrm{tr}\,u)} = e^{-\beta(1-\frac{1}{2}\mathrm{tr}\,u)} e^{-\beta\,\mathrm{tr}\,u}$, so the twist is a family of functions of cells and the periodic bound applies. After the fold the reversed plaquettes of the q^3 copies form, q being even, the boundary $d\gamma$ of a sign field γ on the bonds of Q, and $U \mapsto \gamma U$ removes them: the folds are those of the periodic class. (b) By the trace formula with the time 1, $\omega(\prod_u \phi_u^{(u)})$ is λ_0^{-w} times the mean, in the top eigenvector of T_1, of the product of the operators of the slabs of cells, w the number of their layers; a slab with the function 1 has the operator $T_1^{\lambda'}$, of norm $\lambda_0^{\lambda'}$.

The norm of the operator of an occupied slab is at most its norm of order q, whose q-th power is, as in (a), a twisted integral over the box $(\mathsf{q}\lambda', N, N)$ with the mirror copies of the functions of the slab, complex conjugated at every other place in time, bounded by (a), the fold of a mirror copy being the fold of the function itself (Prop. YM3_I:86.4). At $\phi \equiv 1$ this is $\lambda_0^{\lambda'} \le Z_Q[0]^{k^2/\mathsf{q}^3}$, that is $\Xi \ge 1$. By (a), $\mathrm{Tr}\,T_1^{j\lambda'} = Z_{(j\lambda',N,N)}[n_{23}] \le Z_Q[0]^{jk^2/\mathsf{q}^3}$ for $j \ge \mathsf{q}$, which is the last assertion, and with $\mathrm{Tr}\,T_1^{k_0\lambda'} \le \lambda_0^{(k_0-\mathsf{q})\lambda'} \mathrm{Tr}\,T_1^{\mathsf{q}\lambda'}$ it gives $\Xi^{k_0-\mathsf{q}} \le e^{\Phi'}$.

The linearity of $\log \Xi$ in λ' is the identity $k^2/\mathsf{q}^3 = N^2\lambda'/N'^3$, λ_0 and $Z_Q[0]$ reading N, N' and β alone. □

Lemma 11.9 (A box with flux as a sector of a periodic torus)**.** Let $\ell = 2L^m$ with $m \ge 1$, $K \ge 2$, $s_\mu \in \{L^j, 2L^j : 0 \le j \le m\}$, $n_\mu = \ell/s_\mu$, and $w_{\mu\nu} \in \{0, 1\}$ for $\mu \ne \nu$ with $w_{\mu\nu} = 1$ only if n_μ is even; let $z_{\mu\nu} = w_{\mu\nu} + w_{\nu\mu} \bmod 2$, $\mathfrak{T} = \{\sum_\mu k_\mu s_\mu e_\mu\} \subset \mathbb{Z}^3/\ell\mathbb{Z}^3$ and $n = |\mathfrak{T}| = \ell^3/|s|$. On the lattice $L^{-j}\mathbb{Z}^3$, $j \ge 0$, which covers the lattice of T_ℓ of spacing L^{-j}, let $n_\nu(x) = \lfloor [x]_\nu/s_\nu \rfloor$, $[x]$ the nearest point of $\mathbb{Z}^3$, $\varepsilon_\nu(x) = (-1)^{n_\nu(x)}$ and $\varsigma_\nu(\langle x, x'\rangle) = \varepsilon_\nu(x)\varepsilon_\nu(x')$; ς_ν is invariant under the shifts by $s_\mu e_\mu$, $\mu = 1, 2, 3$, and so it is a sign field on the bonds of the lattice of T_ℓ; and for $v = \sum_\mu k_\mu s_\mu e_\mu$

$$(\gamma_v V)(b) = \chi_v(b)\, V(b - v), \qquad \chi_v = \prod_\mu \prod_{\nu \ne \mu} \varsigma_\nu^{w_{\mu\nu} k_\mu}.$$

A twist triple of the class z is $\Omega \in Q_8^3$, $Q_8 = \{\pm\mathbf{1}, \pm i\sigma_1, \pm i\sigma_2, \pm i\sigma_3\}$, with $\Omega_\mu\Omega_\nu = (-1)^{z_{\mu\nu}}\Omega_\nu\Omega_\mu$ for $\mu \ne \nu$ (for the class whose one twisted plane is 12, $\Omega = (i\sigma_1, i\sigma_2, \mathbf{1})$); as every element of Q_8 is its own inverse up to the sign, the conjugations by $\Omega_1, \Omega_2, \Omega_3$ commute. The twisted fields of a box N of the class z are the fields $\tilde{U}$ on the bonds of $\mathbb{Z}^3$ with $\tilde{U}(b + N_\mu e_\mu) = \Omega_\mu \tilde{U}(b) \Omega_\mu^{-1}$, $\mu = 1, 2, 3$, with the Haar measure $d\tilde{U}$ at the bonds that begin in one period box and the action $A'(\tilde{U})$, the sum of $1 - \frac{1}{2}\mathrm{tr}\,\partial\tilde{U}(p)$ over the $3|N|$ plaquettes whose least corner lies in that box; neither depends

on the period box. Then $\int e^{-\beta A'(\tilde U)} d\tilde U = Z[z]$ on the box N for every $\beta \ge 0$; $v \mapsto \gamma_v$ is an action of $\mathfrak{T}$ on configurations that keeps the Haar measure, and the underlying translations act freely on sites, bonds and plaquettes; every operation of the small-field chain on T_ℓ (averaging, gauge fixing, the minimisers about which it expands and the backgrounds, the charts and their Jacobians, the small-field tests and the cuts) commutes with each γ_v; and, with $dU^{\mathfrak{T}}$ the Haar measure at representatives of the bonds modulo $\mathfrak{T}$,

$$\int_{\{U:\ \gamma_v U = U\ \forall v\}} e^{-\beta A(U)/n}\, \mathrm{d}U^{\mathfrak{T}} = Z^z_K(s;b) \qquad \text{on } \Lambda_{K,\ell}.$$

Every box that is mixed for z is of this form, for some m and w, after a dilation by a power of L. The construction holds on every periodic torus with multiplicities n_μ: for a box s of the lattice $\mathbb{Z}^3$, integers $n_\mu \ge 1$ and $w_{\mu\nu} \in \{0,1\}$ with $w_{\mu\nu} = 1$ only where n_μ is even, and z formed from w as above, the same ς_ν, χ_v and γ_v, formed on $\mathbb{Z}^3$ with $n_\nu(x) = \lfloor x_\nu / s_\nu \rfloor$, give on the periodic torus of the sides $n_\mu s_\mu$ an action on configurations of $\mathfrak{T} = \{\sum_\mu k_\mu s_\mu e_\mu\}$, of order $n = \prod_\mu n_\mu$, that keeps the Haar measure, with free underlying translations on sites, bonds and plaquettes, and $\int_{\{U:\ \gamma_v U = U\ \forall v\}} e^{-\beta A(U)/n}\, \mathrm{d}U^{\mathfrak{T}} = Z[z]$ on the box s for every $\beta \ge 0$. ■

Proof sketch. $\varsigma_\nu = -1$ exactly on the bonds that cross the walls $x_\nu \in s_\nu \mathbb{Z} - \frac{1}{2}$; as L is odd, ε_ν is constant on the blocks of every order and the walls lie between blocks; $n_\nu(x + s_\mu e_\mu) = n_\nu(x) + \delta_{\mu\nu}$, so the two ends of a bond change sign together under these shifts. k_μ is given modulo n_μ, and χ_v reads it modulo 2 only where $w_{\mu\nu} = 1$, that is, where n_μ is even; so χ_v is well defined, $\chi_{v+v'}(b) = \chi_v(b)\chi_{v'}(b-v)$ makes $v \mapsto \gamma_v$ an action, and $\mathfrak{T}$ acts freely on the sites, the bonds and the plaquettes, its least displacement being the least side $s_\mu < \ell$. So γ_v is the translation by v followed by the multiplication by χ_v, a closed central sign field which on the lifts is $\varepsilon_v(x)\varepsilon_v(x')$, $\varepsilon_v = \prod_\mu \prod_{\nu \ne \mu} \varepsilon_\nu^{w_{\mu\nu} k_\mu} = \pm 1$ constant on the blocks of every order: near every block it is a gauge transformation by $\pm\mathbf{1}$.

Every operation of the chain commutes with the translations (its covariance under the lattice isometries), and with the multiplication by $\chi = \chi_v$ for the following reasons. (1) The contours that define $V(y,x)$, x in the block of y, lie inside that block, where $\chi = 1$: $V(y,x)$ is kept, and with it the gauge factors and the axial gauge, which read the family $(V(y,x))$ alone; the straight path $[x,x']$ over a block bond c takes the sign $\chi(c)$, which stands twice in each W_x and each $Y_{x,\Gamma,\Gamma'}$ of the definition of $\mathcal{M}$; so the same case of that definition applies at c, and $\mathcal{M}(\chi V) = \chi\, \mathcal{M}(V)$. (2) χ keeps every plaquette variable and, near every bond, is a gauge transformation by $\pm\mathbf{1}$, which keeps the current; with (1) iterated, the critical regular configurations over V go onto those over χV: the minimisers are covariant, and the cuts and the inner balls, which read the plaquette variables of V and of its minimiser, are invariant. (3) The backgrounds are covariant, $V^{\mathrm{bf}}(\chi W) = \chi V^{\mathrm{bf}}(W)$, defined and stably defined together. (4) In the argument $V(b)V^{\mathrm{bf}}(\mathcal{M}V)(b)^{-1}$ of a small-field test the sign $\chi(b)^2 = 1$ cancels: the tests are invariant; and the charts are covariant, $V_{\gamma_v W}(\gamma_v B) = \gamma_v V_W(B)$, γ_v acting on the fluctuation fields B by the translation alone. (5) The Jacobians of the charts are invariant, $\varrho_{c+v}(\gamma_v V) = \varrho_c(V)$.

Hence every density of the chain is invariant under $\mathfrak{T}$, and the chain of T_ℓ restricted to the fixed fields is a chain for them.

For the class, put $U(b) = \tilde U(b)\Omega_\mu$ at the bonds b of the direction μ that begin in the period box $\prod_\mu [0, N_\mu)$ and end outside it, and $U(b) = \tilde U(b)$ at the other bonds that begin in it: a bijection of the twisted fields onto the periodic fields of the box N that keeps the Haar measure. In a plaquette the factors $\Omega_\mu^{\pm 1}$ cancel, except in the plaquettes of a plane $\mu\nu$ whose least corner has $x_\mu = N_\mu - 1$ and $x_\nu = N_\nu - 1$, where they leave the commutator $\Omega_\nu^{-1}\Omega_\mu^{-1}\Omega_\nu\Omega_\mu = (-1)^{z_{\mu\nu}}$; so $\operatorname{tr} \partial\tilde U(p) = \eta^z(p) \operatorname{tr} \partial U(p)$, η^z being -1 exactly on one column of each twisted plane, and $\int e^{-\beta A'} d\tilde U = Z[z]$. With a twist

triple Ω of z and $g(x) = (-1)^{\sum_{\nu<\mu} w_{\mu\nu} n_\mu(x) n_\nu(x)} \Omega_1^{n_1(x)} \Omega_2^{n_2(x)} \Omega_3^{n_3(x)}$ on the lifts,

$$g(x + s_\mu e_\mu) = \textstyle\prod_{\nu\neq\mu} \varepsilon_\nu(x)^{w_{\mu\nu}}\, g(x)\, \Omega_\mu;$$

so the gauge transformation by g^{-1}, constant on all blocks, maps the fixed fields bijectively onto the fields with $\tilde V(b + s_\mu e_\mu) = \Omega_\mu \tilde V(b) \Omega_\mu^{-1}$, the twisted fields of the class z of the box s measured in the spacing of the lattice, and $dU^{\mathfrak T}$ onto $d\tilde U$, the action of a fixed field being n times that of its image. On $\Lambda_{K,\ell}$, where that box is s/a_K and $\beta = L^K b$, the identity $\int e^{-\beta A'} d\tilde U = Z[z]$ gives the display; and the flat field $\tilde V \equiv \mathbf 1$ gives a flat fixed field. n_μ is even exactly when $s_\mu \in L^{\mathbb Z}$, and $z_{\mu\nu} = 1$ needs $w_{\mu\nu} = 1$ or $w_{\nu\mu} = 1$, that is, a side of the plane in $L^{\mathbb Z}$.

On a periodic torus with the sides $n_\mu s_\mu$ the same computations, on the lattice $\mathbb Z^3$ with $n_\nu(x) = \lfloor x_\nu / s_\nu \rfloor$, read the box s and the multiplicities alone: $n_\nu(x + s_\mu e_\mu) = n_\nu(x) + \delta_{\mu\nu}$, so that ς_ν is a sign field on that torus; χ_v is well defined, as $w_{\mu\nu} = 1$ only where n_μ is even, and $\mathfrak T$ acts freely on sites, bonds and plaquettes; the identity for g gives the bijection of the fixed fields onto the twisted fields of the class z of the box s, the action of a fixed field being n times that of its image; and $\int e^{-\beta A'} d\tilde U = Z[z]$ gives the identity of the integrals. The statements on the blocks and on the chain are those of T_ℓ. □

Proposition 11.10 (The root of an invariant integral)**.** Let $\mathfrak Y$ be a finite set with a pseudo-distance d, $\mathcal A = (\mathbb R^3)^{\mathfrak Y}$, and $\mathfrak T$ a group of order n acting freely on $\mathfrak Y$, keeping d, and on $\mathcal A$ by $(\gamma x)_{\gamma y} = \varrho(\gamma) x_y$, $\varrho(\gamma) \in \mathrm O(3)$; $\sigma = \min\{\mathrm d(y, \gamma y) : \gamma \neq 1\}$ when $n > 1$. For $n = 1$ the two integrals below agree exactly; assume $n > 1$ for the estimate. Let $c_- > 0$, $c_+ > 0$, $\mu > 0$, $g_0 \geq 0$, $r > 0$, $\rho = \frac 3 2 r$, $D_a = \{|x_y| < a \ \forall y\}$, and $H \in C^\infty(D_{2r})$ real and $\mathfrak T$-invariant with $\nabla^2 H \geq c_-$ on D_{2r}, $\sup_y \sum_{y'} \sup_{D_{2r}} |\nabla_y \nabla_{y'} H|\, e^{\mu \mathrm d(y,y')} \leq c_+$, and $|\nabla_y H(0)| \leq g_0$. With $\alpha = \mu c_- / 2c_+$ and $S_\nu = \sup_y \sum_{y'} e^{-\nu \mathrm d(y,y')}$,

$$\Big|\log \int_{D_r \cap \mathcal A^{\mathfrak T}} e^{-H/n} d^{\mathfrak T} x - \frac 1 n \log \int_{D_r} e^{-H} dx\Big| \leq \frac{\#\mathfrak Y}{n} \Big[\frac{4\rho c_+}{c_-} (g_0 + \tfrac 5 2 \rho c_+) S_{\alpha/2} e^{-\alpha\sigma/4} + \rho^2 c_+ e^{-\mu\sigma/2}\Big],$$

$\mathcal A^{\mathfrak T}$ the fixed points and $d^{\mathfrak T} x$ Lebesgue measure at representatives of the orbits. ■

Proof sketch. Interpolate $H_\tau = \frac{c_-}{2}|x|^2 + \tau(H - \frac{c_-}{2}|x|^2)$; at $\tau = 0$ both integrals factor over the sites and agree. The τ-derivatives of the two free energies are sums of means of the same local pieces $\hat u_y(x) = x_y \cdot \int_0^1 \nabla_y H(sx)\, \mathrm d s - \frac{c_-}{2}|x_y|^2$, under the full measure and under the product of n copies of the quotient measure. Fix y_0 and representatives containing the ball of radius $\sigma/2$ about y_0; the gradients of the two Hamiltonians differ only through interactions that leave this ball, and in the norm weighted by $\mathsf W = \mathrm{diag}(e^{-\alpha \mathrm d(y_0, y)})$, in which the Hessian stays $\geq \frac 1 2 c_-$ by Schur's test,

$$\sup |\mathsf W \nabla (h_1 - h_2)| \leq 2\rho c_+ S_{\alpha/2} e^{-\alpha\sigma/4}.$$

Two Langevin dynamics for h_1 and h_2 driven by one Brownian path stay within $2/c_-$ times this in that norm, which bounds the difference of the means of $\hat u_{y_0}$; the walls of D_r are softened by a convex penalty and removed at the end, and the choice of representatives costs the second term. Integrate over τ and sum over the $\#\mathfrak Y / n$ representatives. □

Lemma 11.11 (Block functions at two cutoffs)**.** (a) For $b \geq \gamma^{-2}$, a cube $\ell = 2L^m$ with $MR(b) \leq L^m$, $K \geq 2$ at which the conditions (k0)–(k4) and (k6) of Cor. 2.8(a) hold (§2.3.1 at $n = 1$, whose two depths are $u = \lceil K/1260 \rceil$ and $u' = \lceil K/9 \rceil$, the latter written s there), $0 \leq d \leq u$ and every gauge invariant Borel function F with values in $[0,1]$ of the block field of depth d: $\mu_d^{K+1}(F) \leq e^{m_K} \mu_d^K(F) + \epsilon_K$ and $\mu_d^K(F) \leq e^{m_K} \mu_d^{K+1}(F) + \epsilon_K$, where, with the numbers ϵ_{fg}, ϵ_{cl}, ϵ_{sh} of YM3_I: (2192)–(2194) and $\mathsf R_{t_1}$ of §2.3.1,

$$m_K = 4\epsilon_{\mathrm{fg}} + 2\epsilon_{\mathrm{cl}} + 2\epsilon_{\mathrm{sh}} + 2\mathsf R_{t_1}, \qquad \epsilon_K = \tfrac 4 3 \epsilon_{\mathrm{sh}} + \mathsf R_{t_1}.$$

These are numbers of K, ℓ, b and the constants, free of F and d: $\epsilon_{\rm fg}$ is the error of forgetting the start of the small-field flow, $\epsilon_{\rm cl}$ that of the classical start, $\epsilon_{\rm sh}$ the relative mass where the small-field tests of the two cutoffs differ, and R_{t_1} the relative mass of the large fields of the periodic cube at the depth u, of the form of $\mathsf{R}(u)$ of Thm. AH(7). Up to factors formed from the constants, $\epsilon_{\rm fg}$ is $\ell^3 L^{4u-u'}$, $\epsilon_{\rm cl}$ is $(\log L^K b)^{2\mathfrak{p}_0}\ell^3 K^{1/2}L^{3u'-(K-u')/2}$, $\epsilon_{\rm sh}$ is $(u'-u)\ell^3 L^{3u'}e^{-c_{\rm sh}(\log L^u b)^{2\mathfrak{p}_0}}$ and R_{t_1} is $\ell^3 L^{3u}e^{-c'A_0^2(\log L^u b)^{2\mathfrak{p}_0}}e^{(C+\frac{9}{2}\log b)\ell^3}$, with numbers $c_{\rm sh}, c', A_0, C>0$ formed from the constants. (b) There are $A, c>0$ and $b_0 \ge \gamma^{-2}$, formed from the constants, such that for $b\ge b_0$, $\ell = 2L^m$ with $MR(b)\le L^m$ and $\ell \le 2ML^2(\log b)^{r_\star}$, $K_\sharp = \lceil A\log_L \log b\rceil$, every $K\ge K_\sharp$ and every such F of the block field of depth one,

$$e^{-E}\mu_1^{K_\sharp}(F) - \epsilon \le \mu_1^K(F) \le e^E(\mu_1^{K_\sharp}(F)+\epsilon), \qquad E\le 1/\log b, \quad \epsilon \le e^{-c(\log b)^{2\mathfrak{p}_0}}.$$

As $2\mathfrak{p}_0 > 3r_\star+2>1$ and $t=\ell/b \ge 2MR(b)/b \ge 2M(\log b)^{r_\star}/b$, for every $k>0$ there is b_k, formed from k and the constants, with $\epsilon \le t^k$ for $b\ge b_k$: on these cubes $\epsilon = o(t^k)$ as $b\to\infty$, for every k. ■

Proof sketch. (a) The comparison of two consecutive cutoffs (Section YM3_I:116) restarts the small-field flows of the two lattices from the classical action at the depth u' and compares them at the depth u. For F as in (a), $\mu_d^K(F)$ is the periodic mean of $F(\mathcal{M}^{K-d}U)$, and $F\circ\mathcal{M}^{u-d}$ is a gauge invariant function of the block field of depth u. Split the periodic measure into the small-field measure of the chain at the depth u, carried by the inner ball, and the rest, whose relative mass is at most R_{t_1} by the control of large fields (Prop. 4.54); the gauge factors integrate to at most 1 against gauge invariant functions. The small-field masses at the cutoffs K and $K+1$ are compared pointwise at the fields of depth u': each density differs from that of its classical run by a factor $e^{\pm\epsilon_{\rm fg}}$, the flow forgetting its start; the two classical starts differ by a factor $e^{\pm\epsilon_{\rm cl}}$; and the two systems of small-field tests differ on a set whose relative mass is at most $\epsilon_{\rm sh}$. Putting the factors and the masses together for F and for $F=1$, which compares the two normalisations, gives (a) with the stated m_K and ϵ_K.

(b) Let $\ell \le 2ML^2(\log b)^{r_\star}$, so that $\ell^3 \le C(\log b)^{3r_\star}$. As $4u-u' \le -\frac{K}{10}+4$ and $3u'-\frac12(K-u') \le -\frac{K}{9}+4$, $\sum_{K\ge K_\sharp}(\epsilon_{\rm fg}+\epsilon_{\rm cl}) \le C(\log b)^{2\mathfrak{p}_0+3r_\star}K_\sharp^{2\mathfrak{p}_0+1}L^{-K_\sharp/10}$, which is at most $1/(8\log b)$ for $K_\sharp = \lceil A\log_L\log b\rceil$ with A large and b large. In $\epsilon_{\rm sh}$ and R_{t_1}, $(u\log L+\log b)^{2\mathfrak{p}_0} \ge (\log b)^{2\mathfrak{p}_0} + 2\mathfrak{p}_0(\log b)^{2\mathfrak{p}_0-1}u\log L$, as $2\mathfrak{p}_0\ge 1$; with $u\ge K/1260$ the second term gives a factor L^{-K} for large b, which beats $L^{3u'}\le L^{K/3+3}$ and makes the sum over K converge, and the first leaves $e^{-c(\log b)^{2\mathfrak{p}_0}}$.

The volume factor of R_{t_1} is $e^{(C+\frac92\log b)\ell^3} \le e^{C(\log b)^{3r_\star+1}}$, beaten by $e^{-\frac12 c'A_0^2(\log b)^{2\mathfrak{p}_0}}$ because $2\mathfrak{p}_0 > 3r_\star+1$: this is why ℓ is bounded above. So $\sum_{K\ge K_\sharp}(\epsilon_{\rm sh}+\mathsf{R}_{t_1}) \le e^{-2c(\log b)^{2\mathfrak{p}_0}}$ for large b, $E := \sum_{K\ge K_\sharp} m_K \le 1/\log b$ and $\epsilon := \sum_{K\ge K_\sharp}\epsilon_K \le e^{-c(\log b)^{2\mathfrak{p}_0}}$. The same bounds give the conditions (k0)–(k4) and (k6) at every $K\ge K_\sharp$ for large b ((k0) asks $K\ge 10$, (k1) $b\ge\Theta_\star$, and (k4) a power $\ell^3L^{-5K/9+4}$ that is small). Now iterate (a) at $d=1$: $\mu_1^K \le e^{\sum_j m_j}\mu_1^{K_\sharp} + \sum_j \epsilon_j e^{\sum_{i>j}m_i} \le e^E(\mu_1^{K_\sharp}+\epsilon)$, and $\mu_1^{j+1} \ge e^{-m_j}(\mu_1^j - \epsilon_j)$ gives $\mu_1^K \ge e^{-E}\mu_1^{K_\sharp} - \epsilon$, the sums running over $K_\sharp \le j < K$.

Finally $t \ge 2M(\log b)^{r_\star}/b \ge 1/b$ for large b, and $\epsilon \le b^{-k}$ once $c(\log b)^{2\mathfrak{p}_0-1} \ge k$, which holds for large b as $2\mathfrak{p}_0>1$. □

Proposition 11.12 (Rigidity)**.** Let $q\ge 2$ be an integer, $\tau\in\mathbb{D}^3$, $\mathfrak{S} = \prod_\mu \tau_\mu q^{\mathbb{N}_0}$, and $s^{[\mu]}$ the box s with s_μ multiplied by q. For $b\in(0,\infty)$ let $\mathscr{S}_b$ be the class of the sequences $(K_j,b_j)_{j\ge1}$ of cutoffs and labels with $K_j\to\infty$ and $b_j\to b$; it is closed under subsequences, and the full sequence of the cutoffs at the fixed label b, $K_j=j$ and $b_j=b$, belongs to it. Along a sequence of $\mathscr{S}_b$ a number written with the index K at the label b, as $r_K[n](s;b)$ or $E_i^{K,m,e}$, is taken at the cutoff K_j and the label b_j, that is at $\beta = L^{K_j}b_j$, and it converges if it converges as $j\to\infty$. Along every sequence of $\mathscr{S}_b$ the following are equivalent: **(R1)** for every $s\in\mathfrak{S}$, $r_K[n](s;b)$ converges for $n\ne0$, and $q\log Z_K[0](s;b) - \log Z_K[0](s^{[\mu]};b)$ converges for each μ; **(R2)** for every $s\in\mathfrak{S}$ and every n, $\delta_K^n(s;b)$

converges in $[-\infty,\infty)$; **(R3)** for each time, every L-adic section and all m,e,i, $E_i^{K,m,e}$ converges in $[0,\infty]$; **(R4)** (R1) with every integer $q'\ge 2$ in the place of q, and (R2), hold on every box with L-adic sides.

If they hold along a sequence of $\mathscr{S}_b$, operators $H_{l,m}$ are formed from the limits along it as in Thm. AH(3), and for each time, L-adic section l, direction ν of it and integer $q'\ge 2$, the difference of the norms $a_K^{-1}[q'\log\|T_0(l)\|-\log\|T_0(l^{[\nu]})\|]$ converges along it to a finite number ≥ 0, $l^{[\nu]}$ the section with l_ν multiplied by q'. If they hold along every sequence of $\mathscr{S}_b$, then all levels, $r_K[z]$ on every L-adic box, δ_K^0 and the differences of the norms converge along each of these sequences, to limits that are the same for all of them and so depend on b alone; and if this is so for every b of an open set $I\subset(0,\infty)$, these limits are continuous in $b\in I$ and are reached uniformly for the labels in every compact subset of I, the levels in a metric of $[0,\infty]$. ■

Proof sketch. Fix a sequence of $\mathscr{S}_b$. Its labels and b form a compact set $\mathsf{J}\subset(0,\infty)$, at which the window of Thm. AH(1) holds with one constant C at all large cutoffs, uniformly along the sequence; the window is the one input of the argument that reads the label. (R2) gives (R1): δ_K^0 is bounded (Thm. AH(1)), $r_K[n]=e^{\delta_K^n-\delta_K^0}$, and $q\log Z_K[0](s)-\log Z_K[0](s^{[\mu]})=q\delta_K^0(s)-\delta_K^0(s^{[\mu]})$. Assume (R1). Over a fixed section and with $\Theta_K(t)=\operatorname{Tr}T_0^{t/a_K}/\|T_0\|^{t/a_K}\in[1,B]$ (Thm. AH(1)), the powers of $\|T_0\|$ cancel in $x_p^K=q\log Z_K[0](q^pt_0,l)-\log Z_K[0](q^{p+1}t_0,l)=q\log\Theta_K(q^pt_0)-\log\Theta_K(q^{p+1}t_0)\in[-\log B,q\log B]$, and

$$\log\Theta_K(t_0)=\sum_{p\le J}q^{-p-1}x_p^K+O(q^{-J})$$

uniformly in K; so Θ_K converges at the heights t_0q^p, and with it the eight sector traces $\Theta_K\cdot\frac14\sum_k(-1)^{e\cdot k}r_K[(k,m)]$. Two limits of the levels along subsequences (Lem. 11.2(b)) have Laplace transforms that agree along $t_0q^p\to\infty$, hence are equal (Lem. 11.2(a), applied to the sector traces, whose coefficients are positive): all levels over the section converge. To turn the box, $q\log Z_K[0](s_0,l)-\log Z_K[0](s_0,l')=s_0\kappa_K+q\log\Theta_K(s_0;l)-\log\Theta_K(s_0;l')$ with $\kappa_K=a_K^{-1}[q\log\|T_0(l)\|-\log\|T_0(l')\|]$, which does not depend on s_0: its convergence at one height gives it at all; three turns of the box reach every L-adic box, which gives (R3).

From (R3) the quotients and the differences of (R1) converge at every L-adic box and every q', and $\delta_K^0(s)/|s|$, telescoped over the boxes q^ps with a tail $\le 6Cq^{-J}$ by the last inequality of Thm. AG(2) and the window, converges: (R4). $\kappa_K\ge 0$ because $\log\|T_0(\Sigma)\|/|\Sigma|=\lim_t F_0(t,\Sigma)/(t|\Sigma|)$ does not increase in a side of Σ. Let now (R1)–(R4) hold along every sequence of $\mathscr{S}_b$. Two of these sequences interlace into a third one, which belongs to $\mathscr{S}_b$; along it the limits exist, so they are the same along the two. Let this be so for every $b\in I$, and $b_i\to b$ in I: choose $K_i\to\infty$ with the values at (K_i,b_i) within $1/i$ of the limits at the label b_i; the sequence (K_i,b_i) belongs to $\mathscr{S}_b$, so the values along it, and with them the limits at the labels b_i, tend to the limits at b, and the limits are continuous on I.

If the convergence were not uniform on a compact set $\mathsf{J}'\subset I$, there would be $K_j\to\infty$ and $b_j\in\mathsf{J}'$, $b_j\to b\in\mathsf{J}'$, with the values at (K_j,b_j) at a distance at least $\epsilon>0$ from the limits at the labels b_j; these limits tend to those at b, by the continuity, and so do the values along (K_j,b_j), a sequence of $\mathscr{S}_b$, so that the distance tends to 0. □

Lemma 11.13 (One-sided recursions)**.** Let $0\le x_K\le 1$ with $x_{K+1}\ge e^{-\varepsilon_K}x_K-x'_K$, $\varepsilon_K,x'_K\ge 0$, $\sum_K(\varepsilon_K+x'_K)=D'<\infty$. Then $\sum_K(x_K-x_{K+1})_+\le D'$, $\sum_K|x_{K+1}-x_K|\le 1+2D'$, and x_K converges. By Thm. AH(6), if $\lim_j\phi_j(t)$ exists for every t of a set with a limit point in $(0,\infty)$, it exists at every $t>0$, locally uniformly with all derivatives. ■

Proof sketch. $x_K-x_{K+1}\le(1-e^{-\varepsilon_K})x_K+x'_K\le\varepsilon_K+x'_K$; the positive and the negative parts of the increments differ by at most 1 in their sums. The second assertion is Vitali's theorem for the normal family of Thm. AH(6). □

Proposition 11.14 (Laplace asymptotics on one lattice)**.** For a box N and $\varphi\in[-\pi,\pi]^3$ let $S_\varphi(N)=\sum\log\hat k^2$ over the momenta $k_\mu\in(2\pi\mathbb{Z}+\varphi_\mu)/N_\mu$ mod 2π with $\hat k^2=\sum_\mu(2-2\cos k_\mu)\ne 0$,

$c_\infty = (2\pi)^{-3}\int_{[-\pi,\pi]^3}\log\hat k^2\,\mathrm{d}k$, and $\vartheta(u,\varphi)=\sum_{n'\in\mathbb{Z}}e^{-u(2\pi n'+\varphi)^2}$. $Z^z_\beta(N)$ is $Z[z]$ on the box N at β, so that $\rho^z(N,\beta)=Z^z_\beta(N)/Z^0_\beta(N)$. For $z\neq 0$ let $\theta^a(z)\in\{0,\pi\}^3$, $a=1,2,3$, be the phases of the three colours under a twist triple Ω of the class z (Lem. 11.9) ($\theta^a_\mu=\pi$ where the a-th diagonal entry of the adjoint action of Ω_μ is -1; for one twisted plane 12 they are $(0,\pi,0)$, $(\pi,0,0)$, $(\pi,\pi,0)$, and none is 0), and $\mathscr{D}^z(N)=\exp[-\frac12\sum_a S_{\theta^a(z)}(N)]$. There are absolute constants C_T, C_P, C_S with the following properties.

(a) For every box, every class $z\neq 0$ and $\beta\geq e$ with $\beta/(\log\beta)^3\geq C_T\bar N^{36}$, $\bar N=\max_\mu N_\mu$:

$$\big|\log Z^z_\beta(N)-|N|\log\frac{2}{\pi\beta^3}-\log\mathscr{D}^z(N)\big|\leq C_T\bar N^{18}(\log\beta)^{3/2}\beta^{-1/2}.$$

(b) For every $N\geq 3$ and $\beta\geq e$ with $\beta^{1/4}\geq C_P(N\log\beta)^{C_P}$:

$$C_P^{-1}\beta/N\leq Z^0_\beta(N^{\times 3})(\pi\beta^3/2)^{N^3}e^{\frac32 N^3 c_\infty}\leq C_P\beta/N.$$

So for every $c\in(0,1]$ there are β_c and C_c with $C_c^{-1}N/\beta\leq Z^z_\beta(N^{\times3})/Z^0_\beta(N^{\times3})\leq C_cN/\beta$ for every $N\geq3$, $z\neq0$ and $\beta\geq\max\{\beta_c,\exp(N^c)\}$.

(c) For every cube $N\geq 2$ and $\varphi\in[-\pi,\pi]^3\setminus\{0\}$, $|S_\varphi(N^{\times3})-N^3c_\infty-\log|\varphi|^2|\leq C_S$ and $|S_0(N^{\times3})-N^3c_\infty-2\log N|\leq C_S$. The limits $f(\varphi)=\lim_N[S_\varphi(N^{\times3})-N^3c_\infty]$, $\varphi\neq0$, and $f_0=\lim_N[S_0(N^{\times3})-N^3c_\infty-2\log N]$ exist; $f(\varphi)=\log|\varphi|^2+h(\varphi)$ and $f_0=h(0)$ with the continuous bounded function

$$h(\varphi)=\int_0^\infty\Big[(4\pi u)^{-3/2}-\prod_\mu\vartheta(u,\varphi_\mu)+e^{-u|\varphi|^2}-e^{-u}\Big]\frac{du}{u},$$

even in each φ_μ; and $\lim_N[S_\theta(N^{\times3})-S_{\theta'}(N^{\times3})]=\int_0^\infty[\prod_\mu\vartheta(u,\theta'_\mu)-\prod_\mu\vartheta(u,\theta_\mu)]\frac{du}{u}$ for $\theta,\theta'\in\{0,\pi\}^3\setminus\{0\}$.

(d) For $\lambda,R>0$, $\int_{|c_\mu|\leq R}e^{-\lambda\sum_{\mu<\nu}|c_\mu\times c_\nu|^2}d^9c\leq 48\pi^3R\lambda^{-2}$, and the integral is $\geq\pi^3e^{-5}R\lambda^{-2}$ if $\lambda R^4\geq4$.

(e) With

$$\mathscr{G}_z=\frac{8\pi^2\exp[\frac12 f_0-\frac12\sum_a f(\theta^a(z))]}{\int_{[-\pi,\pi]^3}e^{-f(\varphi)}d\varphi},\tag{11.iii}$$

$\frac{\beta}{N}Z^z_\beta(N^{\times3})/Z^0_\beta(N^{\times3})\to\mathscr{G}_z$ as $N\to\infty$ with $\beta\geq\exp(N^c)$, for every $c\in(0,1]$; so $C^z(N)/N\to\mathscr{G}_z$, $C^z(N)=\lim_{\beta\to\infty}\beta\rho^z(N^{\times3},\beta)$ the number of (g).

(f) (Free values.) For $n\neq0$ put $d_n(N)=-\frac12\sum_a[S_{\theta^a(n)}(N)-|N|c_\infty]$, so that $\log\mathscr{D}^n(N)=d_n(N)-\frac32|N|c_\infty$. On the boxes $N=N's$ of a fixed integer shape s, $d_n(N's)=\log D_n(s)+d_n^{(1)}(s)N'^{-2}+O(N'^{-4}(\log N')^{7/2})$ with coefficients in closed form; on cubes $\log D_n=-\frac12\sum_a f(\theta^a(n))$ is -0.127453, -0.402863, -0.488493 and $d_n^{(1)}$ is 0.476777, -0.755058, -0.834736 for one, two, three twisted planes; and $\sum_{n\neq0}\log D_n(s)=-3\log2$ on every shape. With Thm. AG(14)(b), in the range of (a) with $\beta\geq e^{148}$,

$$\delta^n(N)\geq d_n(N)-C_T\bar N^{18}(\log\beta)^{3/2}\beta^{-1/2}-2.2\cdot10^5\,|N|(\log\beta)\beta^{-1/6}.$$

(g) (A fixed cube.) On every cube $N\geq3$, as $\beta\to\infty$, $\beta^{3N^3}Z^z_\beta(N^{\times3})$ tends to $\mathscr{Z}^z(N)=(\frac{2}{\pi})^{N^3}\mathscr{D}^z(N^{\times3})$ for $z\neq0$, and $\beta^{3N^3-1}Z^0_\beta(N^{\times3})$ tends to

$$\mathscr{Z}(N)=\Big(\frac{2}{\pi}\Big)^{N^3}\frac{e^{-\frac12 S_0(N^{\times3})}}{8\pi^2}\int_{[-\pi,\pi]^3}e^{-S_\varphi(N^{\times3})}\,\mathrm{d}\varphi\in(0,\infty);$$

so for every $z \neq 0$, $\beta\rho^z(N^{\times 3},\beta)$ tends to $C^z(N) = \mathcal{Z}^z(N)/\mathcal{Z}(N) \in (0,\infty)$. There are absolute constants c_3, $x_0 > 0$ such that for $\beta \geq 3$

$$\Big|\frac{\beta\rho^z(N^{\times 3},\beta)}{C^z(N)} - 1\Big| \leq 3c_3\Big(\frac{N^{66}\log\beta}{\beta}\Big)^{1/8} \qquad \text{whenever} \quad c_3\Big(\frac{N^{66}\log\beta}{\beta}\Big)^{1/8} \leq x_0.$$

■

Proof sketch. (a) In a class $z \neq 0$ the minima of the twisted action form exactly two orbits under the gauge group, with stabiliser ± 1 and without zero modes. The Gaussian integral about them is the determinant of the lattice Laplacian with momenta shifted by the colour phases, which is $\mathscr{D}^z$; the cubic and quartic remainders, cut at fields of size $(\log\beta/\beta)^{1/2}$, give the error, and $(2\pi)^3/(2\pi^2)^2 = 2/\pi$ is the Haar normalisation. (b) In the periodic class the nine constant modes $c \in (\mathbb{R}^3)^3$ are kept as coordinates; by

$$1 - \tfrac{1}{2}\mathrm{tr}(ABA^{-1}B^{-1}) = 2\sin^2\alpha\sin^2\alpha'|\mathsf{a}\times\mathsf{a}'|^2 \qquad (A = e^{i\alpha\mathsf{a}\cdot\sigma},\ B = e^{i\alpha'\mathsf{a}'\cdot\sigma},\ |\mathsf{a}| = |\mathsf{a}'| = 1)$$

their action is quartic, between $\frac{32}{\pi^4 N}\sum_{\mu<\nu}|c_\mu \times c_\nu|^2$ and $\frac{2}{N}\sum_{\mu<\nu}|c_\mu\times c_\nu|^2$ for $|c_\mu| \leq \frac{\pi}{2}N$, and (d) gives one power of β/N more than in the twisted classes; the linear term of the fluctuation vanishes, and the determinant of the remaining modes is bounded by (c). The two-sided bound of the quotient follows from (a) and (b) with $\log\beta \geq N^c$. (c) $S_\varphi - N^3 c_\infty$ is a Riemann sum minus its integral; the theta-function representation $\log x = \int_0^\infty (e^{-u} - e^{-ux})\frac{du}{u}$, applied to every momentum, with Poisson summation in each direction, gives the bounds, the limits and the formula for h. (e) Over the moduli of the flat periodic fields, the orbit volume, the three directions $N^{3/2}d^3\theta$ and the determinants $e^{-\frac{1}{2}S_0 - S_\varphi}$ give

$$Z^0_\beta(N^{\times 3}) \sim \Big(\frac{2}{\pi\beta^3}\Big)^{N^3} e^{-\frac{3}{2}N^3 c_\infty}\,\frac{\beta}{N}\,\frac{e^{-f_0/2}}{8\pi^2}\int_{[-\pi,\pi]^3} e^{-f}\,\mathrm{d}\varphi$$

in the range of the limit, and with (a) the quotient tends to $\mathscr{G}_z$; the numerical evaluation of eq. 11.iii uses the integral representation of f and f_0. (f) The expansion is Poisson summation with the lattice factor $\hat{k}^2$ expanded to the fourth order. The eight phase sets $\{0,\pi\}^3$ of the box N together give the momenta of the box $2N$: $\sum_{\varphi\in\{0,\pi\}^3} S_\varphi(N) = S_0(2N)$; so $\sum_{\varphi\neq 0}[S_\varphi(N) - |N|c_\infty] = [S_0(2N) - 8|N|c_\infty - 2\log(2N)] - [S_0(N) - |N|c_\infty - 2\log N] + 2\log 2$ tends to $2\log 2$, and each non-zero φ occurs three times among the colour phases of the seven classes, whence $\sum_{n\neq 0}\log D_n = -\frac{3}{2}\cdot 2\log 2$.

The last inequality is (a) with $P \leq P^{\mathrm{G}} + 2.2\cdot 10^5(\log\beta)\beta^{-1/6}$. (g) Put $y = N^{66}\beta^{-1}\log\beta$. The twisted half is (a) at the fixed cube: for $y \leq 1$ and $\beta \geq 3$, $N^{36}(\log\beta)^3/\beta = y^{6/11}(\log\beta)^{27/11}\beta^{-5/11} \leq 5.4\,y^{6/11}$, the maximum of $(\log\beta)^{27/11}\beta^{-5/11}$ being $(5.4/e)^{27/11} < 5.4$, at $\log\beta = 5.4$; so for $y \leq y_0$, $y_0 \in (0,1]$ absolute, the range of (a) holds and $|\log(\beta^{3N^3}Z^z_\beta(N^{\times 3})/\mathcal{Z}^z(N))| \leq C_T(5.4\,y^{6/11})^{1/2} \leq C_T\sqrt{5.4}\,y^{1/8} \leq 1$, which gives the limit $\mathcal{Z}^z(N)$ and a relative error at most $2C_T\sqrt{5.4}\,y^{1/8}$. In the periodic class the minima of the action are the flat fields; at the flat field with the commuting holonomies $e^{i\phi_\mu \mathsf{a}\cdot\sigma}$ the Gaussian integral over the modes transverse to the flat fields carries the factor $e^{-\frac{1}{2}S_0 - S_\theta}$, $\theta = 2\phi$, from one neutral colour and two charged ones of the phases $\pm\theta$, and the integral over the moduli of the flat fields, in the variable $\theta \in [-\pi,\pi]^3$, with the orbit volume and the quartic action of the constant modes of (b), gives the limit $\mathcal{Z}(N)$.

Its rate comes in four steps; A is the action, $|M|^2 = \frac{1}{2}\mathrm{tr}\,M^*M$ for 2×2 matrices, and $a_\beta = 4N^3\beta^{-1}\log\beta$. *The split.* A gauge transformation sets to $\mathbf{1}$ the bonds of the spanning tree formed by the bonds $\langle x, x+e_3\rangle$ with $x_3 \leq N-2$, $\langle x, x+e_2\rangle$ with $x_3 = 0$, $x_2 \leq N-2$, and $\langle x, x+e_1\rangle$ with $x_2 = x_3 = 0$, $x_1 \leq N-2$, the sites being $x \in \{0,\dots,N-1\}^3$; in integrals of gauge invariant functions this removes the gauge group exactly, without a Jacobian. Then the variable h_μ of the bond $\langle (N-1)e_\mu, Ne_\mu\rangle$ is the holonomy along the axis μ, and the field is bounded by its curvature, through chains of plaquettes along the tree: the variable of every bond of the direction μ outside

the tree lies within $2NA^{1/2}$ of h_μ if the bond starts at $x_\mu = N-1$, and of $\mathbf{1}$ otherwise, and $|h_\mu h_\nu h_\mu^{-1} h_\nu^{-1} - \mathbf{1}| \le N(2A)^{1/2}$.

Let $\mathsf{s}_\mu = 1 - (\frac{1}{2}\mathrm{tr}\, h_\mu)^2$, which is $\sin^2 \frac{\theta_\mu}{2}$ on the flat field of the moduli θ; for $0 < r \le \frac{1}{4}$ let $\psi_\mu = \min\{1, \max\{0, \frac{1}{3}(\mathsf{s}_\mu/r^2 - 1)\}\}$, $\chi_1 = \psi_1$, $\chi_2 = (1-\psi_1)\psi_2$, $\chi_3 = (1-\psi_1)(1-\psi_2)\psi_3$ and $\chi_\mathrm{s} = \prod_\mu (1-\psi_\mu)$, so that $\chi_1 + \chi_2 + \chi_3 + \chi_\mathrm{s} = 1$. χ_i vanishes unless $\mathsf{s}_i \ge r^2$: this is the regular part, at a distance at least $2r$ from $\theta = 0$; χ_s vanishes unless $\mathsf{s}_\mu < 4r^2$ for every μ: this is the part within a distance of the order r of $\theta = 0$, the image of the eight central flat fields $h_\mu = \pm\mathbf{1}$.

The fields with $A \ge a_\beta$ carry at most $e^{-\beta a_\beta} = \beta^{-4N^3}$, which is at most $(C/\beta)^{N^3} \beta^{1-3N^3} \mathcal{Z}(N)$, as $\mathcal{Z}(N) \ge c^{N^3}$ with an absolute $c > 0$. *The regular part.* On the support of χ_1 write $h_1 = e^{i\phi_1 \mathsf{a}\cdot\sigma}$, $|\mathsf{a}| = 1$, $h_\nu = e^{i\alpha_\nu \mathsf{a}\cdot\sigma} e^{iy_\nu\cdot\sigma}$ with $y_\nu \perp \mathsf{a}$, $|y_\nu| < \frac{\pi}{2}$, for $\nu = 2, 3$, and $e^{iX(b)\cdot\sigma}$, $|X(b)| < \pi$, for the variable of every other bond b outside the tree divided by h_μ or by $\mathbf{1}$ as above; the flat fields are $y_2 = y_3 = 0$, $X = 0$, with the coordinates $\phi_1\mathsf{a}$, α_2, α_3 along them, and $\nu = (y_2, y_3, X)$ are the $6N^3 - 2$ coordinates transverse to them.

As $|h_1 h_\nu h_1^{-1} h_\nu^{-1} - \mathbf{1}| = 2|\sin\phi_1| \sin|y_\nu|$ and $|\sin\phi_1| = \mathsf{s}_1^{1/2} \ge r$, the bounds by the curvature give $\|\nu\|^2 \le \Lambda_r A$ with $\Lambda_r = 2\pi^2 N^5 + \frac{\pi^2}{4} N^2 r^{-2}$; the third derivatives of the plaquette terms along the rays through the flat field give $|A - Q| \le 43N^2\|\nu\|^3$, Q the quadratic form of the transverse modes at the flat field, and so $Q \ge \|\nu\|^2/\Lambda_r$. On $\{A < a_\beta\}$ then $\|\nu\| \le \varrho_r = (\Lambda_r a_\beta)^{1/2}$ and $|A - Q| \le \epsilon_r Q$, $\epsilon_r = 43N^2\Lambda_r\varrho_r$: there the integral of $\chi_1 e^{-\beta A}$ lies between the Gaussian integrals of $(1+\epsilon_r)Q$ and of $(1-\epsilon_r)Q$, which are those of Q times $(1\pm\epsilon_r)^{1-3N^3}$, while over $\|\nu\| \le \varrho_r$ the Haar density changes by a factor within $e^{\pm 1.2\varrho_r}$ and χ_1 by at most $r^{-2}\varrho_r^2$, and the Gaussian measures give the outside of $\{A < a_\beta\}$ a relative mass at most β^{-N^3} for $\log\beta \ge 3$.

Let $\mathcal{I}_1$ be the Gaussian integral of Q against χ_1 over the flat fields, which is the limit of $\beta^{3N^3-1}\int \chi_1 e^{-\beta A}\,\mathrm{d}U$ as $\beta\to\infty$ at fixed r, and $\mathcal{I}_2$, $\mathcal{I}_3$ the same for χ_2, χ_3; the same Gaussian integral without χ_i over $\{\mathsf{s}_i \ge r^2\}$ is at most $\mathcal{Z}(N)$ by the limit, and so

$$\Big|\beta^{3N^3-1}\int_{\{A<a_\beta\}} \chi_i e^{-\beta A} dU - \mathcal{I}_i\Big| \le \big[e^{6N^3\epsilon_r + 1.2\varrho_r} - 1 + \beta^{-N^3} + 2r^{-2}\varrho_r^2\big]\mathcal{Z}(N),$$

where, if $\Lambda_r \le 3N^2 r^{-2}$, $6N^3\epsilon_r \le 2682\, N^{19/2} r^{-3} (\beta^{-1}\log\beta)^{1/2}$ and $r^{-2}\varrho_r^2 \le 12N^5 r^{-4}\beta^{-1}\log\beta$: the relative error of Laplace's method on the regular part is $N^{19/2} r^{-3}(\beta^{-1}\log\beta)^{1/2}$ up to an absolute factor. *The part about $\theta = 0$.* Where $\chi_\mathrm{s} \ne 0$, one of the eight changes of sign of the bonds of some directions μ that start at $x_\mu = N-1$, which keep the tree, the action and χ_s and change the signs of these h_μ, makes $\mathrm{tr}\, h_\mu > 0$ for every μ, and then $|h_\mu - \mathbf{1}| \le \arcsin 2r \le \pi r$; if besides $A < a_\beta$, the bounds by the curvature give $\sum_b |U(b) - \mathbf{1}|^2 \le \varrho_\mathrm{s}^2 = 6\pi^2 N^2 r^2 + 24N^5 a_\beta$; let $\varrho_\mathrm{s} \le \frac{1}{2}$.

A gauge transformation that maximises $\sum_b \frac{1}{2}\mathrm{tr}\, U(b)$ (Landau's gauge) keeps this bound and writes the field as the transform, by a gauge transformation u with $u(0) = \mathbf{1}$, of the field $(1 - |Y(b)|^2)^{1/2}\mathbf{1} + iY(b)\cdot\sigma$ with $\sum_\mu [Y(\langle x, x+e_\mu\rangle) - Y(\langle x - e_\mu, x\rangle)] = 0$ at every site; $Y = c + w$, $c = (c_\mu)$ the nine constant modes, with $|c_\mu| \le \varrho_\mathrm{s} N^{-3/2}$, and w the $6N^3 - 6$ others. By Kirchhoff's matrix-tree theorem the Jacobian of $(u, Y) \mapsto U$ is at most $(e^{S_0(N^{\times 3})}/N^3)^{3/2}(1+\epsilon_J)^{3N^3}$, $\epsilon_J = 7N^2\varrho_\mathrm{s}$. The action of a plaquette is at least half the square of the vector part of the difference of the products along its two halves, which at constant fields is exactly $-2c_\mu \times c_\nu$; so $A \ge \frac{1}{2}(1-\epsilon_A)\|d_1 w\|^2 + N^3 \sum_{\mu<\nu} |c_\mu \times c_\nu|^2$, $\|d_1 w\|^2$ the sum over the plaquettes of the squares of the circulations of w, $\epsilon_A = 4\mathsf{k}N\varrho_\mathrm{s} + 3\mathsf{k}^2\varrho_\mathrm{s}^2$ with an absolute k.

The area formula, the Gaussian integral over w, whose determinant is $e^{6S_0(N^{\times 3})}$, and (d) at $\lambda = \beta N^3$ and $R = \varrho_\mathrm{s} N^{-3/2}$, which gives the factor $48\pi^3 R\lambda^{-2}$, together give, the powers of $2\pi^2$, of 2π and of $e^{S_0(N^{\times 3})}$ cancelling against those of $\mathcal{Z}(N)$,

$$\beta^{3N^3-1}\int_{\{A<a_\beta\}} \chi_\mathrm{s} e^{-\beta A} dU \le \mathsf{E}_\mathrm{s}\, \mathcal{Z}(N), \qquad \mathsf{E}_\mathrm{s} = \frac{192}{\mathcal{J}_N} N^{-1/2}\varrho_\mathrm{s} \exp\big[21N^5\varrho_\mathrm{s} + \varrho_\mathrm{s}^2 + 6N^3\epsilon_A\big],$$

$\mathcal{J}_N = N^{-2}\int_{[-\pi,\pi]^3} e^{S_0(N^{\times 3})-S_\varphi(N^{\times 3})}d\varphi \geq e^{-2C_S}\int_{[-\pi,\pi]^3}|\varphi|^{-2}d\varphi$ by (c). So, where $N^5\varrho_{\mathrm{s}}$ is bounded and $r^2 \geq N^3 a_\beta$, the mass about $\theta = 0$ is of the order $N^{1/2}r$ against $\beta^{1-3N^3}\mathcal{Z}(N)$, the one power of r being the one power of R in (d). *The choice of* r. As r decreases, the error $N^{19/2}r^{-3}(\beta^{-1}\log\beta)^{1/2}$ of the regular part grows and the mass $N^{1/2}r$ about $\theta = 0$ shrinks; the two are equal at $r^4 = N^9(\beta^{-1}\log\beta)^{1/2}$, that is at $r = (N^{18}\beta^{-1}\log\beta)^{1/8} = N^{-6}y^{1/8}$. The root of order 8 is this balance between the third power of $1/r$ beside $(\beta^{-1}\log\beta)^{1/2}$ and the one power of r from (d); and at this r, $N^5\varrho_{\mathrm{s}}$, of the order $N^6 r = y^{1/8}$, stays bounded, which is why the power in y is $66 = 8\cdot 6 + 18$.

Indeed, with $r = N^{-6}y^{1/8}$, $\beta^{-1}\log\beta = N^{-66}y$ and $y \leq 1$: $\Lambda_r \leq 3N^2r^{-2} = 3N^{14}y^{-1/4}$, $\varrho_r^2 \leq 12N^{-49}y^{3/4}$, $\epsilon_r \leq 447N^{-17/2}y^{1/8}$, $6N^3\epsilon_r \leq 2682N^{-11/2}y^{1/8} \leq 7y^{1/8}$, $r^{-2}\varrho_r^2 \leq 12N^{-37}y^{1/2}$, $\varrho_{\mathrm{s}}^2 \leq 60N^{-10}y^{1/4}$ and $N^5\varrho_{\mathrm{s}} \leq 8y^{1/8}$; so the exponent in E_{s} is at most an absolute multiple of $y^{1/8}$, and $\mathsf{E}_{\mathrm{s}} \leq CN^{-11/2}y^{1/8}$. As $\beta\to\infty$ at fixed r, $\beta^{3N^3-1}\int\chi_{\mathrm{s}}e^{-\beta A}\,\mathrm{d}U$ tends to $\mathcal{I}_{\mathrm{s}} = \mathcal{Z}(N) - \mathcal{I}_1 - \mathcal{I}_2 - \mathcal{I}_3$, by the limit $\mathcal{Z}(N)$, and $\mathcal{I}_{\mathrm{s}} \leq \mathsf{E}_{\mathrm{s}}\mathcal{Z}(N)$, E_{s} being increasing in a_β.

Summing over the four parts and the fields with $A \geq a_\beta$,

$$\Big|\frac{\beta^{3N^3-1}Z_\beta^0(N^{\times 3})}{\mathcal{Z}(N)} - 1\Big| \leq 3\big[e^{6N^3\epsilon_r+1.2\varrho_r} - 1 + \beta^{-N^3} + 2r^{-2}\varrho_r^2\big] + 2\mathsf{E}_{\mathrm{s}} + \Big(\frac{C}{\beta}\Big)^{N^3} \leq c_3 y^{1/8}$$

for $y \leq y_0$, with an absolute c_3, any larger number serving as well, and y_0 taken smaller if need be, so that $\epsilon_r, \epsilon_A \leq \frac12$, $\varrho_r < 1$, $\varrho_{\mathrm{s}} \leq \frac12$, $r \leq \frac14$ and $\beta^{-N^3} + (C/\beta)^{N^3} \leq y^{1/8}$. So $C^z(N) = \mathcal{Z}^z(N)/\mathcal{Z}(N) = 8\pi^2 e^{\frac12 S_0(N^{\times 3})}\mathscr{Z}^z(N^{\times 3})/\int_{[-\pi,\pi]^3} e^{-S_\varphi(N^{\times 3})}d\varphi$, and with (c) and dominated convergence $C^z(N)/N \to \mathscr{G}_z$. Take $c_3 \geq 2C_T\sqrt{5.4}$ and $x_0 \leq \min\{\frac13, c_3 y_0^{1/8}\}$: where the second right member $x = c_3 y^{1/8}$ is at most x_0, $y \leq y_0$, so that the twisted relative error is at most x as well; and a quotient of two numbers within $x \leq \frac13$ of 1 lies within $3x$ of 1. □

Quantitative estimates for the explicit windows. The windows are constructed below. Their moving-window Laplace bounds and the following numerical estimates remain to be proved. The required estimates, referred to as **(Window estimates)**, would provide absolute constants $0 < c_\flat \leq C_\flat$, a number $x_\flat(L) > 0$, $t_* > 0$, and for $0 < t \leq t_*$ Borel functions F_t, F_t' with values in $[0,1]$ of the block field of depth one on a half slab of normal 3, invariant under the flip c^k of the class $n_{13} = ((1,0),0)_3$, and F_t'' invariant under the flip of the class $((1,1),0)_3$, such that on every cube of $N = \mathsf{n}L^j$ sites, $\mathsf{n} \in 2L^{\mathbb{N}}$, $j \geq 1$, at $\beta = N/t \geq 3$, where they are read on the block field $\mathcal{M}^j U$ of n sites a side (at the cutoff K on the cube ℓ, $\mathsf{n} = L\ell$ and $j = K-1$, and $\mathcal{M}^{K-1}U$ is the block field of depth one), the numbers of Thm. AG(12) obey: for F_t, if $N^{66}\beta^{-1}\log\beta \leq x_\flat(L)$, $c_\flat t \leq w_F \leq C_\flat t$ and $\log(2^{L-1}\mathfrak{q}_L) \geq \lambda_\flat = 0.5516L - 0.7162$; for F_t', if $N^{68}\beta^{-1}\log\beta \leq x_\flat(L)$, $w_F \geq c_\flat t$, $\mathfrak{q}_i \geq 0.84\cdot 0.9356^i$ for $i \leq L$, and $\log(2^{L-1}\mathfrak{q}_L) \geq \lambda_\flat = 0.6266L - 0.873$; and for F_t'' in the same range, $w_F \geq c_\flat t$ and $\mathfrak{q}_i \geq 0.84\cdot 0.944^i$ for $i \leq L$.

On a cube $\ell = 2L^m$ with $MR(b) \leq L^m \leq ML^2(\log b)^{r_\star}$, w_K and $\mathfrak{q}_L(K)$ are these numbers at the cutoff K, and $\varsigma_K = w_K^L\mathfrak{q}_L(K)$ is an integral over the box $(\ell,\ell,L\ell)$ divided by $Z_{K,\ell}^L$.

Construction and elementary properties of the windows. The windows. On the cube of $N = \mathsf{n}L^j$ sites with the time 3 let $h = N/2$; the half slab holds the layers $0 \leq x_3 \leq h$, and the block layers inside it are those of the block lattice of $\mathcal{M}^j U$ at the heights $c = L^j y_3$, $1 \leq y_3 \leq \frac{\mathsf{n}}{2} - 1$. In the block layer at the height c, for a block site y of it let $\bar h_1(y)$, $\bar h_2(y)$ be the products of the block field along the straight closed block lines from y in the directions 1 and 2, and let $\bar s_\pm(c)$, $\bar s_1(c)$, $\bar s_2(c)$ be the means over the n^2 sites y of the layer of $\frac12\mathrm{tr}[\bar h_1(y)\bar h_2(y)^{\pm 1}]$, $\frac12\mathrm{tr}\,\bar h_1(y)$ and $\frac12\mathrm{tr}\,\bar h_2(y)$: $\bar s_\pm$ are the means of the half traces of the closed block lines that wind once in the direction 1 and once, forwards or backwards, in the direction 2. With $c_1 = L^j$, the block layer next to the lower end of the half slab, and $c_2 = h - L^j$, the one next to its upper end, $F_t = \Phi_t \circ \mathcal{M}^j$, $F_t' = \Phi_t' \circ \mathcal{M}^j$,

$F_t'' = \Phi_t'' \circ \mathcal{M}^j$ with

$$\Phi_t = \mathbf{1}\{\bar s_2(c_1) > \tfrac12\}\, e^{-\frac{4}{t}(\bar s_+(c_1)^2 + \bar s_-(c_1)^2)}, \qquad \Phi_t' = \prod_{c\in\{c_1,c_2\}} \mathbf{1}\{\bar s_2(c) > \tfrac12\}\, e^{-\frac{2}{t}(\bar s_+(c)^2 + \bar s_-(c)^2)},$$

$$\Phi_t'' = \prod_{c\in\{c_1,c_2\}} \mathbf{1}\{\bar s_+(c) < -\tfrac12\}\, e^{-\frac{4}{t}(\bar s_1(c)^2 + \bar s_2(c)^2)} :$$

one pin for F_t, two pins for F_t' and F_t''. They are Borel, with values in $[0,1]$, and gauge invariant. The flip of the class n_{13}, taken across a line in a wall of the blocks of every order (L is odd), passes the averaging map and becomes the change of sign of the block bonds of the direction 1 that cross that line: $\bar h_1 \mapsto -\bar h_1$ and $\bar h_2$ is kept, so $\bar s_\pm$ change sign, $\bar s_2$ is kept, and F_t, F_t' are invariant; the flip of $((1,1),0)_3$ changes the signs of $\bar h_1$ and of $\bar h_2$, so $\bar s_1$, $\bar s_2$ change sign, $\bar s_+$ is kept, and F_t'' is invariant. On a flat field with the holonomies $e^{i\varphi_\mu \mathsf{a}\cdot\sigma}$, $|\mathsf{a}| = 1$, the means are $\bar s_\pm = \cos(\varphi_1 \pm \varphi_2)$ and $\bar s_\mu = \cos\varphi_\mu$.

The flip of n_{13} acts by $\varphi_1 \mapsto \varphi_1 + \pi$; up to $\varphi \mapsto -\varphi$ its fixed points are $\varphi_1 = \frac{\pi}{2}$, $\varphi_2 \in \{0,\pi\}$, and inside $\{\bar s_2 > \frac12\}$ the common zeros of $\bar s_+$ and $\bar s_-$ are the one fixed point $(\frac{\pi}{2},0)$, at which they vanish transversally and independently: at $(\frac{\pi}{2}+a, b)$, $\bar s_+^2 + \bar s_-^2 = 2(a^2+b^2) + O(|(a,b)|^4)$. Thus both transverse angle variables have scale $t^{1/2}$ in this flat parametrisation. Likewise the flip of $((1,1),0)_3$ acts by $\varphi_\mu \mapsto \varphi_\mu + \pi$, $\mu = 1,2$; the indicator $\{\bar s_+ < -\frac12\}$ singles out its fixed point $(\frac{\pi}{2},\frac{\pi}{2})$, about which $\bar s_1^2 + \bar s_2^2$ cuts both angles at the width $t^{1/2}$: at $(\frac{\pi}{2}+a, \frac{\pi}{2}+b)$, $\bar s_1^2 + \bar s_2^2 = \sin^2 a + \sin^2 b = a^2 + b^2 + O(|(a,b)|^4)$.

The numbers of Thm. AG(12) are $w_F = \mathcal{Z}_1/Z[0]$ and $\mathfrak{q}_i = \mathcal{Z}_i/\mathcal{Z}_1^i$, where $\mathcal{Z}_i$ is the periodic integral over the box (N,N,iN) of the product of the $2i$ mirrored copies of the window. The integral identity in the definition follows from the slab trace formula. The bounds called **(Window estimates)** are the uniform estimates on these inserted integrals that remain to be established.

The uninserted periodic Laplace estimate of Prop. 11.14 does not establish these bounds for the moving pins. In particular, **(Window estimates)** includes the stated weight bounds, numerical flat ratios and growing-lattice ranges. □

Proposition 11.15 (Winding block loops on small tori)**.** Let $\ell = 2L^m$, $b \ge \gamma^{-2}$ with $MR(b) \le L^m$, and $d \ge 1$ an integer with $L^d b \ge \mathsf{t}_\natural$, $\delta_d = \exp[7(\ell L^d)^2 \varepsilon(L^d b)] - 1 \le \frac{1}{20}$ and $\mathsf{R}(d) \le \frac{1}{40}$, where $\mathsf{t}_\natural$, $\varepsilon(x) = A_0 x^{-1/2}(\log x)^{\mathfrak{p}_0}$ and $\mathsf{R}(d)$ (of Thm. AH(7)) are formed from the constants. For a site y of the block lattice of depth d let $\gamma_1, \gamma_2, \gamma_{12}$ be the closed block lines through y that wind once in the direction 1, once in the direction 2, and once in 1 and then once in 2, W_γ the half trace of the block field along γ, t the translation by $\frac{\ell}{2}e_3$, and $\mathsf{m}_\gamma = \langle W_\gamma \cdot \mathsf{t} W_\gamma\rangle$ at the cutoff $K > d$. Then $\mathsf{m}_{\gamma_2} = \mathsf{m}_{\gamma_1}$ and

$$f^K = 4\mathsf{m}_{\gamma_1} + \mathsf{m}_{\gamma_{12}} \ge (1 - \tfrac14\delta_d^2)(1 - \mathsf{R}(d)) - 5(\delta_d + 2\mathsf{R}(d)) \ge 0.4743,$$

while $|f^K| \le 10[\frac14(1-r_1^2)]^{\frac12 - \frac{1}{\ell L^d}}$ with r_1 on the cube ℓ; so $r_1 \le 1 - 2\,(0.045)^{2/(1-2L^{-d}/\ell)} \le 0.99596$ at every cutoff $K > d$, and there $\delta_K^0 \ge \mathsf{G} \ge 0.003034$ by Thm. AG(13). These conditions hold at $\ell = 2L^m$, $b_m = \exp[L^{(m-m')/r_\star}]$, $d = 1$ for all large m, $M = L^{m'}$, where $\ell/b_m \to 0$. For every $b > 0$, $f^K(b)$ converges as $K \to \infty$ to a number $f(b)$, and, by Thms. YM3_I:AJ and YM3_I:AD, f is real-analytic on $(0,\infty)$. ■

Proof sketch. W_{γ_1} and $W_{\gamma_{12}}$ are real, of modulus ≤ 1, depend only on the bonds of the slab of L^{K-d} layers that holds the block layer of y, and are odd under the flip c^k of the class n_{13} applied to all layers of that slab; Thm. AG(6), with $2k' = \ell L^d$ slabs of that thickness and $j = k'$, gives the upper bound. By the small-field chain and the rarity of its large-field regions on the periodic torus (Prop. 4.54), off an event of periodic probability at most $\mathsf{R}(d)$ every plaquette of the block field V of depth d is within 2ε of $\mathbf{1}$, $\varepsilon = \varepsilon(L^d b)$, in the operator norm $|\cdot|$; let B be the event that this holds, and $\mathsf{n} = \ell L^d$ the number of block bonds of a straight closed block line.

On B the lemma on rearranged paths (Lem. YM3_I:4.1), read on the block lattice of depth d, where a step sequence may wind about the torus, applies to every pair of paths Γ, Γ' from one site whose step sequences, of a length q, are rearrangements of one another: their transports obey $|V(\Gamma')V(\Gamma)^{-1} - \mathbf{1}| \le (1+2\varepsilon)^{q(q-1)/2} - 1 \le e^{\varepsilon q^2} - 1$. (i) Let $\mathsf{H}_i = V(\gamma_i)$, the transport along γ_i from y, so that $W_{\gamma_1} = \frac{1}{2}\mathrm{tr}\,\mathsf{H}_1$, $W_{\gamma_2} = \frac{1}{2}\mathrm{tr}\,\mathsf{H}_2$ and $W_{\gamma_{12}} = \frac{1}{2}\mathrm{tr}\,\mathsf{H}_1\mathsf{H}_2$. The sequences $(e_1^{\mathsf{n}}, e_2^{\mathsf{n}})$ and $(e_2^{\mathsf{n}}, e_1^{\mathsf{n}})$, of the length $q = 2\mathsf{n}$, have the transports $\mathsf{H}_1\mathsf{H}_2$ and $\mathsf{H}_2\mathsf{H}_1$, so that $|\mathsf{H}_2\mathsf{H}_1\mathsf{H}_2^{-1}\mathsf{H}_1^{-1} - \mathbf{1}| \le e^{4\mathsf{n}^2\varepsilon} - 1 \le \delta_d$.

On SU(2), $|g-\mathbf{1}|^2 = 2 - \mathrm{tr}\,g$, and $\mathrm{tr}(ABA^{-1}B^{-1}) = (\mathrm{tr}\,A)^2 + (\mathrm{tr}\,B)^2 + (\mathrm{tr}\,AB)^2 - \mathrm{tr}\,A\,\mathrm{tr}\,B\,\mathrm{tr}\,AB - 2$ (by $X + X^{-1} = (\mathrm{tr}\,X)\mathbf{1}$); so the half traces $x_1 = W_{\gamma_1}$, $x_2 = W_{\gamma_2}$, $x_{12} = W_{\gamma_{12}}$ obey $x_1^2 + x_2^2 + x_{12}^2 - 2x_1x_2x_{12} \ge 1 - \frac{1}{4}\delta_d^2$, and, as $|x_{12}| \le 1$ and $2|x_1x_2| \le x_1^2 + x_2^2$,

$$2W_{\gamma_1}^2 + 2W_{\gamma_2}^2 + W_{\gamma_{12}}^2 \ge 1 - \tfrac{1}{4}\delta_d^2 \qquad \text{on } \mathsf{B}.$$

In the case $\delta_d = 0$, that of two commuting transports $\mathsf{H}_\mu = e^{i\theta_\mu \mathsf{a}\cdot\sigma}$ with one axis a, this inequality is the identity

$$2\cos^2\theta_1 + 2\cos^2\theta_2 + \cos^2(\theta_1+\theta_2) = 1 + (\cos\theta_1 + \cos\theta_2\cos(\theta_1+\theta_2))^2 + \cos^2\theta_2\sin^2(\theta_1+\theta_2) \ge 1.$$

(ii) Let γ be one of the three lines, $\mathfrak{s}$ its step sequence, of the length n or $2\mathsf{n}$, Γ_3 the straight block path of $\frac{\mathsf{n}}{2}$ steps in the direction 3 from y to $\mathsf{t}y$, and v the transport along it. The sequences $(\mathfrak{s}, e_3^{\mathsf{n}/2})$ and $(e_3^{\mathsf{n}/2}, \mathfrak{s})$, of a length $q \le \frac{5}{2}\mathsf{n}$, have the paths γ followed by Γ_3, and Γ_3 followed by $\mathsf{t}\gamma$; so $|\mathsf{v}\,V(\mathsf{t}\gamma)\,\mathsf{v}^{-1}V(\gamma)^{-1} - \mathbf{1}| \le e^{6.25\,\mathsf{n}^2\varepsilon} - 1 \le \delta_d$, which is the place of the numeral 7 in δ_d, and, with $|\frac{1}{2}\mathrm{tr}\,X| \le |X|$, $|VV'^{-1} - \mathbf{1}| = |V - V'|$ for unitary V, V' and the invariance of the trace under conjugation, $|W_{\mathsf{t}\gamma} - W_\gamma| \le \delta_d$ on B. (iii) As $\mathsf{t}W_\gamma = W_{\mathsf{t}\gamma}$ and $|W_\gamma| \le 1$, $W_\gamma\,\mathsf{t}W_\gamma \ge W_\gamma^2 - |W_\gamma - W_{\mathsf{t}\gamma}|$, and $|W_\gamma - W_{\mathsf{t}\gamma}|$ is at most δ_d on B and at most 2 off it; so $\mathsf{m}_\gamma \ge \langle W_\gamma^2\rangle - \delta_d - 2\mathsf{R}(d)$. With the weights $2, 2, 1$, the inequality of (i) on B, $W_\gamma^2 \ge 0$ off B and the periodic probability of B at least $1 - \mathsf{R}(d)$,

$$f^K = 2\mathsf{m}_{\gamma_1} + 2\mathsf{m}_{\gamma_2} + \mathsf{m}_{\gamma_{12}} \ge (1 - \tfrac{1}{4}\delta_d^2)(1 - \mathsf{R}(d)) - 5(\delta_d + 2\mathsf{R}(d)),$$

which decreases in δ_d and in $\mathsf{R}(d)$ and equals $\frac{1599}{1600}\cdot\frac{39}{40} - \frac{1}{2} = 0.47439\ldots$ at $\delta_d = \frac{1}{20}$, $\mathsf{R}(d) = \frac{1}{40}$; $\mathsf{m}_{\gamma_2} = \mathsf{m}_{\gamma_1}$ is the symmetry of the axes, the exchange of the directions 1 and 2 about y keeping the periodic law of the block field and commuting with t. The limit in K is the convergence in total variation of the laws of the block fields on T_ℓ (Thm. YM3_I:D); its analyticity is Thms. YM3_I:AJ and YM3_I:AD, f^K being at every label the mean of one bounded Borel function of the block field. □

Proof of Thms. AG, AH and AI. In this proof A(k) and B(k) denote part (k) of Thms. AG and AH. Thm. AG is proved on one lattice. In Thms. AH and AI, the first sentence of B(1), the inequalities of B(8), the reflection positivity of B(7) at a fixed cutoff and Thm. AI, under its hypothesis (H), at each cutoff follow from Thm. AG and the properties of the averaging map; the rest of B(1) uses Lem. 11.1; everything else uses Thm. YM3_I:D, and B(6), B(9)(c) and the statements on analyticity use Thms. YM3_I:AJ and YM3_I:AD as well, B(9)(c) with the radius and the constants of Thms. YM3_I:AJ and YM3_I:AD, through [7, Lem. N.2] and Thm. YM3_I:AV and, for its last assertion, through the normalisers at the first-passage data (Prop. YM3_I:91.12 (1)), the transfer principle for one law and for two laws under one normaliser (Prop. YM3_I:91.11), the first facts on states ([7, §K.1]), the device of sign averaging over a packing with its count (the packing and sign-averaging argument in the proof of Prop. YM3_I:127.1) and the windows of first passage (Cor. YM3_I:74.2).

Step 1: the trace and the order (A(1)). For a sign field γ on the bonds, -1 being central and Haar measure invariant, $Z_\beta[\eta\,\mathrm{d}\gamma; F] = Z_\beta[\eta; F\circ\gamma]$: columns move, two columns over one plane cancel, and on centre-blind F the places of the columns do not matter. Cut the box along μ into

layers V^τ and crossings S^τ, with the columns through μ in the crossing τ_*:

$$A_{\eta^n}(U) = \sum_\tau A^m_\Sigma(V^\tau) + \sum_{\tau\neq\tau_*} A_\times(V^\tau, S^\tau, V^{\tau+1}) + A_\times(V^{\tau_*}, S^{\tau_*}, c^k V^{\tau_*+1}),$$

$A_\times$ the action of the plaquettes of one crossing; integrating the S^τ gives the trace formula. As $\operatorname{tr}(vw^{-1}) = \sum_{ij} v_{ij}\bar w_{ij}$, $\prod_b e^{-\beta(1-\frac12 \operatorname{tr} V_b W_b^{-1})}$ is a series of squares of monomials with coefficients ≥ 0, so $\mathsf K = \Pi \mathsf K_0 \Pi \ge 0$, Π the gauge average, strictly on gauge invariant functions if $\beta > 0$ (Stone–Weierstrass). A continuous kernel > 0 gives a simple top with a positive eigenfunction, which C^k fixes. $Z[(0,m)_\mu] - Z[(k,m)_\mu] = 2\sum_{e\cdot k=1} Z_{e,m} \ge 0$ removes the planes through μ, and the same with the time in the remaining twisted plane removes it; dually $Z_{00,m} - Z_{e,m} = \frac12 \sum_{e\cdot k=1} Z[(k,m)_\mu] > 0$ for $e \neq 0$, every $Z[n]$ being positive; $A_{\eta^n} \le A + 2\#\{\eta^n = -1\}$ gives the lower bound. $Z_{e,m} > 0$: the product of the traces of straight closed lines of the directions ν, ρ with the exponents e lies in the range of P_e, and Jensen's inequality for its spectral measure bounds $\operatorname{Tr}(P_e T_m^N)$ below. $\|T_1\| \le \|T_0\|$ follows from $\operatorname{Tr} T_1^N \le \operatorname{Tr} T_0^N$ by Lem. 11.2.

A step in the direction ν moves the reversed plaquette of the section by one site, and reversing the bond that it shares with its image moves it back: $\mathcal T_\nu$ keeps dV and A^1_Σ and commutes with $\mathsf K$, so with T_1, and it maps the gauge invariant functions onto themselves, the step and the sign changes carrying gauge transformations to gauge transformations. For a sign field γ on the bonds, $\mathsf S_\nu(\gamma V) = \gamma' \mathsf S_\nu V$ with $\gamma'(\xi, j) = \gamma(\xi + e_\nu, j)$; so the map of $\mathcal T_\nu$ is the step followed by the reversal of the bond (ξ_*, ρ), and by induction $\mathcal T_\nu^q$ is the composition with the step by q sites followed by the reversal of the q bonds $(\xi_* - ie_\nu, \rho)$, $0 \le i < q$, of the direction ρ on the line of the section through ξ_* normal to ρ.

At $q = N_\nu$ the step is the identity and these are the bonds that C_ρ reverses: $\mathcal T_\nu^{N_\nu} = C_\rho$, and likewise $\mathcal T_\rho^{N_\rho} = C_\nu$. In the same way $\mathcal T_\nu \mathcal T_\rho$ and $\mathcal T_\rho \mathcal T_\nu$ differ by the reversal of the four bonds at the site $\xi_* + e_\nu + e_\rho$, the gauge transformation that is -1 at that site. Let $c_\nu^{(s)}$ reverse the bonds (ξ, ν) with $\xi_\nu = s$, so that C_ν is the composition with $c_\nu^{(s)}$ at $s = (\xi_*)_\nu$; then $\mathsf S_\nu c_\nu^{(s)} = c_\nu^{(s-1)} \mathsf S_\nu$, and $c_\nu^{(s-1)} c_\nu^{(s)}$ is the gauge transformation that is -1 at the sites of the line $\xi_\nu = s$ and 1 elsewhere: $\mathcal T_\nu C_\nu$ and $C_\nu \mathcal T_\nu$ differ by it, and $\mathcal T_\rho C_\rho$ and $C_\rho \mathcal T_\rho$ likewise.

Gauge transformations act as the identity on the gauge invariant functions, off which $T_1 = D_1 \mathsf K D_1$ vanishes; for $\beta > 0$ these are the closure of the range of T_1, as $\mathsf K$ is strictly positive on them and D_1 is invertible and gauge invariant. On them T_1, $\mathcal T_\nu$, $\mathcal T_\rho$, C_ν, C_ρ commute pairwise, and on the range of P_e there, where $C_\nu = (-1)^{e_\nu}$ and $C_\rho = (-1)^{e_\rho}$, $\mathcal T_\nu^{N_\nu} = (-1)^{e_\rho}$ and $\mathcal T_\rho^{N_\rho} = (-1)^{e_\nu}$: the commuting unitaries $\mathcal T_\nu$, $\mathcal T_\rho$ have finitely many eigenvalues there, N_ν-th roots of $(-1)^{e_\rho}$ and N_ρ-th roots of $(-1)^{e_\nu}$, which gives the momenta of A(1); their joint eigenspaces are invariant under the compact self-adjoint T_1, and an orthonormal eigenbasis of T_1 in each of them is a joint eigenbasis.

Finally $C^k = C_\nu^{k_1} C_\rho^{k_2} = \mathcal T_\nu^{k_2 N_\nu} \mathcal T_\rho^{k_1 N_\rho}$ gives the trace formula with the momenta. For the crossing relation, expand $Z^{[1]}_{e,m} = \frac14 \sum_k (-1)^{e\cdot k} Z[(k,m)_1]$ and $Z[n] = \sum_{e'} (-1)^{e'\cdot k'} Z^{[2]}_{e',m'}$, $k' = (n_{12}, n_{23})$, $m' = n_{13}$: the sum over n_{12} forces the first entries to agree, and what is left is the display. All $Z^{[2]} \ge 0$, so its modulus is at most its value at $e_2 = 0$, $m = 0$, which is $Z^{[1]}_{e_1 0,0}$; the time 3 in the place of 2 gives $Z_{0e_2,0}$; N_μ-th roots give the norms. $\varepsilon_{\mathrm{mag}}$ and $\varepsilon_{\mathrm{el}}$ are Lem. 11.2 for the sector traces, the top of T_0 being simple with its eigenfunction in the range of P_{00}.

Step 2: lengths (A(2)–(4)). The class n_{12} is electric for the times 1 and 2, so $\frac12(Z[0] \pm Z[n_{12}]) = \operatorname{Tr}(\frac12(1 \pm C) T_0^{N_\mu})$ with either direction of the plane as the time, and Lem. 11.2 applies in each length. $N_\mu^{-1} F_0$ decreases to $\log \|T_0\| \ge |\Sigma| P$, $P \ge -6\beta$ because $A \le 6|N|$, and $\delta(t, \Sigma) = t[\log \|T_0\| - |\Sigma| P] + \mathsf G(t)$ gives $\mathsf G \le \delta$, the identities for $\mathsf d$ and, the bulk cancelling, $qF_0(N) - F_0(N^{[\mu]}) = q\delta(N) - \delta(N^{[\mu]})$; $F_0/|N|$ minus its limit in one side is $\mathsf G/|N|$, and the last inequality of A(2) follows side after side, $\mathsf G$ not increasing in its height.

For A(3), $q(t,\lambda)$ is a power sum in t and in λ, and $p \ge \|T_0\|^t \ge e^{-t\lambda s_0}$, the top eigenfunction lying in the range of $\frac12(1+C)$ and $E_0(\lambda) \le s_0\lambda$; so $\mathfrak{b} = q\,e^{t\lambda s_0} \le \mathfrak{b}(t_0,\lambda_0)^{t\lambda/t_0\lambda_0}$, $\mathfrak{b} \le Z[0]e^{t\lambda s_0} - 1 = e^{\Phi} - 1$ and $q/p \le \mathfrak{b}$; $-t^{-1}\log$ and $t \to \infty$ carry the log-convexity and the submultiplicativity in λ to E_0 and E_{e}, and the monotone increments give s_f and the bounds on s_f and on σ. The roots of the power sums in both lengths give the monotonicity of $-(t\lambda)^{-1}\log\mathfrak{b}$ and its limit $s_{E_{\mathrm{e}}} - s_0 = \sigma$; the lower bound on $-\log\mathfrak{b}(t,\lambda)$ follows from the convexity in t continued from t_0, with the Casimir term $R(\lambda_0) \ge R(\lambda)$; and $q(2t) \le q(t)^2$ gives the bound on ϱ.

For A(4)(a), $g(h) = \log(q(h)/\|T_0\|^h)$ is convex in the height h and $g(h)/h \to -\varepsilon(\lambda)$; its chord from h_0 to h_1, continued beyond h_1, lies below it, so $(h_1-h_0)\varepsilon \le g(h_0) - g(h_1)$, with $g(h_0) \le \log(e^{\mathsf{G}(h_0)}-1)$, the top eigenvalue lying in the even part, and $-g(h_1) \le \log(2/(1-r(h_1)))$; (a) in the two lengths, with A(3), gives the bound on σ. For (b), $\Lambda(a,b') = \log\|T_0\|$ over the section (a,b') is symmetric in its two arguments, $F_0(a,b',c) = c\Lambda(a,b') + \mathsf{G}(c;a,b')$ with any side as the time, $\Lambda_1(b') = \lim_a \Lambda(a,b')/a$, and $\varphi(a) = \Lambda(a,b') - a\Lambda_1(b') = R(a) \ge 0$ is convex and does not increase; writing $F_0(a,b',k'a)$ with the time along a gives $\varphi(a) \le \varphi(a)/k' + \mathsf{G}(a;b',k'a)/(k'a)$, which is the bound; the same loop in the second transverse side and $\delta(M^{\times 3}) = \mathsf{G}(M;M,M) + M\varphi(M;M) + M^2[\Lambda_1(M) - MP]$ give the bound on $\delta(M,M,M)$. (c) is the left bound on σ in A(3) with $q(t) \ge e^{-tE_{\mathrm{e}}}$, $Z[0](2t) \le \|T_0\|^t Z[0](t)$ and (b) at $k' = 2$.

For (d), let p run over the plaquettes of the planar rectangle and let $\gamma_{\mathsf{p}} = -1$ exactly on the planar edges crossed by the straight dual path from p to the free boundary, lifted to the bonds of the directions $1,2$ at all heights: $d\gamma_{\mathsf{p}} = -1$ exactly on the stack of the N' plaquettes over p, the plaquettes through the third direction keep their weight, and $\operatorname{tr} U(C)$ takes the product of the signs of the enclosed p. Averaging over all products of these changes, with $h_{\mathsf{p}} = \frac{\beta}{2}\sum_{\text{stack}} \operatorname{tr}\partial U$, $|h_{\mathsf{p}}| \le \beta N'$, gives $\langle \operatorname{tr} U(C)\rangle = \int \operatorname{tr} U(C) \prod_{\text{inside}} \tanh h_{\mathsf{p}}\, \mathrm{d}\mu'$ with a probability measure μ'. (e) is the same averaging over the reversal of one column inside each container, at fixed walls.

Step 3: how bounds travel; the overlap (A(5), A(6)). A(5)(a): with $B_\pm = \frac12(1 \pm C^k)T_m^N \ge 0$, $B_+B_- = 0$, $\operatorname{Tr} T_m^{2N} = \sum_\pm \operatorname{Tr} B_\pm^2 \le \sum_\pm (\operatorname{Tr} B_\pm)^2 = \frac12(x^2+y^2)$, $x = \operatorname{Tr} T_m^N$, $y = \operatorname{Tr}(T_m^N C^k)$. With $z_e = Z_{e,m} \ge 0$ and $z_{00} \ge \|T_m\|^N$: $Z[(k,m)] = \sum_e (-1)^{e\cdot k} z_e \ge Z' - 2(Z' - z_{00})$, $\sum_k Z[(k,m)] = 4z_{00}$, $\operatorname{Tr} T_m^{2N} \le \sum_e z_e^2 = \frac14 \sum_k Z[(k,m)]^2$, and $Z[((1,0),m)] + Z[((0,1),m)] - Z' = Z[((1,1),m)] - 4z_{11}$. (b): $q(N) = \frac12(Z[0] - Z[n])(N) = \operatorname{Tr}(\frac12(1-C^k)T_0^N)$ is log-convex in N (Lem. 11.2), $q = \frac12 Z[0](1-r)$ with $0 \le 1 - r \le 1$, and the volumes cancel in $Z[0](N)^2/Z[0](N-j)Z[0](N+j) = e^{2\delta(N) - \delta(N-j) - \delta(N+j)}$. (c): $\operatorname{Tr}(P_S T_m^{N'}) \ge \lambda^{N'}$, and $\operatorname{Tr}(P_S T_m^{N'}) \le \lambda^{N'-N''}\operatorname{Tr}(P_S T_m^{N''}) \le \lambda^{N'-N''}\operatorname{Tr} T_0^{N''}$ by Lem. 11.2 and A(1), with $\operatorname{Tr} T_0^{N'} = \|T_0\|^{N'}\Theta(N')$. (d): the first inequality is the log-convexity of q between two heights, the second the monotonicity of $q(N)^{1/N}$ (Lem. 11.2) with $Z[0](N) \ge \|T_0\|^N$, and the third is Lem. 11.3 for the class z with the time μ, divided by the periodic numbers, the volumes cancelling: $k'(n'+m) - (k'-1)m - (k'n'+m) = 0$; the denominator $\mathsf{Z} = Z[(0,m')] \le Z[0]$ stands the right way. (e): put $k' = 2L'+1$, $n' = L'^{j-1}(L'-1)$, $m = L'^{j-1}(L'+1)$ and $m_1 = m_2 = m/2$, so that $n' + m = N'$ and $k'n' + m = L'N'$, and enlarge the three sides of the cube N' one after the other.

In a direction that lies in a twisted plane of z the third inequality of (d) applies; for a class with one twisted plane, in the normal of that plane (c) applies at $M_1 = N'$, $M_0 = N'/L'$, with the exponent $L'^2/(L'-1) \le 2L'+1$. A(1) passes between the three expressions, and all excesses of the auxiliary boxes are bounded against the cube N'/L' by the last inequality of A(2). A(6): $T_m^\tau C^k$ has the kernel $\Psi \circ g$, so $Z[(k,m)_\mu] = \iint \Psi\,\Psi\circ g$ by A(1), and μ_2 is the joint law of the layers 0 and τ; $w = 2\Psi\,\Psi\circ g/(\Psi^2 + \Psi^2\circ g)$ is g-invariant, which gives the three expressions of r'. $\int\sqrt{f_1 f_2} \ge \int \min(f_1,f_2) = 1 - \mathrm{TV}$ and the Cauchy–Schwarz inequality give the bracket; on E_c the integrand $\Psi\circ g/\Psi$ is $\ge c$, off it $< c$, and $\langle(\Psi\circ g/\Psi)^2\rangle_2 = 1$; if $gE = E^c$, $r' = 2\int_E \Psi\,\Psi\circ g/Z'$; for odd Φ, $\langle\Phi\rangle_2 = E_{\bar\nu}[\Phi\mathsf{m}]$, $\bar\nu = \frac12(\mu_2 + g_*\mu_2)$, and the Cauchy–Schwarz inequality with $1 - \sqrt{1-x} \le x$ gives the next bracket; $g_*(p_1\otimes p_1) = p_1\otimes p_1$ gives the bound by $\mathrm{TV}(\mu_2, p_1\otimes p_1)$; the bound on $\mathcal{G}\cap g\mathcal{G}$ is pointwise; Jensen's inequality for $-\log$ gives the entropy bound; and an affinity does not

decrease under a map, whereas relative entropy does not increase, while Φ intertwines the projection on the two layers with g.

For the odd functions, cut the period into $2k'$ slabs of ϑ layers: $Z[0] = \operatorname{Tr} Y^{2k'}$ and $Z[n] = \operatorname{Tr}(\mathsf{Z}Y^{2k'})$ with the transfer operator Y over ϑ layers and the flip Z in one layer, $\mathsf{Z}Y = Y\mathsf{Z}$; the projections $\frac{1}{2}(1 \pm \mathsf{Z})$ carry the weights $\frac{1}{2}(1 \pm r[n])$, the operator of D' maps each of their ranges into the other, and Hölder's inequality for traces over the $j-1$ and the $2k'-j-1$ slabs between the two functions gives the bound.

Step 4: reflection positivity, chessboard bounds, the mixture (A(7)–(9)). For a slab with α, α' crossings on its two sides and the columns through μ on the side of α', the trace formula gives

$$Z[(k,m)_\mu; \overline{\theta F} F] = \sum_e (-1)^{e\cdot k} \| T_m^{\alpha/2} \hat{F}\, T_m^{\alpha'/2} P_e \|_2^2,$$

$\hat{F}$ the slab operator; this is ≥ 0 at $k = 0$, $\sum_k (-1)^{e\cdot k}$ of it is $4\|T_m^{\alpha/2}\hat{F}T_m^{\alpha'/2}P_e\|_2^2 \geq 0$, and summed over the classes allowed by $\mathfrak{s}$ the signs add up to $\prod_j (1 + (-1)^{e_j}) \geq 0$. For $k \neq 0$, $\beta > 0$, write $e^{\frac{\beta}{2}\operatorname{tr} u} = \cosh \cdot (1 + \tanh)$ at $\frac{\beta}{2}\operatorname{tr} u$, and let one plaquette of the flipped row of one crossing carry $\chi_J/\cosh$, χ_J the character of spin J, and the others a factor concentrated at $\operatorname{tr} u = 0$, through which integer spins alone pass; the part even under the flip falls like $c^J/(2J)!$, while the odd part is a Fourier coefficient of a function with the pole of $\tanh(\beta \cos\theta)$ and is $\geq e^{-cJ}$ for infinitely many J: the form is negative. $\sum_n Z[n] \leq 8Z[0]$ is A(1).

For the last assertion of A(7), $Z[(k,m)_\mu; F] = \operatorname{Tr}(\hat{F} T_m^{N_\mu - \vartheta + 1} C^k)$ with one slab, and the Cauchy–Schwarz inequality for traces. On a board a reversed plaquette in a mirror is a unitary flip commuting with $\mathsf{K}^{1/2}$, so $Z_\beta[\eta; \prod F_\kappa] = \operatorname{Tr}(C_r \mathfrak{Y}_r \cdots C_1 \mathfrak{Y}_1)$, and Hölder's inequality for traces does not see unitaries; the adjoint of a slab operator is the operator of the mirror copy of the slab that carries the complex conjugate function, so the norms of the slab operators are fold integrals with the copies complex conjugated in the cells odd relative to the given one (equation YM3_I:(1749)), which are ≥ 0.

Put each transverse electric-flip cut on a wall of the board. A cell function reads only bonds with both endpoints in that cell; the flip changes only bonds crossing the wall, so it fixes every inserted cell function, including a complex one. Applying the flip on all layers of a slab preserves its action and Haar measure, hence its inserted slab operator commutes with the flip. The crossing operator and its square root commute as well. Thus each $\mathfrak{Y}_i$, and each $\mathfrak{Y}_i^* \mathfrak{Y}_i$, commutes with the electric-sector projections. In Z_w the flips add up to $\mathsf{D}_m^w = \sum_e \hat{w}(e,m) P_e \geq 0$, which commutes with all $\mathfrak{Y}_i$: $Z_w[\prod F_\kappa] = \sum_m \operatorname{Tr}(\mathsf{D}_m^w \mathfrak{Y}_r \cdots \mathfrak{Y}_1)$, and $|Z_w| = Z[0] \sum w(n) r[n]$ lies between $w(0)Z[0]$ and $8w(0)Z[0]$ by A(1). Comparing $Z_w[\operatorname{Fold} F]$ with $\sum_n w(n) Z[n; \operatorname{Fold} F]$ and with its term $n = 0$ gives the two bounds of $\|\cdot\|_w$. A covariant function of a cell is $G(U,s) = F(\gamma U)$ with $d\gamma = s$ on the cell and F invariant under the gauge transformations with values ± 1, and $\mathsf{M}_n[\prod G_\kappa] = Z[n; \prod F_\kappa]$ with the columns in the mirrors. The functions of the cells that cancel the weight of the rods of plaquettes through the centres of the cells make $Z[n; \operatorname{Fold} F] = Z[0; \operatorname{Fold} F]$ for every n, which gives equality; the functions that remove the reversed columns from the weight give the failure of the fold in a class, since $Z[n] < Z[0]$.

In a class: Hölder's inequality with the first bound for $|G|^q$; the density bound by the first bound for $h_n^{\mathsf{N}/(\mathsf{N}-1)}$ and the entropy inequality, and Pinsker's inequality. Without a board, one mirror and the Cauchy–Schwarz inequality for $\mathsf{M}_{(k,m)}[\overline{\theta G} G'] = \operatorname{Tr}(Y_m[G]^* C^k Y_m[G'])$ give $r[n]|\mu_n(G)| \leq \langle \overline{\theta F} F \rangle^{1/2}$ when $n = (k,0)_\mu$, that is, when μ lies in every twisted plane of n; the same for $G - \langle F \rangle$ gives the two-sided bound; a second mirror gives the quarter; and the event $\{\prod_{\text{plane}} s = -1\}$ of a plane normal to μ inside the half has periodic probability 0 and probability 1 in a class twisted in that plane.

For pairs, summing over the signs inside each half gives $\mathbb{M}_{\delta_{(k,m)_\mu}}[\overline{\theta G} G'] = \operatorname{Tr}(Y_m[G]^* C^k Y_m[G'])$, and $\sum_k w((k,m)_\mu) C^k = \sum_e \hat{w}(e,m) P_e \geq 0$. On $a_K \mathbb{Z}^3$ a closed sign field is $d\chi$, χ on the bonds, closed sign fields being exact on every box of sites: a class is the image of its measure under

$U \mapsto \chi \operatorname{per}_\ell U$, the plain Wilson measure of a field periodic up to gauge transformations with values ± 1, and its law on the invariant local functions depends neither on s nor on χ; the proof of [7, Cor. B.1] carries over, and for other functions the columns are put on the edges of the board.

Its clause (e) holds with p^w_ℓ: with the time μ and n_μ cells in the direction μ, $Z_w[\operatorname{Fold} F] = \sum_{e,m} \hat w(e,m) \operatorname{Tr} (P_e B_m P_e)^{\mathsf{n}_\mu/2}$, where $B_m = \mathfrak{Y}^*_m \mathfrak{Y}_m \ge 0$ is the operator of a slab of cells and its mirror copy and commutes with P_e, the same operators for every number of cells in the direction μ. For $\mathsf{n}_\mu = q\mathsf{n}'_\mu$, $\operatorname{Tr} B^{q\mathsf{n}'_\mu/2} \le (\operatorname{Tr} B^{\mathsf{n}'_\mu/2})^q$ for $B \ge 0$, $\omega \le \omega^q$ for $\omega = \hat w(e,m)/\underline w \in \{0\} \cup [1,\infty)$, and $\sum_i x_i^q \le (\sum_i x_i)^q$ for $x_i \ge 0$; so $Z_w[\operatorname{Fold} F]/\underline w$ over n_μ cells is at most the q-th power of the same number over n'_μ cells, and its root of the order of the number of cells does not increase when the board grows in one direction.

The three directions in turn carry this from the board of $\Lambda_{K,\ell'}$ to that of $\Lambda_{K,\ell}$; at $F = 1$ it is $p^w_\ell \le p^w_{\ell'}$, and the inequality at F and at $F = 1$ gives $\|F\|_{w,\ell} \le \|F\|_{w,\ell'} e^{|3|(p^w_{\ell'} - p^w_\ell)}$. At $w \equiv 1$, $\hat w(e,m) \in \{0,4\}$, and the same holds without the division by $\underline w$. A(9): for a closed $s = \eta^n d\gamma$, γ is fixed up to the closed sign fields on the bonds and $e^{-\beta A_s}(U) = e^{-\beta A_{\eta^n}}(\gamma U)$; integrating against a centre-blind F gives the first identity, $2^{2|N|-2}$ being the number of the closed sign fields of one class. The mean of $\sum_n e^{-\beta A_{\eta^n}}$ over all sign changes γ is $\hat{\mathsf{w}}_{\mathrm{mix}}$, which is therefore ≤ 8, and centre-blind, gauge invariant and invariant under the isometries.

Writing $e^{-\beta A_s} = e^{-\beta \bar A} \prod_{p:\, s_p \ne \operatorname{sign} t_p} z_p$, the sum over the closed s becomes the sum over the sets $\Gamma = \{p : s_p \ne \operatorname{sign} t_p\}$ with boundary $\mathsf{M}(U)$. At the constant field $U^0(x,\mu) = \cos\alpha + \sin\alpha\, i\sigma_\mu$, $\sin^2\alpha = 2^{-1/2}$, all t_p vanish, and a sixth mixed derivative of $\log \mathsf{w}_{\mathrm{mix}}$ along a hexagon of edges of one cube, two of which lie in no common plaquette, has the modulus $\beta^6(2\sqrt2 - 2)^{1/2} = 0.91018\,\beta^6$, while a sum of functions of single plaquettes has none. On one cube the plaquette variables of the faces, carried to a common corner, multiply to $\mathbf{1}$; if the angles add up to less than π, the product of the signs of the traces is 1, and a cube with six angles $\frac\pi6$ and the product -1 shows that π is sharp.

On $\mathfrak{S}$ the sign field $s(U)$ is locally constant and closed, $\mathsf{w}_{\mathrm{mix}} = e^{-\beta A_{s(U)}} \Xi(z)$ with $\Xi(z) = \sum_\Gamma \prod_{p\in\Gamma} z_p$ over the closed sets Γ, and the expansion of $\log \Xi$ in connected sets at $|z_p| \le \frac{1}{200}$ gives the $E(X)$, which vanish unless X is connected and a union of vortices, connected for the relation of A(9): two plaquettes are neighbours if they are faces of one unit cube, that is, dual bonds with a common dual site. For this relation the faces of one cube are pairwise neighbours, so each cube meets one connected component of a closed set only, the components of a closed set are closed, the closed sets are the unions of vortices pairwise apart, and Ξ is a gas of vortices; for a vortex X that holds no smaller one the only clusters with the support X are the repetitions of X, whose terms add up to $\log(1 + \prod_{p\in X} z_p)$.

The last bound is the smallness, on $\mathfrak{S}_n$, of the weights $\sum_{s\in[n']} e^{-\beta A_s}/\mathsf{w}_{\mathrm{mix}}$ of the other classes, whose sets Γ wind about the torus and so have at least $\min_\mu N_\mu$ plaquettes.

Step 5: one inequality; loops (A(10), A(11)). Under the hypothesis $D_2(l_0) \le \frac{1}{20}$ of A(10), one inequality between the partition functions of the periodic cubes l_0 and $2l_0$, Lem. 11.4 gives $18D_2(2^i l_0) \le (18D_2(l_0))^{2^i}$, hence $\delta(s,u,w) \le 0.14108\,(suw/l^3)D_2(l)$ for $s,u,w \ge l \in \{2^i l_0\}$ and $\mathsf{r}_h(l,l) \le \frac{1}{18} e^{-m_* h}$. Lem. 11.5 then gives, for $\bar\varepsilon \ge \varepsilon_{\mathrm{v}}(l)$ and an integer $h \ge 2l$ with $e^{-m_* h} \le (h\bar\varepsilon)^4$, $h\bar\varepsilon \le \frac{1}{10}$ and $2(h/l)D_2(l) \le \frac14$,

$$2l\,\varepsilon_{\mathrm{v}}(2l) \le 4.84\,(h/l)^3 (l\bar\varepsilon)^4,$$

whose iteration from l_1 gives $l\varepsilon_{\mathrm{v}}(l) \le e^{-\sigma_* l^2}$; the bound $\sigma_* \ge \min\{\frac16 m_*^2, m_*/l_0\}$ is the definition of l_1. With $2\zeta \le h\varepsilon_{\mathrm{v}} + \mathsf{r}_h$ and the doubling of the short sides (Lem. 11.5) this is the bound on the boxes (t,λ,h) and on the cubes; the bound on $\Lambda_-(N)$ is the same chain for the norm of T_0 on the odd functions, which is $\|T_0 P_{10}\|$ by A(1). The loop bounds are Lem. 11.5 on the even cubes N, the factor $8^{T/N}$ tending to 1 as $N \to \infty$; loops made of rows go the same way, row by row, with the norms of the transfer operators with static charges, the excess entering through $e^{1.13 D_2(N/2)}$.

Uniqueness and convergence are Lem. 11.7. Under the hypothesis $\mathsf{e}(l) \le 0.069$, Lem. 11.4 gives $\theta \le 0.8901$, $\mathsf{e}(4l) \le \theta^4/12.9$ and $D_2(8l) \le 20\,\mathsf{e}(4l)^2 \le 20\theta^8/166.41 \le 0.047355$; for a cube $N \ge 2l$ and the largest $M \in \{2^j l\}$, $j \ge 1$, with $M \le N$, $\delta(N^{\times 3}) \le (N/M)^3 \delta(M^{\times 3}) \le 8 \cdot 0.01717\,\theta^{M/l} \le 0.13736\,\theta^{N/2l}$; and $1 - r[n] \le 3(1 - e^{-\mathsf{G}}) \le 3\delta$ (Lem. 11.4), with $3 \cdot 0.13736 \le 0.4121$. A(11): ν is reflection positive at site mirrors and at bond mirrors, the exact statements on periodic Gibbs states and translation invariance comparing it with the periodic tori, where A(7) holds at $k = 0$.

For a rectangle of height T, the paths from a point of the mirror that run k bonds up, R across and k down, with $k = h + 1, h$ at a site mirror for $T = 2h + 1$ and $k = h, h - 1$ at a bond mirror for $T = 2h$, form half loops whose Gram matrix has the entries $2\mathsf{m}(R, T + 1)$, $2\mathsf{m}(R, T)$, $2\mathsf{m}(R, T - 1)$; its positivity is the log-convexity. So $\mathsf{m}(R, T)/\mathsf{m}(R, T - 1)$ does not decrease in T, $\mathsf{m}(R, T) \ge \mathsf{m}(R, 1)^T \ge \mathsf{w}^{RT}$ by the symmetry, V exists, is concave in R by the log-convexity in R, and $V(R)/R$ does not increase; the bounds on w, on m for $R, T \ge 2$ and on $V(1)$, $V(R)$ come from integrating the bonds of the loop in their plaquettes.

Step 6: purity; small cubes; constructions (A(12)–(14)). A(12): $\mathsf{P} = \mathsf{Y}^*\mathsf{Y} \ge 0$ commutes with C^k; with $B_\pm = \frac{1}{2}(1 \pm C^k)\mathsf{P} \ge 0$ and $d = \operatorname{Tr}(C^k\mathsf{P})/\operatorname{Tr}\mathsf{P} \in [0, 1]$,

$$\operatorname{Tr}\mathsf{P}^i = \sum_{\pm} \operatorname{Tr} B_\pm^i \le \sum_{\pm} (\operatorname{Tr} B_\pm)^i = (\operatorname{Tr}\mathsf{P})^i \sum_{\pm} (\tfrac{1 \pm d}{2})^i;$$

convexity in d gives $d \ge (\mathfrak{q}_i - 2^{1-i})/(1 - 2^{1-i})$, at $i = 2$ the display gives $d^2 \ge 2\mathfrak{q}_2 - 1$, and as $i \to \infty$ it gives $\|\mathsf{P}\| \le \frac{1}{2}(1 + d) \operatorname{Tr}\mathsf{P}$; and $r[n] \ge Z[n; F \otimes \theta F]/Z[0] = d\,w_F$. For block fields, the averaging map commutes with the flip in a wall of the blocks of all orders, L being odd. A(13): with the time μ, $Z_{e,0}/Z[0] = \frac{1}{4}\sum_k (-1)^{e \cdot k} r[(k, 0)_\mu]$, and the three classes with $k \ne 0$ have the quotients r_1, r_1, r_2: the first bound; $e^{-\mathsf{G}} = \|T_0\|^N/Z[0] \le Z_{00,0}/Z[0]$; $D_q(N)$ is a sum of defects of q-fold enlargement, the first $q^2[qF_0(N^{\times 3}) - F_0(qN, N, N)] \ge q^2(q - 1)\mathsf{G}(N)$; the bound on $Z[0](N^{\times 3})^2/Z[0](2N, N, N)$ is A(5)(a); if $r_1 \le \frac{1}{3}$, each sector trace at the height N is at most $\frac{1}{2}Z[0]$, and with $\operatorname{Tr}(P_e T_0^N) \le (\operatorname{Tr} P_e T_0^{N/2})^2$ this gives $(\operatorname{Tr} T_0^{N/2})^2 \ge 2 \operatorname{Tr} T_0^N$, that is $\mathsf{e}(N/2) \ge \log 2$; and $\Phi \ge F_0 - N \log \|T_0\| = \mathsf{G}(N)$ on the cube gives the bound on $\mathfrak{b}$.

A(14)(a): there are two Haar-preserving, gauge covariant bijections $\Phi_\pm$ of the fields, moving only the bonds of the directions μ, ν inside a tube of radius h' about the reversed column, which spread the flux of the column over the cross-section of the tube; pointwise, $\frac{1}{2}[A_\eta(\Phi_+ U) + A_\eta(\Phi_- U)] - A(U)$ is at most a sum over the layers of the squares of the spread curvature, of order $(h' - \frac{1}{2})^{-2}$ in total, and of the actions of ladders of plaquettes of length of order h' weighted by the angles of rotation. As $\Phi_\pm$ keep dU, $Z[\eta] = \int e^{-\beta A_\eta(\Phi_\pm U)}\,\mathrm{d}U$ for both signs; the arithmetic mean of the two integrands is at least their geometric mean, and Jensen's inequality in the periodic measure gives $\log r \ge -\beta\langle \frac{1}{2}[A_\eta(\Phi_+ U) + A_\eta(\Phi_- U)] - A(U)\rangle$.

For $\beta \ge 1$ the mean actions $\bar a_{12}, \bar a_{13}, \bar a_{23}$ of the plaquettes of the three orientations in the periodic measure have $\bar a_{12} + \bar a_{13} + \bar a_{23} \le \lambda(\beta)/\beta$, by $\log Z[0] \ge -|N|[\frac{9}{2}\log\beta + 30]$ and the convexity of $\log Z[0]$ in β; and the mean action of a ladder of $1 \times k$ plaquettes of one orientation is k times the mean of that orientation. The second form is the choice of h' of the order of $(\beta/\lambda(\beta))^{1/3}$, and for several twisted planes the tubes are disjoint. (b): $\log Z[0] \ge -|N|[\frac{9}{2}\log\beta + 30]$ by an elementary estimate; the chessboard estimate for $e^{\frac{\beta}{2}(1 - \frac{1}{2}t_p)}$ in one cell bounds its periodic mean by $\beta^{121.5} e^{810}$, so up to a factor 2 in the partition function of the cube all plaquettes lie within $(130\pi^2 \log\beta/\beta)^{1/2}$ of $\mathbf{1}$ once $\log\beta \ge 147.6$; cut the cube into eight free blocks, fix the gauge on a comb in each, and bound the action from below by a Gaussian whose determinant is the product over the inner Neumann momenta, the blocks having a side of the order $\beta^{1/6}$.

Step 7: the window (B(1)). The first sentence of B(1) is A(1) and A(2) at $N_\mu = s_\mu/a_K$, $\beta = L^K b$, with $|s|p_\infty = |N|P$; $\delta_K^0 \ge 0$ is the definition of P. On the cubes $2L^m$, $m \ge m_0(b)$, $b \ge \gamma^{-2}$, Lem. 11.1 places $\log Z_K[0]$ within $C_*(b)\ell^3$ of a vacuum energy whose density varies boundedly from cube to cube, no constant depending on K or m; the monotone density squeezes a box with sides

$\geq 2L^{m_0(b)}$ between two cubes, so $|\log Z_K[0](s) - \epsilon_K|s|| \leq C_*(b)|s|$ and $p_\infty \geq \epsilon_K - C_*(b)$, whence $\delta^0_K \leq 2C_*(b)|s|$ and, on the cubes, the explicit bound. For a compact set $\mathsf{J} \subset [b_-, b_+]$ of labels and $s_{\min} > 0$, all labels of J and all sides $\geq s_{\min}$ are brought into this range at once by

$$Z_K[n](s;b) = Z_{K-j}[n](L^j s; L^j b),$$

with the least j for which

$$L^j b_- \geq \gamma^{-2}$$

and

$$L^j s_{\min} \geq 2L^{m_0(L^j b_+)},$$

which exists because

$$L^{-j} 2L^{m_0(L^j b_+)} \to 0$$

: for $b \in \mathsf{J}$, $L^j b \geq \gamma^{-2}$ and, m_0 not decreasing, $L^j s_{\min} \geq 2L^{m_0(L^j b)}$; and, C_* not decreasing, $C_*(L^j b) \leq C_*(L^j b_+)$, so that $C = L^{3j} C_*(L^j b_+)$ serves for all $b \in \mathsf{J}$ and $K > j$. The bound on

$$\operatorname{Tr} T_m^{t/a_K} / \|T_0\|^{t/a_K}$$

follows from $\operatorname{Tr} T_1^M \leq \operatorname{Tr} T_0^M$ and $\mathsf{G} \leq \delta$. A lattice with N_μ/β in a compact set and β large is a box of a cutoff K at a label in $[1, L)$ with sides in a compact set, where these bounds hold uniformly, by the above at $\mathsf{J} = [1, L]$; for bounded β, $\delta^0 \leq 6\beta|N|$. On a box with L-adic sides the remaining cutoffs are finitely many, each with a finite δ^0_K and $r_K[n] > 0$, and $\delta^n_K - \log r_K[n] = \delta^0_K$ is bounded.

Step 8: floor and two cutoffs on mixed boxes (B(2)(a), B(4)). After a dilation, s is a centre sector of some T_ℓ (Lem. 11.9), and the chain of T_ℓ restricted to the fixed fields is a chain for $Z^z_K(s;b)$. At the level i the fluctuation integral over the fixed fields is, by Prop. 11.10 in the coordinates (requiring **(Hess)** for the full fluctuation Hamiltonian, and with least displacement $s_{\min} L^{K-i}$ in the units of that level), the n-th root of that of the cover up to $\epsilon_i \leq C|s|L^{3(K-i)} e^{-c s_{\min} L^{K-i}}$. With $E(d) = \sum_{i \leq K-d} \epsilon_i$, bounded independently of K, Σ the small-field mass of the cover at the depth d and C_0 formed from the constants,

$$\log Z^z_K(s;b) \geq \frac{\log \Sigma}{n} - E(d) - |s| L^{3d}[C_0 + \tfrac{9}{2}\log(L^d b)], \qquad \Sigma \geq \tfrac{1}{2} Z_{K,\ell}, \qquad \frac{|s|}{\ell^3} \log Z_{K,\ell} \geq |s|\, p_\infty(K,b),$$

the second when the large-field part of the cover is at most $\frac{1}{2}$, the third by A(2); finitely many K remain, at which the integrand is positive. Against the small periodic cube $\ell_\flat(b)$, box and cube being two sectors of one torus, the same chain gives the bound with C_G: the cover drops out. For two cutoffs, the small-field masses at K and $K+1$ are compared pointwise at fixed fields as on T_ℓ, the sector taking $|s|/\ell^3$ times the error of the cover; large fields are rare in the box, since an event of a slab passes by the Cauchy–Schwarz inequality at mirrors, with the power $\frac{1}{8}$, to the small periodic cube $\ell_\flat(b)$, where Prop. 4.54 controls it; box and cube are two sectors of one torus, whose partition function cancels.

The rate $L^{-K/10}$ and the uniformity in the label are those of the cover, with the arithmetic of the constants, which makes the condition on K void for $|s| \leq (\log b)^{2\mathfrak{p}_0 - 1}/C_V$. For two classes the terms $\frac{|s|}{\ell^3}\Delta_K$ cancel in the difference of the two laws, and the increments are summable. So, for a compact set J of labels, B(4) at J for z and for the class 0, for which s is mixed too, gives $|\log r_{K+1}[z] - \log r_K[z]| \leq 2CL^{-K/10}$ for $K \geq K_0(\mathsf{J})$; the first sentence of B(1) bounds $r_K[z]$ from below, uniformly on J, at the finitely many $K < K_0$ at which s is a box of the cutoff K; and as $\delta^0_K \geq 0$, $\log Z^z_K(s;b) - |s|\, p_\infty(K,b) = \log r_K[z] + \delta^0_K \geq \log r_K[z]$: the floor of B(2)(a) holds with $C(s,b)$ bounded on J, and $\inf_K r_K[z] > 0$ uniformly on J.

On the boxes with sides in $2L^{\mathbb{Z}}$ the left member is the increment of $\log r_K[z]$, which tends to 0 by B(2)(c) and B(3), plus the left member of the periodic class, for which every such box is mixed.

Step 9: compactness, no zero, the full sequence (B(2)(b), B(3)). By B(1) at a compact set J of labels that holds the b_j, $i\, e^{-t_0 E_i^K} \leq e^{2Ct_0 l_1 l_2}$: Lem. 11.2(b) applies, so every sequence (K_j, b_j), $b_j \to b$, has a subsequence along which all levels converge, the sector traces divided by $\|T_0\|^{t/a_K}$ tend

to $\mathcal{W}(e;m)(t;l)=\sum_i e^{-tE_i^{m,e}}$, holomorphic in $\operatorname{Re} t>0$, and $v_{(k,m)_\mu}=\sum_e(-1)^{e\cdot k}\mathcal{W}(e;m)/\mathcal{D}$, $\mathcal{D}=\sum_e\mathcal{W}(e;0)\geq 1$, in each of the three expressions. The order of A(1) and A(5)(a) pass to the limit, and so does the crossing relation, for the sector weights $w^{[\mu]}_{e,m}=\frac14\sum_k(-1)^{e\cdot k}v_{(k,m)_\mu}=\mathcal{W}(e;m)/\mathcal{D}$, each the sector trace divided by the periodic trace of its own time and section: on each lattice the relation of A(1) divided by $Z[0]$ is the same relation between the numbers $Z^{[\mu]}_{e,m}/Z[0]=\frac14\sum_k(-1)^{e\cdot k}r[(k,m)_\mu]$, which converge to the $w^{[\mu]}_{e,m}$. A weight $w=\mathcal{W}(e;m)/\mathcal{D}$ vanishes at one height if and only if at all; it is real-analytic in a side x of the section, and if it vanishes at x_0 then, for $q=L^p$ and $\mathsf{c}_q\geq\mathcal{D}(t/q;l_x)$ from B(1),

$$w_t(x)\leq\mathcal{W}(t;l_x)\leq\mathcal{W}(t/q;l_x)^q\leq\left[\mathsf{c}_q\,w_{t/q}(x)\right]^q=O(|x-x_0|^q),$$

so $w\equiv 0$ in that side, and, side after side, on all boxes: this is the alternative of B(3), which passes to the limits of Step 10, and for a class with one twisted plane, whose quotient is the sum of its four sector weights with the time normal to its plane, it holds for v_z. With the floor on the box $(1,1,1)$, which is mixed for every class, this shows that the space of $H_{l,1}$ is not $\{0\}$ and, with the floor along the normal of A(5)(c) and compactness, B(2)(b); with B(5), which makes each electric weight on a small cube at least $\frac14(1-2C_\square t)$, each electric sector is not $\{0\}$, so $v_z<1$. On boxes with sides in $L^{\mathbb{Z}}$, B(4) makes the increments of $\log r_K[z]$ and of $L\log Z_K[0](s)-\log Z_K[0](s^{\mu+})$, $s^{\mu+}$ the box with s_μ multiplied by L, summable uniformly on compacts of labels, the terms Δ_K cancelling in the second: this is (R1) of Prop. 11.12 on the ray of the boxes with sides in $L^{\mathbb{Z}}$, and Prop. 11.12 gives the convergence of all levels along the full sequence, of $r_K[z]$ on every L-adic box, of δ^0_K and of the differences of the norms.

The same holds along every sequence (K_j,b_j) of cutoffs and labels with $K_j\to\infty$ and $b_j\to b\in(0,\infty)$, in the place of the full sequence at the fixed label b; these sequences form a class that is closed under subsequences. (R1) holds along each of them: each $r_K[z](s;\cdot)$ and each difference $L\log Z_K[0](s)-\log Z_K[0](s^{\mu+})$ is continuous in the label at a fixed K, and their increments in K are summable uniformly for the labels in a compact set, so that they converge uniformly on compacts of labels to limits that are continuous in the label, and so converge along (K_j,b_j). The proof of Prop. 11.12 then runs along such a sequence as it stands: its only input, the window, is uniform for the labels in a compact set (B(1): the supremum of δ^0 over the lattices with N_μ/β in a compact set is finite), and Lem. 11.2(a), (b) and the turning of the box do not read the label.

So all levels, $r_K[z]$ on every L-adic box, δ^0_K and the differences of the norms converge along every such sequence. Two such sequences with one limit label b interlace into a third one, along which the limits exist: the limits depend on b alone. They are continuous in b: for $b_i\to b$ a choice of $K_i\to\infty$ with the values at (K_i,b_i) within $1/i$ of the limits at b_i gives a sequence of the class. And the convergence is uniform for the labels in a compact set, since otherwise a compact set of labels would hold a sequence (K_j,b_j) with $b_j\to b$ whose values stay at a fixed distance from the limits at the labels b_j, which tend to those at b, against the convergence along it. E_{mag} and E_{el} are Lem. 11.2 for the limit lists, the space of $H_{l,1}$ not being $\{0\}$.

On the cubes, $L^3\log Z_{K,\ell}-\log Z_{K,L\ell}$ is a difference of periodic numbers of two cubes in which the terms Δ_K of B(4) cancel, so $D(t)$ exists; it depends on t alone by the argument of Step 10; $\delta^0_K(\ell^{\times 3})$, telescoped over the cubes $L^r\ell$, gives $\delta(t)=\sum_r L^{-3(r+1)}D(L^rt)$; and the behaviour of D as $t\to 0$ is the asymptotics of Prop. 11.14(b), (e) for the periodic cubes, the bulk cancelling, carried to every cutoff by B(4) as in Step 11. By Thms. YM3_I:AJ and YM3_I:AD, through Thm. YM3_I:AD, the periodic quotients are normal families in the label, and B(6) gives the analyticity.

Step 10: every lattice; two blocking factors (B(2)(c), B(3)). By Lem. 11.6 and B(1) the quotients are equicontinuous in the sides for N_μ/β in a compact set; with Step 9, in its form for the sequences (K_j,b_j) with a moving label $b_j=L^{-K_j}\beta_j\to b$, this gives a continuous $V^L_z(s;b)$, the limit along every sequence with $L^{-K_j}N_j\to s$, $L^{-K_j}\beta_j\to b$, with the Lipschitz bound of B(3); in each side, at real values of the others, it is $\sum_e(-1)^{e\cdot k}\mathcal{W}(e;m)/\mathcal{D}$, a quotient of Laplace transforms, and

real-analytic. The constants may be chosen with the blocking factors $L_1 = L$ and $L_2 = L+2$, and $\log L_1/\log L_2 \notin \mathbb{Q}$; so for $b_1, b_2 > 0$ there are $K_j, M_j \to \infty$ with $K_j \log L_1 - M_j \log L_2 \to \log(b_2/b_1)$, and $\beta_j = L_1^{K_j} b_1$, $N_j \approx \tau\beta_j$ is a sequence of both scales, of the label b_1 at the scale L_1 and of the moving label $L_2^{-M_j}\beta_j \to b_2$ at the scale L_2, to which Step 9 applies in its form for sequences:

$$V_z^{L_1}(\tau b_1; b_1) = V_z^{L_2}(\tau b_2; b_2),$$

so $V_z(s;b) = v_z(s/b)$, every sequence with $\beta_j \to \infty$, $N_j/\beta_j \to \tau$ converges ($\beta_j = L^{K_j} b_j$, $b_j \in [1, L)$), and the limits do not depend on the choice of the constants. For $s \in (2L^{\mathbb{Z}})^3$, $V_z(s;b) = V_z(\frac{1}{2}s; \frac{1}{2}b) > 0$, the box $\frac{1}{2}s$ being mixed for every class; positivity of the limit and of each r_K is the floor, uniform for the labels in a compact set, since the convergence is uniform there (Step 9), the limits are continuous and positive, and the first sentence of B(1) bounds the finitely many remaining cutoffs; and an infimum 0 over the cubes with $N/\beta \in I$, or over the lattices with N/β in a compact subset of $\{v_z > 0\}$, would give, along $\beta_j \to \infty$, a point of that set with $v_z = 0$. The same for the levels gives their extension to real sections and their homogeneity. The weights of the mixture converge with the $r_K[n]$, and $1 \le \sum_n v_n \le 8$ with $v_0 = 1 \ge v_n$, so that their limits lie in $[v_n/8, \frac{1}{2}]$ for $n \neq 0$.

Step 11: small cubes (B(5)). (a), the ceiling. By A(1), $r_2, r_3 \le r_1$, and by the symmetry of the axes it is enough to bound $r[n]$ for $n = n_{13} = ((1,0),0)_3$, whose twisted plane holds the time 3, on the cubes $\ell = 2L^m$. A dilation by a power of L keeps t and the lattices and brings (ℓ, b) into the range $b \ge b_0$, $MR(b) \le L^m$, $\ell \le 2ML^2(\log b)^{r_\star}$ of Lem. 11.11(b), where small t is large b. In a block layer normal to 3 of the block field of depth one, with $\bar h_1(y)$, $\bar h_2(y)$ the products of the block field along the straight closed block lines from y in the directions 1 and 2 (as in §11.1.5), let $\bar s_1$ and $\bar s_+$ be the means over the sites y of the layer of $\frac{1}{2}\mathrm{tr}\,\bar h_1(y)$ and of $\frac{1}{2}\mathrm{tr}[\bar h_1(y)\bar h_2(y)]$. The flip of n changes the signs of both; on a flat field with the holonomies $e^{i\varphi_\mu \mathsf{a}\cdot\sigma}$ they are $\cos\varphi_1$ and $\cos(\varphi_1+\varphi_2)$, which vanish together exactly at the fixed points $\varphi_1 = \pm\frac{\pi}{2}$, $\varphi_2 \in \{0, \pi\}$ of the flip $\varphi_1 \mapsto \varphi_1 + \pi$, taken up to $\varphi \mapsto -\varphi$. Let

$$F = \mathbf{1}\{\max(|\bar s_1|, |\bar s_+|) < m_0 t^{1/2}\},$$

with an absolute m_0, be read at the block layer next to a block mirror normal to 3 at the cutoffs $K \ge K_\sharp$ of Lem. 11.11(b), and at the bare layer next to a mirror between two layers below $K_\sharp$; F is gauge invariant, Borel with values in $[0,1]$ and flip-blind, and its mirror image θF is F read at the layer on the other side of the mirror. Establishing **(Flux-window)** would give, with absolute C_2 and ϵ_F, at every cutoff and with the columns of n away from the layers of F and θF: **(F1)** $\mu(F) \le C_2 t$, and **(F2)** $Z[n; 1-F] \le \epsilon_F\, t\, Z[0]$.

Both estimates remain to be established in **(Flux-window)**. In particular, (F1) is a bound for a shrinking event of the block means, and does not follow from the uninserted periodic Laplace estimate of Prop. 11.14. The sign reversal of the plane means under the twist likewise does not by itself establish (F2).

Now

$$r[n] = \frac{Z[n; \theta F\, F]}{Z[0]} + \frac{Z[n; 1 - \theta F\, F]}{Z[0]}.$$

The first term is bounded through the overlap representation of the class with flux, in the form that A(6) takes for a function of a slab and its mirror image, the trace formula of Step 4 with α, α' as there: $Z[n; \theta F\, F] = \sum_e (-1)^{e\cdot k} \|T_0^{\alpha/2} \hat F\, T_0^{\alpha'/2} P_e\|_2^2 \le Z[0; \theta F\, F] \le Z[0; F] = \mu(F)\, Z[0]$. In the second, $1 - \theta F\, F \le (1-F) + (1-\theta F)$, and $Z[n; 1-\theta F] = Z[n; 1-F]$ by the reflection, the columns being moved away from the layers of F (Step 1). So $r_1 \le (C_2 + 2\epsilon_F)\, t$ at every cutoff, with the absolute $C_\square = C_2 + 2\epsilon_F$; t_0 and both bounds are as specified in (Flux-window). (b) and (c): on the odd cube $s = \frac{1}{2}\ell_\sharp(b) = LMR(b)$, $b \ge b_\sharp$, $S = s$, B(4) holds uniformly in the label from $K_0 = \lceil c_2 \log_L S \rceil$ on, $c_2 = 20\varpi$: its second condition is void on this cube, since $|s| \le L^6 M^3 (\log b)^{3r_\star} \le (\log b)^{2\mathfrak{p}_0 - 1}/C_V$ for large b, as $2\mathfrak{p}_0 > 3r_\star + 1$, so that the low cutoff of B(4) is K_0.

At K_0 the lattice has $N_0 = SL^{K_0} \in [S^{1+c_2}, LS^{1+c_2}]$ sites a side, so $N_0 \le C(\log b)^{r_\star(1+c_2)}$, at $\beta_0 = bL^{K_0} \ge b$; this lattice and those of the cutoffs $K < K_0$, which are smaller at $\beta \ge b$, lie in the range of Prop. 11.14 for $c < 1/(r_\star(1+c_2))$ and large b. From K_0 on, B(4), uniformly in the label, carries the lattice value to every cutoff with a relative error $o(1)$ as $b \to \infty$, the sum of $c_1 S^{\varpi} L^{-K/10}$ over $K \ge K_0$ being at most a multiple of $S^{\varpi - c_2/10} = S^{-\varpi}$; this is (c), the values $t = LMR(b)/b$ filling $(0, t_0]$; v_z depends on t alone (B(3)), which gives $C^{-1}t \le v_z \le Ct$ and, with Prop. 11.14(e), $v_z(t,t,t)/t \to \mathscr{G}_z$.

The ratios are Prop. 11.14(c), (e) for two classes, the periodic denominators cancelling. (d): with the ceiling, $r_1 \le \frac{1}{5}$ on the cubes $2L^m$ with $t \le t_1 = \min\{t_0, 1/5C_\square\}$ at every cutoff, and A(13) applies; $\delta_K^0 \ge \mathsf{G} \ge \log(4/(1+3r_1)) \ge \log(4/(1+3C_\square t))$.

Step 12: the label (B(6)). A source is a complex function h on T_ℓ in the ball of the norm of Thm. YM3_I:AD (the source w there); for a class n, with $x(p)$ the barycentre of the plaquette p, put

$$Z_K[n;\mathsf{h}] = \int \exp\Big[-\sum_p L^K\big(b_0 - \mathsf{h}(x(p))\big)\big(1 - \tfrac{1}{2}\eta^n(p)\,\mathrm{tr}\,\partial U(p)\big)\Big]\,\mathrm{d}U;$$

at $n = 0$ this is the left member of the identity of Thm. YM3_I:AD at $\beta = L^K b_0$, and at a constant source $\mathsf{h} \equiv c$ it is $Z_K[n]$ at the label $b_0 - c$. Let n have one or two twisted planes, μ a direction that lies in each of them, and θ the reflection $x_\mu \mapsto -x_\mu$, with the mirror planes $x_\mu \in \{0, \frac{\ell}{2}\}$, or $x_\mu \mapsto \frac{\ell}{2} - x_\mu$, with the mirror planes $x_\mu \in \{\frac{\ell}{4}, \frac{3\ell}{4}\}$; the two mirror planes cut T_ℓ into the closed halves H_- and $H_+ = \theta H_-$. Let h be symmetric under θ and real within $\frac{\ell}{16}$ of the two mirror planes, and let its conjugated fold $\mathsf{h}^\flat$ be h on H_- and $\bar{\mathsf{h}}$ on H_+.

With μ as the time and the mirror layers $x_\mu \in \{0, \frac{\ell}{2}\}$, $Z_K[n;\mathsf{h}] = \iint \Psi_-(V,V')\,\Psi_+(V, c^k V')\,\mathrm{d}V\,\mathrm{d}V'$, where V, V' are the fields of the two mirror layers, on whose plaquettes the weights $L^K(b_0 - \mathsf{h}(x(p)))$ are real and positive and give the square roots of their factors to each half, $\Psi_\mp$ are the integrals over the bonds inside $H_\mp$, and c^k is the flip of the electric part of n, whose magnetic part vanishes because μ lies in every twisted plane. The reflection θ keeps the Haar measure and the traces of the plaquettes, so the symmetry of h gives $\Psi_+ = \Psi_-$, while $\overline{\Psi_-}$ is the integral over H_- at the source $\bar{\mathsf{h}}$, that is, by θ, over H_+ at $\bar{\mathsf{h}}$; the Cauchy–Schwarz inequality in the fields of the mirror layers, c^k keeping their Haar measure, gives

$$|Z_K[n;\mathsf{h}]| \le \|\Psi_-\|_2^2 = Z_K[0;\mathsf{h}^\flat].$$

At mirrors midway between two layers the same holds, the crossing kernel at the mirror being positive semidefinite at real positive weights (Step 1). For three twisted planes the same with the time 3, which lies in the planes 13 and 23 and whose section is twisted, gives $|Z_K[n;\mathsf{h}]| \le Z_K[n_{12};\mathsf{h}^\flat]$, the right member being ≥ 0, and a second fold with the time 1, which lies in the plane 12, at mirrors normal to the direction 1 at which the source is symmetric and real as well, gives $Z_K[n_{12};\mathsf{h}^\flat] \le Z_K[0;\mathsf{h}^{\flat\flat}]$, $\mathsf{h}^{\flat\flat}$ being h or $\bar{\mathsf{h}}$ according to the product of the signs of the two halves. The identity of Thm. YM3_I:AD writes $Z_K[0;\mathsf{h}^\flat]/|Z_K[0;\mathsf{h}]|$ as the exponential of the sum of the differences of the real parts of the local energies $e_j(X;\cdot)$ at $\mathsf{h}^\flat$ and at h, times the quotient of the moduli of the two remainders $\widetilde{Z}$.

An energy at $\bar{\mathsf{h}}$ is the conjugate of the energy at h, and it reads the source near its domain; so the real parts agree on every localization domain that does not meet both $H_- \cap \{\mathrm{Im}\,\mathsf{h} \ne 0\}$ and $H_+ \cap \{\mathrm{Im}\,\mathsf{h} \ne 0\}$, two sets at a distance $\ge \frac{\ell}{8}$. A domain of the level j that meets both has an extent of at least $\ell L^{K-j}/8M - 4$ ([7, eq. 321]); the number of the localization domains ([7, Lem. O.1]) and the decay of the energies in the extent bound their sum by a number that does not depend on K, and with the bounds on $\widetilde{Z}$, which also give $Z_K[0;\mathsf{h}] \ne 0$ for small sources,

$$|Z_K[n;\mathsf{h}]| \le Z_K[0;\mathsf{h}^\flat] \le \mathsf{M}_\mathrm{S}|Z_K[0;\mathsf{h}]|, \qquad Z_K[0;\mathsf{h}] \ne 0,$$

and the same with $\mathsf{h}^{\flat\flat}$ for three twisted planes. For the constant source $b_0 \mapsto b_0 - c$, written as $c\chi_1 + c\chi_2$ with a partition of unity $\chi_1 + \chi_2 = 1$ in x_μ, χ_1 vanishing within $\frac{\ell}{16}$ of the planes $x_\mu \in \{0, \frac{\ell}{2}\}$ and χ_2 within $\frac{\ell}{16}$ of the planes $x_\mu \in \{\frac{\ell}{4}, \frac{3\ell}{4}\}$, the quotient $r_K = Z_K[n;\cdot]/Z_K[0;\cdot]$ at the source $\zeta_1\chi_1 + \zeta_2\chi_2$, in which one of the two parts may be complex while the other is real, is bounded by M_S on the two crosses of a bidisc in (ζ_1, ζ_2), and the following lemma bounds it on a smaller bidisc. If F is holomorphic near $\bar{D}_R^2$ and $|F| \le \mathsf{M}$ wherever one variable lies in $\bar{D}_R$ and the other in $[-R, R]$, then $|F| \le 27\,\mathsf{M}$ on $\bar{D}_{R/8}^2$; if F is holomorphic near $\bar{D}_R^4$ and $|F| \le \mathsf{M}$ wherever one variable lies in $\bar{D}_R$ and the three others in $[-R, R]$, then $|F| \le 12600\,\mathsf{M}$ on $\bar{D}_{R/15}^4$ (this form serves for three twisted planes).

For the proof put $\rho_\circ = 2 + \sqrt{3}$ and $\mathsf{j}(u) = \frac{R}{4}(u + u^{-1})$, which maps the annulus $\rho_\circ^{-1} \le |u| \le \rho_\circ$ into $\bar{D}_R$ and the unit circle onto $[-\frac{R}{2}, \frac{R}{2}]$. Moving the circle of one variable of the Cauchy integral to $|u| = \rho_\circ^{\pm 1}$, the other variables staying on the unit circle, where j is real, bounds the Laurent coefficients of $F \circ \mathsf{j}$ by $\mathsf{M}\rho_\circ^{-\max_i |k_i|}$; summing them on the annuli $1.3^{-1} \le |u_i| \le 1.3$, whose images hold $\bar{D}_{R/8}$, gives at most $26.2\,\mathsf{M}$, and on $1.15^{-1} \le |u_i| \le 1.15$, whose images hold $\bar{D}_{R/15}$, at most $12563\,\mathsf{M}$ in four variables. This bounds r_K on a disc whose radius does not depend on K, and Cauchy's estimate gives the derivatives.

In t this is the normal family of the ϕ_j, and Vitali's theorem gives the analyticity of the limit. With $m_0 = \inf_K r_K(b_0) > 0$, Cauchy's estimate keeps r_K away from 0 on a smaller disc, which, $Z[0]$ having no zero there, gives the zero-free discs and Ω; the first two derivatives of $\log r_K$ in the label are differences of means and of variances of the total action, bounded by Cauchy's estimate.

Step 13: block pairs (B(7)). A block mirror lies midway between two layers, L being odd; for G bounded Borel in the block pairs on one side, $(U, s) \mapsto G(\mathcal{M}_s^{K-d} U, s^{(d)})$ depends on the bonds and signs on that side, and the reflection commutes with $\mathcal{M}_s$: the clause on pairs in A(8) gives the positivity, which weak limits keep; at each cutoff $\mathcal{M}_s^{K-d} = \mathcal{M}_s \circ \mathcal{M}_s^{K-d-1}$ makes the tower consistent. Under the conditions of B(7), the small-field chain among the construction results on the periodic torus, with the rarity of its large-field regions (Prop. 4.54), bounds the periodic probability of a rough block plaquette of some depth $d' \in [d, K]$ by $\mathsf{R}(d)$, uniformly in K; the dilation carries this to the other pairs (ℓ, b), the depth d becoming $d - j$.

With γ a chart of s near a block plaquette P, the sign of P in the block sign field times $\partial(\mathcal{M}_s^k U)(P)$ is $\partial(\mathcal{M}^k(\gamma U))(P)$: the event of a rough plaquette is covariant and local, and off it the signs of the block traces are the block sign field. In $\mathbb{Z}/2\lambda$, λ odd, every arc of at most $\frac{1}{2}(\lambda + 1)$ points lies in one of four arcs of λ points, so each local event lies in one of sixteen quarters of the torus cut by mirrors in two directions; the quarter bound of A(8) carries the union of the events of one quarter into the class with the root $\frac{1}{4}$, which gives $16\mathsf{R}(d)^{1/4}$, and the bound for the mixtures.

In a weak limit the class is then, up to null sets, the class of the block sign fields of the tower. For the other statements on limit points pass to a subsequence along which the laws of the towers converge in every class. The quarter bound of A(8) at each cutoff, $r_K[n]\mu_n(B) \le \mu_d^K(B_1)^{1/4}$, and the convergence in total variation of μ_d^K to μ_d (through Thm. YM3_I:D) give $r_n \mathbb{P}^n(B) \le \mu_d(B_1)^{1/4}$ for open B by the portmanteau theorem, and for Borel B by outer regularity, the group of the sign changes being finite; the half bound of A(8) gives the exponent $\frac{1}{2}$. For the consistency of the limit towers the identity at each cutoff is joined to Lusin's theorem, $\mathcal{M}_s$ being a Borel map and not continuous.

The relation between the block sign fields is a closed condition and passes to the limit. For the fields fix a class with $r_n > 0$, a depth $d \ge 1$, a block bond c of depth d and $\epsilon > 0$; $\mathcal{M}(\cdot)(c)$ is a Borel function of the block field of depth $d+1$ near c, and by Lusin's theorem it is continuous on a compact set $\mathcal{K}$ of these local fields of μ_{d+1}-measure at least $1 - \epsilon$. The towers with $V_d(c) = (\mathcal{M}_{s_{d+1}} V_{d+1})(c)$ whose local field near c, read in a chart of s_{d+1}, lies in $\mathcal{K}$ form a closed set; at each cutoff it holds

the tower off a covariant local event whose probability under μ_n is at most $r_K[n]^{-1}\mu_{d+1}^K(\mathcal{K}^c)^{1/4}$ by the quarter bound.

By the portmanteau theorem and the convergence of μ_{d+1}^K in total variation, $\mathbb{P}^n$ gives this closed set a probability at least $1-\epsilon^{1/4}/r_n$, and $\epsilon\to 0$ gives the consistency at c, almost surely; there are finitely many c and countably many d.

Step 14: tension of a slab (B(8)). On one lattice, A(4)(a) at the heights and lengths N/L and N, $N=\ell/a_K$, with $\mathsf{G},\Phi\le\delta$ and the last inequality of A(2), $\delta(N/L,N,N)\le L^2\delta((N/L)^{\times 3})$ and $\delta(N/L,N/L,N)\le L\delta((N/L)^{\times 3})$, gives the two inequalities in lattice units, the three terms of the bound on σ adding up to $(L/(L-1))^2L^3\delta((N/L)^{\times 3})$ exactly; in physical units a product of two lattice lengths carries L^{2K}. The right members are bounded in K: δ_K^0 by B(1), and $\kappa_K\to\log(2/(1-v_1(t,t,t)))<\infty$ by B(3), since $v_1<1$ on the cubes with L-adic sides. In the limit the least level of the sector $e=10$ is $\varepsilon_{\mathrm{el}}$ (A(1)) in physical units. The lower bound on σ_K is A(3) in physical units, and the bound on $-\log\mathfrak{b}_K$ is the last bound of A(3) with $R_K(\lambda_0)\le\mathsf{G}_K(\lambda_0;l,2\lambda_0)/\lambda_0\le\delta_K^0(\lambda_0,l,2\lambda_0)/\lambda_0\le(2\lambda_0 l/s'^3)\bar\delta(s')$ by A(4)(b) and A(2), and $\Phi_K(t_0,\lambda_0)\le(t_0\lambda_0 l/s'^3)\bar\delta(s')$; the equivalence follows by choosing first λ, then t, large.

Step 15: the small twisted section (B(9)). The levels are read in the normalisation of B(9): they are the numbers $a_K^{-1}\log(\lambda_0/\lambda_i)$, $\lambda_0=\|T_1\|$, the four sectors together, so that the least of them, the vacuum, is $E_0=0$; they are the levels of the setting diminished by $a_K^{-1}\log(\|T_0\|/\|T_1\|)$, and their limits are the eigenvalues of $H_{(l,l),1}$ diminished by $E_{\mathrm{mag}}(l,l)$. Over the twisted section (N,N), $N=l/a_K$, at the time 1, $\operatorname{Tr}T_1^h=Z[n_{23}](h,N,N)$; with $x(h)=\sum_{i\ge1}(\lambda_i/\lambda_0)^h$, in

$$L\log\operatorname{Tr}T_1^h-\log\operatorname{Tr}T_1^{Lh}=L\log(1+x(h))-\log(1+x(Lh))$$

the bulk and all that is proportional to the time cancel. The chain that bounds it has these links. (i) A dilation by a power of L keeps t and the lattices and takes (l,b) to (l',b') with $l'=\frac12\ell_\sharp(b')=LMR(b')\ge\ell_\flat(b')$, the values $LMR(b')/b'$ filling $(0,t_0]$ (Step 11); small t is large b'. (ii) The odd boxes (l',l',l'), (Ll',l',l'), (L^2l',l',l') and, for each fixed $j\ge1$, (L^jl',l',l') have their sides in $L^{\mathbb{N}}$ and are mixed for n_{23}; on them B(4) holds uniformly in the label from its low cutoff $\lceil c_2\log_L S\rceil$ on, its second condition being void for large b', since $|s|\le L^{j+6}M^3(\log b')^{3r_\star}\le(\log b')^{2\mathfrak{p}_0-1}/C_V$ as $2\mathfrak{p}_0>3r_\star+1$.

In the left member of the display at $h=N$, which in the dilated units is $L\log Z[n_{23}](l'^{\times 3})-\log Z[n_{23}](Ll',l',l')$, and at $h=LN$, the terms $\frac{|s|}{\ell^3}\Delta_K$ of B(4) cancel, so that B(4) carries its value at the low cutoff to every cutoff with an error summable in K. (iii) At and below the low cutoff the lattices lie in the range of the twisted clause of Prop. 11.14(a), whose volume terms cancel in the display and whose Gaussian term gives the free value: with θ^a the phases of the three colours a of n_{23} and the momenta $p\in((2\pi\mathbb{Z}+\theta_2^a)/N)\times((2\pi\mathbb{Z}+\theta_3^a)/N)$ mod 2π, none of them 0, $\prod_{j<h}(2\cosh\omega-2\cos\frac{2\pi j}{h})=2\cosh h\omega-2$ gives $\log\mathscr{D}^{n_{23}}(h,N,N)=-hE_{\mathrm{free}}+\log(1+x_{\mathrm{free}}(h))$, $E_{\mathrm{free}}=\frac12\sum_{a,p}\omega(p)$, with

$$1+x_{\mathrm{free}}(h)=\prod_{a,p}(1-e^{-h\omega(p)})^{-1},\qquad 2\cosh\omega(p)-2=\hat p^2,$$

and the free value of the display is that of x_{free}. (iv) With $L\ge13$ of the setting, the free value at $h=N$ is at most $0.26(L-1)$: at $N=L=13$ it is 3.0917 against $0.26(L-1)=3.12$, the margin 0.028; at $h=LN$ it is about $4Le^{-\pi L}$. (v) c_0, formed from the constants, is chosen so that for $t\le c_0$ the errors of Laplace's method and of B(4) stay below that margin at $h=N$, and are small enough against the free value at $h=LN$ that $x(LN)\ge0.29693\,e^{-\pi L}$ follows. Then at every cutoff the display at $h=N$ is at most $0.26(L-1)$; as x does not increase in h, $(L-1)\log(1+x(N))\le0.26(L-1)$, so $x(N)\le e^{0.26}-1=0.29693$ and $N\log(\lambda_0/\lambda_1)\ge\log(1/0.29693)\ge1.2142$; at $h=LN$, with this bound on $x(N)$ and $x(LN)\le(\lambda_1/\lambda_0)^{(L-1)N}x(N)$, $N\log(\lambda_0/\lambda_1)\le\pi L/(L-1)$.

The sharper bounds use the same comparison at smaller t with the free values over the twisted section, whose least excitation is ω_N/N with $3.1264 \le \omega_N < \pi$ for $N \ge 13$. Likewise B(4) on the odd boxes $(L^j l', l', l')$, $j \ge 1$, carries the quotients $Z[(k,1)_1]/Z[(0,1)_1]$ to their free values; the sector traces $\frac14 \sum_k (-1)^{e\cdot k} Z[(k,1)_1]$ at the heights $L^j l$, which are $L^j l'$ in the dilated units, fix the levels (Lem. 11.2); in the normalisation of B(9) the least free level of the sector $e \neq 00$ is $\pi|e|/l$, its momenta lying in $\frac{2\pi}{l}(\mathbb{Z} + \frac{e_\rho}{2}) \times \frac{2\pi}{l}(\mathbb{Z} + \frac{e_\nu}{2})$ by A(1), while in the sector 00, whose least level is the vacuum $E_0 = 0$, the least positive free level, the second level of 00 minus the vacuum, is $2\pi/l$.

For (c), the state of the cylinder has $\omega_K(\overline{\theta_K F}\, \tau_{n'} G) = \langle \psi(G), (T_1/\lambda_0)^{n'} \psi(F) \rangle$ for plaquette functions of finitely many layers, with vectors $\psi(F)$ formed from the slab operators and the top eigenvector of T_1. Lem. 11.8(b) with cells of side l/L, that is $\lambda' = L^{J-1}$ sites and $k = L$ cells a side of the section, and with $\mathsf{q} = 2$, compares the cylinder with the periodic cube Q of side $2l/L$, at the cost of one factor Ξ for each slab of cells in time that carries a function; at $k_0 = L$, $\Xi \le \exp[\Phi'/(L-2)]$, and $\Phi' = \frac{L^3}{8}\delta^0_K(Q) - \delta^{n_{23}}_K(l^{\times 3})$, the bulk cancelling, is bounded uniformly in K by the window of B(1) on the cube $2l/L$ and the floor of B(2) on the odd cube l.

A partition of unity of each f_q into pieces supported inside cells of translates of the board cuts a tuple into finitely many tuples placed in cells, at a cost formed from n', R and l; for a placed tuple the product of the centred observables of one cell is a polynomial in the numbers $1 - \frac12 \mathrm{tr}\, \partial U(p)$ of its plaquettes, and its seminorm on Q is a centred moment of the observables of the eight mirror copies in the periodic measure of Q, the means e_K being those of Q; the bounds on the joint cumulants of the action density on that periodic cube ([7, Lem. N.2]) then give $C_{n'}(d, R)$. Compactness (Lem. 11.2(b) for the levels, a diagonal choice for the $S_{n'}$) gives the limit points; reflection positivity is A(7) at the mirrors normal to the time, which lies in no twisted plane of n_{23}.

In every limit, $S(\theta F, \tau_s G) = \sum_i e^{-sE_i} c_i(F, G)$ with a positive semidefinite matrix of coefficients, which defines an isometry of $\mathcal{H}$ into the ℓ^2 space of the limit levels carrying e^{-sH} to multiplication by e^{-sE_i}; $E_0 = 0$ is simple and belongs to Ω, and the gap gives the spectral bound and the clustering. The action density is invariant under the flips C^k, so its vectors lie in the sector 00, whose least positive level exceeds $6.1/l$ at small t when the law of (4) is used on the boxes $(L^3 l, l, l)$. Finally the lower bound on $S_2(\theta f^\circ, f^\circ) - S_1(f^\circ)^2$. Let $j(t)$ be the least integer for which $g = (L^{j(t)}/2t)^{-1/2}$ is at most the ceiling γ_W of the reference couplings of Thm. YM3_I:AV and $2L^{j(t)-1}$ is a side at g (equation YM3_I:(1528)); it is formed from t and the constants.

Put $\hat L = L^{j(t)-m}$ and read everything in the units dilated by $\hat L$, which name the same lattice and the same β: the section has the side $\hat l = \hat L l = L^{j(t)}$, the label is $\hat b = \hat L b = L^{j(t)}/t = 2g^{-2}$, the cutoff is $\hat K = K - j(t) + m$, the bump has the radius $d_* = \hat L d^\circ$, and the partner of the cylinder is the periodic cube of side $2\hat l/L$, which is the cube Q of side $2l/L$ above; by Cor. YM3_I:74.2 and [7, Lem. G.4] every $(\hat K, L^{\hat K}\hat b)$, $\hat K \ge 1$, is a first-passage datum at g with the label $\hat b$. (1) The state of the cylinder defines a law $\tilde\omega$ on the configurations of $a_{\hat K}\mathbb{Z}^3$, carried by the fields that are periodic up to the twist in the two directions of the section, with $\tilde\omega(G) = \omega_K(G^{\mathrm{per}})$ for the continuous plaquette functions G of finitely many plaquettes, G^{per} reading each plaquette at its class modulo the section; on these functions $\tilde\omega$ is invariant under the translations of $a_{\hat K}\mathbb{Z}^3$. (2) Let F_1, F_2 be Lipschitz functions supported in one ball of radius 2 whose centre lies at the distance ≥ 3 from the walls of a board of side L, and $\mathsf{u} = \zeta_1 F_1 + \zeta_2 F_2$, $\zeta \in \mathbb{C}^2$.

Lem. 11.8(b), with cells of side L, that is of $L^{\hat K+1}$ sites a side, one slab of cells in time that carries a function, and the same partner Q, gives $|\tilde\omega(e^{\mathcal{O}_{\hat K}(\mathsf{u})})| \le \hat\Xi\, \|e^{\mathcal{O}_{\hat K}(\mathsf{u})}\|_{\mathrm{ch};Q}$, where $\hat\Xi = \Xi^{L^{2-j(t)}} \le \Xi$, because at a fixed partner $\log \Xi$ is linear in the number of sites of a side of a cell; so $\hat\Xi$ is bounded uniformly in K by a number Ξ_* formed from t and the constants. (3) By the normalisers among the construction results at the first-passage data (Prop. YM3_I:91.12 (1)) there is a function $\mathcal{E}$, holomorphic in ζ on the domain $\mathbb{D}_\varrho(F_1, F_2)$ among the construction results and with $\mathcal{E}(0) = 0$, with $\|e^{\mathcal{O}_{\hat K}(\mathsf{u})}\|_{\mathrm{ch};\ell''} \le |e^{\mathcal{E}(\zeta)}|\, \mathsf{Q}_1(L)$ for all sides ℓ'' at g, the side $2\hat l/L$ among them; so $|e^{-\mathcal{E}(\zeta)}\mathfrak{s}(e^{\mathcal{O}_{\hat K}(\mathsf{u})})| \le$

$Q' = \Xi_* \mathsf{Q}_1(L)$ for $\mathfrak{s} = \tilde{\omega}$ and, by the transfer principle (Prop. YM3_I:91.11), for the periodic state $\tilde{\mu}$ of Q, read as a law on $a_{\hat{K}}\mathbb{Z}^3$. (4) The transfer between two laws under one normaliser (Prop. YM3_I:91.11) gives $|S_2^T[\tilde{\omega}](F_1,F_2) - S_2^T[\tilde{\mu}](F_1,F_2)| \le C_D \|F_1\|_\sharp \|F_2\|_\sharp$, $C_D = 2\log 2\,(2Q'+2)^2(2/\varrho)^2$, ϱ the radius of that domain, with no separation of the supports. (5) Put

$$D(f,h) = S_2^T[\tilde{\omega}](f,h) - S_2^T[\tilde{\mu}](f,h).$$

This is a bilinear form invariant under lattice translations, by (1) and the translation invariance of the periodic law. We apply the packing argument in the proof of Prop. YM3_I:127.1; only these two properties and the unit-ball bound in (4) enter that argument. In particular, it does not require the cylinder law to be a periodic Gibbs state.

Let $r = 5\mathsf{d}$, with $a_{\hat{K}} \le r \le 1/20$, and suppose that f_1, f_2 vanish outside one ball of radius r. Set $\delta = a_{\hat{K}}\lceil 2r/a_{\hat{K}}\rceil$, so that $2r \le \delta < 3r$. Translates of this ball whose centres have all coordinates of modulus at most $0.575 - r$ lie in the unit ball and have disjoint interiors when their translation vectors lie in $\delta\mathbb{Z}^3$. Each coordinate interval contains at least $0.3/r$ points of the required translate of $\delta\mathbb{Z}$. We thus obtain a set P of lattice translations with $N = \#P \ge (0.3/r)^3$. For independent signs $\epsilon_v = \pm 1$ put

$$F_{i,\epsilon} = \sum_{v\in P} \epsilon_v \tau_v f_i, \qquad i = 1,2.$$

Their supremum and Lipschitz norms do not exceed those of f_i. For the Lipschitz assertion, each function vanishes on the boundary of its supporting ball; on a segment joining different balls, bound the endpoint values by the distances to the respective exit and entry points. The sum of these distances is no greater than the length of the segment. Consequently $\|F_{i,\epsilon}\|_\sharp \le \|f_i\|_\sharp$. Bilinearity and sign averaging give

$$\mathbb{E} D(F_{1,\epsilon}, F_{2,\epsilon}) = \sum_{v\in P} D(\tau_v f_1, \tau_v f_2) = N D(f_1,f_2).$$

Applying (4) and dividing by N yields $|D(f_1,f_2)| \le C_D (r/0.3)^3 \|f_1\|_\sharp \|f_2\|_\sharp$. If $\operatorname{Lip} f_i \le 2\|f_i\|_\infty/\mathsf{d}$, then $\|f_i\|_\sharp \le 80M\|f_i\|_\infty/\mathsf{d}$, and hence

$$|S_2^T[\tilde{\omega}](f_1,f_2) - S_2^T[\tilde{\mu}](f_1,f_2)| \le C_2' \mathsf{d}\, \|f_1\|_\infty \|f_2\|_\infty, \qquad C_2' = C_D (50/3)^3 (80M)^2.$$

(6) Let f_+ be the bump of radius d_* about the point $x_+ = (2d_*, \hat{L}\xi)$, the dilate of f°, and f_- its mirror image in the plane at the time $a_{\hat{K}}/2$, the bump about the mirror point x_- of x_+.

When $a_{\hat{K}} \le d_*/4$, $3d_* \le |x_+ - x_-| \le 8d_*$, both bumps vanish off one ball of radius $5d_*$ and have Lipschitz constants $\le 2/d_*$, and (f_-, f_+) is a pair of the law of two points at $\vartheta = \frac12$. For the state $\tilde{\mu}$ of the torus of side $2\hat{l}/L$, a state at g, Thm. YM3_I:AV gives, for $d_* \le \mathsf{d}_0(g,\frac12)$ and $s(d_*) < \hat{K}$, $s(\cdot)$ the witness depth (equation YM3_I:(2321)), $S_2^T[\tilde{\mu}](f_-,f_+) = c_{\mathrm{sd}}\hat{b}^{-2}\mathcal{K}_{\mathrm{free}}(1+\varepsilon')$ with $|\varepsilon'| \le \frac13$, where the free kernel of the pair has $\mathcal{K}_{\mathrm{free}} = \mathcal{K}_{\mathrm{free}}(f_-,f_+) \ge \frac14\, 10^{-6}$. (7) With $d_* = \min\{\mathsf{d}_0(g,\frac12),\, c_{\mathrm{sd}}/(24\cdot 10^6 C_2'\hat{b}^2)\}$, a number formed from t and the constants, $C_2' d_* \le \frac16 c_{\mathrm{sd}}\hat{b}^{-2}\cdot\frac14\,10^{-6} \le \frac16 c_{\mathrm{sd}}\hat{b}^{-2}\mathcal{K}_{\mathrm{free}}$, and so, by (5) at $\mathsf{d} = d_*$,

$$S_2^T[\tilde{\omega}](f_-,f_+) \ge (1 - \tfrac13 - \tfrac16) c_{\mathrm{sd}}\hat{b}^{-2}\mathcal{K}_{\mathrm{free}} \ge \tfrac18\, 10^{-6} c_{\mathrm{sd}}\hat{b}^{-2}$$

at every cutoff with $\hat{K} > s(d_*)$ and $a_{\hat{K}} \le d_*/4$. Back in the original units, $\mathcal{O}_{\hat{K}}(f^{[\hat{L}]}) = \hat{L}^{-1}\mathcal{O}_K(f)$ for $f^{[\hat{L}]}(x) = f(x/\hat{L})$, the barycentres being multiplied by $\hat{L}$ and the factor a^{-1} divided by it; so the covariance of $\mathcal{O}_K(\theta_K f^\circ)$ and $\mathcal{O}_K(f^\circ)$ under ω_K is $\hat{L}^2$ times this one, and $\hat{L}^2\hat{b}^{-2} = b^{-2}$: the bound $\frac18\,10^{-6}c_{\mathrm{sd}}b^{-2}$ holds at all large cutoffs. A covariance does not see the centring by e_K, and $\theta_K f^\circ$ is the translate of θf° by a_K in the time; so the bound holds for $S_2(\theta f^\circ, f^\circ) - S_1(f^\circ)^2$ in every limit.

Step 16: even cubes (B(10)). The reduction is arithmetic in $\ell_\sharp$ and R: for $t \le t_{\mathrm{w}}$ the label $b'(t)$ exists, and the cube $2L^j$ at $\beta = 2L^j/t$ is the cube $\ell_\sharp(b')L^{j-\mu(t)}$ at $\beta = L^{j-\mu(t)}b'$. (a) follows for $N \ge N(t)$ from B(3), B(5) and $C^n(N)/N \to \mathscr{G}_n$ (Prop. 11.14(e)); and up to $K_1(b')$ the cube

$\ell_\sharp(b')$ has $N \le \ell_\sharp b'^{1/68}$ sites a side at $\beta = L^K b'$, so that $N^{66}\log\beta/\beta \le 2\ell_\sharp^{66} b'^{-3/68}\log b'$, which tends to 0 as $b' \to \infty$, and Prop. 11.14(g) gives the lattice value. Under (R), $\sum_{K\ge K_1(b)} c_1 \ell_\sharp^{\varpi} L^{-\kappa K} \le C(\log b)^{r_\star \varpi} b^{-\kappa/68} \to 0$, also under $\sup_{b'\ge b}$; summing from $K_1(b')$ gives the bound of (a) at every $N \in 2L^{\mathbb{N}}$, with its lower bound on $C^n(N)/N$ the floor ct, and the bounded variation, $\phi_{j+1} - \phi_j$ being a difference of two consecutive cutoffs on the cubes $\ell_\sharp(b'(t))L^{j-\mu(t)}$; with the one-sided form, $\log q_K \ge \log q_{K_1(b')} - o(1)$ for $K \ge K_1(b')$ gives $ct \le r_K$, and Lem. 11.13 gives the bounded variation.

The proposed transport argument still requires (W) and (Window estimates). At $K_\sharp$ the lattice lies in the range of §11.1.5: $w_{K_\sharp} \ge c_\flat t$ and $\varsigma_{K_\sharp} \ge 2^{1-L} e^{\lambda_\flat}(c_\flat t)^L$, beside which $e^{-c(\log b)^{1+\eta}}$ is $o(1)$; Lem. 11.11(b) puts w_K within $e^{\pm E}$ of $w_{K_\sharp}$ up to ϵ, which is $o(t)$, indeed $o(t^k)$ for every k, as Lem. 11.11(b) states (m_K and ϵ_K being formed from the numbers YM3_I:(2192)–(2194) and R_{t_1} of §2.3.1), and so negligible beside $w_{K_\sharp} \ge c_\flat t$; so $\log(2^{L-1}\mathfrak{q}_L(K)) \ge \lambda_\flat - \Lambda - o(1) > 0$, and A(12) gives

$$r_K[n_{13}] \ge w_K \,\frac{\mathfrak{q}_L(K) - 2^{1-L}}{1 - 2^{1-L}} \ge c' t;$$

smaller lattices lie in the window. On the box $(\ell, \ell, i\ell)$ the same holds with $2^{i-1}\mathfrak{q}_i \ge 0.42 \cdot 1.8712^i$ for F_t'. Under (W′), A(12) gives $r_K[n_{13}] \ge w_K(2\|\mathsf{P}_K\|/\mathrm{Tr}\,\mathsf{P}_K - 1) \ge w_K(e^{\gamma'} - 1)$, with $w_K \ge \frac{1}{2} c_\flat t$ for small t by Lem. 11.11(b), its ϵ being $o(t)$. With F_t'' the same gives the quotient of the class $((1,1),0)_3$, which on the cube is r_2. $0.55L - 1.1 < 0.5516L - 0.7162$, and $\frac{L}{2} < 0.6266L - 0.873$, as $L \ge 13$. In (d), the consistency is that of B(7); as $\{n(s) = n\}$ is invariant under $s \mapsto s\,\mathrm{d}\gamma$, (M) makes the unnormalised laws of each class converge, and B(2)(c) normalises them.

Step 17: Thm. AI. Under the hypothesis (H), at each $K \ge K_0$, Lem. 11.4 at $l = l_K$ gives $\theta \le 0.8901$, $\mathsf{e}(4l_K) \le \theta^4/12.9$ and $D_2(8l_K) \le 20\,\mathsf{e}(4l_K)^2 \le 0.047355$, so A(10) applies at $l_0 = 8l_K$; $m_* l_0 \ge \log\frac{1}{18\cdot 0.047355} \ge 0.1597$, and if $m_* l_0 < 3$ then $l_1 < 6/m_* \le 37.6\, l_0$ and $\sigma_* > m_*^2/6 \ge 0.00425\, l_0^{-2}$, while if $m_* l_0 \ge 3$ then $\sigma_* = m_*/l_0 \ge 3 l_0^{-2}$; with $l_0 a_K = 4s_0$ this is the stated bound in physical length. The cube and loop bounds are A(10) in physical length, with $4N/l_1 \le \ell/s_0$ and $N \ge 4l_1$ for $\ell \ge 602\, s_0$; the uniqueness of ν_K is Lem. 11.7. On a cube of side $N \ge 2l_K$, A(10) gives $\Phi \le \delta \le 0.13736\,\theta^{N/2l_K}$ and $1 - r[n] \le 0.4121\,\theta^{N/2l_K}$; so $\mathfrak{b} \le e^{0.13736} - 1 < 0.1473$ on that cube, and A(3) gives

$$\sigma(N) \ge N^{-2}\log(1/0.1473) \ge 1.9\, N^{-2}, \qquad \text{that is} \qquad \sigma_K(\ell) \ge 1.9\,\ell^{-2}.$$

For the proposed final exclusion, the following reduction requires the separate estimates (Hess) and (Flux-window). Let $s_0 \le t_1 b$. On the cube of physical side $s_0 = 2l_K a_K$, $r_1 \to v_1(s_0/b, s_0/b, s_0/b) \le C_\square t_1 \le \frac{1}{5}$ by B(3) and B(5), so $r_1 \le \frac{1}{3}$ for all large K, and A(13) gives $\mathsf{e}(l_K) \ge \log 2 > 0.069$ there, against the hypothesis (H). □

11.1.6. *Scope.* Thm. AG is a statement on one lattice, for every box and every $\beta \ge 0$, in lattice units; besides the periodic boxes it treats the infinite lattice at one β, in the last assertion of (8), in (10), under its hypothesis, one inequality between the partition functions of two periodic cubes, with its periodic Gibbs state and its loops, and in (11), for every periodic Gibbs state, and, in (4)(d), a slab with free boundaries in two directions. Thm. AH concerns the periodic boxes of a fixed physical size, at a fixed label or at labels in a compact set, at every cutoff and as $K \to \infty$, the Hamiltonians $H_{l,m}$ over fixed sections, and, in (9)(c), the cylinder over one small twisted section.

Thm. AI concerns the infinite lattice $a_K\mathbb{Z}^3$ at each cutoff $K \ge K_0$: under the hypothesis (H), the inequality $\mathsf{e}(l_K) \le 0.069$ between the partition functions of the two periodic boxes $(l_K, 2l_K, 2l_K)$ and $(2l_K)^{\times 3}$, it gives the uniqueness of the periodic Gibbs state ν_K, an area law for the loops of every rectangle with a physical tension at least $2.65 \cdot 10^{-4} s_0^{-2}$, and bounds on the cubes of every physical side $\ge s_0$, resp. $\ge 602\, s_0$, all uniform in K; it is a statement of the kind of confinement at each cutoff, conditional on the hypothesis (H). Completing (Hess) and (Flux-window) would make the final exclusion show that (H) can hold only at physical sides s_0 of the order of the label or more.

The vacuum, the gap and the clustering of Thm. AH(9) belong to one small twisted section, the uniqueness of ν_K is that of a Gibbs state of the lattice of one cutoff, and the tension $\sigma_K(l)$ of Thms. AH(8) and AI is that of a slab of fixed width l. Thm. AH(10)(b)–(d) and Thm. AI remain sketches where they use (R), (W), (W′) or (M), with the genuine box-inequality premise (H) retained for the area-law criterion. The window estimates specified in (10)(c), and (Hess) and (Flux-window) in the final small-side exclusion, are the remaining tasks; and Thm. AG(7), (8) give reflection positivity for the eight sums over sets of coordinate planes and for the admissible mixtures. A box of the class z is a centre sector of a periodic torus, that is, its fields of the class z are the fields of the torus fixed by a group generated by the shifts by the sides of the box, each followed by the multiplication by a sign field χ on the bonds with $d\chi \equiv 1$, if and only if $n_\mu n_\nu$ is even in every twisted plane $\mu\nu$ of z, n_μ being the side of the torus divided by that of the box.

Indeed, if it is such a sector, a fixed field is determined by its values at representatives of the bonds, which are free, its plaquette variables are periodic under the shifts, and on one copy of the box they are those, $\eta^z \partial U'$, of a field U' of the box; for values in the diagonal subgroup of SU(2), which holds $-\mathbf{1}$, the product of the plaquette variables over a plane $\mu\nu$ of the torus is $\mathbf{1}$, each bond entering it twice in opposite senses, and it is $(-1)^{z_{\mu\nu} n_\mu n_\nu}$, the product over the $n_\mu n_\nu$ copies of a plane of the box; the converse is the construction of Lem. 11.9, read on that torus, which asks $w_{\mu\nu} = 1$ only where n_μ is even.

On the tori $T_{\ell'}$, $\ell' = 2L^{m+i}$ with $i \geq 0$, the multiplicities of the cube $2L^m$ are L^i, odd, and a dilation by a power of L does not change them: the cube $2L^m$ with flux is a centre sector of no torus $T_{\ell'}$, and no box of a class with a twisted plane whose two sides lie in $2L^{\mathbb{Z}}$ is a centre sector of a torus $T_{\ell'}$. The cube $2L^m$ of the class z is a centre sector of the periodic tori with one doubled side in each twisted plane of z: of $(4L^m, 2L^m, 2L^m)$, the doubled side lying in the twisted plane, for one twisted plane, and in both twisted planes for two; of $(4L^m, 4L^m, 2L^m)$ or of the cube $4L^m$ for three. The chain and the two-cutoff comparison are stated on the tori T_ℓ, and Lem. 11.11 on periodic cubes.

For the pairs (U, s) of the mixture μ_{mix} of (M), let a chain be run with the averaging map $\mathcal{M}_s$ of Thm. AH(7) and with the small-field tests applied to the signed plaquette variables, $s(p)\,\partial U(p)$ on the lattice and $s^{(d)}(P)\,\partial(\mathcal{M}_s^{K-d}U)(P)$ on the block lattice of depth d. These variables, and so the small-field and the large-field regions of this chain, do not change under $(U, s) \mapsto (\gamma U, s\,\mathrm{d}\gamma)$, γ a sign field on the bonds: it multiplies $\partial U(p)$ and $s(p)$ by $d\gamma(p)$, and, at each depth, $\mathcal{M}_s^{K-d}U$ by the sign field $\bar\gamma$ on the block bonds, $\bar\gamma(c)$ the product of γ over the straight path of the block bond c, and $s^{(d)}$ by $d\bar\gamma$.

When the small-field region of this chain does not wind about the torus, the closed sign field s is $d\gamma$ on it for a sign field γ on the bonds, and the change of variables $(U, s) \mapsto (\gamma U, s\,\mathrm{d}\gamma)$, under which $A_{s\,\mathrm{d}\gamma}(\gamma U) = A_s(U)$ and which the functions H of (M) do not see, makes the integrand on that region the periodic one; $s\,\mathrm{d}\gamma$ is then a closed sign field of the class $n(s)$ that is 1 on that region, and so, for $n(s) \neq 0, -1$ on some plaquette of the large-field region of this chain (the probability estimate for such large-field events is Prop. 4.54).

12. Vanishing latent heat

Let $P(\beta)$ be the pressure per lattice site and let $J(\beta) = P'(\beta+) - P'(\beta-)$ be its derivative jump. Convexity identifies J with the possible variation of the mean action among periodic Gibbs states. The normalization of the physical action density multiplies this variation by a_K^{-4}.

Sketch K (Decay of the pressure jump). For every $c < 1/180$,

$$J(\beta) = O(\beta^{-4-c}) \qquad (\beta \to \infty).$$

Consequently, along the continuum cutoffs, periodic Gibbs states have asymptotically equal one-point action densities. In particular,

$$3a_K^{-4}|\rho(a_p) - \nu(a_p)| \le a_K^{-4} J(\beta) \longrightarrow 0.$$

This does not assert that J vanishes at a finite coupling. ■□

Sketch of the argument. Compare the pressure of the infinite lattice with that on a fixed physical torus. Reflection positivity bounds their difference, and local source estimates control its variations in the physical label. The part of the second derivative concentrated at one label is precisely the rescaled latent heat. It is therefore enough to prevent the limiting pressure response from accumulating an atom.

Repeat the construction with two blocking factors whose logarithms are incommensurable. They act as two translations of the logarithmic coupling variable. The two resulting descriptions concern the same lattice pressure. The two-period argument shows that the limiting response cannot retain a nonconstant periodic contribution common to both descriptions. In its quantitative form, it bounds the mass of the response on a short logarithmic interval by its length and an error inherited from the cutoff comparison.

The error must be estimated with its actual dependence on the torus. For an n-point function the comparison has the form

$$|S_{K'} - S_K| \le C\ell^{3(n+1)} K^{p_n} L^{-K/10}, \qquad K' \ge K \ge k + 54m, \quad \ell = 2L^m.$$

An improved bound for the discarded large-field mass supplies this rate. Choosing the relation between m and K makes the box error decrease while retaining a sufficiently fine cutoff. For every slope $A > 90$, the two-period estimate gives $\beta^4 J(\beta) \le C_A \beta^{-1/(2A)}$. Taking A sufficiently close to 90 yields every exponent $c < 1/180$.

The derivative-jump interpretation then gives the one-point conclusion directly. For a bounded compactly supported test, the difference of its means is bounded by the displayed plaquette estimate times the volume of its supporting cube and its supremum norm. Thus conditioning on translation-invariant events cannot leave different continuum one-point constants.

The threshold $k + 54m$ grows with the box side and must remain in the argument. Replacing it by a cutoff independent of the side would be a stronger estimate, with a stronger conclusion about J. The present reasoning controls the jump asymptotically; it does not prove uniqueness of all finite-coupling Gibbs states.

12.1. **Latent heat and one-point constants.** Let $P = \lim_{N\to\infty} N^{-3} \log Z_N$ be the lattice pressure and $J(\beta) = P'(\beta+) - P'(\beta-)$ its derivative jump. We prove $J(\beta) = O(\beta^{-4-c})$ for every $c < 1/180$. The proof combines a reflection-positive lower bound, comparison with finite tori and an oscillation estimate for two incommensurable blocking factors.

The cutoff threshold depends on the size of the box. We retain this dependence: the result is a decay estimate for the jump and its continuum one-point consequences. The two-period lemma and the improved discarded-mass estimate are the main quantitative steps.

12.1.1. *Notation and hypotheses. One lattice.* For a plaquette p of $\mathbb{Z}^3$ with barycentre $x(p)$ and holonomy $\partial U(p)$ we put $a_p = 1 - \frac{1}{2}\operatorname{tr} \partial U(p) \in [0, 2]$, and $\mathcal{O}_1(f) = \sum_p f(x(p))a_p$ for bounded f of compact support. For an integer $N \ge 3$, S_N is the sum of the a_p over the $3N^3$ plaquettes of the periodic cube with N^3 sites, $Z_N(\beta) = \int e^{-\beta S_N}\, \mathrm{d}U$ with the Haar probability measure of SU(2) on every bond, $\mu_{N,\beta}$ is the Wilson measure of the cube, read as a law on the configurations of $\mathbb{Z}^3$ that is carried by the $N\mathbb{Z}^3$-periodic configurations (the periodic Wilson law in equation YM3_I:(1817), with $\Gamma = N\mathbb{Z}^3$), and $P_N = N^{-3} \log Z_N$, so $P = \lim_N P_N$.

For $x \in \mathbb{Z}^3$ let $c_x = \frac{1}{2}\sum a_p$, the sum over the six faces of the unit cube $x + [0, 1]^3$. A Gibbs state of Wilson's action at the spacing $a > 0$ and the coupling β is a Borel probability measure ν on the configurations of $a\mathbb{Z}^3$ such that $\nu(\gamma_B F) = \nu(F)$ for every finite set B of bonds and every continuous

function F of finitely many bonds; here the Gibbs kernel γ_B is the normalised average over the configurations on B, with the Haar measure of SU(2) on every bond of B and the weight $e^{-\beta\sum_p a_p}$, the sum over the plaquettes p that contain a bond of B, the configuration off B being kept fixed (Section YM3_I:91).

A Gibbs state is periodic if it is invariant under the translations by a subgroup of finite index of $a\mathbb{Z}^3$ (Section YM3_I:91); by Prop. YM3_I:103.4 every periodic Gibbs state is invariant under all isometries of $a\mathbb{Z}^3$, so the periodic Gibbs states are exactly the Gibbs states that are invariant under $a\mathbb{Z}^3$. $\mathcal{G}(\beta)$ is the set of all periodic Gibbs states on $\mathbb{Z}^3$ at β, that is, at the spacing 1 (the set $\mathcal{G}^{\rm per}_{1,\beta}$ of Section YM3_I:91), and $e_\nu = \nu(a_p)$ for $\nu \in \mathcal{G}(\beta)$; this is one number for all p (Lem. 12.1). For a law $\mathfrak{s}$ that is invariant under $\mathbb{Z}^3$ we write $\langle f; h\rangle_{\mathfrak{s}} = \mathrm{Cov}_{\mathfrak{s}}(\mathcal{O}_1(f), \mathcal{O}_1(h))$ and $\Gamma_{\mathfrak{s}}(z) = \mathrm{Cov}_{\mathfrak{s}}(c_0, c_z)$.

Tents and the reference. $\Lambda_\eta(u) = (1 - |u|/\eta)_+$ for real u and $\eta > 0$. For an integer $\eta \geq 1$ and $X \in \mathbb{Z}^3$ the tent is $\varphi^\eta_X(z) = \prod_{i\leq 3} \Lambda_\eta(z_i - \frac{1}{2} - \eta X_i)$, the hull is $h^\eta = \sum_{|X|_\infty \leq 2} \varphi^\eta_X$, the sum of the 5^3 tents nearest to φ^η_0, and $\varphi^\eta_c(x) = \prod_i \Lambda_\eta(x_i)$ for $x \in \mathbb{Z}^3$. For a law $\mathfrak{s}$ invariant under $\mathbb{Z}^3$ we put $s^\eta_{\mathfrak{s}}(X) = \langle \varphi^\eta_0; \varphi^\eta_X\rangle_{\mathfrak{s}}$, and the near part of the specific heat is $n_\eta(\mathfrak{s}) = \eta^{-3}\langle \varphi^\eta_0; h^\eta\rangle_{\mathfrak{s}}$. For $r_1 > 0$ the reference is

$$\hat{n}(\beta) = \liminf_{N\to\infty} n_{\lceil r_1\beta\rceil}(\mu_{N,\beta}),$$

the lower limit along all integers $N \geq 3$; it is formed from Wilson's action and r_1 alone. $P''(d\beta)$ is the second derivative of the convex function P, a positive measure, and $\mathbb{M}$ is the image under $s = \log\beta$ of the signed measure $\beta^4[P''(d\beta) - \hat{n}(\beta)d\beta]$. Thus $\mathbb{M}(\{s\}) = e^{4s}J(e^s)$, and

$$|\mathbb{M}|([s, s+\delta]) = \int_{[\beta, e^\delta\beta]} x^4\, |P''(dx) - \hat{n}(x)\,\mathrm{d}x| \qquad (\beta = e^s).$$

Settings. A setting consists of a radius $\varrho > 0$, a common choice $\varsigma^{\rm S}$ of the real and source construction constants and auxiliary data in Section YM3_I:82 at the radius required by Thms. YM3_I:AJ and YM3_I:AD, with eq. 2.iv, and the associated Wilson measures and source normalization. Write $\mathcal{T}^{\rm S}[2\varrho]$ for this ordered choice and $\varsigma = \pi^+(\varsigma^{\rm S})$ for its restriction to the real construction, denoted by $\mathcal{T}^{\rm F+}$. The numerical constants retain their values: the real hypotheses are those of Prop. YM3_I:69.1, Prop. YM3_I:69.3 and Cor. YM3_I:69.4; the additional source floors are imposed in their stated order. The coupling ceiling $\gamma \in (0, \frac{1}{2}]$ is chosen last.

Among the constants chosen in Prop. YM3_I:69.1 are the odd block size $L \geq 13$, the cell size M, a power of L, the flow bound $B_\sharp$ with $\bar{B} = B_\sharp L/(L-1)$, the exponent $\mathfrak{p}_0 \in \mathbb{N}$ of the large-field threshold, the exponent r of the cube sizes, with which the cube size at a coupling $x > 1$ is $\mathcal{R}(x) = \min\{L^c : c \in \mathbb{Z}_{\geq 0},\ L^c \geq (\log x)^r\}$ (equation YM3_I:(634); below it is read at couplings $x \geq \gamma^{-2}$ alone), and the amplitudes A_1 and A_0. A lowered choice ς' of a choice ς of $\mathcal{T}^{\rm F+}$ (Def. 4.47) keeps the values of the constants before A_1 and of A_0, has $B'_\sharp \geq B_\sharp$, and has $A'_1 = \lambda'_\natural A_1$ with $\lambda'_\natural = \lambda_\natural/(2 + 30L)$, where $\lambda_\natural \in (0, \frac{1}{8\pi}]$ is a number of ς; so the small-field thresholds δ_j of the two choices satisfy $\delta'_j = \lambda'_\natural\delta_j$.

For the ordered choices constructed in Lem. 4.48, lowered choices exist: decreasing A_1 preserves the eight inequalities at A_0, and the later constants are reselected in the order proved there. Def. 4.47 attaches to the pair (ς, ς') a floor $\mathsf{Y}^\natural$ for the logarithm of the last coupling and the rate $c_2^\natural = \frac{1}{2}\min\{\frac{1}{8}c_1^\natural, \lambda_\natural^2 A_1^2/2A_0^2\}$, with a number $c_1^\natural > 0$ of the pair. We put $a_K = L^{-K}$, $\mathbb{L}_K = a_K\mathbb{Z}^3$, $\lambda = \log L$, $\ell = 2L^m$ and $T_\ell = \mathbb{R}^3/\ell\mathbb{Z}^3$; the label of a pair (K, β) is $b = a_K\beta$, and

$$a_K^{-4}J(\beta) = b^{-4}\beta^4 J(\beta).$$

The physical scale. For the plaquettes p of $\mathbb{L}_K$ we put $\mathcal{O}_K(f) = a_K^{-1}\sum_p f(x(p))a_p$ and $r_K(f) = \frac{1}{3}a_K^3\sum_p f(x(p))$, a Riemann sum of $\int f$; and $\|f\|_\sharp = \max\{\|f\|_\infty, 40M\,\mathrm{Lip}\, f\}$. $\mathcal{G}_{K,\beta}$ is the set of all periodic Gibbs states on $\mathbb{L}_K$ at β, that is, at the spacing a_K (the set $\mathcal{G}^{\rm per}_{K,\beta}$ of Section YM3_I:91), a convex set (Props. YM3_I:91.1 and YM3_I:103.4), and its extreme points are the pure phases

(Prop. YM3_I:106.1). $S_n^T[\nu](f_1,\dots,f_n)$ is the joint cumulant of $\mathcal{O}_K(f_1),\dots,\mathcal{O}_K(f_n)$ in a law ν on $\mathbb{L}_K$; for real Lipschitz f_i on T_ℓ, $S^T_{n;K,\ell,\beta}(f_1,\dots,f_n)$ is the joint cumulant under the Wilson measure of T_ℓ at the spacing a_K, that is, of the periodic cube with ℓL^K sites a side. $f^{\langle\ell\rangle}=\sum_{v\in\ell\mathbb{Z}^3} f(\cdot-v)$ is the periodisation. The spread of $(f_1,\dots,f_n)$ is the least sup-diameter of a set of points taken one from each support; for $n=2$ it is the sup-distance of the two supports. On T_ℓ one has $|\mathcal{O}_K(f)|\le 6\ell^3L^{4K}\|f\|_\infty$, so that

$$|S^T_{n;K,\ell,\beta}(f_1,\dots,f_n)|\le \mathsf{op}_n\prod_{i\le n}6\ell^3L^{4K}\|f_i\|_\infty,$$

with a combinatorial number op_n of n; we call this the crude bound.

Labels and windows. For $0<g\le\gamma$ let $X_g=L(g^{-2}+B_\sharp)$ and $I(g)=(g^{-2}+\bar B, X_g-\bar B]$ (Section YM3_I:74), and let $\mathfrak{d}(g)$ be the canonical family at g ([7, §K.1]), the set of the first-passage data at g (Cor. YM3_I:74.2): the pairs (K,β), $K\in\mathbb{N}$, for which K is the least index k at which the running coupling $x_k(\beta)$, the member of index k of the sequence of the increments from $x_0(\beta)=\beta$, is defined, real and at most X_g; in words, the running coupling arrives at the scale of g at the last step. We use two facts on $\mathfrak{d}(g)$. By [7, §K.1], the labels $a_K\beta$ of the pairs of $\mathfrak{d}(g)$ lie in the compact interval $I^{\mathrm{lab}}(g)=[g^{-2}+B_\sharp\frac{L-2}{L-1},\,L(g^{-2}+\bar B)]$ of [7, §K.1], whose lower end is at least g^{-2}; so $a_K\beta\ge g^{-2}$ on $\mathfrak{d}(g)$. By [7, §K.1], $(K,L^Kb)\in\mathfrak{d}(g)$, with the label b, for every $b\in I(g)$ and $K\in\mathbb{N}$. A window is an interval of labels; in a setting we use the window

$$\mathsf{I}=[b_-,b_+],\qquad b_-=\gamma^{-2}+\bar B+\tfrac12,\qquad b_+=L(\gamma^{-2}+3\bar B+1)-2\bar B,$$

and $g_2=(\gamma^{-2}+2\bar B+1)^{-1/2}$. $m_0\ge1$ is the least integer such that the sides $2L^m$, $m\ge m_0$, are admitted at γ and at g_2 (the sides that the section of Section YM3_I:74 attaches to a reference coupling), and $\ell_0=2L^{m_0}$. With $\varepsilon=L^{-1}$, the tent $\mathsf{f}(y)=\prod_i\Lambda_\varepsilon(y_i-\frac12)$ and its hull $\mathsf{h}=\sum_{|X|_\infty\le2}\mathsf{f}(\cdot-\varepsilon X)$, the far part is

$$y_{m,K}(\upsilon)=b^5\varepsilon^{-3}S^T_{2;K,\ell,L^Kb}(\mathsf{f}^{\langle\ell\rangle},1-\mathsf{h}^{\langle\ell\rangle})\big|_{b=e^\upsilon},\qquad \ell=2L^m.$$

When two settings (S_1), (S_2) are used, an index i marks the objects of (S_i): L_i, $\lambda_i=\log L_i$, M_i, $\mathsf{I}_i=[b_{-,i},b_{+,i}]$, $m_{0,i}$, $\ell_{0,i}$, ε_i, $y^{(i)}_{m,K}$, and so on.

Tuples and functionals. A separated tuple is $X=(f_1,\dots,f_n)$ with real $f_i\in C^1_c(\mathbb{R}^3)$ and pairwise disjoint supports; the empty tuple is one of them, (f) is the tuple with the one entry f, $|X|=n$ is the number of entries, $X_A=(f_i)_{i\in A}$ for $A\subset\{1,\dots,n\}$, $(\tau_vf)(x)=f(x-v)$, $\tau_vX=(\tau_vf_i)_i$, and $X\cdot Y$ is the joined tuple, formed when its entries have pairwise disjoint supports. A symmetric functional is a map σ on the separated tuples that is multilinear and symmetric in the entries, with σ of the empty tuple equal to 1. For a law ν on $\mathbb{L}_K$ we put

$$X^\nu_K=\prod_{i\le n}[\mathcal{O}_K(f_i)-\nu(\mathcal{O}_K(f_i))],\qquad \sigma_\nu(X)=\nu(X^\nu_K).$$

A Gibbs sequence at g is $(K_j,\beta_j,\nu_j)_j$ with $(K_j,\beta_j)\in\mathfrak{d}(g)$, $K_j\to\infty$ and $\nu_j\in\mathcal{G}_j=\mathcal{G}_{K_j,\beta_j}$. A limit in its frame is $\sigma'(X)=\lim_j\rho_j(X^{\nu_j}_{K_j})$, for all separated X, with approximants $\rho_j\in\mathcal{G}_j$. The sequence converges if $\sigma(X)=\lim_j\sigma_{\nu_j}(X)$ exists for every separated X; then σ is its limit, and $S^\sigma_2(f,h)=\sigma((f,h))-\sigma((f))\sigma((h))$. For separated $X^1,\dots,X^q$ the iterated plateau of a symmetric functional σ is the iterated limit

$$\lim_{|v_q|\to\infty}\cdots\lim_{|v_2|\to\infty}\sigma\big(X^1\cdot\tau_{v_2}X^2\cdots\tau_{v_q}X^q\big),$$

the translations receding one after the other, v_2 first; at $q=1$ it is $\sigma(X^1)$, and at $q=2$ it is $\lim_{|v|\to\infty}\sigma(X^1\cdot\tau_vX^2)$. A canonical decomposition $(\mathsf{X},\varpi,(\sigma_\xi)_{\xi\in\mathsf{X}})$ of σ consists of (Can1) a compact metrisable space X, (Can2) a Borel probability measure ϖ of full support on it, (Can3) a map $\xi\mapsto\sigma_\xi$ into the symmetric functionals that is continuous for the pointwise convergence and injective, such

that (Can4) σ has iterated plateaux and $\int \prod_{i\le q} \sigma_\xi(X^i)\,\mathrm{d}\varpi$ is the iterated plateau of $(X^1,\dots,X^q)$, for all $q\ge 1$ and all separated $X^1,\dots,X^q$; at $q=1$, $\sigma=\int \sigma_\xi\,\mathrm{d}\varpi$.

The σ_ξ are the components. A symmetric functional has one vacuum if it has a canonical decomposition whose space is a point. By Thm. YM3_I:AP, the limit σ of a convergent Gibbs sequence at g has a canonical decomposition, unique up to isomorphism; every component σ_ξ is a limit in the frame along the full sequence, with the approximants $\nu_j(\cdot\,|B_j)$, B_j invariant with $\nu_j(B_j)>0$; $\sigma_\xi((f))=\mathfrak{c}_\xi\int f$ with a number $\mathfrak{c}_\xi$; $|\sigma_\xi(X\cdot\tau_v Y)-\sigma_\xi(X)\sigma_\xi(Y)|\le C^g(X,Y)/|v|$ for $|v|\ge r(X,Y)$, with the numbers C^g, r of the paragraph on one cutoff below; and $\#\mathsf{X}=\dim\ker H_\sigma$, both finite and equal or both infinite, where $H_\sigma\ge 0$ is the Hamiltonian of the reconstruction of σ in any of the six coordinate directions and $H_\sigma\Omega_\sigma=0$ for its vacuum vector Ω_σ.

So such a limit has one vacuum if and only if $\ker H_\sigma=\mathbb{C}\Omega_\sigma$. We put $\kappa=\int \mathfrak{c}_\xi^2\,\mathrm{d}\varpi$.

Limits along sequences of pairs. A sequence of pairs is $\mathfrak{p}=(K_j,\beta_j)_j$ with $K_j\in\mathbb{N}$, $\beta_j>0$ and $K_j\to\infty$; it is at the label $b>0$ if $a_{K_j}\beta_j\to b$. $\mathsf{K}_\mathfrak{p}$ is the set of the functionals σ for which there are $\nu_j\in\mathcal{G}_{K_j,\beta_j}$ with $\sigma(X)=\lim_j\sigma_{\nu_j}(X)$ for every separated X, along the full sequence; it carries the topology of pointwise convergence. $\mathsf{K}^\bullet_\mathfrak{p}$ is the set of its members with one vacuum. $\mathrm{Lim}[b]$ is the union of the sets $\mathsf{K}_\mathfrak{p}$ over the sequences of pairs at the label b, and $\mathrm{Lim}^\bullet[b]$ the set of its members with one vacuum; for a set $\mathfrak{q}$ of pairs, $\mathrm{Lim}_\mathfrak{q}[b]$ and $\mathrm{Lim}^\bullet_\mathfrak{q}[b]$ are the same unions over the sequences all of whose pairs lie in $\mathfrak{q}$. A sequence of pairs $\mathfrak{p}$ is saturated if $\mathsf{K}_\mathfrak{p}\ne\emptyset$ and $\mathsf{K}_{\mathfrak{p}'}=\mathsf{K}_\mathfrak{p}$ for every subsequence $\mathfrak{p}'$. For a Borel probability measure μ on a compact metrisable set of symmetric functionals, $\mathrm{bar}(\mu)=\int\psi\,\mu(\,\mathrm{d}\psi)$ is its barycentre.

Dilations; the reconstruction at every label. For $z\in\mathbb{Z}$ and a separated tuple $X=(f_1,\dots,f_n)$ put $X^{[z]}=(f_i(L^z\cdot))_i$, again a separated tuple, and for a symmetric functional σ put $\sigma^{[z]}(X)=L^{-z|X|}\sigma(X^{[z]})$. The dilation $x\mapsto L^zx$ carries a law ν on $\mathbb{L}_K$ to a law $\nu^{(z)}$ on $\mathbb{L}_{K-z}$, maps $\mathcal{G}_{K,\beta}$ onto $\mathcal{G}_{K-z,\beta}$ ([7, Lem. K.1]) and turns $\mathcal{O}_{K-z}(f)$ into $L^{-z}\mathcal{O}_K(f(L^z\cdot))$ ([7, Lem. K.1], applied to the function $f(L^z\cdot)$: the dilated function $f^{[z]}$ of that Proposition is $f(L^{-z}\cdot)$, whereas the tuple $X^{[z]}$ above has the entries $f_i(L^z\cdot)$); so $\sigma_{\nu^{(z)}}=(\sigma_\nu)^{[z]}$, and the pair $(K-z,\beta)$ has the label $L^za_K\beta$.

Hence, if $\mathfrak{p}=(K_j,\beta_j)_j$ is a sequence of pairs at the label b and $\sigma\in\mathsf{K}_\mathfrak{p}$, then $\sigma^{[z]}\in\mathsf{K}_{\mathfrak{p}'}$ for the sequence of pairs $\mathfrak{p}'=(K_j-z,\beta_j)_{j\ge j_0}$ at the label L^zb, where $K_j>z$ for $j\ge j_0$. As $\sigma^{[z]}(X\cdot\tau_vY)=L^{-z(|X|+|Y|)}\sigma(X^{[z]}\cdot\tau_{L^{-z}v}Y^{[z]})$, the conditions (Can1)–(Can4) pass through the dilation, the receding translations by v becoming translations by $L^{-z}v$: $(\mathsf{X},\varpi,(\sigma_\xi)_{\xi\in\mathsf{X}})$ is a canonical decomposition of σ if and only if $(\mathsf{X},\varpi,(\sigma^{[z]}_\xi)_{\xi\in\mathsf{X}})$ is one of $\sigma^{[z]}$, with the same space X.

For every $b>0$ there are an integer $z\ge 0$ and $0<g\le\gamma$ with $g^{-2}+\bar B<L^zb<X_g-\bar B$, because $L^zb>\gamma^{-2}+\bar B$ for large z and these open intervals overlap one another as g decreases and cover $(\gamma^{-2}+\bar B,\infty)$; we fix such a pair (z,g) for b, with $z=0$ if b itself lies in one of these intervals. Then $L^za_{K_j}\beta_j\in I(g)$ for large j, so $(K_j-z,\beta_j)\in\mathfrak{d}(g)$ by [7, §K.1], and $\sigma^{[z]}$ is the limit of a convergent Gibbs sequence at g over these pairs, to which Thm. YM3_I:AP applies. For $\sigma\in\mathsf{K}_\mathfrak{p}$ we put $H_\sigma=L^zH_{\sigma^{[z]}}$ and $\Omega_\sigma=\Omega_{\sigma^{[z]}}$, on the space of the reconstruction of $\sigma^{[z]}$, in each of the six coordinate directions, which the dilation keeps, as it commutes with the coordinate mirrors: the time t of $\sigma^{[z]}$ is the time $L^{-z}t$ of σ, and the kernel is that of $H_{\sigma^{[z]}}$.

As the canonical decompositions of σ and of $\sigma^{[z]}$ correspond to one another with the same space X, which is unique up to homeomorphism (Thm. YM3_I:AP at $\sigma^{[z]}$), $\dim\ker H_\sigma=\#\mathsf{X}$, both finite and equal or both infinite, and $\ker H_\sigma=\mathbb{C}\Omega_\sigma$ if and only if σ has one vacuum; these statements read X alone, so they do not depend on the choice of (z,g), and at $z=0$ they are those of Thm. YM3_I:AP for σ itself.

One cutoff. $\mathcal{I}$ is the σ-field of the Borel sets invariant under the translations by $a_K\mathbb{Z}^3$. By Prop. YM3_I:106.3, for $(K,\beta)\in\mathfrak{d}(g)$ the set $\mathcal{G}_{K,\beta}$ is a class at the cutoff K that obeys the bound system at g: it is not empty; its members are invariant under the isometries of $\mathbb{L}_K$ and reflection positive at every lattice mirror; with ν it contains $h\nu/\nu(h)$ for every bounded $\mathcal{I}$-measurable $h\ge 0$

with $\nu(h) > 0$, so in particular the conditioned states $\nu(\cdot\,|B)$, $B \in \mathcal{I}$, $\nu(B) > 0$; and it obeys the bound system $\mathsf{C}^g = (\hat{C}_1^g, (\hat{C}_n^g)_{n\ge 2})$ at g (equation YM3_I:(2030); Section YM3_I:100), which has two members: for $n \ge 2$ the cumulants of separated tuples of type $(\mathsf{d}; R)$, that is, with supports at mutual distances $\ge \mathsf{d}$ and each in a ball of radius R, satisfy $|S_n^T[\rho](f_1, \dots, f_n)| \le \hat{C}_n^g(\mathsf{d}; R) \prod_i \|f_i\|_\sharp$ for $\rho \in \mathcal{G}_{K,\beta}$; and the one-point functions of two members of the class satisfy $|\rho(\mathcal{O}_K(f)) - \nu(\mathcal{O}_K(f))| \le \hat{C}_1^g(R)\|f\|_\sharp$ for $\rho, \nu \in \mathcal{G}_{K,\beta}$ and every $f \in C_c^1(\mathbb{R}^3)$ with support in a closed ball of radius R.

These numbers are free of K, β and the states. For such a class the plateau theorem, Prop. YM3_I:100.1, provides for separated X and $\bar{R} \in a_K\mathbb{N}$ numbers $\delta(X; \bar{R}) > 0$, formed from X and $\bar{R}$, and $\mathsf{N}(X; \bar{R})$, formed from X, $\bar{R}$ and both members of the bound system at g, and a constant $\mathfrak{k}_0$ (equations YM3_I:(1976)–(1980)); none of them depends on the cutoff, the coupling or the state. Its clustering inequality says: if ν belongs to the class, X, Y are separated with supports in $(-\bar{R}, \bar{R})^3$, $a_K \le 2\min\{\delta(X; \bar{R}), \delta(Y; \bar{R})\}$, $v \in a_K\mathbb{Z}^3$ and $|v| \ge 4\sqrt{3}\bar{R}$, then $X \cdot \tau_v Y$ is formed and

$$\big|\sigma_\nu(X \cdot \tau_v Y) - \nu\big(E_\nu[X_K^\nu|\mathcal{I}]\, E_\nu[Y_K^\nu|\mathcal{I}]\big)\big| \le \mathfrak{k}_0\, [\mathsf{N}(X; \bar{R})\mathsf{N}(Y; \bar{R})]^{1/2}\, \bar{R}/|v|.$$

With it comes the statement on iterated plateaux (Prop. YM3_I:100.2, with the conditional means defined immediately before it): the limit σ of a convergent Gibbs sequence at g has iterated plateaux, and the plateau of $(X^1, \dots, X^q)$ is $\lim_j \nu_j\big(\prod_{i\le q} E_{\nu_j}[(X^i)_{K_j}^{\nu_j}|\mathcal{I}]\big)$. We put $r(X, Y) = 4\sqrt{3}\bar{R}$ and $C^g(X, Y) = \mathfrak{k}_0[\mathsf{N}(X; \bar{R})\mathsf{N}(Y; \bar{R})]^{1/2}\bar{R}$ where $\bar{R}(X, Y)$ is the least L^k, $k \in \mathbb{Z}$, for which $\operatorname{supp} X \cup \operatorname{supp} Y \subset (-L^k/2, L^k/2)^3$, and is 1 when both supports are empty; these are the numbers of the clustering of components above, and they depend on X, Y, the bounds at g, L and M alone.

12.1.2. *Pressure curvature and the two blocking factors.* For the unit lattice, write $P_N(\beta) = N^{-3}\log Z_N(\beta)$ and $P = \lim_N P_N$. The finite-volume identity $P_N'' = N^{-3}\operatorname{Var}(\sum_p a_p) \ge 0$ gives convexity. The one-sided derivatives of P describe the extremal mean action densities of periodic Gibbs states. Its derivative jump

$$J(\beta) = P'(\beta+) - P'(\beta-)$$

is therefore the quantity that bounds differences between those one-point expectations. The problem is to control this possible atomic part of the pressure curvature at large β.

Comparison with finite tori separates the pressure response that is already represented by local covariances from a possible residual response. The proof treats the second derivative of a convex function as a measure; differentiability is not assumed at the point where a jump is to be estimated. After multiplying by β^4 and changing variable to $s = \log\beta$, a jump of size $J(\beta)$ is an atom of size $\beta^4 J(\beta)$ in the corresponding response measure. This normalization matches the physical one-point scale.

One blocking factor compares the response at values of s separated by $\log L$. Such a comparison still permits an oscillation tied to that one logarithmic period. The second admitted construction uses a factor with an incommensurable logarithm. Both describe the same Wilson pressure, so their comparisons constrain the same residual response. The two-period estimate controls the possible concentration that a single period leaves open. In particular, the argument does not infer continuity merely from a bounded second derivative at finite volume; it estimates how a concentrating part could survive both comparisons.

The comparison error has to be made uniform on the boxes used in this argument. The discarded large-field mass is improved before the box side is allowed to grow. A typical cutoff comparison retained in the detailed proof is

$$|S_{K'} - S_K| \lesssim \ell^{3(n+1)} K^{p_n} L^{-K/10}, \qquad \ell = 2L^m, \qquad K \ge k + 54m.$$

Here S_K is the n-point source cumulant at the cutoff under comparison, and p_n depends on n. The power of ℓ competes with the decay in K. The restriction on K is consequently part of using the

estimate, rather than a threshold that may be fixed independently of the box. The two-period argument and the scale balance give

$$J(\beta) = O(\beta^{-4-c}), \qquad c < 1/180.$$

To read this in physical units, set $\beta_K = L^K b$ and $a_K = L^{-K}$. Differences of periodic-state plaquette means are bounded by the pressure jump. Hence

$$3a_K^{-4}|\rho(a_p) - \nu(a_p)| \le a_K^{-4} J(\beta_K) \longrightarrow 0.$$

The strict improvement over β^{-4} is what makes the last limit vanish. It gives agreement of the physical one-point constants in this comparison. It does not assert that J vanishes identically beyond a finite coupling or that all Gibbs states at that coupling coincide. The state-space and subsequence statements in the detailed argument retain these distinctions.

12.1.3. *Principal results.*

Theorem AJ.

(A) *The limit.*

(1) $\beta^4 J(\beta) \to 0$ as $\beta \to \infty$ through the reals; equivalently, $\beta^4 \sup\{|\nu(a_p) - \nu'(a_p)| : \nu, \nu' \in \mathcal{G}(\beta)\} \to 0$.

(2) Let (S_1), (S_2) be settings at one radius with $L_2 = L_1 + 2$ (such pairs exist at every radius), and let $r_1 > 0$ with $13 r_1 b_{+,i} \le 1$, $i = 1, 2$. There is $C < \infty$, formed from the two settings and r_1, such that $\limsup_{s\to\infty} |\mathbb{M}|([s, s+\delta]) \le C\delta$ for every $\delta > 0$.

(3) Let the two settings of (2) have one common value of $\mathfrak{p}_0$; for every radius and every integer $a \ge 1$ there are such pairs with $\mathfrak{p}_0 \ge a$. There are $C_{\mathrm{A}}, s_{\mathrm{A}} < \infty$, formed from the two settings and r_1, such that $\beta^4 J(\beta) \le C_{\mathrm{A}} (\log \beta)^{-\mathfrak{p}_0/3}$ for $\beta \ge 3$, and $|\mathbb{M}|([s, s+\delta]) \le C_{\mathrm{A}}\delta$ for $s \ge s_{\mathrm{A}}$ and $C_{\mathrm{A}} s^{-\mathfrak{p}_0/3} \le \delta \le 1$. Hence for every real q_0 a pair of settings with $\mathfrak{p}_0 \ge 3q_0$ gives $\sup_{\beta\ge 3} (\log\beta)^{q_0} \beta^4 J(\beta) < \infty$, and then, in every setting and for every $0 < g \le \gamma$, $a_K^{-4} J(\beta) \le C' K^{-q_0}$ for all $(K, \beta) \in \mathfrak{d}(g)$, with a number C' free of K and β.

(4) In either of the two settings of (2) put, for $K \in \mathbb{N}$ and $b \in \mathsf{I}$,

$$u_K(b) = a_K^{-3}[P(L^K b) - P_{N_0}(L^K b)], \qquad N_0 = \ell_0 L^K,$$

the free energy per physical volume of the infinite lattice minus that of the torus T_{ℓ_0}. Then $-3C_{\mathrm{v}} \le u_K \le 0$ on I, with the number C_{v} of Prop. 12.6, and there are $C_{\mathrm{sc}}, C_u < \infty$, formed from the two settings and r_1, such that $u_K + \frac{1}{2} C_{\mathrm{sc}} b^2$ is convex on (b_-, b_+) for every K. Every sequence of cutoffs has a subsequence along which u_K converges uniformly on the compact subsets of (b_-, b_+); every such limit u is continuously differentiable, $|u'(b) - u'(b')| \le C_u |b - b'|$, $-3C_{\mathrm{v}} \le u \le 0$, $u + \frac{1}{2} C_{\mathrm{sc}} b^2$ is convex; and along the subsequence the one-sided derivatives $u_K'(b\pm)$ converge to $u'(b)$ uniformly on the compact subsets. For $\nu \in \mathcal{G}_{K, L^K b}$ and bounded f of compact support,

$$\nu(\mathcal{O}_K(f)) - \tilde{\mu}_{K,\ell_0,L^K b}(\mathcal{O}_K(f)) = -\mathsf{u}\, r_K(f) \qquad \text{with a number } \mathsf{u} \in [u_K'(b-), u_K'(b+)],$$

where $\tilde{\mu}_{K,\ell_0,\beta}$ is the Wilson measure of T_{ℓ_0} at the spacing a_K: along the subsequence the energy density of the periodic Gibbs states, counted from that of the torus T_{ℓ_0}, converges to $-u'(b)$, uniformly in the state and in the label on compact sets.

(B) *The power.* Let (S_1), (S_2) and r_1 be as in (A)(2), and let ς_i' be a lowered choice of ς_i, $i = 1, 2$.

(1) For every real $A > 90$ there are $C_{\mathrm{h}}, C_{\mathrm{r}}, \beta_1 < \infty$, given by eq. 12.xi, which depend only on A, r_1, the two settings and the pairs $(\varsigma_i, \varsigma_i')$, such that for $\beta \ge \beta_1$, $s = \log\beta$ and $0 \le \delta \le 1$

$$|\mathbb{M}|([s, s+\delta]) \le C_{\mathrm{h}}\delta + C_{\mathrm{r}} \beta^{-1/2A}, \qquad \beta^4 J(\beta) \le C_{\mathrm{r}} \beta^{-1/2A}.$$

At $A = 91$ the power is $\beta^{-1/182}$.

(2) Consequently $c_*(c) = \sup_{\beta\ge1} \beta^{4+c} J(\beta)$ is finite for every real $c < \frac{1}{180}$; it is a number of Wilson's action alone.
(3) Consequently $\sup_{\beta\ge3} (\log\beta)^q \beta^4 J(\beta) < \infty$ for every real q, at this one pair of settings.
(4) The two bounds of (1) hold, with other constants formed from the same objects, also when only ς_1' is given; then Thm. YM3_I:D is used at the three choices ς_1, ς_1', ς_2.
(5) Let $A > 90$ and $C_{\rm h}$, $C_{\rm r}$, β_1 be as in (1). For $i = 1, 2$, all $m \ge m_{0,i}$, all $K \in \mathbb{N}$ with $L_i^K b_{-,i} \ge \beta_1$ and all $G \in C^2(\mathbb{R})$ with support in $\log \mathsf{I}_i$,

$$\Big| \int G(v)\, y^{(i)}_{m,K}(v)\, \mathrm{d}v \Big| \le (C_{\rm h} + C_{\rm r} + C_N) \|G\|_1 + (C_{\rm h} + C_{\rm r})(L_i^K b_{-,i})^{-1/2A} \|G'\|_1 + c_{{\rm P},i} L_i^{-m} \mathsf{Q}(G),$$

with $\mathsf{Q}(G) = \|G''\|_1 + 7\|G'\|_1 + 12\|G\|_1$, $c_{{\rm P},i}$ of Prop. 12.6(c) and $C_N = \max_i C_i^{\rm ref}$ of Prop. 12.6(d): in the mean over the labels the far part is bounded on all sides, beyond a cutoff that does not depend on the side.

(C) *The physical scale.* In any setting let $0 < g \le \gamma$, $(K, \beta) \in \mathfrak{d}(g)$, $\rho, \nu \in \mathcal{G}_{K,\beta}$ and $0 \le c < \frac{1}{180}$; $e_\rho = \rho(a_p)$ is one number for all plaquettes p of $\mathbb{L}_K$. For every plaquette p of $\mathbb{L}_K$ and every bounded f with support in $[-R, R]^3$,

$$3a_K^{-4} |\rho(a_p) - \nu(a_p)| \le a_K^{-4} J(\beta) \le g^{8+2c} c_*(c) L^{-cK},$$
$$|\rho(\mathcal{O}_K(f)) - \nu(\mathcal{O}_K(f))| = 3a_K^{-4} |e_\rho - e_\nu|\, |r_K(f)| \le a_K^{-4} J(\beta) (2R + a_K)^3 \|f\|_\infty.$$

(D) *One-point functions.* In any setting let $0 < g \le \gamma$ and let $(K_j, \beta_j, \nu_j)_j$ be a Gibbs sequence at g.
(1) For $(K, \beta) \in \mathfrak{d}(g)$, $a_K^{-4} J(\beta) \le g^8 \sup_{\beta' \ge L^K g^{-2}} \beta'^4 J(\beta')$, which tends to 0 as $K \to \infty$. So $\rho_j(\mathcal{O}_{K_j}(f)) - \nu_j(\mathcal{O}_{K_j}(f)) \to 0$ for all $\rho_j \in \mathcal{G}_j$ and all bounded f of compact support.
(2) Every limit σ' in the frame of the sequence has $\sigma'((f)) = 0$ for all $f \in C_c^1(\mathbb{R}^3)$.
(3) If the sequence converges, with the limit σ, then $\mathfrak{c}_\xi = 0$ for every canonical decomposition of σ and every ξ, so $\kappa = 0$; every component σ_ξ is the limit of the Gibbs sequence $(K_j, \beta_j, \nu_j(\cdot\,|B_j))_j$, along the full sequence, with $B_j \in \mathcal{I}$, $\nu_j(B_j) > 0$, so that Thm. YM3_I:AP applies to it, with its reconstruction, and it has one vacuum. For a component of such a limit the three properties $\mathfrak{c}_\xi = 0$, "σ_ξ is the limit of a convergent Gibbs sequence at g with the pairs (K_j, β_j)" and "σ_ξ is such a limit and has one vacuum" are equivalent, by an argument that does not use (A).
(4) In (3), $|S_2^\sigma(f, \tau_v h)| \le C^g((f), (h))/|v|$ for $f, h \in C_c^1(\mathbb{R}^3)$ and $|v| \ge r((f), (h))$.

(E) *The rate on a torus.* Let ς be a choice of $\mathcal{T}^{\rm F+}$, ς' a lowered choice of it, $n \ge 2$ an integer, $0 < b' \le b''$ and $\mathsf{d} > 0$. There are $k^{\rm rt} \ge 0$ and $C^{\rm rt}, C^{\rm pol} < \infty$, depending only on the values of the constants at ς, on $c_2^\natural$, $\Upsilon^\natural$, n, b', b'' and d, with the following property. Let $m \in \mathbb{N}$, $\ell = 2L^m$, $b \in [b', b'']$, and let $f_1, \dots, f_n$ be real Lipschitz functions on T_ℓ, none of them 0, of spread at least d; put $S_K = S^T_{n;K,\ell,L^K b}(f_1, \dots, f_n)$, $\Pi = \prod_i \|f_i\|_\sharp$ and $\mathsf{p}_n = 2\mathsf{p}_0(n+1) + \frac{1}{2}$. Then:
(a) $|S_{K'} - S_K| \le C^{\rm rt} \ell^{3(n+1)} K^{\mathsf{p}_n} L^{-K/10} \Pi$ for all integers $K' \ge K \ge k^{\rm rt} + 54m$; so $S_\infty = \lim_K S_K$ exists, and $|S_\infty - S_K|$ obeys the same bound;
(b) $|S_K| \le C^{\rm pol} \ell^{219n} \Pi$ for all $K \in \mathbb{N}$, and $|S_\infty| \le C^{\rm pol} \ell^{219n} \Pi$;
(c) for every real $A > 30(n+1)$ and all integers $K' \ge K \ge k^{\rm rt} + Am$,

$$|S_{K'} - S_K| \le 2^{3(n+1)} C^{\rm rt} K^{\mathsf{p}_n} L^{-(\frac{1}{10} - \frac{3(n+1)}{A})K} \Pi,$$

a bound that does not depend on the side and tends to 0;
(d) at $n = 2$, $[b', b''] = [2\gamma^{-2}, L^2\gamma^{-2}]$ and $\mathsf{d} = \frac{1}{4}$, with $k^{\rm rt}$, $C^{\rm rt}$, $C^{\rm pol}$ of these data and

$$C = \max\{11,\ 54/\lambda + k^{\rm rt}/\log 2,\ C^{\rm rt}(110\mathsf{p}_2/e\lambda)^{\mathsf{p}_2},\ C^{\rm pol}\} :$$

for all $m \in \mathbb{N}$, $b \in [2\gamma^{-2}, L^2\gamma^{-2}]$, real Lipschitz f_1, f_2 on T_ℓ whose supports are at sup-distance at least $\frac{1}{4}$, and all integers $K \geq C \log \ell$, $|S_K - S_\infty| \leq C\ell^{438} L^{-K/C} \|f_1\|_\sharp \|f_2\|_\sharp$ and $|S_\infty| \leq C\ell^{438} \|f_1\|_\sharp \|f_2\|_\sharp$.

(F) *The plateau at one cutoff.* In a setting let $0 < g \leq \gamma$, $(K, \beta) \in \mathfrak{d}(g)$ and $\nu \in \mathcal{G}_{K,\beta}$.

(1) There is one bounded $\mathcal{I}$-measurable function e with $\mathsf{e} = E_\nu[a_p|\mathcal{I}]$ ν-almost surely for every plaquette p of $\mathbb{L}_K$; and $-3\mathsf{e} \in [P'(\beta-), P'(\beta+)]$ ν-almost surely.

(2) Let $f, h \in C^1_c(\mathbb{R}^3)$, $\bar{R} \in a_K\mathbb{N}$ with

$$\operatorname{supp} f \cup \operatorname{supp} h \subset (-\bar{R}, \bar{R})^3$$

and

$$a_K \leq 2 \min\{\delta((f); \bar{R}), \delta((h); \bar{R})\},$$

and $v \in a_K\mathbb{Z}^3$ with $|v| \geq 4\sqrt{3}\bar{R}$. Then

$$|S_2^T[\nu](f, \tau_v h) - 9a_K^{-8} r_K(f) r_K(h) \operatorname{Var}_\nu \mathsf{e}| \leq \mathfrak{k}_0 [\mathsf{N}((f); \bar{R}) \mathsf{N}((h); \bar{R})]^{1/2} \bar{R}/|v|,$$

$$0 \leq 9a_K^{-8} \operatorname{Var}_\nu \mathsf{e} \leq \tfrac{1}{4}(a_K^{-4} J(\beta))^2 \leq \tfrac{1}{4}(g^{8+2c} c_*(c) L^{-cK})^2 \qquad (0 \leq c < \tfrac{1}{180}).$$

(3) There are $\nu_\pm \in \mathcal{G}_{K,\beta}$ with $-3e_{\nu_\pm} = P'(\beta\pm)$. For any two such states and $\nu = \frac{1}{2}(\nu_+ + \nu_-)$, $9a_K^{-8} \operatorname{Var}_\nu \mathsf{e} = \frac{1}{4}(a_K^{-4} J(\beta))^2$. If ν is a pure phase, or $-3e_\nu \in \{P'(\beta-), P'(\beta+)\}$, then $\operatorname{Var}_\nu \mathsf{e} = 0$.

(4) Let X, Y be separated tuples, and $\bar{R}$, v as in (2) with $\delta(X; \bar{R})$, $\delta(Y; \bar{R})$ in the place of $\delta((f); \bar{R})$, $\delta((h); \bar{R})$. Then

$$|\sigma_\nu(X \cdot \tau_v Y) - \sigma_\nu(X)\sigma_\nu(Y) - \operatorname{Cov}_\nu(E_\nu[X_K^\nu|\mathcal{I}], E_\nu[Y_K^\nu|\mathcal{I}])| \leq \mathfrak{k}_0 [\mathsf{N}(X; \bar{R}) \mathsf{N}(Y; \bar{R})]^{1/2} \bar{R}/|v|,$$

and the modulus of the covariance is at most $\frac{1}{4} \operatorname{osc}(X_K^\nu) \operatorname{osc}(Y_K^\nu)$, where $\operatorname{osc}(F) = \sup_\rho \rho(F) - \inf_\rho \rho(F)$, ρ running through $\mathcal{G}_{K,\beta}$. For every real continuous local function F of the configuration, $\operatorname{osc}(F) \leq D^+\bar{p}^F(0) - D^-\bar{p}^F(0)$, the kink at $t = 0$ of the pressure $\bar{p}^F(t)$ perturbed by tF (Lem. 12.1(b)); at $X = (f)$, $\operatorname{osc}((f)_K^\nu) = a_K^{-4} J(\beta) |r_K(f)|$.

(G) *The limits at a label.* In a setting let $b > 0$, let $\mathfrak{p} = (K_j, \beta_j)_j$ be a sequence of pairs at the label b, $\mathsf{K} = \mathsf{K}_\mathfrak{p}$ and $\mathsf{K}^\bullet = \mathsf{K}^\bullet_\mathfrak{p}$; for $\sigma \in \mathsf{K}$, H_σ and Ω_σ are those of the paragraph on dilations.

(1) K is convex, compact and metrisable; the empty set is not excluded.

(2) Let $\sigma = \lim_j \sigma_{\nu_j} \in \mathsf{K}$. For every canonical decomposition of σ and every ξ there are $B_j \in \mathcal{I}$ with $\nu_j(B_j) > 0$ such that $\sigma_\xi = \lim_j \sigma_{\nu_j(\cdot|B_j)}$ along the full sequence, and σ_ξ has one vacuum; so $\sigma_\xi \in \mathsf{K}^\bullet$. The set $\mathsf{K}^\bullet$ is compact.

(3) $\operatorname{bar} : \operatorname{Prob}(\mathsf{K}^\bullet) \to \mathsf{K}$ is an affine homeomorphism onto K. For $\sigma \in \mathsf{K}$ and every canonical decomposition of σ, the map $\xi \mapsto \sigma_\xi$ is a homeomorphism of X onto the support of the one measure μ_σ with $\operatorname{bar} \mu_\sigma = \sigma$, and it carries ϖ to μ_σ; and $\dim \ker H_\sigma = \#\operatorname{supp} \mu_\sigma$, both finite and equal or both infinite.

(4) For $\sigma \in \mathsf{K}$ these say the same: σ has one vacuum; $\ker H_\sigma = \mathbb{C}\Omega_\sigma$ in one, and then in each, of the six directions; σ is an extreme point of K; $\sigma(X \cdot \tau_v X) \to \sigma(X)^2$ as $|v| \to \infty$, for every separated X; $|\sigma(X \cdot \tau_v Y) - \sigma(X)\sigma(Y)| \leq c_{X,Y}/|v|$ for $|v| \geq r_{X,Y}$, with numbers $c_{X,Y}$, $r_{X,Y}$ free of σ (in a window they are $C^g(X, Y)$, $r(X, Y)$); σ is the limit, along the full sequence, of σ_{ω_j} with pure phases $\omega_j \in \mathcal{G}_{K_j,\beta_j}$; $\operatorname{Var}_{\nu_j}(E_{\nu_j}[X_{K_j}^{\nu_j}|\mathcal{I}]) \to 0$ for every separated X, for one, and then for every, choice of states $\nu_j \in \mathcal{G}_{K_j,\beta_j}$ with $\sigma_{\nu_j} \to \sigma$.
For a subsequence $\mathfrak{p}'$ of $\mathfrak{p}$, K is a closed face of $\mathsf{K}_{\mathfrak{p}'}$; if $\tilde{\mathfrak{p}}$ is a second sequence of pairs at the label b and $\mathfrak{p}''$ interlaces $\mathfrak{p}$ and $\tilde{\mathfrak{p}}$, then $\mathsf{K}_{\mathfrak{p}''} = \mathsf{K} \cap \mathsf{K}_{\tilde{\mathfrak{p}}}$. If $\mathsf{K} \neq \emptyset$, these say the same: K is a point; $\mathsf{K}^\bullet$ is a point; $\mathsf{K} = \mathsf{K}^\bullet$; $\operatorname{Cov}_{\nu_j}(E_{\nu_j}[X_{K_j}^{\nu_j}|\mathcal{I}], E_{\nu_j}[Y_{K_j}^{\nu_j}|\mathcal{I}]) \to 0$ for all separated X, Y and all choices of $\nu_j \in \mathcal{G}_{K_j,\beta_j}$ for which σ_{ν_j} converges. At single fields this covariance tends to 0 in every case:

$$|\operatorname{Cov}_{\nu_j}(E_{\nu_j}[(f)_{K_j}^{\nu_j}|\mathcal{I}], E_{\nu_j}[(h)_{K_j}^{\nu_j}|\mathcal{I}])| \leq \tfrac{1}{4}(a_{K_j}^{-4} J(\beta_j))^2 |r_{K_j}(f) r_{K_j}(h)| \to 0.$$

(5) $\mathrm{Lim}[b]$ has exactly one element if and only if $\mathrm{Lim}^\bullet[b]$ has exactly one element. The same holds for $\mathrm{Lim}_{\mathfrak{q}}[b]$ and $\mathrm{Lim}^\bullet_{\mathfrak{q}}[b]$ for every set $\mathfrak{q}$ of pairs that contains the pairs of a sequence of pairs at the label b; in particular for $\mathfrak{q} = \{(K, L^K b) : K \in \mathbb{N}\}$. If $g^{-2} + \bar{B} < b < X_g - \bar{B}$ for some $0 < g \le \gamma$, a third equivalent statement is that every Gibbs sequence at g with $a_{K_j}\beta_j \to b$ converges; and then the one limit has one vacuum.

(6) $\mathrm{Lim}[b]$ and $\mathrm{Lim}^\bullet[b]$ are non-empty, compact and metrisable. Every sequence of pairs at the label b has a saturated subsequence, and every subsequence of a saturated sequence is saturated, with the same set of limits; $\mathrm{Lim}[b]$ is the union of the sets $\mathsf{K}_{\mathfrak{p}}$, and $\mathrm{Lim}^\bullet[b]$ the union of their sets of extreme points, over the saturated sequences $\mathfrak{p}$ at the label b. Every member of $\mathrm{Lim}[b]$ is the barycentre of exactly one Borel probability measure on $\mathrm{Lim}^\bullet[b]$. The map bar is an affine homeomorphism of $\mathrm{Prob}(\mathrm{Lim}^\bullet[b])$ onto the closed convex hull $\overline{\mathrm{co}}\,\mathrm{Lim}[b]$ of $\mathrm{Lim}[b]$ for the pointwise convergence, and for $\sigma \in \mathrm{Lim}[b]$ these say the same: σ has one vacuum; σ is an extreme point of $\overline{\mathrm{co}}\,\mathrm{Lim}[b]$; σ is an extreme point of $\mathrm{Lim}[b]$.
The following four statements are equivalent, and they hold if $\mathrm{Lim}[b]$ has exactly one element: $\mathrm{Lim}[b]$ is convex; $\mathrm{Lim}[b] = \mathsf{K}_{\mathfrak{q}}$ for one sequence of pairs $\mathfrak{q}$ at the label b; $\mathrm{Lim}[b] = \mathrm{bar}(\mathrm{Prob}(\mathrm{Lim}^\bullet[b]))$; every finite subset of $\mathrm{Lim}^\bullet[b]$ lies in $\mathsf{K}^\bullet_{\mathfrak{p}}$ for one sequence of pairs $\mathfrak{p}$ at the label b.

(7) Let $0 < g \le \gamma_W$, where γ_W is the ceiling of the law of two points, the short-distance law for the two-point function (equation YM3_I:(2346)); $\gamma_W < \gamma$ and $\gamma_W^{-2} \ge L^2\gamma^{-4}$. Let $\mathfrak{p}$ be a sequence of pairs in $\mathfrak{d}(g)$; no label is supposed. If $\mathsf{K}_{\mathfrak{p}} \ne \emptyset$, then $a_{K_j}\beta_j$ converges, to a label b in the interval $I^{\mathrm{lab}}(g)$ of [7, §K.1], and for every $\sigma \in \mathsf{K}_{\mathfrak{p}}$, every $\vartheta \in (0,1)$ and every pair (f,h) of the set $\mathfrak{T}^1_{\mathbb{R}^3}(\mathsf{d}_0(g,\vartheta);\vartheta)$, which is not empty,

$$b^{-2} = \lim_{t\to 0} \frac{S_2^\sigma(f_t, h_t)}{\mathsf{c}_2\,\mathcal{K}_{\mathrm{free}}(f,h)}, \qquad f_t(x) = f(x/t), \qquad h_t(x) = h(x/t).$$

Here $\mathfrak{T}^1_{\mathbb{R}^3}(\mathsf{d};\vartheta)$ is the set of the test pairs of class C^1 at the scale d with the shape parameter ϑ ([7, Lem. I.1] and the test-pair convention preceding equation YM3_I:(2343)): pairs of non-negative C^1 functions f, h, neither of them 0, with supports in balls of radius d at a distance between d and $6\mathsf{d}$, $\int f \ge \vartheta\|f\|_\infty \mathsf{d}^3$ and $\mathrm{Lip}\, f \le \|f\|_\infty/\vartheta\mathsf{d}$, and the same for h; $\mathsf{d}_0(g,\vartheta) > 0$ is the threshold of equations YM3_I:(2349)–(2350); and $\mathsf{c}_2 = \frac{9}{16\pi^2}$ and $\mathcal{K}_{\mathrm{free}}(f,h) = \iint f(x)h(y)|x-y|^{-6}\,\mathrm{d}x\,\mathrm{d}y > 0$ are the constant and the free kernel of Thm. YM3_I:C and equation YM3_I:(2240). So the sets $\mathrm{Lim}_{\mathfrak{d}(g)}[b]$, $b > 0$, are pairwise disjoint and their union is the set of the limits of all convergent Gibbs sequences at g; a sequence of pairs in $\mathfrak{d}(g)$ that interlaces two sequences with different labels has $\mathsf{K}_{\mathfrak{p}} = \emptyset$; and a convex combination $t\sigma + (1-t)\sigma'$, $0 < t < 1$, of limits at two different labels is the limit of no Gibbs sequence at g.

■□

The second theorem has a hypothesis on two periodic boxes. Let $Z(n_1, n_2, n_3)$ be the partition function of Wilson's action at the coupling β on the periodic lattice with n_1, n_2, n_3 sites in the three directions; it does not depend on the spacing. In a setting let $b > 0$, $\ell_* = 2L^{m_*}$ with $m_* \in \mathbb{Z}$, K_* an integer with $K_* + m_* \ge 1$, and $0 < x_0 \le 0.069$. The box inequality $\mathrm{BI}[b, \ell_*, K_*; x_0]$ says: for every integer $K \ge K_*$, with $l_K = \ell_*/2a_K = L^{K+m_*}$,

$$\mathcal{E}(l_K) = 2\log Z(l_K, 2l_K, 2l_K) - \log Z(2l_K, 2l_K, 2l_K) \le x_0 \qquad \text{at } \beta = L^K b.$$

Theorem AK. In a setting suppose $\mathrm{BI}[b, \ell_*, K_*; x_0]$, and let $\underline{K} = \max\{K_*, 1\}$. Then:

(1) for every $K \ge \underline{K}$ the set $\mathcal{G}_{K,L^K b}$ has exactly one element ν_K, which is a pure phase, and $J(L^K b) = 0$;

(2) for every $n \geq 2$ and every tuple X of n real Lipschitz functions on $\mathbb{R}^3$ with pairwise disjoint compact supports, $\lim_K S_n^T[\nu_K](X)$ exists along all $K \geq \underline{K}$ and equals $\lim_{m'\to\infty} \lim_{K\to\infty} S^T_{n;K,\ell',L^K b}(X^{\langle \ell'\rangle})$, $\ell' = 2L^{m'}$;

(3) for $\mathfrak{p} = (K, L^K b)_{K\geq \underline{K}}$ the set $\mathsf{K}_\mathfrak{p}$ is one point $\sigma = \lim_K \sigma_{\nu_K}$, and σ has one vacuum.

For every $z \in \mathbb{Z}$, BI$[L^z b, L^z \ell_*, K_* - z; x_0]$ is the same statement as BI$[b, \ell_*, K_*; x_0]$, so (1)–(3) hold at the label $L^z b$ as well. ■□

12.1.4. *Construction estimates.* The proof starts from Thm. YM3_I:D, with the common numerical choice of Prop. YM3_I:69.1 and its actual-history application in Cor. YM3_I:69.4, and from what is derived from it on the tori: the discarding of the histories with a large-field region (Prop. YM3_I:115.2), with its weight (the definitions preceding Prop. YM3_I:115.2) and with the statement on two nearby positive functionals and their cumulants ([7, Lem. K.2]), by which the discarding holds at every bound on the lost mass that is at most $\frac{1}{2}$; the comparison of two consecutive cutoffs (Thm. YM3_I:AR); and the convergence on each torus (Thm. YM3_I:AS). The last says, at every choice: let $\ell = 2L^m$, $n \geq 2$, let $f_1, \dots, f_n$ be real Lipschitz functions on T_ℓ, none of them 0, whose supports have no common point, let $0 < b' \leq b''$, let $\mathsf{d} > 0$ be at most the spread of $(f_1, \dots, f_n)$, and let $s \geq 0$ be an integer with $L^s b' \geq \gamma^{-2}$ and $M\mathcal{R}(L^s b'') \leq L^{m+s}$; such integers exist (Lem. YM3_I:118.2, equation YM3_I:(2210)), and $s = 0$ is one of them when $b' \geq \gamma^{-2}$ and $M\mathcal{R}(b'') \leq L^m$. Then for all $b \in [b', b'']$ and all integers $K' \geq K \geq K_{1,s}(\mathsf{d}) + s$

$$|S^T_{n;K',\ell,L^{K'}b}(f_1, \dots, f_n) - S^T_{n;K,\ell,L^K b}(f_1, \dots, f_n)| \leq L^{sn} C'_s \, (L^s \ell)^{3(n+1)} (K - s)^{\mathsf{p}_n} L^{-(K-s)/10} \prod_i \|f_i\|_\sharp,$$

so that $S^T_{n;K,\ell,L^K b}(f_1, \dots, f_n)$ converges as $K \to \infty$, uniformly in $b \in [b', b'']$. Here $C'_s = \mathsf{C}'_n([L^s b', L^s b''])$ and $K_{1,s}(\mathsf{d}) = K_1(L^s \ell, n, [L^s b', L^s b''], L^s \mathsf{d})$ are numbers of the constants and thresholds in Thm. YM3_I:AR, which attaches to a side ℓ, to n, to an interval J of labels with $\min \mathsf{J} \geq \gamma^{-2}$ and $M\mathcal{R}(\max \mathsf{J}) \leq L^m$, and to $\mathsf{d} > 0$ the thresholds $K_0(\ell, n, \mathsf{J}) \leq K_1(\ell, n, \mathsf{J}, \mathsf{d})$ and the rate constant $\mathsf{C}'_n(\mathsf{J})$. At $s = 0$ the bound reads $\mathsf{C}'_n([b', b'']) \, \ell^{3(n+1)} K^{\mathsf{p}_n} L^{-K/10} \prod_i \|f_i\|_\sharp$ for $K' \geq K \geq K_1(\ell, n, [b', b''], \mathsf{d})$; for other sides and windows it carries the dilation exponent s.

For a choice and a lowered choice, the setting (Th) gives an admissible lattice by Lem. YM3_I:118.1. §4.3 supplies the common lattice and background definitions, the explicit threshold, the witnesses for each creation, the gauge integration, and the chessboard estimate. Its Prop. 4.52 gives the volume-uniform probability bound. The history weights and masses are those of Thm. YM3_I:D.

From Thms. YM3_I:AJ and YM3_I:AD, it uses the zero-source stability bounds (ii), (iii) of Thm. YM3_I:AD; the comparison of two laws in one ball of radius 1 at a first-passage datum, the common-normalizer comparison in the proof of Prop. YM3_I:91.13, whose hypothesis (Nr1) Prop. YM3_I:91.12 supplies at every pair of $\mathfrak{d}(g)$; the bound at separated supports in every periodic Gibbs state, Prop. YM3_I:91.13; and the family of bounds at g (Prop. YM3_I:106.3, with equation YM3_I:(2030)); for the far part on all sides also the bound for connected functions on the tori at first-passage data, Prop. YM3_I:89.5.

For the limits it uses Thm. YM3_I:AP, Prop. YM3_I:100.1 with equations YM3_I:(1976)–(1980), the statement on iterated plateaux (Prop. YM3_I:100.2), and Prop. YM3_I:106.1 (pure phases drawn from a regular conditional law given $\mathcal{I}$); for (G)(7) the law of two points: its test pairs, its constant and its free kernel (Thm. YM3_I:C and equation YM3_I:(2240), [7, Lem. I.1] and the convention in equation YM3_I:(2343)), with non-empty sets of test pairs ([7, Lem. I.1](c)), its ceiling and its threshold (equations YM3_I:(2346) and YM3_I:(2349)–(2350)), the law in the limits through periodic Gibbs states, with the convergence of the labels (Prop. YM3_I:127.8), and the law in every component (Cor. YM3_I:127.11); for the box inequality the domination theorem (Prop. YM3_I:91.9), the seminorm as a mean over a period box (equation [7, eq. 189]), the chessboard estimate on the tori of $\mathbb{L}_K$ and the comparison of two sides ([7, Cor. B.1](b), (c), (e)), a coarse and a fine

board (the proof of Lem. YM3_I:104.1), the functions of a cell (the plaquette-support argument in [7, §B.1]), the chessboard estimate (Prop. YM3_I:86.3), and the standing choice of an odd blocking factor $L \geq 13$.

The choices used here are supplied by the ordered proof of Prop. YM3_I:69.1 and its sourced extension in Section YM3_I:82. The exponents are fixed before the block size; every sufficiently large odd block size is then admitted, and the remaining constants are chosen in the stated order. Besides, we use the first-passage data at g and the canonical family $\mathfrak{d}(g)$ (Cor. YM3_I:74.2 and [7, result K.1]) with the two facts on its labels ([7, §K.1]), the cube size $\mathcal{R}$ (equation YM3_I:(634)), the dilation of the tori, of their partition functions and connected functions and of the labels (Lem. YM3_I:118.2), the dilation of the lattices $\mathbb{L}_K$, of the laws on them, of their observables and of their periodic Gibbs states ([7, Lem. K.1]), the regions of a history, with $Z_t = \Lambda_t^c$ defined by equation YM3_I:(644), and statements on one lattice that use neither Thm. YM3_I:D nor Thms. YM3_I:AJ and YM3_I:AD: the Gibbs states, the periodic Gibbs states and the pure phases (Section YM3_I:91 and Prop. YM3_I:106.1); periodic laws and their bond representatives (equation YM3_I:(1817) and the construction immediately preceding it); the lower tangent inequality for periodic Gibbs states (Lem. YM3_I:91.6); weak limit points of torus measures are periodic Gibbs states (Prop. YM3_I:91.2), and weak limits of Gibbs states at couplings $\beta_k \to \beta$ are Gibbs states at β (Prop. YM3_I:91.1, with the family of the functions $-a_p$); every periodic Gibbs state is invariant under the isometries of the lattice and reflection positive at every lattice mirror (Prop. YM3_I:103.4), and the periodic Gibbs states at one coupling form a convex set (Props. YM3_I:91.1 and YM3_I:103.4) whose members are integrals of its ergodic members (Prop. YM3_I:106.2); reflection positivity of the torus measures at the coordinate mirrors (Prop. YM3_I:91.4, with $\Gamma = a(N_1\mathbb{Z} \times N_2\mathbb{Z} \times N_3\mathbb{Z})$ and height period $N_\mu a$ in direction e_μ); the reconstruction of the transfer operator with its spectral measure (Lem. YM3_I:99.1); the trace formula for boxes of slabs (Prop. YM3_I:85.2, with the spectral argument in the proof of Prop. YM3_I:86.5 and the slab operators of equation YM3_I:(1744)); and the crossing operator on the Hilbert space of one layer (Section YM3_I:85), which is positive and of trace class (Prop. YM3_I:85.1, with the rank-one expansion in its proof).

Lems. 12.1, 12.2, 12.10, Props. 12.3, 12.5, 12.11, Lem. 12.12(a), (c) and Prop. 12.16(a)–(e) use neither Thm. YM3_I:D nor Thms. YM3_I:AJ and YM3_I:AD, and Prop. 12.16(a)–(e) does not use the supply of choices either. Clause (E) uses Thm. YM3_I:D alone, at a choice and at a lowered choice of it. Clause (A) uses Thms. YM3_I:D, YM3_I:AJ and YM3_I:AD at the choices of its two settings, Thm. YM3_I:D through the convergence on each torus; it uses no lowered choice and no statement on the rarity of a large field. Clause (B), in (1)–(3) and (5) and with the constants eq. 12.xi, uses Thms. YM3_I:AJ and YM3_I:AD at the two choices ς_1^{S}, ς_2^{S} of its settings and Thm. YM3_I:D at the four choices ς_1, ς_2, ς_1', ς_2', at each pair $(\varsigma_i, \varsigma_i')$ through (E).

With other constants, (B)(4) uses Thms. YM3_I:AJ and YM3_I:AD at the same two choices and Thm. YM3_I:D at the three choices ς_1, ς_1', ς_2 alone: at $(\varsigma_1, \varsigma_1')$ through (E), and at ς_2 through the convergence on each torus, by Prop. 12.5. In (A) and (B) Thms. YM3_I:AJ and YM3_I:AD enter only through the stability bounds at zero source (Thm. YM3_I:AD), the comparison of two laws in one unit ball (the common-normalizer comparison in the proof of Prop. YM3_I:91.13, with Prop. YM3_I:91.12) and the bound at separated supports in every periodic Gibbs state (Prop. YM3_I:91.13).

Clause (C) uses, besides (B)(2), only that a setting forms $\mathfrak{d}(g)$; its setting need not be one of the settings of (B). Clauses (D), (F) and (G) use Thms. YM3_I:D, YM3_I:AJ and YM3_I:AD at the choice of the setting, together with (A) where the decay of $a_K^{-4} J(\beta)$ to 0 enters and with (B) where its rate enters; the bound by $\frac{1}{4}(a_K^{-4} J(\beta))^2$ in (F) uses the choice of the setting alone. In Thm. AK, clause (1) uses neither Thm. YM3_I:D nor Thms. YM3_I:AJ and YM3_I:AD nor the supply of choices; clause (2) uses besides Thm. YM3_I:D at the choice of the setting, through the convergence on each torus in Prop. 12.16(f); and clause (3), one vacuum, uses besides Thm. YM3_I:

AP and the family of bounds at g (Prop. YM3_I:106.3) through (G)(4), that is, Thms. YM3_I:D, YM3_I:AJ and YM3_I:AD at the choice of the setting.

It does not use (A): by (1), $J(L^K b) = 0$, so that $a_K^{-4} J(L^K b) = 0$ along the sequence.

12.1.5. *Auxiliary results.*

Lemma 12.1 (One lattice)**.**

(a) For $\beta \geq 0$ the limit $P(\beta) = \lim_N P_N(\beta)$ exists along all integers $N \geq 3$. Each P_N is smooth and convex, $-6\beta \leq P_N \leq 0$, $P_N' = -N^{-3}\mu_{N,\beta}(S_N) \in [-6, 0]$ and $P_N'' = N^{-3}\operatorname{Var}_{\mu_{N,\beta}}(S_N)$. P is finite and convex, its difference quotients lie in $[-6, 0]$, and $0 \leq J \leq 6$.

(b) Let $\beta > 0$ and $\nu \in \mathcal{G}(\beta)$. Then $\nu(a_p)$ is one number e_ν for all p, and $P(\beta+t) - P(\beta) \geq -3e_\nu t$ for $t > -\beta$; so $-3e_\nu \in [P'(\beta-), P'(\beta+)]$, and $3|e_\rho - e_\nu| \leq J(\beta)$ for $\rho, \nu \in \mathcal{G}(\beta)$. There are $\nu_\pm \in \mathcal{G}(\beta)$ with $-3e_{\nu_\pm} = P'(\beta\pm)$; hence $J(\beta) = 3\max\{|e_\rho - e_\nu| : \rho, \nu \in \mathcal{G}(\beta)\}$. More generally, let F be a real continuous function of the bonds in a cube of side $k \in \mathbb{N}$, let F_N be the sum of its translates over a period cell of the cube with N^3 sites, and $p_N^F(t) = N^{-3}\log \mu_{N,\beta}(e^{tF_N})$, $\bar{p}^F = \limsup_N p_N^F$. These functions are convex, $|p_N^F(t)| \leq |t|\|F\|_\infty$, and for every $\nu \in \mathcal{G}(\beta)$, $N \geq 3$ and real t

$$p_N^F(t) \geq t\,\nu(F) - \big[144\beta + 12(2k-1)^3|t|\,\|F\|_\infty\big]/N;$$

so $D^-\bar{p}^F(0) \leq \nu(F) \leq D^+\bar{p}^F(0)$. The same holds on $\mathbb{L}_K$ for $\mathcal{G}_{K,\beta}$.

(c) For laws $\mathfrak{s}$ invariant under $\mathbb{Z}^3$ and integers $\eta \geq 1$: $0 \leq c_x \leq 6$, $|\Gamma_\mathfrak{s}| \leq 9$, $|s_\mathfrak{s}^\eta| \leq 9\eta^6$ and $|n_\eta(\mathfrak{s})| \leq 1125\eta^3$; on the periodic configurations $S_N = \sum_x c_x$ over a period cell, and $P_N'' = \sum_z \Gamma_{\mu_{N,\beta}}(z)$ over a period cell. If η divides N and $N \geq 7\eta$, then, with $\mathcal{D} = \{X \in \mathbb{Z}^3 : -\frac{N}{2\eta} \leq X_i < \frac{N}{2\eta}\}$,

$$P_N''(\beta) = n_\eta(\mu_{N,\beta}) + \eta^{-3} \sum_{X \in \mathcal{D},\ |X|_\infty \geq 3} s^\eta_{\mu_{N,\beta}}(X). \tag{12.i}$$

■

Proof sketch. (a) Removing n plaquette terms changes $\log Z$ by at most $2\beta n$, and a bond that no remaining term reads integrates to 1. A torus and the free box of the same side differ by at most $6N^2$ plaquettes, so their logarithms differ by at most $12\beta N^2$. A free box of side $N' = qN + r'$, $0 \leq r' < N$, holds q^3 disjoint free boxes of side N and at most $9N'^2N + 6N'^3/N$ further plaquettes; so the free energies per site p^{f}_N of free boxes satisfy $|p^{\mathrm{f}}_{N'} - (qN/N')^3 p^{\mathrm{f}}_N| \leq 2\beta(9N/N' + 6/N)$, hence $\limsup_{N'} |p^{\mathrm{f}}_{N'} - p^{\mathrm{f}}_N| \leq 12\beta/N$, and (P_N) is a Cauchy sequence. As $0 \leq a_p \leq 2$, $e^{-6\beta N^3} \leq Z_N \leq 1$; the derivatives are those of the logarithm of a Laplace transform on a compact space; convexity and the bounds on the difference quotients pass to the limit. (b) Periodic Gibbs states are invariant under the lattice isometries, which act transitively on the plaquettes. The lower tangent inequality (Lem. YM3_I:91.6), applied on the cube with N^3 sites, whose boundary layer holds at most $6N^2$ sites, gives $\log \mu_{N,\beta}(e^{tF_N}) \geq tN^3\nu(F) - 144\beta N^2 - 12(2k-1)^3|t|\|F\|_\infty N^2$; divide by N^3. Convexity follows from Hölder's inequality, and the bound by $|t|\|F\|_\infty$ from $|F_N| \leq N^3\|F\|_\infty$; letting $N \to \infty$ gives $\bar{p}^F(t) \geq t\nu(F)$, which is the statement on the one-sided derivatives. At $F = -(a_{p_{12}(0)} + a_{p_{13}(0)} + a_{p_{23}(0)})$, the three plaquettes at the site 0, we have $k = 1$, $\|F\|_\infty \leq 6$, $F_N = -S_N$ and $\mu_{N,\beta}(e^{tF_N}) = Z_N(\beta + t)/Z_N(\beta)$, so $p_N^F(t) = P_N(\beta + t) - P_N(\beta)$ and $\nu(F) = -3e_\nu$:

$$\log Z_N(\beta + t) - \log Z_N(\beta) \geq -3e_\nu t N^3 - 144\beta N^2 - 72|t|N^2,$$

and $-3e_\nu$ is a subgradient of P at β. For the ends take points $\beta_k \downarrow \beta$ (or $\beta_k \uparrow \beta$) at which P is differentiable and $\nu_k \in \mathcal{G}(\beta_k)$ invariant under $\mathbb{Z}^3$, weak limit points of torus measures (Prop. YM3_I:91.2): $-3e_{\nu_k} = P'(\beta_k) \to P'(\beta\pm)$, and weak limit points of (ν_k) are Gibbs states at β (Prop. YM3_I:91.1) invariant under $\mathbb{Z}^3$, so periodic Gibbs states at β. On $\mathbb{L}_K$ use the dilation by L^K, which maps $\mathcal{G}_{K,\beta}$ onto $\mathcal{G}(\beta)$ ([7, Lem. K.1] at $z = K$). (c) $0 \leq c_x \leq 6$ since $0 \leq a_p \leq 2$, and a variable

with values in $[0,6]$ has variance at most 9. Next, $\sum_j \Lambda_\eta(u-\eta j)=1$, $\sum_{x\in\mathbb{Z}}\Lambda_\eta(x)=\eta$, and $\sum_{x\in\mathbb{Z}}\Lambda_\eta(x)\Lambda_1(u-x)=\Lambda_\eta(u)$, the left member being the piecewise affine interpolation of Λ_η at the integers; so $\mathcal{O}_1(\varphi^\eta_X)=\sum_x \varphi^\eta_{\mathfrak{c}}(x-\eta X)c_x$, which gives $|s^\eta_{\mathfrak{s}}|\le 9\eta^6$ and, with the 125 points $|X|_\infty\le 2$, $|n_\eta|\le 1125\eta^3$. Every plaquette is a face of exactly two unit cubes, so $S_N=\sum_x c_x$ and $P''_N=N^{-3}\operatorname{Var}(S_N)=\sum_z\Gamma(z)$. On the periodic configurations $\sum_{X\in\mathcal{D}}\mathcal{O}_1(\varphi^\eta_X)=S_N$, $\mathcal{D}$ being a set of representatives of $\mathbb{Z}^3$ modulo N/η; by invariance $\operatorname{Cov}(\mathcal{O}_1(\varphi^\eta_0),S_N)=\eta^3\operatorname{Cov}(c_0,S_N)=\eta^3P''_N$. Split $\mathcal{D}$ into $|X|_\infty\le 2$, which gives the hull (all these points lie in $\mathcal{D}$ as $N/2\eta\ge 3.5$), and the rest. □

Lemma 12.2 (Reflection positivity: the specific heat is at least its near part)**.**

(a) Let $\beta\ge 0$, $\nu\in\mathcal{G}(\beta)$ and $\eta\ge 1$ an integer. There is a finite positive Borel measure m on $[0,1]\times S^1\times S^1$ with

$$s^\eta_\nu(n+2,w)=\int t^{\eta n}\zeta_2^{\eta w_2}\zeta_3^{\eta w_3}\,\mathrm{d}\mathsf{m}(t,\zeta_2,\zeta_3)\qquad(n\ge 0,\ w\in\mathbb{Z}^2);$$

so $s^\eta_\nu(3e_1)\ge 0$, and $\sum_{|X|_\infty\ge 3}e^{-\epsilon|X|_1}s^\eta_\nu(X)\ge -870\,s^\eta_\nu(3e_1)$ for every $\epsilon>0$.

(b) Let $I\subset(0,\infty)$ be an open interval, $\eta\ge 1$ an integer and q a Borel function, bounded on the compact subsets of I, with $q(\beta)\le n_\eta(\nu)-870\eta^{-3}s^\eta_\nu(3e_1)$ for all $\beta\in I$ and $\nu\in\mathcal{G}(\beta)$. Then $P''(d\beta)\ge q(\beta)\,\mathrm{d}\beta$ on I.

(c) Let $\mathfrak{s}$ be invariant under $\mathbb{Z}^3$, $\eta,\eta'\ge 1$ integers, $\bar\eta=\max\{\eta,\eta'\}$, $w=\varphi^\eta_{\mathfrak{c}}/\eta^3$, $w'=\varphi^{\eta'}_{\mathfrak{c}}/\eta'^3$ (non-negative, each of sum 1), and $\Theta(y)=\prod_i\min\{1,(5-|y_i-\frac12|/\bar\eta)_+\}$. Then

$$n_\eta-n_{\eta'}=\sum_v\Big[w'(v)\frac{\langle\varphi^\eta_0;\Theta-\Theta(\cdot+v)\rangle}{\eta^3}-w(v)\frac{\langle\varphi^{\eta'}_0;\Theta-\Theta(\cdot+v)\rangle}{\eta'^3}\Big]-\frac{\langle\varphi^\eta_0;\Theta-h^\eta\rangle}{\eta^3}+\frac{\langle\varphi^{\eta'}_0;\Theta-h^{\eta'}\rangle}{\eta'^3},\tag{12.ii}$$

all brackets in $\mathfrak{s}$; in each bracket the second entry vanishes within the sup-distance $\min\{\eta,\eta'\}$ of the support of the first, and off $\{|y-(\frac12,\frac12,\frac12)|_\infty\le 6\bar\eta\}$.

(d) Let $N\ge 3$, $\beta\ge 0$, $\delta\ge 0$, and let $\omega_1,\omega_2,\omega_3$ be real, even, summable functions on $\mathbb{Z}$ with non-negative Fourier transforms and $\sum_j\omega_i(jN)\le 1+\delta$. Then $P''_N(\beta)\ge(1+\delta)^{-3}\sum_{z\in\mathbb{Z}^3}\omega_1(z_1)\omega_2(z_2)\omega_3(z_3)\Gamma_{\mu_{N,\beta}}(z)$.

■

Proof sketch. (d) Put $\omega'_i(t)=(1+\delta)^{-1}\sum_j\omega_i(t+jN)$ on $\mathbb{Z}/N$; its Fourier coefficients are non-negative, so $|\omega'_i|\le\omega'_i(0)\le 1$. Reflection positivity of the torus measure at the coordinate mirrors makes every slice sum $t\mapsto\sum_u e^{2\pi i(k_1u_1+k_2u_2)/N}\Gamma(t,u)$, the sum over the two other coordinates, non-negative: for $1\le t\le N-1$ the mirror $\{x_3=\frac{t+1}{2}\}$ carries the cubes of the layer t to those of the layer 0, and at $t=0$ the sum is a square. Writing $1-\omega'_1\omega'_2\omega'_3=(1-\omega'_3)+\omega'_3(1-\omega'_2)+\omega'_3\omega'_2(1-\omega'_1)$ and expanding the factors ω' in characters with non-negative coefficients, each term gives a non-negative sum against Γ; so $\sum_z\Gamma(z)\ge\sum_z\omega'_1\omega'_2\omega'_3\Gamma(z)$, and the left member is P''_N by Lem. 12.1(c). (a) ν is invariant and reflection positive at the mirrors $\{x_1=\frac12\}$ and $\{x_1=1\}$. The reconstruction of the transfer operator T, $0\le T\le 1$, with the step e_1, and of the tangential translations, with their joint spectral measure, gives $\mathsf{m}=\langle[F],\mathfrak{E}(\cdot)[F]\rangle$ for $F=\mathcal{O}_1(\varphi^\eta_{e_1})-\nu(\mathcal{O}_1(\varphi^\eta_{e_1}))$, which reads only bonds in $\{x_1\ge 1\}$ and whose mirror image is the centred $\mathcal{O}_1(\varphi^\eta_{-e_1})$; this is the representation, and $s^\eta_\nu(3e_1)=\int t^\eta d\mathsf{m}\ge 0$. Distribute the X with $|X|_\infty\ge 3$ over the six cones about the coordinate half-axes, with the weights $\kappa(k,w)=1$ inside a cone, $\frac12$ on a face and $\frac13$ on an edge, which sum to one.

By the invariance under the permutations and reflections of the axes the left member becomes $6e^{-2\epsilon}\int\Sigma\,\mathrm{d}\mathsf{m}$, where, with $x=e^{-\epsilon}t^\eta$, $e^{iq_i}=\zeta_i^\eta$ and $\Psi_x(w)=\sum_{k\ge3}x^{k-2}\kappa(k,w)$, $\Sigma=\sum_w e^{-\epsilon(|w_2|+|w_3|)}e^{iq\cdot w}\Psi_x(w)$. For $x\le\frac12$, $|\Sigma|\le\sum_w\Psi_x(w)=x\sum_{i\ge0}x^i(4(i+3)^2+\frac13)\le 145x$. For $x\ge\frac12$, $x^2\Psi_x=\Phi_x-\mathsf{R}_x$ with $\Phi_x(w)=x^{|w|_\infty}\big(\frac{1+x}{2(1-x)}-\frac16\mathbf{1}_{\{|w_2|=|w_3|\}}\big)$ and a remainder $\mathsf{R}_x\ge 0$

supported in $|w|_\infty \le 2$ of total mass $\frac{1}{3}+\frac{13}{3}x+\frac{49}{3}x^2$. Two elementary Fourier inequalities for the transforms $\hat E_x$ of $x^{|w|_\infty}$ on $\mathbb{Z}^2$ and $\hat D_x$ of its restriction to the two diagonals, namely $\hat D_x \ge -1$ and $\hat E_x \ge \frac{1-x}{2(1+x)}(\hat D_x+1)$, give $\hat\Phi_x = \frac{1+x}{2(1-x)}\hat E_x - \frac16 \hat D_x \ge \frac{1}{12}\hat D_x + \frac14 \ge \frac16$; the transform of $e^{-\epsilon(|w_2|+|w_3|)}\Phi_x$ is then a convolution of non-negative functions, so $\Sigma \ge -x^{-2}\sum_w \mathsf{R}_x(w) \ge -53x$.

In both cases $\Sigma \ge -145x$, so the left member is at least $-870e^{-3\epsilon}\int t^\eta d\mathsf{m} \ge -870\, s^\eta_\nu(3e_1)$; and $6\cdot 145 = 870$. (b) The window $\tilde w_1(z) = \eta^{-1}\sum_X e^{-\epsilon|X|}\sum_x \Lambda_\eta(x)\Lambda_\eta(x+z-\eta X)$ is even, has values in $[0,1]$ and the transform $\eta^{-1}\sum_u e^{-\epsilon|u|}e^{iu\eta\theta}\,|\sum_x \Lambda_\eta(x)e^{-ix\theta}|^2 \ge 0$, and $\delta_N = \sum_{j\ne 0}\tilde w_1(jN) \to 0$. By (d) with $\omega_i = \tilde w_1$, $P_N''(\beta) \ge (1+\delta_N)^{-3}\eta^{-3}\sum_X e^{-\epsilon|X|_1} s^\eta_{\mu_{N,\beta}}(X)$. Integrate against $0 \le G \in C_c^\infty(I)$, let $N\to\infty$ (dominated convergence for $\int G''P_N$, Fatou's lemma for the right member, and weak limit points of $\mu_{N,\beta}$ lie in $\mathcal{G}(\beta)$), use (a) to bound the terms with $|X|_\infty \ge 3$, and let $\epsilon\to 0$. (c) $\sum_v \varphi_c^{\eta'}(v)\varphi_0^\eta(\cdot - v) = \sum_{v,x}\varphi_c^{\eta'}(v)\varphi_c^\eta(x)\varphi_0^1(\cdot - v - x)$ is symmetric in (η,η'); expand $\eta'^3\langle\varphi_0^\eta;\Theta\rangle$ with it, exchange η and η', divide by $\eta^3\eta'^3$, and write $h = \Theta - (\Theta - h)$. The vectors v with a weight $\ne 0$ have $|v|_\infty < \bar\eta$, so $\Theta - \Theta(\cdot + v)$ vanishes where $|y - (\frac12,\frac12,\frac12)|_\infty \le 3\bar\eta$, and $\Theta - h^\eta$ vanishes where this distance is at most 2η. □

Proposition 12.3 (For data: two incommensurable periods)**.** Let $\beta_0 > 0$, let P be convex on (β_0,∞), $\hat n$ Borel and bounded on the compact subsets of (β_0,∞), and $C_{\rm low} \ge 0$ with

$$P''(d\beta) \ge \big[\hat n(\beta) - C_{\rm low}\beta^{-5}\big]d\beta \qquad \text{on } (\beta_0,\infty); \tag{12.iii}$$

let $\mathbb{M}$ be the image under $s = \log\beta$ of $\beta^4[P''(d\beta) - \hat n\,\mathrm{d}\beta]$, and $J_P(\beta) = P'(\beta+) - P'(\beta-)$. For $i = 1,2$ let $\lambda_i > 0$ with $\lambda_1/\lambda_2 \notin \mathbb{Q}$, $\alpha_i \in \mathbb{R}$, $\varkappa_i > 0$, $\mathsf{J}_i = [\alpha_i, \alpha_i + \lambda_i + 2\varkappa_i]$, integers k_i with $\alpha_i + k_i\lambda_i > \log\beta_0$, and integers $m_i^0 \ge 0$; for integers $m \ge m_i^0$ and $K \ge k_i$ let $P^{(i)}_{m,K}$ be a C^2 function and $n^{(i)}_{m,K}$ a bounded Borel function on $W_K^{(i)} = \exp(\mathsf{J}_i + K\lambda_i)$, with numbers $\epsilon_m^{(i)}, C_N \ge 0$ such that on $W_K^{(i)}$

$$\beta^3|P - P^{(i)}_{m,K}| \le \epsilon_m^{(i)}, \qquad \beta^5|n^{(i)}_{m,K} - \hat n| \le C_N; \tag{12.iv}$$

put $y^{(i)}_{m,K}(\upsilon) = \beta^5[(P^{(i)}_{m,K})'' - n^{(i)}_{m,K}](\beta)$ at $\beta = e^{\upsilon + K\lambda_i}$, $\upsilon \in \mathsf{J}_i$, and $\varkappa = \min\{\varkappa_1,\varkappa_2,1\}$.

(a) Let $\sup_{K\ge k^0}\sup_{\mathsf{J}_1}|y^{(1)}_{m^0,K}| \le Y^0$ for some $m^0 \ge m_1^0$ and some integer $k^0 \ge k_1$; let $m_i \ge m_i^0$, integers $\bar K_i \ge k_i$ with $\bar K_1 \ge k^0$, and $\omega_i \ge 0$ satisfy $\sup_{\mathsf{J}_i}|y^{(i)}_{m_i,K} - y^{(i)}_{m_i,K'}| \le \omega_i$ for $K,K' \ge \bar K_i$. Let $\epsilon = \max_i \epsilon^{(i)}_{m_i} \le \varkappa^2/4$,

$$\mathsf{M}_1 = (Y^0 + C_N + C_{\rm low})\varkappa_1 + \epsilon^{(1)}_{m^0}(15/\varkappa_1 + 14 + 12\varkappa_1), \qquad \mathsf{A}_0 = 7\mathsf{M}_1/\varkappa_1 + C_{\rm low} + 4C_N + \omega_1 + \omega_2.$$

Then, with $\mathbb{M}^+ = \mathbb{M} + C_{\rm low}ds$, for $s \ge s_* = \max_i(\alpha_i + 2\varkappa_i + \bar K_i\lambda_i)$: $\mathbb{M}^+([s,s+a]) \le \mathsf{A}_0 a + (\mathsf{A}_0 + 134)\epsilon^{1/2}$ for $0 \le a \le \varkappa/2$; and for $0 \le \delta \le 1$

$$e^{4s}J_P(e^s) \le (\mathsf{A}_0 + 134)\epsilon^{1/2}, \qquad |\mathbb{M}|([s,s+\delta]) \le (2\mathsf{A}_0 + C_{\rm low} + 134)\delta + (\mathsf{A}_0 + 134)\epsilon^{1/2}.$$

Hence, if such thresholds $\bar K_i$, of whatever size, exist for all $m_i \ge m_i^0$ with one pair ω_1,ω_2, and $\inf_m \epsilon_m^{(i)} = 0$, then $e^{4s}J_P(e^s) \to 0$ and $\limsup_{s\to\infty}|\mathbb{M}|([s,s+\delta]) \le (2\mathsf{A}_0 + C_{\rm low} + 134)\delta$.

(b) Let $\epsilon_m^{(i)} = c_{{\rm P},i}e^{-\lambda_i m}$ with $c_{{\rm P},i} > 0$, and let there be $\mathsf{a}_i \ge 0$, $A_i > 0$, $\omega_i \ge 0$ with $\sup_{\mathsf{J}_i}|y^{(i)}_{m,K} - y^{(i)}_{m,K'}| \le \omega_i$ for all $m \ge m_i^0$ and all integers $K,K' \ge \bar K_i(m) = \max\{k_i, \lceil \mathsf{a}_i + A_i m\rceil\}$. Then the hypothesis of (a) on Y^0 holds at $m^0 = m_1^0$, $k^0 = \bar K_1(m_1^0)$, with $Y^0 = \sup_{\mathsf{J}_1}|y^{(1)}_{m_1^0,k^0}| + \omega_1$. Let A_0 be that of (a) at these numbers, $A = \max_i A_i$,

$$c_{3,i} = |\alpha_i| + 2\varkappa_i + \lambda_i(\max\{k_i,0\} + \mathsf{a}_i + 1), \quad c_6 = \max_i c_{{\rm P},i}e^{\lambda_i + c_{3,i}/A_i}, \quad T_i(m) = \alpha_i + 2\varkappa_i + \lambda_i\bar K_i(m),$$

$s_\sharp = \max\{0, T_1(m_1^0), T_2(m_2^0), A\log^+(4c_6/\varkappa^2)\}$, $C_{\rm h} = 2\mathsf{A}_0 + C_{\rm low} + 134$ and $C_{\rm r} = (\mathsf{A}_0+134)c_6^{1/2}$. Then for $s \ge s_\sharp$, $\beta = e^s$ and $0 \le \delta \le 1$: $\beta^4 J_P(\beta) \le C_{\rm r}\beta^{-1/2A}$ and $|\mathbb{M}|([s, s+\delta]) \le C_{\rm h}\delta + C_{\rm r}\beta^{-1/2A}$.

(c) (The cut.) Let the hypothesis of (a) on Y^0 hold, and let there be one integer $K_* \ge \max\{k_1, k_2, k^0\}$ and numbers ω_i such that $\sup_{\mathsf{J}_i} |y^{(i)}_{m,K} - y^{(i)}_{m,K'}| \le \omega_i$ for $K, K' \ge K_*$ and every $m \ge m_i^0$, and $\inf_m \epsilon_m^{(i)} = 0$, $i = 1, 2$. Then $J_P(e^s) = 0$ for $s \ge s_* = \max_i(\alpha_i + 2\varkappa_i + K_*\lambda_i)$, and on $[s_*, \infty)$ the measure $\mathbb{M}$ has a density with values in $[-C_{\rm low}, \mathsf{A}_0 - C_{\rm low}]$, A_0 being that of (a).

(d) (The converse.) Let $\mathbb{M}$ be any signed Radon measure on (s_0, ∞), $s_2 > s_0$, $\epsilon_1 : [s_2, \infty) \to (0, 1]$ non-increasing, and $C_{\rm h}, C_{\rm r} \ge 0$ with $|\mathbb{M}|([s, s+\delta]) \le C_{\rm h}\delta + C_{\rm r}\epsilon_1(s)$ for $s \ge s_2$ and $0 \le \delta \le 1$. Then for $s_3 \ge s_2$ and $G \in C^1(\mathbb{R})$ with compact support in $[s_3, \infty)$, $|\langle \mathbb{M}, G\rangle| \le (C_{\rm h} + C_{\rm r})(\|G\|_1 + \epsilon_1(s_3)\|G'\|_1)$; so $|\langle \mathbb{M}, G(\cdot - t)\rangle - \langle \mathbb{M}, G\rangle| \le 2(C_{\rm h} + C_{\rm r})(\|G\|_1 + \epsilon_1(s_3)\|G'\|_1)$ for every $t \ge 0$.

(e) (The exponent is attained.) For all $\lambda_1, \lambda_2 > 0$ with $\lambda_1/\lambda_2 \notin \mathbb{Q}$, all $\alpha_i \in \mathbb{R}$, $\varkappa_i > 0$, $\mathsf{a}_i \ge 0$ and every $A > 0$ there are data as in (b), with $\beta_0 = 1$, $m_i^0 = 0$, $A_1 = A_2 = A$, $C_N = 0$, $C_{\rm low} = 7$, $\omega_1 = \omega_2 = 0$ and $c_{{\rm P},i} = \frac14 e^{(1+|\alpha_i|)/A}$, such that $\beta_k^4 J_P(\beta_k) = \beta_k^{-1/2A}$ at $\beta_k = e^{2k}$, $k \in \mathbb{N}$. ∎

Proof sketch. $\mathbb{M}^+$ is a positive measure by eq. 12.iii; it has the same atoms as $\mathbb{M}$, $\mathbb{M}(\{s\}) = e^{4s}J_P(e^s)$, and $\mathbb{M} \le \mathbb{M}^+$, $|\mathbb{M}| \le \mathbb{M}^+ + C_{\rm low}ds$. For $G \in C^2$ supported in $\mathsf{J}_i + K\lambda_i$ put $g(\beta) = \beta^4 G(\log\beta)$, so that $g'' = \beta^2(G'' + 7G' + 12G)(\log\beta)$. Then $\langle \mathbb{M}, G\rangle = \int g''(P - P^{(i)}_{m,K})\,{\rm d}\beta + \int g\,(n^{(i)}_{m,K} - \hat n)\,{\rm d}\beta + \int g\beta^{-5} y^{(i)}_{m,K}(\log\beta - K\lambda_i)\,{\rm d}\beta$, by two integrations by parts against $P^{(i)}_{m,K}$; with eq. 12.iv this is the identity

$$\langle \mathbb{M}, G\rangle = \int G(x)\, y^{(i)}_{m,K}(x - K\lambda_i)\,{\rm d}x + r, \qquad |r| \le \epsilon_m^{(i)}\mathsf{Q}(G) + C_N\|G\|_1, \quad \mathsf{Q}(G) = \|G''\|_1 + 7\|G'\|_1 + 12\|G\|_1. \tag{12.v}$$

Let G be supported in $[-\varkappa, \varkappa]$ and $A_G(s) = \langle \mathbb{M}, G(\cdot - s)\rangle$. For $s \ge \alpha_i + \varkappa_i + \bar K_i\lambda_i$ write $s = s_0 + K\lambda_i$ with $s_0 \in [\alpha_i + \varkappa_i, \alpha_i + \varkappa_i + \lambda_i)$, so $K \ge \bar K_i$ and $G(\cdot - s)$ is supported in $\mathsf{J}_i + K\lambda_i$; a shift of s by a multiple of λ_i changes K alone, and eq. 12.v at the two cutoffs, with the oscillation bound, gives $|A_G(s) - A_G(s')| \le 2\epsilon^{(i)}_{m_i}\mathsf{Q}(G) + (2C_N + \omega_i)\|G\|_1$ whenever $s' - s \in \lambda_i\mathbb{Z}$, whatever the number of periods. As $\lambda_1/\lambda_2 \notin \mathbb{Q}$, the set $\{k\lambda_1 - k'\lambda_2 : k, k' \ge 0\} \cap [0, \infty)$ is dense in $[0, \infty)$ (by the pigeon-hole principle), and A_G is continuous; moving first forward by $k\lambda_1$ and then back by $k'\lambda_2$ we get, for $s \ge s_*$ and every $t \ge 0$,

$$|A_G(s+t) - A_G(s)| \le 2(\epsilon^{(1)}_{m_1} + \epsilon^{(2)}_{m_2})\mathsf{Q}(G) + (4C_N + \omega_1 + \omega_2)\|G\|_1;$$

only the irrationality of λ_1/λ_2 is used. At the exponent m^0, eq. 12.v with Y^0 gives $\mathbb{M}^+([s, s+\frac12\varkappa_1]) \le \mathsf{M}_1$ for $s \ge \alpha_1 + \varkappa_1 + k^0\lambda_1$. Now let $G = \Phi_{a,\flat}$ be a plateau function: the indicator of $[-\frac\flat2, a + \frac\flat2]$ smoothed with the kernel $\frac{15}{16}(1-y^2)_+^2$ scaled to the width $\frac\flat2$, which is C^2, has values in $[0, 1]$, equals 1 on $[0, a]$, is supported in $[-\flat, a + \flat]$ and has $\mathsf{Q}(G) \le \frac{15}{2\flat} + 14 + 12(a + \flat)$. Averaging the last display over $t \in [0, \varkappa_1]$ and using $\mathbb{M} \le \mathbb{M}^+$ bounds $A_G(s)$ by $7\mathsf{M}_1\|G\|_1/\varkappa_1$ plus the error; hence, for $0 \le a \le \varkappa/2$ and $\flat = \epsilon^{1/2}$,

$$\mathbb{M}^+([s, s+a]) \le \mathsf{A}_0(a + \flat) + 4\epsilon(\tfrac{15}{2\flat} + 26) \le \mathsf{A}_0 a + (\mathsf{A}_0 + 134)\epsilon^{1/2}.$$

At $a = 0$ this bounds the atom; covering $[s, s+\delta]$ by at most $2\delta/\varkappa + 1$ intervals of length $\le \varkappa/2$ gives the bound for $|\mathbb{M}|$. (b) $y^{(1)}_{m_1^0,k^0}$ is bounded on J_1, and the oscillation bound at m_1^0 bounds $y^{(1)}_{m_1^0,K}$, $K \ge k^0$, by it plus ω_1. At a given $s \ge s_\sharp$ let m_i be the largest $m \ge m_i^0$ with

$T_i(m) \le s$; as $\bar K_i(m) \le \max\{k_i,0\} + \mathsf{a}_i + A_i m + 1$, $T_i(m_i+1) > s$ gives $\lambda_i m_i > (s - c_{3,i})/A_i - \lambda_i$, so $\epsilon \le c_6 e^{-s/A} \le \varkappa^2/4$, and (a) applies with $\bar K_i = \bar K_i(m_i)$ and $s_* \le s$; A_0 does not depend on m_1, m_2. (c) A_0 and s_* do not depend on m_1, m_2, and the lower bound of ϵ over the admitted pairs is 0; so $\mathbb{M}^+([s, s+a]) \le \mathsf{A}_0 a$ for $s \ge s_*$: there are no atoms, and by outer regularity $\mathbb{M}^+(B) \le \mathsf{A}_0 |B|$ for Borel sets $B \subset [s_*, \infty)$. (d) Cut $[s_3,\infty)$ into intervals I_j of length $\delta = \epsilon_1(s_3)$; then $|\mathbb{M}|(I_j) \le (C_{\rm h} + C_{\rm r})\delta$, and $\delta \sup_{I_j} |G| \le \int_{I_j} |G| + \delta \int_{I_j} |G'|$; sum over j. The translate $G(\cdot - t)$ has the same norms and its support lies in $[s_3,\infty)$. (e) Let $s_k = 2k$, $\theta_k = e^{-s_k/2A}$, $\Phi(x) = -\frac12(1-|x|)_+^2$, so that $\Phi'' = 2\delta_0 - \mathbf{1}_{(-1,1)}$, and $\varphi = \sum_k \frac12 \theta_k^2 \Phi((\cdot - s_k)/\theta_k)$. Put $P(\beta) = \beta^{-3}[\varphi(\log\beta) + \frac{7}{12}]$ and $\hat n(\beta) = 7\beta^{-5}$. For continuous ϕ the image of $\beta^4\, d^2(\beta^{-3}\phi(\log\beta))$ is the distribution $\phi'' - 7\phi' + 12\phi$; so the image of $\beta^4 P''(d\beta)$ is $\sum_k \theta_k \delta_{s_k} + (\rho + 7)dx$ with a density $\rho \ge -\frac12 - \frac72 - 3 = -7$: P is convex, $\mathbb{M} = \sum_k \theta_k \delta_{s_k} + \rho\, \mathrm{d}x$, eq. 12.iii holds with $C_{\rm low} = 7$, and $\beta_k^4 J_P(\beta_k) = \theta_k = \beta_k^{-1/2A}$. As approximants take $n^{(i)}_{m,K} = \hat n$ and $P^{(i)}_{m,K} = \beta^{-3}[\varphi^{[i,m]}(\log\beta) + \frac{7}{12}]$, where $\varphi^{[i,m]}$ keeps, mollified to within $\epsilon^{(i)}_m$, only the bumps with $s_k + 1 \le \alpha_i + \lambda_i \bar K_i(m)$: the bumps left out have $|\varphi_k| \le \frac14 \theta_k^2 \le \epsilon^{(i)}_m$, and $y^{(i)}_{m,K} = 0$ on J_i for $K \ge \bar K_i(m)$. □

Remark 12.4 (One period does not suffice, and the lower bound cannot be dropped). (1) Let $\lambda > 0$, $0 < \theta < \min\{1, \lambda/2\}$, Φ as in the proof of (e), $\varphi(x) = \sum_{k\in\mathbb{Z}} \frac12 \theta^2 \Phi((x - k\lambda)/\theta)$, $P(\beta) = \beta^{-3}[\varphi(\log\beta) + \frac{7}{12}]$ and $\hat n = 7\beta^{-5}$. With λ-periodic smooth $\varphi^{[m]}$, $\|\varphi - \varphi^{[m]}\|_\infty \le e^{-\lambda m}$, the approximants $\beta^{-3}[\varphi^{[m]}(\log\beta) + \frac{7}{12}]$ and $n_{m,K} = \hat n$ satisfy eq. 12.iii with $C_{\rm low} = 7$, eq. 12.iv with $C_N = 0$ and $\epsilon_m = e^{-\lambda m}$, the bound on Y^0 at every m^0, and the oscillation bound with $\omega = 0$ at every cutoff, a threshold free of m, since $y_{m,K}$ does not depend on K; yet $\beta^4 J_P(\beta) = \theta$ at every $\beta = e^{k\lambda}$. So the hypotheses at one period, or at two commensurable periods $p\lambda$, $q\lambda$ (with $\varphi^{[pm]}$, $\varphi^{[qm]}$ in the place of $\varphi^{[m]}$, so that $\epsilon^{(i)}_m = e^{-p\lambda m}$, $e^{-q\lambda m}$), give neither (a) nor (b) nor (c). (2) The lower bound eq. 12.iii cannot be dropped from (a), nor from (b), at any two periods. Let $\beta_j = 4^j$ for the integers $j \ge j_0 \ge 1$, numbers w_j with $2\beta_j^{-2} \le w_j \le \frac14 \beta_j$ and $w_j/\beta_j \to 0$, $c_j = \frac16 w_j \beta_j^{-4}$, $\tau(x) = -(1-|x|)_+^3$, $\beta_0 = 1$,

$$P(\beta) = -\log\beta + \sum_{j \ge j_0} c_j\, \tau\Big(\frac{\beta - \beta_j}{w_j}\Big) \qquad \text{and} \qquad \hat n(\beta) = \beta^{-2}.$$

The j-th term vanishes off $[\beta_j - w_j, \beta_j + w_j] \subset [\frac34\beta_j, \frac54\beta_j]$, and these intervals are disjoint. The function τ is C^2 off 0, $\tau'(0\pm) = \pm 3$, and $\tau''(x) = -6(1-|x|)$ for $0 < |x| < 1$; so on the j-th interval, off β_j, $P'' \ge \beta^{-2} - 6c_j w_j^{-2} = \beta^{-2} - \beta_j^{-4} w_j^{-1} \ge (\frac{16}{25} - \frac12)\beta_j^{-2} > 0$, while at β_j the slope of P jumps up by $6c_j/w_j = \beta_j^{-4}$. So P is convex on (β_0,∞), and $\beta_j^4 J_P(\beta_j) = 1$ for every j. Moreover $\beta^3 c_j |\tau((\beta - \beta_j)/w_j)| \le (\frac54\beta_j)^3 c_j < \frac12 w_j/\beta_j$. For given λ_i, α_i, $\varkappa_i$, k_i as in Prop. 12.3, put $m_i^0 = 0$ and $n^{(i)}_{m,K} = \hat n$, so that $C_N = 0$, and, for all K,

$$P^{(i)}_{m,K}(\beta) = -\log\beta + \sum_{j_0 \le j < j(m)} c_j\, \tau_m\Big(\frac{\beta - \beta_j}{w_j}\Big),$$

with integers $j(m) \to \infty$ and C^2 functions τ_m with support in $[-1,1]$ and $\sup|\tau_m - \tau| \le \epsilon_m := \sup_{j \ge j(m)} w_j/\beta_j$. Then eq. 12.iv holds with $\epsilon^{(i)}_m = \epsilon_m$, and $\epsilon_m \to 0$. On $W^{(i)}_K$ the function $y^{(i)}_{m,K}$ vanishes as soon as $e^{\alpha_i + K\lambda_i} > \frac54 \beta_{j(m)-1}$, because there $(P^{(i)}_{m,K})'' = \beta^{-2} = \hat n$; so the oscillation bound holds with $\omega_i = 0$ beyond a finite threshold at each m, and the bound on Y^0 holds at every m^0, as the cutoffs below the threshold are finitely many. Thus every condition of (a) and of its last sentence holds except eq. 12.iii: near $\log\beta_j$ the measure $\mathbb{M}$ has the density $\beta^5 c_j w_j^{-2} \tau''((\beta - \beta_j)/w_j)$, which tends to $-\beta_j/w_j$ at $\log\beta_j$, so eq. 12.iii fails for every $C_{\rm low}$; and the conclusion of (a) fails, as $e^{4s} J_P(e^s) = 1$ at every $s = \log\beta_j$.

Let now $w_j = \min\{\frac14\beta_j, \beta_j^{1-1/A}\}$ with $A > \frac13$ and j_0 so large that $w_j \ge 2\beta_j^{-2}$ for $j \ge j_0$, let $\mathsf{a}_i \ge 0$ and $\bar K_i(m) = \max\{k_i, \lceil \mathsf{a}_i + Am\rceil\}$, and at the period λ_i keep in $P^{(i)}_{m,K}$ the terms with $2\beta_j \le e^{\alpha_i+\lambda_i\bar K_i(m)}$, with τ_m within $c_{\mathrm{P},i}e^{-\lambda_i m}$ of τ, where $c_{\mathrm{P},i} = 2^{1/A}e^{|\alpha_i|/A}$. A term left out has $\beta_j > \frac12 e^{\alpha_i+\lambda_i\bar K_i(m)} \ge \frac12 e^{-|\alpha_i|}e^{\lambda_i Am}$, so that $w_j/\beta_j \le \beta_j^{-1/A} < c_{\mathrm{P},i}e^{-\lambda_i m}$; and $y^{(i)}_{m,K} = 0$ on J_i for $K \ge \bar K_i(m)$. So every condition of (b) holds, with $A_1 = A_2 = A$, $\omega_1 = \omega_2 = 0$, $C_N = 0$ and $\epsilon^{(i)}_m = c_{\mathrm{P},i}e^{-\lambda_i m}$, except eq. 12.iii, and the conclusion of (b) fails.

In the form in which the proofs of (a) and (b) use the data, namely a measure with bounded variation on unit intervals whose smoothed translates are nearly invariant, a lower bound is needed as well: $\mathbb{M} = \sum_j(\delta_{s_j} - \delta_{s_j+\eta_j})$, with $s_{j+1} - s_j \ge 1$, $0 < \eta_j < 1$ and $\eta_j \to 0$, has $|\mathbb{M}|(I) \le 4$ on every interval I of length 1 and $\langle \mathbb{M}, G(\cdot - s)\rangle \to 0$ as $s \to \infty$ for every $G \in C^1_c(\mathbb{R})$, but atoms 1 at the points s_j. Its atoms -1 at the points $s_j + \eta_j$ set it apart from the measures $\mathbb{M}$ of data, whose atoms $e^{4s}J_P(e^s)$ are non-negative; in the instance above the density $-\beta_j/w_j$ takes their place. (3) By (d), a bound $|\mathbb{M}|([s, s+\delta]) \le C\delta + Ce^{-cs}$ gives the uniform translation bound of Prop. 12.3(d), with its persistent L^1 term and an exponentially small derivative correction; the proofs of (a) and (b) turn near invariance at such a rate under two incommensurable periods, with eq. 12.iii, into the same bound at the rate $e^{-cs/2}$; and by (e) this loss of one half is real for data.

Proposition 12.5 (For data: only one slope in the exponent)**.** Let the data of Prop. 12.3 be given, with $\epsilon^{(i)}_m = c_{\mathrm{P},i}e^{-\lambda_i m}$, $c_{\mathrm{P},i} > 0$. Suppose:

(i) there are $\mathsf{a}_1 \ge 0$, $A_1 > 0$, $\omega_1 \ge 0$ with $\sup_{\mathsf{J}_1} |y^{(1)}_{m,K} - y^{(1)}_{m,K'}| \le \omega_1$ for all $m \ge m_1^0$ and all integers $K, K' \ge \max\{k_1, \mathsf{a}_1 + A_1 m\}$, and $\sup_{K\ge k_1}\sup_{\mathsf{J}_1}|y^{(1)}_{m_1^0,K}| \le Y^0$;

(ii) there is $\omega_2 \ge 0$ such that for every m in an unbounded set of integers $\ge m_2^0$ there is an integer $\bar K_2(m) \ge k_2$ with $\sup_{\mathsf{J}_2}|y^{(2)}_{m,K} - y^{(2)}_{m,K'}| \le \omega_2$ for $K, K' \ge \bar K_2(m)$.

Put $\mathsf{M}_0 = \varkappa_1(Y^0 + C_N + C_{\mathrm{low}}) + c_{\mathrm{P},1}(4/\varkappa_1 + 14 + 12\varkappa_1)$, $\mathsf{H}_* = \omega_1 + \omega_2 + 4C_N + 120 + 16\mathsf{M}_0/\varkappa_1 + 12\mathsf{M}_0/\lambda_2$ and $C_\vartheta = \max\{1, c_{\mathrm{P},1}^{1/2}\}\exp[(\alpha_1 + \varkappa_1 + \lambda_1(\mathsf{a}_1+1))/2A_1 + \lambda_1/2]$. There is $s_\sharp$ such that for $s \ge s_\sharp$, $\beta = e^s$ and $\delta \ge 0$

$$|\mathbb{M}|([s, s+\delta]) \le (2\mathsf{H}_* + C_{\mathrm{low}})\delta + 2\mathsf{H}_* C_\vartheta \beta^{-1/2A_1}, \qquad \beta^4 J_P(\beta) \le \mathsf{H}_* C_\vartheta \beta^{-1/2A_1}.$$

Only the slope A_1 of the first period enters the exponent; of the second period only one bound ω_2, some finite threshold at each m, and $\epsilon^{(2)}_m \to 0$ are used. ■

Proof sketch. The identity for tents. For $\varphi_{c,\vartheta}(x) = (1 - |x-c|/\vartheta)_+$ with $[c-\vartheta, c+\vartheta] \subset \mathsf{J}_i + K\lambda_i$ the identity eq. 12.v holds with $\mathsf{Q}(G)$ replaced by $4/\vartheta + 14 + 12\vartheta$: with $Q(s) = e^{3s}P(e^s)$ and $Q_{m,K}$ formed in the same way from $P^{(i)}_{m,K}$, the image of $\beta^4 P''(d\beta)$ is the distribution $Q'' - 7Q' + 12Q$, $D = Q - Q_{m,K}$ has $|D| \le \epsilon^{(i)}_m$, and $\int \varphi_{c,\vartheta}\, d(D'' - 7D' + 12D) = [D(c-\vartheta) - 2D(c) + D(c+\vartheta)]/\vartheta + 7\int \varphi'_{c,\vartheta}D + 12\int \varphi_{c,\vartheta}D$. *Step 1 (mass).* With $c_1 = \alpha_1 + k_1\lambda_1 + \varkappa_1$, every $c \ge c_1$ is $\upsilon_0 + K\lambda_1$ with $K \ge k_1$ and $\upsilon_0 \in [\alpha_1 + \varkappa_1, \alpha_1 + \varkappa_1 + \lambda_1]$; the identity at m_1^0, $\vartheta = \varkappa_1$, with (i), gives $\int \varphi_{c,\varkappa_1} d\mathbb{M}^+ \le \mathsf{M}_0$, so $\mathbb{M}^+([c - \frac{\varkappa_1}{2}, c + \frac{\varkappa_1}{2}]) \le 2\mathsf{M}_0$, and an interval of length l in $[c_1, \infty)$ has $\mathbb{M}^+$-mass at most $2\mathsf{M}_0(l/\varkappa_1 + 1)$. *Step 2 (one period).* For an admitted m, $0 < \vartheta \le \min\{\varkappa_i, 1\}$ with $\epsilon^{(i)}_m \le \vartheta^2$, $\upsilon_0 \in [\alpha_i + \varkappa_i, \alpha_i + \varkappa_i + \lambda_i]$ and cutoffs K, K' beyond the threshold of m, the identity at the two cutoffs gives $|\int \varphi_{\upsilon_0 + K\lambda_i,\vartheta} d\mathbb{M}^+ - \int \varphi_{\upsilon_0 + K'\lambda_i,\vartheta} d\mathbb{M}^+| \le t_i\vartheta$ with $t_i = \omega_i + 2C_N + 60$. *Step 3 (two periods).* Let $m \ge m_1^0$, $\vartheta \le \min\{\varkappa_1, \varkappa_2, 1, \lambda_2/6\}$ with $\vartheta^2 \ge c_{\mathrm{P},1}e^{-\lambda_1 m}$, $c = \upsilon_0 + K\lambda_1$ with $K \ge \max\{k_1, \mathsf{a}_1 + A_1 m\}$, and $\mathsf{H} = \vartheta^{-1}\int \varphi_{c,\vartheta} d\mathbb{M}^+$. Take m_2 in the set of (ii) with $\epsilon^{(2)}_{m_2} \le \vartheta^2$, an integer $K_2 \ge \bar K_2(m_2)$ with $S = \alpha_2 + \varkappa_2 + K_2\lambda_2 \ge c_1 + 1$, $n = \lfloor \lambda_2/3\vartheta \rfloor$, and the intervals $I_r = (S + 3\vartheta(r-1), S + 3\vartheta(r-1) + \vartheta) \subset [S, S + \lambda_2)$, $r \le n$. As $\lambda_1/\lambda_2 \notin \mathbb{Q}$, the points $c + k\lambda_1$, $k \ge k_\flat$, are dense modulo λ_2 for every $k_\flat$; so there are integers $k_r \ge 0$ and $j_r \ge 1$ with $p_r = c + k_r\lambda_1 - j_r\lambda_2 \in I_r$.

Step 2 at $i=1$, with the cutoffs K and $K+k_r$, both beyond the threshold however large k_r is, and then at $i=2$, with the cutoffs K_2+j_r and K_2, gives $\int \varphi_{p_r,\vartheta}d\mathbb{M}^+ \geq \vartheta(\mathsf{H}-t_1-t_2)$. The p_r are at least 2ϑ apart, so $\sum_r \varphi_{p_r,\vartheta}\leq 1$, and by Step 1, $n\vartheta(\mathsf{H}-t_1-t_2)\leq \mathbb{M}^+([S-\vartheta,S+\lambda_2+\vartheta])\leq 2\mathsf{M}_0(\frac{4\lambda_2}{3\varkappa_1}+1)$; with $n\vartheta\geq\lambda_2/6$ this gives $\mathsf{H}\leq\mathsf{H}_*$. *Step 4.* For large s let $m(s)=\lfloor((s-\alpha_1-\varkappa_1)/\lambda_1-\mathsf{a}_1-1)/A_1\rfloor$ and $\vartheta(s)=\max\{1,c_{\mathrm{P},1}^{1/2}\}e^{-\lambda_1 m(s)/2}\leq C_\vartheta e^{-s/2A_1}$; choose $s_\sharp$ such that for $s\geq s_\sharp$, $m(s)\geq m_1^0$, $(s-\alpha_1-\varkappa_1)/\lambda_1\geq k_1+1$ and $\vartheta(s)\leq\min\{\varkappa_1,\varkappa_2,1,\lambda_2/6\}$. Then every $c\geq s$ is admitted in Step 3 with $m(s)$ and $\vartheta(s)$. So the atom at s is at most $\int\varphi_{s,\vartheta}d\mathbb{M}^+\leq\mathsf{H}_*\vartheta$; $[s,s+\delta]$ is covered by at most $\delta/\vartheta+1$ intervals $[c-\frac{\vartheta}{2},c+\frac{\vartheta}{2}]$, $c\geq s$, each of $\mathbb{M}^+$-mass at most $2\mathsf{H}_*\vartheta$; and $|\mathbb{M}|\leq\mathbb{M}^+ + C_{\mathrm{low}}ds$. □

Proposition 12.6 (One block size)**.** In a setting:

(a) $(K,L^Kb)\in\mathfrak{d}(\gamma)\cup\mathfrak{d}(g_2)$ for $b\in\mathsf{I}$ and $K\in\mathbb{N}$; $b_+-Lb_-=(2L-2)\bar B+\frac{L}{2}>0$, so $\bigcup_K L^K\mathsf{I}=[Lb_-,\infty)$.

(b) Let $K,m\in\mathbb{N}$, $\ell=2L^m$, $N=2L^{m+K}$, $\beta>0$ and $b=a_K\beta$. The partition function of T_ℓ at the spacing a_K is $Z_N(\beta)$, so its free energy per physical volume is $a_K^{-3}P_N(\beta)$; the dilation by L^K maps $\mathcal{G}_{K,\beta}$ onto $\mathcal{G}(\beta)$ and the Wilson measure of T_ℓ to $\mu_{N,\beta}$; for a law ν on $\mathbb{L}_K$ invariant under $a_K\mathbb{Z}^3$, with image $\nu^{(K)}$, and an integer $\eta\geq1$, $\beta^5 n_\eta(\nu^{(K)})=b^5(a_K\eta)^{-3}S_2^T[\nu](\mathsf{f}_\eta,\mathsf{h}_\eta)$, where f_η is the tent of half width $a_K\eta$ about a point of $a_K(\mathbb{Z}+\frac12)^3$ and h_η its hull; and, with $\eta_K=L^{K-1}$,

$$\beta^5P_N''(\beta)=\beta^5 n_{\eta_K}(\mu_{N,\beta})+y_{m,K}(\log b). \tag{12.vi}$$

(c) For $b\in\mathsf{I}$, $K\in\mathbb{N}$, $\beta=L^Kb$, $m\geq m_0$ and $N=2L^{m+K}$,

$$0\leq a_K^{-3}[P_N(\beta)-P(\beta)]\leq 3\ell_0C_{\mathrm{v}}/\ell,\qquad\text{so}\qquad 0\leq\beta^3[P_N(\beta)-P(\beta)]\leq c_{\mathrm{P}}L^{-m},$$

with $c_{\mathrm{P}}=\frac32 b_+^3\ell_0C_{\mathrm{v}}$ and $C_{\mathrm{v}}=2C_{\mathrm{S}}+\max\{C_\gamma,C_{g_2}\}$, where C_{S} is the constant in Thm. YM3_I: AD(ii), (iii) and $C_g=C_{\mathrm{S}}(1+\log g^{-2})$ is its lower-bound constant at fixed g; C_{S} does not depend on ϱ, γ, g, K, β, ℓ, and C_g not on K, β, ℓ.

(d) Let $r_1>0$ with $13r_1b_+\leq1$. Then $\hat n$ is Borel, $|\hat n(\beta)|\leq1125\lceil r_1\beta\rceil^3$, and $\hat n(\beta)=n_{\lceil r_1\beta\rceil}(\nu_\beta)$ for some $\nu_\beta\in\mathcal{G}(\beta)$. With the constant C_{nr} of the comparison of two laws in one ball of radius 1 (the common-normalizer comparison in the proof of Prop. YM3_I:91.13) and the constants $C_{\mathrm{sp}}(\mathsf{d})$ of the bound at separated supports in every periodic Gibbs state (Prop. YM3_I: 91.13), both taken at $g\in\{\gamma,g_2\}$ as in the proof, $\mathsf{d}_1=\min\{r_1b_-,\varepsilon\}$ and

$$C^{\mathrm{ref}}=b_+^5\mathsf{d}_1^{-3}(40\sqrt3M/\mathsf{d}_1)^2[C_{\mathrm{nr}}+8C_{\mathrm{sp}}(\mathsf{d}_1)+C_{\mathrm{sp}}(r_1b_-)],$$

for $b\in\mathsf{I}$, $K\in\mathbb{N}$, $\beta=L^Kb$, $m\geq m_0$ and $N=2L^{m+K}$:

$$|\beta^5P_N''(\beta)-y_{m,K}(\log b)-\beta^5\hat n(\beta)|\leq C^{\mathrm{ref}};$$

$\beta^5|n_{\lceil r_1\beta\rceil}(\nu)-\hat n(\beta)|\leq C^{\mathrm{ref}}$ and $0\leq\beta^5\lceil r_1\beta\rceil^{-3}s_\nu^{\lceil r_1\beta\rceil}(3e_1)\leq C^{\mathrm{ref}}$ for every $\nu\in\mathcal{G}(\beta)$; and $P''(d\beta)\geq[\hat n(\beta)-871C^{\mathrm{ref}}\beta^{-5}]d\beta$ on (Lb_-,∞).

(e) For all $K\in\mathbb{N}$, $m\geq m_0$ and $\upsilon\in\log\mathsf{I}$, $|y_{m,K}(\upsilon)|\leq b_+^5L^3C_3(\varepsilon)(40\sqrt3ML)^2\ell^3$, where $C_3(\mathsf{d})=(40\sqrt3M+1)C_{\mathrm{tt}}(\mathsf{d})$ and $C_{\mathrm{tt}}(\mathsf{d})$ is the larger, over $g\in\{\gamma,g_2\}$, of the numbers $A(\mathsf{d})B(1)^2\,2^{\varkappa_\star}$ of Prop. YM3_I:89.5 at g: the constant of the bound for connected two-point functions on the tori at first-passage data, for two functions each supported in a closed sup-ball of radius 1, at distance at least d; it does not depend on K, β or the side. This is better than the bound by $C^{\mathrm{pol}}\ell^{438}$ that (E)(b) gives.

■

Proof sketch. (a) With $y_1=\gamma^{-2}+\bar B\geq4+\bar B$ and $L\geq13$, $(L-1)y_1\geq4\bar B+1$, which says $y_2=g_2^{-2}+\bar B=y_1+2\bar B+1\leq Ly_1-2\bar B$; so $I(\gamma)\cup I(g_2)=(y_1,Ly_2-2\bar B]\supset\mathsf{I}$, as $b_+=Ly_2-2\bar B$. And $b_+-Lb_-=L(y_2-y_1)-2\bar B-\frac{L}{2}=(2L-2)\bar B+\frac{L}{2}$, so consecutive intervals $L^K\mathsf{I}$ overlap. (b) The dilation by L^K carries bonds, plaquettes and the Haar measures of $\mathbb{L}_K$ to those of $\mathbb{Z}^3$ and $\ell\mathbb{Z}^3$

to $N\mathbb{Z}^3$, and $\mathcal{O}_K(f)$ becomes $L^K\mathcal{O}_1(f(L^{-K}\cdot))$; so $S_2^T[\nu](f,h) = L^{2K}\langle f(L^{-K}\cdot); h(L^{-K}\cdot)\rangle_{\nu^{(K)}}$, and $\beta^5\eta^{-3}L^{-2K} = b^5(a_K\eta)^{-3}$. As L^K is odd, $\frac{1}{2} \in a_K(\mathbb{Z}+\frac{1}{2})$, and $\varepsilon = a_K\eta_K$. At $\eta = \eta_K$, which divides N with $N/\eta_K = 2L^{m+1} \geq 7$, eq. 12.i becomes eq. 12.vi: the rescaled tents are the translates $\mathsf{f}(\cdot - \varepsilon X)$, their sum over the period cell is 1 on T_ℓ, so the far sum is $1 - \mathsf{h}^{\langle\ell\rangle}$. (c) By (a), $(K,\beta) \in \mathfrak{d}(g)$ for some $g \in \{\gamma, g_2\}$. Let $p(j)$, $j \in \mathbb{N}^3$, be the free energy per physical volume of the periodic box with $j_\mu\ell_0/a_K$ sites in the direction μ, at the spacing a_K and the coupling β. With $Z_{K,\ell,\beta} = Z_{\ell L^K}(\beta)$, the partition function of T_ℓ at the spacing a_K, (b) gives $p(k,k,k) = (k\ell_0)^{-3}\log Z_{K,k\ell_0,\beta} = a_K^{-3}P_{k\ell_0 L^K}(\beta)$ for every power k of L. In the first direction such a box consists of $2j_1$ slabs, and the trace formula (Prop. YM3_I:85.2) writes its partition function as $\operatorname{Tr} B^{j_1}$ with one positive trace-class operator B for all j_1, formed at j_2, j_3.

With the top eigenvalue $\|B\|$, $u_k = \frac{1}{k}\log\operatorname{Tr} B^k$ satisfies $0 \leq k(u_k - \log\|B\|) \leq u_1 - \log\|B\|$ and does not increase in k; hence p does not increase in j_1 and $0 \leq p(k',j_2,j_3) - p(k,j_2,j_3) \leq \frac{1}{k'}[p(1,j_2,j_3) - \lim_n p(n,j_2,j_3)]$ for $k' \leq k$. The same argument in the second and in the third direction shows that p does not increase in any of its three arguments, with the same bound in each direction. The bracket is at most C_{v}, for all j_2, j_3. By the monotonicity in each argument, $p(1,j_2,j_3) \leq p(1,1,1)$, and $p(n,j_2,j_3) \geq p(L^i,L^i,L^i)$ whenever $L^i \geq \max\{n,j_2,j_3\}$; so the bracket is at most $\sup_{i\geq 0}[p(1,1,1) - p(L^i,L^i,L^i)]$, where $p(1,1,1)$ and $p(L^i,L^i,L^i)$ belong to the tori with the sides $\ell_0 = 2L^{m_0}$ and $L^i\ell_0 = 2L^{m_0+i}$, both admitted at γ and at g_2.

To (K,β) and to each side ℓ admitted at g, Thm. YM3_I:AD attaches a real number $E_{K,\ell,\beta}$, and its stability bounds say: by the zero-source factorisation and (ii), $-C_g \leq \ell^{-3}\log Z_{K,\ell,\beta} - \ell^{-3}E_{K,\ell,\beta} \leq C_{\mathrm{S}}$, that is, $\ell^{-3}\log Z_{K,\ell,\beta}$ lies within C_{S} above and C_g below the energy per volume $\ell^{-3}E_{K,\ell,\beta}$; by (iii) the energies per volume at two such sides differ by at most C_{S}; and by the sentence following (iii), C_{S} is free of $\varrho, \gamma, g, K, \beta, \ell$. At the two tori this gives $p(1,1,1) - p(L^i,L^i,L^i) \leq C_{\mathrm{S}} + C_{\mathrm{S}} + C_g \leq C_{\mathrm{v}}$ for every i; and the brackets of the directions 2 and 3 obey the same bound.

Changing one direction at a time, from (k',k',k') through (k,k',k') and (k,k,k') to (k,k,k), with $k' = \ell/\ell_0 = L^{m-m_0}$ and a power $k \geq k'$ of L, each step changes p by at most C_{v}/k', so $0 \leq p(k',k',k') - p(k,k,k) \leq 3C_{\mathrm{v}}/k'$, that is, $0 \leq a_K^{-3}[P_N - P_{k\ell_0 L^K}] \leq 3\ell_0 C_{\mathrm{v}}/\ell$, as $N = \ell L^K$; let $k \to \infty$ through the powers of L (Lem. 12.1(a)), and multiply by $b^3 = \beta^3 a_K^3$. (d) Thms. YM3_I:AJ and YM3_I:AD enter through two bounds at first-passage data, with constants free of K, β, m and the state; by (a) the pair (K,β) lies in $\mathfrak{d}(g)$ for some $g \in \{\gamma, g_2\}$. By the common-normalizer comparison in the proof of Prop. YM3_I:91.13 at $n = 2$, whose hypothesis (Nr1) Prop. YM3_I:91.12 supplies at every pair of $\mathfrak{d}(g)$, two laws, each a torus measure on an admitted side $2L^m$, $m \geq m_0$, or a periodic Gibbs state at (K,β), satisfy $|S_2^T[\mathfrak{s}](F_1,F_2) - S_2^T[\mathfrak{s}'](F_1,F_2)| \leq C_{\mathrm{nr}}\|F_1\|_\sharp\|F_2\|_\sharp$ for F_1, F_2 supported in one closed ball of radius 1 about a lattice point; here C_{nr} is the larger, over $g \in \{\gamma, g_2\}$, of the numbers $8\log 2\,(2\mathsf{Q}_1(L)+2)^2\varrho^{-2}$, with $\mathsf{Q}_1(L)$ of equation eq. 5.xxvii at g.

By Prop. YM3_I:91.13 at $n = 2$ and $R = 2$, in every periodic Gibbs state $|S_2^T[\nu](F_1,F_2)| \leq C_{\mathrm{sp}}(\mathsf{d})\|F_1\|_\sharp\|F_2\|_\sharp$ for supports in balls of radius 2 at distance at least d; here $C_{\mathrm{sp}}(\mathsf{d})$ is the larger, over $g \in \{\gamma, g_2\}$, of the numbers $C_2^{\mathrm{per}}(\mathsf{d};2)$ of equation eq. 5.xxvii at g. The physical half widths $a_K\lceil r_1\beta\rceil \in [r_1 b, r_1 b + a_K]$ and $a_K\eta_K = \varepsilon$ lie in $[\mathsf{d}_1, \frac{2}{13}]$, so all tents and hulls involved, and the function Θ of eq. 12.ii rescaled, have $\|\cdot\|_\sharp \leq 40\sqrt{3}M/\mathsf{d}_1$ and live in a ball of radius 1 or 2 about one lattice point.

On each interval on which $\lceil r_1\beta\rceil$ is constant, $\beta \mapsto n_{\lceil r_1\beta\rceil}(\mu_{N,\beta})$ is continuous; so $\hat{n}$ is Borel, and it is attained along a subsequence $\mu_{N_k,\beta} \to \nu_\beta \in \mathcal{G}(\beta)$, the near part being weakly continuous. Write

$$n_{\eta_K}(\mu_{N,\beta}) - \hat{n}(\beta) = [n_{\eta_K}(\mu_{N,\beta}) - n_{\eta_K}(\nu_\beta)] + [n_{\eta_K}(\nu_\beta) - n_{\lceil r_1\beta\rceil}(\nu_\beta)]\,:$$

the first bracket compares two states in one unit ball, and contributes at most $b_+^5\mathsf{d}_1^{-3}(40\sqrt{3}M/\mathsf{d}_1)^2 C_{\mathrm{nr}}$ after multiplication by β^5; this is the one place where m enters, and the bound does not see it. By eq. 12.ii the second bracket is a sum of four means, with weights summing to 1, of covariances in the one state ν_β between a tent and a function H with $\|H\|_\sharp \leq 2\cdot 40\sqrt{3}M/\mathsf{d}_1$ which vanishes

within the distance d_1 of the support of the tent; after multiplication by β^5 each mean is at most $2b_+^5\mathsf{d}_1^{-3}(40\sqrt{3}M/\mathsf{d}_1)^2C_{\mathrm{sp}}(\mathsf{d}_1)$. With eq. 12.vi this gives the first inequality. In the same way $\beta^5|n_{\lceil r_1\beta\rceil}(\nu)-\hat n|$ compares two states in a unit ball, and $\beta^5\lceil r_1\beta\rceil^{-3}s_\nu(3e_1)$ is a covariance at the distance $a_K\lceil r_1\beta\rceil\ge r_1b_-$, non-negative by Lem. 12.2(a). Finally, every $\beta>Lb_-$ is L^Kb with $b\in\mathsf{I}$ by (a); on each open interval of (Lb_-,∞) on which $\lceil r_1\beta\rceil$ is constant, Lem. 12.2(b) applies with $q=\hat n-(1+870)C^{\mathrm{ref}}\beta^{-5}$; the end points of these intervals are countably many, and there P'' has atoms ≥ 0 while $q\,\mathrm{d}\beta$ has none. (e) Let χ_k, $k\in(\mathbb{Z}/\ell\mathbb{Z})^3$, be the periodised products $\prod_i(1-|y_i-k_i|)_+$: they sum to 1 on T_ℓ, each is supported in a closed sup-ball of radius 1, $\mathrm{Lip}\,\chi_k\le\sqrt{3}$, and so $\|\chi_kw\|_\sharp\le(40\sqrt{3}M+1)\|w\|_\sharp$ and $\mathrm{supp}(\chi_kw)\subset\mathrm{supp}\,w$. By bilinearity $S_2^T(u,w)=\sum_kS_2^T(u,\chi_kw)$, a sum of ℓ^3 terms, each at most $C_{\mathrm{tt}}(\mathsf{d})(40\sqrt{3}M+1)\|u\|_\sharp\|w\|_\sharp$ when u is supported in a sup-ball of radius 1 and the supports of u and w are at distance at least d. Apply this to $u=\mathsf{f}^{\langle\ell\rangle}$, $w=1-\mathsf{h}^{\langle\ell\rangle}$, $\mathsf{d}=\varepsilon$, with $\|u\|_\sharp,\|w\|_\sharp\le40\sqrt{3}ML$ and $b^5\varepsilon^{-3}\le b_+^5L^3$. □

Lemma 12.7 (Two block sizes)**.** For every $\varrho>0$ and every integer $a\ge1$ there are two settings at the radius ϱ with the block sizes L_1 and $L_2=L_1+2$ and one common value $\mathfrak{p}_0\ge a$. For every odd $L_1\ge3$, $\log L_1/\log(L_1+2)\notin\mathbb{Q}$. ■

Proof sketch. Use the ordered construction in the proof of Prop. YM3_I:69.1: choose $r\ge2$, then an integer $\mathfrak{p}_0\ge\max\{a,6r+1\}$ and $q\ge41\mathfrak{p}_0+1$, followed by the reserve fractions and the other constants preceding L. The proof admits every sufficiently large odd L with these values fixed. The additional floors in Section YM3_I:82 have the same property. Choose an odd L_1 above their common threshold and complete the prescribed construction separately at L_1 and L_1+2; with ϱ and the associated Wilson measures and source normalization, each is a setting. If $\log L_1/\log L_2=u/v$ with positive integers, then $L_1^v=L_2^u$, and a prime factor of L_1 divides $L_2-L_1=2$, which is impossible for odd L_1. Every statement that we use is made for every radius and every choice, and no object of one choice is compared with an object of the other; the two settings meet in P and $\hat n$ alone. □

Lemma 12.8 (The lost mass on tori, without an exponential of the volume)**.** Let ς' be a lowered choice of a choice ς of $\mathcal{T}^{\mathrm{F}+}$, and let the setting (Th) of §2.3.1 hold at ς and at a label $b\ge\gamma^{-2}$ with $\log b\ge\mathsf{Y}^\natural$: $K\ge2$, $\ell=2L^m$ with $M\mathcal{R}(b)\le L^m$, where $\mathcal{R}(x)=\min\{L^c:c\in\mathbb{Z}_{\ge0},\ L^c\ge(\log x)^r\}$ (equation YM3_I:(634)), the line of couplings $x_j=L^{K-j}b$, and a top level. Let $Z=\sum_h\mathsf{t}(h)$ be the sum of the masses $\mathsf{t}(h)$ of the histories (Thm. YM3_I:D), $Z_t(h)=\Lambda_t(h)^c$ the large-field region defined by equation YM3_I:(644), and $N_t=\ell^3L^{3(K-t)}$. Then for $1\le t\le K$

$$\sum_{h:\,Z_t(h)\neq\emptyset}\mathsf{t}(h)\le\mathsf{R}_tZ,\qquad\mathsf{R}_t=\tfrac{L}{L-1}M^{-3}N_t\,e^{-c_2^\natural A_0^2(\log x_t)^{2\mathfrak{p}_0}}.$$

The side enters R_t only through the number N_t of the sites of the level t; and R_t is smaller by the factor 25000 than the number $2\cdot24^3M^{-3}N_te^{-c_2^\natural A_0^2(\log x_t)^{2\mathfrak{p}_0}}$ that the rarity of a large field at a point, Prop. 4.52, gives. ■

Proof. The line $x_j=L^{K-j}b$ is admissible under the stated side and coupling conditions, by Lem. YM3_I:118.1. Def. 4.47 and Lem. 4.48 put it in the common setting of Prop. 4.52. That proposition bounds the probability of a prescribed creation at level t' by $e^{-c_2^\natural p_0(g_{t'})^2}$.

If $Z_t\neq\emptyset$, Prop. YM3_I:67.2 gives a creation at some level $t'\le t$. There are at most $M^{-3}N_{t'}=M^{-3}N_tL^{3(t-t')}$ cells of that level, since $\check R_{t'}\ge1$. Write $a=t-t'$. On this line, $y_{t'}=y_t+a\log L$, and convexity gives

$$y_{t'}^{2\mathfrak{p}_0}-y_t^{2\mathfrak{p}_0}\ge2\mathfrak{p}_0y_t^{2\mathfrak{p}_0-1}a\log L.$$

Condition (Y.7) of eq. 4.xxi therefore bounds the contribution of level $t-a$ by L^{-a} times the contribution of level t. The union bound and the geometric series yield

$$\sum_{h:Z_t(h)\neq\emptyset} \mathsf{t}(h) \le M^{-3}N_t e^{-c_2^\natural p_0(g_t)^2} Z \sum_{a=0}^{t-1} L^{-a} \le \frac{L}{L-1} M^{-3}N_t e^{-c_2^\natural p_0(g_t)^2} Z.$$

Finally $L/(L-1) \le 13/12 < 2\cdot 24^3/25000$. □

Lemma 12.9 (The threshold on a torus is of exact order $\ell^{3/2\mathfrak{p}_0}$)**.** In a setting let $K_0(\ell) = K_0(\ell, 2, \mathsf{I})$ and $K_1(\ell) = K_1(\ell, 2, \mathsf{I}, \varepsilon)$ be the thresholds of the convergence on the torus T_ℓ (Thm. YM3_I:AS at $s=0$, with the constants and thresholds in Thm. YM3_I:AR) at $n=2$, for the labels of I and the spread ε, as they are set there. There are $c_1, c_2 < \infty$ such that for all $m \ge m_0$ with $M\mathcal{R}(b_+) \le L^m$, $\ell = 2L^m$,

$$\frac{1800}{\lambda}\Big[\Big(\frac{\Xi\ell^3}{c'A_0^2}\Big)^{1/2\mathfrak{p}_0} - \log(Lb_-)\Big] \le K_0(\ell) \le K_1(\ell) \le c_1 + c_2\ell^{3/2\mathfrak{p}_0},$$

where $\Xi > 0$ is the number in the factor $e^{\Xi\ell^3}$ that the bound on the lost mass, with which these thresholds are formed, carries, and $c' > 0$ the number in its large-field weight $e^{-c'A_0^2(\log x)^{2\mathfrak{p}_0}}$. ■

Proof sketch. The conditions that define the thresholds read the side only through $N_{[d]} = \ell^3 L^{3d}$ at depths $d \le K/9+1$ and through the factor $e^{\Xi\ell^3}$. At $n=2$ the depth u of the comparison of the cutoffs satisfies $K/1800 \le u \le K/1800+1$. One condition asks that the bound on the lost mass, $2M^{-3}\ell^3 L^{3u}\exp[\Xi\ell^3 - c'A_0^2(\log(L^u b_-))^{2\mathfrak{p}_0}]$, be at most $\frac12$; as $2M^{-3}\ell^3 \ge 16$ and $\log(L^u b_-) \le K\lambda/1800 + \log(Lb_-)$, it fails unless $(K\lambda/1800 + \log(Lb_-))^{2\mathfrak{p}_0} \ge \Xi\ell^3/c'A_0^2$, which is the lower bound. For the upper bound, each condition says that a quantity $c\,\ell^3 L^{-cK}$, or $c\exp[c\log K + c\log\ell + cK + \Xi\ell^3 - c''K^{2\mathfrak{p}_0}]$ with $c'' > 0$ formed from A_0, A_1 and the constants that enter the definition of the thresholds, is at most a fixed number. For $K \ge c_2\ell^{3/2\mathfrak{p}_0}$ we have $\Xi\ell^3 \le \frac12 c''K^{2\mathfrak{p}_0}$ and $\log\ell \le \frac{2\mathfrak{p}_0}{3c_2}K$, and all conditions hold once also $K \ge c_1$, because $2\mathfrak{p}_0 \ge 2$. □

Lemma 12.10 (One cutoff: conditional expectations)**.** Let $K \in \mathbb{N}$, and let $\mathcal{G}$ be a set of laws on $\mathbb{L}_K$, each invariant under the isometries of $\mathbb{L}_K$, such that $h\nu/\nu(h) \in \mathcal{G}$ for $\nu \in \mathcal{G}$ and every bounded $\mathcal{I}$-measurable $h \ge 0$ with $\nu(h) > 0$. For $\rho \in \mathcal{G}$, $e_\rho = \rho(a_p)$ is one number for all plaquettes p; put $e_+ = \sup_{\rho\in\mathcal{G}} e_\rho$, $e_- = \inf_{\rho\in\mathcal{G}} e_\rho$, and let $\nu \in \mathcal{G}$.

(a) There is one bounded $\mathcal{I}$-measurable e with $\mathsf{e} = E_\nu[a_p|\mathcal{I}]$ ν-almost surely for every plaquette p; $e_- \le \mathsf{e} \le e_+$ ν-almost surely, and $\mathrm{Var}_\nu \mathsf{e} \le \frac14(e_+ - e_-)^2$.

(b) For bounded f of compact support, $E_\nu[\mathcal{O}_K(f) - \nu(\mathcal{O}_K(f))\,|\,\mathcal{I}] = 3a_K^{-4}r_K(f)(\mathsf{e} - e_\nu)$ ν-almost surely.

(c) For every bounded real Borel function F, $E_\nu[F|\mathcal{I}]$ takes ν-almost surely values in $[\inf_{\rho\in\mathcal{G}}\rho(F), \sup_{\rho\in\mathcal{G}}\rho(F)]$.

(d) If ν is ergodic, or $e_\nu \in \{e_-, e_+\}$, then e is ν-almost surely constant.

■

Proof sketch. Let k be bounded and $\mathcal{I}$-measurable, $c = 1 + \sup|k|$ and $h = k + c \ge 1$. The laws $h\nu/\nu(h)$ and ν lie in $\mathcal{G}$, so they are invariant, and the isometries act transitively on the plaquettes; so $\nu(ha_p) = \nu(ha_{p'})$ and $\nu(ca_p) = \nu(ca_{p'})$ for all p, p', and subtracting, $\nu(ka_p) = \nu(ka_{p'})$: all a_p have one conditional expectation given $\mathcal{I}$. (c): if $\{E_\nu[F|\mathcal{I}] > \sup_\rho\rho(F) + \epsilon\}$ had positive mass, conditioning on it would give a member of $\mathcal{G}$ in which the mean of F exceeds $\sup_\rho\rho(F)$; likewise below. At $F = a_p$ this gives the range of e, and a variable with values in an interval of length $e_+ - e_-$ has variance at most $\frac14(e_+ - e_-)^2$. (b): linearity, and $a_K^{-1}\sum_p f(x(p)) = 3a_K^{-4}r_K(f)$. (d): if ν is ergodic, $\mathcal{I}$ is trivial modulo ν; if $\nu(\mathsf{e}) = e_\nu = e_+$, the function $\mathsf{e} \le e_+$ has the mean e_+, so it is constant; likewise at e_-. □

Proposition 12.11 (One cutoff: the plateau for several fields)**.** Let $\mathcal{G}$ be a class at the cutoff K (Section YM3_I:100) that obeys the bound system $\mathsf{C}^g = (\hat{C}_1^g, (\hat{C}_n^g)_{n\ge 2})$ at g, with the properties listed in the paragraph on one cutoff: for $n \ge 2$ and every tuple $X = (f_1, \dots, f_n)$ of type $(\mathsf{d}; R)$, $|S_n^T[\rho](X)| \le \hat{C}_n^g(\mathsf{d}; R) \prod_i \|f_i\|_\sharp$ for $\rho \in \mathcal{G}$, and $|\rho(\mathcal{O}_K(f)) - \rho'(\mathcal{O}_K(f))| \le \hat{C}_1^g(R)\|f\|_\sharp$ for $\rho, \rho' \in \mathcal{G}$ and $f \in C_c^1(\mathbb{R}^3)$ with support in a closed ball of radius R; the numbers $\delta(X; \bar{R})$ and $\mathsf{N}(X; \bar{R})$ are those of the plateau theorem, the second formed with this bound system. Let $\nu \in \mathcal{G}$. Let X, Y be separated, $\bar{R} \in a_K\mathbb{N}$ with all supports in $(-\bar{R}, \bar{R})^3$, $a_K \le 2\min\{\delta(X; \bar{R}), \delta(Y; \bar{R})\}$, $v \in a_K\mathbb{Z}^3$ and $|v| \ge 4\sqrt{3}\bar{R}$. Then

$$\big|\sigma_\nu(X \cdot \tau_v Y) - \sigma_\nu(X)\sigma_\nu(Y) - \mathrm{Cov}_\nu(E_\nu[X_K^\nu|\mathcal{I}], E_\nu[Y_K^\nu|\mathcal{I}])\big| \le \mathfrak{k}_0[\mathsf{N}(X; \bar{R})\mathsf{N}(Y; \bar{R})]^{1/2}\bar{R}/|v|,$$
$$\big|\mathrm{Cov}_\nu(E_\nu[X_K^\nu|\mathcal{I}], E_\nu[Y_K^\nu|\mathcal{I}])\big| \le \tfrac{1}{4}\,\mathrm{osc}_{\mathcal{G}}(X_K^\nu)\,\mathrm{osc}_{\mathcal{G}}(Y_K^\nu), \qquad \mathrm{osc}_{\mathcal{G}}(F) = \sup_{\rho\in\mathcal{G}} \rho(F) - \inf_{\rho\in\mathcal{G}} \rho(F);$$

and $\mathrm{osc}_{\mathcal{G}}((f)_K^\nu) = 3a_K^{-4}|r_K(f)|(e_+ - e_-)$. ∎

Proof sketch. Clause (b) of the plateau theorem at ν compares $\sigma_\nu(X \cdot \tau_v Y)$ with

$$\nu(E_\nu[X_K^\nu|\mathcal{I}]E_\nu[Y_K^\nu|\mathcal{I}]),$$

which is $\sigma_\nu(X)\sigma_\nu(Y)$ plus the covariance, as $\nu(E_\nu[A|\mathcal{I}]) = \nu(A)$. By Lem. 12.10(c) each conditional expectation lies almost surely in an interval of length $\mathrm{osc}_{\mathcal{G}}$, so its variance is at most $\frac{1}{4}\mathrm{osc}_{\mathcal{G}}^2$; then the Cauchy–Schwarz inequality. At $X = (f)$, $\rho((f)_K^\nu) = 3a_K^{-4}r_K(f)(e_\rho - e_\nu)$. □

Lemma 12.12 (Recentring and clustering)**.**

(a) For laws ρ, ν, ν' on $\mathbb{L}_K$ and $X = (f_1, \dots, f_n)$,

$$\rho(X_K^{\nu'}) = \sum_{A\subset\{1,\dots,n\}} \prod_{i\in A} [\nu(\mathcal{O}_K(f_i)) - \nu'(\mathcal{O}_K(f_i))] \\ \rho((X_{A^c})_K^\nu). \tag{12.vii}$$

(b) Let $0 < g \le \gamma$ and $B_n^g(\mathsf{d}; R) = \sum_\pi \prod_{A\in\pi} \hat{C}_{|A|}^g(\mathsf{d}; R)$, the sum over the partitions of $\{1, \dots, n\}$ into blocks of at least two elements ($B_0^g = 1$, $B_1^g = 0$). The set $\mathcal{E}^g$ of the symmetric functionals Φ with $|\Phi(X)| \le B_n^g(\mathsf{d}; R)\prod_i \|f_i\|_\sharp$ for every X of type $(\mathsf{d}; R)$ is convex, and compact and metrisable for pointwise convergence; it contains σ_ν for $(K, \beta) \in \mathfrak{d}(g)$ and $\nu \in \mathcal{G}_{K,\beta}$.

(c) Let T be a compact metrisable set of symmetric functionals that are real-valued on real tuples, with pointwise convergence, such that for all separated X, Y there are $r_{X,Y}, c_{X,Y} < \infty$ with $\sup_{\psi\in\mathsf{T}} |\psi(X \cdot \tau_v Y) - \psi(X)\psi(Y)| \le c_{X,Y}/|v|$ for $|v| \ge r_{X,Y}$. Then for every Borel probability measure μ on T, $(\mathrm{supp}\,\mu, \mu, (\psi)_{\psi\in\mathrm{supp}\,\mu})$ is a canonical decomposition of $\mathrm{bar}(\mu)$, and $\mu \mapsto \mathrm{bar}(\mu)$ is affine, continuous and injective.

(d) Let $0 < g \le \gamma$, $(K, \beta) \in \mathfrak{d}(g)$, $\nu^1, \nu^2 \in \mathcal{G}_{K,\beta}$, $0 \le t \le 1$, $\nu = t\nu^1 + (1-t)\nu^2$, X of n entries and of type $(\mathsf{d}; R)$, and $\eta = a_K^{-4}J(\beta)(2R+1)^3$. Then

$$|\sigma_\nu(X) - t\sigma_{\nu^1}(X) - (1-t)\sigma_{\nu^2}(X)| \le \sum_{k=1}^n \binom{n}{k} \eta^k B_{n-k}^g(\mathsf{d}; R) \prod_i \|f_i\|_\sharp,$$

and at $n = 2$ exactly

$$\sigma_\nu((f,h)) = t\sigma_{\nu^1}((f,h)) + (1-t)\sigma_{\nu^2}((f,h)) + 9t(1-t)a_K^{-8}r_K(f)r_K(h)(e_{\nu^1} - e_{\nu^2})^2.$$

With (C), $\eta \le (2R+1)^3 g^{8+2c} c_*(c) L^{-cK}$ for $0 \le c < \frac{1}{180}$: the defect of convexity of $\nu \mapsto \sigma_\nu$ at one cutoff falls like a power of the spacing.

∎

Proof sketch. (a) Write $\mathcal{O}_K(f_i)-\nu'(\mathcal{O}_K(f_i)) = [\mathcal{O}_K(f_i)-\nu(\mathcal{O}_K(f_i))]+[\nu-\nu'](\mathcal{O}_K(f_i))$ and multiply out. (b) A centred moment $\sigma_\nu(X)$ is the sum, over the partitions without singletons, of the products of the cumulants of the blocks, which the family of bounds at g bounds; the defining conditions are inequalities and equalities at finitely many tuples, so they pass to convex combinations and to pointwise limits, and Tychonoff's theorem gives compactness. Approximating each entry of a tuple in $\|\cdot\|_\sharp$ by functions from a countable family supported in finite unions of rational cubes, and using multilinearity with the bound, one finds a countable set of tuples on which two members of $\mathcal{E}^g$ agree only if they are equal, and on which pointwise convergence implies convergence at every tuple; so $\mathcal{E}^g$ is metrisable. (c) By induction on q, the uniform clustering shows that $\int\prod_{i\le q}\psi(X^i)\,\mathrm{d}\mu$ is the iterated plateau of $\mathrm{bar}(\mu)$ at $(X^1,\dots,X^q)$: this is (Can4); the other conditions are plain.

So $\mathrm{bar}(\mu)$ determines the integrals of the products of the functions $\psi\mapsto\psi(X)$, which are real-valued and separate the points of T; their products and constants form a real unital algebra, so by the Stone–Weierstrass theorem it determines μ. The barycentre is affine in μ, and continuous since the integrands are continuous. (d) $\sigma_\nu(X) = t\nu^1(X_K^\nu)+(1-t)\nu^2(X_K^\nu)$, and by eq. 12.vii with $\rho=\nu^i$, $\nu'=\nu$ and the centring ν^i, $\nu^i(X_K^\nu)-\sigma_{\nu^i}(X)$ is a sum over $A\neq\emptyset$ of products of $|A|$ one-point differences, each at most $a_K^{-4}J(\beta)(2R+a_K)^3\|f\|_\infty\le\eta\|f\|_\sharp$ by (C), times $\sigma_{\nu^i}(X_{A^c})$, which is at most $B^g_{n-|A|}(\mathsf{d};R)\prod_{A^c}\|f_i\|_\sharp$ by (b).

At $n=2$, $\mathrm{Cov}_\nu(F,H) = t\,\mathrm{Cov}_{\nu^1}(F,H)+(1-t)\,\mathrm{Cov}_{\nu^2}(F,H)+t(1-t)[\nu^1-\nu^2](F)\,[\nu^1-\nu^2](H)$ for $F=\mathcal{O}_K(f)$, $H=\mathcal{O}_K(h)$, and $[\nu^1-\nu^2](\mathcal{O}_K(f)) = 3a_K^{-4}r_K(f)(e_{\nu^1}-e_{\nu^2})$. □

Remark 12.13 (The one-point differences and the convexity of the limits). In the proof of (G)(1) the vanishing of the one-point differences, (D)(1), makes the limits of the mixtures the mixtures of the limits; the identity at $n=2$ in (d) shows, for data, what can happen without it. Let, along a sequence of pairs, the class at the j-th pair be the segment $\{s\nu_j^+ + (1-s)\nu_j^- : 0\le s\le 1\}$ between two ergodic laws $\nu_j^\pm$ whose one-point functions differ by an amount that tends to $c\int f$ with $c\neq 0$, and let $\sigma_{\nu_j^\pm}\to\sigma^\pm$. Then the mixtures $\nu_j^s = s\nu_j^+ + (1-s)\nu_j^-$ have

$$\sigma_{\nu_j^s}((f,h)) \to \sigma_s((f,h)) = s\sigma^+((f,h)) + (1-s)\sigma^-((f,h)) + s(1-s)c^2\int f\int h.$$

If moreover $(f,h)\mapsto\sigma^+((f,h))-\sigma^-((f,h))$ is not a multiple of $(f,h)\mapsto\int f\int h$, the limits σ_s, $0\le s\le 1$, form an arc and not a segment, and the set of the limits along the sequence, which is $\{\sigma_s : 0\le s\le 1\}$, does not contain the midpoint of σ^+ and σ^-: it is not convex. For the model, (D)(1) excludes such sequences among the periodic Gibbs states at first-passage data.

Proposition 12.14 (What a rate on the tori gives)**.** Let (S$_1$), (S$_2$) and r_1 be as in (A)(2); no lowered choice is given. Suppose that for $i=1,2$ there are numbers $C_i\ge 1$, $c_{1,i}\ge 0$, integers $m_{1,i}\ge m_{0,i}$ and, for $m\ge m_{1,i}$, $\ell = 2L_i^m$ and $b\in\mathsf{l}_i$, real numbers $S^{(\ell,b)}$ such that for all integers $K\ge C_i\log\ell$

$$|S^T_{2;K,\ell,L_i^K b}(f_1,f_2) - S^{(\ell,b)}| \le C_i\ell^{c_{1,i}}L_i^{-K/C_i}\|f_1\|_\sharp\|f_2\|_\sharp \qquad \text{at the one pair } f_1=\mathsf{f}^{\langle\ell\rangle},\ f_2=1-\mathsf{h}^{\langle\ell\rangle}.$$

Then the oscillation hypothesis of Prop. 12.3(b) holds for $m\ge m_{1,i}$, with $\mathsf{a}_i = C_i\log 2$, $A_i = C_i\max\{\lambda_i,c_{1,i}\}$ and $\omega_i = 2^{1+c_{1,i}}\cdot 4800 b_{+,i}^5 M_i^2 L_i^5 C_i$, and the two bounds of (B)(1) hold with the power $\beta^{-1/2A}$, $A=\max_i A_i$, beyond some β_1. More generally, any oscillation bound for the far parts with thresholds $\mathsf{a}_i + A_i m$ on all sides $m\ge m_{1,i}$ gives $\beta^4 J(\beta)\le C\beta^{-1/2\max_i A_i}$ for large β. ■

Proof sketch. Index i dropped. For $m\ge m_1$ and integers $K,K'\ge \mathsf{a}+Am$ we have $K\ge C\log 2 + C\lambda m = C\log\ell$, the same for K', and $\ell^{c_1}L^{-K/C} = 2^{c_1}L^{c_1 m - K/C}\le 2^{c_1}$, since $K/C\ge c_1 m$. So the two torus functions differ by at most $2C2^{c_1}\|f_1\|_\sharp\|f_2\|_\sharp$; with $b^5\varepsilon^{-3}\le b_+^5 L^3$ and $\|f_1\|_\sharp,\|f_2\|_\sharp\le 40\sqrt{3}ML$, $\sup_{\log\mathsf{l}}|y_{m,K}-y_{m,K'}|\le\omega$. Enter in Prop. 12.3(b) as in Step 1 of the proof of Thm. AJ below, with $m_i^0 = m_{1,i}$; the second sentence is the same argument with the given thresholds. □

Proposition 12.15 (A threshold that does not grow with the side gives $J = 0$ on a half line)**.** Let (S_1), (S_2) and r_1 be as in (A)(2). Suppose that there are an integer K_* and numbers ω_i such that $\sup_{\log \mathsf{I}_i} |y^{(i)}_{m,K} - y^{(i)}_{m,K'}| \le \omega_i$ for all integers $K, K' \ge K_*$ and all $m \ge m_{0,i}$, $i = 1, 2$. Then there is β_* with $J(\beta) = 0$ for $\beta \ge \beta_*$, and on (β_*, ∞) the measure $\beta^4[P''(d\beta) - \hat{n}\,\mathrm{d}\beta]$ has a bounded density. In terms of one lattice: with $Y_N(\beta) = \beta^5[P''_N(\beta) - \hat{n}(\beta)]$, Prop. 12.6(d) puts $Y_N(\beta)$ within C^{ref} of $y_{m,K}(\log b)$ for $N = 2L^{m+K}$, $\beta = L^K b$, $b \in \mathsf{I}$, $m \ge m_0$. Given lowered choices, the oscillation bound of Step 5 of the proof of Thm. AJ gives $|Y_{L^k N}(L^k\beta) - Y_N(\beta)| \le \omega_A + 2C^{\mathrm{ref}}$ for all $k \ge 0$ at these N, β with $N/\beta \le 2b_+^{-1-1/A} L^{-k^{\mathrm{rt}}/A} \beta^{1/A}$; the hypothesis here asks such a bound at all of them. ■

Proof sketch. Enter in Prop. 12.3(c) with the data of Step 1 of the proof of Thm. AJ and with $\max\{K_*, k_1, k_2\}$ in the place of K_*: $\epsilon^{(i)}_m = c_{\mathrm{P},i} L_i^{-m}$ has the lower bound 0, and the hypothesis on Y^0 holds at $m^0 = m_{0,1}$ with $k^0 = \max\{K_*, k_1\}$, since $|y^{(1)}_{m_{0,1},K}| \le |y^{(1)}_{m_{0,1},k^0}| + \omega_1$ for $K \ge k^0$. Since P is the pressure, $J_P = J$. For the last sentence: $L^k N = 2L^{m+K+k}$ and $L^k\beta = L^{K+k}b$, and $K \ge k^{\mathrm{rt}} + Am$ reads $N/\beta \le 2b^{-1-1/A} L^{-k^{\mathrm{rt}}/A} \beta^{1/A}$. □

Proposition 12.16 (The inequality of two boxes at one pair, and along a label)**.** Let $a > 0$ and $\beta > 0$. For the periodic boxes $\underline{n} = (n_1, n_2, n_3)$, $n_\mu \ge 3$, of the lattice $a\mathbb{Z}^3$ at the coupling β let $F(\underline{n}) = \log Z(\underline{n})$, $|\underline{n}| = n_1 n_2 n_3$, $\Delta(\underline{n}) = F(\underline{n}) - |\underline{n}|P(\beta)$, write $F(N^3)$ and $\Delta(N^3)$ for the cube of side N, and $\mathcal{E}(l) = 2F(l, 2l, 2l) - F(2l, 2l, 2l)$.

(a) $\Delta(\underline{n}) \ge 0$; $\Delta(\underline{n}')/|\underline{n}'| \le \Delta(\underline{n})/|\underline{n}|$ for $\underline{n}' \ge \underline{n}$ in each component; and $\mathcal{E}(N) \le 8\Delta(N^3) - \Delta((2N)^3) \le 8\Delta(N^3)$ for $N \ge 3$.

(b) If $\mathcal{E}(l) \le x_0$ for one integer $l \ge 3$ and one number $0 < x_0 \le 0.069$, then, with $\theta = 12.9x_0 \le 0.8901$: $\mathcal{E}(2l) \le 12.9\,\mathcal{E}(l)^2$; $\Delta(N'^3) < 0.13736\,\theta^{N'/2l}$ for every $N' \ge 2l$; and $\mathcal{E}(l') \le 0.743\,\theta^{l'/2l}$ for every $l' \ge l$.

(c) If $\mathcal{E}(l) \le 0.069$ for one integer $l \ge 3$, the lattice $a\mathbb{Z}^3$ has exactly one periodic Gibbs state at β; it is a pure phase, and $J(\beta) = 0$.

(d) If the lattice has two periodic Gibbs states at β, then $\mathcal{E}(l) > 0.069$ for every $l \ge 3$, $\Delta(N^3) > 0.0086$ for every $N \ge 3$, and $\liminf_{j\to\infty} \Delta((2L^j)^3) \ge \log 2$ for every odd integer $L \ge 13$.

(e) At every $\beta > 0$, either $\mathcal{E}(l) > 0.069$ for every $l \ge 3$, or $\mathcal{E}(l') \to 0$ exponentially in l'.

(f) In a setting suppose $\mathrm{BI}[b, \ell_*, K_*; x_0]$, let $\underline{K} = \max\{K_*, 1\}$, $\theta = 12.9x_0$, and let ν_K be the one periodic Gibbs state of (c) at $a = a_K$, $\beta = L^K b$. Let $n \ge 2$, and let $X = (f_1, \dots, f_n)$ be real Lipschitz functions, none of them 0, with pairwise disjoint supports in $(-\frac{\ell''}{4}, \frac{\ell''}{4})^3$ for a side $\ell'' = 2L^{m''} \ge \ell_*$, $m'' \in \mathbb{N}$. There is $C(X) < \infty$, formed from X, ℓ'' and b and from no cutoff and no further side, such that for all $K \ge \underline{K}$ and all sides $\ell' = 2L^{m'} \ge \ell''$

$$|S^T_n[\nu_K](X) - S^T_{n;K,\ell',L^K b}(X^{\langle \ell' \rangle})| \le C(X)\,\theta^{\ell'/4\ell_*}.$$

■

Proof sketch. (a) Fix a direction of a box as the time and the two other sides, $n_\nu \ge 3$, the section Σ, and write $F(t, \Sigma)$ and $Z(t, \Sigma)$ for the box with $t \ge 3$ layers. On the Hilbert space of the configurations of the bonds of one layer (Section YM3_I:85) let C_Σ be the crossing operator of Section YM3_I:85, written K there, and D_Σ the multiplication by $e^{-\beta A_\Sigma}$, the weight of the plaquettes of one layer. C_Σ is positive (Prop. YM3_I:85.1), and it is of trace class: the series the rank-one expansion in the proof of Prop. YM3_I:85.1 in the proof of that clause writes it as $\sum_\alpha c_\alpha |\chi_\alpha\rangle\langle\chi_\alpha|$ with $\|\chi_\alpha\| \le 1$ and $c_\alpha \ge 0$, $\sum_\alpha c_\alpha < \infty$ the rank-one expansion in the proof of Prop. YM3_I:85.1.

So the transfer operator of one layer,

$$T_\Sigma := \mathsf{C}_\Sigma^{1/2} \mathsf{D}_\Sigma \mathsf{C}_\Sigma^{1/2},$$

is positive, as

$$\langle\psi, T_\Sigma\psi\rangle = \int e^{-\beta A_\Sigma}|\mathsf{C}_\Sigma^{1/2}\psi|^2\,\mathrm{d}V,$$

and of trace class; like C_Σ and D_Σ it does not depend on the number of layers (Prop. YM3_I:85.2). Integrating the variables of the bonds between two layers turns the weight of a slab into a chain of kernels: the slab operator of the function 1 on $\vartheta \geq 2$ layers is $\mathsf{D}_\Sigma(\mathsf{C}_\Sigma\mathsf{D}_\Sigma)^{\vartheta-1}$, so that

$$\mathfrak{Y}[1] = \mathsf{C}_\Sigma^{1/2}\mathsf{D}_\Sigma(\mathsf{C}_\Sigma\mathsf{D}_\Sigma)^{\vartheta-1}\mathsf{C}_\Sigma^{1/2} = T_\Sigma^\vartheta$$

(equation YM3_I:(1744)).

The trace formula, Prop. YM3_I:85.2 with the function 1 on two slabs of 2 and $t-2$ layers, gives $Z(t,\Sigma) = \operatorname{Tr} T_\Sigma^t$ for $t \geq 4$; at $t=3$ the integral over the three layers is $\operatorname{Tr}(\mathsf{C}_\Sigma\mathsf{D}_\Sigma)^3$, by the Hilbert–Schmidt kernel identity $\operatorname{Tr}(AB) = \iint A(y,x)B(x,y)\,\mathrm{d}x\,\mathrm{d}y$ used in the proof of Prop. YM3_I:85.2, applied to the operators $(\mathsf{C}_\Sigma\mathsf{D}_\Sigma)^2$ and $\mathsf{C}_\Sigma\mathsf{D}_\Sigma$, which have bounded kernels, and this is $\operatorname{Tr} T_\Sigma^3$ by the cyclicity of the trace. With the eigenvalues $\mu_i \geq 0$ of the positive compact operator T_Σ, counted with their multiplicities, the trace of a power is the sum of the powers of the eigenvalues, by computing the trace in an orthonormal eigenbasis, and

$$e^{F(t,\Sigma)} = Z(t,\Sigma) = \operatorname{Tr} T_\Sigma^t = \sum_i \mu_i^t \qquad \text{for every integer } t \geq 3,$$

with numbers $\mu_i \geq 0$ that do not depend on t, odd t and $t=3$ included. By the spectral argument in the proof of Prop. YM3_I:86.5, $(\operatorname{Tr} T_\Sigma^t)^{1/t}$ does not increase in t; so $F/|\underline{n}|$ does not increase when a side grows, and as $F(N^3)/N^3 \to P$ (Lem. 12.1(a)), $P = \inf F/|\underline{n}|$, and $\Delta \geq 0$ with the monotonicity per site. In particular $F(N,2N,2N) \leq 4F(N^3)$, so $\mathcal{E}(N) \leq 8F(N^3) - F((2N)^3) = 8\Delta(N^3) - \Delta((2N)^3)$. (b) *One direction.* Fix a direction of a box as the time and the two other sides, the section Σ, and write $F(t,\Sigma)$ for the box with $t \geq 3$ layers. By (a), with $\lambda_0 > 0$ the largest of the numbers μ_i and $\mathsf{y}_j \in [0,1]$ the quotients μ_i/λ_0 of the others, $F(t,\Sigma) = t\log\lambda_0 + \log(1+\mathsf{r}_t)$ with $\mathsf{r}_t = \sum_j \mathsf{y}_j^t$. So r_t does not increase in t, $\mathsf{r}_{2t} \leq \mathsf{r}_t^2$, and

$$\mathsf{d}(t) := 2F(t,\Sigma) - F(2t,\Sigma) = 2\log(1+\mathsf{r}_t) - \log(1+\mathsf{r}_{2t}) \;\geq\; \log(1+\mathsf{r}_t) \;\geq 0.$$

Let $\mathsf{d}(t) \leq x < \log 2$, $\eta = e^x - 1 < 1$ and $\rho(\eta) = \eta/(1+\sqrt{1-\eta^2})$. Then $\mathsf{r}_t \leq \eta$, and $1 + 2\mathsf{r}_t/(1+\mathsf{r}_t^2) \leq (1+\mathsf{r}_t)^2/(1+\mathsf{r}_{2t}) = e^{\mathsf{d}(t)} \leq 1+\eta$; as $r \mapsto 2r/(1+r^2)$ increases on $[0,1]$ and takes the value η at $r = \rho(\eta)$, we get $\mathsf{r}_t \leq \rho(\eta)$, and so

$$\mathsf{d}(2t) \leq 2\log(1+\mathsf{r}_{2t}) \leq 2\log(1+\mathsf{r}_t^2) \leq \psi(x) := 2\log(1+\rho(e^x-1)^2).$$

The function ψ increases on $[0,\log 2)$, and, as $(e^y-1)/y$ increases, $\psi(y) \leq 2\kappa(y_0)^2y^2$ for $0 \leq y \leq y_0 < \log 2$, with $\kappa(y_0) = \frac{e^{y_0}-1}{y_0}\big(1+\sqrt{1-(e^{y_0}-1)^2}\big)^{-1}$; and $\kappa(0.069) \leq 0.51832$, $\kappa(0.14056) \leq 0.53995$, $\kappa(0.29264) \leq 0.59869$.

Doubling. For a box $\underline{n}$ and a direction μ let $\mathsf{d}^{(\mu)}(\underline{n})$ be $2F(\underline{n})$ minus F of the box with n_μ doubled. By the above, read in the direction μ, $\mathsf{d}^{(\mu)} \geq 0$, and $\mathsf{d}^{(\mu)}$ of the box with n_μ doubled is at most $\psi(\mathsf{d}^{(\mu)}(\underline{n}))$ when $\mathsf{d}^{(\mu)}(\underline{n}) < \log 2$. Let $\epsilon = \mathcal{E}(l) \leq 0.069$. As F does not depend on the order of the three sides, $\epsilon = \mathsf{d}^{(1)}(l,2l,2l) = \mathsf{d}^{(2)}(2l,l,2l) = \mathsf{d}^{(3)}(2l,2l,l)$. Then:

(i) $\mathsf{a} := \mathsf{d}^{(2)}(2l,2l,2l) \leq \psi(\epsilon) \leq 2\kappa(0.069)^2\epsilon^2 \leq 0.53732\,\epsilon^2 \leq 0.03708\,\epsilon$;

(ii) since $F(2l,4l,l) \leq 2F(2l,2l,l)$ and $F(2l,4l,2l) = 2F(2l,2l,2l) - \mathsf{a}$, we have $\mathsf{d}^{(3)}(2l,4l,l) \leq 2\,\mathsf{d}^{(3)}(2l,2l,l) + \mathsf{a} = 2\epsilon + \mathsf{a} \leq 2.03708\,\epsilon \leq 0.14056$, so $\mathsf{b} := \mathsf{d}^{(3)}(2l,4l,2l) \leq \psi(2\epsilon+\mathsf{a}) \leq 2.41970\,\epsilon^2 \leq 0.16696\,\epsilon$;

(iii) since $F(l,4l,4l) \leq 4F(l,2l,2l)$ and $F(2l,4l,4l) = 2F(2l,4l,2l) - \mathsf{b} = 4F(2l,2l,2l) - 2\mathsf{a} - \mathsf{b}$, we have $\mathsf{c} := \mathsf{d}^{(1)}(l,4l,4l) \leq 4\epsilon + 2\mathsf{a} + \mathsf{b} \leq 4.24112\,\epsilon \leq 0.29264 < \log 2$;

(iv) $\mathcal{E}(2l) = \mathsf{d}^{(1)}(2l,4l,4l) \leq \psi(\mathsf{c}) \leq 2\cdot 0.59869^2 \cdot 4.24112^2\,\epsilon^2 \leq 12.8943\,\epsilon^2$.

So $\mathcal{E}(2l) \le 12.9\,\mathcal{E}(l)^2$ when $\mathcal{E}(l) \le 0.069$. As $12.9 \cdot 0.069 = 0.8901 < 1$, induction shows that $\epsilon_k = \mathcal{E}(2^k l)$ does not increase in k and that $12.9\,\epsilon_{k+1} \le (12.9\,\epsilon_k)^2$; so $12.9\,\epsilon_k \le (12.9\,\epsilon_0)^{2^k} \le \theta^{2^k}$, that is, $\mathcal{E}(2^k l) \le x_k := \theta^{2^k}/12.9 \le x_0 \le 0.069$.

Cubes. Put $D(N) = 8F(N^3) - F((2N)^3) = 8\Delta(N^3) - \Delta((2N)^3)$. At $n = 2^k l$, with a and b formed at n in the place of l, the three brackets of

$$D(2n) = \big[8F((2n)^3) - 4F(2n,4n,2n)\big] + \big[4F(2n,4n,2n) - 2F(2n,4n,4n)\big] + \big[2F(2n,4n,4n) - F((4n)^3)\big]$$

are $4\mathsf{a}$, $2\mathsf{b}$ and $\mathcal{E}(2n)$; so $D(2n) \le (2.14928 + 4.83940 + 12.8943)\,\epsilon_k^2 \le 20\,\epsilon_k^2$. By (a) the density $F/|\underline{n}|$ decreases to P along the cubes of the sides $2^s M$, $s \to \infty$, so that $\Delta(M^3) = \sum_{s\ge 0} 8^{-(s+1)} D(2^s M)$. At $M = 2^j l$, $j \ge 1$, every $D(2^s M)$ is at most $20\,\epsilon_{s+j-1}^2 \le 20\,\epsilon_{j-1}^2$, and

$$\Delta(M^3) \le \tfrac{20}{7}\,\epsilon_{j-1}^2 \le \tfrac{20}{7\cdot 12.9^2}\,\theta^{2^j} \le 0.017170\,\theta^{M/l}.$$

In particular $\Delta((2n)^3) \le \frac{20}{7}\,\mathcal{E}(n)^2 \le \frac{20}{7}x_k^2$ at $n = 2^k l$. With $\mathcal{E}(N) \le 8\Delta(N^3)$ from (a) and $\frac{20}{7}\cdot 64 < 182.9$ this gives $\Delta((2N)^3) \le 183\,\Delta(N^3)^2$ when $\Delta(N^3) \le 0.008625$, for then $\mathcal{E}(N) \le 0.069$; if moreover $\Delta(N^3) > 0$, the hypothesis of (b) holds at $l = N$ with $x_0 = 8\Delta(N^3)$, so with $\theta = 103.2\,\Delta(N^3)$, as $12.9 \cdot 8 = 103.2$. For $N' \ge 2l$ let $j \ge 1$ with $M = 2^j l \le N' < 2M$; by (a), $\Delta(N'^3) \le (N'/M)^3 \Delta(M^3) < 8 \cdot 0.017170\,\theta^{M/l} \le 0.13736\,\theta^{N'/2l}$, since $M/l > N'/2l$ and $0 < \theta < 1$.

All sides $l' \ge l$. At $n = 2^k l$, the identity $\mathcal{E}(n) = 2\Delta(n,2n,2n) - \Delta((2n)^3)$ and the two bounds $\mathcal{E}(n) \le x_k$, $\Delta((2n)^3) \le \frac{20}{7}x_k^2$ give $\Delta(n,2n,2n) \le \frac{x_k}{2} + \frac{10}{7}x_k^2$. Let $2^k l \le l' < 2^{k+1} l$. As $\Delta \ge 0$, and by (a) applied to the boxes $(2^k l, 2^{k+1} l, 2^{k+1} l) \le (l', 2l', 2l')$, whose volumes have the ratio $(l'/2^k l)^3 < 8$,

$$\mathcal{E}(l') \le 2\Delta(l',2l',2l') \le 16\,\Delta(2^k l, 2^{k+1} l, 2^{k+1} l) \le 8x_k + \tfrac{160}{7}x_k^2 \le \Big(\frac{8}{12.9} + \frac{160\cdot 0.8901}{7\cdot 12.9^2}\Big)\theta^{2^k} \le 0.743\,\theta^{2^k},$$

using $x_k^2 = \theta^{2^{k+1}}/12.9^2 \le 0.8901\,\theta^{2^k}/12.9^2$; and $\theta^{2^k} \le \theta^{l'/2l}$, since $2^k > l'/2l$. (c) Let L be an odd integer ≥ 13, an allowed blocking factor, and $K \in \mathbb{N}$. Wilson's action reads the lattice only as a graph, so the dilation $x \mapsto (a_K/a)x$ carries the periodic boxes and the periodic Gibbs states of $a\mathbb{Z}^3$ at β onto those of $\mathbb{L}_K$, with the same partition functions, and we take $a = a_K$. The cube with $(2L^j)^3$ sites, $j > K$, is then the torus T_ℓ, $\ell = 2L^{j-K}$, at the spacing a_K; we put $p_\ell = \ell^{-3}\log Z_{K,\ell,\beta}$. The domination theorem (Prop. YM3_I:91.9) bounds the number n of pairwise mutually singular periodic Gibbs states by $\liminf_j e^{\Delta((2L^j)^3)}$. Indeed, let Ψ be a function of the cell $[-\frac{\ell}{4}, \frac{\ell}{4}]^3$ of the board of $\mathbb{L}_K$ with the side $\frac{\ell}{2}$, half the side of T_ℓ, and the offset 0, and let ν be a periodic Gibbs state. Prop. YM3_I:91.9 gives $|\nu(\Psi)| \le \liminf_{\ell'\to\infty} \|\Psi\|_{\mathrm{ch},\ell'}$, and the comparison of two sides, [7, Cor. B.1](e), gives $\|\Psi\|_{\mathrm{ch},\ell'} \le \|\Psi\|_{\mathrm{ch},\ell}\, e^{(\ell/2)^3(p_\ell - p_{\ell'})}$ for $\ell' \ge \ell$, where $(\ell/2)^3(p_\ell - p_{\ell'}) \to \frac{1}{8}\Delta$, Δ the excess of the cube with ℓ/a_K sites a side, as $p_{\ell'} \to a_K^{-3}P$; so

$$|\nu(\Psi)| \le e^{\Delta/8}\,\|\Psi\|_{\mathrm{ch},\ell},$$

and $\|\Psi\|_{\mathrm{ch},\ell}^8$ is the mean, in the torus measure $\tilde{\mu}$ of T_ℓ, of the product of the eight mirror copies of Ψ (equation [7, eq. 189]). Mutually singular laws $\nu_1, \dots, \nu_n$ have pairwise disjoint carriers, which local events approximate: for $\epsilon > 0$ there are pairwise disjoint local events A_i with $\nu_i(A_i) \ge 1 - n\epsilon$, and for large ℓ they are events of the bonds of the cell. At their indicators the last bound reads $\nu_i(A_i) \le e^{\Delta/8}\tilde{\mu}(E_i)^{1/8}$, where E_i is the intersection of the eight mirror images of A_i; the E_i are pairwise disjoint, so $n(1-n\epsilon)^8 \le e^{\Delta}$. By (b) the right member tends to 1 as $\ell \to \infty$; as ϵ is arbitrary, $n \le 1$. Every periodic Gibbs state is an integral of ergodic periodic Gibbs states, and two different ergodic ones are mutually singular; so there is one, it is extreme, and $J(\beta) = 0$ by Lem. 12.1(b). (d) The first statement is (c); the second follows from (a), as $8 \cdot 0.008625 = 0.069$; the third from the bound of (c) with $n = 2$, at every odd $L \ge 13$, two different ergodic periodic Gibbs states being

mutually singular. (e) is (b) and (c). (f) Let G be a product of centred $\mathcal{O}_K(f_i)$, $i \in I \subset \{1,\dots,n\}$, centred at the means on $T_{\ell''}$. Its chessboard seminorm $\|G\|_{\mathrm{ch},\ell''}$, on the board of $T_{\ell''}$ whose cell holds the supports, has as its eighth power a centred moment, on the fixed torus $T_{\ell''}$, of $8|I|$ observables with pairwise disjoint supports, the mirror copies of the $\mathcal{O}_K(f_i)$; it is a polynomial in cumulants of orders ≥ 2, each of which converges as $K \to \infty$ by Thm. YM3_I:AS; so it is bounded in K. Let μ' be the measure of $T_{\ell'}$ at the spacing a_K, read on $\mathbb{L}_K$, $Q' = e^{\Delta'/8}$ with Δ' the excess of the cube with ℓ'/a_K sites a side, $Q'' = e^{\Delta''/8}$ likewise at ℓ'', $p_\ell = \ell^{-3} \log Z_{K,\ell,L^K b}$ as in the proof of (c), and $\mathsf{g} = Q'' \|G\|_{\mathrm{ch},\ell''}$. Then

$$|\nu_K(G) - \mu'(G)| \leq 10.7\,[Q'(Q'-1)]^{1/2}\,\mathsf{g} \qquad \text{if } \Delta' \leq 1.81. \tag{12.viii}$$

Four facts enter. (i) For a real function Ψ of the cell $[-\frac{\ell'}{4}, \frac{\ell'}{4}]^3$ of the board of $\mathbb{L}_K$ with the side $\frac{\ell'}{2}$ and the offset 0, the coarse board, $|\nu_K(\Psi)| \leq Q' \|\Psi\|_{\mathrm{ch},\ell'}$, by Prop. YM3_I:91.9 and [7, Cor. B.1](e), as in the proof of (c); G is such a function (the plaquette-support argument in [7, §B.1]). (ii) The board with the side $\frac{\ell''}{2}$ and the offset 0, the fine board, has the cell $[-\frac{\ell''}{4}, \frac{\ell''}{4}]^3$, which holds the supports. As $\ell'/\ell'' = L^{m'-m''}$ is odd, the coarse board is coarser than the fine one, and each of its mirror maps carries the cells of the fine board onto cells of the fine board, with the same parity (the proof of Lem. YM3_I:104.1).

So the mirror copies $G^{[k]}$ of G in the eight cells k of the coarse board of $T_{\ell'}$ are mirror copies of G in eight different cells of a period box of the fine board, and the chessboard estimate on the fine board of $T_{\ell'}$ ([7, Cor. B.1](b), (c), which rests on Prop. YM3_I:86.3) gives $|\mu'(\prod_{k\in S} G^{[k]})| \leq \|G\|_{\mathrm{f}}^{|S|}$ for every set S of these cells, $\|G\|_{\mathrm{f}}$ being the seminorm of G at the side ℓ' on the fine board. (iii) By [7, Cor. B.1](e) on the fine board, at the sides $\ell' \geq \ell''$, and as (a) gives $\Delta \geq 0$ at the cube with ℓ'/a_K sites a side, that is, $p_{\ell'} \geq a_K^{-3} P$: $\|G\|_{\mathrm{f}} \leq \|G\|_{\mathrm{ch},\ell''}\, e^{(\ell''/2)^3 (p_{\ell''} - p_{\ell'})} \leq \mathsf{g}$. (iv) μ' is invariant under the isometries of $T_{\ell'}$, so $\mu'(G^{[k]}) = \mu'(G)$, and $|\mu'(G)| \leq \|G\|_{\mathrm{f}} \leq \mathsf{g}$.

Now let t be real, $\Psi = 1 + tG$, $x = |t|\mathsf{g}$ and $R(x) = (1+x)^8 - 1 - 8x$. By equation [7, eq. 189] on the coarse board, (ii) and (iv),

$$\|\Psi\|_{\mathrm{ch},\ell'}^8 = \mu'\Big(\prod_k (1 + tG^{[k]})\Big) = 1 + 8t\mu'(G) + \sum_{|S|\geq 2} t^{|S|} \mu'\Big(\prod_{k\in S} G^{[k]}\Big) \leq 1 + 8t\mu'(G) + R(x),$$

so $\|\Psi\|_{\mathrm{ch},\ell'} \leq 1 + t\mu'(G) + \frac{1}{8}R(x)$, as $(1+y)^{1/8} \leq 1 + \frac{y}{8}$ for $y \geq -1$; and $1 + t\nu_K(G) = \nu_K(\Psi) \leq Q' \|\Psi\|_{\mathrm{ch},\ell'}$ by (i). Hence $t(\nu_K(G) - \mu'(G)) \leq (Q'-1)(1+x) + \frac{1}{8}Q'R(x)$ for both signs of t, with $Q' \geq 1$ by (a). At $\mathsf{g} = 0$ this gives $\nu_K(G) = \mu'(G)$; at $\mathsf{g} > 0$

$$|\nu_K(G) - \mu'(G)| \leq \mathsf{g}\Big[\frac{(Q'-1)(1+x)}{x} + \frac{Q'R(x)}{8x}\Big] \qquad \text{for every } x > 0.$$

For $Q' > 1$ and $x^2 = (Q'-1)/Q'$ the bracket is $[Q'(Q'-1)]^{1/2}(1 + x + R(x)/8x^2)$, and $1 + x + R(x)/8x^2 \leq 10.7$ for $x \leq 0.45$, that is, for $\Delta' \leq 1.81$; for $Q' = 1$ let $x \to 0$. This is eq. 12.viii. By (b) at $l = l_K = \ell_*/2a_K$, $\Delta' < 0.13736\,\theta^{\ell'/\ell_*}$ and $\Delta'' < 0.13736$; so $Q'' < 1.02$, $[Q'(Q'-1)]^{1/2} \leq 0.14\,\theta^{\ell'/2\ell_*}$, and eq. 12.viii gives $|\nu_K(G) - \mu'(G)| \leq 2\,\theta^{\ell'/2\ell_*} \|G\|_{\mathrm{ch},\ell''}$. A cumulant of order $n \geq 2$ is one polynomial in the means $\mathfrak{s}(G_I)$, $I \subset \{1,\dots,n\}$, for both laws $\mathfrak{s} \in \{\nu_K, \mu'\}$; by (iv) and the last bound these means are bounded in K and ℓ', and as $\theta^{\ell'/2\ell_*} \leq \theta^{\ell'/4\ell_*}$ this gives the display. □

12.1.6. *Proof of Thm. AJ. Step 1: (A)(1) and (A)(2).* Take two settings at one radius with $L_2 = L_1 + 2$ (Lem. 12.7) and $r_1 > 0$ with $13 r_1 b_{+,i} \leq 1$. We enter in Prop. 12.3: $\beta_0 = \max_i L_i b_{-,i}$; $C_{\mathrm{low}} = 871 C_1^{\mathrm{ref}}$; $\lambda_i = \log L_i$, so $\lambda_1/\lambda_2 \notin \mathbb{Q}$ by Lem. 12.7; $\alpha_i = \log b_{-,i}$ and $\varkappa_i = \frac{1}{2}[\log(b_{+,i}/b_{-,i}) - \lambda_i] > 0$, so that $\mathsf{J}_i = \log \mathsf{I}_i$; k_i the least integer ≥ 1 with $L_i^{k_i} b_{-,i} > \beta_0$; $m_i^0 = m_{0,i}$; $P_{m,K}^{(i)} = P_N$ and $n_{m,K}^{(i)} = n_{\eta_K}(\mu_{N,\beta})$ with $N = 2L_i^{m+K}$ and $\eta_K = L_i^{K-1}$. By Prop. 12.6(b)–(d), eq. 12.iv holds with

$\epsilon_m^{(i)} = c_{\mathrm{P},i} L_i^{-m}$ and $C_N = \max_i C_i^{\mathrm{ref}}$, eq. 12.iii holds on (β_0, ∞), and by eq. 12.vi the function $y_{m,K}^{(i)}$ of Prop. 12.3 is the far part of the setting (S_i). By the convergence on each torus (Thm. YM3_I:AS at ς_i; the supports of $\mathsf{f}^{\langle\ell\rangle}$ and $1-\mathsf{h}^{\langle\ell\rangle}$ have no common point), for each m the far part converges as $K \to \infty$, uniformly on J_i; so there are non-decreasing integers $K^{(i)}(m) \ge k_i$, the maxima over $m_{0,i} \le m' \le m$ of the thresholds at the single sides, with

$$\sup_{\mathsf{J}_i} |y_{m,K}^{(i)} - y_{m,K'}^{(i)}| \le 1 \qquad (K, K' \ge K^{(i)}(m)). \tag{12.ix}$$

At $m^0 = m_{0,1}$ and $k^0 = K^{(1)}(m^0)$ this gives $Y^0 = \sup_{\mathsf{J}_1} |y^{(1)}_{m^0,k^0}| + 1 < \infty$, the far part at one cutoff being bounded on J_1. Prop. 12.3(a) with $\omega_i = 1$, large m_i and $\bar K_i = K^{(i)}(m_i)$ gives, with A_0 independent of m_1, m_2 and for $0 \le \delta \le 1$,

$$\limsup_{s\to\infty} e^{4s} J(e^s) \le (\mathsf{A}_0 + 134) \max_i (c_{\mathrm{P},i} L_i^{-m_i})^{1/2}, \quad \limsup_{s\to\infty} |\mathbb{M}|([s, s+\delta]) \le (2\mathsf{A}_0 + C_{\mathrm{low}} + 134)\delta + (\mathsf{A}_0 + 134) \max_i (c_{\mathrm{P},i} L_i^{-m_i})^{1/2}.$$

Letting $m_1, m_2 \to \infty$ gives (A)(1), and (A)(2) with $C = 2\mathsf{A}_0 + C_{\mathrm{low}} + 134$, first for $\delta \le 1$ and then for all $\delta > 0$ by cutting $[s, s+\delta]$ into intervals of length at most 1. The equivalent form of (A)(1) is Lem. 12.1(b).

Step 2: (A)(3). Let the two settings have one value of $\mathfrak{p}_0$; such pairs exist with $\mathfrak{p}_0 \ge a$ by Lem. 12.7. The pair $(\mathsf{f}^{\langle\ell\rangle}, 1-\mathsf{h}^{\langle\ell\rangle})$ has the spread ε and norms $\|\cdot\|_\sharp \le 40\sqrt{3}ML$ (values in $[0,1]$, Lipschitz constants at most $\sqrt{3}L$), and $b^5\varepsilon^{-3} \le b_+^5 L^3$. By the rate of the convergence on each torus and Lem. 12.9, in each setting and for the sides with $M\mathcal{R}(b_+) \le L^m$,

$$\sup_{\log \mathsf{I}} |y_{m,K} - y_{m,K'}| \le 2b_+^5 L^3 (40\sqrt{3}ML)^2 C' \ell^9 K^{\mathfrak{p}_2} L^{-K/10} \le 1 \qquad \text{for } K, K' \ge c(1+\ell^{3/2\mathfrak{p}_0}),$$

with the rate constant $C' = \mathsf{C}_2'(\mathsf{I})$ of Thm. YM3_I:AS at $n = 2$ and $s = 0$, which is admitted as $b_- \ge \gamma^{-2}$ and $M\mathcal{R}(b_+) \le L^m$, and a number c independent of m, because $\log \ell \le \frac{2\mathfrak{p}_0}{3}\ell^{3/2\mathfrak{p}_0}$. We may therefore take $K^{(i)}(m) \le c(1 + (2L_i^m)^{3/2\mathfrak{p}_0})$ in Step 1. At a large s let m_i be the largest integer with $\alpha_i + 2\varkappa_i + \lambda_i K^{(i)}(m_i) \le s$; then $s < c'(1 + L_i^{3(m_i+1)/2\mathfrak{p}_0})$, so $L_i^{-m_i} \le c'' s^{-2\mathfrak{p}_0/3}$, and Prop. 12.3(a) gives $e^{4s}J(e^s) \le Cs^{-\mathfrak{p}_0/3}$ and $|\mathbb{M}|([s,s+\delta]) \le (2\mathsf{A}_0 + C_{\mathrm{low}} + 134)\delta + Cs^{-\mathfrak{p}_0/3}$ for large s; the second is at most $C_{\mathrm{A}}\delta$ when $C_{\mathrm{A}} \ge \max\{2\mathsf{A}_0 + C_{\mathrm{low}} + 135, C\}$ and $C_{\mathrm{A}} s^{-\mathfrak{p}_0/3} \le \delta$. On bounded sets of $\beta \ge 3$ use $J \le 6$. For a real q_0 choose $\mathfrak{p}_0 \ge 3q_0$; as $\log\beta \ge 1$ for $\beta \ge 3$, $(\log\beta)^{-\mathfrak{p}_0/3} \le (\log\beta)^{-q_0}$. In any setting, for $(K,\beta) \in \mathfrak{d}(g)$ we have $b = a_K\beta \ge g^{-2} \ge 4$, so $\log\beta \ge K\lambda$ and $a_K^{-4} J(\beta) = b^{-4}\beta^4 J(\beta) \le g^8 C (K\lambda)^{-q_0}$.

Step 3: (A)(4). In one of the two settings let $b \in \mathsf{I}$, $a = a_K$, $\beta = L^K b$. By Prop. 12.6(c) at $m = m_0$, $0 \le a^{-3}[P_{N_0}(\beta) - P(\beta)] \le 3C_{\mathrm{v}}$, which is $-3C_{\mathrm{v}} \le u_K \le 0$. u_K is the difference of a convex and a smooth function of b; $u_K'(b\pm) = a^{-4}[P'(\beta\pm) - P'_{N_0}(\beta)]$, and as measures on (b_-, b_+), by eq. 12.vi at $m = m_0$,

$$u_K''(db) = b^{-4}\mathbb{M}_K(db) + \vartheta_K(b)\,\mathrm{d}b, \qquad \vartheta_K = -b^{-5}[\beta^5(n_{\eta_K}(\mu_{N_0,\beta}) - \hat n(\beta)) + y_{m_0,K}(\log b)],$$

where $\mathbb{M}_K$ is the image of $\mathbb{M}$ under $s \mapsto e^{s-K\lambda}$. The far parts at m_0 are bounded at each cutoff and converge by eq. 12.ix, so $Y_0^* = \sup_{K\ge1}\sup_{\log\mathsf{I}} |y_{m_0,K}| < \infty$, and by Prop. 12.6(d), $|\vartheta_K| \le b_-^{-5}(C^{\mathrm{ref}} + Y_0^*)$. By the last assertion of Prop. 12.6(d), in the setting at hand, $P''(d\beta) \ge [\hat n(\beta) - 871 C^{\mathrm{ref}}\beta^{-5}]d\beta$ on (Lb_-, ∞), which contains $L^K(b_-, b_+)$ for every $K \ge 1$; so $\mathbb{M} \ge -871 C^{\mathrm{ref}} ds$ on $(\log(Lb_-), \infty)$, and, as $ds = db/b$, $b^{-4}\mathbb{M}_K(db) \ge -871 C^{\mathrm{ref}} b_-^{-5} db$ on (b_-, b_+). So $u_K'' \ge -C_{\mathrm{sc}}\,\mathrm{d}b$ with $C_{\mathrm{sc}} = b_-^{-5}(871C^{\mathrm{ref}} + C^{\mathrm{ref}} + Y_0^*)$, for every $K \ge 1$, the numbers C^{ref}, b_- and Y_0^* being those of the setting at hand.

As $\log(b+h) - \log b \le h/b_-$, (A)(2) gives $\limsup_K \sup\{|u_K''|([b, b+h]) : b_- < b,\ b+h < b_+\} \le C_u h$ for every $h > 0$, with $C_u = b_-^{-5}(C + C^{\mathrm{ref}} + Y_0^*)$, the upper limit of (A)(2) being over the real

$s \to \infty$ and so uniform in b. Let $I' = [\alpha, \alpha'] \subset (b_-, b_+)$. By the last bound and a covering, $\sup_{K \geq K_1} |u_K''|(I') < \infty$ for some K_1; the mean of u_K' over I' is $(u_K(\alpha') - u_K(\alpha))/|I'|$, of modulus at most $3C_{\mathrm{v}}/|I'|$, and $u_K'(\cdot\pm)$ differs from it by at most $|u_K''|(I')$; so the u_K, $K \geq K_1$, are bounded and have one Lipschitz constant on I'. The theorem of Arzelà and Ascoli and a diagonal sequence give the subsequence.

For $b, b+h \in I'$, $|u_K'(b\pm) - h^{-1}(u_K(b+h) - u_K(b))| \leq |u_K''|([b, b+h])$; so along the subsequence $\limsup_{K,K'} \sup_{I'} |u_K'(\cdot\pm) - u_{K'}'(\cdot\pm)| \leq 2C_u h$ for every $h > 0$: the one-sided derivatives converge uniformly on I' to a function w, $u(b') - u(b) = \int_b^{b'} w$, and $|w(b') - w(b)| \leq \limsup_K |u_K''|([b, b']) \leq C_u|b' - b|$; so $u' = w$. The other properties pass to uniform limits. Last, by Lem. 12.1(b) on $\mathbb{L}_K$, $\nu(\mathcal{O}_K(f)) = 3a^{-4}e_\nu r_K(f)$ with $-3e_\nu \in [P'(\beta-), P'(\beta+)]$; the torus measure $\tilde{\mu} = \tilde{\mu}_{K,\ell_0,\beta}$ is invariant under the isometries of $\mathbb{L}_K$, so $\tilde{\mu}(\mathcal{O}_K(f)) = 3a^{-4}e_* r_K(f)$ with $-3e_* = P_{N_0}'(\beta)$; subtract.

Step 4: (E). (i) Let first $b' \geq \max\{\gamma^{-2}, e^{\mathsf{Y}^\natural}\}$ and $M\mathcal{R}(b'') \leq L^m$. Discarding (Prop. YM3_I: 115.2, its lost-mass and cumulant bounds) holds with any number ϵ_{R} that bounds the lost mass, $\sum_{h: Z_t(h) \neq \emptyset} \mathsf{t}(h) \leq \epsilon_{\mathrm{R}} Z$, in the place of the bound on the lost mass that its statement takes from the definitions preceding Prop. YM3_I:115.2: if $\epsilon_{\mathrm{R}} \leq \frac{1}{2}$, then the generating function T of the kept histories is entire, $T(0) \geq \frac{1}{2}Z$, and

$$|S^T_{n;K,\ell,\beta}(\underline{f}) - \partial_z^n \log T(0)| \leq 4\, \mathsf{op}_n \epsilon_{\mathrm{R}} \prod_i 6\ell^3 L^{4K} \|f_i\|_\infty.$$

Indeed, with the bare density ρ_0 and the weight $\mathfrak{w}_t^{\mathrm{sf}} \geq 0$ of the definitions preceding Prop. YM3_I: 115.2, $\Lambda(F) = \int F \rho_0 \, \mathrm{d}U$ and $\Lambda'(F) = \int F\, \mathfrak{w}_t^{\mathrm{sf}} \rho_0 \, \mathrm{d}U$ are positive functionals on the gauge-invariant bounded Borel functions, $\Lambda - \Lambda'$ is positive by Prop. YM3_I:115.2, its positive-domination inequality, $\Lambda(1) = Z$, and $T(z) = \Lambda'(e^{\sum_i z_i \mathcal{O}_K(f_i)})$. The proof of equation YM3_I:(2190) first bounds the lost mass by the sum of the histories with $Z_t(h) \neq \emptyset$, before estimating that sum. It therefore gives $\Lambda(1) - \Lambda'(1) \leq \epsilon_{\mathrm{R}} \Lambda(1)$; and the statement on two nearby positive functionals and their cumulants ([7, Lem. K.2], with $|\mathcal{O}_K(f_i)| \leq 6\ell^3 L^{4K} \|f_i\|_\infty$) gives $T(0) = \Lambda'(1) \geq \frac{1}{2}Z$ and the display, the cumulants of Λ and of Λ' being $S^T_{n;K,\ell,\beta}(\underline{f})$ and $\partial_z^n \log T(0)$.

These are the last two steps of the proof of Prop. YM3_I:115.2, made with ϵ_{R} in the place of that bound. The proof of Thm. YM3_I:AR (the comparison of the cutoffs K and $K+1$, with its three lemmas) uses of the bound ϵ_{R} only these two facts, that it bounds the lost mass and that it is at most $\frac{1}{2}$. Its error terms are of three kinds: terms exponentially small in K times $N_{[d]} = \ell^3 L^{3d}$ at depths $d \leq K/9 + 1$; the term $8\, \mathsf{op}_n \epsilon_{\mathrm{R}} \prod_i 6\ell^3 L^{4(K+1)} \|f_i\|_\infty$; and, for $n \geq 2$ and spread d, a contact term at most $c\, \ell^3 L^{3u} e^{-\mathsf{d} L^u / M}$, with the depth $u = \lceil K/180(3n+4) \rceil$.

The bound on the lost mass with which the thresholds of Thm. YM3_I:AS are formed carries a factor $e^{\mathrm{const}\, \ell^3}$. We take instead $\epsilon_{\mathrm{R}} = \mathsf{R}_{t_1}$ of Lem. 12.8 at the level $t_1 = K - u$: since $\log(L^u b) \geq u\lambda \geq K\lambda/180(3n+4)$ and $2\mathfrak{p}_0 \geq 2$, $\mathsf{R}_{t_1} \leq c\, \ell^3 L^{3u} e^{-c'' K^{2\mathfrak{p}_0}}$, which beats the factor ℓ^3 and the products of the crude bound. The most demanding of the remaining conditions is $\ell^3 L^{-K/18} \leq c$. Hence there is an integer $k_\flat$, free of m, ℓ and K, such that for $K \geq k_\flat + 54m$ all conditions hold and

$$|S_{K+1} - S_K| \leq C_n \ell^{3(n+1)} K^{\mathsf{p}_n} L^{-K/10} \Pi.$$

Summing over the cutoffs, with $C_n \sum_{k \geq K} k^{\mathsf{p}_n} L^{-k/10} \leq C_n' K^{\mathsf{p}_n} L^{-K/10}$, which defines C_n', gives (a) with $k_\flat$ and C_n' in this range. For $K \leq k_\flat + 54m$ the crude bound and $L^{4n(k_\flat + 54m)} \leq L^{4nk_\flat} \ell^{216n}$ give $|S_K| \leq \mathsf{op}_n 6^n L^{4nk_\flat} \ell^{219n} \Pi$, and for larger K (a) compares S_K with $S_{k_\flat + 54m}$; this gives (b) with $C_n^{\mathrm{pol}} = \mathsf{op}_n 6^n L^{4nk_\flat} + C_n' \sup_k k^{\mathsf{p}_n} L^{-k/10}$, as $3(n+1) \leq 219n$. (ii) Let $\jmath \geq 0$ be the least integer with $L^\jmath \geq \max\{\gamma^{-2}/b', e^{\mathsf{Y}^\natural}/b', b''\}$ and $L^{\jmath/2} \geq M(4r/e)^r$. For $K > \jmath$, Lem. YM3_I:118.2 gives, with $f^{(\jmath)} = f(L^{-\jmath}\cdot)$,

$$S^T_{n;K,\ell,L^K b}(\underline{f}) = L^{\jmath n} S^T_{n;K-\jmath,L^\jmath \ell, L^{K-\jmath}(L^\jmath b)}(\underline{f}^{(\jmath)}) :$$

both members are cumulants of the same variables on the same periodic cube with ℓL^K sites a side, since $\mathcal{O}_K(f) = L^\jmath \mathcal{O}_{K-\jmath}(f^{(\jmath)})$ under the dilation by $L^\jmath$. The dilation keeps the supremum, divides the Lipschitz constant by $L^\jmath$ and multiplies the spread by $L^\jmath$, so $\|f^{(\jmath)}\|_\sharp \le \|f\|_\sharp$. The window $L^\jmath[b', b'']$ lies above both floors; and as $\mathcal{R}(x) < L(\log x)^r$, $\log(L^\jmath b'') \le 2\jmath\lambda$ and $(2\jmath\lambda)^r \le (4r/e)^r L^{\jmath/2}$, the sides $2L^{m+\jmath}$, $m \ge 1$, satisfy $M\mathcal{R}(L^\jmath b'') \le L^{m+\jmath}$. So (i) applies on the dilated data, with numbers $k_{\flat,\jmath}$, $C'_{n,\jmath}$, $C^{\rm pol}_{n,\jmath}$; as $n + 3(n+1) + \frac{1}{10} \le 4n+4$ and $K - \jmath \ge k_{\flat,\jmath} + 54(m+\jmath)$ for $K \ge k_{\flat,\jmath} + 55\jmath + 54m$, this gives (a) with $k^{\rm rt} = k_{\flat,\jmath} + 55\jmath$ and $C^{\rm rt} = L^{\jmath(4n+4)} C'_{n,\jmath}$, and $|S_K| \le L^{220n\jmath} C^{\rm pol}_{n,\jmath} \ell^{219n} \Pi$ for $K > \jmath$.

For $1 \le K \le \jmath$ the crude bound gives $|S_K| \le \mathsf{op}_n 6^n L^{4n\jmath} \ell^{3n} \Pi$; so (b) holds with $C^{\rm pol} = \max\{L^{220n\jmath} C^{\rm pol}_{n,\jmath}, \mathsf{op}_n 6^n L^{4n\jmath}\}$. (c): $A > 30(n+1) \ge 90 > 54$, so (a) applies for $K \ge k^{\rm rt} + Am$, and $m \le K/A$ gives $\ell^{3(n+1)} \le 2^{3(n+1)} L^{3(n+1)K/A}$. (d): $K \ge C \log \ell$ with $C \ge 54/\lambda + k^{\rm rt}/\log 2$ gives $K \ge 54m + k^{\rm rt}$, as $\log \ell \ge \log 2$ and $\log \ell = \log 2 + m\lambda$; with $C \ge 11$, $\frac{1}{10} - \frac{1}{C} \ge \frac{1}{110}$ and $K^{\mathsf{p}_2} L^{-K/110} \le (110\mathsf{p}_2/e\lambda)^{\mathsf{p}_2}$, so (a) gives the first bound, as $\ell^9 \le \ell^{438}$; (b) gives the second.

Step 5: the far part beyond $k^{\rm rt} + Am$. In a setting apply (E)(a) at $n = 2$ to ς and a lowered choice ς', with $[b', b''] = \mathsf{I}$, $\mathsf{d} = \varepsilon$ and the pair of Step 2. With $(40\sqrt{3}ML)^2 = 4800M^2L^2$, for all $m \in \mathbb{N}$, $\upsilon \in \log \mathsf{I}$ and $K' \ge K \ge k^{\rm rt} + 54m$,

$$|y_{m,K'}(\upsilon) - y_{m,K}(\upsilon)| \le c_{\rm y} \ell^9 K^{\mathsf{p}_2} L^{-K/10}, \qquad c_{\rm y} = 4800 b_+^5 M^2 L^5 C^{\rm rt}. \tag{12.x}$$

For real $A > 90$ and $K, K' \ge k^{\rm rt} + Am$ we have $\ell^9 = 2^9 L^{9m} \le 2^9 L^{9K/A}$ and $\frac{1}{10} > \frac{9}{A}$; so

$$\sup_{\log \mathsf{I}} |y_{m,K} - y_{m,K'}| \le \omega_A = 2^9 c_{\rm y} \sup_{k \ge 1} k^{\mathsf{p}_2} L^{-(\frac{1}{10} - \frac{9}{A})k} < \infty.$$

This is the oscillation hypothesis of Prop. 12.3(b) with the slope A; and $\frac{1}{10} > \frac{9}{A}$ holds exactly for $A > 90$. With it, $\sup_K \sup_{\log \mathsf{I}} |y_{m,K}| < \infty$ at each m, the finitely many cutoffs below the threshold being bounded one by one. (B) uses (E) here and nowhere else.

Step 6: (B). Enter in Prop. 12.3(b) with the data of Step 1, $\mathsf{a}_i = k_i^{\rm rt}$, $A_i = A$ and $\omega_i = \omega_{A,i}$ of Step 5 at $(\varsigma_i, \varsigma_i')$. At each single (m, K), by eq. 12.vi and Lem. 12.1(c), $|y_{m,K}| \le \beta^5 (P''_N + |n_{\eta_K}(\mu_{N,\beta})|) \le (L^K b_+)^5 (9N^3 + 1125\eta_K^3)$. This gives (B)(1) with the constants

$$\begin{aligned}
k^0 &= \max\{k_1, \lceil k_1^{\rm rt} + Am_{0,1} \rceil\},\\
Y^0 &\le (L_1^{k^0} b_{+,1})^5 (72 L_1^{3(m_{0,1}+k^0)} + 1125 L_1^{3(k^0-1)}) + \omega_{A,1},\\
\mathsf{M}_1 &= (Y^0 + C_N + C_{\rm low})\varkappa_1 + c_{{\rm P},1} L_1^{-m_{0,1}} (15/\varkappa_1 + 14 + 12\varkappa_1),\\
\mathsf{A}_0 &= 7\mathsf{M}_1/\varkappa_1 + C_{\rm low} + 4C_N + \omega_{A,1} + \omega_{A,2},\\
c_{3,i} &= |\log b_{-,i}| + 2\varkappa_i + \lambda_i (k_i + k_i^{\rm rt} + 1),\\
c_6 &= \max_i c_{{\rm P},i} e^{\lambda_i + c_{3,i}/A},\\
T_i(m) &= \log b_{-,i} + 2\varkappa_i + \lambda_i \max\{k_i, \lceil k_i^{\rm rt} + Am \rceil\},\\
C_{\rm h} &= 2\mathsf{A}_0 + C_{\rm low} + 134,\\
C_{\rm r} &= (\mathsf{A}_0 + 134) c_6^{1/2},\\
\log \beta_1 &= \max\{0, T_1(m_{0,1}), T_2(m_{0,2}), A \log^+(4c_6/\varkappa^2)\},
\end{aligned} \tag{12.xi}$$

since P is the pressure, $J_P = J$, and $|\mathbb{M}|([s, s+\delta])$ is the integral of (B)(1). (2): for $0 < c < \frac{1}{180}$ take $A = \frac{1}{2c} > 90$; then $\beta^{4+c} J(\beta) \le C_{\rm r}$ for $\beta \ge \beta_1$, and $\beta^{4+c} J \le 6 \max\{1, \beta_1\}^{4+c}$ on $[1, \beta_1]$; for $c \le 0$, $\beta^{4+c} J \le \beta^{4+1/182} J$ on $[1, \infty)$. $c_*(c)$ is defined by P alone. (3): $(\log \beta)^q \beta^{-1/182}$ is bounded on $[3, \infty)$. (4): enter in Prop. 12.5 with the data of Step 1. At $i = 1$, Step 5 at $(\varsigma_1, \varsigma_1')$ gives (i) with $\mathsf{a}_1 = k_1^{\rm rt}$, $A_1 = A$ and $\omega_1 = \omega_{A,1}$, and $\sup_{K \ge k_1} \sup |y^{(1)}_{m_{0,1},K}|$ is finite, the cutoffs below the

threshold being finitely many; at $i=2$, eq. 12.ix at ς_2 gives (ii) with $\omega_2=1$ for every $m\ge m_{0,2}$, and $\epsilon^{(2)}_m = c_{\mathrm{P},2}L_2^{-m}\to 0$.

Thm. YM3_I:D is used at ς_1 and ς_1' through (E), and at ς_2 through the convergence on each torus. (5): by eq. 12.v with the data of Step 1 and $G_K = G(\cdot - K\lambda_i)$, which is supported in $\mathsf{J}_i + K\lambda_i \subset [s_3,\infty)$, $s_3 = \log(L_i^K b_{-,i}) \ge \log\beta_1$ (and $K\ge k_i$, as $\beta_1>\beta_0$), $\int G\, y^{(i)}_{m,K} = \langle \mathbb{M}, G_K\rangle - r$ with $|r| \le c_{\mathrm{P},i}L_i^{-m}\mathsf{Q}(G) + C_N\|G\|_1$; by (1), the hypothesis of Prop. 12.3(d) holds with $s_2=\log\beta_1$ and $\epsilon_1(s) = e^{-s/2A}$, and it bounds $|\langle\mathbb{M},G_K\rangle|$ by $(C_{\mathrm{h}}+C_{\mathrm{r}})(\|G\|_1 + (L_i^K b_{-,i})^{-1/2A}\|G'\|_1)$.

Step 7: (C). By the dilation, $\rho^{(K)},\nu^{(K)}\in\mathcal{G}(\beta)$, and $\rho(a_p) = e_{\rho^{(K)}}$; Lem. 12.1(b) gives $3|e_\rho - e_\nu| \le J(\beta)$, and invariance gives $\nu(\mathcal{O}_K(f)) = 3a_K^{-4}e_\nu r_K(f)$. For each of the three orientations the barycentres form a translate of $a_K\mathbb{Z}^3$, of which at most $(2R/a_K+1)^3$ points lie in $[-R,R]^3$; so $|r_K(f)| \le (2R+a_K)^3\|f\|_\infty$. On $\mathfrak{d}(g)$, $b = a_K\beta \ge g^{-2}\ge 4$, so $\beta\ge 1$, and, as $4+c>0$,

$$a_K^{-4}J(\beta) = b^{-4}\beta^4 J(\beta) \le c_*(c)\, b^{-4}\beta^{-c} = c_*(c)\, b^{-4-c}L^{-cK} \le g^{8+2c}c_*(c)L^{-cK}.$$

The setting is used only to form $\mathfrak{d}(g)$; $c_*(c)$ is a number of Wilson's action.

Step 8: (D). (1) On $\mathfrak{d}(g)$, $b\ge g^{-2}$ and $\beta = L^K b \ge L^K g^{-2}$, so $a_K^{-4}J(\beta) = b^{-4}\beta^4 J(\beta) \le g^8 \sup_{\beta'\ge L^K g^{-2}} \beta'^4 J(\beta')$, which tends to 0 by (A)(1); and by (C) with $a_K\le 1$ the difference of the one-point functions is at most $a_K^{-4}J(\beta)(2R+1)^3\|f\|_\infty$. (2) At $X=(f)$ a limit in the frame is $\lim_j[\rho_j(\mathcal{O}_{K_j}(f)) - \nu_j(\mathcal{O}_{K_j}(f))] = 0$. (3) Each component is a limit in the frame along the full sequence, with the approximants $\rho_j = \nu_j(\cdot|B_j)$ (Thm. YM3_I:AP); by (2), $0 = \sigma_\xi((f)) = \mathfrak{c}_\xi\int f$; take $\int f = 1$. As the one-point values of the component vanish, eq. 12.vii with $\rho=\nu'=\rho_j$ and $\nu=\nu_j$ gives $\rho_j(X^{\rho_j}_{K_j}) - \rho_j(X^{\nu_j}_{K_j})\to 0$, every term with $A\ne\emptyset$ having a factor that tends to 0 and a factor that converges; so the Gibbs sequence (K_j,β_j,ρ_j) converges, with the limit σ_ξ.

The component clusters with the numbers C^g, r (Thm. YM3_I:AP); so for its own canonical decomposition $(\mathsf{X}',\varpi',(\sigma'_\zeta))$, (Can4) at $q=2$ gives $\int\sigma'_\zeta(X)^2 d\varpi' = \lim_{|v|\to\infty}\sigma_\xi(X\cdot\tau_v X) = \sigma_\xi(X)^2 = (\int\sigma'_\zeta(X)d\varpi')^2$; so $\zeta\mapsto\sigma'_\zeta(X)$ is constant on the support of ϖ', which is X', and the injectivity in (Can3) makes X' a point. This argument used of (A) only that $\mathfrak{c}_\xi = 0$; conversely, a limit of a convergent Gibbs sequence (K_j,β_j,ρ'_j) has the one-point values $\lim_j \rho'_j((f)^{\rho'_j}_{K_j}) = 0$, so $\mathfrak{c}_\xi = 0$; this is the equivalence. (4) The one-point values vanish; with (Can4) at $q=1$ and the clustering of the components,

$$S_2^\sigma(f,\tau_v h) = \sigma((f)\cdot\tau_v(h)) = \int \sigma_\xi((f)\cdot\tau_v(h))\,\mathrm{d}\varpi, \qquad |\sigma_\xi((f)\cdot\tau_v(h))| \le C^g((f),(h))/|v|.$$

Step 9: (F). (1) By Prop. YM3_I:106.3, $\mathcal{G}_{K,\beta}$ satisfies the hypotheses of Lem. 12.10, and by Lem. 12.1(b) on $\mathbb{L}_K$, $-3e_+ = P'(\beta-)$, $-3e_- = P'(\beta+)$ and $e_+ - e_- = \frac13 J(\beta)$. (2) The clustering inequality of the plateau theorem at $X=(f)$, $Y=(h)$ compares $\sigma_\nu((f)\cdot\tau_v(h)) = S_2^T[\nu](f,\tau_v h)$ with $\nu(E_\nu[(f)^\nu_K|\mathcal{I}]\,E_\nu[(h)^\nu_K|\mathcal{I}])$, which is $9a_K^{-8}r_K(f)r_K(h)\operatorname{Var}_\nu \mathsf{e}$ by Lem. 12.10(b), since $\nu(\mathsf{e}) = e_\nu$; and $\operatorname{Var}_\nu\mathsf{e} \le \frac14(J/3)^2$; the last bound is (C). (3) $\nu_\pm$ exist by Lem. 12.1(b) on $\mathbb{L}_K$. By Lem. 12.10(d), e is almost surely constant in $\nu_\pm$, in every ν with $-3e_\nu$ at an end of the interval, and in a pure phase, which is ergodic. For $\nu = \frac12(\nu_+ + \nu_-)$, which lies in the convex set $\mathcal{G}_{K,\beta}$, and bounded F,F',

$$\operatorname{Cov}_\nu(F,F') = \tfrac12\operatorname{Cov}_{\nu_+}(F,F') + \tfrac12\operatorname{Cov}_{\nu_-}(F,F') + \tfrac14(\nu_+(F)-\nu_-(F))(\nu_+(F')-\nu_-(F')).$$

Take $F=\mathcal{O}_K(f)$ and $F'=\mathcal{O}_K(\tau_v f)$, $v\in a_K\mathbb{Z}^3$, with $0\le f\in C^1_c((-\frac12,\frac12)^3)$ and $r_K(f)>0$, for which (2) applies at $\bar R = 1$. Then $\nu_+(F)-\nu_-(F) = 3a_K^{-4}r_K(f)(e_{\nu_+}-e_{\nu_-}) = -a_K^{-4}J(\beta)r_K(f)$, the same for F', as $r_K(\tau_v f) = r_K(f)$; (2) in the three states, with $|v|\to\infty$, gives $9a_K^{-8}r_K(f)^2\operatorname{Var}_\nu\mathsf{e} = \frac14(a_K^{-4}J(\beta)r_K(f))^2$. (4) Prop. 12.11 with $\mathcal{G}=\mathcal{G}_{K,\beta}$; X^ν_K is a real continuous local function, a polynomial in finitely many a_p, and by Lem. 12.1(b) every $\rho\in\mathcal{G}_{K,\beta}$ has $D^-\bar p^F(0)\le\rho(F)\le D^+\bar p^F(0)$ for such F; at $X=(f)$, $3a_K^{-4}|r_K(f)|(e_+-e_-) = a_K^{-4}J(\beta)|r_K(f)|$.

Step 10: (G) in a window. Let $g^{-2}+\bar B<b<X_g-\bar B$ for some $0<g\le\gamma$. From some index on $a_{K_j}\beta_j\in I(g)$, so $(K_j,\beta_j)\in\mathfrak d(g)$; we drop the first members, which changes no set of limits. Let $\mathsf S_j=\{\sigma_\nu:\nu\in\mathcal G_{K_j,\beta_j}\}\subset\mathcal E^g$ and fix a metric of $\mathcal E^g$.

(1) $\mathsf K=\{\sigma\in\mathcal E^g:\operatorname{dist}(\sigma,\mathsf S_j)\to0\}$, a closed subset of the compact metrisable space $\mathcal E^g$ (Lem. 12.12(b)). Convexity: if $\sigma^i=\lim_j\sigma_{\nu_j^i}$ and $\nu_j=t\nu_j^1+(1-t)\nu_j^2\in\mathcal G_{K_j,\beta_j}$, then

$$\sigma_{\nu_j}(X)=t\,\nu_j^1(X_{K_j}^{\nu_j})+(1-t)\,\nu_j^2(X_{K_j}^{\nu_j}),$$

and by eq. 12.vii with the centrings ν_j^i and ν_j each bracket is at most $a_{K_j}^{-4}J(\beta_j)(2R+1)^3\|f\|_\infty\to0$ by (D)(1), while the moments $\nu_j^i((X_A)_{K_j}^{\nu_j^i})$ converge; so $\sigma_{\nu_j}\to t\sigma^1+(1-t)\sigma^2$.

(2) $(K_j,\beta_j,\nu_j)_j$ is a convergent Gibbs sequence at g; by (D)(3) each component is $\lim_j\sigma_{\nu_j(\cdot|B_j)}$ and has one vacuum. Components cluster at the rate $C^g(X,Y)/|v|$. A member of $\mathsf K$ that clusters, in the sense $\sigma(X\cdot\tau_vX)\to\sigma(X)^2$ for all X, has one vacuum, by (Can4) at $q=2$ as in Step 8; conversely, a member with one vacuum is its own component and clusters at the rate $C^g/|v|$. As C^g and r do not depend on the member, pointwise limits of members of $\mathsf K^\bullet$ cluster: $\mathsf K^\bullet$ is closed in $\mathsf K$, so compact.

(3) Barycentres of measures on $\mathsf K^\bullet$ lie in $\mathsf K$: for finite supports by (1), and in general by Riemann sums, the uniform continuity of $\psi\mapsto\psi(X)$ on the compact $\mathsf K^\bullet$, and the closedness of $\mathsf K$. Onto: for $\sigma\in\mathsf K$ the image of ϖ under $\xi\mapsto\sigma_\xi$ is a measure on $\mathsf K^\bullet$ by (2), with the barycentre σ by (Can4) at $q=1$. Injective: Lem. 12.12(c) with $\mathsf T=\mathsf K^\bullet$. A continuous bijection of the compact $\operatorname{Prob}(\mathsf K^\bullet)$ onto the Hausdorff space $\mathsf K$ is a homeomorphism; by injectivity the image of ϖ is μ_σ; $\xi\mapsto\sigma_\xi$ is continuous and injective on the compact $\mathsf X$, and its image is the support of μ_σ, ϖ having full support. $\dim\ker H_\sigma=\#\mathsf X=\#\operatorname{supp}\mu_\sigma$ by Thm. YM3_I:AP.

(4) One vacuum is $\ker H_\sigma=\mathbb C\Omega_\sigma$ by $\#\mathsf X=\dim\ker H_\sigma$, in each of the six directions. By (3), the extreme points of $\mathsf K$ are the barycentres of the extreme points of $\operatorname{Prob}(\mathsf K^\bullet)$, the point masses: extreme points are the members with one vacuum. The two forms of clustering are (2). *Pure phases.* If $\sigma=\lim_j\sigma_{\omega_j}$ with pure phases ω_j, each ω_j is ergodic, so $E_{\omega_j}[\cdot\,|\mathcal I]$ is almost surely constant, and by the statement on iterated plateaux $\int\sigma_\xi(X)^2d\varpi-\sigma(X)^2=\lim_j\operatorname{Var}_{\omega_j}(E_{\omega_j}[X_{K_j}^{\omega_j}|\mathcal I])=0$ for every X; so $\xi\mapsto\sigma_\xi(X)$ is constant on $\mathsf X$, and $\mathsf X$ is a point.

Conversely, let $\sigma\in\mathsf K^\bullet$ be the limit of σ_{ν_j}. By the variance statement below, $E_{\nu_j}[X_{K_j}^{\nu_j}|\mathcal I]\to\sigma(X)$ in $L^2(\nu_j)$ for every X. Let $(\nu_j^U)_U$ be a regular conditional law of ν_j given $\mathcal I$; by Prop. YM3_I:106.1 its members are, off a ν_j-null set, pure phases in $\mathcal G_{K_j,\beta_j}$, and $\nu_j^U(X_{K_j}^{\nu_j})=E_{\nu_j}[X_{K_j}^{\nu_j}|\mathcal I](U)$ for ν_j-almost every U. Take a countable family $X_1,X_2,\dots$ of tuples on which pointwise convergence in $\mathcal E^g$ implies convergence at every tuple (Lem. 12.12(b)). By Chebyshev's inequality there are, for each j, a point U_j outside the null sets and numbers $\epsilon_j\to0$, $k_j\to\infty$ with $|\nu_j^{U_j}((X_k)_{K_j}^{\nu_j})-\sigma(X_k)|\le\epsilon_j$ for $k\le k_j$.

The pure phases $\omega_j=\nu_j^{U_j}$ then have $\omega_j((X_k)_{K_j}^{\nu_j})\to\sigma(X_k)$ for every k; by eq. 12.vii and (D)(1), $\sigma_{\omega_j}(X_k)\to\sigma(X_k)$; and as $\sigma_{\omega_j}\in\mathcal E^g$, $\sigma_{\omega_j}\to\sigma$. *Variances.* By the statement on iterated plateaux, $\operatorname{Var}_{\nu_j}(E_{\nu_j}[X_{K_j}^{\nu_j}|\mathcal I])\to\int\sigma_\xi(X)^2d\varpi-\sigma(X)^2$, which does not depend on the choice of the ν_j, and which vanishes for every X if and only if $\mathsf X$ is a point. *Faces.* For a subsequence $\mathfrak p'$, $\mathsf K\subset\mathsf K_{\mathfrak p'}$. If $\sigma\in\mathsf K$ is $t\sigma_1+(1-t)\sigma_2$ with $\sigma_i\in\mathsf K_{\mathfrak p'}$, $0<t<1$, the uniqueness of the representing measure on $\operatorname{Lim}^\bullet[b]$ (Lem. 12.12(c), with the compact set $\operatorname{Lim}^\bullet[b]$ of (6)) gives $\mu_\sigma=t\mu_{\sigma_1}+(1-t)\mu_{\sigma_2}$; so the μ_{σ_i} live on $\operatorname{supp}\mu_\sigma\subset\mathsf K^\bullet$, and $\sigma_i\in\mathsf K$ by (3).

For the interlacing $\mathfrak p''$, $\mathsf K_{\mathfrak p''}\subset\mathsf K\cap\mathsf K_{\tilde{\mathfrak p}}$, as $\mathfrak p$ and $\tilde{\mathfrak p}$ are subsequences of $\mathfrak p''$; conversely, two Gibbs sequences that converge to σ along $\mathfrak p$ and along $\tilde{\mathfrak p}$ interlace to one that converges to σ along $\mathfrak p''$. *One point.* Let $\mathsf K\neq\emptyset$. If $\mathsf K$ is a point, $\mathsf K^\bullet$ is not empty, as it holds the components of that point, and lies in $\mathsf K$; if $\mathsf K^\bullet$ is a point, so is $\operatorname{Prob}(\mathsf K^\bullet)$ and hence $\mathsf K$; if $\mathsf K$ is a point, $\mathsf K=\mathsf K^\bullet$; if $\mathsf K=\mathsf K^\bullet$ had two members $\sigma_1\neq\sigma_2$, then $\frac12(\sigma_1+\sigma_2)\in\mathsf K=\mathsf K^\bullet$ would be an extreme point.

By the statement on iterated plateaux, $\mathrm{Cov}_{\nu_j}(E_{\nu_j}[X^{\nu_j}_{K_j}|\mathcal{I}], E_{\nu_j}[Y^{\nu_j}_{K_j}|\mathcal{I}]) \to \int \sigma_\xi(X)\sigma_\xi(Y)d\varpi - \sigma(X)\sigma(Y)$, which vanishes for all X, Y if and only if X is a point; for all convergent choices this says $\mathsf{K} = \mathsf{K}^\bullet$. At single fields, Lem. 12.10(b) and (F)(2) give the displayed bound, with $|r_K(f)| \le (2R+1)^3\|f\|_\infty$, and it tends to 0 by (D)(1).

(6) The pairs $(K, L^K b)$, $K \in \mathbb{N}$, lie in $\mathfrak{d}(g)$; as $\mathcal{E}^g$ is compact, a subsequence of (σ_{ν_K}) converges, so $\mathrm{Lim}[b] \neq \emptyset$. $\mathrm{Lim}[b] = \bigcap_k \overline{\bigcup\{\mathsf{S}_{K,\beta} : K \ge k,\ |a_K\beta - b| \le \frac{1}{k}\}}$, with $\mathsf{S}_{K,\beta} = \{\sigma_\nu : \nu \in \mathcal{G}_{K,\beta}\}$, is compact, and $\mathrm{Lim}^\bullet[b]$ is closed in it by the uniform clustering. A saturated subsequence: with a countable base (U_r) of $\mathcal{E}^g$, pass to nested subsequences along which, for each r, $\mathsf{S}_j \cap U_r$ is either non-empty for all members or empty for all members, and take the diagonal; for it and for each of its subsequences a functional is a limit if and only if each U_r that holds it meets S_j for all large j, so all these sets of limits agree, and they are not empty by compactness. Every $\sigma \in \mathsf{K}_\mathfrak{p}$ lies in $\mathsf{K}_{\mathfrak{p}'}$ for a saturated subsequence $\mathfrak{p}'$, and the extreme points are the members with one vacuum by (4); this gives the two unions. Uniqueness of the representing measure on $\mathrm{Lim}^\bullet[b]$: Lem. 12.12(c) with $\mathsf{T} = \mathrm{Lim}^\bullet[b]$; existence: (3).

The hull. $\mathrm{Lim}^\bullet[b]$ is compact and its members cluster uniformly, with the numbers C^g, r; so by Lem. 12.12(c) the map bar is affine, continuous and injective on the compact convex set $\mathrm{Prob}(\mathrm{Lim}^\bullet[b])$, hence an affine homeomorphism onto its image, a compact convex set. The image contains $\mathrm{Lim}[b]$, by the existence just shown, and it is the closure of the image of the finitely supported measures, which is the convex hull of $\mathrm{Lim}^\bullet[b] \subset \mathrm{Lim}[b]$; so it is $\overline{\mathrm{co}}\,\mathrm{Lim}[b]$. Its extreme points are the images of the point masses, that is, the members of $\mathrm{Lim}^\bullet[b]$. Hence a member of $\mathrm{Lim}[b]$ with one vacuum is an extreme point of $\overline{\mathrm{co}}\,\mathrm{Lim}[b]$, and so of $\mathrm{Lim}[b]$; and a member σ that is an extreme point of $\mathrm{Lim}[b]$ is an extreme point of every $\mathsf{K}_\mathfrak{p}$ that contains it, so it has one vacuum by (4).

Convexity. If $\mathrm{Lim}[b] = \mathsf{K}_\mathfrak{q}$, it is convex by (1). If $\mathrm{Lim}[b]$ is convex, it is compact and convex, so it equals $\overline{\mathrm{co}}\,\mathrm{Lim}[b] = \mathrm{bar}(\mathrm{Prob}(\mathrm{Lim}^\bullet[b]))$. If this equality holds and $\psi_1, \dots, \psi_n \in \mathrm{Lim}^\bullet[b]$, then $\frac{1}{n}\sum_i \psi_i$ lies in some $\mathsf{K}_\mathfrak{p}$; by (3) it is the barycentre of a measure on $\mathsf{K}^\bullet_\mathfrak{p}$, which by the uniqueness on $\mathrm{Lim}^\bullet[b]$ is $\frac{1}{n}\sum_i \delta_{\psi_i}$, so that $\psi_1, \dots, \psi_n \in \mathsf{K}^\bullet_\mathfrak{p}$. If every finite subset of $\mathrm{Lim}^\bullet[b]$ lies in one $\mathsf{K}^\bullet_\mathfrak{p}$, let $\psi_1, \psi_2, \dots$ be dense in $\mathrm{Lim}^\bullet[b]$ and $\psi_1, \dots, \psi_n \in \mathsf{K}^\bullet_{\mathfrak{p}_n}$; for each n there is a pair (K'_n, β'_n) of $\mathfrak{p}_n$ with $K'_n \ge n$ and $|a_{K'_n}\beta'_n - b| \le \frac{1}{n}$ which carries, for each $i \le n$, a state ν with $\mathrm{dist}(\sigma_\nu, \psi_i) \le \frac{1}{n}$.

Along $\mathfrak{q} = (K'_n, \beta'_n)_n$, a sequence of pairs at the label b, every ψ_i is a limit; so the closed set $\mathsf{K}_\mathfrak{q}$ contains $\mathrm{Lim}^\bullet[b]$, and by (3) for $\mathfrak{q}$ it contains $\mathrm{bar}(\mathrm{Prob}(\mathrm{Lim}^\bullet[b])) \supset \mathrm{Lim}[b] \supset \mathsf{K}_\mathfrak{q}$. Finally, a set with one element is convex.

(5) Every $\sigma \in \mathrm{Lim}[b]$ lies in some $\mathsf{K}_\mathfrak{q}$; its components lie in $\mathsf{K}^\bullet_\mathfrak{q} \subset \mathrm{Lim}^\bullet[b]$, and $\sigma = \int \sigma_\xi \,\mathrm{d}\varpi$. If $\mathrm{Lim}^\bullet[b] = \{\tau\}$, all components are τ and $\sigma = \tau$; if $\mathrm{Lim}[b] = \{\sigma\}$, then $\mathrm{Lim}^\bullet[b]$ is not empty by (6) and lies in $\{\sigma\}$. The same lines serve for $\mathrm{Lim}_\mathfrak{q}[b]$, the components of a limit along pairs of $\mathfrak{q}$ being limits along the same pairs, and $\mathrm{Lim}_\mathfrak{q}[b]$ being non-empty by (6) applied to a sequence with pairs in $\mathfrak{q}$. For the third statement: a Gibbs sequence at g with the label b has convergent subsequences (Thm. YM3_I:AP), whose limits lie in $\mathrm{Lim}[b]$; if this is one point, every subsequence has a subsequence that converges to it, so the sequence converges; conversely, interlacing two convergent Gibbs sequences with the label b gives one, which converges, so their limits agree. The one limit then has one vacuum, as $\mathrm{Lim}^\bullet[b] \neq \emptyset$.

The clauses (1)–(4) use of (A) only that $a_{K_j}^{-4} J(\beta_j) \to 0$ along $\mathfrak{p}$. Along $\mathfrak{p} = (K, L^K b)_K$ this holds without (A) for every b outside the countable set $\bigcup_K L^{-K}\{\beta : J(\beta) > 0\}$, as the convex function P is differentiable off a countable set; at such labels these clauses use Thms. YM3_I:D, YM3_I:AJ and YM3_I:AD at the choice of the setting alone.

Step 11: (G) at every label. For $z \in \mathbb{Z}$ put, as in the paragraph on dilations, $X^{[z]} = (f_i(L^z\cdot))_i$ and $\sigma^{[z]}(X) = L^{-z|X|}\sigma(X^{[z]})$. The dilation $x \mapsto L^z x$ carries a law ν on $\mathbb{L}_K$ to a law $\nu^{(z)}$ on $\mathbb{L}_{K-z}$, maps $\mathcal{G}_{K,\beta}$ onto $\mathcal{G}_{K-z,\beta}$ and conditioned states to conditioned states, as it carries the invariant events to the invariant events ([7, Lem. K.1]), and turns $\mathcal{O}_{K-z}(f)$ into $L^{-z}\mathcal{O}_K(f(L^z\cdot))$ ([7, Lem. K.1], applied

to the function $f(L^z\cdot)$); so

$$\sigma_{\nu^{(z)}} = (\sigma_\nu)^{[z]}, \qquad a_{K-z}^{-4} J(\beta) = L^{-4z} a_K^{-4} J(\beta).$$

So $\sigma \mapsto \sigma^{[z]}$ is a linear homeomorphism that carries $\mathsf{K}_\mathfrak{p}$, $\mathrm{Lim}[b]$, $\mathrm{Lim}^\bullet[b]$ onto the same sets at the label $L^z b$ for the pairs $(K_j - z, \beta_j)$, and canonical decompositions to canonical decompositions, component by component, with the same space X. It carries the reconstruction as well. Let (z, g) be a pair for b, that is, $z \in \mathbb{Z}$ and $0 < g \le \gamma$ with $g^{-2} + \bar{B} < L^z b < X_g - \bar{B}$; the pair fixed for b in the paragraph on dilations is one, with $z \ge 0$. For $\sigma \in \mathsf{K}_\mathfrak{p}$, $\sigma^{[z]}$ is then, as in that paragraph, the limit of a convergent Gibbs sequence at g over the pairs $(K_j - z, \beta_j)$ of large index, and Thm. YM3_I: AP gives its reconstruction, with the Hamiltonian $H_{\sigma^{[z]}}$ and the vacuum vector $\Omega_{\sigma^{[z]}}$, in each of the six coordinate directions.

For a coordinate direction u, the reflection θ in the plane $\{x \cdot u = 0\}$ commutes with the dilation, so that $\sigma^{[z]}(\theta X \cdot Y) = L^{-z(|X|+|Y|)}\sigma(\theta X^{[z]} \cdot Y^{[z]})$ for the tuples X, Y in the half space $\{x \cdot u > 0\}$; hence the form $\langle [X], [Y] \rangle_\sigma = \sigma(\theta X \cdot Y)$ is positive together with that of $\sigma^{[z]}$, and $W : [X] \mapsto L^{-z|X|}[X^{[z]}]$ is a unitary map of the space of the reconstruction of $\sigma^{[z]}$ onto the space that this form builds from σ, with $W[\emptyset] = [\emptyset]$. As $(\tau_v X)^{[z]} = \tau_{L^{-z}v} X^{[z]}$, W carries the translations by tu to the translations by $L^{-z}tu$; so, with $\hat{H}_\sigma$ the generator of the semigroup $e^{-t\hat{H}_\sigma}[X] = [\tau_{tu} X]$, $t \ge 0$, in the space that the form builds from σ, $\hat{\Omega}_\sigma = [\emptyset]$ the class of the empty tuple there, and $e^{-tH_{\sigma^{[z]}}}[X] = [\tau_{tu} X]$, $\Omega_{\sigma^{[z]}} = [\emptyset]$ in the reconstruction of $\sigma^{[z]}$,

$$H_{\sigma^{[z]}} = L^{-z}\, W^* \hat{H}_\sigma W, \qquad W\Omega_{\sigma^{[z]}} = \hat{\Omega}_\sigma, \qquad \ker \hat{H}_\sigma = W \ker H_{\sigma^{[z]}},$$

in each of the six directions: the kernels and the vacua correspond. A linear homeomorphism carries convex sets, closed sets, extreme points, faces and one-point sets to the same, and the dilation carries the clustering numbers to dilated ones. For every $b > 0$ there are pairs for b, already with $z \ge 0$, because $L^z b > \gamma^{-2} + \bar{B}$ for large z and the open intervals $(g^{-2} + \bar{B}, X_g - \bar{B})$ overlap one another as g decreases and cover $(\gamma^{-2} + \bar{B}, \infty)$. So W carries the operator $L^z H_{\sigma^{[z]}}$ and the vector $\Omega_{\sigma^{[z]}}$, which are H_σ and Ω_σ of the paragraph on dilations at the pair (z, g) fixed for b, to $\hat{H}_\sigma$ and $\hat{\Omega}_\sigma$, the generator of the translations and the class of the empty tuple in the space that the form builds from σ.

These read no window, so another pair (z, g) gives a unitarily equivalent operator with the corresponding vector, and $\dim\ker H_\sigma = \dim\ker H_{\sigma^{[z]}} = \#\mathsf{X}$, both finite and equal or both infinite. Step 10 at $(L^z b, (K_j - z, \beta_j))$ gives (1)–(6) at $(b, \mathfrak{p})$. In particular the kernel clauses of (3) and (4) pass to $(b, \mathfrak{p})$, with $H_\sigma = L^z H_{\sigma^{[z]}}$ and $\Omega_\sigma = \Omega_{\sigma^{[z]}}$ of the paragraph on dilations at the pair (z, g) fixed for b: the dilation carries μ_σ to $\mu_{\sigma^{[z]}}$, so that $\dim\ker H_\sigma = \dim\ker H_{\sigma^{[z]}} = \#\mathsf{X} = \#\operatorname{supp}\mu_\sigma$, and $\ker H_\sigma = \mathbb{C}\Omega_\sigma$ in one, and then in each, of the six directions if and only if σ has one vacuum.

Step 12: (G)(7). Let $0 < g \le \gamma_W$ and let $\mathfrak{p}$ be a sequence of pairs in $\mathfrak{d}(g)$ with $\mathsf{K}_\mathfrak{p} \ne \emptyset$; $\sigma \in \mathsf{K}_\mathfrak{p}$ is the limit of a convergent Gibbs sequence over $\mathfrak{p}$. By Prop. YM3_I:127.8, applied with $\rho_j = \nu_j$ (see also Cor. YM3_I:127.11), along a convergent Gibbs sequence at g the labels $a_{K_j}\beta_j$ converge, to a number $b \in I^{\mathrm{lab}}(g)$, and the two-point function of the limit, $S_2^\sigma = \lim_j S_2^T[\nu_j]$, has the displayed limit with this b, for every $\vartheta \in (0, 1)$ and every pair of $\mathfrak{T}^1_{\mathbb{R}^3}(\mathsf{d}_0(g, \vartheta); \vartheta)$; this set is not empty by [7, Lem. I.1](c).

The right member of the formula reads σ alone; so the sets $\mathrm{Lim}_{\mathfrak{d}(g)}[b]$ are pairwise disjoint, and every convergent Gibbs sequence at g is over a sequence of pairs with a non-empty set of limits, which has a label. An interlacing of two sequences with different labels has no label, so its set of limits is empty. Suppose that $\sigma'' = t\sigma + (1-t)\sigma'$, $0 < t < 1$, with $\sigma \in \mathrm{Lim}_{\mathfrak{d}(g)}[b]$, $\sigma' \in \mathrm{Lim}_{\mathfrak{d}(g)}[b']$, were the limit of a Gibbs sequence over $\mathfrak{p}''$; then $\mathfrak{p}''$ has a label b''. The sets of the limits with one vacuum at the labels b and b', along sequences in $\mathfrak{d}(g)$, are compact, by the diagonal argument of Step 10 (6), and their members cluster uniformly, with the numbers C^g, r; with the representing measures

μ_σ, $\mu_{\sigma'}$ of (G)(6), carried by the components of σ and of σ', Lem. 12.12(c) makes $(\operatorname{supp}\mu'', \mu'', (\psi))$, $\mu'' = t\mu_\sigma + (1-t)\mu_{\sigma'}$, a canonical decomposition of σ''.

By Cor. YM3_I:127.11, the formula in every component of every canonical decomposition, each component of σ'' has the coefficient b''^{-2}, whereas the members of $\operatorname{supp}\mu_\sigma$, which are the components of σ by (G)(3), have b^{-2}, and those of $\operatorname{supp}\mu_{\sigma'}$ have b'^{-2}; so $b = b'' = b'$. □

Proof of Thm. AK. (1) is Prop. 12.16(c) at $a = a_K$, $\beta = L^K b$ and $l = l_K = L^{K+m_*} \geq L$, as $K + m_* \geq K_* + m_* \geq 1$; that P is differentiable at β is $J(\beta) = 0$. (2) Let first the entries be none 0 and their supports lie in $(-\frac{\ell''}{4}, \frac{\ell''}{4})^3$ for a side $\ell'' = 2L^{m''} \geq \ell_*$, $m'' \in \mathbb{N}$; an entry 0 makes every cumulant 0, and every compact set lies in such a cube. The torus cumulants converge as $K \to \infty$ (Thm. YM3_I:AS), so Prop. 12.16(f) gives $\limsup_K S_n^T[\nu_K](X) - \liminf_K S_n^T[\nu_K](X) \leq 2C(X)\theta^{\ell'/4\ell_*}$ for every side $\ell' \geq \ell''$; let $\ell' \to \infty$. The same bound shows that the limit is within $C(X)\theta^{\ell'/4\ell_*}$ of $\lim_K S^T_{n;K,\ell',L^K b}(X^{\langle \ell' \rangle})$. (3) Since $\mathcal{G}_{K,L^K b} = \{\nu_K\}$, the set $\mathsf{K}_\mathfrak{p}$ has at most one point, the limit of σ_{ν_K} if it exists. For a separated X, $\sigma_{\nu_K}(X)$ is a centred moment, so

$$\sigma_{\nu_K}(X) = \sum_\pi \prod_{A \in \pi} S^T_{|A|}[\nu_K](X_A),$$

the sum over the partitions of $\{1, \dots, |X|\}$ into blocks of at least two elements; each factor converges by (2). So $\mathsf{K}_\mathfrak{p} = \{\sigma\}$; the point of a one-point convex set is extreme, so σ has one vacuum by Thm. AJ(G)(4). The last sentence: $l_K = \ell_*/2a_K$ and $\beta = L^K b$ do not change when (b, ℓ_*, K) is replaced by $(L^z b, L^z \ell_*, K - z)$, and $K \geq K_*$ says $K - z \geq K_* - z$. □

12.1.7. *Scope of the conclusions.* The bounds (A)–(C) are upper bounds; they do not show that J vanishes at any finite β. The numbers 54, 90 and $\frac{1}{180}$ are those of sufficient conditions in the estimates: $54 = 3 \cdot 18$ comes from the condition $\ell^3 L^{-K/18} \leq c$ of Step 4 and stands in the threshold $k^{\mathrm{rt}} + 54m$ of (E), which the slope $A > 90$ of Step 5 covers in (B); $90 = 9 \cdot 10$ comes from the comparison of ℓ^9 with $L^{-K/10}$ in Step 5; and $\frac{1}{180} = \sup_{A>90} \frac{1}{2A}$ comes from Prop. 12.3(b) in Step 6. Prop. 12.3(e) is a statement on data and on the map from the slope to the exponent: for every slope $A > 0$ there are data as in (b), with $A_1 = A_2 = A$, whose atoms are $\beta_k^{-1/2A}$, so that the loss of one half is real for data (Remark 12.4(3)); and the data of (e) at $A = 90$ satisfy the hypothesis of (b) at every slope $A' \geq 90$, a larger threshold being a weaker hypothesis, and have the atoms $\beta_k^{-1/180}$.

By Prop. 12.14 the exponent follows the slope that a rate on the tori gives, and by Prop. 12.15 a threshold that does not grow with the side gives $J = 0$ on a half line. Thm. AK assumes the box inequality and speaks of the couplings $L^K b$ of one label. Clause (G)(5) is an equivalence and asserts neither of its two sides; clause (G)(6) asserts none of its four equivalent statements on the convexity of $\mathrm{Lim}[b]$; and (G)(1)–(4) allow $\mathsf{K}_\mathfrak{p} = \emptyset$ for the given sequence, whereas (G)(6) gives non-empty sets along saturated subsequences. Clause (E) concerns $n \geq 2$ and tuples of positive spread, with a threshold $k^{\mathrm{rt}} + 54m$ that grows with the side, and (A)(4) concerns limit points along subsequences of the cutoffs.

13. Infrared bounds

Let S_2 be the two-point function of the centred action density in a continuum limit through periodic Gibbs states, or in one of its components. Euclidean invariance writes its kernel off the diagonal as $s(|x - y|)$. Large cube averages test the small spectral masses of this field without requiring pointwise control of the underlying connection.

Sketch L (Polynomial infrared decay). The radial kernel has a positive representation

$$s(r) = \frac{1}{4\pi r} \int_0^\infty e^{-mr} \, \mathrm{d}\rho(m), \qquad r > 0,$$

and, at large r and small m, respectively,

$$s(r) = O(r^{-2}), \qquad \rho([0,m]) = O(m), \qquad \rho(\{0\}) = 0.$$

These conclusions exclude a zero-mass atom, while allowing spectral mass arbitrarily close to zero. ■□

Sketch of the argument. Average the lattice action density over a cube of side q, normalizing by its volume. Keep a collar of width T at its boundary, so that sources in distinct cells can be reflected and folded without meeting the walls. Source-pressure bounds and the chessboard estimate give, for the resulting cube means,

$$|\operatorname{Cov}(x_c, x_{c'})| \le C(T/q)^2 \qquad (c \ne c').$$

The constants are uniform in the fine cutoff once its threshold for the selected cube size has been reached.

The source-pressure argument first gives a semiconvex response measure. A concentration of this measure would obstruct the uniform cube estimate. Compare the pressure using a second blocking factor with an incommensurable logarithm, as in the latent-heat argument. This comparison excludes concentration and permits the bound to pass to continuum limits. The vanishing one-point differences remove the plateau that could otherwise come from varying component means.

Reflection positivity gives a positive spectral representation of the radial two-point function. Test it with two reflected large-cube profiles. Their positive kernels compare the cube estimate with the value of s at their separation, yielding $s(r) \le C/r^2$. In the positive integral representation, the part with $0 \le m \le 1/r$ is bounded below by $e^{-1}\rho([0,1/r])/(4\pi r)$. Consequently $\rho([0,m]) \le Cm$ near zero. A zero-mass atom would instead contribute a nonzero multiple of $1/r$, so it is absent.

For separated compact tests this yields a two-point bound of order the inverse square of their separation. Reflection positivity and the corresponding reconstructed vectors also bound a connected function with one insertion moved far away by the inverse distance. These are infrared estimates for the Euclidean-invariant Gibbs limits. General growing-torus limits retain weaker direction-dependent conclusions until their rotation problem is settled.

Nothing in this argument supplies a positive mass threshold below the support of ρ. The stronger exponential decay and spectral gap follow from the finite-size criterion in the last sketch. They should not be inferred from the absence of the atom proved here.

13.1. **Cube means and infrared bounds.** We average the action density over cubes with a boundary collar. Chessboard estimates turn control of the source pressure into bounds on correlations of distinct cube means. Comparing two blocking factors with incommensurable logarithms excludes concentration of the pressure response.

The resulting continuum bounds give an $O(r^{-2})$ radial two-point kernel and at most linear spectral mass near zero. They exclude an atom at zero without giving a positive gap. The growing-torus conclusions use only the positivity and symmetry established for those limits. The stronger exponential conclusions explicitly assume the box criterion.

13.1.1. *Notation and hypotheses.* Lengths are measured on the physical scale, on which the bare lattice spacing is $a_K = L^{-K}$; thus ℓ, M, I, T, q and r are lengths on this scale. A floor is a lower threshold, a window a compact set of labels; a quantity is free of another if it does not depend on it. $|\cdot|$ is the Euclidean norm, $|\cdot|_\infty$ the sup-norm and $\operatorname{dist}_\infty$ its distance (on a torus, the distance of the torus); $[t]_+ = \max\{t, 0\}$; $\operatorname{int} I$ and $|I|$ are the interior and the length of an interval I; Lip is the Lipschitz constant for the Euclidean distance.

Definition 13.1 (Frames; the numbers at a reference coupling)**.**

(1) A *frame* consists of a radius $\varrho > 0$, a common real and sourced choice of constants obtained by the ordered construction in the proof of Prop. YM3_I:69.1 and Section YM3_I:82, and the corresponding Wilson measures and source normalization of Thms. YM3_I:AJ

and YM3_I:AD. The choice retains the source-room condition eq. 2.iv and the array radius [7, eq. 194]. A frame determines an odd integer $L \geq 13$; $M = L^{m'}$ with an integer m'; numbers $B_\sharp \geq 0$ and $\bar{B} = B_\sharp L/(L-1)$; $\gamma \in (0, \frac{1}{2}]$; and the upper stability constant C_S and decay rate $\kappa > \log 9216 + 2$ of Thm. YM3_I:AD. Choose C_{st} and $C_S \geq C_{\mathrm{fp}}^{\mathrm{st}}$ as in [7, eq. 288]. [7, Lem. G.2] gives the lower stability bound with $C_g = C_{\mathrm{st}} + \frac{9}{2}\log X_g$ and the estimate $C_g \leq C_S(1 + \log g^{-2})$. We put $S(\kappa) = 2^{12}(1 - 9216\, e^{-\kappa})^{-1}$, the number of the count of domains.

(2) At a reference coupling $0 < g \leq \gamma$ we put

$$X_g = L(g^{-2} + B_\sharp), \quad I(g) = (g^{-2} + \bar{B},\, X_g - \bar{B}], \quad C_g = C_{\mathrm{st}} + \tfrac{9}{2}\log X_g, \quad B' = 2C_S + C_g,$$

and $I^{\mathrm{lab}}(g) = [g^{-2} + B_\sharp \frac{L-2}{L-1},\, L(g^{-2} + \bar{B})]$. A first-passage datum at g is a pair (K, β) of a cutoff and a bare coupling such that K is the first index at which the running inverse coupling started at β is $\leq X_g$, as in Prop. YM3_I:74.1; $\mathfrak{d}(g)$ is the set of these pairs. $\mathfrak{L}(g) = \{2L^m : m \geq m_{\mathfrak{L}}(g)\}$ is the set of the sides at g of equation YM3_I:(1528), $m_{\mathfrak{L}}(g)$ being the integer YM3_I:(1528) there, which the frame and g fix. The set $\mathfrak{L}(g)$ is closed upward among the sides $2L^m$, and by [7, Lem. G.4] and equation YM3_I:(1533), for $0 < g' \leq g \leq \gamma$, $m_{\mathfrak{L}}(g) \leq m_{\mathfrak{L}}(g')$, so that $\mathfrak{L}(g') \subset \mathfrak{L}(g)$. These are the sides of which Thm. YM3_I:AD and the transfer (M7) speak.

(3) A second frame, whose objects wear a hat, is a *partner* of the first if $\log L/\log \hat{L} \notin \mathbb{Q}$.

■

Definition 13.2 (The lattice, the pressures, the sources)**.**

(1) For an integer $K \geq 0$, $a_K = L^{-K}$ and $\mathbb{L}_K = a_K\mathbb{Z}^3$; for $\ell = 2L^m$, $T_\ell = (\mathbb{R}/\ell\mathbb{Z})^3$ carries the lattice of spacing a_K, with $\mathsf{N} = \ell/a_K$ sites a side. For a plaquette p, $a_p = 1 - \frac{1}{2}\operatorname{tr}\partial U(p) \in [0,2]$, and $x(p)$ is its barycentre. For a function f we put $\mathcal{O}(f) = \mathcal{O}_K(f) = a_K^{-1}\sum_p f(x(p))a_p$ and $\mathrm{Rie}_K(f) = \frac{1}{3}a_K^3\sum_p f(x(p))$, the sums over the plaquettes of $\mathbb{L}_K$ or of the torus.

(2) $Z_{K,\ell,\beta}[w] = \int e^{-\beta\sum_p a_p + \mathcal{O}(w)}\,\mathrm{d}U$, with the Haar measure of mass 1; $Z_{K,\ell,\beta} = Z_{K,\ell,\beta}[0]$ defines the law $\langle\cdot\rangle_{K,\ell,\beta}$. $S^T_{n;K,\ell,\beta}(f_1,\dots,f_n)$ and $S^T_n[\nu](f_1,\dots,f_n)$ are the joint cumulants of $\mathcal{O}(f_1),\dots,\mathcal{O}(f_n)$ under the law of the torus and under a law ν; Cov and Var are covariance and variance. $\mathcal{G}^{\mathrm{per}}_{K,\beta}$ is the set of the periodic Gibbs states on $\mathbb{L}_K$ at β, in the sense of the Definitions of kernels and Gibbs states (Section YM3_I:91).

(3) The label of (K,β) is $b = a_K\beta$. We put $p_{K,\ell}(b) = \ell^{-3}\log Z_{K,\ell,L^K b}$ and $p_{K,\infty}(b) = \inf_\ell p_{K,\ell}(b)$ over the sides $\ell = 2L^m$. For a reference side ℓ': $D_{K,\ell} = p_{K,\ell} - p_{K,\ell'}$, $D_K = p_{K,\infty} - p_{K,\ell'}$, and $\bar{\eta}(\ell) = 3\ell' B'/\ell$. P is the pressure of the Wilson model on $\mathbb{Z}^3$ (Lem. 13.11) and $J(\beta) = P'(\beta+) - P'(\beta-)$. For a function D on an interval such that $D + \frac{1}{2}Cy^2$ is convex, D'' is the signed measure $(D + \frac{1}{2}Cy^2)'' - C\,\mathrm{d}y$; $\Delta_s^2 D(b) = D(b+s) + D(b-s) - 2D(b)$.

(4) A *source* is a complex Lipschitz function w on T_ℓ with $\|w\|_\sharp = \max\{\sup|w|, 40M\operatorname{Lip} w\} < 2\varrho$; this class lies in the class of the sources of Thm. YM3_I:AD. A real number x in the place of a source is the constant function $x\cdot 1$. Since $\mathcal{O}(x\cdot 1) = L^K x\sum_p a_p$, we have the *shift of the label*

$$Z_{K,\ell,L^K b}[w + x] = Z_{K,\ell,L^K(b-x)}[w].$$

(5) For $(K,\beta) \in \mathfrak{d}(g)$ and $\ell \in \mathfrak{L}(g)$, Thm. YM3_I:AD provides the source energies: a number E_ℓ and local terms $e_j(X; w)$, $1 \leq j \leq K$, X a localisation domain of level j with linear size $d_j(X)$. We put $\mathcal{E}_\ell = \sum_{j,X} e_j(X;\cdot)$ and $\widetilde{Z}_\ell(w) = e^{-E_\ell - \mathcal{E}_\ell(w)}Z_{K,\ell,\beta}[w]$; the index ℓ is dropped where one torus is fixed. G is the finite group generated by the translations of T_ℓ by $a_K\mathbb{Z}^3$ and the reflections $x_\mu \mapsto -x_\mu$, acting on sources by $(\mathsf{r}w)(y) = w(\mathsf{r}^{-1}y)$; it contains the reflections at all mirrors of all boards. We put $\mathcal{E}^{\mathrm{av}}(w) = |\mathsf{G}|^{-1}\sum_{\mathsf{r}\in\mathsf{G}}\mathcal{E}(\mathsf{r}w)$ and $\widetilde{Z}^{\mathrm{av}} = e^{-E-\mathcal{E}^{\mathrm{av}}}Z$.

■

Definition 13.3 (Boards, profiles, cube means)**.**

(1) l and l_1 are the least powers of L with $\mathsf{l} \geq 32M$ and $\mathsf{l}_1 \geq 8M + 16$; $T_0 = 640\sqrt{3}M$. The width of the collar is a real number $T \geq \max\{T_0, \mathsf{l}\}$; the side of a board that carries profiles and cube means is a power $q = L^i \geq 2T$; we write $q_i = L^i$.

(2) For a power $\mathsf{j} = L^i$ with $\mathsf{j} \geq La_K$, a *board* of side j with offset $o \in a_K\mathbb{Z}^3$ is the partition of $\mathbb{L}_K$, or of T_ℓ when $\ell \geq 2\mathsf{j}$, into the cubes $\square_c$ of side j centred at the points $o + \mathsf{j}c$, $c \in \mathbb{Z}^3$ (on T_ℓ, c modulo ℓ/j), the *cells*; c is the *index* of the cell. Since j/a_K is odd, the centres are sites and the *mirrors*, the planes half way between neighbouring centres, pass through midpoints of bonds. On T_ℓ there are $N = (\ell/\mathsf{j})^3$ cells, an even number in each direction, and every cell is even or odd, by the parity of $c_1 + c_2 + c_3$. The mirror copy $F^{[c']}$ of a function F of the cell c in the cell c' is its image under the reflections at the mirrors that carry c to c', complex conjugated if c and c' have different parities; $\|F\|_{\mathrm{ch}} = \langle \prod_{c'} F^{[c']} \rangle^{1/N}$ is the chessboard seminorm (equation YM3_I:(1756)). Profiles and cube means are formed on boards of the sides q of (1); boards of the sides l_1 and l enter the proofs as well.

(3) For a cell $\square_c$ of a board of side q, $\mathfrak{P}_T(\square_c)$ is the set of the real C^1 functions f with $0 \leq f \leq 1$, $f = 1$ at sup-distance $\geq T/2$ from $\partial\square_c$, $f = 0$ off $\square_c$ and at sup-distance $\leq T/8$ from $\partial\square_c$, and $\operatorname{Lip} f \leq 16\sqrt{3}/T$; then $\|f\|_\sharp = 1$. Its members are the *profiles* of the cell.

(4) The *cube mean* of the cell c with the profile f is $x_c = q^{-3}\mathcal{O}(f)$.

(5) $\mathsf{c} = \max\{1, 40\sqrt{3}M(16/T + 4/\mathsf{l})\} \in [1, 1 + 5\sqrt{3}]$; $C_w = 96\mathsf{c}(C_S + C_g)/\varrho$, and C_w^-, C_w^+ are the same number with 1, respectively $1 + 5\sqrt{3}$, in the place of c; $C_2 = 72e^4C_w^2$; $C_I' = 16(C_S + C_g)/(9\varrho^2)$; $x_q = 1/(3C_wTq^2)$ and $I_q(b) = [b - x_q, b + x_q]$.

(6) On T_ℓ, $\ell \geq 2q$, for a profile f of a cell c: F_e and F_o are the sums of the mirror copies of f over the cells of the parity of c and over the others, $F = F_e + F_o$, and $1 - F$ are the *walls*. Copies in different cells have supports at sup-distance $\geq T/4$, and $\|F\|_\sharp = \|F_e - F_o\|_\sharp = 1$. For complex z, $w_z = zF_e + \bar{z}F_o$ is the source of the source formula for the source zf of the cell, and $\Lambda(z) = \log \|e^{z\mathcal{O}(f)}\|_{\mathrm{ch}} = N^{-1} \log(Z[w_z]/Z)$.

(7) The model profile is $f_0 = \prod_\mu \varphi_0(x_\mu)$, with φ_0 even, C^1, $0 \leq \varphi_0 \leq 1$, non-increasing in $|s|$, $\varphi_0 = 1$ for $|s| \leq q/2 - 3T/8$, $\varphi_0 = 0$ for $|s| \geq q/2 - T/4$, $|\varphi_0'| \leq 16/T$; then $f_0(\cdot - y) \in \mathfrak{P}_T(\square_c)$ whenever $|y - y_c|_\infty \leq T/8$, y_c the centre of $\square_c$. With f_0 formed at the side q_i: $w_i = q_i^{-3} f_0(\cdot - \frac{1}{2}q_i e_3)$ and $\theta : x_3 \mapsto -x_3$; for $\mu = 1, 2, 3$, $w_i^{(\mu)} = q_i^{-3} f_0(\cdot - \frac{1}{2}q_i e_\mu)$ and $\theta_\mu : x_\mu \mapsto -x_\mu$.

■

Definition 13.4 (Limits)**.**

(1) $\mathfrak{S}_n(O)$ is the set of the n-tuples of real functions of $C_c^1(O)$ with pairwise disjoint supports, and $\mathfrak{S}(O) = \bigcup_n \mathfrak{S}_n(O)$, the separated tuples of [7, §K.1]. $X \cdot Y$ is the concatenation of tuples, τ_a the translation by a, (h) the tuple with the one entry h; translations and reflections act entry by entry. All assertions below on limits concern such tuples.

(2) A *Gibbs sequence* at g is a sequence (K_j, β_j, ν_j) with $(K_j, \beta_j) \in \mathfrak{d}(g)$, $K_j \to \infty$ and $\nu_j \in \mathcal{G}^{\mathrm{per}}_{K_j,\beta_j}$; its labels are $b_j = a_{K_j}\beta_j$. It *converges* if $\sigma(X) = \lim_j \nu_j(\prod_i(\mathcal{O}(f_i) - \nu_j(\mathcal{O}(f_i))))$ exists for every $X = (f_1, \dots, f_n) \in \mathfrak{S}(\mathbb{R}^3)$. By the theorem on their limits (Thm. YM3_I:AP), σ is invariant under $\mathbb{R}^3 \rtimes \mathrm{O}(3)$ and reflection positive at every mirror; its connected functions S_n^σ are the limits of the cumulants $S_n^T[\nu_j]$; $\sigma((f)) = 0$; and in a canonical decomposition $(\mathsf{X}, \lambda, (\sigma_\xi))$ of σ, in the sense of Thm. YM3_I:AL, applied to the classes of periodic Gibbs states in Thm. YM3_I:AP, every component is a limit along the full sequence, $\sigma_\xi(X) = \lim_j \rho_j(\prod_i(\mathcal{O}(f_i) - \nu_j(\mathcal{O}(f_i))))$ with states $\rho_j \in \mathcal{G}^{\mathrm{per}}_{K_j,\beta_j}$, is invariant and reflection positive likewise, and has $\sigma_\xi((f)) = \mathfrak{c}_\xi \int f$ with a number $\mathfrak{c}_\xi$, its one-point constant. Both obey the bound tables (Prop. YM3_I:106.3) of Thm. YM3_I:AP.

(3) For $\sigma' \in \{\sigma, \sigma_\xi\}$ there is a real-analytic function $\mathbf{s} \geq 0$ on $(0,\infty)$ such that

$$\sigma'(f,h) = \iint f(x)h(y)\mathbf{s}(|x-y|)\,\mathrm{d}x\,\mathrm{d}y, \qquad (f,h) \in \mathfrak{S}_2(\mathbb{R}^3).$$

Indeed, Thm. YM3_I:AP supplies the reconstruction data, and Prop. YM3_I:108.3 gives their real-analytic two-point kernel off the diagonal. Translation and rotation invariance identify this kernel with $\mathbf{s}(|x-y|)$. At separation re_3, its point-operator formula is

$$\mathbf{s}(r) = \langle \Omega, \mathsf{B}(e_3, \tfrac{r}{2}e_3)\mathsf{B}(\tfrac{r}{2}e_3, e_3)\Omega\rangle = \|\mathsf{B}(\tfrac{r}{2}e_3, e_3)\Omega\|^2 \geq 0.$$

Here the first equality follows by joining the two inner semigroup intervals and using that the vacuum is fixed by translations; the second is the adjoint identity for the point operators. Holomorphy of these operators for $\operatorname{Re} r > 0$ proves the stated real analyticity.

(4) The reflection Hilbert space and translation construction in Prop. YM3_I:94.1 gives, for such a σ' and the direction e_3, a Hilbert space $\mathcal{H}(\sigma', e_3)$ with vectors $[X]$, $X \in \mathfrak{S}(\mathbb{E}_+)$, $\mathbb{E}_+ = \{x_3 > 0\}$, $\langle [X], [Y]\rangle = \sigma'(\theta X \cdot Y)$; a self-adjoint $H \geq 0$ and unitary operators $\mathcal{U}(v)$, $v \perp e_3$, commuting and strongly continuous, with $[\tau_{te_3+v}X] = e^{-tH}\mathcal{U}(v)[X]$; and the joint spectral measure $\mathcal{P}$ of (H, P_1, P_2), carried by $[0,\infty)\times\mathbb{R}^2$, with $e^{-tH}\mathcal{U}(v) = \int e^{-tE+iv\cdot\mathfrak{q}}\,\mathrm{d}\mathcal{P}(E,\mathfrak{q})$. For real $h \in C_c^1(\mathbb{E}_+)$ we put $\psi_h = [(h)]$ and $\mu_h = \langle \psi_h, \mathcal{P}(\cdot)\psi_h\rangle$; $Q_r = [0,r]\times[-r,r]^2$.

(5) A continuum limit S at g *along growing tori* has a defining sequence (K_j, ℓ_j, β_j) with $(K_j, \beta_j) \in \mathfrak{d}(g)$, $\ell_j \in \mathfrak{L}(g)$, $K_j \to \infty$ and $\ell_j \to \infty$, no relation between K_j and ℓ_j being asked, and $S_n(X) = \lim_j S^T_{n;K_j,\ell_j,\beta_j}(X^{\langle \ell_j\rangle})$ for $n \geq 2$, $X^{\langle \ell\rangle}$ the periodisation; σ_S is its centred functional, formed from the S_n with $S_1 = 0$. By Props. YM3_I:92.1, YM3_I:92.3 and YM3_I:92.4, σ_S obeys a bound table with one-point constant 0, is invariant under the translations of $\mathbb{R}^3$ and the cubic group, and is reflection positive at every coordinate mirror; the reconstruction applies to it in each of the directions e_1, e_2, e_3.

■

Lemma 13.5 (Every frame has a partner)**.** For every frame there is a partner frame with the same radius and the same values of the constants that precede L in the list $\mathcal{T}^{\mathrm{S}}[2\varrho]$: one may take $\hat{L} = L^k + 2$ for every large integer k. For every radius there are also partner frames whose block sizes are L_1 and $L_1 + 2$. The cited construction theorems apply to each of these common choices, with its own subsequent constants and coupling ceiling. ■

Sketch of proof. Keep the exponents, reserve fractions and other constants preceding L in the ordered construction that defines the frame. The proof of Prop. YM3_I:69.1 admits every sufficiently large odd value of the new block size with these preceding values fixed. The further floors of Section YM3_I:82 preserve this property. Thus $\hat{L} = L^k + 2$ is admitted for every sufficiently large k. Complete the same construction at $\hat{L}$, choosing all subsequent constants anew and retaining the source-room and array-radius conditions at their respective choices; choose the coupling ceiling last. This gives a second frame with the same radius and the same pre-L data. Since L is odd, $L^k + 2$ is odd and $\gcd(L^k + 2, L) = \gcd(2, L) = 1$; if $L^p = \hat{L}^{p'}$ with positive integers p, p', a prime factor of L would divide $\hat{L}$, so

$$\log L / \log \hat{L} \notin \mathbb{Q}.$$

For an odd L_1 above the threshold of admission, L_1 and $L_1 + 2$ are both admitted, and a common prime factor would divide 2. Thm. YM3_I:AJ and the source-energy bounds of Thm. YM3_I:AD apply separately at both frames. The argument does not require equality of constants chosen after L. □

Definition 13.6 (The Setting; the constants; the floors)**.**

(1) *Frames.* Two frames, the second a partner of the first (Lem. 13.5); a hat marks the objects of the second.

(2) *Windows and reference sides.* In the first frame: $0 < g \le \gamma$; a nonempty compact set $W \subset I(g)$; a number $g_1 \in (0, g]$ with $W \subset \operatorname{int} I(g_1)$, which exists by Lem. 13.24; a number $\varepsilon > 0$ with $W_1 = [\min W - 3\varepsilon, \max W + 3\varepsilon] \subset I(g_1)$; the interval $W' = [\min W - 2\varepsilon, \max W + 2\varepsilon]$; and a reference side $\ell' \in \mathfrak{L}(g_1)$ with $\ell' \ge 2\mathsf{l}_1$. In the second frame: $0 < \hat g \le \hat\gamma$; a compact interval $\hat W = [\hat w_-, \hat w_+] \subset \hat I(\hat g)$ of positive length; and a reference side $\hat\ell' \in \hat{\mathfrak{L}}(\hat g)$ with $\hat\ell' \ge 2\hat{\mathsf{l}}_1$. All numbers of the first frame are formed at g_1 (so C_g stands for C_{g_1}) with the reference side ℓ', and those of the second at $\hat g$ with $\hat\ell'$.

(3) *The constants of the argument.*

- $C_\times(y) = 2592\,[65y^3 + 2]\,C_S S(\kappa)$ for $y > 0$, and $M_\times = 1348488\,C_S S(\kappa)$.
- Q_1 is the constant of the transfer (Subsection 13.1.4, item (M7)); it depends on L and on g_1 and on none of K, β, ℓ.
- $\mathsf{c}_\chi = 40\sqrt{3}ML$; $\mathsf{Q}_2 = \exp[(C_S + C_g)\mathsf{l}_1^3 + C_\times(1/8)]$; $\mathsf{Q}_* = 2\max\{\mathsf{Q}_1, \mathsf{Q}_2\} + 2$; $C_{\mathrm{lg}} = 54L^3 C_S S(\kappa)$; $C'_{\mathrm{lg}} = 27(10m' + 75)C_S S(\kappa)$;

$$h_1 = \frac{\varrho}{250e\,\mathsf{c}_\chi \mathsf{Q}_*}, \qquad \mathsf{A}_1 = 2C_{\mathrm{lg}} + \frac{L^3}{125}(4\log 2 + 2C'_{\mathrm{lg}}).$$

- $r_2 = \varrho/(e^{(C_S+C_g)\ell'^3}+1)$; $C_\omega = 2\log 2/(\ell'^3 r_2^2)$; $C'_2 = \mathsf{c}^2\varrho^{-2}[1152\,C_\times(7/2)\,\mathsf{l}^{-3} + 65536\,(C_S + C_g)]$; $r_1 = \min\{\varrho/(256\mathsf{c}),\, r_2/2,\, h_1/2\}$;

$$C_I = C'_I + C'_2 + C_\omega + 2\mathsf{A}_1/h_1^2 + M_\times T^{-3}\varrho^{-2}, \qquad C_0 = 2C'_I + C_\omega + 2\mathsf{A}_1/h_1^2;$$

  r_1^- and C_I^+ are r_1 and C_I with $1 + 5\sqrt{3}$ in the place of c and T_0 in the place of T.
- With $w_\pm$ the ends of W_1: $\lambda_- = \hat w_-/w_+$, $\lambda_+ = \hat w_+/w_-$, and

$$\delta_{\max} = \min\{\varepsilon/2,\, |\hat W|/(48\lambda_+)\}.$$

- k_7 is the least integer ≥ 1 with $\varsigma_7 = L^{-k_7} \le 2/(3\sqrt{3}\max\{1, \lambda_+\})$; the *tent* is $\chi(y) = \prod_\mu (1 - |y_\mu|/\varsigma_7)_+$ and the *plateau* $\eta(y) = \prod_\mu \min\{1, (3 - |y_\mu|/\varsigma_7)_+\}$; $\mathsf{c}_7 = 40\sqrt{3}M/\varsigma_7$; $c_T = 2\log 2\,(2\mathsf{Q}_1 + 2)^2(2/\varrho)^2$; $c_N = \varsigma_7^{-3} c_T \mathsf{c}_7^2$; $\hat c_N = \hat L^3 \hat c_T (40\sqrt{3}\hat M\hat L)^2$; $\hat{\mathsf{c}}_8 = \max\{1, 40\sqrt{3}\hat M/(\lambda_-\varsigma_7)\}$;

$$c_F = \sup_{K \ge k_7} \sup_{b \in W_1} |S^T_{2;K,\ell',L^K b}(\chi, 1 - \eta)|;$$

  $\hat c_F$ is the bound that the convergence on fixed tori (item (M8)) gives in the second frame at the side $\hat\ell'$ for $n = 2$, the interval $\hat W$, the spread $\lambda_-\varsigma_7$ and norms $\le \hat{\mathsf{c}}_8$; both are finite by (M8);

$$c_2 = \varsigma_7^{-3}[c_T\mathsf{c}_7^2 + \lambda_+^2 \hat c_T \hat{\mathsf{c}}_8^2 + c_F + \lambda_+^2 \hat c_F], \qquad \hat{\mathfrak{m}} = 16\hat B'/|\hat W| + \hat C_0|\hat W|,$$

$$C_\flat = \lambda_+^4[24\lambda_+\hat{\mathfrak{m}}/|\hat W| + 10\lambda_+\hat c_N + 1] + 6c_N + 3(C_0 + c_2) + 1.$$

- $\mu_K = (D_K + \frac{1}{2}C_0 y^2)''$ on $\operatorname{int} I(g_1)$ (a positive measure by Prop. 13.16). For $0 < \delta \le \delta_{\max}$, $K_1(\delta)$ is the least integer $K_1 \ge 1$ such that $\mu_K(I) \le C_\flat(|I| + \delta)$ for all $K \ge K_1$ and all closed intervals or points $I \subset W'$; such integers exist by Prop. 13.23, and $K_1(\delta)$ does not increase in δ.

(4) *The floors.*

$$q_0 = \max\Big\{\frac{4C_\flat + C_I^+}{9(C_w^-)^2 T_0^2},\ (3C_w^- T_0 \min\{r_1^-, \varepsilon/2\})^{-1/2}\Big\}, \qquad \delta^\circ(q) = \min\Big\{\delta_{\max},\ \frac{3C_w^- T_0}{2C_\flat q}\Big\},$$

$$K_0(q) = K_1(\delta^\circ(q)), \qquad \ell_0(q) = \max\Big\{2q,\ \ell',\ \frac{12\ell' B'}{C_\flat \varepsilon^2},\ \frac{16\,C_\flat \ell' B' q^2}{3(C_w T)^2}\Big\}.$$

(5) *Dependence.* All these numbers are formed from the two frames, g_1, $\hat{g}$, W, ε, $\hat{W}$, ℓ', $\hat{\ell}'$, and those that contain c or T (namely c, C_w, C_2, C_2', C_I, r_1, x_q and $\ell_0(\cdot)$) also from T; none depends on K, ℓ, b, a profile, an offset or a state. q_0, C_0, $C_\flat$, $\delta_{\max}$, $K_1(\cdot)$ and $K_0(\cdot)$ do not depend on T, and $K_0(q)$ and $\ell_0(q)$ do not depend on each other. $K_0(q)$ does not decrease in q.

■

13.1.2. *From cube means to low spectral mass.* The infrared argument measures an observable on a large cube while leaving a collar of fixed physical width. Let q be the cube side and T that width. Profiles in different cubes have separated supports, so their source insertions can be compared by chessboard estimates. The normalization by the cube volume turns the pressure estimates into a bound of order $(T/q)^2$ for the covariance of distinct cube means. The collar is retained throughout this comparison. Removing it prematurely would place insertions against the reflection boundaries used to prove the estimate.

There are two uses of positivity. On the lattice, reflection positivity controls the repeated source insertion by one periodic board. In the Euclidean invariant continuum limit, it supplies a positive representation of the radial two-point kernel. The two blocking factors enter between these steps: their pressure comparison prevents a concentrating response from defeating the cube-mean estimate. The pressure mechanism is shared with the preceding section, but its use here is a spatial covariance bound rather than a bound on a one-point jump.

Write $\mathbf{s}(r)$ for the radial two-point kernel after the constant part appropriate to the component has been removed. The representation proved in the detailed argument is

$$\mathbf{s}(r) = \frac{1}{4\pi r} \int_{[0,\infty)} e^{-mr} \, \mathrm{d}\rho(m).$$

Before the cube bound is imposed, the source also keeps a possible nonnegative constant κ' in the kernel. The cube-mean estimate excludes this constant and yields $\mathbf{s}(r) = O(r^{-2})$. To see the role of the cube geometry, take the reflected profiles in adjacent cells. Their separation is bounded above by a constant times q, their normalized integrals stay bounded below, and $\mathbf{s}$ is nonincreasing. Their covariance therefore bounds $\mathbf{s}(Cq)$ from above, up to those integral factors. The $(T/q)^2$ estimate passes to the claimed radial decay. The detailed calculation below keeps the collar correction and interpolates between successive admitted values of q.

The estimate on ρ then follows directly from its positivity. At $r = 1/m$, the integrand is at least e^{-1} on $[0, m]$, so the source obtains

$$\rho([0, m]) \le 4\pi e \, m^{-1} \mathbf{s}(1/m) = O(m).$$

In particular $\rho(\{0\}) = 0$. This is a bound on how much spectral mass can lie near zero. It permits nonzero mass in every interval $(0, m)$.

The source provides an example that attains both orders: $\rho = c \, \mathrm{d}m$ gives $\mathbf{s}(r) = c/(4\pi r^2)$. Thus neither the power bound nor the absence of an atom implies a positive mass gap. The conditional gap estimate belongs to the finite-size criterion later in the manuscript. In addition, the radial representation uses the Euclidean invariant Gibbs limits and their components. The weaker statements for arbitrary growing-torus limits are retained with their actual directional positivity; they are not silently identified with the radial case.

13.1.3. *Principal results.*

Theorem AL (Cube means in every periodic Gibbs state, on large tori and in every continuum limit)**.** Let Thm. YM3_I:D, Thms. YM3_I:AJ and YM3_I:AD and the statements (M1)–(M12) among the construction results listed in Subsection 13.1.4 be granted at each of the two frames of a Setting (Def. 13.6); by Lem. 13.5 every frame is the first frame of a Setting. Let $T \ge \max\{T_0, \mathsf{l}\}$. Then for every $q = L^i \ge \max\{2T, q_0\}$ the following hold.

(a) *Periodic Gibbs states.* Let $K \ge K_0(q)$, $b \in W$ and $\nu, \nu' \in \mathcal{G}^{\mathrm{per}}_{K,L^K b}$. For every board of $\mathbb{L}_K$ of side q, with any offset, and profiles of $\mathfrak{P}_T$ in distinct cells $c, c', c_1, \dots, c_n$ of it:

$$|\operatorname{Cov}_\nu(x_c, x_{c'})| \le C_2 (T/q)^2, \qquad |\nu(x_c) - \nu'(x_c)| \le 12 e^2 C_w T/q, \qquad \Big|\nu\Big(\prod_{i=1}^n (x_{c_i} - \nu(x_{c_i}))\Big)\Big| \le (12 e^2 C_w T/q)^n .$$

(b) *Tori.* Let $K \ge K_0(q)$, $b \in W$ and $\ell \in \mathfrak{L}(g)$ with $\ell \ge \ell_0(q)$. For the law $\langle\cdot\rangle_{K,\ell,L^K b}$ and every board of T_ℓ of side q the first and the third bound of (a) hold.

(c) *Limits of Gibbs sequences.* Let (K_j, β_j, ν_j) be a convergent Gibbs sequence at g whose labels lie in W from some index on (they need not converge), σ its limit, $(\mathsf{X}, \lambda, (\sigma_\xi))$ a canonical decomposition of σ, $\sigma' = \sigma$ or $\sigma' = \sigma_\xi$ for any ξ, $\mathbf{s}$ the radial kernel of σ' and $\mathcal{H}(\sigma', e_3)$ its reconstruction. Let j_0 be an index with $L^{j_0} \ge \max\{2T, q_0\}$, $r_0 = \sqrt{6} L^{j_0}$ and $C_{\mathbf{s}} = 6L^2 (1 - 3T/(4L^{j_0}))^{-6} C_2$. Then:

1. $\mathfrak{c}_\xi = 0$ for every ξ.
2. $S_2^{\sigma'}(\theta w_i, w_i) \le C_2 (T/q_i)^2$ for all $i \ge j_0$.
3. There is exactly one positive Borel measure ρ on $[0, \infty)$ with $\mathbf{s}(r) = \int e^{-mr} (4\pi r)^{-1} \,\mathrm{d}\rho(m)$ for all $r > 0$.
4. $\mathbf{s}(r) \le C_{\mathbf{s}} T^2 / r^2$ for $r \ge r_0$; in particular $r\mathbf{s}(r) \to 0$ as $r \to \infty$.
5. $|S_2^{\sigma'}(f, \tau_a h)| \le C_{\mathbf{s}} T^2 \|f\|_1 \|h\|_1 (|a| - R_f - R_h)^{-2}$ for all $f, h \in C^1_c(\mathbb{R}^3)$ with supports in the balls of radii R_f, R_h about 0 and all a with $|a| \ge R_f + R_h + r_0$.
6. $\rho([0, m]) \le 4\pi e C_{\mathbf{s}} T^2 m$ for $0 < m \le 1/r_0$.
7. $\rho(\{0\}) = 0$.
8. For real $h \in C^1_c(\mathbb{E}_+)$: $\|e^{-tH} \psi_h\|^2 \le C_{\mathbf{s}} T^2 \|h\|_1^2 / (4t^2)$ for $t \ge r_0/2$; ψ_h is orthogonal to $\ker H$; $\int_0^\infty \sigma'(\theta h, \tau_{te_3} h) \,\mathrm{d}t = \|H^{-1/2} \psi_h\|^2 < \infty$; and $\langle \psi_h, 1_{[0,\epsilon]}(H) \psi_h \rangle \le \frac{e^2}{4} C_{\mathbf{s}} T^2 \|h\|_1^2 \epsilon^2$ for $0 < \epsilon \le 2/r_0$.
9. Let $X = (f_1, \dots, f_n) \in \mathfrak{S}_n(\mathbb{R}^3)$ have its supports in the ball $B(0, R_X)$ and $h \in C^1_c(\mathbb{R}^3)$ its support in $B(0, R_h)$; let $\mathsf{N}(X)^2 = \sup_u \sigma'(X \cdot \theta_u X)$, the supremum over the unit vectors u, θ_u the reflection in the plane $\{u \cdot x = R_X + 1\}$; it is finite. Then for $|a| \ge R_X + R_h + 2 + r_0$

$$|\sigma'(X \cdot \tau_a h)| \le \frac{\mathsf{N}(X) \, \|h\|_1 \, C_{\mathbf{s}}^{1/2} T}{2(|a| - R_X - R_h - 2)}, \qquad |S^{\sigma'}_{n+1}(f_1, \dots, f_n, \tau_a h)| \le \frac{C(X) \, \|h\|_1 \, T}{|a| - R_X - R_h - 2},$$

with a number $C(X)$ formed from $C_{\mathbf{s}}$ and from the values of σ' at the subtuples of X and at their concatenations with their reflections: a connected function with one argument moved away tends to 0 at the rate $|a|^{-1}$, in every direction.

10. The converse and the measure that attains the bounds (Prop. 13.25 (d), (e)): for the limit of every convergent Gibbs sequence and every component of it, with the constant κ' and the measure ρ of Prop. 13.25, a bound $\sigma'(\theta w_i, w_i) \le C(T/q_i)^2$ for all large i, a bound $\mathbf{s}(r) = O(r^{-2})$ as $r \to \infty$, the pair $\kappa' = 0$, $\rho([0, m]) = O(m)$ as $m \to 0$, and a bound $\mu_h(Q_r) \le A_\mu \|h\|_1^2 r^2$ for small r are equivalent; the measure $\rho = c_\rho \,\mathrm{d}m$, whose kernel is $\mathbf{s}(r) = c_\rho/(4\pi r^2)$, satisfies them and attains 4 and 6 in order, so that these conclusions do not exclude it; and with no hypothesis on the labels the cube means of σ' obey the bound of the order $1/q$ of Prop. 13.25 (d), and q times the quantity that it bounds tends to 0 if and only if $\rho(\{0\}) = 0$.

(d) *Growing tori.* Let S be a continuum limit at g along growing tori, with no relation between K_j and ℓ_j, whose defining sequence has its labels in W from some index on, and let j_0 be as in (c). Then:

1. $\sigma_S(\theta_\mu w_i^{(\mu)}, w_i^{(\mu)}) \le C_2 (T/q_i)^2$ for all $i \ge j_0$ and $\mu = 1, 2, 3$.

2. For real $h \in C^1_c(\mathbb{E}_+)$, in $\mathcal{H}(\sigma_S, e_3)$:
$$\mu_h(Q_r) \le 12.46\,(1 - 3T/(4q_{j_0}))^{-6} L^2 C_2 \|h\|_1^2 T^2 r^2$$
for $0 < r \le 1/q_{j_0}$, and $\mathcal{P}(\{0\})\psi_h = 0$.
3. For
$$X = (f_1, \dots, f_n) \in \mathfrak{S}_n(\mathbb{R}^3)$$
and $h \in C^1_c(\mathbb{R}^3)$: $\sigma_S(X \cdot \tau_a h) \to 0$ and $S_{n+1}(f_1, \dots, f_n, \tau_a h) \to 0$ as $\min_\mu |a_\mu| \to \infty$.

(e) *Every label and every reference coupling.*

1. For every $b_\infty > 0$ there are an integer $s \ge 0$ and $g_2 \in (0, \gamma]$ with $L^s b_\infty \in \operatorname{int} I(g_2)$.
2. Let $s \ge 0$ be an integer, $0 < g_2 \le \gamma$, $W^{(2)} \subset I(g_2)$ compact, and form the numbers of Def. 13.6 for a Setting built on $(g_2, W^{(2)})$ in the place of (g, W), with the same second frame, and with the width $L^s T$. If $L^s b \in W^{(2)}$, $L^s T \ge \max\{T_0, \mathsf{l}\}$, $L^s q \ge \max\{2L^s T, q_0\}$ and $K - s \ge K_0(L^s q)$, then at $(K, L^K b)$ the three bounds of (a) hold multiplied by L^{8s}, L^{4s} and L^{4ns} respectively, T/q standing as it is; if besides $L^s \ell \in \mathfrak{L}(g_2)$ and $L^s \ell \ge \ell_0(L^s q)$, the bounds of (b) hold on T_ℓ multiplied likewise.
3. For every compact set $\mathsf{B} \subset (0, \infty)$ there are a constant C_B and floors q_B, $K_\mathsf{B}(\cdot)$, $\ell_\mathsf{B}(\cdot)$, none of them depending on T, such that
$$|\operatorname{Cov}_\nu(x_c, x_{c'})| \le C_\mathsf{B}(T/q)^2, \qquad |\nu(x_c) - \nu'(x_c)| \le (2C_\mathsf{B})^{1/2} T/q$$
for all $b \in \mathsf{B}$, $T \ge \max\{T_0, \mathsf{l}\}$, $q = L^i \ge \max\{2T, q_\mathsf{B}\}$, $K \ge K_\mathsf{B}(q)$, $\nu, \nu' \in \mathcal{G}^{\mathrm{per}}_{K, L^K b}$ and cells and profiles as in (a); and the covariance bound alone holds for the law $\langle\cdot\rangle_{K,\ell,L^K b}$ on boards of T_ℓ for all sides $\ell = 2L^m \ge \ell_\mathsf{B}(q)$.
4. At every $g' \in (0, \gamma]$, clauses (c) and (d) hold for every convergent Gibbs sequence at g' and for every continuum limit at g' along growing tori, with no condition on the labels, with C_B for $\mathsf{B} = I^{\mathrm{lab}}(g')$ in the place of C_2 and with q_B in the place of q_0.

■□

Theorem AM (The curvature of the pressure at large cutoffs; mean actions; the second bound of differences of means)**.** Assume the construction inputs listed in Thm. AL, and let a Setting be given.

(a) *No pile of specific heat.* For $0 < \delta \le \delta_{\max}$ and $K \ge K_1(\delta)$: $\mu_K(I) \le C_\flat(|I| + \delta)$ for every closed interval or point $I \subset W'$; $|D''_K|(I) \le (C_\flat + C_0)(|I| + \delta)$; and
$$-C_0 s^2 \le \Delta_s^2 D_K(b) \le C_\flat s^2 + C_\flat \delta s \qquad \text{whenever } [b - s, b + s] \subset W'.$$
With $\delta_0(K) = \inf\{\delta \in (0, \delta_{\max}] : K_1(\delta) \le K\}$, which tends to 0 as $K \to \infty$, these bounds hold with $\delta_0(K)$ in the place of δ. For $\delta' > 0$ let $\mathsf{p} = D_K * \phi_{\delta'}$, $\phi_{\delta'}(t) = \delta'^{-2}(\delta' - |t|)_+$; at every b with $[b - \delta', b + \delta'] \subset W'$: $-C_0 \le \mathsf{p}''(b) \le C_\flat(1 + \delta_0(K)/\delta')$ and $-C_0\delta'^2/12 \le \mathsf{p}(b) - D_K(b) \le C_\flat(\delta'^2/12 + \delta_0(K)\delta'/6)$; at $\delta' = \delta_0(K) > 0$: $-C_0 \le \mathsf{p}'' \le 2C_\flat$ and $|\mathsf{p} - D_K| \le \frac{1}{4}\max\{C_\flat, C_0/3\}\delta_0(K)^2$.

(b) *The atom at one label.* For $0 < \delta \le \delta_{\max}$, $b \in W'$ and $K \ge K_1(\delta)$:
$$\mu_K(\{b\}) = p'_{K,\infty}(b+) - p'_{K,\infty}(b-) = L^{4K} J(L^K b) \le C_\flat \delta.$$
And $\beta^4 J(\beta) \to 0$ as $\beta \to \infty$ through the reals.

(c) *Mean actions.* For every integer $K \ge 0$, every $b > 0$ and $\nu \in \mathcal{G}^{\mathrm{per}}_{K, L^K b}$, the number $\nu(a_p) = \mathsf{a}(\nu)$ is the same for all plaquettes p; with $e(\nu) = 3a_K^{-4}\mathsf{a}(\nu)$ we have $\nu(\mathcal{O}(f)) = \mathrm{Rie}_K(f)\, e(\nu)$ for every bounded f with compact support, and $p_{K,\infty}(b') \ge p_{K,\infty}(b) - (b' - b)e(\nu)$ for all $b' \ge 0$. There are $\nu_\pm \in \mathcal{G}^{\mathrm{per}}_{K, L^K b}$ with $-e(\nu_\pm) = p'_{K,\infty}(b\pm)$. Hence
$$\sup\{|e(\rho) - e(\nu)| : \rho, \nu \in \mathcal{G}^{\mathrm{per}}_{K, L^K b}\} = J_K(b) := p'_{K,\infty}(b+) - p'_{K,\infty}(b-) = L^{4K} J(L^K b),$$
and $J_K(b) = 0$ if and only if all periodic Gibbs states at $(K, L^K b)$ have the same mean action.

(d) *The second bound of differences of means.* For every integer $K \ge 0$, $b > 0$, $\nu, \nu' \in \mathcal{G}^{\mathrm{per}}_{K, L^K b}$, every board of $\mathbb{L}_K$ of side q and every f with $0 \le f \le 1$ that vanishes off a cube of side $\le q - a_K$ (in particular every profile, every q and every cell): $|\nu(x_c) - \nu'(x_c)| \le J_K(b)$. In the Setting, $J_K(b) = \mu_K(\{b\}) \le C_\flat \delta$ for $b \in W'$ and $K \ge K_1(\delta)$. So in Thm. AL (a) the second bound may be read as $\min\{12e^2 C_w T/q,\ \mu_K(\{b\})\}$, and $\sup_{b \in W} |\nu(x_c) - \nu'(x_c)|$ tends to 0 as $K \to \infty$ uniformly in q, in the cells and in the profiles.

(e) *The plateau.* In any frame let $0 < g \le \gamma$, $(K, \beta) \in \mathfrak{d}(g)$, $b = a_K \beta$, $\nu \in \mathcal{G}^{\mathrm{per}}_{K,\beta}$, and let $\mathcal{I}$ be the σ-field of the Borel sets of configurations fixed by all translations by $a_K \mathbb{Z}^3$. There is one bounded real $\mathcal{I}$-measurable function e with $\nu(k a_p) = \nu(k\mathsf{e})$ for every plaquette p and every bounded $\mathcal{I}$-measurable k. For $f, h \in C^1_c(\mathbb{R}^3)$, $\bar R \in a_K \mathbb{N}$ with $\operatorname{supp} f \cup \operatorname{supp} h \subset (-\bar R, \bar R)^3$ and $a_K \le 2 \min\{\delta((f); \bar R), \delta((h); \bar R)\}$, and $v \in a_K \mathbb{Z}^3$ with $|v| \ge \mathfrak{k}_1 \bar R$:

$$\left| S_2^T[\nu](f, \tau_v h) - 9 a_K^{-8} \operatorname{Rie}_K(f) \operatorname{Rie}_K(h) \operatorname{Var}_\nu(\mathsf{e}) \right| \le \mathfrak{k}_0 \, \mathsf{N}((f); \bar R)^{1/2} \, \mathsf{N}((h); \bar R)^{1/2} \, \frac{\bar R}{|v|},$$

where $\mathfrak{k}_0$, $\mathfrak{k}_1 = 4\sqrt{3}$, $\delta(\cdot; \bar R)$ and $\mathsf{N}(\cdot; \bar R)$ are the numbers of the plateau statement (Prop. YM3_I:100.1 and equations YM3_I:(1976)–(1980)); and $0 \le 9 a_K^{-8} \operatorname{Var}_\nu(\mathsf{e}) \le \frac{1}{4} J_K(b)^2$, with equality at $\nu = \frac{1}{2}(\nu_+ + \nu_-)$, while $\operatorname{Var}_\nu(\mathsf{e}) = 0$ if ν is ergodic or $\nu = \nu_\pm$.

(f) *A power of the spacing, under a rate in the cutoff.* Let two frames at one radius be given whose block sizes are L_1 and $L_2 = L_1 + 2$, which exist by Lem. 13.5; an index $i \in \{1, 2\}$ marks the objects of the frame i. Let Thms. YM3_I:AJ and YM3_I:AD be granted at both frames, and Thm. YM3_I:D at the choices of its constants that the two frames determine and at a lowered choice of each (Def. 4.47). In the frame i let $\mathcal{W}_i = [\gamma_i^{-2} + \bar B_i + \frac{1}{2},\ L_i(\gamma_i^{-2} + 3\bar B_i + 1) - 2\bar B_i]$, let $\chi^{(i)}(y) = \prod_\mu (1 - L_i |y_\mu - \frac{1}{2}|)_+$ and $\eta^{(i)}(y) = \prod_\mu \min\{1, (3 - L_i |y_\mu - \frac{1}{2}|)_+\}$ be the tent and the plateau of half-width $1/L_i$ about the point $(\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$, and for integers $m, K \ge 1$, $\ell = 2L_i^m$ and $b > 0$ let

$$\mathsf{y}^{(i)}_{m,K}(b) = b^5 L_i^3 \, S^T_{2; K, \ell, L_i^K b}((\chi^{(i)})^{\langle \ell \rangle}, 1 - (\eta^{(i)})^{\langle \ell \rangle})$$

be the far part of the specific heat of T_ℓ. Let the far part settle in the cutoff at a rate on every torus: there are numbers $\mathsf{k}_i \ge 0$ and $\mathsf{c}_{\mathrm{y},i} < \infty$, free of m, K and b, such that for all integers $m \ge 1$, $\ell = 2L_i^m$, all $b \in \mathcal{W}_i$ and all integers $K' \ge K \ge \mathsf{k}_i + 54m$

$$|\mathsf{y}^{(i)}_{m,K'}(b) - \mathsf{y}^{(i)}_{m,K}(b)| \le \mathsf{c}_{\mathrm{y},i} \, \ell^9 K^{\mathsf{p}_{2,i}} L_i^{-K/10}, \qquad \mathsf{p}_{2,i} = 6\mathfrak{p}_{0,i} + \tfrac{1}{2},$$

$\mathfrak{p}_{0,i}$ the exponent of the theorem on two consecutive cutoffs (Thm. YM3_I:AR) in the frame i. Then for every real $A > 90$ there are numbers C_{r} and β_1 with $\beta^4 J(\beta) \le \mathsf{C}_{\mathrm{r}} \beta^{-1/(2A)}$ for all $\beta \ge \beta_1$. Hence for every $0 \le \alpha < 1/180$ the number $\mathsf{c}_*(\alpha) = \sup_{\beta \ge 1} \beta^{4+\alpha} J(\beta)$, a number of the Wilson model on $\mathbb{Z}^3$, is finite; and in every frame, for $0 < g \le \gamma$, $(K, \beta) \in \mathfrak{d}(g)$ and $\rho, \nu \in \mathcal{G}^{\mathrm{per}}_{K,\beta}$:

$$3 a_K^{-4} |\rho(a_p) - \nu(a_p)| \le a_K^{-4} J(\beta) \le g^{8+2\alpha} \, \mathsf{c}_*(\alpha) \, L^{-\alpha K};$$

$|\rho(\mathcal{O}_K(f)) - \nu(\mathcal{O}_K(f))| \le a_K^{-4} J(\beta)(2R + a_K)^3 \|f\|_\infty$ for every bounded f with support in $[-R, R]^3$; and $|\rho(x_c) - \nu(x_c)| \le g^{8+2\alpha} \mathsf{c}_*(\alpha) L^{-\alpha K}$ for every cube mean, free of q and of T, with no floor in the cutoff.

■□

The hypothesis K1 is the following inequality between the free energies of two periodic boxes. Its data are a label $\mathsf{b} > 0$, a side $\ell_\circ = 2L^{m_\circ}$ with $m_\circ \in \mathbb{Z}$, an integer $K_\circ$ with $K_\circ + m_\circ \ge 1$ and a number $0 < x_\circ \le 0.069$; with $Z(n_1, n_2, n_3; \beta)$ the partition function of Wilson's action on the periodic lattice with n_1, n_2, n_3 sites a side, $l_K = \ell_\circ/(2a_K) = L^{K + m_\circ}$ and $\beta_K = L^K \mathsf{b}$, the hypothesis K1$[\mathsf{b}, \ell_\circ, K_\circ; x_\circ]$ requires

$$2 \log Z(l_K, 2l_K, 2l_K; \beta_K) - \log Z(2l_K, 2l_K, 2l_K; \beta_K) \le x_\circ \qquad \text{for every integer } K \ge K_\circ.$$

We put $\theta_\circ = 12.9\, x_\circ \le 0.8901$ and $m_* = 2\log(1/\theta_\circ)/\ell_\circ \ge 0.2328/\ell_\circ$.

Theorem AN (The two-point function of the action density under the hypothesis K1 at one label)**.** Assume the construction inputs listed in Thm. AL; let $0 < g \le \gamma$, $\mathsf{b} \in I(g)$, and assume K1$[\mathsf{b}, \ell_\circ, K_\circ; x_\circ]$.

(a) *The gap* (from (M13) of Subsection 13.1.4). For each $K \ge \max\{K_\circ, 1\}$ there is exactly one periodic Gibbs state ν_K at $(K, L^K\mathsf{b})$; the Gibbs sequence $(K, L^K\mathsf{b}, \nu_K)_{K\ge\max\{K_\circ,1\}}$ converges; its limit σ is its own single component, has a unique vacuum Ω and is invariant under $\mathbb{R}^3 \rtimes \mathrm{O}(3)$; the limits at this label in every order of cutoff and volume, that is, the limits of $S^T_{n;K,\ell,L^K\mathsf{b}}(X^{\langle\ell\rangle})$ for $n \ge 2$ and $X \in \mathfrak{S}_n(\mathbb{R}^3)$, at the couplings $L^K\mathsf{b}$ themselves and with ℓ running through the sides $2L^m$, as K and ℓ tend to infinity jointly in any relation, or ℓ first and then K, or K first and then ℓ, are the connected functions $S^\sigma_n(X)$; in particular every continuum limit along growing tori at this label, that is, with a defining sequence in which $\beta_j = L^{K_j}\mathsf{b}$ from an index on, has the centred functional σ; the spectrum of H in $\mathcal{H}(\sigma, e_3)$ lies in $\{0\} \cup [m_*, \infty)$, and for $X, Y \in \mathfrak{S}(\mathbb{E}_+)$ and $t \ge 0$

$$\big|\sigma(\theta X \cdot \tau_{te_3} Y) - \sigma(X)\sigma(Y)\big| \le e^{-m_* t}\, \|(1 - P_\Omega)[X]\|\, \|(1 - P_\Omega)[Y]\|,$$

P_Ω the projection on $\mathbb{C}\Omega$, and likewise in every direction.

(b) *The kernel and the spectral measure.* Let $\mathbf{s}$ and ρ be the radial kernel of σ and its measure of Thm. AL (c) 3, which applies to σ by Thm. AL (e) 4. Then $\mathbf{s}(r) \le e^{-m_*(r-r')}\mathbf{s}(r')$ for $0 < r' \le r$; $\rho([0, m_*)) = 0$; and $\mathbf{s}(r) \le (r'/r)\, e^{-m_*(r-r')}\, \mathbf{s}(r')$ for $0 < r' \le r$. So $\mathbf{s}(r) = O(r^{-3})$ as $r \to \infty$ and $\rho([0, m]) = O(m^2)$ as $m \to 0$.

(c) *Cube means in the limit.* For $T \ge \max\{T_0, 1\}$, $q = L^i \ge 2T$, a board of $\mathbb{R}^3$ of side q (a partition of $\mathbb{R}^3$ into the cubes $\square_c$ of side q centred at the points $o + qc$, $c \in \mathbb{Z}^3$, with an offset $o \in \mathbb{R}^3$) and profiles $f_c \in \mathfrak{P}_T(\square_c)$, $f_{c'} \in \mathfrak{P}_T(\square_{c'})$ of distinct cells:

$$\big|S^\sigma_2(q^{-3} f_c, q^{-3} f_{c'})\big| \le \chi_\circ\, q^{-3}, \qquad \chi_\circ = 4\pi \int_{T/4}^\infty r^2 \mathbf{s}(r)\, \mathrm{d}r < \infty;$$

if the closures of the two cells have no common point, that is $|c - c'|_\infty \ge 2$, the supports of the two profiles are at distance $\ge q + T/4$, and the left member is at most $\mathbf{s}(q)$, which is exponentially small in q by (b).

■□

Proposition 13.7 (Two fixed tori: means, and the convergence of the difference of the pressures)**.** Assume the construction inputs listed in Thm. AL; let one frame and $0 < g \le \gamma$ be given.

(a) *An exact identity.* For $K \ge 1$, a side $\lambda = 2L^m$, $b \ge 0$ and a bounded f with compact support on $\mathbb{R}^3$: $\langle \mathcal{O}(f^{\langle\lambda\rangle})\rangle_{K,\lambda,L^K b} = -\operatorname{Rie}_K(f)\, p'_{K,\lambda}(b)$. For two sides $\ell > \ell'$,

$$A_K(f; b) := \langle \mathcal{O}(f^{\langle\ell\rangle})\rangle_{K,\ell,L^K b} - \langle \mathcal{O}(f^{\langle\ell'\rangle})\rangle_{K,\ell',L^K b} = -\operatorname{Rie}_K(f)\, D'_{K,\ell}(b),$$

where $D'_{K,\ell}(b) = 3a_K^{-4}[\langle a_p\rangle_{K,\ell',L^K b} - \langle a_p\rangle_{K,\ell,L^K b}] = -L^3 A_K(\chi_0; b)$, $\chi_0(y) = \prod_\mu (1 - L|y_\mu|)_+$.

(b) *Bounded, uniformly in the cutoff and in both sides.* For $b \in I(g)$, $K \ge 1$, $\ell, \ell' \in \mathfrak{L}(g)$ and f with support in a ball of radius ≤ 2: $|A_K(f; b)| \le 2\log 2\, (2\mathsf{Q}_1 + 2)\varrho^{-1}\|f\|_\sharp$; so $|D'_{K,\ell}(b)| \le 2\log 2\, (2\mathsf{Q}_1 + 2)\varrho^{-1} L^3 \mathsf{c}_\chi$, free of K, b, ℓ, ℓ'. For a cube mean whose cell lies in a ball of radius $\le \ell'/8$, its expectations on the two tori differ by $-q^{-3}\operatorname{Rie}_K(f) D'_{K,\ell}(b)$, and $0 \le q^{-3}\operatorname{Rie}_K(f) \le 1$.

(c) *Convergence of the difference, with a rate.* Let $\ell' < \ell$ be two sides of $\mathfrak{L}(g)$, $\mathsf{J} \subset I(g)$ a compact interval and $U_K = p_{K,\ell} - p_{K,\ell'}$. For $b \in \mathsf{J}$ and $K \geq \max\{K_{\mathrm{bf}}(\ell, \mathsf{J}), K_{\mathrm{bf}}(\ell', \mathsf{J})\}$:

$$|U_{K+1}(b) - U_K(b)| \leq 2\,\mathsf{C}_{\mathrm{bf}}(\mathsf{J})\,K^{\mathsf{p}_0} L^{-K/10} + 2\mathsf{v}_K, \qquad \mathsf{v}_K = 2C_{\mathrm{tt}} E_0 \sum_{s \geq u} L^{3s} \exp\Big[- \frac{\kappa \ell' L^s}{5M}\Big], \quad u = \lceil K/1260 \rceil,$$

with $\mathsf{p}_0 = 2\mathfrak{p}_0 + \frac{1}{2}$; here the number $\mathsf{C}_{\mathrm{bf}}(\mathsf{J})$, formed from the constants of the frame and the ends of J and free of the sides, the exponent p_0 and the thresholds $K_{\mathrm{bf}}(\cdot, \mathsf{J})$ are those of the theorem on two consecutive cutoffs (Thm. YM3_I:AR, Def. 2.7 (5), Cor. 2.8 (a)), and C_{tt}, E_0 those of the comparison of the vacuum energies of two tori (Lem. 13.31). So U_K converges as $K \to \infty$, uniformly on J.

(d) *Convergence of the derivative.* Let moreover $\ell' \geq 2\mathsf{l}_1$, and put $r_3 = \min\{h_1, \varrho/(e^{(C_S + C_g)\ell^3} + 1)\}$ and $M_3 = \mathsf{A}_1 + (\ell^{-3} + \ell'^{-3}) \log 2$, both free of K and of b. Then $|\partial_b^n U_K(b)| \leq n!\, M_3 r_3^{-n}$ for all $b \in I(g)$, $K \geq 1$ and $n \geq 1$. Let $\mathsf{W} = [b_-, b_+]$ with $\mathsf{J} = [b_- - r_3, b_+ + r_3] \subset I(g)$, and $\mathsf{o}_K = 4\mathsf{C}_{\mathrm{bf}}(\mathsf{J}) K^{\mathsf{p}_0} L^{-K/10} + 4\mathsf{v}_K$. For $K \geq \max\{K_{\mathrm{bf}}(\ell, \mathsf{J}), K_{\mathrm{bf}}(\ell', \mathsf{J})\}$

$$\sup_{\mathsf{W}} |U'_{K+1} - U'_K| \leq \frac{4\mathsf{o}_K}{r_3}\Big[1 + \log_2^+ \frac{2M_3}{\mathsf{o}_K}\Big].$$

Hence $\sum_K K \sup_{\mathsf{W}} |U'_{K+1} - U'_K| < \infty$, U'_K converges uniformly on W, and $\sup_K \sup_{\mathsf{W}} |U'_K| < \infty$; and for every real Lipschitz $f \neq 0$ with support in a ball of radius $\leq \ell'/8$, the difference between the two tori of the derivatives at zero source of the contact terms that the theorem on two consecutive cutoffs splits off at the cutoffs K and $K+1$ is summable in K, uniformly on W.

■

Proposition 13.8 (Variances, the cost of a source on the walls, and covariances)**.** Assume the construction inputs listed in Thm. AL; let one frame, $0 < g \leq \gamma$, $T \geq \max\{T_0, \mathsf{l}\}$ and $q = L^i \geq 2T$ be given.

(a) *Variances from below.* Let $b \in I(g)$, $K \geq 1$, a board of $\mathbb{L}_K$ of side q with any offset, a profile f of one of its cells c, and $X_\kappa = \mathcal{O}(f^{[\kappa]})$ for the cells κ. For $\ell \in \mathfrak{L}(g)$, $\ell \geq 2q$, with the board of T_ℓ of the same offset, let $\mathsf{e}_\ell = N^{-1}\partial_s^2 \mathcal{E}_\ell^{\mathrm{av}}(s(F_e - F_o))|_{s=0}$ and $\bar{\mathsf{e}} = \limsup_{\ell \to \infty} \mathsf{e}_\ell$, which is finite. Then for every $\nu \in \mathcal{G}^{\mathrm{per}}_{K, L^K b}$ and real numbers y_κ, finitely many of them non-zero,

$$\mathrm{Var}_\nu\Big(\sum_\kappa y_\kappa X_\kappa\Big) \geq (\bar{\mathsf{e}} - 2C'_I q^3) \sum_\kappa y_\kappa^2.$$

Consequently $\mathsf{V}_\nu(\kappa) := \mathrm{Var}_\nu(X_\kappa) - \bar{\mathsf{e}} \geq -2C'_I q^3$, and $|\operatorname{Cov}_\nu(X_\kappa, X_{\kappa'})| \leq \frac{1}{2}[\mathsf{V}_\nu(\kappa) + \mathsf{V}_\nu(\kappa')] + 2C'_I q^3$ for $\kappa \neq \kappa'$.

(b) *The cost of a source on the walls controls the cube means.* Let $b \in I(g)$, $K \geq 1$, $\ell \in \mathfrak{L}(g)$ with $\ell \geq 2q$, a board of T_ℓ of side q, a profile f of a cell c, and for $0 < r \leq \varrho/2$

$$\mathsf{S}_f(r) = N^{-1} \log \frac{\widetilde{Z}^{\mathrm{av}}(rF)\, \widetilde{Z}^{\mathrm{av}}(-rF)}{\widetilde{Z}^{\mathrm{av}}(0)^2}.$$

Then $\mathsf{S}_f(r) \geq -2C'_I q^3 r^2$, and there is a real number m with

$$\|x_c - m\|_{\mathrm{ch}} \leq (rq^3)^{-1} \exp\{\tfrac{1}{2}\mathsf{S}_f(r) + (2C'_I + M_\times T^{-3}\varrho^{-2}) q^3 r^2\}.$$

So for profiles f, f' of distinct cells c, c': $|\operatorname{Cov}_{K,\ell,L^K b}(x_c, x_{c'})| \leq 2(rq^3)^{-2} \exp\{\frac{1}{2}\mathsf{S}_f(r) + \frac{1}{2}\mathsf{S}_{f'}(r) + 2(2C'_I + M_\times T^{-3}\varrho^{-2}) q^3 r^2\}$; in particular, if $\mathsf{S}_f(r), \mathsf{S}_{f'}(r) \leq \mathsf{C}$ at $r = \hat{r} q^{-3+p/2} \leq \varrho/2$ with $p \in [2, 3]$ and $0 < \hat{r} \leq 1$, then $|\operatorname{Cov}_{K,\ell,L^K b}(x_c, x_{c'})| \leq 2\hat{r}^{-2} e^{\mathsf{C} + 2(2C'_I + M_\times T^{-3}\varrho^{-2})\hat{r}^2} q^{-p}$.

(c) *The cost from above.* In a Setting, with the numbers at g_1: for $b \in I(g_1)$, $K \geq 1$, $\ell \in \mathfrak{L}(g_1)$ with $\ell \geq \max\{2q, \ell'\}$, and $0 < r \leq \min\{r_1, b\}$,

$$\mathsf{S}_f(r) \leq q^3 \Delta_r^2 D_{K,\ell}(b) + 6C_w T q^2 r + 2C_I q^3 r^2.$$

In particular, for $b \in W$, $\ell \geq \max\{2q, \ell', 6\ell' B' q^3\}$, $0 < \hat{r} \leq 1$ with $\hat{r} q^{-3/2} \leq \min\{r_1, \varepsilon/2\}$, and $K \geq K_1(\min\{\delta_{\max}, q^{-3/2}\})$:

$$-2C_I' \hat{r}^2 \leq \mathsf{S}_f(\hat{r} q^{-3/2}) \leq 6C_w T \hat{r}\, q^{1/2} + 2C_I \hat{r}^2 + 2C_\flat + 1.$$

(d) *A power of q passes to the kernel.* Let $0 < g' \leq \gamma$, $p \in [2,3]$, and let a convergent Gibbs sequence at g' satisfy the hypothesis of Lem. 13.27 with numbers C_{LH}, c_{LH}, q_{LH} and, besides, for every $q = L^i \geq \max\{2T, q_{\mathrm{LH}}\}$ and $j \geq j(q)$, $|\operatorname{Cov}_\nu(x_c, x_{c'})| \leq C_p q^{-p}$ for all $\nu \in \mathcal{G}^{\mathrm{per}}_{K_j,\beta_j}$ and profiles in two distinct cells of one board of side q. Then for σ and every σ_ξ, with $L^{j_0} \geq \max\{2T, q_{\mathrm{LH}}\}$ and $C_p' = (\sqrt{6}L)^p (1 - 3T/(4L^{j_0}))^{-6} C_p$: $\mathfrak{s}(r) \leq C_p' r^{-p}$ for $r \geq \sqrt{6} L^{j_0}$, and $\rho([0,m]) \leq 4\pi e C_p' m^{p-1}$ for $0 < m \leq 1/(\sqrt{6}L^{j_0})$.

(e) *One frame and a growth bound of the specific heat.* Let $W \subset \operatorname{int} I(g)$ be a compact interval, $\ell' \in \mathfrak{L}(g)$ with $\ell' \geq 2\mathsf{l}_1$, $C_{\mathrm{FS}} = 3\ell' B'$, and suppose that there are Y_0 and integers $K_Y(\ell)$ with $D''_{K,\ell} \leq Y_0 \ell^{1/3}$ on W for all $\ell \in \mathfrak{L}(g)$ with $\ell \geq \ell'$ and all $K \geq K_Y(\ell)$. Let $q \geq q_1 = \max\{2T, C_I/(9C_w^2 T^2), (3C_w T r_1)^{-1/2}\}$ with $6C_{\mathrm{FS}} q^3 \geq \max\{2q, \ell', 2L^{m_{\mathfrak{L}}(g)}\}$, let ℓ_q be the least side of $\mathfrak{L}(g)$ with $\ell_q \geq 6C_{\mathrm{FS}} q^3$, $K \geq K_Y(\ell_q)$ and $[b - x_q, b + x_q] \subset W$. Then for all $\nu, \nu' \in \mathcal{G}^{\mathrm{per}}_{K,L^K b}$ and profiles of $\mathfrak{P}_T$ in distinct cells $c, c', c_1, \dots, c_n$ of one board of side q: $|\operatorname{Cov}_\nu(x_c, x_{c'})| \leq 2e^4 \delta^2$, $|\nu(x_c) - \nu'(x_c)| \leq 2e^2 \delta$ and $|\nu(\prod_i (x_{c_i} - \nu(x_{c_i})))| \leq (2e^2\delta)^n$, with

$$\delta \leq \frac{T}{q}\Big[4C_w + \frac{(6LC_{\mathrm{FS}})^{1/3} Y_0}{3C_w T^2}\Big] + C_0 x_q.$$

(f) *The cost bounds the variances in every periodic Gibbs state.* Let $b \in I(g)$, $K \geq 1$, a board of $\mathbb{L}_K$ of side q, a profile f of one of its cells c, $X = \mathcal{O}(f)$, and sides $\ell_k \in \mathfrak{L}(g)$ with $\ell_k \geq 2q$ and $\ell_k \to \infty$ be given, the boards of T_{ℓ_k} having the same offset; let $\bar{\mathsf{e}}$ be the number of (a), $\mathsf{S}_{f,\ell_k}(r)$ the cost of (b) on T_{ℓ_k}, $c_X = M_\times q^3 T^{-3}$ the constant of the cross terms between the cells, and $c_3 = 3C_I' q^3 + 4c_X/(9\varrho^2)$. Then for every $\nu \in \mathcal{G}^{\mathrm{per}}_{K,L^K b}$ and $0 < r \leq \varrho/2$

$$\operatorname{Var}_\nu(X) \leq \bar{\mathsf{e}} + 2c_X \varrho^{-2} + 2r^{-2} \exp\Big[\limsup_{k\to\infty} \mathsf{S}_{f,\ell_k}(r) + c_3 r^2\Big].$$

If, with V_ν as in (a), $\mathsf{V}_\nu(\kappa) \leq C_{\mathrm{var}} q^{6-p}$ and $\mathsf{V}_\nu(\kappa') \leq C_{\mathrm{var}} q^{6-p}$ at two distinct cells κ, κ', with $p \in [2,3]$, then $|\operatorname{Cov}_\nu(q^{-3} X_\kappa, q^{-3} X_{\kappa'})| \leq (C_{\mathrm{var}} + 2C_I') q^{-p}$; at $p = 3$ this is the bound $(C_{\mathrm{var}} + 2C_I') q^{-3}$.

(g) *A one-sided bound on the walls bounds the cost.* In the situation of (b) let

$$\Upsilon_f(x) = N^{-1}\big[\log \widetilde{Z}^{\mathrm{av}}(xF) - \log \widetilde{Z}^{\mathrm{av}}(x)\big], \qquad \upsilon_f(x) = N^{-1} \partial_t \log \widetilde{Z}^{\mathrm{av}}(xF + t(1-F))\big|_{t=0},$$

the second being the derivative at the source xF in the direction of the walls. Then $\Upsilon_f(x) \leq -x\, \upsilon_f(x) + C_I' q^3 x^2$ for real $|x| \leq \varrho/2$. In a Setting, with the numbers at g_1, let $b \in W$, $\ell \in \mathfrak{L}(g_1)$ with $\ell \geq \max\{2q, \ell'\}$, $0 < \delta \leq \delta_{\max}$ and $K \geq K_1(\delta)$, and let $\Upsilon_f(x) \leq \mathsf{s}_1 x + C_\Upsilon$ for all real $|x| \leq \bar{x}$, with real numbers s_1, C_Υ and $\bar{x} > 0$. Then for $0 < r \leq \min\{\bar{x}, r_1, \varepsilon/2\}$

$$\mathsf{S}_f(r) \leq 2C_\Upsilon + q^3\big[C_\flat r^2 + C_\flat \delta r + 2\bar{\eta}(\ell)\big] + 2(C_\omega + 2\mathsf{A}_1/h_1^2) q^3 r^2.$$

■

13.1.4. *Construction estimates.* The proofs use Thm. YM3_I:D and Thms. YM3_I:AJ and YM3_I:AD among the construction results and the following statements (M1)–(M12) among the construction results, each taken at both frames of the Setting. Thms. YM3_I:D, YM3_I:AJ and YM3_I:AD are granted together with these statements, several of which rest on them: among them, the convergence on fixed tori (M8) and the theorem on two consecutive cutoffs in (M12) rest on Thm. YM3_I:D, and the theorems on limits (M10) and Prop. YM3_I:106.3 in (M9) on Thms. YM3_I:D, YM3_I:AJ and YM3_I:AD.

Statements among the construction results that are not cited by number are cited by the names they carry there. Thm. AN uses besides (M13), and clause (f) of Thm. AM uses (M14). (M13) is proved in §17.2 on the hypothesis K1, under that hypothesis. (M14) is proved in §12.1 on the latent heat of the Wilson model: its power is derived there from Thms. YM3_I:AJ and YM3_I:AD at the two frames of (M14), Thm. YM3_I:D at their choices and at a lowered choice of each, and the rate at which the far part of the specific heat settles in the cutoff on every torus, the rate that clause (f) of Thm. AM assumes and that (M14) carries among its hypotheses: for $i = 1, 2$, numbers $\mathsf{k}_i \geq 0$ and $\mathsf{c}_{\mathrm{y},i} < \infty$, free of m, K and b, with $|\mathsf{y}^{(i)}_{m,K'}(b) - \mathsf{y}^{(i)}_{m,K}(b)| \leq \mathsf{c}_{\mathrm{y},i}\, \ell^9 K^{\mathsf{p}_{2,i}} L_i^{-K/10}$ for all integers $m \geq 1$, $\ell = 2L_i^m$, all $b \in \mathcal{W}_i$ and all integers $K' \geq K \geq \mathsf{k}_i + 54m$.

The list is followed by the four statements (i)–(iv) that the proof of Prop. 13.7 (c) derives from (M12).

(M1) *Source energies.* Thm. YM3_I:AD, Prop. YM3_I:83.2, and Prop. YM3_I:74.4 give the following at $(K, \beta) \in \mathfrak{d}(g)$ and $\ell \in \mathfrak{L}(g)$. Each $e_j(X; \cdot)$ is holomorphic along finite families $z \mapsto w_0 + \sum_q z_q f_q$, vanishes at zero, depends on the source only on X with one layer of its level cubes, obeys conjugation reality, and has modulus at most $C_S e^{-\kappa d_j(X)}$. Moreover,

$$|\widetilde{Z}_\ell(w)| \leq e^{C_S \ell^3}, \qquad \widetilde{Z}_\ell(0) \geq e^{-C_g \ell^3}, \qquad |E_\ell \ell^{-3} - E_{\ell'} \ell'^{-3}| \leq C_S.$$

The vacuum splitting is equation YM3_I:(1737). The translation law in Thm. YM3_I:AJ, summed in equation YM3_I:(1736), makes $\mathcal{E}_\ell$ invariant under translations by $M\mathbb{Z}^3$.

(M2) *The stability functional.* Thm. YM3_I:AJ gives the finite history decomposition, its source-independent reference measures and evaluation domains, and one null set outside which both exponents are defined for all sources. Its equation YM3_I:(1728) is the density $\mathsf{Stay}_h^{\mathrm{T}} \chi_K(\Omega_K) e^{\mathbf{\Phi}_h^w + A^w_{K,h}}$. The local, boundary and retained terms are holomorphic in the source on these domains and obey conjugation reality. The real construction supplies the nonnegative Borel test factors and the reference measure; at a real source, the resulting history measure is positive because it is the pushforward of the positive bare density. The bare action is linear in its weight:

$$\rho_0^w(U) = e^{-A(\beta - L^K w, U)} = e^{-\beta \sum_p a_p + \mathcal{O}(w)}, \qquad A(c, U) = \sum_p c(x(p)) a_p(U), \qquad Z[w] = \int \rho_0^w \, \mathrm{d}U.$$

The source-independent coordinate maps and the measure identity are those of Thm. YM3_I:AI and Lem. YM3_I:82.5. Prop. YM3_I:83.2 identifies $E_\ell = E_{(K)}$, the source energies in equation YM3_I:(1736), and $\widetilde{Z}_\ell = \widetilde{Z}_K$ in equation YM3_I:(1737). Its equation YM3_I:(1738) bounds the sum of the normalized total variations of the history measures. This stronger estimate, not merely a bound on $|\widetilde{Z}|$, is used below.

(M3) *The count of domains.* For a domain X of level j and depth $s = K - j$, and sets Y, Y': (a) if X with one layer of cubes meets Y and Y', then $d_j(X) \geq L^s \operatorname{dist}_\infty(Y, Y')/M - 4$; (b) for $\kappa' > \log 9216$ and real D', the sum of $e^{-\kappa' d_j(X)}$ over the X that with one layer meet Y and have $d_j(X) \geq D'$ is $\leq 27 N_s(Y) S(\kappa') e^{-(\kappa' - \log 9216)[D']_+}$; (c) $N_s(Y) \leq (\ell L^s/M)^3$, and $N_s(Y) \leq (2RL^s/M + 2)^3$ if Y lies in a sup-ball of radius R, $N_s(Y)$ being the number of the cubes of side ML^{-s} of the standard partition that meet Y.

(M4) *The isometries.* The translations by $a_K\mathbb{Z}^3$ and the reflections carry the Wilson weight and the Haar measure into themselves ($\operatorname{tr} g^{-1} = \operatorname{tr} g$ on SU(2)), and $\mathcal{O}(\mathsf{r}w) = \mathcal{O}(w) \circ \mathsf{r}^{-1}$; so $Z[\mathsf{r}w] = Z[w]$ and $\|\mathsf{r}w\|_\sharp = \|w\|_\sharp$.

(M5) *The passage and labels.* For $b \in I(g)$ and $K \geq 1$, the pair $(K, L^K b)$ belongs to $\mathfrak{d}(g)$, by Cor. YM3_I:74.2. Prop. YM3_I:74.1 gives the label bounds, in particular $a_K\beta \geq g^{-2}$; their fixed compact interval is $I^{\mathrm{lab}}(g)$, equation YM3_I:1854. *The sides.* The family $\mathfrak{L}(g)$ is closed upward, and $\mathfrak{L}(g') \subset \mathfrak{L}(g)$ for $0 < g' \leq g$, as follows from its lower bound on the side.

(M6) *The chessboard estimate*, at every $\beta \geq 0$, on boards of every side $\mathsf{j} = L^i \leq \ell/2$ (Def. 13.3 (2)): $\|cF\|_{\mathrm{ch}} = |c|\,\|F\|_{\mathrm{ch}}$, $\|1\|_{\mathrm{ch}} = 1$, $\|F\|_{\mathrm{ch}} \leq \sup|F|$, the triangle inequality, and $|\langle \prod_s F_s\rangle| \leq \prod_s \|F_s\|_{\mathrm{ch}}$ for functions F_s of different cells. *The source formula*: for a complex source w that vanishes outside the interior of a cell, $\|e^{\mathcal{O}(w)}\|_{\mathrm{ch}}^N = Z_\beta[w^{\mathrm{ch}}]/Z_\beta \geq 0$, wch the sum of the mirror copies of w (conjugated in the cells of the other parity). *The domination*: for $K \in \mathbb{N}$, $\beta > 0$, $\nu \in \mathcal{G}^{\mathrm{per}}_{K,\beta}$, a board of $\mathbb{L}_K$ of any side $\geq La_K$ and any offset, functions $F_{\mathbf{k}}$ of finitely many different cells and every sequence of sides $\ell_k = 2L^{m_k} \to \infty$: $|\nu(\prod F_{\mathbf{k}})| \leq \liminf_k \prod \|F_{\mathbf{k}}\|_{\mathrm{ch},\ell_k}$.

(M7) *The transfer.* There is Q_1, depending on L and on g and on none of K, β, ℓ, such that for $(K,\beta) \in \mathfrak{d}(g)$, all $n \geq 1$, all $f_1, \dots, f_n \in \mathrm{Lip}_c(\mathbb{R}^3)$ with supports in one ball of radius 2, in any mutual position, and any two of the laws $\nu \in \mathcal{G}^{\mathrm{per}}_{K,\beta}$ and $\langle\cdot\rangle_{K,\ell,\beta}$, $\ell \in \mathfrak{L}(g)$ (the latter carried to $\mathbb{L}_K$, so that its cumulants at f are $S^T_{n;K,\ell,\beta}(f^{\langle\ell\rangle})$), the cumulants of order n differ by at most $2\log 2\,(2\mathsf{Q}_1 + 2)^n (n/\varrho)^n \prod_q \|f_q\|_\sharp$.

(M8) *The convergence on fixed tori.* For every side $\ell = 2L^m$, $n \geq 2$, real Lipschitz $f_1, \dots, f_n$ on T_ℓ, none of them 0, whose supports have no common point and the spread $\geq \mathsf{d} > 0$ (for $n = 2$: the sup-distance of the supports), and every compact interval $\mathsf{J} \subset (0,\infty)$: $S^T_{n;K,\ell,L^K b}(f_1, \dots, f_n)$ converges as $K \to \infty$ uniformly in $b \in \mathsf{J}$, and $\sup_K \sup_{b\in\mathsf{J}} |S^T_{n;K,\ell,L^K b}(f)| \leq \mathsf{c}'(\ell, n, \mathsf{J}, \mathsf{d}) \prod_i \|f_i\|_\sharp$, the right member reading of the functions the spread and the norms alone.

(M9) *Periodic Gibbs states.* The Definitions of kernels and Gibbs states: the Gibbs kernel γ_B of a finite set B of bonds is formed with the exponent $H_B = -\beta \sum_{p\in \mathrm{Pl}[B]} a_p$, $\mathrm{Pl}[B]$ the plaquettes containing a bond of B; ν is a Gibbs state if $\nu(\gamma_B F) = \nu(F)$ for all B and bounded Borel F, and periodic if it is invariant under a subgroup of finite index of the lattice translations. There is at least one periodic Gibbs state at every (a,β), and weak limits of the laws of the tori are such states; every periodic Gibbs state is invariant under the translations and the cubic group of its lattice; weak limits of periodic Gibbs states at couplings $\beta_k \to \beta$ that are invariant under the lattice translations are periodic Gibbs states at β; the tangent inequality for periodic Gibbs states; the exact dilation facts: for real $s > 0$ the dilation $\mathrm{dil}_s : x \mapsto sx$ carries the bonds, the plaquettes, the functions a_p, the product Haar measures and the glued configurations of $a\mathbb{Z}^3$ to those of $sa\mathbb{Z}^3$, with $\mathcal{O}_a(f)(U) = s\,\mathcal{O}_{sa}(f(\cdot/s))(\mathrm{dil}_s U)$, and for s a power of L it maps the periodic Gibbs states onto the periodic Gibbs states; the classes at a cutoff and the bound system at g (equation YM3_I:(2030); Section YM3_I:100), whose two members bound the cumulants of separated tuples in the members of a class and the differences of the one-point functions of two members, free of the cutoff, the coupling and the state; Prop. YM3_I:106.3: for $(K,\beta) \in \mathfrak{d}(g)$ the set $\mathcal{G}^{\mathrm{per}}_{K,\beta}$ is a class at the cutoff K that obeys the bound system at g, and with ν it contains $k\nu/\nu(k)$ for every bounded $\mathcal{I}$-measurable $k \geq 0$ with $\nu(k) > 0$; and the plateau statement for such a class (Prop. YM3_I:100.1, with the numbers of equations YM3_I:(1976)–(1980)), with its statements on conditional expectations given the invariant σ-field $\mathcal{I}$.

(M10) *Limits.* We use the Gibbs limits and their component laws in Thms. YM3_I:AP and YM3_I:AL, and the compactness, translation and cubic invariance, and coordinate reflection positivity of growing-torus limits in Props. YM3_I:92.1, YM3_I:92.3 and YM3_I:92.4. The

radial kernel and the reconstruction, including their bounds and continuity, are those stated in Def. 13.4.

(M11) *The ordered choice of constants.* Use the order in the proof of Prop. YM3_I:69.1, with the source floors of Section YM3_I:82 inserted at their stated places. Every initial segment selected by this procedure extends to a complete choice: the next free constant has either a finite lower bound, a positive upper bound or the nonempty interval specified at that step, and each determined constant is then fixed by its formula. Continue through the finite list and choose γ last. With the constants before L fixed, all sufficiently large odd L are allowed; the remaining constants are subsequently chosen in that same order.

(M12) For Prop. 13.7 alone, the following, on one torus T_λ, $\lambda = 2L^m$, at one label (a hat marks the objects of the classical starts). The theorem among the construction results on two consecutive cutoffs (Thm. YM3_I:AR), with its read-out of the one-point functions, for derivatives in the source and read here at the order 1, up to the derivative at the source 0 of the difference of its contact terms (Section YM3_I:116), and with the bound of its error, summable in K (its clause (3)). The objects of its comparison (Section YM3_I:116) for the lattices of the cutoffs K and $K+1$, marked A and B, with the small numbers $\epsilon_{\rm fg}$, $\epsilon_{\rm cl}$, $\epsilon_{\rm sh}$, $\epsilon_3 = 4\epsilon_{\rm cl} + 6\epsilon_{\rm sh}$ and the depth $u = \lceil K/1260 \rceil$ of §2.3.1 at $n = 1$, where $\epsilon_{\rm cl}, \epsilon_{\rm sh} \le \frac{1}{20}$ under the conditions of the theorem: on the inner ball In, a set of block fields of the depth u that contains the trivial block field $\mathbf{1}$, the measures of the chains and of the classical starts are a constant times the exponential of an exponent times the Haar measure, and their masses, integrals and constants satisfy $T^\bullet = e^{c^\bullet} I^\bullet$, $\hat T^\bullet = e^{\hat c^\bullet} \hat I^\bullet$ and are holomorphic in the source (Prop. YM3_I:114.2 and the proof of Prop. YM3_I:116.3); the measures of the classical starts are integrals over the block fields of a larger depth, against the weight of the run (§2.3.2), and at real sources the functions so integrated and the measures are non-negative (the proof of Prop. YM3_I:116.1).

The data of its read-out (Section YM3_I:116): $|I^\bullet/\hat I^\bullet - 1| \le 2\epsilon_{\rm fg} \le \frac{1}{4}$ and a bound of the constants of the classical starts (the proof of Prop. YM3_I:116.3), and at real sources $\hat T^{\rm A}, \hat T^{\rm B} > 0$ with $|\hat T^{\rm B}/\hat T^{\rm A} - 1| \le \epsilon_3 \le \frac{1}{2}$ (Prop. YM3_I:116.1). The discarding bound $0 \le Z - T^\bullet(0) \le \mathsf{R}_{t_1} Z$, Z the partition function of the torus at the cutoff of $\bullet$ and R_{t_1} the discarding number, one number for both cutoffs (Prop. YM3_I:115.2, the proof of Thm. YM3_I:AR), with $\mathsf{R}_{t_1} \le \frac{1}{2}$ under the conditions of the theorem.

The number $\mathfrak{o}_K = 8\mathsf{R}_{t_1} + 12\epsilon_{\rm fg} + 8\epsilon_{\rm cl} + 12\epsilon_{\rm sh}$ and, for an interval J of labels of the theorem on the side λ (§2.3.1; every compact interval $\mathsf{J} \subset I(g)$ is one on every side of $\mathfrak{L}(g)$, as $\min \mathsf{J} > g^{-2} + \bar B \ge \gamma^{-2}$, $\max \mathsf{J} \le X_g - \bar B \le L(g^{-2} + \bar B)$ and $M\,R(\max \mathsf{J}) \le M\,R(L(g^{-2} + \bar B)) < L^{m_{\mathfrak{L}}(g)}$ by equation YM3_I:(1528), the cube-size function R being non-decreasing), its number $\mathsf{C}_{\rm bf}(\mathsf{J})$, formed from the constants and the ends of J, its exponent $\mathfrak{p}_0$ and its thresholds $K_{\rm bf}(\lambda, \mathsf{J})$: for every label of J and every $K \ge K_{\rm bf}(\lambda, \mathsf{J})$ the conditions of the theorem hold and $\mathfrak{o}_K \le \mathsf{C}_{\rm bf}(\mathsf{J})\lambda^3 K^{\mathfrak{p}_0} L^{-K/10}$, $\mathfrak{p}_0 = 2\mathfrak{p}_0 + \frac{1}{2}$ (Def. 2.7 (4), (5), Cor. 2.8 (a)); with these conditions the setting (Th) among the construction results holds for the run of each of the two cutoffs on the torus (§2.3.1).

Its comparison of the two classical runs: their masses and their densities in L^1 (Prop. YM3_I:116.1 and its proof); the pointwise quotient of the two classical starts at a real source, a factor in $[e^{-\epsilon_{\rm cl}}, e^{\epsilon_{\rm cl}}]$ where both weights are positive, the two weights being equal there (Prop. YM3_I:116.1); and the bound, by $\epsilon_{\rm sh}$ times its mass, of the part of one classical run where the weight of the other vanishes (the positive-kernel argument in the proof of Prop. YM3_I:116.1), which bounds the tails of a classical run by multiples of its measures as an order of measures, against every bounded Borel function of the block field. The exponent and the constant of a step of a run (Prop. YM3_I:114.2), and the action of a bare array, in which every member enters less its value at the trivial pair (equation YM3_I:(874)); the

one value and the continuity of the weighted action of the background (the gauge-invariance argument preceding equation YM3_I:(2133) and the local analytic-background argument in the proof of Prop. YM3_I:52.3), the continuous choice of the background near a regular block field (Thm. YM3_I:E and its real-analytic continuation), and the one value of the action of the array on the minimal orbit (Props. YM3_I:114.2 and YM3_I:113.13); a local term is a holomorphic, so continuous, function of the pair that it reads, on an open set of pairs (Lem. 5.26), and the pair of a configuration on a domain is a polynomial map of its matrix entries (§5.2, YM3_I:(70), YM3_I:(69)).

The line $x_j = L^{K-j}b$, $0 \le j \le K$, of a label, whose increments vanish, is an admissible finite sequence with last index K, which with the integer m of a side $2L^m$ forms admissible data, and its cut $(x_0, \dots, x_{K-1})$ with the lattice (λ, K) and these couplings forms a frame of level $K-1$ among the construction results (the proof of Lem. YM3_I:118.1). The family of the level j of the chain at the source 0 is the family of the level j of the canonical progression of the cut $(x_0, \dots, x_{K-1})$ of the line $x_j = L^{K-j}b$ of the label (the proof of Lem. YM3_I:118.1); at that frame the members of the levels $j \le K$ of this progression are those of the progression of the line, and they are regular families on every lattice of their level, with their companions (Thm. YM3_I:Y).

The exact step identity eq. 13.ii identifies the logarithm of the Gaussian normalizing factor with the extensive term and the sum of its local Gaussian constants. Adding the values $E^{(j+1)}(X, \mathbf{1}, z)$ of the regular family gives the vacuum number $\mathsf{E}^{(j)}(T_\ell, T_\ell)$ of that step. Their sum through step n is $E_{(n)}$, as in eq. 13.i; these definitions agree with equations YM3_I:(1406)–(1407). Lem. 13.31 proves the exact identity and the comparison needed below.

For the comparison on two tori we use the same admissible sequence and its regular families, with their common companions, on both sides. The real vacuum families and the rate conditions are those stated in Lem. 13.31. The pair of sides is admitted once $LMR(x_K) \le L^{m_i}$ for $i = 1, 2$: the retained room condition [7, eq. 215] gives $C_L \le 2LR(\gamma^{-2}) \le 2LR(x_K)$, so $\ell_i = 2L^{m_i} \ge 2LMR(x_K) \ge C_L M$. For the line $x_j = L^{K-j}b$ of a label $b \in \mathsf{J}$, the last radius is $R(b)$ (equation YM3_I:(634)), and on the sides of $\mathfrak{L}(g)$ the required room is $LMR(b) \le LMR(L(g^{-2} + \bar{B})) \le L^{m_{\mathfrak{L}}(g)} \le L^{m_i}$, by equation YM3_I:(1528).

(M13) For Thm. AN alone: the consequences of the hypothesis K1 proved in §17.2 on one inequality between the partition functions of two periodic boxes. Let $0 < g \le \gamma$, $\mathsf{b} \in I(g)$, let K1$[\mathsf{b}, \ell_\circ, K_\circ; x_\circ]$ hold, and let $K \ge \max\{K_\circ, 1\}$ and $\beta = L^K \mathsf{b}$. (i) For every side $\ell = 2L^m \ge \ell_\circ$: $0 \le \log Z_{K,\ell,\beta} - \ell^3 p_{K,\infty}(\mathsf{b}) < 0.13736\, \theta_\circ^{\ell/\ell_\circ}$, the bound being the same at every such cutoff. (ii) There is exactly one periodic Gibbs state ν_K at (K, β). (iii) Over every cross-section of physical sides $\ge \ell_\circ$ the largest eigenvalue λ_0 of the transfer operator defined in §17.2 is simple, and the next one, λ_1, has $-a_K^{-1} \log(\lambda_1/\lambda_0) > m_*$. (iv) For $n \ge 2$, a side $\ell_1 = 2L^{m_1} \ge \ell_\circ$ and $X = (f_1, \dots, f_n)$ of real Lipschitz functions, none of them 0, with pairwise disjoint compact supports in $(-\ell_1/4, \ell_1/4)^3$, there is a number $C(X)$ with $|S_n^T[\nu_K](X) - S^T_{n;K,\ell_2,\beta}(X^{\langle \ell_2 \rangle})| \le C(X)\, \theta_\circ^{\ell_2/(2\ell_\circ)}$ for all $K \ge \max\{K_\circ, 1\}$ and all sides $\ell_2 = 2L^{m_2} \ge \ell_1$. (v) The Gibbs sequence $(K, L^K \mathsf{b}, \nu_K)_{K \ge \max\{K_\circ, 1\}}$ converges; its limit σ is its own single component; $\ker H = \mathbb{C}\Omega$ in $\mathcal{H}(\sigma, e_3)$; and the spectrum of H lies in $\{0\} \cup [m_*, \infty)$.

(M14) For clause (f) of Thm. AM alone: the power of the jump of the slope of the pressure that §12.1 on the latent heat of the SU(2) Wilson model on $\mathbb{Z}^3$ proves for two frames at one radius with the block sizes L_1 and $L_2 = L_1 + 2$, under the following hypotheses. Thms. YM3_I:AJ and YM3_I:AD are granted at both frames, and Thm. YM3_I:D at their choices and at a lowered choice of each. In each frame $i \in \{1, 2\}$ the far part of the specific heat settles in the cutoff at a rate on every torus: with the window $\mathcal{W}_i$, the tent $\chi^{(i)}$ and the plateau $\eta^{(i)}$ of half-width $1/L_i$ about $(\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$ of clause (f), and the far part

$\mathsf{y}^{(i)}_{m,K}(b) = b^5 L_i^3\, S^T_{2;K,\ell,L_i^K b}((\chi^{(i)})^{\langle\ell\rangle}, 1-(\eta^{(i)})^{\langle\ell\rangle})$ of the specific heat of T_ℓ, $\ell = 2L_i^m$, there are numbers $\mathsf{k}_i \geq 0$ and $\mathsf{c}_{\mathrm{y},i} < \infty$, free of m, K and b, such that for all integers $m \geq 1$, all $b \in \mathcal{W}_i$ and all integers $K' \geq K \geq \mathsf{k}_i + 54m$

$$|\mathsf{y}^{(i)}_{m,K'}(b) - \mathsf{y}^{(i)}_{m,K}(b)| \leq \mathsf{c}_{\mathrm{y},i}\, \ell^9 K^{\mathsf{p}_{2,i}} L_i^{-K/10}, \qquad \mathsf{p}_{2,i} = 6\mathfrak{p}_{0,i} + \tfrac{1}{2},$$

$\mathfrak{p}_{0,i}$ the exponent of the theorem on two consecutive cutoffs (Thm. YM3_I:AR) in the frame i. Under these hypotheses, for every real $A > 90$ there are numbers C_{r} and β_1 with $\beta^4 J(\beta) \leq \mathsf{C}_{\mathrm{r}} \beta^{-1/(2A)}$ for all $\beta \geq \beta_1$.

The proof of Prop. 13.7 (c) derives from (M12), in its Step 3 and on one torus T_λ, four statements: (i) the read-out at the order 0; (ii) the comparison of the two classical starts as measures on the inner ball; (iii) the cancelling of the constants of the two classical starts, from the value 0 and the continuity of their exponents at the trivial configuration; and (iv) the identity of the sum of the constants of the chain at the source 0 with a vacuum energy of the comparison of two tori, a real number. We use the following standard results: Cauchy's formula and estimates, Schwarz's lemma, the maximum principle, Poisson's formula, the inequalities of Jensen, Hölder and Cauchy–Schwarz, the theorems of Tonelli and Fubini, Morera's theorem, Kronecker's theorem (the multiples of an irrational number are dense modulo 1), Helly's selection theorem, Bernstein's theorem on completely monotone functions and his inequality for the derivatives of polynomials, the spectral theorem, Stolz's theorem, and Hölder's inequality in the Schatten classes, which in the one case used is proved in Prop. 13.15.

13.1.5. *Estimates and proofs.* Up to the window inequality the argument runs in one frame. The energies averaged over the isometries and their cross terms (Lem. 13.9) give the energies of a constant source on two sides on a complex disc (Lem. 13.10); finite size (Lem. 13.11) is exact; the stability functional (Lem. 13.12) gives the imaginary direction (Prop. 13.13), which with the cross terms and the chessboard estimate gives the walls (Prop. 13.14); these four give the main inequality and semi-convexity (Prop. 13.16); with Cauchy's estimate in the chessboard seminorm (Lem. 13.17) the main inequality gives the reduction on tori (Prop. 13.18), and with finite size and the domination the window inequality in every periodic Gibbs state (Prop. 13.19). The bound against piles of specific heat (Prop. 13.23) uses semi-convexity and finite size in both frames, the transfer, the convergence on fixed tori and dilations by real factors (Lems. 13.20–13.22). Clauses (a), (b) of Thm. AL follow from the reduction and this bound; clauses (c), (d) from (a), (b) and the theorems on limits through Prop. 13.25 and Lems. 13.26–13.28; clause (e) through Lem. 13.24.

Lemma 13.9 (The energies averaged over the isometries; cross terms)**.** Let $(K, \beta) \in \mathfrak{d}(g)$ and $\ell \in \mathfrak{L}(g)$.

(A1) $\mathcal{E}^{\mathrm{av}}$ is holomorphic along finite families, $\mathcal{E}^{\mathrm{av}}(0) = 0$, $\mathcal{E}^{\mathrm{av}}(\bar{w}) = \overline{\mathcal{E}^{\mathrm{av}}(w)}$, $\mathcal{E}^{\mathrm{av}}(\mathsf{r}w) = \mathcal{E}^{\mathrm{av}}(w)$ for $\mathsf{r} \in \mathsf{G}$, $\mathcal{E}^{\mathrm{av}}(z \cdot 1) = \mathcal{E}(z \cdot 1)$ for complex $|z| < 2\varrho$, and $|\widetilde{Z}^{\mathrm{av}}(w)| = \prod_{\mathsf{r}\in\mathsf{G}} |\widetilde{Z}(\mathsf{r}w)|^{1/|\mathsf{G}|} \leq e^{C_S \ell^3}$.

(A2) Let $\mathcal{F} \in \{\mathcal{E}, \mathcal{E}^{\mathrm{av}}\}$, and let $u, w_1, \dots, w_n$ be complex Lipschitz functions on T_ℓ such that the support of w_r is covered by n_r closed sup-balls of radius R_0, the supports of different w_r are at sup-distance $\geq \Delta \geq 4M$, and $\|u\|_\sharp$, $\|u + w_r\|_\sharp$, $\|u + \sum_r w_r\|_\sharp < 2\varrho$. With $\mathcal{F}'(w) = \mathcal{F}(u+w) - \mathcal{F}(u)$, the cross term $\mathrm{cr} = \mathcal{F}'(\sum_r w_r) - \sum_r \mathcal{F}'(w_r)$ satisfies $|\mathrm{cr}| \leq C_\times (R_0/\Delta) \sum_r n_r$.

(A3) If the w_r depend linearly on $\zeta \in \mathbb{C}^k$ and (A2) applies for all ζ with $\max_i |\zeta_i| < \rho_3$, then $|\mathrm{cr}(\zeta)| \leq C_\times (R_0/\Delta)(\sum_r n_r)(\max_i |\zeta_i|/\rho_3)^2$.

These bounds are free of K; what grows with K is only the bound of $\mathcal{E}$ itself. ■

Sketch of proof. (A1) By (M4), $Z[w] = Z[\mathsf{r}w] = e^{E + \mathcal{E}(\mathsf{r}w)} \widetilde{Z}(\mathsf{r}w)$ for every $\mathsf{r} \in \mathsf{G}$; the geometric mean of the moduli over G gives $|Z[w]| = e^{E + \mathrm{Re}\, \mathcal{E}^{\mathrm{av}}(w)} \prod_{\mathsf{r}} |\widetilde{Z}(\mathsf{r}w)|^{1/|\mathsf{G}|}$, which is the formula for $|\widetilde{Z}^{\mathrm{av}}(w)|$, bounded by (M1). Holomorphy, the value at 0 and the conjugation pass from $\mathcal{E}$ to its mean; the

invariance holds because G is a group; constants are fixed by every r. (A2) It suffices to take $\mathcal{F}=\mathcal{E}$, since the isometries preserve the hypotheses; then $\mathrm{cr}=\sum_{j,X}\mathsf{b}(X)$ with

$$\mathsf{b}(X)=e_j(X;u+\textstyle\sum_r w_r)-e_j(X;u)-\sum_r\big[e_j(X;u+w_r)-e_j(X;u)\big].$$

By locality $\mathsf{b}(X)=0$ unless X with one layer of cubes meets at least two of the supports; then $\mathsf{b}(X)$ has at most $3\#R(X)$ terms that do not cancel, $R(X)$ the set of the supports met, each $\le C_S e^{-\kappa d_j(X)}$, and $d_j(X)\ge y_s-4$ with $y_s=L^s\Delta/M\ge 4L^s$ by (M3)(a). The count of domains at Y one of the n_r balls, with $\kappa-\log 9216>2$, bounds the sum over such X by $n_r\cdot 27S(\kappa)(2R_0L^s/M+2)^3e^{-2(y_s-4)}$; with $(2R_0L^s/M+2)^3\le 32[(R_0/\Delta)^3y_s^3+1]$, $\sum_{s\ge0}y_s^3e^{-2(y_s-4)}\le 65$ and $\sum_s e^{-2(y_s-4)}\le 2$ one obtains $|\mathrm{cr}|\le 3\cdot 27\cdot 32\,[65(R_0/\Delta)^3+2]C_SS(\kappa)\sum_r n_r$, and $3\cdot27\cdot32=2592$. (A3) $\mathrm{cr}(\zeta)$ is holomorphic on the polydisc, vanishes at 0 with its first derivatives, since $D\mathcal{F}'(0)$ is linear; the maximum principle for $t\mapsto \mathrm{cr}(t\zeta^0)/t^2$ gives the claim. □

Lemma 13.10 (Two sides at a constant source, on a complex disc)**.** Let $(K,\beta)\in\mathfrak{d}(g)$, $K\ge1$, and let $\ell_1,\ell_2\in\mathfrak{L}(g)$ with $\ell_1,\ell_2\ge 2\mathsf{l}_1$; let h_1, A_1 be the numbers of Def. 13.6 formed at g. The function

$$d(z)=\ell_1^{-3}\mathcal{E}_{\ell_1}(z\cdot1)-\ell_2^{-3}\mathcal{E}_{\ell_2}(z\cdot1)$$

is holomorphic on $|z|<2\varrho$, real on the real axis, $d(0)=0$, and $|d(z)|\le\mathsf{A}_1$ for all complex $|z|\le h_1$. Hence, with $\mu'=d'(0)\in\mathbb{R}$, $|d(x)-\mu' x|\le 2\mathsf{A}_1x^2/h_1^2$ for real $|x|\le h_1/2$ and $|d''(0)|\le 2\mathsf{A}_1/h_1^2$. The numbers h_1, A_1 are free of K, β, ℓ_1 and ℓ_2. ■

Sketch of proof. Write $z\cdot1=\sum_u z\chi_u$ with the tents χ_u of half-width $\varsigma=1/L$ about the points $u\in\varsigma\mathbb{Z}^3$; $\|\chi_u\|_\sharp\le\mathsf{c}_\chi$. Call a domain of depth s large if $d_j(X)\ge A_s-4$, $A_s=L^{s-1-m'}$, and small otherwise; by (M3)(a) a small domain cannot meet two sets at sup-distance $\ge\varsigma$. The large part has a density bounded on the whole disc, torus by torus: $\ell^{-3}|\mathcal{E}_\ell^{\mathrm{lg}}(z\cdot1)|\le 27C_SS(\kappa)L^3\sum_s A_s^3e^{-2[A_s-4]_+}\le C_{\mathrm{lg}}$. For the small part, the invariance of $\mathcal{E}^{\mathrm{av}}$ under the translations by $\varsigma\mathbb{Z}^3$ and the chain rule give

$$\ell^{-3}\partial_z^n\mathcal{E}_\ell^{\mathrm{sm}}(z\cdot1)\big|_0=L^3\sum_{\mathrm{near}}D^n\mathcal{E}_\ell^{\mathrm{sm,av}}[\chi_0,\chi_{v_2},\dots,\chi_{v_n}],$$

the sum over the at most 125^{n-1} near tuples, $|v_q-v_{q'}|_\infty\le 2\varsigma$, the same index set on both tori (here $\ell_1,\ell_2\ge2\mathsf{l}_1$ is used: the 125 points are distinct points of each torus). At a near tuple $D^n\mathcal{E}^{\mathrm{av}}$ is a joint cumulant minus a derivative of $\log G_\ell$, $G_\ell(\zeta)=\widetilde{Z}_\ell^{\mathrm{av}}(w)/\widetilde{Z}_\ell(0)$, $w=\sum\zeta_q\chi_{v_q}$. The chessboard estimate on the board of side l_1 (an even number ≥2 of cells in each direction), with the source formula and the cross terms of Lem. 13.9 at $R_0/\Delta\le1/8$, spends the extensive bounds of (M1) on one cell and gives $|G_\ell|\le\mathsf{Q}_2$; Schwarz's lemma and Cauchy's estimate give $|\partial^n\log G_\ell(0)|\le\log2\,(n\mathsf{c}_\chi(2\mathsf{Q}_2+2)/\varrho)^n$; the large part at a near tuple is $\le C'_{\mathrm{lg}}(n\mathsf{c}_\chi/\varrho)^n$ by Cauchy; and the cumulants on the two tori differ by the transfer (M7) by at most $2\log2\,(2\mathsf{Q}_1+2)^n(n\mathsf{c}_\chi/\varrho)^n$. Altogether, with $n^n\le e^nn!$,

$$\frac{|\partial_z^nd^{\mathrm{sm}}(0)|}{n!}\le\frac{L^3}{125}(4\log2+2C'_{\mathrm{lg}})\Big(\frac{125e\,\mathsf{c}_\chi\mathsf{Q}_*}{\varrho}\Big)^n=\frac{L^3}{125}(4\log2+2C'_{\mathrm{lg}})(2h_1)^{-n},$$

which sums on $|z|\le h_1$; with the large part, $|d|\le\mathsf{A}_1$. The forms at real x follow from Cauchy's estimate on $|z|=h_1$; d is real on the real axis by the conjugation in (M1). □

Lemma 13.11 (Finite size, exactly)**.**

(a) For even integers $n_1,n_2,n_3\ge2$ and $\beta\ge0$ let $Z(n_1,n_2,n_3)$ be the partition function of Wilson's action on the periodic box $\mathbb{Z}^3/(n_1\mathbb{Z}\times n_2\mathbb{Z}\times n_3\mathbb{Z})$ and $\mathsf{g}=\log Z/(n_1n_2n_3)$. Then g does not increase in each argument; $P(\beta)=\inf\mathsf{g}=\lim_n\mathsf{g}(n,n,n)$, the limit along the even n, is convex in β and $P\ge-6\beta$; and for even $\mathsf{N}\ge e_0\ge2$

$$0\le\mathsf{g}(\mathsf{N},\mathsf{N},\mathsf{N})-P\le3(e_0/\mathsf{N})\big[\mathsf{g}(e_0,e_0,e_0)-P\big].$$

(b) For every block size L, every $K \geq 0$, $b \geq 0$ and sides $\ell \geq \ell'$ of the form $2L^m$: $p_{K,\infty}(b) = \lim_\ell p_{K,\ell}(b) = L^{3K} P(L^K b)$ with the one function P of (a), $D_K \leq D_{K,\ell} \leq 0$, and $0 \leq p_{K,\ell} - p_{K,\infty} \leq 3(\ell'/\ell)(p_{K,\ell'} - p_{K,\infty})$.

(c) In a frame, for $b \in I(g)$, $K \geq 1$ and $\ell \geq \ell'$ in $\mathfrak{L}(g)$: $-B' \leq D_K(b) \leq D_{K,\ell}(b) \leq 0$ and $0 \leq D_{K,\ell}(b) - D_K(b) \leq \bar\eta(\ell)$.

Parts (a) and (b) are exact statements on one lattice, at every $\beta \geq 0$. ■

Sketch of proof. Fix even n_2, n_3 and let $K_n(\phi, \phi')$ be the slab kernel: the integral, over the bonds of the slab $\{0 \leq x_1 \leq n\}$ that lie in neither boundary plane, of $\exp[-\beta(\sum' a_p + \frac12 \sum_{x_1=0} a_p + \frac12 \sum_{x_1=n} a_p)]$, the bonds of the two planes being fixed to ϕ, ϕ'. It is continuous with values in $(0,1]$, symmetric (the reflection $\vartheta : x_1 \mapsto n - x_1$ maps the slab onto itself, preserves the Haar measure and carries a_p to $a_{\vartheta p}$, because $\operatorname{tr} g^{-1} = \operatorname{tr} g$ on SU(2)) and multiplicative, $K_{n+m}(\phi, \phi'') = \int K_n(\phi, \phi') K_m(\phi', \phi'')\, \mathrm{d}\phi'$ (Fubini; the plaquettes of the middle plane receive $\frac12 + \frac12$). With $z(n) = Z(n, n_2, n_3) = \int K_n(\phi, \phi)\, \mathrm{d}\phi$ one has $z(n+m) = \langle K_n, K_m \rangle_{L^2}$ and $\|K_n\|_2^2 = z(2n)$, so the Cauchy–Schwarz inequality gives

$$z(n+m)^2 \leq z(2n) z(2m), \qquad z(4k) \leq z(2k)^2.$$

The first, at $n = e/2 - 1$, $m = e/2 + 1$, makes $\varphi(e) = \log z(e) \leq 0$ convex on the even e; its slopes increase to a finite $\Lambda_\infty \leq 0$, and $\varphi(e)/e \to \Lambda_\infty$ by Stolz. The excess $\operatorname{exc}(e) = \varphi(e) - e\Lambda_\infty$ does not increase, and it is ≥ 0 because $\varphi(2^k e)/2^k e$ does not increase by the second inequality; so $\varphi(e)/e = \Lambda_\infty + \operatorname{exc}(e)/e$ does not increase. This in each direction gives the monotonicity and $P = \inf \mathsf{g} = \lim \mathsf{g}(n,n,n)$, convex as a limit of convex functions and $\geq -6\beta$ since $a_p \leq 2$ and there are three plaquettes to a site. For the rate, $\operatorname{exc}(\mathsf{N}) \leq \operatorname{exc}(e_0) \leq e_0 n_2 n_3 [\mathsf{g}(e_0, e_0, e_0) - P]$ gives $\mathsf{g}(\mathsf{N}, n_2, n_3) \leq \mathsf{g}(n, n_2, n_3) + e_0[\mathsf{g}(e_0, e_0, e_0) - P]/\mathsf{N}$ for all even n; applied in the three directions in turn, and letting $n \to \infty$, this is (a).

For (b), T_ℓ at the cutoff K is the box with $\mathsf{N} = \ell L^K$ sites a side, $p_{K,\ell}(b) = L^{3K} \mathsf{g}(\mathsf{N}, \mathsf{N}, \mathsf{N})$ at $\beta = L^K b$, and $e_0/\mathsf{N} = \ell'/\ell$ for $e_0 = \ell' L^K$; the limit P names no block size. For (c), by the passage and (M1), $p_{K,\ell} = \ell^{-3}[E_\ell + \log \widetilde{Z}_\ell(0)] \in \ell^{-3} E_\ell + [-C_g, C_S]$, and $|\ell^{-3} E_\ell - \ell'^{-3} E_{\ell'}| \leq C_S$ gives $D_{K,\ell} \geq -B'$ for all $\ell \in \mathfrak{L}(g)$, hence $D_K \geq -B'$; enter $-D_K \leq B'$ in (b). □

Lemma 13.12 (The third member of the stability chain as a functional of the source)**.** Let $(K, \beta) \in \mathfrak{d}(g)$ and $\ell \in \mathfrak{L}(g)$. There are a positive σ-finite measure space $(\Xi, \mathfrak{M})$ and functions Ψ_ϖ, $\varpi \in \Xi$, on the class of the sources, neither Ξ nor $\mathfrak{M}$ depending on the source, such that with $h_\varpi = \operatorname{Re} \Psi_\varpi$:

(T1) for every source w, $\varpi \mapsto e^{\Psi_\varpi(w)}$ is measurable for the completion of $\mathfrak{M}$, $\widetilde{Z}(w) = \int e^{\Psi_\varpi(w)}\, \mathrm{d}\mathfrak{M}(\varpi)$, and

$$|\widetilde{Z}(w)| \leq \mathsf{T}(w) := \int e^{h_\varpi(w)}\, \mathrm{d}\mathfrak{M}(\varpi) \leq e^{C_S \ell^3};$$

(T2) $\mathsf{T}(u) = \widetilde{Z}(u) > 0$ at every real source u;

(T3) for every ϖ and every finite family $f_1, \dots, f_n$, the map $z \mapsto e^{\Psi_\varpi(\sum_q z_q f_q)}$ is holomorphic and without zeros on the convex open set $\{z : \|\sum_q z_q f_q\|_\sharp < 2\varrho\}$; so $z \mapsto h_\varpi(\sum_q z_q f_q)$ is the real part of a holomorphic function there;

(T4) for real u, v there is an $\mathfrak{M}$-null set, depending on the pair, off which $h_\varpi(u + \bar\zeta v) = h_\varpi(u + \zeta v)$ for all ζ with $\|u + \zeta v\|_\sharp < 2\varrho$; and $(\varpi, z) \mapsto h_\varpi(w(z))$ is jointly measurable.

The functional T is the normalized total-variation sum in equation YM3_I:(1738), at $k = K$. ■

Sketch of proof. Take $k = K$. By Thm. YM3_I:AJ, there are finitely many histories and a common source-independent null set N_h for each history. Outside N_h, let D_w be its density: it is zero where $\mathsf{Stay}_h^{\mathrm{T}} \chi_K(\Omega_K) = 0$ and otherwise equals $\mathsf{Stay}_h \exp(\mathbf{\Phi}_h^w + A_{K,h}^w)$. Put $\Xi_h = \{\mathsf{D}_0 \neq 0\}$. This set is the same for all sources, since the tests are unchanged and an exponential has no zero. On it put $d\mathfrak{M}_h = \mathsf{Stay}_h\, \mathrm{d}\mathcal{M}_h$. The reference measure is a finite product of Haar and Lebesgue measures, hence

is σ-finite; multiplication by the nonnegative finite Borel test factor preserves this property. Neither this measure nor Ξ_h depends on the source.

Histories with $\mathfrak{M}_h(\Xi_h) = 0$ contribute zero at every source and may be omitted. Take the disjoint union of the remaining Ξ_h, with measure $\mathfrak{M}_h$ on each piece, and set

$$\Psi_{(h,\phi)}(w) = \mathbf{\Phi}^w_h(\phi) + A^w_{K,h}(\phi) - E^w_{(K)}.$$

The total variation of $\mathsf{D}_w\mathcal{M}_h$ is $|\mathsf{D}_w|\mathcal{M}_h$. Thus the integral of $e^{\operatorname{Re}\Psi}$ is exactly the normalized total-variation sum in equation YM3_I:(1738), which proves the two inequalities in (T1). Without moduli, the history decomposition in Thm. YM3_I:AJ and the vacuum splitting in equation YM3_I:(1737) give $\int e^{\Psi(w)} d\mathfrak{M} = e^{-E^w_{(K)}} Z[w] = \widetilde{Z}(w)$.

For a real source, the bare density is positive, so each history measure is positive; the vacuum constant is real by conjugation. This proves (T2). To prove (T3), fix a history point and a source-independent evaluation representative, as supplied by Thm. YM3_I:AJ. Every argument in its exponent is fixed as the source varies. The active action is linear in w, and the regular, boundary and retained terms are finite sums of the holomorphic families in that theorem. The whole-torus vacuum constant has the holomorphic finite-sum splitting in Prop. YM3_I:83.2.

The one member that no clause covers, the constant $c_h(w) = \exp(-\mathcal{E}^w_{K,\mathcal{S}(h)})$ of the label, which does not depend on ϕ, is holomorphic as a quotient: on measurable sets $B_n \uparrow \Xi_h$ of finite measure on which the rest $e^{\Psi^\circ_\varpi}$ of the exponential is bounded near z_0, the map $z \mapsto \int F_0 1_{B_n}\, \mathrm{d}\hat{\tau}^{\mathrm{co}\prime}_h[\rho_0^{w(z)}]$ is holomorphic by the bare-integral formula and holomorphy assertion of Prop. YM3_I:73.5, and it equals $e^{E^{w(z)}_{(K)}}\, c_h(w(z)) I_n(z)$ with $I_n(z) = \int_{B_n} F_0 e^{\Psi^\circ_\varpi(w(z))} d\mathfrak{M}$ holomorphic by Morera and Fubini, and $I_n(z_0) = \int_{B_n} |e^{\Psi^\circ_\varpi(w(z_0))}|\, \mathrm{d}\mathfrak{M} > 0$ for all large n when F_0 is the conjugate phase of the integrand at z_0, because the integrand has no zero, $B_n \uparrow \Xi_h$ and $\mathfrak{M}(\Xi_h) > 0$.

So $e^{\Psi_\varpi(w(z))}$ is holomorphic without zeros on a convex set and has a holomorphic logarithm, whose real part is $h_\varpi(w(z))$. (T4) follows from conjugation of the history measures in Thm. YM3_I:AJ and of the vacuum splitting in Prop. YM3_I:83.2, first at the points ζ with rational coordinates and then by continuity; joint measurability from measurability in ϖ and continuity in z. □

Proposition 13.13 (The imaginary direction; a floor; Harnack; Hölder over several directions)**.** Let $(K,\beta) \in \mathfrak{d}(g)$ and $\ell \in \mathfrak{L}(g)$, and let $u, v, v_1, \dots, v_n$ be real Lipschitz functions on T_ℓ; $\|\cdot\| = \|\cdot\|_\sharp$.

(P1) If $\|u\| + \|v\| < 2\varrho$: $|\widetilde{Z}(u+iv)| \le \mathsf{T}(u+iv) \le \widetilde{Z}(u)^{1-\vartheta} e^{\vartheta C_S \ell^3}$ with $\vartheta = 2\|v\|^2/((2\varrho - \|u\|)^2 + \|v\|^2)$.

(P2) If $\|u\| + \|v\| < 2\varrho$: $\widetilde{Z}(u+v) \le \widetilde{Z}(u)^{1-\vartheta_1} e^{\vartheta_1 C_S \ell^3}$ with $\vartheta_1 = 2\|v\|/(2\varrho - \|u\| + \|v\|)$.

(P3) If $\rho > 0$, $\|u\| + \rho \sum_i \|v_i\| < 2\varrho$ and $0 \le nt \le \rho/4$: $\widetilde{Z}(u + t\sum_i v_i) \le \prod_i \widetilde{Z}(u + ntv_i)^{(1-\vartheta'')/n} e^{\vartheta'' C_S \ell^3}$ with $\vartheta'' = 8n^2t^2/\rho^2$.

For $\|u\| \le \varrho/2$: $\log \widetilde{Z}(u) \ge -(2C_g + C_S)\ell^3$ (the floor); if moreover $\|u\| + \|v\| < 2\varrho$, then

$$\log|\widetilde{Z}(u+iv)| \le \log \widetilde{Z}(u) + C_I' \|v\|^2 \ell^3 \quad \text{(IM)}, \qquad \log \widetilde{Z}(u+v) - \log \widetilde{Z}(u) \le \tfrac{8}{3}(C_S + C_g)\varrho^{-1}\|v\|\ell^3 \quad \text{(the Harnack bound)},$$

and $\partial_t^2 \log \widetilde{Z}(u+tv)|_{t=0} \ge -2C_I'\|v\|^2\ell^3$ (semi-convexity). All of this holds for $\widetilde{Z}^{\mathrm{av}}$ in the place of $\widetilde{Z}$. C_I' is free of K, β, ℓ and of the sources. ■

Sketch of proof. For the Poisson kernel $\mathsf{P}_\rho(\zeta_0;\phi) = (\rho^2 - |\zeta_0|^2)/|\rho e^{i\phi} - \zeta_0|^2$ one has (i) $\frac{1}{2}[\mathsf{P}_\rho(iy;\phi) + \mathsf{P}_\rho(iy;-\phi)] \ge 1 - 2y^2/(\rho^2+y^2)$ for $0 \le y < \rho$; (ii) $\mathsf{P}_\rho(s;\phi) \ge 1 - 2s/(\rho+s)$ for $0 \le s < \rho$; (iii) $\prod_{i\le n} \mathsf{P}_\rho(t;\phi_i) \ge (1 - 8n^2t^2/\rho^2)\, n^{-1} \sum_i \mathsf{P}_\rho(nt;\phi_i)$ for $0 \le nt \le \rho/4$; (i), (ii) are elementary, and (iii) follows from $\log \mathsf{P}_\rho(t;\phi) \ge \log(1-\tau^2) + 2\tau\cos\phi - \tau^2/(1-\tau)$ and $n^{-1}\sum_i \mathsf{P}_\rho(nt;\phi_i) \le (1+\mathsf{y})[1 + \frac{16}{3}n^2\tau^2]$, $\tau = t/\rho$, $\mathsf{y} = 2\tau\sum_i \cos\phi_i$. Thus each kernel is $(1-\vartheta)$ times the reference kernel plus ϑR with $R \ge 0$ of mean 1. For (P1) take $1 < \rho < (2\varrho - \|u\|)/\|v\|$; by (T3), (T4) of Lem. 13.12, off a null

set $G(\zeta) = h_\varpi(u + \zeta v)$ is harmonic near $|\zeta| \le \rho$ and $G(\bar\zeta) = G(\zeta)$. Poisson's formula at $\zeta = i$, made symmetric, gives $G(i) = (1-\vartheta)G(0) + \vartheta \int R G(\rho e^{i\phi})\,\mathrm{d}m$; Jensen's inequality, integration in ϖ, Hölder's inequality with the exponents $1/(1-\vartheta)$, $1/\vartheta$ and Tonelli give

$$\mathsf{T}(u+iv) \le \mathsf{T}(u)^{1-\vartheta}\Big(\int R\,\mathsf{T}(u+\rho e^{i\phi}v)\,\mathrm{d}m\Big)^{\vartheta} \le \widetilde{Z}(u)^{1-\vartheta} e^{\vartheta C_S \ell^3}$$

by (T1), (T2); let ρ increase to its bound. (P2) is the same at $\zeta = 1$ with (ii), and (P3) the same in n variables with (iii), Hölder's inequality now with n exponents $n/(1-\vartheta'')$ and one $1/\vartheta''$. The floor is (P2) at the base u and the direction $-u$, where $\vartheta_1 = \|u\|/\varrho \le \frac12$, with $\widetilde{Z}(0) \ge e^{-C_g\ell^3}$ from (M1). (IM): $\vartheta \le 8\|v\|^2/(9\varrho^2)$ and $C_S\ell^3 - \log\widetilde{Z}(u) \le 2(C_S + C_g)\ell^3$ by the floor; the Harnack bound likewise with $\vartheta_1 \le 4\|v\|/(3\varrho)$. Semi-convexity: $\mathsf{h}(\zeta) = \log\widetilde{Z}(u+\zeta v)$ is holomorphic near 0 and real on the real axis, and $\operatorname{Re}\mathsf{h}(iy) - \mathsf{h}(0) = -\frac12\mathsf{h}''(0)y^2 + O(y^4) \le C_I'\|v\|^2y^2\ell^3$ by (IM). For $\widetilde{Z}^{\mathrm{av}}$ each inequality is one between products of powers of values of T whose exponents read norms alone; write it at the images under $\mathsf{r} \in \mathsf{G}$ and take the geometric mean, using (A1) of Lem. 13.9 and equality at real sources. □

Proposition 13.14 (The walls)**.** Let $b \in I(g)$, $K \ge 1$, $T \ge \max\{T_0, \mathsf{l}\}$, $q = L^i \ge 2T$, $\ell \in \mathfrak{L}(g)$ with $\ell \ge 2q$, a board of T_ℓ of side q with any offset, and $f \in \mathfrak{P}_T$ a profile of one of its cells, with F as in Def. 13.3. For real x with $|x| \le \varrho/(256\mathsf{c})$ and $x \le b$:

$$\log\widetilde{Z}^{\mathrm{av}}(xF) - \log\widetilde{Z}^{\mathrm{av}}(x) \le N\big[3C_wTq^2|x| + C_2'q^3x^2\big].$$

■

Sketch of proof. By the passage, $(K, L^K b) \in \mathfrak{d}(g)$. Fix $o_0 \in a_K\mathbb{Z}$ and take eight boards of side l: the board $\iota \in \{0,1\}^3$ has the offset o_ι with $(o_\iota)_\mu = o_0 + \iota_\mu(\mathsf{l} + a_K)/2$, a point of $a_K\mathbb{Z}^3$ as l/a_K is odd. Take the partition of unity $\Theta_\iota = \prod_\mu \mathsf{t}_{\iota_\mu}(y_\mu)$, $\mathsf{t}_1 = 1 - \mathsf{t}_0$, t_0 the l-periodic trapezoid equal to 1 within $\mathsf{l}/8$ of $o_0 + \mathsf{l}\mathbb{Z}$ and to 0 beyond $3\mathsf{l}/8$, with $\operatorname{Lip}\Theta_\iota \le 4\sqrt{3}/\mathsf{l}$; in a cell of the board ι the support of Θ_ι lies in a sup-ball of radius $R_0 \le 7\mathsf{l}/16$ at sup-distance $\ge \mathsf{l}/16$ from the walls of that board.

The walls $\mathsf{g} = 1 - F$ split as $\mathsf{g}\Theta_\iota = \sum_r h_r$ into pieces of the cells of the board ι, with $\|h_r\|_\sharp \le \mathsf{c}$ and mutual sup-distances $\ge \Delta = \mathsf{l}/8 \ge 4M$; since $\mathsf{g} \ne 0$ only within $T/2$ of a mirror of the board of side q, at most $n_\iota \le 9NTq^2/\mathsf{l}^3$ pieces are non-zero. Hölder's inequality (P3) at $u = x$, $n = 8$, $v_\iota = -\operatorname{sgn}(x)\mathsf{g}\Theta_\iota$, $t = |x|$, $\rho = \varrho/(8\mathsf{c})$, with the floor, costs $\vartheta''(2C_S + 2C_g)\ell^3$, $\vartheta'' = 32768\mathsf{c}^2x^2/\varrho^2$ (this is the term in x^2, paid on the whole torus), and leaves the eight numbers $\mathsf{B}_\iota = \log[\widetilde{Z}^{\mathrm{av}}(x + z\mathsf{g}\Theta_\iota)/\widetilde{Z}^{\mathrm{av}}(x)]$, $z = -8x$.

Write $Z_b[w] = Z_{K,\ell,L^Kb}[w]$. On the board ι, the shift of the label, $Z_b[x+w]/Z_b[x] = Z_{K,\ell,\beta'}[w]/Z_{K,\ell,\beta'}$ with $\beta' = L^K(b-x) \ge 0$, and the chessboard estimate at β' give $Z_b[x + z\mathsf{g}\Theta_\iota]/Z_b[x] \le \prod_r \|e^{z\mathcal{O}(h_r)}\|_{\mathrm{ch}}$, and by the source formula $\|e^{z\mathcal{O}(h_r)}\|_{\mathrm{ch}}^{N_f} = Z_b[x + zh_r^{\mathrm{per}}]/Z_b[x]$, where h_r^{per}, the sum of the $N_f = (\ell/\mathsf{l})^3$ mirror copies of h_r, is a dense array. The energies, whose bounds grow with K, cancel exactly up to cross terms. Put $\mathcal{E}'(w) = \mathcal{E}^{\mathrm{av}}(x+w) - \mathcal{E}^{\mathrm{av}}(x)$, the function $\mathcal{F}'$ of Lem. 13.9 (A2) for $\mathcal{F} = \mathcal{E}^{\mathrm{av}}$ at the base source $u = x$; then $Z_b[x+w]/Z_b[x] = e^{\mathcal{E}'(w)}\widetilde{Z}^{\mathrm{av}}(x+w)/\widetilde{Z}^{\mathrm{av}}(x)$, and so, the sums being over the r with $h_r \ne 0$,

$$\mathsf{B}_\iota \le -\mathcal{E}'(z\mathsf{g}\Theta_\iota) + \sum_r N_f^{-1}\Big[\mathcal{E}'(zh_r^{\mathrm{per}}) + \log\frac{\widetilde{Z}^{\mathrm{av}}(x + zh_r^{\mathrm{per}})}{\widetilde{Z}^{\mathrm{av}}(x)}\Big].$$

The mirror copies of h_r are its images $\mathsf{r}h_r$ under reflections $\mathsf{r} \in \mathsf{G}$ at mirrors of the board ι, which fix the constant x, so that $\mathcal{E}^{\mathrm{av}}(x + z\mathsf{r}h_r) = \mathcal{E}^{\mathrm{av}}(\mathsf{r}(x + zh_r)) = \mathcal{E}^{\mathrm{av}}(x + zh_r)$ by (A1). By Lem. 13.9 (A3), applied twice with the base source $u = x$, $\mathcal{E}'(z\mathsf{g}\Theta_\iota) = \sum_r \mathcal{E}'(zh_r) + \mathrm{cr}_\iota$ and $\mathcal{E}'(zh_r^{\mathrm{per}}) = N_f\mathcal{E}'(zh_r) + \mathrm{cr}_r$; so the terms in $\mathcal{E}'$ of the bound of B_ι are $-\mathrm{cr}_\iota + \sum_r N_f^{-1}\mathrm{cr}_r$, at most $2C_\times(7/2)n_\iota|z|^2\mathsf{c}^2/\varrho^2$. The dense arrays are bounded by the Harnack bound, $N_f^{-1}\log[\widetilde{Z}^{\mathrm{av}}(x + zh_r^{\mathrm{per}})/\widetilde{Z}^{\mathrm{av}}(x)] \le 4\mathsf{c}(C_S + C_g)\varrho^{-1}\mathsf{l}^3|z|$.

Summing,

$$\mathsf{B}_\iota \le n_\iota\Big[\frac{32\mathsf{c}(C_S+C_g)}{\varrho}\mathsf{I}^3|x| + \frac{128\,C_\times(7/2)\mathsf{c}^2x^2}{\varrho^2}\Big] \le N\Big[3C_wTq^2|x| + \frac{1152\,C_\times(7/2)\mathsf{c}^2}{\varrho^2\mathsf{I}^3}Tq^2x^2\Big],$$

since $72\cdot 4\mathsf{c}(C_S+C_g)/\varrho = 3C_w$; with $Tq^2 \le q^3$ and the cost of Hölder's inequality this is the claim. □

Proposition 13.15 (Slabs; the walls without a term in x^2 for products)**.**

(a) *The slab inequality (exact).* On a periodic box $(\mathbb{Z}/\mathsf{N}\mathbb{Z})^3$ let $c(p)$ be any real weights on the plaquettes and $Z(c) = \int e^{-\sum_p c(p)a_p}\,\mathrm{d}U$. Let $\mathsf{N} = 2k\,t$ with integers $k, t \ge 1$, cut the box in one direction by the $2k$ planes of sites $x_\mu = n_0 + jt$ into the closed slabs S_j, and let c_j^{arr} be the weights that agree with c on S_j and are symmetric under the reflections at all $2k$ planes. Then $Z(c) \le \prod_j Z(c_j^{\mathrm{arr}})^{1/(2k)}$; no sign of c is supposed. For $c(p) = \beta - a^{-1}w(x(p))$ with w real this reads $Z[w] \le \prod_j Z[w_j^{\mathrm{arr}}]^{1/(2k)}$, w_j^{arr} the function that agrees with w on S_j and is even at all the planes.

(b) *The walls for products.* Let $(K,\beta) \in \mathfrak{d}(g)$, $\ell \in \mathfrak{L}(g)$, $T \ge T_0$, $q = L^i \ge 2T$, $\ell \ge 2q$, a board of T_ℓ of side q with mirrors $y_\mu \in s_\mu + q\mathbb{Z}$, and T' the least power of L that is $\ge 2T$. Let $\varphi_\mu : \mathbb{R}/\ell\mathbb{Z} \to [0,1]$ have $\operatorname{Lip}\varphi_\mu \le 16/T$ and $\varphi_\mu = 1$ at distance $\ge T/2$ from $s_\mu + q\mathbb{Z}$, and $F(y) = \prod_\mu \varphi_\mu(y_\mu)$; let x_0, x be real with $|x_0| + |x| \le \varrho/2$, and $C_V = 16L(C_S+C_g)/\varrho + 3C_\times(1)/(4\varrho T_0^3)$. Then

$$\log\widetilde{Z}^{\mathrm{av}}(x_0 + xF) - \log\widetilde{Z}^{\mathrm{av}}(x_0 + x) \le C_V|x|\,Tq^2N;$$

if every φ_μ has the period q and is even at s_μ (as for the mirror copies of the centred f_0), the same holds for $\widetilde{Z}$ itself. No condition $x \le b$ is asked.

■

Sketch of proof. (a) Cut in the direction $\mu = 1$. The kernel $K_j(\phi,\phi')$ of the slab S_j, with half weights on its two faces and the bonds of the faces fixed to ϕ, ϕ', is real, continuous and positive on a compact space, so a Hilbert–Schmidt operator, and Fubini gives $Z(c) = \operatorname{Tr}(K_0K_1\cdots K_{2k-1})$. The reflection at the mid-plane of a slab shows that K_j^T is the kernel of the slab with mirrored weights, so $\operatorname{Tr}((K_jK_j^T)^k) = Z(c_j^{\mathrm{arr}})$; Hölder's inequality in the Schatten classes, $|\operatorname{Tr}(A_1\cdots A_{2k})| \le \prod_i \|A_i\|_{2k}$, is the claim. In the case used it is the classical chessboard argument: $\mathsf{F}(C_1,\dots,C_{2k}) = \operatorname{Tr}(C_1C_2^TC_3C_4^T\cdots C_{2k-1}C_{2k}^T)$ is multilinear, invariant under a cyclic shift by one place with all letters transposed, and satisfies $|\mathsf{F}(C)|^2 \le \mathsf{F}(C_1,\dots,C_k,C_k,\dots,C_1)\,\mathsf{F}(C_{2k},\dots,C_{k+1},C_{k+1},\dots,C_{2k})$ by Cauchy–Schwarz; after the normalisation $\mathsf{F}(C_i,\dots,C_i) = 1$, among the words that maximise $|\mathsf{F}|$ one with the longest run of equal letters must have a run of length $\ge k$, whence the maximum is ≤ 1. (b) Three steps, one direction at a time: $u_0 = x_0 + x$ and $u_\mu = x_0 + x\prod_{\nu\le\mu}\varphi_\nu(y_\nu)$, so that $u_3 = x_0 + xF$.

In the direction μ the planes $y_\mu \in s_\mu + T'/2 + T'\mathbb{Z}$ are planes of sites, because q/a_K and T'/a_K are odd, and they cut T_ℓ into the ℓ/T' closed slabs $\mathsf{S}_j = \{|y_\mu - s_\mu - jT'| \le T'/2\}$, $j \in \mathbb{Z}/(\ell/T')\mathbb{Z}$, an even number of them; as q/T' is an integer, the mirrors are the mid-planes of the ℓ/q *wall slabs*, those with $jT' \in q\mathbb{Z}$. The function $u_{\mu-1}$ does not read y_μ, and $u_\mu - u_{\mu-1} = -x\prod_{\nu<\mu}\varphi_\nu(y_\nu)\,(1-\varphi_\mu(y_\mu))$ vanishes at distance $\ge T/2$ from the mirrors, hence off the wall slabs and near their faces, as $T'/2 \ge T$. For a wall slab S_j put

$$w_j = (u_\mu - u_{\mu-1})\mathbf{1}_{\mathsf{S}_j} = -x\prod_{\nu<\mu}\varphi_\nu(y_\nu)\,(1-\varphi_\mu(y_\mu))\mathbf{1}_{\mathsf{S}_j},$$

so that $u_\mu = u_{\mu-1} + \sum_{\text{wall slabs}} w_j$, and let w_j^{arr}, the sum of the ℓ/T' mirror images of w_j at the planes, one in each slab, be the even continuation of w_j from S_j. The even continuation of u_μ from

a slab is $u_{\mu-1}$ for a slab that is not a wall slab and $u_{\mu-1}+w_j^{\mathrm{arr}}$ for a wall slab S_j; so (a) at the real source u_μ gives

$$\log\frac{Z[u_\mu]}{Z[u_{\mu-1}]}\le\frac{T'}{\ell}\sum_{\text{wall slabs}}\log\frac{Z[u_{\mu-1}+w_j^{\mathrm{arr}}]}{Z[u_{\mu-1}]}.$$

The normalisers cancel exactly. Let $\mathcal{F}=\mathcal{E}^{\mathrm{av}}$ for the first assertion and $\mathcal{F}=\mathcal{E}$ in the periodic even case, $\widetilde{Z}^{\mathcal{F}}=e^{-E-\mathcal{F}}Z$ (that is, $\widetilde{Z}^{\mathrm{av}}$, respectively $\widetilde{Z}$), and $\mathcal{F}'(w)=\mathcal{F}(u_{\mu-1}+w)-\mathcal{F}(u_{\mu-1})$ as in Lem. 13.9 (A2) at the base $u_{\mu-1}$, so that $Z[u_{\mu-1}+w]/Z[u_{\mu-1}]=e^{\mathcal{F}'(w)}\widetilde{Z}^{\mathcal{F}}(u_{\mu-1}+w)/\widetilde{Z}^{\mathcal{F}}(u_{\mu-1})$. The pieces of w_j^{arr} are images $\mathsf{r}w_j$ under reflections $\mathsf{r}\in\mathsf{G}$ at the planes, which fix $u_{\mu-1}$, so $\mathcal{F}'(\mathsf{r}w_j)=\mathcal{F}'(w_j)$ for $\mathcal{F}=\mathcal{E}^{\mathrm{av}}$ by (A1); in the periodic even case $1-\varphi_\mu$ is even at the mid-plane of S_j, the pieces are the translates of w_j by $T'\mathbb{Z}e_\mu$, which lies in $M\mathbb{Z}^3$ as T' and M are powers of L with $T'>M$, and the same holds for $\mathcal{E}$ itself by (M1).

Hence $\mathcal{F}'(w_j^{\mathrm{arr}})=(\ell/T')\mathcal{F}'(w_j)+\mathrm{cr}_j$ and $\mathcal{F}'(\sum_j w_j)=\sum_j\mathcal{F}'(w_j)+\mathrm{cr}$, the sum over the wall slabs, and in $\log[\widetilde{Z}^{\mathcal{F}}(u_\mu)/\widetilde{Z}^{\mathcal{F}}(u_{\mu-1})]=\log(Z[u_\mu]/Z[u_{\mu-1}])-\mathcal{F}'(\sum_j w_j)$ the terms $\mathcal{F}'(w_j)$ cancel, leaving the cross terms. These are bounded by Lem. 13.9 (A3), with x replaced by ζ, $|\zeta|<\varrho$, in the w_j alone: each support lies within $T/2$ of the mid-plane of its slab and is covered by $(\ell/T')^2$ closed sup-balls of radius $R_0=T'/2$, and different supports are at sup-distance $\ge\Delta=T'-T\ge T'/2\ge 4M$, so that $R_0/\Delta\le 1$.

The Harnack bound at the base $u_{\mu-1}$, with $\|u_{\mu-1}\|_\sharp\le|x_0|+|x|\le\varrho/2$, and in the direction w_j^{arr}, with $\|w_j^{\mathrm{arr}}\|_\sharp\le|x|$ as $40M\cdot16\sqrt{3}/T\le1$, costs $\frac{8}{3}(C_S+C_g)\varrho^{-1}|x|\ell^3$ for each wall array, and $\#\{\text{wall slabs}\}\,T'/\ell=T'/q$; a slab needs no partition of unity across it, so (P3) and its term in x^2 do not enter. Summing the three steps with $2T\le T'<2LT$ and $|x|\le\varrho/2$ gives the constant C_V. □

Proposition 13.16 (The main inequality; semi-convexity)**.** Let $\ell'\in\mathfrak{L}(g)$ with $\ell'\ge2\mathsf{l}_1$ be a reference side, $T\ge\max\{T_0,\mathsf{l}\}$, $q=L^i\ge2T$, $\ell\in\mathfrak{L}(g)$ with $\ell\ge\max\{2q,\ell'\}$, $K\ge1$ and $b\in I(g)$, all numbers of Def. 13.6 being formed at g with ℓ'. Put $\omega(z)=\ell'^{-3}\log[\widetilde{Z}_{\ell'}(z)/\widetilde{Z}_{\ell'}(0)]$, let $\mu'=d'(0)$ for the d of Lem. 13.10 at $(\ell_1,\ell_2)=(\ell,\ell')$, and for a profile f let $\epsilon(z)=\mathcal{E}^{\mathrm{av}}_\ell(zf)+q^3[\omega'(0)-\mu']z$, a function holomorphic on $|z|<2\varrho$ with $\epsilon(0)=0$ and $\epsilon'(0)$ real.

(a) For every board of T_ℓ of side q, every offset and every profile $f\in\mathfrak{P}_T$, for $|z|\le r_1$ with $\operatorname{Re}z\le b$:

$$\Lambda(z)-\operatorname{Re}\epsilon(z)\le q^3\big[D_{K,\ell}(b-\operatorname{Re}z)-D_{K,\ell}(b)\big]+3C_wTq^2|\operatorname{Re}z|+C_Iq^3|z|^2.$$

(b) $D''_{K,\ell}(b)\ge-C_0$ at every $b\in I(g)$. So $D_{K,\ell}+\frac12C_0y^2$ and $D_K+\frac12C_0y^2$ are convex on $I(g)$; $\mu_K=(D_K+\frac12C_0y^2)''$ and $\mu_{K,\ell}=(D_{K,\ell}+\frac12C_0y^2)''$ are positive Radon measures on $\operatorname{int}I(g)$; and for both, $\mu([y_1,y_2])\le2B'/t+C_0(y_2-y_1+t)$ whenever $[y_1-t,y_2+t]\subset I(g)$.

■

Sketch of proof. By the passage $(K,L^Kb)\in\mathfrak{d}(g)$, so Lems. 13.9, 13.10, 13.12 and Props. 13.13, 13.14 hold on both tori. The shift of the label gives on each torus, for real $|x|<2\varrho$, $\log\widetilde{Z}(x)-\log\widetilde{Z}(0)=\ell^3[p_{K,\ell}(b-x)-p_{K,\ell}(b)]-\mathcal{E}(x\cdot1)$; dividing by the volumes and subtracting,

$$\ell^{-3}\big[\log\widetilde{Z}_\ell(x)-\log\widetilde{Z}_\ell(0)\big]=\big[D_{K,\ell}(b-x)-D_{K,\ell}(b)\big]+\omega(x)-d(x).$$

$H(z)=\widetilde{Z}_{\ell'}(z)/\widetilde{Z}_{\ell'}(0)$ is holomorphic on $|z|<2\varrho$, $H(0)=1$ and $|H|\le e^{(C_S+C_g)\ell'^3}$ by (M1); Schwarz's lemma gives $|H-1|\le\frac12$ on $|z|\le r_2$, so $|\omega(x)-\omega'(0)x|\le C_\omega x^2$ for $|x|\le r_2/2$ and $|\omega''(0)|\le C_\omega$. For (a), with $z=x+i\tau$, the source formula gives $N\Lambda(z)=\operatorname{Re}\mathcal{E}^{\mathrm{av}}(w_z)+\log|\widetilde{Z}^{\mathrm{av}}(w_z)|-\log\widetilde{Z}(0)$. Then (i) (IM) for $\widetilde{Z}^{\mathrm{av}}$ at $u=xF$, $v=\tau(F_e-F_o)$ removes τ at the cost $C'_I\tau^2\ell^3$; (ii) the walls (Prop. 13.14) replace xF by the constant x at the cost $N[3C_wTq^2|x|+C'_2q^3x^2]$; (iii) the identity, the bound of ω and Lem. 13.10 turn $\log\widetilde{Z}(x)-\log\widetilde{Z}(0)$ into at most $Nq^3\{[D_{K,\ell}(b-x)-D_{K,\ell}(b)]+[\omega'(0)-\mu']x+(C_\omega+2\mathsf{A}_1/h_1^2)x^2\}$; (iv) the cross terms between the cells, Lem. 13.9 (A3) with $R_0=q/2$, $\Delta=T/4$, $\rho_3=\varrho$ and $C_\times(2q/T)\le M_\times(q/T)^3$, give $\operatorname{Re}\mathcal{E}^{\mathrm{av}}(w_z)\le N\operatorname{Re}\mathcal{E}^{\mathrm{av}}(zf)+NM_\times T^{-3}\varrho^{-2}q^3|z|^2$.

Adding and dividing by N gives (a). For (b), semi-convexity in Prop. 13.13 at $u=0$, $v=1$ gives $\mathsf{h}''(0) \ge -2C_I'\ell^3$ for $\mathsf{h}(x) = \log \widetilde{Z}_\ell(x)$; differentiating the identity twice at $x=0$,

$$D''_{K,\ell}(b) = \ell^{-3}\mathsf{h}''(0) - \omega''(0) + d''(0) \ge -2C_I' - C_\omega - 2\mathsf{A}_1/h_1^2 = -C_0.$$

$D_{K,\ell}$ is real-analytic in the label, and D_K is its pointwise limit by Lem. 13.11. For $\Phi = D + \frac{1}{2}C_0 y^2$ with $D \in \{D_K, D_{K,\ell}\}$, $\mu([y_1,y_2]) = \Phi'(y_2+) - \Phi'(y_1-)$ is bounded by difference quotients over the length t, and $D \in [-B', 0]$ by Lem. 13.11 (c). □

Lemma 13.17 (Cauchy's estimate in the chessboard seminorm; moments from the seminorm)**.** On a board of T_ℓ, at any $\beta \ge 0$:

(a) If $X = \mathcal{O}(f)$ is a real function of one cell and $\tilde\epsilon$ is holomorphic near the disc $|z| \le r$ with $\tilde\epsilon(0) = 0$, then $\|X - \tilde\epsilon'(0)\|_{\mathrm{ch}} \le r^{-1} \max_{|z|=r} \exp[\Lambda(z) - \operatorname{Re}\tilde\epsilon(z)]$.

(b) For functions $X_1, \dots, X_n$ of n different cells and complex numbers m_i: $|\langle \prod_i (X_i - m_i)\rangle| \le \prod_i \|X_i - m_i\|_{\mathrm{ch}}$; $|\langle X_i\rangle - m_i| \le \|X_i - m_i\|_{\mathrm{ch}}$; $|\operatorname{Cov}(X_1, X_2)| \le 2\|X_1 - m_1\|_{\mathrm{ch}}\|X_2 - m_2\|_{\mathrm{ch}}$; and the centred moments satisfy $|\langle \prod_i (X_i - \langle X_i\rangle)\rangle| \le \prod_i 2\|X_i - m_i\|_{\mathrm{ch}}$.

■

Sketch of proof. (a) Pointwise on the configurations, $X - \tilde\epsilon'(0) = (2\pi i)^{-1} \oint_{|z|=r} e^{zX - \tilde\epsilon(z)} z^{-2}\,\mathrm{d}z$; the Riemann sums converge uniformly since X is bounded; the triangle inequality and the homogeneity of the seminorm, with $\|F\|_{\mathrm{ch}} \le \sup|F|$ for the remainder, give $\|X - \tilde\epsilon'(0)\|_{\mathrm{ch}} \le r^{-1}\max_{|z|=r}\|e^{zX-\tilde\epsilon(z)}\|_{\mathrm{ch}}$, and $\|e^{zX-\tilde\epsilon(z)}\|_{\mathrm{ch}} = e^{-\operatorname{Re}\tilde\epsilon(z)}(Z[w_z]/Z)^{1/N}$ by the source formula. (b) The first inequality is the chessboard estimate; the second is the first with one factor and $\|1\|_{\mathrm{ch}} = 1$; the third follows from $\operatorname{Cov}(X_1,X_2) = \langle (X_1 - m_1)(X_2 - m_2)\rangle - (\langle X_1\rangle - m_1)(\langle X_2\rangle - m_2)$; and $\|X_i - \langle X_i\rangle\|_{\mathrm{ch}} \le \|X_i - m_i\|_{\mathrm{ch}} + |\langle X_i\rangle - m_i| \le 2\|X_i - m_i\|_{\mathrm{ch}}$. □

Proposition 13.18 (From a bound on increments of the pressure to the cube means, on a torus)**.** In the setting of Prop. 13.16, let s, δ_2, $\tilde C$, η_+, x_0 be real numbers with $\delta_2 \ge 0$, $\tilde C \ge 0$, $\eta_+ \ge 0$ and $0 < x_0 \le b$, such that

$$D_{K,\ell}(b-x) - D_{K,\ell}(b) \le \mathsf{s}x + \delta_2|x| + \tilde C x^2 + \eta_+ \qquad \text{for all real } |x| \le x_0.$$

Put $\delta = 3C_wT/q + \delta_2$, and let $q \ge \max\{(\tilde C + C_I)/(9C_w^2T^2), (3C_wT\min\{r_1, x_0\})^{-1/2}\}$. Then on every board of T_ℓ of side q, $\|x_c - m'\|_{\mathrm{ch}} \le e^{2+q^3\eta_+}\delta$ with the real number $m' = q^{-3}\epsilon'(0) + \mathsf{s}$, which depends on ℓ and on the profile. Hence, for profiles of $\mathfrak{P}_T$ in distinct cells,

$$|\operatorname{Cov}_{K,\ell,L^K b}(x_c, x_{c'})| \le 2e^{4+2q^3\eta_+}\delta^2, \qquad \left|\langle \textstyle\prod_{i=1}^n (x_{c_i} - \langle x_{c_i}\rangle)\rangle_{K,\ell,L^K b}\right| \le \left(2e^{2+q^3\eta_+}\delta\right)^n.$$

■

Sketch of proof. Put $\tilde\epsilon(z) = \epsilon(z) + q^3\mathsf{s}z$. For $|z| = r \le \min\{r_1, x_0\}$, so that $\operatorname{Re} z \le x_0 \le b$, the main inequality and the hypothesis at $x = \operatorname{Re} z$ give, since $\delta_2 \ge 0$ and $\tilde C \ge 0$ allow $\delta_2|x| \le \delta_2 r$ and $\tilde C x^2 \le \tilde C r^2$,

$$\Lambda(z) - \operatorname{Re}\tilde\epsilon(z) \le q^3\delta r + q^3(\tilde C + C_I)r^2 + q^3\eta_+.$$

Take $r = 1/(q^3\delta)$; since $\delta \ge 3C_wT/q$, $r \le 1/(3C_wTq^2) \le \min\{r_1, x_0\}$ by the second floor; $q^3\delta r = 1$; and $q^3(\tilde C + C_I)r^2 = (\tilde C + C_I)/(q^3\delta^2) \le (\tilde C + C_I)/(9C_w^2T^2q) \le 1$ by the first. Lem. 13.17 (a) gives $\|\mathcal{O}(f) - \tilde\epsilon'(0)\|_{\mathrm{ch}} \le q^3\delta e^{2+q^3\eta_+}$; divide by q^3; for a profile of another cell the same holds with its own ϵ and m'; Lem. 13.17 (b) gives the rest. □

Proposition 13.19 (The window inequality, in one frame, at every cutoff)**.** In the setting of Prop. 13.16, let b be a label with $I_q(b) \subset \operatorname{int} I(g)$, $q \ge q_1 = \max\{2T, C_I/(9C_w^2T^2), (3C_wTr_1)^{-1/2}\}$, $K \ge 1$, and

$$\delta_K(q,b) = 3C_wT/q + \mu_K(I_q(b)).$$

(a) For all $\nu,\nu' \in \mathcal{G}^{\mathrm{per}}_{K,L^K b}$, every board of $\mathbb{L}_K$ of side q with any offset and profiles of $\mathfrak{P}_T$ in distinct cells $c, c', c_1, \dots, c_n$:

$$|\operatorname{Cov}_\nu(x_c, x_{c'})| \le 2e^4 \delta_K(q,b)^2, \qquad |\nu(x_c) - \nu'(x_c)| \le 2e^2 \delta_K(q,b), \qquad \Big|\nu\Big(\prod_i (x_{c_i} - \nu(x_{c_i}))\Big)\Big| \le (2e^2\delta_K(q,b))^n.$$

(b) The same holds with $\delta = 3C_w T/q + C_0 x_q + x_q^{-1}\Delta^2_{x_q} D_K(b)$, which is $\le \delta_K(q,b)$; and for every $\ell \in \mathfrak{L}(g)$ with $\ell \ge \ell'$,

$$x_q^{-1}\Delta^2_{x_q} D_K(b) \le x_q^{-1}\Delta^2_{x_q} D_{K,\ell}(b) + 6C_w C_{\mathrm{FS}} T q^2/\ell \le x_q \sup_{|y-b|\le x_q} D''_{K,\ell}(y) + 6C_w C_{\mathrm{FS}} T q^2/\ell, \qquad C_{\mathrm{FS}} = 3\ell' B' :$$

the bounds in every periodic Gibbs state are governed by the pressures of the two tori T_ℓ, $T_{\ell'}$ at a side ℓ of the order q^3.

■

Sketch of proof. (a) *A window gives the hypothesis of Prop. 13.18.* $\Phi = D_K + \frac12 C_0 y^2$ is convex on $\operatorname{int} I(g) \supset I_q(b)$; with $\sigma_0 = \Phi'(b-)$, for $0 < x \le x_q$, $\Phi(b+x) - \Phi(b) \le \sigma_0 x + x\mu_K([b, b+x))$ and $\Phi(b-x) - \Phi(b) \le -\sigma_0 x + x\mu_K((b-x, b))$; since $D_K(b-x) - D_K(b) = \Phi(b-x) - \Phi(b) + C_0 bx - \frac12 C_0 x^2$, the function D_K has the hypothesis with $\mathsf{s} = C_0 b - \sigma_0$, $\delta_2 = \mu_K(I_q(b)) \ge 0$, $\tilde C = 0$, $\eta_+ = 0$, $x_0 = x_q$. *From the tori to the states.* By Lem. 13.11, $D_{K,\ell} = D_K + \mathsf{w}_\ell$ with $0 \le \mathsf{w}_\ell \le \bar\eta(\ell)$, so $D_{K,\ell}$ has it with the tolerance $\eta_+ = \bar\eta(\ell)$, and Prop. 13.18 gives $\|x_c - m'_\ell\|_{\mathrm{ch},\ell} \le e^{2+q^3\bar\eta(\ell)}\delta_K(q,b)$ for all $\ell \in \mathfrak{L}(g)$, $\ell \ge \max\{2q, \ell'\}$. The real numbers m'_ℓ are bounded at fixed K, since $|m'_\ell - \langle x_c\rangle_\ell| \le e^3\delta_K$ for large ℓ and $|x_c|$ is bounded. For finitely many profiles choose $\ell_k \to \infty$ with $m'_{\ell_k,i} \to m_{*,i}$; then $\limsup_k \|x_{c_i} - m_{*,i}\|_{\mathrm{ch},\ell_k} \le e^2\delta_K$, and the domination gives

$$\Big|\nu\Big(\prod_i (x_{c_i} - m_{*,i})\Big)\Big| \le (e^2\delta_K)^n \qquad \text{for every } \nu \in \mathcal{G}^{\mathrm{per}}_{K,L^K b};$$

at $n = 1$, $|\nu(x_c) - m_*| \le e^2\delta_K$ for ν and for ν', one common centre serving both; expanding as in Lem. 13.17 (b) gives the three bounds. (b) In Prop. 13.18 the hypothesis enters only at $|x| \le r = 1/(q^3\delta) \le x_q$. On $[b-r, b+r]$ the convex Φ lies below its chord, so $\Phi(b-x) - \Phi(b) \le -\mathsf{s}_r x + \frac12\Delta^2_r\Phi(b)$ with the slope s_r of the chord; and $\Delta^2_r\Phi(b)/r = r^{-1}\int_0^r \mu_K((b-u, b+u))\,du$ does not decrease in r, so $\frac12\Delta^2_r\Phi(b) \le r\, x_q^{-1}\Delta^2_{x_q}\Phi(b)$, which is $r(C_0 x_q + x_q^{-1}\Delta^2_{x_q} D_K(b))$ and at most $r\mu_K(I_q(b))$. The comparison with the torus follows from $D_K \le D_{K,\ell} \le D_K + \bar\eta(\ell)$, $2\bar\eta(\ell)/x_q = 6C_w C_{\mathrm{FS}} T q^2/\ell$, and $\Delta^2_s D \le s^2 \sup D''$ for the real-analytic $D_{K,\ell}$. □

Lemma 13.20 (A sum rule; near and far parts; a pile recurs at all large cutoffs)**.** Let one frame, $0 < g \le \gamma$, a compact interval $W \subset I(g)$ of positive length and $\ell' \in \mathfrak{L}(g)$ with $\ell' \ge 2\mathsf{l}_1$ be given, let $k_7 \ge 1$ be an integer with $\varsigma_7 = L^{-k_7} \le 2/(3\sqrt3)$, and let χ, η, c_7, c_T, c_N be formed with this ς_7 as in Def. 13.6.

(i) *Exact.* For $K \ge k_7$, every side $\ell > 6\varsigma_7$ and $b \ge 0$:

$$p''_{K,\ell}(b) = \varsigma_7^{-3} S^T_{2;K,\ell,L^K b}(\chi, 1) = V^{\mathrm{near}}_{K,\ell}(b) + V^{\mathrm{far}}_{K,\ell}(b), \qquad V^{\mathrm{near}} = \varsigma_7^{-3} S^T_2(\chi,\eta), \quad V^{\mathrm{far}} = \varsigma_7^{-3} S^T_2(\chi, 1-\eta).$$

(ii) $|V^{\mathrm{near}}_{K,\ell} - V^{\mathrm{near}}_{K,\ell'}| \le c_N$ on $I(g)$ for all $K \ge k_7$ and $\ell \in \mathfrak{L}(g)$; and for each side ℓ, $\omega_\ell(K, K'') = \sup_W |V^{\mathrm{far}}_{K,\ell} - V^{\mathrm{far}}_{K'',\ell}| \to 0$ as $K, K'' \to \infty$.

(iii) For every closed interval I and $\delta > 0$ with $I^{+\delta} = [\min I - \delta, \max I + \delta] \subset \operatorname{int} W$, all $K, K'' \geq k_7$ and all $\ell \in \mathfrak{L}(g)$ with $\ell \geq \ell'$:

$$\mu_K(I) \leq \mu_{K''}(I^{+\delta}) + (|I| + 2\delta)\big[2c_N + \omega_\ell(K, K'') + \omega_{\ell'}(K, K'')\big] + \frac{24\ell' B'}{\delta \ell}.$$

■

Sketch of proof. (i)

$$p''_{K,\ell} = \ell^{-3} L^{2K} \operatorname{Var}(\sum_p a_p) = a_K^{-5} \mathsf{N}^{-3} \operatorname{Var}(\sum_p a_p),$$

while $S_2^T(\chi, 1) = a_K^{-2} \sum_p \chi(x(p)) \operatorname{Cov}(a_p, \sum_{p'} a_{p'})$. By translation invariance the covariance depends on the orientation of p alone, the three orientations summing to $\mathsf{N}^{-3} \operatorname{Var}(\sum_p a_p)$, and for each orientation $\sum_p \chi(x(p)) = (\varsigma_7/a_K)^3$, the sums of tents of the integer half-width ς_7/a_K; the splitting is $1 = \eta + (1-\eta)$. (ii) The supports of χ and η lie in a ball of radius $3\sqrt{3}\varsigma_7 \leq 2$, and $\|\chi\|_\sharp, \|\eta\|_\sharp, \|1-\eta\|_\sharp \leq \mathsf{c}_7$, so the transfer (M7) at $n = 2$ between the two tori, at the first-passage data given by the passage, bounds the near parts; the supports of χ and $1 - \eta$ are at sup-distance ς_7, so the convergence on fixed tori (M8) gives the far parts. (iii) Take a C^2 function $1_I \leq \eta_I \leq 1_{I^{+\delta}}$ with $\|\eta_I''\|_1 \leq 8/\delta$, the indicator of $[\min I - \delta/2, \max I + \delta/2]$ smoothed by the kernel $\frac{15}{16}(1-t^2)_+^2$ at the scale $\delta/2$. Since $\mu_K \geq 0$, $\mu_K(I) \leq \int \eta_I \, \mathrm{d}\mu_K$ and $\int \eta_I \, \mathrm{d}\mu_{K''} \leq \mu_{K''}(I^{+\delta})$, and

$$\int \eta_I \, d(\mu_K - \mu_{K''}) = \int \eta_I'' \big[(D_K - D_{K,\ell}) - (D_{K''} - D_{K'',\ell})\big] + \int \eta_I \, (D''_{K,\ell} - D''_{K'',\ell}),$$

where the bracket lies in $[-\bar\eta(\ell), \bar\eta(\ell)]$ by Lem. 13.11, and by (i) $D''_{K,\ell} = (V^{\mathrm{near}}_{K,\ell} - V^{\mathrm{near}}_{K,\ell'}) + V^{\mathrm{far}}_{K,\ell} - V^{\mathrm{far}}_{K,\ell'}$, so that $|D''_{K,\ell} - D''_{K'',\ell}| \leq 2c_N + \omega_\ell + \omega_{\ell'}$ on W. In words: a concentration of μ_K at one large cutoff recurs at all large cutoffs. □

Lemma 13.21 (Dilations by real factors)**.** For real $s > 0$, $a > 0$ and $\beta > 0$ the image under dil_s maps the periodic Gibbs states at (a, β) onto the periodic Gibbs states at (sa, β), and $S_n^T[\nu](f_1, \dots, f_n) = s^n \, S_n^T[(\mathrm{dil}_s)_*\nu](f_1(\cdot/s), \dots, f_n(\cdot/s))$. ■

Sketch of proof. The exponent $H_B = -\beta \sum_{p \in \mathrm{Pl}[B]} a_p$ of a Gibbs kernel contains no spacing. By the exact dilation facts (M9), dil_s carries $\mathrm{Pl}[B]$ onto $\mathrm{Pl}[\mathrm{dil}_s B]$, $a_{\mathrm{dil}_s p}(\mathrm{dil}_s U) = a_p(U)$, the product Haar measure on the bonds of B to that on the bonds of $\mathrm{dil}_s B$, and the configuration that is V on B and U off B to the one that is $\mathrm{dil}_s V$ on $\mathrm{dil}_s B$ and $\mathrm{dil}_s U$ off it; hence

$$\mathrm{dil}_s(\gamma_B F) = \gamma_{\mathrm{dil}_s B}(\mathrm{dil}_s F),$$

and $((\mathrm{dil}_s)_*\nu)(\gamma_{\mathrm{dil}_s B}\mathrm{dil}_s F) = \nu(\gamma_B F) = \nu(F)$; as $B \mapsto \mathrm{dil}_s B$ and $F \mapsto \mathrm{dil}_s F$ are onto, the image is a Gibbs state. Invariance under the translations by a subgroup Γ of finite index passes to $s\Gamma$, of the same index; $\mathrm{dil}_{1/s}$ is the inverse. The formula for the cumulants follows from $\mathcal{O}_a(f)(U) = s \, \mathcal{O}_{sa}(f(\cdot/s))(\mathrm{dil}_s U)$. □

Lemma 13.22 (One pressure for both frames; the reference tori compared)**.** In a Setting, let $K \geq k_7$, $\hat K \geq 1$, $\lambda = L^K/\hat L^{\hat K}$ and J a closed interval with $J \subset W_1$ and $\lambda J \subset \hat W$, so that $\lambda \in [\lambda_-, \lambda_+]$. Then $p''_{K,\infty}(J) = \lambda^4 \hat p''_{\hat K,\infty}(\lambda J)$ and

$$D''_K(J) \leq \lambda^4 \hat D''_{\hat K}(\lambda J) + c_2 |J|.$$

■

Sketch of proof. By Lem. 13.11 (b), $p_{K,\infty}(b) = L^{3K} P(L^K b)$ and $\hat p_{\hat K,\infty}(\hat b) = \hat L^{3\hat K} P(\hat L^{\hat K} \hat b)$ with one convex P; for $J = [y_1, y_2]$, $p''_{K,\infty}(J) = L^{4K}[P'(L^K y_2+) - P'(L^K y_1-)] = \lambda^4 \hat p''_{\hat K,\infty}(\lambda J)$, since $\hat L^{\hat K} \lambda = L^K$. With $\mathsf{r}_1 = p''_{K,\ell'}$, $\mathsf{r}_2 = \hat p''_{\hat K,\hat\ell'}$ this gives $D''_K(J) - \lambda^4 \hat D''_{\hat K}(\lambda J) = \int_J [\lambda^5 \mathsf{r}_2(\lambda b) - \mathsf{r}_1(b)] \, \mathrm{d}b$, and both $\mathsf{r}_1(b)$ and $\lambda^5 \mathsf{r}_2(\lambda b)$ are $a_K^{-5}(\mathrm{box})^{-3}$ times a variance of the action at the one coupling

$\beta = L^K b = \hat{L}^{\hat{K}} \lambda b$. In the units of the second frame the tent becomes $\chi(\cdot/\lambda)$, $\hat{\mathcal{O}} = \lambda^{-1}\mathcal{O}$ and $S_2 = \lambda^2 \hat{S}_2$; the sum rule of Lem. 13.20 holds on both reference boxes, as $6\lambda\varsigma_7 < \hat{\ell}'$.

The far parts are bounded by c_F and $\hat{c}_F$, the bound of (M8) reading of the functions only the spread $\lambda\varsigma_7 \geq \lambda_-\varsigma_7$ and the norms $\leq \hat{\mathsf{c}}_8$. For the near parts take $\nu \in \mathcal{G}^{\mathrm{per}}$ at (a_K, β), which exists by (M9), and its image $\hat{\nu}$ under the dilation by the real factor λ, a periodic Gibbs state at $(\hat{a}_{\hat{K}}, \beta)$ by Lem. 13.21, with $S_2[\nu](\chi, \eta) = \lambda^2 \hat{S}_2[\hat{\nu}](\chi(\cdot/\lambda), \eta(\cdot/\lambda))$; by the passage in each frame both pairs are first-passage data, and the transfer in each frame between the reference torus and the state $(3\sqrt{3}\lambda\varsigma_7 \leq 2)$ gives

$$|\lambda^5 \mathsf{r}_2(\lambda b) - \mathsf{r}_1(b)| \leq \varsigma_7^{-3}\big[c_T \mathsf{c}_7^2 + \lambda_+^2 \hat{c}_T \hat{\mathsf{c}}_8^2 + c_F + \lambda_+^2 \hat{c}_F\big] = c_2. \qquad \square$$

Proposition 13.23 (No pile of specific heat)**.** In a Setting, with (g_1, W_1, W') and $\mathrm{dist}(W', \partial W_1) = \varepsilon$: for every $0 < \delta \leq \delta_{\max}$ there is an integer K° with

$$\mu_K(I) \leq C_\flat(|I| + \delta) \qquad \text{for all } K \geq K^\circ \text{ and all closed intervals or points } I \subset W'.$$

So $K_1(\delta)$ of Def. 13.6 exists, and it does not increase in δ. Of the convergence on fixed tori (item (M8)) the proof reads two clauses, both at $n = 2$, for pairs with disjoint supports and without a rate: (1) the convergence in the cutoff, uniformly in the label, of the cumulant S_2^T at the pair $(\chi, 1-\eta)$ on the tori T_ℓ, $\ell \in \mathfrak{L}(g_1)$, $\ell \geq \ell'$, over W_1, and at the pair of the tent and the plateau of half-width $1/\hat{L}$ on the tori $T_{\hat{\ell}}$, $\hat{\ell} \in \hat{\mathfrak{L}}(\hat{g})$, $\hat{\ell} \geq \hat{\ell}'$, over $\hat{W}$; (2) the bound clause, which gives c_F, for the one pair $(\chi, 1-\eta)$ on $T_{\ell'}$ over W_1, and $\hat{c}_F$ on $T_{\hat{\ell}'}$, uniformly in $\hat{K} \geq 1$, in $\hat{b} \in \hat{W}$ and in the pairs $(\chi(\cdot/\lambda), 1-\eta(\cdot/\lambda))$, $\lambda \in [\lambda_-, \lambda_+]$, whose spread $\lambda\varsigma_7 \geq \lambda_-\varsigma_7$ and whose norms $\leq \hat{\mathsf{c}}_8$ are all that the clause reads of them. Thus on $T_{\hat{\ell}'}$ two kinds of pairs are read: the pair of half-width $1/\hat{L}$ through the convergence, and the family of the dilated pairs through the bound. ■

Sketch of proof. Thresholds. In the first frame Lem. 13.20 is used with g_1, W_1, ℓ' and k_7 of Def. 13.6. Take $\ell(\delta) \in \mathfrak{L}(g_1)$, $\ell(\delta) \geq \ell'$, with $24\ell' B'/(\delta\ell(\delta)) \leq \delta/2$, and an integer $K^\circ \geq k_7$ with $\omega_{\ell(\delta)}(K, K'') + \omega_{\ell'}(K, K'') \leq 1/6$ for all $K, K'' \geq K^\circ$ (Lem. 13.20 (ii)). In the second frame Lem. 13.20 is used with $\hat{g}$, $\hat{W}$, $\hat{\ell}'$ and $\hat{k}_7 = 1$, so that the tent and the plateau have the half-width $1/\hat{L} \leq 2/(3\sqrt{3})$ there and the constant of the near parts is $\hat{c}_N$. Put $\hat{\delta} = \lambda_+\delta$; take $\hat{\ell} \in \hat{\mathfrak{L}}(\hat{g})$, $\hat{\ell} \geq \hat{\ell}'$, with

$$\frac{24\hat{\ell}'\hat{B}'}{\hat{\delta}\hat{\ell}} \leq \frac{\delta}{2},$$

and an integer $\hat{K}_* \geq 1$ with $\hat{\omega}_{\hat{\ell}}(\hat{K}, \hat{K}'') + \hat{\omega}_{\hat{\ell}'}(\hat{K}, \hat{K}'') \leq 1/(10\lambda_+)$ for all $\hat{K}, \hat{K}'' \geq \hat{K}_*$. *One interval.* Let $I = [y - \delta/2, y + \delta/2]$, $y \in W'$, $J = I^{+\delta} \subset \mathrm{int}\, W_1$, $K \geq K^\circ$. Let $R = \lfloor |\hat{W}|/(12\lambda_+\delta) \rfloor \geq |\hat{W}|/(24\lambda_+\delta)$ and $t_r = \hat{w}_- + |\hat{W}|/4 + (r - \frac{1}{2})6\lambda_+\delta$, $r = 1, \dots, R$. As $\log L/\log \hat{L}$ is irrational, Kronecker's theorem gives integers $K_r'' \geq K^\circ$ and $\hat{K}_r \geq \hat{K}_*$ with

$$|\lambda_r y - t_r| < \lambda_+\delta/2, \qquad \lambda_r = L^{K_r''}/\hat{L}^{\hat{K}_r};$$

then $\lambda_r \in [\lambda_-, \lambda_+]$, and the intervals $\mathsf{I}_r = (\lambda_r J)^{+\hat{\delta}}$ are pairwise disjoint and lie in the middle half of $\hat{W}$. *The chain.* Lem. 13.20 (iii) in the first frame, for I with the margin δ and the side $\ell(\delta)$, from K to K_r'': $\mu_K(I) \leq \mu_{K_r''}(J) + 3\delta(2c_N + \frac{1}{6}) + \frac{\delta}{2} = \mu_{K_r''}(J) + (6c_N + 1)\delta$. Lem. 13.22 between the frames, with $\hat{D}'' \leq \hat{\mu}$ (as $\hat{C}_0 \geq 0$), $\hat{\mu} \geq 0$ and $\lambda_r \leq \lambda_+$: $\mu_{K_r''}(J) = D''_{K_r''}(J) + 3C_0\delta \leq \lambda_+^4 \hat{\mu}_{\hat{K}_r}(\lambda_r J) + 3(c_2 + C_0)\delta$. Lem. 13.20 (iii) in the second frame, for the interval $\lambda_r J$, of length $3\lambda_r\delta \leq 3\lambda_+\delta$, with the margin $\hat{\delta}$ and the side $\hat{\ell}$, from $\hat{K}_r$ to the one cutoff $\hat{K}_*$, whose added terms are at most $5\lambda_+\delta\,(2\hat{c}_N + \frac{1}{10\lambda_+}) + \frac{\delta}{2} = 10\lambda_+\hat{c}_N\delta + \delta$: $\hat{\mu}_{\hat{K}_r}(\lambda_r J) \leq \hat{\mu}_{\hat{K}_*}(\mathsf{I}_r) + 10\lambda_+\hat{c}_N\delta + \delta$. Together:

$$\mu_K(I) \leq \lambda_+^4\,[\hat{\mu}_{\hat{K}_*}(\mathsf{I}_r) + 10\lambda_+\hat{c}_N\delta + \delta] + [6c_N + 3(c_2 + C_0) + 1]\delta \qquad \text{for every } r.$$

Take the mean over r: the l_r are pairwise disjoint and lie in the middle half of $\hat{W}$, whose mass for $\hat{\mu}_{\hat{K}_*}$ is at most $8\hat{B}'/|\hat{W}| + \frac{3}{4}\hat{C}_0|\hat{W}| \le \hat{\mathfrak{m}}$ by the mass bound of Prop. 13.16 (b) in the second frame at $t = |\hat{W}|/4$; so $R^{-1}\sum_r \hat{\mu}_{\hat{K}_*}(\mathsf{l}_r) \le \hat{\mathfrak{m}}/R \le 24\lambda_+\hat{\mathfrak{m}}\delta/|\hat{W}|$. This is $\mu_K(I) \le C_\flat\delta$. *Any closed* $I \subset W'$ is covered by at most $|I|/\delta + 1$ such intervals. The property at δ implies it at every $\delta' > \delta$, so K_1 does not increase. □

Lemma 13.24 (The labels; the exact dilation)**.**

(i) If $W \subset I(g)$ is nonempty and compact, then $W \subset \operatorname{int} I(g_1)$ for every $g_1 < g$ with $g_1^{-2} + \bar{B} < \min W$, and such g_1 exist.

(ii) Every $b > \gamma^{-2} + \bar{B}$ lies in $\operatorname{int} I(g_2)$ for some $g_2 \le \gamma$; hence for every $b_\infty > 0$ there are an integer $s \ge 0$ and $g_2 \le \gamma$ with $L^s b_\infty \in \operatorname{int} I(g_2)$.

(iii) For integers $0 \le s \le K$ the dilation $x \mapsto L^s x$ carries $\mathbb{L}_K$ onto $\mathbb{L}_{K-s}$, $\mathcal{G}^{\mathrm{per}}_{K,\beta}$ onto $\mathcal{G}^{\mathrm{per}}_{K-s,\beta}$, T_ℓ at the cutoff K with its law onto $T_{L^s\ell}$ at the cutoff $K-s$, boards of side q onto boards of side $L^s q$, $\mathfrak{P}_T(\square_c)$ onto $\mathfrak{P}_{L^sT}(L^s\square_c)$ by $\tilde{f} = f(\cdot/L^s)$, and the label b into $L^s b$; and $x_c = L^{4s}\tilde{x}_c$.

■

Sketch of proof. (i) $I(g)$ is open on the left and W is compact, so $\min W > g^{-2} + \bar{B}$; for $g_1^{-2} \in (g^{-2}, \min W - \bar{B})$ one has $g_1 < g$, $g_1^{-2} + \bar{B} < \min W$ and $X_{g_1} - \bar{B} > X_g - \bar{B} \ge \max W$. (ii) Put $g_2^{-2} = b - \bar{B} - \epsilon$ with $\epsilon = \min\{1, b - \bar{B} - \gamma^{-2}\}$; then $g_2 \le \gamma$ and $g_2^{-2} + \bar{B} < b$, and since $L(\bar{B} - B_\sharp) = \bar{B}$,

$$X_{g_2} - \bar{B} > b \iff (L-1)b > 2\bar{B} + L\epsilon;$$

as $b > 4 + \bar{B}$ (because $\gamma \le \frac{1}{2}$), $(L-1)b > 4(L-1) + (L-1)\bar{B} \ge L + 2\bar{B} \ge L\epsilon + 2\bar{B}$. For $b_\infty > 0$ take s with $L^s b_\infty > \gamma^{-2} + \bar{B}$. (iii) The dilation is a graph isomorphism of the lattices, the Wilson weight reads the graph and β alone, so the laws of the tori correspond, and the periodic Gibbs states correspond by (M9); $\beta = L^K b = L^{K-s}(L^s b)$; $\tilde{\mathcal{O}}_{K-s}(\tilde{f}) = a_{K-s}^{-1}\sum_p f(x(p))a_p = L^{-s}\mathcal{O}_K(f)$, hence $\tilde{x}_c = (L^s q)^{-3}\tilde{\mathcal{O}}(\tilde{f}) = L^{-4s}x_c$; the collars and the Lipschitz constant scale with L^s, and offsets in $a_K\mathbb{Z}^3$ go to offsets in $a_{K-s}\mathbb{Z}^3$. □

Proposition 13.25 (The Källén–Lehmann form)**.** Let σ' be the limit σ of a convergent Gibbs sequence or a component σ_ξ of it, and $\mathbf{s}$ its radial kernel.

(a) $\mathbf{s}$ is completely monotone on $(0,\infty)$; so $\mathbf{s} \ge 0$ does not increase, and $\kappa' = \lim_{r\to\infty}\mathbf{s}(r) \in [0,\infty)$ exists.

(b) $r \mapsto r(\mathbf{s}(r) - \kappa')$ is completely monotone; there is exactly one positive Borel measure ρ on $[0,\infty)$ with $\mathbf{s}(r) = \kappa' + \int e^{-mr}(4\pi r)^{-1}\,\mathrm{d}\rho(m)$.

(c) $\rho(\{0\}) = \lim_{r\to\infty} 4\pi r(\mathbf{s}(r) - \kappa')$, and $\rho([0,m]) \le 4\pi e\, m^{-1}(\mathbf{s}(1/m) - \kappa')$ for $m > 0$.

(d) *Cube means, with no further hypothesis.* Let $T \ge \max\{T_0, \mathsf{l}\}$, $q = L^i \ge 2T$, a board of $\mathbb{R}^3$ of side q (a partition of $\mathbb{R}^3$ into the cubes $\square_c$ of side q centred at the points $o + qc$, $c \in \mathbb{Z}^3$, with an offset $o \in \mathbb{R}^3$; c is the index of the cell) and profiles $f_c \in \mathfrak{P}_T(\square_c)$, $f_{c'} \in \mathfrak{P}_T(\square_{c'})$ of two distinct cells be given. For every $0 < r_0 \le T/4$

$$0 \le q^{-6}\Big[\sigma'(f_c, f_{c'}) - \kappa' \int f_c \int f_{c'}\Big] \le 24\pi\, r_0(\mathbf{s}(r_0) - \kappa')/q,$$

and if the two cells share a face, the middle member is $\ge \rho(\{0\})(1 - T/q)^6/(4\sqrt{6}\,\pi q)$. So q times the middle member tends to 0 as $q \to \infty$, uniformly in the pairs of distinct cells and in the profiles, if and only if $\rho(\{0\}) = 0$.

(e) *The converse; the measure that attains the bounds.* The following are equivalent: (i) there are $T \ge \max\{T_0, \mathsf{l}\}$, C and j_0 with $L^{j_0} \ge 2T$ and $\sigma'(\theta w_i, w_i) \le C(T/q_i)^2$ for all $i \ge j_0$, the w_i of Def. 13.3 (7) being formed with this T; (ii) $\mathbf{s}(r) = O(r^{-2})$ as $r \to \infty$; (iii) $\kappa' = 0$ and $\rho([0,m]) = O(m)$ as $m \to 0$; (iv) there are A_μ and $r_* > 0$ with $\mu_h(Q_r) \le A_\mu \|h\|_1^2 r^2$ for every real $h \in C^1_c(\mathbb{E}_+)$ and $0 < r \le r_*$, μ_h formed in $\mathcal{H}(\sigma', e_3)$. Under them $\rho(\{0\}) = 0$, so that $\rho \ge \mathsf{Z}\delta_0$ with $\mathsf{Z} > 0$ is excluded, and so is $\rho([0,m]) \ge c_\rho m^\vartheta$ for all small m with $c_\rho > 0$ and

$\vartheta < 1$; the measure $\rho = c_\rho \, \mathrm{d}m$ on $[0,\infty)$, $c_\rho > 0$, whose kernel is $\mathbf{s}(r) = c_\rho/(4\pi r^2)$, satisfies (ii) and (iii) and attains both bounds in order.

■

Sketch of proof. We use Bernstein's theorem (a function on $(0,\infty)$ is completely monotone if and only if it is $\int e^{-tE}\,\mathrm{d}\mu(E)$ with a unique positive μ), that $-c'$ is completely monotone with c, and that a pointwise finite limit of completely monotone functions is completely monotone (Helly's theorem applied to the measures $e^{-t_0 E} d\mu_n$). (a) By (M10), σ' is invariant under $\mathbb{R}^3 \rtimes \mathrm{O}(3)$ and reflection positive, so the reconstruction applies: for $f_\epsilon(x) = \eta_\epsilon(x_3)\mathsf{g}(x_1,x_2)$ with $0 \le \eta_\epsilon \in C^1_c((s-\epsilon, s+\epsilon))$ of mass 1 and real $\mathsf{g} \in C^1_c(\mathbb{R}^2)$, $c_\epsilon(t) = \sigma'(\theta f_\epsilon, \tau_{te_3} f_\epsilon) = \int e^{-tE} d\mu_{f_\epsilon}$ is completely monotone, and by the kernel it tends to $\mathsf{P}_{\mathsf{g}}(2s+t)$, $\mathsf{P}_{\mathsf{g}}(u) = \iint \mathsf{g}(y)\mathsf{g}(y')\mathbf{s}((u^2+|y-y'|^2)^{1/2})\,\mathrm{d}y\,\mathrm{d}y'$; letting g tend to a point mass gives (a). (b) For a radial $\mathsf{g} \ne 0$ put $G = \mathsf{g} * \mathsf{g}$, with radial profile $\hat G$, and $\mathsf{g}_R = R^{-1}\mathsf{g}(\cdot/R)$, so that $\mathsf{P}_R(u) = 2\pi \int_0^\infty \hat G(\varrho'/R)\mathbf{s}((u^2+\varrho'^2)^{1/2})\varrho' d\varrho'$; with $\partial_u \mathbf{s}((u^2+\varrho'^2)^{1/2}) = (u/\varrho')\partial_{\varrho'}\mathbf{s}((u^2+\varrho'^2)^{1/2})$ one integration by parts gives

$$-\mathsf{P}'_R(u) = 2\pi u\Big[\hat G(0)\mathbf{s}(u) + \int_0^\infty \hat G'(v)\,\mathbf{s}((u^2+R^2v^2)^{1/2})\,\mathrm{d}v\Big] \;\longrightarrow\; 2\pi\hat G(0)\,u\,(\mathbf{s}(u)-\kappa') \qquad (R\to\infty),$$

and $-\mathsf{P}'_R$ is completely monotone; Bernstein's theorem gives ρ, normalised so that $4\pi r(\mathbf{s}(r)-\kappa') = \int e^{-mr}\,\mathrm{d}\rho(m)$. (c) Monotone convergence, and $\int e^{-mr}\,\mathrm{d}\rho(m) \ge e^{-1}\rho([0,1/r])$. (d) As $\mathbf{s} \ge \kappa'$ and $f_c, f_{c'} \ge 0$, the middle member is $q^{-6}\iint f_c(x) f_{c'}(y)(\mathbf{s}(|x-y|)-\kappa')\,\mathrm{d}x\,\mathrm{d}y \ge 0$. The supports are at distance $\ge T/4 \ge r_0$, where $\mathbf{s}(|x-y|)-\kappa' \le r_0(\mathbf{s}(r_0)-\kappa')/|x-y|$, since $r(\mathbf{s}(r)-\kappa')$ does not increase by (b); and with $0 \le f_c, f_{c'} \le 1$ on the cells, $u = (y-x)/q$ and the tent $\chi_{\mathbf{d}}(u) = \prod_\mu (1-|u_\mu - d_\mu|)_+$ about the difference $\mathbf{d} \ne 0$ of the indices of the two cells, $q^{-6}\iint_{\square_c\times\square_{c'}} |x-y|^{-1}\,\mathrm{d}x\,\mathrm{d}y = q^{-1}\int \chi_{\mathbf{d}}(u)|u|^{-1}\,\mathrm{d}u$, which is $\le q^{-1}$ if $|\mathbf{d}|_\infty \ge 2$ and $\le q^{-1}\int_{|u|\le 2\sqrt3} |u|^{-1} du = 24\pi/q$ otherwise.

For cells that share a face, $4\pi r(\mathbf{s}(r)-\kappa') = \int e^{-mr}\,\mathrm{d}\rho \ge \rho(\{0\})$, the profiles are 1 on the concentric cubes of side $q-T$, and two points of these cubes are at distance $\le \sqrt6 q$. If $\rho(\{0\}) = 0$, then $r(\mathbf{s}(r)-\kappa') \le \epsilon_1$ for $r \ge r_{\epsilon_1}$, and the pairs of points of $\square_c \times \square_{c'}$ at distance $< r_{\epsilon_1}$ have measure $\le 8\pi q^2 r_{\epsilon_1}^4$; so q times the middle member is $\le 24\pi\epsilon_1 + 8\pi(\mathbf{s}(T/4)-\kappa')r_{\epsilon_1}^4 q^{-3}$, for every $\epsilon_1 > 0$. (e) (i) gives (ii): $\theta w_i, w_i \ge 0$ have disjoint supports in a box of diameter $\sqrt6 q_i$, $\int w_i \ge (1-3T/(4q_i))^3$, and $\mathbf{s}$ does not increase, so $C(T/q_i)^2 \ge \sigma'(\theta w_i, w_i) \ge \mathbf{s}(\sqrt6 q_i)(1-3T/(4q_i))^6$; for $\sqrt6 q_i \le r < \sqrt6 q_{i+1}$, $\mathbf{s}(r) \le \mathbf{s}(\sqrt6 q_i)$ and $q_i^{-2} < 6L^2/r^2$. (ii) gives (i): if $\mathbf{s}(r) \le A_{\mathbf{s}} r^{-2}$ for $r \ge r_0$, take $T \ge \max\{T_0, \mathsf{l}, 4r_0\}$; the supports of θw_i and w_i are at distance $\ge T/4 \ge r_0$, so $\sigma'(\theta w_i, w_i) \le A_{\mathbf{s}} q_i^{-2}\int \chi_{e_3}(u)|u|^{-2}\,\mathrm{d}u \le 3\pi A_{\mathbf{s}} q_i^{-2}$, as $\chi_{e_3}(u) \le u_3$ on its support, which lies in $\{|u| \le \sqrt6,\ u_3 > 0\}$. (ii) gives (iii) by (c). (iii) gives (ii): if $\kappa' = 0$ and $\rho([0,m]) \le C_\rho m$ for $0 < m \le m_0$, then $e^{-mr} = e^{-m_0 r} + r\int_m^{m_0} e^{-ur} du$ and Fubini give $\int_{[0,m_0]} e^{-mr} d\rho(m) \le C_\rho m_0 e^{-m_0 r} + C_\rho/r$, and for $r \ge 1$ the rest of $4\pi r\mathbf{s}(r)$ is $\le e^{-m_0(r-1)} 4\pi \mathbf{s}(1)$. (iii) and (iv): for real $h \in C^1_c(\mathbb{E}_+)$ the kernel, the form (b) and the formula

$$\frac{e^{-m\sqrt{\tau^2+|w|^2}}}{4\pi\sqrt{\tau^2+|w|^2}} = \int_{\mathbb{R}^2} e^{i\mathsf{q}\cdot w}\,\frac{e^{-\tau\omega_m(\mathsf{q})}}{2\omega_m(\mathsf{q})}\,\frac{d^2\mathsf{q}}{(2\pi)^2}, \qquad \omega_m(\mathsf{q}) = (m^2+|\mathsf{q}|^2)^{1/2},\ \tau > 0,\ w \in \mathbb{R}^2,$$

give for $t \ge 0$ and $v \perp e_3$

$$\langle \psi_h, e^{-tH}\mathcal{U}(v)\psi_h\rangle = \sigma'(\theta h, \tau_{te_3+v}h) = \kappa'\Big(\int h\Big)^2 + \int \mathrm{d}\rho(m)\int e^{-t\omega_m(\mathsf{q})-iv\cdot\mathsf{q}}\,\frac{|\hat h_m(\mathsf{q})|^2}{2\omega_m(\mathsf{q})}\,\frac{d^2\mathsf{q}}{(2\pi)^2},$$

with $\hat h_m(\mathsf{q}) = \int h(y) e^{-\omega_m(\mathsf{q})y_3 + i\mathsf{q}\cdot y_\perp}\,\mathrm{d}y$; as a finite measure on $[0,\infty)\times\mathbb{R}^2$ is determined by its transform, μ_h is $\kappa'(\int h)^2\delta_{(0,0)}$ plus the images under $\mathsf{q} \mapsto (\omega_m(\mathsf{q}), -\mathsf{q})$ of the measures $|\hat h_m|^2(2\omega_m)^{-1}d^2\mathsf{q}/(2\pi)^2$, integrated in $d\rho(m)$. As $|\hat h_m| \le \|h\|_1$ and $\int_{\omega_m(\mathsf{q})\le r}(2\omega_m(\mathsf{q}))^{-1}d^2\mathsf{q}/(2\pi)^2 = (r-m)_+/(4\pi)$, this gives $\mu_h(Q_r) \le \kappa'\|h\|_1^2 + (4\pi)^{-1} r\,\rho([0,r])\|h\|_1^2$, so (iii) gives (iv). Conversely, take $0 \le h \ne 0$ with support in $\{y_3 \le R_h,\ |y_\perp| \le R_h\}$ and $r \le \min\{r_*, 1/(2R_h)\}$: the points $(\omega_m(\mathsf{q}), -\mathsf{q})$ with $m \le r/2$ and $|\mathsf{q}| \le r/2$ lie in Q_r, and there $|\hat h_m(\mathsf{q})| \ge e^{-1}\cos(\frac12)\|h\|_1$; so

$A_\mu\|h\|_1^2 r^2 \ge \mu_h(Q_r) \ge \kappa'\|h\|_1^2 + e^{-2}\cos^2(\frac12)(32\pi)^{-1} r\,\rho([0,r/2])\|h\|_1^2$, which gives (iii). Under (iii), $\rho(\{0\}) \le \rho([0,m]) \to 0$ as $m \to 0$; and $\int_0^\infty e^{-mr} c_\rho\,\mathrm{d}m/(4\pi r) = c_\rho/(4\pi r^2)$. □

Lemma 13.26 (Three mirrors; the spectral form)**.**

(a) Let σ' be a real, symmetric, normalized functional on $\mathfrak{S}(\mathbb{R}^3)$, multilinear on products of test spaces with disjoint closures, whose one-point function is $\sigma'(f) = c\int f$ and whose connected functions obey a bound table as in Section YM3_I:90. Assume that σ' is invariant under the translations of $\mathbb{R}^3$ and reflection positive at the three coordinate mirrors through 0; let $X, Y \in \mathfrak{S}(\mathbb{R}^3)$ have their supports in $\{|x|_\infty < R\}$ and $G(a) = \sigma'(X \cdot \tau_a Y)$. Then $G(a)$ converges as $\min_\mu a_\mu \to \infty$, jointly in the three coordinates, to $\langle[\theta X_0], \mathcal{P}(\{0\})[Y_0]\rangle$ in $\mathcal{H}(\sigma', e_3)$, with $X_0 = \tau_{-(R+\frac12)e_3} X$ and $Y_0 = \tau_{(R+\frac12)e_3} Y$; likewise in each of the eight orthants, X and Y being exchanged in the directions in which $a_\mu \to -\infty$.

(b) Let σ' obey a bound table with one-point constant 0, be invariant under the translations of $\mathbb{R}^3$ and reflection positive at $(e_3, 0)$, and let $\sigma'(\theta w_i, w_i) \le C_{\mathrm{LH}}(T/q_i)^2$ for $i \ge j_0$, with $q_{j_0} \ge 2T$. Then for real $h \in C_c^1(\mathbb{E}_+)$

$$\mu_h(Q_r) \le 12.46\,(1 - 3T/(4q_{j_0}))^{-6} L^2 C_{\mathrm{LH}} \|h\|_1^2 T^2 r^2 \quad (0 < r \le 1/q_{j_0}), \qquad \mathcal{P}(\{0\})\psi_h = 0.$$

Neither the cubic group nor O(3) nor a kernel is used. ■

Sketch of proof. (a) In the picture e_3, for $t = a_3 - 2R - 1 \ge 0$, translation invariance and the reconstruction give $G(a) = \langle[\theta X_0], e^{-tH}\mathcal{U}(a_\perp)[Y_0]\rangle = \int e^{-tE + i a_\perp\cdot \mathfrak{q}}\,\mathrm{d}m_3$, $m_3 = \langle[\theta X_0], \mathcal{P}(\cdot)[Y_0]\rangle$, so $\sup_{a_\perp}|G(a) - G_3(a_\perp)| \le \varepsilon_3(t) = \int_{E>0} e^{-tE} d|m_3| \downarrow 0$, with $G_3(a_\perp)$ the part at $E = 0$; the same in the pictures e_1, e_2. Changing one coordinate at a time between two points with all coordinates $\ge A$,

$$|G(a) - G(a')| \le 2\sum_\mu \varepsilon_\mu(A - 2R - 1) \to 0,$$

so G converges; its limit is the limit of the Cesàro means $A^{-3}\int_{[A,2A]^3} G$, whose three brackets in the picture e_3 tend to $1_{E=0}$, $1_{\mathfrak{q}_1=0}$, $1_{\mathfrak{q}_2=0}$, giving $m_3(\{(0,0,0)\})$. In the other orthants write $G(a) = \sigma'(\tau_{-a}X \cdot Y)$. (b) Let $w = w_i$, $q = q_i$, $i \ge j_0$; $w \ge 0$ has its support in $\{T/4 \le x_3 \le q - T/4\}$. The convolution $w * h = \int w(y)\tau_y h\,\mathrm{d}y = \int h(y)\tau_y w\,\mathrm{d}y$ is read twice, the Riemann sums converging in $\|\cdot\|_\sharp$ and $h \mapsto \psi_h$ being linear and bounded by the bound table:

$$\psi_{w*h} = \mathsf{G}_w(H,P)\psi_h, \quad \mathsf{G}_w(E,\mathfrak{q}) = \int w(y) e^{-y_3 E + i y_\perp\cdot\mathfrak{q}}\,\mathrm{d}y, \qquad \psi_{w*h} = \int h(y) e^{-y_3 H}\mathcal{U}(y_\perp)\psi_w\,\mathrm{d}y.$$

The operators $e^{-tH}\mathcal{U}(v)$ are contractions, so

$$\int |\mathsf{G}_w|^2 d\mu_h = \|\psi_{w*h}\|^2 \le \|h\|_1^2 \sigma'(\theta w, w) \le \|h\|_1^2 C_{\mathrm{LH}}(T/q)^2.$$

On $Q_{1/q}$, as φ_0 is even,

$$\mathsf{G}_w = q^{-3}\int \varphi_0(y_3 - \frac{q}{2}) e^{-y_3 E}\,\mathrm{d}y_3 \prod_{k=1,2}\int \varphi_0(y_k)\cos(y_k\mathfrak{q}_k)\,\mathrm{d}y_k \ge e^{-1}\cos^2(\tfrac12)(1 - 3T/(4q))^3;$$

and $e^2/\cos^4(\frac12) \approx 12.4577$. For $1/(Lq_i) < r \le 1/q_i$, $\mu_h(Q_r) \le \mu_h(Q_{1/q_i})$ and $q_i^{-2} < L^2 r^2$. Finally $\|\mathcal{P}(\{0\})\psi_h\|^2 = \mu_h(\{0\}) \le \mu_h(Q_r) \to 0$. □

Lemma 13.27 (Limits of Gibbs sequences, with free constants)**.** Let a frame, $0 < g' \le \gamma$, $T \ge \max\{T_0, \mathrm{l}\}$ and numbers $C_{\mathrm{LH}}, c_{\mathrm{LH}}, q_{\mathrm{LH}} > 0$ be given, and let (K_j, β_j, ν_j) be a convergent Gibbs sequence at g' with the following property: for every $q = L^i \ge \max\{2T, q_{\mathrm{LH}}\}$ there is $j(q)$ such that for all $j \ge j(q)$, all $\nu, \nu' \in \mathcal{G}^{\mathrm{per}}_{K_j,\beta_j}$, every board of $\mathbb{L}_{K_j}$ of side q with any offset and profiles of $\mathfrak{P}_T$ in two distinct cells c, c': $|\operatorname{Cov}_\nu(x_c, x_{c'})| \le C_{\mathrm{LH}}(T/q)^2$ and $|\nu(x_c) - \nu'(x_c)| \le c_{\mathrm{LH}} T/q$.

Then the conclusions 1–9 of Thm. AL (c) hold for σ and every σ_ξ, with $C_{\rm LH}$ in the place of C_2, $L^{j_0} \ge \max\{2T, q_{\rm LH}\}$, $r_0 = \sqrt{6}L^{j_0}$ and $C_{\mathbf{s}} = 6L^2(1-3T/(4L^{j_0}))^{-6}C_{\rm LH}$. ■

Sketch of proof. The pair of cells. Fix $q = q_i \ge \max\{2T, q_{\rm LH}\}$. On the board of $\mathbb{L}_{K_j}$ of side q with the offset $(0,0,\frac12 q + \frac12 a_{K_j})$, which lies in $a_{K_j}\mathbb{Z}^3$ because q/a_{K_j} is odd, the two cells with the centres $(0,0,\pm\frac12 q + \frac12 a_{K_j})$ have a common face, and $f_0(\cdot \mp \frac12 q e_3)$ is moved from them by $\frac12 a_{K_j} \le T/8$ for large j; so, φ_0 being even, $\mathcal{O}(w_i)$ and $\mathcal{O}(\theta w_i)$ are cube means x_c, $x_{c'}$ of two distinct cells. *The lattice bounds pass to σ and to every σ_ξ*, because they name the pair (K_j, β_j) alone and so hold for the ν_j and for the approximants ρ_j of σ_ξ:

$$\mathfrak{c}_\xi \int w_i = \sigma_\xi((w_i)) = \lim_j (\rho_j - \nu_j)(x_c), \qquad |(\rho_j - \nu_j)(x_c)| \le c_{\rm LH} T/q, \qquad \int w_i \ge (1 - \tfrac{3T}{4q})^3 \ge (\tfrac58)^3,$$

for every q above the floor, so $\mathfrak{c}_\xi = 0$; this is shown before the kernel is used, so that $S_2^{\sigma'}(f,h) = \sigma'(f,h)$. Further $S_2^{\sigma}(\theta w_i, w_i) = \lim_j \operatorname{Cov}_{\nu_j}(x_{c'}, x_c)$ and $\sigma_\xi(\theta w_i, w_i) = \lim_j [\operatorname{Cov}_{\rho_j}(x_{c'}, x_c) + (\rho_j - \nu_j)(x_{c'})(\rho_j - \nu_j)(x_c)]$, whose second term tends to 0; this is conclusion 2. *From two cells with a common face to the kernel.* By Prop. 13.25, $\mathbf{s} \ge 0$ does not increase; θw_i, $w_i \ge 0$ have disjoint supports in a box of diameter $\sqrt{6}q$, so $C_{\rm LH}(T/q_i)^2 \ge \sigma'(\theta w_i, w_i) \ge \mathbf{s}(\sqrt{6}q_i)(1 - 3T/(4q_i))^6$; for $r \ge r_0$ take $i \ge j_0$ with $\sqrt{6}q_i \le r < \sqrt{6}Lq_i$, so $q_i^{-2} < 6L^2/r^2$ and $\mathbf{s}(r) \le C_{\mathbf{s}} T^2/r^2$.

Hence $\kappa' = 0$, conclusion 3 is Prop. 13.25 (b), conclusion 5 follows from $|x-y| \ge |a| - R_f - R_h \ge r_0$ on the supports, and 6, 7 from Prop. 13.25 (c). *One field.* For real $h \in C_c^1(\mathbb{E}_+)$, $\|e^{-tH}\psi_h\|^2 = \sigma'(\theta h, \tau_{2te_3} h) \le \|h\|_1^2 \mathbf{s}(2t)$, the supports being at distance $\ge 2t$; so $e^{-tH}\psi_h \to 0$, which kills the projection on $\ker H$; $\int_0^\infty \langle \psi_h, e^{-tH}\psi_h\rangle \, \mathrm{d}t = \|H^{-1/2}\psi_h\|^2$ is finite by the bound at large t; and $e^{-2}\langle \psi_h, 1_{[0,\epsilon]}(H)\psi_h\rangle \le \|e^{-H/\epsilon}\psi_h\|^2$. For conclusion 9 rotate a to the e_3 axis by O(3) and put $X_0 = \tau_{-(R_X+1)e_3} X$, $h_0 = \tau_{(R_h+1)e_3} h$, $t = |a| - R_X - R_h - 2$:

$$\sigma'(X \cdot \tau_a h) = \langle [\theta X_0], e^{-tH}\psi_{h_0}\rangle, \qquad |\sigma'(X \cdot \tau_a h)| \le \|[\theta X_0]\| \, \|e^{-tH}\psi_{h_0}\| \le \mathsf{N}(X)\|h\|_1 C_{\mathbf{s}}^{1/2} T/(2t),$$

where $\|[\theta X_0]\|^2 = \sigma'(X \cdot \theta_{e_3} X) \le \mathsf{N}(X)^2$, finite by the bound tables since the supports of $X \cdot \theta_u X$ lie in one ball and those of X and $\theta_u X$ are at distance ≥ 2; in the moment–cumulant formula for $S_{n+1}^{\sigma'}(f_1, \dots, f_n, \tau_a h)$ exactly one factor holds $\tau_a h$, and it is $\sigma'(X_J \cdot \tau_a h)$ for a subtuple X_J or $\sigma'((\tau_a h)) = 0$. □

Lemma 13.28 (Growing tori, with free constants)**.** Let a frame, $0 < g' \le \gamma$, $T \ge \max\{T_0, \mathsf{l}\}$, numbers $C_{\rm LH}, q_{\rm LH} > 0$ and j_0 with $L^{j_0} \ge \max\{2T, q_{\rm LH}\}$ be given, and let S be a continuum limit at g' along growing tori with a defining sequence (K_j, ℓ_j, β_j) that has the following property: for every $q = L^i \ge \max\{2T, q_{\rm LH}\}$ there is $j(q)$ such that for all $j \ge j(q)$, $\ell_j \ge 2q$ and $|\operatorname{Cov}_{K_j,\ell_j,\beta_j}(x_c, x_{c'})| \le C_{\rm LH}(T/q)^2$ for profiles of $\mathfrak{P}_T$ in two distinct cells of every board of T_{ℓ_j} of side q. Then the conclusions 1–3 of Thm. AL (d) hold with $C_{\rm LH}$ in the place of C_2. ■

Sketch of proof. By (M10), σ_S obeys a bound table with one-point constant 0, is invariant under the translations and the cubic group, and is reflection positive at the coordinate mirrors, so Lem. 13.26 applies at $u = e_1, e_2, e_3$. For $q = q_i$ and large j, as in the proof of Lem. 13.27 on the board of T_{ℓ_j} of side q with the offset $(0,0,\frac12 q + \frac12 a_{K_j})$ ($2q$ divides ℓ_j, and the supports lie in $(-\ell_j/4, \ell_j/4)^3$), θw_i and w_i are q^{-3} times profiles of two distinct cells, so

$$\sigma_S(\theta w_i, w_i) = S_2(\theta w_i, w_i) = \lim_j S^T_{2;K_j,\ell_j,\beta_j}(\theta w_i, w_i) \le C_{\rm LH}(T/q_i)^2;$$

the cubic group carries this to the mirrors $(e_1, 0)$, $(e_2, 0)$. The spectral clauses are Lem. 13.26 (b). For conclusion 3, Lem. 13.26 (a) at $Y = (h)$ gives the limit $\langle [\theta X_0], \mathcal{P}(\{0\})\psi_{h_0}\rangle = 0$, $h_0 = \tau_{(R+\frac12)e_3} h \in C_c^1(\mathbb{E}_+)$, and 0 likewise in the other orthants; in the moment–cumulant formula exactly one factor holds $\tau_a h$, and it is $\sigma_S(X_J \cdot \tau_a h) \to 0$ or $\sigma_S((\tau_a h)) = 0$. □

Proof of Thm. AL. Step 1: the Setting is consistent. g_1 exists by Lem. 13.24 (i), hence ε exists; Prop. 13.23 is applied to (g_1, W_1, W'), with $\operatorname{dist}(W', \partial W_1) = \varepsilon$. Since $C_w^- \le C_w$, $r_1^- \le r_1$, $C_I \le C_I^+$ and $T \ge T_0$, the floor $q \ge q_0$ gives

$$\text{(f1)} \quad q \ge \frac{4C_\flat + C_I}{9C_w^2 T^2}, \qquad \text{(f2)} \quad x_q = \frac{1}{3C_w T q^2} \le \min\{r_1, \varepsilon/2\},$$

so $I_q(b) \subset W' \subset \operatorname{int} I(g_1)$ for $b \in W$, and $\delta = \delta^\circ(q) \le \delta_{\max}$ has $C_\flat \delta \le \frac{3}{2} C_w^- T_0/q \le \frac{3}{2} C_w T/q$. The number $C_\flat$ is formed from λ_+, $\hat{\mathfrak{m}}$, $\hat{c}_N$, c_N, c_2 and C_0, none of which depends on T.

Step 2: clause (a). Let $b \in W$ and $K \ge K_0(q) = K_1(\delta)$. Prop. 13.23 gives $\mu_K(I_q(b)) \le C_\flat(2x_q + \delta)$, and $2C_\flat x_q \le \frac{3}{2} C_w T/q$ is (f1); so

$$\delta_K(q, b) = 3C_w T/q + \mu_K(I_q(b)) \le 6C_w T/q.$$

Prop. 13.19 (a) at g_1 applies, as $q \ge q_1$ by $q \ge 2T$, (f1) and (f2): $2e^4 \cdot 36 = 72e^4$ and $2e^2 \cdot 6 = 12e^2$. The clause names the states at $(K, L^K b)$, which do not depend on the reference coupling.

Step 3: clause (b). Let $\ell \in \mathfrak{L}(g)$, $\ell \ge \ell_0(q) \ge \ell'$; as $\mathfrak{L}(g_1)$ is closed upward, $\ell \in \mathfrak{L}(g_1)$. $\Psi = D_K + \frac{1}{2} C_0 y^2$ and $\Phi = D_{K,\ell} + \frac{1}{2} C_0 y^2$ are convex on W_1 and $0 \le \Phi - \Psi \le \bar\eta = \bar\eta(\ell)$. With $\sigma_0 = \Psi'(b-)$, $t = (\bar\eta/C_\flat)^{1/2} \le \varepsilon/2$ (this is $\ell \ge 12\ell' B'/(C_\flat \varepsilon^2)$) and $0 < x \le \varepsilon/2$, the slopes of a convex function do not decrease:

$$\frac{\Phi(b+x) - \Phi(b)}{x} \le \frac{\Phi(b+x+t) - \Phi(b+x)}{t} \le \frac{\Psi(b+x+t) - \Psi(b+x) + \bar\eta}{t} \le \sigma_0 + C_\flat(x + t + \delta) + \frac{\bar\eta}{t},$$

by Prop. 13.23 on $[b, b+x+t] \subset W'$, and the mirror bound holds on the left; $C_\flat t + \bar\eta/t = 2(C_\flat \bar\eta)^{1/2}$. So $D_{K,\ell}$ has the hypothesis of Prop. 13.18 with $\mathsf{s} = C_0 b - \sigma_0$, $\delta_2 = 2(C_\flat \bar\eta)^{1/2} + C_\flat \delta \ge 0$, $\tilde C = C_\flat \ge 0$, $\eta_+ = 0$, $x_0 = \varepsilon/2 < b$. Its floors are (f1), (f2); $2(3C_\flat \ell' B'/\ell)^{1/2} \le \frac{3}{2} C_w T/q$ is $\ell \ge \frac{16}{3} C_\flat \ell' B' q^2/(C_w T)^2$; so $\delta_2 \le 3C_w T/q$, $\delta \le 6C_w T/q$, and the bounds follow as in Step 2.

Step 4: clause (c). Take $g' = g$ in Lem. 13.27. For $q = L^i \ge \max\{2T, q_0\}$ and large j we have $K_j \ge K_0(q)$ and $b_j \in W$; clause (a), which names the pair (K_j, β_j) alone, gives the hypothesis of Lem. 13.27 with

$$(C_{\mathrm{LH}}, c_{\mathrm{LH}}, q_{\mathrm{LH}}) = (C_2,\ 12e^2 C_w,\ q_0),$$

and its conclusions are clause (c).

Step 5: clause (d). Take $g' = g$ in Lem. 13.28. For $q = L^i \ge \max\{2T, q_0\}$ and large j: $K_j \ge K_0(q)$, $b_j \in W$, and $\ell_j \in \mathfrak{L}(g)$, $\ell_j \ge \ell_0(q) \ge 2q$. The two floors do not depend on each other, so no relation between K_j and ℓ_j is used, and clause (b) gives the hypothesis of Lem. 13.28 with $(C_{\mathrm{LH}}, q_{\mathrm{LH}}) = (C_2, q_0)$; its conclusions are clause (d).

Step 6: clauses (e) 1 and (e) 2. (e) 1 is Lem. 13.24 (ii). For (e) 2, Lem. 13.24 (iii) carries the states at $(K, L^K b)$ and their boards of side q to states at $(K - s, L^{K-s} \cdot L^s b)$ and boards of side $L^s q$, with

$$\operatorname{Cov}_\nu(x_c, x_{c'}) = L^{8s} \operatorname{Cov}_{\tilde\nu}(\tilde x_c, \tilde x_{c'}), \qquad \nu(x_c) - \nu'(x_c) = L^{4s}(\tilde\nu - \tilde\nu')(\tilde x_c),$$

and n-cell moments carrying L^{4ns}; $(L^s T)/(L^s q) = T/q$; apply (a), (b) in the Setting built on $(g_2, W^{(2)})$ at $(L^s T, L^s q, K - s, L^s \ell)$.

Step 7: clause (e) 3. Each $b_\infty \in \mathsf{B}$ has, by (e) 1, a pair (s, g_2) and a compact interval $W^{(2)} \subset \operatorname{int} I(g_2)$ holding $L^s b_\infty$ in its interior; finitely many of the open sets $L^{-s} \operatorname{int} W^{(2)}$ cover B, say $(s_i, g^{(i)}, W^{(i)})$, $i \le n_{\mathsf{B}}$, each with its Setting in the frame L and one second frame for all. Put

$$C_{\mathsf{B}} = \max_i L^{8s_i} 72e^4 \big(C_w^{+,(i)}\big)^2, \quad q_{\mathsf{B}} = \max_i L^{-s_i} q_0^{(i)}, \quad K_{\mathsf{B}}(q) = \max_i [s_i + K_0^{(i)}(L^{s_i} q)],$$

and let $\ell_{\mathsf{B}}(q)$ be the maximum over i of L^{-s_i} times $\ell_0^{(i)}(L^{s_i} q)$ formed with $C_w^{-,(i)} T_0$ in the place of $C_w T$, which bounds $\ell_0^{(i)}(L^{s_i} q)$ at the width $L^{s_i} T$ because $(L^{s_i} q)^2/(C_w L^{s_i} T)^2 = q^2/(C_w T)^2$. None of these depends on T. A side $L^{s_i} \ell = 2L^{m+s_i} \ge \ell'^{(i)}$ lies in $\mathfrak{L}(g_1^{(i)}) \subset \mathfrak{L}(g^{(i)})$ by (M5), so (b) at $g^{(i)}$

speaks of it; $T \geq \max\{T_0, \mathsf{l}\}$ gives $L^{s_i}T \geq \max\{T_0, \mathsf{l}\}$; and $(2C_{\mathsf{B}})^{1/2} \geq L^{4s_i} 12 e^2 C_w^{(i)}$. Step 6 gives (e) 3.

Step 8: clause (e) 4. At $g' \leq \gamma$ the labels of every Gibbs sequence, and of every defining sequence of a limit along growing tori, lie in the compact set $\mathsf{B} = I^{\mathrm{lab}}(g')$ by (M5). For $q = L^i \geq \max\{2T, q_{\mathsf{B}}\}$ and large j, $K_j \geq K_{\mathsf{B}}(q)$ and $\ell_j \geq \ell_{\mathsf{B}}(q) \geq 2q$, so (e) 3 gives the hypotheses of Lems. 13.27 and 13.28 with

$$(C_{\mathrm{LH}}, c_{\mathrm{LH}}, q_{\mathrm{LH}}) = (C_{\mathsf{B}}, (2C_{\mathsf{B}})^{1/2}, q_{\mathsf{B}}),$$

and these lemmas, in which the theorems on limits are taken at g' itself, give (c) and (d) at g'. The lemmas with free constants are needed here: no single window, dilated or not, holds $I^{\mathrm{lab}}(g')$, whose ends have a quotient $\geq L$, while the quotient of the ends of $I(g)$ is at most L with the lower end open; so (e) 4 does not follow from (c), (d) applied to the sequence taken whole, and the only lattice input of (c), (d) is a bound expressed through (K_j, β_j) and ℓ_j. □

Lemma 13.29 (Mean actions are slopes of the pressure; both ends are taken; the plateau)**.** Let $a > 0$ and $\beta > 0$.

(i) For $\nu \in \mathcal{G}^{\mathrm{per}}_{a,\beta}$ the number $\mathsf{a}(\nu) = \nu(a_p)$ is the same for all plaquettes p, and $P(\beta+t) - P(\beta) \geq -3\mathsf{a}(\nu)t$ for all $t > -\beta$; so $-3\mathsf{a}(\nu) \in [P'(\beta-), P'(\beta+)]$.

(ii) There are $\nu_\pm \in \mathcal{G}^{\mathrm{per}}_{a,\beta}$ with $-3\mathsf{a}(\nu_\pm) = P'(\beta\pm)$; so $3\sup\{|\mathsf{a}(\rho) - \mathsf{a}(\nu)| : \rho, \nu \in \mathcal{G}^{\mathrm{per}}_{a,\beta}\} = J(\beta)$.

(iii) At $a = a_K$, for $(K, \beta) \in \mathfrak{d}(g)$ and $\nu \in \mathcal{G}^{\mathrm{per}}_{K,\beta}$, the function e of Thm. AM (e) exists, takes its values ν-almost surely in $[\inf_\rho \mathsf{a}(\rho), \sup_\rho \mathsf{a}(\rho)]$, an interval of length $J(\beta)/3$, and $3a_K^{-4}\,\mathrm{Rie}_K(f)(\mathsf{e} - \mathsf{a}(\nu))$ is a conditional expectation of $\mathcal{O}_K(f) - \nu(\mathcal{O}_K(f))$ given $\mathcal{I}$, for every bounded f with compact support.

■

Sketch of proof. (i) The lattice isometries act transitively on the plaquettes, carry a_p to a_{rp}, and leave every periodic Gibbs state invariant (M9); so $\nu(a_p)$ is one number. The tangent inequality (M9) for the function $A = -(a_{p_{12}(0)} + a_{p_{13}(0)} + a_{p_{23}(0)})$ of the three plaquettes at one site, whose array over the periodic box of N sites a side is $-\sum_p a_p$, reads

$$\log Z_N(\beta + t) - \log Z_N(\beta) \geq -3\mathsf{a}(\nu)\, tN^3 - 144\,\beta N^2 - 72\,|t|N^2,$$

where Z_N is the partition function of that box, which does not depend on the spacing; divide by N^3 and let $N \to \infty$ (Lem. 13.11). (ii) Take points $\beta_k \downarrow \beta$ at which P is differentiable and $\nu_k \in \mathcal{G}^{\mathrm{per}}_{a,\beta_k}$, which exist by (M9); by (i), $-3\mathsf{a}(\nu_k) = P'(\beta_k)$; a subsequence converges weakly, its limit is a periodic Gibbs state at β by the closedness in (M9), and a_p being continuous, $-3\mathsf{a}(\nu_+) = \lim_k P'(\beta_k) = P'(\beta+)$; likewise $\beta_k \uparrow \beta$ gives ν_-. (iii) By Prop. YM3_I:106.3 (M9), at a first-passage datum the set $\mathcal{G}^{\mathrm{per}}_{K,\beta}$ is a class at the cutoff K that obeys the bound system at g, and with ν it contains $k\nu/\nu(k)$ for every bounded $\mathcal{I}$-measurable $k \geq 0$ with $\nu(k) > 0$; with the invariance used in (i) this gives $\nu(ka_p) = \nu(ka_{p_0})$ for all plaquettes p, so one $\mathcal{I}$-measurable e serves all plaquettes, and the statements on conditional expectations given $\mathcal{I}$ (M9) place its values between the infimum and the supremum of the $\mathsf{a}(\rho)$, an interval of length $J(\beta)/3$ by (ii); linearity gives the conditional expectation of $\mathcal{O}_K(f)$, since $a_K^{-1}\sum_p f(x(p)) = 3a_K^{-4}\,\mathrm{Rie}_K(f)$. □

Proof of Thm. AM. Step 1: clause (a). The first bound is Prop. 13.23; $|D_K''|(I) \leq \mu_K(I) + C_0|I|$. For the convex $\Phi = D_K + \frac{1}{2}C_0 y^2$,

$$\Delta_s^2\Phi(b) = \int (s - |y - b|)_+ \,\mathrm{d}\mu_K(y) = \int_0^s \mu_K((b-u, b+u))\,\mathrm{d}u \leq C_\flat(s^2 + \delta s),$$

and $\Delta_s^2 D_K = \Delta_s^2\Phi - C_0 s^2 \geq -C_0 s^2$ because $\mu_K \geq 0$. Since K_1 does not increase, the bounds hold for every $\delta > \delta_0(K)$, hence at $\delta_0(K)$, and $\delta_0(K) \leq \delta$ once $K \geq K_1(\delta)$. For the tent $\phi_{\delta'}$, $\mathsf{p}''(b) = \int \phi_{\delta'}(b - y)\,\mathrm{d}\mu_K(y) - C_0$, and the layer-cake formula with the bound on the open intervals

$(b-\delta'(1-\delta' t), b+\delta'(1-\delta' t))$ gives $\int \phi_{\delta'}(b-y)\,\mathrm{d}\mu_K(y) \le C_\flat(1+\delta_0/\delta')$; and $\mathsf{p}(b) - D_K(b) = \int_0^{\delta'} \phi_{\delta'}(t)\Delta_t^2 D_K(b)\,\mathrm{d}t$ with $\int_0^{\delta'} \phi_{\delta'}(t)t^2\,\mathrm{d}t = \delta'^2/12$ and $\int_0^{\delta'} \phi_{\delta'}(t)t\,\mathrm{d}t = \delta'/6$.

Step 2: clause (b). $p_{K,\ell'}$ is real-analytic, so the atom of $\mu_K = (D_K + \frac12 C_0 y^2)''$ at b is the jump of $p'_{K,\infty}$ there, and $p_{K,\infty}(b) = L^{3K}P(L^K b)$ by Lem. 13.11:

$$\mu_K(\{b\}) = p'_{K,\infty}(b+) - p'_{K,\infty}(b-) = L^{4K}J(L^K b) \le C_\flat\delta.$$

For the limit of $\beta^4 J(\beta)$: by Lem. 13.24 (ii) every point of $[1,L]$ has a dilate $L^s b_\infty$ in the interior of a window $W^{(\iota)} \subset \operatorname{int} I(g^{(\iota)})$; finitely many of the sets $L^{-s_\iota}\operatorname{int} W^{(\iota)}$ cover $[1,L]$, each window with its Setting. Every $\beta \ge L^{1+\max s_\iota}$ is $L^{K'}y$ with $y \in W^{(\iota)}$ for some ι and an integer K' that tends to ∞ with β; so $\beta^4 J(\beta) = y^4 \mu^{(\iota)}_{K'}(\{y\}) \le (\max W^{(\iota)})^4 C_\flat^{(\iota)}\delta$ once $K' \ge K_1^{(\iota)}(\delta)$, for every δ.

Step 3: clause (c). Lem. 13.29 (i), (ii) at $a = a_K$, $\beta = L^K b$. With $e(\nu) = 3a_K^{-4}\mathsf{a}(\nu)$: $\nu(\mathcal{O}(f)) = a_K^{-1}\mathsf{a}(\nu)\sum_p f(x(p)) = \mathrm{Rie}_K(f)e(\nu)$, and the tangent inequality in the label reads

$$p_{K,\infty}(b') - p_{K,\infty}(b) = L^{3K}\big[P(L^K b') - P(L^K b)\big] \ge -3\mathsf{a}(\nu)L^{4K}(b'-b) = -(b'-b)e(\nu),$$

since $p'_{K,\infty}(b\pm) = L^{4K}P'(L^K b\pm)$; the end points are taken by $\nu_\pm$.

Step 4: clause (d). By (c), $\nu(x_c) - \nu'(x_c) = q^{-3}\,\mathrm{Rie}_K(f)(e(\nu)-e(\nu'))$, and for $0 \le f \le 1$ vanishing off a cube of side $\le q - a_K$, each of the three orientations contributes at most $(q/a_K)^3$ plaquettes, so $0 \le q^{-3}\,\mathrm{Rie}_K(f) \le 1$:

$$|\nu(x_c) - \nu'(x_c)| \le |e(\nu) - e(\nu')| \le J_K(b) = \mu_K(\{b\}).$$

Uniformity in $b \in W \subset W'$ follows from (b) at every δ.

Step 5: clause (e). Prop. YM3_I:100.1, applied to ν and the tuples (f), (h), bounds the difference between $S_2^T[\nu](f, \tau_v h)$ and $\nu(E_\nu[\mathcal{O}(f) - \nu(\mathcal{O}(f))\,|\,\mathcal{I}]\,E_\nu[\mathcal{O}(h) - \nu(\mathcal{O}(h))\,|\,\mathcal{I}])$ by the right member; by Lem. 13.29 (iii) the latter is $9a_K^{-8}\,\mathrm{Rie}_K(f)\,\mathrm{Rie}_K(h)\,\mathrm{Var}_\nu(\mathsf{e})$, and a random variable with values in an interval of length $J(\beta)/3$ has variance $\le \frac14(J(\beta)/3)^2$, so $9a_K^{-8}\,\mathrm{Var}_\nu(\mathsf{e}) \le \frac14(a_K^{-4}J(\beta))^2 = \frac14 J_K(b)^2$. As $-3\mathsf{a}(\nu_\pm) = P'(\beta\pm)$ and $P'(\beta+) \ge P'(\beta-)$, $\mathsf{a}(\nu_+) = \inf_\rho \mathsf{a}(\rho)$ and $\mathsf{a}(\nu_-) = \sup_\rho \mathsf{a}(\rho)$; the mean of e in a state ν is $\nu(a_p) = \mathsf{a}(\nu)$. So at ν_+ the function e is $\ge$ its mean, and at ν_- it is $\le$ its mean; in both cases the mean is an end of the interval, so e is almost surely constant. An ergodic ν makes the $\mathcal{I}$-measurable e constant. At $\nu = \frac12(\nu_+ + \nu_-)$, a periodic Gibbs state,

$$\mathrm{Cov}_\nu(F,F') = \tfrac12\,\mathrm{Cov}_{\nu_+}(F,F') + \tfrac12\,\mathrm{Cov}_{\nu_-}(F,F') + \tfrac14(\nu_+(F) - \nu_-(F))(\nu_+(F') - \nu_-(F'));$$

at $F = \mathcal{O}(f)$, $F' = \mathcal{O}(\tau_v f)$ with $\mathrm{Rie}_K(f) > 0$ the covariances in $\nu_\pm$ are $O(1/|v|)$ by the plateau at $\nu_\pm$, $\nu_+(F) - \nu_-(F) = -a_K^{-4}J(\beta)\,\mathrm{Rie}_K(f)$, and $|v| \to \infty$ gives the equality.

Step 6: clause (f). (M14) is taken with what the clause supposes, which is what (M14) asks: two frames at one radius with the block sizes L_1 and $L_2 = L_1 + 2$, Thms. YM3_I:AJ and YM3_I:AD granted at both, Thm. YM3_I:D at the choices of its constants that they determine and at a lowered choice of each, and, in each frame i, the rate of the far part, that is, numbers $\mathsf{k}_i \ge 0$ and $\mathsf{c}_{\mathrm{y},i} < \infty$, free of m, K and b, with $|\mathsf{y}^{(i)}_{m,K'}(b) - \mathsf{y}^{(i)}_{m,K}(b)| \le \mathsf{c}_{\mathrm{y},i}\,\ell^9 K^{\mathsf{p}_{2,i}} L_i^{-K/10}$ for all integers $m \ge 1$, $\ell = 2L_i^m$, all $b \in \mathcal{W}_i$ and all integers $K' \ge K \ge \mathsf{k}_i + 54m$.

The power of (M14) is derived, in §12.1 on the latent heat of the SU(2) Wilson model on $\mathbb{Z}^3$, from these data, the rate among them; so (M14) gives for every real $A > 90$ numbers C_r and β_1 with $\beta^4 J(\beta) \le \mathsf{C}_\mathrm{r}\beta^{-1/(2A)}$ for $\beta \ge \beta_1$. That proof, measured on the physical scale, compares the specific heats of the Wilson model at the block sizes L_1 and $L_2 = L_1 + 2$, whose logarithms are rationally independent (Lem. 13.5), through the exact finite-size bounds, a common near part and the far parts $\mathsf{y}^{(i)}_{m,K}$, as in Prop. 13.23 made quantitative. The rate enters at the cutoffs $K \ge \mathsf{k}_i + Am$: there $\ell^9 = 2^9 L_i^{9m} \le 2^9 L_i^{9K/A}$, so that for $K' \ge K$

$$\sup_{b\in\mathcal{W}_i} |\mathsf{y}^{(i)}_{m,K'}(b) - \mathsf{y}^{(i)}_{m,K}(b)| \le 2^9 \mathsf{c}_{\mathrm{y},i} \sup_{k\ge1} k^{\mathsf{p}_{2,i}} L_i^{-(\frac{1}{10} - \frac{9}{A})k},$$

a number free of m and K, finite as $A > 90$. With $A = 1/(2\alpha)$ for $0 < \alpha < 1/180$, and $J \le 6$ (the difference quotients of P lie in $[-6, 0]$, since $0 \le \sum_p a_p \le 6$ per site), $\mathsf{c}_*(\alpha) < \infty$; for $\alpha = 0$ use $\alpha = 1/182$. On $\mathfrak{d}(g)$ the label is $b = a_K \beta \ge g^{-2}$ by (M5), so $\beta \ge 1$ and

$$a_K^{-4} J(\beta) = b^{-4}\beta^4 J(\beta) \le b^{-4}\mathsf{c}_*(\alpha)\beta^{-\alpha} = \mathsf{c}_*(\alpha)\, b^{-4-\alpha} L^{-\alpha K} \le g^{8+2\alpha}\mathsf{c}_*(\alpha) L^{-\alpha K}.$$

By (c) at b, $3a_K^{-4}|\rho(a_p) - \nu(a_p)| = |e(\rho) - e(\nu)| \le a_K^{-4} J(\beta)$; for f with support in $[-R, R]^3$ each orientation has at most $(2R/a_K + 1)^3$ barycentres there, which gives the bound for $\mathcal{O}_K(f)$; and (d) gives the bound for the cube means. □

Proof of Thm. AN. Step 1: clause (a) and the limit at the label. By the passage, $(K, L^K \mathsf{b}) \in \mathfrak{d}(g)$ for $K \ge 1$, so $(K, L^K \mathsf{b}, \nu_K)_{K \ge \max\{K_\circ, 1\}}$ is a Gibbs sequence at g, with the constant label b. By (M13) (ii) and (v), ν_K is the one periodic Gibbs state at $(K, L^K \mathsf{b})$, the sequence converges, its limit σ is its own single component with $\ker H = \mathbb{C}\Omega$, and the spectrum of H lies in $\{0\} \cup [m_*, \infty)$; σ is invariant under $\mathbb{R}^3 \rtimes \mathrm{O}(3)$, as every limit of a convergent Gibbs sequence (M10). We use two terms. A *limit at the label* b is a limit, for $n \ge 2$ and $X \in \mathfrak{S}_n(\mathbb{R}^3)$, of $S^T_{n;K,\ell,L^K\mathsf{b}}(X^{\langle \ell \rangle})$ as K and $\ell = 2L^m$ tend to infinity, together or in either order, the bare coupling being $L^K \mathsf{b}$ at every cutoff K.

A *continuum limit along growing tori at the label* b is a continuum limit at g along growing tori (Def. 13.4 (5)) whose defining sequence (K_j, ℓ_j, β_j) has $\beta_j = L^{K_j}\mathsf{b}$, that is, all its labels equal to b, from an index on. Let $X = (f_1, \dots, f_n) \in \mathfrak{S}_n(\mathbb{R}^3)$, $n \ge 2$ (an entry 0 makes both sides 0), and take a side $\ell_1 = 2L^{m_1} \ge \ell_\circ$ with the supports in $(-\ell_1/4, \ell_1/4)^3$. By (M13) (iv), for all $K \ge \max\{K_\circ, 1\}$ and all sides $\ell \ge \ell_1$,

$$|S^T_{n;K,\ell,L^K\mathsf{b}}(X^{\langle \ell \rangle}) - S^T_n[\nu_K](X)| \le C(X)\, \theta_\circ^{\ell/(2\ell_\circ)},$$

a bound free of K, while $S^T_n[\nu_K](X) \to S^\sigma_n(X)$ as $K \to \infty$, the Gibbs sequence converging. So $S^T_{n;K,\ell,L^K\mathsf{b}}(X^{\langle \ell \rangle}) \to S^\sigma_n(X)$ when K and ℓ tend to infinity together; in the order ℓ first the inner limit is $S^T_n[\nu_K](X)$; in the order K first the inner limit exists at every fixed side $\ell \ge \ell_1$ by (M8), the supports of the periodised functions being disjoint, and it lies within $C(X)\theta_\circ^{\ell/(2\ell_\circ)}$ of $S^\sigma_n(X)$. Along a defining sequence with $\beta_j = L^{K_j}\mathsf{b}$ from an index on, $K_j \to \infty$ and $\ell_j \to \infty$, the same bound gives $S_n(X) = \lim_j S^T_{n;K_j,\ell_j,\beta_j}(X^{\langle \ell_j \rangle}) = S^\sigma_n(X)$ for $n \ge 2$; as σ_S and σ are formed from their connected functions with the one-point functions 0 ($S_1 = 0$, and $\sigma((f)) = 0$ by Def. 13.4 (2)), $\sigma_S = \sigma$.

So every limit at the label b, in any order of cutoff and volume, and every continuum limit along growing tori at the label b is σ. The class $[\varnothing]$ of the empty tuple has norm $\sigma(\varnothing)^{1/2} = 1$ and is fixed by every e^{-tH}, so it spans $\ker H$; take $\Omega = [\varnothing]$. For $X, Y \in \mathfrak{S}(\mathbb{E}_+)$, $\langle [X], \Omega \rangle = \sigma(\theta X) = \sigma(X)$ by the invariance, and P_Ω commutes with H, so

$$\sigma(\theta X \cdot \tau_{te_3} Y) - \sigma(X)\sigma(Y) = \big\langle (1 - P_\Omega)[X], e^{-tH}(1 - P_\Omega)[Y] \big\rangle,$$

and $\|e^{-tH}(1 - P_\Omega)\| \le e^{-m_* t}$ by the spectral theorem; the rotations carry this to every direction. By Thm. AL (e) 4 at g, clause (c) of that theorem applies to σ: $\sigma((f)) = 0$, the kernel $\mathbf{s} \ge 0$ does not increase, and

$$4\pi r\, \mathbf{s}(r) = \int e^{-mr}\, \mathrm{d}\rho(m), \qquad \rho \ge 0, \qquad \rho(\{0\}) = 0.$$

Step 2: the gap passes to the kernel. For real $f \in C^1_c(\mathbb{E}_+)$ the vector ψ_f is orthogonal to Ω, as $\langle \Omega, \psi_f \rangle = \sigma((f)) = 0$; so by the gap $\sigma(\theta f, \tau_{te_3} f) = \langle \psi_f, e^{-tH}\psi_f \rangle \le e^{-m_* t}\|\psi_f\|^2$. With $f_\epsilon(x) = \eta_\epsilon(x_3)\mathsf{g}(x_1, x_2)$ as in the proof of Prop. 13.25, this reads $c_\epsilon(t) \le e^{-m_* t} c_\epsilon(0)$; letting $\epsilon \to 0$ and then g tend to a point mass,

$$\mathbf{s}(2s + t) \le e^{-m_* t}\, \mathbf{s}(2s) \qquad (s > 0,\ t \ge 0),$$

that is $\mathbf{s}(r) \le e^{-m_*(r - r')}\mathbf{s}(r')$ for $0 < r' \le r$.

Step 3: no spectral mass below the gap. Fix $r' > 0$. For $0 \le m < m_*$ and $r \ge r'$, $\int e^{-m''r}\,\mathrm{d}\rho(m'') \ge e^{-mr}\rho([0,m])$, so

$$\rho([0,m]) \le e^{mr}\,4\pi r\,\mathbf{s}(r) \le 4\pi r\,e^{-(m_*-m)r}e^{m_*r'}\mathbf{s}(r') \to 0 \qquad (r\to\infty),$$

and $\rho([0,m_*)) = 0$.

Step 4: the sharper decay. As ρ is carried by $[m_*,\infty)$, for $0 < r' \le r$

$$4\pi r\,\mathbf{s}(r) = \int_{[m_*,\infty)} e^{-m''r}\,\mathrm{d}\rho(m'') \le e^{-m_*(r-r')}\int e^{-m''r'}\,\mathrm{d}\rho(m'') = e^{-m_*(r-r')}\,4\pi r'\mathbf{s}(r');$$

so $\mathbf{s}(r) = O(r^{-3})$, and $\rho([0,m]) = 0$ for $m < m_*$ gives $\rho([0,m]) = O(m^2)$.

Step 5: cube means in the limit. Profiles of distinct cells have supports at distance $\ge T/4$; as $\sigma((f)) = 0$, $S_2^\sigma = \sigma$ on pairs, and with $0 \le f_c, f_{c'} \le 1$ and $\int f_c \le q^3$,

$$\bigl|S_2^\sigma(q^{-3}f_c, q^{-3}f_{c'})\bigr| \le q^{-6}\int f_c(x)\int_{|y-x|\ge T/4}\mathbf{s}(|x-y|)\,\mathrm{d}y\,\mathrm{d}x \le q^{-3}\,4\pi\int_{T/4}^{\infty} r^2\mathbf{s}(r)\,\mathrm{d}r,$$

which is finite by Step 4 and the continuity of $\mathbf{s}$ on $[T/4,\infty)$. The cells $\square_c$ and $\square_{c'}$ are centred at $o+qc$ and $o+qc'$, $c,c'\in\mathbb{Z}^3$, as in clause (c); let $\mathbf{d} = c'-c$ be the difference of their indices, as in the proof of Prop. 13.25 (d). If the two cells have no common point, that is $|\mathbf{d}|_\infty \ge 2$, then the cells are at sup-distance $q(|\mathbf{d}|_\infty - 1) \ge q$, each support lies at sup-distance $\ge T/8$ from the boundary of its cell, so the supports are at distance $\ge q + T/4$, and as $\mathbf{s}$ does not increase the left member is at most $q^{-6}\|f_c\|_1\|f_{c'}\|_1\mathbf{s}(q) \le \mathbf{s}(q)$, which by Step 4 is at most $(r'/q)\,e^{-m_*(q-r')}\mathbf{s}(r')$ for $0 < r' \le q$. For the 26 cells with $|\mathbf{d}|_\infty = 1$, which share a face, an edge or a corner with the first, the bound is $\chi_\circ q^{-3}$ of the display. □

Lemma 13.30 (A derivative from the values on a diameter)**.** Let u be holomorphic on $|z| < r$ with $|\mathsf{u}| \le M$ there, and $|\mathsf{u}(x)| \le o$ for real $|x| \le r/2$, where $o > 0$. Then $|\mathsf{u}'(0)| \le (4o/r)[1 + \log_2^+(M/o)]$. ■

Sketch of proof. Write $\mathsf{u}(z) = \sum_m c_m z^m$ with $|c_m| \le Mr^{-m}$. If $o \ge M$, $|\mathsf{u}'(0)| \le M/r \le o/r$. Otherwise let $N = \lceil \log_2(M/o)\rceil \ge 1$ and $Q(x) = \sum_{m\le N} c_m x^m$; for real $|x| \le r/2$, $|\mathsf{u}(x) - Q(x)| \le M\sum_{m>N} 2^{-m} \le o$, so $|Q| \le 2o$ there. Bernstein's inequality for a real polynomial $Q_1(t) = \sum_{k\le N} c'_k t^k$ with $|Q_1| \le 1$ on $[-1,1]$ gives $|Q_1'(0)| \le N$; applied to $Q_1(t) = \mathrm{Re}[e^{i\varphi}Q(rt/2)]/(2o)$ with the phase φ that makes $e^{i\varphi}Q'(0) \ge 0$, it gives $|\mathsf{u}'(0)| = |Q'(0)| \le 4oN/r$. □

The comparison of two cutoffs stops before the terminal scale. We therefore need the vacuum constant of the steps already performed, with the stopping index retained in the two-torus estimate.

Lemma 13.31 (Vacuum constants before the terminal scale)**.** Fix the common construction choice and the real regular progression of Thm. YM3_I:Y for one admissible sequence $x_0,\dots,x_K$. Let T_{ℓ_1} and T_{ℓ_2} be admitted tori for this sequence, where $\ell_i = 2L^{m_i}$, and assume

$$LMR(x_K) \le L^{m_i}, \qquad \ell_i \ge C_L M, \qquad \kappa \ge 2\log 9216 + 5, \qquad \delta_N M \ge \kappa, \qquad E_0 \ge 2E_N.$$

Here C_L is the room constant of Prop. YM3_I:74.5, and E_N, δ_N are the normalizer constants of equations YM3_I:(1399)–(1401). At step $j+1$ put

$$\mathsf{vl}^{(j+1)}(X,z) = E^{(j+1)}(X,\mathbf{1},z), \qquad \mathsf{vn}^{(j+1)}(X,z) = \mathsf{n}^{(j+1)}(X,z).$$

The second family is the local Gaussian constant in equation YM3_I:(1399). Both families are real and bounded by $E_0 e^{-\kappa d_{j+1}(X)}$. With the extensive term $\mathsf{N}^{(j)}$ of equations YM3_I:(1402), (1405),

define

$$\begin{aligned}\mathsf{E}^{(j)}(T_\ell,T_\ell)&=\mathsf{N}^{(j)}(T_\ell,T_\ell)+\sum_X\sum_{z\in X}\bigl(\mathsf{vl}^{(j+1)}(X,z)+\mathsf{vn}^{(j+1)}(X,z)\bigr),\\ E_{(n)}[T_\ell]&=\sum_{j=0}^{n-1}\mathsf{E}^{(j)}(T_\ell,T_\ell),\qquad 0\le n\le K.\end{aligned}\tag{13.i}$$

The domains and sites in the first sum have level $j+1$. These are the constants in equations YM3_I:(1406)–(1407): on a whole torus there is no boundary layer to remove from the set of base points.

The normalizing factor of each real step satisfies

$$\log\bigl(\mathsf{pf}_j(x_j)\mathsf{Z}_{\mathbf{1}}^{(j)}\bigr)=\mathsf{N}^{(j)}(T_\ell,T_\ell)+\sum_X\sum_{z\in X}\mathsf{vn}^{(j+1)}(X,z).\tag{13.ii}$$

Consequently the constant accumulated by the zero-source chain through step n is $E_{(n)}[T_\ell]$.

Put $\ell_*=\min(\ell_1,\ell_2)$ and $C_{\mathrm{tt}}=4096\sum_{q\ge0}e^{-q}$, as in Prop. YM3_I:74.5. Then

$$\left|\frac{E_{(n)}[T_{\ell_1}]}{\ell_1^3}-\frac{E_{(n)}[T_{\ell_2}]}{\ell_2^3}\right|\le 2C_{\mathrm{tt}}E_0\sum_{s=K-n}^{K-1}L^{3s}e^{-\kappa\ell_*L^s/(5M)}.\tag{13.iii}$$

The sum is empty when $n=0$. For every integer $u\ge0$,

$$\sum_{s\ge u}L^{3s}e^{-\alpha L^s}\le 2L^{3u}e^{-\alpha L^u},\qquad \alpha=\frac{\kappa\ell_*}{5M}>4.\tag{13.iv}$$

■

Proof. We first identify the constant of one step. Write $N_j=\#\mathbb{T}^{(j)}$, $D=N_j-N_{j+1}$ and $\nu=\frac{3}{2}\log(2\pi)$. There are $3D$ positive nonpivot bonds and $3N_{j+1}$ pivot bonds. At the trivial triple the fluctuation matrix $\mathsf{A}_{\mathbf{1}}$ is positive on a real space of dimension $9D$, and the elimination map at every pivot is $L^{-3}\mathrm{id}$ on the three-dimensional colour space. The real Gaussian integral and these pivot determinants give

$$\log\mathsf{Z}_{\mathbf{1}}^{(j)}=27N_{j+1}\log L+3D\nu-\tfrac{1}{2}\log\det\mathsf{A}_{\mathbf{1}}.$$

The Haar prefactor of the step, at $t=x_j$, is

$$\log\mathsf{pf}_j(t)=-D\log\mathsf{z}(t,\varepsilon(t))+3D\bigl(\log\sigma_{\mathrm{H}}-\tfrac{3}{2}\log t\bigr)-9N_{j+1}\log L.$$

Since $D=(L^3-1)N_{j+1}$, the definition of $\mathsf{N}^{(j)}$ yields

$$\log\bigl(\mathsf{pf}_j(t)\mathsf{Z}_{\mathbf{1}}^{(j)}\bigr)=\mathsf{N}^{(j)}(T_\ell,T_\ell)+18N_{j+1}\log L-\tfrac{1}{2}\log\det\mathsf{A}_{\mathbf{1}}.$$

It remains to identify the last two terms with the local Gaussian constants. The logarithm series defining the local normalizer in the proof of Prop. YM3_I:50.2 has $T=1-c_N^{-1}\mathsf{A}_{\mathbf{1}}$, with $\|T\|<1$. Adding its diagonal terms over the nonpivot bonds gives

$$\sum_b f_b(T_\ell)=9D\log c_N-\sum_{q\ge1}\frac{1}{q}\operatorname{Tr}T^q=\log\det\mathsf{A}_{\mathbf{1}}.$$

The series is absolutely convergent. The last equality follows by applying $-\sum_{q\ge1}(1-v)^q/q=\log v$ to each positive eigenvalue v of $c_N^{-1}\mathsf{A}_{\mathbf{1}}$, which lies in $(0,2)$. Summing the local finite differences of f_b recovers $f_b(T_\ell)$, and the tents sum to one for each bond. Thus the nonpivot constants sum to $-\frac{1}{2}\log\det\mathsf{A}_{\mathbf{1}}$. Each of the $3N_{j+1}$ pivot bonds contributes $6\log L$, by equation YM3_I:(1399). This proves eq. 13.ii. The chain identity in Prop. YM3_I:114.2 adds to this logarithm the values $E^{(j+1)}(X,\mathbf{1},z)$ of the new local terms. Summing over the steps proves the assertion about its accumulated constant.

For the comparison, we retain the level in the proof of Prop. YM3_I:74.5. At level j, the localization cubes have side ML^{j-K}. A domain is common if it has a lift contained in an axis-parallel box of side less than $\ell_*/2$. The regular families have the same value on these lifts by Thm. YM3_I:Y. The Gaussian constants have the same property: the local logarithm series, pivot bond and tent are transported by the same chart, as in the proof following equation YM3_I:(1400). Thus both families define one function over the common lifts.

The sum of this function over a period of either torus, divided by its volume, has the same value. Indeed, the larger side is an integer multiple of the smaller one; a period of the larger torus contains that multiple cubed of the lifts of each site of the smaller torus. The extensive term also cancels, since

$$\ell_i^{-3}\mathsf{N}^{(j-1)}(T_{\ell_i},T_{\ell_i}) = L^{3(K-j)}\left[3(L^3-1)(\mathsf{co}^{\mathrm{b}}_{j-1}+\mathsf{co}^{\mathrm{i}}_{j-1})+\mathsf{co}^{\mathrm{s}}_{j-1}\right]$$

depends only on the common sequence. Therefore, writing $\mathsf{S}_i^{(j),\mathrm{nc}}$ for the sum of the two local families over the non-common domains on T_{ℓ_i}, we have

$$\frac{E_{(n)}[T_{\ell_1}]}{\ell_1^3}-\frac{E_{(n)}[T_{\ell_2}]}{\ell_2^3}=\sum_{j=1}^{n}\left(\ell_1^{-3}\mathsf{S}_1^{(j),\mathrm{nc}}-\ell_2^{-3}\mathsf{S}_2^{(j),\mathrm{nc}}\right).$$

The lifting argument in that proof gives $d_j(X)\ge \ell_* L^{K-j}/(2M)-2$ for each non-common domain. For a fixed site z, its domain-counting estimate is

$$\sum_{X\ni z\,;\ d_j(X)\ge D_0} e^{-\kappa d_j(X)} \le 2048\sum_{q\ge D_0} e^{-(\kappa-\log 9216)q}\le \frac{C_{\mathrm{tt}}}{2}e^{-\kappa D_0/2}.$$

The hypotheses give $t_j=\ell_* L^{K-j}/M\ge 2L^2\ge 338$, so $\frac12(t_j/2-2)\ge t_j/5$. Multiplying the last bound by $2E_0$ and by the number $\ell_i^3 L^{3(K-j)}$ of level-j sites gives

$$\ell_i^{-3}|\mathsf{S}_i^{(j),\mathrm{nc}}|\le C_{\mathrm{tt}}E_0 L^{3(K-j)}e^{-\kappa\ell_* L^{K-j}/(5M)}.$$

Sum only over $1\le j\le n$ and put $s=K-j$ to obtain eq. 13.iii. Finally the ratio of successive terms in eq. 13.iv is

$$\exp\bigl(3\log L-\alpha L^s(L-1)\bigr)\le \tfrac12,$$

because $L\ge 13$, $\alpha\ge 4$ and $\log L\le L-1$. Summing this geometric majorant from $s=u$ proves the tail bound. □

Proof of Prop. 13.7. Step 1: the identity. The translations by $a_K\mathbb{Z}^3$ and the permutations of the axes act on T_λ, preserve the weight and the Haar measure and are transitive on the plaquettes, so $\langle a_p\rangle_{K,\lambda,L^K b}=\mathsf{a}_\lambda$ is one number; there are $3(\lambda/a_K)^3$ plaquettes, and $p'_{K,\lambda}(b)=-\lambda^{-3}L^K\langle\sum_p a_p\rangle = -3a_K^{-4}\mathsf{a}_\lambda$. As each plaquette of $\mathbb{L}_K$ has exactly one image on the torus, $\sum_{p\subset T_\lambda} f^{\langle\lambda\rangle}(x(p))=\sum_{p\subset\mathbb{L}_K} f(x(p))$, so

$$\langle\mathcal{O}(f^{\langle\lambda\rangle})\rangle_{K,\lambda,L^K b}=a_K^{-1}\mathsf{a}_\lambda\sum_p f(x(p))=-\operatorname{Rie}_K(f)\,p'_{K,\lambda}(b);$$

subtract the two sides; for the tent χ_0 of half-width $1/L$ about a site, $\operatorname{Rie}_K(\chi_0)=L^{-3}$.

Step 2: the bound. By the passage both laws are at a first-passage datum, and the transfer (M7) at $n=1$, for f with support in a ball of radius 2, gives $|A_K(f;b)|\le 2\log 2\,(2\mathsf{Q}_1+2)\varrho^{-1}\|f\|_\sharp$; at $f=\chi_0$, $\|\chi_0\|_\sharp\le \mathsf{c}_\chi$. For a cube mean, $0\le q^{-3}\operatorname{Rie}_K(f)\le 1$ as in the proof of Thm. AM (d).

Step 3: the read-out at the order 0 *on one torus.* Let $\lambda\in\{\ell,\ell'\}$, $b\in\mathsf{J}$ and $K\ge K_{\mathrm{bf}}(\lambda,\mathsf{J})$. The theorem on two consecutive cutoffs (Thm. YM3_I:AR) compares on T_λ, at the label b, the run A at the cutoff $K^{\mathrm{A}}=K$ with the run B at the cutoff $K^{\mathrm{B}}=K+1$; the mark $\bullet$ stands for A or B. Its objects are taken at the order 1, the tuple being any one real Lipschitz function on T_λ that is not 0, and at the source 0, where no object reads that function (§2.3.1 and Section YM3_I:116); its conditions hold at every such K (Cor. 2.8 (a)).

The objects are: the depth $u = \lceil K/1260 \rceil$ and the level $t_1^\bullet = K^\bullet - u$ of $\bullet$ at this depth; the space of the configurations at the depth u, one space for both runs, with its Haar measure dV and its inner ball In, an open set that holds the trivial configuration $\mathbf{1}$; the chain of $\bullet$, with its exponent $\Phi^\bullet$ on In, the sum $c^\bullet(0)$ of the constants split off along it at the steps $1, \dots, t_1^\bullet$, the integral $I^\bullet = \int_{\mathsf{In}} e^{\Phi^\bullet} dV$ and the mass $T^\bullet = e^{c^\bullet(0)} I^\bullet$; and the classical start of $\bullet$ at the depth $\lceil K/9 \rceil$, with its exponent $\hat\Phi^\bullet$ on In, the sum $\hat c^\bullet(0)$ of its constants, $\hat I^\bullet = \int_{\mathsf{In}} e^{\hat\Phi^\bullet} dV$, its measure $\hat\nu^\bullet = e^{\hat c^\bullet(0)} e^{\hat\Phi^\bullet} dV$ on In and its mass $\hat T^\bullet = \hat\nu^\bullet(\mathsf{In}) = e^{\hat c^\bullet(0)} \hat I^\bullet$ (the form of the theorem, the proof of Prop. YM3_I:116.1).

The small numbers $\epsilon_{\mathrm{fg}} \le \frac{1}{8}$, $\epsilon_{\mathrm{cl}} \le \frac{1}{20}$, $\epsilon_{\mathrm{sh}} \le \frac{1}{20}$ and $\mathsf{R}_{t_1} \le \frac{1}{2}$ are those of §2.3.1 on T_λ, and $\epsilon_3 = 4\epsilon_{\mathrm{cl}} + 6\epsilon_{\mathrm{sh}}$. We write $\mathsf{R}_K[T_\lambda]$ for the number $\mathfrak{o}_K$ of (M12) on the torus T_λ (Def. 2.7 (4)); with its bound of Cor. 2.8 (a),

$$\mathsf{R}_K[T_\lambda] = \mathfrak{o}_K = 8\mathsf{R}_{t_1} + 12\epsilon_{\mathrm{fg}} + 8\epsilon_{\mathrm{cl}} + 12\epsilon_{\mathrm{sh}} \le \mathsf{C}_{\mathrm{bf}} \lambda^3 K^{\mathsf{p}_0} L^{-K/10}.$$

The links (i)–(iv) below are the four statements of Subsection 13.1.4 that the proof of clause (c) uses, each derived here from the cited construction estimates. The links (i)–(iii) give

$$\left|\log Z_{K+1,\lambda,L^{K+1}b} - \log Z_{K,\lambda,L^K b} - \mathrm{Re}(c^{\mathrm{B}} - c^{\mathrm{A}})(0)\right| \le \mathsf{R}_K[T_\lambda] \le \mathsf{C}_{\mathrm{bf}} \lambda^3 K^{\mathsf{p}_0} L^{-K/10},$$

so that $\mathsf{R}_K[T_\lambda]$ bounds the error of the read-out at the order 0 on the torus T_λ; the link (iv) identifies $c^\bullet(0)$ for Step 4.

(i) *The read-out at the order* 0, from the discarding estimate, the form and the data of the read-out at $z = 0$. With $Z^\bullet = Z_{K^\bullet,\lambda,L^{K^\bullet}b}$, the discarding estimate (Prop. YM3_I:115.2, at the level $t_1^\bullet$) gives $(1 - \mathsf{R}_{t_1}) Z^\bullet \le T^\bullet \le Z^\bullet$. The data of the read-out at $z = 0$ (Section YM3_I:116; the proofs of Props. YM3_I:116.1 and YM3_I:116.3) give $|I^\bullet/\hat I^\bullet - 1| \le 2\epsilon_{\mathrm{fg}}$, and $\hat T^{\mathrm{A}} > 0$, $\hat T^{\mathrm{B}} > 0$ with $|\hat T^{\mathrm{B}}/\hat T^{\mathrm{A}} - 1| \le \epsilon_3 \le \frac{1}{2}$. As $T^\bullet = e^{(c^\bullet - \hat c^\bullet)(0)} (I^\bullet/\hat I^\bullet) \hat T^\bullet$ with $T^\bullet > 0$ and $\hat T^\bullet > 0$, moduli and logarithms, with $-\log(1-t) \le 2t \log 2$ for $0 \le t \le \frac{1}{2}$ and $\log \frac{1+y}{1-y} \le \frac{32}{15} y$ for $0 \le y \le \frac{1}{4}$, give

$$\left|\log \frac{Z^{\mathrm{B}}}{Z^{\mathrm{A}}} - \mathrm{Re}\big[(c^{\mathrm{B}} - \hat c^{\mathrm{B}}) - (c^{\mathrm{A}} - \hat c^{\mathrm{A}})\big](0)\right| \le 1.39\,\mathsf{R}_{t_1} + 4.27\,\epsilon_{\mathrm{fg}} + 1.39\,\epsilon_3 = 1.39\,\mathsf{R}_{t_1} + 4.27\,\epsilon_{\mathrm{fg}} + 5.56\,\epsilon_{\mathrm{cl}} + 8.34\,\epsilon_{\mathrm{sh}}.$$

(ii) *The two classical starts are comparable as measures on the inner ball*, from the proof of the comparison of their masses. Let Y run through the configurations at the depth $\lceil K/9 \rceil$, with the Haar measure dY, Y_{t_1} being the block average of Y at the depth u, and let $\mathsf{k}^\bullet \ge 0$ and $\hat F^\bullet \ge 0$ be the weights and the starts of Section YM3_I:116 at the source 0, so that $\int_{\mathsf{In}} H \, \mathrm{d}\hat\nu^\bullet = \int H(Y_{t_1})\, \mathsf{k}^\bullet \hat F^\bullet \, \mathrm{d}Y$ for every bounded Borel function H on the configurations at the depth u (§2.3.2). The positive-kernel argument in the proof of Prop. YM3_I:116.1 runs with a bounded Borel function $H \ge 0$ of the block field at the depth u as a further factor, since its kernel inequalities hold against every bounded nonnegative Borel function; it gives

$$\int H(Y_{t_1})\, \mathsf{k}^\bullet \mathbf{1}\{\mathsf{k}^\circ = 0\}\, \hat F^\bullet \, \mathrm{d}Y \le \epsilon_{\mathrm{sh}} \int_{\mathsf{In}} H \, \mathrm{d}\hat\nu^\bullet \qquad (\{\bullet, \circ\} = \{\mathrm{A}, \mathrm{B}\}).$$

Where both weights are positive they are equal and

$$\hat F^{\mathrm{B}}/\hat F^{\mathrm{A}} \in [e^{-\epsilon_{\mathrm{cl}}}, e^{\epsilon_{\mathrm{cl}}}]$$

(Prop. YM3_I:116.1). Dropping from $\int_{\mathsf{In}} H \, \mathrm{d}\hat\nu^{\mathrm{B}}$ the part where $\mathsf{k}^{\mathrm{A}} = 0$, comparing the starts where both weights are positive, and using the display for $(\bullet, \circ) = (\mathrm{A}, \mathrm{B})$, and then the same with A and B exchanged, one obtains for every bounded Borel $H \ge 0$

$$e^{-\epsilon_{\mathrm{meas}}} \int_{\mathsf{In}} H \, \mathrm{d}\hat\nu^{\mathrm{A}} \le \int_{\mathsf{In}} H \, \mathrm{d}\hat\nu^{\mathrm{B}} \le e^{\epsilon_{\mathrm{meas}}} \int_{\mathsf{In}} H \, \mathrm{d}\hat\nu^{\mathrm{A}},$$
$$\epsilon_{\mathrm{meas}} = \epsilon_{\mathrm{cl}} - \log(1 - \epsilon_{\mathrm{sh}}) \le \epsilon_{\mathrm{cl}} + 1.06\,\epsilon_{\mathrm{sh}}.$$

At the real source 0 the density $\mathsf{D}^\bullet = e^{\hat{c}^\bullet(0)} e^{\hat{\Phi}^\bullet}$ of $\hat{\nu}^\bullet$ is positive at every point of In (the positive classical density in the proof of Prop. YM3_I:116.1), so $\log \mathsf{D}^\bullet = \operatorname{Re} \hat{c}^\bullet(0) + \operatorname{Re} \hat{\Phi}^\bullet$; taking for H the indicator functions of the sets where $\mathsf{D}^{\mathrm{B}} < e^{-\epsilon_{\mathrm{meas}}} \mathsf{D}^{\mathrm{A}}$ or $\mathsf{D}^{\mathrm{A}} < e^{-\epsilon_{\mathrm{meas}}} \mathsf{D}^{\mathrm{B}}$, the display gives

$$\big|[\operatorname{Re} \hat{c}^{\mathrm{B}}(0) + \operatorname{Re} \hat{\Phi}^{\mathrm{B}}] - [\operatorname{Re} \hat{c}^{\mathrm{A}}(0) + \operatorname{Re} \hat{\Phi}^{\mathrm{A}}]\big| \le \epsilon_{\mathrm{meas}} \qquad \text{Haar-almost everywhere on } \mathsf{In}.$$

(iii) *The constants of the two classical starts cancel*, from the value 0 and the continuity of the exponent of the classical start at the trivial configuration. At the source 0 the exponent $\hat{\Phi}^\bullet(V)$ of the classical start of $\bullet$ at the level $t_1^\bullet$ is the action of its bare array at a background $U(V)$ over V less the Wilson action at that background weighted by the coupling of the level (Prop. YM3_I:114.2). It vanishes at $V = \mathbf{1}$ and is continuous there. Indeed the trivial configuration of the bare lattice has the action 0 and the block average $\mathbf{1}$, so it lies on the minimal orbit over $\mathbf{1}$, on which both terms take one value (the gauge-invariance argument preceding equation YM3_I:(2133), Props. YM3_I:114.2 and YM3_I:113.13); the weighted action vanishes at it, and so does the action of the bare array, every member of which enters less its value at the trivial pair (equation YM3_I:(874)).

The actual analytic-background argument in the proof of Prop. YM3_I:52.3 supplies a continuous representative $\Upsilon(V)$ near $\mathbf{1}$. The weighted action is the finite sum $\eta_t^{-1} \sum_p c(p) a_p(\Upsilon(V))$ of continuous plaquette functions; gauge invariance identifies this sum with the action of the background. Thus it is continuous at $\mathbf{1}$. The action of the array is continuous there as well: its members are local terms, holomorphic in the pair that they read (Lem. 5.26), the pair being a polynomial map of the configuration (§5.2, YM3_I:(70), YM3_I:(69)); so the action of the array is continuous at $\mathbf{1}$ along the continuous choice of Thm. YM3_I:E and its real-analytic continuation. So for every $\delta > 0$ there is an open set $\mathcal{N} \subset \mathsf{In}$ with $\mathbf{1} \in \mathcal{N}$ on which $|\operatorname{Re} \hat{\Phi}^{\mathrm{A}}| \le \delta$ and $|\operatorname{Re} \hat{\Phi}^{\mathrm{B}}| \le \delta$.

The comparison (ii) holds Haar-almost everywhere, so the argument reads the open set $\mathcal{N}$, which has positive Haar measure: at a point of $\mathcal{N}$ off the null set of (ii), $|\operatorname{Re} \hat{c}^{\mathrm{B}}(0) - \operatorname{Re} \hat{c}^{\mathrm{A}}(0)| \le \epsilon_{\mathrm{meas}} + 2\delta$. As $\delta > 0$ is arbitrary, $|\operatorname{Re}(\hat{c}^{\mathrm{B}} - \hat{c}^{\mathrm{A}})(0)| \le \epsilon_{\mathrm{meas}}$, and with (i)

$$\big|\log Z^{\mathrm{B}} - \log Z^{\mathrm{A}} - \operatorname{Re}(c^{\mathrm{B}} - c^{\mathrm{A}})(0)\big| \le 1.39\, \mathsf{R}_{t_1} + 4.27\, \epsilon_{\mathrm{fg}} + 6.56\, \epsilon_{\mathrm{cl}} + 9.40\, \epsilon_{\mathrm{sh}} \le \mathsf{R}_K[T_\lambda],$$

which is the display above.

(iv) *The constants at zero source.* We claim that

$$c^\bullet(0) = E_{(n)}[T_\lambda], \qquad n = t_1^\bullet = K^\bullet - u,$$

where the right side is the vacuum constant of the first n steps of the actual progression at cutoff $K^\bullet$. The hypotheses at $K \ge \max\{K_{\mathrm{bf}}(\ell, \mathsf{J}), K_{\mathrm{bf}}(\ell', \mathsf{J})\}$ give the admitted setting of §2.3.1 on both tori.

The couplings $x_j^\bullet = L^{K^\bullet - j} b$ have zero increments and form an admissible finite sequence. With either side $2L^m$ they give admitted data, by the proof of Lem. YM3_I:118.1. The stronger room required in Lem. 13.31 follows from $LMR(b) \le LMR(L(g^{-2} + \bar{B})) \le L^{m_{\mathfrak{L}}(g)} \le L^m$; its proof above also gives $2L^m \ge C_L M$.

At zero source, the family of level j in the chain is the family of the canonical progression of the cut $(x_0^\bullet, \dots, x_{K^\bullet - 1}^\bullet)$, by the same proof of Lem. YM3_I:118.1. Thm. YM3_I:Y identifies these with the first $K^\bullet$ families of the progression of the full line, with their common companions on all admitted tori. They are real at the trivial configuration and satisfy the bound in Lem. 13.31; the common choice retains $\delta_N M \ge \kappa$, $E_0 \ge 2E_N$ and $\kappa \ge 2 \log 9216 + 5$.

By Prop. YM3_I:114.2, step i adds the logarithm of its normalizing factor and the sum of the local values $E^{(i)}(X, \mathbf{1}, z)$. Equation eq. 13.ii identifies the logarithm with $\mathsf{N}^{(i-1)}$ and the sum of $\mathsf{vn}^{(i)}$. Since a whole torus has no boundary, every site in each domain is a base point. Thus the constant added at step i is exactly $\mathsf{E}^{(i-1)}(T_\lambda, T_\lambda)$. Summation over $1 \le i \le n$ proves the claimed identity. In particular these accumulated constants are real.

Step 4: two tori. Equation eq. 13.iii gives, per volume,

$$\left|E_{(n)}[T_\ell]\ell^{-3} - E_{(n)}[T_{\ell'}]\ell'^{-3}\right| \le 2C_{\mathrm{tt}}E_0 \sum_{s=K-n}^{K-1} L^{3s}\exp[-\kappa\ell' L^s/(5M)];$$

divide Step 3 by the volumes and subtract the two tori:

$$|U_{K+1}(b) - U_K(b)| \le \ell^{-3}\mathsf{R}_K[T_\ell] + \ell'^{-3}\mathsf{R}_K[T_{\ell'}] + 2\mathsf{v}_K \le 2\mathsf{C}_{\mathrm{bf}}K^{\mathsf{p}_0}L^{-K/10} + 2\mathsf{v}_K.$$

The right member is summable in K, so U_K converges uniformly on J: clause (c).

Step 5: a holomorphic extension free of the cutoff. For $\lambda \in \{\ell, \ell'\}$, $\phi_\lambda(z) = \widetilde{Z}_\lambda(z)/\widetilde{Z}_\lambda(0)$ is holomorphic on $|z| < 2\varrho$ with $\phi_\lambda(0) = 1$ and $|\phi_\lambda| \le e^{(C_S+C_g)\lambda^3} \le e^{(C_S+C_g)\ell^3}$ by (M1); by Schwarz's lemma $|\phi_\lambda - 1| \le \frac12$, so $\log\phi_\lambda$ is holomorphic with modulus $\le \log 2$, on $|z| < \varrho/(e^{(C_S+C_g)\ell^3}+1)$. By the shift of the label, $\lambda^{-3}\log\phi_\lambda(x) = p_{K,\lambda}(b-x) - p_{K,\lambda}(b) - \lambda^{-3}\mathcal{E}_\lambda(x\cdot 1)$ for real x, the logarithm being real there by the conjugation in (M1); so, with d of Lem. 13.10,

$$\mathsf{u}_{K,b}(z) = \ell^{-3}\log\phi_\ell(z) - \ell'^{-3}\log\phi_{\ell'}(z) + d(z)$$

is holomorphic on $|z| < r_3$, bounded there by M_3, and equals $U_K(b-x) - U_K(b)$ for real $|x| < r_3$; Cauchy's estimate gives $|\partial_b^n U_K(b)| \le n! M_3 r_3^{-n}$.

Step 6: the derivative. For $b \in \mathsf{W}$ the function $\mathsf{h} = \mathsf{u}_{K+1,b} - \mathsf{u}_{K,b}$ is holomorphic on $|z| < r_3$ with $|\mathsf{h}| \le 2M_3$, and for real $|x| \le r_3/2$ both b and $b-x$ lie in J, so by Step 4, with the number $\mathsf{o}_K = 4\mathsf{C}_{\mathrm{bf}}(\mathsf{J})K^{\mathsf{p}_0}L^{-K/10} + 4\mathsf{v}_K$ of (d),

$$|\mathsf{h}(x)| = \left|(U_{K+1} - U_K)(b-x) - (U_{K+1} - U_K)(b)\right| \le \mathsf{o}_K.$$

Lem. 13.30 gives $|U'_{K+1}(b) - U'_K(b)| = |\mathsf{h}'(0)| \le (4\mathsf{o}_K/r_3)[1 + \log_2^+(2M_3/\mathsf{o}_K)]$. As o_K decays like $K^{\mathsf{p}_0}L^{-K/10}$ and its logarithm grows like K, the series $\sum_K K \sup_{\mathsf{W}}|U'_{K+1} - U'_K|$ converges; this gives the uniform convergence of U'_K and the bound of $\sup_K \sup_{\mathsf{W}}|U'_K|$.

Step 7: the contact terms. The read-out of the theorem on two consecutive cutoffs at the order 1 in the source (Thm. YM3_I:AR, equation YM3_I:(2202), at $n = 1$), on each torus with $f^{\langle\lambda\rangle}$, gives $S^T_{1;K+1}(f) - S^T_{1;K}(f)$ up to the derivative at zero source of the difference of the contact terms at the cutoffs $K+1$ and K and an error that is summable in K; subtracting the two tori, by Step 1, $A_{K+1}(f) - A_K(f) = -\operatorname{Rie}_{K+1}(f)(U'_{K+1} - U'_K)(b) - [\operatorname{Rie}_{K+1}(f) - \operatorname{Rie}_K(f)]U'_K(b)$, with $|\operatorname{Rie}_{K+1}(f) - \operatorname{Rie}_K(f)| \le \sqrt{3}a_K \operatorname{Lip}(f)(2R+2)^3$ for R the radius of a ball about the origin that holds the support of f, and $|U'_K|$ bounded; so the difference between the two tori of the derivatives of the contact terms is summable in K, uniformly on W. □

Proof of Prop. 13.8. Step 1: characteristic functions from the imaginary axis. In (a), by the passage $(K, L^K b) \in \mathfrak{d}(g)$. For real y and t the function e^{ityX_κ} of the cell κ has, by the source formula, $\|e^{ityX_\kappa}\|^N_{\mathrm{ch},\ell} = Z[ity(F_e - F_o)]/Z$, a number ≥ 0, the same for both parities of κ; with $v = y(F_e - F_o)$, $\|v\|_\sharp = |y|$, and (IM) for $\widetilde{Z}^{\mathrm{av}}$ at $u = 0$,

$$\|e^{ityX_\kappa}\|_{\mathrm{ch},\ell} \le \exp[N^{-1}\operatorname{Re}\mathcal{E}^{\mathrm{av}}_\ell(itv) + C'_I t^2 y^2 q^3].$$

The Taylor coefficients of $s \mapsto N^{-1}\mathcal{E}^{\mathrm{av}}_\ell(s(F_e - F_o))$ are real by (A1) of Lem. 13.9, and bounded uniformly in ℓ at fixed K, by the count of domains, which bounds the energies of the domains that meet one cell; so $N^{-1}\operatorname{Re}\mathcal{E}^{\mathrm{av}}_\ell(itv) = -\frac12 t^2 y^2 \mathsf{e}_\ell + O(t^4)$ uniformly in ℓ, and $\bar{\mathsf{e}}$ is finite.

Step 2: the variance from below. Choose sides $\ell_k \to \infty$ with $\mathsf{e}_{\ell_k} \to \bar{\mathsf{e}}$. The domination, for the functions $e^{ity_\kappa X_\kappa}$ of different cells, gives

$$\left|\nu(e^{it\sum_\kappa y_\kappa X_\kappa})\right| \le \exp\Big[-\tfrac12 t^2(\bar{\mathsf{e}} - 2C'_I q^3)\sum_\kappa y_\kappa^2 + O(t^4)\Big],$$

while $|\nu(e^{itY})| \ge \operatorname{Re}\nu(e^{it(Y-\nu(Y))}) \ge 1 - \frac12 t^2 \operatorname{Var}_\nu(Y)$; letting $t \to 0$ gives (a).

Step 3: covariances by variances. Apply (a) to $X_\kappa \pm X_{\kappa'}$:

$$\mathrm{Var}_\nu(X_\kappa) + \mathrm{Var}_\nu(X_{\kappa'}) \pm 2\,\mathrm{Cov}_\nu(X_\kappa, X_{\kappa'}) \ge 2(\bar{\mathsf{e}} - 2C_I' q^3),$$

which is the second assertion of (a).

Step 4: Cauchy's estimate with the chord. In (b), for $z = x + i\tau$ with $|z| = r \le \varrho/2$, the source formula gives

$$N\Lambda(z) = \mathrm{Re}\,\mathcal{E}^{\mathrm{av}}(w_z) + \log|\widetilde{Z}^{\mathrm{av}}(w_z)| - \log\widetilde{Z}(0).$$

(IM) at $u = xF$, $v = \tau(F_e - F_o)$ costs $C_I'\tau^2\ell^3$. By semi-convexity, $\mathsf{g}(t) = \log\widetilde{Z}^{\mathrm{av}}(tF)$ has $\mathsf{g}'' \ge -2C_I'\ell^3$ on $[-r, r]$, so $\mathsf{g} + C_I'\ell^3 t^2$ lies below its chord:

$$\mathsf{g}(x) \le \tfrac{1}{2}[\mathsf{g}(r) + \mathsf{g}(-r)] + C_I'\ell^3 r^2 + \mathsf{m}_r x, \qquad \mathsf{m}_r = [\mathsf{g}(r) - \mathsf{g}(-r)]/(2r).$$

The cross terms between the cells give $\mathrm{Re}\,\mathcal{E}^{\mathrm{av}}(w_z) \le N\,\mathrm{Re}\,\mathcal{E}^{\mathrm{av}}(zf) + NM_\times T^{-3}\varrho^{-2}q^3|z|^2$, as in Prop. 13.16. With $\tilde\epsilon(z) = \mathcal{E}^{\mathrm{av}}(zf) + N^{-1}\mathsf{m}_r z$, holomorphic with $\tilde\epsilon(0) = 0$ and $\tilde\epsilon'(0)$ real, $\Lambda(z) - \mathrm{Re}\,\tilde\epsilon(z) \le \frac{1}{2}\mathsf{S}_f(r) + (2C_I' + M_\times T^{-3}\varrho^{-2})q^3 r^2$ on $|z| = r$, and Lem. 13.17 (a) gives the bound, with $m = q^{-3}\tilde\epsilon'(0)$; Lem. 13.17 (b) gives the covariance. At $r = \hat r q^{-3+p/2}$ one has $(rq^3)^{-2} = \hat r^{-2}q^{-p}$ and $q^3r^2 = \hat r^2 q^{p-3} \le \hat r^2$.

Step 5: the cost from below. By semi-convexity at the bases $\pm tF$, $|t| \le r \le \varrho/2$,

$$\mathsf{g}(r) + \mathsf{g}(-r) - 2\mathsf{g}(0) = \int_0^r (r-t)[\mathsf{g}''(t) + \mathsf{g}''(-t)]\,dt \ge -2C_I'\ell^3 r^2,$$

and $\ell^3/N = q^3$.

Step 6: the cost from above. In (c), the walls (Prop. 13.14) at $x = \pm r$ ($r \le b$, $r \le \varrho/(256\mathsf{c})$), the identity in the proof of Prop. 13.16, the bounds $\omega(r)+\omega(-r) \le 2C_\omega r^2$ and $|d(r)+d(-r)| \le 4\mathsf{A}_1 r^2/h_1^2$ give

$$\mathsf{S}_f(r) \le q^3\Delta_r^2 D_{K,\ell}(b) + 6C_w T q^2 r + 2q^3r^2[C_2' + C_\omega + 2\mathsf{A}_1/h_1^2] \le q^3\Delta_r^2 D_{K,\ell}(b) + 6C_w T q^2 r + 2C_I q^3 r^2.$$

At $r = \hat r q^{-3/2}$: $q^3\Delta_r^2 D_{K,\ell}(b) \le q^3[\Delta_r^2 D_K(b) + 2\bar\eta(\ell)]$ by Lem. 13.11, $q^3\Delta_r^2 D_K(b) \le C_\flat \hat r^2 + C_\flat \hat r q^{3/2}\delta \le 2C_\flat$ by Thm. AM (a) with $\delta \le q^{-3/2}$ and $[b \pm r] \subset W'$, and $2q^3\bar\eta(\ell) = 6\ell' B' q^3/\ell \le 1$; the lower bound is Step 5.

Step 7: a power of q. In (d) the proof of Lem. 13.27 applies with $C_p q_i^{-p}$ in the place of $C_{\mathrm{LH}}(T/q_i)^2$: $C_p q_i^{-p} \ge \sigma'(\theta w_i, w_i) \ge \mathbf{s}(\sqrt{6}q_i)(1 - 3T/(4q_i))^6$, and for $\sqrt{6}q_i \le r < \sqrt{6}Lq_i$, $q_i^{-p} < (\sqrt{6}L)^p r^{-p}$; then $\rho([0,m]) \le 4\pi e\, m^{-1}\mathbf{s}(1/m)$ by Prop. 13.25.

Step 8: one frame and a growth bound. In (e), Prop. 13.19 (b) at the side ℓ_q, which lies below $6LC_{\mathrm{FS}}q^3$, gives

$$\delta \le \frac{3C_wT}{q} + C_0 x_q + x_q Y_0 \ell_q^{1/3} + \frac{6C_wC_{\mathrm{FS}}Tq^2}{\ell_q} \le \frac{3C_wT}{q} + C_0 x_q + \frac{Y_0(6LC_{\mathrm{FS}})^{1/3}q}{3C_wTq^2} + \frac{C_wT}{q},$$

which is the display; the window inequality gives the three bounds. Only one frame enters.

Step 9: the variances from above. In (f), by the passage $(K, L^K b) \in \mathfrak{d}(g)$. Let ν and $0 < r \le \varrho/2$ be given, and on T_{ℓ_k}, with N_k cells, let $\tilde\epsilon_k(z) = \mathcal{E}^{\mathrm{av}}_{\ell_k}(zf) + N_k^{-1}\mathsf{m}_{r,k}z$, $\mathsf{m}_{r,k}$ the slope of the chord of Step 4. For $z = x + i\tau$ with $|z| = r$, the source formula, (IM) at $u = xF$, $v = \tau(F_e - F_o)$, the chord of Step 4, and the cross terms between the cells (Lem. 13.9 (A3) with $R_0 = q/2$, $\Delta = T/4$, $\rho_3 = 3\varrho/2$, so that $C_\times(2q/T) \le c_X$) give

$$\log\|e^{zX}\|_{\mathrm{ch},\ell_k} - \mathrm{Re}\,\tilde\epsilon_k(z) \le \tfrac{1}{2}\mathsf{S}_{f,\ell_k}(r) + 2C_I'q^3r^2 + \tfrac{4}{9}c_X\varrho^{-2}r^2 \le \mathsf{S}_{f,\ell_k}(r) + c_3 r^2,$$

the last step by $\mathsf{S}_{f,\ell_k}(r) \ge -2C_I'q^3r^2$ (Step 5). The functions $\tilde\epsilon_k$ are holomorphic on $|z| < 2\varrho$, real on the real axis and 0 at 0, and they are bounded uniformly in k, by the count of domains for the domains that meet one cell and by (M1) and the floor for the slopes; along a subsequence they converge, locally uniformly, to such a function $\tilde\epsilon$. The domination gives $|\nu(e^{zX})| \le \liminf_k \|e^{zX}\|_{\mathrm{ch},\ell_k}$,

so $G(z)=\nu(e^{zX})e^{-\tilde\epsilon(z)}$ is holomorphic on $|z|<2\varrho$, $G(0)=1$, and $|G|\le\exp[\limsup_k \mathsf{S}_{f,\ell_k}(r)+c_3r^2]$ on $|z|=r$. The coefficient of z^2 in G is $\frac12[\nu((X-\tilde\epsilon'(0))^2)-\tilde\epsilon''(0)]$, and Cauchy's estimate gives

$$\mathrm{Var}_\nu(X)\le\nu((X-\tilde\epsilon'(0))^2)\le\tilde\epsilon''(0)+2r^{-2}\exp\Big[\limsup_k \mathsf{S}_{f,\ell_k}(r)+c_3r^2\Big].$$

Finally $\tilde\epsilon''(0)$ is the limit of $\partial_s^2\mathcal{E}^{\mathrm{av}}_{\ell_k}(sf)|_{s=0}$ along the subsequence. By (A1) of Lem. 13.9 and the cross terms at $\rho_3=\varrho$, $N^{-1}\mathcal{E}^{\mathrm{av}}_\ell(s(F_e-F_o))=\frac12[\mathcal{E}^{\mathrm{av}}_\ell(sf)+\mathcal{E}^{\mathrm{av}}_\ell(-sf)]+N^{-1}\mathrm{cr}(s)$ with $|\mathrm{cr}(s)|\le Nc_X|s|^2/\varrho^2$ for complex $|s|<\varrho$, so Cauchy's estimate gives $|\mathrm{cr}''(0)|\le 2Nc_X/\varrho^2$, hence $\partial_s^2\mathcal{E}^{\mathrm{av}}_\ell(sf)|_{s=0}\le \mathsf{e}_\ell+2c_X\varrho^{-2}$ and $\tilde\epsilon''(0)\le\bar{\mathsf{e}}+2c_X\varrho^{-2}$. The second assertion of (f) is (a): $|\mathrm{Cov}_\nu(X_\kappa,X_{\kappa'})|\le C_{\mathrm{var}}q^{6-p}+2C_I'q^3\le(C_{\mathrm{var}}+2C_I')q^{6-p}$, as $p\le3$; divide by q^6.

Step 10: the walls and the cost. In (g), for t between 0 and x the source $xF+t(1-F)=t+(x-t)F$ has the norm $\le|x|\le\varrho/2$, and $\|1-F\|_\sharp\le1$; so semi-convexity gives $\partial_t^2\log\widetilde{Z}^{\mathrm{av}}(xF+t(1-F))\ge-2C_I'\ell^3$, and at $t=x$

$$\log\widetilde{Z}^{\mathrm{av}}(x)\ge\log\widetilde{Z}^{\mathrm{av}}(xF)+xN\upsilon_f(x)-C_I'\ell^3x^2,$$

which divided by N is the first assertion. Next, by (A1) of Lem. 13.9 at constant sources,

$$\mathsf{S}_f(r)=\Upsilon_f(r)+\Upsilon_f(-r)+N^{-1}\big[\log\widetilde{Z}(r)+\log\widetilde{Z}(-r)-2\log\widetilde{Z}(0)\big];$$

the first two terms are $\le2C_\Upsilon$; by the identity in the proof of Prop. 13.16 and the bounds of ω and of d there, the bracket is $q^3[\Delta_r^2D_{K,\ell}(b)+\omega(r)+\omega(-r)-d(r)-d(-r)]\le q^3[\Delta_r^2D_{K,\ell}(b)+2(C_\omega+2\mathsf{A}_1/h_1^2)r^2]$, as $r\le r_1\le\min\{r_2/2,h_1/2\}$; and $\Delta_r^2D_{K,\ell}(b)\le\Delta_r^2D_K(b)+2\bar\eta(\ell)$ by Lem. 13.11, while $\Delta_r^2D_K(b)\le C_\flat r^2+C_\flat\delta r$ by Thm. AM (a), since $[b-r,b+r]\subset W'$. □

14. The Haag net of the action density

Write $\phi(f)$ for the reconstructed action-density field on Minkowski space $\mathbb{R}^{1+2}$, and Ω for its vacuum vector. The polynomial energy bound gives a common invariant domain and closed smeared fields $\bar\phi(f)$. For an open region O, let $\mathcal{C}(O)$ be the bounded weak commutant of these local fields and their adjoints, and set $\mathcal{A}(O)=\mathcal{C}(O)'$. Here weak commutation is tested on their common energy domain. Write $O^\times$ for the open spacelike complement of O, and generate $\mathcal{A}(O^\times)$ from the algebras of regions contained in that complement.

Sketch M (Local algebras and wedge duality). For a volume-first limit and each of its components, $O\mapsto\mathcal{A}(O)$ is an isotonic, covariant local net. The field $\bar\phi(f)$ is affiliated with $\mathcal{A}(O)$ when $\mathrm{supp}\,f\subset O$. For a wedge W and its spacelike complement W', the vacuum is cyclic and separating, and

$$\mathcal{A}(W)'=\mathcal{A}(W'),\qquad \Delta^{it}=U(B(-2\pi t)).$$

Here U implements Poincaré transformations, $B(s)$ is the wedge-preserving boost of rapidity s, and Δ is the vacuum modular operator of $\mathcal{A}(W)$. For a double cone O, the dual completion $\mathcal{A}^d(O)=\mathcal{A}(O^\times)'$, extended to other open regions by joining the double-cone algebras, is a local Haag-dual net. Equality of $\mathcal{A}$ and $\mathcal{A}^d$ for all double cones is not asserted. ■□

Sketch of the argument. The reconstruction supplies local unbounded fields. Passing from their commutation relations to commutation of bounded operator algebras requires an additional argument: weak commutation on a common domain does not by itself identify the commutants of the closures. The lattice provides bounded operators with which to make this passage.

Use the algebra of bounded operators on the right half of the lattice time-zero plane. Reflection positivity embeds its vectors in the lattice Hilbert space. A quarter turn relates the two Euclidean quadrants. In the continuum this relation becomes the quarter power $\Delta^{1/4}$ of the boost modular operator. Convergence of the reflection-positive forms and functional calculus then carry the lattice modular translations to the boost group.

Every bounded weak limit from the lattice half-plane algebra commutes with the fields in the opposite wedge. To show that there are enough such limits, regularize the lattice insertion of a field and use the quarter-turn relation. The resulting bounded approximants reach a dense family of vectors $\phi(f)\Omega$ in the closure of the opposite-wedge commutant acting on Ω. Their error is controlled by a separated Euclidean four-point function. The modular relation

$$J\Delta^{1/2}Y\Omega = Y^*\Omega$$

identifies the two wedge algebras and yields locality and wedge duality. The weak-commutant definition then gives affiliation of the closed local fields. Intersecting the containing wedge algebras produces the stated dual completion.

The argument also passes to components. Translation-invariant weights can be approximated by distant averages, so they preserve the relevant positivity forms. The reconstructed Hilbert space and wedge algebra decompose over the component measure; their fibres agree with the component wedge algebras almost everywhere.

The periodic lattice approximants are essential to this supply of bounded operators. For a general limit of periodic Gibbs states the source establishes a supply of half-plane operators, but does not assert the whole locality theorem by this argument. Nor should wedge duality be renamed Haag duality of the original net on arbitrary bounded regions.

14.1. **Local nets and duality.** We construct the local net of the action density from the continuum limits and their components. The lattice quarter turn determines the modular data of a wedge. Passing its bounded operators to the limit gives wedge duality and the affiliation of the closed field operators with their local algebras.

We distinguish the original net from its dual completion: the latter is Haag dual, while equality with the original net is an additional assertion. Energy bounds, Gaussian examples and the lattice ring calculation clarify the scope of this distinction. The fixed-depth block-loop arguments use §10.1.

14.1.1. *Notation and hypotheses.*
Constants. We fix the radius $\varrho > 0$ and the constants (σ^-, γ) of the list $\mathcal{T}^{\mathrm{S}}[2\varrho]$ of the common sourced construction in Section YM3_I:82, among them the integers $L, M, \mathfrak{p}_0$ and the number $B_\sharp \geq 0$ that bounds the increments of the flow of the running coupling (see the objects below), and a coupling $0 < g \leq \gamma$; $\bar{B} = B_\sharp L/(L-1)$ is formed from σ^- alone, with no radius, γ or g, and $\bar{B} \leq \frac{13}{24}\gamma^{-2} < \gamma^{-2}$ (indeed, $B_\sharp\gamma^2 \leq \frac{1}{2}$ and $L \geq 13$ give $\bar{B}\gamma^2 \leq L/[2(L-1)] \leq 13/24$). C^g_{pt} is the constant of the point bound at g of equation YM3_I:(2097) among the construction results, and $\bar{\mathfrak{c}}(\mathsf{C}^g)$, with the bound system C^g at g of equation YM3_I:(2030), is the bound of the one-point constants of the components of equation YM3_I:(1993). The short-distance law of the two-point function among the construction results, the law of two points of Thm. YM3_I:AV as it holds in every component (Cor. YM3_I:127.11), holds at the couplings $0 < g \leq \gamma_{\mathrm{sd}}$, where $\gamma_{\mathrm{sd}} = \gamma_W \in (0, \gamma]$ is its ceiling equation YM3_I:(2346), a number formed from (σ^-, γ) alone. For $0 < g \leq \gamma_{\mathrm{sd}}$ we put

$$c_{\mathrm{sd}} = c_{\mathrm{sd}}(g) = 4.6 \cdot 10^{-9} L^{-2}(g^{-2} + \bar{B})^{-2}, \qquad r_{\mathrm{sd}} = r_{\mathrm{sd}}(g) = 2.2\,\mathsf{d}_0(g, \tfrac{1}{10}),$$

with $\bar{B} = B_\sharp L/(L-1)$ and the threshold d_0 of the law, equation YM3_I:(2350): two numbers that depend on g and on no sequence, limit, decomposition or ξ, and $c_{\mathrm{sd}}(g) \leq 4.6 \cdot 10^{-9} g^4$. Then for every component σ'_ξ of a limit of periodic Gibbs states at g the kernel s of its two-point function, $\sigma'_\xi((f,h)) = \iint f(x)h(y)\,\mathsf{s}(|x-y|)\,\mathrm{d}x\,\mathrm{d}y$ for f, h of disjoint supports, is ≥ 0 and non-increasing, and $\mathsf{s}(r) \geq c_{\mathrm{sd}} r^{-6}$ for $0 < r \leq r_{\mathrm{sd}}$. Indeed, with the point vectors Ψ_1 of the reconstruction recalled below and $0 < \epsilon < r/2$, $\mathsf{s}(r) = \|e^{-(r/2-\epsilon)H}\Psi_1(\epsilon e_3)\|^2$, and the truncated kernel, s minus the square of the one-point constant, is $\|e^{-(r/2-\epsilon)H}(1 - |\Omega\rangle\langle\Omega|)\Psi_1(\epsilon e_3)\|^2$, the vacuum of a component being unique; both are ≥ 0 and non-increasing in r.

For $0 < d \le \mathsf{d}_0(g, \frac{1}{10})$ let $f \ge 0$ be of class C^1, equal to 1 on the ball of radius $0.3d$ about $2de_3$ and to 0 off the ball of radius $0.9d$ about it, with $\sup f = 1$ and $\operatorname{Lip} f < 3/d$: then $(\theta f, f)$ is a test pair at the scale d with the shape parameter $\frac{1}{10}$ ([7, Lem. I.1]), its two supports lie at distances between $2.2d$ and $5.8d$, and $\int f < 3.06\, d^3$. The lower bound of Cor. YM3_I:127.11 makes the truncated two-point function at this pair at least $\frac{2}{3}\, 10^{-8} \mathsf{c}_2 \mathsf{b}^{-2}$, with $\mathsf{c}_2 = 9/16\pi^2$ YM3_I:C and the label $\mathsf{b} \le L(g^{-2} + \bar{B})$ (equation YM3_I:(1854)); as the truncated kernel is non-increasing, its value at $2.2d$ is at least $\frac{2}{3}\, 10^{-8} \mathsf{c}_2 \mathsf{b}^{-2} (3.06\, d^3)^{-2}$, and $\frac{2}{3}\, 10^{-8}\, \mathsf{c}_2\, (2.2)^6 (3.06)^{-2} > 4.6 \cdot 10^{-9}$ gives the bound at $r = 2.2d$, for the truncated kernel and so for s.

Space-time. In $\mathbb{R}^3 \ni x = (x_1, x_2, x_3) = (\vec{x}, x_3)$ the coordinate x_3 is Euclidean time and $\mathbb{E}_+ = \{x_3 > 0\}$; θ, θ_1 are the reflections at $x_3 = 0$ and at $x_1 = 0$, τ_v is the translation by v, $\mathsf{d}(x_1, x_2, x_3) = (x_3, x_2, x_1)$, and $R_\tau x = (x_1 \cos\tau - x_3 \sin\tau,\, x_2,\, x_1 \sin\tau + x_3 \cos\tau)$, also on $\mathbb{C}^3$, with $R = R_{\pi/2} = \theta_1 \mathsf{d}$; $\mathsf{I} = \{x_1 > 0, x_3 > 0\}$ and $\mathsf{II} = R\,\mathsf{I} = \{x_1 < 0, x_3 > 0\}$. An isometry r acts on functions by $\mathsf{r}h = h \circ \mathsf{r}^{-1}$ and on tuples entry by entry. In $\mathbb{R}^{1+2} \ni x = (x^0, x^1, x^2)$, $t = x^0$: $V_+ = \{x^0 > |\vec{x}|\}$, $W = \{x^1 > |x^0|\}$, $W' = -W$, $\mathsf{j}(x) = (-x^0, -x^1, x^2)$, and $B(s)(t, x^1, x^2) = (t\cosh s + x^1 \sinh s,\, t \sinh s + x^1 \cosh s,\, x^2)$ are the boosts that keep W. A wedge is a region $\mathfrak{g}W$, $\mathfrak{g}$ in the Poincaré group; a double cone is a Poincaré image of $\{|x^0| + |\vec{x}| < r\}$, upright if it is a translate of one; $O^\times$ is the union of the open sets that are spacelike separated from O; $\mathcal{F}(O) = C_c^\infty(O; \mathbb{C})$, $\mathcal{F}_\mathbb{R}(O)$ its real elements, $\mathcal{S} = \mathcal{S}(\mathbb{R}^{1+2})$, $\hat{f}$ the Fourier transform, and $f^{(a,\Lambda)} = f \circ (a, \Lambda)^{-1}$.

Data (Rec) and the reconstruction. A separated tuple is a tuple of real C_c^1 functions on $\mathbb{R}^3$ with pairwise disjoint supports; F, F' denote finite complex combinations of separated tuples, over a set if all supports lie in it, and $\bar{F}$ has the conjugate coefficients. Data (Rec) with a triple $(C_\wp, \wp, c_0)$ are the symmetric functionals σ on separated tuples, $\sigma(\varnothing) = 1$, with the following conditions (Section YM3_I:107): (Rec1) σ is invariant under the isometries of $\mathbb{R}^3$; (Rec2) σ is reflection positive at $x_3 = 0$, the classes $[F]$ of the combinations over $\mathbb{E}_+$ span the Hilbert space $\mathcal{H}$, $\Omega = [\varnothing]$, and $e^{-tH}[F] = [\tau_{te_3} F]$ with $H \ge 0$; (Rec3) the joint spectrum of H and the momenta lies in $\{E \ge |q|\}$; (Rec4) $|\sigma(\theta\bar{F} \cdot (h) \cdot F')| \le C_\wp \eta^{-\wp} \|h\|_{L^1} \|[\tau_{-\eta e_3} F]\| \|[\tau_{-\eta e_3} F']\|$ for $0 < \eta \le 1$, F, F' over $\{x_3 > \eta\}$ and $h \in C_c^1(\{|x_3| < c_0\eta\})$. (Rec5) is the uniqueness of the vacuum, $\ker H = \mathbb{C}\Omega$.

The reconstruction in Sections YM3_I:109 and YM3_I:110 gives $N = 1 + H$, $\mathcal{D}_\infty = \bigcap_k D(H^k)$, a representation U of the Poincaré group, and the field $\phi(f)$, $f \in \mathcal{S}$, on $\mathcal{D}_\infty$, with $\phi(f)^\dagger = \phi(\bar{f})$ and $[H, \phi(f)] = i\phi(\partial_t f)$ on $\mathcal{D}_\infty$, and with Lorentz covariance and locality on the span $\mathcal{D}_0$ of the vectors $\phi(f_1) \cdots \phi(f_n)\Omega$ ($\mathcal{D}_0(O)$: all $f_k \in \mathcal{F}(O)$); for $Y = \phi(f_1) \cdots \phi(f_n)$ we write $Y^\dagger = \phi(\bar{f}_n) \cdots \phi(\bar{f}_1)$. It gives the field $\mathsf{B}(s, s')$ at the point 0 between the Euclidean times $s, s' > 0$, with $e^{-tH} \mathsf{B}(s, s') e^{-t'H} = \mathsf{B}(s + t, s' + t')$, $\mathsf{B}(s, s')^* = \mathsf{B}(s', s)$ and

$$\|\mathsf{B}(s, s')\| \le C_\wp \min\{s, s', 1\}^{-\wp}, \qquad e^{-sH} \phi(f) e^{-sH} = \int f(x)\, U(x) \mathsf{B}(s, s) U(x)^{-1}\, \mathrm{d}x, \tag{14.i}$$

the insertion operators of Lem. YM3_I:107.1: for $0 < \eta \le 1$ and $h \in L^1(\{|x_3| < c_0\eta\})$, $\mathsf{B}^{\mathrm{in}}_\eta(h)$ is the bounded operator, linear in h and of norm $\le C_\wp \eta^{-\wp} \|h\|_{L^1}$, with $\langle [\tau_{-\eta e_3} F], \mathsf{B}^{\mathrm{in}}_\eta(h) [\tau_{-\eta e_3} F'] \rangle = \sigma(\theta\bar{F} \cdot (h) \cdot F')$ for F, F' over $\{x_3 > \eta\}$ when $h \in C_c^1(\{|x_3| < c_0\eta\})$, and by the insertion formula YM3_I:(2039) $\langle \psi, \mathsf{B}^{\mathrm{in}}_\eta(h) \psi' \rangle = \int h(x) \langle \psi, U(\vec{x}) \mathsf{B}(\eta + x_3, \eta - x_3) U(\vec{x})^{-1} \psi' \rangle\, \mathrm{d}x$; the form q_0 of the field at the point 0 and the time 0, $q_0(e^{-sH}\psi', e^{-s'H}\psi) = \langle \psi', \mathsf{B}(s, s')\psi \rangle$, and point vectors $\Psi_n(z)$, holomorphic on the tube $\mathsf{Tb}^{(n)} = \{z \in (\mathbb{C}^3)^n : d(z_1) > 0,\ d(z_{k+1} - z_k) > 0\}$, $d(z) = \operatorname{Re} z_3 - |\operatorname{Im}(z_1, z_2)|$, of norm $\le (C_\wp \eta^{-\wp})^n$ where these numbers are $\ge 2\eta$, $\eta \le 1$, with $[G] = \int \prod_k h_k(x_k) \Psi_n(x)\, \mathrm{d}x$ for ordered tuples $G = (h_k)$ in $\mathbb{E}_+$ whose supports lie in balls of radii below c_0 times the gaps of the heights.

With the continued translations $\mathcal{T}(\mathfrak{w})$, $\mathcal{T}(\eta e_3) = e^{-\eta H}$, and $\iota(t, \vec{x}) = (\vec{x}, -it)$, Lem. YM3_I:109.2, which links the field operators with the point vectors, reads

$$\text{(Br)} \qquad \mathcal{T}(\mathfrak{w}_1) \phi(f_1) \cdots \mathcal{T}(\mathfrak{w}_n) \phi(f_n) \Omega = \int \prod_k f_k(x_k)\, \Psi_n\big((\iota x_k + \textstyle\sum_{i \le k} \mathfrak{w}_i)_k\big)\, \mathrm{d}x \qquad (d(\mathfrak{w}_i) > 0),$$

and the products of fields are the limits of these vectors as the $\mathfrak{w}_i$ tend to 0.

The net and the conditions. $P_n = \mathbf{1}_{[n,n+1)}(H)$, $b_{mn}(f) = \|P_m\phi(f)P_n\|$, $\bar\phi(f)$ is the closure of $\phi(f)$, $\mathcal{P}$ is the joint spectral measure of the translations, $U(B(s)) = e^{isK}$ and $\Delta^s = e^{-2\pi sK}$. For open O

$$\mathcal{C}(O) = \{Q \in \mathcal{B}(\mathcal{H}) : \langle \phi(\bar f)\psi, Q\psi'\rangle = \langle \psi, Q\phi(f)\psi'\rangle \ \ \forall f \in \mathcal{F}(O),\ \psi,\psi' \in \mathcal{D}_0\}, \qquad \mathcal{A}(O) = \mathcal{C}(O)';$$

$\mathcal{L}(O) = \mathcal{A}(O)\cap\mathcal{C}(O^\times)$; $\mathcal{H}_0$ is the closure of $\mathcal{C}(W)\Omega$; for bounded open convex O, $\mathcal{A}^d(O) = \bigcap\{\mathcal{A}(\mathfrak{g}W) : \mathfrak{g}W \supset O\}$, and for other open G, $\mathcal{A}^d(G) = \bigvee\{\mathcal{A}^d(O) : O \subset G \text{ a double cone}\}$. $\mathfrak{P}$ is the unital $*$-algebra generated linearly by $\mathcal{F}(\mathbb{R}^{1+2})$, with $f^* = \bar f$ and $gh = hg$ for real spacelike separated g, h; π is its representation by the fields on $\mathcal{D}_0$ and $\mathcal{W}(f_1\cdots f_n) = \langle\Omega, \phi(f_1)\cdots\phi(f_n)\Omega\rangle$ its Wightman functional; for a subalgebra $\mathfrak{A}$, $\mathfrak{A}^+$ is the set of the finite sums $\sum Z_i^* Z_i$, $Z_i \in \mathfrak{A}$; $\mathfrak{P}(W)$ is generated by 1 and $\mathcal{F}(W)$.

(W) $\mathcal{C}(O_1)' \subset \mathcal{C}(O_2)$ for all non-empty open spacelike separated O_1, O_2.

(CP) For every upright double cone K, real $f \in \mathcal{F}(K)$ and $Y_0, \dots, Y_r$ in the algebra $\mathfrak{A}$ generated by 1, f and $\mathcal{F}(K^\times)$: if $\sum_n \lambda^n Y_n \in \mathfrak{A}^+$ for all real λ, then $\mathcal{W}(\sum_n f^n Y_n) \ge 0$.

(QCP$_f$) For real $f \in \mathcal{F}(K)$, K an upright double cone with $\bar K \subset W'$, and $\eta = \phi(f)\Omega$: if $Y_0, Y_1, Y_2 \in \mathfrak{P}(W)$ and $Y_0 + \lambda Y_1 + \lambda^2 Y_2 \in \mathfrak{P}(W)^+$ for all real λ, then $\langle\Omega, Y_0\Omega\rangle + \langle\Omega, Y_1\eta\rangle + \langle\eta, Y_2\eta\rangle \ge 0$. This is the second order of (CP) at f; its first order is the same statement with $Y_2 = 0$.

(LB$_{u,c}$) $|\langle\psi, \phi(u)\psi\rangle| \le c\langle\psi, N\psi\rangle$ on $\mathcal{D}_0$; (LB$^\pm_{u,c}$) are its halves $\pm\langle\psi, \phi(u)\psi\rangle \le c\langle\psi, N\psi\rangle$; (LB$_u$) holds if (LB$_{u,c}$) holds for some c, and $c_*(u) \in [0,\infty]$ is the least such c.

(QA$_f$) $\sum_n \|\phi(f)^n\Omega\|^{-1/n} = \infty$.

The objects. For a state ν on $L^{-K}\mathbb{Z}^3$ or on a torus and a tuple $X = (f_i)$ let $(X)^\nu_K = \prod_i(\mathcal{O}_K(f_i) - \nu(\mathcal{O}_K(f_i)))$, extended linearly to combinations; θ and the bar act on lattice functions by reflection and conjugation. A pair (K,β) is admitted at g if it lies in the canonical family $\mathfrak{d}(g)$ of equation YM3_I:(1843) ([7, §K.1]), that is, if it is a first-passage datum at g (Cor. YM3_I:74.2), formed at the fixed (σ^-, γ): along the flow $x_0 = \beta$, $x_k = L^{-1}x_{k-1} - b_k$ of the inverse squared running coupling, whose increments obey $|b_k| \le B_\sharp$, K is the least index with $x_K \le X_g = L(g^{-2} + B_\sharp)$.

Then $x_0 > x_1 > \cdots > x_K \ge \gamma^{-2}$, and the label $L^{-K}\beta$ lies in the outer interval of the labels $I^{\mathrm{lab}}(g) = [g^{-2} + B_\sharp - L^{-1}\bar B,\ L(g^{-2} + \bar B)]$ of (equation YM3_I:(1854)) (Cor. YM3_I:74.2). A periodic Gibbs state at (K,β) is a Gibbs state of the Wilson weight $e^{-\beta\sum_p a_p}$ on $L^{-K}\mathbb{Z}^3$ that is invariant under the translations of a subgroup of finite index of $L^{-K}\mathbb{Z}^3$; a torus state at (K,β) is a weak limit point, as $m \to \infty$, of the Wilson measures of the tori $L^{-K}\mathbb{Z}^3/2L^m\mathbb{Z}^3$ (Section YM3_I:91), and it is a periodic Gibbs state (Prop. YM3_I:91.2). A volume-first limit at g ([7, §K.1]) has a defining sequence (K_j, β_j, ν_j): $K_j \to \infty$, pairs (K_j,β_j) admitted at g, and torus states ν_j at (K_j,β_j) whose connected functions converge; its centred functional is $\sigma(X) = \lim_j \nu_j((X)_j)$, $(X)_j = (X)^{\nu_j}_{K_j}$.

A limit of periodic Gibbs states at g is $\sigma'(X) = \lim_j \rho_j((X)^{\rho_j}_{K_j})$ for periodic Gibbs states ρ_j of the Wilson weight at admitted pairs (K_j,β_j), $K_j \to \infty$. In general $(\cdot)_j$ is the product at K_j centred by the j-th state of the sequence at hand: $(X)_j = (X)^{\nu_j}_{K_j}$ for the centred functional, $(X)_j = (X)^{\rho_j}_{K_j}$ for a limit of periodic Gibbs states ρ_j, also where the periodic Gibbs states $\rho_{j,\xi}$ that approximate a component integrate, the ρ_j giving the centring, and, in Steps 2–7 of the proof of Thm. AP, the product on the torus Λ_j centred by the Wilson measure μ_j there. A canonical decomposition $(\mathsf{X}, \lambda, (\sigma_\xi))$ of σ has X compact metrisable, λ a probability measure of full support, $\xi \mapsto \sigma_\xi(X)$ continuous, $\xi \mapsto \sigma_\xi$ injective, and iterated cluster limits $\lim_{|v_q|\to\infty}\cdots\lim_{|v_1|\to\infty}\sigma(X^0\cdot\tau_{v_1}X^1\cdots\tau_{v_q}X^q) = \int\prod_i\sigma_\xi(X^i)\,\lambda(\mathrm{d}\xi)$; the σ_ξ are its components, and objects of σ_ξ bear the index ξ.

Lattice, norms, slabs. On $\Lambda = a\mathbb{Z}^3/\ell\mathbb{Z}^3$, ℓ/a even and ≥ 4: $G = \mathrm{SU}(2)$ with Haar measure, $t_p = \frac12\mathrm{tr}\,\partial U(p)$, $a_p = 1 - t_p$, the Wilson measure $\mu_\beta \propto e^{-\beta\sum_p a_p}dU$, and $\mathcal{O}(f) = a^{-1}\sum_p f(x(p))a_p$, $x(p)$ the barycentre of p ($\mathcal{O}_K$ at $a = L^{-K}$). $\|h\|_\sharp = \max\{\sup|h|, 40M\,\mathrm{Lip}\,h\}$ and $\|h\|_{\sharp,\mathsf{I}} = \max\{\sup|h|, 40M|\,\mathrm{Lip}\,h\}$; on $\mathbb{R}^2$, $|h|_{C^1} = \sup|h| + \sup|\nabla h|$; $\|f\|_\star = \int\mathrm{d}t\sum_{\kappa\in\mathbb{Z}^2}\sup_{\vec x\in\kappa+[-1,1]^2}(|f| +$

$|\nabla_{\vec{x}} f|)(t,\vec{x})$. A board (i,o) of $L^{-K}\mathbb{Z}^3$, $i \le K-1$ and o a site of $L^{-K}\mathbb{Z}^3$, is its partition into the cells $\square_{\mathbf{k}} = o + \mathsf{l}\mathbf{k} + [-\frac{\mathsf{l}}{2}, \frac{\mathsf{l}}{2}]^3$, $\mathbf{k} \in \mathbb{Z}^3$, cubes of side $\mathsf{l} = L^{-i}$ centred at the sites $o + \mathsf{l}\mathbf{k}$; the middle of $\square_{\mathbf{k}}$ is $\{|x - o - \mathsf{l}\mathbf{k}|_\infty \le \frac{\mathsf{l}}{4}\}$; for odd L the walls of the cells lie midway between sites and the middle planes of the cells pass through sites; and $\|\cdot\|_{\mathrm{ch}}$ is the seminorm of the chessboard estimate.

For $h \in L^1(\mathbb{R}^2)$, $\mathsf{B}_h(s,s') = \int h(\vec{x}) U(\vec{x}) \mathsf{B}(s,s') U(\vec{x})^{-1} d^2x$ is the field at time 0 smeared in space; $T_{mn}(h) = P_m e^{sH} \mathsf{B}_h(s,s') e^{s'H} P_n$ does not depend on s, s', and $\mathsf{T}(h) = \sum_{m,n} T_{mn}(h)$ is the form of the field at time 0 on the vectors of bounded energy. With $\rho_{\mathsf{l}}(\varsigma) = \frac{5}{\mathsf{l}} \cos^2 \frac{5\pi\varsigma}{2\mathsf{l}}$ on $|\varsigma| \le \frac{\mathsf{l}}{5}$ and 0 elsewhere, the plate is $\mathsf{l}_{\mathsf{l}}(h) = \int \rho_{\mathsf{l}}(\varsigma) \mathsf{B}_h(\frac{\mathsf{l}}{2} + \varsigma, \frac{\mathsf{l}}{2} - \varsigma)\, \mathrm{d}\varsigma$.

14.1.2. *From fields to bounded local operators.* There are two locality questions in this argument. The reconstructed field satisfies a commutation relation on its common invariant domain. A local net requires commuting algebras of bounded operators. Passing from the first statement to the second requires control of closures and their commutants. The energy estimates make the closures available; the lattice approximation supplies the bounded operators. Keeping these two tasks separate explains why reconstruction alone does not end the argument.

For an open region O in Minkowski space, write $\mathcal{F}(O) = C_c^\infty(O;\mathbb{C})$ and let $\mathcal{D}_0$ be the span of field polynomials applied to the vacuum. A bounded operator Q belongs to the weak commutant $\mathcal{C}(O)$ when

$$\langle \phi(\bar{f})\psi, Q\psi' \rangle = \langle \psi, Q\phi(f)\psi' \rangle, \qquad f \in \mathcal{F}(O), \quad \psi, \psi' \in \mathcal{D}_0.$$

Thus the definition involves bounded Q, although the field on either side is initially defined on $\mathcal{D}_0$. The proposed local algebra is $\mathcal{A}(O) = \mathcal{C}(O)'$. To establish locality one must show that these algebras commute for spacelike separated regions. The relation of field polynomials on $\mathcal{D}_0$ does not itself give that assertion about their bounded commutants.

The construction first uses the wedges

$$W = \{(t, x^1, x^2) : x^1 > |t|\}, \qquad W' = -W.$$

Their Euclidean antecedents are quadrants. A quarter turn exchanges the two boundaries of a quadrant, so the two Euclidean reflection forms put a finite-lattice operator on the two sides of the same Hilbert-space pairing. The finite-lattice four-square construction in proposition 14.13 is the source of the required bounded approximation. It retains the operator norm bound while keeping track of what the quarter turn does to vectors. Both pieces of information are needed: convergence of correlation functions would retain the vectors without supplying a bounded operator in the desired commutant.

In the limit, the quarter-turn operator is identified with $\Delta^{1/4}$, where the boost generator K is normalized by $\Delta^s = e^{-2\pi s K}$. This is an assertion about functional calculus, rather than a substitution of an imaginary angle into an unbounded operator. First prove convergence for the quarter turn, applies bounded functions of it, and only then reads off the corresponding bounded functions of K. This order permits the approximation to pass through the lattice Hilbert spaces.

Write $\mathsf{l} = \{x_1 > 0, x_3 > 0\}$ in Euclidean space. For real $u \in C_c^\infty(\mathsf{l})$, define the Schwartz function

$$f_u(B(s)(0,\rho,x_2)) = \int_0^{\pi/2} u(\rho\cos\alpha, x_2, \rho\sin\alpha) e^{-(s+i(\pi/4-\alpha))^2}\, \mathrm{d}\alpha$$

on W, and set it to zero off W. Here $\rho > 0$ and s is the boost parameter. This Gaussian smoothing is the bounded functional calculus just described. Let θ_1 reflect the first Euclidean coordinate. The construction gives bounded $T_u^{(\epsilon)} \in \mathcal{C}(W')$ such that

$$\|T_u^{(\epsilon)}\Omega - \phi(f_u)\Omega\| \le \frac{\sqrt{\epsilon}}{2} \pi e^{5\pi^2/8} \|[\theta_1 u \cdot u]\|.$$

The norm on the right is determined by a separated Euclidean four-point function. It is finite before ϵ is sent to zero. Hence the field vectors $\phi(f_u)\Omega$ lie in $\overline{\mathcal{C}(W')\Omega}$. The Fourier transforms of the f_u

have no common real zero, which gives density. This is the point at which a supply of bounded lattice limits becomes sufficient to determine the wedge algebra.

Let J be the antiunitary implementing the wedge reflection in the modular construction. The modular relation on this dense family is

$$J\Delta^{1/2}Y\Omega = Y^{*}\Omega.$$

Boost invariance makes the family a core for the operator on the left. Thus a smaller algebra generated by the lattice limits and the candidate wedge algebra have the same modular conjugation. The inclusion between the algebras gives the reverse inclusion between their commutants; conjugation by J closes the inclusion back on the algebras. The detailed argument below uses this to identify the wedge algebra, prove wedge duality, and obtain

$$\mathcal{A}(W)' = \mathcal{A}(W'), \qquad \Delta^{it} = U(B(-2\pi t)).$$

Double cones require a further distinction. For a bounded convex region O, intersect the wedge algebras containing O; for general open regions, take joins over contained double cones. This defines the dual completion $\mathcal{A}^{d}$. The argument establishes its locality and duality, but does not identify it with $\mathcal{A}$ on every double cone. Wedge equality supplies the construction of $\mathcal{A}^{d}$ without settling that additional equality.

Finally, the component argument has two levels. Positivity inequalities pass to every component through the continuous decomposition of the Euclidean functional. The identification of the fibre of a decomposable wedge algebra is a direct-integral statement and holds almost everywhere. In particular, a fibre identity is not promoted to an identity at every point of the component space.

14.1.3. *Principal results.*

Theorem AO (The net of data (Rec))**.** Let σ be data (Rec) with the triple $(C_{\wp}, \wp, c_0)$; uniqueness of the vacuum is not assumed. Every statement holds also with W and W' exchanged.

(a) *Frame.* For every n the spaces $N^{n}\mathcal{D}_0$ and $N^{n}\mathcal{D}_0(O)$, $O \neq \emptyset$ open, are dense, and $\phi(f)$ on $\mathcal{D}_0$ and on $\mathcal{D}_\infty$ have one closure $\bar{\phi}(f)$. $\mathcal{C}(O) = \{Q : Q\bar{\phi}(f) \subset \bar{\phi}(f)Q,\ f \in \mathcal{F}(O)\}$ is a von Neumann algebra, $U(\mathfrak{g})\mathcal{C}(O)U(\mathfrak{g})^{-1} = \mathcal{C}(\mathfrak{g}O)$, and $\mathcal{A}$ is isotonous and covariant. For $O \neq \emptyset$, Ω is separating for $\mathcal{C}(O)$ and cyclic for $\mathcal{A}(O)$; $\bar{\phi}(f)$, $f \in \mathcal{F}(O)$, is affiliated with $\mathcal{A}(O)$, and $\mathcal{A}$ is the least net with this property; $\mathcal{C}(\bigcup_i O_i) = \bigcap_i \mathcal{C}(O_i)$ and $\mathcal{A}(\bigcup_i O_i) = \bigvee_i \mathcal{A}(O_i)$. $\mathcal{L}$ is isotonous, covariant and local; (W) holds if and only if $\mathcal{L} = \mathcal{A}$, and if and only if it holds for spacelike separated upright double cones.

(b) *The order $\wp$.* For $f \in \mathcal{S}$ and $m, n \in \mathbb{N}_0$: (b1) $b_{mn}(f) \leq eC_{\wp}\|f\|_{L^1}(m+n+2)^{\wp}$; (b2) $(|m-n|-1)^{k}b_{mn}(f) \leq b_{mn}(\partial_t^{k}f)$ if $|m-n| \geq 2$, $k \in \mathbb{N}$; (b3) for $\epsilon \in (0,1]$, $k_1 = \max\{0, \lceil(\wp-1)/\epsilon\rceil\}$ and $|m-n| > (m+n+2)^{\epsilon}$: $b_{mn}(f) \leq 3^{k_1}eC_{\wp}\|\partial_t^{k_1}f\|_{L^1}(m+n+2)$, so that the blocks are linearly bounded off every band $|m-n| \leq (m+n+2)^{\epsilon}$, $\epsilon > 0$; (b4) $b_{mn}(h*f) \leq \|h\|_{L^1}b_{mn}(f)$ for $h \in L^1(\mathbb{R}^{1+2})$ with $h*f \in \mathcal{S}$; (b5) with $\mathsf{s}_0(q) = 1+2\cdot 2^{q}+2\cdot 3^{q}$ and $\mathsf{s}(q,k) = \sum_{l\geq 1}(l+3)^{q}l^{-k}$, for real $j \geq 0$ and $\psi, \psi' \in \mathcal{D}_\infty$:

$$\|N^{j}\phi(f)\psi\| \leq 2^{j}e^{2}C_{\wp}\big[\mathsf{s}_0(\wp+j)\|f\|_{L^1} + 2\mathsf{s}(\wp+j,k_j)\|\partial_t^{k_j}f\|_{L^1}\big]\|N^{\wp+j}\psi\|, \qquad k_j = \lfloor\wp+j\rfloor+2,$$

$$|\langle\psi', \phi(f)\psi\rangle| \leq e^{2}C_{\wp}\big[\mathsf{s}_0(\tfrac{\wp}{2})\|f\|_{L^1} + 2\mathsf{s}(\tfrac{\wp}{2},k_\flat)\|\partial_t^{k_\flat}f\|_{L^1}\big]\|N^{\wp/2}\psi'\|\,\|N^{\wp/2}\psi\|, \qquad k_\flat = \lfloor\tfrac{\wp}{2}\rfloor+2,$$

and $D(H^{\wp}) \subset D(\bar{\phi}(f))$; at $\wp = 4$ the two brackets are bounded above by $195\|f\|_{L^1} + 537.82\|\partial_t^{6}f\|_{L^1}$ and $27\|f\|_{L^1} + 37.20\|\partial_t^{4}f\|_{L^1}$.

(c) *Wedge relation.* There is exactly one antiunitary J with $J\Omega = \Omega$, $J\mathcal{D}_0 = \mathcal{D}_0$ and $J\phi(f)\psi = \phi(\bar{f}\circ\mathrm{j})J\psi$ for $\psi \in \mathcal{D}_0$; $J^2 = 1$, $JU(a)J = U(\mathrm{j}a)$, $JU(B(s))J = U(B(s))$ and $J\mathcal{C}(O)J = \mathcal{C}(\mathrm{j}O)$. For $f_1, \ldots, f_n \in \mathcal{F}(W)$ and $Y = \phi(f_1)\cdots\phi(f_n)$: $Y\Omega \in D(\Delta^{1/2})$ and $J\Delta^{1/2}Y\Omega = Y^{\dagger}\Omega$; every boost-invariant dense subspace of $D(\Delta^{1/2})$ is a core of $\Delta^{1/2}$, and $\mathcal{D}_0(W)$ is one. For $Q \in \mathcal{C}(W)$: $JQ\Omega \in D(\Delta^{1/2})$ and $\Delta^{1/2}JQ\Omega = Q^{*}\Omega$. Moreover $\mathcal{C}(W') \subset \mathcal{C}(W)'$ on all of $\mathcal{H}$;

hence $O \mapsto \mathcal{C}(O^\times)$ is isotonous, covariant and local on double cones, $\mathcal{C}(O^\times) \subset \mathcal{A}^d(O)$, and (W) holds if and only if $\mathcal{C}(O^\times) = \mathcal{A}^d(O)$ for all double cones O.

(d) *What locality says.* For every $\epsilon > 0$ there is a real $f_* \in C_c^\infty(\mathbb{R}^{1+2})$, $f_* \geq 0$, $\int f_* = 1$, even in each coordinate, with support in $[-\epsilon,\epsilon]^3$ and $\hat{f}_*(p) > 0$ for all real p; it may be chosen with $\|\partial_t^k f_*\|_{L^1} \leq A_0 R_0^k (k!)^2$ for all k, $A_0 = \frac{e^3\pi}{2}$, $R_0 = \frac{3e\pi^2}{2\epsilon}$. The following are equivalent: (W); $\mathcal{C}(W)\Omega$ is dense; $\mathcal{C}(W)' = \mathcal{C}(W')$; $J\mathcal{A}(W)J \subset \mathcal{C}(W)$; (CP); $\phi(f)\Omega \in \mathcal{H}_0$ for all $f \in \mathcal{F}(W)$; $\phi(f)\Omega \in \mathcal{H}_0$ for all f in a set $\mathfrak{F} \subset \mathcal{S}$ whose Fourier transforms have no common real zero (for instance $\mathfrak{F} = \{f_s\}$, one real f_s with $\hat{f}_s$ free of real zeros); $\inf_{s\in\mathbb{R}} \operatorname{dist}(U(B(s))\phi(f_s)\Omega, \mathcal{H}_0) = 0$ for one such f_s; for one real $f_0 \geq 0$, $f_0 \neq 0$, in C_c^∞, every Lorentz transformation Λ has a translation a_Λ with $\phi(f_0^{(a_\Lambda,\Lambda)})\Omega \in \mathcal{H}_0$; ($\mathrm{QCP}_{f_s}$) for one such f_s with support in an upright double cone K, $\bar{K} \subset W'$.

The first order of (QCP_f) holds for all data (Rec), and (QCP_f) gives operators $Q_n \in \mathcal{C}(W)$ with $\|Q_n\| \leq n$ and $Q_n\Omega \to \phi(f)\Omega$. $\mathsf{D}(f) = \operatorname{dist}(\phi(f)\Omega, \mathcal{H}_0)$ is a seminorm on $\mathcal{S}$, invariant under translations and under the boosts $B(s)$, with $\mathsf{D}(h * f) \leq \|h\|_{L^1}\mathsf{D}(f)$; (W) holds if and only if $\mathsf{D} \equiv 0$. Under (W), $\mathcal{A}(W) = \mathcal{C}(W')$, Ω is cyclic and separating for it, its modular group is $\Delta^{it} = U(B(-2\pi t))$ and its modular conjugation is J. (CP) implies, for real f, g_1, g_2 with pairwise spacelike separated compact supports,

$$|\mathcal{W}(fg_1g_2)|^4 \leq \mathcal{W}(f^2g_1^2)\,\mathcal{W}(f^2g_2^2)\,\mathcal{W}(g_1^2g_2^2),$$

and $\mathcal{W}(p(f_1,f_2)) \geq 0$ for every real polynomial $p \geq 0$ on $\mathbb{R}^2$ and real spacelike separated f_1, f_2; Motzkin's polynomial $x^4y^2 + x^2y^4 - 3x^2y^2 + 1$, which is no sum of squares, is among them.

(e) *Dual net.* Assume (W), and let O, O_1, O_2 be bounded, open and convex. Then $\mathcal{A}^d(O) = \mathcal{C}(O^\times) = \mathcal{A}(O^\times)' \supset \mathcal{A}(O)$ is a von Neumann algebra; $\mathcal{A}^d$ is isotonous and covariant, $J\mathcal{A}^d(O)J = \mathcal{A}^d(\mathrm{j}O)$, Ω is cyclic and separating for $\mathcal{A}^d(O)$, and $\bar{\phi}(f)$, $f \in \mathcal{F}(O)$, is affiliated with it; $\mathcal{A}^d(O_1) \subset \mathcal{A}^d(O_2)'$ for spacelike separated O_1, O_2; $\mathcal{A}^d(O)' = \mathcal{A}(O^\times) = \mathcal{A}^d(O^\times) = \bigvee\{\mathcal{A}(\tilde{W}') : \tilde{W} \supset O \text{ a wedge}\}$; every family $O \mapsto \mathfrak{F}(O)$ of bounded operators with $\mathfrak{F}(O) \subset \mathcal{A}(O_1)'$ for all double cones $O_1 \subset O^\times$ has $\mathfrak{F}(O) \subset \mathcal{A}^d(O)$; $\mathcal{A}^d(\tilde{W}) = \mathcal{A}(\tilde{W})$ for wedges $\tilde{W}$, so $(\mathcal{A}^d)^d = \mathcal{A}^d$; and $\mathcal{A}(O) = \mathcal{A}^d(O)$ if and only if $\mathcal{C}(O^\times) \subset \mathcal{C}(O)'$. $\mathcal{A}^d$ is outer regular: for a double cone $O = (a + V_+) \cap (b - V_+)$ and $O_\epsilon = (a - \epsilon e_0 + V_+) \cap (b + \epsilon e_0 - V_+)$, $\mathcal{A}^d(O) = \bigcap_{\epsilon>0} \mathcal{A}^d(O_\epsilon)$. Without (W): $O \mapsto \bigcap_{\mathfrak{g}W\supset O} \mathcal{A}(\mathfrak{g}W)$ is local if and only if (W) holds.

(f) *Sufficient conditions.* Let $k_0 = \max\{1, \lceil \wp \rceil\}$. If f is real and $|\langle\psi, \phi(\partial_t f)\psi\rangle| \leq c\langle\psi, N\psi\rangle$ on $\mathcal{D}_0$, then $\phi(f)$ is essentially self-adjoint on $\mathcal{D}_0$; if besides $\sup_{m,n} b_{mn}(\partial_t^j f)/(m+n+2) < \infty$ for $1 \leq j \leq k_0 + 3$, the resolvents of $\bar{\phi}(f)$ keep $D(H^k)$, $k \leq k_0$, off a strip about $\mathbb{R}$, and all of them lie in $\mathcal{C}(O)$ for every open O spacelike separated from $\operatorname{supp} f$. Each of the following implies (W) and (CP). (f1) There are real $f_i \in C_c^\infty$, $\operatorname{supp} f_i \subset \{|x| \leq \delta_i\}$, $\delta_i \to 0$, $\int f_i = 1$, $\sup_i \|f_i\|_{L^1} < \infty$, and numbers $c_i, \epsilon_i > 0$, nothing uniform in i, with $b_{mn}(\partial_t f_i) \leq c_i(m+n+2)(1 + \log(m+n+2))$ for $|m-n| \leq (m+n+2)^{\epsilon_i}$. (f2) There is a real $f_0 \in C_c^\infty$, $f_0 \geq 0$, $f_0 \neq 0$, even in t, with $\langle\psi, \phi(\partial_t f_0)\psi\rangle \leq c\langle\psi, N\psi\rangle$ on $\mathcal{D}_0$. (f3) (QA_{f_0}) holds for one real $f_0 \in C_c^\infty$, $f_0 \geq 0$, $f_0 \neq 0$. (f4) $\wp \leq 1$, or $\|\mathsf{B}(s,s)\| \leq Cs^{-1}(1 + \log s^{-1})$ for $0 < s \leq 1$; then every $\phi(f)$, $f \in \mathcal{S}$ real, is essentially self-adjoint on $\mathcal{D}_0$, and the closures commute strongly at spacelike separation. (f5) There are bounded operators $\mathsf{B}_h^e(s,s')$, $s, s' > 0$, real linear in $h \in C_c^1(\mathbb{R}^2)$, with $e^{-tH}\mathsf{B}_h^e(s,s')e^{-t'H} = \mathsf{B}_h^e(s+t, s'+t')$, $\mathsf{B}_h^e(s,s')^* = \mathsf{B}_h^e(s',s)$ and $\|\int \rho_e(\varsigma/\eta)\mathsf{B}_h^e(\eta+\varsigma, \eta-\varsigma)\,\mathrm{d}\varsigma\| \leq C_e|h|_{C^1}$ for all $\eta = \frac{1}{4(k+2)}$, $k \in \mathbb{N}_0$, for one measurable $\rho_e \geq 0$ on $[-\frac{1}{2}, \frac{1}{2}]$ with $\bar{\rho}_e = \int \rho_e > 0$; and with the time average $\mathsf{e}(f) = \int q_{f_t}(e^{-itH}\cdot, e^{-itH}\cdot)\,\mathrm{d}t$ of the sharp-time forms q_h of B^e (Lem. 14.10), the boosts

$B_j(s)$ in the direction e_j, one $s>0$ and real c,c':

$$\langle\psi,\phi(f)\psi\rangle=\mathsf{c}\Big[(1+2\coth^2 s)\,\mathsf{e}(f)(\psi,\psi)-\frac{1}{2\sinh^2 s}\sum_{j=1,2}\sum_{\pm}\mathsf{e}\big(f\circ B_j(\pm s)\big)(\psi_{j,\pm},\psi_{j,\pm})\Big]+\mathsf{c}'\int f\,\|\psi\|^2$$

for real $f\in C_c^\infty$ and $\psi\in\mathcal{D}_0$, $\psi_{j,\pm}=U(B_j(\pm s))^{-1}\psi$. Then (LB_f) holds for every real $f\in C_c^\infty$. If the bound of (f2) holds at $f_0=f_*$, the function of (d), then (f1) holds with the functions $f_i=f_*\ast\mu_i$, $\mu_i\in\mathcal{S}$; this route to (W) does not use the wedge relation (c).

(g) *Mixtures and limits.* Let $(\mathsf{X},\lambda,(\sigma_\xi))$ be a canonical decomposition of σ with all σ_ξ data (Rec) with one triple. For real $u\in\mathcal{S}$ and $c\ge 0$, $(\mathrm{LB}_{u,c})$, and each of $(\mathrm{LB}^{\pm}_{u,c})$, holds for σ if and only if it holds for every σ_ξ; so do (CP) and (W); and every inequality $\mathcal{W}(a)\ge 0$, $a\in\mathfrak{P}$ fixed, passes from all σ_ξ to σ. If data (Rec) σ_j with one triple converge to σ on separated tuples, then $\mathcal{W}^{\sigma_j}(f_1\cdots f_n)\to\mathcal{W}^{\sigma}(f_1\cdots f_n)$ for all $f_k\in\mathcal{S}$: the real-time functions converge; every inequality $\mathcal{W}(a)\ge 0$, $a\in\mathfrak{P}$, passes to σ, and $(\mathrm{LB}^{\pm}_{u,c})$, (CP) and (W) pass from all σ_j to σ.

(h) *Two bounds towards a linear one.* (h1) Let $\mathsf{l},r>0$, $Q\ge 1$, $c_2\ge 0$, $\mathfrak{m}\in\mathbb{R}$, $h'\in C_c^\infty(\mathbb{R}^2)$ real, $\rho_1\ge 0$ even, of class C^1, carried by $(-\frac14,\frac14)$, $\int\rho_1=1$, and let $\hat D(x)$, $x\in[-r,r]$, be bounded operators with: (i) $x\mapsto\hat D(x)$ is the restriction of a function holomorphic in norm on a complex neighbourhood of $[-r,r]$; (ii) $\hat D(x)=\hat D(x)^*\ge 0$ and $\|\hat D(x)\|\le Q$; (iii) $\hat D(0)=e^{-\mathsf{l}H}$; (iv) $\hat D'(0)=\mathsf{l}-\mathfrak{m}e^{-\mathsf{l}H}$ with $\mathsf{l}=\int\mathsf{l}^{-1}\rho_1(\varsigma/\mathsf{l})\mathsf{B}_{h'}(\frac{\mathsf{l}}{2}+\varsigma,\frac{\mathsf{l}}{2}-\varsigma)\,\mathrm{d}\varsigma$; (v) for every $\Lambda<\infty$ and every ψ in $\mathcal{H}_\Lambda=\mathbf{1}_{[0,\Lambda]}(H)\mathcal{H}$: $\limsup_{\epsilon\downarrow 0}\sup_{|x|\le r}\frac{d^2}{dx^2}\langle\psi,-\log(\hat D(x)+\epsilon)\psi\rangle\le c_2\langle\psi,N\psi\rangle$. Then for $\psi\in\mathcal{D}_\infty$

$$\big|\langle\psi,\phi(g_{\mathsf{l}}\otimes h')\psi\rangle-\mathfrak{m}\|\psi\|^2\big|\le\frac{\mathsf{l}\langle\psi,H\psi\rangle+\log(Q+1)\|\psi\|^2}{r}+\frac{r}{2}\,c_2\langle\psi,N\psi\rangle,$$

where $\hat g_{\mathsf{l}}(E)=\mathsf{G}(\mathsf{l}E)$, $\mathsf{G}(y)=\int\rho_1(s)\cosh(sy)\,\mathrm{d}s\,\frac{y/2}{\sinh(y/2)}$. No injectivity of $\hat D(x)$ is assumed, and under (i)–(iv) the slope is the real-time average of the field: $\lim_{\epsilon\downarrow0}\frac{d}{dx}\langle\psi',\log(\hat D(x)+\epsilon)\psi\rangle|_{x=0}=\langle\psi',\phi(g_{\mathsf{l}}\otimes h')\psi\rangle-\mathfrak{m}\langle\psi',\psi\rangle$ for $\psi,\psi'\in\mathcal{H}_\Lambda$. If such bounds hold at all $\mathsf{l}=L^{-i}$ with constants $\le C\mathsf{l}^{-N_0}$, then $(\mathrm{LB}_{k\otimes h'})$ holds for every real $k\in\mathcal{S}(\mathbb{R})$. (h2) Let $\wp<6$. For a unit timelike vector $n\in V_+$, $\mu\ge1$, Borel sets $\Delta,\Delta'\subset\{p\in\bar V_+:n\cdot p\le\mu\}$ and $u\in\mathcal{S}$: $\|\mathcal{P}(\Delta')\phi(u)\mathcal{P}(\Delta)\|\le e^2C_\wp\|u\|_{L^1}\mu^\wp$. For $\mu\ge1$, $u\in\mathcal{S}$ and vectors ψ,ψ' of bounded energy in the spectral subspace of $\{p\cdot p\le\mu^2\}$:

$$|\langle\psi',\phi(u)\psi\rangle|\le C_{\mathrm{L}}(\|u\|_{L^1}+\|\mathcal{L}^4u\|_{L^1})\mu^\wp\|\psi'\|\|\psi\|,\qquad\mathcal{L}=-\partial_t^2-\partial_1^2-\partial_2^2,$$

with $C_{\mathrm{L}}=100e^2C_\wp\max\{18^\wp,6^\wp c\,\Sigma_\wp\}$, $\Sigma_\wp=\sum_{r\ge1}(r+1)r^{-8}(2r+3)^\wp$ and a numeral c, formed from $C_\wp,\wp$ alone; for the following two consequences assume additionally $\wp>0$. Then for all m,n, with $\mathcal{P}_<$ the spectral projection of $\{p\cdot p\le(m+n+2)^{2/\wp}\}$: $\|\mathcal{P}_<P_m\phi(u)P_n\mathcal{P}_<\|\le C_{\mathrm{L}}(\|u\|_{L^1}+\|\mathcal{L}^4u\|_{L^1})(m+n+2)$. With Π the spectral projection of $\{p:p\cdot p\le(1+p^0)^{2/\wp}\}$, $\|\Pi e^{-sH}\phi(u)e^{-sH}\Pi\|\le c(u)s^{-1}$ for $0<s\le1$, $c(u)$ formed from C_{L} and the norms $\|\partial_t^ju\|_{L^1}+\|\mathcal{L}^4\partial_t^ju\|_{L^1}$, $j\le2$. For data (Rec) of any order $\wp$ and real $f\in\mathcal{S}$, energy and invariant mass together bound the field: $\pm\phi(f)\le c(f)(1+H+M^2)^{\wp/2}$ as forms on the vectors of bounded energy, $M^2=\int(E^2-|q|^2)\,\mathrm{d}\mathcal{P}$ the operator of the invariant mass squared, $c(f)$ formed from $C_\wp$, $\wp$ and finitely many of the norms $\|\partial_\nu^kf\|_{L^1}$, $\nu=0,1,2$.

■□

Theorem AP (Locality of the continuum limits)**.** Let $0<g\le\gamma$, let σ be the centred functional of a volume-first limit at g, and let τ be σ or a component of a canonical decomposition of σ; every such limit, decomposition and component is admitted. On $\mathcal{H}(\tau)$:

(a) $\mathcal{C}_\tau(W')\Omega$ is dense, and (W) holds: $O\mapsto\mathcal{A}_\tau(O)$ is an isotonous, covariant and local net of von Neumann algebras, $\mathcal{L}_\tau=\mathcal{A}_\tau$, and $\mathcal{H}_0=\mathcal{H}(\tau)$.

(b) $\mathcal{A}_\tau(W) = \mathcal{C}_\tau(W') = \mathcal{C}_\tau(W)'$ and $\mathcal{A}_\tau(W)' = \mathcal{A}_\tau(W')$; Ω is cyclic and separating for $\mathcal{A}_\tau(W)$, with modular group $\Delta^{it} = U(B(-2\pi t))$ and modular conjugation J; in particular $\mathcal{A}_\tau(W) \neq \mathbb{C}1$ unless $\mathcal{H}(\tau) = \mathbb{C}\Omega$.

(c) For open O and $f \in \mathcal{F}(O)$, $\bar\phi(f)$ is affiliated with $\mathcal{A}_\tau(O)$.

(d) (CP) holds, with the two inequalities of Thm. AO(d).

(e) The dual net $\mathcal{A}^d_\tau(O) = \mathcal{C}_\tau(O^\times) = \mathcal{A}_\tau(O^\times)'$ has all the properties of Thm. AO(e): it is local, Haag dual, the largest family local relative to $\mathcal{A}_\tau$, and its own dual net.

(f) Let $\tau = \sigma$, let Λ_j be the tori of its truncated functions and $\mathcal{M}_{\Lambda_j}$ the algebras of all bounded operators on the bonds of the right halves of their time-zero planes (Prop. 14.13). Every weak limit Q, along a subsequence, of operators $Q_j \in \mathcal{M}_{\Lambda_j}$ lies in $\mathcal{C}_\sigma(W')$; the lattice modular flows converge to the boosts, $\Delta^{it}_{\Lambda_j}\psi_j \Rightarrow U(B(-2\pi t))\psi$ in the sense of Lem. 14.1; for real $u \in C^\infty_c(\mathsf{I})$ and $\epsilon > 0$ there is $T^{(\epsilon)}_u \in \mathcal{C}_\sigma(W')$, a weak limit of operators of $\mathcal{M}_{\Lambda_j}$, with $\|T^{(\epsilon)}_u\| \le (2\sqrt{\epsilon})^{-1}$ and $\|T^{(\epsilon)}_u\Omega - \phi(f_u)\Omega\| \le \frac{\sqrt{\epsilon}}{2}\pi e^{5\pi^2/8}\|[\theta_1 u \cdot u]\|$, f_u the function of Lem. 14.2(c); and $\mathcal{A}_\sigma(W)$ is the von Neumann algebra generated by the $U(\mathfrak{g})QU(\mathfrak{g})^{-1}$, $\mathfrak{g}W \subset W$, with Q or Q^* such a weak limit.

(g) For $\tau = \sigma$ and a canonical decomposition: $[G] \mapsto ([G]_\xi)_\xi$ is a unitary of $\mathcal{H}(\sigma)$ onto $\int^\oplus \mathcal{H}(\sigma_\xi)\,\lambda(\mathrm{d}\xi)$, by which we identify the two, $[G] = \int^\oplus [G]_\xi$; the semigroup, the translations, the insertions, the operators $\mathsf{B}(s,s')$ and the fields decompose; the continuous functions on X act by multiplication (the operators of Lem. 14.7(i)), generate the diagonal algebra and lie in the centre of $\mathcal{A}_\sigma(W)$; and $\mathcal{A}_\sigma(W) = \int^\oplus \mathcal{A}_{\sigma_\xi}(W)\,\lambda(\mathrm{d}\xi)$: for λ-almost every ξ, among them every ξ with $\lambda(\{\xi\}) > 0$, the fibre at ξ of the algebra of (f) is the wedge algebra of the component σ_ξ.

(h) *A supply.* Let σ' be a limit of periodic Gibbs states ρ_j at g, $c \in \bigcup_K L^{-K}\mathbb{Z}$, and B_j Borel functions, $|B_j| \le \mathsf{b}$, of finitely many bonds of $L^{-K_j}\mathbb{Z}^3$ in $\{x_3 = 0,\ x_1 \le c\}$. Then $|\rho_j(\overline{\theta(F')_j}B_j(F)_j)| \le \mathsf{b}\,\rho_j(\overline{\theta(F')_j}(F')_j)^{1/2}\rho_j(\overline{\theta(F)_j}(F)_j)^{1/2}$ for F, F' over $\mathbb{E}_+$ and large j; along a subsequence $\rho_j(\overline{\theta(F')_j}B_j(F)_j) \to \langle [F'], Q[F]\rangle$ for all such F, F', with $\|Q\| \le \mathsf{b}$; and every such Q lies in $\mathcal{C}_{\sigma'}(W + ce_1)$. The same holds for every component σ'_ξ of a canonical decomposition of σ', with ρ_j replaced, as the states that integrate, by the periodic Gibbs states $\rho_{j,\xi}$ that approximate the component (Thm. YM3_I:AP among the construction results, recalled in the results used), the ρ_j still giving the centring.

By contrast, on the space of a component an operator $e^{-sH}V_1e^{-sH}$, $s > 0$, V_1 bounded, lies in no $\mathcal{C}(O)$, $O \neq \emptyset$ open, unless it is 0 or the space is $\mathbb{C}\Omega$: weak limits of the operators $[G] \mapsto [D_j \cdot \tau_{te_3}G]$ of lattice functions D_j placed strictly between two times, in $\{s < x_3 < t - s\}$ with $s, t \in \bigcup_K L^{-K}\mathbb{Z}$, whose inner operators $[G] \mapsto [\tau_{-se_3}D_j \cdot \tau_{(t-2s)e_3}G]$ are bounded uniformly in j, on a nontrivial component space are nonlocal unless they vanish (Prop. 14.3(a)).

■□

Theorem AQ (Energy bounds of the field)**.** Let $0 < g \le \gamma$ and let σ' be a limit of periodic Gibbs states at g or a component of a canonical decomposition of one. Let $f \in \mathcal{S}$, $m, n \in \mathbb{N}_0$, $\psi, \psi' \in \mathcal{D}_\infty$, and let $c_V, \bar\Phi, c'_V, \Phi'$ be the numbers of Lem. 14.15; none of them depends on the cutoff, the sequence, σ' or ξ.

(a) *Order* 3. With $\mathsf{C}_3 = \frac{4\sqrt{8}}{\varrho}L^4\,40M(40 + 8L^2)e^{\frac{8}{3}c_V + \frac{1}{2}\bar\Phi}$ and $\mathsf{c}_k = 27L^3\mathsf{C}_3\|\partial_t^k f\|_\star$: $b_{mn}(f) \le \mathsf{c}_0(m + n + 2)^3$ for all m, n, and $b_{mn}(f) \le \mathsf{c}_k(m + n + 2)^3(|m - n| - 1)^{-k}$ if $|m - n| \ge 2$, $k \in \mathbb{N}$; $\|\phi(f)\psi\| \le 8(71\mathsf{c}_0 + 6.3\mathsf{c}_5)\|N^3\psi\|$, and $\|N^a\phi(f)\psi\| \le c_a(f)\|N^{a+3}\psi\|$ for real a, so that $D(N^3) \subset D(\bar\phi(f))$; $|\langle\psi', \phi(f)\psi\rangle| \le 8(17.1\mathsf{c}_0 + 5.7\mathsf{c}_3)\|N^{3/2}\psi'\|\|N^{3/2}\psi\|$; and $\|e^{-sH}\phi(f)e^{-sH}\| \le c(f)s^{-3}$ for $0 < s \le 1$. If f is real and $\|\partial_t^k f\|_\star \le A^{k+1}(k!)^{\mathsf{s}}$ for all $k \ge 0$, with $\mathsf{s} \ge 1$ (such $f \ge 0$, $f \neq 0$, of compact support exist for every $\mathsf{s} > 1$), then $\|\phi(f)^n\Omega\| \le C^n(n!)^{3\mathsf{s}}$ for all n, with $C = C(A, \mathsf{s}, L^3\mathsf{C}_3)$.

(b) *Order* 2. With $\mathsf{C}_2 = \frac{4(\sqrt{8}+1)}{\varrho} L^4\, 40M(40+8L^2) e^{\frac{8}{3}c'_V + \frac{1}{4}\Phi'}$: $b_{mn}(f) \le 27L^2\mathsf{C}_2 \|f\|_\star (m+n+2)^2$, and $b_{mn}(f) \le 27L^2\mathsf{C}_2 \|\partial_t^k f\|_\star (m+n+2)^2 (|m-n|-1)^{-k}$ if $|m-n| \ge 2$; $\|\phi(f)\psi\| \le c(f)\|N^2\psi\|$, $|\langle \psi', \phi(f)\psi\rangle| \le c'(f)\|N\psi'\|\|N\psi\|$ and $\|e^{-sH}\phi(f)e^{-sH}\| \le c''(f)s^{-2}$ for $0 < s \le 1$; and for f as at the end of (a), $\|\phi(f)^n\Omega\| \le C^n (n!)^{2\mathsf{s}}$ with $C = C(A, \mathsf{s}, L^2\mathsf{C}_2)$.

(c) *At a point; at one time.* Here Thms. YM3_I:AJ and YM3_I:AD are used at every radius and for every admitted choice of the constants (σ^-, γ), at all couplings at once. With $q = \mathfrak{p}_0 + \frac{1}{6}$ and C formed from $\sigma^-, \gamma, \varrho, g$: $\|\mathsf{B}(s,s')\| \le C\mu^{-7/2}(1+\log\mu^{-1})^q$, $\mu = \min\{s, s', \frac{1}{2}\}$; so for every $\epsilon > 0$, σ' is data (Rec) with a triple $(C_\epsilon, \frac{7}{2}+\epsilon, \frac{1}{2})$, and Thm. AO applies at the order $\frac{7}{2}+\epsilon$. For real $k \in C^1_c(\mathbb{R}^2)$ carried by a square of side 2: $\|\mathsf{B}_k(s,s')\| \le C|k|_{C^1}\mu^{-5/2}(\log\frac{e}{\mu})^q$, $\mu = \min\{s,s',1\}$; so $b_{mn}(f) \le c(f)(m+n+2)^{5/2}(1+\log(m+n+2))^q$ for all m, n.

(d) *Sharpness.* Let $0 < g \le \gamma_{\rm sd}$ and let σ' be a component. For continuous $k \ge 0$, $k \ne 0$, of compact support, $\liminf_{u\downarrow 0} u^2\|\mathsf{B}_k(u,u)\| \ge (\pi c_{\rm sd}/32)^{1/2}\|k\|_{L^2}$; so the field at one time, the form $\mathsf{T}(k)$ on the vectors of bounded energy, has no bound $\pm\mathsf{T}(k) \le cN$. $\|\mathsf{B}(s,s)\| \ge \|\mathsf{B}(s,s)\Omega\| \ge \frac{1}{8}c_{\rm sd}^{1/2}s^{-3}$ for $0 < 2s \le r_{\rm sd}$: the infimum of admissible pure-power exponents in the point bound lies in $[3, \frac{7}{2}]$. For $0 \ne h \ge 0$ in $C^1_c(\mathbb{R}^2)$, $\|\mathsf{I}_\mathsf{l}(h)\Omega\| \ge c(h)\mathsf{l}^{-2}$ for small l; so no bound $\|\mathsf{I}_\mathsf{l}(h)\| \le C\mathsf{l}^{-p}|h|_{C^1}$ holds with $p < 2$, and the exponent 2 of the plates in Lem. 14.15(c) is exact. For $0 \ne k \ge 0$ in $C^1_c(\mathbb{R}^2)$, $\epsilon \in (0,1]$ and $q' < \frac{3}{2}\epsilon$:

$$\sup\{\|T_{mn}(k)\|(m+n+2)^{-q'} :\ m,n \ge 0,\ |m-n| \le (m+n+2)^\epsilon\} = \infty;$$

in particular no bound $\|T_{mn}(k)\| \le c(m+n+2)$ holds on a band with $\epsilon > \frac{2}{3}$.

(e) *Slab families.* Let σ' be a component or the centred functional of a volume-first limit, $\mathsf{l} = L^{-i}$, $i \ge 1$, $\vec{y}^* \in L^{-i-2}\mathbb{Z}^2$, $h' \ne 0$ real and carried by $\{|\vec{x} - \vec{y}^*|_\infty \le \frac{\mathsf{l}}{4}\}$, ρ_1 as in Thm. AO(h1), and $f(x) = \mathsf{l}^{-1}\rho_1(x_3/\mathsf{l})h'(\vec{x})$. The limits $\mathbb{D}(z)$, along a subsequence, of the lattice operators $e^{z\mathcal{O}(f) - \mathcal{E}(z)}$ between reflection-positive vectors, $\mathcal{E}$ the normaliser of the one-cell bound (Prop. YM3_I:89.1), are holomorphic on $|z| < \varrho/\|f\|_\sharp$ with $\|\mathbb{D}(z)\| \le \bar{\mathsf{Q}}$, a number free of l, f, ξ, $\mathbb{D}(0) = e^{-\mathsf{l}H}$ and $\mathbb{D}'(0)$ the insertion of f plus a multiple of $e^{-\mathsf{l}H}$, and $\hat{D}(x) = \mathbb{D}(x)/\langle\Omega, \mathbb{D}(x)\Omega\rangle$, $|x| \le \varrho/(2(\bar{\mathsf{Q}}+1)\|f\|_\sharp)$, has (i)–(iv) of Thm. AO(h1) with $Q = 2\bar{\mathsf{Q}}$ and $\mathfrak{m} = q_0(\Omega,\Omega)\int h'$. ■□

Theorem AR (Examples: Wick powers of free fields; free photons)**.**

(a) *Gaussian families.* Let (M, μ) be a σ-finite measure space with $L^2(\mathsf{M}, \mu)$ separable, $\omega > 0$ and $\vec{p}$ measurable on it with $|\vec{p}| \le \omega$, κ_α, α in a finite set A, measurable, with signs ε_α, such that $\|\omega^{-1/2}\kappa_\alpha e^{-s\omega}\| \le c_\kappa s^{-\delta}$ for $0 < s \le 1$, $\delta \ge 0$, and $G = (G_{\alpha\beta})$ real and continuous on $\mathbb{R}^3 \setminus \{0\}$ with $G_{\alpha\beta}(-x) = G_{\beta\alpha}(x)$, such that the vectors $e^\alpha_x = \kappa_\alpha e^{-x_3\omega - i\vec{x}\cdot\vec{p}}$, $x \in \mathbb{E}_+$, have the inner products $\langle e^\alpha_x, e^\beta_y\rangle = \varepsilon_\alpha G_{\alpha\beta}(\theta x - y)$.

Let $\mathrm{O}(3)$ act on $\mathbb{R}^\mathsf{A}$ by a homomorphism ρ_A into the orthogonal matrices, with $G(\mathsf{R}x) = \rho_\mathsf{A}(\mathsf{R})G(x)\rho_\mathsf{A}(\mathsf{R})^{\rm T}$ for $\mathsf{R} \in \mathrm{O}(3)$ and $\rho_\mathsf{A}(\theta) = \mathrm{diag}(\varepsilon_\alpha)$: the reflection acts on the components by the signs (for photons A indexes the three derivatives, times the colours, and $\rho_\mathsf{A}(\mathsf{R}) = \mathsf{R}\otimes 1$; for a scalar family ρ_A is trivial). Let

$$\mathsf{P}(t) = \sum_{\underline{\alpha}} \mathsf{p}_{\underline{\alpha}} t_{\alpha_1}\cdots t_{\alpha_k} \ne 0$$

be real and k-homogeneous, $k \ge 1$, with a symmetric real tensor p, and invariant, $\mathsf{P}(\rho_\mathsf{A}(\mathsf{R})t) = \mathsf{P}(t)$ for $\mathsf{R} \in \mathrm{O}(3)$; $|\mathsf{p}|_1 = \sum|\mathsf{p}_{\underline{\alpha}}|$.

The Schwinger functions of $:\mathsf{P}(\Phi):$ for the Gaussian field $\Phi = (\Phi_\alpha)$ of the covariance G, that is, the sums over the complete pairings of n vertices with k legs each, no leg paired within its vertex, of the products of the $G_{\alpha\beta}(x_i - x_l)$ weighted by p, are data (Rec) with $C_\wp = 2^\wp 2^k|\mathsf{p}|_1(2^\delta c_\kappa)^k(k/e)^{k/2}$, $\wp = k(\delta + \frac{1}{2})$, $c_0 = \frac{1}{2}$, and a unique vacuum. A unitary V

carries $\mathcal{H}$ onto a closed subspace of the Fock space over $L^2(\mathsf{M},\mu)$ that reduces $d\Gamma(\omega)$, $d\Gamma(\vec{p})$ and the Fock operators of the field at a point, with $V\Omega=\Omega_{\mathrm{F}}$ and $VHV^{-1}=d\Gamma(\omega)$ there, and $\langle\chi,V\phi(f)\psi\rangle=\Phi_f(\chi,V\psi)$ for every $\chi\in\mathcal{D}_\infty(d\Gamma(\omega))$ and $\psi\in\mathcal{D}_\infty$, Φ_f the form of the smeared Wick polynomial.

(b) *Growth.* If $k\geq 3$ and for every $u\in C_c^\infty\setminus\{0\}$ the function

$$w_u(m_1,\dots,m_k)=\hat{u}(\sum_i\omega(m_i),\sum_i\vec{p}(m_i))\sum_{\underline{\alpha}}\mathsf{p}_{\underline{\alpha}}\prod_i\kappa_{\alpha_i}(m_i)$$

lies in $L^2(\mathsf{M}^k)\setminus\{0\}$, then $\|\phi(u)^n\Omega\|\geq c_u^n((kn)!)^{1/2}$ for real $u\in C_c^\infty\setminus\{0\}$; so (QA_u) fails, no bound $|\langle\psi,\phi(u)\psi\rangle|\leq c\langle\psi,N\psi\rangle$ holds at any real $u\neq 0$ in C_c^∞, no bound $b_{mn}(u)\leq\beta(m+n+2)^q$ with $q<\frac{k}{2}$ holds on any band $|m-n|\leq(m+n+2)^\epsilon$, and no triple of the datum has $\wp<\frac{k}{2}$.

(c) *Wick powers.* For the free field χ of mass $\mathsf{m}>0$ and $\mathsf{P}=t^k$, (a) holds with every $\delta\in(0,\frac12]$ and $c_\kappa=(8\pi\delta)^{-1/2}(2\mathsf{m})^{-\delta}$, so $:\chi^k:$ is data (Rec) of every order $\wp>\frac{k}{2}$; for $k\geq 2$, $w_u\neq 0$ as soon as $\hat{u}$ does not vanish identically on $\{E>(|\vec{k}|^2+k^2\mathsf{m}^2)^{1/2}\}$, which holds for every $u\in C_c^\infty\setminus\{0\}$, for every $u\in\mathcal{S}\setminus\{0\}$ that vanishes off $\bar{W}$, and for $u=\mathsf{g}\otimes h'$ with $\hat{\mathsf{g}}$ free of real zeros and $h'\in C_c^\infty(\mathbb{R}^2)\setminus\{0\}$; so (b) holds for $k\geq 3$. $:\chi^6:$ is data (Rec) with $(93312(\pi e)^{-3}\mathsf{m}^{-1},4,\frac12)$, a unique vacuum, $\operatorname{spec}H\subset\{0\}\cup[2\mathsf{m},\infty)$ and $S_2(x)=720(4\pi|x|)^{-6}e^{-6\mathsf{m}|x|}$. (W) and (CP) hold for every $:\chi^k:$, $k\geq 1$, $\mathsf{m}>0$; for every real homogeneous Wick polynomial in the first derivatives $\partial_\alpha\varphi^a$ of n massless free fields $\varphi=(\varphi^a)_{a\leq\mathsf{n}}$ that is invariant under $\mathrm{O}(3)$ and $\mathrm{O}(\mathsf{n})$, for instance $:(\partial\varphi\cdot\partial\varphi)^2:$ with $\wp=6$; and for the Wick cube of the generalised free field with the kernel r^{-2} ($\wp=3$, $S_2=6r^{-6}$, no gap).

So neither (LB_u) nor (QA_u), at any u, is necessary for (W) among data (Rec) with $\wp=4$, $c_0=\frac12$, a unique vacuum, a gap and $S_2\sim r^{-6}$; and for every $\wp>\frac32$ there are data (Rec) of order $\wp$ without (LB_u) at any $u\neq 0$.

(d) *Free photons.* Let φ be the free massless field on $\mathbb{R}^{1+2}$ and $\Theta=:\partial_\lambda\varphi\,\partial^\lambda\varphi:$. The Schwinger functions of $:\partial_\alpha\Phi\,\partial_\alpha\Phi:$ for the Euclidean massless free field Φ on $\mathbb{R}^3$ (the pairings of $-\partial_a\partial_b(4\pi|x|)^{-1}$) are data (Rec) with $(24/\pi e,3,\frac12)$, a unique vacuum and $S_2=12/16\pi^2r^6$; their space is the whole even Fock space, with $V\phi(f)V^{-1}=-\Theta(f)$. For real $f\in\mathcal{S}$: $\pm\Theta(f)\leq 16\pi|f|_8\,N$, $|f|_8$ the Schwartz seminorm in equation YM3_I:(1873) with $|\hat{f}(k)|\leq 16\pi^2|f|_8(1+|k|^2)^{-2}$; $\|\Theta(f)\psi\|\leq c(f)\|N\psi\|$; $\|\Theta(f_1)\cdots\Theta(f_n)\Omega\|\leq 2^n n!\prod_k c(f_k)$; and $\Theta(f)$ is essentially self-adjoint on every core of H; (W) and (CP) hold. $\|\mathsf{B}(s,s)\|\geq\|\mathsf{B}(s,s)\Omega\|=\frac{\sqrt{3}}{16\pi}s^{-3}$, so no triple has $\wp<3$, while on its own space the order is exactly 1: for real $f\neq 0$, $\alpha<1$ and every c some $\psi\in\mathcal{D}_\infty$ has $|\langle\psi,\phi(f)\psi\rangle|>c\langle\psi,N^\alpha\psi\rangle$. At one time, with $\Theta_k=\int k(\vec{x})\Theta(0,\vec{x})d^2x$, $k\in\mathcal{S}(\mathbb{R}^2)$ real, and $G=m+n+2$:

$$\|P_m\Theta_kP_n\|\leq(1+\sqrt{3})\|k\|_\infty G+2\mathsf{n}_k(|m-n|+2)\sqrt{G},\qquad \mathsf{n}_k=\tfrac{1}{2\pi}\big[10.4\|k\|_2^2+1.6\|k\|_2\|\nabla k\|_2\big]^{1/2},$$

and the second term is attained: for $k\neq 0$, every integer $\omega\geq\max\{3,7\xi_0(k)+1\}$ and every $j\in\mathbb{N}$ there are m,n with $|n-m-\omega|\leq 1$, $2j(\omega-\frac12)+\omega-1<G\leq(j+1)(2\omega+3)$ and $\|P_n\Theta_kP_m\|\geq 3\cdot10^{-3}\|k\|_2\,\omega\sqrt{G}-\kappa_k(\omega-2)G$, where $\kappa_k(R)=(2\pi)^{-2}\int_{|\xi|>R}|\tilde{k}|$ and $\xi_0(k)$ is the least R with $\int_{|\xi|\leq R}|\tilde{k}|^2\geq\frac12\|\tilde{k}\|_2^2$. So for real $k\neq 0$ in C_c^∞ the band bound $\|P_me^{-H/G}\Theta_ke^{-H/G}P_n\|\leq CG$ on $|m-n|\leq G^\epsilon$ holds if and only if $\epsilon\leq\frac12$.

(e) n *colours.* Let $\Theta=\sum_{a\leq\mathsf{n}}:\partial_\lambda\varphi^a\partial^\lambda\varphi^a:$, $\mathsf{n}\geq 1$. The closed span of Ω and the $\Theta(f_1)\cdots\Theta(f_n)\Omega$ ($f_k\in\mathcal{S}$, or $f_k\in C_c^\infty$ real) is the space of all $\mathrm{O}(\mathsf{n})$-invariant vectors, and it is the closure of the Euclidean vectors; with $\mathrm{SO}(3)$ in the place of $\mathrm{O}(3)$ at $\mathsf{n}=3$ this is false. $\|\mathsf{B}(s,s)\Omega\|=\frac{\sqrt{3\mathsf{n}}}{16\pi}s^{-3}$. For real $f\neq 0$, $\alpha<1$, every c and either sign there is $\psi\in\mathcal{D}_0$ with $\pm\langle\psi,\phi(f)\psi\rangle>c\langle\psi,N^\alpha\psi\rangle$, while $\pm\phi(f)\leq\mathsf{n}\,16\pi|f|_8N$: on the invariant space the order is exactly 1, and $\sup_{m,n}b_{mn}(f)(1+\max\{m,n\})^{-p}=\infty$ for every $p<1$.

(f) *The net of the action density of free photons.* Let $\mathsf{n} \geq 1$ and O be a double cone. For a family Φ of Wightman fields on the Fock space of n free fields $\varphi = (\varphi^a)_{a \leq \mathsf{n}}$ of a mass $\mathsf{m} \geq 0$, among them $\Theta = \sum_a :\partial_\lambda \varphi^a \partial^\lambda \varphi^a:$, $S = :\varphi \cdot \varphi: = \sum_a :\varphi^a \varphi^a:$ and the multiplet $(:\partial_\mu \varphi \cdot \partial_\nu \varphi:)_{\mu\nu}$: $\mathcal{P}_\Phi(O)$ is the algebra of the polynomials in the $\Phi(f)$, $f \in \mathcal{F}(O)$, $\mathcal{D}_0^\Phi = \mathcal{P}_\Phi(\mathbb{R}^{1+2})\Omega$ and $\mathcal{H}_\Phi$ its closure, $\mathcal{C}_\Phi(O)$ the set of the bounded Q on $\mathcal{H}_\Phi$ with $\langle \Phi(\bar{f})\psi, Q\psi' \rangle = \langle \psi, Q\Phi(f)\psi' \rangle$ for $f \in \mathcal{F}(O)$ and $\psi, \psi' \in \mathcal{D}_0^\Phi$, $\mathcal{A}_\Phi(O) = \mathcal{C}_\Phi(O)'$, and $\mathcal{A}_\Phi^d(O) = \bigcap\{\mathcal{A}_\Phi(\tilde{W}) : \tilde{W} \supset O \text{ a wedge}\}$ its dual net.

At the mass 0 the unitary V of (a), for the data of (d), (e), carries $\mathcal{D}_0$, $\mathcal{C}(O)$, $\mathcal{A}(O)$ and $\mathcal{A}^d(O)$ of the datum onto $\mathcal{D}_0^\Theta$, $\mathcal{C}_\Theta(O)$, $\mathcal{A}_\Theta(O)$ and $\mathcal{A}_\Theta^d(O)$, so that Thm. AO applies to Θ; there $\mathcal{H}_\Theta$, $\mathcal{H}_S$ and the space of the multiplet are all the space of the $\mathrm{O}(\mathsf{n})$-invariant vectors, and S is no datum (Rec), its point operators being unbounded in the infrared: its nets are those of the definition alone. For every real $f \in \mathcal{F}$, the massless square $S(f)$ is essentially self-adjoint on its polynomial cyclic domain $\mathcal{D}_0^S$. At a mass $\mathsf{m} > 0$, Θ and S are data (Rec) by (a), and the unitary V of (a) carries the nets of each onto those of the definition.

At the mass 0: $\mathcal{A}_\Theta(O)$ is the $\mathrm{O}(\mathsf{n})$-invariant part of the net of the field strengths, restricted to the invariant vectors (for $\mathsf{n} = 1$ it is generated by the $\cos \bar{\varphi}(h)$, $h \in \mathcal{F}_\mathbb{R}(O)$, $\int h = 0$, where $\bar{\varphi}(h)$ is the closure of the field smeared with the real $h \in C_c^\infty(O)$ and $\int h$ is the integral over space-time); the multiplet $:\partial_\mu \varphi \cdot \partial_\nu \varphi:$ has the same net, and for $\mathsf{n} \in \{1, 3\}$ the closures of the $T_{\mu\nu}(f)$, $T_{\mu\nu} = :\partial_\mu \varphi \cdot \partial_\nu \varphi: - \frac{1}{2}\eta_{\mu\nu}\Theta$ with $\eta = \mathrm{diag}(1, -1, -1)$, $T_{00}(f)$ among them, $f \in \mathcal{F}_\mathbb{R}(O)$, are self-adjoint and affiliated with $\mathcal{A}_\Theta(O)$; (W) holds. $\mathcal{A}_\Theta(O) \neq \mathcal{A}_\Theta^d(O)$: for real h carried by O with $\int h \neq 0$ the mean over $\mathrm{O}(\mathsf{n})$ of $e^{i\bar{\varphi}^1(h)}$ lies in $\mathcal{C}_\Theta(O^\times) = \mathcal{A}_\Theta^d(O)$ and not in $\mathcal{A}_\Theta(O)$; for $\mathsf{n} = 1$, $\cos(\mathsf{s}\bar{\pi}(g))$, $\bar{\pi}(g)$ the closure of the momentum at time 0 smeared with a real $g \in C_c^\infty(\mathbb{R}^2)$ that is 1 near the closed base of O, lies in $\mathcal{C}_\Theta(O)$ and does not commute with $\cos \bar{\varphi}(h)$ when $\mathsf{s} \int h \notin 2\pi\mathbb{Z}$.

For $\mathsf{n} = 1$ and closed subgroups $\Lambda \subset \mathbb{R}$ the even parts of the nets $\{e^{i\bar{\varphi}(f)} : f \in \mathcal{F}_\mathbb{R}(O), \int f \in \Lambda\}''$ are pairwise different local nets between $\mathcal{A}_\Theta$ and $\mathcal{A}_\Theta^d$ with the same wedge algebras. At a mass $\mathsf{m} > 0$, where $\varphi^a(h) = \varphi^a(-\mathsf{m}^{-2}\Box h)$ makes every smeared field one of a test function of integral 0 and $\Theta = \frac{1}{2}(\Box + 2\mathsf{m}^2)S$, and where, as at the mass 0 and by the same steps, which hold at every $\mathsf{m} \geq 0$, the commutators of the $\Theta(f)$, $f \in \mathcal{F}(O)$, reach the colour sums $\sum_a \varphi^a(h)\varphi^a(h')$, whose commutators have the same structure constants at every mass, for real $h, h' \in \mathcal{F}(O)$ of integral 0, hence for all real $h, h' \in \mathcal{F}(O)$ (every $Q \in \mathcal{C}_\Theta(O)$ commutes weakly with them), among them the squares $\sum_a \varphi^a(g)^2$, whose closures generate $\mathcal{A}_S(O)$, the nets of Θ and of S are one, $\mathcal{A}_\Theta = \mathcal{A}_S$ on $\mathcal{H}_\Theta = \mathcal{H}_S$, for every $\mathsf{n} \geq 1$, and the net of Θ, as that of S, is its own dual net.

The net of $S = :\varphi \cdot \varphi:$ is its own dual net at every mass; at the mass 0, where $\Theta(f) = \frac{1}{2}S(\Box f)$, $\mathcal{A}_\Theta(O) \subsetneq \mathcal{A}_S(O) = \mathcal{A}_S^d(O) = \mathcal{A}_\Theta^d(O)$, and the two nets have the same wedge algebras: on one space, with one vacuum and one family of wedge algebras, the field S is Haag dual and the field $\frac{1}{2}\Box S$ is not.

■□

Theorem AS (Lattice algebras at one time; the defect $\mathbb{Z}_2$ and its Euclidean half)**.** (a) Let G be a compact group, (Σ, B) a finite graph, $\mathcal{H}_\mathrm{ph} \subset L^2(G^\mathsf{B})$ the gauge-invariant subspace, and $\mathsf{A}(Y)$, $Y \subset \mathsf{B}$, the algebra of all gauge-invariant bounded operators of the bonds of Y on $\mathcal{H}_\mathrm{ph}$. Then $\mathsf{A}(Y)$ is a von Neumann algebra, $\mathsf{A}(Y)' = \mathsf{A}(\mathsf{B} \setminus Y)$ for every Y, $\mathsf{A}(\mathsf{B}) = \mathcal{B}(\mathcal{H}_\mathrm{ph})$, and $\mathsf{A}(Y_1 \cup Y_2) = \mathsf{A}(Y_1) \vee \mathsf{A}(Y_2)$ if Y_1, Y_2 have no common site.

In (b)–(i), $G = \mathrm{SU}(2)$. On a layer $\mathbb{Z}/N_1 \times \mathbb{Z}/N_2$, $N_i \geq 4$, let, for a set S of plaquettes, B_S be their bonds, $\mathsf{A}_\bullet(S)$ the von Neumann algebra generated by the traces M_{t_p}, $p \in S$, and the spin projections Π_b^j, $b \in \mathsf{B}_S$; $Z^1(S)$ the sign fields c on B_S with $\prod_{b \in \partial p} c_b = 1$ for $p \in S$; B^1 the coboundaries of the whole layer, $c_b = z(b_-)z(b_+)$ for a sign field z on all its sites, and $B^1(S) \subset Z^1(S)$ the coboundaries

on B_S, $c_b = z(b_-)z(b_+)$, $b \in \mathsf{B}_S$, for a sign field z on the sites of B_S; $H^1(S) = Z^1(S)/B^1(S)$; $\mathsf{V}(c)$ the unitary of $V_b \mapsto c_b V_b$, a sign field on a set of bonds being continued by 1 to the other bonds; and for a dual path γ, $\mathsf{V}_\gamma = \mathsf{V}(c)$ with $c = -1$ exactly on the bonds that γ crosses. S is admissible if it has at least three plaquettes, any two of them are joined by a chain of plaquettes of S each sharing a bond with the next, and the plaquettes of S at each site are so joined through bonds at that site.

For every closed path C' and spin j: $\mathsf{V}(c) M_{\chi_j(U(C'))} = c(C')^{2j} M_{\chi_j(U(C'))} \mathsf{V}(c)$, $\mathsf{V}(c)\mathsf{V}(c') = \mathsf{V}(cc')$; $\mathsf{V}(c) = 1$ on $\mathcal{H}_{\mathrm{ph}}$ for $c \in B^1$, $\mathsf{V}(c)$ being then the gauge transformation by the central field z; and for $c \in B^1(S)$, with z continued by 1 to the other sites, $\mathsf{V}(c) = \mathsf{V}(c^\circ)$ on $\mathcal{H}_{\mathrm{ph}}$, c° the restriction of $(z(b_-)z(b_+))_b$ to the bonds off B_S, so that $\mathsf{V}(c)|_{\mathcal{H}_{\mathrm{ph}}}$, a function of spin projections of bonds off B_S, lies in $\mathsf{A}(\mathsf{B} \setminus \mathsf{B}_S) = \mathsf{A}(\mathsf{B}_S)'$ and $\operatorname{Ad} \mathsf{V}(c)$ is trivial on $\mathsf{A}(\mathsf{B}_S)$; on a periodic layer the plaquettes on which a sign field is not closed are even in number: there is no single point operator.

(b) For admissible S: $\mathsf{A}_\bullet(S) = \mathsf{A}(\mathsf{B}_S) \cap \{\mathsf{V}(c) : c \in Z^1(S)\}'$ and $\mathsf{A}_\bullet(S)' = \mathsf{A}(\mathsf{B} \setminus \mathsf{B}_S) \vee \{\mathsf{V}(c) : c \in Z^1(S)\}$; the $\mathsf{V}(c)$, $c \in Z^1(S)$, lie in the centre of $\mathsf{A}_\bullet(S)$; $\operatorname{Ad} \mathsf{V}(c)$ on $\mathsf{A}(\mathsf{B}_S)$ depends on the class of c in $H^1(S) = Z^1(S)/B^1(S)$ alone, so that one c of each class serves in both identities, and $\mathsf{A}(\mathsf{B}_S) = \bigoplus_\kappa \mathsf{A}(\mathsf{B}_S)_\kappa$ over the characters κ of $H^1(S)$, where $\mathsf{A}(\mathsf{B}_S)_\kappa = \{X \in \mathsf{A}(\mathsf{B}_S) : \mathsf{V}(c) X \mathsf{V}(c)^* = \kappa([c])X$ for all $c \in Z^1(S)\}$ is a character subspace. This is a linear grading; every summand $\neq 0$, the summand $\kappa = 1$ being $\mathsf{A}_\bullet(S)$; if every $\mathbb{Z}_2$-cycle of B_S is a sum of boundaries ∂p, $p \in S$, then $\mathsf{A}_\bullet(S) = \mathsf{A}(\mathsf{B}_S)$. $\mathsf{A}_\bullet(S)$ is the join of the $\mathsf{A}(\mathsf{B}_T)$ over the chains $T \subset S$ of three plaquettes, and also of the $\mathsf{A}(\partial p)$, $p \in S$. The traces and spin projections of one plaquette, or of two adjacent plaquettes, do not generate all gauge-invariant operators of their bonds; the traces alone generate an abelian algebra, while $\mathsf{A}(\mathsf{B}_S)$ is not abelian.

(c) *The ring.* Let $a_i < b_i < b_i' < a_i'$ be integers with $a_i' - a_i \leq N_i - 1$, $Q = [a_1, a_1'] \times [a_2, a_2']$, $D = [b_1, b_1'] \times [b_2, b_2']$, R the plaquettes of Q off D, $C = \partial D$, $W_C = M_{\frac{1}{2}\operatorname{tr} U(C)}$, $u = M_{\operatorname{sign}\operatorname{tr} U(C)}$, γ a straight dual path from a plaquette of D to a plaquette outside Q, $c_* = -1$ on the bonds of B_R that γ crosses, $v = \mathsf{V}(c_*)$, and $D_1, \dots, D_4$ the four rectangles $[a_1, a_1'] \times [a_2, b_2]$, $[b_1', a_1'] \times [a_2, a_2']$, $[a_1, a_1'] \times [b_2', a_2']$, $[a_1, b_1] \times [a_2, a_2']$. Then R is admissible, $H^1(R) = \{1, [c_*]\} \cong \mathbb{Z}_2$, and on $\mathcal{H}_{\mathrm{ph}}$

$$\begin{aligned} \mathsf{A}(\mathsf{B} \setminus \mathsf{B}_R)' = \mathsf{A}(\mathsf{B}_R) &= \mathsf{A}_\bullet(R) \vee \{W_C\} \\ &= \mathsf{A}_\bullet(R) \oplus \mathsf{A}_\bullet(R) u, \\ \mathsf{A}_\bullet(R)' = \mathsf{A}(\mathsf{B} \setminus \mathsf{B}_R) &\vee \{\mathsf{V}_\gamma\} \\ = \mathsf{A}(\mathsf{B} \setminus \mathsf{B}_R) &\oplus \mathsf{A}(\mathsf{B} \setminus \mathsf{B}_R)\mathsf{V}_\gamma, \end{aligned}$$

with the normal conditional expectation $X \mapsto \frac{1}{2}(X + vXv)$ of $\mathsf{A}(\mathsf{B}_R)$ onto $\mathsf{A}_\bullet(R)$; $vW_C v = -W_C$, $W_C \notin \mathsf{A}_\bullet(R)$, $\mathsf{V}_\gamma \notin \mathsf{A}(\mathsf{B} \setminus \mathsf{B}_R)$, $\mathsf{A}(\mathsf{B} \setminus \mathsf{B}_R) = \mathsf{A}_\bullet(R)' \cap \{u\}'$, $u\mathsf{V}_\gamma = -\mathsf{V}_\gamma u$; each inclusion has the index two. v is central in $\mathsf{A}_\bullet(R)$, and for a closed path C' in B_R, $M_{\chi_j(U(C'))} \in \mathsf{A}_\bullet(R)$ if and only if j is an integer or $c_*(C') = 1$: the adjoint loop around the hole lies in $\mathsf{A}_\bullet(R)$, the fundamental one does not. So over a ring duality and additivity exclude each other: $Y \mapsto \mathsf{A}(Y)$ is dual and not additive at B_R, $S \mapsto \mathsf{A}_\bullet(S)$ is additive and not dual at R, and no assignment of von Neumann algebras that equals A on the closed and on the open rectangles is both.

At every spacing $\mathsf{A}_\bullet(R) = \bigvee_{i \leq 4} \mathsf{A}(\mathsf{B}_{D_i})$; if no corner $D_i \cap D_{i+1}$ is one plaquette, the multiplications by flip-invariant gauge-invariant functions of the bonds of R are generated by those of the D_i, and $M_{\chi_1(U(C))} = \sum_a M_{(T_1)_{a_4 a_1}} M_{(T_2)_{a_1 a_2}} M_{(T_3)_{a_2 a_3}} M_{(T_4)_{a_3 a_4}}$ with gauge-invariant functions $(T_i)_{ab}$ of the bonds of D_i of modulus ≤ 1.

(d) *Two discs; the whole layer.* Let D_1, D_2 be rectangles without a common site, with a row and a column of plaquettes that meet neither, S the plaquettes off $D_1 \cup D_2$, $\mathsf{B}^\circ_{D_i}$ the bonds inside D_i, γ a dual path from D_1 to D_2 whose other plaquettes lie in S off that row and column, $\mathsf{V}_1, \mathsf{V}_2$ the two fluxes of the torus, and $\mathsf{N} = \mathsf{A}(\mathsf{B}^\circ_{D_1}) \vee \mathsf{A}(\mathsf{B}^\circ_{D_2})$. Then S is admissible, $H^1(S) = \mathbb{Z}_2^3$, $\mathsf{A}(\mathsf{B}_S)' = \mathsf{N}$, $\mathsf{A}_\bullet(S)' = \mathsf{N} \vee \{\mathsf{V}_\gamma, \mathsf{V}_1, \mathsf{V}_2\}$, $\mathsf{V}_\gamma \notin \mathsf{N} \vee \{\mathsf{V}_1, \mathsf{V}_2\}$ and $\mathsf{V}_i \notin \mathsf{N} \vee \{\mathsf{V}_\gamma\}$.

On the whole layer $\mathsf{A}_\bullet(\mathsf{P})' = \mathrm{span}\{1, \mathsf{V}_1, \mathsf{V}_2, \mathsf{V}_1\mathsf{V}_2\}$, $\mathcal{H}_{\rm ph}$ is the sum of four flux sectors, none 0, and $\mathsf{A}_\bullet(\mathsf{P})$ is the sum of all bounded operators of the sectors; one disc has no defect but the fluxes: $\mathsf{A}_\bullet(\mathsf{P} \setminus D)' = \mathsf{A}(\mathsf{B}^\circ_D) \vee \{\mathsf{V}_1, \mathsf{V}_2\}$.

(e) *Three space dimensions.* On a finite part of $\mathbb{Z}^3$, or on a periodic one with sides ≥ 4, at one time: for all the plaquettes S of a box $[0, n_1] \times [0, n_2] \times [0, n_3]$, $\bigvee_{p\in S} \mathsf{A}(\partial p) = \mathsf{A}(\mathsf{B}_S)$; for the plaquettes S of the eight unit cubes of $[0, 3]^2 \times [0, 1]$ off $(1, 2)^2$, $\bigvee_{p\in S} \mathsf{A}(\partial p) = \mathsf{A}(\mathsf{B}_S) \cap \{\mathsf{V}(c_*)\}'$ and $(\bigvee_p \mathsf{A}(\partial p))' = \mathsf{A}(\mathsf{B} \setminus \mathsf{B}_S) \vee \{\mathsf{V}(c_*)\}$, $c_* = -1$ on the four bonds of $[1, 2] \times [2, 3] \times [0, 1]$ parallel to e_1, with $\mathsf{V}(c_*)W_C = -W_C\mathsf{V}(c_*)$ for the fundamental loop C around the ring; for a dual surface $\mathfrak{s}$ that meets the ring in that section, $\mathsf{V}(c_\mathfrak{s}) = \mathsf{V}(c_*)\,\mathsf{V}(c_\mathfrak{s}|_{\mathsf{B}\setminus\mathsf{B}_S})$ with the second factor in $\mathsf{A}(\mathsf{B} \setminus \mathsf{B}_S)$: the defect $\mathbb{Z}_2$ is carried by the flip across a surface whose rim is a closed dual line in the complement that links the ring, a closed vortex line.

For a set S of plaquettes let $Z^1(S)$, $B^1(S)$ be formed as on a layer, and call $H^1(S) = Z^1(S)/B^1(S)$ its defect; call an order $p_1, \dots, p_n$ of S a shelling if for each $k \geq 2$, with X_{k-1} the set of the bonds of $p_1, \dots, p_{k-1}$, either all bonds of p_k lie in X_{k-1}, or p_k has exactly one bond in common with some p_j, $j < k$, the set $\partial p_j \cup (\partial p_k \cap X_{k-1})$ is connected, and every site of a bond of p_k off X_{k-1} that is a site of X_{k-1} lies on that set. If S has a shelling, $\bigvee_{p\in S} \mathsf{A}(\partial p) = \mathsf{A}(\mathsf{B}_S)$; a box has one. If $S = S_1 \cup S_2$, both with shellings, $I = \mathsf{B}_{S_1} \cap \mathsf{B}_{S_2}$ has r connected components, each with a cycle, and every site common to B_{S_1} and B_{S_2} is a site of I, then $\bigvee_{p\in S} \mathsf{A}(\partial p) = \mathsf{A}(\mathsf{B}_S) \cap \{\mathsf{V}(c) : c \in Z^1(S)\}'$ and the defect is $\mathbb{Z}_2^{r-1}$; the solid ring above is such a union, with $r = 2$.

(f) *The sign rule.* On every periodic lattice in three dimensions and at every β, for sign fields η on the plaquettes and s on the bonds, $(ds)_p = \prod_{b\in\partial p} s_b$, bounded Borel Φ, closed paths C_i and spins j_i:

$$Z[\eta\,\mathrm{d}s; \Phi; C_1, \dots, C_r] = \prod_i s(C_i)^{2j_i}\, Z[\eta; \Phi; C_1, \dots, C_r],$$

$$Z[\eta; \Phi; C] = \int \Phi((\eta_p t_p)_p) \prod_i \chi_{j_i}(U(C_i))\, e^{-\beta\sum_p (1-\eta_p t_p)}\, \mathrm{d}U.$$

Carrying a dual string once across a fundamental loop changes the sign, across an adjoint loop it does not; the rule holds for bounded fat odd loops F, $F(sU) = s(C)F(U)$ for s closed on the plaquettes of a tube around C; and every limit point of renormalised mixed functions of loops, pairs of monopoles and twisted action densities has it.

(g) *Block loops.* For the constants of Thm. YM3_I:D, L odd, a label $\mathsf{b} \geq \gamma^{-2}$, a torus of side $\ell = 2L^m$ with $MR(\mathsf{b}) \leq L^m$ (R the function of Thm. YM3_I:D), $\beta = L^K\mathsf{b}$ on $\Lambda_{K,\ell} = L^{-K}\mathbb{Z}^3/\ell\mathbb{Z}^3$, $\langle\cdot\rangle_K$ the Wilson measure there and $\mathcal{M}^{K-d}U$ the block field on $\Lambda_{d,\ell}$ with law μ_d^K: the block map is covariant under the centre, $\mathcal{M}^n(sU)(c) = s[c]\,\mathcal{M}^n(U)(c)$ for every U if $(ds)_p = 1$ at every plaquette with all corners in the double block of c, $s[c]$ the product of s along the straight path from c_- to c_+; for every closed path C of $\Lambda_{d,\ell}$ and bounded Borel h, $\langle h(W_C(\mathcal{M}^{K-d}U))\rangle_K$ converges along the full sequence of cutoffs, consecutive terms differing by $\leq C\ell^3 K^{\mathfrak{p}_0} L^{-K/10} \sup|h|$ for $K \geq K_{\rm bf}$ with $\lceil K/1260\rceil \geq d$, $K_{\rm bf}$ the threshold of the rate $\mathfrak{o}_K$ of Thm. YM3_I:D (Cor. 2.8 among the construction results), and so do all mixed moments of loops of block fields; the flips $\bar\xi$, $d\bar\xi = 1$, of the torus are symmetries of the limit laws μ_d, so that odd and even functions are orthogonal in $L^2(\mu_d)$; and a closed vortex line that bounds passes as a flip: for a finite union S of closed dual plaquettes of $\Lambda_{d_0,\ell}$, $d \geq d_0$, η_K the sign field that is -1 exactly on the plaquettes of $\Lambda_{K,\ell}$ pierced by $\partial\mathsf{S}$, and $\bar\xi_c = (-1)^{\#(c\cap\mathsf{S})}$: $Z_K[\eta_K; F(\mathcal{M}^{K-d}\cdot)]/Z_K[1; 1] = \int F(\bar\xi V)\,\mathrm{d}\mu_d^K(V) \to \int F(\bar\xi V)\,\mathrm{d}\mu_d(V)$ for bounded Borel F of the block bonds whose blocks miss $\partial\mathsf{S}$, the equality holding at every K and the error being $\leq \sup|F| \sum_{K'\geq K} \mathfrak{o}_{K'}$ for $K \geq K_{\rm bf}$ and $\lceil K/1260\rceil \geq d$, $\mathfrak{o}_K$ the rate of Thm. YM3_I:D; for $F = W_C F'$, with F' a bounded Borel function of such block bonds and

$F'(\bar\xi V)=F'(V)$, $W_C(\bar\xi V)=(-1)^{\#(C\cap\mathsf S)}W_C(V)$ gives the factor $(-1)^{\#(C\cap\mathsf S)}$, the linking number modulo 2 of C and $\partial\mathsf S$.

(h) *Sharp loops; flips.* In (g) let C be a lattice polygon without self-intersection that does not wind and has a neighbourhood in a cube of side $\le \ell/2$, h bounded on $[-1,1]$ and continuous off a Lebesgue null set, $\bar h=\frac{2}{\pi}\int h(w)\sqrt{1-w^2}\,\mathrm dw$, and F_K Borel functions, $|F_K|\le 1$, of the bonds of $\Lambda_{K,\ell}$ at a distance $\ge \mathsf d>0$ from C. Then $\langle F_K h(W_C)\rangle_K-\bar h\langle F_K\rangle_K\to 0$, $W_C=\frac12\mathrm{tr}\,U(C)$. For continuous h this is Lem. 14.9, Thm. AF of §10.1 read at one polygon (see the results used): every character of non-zero spin of the holonomy of C, times a function of modulus ≤ 1 of the bonds at a fixed distance from C, has a mean of modulus at most a constant times a positive power of $L^{-K}\log^3 L^K$ eq. 14.iii, through a rotation of the bonds of a tube of radius ρL^{-K} about C, ρ of the order of $\beta^{1/4}$ up to logarithms, so that the class of the holonomy tends to the Haar law, whose image under $\frac12\mathrm{tr}$ has the density $\frac{2}{\pi}\sqrt{1-w^2}$; a bounded h continuous off a null set lies between continuous functions $h_-\le h\le h_+$ with $\bar h_+-\bar h_-$ as small as we please.

So, with C in a site mirror, the odd unitary u_K of (c) and every character $\chi_j(U(C))$, $j\ge\frac12$, the adjoint one included, tend weakly to 0 between the vectors of such functions in the space of reflection positivity, while $u_K^2=1$ and $\langle W_C^2\rangle_K\to\frac14$. For a rectangular path γ whose sides have at least 3 and at most $N-2$ bonds, $N=\ell L^K$ the number of sites on a side of the torus, γ' its bonds but the two at each corner (no two of them lie in one plaquette, which a side of $N-1$ bonds would allow), $n'=\#\gamma'$, and bounded Borel F of the other bonds: $|\langle\frac12\mathrm{tr}\,U(\gamma)F\rangle_K|\le \mathsf r(4\beta)^{n'}\sup|F|$, $\mathsf r(x)=I_2(x)/I_1(x)$ the ratio of modified Bessel functions; the bound tends to $\exp(-3|\gamma|/8\mathsf b)$.

The flips fade on bounded vectors: let ω_K be the state on the layer of $\Lambda_{K,\ell}$ given by the transfer operator of Section YM3_I:85 (with its trace formula, Prop. YM3_I:85.2), ζ a dual path of the layer with an end plaquette p_0, $Q_0=M_{\mathbf 1\{a_{p_0}<1\}}$, and X,Y bounded operators commuting with Q_0; then

$$|\omega_K(X\mathsf V_\zeta Y)|\le 2\|X\|\|Y\|\langle a_p\rangle_K^{1/2},\qquad \langle a_p\rangle_K\le(21+3\log\beta)/\beta\quad(\beta\ge1).$$

(i) *The Euclidean $\mathbb Z_2$.* In (g) let $\mathcal R$ be a solid ring, a union of closed cubes of side L^{-d} centred at sites of $\Lambda_{d,\ell}$, and $\bar\xi$ a flip, the sign change of the block variables crossing a dual surface $\mathsf S$ whose rim misses $\mathcal R$; $\mathsf h_D$ is Haar measure on the block fields at the depth D and $\mathsf{In}_{[D]}=\{V:\max_p|\partial V(p)-\mathbf 1|<2\varepsilon(x_{[D]})\}$, $|\cdot|$ the operator norm, the set of the small fields of Thm. YM3_I:D at the depth D, its inner ball (the common-depth definitions in Section YM3_I:116, the definitions preceding Lem. YM3_I:113.10), with $x_{[D]}=L^D\mathsf b$ (§2.3.2) and $\varepsilon(\mathsf t)=A_0\mathsf t^{-1/2}(\log\mathsf t)^{\mathfrak p_0}$ YM3_I:(634), A_0 a constant of the list of Thm. YM3_I:D.

There are $D>d$ and numbers $\kappa_D,\bar\kappa_D>0$, $\mathsf R_{[D]}\to0$, free of K, with $\mu_D^K(E)\ge\kappa_D\,\mathsf h_D(E\cap\mathsf{In}_{[D]})$ and $\mu_D^K(E)\le\bar\kappa_D\,\mathsf h_D(E\cap\mathsf{In}_{[D]})+\mathsf R_{[D]}$ for all Borel E and $\mu_D^K(\mathsf{In}_{[D]})\ge1-\mathsf R_{[D]}$, for all $K\ge D+1$ and for the limit law μ_D. A thick vortex is a small field: for every $D\ge d+2$ there is a block field V^1 at the depth D with values in an abelian subgroup, equal to $\bar\xi_c$ on the bonds of $\mathcal R$, whose plaquette angles are $\le 16\pi n_{\mathsf S}L^{-2(D-d)}$, $n_{\mathsf S}$ the number of dual bonds in $\partial\mathsf S$. Hence one positive measure θ_0 of mass $\varkappa_D>0$, which depends on no cutoff and no surface, lies below the law of the block fields of $\mathcal R$ and below all its flips, for all $K\ge D+1$ and in the limit; so the law and each flip differ by at most $1-\varkappa_D$ in total variation, every odd function of the block fields of $\mathcal R$ with values in $[-1,1]$ has its mean in $[\varkappa_D-1,1-\varkappa_D]$, and for a closed path C in $\mathcal R$ with $\#(C\cap\mathsf S)$ odd the fundamental block loop W_C and its sign keep an L^1-distance $\ge\mathsf m_C=\int|W_C|\,\mathrm d\theta_0>0$ from every flip-invariant function, at every $K\ge D+1$ and in the limit.

The limit laws have densities for Haar measure by Lem. 14.18. Conversely, let $\mathcal{R}$ be a rectangular solid ring of $\Lambda_{d,\ell}$: with $N = \ell L^d$ sites on a side of the torus, integers $a_i < b_i < b_i' < a_i'$ $(i = 1, 2)$ and $t_1 \le t_2$, in the units of $\Lambda_{d,\ell}$, $\mathcal{R}$ is the union of the four bricks $[a_1, a_1'] \times [a_2, b_2] \times [t_1, t_2]$, $[b_1', a_1'] \times [a_2, a_2'] \times [t_1, t_2]$, $[a_1, a_1'] \times [b_2', a_2'] \times [t_1, t_2]$ and $[a_1, b_1] \times [a_2, a_2'] \times [t_1, t_2]$ of sites, where $t_2 > t_1$ or no corner of two consecutive bricks is one plaquette, and where $b_i' - b_i \ge 2$, $a_i' - a_i \le N - 2$ and $t_2 - t_1 \le N - 2$, so that the hole holds a site and the four bricks hold all the bonds between the sites of $\mathcal{R}$; let α be its flip, the sign change of the block variables of the bonds $\langle x, x + e_1\rangle$ of $\mathcal{R}$ with $x_1 = b_1$ and $x_2 \le b_2$, which cross one section of the ring, $\mu = \rho \, \mathrm{d}V$ the limit law of its block fields, with a gauge-invariant density ρ, $u = \operatorname{sign} W_C$ for the loop C around the hole, $\mathfrak{L}$ the classes of all bounded gauge-invariant Borel functions of these block fields and $\mathfrak{E}$ those of the flip-invariant ones.

Then every $F \in \mathfrak{L}$ is $F = \Phi_0 + \Phi_1 u$ μ-almost everywhere with $\Phi_0, \Phi_1 \in \mathfrak{E}$ measurable for the four bricks (for the cubes of three sites if the bricks are three sites thick), $\sup|\Phi_i| \le \sup|F|$, and the same holds for the block fields of all coarser depths in $\mathcal{R}$; $\mathfrak{E} = \bigvee_i \mathfrak{L}(\mathrm{brick}_i)$ and $\mathfrak{L} = \mathfrak{E} + \mathfrak{E}u$; the adjoint loop is $\chi_1(V(C)) = \operatorname{tr}(T_1T_2T_3T_4)$ with entries $(T_i)_{ab}$ functions of the i-th brick of modulus ≤ 1. On the closed subspace $\mathfrak{h}$ of $L^2(\mu)$ of the classes of the gauge-invariant functions, on which $\mathfrak{L}$ and $\mathfrak{E}$ act by multiplication, the commutant of $\mathfrak{E}$ is $\mathfrak{L} + \mathfrak{L}v$, of type $\mathrm{I}_1 \oplus \mathrm{I}_2$, where $(vF)(V) = e(V)(\rho(\alpha V)/\rho(V))^{1/2} F(\alpha V)$ and e is the indicator of $\{\rho > 0\} \cap \alpha\{\rho > 0\}$: $v = v^*$, $v^2 = e$, $vu = -uv$, and $\int e \, \mathrm{d}\mu \ge \varkappa_D$, so that μ and its image under α are not mutually singular and $v \ne 0$: L being odd, $\mathcal{R}$ is also a union of closed cubes of side $L^{-(d+1)}$ centred at sites of $\Lambda_{d+1,\ell}$, and α agrees on the bonds of $\mathcal{R}$, up to a central gauge transformation, with the sign field into which the block map from the depth $d+1$ to the depth d turns the flip of a flat sheet S_1 of $\Lambda_{d+1,\ell}$ between two straight closed dual lines in the direction e_3 that lie over the open unit squares about a site of the hole and about a site of the same row outside $[a_1, a_1'] \times [a_2, a_2']$, so that its rim misses $\mathcal{R}$; here D and $\varkappa_D$ are those of the first half at the depth $d+1$.

And v is the limit of the self-adjoint unitaries v_K of the closed vortex line at the cutoffs: for all bounded Borel functions F, F' of the block fields of $\mathcal{R}$, with μ^K their law at the cutoff K, $|\langle F, v_K F'\rangle_{L^2(\mu^K)} - \langle F, vF'\rangle_{L^2(\mu)}| \le 2\sqrt{2} \sup|F| \sup|F'| (\sum_{K' \ge K} \mathfrak{o}_{K'})^{1/2}$ for $K \ge K_{\mathrm{bf}}$ and $\lceil K/1260\rceil \ge d$. If besides the hole is at least three bonds wide in both directions, the ring has positive height and it lies in an open ball of radius $< \ell/4$, then the law and its flip have the same null sets, $e = 1$, v is a self-adjoint unitary, and $\mathfrak{L} = \mathfrak{E} \oplus \mathfrak{E}u$ is a grading by $\mathbb{Z}_2$.

For a band $\mathbb{Z}/N \times I_2 \times I_3$ that winds around the torus, I_2, I_3 intervals of 3 to $N - 1$ sites, the defect is exactly $\mathbb{Z}_2$: $\mathfrak{L} = \mathfrak{E} \oplus \mathfrak{E}u_1$, orthogonally, $\mathfrak{E}u_1 \ne 0$, u_1 the sign of the straight loop that winds, $\mathfrak{E}$ is generated by the cubes of the band, and the flip is a unitary v with $\mathfrak{E}' = \mathfrak{L} \vee \{v\}$ on the space of the gauge-invariant classes for the law of the block fields of the band.

■□

14.1.4. *Construction estimates.* We use the reconstruction from data (Rec) of Section YM3_I:107 and Sections YM3_I:109 and YM3_I:110, as recalled in the setting: the lemma on the smeared insertion (Lem. YM3_I:107.1) with the insertion formula YM3_I:(2039), the point bound eq. 14.i (Prop. YM3_I:107.2 and Lem. YM3_I:109.2), the uniqueness of the family of point operators (Prop. YM3_I:107.2), the lemma on the energy bound (Lem. YM3_I:107.3) and the definition of the energy form q_0 (Lem. YM3_I:107.3), the field operators of Prop. YM3_I:109.1, the boundary distributions and polynomial cyclicity proved in Section YM3_I:109, and the spectrum condition in Thm. YM3_I:AO, the lemma (Br) that links the field operators with the point vectors (Lem. YM3_I:109.2), the propositions on Lorentz covariance and on locality (Props. YM3_I:110.1 and YM3_I:110.2), and the domain argument preceding Prop. YM3_I:109.1; [7, Lem. G.5], by

which a holomorphic function on a connected open subset of $\mathbb{C}^N$ that vanishes on a non-empty open set of real points vanishes; Prop. YM3_I:97.2, which turns an insertion bound for sources in small balls with a constant C_{I} into the point bound of order 4 with the constant $C_{\mathrm{pt}} = c_{\mathrm{pt}} C_{\mathrm{I}}$, $c_{\mathrm{pt}} = 320 \cdot 8^4\, \mathsf{mv}_3(40L)^4 M$ the coefficient of C_{I} in YM3_I:(1919) at $c_{\mathrm{I}} = \frac{1}{40L}$, with $\mathsf{mv}_3 = m_3$ as defined immediately after that equation, and the insertion bound for sources in small balls of one cell, Prop. YM3_I:105.1, with its proof, which the proof of Thm. AQ(c) uses up to its last step together with the insertion bound on one lattice for a pair that carries systems (Lem. YM3_I:104.4, with its normalizer YM3_I:(2017)), the systems of localised source energies of a pair and their constants ([7, Lem. N.1], with [7, eq. 288]), the window YM3_I:(1531) of the labels (Prop. YM3_I:74.1), the two source-radius conditions of Prop. YM3_I:82.2, at the same radius-independent numerical prefix, and the dilation among the construction results ([7, Lem. K.1]); and Thm. YM3_I:AP among the construction results on the limits of periodic Gibbs states and their canonical decompositions, which follows from Thm. YM3_I:D, at the constants of Prop. YM3_I:69.1 and Thms. YM3_I:AJ and YM3_I:AD.

By that theorem every limit of periodic Gibbs states at g, volume-first ones included, is data (Rec) with $(16C_{\mathrm{pt}}^g, 4, \frac{1}{2})$ and has a canonical decomposition, every component of which is data (Rec) with $(16C_{\mathrm{pt}}^g + \bar{\mathfrak{c}}(\mathsf{C}^g), 4, \frac{1}{2})$ and a unique vacuum and is approximated by lattice states: if $\sigma'(X) = \lim_j \rho_j((X)_{K_j}^{\rho_j})$, every component is $\sigma'_\xi(X) = \lim_j \rho_{j,\xi}((X)_{K_j}^{\rho_j})$ along the full sequence, with the periodic Gibbs states $\rho_{j,\xi}(F) = \rho_j(\mathbf{1}_{E_j} F)/\rho_j(E_j)$ at the same couplings, E_j events with $\rho_j(E_j) > 0$ (Thm. YM3_I:AP); these are the lattice approximants of the component; the numbers depend on no sequence, limit, decomposition or ξ; and a volume-first σ has $S_1^\sigma = 0$, its truncated functions S_n^σ, $n \geq 2$, being the limits of the joint cumulants $S_{n;j}^T$ of the $\mathcal{O}_{K_j}(f)$ under the Wilson measures μ_j on tori $\Lambda_j = L^{-K_j}\mathbb{Z}^3/\ell_j\mathbb{Z}^3$, $\ell_j = 2L^{m_j} \to \infty$.

We use besides the reflection positivity of periodic Gibbs states at the nine families of lattice mirrors (Prop. YM3_I:103.4), the chessboard estimate (Prop. YM3_I:86.3, with equation YM3_I:(1759)) and the multiple reflections (Prop. YM3_I:104.2), the one-cell bound with a source (the cell bound, Prop. YM3_I:89.1), the systems of localised source energies (Section YM3_I:88) and the one-lattice source recursion of Lem. 14.14, with its identification of the source energies and its proof on smaller cores, together with the modification of multiple reflections stated and proved in Lem. 14.15(e), and, for Thm. AQ(c), Thms. YM3_I:AJ and YM3_I:AD at every radius and every admitted choice of the constants (σ^-, γ); from Thm. YM3_I:D, that on one torus the laws of the block fields at a fixed depth converge, at a rate (Cor. 2.8), with a floor, an upper bound and a tail on the small fields, and a tail at a fraction of the threshold, read off the small-field chain (the setting of the chain, §2.3.1; Prop. YM3_I:115.2 with the definition preceding Prop. YM3_I:115.2, YM3_I:(2187) and YM3_I:(2188)–(2190); Props. YM3_I:114.2 and YM3_I:113.13; Lem. YM3_I:113.10 with the definitions preceding Lem. YM3_I:113.10, YM3_I:(2159) and YM3_I:(2161); equation YM3_I:(2180) with YM3_I:(2179); Thm. YM3_I:E and the curvature comparison in the proof of Lem. YM3_I:113.10; the condition $A_0 \geq 4B_3 C_H A_1$ in the ordered choice, written explicitly in the proof of Lem. 4.48; and YM3_I:(634), (§2.3.2), the common-depth definitions in Section YM3_I:116); the block-averaging map (equations YM3_I:(37)–(39)) with its locality, its real-analyticity on the small fields and its linear part at the unit (Prop. YM3_I:4.5, YM3_I:4.5 and YM3_I:4.6); the transfer operator of a periodic lattice in one direction and its trace formula (Section YM3_I:85, Prop. YM3_I:85.2); and the short-distance law of the two-point function recalled in the setting (Thm. YM3_I:AV, Cor. YM3_I:127.11).

The theorem on sharp loops on tori on which Thm. AS(h) rests is Thm. AF of §10.1, in its limit, as the spacing tends to 0, for polygons with a further factor G, read at one polygon with $G = F_K$, together with the bound for the characters behind that limit; it is stated with a sketch of its proof as Lem. 14.9 below: at every label $\mathsf{b} > 0$, with $\beta = L^K \mathsf{b}$ and $\langle\cdot\rangle_K$ the Wilson measure on $\Lambda_{K,\ell}$, for a lattice polygon C without self-intersection that does not wind and has a neighbourhood in

a cube of side $\le \ell/2$, and continuous h on $[-1,1]$, $\langle F_K h(\frac{1}{2}\mathrm{tr}\, U(C))\rangle_K - \bar{h}\langle F_K\rangle_K \to 0$ uniformly in Borel functions $|F_K| \le 1$ of the bonds at a fixed positive distance from C, $\bar{h} = \frac{2}{\pi}\int h(w)\sqrt{1-w^2}\,\mathrm{d}w$; its proof reads the Wilson weight on one torus, the Schwarz inequality at site mirrors and Haar measure alone.

The textbook theorems used are the spectral theorem, Stone's theorem on strongly continuous unitary groups, polar decomposition, the bicommutant theorem, von Neumann's theorem on $1+T^*T$, modular theory, reduction theory (direct integrals of separable Hilbert spaces and of von Neumann algebras over a compact metrisable space with a Borel measure: the decomposable operators are the commutant of the diagonal algebra, a von Neumann algebra between the diagonal algebra and its commutant is the direct integral of a measurable field of von Neumann algebras, whose fibres are fixed up to a null set by a countable generating family, and the commutant of a direct integral is the direct integral of the commutants), Hölder's inequality in Schatten classes, Jensen's inequality, Gronwall's lemma, the three-lines theorem, the theorems of Stone–Weierstrass, Montel, Vitali and Liouville, the extension theorem of M. Riesz, the theorems of Peter–Weyl, Clebsch–Gordan and Burnside, the theorem on quasi-analytic vectors, the duality theorem for the free field, the first fundamental theorem for orthogonal groups, Fubini's theorem, the implicit function theorem, and the fact that the zeros of a real-analytic function that does not vanish identically on a connected open set form a null set.

14.1.5. *Auxiliary estimates.*

Lemma 14.1 (Frames)**.** Let $\iota : \mathcal{V} \to \mathcal{H}$ be linear with dense range in a separable Hilbert space, and $\iota_j : \mathcal{V} \to \mathcal{H}_j$ linear, $\iota_j G$ defined for $j \ge j(G)$, with $\langle \iota_j G', \iota_j G\rangle \to \langle \iota G', \iota G\rangle$. Write $\psi_j \Rightarrow \psi$ if for every $\delta > 0$ some G has $\|\psi - \iota G\| < \delta$ and $\limsup_j \|\psi_j - \iota_j G\| < \delta$; $\psi_j \rightharpoonup \psi$ if $\sup_j \|\psi_j\| < \infty$ and $\langle \iota_j G, \psi_j\rangle \to \langle \iota G, \psi\rangle$ for all G; and, for bounded operators, $Q_j \rightharpoonup Q$ if $\sup_j \|Q_j\| < \infty$ and $\langle \iota_j G', Q_j \iota_j G\rangle \to \langle \iota G', Q \iota G\rangle$ for all G, G'. Then: (i) $\iota_j G \Rightarrow \iota G$; $\psi_j \Rightarrow \psi$ and $\psi_j' \rightharpoonup \psi'$ give $\langle \psi_j, \psi_j'\rangle \to \langle \psi, \psi'\rangle$; $\Rightarrow$ implies $\rightharpoonup$ and $\|\psi_j\| \to \|\psi\|$; and $\psi_j \rightharpoonup \psi$ gives $\|\psi\| \le \liminf_j \|\psi_j\|$. (ii) Bounded sequences of vectors and of operators have subsequences that converge in the sense of $\rightharpoonup$, the limits with $\|Q\| \le \liminf_j \|Q_j\|$. (iii) $Q_j \rightharpoonup Q$ and $\psi_j \Rightarrow \psi$ give $Q_j\psi_j \rightharpoonup Q\psi$, and $Q_j^* \rightharpoonup Q^*$. (iv) If π_j are projections with $\|(1-\pi_j)\iota_j G\| \to 0$ for all G, then $(\pi_j \mathcal{H}_j, \pi_j \iota_j)$ has the same properties, and for $\psi_j = \pi_j \psi_j$ the two meanings of $\Rightarrow$ and of $\rightharpoonup$ agree. (v) Let P_j and A be self-adjoint, $\mathcal{V}_0 \subset \mathcal{V}$ a subspace, $\mathsf{r} : \mathcal{V}_0 \to \mathcal{V}$ linear, with $\iota\mathcal{V}_0 \subset D(\mathsf{A})$, $(\mathsf{A} \pm i)\iota\mathcal{V}_0$ dense, $\mathsf{A}\iota G = \iota \mathsf{r} G$ and $\mathsf{P}_j \iota_j G = \iota_j \mathsf{r} G$ for $G \in \mathcal{V}_0$ and large j.

Then $\Phi(\mathsf{P}_j)\psi_j \Rightarrow \Phi(\mathsf{A})\psi$ for every $\Phi \in C_0(\mathbb{R})$ and all $\psi_j \Rightarrow \psi$. ∎

Proof sketch. (i) The triangle inequality and the convergence of the Gram matrices, as in $|\langle \psi_j, \psi_j'\rangle - \langle \psi, \psi'\rangle| \le \|\psi_j - \iota_j G\| \sup_j \|\psi_j'\| + |\langle \iota_j G, \psi_j'\rangle - \langle \iota G, \psi'\rangle| + \|\iota G - \psi\|\|\psi'\|$. (ii) On a countable $\mathbb{Q}(i)$-subspace with dense image the forms converge along a diagonal subsequence; their limits are bounded by $\sup_j \|Q_j\|\,\|\iota G'\|\|\iota G\|$, Riesz's theorem gives Q, and the convergence of the Gram matrices passes to all of $\mathcal{V}$. (iii), (iv) The triangle inequality. (v) Call a bounded sequence (X_j) with a bounded X good if $\psi_j \Rightarrow \psi$ implies $\mathsf{X}_j\psi_j \Rightarrow \mathsf{X}\psi$; sums, products and limits uniform in j of good sequences are good. Given δ, choose $G \in \mathcal{V}_0$ with $\|\psi - (\mathsf{A} \pm i)\iota G\| < \delta$; as $(\mathsf{P}_j \pm i)\iota_j G = \iota_j(\mathsf{r}G \pm iG)$,

$$\limsup_j \|(\mathsf{P}_j \pm i)^{-1}\psi_j - \iota_j G\| \le \limsup_j \|\psi_j - \iota_j(\mathsf{r}G \pm iG)\| < \delta,$$

so the resolvents are good; the polynomials without constant term in $(x+i)^{-1}, (x-i)^{-1}$ are dense in $C_0(\mathbb{R})$, and $\|\Phi(\mathsf{P}_j)\| \le \sup|\Phi|$. □

Lemma 14.2 (Rotations and boosts of point vectors)**.** Let σ be data (Rec), $\tilde{B}(\zeta) = \iota B(\zeta)\iota^{-1}$ on $\mathbb{C}^3$, applied to each point of a $z \in (\mathbb{C}^3)^n$, so that $\tilde{B}(i\tau) = R_\tau$; for $0 < \delta \le \frac{\pi}{2}$ let $\mathsf{I}_\delta = \{x_1 > 0,\ 0 < x_3 < x_1 \tan\delta\}$, $\mathsf{I}_{\pi/2} = \mathsf{I}$, and

$$\mathsf{I}_\delta^{(n)} = \{x \in (\mathbb{R}^3)^n :\ x_1 \in \mathsf{I}_\delta,\ x_{k+1} - x_k \in \mathsf{I}_\delta\ (1 \le k < n)\},$$

an open convex cone. (a) $\tilde B(s)\mathsf{Tb}^{(n)} = \mathsf{Tb}^{(n)}$ and $U(B(s))\Psi_n(z) = \Psi_n(\tilde B(s)z)$ for real s; if $R_\tau z \in \mathsf{Tb}^{(n)}$ for $0 \le \tau \le \tau_0$, then $\Psi_n(z) \in D(e^{-\tau_0 K})$ and $e^{-\tau K}\Psi_n(z) = \Psi_n(R_\tau z)$ for $0 \le \tau \le \tau_0$; and for $x \in \mathsf{I}_\delta^{(n)}$, $R_\tau x \in \mathsf{Tb}^{(n)}$ for $0 \le \tau \le \pi - \delta$. (b) A small tuple in I_δ is a tuple $G = (h_1, \dots, h_n)$ of real $h_k \in C_c^1(\mathbb{B}(x_k^\circ, \epsilon))$, $\mathbb{B}(y,\epsilon)$ the open ball of $\mathbb{R}^3$, with centres $x^\circ = (x_k^\circ)_k$ such that $\prod_k \mathbb{B}(x_k^\circ, \epsilon) \subset \mathsf{I}_\delta^{(n)}$ and $\epsilon < c_0\eta_{x^\circ}$, where

$$\eta_{x^\circ} = \min\Big\{1,\ \min_{0\le\tau\le\pi-\delta}(R_\tau x_1^\circ)_3,\ \tfrac12 \min_{k<n}\ \min_{0\le\tau\le\pi-\delta}\big(R_\tau(x_{k+1}^\circ - x_k^\circ)\big)_3\Big\} > 0;$$

For $n = 1$ the member involving $\min_{k<n}$ is omitted from this minimum; the small tuples in I are those in $\mathsf{I}_{\pi/2}$. For a small tuple G in I_δ and $0 \le \tau \le \pi - \delta$: $[G] \in D(e^{-\tau K})$ and $e^{-\tau K}[G] = [R_\tau G]$; in particular $\Delta^{1/4}[G] = [RG]$ for the small tuples in I. The span $\mathcal{V}_0$ of $\varnothing$ and of the small tuples in I gives a core $[\mathcal{V}_0]$ of $\Delta^{1/4}$, with $(\Delta^{1/4} \pm i)[\mathcal{V}_0]$ dense. (c) For real $u \in C_c^\infty(\mathsf{I})$ let $f_u(B(s)(0,\rho,x_2)) = \int_0^{\pi/2} u(\rho\cos\alpha, x_2, \rho\sin\alpha)\, e^{-(s+i(\pi/4-\alpha))^2} d\alpha$ on W, and $f_u = 0$ off W. Then $f_u \in \mathcal{S}$, $\operatorname{supp} f_u \subset W \cap \{r_1 \le ((x^1)^2 - (x^0)^2)^{1/2} \le r_2\}$ for some $0 < r_1 < r_2$,

$$\mathsf{F}(K)[u] = \phi(f_u)\Omega, \qquad \mathsf{F}(k) = \sqrt{\pi}\, e^{-k^2/4 + \pi k/4}, \tag{14.ii}$$

and for every $p \in \mathbb{R}^3$ some u has $\hat f_u(p) \ne 0$. ∎

Proof sketch. (a) The analytic Wightman functions are invariant under the complexified Lorentz transformations that keep the tube, and the inner products $\langle \Psi_m(w), \Psi_n(z)\rangle$ are such functions; so $\Omega \mapsto \Omega$, $\Psi_n(z) \mapsto \Psi_n(\tilde B(s)z)$ extends to an isometry. By (Br) at $\mathfrak{w}_k = \eta e_3$, then at $\mathfrak{w}_k = \eta\tilde B(s)e_3$, and $\eta \downarrow 0$, it carries $\phi(f_1)\cdots\phi(f_n)\Omega$ to $\phi(f_1 \circ B(-s))\cdots\phi(f_n \circ B(-s))\Omega = U(B(s))\phi(f_1)\cdots\phi(f_n)\Omega$; $\mathcal{D}_0$ is dense, so it is $U(B(s))$. The set of ζ with $\tilde B(\zeta)z \in \mathsf{Tb}^{(n)}$ is invariant under real shifts and holds $i[0,\tau_0]$, so it holds a strip; there $\zeta \mapsto \Psi_n(\tilde B(\zeta)z)$ is holomorphic in norm and equals $e^{i\zeta K}\Psi_n(z)$ on $\mathbb{R}$; comparison with $\int_{[-\mathsf{M},\mathsf{M}]} e^{i\zeta k} dm^K_{\chi,\Psi_n(z)}$, where $m^K_{\chi,\psi} = \langle \chi, \mathcal{P}_K(\cdot)\psi\rangle$, $\mathcal{P}_K$ the spectral measure of K, and $m^K_\psi = m^K_{\psi,\psi}$, on bounded parts of the spectrum of K at $\zeta = i\tau$ gives $\int e^{-2\tau k}\, \mathrm{d}m^K_{\Psi_n(z)} \le \|\Psi_n(R_\tau z)\|^2$ and the identity.

A point $y \in \mathsf{I}_\delta$ is $(\rho\cos\alpha, y_2, \rho\sin\alpha)$ with $\rho > 0$, $0 < \alpha < \delta$, and $(R_\tau y)_3 = \rho\sin(\alpha + \tau) > 0$ for $0 \le \tau \le \pi - \delta$; at real points $d(x) = x_3$, and R_τ is real and linear; so for $x \in \mathsf{I}_\delta^{(n)}$ the points $y = x_1$ and $y = x_{k+1} - x_k$ give $R_\tau x \in \mathsf{Tb}^{(n)}$. (b) Let G be a small tuple in I_δ with the centres x°, and $0 \le \tau \le \pi - \delta$. The balls $\mathbb{B}(R_\tau x_k^\circ, \epsilon)$ that carry the entries of $R_\tau G$ have a first height $\ge \eta_{x^\circ}$ and heights that grow by $\ge 2\eta_{x^\circ}$, and $\epsilon < c_0\eta_{x^\circ}$; so $[R_\tau G] = \int \prod_k h_k(x_k)\, \Psi_n(R_\tau x)\, \mathrm{d}x$, and on $\prod_k \mathbb{B}(x_k^\circ, \epsilon) \subset \mathsf{I}_\delta^{(n)}$, $\Psi_n(R_\tau x) = e^{-\tau K}\Psi_n(x)$ by (a); both integrands are continuous in norm and $e^{-\tau K}$ is closed, so $[G] \in D(e^{-\tau K})$ and $e^{-\tau K}[G] = [R_\tau G]$; at $\delta = \tau = \frac{\pi}{2}$ this is $\Delta^{1/4}[G] = [RG]$; and $\Delta^{1/4}\Omega = \Omega$.

Let $\chi \perp (\Delta^{1/4} + 1)[\mathcal{V}_0]$; then $\chi \perp \Omega$. Let $0 < \delta < \frac{\pi}{2}$ and G' a small tuple in I_δ with the centres x°. For $0 \le \beta \le \frac{\pi}{2} - \delta$ the tuple $R_\beta G'$ is a small tuple in I with the centres $R_\beta x^\circ$: $R_\beta \mathsf{I}_\delta \subset \mathsf{I}$, so that $R_\beta \mathsf{I}_\delta^{(n)} \subset \mathsf{I}_{\pi/2}^{(n)}$, and the heights of $R_\tau R_\beta x^\circ$, $0 \le \tau \le \frac{\pi}{2}$, are those of $R_{\beta+\tau}x^\circ$ with $\beta + \tau \in [0, \pi - \delta]$, so that $\eta_{R_\beta x^\circ} \ge \eta_{x^\circ} > \epsilon/c_0$. The function $\mathsf{f}(z) = \int e^{-zk}\, \mathrm{d}m^K_{\chi,[G']}(k)$ is continuous on $0 \le \operatorname{Re} z \le \pi - \delta$, as $\int e^{-\operatorname{Re} z\, k} d|m^K_{\chi,[G']}| \le \|\chi\|(\|[G']\| + \|e^{-(\pi-\delta)K}[G']\|)$, holomorphic inside, and $\mathsf{f}(\beta) = \langle \chi, e^{-\beta K}[G']\rangle = \langle \chi, [R_\beta G']\rangle$.

As $\chi \perp (\Delta^{1/4} + 1)[R_\beta G'] = [R_\beta G'] + [R_{\beta+\pi/2}G']$, $\mathsf{f}(\beta) + \mathsf{f}(\beta + \frac{\pi}{2}) = 0$ for $0 \le \beta \le \frac{\pi}{2} - \delta$, hence on the strip $0 < \operatorname{Re} z < \frac{\pi}{2} - \delta$; at $z = \beta_0 + it$ the finite measure $e^{-\beta_0 k}(1 + e^{-\pi k/2}) m^K_{\chi,[G']}$ has Fourier transform 0, so it vanishes, and $\chi \perp [G']$. Small tuples in I_δ, with arbitrary entries, are had about every point of the open set $\mathsf{I}_\delta^{(n)}$, and $\langle \chi, \Psi_n(\cdot)\rangle$ is continuous; so it vanishes on $\mathsf{I}_\delta^{(n)}$, an open set of real points of the tube, hence on the tube ([7, Lem. G.5]), and $\chi \perp \mathcal{D}_0$ by (Br): $\chi = 0$.

The closure of $\Delta^{1/4}$ on $[\mathcal{V}_0]$ is symmetric, ≥ 0, lies in $\Delta^{1/4}$, and the range $(\Delta^{1/4}+1)[\mathcal{V}_0]$ is dense; so it is self-adjoint, and it is $\Delta^{1/4}$. (c) f_u is smooth with all Schwartz seminorms finite because $|\partial_s^a g(s+i\tau)| \leq c_a(1+|s|)^a e^{-s^2+\tau^2}$, $g(s)=e^{-s^2}$. As $\int g(s+i\beta')e^{isk}\,\mathrm{d}s = \sqrt{\pi}e^{-k^2/4+\beta' k}$, (a) used backwards gives for $y=(0,\rho,x_2)$ and $0<\gamma\leq\alpha$

$$\mathsf{F}(K)\Psi_1(\iota B(i\alpha)y) = \int g(s+i(\gamma+\tfrac{\pi}{4}-\alpha))\,\Psi_1(\iota B(s+i\gamma)y)\,\mathrm{d}s;$$

integrate against u in polar coordinates of the (x_1,x_3)-plane, pair with $e^{-\eta H}\chi$, and let $\gamma\downarrow 0$ by dominated convergence, the norms being $\leq C_\wp \min\{1,\eta\}^{-\wp}$ because $d(\iota B(i\gamma)x+\eta e_3)\geq\eta$ on W; (Br) at $n=1$ gives $\langle e^{-\eta H}\chi, \mathsf{F}(K)[u]\rangle = \langle e^{-\eta H}\chi, \phi(f_u)\Omega\rangle$, and $e^{-\eta H}\mathcal{H}$ is dense. If $\hat{f}_u(p)=0$ for all u, the entire function $\beta\mapsto\int g(s+i\beta)e^{ip\cdot B(s)y}\,\mathrm{d}s$ would vanish, so $g * e^{ip\cdot B(\cdot)y}=0$, which is absurd as $\hat{g}$ has no zero. □

Proposition 14.3 (Operators between two Euclidean times)**.** Let σ be data (Rec) with a unique vacuum. (a) If $O\neq\emptyset$ is open, $Q\in\mathcal{C}(O)$, $s>0$ and $Q\Omega, Q^*\Omega\in D(e^{sH})$, then $Q\in\mathbb{C}1$. So a bounded operator $e^{-sH}V_1e^{-sH}$, V_1 bounded and $s>0$, lies in no $\mathcal{C}(O)$ unless it is 0 or H is bounded, that is, unless $\mathcal{H}=\mathbb{C}\Omega$; this covers $\mathsf{B}(s,s')$, every product $\mathcal{T}(\mathfrak{z})\mathsf{B}(s,s')\mathcal{T}(\mathfrak{z}')\cdots$, and every weak limit V, along a subsequence and in the sense of Lem. 14.1, of lattice operators $V_j:[G]\mapsto[D_j\cdot\tau_{te_3}G]$ on the spaces of reflection positivity at $x_3=0$ of lattice states ρ_j that converge to σ and are reflection positive at the site planes orthogonal to e_3, as periodic Gibbs states are (Prop. YM3_I:103.4), the lattice approximants of a component among them, with $s,t\in\bigcup_K L^{-K}\mathbb{Z}$ and D_j placed in $\{s<x_3<t-s\}$, whose inner operators $V_{1,j}:[G]\mapsto[\tau_{-se_3}D_j\cdot\tau_{(t-2s)e_3}G]$ are bounded uniformly in j on the spaces of reflection positivity at $x_3=0$ of the shifted states $\rho_j\circ\tau_{se_3}$, $(\rho_j\circ\tau_{se_3})(F)=\rho_j(\tau_{se_3}F)$, which are reflection positive there because ρ_j is so at the site plane $x_3=s$. (b) For every open $O\neq\emptyset$ the von Neumann algebra generated by the e^{-tH} and the $U(x)\mathsf{B}(s,s)U(x)^{-1}$, $t,s>0$, $x\in O$, is $\mathcal{B}(\mathcal{H})$. ■

Proof sketch. (a) Take $O_1\neq\emptyset$ open and $\delta>0$ with $O_1+te_0\subset O$ for $|t|<\delta$, and a product X of fields of O_1. Weak commutation gives $\langle Q^*\Omega, e^{itH}X\Omega\rangle = \langle X^\dagger\Omega, e^{-itH}Q\Omega\rangle$ for $|t|<\delta$. The left member continues to a bounded holomorphic function on $\operatorname{Im}\zeta>-s$, the right member to one on $\operatorname{Im}\zeta<s$; they agree on an interval, so they form one bounded entire function, a constant (Liouville). So the spectral measure of H for $(Q^*\Omega, X\Omega)$ sits at 0; $\mathcal{D}_0(O_1)$ is dense (Thm. AO(a)), so $Q^*\Omega\in\ker H=\mathbb{C}\Omega$, and $Q^*-\bar{c}\in\mathcal{C}(O)$ annihilates Ω, which separates $\mathcal{C}(O)$. The operators named are of the form $V=e^{-sH}V_1e^{-sH}$ with a bounded V_1: $\mathsf{B}(s,s')$ and the products by the semigroup law; for a weak limit V of the V_j, se_3 and te_3 are lattice vectors for large j, and $\tau_{se_3}\theta\tau_{se_3}=\theta$ gives, for every state ρ_j,

$$\rho_j(\theta\bar{G}'\cdot D_j\cdot\tau_{te_3}G) = (\rho_j\circ\tau_{se_3})(\theta\tau_{se_3}\bar{G}'\cdot\tau_{-se_3}D_j\cdot\tau_{(t-2s)e_3}\tau_{se_3}G),$$

that is $\langle[G'],V_j[G]\rangle = \langle[\tau_{se_3}G'],V_{1,j}[\tau_{se_3}G]\rangle$, the right member in the space of $\rho_j\circ\tau_{se_3}$; the shifted states converge to $\sigma\circ\tau_{se_3}=\sigma$, as σ is invariant under the translations, so that their spaces, with the classes $[G]$, are frames for $\mathcal{H}(\sigma)$ as in Lem. 14.1; along a further subsequence $V_{1,j}\rightharpoonup V_1$ in these frames, V_1 bounded (Lem. 14.1(ii)), and $[\tau_{se_3}G]=e^{-sH}[G]$ gives $\langle[G'],V[G]\rangle=\langle e^{-sH}[G'],V_1e^{-sH}[G]\rangle$ for all G,G'. Then $V\Omega, V^*\Omega\in e^{-sH}\mathcal{H}\subset D(e^{sH})$, and $V=c\neq 0$ would make e^{sH} bounded. If H is bounded, the joint spectrum of the translations, which is invariant under the Lorentz transformations and lies in $\{E\geq|q|\}$, is bounded, so it is $\{0\}$; then $H=0$ and $\mathcal{H}=\ker H=\mathbb{C}\Omega$. (b) An operator Y commuting with the generators commutes with $\lim_t e^{-tH}=|\Omega\rangle\langle\Omega|$, so $Y\Omega=y\Omega$ and $Y=y$ on the vectors $e^{-t_0H}\mathsf{b}(x_1)e^{-t_1H}\cdots\mathsf{b}(x_n)\Omega$, $\mathsf{b}(x)=U(x)\mathsf{B}(s,s)U(x)^{-1}$, $x_k\in O$; a vector orthogonal to all of them is, by analyticity in the differences of the points and in the times, orthogonal to all point vectors, hence to $\mathcal{D}_0$ by (Br). □

Lemma 14.4 (A second reflection)**.** Let σ be data (Rec), Q bounded and $\tilde{q}\in\mathcal{H}$ with $\langle[G_1],Q[G_2]\rangle = \langle\tilde{q},[R^{-1}(\theta G_1\cdot G_2)]\rangle$ for all tuples G_1,G_2 over II. Then $Q\in\mathcal{C}(W')$. ■

Proof sketch. For $\mathfrak{z} \in \mathbb{C}^3$ put $\mathfrak{z}^\flat = R^{-1}\mathfrak{z} = (\mathfrak{z}_3, \mathfrak{z}_2, -\mathfrak{z}_1)$ and $\mathfrak{z}^\natural = (-\overline{\mathfrak{z}_3}, \overline{\mathfrak{z}_2}, -\overline{\mathfrak{z}_1})$, so that $\mathfrak{z}^\natural = R^{-1}\theta\mathfrak{z}$ for real $\mathfrak{z}$; for a point x of II the third coordinate of $x^\flat$ and of $x^\natural$ is $-x_1 > 0$. (1) Let $w \in \mathsf{II}^m$ and $z \in \mathsf{II}^n$ be Euclidean points with growing heights, $w_{1,3} < \cdots < w_{m,3}$ and $z_{1,3} < \cdots < z_{n,3}$, whose $N = m + n$ numbers $-w_{k,1}$, $-z_{k,1}$ are pairwise different. Then the identity of the lemma holds for the kernels: $\langle \Psi_m(w), Q\Psi_n(z)\rangle = \langle \tilde{q}, \Psi_N(\mathsf{s}(w,z))\rangle$, $\mathsf{s}(w,z)$ the points $w_k^\natural$, $z_k^\flat$ in the order of their third coordinates.

Indeed, integrated against real entries g_k, h_k carried by small balls about the w_k, z_k in II, the two members are the two members of that identity at $G_1 = (g_k)$, $G_2 = (h_k)$, whose classes, and the class of $R^{-1}(\theta G_1 \cdot G_2)$ after a permutation of its entries, are integrals of point vectors; and the kernels are continuous. (2) Let $0 < r_1 < \cdots < r_N$ be these numbers, which are the heights of the points of $\mathsf{s}(w,z)$, and y_i the second coordinates, and continue in the third coordinates alone: a ket point is $(-r_i, y_i, \omega_i)$, a bra point $(-r_i, y_i, -\overline{\omega_i})$, so that $\mathsf{s}(w,z)$ has the points $\mathsf{s}_i = (\omega_i, y_i, r_i)$. The left member is holomorphic in ω on the convex open set D_1 where the real parts of the ω_i of the ket points increase from 0 in the order of the ket entries and those of the bra points decrease from 0 in the order of the bra entries; the right member is holomorphic on the convex open set $D_2 = \{|\mathrm{Im}\,\omega_1| < r_1,\ |\mathrm{Im}(\omega_{i+1} - \omega_i)| < r_{i+1} - r_i\}$, as $d(\mathsf{s}_1) = r_1 - |\mathrm{Im}\,\omega_1|$ and $d(\mathsf{s}_{i+1} - \mathsf{s}_i) = r_{i+1} - r_i - |\mathrm{Im}(\omega_{i+1} - \omega_i)|$, whatever the real parts are. $D_1 \cap \mathbb{R}^N \subset D_2$, and the two members agree there by (1); by [7, Lem. G.5] they agree on $D_1 \cap D_2$. (3) Let $p \in W'$ and $\mathsf{r} > 0$ with $\overline{\mathbb{B}(p,\mathsf{r})} \subset W'$, and $\mathbb{B}_k = \mathbb{B}(p - 8k\mathsf{r}e_1, \mathsf{r}) \subset W'$, $k \in \mathbb{N}_0$, balls of $\mathbb{R}^{1+2}$, e_1 the unit vector of the x^1-axis.

For $x \in \mathbb{B}_k$ and $x' \in \mathbb{B}_{k'}$, $k < k'$: $-x^1 > |x^0|$ and $(-x'^1) - (-x^1) \geq 6\mathsf{r} > 2\mathsf{r} \geq |x'^0 - x^0|$. Take N functions $f_i \in \mathcal{F}(\mathbb{B}_{k_i})$, one in each of N balls of pairwise different indices, and let Y be a product of some of the $\phi(f_i)$ and X a product of the others. By (Br) at $\mathfrak{w}_a = \eta e_3$ the vector X_η obtained from $X\Omega$ by putting $e^{-\eta H}$ before each field is the integral of Ψ_n at the points $\iota x + a\eta e_3 = (x^1, x^2, a\eta - ix^0)$, a counting the fields of X from the left, and the like holds for Y_η; so $r_i = -x_i^1$, $y_i = x_i^2$, $\mathrm{Im}\,\omega_i = -x_i^0$ and $\mathrm{Re}\,\omega_i = \pm a\eta$, and $\omega \in D_1 \cap D_2$ by the inequalities just written.

By (2) and as $\eta \downarrow 0$,

$$\langle Y\Omega, QX\Omega\rangle = \int \prod \bar{f} \prod f\, \mathsf{L}(x)\, \mathrm{d}x$$

with the one kernel

$$\mathsf{L}(x) = \langle \tilde{q}, \Psi_N((-ix_i^0, x_i^2, -x_i^1)_i)\rangle,$$

whichever of the fields stand in the bra Y or in the ket X. (4) Hence

$$\langle \phi(\bar{h})\psi, Q\psi'\rangle = \langle \psi, Q\phi(h)\psi'\rangle$$

for $h \in \mathcal{F}(\mathbb{B}_0)$, ψ in the span of Ω and of the products of fields, one in each of distinct balls $\mathbb{B}_k$ of odd index, applied to Ω, and ψ' in the like span with even indices $k \geq 2$. (5) These spans stay dense after every power of $1 + H$ (analyticity in the translations), and the polynomial bound of the field carries the relation to $\mathcal{D}_0$: Q commutes weakly with every $\phi(h)$, $h \in \mathcal{F}(\mathbb{B}(p,\mathsf{r}))$.

Every $p \in W'$ and every $\mathsf{r} > 0$ with $\overline{\mathbb{B}(p,\mathsf{r})} \subset W'$ are admitted, these balls cover W', and a partition of unity on the support of $h \in \mathcal{F}(W')$ gives $Q \in \mathcal{C}(W')$. □

Lemma 14.5 (Two mirrors)**.** Let σ be data (Rec) and Q a bounded operator. For combinations F, F' over $\mathsf{Q}_L = \{x_1 < 0, x_3 > 0\}$ and $h = h_+ + h_-$, $h_\pm \in C_c^1(\{x_1 > 0,\ \pm x_3 > 0\})$, put $\ell^Q_{F',F}(h) = \langle [F'], Q[F \cdot h_+]\rangle + \langle [F' \cdot \theta\bar{h}_-], Q[F]\rangle$. If for all F, F' there is $C < \infty$ with $|\ell^Q_{F',F}(h)| \leq C\,\sigma(\theta_1\bar{h} \cdot h)^{1/2}$ for all such h, then $Q \in \mathcal{C}(W)$. ■

Proof sketch. $\sigma(\theta_1\bar{h} \cdot h) = \|[\mathsf{d}h]\|^2$ is the norm of reflection positivity at the mirror $x_1 = 0$. Step 1: by Riesz, $\ell(h) = \langle \Psi, [\mathsf{d}h]\rangle$ with some $\Psi \in \mathcal{H}$; with $v(y) = \mathsf{B}(y,1)\Omega$ and $[k] = \int k(x)U(\vec{x})v(x_3)\,\mathrm{d}x$ for $k \in C_c^1(\mathbb{E}_+)$, $\ell(h) = \int h(x)\Phi(x_1, x_2; x_3)\,\mathrm{d}x$, $\Phi(x_1, x_2; \zeta) = \langle \Psi, U((\zeta, x_2))v(x_1)\rangle$, and by the spectrum condition Φ continues holomorphically to $|\mathrm{Im}\,\zeta| < x_1$: its domain holds all real ζ of both signs. Step 2: with F, F' over $\{x_3 > \tau_0\}$, $\psi = [\tau_{-\tau_0 e_3}F]$, $\psi' = [\tau_{-\tau_0 e_3}F']$ and $\mathsf{B}(\zeta, \tau_0 - \zeta) = e^{-i\mathrm{Im}\zeta H}\mathsf{B}(\mathrm{Re}\,\zeta, \tau_0 -$

$\mathrm{Re}\,\zeta)e^{i\mathrm{Im}\zeta H}$, the first mirror gives the two functions $\mathsf{G}_+(\vec{x};\zeta)=\langle Q^*[F'],U(\vec{x})\mathsf{B}(\zeta,\tau_0-\zeta)U(\vec{x})^{-1}\psi\rangle$, $0<\mathrm{Re}\,\zeta<\tau_0$, and $\mathsf{G}_-(\vec{x};\zeta)=\langle\psi',U(\vec{x})\mathsf{B}(\tau_0+\zeta,-\zeta)U(\vec{x})^{-1}Q[F]\rangle$, $-\tau_0<\mathrm{Re}\,\zeta<0$; they are Φ on real segments, so on $0<\pm\mathrm{Re}\,\zeta<\tau_0$, $|\mathrm{Im}\,\zeta|<x_1$.

Step 3: for $f\in\mathcal{F}(W)$, $|t|\le x_1-\delta$ on $\mathrm{supp}f$; by eq. 14.i,

$$\int f\,\mathsf{G}_+(\vec{x};s-it)=\langle e^{-sH}Q^*[F'],\phi(f)e^{-(\tau_0-s)H}\psi\rangle\to\langle[F'],Q\phi(f)[F]\rangle$$

and $\int f\,\mathsf{G}_-(\vec{x};-s-it)\to\langle\phi(\bar f)[F'],Q[F]\rangle$ as $s\downarrow 0$, and both tend to $\int f\,\Phi(x_1,x_2;-it)$. Step 4: the classes over Q_L are dense after every power of N (a translation along e_1 carries every class into them, and a vector orthogonal to them gives a function of the translation that vanishes on a half-line and, as $|q_1|\le E$ on the spectrum and every class is $e^{-\varepsilon H}$ of a class, continues holomorphically to a strip, hence vanishes), and the polynomial bound of the field carries the relation to $\mathcal{D}_0$. □

Lemma 14.6 (Central positivity)**.** (a) Let $\mathfrak{A}$ be a unital $*$-algebra, ω a state on it with GNS data $(\pi,\mathcal{D},\Omega)$, and $X_0=X_0^*$ central. The following are equivalent: (CP) $\omega(\sum X_0^nY_n)\ge 0$ whenever $\sum\lambda^nY_n\in\mathfrak{A}^+$ for all real λ; (SA) $\pi(X_0)$ has a self-adjoint extension on a larger Hilbert space whose bounded Borel functions F satisfy $\langle\pi(Y^*)\psi,F\psi'\rangle=\langle\psi,F\pi(Y)\psi'\rangle$ for $Y\in\mathfrak{A}$, $\psi,\psi'\in\mathcal{D}$; ($\mathrm{CP}^\sharp$) $\omega(a)\ge 0$ whenever $(1+X_0^2)^ka\in\mathfrak{A}^+$ for some $k\ge 0$; (Ext) with $\mathscr{Q}$ the polynomials $\sum_n\lambda^nY_n$ in a real variable λ with coefficients $Y_n\in\mathfrak{A}$, $\mathrm{ev}:\mathscr{Q}\to\mathfrak{A}$ the $*$-homomorphism $\sum_n\lambda^nY_n\mapsto\sum_nX_0^nY_n$, $\mathfrak{r}=\mathbb{C}[\lambda,(1+\lambda^2)^{-1}]$ and $\mathscr{Q}_\mathfrak{r}$ the $*$-algebra of the functions $\sum_k g_k(\lambda)Y_k$, $g_k\in\mathfrak{r}$, $Y_k\in\mathfrak{A}$: $\omega\circ\mathrm{ev}$ has a linear extension $\tilde\omega$ to $\mathscr{Q}_\mathfrak{r}$ with $\tilde\omega(F^*F)\ge 0$ for all $F\in\mathscr{Q}_\mathfrak{r}$. (b) A closed symmetric operator T affiliated with a von Neumann algebra $\mathfrak{M}$ has a self-adjoint extension on $\mathcal{H}\otimes\mathbb{C}^2$ affiliated with $\mathfrak{M}\otimes M_2(\mathbb{C})$. (c) Two closed symmetric operators affiliated with commuting von Neumann algebras have strongly commuting self-adjoint extensions on $\mathcal{H}\otimes\mathbb{C}^2\otimes\mathbb{C}^2$. (d) For a linear functional L on $\mathbb{C}[x,y]$ with $L(|q|^2)\ge 0$: $L(p)\ge 0$ for all real $p\ge 0$ on $\mathbb{R}^2$ if and only if $L(a)\ge 0$ for all real a such that $(1+x^2)^ka$ is a sum of squares for some k; Motzkin's polynomial M has the level $k=1$: $(1+x^2)M=(x^2y^2-1)^2+(xy(x^2-1))^2+(x(y^2-1))^2$. ■

Proof sketch. (a) (SA)⇒(CP), ($\mathrm{CP}^\sharp$): with E the spectral measure of the extension let $\mu_Y=\langle\Omega,E(\cdot)\pi(Y)\Omega\rangle$, $Y\in\mathfrak{A}$, a complex measure, linear in Y. By (SA), $\mu_{Z^*Z}=\langle\pi(Z)\Omega,E(\cdot)\pi(Z)\Omega\rangle\ge 0$, so $\mu_b\ge 0$ for $b\in\mathfrak{A}^+$; and $d\mu_{X_0^nY}=\lambda^nd\mu_Y$, as the extension continues $\pi(X_0)$ on $\pi(\mathfrak{A})\Omega$. If $(1+X_0^2)^ka=b\in\mathfrak{A}^+$, then $(1+\lambda^2)^kd\mu_a=d\mu_b\ge 0$, so $\mu_a\ge 0$ and $\omega(a)=\mu_a(\mathbb{R})\ge 0$: ($\mathrm{CP}^\sharp$). If $\sum_n\lambda^nY_n\in\mathfrak{A}^+$ for all real λ, then for each real λ_0 the measure $\sum_n\lambda_0^n\mu_{Y_n}$ is positive, so the densities ρ_n of the μ_{Y_n} with respect to the sum of their variations satisfy $\sum_n\lambda_0^n\rho_n\ge 0$ for all rational λ_0 off one null set, hence $\sum_n\lambda^n\rho_n(\lambda)\ge 0$, and $\omega(\sum X_0^nY_n)=\int\sum_n\lambda^n\rho_n(\lambda)\ge 0$: (CP). (CP)⇒(Ext): the extension theorem of M. Riesz, for the real space of the hermitian $F\in\mathscr{Q}_\mathfrak{r}$, its cone K of the F with $F(\lambda)\in\mathfrak{A}^+$ for every real λ, which holds every F^*F, and the subspace of the hermitian polynomials, on whose part in K the functional $\omega\circ\mathrm{ev}$ is ≥ 0 by (CP), gives an extension $\tilde\omega\ge 0$ on K, continued complex-linearly to $\mathscr{Q}_\mathfrak{r}$, because every hermitian F is majorised by a hermitian polynomial: F is a finite sum of terms gS, $g\in\mathfrak{r}$ real, $S=S^*\in\mathfrak{A}$ (as $gY+\bar gY^*=(\mathrm{Re}\,g)(Y+Y^*)+(\mathrm{Im}\,g)\,i(Y-Y^*)$), and for a real polynomial $q\ge|g|$

$$\tfrac12q(S^2+1)-gS=\tfrac14(q-g)(S+1)^2+\tfrac14(q+g)(S-1)^2\in\mathsf{K};$$

for $|g|\le C(1+\lambda^2)^r$ one may take $q=C(1+\lambda^2)^r$, so that the extension holds at each fixed $r\in\mathbb{N}_0$, from the polynomials $\sum_{n\le 2r}\lambda^nY_n$ to the functions whose coefficients are $O((1+\lambda^2)^r)$. ($\mathrm{CP}^\sharp$)⇒(Ext): the same theorem, with the cone of the finite sums of the F^*F, $F\in\mathscr{Q}_\mathfrak{r}$, in the place of K. If a polynomial p is such a sum $\sum_iF_i^*F_i$, then the $G_i=(1+\lambda^2)^kF_i$ are polynomials for some k, $(1+\lambda^2)^{2k}p=\sum_iG_i^*G_i$, and the $*$-homomorphism ev gives $(1+X_0^2)^{2k}\mathrm{ev}(p)\in\mathfrak{A}^+$, so $\omega(\mathrm{ev}\,p)\ge 0$ by ($\mathrm{CP}^\sharp$). The majorisation: for $g=p_0/(1+\lambda^2)^k$, p_0 a real polynomial, $q=\frac12(1+p_0^2)\ge|g|$, and $q\mp g=n_\mp/(1+\lambda^2)^k$ with real polynomials $n_\mp\ge 0$ on $\mathbb{R}$, which are $|m_\mp|^2$ for complex polynomials $m_\mp$, a polynomial that is ≥ 0 on $\mathbb{R}$ being a sum of two squares; so $q\mp g=|u_\mp|^2$ with $u_\mp=m_\mp(\lambda-i)^k(1+\lambda^2)^{-k}\in\mathfrak{r}$, and the

identity above reads $\frac12 q(S^2+1)-gS=(\frac12 u_-(S+1))^*(\frac12 u_-(S+1))+(\frac12 u_+(S-1))^*(\frac12 u_+(S-1))$. (Ext)⇒(SA): $\tilde\omega(F^*)=\overline{\tilde\omega(F)}$ by polarisation; in the GNS space of $\tilde\omega$, which holds that of ω isometrically, multiplication by λ is symmetric with $(\lambda\pm i)\mathscr{Q}_{\mathfrak r}=\mathscr{Q}_{\mathfrak r}$, as $(\lambda\pm i)^{-1}\in\mathfrak r$, so its closure is self-adjoint and extends $\pi(X_0)$; the multiplications by the $Y\in\mathfrak A$ commute with its resolvents, and the bounded C with $\langle\pi(Y^*)\psi,C\psi'\rangle=\langle\psi,C\pi(Y)\psi'\rangle$ for $Y\in\mathfrak A$, $\psi,\psi'\in\mathcal D$, form a weakly closed space that holds the polynomials in the two resolvents, hence, by the bicommutant theorem, all bounded Borel functions of the closure. (b) $T\oplus(-T)$ has the deficiency spaces $\mathcal N_+\oplus\mathcal N_-$ and $\mathcal N_-\oplus\mathcal N_+$, which the flip $(x,y)\mapsto(y,x)\in 1\otimes M_2$ exchanges isometrically; the extension that this flip defines commutes with the unitaries of $\mathfrak M'\otimes 1$. (c) Apply (b) in two tensor factors. (d) This is (CP)⇔(CP$^\sharp$) of (a) for $\mathfrak A=\mathbb C[x,y]$, $X_0=x$ and $\omega=L/L(1)$ (if $L(1)=0$, then $L=0$ by the Schwarz inequality), where $\mathfrak A^+$ is the set of the sums of squares of real polynomials, as $|q|^2=(\operatorname{Re}q)^2+(\operatorname{Im}q)^2$: there (CP) says $L(p)\ge 0$ for every real $p\ge 0$ on $\mathbb R^2$, since $\sum_n\lambda^nY_n\in\mathfrak A^+$ for all real λ gives $\sum_n x^nY_n\ge 0$ on $\mathbb R^2$ (put $\lambda=x$), and a real $p=\sum_n x^np_n(y)\ge 0$ has $\sum_n\lambda^np_n\in\mathfrak A^+$ for every real λ, a polynomial in one variable that is ≥ 0 being a sum of two squares; and if $(1+x^2)^ka$ is a sum of squares, a is real.

The identity is checked by expanding, the three squares summing to $x^6y^2+x^4y^4-2x^4y^2+x^2y^4-3x^2y^2+x^2+1$. □

Lemma 14.7 (Decomposition and limits)**.** Let $(\mathsf X,\lambda,(\sigma_\xi))$ be a canonical decomposition of data (Rec) σ, all σ_ξ data (Rec) with one triple, and $\mathcal W_\xi$ their Wightman functionals. (i) There is exactly one unital $*$-representation $g\mapsto M_g$ of $C(\mathsf X)$ on $\mathcal H(\sigma)$ with $\langle[F'],M_g[F]\rangle=\int g(\xi)\sigma_\xi(\theta\bar F'\cdot F)\lambda(\mathrm d\xi)$; $\|M_g\|\le\sup|g|$, and M_g commutes with e^{-tH}, the $U(x)$, the insertions and the $\mathsf B(s,s')$, keeps every $D(H^a)$, and commutes with every $\phi(f)$ on $\mathcal D_\infty$; so $M_g\in\mathcal C(O)$ for every O. (ii) For $u\in L^1(\mathbb R^{1+2})$ let $\phi_u(\psi',\psi)=\int u(x)\,q_0(U(x)^{-1}\psi',U(x)^{-1}\psi)\,\mathrm dx$ on the vectors $\psi=e^{-\eta H}\chi$, $\psi'=e^{-\eta H}\chi'$, $\eta>0$, among them the classes $[F]$ and the $M_g[F]$, F over $\mathbb E_+$: the form of the field smeared with u, with $|\phi_u(\psi',\psi)|\le\|u\|_{L^1}C_\wp\min\{\eta,1\}^{-\wp}\|\chi'\|\|\chi\|$ and $\phi_u(\psi',\psi)=\langle\psi',\phi(u)\psi\rangle$ for $u\in\mathcal S$, by eq. 14.i; ϕ_u^ξ is the same form for σ_ξ. For combinations F,F' over $\mathbb E_+$ and $u\in L^1(\mathbb R^{1+2})$ the functions $\xi\mapsto\langle[F']_\xi,[F]_\xi\rangle$, $\langle[F']_\xi,H_\xi[F]_\xi\rangle$ and $\phi_u^\xi([F']_\xi,[F]_\xi)$ are continuous, and $\phi_u([F'],M_g[F])=\int g\,\phi_u^\xi([F']_\xi,[F]_\xi)\,\mathrm d\lambda$; the like for 1 and for H. (iii) For $a\in\mathfrak P$, $\xi\mapsto\mathcal W_\xi(a)$ is continuous and $\langle\Omega,M_g\pi(a)\Omega\rangle=\int g\,\mathcal W_\xi(a)\,\mathrm d\lambda$; in particular $\mathcal W_\sigma=\int\mathcal W_\xi\,\mathrm d\lambda$. For $a,b_1,\dots,b_q\in\mathfrak P$ and spatial translations v_i, $\lim_{|v_q|\to\infty}\cdots\lim_{|v_1|\to\infty}\mathcal W_\sigma(a\,\tau_{v_1}b_1\cdots\tau_{v_q}b_q)=\int\mathcal W_\xi(a)\prod_i\mathcal W_\xi(b_i)\,\lambda(\mathrm d\xi)$, each limit existing in turn. (iv) Let $a\in\mathfrak P$, $a^*=a$, and suppose that for all $b_1,\dots,b_r\in\mathfrak P$ there is R such that $\mathcal W_\sigma(b^*ab)\ge 0$ for every b obtained from a complex polynomial $\mathsf p$ in r variables by putting, at each occurrence of its i-th variable, a mean $\frac{1}{\mathsf n}\sum_{\nu=1}^{\mathsf n}\tau_{v_\nu}b_i$ over spatial translations $v_\nu=(0,\vec v_\nu)$ of length $\ge R$, all the translations that occur in b being pairwise different. Then $\mathcal W_\xi(a)\ge 0$ for every ξ. (v) If data (Rec) σ_j with one triple converge to σ on separated tuples, the forms of the fields and of the energy on the classes of combinations converge, and $\mathcal W^{\sigma_j}(f_1\cdots f_n)\to\mathcal W^\sigma(f_1\cdots f_n)$ for all $f_k\in\mathcal S$. ■

Proof sketch. (i) On $C(\mathsf X)\odot\{F\}$ the form $\int\bar g'g\langle[F']_\xi,[F]_\xi\rangle\,\mathrm d\lambda$ is positive; multiplication by g acts on its Hilbert space $\mathfrak K$, and $[F]\mapsto 1\otimes F$ is an isometry $\mathsf W$ of $\mathcal H(\sigma)$ into $\mathfrak K$. For a tuple Y over $\mathbb E_+$ and far horizontal translations v the vectors $[F\cdot\tau_vY]$ tend weakly to $\mathsf W^*\mathsf m(g_Y)\mathsf W[F]$, $g_Y(\xi)=\sigma_\xi(Y)$, and the iterated cluster limits give $\|\mathsf W^*\mathsf m(g_Y)\mathsf W[F]\|=\|\mathsf m(g_Y)\mathsf W[F]\|$, so $\mathsf m(g_Y)$ keeps the range of $\mathsf W$. The g_Y separate the points of $\mathsf X$ ($\xi\mapsto\sigma_\xi$ is injective); by Stone–Weierstrass every $\mathsf m(g)$ keeps the range, which is then all of $\mathfrak K$, and $M_g=\mathsf W^{-1}\mathsf m(g)\mathsf W$.

It commutes with e^{-tH} because each e^{-tH_ξ} is symmetric, with the translations, and with the insertions because the vector $\mathsf B^{\mathrm{in}}_\eta(h)[G]$ is a class given by one formula for σ and for all σ_ξ; hence with all $\mathsf B(s,s')$ and with the fields. (ii) For an insertion smeared in space, $\mathsf G_{\mathsf k}(z)=\int\mathsf k(\vec x)\langle e^{-(\frac\eta2+\bar z)H}\chi',U(\vec x)\mathsf B(\frac\eta2,\frac\eta2)U(\vec x)^{-1}e^{-(\frac\eta2-z)H}\chi\rangle d^2x$ is holomorphic on the strip $|\operatorname{Re}z|<\frac\eta2$ and

bounded by $\|\mathsf{k}\|_{L^1} C_\wp (\eta/2)^{-\wp} \|\chi'\| \|\chi\|$, the same numbers for all ξ; at real z it gives the Euclidean insertion, at $z = -it$ the real-time field.

Montel's theorem and the identity theorem carry the continuity in ξ of the Euclidean values to the imaginary axis; the same with the weight M_g. (iii) In n variables:

$$\mathcal{W}_n^{(\varepsilon)}(f_1, \dots, f_n) = \langle \Omega, \phi(f_1) e^{-\varepsilon H} \cdots e^{-\varepsilon H} \phi(f_n) \Omega \rangle$$

is a value on the imaginary axes of a function holomorphic and bounded on a polystrip, uniformly in ξ, and $|\mathcal{W}_n - \mathcal{W}_n^{(\varepsilon)}| \le \varepsilon\, \mathsf{C}(f_1, \dots, f_n)$ with C formed from the triple and the $\|\partial_t^k f_i\|_{L^1}$; Montel and the identity theorem give the continuity. For the cluster limits, the analytic Wightman functions are bounded by $(C_\wp \eta^{-\wp})^n$ uniformly in ξ and in horizontal translations, and the rate

$$\begin{aligned} &|\mathcal{W}(f_1 \cdots f_n) - \langle \Omega, e^{-\eta H} \phi(f_1) \cdots e^{-\eta H} \phi(f_n) \Omega \rangle| \\ &\qquad \le c \int_0^\eta (1 + |\log s|)\, \mathrm{d}s, \\ &\qquad c = 2^{\mathsf{m}-1} (2^\wp C_\wp)^n \max_{l \le \mathsf{m}+1} \Big\| \mathsf{D}^l \prod_k f_k \Big\|_{L^1}, \end{aligned}$$

$\mathsf{D} = \sum_k k\, \partial/\partial x_k^0$, m an integer $\ge \max\{2, n\wp\}$ (by (Br) at $\mathfrak{w}_k = \eta e_3$ the integrand is holomorphic in the $x_k^0 + ik\eta$, and $\mathsf{m}+1$ integrations by parts in the real times, with the bound of the point vectors at $\frac{\eta}{2}$, bound the derivatives in η), with c formed from the triples ($C_\wp$, $\wp$ the larger of the numbers of the triple of σ and of the one triple of the σ_ξ), n and norms that translations of the f_k do not change, so free of ξ and of the translations, exchanges the cluster limits with $\eta \downarrow 0$. (iv) Expand $b^* a b$ in the means and let the translations tend to infinity one after the other, a factor whose translation goes off commuting in $\mathfrak{P}$ with the others once it is spacelike to them.

By (iii) the terms in which no translation of b^* is one of b tend to $\int \mathcal{W}_\xi(a)$ times the product of the $\mathcal{W}_\xi(b_i)$ and $\mathcal{W}_\xi(b_i^*) = \overline{\mathcal{W}_\xi(b_i)}$ that they carry, and the others, a share $O(\frac{1}{\mathsf{n}})$ of the terms, are bounded uniformly, $\xi \mapsto \mathcal{W}_\xi(c')$ being continuous on the compact X for each $c' \in \mathfrak{P}$. So $0 \le \int \mathcal{W}_\xi(a)(|\mathsf{q}(\xi)|^2 + O(\frac{1}{\mathsf{n}}))\, \mathrm{d}\lambda$, $\mathsf{q} = \mathsf{p}(\mathcal{W}_\xi(b_1), \dots, \mathcal{W}_\xi(b_r))$, and $\int \mathcal{W}_\xi(a) |\mathsf{q}|^2\, \mathrm{d}\lambda \ge 0$. These q form a unital $*$-algebra of continuous functions that separates the points of X, as $\xi \mapsto \sigma_\xi$ is injective and $\mathcal{W}_\xi$ fixes σ_ξ (its analytic continuation to Euclidean points gives the Schwinger functions of σ_ξ); it is dense in $C(\mathsf{X})$ (Stone–Weierstrass). $\mathcal{W}_\xi(a)$ is real and continuous and λ has full support: $\mathcal{W}_\xi(a) \ge 0$ for every ξ. (v) The proofs of (ii) and (iii) use of the family only one triple and the convergence on separated tuples. □

Lemma 14.8 (Bump functions with transforms free of zeros; division)**.** (a) For $\epsilon_k > 0$ with $\sum_k \epsilon_k = \epsilon < \infty$ let u be the infinite convolution of the probability densities $\frac{e^{-x/\epsilon_k}}{\epsilon_k(1-e^{-1})}$ on $[0, \epsilon_k]$, and $w = \mathsf{u} * \mathsf{u}(-\cdot)$. Then $w \in C_c^\infty(\mathbb{R})$, $w \ge 0$, w is even, $\int w = 1$, $\operatorname{supp} w \subset [-\epsilon, \epsilon]$, $\hat{w}$ is entire, $\hat{w}(s) = |\hat{\mathsf{u}}(s)|^2 > 0$ on $\mathbb{R}$, and $\hat{w}$ has no zero in $|\operatorname{Im}\zeta| < 1/\epsilon$; $f_* = w \otimes w \otimes w$ is the function of Thm. AO(d). At $\epsilon_k = \frac{6\epsilon}{\pi^2 k^2}$: $\|w^{(k)}\|_{L^1} \le A_0 R_0^k (k!)^2$, $A_0 = \frac{e^3\pi}{2}$, $R_0 = \frac{3e\pi^2}{2\epsilon}$. (b) For $\delta > 0$, $\gamma > 1$ and $a_k = \delta k^{-\gamma} / \sum_l l^{-\gamma}$ let $g_{\delta,\gamma}$ be the infinite convolution of the uniform probability densities on $[-\frac{a_k}{2}, \frac{a_k}{2}]$: $g_{\delta,\gamma} \in C_c^\infty(\mathbb{R})$ is ≥ 0, even, $\int g_{\delta,\gamma} = 1$, $\operatorname{supp} g_{\delta,\gamma} \subset [-\frac{\delta}{2}, \frac{\delta}{2}]$.

If $\epsilon_k = \epsilon 2^{-k}$ and $\gamma > 1$, or $\epsilon_k = 6\epsilon/\pi^2 k^2$ and $1 < \gamma < 2$, then $g_{\delta,\gamma} = w * \mu$ with a real even $\mu \in \mathcal{S}(\mathbb{R})$, whatever the sizes of δ and ϵ; so $g_{\delta,\gamma}^{\otimes 3} = f_* * \mu^{\otimes 3}$. (c) For data (Rec), real $u \in \mathcal{S}$ and real $\mu \in L^1(\mathbb{R}^{1+2})$ with $u * \mu \in \mathcal{S}$: $\langle \psi, \phi(u*\mu)\psi' \rangle = \int \mu(a) \langle U(a)^{-1}\psi, \phi(u) U(a)^{-1} \psi' \rangle\, \mathrm{d}a$ on $\mathcal{D}_\infty$, so $c_*(u * \mu) \le \|\mu\|_{L^1} c_*(u)$ and $b_{mn}(u*\mu) \le \|\mu\|_{L^1} b_{mn}(u)$; and $c_*(u \circ \Lambda^{-1}) \le ((\Lambda e_0)^0 + |\overrightarrow{\Lambda e_0}|) c_*(u)$ for Lorentz transformations Λ. (d) For $f_0 \in C_c^\infty$, $f_0 \ge 0$, $f_0 \ne 0$, every real p has a Lorentz transformation Λ with $\hat{f}_0(\Lambda^{\mathrm{T}} p) \ne 0$. ■

Proof sketch. (a) $\hat{\mathsf{u}}(\zeta) = \prod_k \frac{1 - e^{-(1 - i\zeta\epsilon_k)}}{(1-e^{-1})(1-i\zeta\epsilon_k)}$, a convergent product since $\sum \epsilon_k < \infty$; no factor vanishes for $|\operatorname{Im}\zeta| < 1/\epsilon_k$, as $|e^{-(1-i\zeta\epsilon_k)}| < 1$ there. The decay of $\hat{\mathsf{u}}$ faster than every power gives smoothness;

on $|\operatorname{Im}\zeta| \le \frac{1}{4\epsilon}$, $|\hat w(s+iq)| \ge e^{-2\epsilon|q|}\Pi(s)^{-2}$, $\Pi(s) = \prod_k (1+\frac{16}{9}s^2\epsilon_k^2)^{1/2}$, with $\Pi(s) \le e^{1/6}(1+y^2)^{(1+\log_2^+ y)/2}$, $y = \frac{4}{3}\epsilon|s|$, at the dyadic choice and $\Pi(s) \le \exp(2.6(8\epsilon|s|/\pi^2)^{1/2})$ at the other; there also $\hat w(s) \le \exp(2-2(2\epsilon|s|/e\pi^2)^{1/2})$, which gives the Gevrey bound. (b) $\hat g_{\delta,\gamma}(\zeta) = \prod_k \frac{\sin(a_k\zeta/2)}{a_k\zeta/2}$ has $|\hat g_{\delta,\gamma}(s+iq)| \le \exp(\frac{\delta}{2}|q| + 1 - (\delta|s|/2e\zeta_\gamma)^{1/\gamma})$, $\zeta_\gamma = \sum k^{-\gamma}$: it falls faster than $|\hat w|$ falls, and $\hat w$ has no zero on a strip; so $\hat g_{\delta,\gamma}/\hat w$ is holomorphic on the strip and its restriction to $\mathbb{R}$ is a Schwartz function. (c) $\phi(u*\mu) = \int \mu(a)U(a)\phi(u)U(a)^{-1}\,\mathrm{d}a$, and $U(a)$ commutes with N and with the P_m; for a Lorentz transformation $U(\Lambda)HU(\Lambda)^{-1} = \int(\Lambda^0{}_0 E + \Lambda^j{}_0 q_j)d\mathcal{P} \le ((\Lambda e_0)^0 + |\overrightarrow{\Lambda e_0}|)H$ on the spectrum $\{E \ge |q|\}$. (d) Rotate p to p' with $p'_2 = 0$; $s \mapsto \hat f_0(B(s)^{\mathrm{T}}p')$ is entire, and at $s = i\frac{\pi}{2}$ it is $\int e^{p'_1x^0 + p'_0x^1} f_0(x)\,\mathrm{d}x > 0$; so it does not vanish at some real s. □

Lemma 14.9 (Sharp loops on a torus)**.** Let $\mathsf{b} > 0$, $\ell = 2L^m$, $a_K = L^{-K}$, $\beta = L^K\mathsf{b}$, and let $\langle\cdot\rangle_K$ be the Wilson measure μ_β on $\Lambda_{K,\ell} = a_K\mathbb{Z}^3/\ell\mathbb{Z}^3$. For $j \in \frac{1}{2}\mathbb{N}$ let D^j be the irreducible representation of SU(2) of dimension $2j+1$ and χ_j its character. Let C be a closed path of $\Lambda_{K_0,\ell}$ for some K_0, read as a path of every $\Lambda_{K,\ell}$, $K \ge K_0$, that has no self-intersection, does not wind (the vectors of its bonds add up to 0 in $\mathbb{R}^3$) and has a neighbourhood in a cube of side $\le \ell/2$; $|C|$ is its length. Let $\mathsf{d} > 0$ and let F_K be Borel functions, $|F_K| \le 1$, of the bonds of $\Lambda_{K,\ell}$ at a distance $\ge \mathsf{d}$ from C. Then for every $j \ge \frac{1}{2}$ there are K_j^{sl}, formed from j, C, b, ℓ and d, and A_j^{sl}, formed from j, C, b and ℓ, with

$$\big|\langle F_K\,\chi_j(U(C))\rangle_K\big| \le A_j^{\mathrm{sl}}\,(a_K\log^3 a_K^{-1})^{j(j+1)|C|/4\pi\mathsf{b}} \qquad (K \ge K_j^{\mathrm{sl}}). \tag{14.iii}$$

Hence, for every continuous h on $[-1,1]$, $\langle F_K h(W_C)\rangle_K - \bar h\langle F_K\rangle_K \to 0$ uniformly in such F_K, where $W_C = \frac{1}{2}\operatorname{tr}U(C)$ and $\bar h = \frac{2}{\pi}\int_{-1}^1 h(w)\sqrt{1-w^2}\,\mathrm{d}w$. ■

Proof sketch. We count in units of a_K: C has $n = |C|/a_K$ steps and p_C sides; $\mathsf{c}_j = j(j+1)$, $\Lambda_C = \max\{1, |C|/\mathsf{b}\}$ and $\rho = \left\lfloor \left(\frac{\beta}{\Lambda_C\log^3\beta}\right)^{1/4} \right\rfloor$. *Slabs.* For a step c of C with the first point x_- and the vector e let $\Pi(c)$ be the lattice plane through x_- orthogonal to c, $\mathsf{T}_\rho(c)$, the slab of c, the set of the translates $c(y)$ of c by the vectors y of $\Pi(c) - x_-$ with $|y| < \rho$, and $\{x_- + y + s'\mathsf{e} : |y|_\infty \le \rho,\ 0 \le s' \le 1\}$ its box. Let $K \ge K_j^{\mathrm{sl}}$, where K_j^{sl} is so large that for $K \ge K_j^{\mathrm{sl}}$: $\ell/a_K \ge 10$, $\beta \ge e^3$, $\rho \ge 2$, $\frac{4}{3}\mathsf{c}_j \le 20\log\beta$, every side of C has at least $4\rho+1$ steps, two sides without a common corner are at a sup-distance $\ge 2\rho+1$, the points at a sup-distance $\le \rho+1$ from C lie in a cube of side $\le \ell/2a_K$, and $\rho a_K < \mathsf{d}$.

The steps c_i of C at least 2ρ steps away from its corners form a set I of $n_I \ge n - 4\rho\mathsf{p}_C$ steps; no step of C but c_i runs through a bond of $\mathsf{T}_\rho(c_i)$, the box of c_i holds no bond of another slab of I, and no plaquette contains bonds of two of these slabs. So, given the bonds off $B = \bigcup_{i\in I}\mathsf{T}_\rho(c_i)$, the slabs are independent under the Wilson measure, and F_K and the paths δ_i of C from c_i to the next step of I read no bond of B; with Q_i the conditional mean of $D^j(U(c_i))$, $\langle F_K\chi_j(U(C))\rangle_K = \langle F_K \operatorname{tr}\prod_i Q_i D^j(U(\delta_i))\rangle_K$, and $|\langle F_K\chi_j(U(C))\rangle_K| \le (2j+1)\langle\prod_i\|Q_i\|\rangle_K$. *One slab.* Given the outer bonds, the variables $v_y = U(c(y))$, $|y| < \rho$, of the slab of $c = c_i$ have a density $\varrho_\circ \propto \exp(-\beta\sum_e a_{p(e)})$ for Haar measure, $p(e)$ the plaquette spanned by a bond e of $\Pi(c)$ and the direction of c. Move $v_y \mapsto \exp(is\varphi_\rho(y)\,\mathsf{n}_y\cdot\frac{\sigma}{2})v_y$, with the logarithmic weight $\varphi_\rho(y) = \min\{1, \log(\rho/|y|)/\log\rho\}$, $\varphi_\rho(0) = 1$, $\varphi_\rho(y) = 0$ for $|y| \ge \rho$, and the colour $\mathsf{n}_y\cdot\sigma = P_y^{-1}(\mathsf{t}\cdot\sigma)P_y$, t a unit vector of $\mathbb{R}^3$, $P_y = U(\pi_y)$ for the path π_y of $\Pi(c)$ from x_- along the first axis and then parallel to the second axis to $x_- + y$.

These moves are left translations and keep Haar measure; integration by parts along them and the Schwarz inequality give the information inequality for the three colours at once, $\|Q_i\|^2 \le (1+\mathsf{c}_j/\bar{\mathcal{I}})^{-1}$, $\bar{\mathcal{I}}$ the largest, over t, conditional mean of $-\partial_s^2\log\varrho_\circ$ at $s=0$. Along a move, for a bond e of $\Pi(c)$ from $x_- + y$ to $x_- + y'$, $\partial_s^2 a_{p(e)} \le \frac{1}{4}(\varphi_\rho(y) - \varphi_\rho(y'))^2$, plus $\varphi_\rho(y)\varphi_\rho(y')(3a(H_e) + a_{p(e)})$, $a(g) = 1 - \frac{1}{2}\operatorname{tr}g$, if e is a rung of the comb of the π_y, a bond parallel to the first axis at a distance $t_e \ge 1$ from it; there $H_e = P_{y'}U(e)^{-1}P_y^{-1}$ is a conjugate of the holonomy around the strip S_e of the t_e plaquettes of $\Pi(c)$ between e and the first axis, and $a(H_e) \le t_e\sum_{q\in S_e} a_q$ (Stokes for a strip).

With $\sum(\varphi_\rho(y)-\varphi_\rho(y'))^2 \le \bar E_\rho = (2\pi\log\rho + 7.63)/\log^2\rho$, the sum over the bonds $\langle y,y'\rangle$ of $\mathbb{Z}^2$: $\bar{\mathcal{I}} \le \frac{\beta}{4}(\bar E_\rho + X_i)$, $X_i = \sum_e \varphi_\rho(y)\varphi_\rho(y')(12a(H_e) + 4\,\mathbb{E}_i(a_{p(e)}))$ over the rungs, $\mathbb{E}_i$ the conditional mean, and

$$\log\|Q_i\| \le -\frac{2\mathsf{c}_j}{\beta\bar E_\rho}\Big(1-\frac{2\mathsf{c}_j}{\beta\bar E_\rho}\Big) + \frac{2\mathsf{c}_j}{\beta\bar E_\rho^2}\,X_i.$$

The mean. Jensen's inequality for the conditional means, the independence of the slabs and Hölder's inequality bound $\langle\exp\sum_i \frac{2\mathsf{c}_j}{\beta\bar E_\rho^2}X_i\rangle_K$ by $\mathsf{M}_\beta^{\varpi}$, $\mathsf{M}_\beta = \sup_p\langle e^{\beta a_p}\rangle_K$, $\varpi = 40\rho^4\mathsf{c}_j n_I/(\beta\bar E_\rho)^2 \le 1$. And $\log\mathsf{M}_\beta \le 192\mathsf{b}/\ell + 4(14.15+3\log\beta)$: the Schwarz inequality at site mirrors, used $3\mathsf{k}$ times, gives $\mathsf{M}_\beta \le \langle\prod_q e^{\beta a_q}\rangle_K^{\mathsf{w}^{-3}}$, the product over the w^3 plaquettes q parallel to one plaquette in $[0,\mathsf{w}]^2\times\{0,2,\dots,2\mathsf{w}-2\}$, $\mathsf{w}=2^{\mathsf{k}}\in(\frac{\ell}{4a_K},\frac{\ell}{2a_K}]$; and the equation of the Wilson measure for the bonds of the box $[0,\mathsf{w}]^2\times[0,2\mathsf{w}-2]$ bounds the conditional mean of that product by $e^{2\beta|\mathsf{P}_\partial|}$, $|\mathsf{P}_\partial|\le 24\mathsf{w}^2$ the number of the plaquettes that stick out of the box, divided by the partition function of the box with free boundary, which is $\ge (1.4\cdot 10^6\beta^3)^{-(\mathsf{w}+1)^2(2\mathsf{w}-1)}$ (a spanning tree of the box, and a ball of radius $(4\beta)^{-1/2}$ about $\mathbf{1}$ for the other bonds).

As $2\mathsf{c}_j/\bar E_\rho \ge \frac{\mathsf{c}_j}{\pi}(\log\rho - 1.2144)$, $n_I/\beta \ge |C|/\mathsf{b} - 4\rho\mathsf{p}_C/\beta$, $\rho \ge \frac12\Big(\frac{\beta}{\Lambda_C\log^3\beta}\Big)^{1/4}$ and $\varpi\log\mathsf{M}_\beta$ stays bounded, the product of the $\|Q_i\|$ gives eq. 14.iii. *The law.* $\chi_{l/2}(g) = \mathsf{U}_l(\frac12\mathrm{tr}\,g)$ with the Chebyshev polynomials U_l of the second kind, $\int\chi_j(g)\,\mathrm{d}g = 0$ for $j>0$, and the law of $\frac12\mathrm{tr}\,g$ under Haar measure is $\frac{2}{\pi}\sqrt{1-w^2}\,\mathrm{d}w$. For a polynomial $h=\sum_l\hat h_l\mathsf{U}_l$, $\langle F_K h(W_C)\rangle_K - \bar h\langle F_K\rangle_K = \sum_{l\ge1}\hat h_l\langle F_K\chi_{l/2}(U(C))\rangle_K \to 0$ by eq. 14.iii, uniformly in F_K; the polynomials are dense in $C[-1,1]$, and $|W_C|\le 1$. □

Lemma 14.10 (From slabs to linear blocks)**.** Let $H\ge0$ be self-adjoint, V a real vector space with a seminorm $|\cdot|_V$, and $\mathsf{B}_h(s,s')$, $h\in V$, $s,s'>0$, bounded operators, linear in h, with $e^{-tH}\mathsf{B}_h(s,s')e^{-t'H} = \mathsf{B}_h(s+t,s'+t')$, $\mathsf{B}_h(s,s')^* = \mathsf{B}_h(s',s)$, and $\|\int\rho_e(\varsigma/\eta)\mathsf{B}_h(\eta+\varsigma,\eta-\varsigma)\,\mathrm{d}\varsigma\| \le C_e|h|_V$ for $\eta = \frac{1}{4(k+2)}$, $k\in\mathbb{N}_0$, where $\rho_e\ge0$ is measurable on $[-c_1,c_1]$, $c_1\le\frac12$, $\bar\rho_e = \int\rho_e\in(0,\infty)$. Then: (a) there is exactly one family of hermitian forms q_h on the vectors of bounded energy with $q_h(e^{-sH}\chi, e^{-s'H}\chi') = \langle\chi,\mathsf{B}_h(s,s')\chi'\rangle$; (b) $|q_h(P_m\psi,P_n\psi')| \le C_\flat|h|_V(m+n+2)\|P_m\psi\|\|P_n\psi'\|$ with $C_\flat = 8C_e/\bar\rho_e$; (c) for $t\mapsto f_t\in V$ three times continuously differentiable for $|\cdot|_V$ and of compact support, $\nu_j(f) = \int|\partial_t^j f_t|_V dt$ and $\nu^*(f) = \sum_{j\le3}\binom{3}{j}2^{3-j}\nu_j(f)$, the time average $\mathsf{e}(f)(\psi,\psi') = \int q_{f_t}(e^{-itH}\psi,e^{-itH}\psi')\,\mathrm{d}t$ has $|\mathsf{e}(f)(\psi,\psi)| \le c_e(f)\langle\psi,(1+H)\psi\rangle$, $c_e(f) = 2C_\flat[\nu_0(f)+2(\zeta(2)+\zeta(3))\nu^*(f)]$, and extends to the form domain of H with the same bound; (d) if, for data (Rec), the forms e of such a family with $V = C^1_c(\mathbb{R}^2)$ and the field satisfy the identity of Thm. AO(f5), then $(\mathrm{LB}_{f,c(f)})$ holds with $c(f) = |\mathsf{c}|[(1+2\coth^2 s)c_e(f) + \frac{e^s}{2\sinh^2 s}\sum_{j,\pm}c_e(f\circ B_j(\pm s))] + |\mathsf{c}'||\int f|$.

For the free massless field of Thm. AR(d), with $T_{\mu\nu} = {:}\partial_\mu\varphi\,\partial_\nu\varphi{:} - \frac12\eta_{\mu\nu}{:}\partial_\lambda\varphi\,\partial^\lambda\varphi{:}$ and $\eta = \mathrm{diag}(1,-1,-1)$, one has $T_{00} = \frac12{:}(\dot\varphi^2+|\nabla\varphi|^2){:}$ and $\Theta = -2\eta^{\mu\nu}T_{\mu\nu}$, so that the field of the datum is $-\Theta = 2\eta^{\mu\nu}T_{\mu\nu}$; the identity holds with e the form of T_{00}, $\mathsf{c}=2$ and $\mathsf{c}'=0$, because every symmetric tensor satisfies $\eta^{\mu\nu}T_{\mu\nu} = (1+2\coth^2 s)T_{00} - \frac{1}{2\sinh^2 s}\sum_{j=1,2}\sum_\pm T_{uu}$, $u = \cosh s\,e_0 \pm \sinh s\,e_j$. ■

Proof sketch. (a) Two pairs of times are compared through a pair larger than both. (b) Let X be the operator $P_n\mathcal{H}\to P_m\mathcal{H}$ of q_h, and write $H = m+\delta_m$ on $P_m\mathcal{H}$, $0\le\delta_m<1$. Then $P_m\mathsf{B}_h(\eta+\varsigma,\eta-\varsigma)P_n = e^{-(\eta+\varsigma)m-(\eta-\varsigma)n}e^{-(\eta+\varsigma)\delta_m}Xe^{-(\eta-\varsigma)\delta_n}$ and $\|e^{-\alpha\delta_m}Xe^{-\alpha'\delta_n}-X\| \le (\alpha+\alpha')\|X\|$; with $\kappa = \int\rho_e(\varsigma/\eta)e^{-(\eta+\varsigma)m-(\eta-\varsigma)n}\,\mathrm{d}\varsigma$ the slab operator S_η has $\|P_m\mathsf{S}_\eta P_n - \kappa X\| \le 2\eta\kappa\|X\|$, so $\kappa(1-2\eta)\|X\| \le C_e|h|_V$; at $\eta = \frac{1}{4(m+n+2)}$, $\kappa\ge\eta\bar\rho_e e^{-3/8}$, and $\frac{16}{3}e^{3/8}<8$. (c) The block (m,n) of $\mathsf{e}(f)$ is $\int e^{it(m-n)}e^{it\delta_m}X_{mn}(f_t)e^{-it\delta_n}\,\mathrm{d}t$; three integrations by parts for $m\ne n$ give the decay $|m-n|^{-3}\nu^*(f)$ times the bound (b); with $A+B\le2\sqrt{AB}(1+|A-B|)$, $A=m+1$, $B=n+1$, Schur's test gives the form bound. (d) In a boosted frame $U(\Lambda)HU(\Lambda)^{-1}\le e^sH$ on the spectrum, so each of the five terms is bounded by its constant times $e^s\langle\psi,N\psi\rangle$.

The free identity: $T_{uu} = \cosh^2 s\, T_{00} \pm 2\cosh s \sinh s\, T_{0j} + \sinh^2 s\, T_{jj}$; the sum over the signs and over j is $4\cosh^2 s\, T_{00} + 2\sinh^2 s(T_{11}+T_{22})$, and the combination is $T_{00}-T_{11}-T_{22} = \eta^{\mu\nu}T_{\mu\nu}$. Since $\eta^{\mu\nu}\eta_{\mu\nu} = 3$, $\eta^{\mu\nu}T_{\mu\nu} = \Theta - \frac{3}{2}\Theta = -\frac{1}{2}\Theta$, and $V\phi(f)V^{-1} = -\Theta(f)$ (Thm. AR(d)) gives $\mathsf{c} = 2$; by covariance $\mathsf{e}(f\circ B_j(\pm s))(\psi_{j,\pm},\psi_{j,\pm})$ is the form of $T_{uu}(f)$ at ψ. □

Lemma 14.11 (Locality through a dilation)**.** (a) Let $\mathcal{R}_0$ be a $*$-algebra of bounded operators on a Hilbert space $\mathcal{F}$, $\Omega_{\rm F}$ cyclic for it, P a projection with $P\Omega_{\rm F} = \Omega_{\rm F}$, and $\mathcal{K} = P\mathcal{F}$. Every bounded T on $\mathcal{K}$ that commutes with all $PAP|_{\mathcal{K}}$, $A\in\mathcal{R}_0$, is $PT_1P|_{\mathcal{K}}$ with $T_1\in\mathcal{R}_0'$, $\|T_1\|\le\|T\|$. (b) For data (Rec) and non-empty open O_1, O_2 the following say the same: $\mathcal{C}(O_1)'\subset\mathcal{C}(O_2)$; there are a Hilbert space $\mathcal{F}$, an isometry $V:\mathcal{H}\to\mathcal{F}$ and a $*$-algebra $\mathcal{R}_0\subset\mathcal{B}(\mathcal{F})$ with $\mathcal{R}_0V\Omega$ dense, $V^*\mathcal{R}_0V\subset\mathcal{C}(O_1)$ and $V^*\mathcal{R}_0'V\subset\mathcal{C}(O_2)$. ■

Proof sketch. (a) With T also T^*T commutes with $X = PA^*AP|_{\mathcal{K}} \ge 0$, so $\|AT\Omega_{\rm F}\|^2 = \langle T\Omega_{\rm F}, XT\Omega_{\rm F}\rangle = \langle X^{1/2}\Omega_{\rm F}, T^*TX^{1/2}\Omega_{\rm F}\rangle \le \|T\|^2\|A\Omega_{\rm F}\|^2$; so $T_1A\Omega_{\rm F} = AT\Omega_{\rm F}$ defines a bounded T_1, and $T_1\in\mathcal{R}_0'$. For $\xi\in\mathcal{K}$ and $A_i\Omega_{\rm F}\to\xi$: $PT_1A_i\Omega_{\rm F} = (PA_iP)T\Omega_{\rm F} = T(PA_iP)\Omega_{\rm F}\to T\xi$, so $PT_1P|_{\mathcal{K}} = T$. (b) If the dilation exists and $T\in\mathcal{C}(O_1)'$, then VTV^* commutes on $V\mathcal{H}$ with the compressions of $\mathcal{R}_0$, so by (a) $T = V^*T_1V$ with $T_1\in\mathcal{R}_0'$, and $T\in\mathcal{C}(O_2)$. Conversely take $\mathcal{F}=\mathcal{H}$, $V=1$, $\mathcal{R}_0 = \mathcal{C}(O_1)$; it holds $\mathcal{A}(O_2) = \mathcal{C}(O_2)'\subset\mathcal{C}(O_1)$, and $\mathcal{A}(O_2)\Omega$ is dense. □

Lemma 14.12 (Weyl operators and Wick forms)**.** Let $\mathfrak{h} = L^2(\mathbb{R}^2)$, $\omega = (p^2+\mathsf{m}^2)^{1/2}$ with $\mathsf{m}>0$, $\mathcal{F}$ the Fock space over $\mathfrak{h}$, $H_{\rm F} = d\Gamma(\omega)$, $\kappa^x = (2\pi)^{-1}(2\omega)^{-1/2}e^{i(t\omega-\vec{x}\cdot p)}$, $u_h = \int h(x)\kappa^x dx$, $A(u) = a^*(u)+a(u)$, $\mathsf{W}[u] = e^{i\overline{A(u)}}$, $\mathcal{R}_0(O)$ the span of the $\mathsf{W}[u_h]$, $h\in\mathcal{F}_{\mathbb{R}}(O)$, and Φ_f the form of the smeared Wick power $:\chi^k:(f)$ of the free field $\chi(h) = A(u_h)$. (a) For $u\in\bigcap_r D(\omega^r)$ and every n: $\mathsf{W}[u]D(H_{\rm F}^n) = D(H_{\rm F}^n)$ and $(1+H_{\rm F})^n\mathsf{W}[u] = \mathsf{W}[u](1+H_{\rm F}+A(i\omega u)+\langle u,\omega u\rangle)^n$ there. (b) $\Phi_f(\zeta,\zeta') = \sum_j\binom{k}{j}\int f(x)\langle a(\kappa^x)^j\zeta, a(\kappa^x)^{k-j}\zeta'\rangle\,{\rm d}x$ on $\mathcal{D}_\infty(H_{\rm F})$, with $|\Phi_f(\zeta,\zeta')|\le 2^k(4\pi\mathsf{m})^{-k/2}\|f\|_{L^1}\|(1+H_{\rm F})^k\zeta\|\|(1+H_{\rm F})^k\zeta'\|$; it is the form of an operator $\Phi(f)$ on $\mathcal{D}_\infty(H_{\rm F})$. (c) For real $h\in C_c^\infty$ and x spacelike to $\operatorname{supp}h$, $\langle\kappa^x,u_h\rangle = \int h(y)\frac{e^{-\mathsf{m}\mathsf{d}_{xy}}}{4\pi\mathsf{d}_{xy}}\,{\rm d}y$, $\mathsf{d}_{xy} = (|\vec{x}-\vec{y}|^2-(x^0-y^0)^2)^{1/2}$: a real number; so Weyl operators of spacelike separated regions commute, and $\Phi_f(\mathsf{W}[u_h]\zeta,\mathsf{W}[u_h]\zeta') = \Phi_f(\zeta,\zeta')$ if $\operatorname{supp}f$ and $\operatorname{supp}h$ are spacelike separated. (d) For every non-empty open O the vectors $\mathsf{W}[u_h]\Omega_{\rm F}$, $h\in\mathcal{F}_{\mathbb{R}}(O)$, are total in $\mathcal{F}$. (e) For $f\in\mathcal{F}(O)$ and $B\in\mathcal{R}_0(O)'$: $\langle\Phi(\bar f)\zeta, B\zeta'\rangle = \langle\zeta, B\Phi(f)\zeta'\rangle$ for $\zeta,\zeta'\in\mathcal{D}_\infty(H_{\rm F})$. (f) *Gaussian families.* Let $(\kappa_\alpha,\varepsilon_\alpha,\mathsf{P})$, $\alpha\in\mathsf{A}$, be a Gaussian family of Thm. AR(a) on (M,μ), and now let $\mathfrak{h} = L^2(\mathsf{M},\mu)$, $\mathcal{F}$ the Fock space over it, $H_{\rm F} = d\Gamma(\omega)$, $\kappa_\alpha^x = e^{i(t\omega-\vec{x}\cdot\vec{p})}\kappa_\alpha$, and Φ_f the form of the smeared Wick polynomial $:\mathsf{P}(\Phi):(f)$.

Put $\varsigma_\alpha = 1$ if $\varepsilon_\alpha = +1$ and $\varsigma_\alpha = i$ if $\varepsilon_\alpha = -1$, $u_h^\alpha = \varsigma_\alpha\hat h(\omega,\vec p)\kappa_\alpha$ for $h\in\mathcal{S}$, let $\mathfrak{k}(O)$ be the real span of the u_h^α, $h\in\mathcal{F}_{\mathbb{R}}(O)$, $\alpha\in\mathsf{A}$, and let $\mathcal{R}_0(O)$ now be the span of the $\mathsf{W}[u]$, $u\in\mathfrak{k}(O)$. Suppose (f1) $\mathfrak{k}(O)+i\,\mathfrak{k}(O)$ is dense in $\mathfrak{h}$ for every non-empty open O, and (f2) $\overline{\langle\kappa_\alpha^x,u\rangle} = \varepsilon_\alpha\langle\kappa_\alpha^x,u\rangle$ for $\alpha\in\mathsf{A}$, $u\in\mathfrak{k}(O)$ and x spacelike to O. Then (a)–(e) hold with the $u\in\mathfrak{k}(O)$ in the place of the u_h, $h\in\mathcal{F}_{\mathbb{R}}(O)$, and with three changes.

In (a), $u\in\bigcap_r D(\omega^r)$ with $\omega^{-1/2}u\in\mathfrak{h}$, which holds for every $u\in\mathfrak{k}(O)$. In (b), the form is

$$\Phi_f(\zeta,\zeta') = \sum_j\binom{k}{j}\sum_{\underline\alpha}\mathsf{p}_{\underline\alpha}\prod_{i>j}\varepsilon_{\alpha_i}\int f(x)\Big\langle\prod_{i\le j}a(\kappa_{\alpha_i}^x)\zeta,\prod_{i>j}a(\kappa_{\alpha_i}^x)\zeta'\Big\rangle\,{\rm d}x,$$

and with the weight $\varpi = \omega^{1/2}(1+\omega)^{r'}$, $r'>\delta$, the number $\mathsf{n}_* = \max_\alpha\|\varpi^{-1}\kappa_\alpha\|<\infty$ and the order $L_* = \lceil k(r'+\frac12)\rceil$ it obeys $|\Phi_f(\zeta,\zeta')|\le 2^k|\mathsf{p}|_1\mathsf{n}_*^k\|f\|_{L^1}\|(1+H_{\rm F})^{L_*}\zeta\|\|(1+H_{\rm F})^{L_*}\zeta'\|$, the operator $\Phi(f)$ being bounded by a multiple of $\|(1+H_{\rm F})^{2L_*}\cdot\|$. In (c), the invariance holds for $u\in\mathfrak{k}(O)$ and $\operatorname{supp}f$ spacelike to O, and Weyl operators of $\mathfrak{k}(O_1)$, $\mathfrak{k}(O_2)$ commute for spacelike separated O_1, O_2. The hypotheses (f1), (f2) hold for the first derivatives $\partial_\alpha\varphi^a$ of n massless free fields, the free photons with n colours of Thm. AR(c)–(e), and for the generalised free fields of Thm. AR(c), the free fields of a mixture $d\rho(\mathsf{m})$ of masses with $\varepsilon=+$, the kernel r^{-2} being the mixture $d\rho = 4\pi\,{\rm d}\mathsf{m}$. ■

Proof sketch. (a) On exponential vectors $\mathsf{W}[u]\mathrm{Exp}(v) = e^{-\frac{1}{2}\|u\|^2+i\langle u,v\rangle}\mathrm{Exp}(v+iu)$, whence $H_{\mathrm{F}}\mathsf{W}[u] = \mathsf{W}[u](H_{\mathrm{F}} + A(i\omega u) + \langle u, \omega u\rangle)$ there; exponential vectors are dense in every $D(H_{\mathrm{F}}^n)$, H_{F} is closed, and an induction gives the domains. (b) Wick's theorem, with $\|a(\kappa^x)^j\zeta\| \le \|\omega^{-1}\kappa^x\|^j\|H_{\mathrm{F}}^j\zeta\|$ and $\|\omega^{-1}\kappa^x\|^2 = 1/4\pi\mathsf{m}$; the operator by Riesz after the commutator identity $\Phi_f(H_{\mathrm{F}}\zeta,\zeta') - \Phi_f(\zeta, H_{\mathrm{F}}\zeta') = i\Phi_{\partial_t f}(\zeta,\zeta')$. (c) The Euclidean two-point function $e^{-\mathsf{m}r}/4\pi r$ continued to real time is real at spacelike separation; on exponential vectors the conjugation by $\mathsf{W}[u_h]$ shifts each annihilator $a(\kappa^x)$ by $i\langle\kappa^x, u_h\rangle$ in the ket and conjugately in the bra, and the reality of the shift makes the two changes cancel. (d) $\{u_h : h \in \mathcal{F}(O)\}$, h complex, is dense in $\mathfrak{h}$ by the reflection principle and uniqueness of Fourier transforms: for $g \perp u_h$, $h \in \mathcal{F}(O)$, and η carried by a small ball, $x \mapsto \langle g, u_{\eta(\cdot-x)}\rangle = \int \bar{g}\kappa\hat{\eta}(\omega,p)e^{i(t\omega-\vec{x}\cdot p)}d^2p$ vanishes near a point and continues boundedly into every timelike direction, so it vanishes, and its Fourier transform at $t=0$ gives $\bar{g}\kappa\hat{\eta} = 0$; along an approximate unit $\hat{\eta}\to 1$, and $\kappa > 0$, so $g = 0$.

The vectors of (d) are $\mathsf{W}[u_h]\Omega_{\mathrm{F}} = e^{-\frac{1}{2}\|u_h\|^2}\mathrm{Exp}(iu_h)$ with real h. Let Ψ be orthogonal to all of them. For real $h_1,\dots,h_n \in \mathcal{F}_{\mathbb{R}}(O)$ the function $z \mapsto \langle\Psi, \mathrm{Exp}(\sum_j z_j u_{h_j})\rangle$ is entire on $\mathbb{C}^n$ and vanishes on $i\mathbb{R}^n$, since $\sum_j iy_j u_{h_j} = iu_{\sum_j y_j h_j}$; so all its Taylor coefficients at 0 vanish: $\Psi \perp \Omega_{\mathrm{F}}$ and $\Psi \perp \mathrm{Sym}(u_{h_1}\otimes\cdots\otimes u_{h_n})$. This last product is complex-linear in each u_{h_j}, and $h \mapsto u_h$ is complex-linear; so Ψ is orthogonal to it for all $h_j \in \mathcal{F}(O)$, and these symmetric tensors are total in each n-particle space by the density; hence $\Psi = 0$. (e) The Wick power of one smeared field is a real polynomial in it: $:A(u)^k: = \mathsf{h}_k(A(u))$ with $\mathsf{h}_0 = 1$, $\mathsf{h}_1 = \lambda$, $\mathsf{h}_{k+1} = \lambda\mathsf{h}_k - k\|u\|^2\mathsf{h}_{k-1}$; for $u = u_{\delta_\epsilon(\cdot-x)}$, δ_ϵ an approximate unit and $x \in \mathrm{supp} f$, the unitary group of $\overline{A(u)}$ lies in $\mathcal{R}_0(O)$, so B commutes weakly with $:A(u)^k:$ on $\mathcal{D}_\infty(H_{\mathrm{F}})$; integrate against f and let $\epsilon\downarrow 0$, the regularised Wick powers being bounded by $c(f)\|(1+H_{\mathrm{F}})^{2k}\cdot\|$ uniformly in ϵ and converging weakly. (f) The changes, clause by clause. (a): $\|a(w)\zeta\| \le \|\omega^{-1/2}w\|\|H_{\mathrm{F}}^{1/2}\zeta\|$ and $\|a^*(w)\zeta\|^2 = \|a(w)\zeta\|^2 + \|w\|^2\|\zeta\|^2$ on each n-particle space take the place of the bound by the number of particles, which uses $\mathsf{m} > 0$; so $A(u)$ is defined on $D(H_{\mathrm{F}}^{1/2})$ for $u, \omega^{-1/2}u \in \mathfrak{h}$, it is essentially self-adjoint on the vectors of finitely many particles in $\mathcal{D}_\infty(H_{\mathrm{F}})$, and the proof of (a) runs as it stands.

Every u_h^α lies in all $D(\omega^r)$ and has $\omega^{-1/2}u_h^\alpha \in \mathfrak{h}$, because $\hat{h}$ decays faster than every power and $\varpi^{-1}\kappa_\alpha \in \mathfrak{h}$. (b): in the bound of the annihilators the weight ϖ takes the place of ω,

$$\|a(g_1)\cdots a(g_m)\zeta\| \le \prod_i \|\varpi^{-1}g_i\|\|(1+H_{\mathrm{F}})^{m(r'+\frac{1}{2})}\zeta\|,$$

and

$$\|\varpi^{-1}\kappa_\alpha\|^2 = \Gamma(2r')^{-1}\int_0^\infty s^{2r'-1}e^{-s}\|\omega^{-1/2}\kappa_\alpha e^{-s\omega/2}\|^2 ds < \infty$$

by the bound $c_\kappa s^{-\delta}$ of Thm. AR(a) and $r' > \delta$; the form is hermitian, $\overline{\Phi_{\bar{f}}(\zeta',\zeta)} = \Phi_f(\zeta,\zeta')$, because $\mathsf{P}(\varepsilon t) = \mathsf{P}(t)$; the operator comes as before, from the commutator with H_{F}. (c): on exponential vectors

$$\Phi_f(\mathrm{Exp}\, v, \mathrm{Exp}\, v') = e^{\langle v,v'\rangle}\int f(x)\,\mathsf{P}\big(\bar{z}(x) + \varepsilon z'(x)\big)\,\mathrm{d}x,$$

$z_\alpha = \langle\kappa_\alpha^x, v\rangle$, $z'_\alpha = \langle\kappa_\alpha^x, v'\rangle$; under $v \to v+iu$, $v' \to v'+iu$ the argument of P changes by $-i\bar{\zeta}_\alpha + i\varepsilon_\alpha\zeta_\alpha$, $\zeta_\alpha = \langle\kappa_\alpha^x, u\rangle$, which (f2) makes 0; and $\langle u_{h'}^\alpha, u_h^\beta\rangle = \bar{\varsigma}_\alpha\int h'(x)\langle\kappa_\alpha^x, u_h^\beta\rangle\,\mathrm{d}x$ is real by (f2), so the Weyl operators commute. (d): the argument of (d) with the $u_j \in \mathfrak{k}(O)$ in the place of the u_{h_j}, and (f1) in the place of the density of the u_h. (e): with $g_\alpha = \rho_\epsilon\kappa_\alpha^x$, $\rho_\epsilon = \hat{\delta}_\epsilon(\omega,\vec{p})$, the regularised fields $\Phi_\alpha = a^*(g_\alpha) + \varepsilon_\alpha a(g_\alpha)$ are $\bar{\varsigma}_\alpha A(\varsigma_\alpha g_\alpha)$, as $\bar{\varsigma}_\alpha^2 = \varepsilon_\alpha$, and $\varsigma_\alpha g_\alpha = u^\alpha_{\delta_\epsilon(\cdot-x)} \in \mathfrak{k}(O)$ for $x \in \mathrm{supp} f$ and small ϵ, so the unitary groups of the closures of the $A(\varsigma_\alpha g_\alpha)$ lie in $\mathcal{R}_0(O)$.

The recursion

$$\Phi_{\alpha_0}{:}\Phi_{\underline{\alpha}}{:} = {:}\Phi_{(\alpha_0,\underline{\alpha})}{:} + \sum_i \varepsilon_{\alpha_0}\langle g_{\alpha_0}, g_{\alpha_i}\rangle {:}\Phi_{\underline{\alpha}\setminus i}{:}$$

makes every Wick monomial a polynomial with complex coefficients in several non-commuting $A(\varsigma_\alpha g_\alpha)$; B passes through one factor at a time, so $\langle\zeta, BZ\zeta'\rangle = \langle Z^\dagger\zeta, B\zeta'\rangle$ for such a polynomial Z and $Z^\dagger$ the polynomial with the conjugate coefficients and the products reversed; and $Z^\dagger = Z$ on $\mathcal{D}_\infty(H_{\mathrm{F}})$ for the regularised Wick polynomial $Z = \sum_{\underline{\alpha}} \mathsf{p}_{\underline{\alpha}} {:}\Phi_{\underline{\alpha}}{:}$, by the hermiticity of (b) at $B = 1$.

Then integrate against f and let $\epsilon \downarrow 0$ as before, with ϖ for ω in the weights and L_* for k in the norms. *The families.* For the photons, $\kappa_0 = (2\pi)^{-1}(2|p|)^{-1/2}$, $\kappa_j = -ip_j\kappa_0$, $\kappa_3 = -\omega\kappa_0$ and $\varepsilon = (+,+,-)$ (the proof of Thm. AR(d)), so $u_h^j = \hat{h}'(\omega,p)\kappa_0$ with $h' = -\partial_j h$, and $u_h^3 = \hat{h}'(\omega,p)\kappa_0$ with $h' = \partial_t h$. For real $h' \in C_c^\infty$ the function $F_{h'}(x) = \langle\kappa_0^x, \hat{h}'(\omega,p)\kappa_0\rangle$ is smooth and, by the computation of (c) at $\mathsf{m} = 0$, real on the open set of the x spacelike to $\operatorname{supp} h'$; there $\langle\kappa_j^x, \hat{h}'\kappa_0\rangle = \partial_j F_{h'}(x)$ is real and $\langle\kappa_3^x, \hat{h}'\kappa_0\rangle = -i\partial_t F_{h'}(x)$ is imaginary: (f2). $\mathfrak{k}(O) + i\,\mathfrak{k}(O)$ holds the vectors $\hat{h}(\omega,p)\,\omega\kappa_0$, $h \in \mathcal{F}(O)$, and the reflection argument of (d) with $\omega\kappa_0 > 0$ in the place of κ, $\omega - \vec{v}\cdot p \geq (1 - |\vec{v}|)|p| > 0$, gives (f1).

For n colours $\mathfrak{h}$ is the sum of n copies, $\langle\kappa_{(a,\alpha)}^x, u_h^{(b,\beta)}\rangle = \delta_{ab}\langle\kappa_\alpha^x, u_h^\beta\rangle$, and both hypotheses hold copy by copy. For a mixture of masses, $\mathsf{M} = \mathbb{R}^2 \times (0,\infty)$, $d\mu = d^2p\,\mathrm{d}\rho(\mathsf{m})$, $\omega = (p^2 + \mathsf{m}^2)^{1/2}$ and one field: $\langle\kappa^x, u_h\rangle = \int h(y) \int \frac{e^{-\mathsf{m}\mathsf{d}_{xy}}}{4\pi\mathsf{d}_{xy}}\,\mathrm{d}\rho(\mathsf{m})\,\mathrm{d}y$ is real at spacelike separation, by (c) and a ρ-integrable majorant, and the reflection argument, applied to the image of the finite measure $\bar{g}\kappa\hat{\eta}(\omega,\vec{p})\,\mu$ under $(\vec{p},\mathsf{m}) \mapsto (\omega,\vec{p})$, a homeomorphism onto $\{E > |\vec{p}|\}$, gives the density; only the weight changes. □

Proposition 14.13 (Four squares)**.** Let $\Lambda = a\mathbb{Z}^3/\ell\mathbb{Z}^3$, ℓ/a even and ≥ 4, $\beta \geq 0$, $\mu = \mu_\beta$ with the normalisation Z. The planes $\mathsf{P}_\mu = \{x_\mu \in \{0, \frac{\ell}{2}\}\}$, $\mu = 1,3$, pass through sites and cut the torus into $\mathsf{Q}_1 = \{0 < x_1, x_3 < \frac{\ell}{2}\}$, $\mathsf{Q}_2 = \theta_1\mathsf{Q}_1$, $\mathsf{Q}_3 = \theta\theta_1\mathsf{Q}_1$, $\mathsf{Q}_4 = \theta\mathsf{Q}_1$; giving each plaquette in a plane half of its weight in each of the two adjacent closed quadrants, $e^{-\beta\sum_p a_p} = w_1w_2w_3w_4$, w_i depending only on the bonds of $\bar{\mathsf{Q}}_i$. Let B_c be the bonds on the four corner lines $\mathsf{P}_1 \cap \mathsf{P}_3$; B_R the other bonds of $\mathsf{P}_3 \cap \{0 \leq x_1 \leq \frac{\ell}{2}\}$, in both planes $x_3 \in \{0, \frac{\ell}{2}\}$; $\mathsf{B}_L = \theta_1\mathsf{B}_R$; $\mathsf{B}_U = \mathsf{d}\mathsf{B}_R$, the bonds of $\mathsf{P}_1 \cap \{0 \leq x_3 \leq \frac{\ell}{2}\}$ off the corner lines; $\mathsf{X} = G^{\mathsf{B}_R}$, $\mathsf{C} = G^{\mathsf{B}_c}$, κ the map of C induced by d, $\mathcal{K} = L^2(\mathsf{X})$, and $\mathcal{H}_\Lambda = L^2(\mathsf{C}\times\mathsf{X}\times\mathsf{X})$, regarded as the space of the fields $c \mapsto \hat{\xi}_c$ of Hilbert–Schmidt operators on $\mathcal{K}$ whose left index is the θ_1-image x_L and whose right index is x_R.

For a bounded Borel function F of the bonds of $\bar{\mathsf{Q}}_1$ let A_c^F be the Hilbert–Schmidt operator on $\mathcal{K}$ with the kernel $Z^{-1/4}\int F w_1$, integrated over the bonds of $\bar{\mathsf{Q}}_1$ in no plane, between the configuration x of B_R and the configuration x' of B_U carried to B_R by d, at the corner configuration c; $Z_c = \mathsf{A}_c^1$. For a bounded Borel F of the bonds of $\{0 \leq x_3 \leq \frac{\ell}{2}\}$ let $\iota_\Lambda F = Z^{-1/2}\int F w_1 w_2$ over the bonds of that half off $\mathsf{B}_c \cup \mathsf{B}_R \cup \mathsf{B}_L$, $\Omega_\Lambda = \iota_\Lambda 1$; and $\mathcal{M}_\Lambda = \{\hat{\xi}_c \mapsto \hat{\xi}_c q_c\}$, $\mathcal{M}'_\Lambda = \{\hat{\xi}_c \mapsto b_c\hat{\xi}_c\}$ with bounded measurable fields q_c, b_c of operators of $\mathcal{K}$.

Then:

(L1) $\langle\iota_\Lambda F', \iota_\Lambda F\rangle = \mu(\theta\bar{F}'\cdot F)$; $J_\Lambda\hat{\xi}_c = \hat{\xi}_c^*$ has $J_\Lambda\iota_\Lambda F = \iota_\Lambda(\theta_1\bar{F})$ and $J_\Lambda\mathcal{M}_\Lambda J_\Lambda = \mathcal{M}'_\Lambda$; $\mathcal{M}_\Lambda$ and $\mathcal{M}'_\Lambda$ are von Neumann algebras, each the commutant of the other.

(L2) For F_1 of $\bar{\mathsf{Q}}_1$ and F_2 of $\bar{\mathsf{Q}}_2$: $(\iota_\Lambda(F_2F_1))_c^\wedge = \mathsf{A}_{\kappa c}^{R^{-1}F_2}\mathsf{A}_c^{F_1}$; $(\mathsf{A}_c^F)^* = \mathsf{A}_{\kappa c}^{\mathsf{d}\bar{F}}$, $Z_c^* = Z_{\kappa c}$, and $\hat{\Omega}_c = Z_c^*Z_c = \rho_c^{1/2}$ with ρ_c of trace class, $\int \operatorname{Tr}\rho_c\,\mathrm{d}c = 1$.

(L3) For $Q \in \mathcal{M}_\Lambda$ there is $\tilde{q} \in \mathcal{H}_\Lambda$, $\|\tilde{q}\| \leq \|Q\|$, with $\langle\iota_\Lambda F', Q\iota_\Lambda F\rangle = \langle\tilde{q}, \iota_\Lambda(R^{-1}(\theta\bar{F}'\cdot F))\rangle$ for F, F' of the bonds of $\bar{\mathsf{Q}}_2$.

(L4) Let e_c be the projection on $(\ker Z_c)^\perp$, $e_R : \hat{\xi}_c \mapsto \hat{\xi}_c e_c$ in $\mathcal{M}_\Lambda$, $e_L = J_\Lambda e_R J_\Lambda$, $\pi = e_L e_R$, and $e^{-2\pi s K_\Lambda}\hat{\xi}_c = \rho_c^{-s}\hat{\xi}_c\rho_c^s$ on $\pi\mathcal{H}_\Lambda$: $\Delta_\Lambda = e^{-2\pi K_\Lambda}$ is the modular operator of $(\mathcal{M}_\Lambda, \Omega_\Lambda)$. There is a self-adjoint involution Σ of $\pi\mathcal{H}_\Lambda$ commuting with K_Λ such that $\mathsf{P}_\Lambda = \Sigma e^{-\frac{\pi}{2}K_\Lambda}$ is self-adjoint, $|\mathsf{P}_\Lambda| = e^{-\frac{\pi}{2}K_\Lambda}$, and $\pi\iota_\Lambda F \in D(\mathsf{P}_\Lambda)$, $\mathsf{P}_\Lambda\pi\iota_\Lambda F = \pi\iota_\Lambda(RF)$ for F of $\bar{\mathsf{Q}}_1$: the quarter turn is the fourth root of the modular operator, up to a sign.

(L5) For such F and $\xi = \mathsf{F}(K_\Lambda)\pi\iota_\Lambda F$ there is a closed, densely defined T commuting with the unitaries of $\mathcal{M}'_\Lambda$, with $T\Omega_\Lambda = \xi$ and $\|T^*T\Omega_\Lambda\| \leq \pi e^{5\pi^2/8}\|\iota_\Lambda(\theta_1\bar{F}\cdot F)\| = \pi e^{5\pi^2/8}\mu(\theta\theta_1 F\cdot$

$\theta\bar F\cdot\theta_1\bar F\cdot F)^{1/2}$; so $T^{(\epsilon)}=T(1+\epsilon T^*T)^{-1}\in\mathcal{M}_\Lambda$, $\|T^{(\epsilon)}\|\le(2\sqrt{\epsilon})^{-1}$, and $\|T^{(\epsilon)}\Omega_\Lambda-\xi\|\le\frac{\sqrt{\epsilon}}{2}\|T^*T\Omega_\Lambda\|$.

No coupling enters, and no injectivity of ρ_c is assumed: gauge invariance destroys it. ■

Proof sketch. The maps $\theta,\theta_1,\mathsf{d}$ keep Λ, the Haar measure and the Wilson weight ($a_{rp}(\mathsf{r}U)=a_p(U)$, as $\operatorname{tr}g^{-1}=\operatorname{tr}g$ on SU(2)). (L1) $\mu(\theta\bar F'\cdot F)=Z^{-1}\int\overline{F'(\theta U)}F(U)w_1w_2(U)w_1w_2(\theta U)\,\mathrm{d}U$; at fixed (c,x_L,x_R), which θ fixes, the integrand is a function of the upper bonds times one of the lower bonds, and the two integrals are $Z^{1/2}\iota_\Lambda F$ and $Z^{1/2}\overline{\iota_\Lambda F'}$; θ_1 exchanges w_1,w_2 and x_L,x_R; $\mathcal{H}_\Lambda=L^2(\mathsf{C})\otimes\mathcal{K}\otimes\mathcal{K}$ and $\mathcal{M}_\Lambda=L^\infty(\mathsf{C})\otimes1\otimes\mathcal{B}(\mathcal{K})$. (L2) At $F=F_2F_1$ the integral over the bonds of $\mathsf{Q}_1\cup\mathsf{Q}_2$ factorises at fixed configuration of B_U; the substitution $U\to\theta_1U$ in the factor of Q_2 and the diagonal reflection d, which gives $(\mathsf{A}_c^F)^{\mathrm{T}}=\mathsf{A}_{\kappa c}^{\mathsf{d}F}$, yield the product formula; the kernel of Z_c is real. (L3) With $E=R^{-1}F$, $E'=R^{-1}F'$ of $\bar{\mathsf{Q}}_1$: $\langle\iota_\Lambda F',Q\iota_\Lambda F\rangle=\int\operatorname{Tr}((Z_cq_c^*Z_c^*)^*(\iota_\Lambda(\theta_1\bar E'\cdot E))^\wedge_{\kappa c})\,\mathrm{d}c$ and $\theta_1\bar E'=R^{-1}\theta\bar F'$; put $\hat{\hat q}_{\kappa c}=Z_cq_c^*Z_c^*$; Hölder gives $\|Z_cq_c^*Z_c^*\|_2\le\|q_c\|\|Z_c\|_4^2$, and $\int\|Z_c\|_4^4\,\mathrm{d}c=\|\Omega_\Lambda\|^2=1$. (L4) With the polar decomposition $Z_c=W_c\rho_c^{1/4}$, $W_{\kappa c}=W_c^*$; put $(\Sigma\xi)^\wedge_c=W_c^*\hat\xi_{\kappa c}W_c$, a symmetric involution of $\pi\mathcal{H}_\Lambda$ commuting with the modular flow; then $(\pi\iota_\Lambda F)^\wedge_c=\rho_c^{1/4}\mathsf{c}_c$ with $\mathsf{c}_c=W_c^*\mathsf{A}_c^Fe_c=e_c\mathsf{c}_ce_c$, while $(\Sigma\pi\iota_\Lambda RF)^\wedge_c=\mathsf{c}_c\rho_c^{1/4}=(e^{-\frac{\pi}{2}K_\Lambda}\pi\iota_\Lambda F)^\wedge_c$. (L5) The Schur multipliers $h_\gamma(\omega)=\sqrt{\pi}e^{-\omega^2/16\pi^2+\gamma\omega}$ of the eigenvalue differences ω of $\log\rho_c$ have the norm $\sqrt{\pi}e^{4\pi^2\gamma^2}$ on every Schatten class; $\mathsf{F}(\omega/2\pi)=h_{1/8}(\omega)$, so $\hat\xi_c=\rho_c^{1/4}h_{1/8}(\mathsf{c}_c)$, and with $\hat\xi_c^\sharp=\rho_c^{1/4}h_{1/8}(\mathsf{c}_c^*)$, $\hat\zeta_c=h_{3/8}(\mathsf{c}_c)h_{1/8}(\mathsf{c}_c^*)$,

$$\rho_c^{1/2}\hat\xi_c^*=\hat\xi_c^\sharp\rho_c^{1/2},\qquad\hat\xi_c\hat\xi_c^*=\hat\zeta_c\rho_c^{1/2}.$$

The first makes $Q'\Omega_\Lambda+\eta\mapsto Q'\xi$, $Q'\in\mathcal{M}'_\Lambda$, $\eta\in(1-e_R)\mathcal{H}_\Lambda$, closable, with a closure T that commutes with the unitaries of $\mathcal{M}'_\Lambda$; the second gives $T^*\xi=\zeta$ with $\|\hat\zeta_c\|_2\le\pi e^{5\pi^2/8}\|\mathsf{A}_c^F\|_4^2$ and $\|\mathsf{A}_c^F\|_4^4=\|(\mathsf{A}_c^F)^*\mathsf{A}_c^F\|_2^2=\|(\iota_\Lambda(\theta_1\bar F\cdot F))^\wedge_c\|_2^2$; (L1) turns $\|\iota_\Lambda(\theta_1\bar F\cdot F)\|$ into the four-point function. Von Neumann's theorem and the bicommutant theorem give $T^{(\epsilon)}\in\mathcal{M}_\Lambda$, and $T^{(\epsilon)}\Omega_\Lambda-\xi=-\epsilon T^{(\epsilon)}T^*T\Omega_\Lambda$. □

The one-lattice source recursion. We need more than the bound on the source energies on their original source ball: the two-cell estimate uses their continuation to larger source discs at the earlier levels. We give the induction that provides this continuation, retaining the dependence on the background radius. The local maps are those of Thm. YM3_I:W, Prop. YM3_I:51.16, Lem. YM3_I:114.1 and Prop. YM3_I:77.2.

Here is the notation used for those maps. On a domain Y at level j, P is its background triple, B is its fluctuation field, and $|B|<\mathsf{C}_{\mathcal{A}}\mathfrak{b}(x_j)$ is the field polydisc. For a family V vanishing at $B=0$, its norm at a fixed background is

$$\|V\|_{\mathrm{f},P}=\sup_{Y,B}\frac{e^{\lambda_V d_j(Y)}|V(Y;P,B)|}{1+q_Y(B)},\qquad q_Y(B)=M^{-3}\sum_b|B(b)|^2.$$

The norm $\mathsf{n}^{\mathrm{f}}(V)$ also takes the supremum over the specified background cores. For a pointed family E of level j, $\|E\|_{j,\kappa+1}$ is the supremum of $e^{(\kappa+1)d_j(X)}|E(X,P,z)|$ over its domains, background cores and marked sites. The local old-potential map is denoted by $\mathbb{V}'[\mathfrak{f}]$, the pointed integration map by $\mathbf{F}$, and the source-independent normalizer by $\mathbf{N}$.

For the admitted run $x_0>\cdots>x_K$ put $b_i=L^{-1}x_{i-1}-x_i$, so that $|b_i|\le B_\sharp$, and $v_j^w=-L^{K-j}w/x_j$. The recursion starts with the empty array and is

$$\mathfrak{f}_j[w]=((E^{(i)}[w],b_i)_{1\le i\le j},\emptyset),\qquad E^{(j+1)}[w]=\mathbf{N}^{(j+1)}+\mathbf{F}[\mathbb{V}'[\mathfrak{f}_j[w]]-\mathbb{W}^{\mathrm{src}}[v_j^w]].$$

The direct source family is equation YM3_I:2165:

$$\mathbb{W}^{\mathrm{src}}[v](Y;P,B)=x_j\mathbb{V}_v^{\mathrm{act}}(Y;P,x_j^{-1/2}B)-x_j\sum_b v(p_b)D_A\mathbb{V}_1^{\mathrm{act}}(Y;P,0)[x_j^{-1/2}B(b)\delta_b].$$

Here p_b is the fixed reference plaquette in the cube containing b_-, and $\mathbb{V}_v^{\mathrm{act}}$ is the localized weighted Wilson action. At level zero it is the finite plaquette sum with the pivot completion. All these maps use the fixed auxiliary data. Changing a core radius restricts their domains; it does not change their formulas.

This recursion gives the regular terms used by the source energies. Indeed, its negative direct source family is the one-lattice escort of Prop. YM3_I:78.6: their weighted action and reference subtraction are identical, with opposite signs. On the history with no large-field region every bond is an integration bond, so the pointed integration map is the same. Thus the regular recursion is equation YM3_I:1683, and equation YM3_I:1736 gives

$$e_i(X;w) = \sum_{z\in X\cap\mathbb{T}^{(i)}} \big[E^{(i)}[w](X,\mathbf{1},z) - E^{(i)}[0](X,\mathbf{1},z)\big].$$

In particular, the continuation constructed below continues the source energies in use, not a second choice of their factorization.

Lemma 14.14 (The source recursion on smaller cores)**.** Let (K,β) be admitted at g, with run $x_0 > \cdots > x_K \ge \gamma^{-2}$. Use the fixed one-lattice data above and the reference torus $T_{\ell'}$, where ℓ' is the least side in $\mathfrak{L}(g)$; in particular $\ell' \ge 2L^7$. Write $\bar B = B_\sharp L/(L-1)$ and $\delta(x) = x^{-1/2}\mathfrak{b}(x)$, $\mathfrak{b}(x) = A_1(\log x)^{\mathfrak{p}_0}$. The fixed core radii and size constants obey

$$\alpha_0+\alpha_1 \le a_\star^{\mathrm{S}}, \qquad c_{\mathrm{sh}} M\alpha_0 < \alpha_1, \qquad \alpha_1^2 \le c_\natural \alpha_0, \qquad s_\gamma := \frac{(1+\beta_{\mathrm{s}})\delta(\gamma^{-2})}{\alpha_0} \le s_\star^{\mathrm{S}},$$

$$E_0 \ge 2E_{\mathsf{N}}, \qquad N_\star^{\mathrm{s}} \ge \frac{L}{L-1}(E_0 + \tfrac{3}{2}\alpha_0^2 B_\sharp),$$

$$K_{\mathbf{M}}\big(\delta(\gamma^{-2}) + s_\gamma N_\star^{\mathrm{s}} + e^{-\gamma_2 \mathfrak{b}(\gamma^{-2})^2/2}\big) \le E_0/4, \qquad 8K_\Lambda s_\gamma N_\star^{\mathrm{s}} \le w_\star, \qquad 8K_P s_\gamma \le w_\star.$$

These are the absolute-core and source-recursion conditions, with their size bounds as in equations YM3_I:2169 and YM3_I:2170. Fix uniform bounds $K_{\mathrm{Q}}^\vee, K_{\mathrm{L}}^\vee$ for the quadratic and linear source coefficients, let

$$\mu_\star = \min\{w_\star/32, E_0/(16K_{\mathbf{M}}^{\mathrm{f}})\}, \qquad \sigma_{\mathrm{src}} = \mu_\star/K_{\mathrm{Q}}^\vee, \qquad K^{\mathrm{d}} \le K_{\mathbf{M}}^{\mathrm{f}},$$

and retain $4\beta_{\mathrm{s}}\kappa \ge 1$ and $\log\gamma^{-2} > 2\mathfrak{p}_0$. Then the following statements hold.

(i) On cores of radius $\bar a_0 \le 1$, at every level $j \ge 0$, the part of the direct source linear in B satisfies

$$|\mathbb{W}^{\mathrm{lin}}[v](Y;P,B)| \le \bar a_0 K_{\mathrm{L}}^\vee x_j^{1/2}\, \mathrm{sw}_j(v)(1+q_Y(B))e^{-\lambda_V d_j(Y)}.$$

The constants may be fixed uniformly over all levels and all the smaller cores in (ii). Consequently the one-step estimates apply when the oscillation allowance is divided by $\bar a_0$.

(ii) Fix $1 \le k \le K$, and put $\theta_k = \delta(x_{k-1})/\delta(\gamma^{-2})$. For every $\theta \in [\theta_k, 1]$ the recursion through level k is defined on the cores with radii $(\theta\alpha_0, \theta\alpha_1)$ for sources satisfying

$$|w|_\infty < \varrho_0^\star := \sigma_{\mathrm{src}}(x_K - \bar B), \qquad \frac{40}{\sqrt{3}} M \operatorname{Lip}_\infty w < \varrho_1^{[k]}(\theta) := \frac{40}{\sqrt{3}}\frac{\mu_\star}{K_{\mathrm{L}}^\vee}\frac{x_{k-1}^{1/2}}{\theta\alpha_0}.$$

It is jointly holomorphic in the source and the open background slices, local in the source, and its pointed families satisfy $\|E^{(i)}[w]\|_{i,\kappa+1} \le E_0$ for $1 \le i \le k$. At $\theta = 1$ the estimate without the factor α_0 in (i) also gives the same conclusions with second source radius $(40/\sqrt{3})(\mu_\star/K_{\mathrm{L}}^\vee)x_{k-1}^{1/2}$.

(iii) The source class at $\theta = 1$ contains the original class with radii $2\varrho, 2\varrho$. On common source classes, the families constructed at θ and at 1 agree on the smaller cores.

(iv) The pointed integration at one step is a holomorphic function of its fourth-family entry. On the admitted norm ball it is Lipschitz, with bound $K^{\mathrm{d}} e^{-\lambda_{\mathrm{out}} d}$.

(v) At the trivial background the linear source potential vanishes. The old potential vanishes to second order in the fluctuation field. If N is its weighted array size and s its absolute-core factor, then

$$|\mathbb{V}'(Y;\mathbf{1},B)| \le K_{\Lambda} s N e^{-\lambda_V d_j(Y)} \left(\frac{\sup|B|}{\mathsf{C}_{\mathcal{A}}\mathfrak{b}(x_j)} \right)^2 .$$

For the consequence used below, let $f \neq 0$ be a real Lipschitz function vanishing off the middle of a cell of side l, put $r = \varrho/\|f\|_\sharp$, and let v'^{+}, v'^{-} be its even and odd mirror sums on $T_{\ell'}$. For $1 \le k \le K$, put $s = K - k$ and

$$\lambda_s = (1 + (s+1)\log L)^{\mathfrak{p}_0},$$
$$\epsilon_s = \max\left\{ \frac{\mathsf{l}}{160M}, \min\left\{ \left(\frac{24}{11}L^{-s-1}\right)^{1/2}, \frac{24\alpha_0}{11}L^{-s-1}\lambda_s \right\} \right\},$$
$$\hat{\epsilon}_s = \max\left\{ \frac{\mathsf{l}}{160M}, \frac{24\alpha_0}{11}L^{-s-1} \right\}.$$

If X is a level-k domain that sees two cells, then $G_{k,X}(z,z') = e_k(X; zv'^{+} + z'v'^{-})$ continues holomorphically to a bidisc of radius $R_k \ge 4r/\epsilon_s$, with

$$|G_{k,X}(z,z')| \le 2M^3 E_0 e^{-\kappa d_k(X)}, \qquad |\partial_z \partial_{z'} G_{k,X}(0,0)| \le \frac{M^3 E_0 c_2}{32}\frac{\hat{\epsilon}_s^2}{r^2} e^{-\kappa d_k(X)}.$$

Here c_2 depends only on $M, \mathsf{C}_{\mathcal{A}}, A_1, \mathfrak{p}_0, \gamma$. Using the last assertion of (ii) instead gives a bidisc of radius at least $4r/\epsilon_s^{\circ}$, where

$$\epsilon_s^{\circ} = \max\left\{ \frac{\mathsf{l}}{160M}, \left(\frac{24}{11}L^{-s-1}\right)^{1/2} \right\}.$$

■

Proof. We first retain the core-radius factor in the direct source calculation. At positive level write $a = \bar{a}_0$ and $\varepsilon = \min\{\varepsilon_{\natural}, c_{\mathrm{os}} a/(2c_{\natural})\}$. The proof of Lem. YM3_I:114.1 gives, with $N = \#_{\pi_j} Y$,

$$e^{r' d_j(Y)} N \sup_{|A|<\varepsilon} |\mathbb{V}_v^{\mathrm{act}}(Y;P,A)| \le 672 M^3 a^2 |v|_Y \qquad (0 \le r' \le \bar{\kappa} - 1).$$

For its linear Taylor term Cauchy's formula divides this bound by ε. Retain $a^2/\varepsilon = a(a/\varepsilon)$, instead of replacing it by a/ε. The bond sum obeys $\sum_b |B(b)| \le (3/2)M^3 N(1 + q_Y(B))$. The path from a plaquette to its reference plaquette has at most $d_j(Y)+2$ links, and $e^{-d}(d+2) \le 2$. These estimates give (i). The quotient a/ε is bounded by its value at the original radius, so the constant is uniform on smaller cores. At level zero the potential is zero unless Y is the domain of one plaquette. On the polydisc scaled by a, each shifted plaquette angle is bounded by a fixed multiple of a. Its Wilson potential is therefore bounded by a fixed multiple of a^2; Cauchy's estimate for its linear term loses one factor a. The pivot calculation in Prop. YM3_I:51.4 gives (i) at this level as well, with one fixed $K_{\mathrm{L}}^{\vee}$.

We next verify that the smaller cores admit every required step. The first three absolute-core conditions become

$$\theta(\alpha_0 + \alpha_1) \le a_\star^{\mathrm{S}}, \qquad c_{\mathrm{sh}} M \theta \alpha_0 < \theta \alpha_1, \qquad \theta^2 \alpha_1^2 \le c_{\natural} \theta \alpha_0.$$

They follow from $0 < \theta \le 1$. For $j < k$, monotonicity of δ gives

$$\delta(x_j) \le \delta(x_{k-1}) = \theta_k \delta(\gamma^{-2}) \le \theta \delta(\gamma^{-2}), \qquad s_j^{\theta} := \frac{(1+\beta_{\mathrm{s}})\delta(x_j)}{\theta\alpha_0} \le s_\gamma.$$

This is the fourth absolute-core condition. It also preserves the field-room inequalities: the allowed Taylor radius at $\theta\alpha_0$ is at least θ times the original one. The fixed auxiliary maps restrict to these cores.

For the relative weight of a source in (ii), Prop. YM3_I:73.3 gives

$$x_j \ge L^{K-j}(x_K - \bar{B}), \qquad |v_j^w| \le \frac{|w|}{x_K - \bar{B}} < \sigma_{\mathrm{src}}, \qquad \mathrm{sw}_j(v_j^w) \le x_j^{-1} M \operatorname{Lip}_\infty w.$$

Hence

$$\theta\alpha_0 x_j^{1/2}\,\mathrm{sw}_j(v_j^w) < \frac{\mu_\star}{K_{\mathrm{L}}^\vee}(x_{k-1}/x_j)^{1/2} \le \frac{\mu_\star}{K_{\mathrm{L}}^\vee}.$$

The quadratic and higher source terms cost at most $2K_{\mathrm{Q}}^\vee|v_j^w|$, by the same half-radius Taylor estimate as in Lem. YM3_I:114.1. With (i), the full source norm is therefore less than $3\mu_\star$.

We can now carry out the induction, including its counterterms. If the earlier pointed families have norm at most E_0, their weighted array has size

$$N_j = \sum_{i=1}^{j} L^{-(j-i)}(\|E^{(i)}[w]\|_{i,\kappa} + \tfrac{3}{2}\theta^2\alpha_0^2|b_i|) \le \frac{L}{L-1}(E_0 + \tfrac{3}{2}\alpha_0^2 B_\sharp) \le N_\star^{\mathrm{s}}.$$

The old-potential rule of Prop. YM3_I:51.16 thus has norm at most $K_\Lambda s_j^\theta N_j \le w_\star/8$. Adding the source gives at most $w_\star/8 + 3w_\star/32 = 7w_\star/32 < w_\star/4$. The internal potentials have total size at most $3K_P s_j^\theta \le 3w_\star/8$. Thm. YM3_I:W bounds the unsourced output by $3E_0/4$. Prop. YM3_I:77.2 bounds the added source contribution by $3K^{\mathrm{d}}\mu_\star \le 3E_0/16$. The new pointed family therefore has norm at most $15E_0/16$. At the first step the old array is empty, so the same argument starts the induction. This is the proof of Prop. YM3_I:114.2 with the displayed counterterms and relative weights retained. Holomorphy follows from the parameter version of the pointed expansion and the linearity of the old-potential rule. Locality follows inductively because a term over X reads only earlier terms and source values inside X. At $\theta = 1$, omit the factor α_0 from (i) and repeat the same argument with the smaller source radius stated in (ii).

The source-radius conditions give, for $x \ge \gamma^{-2}$,

$$4\varrho \le \sigma_{\mathrm{src}}(x - \bar{B}), \qquad 4K_{\mathrm{L}}^\vee \varrho x^{-1/2} \le 40\mu_\star/\sqrt{3}.$$

At $x = x_K$ and $x = x_{k-1}$ these imply $\varrho_0^\star \ge 4\varrho$ and $\varrho_1^{[k]}(1) \ge 4\varrho$. The recursion contains no core radius except through its domain; induction on the level therefore identifies the constructions on their common domains. This proves (iii). Assertion (iv) is the holomorphic parameter statement of Lem. YM3_I:77.1 and the entry estimate of Prop. YM3_I:77.2, at the norm reserves just checked.

For (v), the old potential is zero at $B = 0$. Its differential there is invariant under all constant gauge transformations. Averaging a traceless matrix over these transformations gives zero, so the differential vanishes. Schwarz's lemma on the field polydisc of radius $\mathsf{C}_{\mathcal{A}}\mathfrak{b}(x_j)$ gives the stated square. The same invariance makes the linear source term zero at the trivial background.

We turn to the two-variable consequence. The two mirror sums have disjoint supports in cell interiors. Thus their source belongs to the class in (ii) at θ_k whenever $|z|, |z'| < R_k$, with

$$R_k = \min\left\{ \frac{\varrho_0^\star}{\sup|f|}, \frac{\varrho_1^{[k]}(\theta_k)}{\|f\|_\sharp} \right\}.$$

The first denominator is controlled by $\sup|f| \le \mathsf{l}\operatorname{Lip} f/4 = \mathsf{l}\|f\|_\sharp/(160M)$. For the second, the run and $\bar{B} \le (13/24)\gamma^{-2}$ give

$$x_{k-1} \ge L^{s+1}(x_K - \bar{B}) \ge (11/24)L^{s+1}\gamma^{-2}, \qquad \frac{40}{\sqrt{3}}\frac{\mu_\star}{K_{\mathrm{L}}^\vee} \ge 4\varrho\gamma.$$

Using first $\theta_k\alpha_0 \le 1$ and then the formula for θ_k gives respectively

$$\frac{\gamma x_{k-1}^{1/2}}{\theta_k\alpha_0} \ge \left(\frac{11}{24}L^{s+1}\right)^{1/2}, \qquad \frac{\gamma x_{k-1}^{1/2}}{\theta_k\alpha_0} = \frac{\gamma^2 x_{k-1}}{\alpha_0}\frac{\mathfrak{b}(\gamma^{-2})}{\mathfrak{b}(x_{k-1})} \ge \frac{11}{24\alpha_0}L^{s+1}\lambda_s^{-1}.$$

For the last inequality use the decrease of $x^{-1}(\log x)^{\mathfrak{p}_0}$ and $\log((11/24)L^{s+1}\gamma^{-2})/\log\gamma^{-2} \le 1+(s+1)\log L$. These are exactly the bounds $R_k \ge 4r/\epsilon_s$. At $\theta=1$, using the last class in (ii) gives $R_k \ge 4r/\epsilon_s^\circ$. Summing the pointed terms at the trivial background, and subtracting their values at source zero, gives $|G_{k,X}| \le 2M^3E_0e^{-\kappa d_k(X)}$.

It remains to improve the mixed coefficient. Put $x=x_{k-1}$, $b_x=\mathfrak{b}(x)$ and $R=R_k$. Use the fourth-family norm at the trivial background, with its actual denominator $1+q_Y(B)$. Assertion (v) gives

$$\|V_{\mathrm{old}}(z,z')\|_{\mathrm{f}} \le C\mathsf{A}b_x^{-2}, \qquad \|(V_{\mathrm{old}})_z\|_{\mathrm{f}} \le C\mathsf{A}b_x^{-2}R^{-1}, \qquad \|(V_{\mathrm{old}})_{zz'}\|_{\mathrm{f}} \le C\mathsf{A}b_x^{-2}R^{-2},$$

where A is the earlier-family majorant from the induction; C here depends on M and $\mathsf{C}_{\mathcal{A}}$. The direct source is linear in (z,z'), so the last entry is

$$V(z,z') = V_{\mathrm{old}}(z,z') + zW_+ + z'W_-.$$

Its part linear in B vanishes at the trivial background, and the quadratic and higher direct-source estimates give $\|W_\pm\|_{\mathrm{f}} \le C\mu_\star \mathsf{l}/r$. Here one uses $L^{s+1}/x \le (x_K-\bar B)^{-1}$, $4\varrho \le \sigma_{\mathrm{src}}(x_K-\bar B)$ and $\sup|f| \le \mathsf{l}\varrho/(160Mr)$. The direct source has zero mixed entry derivative; its first derivatives do not receive a factor b_x^{-2}.

Let $\mathfrak{F}_X$ denote the one-entry map with its source-independent terms omitted. At $V_0=V(0,0)$ the mixed coefficient is

$$D\mathfrak{F}_X(V_0)[(V_{\mathrm{old}})_{zz'}] + D^2\mathfrak{F}_X(V_0)[(V_{\mathrm{old}})_z+W_+,(V_{\mathrm{old}})_{z'}+W_-].$$

The bounds $\mathsf{A}\le w_\star/8$, $\mu_\star \le w_\star/32$, $K^{\mathrm{d}}\mathsf{A} \le 13E_0/72$ and $K^{\mathrm{d}}\mu_\star \le E_0/16$ leave a fixed norm reserve. The Lipschitz bound in (iv) and Cauchy's estimate in that reserve bound the first and second derivatives by $CK^{\mathrm{d}}e^{-\lambda_{\mathrm{out}}d_k(X)}$ and $CK^{\mathrm{d}}w_\star^{-1}e^{-\lambda_{\mathrm{out}}d_k(X)}$. Bounding the cross products by the two squares therefore gives

$$|\partial_z\partial_{z'}G_{k,X}(0,0)| \le CM^3E_0\big[b_x^{-2}R^{-2}+\mathsf{l}^2/r^2\big]e^{-\kappa d_k(X)}.$$

Finally λ_s/b_x is bounded in terms of $A_1,\mathfrak{p}_0,\gamma$, since $x \ge (11/24)L^{s+1}\gamma^{-2}$ and $\log\gamma^{-2} > 2\mathfrak{p}_0$. Together with $R\ge 4r/\epsilon_s$, this bounds the bracket by $C\hat\epsilon_s^2/r^2$. Choosing c_2 as in the statement proves the mixed-coefficient estimate. □

Lemma 14.15 (Cell bounds)**.** Let (K,β) be admitted at g, (i,o) a board of $L^{-K}\mathbb{Z}^3$ with cells of side $\mathsf{l}=L^{-i}$, $1\le i\le K-1$, f real Lipschitz, $f\neq 0$, vanishing off the middle of one cell, $r=\varrho/\|f\|_\sharp$ and $t=|z|/r$. Let $c_V=E_+ + C_{\mathrm{low}}+C_{\mathrm{gap}}$ be the constants of the systems of localised source energies that the pairs admitted at g carry on the sides of the family $\mathfrak{L}(g)$ of Lem. 14.14 (Thm. YM3_I:AD), and $\bar\Phi = \sup_{0<\mathsf{l}\le 1}\Phi_1(\frac{\mathsf{l}}{4},\frac{\mathsf{l}}{4};\mathsf{l})$ the supremum in l of the second-order constant of the one-cell bound, Φ_1 the function of equation YM3_I:(1786) at the constants of these systems. There are numbers c_V', formed from (σ^-,γ) and g, and Φ', $\Phi^{(1)}$, formed from (σ^-,γ), none of them depending on the cutoff, l, the cell, f or the volume, and holomorphic normalisers $\mathcal{E}^\flat,\mathcal{E}^\natural$, formed on the reference torus $T_{\ell'}$ of Lem. 14.14 from its system, the board, the cell and f, and free of the side of the torus on which the seminorm is taken, such that on every torus T_ℓ, $\ell\in\mathfrak{L}(g)$, $\ell\ge 2\mathsf{l}$:

(a) $\|e^{\mathcal{O}_K(zf)}\|_{\mathrm{ch}} \le |e^{\mathcal{E}^\flat(z)}|\exp[c_V\mathsf{l}^3+\bar\Phi\,t^2/(1-t)]$ for $|z|<r$;

(a′) $\|e^{\mathcal{O}_K(zf)}\|_{\mathrm{ch}} \le |e^{\mathcal{E}^\natural(z)}|\exp[c_V\mathsf{l}^3+\Phi^{(1)}\mathsf{l}\,t^2]$ for $|z|\le r$, using the last source class in Lem. 14.14(ii);

(b) $\|e^{\mathcal{O}_K(zf)}\|_{\mathrm{ch}} \le |e^{\mathcal{E}^\natural(z)}|\exp[c_V'\mathsf{l}^3+\Phi'(|z|\|f\|_{\sharp,\mathsf{l}}/\varrho)^2]$ for $|z|\le r$;

(c) for every σ' as in Thm. AQ, $i\ge 1$ and real $h\in C^1_c(\mathbb{R}^2)$ carried by a square of side 2: $\|\mathsf{I}_{\mathsf{l}}(h)\| \le \mathsf{C}_3\mathsf{l}^{-3}|h|_{C^1}$ and $\|\mathsf{I}_{\mathsf{l}}(h)\| \le \mathsf{C}_2\mathsf{l}^{-2}|h|_{C^1}$, with the numbers of Thm. AQ(a), (b); from (a′), $\|\mathsf{I}_{\mathsf{l}}(h)\| \le C\mathsf{l}^{-5/2}|h|_{C^1}$;

(d) for σ' a limit of periodic Gibbs states ν_j, $i\ge 1$, a board (i,o) with $o\in L^{-i-2}\mathbb{Z}^3$ and $o_3=0$, a finite set $\mathcal{K}\subset\mathbb{Z}^2\times\{0\}$, so that the cells $\square_{\mathbf{k}}$, $\mathbf{k}\in\mathcal{K}$, lie in the layer $\{|x_3|\le\frac12\}$, real $f_{\mathbf{k}}\in$

$C^1_c(\mathbb{R}^3)$, $f_{\mathbf{k}} \neq 0$, vanishing off the middles of the cells $\square_{\mathbf{k}}$, $r_{\mathbf{k}} = \varrho/\|f_{\mathbf{k}}\|_\sharp$, $t_{\mathbf{k}} = |z_{\mathbf{k}}|/r_{\mathbf{k}}$, and $D_j[z] = \prod_{\mathbf{k}} \exp(z_{\mathbf{k}} \mathcal{O}_{K_j}(f_{\mathbf{k}}) - \mathcal{E}^\flat_{\mathbf{k},j}(z_{\mathbf{k}}))$: along a subsequence $\nu_j \overline{(\theta(\tau_{\frac{1}{2}e_3} G')_j} D_j[z](\tau_{\frac{1}{2}e_3} G)_j) \to \langle [G'], \mathbb{D}[z][G] \rangle$ locally uniformly, with bounded operators $\mathbb{D}[z]$, $|z_{\mathbf{k}}| < r_{\mathbf{k}}$, holomorphic in norm, $\|\mathbb{D}[z]\| \leq \exp[c_V \# \mathcal{K} \mathsf{l}^3 + \bar\Phi \sum_{\mathbf{k}} t^2_{\mathbf{k}}/(1-t_{\mathbf{k}})]$, $\mathbb{D}[0] = e^{-\mathsf{l}H}$ and $\partial_{z_{\mathbf{k}}} \mathbb{D}[0] = \mathsf{B}^{\mathrm{in}}_{\mathsf{l}/2}(f_{\mathbf{k}}) + c_{\mathbf{k}} e^{-\mathsf{l}H}$; for one cell and f even under the reflection at the middle plane of the cell, $\mathbb{D}(x)$ is at real x a number times a positive operator, and $\mathbb{D}(x)/\langle \Omega, \mathbb{D}(x)\Omega \rangle \geq 0$ where $\langle \Omega, \mathbb{D}(x)\Omega \rangle \neq 0$.

For a component σ'_ξ, with its lattice approximants $\rho_{j,\xi}$ (recalled in the results used) as the states that integrate, the family is had for one cell: bounded operators $\mathbb{D}(z)$ on $\mathcal{H}(\sigma'_\xi)$, limits along a subsequence, holomorphic on $|z| < \varrho/\|f\|_\sharp$, with $\|\mathbb{D}(z)\| \leq \bar{\mathsf{Q}}$, $\bar{\mathsf{Q}}$ the number of Thm. AQ(e), formed from the cell bound in Prop. YM3_I:89.1, as in the proof of Thm. AQ(e), and free of l, f and ξ, with $\mathbb{D}(0) = e^{-\mathsf{l}H}$, $\mathbb{D}'(0)$ the insertion of f plus a multiple of $e^{-\mathsf{l}H}$, and with the same positivity at real x for even f; and $\|\mathsf{B}^{\mathrm{in},\xi}_\eta(h)\| \leq \|\mathsf{B}^{\mathrm{in}}_\eta(h)\|$ for every insertion.

(e) *Reflections for many cells of one layer.* Let ν be a law on $L^{-K}\mathbb{Z}^3$ invariant under lattice isometries and reflection positive at every coordinate bond mirror. Fix a sequence $\ell_\bullet$ of sides tending to ∞ and $\mathfrak{C} \in \{\mathcal{F}_{\mathrm{loc}}, \mathcal{C}_{\mathrm{loc}}\}$, the bounded Borel or continuous local functions, respectively. Assume also chessboard domination: on every board and for every finite family of cell functions $F_{\mathbf{k}} \in \mathfrak{C}$,

$$\left| \nu \left(\prod_{\mathbf{k}} F_{\mathbf{k}} \right) \right| \leq \liminf_{m \to \infty} \prod_{\mathbf{k}} \|F_{\mathbf{k}}\|_{\mathrm{ch},\ell_m}.$$

These are the three properties needed below, as every periodic Gibbs state of the Wilson weight has them (Prop. YM3_I:103.4, Prop. YM3_I:91.9); let $\{x_3 = c\}$ be a wall plane of the board, $\mathcal{K} \subset \mathbb{Z}^3$ finite and non-empty with $\square_{\mathbf{k}} \subset \{c \leq x_3 \leq c + \mathsf{l}\}$ for $\mathbf{k} \in \mathcal{K}$, $D_{\mathbf{k}} \in \mathcal{A}(\square_{\mathbf{k}}) \cap \mathfrak{C}$ functions of the bonds of the cells, and $\mathsf{N} \geq 0$ with $\prod_{\mathbf{k} \in \mathcal{K}} \|D_{\mathbf{k}}\|_{\mathrm{ch},\ell_m} \leq \mathsf{N}$ for all large m. Then for all $F, F' \in \mathfrak{C}(\mathbb{H}^+_c)$, functions of the bonds of $\mathbb{H}^+_c = \{x_3 \geq c\}$,

$$\left| \nu \Big(\overline{\theta_c F'} \cdot \prod_{\mathbf{k} \in \mathcal{K}} D_{\mathbf{k}} \cdot \tau_{\mathsf{l}e_3} F \Big) \right| \leq \mathsf{N} \, \|F'\|_{\nu,c} \|F\|_{\nu,c}, \qquad \|F\|^2_{\nu,c} = \nu(\overline{\theta_c F}\, F).$$

For one cell the proof of Prop. YM3_I:104.2 uses exactly these three properties and gives this estimate.

■

Proof sketch. (a) By the root identity of equation YM3_I:(1759), the sorting identity and the two counting estimates in the proof of the cell bound, Prop. YM3_I:89.1 among the construction results, the source energies that see two cells, $\mathscr{T}(z, z') = (\mathsf{l}/\ell')^3 \sum_j \sum_X e_j(X; z v'^+ + z' v'^-)$, the sum over the levels $0 \leq j \leq K$ and the domains X of the level j on $T_{\ell'}$ that see at least two cells, with ℓ', $T_{\ell'}$, e_j and the mirror sums $v'^\pm$ of Lem. 14.14, are holomorphic on $|z| + |z'| < 2r$ and bounded by $\bar\Phi$; their part linear in z goes into the normaliser, $\mathcal{E}^\flat(z) = \mathcal{E}^\circ(z) + \mathsf{a} z$, $\mathcal{E}^\circ$ the normaliser of the cell bound (Prop. YM3_I:89.1), which is the normaliser $\mathcal{E}$ of Thm. AQ(e), and Cauchy's estimate bounds the rest of $\mathrm{Re}\, \mathscr{T}(z, \bar z)$ by $\bar\Phi \sum_{k \geq 2} t^k$. $\mathscr{T}$, $\mathcal{E}^\circ$, a and the normalisers are formed on $T_{\ell'}$ and read no side ℓ; the root identity carries the bound from $T_{\ell'}$ to every T_ℓ, $\ell \in \mathfrak{L}(g)$, $\ell \geq 2\mathsf{l}$, the difference of the two energy densities being at most C_{gap}. (a′), (b) The normaliser $\mathcal{E}^\natural = \mathcal{E}^\circ + \mathscr{T}(z, 0) + \overline{\mathscr{T}(0, \bar z)}$ absorbs all unmixed terms, and only the mixed part $\mathscr{T}^{\mathrm{mx}}$ counts.

The two counting estimates for domains that see two cells, in the proof of Prop. YM3_I:89.1, give at the level $k = K - s$, with $u_s = \mathsf{l} L^s/M$, $(\mathsf{l}/\ell')^3 \sum_X e^{-\kappa d_k(X)} \leq S(\kappa)\, m_s$, the sum over the domains X of the level k that see at least two cells and on which the mixed part of $G_{k,X}$ is not 0, with $m_s \leq \min\{u_s^3, 27(\frac{u_s}{2} + 2)^3 e^{-\delta_e [u_s/2 - 4]_+}\}$; here $\delta_e = \kappa - \kappa_{\mathcal{D}} > 0$ is the rate that the count of the

domains leaves (Section YM3_I:89, at $\kappa_e = \kappa$), and $\sum_s m_s \le \sigma_\Phi$, σ_Φ the number, formed from L, M and κ, that the sum of the two counting bounds in that proof gives at $R = D = \frac{1}{4}$.

Cauchy's estimate on the bidiscs of Lem. 14.14, summed over the levels with these counts, gives $|\mathscr{T}^{\mathrm{mx}}(z,\bar z)| \le \Phi^{(1)}\mathsf{l}\, t^2$ from the radius $4r/\epsilon_s^\circ$, using the final assertion of Lem. 14.14(ii), whose source radius grows with the depth, and $|\mathscr{T}^{\mathrm{mx}}(z,\bar z) - \tau_{11}|z|^2| \le \Phi^{(3)}\mathsf{l}^3\lambda(\mathsf{l})^4 t^3$, $\lambda(\mathsf{l}) = (1+\log\frac{LM}{\mathsf{l}})^{\mathfrak{p}_0}$, from the radius $4r/\epsilon_s$, that is, from (i)–(iii); the coefficient $\tau_{11} = \partial_z\partial_{z'}\mathscr{T}(0,0)$ obeys $|\tau_{11}| \le \Phi^{(4)}\mathsf{l}^2 r^{-2}$ by the last bound of Lem. 14.14; and $\mathsf{l}\, t = |z|\|f\|_{\sharp,\mathsf{l}}/\varrho$ for f in the middle of a cell, where $\sup|f| \le \frac{\mathsf{l}}{4}\mathrm{Lip} f$.

So $\Phi' = \Phi^{(4)} + \Phi^{(3)}\sup_{\mathsf{l}\le 1}\mathsf{l}\lambda(\mathsf{l})^4$. (c) Tents of side l/L^2 cut the plate function $\rho_{\mathsf{l}}(x_3)h(\vec x)$ into pieces $f_{\mu,k}$, sorted into L^4 classes μ so that the pieces of one class lie in the middles of distinct cells of one layer; $\|f_{\mu,k}\|_\sharp \le F = 40M(40+8L^2)\mathsf{l}^{-2}|h|_{C^1}$ and at most $n \le 8\mathsf{l}^{-2}$ pieces in a class. For one class the reflections for many cells of one layer, (e), bound the expectation of the product of the exponentials between reflected vectors by the product of their chessboard seminorms; with (a), Cauchy's estimate at the radius $R = \varrho/2S_\mu$, $S_\mu^2 = \sum_k \|f_{\mu,k}\|_\sharp^2$, gives the insertion bound $\frac{4}{\varrho}e^{c_V n\mathsf{l}^3 + \bar\Phi/2}S_\mu$: n cells of a layer cost $\sqrt{n}$, not n; summing the classes gives $\mathsf{C}_3\mathsf{l}^{-3}$.

With (b), Cauchy at $R = \min\{\varrho/2\mathsf{l}S_\mu, \min_k \varrho/\|f_{\mu,k}\|_\sharp\}$, where the quadratic terms add up to $\le \frac{1}{4}\Phi'$ and $R^{-1} \le (2\mathsf{l}S_\mu + F)/\varrho \le 2(\sqrt{8}+1)F/\varrho$, gives $\mathsf{C}_2\mathsf{l}^{-2}$. With (a′), Cauchy at $R = \min\{\varrho/(S_\mu(\Phi^{(1)}\mathsf{l})^{1/2}), \min_k \varrho/\|f_{\mu,k}\|_\sharp\}$, where the quadratic terms add up to ≤ 1 and $R^{-1} \le ((8\Phi^{(1)})^{1/2}\mathsf{l}^{-1/2} + 1)F/\varrho$, gives the exponent $\frac{5}{2}$. The cutoff is removed with bounds uniform in K, through the lemma on the smeared insertion, Lem. YM3_I:107.1. (d) The reflections (e) with (a) for each cell, Montel's theorem in $\#\mathcal{K}$ variables and a diagonal choice over a countable set of combinations with dense classes give the limit; at $z = 0$ the semigroup, and the derivative is the insertion plus a number times $e^{-\mathsf{l}H}$.

For real x and f even under the reflection at the middle plane, which passes through sites, $e^{x\mathcal{O}(f)}$ is a function of the mirror bonds, positive, times a product of a function of one side and its mirror image, so reflection positivity makes the lattice operators the numbers $e^{-\mathcal{E}_j^\flat(x)}$ times positive operators; as $\mathcal{E}_j^\flat = \mathcal{E}_j^\circ + \mathsf{a}_j z$ with complex a_j, these numbers need not be positive, but along a further subsequence their phases converge, so $\mathbb{D}(x)$ is a number times a positive operator, and the division by $\langle\Omega, \mathbb{D}(x)\Omega\rangle$ removes the number. For a component the family of one cell is formed in the same way with its lattice approximants, the one-cell bound among the construction results giving the bound $\bar{\mathsf{Q}}$, free of l, f and ξ, as in the proof of Thm. AQ(e): the approximants $\rho_{j,\xi}$ are periodic Gibbs states at the couplings of the sequence (Thm. YM3_I:AP), so they have the three conditions of (e) and the bounds of the cell bound, and their moments, centred by the ρ_j, converge to those of σ'_ξ along the full sequence; so the reflections, Montel's theorem, the diagonal choice and the positivity at the middle plane run on $\mathcal{H}(\sigma'_\xi)$ as on $\mathcal{H}(\sigma')$.

For a component,

$$\|M_g \mathsf{B}_\eta^{\mathrm{in}}(h)[G]\|^2 = \int g^2 \|[X]_\xi\|^2 \,\mathrm{d}\lambda$$

with X the tuple of the insertion (Lem. 14.7(i)), and full support gives the bound at every ξ. (e) The multiple-reflection proof of Prop. YM3_I:104.2, with $D = \prod_{\mathbf{k}\in\mathcal{K}} D_{\mathbf{k}}$ in the place of its function of one cell. All the cells lie in the one layer $c \le x_3 \le c + \mathsf{l}$, and the mirror θ' at its middle plane keeps every cell; so D is a function of $\mathfrak{C}(\mathbb{H}_c^+)$,

$$D^\sharp = \overline{\theta' D} = \prod_{\mathbf{k}} \overline{\theta' D_{\mathbf{k}}},$$

and the operator $\hat D X = D \cdot \tau_{\mathsf{l}e_3} X$ has the adjoint $\widehat{D^\sharp}$ for the form of the mirror, which, with the convexity of the moments $\mathsf{s}_n = \langle F, (\widehat{D^\sharp}\hat D)^n F\rangle_{\nu,c}$, holds as in that proof.

The moment s_n is the mean of a product over $2n$ layers of the translates of the $D_{\mathbf{k}}$ and of the $D_{\mathbf{k}}^\sharp$, each with the chessboard seminorm of $D_{\mathbf{k}}$. The coarser board is chosen so that one of its

cells holds the bonds of F and all the cells $\square_{\mathbf{k}}$, $\mathbf{k} \in \mathcal{K}$; the chessboard domination on it, with the estimate from fine to coarse cells (Lem. YM3_I:104.1), the function 1 standing in the other fine cells, gives $\mathsf{s}_n \le (\sup|F|)^2 \mathsf{N}^{2n}$; the n-th root as $n \to \infty$ and the Schwarz inequality of the form give the claim. □

Lemma 14.16 (Gluing)**.** Let $G = \mathrm{SU}(2)$ and (Σ, B) a finite graph. (a) Let $Y_1, Y_2 \subset \mathsf{B}$, $I = Y_1 \cap Y_2$, $\mathsf{R}_2 = Y_2 \setminus I$, every site of Y_1 on R_2 a site of I, and every connected component of I with a cycle. Then the gauge-invariant operators of Y_1 and of Y_2 generate those gauge-invariant operators of $Y_1 \cup Y_2$ that commute with the gauge transformations of R_2 by central elements that are constant on each component of I and 1 off I; for connected I, all of them. For two plaquettes with exactly one common bond (seven bonds, six sites), the gauge-invariant operators of the two plaquettes generate all those of their union. (b) Let connected subgraphs Γ_i (boxes) cover the graph, such that at every site x any two boxes at x are joined by a chain of boxes at x in which consecutive boxes meet in a component at x with two independent cycles.

Let Z be the sign fields c with $c(C') = 1$ for every closed path C' in one box, B the coboundaries and $\mathsf{H} = Z/B$. On an open set X_0 of full Haar measure, invariant under gauge transformations and sign fields: for $V \in X_0$, a configuration V' looks as V in every box up to the gauge transformations of the box if and only if $V' = g \cdot (cV)$ with a gauge transformation g and $c \in Z$; the Borel sets invariant under gauge transformations and under Z agree on X_0 with the sets of the σ-algebra $\mathcal{S}_{\mathrm{add}}$ that the boxes generate; and for every character κ of H there is a closed path C_κ with $c(C_\kappa) = \kappa([c])$, such that every bounded gauge-invariant Borel F is $\sum_\kappa \Phi_\kappa s_\kappa$ on a set of full Haar measure in $\mathcal{S}_{\mathrm{add}}$, $s_\kappa = \operatorname{sign} \operatorname{tr} V(C_\kappa)$, Φ_κ $\mathcal{S}_{\mathrm{add}}$-measurable with $\sup|\Phi_\kappa| \le \sup|F|$. (c) For a cyclic chain of boxes $\Gamma_1, \dots, \Gamma_n$ with components $\Delta_i \subset \Gamma_i \cap \Gamma_{i+1}$ that hold two independent cycles, frames $\mathfrak{f}_i(V) \in \mathrm{SO}(3)$ formed from two holonomies of Δ_i at a site y_i, and paths C_i in Γ_i from y_{i-1} to y_i: the entries of $T_i = \mathfrak{f}_{i-1}^{\mathrm{T}} \mathrm{Ad}(V(C_i)) \mathfrak{f}_i$ are invariant under gauge transformations and under Z, measurable for Γ_i and of modulus ≤ 1, and $\chi_1(V(C_1 \cdots C_n)) = \operatorname{tr}(T_1 \cdots T_n)$ on X_0. ■

Proof sketch. (a) Write $\mathfrak{F}(Y)$ for the bounded operators of the bonds of Y, acting on $L^2(G^Y)$ and tensored with 1, $\Gamma_Y(g)$ for the gauge transformation by g on the bonds of Y alone, Γ for Γ_{B}, and $\Sigma_* = \Sigma(Y_1) \cap \Sigma(\mathsf{R}_2)$ for the common sites, which lie in $\Sigma(I)$ by hypothesis. *Fullness.* A spanning tree of a connected set J of bonds with $n \ge 1$ independent cycles fixes the gauge at all but one site, so the gauge action Γ_J on $L^2(G^J)$ is induced from the conjugation action of G on $L^2(G^n)$; by Peter–Weyl every irreducible representation of the gauge group of the sites of J with trivial total central character occurs in it, since every irreducible representation of SO(3) occurs in the conjugation action of SU(2) on $L^2(\mathrm{SU}(2))$: J is full.

Contained in. $\mathfrak{F}(Y_1)$ commutes with $\Gamma_{\mathsf{R}_2}(\zeta)$; on $\mathfrak{F}(Y_2)$ the adjoint action of $\Gamma_{\mathsf{R}_2}(\zeta)$ is that of $\Gamma(\zeta)$, because $\Gamma_I(\zeta) = 1$, both ends of a bond of I lying in one component, on which ζ is a constant central element; so the invariant operators of Y_2 commute with $\Gamma_{\mathsf{R}_2}(\zeta)$. *Contains,* in three steps. (1) The bonds of $Y_1 \cup Y_2$ are those of Y_1 and of R_2, so its operators are weak limits of sums of products AC, $A \in \mathfrak{F}(Y_1)$, $C \in \mathfrak{F}(\mathsf{R}_2)$; averaging over the gauge group and over the $\Gamma_{\mathsf{R}_2}(\zeta)$, and writing C as a weak limit of sums of members of multiplets, the right member is the weakly closed span of the sums $\sum_k A_k D_k^*$, where (A_k) in $\mathfrak{F}(Y_1)$ and (D_k) in $\mathfrak{F}(\mathsf{R}_2)$ are multiplets of one irreducible representation σ of G^{Σ_*}, invariant under the gauge transformations at the other sites, σ of trivial central character on the sites of each component of I, the mean over the ζ removing the other σ. (2) The common sites lie in $\Sigma(I)$, so the gauge group of Σ_* acts on the bonds of I (here the hypothesis on the common sites is read), and each component of I is full; so there are multiplets $\chi^{(\mathsf{r})}$ of σ in $\mathfrak{F}(I)$, invariant at the sites of I off Σ_*, with $\sum_{\mathsf{r}} \chi_k^{(\mathsf{r})*} \chi_{k'}^{(\mathsf{r})} = \delta_{kk'}$, the partial sums bounded by 1: a multiplet of σ in $\mathfrak{F}(I)$ is an intertwiner from $V_\sigma \otimes L^2(G^I)$ to $L^2(G^I)$, every type in $\sigma \otimes \Gamma_I$ has trivial central character on each component of I and so occurs in Γ_I, and partial isometries between the copies of each type give the $\chi^{(\mathsf{r})}$. (3) Then $\sum_k A_k D_k^* = \sum_{\mathsf{r}} a^{(\mathsf{r})} b^{(\mathsf{r})}$ with $a^{(\mathsf{r})} = \sum_k A_k \chi_k^{(\mathsf{r})*}$ and $b^{(\mathsf{r})} = \sum_k \chi_k^{(\mathsf{r})} D_k^*$; $a^{(\mathsf{r})}$ is

gauge invariant of Y_1, by the unitarity of σ, and $b^{(\mathsf{r})}$ is gauge invariant of Y_2, because the sites of I off Σ_* are no sites of R_2 and the sites of R_2 off Σ_* no sites of I, $\Sigma(I)$ lying in $\Sigma(Y_1)$.

For connected I every σ that occurs is trivial on the constant central elements of Σ_*, a constant central gauge transformation of all the sites of R_2 acting trivially on $\mathfrak{F}(\mathsf{R}_2)$; so the condition of commutation is void. *The two plaquettes.* On the space of the gauge-invariant vectors of a periodic layer with $N_1, N_2 \geq 4$ the invariant operators of two plaquettes with one common bond generate those of their seven bonds, by the argument with the full algebras of single plaquettes as cells that the last paragraph of the proof of Thm. AS(b), on two plaquettes with one common bond, gives on $L^2(G^7)$ itself (spin labels, Lem. 14.19, one plaquette made large, a monotone path, and the identity for the cross term at the ends of the common bond), every closed sign field of the seven bonds being a coboundary.

This is a statement on the abstract graph of the seven bonds: if the complement of a set Y of bonds of a finite graph is connected, holds a cycle and meets every site, then $X \mapsto XP$, P the projection on the gauge-invariant vectors, is injective on the invariant operators of Y, because such an operator is a sum of blocks $1 \otimes \mathcal{B}(\mathsf{K}_\tau)$ over the types τ of Γ_Y, all of trivial total central character, and the block of τ is killed by P only if the conjugate type does not occur in the gauge action on the complement, which, full and meeting every site, holds it. In a periodic layer with $N_1, N_2 \geq 4$ the complement of two plaquettes with a common bond is connected, holds cycles and meets each of their six sites. (b) Two elements of SU(2) that do not commute have the centraliser $\{\pm 1\}$: in a component with two independent cycles the two holonomies at a site fix the gauge transformation there up to the centre, and the chains carry this along; the quotients by the compact gauge groups are standard Borel spaces, and the sign of the trace of a loop that detects a character separates the classes. (c) Under a gauge transformation the frames turn by $\mathrm{Ad}(g(y_i))$, which keeps scalar and vector products; sign fields do not change Ad; and $\chi_1(V(C)) = \operatorname{tr} \mathrm{Ad}(V(C))$. □

Lemma 14.17 (Sharp loops and flips on one lattice)**.** On a torus $\Lambda_{K,\ell}$ at the coupling β, with $N = \ell L^K$ sites on a side and $\langle\cdot\rangle$ the Wilson measure: (a) for a rectangular path γ with sides of at least three and at most $N-2$ bonds, γ' its bonds but the two at each corner, $n' = \#\gamma'$, and bounded Borel F of the other bonds, $|\langle \frac{1}{2}\operatorname{tr} U(\gamma) F\rangle| \leq (I_2(4\beta)/I_1(4\beta))^{n'} \sup|F|$; (b) on a cubic torus at $\beta \geq 1$, $\langle a_p\rangle \leq (21 + 3\log\beta)/\beta$ for every plaquette; (c) for a state ω, a bounded V and a projection Q_0 with $Q_0 V Q_0 = 0$, and bounded X, Y commuting with Q_0: $|\omega(XVY)| \leq 2\|X\|\|V\|\|Y\|\omega(1-Q_0)^{1/2}$; for the flip V_ζ of a dual path ζ of a layer with an end plaquette p_0 and $Q_0 = M_{\mathbf{1}\{a_{p_0}<1\}}$, $Q_0 \mathsf{V}_\zeta Q_0 = 0$ and $\omega(1-Q_0) \leq \omega(M_{a_{p_0}})$. ■

Proof sketch. (a) Two bonds of γ' lie in no common plaquette: collinear neighbours share none; bonds of two adjacent sides could share one only at a corner, and the corners are removed; and two parallel bonds lie in a common plaquette only at the distance of one bond, while two opposite sides of γ are at least three bonds apart in the plane of γ and, as each side has at most $N-2$ bonds, at least two bonds apart also across the seam of the torus. Given all other bonds, the weight factorises over the bonds of γ', and $\frac{1}{2}\operatorname{tr} U(\gamma)$ is linear in each of them. For a bond b the sum A_b of the four staples of its plaquettes is $a\,g$ with $0 \leq a \leq 4$ and $g \in \mathrm{SU}(2)$, and by the character expansion $\int U e^{\frac{\beta}{2}\operatorname{tr}(UA_b)}\,\mathrm{d}U / \int e^{\frac{\beta}{2}\operatorname{tr}(UA_b)}\,\mathrm{d}U = \frac{I_2(\beta a)}{I_1(\beta a)} g^{-1}$, a ratio that increases with its argument; the product over γ' gives the bound. (b) $\log Z(\beta)$ is convex with $Z(0) = 1$, so $\langle \sum_p a_p\rangle_\beta \leq -\log Z(\beta)/\beta$; restricting the integral to $|U_b - 1| \leq \epsilon$ at all bonds, where $a_p = \frac{1}{2}|\partial U(p) - 1|^2 \leq 8\epsilon^2$ and the Haar measure of $\{|U-1| \leq \epsilon\}$ is $\geq 8\epsilon^3/3\pi^3$, and using that a cubic torus has as many bonds as plaquettes, $\langle a_p\rangle \leq [\log(3\pi^3/8) + 3\log\epsilon^{-1}]/\beta + 8\epsilon^2$; at $\epsilon^2 = 1/8\beta$ this is $\leq (7 + \frac{3}{2}\log\beta)/\beta$. (c) $XVY = X(1-Q_0)VY + XQ_0V(1-Q_0)Y$; the Schwarz inequality for ω, with $X(1-Q_0)X^* \leq \|X\|^2(1-Q_0)$ and $Y^*(1-Q_0)Y \leq \|Y\|^2(1-Q_0)$, bounds each term by $\|X\|\|V\|\|Y\|\omega(1-Q_0)^{1/2}$. V_ζ reverses t_{p_0}, the end plaquette having exactly one crossed bond, so it carries $\{a_{p_0} < 1\}$ onto $\{a_{p_0} > 1\}$; and $\mathbf{1}\{a_{p_0} \geq 1\} \leq a_{p_0}$. □

Lemma 14.18 (Haar-null sets under the block map)**.** Let $0 \le k < K$ and let $\mathcal{M}$ be the block-averaging map of one step from the configurations of the fine lattice $\mathbb{T}^{(k)}$ to those of the coarse lattice $\mathbb{T}^{(k+1)}$ (equations YM3_I:(37)–(39)). For every Haar-null Borel set N of coarse configurations, $\mathcal{M}^{-1}(N)$ is a Haar-null Borel set of fine configurations; so the same holds for every iterate $\mathcal{M}^n$. Hence, in the setting of Thm. AS(g), the laws μ_d^K of the block fields, $K \ge \max\{d, 1\}$, are absolutely continuous for the Haar measure h_d, and so are the limit laws μ_d. ■

Proof. The positively oriented fine bonds are the disjoint union of the bonds with both ends in one block and of the sets X_c, c a positively oriented coarse bond, where X_c is made of the L^2 fine bonds that lie over c, parallel to c, one at each offset across c: each coarse bond has L^2 fine bonds of its own. $\mathcal{M}(U)(c)$ depends only on the values of U on the bonds inside the two blocks of c and on X_c (Prop. YM3_I:4.5). At fixed values ι of the bonds inside the blocks, $\mathcal{M}(\iota, \cdot)$ is thus a product over c of Borel maps Ψ_c^ι of the bonds of X_c, and by Fubini's theorem it suffices that, for ι off a null set, every Ψ_c^ι pulls null sets back to null sets, a property that passes to products.

A contour variable is the ordered product of the U along a path; the intrinsic logarithm is the inverse of exp on $\{|X| < \pi\}$, real-analytic on $\mathrm{SU}(2) \setminus \{-\mathbf{1}\}$; $U[c]$ is the contour variable of the straight path from c_- to c_+, and $x' = x + (c_+ - c_-)$ for $x \in B(c_-)$. Equations YM3_I:(37)–(39) give $\mathcal{M}(U)(c)$ by three cases, with disjoint domains that exhaust all configurations (Prop. YM3_I:4.5). *First case:* on the open set $\mathcal{S}_c$ of the configurations whose plaquettes with all corners in the double block of c lie within $\omega_M = \frac{1}{200L^2}$ of $\mathbf{1}$ (the small fields of the double block, a condition on its bonds alone), $\mathcal{M}(U)(c) = \exp(L^{-3}\sum_{x\in B(c_-)} \log W_x) U[c]$ with $W_x = U(c_-, x)U([x, x'])U(x', c_+)U[c]^{-1}$, the $U(y, x)$ being the group means of the contour variables along the staircase paths of a block from its centre y to x, and log the series; the first case applies exactly on $\mathcal{S}_c$.

Second case: off $\mathcal{S}_c$, where none of the closed contour variables

$$Y_{x,\Gamma,\Gamma'} = U(\Gamma)U([x, x'])U(\Gamma')^{-1}U[c]^{-1},$$

$x \in B(c_-)$, Γ and Γ' staircase paths from c_- to x and from c_+ to x', equals $-\mathbf{1}$,

$$\mathcal{M}(U)(c) = \exp\big(L^{-3}\sum_x \mathrm{mean}_{\Gamma,\Gamma'} \log Y_{x,\Gamma,\Gamma'}\big) U[c]$$

with the intrinsic logarithm. *Third case:* at all other configurations $\mathcal{M}(U)(c) = U[c]$. A contour variable $Y_{x,\Gamma,\Gamma'}$ with x on the line through c_- parallel to c runs forth and back on that line and is identically $\mathbf{1}$; every other one passes once through the bond of X_c on $[x, x']$ and once through the bond of X_c on $[c_-, c_+]$, two different bonds, all its other bonds lying inside the blocks; so the set where it equals $-\mathbf{1}$ is the graph of one bond of X_c over the other bonds, a closed real-analytic submanifold of codimension 3.

The finite union $\mathcal{N}_c$ of these graphs is closed and null, and its section at every value ι of the bonds inside the blocks is null; its complement $\hat{\mathcal{O}}$ is open, connected, as the complement of finitely many closed submanifolds of codimension 3 of a connected manifold, and holds $\mathbf{1}$. The domain of the second case lies in $\hat{\mathcal{O}}$, that of the third in $\mathcal{N}_c$. In the formula of the second case every logarithm is taken of one of these closed contour variables, and no group mean occurs: products of bond variables, the intrinsic logarithm on $\mathrm{SU}(2) \setminus \{-\mathbf{1}\}$ and the exponential make it a real-analytic map F_c on $\hat{\mathcal{O}}$, jointly in the bonds inside the blocks and in X_c.

Its differential in the variables of X_c at $\mathbf{1}$: the factors $U(\Gamma)$, $U(\Gamma')$ do not read X_c, the differential of the logarithm at $\mathbf{1}$ is the identity, so $dY_{x,\Gamma,\Gamma'} = dU([x, x']) - dU[c]$ there, and the differential of F_c is $L^{-3}\sum_x dU([x, x']) = L^{-2}\sum_{b\in X_c} du_b$, each bond of X_c lying on L of the L^3 straight paths $[x, x']$: the mean over the L^2 bonds, the value that Lem. YM3_I:4.6 gives for the formula of the first case. The Gram determinant of this differential is a real-analytic function on the connected set $\hat{\mathcal{O}}$ with the value L^{-6} at $\mathbf{1}$; so its zero set $\mathcal{Z}_c$ is null, and off it F_c is a submersion in the variables of X_c.

We now prove that the first branch is a submersion at every point of $\mathcal{S}_c$. This avoids any assumption about its connected components. Let ν be the direction of c. Index the L^2 lines parallel

to c in its double block by their transverse coordinates z. Each line has one crossing bond $b_z \in X_c$; write $u_z = U(b_z)$ and let z_0 index the line through c_-. For $x \in B(c_-)$ on the line z, factor the straight transport as

$$U([x,x']) = P_x u_z Q_x, \qquad U[c] = g = P_0 u_{z_0} Q_0.$$

The factors P_x, Q_x use only bonds inside the blocks. The contour means also use only those bonds. Thus, with

$$a_x = U(c_-,x)P_x, \qquad b_x = Q_x U(c_+,x')^{-1},$$

the first-branch variables are $W_x = a_x u_z b_x g^{-1}$, and a_x, b_x stay fixed when the crossing variables vary.

Let t_z be the point of the line z whose ν-coordinate is that of c_-, so $t_{z_0} = c_-$ and $a_{c_-} = P_0$. Given $\xi \in \mathfrak{g}$, vary all crossing bonds together by

$$u_z(\epsilon) = e^{i\epsilon X_z} u_z, \qquad X_z = \mathrm{Ad}(a_{t_z})^{-1}\xi.$$

The norm of this tangent vector in the Euclidean product of the crossing variables is $L|\xi|$, and $g(\epsilon) = e^{i\epsilon\xi} g$. Moreover,

$$W_x(\epsilon) = e^{i\epsilon(\xi+\rho_x)} W_x e^{-i\epsilon\xi}, \qquad \xi + \rho_x = \mathrm{Ad}(a_x a_{t_z}^{-1})\xi.$$

The error ρ_x is small because the contour mean is close to every contour in its family. Indeed,

$$a_x a_{t_z}^{-1} = U(c_-,x)U([x,t_z])U(c_-,t_z)^{-1} = h_1 h_2.$$

To obtain the last factorization, choose the contour from c_- to x which traverses the ν direction last. It is a contour $\Gamma_\perp$ from c_- to t_z followed by $[t_z,x]$. Then

$$h_1 = U(c_-,x)U(\Gamma)^{-1}, \qquad h_2 = U(\Gamma_\perp)U(c_-,t_z)^{-1}.$$

Lem. YM3_I:4.4 gives $|h_i - \mathbf{1}| \le 1/40$ and $|W_x - \mathbf{1}| < 1/14$. Unitarity and $|\mathrm{Ad}(h)\xi - \xi| \le 2|h-\mathbf{1}||\xi|$ therefore give

$$|h_1 h_2 - \mathbf{1}| \le \frac{1}{20}, \qquad |\rho_x| \le \frac{1}{10}|\xi|.$$

Write $W_x = e^{iw_x}$ with $w_x \in \mathfrak{g}$ and $|w_x| < 0.0715$. The differential of the logarithm gives

$$\frac{d}{d\epsilon}\log W_x(\epsilon)\Big|_{\epsilon=0} = i\eta_x, \qquad |\eta_x| \le \left(1.001\cdot\frac{1}{10} + 2|w_x|\right)|\xi| < \frac{1}{4}|\xi|.$$

Here the first term bounds the logarithmic differential of the left perturbation ρ_x, and the second is the commutator from conjugating w_x by $e^{i\epsilon\xi}$.

Put $\mathsf{E} = L^{-3}\sum_x w_x$ and $\bar\eta = L^{-3}\sum_x \eta_x$. The formula for the average now yields

$$-i\frac{d}{d\epsilon}\mathcal{M}(U(\epsilon))(c)\mathcal{M}(U)(c)^{-1}\Big|_{\epsilon=0} = \mathsf{T}_U\xi := \mathsf{f}(\mathrm{ad}(i\mathsf{E}))\bar\eta + \mathrm{Ad}(e^{i\mathsf{E}})\xi, \qquad \mathsf{f}(s) = \frac{e^s-1}{s}.$$

The operator $\mathsf{f}(\mathrm{ad}(i\mathsf{E}))$ is the average, for $0 \le t \le 1$, of the isometries $\mathrm{Ad}(e^{it\mathsf{E}})$. Consequently $|\mathsf{T}_U\xi| \ge 3|\xi|/4$. It is an invertible map of $\mathfrak{g}$, and

$$\zeta \longmapsto \left(\mathrm{Ad}(a_{t_z})^{-1}\mathsf{T}_U^{-1}\zeta\right)_z$$

is a right inverse of the crossing-variable differential, of norm at most $4L/3$.

A submersion pulls Haar-null sets back to null sets, since in local coordinates it is a projection and Haar measure has a smooth positive density. The first branch is a submersion everywhere on $\mathcal{S}_c$. For the second branch, the exceptional set $\mathcal{Z}_c$ found above is null; the third branch is contained in $\mathcal{N}_c$. Fubini's theorem therefore gives, for almost every fixed internal configuration ι, the null-set property for Ψ_c^ι. There are finitely many coarse bonds, so these properties hold simultaneously outside one exceptional set of ι of measure zero. The product map has the same property, again by Fubini.

So $\mathcal{M}^{-1}(N)$ is null. Then $\mu_d^K(N) = Z_K^{-1}\int \mathbf{1}_N(\mathcal{M}^{K-d}U)\,e^{-\beta\sum_p a_p}\,\mathrm{d}U = 0$ for a null N, and $\mu_d(N) = \lim_K \mu_d^K(N) = 0$, the laws converging uniformly on all Borel sets (Cor. 2.8). □

Lemma 14.19 (One vertex)**.** Let $V_{s_1}, V_{s_2}, V_{s_3}$ be irreducible representations of SU(2), $J_i = (J_i^a)_{a\le 3}$ the spins acting in the i-th factor of $V_{s_1}\otimes V_{s_2}\otimes V_{s_3}$, $[J_i^a, J_i^b] = i\epsilon_{abc}J_i^c$, and $t_{ik} = \sum_a J_i^a J_k^a$. For every spin J the algebra generated by 1, t_{12} and t_{23} acts irreducibly on the multiplicity space $\mathsf{M}^J = \mathrm{Hom}_{\mathrm{SU}(2)}(V_J, V_{s_1}\otimes V_{s_2}\otimes V_{s_3})$: it is the algebra of all operators of M^J. ■

Proof sketch. The enveloping algebra $\mathfrak{U}$ of three copies of the Lie algebra of SU(2) maps onto the algebra of all operators of the irreducible module $V_{s_1}\otimes V_{s_2}\otimes V_{s_3}$ (Burnside), equivariantly for the diagonal adjoint action, which is locally finite; averaging over the group shows that the invariant part $\mathfrak{U}^{\mathrm{inv}}$ maps onto the commutant of the diagonal action, the sum of the algebras of all operators of the spaces M^J. By symmetrisation the graded algebra of $\mathfrak{U}^{\mathrm{inv}}$ is the algebra of the polynomial invariants of three vectors of $\mathbb{C}^3$ under SO(3), which is generated by the scalar products and the determinant (first fundamental theorem); by induction along the filtration of $\mathfrak{U}$ by the order, $\mathfrak{U}^{\mathrm{inv}}$ is generated by the J_i^2, the t_{ik} and $T_3 = \sum \epsilon_{abc}J_1^aJ_2^bJ_3^c$, and $[t_{12}, t_{23}] = \sum J_1^aJ_3^c[J_2^a, J_2^c] = -iT_3$. On the isotypic part $V_J\otimes\mathsf{M}^J$ the $J_i^2 = s_i(s_i+1)$ are numbers and, the square of the total spin being $J(J+1)$, $t_{13} = \frac12(J(J+1) - \sum_i s_i(s_i+1)) - t_{12} - t_{23}$. So 1, t_{12}, t_{23} generate the image of $\mathfrak{U}^{\mathrm{inv}}$ in the operators of M^J, which is all of them. □

14.1.6. *Proof of Thm. AO. (a)* If $\eta \perp N^n\mathcal{D}_0$, then $\langle\eta, U(x)N^n\chi\rangle = 0$ for all x and $\chi\in\mathcal{D}_0$, as $U(x)$ keeps $\mathcal{D}_0$ and commutes with N; so the spectral measure of the pair $(\eta, N^n\chi)$ vanishes and $\langle\eta,\chi\rangle = \int(1+E)^{-n}\,\mathrm{d}m_{\eta,N^n\chi} = 0$. If $\psi\perp N^n\mathcal{D}_0(O)$, the function $(a_k)_{k\le q}\mapsto\langle\psi, N^nU(a_1)\phi(f_1)\cdots U(a_q)\phi(f_q)\Omega\rangle$, $f_k\in\mathcal{F}(O_1)$, with arbitrary $q\ge 1$ with $O_1 + \{|a|<\delta\}\subset O$, vanishes for small a_k and is the boundary value of a function holomorphic in $a_k + iV_+$ (spectrum condition); by the reflection principle it vanishes for all a_k, and a partition of unity gives $\psi\perp N^n\mathcal{D}_0$.

By the bound (b5) and the core $\mathcal{D}_0$ of N^n, $n\ge\wp$, the closures from $\mathcal{D}_0$ and from $\mathcal{D}_\infty$ agree, $\langle\phi(\bar f)\chi,\eta\rangle = \langle\chi,\bar\phi(f)\eta\rangle$ for $\chi\in\mathcal{D}_\infty$, $\eta\in D(\bar\phi(f))$, and the relation that defines $\mathcal{C}(O)$ passes from $\mathcal{D}_0$ to $\mathcal{D}_\infty$. A smoothing in time turns it into $Q\bar\phi(f)\subset\bar\phi(f)Q$. Let $Q\in\mathcal{C}(O)$, $f\in\mathcal{F}(O)$, $\delta>0$ with $\mathrm{supp} f + (-\delta,\delta)e_0\subset O$, $g\in C_c^\infty((-\delta,\delta))$ and $Q(g) = \int g(t)e^{itH}Qe^{-itH}\,\mathrm{d}t$. As $e^{-itH}\phi(f) = \phi(f(\cdot + te_0))e^{-itH}$ on $\mathcal{D}_\infty$ and $f(\cdot+te_0)\in\mathcal{F}(O)$ for $|t|<\delta$, $Q(g)$ has the same relation for this f on $\mathcal{D}_\infty$; and $\langle N^n\chi, Q(g)\psi\rangle = \sum_{k\le n}\binom{n}{k}i^k\langle\chi, Q(g^{(k)})N^{n-k}\psi\rangle$ for $\chi,\psi\in\mathcal{D}_\infty$ gives $Q(g)\mathcal{D}_\infty\subset\mathcal{D}_\infty$.

With $\phi(f)^\dagger = \phi(\bar f)$: $\phi(f)Q(g) = Q(g)\phi(f)$ on $\mathcal{D}_\infty$; as $\bar\phi(f)$ is closed, $Q(g)\bar\phi(f)\subset\bar\phi(f)Q(g)$; for an approximate unit g_k, $Q(g_k)\to Q$ strongly, and closedness once more gives $Q\bar\phi(f)\subset\bar\phi(f)Q$. Conversely a bounded Q with $Q\bar\phi(f)\subset\bar\phi(f)Q$ for all $f\in\mathcal{F}(O)$ lies in $\mathcal{C}(O)$, by the relation between $\phi(\bar f)$ and $\bar\phi(f)$ above. So $\mathcal{C}(O)$ is closed under products; it holds 1, is closed under adjoints and weakly closed: a von Neumann algebra; covariance is $U(\mathfrak{g})\phi(f)U(\mathfrak{g})^{-1} = \phi(f^{\mathfrak{g}})$ on $\mathcal{D}_0$. If $Q\in\mathcal{C}(O)$ and $Q\Omega = 0$, then $\langle\psi, QX\Omega\rangle = \langle X^\dagger\psi, Q\Omega\rangle = 0$ for products X of fields of O, so $Q = 0$.

The projection onto the closure of $\mathcal{A}(O)\Omega$ lies in $\mathcal{A}(O)' = \mathcal{C}(O)$, commutes with every $\bar\phi(f)$, $f\in\mathcal{F}(O)$, and fixes Ω, so that closure holds $\mathcal{D}_0(O)$. The unitaries of $\mathcal{A}(O)' = \mathcal{C}(O)$ commute with $\bar\phi(f)$; and a net $\mathfrak{N}$ with this affiliation has $\mathfrak{N}(O)'\subset\mathcal{C}(O)$, so $\mathfrak{N}(O)\supset\mathcal{A}(O)$. A partition of unity writes $f\in\mathcal{F}(\bigcup O_i)$ as a finite sum of $f_i\in\mathcal{F}(O_i)$: $\mathcal{C}(\bigcup O_i) = \bigcap\mathcal{C}(O_i)$, and the commutants give $\mathcal{A}(\bigcup O_i) = \bigvee\mathcal{A}(O_i)$. For $O_2\subset O_1^\times$, $\mathcal{L}(O_1)\subset\mathcal{C}(O_1^\times)\subset\mathcal{C}(O_2)$ commutes with $\mathcal{A}(O_2)\supset\mathcal{L}(O_2)$. (W) says $\mathcal{A}(O_1)\subset\mathcal{C}(O_2)$ for all $O_2\subset O_1^\times$, that is $\mathcal{A}(O_1)\subset\mathcal{C}(O_1^\times)$ by additivity, that is $\mathcal{L} = \mathcal{A}$; and an open set is the union of the upright double cones in it, so additivity reduces (W) to such cones.

(b) (b1) $P_m\phi(f)P_n = P_me^{sH}(e^{-sH}\phi(f)e^{-sH})e^{sH}P_n$ with $\|P_me^{sH}\|\le e^{s(m+1)}$; at $s = (m+n+2)^{-1}$, eq. 14.i gives $b_{mn}(f)\le e\,C_\wp\|f\|_{L^1}(m+n+2)^\wp$. (b2) With $X = P_m\phi(f)P_n$ and $[H,\phi(f)] = i\phi(\partial_t f)$,

$$(m-n)X = iP_m\phi(\partial_t f)P_n - (H-m-\tfrac12)X + X(H-n-\tfrac12),\qquad \|(H-m-\tfrac12)P_m\|\le\tfrac12,$$

so $(|m-n|-1)b_{mn}(f) \le b_{mn}(\partial_t f)$; iterate. (b3) If $\wp \le 1$, then $k_1 = 0$ and the claim follows directly from (b1). Suppose $\wp > 1$, so $k_1 \ge 1$. For $|m-n| > (m+n+2)^\epsilon \ge 2^\epsilon$, $|m-n| \ge 2$ and $|m-n|-1 \ge \frac{1}{3}|m-n|$, so (b2) at k_1 and (b1) at $\partial_t^{k_1} f$ give $b_{mn}(f) \le 3^{k_1}|m-n|^{-k_1} e C_\wp \|\partial_t^{k_1} f\|_{L^1}(m+n+2)^\wp$, and $|m-n|^{k_1} > (m+n+2)^{\epsilon k_1} \ge (m+n+2)^{\wp-1}$. (b4) $\phi(h*f) = \int h(a)U(a)\phi(f)U(a)^{-1}\,\mathrm{d}a$, and $U(a)$ commutes with P_m. (b5) At $s = (1+\max\{m,n\})^{-1}$ the argument of (b1) gives $b_{mn}(f) \le e^2 C_\wp \|f\|_{L^1}(1+\max\{m,n\})^\wp$, and with $\|(H-m)P_m\| \le 1$ the argument of (b2) gives $(|m-n|-2)^k b_{mn}(f) \le b_{mn}(\partial_t^k f)$ for $|m-n| \ge 3$. With $(1+\max\{m,n\})^2 \le (1+m)(1+n)(1+|m-n|)$: $b_{mn}(f) \le e^2 C_\wp (1+m)^{\wp/2}(1+n)^{\wp/2}\gamma(m-n)$, $\gamma(d) = (1+|d|)^{\wp/2}\|f\|_{L^1}$ for $|d| \le 2$ and $\gamma(d) = (1+|d|)^{\wp/2}(|d|-2)^{-k_\flat}\|\partial_t^{k_\flat} f\|_{L^1}$ for $|d| \ge 3$; $\sum_d \gamma(d)$ is the bracket of the form bound, and Schur's test with $\sum_a (1+a)^{2q}\|P_a\psi\|^2 \le \|N^q\psi\|^2$ gives it. For the operator bound, $(2+a)^j \le 2^j(1+b)^j(1+|a-b|)^j$ and $(1+\max\{a,b\})^\wp \le (1+b)^\wp(1+|a-b|)^\wp$, and the second form of Schur's test. At $\wp = 4$: $\mathsf{s}_0(4) = 195$, $2\mathsf{s}(4,6) < 537.82$, $\mathsf{s}_0(2) = 27$, $2\mathsf{s}(2,4) < 37.20$. $\mathcal{D}_\infty$ is a core of $N^\wp$.

(c) Put $J\phi(f_1)\cdots\phi(f_n)\Omega = \phi(\bar{f}_1\circ\mathsf{j})\cdots\phi(\bar{f}_n\circ\mathsf{j})\Omega$; it is antiunitary because $\mathcal{W}(F_1\circ\mathsf{j}\otimes\cdots\otimes F_N\circ\mathsf{j}) = \mathcal{W}(F_N\otimes\cdots\otimes F_1)$, which at Euclidean points is the invariance of σ under the rotation R_π by π in the (x_1,x_3)-plane, the image of $\mathsf{j} = B(i\pi)$, which reverses the order of the Euclidean times, continued by [7, Lem. G.5]; uniqueness holds because $\mathcal{D}_0$ is dense. Call $X = \phi(f_1)\cdots\phi(f_n)$, $f_k \in C_c^\infty(\mathbb{R}^{1+2})$, an ordered product in W if $\mathrm{supp} f_1 \times\cdots\times \mathrm{supp} f_n$ lies in $W^{(n)} = \{(x_k)_k :\ x_1 \in W,\ x_{k+1}-x_k \in W\ (1\le k<n)\}$ (an order in real time along the wedge, not the order of the Euclidean heights of the setting), and put $X^{\mathsf{j}} = \phi(f_1\circ\mathsf{j})\cdots\phi(f_n\circ\mathsf{j})$, in the same order, and $X^\dagger = \phi(\bar{f}_n)\cdots\phi(\bar{f}_1)$.

As W is an open convex cone, $x_k \in W$ and $x_l - x_k \in W$ for $k<l$ on the supports: all f_k lie in $\mathcal{F}(W)$, and their supports are pairwise spacelike separated. With $\mathsf{m}(y) = y^1 - |y^0|$, so that $W = \{\mathsf{m} > 0\}$ and $\mathsf{m}(B(s)y) \ge e^{-|s|}\mathsf{m}(y)$, one has $d(\iota B(s+i\tau)y) = \sin\tau\,\mathsf{m}(B(s)y) \ge \sin\tau\, e^{-|s|}\mathsf{m}(y)$ for $y \in W$, $0 \le \tau \le \pi$; read at $y = x_1$ and at $y = x_{k+1}-x_k$, this puts $(\iota B(\zeta)x_k)_k$ into $\mathsf{Tb}^{(n)}$ for $0 < \mathrm{Im}\,\zeta < \pi$, and $(\iota B(\zeta)x_k + k\eta e_3)_k$ for $0 \le \mathrm{Im}\,\zeta \le \pi$, $\eta > 0$. So the function $\zeta \mapsto \langle\chi, U(B(\zeta))X\Omega\rangle$ continues through the point vectors, by (Br) and $\eta \downarrow 0$ at the edges, to a function continuous on the closed strip $0 \le \mathrm{Im}\,\zeta \le \pi$ and holomorphic inside, with the boundary value $\langle\chi, U(B(s))X^{\mathsf{j}}\Omega\rangle$ at $\mathrm{Im}\,\zeta = \pi$, $B(s+i\pi) = B(s)\mathsf{j}$; the comparison with the spectral integrals of K on its bounded spectral sets gives $X\Omega \in D(\Delta^{1/2})$ and $\Delta^{1/2}X\Omega = X^{\mathsf{j}}\Omega$.

By the definition of J ($\mathsf{j}^2 = 1$), $JX^{\mathsf{j}}\Omega = \phi(\bar{f}_1)\cdots\phi(\bar{f}_n)\Omega$, and this is $X^\dagger\Omega$, the order being reversed by locality on $\mathcal{D}_0$, as the supports are pairwise spacelike separated. A boost-invariant dense subspace $\mathcal{D}$ of $D(\Delta^{1/2})$ is a core: if $\langle\xi,\chi\rangle + \langle\Delta^{1/2}\xi, \Delta^{1/2}\chi\rangle = 0$ for $\chi \in \mathcal{D}$, then the measure $(1+e^{-2\pi k})dm^K_{\xi,\chi}$ has Fourier transform 0, and $\langle\xi,\chi\rangle = 0$ follows. The span of Ω and of the $X\Omega$, X ordered in W, is such a subspace: it is boost invariant, as $B(s)^{\times n}W^{(n)} = W^{(n)}$, and it is dense by the argument of (a): products of fields in the balls of radius $\frac{1}{4}$ about the points $(0,2k,0)$ stay ordered in W under small translations of each factor, the reflection principle carries the orthogonality to all translations, and a partition of unity to all products of fields.

Locality removes the ordering: for $Y = \phi(h_1)\cdots\phi(h_r)$, $h_k \in \mathcal{F}(W)$ in any order, and $\xi = X\Omega$, X ordered in W, the fields of $(X^{\mathsf{j}})^\dagger = \phi(\bar{f}_n\circ\mathsf{j})\cdots\phi(\bar{f}_1\circ\mathsf{j})$ lie in $\mathcal{F}(W')$, so they commute on $\mathcal{D}_0$ with those of Y, and $(X^{\mathsf{j}})^\dagger\Omega = J\xi$; hence $\langle\Delta^{1/2}\xi, Y\Omega\rangle = \langle\Omega, (X^{\mathsf{j}})^\dagger Y\Omega\rangle = \langle Y^\dagger\Omega, (X^{\mathsf{j}})^\dagger\Omega\rangle = \langle Y^\dagger\Omega, J\xi\rangle = \langle\xi, JY^\dagger\Omega\rangle$, also at $\xi = \Omega$; on the core this gives $Y\Omega \in D(\Delta^{1/2})$ and $J\Delta^{1/2}Y\Omega = Y^\dagger\Omega$ for all $h_k \in \mathcal{F}(W)$.

For $Q \in \mathcal{C}(W)$ the weak commutation, used n times, gives $\langle Q^*\Omega, X\Omega\rangle = \langle X^\dagger\Omega, Q\Omega\rangle = \langle J\Delta^{1/2}X\Omega, Q\Omega\rangle$ on the core, so $\Delta^{1/2}JQ\Omega = Q^*\Omega$. Opposite wedges: let $Q \in \mathcal{C}(W)$, $Q' \in \mathcal{C}(W')$ and $Q'_g = \int g(a)U(a)Q'U(a)^{-1}\,\mathrm{d}a$, $g \in C_c^\infty(W)$; as $W' \subset W' + a$ for $a \in W$, $Q'_g \in \mathcal{C}(W')$; it keeps $\mathcal{D}_\infty$ (smoothness in the time translation) and commutes there with the fields of W'. With $S = J\Delta^{1/2}$, J carries the graph of $\Delta^{1/2}$ onto that of $\Delta^{-1/2}$, so $\mathcal{D}_0(W')$ is a core of $S^* = J\Delta^{-1/2}$, and every R on $\mathcal{D}_\infty$ with an adjoint there that commutes with the fields of W' has $SR\Omega = R^\dagger\Omega$; and $S^*Q\Omega = Q^*\Omega$.

For products X, Y of fields of W and $R = Y^\dagger Q'_g X$:

$$\langle Y\Omega, QQ'_g X\Omega\rangle = \langle S^*Q\Omega, R\Omega\rangle = \langle SR\Omega, Q\Omega\rangle = \langle Y\Omega, Q'_g QX\Omega\rangle;$$

so $QQ'_g = Q'_g Q$, and $g \to \delta$ gives $QQ' = Q'Q$. For a wedge $\tilde{W} \supset O$, $\tilde{W}' \subset O^\times$, so $\mathcal{C}(O^\times) \subset \mathcal{C}(\tilde{W}') \subset \mathcal{C}(\tilde{W})' = \mathcal{A}(\tilde{W})$, whence $\mathcal{C}(O^\times) \subset \mathcal{A}^d(O)$; two spacelike separated double cones are parted by a wedge, which gives the locality; if $\mathcal{C}(O^\times) = \mathcal{A}^d(O)$ for all double cones, then $\mathcal{A}(O) \subset \mathcal{C}(O^\times)$ and (W) holds by (a), and the converse is (e).

(d) The function f_* is that of Lem. 14.8(a). (W) implies density: $\mathcal{A}(W') \subset \mathcal{C}(W)$ and $\mathcal{A}(W')\Omega$ is dense. Density implies wedge duality: Ω is then cyclic and separating for $\mathcal{N} = \mathcal{C}(W)$; by (c), $Q\Omega \mapsto Q^*\Omega$ is the restriction of $\Delta^{1/2} J = J\Delta^{-1/2}$ ($JKJ = -K$), and $J\mathcal{N}\Omega$ is a boost-invariant dense subspace of $D(\Delta^{1/2})$, hence a core; so modular theory gives the conjugation J, the modular operator Δ^{-1}, and $\mathcal{N}' = J\mathcal{N}J = \mathcal{C}(W')$. Wedge duality implies (W): spacelike separated upright double cones lie in some $\mathfrak{g}W'$ and $\mathfrak{g}W$, and $\mathcal{C}(K_1)' \subset \mathcal{C}(\mathfrak{g}W')' = \mathcal{C}(\mathfrak{g}W) \subset \mathcal{C}(K_2)$. $J\mathcal{A}(W)J = \mathcal{C}(W')'$, so $J\mathcal{A}(W)J \subset \mathcal{C}(W)$ is (W) for the pair (W', W), equivalent to (W) by the same chain with W and W' exchanged. (W) implies (CP): $\bar{\phi}(f)$, $f \in \mathcal{F}(K)$ real, is affiliated with $\mathcal{A}(K) \subset \mathcal{C}(K^\times)$; the extension of Lem. 14.6(b) has its bounded Borel functions in $\mathcal{A}(K) \otimes M_2$, which commute weakly with the fields of $K^\times$; Lem. 14.6(a) at $\mathfrak{A}$ generated by 1, f and $\mathcal{F}(K^\times)$, $X_0 = f$. (CP) implies (QCP_f), as $W \subset K^\times$ for $\bar{K} \subset W'$. (QCP_f) gives $\phi(f)\Omega \in \mathcal{H}_0$.

Let $\mathsf{L}(Y_0 + \lambda Y_1 + \lambda^2 Y_2) = \langle \Omega, Y_0\Omega\rangle + \langle \Omega, Y_1\eta\rangle + \langle \eta, Y_2\eta\rangle$ on the quadratic polynomials in λ with coefficients in $\mathfrak{P}(W)$; f commutes with $\mathfrak{P}(W)$ in $\mathfrak{P}$, as $\bar{K}$ and W are spacelike separated. On a square $(Z_0 + \lambda Z_1)^*(Z_0 + \lambda Z_1)$, L is $\|Z_0\Omega + Z_1\eta\|^2 \geq 0$ for all data; (QCP_f) is the positivity of L on the larger cone of the quadratic polynomials whose values at all real λ lie in $\mathfrak{P}(W)^+$, and the extension below is positive on this cone. Let $\mathscr{C}$ be the space of the functions $G = \sum_k g_k(\lambda) Y_k$, $Y_k \in \mathfrak{P}(W)$, g_k complex Borel with $|g_k| \leq C(1 + \lambda^2)$, and $\mathscr{C}^+$ the cone of those whose values at all real λ lie in $\mathfrak{P}(W)^+$.

Every hermitian element of $\mathscr{C}$ is a sum of terms gS, g real, $S = S^*$, and $\frac{1}{2}q(S^2+1) - gS = \frac{1}{4}(q-g)(S+1)^2 + \frac{1}{4}(q+g)(S-1)^2 \in \mathscr{C}^+$ for $q = C(1+\lambda^2)$, $|g| \leq q$, with $\frac{1}{2}q(S^2+1)$ a quadratic polynomial: the step of the proof of Lem. 14.6(a) at $r = 1$. The extension theorem of M. Riesz extends L to a linear $\hat{\omega}$ on $\mathscr{C}$ with $\hat{\omega} \geq 0$ on $\mathscr{C}^+$. For G, G' with $|g_k| \leq C(1 + |\lambda|)$, $G^*G \in \mathscr{C}^+$, so $\langle [G], [G']\rangle = \hat{\omega}(G^*G')$ gives a Hilbert space; $V\pi(Y)\Omega = [Y]$ on the dense $\mathcal{D}_0(W)$ is an isometry V of $\mathcal{H}$ into it; $\hat{\pi}(g)[G] = [gG]$, g bounded Borel, is bounded by $\sup|g|$, as $(\sup|g|^2 - |g|^2)G^*G \in \mathscr{C}^+$; and $Q_g = V^*\hat{\pi}(g)V$ commutes weakly with the fields of W on $\mathcal{D}_0(W)$, so on $\mathcal{D}_0$ by (a) and (b5): $Q_g \in \mathcal{C}(W)$, $\|Q_g\| \leq \sup|g|$.

With $g_n(\lambda) = \lambda \mathbf{1}_{|\lambda| \leq n}$, $Q_{g_n}\Omega \to \phi(f)\Omega$, because $\|[\mathbf{1}_{|\lambda|>n} Y]\|^2 \leq n^{-2}\mathcal{W}(f^2 Y^* Y)$ and $V^*[\lambda] = \phi(f)\Omega$. The first order holds for all data: if $Y_0 + \lambda Y_1 \in \mathfrak{P}(W)^+$ for all real λ, then $\langle \psi, \pi(Y_1)\psi\rangle = 0$ for $\psi \in \mathcal{D}_0$, so $\pi(Y_1) = 0$ there by polarisation, and $\mathcal{W}(Y_0 + fY_1) = \langle \Omega, Y_0\Omega\rangle \geq 0$. One vector decides: $\mathcal{H}_0$ is invariant under $\mathcal{C}(W)$, under the boosts, and under the translations $U(-a)$, $a \in \bar{W}$ (as $\mathcal{C}(W) \subset \mathcal{C}(W + a)$), hence, by the spectrum condition and the reflection principle along the light rays $(\pm 1, 1, 0)$ and along e_2, under all translations; so its projection P_0 commutes with $\mathcal{P}$, and $\mathsf{D}(f) = \|(1 - P_0)\phi(f)\Omega\|$ is a seminorm invariant under translations and boosts, with $\mathsf{D}(h * f) \leq \|h\|_{L^1} \mathsf{D}(f)$.

For $\chi \perp \mathcal{H}_0$ the tempered distribution $\mathsf{T}(h) = \langle \chi, \phi(h)\Omega\rangle$ vanishes on all $f * k$, $f \in \mathfrak{F}$, $k \in \mathcal{S}$; if the $\hat{f}$, $f \in \mathfrak{F}$, have no common real zero, a partition of unity shows that T vanishes on all h with $\hat{h} \in C_c^\infty$, so $\phi(h)\Omega \in \mathcal{H}_0$ for all $h \in \mathcal{S}$. Then for $\chi \in \mathcal{D}_\infty \cap \mathcal{H}_0^\perp$ and $h \in \mathcal{F}(W)$, $\langle Q\Omega, \phi(\bar{h})\chi\rangle = \langle \bar{\phi}(h)Q\Omega, \chi\rangle = \langle Q\phi(h)\Omega, \chi\rangle = 0$ for $Q \in \mathcal{C}(W)$; for $\psi \in \mathcal{H}_0 \cap \mathcal{D}_\infty$ the vector $\chi = (1 - P_0)\phi(h)\psi$ has $\|\chi\|^2 = \langle \phi(\bar{h})\chi, \psi\rangle = 0$; by induction $\mathcal{D}_0(W) \subset \mathcal{H}_0$, and $\mathcal{H}_0$ is dense.

The infimum over the boosts is the distance itself, by the boost invariance of $\mathcal{H}_0$; for one $f_0 \geq 0$ the transforms of its Lorentz images have no common real zero by Lem. 14.8(d), and translations do not change zero sets. So all the conditions are equivalent, and the modular data under (W)

are those of the density step, applied to $\mathcal{A}(W) = \mathcal{C}(W')$. (CP) gives (W) also without the wedge relation: for spacelike separated upright double cones K_1, K_2, $Q \in \mathcal{C}(K_1)'$ and real $g \in \mathcal{F}(K_2)$, the extension of Lem. 14.6(a) for (K_2, g) gives bounded $Q_h = V^* h(\hat{T}) V \in \mathcal{C}(K_2^\times) \subset \mathcal{C}(K_1)$, which commute with Q, and $Q_{g_n}\psi \to \phi(g)\psi$ for ψ in the domain of the GNS representation, which holds $\mathcal{D}_0(K_2^\times)$; so Q commutes weakly with $\phi(g)$ there, and the polynomial bound carries this to $\mathcal{D}_0$.

The inequalities: under (W) the closures of $\phi(g_1), \phi(g_2)$ are affiliated with commuting algebras, and Lem. 14.6(c) gives strongly commuting self-adjoint extensions $\hat{A}_1, \hat{A}_2$; with $\psi = \phi(f)\Omega$ and their joint spectral measure,

$$\begin{aligned} |\mathcal{W}(f g_1 g_2)| &= \Big| \int \lambda_1 \lambda_2 \, \mathrm{d}\mu_{\psi,\Omega} \Big| \\ &\le \Big(\int |\lambda_1 \lambda_2| d\mu_\psi \Big)^{1/2} \Big(\int |\lambda_1 \lambda_2| d\mu_\Omega \Big)^{1/2} \\ &\le \big(\|\hat{A}_1 \psi\| \|\hat{A}_2 \psi\| \big)^{1/2} \|\hat{A}_1 \hat{A}_2 \Omega\|^{1/2}, \end{aligned}$$

and locality on $\mathcal{D}_0$ identifies the squares of the three norms with $\mathcal{W}(f^2 g_1^2)$, $\mathcal{W}(f^2 g_2^2)$, $\mathcal{W}(g_1^2 g_2^2)$; likewise $\mathcal{W}(p(f_1, f_2)) = \int p \, \mathrm{d}\mu_\Omega \ge 0$, and Motzkin's polynomial is covered by Lem. 14.6(d).

(e) Under (W), $\mathcal{A}(\tilde{W}) = \mathcal{C}(\tilde{W}')$ for every wedge by (d) and covariance. Two spacelike separated open convex sets are parted by a wedge (Hahn–Banach), so $O^\times = \bigcup\{\tilde{W}' : \tilde{W} \supset O\}$, and additivity gives $\mathcal{A}^d(O) = \bigcap \mathcal{C}(\tilde{W}') = \mathcal{C}(O^\times)$. Locality: a wedge parts O_1 from O_2. Haag duality: $\mathcal{A}^d(O)' = \mathcal{A}(O^\times) = \bigvee \mathcal{A}(O_3) \subset \bigvee \mathcal{A}^d(O_3) = \mathcal{A}^d(O^\times) \subset \mathcal{A}^d(O)'$, the joins over the double cones $O_3 \subset O^\times$; and as $O^\times = \bigcup\{\tilde{W}' : \tilde{W} \supset O\}$, (a) gives $\mathcal{A}(O^\times) = \bigvee\{\mathcal{A}(\tilde{W}') : \tilde{W} \supset O\}$.

Outer regularity: a point with a neighbourhood spacelike to O is spacelike to the compact set $\bar{O}$, hence to $\bar{O}_\epsilon$ for small ϵ, the compact sets $\bar{O}_\epsilon$ decreasing to $\bar{O}$; so $O^\times = \bigcup_{\epsilon>0} O_\epsilon^\times$, and (a) gives $\mathcal{A}^d(O) = \mathcal{C}(O^\times) = \bigcap_\epsilon \mathcal{C}(O_\epsilon^\times) = \bigcap_\epsilon \mathcal{A}^d(O_\epsilon)$. An operator commuting with all $\mathcal{A}(O_1)$, $O_1 \subset O^\times$, commutes with $\mathcal{A}(O^\times)$, so it lies in $\mathcal{A}^d(O)$. For $O \subset \tilde{W}$: $\mathcal{A}(O) \subset \mathcal{A}^d(O) \subset \mathcal{A}(\tilde{W})$, so the wedge algebras of $\mathcal{A}^d$ are those of $\mathcal{A}$, and $(\mathcal{A}^d)^d = \mathcal{A}^d$. $\mathcal{A}(O) = \mathcal{A}^d(O)$ if and only if $\mathcal{C}(O^\times) \subset \mathcal{A}(O) = \mathcal{C}(O)'$.

Covariance, J, cyclicity and affiliation follow from (a) and (c). Without (W): if the intersections are local, then $\mathcal{A}(O_1) \subset \mathcal{A}^d(O_1) \subset \mathcal{A}^d(O_2)' \subset \mathcal{A}(O_2)' = \mathcal{C}(O_2)$ for spacelike separated double cones.

(f) Let $A_j = \phi(\partial_t^j f)$ and $R_\epsilon = (1 + \epsilon N)^{-1}$. From $[H, A_0] = iA_1$: $R_\epsilon^r A_0 \chi = A_0 R_\epsilon^r \chi - i\epsilon \sum_{j=1}^r R_\epsilon^j A_1 R_\epsilon^{r-j+1} \chi$. For $A_0^* \eta = \pm i\eta$ and $\eta_\epsilon = R_\epsilon^{k_0}\eta$, the imaginary part of $\langle \eta_\epsilon, \bar{A}_0 \eta_\epsilon \rangle$ and the form bound of A_1 give $\|\eta_\epsilon\|^2 \le c \sum_j \epsilon^{1/2} \|N^{1/2} R_\epsilon^{k_0 - j+1} \eta_\epsilon\| \, \epsilon^{1/2} \|N^{1/2} R_\epsilon^j \eta\| \to 0$, while $\eta_\epsilon \to \eta$; so $\bar{A}_0$ is self-adjoint. $N^k \phi(f) = \sum_j \binom{k}{j} i^j \phi(\partial_t^j f) N^{k-j}$ gives $\bar{A}_0 N^{-k} = N^{-k}(\bar{A}_0 + K_k)$ with K_k bounded under the block bounds, whence the resolvents keep $D(H^k)$ off a strip and, through cores, commute with $\bar{\phi}(h)$ for h spacelike to f; Neumann series reach all $z \notin \mathbb{R}$; so the resolvents lie in $\mathcal{C}(O)$. (f1) With the logarithm the self-adjointness and the commutation come from unitary groups along which the exponent of the energy falls.

Put $\mathsf{w}(x) = x(1 + \log x)$, and for real $h \in C_c^\infty$ let $v = f_i * h$. Off the band (b3) bounds the blocks of $\partial_t f_i$ linearly, so $b_{mn}(\partial_t f_i) \le c\, \mathsf{w}(m + n + 2)$ for all m, n; by (b4), $\partial_t^j v = (\partial_t f_i) * \partial_t^{j-1} h$ has the same bound with $c\|\partial_t^{j-1} h\|_{L^1}$, $j \ge 1$, and (b2) adds the decay $(|m - n| - 1)^{-3}$ for $|m - n| \ge 2$; as $\mathsf{w}(m + n + 2) \le 3.4(1 + |m - n|)\, \mathsf{w}(m + 1)^{1/2} \mathsf{w}(n + 1)^{1/2}$, Schur's test with the weights $\mathsf{w}(n + 1)$ gives $|\langle \psi', \phi(\partial_t^j v)\psi \rangle| \le c_j \|\mathsf{w}(N)^{1/2} \psi'\| \|\mathsf{w}(N)^{1/2} \psi\|$ on $\mathcal{D}_\infty$, $j \ge 1$. By $\log x = \int_0^\infty [(1 + t)^{-1} - (x + t)^{-1}] dt$, $[N + t, \phi(\partial_t^j v)] = i\phi(\partial_t^{j+1} v)$ and $\int_0^\infty \mathsf{w}(x)(x + t)^{-2} dt = 1 + \log x$, this turns into $|\langle \log N\, \psi, \phi(\partial_t^j v)\psi \rangle - \langle \phi(\partial_t^j v)\psi, \log N\, \psi \rangle| \le c_{j+1} \langle \psi, (1 + \log N)\psi \rangle$, $j \ge 0$, the left member being $|\mathfrak{l}^{(j)}(\psi, \psi)|$ for the hermitian form $\mathfrak{l}^{(j)}(\psi', \psi) = \int_0^\infty \langle (N + t)^{-1} \psi', \phi(\partial_t^{j+1} v)(N + t)^{-1} \psi \rangle dt$.

The three-lines theorem, on vectors of bounded energy, carries the bound to $\mathfrak{l}^{(j)}(N^\tau \psi', N^{-\tau}\psi)$ with the constant $\sum_{l \le \lceil |\tau| \rceil} \binom{\lceil |\tau| \rceil}{l} c_{j+1+l}$; as $\frac{d}{d\tau} \langle N^\tau \chi', A N^{-\tau} \chi \rangle = i\, \mathfrak{l}^{(0)}(N^\tau \chi', N^{-\tau} \chi)$ for $A = \phi(v)$,

and a second derivative brings in $\mathfrak{l}^{(1)}$, integration over $|\tau| \le \frac{\rho}{2}$ at $\chi' = \chi$ gives

$$|\langle N^\rho\chi, A\chi\rangle - \langle\chi, AN^\rho\chi\rangle| \le \rho\, c_1\langle\chi, N^\rho(1+\log N)\chi\rangle + \rho^2 c'(\rho)\langle\chi, N^\rho\chi\rangle \qquad (\chi\in\mathcal{D}_\infty,\ \rho\ge 0),$$

$c'(\rho) = \sum_{l\le\lceil\rho/2\rceil}\binom{\lceil\rho/2\rceil}{l}c_{l+2}$. The bounded $A_s = e^{-sH}Ae^{-sH}$ satisfy the same with the same numbers. For $\psi\in\mathcal{D}_\infty$ and $y(\lambda) = \|N^{\rho(\lambda)/2}e^{i\lambda A_s}\psi\|^2$ with the falling exponent $\rho(\lambda) = \rho_0 e^{-c_1|\lambda|}$, the term $\rho'\langle N^\rho\log N\rangle$ of y' cancels the term with the logarithm: $y' \le (\rho_0 c_1 + \rho_0^2 c'(\rho_0))\, y$ for $\lambda\ge 0$, and Gronwall's lemma bounds $\|N^{\rho(\lambda)/2}e^{i\lambda A_s}\psi\|$ uniformly in s. As $s\downarrow 0$, $\|(A_s - A_{s'})\chi\| \le c|s-s'|\|N^{q+1}\chi\|$ ($q\ge\wp$ an integer) makes the groups $e^{i\lambda A_s}$ converge strongly to a unitary group V with $V(\lambda)D(N^r)\subset D(N^{re^{-c_1|\lambda|}})$ for $r\ge 0$, so $V(\lambda)\mathcal{D}_\infty = \mathcal{D}_\infty$, and $\lambda\mapsto V(\lambda)\psi$ is continuously differentiable in every norm $\|N^a\cdot\|$ with the derivative $iAV(\lambda)\psi$; its generator is the closure of A on $\mathcal{D}_\infty$ (for η with $(A|_{\mathcal{D}_\infty})^*\eta = \pm i\eta$, $\lambda\mapsto\langle V(\lambda)\psi,\eta\rangle$ is bounded with the derivative $\pm$ itself, hence 0): $\phi(v)$ is essentially self-adjoint on $\mathcal{D}_\infty$ and on $\mathcal{D}_0$.

The group loses moments of the energy and keeps $\mathcal{D}_\infty$. For $h'\in\mathcal{S}$ whose support is spacelike separated from $\operatorname{supp} v$, $\phi(v)\phi(h') = \phi(h')\phi(v)$ on $\mathcal{D}_\infty$ (locality on $\mathcal{D}_0$, the core $\mathcal{D}_0$ of N^n, (b5)), and $\frac{d}{d\lambda}\langle\phi(\bar h')V(\lambda)\psi', V(\lambda)\psi\rangle = 0$ gives $V(\lambda)\bar\phi(h') \subset \bar\phi(h')V(\lambda)$: by (a) the bounded Borel functions of $\bar\phi(v)$ lie in $\mathcal{C}(O)$ for every open O spacelike separated from $\operatorname{supp} v$. For real $h\in\mathcal{F}(O)$ the $f_i * h$ lie in $\mathcal{F}(O)$ for large i and tend to h in $\mathcal{S}$; so $\mathcal{C}(O)$ is the commutant of the groups of the $\bar\phi(f_i*h)$ with $\operatorname{supp}(f_i*h)\subset O$ (by (a) every $Q\in\mathcal{C}(O)$ commutes with them; conversely weak commutation with the $\phi(f_i*h)$ passes to $\phi(h)$ by (b5)), and for spacelike separated O_1, O_2 the groups of O_1 lie in $\mathcal{C}(O_2)$: $\mathcal{C}(O_1)'\subset\mathcal{C}(O_2)$, which is (W) without a wedge, and (CP) by (d). (f2) The time reversal $[F]\mapsto[\bar F]$ gives the other sign for f_0 even in t; $\partial_t^j(h*f_0) = (\partial_t^{j-1}h)*\partial_t f_0$ gives the hypotheses of the first part at every $h*f_0$; by Lem. 14.8(d) the images of f_0 under rotations, boosts and translations into W' have no common real zero of their transforms, and the resolvents of the corresponding operators lie in $\mathcal{C}(W)$ and produce from Ω a total set; (d) applies. (f3) For products X of fields spacelike to f_0, $\|\phi(f_0)^n X\Omega\|^2 = \langle\phi(f_0)^{2n}\Omega, X^\dagger X\Omega\rangle \le \|\phi(f_0)^{2n}\Omega\|\|X^\dagger X\Omega\|$, so these total vectors are quasi-analytic, $\bar\phi(f_0)$ is self-adjoint with resolvents commuting with the outside fields, and the argument of (f2) applies. (f4) At $s = (m+n+2)^{-1}$ the argument of (b1) gives $b_{mn}(f)\le e\,C\|f\|_{L^1}(m+n+2)(1+\log(m+n+2))$ for every $f\in\mathcal{S}$ (for $\wp\le 1$ (b1) gives it without the logarithm), so for every $\partial_t^j f$: the hypothesis of (f1) holds for every approximate unit f_i, and the argument of (f1) runs with $v = f$ itself; so every real $\phi(f)$ is essentially self-adjoint on $\mathcal{D}_0$, and $e^{i\lambda\bar\phi(f)}\bar\phi(h) \subset \bar\phi(h)e^{i\lambda\bar\phi(f)}$ for real h whose support is spacelike separated from that of f; $\bar\phi(h)$ being self-adjoint, the group of $\bar\phi(f)$ commutes with its spectral projections: the closures commute strongly. (f5) Lem. 14.10(b)–(d) gives (LB_f) for every real $f\in C_c^\infty$, so (f2) at every f_0.

The last assertion of (f): if (f2) holds at f_*, time reversal gives the two-sided bound $c_*(\partial_t f_*) < \infty$, and with Lem. 14.8(b), (c) the functions $f_i = g^{\otimes 3}_{1/i,3/2} = f_* * \mu_i^{\otimes 3}$ have $c_*(\partial_t f_i)\le\|\mu_i\|_{L^1}^3 c_*(\partial_t f_*)$, hence $b_{mn}(\partial_t f_i)\le\|\mu_i\|_{L^1}^3 c_*(\partial_t f_*)(m+n+2)$ by polarisation; these functions have the properties that (f1) asks of the f_i.

(g) By Lem. 14.7(ii), $\phi_u([F], M_{g^2}[F]) = \int g^2\phi_u^\xi\,\mathrm{d}\lambda$ and $\langle[F], M_{g^2}N[F]\rangle = \int g^2\langle[F]_\xi, N_\xi[F]_\xi\rangle\,\mathrm{d}\lambda$; at $g=1$ the bound passes from all σ_ξ to σ; conversely, the bound for σ at the vectors $M_g[F]$ gives $\int g^2(c\,\mathsf{n}-\mathsf{a})\,\mathrm{d}\lambda\ge 0$ for all real g, with $\mathsf{a}(\xi) = \phi_u^\xi([F]_\xi,[F]_\xi)$ and $\mathsf{n}(\xi) = \langle[F]_\xi, N_\xi[F]_\xi\rangle$ continuous, and full support gives $\mathsf{a}\le c\,\mathsf{n}$ at every ξ; the classes $[F]_\xi$ form a core of N_ξ. The premise of (CP) speaks of $\mathfrak{P}$ alone, and $\mathcal{W}_\sigma = \int\mathcal{W}_\xi\,\mathrm{d}\lambda$ (Lem. 14.7(iii)): (CP) passes from all σ_ξ to σ, and so does every fixed inequality $\mathcal{W}(a)\ge 0$.

Conversely, for $a = \sum f^n Y_n$ as in (CP), which is hermitian, and b formed as in Lem. 14.7(iv), a polynomial in means of translates of elements b_i of $\mathfrak{P}$ over pairwise different spatial translations of length $\ge R$, with R so large that the translated b_i are spacelike to K, which puts b in the algebra of (CP), $\sum\lambda^n b^* Y_n b\in\mathfrak{A}^+$ and f is central, so $\mathcal{W}_\sigma(b^*ab)\ge 0$, and Lem. 14.7(iv) gives (CP) for every σ_ξ; (d) turns (CP) into (W). The limits: Lem. 14.7(v); inequalities $\mathcal{W}(a)\ge 0$ pass to the limit.

(h) (h1) Fix $\Lambda < \infty$; on $\mathcal{H}_\Lambda$ let q be the bounded operator of the form

$$\int h'(\vec{x}) q_0(U(-\vec{x})\cdot, U(-\vec{x})\cdot) d^2x;$$

then

$$\langle \psi', \mathsf{l}\psi\rangle = \int \mathsf{l}^{-1}\rho_1(\varsigma/\mathsf{l})\langle e^{-(\frac{\mathsf{l}}{2}+\varsigma)H}\psi', q\, e^{-(\frac{\mathsf{l}}{2}-\varsigma)H}\psi\rangle \,\mathrm{d}\varsigma$$

and

$$\langle \psi', \phi(g_\mathsf{l}\otimes h')\psi\rangle = \int g_\mathsf{l}(t)\langle e^{-itH}\psi', q\, e^{-itH}\psi\rangle \,\mathrm{d}t.$$

An operator integral $\int \alpha_\tau(H) q \beta_\tau(H)\,\mathrm{d}\mu(\tau)$ depends on (α,β,μ) only through its symbol

$$\int \alpha_\tau(E)\beta_\tau(E')\,\mathrm{d}\mu(\tau)$$

(approximate the functions by step functions on a fine partition of $[0,\Lambda]$). With $A(x) = \hat{D}(x) + \epsilon$ and $\log A = \int_0^\infty [(1+t)^{-1} - (t+A)^{-1}]dt$, $L_\epsilon(x) = -\log(\hat{D}(x)+\epsilon)$ is C^2 in norm and $L_\epsilon'(0) = -\int_0^\infty R_t^\epsilon \hat{D}'(0) R_t^\epsilon dt$, $R_t^\epsilon = (t+\epsilon+e^{-\mathsf{l}H})^{-1}$. As $\epsilon \downarrow 0$ on $\mathcal{H}_\Lambda$, $-L_\epsilon'(0)$ tends to the operator integral with the symbol

$$e^{-\frac{1}{2}(E+E')}\int \mathsf{l}^{-1}\rho_1(\varsigma/\mathsf{l}) e^{-\varsigma(E-E')}\,\mathrm{d}\varsigma \int_0^\infty \frac{\mathrm{d}t}{(t+e^{-\mathsf{l}E})(t+e^{-\mathsf{l}E'})} = \mathsf{G}(\mathsf{l}(E-E')),$$

minus $\mathfrak{m}$; and $\phi(g_\mathsf{l}\otimes h')$ has the same symbol: this is the slope. Taylor's formula for $u_\epsilon(x) = \langle\psi, L_\epsilon(x)\psi\rangle$ at $x = \pm r$, with $L_\epsilon \geq -\log(Q+1)$ and $L_\epsilon(0) \leq \mathsf{l}H$, gives

$$r|u_\epsilon'(0)| \leq \mathsf{l}\langle H\rangle_\psi + \log(Q+1)\|\psi\|^2 + \frac{r^2}{2}\max\{\sup_{|x|\leq r} u_\epsilon'', 0\};$$

let $\epsilon \downarrow 0$ and use (v); extend from $\bigcup_\Lambda \mathcal{H}_\Lambda$ to $\mathcal{D}_\infty$ by continuity in a norm $\|N^a\cdot\|$. For every real $k \in \mathcal{S}(\mathbb{R})$, a dyadic partition χ_i of the energy transfer and the functions μ_i with $\hat{\mu}_i = \hat{k}\chi_i/\mathsf{G}(\mathsf{l}_i\cdot)$, $\|\mu_i\|_{L^1} \leq C_N \mathsf{l}_i^N$, write $k = \sum_i \mu_i * g_{\mathsf{l}_i}$ in L^1, and Lem. 14.8(c) sums the bounds. (h2) By Lorentz covariance on $\mathcal{D}_\infty$ the bound of (b1) holds in every frame for vectors with $n\cdot p \leq \mu$: boost the two spectral sets to the rest frame of n, where the argument of (b1) gives $e^2 C_\wp \|u\|_{L^1}\mu^\wp$.

The set $\{p\cdot p \leq \mu^2\}$ is cut into cells: $\{p\cdot p \leq \mu^2,\ E < 2\mu\}$ and, for $a \geq 1$, the cells $\{p\cdot p \leq \mu^2,\ 2^a\mu \leq E < 2^{a+1}\mu\}$ on which the direction of $\vec{p}$ lies in one of $\lceil 2\pi 2^a\rceil$ equal sectors, of angle about 2^{-a}. The frame of such a cell has the rapidity $a\log 2$ in the direction of the middle of its sector, and $n\cdot p < 3\mu$ on the cell. For two cells at the distance D in momentum space let n be the midpoint of the hyperbolic segment between their frames, whose length is d: then $n\cdot p \leq 3\mu e^{\mathsf{d}/2}$ on both cells, and $e^{\mathsf{d}} \leq 4(1+2D/\mu)^2$, so $n\cdot p \leq 6\mu + 12D$ there. A test function whose Fourier transform vanishes for $|k| \geq \frac{3}{4}D$ has a vanishing block between the two cells, the spectral projections being smeared translations; so, cutting $\hat{u} = \hat{\kappa}(\cdot/D)\hat{u} + (1-\hat{\kappa}(\cdot/D))\hat{u}$ with a fixed smooth $\hat{\kappa}$ equal to 1 on $\{|k| \leq \frac{1}{2}\}$ and to 0 on $\{|k| \geq \frac{3}{4}\}$, the block is that of a function of L^1 norm $\leq c\, D^{-8}\|\mathcal{L}^4 u\|_{L^1}$.

For $r\mu \leq D < (r+1)\mu$, $r \geq 1$, the block is therefore $\leq e^2 C_\wp\, c\,(r\mu)^{-8}\|\mathcal{L}^4 u\|_{L^1}(6\mu(2r+3))^\wp$, and for $D < \mu$ it is $\leq e^2 C_\wp \|u\|_{L^1}(18\mu)^\wp$. At most $100(r+1)$ cells lie at a distance $< (r+1)\mu$ from a given cell; Schur's test over the cells, with the weight $r+1$ of this count, gives the bound with the constant C_L, the sum over r converging for $\wp < 6$, with no dependence on the bound of the energy. For these two consequences let $\wp > 0$. On $P_m\mathcal{H} \cap \mathcal{P}_<\mathcal{H}$ the invariant mass is $\leq (m+n+2)^{1/\wp}$, which gives the bound of the blocks. For the last statement, $\Pi P_m = \mathcal{P}_<\Pi P_m$ and $P_n\Pi = P_n\Pi\mathcal{P}_<$ with the $\mathcal{P}_<$ of the pair (m,n), since $p\cdot p \leq (1+p^0)^{2/\wp} \leq (m+n+2)^{2/\wp}$ on both ranges; Π commutes with H, so the argument of (b2) gives $(|m-n|-1)^j\|\Pi P_m\phi(u)P_n\Pi\| \leq \|\Pi P_m\phi(\partial_t^j u)P_n\Pi\|$; the blocks of $\Pi e^{-sH}\phi(u)e^{-sH}\Pi$ are then at most $c\,e^{-s(m+n)}(m+n+2)\min\{1,(|m-n|-1)^{-2}\}$, and Schur's test with $\sup_{x\geq 0}(x+2)e^{-sx} \leq e/s$ gives the bound.

The bound by $1+H+M^2$: cut momentum space into unit cubes; for two cubes at the sup-distance d, $D=\sqrt{3}(d+1)$, and a point p_0 of energy E_0 and invariant mass m_0 in one of them, a timelike vector a with $|a|=(\sqrt{2}\mathsf{M}_1)^{-1}$, $\mathsf{M}_1^2\le 2D^2(1+E_0+m_0^2)$, has $a\cdot p\le 1$ on both cubes, and the point bound in the frame of a bounds the block of $\phi(f)$ between the two cubes by $e^2C_\wp\|f\|_{L^1}(2D)^\wp$ times the least value of $(1+p^0+p\cdot p)^{\wp/2}$ on either cube; derivatives in a coordinate along which the cubes are d apart give the decay $(d-1)^{-k}$ as in (b2), and Schur's test with the weights $(1+p^0+p\cdot p)^{\wp/2}$ of the cubes gives the form bound.

14.1.7. *Proof of Thm. AP. Step 1.* By Thm. YM3_I:AP among the construction results on the limits of periodic Gibbs states, σ is data (Rec) with $(16C_{\mathrm{pt}}^g,4,\frac12)$ and $S_1^\sigma=0$, and its truncated functions S_n^σ, $n\ge 2$, are the limits of the joint cumulants $S_{n;j}^T$ of the $\mathcal{O}_{K_j}(f)$ under $\mu_j=\mu_{\beta_j}$ on Λ_j, with $\ell_j/a_j=2L^{K_j+m_j}$ even, $a_j=L^{-K_j}$ the spacing of Λ_j; every component is data (Rec) with one triple.

Step 2 (frames). Let $\mathcal{H}_j,\iota_{\Lambda_j},\mathcal{M}_j,\pi_j,K_{\Lambda_j},\mathsf{P}_j$ be the objects of Prop. 14.13 on Λ_j at β_j, and, for tuples $G=(h_k)$ over $\mathbb{E}_+$, $(G)_j=\prod_k(\mathcal{O}_{K_j}(h_k)-\mu_j(\mathcal{O}_{K_j}(h_k)))$, the functions h_k periodised on Λ_j, and $\iota_jG=\iota_{\Lambda_j}((G)_j)$, both extended linearly. In Steps 2–7 the sign $(\cdot)_j$ is this product on the torus Λ_j, centred by μ_j; it is not the product $(X)_{K_j}^{\nu_j}$ on $L^{-K_j}\mathbb{Z}^3$ of the setting, and for a tuple G over a quadrant it is, for large j, a real function of the bonds of the closed quadrant of Λ_j, with $\mathsf{r}(G)_j=(\mathsf{r}G)_j$ for $\mathsf{r}\in\{\theta,\theta_1,R\}$ ($\mathsf{r}\mathcal{O}(f)=\mathcal{O}(\mathsf{r}f)$, μ_j being invariant).

For large j the supports lie in $(-\frac{\ell_j}{4},\frac{\ell_j}{4})^3$ and farther than $2a_j$ from the time-zero plane. For $X=\theta G'\cdot G=(f_k)$, (L1) and the invariance of the weight give $\langle\iota_jG',\iota_jG\rangle=\mu_j\prod_k(\mathcal{O}(f_k)-\mu_j\mathcal{O}(f_k))=\sum_\varpi\prod_{\mathsf{b}\in\varpi}S_{|\mathsf{b}|;j}^T(X_\mathsf{b})$, the sum over the partitions without singletons (a moment of centred variables, and a cumulant of order ≥ 2 does not see the centring); by Step 1 it tends to $\sum_\varpi\prod S_{|\mathsf{b}|}^\sigma(X_\mathsf{b})=\sigma(X)=\langle[G'],[G]\rangle$. So $(\mathcal{H}_j,\iota_j)$ is a frame for $(\mathcal{H}(\sigma),[\cdot])$, and Lem. 14.1 applies.

Step 3 (limits of half-plane operators lie in $\mathcal{C}(W')$). Let $Q_j\in\mathcal{M}_j$, $Q_j\rightharpoonup Q$ along a subsequence; take the $\tilde q_j$ of (L3), $\|\tilde q_j\|\le\sup\|Q_j\|$, and a further subsequence with $\tilde q_j\rightharpoonup\tilde q$. For tuples G_1,G_2 over II and large j, the $(G_i)_j$ are real functions of the bonds of $\bar{\mathsf{Q}}_2$, and $R^{-1}(\theta(G_1)_j\cdot(G_2)_j)=(R^{-1}(\theta G_1\cdot G_2))_j$; the limit of (L3) is the hypothesis of Lem. 14.4, so $Q\in\mathcal{C}(W')$.

Step 4 (the lattice modular flows converge to the boosts). Every subsequence of the support projections $e_{R,j}\in\mathcal{M}_j$ has a further one with $e_{R,j}\rightharpoonup\mathsf{E}$; by Step 3, $\mathsf{E}\in\mathcal{C}(W')$, and by Lem. 14.1(iii), $\mathsf{E}\Omega$ is the weak limit of $e_{R,j}\Omega_{\Lambda_j}=\Omega_{\Lambda_j}=\iota_j\varnothing$: $\mathsf{E}\Omega=\Omega$. Ω separates $\mathcal{C}(W')$, so $\mathsf{E}=1$ and $\|(1-e_{R,j})\iota_jG\|^2=\langle\iota_jG,(1-e_{R,j})\iota_jG\rangle\to 0$ along the full sequence; by (L1), $\|(1-e_{L,j})\iota_jG\|=\|(1-e_{R,j})\iota_j(\theta_1\bar G)\|\to 0$. So Lem. 14.1(iv) applies to π_j, and by (L4) (for $G\in\mathcal{V}_0$ and large j, $(G)_j$ is a function of $\bar{\mathsf{Q}}_1$ with $R(G)_j=(RG)_j$) and Lem. 14.2(b), Lem. 14.1(v) applies to P_j, $\Delta^{1/4}$, $\mathsf{r}=R$:

$$\Phi(\mathsf{P}_j)\psi_j\Rightarrow\Phi(\Delta^{1/4})\psi\quad(\Phi\in C_0(\mathbb{R})),\qquad e^{-2\pi itK_{\Lambda_j}}\psi_j\Rightarrow U(B(-2\pi t))\psi,$$

the second from the even functions $\Phi(x)=|x|^{4it}\chi_{\mathsf{M}}(x)$, χ_{M} a cut-off near 0 and ∞, since $|\mathsf{P}_j|=e^{-\frac{\pi}{2}K_{\Lambda_j}}$ and $\Delta^{1/4}$ is injective.

Step 5 (the field on the vacuum is reached). For real $u\in C_c^\infty(\mathsf{I})$ let $\Phi(x)=\mathsf{F}(-\frac{2}{\pi}\log|x|)$, $\Phi(0)=0$: an even function of $C_0(\mathbb{R})$ with $\Phi(\mathsf{P}_j)=\mathsf{F}(K_{\Lambda_j})$ and $\Phi(\Delta^{1/4})=\mathsf{F}(K)$. Step 4 and eq. 14.ii give $\xi_j=\mathsf{F}(K_{\Lambda_j})\pi_j\iota_j(u)\Rightarrow\phi(f_u)\Omega$. (L5) with $F=(u)_j$ gives $T_j^{(\epsilon)}\in\mathcal{M}_j$; along a subsequence $T_j^{(\epsilon)}\rightharpoonup T_u^{(\epsilon)}$, which lies in $\mathcal{C}(W')$ by Step 3, and by Lem. 14.1(i), (iii) and Step 2

$$\|T_u^{(\epsilon)}\Omega-\phi(f_u)\Omega\|\le\tfrac{\sqrt{\epsilon}}{2}\,\pi e^{5\pi^2/8}\|[\theta_1u\cdot u]\|,\tag{14.iv}$$

one Euclidean four-point function at separated points. With $\epsilon\to 0$, $\phi(f_u)\Omega$ lies in the closure of $\mathcal{C}(W')\Omega$.

Step 6 (density, locality). By Lem. 14.2(c) the transforms of the f_u have no common real zero; Thm. AO(d), with W and W' exchanged, gives the density of $\mathcal{C}(W')\Omega$, (W), (CP), the duality for

wedges and the modular data; Thm. AO(a), (e) give (a)–(e) for $\tau = \sigma$; $\mathcal{A}(W) = \mathbb{C}1$ would make $\mathcal{H} = \overline{\mathcal{A}(W)\Omega} = \mathbb{C}\Omega$.

Step 7 (the wedge algebra is made of lattice limits). Let $\mathfrak{M}$ be the von Neumann algebra generated by the operators of (f) and $\mathcal{N} = \mathcal{C}(W') = \mathcal{A}(W)$. For $\mathfrak{g}W \subset W$ one has $\mathfrak{g}W' \supset W'$, so $U(\mathfrak{g})\mathcal{C}(W')U(\mathfrak{g})^{-1} = \mathcal{C}(\mathfrak{g}W') \subset \mathcal{C}(W')$, and $\mathfrak{M} \subset \mathcal{N}$ by Step 3. The closure of $\mathfrak{M}\Omega$ is invariant under $\mathfrak{M}$ and under the $U(a)$, $a \in W$, hence under all translations, and it holds every $\phi(f_u)\Omega$ (Step 5); the one-vector argument of Thm. AO(d), with $\mathfrak{M}$ in the place of $\mathcal{C}(W')$, makes $\mathfrak{M}\Omega$ dense. $\mathfrak{M}\Omega$ and $\mathcal{N}\Omega$ are boost-invariant dense subspaces of $D(\Delta^{1/2})$ on which $Q\Omega \mapsto Q^*\Omega$ is the restriction of $J\Delta^{1/2}$ (Thm. AO(c) for W'), hence cores of it; so the two algebras have the one modular conjugation J, and $\mathfrak{M}' = J\mathfrak{M}J \subset J\mathcal{N}J = \mathcal{N}'$: $\mathcal{N} \subset \mathfrak{M}$.

Step 8 (components). All σ_ξ are data (Rec) with one triple. For $a = \sum f^n Y_n$ as in (CP), which is hermitian, and b formed as in Lem. 14.7(iv), a polynomial in means of translates of elements b_i of $\mathfrak{P}$ over pairwise different spatial translations of length $\geq R$, with R so large that the translated b_i are spacelike to K and so in the algebra of (CP), $\sum \lambda^n b^* Y_n b \in \mathfrak{A}^+$ for all real λ and f is central, so $\mathcal{W}_\sigma(b^*ab) \geq 0$ by Step 6; Lem. 14.7(iv) gives (CP) for every σ_ξ, and Thm. AO(d), (a), (e) give (a)–(e) for $\tau = \sigma_\xi$. The same follows from the vectors $M_g\Omega$ of Lem. 14.7(i): $M_g \in \mathcal{C}(O)$ for every O, and the extension of the proof of (W)⇒(CP) to the vector $M_g\Omega$ gives $\int g^2 \mathcal{W}_\xi(a)\,\mathrm{d}\lambda \geq 0$ for all real $g \in C(\mathsf{X})$.

Step 9 (fibres). The operators M_g of Lem. 14.7(i) commute with the semigroup, the translations and the fields, so they lie in $\mathcal{C}_\sigma(O)$ for every O, hence in $\mathcal{C}_\sigma(W') \cap \mathcal{C}_\sigma(W) = \mathcal{A}_\sigma(W) \cap \mathcal{A}_\sigma(W)'$ (Step 6). The rest is reduction theory: on a separable Hilbert space that is a direct integral over (X, λ), X compact metrisable and λ a Borel measure, the decomposable operators are the commutant $\mathcal{Z}'$ of the diagonal algebra $\mathcal{Z}$; a von Neumann algebra $\mathcal{N}$ with $\mathcal{Z} \subset \mathcal{N} \subset \mathcal{Z}'$ is $\int^\oplus \mathcal{N}_\xi\,\mathrm{d}\lambda$ for a measurable field of von Neumann algebras, whose fibres are those of a countable generating family and are fixed up to a λ-null set; and $(\int^\oplus \mathcal{N}_\xi\,\mathrm{d}\lambda)' = \int^\oplus \mathcal{N}'_\xi\,\mathrm{d}\lambda$. (1) The classes depend continuously on the entries, with constants of one triple for σ and for all ξ; so $\mathcal{H}(\sigma)$ is separable, and there are countably many combinations F_n whose classes $[F_n]_\xi$ are total in every $\mathcal{H}(\sigma_\xi)$; and $\xi \mapsto \sigma_\xi(\theta\bar{F}' \cdot F)$ is continuous: the fields $\xi \mapsto [F_n]_\xi$ are a fundamental sequence.

The Hilbert space $\mathfrak{K}$ of the proof of Lem. 14.7(i) is the direct integral $\int^\oplus \mathcal{H}(\sigma_\xi)\lambda(\,\mathrm{d}\xi)$: $g \otimes F \mapsto (g(\xi)[F]_\xi)_\xi$ is isometric by the form of $\mathfrak{K}$, and onto, as a field orthogonal to all $g[F_n]$, $g \in C(\mathsf{X})$, has $\langle [F_n]_\xi, x_\xi\rangle = 0$ for almost every ξ and all n. So $[F] \mapsto ([F]_\xi)_\xi$ is unitary, $\mathcal{H}(\sigma) = \int^\oplus \mathcal{H}(\sigma_\xi)\lambda(\,\mathrm{d}\xi)$, and M_g is the multiplication by g; the diagonal algebra $\mathcal{Z}$ is the weak closure of the M_g. (2) The semigroup, the translations and the insertions decompose, being given by one formula for σ and for all σ_ξ; so do the bounded $e^{-sH}\phi(f)e^{-sH}$, by eq. 14.i; as the e^{-sH_ξ} are injective and the polynomial bound holds with one triple, a subsequence converging fibrewise gives $(\phi(f)\psi)_\xi = \phi_\xi(f)\psi_\xi$ for almost every ξ, $\psi \in \mathcal{D}_\infty$, and by induction the fibres of $\phi(f_1)\cdots\phi(f_n)\Omega$ are $\phi_\xi(f_1)\cdots\phi_\xi(f_n)\Omega_\xi$ almost everywhere. (3) $M_g \in \mathcal{A}_\sigma(W)'$ and weak closure give $\mathcal{Z} \subset \mathcal{A}_\sigma(W)'$, so $\mathcal{A}_\sigma(W) \subset \mathcal{Z}'$ consists of decomposable operators; $M_g \in \mathcal{A}_\sigma(W)$ gives $\mathcal{Z} \subset \mathcal{A}_\sigma(W)$: $\mathcal{Z}$ lies in the centre of $\mathcal{A}_\sigma(W)$, and $\mathcal{A}_\sigma(W) = \int^\oplus \mathfrak{A}_\xi\,\mathrm{d}\lambda$ for a measurable field of von Neumann algebras $\mathfrak{A}_\xi$. (4) A decomposable bounded Q lies in $\mathcal{C}_\sigma(W')$ if and only if $Q_\xi \in \mathcal{C}_{\sigma_\xi}(W')$ for almost every ξ, and the same with W for W': "if" by integration of the relation of the fibres, by (2); "only if" because $\mathcal{C}_\sigma(W')$ is an algebra that holds the M_g, so that $\int g\,[\langle \phi_\xi(\bar{f})\psi_\xi, Q_\xi\psi'_\xi\rangle - \langle \psi_\xi, Q_\xi\phi_\xi(f)\psi'_\xi\rangle]\,\mathrm{d}\lambda = 0$ for all $g \in C(\mathsf{X})$; the bracket vanishes almost everywhere, with one null set for countably many test functions $f \in \mathcal{F}(W')$ and products of fields, and the polynomial bound with one triple carries it to all. (5) A countable family that generates $\mathcal{A}_\sigma(W) = \mathcal{C}_\sigma(W')$ has, by (4), its fibres in the von Neumann algebra $\mathcal{C}_{\sigma_\xi}(W')$ for almost every ξ, and these fibres generate $\mathfrak{A}_\xi$: $\mathfrak{A}_\xi \subset \mathcal{C}_{\sigma_\xi}(W')$ almost everywhere. (6) $\mathcal{A}_\sigma(W)' = \mathcal{C}_\sigma(W) = \int^\oplus \mathfrak{A}'_\xi\,\mathrm{d}\lambda$, and (4) for W, read on a countable generating family, gives $\mathfrak{A}'_\xi \subset \mathcal{C}_{\sigma_\xi}(W)$ almost everywhere; with wedge duality for each σ_ξ (Step 8), $\mathfrak{A}_\xi \supset \mathcal{C}_{\sigma_\xi}(W)' = \mathcal{C}_{\sigma_\xi}(W')$.

So $\mathfrak{A}_\xi = \mathcal{C}_{\sigma_\xi}(W') = \mathcal{A}_{\sigma_\xi}(W)$ for λ-almost every ξ, every ξ with $\lambda(\{\xi\}) > 0$ among them, as such a point is no null set, and $\mathcal{A}_\sigma(W) = \int^\oplus \mathcal{A}_{\sigma_\xi}(W)\,\lambda(\mathrm{d}\xi)$: for λ-almost every ξ the wedge algebra of the component σ_ξ is the fibre of the algebra of (f), the fibres of a direct integral being fixed up to null sets.

Step 10 (the supply). Here $(F)_j = (F)^{\rho_j}_{K_j}$ is the product at K_j centred by ρ_j, as in (h), also for a component, where the states $\rho_{j,\xi}$ integrate and the ρ_j give the centring. Reflection positivity of ρ_j at the site mirror $x_3 = 0$: a bounded Borel function B of bonds in the mirror is fixed by θ, and with $\mathsf{c} = (\sup|B|^2 - |B|^2)^{1/2}$, $\sup|B|^2\rho_j(\overline{\theta G}G) - \rho_j(\overline{\theta(BG)}BG) = \rho_j(\overline{\theta(\mathsf{c}G)}\mathsf{c}G) \geq 0$; the Schwarz inequality gives the first bound. $\mathcal{H}(\sigma')$ is separable; a diagonal subsequence over a countable set of combinations with dense classes, and the first bound, give the limit form and Q, $\|Q\| \leq \mathsf{b}$. For $c = 0$ and combinations F, F' over Q_L put $A_j = B_j\overline{\theta(F')_j}(F)_j$; then $\ell^Q_{F',F}(h) = \lim_j \rho_j(A_j(h)_j)$, and the Schwarz inequality for the reflection positivity at the site mirror $x_1 = 0$ gives

$$|\rho_j(A_j(h)_j)|^2 \leq \rho_j(A_j\,\theta_1\bar{A}_j)\,\rho_j(\overline{\theta_1(h)_j}(h)_j),$$

whose second factor tends to $\sigma'(\theta_1\bar{h}\cdot h)$; in the first, with $Y = F\cdot\theta_1\bar{F}$, $Y' = F'\cdot\theta_1\bar{F}'$ and the function $B_j\theta_1\bar{B}_j$ of the bonds of the mirror $x_3 = 0$, $A_j\theta_1\bar{A}_j = \overline{\theta(Y')_j}\,B_j\theta_1\bar{B}_j\,(Y)_j$, and the first mirror, used a second time, bounds it by $\mathsf{b}^2\|[Y']\|\|[Y]\|$ in the limit. So the hypothesis of Lem. 14.5 holds and $Q \in \mathcal{C}_{\sigma'}(W)$. For $c \neq 0$, ce_1 is a lattice vector for large j; the translate $\tau_{-ce_1}B_j$ is of the case $c = 0$, and by the invariance of ρ_j its limit is $U(ce_1)^{-1}QU(ce_1)$. The proof uses of the states that integrate only reflection positivity at the site mirrors $x_3 = 0$, $x_1 = 0$ and invariance under the lattice translations, which the lattice approximants of a component have; so the same holds for every component.

Thms. YM3_I:D, YM3_I:AJ and YM3_I:AD enter only through Thm. YM3_I:AP.

14.1.8. *Proof of Thm. AQ.* Every bound below passes to the components through the last sentence of Lem. 14.15(d), which carries the norm bound of every insertion from σ' to each component; every component is data (Rec) with one triple.

(a) Step 1: the blocks at one time. Let $h \in C^1_c(\mathbb{R}^2)$ be real and carried by a square of side 2, $q = T_{mn}(h)$, $A = (H-m)P_m$, $B = (H-n)P_n$, so that $0 \leq A, B \leq 1$. As $P_m\mathsf{B}_h(s,s')P_n = e^{-sH}q\,e^{-s'H}$,

$$e^{\frac{\mathsf{l}}{2}(m+n)}P_m\mathsf{I}_\mathsf{l}(h)P_n = \int \rho_\mathsf{l}(\varsigma)e^{-\varsigma(m-n)}\,e^{-(\frac{\mathsf{l}}{2}+\varsigma)A}\,q\,e^{-(\frac{\mathsf{l}}{2}-\varsigma)B}\,\mathrm{d}\varsigma,$$

and $\|e^{-uA}qe^{-u'B} - q\| \leq (u+u')\|q\|$ with $u + u' = \mathsf{l}$; since $\int \rho_\mathsf{l}(\varsigma)e^{-\varsigma(m-n)}\,\mathrm{d}\varsigma = \int \rho_\mathsf{l}(\varsigma)\cosh(\varsigma(m-n))\,\mathrm{d}\varsigma \geq 1$, $\|q\| \leq (1-\mathsf{l})^{-1}e^{\frac{\mathsf{l}}{2}(m+n)}\|\mathsf{I}_\mathsf{l}(h)\|$. Take the least $i \geq 1$ with $L^{-i} \leq 2/(m+n+2)$: then $e^{\frac{\mathsf{l}}{2}(m+n)} \leq e$ and $\mathsf{l}^{-1} \leq L(m+n+2)$, and Lem. 14.15(c) gives

$$\|T_{mn}(h)\| \leq \tfrac{3e}{2}L^3\mathsf{C}_3(m+n+2)^3|h|_{C^1}.$$

Step 2: the field. By eq. 14.i, $\langle\chi', P_m\phi(f)P_n\chi\rangle = \int\langle e^{-itH}\chi', T_{mn}(f_t)e^{-itH}\chi\rangle\,\mathrm{d}t$ with $f_t = f(t,\cdot)$; a partition of unity on $\mathbb{R}^2$ subordinate to the squares $\kappa + [-1,1]^2$, $\kappa \in \mathbb{Z}^2$, splits the real and imaginary parts of f_t into pieces of norm $|\cdot|_{C^1} \leq 3.3\sup_{\kappa+[-1,1]^2}(|f_t| + |\nabla f_t|)$, and $2\cdot 3.3\cdot\frac{3e}{2} < 27$ gives $b_{mn}(f) \leq \mathsf{c}_0(m+n+2)^3$; Thm. AO(b2), which holds for all data, gives the decay. Step 3: the operator and form bounds. $\|P_m\phi(f)\psi\| \leq \sum_n b_{mn}(f)\|P_n\psi\|$, and $m+n+2 \leq 2(1+n)(1+|m-n|)$; so $\|\phi(f)\psi\| \leq \|\kappa\|_{\ell^1}\|N^3\psi\|$ by Young's inequality, $\kappa(d) = 8(1+|d|)^3\min\{\mathsf{c}_0, \mathsf{c}_5(|d|-1)^{-5}\}$, and $\sum_{|d|\leq 2}(1+|d|)^3 = 71$, $\sum_{|d|\geq 3}(1+|d|)^3(|d|-1)^{-5} < 6.3$; with a weight $(1+m)^a \leq (1+n)^a(1+|d|)^{|a|}$ and a larger k the same gives the bound with N^{a+3}.

For the form, $(m+n+2)^2 \leq 4(1+m)(1+n)(1+|d|)$ and Schur's test with $\kappa'(d) = 8(1+|d|)^{3/2}\min\{\mathsf{c}_0, \mathsf{c}_3(|d|-1)^{-3}\}$, whose two sums are less than 17.1 and 5.7, respectively. The blocks of $e^{-sH}\phi(f)e^{-sH}$ are $\leq e^{-s(m+n)}b_{mn}(f)$, and $\sup_{x\geq 0}(x+2)^3e^{-sx} \leq (3/s)^3e^{2s}$. Step 4: growth.

For $|m-n| = d \geq 2$, the bound of Step 2 at $\partial_t^k f$ with $\|\partial_t^k f\|_\star \leq A^{k+1}(k!)^{\mathsf{s}} \leq A^{k+1}k^{\mathsf{s}k}$, at $k = \lfloor((d-1)/e^{\mathsf{s}}A)^{1/\mathsf{s}}\rfloor$, gives

$$b_{mn}(f) \leq \mathsf{G}\,(m+n+2)^3 e^{-\gamma'|m-n|^{1/\mathsf{s}}} \qquad \text{for all } m,n, \qquad \gamma' = \mathsf{s}e^{-1}A^{-1/\mathsf{s}},\ \mathsf{G} = 27L^3\mathsf{C}_3 A e^{\mathsf{s}+\gamma'}.$$

As $\Omega = P_0\Omega$, $\|\phi(f)^n\Omega\| \leq \sum \prod_{k\leq n} b_{m_k m_{k-1}}(f)$ over paths $0 = m_0, m_1, \dots, m_n$; with $\mathsf{D} = \sum_k |m_k - m_{k-1}| \geq \max_k m_k$, at most 2^n paths for each sequence of steps, and $\sum_k |d_k|^{1/\mathsf{s}} \geq \mathsf{D}^{1/\mathsf{s}}$,

$$\|\phi(f)^n\Omega\| \leq (2\mathsf{G})^n \sup_{\mathsf{D}\geq 0}(2\mathsf{D}+2)^{3n} e^{-\frac{\gamma'}{2}\mathsf{D}^{1/\mathsf{s}}} \Big(\sum_{d\geq 0} e^{-\frac{\gamma'}{2}d^{1/\mathsf{s}}}\Big)^n \leq C^n (n!)^{3\mathsf{s}}.$$

Such f exist: $w \otimes b$ with $b \in C_c^\infty(\mathbb{R}^2)$, $b \geq 0$, and w the infinite convolution of the uniform probability densities on $[0, i^{-\mathsf{s}}]$, $i \geq 1$, for which $\|w^{(k)}\|_{L^1} \leq \prod_{i\leq k} 2i^{\mathsf{s}} = 2^k (k!)^{\mathsf{s}}$, because the k-th derivative falls on the first k factors, each becoming a difference of two point masses divided by the width.

(b) The same four steps with the second bound of Lem. 14.15(c), $\|T_{mn}(h)\| \leq \frac{3e}{2} L^2 \mathsf{C}_2 (m+n+2)^2 |h|_{C^1}$, the exponent 2 in the place of 3, the weight $(1+n)^2$ in Step 3 for the operator bound and $[(1+m)(1+n)]^1$ for the form bound, $\sup_x (x+2)^2 e^{-sx} \leq (2/s)^2 e^{2s}$, and $(n!)^{2\mathsf{s}}$ at the end.

(c) The point. Put $\bar{B} = B_\sharp L/(L-1)$, where $B_\sharp \geq 0$ is one of the fixed constants σ^-. Thus $\bar{B}$ is independent of the source radius, γ and g, and $\bar{B} < \gamma^{-2} \leq g^{-2}$. For an integer $z \geq 2$ set

$$v_z^{-2} = L^{z-1}(g^{-2} + B_\sharp - L^{-1}\bar{B}) + L^{-1}\bar{B} - B_\sharp, \qquad G = g^{-2} + 2\bar{B}.$$

Then $L^{z-1}g^{-2} \leq v_z^{-2} \leq L^z G$. Define

$$\Lambda_z = 1 + \log(L^z G), \qquad \varrho_z = \frac{1}{2}\min\{\sigma^{\mathrm{S}}(v_z^{-2} - \bar{B}),\ \sigma_{\mathrm{L}}^{\mathrm{S}} v_z^{-1}(\log v_z^{-2})^{-\mathfrak{p}_0}\}.$$

Here σ^{S} and $\sigma_{\mathrm{L}}^{\mathrm{S}}$ depend only on σ^-. We check explicitly that this radius is allowed. The radius-independent conditions remain valid when the coupling ceiling is decreased from γ to v_z. The two source-radius conditions in Prop. YM3_I:82.2, with source radius $2\varrho_z$, are

$$2\varrho_z \leq \sigma^{\mathrm{S}}(x - \bar{B}), \qquad 2\varrho_z x^{-1/2}(\log x)^{\mathfrak{p}_0} \leq \sigma_{\mathrm{L}}^{\mathrm{S}} \quad (x \geq v_z^{-2}).$$

The first left margin is smallest at $x = v_z^{-2}$. The calibration also gives $\log \gamma^{-2} > 2\mathfrak{p}_0$. Therefore $x^{-1/2}(\log x)^{\mathfrak{p}_0}$ is decreasing for $x \geq v_z^{-2} \geq \gamma^{-2}$, and the second condition is likewise strongest at that endpoint. Both conditions hold by the definition of ϱ_z. This proves the required admissibility and also identifies ϱ_z as the largest radius permitted by these two source conditions at the fixed ceiling v_z.

Let (K, β) be admitted at g, $K > z$. Its label $\mathsf{b} = L^{-K}\beta$ lies in the window YM3_I:(1531) of Prop. YM3_I:74.1, $g^{-2} + B_\sharp \frac{L-2}{L-1} < \mathsf{b} \leq L(g^{-2} + \bar{B})$; so the running coupling g', $g'^{-2} = L^{-1}(L^z\mathsf{b} + \bar{B}) - B_\sharp$, at which the label $L^z\mathsf{b}$ of $(K-z, \beta)$ passes first, has $v_z^{-2} < g'^{-2} \leq L^z G$ and $1 + \log g'^{-2} \leq \Lambda_z$. At the radius ϱ_z and the choice (σ^-, v_z) the dilated pair $(K-z, \beta)$ is a first-passage datum at $g' \leq v_z$, and it carries systems of localised source energies with the constants $(2\varrho_z; C_{\mathrm{S}}, \kappa; C_{\mathrm{S}}, C_{g'}; C_{\mathrm{gap}})$ ([7, Lem. N.1]), where $C_{\mathrm{S}}, \kappa, C_{\mathrm{gap}}$ are the numbers of the pairs admitted at g, formed from σ^- alone, and $C_{g'} \leq C_{\mathrm{S}}(1 + \log g'^{-2}) \leq C_{\mathrm{S}}\Lambda_z$ [7, eq. 288].

As $v_z^{-2} - \bar{B} \geq \frac{12}{13} L^{z-1} g^{-2}$, $v_z \leq g L^{-(z-1)/2}$, $\log v_z^{-2} < \Lambda_z$ and $g \leq \frac{1}{2}$,

$$\frac{1}{\varrho_z} \leq 2c_\sigma\, g\, L^{-(z-1)/2} \Lambda_z^{\mathfrak{p}_0}, \qquad \Lambda_z = 1 + z\log L + \log G, \qquad G = g^{-2} + 2\bar{B},$$

$c_\sigma = \max\{\frac{13}{24\sigma^{\mathrm{S}}}, \frac{1}{\sigma_{\mathrm{L}}^{\mathrm{S}}}\}$, formed from σ^- alone: the radius of analyticity in the source grows like $1/g'$, the square root of the label g'^{-2}, up to logarithms. This is where Thms. YM3_I:AJ and YM3_I:AD are used at every radius and every admitted choice. Fix $0 < \eta \leq 1$, a real $h \in C_c^1$ carried by a ball $B(y, \frac{\eta}{40L})$ with $|y_3| \leq \frac{\eta}{8}$, and F, F' over $\{x_3 > \frac{\eta}{4}\}$. The cell: $\mathsf{l} = L^{-i}$ with $\frac{\eta}{5L} < \mathsf{l} \leq \frac{\eta}{5}$ and a centre $y^* \in L^{-i-2}\mathbb{Z}^3$ within $\frac{\mathsf{l}}{2L^2}$ of y; then the ball lies in the middle of the cell, and the cell in $\{|x_3| < \frac{\eta}{4}\}$.

The dilation by L^z among the construction results ([7, Lem. K.1]) carries periodic Gibbs states, mirrors and boards to the like and $\mathcal{O}_K(h)$ to $\mathcal{O}_{K-z}(L^z h(L^{-z}\cdot))$; it carries the cell to one of side $L^z\mathsf{l} \le L^z\eta/5$, and h to a source of norm $\max\{L^z \sup|h|, 40M \operatorname{Lip} h\}$, which is $\le \|h\|_\sharp$ as long as $L^z\eta \le 5$, since h vanishes on the boundary of its ball, so that $\sup|h| \le \frac{\eta}{40L}\operatorname{Lip} h$. The insertion bound on one lattice for a pair that carries systems, Lem. YM3_I:104.4 among the construction results, read on the dilated lattice at the radius ϱ_z, has the constant $2\mathsf{Q}_{\mathrm{mid}}(L^z\mathsf{l})/\varrho_z$ at the side of the one cell, with $\log \mathsf{Q}_{\mathrm{mid}}(\mathsf{l}') = (C_{\mathrm{S}} + C_{g'} + C_{\mathrm{gap}})\mathsf{l}'^3 + \Phi_1(\frac{\mathsf{l}'}{4}, \frac{\mathsf{l}'}{4}; \mathsf{l}')$ (equation YM3_I:(2017)) and $\Phi_1(\frac{\mathsf{l}'}{4}, \frac{\mathsf{l}'}{4}; \mathsf{l}') \le \bar\Phi$ for $0 < \mathsf{l}' \le 1$, where

$$\bar\Phi = C_{\mathrm{S}} S(\kappa)\left[\frac{27}{26} + 27\{32(C''_\Sigma + \log_L 10) + 4C'_\Sigma\}\right]$$

is the contribution of Φ_1 in equation YM3_I:(2018); so $\mathsf{Q}_{\mathrm{mid}}(L^z\mathsf{l}) \le Q_\star = e^{2C_{\mathrm{S}} + C_{\mathrm{gap}} + \bar\Phi}$, free of z, as long as $(L^z\eta/5)^3\Lambda_z \le 1$.

This is the proof of Prop. YM3_I:105.1 up to its last step, with the side of the one cell in the place of the supremum over the sides. With z fixed and the cutoff removed, $|\sigma'(\theta\bar F \cdot (h) \cdot F')| \le \frac{2Q_\star}{\varrho_z}\|h\|_\sharp \|[\tau_{-\frac{\eta}{4}e_3}F]\| \|[\tau_{-\frac{\eta}{4}e_3}F']\|$ for every $z \ge 2$ with $(L^z\eta/5)^3\Lambda_z \le 1$. As $z \mapsto (L^z\eta/5)^3\Lambda_z$ increases, such a z exists if and only if $\eta \le \eta_1 = 5L^{-2}\Lambda_2^{-1/3}$, a number $< L^{-1}$. At the largest such z: the condition gives $L^z \le 5/\eta$, so $\Lambda_{z+1} = 1 + \log L + z\log L + \log G \le \mathsf{L}(\eta)$, $\mathsf{L}(\eta) = 1 + \log(5LG) + \log\frac{1}{\eta}$; its failure at $z+1$ gives $L^{-z} < \frac{L\eta}{5}\Lambda_{z+1}^{1/3}$; hence $L^{-(z-1)/2} < \frac{L}{\sqrt{5}}\eta^{1/2}\mathsf{L}(\eta)^{1/6}$, and the insertion bound for a source in a ball of radius $\eta/40L$ holds for $\eta \le \eta_1$ with the constant $A_1\eta^{1/2}\mathsf{L}(\eta)^q$, $A_1 = \frac{4}{\sqrt{5}}LQ_\star c_\sigma g$, $q = \mathfrak{p}_0 + \frac{1}{6}$, formed from σ^- and g alone.

For $\eta_1 < \eta \le 1$ Prop. YM3_I:105.1 itself, for the undilated pairs, gives the constant $C_{\mathrm{I}} = 2\bar{\mathsf{Q}}_{\mathrm{mid}}/\varrho$, and $C_{\mathrm{I}} \le C_{\mathrm{I}}\eta_1^{-1/2}\eta^{1/2}\mathsf{L}(\eta)^q$ as $\mathsf{L}(\eta) \ge 1$. So for all $0 < \eta \le 1$ the bound holds with the running constant $\mathsf{c}_*(\eta) = A\,\eta^{1/2}\mathsf{L}(\eta)^q$, $A = \max\{A_1, C_{\mathrm{I}}\eta_1^{-1/2}\}$; the radius ϱ enters A through C_{I} alone. The data dilated by L^k, $\sigma'_k = \sigma' \circ \mathsf{D}_k^{-1}$, where $\mathsf{D}_k(f_1, \dots, f_n) = (L^k f_i(L^{-k}\cdot))_i$, extended linearly, commutes with θ, the conjugation and the product of tuples, has $\mathsf{D}_k\tau_v = \tau_{L^k v}\mathsf{D}_k$ and keeps the norms of the classes, are again data with this property, with $\mathsf{c}_*(L^{-k}\cdot)$, as $\|L^{-k}h''(L^k\cdot)\|_\sharp \le \|h''\|_\sharp$ for $k \ge 0$; and Prop. YM3_I:97.2 among the construction results, which turns an insertion bound for sources in small balls with a constant C_{I} into the point bound of order 4 with the constant $c_{\mathrm{pt}}C_{\mathrm{I}}$, $c_{\mathrm{pt}} = 320 \cdot 8^4\,\mathsf{mv}_3(40L)^4M$ the coefficient of C_{I} in equation YM3_I:(1919) at $c_{\mathrm{I}} = \frac{1}{40L}$ (with $\mathsf{mv}_3 = m_3$ in the notation following that equation), applied to σ'_k at $\eta'' = L^k\eta \in (L^{-1}, 1]$ and carried back, gives

$$\begin{aligned}&|\sigma'(\theta\bar F \cdot (h) \cdot F')| \\ &\le 16\, c_{\mathrm{pt}}\, A\, s_q\, L^{1/2}\, \eta^{-7/2}\mathsf{L}(\eta)^q\, \|h\|_{L^1}\, \|[\tau_{-\eta e_3}F]\|\, \|[\tau_{-\eta e_3}F']\|, \\ &s_q = \sup_{u \ge 0} e^{-u/2}(1+u)^q,\end{aligned}$$

for F, F' over $\{x_3 > \eta\}$ and h carried by $\{|x_3| < \frac{1}{2}\eta\}$; this is (Rec4) with $(C_\epsilon, \frac{7}{2} + \epsilon, \frac{1}{2})$, $C_\epsilon = 16c_{\mathrm{pt}}As_qL^{1/2}\sup_{\eta \le 1}\eta^\epsilon\mathsf{L}(\eta)^q$, and the point bound with the logarithm follows by taking $\epsilon = q/\mathsf{L}(\mu)$.

One time. A chain of dilates $y = 0, 1, \dots, Y$ of one pair bounds the coefficients of the mixed part of the source energies that see two cells. Let (K, β) be admitted at g, with the label $\mathsf{b} = L^{-K}\beta \ge g^{-2}$, (i, o) a board with cells of side $\mathsf{l} = L^{-i}$, $f \ne 0$ a real Lipschitz source vanishing off the middle of one cell, ℓ' the side of a reference torus, $v'^{\pm}$ the sums of the mirror copies of f in the even and in the odd cells of the board on the torus of side ℓ', and $\mathsf{N} = (\ell'/\mathsf{l})^3$. The normalised partition function $F(z, z') = Z_{K,\ell',\beta}[zv'^+ + z'v'^-]/Z_{K,\ell',\beta}$, $Z_{K,\ell',\beta}[w]$ the partition function of the Wilson weight with the source w on the torus $L^{-K}\mathbb{Z}^3/\ell'\mathbb{Z}^3$, is entire with $F(0,0) = 1$, and under the dilation by L^y it is

the same number for the pair $(K-y,\beta)$ on the torus of side $L^y\ell'$ with the arrays $L^y v'^{\pm}(L^{-y}\cdot)$: one function for all $0\le y<K$.

For $y\ge 1$ put $x_y=\frac12 L^y\mathsf{b}$, $g_y=x_y^{-1/2}$, $2\varrho_y=\min\{\sigma^{\mathrm S}(x_y-\bar B),\sigma^{\mathrm S}_{\mathrm L}x_y^{1/2}(\log x_y)^{-\mathfrak p_0}\}$, $\lambda_y=2+\log x_y$, and $\varrho_0=\varrho$, $\lambda_0=2+\log g^{-2}$: at the radius ϱ_y and the choice (σ^-,g_y) the pair $(K-y,\beta)$ is a first-passage datum at g_y and carries systems of localised source energies with the constants $(2\varrho_y;C_{\mathrm S},\kappa;C_{\mathrm S},C_{g_y};C_{\mathrm{gap}})$, $C_{\mathrm S}+C_{g_y}\le C_{\mathrm S}\lambda_y$, as in (c). The system of the y-th dilate gives a factorisation $F=e^{S_y}T_y$ on the bidisc of radius $r_y=\varrho_y/\|f\|_\sharp$ (the dilated source keeps the norm $\|f\|_\sharp$ while $L^y\mathsf l\le 1$): S_y, with $S_y(0,0)=0$, is the sum of its source energies at the dilated arrays, and T_y is the remainder normalised at 0, with $|T_y|\le e^{\mathsf N V_y}$, $V_y=C_{\mathrm S}\lambda_y(L^y\mathsf l)^3$, by the stability bounds of the systems (Section YM3_I:88).

If one function on a bidisc has two factorisations $e^{S_1}T_1=e^{S_2}T_2$ with $S_\iota(0,0)=0$, $T_\iota(0,0)=1$, $|T_\iota|\le e^{B_\iota}$, Jensen's inequality on the torus of the bidisc, $\int|\log|T_\iota||\le 2B_\iota$ for each factor, bounds the Taylor coefficients of S_1-S_2 on a bidisc of radius r: $|(S_1)_{ab}-(S_2)_{ab}|\le 4(B_1+B_2)r^{-(a+b)}$ for $(a,b)\ne(0,0)$, as they are Fourier coefficients of $2\mathrm{Re}(S_1-S_2)=2(\log|T_2|-\log|T_1|)$. For the mixed parts per cell, $\mathsf m_y=\mathsf N^{-1}S_y^{\mathrm{mix}}$, $S^{\mathrm{mix}}(z,z')=S(z,z')-S(z,0)-S(0,z')+S(0,0)$ the sum of the terms $z^a z'^b$ with $a,b\ge1$, $\mathsf m_0$ being the mixed part of the proof of Lem. 14.15, the dilates y and $y+1$ give the chain

$$|\mathsf m_{0;ab}|\le|\mathsf m_{Y;ab}|+\sum_{y=0}^{Y-1}4(V_y+V_{y+1})\min\{r_y,r_{y+1}\}^{-(a+b)}\qquad(a,b\ge1),$$

and in the top dilate Cauchy's estimate on the bidisc of radius r_Y with the two counting estimates in the proof of the cell bound (Prop. YM3_I:89.1) gives $|\mathsf m_{Y;ab}|\le\bar\Phi r_Y^{-(a+b)}$. The radii have $\varrho_y\ge\underline\varrho_y=\frac12\sigma_\wedge x_y^{1/2}(\log x_y)^{-\mathfrak p_0}$, $\sigma_\wedge=\min\{\frac{11}{13}\sigma^{\mathrm S},\sigma^{\mathrm S}_{\mathrm L}\}$, and $\underline\varrho_y/\underline\varrho_{y'}\ge L^{-(y'-y)/2}$ for $1\le y\le y'$. As $V_y\le L^{-3(Y-y)}V_Y$ while $\underline\varrho_y^{-k}\le L^{(Y-y)k/2}\underline\varrho_Y^{-k}$, the terms $1\le y<Y$ of the chain form a geometric sum, at most $5.6L^{k/2}V_Y\underline\varrho_Y^{-k}\|f\|_\sharp^k$ for $k=a+b\le5$, and the term $y=0$ is at most $8L^3C_{\mathrm S}\bar\lambda_1\mathsf l^3\varrho_\wedge^{-k}\|f\|_\sharp^k$, with $\bar\lambda_1=2+\log(\frac12L^2(g^{-2}+\bar B))$ and $\varrho_\wedge=\min\{\varrho,\frac12\sigma_\wedge(\frac12Lg^{-2})^{1/2}(\log(\frac12Lg^{-2}))^{-\mathfrak p_0}\}$.

Take $Y=i-r(\mathsf l)$, $r(\mathsf l)$ the least $r\ge1$ with $L^{3r}\ge\bar\lambda(\mathsf l)=2+\log\frac{L(g^{-2}+\bar B)}{2\mathsf l}$: in the dilate Y the cell has the side $L^{-r(\mathsf l)}$, of the order $\bar\lambda(\mathsf l)^{-1/3}$, so that $V_Y\le C_{\mathrm S}$ and $\underline\varrho_Y^{-2}\le 8L\sigma_\wedge^{-2}g^2\,\mathsf l\,\bar\lambda(\mathsf l)^{2\mathfrak p_0+1/3}$; and $\mathsf lL^2\bar\lambda(\mathsf l)^{1/3}\le1$ gives $1\le Y\le i-1$. The reference tori are those of the sides $2L^m$, $m\ge m_*$, with L^y times the side in the family of sides of the y-th dilate for all $y\le Y$, the threshold m_* being formed from σ^- and g alone. The two factorisations bound the mixed parts themselves,

$$|S_1^{\mathrm{mix}}-S_2^{\mathrm{mix}}|(z,z')\le4(B_1+B_2)\frac{tt'}{(1-t)(1-t')},$$

$t=|z|/r$, $t'=|z'|/r$, and $t\le\frac12$ on all the bidiscs of the chain for $|z|\|f\|_\sharp\le\frac12\varrho_\wedge$; this gives, for a source f in the middle of a cell of side $\mathsf l$ with $\mathsf lL^2\bar\lambda(\mathsf l)^{1/3}\le1$ and $|z|\le\frac12\varrho_\wedge/\|f\|_\sharp$,

$$\|e^{\mathcal O_K(zf)}\|_{\mathrm{ch}}\le|e^{\mathcal E_f(z)}|\exp\big[c_V\mathsf l^3+\mathsf a(\mathsf l)|z|^2\|f\|_\sharp^2\big],$$

$$\mathsf a(\mathsf l)=\big[3\bar\Phi+22.4LC_{\mathrm S}\big]\frac{8Lg^2}{\sigma_\wedge^2}\,\mathsf l\,\bar\lambda(\mathsf l)^{2\mathfrak p_0+\frac13}$$
$$+32L^3C_{\mathrm S}\bar\lambda_1\mathsf l^3\varrho_\wedge^{-2},$$

with $\mathcal E_f$ holomorphic on $|z|<\varrho/\|f\|_\sharp$, $\mathcal E_f(0)=0$, one function for all the reference sides, the normaliser of the one-cell bound that absorbs the unmixed terms. In the multiple reflections for the n cells of one layer, Lem. 14.15(e) (as in the proof of Lem. 14.15(c)), Cauchy's estimate at the radius $\min\{\mathsf a(\mathsf l)^{-1/2}\mathcal N^{-1},\varrho_\wedge/2\mathcal N_\infty\}$, $\mathcal N^2=\sum_k\|f_k\|_\sharp^2$, $\mathcal N_\infty=\max_k\|f_k\|_\sharp$, bounds the insertion of the layer by $2e^{nc_V\mathsf l^3+1}[\mathsf a(\mathsf l)^{1/2}\mathcal N+2\mathcal N_\infty/\varrho_\wedge]$; for the pieces of $\rho_{\mathsf l}\otimes k$ this is $\le C\mathsf l^{-5/2}\bar\lambda(\mathsf l)^{\mathfrak p_0+1/6}|k|_{C^1}$, and

the limit of the cutoffs gives $\|\mathsf{I}_\mathsf{l}(k)\| \le C\mathsf{l}^{-5/2}\bar\lambda(\mathsf{l})^{\mathfrak{p}_0+1/6}|k|_{C^1}$. The operator at one time has no larger norm than the plate: on the vectors of bounded energy $\mathsf{B}_k(\frac{\mathsf{l}}{2}+\varsigma,\frac{\mathsf{l}}{2}-\varsigma) = e^{-\varsigma H}\mathsf{B}_k(\frac{\mathsf{l}}{2},\frac{\mathsf{l}}{2})e^{\varsigma H}$, so the plate is the operator integral of $\mathsf{B}_k(\frac{\mathsf{l}}{2},\frac{\mathsf{l}}{2})$ with the symbol $\tilde\rho(E-E')$,

$$\tilde\rho(x) = \int \rho_\mathsf{l}(\varsigma)\cosh(\varsigma x)\,\mathrm{d}\varsigma = \frac{\sinh(ax)}{ax}\,\frac{\pi^2}{\pi^2+a^2x^2}, \qquad a=\frac{\mathsf{l}}{5}, \qquad \frac{1}{\tilde\rho(x)} = \prod_{j\ge 2}\Big(1+\frac{a^2x^2}{j^2\pi^2}\Big)^{-1},$$

and $1/\tilde\rho$ is the Fourier transform of a probability density g, a product of Fourier transforms of Laplace densities; hence $\mathsf{B}_k(\frac{\mathsf{l}}{2},\frac{\mathsf{l}}{2}) = \int \mathsf{g}(t)e^{itH}\mathsf{I}_\mathsf{l}(k)e^{-itH}\,\mathrm{d}t$ and $\|\mathsf{B}_k(\frac{\mathsf{l}}{2},\frac{\mathsf{l}}{2})\| \le \|\mathsf{I}_\mathsf{l}(k)\|$. This gives the bound at $s=\frac{\mathsf{l}}{2}$, $\mathsf{l}=L^{-i}$, for $\mathsf{l}L^2\bar\lambda(\mathsf{l})^{1/3}\le 1$, Lem. 14.15(c) gives it for the finitely many larger $\mathsf{l}=L^{-i}$, and the semigroup property gives it for all s,s'. The block bound follows from $\|P_m\phi(f)P_n\| \le e^{s(m+n+2)}\|e^{-sH}\phi(f)e^{-sH}\| \le e\int\|\mathsf{B}_{f_t}(s,s)\|\,\mathrm{d}t$ at $s=1/(m+n+2)$ and the partition of unity of (a).

(d) For a component, $\mathsf{B}_k(u,u)\Omega$ is the class of the insertion of k at the height u, and

$$\|\mathsf{B}_k(u,u)\Omega\|^2 = \iint k(\vec x)k(\vec y)\,\mathsf{s}\big((|\vec x-\vec y|^2+4u^2)^{1/2}\big)\,d^2x\,d^2y.$$

With $\mathsf{s}\ge 0$, $\mathsf{s}(r)\ge c_{\mathrm{sd}}r^{-6}$ for $r\le r_{\mathrm{sd}}$, and $u^4(|z|^2+4u^2)^{-3}$, an approximate identity on $\mathbb{R}^2$ of the mass $2\pi u^4\int_0^\infty r(r^2+4u^2)^{-3}dr = \frac{\pi}{32}$, the liminf follows for continuous $k\ge 0$ of compact support. A bound $\pm\mathsf{T}(k)\le cN$ gives $|\langle\psi',\mathsf{T}(k)\psi\rangle| \le c\|N^{1/2}\psi'\|\|N^{1/2}\psi\|$ by polarisation, so $\|\mathsf{B}_k(u,u)\| \le c\sup_E(1+E)e^{-2uE} \le c(1+(2eu)^{-1})$, against the liminf. $\|\mathsf{B}(s,s)\Omega\|^2 = \mathsf{s}(2s) \ge c_{\mathrm{sd}}(2s)^{-6}$; with the upper bound in (c), the infimum of admissible pure-power exponents lies in $[3,\frac{7}{2}]$. For the plates, $\mathsf{I}_\mathsf{l}(h)\Omega = \tilde\rho(H)\mathsf{B}_h(\frac{\mathsf{l}}{2},\frac{\mathsf{l}}{2})\Omega$ with $\tilde\rho(E)=\int\rho_\mathsf{l}(\varsigma)\cosh(\varsigma E)\,\mathrm{d}\varsigma\ge 1$, so $\|\mathsf{I}_\mathsf{l}(h)\Omega\| \ge (1-o(1))(\pi c_{\mathrm{sd}}/32)^{1/2}\|h\|_{L^2}(\mathsf{l}/2)^{-2}$.

For the blocks, put $w_D = T_{D0}(k)\Omega$, and for $0<\eta\le 1$ let $\rho=\rho^{(\eta)}\in C^1_c((-\frac{\eta}{4},\frac{\eta}{4}))$ be even and ≥ 0 with $\int\rho=1$, and $Y_\eta = \int\rho(\varsigma)\mathsf{B}_k(\eta+\varsigma,\eta-\varsigma)\,\mathrm{d}\varsigma$, the insertion of $\rho\otimes k$ at the height η (an admitted insertion, as $\rho\otimes k$ is carried by $\{|x_3|<\frac{1}{2}\eta\}$). As ρ is even, $\|P_mY_\eta P_n\| \le e^{-\eta(m+n)}\cosh(\frac{\eta}{4}|m-n|)\|T_{mn}(k)\|$, and $P_DY_\eta\Omega = e^{-\eta H}\tilde\rho_\eta(H)w_D$ with $\tilde\rho_\eta(E)=\int\rho(\varsigma)\cosh(\varsigma E)\,\mathrm{d}\varsigma\ge 1$. First, for every $D\ge 0$ and every n with $P_n\ne 0$,

$$\sum_{j=-1}^{1}\|T_{n+D+j,n}(k)\|^2 \ge \|w_D\|^2:$$

for a unit vector $\psi\in P_n\mathcal{H}$ and its far translates $\psi_t = U(te_1)\psi$, the spectral measure of H in $Y_\eta\psi_t$ tends from below on open sets to the convolution of those of $Y_\eta\Omega$ and of $e^{-2\eta H}\psi$, by clustering, which the unique vacuum of a component gives ($e^{-tH}\to|\Omega\rangle\langle\Omega|$); so the column w_D of the vacuum reappears, up to the factors $e^{-\eta(\cdot)}$, in the blocks leaving every non-empty energy shell $P_n\mathcal{H}$: on the open set $(n+D-\frac{1}{2},n+D+2)$, with the two bounds above,

$$\sum_{j=-1}^{1}e^{-2\eta(2n+D+j)}\cosh^2(\tfrac{\eta}{4}(D+1))\|T_{n+D+j,n}(k)\|^2 \ge e^{-2\eta(D+1)-4\eta(n+1)}\|w_D\|^2,$$

and $\eta\downarrow 0$ gives the inequality, the blocks and w_D depending neither on η nor on ρ. Secondly, $\|w_D\|$ is large for infinitely many D: by the two-point law as above, the heights $2\eta+\varsigma+\varsigma'$ of the two insertions being $\le\frac{5}{2}\eta$, $\|Y_\eta\Omega\|^2 \ge (1-o(1))\,c_{\mathrm{sd}}\frac{\pi}{2}(\frac{2}{5})^4\eta^{-4}\|k\|^2_{L^2}$ as $\eta\downarrow 0$; while on $[D,D+1)$, $e^{-\eta E}\cosh(\varsigma E)\le e^{-(\eta-|\varsigma|)E}\le e^{-\frac{3}{4}\eta D}$ for $|\varsigma|\le\frac{\eta}{4}$, so that $\|Y_\eta\Omega\|^2=\sum_D\|P_DY_\eta\Omega\|^2\le\sum_D e^{-\frac{3}{2}\eta D}\|w_D\|^2$; if $\|w_D\|^2\le(D+2)^{2q''}$ for all large D with $q''<\frac{3}{2}$, the right member would be $O(\eta^{-2q''-1})=o(\eta^{-4})$. Thirdly, every shell $P_n\mathcal{H}$ with $n\ge E_0$ is non-empty, because the support of the spectral measure of the translations holds a point (E_0,q_0) with $E_0\ge 1$ and is invariant under the boosts, along which the energy runs to infinity. Now let $q'<\frac{3}{2}\epsilon$, choose q'' with $q'/\epsilon<q''<\frac{3}{2}$, and for the infinitely many D with $\|w_D\|^2>(D+2)^{2q''}$ put $n=\lceil(D+1)^{1/\epsilon}\rceil$: the three blocks

$(n+D+j,n)$ lie in the band and $m+n+2\le 6(D+1)^{1/\epsilon}$, so a bound $C(m+n+2)^{q'}$ in the band would give $(D+2)^{2q''}<3C^2 6^{2q'}(D+1)^{2q'/\epsilon}$, false for large D.

(e) The multiple reflections for one cell (Prop. YM3_I:104.2) and the one-cell bound (Prop. YM3_I:89.1), with Montel's theorem and a diagonal choice over a countable set of combinations with dense classes, give $\mathbb{D}(z)$ on the whole disc with $\|\mathbb{D}(z)\|\le \bar{\mathsf{Q}}$, $\mathbb{D}(0)=e^{-\mathsf{I}H}$ and $\mathbb{D}'(0)=\mathsf{I}+c\,e^{-\mathsf{I}H}$, I the insertion of f, which is the slab operator of Thm. AO(h1)(iv) by the insertion formula YM3_I:(2039). $G(x)=\langle\Omega,\mathbb{D}(x)\Omega\rangle$ is holomorphic, $G(0)=1$, $|G|\le\bar{\mathsf{Q}}$; by the lemma of Schwarz $|G(x)-1|\le(1+\bar{\mathsf{Q}})|x|\|f\|_\sharp/\varrho\le\frac12$ on the smaller disc, so $\|\hat D(x)\|\le 2\bar{\mathsf{Q}}$.

At real x the lattice operators are $e^{-\mathcal{E}(x)}$ times positive operators, by reflection positivity at the middle plane of the cell, a site plane, f being even under that reflection; the quotient $\hat D(x)$ removes the factor $e^{-\mathcal{E}(x)}$, so $\hat D(x)\ge 0$. Finally

$$\hat D'(0)=\mathbb{D}'(0)-G'(0)e^{-\mathsf{I}H}=\mathsf{I}-\langle\Omega,\mathsf{I}\Omega\rangle e^{-\mathsf{I}H}$$

, and $\langle\Omega,\mathsf{I}\Omega\rangle=q_0(\Omega,\Omega)\int h'$ because $\langle\Omega,\mathsf{B}(s,s')\Omega\rangle=q_0(\Omega,\Omega)$ for all s,s' and $\int\rho_1=1$. For a component the same holds with its lattice approximants, the periodic Gibbs states $\rho_{j,\xi}$ recalled in the results used.

14.1.9. *Proof of Thm. AR. (a)* Let $e^\alpha_x=\kappa_\alpha e^{-x_3\omega-i\vec x\cdot\vec p}$ for $x\in\mathbb{E}_+$, and for pairwise different points of $\mathbb{E}_+$

$$J_n(x)=\prod_i\mathsf{P}(\partial_{t_i})\Big[\exp\Big(\sum_{i<l}t_iG(x_i-x_l)t_l\Big)\mathrm{Exp}\Big(\sum_i t_i\cdot e_{x_i}\Big)\Big]_{t=0},\qquad J(G)=\int\prod_k h_k(x_k)J_n(x)\,\mathrm{d}x,$$

$\mathrm{Exp}(u)=\sum_m u^{\otimes m}/\sqrt{m!}$. The covariance of G and the invariance of P under $\mathrm{O}(3)$ are, as in the statement, those of the orthogonal representation ρ_{A} of $\mathrm{O}(3)$ on $\mathbb{R}^{\mathsf{A}}$: $G(\mathsf{R}x)=\rho_{\mathsf{A}}(\mathsf{R})G(x)\rho_{\mathsf{A}}(\mathsf{R})^{\mathrm{T}}$, $\mathsf{P}(\rho_{\mathsf{A}}(\mathsf{R})t)=\mathsf{P}(t)$, with $\rho_{\mathsf{A}}(\theta)=\mathrm{diag}(\varepsilon_\alpha)$; and the tensor p is symmetric. So the invariance of P under θ is $\mathsf{P}(\varepsilon t)=\mathsf{P}(t)$, $(\varepsilon t)_\alpha=\varepsilon_\alpha t_\alpha$, that is $\mathsf{p}_{\underline\alpha}\prod_i\varepsilon_{\alpha_i}=\mathsf{p}_{\underline\alpha}$, and the covariance under θ is $\varepsilon G(x)\varepsilon=G(\theta x)$, $\varepsilon=\mathrm{diag}(\varepsilon_\alpha)$.

As $\langle\mathrm{Exp}\,u,\mathrm{Exp}\,v\rangle=e^{\langle u,v\rangle}$ and $\langle e^\alpha_x,e^\beta_y\rangle=\varepsilon_\alpha G_{\alpha\beta}(\theta x-y)$, the substitution $s_i=\varepsilon\tilde s_i$, with $\mathsf{P}(\varepsilon t)=\mathsf{P}(t)$ in the derivatives and $\varepsilon G(x_i-x_l)\varepsilon=G(\theta x_i-\theta x_l)$ in the exponent, gives $\langle J_m(x),J_n(y)\rangle=S_{m+n}(\theta x,y)$, the Schwinger function of $:\mathsf{P}(\Phi):$; so $V[G]=J(G)$ is an isometry, (Rec1) follows from the covariance of G and the invariance of P, $e^{-tH_{\mathrm F}}J_n(x)=J_n(x+te_3)$ gives (Rec2), $|\sum\vec p(m_i)|\le\sum\omega(m_i)$ gives (Rec3), and $\ker d\Gamma(\omega)=\mathbb{C}\Omega_{\mathrm F}$ gives (Rec5).

The point field is the Fock operator

$$\mathsf{B}_{\mathrm F}(s,s')=\sum_j\binom{k}{j}\sum_{\underline\alpha}\mathsf{p}_{\underline\alpha}\prod_{i>j}\varepsilon_{\alpha_i}\,a^*(e^{\alpha_1}_{se_3})\cdots a^*(e^{\alpha_j}_{se_3})\,\Gamma(e^{-(s+s')\omega})\,a(e^{\alpha_{j+1}}_{s'e_3})\cdots a(e^{\alpha_k}_{s'e_3}),$$

with $\mathsf{B}_{\mathrm F}(s,s')J_n(y)=J_{n+1}(se_3,y+(s+s')e_3)$; from

$$\|a(u_1)\cdots a(u_m)e^{-cH_{\mathrm F}}\|\le\prod_i\|\omega^{-1/2}u_i\|(m/2ec)^{m/2}$$

and $\|\omega^{-1/2}e^\alpha_{ue_3}\|\le c_\kappa u^{-\delta}$ one gets $\|\mathsf{B}_{\mathrm F}(s,s')\|\le C_{\mathsf B}\min\{s,s',1\}^{-\wp}$, $C_{\mathsf B}=2^k|\mathsf{p}|_1(2^\delta c_\kappa)^k(k/e)^{k/2}$, hence (Rec4) with $C_\wp=2^\wp C_{\mathsf B}$, $c_0=\frac12$. The uniqueness of the family of point operators, Prop. YM3_I:107.2, identifies $V^{-1}\mathsf{B}_{\mathrm F}V$ with B, and so the field with the form of the Wick polynomial, which on vectors with finitely many particles and bounded energy is

$$\Phi_f(\chi,\chi')=\sum_j\binom{k}{j}\sum_{\underline\alpha}\mathsf{p}_{\underline\alpha}\prod_{i>j}\varepsilon_{\alpha_i}\int f(x)\Big\langle\prod_{i\le j}a(\kappa^x_{\alpha_i})\chi,\prod_{i>j}a(\kappa^x_{\alpha_i})\chi'\Big\rangle\,\mathrm{d}x,\qquad\kappa^x_\alpha=e^{i(t\omega-\vec x\cdot\vec p)}\kappa_\alpha;$$

the projection on $V\mathcal{H}$ commutes with $H_{\mathrm F}$, the translations and $\mathsf{B}_{\mathrm F}$, which gives the identity for every $\chi\in\mathcal{D}_\infty(d\Gamma(\omega))$.

(b) The component with kn particles of $V\phi(u)^n\Omega$ is $((kn)!)^{1/2}\mathrm{Sym}(w_u^{\otimes n})$, and no component has more particles: in $\Phi_u(\chi,\chi')$ the j-th term changes the number of particles by $2j-k$, the top term $j=k$ is the creation of w_u, by Fubini and $\int u(x)\prod_i e^{i(t\omega(m_i)-\vec{x}\cdot\vec{p}(m_i))}\,\mathrm{d}x=\hat{u}(\sum\omega,\sum\vec{p})$, and w_u is a symmetric function, the tensor p being symmetric, so that $\mathrm{Sym}(w_u)=w_u$. The symmetric tensors $v^{\otimes k}$ are total among the symmetric ones, so some unit v has $c_u=|\langle v^{\otimes k},w_u\rangle|>0$, and $\|\phi(u)^n\Omega\|\geq|\langle v^{\otimes kn},\mathrm{Sym}(w_u^{\otimes n})\rangle|((kn)!)^{1/2}=c_u^n((kn)!)^{1/2}$; $m!\geq(m/e)^m$ makes $\sum_n\|\phi(u)^n\Omega\|^{-1/n}$ converge for $k\geq 3$.

Let a real $u_0\neq 0$ in C_c^∞ have $b_{mn}(u_0)\leq\beta(m+n+2)^q$ on a band, $q<\frac{k}{2}$; outside the band Thm. AO(b3) gives a linear bound, so $b_{mn}(u_0)\leq\beta'(m+n+2)^{\max\{q,1\}}$ for all m,n; a linear form bound at u_0 gives $b_{mn}(u_0)\leq c\sqrt{(m+2)(n+2)}$ as well. Choose $\mathsf{s}>1$ with $\max\{q,1\}\mathsf{s}<\frac{k}{2}$, and $u=h*u_0$ with $h=g\otimes g_2$, g a bump of the Gevrey class s (as in the proof of Thm. AQ(a)) and $g_2\geq 0$ in $C_c^\infty(\mathbb{R}^2)$: $u\in C_c^\infty$ is real and $\neq 0$, $\hat{u}=\hat{h}\hat{u}_0$ being a product of entire functions $\neq 0$. Thm. AO(b2), (b4) and the choice of the number of derivatives as in the proof of Thm. AQ(a) give $b_{mn}(u)\leq Ce^{-\mathsf{a}|m-n|^{1/\mathsf{s}}}b_{mn}(u_0)$, and the sum over the paths from Ω gives $\|\phi(u)^n\Omega\|\leq C^n(n!)^{\max\{q,1\}\mathsf{s}}$, against $c_u^n(n!)^{k/2}$.

The same with a triple of order $\wp<\frac{k}{2}$ in the place of the band bound.

(c) For the free field, $\|\omega^{-1/2}\kappa e^{-s\omega}\|^2=\frac{1}{4\pi}\int_{2s\mathsf{m}}^\infty e^{-v}\frac{dv}{v}\leq\frac{1}{8\pi\delta}(2s\mathsf{m})^{-2\delta}$ for $0<\delta\leq\frac{1}{2}$. For $k\geq 2$, the map $(p_1,\dots,p_k)\mapsto(\sum\omega(p_i),\sum p_i)$ has an image that holds $\mathcal{V}_k=\{E>(|\vec{k}|^2+k^2\mathsf{m}^2)^{1/2}\}$, and w_u is continuous with $\kappa>0$; so $w_u\neq 0$ as soon as $\hat{u}\neq 0$ at a point of $\mathcal{V}_k$. For $u\in C_c^\infty\setminus\{0\}$, $\hat{u}$ is entire. For $u\in\mathcal{S}$ vanishing off $\bar{W}$ and every k_2, $(E,k_1)\mapsto\hat{u}(E,k_1,k_2)$ continues boundedly into the directions η with $\eta_1<-|\eta_0|$, since $x^0\eta_0-x^1\eta_1\geq 0$ on $\bar{W}$; if it vanished on the open set where the plane meets $\mathcal{V}_k$, it would vanish on the plane.

For $u=\mathsf{g}\otimes h'$ take $\vec{k}_*$ with $\hat{h}'(\vec{k}_*)\neq 0$ and E large. At $k=6$, $\delta=\frac{1}{6}$: $\wp=4$ and $C_\wp=2^4\cdot 2^6\cdot 2c_\kappa^6\cdot(6/e)^3$ with $c_\kappa^6=(\frac{4\pi}{3})^{-3}(2\mathsf{m})^{-1}$, which is $93312(\pi e)^{-3}\mathsf{m}^{-1}$; the field changes the number of particles by an even number, so the space lies in the even Fock space, where $d\Gamma(\omega)\geq 2\mathsf{m}$ off Ω_F; $S_2=6!\,G^6$, $G=e^{-\mathsf{m}r}/4\pi r$. *Locality.* Let O_1,O_2 be spacelike separated, P the projection on $V\mathcal{H}$ and $\mathcal{R}_0=\mathcal{R}_0(O_2)$. By Lem. 14.12(a), (c), for real $h\in\mathcal{F}(O_2)$ the compression $V^{-1}P\mathsf{W}[u_h]V$ commutes weakly with the fields of O_1 on $\mathcal{D}_0$, so $P\mathcal{R}_0P|_{V\mathcal{H}}\subset V\mathcal{C}(O_1)V^{-1}$; by Lem. 14.12(d) Ω_F is cyclic for $\mathcal{R}_0$, and Lem. 14.11(a) gives $(P\mathcal{R}_0P|)'\subset P\mathcal{R}_0'P|$; by Lem. 14.12(e), $V^{-1}P\mathcal{R}_0'PV\subset\mathcal{C}(O_2)$.

So

$$\mathcal{C}(O_1)'\subset V^{-1}(P\mathcal{R}_0P|)'V\subset V^{-1}P\mathcal{R}_0'PV\subset\mathcal{C}(O_2),$$

which is (W), and (CP) follows by Thm. AO(d). For the derivatives $\partial_\alpha\varphi^a$ of massless fields and for the generalised free fields the same proof runs with Lem. 14.12(f), whose hypotheses (f1), (f2) hold for them, in the place of Lem. 14.12(a)–(e): $\mathcal{R}_0(O)$ is the span of the $\mathsf{W}[u]$, $u\in\mathfrak{k}(O)$, the real span of the $u_h^\alpha=\varsigma_\alpha\hat{h}(\omega,\vec{p})\kappa_\alpha$, $h\in\mathcal{F}_\mathbb{R}(O)$, with the weight $\omega^{1/2}(1+\omega)^{r'}$, $r'>\delta$, in the place of ω in the bounds of Lem. 14.12, the reality of $\langle\kappa^x,u_h\rangle$ replaced by the reality or pure imaginarity of $\langle\kappa_\alpha^x,u\rangle$, $u\in\mathfrak{k}(O)$, according to ε_α (the derivatives in space and in time of a kernel that is real at spacelike separation), and the density of the one-particle vectors of a region by the same reflection argument with $\omega\kappa_0$ in the place of κ.

For $:(\partial\varphi\cdot\partial\varphi)^2::$ $k=4$, $\delta=1$, $\wp=6$; for the kernel r^{-2}, the mixture of masses $4\pi\,\mathrm{d}\mathsf{m}$ has $\|\omega^{-1/2}\kappa e^{-s\omega}\|^2=\frac{1}{2s}$, so $\delta=\frac{1}{2}$ and $\wp=3$ at $k=3$, $S_2=3!\,r^{-6}$, and the spectrum reaches 0. The last two assertions of (c) follow from (b), applied to $:\chi^6:$ and to $:\chi^3:$.

(d) The photons are the Gaussian family of (a) with $\mathsf{M}=\mathbb{R}^2\setminus\{0\}$, $d\mu=d^2p$, $\omega=|p|$, $\mathsf{P}=\sum_\alpha t_\alpha^2$, $\kappa_0=(2\pi)^{-1}(2|p|)^{-1/2}$, $\kappa_j=-ip_j\kappa_0$, $\kappa_3=-\omega\kappa_0$, $\varepsilon=(+,+,-)$, $G_{\alpha\beta}=-\partial_\alpha\partial_\beta(4\pi|x|)^{-1}$; the form of (a) is $:|\nabla\varphi|^2-\dot{\varphi}^2:=-\Theta$. With $\|\omega^{-1/2}e^3_{ue_3}\|^2=(16\pi u^2)^{-1}$ and $\|\omega^{-1/2}e^j_{ue_3}\|^2=(32\pi u^2)^{-1}$ the three terms of B_F are bounded by $(\pi e)^{-1}\min\{s,s'\}^{-3}$ each, so $C_\wp=2^3\cdot 3/\pi e=24/\pi e$; $S_2=2\sum_{\alpha\beta}G_{\alpha\beta}^2=12/16\pi^2r^6$ and $\|\mathsf{B}(s,s)\Omega\|^2=S_2(2s)$.

The span of the Θ-vectors is the even Fock space, as in (e) at $\mathsf{n}=1$. *The linear bound.* Let $\tilde f(k)=\int f(x)e^{i(k^0t-\vec k\cdot\vec x)}\,\mathrm{d}x$ (the transform $\hat f$ of the setting) and $\kappa_f(\vec k)=\frac{1}{2\pi^2}\sup_{k^0}|\tilde f(k^0,\vec k)|\le 8|f|_8(1+|\vec k|^2)^{-2}$, $\theta(f)=\int\kappa_f\,d^2k\le 8\pi|f|_8$. The field $\phi(f)=-\Theta(f)$, f real, has a part with one creator and one annihilator, with the kernel $t_f(p,q)=2\kappa_0(p)\kappa_0(q)(\vec p\cdot\vec q-\omega_p\omega_q)\tilde f(\omega_p-\omega_q,\vec p-\vec q)$, of modulus $\le\sqrt{\omega_p\omega_q}\,\kappa_f(\vec p-\vec q)$, bounded by $\theta(f)H$ by Young's inequality; and the two pair parts $\mathsf{Z}^+(w_f)+\mathsf{Z}^+(w_f)^\dagger$, $\mathsf{Z}^+(c)=\iint c(p,q)a_p^*a_q^*\,\mathrm{d}p\,\mathrm{d}q$, with the kernel $w_f(p,q)=\sum_\alpha\kappa_\alpha(p)\kappa_\alpha(q)\tilde f(\omega_p+\omega_q,\vec p+\vec q)$, of modulus $\le\frac12\sqrt{\omega_p\omega_q}\,\kappa_f(\vec p+\vec q)$, so that $w_f/\sqrt{\omega_p\omega_q}$ has the operator norm $\le\frac12\theta(f)$.

For them a square is completed: with $\epsilon>0$ and $g_p=\pm w_f(p,\cdot)/\epsilon\omega_p$, $0\le\int\omega_p\|(a_p+a^*(g_p))\psi\|^2\,\mathrm{d}p$ is $\langle\psi,H\psi\rangle$, plus the pair form at ψ divided by $\pm\epsilon$, plus $\int\omega_p\|a(g_p)\psi\|^2\,\mathrm{d}p\le\frac{\theta(f)^2}{4\epsilon^2}\langle\psi,H\psi\rangle$, plus $\int\omega_p\|g_p\|^2\,\mathrm{d}p\,\|\psi\|^2\le\frac{\mathsf{n}(f)}{4\epsilon^2}\|\psi\|^2$, $\mathsf{n}(f)=\frac{1}{4\pi^4}\iint\omega_q|\tilde f(\omega_p+\omega_q,\vec p+\vec q)|^2\le 32\pi^3|f|_8^2$; so $\pm(\mathsf{Z}^+(w_f)+\mathsf{Z}^+(w_f)^\dagger)\le(\epsilon+\frac{\theta(f)^2}{4\epsilon})H+\frac{\mathsf{n}(f)}{4\epsilon}$, and at $\epsilon=\frac{\vartheta}{2}$, $\vartheta\ge\theta(f)$: $\le\vartheta H+\mathsf{n}(f)/2\vartheta$.

At $\vartheta=8\pi|f|_8\ge\theta(f)$: $\pm\Theta(f)\le 16\pi|f|_8H+2\pi^2|f|_8\le 16\pi|f|_8N$. The same bound for $\Theta(\partial_tf)=-i[H,\Theta(f)]$ gives the operator bound and essential self-adjointness on every core of H; $\|\Theta(f)\psi\|\le c(f)(n+2)\|\psi\|$ on vectors with at most n particles gives the growth; (W) and (CP) by (c). *Order exactly one* on the even space: the squeezed vectors of (e) at $\mathsf{n}=1$. *At one time.* For $m\le n$ write $\Theta_k=\Theta^{(0)}+\Theta^{(-)}+\Theta^{(+)}$ (number-preserving, annihilating and creating parts), with $\tilde k(\xi)=\int k(\vec x)e^{-i\xi\cdot\vec x}d^2x$, $\Theta^{(0)}=\iint(4\pi^2)^{-1}\sqrt{\omega_p\omega_q}(1-\hat p\cdot\hat q)\tilde k(p-q)\,a_p^*a_q$, $\Theta^{(-)}=\iint\Xi(p,q)\,a_pa_q$, $\Xi(p,q)=-(8\pi^2)^{-1}\sqrt{\omega_p\omega_q}(1-\hat p\cdot\hat q)\tilde k(-p-q)$, and $\Theta^{(+)}=(\Theta^{(-)})^\dagger$.

The annihilation and creation densities select energies: $\langle\chi,a_pa_q\psi\rangle=0$ unless $\omega_p+\omega_q\in(n-m-1,n-m+1)$ for $\chi\in P_m\mathcal{F}$, $\psi\in P_n\mathcal{F}$. The integral operators with the kernels $\tilde k(q\mp p)$ have norms $\le(2\pi)^2\|k\|_\infty$ (Plancherel), and with the factor $1-\hat p\cdot\hat q$ at most twice that; multiplication of a kernel by a window $\eta(\omega_p+\omega_q-\omega)$, η the inverse Fourier transform of an integrable $\hat\eta$, multiplies its norm by at most $\frac{1}{2\pi}\|\hat\eta\|_{L^1}$, which is at most $\sqrt3$ (Schwarz) for $\eta_2=\mathbf{1}_{[-\frac32,\frac32]}*\mathbf{1}_{[-\frac12,\frac12]}$, a function equal to 1 on $[-1,1]$ and carried by $[-2,2]$.

With $\omega=n-m$ and $\Xi_\eta(p,q)=\Xi(p,q)\eta_2(\omega_p+\omega_q-\omega)$, $P_m\Theta^{(-)}P_n=P_m\iint\Xi_\eta a_pa_qP_n$. So $\|P_m\Theta^{(0)}P_n\|\le 2\|k\|_\infty\sqrt{(m+1)(n+1)}$ and, with $g_q=\overline{\Xi_\eta(\cdot,q)}\omega_q^{-1/2}$,

$$\mathsf{h}(\omega)^2=\iint|\Xi_\eta(p,q)|^2\omega_q^{-1}\,\mathrm{d}p\,\mathrm{d}q=\int\|g_q\|^2\,\mathrm{d}q$$

and $\|a^*(g)\chi\|^2=\|a(g)\chi\|^2+\|g\|^2\|\chi\|^2$,

$$\|P_m\Theta^{(-)}P_n\|\le\sqrt{n+1}\,[\sqrt3\|k\|_\infty\sqrt{m+1}+\mathsf{h}(\omega)],\qquad \mathsf{h}(\omega)^2\le\frac{\omega+2}{(2\pi)^4}\int|\tilde k(\xi)|^2|\mathcal{A}_\xi|\,\mathrm{d}\xi\le(\omega+2)^2\mathsf{n}_k^2,$$

$\mathcal{A}_\xi=\{p:|p|+|\xi-p|\in[\omega-2,\omega+2]\}$, of area $\le 10.4(\omega+2)$ for $|\xi|\le\frac12(\omega+2)$ (confocal ellipses) and $\le\frac{\pi}{2}(\omega+2)|\xi|$ otherwise; $\Theta^{(+)}$ vanishes between these shells for $m<n$. This is the upper bound. For the lower bound let $\Xi_1=\Xi\,\eta_1(\omega_p+\omega_q-\omega)$, $\eta_1=(\frac12-|\cdot|)_+$, $L_1=\iint\Xi_1a_pa_q$; then $[L_1,L_1^*]\ge 2\|\Xi_1\|_2^2$, and the vectors $\chi_j=(L_1^*)^j\Omega$ have $\|\chi_{j+1}\|^2\ge 2(j+1)\|\Xi_1\|_2^2\|\chi_j\|^2$; as L_1^* raises the energy by $\omega\pm\frac12$, one shell P_m in the range $[j(\omega-\frac12),j(\omega+\frac12)]$ has $\|P_nL_1^*P_m\|\ge(\frac{j+1}{3})^{1/2}\|\Xi_1\|_2$ for one of three n; the window costs the factor $\frac12$, and $\|\Xi_1\|_2^2\ge 2.3\cdot10^{-5}\|k\|_2^2\omega^3$ for $\omega\ge\max\{3,7\xi_0(k)+1\}$, because on the ring $\mathcal{A}'_\xi=\{|p|+|\xi-p|\in[\omega-\frac14,\omega+\frac14]\}$, $|\xi|\le\xi_0$, where $|p|\ge 3\xi_0$, the kernel has $|\Xi(p,\xi-p)|^2\ge\frac43|p|^2|\tilde k(\xi)|^2/64\pi^4$; the number-preserving part is at most $\kappa_k(\omega-2)G$ there, as only momentum transfers $|p-q|>\omega-2$ occur. The band statement follows by taking $j\approx(\omega+1)^{1/\epsilon}/\omega$.

(e) The cyclic space. The Θ-vectors are $\mathrm{O}(\mathsf{n})$-invariant. For the converse use the explicit bilocal calculation in steps (1)–(4) of (f) below, first as identities of Wick polynomials, before taking commutants. The timelike matrix $\mathsf{H}_{\mu\nu}=\partial_\mu\partial_\nu\Delta_{\mathrm{c}}$ is invertible. Point splitting gives its traceless symmetric contractions, and adjoining the already available trace Θ gives every local symmetric component. The two contractions $5\mathsf{X}+\mathsf{Ht}$ and $8\mathsf{t}$ in (3) then recover every bilocal component

at timelike separated supports. The commutator identity and overlapping-cone chain in (4) give the colour sums $\sum_a \varphi^a(h)\varphi^a(h')$ for arbitrary real integral-zero tests in a connected double cone, including spacelike separated supports. These are limits of polynomials in the $\Theta(f)$ on the common finite-particle energy-smooth core, with the finite-particle bounds used in those limits. This argument applies to the polynomial vectors themselves and does not use the cyclic-space identification it proves. In particular, for real g of integral zero in a double cone O, the sums of squares $\sum_a \varphi^a(g)^2$ are such limits.

One more commutator gives the squares: let $O_3 \subset O$ be a smaller double cone that holds $\operatorname{supp} g$, so that $g = \partial_\mu \mathsf{G}^\mu$ with $\mathsf{G} \in C_c^\infty(O_3)^3$, take a ball $O_1 \subset O$ spacelike to O_3, a ball $O_2 \subset O$ in the common future of O_1 and O_3, and $g_1, g_3 \in \mathcal{F}_{\mathbb{R}}(O_1)$ of integral 0 with $\Delta_{\mathrm{c}}(g_1, g_3) \neq 0$; then $\sum_a \varphi^a(g_i)\varphi^a(g)$, $i = 1, 3$, are such limits up to numbers, and, as $\Delta_{\mathrm{c}}(g, g_3) = \Delta_{\mathrm{c}}(g_1, g) = \Delta_{\mathrm{c}}(g, g) = 0$,

$$\Big[\sum_a \varphi^a(g_1)\varphi^a(g), \sum_b \varphi^b(g_3)\varphi^b(g)\Big] = i\Delta_{\mathrm{c}}(g_1, g_3) \sum_a \varphi^a(g)^2;$$

by the first fundamental theorem for $\mathrm{O}(\mathsf{n})$ the invariant polynomials in the $\varphi^a(g)$ are generated by the scalar products $\sum_a \varphi^a(g)\varphi^a(g')$, which polarisation reduces to squares; and the invariant polynomial vectors are dense in the invariant vectors. The Euclidean vectors give the same space by (a). With SO(3): $\sum \epsilon_{abc} a_a^*(u)a_b^*(v)a_c^*(w)\Omega \neq 0$ is SO(3)-invariant, has three particles and is orthogonal to the even space. $\|\mathsf{B}(s, s)\Omega\|^2 = \mathsf{n}S_2(2s)$ with the S_2 of one colour. *Order one.* For a bounded unit $u \in \mathfrak{h}$ of compact support put $A^* = \sum_a a_a^*(u)^2$; then $A(A^*)^k\Omega = 2k(2k - 2 + \mathsf{n})(A^*)^{k-1}\Omega$, and the singlet squeezed vectors $\chi_\lambda = (1 - \lambda)^{\mathsf{n}/4} \exp(\frac{\sqrt{\lambda}e^{i\beta}}{2} A^*)\Omega$, $0 \leq \lambda < 1$, are invariant unit vectors of $\mathcal{D}_\infty(H)$.

With the one-particle operator T_f of the kernel $\frac{\omega_p\omega_q - \vec{p}\cdot\vec{q}}{4\pi^2\sqrt{\omega_p\omega_q}} \tilde{f}(\omega_p - \omega_q, \vec{p} - \vec{q})$ (the kernel $-t_f$ of (d)) and $c_f(p, q) = \frac{\omega_p\omega_q - \vec{p}\cdot\vec{q}}{8\pi^2\sqrt{\omega_p\omega_q}} \tilde{f}(\omega_p + \omega_q, \vec{p} + \vec{q})$ (the w_f of (d)), the field of n colours is $\Theta(f) = d\Gamma(T_f \otimes 1) - \mathsf{Z}^+(c_f) - \mathsf{Z}^+(c_{\bar{f}})^\dagger$, $\mathsf{Z}^+(c) = \sum_a \iint c(p, q)a_a^*(p)a_a^*(q)\,\mathrm{d}p\,\mathrm{d}q$, and $\phi(f) = -\Theta(f)$; so

$$\langle \chi_\lambda, \phi(f)\chi_\lambda\rangle = \frac{\mathsf{n}}{1 - \lambda}\big[2\sqrt{\lambda}\operatorname{Re}(e^{i\beta}\mathsf{w}) - \lambda\mathsf{t}\big], \qquad \langle \chi_\lambda, H\chi_\lambda\rangle = \frac{\mathsf{n}\lambda\,\mathsf{e}}{1 - \lambda},$$

$\mathsf{t} = \langle u, T_f u\rangle$, $\mathsf{w} = \langle c_f, u \otimes u\rangle$, $\mathsf{e} = \langle u, \omega u\rangle$. Some u has $|\mathsf{t}| + 2|\mathsf{w}| > 0$: otherwise $\tilde{f}$ would vanish on $\{|k^0| \leq |\vec{k}|\}$ and on $\pm\bar{V}_+$. With Jensen's inequality $\langle \psi, N^\alpha\psi\rangle \leq \langle \psi, N\psi\rangle^\alpha$ the quotient grows like $(1 - \lambda)^{\alpha-1}$. Either sign: the one-particle operator T_f has a continuous kernel that vanishes on the diagonal, so it is 0 or indefinite; choose u with t of the sign needed, or with $\mathsf{w} \neq 0$ if $T_f = 0$, and β. These vectors are limits in every $\|N^l \cdot\|$ of vectors of $\mathcal{D}_0$. The upper bound adds the bounds of the n colours. If $b_{mn}(f) \leq \beta_p(1 + \max\{m, n\})^p$ with $p < 1$, an interpolation of this bound with the decay of Thm. AO(b2) and the crude bound $b_{mn}(f) \leq c(2 + \min\{m, n\})^\wp$, from Thm. AO(b5) at $j = 0$ and $b_{mn}(f) = b_{nm}(\bar{f})$, gives $b_{mn}(f) \leq C(2 + m)^{p'/2}(2 + n)^{p'/2}(1 + |m - n|)^{-2}$ with $p < p' < 1$, hence a form bound with $N^{p'}$, which the squeezed vectors exclude.

(f) For a field Φ among Θ, $S = {:}\varphi\cdot\varphi{:}$, the multiplet $\Theta_{\mu\nu} = {:}\partial_\mu\varphi\cdot\partial_\nu\varphi{:}$ and the $T_{\mu\nu} = \Theta_{\mu\nu} - \frac{1}{2}\eta_{\mu\nu}\Theta$, all on the Fock space, $\mathcal{D}_0^\Phi$ is the span of Ω and of the $\Phi(f_1)\cdots\Phi(f_n)\Omega$, $\mathcal{H}_\Phi$ its closure, and $\mathcal{C}_\Phi(O)$, $\mathcal{A}_\Phi(O) = \mathcal{C}_\Phi(O)'$ and $\mathcal{A}_\Phi^d(O) = \bigcap\{\mathcal{A}_\Phi(\tilde{W}) : \tilde{W} \supset O\}$ are formed as in the setting, with Φ and $\mathcal{D}_0^\Phi$ in the place of ϕ and $\mathcal{D}_0$. Θ at the mass 0 (by (d), (e)), Θ at a mass $\mathsf{m} > 0$ (by (a), with $\delta = 1$, as $\|\omega^{-1/2}\kappa_\alpha e^{-s\omega}\|^2 \leq (16\pi s^2)^{-1}$) and S at $\mathsf{m} > 0$ (by (a), as in (c)) are data (Rec), and V carries their nets onto these; so Thm. AO(e) applies to them. For S at the mass 0, $\mathcal{A}_S^d$ is the net of this definition through wedges. $\mathfrak{R}(\mathsf{K})$ is the von Neumann algebra of the Weyl operators of the vectors of a closed real subspace K of the one-particle space, and K' the symplectic complement.

Let first $\mathsf{n} = 1$, $\mathsf{m} = 0$. With $\mathsf{H}_{\kappa\lambda} = \partial_\kappa\partial_\lambda\Delta_{\mathrm{c}}$ and $\mathsf{X}_{\kappa\lambda}(x, y) = {:}\partial_\kappa\varphi(x)\partial_\lambda\varphi(y){:}$, on vectors with finitely many particles and for timelike separated supports,

$$[\Theta(f_1), \Theta(f_2)] = -4i \iint f_1(x)f_2(y)\,\mathsf{H}^{\kappa\lambda}(x - y)\,\mathsf{X}_{\kappa\lambda}(x, y)\,\mathrm{d}x\,\mathrm{d}y + c\,1,$$

and $\Delta_{\mathrm{c}}(z) = \mp(2\pi)^{-1}(z\cdot z)^{-1/2} \neq 0$ on $\pm V_+$, inside the cone in $1+2$ dimensions; on V_+, $-2\pi(z\cdot z)^{3/2}\mathsf{H}_{\kappa\lambda}(z) = 3z_\kappa z_\lambda/(z\cdot z) - \eta_{\kappa\lambda}$, whose mixed form has the eigenvalues $2,-1,-1$, so that H is invertible at timelike points. For Θ the following identities are used on the common energy-smooth core. First time-smooth the bounded weakly commuting operator Q as in the proof of Thm. AO(a), working in a region with closure in O; the linear energy bounds and essential self-adjointness extend the polynomial-domain identities to this core. Products and commutators are taken there, and the displayed Wick and point-splitting limits pass with their finite-particle bounds. At the end let the smoothing tend to the identity. This is a statement about these cores and limits, not an unrestricted rule for weak commutants of unbounded operators. Let $Q \in \mathcal{C}_\Theta(O)$, O open and connected. (1) The display gives the contractions $\mathsf{X}[\Psi\mathsf{H}] = \iint \Psi\,\mathsf{H}^{\kappa\lambda}\mathsf{X}_{\kappa\lambda}$, Ψ scalar in C_c^∞ of the timelike pairs of $O\times O$: one component of nine. (2) For $f\in\mathcal{F}(O)$, $\chi\in C_c^\infty(V_+)$ and small ϵ, $\Psi(x,y) = -2\pi f(y)\epsilon^{-3}\chi(\frac{x-y}{\epsilon})((x-y)\cdot(x-y))^{3/2}$ makes $\Psi\mathsf{H}$, as $3z_\kappa z_\lambda/(z\cdot z)-\eta_{\kappa\lambda}$ is homogeneous of degree 0, the function $f(y)\epsilon^{-3}\zeta(\frac{x-y}{\epsilon})$ with $\zeta(z) = \chi(z)(3z_\kappa z_\lambda/(z\cdot z)-\eta_{\kappa\lambda})$, and point splitting gives $3\mathsf{N}^{\kappa\nu}[\chi]\Theta_{\kappa\nu}(f) - (\int\chi)\Theta(f)$, $\mathsf{N}^{\kappa\nu}[\chi] = \int\chi\, z^\kappa z^\nu/(z\cdot z)$; these tensors span the traceless symmetric ones, and adjoining the trace Θ, with which Q already commutes, gives all symmetric components: Q commutes weakly with all $\Theta_{\kappa\nu}(f)$, $f\in\mathcal{F}(O)$. (3) $[\Theta_{\kappa\lambda}(f_1),\Theta_{\mu\nu}(f_2)] = -i\iint f_1 f_2\,\mathsf{Y}_{\kappa\lambda,\mu\nu} + c$, $\mathsf{Y}_{\kappa\lambda,\mu\nu} = \mathsf{H}_{\lambda\mu}\mathsf{X}_{\kappa\nu} + \mathsf{H}_{\lambda\nu}\mathsf{X}_{\kappa\mu} + \mathsf{H}_{\kappa\mu}\mathsf{X}_{\lambda\nu} + \mathsf{H}_{\kappa\nu}\mathsf{X}_{\lambda\mu}$, and with $\mathsf{t} = (\mathsf{H}^{-1})^{\lambda\mu}\mathsf{X}_{\lambda\mu}$: $(\mathsf{H}^{-1})^{\lambda\mu}\mathsf{Y}_{\kappa\lambda,\mu\nu} = 5\mathsf{X}_{\kappa\nu} + \mathsf{H}_{\kappa\nu}\mathsf{t}$ and $(\mathsf{H}^{-1})^{\kappa\nu}(\mathsf{H}^{-1})^{\lambda\mu}\mathsf{Y}_{\kappa\lambda,\mu\nu} = 8\,\mathsf{t}$; so $\iint\Psi^{\kappa\nu}\mathsf{X}_{\kappa\nu} = \iint\Upsilon^{\kappa\lambda\mu\nu}\mathsf{Y}_{\kappa\lambda,\mu\nu}$ with $\Upsilon^{\kappa\lambda\mu\nu} = \frac{1}{5}[\Psi^{\kappa\nu}(\mathsf{H}^{-1})^{\lambda\mu} - \frac{1}{8}(\Psi^{\rho\sigma}\mathsf{H}_{\rho\sigma})(\mathsf{H}^{-1})^{\kappa\nu}(\mathsf{H}^{-1})^{\lambda\mu}]$: here the invertibility of H solves for every bilocal field $\mathsf{X}[\Psi]$, Ψ a matrix in C_c^∞ of the timelike pairs. (4) For double cones K, K' in O write $K\approx K'$ if Q commutes weakly with all $\varphi(h)\varphi(h')$, $h\in\mathcal{F}_{\mathbb{R}}(K)$, $h'\in\mathcal{F}_{\mathbb{R}}(K')$ of integral 0; such h are divergences $\partial_\kappa \mathsf{k}^\kappa$, so $\varphi(h)\varphi(h')$ is a bilocal field up to a number, and (3) gives $K\approx K'$ when all pairs of $K\times K'$ are timelike.

In

$$\begin{aligned}&[\varphi(k)\varphi(k'),\varphi(l)\varphi(l')]\\ &= i\Delta_{\mathrm{c}}(k',l)\varphi(k)\varphi(l')\\ &+i\Delta_{\mathrm{c}}(k',l')\varphi(k)\varphi(l)\\ &+i\Delta_{\mathrm{c}}(k,l)\varphi(l')\varphi(k')\\ &+i\Delta_{\mathrm{c}}(k,l')\varphi(l)\varphi(k'),\end{aligned}$$

with k,k',l,l' of integral 0: $k=l$ in K and k',l' in K' give $K\approx K$ when $K\approx K'$ (the left member and the first and fourth terms are known, the third has the factor $\Delta_{\mathrm{c}}(k,k)=0$, and Δ_{c} does not vanish identically on the pairs of functions of integral 0 in K'); k in K, k',l in K'' and l' in K' give $K\approx K'$ from $K\approx K''$, $K''\approx K''$, $K''\approx K'$ (the left member and the second, third and fourth terms are known). Every point of O has a small double cone K with $K\approx K$ (with a translate of K along the time axis), the classes of the relation so induced on O are open, and O is connected: there is one class.

With a Lebesgue number of a cover of a compact connected set that holds the supports, every $h\in\mathcal{F}_{\mathbb{R}}(O)$ of integral 0 is cut into pieces of integral 0 in small double cones along a chain, $h = \sum_i(\chi_i h - \vartheta_i b_i) + \sum_i \vartheta_i(b_i - b_1)$, $\vartheta_i = \int\chi_i h$, $\int b_i = 1$, χ_i a partition of unity and $b_i - b_1$ cut along the chain: Q commutes weakly with all $\varphi(h)\varphi(h')$, $h,h'\in\mathcal{F}_{\mathbb{R}}(O)$ of integral 0. (5) The cosine series converges on vectors with finitely many particles: Q commutes with the $\cos\bar\varphi(h)$, $h\in\mathcal{F}_{\mathbb{R}}(O)$, $\int h = 0$. Conversely, a bounded Q on the even space that commutes with these cosines lies in $\mathcal{C}_\Theta(O)$: $t^{-2}(1-\cos t\bar\varphi(h))\psi \to \frac{1}{2}\varphi(h)^2\psi$ on vectors with finitely many particles, polarisation gives the $\varphi(h)\varphi(h')$, with $h = \partial_\kappa\mathsf{k}^\kappa$ all bilocal fields smeared in $O\times O$, and point splitting the $\Theta_{\kappa\lambda}(f)$ and $\Theta(f)$, $f\in\mathcal{F}(O)$, whose polynomial domain lies in that of these limits.

So $\mathcal{C}_\Theta(O) = \{\cos\bar\varphi(h) : \int h = 0\}'$ on the even space, $\mathcal{A}_\Theta(O)$ is the even part of the algebra of the field strengths, $\mathfrak{R}(\mathsf{K}_0(O))$ restricted to the even space (with $\mathsf{K}_0(O)$ below), and the multiplet has the same weak commutants; the closures of the $T_{\kappa\lambda}(f)$ are self-adjoint by the linear bound and affiliated

by this identification; (W) follows from the Weyl relations. For n colours the same proof gives the net, with the $\mathrm{O}(\mathsf{n})$-invariant parts, the bounded functions of the operators $\sum_a(\sum_i \alpha_i x_{ia} + \beta_i p_{ia})^2$ on $L^2(\mathbb{R}^{n'\times\mathsf{n}})$, for every finite number n' of modes, generating the commutant of the rotations of the colours; at $\mathsf{n}=3$ it gives the affiliation of the $T_{\kappa\lambda}(f)$ as well.

The defect. For a region O let $\mathsf{K}(O)$ and $\mathsf{K}_0(O)$ be the closed real spans of the one-particle vectors of the $\varphi(h)$, $h\in\mathcal{F}_{\mathbb{R}}(O)$, respectively with $\int h=0$; at one time, $\mathsf{K}[D]$ is the closed real span of the one-particle vectors u_f+v_g of $\varphi(0,f)$, $\pi(0,g)$, $f,g\in C_c^\infty(D)$ real, and $\mathsf{K}_0[D]$ that with $\int f=0$; $D_\rho=\{|\vec{x}|<\rho\}$ and $D_\rho^e=\{|\vec{x}|>\rho\}$. For O the double cone over the disc D_ρ the Cauchy data at the time 0 and the finite speed give $\mathsf{K}(O)=\mathsf{K}[D_\rho]$, $\mathsf{K}_0(O)=\mathsf{K}_0[D_\rho]$, $\mathsf{K}(O^\times)=\mathsf{K}[D_\rho^e]$; and outside the charges escape to infinity: for $b\in C_c^\infty(D_\rho^e)$ with $\int b=1$, $b_n=n^{-2}b(\cdot/n)$ has $\|u_{b_n}\|^2=n^{-1}\|u_b\|^2\to 0$, so $u_f=\lim_n u_{f-(\int f)b_n}$ and $\mathsf{K}_0(O^\times)=\mathsf{K}(O^\times)=\mathsf{K}[D_\rho^e]$.

The symplectic complements: $\mathsf{K}[D_\rho^e]'=\overline{\mathsf{K}[D_\rho]}$, because for $w=w_1+iw_2\in\mathsf{K}[D_\rho^e]'$, w_1,w_2 in the real subspace of the w with $w(-\vec{k})=\overline{w(\vec{k})}$, the distributions F,G with $\tilde{F}=w_1$, $\tilde{G}=w_2/\omega$ have $\mathrm{Im}\langle u_f+v_g,w\rangle=\frac{1}{2}(\int fG-\int gF)$, so they vanish on D_ρ^e and are carried by $\bar{D}_\rho$; a dilation by $\lambda<1$, uniformly bounded on $L^2(d^2k/\omega)$ and on $L^2(\omega\,d^2k)$ and strongly continuous as $\lambda\uparrow 1$, and a mollifier make w a limit of vectors u_f+v_g with $f,g\in C_c^\infty(D_\rho)$; so $\mathsf{K}[D_\rho]'=\mathsf{K}[D_\rho^e]''=\mathsf{K}[D_\rho^e]$; and $\mathsf{K}_0[D_\rho]'=\mathsf{K}[D_\rho^e]+\mathbb{R}v_{g_1}$, $g_1=1$ near $\bar{D}_\rho$, because $u_f+v_g\mapsto\int f=2\mathrm{Im}\langle u_f+v_g,v_{g_1}\rangle$ is a continuous functional on $\mathsf{K}[D_\rho]$ with the kernel $\mathsf{K}_0[D_\rho]$.

The duality theorem for the free field, $\mathfrak{R}(\mathsf{K})'=\mathfrak{R}(\mathsf{K}')$, turns these into $\mathcal{C}_\Theta(O^\times)=\mathcal{A}_\Theta(O^\times)'$, the even part of $\mathfrak{R}(\mathsf{K}[D_\rho])$, and $\mathcal{C}_\Theta(O)=\mathcal{A}_\Theta(O)'$, the even part of $\mathfrak{R}(\mathsf{K}[D_\rho^e]+\mathbb{R}v_{g_1})$; the Weyl relation $e^{i\bar{\varphi}(h)}e^{i\mathsf{s}\bar{\pi}(g_1)}=e^{-i\mathsf{s}\int h}e^{i\mathsf{s}\bar{\pi}(g_1)}e^{i\bar{\varphi}(h)}$ gives the non-commuting pair. For $\mathsf{n}\geq 1$ colours, $h\in\mathcal{F}_{\mathbb{R}}(O)$ with $\mathsf{c}=\int h\neq 0$ and $\vec{n}$ uniform on the sphere $S^{\mathsf{n}-1}$, the mean over $\mathrm{O}(\mathsf{n})$ of $e^{i\bar{\varphi}^1(h)}$ is $X=\int e^{i\bar{\varphi}(h\vec{n})}d\vec{n}$, $\varphi(h\vec{n})=\sum_a n_a\varphi^a(h)$, and its restriction to the invariant vectors lies in $\mathcal{C}_\Theta(O^\times)$ by the Weyl relations.

If its restriction to the invariant vectors lay in $\mathcal{A}_\Theta(O)$, then, Ω being separating for the algebra of the field of O, so that the restriction to the invariant vectors is injective on its invariant part, X would lie in the algebra of the field strengths of O, which commutes with the shifts $e^{i\vec{s}\cdot\bar{\pi}(g_1)}$, $\bar{\pi}(g_1)=(\bar{\pi}^a(g_1))_a$; but conjugation by them multiplies $e^{i\bar{\varphi}(h\vec{n})}$ by $e^{i\mathsf{c}\vec{s}\cdot\vec{n}}$, and the one-particle part of the difference on Ω, $ie^{-\|u_h\|^2/2}u_h\otimes\int(e^{i\mathsf{c}\vec{s}\cdot\vec{n}}-1)\vec{n}\,\mathrm{d}\vec{n}$, u_h the one-particle vector of $\varphi(h)$, is $\neq 0$ for small $\vec{s}\neq 0$.

So the restriction of X lies in $\mathcal{A}_\Theta^d(O)=\mathcal{C}_\Theta(O^\times)$ and outside $\mathcal{A}_\Theta(O)$. The even parts, restricted to the even Fock space, of the nets $\mathfrak{F}_\Lambda(O)=\{e^{i\bar{\varphi}(f)}:f\in\mathcal{F}_{\mathbb{R}}(O),\ \int f\in\Lambda\}''$ are local, lie between, are injective in Λ (an h with $\int h\in\Lambda\setminus\Lambda'$ separates), and have the wedge algebras of Θ, because in a wedge the field strengths generate the field, the charges escaping to infinity inside the wedge: for $f\in\mathcal{F}_{\mathbb{R}}(W)$, $f_n=n^{-3}f(\cdot/n)\in\mathcal{F}_{\mathbb{R}}(W)$ has $\int f_n=\int f$ and $\|u_{f_n}\|^2=n^{-1}\|u_f\|^2\to 0$, while $f-f_n$ has the integral 0. At $\mathsf{m}>0$ the cosines of all $h\in\mathcal{F}_{\mathbb{R}}(O)$ are reached, as $\varphi(h)=\varphi(\partial_\kappa(-\mathsf{m}^{-2}\partial^\kappa h))$, and the one-particle duality, which holds with $\omega=(|\vec{k}|^2+\mathsf{m}^2)^{1/2}$, gives $\mathcal{C}_\Theta(O^\times)=\mathcal{A}_\Theta(O)$.

This is the case $\mathsf{n}=1$. For n colours at a mass $\mathsf{m}>0$ the steps (1)–(4) run with the colour sums in the place of the products of one field: the commutator of two $\Theta(f_i)$ is the colour sum of the bilocal fields, weighted by the matrix H of the second derivatives of the commutator function of the mass m, which is invertible at short timelike separations, $\mathsf{m}(z\cdot z)^{1/2}<2$, and differs from that of the mass 0 by terms less singular at $z=0$, which the point splitting of (2) does not see; the double cones of (4) are taken small; and the commutators of the colour sums $\sum_a\varphi^a(h)\varphi^a(h')$ have the structure constants of the display of (4) at every mass.

So every $Q\in\mathcal{C}_\Theta(O)$, O open and connected, commutes weakly with the colour sums of all real $h,h'\in\mathcal{F}(O)$ of integral 0, and, by $\varphi^a(h)=\varphi^a(\partial_\kappa(-\mathsf{m}^{-2}\partial^\kappa h))$, with those of all real $h,h'\in\mathcal{F}(O)$; among them are the squares $\sum_a\varphi^a(g)^2$, whose closures generate $\mathcal{A}_S(O)$, and point splitting gives the $S(f)$, $f\in\mathcal{F}(O)$: $\mathcal{C}_\Theta(O)\subset\mathcal{C}_S(O)$, so $\mathcal{A}_S(O)\subset\mathcal{A}_\Theta(O)$ on $\mathcal{H}_\Theta=\mathcal{H}_S$. $\Theta(f)=\frac{1}{2}S((\Box+2\mathsf{m}^2)f)$ gives the converse. So $\mathcal{A}_\Theta=\mathcal{A}_S$ on every connected open region, wedges included, and Thm. AO(e)

for Θ with the duality theorem for the free field gives, as at the mass 0 below, $\mathcal{A}^d_\Theta(O) = \mathcal{C}_\Theta(O^\times)$, the invariant part of the algebra of the field of O, which is $\mathcal{A}_S(O) = \mathcal{A}_\Theta(O)$: the net of Θ, as that of S, is its own dual net, for every $\mathsf{n} \geq 1$.

The massless square. We first establish the operator core for $S = {:}\varphi \cdot \varphi{:}$. This uses particle number, rather than the energy bound used for Θ. Work in the one-particle realization $L^2(\mathbb{R}^2, d^2p)$, with $\omega_p = |p|$. The Wick expansion of $S(f)$ has a number-preserving part and pair-creation and pair-annihilation parts. Apart from the fixed Fourier normalization and the factor 2 in the number-preserving part, their kernels are

$$K_f(p,q) = \frac{\hat{f}(\omega_p - \omega_q, p-q)}{\sqrt{\omega_p \omega_q}}, \qquad B_f(p,q) = \frac{\hat{f}(\omega_p + \omega_q, p+q)}{\sqrt{\omega_p \omega_q}}.$$

For $f \in \mathcal{S}$, the first kernel defines a bounded operator and the second is Hilbert–Schmidt. To see the first assertion, on $|p|, |q| \geq 1$ use Schur's test and $|\hat{f}(\omega_p - \omega_q, p-q)| \leq C_M(1 + |p-q|)^{-M}$, with $M > 2$. The complementary part is Hilbert–Schmidt: $|p|^{-1}$ is locally integrable in two dimensions, and the same decay controls the other variable when its radius is large. For the pair kernel, decay in the energy sum gives

$$\iint |B_f(p,q)|^2 \, d^2p \, d^2q \leq C_M \int_0^\infty \int_0^\infty (1 + r + s)^{-M} \, \mathrm{d}r \, \mathrm{d}s < \infty \qquad (M > 2).$$

The finitely many colours only change the constants.

Let $\mathcal{F}_{\leq k}$ denote the subspace of vectors with at most k particles. The creation and annihilation formulas therefore give

$$S(f)\mathcal{F}_{\leq k} \subset \mathcal{F}_{\leq k+2}, \qquad \|S(f)\psi\| \leq c_f(k+2)\|\psi\|.$$

These are particle-sector bounds; the operator $N = 1 + H$ retains its earlier meaning. For real f and $\psi \in \mathcal{F}_{\leq k}$, iteration gives

$$\|S(f)^r \psi\| \leq c_f^r \prod_{j=1}^{r} (k + 2j) \, \|\psi\|.$$

The series $\sum_{r \geq 0} t^r \|S(f)^r \psi\|/r!$ converges for $2c_f|t| < 1$. Every vector of $\mathcal{D}_0^S$ has finitely many particles, and $S(f)$ preserves $\mathcal{D}_0^S$. This domain is dense in $\mathcal{H}_S$ by its definition. The analytic-vector theorem, in the form of [6, Thm. 2, pp. 180–181], thus proves that $S(f)$ is essentially self-adjoint on $\mathcal{D}_0^S$, without using any identification of its weak commutant or its net.

The identification of $\mathcal{H}_S$ with the invariant subspace is also made before using weak commutants. The bilocal commutator construction used in (e), with S in place of Θ, gives the colour sums $\sum_a \varphi^a(g)\varphi^a(g')$ and their squares for all real compactly supported tests, with no integral-zero restriction. They are limits of polynomials in the $S(f)$ on vectors of finitely many particles, with bounds uniform on each $\mathcal{F}_{\leq k}$. The third commutator and one further commutator give the squares; polarization gives the colour sums. The first fundamental theorem for $\mathrm{O}(\mathsf{n})$ expresses every invariant field polynomial in these colour sums. Applying their polynomial approximants to the vacuum and using those finite-particle bounds shows that $\mathcal{H}_S$ contains the invariant polynomial vectors. Those vectors are dense in the invariant subspace, while every S-polynomial vector is invariant. Hence the two spaces agree.

On the common finite-particle energy-smooth domain the symmetric Wick operator extends its restriction to $\mathcal{D}_0^S$. Essential self-adjointness identifies its closure with the closure just obtained. If a bounded operator Q commutes weakly with $S(f)$ on $\mathcal{D}_0^S$, the adjoint-domain criterion first gives $Q\mathcal{D}_0^S \subset D(\overline{S(f)})$ and $\overline{S(f)}Q\psi = QS(f)\psi$ there. Approximation in the graph norm then gives $Q\overline{S(f)} \subset \overline{S(f)}Q$; the same holds for Q^*. Thus Q commutes with the bounded functions of $\overline{S(f)}$, and the Wick products and the limits described above can be taken on the common core.

For S, the squares $\sum_a \varphi^a(g)^2$ are reached for every real $g \in \mathcal{F}(O)$ by these commutators. Consequently $\mathcal{A}_S(O)$ is the invariant part of the algebra of the field of O, and $\mathcal{A}_\Theta(O) \subsetneq \mathcal{A}_S(O)$ at

$\mathsf{m}=0$. The field strengths generate the field in every wedge, so $\mathcal{A}_S(\tilde W)=\mathcal{A}_\Theta(\tilde W)$ and $\mathcal{A}_S^d=\mathcal{A}_\Theta^d$. Finally, Thm. AO(e) for Θ and the duality theorem for the free field give $\mathcal{A}_\Theta^d(O)=\mathcal{C}_\Theta(O^\times)$, the invariant part of the algebra of the field of O. This is $\mathcal{A}_S(O)$, and therefore $\mathcal{A}_S=\mathcal{A}_S^d=\mathcal{A}_\Theta^d$.

14.1.10. *Proof of Thm. AS. (a)* Let $\Gamma(g)$, $g\in G^\Sigma$, be the gauge transformations on $L^2(G^{\mathsf B})$, $\mathfrak{F}(Y)=\mathcal{B}(L^2(G^Y))\otimes 1$, and $P=\int\Gamma(g)\,\mathrm{d}g$ the projection on $\mathcal{H}_{\rm ph}$. $\mathfrak{N}=\mathfrak{F}(Y)\cap\Gamma(G^\Sigma)'$ is a von Neumann algebra and $P\in\Gamma(G^\Sigma)''\subset\mathfrak{N}'$, so $\mathsf{A}(Y)=\mathfrak{N}P$ is a von Neumann algebra with the commutant $P\mathfrak{N}'P|_{\mathcal{H}_{\rm ph}}$, and $\mathfrak{N}'=\mathfrak{F}(\mathsf{B}\setminus Y)\vee\Gamma(G^\Sigma)''$ is the weak closure of the span of the $X\Gamma(g)$, $X\in\mathfrak{F}(\mathsf{B}\setminus Y)$. As $P\Gamma(g)=P$, $PX\Gamma(g)P=PXP=PX^\natural P$, $X^\natural=\int\Gamma(g)X\Gamma(g)^{-1}\,\mathrm{d}g\in\mathfrak{F}(\mathsf{B}\setminus Y)\cap\Gamma'$; so $\mathsf{A}(Y)'=\mathsf{A}(\mathsf{B}\setminus Y)$. At $Y=\varnothing$ this is $\mathsf{A}(\mathsf{B})=\mathcal{B}(\mathcal{H}_{\rm ph})$. Without a common site the gauge groups of Y_1 and Y_2 are independent, $(X_1\otimes X_2)^\natural=X_1^\natural\otimes X_2^\natural$, and the products are weakly total.

The flips. $V\mapsto cV$ keeps the Haar measure; $\chi_j(-g)=(-1)^{2j}\chi_j(g)$ gives $\mathsf{V}(c)M_{\chi_j(U(C'))}=c(C')^{2j}M_{\chi_j(U(C'))}\mathsf{V}(c)$. Two groups of coboundaries are told apart. If $c_b=z(b_-)z(b_+)$ on all the bonds of the layer, z a sign field on all its sites, $\mathsf{V}(c)$ is the gauge transformation $\Gamma(z)$ by the central field z, which is 1 on $\mathcal{H}_{\rm ph}$. For a set S of plaquettes, the coboundaries $B^1(S)=\{dz|_{\mathsf{B}_S}\}$, z a sign field on the sites of B_S, form the subgroup of $Z^1(S)$ by which $H^1(S)=Z^1(S)/B^1(S)$ is formed. For $c=dz|_{\mathsf{B}_S}$ and $\tilde z$ the continuation of z by 1 to all sites, $\mathsf{V}(c)=\Gamma(\tilde z)\mathsf{V}(c^\circ)$, where $c^\circ=d\tilde z$ on $\mathsf{B}\setminus\mathsf{B}_S$ and $c^\circ=1$ on B_S, the sign field c° of the statement; so on $\mathcal{H}_{\rm ph}$, $\mathsf{V}(c)=\mathsf{V}(c^\circ)$, a product of the parities $\sum_j(-1)^{2j}\Pi_b^j$ of bonds $b\notin\mathsf{B}_S$, which lies in $\mathsf{A}(\mathsf{B}\setminus\mathsf{B}_S)$; it is not 1 when, say, $z=-1$ at one site x alone and a plaquette $q\notin S$ has at x one bond in B_S and one off it, for then it anticommutes with $M_{t_q}\neq 0$.

As $\Gamma(\tilde z)$ commutes with every gauge-invariant operator and $\mathsf{V}(c^\circ)$ with every operator of the bonds of B_S, $\operatorname{Ad}\mathsf{V}(c)$ is the identity on $\mathsf{A}(\mathsf{B}_S)$ for $c\in B^1(S)$; and on $\mathcal{H}_{\rm ph}$ the flips of one class of $H^1(S)$ differ by elements of $\mathsf{A}(\mathsf{B}\setminus\mathsf{B}_S)$, so that beside $\mathsf{A}(\mathsf{B}\setminus\mathsf{B}_S)$ one flip of each class is enough. On the bonds where $c=-1$, $\mathsf{V}(c)$ acts in the spin-j sector of the bond as $(-1)^{2j}$, so it lies in the algebra of the spin projections of these bonds. On a periodic layer every bond lies in two plaquettes, so $\prod_p\prod_{b\in\partial p}c_b=\prod_b c_b^2=1$, and the plaquettes where c is not closed are even in number.

(b) Let Y commute with the M_{t_p}, $p\in S$, and the Π_b^j, $b\in\mathsf{B}_S$, on $L^2(G^{\mathsf{B}_S})$. Then Y keeps the polynomials in the matrix entries and commutes with the Laplacian $\Delta_b=-\sum_j j(j+1)\Pi_b^j$ of every bond; for functions F,F' with $M_F,M_{F'}$ in the algebra generated,

$$[[\Delta_b,M_F],M_{F'}]=2M_{D_b(F,F')},\qquad D_b(F,F')=\sum_a(J_b^aF)(J_b^aF'),$$

J_b^a the infinitesimal left translations of the bond b, so the real algebra $\mathfrak{W}$ of such functions is closed under every D_b. For two closed paths $\mathsf{M}=V_b\mathsf{a}$, $\mathsf{N}=V_b\mathsf{c}$ that run once through b, $D_b(\operatorname{tr}\mathsf{M},\operatorname{tr}\mathsf{N})=\frac14[\operatorname{tr}\mathsf{M}^{-1}\mathsf{N}-\operatorname{tr}\mathsf{MN}]$ and $\operatorname{tr}\mathsf{M}\operatorname{tr}\mathsf{N}=\operatorname{tr}\mathsf{MN}+\operatorname{tr}\mathsf{M}^{-1}\mathsf{N}$: paths may be fused at a common bond. Four fusions in every chain T of three plaquettes, in a row or around a site, give the seven traces $\frac12$tr of α_i, $\alpha_i\alpha_k$, $\alpha_1\alpha_2\alpha_3$, α_i the boundaries at one site. With $g=\mathsf{s}(g)+i\mathsf{v}(g)\cdot\sigma$,

$$\mathsf{s}(g_1g_2g_3)=\mathsf{s}_1\mathsf{s}_2\mathsf{s}_3-\mathsf{s}_1\mathsf{v}_2\cdot\mathsf{v}_3-\mathsf{s}_2\mathsf{v}_1\cdot\mathsf{v}_3-\mathsf{s}_3\mathsf{v}_1\cdot\mathsf{v}_2+\det(\mathsf{v}_1,\mathsf{v}_2,\mathsf{v}_3),$$

so the seven numbers are the Gram matrix and the determinant of the $\mathsf{v}(V(\alpha_i))$, which fix the triple up to a rotation; every closed path of T is a word in the α_i, and so two configurations of T with the same seven numbers differ by a gauge transformation of T. Two chains that share two plaquettes fix the gauge transformation on their common sites up to the centre, as two non-commuting elements of SU(2) have the centraliser $\{\pm1\}$; along the chains of S the configurations with the same functions in $\mathfrak{W}$ are those that differ by a gauge transformation and a closed sign field. Let $\mathcal{K}_S$ be the compact group of the pairs (g,c), g a gauge transformation of the sites of B_S and $c\in Z^1(S)$, acting by $((g,c)V)_b=c_bg(b_-)V_bg(b_+)^{-1}$, and λ its representation on $L^2(G^{\mathsf{B}_S})$; $\lambda(\mathcal{K}_S)$ commutes with the M_{t_p} and the Π_b^j.

Let $\delta_0 = \prod |\mathsf{v}(V(\alpha_p)) \times \mathsf{v}(V(\alpha_q))|^2$, the product over the pairs of adjacent plaquettes of S, α_p, α_q their boundaries begun at an end of the common bond; it is an invariant polynomial, $\not\equiv 0$ on the connected space G^{B_S}, so its zero set is a null set. Off it the stabiliser of V in $\mathcal{K}_S$ is $\{(z, dz)\}$, z central, which acts trivially (an element of SU(2) that commutes with two elements that do not commute is ± 1), and the seven numbers of the chains separate the orbits; so by Stone–Weierstrass on the compact sets $\{|\delta_0| \geq \epsilon\}$, M_F lies in the generated algebra for every continuous $\mathcal{K}_S$-invariant F, and Y commutes with all of them.

The polynomials are the sum of the isotypic parts Pol_π of the irreducible representations (π, V_π) of $\mathcal{K}_S$ that are trivial on the stabiliser; at a point V off the zeros of δ_0 the functionals $v \mapsto m_v(V)$ of the equivariant linear maps $m : \mathsf{V}_\pi \to \mathrm{Pol}$ fill V_π^*, the orbit of V being $\mathcal{K}_S$ modulo the stabiliser (Peter–Weyl and Stone–Weierstrass on it). With $\mathsf{n} = \dim \pi$ such maps $m^{(i)}$ whose functionals at one point V_1 form a basis of V_π^*, $\mathsf{M} = (m^{(i)}_{e_k})_{ik}$ in a basis (e_k) of V_π and $\delta_\pi = \det \mathsf{M}$, Cramer's rule writes $|\delta_\pi|^2 m_v$ as a combination of the $m_v^{(i)}$ with invariant polynomial coefficients, with which Y commutes; so $Y m_{e_k} = \sum_l m_{e_l} \tau_{lk}$ for every equivariant m on $\{\delta_\pi \neq 0\}$, with $\tau = \mathsf{M}^{-1}(Y m^{(i)}_{e_k})_{ik}$.

The vector fields $\xi_{b,F} = \sum_a (J^a_b F) J^a_b = \frac{1}{2}([\Delta_b, M_F] - M_{\Delta_b F})$, F a real invariant polynomial, commute with Y on the polynomials and with $\lambda(\mathcal{K}_S)$, and carry equivariant maps to equivariant maps; the identity for the equivariant map $\xi_{b,F} m$, set beside $\xi_{b,F}$ applied to the identity for m ($\xi_{b,F}$ is a derivation), gives $\mathsf{M}\, \xi_{b,F} \tau = 0$, so $\xi_{b,F} \tau = 0$ on $\{\delta_\pi \neq 0\}$. The $\xi_{b,F}(V)$ span the tangent space on a dense open set: at a bond b of a plaquette p of a chain $T \subset S$, with α the boundary of p begun with b and w_1, w_2 closed paths at b_- through the two bonds c_1, c_2 that a spanning tree of $\mathsf{B}_T \setminus \{b\}$ leaves out, the traces $\operatorname{tr} V(\alpha)$, $\operatorname{tr} V(\alpha w_i)$ are invariant and their derivatives $(J^a_b \cdot)_a$ are $-\mathsf{v}(V(\alpha))$, $-\mathsf{v}(V(\alpha w_i))$, whose determinant is a polynomial that is not 0 at a configuration with $\mathbf{1}$ on the tree and $V(\alpha) = i\sigma_3$, $V(\alpha w_1) = i\sigma_1$, $V(\alpha w_2) = i\sigma_2$ (the bonds b, c_1, c_2 are free there).

So $d\tau = 0$ on a dense open part of $\{\delta_\pi \neq 0\}$, τ is constant, τ^π, near V_1, and the matrix of polynomials $(Y m^{(i)}_{e_k})_{ik} - \mathsf{M}\tau^\pi$, which vanishes near V_1, vanishes identically: here analyticity enters. Hence $Y m_v = m_{\tau^\pi v}$ for all equivariant m, and with $y_\pi(\kappa) = \dim \pi \operatorname{tr}(\tau^\pi \pi(\kappa)^{-1})$, $Y P_\pi = \lambda(y_\pi) \in \lambda(\mathcal{K}_S)''$, P_π the projection on the isotypic part; summed over π, $Y \in \lambda(\mathcal{K}_S)''$. So the generated algebra is the commutant of the gauge transformations of S and of the $\mathsf{V}(c)$, $c \in Z^1(S)$; restricted to $\mathcal{H}_{\mathrm{ph}}$ and combined with (a), this is the pair of identities for $\mathsf{A}_\bullet(S)$, and it exhibits $\mathsf{A}_\bullet(S)$ as the join of the $\mathsf{A}(\mathsf{B}_T)$ over the chains.

For admissible S the join of the $\mathsf{A}(\partial p)$, $p \in S$, is $\mathsf{A}_\bullet(S)$: the generators M_{t_p} and Π^j_b, $b \in \partial p$, lie in $\mathsf{A}(\partial p)$, and $\mathsf{A}(\partial p) \subset \mathsf{A}(\mathsf{B}_T) \subset \mathsf{A}_\bullet(S)$ for a chain $T \subset S$ of three plaquettes through p, which admissibility provides. The $\mathsf{V}(c)$, $c \in Z^1(S)$, are functions of spin projections of B_S and commute with $\mathsf{A}_\bullet(S)$: they are central. $\operatorname{Ad} \mathsf{V}(c)$ is the identity on $\mathsf{A}(\mathsf{B}_S)$ for $c \in B^1(S)$ (the flips above), so $H^1(S)$ acts, and the decomposition into its characters is the spectral decomposition of this action of a finite abelian group; the summand of a character κ holds $M_{\chi_{1/2}(U(C_\kappa))} \neq 0$ for a closed path C_κ with $c(C_\kappa) = \kappa([c])$.

If every $\mathbb{Z}_2$-cycle is a sum of boundaries, every closed c is a coboundary. For one plaquette, the cyclic shift of the four bonds times the projection on the spin 1 at one corner and 0 at the others commutes with M_{t_p} and all Π^j_b and moves the charge to the next corner; for two adjacent plaquettes use reflection in the perpendicular bisector of the common bond, composed with the projection on charge 1 at one end and 0 at all other vertices. On this projected block the zero charges at the outer vertices force equal bond spins along each of the two outer three-bond paths. Reflection reverses each path and preserves each individual bond-spin projection, as well as each plaquette trace, but moves the endpoint charge; neither lies in the von Neumann algebra generated by the gauge transformations and the closed flips, which is the commutant of all invariant operators, so the traces and spin projections generate less. The traces generate multiplications, and M_{t_p} does not commute with Π^j_b, $b \in \partial p$.

Two plaquettes with one common bond: the full algebras generate. Let p, q share the bond c_0 with the ends x, y; their union has seven bonds and six sites, and at x, as at y, three bonds meet: $a \in \partial p$, c_0 and $a' \in \partial q$. Let $\mathfrak{B}$ be the algebra of all gauge-invariant bounded operators on $L^2(G^7)$, the functions of these bonds, and $\mathfrak{N} \subset \mathfrak{B}$ the von Neumann algebra generated by the gauge-invariant operators of the four bonds of p and those of the four bonds of q; we show $\mathfrak{N} = \mathfrak{B}$, which, restricted to $\mathcal{H}_{\mathrm{ph}}$ by a normal homomorphism, gives $\mathsf{A}(\partial p) \vee \mathsf{A}(\partial q) = \mathsf{A}(\partial p \cup \partial q)$.

By Peter–Weyl, $L^2(G^7) = \bigoplus_{\mathbf{j}} \mathcal{K}(\mathbf{j})$ over the spins $\mathbf{j}$ of the seven bonds, $\mathcal{K}(\mathbf{j}) = \bigotimes_z \mathcal{V}_z(\mathbf{j})$, $\mathcal{V}_z(\mathbf{j})$ the product of the spin-j_b representations of the bonds at the site z, on which the gauge group at z acts; $\mathcal{V}_z(\mathbf{j}) = \bigoplus_J V_J \otimes \mathsf{M}_z(\mathbf{j}, J)$, and a label $(\mathbf{j}, \mathbf{J})$, $\mathbf{J}$ the charges at the six sites, occurs if the polygon inequalities hold at every site. So $\mathfrak{B} = \bigoplus_{\mathbf{J}} \mathcal{B}(\mathsf{M}_{\mathbf{J}})$, $\mathsf{M}_{\mathbf{J}} = \bigoplus_{\mathbf{j}} \bigotimes_z \mathsf{M}_z(\mathbf{j}, J_z)$. (i) *Labels.* Let $Y \in \mathfrak{B}$ commute with $\mathfrak{N}$.

It commutes with the Casimir operators of the bonds, so it keeps every $\mathcal{K}(\mathbf{j})$; being gauge invariant, it keeps the charges $\mathbf{J}$; and for a corner (b_1, b_2) of p or q at z the spectral projections of the Casimir operator of the gauge group at z acting on the two bonds of that plaquette lie in $\mathfrak{N}$, so Y commutes with $t^z_{b_1 b_2} = J_{b_1,z} \cdot J_{b_2,z}$ on every $\mathcal{K}(\mathbf{j})$. At the four sites with two bonds M_z is one-dimensional; at x and at y, t_{ac_0} and $t_{c_0 a'}$ generate all operators of $\mathsf{M}_z(\mathbf{j}, J_z)$ by Lem. 14.19. So $Y = \sum y(\mathbf{j}, \mathbf{J}) P(\mathbf{j}, \mathbf{J})$, $P(\mathbf{j}, \mathbf{J})$ the projection on the label. (ii) *One plaquette made large.* If two labels with the same charges occur and agree off ∂p, y takes one value at them.

Indeed $L^2(G^{\partial p}) = \bigoplus_\pi V_\pi \otimes \mathrm{Mult}_\pi$ over the charges $\pi = (f_z)$ at the four corners of p, Mult_π with the basis $e_{\mathbf{i}}$ indexed by the spins $\mathbf{i}$ of ∂p with $f_z \in i_{b_1} \otimes i_{b_2}$ at every corner (Clebsch–Gordan is free of multiplicities); the gauge-invariant operators of ∂p are the operators $1 \otimes \mathcal{B}(\mathrm{Mult}_\pi)$, and Y, commuting with them, is the identity on every Mult_π; so y takes one value on the spins $\mathbf{i}$ of ∂p of every π that occurs with the given spins off ∂p and the given charges; and with N above all charges that occur and $e_b \in \{0, \frac{1}{2}\}$ with $e_{b_1} + e_{b_2} \equiv f_z$ modulo 1 at the four corners (they exist, the charges at the corners being fixed modulo 1 by the charges $\mathbf{J}$ and the spins off ∂p, with an integer sum), the spins $(N + e_b)_{b \in \partial p}$ belong to every such π, which joins them all. (iii) *A monotone path.* Raising the four spins of one plaquette by $\frac{1}{2}$ keeps the charges and the polygon inequalities, two spins at each corner growing by $\frac{1}{2}$.

By such raisings two labels with the same charges are brought to an integer difference, twice their difference modulo 2 being a cycle and so a sum of ∂p and ∂q; raised $2N$ times more on both plaquettes, N large, they are joined through the label that takes the larger of the two spins on the six bonds that lie on one plaquette and the smaller on c_0, changing first the bonds of p, then those of q; all labels on the way occur. With (ii) at every change, y depends on $\mathbf{J}$ alone. (iv) *The charges.* The projections E^z_J on the charge J at z lie in $\mathfrak{N}$. At the four outer sites they are spectral projections of Casimir operators of one plaquette.

At x, with $A = J_{a,x}$, $B = J_{c_0,x}$, $D = J_{a',x}$, the Casimir operator $(A+B+D)^2$ holds the cross term t_{AD} of the two outer bonds, which belongs to neither plaquette; the identity of three commuting spins

$$(B^2 + t_{AB})\, t_{AD} = A^2 t_{BD} + t_{AB} t_{BD} - [t_{AB}, t_{BD}] - [t_{AB}, [t_{AB}, t_{BD}]]$$

puts $\mathsf{F} t_{AD}$ in $\mathfrak{N}$ on every block of spins, $\mathsf{F} = B^2 + t_{AB} = \frac{1}{2}((A+B)^2 + B^2 - A^2)$, and with it $\mathsf{F}(A+B+D)^2$. As $(A+B+D)^2$ commutes with all of $\mathfrak{B}$, and F is the number $\frac{1}{2}(f(f+1) + s_c(s_c+1) - s_a(s_a+1))$ on the range where the spins of a, c_0 are s_a, s_c and $(A+B)^2 = f(f+1)$, a range given by projections of $\mathfrak{N}$, $(A+B+D)^2$ times such a range lies in $\mathfrak{N}$ where this number is not 0, and with it E^x_J times the range; the same holds where $f = 0$, as $(A+B+D)^2 = D^2$ there. The weakly closed ideal of the $n \in \mathfrak{N}$ with $nE^x_J \in \mathfrak{N}$ is $\mathfrak{N}\mathsf{p}$ with a central projection p of $\mathfrak{N}$ that holds these ranges; $1 - \mathsf{p}$ commutes with $\mathfrak{N}$, so by (i)–(iii) it is the sum of the $P(\mathbf{j}, \mathbf{J})$ over the labels of some charges, and if it were not 0, each of its ranges of the kind above would have $f > 0$ and $f(f+1) = s_a(s_a+1) - s_c(s_c+1)$; but raising p by $\frac{1}{2}$ keeps the charges and f and turns such a range into one where the number is $\frac{1}{2}(s_c - s_a) \neq 0$, a range of p: so $\mathsf{p} = 1$. (v) $\mathfrak{N}$ holds the projections on

the $\mathsf{M}_{\mathbf{J}}$, and an operator on $\mathsf{M}_{\mathbf{J}}$ that commutes with $\mathfrak{N}$ is a number by (i)–(iii); so $\mathfrak{N}$ cut down to $\mathsf{M}_{\mathbf{J}}$ is irreducible, and $\mathfrak{N} = \bigoplus_{\mathbf{J}} \mathcal{B}(\mathsf{M}_{\mathbf{J}}) = \mathfrak{B}$.

This is the two-plaquette case of Lem. 14.16(a).

(c) R deformation retracts to a circle around the hole: $H^1(R) = \mathbb{Z}_2$, generated by $[c_*]$, and $c_*(C) = -1$. $\mathsf{A}(\mathsf{B} \setminus \mathsf{B}_R)' = \mathsf{A}(\mathsf{B}_R)$ by (a). As v represents the generator, (b) gives $\mathsf{A}(\mathsf{B}_R) = \mathsf{A}_\bullet(R) \oplus \mathsf{A}(\mathsf{B}_R)_-$, and $u = \operatorname{sign} \operatorname{tr} U(C)$ is an odd self-adjoint unitary in $\{W_C\}''$, so $\mathsf{A}(\mathsf{B}_R)_- = \mathsf{A}_\bullet(R)u$; $\frac{1}{2}(X + vXv)$ is the projection on the even part. $\mathsf{A}_\bullet(R)' = \mathsf{A}(\mathsf{B} \setminus \mathsf{B}_R) \vee \{v\}$ by (b), v being one flip of the class that is not trivial and the flips of $B^1(R)$ acting on $\mathcal{H}_{\mathrm{ph}}$ through $\mathsf{A}(\mathsf{B} \setminus \mathsf{B}_R)$; and on $\mathcal{H}_{\mathrm{ph}}$, $v = \mathsf{V}_\gamma \mathsf{V}(c')$ with c' carried by $\mathsf{B} \setminus \mathsf{B}_R$ and $\mathsf{V}(c') \in \mathsf{A}(\mathsf{B} \setminus \mathsf{B}_R)$; $\operatorname{Ad} u$ fixes $\mathsf{A}(\mathsf{B} \setminus \mathsf{B}_R)$ and changes the sign of V_γ ($c_\gamma(C) = -1$), which gives the second decomposition and $\mathsf{A}(\mathsf{B} \setminus \mathsf{B}_R) = \mathsf{A}_\bullet(R)' \cap \{u\}'$.

For a closed path C', $vM_{\chi_j(U(C'))}v = c_*(C')^{2j} M_{\chi_j(U(C'))}$. If an assignment $\mathscr{N}$ equal to A on closed and open rectangles were additive and dual, then $\mathscr{N}(\mathsf{B}_R) = \bigvee_i \mathsf{A}(\mathsf{B}_{D_i}) = \mathsf{A}_\bullet(R)$ and $\mathscr{N}(\mathsf{B} \setminus \mathsf{B}_R) = \mathsf{A}(\mathsf{B}_{\mathrm{in}}) \vee \mathsf{A}(\mathsf{B}_{\mathrm{out}}) = \mathsf{A}(\mathsf{B} \setminus \mathsf{B}_R)$, the two pieces of the complement having no common site; duality would give $\mathsf{V}_\gamma \in \mathsf{A}(\mathsf{B} \setminus \mathsf{B}_R)$. The four rectangles: every chain of three plaquettes of R lies in one D_i or in the union of two adjacent ones, and two adjacent rectangles glue by Lem. 14.16(a), their overlap being a connected rectangle with a cycle; so $\mathsf{A}_\bullet(R) = \bigvee_i \mathsf{A}(\mathsf{B}_{D_i})$.

The commutative statement and the formula for the adjoint loop are Lem. 14.16(b), (c) for the four rectangles as boxes, whose overlaps hold two independent cycles when no corner is one plaquette.

(d) S deformation retracts to the torus with two holes: $H^1(S) = \mathbb{Z}_2^3$, generated by the cut γ and the two winding cuts. $\mathsf{A}(\mathsf{B}_S)' = \mathsf{A}(\mathsf{B}^\circ_{D_1} \cup \mathsf{B}^\circ_{D_2}) = \mathsf{N}$ by (a), and (b) gives $\mathsf{A}_\bullet(S)' = \mathsf{N} \vee \{\mathsf{V}_\gamma, \mathsf{V}_1, \mathsf{V}_2\}$, the products of $\mathsf{V}_\gamma, \mathsf{V}_1, \mathsf{V}_2$ meeting every class of $H^1(S)$ and the flips of $B^1(S)$ acting on $\mathcal{H}_{\mathrm{ph}}$ through N. The fundamental loop around D_1 lies in $\mathsf{A}(\mathsf{B}_S)$, commutes with N, V_1, V_2 and anticommutes with V_γ; the winding loops along the free row and column anticommute with one V_i each and commute with N and V_γ: hence the two non-memberships. On the whole layer $H^1 = \mathbb{Z}_2^2$ and $\mathsf{A}(\varnothing) = \mathbb{C}$; the four joint eigenspaces of $\mathsf{V}_1, \mathsf{V}_2$ are carried into one another by the fundamental winding loops, so none is 0. Off one disc, H^1 is generated by the two fluxes, the loop around the disc being a boundary in homology modulo 2 there.

(e) A box has a shelling, an order of its plaquettes in which each new plaquette either has all its bonds already present or is glued to an earlier one along a single bond so that the overlap with what is present is connected, and every site of a new bond of p_k that is already a site of X_{k-1} lies on the boundary ∂p_j of that earlier plaquette together with the overlap $\partial p_k \cap X_{k-1}$; Lem. 14.16(a), the two-plaquette case included, carries the generation from step to step: at a step that brings new bonds it applies to $Y_1 = X_{k-1}$ and $Y_2 = \partial p_j \cup \partial p_k$, with $I = \partial p_j \cup (\partial p_k \cap X_{k-1})$ connected and holding the cycle ∂p_j, $\mathsf{R}_2 = \partial p_k \setminus X_{k-1}$, and the site clause of the shelling as its hypothesis on the common sites, while the gauge-invariant operators of Y_2 are generated by those of ∂p_j and of ∂p_k, by the two-plaquette case.

The solid ring is the union of two shellable halves, the cubes $[i_1, i_1 + 1] \times [i_2, i_2 + 1] \times [0, 1]$ of the ring with $i_1 \in \{0, 1\}$ and those with $i_1 \in \{1, 2\}$, whose intersection has two components, the bonds of the two cubes with $i_1 = 1$, $i_2 \in \{0, 2\}$, each with a cycle; a site common to the two halves has $x_1 \in \{1, 2\}$, and the sites of the first half with $x_1 = 2$, as those of the second with $x_1 = 1$, are corners of these two cubes, so every site common to B_{S_1} and B_{S_2} is a site of I, the hypothesis of Lem. 14.16(a) on the common sites, as in the general union of (e), where a site of B_{S_1} on a bond of B_{S_2} off I is common to both; Lem. 14.16(a) gives the join as the invariant operators that commute with the central gauge transformations constant on each component, that is with $\mathsf{V}(c_*)$, and duality gives the commutant; $\mathsf{V}(c_*)$ anticommutes with W_C because $c_*(C) = -1$; for a dual surface meeting the ring in that section, $c_{\mathfrak{s}} c_*$ is carried by the complement.

With r components the central gauge transformations constant on components, modulo the global ones, form $\mathbb{Z}_2^{r-1}$.

(f) The substitution $U \mapsto sU$ keeps the product Haar measure, $t_p(sU) = (ds)_p t_p(U)$ and $\chi_j(U(C))(sU) = s(C)^{2j}\chi_j(U(C))(U)$. For a fat odd loop the same with $F(sU) = s(C)F(U)$. Both members may be multiplied by any non-zero numbers, which carries the rule to limit points.

(g) The block map is formed from parallel transports along paths in the double block of c and from averages of them, and it is covariant under gauge transformations; if $ds = 1$ on the double block, s is there the coboundary of a central field z, and $\mathcal{M}^n(sU)(c) = z(c_-)\mathcal{M}^n(U)(c)z(c_+) = s[c]\mathcal{M}^n(U)(c)$. The convergence of $\langle h(W_C(\mathcal{M}^{K-d}U))\rangle_K$, with the stated rate, is the convergence of the laws of the block fields at a fixed depth of Thm. YM3_I:D (Cor. 2.8), a bounded Borel function of the block field being tested. A flip $\bar\xi$ of the torus is the image of a closed sign field ξ of the fine lattice, $d\xi = 1$, under which the Wilson measure is invariant; by covariance the block-field laws are invariant under $\bar\xi$. For the bounding vortex line, $\eta_K = ds$ off the plaquettes pierced by $\partial\mathsf{S}$, $s = -1$ on the fine bonds that cross S; the sign rule gives $Z_K[\eta_K; F\circ\mathcal{M}^{K-d}] = Z_K[1; F\circ\mathcal{M}^{K-d}\circ s]$ and covariance gives $\mathcal{M}^{K-d}(sU) = \bar\xi\mathcal{M}^{K-d}(U)$ on the blocks that miss $\partial\mathsf{S}$; the limit and its error come from Thm. YM3_I:D; for $F = W_C F'$ the factor is $\bar\xi(C) = (-1)^{\#(C\cap\mathsf{S})}$.

(h) For continuous h the first statement is Lem. 14.9 (Thm. AF of §10.1, read at one polygon), applied to the polygon C, a closed path of some $\Lambda_{K_0,\ell}$ taken in every $\Lambda_{K,\ell}$, $K \ge K_0$, with the functions F_K of the bonds at the distance $\ge \mathsf{d}$ from C, uniformly in them; a bounded h that is continuous off a null set lies between continuous functions $h_- \le h \le h_+$ with $\overline{h_+} - \overline{h_-}$ as small as we please, and F_K is a combination of four functions with values in $[0,1]$. The law of $\frac12\mathrm{tr}\, g$ under Haar measure is $\frac{2}{\pi}\sqrt{1-w^2}\,\mathrm{d}w$; so $\bar h = 0$ for $h = \mathrm{sign}$ and for the characters χ_j, $j \ge \frac12$, written as functions of $w = \frac12\mathrm{tr}$, and $\bar h = \frac14$ for $h(w) = w^2$.

With C in a site mirror and $F_K = \overline{\theta G'_K}G_K$, $|G_K|, |G'_K| \le 1$ at a fixed distance, this is the fading between the vectors of the space of reflection positivity. The Bessel bound is Lem. 14.17(a), and $\mathsf{r}(x) = 1 - \frac{3}{2x} + O(x^{-2})$, $n' \approx |\gamma|L^K$, $4\beta = 4L^K\mathsf{b}$ give the limit $\exp(-3|\gamma|/8\mathsf{b})$. The fading of the flips is Lem. 14.17(b), (c), with $\omega_K(1-Q_0) \le \omega_K(M_{a_{p_0}}) = \langle a_{p_0}\rangle_K$, the multiplication operators of one layer having the Wilson measure as their law in the state of the transfer operator.

(i) Notation. $x_{[D]} = L^D\mathsf{b}$ (§2.3.2) is the coupling at the depth D; $\varepsilon(\mathsf{t}) = A_0\mathsf{t}^{-1/2}(\log\mathsf{t})^{\mathfrak{p}_0}$ and $\delta(\mathsf{t}) = (A_1/A_0)\varepsilon(\mathsf{t})$ YM3_I:(634), with the constants $A_0, A_1 > 0$, $\mathfrak{p}_0 \in \mathbb{N}$ of the list of Thm. YM3_I:D; $\varpi(V) = \max_p|\partial V(p) - \mathbf{1}|$, $|\cdot|$ the operator norm; and the small fields of the depth D are those of the inner ball $\mathsf{In}_{[D]} = \{V : \varpi(V) < 2\varepsilon(x_{[D]})\}$ of Thm. YM3_I:D (the common-depth definitions in Section YM3_I:116, the definitions preceding Lem. YM3_I:113.10), an open gauge-invariant set (Lem. YM3_I:113.10). $\mathsf{R}_{[D]} = 2M^{-3}(\ell L^D)^3\exp[-c'A_0^2(\log x_{[D]})^{2\mathfrak{p}_0}]\exp[(\mathsf{C}^\Sigma_{\mathrm{maj}} + 2 + C_{\mathrm{low}}(\mathsf{b}))\ell^3]$ is the number YM3_I:(2187) at the level of the depth D; it reads D and not K, and it tends to 0 as $D\to\infty$, since $c'A_0^2 > 0$ and $2\mathfrak{p}_0 \ge 2$. An element g of G with the eigenvalues $e^{\pm i\phi}$, $\phi\in[0,\pi]$, has $|g-\mathbf{1}| = 2\sin\frac{\phi}{2} \le \phi$, so $|e^{iw\sigma_3} - \mathbf{1}| \le |w|$; and replacing the four factors of $\partial(kV)(p)$ by those of $\partial k(p)$ one by one, $|\partial(kV)(p) - \partial k(p)| \le \sum_{b\in\partial p}|V_b - \mathbf{1}|$ for G-valued fields k, V and the bondwise product kV.

The floor. Let D be so large that $x_{[D]} \ge \mathsf{t}_\natural$ and $\mathsf{R}_{[D]} < 1$, let $K \ge D+1$, and let $t = K - D$ be the level of the depth D. The setting of the small-field chain of Thm. YM3_I:D asks of its top level only that it lie between 1 and $K-1$ and carry a coupling $\ge \mathsf{t}_\natural$ (§2.3.1); so it holds with the top level t, at a fixed depth. At the source 0, Prop. YM3_I:115.2 gives a bounded Borel weight $\mathfrak{w}^{\mathrm{sf}}_t \ge 0$ defined before Prop. YM3_I:115.2 such that the image of $\mathfrak{w}^{\mathrm{sf}}_t\rho_0\,\mathrm{d}U$ under $\mathcal{M}^t$ is the measure $\mathbf{1}_{\mathsf{In}_{[D]}}\nu_t$ of the chain YM3_I:(2188), $\int F\mathfrak{w}^{\mathrm{sf}}_t\rho_0\,\mathrm{d}U \le \int F\rho_0\,\mathrm{d}U$ for gauge-invariant bounded Borel $F \ge 0$ YM3_I:(2189), and $0 \le Z - \int\mathfrak{w}^{\mathrm{sf}}_t\rho_0\,\mathrm{d}U \le \mathsf{R}_{[D]}Z$ YM3_I:(2190), $Z = \int\rho_0\,\mathrm{d}U$.

So the measure of the chain lies below Z times the law on gauge-invariant functions (histories with a large field at a finer level, which may end in $\mathsf{In}_{[D]}$, are left out of it), and what it leaves out has a mass $\le \mathsf{R}_{[D]}Z$. On $\mathsf{In}_{[D]}$, which lies in the field cut (Lem. YM3_I:113.10), the measure of the chain has the density $p = e^{\mathsf{c}}e^{\Phi_t}$ for h_D, c a complex number (Prop. YM3_I:114.2), and

there $|\Phi_t| \le \mathsf{O}_D := (\ell L^D)^3[\frac{3}{2}c_\rho^2 A_0^2(\log x_{[D]})^{2\mathfrak{p}_0} + C_{\text{old}}\frac{L}{L-1}E_0]$: the bound YM3_I:(2161) of the action on a field cut, at the radius $c_\rho\varepsilon(x_{[D]})$ and the weight $x_{[D]}$, gives the first term, because $\mathsf{t}\,\varepsilon(\mathsf{t})^2 = A_0^2(\log\mathsf{t})^{2\mathfrak{p}_0}$, and the bound m_t^S YM3_I:(2179) of the added action (equation YM3_I:(2180)) gives the second; O_D reads D and not K.

A complex density of a positive measure is ≥ 0 almost everywhere; so $p = e^{\mathrm{Re}\,\mathsf{c}}e^{\mathrm{Re}\,\Phi_t}$ lies in $e^{\mathrm{Re}\,\mathsf{c}}[e^{-\mathsf{O}_D}, e^{\mathsf{O}_D}]$, and its mass, which lies in $[(1-\mathsf{R}_{[D]})Z, Z]$, puts $e^{\mathrm{Re}\,\mathsf{c}}/Z$ between $(1-\mathsf{R}_{[D]})e^{-\mathsf{O}_D}/\mathsf{h}_D(\mathsf{In}_{[D]})$ and $e^{\mathsf{O}_D}/\mathsf{h}_D(\mathsf{In}_{[D]})$: the constant c, the sum of the vacuum energies of the steps, drops out. For a gauge-invariant Borel F with $0 \le F \le 1$ on the fields of the depth D, $F\circ\mathcal{M}^t$ is gauge invariant, the block map being covariant, and this gives

$$\kappa_D \int_{\mathsf{In}_{[D]}} F\,\mathrm{d}\mathsf{h}_D \;\le\; \int F\,\mathrm{d}\mu_D^K \;\le\; \bar\kappa_D \int_{\mathsf{In}_{[D]}} F\,\mathrm{d}\mathsf{h}_D + \mathsf{R}_{[D]},$$

$$\kappa_D = \frac{(1-\mathsf{R}_{[D]})e^{-2\mathsf{O}_D}}{\mathsf{h}_D(\mathsf{In}_{[D]})},$$

$$\bar\kappa_D = \frac{e^{2\mathsf{O}_D}}{\mathsf{h}_D(\mathsf{In}_{[D]})},$$

the right inequality by YM3_I:(2189) at $1-F$ together with YM3_I:(2190); at $F = \mathbf{1}_{\mathsf{In}_{[D]}}$, $\mu_D^K(\mathsf{In}_{[D]}) \ge 1-\mathsf{R}_{[D]}$. For a Borel set E, $\mu_D^K(E) = \int \bar{\mathbf{1}}_E\,\mathrm{d}\mu_D^K$ with the mean $\bar{\mathbf{1}}_E$ of $\mathbf{1}_E$ over the gauge group, the law being gauge invariant, while $\mathsf{In}_{[D]}$ and h_D are gauge invariant; so the floor $\mu_D^K(E) \ge \kappa_D\mathsf{h}_D(E\cap\mathsf{In}_{[D]})$ and the upper bound $\mu_D^K(E) \le \bar\kappa_D\mathsf{h}_D(E\cap\mathsf{In}_{[D]}) + \mathsf{R}_{[D]}$ hold for all Borel E, free of K, and for the limit law, the laws converging on every Borel set (Cor. 2.8).

The thick vortex. On the bonds c of the depth D let $n(c) = \#(c\cap\mathsf{S})$, let j_0 be an integer cycle carried by the dual bonds of the rim with $j_0 \equiv dn$ modulo 2 (the rim, with an even number of its dual bonds at every dual site, is a union of closed dual paths, which are oriented), and $\mathsf{r} = \lfloor L^{D-d}/4\rfloor$. The translates $\tau_v j_0$ by the $(2\mathsf{r}+1)^3$ lattice vectors v of a box of radius $\mathsf{r}L^{-D}$ differ from j_0 by coboundaries $d\beta_v$, β_v carried within the distance $< L^{-d}$ from the rim: with the prism $(\mathsf{pr}_\lambda j)(x;\mu) = j(x;\lambda\mu)$, which takes functions on plaquettes to functions on bonds and functions on cubes to functions on plaquettes, $d\,\mathsf{pr}_\lambda j + \mathsf{pr}_\lambda dj = \tau_\lambda j - j$ and $dj_0 = 0$, summed along a path to v.

With φ the mean of the $\pi\beta_v$, ω the mean of the $\pi\tau_v j_0$, $\theta = \pi n + \varphi$ and $V_c^1 = e^{i\theta(c)\sigma_3}$: $d\theta = \omega + 2\pi\mathsf{k}$ with k integer, so $\partial V^1(p) = e^{i\omega_p\sigma_3}$, and $|\omega_p| \le \pi n_\mathsf{S}(2\mathsf{r}+1)^{-2} \le 16\pi n_\mathsf{S}L^{-2(D-d)}$, as a segment of the rim normal to p meets at most $2\mathsf{r}+1$ of the translates of p; the rim misses $\mathcal{R}$, so it keeps the distance $\ge L^{-d}$ from it (the closed cubes of side L^{-d} about sites that touch a dual bond of $\Lambda_{d,\ell}$ in the rim cover its neighbourhood of that radius, and none of them belongs to $\mathcal{R}$), and on the bonds of $\mathcal{R}$, $\varphi = 0$ and $V^1 = \bar\xi$.

Counted in $\Lambda_{d,\ell}$, n_S is at most the number $3(\ell L^d)^3$ of all its dual bonds; so for D so large that $48\pi(\ell L^d)^3L^{-2(D-d)} \le \varepsilon(x_{[D]})$, which holds for large D as $\varepsilon(x_{[D]})$ decays only like $L^{-D/2}$ up to logarithms, the thick vortex of every flip has $\varpi(V^1) \le \varepsilon(x_{[D]})$: it is a small field, and one D serves all surfaces. *The common part.* Let $\eta = \frac{1}{4}\varepsilon(x_{[D]})$ and $\mathrm{vol}(\eta)$ the Haar measure of $\{g\in G : |g-\mathbf{1}| < \eta\}$. If every bond of a field W is within η of $\mathbf{1}$, then $\varpi(W) < 4\eta < 2\varepsilon(x_{[D]})$ and $\varpi(V^1W) < \varpi(V^1) + 4\eta \le 2\varepsilon(x_{[D]})$, by the bound of a product of four: W and V^1W (bondwise product) are small fields, and $(V^1W)|_\mathcal{R} = \bar\xi\,W|_\mathcal{R}$.

By the floor, the law $\mu_\mathcal{R}$ of the block fields of $\mathcal{R}$ at the depth D dominates the measure $\kappa_D\mathsf{g}(w)\,\mathrm{d}w$, $\mathsf{g}(w)$ the Haar measure of the continuations of w to small fields; on the configurations w of $\mathcal{R}$ within η of $\mathbf{1}$, $\mathsf{g}(w)$ and $\mathsf{g}(\bar\xi w)$ are $\ge c_\eta = \mathrm{vol}(\eta)^{3(\ell L^D)^3 - n_\mathcal{R}}$, $n_\mathcal{R}$ the number of the bonds of $\mathcal{R}$ at the depth D (continue w by bonds within η of $\mathbf{1}$, and $\bar\xi w$ by the same bonds times those of V^1); so $\theta_0 = \kappa_D c_\eta \mathbf{1}_{\{|w_c-\mathbf{1}|<\eta\ \forall c\}}\,\mathrm{d}w$ lies below $\mu_\mathcal{R}$ and below $\bar\xi_*\mu_\mathcal{R}$, for every flip of a surface whose rim

misses $\mathcal{R}$ and at every cutoff $K \geq D+1$, and in the limit; it reads no surface and no cutoff, and its mass $\varkappa_D = \kappa_D \mathrm{vol}(\eta)^{3(\ell L^D)^3}$ reads D alone.

The block map from the depth D to the depth d, which reads on the bonds of $\mathcal{R}$ the bonds of $\mathcal{R}$ alone and turns $\bar{\xi}$ at the depth D into $\bar{\xi}$ at the depth d by the first clause of (g), carries this down. Then $\mu_{\mathcal{R}} \wedge \bar{\xi}_* \mu_{\mathcal{R}}$ has mass $\geq \varkappa_D$, which is the bound in total variation; for odd F with values in $[-1,1]$, $\int F \, \mathrm{d}\mu_{\mathcal{R}} = \frac{1}{2} \int F \, d(\mu_{\mathcal{R}} - \bar{\xi}_* \mu_{\mathcal{R}})$ lies in $[\varkappa_D - 1, 1 - \varkappa_D]$; and for flip-invariant Φ, since $\mu \geq \frac{1}{2}(\theta_0 + \bar{\xi}_* \theta_0)$ and $W_C \circ \bar{\xi} = -W_C$,

$$\int |W_C - \Phi| \, \mathrm{d}\mu \geq \frac{1}{2} \int (|W_C - \Phi| + |W_C + \Phi|) \, \mathrm{d}\theta_0 \geq \int |W_C| \, \mathrm{d}\theta_0 = \mathsf{m}_C > 0,$$

$|W_C|$ being near 1 on the configurations near 1; the same for $\operatorname{sign} W_C$. *Densities.* If $\mathsf{h}_d(E) = 0$, the preimage of E under the block map from the depth D is Haar-null by Lem. 14.18, and the upper bound found with the floor gives $\mu_d(E) \leq \mathsf{R}_{[D]}$ for every large D, so $\mu_d(E) = 0$. *The ceiling.* On the open set X_0 of full Haar measure of Lem. 14.16(b), which has full μ-measure, the four bricks with their overlaps determine a configuration up to a gauge transformation and one flip; Lem. 14.16(b) with $\mathsf{H} = \mathbb{Z}_2$, $C_\kappa = C$ for the non-trivial character, gives $F = \Phi_0 + \Phi_1 u$ with Φ_i measurable for the bricks; for bricks three sites thick the cubes generate the bricks by the same lemma; the block field of a coarser depth in $\mathcal{R}$ is a function of that of the depth d in $\mathcal{R}$.

Lem. 14.16(c) gives the adjoint loop. *The commutant.* $\mathfrak{L}$ is maximal abelian on $\mathfrak{h}$, the space of the gauge-invariant classes of $L^2(\mu)$, the density ρ being chosen gauge invariant; $\mathfrak{E}$ is its part fixed by the flip α; on $Y = \{\rho > 0\} \cap \alpha\{\rho > 0\}$ the flip is implemented by v, a partial isometry with $v^2 = e$ and $v M_F v = M_{F \circ \alpha} e$, and on the rest of $\{\rho > 0\}$ the flip moves no mass inside; so $\mathfrak{E}' = \mathfrak{L} + \mathfrak{L} v$, isomorphic to $L^\infty(Y_1) \oplus L^\infty(Y \cap \{u = 1\}) \otimes M_2(\mathbb{C})$, $Y_1 = \{\rho > 0\} \setminus Y$; $vu = -uv$, and $v \neq 0$ because $\mu(Y) \geq \varkappa_D$: the measure $\mu \wedge \alpha_* \mu$, of the density $\min\{\rho, \rho \circ \alpha\}$, vanishes off Y and lies above θ_0.

For this last point α is realised as the flip of a surface whose rim misses $\mathcal{R}$, at the depth $d+1$. In the units of $\Lambda_{d,\ell}$ let $y_0 = (b_1 + 1, b_2 + 1)$, a site of the hole ($b'_i - b_i \geq 2$), and $y_1 = (a'_1 + 1, b_2 + 1)$, a site beyond Q ($a'_1 - a_1 \leq N - 2$); let S_1 be the flat sheet of $\Lambda_{d+1,\ell}$ in the plane $x_2 = b_2 + 1 + \frac{1}{2L}$ from the straight closed dual line of $\Lambda_{d+1,\ell}$ in the direction 3 over $y_0 + \frac{1}{2L}(1,1)$ to the one over $y_1 + \frac{1}{2L}(1,1)$, across the arm of the ring at $x_1 = b'_1$. Each line lies over the open square of side 1 about y_0 or y_1, which meets no closed cube of $\mathcal{R}$: the rim misses $\mathcal{R}$; and C crosses S_1 once. $\mathcal{R}$ is a union of closed cubes of side $L^{-(d+1)}$ about sites of $\Lambda_{d+1,\ell}$, L being odd; so the first half of (i) and (g) hold for S_1 at the depth $d+1$.

The block map from the depth $d+1$ to the depth d reads on the bonds of $\mathcal{R}$ the bonds of $\mathcal{R}$ alone, and by the first clause of (g), no plaquette in $\mathcal{R}$ being pierced by the rim, it turns the flip of S_1 into the sign field $(-1)^{\#(c \cap \mathsf{S}_1)}$ on the bonds c of $\mathcal{R}$ at the depth d; this sign field is closed on the plaquettes of the bricks and odd on C, so it differs from α by a coboundary, which acts trivially on the gauge-invariant functions. Hence θ_0, carried down from the depth $d+1$, lies below μ and below $\alpha_* \mu$, and its mass is $\varkappa_D$. This holds for every ring admitted, D and $\varkappa_D$ being those of the first half of (i) at the depth $d+1$.

If besides $b'_i - b_i \geq 3$ ($i = 1, 2$) and $a'_i - a_i \leq N - 3$ for one i, say $i = 1$, a sheet of $\Lambda_{d,\ell}$ itself serves as well, with D and $\varkappa_D$ at the depth d: in the units of $\Lambda_{d,\ell}$, the flat strip $\{x_2 = b_2 + \frac{3}{2},\ b_1 + \frac{3}{2} \leq x_1 \leq a'_1 + \frac{3}{2}\}$ between the straight closed dual lines in the direction 3 over $(b_1 + \frac{3}{2}, b_2 + \frac{3}{2})$ and over $(a'_1 + \frac{3}{2}, b_2 + \frac{3}{2})$. A closed cube of side 1 about a site meets such a line only if the site lies in one of the four columns of sites nearest to the line, and these columns hold no site of $\mathcal{R}$: about the first line they lie over $b_1 + 1, b_1 + 2$ and $b_2 + 1, b_2 + 2$, in the hole, as $b_i + 2 < b'_i$; about the second they lie over $a'_1 + 1, a'_1 + 2$ in the direction 1, beyond Q, as $a'_1 + 2 < a_1 + N$.

So the four cubes about each line, at every height, are free, and the rim of the strip misses $\mathcal{R}$; C crosses the strip once, on its side $x_1 = b'_1$, as $b_1 < b_1 + \frac{3}{2} < b'_1 < a'_1 + \frac{3}{2}$ and $b_2 < b_2 + \frac{3}{2} < b'_2$; the flip of the strip is closed on the plaquettes of the bricks and odd on C, so it differs from α by

a coboundary, and the first half of (i) at the depth d puts the measure θ_0 of that depth below μ and below $\alpha_*\mu$. *The vortex line.* At the cutoff K the flip of the block fields of $\mathcal{R}$ is the insertion of the closed vortex line $\partial\mathsf{S}_1$: (g) at the depth $d+1$, for the functions of the block fields of $\mathcal{R}$ at the depth $d+1$ that factor through the block map, together with the covariance just used, gives $\mu^K\circ\alpha = x_K\mu^K$ with $x_K = \mathbb{E}_K[X_K\,|\,V]$, X_K the ratio of the twisted to the untwisted weight, at every K; so $x_K\cdot x_K\circ\alpha = 1$, $v_KF = x_K^{1/2}F\circ\alpha$ is a self-adjoint unitary, and, with ρ_K the density of μ^K for Haar measure (Lem. 14.18), $|\langle F, v_KF'\rangle - \langle F, vF'\rangle| \le \sup|F|\sup|F'|\int|\sqrt{\rho_K\,\rho_K\circ\alpha} - \sqrt{\rho\,\rho\circ\alpha}|\,\mathsf{dh}$, which the Schwarz inequality bounds by $2\sqrt{2}\sup|F|\sup|F'|$ times the square root of the total variation between μ^K and μ, at most $\sum_{K'\ge K}\mathfrak{o}_{K'}$.

Wide rings. Let moreover the hole be at least three bonds wide in both directions, and the ring be of positive height and lie in an open ball of radius $< \ell/4$. Then a sheet of $\Lambda_{d,\ell}$ itself realises α: in the units of $\Lambda_{d,\ell}$, the flat sheet S in the plane $x_2 = b_2 + \frac{3}{2}$ from the straight closed dual line in the direction 3 over the centre $(b_1+\frac{3}{2}, b_2+\frac{3}{2})$ of four cubes of the hole (here the width three is used) to the one over the centre of four cubes beyond the ring on the same row (for which the ball leaves room) has its rim off $\mathcal{R}$ and crosses C once. Let $\xi^{(D)}$ be its flip at the depth D, π_D the limit law of the block fields of $\mathcal{R}$ at the depth D, so that $\pi_d = \mu$, and Φ_D the block map from these fields to those at the depth d; by the first clause of (g), $\Phi_D\circ\xi^{(D)} = \bar\xi\circ\Phi_D$ with the flip $\bar\xi$ of S at the depth d, which differs from α by a coboundary; as μ is invariant under gauge transformations, we take $\alpha = \bar\xi$ here.

We show that $\mu(\alpha A) = 0$ gives $\mu(A) = 0$ for a Borel set A of the block fields of $\mathcal{R}$ at the depth d.

(1) *From the measure to the fields.* Let $\mathcal{G}_D$ be the set of the $V\in\mathsf{In}_{[D]}$ whose flipped restriction $\xi^{(D)}V|_{\mathcal{R}}$ continues to a field of $\mathsf{In}_{[D]}$. If $\mu(\alpha A) = 0$, then $\mu(A)\le 1-\mu_D(\mathcal{G}_D)+\mathsf{R}_{[D]}$. Indeed, with $\tilde A = \Phi_D^{-1}A$, $\pi_D(\xi^{(D)}\tilde A) = \mu(\alpha A) = 0$; by the floor, $\int_{\xi^{(D)}\tilde A} q_D\,\mathrm{d}w = 0$, $q_D(w)$ the Haar measure of the continuations of w into $\mathsf{In}_{[D]}$, and $q_D > 0$ exactly on the projection Pr_D of the open set $\mathsf{In}_{[D]}$ to the bonds of $\mathcal{R}$; so $\tilde A\cap\xi^{(D)}\mathsf{Pr}_D$ is Haar-null. The fields of $\mathcal{G}_D$ whose restriction to $\mathcal{R}$ lies in $\tilde A$ have it in $\tilde A\cap\xi^{(D)}\mathsf{Pr}_D$; they form a Haar-null set, of μ_D-measure $\le\mathsf{R}_{[D]}$ by the upper bound; the others with the restriction in $\tilde A$ lie off $\mathcal{G}_D$.

(2) *Insertion of a straight line into a small field.* In units of L^{-D} let l_P be the straight closed dual line in the direction 3 over a dual point P of the $(1,2)$-plane, $\varrho_1\ge 1$ an integer with $2\varrho_1+4<\ell L^D$, $\mathcal{T}$ the sites whose horizontal part lies within the sup-distance $\varrho_1+\frac{3}{2}$ of P (the tube), s a sign field with $(ds)_p = -1$ exactly at the plaquettes with corners in $\mathcal{T}$ that l_P pierces, and V a field with $|\partial V(p)-\mathbf{1}|\le\delta_0$ at the plaquettes with corners in $\mathcal{T}$. Then there is a field V' with $V'_c = s_cV_c$ on every bond that has not both ends in $\mathcal{T}$, $|\partial V'(p)-\mathbf{1}|\le 6\varrho_1^{-2}+3\delta_0$ at the plaquettes with all corners in $\mathcal{T}$, and $\partial V'(p) = (ds)_p\partial V(p)$ at all the others.

Idea: take an axial gauge γ of $\mathcal{T}$, $\mathbf{1}$ on the spanning tree made of all the bonds in the direction 3 at a corner o of the unit plaquette centred at P but one, one line in the direction 1 through o in each layer, and all the bonds in the direction 2, the holonomy Υ around the torus along the bonds in the direction 3 at o being made diagonal by a constant gauge transformation; then, x measured from o, a bond in the direction 1 at x is within $|x_2|\delta_0$ of $\mathbf{1}$, and a bond in the direction 3 at x is $\mathsf{Y}_x\Upsilon_x$ with $\Upsilon_x\in\{\mathbf{1},\Upsilon\}$ and Y_x within $|x|_1\delta_0$ of $\mathbf{1}$. For positively oriented bonds put $V'_c = s_c\gamma(c_-)^{-1}e^{-i\varsigma_c\sigma_3}\gamma(c_-)V_c$ on the bonds with both ends in $\mathcal{T}$, with the profile $\varsigma_c = (1-f_c)\vartheta_c$ on the horizontal bonds, ϑ_c half the signed angle that c subtends at P, $f_c = \min\{1,(r_c^2-\frac{1}{4})/\varrho_1^2\}$, r_c the sup-distance of the midpoint of c from P, and $\varsigma_c = 0$ on the vertical bonds; define the values on reversed bonds by inversion.

Then $d\varsigma = \pi$ at the pierced plaquettes, which takes away the sign of ds there, and $|d\varsigma|\le 6\varrho_1^{-2}$ at the other horizontal ones, as $d\vartheta = 0$ there: three terms $\pm(f_{c'}-f_c)\vartheta_c$ remain, each $\le\frac{2r}{\varrho_1^2}\cdot\frac{1}{r}$, $r\ge 1$ the sup-distance of the centre of the plaquette from P; this is the source of the 6. The rotations $e^{-i\varsigma_c\sigma_3}$ commute with one another and with Υ. With $o=(0,0)$ and $P=(\frac{1}{2},\frac{1}{2})$ in a horizontal layer,

the half-angle bound is $|\vartheta_c|\,|c_- - o|_1 \le 1$ for the positively oriented horizontal bonds. It is paired with the gauge error measured from the same o. For a horizontal plaquette the direction-2 bonds are $\mathbf{1}$, leaving one commutator with a direction-1 bond. For a vertical plaquette first factor out the original plaquette product; the remaining error is one commutator with Y_x in $\mathsf{Y}_x\Upsilon_x$, since the inserted rotations commute with Υ_x. Its cost is at most $2\delta_0$ in either case. Adding the original plaquette error δ_0 gives $3\delta_0$. At the boundary $f_c = 1$, so the insertion matches the prescribed exterior field; no boundary value of the gauge at o is required.

(3) *The witness.* Let $\varrho_1 = \frac{1}{2}(L^{D-d} - 5)$. At the depth D the hole holds a square of $(2\varrho_1 + 4)^2$ sites about a dual point P_1 of the $(1,2)$-plane, and beyond the ring, on its row, there is a dual point P_2 whose square of that size misses Q (the ball leaves room); the tubes $\mathcal{T}_1, \mathcal{T}_2$ over them are disjoint and hold no site of $\mathcal{R}$. Let $s = -1$ on the bonds in the direction 2 that cross the flat strip between the vertical lines over P_1 and P_2, and $s = 1$ elsewhere: $(ds)_p = -1$ exactly at the plaquettes that the two lines pierce. On the bonds of $\mathcal{R}$ at the depth D, s and $\xi^{(D)}$ are closed on the plaquettes of the bricks and odd on the loop around the hole, so $s = \xi^{(D)}dz$ there, with signs z on the sites of $\mathcal{R}$ (Lem. 14.16(b) for the four bricks, whose group Z/B is $\mathbb{Z}_2$ and is read on that loop).

Let $\theta' < \frac{2}{3}$ and D be so large that $6\varrho_1^{-2} + 3\theta'\varepsilon(x_{[D]}) < 2\varepsilon(x_{[D]})$, as ϱ_1^{-2} is of the order of $L^{-2(D-d)}$ and $\varepsilon(x_{[D]})$ of the order of $L^{-D/2}$ up to logarithms. If $\varpi(V) < \theta'\varepsilon(x_{[D]})$, V is a small field, and the insertion in both tubes with $\delta_0 = \theta'\varepsilon(x_{[D]})$ gives V' with $V' = sV$ off the tubes, so on the bonds of $\mathcal{R}$, and $\varpi(V') < 2\varepsilon(x_{[D]})$; the gauge transform of V' by z, continued by 1, is a small field equal to $\xi^{(D)}V$ on the bonds of $\mathcal{R}$. So $\{\varpi < \theta'\varepsilon(x_{[D]})\} \subset \mathcal{G}_D$.

(4) *The tail at a fraction of the threshold.* For integers $n \ge 0$ and $D' = D - n$ with $x_{[D']} \ge \mathsf{t}_\natural$, let $\Theta_n = 4B_3L^{-3n/2} + 2\frac{C_H A_1}{A_0}\sum_{j<n} L^{-3j/2}$, where $B_3 \ge 1$ and $C_H \ge 1$ are the constants of the list of Thm. YM3_I:D that enter the bound of the background over a small field and the contraction of the backgrounds from one level to the next (Thm. YM3_I:E and the curvature comparison in the proof of Lem. YM3_I:113.10). Then $\mu_D^K\{\varpi \ge \Theta_n\varepsilon(x_{[D]})\} \le \mathsf{R}_{[D']}$ for $K \ge D+1$, and in the limit. Run the chain with its top level at the depth D'. On the histories with $\mathfrak{w}^{\mathrm{sf}} \ne 0$ the field at the depth D' lies in the inner ball, and the field of every finer level lies in the field cut and is the field over the next coarser one moved by a fluctuation from the ball of the step; so the regularity numbers m of the backgrounds obey $\mathsf{m} < 2B_3\varepsilon(x_{[D']})$ at the top (Thm. YM3_I:E) and $\mathsf{m}_k \le L^{-2}\mathsf{m}_{k+1} + C_H\delta(x_k)$ from the level $k+1$ to the finer level k (the proof of Lem. YM3_I:113.10).

Summed down the n levels, with $\varepsilon(x_{[D-j]}) \le L^{j/2}\varepsilon(x_{[D]})$ and the same for δ, this gives $\mathsf{m} < \frac{1}{2}\Theta_n\varepsilon(x_{[D]})$ at the depth D, and a field whose background has the regularity number below m has its plaquettes below $2\mathsf{m}$ (as in the inclusions YM3_I:(2159)): $\varpi < \Theta_n\varepsilon(x_{[D]})$. The other histories have the mass $\le \mathsf{R}_{[D']}Z$ by YM3_I:(2189) at $1 - F$ and YM3_I:(2190), F the indicator of $\{\varpi(\mathcal{M}^{K-D}U) \ge \Theta_n\varepsilon(x_{[D]})\}$; the limit follows.

(5) *The end.* Let n_0 be the least integer with $L^{3n_0/2} \ge 80B_3$; it is fixed by L and B_3, and chosen so that $\Theta_{n_0} < \frac{3}{5}$: the first term of Θ_{n_0} is $\le \frac{1}{20}$, and the condition $A_0 \ge 4B_3C_HA_1$, one of the conditions on A_0 written in the proof of Lem. 4.48, with $B_3 \ge 1$ and $L \ge 13$, makes the second $\le \frac{1}{2}(1 - 13^{-3/2})^{-1} < 0.52$. For D so large that $x_{[D-n_0]} \ge \mathsf{t}_\natural$, $\mathsf{R}_{[D]} < 1$ and $6\varrho_1^{-2} < \frac{1}{5}\varepsilon(x_{[D]})$, (1) and (3) at $\theta' = \frac{3}{5}$ give $\mu(A) \le \mu_D\{\varpi \ge \frac{3}{5}\varepsilon(x_{[D]})\} + \mathsf{R}_{[D]}$, and (4) at $n = n_0$ bounds the first term by $\mathsf{R}_{[D-n_0]}$. So $\mathsf{R}_{[D-n_0]}$ bounds the mass of the histories that the chain with its top level at the depth $D - n_0$ leaves out, this chain controlling the plaquettes at the depth D, and $\mathsf{R}_{[D]}$ comes from the upper bound in (1). Both tend to 0 as $D \to \infty$, and $\mu(A)$ reads no D: $\mu(A) = 0$. With αA in the place of A the converse holds; so the law and its flip have the same null sets, $e = 1$, v is a self-adjoint unitary, and with the ceiling, whose hypothesis on the corners the positive height meets, $\mathfrak{L} = \mathfrak{E} \oplus \mathfrak{E}u$ is a grading by $\mathbb{Z}_2$.

A band that winds. The flip $\bar{\xi}_1$ of a band $\mathbb{Z}/N \times I_2 \times I_3$ is the restriction of a flip of the torus, a symmetry of the law: v is unitary, odd and even functions are orthogonal, $\mathfrak{E}u_1 \ne 0$ as $u_1^2 = 1$, and

the cubes of the band generate $\mathfrak{E}$ by Lem. 14.16(b) with the cubes as boxes; the only closed sign fields on the cubes of the band modulo coboundaries are 1 and the winding one.

14.1.11. *Scope of the conclusions.* Thm. AP(a)–(g) concern the centred functionals of volume-first limits and their components, for which the proof uses a periodic lattice; for other limits of periodic Gibbs states only (h) is asserted. The proofs of Thms. AO and AP use no energy bound beyond the polynomial one of the reconstruction, no gap, no uniqueness of the vacuum of σ and no rate of clustering. Thm. AQ(d) bounds from below the norms of the field at one instant and of one Euclidean slab. Thm. AS(g)–(i) are statements on the commutative algebras of the Euclidean block fields at a fixed depth on one torus.

15. A continuum Langevin process

Fix the physical torus and the label b, and keep the invariant thread law obtained from the compatible block-field marginals. Put $A_K = \sum_p a_p$, and write μ^K for the Wilson probability measure of [7, §2] on this fixed torus. On the cutoff lattice the stationary Wilson Langevin process has generator and Dirichlet form

$$\mathcal{L}_K = a_K^{-1} \sum_{b,\alpha} (\partial_b^\alpha)^2 - a_K^{-1} \beta\, \partial A_K \cdot \partial, \qquad \mathcal{E}_K(F) = a_K^{-1} \int |\partial F|^2 \, \mathrm{d}\mu^K.$$

The derivatives are along the three orthonormal colour directions at each link.

Sketch N (A nonfrozen reversible Markov limit). Subsequences of the stationary lattice processes converge on gauge-invariant cylinder observables to a nonconstant stationary reversible Markov process on block-field threads. Its closed Dirichlet form is the strong-and-weak limit of the lattice forms, and its semigroup mixes to the invariant thread law. The result is on a fixed torus and does not identify the limits from different subsequences. ■□

Sketch of the argument. Separate the fine field from a fixed block field using the small-field chain. On each fluctuation box the conditional Hamiltonian is strictly convex after the controlled nonlinear remainder is included. The conditional law therefore has Poincaré and logarithmic Sobolev bounds. Differentiate the conditional expectation defining one averaging step. The response of the background transports the coarse gradient, while the fluctuation covariance controls the additional term. The resulting multiplicative losses have the form $1 + C\delta(L^d b)$, and their product is finite because $\sum_d \delta(L^d b) < \infty$.

At a selected coarse depth n, retain a connected region of regular fields. Its finite-dimensional inequality supplies the remaining top-scale bound. The small-field chain omits mass $r_n \to 0$, which appears as a defect rather than a uniform spectral-gap constant:

$$\operatorname{Var}(F) \le C_n \mathcal{E}(F) + r_n \operatorname{osc}(F)^2.$$

Conditional versions control the fine field uniformly in the cutoff; the constant C_n at the top can depend on the torus.

These bounds give compactness for uniformly bounded observables of bounded energy. Along a process subsequence, they provide the lower bound for weakly convergent functions and recovery sequences converging strongly. Resolvents and semigroups then converge strongly, identifying a closed Markovian limiting form and the reversible Markov process it generates. Let Y_n be the selected coarse block field and take its plaquette observable $F = a_p(Y_n)$. Apply the coarse-gradient estimate to a recovery sequence for F. Lower semicontinuity of its Dirichlet integral on the regular coarse-field set gives $\mathcal{E}(F) > 0$. The limiting semigroup therefore moves this observable, so the process is not frozen.

For mixing, fix n in the weak variance inequality and let time increase. Its exponential term disappears, leaving the defect $r_n \operatorname{osc}(F)^2$. Now increase n to remove that defect. The analogous

entropy estimate gives relaxation of bounded probability densities in total variation on the invariant algebra.

This order of limits matters. The argument supplies constants uniform in the fine cutoff, not constants uniform in the torus volume. Stronger gauge estimates across several block levels would make the top-scale constants more explicit; the qualitative conclusion above does not require them.

15.1. **Continuum limits of the lattice Langevin dynamics.** At a fixed physical label and on a fixed torus, the Wilson Langevin processes have subsequential limits on threads of block fields. We prove that each limit is a nonconstant stationary reversible Markov process on the gauge-invariant σ-algebra. The lattice Dirichlet forms converge in the strong and weak sense. Conditional Poincaré and logarithmic Sobolev inequalities control the fine field; a connected coarse-scale region then gives weak inequalities with defects tending to zero. These estimates imply mixing and entropy relaxation. Their constants may depend on the fixed torus, and the result does not identify the processes obtained along different subsequences.

We also study observables regularised by the Wilson gradient flow. The finite-cutoff flow identities and upper bounds use no probabilistic assumption. The moment estimates use the admissibility conclusion (Adm) of Prop. 16.1, concerning conditional layer means and colour covariances. Identifying the continuum flow limits requires the approximation step described below. The nonlinear defect bounds are proved in Lem. 16.4. Lem. 16.8 proves lower propagation on a regular time interval; Prop. 16.9(c) applies it on the interval of the short-time flow theorem and extends the lower bound to every final time by current transport and the integrated curvature. These estimates do not complete the continuum-flow identification, whose remaining work is described under (Flow approximation).

The generator requires separate treatment. Some bounded flowed observables have generator images whose norms grow exponentially in the inverse square of the spacing. We therefore first insert condenser cutoffs, pass to the continuum limit, and then remove the cutoffs. The resulting generator identity is distributional.

15.1.1. *The lattice process and its limiting objects.* $G = \mathrm{SU}(2)$; $\mathfrak{g}$ is the space of hermitian traceless 2×2 matrices, $\langle X, Y\rangle = \frac{1}{2}\mathrm{tr}\, XY$, $|X| = \langle X, X\rangle^{1/2}$, so that $X^2 = |X|^2\mathbf{1}$ and $e^{iX} \in G$; d is the geodesic distance of G, the unit sphere S^3, with $\mathsf{d}(e^{iA}, \mathbf{1}) = |A|$ for $|A| \le \pi$ and $|g - h| = 2\sin\frac{1}{2}\mathsf{d}(g, h)$. The numbers L, $a_d = L^{-d}$, $\ell = 2L^m$, the torus $T_\ell = \mathbb{R}^3/\ell\mathbb{Z}^3$ of side ℓ, the lattices $\Lambda_{d,\ell} = a_d\mathbb{Z}^3/\ell\mathbb{Z}^3 \subset T_\ell$ (Section YM3_I:3) and the averaging map $\mathcal{M}$ (equations YM3_I:(37)–(39)) have the conventions just stated. The label b and setting (E) are those of §2.3.1: the constants of the construction are an admissible choice $\varsigma = (\sigma^-, \gamma)$ of the canonical list of constants and conditions (§2.3.1), subject to the conditions (Th1) and (Th2) of §2.3.1, with γ the fixed reference-coupling ceiling; the label satisfies $\mathsf{b} \ge \gamma^{-2}$ and $M\, R(\mathsf{b}) \le L^m$, with the cube size $M = L^{m'}$ and the function R of §2.3.1, $R(x)$ the least power L^j, $j \ge 0$, that is not smaller than a fixed power of $\log x$; and $\beta = L^K\mathsf{b}$, so that $\mathsf{b} = a_K\beta$.

Ω_d is the set of configurations on $\Lambda_{d,\ell}$, dV (on Ω_K: dU) its Haar probability measure, $V^u(b) = u(b_-)V(b)u(b_+)^{-1}$ the gauge action; invariant means gauge invariant. $\partial V(p)$ is the plaquette variable, $a_p = 1 - \frac{1}{2}\mathrm{tr}\,\partial V(p)$, $\varpi_p = |\partial V(p) - \mathbf{1}| = (2a_p)^{1/2}$, $\varpi(V) = \max_p \varpi_p$, $A_K = \sum_p a_p$, $A^a = a_K^{-1}A_K$, and $\mathcal{O}(f) = a_K^{-1}\sum_p f(x(p))\, a_p$ for real f on T_ℓ, $x(p)$ the barycentre of p. On Ω_K we write $N = \ell/a_K$ for the number of sites along an axis. Tangent vectors are real $\mathfrak{g}$-valued 1-forms X, with the curves $s \mapsto e^{isX}V$; $\langle X, \partial F(V)\rangle = \frac{d}{ds}F(e^{isX}V)|_{s=0}$, $\|X\|^2 = \sum_b |X(b)|^2$; $\partial_b F$ is the value of the 1-form ∂F at the bond b, and $\partial_b^\alpha F = \langle \sigma_\alpha, \partial_b F\rangle = \frac{d}{ds}F(e^{is\sigma_\alpha\delta_b}V)|_{s=0}$, with the Pauli matrices $\sigma_1, \sigma_2, \sigma_3$, a basis of $\mathfrak{g}$ that is orthonormal for $\langle\cdot,\cdot\rangle$, and δ_b the function of the bonds that is 1 at b and 0 at the others; $\Gamma_d(F) = a_d^{-1}|\partial F|^2$ on Ω_d, and $\Gamma_d(F, F')$ is its polarisation.

The vectors $\mathrm{d}_V\lambda$, $(\mathrm{d}_V\lambda)(b) = \lambda(b_-) - \mathrm{Ad}(V(b))\lambda(b_+)$ for $\mathfrak{g}$-valued site functions λ, are the gauge directions at V; a tangent vector orthogonal to them is horizontal, and the gradient of an invariant function is horizontal. $\mathrm{osc} = \sup - \inf$.

Dynamics. $\mu^K = Z_K^{-1} e^{-\beta A_K}\, dU$; $\Delta_K = a_K^{-1}\sum_{b,\alpha}(\partial_b^\alpha)^2$; $\mathcal{L}_K = \Delta_K - a_K^{-1}\beta\sum_b \langle \partial_b A_K, \partial_b \cdot\rangle$; $\mathcal{E}_K(F) = \int \Gamma_K(F)\,\mathrm{d}\mu^K$ on the closure $D(\mathcal{E}_K)$ of $C^\infty(\Omega_K)$, with $\mathcal{E}_K(F) = \infty$ off $D(\mathcal{E}_K)$; $P_t^K = e^{t\mathcal{L}_K}$, and $(U_t)_{t\in\mathbb{R}}$ is the stationary process. $J(U) = a_K^{-3}\partial A_K(U)$ is the current (YM3_I:9.2), $(J,J)_a = a_K^{-3}|\partial A_K|^2$; Φ_s is the gradient flow $\frac{d}{ds}U(b) = -i a_K J(U)(b)U(b)$, $U_s = \Phi_s U$, $v = -a_K^{-2}\sum_b \langle \partial_b A_K, \partial_b \cdot\rangle$ its vector field, and $\Lambda_\sigma(U) = \int_0^\sigma a_K^{-2}\varpi(U_s)\,\mathrm{d}s$.

Threads and limits. $\mathcal{Y} = \prod_{d\geq 0}\Omega_d$ is compact and metrisable; a thread is a point $Y = (Y_d)$ of $\mathcal{Y}$ with $Y_d = \mathcal{M}Y_{d+1}$ for all d; $\pi_{K,d} = \mathcal{M}^{K-d}$. The laws μ_d^K of $\pi_{K,d}U$ under μ^K converge in total variation to measures μ_d, with $\mu_{d+1}\circ\mathcal{M}^{-1} = \mu_d$ (Cor. 2.8), and $\mu = \mu_{\ell,\mathsf{b}}$ is the measure on threads with the marginals μ_d. A cylinder function is $\Psi(Y_d)$, Ψ bounded Borel on Ω_d, and $\Psi_K = \Psi\circ\pi_{K,d}$; L^2_{inv} is the closure in $L^2(\mu)$ of the invariant cylinder functions, which generate the invariant σ-algebra, and Π is the orthogonal projection onto L^2_{inv}.

Along $K_j\to\infty$, $H_j \in L^2(\mu^{K_j})$ converge *strongly* to $F\in L^2(\mu)$ if there are cylinder functions $\Psi_m\to F$ in $L^2(\mu)$ with $\lim_m \limsup_j \|H_j - \Psi_{m,K_j}\|_{L^2(\mu^{K_j})} = 0$, and *weakly* if $\sup_j\|H_j\| < \infty$ and $\int H_j\Psi_{K_j}\,\mathrm{d}\mu^{K_j} \to \int F\Psi\,\mathrm{d}\mu$ for every cylinder function Ψ. A *limit* is a sequence $K_j\to\infty$ along which all means $\mathbb{E}\prod_i \Psi_i(\pi_{K_j,d}U_{t_i})$ of bounded Borel Ψ_i converge, with its limit process (Y_t) on threads; $(F, Q_tF') = \mathbb{E}F(Y_0)F'(Y_t)$, $\mathcal{E}_\star(F) = \sup_{t>0} t^{-1}(F,(1-Q_t)F)$, $A_\star$ is the operator of $\mathcal{E}_\star$, and the limit is frozen if $\mathcal{E}_\star$ vanishes on L^2_{inv}, that is, if $Q_t = 1$ on L^2_{inv} for all t ($\mathcal{E}_\star$ is finite only on L^2_{inv}, Lem. 15.1(b)).

The forms converge *in the strong and weak sense* if (M1) $\mathcal{E}_\star(F) \leq \liminf_j \mathcal{E}_{K_j}(H_j)$ whenever $H_j\to F$ weakly, and (M2) every F has $H_j\to F$ strongly with $\mathcal{E}_{K_j}(H_j)\to\mathcal{E}_\star(F)$.

The chain. We recall the following objects of the small-field chain on one torus (Sections YM3_I:113, YM3_I:114, YM3_I:115), which rests on the results on one step of Prop. 6.32. $\delta(\mathsf{t}) = A_1\mathsf{t}^{-1/2}(\log\mathsf{t})^{\mathfrak{p}_0}$, $\varepsilon = \frac{A_0}{A_1}\delta$, $\mathfrak{b}(\mathsf{t}) = \mathsf{t}^{1/2}\delta(\mathsf{t})$, the threshold functions of equation YM3_I:(634), with the constants $A_0, A_1 > 0$ and $\mathfrak{p}_0\in\mathbb{N}$ of the construction; $\varepsilon_n = \varepsilon(L^n\mathsf{b})$; $\sum_d \delta(L^d\mathsf{b}) < \infty$. For an integer $n\geq 1$ with $L^n\mathsf{b}\geq \mathsf{t}_\natural$, $\mathsf{t}_\natural \geq \gamma^{-2}$ the fixed calibration of §2.3.1, a number formed from the constants alone, and for $K > n$, put $t = K - n$, so that $1\leq t\leq K-1$, as required of a top level in §2.3.1: level k is depth $d = K-k$, $V_k = \mathcal{M}^k U$, $x_k = L^d\mathsf{b}$, so that $x_k^{-1} = a_d/\mathsf{b}$.

The cuts $\mathsf{In}_k \subset \mathsf{Cut}_k \subset \Omega_d$ of the definitions preceding Lem. YM3_I:113.10 are open and invariant (Lem. YM3_I:113.10); $\mathsf{Cut}_k \subset \{\varpi < \theta_d\}$ with θ_d below; $\mathsf{I}_n = \mathsf{In}_t = \{\varpi < 2\varepsilon_n\} \subset \Omega_n$, and D_n is the connected component of $\mathbf{1}$ in I_n. R_n is the number YM3_I:(2187) at the level $t = K-n$, which depends on n, ℓ and b but not on K; it bounds the mass of the large fields (Prop. YM3_I:115.2, lost-mass bound), and by YM3_I:(2187) $\mathsf{R}_n \leq C(\ell,\mathsf{b})\,L^{3n}\exp[-c' A_0^2(\log L^n\mathsf{b})^{2\mathfrak{p}_0}]$ with the rate $c' > 0$ of §2.3.1 and $2\mathfrak{p}_0 \geq 2$, so that it tends to 0 faster than every exponential in n.

The chain measure is $\sigma^K = Z_K^{-1}\mathfrak{w}_t^{\mathrm{sf}} e^{-\beta A_K}\,dU$, with the constant Z_K of μ^K and the weight of the small-field chain of level t, the function defined before Prop. YM3_I:115.2,

$$\mathfrak{w}_t^{\mathrm{sf}}(U) = \mathbf{1}_{\mathsf{In}_t}(V_t)\prod_{k=0}^{t-1}\big[\mathbf{1}_{\mathsf{Cut}_k}\,\mathfrak{G}_k[x_k]\,\chi_{x_k}^{\mathrm{sf}}\big](V_k),$$

where $\mathfrak{G}_k[x_k]$ is the gauge-fixing factor of the axial gauge on the blocks of the level $k+1$ and $\chi_{x_k}^{\mathrm{sf}}$ the fluctuation test of the step $k+1$ ([7, §B.2]); the weight is a bounded Borel function with values in $[0,\infty)$ (Prop. YM3_I:115.2, proof and the definitions preceding Prop. YM3_I:115.2). So σ^K is a positive measure of mass in $[1-\mathsf{R}_n, 1]$ (Lem. 15.3(i)), normalised by the constant of μ^K and not by its own mass; and as its weight carries gauge-fixing factors, σ^K is not gauge invariant, and its comparisons with μ^K are made on invariant functions. σ_d^K is its image at depth d, σ_n^K the top law.

For $0 \le k < t$ and $W \in \mathsf{Cut}_{k+1}$, a background U_+ over W is any element of the minimal orbit over W with $\mathcal{M}^k U_+$ in the axial gauge, and $V_W(B)$ is the chart of the fibre $\{\mathcal{M}V = W\}$ ([7, §B.2]), defined on the convex box $\{|B(b)| < \mathfrak{b}(x_k)\}$ of the bonds that are not pivot bonds, with $V_W(B)(b) = e^{ix_k^{-1/2}B(b)}(\mathcal{M}^k U_+)(b)$ there; it carries the box into Cut_k (Lem. YM3_I:113.10). The fixed radii $0 < \alpha_0 < \alpha_1$ are two constants of the construction (Def. YM3_I:49.1 and Lem. YM3_I:82.1, Lem. YM3_I:82.1), free of n, at which the analyticity cores of the chain are taken (Def. YM3_I:49.1); Cut_{k+1} lies in the class $\mathcal{W}_{k+1}(x_k; \alpha_0, \alpha_1)$ of the sum rule (Lem. YM3_I:113.10).

One fluctuation integral p_W is set at every $W \in \mathsf{Cut}_{k+1}$ by the chart integral I of [7, eq. 191] at $\mathsf{t} = x_k$ ([7, Def. B.2] and equation [7, eq. 191]): it is the measure on the box with

$$\int h(V_W(B))\, p_W(\,\mathrm{d}B) = \frac{\mathsf{I}[hF_k](W)}{\mathsf{I}[F_k](W)} \qquad \text{for every bounded Borel } h \text{ on } \Omega_d,$$

F_k the integrand of the chain at the level k and at the source 0 (Prop. YM3_I:114.2); that is, p_W has on the box the density proportional to $\exp[-x_k G_W(B) + \mathsf{J}_W(B)]\, F_k(V_W(B))$, with the gauge-fixing form G_W and the exponent J_W of the Jacobians of the chart ([7, §B.2]). The denominator vanishes at no $W \in \mathsf{Cut}_{k+1}$: on the class of the sum rule $\mathsf{I}[F_k] = e^{\mathsf{c}_{k+1}(0)} F_{k+1} \neq 0$, $\mathsf{c}_{k+1}(0)$ the constant YM3_I:(2173) of the step (Prop. YM3_I:114.2). p_W depends on K, k and W, and not on the top level; that it is a probability measure with a positive continuous density on the box, and that its images under V_W form a regular conditional law of V_k given $V_{k+1} = W$ under σ^K, is Lem. 15.3(ii).

The tower of H is $h_0 = H$, $h_{k+1}(W) = \int h_k(V_W(B))\, p_W(\,\mathrm{d}B)$ at every $W \in \mathsf{Cut}_{k+1}$. For a finite measure ν, $\mathrm{Ent}_\nu(f) = \inf_{c>0} \int \Xi(f, c)\,\mathrm{d}\nu$, $\Xi(x, c) = x \log \frac{x}{c} - x + c \ge 0$.

Constants. $\mathsf{o}_n = [2C_{\mathrm{nl}} \frac{L}{L-1} E_0 + \frac{3}{2} c_\rho^2 A_0^2 (\log L^n \mathsf{b})^{2\mathfrak{p}_0}](\ell L^n)^3$, with the number $C_{\mathrm{nl}} = C'_{\mathrm{A}} + 2C_{\mathrm{E},0}(C_{\mathrm{lg}} + L^{4l_0})$, with C'_{A}, l_0, C_{lg} and $C_{\mathrm{E},0}$ defined exactly as in equation YM3_I:(2163) and its following sentence, using $K_{\mathrm{A}} = K^\sharp_{\mathrm{tw}}$ and $a_{\mathrm{A}} = a^\sharp$. The proof following that equation gives the bound on the non-local action, that is, the action of a bare array at the background of a field in a cut (equation YM3_I:(2180)). We also use the size $E_0 > 0$ of §15.3.1 and the constant c_ρ of the field cuts §2.3.1 (and C_H defined in [7, §P.1]); $\theta_d = 2c_\rho \varepsilon(L^d \mathsf{b})$; $\gamma_{\mathrm{r}} \in (0, 1]$ and $c_{\mathrm{r}} \ge 1$ are the numbers of Thm. YM3_I:U, entered through §15.3.1, which depend on L alone: the floor and the ceiling of a real reference form of the Gaussian integral of one step, from which the form of the step itself differs, in the ℓ^2 operator norm, by at most the defect $\frac{1}{2}\mathsf{u}_{\mathrm{G}}(LM)^{-3} \le \frac{1}{2}\gamma_{\mathrm{r}}$ (the completeness (St4) of the Gaussian structure of the step, Def. YM3_I:47.2 and Thm. YM3_I:W, with $\mathsf{u}_{\mathrm{G}} = 4C_{\mathrm{Str}}$ of Thm. YM3_I:U and the proof of Prop. YM3_I:113.4); so the form of the step lies between $\frac{1}{2}\gamma_{\mathrm{r}}$ and $c_{\mathrm{r}} + \frac{1}{2}\gamma_{\mathrm{r}}$; $\Lambda_+ = c_{\mathrm{r}} + \gamma_{\mathrm{r}}$; C_Q, c_Q, a_Q are the constants defined in Lem. 5.30, which depend only on L; for a 1-form X on $\Lambda_{K-j,\ell}$, $m_y^j(X) = \sum_{c'} e^{-\delta_0 d(y, c'_-)} |X(c')|$, $\delta_0 > 0$ the rate of decay in the bounds of the response, are the averages that enter these bounds, and B, S are the constants with $|\mathcal{H}_V(\zeta)(b)| \le \mathsf{B}\, m_y^j(\zeta)$ for b_- in the block of y and $\sum_y m_y^j(X)^2 \le \mathsf{S}^2 \|X\|^2$ (§5.3, Lem. 15.9); $\mathsf{M} = \max\{1, \mathsf{BS}\}$.

$C_{\mathrm{P}} = 4C_{\mathrm{ch}}/\gamma_{\mathrm{r}}$ (Lem. 15.3); $\epsilon_d^\flat = [1 + (C_\flat + C_{\mathrm{cov}})\delta(L^d \mathsf{b})]^2 - 1$ and $\Pi_n^\flat = \prod_{d>n}(1 + \epsilon_d^\flat)$, finite and tending to 1 (Prop. 15.5); $C^\sigma = 3\mathsf{M}^2 \Pi^\flat_{n_{\mathrm{a}}}$. Thresholds: n_2 is the least integer ≥ 1 with $L^{n_2}\mathsf{b} \ge \mathsf{t}_\natural$ such that $C_{\mathrm{Hs}}\delta(L^d \mathsf{b}) \le \frac{1}{4}\gamma_{\mathrm{r}}$ and $\theta_d \le \omega_L$ for $d \ge n_2$, with C_{Hs} the constant of Lem. 15.3(iv) and a number $\omega_L \le (3200L^3)^{-1}$ that depends only on L; $n_3 \ge n_2$ is the least integer such that for $d \ge n_3$ the number θ_d lies below the radius of §5.3 and $2\theta_d \le \varepsilon_{\mathcal{W}}\alpha_0$, with the fixed radii $\alpha_0 < \alpha_1$ of the chain and the number $\varepsilon_{\mathcal{W}}$ of Prop. YM3_I:52.3, so that θ_d is at most the threshold $\varepsilon_{\mathcal{W}} \cdot \frac{1}{2}\alpha_0$ of Prop. YM3_I:52.3 at the radii $(\frac{1}{2}\alpha_0, \frac{1}{2}\alpha_1)$; n_{a} is the least integer $\ge \max\{1, n_3\}$ such that for all $d \ge n_{\mathrm{a}}$: $L^d \mathsf{b} \ge \mathsf{t}_\natural$; $\theta_d \le \min\{a_Q, \varpi_{\mathrm{db}}\}$ (that θ_d lies below the radius of §5.3 holds already for $d \ge n_3$); $4\delta(L^d\mathsf{b}) \le r_2$; $4C_{\mathrm{db}}\delta(L^d\mathsf{b}) \le r_1$; and $C_z\delta(L^d\mathsf{b}) \le \min\{\frac{1}{2}r_{\mathrm{h}}, \frac{1}{2}, (36 + 6C_{\mathrm{pz}}^2)^{-1/2}\}$.

Here $\varpi_{\mathrm{db}} = r_{\mathrm{y}}^2/(8\pi L^3)$ and $C_{\mathrm{db}} = 8L^3$ are the numbers of [7, Def. B.2], $r_{\mathrm{h}} = \min\{r_{\mathrm{B}}, \varrho_{\mathrm{R}}^0\}$ is the radius of holomorphy of the response in §5.3, $r_1 = \min\{\frac{c_Q}{4L}, \frac{c_Q}{32C_Q L^4}\}$, $r_2 = \frac{r_1}{8C_Q L^3}$, $C_z = 4C_Q L^3$ and

$C_{\mathrm{pz}} = 24C_QL^3 + 128C_QL^4/c_Q$, so that these numbers depend only on L and the constants of §5.3; n'' is the least integer $\ge n_{\mathrm{a}}$ with $\mathsf{R}_e \le \frac12$ for all $e \ge n''$; n_1 is the least $n \ge 2$ with $L^{n-1}\mathsf{b} \ge \mathsf{t}_\natural$ and $4\theta_d \le (48000L^4)^{-1}$ for $d \ge n$.

A quantitative extension of the gauge construction. The explicit top-scale bounds below require a compatible gauge construction across several block levels. The missing estimate, referred to as *(Multiscale gauge)*, would provide finite constants $R_{\mathrm{S}}, C_{\mathrm{S}} \ge 1$, independent of the odd integer $L \ge 13$, the side ℓ, the integers n, J and the field, with the following property. If $V \in \Omega_n$, $J \ge 1$, $n - J \ge 1$, $L^J \ge R_{\mathrm{S}}$, $\varpi(\mathcal{M}^jV) < (200L^2)^{-1}$ for $0 \le j < J$, and $\frac{\pi}{2}L^J\varpi(\mathcal{M}^JV) \le 1$, there is a periodic site gauge u such that

$$\mathsf{d}(V^u(b), \mathbf{1}) \le 44\pi L^2 \sum_{j=0}^{J-1} \varpi(\mathcal{M}^jV) + C_{\mathrm{S}}L^{-J} \quad \text{at every bond } b.$$

This quantitative estimate remains to be proved. The branches (B1)(e), (B2)(d) and the explicit-top variants of (A4)(f) and (A5) below remain sketches at this estimate; the assertions using j_1, w_n and C_n^{t} retain their stated hypotheses. With $C_{\mathrm{BG}} = 2\lambda_{\mathrm{BG},3}$ of eq. 15.v, $C'' = 1 + 20\pi^2C_{\mathrm{BG}}^2$, $\kappa_0 = (8\pi C_{\mathrm{BG}})^{-2}$, $\vartheta_\infty = C_HA_1/((1 - L^{-3/2})A_0)$, which is < 1 under (E), and j_1 the least $j \ge 1$ with $L^{-3j/2}c_\rho \le \min\{\frac12(1 - \vartheta_\infty), 1/C''\}$: n_{c} is the least integer such that for $n \ge n_{\mathrm{c}}$: $n - j_1 \ge 1$, $L^{n-j_1}\mathsf{b} \ge \mathsf{t}_\natural$, $2\varepsilon_n \le C''\kappa_0$, $\ell L^n \ge \max\{\lambda_{\mathrm{BG},3}, (C''/2\varepsilon_n)^{1/2}\}$; $\mathsf{r}_n = \mathsf{R}_n + \mathsf{R}_{n-j_1}$. For the proposed variants requiring the multiscale gauge estimate, let $\alpha_n = \frac18(2\arcsin\varepsilon_n)^{1/2}$ for $n \ge 2$ (the indices 0 and 1 belong to the fixed radii), and let J_n be the least integer $J \ge 1$ with $L^J \ge \max\{R_{\mathrm{S}}, 8C_{\mathrm{S}}/\alpha_n\}$. Then $J_n \le n/4 + c$, with c independent of n and K.

For these proposed multiscale variants, n_{e} denotes the least integer ≥ 2 such that for $n \ge n_{\mathrm{e}}$: $\varepsilon_n \le \frac{1}{51}$, $\log L^n\mathsf{b} > 2\mathfrak{p}_0$, $n - J_n \ge 1$, $L^{n-J_n}\mathsf{b} \ge \mathsf{t}_\natural$, $\theta_d < (200L^2)^{-1}$ for $d \ge n - J_n$, $44\pi L^2\sum_{j<J_n}\theta_{n-j} \le \frac18\alpha_n$, and $\frac{\pi}{2}L^{J_n}\theta_{n-J_n} \le 1$. No constant of this section depends on K; $n_1, n_2, n_3, n_{\mathrm{a}}, n_{\mathrm{e}}, C^\sigma, \Pi_n^\flat$ depend only on b and the constants of the construction, and R_n, o_n, n'', n_{c} and the constants defined from them (r_n, C_n^{t}, $C_n^\flat$, C_n, C'_n) depend besides on ℓ.

Flowed observables. Fix $0 < \kappa < \frac16$. For $s > 0$ and $V \in \Omega_K$, $\mathsf{m}_s(V)$ is the least

$$\max\{s^{1/4+\kappa/2}\|A\|_\infty, s^{3/4+\kappa/2}\|\nabla A\|_\infty\}$$

over the real 1-forms A with e^{ia_KA} gauge equivalent to V (∇ forward difference quotients);

$$\mathsf{O}_K(C, \bar s) = \{U : \mathsf{m}_s(U_s) \le C \text{ for } 0 < s \le \bar s\}.$$

The *flowed observables* are the functions $H_K^\sigma = \varphi(\mathcal{O}(f_1), \dots, \mathcal{O}(f_r)) \circ \Phi_\sigma$, $\sigma > 0$, $f_i \in C^\infty(T_\ell)$ real, $\varphi \in C^\infty(\mathbb{R}^r)$ bounded with all its derivatives. For $0 < \alpha < 1$ let $\mathcal{O}^\alpha$ be the space of orbits of $\mathfrak{g}^3$-valued $C^{0,\alpha}$ connections $\mathfrak{A}$ on T_ℓ under $C^{1,\alpha}$ gauge transformations; a *flow line* is a continuous map $(0,\infty) \to \mathcal{O}^\alpha$ that near every time is the orbit of a mild solution of $\partial_s\mathfrak{A} = -\mathrm{d}_{\mathfrak{A}}^*F_{\mathfrak{A}} + \mathrm{d}_{\mathfrak{A}}(\partial_\nu\mathfrak{A}_\nu)$, the Yang–Mills gradient flow up to an infinitesimal gauge transformation; $\mathcal{O}^0(f)(\mathfrak{A}) = \frac12\int f\sum_{\mu<\nu}|F_{\mu\nu}(\mathfrak{A})|^2$.

With $P_s = e^{s\Delta_a}$ the heat semigroup of $\Lambda_{K,\ell}$, $a = a_K$, $|f|_{\eta;a} = \sup_{0<s\le1} s^{-\eta/2}\|P_sf\|_\infty$ and $X \wedge Y = \frac{1}{2i}[X, Y]$, the rough regularity quantity of a 1-form $\mathbf{B}$ is

$$\|\mathbf{B}\|_{\kappa;a} = |\mathbf{B}|_{-\frac12-\kappa;a} + \sup_{0<s\le1} s^{\frac34+\kappa}\max\big|(\tau_h\nabla_\lambda P_s\mathbf{B}_\nu) \wedge (\tau_{h'}P_s\mathbf{B}_{\nu'})\big|_{-\frac12;a},$$

the maximum over directions λ, ν, ν' and translations $\tau_h, \tau_{h'}$ in a fixed finite set containing every translation in the quadratic term of the gauge-fixed flow equation below; it bounds $s^{\frac14+\frac\kappa2}\|P_s\mathbf{B}\|_\infty$ and $\frac12 s^{\frac34+\frac\kappa2}\|\nabla P_s\mathbf{B}\|_\infty$. The rough distance of $\mathbf{B}$, $\mathbf{B}'$ adds to $|\mathbf{B} - \mathbf{B}'|_{-\frac12-\kappa;a}$ the same supremum for the products $\nabla P_s(\mathbf{B} - \mathbf{B}') \wedge P_s\mathbf{B}$ and $\nabla P_s\mathbf{B}' \wedge P_s(\mathbf{B} - \mathbf{B}')$.

15.1.2. *Why the process limit is Markov.* At a fixed physical torus, the compatible block fields give a common space on which to compare cutoff processes. An observable of one block level can be evaluated at every finer cutoff by pulling it back along the averaging map. Convergence of these observables defines the one-time thread law. Convergence of their multi-time expectations gives stationary reversible limiting processes along subsequences. There is a further issue: projecting a Markov process onto a smaller space can retain memory. The semigroup identity must therefore be proved after taking the limit.

The source makes that issue explicit. A subsequential transition family has the form $Q_t = J^* R_t J$, where J embeds the limiting L^2 space isometrically into another Hilbert space and R_t is a self-adjoint contraction semigroup there. The possible loss is measured exactly by

$$(F, Q_{2t}F) - \|Q_t F\|^2 = \|(1 - JJ^*)R_t J F\|^2.$$

Consequently, weak convergence of transition operators leaves open whether their limit is a semigroup. Strong convergence will make the two sides of the intermediate scalar-product identity agree and remove this defect. The multiscale estimates are used to obtain that strong convergence.

Fix a coarse depth n and a finer cutoff K. Successively average an invariant observable H under the conditional fluctuation probabilities. Denote the resulting conditional means by h_k. These functions form a martingale when the small-field chain is read from the coarse level to the fine level. Its variance decomposition is a sum of conditional variances. The convexity of each fluctuation integral bounds its variance by the Dirichlet energy at that step. The response estimate transports that energy to the next step with a multiplicative error whose product stays bounded. Summing the physical scale factors gives the principal approximation bound

$$\|H - (\mathbf{1}_{\mathsf{I}_n} h_{K-n}) \circ \pi_{K,n}\|^2_{L^2(\mu^K)} \le \frac{C^\sigma C_{\mathrm{P}}}{L^2 - 1} \frac{a_n^2}{b} \mathcal{E}_K(H) + 4\mathsf{R}_n \|H\|^2_\infty.$$

Here I_n is the admitted open set at depth n, $\pi_{K,n}$ is the block projection, and R_n is the mass discarded by the chain. All these objects are defined precisely below. The two errors have different origins. The factor a_n^2 comes from summing the conditional variances above depth n; R_n accounts for the part of the Wilson measure on which the chain was discarded. Both go to zero as n increases, with the physical torus held fixed.

This estimate reduces compactness to one finite-dimensional block space. For bounded observables with bounded energies, the conditional means have bounded Dirichlet integrals on compact subsets of I_n. Rellich compactness is applied on an exhaustion of that open set; no smoothness of its boundary is required. Comparison of the block law with Haar measure then transfers convergence of the conditional means to the block probability measure. Finally, the displayed approximation permits n to increase and gives strong compactness in the thread law. The order is important: finite-dimensional compactness is used at a fixed n, and the uniform approximation is used afterwards.

Apply this compactness statement to $H_K = P_t^K \Psi_K$, where Ψ is an invariant cylinder function. The lattice spectral theorem supplies

$$\mathcal{E}_K(P_t^K \Psi_K) \le \frac{\|\Psi_K\|_2^2}{2et}, \qquad t > 0.$$

Every subsequence therefore has a strongly convergent subsequence. Its limit must equal the already determined weak limit $Q_t \Psi$. Thus the entire chosen process subsequence converges strongly at this observable and time. Density extends this conclusion to the invariant L^2 space. It gives the semigroup property and the Markov property on the invariant sigma-algebra. It does not identify the limits arising from two different process subsequences.

The form domain is recovered by cylinder functions multiplied by smooth regularity cutoffs at all finer block levels. The large-field tail makes these cutoffs converge to one, and their Dirichlet costs tend to zero. This produces a common core rather than merely a collection of cylinder functions with

no energy control. The lower semicontinuity and recovery-sequence arguments then give convergence of forms in the weak and strong senses and convergence of the resolvents.

Nonconstancy is checked on the limiting form $\mathcal{E}_\star$. The conditional-mean comparison passes a positive coarse Dirichlet integral to every recovery sequence for a nonconstant plaquette observable. The coarse measure has a positive density on its regular open set, and the squared gradient of that observable is positive there away from zero curvature. Hence the limiting energy is positive. The spectral measure of the observable cannot be concentrated at zero, so its transition correlation changes with time. A positive lattice energy alone would not give this conclusion without the comparison with recovery sequences.

For mixing, the connected part of the coarse regular region gives a Poincaré inequality, while its complement contributes a defect. After transport through the chain, its useful form is

$$\mathrm{Var}_\mu(F) \le C_n \mathcal{E}_\star(F) + r_n \,\mathrm{osc}(F)^2, \qquad r_n \longrightarrow 0.$$

At fixed n, apply this inequality along the semigroup and let time increase; then let n increase to remove the defect. The constants C_n may depend on the physical torus. This proves the stated mixing without supplying a spectral gap uniform in volume.

15.1.3. *Markov limits, functional inequalities and flowed observables.*

Theorem AT (Nontrivial Markov limits and mixing)**.** Assume (E).

(A1) *Motion.* For every limit, every integer $n \ge n''$ and every invariant $\Phi \in C^1(\Omega_n)$, with $|\mathsf{l}_n|$ the Haar measure of l_n,

$$\mathcal{E}_\star(\Phi(Y_n)) \ge \frac{e^{-\mathsf{o}_n}}{2C^\sigma |\mathsf{l}_n|} \int_{\mathsf{l}_n} \Gamma_n(\Phi)\, \mathrm{d}V. \tag{15.i}$$

Hence $(\Psi, Q_t\Psi) < \|\Psi\|^2$ for $\Psi = a_p(Y_n)$ and every $t > 0$: no limit is frozen.

(A2) *Compactness.* Along every sequence $K_j \to \infty$, invariant $H_j \in D(\mathcal{E}_{K_j})$ with $\sup_j(\sup|H_j| + \mathcal{E}_{K_j}(H_j)) < \infty$ have a strongly convergent subsequence; its limit G has $|G| \le \sup_j \sup|H_j|$.

(A3) *The Markov property and form convergence.* For every limit, $\mathcal{E}_\star$ is a closed, densely defined Markovian form on L^2_{inv}, and $Q_t = e^{-tA_\star}$ is a strongly continuous symmetric Markov semigroup on L^2_{inv}; $P_t^{K_j} H_j \to Q_t F$ strongly for $t > 0$ whenever $H_j \to F$ strongly, in particular $P_t^{K_j}\Psi_{K_j} \to Q_t\Psi$ for every invariant cylinder function Ψ; (M1) and (M2) hold, and $(\alpha - \mathcal{L}_{K_j})^{-1} W_j \to (\alpha + A_\star)^{-1} W$ strongly for $\alpha > 0$ and invariant $W_j \to W$ strongly; (Y_t) is a stationary reversible Markov process on the invariant σ-algebra with the transition operators Q_t and the invariant measure $\mu_{\ell,\mathsf{b}}$; and $\{G \in L^2_{\mathrm{inv}} : |G| \le 1,\ \mathcal{E}_\star(G) \le 1\}$ is compact.

If invariant $H_j \in C^\infty(\Omega_{K_j})$ converge strongly to F and $\sup_j \|\mathcal{L}_{K_j} H_j\|_{L^2(\mu^{K_j})} < \infty$, then $F \in D(A_\star)$, $\mathcal{L}_{K_j} H_j \to -A_\star F$ weakly and $\mathcal{E}_{K_j}(H_j) \to \mathcal{E}_\star(F)$; under the assumptions of (C7), this hypothesis fails at the flowed observables of (C7)(b), (c), (e). If invariant $u_j \to u$ strongly with $\mathcal{E}_{K_j}(u_j) \to \mathcal{E}_\star(u)$, and invariant $v_j \to v$ weakly with $\sup_j \mathcal{E}_{K_j}(v_j) < \infty$, then $\mathcal{E}_{K_j}(u_j, v_j) \to \mathcal{E}_\star(u, v)$.

(A4) *Variance bounds and mixing.* Let $n \ge \max\{n'', n_{\mathrm{c}}\}$ and $\mathsf{C}_n = C^\sigma[\frac{C_{\mathrm{P}}}{L^2-1}\frac{a_n^2}{\mathsf{b}} + \frac{1}{2}C_n^{\mathrm{t}}]$, with C_n^{t} of (B1). For all $K > n$: (a) $\mathrm{Var}_{\mu^K}(H) \le \mathsf{C}_n \mathcal{E}_K(H) + \mathsf{r}_n \mathrm{osc}(H)^2$ for bounded invariant $H \in D(\mathcal{E}_K)$; (b) $\|P_t^K H - \mu^K(H)\|^2_{L^2(\mu^K)} \le (\mathsf{r}_n + \frac{1}{4}e^{-2t/\mathsf{C}_n})\mathrm{osc}(H)^2$ for bounded Borel invariant H and $t \ge 0$. For every limit: (c) $\mathrm{Var}_\mu(F) \le \mathsf{C}_n \mathcal{E}_\star(F) + \mathsf{r}_n \mathrm{osc}(F)^2$ for bounded $F \in L^2_{\mathrm{inv}}$, so $\mathcal{E}_\star(F) = 0$ only for constant F, and $(F, (1 - Q_t)F) > 0$ for every non-constant bounded $F \in L^2_{\mathrm{inv}}$ and $t > 0$; (d) $0 \le \mathbb{E}F(Y_0)F(Y_t) - (\int F\,\mathrm{d}\mu)^2 \le \inf_n(\mathsf{r}_n + \frac{1}{4}e^{-t/\mathsf{C}_n})\mathrm{osc}(F)^2$ for bounded $F \in L^2_{\mathrm{inv}}$, the infimum over the admitted n, which tends to 0 as $t \to \infty$; and a probability measure $\rho\,\mu$, $\rho \in L^2_{\mathrm{inv}}$, with $Q_t\rho = \rho$ for one $t > 0$ is μ.

(e) If $F_K \to F$ strongly along the full sequence of cutoffs, $\sup_K \sup|F_K| \le M_F$, F is not constant, and H_K are invariant Lipschitz functions with $\|H_K - F_K\|_{L^2(\mu^K)} \to 0$, then

$\liminf_K \mathcal{E}_K(H_K) \ge \sup_n(\mathrm{Var}_\mu F - 4\mathsf{r}_n M_F^2)/\mathsf{C}_n > 0$; and for invariant Lipschitz H_K with $\sup_K \mathrm{osc}(H_K) \le O$, $\liminf_K \mathcal{E}_K(H_K) \ge \sup_n[\liminf_K \mathrm{Var}_{\mu^K}(H_K) - \mathsf{r}_n O^2]/\mathsf{C}_n$. (f) The multiscale gauge extension remains a sketch. Completing that estimate would give all of (a)–(e) as well for $n \ge \max\{n'', n_{\mathrm{e}}\}$ with $\mathsf{R}_n < 1$, with $C_n^\flat$ of (B1)(e) in the place of C_n^{t} and $\mathsf{R}_n + \mathsf{R}_{n-J_n}$ in the place of r_n.

(A5) *Entropy relaxation.* Let $n \ge \max\{n_{\mathrm{a}}, n_{\mathrm{c}}\}$, C'_n and ς^K as in (B2). For all $K > n$: (a) if $G \in C^1(\Omega_K)$ is real and invariant, $M_G = \sup|G|$ and $m_2 = \int G^2 \,\mathrm{d}\mu^K > \mathsf{r}_n M_G^2$, then

$$\mathrm{Ent}_{\mu^K}(G^2) \le \mathsf{C}'_n \mathcal{E}_K(G) + \mathsf{r}_n M_G^2 \Big[1 + \log \frac{M_G^2}{m_2 - \mathsf{r}_n M_G^2}\Big];$$

(b) if $\psi \in C^1(\Omega_K)$ is real and invariant with $\Gamma_K(\psi) \le l^2$ and $\bar\psi = \int \psi \,\mathrm{d}\varsigma^K / \varsigma^K(\Omega_K)$, then $\mu^K(\psi - \bar\psi \ge u) \le e^{-u^2/(\mathsf{C}'_n l^2)} + \mathsf{r}_n$ for $u > 0$; (c) if $f_0 \ge 0$ is bounded Borel invariant, $m_1 = \int f_0 \,\mathrm{d}\mu^K > 0$, $\bar M = \sup f_0$ and $\mathsf{r}_n \bar M \le \frac12 m_1$, then for $\tau \ge 0$

$$\mathrm{Ent}_{\mu^K}(P_\tau^K f_0) \le e^{-4\tau/\mathsf{C}'_n} \mathrm{Ent}_{\mu^K}(f_0) + \mathsf{r}_n \bar M \Big[1 + \log \frac{2\bar M}{m_1}\Big].$$

For every limit, (a) holds for bounded $F \in D(\mathcal{E}_\star)$ with $\mu, \mathcal{E}_\star$ in the places of $\mu^K, \mathcal{E}_K$; and if $F \in L^2_{\mathrm{inv}}$, $0 \le F \le \bar M$, $m_1 = \int F \,\mathrm{d}\mu > 0$ and $\mathsf{r}_n \bar M < \frac12 m_1$, then $\mathrm{Ent}_\mu(Q_\tau F) \le e^{-4\tau/\mathsf{C}'_n} \mathrm{Ent}_\mu(F) + \mathsf{r}_n \bar M[1 + \log \frac{2\bar M}{m_1}]$; so $\mathrm{Ent}_\mu(Q_\tau F) \to 0$ as $\tau \to \infty$, and $m_1^{-1}(Q_\tau F)\,\mu \to \mu$ in total variation on the invariant σ-algebra. The corresponding extension remains a sketch at the multiscale gauge estimate; its intended conclusion is the same for $n \ge \max\{n_{\mathrm{a}}, n_{\mathrm{e}}\}$ with $\mathsf{R}_n < 1$, with the explicit top of (B1)(e).

■□

Theorem AU (Functional inequalities uniform in the cutoff)**.** Assume (E).

(B1) *The coarse-scale law.* For $n \ge n_{\mathrm{c}}$ there are an invariant Borel weight $w_n : \Omega_n \to [0,1]$ and a constant $C_n^{\mathrm{t}} < \infty$ such that for all $K > n$ and every bounded invariant locally Lipschitz ϕ on I_n: (a) $w_n \le \mathbf{1}_{\mathsf{D}_n}$ and $\sigma_n^K(w_n < 1) \le \mathsf{R}_{n-j_1}$; (b) $\mathrm{Ent}_{w_n\sigma_n^K}(\phi^2) \le C_n^{\mathrm{t}} \int \Gamma_n(\phi)\, w_n \,\mathrm{d}\sigma_n^K$; (c) $\int(\phi - c)^2 \,\mathrm{d}\sigma_n^K \le \frac12 C_n^{\mathrm{t}} \int \Gamma_n(\phi)\,\mathrm{d}\sigma_n^K + \mathsf{R}_{n-j_1}\mathrm{osc}(\phi)^2$, c the mean of ϕ under $w_n\sigma_n^K$; (d) $\mathrm{Ent}_{\sigma_n^K}(\phi^2) \le C_n^{\mathrm{t}} \int \Gamma_n(\phi)\,\mathrm{d}\sigma_n^K + \mathsf{R}_{n-j_1} \sup_{\mathsf{I}_n} \Xi(\phi^2, c')$, c' the mean of ϕ^2 under $w_n\sigma_n^K$.

Here $w_n = \min\{1, \rho/\rho_0\}$, $\rho(V) = \int \mathbf{1}_{S'}(V^g)\,\mathrm{d}g$ for a connected open $S' \subset \mathsf{D}_n$ that is a finite union of chart balls, $\rho_0 = \min_{\mathcal{K}_1}\rho > 0$ on a compact invariant set $\mathcal{K}_1 \subset \mathsf{D}_n$, and $C_n^{\mathrm{t}} = a_n\, c(S')\, e^{\mathsf{o}_n}/\rho_0$, $c(S')$ the log-Sobolev constant of Haar measure on S'. (e) (An explicit top.) The following variant remains a sketch at the multiscale gauge estimate. Let $n \ge n_{\mathrm{e}}$, $\mathsf{R}_n < 1$, $N = \ell L^n$, $w_n^{\mathrm{e}}(V) = \int \mathbf{1}\{\mathsf{d}(V^u(b), \mathbf{1}) < \alpha_n \text{ for all } b\}\,\mathrm{d}u$ (du Haar measure of the gauge group), $c_n = [(\frac12\alpha_n - \sin\frac12\alpha_n)/2\pi]^{N^3}$ and $\chi_n = \min\{1, w_n^{\mathrm{e}}/c_n\}$.

Completing that estimate would give (a)–(d) with $\chi_n \mathbf{1}_{\mathsf{I}_n}$ in the place of w_n, R_{n-J_n} in the place of R_{n-j_1}, and the constant

$$C_n^\flat = \frac{a_n e^{2\mathsf{o}_n} C_{\mathscr{D}}}{(1-\mathsf{R}_n)c_n}, \quad C_{\mathscr{D}} = 2\sqrt{e}\Big(\frac{\alpha_n}{\sin\alpha_n}\Big)^{6N^3}\Big(\frac{2\rho_{\max}}{r'}\Big)^{9N^3}\Big(\rho_{\max} + \frac{8\pi\rho_{\max}^2}{r'}\Big)^2, \quad \rho_{\max} = \alpha_n\sqrt{3N^3},\ r' = 64\alpha_n^2,$$

in the place of C_n^{t}; the set $\{\chi_n > 0\} \cap \mathsf{I}_n$ is open, invariant, path connected and contained in D_n; and for bounded invariant $\Phi \in C^1(\Omega_n)$, $\mathrm{Var}_{\mu_n^K}(\Phi) \le \frac12 C_n^\flat \int_{\mathsf{I}_n} \Gamma_n(\Phi)\,\mathrm{d}\mu_n^K + (\mathsf{R}_n + \mathsf{R}_{n-J_n})\mathrm{osc}(\Phi)^2$, and the same with μ_n in the place of μ_n^K.

(B2) *The chain with a connected coarse-scale region.* Let $n \ge \max\{n_{\mathrm{a}}, n_{\mathrm{c}}\}$, $K > n$, $\varsigma^K = w_n(V_t)\,\sigma^K$ and

$$\mathsf{C}'_n = \Big[\frac{2C_{\mathrm{P}}}{L^2 - 1}\frac{a_n^2}{\mathsf{b}} + C_n^{\mathrm{t}}\Big]\, 3\mathsf{M}^2\, \Pi_n^\flat.$$

(a) $\int f\,\mathrm{d}\varsigma^K \le \int f\,\mathrm{d}\sigma^K \le \int f\,\mathrm{d}\mu^K \le \int f\,\mathrm{d}\varsigma^K + \mathsf{r}_n \sup f$ for bounded Borel invariant $f \ge 0$; (b) $\mathrm{Ent}_{\varsigma^K}(G^2) \le \mathsf{C}'_n \int \Gamma_K(G)\,\mathrm{d}\varsigma^K$ for real invariant $G \in C^1(\Omega_K)$; (c) for $\nu \in \{\sigma^K, \mu^K\}$: $\mathrm{Ent}_\nu(G^2) \le \mathsf{C}'_n \int \Gamma_K(G)\,\mathrm{d}\nu + \mathsf{r}_n \sup_U \Xi(G(U)^2, c')$, c' the mean of G^2 under ς^K. (d) This variant remains a sketch at the multiscale gauge estimate. Its intended conclusion is: for Lebesgue-almost every label b, all m and all $n \ge \max\{n_\mathrm{a}, n_\mathrm{e}\}$ with $\mathsf{R}_n < 1$, (a)–(c) hold with $\mathbf{1}_{\mathsf{D}_n}(V_t)\,\sigma^K$ in the place of ς^K, a finite constant $C_n^\sharp$ in the place of C_n^t, and $\mathsf{R}_n + \mathsf{R}_{n-J_n}$ in the place of r_n; for every b and $\eta > 0$ they hold with the component of $\mathbf{1}$ in $\{\varpi < \tilde r\}$, for some $\tilde r \in (\varepsilon_n, 2\varepsilon_n)$ that does not depend on K, in the place of D_n, and $\mathsf{R}_n + \mathsf{R}_{n-J_n} + \eta$ in the place of r_n; with the explicit top of (B1)(e) they hold with $\chi_n(V_t)\sigma^K$, $C_n^\flat$ and $\mathsf{R}_n + \mathsf{R}_{n-J_n}$.

(e) (No weight.) For $n \ge n_\mathrm{a}$, $K > n$, real invariant $G \in C^1(\Omega_K)$, f_t the top of the tower of G^2 and $\epsilon > 0$:

$$\mathrm{Ent}_{\sigma^K}(G^2) \le \frac{6C_\mathrm{P}\mathsf{M}^2\Pi_n^\flat}{L^2-1}\frac{a_n^2}{\mathsf{b}}\int \Gamma_K(G)\,\mathrm{d}\sigma^K + \mathrm{Ent}_{\sigma_n^K}(f_t), \qquad \int \Gamma_n((f_t+\epsilon)^{1/2})\,\mathrm{d}\sigma_n^K \le 3\mathsf{M}^2\Pi_n^\flat \int \Gamma_K(G)\,\mathrm{d}\sigma^K.$$

(B3) *The fine field conditioned on a block field.* Let $n \ge n_\mathrm{a}$, $K > n$, $1 \le t' \le K-n$, $e = K - t'$, $W \in \mathsf{Cut}_{t'}$, and let P_W be the law of V_0 when $V_{t'} = W$ and $V_k = V_{V_{k+1}}(B_k)$, B_k of law $p_{V_{k+1}}$. Then P_W is a regular conditional law of U given $\mathcal{M}^{t'}U = W$ under σ^K and under ς^K, and for real invariant $G \in C^1(\Omega_K)$ with tower (g_k) and every $W \in \mathsf{Cut}_{t'}$

$$\mathrm{Ent}_{\mathsf{P}_W}(G^2) \le C_e^\mathrm{fib}\int \Gamma_K(G)\,\mathrm{d}\mathsf{P}_W, \qquad \Gamma_e(g_{t'})(W) \le 3\mathsf{M}^2\Pi_e^\flat\int \Gamma_K(G)\,\mathrm{d}\mathsf{P}_W, \qquad C_e^\mathrm{fib} = \frac{6\mathsf{M}^2C_\mathrm{P}\Pi_e^\flat}{L^2-1}\frac{a_e^2}{\mathsf{b}},$$

a constant that depends neither on K nor on ℓ. If $\psi \in C^1(\Omega_K)$ is real and invariant and $\Gamma_K(\psi) \le l^2$ P_W-almost everywhere, then $\mathsf{P}_W(\psi - \psi_{t'}(W) \ge u) \le e^{-u^2/(C_e^\mathrm{fib}l^2)}$, $\psi_{t'}$ the member of the tower of ψ; and if $\Gamma_K(\psi) \le l^2$ on all of Ω_K, then, with $\psi_{t'} := 0$ off $\mathsf{Cut}_{t'}$, $\mu^K(|\psi - \psi_{t'}(\mathcal{M}^{t'}U)| \ge u) \le 2e^{-u^2/(C_e^\mathrm{fib}l^2)} + \mathsf{R}_n$.

■□

Admissible top depths. The assertions about flowed observables use the following condition, in the notation of Prop. 16.1.

A top depth d_0 is *admissible with constant* C_adm if $d_0 \ge n_\mathrm{a}$, $d_0 \ge d_\mathrm{num}(\ell, \mathsf{b})$, $c_\rho\varepsilon(L^{d_0}\mathsf{b}) \le C_\mathrm{BG}^{-2}$ and $4c_\mathsf{h}\delta(L^{d_0}\mathsf{b}) \le 1/4$, and, for every $K > d_0$ and $d_0 < d \le K$, (M) and (Col) hold with that same constant. Here d_num is the explicit numerical depth in the proof below. The mean and covariance-error estimates hold at every $W \in \mathsf{Cut}_{K-d+1}$; the Gaussian layer-kernel estimates hold almost surely in the chain gauge, at every pair of bare bonds. (Adm) Under (E), there are d_adm and C_adm, depending on ℓ, b and the construction constants but not on K or the chosen top depth, such that every $d_0 \ge d_\mathrm{adm}$ is admissible with C_adm. This is proved in Prop. 16.1 below.

The proof of Prop. 16.1 establishes (M) and (Col) uniformly in the cutoff and top depth whenever $d_0 \ge d_\mathrm{adm}$. Thus the flowed estimates use (Adm) as a consequence of (E).

15.1.4. *Renormalisation inputs.* We use Thms. YM3_I:D, YM3_I:AJ and YM3_I:AD among the construction results through its results on one step (Prop. 6.32 at zero source) and on the small-field chain on one torus (Sections YM3_I:113, YM3_I:114, YM3_I:115, Section YM3_I:116): setting (E) (§2.3.1), the field cuts (the definitions preceding Lem. YM3_I:113.10, Lem. YM3_I:113.10) and the gauge action (Section YM3_I:3), the chart at the pivots ([7, §B.2]), the iteration from a level with its exponents and the bound of the non-local action (Prop. YM3_I:114.2, Prop. YM3_I:114.2, equation YM3_I:(2163), equation YM3_I:(2180)), the comparison of the chain with the bare

density through the weight $\mathfrak{w}_t^{\mathrm{sf}}$ and the bound R_n (the definitions preceding Prop. YM3_I:115.2, Prop. YM3_I:115.2), and the convergence in total variation of the laws of the block fields (Cor. 2.8).

Further construction results are cited where they are used: the constants and thresholds of the construction (equation YM3_I:(634), §15.3.1, 2.3.1, 2.3.1, 2.3.1, §2.3.1), with the old-term constants defined in equation YM3_I:(2163) and its following sentence; the current (YM3_I:9.2); the lattices and the localisation domains with their linear sizes (Section YM3_I:3, YM3_I:72 (the definitions of $\mathcal{D}_j$ and d_j at its beginning), with YM3_I:72 (the definitions of $\mathcal{D}_j$ and d_j at its beginning)); averages (equations YM3_I:(37)–(39), Prop. YM3_I:51.1 and Lem. 5.30), and of the averaging map its gauge covariance (Prop. YM3_I:4.5, YM3_I:(40)), its real-analyticity on the small fields and its measurability (Prop. YM3_I:4.5), its linear part at the identity, which is the linear average YM3_I:(42) of equation YM3_I:(42) (Lem. YM3_I:4.6), with the k-fold composites $\mathbf{Q}_k^{\mathrm{lin}}$ of the linear average, and the bound for rearranged paths (Lem. YM3_I:4.1); backgrounds over a block field, whose averages give back the block field (Defs. YM3_I:9.3 and YM3_I:9.5, §5.2); the fixed radii and the analyticity cores (Def. YM3_I:49.1 and Lem. YM3_I:82.1); the bounds of the response (Lem. 15.9, §5.3) and its Landau gauge on the kernel of the site averages (§15.3, equation YM3_I:(837)); the constant-frame row map and its response identity, with $C_{\mathcal{Q}} = 34(L^3+2)$ and the counting-measure kernel bounds (Lem. 5.32); the double-block constants of [7, Def. B.2]; the regularity numbers and the classes of covariance ([7, Def. B.2], [7, Lem. B.4], and the curvature comparison in the proof of Lem. YM3_I:113.10, with C_H as fixed in [7, §P.1]); backgrounds and small plaquettes (Prop. YM3_I:52.3, Prop. 15.8); the local potentials and their bounds (Prop. YM3_I:51.4), with their sums in equations YM3_I:(828)–YM3_I:(830); the outer substitution (Prop. YM3_I:51.10); the sum over polymers [7, eq. 320]; §15.3.1; Thm. YM3_I:W and equation YM3_I:(925); the Gaussian structure of a step and its numbers (Def. YM3_I:47.2, Thm. YM3_I:U and §15.3.1); the gauge-fixing factor, fluctuation test and chart integral of [7, §B.2], with the measure identity of [7, Lem. B.3]; the number $\varepsilon_{\mathcal{W}}$ Prop. YM3_I:52.3; and the consequences of the setting (§2.3.1).

These are the construction results used in this section. The admissibility conclusion (Adm) follows from (E). The multiscale-gauge and flow-approximation arguments require the separate tasks specified below. In particular, the continuum-flow identification in Prop. 16.1(b) still requires the compactness, equation identification, initial holonomies and uniqueness described under (Flow approximation).

Within the section, the proofs of (C2)(b), (C2)(c) and (C3)–(C7) start besides from the moments of the gauge-fixed field $\mathbf{B}_K$ of Prop. 16.1(a), gathered before Step 6, and from its proved conclusion (Adm) on the top depths. The moment bound under the chain measure from an admissible top depth rests on Lem. 15.3(iv), the bounds of the response and of the heat kernel, the grid lemma, and three statements on one fluctuation integral, set in the proof of Prop. 16.1(a): the bounds (M) of the conditional mean of a layer, the bounds (Col) of the part of its conditional covariance that is not scalar in colour, and the exponential bound (Q) of centred quadratic forms under the log-concave law of one integral.

A top depth d_0 is admissible when it meets the numerical and uniform-estimate conditions stated before Thm. AV; under (Adm) the chains of all the large top depths give the moments under μ^K. The statements made for all K, m and $\beta > 0$, namely Lem. 15.1(a), (C1), the short-time theory of the gauge-fixed flow in Prop. 16.1(a) (the flow on $[0, s_0(R)]$ with its four bounds, the Lipschitz bound in the rough distance and the bound of the flowed action), Lems. 16.2, 16.3 and 16.5, Lem. 16.4, Prop. 16.6(a), (b) and Prop. 16.9(a), (b) and the upper bound in (c), use no continuum-limit result. The lower bound in Prop. 16.9(c) follows from the adjoint frequency estimate and exact current transport, with the integrated curvature and current ratio displayed there. The continuum identification requires the separate (Flow approximation) step.

15.1.5. *The lattice dynamics and a common core.*

Lemma 15.1 (Lattice dynamics, subsequential limits and a common core)**.** (a) For all K, m, $\beta > 0$: $\int G\,\mathcal{L}_K F\,\mathrm{d}\mu^K = -\int \Gamma_K(G,F)\,\mathrm{d}\mu^K$ for $F, G \in C^\infty(\Omega_K)$; $\beta\,\partial_b A_K = \mathsf{b}\,a_K^2 J(U)(b)$, so $\mathcal{L}_K = \Delta_K + \mathsf{b}\,v$ is the generator of $dA = -\mathsf{b}\,J(U)\,\mathrm{d}t + \sqrt{2}\,\mathrm{d}\mathcal{W}$, $dU(b) = ia_K \circ dA(b)\,U(b)$, with $\mathcal{W}$ the Brownian motion of the 1-forms for the scalar product $a_K^3 \sum_b \langle\cdot,\cdot\rangle$: no constant is tuned. P_t^K is a symmetric Markov semigroup on $L^2(\mu^K)$ that commutes with gauge transformations; (U_{-t}) has the law of (U_t); $\mathbb{E}|F(U_t) - F(U_0)|^2 = 2(F, (1 - P_t^K)F) \le 2t\,\mathcal{E}_K(F)$; for $F \in D(\mathcal{L}_K)$, $\mathbb{E}|F(U_t) - F(U_0)|^2 = 2t\,\mathcal{E}_K(F) - \mathbb{E}|\int_0^t \mathcal{L}_K F(U_s)\,\mathrm{d}s|^2 \ge 2t\,\mathcal{E}_K(F) - t^2\|\mathcal{L}_K F\|^2$; and $\mathcal{E}_K(P_t^K F) \le \|F\|^2/2et$; $v(G \circ \Phi_\sigma) = \partial_\sigma(G \circ \Phi_\sigma)$.

With Π_y the Haar average over the gauge transformations at a site y and $\Pi_S = \prod_{y\in S}\Pi_y$: $\mathcal{E}_K(F) \ge (2a_K)^{-1}\|F - \Pi_y F\|^2$ and $\|P_t^K(F - \Pi_S F)\| \le e^{-t/2a_K}\|F - \Pi_S F\|$.

(b) Assume (E). Every sequence of cutoffs has a subsequence that is a limit. For every limit: (Y_t) is stationary, reversible and positive under time reflection ($\mathbb{E}[\overline{\theta\mathsf{F}}\,\mathsf{F}] \ge 0$ for finite sums F of products of cylinder functions at times ≥ 0, θ the reflection of the times), with one-time law μ; its multi-time means are real-analytic functions of the strictly positive consecutive time differences; its finite-dimensional laws are invariant under gauge transformations chosen independently at different times; Q_t is a symmetric positivity preserving contraction of $L^2(\mu)$, $Q_t 1 = 1$, $Q_t = \Pi Q_t \Pi$, and $\|Q_t F\|^2 \le (F, Q_{2t}F)$.

If $H_j \to F$ strongly, the spectral measures of H_j for $-\mathcal{L}_{K_j}$, taken as measures on $[0,\infty]$, converge weakly to a measure ν_F of mass $\|F\|^2$ with $(F, Q_t F) = \int e^{-t\lambda} d\nu_F$ for $t > 0$; $\mathcal{E}_\star(F) = \int \lambda\,\mathrm{d}\nu_F$; $\mathcal{E}_\star$ is closed, Markovian and finite only on L^2_{inv}; (M2) holds, and $\mathcal{E}_\star(F) \le \liminf_j \mathcal{E}_{K_j}(H_j)$ whenever $H_j \to F$ strongly. The following are equivalent: $\|Q_t\Psi\|^2 = (\Psi, Q_{2t}\Psi)$ for invariant cylinder functions Ψ and $t > 0$; $Q_{t+s} = Q_t Q_s$; $P_t^{K_j}\Psi_{K_j} \to Q_t\Psi$ strongly for these Ψ; (Y_t) is a Markov process on the invariant σ-algebra with the transition operators Q_t.

They imply (M1), and $P_t^{K_j} H_j \to Q_t F$ strongly whenever $H_j \to F$ strongly. In general $Q_t = J^* R_t J$, where J is an isometry of $L^2(\mu)$ into a Hilbert space $\mathcal{K}$ and (R_t) is a semigroup of positive self-adjoint contractions of $\mathcal{K}$, and $(F, Q_{2t}F) - \|Q_t F\|^2 = \|(1 - JJ^*)R_t J F\|^2$; the four properties hold if and only if $R_t J L^2_{\mathrm{inv}} \subset J L^2_{\mathrm{inv}}$ for all t.

(c) Assume (E). For every integer $n \ge 1$ with $L^n\mathsf{b} \ge \mathsf{t}_\natural$ and every $K > n$, $\mu^K(\varpi(\pi_{K,d}U) < \theta_d$ for all $n \le d \le K) \ge 1 - \mathsf{R}_n$, and $\mu(\varpi(Y_d) < \theta_d$ for all $d \ge n) \ge 1 - \mathsf{R}_n$. Fix $\varphi_{\mathrm{cut}} \in C^\infty([0,\infty); [0,1])$ with $\varphi_{\mathrm{cut}} = 1$ on $[0,1]$, $\varphi_{\mathrm{cut}} = 0$ on $[4,\infty)$ and $|\varphi'_{\mathrm{cut}}| \le 1$, and put

$$\chi_d = \prod_p \varphi_{\mathrm{cut}}(\varpi_p^2/(2\theta_d)^2) \quad \text{on } \Omega_d, \qquad \mathsf{k}_d = 24 L^{d/2}\theta_d^{-1}(\ell L^d)^3,$$

the product over the plaquettes of $\Lambda_{d,\ell}$: χ_d is smooth and invariant, takes its values in $[0,1]$, is 1 on $\{\varpi \le 2\theta_d\}$ and 0 on $\{\varpi \ge 4\theta_d\}$, and $\Gamma_d(\chi_d)^{1/2} \le \mathsf{k}_d \mathbf{1}\{\varpi \ge \theta_d\}$; the number k_d depends on ℓ and not on K, and it is at most a constant times $\ell^3 L^{4d}$. The span $\mathcal{C}$ of the functions $F = \Phi(Y_n)\prod_{d\ge n}\chi_d(Y_d)$, $n \ge n_1$, $\Phi \in C^\infty(\Omega_n)$ invariant, is dense in L^2_{inv}; the lattice forms $F_K = \Phi \circ \pi_{K,n} \prod_{n\le d\le K} \chi_d \circ \pi_{K,d}$ are smooth and invariant, $|F_K| \le \sup|\Phi|$, $F_K \to F$ strongly along the full sequence, and $\sup_K \int \Gamma_K(F_K)^p d\mu^K < \infty$ for every $p < \infty$; the cutoffs $\chi_n^K = \prod_{n\le d\le K}\chi_d \circ \pi_{K,d}$ have $\mu^K(\chi_n^K < 1) \le \mathsf{R}_n$ and $\|\Gamma_K(\chi_n^K)^{1/2}\|_{L^{2p}(\mu^K)} \to 0$ as $n \to \infty$, uniformly in K.

Consequently, for every limit, $\mathcal{E}_\star$ is densely defined, $\nu_F(\{\infty\}) = \|F - \Pi F\|^2$ for $F \in L^2(\mu)$, $(F, Q_t F) \to \|\Pi F\|^2$ and $Q_t \to 1$ strongly on L^2_{inv} as $t \to 0$, and $\frac{1}{2}\mathbb{E}|F(Y_t) - F(Y_0)|^2 \le t \sup_K \mathcal{E}_K(F_K)$ for $F \in \mathcal{C}$. ∎

Proof sketch. (a) Haar measure is invariant under left translations in each factor, so

$$\int \partial_b^\alpha (F G e^{-\beta A_K})\,\mathrm{d}U = 0,$$

which is the integration by parts; (YM3_I:9.2) and $\beta = \mathsf{b}/a_K$ give the drift. Writing a tangent vector as $X = a_K B$ with a 1-form B per unit length, $a_K\|X\|^2 = (B,B)_{a_K}$, and $\Gamma_K(F)$ is the square

of the gradient for $(\cdot,\cdot)_{a_K}$: this is the stochastic equation. The closure of $(\mathcal{E}_K, C^\infty)$ is a Dirichlet form, gauge transformations act unitarily and preserve it, so they commute with P_t^K; the two bounds are $1-e^{-t\lambda}\le t\lambda$ and $\lambda e^{-2t\lambda}\le 1/2et$ under the spectral measures of the spectral theorem, and the identity is

$$2\int(1-e^{-t\lambda})\,\mathrm{d}\nu = 2t\int\lambda\,\mathrm{d}\nu - 2\int(t\lambda-1+e^{-t\lambda})\,\mathrm{d}\nu,$$

with

$$\mathbb{E}|\int_0^t \mathcal{L}_K F(U_s)\,\mathrm{d}s|^2 = 2\int(t\lambda-1+e^{-t\lambda})\,\mathrm{d}\nu \le t^2\|\mathcal{L}_K F\|^2$$

by stationarity, ν the spectral measure of F.

The generators X_α of the rotations at y act on six bonds, $\sum_\alpha |X_\alpha F|^2 \le 6|\partial F|^2$, and the first non-zero eigenvalue 3 of the Laplacian of the unit sphere $S^3 = G$ gives $\int\sum_\alpha |X_\alpha F|^2\,\mathrm{d}\mu^K \ge 3\|F-\Pi_y F\|^2$. The commuting projections Π_y split $L^2(\mu^K)$ into joint eigenspaces invariant under P_t^K, on which $\frac{d}{dt}\|P_t^K F\|^2 = -2\mathcal{E}_K(P_t^K F)$. Since v is the field of Φ, $v(G\circ\Phi_\sigma)(U) = \frac{d}{ds}G(\Phi_\sigma\Phi_s U)|_{s=0}$.

(b) The laws of the block fields at rational times lie in a compact set of measures on the compact metrisable space $\mathcal{Y}^{\mathbb{Q}}$. The means $\langle\Psi_0, P_{s_1}^K\Psi_1\cdots P_{s_n}^K\Psi_n\rangle$ extend holomorphically and boundedly to $\operatorname{Re} s_i > 0$, so normal families carry convergence from rational to all times and give the real analyticity; convergence in total variation of the laws of the block fields carries it from continuous to Borel functions. Stationarity, reversibility and positivity under reflection hold at every cutoff, where $\mathbb{E}[\overline{\theta\mathsf{F}}\mathsf{F}]$ is the square of a norm by the Markov property, and pass to the limit.

By (a) the non-invariant parts decay in a time $O(a_K)$; this gives the independent gauges and $Q_t = \Pi Q_t \Pi$. Since 1 and the $e^{-t\lambda}$ span a dense algebra of $C([0,\infty])$, the spectral measures converge; $\frac{1}{t}(1-e^{-t\lambda})\uparrow\lambda$ gives $\mathcal{E}_\star(F) = \int\lambda\,\mathrm{d}\nu_F$, lower semicontinuity and closedness, and $(F,(1-Q_t)F) = \frac{1}{2}\mathbb{E}(F(Y_t)-F(Y_0))^2$ the Markov property of the form. $\|Q_t F\|^2 \le (F, Q_{2t}F)$ is Cauchy–Schwarz at the cutoff; recovery sequences are $\Lambda(\Lambda-\mathcal{L}_{K_j})^{-1}G_j$ with $\Lambda\to\infty$ slowly. In the equivalences, weak convergence with convergence of the norms is strong convergence, and $\|P_t^K\Psi_K\|^2 = (\Psi_K, P_{2t}^K\Psi_K)$; the strong convergence of $P_t^{K_j}\Psi_{K_j}$ gives (M1) through the quadratic forms $\langle H,(1-P_t^{K_j})H\rangle \ge 2\langle H,(1-P_t^{K_j})G_j\rangle - \langle G_j,(1-P_t^{K_j})G_j\rangle$, and the non-invariant parts are handled by (a).

For the last statement, the form $\mathbb{E}[\overline{\theta\mathsf{F}}\,\mathsf{G}]$ is hermitian and positive; $\mathcal{K}$ is its completion, $JF = [F(Y_0)]$, R_s the shift of the times by s, self-adjoint because θ reverses the shift and contractive by repeated Cauchy–Schwarz; $(F, Q_t F') = \langle JF, R_t JF'\rangle$, and Pythagoras gives the identity.

(c) The weight defined before Prop. YM3_I:115.2 of the chain with top depth n vanishes unless all $\pi_{K,d}U$, $n\le d\le K$, lie in their cuts, and by the comparison of the chain with the bare density (Prop. YM3_I:115.2, domination and lost-mass bounds) its integral against $e^{-\beta A_K}dU$ falls short of Z_K by at most $\mathsf{R}_n Z_K$; so off a μ^K-mass $\le \mathsf{R}_n$ all $\pi_{K,d}U$ lie in the cuts, and $\varpi(\pi_{K,d}U) < \theta_d$; for μ by total variation at finitely many depths (Cor. 2.8) and a monotone limit. Four statements among the construction results on the averaging map $\mathcal{M}$ enter: its gauge covariance $\mathcal{M}(U^u) = \mathcal{M}(U)^{u|}$, $u|$ the restriction of a gauge transformation u to the sites of the coarser lattice (Prop. YM3_I:4.5, YM3_I:(40)); $\mathcal{M}$ is Borel, and on the small fields, where the plaquette variables of the double block of a coarse bond lie within $(200L^2)^{-1}$ of $\mathbf{1}$, it is given at that bond by the first of the three formulas of its definition and is real-analytic (Prop. YM3_I:4.5); its linear part at $\mathbf{1}$ (Lem. YM3_I:4.6, YM3_I:(42)) is the straight average along the bonds plus the difference of the averages along the contours at the two ends of the coarse bond, which is the coarse gradient of a site function; and the bound for rearranged paths (Lem. YM3_I:4.1), which gives on the triple block of a coarse bond c, the union of the blocks of c_-, of c_+ and of the end point of the next bond c^+ in the direction of c, a gauge in which every bond variable is e^{iA_b} with $A_b\in\mathfrak{g}$, $|A_b|\le 15L^2\epsilon$, if all plaquette variables there lie within $\epsilon\le(200L^2)^{-1}$ of $\mathbf{1}$.

With them the averaging map acts on gradients of invariant functions as follows: if $\varpi(\mathcal{M}^j U) \le \epsilon'_j \le (48000L^4)^{-1}$ for $j<k$, then $\mathcal{M}^k$ is real-analytic near U and

$$\Gamma_{d+k}(\Psi\circ\mathcal{M}^k)(U) \le \prod_{j<k}(1+C_D\epsilon'_j)^2\,\Gamma_d(\Psi)(\mathcal{M}^k U), \qquad C_D = C_D(L),$$

for invariant $\Psi\in C^1(\Omega_d)$: the transpose of the differential of $\mathcal{M}$ maps horizontal vectors to horizontal vectors with norm at most $L^{-1/2}(1+C_D\epsilon)$, because the linear average is the straight average along the bonds, of norm $L^{-1/2}$ by Cauchy–Schwarz, plus the coarse gradient of the contour averages, which is a gauge direction of the block field and is orthogonal to the horizontal vectors, and the rest has norm at most $37L^2C_L\epsilon$, by the gauge covariance in the gauge of the triple block and the mean value inequality on the ball $\{|A_b|\le(3200L^2)^{-1}\}$, C_L the largest norm there of the differentials of $A\mapsto\mathcal{M}(e^{iA})(c)$ and of the rest, so that $C_D = 37L^{5/2}C_L$ depends on L alone; and $a_{d+k}^{-1}L^{-k} = a_d^{-1}$.

The cutoffs: $|\partial(\varpi_p^2)|\le 4\varpi_p$ by the computation of Lem. 16.2(a) on Ω_d, so each factor of χ_d has a gradient of norm at most $4\varpi_p/(2\theta_d)^2 \le 4/\theta_d$, which vanishes unless $2\theta_d\le\varpi_p\le 4\theta_d$, and there are $3(\ell L^d)^3$ plaquettes; this gives $\Gamma_d(\chi_d)^{1/2}\le \mathsf{k}_d\mathbf{1}\{\varpi\ge\theta_d\}$. F_K is smooth: by induction from the depth K down, the closure of the set where $\prod_{e<e'\le K}\chi_{e'}\circ\pi_{K,e'}\neq 0$ lies in $\{\varpi(\pi_{K,e'}U)\le 4\theta_{e'}$ for $e<e'\le K\}$, where $4\theta_{e'}\le(48000L^4)^{-1}$ as $e'>e\ge n\ge n_1$, so that $\pi_{K,e}$ is real-analytic near it; on that open neighbourhood the product with $\chi_e\circ\pi_{K,e}$ is smooth, it is 0 on the open complement of the closure, and the two open sets cover Ω_K; the same holds for the factor $\Phi\circ\pi_{K,n}$.

This is the role of the cutoffs: they keep every block field that enters F_K where $\mathcal{M}$ is given by its real-analytic formula. By the product rule and the averaging bound at $\epsilon'_j = 4\theta_{K-j}$, with $\prod_j(1+C_D\epsilon'_j)\le \mathsf{c}_* = \exp(4C_D\sum_{e\ge n_1}\theta_e)$,

$$\Gamma_K(F_K)^{1/2} \le \mathsf{c}_*\Big[\sup\Gamma_n(\Phi)^{1/2} + \sup|\Phi|\sum_{n\le d\le K}\mathsf{k}_d\,\mathbf{1}\{\varpi(\pi_{K,d}U)\ge\theta_d\}\Big],$$

and the event in the sum has μ^K-mass $\le\max\{\mathsf{R}_d,\mathsf{R}_{d-1}\}$, by the first statement of (c) at the depth d, or at the depth $d-1$ if $d=K$. Minkowski's inequality and the decay of R_d faster than every exponential, against $\mathsf{k}_d = O(L^{4d})$, bound every L^p norm uniformly in K, and $\|\Gamma_K(\chi_n^K)^{1/2}\|_{L^{2p}(\mu^K)}\le \mathsf{c}_*\sum_{d\ge n}\mathsf{k}_d\max\{\mathsf{R}_d,\mathsf{R}_{d-1}\}^{1/2p}\to 0$ as $n\to\infty$. Density: a bounded Borel invariant function of a small depth is μ-almost surely a Borel function of a large depth, it is approximated in $L^2(\mu_d)$ by invariant polynomials, and the cutoffs change it by $\le\mathsf{R}_d^{1/2}$.

Then $\mathcal{E}_\star(F)\le\liminf_K\mathcal{E}_K(F_K)$, $\|Q_tF-F\|^2\le 2(F,(1-Q_t)F)\le 2t\sup_K\mathcal{E}_K(F_K)$, and $\nu_F(\{\infty\}) = \|F\|^2-\lim_{t\to0}(F,Q_tF)$. Each property of (Y_t) and of $\mathcal{E}_\star$ stated in (b) and (c) is shared by the frozen process, which keeps Y_0 up to gauge transformations of the threads drawn from the Haar measure independently at different times, so that $Q_t=\Pi$ for $t>0$ and $\mathcal{E}_\star=0$ on L^2_{inv}: motion is the content of (A1). □

Proposition 15.2 (The limiting process on the common core)**.** Assume (E); let $\mathcal{C}$ and F_K be as in Lem. 15.1(c), and $T>0$.

(a) (Paths.) For $F\in\mathcal{C}$ and $1\le p<\infty$, $\mathbb{E}|F_K(U_t)-F_K(U_s)|^{2p}\le C_p(F)|t-s|^p$ uniformly in K; the processes $(F_K(U_t))_{t\le T}$ are tight in $C[0,T]$, and along every limit $F(Y_t)$ has a continuous modification, locally Hölder continuous of every exponent $<\frac12$. For $d\ge n_1$ there is $\mathsf{p}_d(T)$, which does not depend on K and tends to 0 faster than every exponential in d, such that for all $K>d$, with probability at least $1-\mathsf{p}_d(T)$, $\varpi(\pi_{K,e}U_t)<2\theta_e$ for all $d\le e\le K$ and all $t\in[0,T]$.

(b) (Energies.) For $F\in\mathcal{C}$ the limit $\mathcal{E}_\infty(F)=\lim_K\mathcal{E}_K(F_K)$ exists along the full sequence of cutoffs, and $\mathcal{E}_\infty(F)=\int\Gamma_\infty(F)\,\mathrm{d}\mu$ with

$$\Gamma_\infty(F)(Y) = \lim_{d\to\infty}a_d^{-1}\,\mathsf{t}[\tfrac23,\tfrac16;Y_d](\partial F^{[d]}(Y_d)) \quad \mu\text{-almost surely and in } L^1(\mu),$$

where $F^{[d]}$ is the lattice form of F at depth d and

$$\mathsf{t}[\alpha,\beta;V](B)=\alpha\|B\|^2+2\beta\sum_{c}\langle B(c),\mathrm{Ad}(V(c))B(c^+)\rangle,$$

c^+ the bond that follows c in its direction. $\mathcal{E}_\infty(F)$ depends only on the class of F in $L^2(\mu)$, $\mathcal{E}_\star\le\mathcal{E}_\infty$ on $\mathcal{C}$ for every limit, and for $F=\Phi(Y_d)\prod_{e\ge d}\chi_e(Y_e)$ with $\mathsf{R}_d<1$

$$\mathcal{E}_\infty(F)\ge(\tfrac13-\tfrac12\bar\eta_d)(1-\mathsf{R}_d)\,e^{-\mathsf{o}_d}\,a_d^{-1}|\mathsf{I}_d|^{-1}\int_{\mathsf{I}_d}|\partial\Phi|^2dV,\qquad\bar\eta_d\to0.$$

(c) (The laws of the block fields.) For an integer $d\ge1$ with $L^d\mathsf{b}\ge\mathsf{t}_\natural$ and every bounded Borel invariant $f\ge0$ on Ω_d,

$$(1-\mathsf{R}_d)\,e^{-\mathsf{o}_d}|\mathsf{I}_d|^{-1}\int_{\mathsf{I}_d}f\,\mathrm{d}V\le\int f(Y_d)\,\mathbf{1}\{\varpi(Y_e)<\theta_e\text{ for all }e\ge d\}\,\mathrm{d}\mu\le\int f\,\mathrm{d}\mu_d$$
$$\le e^{\mathsf{o}_d}|\mathsf{I}_d|^{-1}\int_{\mathsf{I}_d}f\,\mathrm{d}V+\mathsf{R}_d\sup f,$$

and the same at every cutoff $K>d$, with μ_d^K in the place of μ_d and with the middle member $\int f(\pi_{K,d}U)\,\mathbf{1}\{\varpi(\pi_{K,e}U)<\theta_e\text{ for }d\le e\le K\}\,\mathrm{d}\mu^K$. So an invariant Haar-null set has μ_d- and μ_d^K-mass $\le\mathsf{R}_d$; and if $\mathsf{R}_d<1$, every non-empty open subset of I_d has positive μ_d-mass and μ_d^K-mass bounded below uniformly in K.

(d) (Square field and martingales.) Let (K_j) be a limit and $F\in\mathcal{C}$. There is $\Gamma_\star(F)\in L^2_{\mathrm{inv}}$ with $(2t)^{-1}\mathbb{E}[\phi(Y_0)(F(Y_t)-F(Y_0))^2]\to\int\phi\,\Gamma_\star(F)\,\mathrm{d}\mu$ as $t\to0$ for bounded invariant ϕ; $0\le\Gamma_\star(F)\le\Gamma_\infty(F)$, $\int\Gamma_\star(F)\,\mathrm{d}\mu=\mathcal{E}_\star(F)$, and $\mathcal{E}_\star(\psi(F))=\int\psi'(F)^2\Gamma_\star(F)\,\mathrm{d}\mu$ for $\psi\in C^2(\mathbb{R})$. Along a further subsequence the lattice martingales converge, and $F(Y_t)=F(Y_0)+M_t^F+D_t^F$, with a continuous martingale M^F with $\langle M^F\rangle_t=2\int_0^t\Gamma_\infty(F)(Y_s)\,\mathrm{d}s$, and a continuous process D^F with stationary increments, $\mathbb{E}|D_t^F|^2=2[t\,\mathcal{E}_\infty(F)-(F,(1-Q_t)F)]$, $\mathbb{E}[D_t^F(F(Y_t)-F(Y_0))]=0$, and $\lim_{t\to0}t^{-1}\mathbb{E}|D_t^F|^2=2[\mathcal{E}_\infty(F)-\mathcal{E}_\star(F)]$; in reversed time, $F(Y_t)-F(Y_0)=\frac12M_t^F-\frac12(\hat M_T^F-\hat M_{T-t}^F)$ with a martingale $\hat M^F$ of the law of M^F.

The following are equivalent: $\mathcal{E}_\star(F)=\mathcal{E}_\infty(F)$; $\Gamma_\star(F)=\Gamma_\infty(F)$ μ-almost everywhere; D^F has zero energy; $\lim_{\Lambda\to\infty}\limsup_j\int_{(\Lambda,\infty)}\lambda\,\mathrm{d}\nu_F^{K_j}=0$, ν_F^K the spectral measure of F_K. And F is frozen ($Q_tF=F$ for all t) if and only if $D^F=-M^F$.

(e) (Motion from fields with a divergence.) A vector field with a divergence at the cutoff K is a $\zeta\in L^2(\mu^K;\mathfrak{g}^{\mathrm{bonds}})$ with $\delta_K\zeta\in L^2(\mu^K)$ such that $a_K^{-1}\int\langle\zeta,\partial H\rangle\,\mathrm{d}\mu^K=-\int H\,\delta_K\zeta\,\mathrm{d}\mu^K$ for $H\in C^\infty(\Omega_K)$. If $a_{K_j}^{-1}\int|\zeta_j|^2\,\mathrm{d}\mu^{K_j}\le\mathsf{A}^2$, $\|\delta_{K_j}\zeta_j\|\le\mathsf{D}$ and $\delta_{K_j}\zeta_j\to g$ weakly, then for $F\in L^2(\mu)$ and $t>0$

$$\mathcal{E}_\star(F)\ge\frac{(F,g)^2}{\mathsf{A}^2},\qquad(F,(1-Q_t)F)\ge\frac{t\,(F,g)^2}{\mathsf{A}^2+t\,\mathsf{D}^2};$$

conversely, if $\mathcal{E}_\star(F)>0$, such fields with $(F,g)\ne0$ exist. A field that does not depend on the configuration pairs to zero with every invariant gradient: $\int\langle\partial F,z\rangle\,\mathrm{d}\mu^K=0$ for constant z and invariant F.

(f) (An example.) In a limit of reversible diffusions observed through a block variable whose law converges in total variation, the inequality $\mathcal{E}_\star\le\mathcal{E}_\infty$ can be strict: for the reversible diffusion on $\mathbb{R}^2$ with the weight $e^{-\frac{\kappa}{2}(x-w)^2-\frac12w^2}$, the form $\int(|\partial_xF|^2+s^2|\partial_wF|^2)$ with fixed $s>0$ and the block variable x, as $\kappa\to\infty$ the law of x converges in total variation, the square field of x is 1, the limit is an Ornstein–Uhlenbeck process, Markov and moving, and $\mathcal{E}_\star(x)=s^2/(1+s^2)<1=\mathcal{E}_\infty(x)$.

■

Proof sketch. (a) At every cutoff, for $H \in C^\infty(\Omega_K)$ and $0 \le t \le T$, $H(U_t) - H(U_0) = \frac{1}{2}M_t - \frac{1}{2}(\hat{M}_T - \hat{M}_{T-t})$ with forward and backward martingales of brackets $2\int \Gamma_K(H)$ (the drift drops out, and the square field alone enters); the martingale moment inequalities and stationarity give $\mathbb{E}|H(U_t) - H(U_s)|^{2p} \le c_p 2^p |t-s|^p \int \Gamma_K(H)^p d\mu^K$, and $\mathbb{P}(\sup_{t\le T}|H(U_t) - H(U_0)| \ge \lambda) \le 10T\lambda^{-2}\mathcal{E}_K(H)$. With the bounds of Lem. 15.1(c) this is uniform in K, and the criterion of Kolmogorov and Chentsov gives tightness and Hölder continuity. For the small fields apply the maximal bound to $H = \prod_{d\le e\le K} \chi'_e \circ \pi_{K,e}$, built from a second family of cutoffs, $\chi'_e = \prod_p \varphi_{\mathrm{cut}}(\varpi_p^2/\theta_e^2)$ on Ω_e with the function φ_{cut} of Lem. 15.1(c): χ'_e is 1 where $\varpi \le \theta_e$ and 0 where $\varpi \ge 2\theta_e$, and, as there, each factor has a gradient of norm at most $4\varpi_p/\theta_e^2 \le 8/\theta_e$, so that $\Gamma_e(\chi'_e)^{1/2} \le \mathsf{k}_e \mathbf{1}\{\varpi \ge \theta_e\}$ with the same numbers $\mathsf{k}_e = 24L^{e/2}\theta_e^{-1}(\ell L^e)^3$, which thus bound $\Gamma_e^{1/2}$ of both families.

As in the proof of Lem. 15.1(c), the energy of H is at most $\mathsf{c}_*^2\Sigma_d^2$, $\Sigma_d = \sum_{e\ge d} \mathsf{k}_e \max\{\mathsf{R}_e, \mathsf{R}_{e-1}\}^{1/2}$, with $\mathsf{c}_* = \exp(4C_D \sum_{e\ge n_1} \theta_e)$; $H(U_0) = 1$ off a mass $\le \mathsf{R}_d$, and $H(U_t) > 0$ forces $\varpi(\pi_{K,e}U_t) < 2\theta_e$ for $d \le e \le K$; so $\mathsf{p}_d(T) = \mathsf{R}_d + 10T\mathsf{c}_*^2\Sigma_d^2$.

(b) On the small fields $F_K = F^{[d]} \circ \pi_{K,d}$, $\partial F_K = \mathcal{D}^{(k)}(U)^* \partial F^{[d]}$ with $\mathcal{D}^{(k)}$ the differential of $\mathcal{M}^k$, and the transpose of the differential carries the quadratic forms along: $L^k \mathsf{t}[\alpha,\beta;U](\mathcal{D}^{(k)*}B) = \mathsf{t}[\mathsf{A}_L^k(\alpha,\beta);V](B) + O(\sum_j \epsilon'_j)\|B\|^2$ for horizontal B at $V = \mathcal{M}^k U$, where A_L is the linear map of the flat average, with the eigenvector $(\frac{2}{3}, \frac{1}{6})$ of eigenvalue 1 (the Gram matrix of the kernels that are a tent along the bond and a box across it) and $(\frac{1}{3}, -\frac{1}{6})$ of eigenvalue L^{-2}.

So $\Gamma_K(F_K)$ is close in $L^1(\mu^K)$, uniformly in K, to $a_d^{-1}\mathsf{t}[\frac{2}{3}, \frac{1}{6}; \cdot](\partial F^{[d]}) \circ \pi_{K,d}$, and total variation gives the limit along the full sequence; the same estimate between two depths along a thread gives Γ_∞. A function of $\mathcal{C}$ that vanishes μ-almost everywhere has a gradient that vanishes off an invariant Haar-null set, whose mass is controlled by (c); this gives the dependence on the class. $\mathcal{E}_\star \le \mathcal{E}_\infty$ is the lower semicontinuity of Lem. 15.1(b). The lower bound: on the threads with $\varpi(Y_e) < 2\theta_e$ for all $e \ge d$, where all cutoffs χ_e equal 1 near Y_e and $F^{[e]} = \Phi \circ \mathcal{M}^{e-d}$, the same estimate between the depths d and $e \to \infty$ gives $|\Gamma_\infty(F)(Y) - a_d^{-1}\mathsf{t}[\frac{2}{3}, \frac{1}{6}; Y_d](\partial\Phi(Y_d))| \le \frac{1}{2}\bar{\eta}_d a_d^{-1}|\partial\Phi(Y_d)|^2$, with $\bar{\eta}_d = 4C_{\mathrm{T}}\mathsf{c}_*^2 \sum_{e>d}\theta_e$, $C_{\mathrm{T}} = C_{\mathrm{T}}(L)$ the constant of the term $O(\sum_j \epsilon'_j)$ above and c_* that of the proof of Lem. 15.1(c), so $\Gamma_\infty(F)(Y) \ge (\frac{1}{3} - \frac{1}{2}\bar{\eta}_d)a_d^{-1}|\partial\Phi(Y_d)|^2$ as $\mathsf{t}[\frac{2}{3}, \frac{1}{6}; V] \ge \frac{1}{3}\|\cdot\|^2$; elsewhere $\Gamma_\infty(F) \ge 0$.

The event of the middle member of (c) lies in that set of threads, and (c) at the invariant function $f = |\partial\Phi|^2$ gives the bound.

(c) At the top depth d the chain from level 0 has the density $e^{\Sigma + \Phi_t}$ with respect to Haar measure on I_d, Σ a constant (Prop. YM3_I:115.2, image identity, Prop. YM3_I:114.2), its non-classical part bounded by $C_{\mathrm{nl}}\frac{L}{L-1}E_0(\ell L^d)^3$ (equation YM3_I:(2180)) and the classical part between 0 and $\frac{3}{2}c_\rho^2 A_0^2(\log L^d \mathsf{b})^{2\mathfrak{p}_0}(\ell L^d)^3$ (by YM3_I:(2161) at the weight x_t); its mass lies in $[1 - \mathsf{R}_d, 1]$; the comparison of the chain with the bare density (Prop. YM3_I:115.2, domination and lost-mass bounds) gives the sandwich for μ_d^K, and total variation (Cor. 2.8) for μ_d.

The middle member: the weight of the chain with top depth d vanishes unless $\pi_{K,e}U$ lies in its cut, so in $\{\varpi < \theta_e\}$, for $d \le e \le K$; the function $f(\pi_{K,d}U)\mathbf{1}\{\varpi(\pi_{K,e}U) < \theta_e \text{ for } d \le e \le K\}$ is invariant, by the gauge covariance of $\mathcal{M}$; so, σ^K being at most μ^K on invariant functions (Lem. 15.3(i)), $\int f\,\mathrm{d}\sigma_d^K = \int f(\pi_{K,d}U)\mathbf{1}\{\cdots\}\,\mathrm{d}\sigma^K \le \int f(\pi_{K,d}U)\mathbf{1}\{\cdots\}\,\mathrm{d}\mu^K$, and $\int f\,\mathrm{d}\sigma_d^K \ge (1 - \mathsf{R}_d)e^{-\mathsf{o}_d}|\mathsf{I}_d|^{-1}\int_{\mathsf{I}_d} f\,\mathrm{d}V$ by the lower bound of the density of σ_d^K (Lem. 15.3(iii)).

For μ: at a depth $D > d$ the event $\{\varpi(Y_e) < \theta_e \text{ for } d \le e \le D\}$ is an event of the block field of depth D, it contains the event at the cutoff $K > D$, and total variation at the depth D carries the bound to μ; then $D \to \infty$, a monotone limit. Positivity: for a non-empty open $O \subset \mathsf{I}_d$ and a smaller non-empty open O' with closure in O, the gauge orbit of O' is open, invariant and of positive Haar measure, and it is covered by finitely many gauge translates of O, which all have the mass of O.

(d) For bounded invariant ϕ, the forward–backward decomposition gives $(2t)^{-1}\mathbb{E}[\phi(U_0)(F_K(U_t) - F_K(U_0))^2] \le \int \phi\,\Gamma_K(F_K) + c\,\langle\phi, (1 - P_t)\phi\rangle^{1/2}$ for $\phi \ge 0$; the limit in K and then in t gives

$\Gamma_\star \le \Gamma_\infty$, and, with $\mathsf{e}_t(\phi;F) = (2t)^{-1}\mathbb{E}[\phi(Y_0)(F(Y_t)-F(Y_0))^2]$, the identity $2t\,\mathsf{e}_t(\phi;F) = 2\langle \phi F, (1-Q_t)F\rangle - \langle \phi, (1-Q_t)F^2\rangle$ gives the existence. The chain rule follows from fourth moments of the increments, which are $O(t^2)$. The lattice processes $F_K(U_t)$, their martingales $M^{K,F}$ and brackets are tight in $C[0,T]$, limits of L^4-bounded martingales are martingales, and the brackets converge to $2\int \Gamma_\infty(F)(Y_s)\,\mathrm{d}s$ by (b).

At the cutoff, with the spectral measure ν of F_K, $\mathbb{E}|\int_0^t \mathcal{L}_K F_K(U_s)\,\mathrm{d}s|^2 = 2\int (t\lambda - 1 + e^{-t\lambda})\,\mathrm{d}\nu = 2[t\,\mathcal{E}_K(F_K) - (F_K,(1-P_t^K)F_K)]$ and the drift is orthogonal to the increment; these pass to the limit. The equivalences: $\mathcal{E}_\infty(F) - \mathcal{E}_\star(F) = \lim_\Lambda \lim_j \int(\lambda-\Lambda)_+\,\mathrm{d}\nu_F^{K_j}$, the energy that escapes to infinite frequency, is the energy of the drift and the integral of $\Gamma_\infty - \Gamma_\star$.

(e) Let $H_j \to F$ strongly with $\mathcal{E}_{K_j}(H_j) \to \mathcal{E}_\star(F)$; then $|\langle H_j, \delta_{K_j}\zeta_j\rangle| \le \mathsf{A}\,\mathcal{E}_{K_j}(H_j)^{1/2}$, and the left side tends to $|(F,g)|$. At a positive time, $1-e^{-x} \ge \frac{x}{1+x}$ for $x \ge 0$ (as $e^x \ge 1+x$) and $\int \frac{t\lambda}{1+t\lambda}\,\mathrm{d}\nu = \min_G\{\|H-G\|^2 + t\,\mathcal{E}_K(G)\}$, which is bounded below, for every real s, by $2s\langle H, \delta_K\zeta\rangle - s^2(\|\delta_K\zeta\|^2 + t^{-1}a_K^{-1}\int|\zeta|^2)$. Conversely, $\zeta_j = \partial(1-\mathcal{L}_{K_j})^{-1}H_j$ has $\delta_{K_j}\zeta_j = (1-\mathcal{L}_{K_j})^{-1}H_j - H_j \to (1+A_\star)^{-1}F - F$ by (A3), and $(F,(1+A_\star)^{-1}F - F) = -\int \frac{\lambda}{1+\lambda} d\nu_F \ne 0$. For constant z, $\int \partial_b F\,\mathrm{d}\mu^K$ is invariant under $\mathrm{Ad}(g)$ for all g, hence 0.

(f) The operator of the diffusion on linear functions has the trace $\kappa(1+s^2)+s^2$ and the determinant κs^2; its eigenvalues are $\lambda_+ \sim \kappa(1+s^2)$ and $\lambda_- \to s^2/(1+s^2)$, and the spectral measure of x puts a mass asymptotic to $\frac{1}{1+s^2}\lambda_+^{-1}$ at λ_+, so $\int \lambda\,\mathrm{d}\nu_x^\kappa = 1$ for all κ while $\mathcal{E}_\star(x) = s^2/(1+s^2)$. □

15.1.6. *Conditional fluctuation laws and transport estimates.*

Lemma 15.3 (Conditional laws along the renormalisation chain)**.** Assume (E), let $n \ge 1$ be an integer with $L^n\mathsf{b} \ge \mathsf{t}_\natural$, and let $K > n$. (i) $1 - \mathsf{R}_n \le \sigma^K(\Omega_K) \le 1$, and $\int f\,\mathrm{d}\sigma^K \le \int f\,\mathrm{d}\mu^K \le \int f\,\mathrm{d}\sigma^K + \mathsf{R}_n \sup f$ for bounded Borel invariant $f \ge 0$. (ii) For $0 \le k < t$, $d = K-k$, and every $W \in \mathsf{Cut}_{k+1}$, p_W is a probability measure with a positive continuous density on the box; σ_{d-1}^K is carried by Cut_{k+1}, and

$$\int \Big[\int f(V_W(B))\,p_W(\,\mathrm{d}B)\Big]\sigma_{d-1}^K(\,\mathrm{d}W) = \int f\,\mathrm{d}\sigma_d^K \qquad \text{for every Borel } f \ge 0 \text{ on } \Omega_d;$$

so the images of the p_W under the V_W form a regular conditional law of V_k given $V_{k+1} = W$ under σ^K, and under σ^K the sequence $V_t, \dots, V_0$ is a Markov chain from coarse to fine whose kernel at every $W \in \mathsf{Cut}_{k+1}$ is the image under V_W of p_W; the kernels carry bounded continuous functions on Cut_k to continuous functions on Cut_{k+1}, that is, $W \mapsto \int h(V_W(B))\,p_W(\,\mathrm{d}B)$ is continuous on Cut_{k+1} for every bounded continuous h on Cut_k; kernels and towers do not depend on the top level; $h_k(V_k)$ is the conditional mean of H given V_k; the tower of an invariant H is invariant at every point of Cut_k, with $\inf H \le h_k \le \sup H$; and a density that is a function of the first member V_t changes no kernel and no tower.

(iii) $\sigma_n^K = \mathsf{g}\,\mathrm{d}V$ on I_n, $\sup \mathsf{g} \le e^{\mathsf{o}_n} \inf \mathsf{g}$, and $(1-\mathsf{R}_n)e^{-\mathsf{o}_n} \le \mathsf{g}\,|\mathsf{I}_n| \le e^{\mathsf{o}_n}$ at every point. (iv) For $n \ge n_2$, $0 \le k < t$, $W \in \mathsf{Cut}_{k+1}$: $p_W = e^{-\Psi_W}dB$ with Ψ_W real-analytic and $\frac14\gamma_{\mathrm{r}} \le D^2\Psi_W \le \Lambda_+$; every tangent vector X of the fibre $\{\mathcal{M}V = W\}$ at $V_W(B)$ has $\|X\|^2 \le C_{\mathrm{ch}}\sum|X(b)|^2$, the sum over the bonds that are not pivot bonds; and for bounded $g \in C^1(\mathsf{Cut}_k)$

$$\mathrm{Var}_{p_W}(g\circ V_W),\quad \tfrac12\mathrm{Ent}_{p_W}\big((g\circ V_W)^2\big) \;\le\; C_{\mathrm{P}}\frac{a_d^2}{\mathsf{b}}\int \Gamma_d(g)(V_W(B))\,p_W(\,\mathrm{d}B),$$

with $C_{\mathrm{P}} = 4C_{\mathrm{ch}}/\gamma_{\mathrm{r}}$, which depends only on the constants of the construction, and not on K, k, W, ℓ or b. ■

Proof sketch. (i) is Prop. YM3_I:115.2, domination and lost-mass bounds, the comparison of the small-field chain with the bare density, with the weight $\mathfrak{w}_t^{\mathrm{sf}}$ and the bound R_n of the definitions preceding Prop. YM3_I:115.2; (ii): the weight defined before Prop. YM3_I:115.2 vanishes unless every V_j lies in its cut, so σ_{d-1}^K is carried by Cut_{k+1}. Positivity at every point, by induction over

the scales: with Σ_k the sum of the constants of the steps $1,\dots,k$ at the source 0, $e^{\Sigma_0}F_0 = F_0$ is the bare density on Cut_0 (the definition preceding Prop. YM3_I:114.2), and if $e^{\Sigma_k}F_k > 0$ on Cut_k, then $e^{\Sigma_{k+1}}F_{k+1} = \mathsf{I}[e^{\Sigma_k}F_k] > 0$ on Cut_{k+1} YM3_I:(2174), because the prefactors of the chart integral are positive, G_W and J_W are real ([7, Lem. B.3]) and the chart carries the box into Cut_k (Lem. YM3_I:113.10); so the density of p_W, a positive multiple of $e^{-x_k G_W + \mathsf{J}_W}(e^{\Sigma_k}F_k)\circ V_W$, is positive at every point of the box, and it is continuous, as $(W,B)\mapsto V_W(B), G_W(B), \mathsf{J}_W(B)$ are continuous and the chart integral of a continuous function is continuous ([7, Lem. B.3]), which makes every F_k continuous on Cut_k by the same induction.

The kernels carry bounded continuous functions to continuous functions, in the whole range $L^n\mathsf{b} \geq \mathsf{t}_\natural$: at a bounded continuous h on Cut_k the kernel at W is $\mathsf{I}[hF_k](W)/\mathsf{I}[F_k](W)$, the normalised chart integral [7, eq. 191] of the integrand of the chain; Cut_{k+1} lies in the set O^{im} (Lem. YM3_I:113.10), where the chart, its gauge-fixing form and the exponent of its Jacobians are continuous in (W,B) ([7, Lem. B.3]), and F_k is continuous on Cut_k, into which the chart carries the box; so the chart integrals of the non-negative functions $(\sup|h| \pm h)F_k$ integrate over the fixed box functions that are continuous in (W,B), Fatou's lemma on the box makes them lower semicontinuous in W, and as their sum $2\sup|h|\,\mathsf{I}[F_k] = 2\sup|h|\,e^{\mathsf{c}_{k+1}(0)}F_{k+1}$ is continuous by the induction above, both are continuous; the denominator $\mathsf{I}[F_k]$ is continuous and positive.

The identity: the factors of the weight of the levels $> k$ are functions of V_{k+1}; the measure of the chain of the level k is $\nu_k = e^{\Sigma_k}\mathbf{1}_{\mathsf{Cut}_k}F_k\,\mathrm{d}V$ the measure recursion in Prop. YM3_I:114.2, YM3_I:(2175); and by the identity of measures of the step ([7, Lem. B.3]) the image under $\mathcal{M}$ of $\mathfrak{G}_k[x_k]\chi^{\mathrm{sf}}_{x_k}\,h\,\nu_k$ has the density $e^{\Sigma_k}\mathsf{I}[h\mathbf{1}_{\mathsf{Cut}_k}F_k]$ on the open set O^{im}, which contains $\mathcal{W}_{k+1}(x_k;\alpha_0,\alpha_1)\supset \mathsf{Cut}_{k+1}$ (Lem. YM3_I:113.10), for every bounded Borel h on Ω_d, and on Cut_{k+1} this density is $e^{\Sigma_k}\mathsf{I}[hF_k]$, the chart carrying the box into Cut_k; dividing by $e^{\Sigma_k}\mathsf{I}[F_k]$ gives the identity for bounded Borel $f\geq 0$, and monotone convergence for every Borel $f\geq 0$.

As $V_{k+2},\dots,V_t$ are functions of V_{k+1}, the identity at $f = h\cdot(g\circ\mathcal{M})$ is the Markov property with the kernel p_W. The tower of an invariant H is invariant at every point by the covariance of the chain under the gauge transformations that are constant on blocks; (iii) follows from the statements on the iteration from a level: by Prop. YM3_I:115.2, image identity and Prop. YM3_I:114.2, at the top the density is $e^{\Sigma+\Phi_t}$ on I_n, Σ a constant, with a non-classical part bounded by $C_{\mathrm{nl}}\frac{L}{L-1}E_0(\ell L^n)^3$ (equation YM3_I:(2180)) and a classical part between 0 and $\frac{3}{2}c_\rho^2 A_0^2(\log L^n\mathsf{b})^{2\mathfrak{p}_0}(\ell L^n)^3$ (by YM3_I:(2161) at the weight x_t), so it oscillates by at most o_n, and its mass is that of (i).

(iv): at every $W\in\mathsf{Cut}_{k+1}$, which lies in the class $\mathcal{W}_{k+1}(x_k;\alpha_0,\alpha_1)$ of the sum rule (Lemma YM3_I:113.10), the sum identities in equations YM3_I:(828)–YM3_I:(830) and their preceding Jacobian calculation and the step (Prop. 6.32) give the density of p_W the form $e^{\mathcal{E}(P_W;B)}$, normalised, of Prop. 15.4(b). This form holds at every depth, at every W of the class of the sum rule, where the restrictions of P_W to the polymers lie in the analyticity cores and the sum rules hold, so that every term of $\mathcal{E}(P_W;\cdot)$ is formed and holomorphic in B on $\{|B(b)|<C_A\mathfrak{b}(x_k)\}$ (Thm. YM3_I:W, Prop. YM3_I:51.4 and its proof, the potential definitions in Section YM3_I:51, and §15.3.1); the conditions of n_3 enter Prop. 15.4 only through the ball of complex triples of (a) and through the response.

So the form holds for every integer $n\geq 1$ with $L^n\mathsf{b}\geq\mathsf{t}_\natural$, from n_2 on in particular, as Prop. 15.4(b) states. The density being positive by (ii), $e^{\mathcal{E}(P_W;\cdot)}$ is a constant multiple of a positive function, so the imaginary part of $\mathcal{E}(P_W;\cdot)$ is constant on the connected box, and $\Psi_W = -\mathcal{E}(P_W;\cdot) + \log\int e^{\mathcal{E}(P_W;B')}\,\mathrm{d}B'$, with the branch of the logarithm that makes it real, is real-analytic near the closed box. Its Hessian is $D^2\Psi_W = \mathsf{A}_W - D^2\sum_Y v_Y$, with the form $\mathsf{A}_W = \sum_Y \mathsf{b}_Y(P_W) + \sum_\square \hat{\mathsf{G}}_\square(P_W)$ of the Gaussian integral of one step at P_W, whose inverse is the whole covariance of the step at P_W.

It lies within $\frac{1}{2}\mathsf{u}_{\mathrm{G}}(LM)^{-3}\leq\frac{1}{2}\gamma_{\mathrm{r}}$, in the ℓ^2 operator norm, of a real reference form whose floor and ceiling are γ_{r} and c_{r} (the completeness (St4) of Def. YM3_I:47.2, Thm. YM3_I:W, the proof of Prop. YM3_I:113.4), so $\frac{1}{2}\gamma_{\mathrm{r}}\leq\mathsf{A}_W\leq c_{\mathrm{r}}+\frac{1}{2}\gamma_{\mathrm{r}}$. The potentials are bounded by a constant times

$\delta(x_k)\mathfrak{b}(x_k)^2 e^{-\lambda d(Y)}$ on the whole product of the cores and the polydisc of radius $C_A\mathfrak{b}(x_k)$, so at P_W for every W of the class of the sum rule (Thm. YM3_I:W, Prop. YM3_I:51.4 and its proof, as in the proof of Prop. 15.4(d)), and by Cauchy's estimate their Hessians are at most $C_{\mathrm{Hs}}\delta(x_k) \le \frac{1}{4}\gamma_{\mathrm{r}}$ for $n \ge n_2$; hence $\frac{1}{4}\gamma_{\mathrm{r}} \le D^2\Psi_W \le c_{\mathrm{r}} + \frac{3}{4}\gamma_{\mathrm{r}} \le \Lambda_+$.

The log-Sobolev inequality $\mathrm{Ent}_p(f^2) \le \frac{2}{\kappa}\int |\nabla f|^2 \,\mathrm{d}p$ of a probability measure $p = e^{-\Psi}dx$ on an open convex set with $D^2\Psi \ge \kappa$, at $\kappa = \frac{1}{4}\gamma_{\mathrm{r}}$ on the convex box, gives $\mathrm{Ent}_{p_W}(F^2) \le \frac{8}{\gamma_{\mathrm{r}}}\int |\nabla_B F|^2 \,\mathrm{d}p_W$, and its linearisation the Poincaré inequality with $\frac{4}{\gamma_{\mathrm{r}}}$; since $x_k^{-1} = a_d/\mathfrak{b}$, the chart gives $|\nabla_B(g\circ V_W)|^2 \le C_{\mathrm{ch}}x_k^{-1}|\partial g|^2 = C_{\mathrm{ch}}\frac{a_d^2}{\mathfrak{b}}\Gamma_d(g)\circ V_W$. □

Proposition 15.4 (Log-concavity of the fluctuation law)**.** Assume (E). For $0 \le k < K$, lengths at the level $k+1$ are measured in its units, in which the bare spacing is $\eta_{k+1} = L^{-(k+1)}$; $J^{(k+1)}(U) = \eta_{k+1}^{-3}\partial A_K(U) = L^{-3(K-k-1)}J(U)$ is the current (YM3_I:9.2) formed with this spacing; for a 1-form A and a configuration U, $(\nabla_\mu^U A_\nu)(x) = \eta_{k+1}^{-1}[\mathrm{Ad}(U(\langle x, x+\eta_{k+1}e_\mu\rangle))A_\nu(x+\eta_{k+1}e_\mu) - A_\nu(x)]$, and $|\nabla_U A|(b) = \max_\mu |(\nabla_\mu^U A_\nu)(x)|$ at the bond $b = \langle x, x+\eta_{k+1}e_\nu\rangle$. π_k^M is the partition of the torus into the cubes of side M in the units of level k (the partition π_k of Section YM3_I:72 (the definitions of $\mathcal{D}_j$ and d_j at its beginning)); $\mathcal{D}_k$ is the set of its localisation domains, the unions of cubes of π_k^M that are connected through their faces, which we call polymers; $d(Y) = d_k(Y)$ is the linear size YM3_I:72 (the definitions of $\mathcal{D}_j$ and d_j at its beginning) of a polymer Y, the least number of edges of a tree of edges of its cubes that meets all of them; $\mathrm{Bd}(Y)$ is the set of the bare bonds with an end point in Y; and for a function f of triples and a pair $(a, \mathbf{j})$ of complex 1-forms, $D_{(a,\mathbf{j})}f(P_W) = \frac{d}{dz}f(U_+, za, \mathbf{J}_+ + z\mathbf{j})|_{z=0}$.

There are $r_{\mathrm{c}} \in (0, \alpha_0/256]$, determined by the fixed radii $\alpha_0 < \alpha_1$ of the notation and the constants of Prop. YM3_I:51.10, and $C_{\mathrm{mx}} < \infty$, depending only on the constants of the construction, such that for $K > n \ge n_3$, $0 \le k < K - n$ ($k = 0$ included), $W \in \mathsf{Cut}_{k+1}$, any background U_+ over W, $\mathbf{J}_+ = J^{(k+1)}(U_+)$ and $P_W = (U_+, 0, \mathbf{J}_+)$:

(a) for every polymer $Y \in \mathcal{D}_k$ and complex 1-forms $A', \mathbf{j}$ on $\mathrm{Bd}(Y)$ with $|A'|, |\nabla_{U_+}A'|, |\mathbf{j}| < r_{\mathrm{c}}$ there, the triple $(U_+, A', \mathbf{J}_+ + \mathbf{j})$ on $\mathrm{Bd}(Y)$ lies in the analyticity core $\mathcal{U}_{k+1}^\circ(Y; \alpha_0, \alpha_1)$ (Def. YM3_I:49.1, Prop. YM3_I:51.10, first assertion);

(b) $p_W \propto e^{\mathcal{E}(P_W;B)}dB$, where

$$\mathcal{E}(P;B) = \sum_{Y\in\mathcal{D}_k}\left[-\tfrac{1}{2}(B, \mathsf{b}_Y(P)B) + v_Y(P;B)\right] - \tfrac{1}{2}\sum_{\square\in\pi_k^M}(B, \hat{\mathsf{G}}_\square(P)B),$$

$\mathsf{b}_Y = x_k D_B^2\Psi^{\mathrm{A}}(Y; P, 0)$ with Ψ^{A} of equations YM3_I:(815) and YM3_I:(816), v_Y the potentials of Thm. YM3_I:W and of the fourth family at zero source (Prop. 6.32), $\hat{\mathsf{G}}_\square$ of §15.3.1; the quadratic form, too, is a sum of local terms; and $(A', \mathbf{j}, B) \mapsto \mathcal{E}$ is holomorphic on the product of the ball of (a) with $\{|B(b)| < C_A\mathfrak{b}(x_k)\}$, $C_A \ge 3$, which contains the closed box. The formula for p_W, the locality of its quadratic form and the holomorphy of $B \mapsto \mathcal{E}(P_W; B)$ on $\{|B(b)| < C_A\mathfrak{b}(x_k)\}$ hold as well for every integer $n \ge 1$ with $L^n\mathfrak{b} \ge \mathfrak{t}_\natural$, so from n_2 on, with $K > n$, $0 \le k < K - n$, at every $W \in \mathsf{Cut}_{k+1}$ and any background U_+ over W: they need of W only that it lies in the class $\mathcal{W}_{k+1}(x_k; \alpha_0, \alpha_1)$ of the sum rule (Lem. YM3_I:113.10), at which the restrictions of P_W to the polymers lie in the analyticity cores;

(c) for W at a depth $\ge n_{\mathrm{a}}$, with $\mathcal{H}^+$ the response of U_+ at level $k+1$ (§5.3), $\mathbf{U}_\xi = e^{iL^{-k-1}\mathcal{H}^+(\xi)}U_+$, $W'_\xi = \mathcal{M}^{k+1}\mathbf{U}_\xi$ and $P_\xi = (U_+, \mathcal{H}^+(\xi), J^{(k+1)}(\mathbf{U}_\xi))$, there is $r_0 > 0$, which may depend on W and K, such that for real ξ with $\sup|\xi| < r_0$: $W'_\xi \in \mathsf{Cut}_{k+1}$ is a gauge transform of $e^{i\xi}W$, and $\mathbf{U}_\xi$ lies in the minimal orbits over W'_ξ and over $\mathcal{M}^k\mathbf{U}_\xi$; there is a chart $\check{V}_\xi(B)$ of the fibre over W'_ξ, real-analytic in (ξ, B) near the closed box, $\check{V}_0 = V_W$, with values in a compact invariant

subset of Cut_k; for every bounded Borel invariant H, $h_{k+1}(e^{i\xi}W) = \int h_k(\check V_\xi(B))\,\check p_\xi(\mathrm dB)$, where $\check p_\xi$ has the density $e^{\mathcal E(P_\xi;B)}/\int e^{\mathcal E(P_\xi;B')}\,\mathrm dB'$, real-analytic in ξ uniformly on the closed box, and $\check p_0 = p_W$; the frame, $\check p_\xi$ and $\psi_\xi = \frac{d}{ds}\mathcal E(P_{s\xi};\cdot)|_{s=0}$ do not depend on the choice of the axial element;

(d) for real ζ on the box and every pair $(a,\mathbf j)$ of complex bare 1-forms, with $\mu_y(a,\mathbf j)$ the largest of $|a|$, $|\nabla_{U_+}a|$, $|\mathbf j|$ at the bare bonds with an end point in the block of $y \in \Lambda_{K-k-1,\ell}$,

$$|\partial_\zeta D_{(a,\mathbf j)}\mathcal E(P_W;B)| \le C^\sharp_{\mathrm{mx}}(x_k)\,\mathfrak b(x_k)\,\|\zeta\|\Big(\sum_y \mu_y(a,\mathbf j)^2\Big)^{1/2},$$

with $C^\sharp_{\mathrm{mx}}$ depending on x_k only through $\delta(x_k) \le 1$; and for every real ξ on $\Lambda_{K-k-1,\ell}$, $\sup_B |\partial_\zeta\partial_s\mathcal E(P_{s\xi};B)|_{s=0}| \le C_{\mathrm{mx}}\mathfrak b(x_k)\|\zeta\|\|\xi\|$, $C_{\mathrm{mx}} = \mathsf B\sup_{\mathfrak t\ge\mathfrak t_\natural} C^\sharp_{\mathrm{mx}}(\mathfrak t)$.

■

Proof sketch. (a) Prop. YM3_I:52.3 holds for free radii $b_0, b_1 \in (0,1]$, with the threshold $\varepsilon_{\mathcal W}b_0$ for $\varpi(W)$; the threshold that enters n_3 is the one at the radii $b_0 = \frac12\alpha_0$, $b_1 = \frac12\alpha_1$, that is, $2\theta_d \le \varepsilon_{\mathcal W}\alpha_0$ for $d \ge n_3$. So every $W \in \mathsf{Cut}_{k+1}$ with $K-k-1 \ge n \ge n_3$ has $\varpi(W) < \theta_{K-k-1} \le \frac12\varepsilon_{\mathcal W}\alpha_0$, and by the plain-core verification in its proof P_W lies in the core $\mathcal U^\circ_{k+1}(Y;\frac12\alpha_0,\frac12\alpha_1)$ with half the radii. The outer substitution from half the radii (Prop. YM3_I:51.10 (first assertion) at the radii α_0,α_1) carries this triple, moved by $(A',\mathbf j)$ with $|A'|, |\nabla_{U_+}A'|, |\mathbf j| < r_{\mathrm c}$, into the core with the radii α_0,α_1: the halved radii leave the room $r_{\mathrm c}$.

(b) The density in the chart comes from the sum identities in equations YM3_I:(828)–YM3_I:(830) and their preceding Jacobian calculation and the step (Prop. 6.32); the quadratic form is a sum of local terms, obtained by subtracting polynomials that agree on an open set. The formula for p_W, the locality of its quadratic form and the holomorphy of $B \mapsto \mathcal E(P_W;B)$ on $\{|B(b)| < C_A\mathfrak b(x_k)\}$ need of W only that it lies in the class $\mathcal W_{k+1}(x_k;\alpha_0,\alpha_1)$ of the sum rule (Lem. YM3_I:113.10), where the restrictions of P_W to the polymers lie in the analyticity cores and the sum rules hold, so that every term of $\mathcal E(P_W;\cdot)$ is formed and holomorphic in B there (Thm. YM3_I:W, Prop. YM3_I:51.4 and its proof); so they hold for every integer $n \ge 1$ with $L^n\mathfrak b \ge \mathfrak t_\natural$, from n_2 on in particular.

Only the joint holomorphy in $(A',\mathbf j)$ on the ball of (a) uses (a), and with it the condition $2\theta_d \le \varepsilon_{\mathcal W}\alpha_0$ of n_3. (c) Moving W along the response keeps the background in the minimal orbit, so no control of the axial gauge is needed. With the residual gauge h_ξ such that the background over W'_ξ is $\mathbf U_\xi^{h_\xi}$, $\check V_\xi(B)$ is $V_{W'_\xi}(\mathrm{Ad}(h_\xi)B)$ carried back by h_ξ; its values at the pivots follow from the contraction there (the contraction leading to YM3_I:(800), with the inverse of the one-step pivot derivative bounded by $2L^2$ in YM3_I:(798)) and the implicit function theorem. Every term of $\mathcal E$ is invariant under $(P,B) \mapsto (P^u, \mathrm{Ad}(u)B)$ and depends on $P = (U,A',\mathbf J)$ only through $(e^{iL^{-k-1}A'}U,\mathbf J)$ (Thm. YM3_I:W, Def. YM3_I:52.1, Prop. YM3_I:51.4 and its proof, §15.3.1, and the pair and gauge covariance of Prop. 6.32), and $\mathrm{Ad}(h_\xi)$ keeps dB and the box: so the density depends on P_ξ alone, and no derivative of h_ξ enters.

The values lie in a compact subset because the regularity number of the backgrounds stays strictly below the cut. (d) Each local term is bounded on the whole product of the core and the polydisc (Prop. YM3_I:51.4 and its proof, Thm. YM3_I:W), for instance $|v_Y| \le 32C_V\delta(x_k)(d(Y)+1)C_A^2\mathfrak b(x_k)^2e^{-\lambda d(Y)}$ with constants C_V, λ of the step; Cauchy's estimate on a bidisc, the Cauchy–Schwarz inequality and the sum over polymers [7, eq. 320] give the first bound. At $s=0$, the derivative of $P_{s\xi}$ near y is at most $\mathsf B\,m_y(\xi)$ by Lem. 15.9, which also gives $\sum_y m_y(\xi)^2 \le \mathsf S^2\|\xi\|^2$. □

Proposition 15.5 (One-step transport of the Dirichlet energy)**.** For a level j and a $\mathfrak g$-valued function w of the bare bonds let $[w]_j^2 = \sum_y (L^j \max\{|w(b)| : b_- \in B^j(y)\})^2$, $B^j(y)$ the cube of L^{3j} bare sites centred at $y \in \Lambda_{K-j,\ell}$; then $L^{j/2}[w]_j \le L^{(j+1)/2}[w]_{j+1}$, $\|w\|^2 \le 3L^j[w]_j^2$, $[w]_0 \le \|w\|$, and $[\mathrm{Ad}(g)w]_j = [w]_j$. For $V \in \mathsf{Cut}_j$ at a depth $\ge n_{\mathrm a}$, U in the minimal orbit over V and a real 1-form

X on $\Lambda_{K-j,\ell}$, let $N_V^\flat(X) = L^{j/2}\min_\lambda[L^{-j}D\mathcal{H}_U(0)X - \mathrm{d}_U\lambda]_j$ for $j \geq 1$, $N_V^\flat(X) = \min_\lambda[X - \mathrm{d}_V\lambda]_0$ for $j = 0$, and $\Gamma_{K-j}^\flat(g)(V) = L^K \sup_{X:\, N_V^\flat(X)>0}\langle X, \partial g(V)\rangle^2/N_V^\flat(X)^2$.

$N_V^\flat$ is a seminorm that does not depend on the choice of U and vanishes exactly on the gauge directions; $N_V^\flat(X) \leq L^{j/2}\mathsf{M}\|X\|$, so $\Gamma_d \leq \mathsf{M}^2\Gamma_d^\flat$; $\Gamma_K^\flat(H) \leq 3\Gamma_K(H)$ for invariant H; and $\Gamma_d^\flat(g)$ is Borel for invariant $g \in C^1$. There are n_{a} as in the notation and $C_\flat$, depending only on L, C_Q, M and the constants of §5.3, such that, with $C_{\mathrm{cov}} = \frac{4}{\gamma_{\mathrm{r}}}C_{\mathrm{ch}}^{1/2}\mathsf{M}\sqrt{6}C_Q L^{-1/2}C_{\mathrm{mx}}$, for $K > n \geq n_{\mathrm{a}}$, $0 \leq k < K - n$, $W \in \mathsf{Cut}_{k+1}$:

(a) every real ξ' on $\Lambda_{K-k-1,\ell}$ has a representative ξ modulo gauge at W, depending on W, U_+, ξ' and not on B, with $\|\xi\| \leq \sqrt{6}C_Q L^{-(k+1)/2}N_W^\flat(\xi')$ and $N_{V_W(B)}^\flat(\check{\mathsf{d}}_k(W,B)\xi) \leq (1 + C_\flat\delta(x_k))N_W^\flat(\xi')$ for all B in the box, $\check{\mathsf{d}}_k(W,B)\xi = \frac{d}{ds}\check{V}_{s\xi}(B)|_{s=0}$; more precisely, for every bare velocity w of backgrounds at U_+ of level $k+1$ and $\xi = D\mathcal{M}^{k+1}(U_+)w$, $N_{V_W(B)}^\flat(\check{\mathsf{d}}_k(W,B)\xi) \leq (1 + C_\flat\delta(x_k))L^{(k+1)/2}[w]_{k+1}$;

(b) for real invariant $H \in C^1(\Omega_K)$ the tower has $h_k \in C^1(\mathsf{Cut}_k)$, and for $W \in \mathsf{Cut}_{k+1}$ and real ξ, with ψ_ξ of Prop. 15.4(c),

$$\langle \xi, \partial h_{k+1}(W)\rangle = \int \langle \check{\mathsf{d}}_k(W,B)\xi, \partial h_k(V_W(B))\rangle\, p_W(\,\mathrm{d}B) + \mathrm{Cov}_{p_W}(h_k \circ V_W, \psi_\xi),$$

$$\Gamma_{d-1}^\flat(h_{k+1})(W)^{1/2} \leq (1 + C_\flat\delta(x_k))\int \Gamma_d^\flat(h_k)^{1/2} \circ V_W\,\mathrm{d}p_W + C_{\mathrm{cov}}\,\delta(x_k)\Big(\int \Gamma_d^\flat(h_k) \circ V_W\,\mathrm{d}p_W\Big)^{1/2},$$

so $\Gamma_{d-1}^\flat(h_{k+1})(W) \leq (1 + \epsilon_d^\flat)\int \Gamma_d^\flat(h_k) \circ V_W\,\mathrm{d}p_W$ at every W; the same holds for $\phi_k = f_k^{1/2}$, (f_k) the tower of $G^2 + \epsilon$, G real invariant, $\epsilon > 0$;

(c) for $0 \leq k \leq K - n$,

$$\int \Gamma_{K-k}(h_k)\,\mathrm{d}\sigma_{K-k}^K \leq 3\mathsf{M}^2\Pi_n^\flat \int \Gamma_K(H)\,\mathrm{d}\sigma^K, \tag{15.ii}$$

and, pointwise, $\Gamma_e(g_{t'})(W) \leq 3\mathsf{M}^2\Pi_e^\flat\int \Gamma_K(G)\,\mathrm{d}\mathsf{P}_W$ for the law P_W of (B3) and every $W \in \mathsf{Cut}_{t'}$.

■

Proof sketch. (a) Three facts: the monotonicity of $L^{j/2}[w]_j$ in j, with constant one; $\|D\mathcal{M}^j(U)w\| \leq \sqrt{6}C_Q[w]_j$ when the plaquettes of U lie within $a_Q L^{-2j}$ of $\mathbf{1}$ (equation eq. 5.xxx); and

$$[L^{-j}D\mathcal{H}_V(\zeta)X]_j \leq \mathsf{BS}\|X\|$$

(§5.3). The key is that the gauge compensator is bounded pointwise: every bare velocity $w = L^{-j}D\mathcal{H}_U(0)X - \mathrm{d}_U\lambda$ is also $L^{-j}D\mathcal{H}_U(0)X^w + \mathrm{d}_U\nu$, $X^w = D\mathcal{M}^j(U)w$, with

$$|\nu(x)| \leq (k_6 + \tfrac{3}{2})\,\mathsf{B}\,m_y^j(X^w) + \tfrac{3}{2}L^j \max_{b_- \in B^j(y)}|w(b)| \qquad (x \in B^j(y)),$$

at the block centres by equation YM3_I:(996), with $k_6 = 4\kappa_{\mathrm{av}}$ as fixed in §5.3, and inside a block along a path, by $|\lambda(x)| \leq |\lambda(y)| + \frac{3}{2}L^j\max_{B^j(y)}|\mathrm{d}_U\lambda|$. Lift the velocity of level $k+1$ to a bare velocity w; the fluctuation B moves the background of level k by $O(\delta(x_k))$, so there is a curve of backgrounds of level k over $\check{V}_{s\xi}(B)$, up to a gauge, whose bare velocity differs from w by at most $C_\flat\delta(x_k)L^{1/2}[w]_{k+1}$ in $[\cdot]_k$; the monotonicity then costs nothing. The minimum in $N_W^\flat(\xi')$ is attained at some w; $\xi = X^w$ is the representative, and $\|\xi\| \leq \sqrt{6}C_Q[w]_{k+1}$.

The argument of (a) is deterministic. The properties of $N_V^\flat$: a curve of backgrounds whose block field moves with velocity X has a bare velocity w with $L^{j/2}[w]_j \geq N_V^\flat(X)$; $\Gamma_K^\flat \leq 3\Gamma_K$ because $\langle X, \partial H\rangle^2 = \langle X - \mathrm{d}_V\lambda, \partial H\rangle^2 \leq \|X - \mathrm{d}_V\lambda\|^2|\partial H|^2 \leq 3[X - \mathrm{d}_V\lambda]_0^2|\partial H|^2$ for invariant H and every λ; upper semicontinuity of $(V, X) \mapsto N_V^\flat(X)$ gives the Borel property.

(b) Differentiate Prop. 15.4(c) at $\xi = 0$; domination holds on the closed box, as the moving chart takes its values in a compact subset of Cut_k. The left side depends on ξ only modulo gauge: take the representative of (a); the first term is then at most $L^{-K/2}(1 + C_\flat\delta(x_k))N_W^\flat(\xi')\int\Gamma_d^\flat(h_k)^{1/2}\,\mathrm{d}p_W$. For the plain tower, the Poincaré inequality of p_W used twice gives $|\mathrm{Cov}_p(f,\psi)| \le \frac{4}{\gamma_\mathrm{r}}\|\nabla_B f\|_{L^2(p)}\sup|\nabla_B\psi|$. For the square roots, the log-Sobolev inequality of p_W with $\kappa = \frac14\gamma_\mathrm{r}$ gives $\log\int e^{s(\psi-\int\psi)}\,\mathrm{d}p \le s^2l^2/2\kappa$ when $|\nabla_B\psi| \le l$, and the variational formula of the entropy at ϕ^2 and $s(\psi - \int\psi)$, optimised in s, gives

$$|\mathrm{Cov}_p(\phi^2,\psi)| \le \frac{2l}{\kappa}\,\|\phi\|_{L^2(p)}\|\nabla_B\phi\|_{L^2(p)},$$

with equality for a Gaussian p, $\phi = e^{ax/2}$, $\psi = x$. By Prop. 15.4(d), $\sup|\nabla_B\psi_\xi| \le C_\mathrm{mx}\mathfrak{b}(x_k)\|\xi\|$; the chart factor $x_k^{-1/2}$ turns $\mathfrak{b}(x_k)$ into $\delta(x_k)$, and $L^{K/2}L^{-d/2}L^{-(k+1)/2} = L^{-1/2}$. Directions with $N_W^\flat(\xi') = 0$ contribute zero by gauge invariance; for the other directions divide by $L^{-K/2}N_W^\flat(\xi')$ (for the square roots by $2\phi_{k+1}(W) = 2\|\phi_k\circ V_W\|_{L^2(p_W)}$) and use Jensen's inequality for the probability p_W.

(c) Integrate the one-step bound against σ_{d-1}^K (Lem. 15.3(ii)) and iterate, using $\Gamma_d \le \mathsf{M}^2\Gamma_d^\flat$ at the top and $\Gamma_K^\flat \le 3\Gamma_K$ at the bottom; for the pointwise form iterate at the fixed block field along the kernels. The constant does not depend on K because $\sum_d\delta(L^d\mathfrak{b}) < \infty$. □

Proposition 15.6 (Transport in the ℓ^2 metric)**.** Assume (E). For $V \in \mathsf{Cut}_j$ with $\varpi(V)$ below the radius of §5.3, U in the minimal orbit over V and $\mathsf{Z} = L^{-j}D\mathcal{H}_U(0)$, let $N_V(\xi) = \min_\lambda\|\mathsf{Z}\xi - \mathrm{d}_U\lambda\|$ ($N_V(\xi) = \min_\lambda\|\xi - \mathrm{d}_V\lambda\|$ at $j = 0$). N_V is a seminorm that does not depend on the choice of U and vanishes exactly on the gauge directions; $N_V(\xi)^2 \le \|\mathsf{Z}\xi\|^2 \le C_\mathrm{rs}L^j\|\xi\|^2$, with C_rs depending only on L and the constants of §5.3. Every real ξ is $\xi_* + \mathrm{d}_V\phi$ with exactly one horizontal datum ξ_*, that is, with $\mathrm{d}_U^\mathsf{T}\mathsf{Z}\xi_* = 0$; on horizontal data $N_V(\xi_*) = \|\mathsf{Z}\xi_*\|$; every real ξ has $\|\xi\| \le C_\mathcal{Q}L^{-j/2}\|\mathsf{Z}\xi\|$, so $\|\xi_*\| \le C_\mathcal{Q}L^{-j/2}N_V(\xi_*)$, with $C_\mathcal{Q} = 34(L^3+2)$ as in Lem. 5.32 (a number other than the C_Q and C_Q' of Lem. 5.30); and the two metrics are equivalent: $3^{-1/2}N_V \le N_V^\flat \le \mathsf{M}C_\mathcal{Q}N_V$.

There are $C_\sharp$ and a depth $n_4 \ge n_3$, depending only on b and the constants of the construction, such that for $K - k > n_4$, $W \in \mathsf{Cut}_{k+1}$ and U_+ a background over W the following hold. All clauses of Prop. 15.4(c) hold at W, with the frame $\check V_\xi(B)$ real-analytic in (ξ, B) on $\{\sup|\xi| < r_0\}$ times the open box and with first derivatives in ξ that are bounded there. For every horizontal datum ξ at W, that is, every real ξ on $\Lambda_{K-k-1,\ell}$ with $\mathrm{d}_{U_+}^\mathsf{T}\mathsf{Z}_{k+1,U_+}\xi = 0$, $\mathsf{Z}_{k+1,U_+} = L^{-k-1}D\mathcal{H}^+(0)$, and all B in the box, $s \mapsto \check V_{s\xi}(B)$ is defined and real-analytic near 0 and

$$N_{V_W(B)}(\check{\mathsf{d}}_k(W,B)\xi) \le (1 + C_\sharp\,\delta(L^{K-k}\mathsf{b}))N_W(\xi).$$

Consequently Prop. 15.5(b), (c) hold as well in the metric N for $K > n \ge n_4$, (B3) for $K > e \ge n \ge n_4$, and (B2) in (a)–(c) for $n \ge \max\{n_4, n_\mathrm{c}\}$, in the proposed variant (d), still requiring the multiscale gauge estimate, for $n \ge \max\{n_4, n_\mathrm{e}\}$ with $\mathsf{R}_n < 1$, and in (e) for $n \ge n_4$, with $C_\mathrm{rs}\Pi_n$, $\Pi_n = \prod_{d>n}(1+\epsilon_d)$, $1 + \epsilon_d = [1 + (C_\sharp + \frac{4}{\gamma_\mathrm{r}}(C_\mathrm{ch}C_\mathrm{rs})^{1/2}C_\mathcal{Q}L^{-1/2}C_\mathrm{mx})\delta(L^d\mathfrak{b})]^2$, in the place of $3\mathsf{M}^2\Pi_n^\flat$ (in (B3): $C_\mathrm{rs}\Pi_e$ in the place of $3\mathsf{M}^2\Pi_e^\flat$). ■

Proof sketch. The range of Z. The response is in the Landau gauge on the kernel of the site averages: by equation YM3_I:(837), $\mathsf{R}^\mathrm{La}D^\mathsf{T}\mathcal{H} = 0$, $D = \mathrm{d}_U$, R^La the orthogonal projection onto $D^\mathsf{T}D\mathcal{N}$ and $\mathcal{N}$ the site functions whose site averages Q_q' vanish at the tops (§15.3); so $\mathsf{R}^\mathrm{La}\mathrm{d}_U^\mathsf{T}\mathsf{Z}\xi = 0$ for every ξ. The response is covariant, as the averaging map is, so Z carries the gauge directions $\mathrm{d}_V\psi$ of the block field to gauge directions at U; these meet $\mathrm{d}_U\mathcal{N}$ only in 0 (a $\mathrm{d}_U\mu$, $\mu \in \mathcal{N}$, in the range of Z has $\mathsf{R}^\mathrm{La}\mathrm{d}_U^\mathsf{T}\mathrm{d}_U\mu = 0$ while $\mathrm{d}_U^\mathsf{T}\mathrm{d}_U\mu$ lies in the range of R^La, so $\mathrm{d}_U\mu = 0$), and by the count of dimensions the two together span all the gauge directions at U.

A horizontal velocity of backgrounds at U, $h = \mathsf{Z}X - \mathrm{d}_U\lambda$ with $\mathrm{d}_U^\mathsf{T} h = 0$, is then $\mathsf{Z}X' - \mathrm{d}_U\nu$ with $\nu \in \mathcal{N}$, and $\mathsf{R}^{\mathrm{La}}\mathrm{d}_U^\mathsf{T}\mathrm{d}_U\nu = \mathsf{R}^{\mathrm{La}}\mathrm{d}_U^\mathsf{T}\mathsf{Z}X' = 0$: so $\mathrm{d}_U\nu = 0$ and $h = \mathsf{Z}X'$. The range of Z holds every horizontal velocity of backgrounds, with no compensator.

The decomposition. The part h of $\mathsf{Z}\xi$ orthogonal to the range of d_U is a horizontal velocity of backgrounds, $h = \mathsf{Z}\xi_*$; $\mathsf{Z}(\xi - \xi_*)$ is a gauge direction at U, so $\xi - \xi_*$ is a gauge direction at V; ξ_* is unique, as Z is injective (below) and a horizontal vector that is a gauge direction vanishes.

The bound. Lem. 5.32 applies in the constant frame, with bare spacing L^{-j} and terminal spacing one. It gives $\mathcal{Q}(\mathcal{H}_U(\zeta)) = \zeta$, and hence $L^j K_0 \mathsf{Z}\xi = \xi$, together with $\|K_0\| \le 2L^{-3j/2} \le C_\mathcal{Q} L^{-3j/2}$ on counting-measure ℓ^2. Thus $\|\xi\| \le C_\mathcal{Q} L^{-j/2}\|\mathsf{Z}\xi\|$; on horizontal data $\|\mathsf{Z}\xi_*\| = N_V(\xi_*)$. The equivalence follows from $\|w\|^2 \le 3L^j[w]_j^2$ and $N_V^\flat(\xi) = N_V^\flat(\xi_*) \le L^{j/2}\mathsf{M}\|\xi_*\|$.

The estimate. U_+ is a background of the level k as well, and the same argument at the level k shows that the velocity of a horizontal datum ξ at W is a response one level finer, with no compensator: $\mathsf{Z}_{k+1,U_+}\xi = \mathsf{Z}_{k,U_+}\zeta$, $\zeta = D\mathcal{M}^k(U_+)\mathsf{Z}_{k+1,U_+}\xi$; the fluctuation moves the background of level k by $O(\delta(x_k))$, which costs the factor $1 + C_\sharp\delta$.

The frame from the depth n_4 on. As $n_4 \ge n_3$, θ_d lies below the radius of §5.3; U_+ is a reference of the levels $k+1$ and k; for small real ξ, W'_ξ is a gauge transform of $e^{i\xi}W$ in Cut_{k+1}, $\mathbf{U}_\xi$ lies in the minimal orbits over W'_ξ and over $\mathcal{M}^k\mathbf{U}_\xi$, and $\xi \mapsto \mathcal{M}^k\mathbf{U}_\xi$, $\xi \mapsto W'_\xi$ are real-analytic. The value of $\mathcal{M}(V)$ at a bond c of $\Lambda_{K-k-1,\ell}$ reads V on the bonds of the double block of c alone, of the pivot bonds the pivot bond of c alone, and its derivative in the value at that pivot bond is invertible, with an inverse bounded by $2L^2$, on the compact closure of the set of the restrictions of the $V_W(B)$, B in the box.

The implicit function theorem, with radii that are uniform over this compact set, gives about every base point $V_W(B)$ exactly one solution at each pivot near the value of the base point, real-analytic in the data, with bounded first derivatives. The frame $\check{V}_\xi(B)$ is $e^{ix_k^{-1/2}B(b)}(\mathcal{M}^k\mathbf{U}_\xi)(b)$ at the bonds that are not pivot bonds and this solution at the pivots; by the uniqueness at the pivots and continuation in B along τB, $\tau \in [0,1]$, from $B = 0$, it is $V_{W'_\xi}(\mathrm{Ad}(h_\xi)B)$ carried back by h_ξ. Hence, as in Prop. 15.4(c): the tower identity; the density $e^{\mathcal{E}(P_\xi;B)}/\int e^{\mathcal{E}(P_\xi;B')}\,\mathrm{d}B'$ of $\check{p}_\xi$, which reads the depth only through n_3; the values in a compact invariant subset of Cut_k; analyticity on small ξ times the open box, with ξ-derivatives bounded there, which carries the domination in the differentiation of Prop. 15.5(b); and at the depths $\ge n_\mathrm{a}$ the frame is that of Prop. 15.4(c).

In the differentiated step the representative is used only through $\|\xi\| \le c\,L^{-(k+1)/2}N_W(\xi)$, which the horizontal datum of the class satisfies with $c = C_\mathcal{Q}$; the rest of the proof of Prop. 15.5 is unchanged, with the square field Γ^N built from N as $\Gamma^\flat$ is built from $N^\flat$, $N_V(X) \le C_\mathrm{rs}^{1/2}L^{j/2}\|X\|$ at the top and $\Gamma_K^N \le \Gamma_K$ at the bottom. □

Lemma 15.7 (Connected regions at the coarse scale)**.**

(a) Let $\mathsf{n} \ge \lambda_{\mathrm{BG},3}$, $0 < \kappa' \le \kappa_0$, $\mathsf{n} \ge \kappa'^{-1/2}$, and V a configuration on $(\mathbb{Z}/\mathsf{n})^3$ with $\varpi(V) \le \kappa'$. Some gauge has $|V^u(b) - \mathbf{1}| \le C_\mathrm{BG}\sqrt{\kappa'}$ at every bond; $\{\varpi < \kappa'\}$ lies in the path component of $\mathbf{1}$ in $\{\varpi < C''\kappa'\}$; so, at $\kappa' = r/C''$, for $0 < r \le C''\kappa_0$ and $\mathsf{n} \ge \max\{\lambda_{\mathrm{BG},3}, (C''/r)^{1/2}\}$, every other component of $\{\varpi < r\}$ lies in the shell $\{r/C'' \le \varpi < r\}$.

The estimate involving J block levels remains to be proved, as described under (Multiscale gauge). The proposed estimate requires finite constants; no numerical values for them are asserted here.

(b) For a finite set E of bonds, $0 < \alpha < \pi$, $\mathsf{K} \subset \{A \in \mathfrak{g}^E : |A(b)| < \alpha\}$ open and convex and $V_0 \in G^E$, $A \mapsto e^{iA}V_0$ maps K diffeomorphically onto its image B; there Haar measure is $\prod_b (2\pi^2)^{-1}\mathsf{j}(A(b))\,\mathrm{d}A(b)$, $\mathsf{j}(X) = (\sin|X|/|X|)^2$; $-\log\prod_b \mathsf{j}$ has Hessian $\ge \frac{2}{3}$; and $\mathrm{Ent}_{dV|\mathsf{B}}(\psi^2) \le 3\int_\mathsf{B} |\partial\psi|^2 dV$ for bounded locally Lipschitz ψ, whatever the number of bonds.

(c) If $\nu|_{S_i}$ have log-Sobolev constants c_i on open S_1, S_2 with $\nu(S_1 \cap S_2) > 0$, then $\nu|_{S_1\cup S_2}$ has $2\max_i c_i(1 + 16\nu(S_1 \cup S_2)/\nu(S_1 \cap S_2))$. If Haar measure on an open S' has the constant c,

$\rho(V) = \int \mathbf{1}_{S'}(V^g)\,\mathrm{d}g$, $\rho_0 > 0$ and $w = \min\{1, \rho/\rho_0\}$, then $\mathrm{Ent}_{w\mathsf{g}\,\mathrm{d}V}(\phi^2) \le \frac{c\,\mathsf{g}_+}{\rho_0 \mathsf{g}_-} \int |\partial\phi|^2 w\,\mathsf{g}\,\mathrm{d}V$ for invariant ϕ and Borel $0 < \mathsf{g}_- \le \mathsf{g} \le \mathsf{g}_+$, whatever the boundary of S'. For an open invariant $\mathcal{O}$, a component C of it and a compact invariant $\mathcal{K} \subset \mathsf{C}$ there is a connected finite union S' of sets $\{e^{iA}V_i : |A(b)| < \delta_i\}$ with $\mathcal{K} \subset S' \subset \mathsf{C}$.

(d) The set $\mathcal{N} \subset (0,2)$ of the $r = \varpi(V)$ at which $\sum_p c_p \partial a_p(V) = 0$ for some $c \ge 0$, $c \ne 0$, carried by the plaquettes with $\varpi_p(V) = r$, is Lebesgue-null; for $r \notin \mathcal{N}$, $\{\varpi < r\}$ has finitely many components, each a connected open set with a Lipschitz boundary. The abelian fields whose $(1,2)$ plaquettes all equal $e^{if\sigma_3}$, $f = 2\pi m/\mathsf{n}^2$, and whose other plaquettes are $\mathbf{1}$, are such critical points: $2\sin(\pi m/\mathsf{n}^2) \in \mathcal{N}$ for integers $0 < m < \mathsf{n}^2/2$.

(e) For $0 < r < 2$, every connected component of $\{\varpi < r\}$ is open, invariant, and contains a local minimiser of ϖ. For $\phi_1 \in C^2[0,c)$, $c = \frac{1}{2}r^2$, with $\phi_1' > 0$ and $\phi_1 \to \infty$ at c, the function $\Phi = \sum_p \phi_1(a_p)$ has compact sublevel sets in $\{\varpi < r\}$, every component contains a point where Φ restricted to it is least, and there $\sum_p \phi_1'(a_p)\,\partial a_p = 0$ with weights $\phi_1'(a_p) > 0$ at all plaquettes and $\nabla^2\Phi \ge 0$; $\{\varpi < r\}$ is connected if and only if every local minimiser of Φ lies in the component of $\mathbf{1}$; and $\{\varpi < r\}$ is the increasing union of the sets $\{\sum_p a_p^q < c^q\}$, $q \to \infty$.

(f) Let $0 < \alpha \le \frac{1}{32}$, $r' = 64\alpha^2$, N the number of sites along an axis, and $\mathscr{D} = \{A : |A(b)| < \alpha\ \forall b,\ |Y_p(A)| < r'\ \forall p\}$, $e^{iY_p(A)} = \partial(e^{iA})(p)$, $|Y_p| \le \pi$. Then $\mathscr{D}$ is star-shaped, $\mathscr{D} = \{s\omega : |\omega| = 1,\ 0 \le s < \varrho(\omega)\}$ with $\frac{1}{2}r' \le \varrho \le \rho_{\max} = \alpha\sqrt{3N^3}$ and $|\varrho(\omega) - \varrho(\omega')| \le \frac{4\pi\rho_{\max}^2}{r'}|\omega - \omega'|$; $A \mapsto e^{iA}$ maps $\mathscr{D}$ diffeomorphically onto $\exp\mathscr{D} = \{V : \mathsf{d}(V(b), \mathbf{1}) < \alpha,\ \mathsf{d}(\partial V(p), \mathbf{1}) < r'\}$, and $\mathrm{Ent}_{\mathbf{1}_{\exp\mathscr{D}}dV}(\phi^2) \le C_{\mathscr{D}} \int_{\exp\mathscr{D}} |\partial\phi|^2 dV$ for bounded locally Lipschitz ϕ, with $C_{\mathscr{D}}$ of (B1)(e) at $\alpha_n = \alpha$.

With $w(V) = \int \mathbf{1}\{\mathsf{d}(V^u(b), \mathbf{1}) < \alpha\ \forall b\}\,\mathrm{d}u$ and $\mathsf{In} = \{V : \mathsf{d}(\partial V(p), \mathbf{1}) < r'\ \forall p\}$: $\int_{\mathsf{In}} f\,w\,\mathrm{d}V = \int_{\exp\mathscr{D}} f\,\mathrm{d}V$ for invariant Borel $f \ge 0$; $w(V) \ge [(\frac{1}{2}\alpha - \sin\frac{1}{2}\alpha)/2\pi]^{N^3}$ if some gauge brings all bonds of V within $\frac{1}{4}\alpha$ of $\mathbf{1}$; and $\mathsf{In} \cap \{w > 0\}$ is path connected.

■

Proof sketch. (a) The gauge is built cell by cell on a periodic grid of mesh $\kappa'^{-1/2}$, along the proof of Prop. 15.8; along the ray to 0 in exponential coordinates, $||\partial e^{itA}(p) - \mathbf{1}| - |t\,\mathrm{d}A(p)|| \le 10\alpha^2$, with $dA(p) = A(b_1) + A(b_2) - A(b_3) - A(b_4)$ the plain sum of the bond function A around p, its bonds $b_1, \dots, b_4$ taken with the signs $(+,+,-,-)$ of Lem. 16.2, when $|A| \le \alpha \le \frac{1}{8}$, which keeps the ray in $\{\varpi < C''\kappa'\}$. The multilevel estimate (Multiscale gauge) is a separate task; this construction concerns one scale.

For U(1) the flux sectors are distinct components. (b) The radial function $h(\rho) = -2\log(\sin\rho/\rho)$ on $\mathbb{R}^3$ has

$$h''(\rho) = 2\sum_{m\ge1}\left[(m\pi-\rho)^{-2} + (m\pi+\rho)^{-2}\right] \ge 4\sum_{m\ge1}(m\pi)^{-2} = \tfrac{2}{3}, \qquad \frac{h'(\rho)}{\rho} = 4\sum_{m\ge1}(m^2\pi^2-\rho^2)^{-1} \ge \tfrac{2}{3};$$

$|\nabla(\psi\circ\exp)| \le |\partial\psi|$ since $\frac{d}{ds}e^{i(X+sY)} = iZe^{iX}$ with $Z = \int_0^1 \mathrm{Ad}(e^{iuX})Y\,\mathrm{d}u$, $|Z| \le |Y|$; then the log-Sobolev inequality on a convex set, and locally Lipschitz ψ by the dilations $A_0 + \vartheta(A - A_0)$, $\vartheta \to 1$. (c) The first bound by comparing means on the overlap $S_1 \cap S_2$; the second by applying the inequality on S' in each orbit through the orbit average ρ, capped at ρ_0, and the bounded change of density g; the cover of $\mathcal{K}$ by chart balls inside C is joined along paths in C. (d) Sard's theorem on the strata indexed by the set of the maximal plaquettes; off $\mathcal{N}$ Gordan's alternative gives a common descent direction of all maximal plaquettes, which makes the boundary locally a Lipschitz graph.

At the abelian flux fields $\sum_p \partial a_p$ over the $(1,2)$ plaquettes vanishes. (e) Compactness of sublevel sets in the open set; a component is closed in $\{\varpi < r\}$, so a minimising sequence in it has its limit points in it. (f) Along rays the plaquette angle grows: $\frac{d}{dt}|Y(t)| \ge |Y| - 32\alpha^2$, which gives the star shape and the Lipschitz radial function; the log-Sobolev inequality on a star-shaped domain with

Lipschitz radial function follows by mapping it to a ball, at the cost of the Jacobian and of the Lipschitz constant, which is $C_{\mathscr{D}}$. The identity is Fubini over the gauge group; the lower bound counts, at each site, the gauge transformations within $\frac14\alpha$ of the given one, a ball of normalised volume $(\frac12\alpha-\sin\frac12\alpha)/2\pi$. □

15.1.7. *Completion of the limit and flow arguments. Step 1: compactness, motion and the Markov property, (A1)–(A3).* Let $K>n\ge n''$ and let $H\in C^1(\Omega_K)$ be real and invariant, with tower (h_k). Under σ^K, $h_k(V_k)$ is a martingale from the top to the bare level (Lem. 15.3(ii)), so $\int|H-h_t(V_t)|^2\,\mathrm{d}\sigma^K=\sum_{k<t}\int\mathrm{Var}_{p_W}(h_k\circ V_W)\,\sigma^K_{d-1}(\,\mathrm{d}W)$; Lem. 15.3(iv) at $g=h_k$, eq. 15.ii, $\sigma^K\le\mu^K$ on invariant functions, $\sum_{d>n}a_d^2=\frac{a_n^2}{L^2-1}$, and Lem. 15.3(i) for the passage to μ^K give

$$\|H-(\mathbf{1}_{\mathsf{l}_n}h_t)\circ\pi_{K,n}\|^2_{L^2(\mu^K)}\le\frac{C^\sigma C_{\mathrm P}}{L^2-1}\frac{a_n^2}{\mathsf b}\,\mathcal{E}_K(H)+4\mathsf R_n\sup|H|^2,\qquad\int\Gamma_n(h_t)\,\mathrm{d}\sigma_n^K\le C^\sigma\mathcal{E}_K(H).$$

(A2): after replacing H_j by smooth invariant functions with the same bounds (closure, a smooth contraction, the gauge average), let $c=\sup_j\sup|H_j|$, $e=\sup_j\mathcal{E}_{K_j}(H_j)$, and for $n\ge n''$ and $K_j>n$ let h_n^j be the top of the tower of H_j at $t=K_j-n$ and $g_n^j=\mathbf{1}_{\mathsf{l}_n}h_n^j$. The tops are bounded by c and, by the display and Lem. 15.3(iii), have Dirichlet integrals on l_n for Haar measure bounded uniformly in j. Rellich's theorem on compact subsets of the open set l_n, which needs no regularity of its boundary, the bound c, and diagonal sequences over an exhaustion of l_n by compact sets and over n, give one subsequence along which $h_n^j\to\phi_n$ in $L^2(\mathsf{l}_n,dV)$ for every $n\ge n''$.

The limit ϕ_n is a Haar class; as a function under μ_n we take a representative: let Ψ_n be $\mathbf{1}_{\mathsf{l}_n}$ times the gauge average of a Borel function in the class of ϕ_n with values in $[-c,c]$; it is Borel, invariant at every point, $|\Psi_n|\le c$, and $\Psi_n=\phi_n$ Haar-almost everywhere on l_n, as ϕ_n is invariant almost everywhere. Prop. 15.2(c) for $\mu_n^{K_j}$ at $f=|\Psi_n-g_n^j|^2\le4c^2$, which vanishes off l_n, gives $\int f\,\mathrm{d}\mu_n^{K_j}\le e^{\mathsf o_n}|\mathsf{l}_n|^{-1}\int_{\mathsf{l}_n}|\phi_n-h_n^j|^2\,\mathrm{d}V+4c^2\mathsf R_n$, so with the display

$$\limsup_j\|H_j-\Psi_n\circ\pi_{K_j,n}\|_{L^2(\mu^{K_j})}\;\le\;2c\,\mathsf R_n^{1/2}+\Big(\frac{C^\sigma C_{\mathrm P}}{L^2-1}\frac{a_n^2}{\mathsf b}\,e+4\mathsf R_nc^2\Big)^{1/2}=:\epsilon_n,\qquad\epsilon_n\to0.$$

Total variation enters only here: $\|\Psi_n(Y_n)-\Psi_{n'}(Y_{n'})\|_{L^2(\mu)}=\lim_j\|\Psi_n\circ\pi_{K_j,n}-\Psi_{n'}\circ\pi_{K_j,n'}\|_{L^2(\mu^{K_j})}\le\epsilon_n+\epsilon_{n'}$, so the invariant cylinder functions $\Psi_n(Y_n)$ converge in $L^2(\mu)$ to some G, $|G|\le c$, and $\lim_n\limsup_j\|H_j-\Psi_n\circ\pi_{K_j,n}\|_{L^2(\mu^{K_j})}=0$: $H_j\to G$ strongly. This holds along every sequence of cutoffs. (A1): let $H_j\to\Phi(Y_n)$ be a recovery sequence of Lem. 15.1(b), smooth, invariant, $\sup|H_j|\le\sup|\Phi|$ (truncation and the gauge average do not raise the energy). The conditional mean under σ^K given V_t is an $L^2(\sigma^K)$-contraction that fixes $\Phi(V_t)$, so the tops converge to Φ in $L^2(\sigma_n^K)$, hence in $L^2(\mathsf{l}_n,dV)$ by the lower bound of the density; the bound on $\int\Gamma_n(h_t)\,\mathrm{d}\sigma_n^K$, $\inf\mathsf g\ge(1-\mathsf R_n)e^{-\mathsf o_n}/|\mathsf{l}_n|\ge e^{-\mathsf o_n}/2|\mathsf{l}_n|$ and the lower semicontinuity of the Dirichlet integral give eq. 15.i.

As $\Gamma_n(a_p)=4a_n^{-1}|\mathsf s_p|^2$, by the computation of Lem. 16.2(a) on Ω_n, is positive at the fields of l_n with $\varpi_p>0$, $\mathcal{E}_\star(a_p(Y_n))>0$; since $\nu_\Psi(\{\infty\})=0$ for invariant Ψ, $(\Psi,(1-Q_t)\Psi)=\int(1-e^{-t\lambda})d\nu_\Psi>0$. (A3): for an invariant cylinder function Ψ and $t>0$, $H_j=P_t^{K_j}\Psi_{K_j}$ is invariant, $|H_j|\le\sup|\Psi|$, $\mathcal{E}_{K_j}(H_j)\le\sup|\Psi|^2/2et$ (Lem. 15.1(a)), and $H_j\to Q_t\Psi$ weakly; by (A2) every subsequence has a strongly convergent subsequence, whose limit is the weak limit; so $H_j\to Q_t\Psi$ strongly, which is one of the equivalent properties of Lem. 15.1(b), and with it all of them, (M1), and the convergence of $P_t^{K_j}H_j$ for all $H_j\to F$.

With (M2) of Lem. 15.1(b) the forms converge in the strong and weak sense, which gives the strong convergence of the resolvents by the theorem on the convergence of resolvents under convergence of forms in the strong and weak sense; Lem. 15.1(c) gives a dense domain and strong continuity. Compactness: fix a countable dense set (Φ_i) of invariant cylinder functions; for G_m in the set choose a cylinder function Ψ_m with $\|\Psi_m-G_m\|\le\frac1m$ and, by (M2), truncation at ±1 and the gauge average,

smooth invariant H_m on $\Omega_{K_{j(m)}}$ with $|H_m| \le 1$, $\mathcal{E}_{K_{j(m)}}(H_m) \le 1 + \frac{1}{m}$ and $\|H_m - \Psi_{m,K_{j(m)}}\| \le \frac{1}{m}$, where $j(m)$ increases so fast that $\|\Psi_{m,K} - \Phi_{i,K}\|$ is within $\frac{1}{m}$ of $\|\Psi_m - \Phi_i\|$ for $i \le m$ and $K \ge K_{j(m)}$; by (A2) a subsequence of (H_m) converges strongly to some G, then $G_m \to G$ in $L^2(\mu)$ along it, and $|G| \le 1$, $\mathcal{E}_\star(G) \le 1$ by (M1).

The criterion without a cutoff: $W_j = (1 - \mathcal{L}_{K_j})H_j$ is bounded in L^2, so a subsequence converges weakly to some W (a diagonal sequence over a countable dense set of cylinder functions); for invariant cylinder Ψ, $\int H_j \Psi_{K_j} \, \mathrm{d}\mu^{K_j} = \int W_j (1 - \mathcal{L}_{K_j})^{-1} \Psi_{K_j} \, \mathrm{d}\mu^{K_j} \to (W, (1 + A_\star)^{-1}\Psi)$, the resolvents being symmetric and strongly convergent, so $F = (1 + A_\star)^{-1} W \in D(A_\star)$; the weak limit $F - W = -A_\star F$ of $\mathcal{L}_{K_j} H_j$ does not depend on the subsequence, and $\mathcal{E}_{K_j}(H_j) = \langle H_j, W_j \rangle - \|H_j\|^2 \to (F, W) - \|F\|^2 = \mathcal{E}_\star(F)$. The bilinear forms: with $T_j^\alpha = \alpha(\alpha - \mathcal{L}_{K_j})^{-1}$ and $T^\alpha = \alpha(\alpha + A_\star)^{-1}$, $\mathcal{E}_{K_j}(T_j^\alpha u_j, v_j) = \alpha \langle u_j - T_j^\alpha u_j, v_j \rangle \to \alpha(u - T^\alpha u, v) = \mathcal{E}_\star(T^\alpha u, v)$, as $u_j - T_j^\alpha u_j \to u - T^\alpha u$ strongly; by the spectral theorem $\mathcal{E}_{K_j}(u_j - T_j^\alpha u_j) \le \mathcal{E}_{K_j}(u_j) - \alpha \langle u_j, u_j - T_j^\alpha u_j \rangle$, whose limit $\mathcal{E}_\star(u) - \int \frac{\alpha\lambda}{\alpha + \lambda} d\nu_u$ tends to 0 as $\alpha \to \infty$; with $|\mathcal{E}_{K_j}(u_j - T_j^\alpha u_j, v_j)| \le \mathcal{E}_{K_j}(u_j - T_j^\alpha u_j)^{1/2} \mathcal{E}_{K_j}(v_j)^{1/2}$, $\mathcal{E}_\star(v) \le \liminf_j \mathcal{E}_{K_j}(v_j)$ by (M1) and $\mathcal{E}_\star(T^\alpha u, v) \to \mathcal{E}_\star(u, v)$ this gives the convergence.

Step 2: the connected coarse-scale law, (B1). The regularity number of a background U of level k is $L^{2k} \max_p |\partial U(p) - \mathbf{1}|$, the maximum over the bare plaquettes ([7, Def. B.2]), and Cut_k is the set of the block fields over which it is $< c_\rho \varepsilon(x_k)$ (the definitions preceding Lem. YM3_I:113.10). By the curvature comparison in the proof of Lem. YM3_I:113.10, these numbers contract from level to level, by the factor L^{-2} up to $C_H \delta(x_k)$: for $W \in \mathsf{Cut}_{k+1}$ and B in the box, the regularity number of every background over $V_W(B)$ is at most L^{-2} times that of a background over W plus $C_H \delta(x_k)$; starting the chain j_1 levels higher, at every U in the support of its small-field weight,

$$\varpi(V_t) \le 2\vartheta\varepsilon_n, \qquad \vartheta = \vartheta_\infty + L^{-3j_1/2} c_\rho < 1.$$

The set $\mathcal{K}_0 = \{\varpi \le 2\vartheta\varepsilon_n\}$ is compact and lies in I_n; as D_n and $\mathsf{I}_n \setminus \mathsf{D}_n$ are open, it is the disjoint union of the compact invariant sets $\mathcal{K}_1 = \mathcal{K}_0 \cap \mathsf{D}_n$ and $\mathcal{K}_2 = \mathcal{K}_0 \setminus \mathsf{D}_n$, which are free of K. The convexity of one box joins $V_W(B)$ to $V_W(0)$ at a fixed W only; the j_1 levels are passed with the tower of a continuous function. Let $\mu_{t+j_1} = c_\rho \varepsilon(x_{t+j_1})$ and $\mu_k = L^{-2}\mu_{k+1} + C_H \delta(x_k)$ for $t \le k < t + j_1$, so that $\mu_t \le \vartheta\varepsilon_n$, and let D'_k be the set of the $W \in \mathsf{Cut}_k$ over which the regularity number is $< \mu_k$: $\mathsf{D}'_{t+j_1} = \mathsf{Cut}_{t+j_1}$, $V_W(B) \in \mathsf{D}'_k$ for $W \in \mathsf{D}'_{k+1}$ and every B in the box (Lem. YM3_I:113.10, the curvature comparison in the proof of Lem. YM3_I:113.10), and $\mathsf{D}'_t \subset \mathcal{K}_0$ (the small-curvature averaging bound of Prop. YM3_I:14.1).

Let h be continuous on Ω_n with values in $[0,1]$, $h = 1$ on $\mathcal{K}_1$ and $h = 0$ on $\mathcal{K}_2$ (Urysohn), and let $h_t = h$, $h_{k+1}(W) = \int h_k(V_W(B)) \, p_W(\mathrm{d}B)$, be its tower upward from the level t in the chain from the depth $n - j_1$. The kernels carry bounded continuous functions on Cut_k to continuous functions on Cut_{k+1} (as in the proof of Lem. 15.3(ii), where the integrands F_k are shown continuous on the cuts): the kernel at W is the normalised chart integral [7, eq. 191] of the integrand F_k of the chain at the level k; on Cut_{k+1}, which lies in the set O^{im} of the chart (Lem. YM3_I:113.10), the chart, its gauge-fixing form and the exponent of its Jacobian are continuous in (W, B), and the chart integral of a continuous function is continuous ([7, Lem. B.3]); F_k is continuous on Cut_k, into which the chart carries the box (Lem. YM3_I:113.10), and dominated convergence on the box applies.

By induction on k, h_k is continuous on Cut_k and takes on D'_k the values 0 and 1 alone: for $W \in \mathsf{D}'_{k+1}$ the function $B \mapsto h_k(V_W(B))$ is continuous on the convex box with values in $\{0,1\}$, hence constant, and as p_W is a probability, $h_{k+1}(W) = h_k(V_W(B))$ for every B in the box. Take all the fluctuations 0: for $W \in \mathsf{Cut}_{t+j_1}$ let $W^{(t+j_1)} = W$ and $W^{(k)} = V_{W^{(k+1)}}(0) = \mathcal{M}^k U_+^{(k)}$, $U_+^{(k)}$ a background over $W^{(k+1)}$. The backgrounds are nested: $U_+^{(k)}$ is a background over $W^{(k)}$ as well ([7, Lem. B.4](2)), so the regularity number over $W^{(k)}$ is exactly L^{-2} times that over $W^{(k+1)}$ (the identity $\mathsf{m}_k = L^{-2}\mathsf{m}_{k+1}$), with no $C_H \delta(x_k)$; hence $\varpi(W^{(t)}) < 2L^{-2j_1} c_\rho \varepsilon(x_{t+j_1}) \le 2L^{-3j_1/2} c_\rho \varepsilon_n \le 2\varepsilon_n / C''$, $W^{(t)} \in \mathsf{D}_n$ by Lem. 15.7(a), so $W^{(t)} \in \mathcal{K}_1$ and $h_{t+j_1}(W) = h(W^{(t)}) = 1$.

On the support of the small-field weight of the chain from the depth $n-j_1$, $V_{t+j_1}\in \mathsf{Cut}_{t+j_1}$ and $V_k=V_{V_{k+1}}(B_k)\in \mathsf{D}'_k$, so $h(V_t)=h_{t+1}(V_{t+1})=\cdots=h_{t+j_1}(V_{t+j_1})=1$; as $V_t\in\mathcal{K}_0$ and $h=0$ on $\mathcal{K}_2$, $V_t\in\mathcal{K}_1$. Of the dependence on W this argument uses the continuity of the kernels alone. So $\sigma_n^K(\Omega_n\setminus\mathcal{K}_1)\le \mathsf{R}_{n-j_1}$ by Lem. 15.3(i) for the chain from depth $n-j_1$. Glue finitely many chart balls in D_n to a connected $S'\supset\mathcal{K}_1$ (Lem. 15.7(b), (c)), put $w_n=\min\{1,\rho/\rho_0\}$, $\rho_0=\min_{\mathcal{K}_1}\rho>0$, so that $w_n=1$ on $\mathcal{K}_1$ and $w_n=0$ off D_n: this is (a).

Lem. 15.7(c) with the density of σ_n^K, $\sup \mathsf{g}\le e^{\mathsf{o}_n}\inf\mathsf{g}$ (Lem. 15.3(iii)), and $\Gamma_n=a_n^{-1}|\partial\cdot|^2$ give (b) with $C_n^{\mathrm{t}}=a_nc(S')e^{\mathsf{o}_n}/\rho_0$. (c): the linearisation of (b) gives $\int(\phi-c)^2w_n\,\mathrm{d}\sigma_n^K\le\frac12 C_n^{\mathrm{t}}\int\Gamma_n(\phi)w_n\,\mathrm{d}\sigma_n^K$, and the part where $w_n<1$ costs at most $\mathrm{osc}(\phi)^2\mathsf{R}_{n-j_1}$. (d): $\mathrm{Ent}_{\sigma_n^K}(\phi^2)\le\int\Xi(\phi^2,c')\,\mathrm{d}\sigma_n^K=\mathrm{Ent}_{w_n\sigma_n^K}(\phi^2)+\int\Xi(\phi^2,c')(1-w_n)\,\mathrm{d}\sigma_n^K$. (e): the explicit-top sketch still requires the multiscale gauge estimate. The conditions of n_{e} are those of (Multiscale gauge) at $J=J_n$; a field of I_n whose block fields $\mathcal{M}^jV$, $j\le J_n$, lie in the cuts has a gauge with all bonds within $44\pi L^2\sum_{j<J_n}\theta_{n-j}+C_{\mathrm{S}}L^{-J_n}\le\frac14\alpha_n$ of $\mathbf{1}$, so $w_n^{\mathrm{e}}\ge c_n$ there by Lem. 15.7(f), $\chi_n=1$, and $\sigma_n^K(\chi_n<1)\le\mathsf{R}_{n-J_n}$ by Lem. 15.3(i) for the chain from depth $n-J_n$.

Since $r'_n=2\arcsin\varepsilon_n=64\alpha_n^2$ is the angle of the radius $2\varepsilon_n$, $\mathsf{I}_n=\{\mathsf{d}(\partial V(p),\mathbf{1})<r'_n\ \forall p\}$, and the identity of Lem. 15.7(f) carries the log-Sobolev inequality of Haar measure on $\exp\mathscr{D}$ to invariant functions under $w_n^{\mathrm{e}}dV$, then to χ_ndV at the cost c_n^{-1}, and to $\chi_n\sigma_n^K$ at the cost $e^{2\mathsf{o}_n}/(1-\mathsf{R}_n)$ of the density: this is $C_n^{\flat}$. The set $\{\chi_n>0\}\cap\mathsf{I}_n$ is the path connected set of Lem. 15.7(f); it contains $\mathbf{1}$, so it lies in D_n. For the block-field laws, $\mathrm{Var}_{\mu_n^K}(\Phi)\le\int(\Phi-c)^2\,\mathrm{d}\mu_n^K\le\int(\Phi-c)^2\,\mathrm{d}\sigma_n^K+\mathsf{R}_n\mathrm{osc}(\Phi)^2$ by Lem. 15.3(i), then (c) with χ_n and $\int\Gamma_n(\Phi)\,\mathrm{d}\sigma_n^K\le\int_{\mathsf{I}_n}\Gamma_n(\Phi)\,\mathrm{d}\mu_n^K$; for μ_n by total variation.

Step 3: weak Poincaré bounds and mixing, (A4). Let c be the mean of H under ς^K. The density $w_n(V_t)$ is a function of the first member of the chain, so it changes no kernel and no tower, and

$$\int(H-c)^2\,\mathrm{d}\varsigma^K=\int(H-h_t(V_t))^2\,\mathrm{d}\varsigma^K+\int(h_t-c)^2w_n\,\mathrm{d}\sigma_n^K\le\mathsf{C}_n\mathcal{E}_K(H),$$

the first term by the martingale sum of Step 1, the second by (B1)(b) linearised and eq. 15.ii at $k=t$; passing to μ^K costs $\mathsf{r}_n\mathrm{osc}(H)^2$ by (B2)(a). This is (a), first for smooth H and then by closure. (b): $\frac{d}{dt}\mathrm{Var}(P_t^KH)=-2\mathcal{E}_K(P_t^KH)\le-\frac{2}{\mathsf{C}_n}[\mathrm{Var}(P_t^KH)-\mathsf{r}_n\mathrm{osc}(H)^2]$, as $\mathrm{osc}(P_t^KH)\le\mathrm{osc}(H)$, and $\mathrm{Var}(H)\le\frac14\mathrm{osc}(H)^2$. (c): truncated and gauge-averaged recovery sequences and (a); if $\mathcal{E}_\star(F)=0$, then $\mathrm{Var}_\mu(F)\le\mathsf{r}_n\mathrm{osc}(F)^2$ for all admitted n. For non-constant bounded $F\in L^2_{\mathrm{inv}}$ the spectral measure ν_F is not carried by $\{0\}$, as else $\mathcal{E}_\star(F)=0$, and $\nu_F(\{\infty\})=0$ (Lem. 15.1(c)); so $(F,(1-Q_t)F)=\int(1-e^{-t\lambda})\,\mathrm{d}\nu_F>0$.

(d): $\varphi(t)=\|Q_{t/2}(F-\int F)\|^2$ has $\varphi'(t)=-\mathcal{E}_\star(Q_{t/2}(F-\int F))\le-\mathsf{C}_n^{-1}[\varphi(t)-\mathsf{r}_n\mathrm{osc}(F)^2]$ by (c) and spectral calculus, and $\varphi(0)\le\frac14\mathrm{osc}(F)^2$. If $Q_t\rho=\rho$, then $(\rho,(1-Q_t)\rho)=0$, ν_ρ sits at 0, $\mathcal{E}_\star(\rho)=0$, so $\mathcal{E}_\star(\rho\wedge M)=0$ for every M, and $\rho\wedge M$ is constant by (c): $\rho=1$. (e): apply (a) to H_K truncated at $\pm M_F$, which does not raise the energy and keeps $\|H_K-F_K\|\to0$, so that Var_{μ^K} of the truncation tends to $\mathrm{Var}_\mu F$; the second form is (a) itself.

(f): the same lines with the explicit top of (B1)(e).

Step 4: entropy along the chain and the conditional law, (B2), (B3). (a): Lem. 15.3(i) and $\int f(1-w_n(V_t))\,\mathrm{d}\sigma^K\le\sup f\,\sigma_n^K(w_n<1)$. (b): with f_k the tower of $G^2+\epsilon$ and $\phi_k=f_k^{1/2}$, since the p_W are probabilities the entropy splits along the chain,

$$\mathrm{Ent}_{\varsigma^K}(G^2+\epsilon)=\sum_{k<t}\int\mathrm{Ent}_{p_W}(f_k\circ V_W)\,\varsigma_{d-1}^K(\,\mathrm{d}W)+\mathrm{Ent}_{w_n\sigma_n^K}(f_t);$$

the fibres by Lem. 15.3(iv), $\mathrm{Ent}_{p_W}(\phi_k^2\circ V_W)\le2C_{\mathrm{P}}\frac{a_d^2}{\mathsf{b}}\int\Gamma_d(\phi_k)\circ V_W\,\mathrm{d}p_W$; the top by (B1)(b); and $\int\Gamma_d(\phi_k)\,\mathrm{d}\varsigma_d^K\le3\mathsf{M}^2\Pi_n^{\flat}\int\Gamma_K(G)\,\mathrm{d}\varsigma^K$ (Prop. 15.5(b), (c), as $\Gamma_K((G^2+\epsilon)^{1/2})\le\Gamma_K(G)$); sum over $d>n$ and let $\epsilon\to0$. (c): $\mathrm{Ent}_\nu(G^2)\le\int\Xi(G^2,c')\,\mathrm{d}\nu$ and (a). (d): for the proposed multiscale variant, the function $\mathsf{b}\mapsto2\varepsilon(L^n\mathsf{b})$ is smooth with a non-vanishing derivative where $\log L^n\mathsf{b}>2\mathfrak{p}_0$, so $2\varepsilon_n\notin\mathcal{N}$

for almost every b, for all m and n at once; by Lem. 15.7(d) D_n is then a connected Lipschitz domain, which carries a log-Sobolev inequality for the volume (the extension theorem, with the inequalities of Sobolev and Rellich), and the density bounds give $C_n^\sharp$; $\sigma_n^K(\mathsf{I}_n \setminus \mathsf{D}_n) \le \sigma_n^K(\chi_n < 1) \le \mathsf{R}_{n-J_n}$ by Step 2(e).

For every b, a radius $\tilde r \notin \mathcal{N}$ close to $2\varepsilon_n$ does the same at the cost η: the fields with $\chi_n = 1$ and $\varpi < \tilde r$ lie in the component of $\mathbf{1}$ in $\{\varpi < \tilde r\}$, since along the rays of Lem. 15.7(f) the plaquette angles stay below the larger of $32\alpha_n^2$ and their final values, and the shells $\{\tilde r \le \varpi < 2\varepsilon_n\}$ have σ_n^K-masses at most e^{o_n} times their relative Haar measure, uniformly in K. With χ_n the same lines as in (b). (e) is the splitting of (b) with the top law of σ^K itself. (B3): the splitting of (b), stopped at depth e and taken at a fixed block field W, needs no top: with $\Pi_{(e)} = \prod_{e<d\le K}(1+\epsilon_d^\flat) \le \Pi_e^\flat$,

$$\mathrm{Ent}_{\mathsf{P}_W}(G^2) \le \sum_{e<d\le K} 2C_{\mathrm{P}}\frac{a_d^2}{\mathsf{b}}\,3\mathsf{M}^2\Pi_{(e)}\int \Gamma_K(G)\,\mathrm{d}\mathsf{P}_W \le C_e^{\mathrm{fib}}\int \Gamma_K(G)\,\mathrm{d}\mathsf{P}_W,$$

and the one-step bound of Prop. 15.5(b), iterated pointwise, gives the bound on $\Gamma_e(g_{t'})(W)$; no factor depends on ℓ. P_W is a regular conditional law by Lem. 15.3(ii), also under ς^K, whose weight is a function of the top. Concentration: for real invariant ψ the log-Sobolev inequality, applied to $e^{s\psi/2}$, gives $\int e^{s(\psi-\psi_{t'}(W))}d\mathsf{P}_W \le e^{s^2C_e^{\mathrm{fib}}l^2/4}$, as $\int \psi\,\mathrm{d}\mathsf{P}_W = \psi_{t'}(W)$, then Chebyshev's inequality; for μ^K, the weight $\mathfrak{w}_t^{\mathrm{sf}}$ of σ^K vanishes unless $\mathcal{M}^{t'}U \in \mathsf{Cut}_{t'}$, and Lem. 15.3(i) bounds what μ^K adds to σ^K by R_n; both tails.

Step 5: entropy relaxation, (A5). (a) is (B2)(c) at $\nu = \mu^K$, with $\Xi(x,c) \le \bar M[1+\log(\bar M/c)]$ for $0 \le x \le \bar M$, $0 < c \le \bar M$, and $c' \ge m_2 - \mathsf{r}_n M_G^2$, as $\int G^2\,\mathrm{d}\varsigma^K \ge \int G^2\,\mathrm{d}\mu^K - \mathsf{r}_n M_G^2$ and $\varsigma^K(\Omega_K) \le 1$. (b): (B2)(b) for the normalised measure $\varsigma^K/\varsigma^K(\Omega_K)$ gives Gaussian concentration about $\bar\psi$, and (B2)(a) adds r_n. (c): $P_\tau^K f_0$ is smooth and strictly positive for $\tau > 0$, the kernel of the elliptic diffusion being positive, and

$$\tfrac{d}{d\tau}\mathrm{Ent}_{\mu^K}(P_\tau^K f_0) = -\mathcal{E}_K(P_\tau^K f_0, \log P_\tau^K f_0) \le -4\mathcal{E}_K((P_\tau^K f_0)^{1/2}) \le -\tfrac{4}{\mathsf{C}_n'}\big[\mathrm{Ent}_{\mu^K}(P_\tau^K f_0) - \mathsf{r}_n\bar M(1+\log\tfrac{2\bar M}{m_1})\big]$$

by (a) at $G = (P_\tau^K f_0)^{1/2}$, with $M_G^2 \le \bar M$, $m_2 = m_1$ and $m_1 - \mathsf{r}_n\bar M \ge \frac12 m_1$; Gronwall. For limits: (a) by recovery sequences, as integrals of continuous functions of bounded strongly convergent sequences converge; the decay first for invariant cylinder f_0 with values in $[0,\bar M]$, by (A3), and then by approximation in L^2_{inv} under the strict condition $\mathsf{r}_n\bar M < \frac12 m_1$; $\tau\to\infty$ and then $n\to\infty$ give $\mathrm{Ent}_\mu(Q_\tau F) \to 0$, and Pinsker's inequality gives the convergence in total variation.

The moments of the gauge-fixed field. Steps 6–10 use Prop. 16.1(a) through the following facts. Use the consequence (Adm) of (E) proved in Prop. 16.1, and let d_1 be a depth, free of K, such that every top depth $\ge d_1$ is admissible (the depth d_{adm} of the proof of Prop. 16.1(a)). With $\mathbb{E}^{[d]}$ the normalised chain measure from the top depth d and $\mathbf{B}_K$ its gauge-fixed field: (i) for every admissible d, $K > d$, $q < \infty$ and every index $\kappa' \in (0, \frac16)$,

$$\mathbb{E}^{[d]}\big[\|\mathbf{B}_K\|_{\kappa';a}^q\big]^{1/q} \;\le\; C(q,\kappa')\,L^{20d}, \tag{15.iii}$$

with C free of K and d: given the coarser layers, the layer of depth d' is a function of one fluctuation integral with conditional covariance at most $Ca_{d'}^2/\mathsf{b}$ and Gaussian tails (Lem. 15.3(iv)), the products of layers of two depths in the bilinear part of the rough norm are martingale differences, those of one depth are controlled by the bound (Q) of centred quadratic forms and by the cancellation in colour (Col), the conditional means by (M), all three set in the proof of Prop. 16.1(a), and the top field, of size at most a constant times L^{10d} in its regular gauge, enters to the second power; (ii) the short-time theory of the gauge-fixed flow, which uses no measure: the numbers c_1, c_2, c_3, $s_0(R)$, $a_0(R)$ at every index κ', the four bounds of A_s, ∇A_s, $A_s - P_s\mathbf{B}$, $\nabla(A_s - P_s\mathbf{B})$, monotone in R, the bound of the flowed action and the Lipschitz bound in the rough distance; (iii) for invariant Borel

$F \geq 0$ and $r > 1$,

$$\langle F\rangle_{\mu^K} \;\leq\; \mathbb{E}^{[d_1]}[F] + \sum_{d_1 \leq d \leq K-2} \mathsf{R}_d^{1-1/r}\, \mathbb{E}^{[d+1]}\big[F^r\big]^{1/r} + \mathsf{R}_{K-1} \sup F, \tag{15.iv}$$

from the chains of all the top depths $d \geq d_1$, each compared with the bare density by Lem. 15.3(i), and Hölder's inequality; and $\sum_d \mathsf{R}_d^{1/2} L^{pd}$ and $\sup_K \mathsf{R}_{K-1} L^{pK}$ are finite for every p, as R_d decays faster than every exponential in d; (iv) hence, with $\mathsf{N}' = \mathsf{N}'(U)$ of the head of Prop. 16.1 formed at the index κ', the least $\|\mathbf{B}\|_{\kappa';a}$ over the 1-forms $\mathbf{B}$ with $e^{ia_K\mathbf{B}}$ in the orbit of U, which is at most a fixed power of L^K at every field, $\mathsf{M}_q(\kappa') = \sup_K \langle \mathsf{N}'^q\rangle_{\mu^K} < \infty$, and the moments under μ^K, bounded uniformly in K, of $\sup_{0<s\leq 1} s^{\frac{3}{2}+\kappa'} A^a(U_s)$, of $(J,J)_a(U_\sigma)$ and of $a_K^{-4} \max_p a_p(U_\sigma)$ in Prop. 16.1(a), at every index $\kappa' \in (0, \frac{1}{6})$.

Step 6: flow identities and uniform estimates, (C1), (C2). (C1) is Lem. 15.1(a) with the integration by parts, and $\int \mathcal{L}_K H \,\mathrm{d}\mu^K = 0$. The formula for $P_t^K H^\sigma$: $\frac{d}{ds} P_s^K H^{\sigma+\mathsf{b}(t-s)} = P_s^K(\mathcal{L}_K - \mathsf{b}\,\partial_\sigma) H^{\sigma+\mathsf{b}(t-s)} = P_s^K \Delta_K H^{\sigma+\mathsf{b}(t-s)}$, integrated over $[0,t]$; the martingale is Itô's formula for the time-dependent function $(r,U) \mapsto H^{\sigma+\mathsf{b}(t-r)}(U)$, whose drift is $(\partial_r + \mathcal{L}_K) H^{\sigma+\mathsf{b}(t-r)} = \Delta_K H^{\sigma+\mathsf{b}(t-r)}$. (C2): on $\mathsf{O}_K(\bar C, \bar s)$, for all K and $\sigma \leq \bar s$, $\Lambda_\sigma \leq (8\bar C \sigma^{\frac{1}{4}-\frac{\kappa}{2}} + 4\bar C^2 \sigma^{\frac{1}{2}-\kappa})/(1-2\kappa)$ (Lem. 16.2(e)) and $A^a(U_{\bar s/2}) \leq E_\sharp = 6\ell^3(\bar C^2(\bar s/2)^{-\frac{1}{2}-\kappa} + \bar C(\bar s/2)^{-\frac{3}{4}-\frac{\kappa}{2}})^2$.

For $K \geq K_0$, the least K with $a_K E_\sharp \leq e_0$, Lem. 16.2(f) at $s_1 = \bar s/2$ gives $a_K^{-4} \max_p a_p(U_s) \leq Z_\sharp = \frac{5}{2} E_\sharp \min\{\bar s/2, x_0^4 E_\sharp^{-2}\}^{-3/2}$ for $s \geq \bar s$, so $\Lambda_\sigma \leq \Lambda_{\bar s} + (2Z_\sharp)^{1/2}$ for $\bar s < \sigma \leq 1$. With $G = \varphi(\mathcal{O}(f_1), \dots, \mathcal{O}(f_r))$,

$$\Gamma_K(H_K^\sigma) \leq e^{24\Lambda_\sigma} \|\nabla\varphi\|_\infty^2 \sum_i \Gamma_K(\mathcal{O}(f_i))(U_\sigma), \qquad |\partial_\sigma H_K^\sigma| \leq (\Gamma_K(G)(J,J)_a)^{1/2}(U_\sigma),$$

with $\frac{\sigma}{2}(J,J)_a(U_\sigma) \leq e^{24\Lambda_\sigma} A^a(U_{\sigma/2})$, $A^a(U_{\sigma/2}) \leq 24\ell^3\Lambda_\sigma^2\sigma^{-2}$ for $\sigma \leq \bar s$ and $\leq E_\sharp$ beyond, and Lem. 16.2(d). (b): the same pointwise inequalities without a set, with the moments (iv) of Prop. 16.1(a) above, by (Adm); the Lipschitz bound in σ is the integral of ∂_σ. (c): the second flow-time derivative involves $v^2\mathcal{O}(f)$ along the flow, which is bounded by $\mathsf{h}_s \Gamma_K(\mathcal{O}(f))^{1/2}$ plus $\|f\|_\infty$ times the square of the L^2 norm of the covariant gradient of the current and a term of the first level, $\mathsf{h}_s = a_K^{-7/2} \|\nabla^2 A_K(U_s)\, \partial A_K(U_s)\|$; a second level of the energy estimates rests on a floor of the Hessian on horizontal vectors by the action alone: Prop. 16.6(a) at $k = N-1$, where $\mathsf{e}_N^* = 2A^a$ and $a_K(N-1) \geq \frac{1}{2}\ell$, gives $a_K^{-2}\nabla^2 A_K(V)[X,X] \geq -c(A^a(V)^2 + \ell^{-2})\|X\|^2$ for horizontal X when $\max_p a_p(V) \leq \frac{1}{2}$, with an absolute c.

With it, for every field, $0 \leq \sigma_0 < \sigma$ and $E = A^a(U_{\sigma_0})$,

$$(J,J)_a(U_\sigma) \leq e\, E\, [(\sigma - \sigma_0)^{-1} + 2c(E^2 + \ell^{-2})] + 3072\, \ell^3 E^6.$$

Where $a_K E \leq \frac{1}{2}$, $\max_p a_p(U_s) \leq a_K A^a(U_s) \leq \frac{1}{2}$ for $s \geq \sigma_0$, as the action does not increase along the flow, so the floor applies to the current, which is horizontal and is carried by the linearised flow: $\partial_s (J,J)_a(U_s) \leq 2c(E^2+\ell^{-2})(J,J)_a(U_s)$, and the mean of $(J,J)_a(U_s)$ over the last interval before σ of length $\min\{\sigma - \sigma_0, (2c(E^2+\ell^{-2}))^{-1}\}$ is at most E divided by that length, as $\partial_s A^a(U_s) = -(J,J)_a(U_s)$, while $(J,J)_a(U_\sigma)$ is at most e times that mean; where $a_K E > \frac{1}{2}$, $a_K^{-1} < 2E$, and the crude bound $(J,J)_a \leq 48\ell^3 a_K^{-6}$ gives the last member. The second level needs more than this floor: with $g = \partial A_K(U_s)$, which is horizontal, and $\mathsf{Q}_s = a_K^{-5}\nabla^2 A_K(U_s)[g,g]$, $\partial_s (J,J)_a(U_s) = -2\mathsf{Q}_s$ and

$$\partial_s \mathsf{Q}_s = -2\mathsf{h}_s^2 - a_K^{-7}\nabla^3 A_K(U_s)[g,g,g],$$

and the cubic member has no sign. Plaquette by plaquette, with $|X|_p^2 = \sum_{b\in p} |X(b)|^2$ and the covariant forward differences ∇ of the field, $|\frac{d^3}{dt^3} a_p(e^{itX}V)|_{t=0}| \leq 6|X|_p^2(|\nabla_\nu X_\mu(x)| + |\nabla_\mu X_\nu(x)|) + 25\varpi_p |X|_p^3$ for $p = p_{\mu\nu}(x)$, since the cubic term of the plaquette is a sum of triple products of the four values of X at p, carried to its base point with their signs, each of which holds the sum of

the values at two parallel bonds, a covariant difference up to $2\varpi_p|X|_p$; so $|\nabla^3 A_K(V)[X,X,X]| \le 24\sqrt{2}\,\|X\|_{\ell^4}^2 \mathsf{D}(X) + 200(2A_K(V))^{1/2}\|X\|_{\ell^6}^3$, $\mathsf{D}(X)$ the ℓ^2 norm of ∇X.

Kato's inequality, $||X_\mu(x+a_K e_\nu)| - |X_\mu(x)|| \le |\nabla_\nu X_\mu(x)|$, and the Sobolev inequality of the periodic lattice with constants free of N, $\|u\|_{\ell^6} \le N^{-1}\|u\|_{\ell^2} + \frac{4}{\sqrt{3}}\|\nabla u\|_{\ell^2}$ and $\|u\|_{\ell^4}^4 \le \|u\|_{\ell^2}\|u\|_{\ell^6}^3$, bound these norms through $\|X\|$ and $\mathsf{D}(X) + N^{-1}\|X\|$. The same tools bound the remainder of the Weitzenböck identity of Lem. 16.2(h), at most a constant times $\sum_p \varpi_p |X|_p^2$, by $\frac{1}{8}\mathsf{D}(X)^2 + c(A_K(V)^2 + N^{-2})\|X\|^2$, c taken large enough: this is the floor in the form that keeps the covariant gradient, $\nabla^2 A_K(V)[X,X] \ge \frac{1}{8}\mathsf{D}(X)^2 - c(A_K(V)^2 + N^{-2})\|X\|^2$ for horizontal X when $\max_p a_p(V) \le \frac{1}{2}$, whence at $X = g$, $\mathsf{D}(g)^2 + N^{-2}\|g\|^2 \le 8\|g\|\,\|\nabla^2 A_K(U_s)g\| + 9c(A_K(U_s)^2 + N^{-2})\|g\|^2$.

Let $\sigma_1 = \min I$, $\sigma_0 = \frac{1}{4}\sigma_1$, $\Delta = \sigma_1 - \sigma_0$ and $E = A^a(U_{\sigma_0})$. Where $a_K E \le \frac{1}{2}$, the display bounds $(J,J)_a(U_s)$ by $\bar{\mathsf{y}} = \mathrm{e}\,E[2\Delta^{-1} + 2c(E^2 + \ell^{-2})]$ for $s \ge \sigma_0 + \frac{1}{2}\Delta$, and Young's inequality absorbs the cubic member into h_s^2 ($N^{-1} = a_K/\ell$, $A_K = a_K A^a$):

$$\partial_s \mathsf{Q}_s \le -\mathsf{h}_s^2 + \mathsf{B}, \qquad \mathsf{B} \le C\Big[\bar{\mathsf{y}}^{7/3} + E^2\bar{\mathsf{y}}^3 + ((E^2+\ell^{-2})^{5/4} + (E^2+\ell^{-2})^{3/2}E^{1/2})\bar{\mathsf{y}}^{3/2}\Big],$$

C absolute. Above, Q_s is at most its mean over the interval of length $\frac{1}{2}\Delta$ before s, that is half the fall of $(J,J)_a$ over that interval divided by its length, plus $\frac{1}{2}\mathsf{B}\Delta$; below, the floor gives $\mathsf{Q}_s \ge -c(E^2+\ell^{-2})(J,J)_a(U_s)$. So for $\sigma_1 \le r_1 < r_2$, $\int_{r_1}^{r_2} \mathsf{h}_s^2 ds \le \mathsf{Q}_{r_1} - \mathsf{Q}_{r_2} + \mathsf{B}(r_2 - r_1)$, and $a_K^{-5}\mathsf{D}(g)^2 \le 8\mathsf{Q}_s + 8c(E^2+\ell^{-2})(J,J)_a(U_s)$ bounds the square of the L^2 norm of the covariant gradient of the current. With $A_f = \sum_p f(x(p))a_p$, $v^2\mathcal{O}(f) = a_K^{-5}[\langle \nabla^2 A_K\, g, \partial A_f\rangle + \nabla^2 A_f[g,g]]$, and $|v^2\mathcal{O}(f)|(U_s) \le \mathsf{h}_s \Gamma_K(\mathcal{O}(f))(U_s)^{1/2} + \|f\|_\infty[3a_K^{-5}\mathsf{D}(g)^2 + 4c(E^2+\ell^{-2})(J,J)_a(U_s)]$; integrated over s between two points of I, with the Cauchy–Schwarz inequality in s, these are polynomials in E with powers at most 11, times the square root of the length.

Where $a_K E > \frac{1}{2}$, $a_K^{-1} < 2E$, and the crude bounds $|v^2\mathcal{O}(f)| \le 1536\|f\|_\infty \ell^3 a_K^{-8}$ and $|v\mathcal{O}(f)| \le 48\|f\|_\infty \ell^3 a_K^{-6}$ become powers of E, the square of the second one the power 12. As $E \le (4/\sigma_1)^{5/3}\sup_{0<s\le 1} s^{\frac{3}{2}+\kappa}A^a(U_s)$, this gives $|\partial_\sigma H_K^{\sigma''} - \partial_\sigma H_K^{\sigma'}| \le C_I|\sigma'' - \sigma'|^{1/2}(1 + \sup_{0<s\le 1} s^{\frac{3}{2}+\kappa}A^a(U_s))^{12}$ at every field and every cutoff, and the moments of Prop. 16.1(a), by (Adm), give (c).

Step 7: convergence of flowed observables along the full sequence, (C3). Bounded strongly convergent sequences are stable under continuous functions, so $\varphi(\Theta_1,\dots,\Theta_r)$, Θ_i of Prop. 16.1(c), under the proved conclusion (Adm), would converge strongly for f_i applied to the flowed loops once the flow-approximation step is established, and $|H_K^\sigma - \varphi(\Theta_1,\dots,\Theta_r)|$ is at most $\mathrm{Lip}(\varphi)\, r\, \eta$ on $\bigcap_i \mathsf{G}_K^{(i)}$ and at most $2\sup|\varphi|$ elsewhere; let $\eta, \epsilon \to 0$. The joint convergence in law identifies the limit as G_σ, an expression independent of the cutoff; only the Lipschitz constants of the f_i and of φ were used.

By (C2)(b), $\|\mathsf{G}_\sigma - \mathsf{G}_{\sigma'}\| = \lim_K \|H_K^\sigma - H_K^{\sigma'}\| \le C|\sigma - \sigma'|$. For $\dot{\mathsf{G}}_\sigma$: by (C2)(b), (c) the functions $\sigma \mapsto \int \Psi_K \partial_\sigma H_K^\sigma \,\mathrm{d}\mu^K$ are bounded and locally Hölder continuous of exponent $\frac{1}{2}$, uniformly in K; their primitives $\int \Psi_K H_K^\sigma \,\mathrm{d}\mu^K$ converge to $\int \Psi \mathsf{G}_\sigma \,\mathrm{d}\mu$; so the derivatives converge, locally uniformly, to the derivative of the limit, which is then continuously differentiable with a Hölder continuous derivative; the bound in L^2 and the density of the cylinder functions represent the limit as $\int \Psi \dot{\mathsf{G}}_\sigma \,\mathrm{d}\mu$ with $\dot{\mathsf{G}}_\sigma \in L^2_{\mathrm{inv}}$, along the full sequence.

Step 8: condenser cutoffs and the generator limit, (C4), (C5). Let $c_3' = c_3(\kappa/2)$, $R_\flat = 2(2 + c_3')$, $\rho = R/2(2+c_3')$, $q = 4p/\kappa$; $\bar{s}(R) = 2^{-j_0(R)}$ is the largest dyadic number $\bar{s}$ with $\bar{s} \le (c_1'\rho^{-2})^8$, which lies below the time $s_0(\rho)$ of Prop. 16.1(a) at the index $\frac{\kappa}{2}$, $c_1' = c_1(\frac{\kappa}{2})$; $c_{\mathrm{st}} R^2 \bar{s}^{\frac{1}{4}-\frac{\kappa}{2}} \le 1$; and $486R^2\bar{s}^{\frac{1}{2}-\kappa} + 108R\bar{s}^{\frac{1}{4}-\frac{\kappa}{2}} \le 1$, which by Lem. 16.2(e) at $\bar{C} = 9R$ gives $\Lambda_s \le 1$ for $s \le \bar{s}$ on $\mathsf{O}_K(9R,\bar{s})$. $\bar{K}(R)$ is the least integer above the depth d_1 of the conclusion (Adm) at which $a = a_K$ has $a \le (c_2'/\rho)^3$, which lies below $a_0(\rho)$ at the index $\frac{\kappa}{2}$, $c_2' = c_2(\frac{\kappa}{2})$, $c_{\mathrm{st}} R a^{\frac{1}{2}-\kappa} \le 1$, and $4a^2 \le \bar{s}(R)$.

Here c_{st} is an absolute constant of the short-time theory of the gauge-fixed flow; all these conditions are upper bounds of $\bar s$ and of a that decrease in R. Let $g_j = \sup\{\mathsf{m}_s(U_s) : s \in (2^{-j-1}, 2^{-j}] \cap [a_K^2, \bar s(R)]\}$, $g_\flat = a_K^{1/2+\kappa}$ min $\|\mathbf{B}\|_\infty$ over $e^{ia_K\mathbf{B}}$ in the orbit of U, $\mathsf{G}^* = \max\{g_\flat, \max_j g_j\}$, and $\chi_{K,R}$ the average of $\min\{1, \max\{0, 4 - 2\mathsf{G}^*/R\}\}$ against a product of smooth conjugation-invariant approximate identities supported on small rotations of the bond groups. On $\{\mathsf{G}^* \le 3R\}$, $\mathsf{G}^* \le c$ implies $U \in \mathsf{O}_K(3c, \bar s)$ and $\Lambda_s \le 1$ for $s \le \bar s$; there the pieces are locally Lipschitz for the L^2 metric, $\operatorname{lip} g_j \le C_L 2^{(j+1)(1-\kappa)/2}$, $\operatorname{lip} g_\flat \le 2a_K^{-1+\kappa}$, $C_L = 24e^{24}$, because the linearised flow smooths from L^2 modulo gauge: $\mathrm{Ad}(u)D\Phi_s(U)Y = \zeta + a_K^{-1}\mathrm{d}\vartheta$ with $\|\zeta\|_\infty \le 9e^{24}s^{-3/4}\|Y\|_2$ and $\|\nabla\zeta\|_\infty \le 17e^{24}s^{-5/4}\|Y\|_2$.

The rotations are chosen so small that G^* changes by at most $\frac{R}{2}$ under them; so $\chi_{K,R} = 1$ on $\{\mathsf{G}^* \le R\}$, $\chi_{K,R} = 0$ off $\{\mathsf{G}^* \le \frac52 R\} \subset \mathsf{O}_K(8R, \bar s(R))$, and the piece j counts in $\Gamma_K(\chi_{K,R})$ only where $g_j \ge R$. The short-time theory at the index $\frac\kappa2$, which gains $s^{\kappa/4}$, and $\mathsf{M}_q = \mathsf{M}_q(\frac\kappa2) < \infty$, the number of Prop. 16.1(a) at the index $\frac\kappa2$ under (Adm) ((iv) of the moments of the gauge-fixed field), give $\mu^K(g_j \ge R) \le \mathsf{M}_q\rho^{-q}2^{-pj}$ and $\mu^K(g_\flat \ge R) \le \mathsf{M}_q(2e^{12}/R)^q a_K^{2p}$; with $2^{(j+1)(1-\kappa)p - pj} \le 2^p 2^{-\kappa pj}$,

$$\mu^K(\chi_{K,R} < 1) \le \mathsf{M}_q[2\rho^{-q} + (2e^{12}/R)^q], \qquad \int \Gamma_K(\chi_{K,R})^p d\mu^K \le (\tfrac{2}{R})^{2p}\mathsf{M}_q\Big[\frac{C_L^{2p}2^p\rho^{-q}}{1-2^{-\kappa p}} + 4^p(\tfrac{2e^{12}}{R})^q\Big],$$

and we call the first of these bounds $\epsilon_p(R)$ and the second $C_\chi(p,R)$, $q = 4p/\kappa$. As $\Gamma_K(\chi_{K,R})$ vanishes off the shell $\{0 < \chi_{K,R} < 1\}$, Hölder's inequality on the shell bounds the energy, $\mathcal{E}_K(\chi_{K,R}) \le C_\chi(p,R)^{1/p}\epsilon_p(R)^{1-1/p} \le c(p)\mathsf{M}_q R^{-2-q}$, the exponents of R adding up to $-(2+\frac{q}{p}) - (q - \frac{q}{p})$, while the mass is only $\int(1-\chi_{K,R})^2\,\mathrm{d}\mu^K \le \mu^K(\chi_{K,R} < 1) \le \epsilon_p(R) \le c(p)\mathsf{M}_q R^{-q}$. The cutoff $\chi_{K,R}$ does not depend on p, so one cutoff serves all p, and

$$\bar\varkappa(R) := \min_{p\in\mathbb{N},\ p\le R}\Big[C_\chi(p,R)^{1/p}\epsilon_p(R)^{1-1/p} + \epsilon_p(R)\Big] > 0$$

bounds $\mathcal{E}_K(\chi_{K,R}) + \int(1-\chi_{K,R})^2\,\mathrm{d}\mu^K$ for $K \ge \bar K(R)$ and is at most $c(p)\mathsf{M}_{4p/\kappa}R^{-4p/\kappa}$ for every $p \le R$; so $R^{q'}\bar\varkappa(R) \to 0$ for every q'. The numbers $\bar s(R)$, $\bar K(R)$ are monotone in R, and G^* does not increase when $\bar s$ decreases; so for $R' \ge \frac52 R$, $\{\chi_{K,R} > 0\} \subset \{\mathsf{G}^*_{(R)} \le \frac52 R\} \subset \{\mathsf{G}^*_{(R')} \le R'\}$, on which $\chi_{K,R'} = 1$. The condenser cutoff: Lem. 16.5 at $\eta = 1 - \chi_{K,R}$, with $\varkappa \le \mathcal{E}_{K,1}(\eta,\eta) \le \bar\varkappa(R)$. (C5): by (C1) and Lem. 16.5, $\int \chi^{\mathrm{c}}_{K,R}\mathcal{L}_K H\,\mathrm{d}\mu^K = \mathcal{E}_K(w_{K,R}, H)$, so $|\int \chi^{\mathrm{c}}_{K,R}\mathcal{L}_K H^\sigma_K| \le \mathrm{osc}(\varphi)\bar\varkappa(R)$; moreover $|\int \chi^{\mathrm{c}}\mathsf{b}\,\partial_\sigma H - \mathsf{b}\,\partial_\sigma\langle H\rangle| = \mathsf{b}|\int w\,\partial_\sigma H| \le \mathsf{b}C_\partial\bar\varkappa(R)^{1/2}$, which gives the second line of (a).

(b) is the last part of Lem. 16.5, as $q(1-u) = 1 - q(u)$. (c): $\int\chi^{\mathrm{c}}(H'\mathcal{L}_K H - H\mathcal{L}_K H') = \mathcal{E}_K(w, HH') - 2\int H\,\Gamma_K(w,H')$, and $\Gamma_K(w,\cdot)$ vanishes off the shell. (d): by (C1) and Lem. 16.5,

$$\Big|\int\chi^{\mathrm{c}}F\mathcal{L}_K H - \int H\mathcal{L}_K F\Big| \le \sup|H|\,[2\|F\|_\infty\varkappa + (2\mathcal{E}_K(F)^{1/2} + \|\mathcal{L}_K F\|)\varkappa^{1/2}];$$

with a test sequence, $\int H^\sigma_{K_j}\mathcal{L}_{K_j}F_j\,\mathrm{d}\mu^{K_j} \to -\int(A_\star F)\mathsf{G}_\sigma\,\mathrm{d}\mu$ by (C3); the convergence of resolvents in (A3) gives $\alpha(\alpha - \mathcal{L}_{K_j})^{-1}W_j \to \alpha(\alpha + A_\star)^{-1}W$ strongly, bounded by $\sup_j\|W_j\|_\infty$, hence test sequences; c_F uses $\bar\varkappa \le \bar\varkappa^{1/2}$. The flow-time half: $\int\chi^{\mathrm{c}}F_j\partial_\sigma H^\sigma_{K_j}$ differs from $\partial_\sigma\int F_j H^\sigma_{K_j}$ by at most $\sup_j\|F_j\|_\infty C_\partial\bar\varkappa(R)^{1/2}$, and $\int F_j H^\sigma_{K_j} \to (F, \mathsf{G}_\sigma)$, locally Lipschitz in σ uniformly in j; this gives the identity of distributions.

Reversibility: take $F_j = \alpha(\alpha - \mathcal{L}_{K_j})^{-1}H'_{K_j}$, a test sequence for $F = \alpha(\alpha + A_\star)^{-1}\mathsf{G}'$, and note that $A_\star\alpha(\alpha + A_\star)^{-1}$ is self-adjoint. The criteria: for $F^\Lambda = \Lambda(\Lambda + A_\star)^{-1}\mathsf{G}_\sigma$ with its test sequence, $(A_\star F^\Lambda, \mathsf{G}_\sigma) = \int\frac{\Lambda\lambda}{\Lambda+\lambda}d\nu_{\mathsf{G}_\sigma}$ increases to $\mathcal{E}_\star(\mathsf{G}_\sigma)$ as $\Lambda\to\infty$, and is at most $\sup|\varphi|\,C_1$ after $R\to\infty$; in L^2 the same bound with Cauchy–Schwarz holds on a core of $A_\star$.

Step 9: lower energy bounds, (C6). By (Adm), let $\mathbb{P}_K$ be the normalised chain measure from the least top depth d_0 with $\mathsf{R}_{d_0} \le \frac12$ such that every top depth $\ge d_0$ is admissible, as in the head of Lem. 16.7. Take R with $\mathbb{P}_K(\|\mathbf{B}_K\|_{\kappa;a} > R) \le \frac14 p_*$ (by eq. 15.iii at $q = 1$ and Markov's

inequality, $R = \max\{1, 4C(1,\kappa)L^{20d_0}/p_*\}$), and d with $\theta_0 a_{d+1}^2 < \frac{3}{2}\sigma \le \theta_0 a_d^2$. On the complement $\{\|\mathbf{B}_K\|_{\kappa;a} \le R\}$ of this event and on the event of Lem. 16.7(a) at $s = \frac{3}{2}\sigma$, which meet in a $\mathbb{P}_K$-probability at least $\frac{3}{4}p_*$, Lem. 16.7(b) and the flow bounds of Prop. 16.1(a) give, for $\sigma \le \sigma_*$, where $\kappa < \frac{1}{6}$ makes the error of lower order,

$$\Gamma_K(A^a \circ \Phi_\sigma) \ge cR^{-2}\mathsf{b}^{-2}\ell^3\sigma^{-5/2+\kappa}, \qquad A^a \circ \Phi_\sigma \le S_*\sigma^{-3/2-\kappa},$$

on a closed invariant L_K with $\mathbb{P}_K(\mathsf{L}_K) \ge \frac{3}{4}p_*$, so $\mu^K(\mathsf{L}_K) \ge \frac{3}{8}p_*$ (Lem. 15.3(i), $\mathsf{R}_{d_0} \le \frac{1}{2}$); hence $\mu^K(\mathsf{L}_K \cap \{\chi_K = 1\}) \ge \frac{1}{4}p_*$, and $\Gamma_K(\varphi(A^a) \circ \Phi_\sigma) = \varphi'(A^a \circ \Phi_\sigma)^2\,\Gamma_K(A^a \circ \Phi_\sigma)$. The mean flowed action from below is the same argument at $s = \sigma$ with the last inequality of Lem. 16.7(b): $A^a \circ \Phi_\sigma \ge \frac{1}{8}k''_\Sigma \mathsf{b}^{-1}\ell^3\sigma^{-3/2}$, $k''_\Sigma = k_\Sigma\theta_0^{3/2}L^{-3}$, with μ^K-probability $\ge \frac{3}{8}p_*$, so $c_\circ = \frac{3}{64}p_*k''_\Sigma$.

From above, under μ^K and not only under a chain measure: on invariant functions the chain measure σ^K from a top depth d has a density $\bar{\mathfrak{w}} \in [0,1]$ with respect to μ^K, with $\int(1-\bar{\mathfrak{w}})\,\mathrm{d}\mu^K \le \mathsf{R}_d$ (Lem. 15.3(i)), so for invariant $F \ge 0$, by the Cauchy–Schwarz inequality, $\langle F\rangle_{\mu^K} \le \mathbb{E}^{[d]}[F] + \mathsf{R}_d^{1/2}\langle F^2\rangle_{\mu^K}^{1/2}$. Take $F = A^a \circ \Phi_\sigma$, with $\langle F^2\rangle_{\mu^K}^{1/2} \le \mathsf{C}\sigma^{-3/2-\kappa}$ by the moments (iv) of Prop. 16.1(a), under (Adm), and $d = d_\sigma$, the least depth $d \ge d_1$ with $\mathsf{R}_d \le \sigma^{2\kappa}$, an admissible top depth by (Adm); then the second member is at most $\mathsf{C}\sigma^{-3/2}$, and as R_d decays faster than every exponential in d, $L^{d_\sigma} = \sigma^{-o(1)}$ as $\sigma \to 0$.

Under $\mathbb{E}^{[d]}$, d admissible, the centred layers of all depths, with conditional covariances at most $Ca_{d'}^2/\mathsf{b}$, give $\mathbb{E}^{[d]}\|\mathrm{d}_a P_\sigma \mathbf{B}_K\|_2^2 \le c\,\mathsf{b}^{-1}\ell^3\sigma^{-3/2}$ against the heat kernel, with c free of d and K; the top field, the conditional means of the layers and the remainder of the logarithm add a lower power of σ times a number that grows at most like a power of L^d; on $\{\|\mathbf{B}_K\|_{\kappa;a} \le \sigma^{-\epsilon}\}$, $\epsilon > 0$ small, the last inequality of Lem. 16.7(b) passes from $\|\mathrm{d}_a P_\sigma \mathbf{B}_K\|_2^2$ to $2A^a \circ \Phi_\sigma$ at the cost of a lower power of σ; and off this event the bound $A^a \circ \Phi_\sigma \le c_4\ell^3(1 + \|\mathbf{B}_K\|_{\kappa;a})^{24}\sigma^{-3/2-\kappa}$ of Prop. 16.1(a) with eq. 15.iii at a large q costs at most $CL^{pd}\sigma^{-1}$.

So $\mathbb{E}^{[d]}[F] \le c\,\mathsf{b}^{-1}\ell^3\sigma^{-3/2} + CL^{pd}\sigma^{-3/2+\epsilon'}$ for large K, with some $\epsilon' > 0$, and at $d = d_\sigma$ the second member is $o(\sigma^{-3/2})$: with the bound $\mathsf{R}_{d_\sigma}^{1/2}\langle F^2\rangle_{\mu^K}^{1/2} \le \mathsf{C}\sigma^{-3/2}$ this gives $C_\circ$, for $0 < \sigma \le \sigma_{**}$. For $\vartheta < 1$ and for the probability that tends to 1: let $\psi \in C^\infty(\mathbb{R})$ take values in $[0,1]$. By Prop. 16.6(b) the square field of $\psi(\sigma^{3/2}A^a) \circ \Phi_\sigma$ is bounded on $\{\mathscr{N} \le R\}$ uniformly in K and in small σ, so by Prop. 16.6(d), that is, by the key estimate of Step 1, which rests on eq. 15.ii, the function $\chi'_K\,\psi(\sigma^{3/2}A^a \circ \Phi_\sigma)$, χ'_K the cutoff of Prop. 16.6(c), is within a distance in $L^2(\mu^K)$ that is small for large $\hat{d}_0$, uniformly in K and in small σ, of a function of the block field at depth $\hat{d}_0$; under (Adm), every $\hat{d}_0 \ge d_1$ is an admissible top depth, and $\mathsf{R}_{\hat{d}_0} \le \frac{1}{2}$ for large $\hat{d}_0$.

By the last clause of Lem. 16.7(a), at the top depth $\hat{d}_0$ in the place of d_0, off an event of probability τ_d the lower bounds hold, given the block field at depth $\hat{d}_0$, with a conditional probability that is bounded below by a positive number free of K, σ and $\hat{d}_0$; a function that is close to a function of the block field and exceeds a level with conditional probability bounded below exceeds a slightly smaller level with high probability. So the flowed action exceeds a fixed multiple of $\mathsf{b}^{-1}\ell^3\sigma^{-3/2}$ with a probability that tends to 1 as $\sigma \to 0$, uniformly in K; at the flow time $\frac{3}{2}\sigma$ this bounds $\|\mathrm{d}_a P_{3\sigma/2}\mathbf{B}_K\|_2^2$ from below, by the last inequality of Lem. 16.7(b), and the first inequality there, on $\{\|\mathbf{B}_K\|_{\kappa;a} \le R\}$ for a large R, gives the same for $\Gamma_K(A^a \circ \Phi_\sigma)$ and $\sigma^{-5/2+\kappa}$.

By Markov's inequality and the upper bound of the mean flowed action, $\mu^K(A^a \circ \Phi_\sigma > S_*\sigma^{-3/2-\kappa}) \le 2C_\circ\mathsf{b}^{-1}\ell^3\sigma^\kappa/S_*$ for $0 < \sigma \le \sigma_{**}$ and all large K, which tends to 0 with σ. Given $\vartheta < 1$, choose σ_ϑ so that the probability of these lower bounds, together with $A^a \circ \Phi_\sigma \le S_*\sigma^{-3/2-\kappa}$, exceeds $\frac{1}{2}(1+\vartheta)$ for all large K; the set $\{\chi_K = 1\}$, of mass $\ge 1-\vartheta$, meets this event in a mass $\ge \frac{1}{2}(1-\vartheta)$. The current: $\Gamma_K(A^a \circ \Phi_\sigma)(U) \le e^{24\Lambda_\sigma(U)}(J,J)_a(U_\sigma)$ by Lem. 16.2(b), since $\Gamma_K(A^a) = (J,J)_a$; by the tightness of Λ_σ (Prop. 16.1(a), under (Adm)) there is λ with $\mu^K(\Lambda_\sigma > \lambda) \le \frac{1}{8}p_*$ for large K, and on $\mathsf{L}_K \cap \{\Lambda_\sigma \le \lambda\}$, of mass $\ge \frac{1}{4}p_*$, $(J,J)_a(U_\sigma) \ge e^{-24\lambda}cR^{-2}\mathsf{b}^{-2}\ell^3\sigma^{-5/2+\kappa}$.

For small σ the same property of the current follows by a second route from the lower bound of the mean flowed action, through Prop. 16.9(d)(iii); and, by Prop. 16.9(d)(i) it gives back, conversely, the positivity in the limit of the energies $\int \chi_K \Gamma_K(\varphi(A^a)\circ\Phi_\sigma)\,\mathrm{d}\mu^K$, φ as there, for every weight with $\limsup_K \mu^K(\chi_K<1) < \frac14 p_*$.

Step 10: growth of uncut generator images, (C7). Let $U \in \mathsf{O}_K(\bar C,\bar s)$, of μ^K-mass $\ge 1-\epsilon$ (Prop. 16.1(a), by (Adm)). For large K its flow in lattice time satisfies the hypotheses of Lem. 16.4 with $T=\sigma/a_K^2$: by Lem. 16.2(e) at the flow times $s \ge a_K^{9/4}$, where $\kappa<\frac16$ gives $\varpi(U_s)\le (2\bar C+2\bar C^2)a_K^{1/8}$, and, at smaller flow times, by the bound 16 of the speed of the plaquette variables in lattice time, $\sup_{t\le T}\varpi(\Phi_{a_K^2 t}U)\le (4\bar C^2+20)a_K^{1/8}$, $\int_0^T \varpi\,\mathrm{d}t=\Lambda_\sigma$ is bounded, and the action at $T/2$ is small.

Given the other bonds, the law of the three bonds of Lem. 16.4 under μ^K has a density $\ge e^{-20\beta}$ with respect to Haar measure, as they lie in ten plaquettes, and $20\beta = 20\mathsf{b}\,a_K T/\sigma = o(T)$. So, with $\varepsilon=0.002$,

$$\int |\partial(A_K\circ\Phi_\sigma)|^q\,\mathrm{d}\mu^K \ge e^{-20\beta}(1-\epsilon)\,c\,e^{[(q-3)\lambda_1-6\lambda_2-(q+1)\varepsilon]T},$$

and the enclosures of λ_1,λ_2 give (a); at $q=4$, $\Gamma_K(A^a\circ\Phi_\sigma)^2 = a_K^{-6}|\partial(A_K\circ\Phi_\sigma)|^4$ and $\lambda_1-6\lambda_2>1.8$. (b): $G=A^a\circ\Phi_\sigma$ takes its values in $[0,6\ell^3 a_K^{-4}]$, as $A_K\le 6N^3$, and $H=\arctan G$ has $|\partial H| = a_K^{-1}|\partial(A_K\circ\Phi_\sigma)|/(1+G^2)$; so (a) at $q=4$ gives $\int|\partial H|^4\,\mathrm{d}\mu^K \ge c(\ell)\,a_K^{28}e^{1.76\,\sigma/a_K^2}$. Lem. 16.2(g) with $\sup|H|\le\frac\pi2$ gives $\frac\pi2(3\mathsf{m}+14\beta^{1/2}\mathsf{m}^{1/2}) \ge c(\ell)^{1/2}a_K^{14}e^{0.88\,\sigma/a_K^2}$; the right side tends to ∞ and $\beta^{1/2}=(\mathsf{b}/a_K)^{1/2}$, so $\mathsf{m}\ge 1$ for large K and then $\mathsf{m}\ge c\,a_K^{14.5}e^{0.88\,\sigma/a_K^2}$; and $\|\mathcal{L}_K H\| = \mathsf{m}/a_K \ge e^{0.8\,\sigma/a_K^2}$ for large K.

(c): $f=\mathfrak{f}^2$ with $\mathfrak{f}(x)=\sqrt2|\sin(\pi x_1/\ell)|$, of Lipschitz constant $\sqrt2\pi/\ell$. Let b_0 be the bond in the direction 3 at the site $a_K(k,0,0)$, $k\ge1$ an integer fixed below, $\alpha=2\pi a_K/\ell$, $\mathsf{f}_p=f(x(p))$ and $\mathsf{f}_0=1-\cos(\alpha k)\ge\frac{2}{\pi^2}\alpha^2k^2$. For p at the distance j from b_0, $|\mathsf{f}_p-\mathsf{f}_0|\le\alpha^2(j+3)(k+j+3)$, so $(\sum_p|\mathsf{f}_p-\mathsf{f}_0|^2e^{-2\theta|p|})^{1/2}\le C(\theta)\alpha^2(k+1)$; fix k with $C_\sharp C(\theta)(k+1)\le c_\sharp k^2/\pi^2$. Then Lem. 16.4(b) gives $|\partial_{b_0}(\mathcal{O}(f)\circ\Phi_\sigma)(W_\zeta)|\ge a_K^{-1}\frac{c_\sharp}{\pi^2}\alpha^2k^2e^{(\lambda_1-\varepsilon)T}$ for $\zeta\in\mathcal{S}$.

The value: with $\tilde V_t$ of Lem. 16.4 and $\mathcal{A}_f(V)=\sum_p \mathsf{f}_p\varsigma_p a_p(V)$,

$$\mathcal{O}(f)(\Phi_\sigma W_\zeta) = a_K^{-1}\Big[\mathcal{A}_f(\tilde V_T) + 2\sum_{p\ni b_0}\mathsf{f}_p\Big]\ge 0,$$

and the second term is at most $4a_K^{-1}\alpha^2(k+3)^2$, which tends to 0. For the first, a local energy inequality holds along the flow of $\tilde A$ in lattice time: with $\nu=-\partial\tilde A(\tilde V_t)$ its velocity, $f_b=f(a_Kb_-)$ and $a'=3a_K\sqrt2\pi/\ell$,

$$\tfrac{d}{dt}\mathcal{A}_f(\tilde V_t) = -\sum_b f_b|\nu(b)|^2 + \sum_b\Big\langle\nu(b),\sum_{p\ni b}(\mathsf{f}_p-f_b)\varsigma_p\,\partial_b a_p(\tilde V_t)\Big\rangle \le 64a'^2\sum_p a_p(\tilde V_t),$$

since $|\mathsf{f}_p-f_b|\le a'(2|\mathfrak{f}(a_Kb_-)|+a')$ for $b\in p$, $|\partial_b a_p|\le\varpi_p$, $|\nu(b)|\le\sum_{p\ni b}\varpi_p$, and $\sum_b(\sum_{p\ni b}\varpi_p)^2\le 32\sum_p a_p$. So $\mathcal{A}_f(\tilde V_T)\le\mathcal{A}_f(\tilde V_t)+64a'^2\int_t^T\sum_p a_p(\tilde V_u)\,\mathrm{d}u$ for $t\le T$; the mean over $t\in[\frac T2,T]$, $|\mathcal{A}_f|\le 2\sum_p a_p$, Lem. 16.4(c) and $A_K(U^\circ_{T/2})\le a_KE^{\mathrm{a}}_\sigma$ (Lem. 16.2(c), (e)) give

$$\mathcal{A}_f(\tilde V_T)\le\Big(\frac4T+64a'^2\Big)\Big(\frac32 T\,a_KE^{\mathrm{a}}_\sigma+C_\sharp\Big)\le a_K\Big(\frac4\sigma+\frac{11400}{\ell^2}\Big)\Big(\frac32\sigma E^{\mathrm{a}}_\sigma+C_\sharp a_K\Big),$$

as $64a'^2=1152\pi^2a_K^2/\ell^2\le 11400\,a_K^2/\ell^2$. So $0\le\mathcal{O}(f)\circ\Phi_\sigma\le M_\sigma$ on the fields W_ζ, $\zeta\in\mathcal{S}$, for large K, and $\int|\partial\varphi(\mathcal{O}(f)\circ\Phi_\sigma)|^4\,\mathrm{d}\mu^K\ge(\inf_{[0,M_\sigma]}|\varphi'|)^4e^{-20\beta}(1-\epsilon)\,c\,a_K^4e^{(\lambda_1-6\lambda_2-5\varepsilon)T}$; Lem. 16.2(g) as in (b). (d): the last identity of Lem. 16.2(g) gives $|\mathcal{L}_K\cos G|+|\mathcal{L}_K\sin G|\ge\Gamma_K(G)$, and (a) at $q=2p$, whose exponent $(2p-3)\lambda_1-6\lambda_2$ is positive for $p>\frac32+3\lambda_2/\lambda_1$, which is less than 1.8637; for $p>4$ the L^p norm dominates the L^4 norm.

(e): by Lem. 16.2(g) and $\nabla^2 A_K \geq -24$, $a_K^2 M^2 + 24\mathsf{b}\, M \geq \int \|\nabla^2 H\|_{\mathrm{HS}}^2 \,\mathrm{d}\mu^K$ for $M = \|\mathcal{L}_K H\|$ and $\sup|H| \leq 1$, since $\int |\partial H|^2 \,\mathrm{d}\mu^K = -a_K \int H \mathcal{L}_K H \,\mathrm{d}\mu^K$. Let $H = \sin(\mathcal{O}(f) \circ \Phi_\sigma)$; as sin is odd we may replace f by $-f$, so that $f(x_*) = \max|f| > 0$ at some x_*, and we base b_0 at a site nearest to x_*. For large K every $U \in \mathsf{O}_K(\bar{C}, \bar{s})$ satisfies the hypotheses of Lem. 16.4(d) with $\Lambda_0 = \max\{1, \sigma/\ell^2 + (8\bar{C}\sigma^{\frac{1}{4}-\frac{\kappa}{2}} + 4\bar{C}^2\sigma^{\frac{1}{2}-\kappa})/(1-2\kappa)\}$ (Lem. 16.2(e)): the supremum of ϖ along the flow tends to 0 as above, $T/N^2 = \sigma/\ell^2$, and $N^3 T^{-2} = \ell^3 a_K \sigma^{-2} \to 0$.

The fibre $g \mapsto W^{(g)}$ of the bond b_0 is totally geodesic in Ω_K, so $\mathfrak{h}(g) = H(W^{(g)})$ has $\|\nabla^2_{S^3}\mathfrak{h}\|_{\mathrm{HS}}(g) \leq \|\nabla^2 H\|_{\mathrm{HS}}(W^{(g)})$. For $(y_1, y_2) \in \mathsf{Y}$ of Lem. 16.4(d), the function $r \mapsto \mathcal{O}(f)(\Phi_\sigma W^{(g_r)}) = a_K^{-1}\mathsf{v}_f(r)$ falls by at least $f(x_*)\rho_2^2/a_K \geq 2\pi$ on $[0, r_*]$ for large K, so $\mathfrak{h}$ takes the values 1 and -1 at two points g', g'' with $\mathsf{d}(g', g'') \leq r_* \leq \frac{4}{3}\rho_2 e^{-\lambda_1' T}$; the inequalities of Sobolev and Morrey on S^3 give $|\mathfrak{h}(g') - \mathfrak{h}(g'')| \leq C\|\nabla^2_{S^3}\mathfrak{h}\|_{L^2(S^3)}\mathsf{d}(g', g'')^{1/2}$, so $\int \|\nabla^2 H\|_{\mathrm{HS}}^2(W^{(g)})\,\mathrm{d}g \geq c\, e^{\lambda_1' T}$ for the Haar probability measure dg.

The set Y has Lebesgue measure $\geq c_{\mathrm{d}} e^{-6\lambda_2'' T}$, the Haar measure in the chart $e^{iy_j} U(b_0 + \hat{e}_j)$ has a density $\geq (4\pi^2)^{-1}$ for $|y_j| \leq 1$, and given the other bonds the law of the three bonds has a density $\geq e^{-20\beta}$; so $\int \|\nabla^2 H\|_{\mathrm{HS}}^2 \,\mathrm{d}\mu^K \geq c(1-\epsilon)\exp[(\lambda_1' - 6\lambda_2'')T - 20\beta]$, with $\lambda_1' - 6\lambda_2'' > \frac{2}{5}$. This gives $a_K^2 M^2 + 24\mathsf{b} M \geq c\exp[\frac{2}{5}\sigma/a_K^2 - 20\mathsf{b}/a_K]$, hence $M \geq \exp[\frac{\sigma}{5a_K^2} - \frac{10\,\mathsf{b}}{a_K} - C]$ for sin and every $f \not\equiv 0$. For a non-constant periodic φ the same steps hold with φ in the place of sin (and $\varphi(-\,\cdot\,)$ in the place of φ if f was replaced by $-f$): for large K the fall $f(x_*)\rho_2^2/a_K$ exceeds a period of φ, so that φ takes its maximum and its minimum on the fibre within the distance r_*, $\operatorname{osc}\varphi$ replaces 2, and $\sup|\varphi|$ replaces 1 in the first inequality; this gives the exponent $\sigma/20a_K^2$ for large K.

(f): on the fields W_ζ of Lem. 16.4 the four plaquettes at b_0 stay within $\delta \leq \frac{1}{10}$ of $-\mathbf{1}$ during the flow, so $\Lambda_\sigma(W_\zeta) \geq (2-\delta)\sigma/a_K^2$, while their mass is at least $(1-\epsilon)e^{-20\beta} c_\sharp e^{-(3\lambda_1 + 6\lambda_2 + \varepsilon)\sigma/a_K^2}$; and $13(2-\delta) \geq 24.7 > 3\lambda_1 + 6\lambda_2 + \varepsilon$. □

15.1.8. *Uniformity and the order of limits.* Thm. AT holds for every limit, with constants and rates that do not depend on the subsequence. The inequalities of (A4), (A5) and (B2)(c) carry a defect, and their constants depend on ℓ; the constants of Thm. AU that are free of the volume are those of the fine field given a block field: C_e^{fib} and $3\mathsf{M}^2\Pi_e^\flat$ in (B3), in the log-Sobolev inequality of P_W, in the pointwise bound of $\Gamma_e(g_{t'})$ and in the tail under P_W, and the same two numbers at $e = n$ in (B2)(e), whose first inequality is (B3) at $t' = K - n$ integrated over the top law by the chain rule of the entropy.

The tail under μ^K in (B3) adds R_n; the first inequality of (B2)(e) keeps the entropy of the top law, which (B1) bounds at the cost C_n^{t}; and the constants, defects and thresholds of (B1) and (B2)(a)–(c), C_n^{t}, $C_n^\flat$, C_n', $\mathsf{R}_\cdot$, r_n, n_{c}, depend on ℓ. The flowed observables are taken at a fixed positive flow time; the proposed conclusion (C5) would identify the generator on them as a distribution once the flow-approximation step is completed, in the tested outer-R, inner-$\limsup_{K\to\infty}$ error sense of (C5)(d), and (C7) shows that without cutoffs the images under $\mathcal{L}_K$ of some of them are not bounded uniformly in K: for $p > 1.8637$, $\sup_K[\|\mathcal{L}_K \cos(A^a \circ \Phi_\sigma)\|_{L^p(\mu^K)} + \|\mathcal{L}_K \sin(A^a \circ \Phi_\sigma)\|_{L^p(\mu^K)}] = \infty$, so the norm is unbounded at cos or at sin, and at $\cos(\cdot + \theta)$ for all $\theta \in [0, \pi)$ but at most one, since cos and sin are linear combinations of any two of these functions.

15.2. **A uniform gauge on a box.** A comb gauge gives a bond bound proportional to the side of the box. To obtain a bound independent of that side, divide the box into cells of side comparable to the inverse square root of the plaquette bound. First fix the gauge on their edges. Extend it over each face and then each cell, keeping its already prescribed boundary values. For $SU(2)$, the required extensions can be constructed quantitatively on the unit sphere in the quaternions.

The following constants will also fix the numerical conventions in Lem. 15.7:

$$\lambda_{\mathrm{BG},1} = \pi + 2, \qquad \rho_{\mathrm{BG},1} = 1/250, \qquad \lambda_{\mathrm{BG},2} = 2 + 12\sqrt{3}\lambda_{\mathrm{BG},1}\rho_{\mathrm{BG},1}^{-2},$$
$$\lambda'_{\mathrm{BG},2} = \lambda_{\mathrm{BG},2} + 4, \qquad \rho_{\mathrm{BG},2} = (250000\lambda'^{2}_{\mathrm{BG},2})^{-1}, \tag{15.v}$$
$$\lambda_{\mathrm{BG},3} = 4 + 12\sqrt{3}\lambda'_{\mathrm{BG},2}\rho_{\mathrm{BG},2}^{-2}, \qquad C_{\mathrm{BG}} = 2\lambda_{\mathrm{BG},3}.$$

Proposition 15.8. Let U be an $SU(2)$ configuration on the bonds of $\{0,\dots,n\}^3$, where $n \geq 0$. If $0 < \kappa \leq C_{\mathrm{BG}}^{-2}$ and $|\partial U(p) - \mathbf{1}| \leq \kappa$ at every plaquette, then some gauge u satisfies

$$|U^u(b) - \mathbf{1}| \leq C_{\mathrm{BG}}\sqrt{\kappa} \quad \text{at every bond.} \tag{15.vi}$$

The same conclusion holds on $(\mathbb{Z}/n\mathbb{Z})^3$ if $n \geq \kappa^{-1/2}$. ■

Proof. We use the operator norm, which on quaternions is their Euclidean norm, and identify $SU(2)$ with the unit sphere S^3. Unitary multiplication preserves this norm. In particular, reversing a bond does not change its distance from $\mathbf{1}$.

Avoiding a point of the sphere. If a subset $S \subset S^3$ is covered by m closed balls of radius $0 < \rho < 1/8$ and $m < (8\rho)^{-3}$, there is $q \in S^3$ such that $|z + q| > \rho$ for every $z \in S$. Indeed, put $k = \lceil (8\rho)^{-1} \rceil - 1 \geq 1$. The points

$$p_v = \frac{(1, v)}{|(1, v)|}, \qquad v \in \{-1 + 2a/k : 0 \leq a \leq k\}^3,$$

number $(k+1)^3 \geq (8\rho)^{-3} > m$. For distinct grid points v, w, the wedge product of $(1, v)$ and $(1, w)$ has norm at least $|v - w|$, whereas both vectors have norm at most 2. Consequently

$$|p_v - p_w| \geq \sin\angle((1, v), (1, w)) \geq \tfrac{1}{4}|v - w| \geq \frac{1}{2k} > 4\rho.$$

Each of the m ball centres is within distance 2ρ of at most one p_v. Choose a remaining p_v and put $q = -p_v$.

Extending a map which avoids that point. Let Y be a nonempty subset of a finite metric space (X, d). Suppose $h : Y \to S^3$ is a-Lipschitz and $|h + q| \geq \rho$. Then it extends to $H : X \to S^3$ with Lipschitz constant $4\sqrt{3}\rho^{-2}a$. To see this, use the inversion

$$I_q(z) = -q + 2\frac{z + q}{|z + q|^2}, \qquad |I_q(z) - I_q(w)| = \frac{2|z - w|}{|z + q|\,|w + q|}.$$

It is its own inverse, maps $S^3 \setminus \{-q\}$ to $q^\perp$, and maps $q^\perp$ to S^3. Its Lipschitz constants on the two sets in use are $2\rho^{-2}$ and 2, respectively. Each coordinate f of $I_q h$ in an orthonormal basis of $q^\perp$ extends by

$$F(x) = \min_{y \in Y}\{f(y) + 2\rho^{-2}a\, d(x, y)\}.$$

The Lipschitz inequality for f shows that $F = f$ on Y; the triangle inequality shows that F has the same Lipschitz constant. The three coordinate extensions together have Lipschitz constant $2\sqrt{3}\rho^{-2}a$. Composition with I_q gives H.

The cell geometry and the comb. Put $N = \lfloor \kappa^{-1/2} \rfloor$. Then

$$N^2\kappa \leq 1, \qquad N\kappa \leq N^{-1}, \qquad N \geq \lambda_{\mathrm{BG},3}, \qquad N^{-1} \leq 2\sqrt{\kappa}.$$

For a coordinate rectangle or box with side lengths between N and $2N - 1$, any two boundary sites are joined along its boundary by a path of length at most three times their ℓ^1 distance. For sites on one face, or on intersecting faces, a monotone path has exactly that distance. For opposite faces, use either one of the two intervening faces. If their separating direction is μ and the chosen other direction is ν, the shorter detour costs at most $n_\nu + n_\mu$ in those directions; this is at most $3n_\mu$, since

$n_\nu < 2n_\mu$. The remaining coordinate contributes its ordinary difference. The rectangle proof is the same argument with two coordinates.

In a cell with lowest corner y, let $r_y(x)$ be the transport from y to x, first in direction 1, then 2, then 3. For $b = \langle x, x + e_\mu\rangle$ put $m_b = r_y(x)U(b)r_y(x + e_\mu)^{-1}$. Moving the last step through the later directions in the comb exchanges one plaquette at a time. The holonomy of a strip of k plaquettes is a product of k unitary conjugates of those plaquettes, so its distance from $\mathbf{1}$ is at most $k\kappa$. Thus, writing $x - y = (k_1, k_2, k_3)$,

$$|m_b - \mathbf{1}| \le \begin{cases} (k_2 + k_3)\kappa, & \mu = 1, \\ k_3\kappa, & \mu = 2, \\ 0, & \mu = 3. \end{cases}$$

This is the comb calculation of Lem. YM3_I:14.4, with the coordinate order reversed. It proves the estimate on an ordinary coordinate box and on the lift of a periodic cell, without any condition on the ambient torus side. On a face the bound is $2/N$, and on a cell it is $4/N$. If $n < N$ in the nonperiodic case, the same comb on the whole box gives $2n\kappa \le 4\sqrt{\kappa}$, which proves the assertion.

Suppose now $n \ge N$. Divide each coordinate interval into lengths between N and $2N - 1$: take $t_i = iN$ for $0 \le i < \lfloor n/N\rfloor$ and the last endpoint n. Their products give nodes, edges, faces and cells. Each new piece meets the union of lower-dimensional pieces exactly in its boundary; different interiors are disjoint. Every bond is contained in a cell. We can therefore define the gauge successively on nodes, edges, faces and cells.

Edges. Set $u = \mathbf{1}$ at the nodes. On an edge from y to $y' = y + n_c e_\mu$, write $U([y, y']) = e^{iY}$ with $|Y| \le \pi$, and set

$$u(y + ke_\mu) = e^{-ikY/n_c}U([y, y + ke_\mu]), \qquad 0 \le k \le n_c.$$

Both endpoint values are $\mathbf{1}$. At an edge bond the transformed field is e^{iY/n_c}, so its distance from $\mathbf{1}$ is at most π/N.

For the next two steps, write $h = ur_y^{-1}$ on the prescribed part of a piece and $w_b = U^u(b)$. Since $h(b_-)m_b h(b_+)^{-1} = w_b$, we have

$$\begin{aligned} |h(b_+) - h(b_-)| &\le |w_b - \mathbf{1}| + |m_b - \mathbf{1}|, \\ |w_b - \mathbf{1}| &\le |m_b - \mathbf{1}| + |h(b_+) - h(b_-)|. \end{aligned} \tag{15.vii}$$

Faces. On the boundary of a face, the bond differences of h are at most $\lambda_{\mathrm{BG},1}/N$; the boundary-path bound makes h $3\lambda_{\mathrm{BG},1}/N$-Lipschitz in ℓ^1 distance. The boundary cycle has at most $8N$ bonds. Sample it at intervals $s_1 = \lfloor \rho_{\mathrm{BG},1}N/\lambda_{\mathrm{BG},1}\rfloor$. The constants ensure $s_1 \ge \rho_{\mathrm{BG},1}N/(2\lambda_{\mathrm{BG},1}) \ge 1$. The image is covered by balls of radius $\rho_{\mathrm{BG},1}$ whose number is at most

$$16\lambda_{\mathrm{BG},1}/\rho_{\mathrm{BG},1} + 1 \le 17\lambda_{\mathrm{BG},1}/\rho_{\mathrm{BG},1} < 21853 < (8\rho_{\mathrm{BG},1})^{-3}.$$

The avoidance and extension arguments give an extension H over the face with bond differences bounded by $12\sqrt{3}\lambda_{\mathrm{BG},1}\rho_{\mathrm{BG},1}^{-2}/N$. Set $u = Hr_y$ there. It retains the prescribed edge values, and eq. 15.vii gives $|U^u(b) - \mathbf{1}| \le \lambda_{\mathrm{BG},2}/N$ on every face bond.

Cells. On the boundary of a cell, the bond differences of h are at most $\lambda'_{\mathrm{BG},2}/N$, and its ℓ^1 Lipschitz constant is $3\lambda'_{\mathrm{BG},2}/N$. On each face sample both coordinates at multiples of $s_2 = \lfloor \rho_{\mathrm{BG},2}N/(2\lambda'_{\mathrm{BG},2})\rfloor$. Here $s_2 \ge \rho_{\mathrm{BG},2}N/(4\lambda'_{\mathrm{BG},2}) \ge 1$. Lowering the two coordinates to these multiples uses fewer than $2s_2$ steps, so the image lies within $\rho_{\mathrm{BG},2}$ of a sampled value. The number of samples on all six faces is at most

$$m = 486\lambda'^2_{\mathrm{BG},2}\rho_{\mathrm{BG},2}^{-2}, \qquad m(8\rho_{\mathrm{BG},2})^3 \le 248832/250000 < 1.$$

The same extension gives H over the cell, with bond differences at most $12\sqrt{3}\lambda'_{\mathrm{BG},2}\rho_{\mathrm{BG},2}^{-2}/N$. Again set $u = Hr_y$. The boundary values agree, and eq. 15.vii gives

$$|U^u(b) - \mathbf{1}| \le \lambda_{\mathrm{BG},3}/N \le C_{\mathrm{BG}}\sqrt{\kappa}.$$

Every bond belongs to a cell, proving the box assertion.

The periodic lattice. Divide each coordinate circle of length n into the same integer lengths, now identifying its last endpoint with its first. Work on each piece in its coordinate lift. Put $u = \mathbf{1}$ at the periodic nodes and perform the edge, face and cell constructions just given once for each periodic piece. The edge formula agrees at both endpoints even when an edge is a full circle. At every later stage the boundary values are the values of the already defined periodic function u. The auxiliary function $h = ur_y^{-1}$ is defined on the lifted boundary; it need not have equal values on opposite lifted faces, since the comb transports there may differ. After extension, however, Hr_y equals the prescribed u at every boundary point. Thus those values agree at all identified points. No new interior point has two preimages, even when a cell spans an entire coordinate circle. The extensions therefore descend to a single gauge on the torus. Every periodic bond has a lift in one cell, with the same plaquette and comb bounds as above. This proves the periodic assertion. □

For completeness, this also gives the first component assertion of Lem. 15.7(a). Let $0 < \kappa' \le (8\pi C_{\mathrm{BG}})^{-2}$ and use the periodic gauge. Write $V^u = e^{iA}$, choosing the principal logarithm at positive bonds and extending by reversal. Then

$$|A| \le \alpha := \tfrac{\pi}{2} C_{\mathrm{BG}}\sqrt{\kappa'} \le 1/16.$$

For the product of four exponentials around a plaquette, Taylor's formula in t gives $|\partial e^{itA}(p) - \mathbf{1} - it\,\mathrm{d}A(p)| \le 8\alpha^2$: the norm of its second derivative is bounded by $(\sum_{b\in\partial p}|A(b)|)^2 \le 16\alpha^2$. Here $dA(p)$ is the unnormalised oriented sum around p. In particular, the bound $10\alpha^2$ used in Lem. 15.7 holds. Applying it at t and at 1 yields

$$|\partial e^{itA}(p) - \mathbf{1}| \le |\partial V^u(p) - \mathbf{1}| + 20\alpha^2 \le (1 + 5\pi^2 C_{\mathrm{BG}}^2)\kappa' < C''\kappa', \qquad 0 \le t \le 1.$$

The gauge group is a finite product of connected copies of $SU(2)$, so a path of gauge transformations joins V to V^u without changing its plaquette norms. Joining this path to e^{itA} places V in the component of $\mathbf{1}$ in $\{\varpi < C''\kappa'\}$. Taking $\kappa' = r/C''$ gives the shell assertion there. This single-scale construction does not supply the separate multilevel estimate (Multiscale gauge).

15.3. **Response and one-step conventions.** We record the response bound needed when differentiating the normalised measure of a step. Work on a constant torus sequence, at level $j \ge 1$, with the real regular background, geometric thresholds and datum ball of §5.3. Lengths in the following lemma are in level-j units, so the bare spacing is $\eta_j = L^{-j}$ and every terminal pillar has scale 1. Let d_j be the periodic ℓ^1 distance on the terminal lattice. In this setting it is the weighted path distance in equation YM3_I:(838). For a field X on its positive bonds put

$$m_y(X) = \sum_{c>0} e^{-\delta_0 d_j(y,c_-)}|X(c)|, \qquad S_0 = \left(\frac{1+e^{-\delta_0}}{1-e^{-\delta_0}}\right)^3, \qquad \mathsf{S} = 3S_0,$$

where $\delta_0 > 0$ is no larger than the rate in that equation.

Lemma 15.9. There is a constant B, depending only on the fixed construction constants, such that the response $\mathcal{H}_U$ obeys

$$\max\{|\mathcal{H}_U(X)(b)|, |\nabla_U\mathcal{H}_U(X)(b)|\} \le \mathsf{B}m_y(X)$$

on its datum ball, whenever b has an endpoint in the pillar of y. At zero datum, put

$$h = D\mathcal{H}_U(0)X, \qquad j_X = \frac{d}{ds}J^{(j)}(e^{i\eta_j\mathcal{H}_U(sX)}U)\Big|_{s=0}.$$

Then

$$\max\{|h(b)|, |\nabla_U h(b)|, |j_X(b)|\} \le \mathsf{B} m_y(X), \qquad \sum_y m_y(X)^2 \le \mathsf{S}^2 \sum_{c>0} |X(c)|^2. \tag{15.viii}$$

All constants are independent of j and of the torus side. ■

Proof. For one input bond, equation YM3_I:(838) bounds h, $\nabla_U h$ and $D_U^{\mathsf{T}} D_U h$ with exponential decay. Sum this bound over the input bonds. Integrating the same estimate along sX, $0 \le s \le 1$, proves the first assertion, since $\mathcal{H}_U(0) = 0$ and the datum ball is convex.

The current requires one further identity. Let $F(A)$ be the local Wilson action at $e^{i\eta_j A}U$ and set $f(z) = (e^z - 1)/z$, with its power-series value at zero. Prop. YM3_I:13.1 gives, on a complete plaquette star,

$$J^{(j)}(e^{i\eta_j A}U)(b) = f(-i\eta_j \operatorname{ad}_{A(b)})^{-1} F'(A)(b), \qquad F''(0) = D_U^{\mathsf{T}} D_U + \Delta'.$$

Since $f(z)^{-1} = 1 - z/2 + O(z^2)$, differentiation at $A = 0$ gives

$$j_X(b) = (D_U^{\mathsf{T}} D_U h)(b) + (\Delta' h)(b) + \tfrac{i\eta_j}{2}[h(b), J^{(j)}(U)(b)].$$

All stars are complete on the torus. The same proposition bounds $\Delta' h$ by $C\varepsilon$ times the supremum of h on this fixed star, where ε is the background regularity radius. Regularity also gives $|J^{(j)}(U)| \le \varepsilon$. The star meets only pillars at a fixed distance from y; translating the exponential weight over that distance costs a fixed factor. Together with $\eta_j \le 1$ and $|[h, J]| \le 2|h||J|$, the three terms therefore have the asserted bound, after increasing B.

Finally the sum of $e^{-\delta_0 d_j(y,z)}$ over a periodic lattice is at most S_0: choose least coordinate representatives and bound their sum by the sum over $\mathbb{Z}^3$. There are three positive bonds starting at each site. The nonnegative kernel defining m_y has row sums at most $3S_0$ and column sums at most S_0. Cauchy–Schwarz, first with this kernel and then after summing in y, gives an operator norm at most $\sqrt{3}S_0 \le \mathsf{S}$. □

The gauge equation used with this lemma is exactly equation YM3_I:(837). Thus the projection denoted by R^{La} in the dynamics section is the companion's R_{m}: the projection onto $D_U^{\mathsf{T}} D_U \mathcal{N}$, where $\mathcal{N}$ consists of site fields with zero prescribed site averages. This is the mean-Landau choice of Thm. YM3_I:K.

15.3.1. *The local input and its global sum.* At height k, the actual input is the one in Section YM3_I:52.4 and Thm. YM3_I:W. Keep its geometric thresholds, admissible radius system, shift hypotheses, octet condition when $k \ge 2$, numerical ceilings and array bound $N \le N_*$. In particular, $E_0 > 0$ is the fixed size of the output family there; it is chosen before the height, lattice and input array. For the density identity, keep in addition the whole-data response neighbourhood, full-torus core membership and bounded Borel integrand required in that theorem.

A family in the fourth place means the fourth potential of Def. YM3_I:52.1. Its members depend on the represented pair, are gauge invariant, vanish at $B = 0$, and are jointly holomorphic on each core slice and the field polydisc $|B| < C_{\mathcal{A}}\mathfrak{b}(\mathsf{t})$. With additional parameters, joint holomorphy is required in those parameters too. Its size is equation YM3_I:(888). For a parameter space consisting of one point this is the same definition without a parameter.

The phrase that this family *sums to S at W* has the following precise meaning. The averaged background is real, admissible, and has invertible pivots; the real cutoff chart is defined throughout its box, every background over a point of that chart exists, and P_W belongs to the full-torus core. The gauge-invariant function S is defined on both critical orbits. For every real cutoff field,

$$\sum_Y \mathbb{V}'(Y; P_W|_Y, B|_Y) = S(U_k(V_W(B))) - S(U_{k+1}(W)). \tag{15.ix}$$

These are the background and chart conditions in §5.3.1. For a bare-array potential the identity is equation YM3_I:(882). For a source it is the additional sum condition imposed at its entry into the step.

The block field of a triple is $V[P] = \mathcal{M}^k(e^{i\eta_{k+1}A'}U)$ on the level-k bonds of its domain, with the actual complex averaging domains retained. The fine-cube gauge form $\widehat{\mathsf{G}}_\square(P)$ is evaluated at this field by equation YM3_I:(823). Equivalently it is the quadratic form obtained from $\mathfrak{t}\Psi^{\mathrm{G}}(\square; P, B)$ in the proof of Prop. YM3_I:51.4. Def. YM3_I:52.1, item 2, records its pair dependence, slice holomorphy, covariance and compatibility with restrictions; its sum on the full torus is equation YM3_I:(828). At real data the inverse whole covariance differs from its real reference form, with floor γ_{r} and ceiling c_{r}, by at most $\frac{1}{2}u_{\mathrm{G}}(LM)^{-3}$. These are the actual constants of Thm. YM3_I:U, used with the choice $u_{\mathrm{G}} = 4C_{\mathrm{Str}}$ and the radius inequalities of Section YM3_I:52.4. Completeness is Def. YM3_I:47.2, condition (St4).

The expansion numbers $w_0, \mathsf{C}_{\mathrm{ex}}, c_G, c_{\mathrm{v}}, c_{\mathrm{w}}, \lambda_1, \lambda_V, w_\star$ are those defined immediately before and in equation YM3_I:(705); in particular $\mathsf{C}_{\mathrm{ex}} \geq 8e^8$. Their one-step substitution is equation YM3_I:(889), including $\lambda_{\mathrm{out}} = (1 + 4\beta_{\mathrm{s}})\kappa$ and $(LM)^3 u = 3 + c_{\mathrm{r}} + u_{\mathrm{G}}$.

16. Positive-time gradient flow

The deterministic Wilson flow Φ_σ decreases the lattice action. It is related to the Langevin generator, but its continuum identification is a separate problem. At positive time the intended limiting object is a connection modulo periodic gauge transformations, specified by its holonomies.

Sketch O (The proposed continuum flow). The sketch describes a positive-time Yang–Mills gradient flow from almost every block-field thread, with strongly convergent flowed gauge-invariant observables. Its identification requires compactness modulo gauges, passage to the limiting equation, recovery of the initial holonomies and uniqueness. The estimates below provide the preparatory bounds; these four steps remain to be established. ■□

Sketch of the argument. For a smooth lattice observable G, put $H_K^\sigma = G \circ \Phi_\sigma$ and $\Delta_K = a_K^{-1} \sum_{b,\alpha} (\partial_b^\alpha)^2$, the second-order part of the Langevin generator. The finite-dimensional identity

$$\mathcal{L}_K H_K^\sigma = (b\partial_\sigma + \Delta_K) H_K^\sigma$$

is exact. The action drift in the stochastic generator is cancelled by moving along the deterministic flow. Integrating against the stationary measure gives the corresponding identity for means; multiplying by a test observable gives its Dirichlet-form version.

The small-field chain decomposes a gauge-fixed potential into layers. Conditional estimates control the layer means and the part of their colour covariance that is not scalar. Their summation gives rough-field moments. Deterministic heat estimates then control the flow at positive times, producing uniform moments of the flowed curvature and current and high-probability regularity sets. Sufficiently deep layers satisfy the admissibility bounds needed for these estimates.

For the continuum identification, start with minimal backgrounds over converging coarse data. One must choose periodic gauges so that their positive-time potentials have convergent subsequences, locally uniformly in time. Their limit must satisfy the gauge-fixed mild equation

$$\partial_\sigma \mathfrak{A} = -\mathrm{d}_{\mathfrak{A}}^* F_{\mathfrak{A}} + \mathrm{d}_{\mathfrak{A}}(\partial_\nu \mathfrak{A}_\nu),$$

recover the prescribed smoothed holonomies as $\sigma \downarrow 0$, and be unique modulo gauge with those initial holonomies. The source explicitly leaves these linked claims as an additional approximation step. The derivative and moment estimates do not alone complete it.

There is another difficulty even after identifying the flowed observables. Positive flow time does not automatically put them in the domain of the limiting stochastic generator. Some bounded flowed observables have $\|\mathcal{L}_K H_K^\sigma\|_2$ growing exponentially in a_K^{-2}. Insert invariant regularization cutoffs $\chi_{K,R}$ whose lost mass and Dirichlet cost tend to zero as $R \to \infty$. First pass $K \to \infty$ at fixed

R; only afterwards remove R. For resolvent test functions this gives a tested generator identity, and after integrating in σ a distributional one. An unregularized operator-domain claim requires a separate uniform generator bound, which those examples show cannot be assumed for every flowed observable.

16.1. **The identification that the flow sketch requires.** Deterministic smoothing and stochastic evolution act on the same finite lattice, but their continuum limits require different arguments. Let Φ_σ be the Wilson gradient flow and let G be a smooth observable of the lattice field. Write $H^\sigma = G \circ \Phi_\sigma$. With the bond Laplacian $\Delta_K = a_K^{-1} \sum_{b,\alpha} (\partial_b^\alpha)^2$, the finite-dimensional chain rule gives

$$\mathcal{L}_K H^\sigma = (b\partial_\sigma + \Delta_K) H^\sigma.$$

Stationarity and integration by parts therefore imply

$$b\partial_\sigma \langle H^\sigma \rangle_{\mu^K} = -\langle \Delta_K H^\sigma \rangle_{\mu^K}.$$

These are finite-cutoff identities. They describe how flow time enters the observable, without yet identifying a random continuum connection or its deterministic flow.

The prospective continuum flow is most naturally specified modulo periodic gauge transformations. For a fixed admitted block depth D and block field V, choose minimal backgrounds at finer and finer cutoffs over V. The unflowed loop comparison identifies limiting holonomies $w_C^D(V)$ for fixed closed paths C. Positive-time regularity estimates then control the smoothed backgrounds in gauges. To finish the identification, one must obtain compactness modulo periodic gauges, pass the lattice equation to the continuum equation, recover these initial holonomies as flow time decreases to zero, and establish uniqueness of the resulting orbit-valued flow. The detailed estimates below provide the preparatory bounds for these four steps, which remain to be established.

The proposed limiting equation, in a local gauge representative $\mathfrak{A}_s$, is

$$\partial_s \mathfrak{A} = -\mathrm{d}_{\mathfrak{A}}^* F_{\mathfrak{A}} + \mathrm{d}_{\mathfrak{A}}(\partial_\nu \mathfrak{A}_\nu).$$

The last term is an infinitesimal gauge transformation. Thus the object to identify is a path of gauge orbits, with local representatives satisfying this equation, rather than one global choice of potentials. The initial condition is expressed by $W_C(\mathfrak{A}_s) \to w_C^D(V)$ as $s \downarrow 0$. Neither the finite-lattice chain rule nor positive-time moment bounds alone establish uniqueness for that initial condition.

A further limit issue concerns the generator. Smoothness at positive flow time does not give a bound for $\|\mathcal{L}_K H_K^\sigma\|_2$ uniform in K. The source gives bounded flowed observables for which that norm grows exponentially in a_K^{-2}. One therefore tests the generator identity with regularity cutoffs whose Dirichlet cost vanishes. First pass to the continuum with a fixed regularization; then remove that regularization. The resulting identity has the tested or distributional meaning established by this procedure. It cannot be replaced by an assertion that every bounded flowed observable lies in the domain of the limiting generator.

16.2. **The background-flow identification.**
Approximation of the fixed-depth background flow. Fix $0 < \alpha < 1$. For D sufficiently large put $\mathcal{K}_D = \{V \in \Omega_D : \varpi(V) \le \theta_D\}$, and let $U_{K,D}(V)$ be any member of the minimal orbit over V on $\Lambda_{K,\ell}$, $K > D$. The unflowed comparison in the proof of Prop. 16.1(b) defines $w_C^D(V) = \lim_{K\to\infty} W_C(U_{K,D}(V))$ for each closed lattice path C. The remaining approximation step, referred to as *(Flow approximation)*, is to prove the following compactness and identification statement. For each fixed D, every sequence $K_\nu \to \infty$, $V_\nu \to V$ in $\mathcal{K}_D$ and arbitrary minimal backgrounds U_ν over V_ν should have a subsequence whose positive-time flows converge modulo periodic gauges, locally uniformly in time in $\mathcal{O}^\alpha$, to an orbit-valued flow line. More explicitly, on each compact positive time interval, periodic gauges give real bond potentials $A_\nu(s)$ representing $\Phi_s U_\nu$ as $e^{ia_{K_\nu} A_\nu(s)}$ and

local representatives $\mathfrak{A}_s$ of the limit for which

$$\sup_{\tau\le s\le T}\max_b |A_\nu(s,b)-\mathfrak{A}_{s,\mu(b)}(x_b)|\longrightarrow 0,\qquad 0<\tau<T<\infty.$$

Here x_b is the bond midpoint and $\mu(b)$ its direction. The limit must have local mild representatives satisfying

$$\partial_s\mathfrak{A}=-\mathrm{d}^*_{\mathfrak{A}}F_{\mathfrak{A}}+\mathrm{d}_{\mathfrak{A}}(\partial_\nu\mathfrak{A}_\nu),\qquad \lim_{s\downarrow 0}W_C(\mathfrak{A}_s)=w_C^D(V),$$

and one must prove uniqueness of this orbit-valued flow line from those initial holonomies. These are the compactness and identification tasks of the sketch. The positive-time derivative bounds used below are proved separately by the single-scale gauge and the short-time regularisation in the proof of Prop. 16.1(c).

Detailed sketch AV (Flowed estimates and the proposed continuum identification). Assume (E) and $0<\kappa<\frac16$; (C1) holds for all K, m, $\beta>0$, with $\mathsf{b}=a_K\beta$. (C2)(b), (C2)(c) and (C3)–(C7), which read the moments, the good sets or the field $\mathbf{B}_K$ of Prop. 16.1(a), use the consequence (Adm) of (E) proved in Prop. 16.1: every top depth from some depth on, a depth free of K, is admissible in the sense of Prop. 16.1. (C1) and (C2)(a) are free of it. The continuum identifications in (C3) and the part of (C5)(d) involving G_σ remain proposed conclusions until the approximation step above is completed. The assertions in (C7) use the nonlinear defect estimates of Lem. 16.4.

(C1) For $G\in C^\infty(\Omega_K)$, $H^\sigma=G\circ\Phi_\sigma$ and $\sigma\ge 0$: $\mathcal{L}_KH^\sigma=(\mathsf{b}\,\partial_\sigma+\Delta_K)H^\sigma$. Hence $\mathsf{b}\,\partial_\sigma\langle H^\sigma\rangle_{\mu^K}=-\langle\Delta_KH^\sigma\rangle_{\mu^K}$; for $\chi\in D(\mathcal{E}_K)\cap L^\infty$, $\int\chi\,(\mathsf{b}\partial_\sigma+\Delta_K)H^\sigma\,\mathrm{d}\mu^K=-\int\Gamma_K(\chi,H^\sigma)\,\mathrm{d}\mu^K=-\int(1-\chi)\,\mathcal{L}_KH^\sigma\,\mathrm{d}\mu^K$; for $t\ge0$, $P_t^KH^\sigma=H^{\sigma+\mathsf{b}t}+\int_0^tP_s^K\Delta_KH^{\sigma+\mathsf{b}(t-s)}\,\mathrm{d}s$; and for the stationary process and $0\le r\le t$, $H^{\sigma+\mathsf{b}(t-r)}(U_r)-H^{\sigma+\mathsf{b}t}(U_0)-\int_0^r(\Delta_KH^{\sigma+\mathsf{b}(t-s)})(U_s)\,\mathrm{d}s$ is a martingale with the bracket $2\int_0^r\Gamma_K(H^{\sigma+\mathsf{b}(t-s)})(U_s)\,\mathrm{d}s$: in the frame that moves along $\sigma+\mathsf{b}t=\text{const}$ the drift of the action is gone.

(C2) (a) For $\bar C\ge1$, $0<\bar s\le1$ there is K_0 such that for every flowed observable, $0<\sigma\le1$ and $K\ge K_0$: $\Gamma_K(H_K^\sigma)\le C_\Gamma$ and $|\partial_\sigma H_K^\sigma|\le C_\partial$ on $\mathsf{O}_K(\bar C,\bar s)$, with C_Γ,C_∂ depending on $\bar C,\bar s,\ell,\kappa,\sigma$ and the norms of $\nabla\varphi$, f_i, ∇f_i, bounded for σ in compact subsets of $(0,1]$; for $\sigma\le\bar s$ this holds for every K. (b) For $0<\sigma\le1$, $1\le p<\infty$ and all K: $\|\partial_\sigma H_K^\sigma\|_{L^p(\mu^K)}\le C_p(\sigma)\sum_i\|\partial_i\varphi\|_\infty(2\|f_i\|_\infty+4\|\nabla f_i\|_\infty)$, $C_p(\sigma)$ bounded on compact subsets of $(0,1]$; so $\|H_K^\sigma-H_K^{\sigma'}\|_{L^p(\mu^K)}\le C|\sigma-\sigma'|$ locally uniformly in $\sigma,\sigma'\in(0,1]$ and in K, and, by (C1), $\sup_K\|\mathcal{L}_KH_K^\sigma\|_{L^p(\mu^K)}<\infty$ if and only if $\sup_K\|\Delta_KH_K^\sigma\|_{L^p(\mu^K)}<\infty$.

(c) For compact $I\subset(0,1]$, $1\le q<\infty$: $\sup_K\|\partial_\sigma H_K^{\sigma''}-\partial_\sigma H_K^{\sigma'}\|_{L^q(\mu^K)}\le C_I(q)|\sigma''-\sigma'|^{1/2}$ for $\sigma',\sigma''\in I$.

(C3) For every flowed observable and $\sigma>0$, $H_K^\sigma\to\mathsf{G}_\sigma$ strongly along the full sequence of cutoffs, where

$$\mathsf{G}_\sigma(Y)=\varphi\big(\mathcal{O}^0(f_1)(\mathfrak{F}(Y)_\sigma),\dots,\mathcal{O}^0(f_r)(\mathfrak{F}(Y)_\sigma)\big),$$

$\mathfrak{F}$ the map of Prop. 16.1(b); the same holds for real Lipschitz f_i and bounded Lipschitz φ, and $\sigma\mapsto\mathsf{G}_\sigma$ is locally Lipschitz from $(0,1]$ into $L^2(\mu)$. For $0<\sigma<1$ there is $\dot{\mathsf{G}}_\sigma\in L^2_{\mathrm{inv}}$ with $\int\Psi_K\,\partial_\sigma H_K^\sigma\,\mathrm{d}\mu^K\to\int\Psi\,\dot{\mathsf{G}}_\sigma\,\mathrm{d}\mu$ along the full sequence for every invariant cylinder function Ψ; $\sigma\mapsto\int\Psi\dot{\mathsf{G}}_\sigma\,\mathrm{d}\mu$ is locally Hölder continuous of exponent $\frac12$ and is the derivative of the continuously differentiable function $\sigma\mapsto\int\Psi\mathsf{G}_\sigma\,\mathrm{d}\mu$.

(C4) *Regularisation and condenser cutoffs.* There are $R_\flat\ge1$ and, for $R\ge R_\flat$, numbers $\bar s(R)\in(0,1]$, $\bar K(R)$, $\bar\varkappa(R)>0$ with $R^q\bar\varkappa(R)\to0$ for every q, and for $K\ge\bar K(R)$ invariant $\chi_{K,R}\in C^\infty(\Omega_K)$ with values in $[0,1]$ such that $\chi_{K,R}=1$ on $\{\mathsf{G}^*\le R\}$ and $\chi_{K,R}=0$ off $\{\mathsf{G}^*\le\frac52R\}\subset\mathsf{O}_K(8R,\bar s(R))$, G^* the functional of Step 8; $\mathcal{E}_K(\chi_{K,R})+\int(1-\chi_{K,R})^2\,\mathrm{d}\mu^K\le\bar\varkappa(R)$; $\sup_{K\ge\bar K(R)}\int\Gamma_K(\chi_{K,R})^p d\mu^K<\infty$ for every $p<\infty$; and $\mathbf{1}_{\{\chi_{K,R}>0\}}\le\chi_{K,R'}$ for $R'\ge\frac52R$, $K\ge\bar K(R')$.

Let $w_{K,R}$ be the potential of Lem. 16.5 for $\eta = 1 - \chi_{K,R}$, and $\chi^{\mathrm{c}}_{K,R} = 1 - w_{K,R}$, the condenser cutoff. Then $\chi^{\mathrm{c}}_{K,R} \in D(\mathcal{E}_K)$ is invariant, takes values in $[0,1]$, is 1 where $\chi_{K,R} = 1$ and 0 where $\chi_{K,R} = 0$, has $\Gamma_K(\chi^{\mathrm{c}}_{K,R}, \cdot) = 0$ off the shell $\{0 < \chi_{K,R} < 1\}$, $\mathcal{E}_K(\chi^{\mathrm{c}}_{K,R}) + \int (1 - \chi^{\mathrm{c}}_{K,R})^2 \,\mathrm{d}\mu^K \le \bar{\varkappa}(R)$, and $|\int \Gamma_K(\chi^{\mathrm{c}}_{K,R}, \psi)\,\mathrm{d}\mu^K| \le \operatorname{osc}(\psi)\,\bar{\varkappa}(R)$ for $\psi \in D(\mathcal{E}_K) \cap L^\infty$. A second family of smooth invariant cutoffs, defined without choosing a gauge, is given by Prop. 16.6.

(C5) *Static invariance and the generator.* (a) For every flowed observable, $\sigma > 0$, $R \ge R_\flat$ and $K \ge \bar{K}(R)$,

$$\Big| \int \chi^{\mathrm{c}}_{K,R}\, (\mathsf{b}\,\partial_\sigma + \Delta_K) H^\sigma_K \,\mathrm{d}\mu^K \Big| \le \operatorname{osc}(\varphi)\,\bar{\varkappa}(R);$$

for $0 < \sigma \le 1$, with $C_\partial = \sup_K \|\partial_\sigma H^\sigma_K\|_{L^2(\mu^K)}$, $|\int \chi^{\mathrm{c}}_{K,R} \Delta_K H^\sigma_K \,\mathrm{d}\mu^K + \mathsf{b}\,\partial_\sigma \langle H^\sigma_K \rangle_{\mu^K}| \le \operatorname{osc}(\varphi)\bar{\varkappa}(R) + \mathsf{b}\, C_\partial\, \bar{\varkappa}(R)^{1/2}$; and $|\int \chi^{\mathrm{c}}_{K,R} \mathcal{L}_K H \,\mathrm{d}\mu^K| \le \operatorname{osc}(H)\bar{\varkappa}(R)$ for every $H \in C^2(\Omega_K)$. (b) With $q(u) = 3u^2 - 2u^3$, $q(\chi^{\mathrm{c}}_{K,R}) \in D(\mathcal{E}_K)$ is invariant, takes values in $[0,1]$, is 1 where $\chi_{K,R} = 1$ and 0 where $\chi_{K,R} = 0$, has $\mathcal{E}_K(q(\chi^{\mathrm{c}}_{K,R})) \le \frac{9}{4}\bar{\varkappa}(R)$, and its generator is a function: $\int q(\chi^{\mathrm{c}}_{K,R})\, \mathcal{L}_K H \,\mathrm{d}\mu^K = \int H\, \mathsf{I}_{K,R} \,\mathrm{d}\mu^K$ for $H \in C^2(\Omega_K)$, $\|\mathsf{I}_{K,R}\|_{L^1(\mu^K)} \le \frac{15}{2}\bar{\varkappa}(R)$. (c) For $H, H' \in C^2(\Omega_K)$,

$$\begin{aligned} \Big| \int \chi^{\mathrm{c}}_{K,R}\, (H' \mathcal{L}_K H - H \mathcal{L}_K H') \,\mathrm{d}\mu^K \Big| &\le \operatorname{osc}(HH')\,\bar{\varkappa}(R) \\ &\quad + 2 \sup|H|\, \bar{\varkappa}(R)^{1/2} \Big(\int_{\{0 < \chi_{K,R} < 1\}} \Gamma_K(H') \,\mathrm{d}\mu^K \Big)^{1/2}. \end{aligned}$$

(d) Let (K_j) be a limit and $F \in D(A_\star)$ with a test sequence, that is, invariant $F_j \in D(\mathcal{L}_{K_j})$ with $\sup_j \|F_j\|_\infty < \infty$, $F_j \to F$ and $\mathcal{L}_{K_j} F_j \to -A_\star F$ strongly; test sequences exist for $F = \alpha(\alpha + A_\star)^{-1} W$, $\alpha > 0$, whenever W is the strong limit of uniformly bounded invariant W_j (for instance an invariant cylinder function or a G_σ). Then for $R \ge R_\flat$ with $\bar{\varkappa}(R) \le 1$

$$\limsup_{j \to \infty} \Big| \int \chi^{\mathrm{c}}_{K_j,R} F_j\, (\mathsf{b}\,\partial_\sigma + \Delta_{K_j}) H^\sigma_{K_j} \,\mathrm{d}\mu^{K_j} + \int (A_\star F)\, \mathsf{G}_\sigma \,\mathrm{d}\mu \Big| \le c_F\, \bar{\varkappa}(R)^{1/2},$$

$c_F = \sup|\varphi| \sup_j [2\|F_j\|_\infty + 2\mathcal{E}_{K_j}(F_j)^{1/2} + \|\mathcal{L}_{K_j} F_j\|_{L^2}]$: the displayed tested error tends to zero as $R \to \infty$. For every $\tau \in C^\infty_c(0,1)$ the corresponding distributional statement is

$$\begin{aligned} \lim_{R \to \infty} \limsup_{j \to \infty} \Big| \int_0^1 \tau(\sigma) \int \chi^{\mathrm{c}}_{K_j,R} F_j\, \Delta_{K_j} H^\sigma_{K_j} \,\mathrm{d}\mu^{K_j} \,\mathrm{d}\sigma \\ + \int_0^1 [\tau(\sigma)(A_\star F, \mathsf{G}_\sigma) - \mathsf{b}\,\tau'(\sigma)(F, \mathsf{G}_\sigma)] \,\mathrm{d}\sigma \Big| = 0. \end{aligned}$$

For two flowed observables H_K, H'_K at σ, σ' with limits G, G' and $\alpha > 0$,

$$\begin{aligned} \lim_R \limsup_j \Big| \int \chi^{\mathrm{c}}_{K_j,R} [\alpha(\alpha - \mathcal{L}_{K_j})^{-1} H'_{K_j}]\, \mathcal{L}_{K_j} H_{K_j} \,\mathrm{d}\mu^{K_j} \\ + (\alpha A_\star (\alpha + A_\star)^{-1} \mathsf{G}', \mathsf{G}) \Big| = 0, \end{aligned}$$

a limit that is symmetric in the two observables.

If $C_1 = \liminf_R \limsup_j \int \chi^{\mathrm{c}}_{K_j,R} |\mathcal{L}_{K_j} H^\sigma_{K_j}| \,\mathrm{d}\mu^{K_j} < \infty$, then $\mathsf{G}_\sigma \in D(\mathcal{E}_\star)$ and $\mathcal{E}_\star(\mathsf{G}_\sigma) \le \sup|\varphi|\, C_1$; if $C_2 = \liminf_R \limsup_j \|\chi^{\mathrm{c}}_{K_j,R} \mathcal{L}_{K_j} H^\sigma_{K_j}\|_{L^2(\mu^{K_j})} < \infty$, then $\mathsf{G}_\sigma \in D(A_\star)$ and $\|A_\star \mathsf{G}_\sigma\| \le C_2$. Without a cutoff, the criterion of (A3) gives $\mathsf{G}_\sigma \in D(A_\star)$, with $\mathcal{L}_{K_j} H^\sigma_{K_j} \to -A_\star \mathsf{G}_\sigma$ weakly and $\mathcal{E}_{K_j}(H^\sigma_{K_j}) \to \mathcal{E}_\star(\mathsf{G}_\sigma)$, whenever $\sup_K \|\mathcal{L}_K H^\sigma_K\|_{L^2(\mu^K)} < \infty$; this fails at the members of (C7)(b), (c), (e).

(C6) *Lower energy bounds.* There is $p_* > 0$, depending only on the constants of the construction, and there are $\sigma_*, c_* > 0$, $S_* < \infty$ such that for $0 < \sigma \le \sigma_*$, every $\varphi \in C^1(\mathbb{R})$ and every invariant Borel $\chi_K : \Omega_K \to [0,1]$ with $\mu^K(\chi_K < 1) \le p_*/8$,

$$\liminf_{K\to\infty} \int \chi_K \, \Gamma_K(\varphi(A^a) \circ \Phi_\sigma) \, \mathrm{d}\mu^K \;\ge\; c_* \, \sigma^{-\frac{5}{2}+\kappa} \inf_{[0, S_* \sigma^{-3/2-\kappa}]} \varphi'^2.$$

For every $\vartheta < 1$ there is $\sigma_\vartheta > 0$ such that the left side is positive for $0 < \sigma \le \sigma_\vartheta$, every φ with $\varphi' > 0$ and every such χ_K with $\mu^K(\chi_K < 1) \le \vartheta$; in particular $\liminf_K \mathcal{E}_K(\varphi(A^a) \circ \Phi_\sigma) > 0$. There are $\sigma_{**} > 0$ and $0 < c_\circ \le C_\circ$, $c_\circ$ depending only on the constants of the construction, with

$$c_\circ \, \mathsf{b}^{-1} \ell^3 \sigma^{-3/2} \le \liminf_K \langle A^a \circ \Phi_\sigma \rangle_{\mu^K} \le \limsup_K \langle A^a \circ \Phi_\sigma \rangle_{\mu^K} \le C_\circ \, \mathsf{b}^{-1} \ell^3 \sigma^{-3/2} \qquad (0 < \sigma \le \sigma_{**});$$

for every $\varrho \in (0,1)$ there are $c_\varrho, \sigma_\varrho > 0$ such that for $0 < \sigma \le \sigma_\varrho$ and all large K, with μ^K-probability at least $1 - \varrho$, the flowed action exceeds $c_\varrho \mathsf{b}^{-1} \ell^3 \sigma^{-3/2}$ and $\Gamma_K(A^a \circ \Phi_\sigma)$ exceeds $c_\varrho \sigma^{-5/2+\kappa}$; and for $0 < \sigma \le \sigma_*$ the flowed current does not vanish in probability: $\liminf_K \mu^K((J,J)_a(U_\sigma) \ge c_J(\sigma)) \ge \frac{1}{4} p_*$ for some $c_J(\sigma) > 0$.

(C7) *The generator without a regularisation cutoff.* Let $\bar{s}, \bar{C}$ be given by Prop. 16.1(a) for some $0 < \epsilon < 1$, and $0 < \sigma \le \bar{s}$. There is K_1 such that for $K \ge K_1$, $a = a_K$: (a) $\int |\partial(A_K \circ \Phi_\sigma)|^q \, \mathrm{d}\mu^K \ge \exp[(6.6\, q - 24.64)\, \sigma/a^2]$ for $q \in [3.74, 8]$, and $\int \Gamma_K(A^a \circ \Phi_\sigma)^2 \, \mathrm{d}\mu^K \ge e^{1.76\, \sigma/a^2}$; (b) $\|\mathcal{L}_K \arctan(A^a \circ \Phi_\sigma)\|_{L^2(\mu^K)} \ge e^{0.8\, \sigma/a^2}$; (c) for $f(x) = 1 - \cos(2\pi x_1/\ell)$, $E^{\mathrm{a}}_\sigma = \frac{3}{2}\ell^3 (2\bar{C}(\frac{\sigma}{2})^{-\frac{3}{4}-\frac{\kappa}{2}} + 8\bar{C}^2(\frac{\sigma}{2})^{-\frac{1}{2}-\kappa})^2$, $M_\sigma = (\frac{4}{\sigma} + \frac{11400}{\ell^2})(\frac{3}{2}\sigma E^{\mathrm{a}}_\sigma + 1) + 1$ and every $\varphi \in C^\infty(\mathbb{R})$, bounded with all its derivatives, with $\inf_{[0,M_\sigma]} |\varphi'| > 0$ (K_1 depending on φ): $\|\mathcal{L}_K \varphi(\mathcal{O}(f) \circ \Phi_\sigma)\|_{L^2(\mu^K)} \ge e^{0.8\, \sigma/a^2}$; (d) for $p \in [1.87, 4]$: $\|\mathcal{L}_K \cos(A^a \circ \Phi_\sigma)\|_{L^p(\mu^K)} + \|\mathcal{L}_K \sin(A^a \circ \Phi_\sigma)\|_{L^p(\mu^K)} \ge e^{0.04\, \sigma/(p a^2)}$, and for every $p > 1.8637$ the left side is unbounded in K; (e) for every real $f \in C^\infty(T_\ell)$, $f \not\equiv 0$, with K_1 and a constant C depending on f but not on K: $\|\mathcal{L}_K \sin(\mathcal{O}(f) \circ \Phi_\sigma)\|_{L^2(\mu^K)} \ge \exp[\frac{\sigma}{5a^2} - \frac{10\,\mathsf{b}}{a} - C]$, and $\|\mathcal{L}_K \varphi(\mathcal{O}(f) \circ \Phi_\sigma)\|_{L^2(\mu^K)} \ge e^{\sigma/20a^2}$ for every non-constant periodic $\varphi \in C^\infty(\mathbb{R})$, K_1 depending besides on φ; (f) $\sup_K \int e^{13\Lambda_\sigma} \, \mathrm{d}\mu^K = \infty$.

■□

16.3. **Gauge-fixed regularity and the admissibility hypothesis.**

Proposition 16.1 (Gauge-fixed fields and the gradient flow)**.** Let $0 < \kappa < \frac{1}{6}$, and let $\mathsf{N}'(U)$ be the least $\|\mathbf{B}\|_{\kappa;a}$ over the 1-forms $\mathbf{B}$ with $e^{i a_K \mathbf{B}}$ in the orbit of U. *The gauge-fixed flow* is the lattice gradient flow of the action with a gauge term: $V_s = e^{i a_K A_s}$, $A_0 = \mathbf{B}$, moves with the velocity of the Wilson flow Φ plus a gauge direction $\mathrm{d}_{V_s}\lambda_s$, λ_s the lattice form of the divergence of A_s that answers to the member $\mathrm{d}_{\mathfrak{A}}(\partial_\nu \mathfrak{A}_\nu)$ of the flow lines, fixed so that the linear part of the equation is the lattice heat operator. In the chart it reads $\frac{d}{ds} A_s(b) = a_K^{-3} \Theta^{\mathrm{ch}}(a_K A_s|_{\mathsf{N}(b)})$, $\mathsf{N}(b)$ the fifteen bonds near b, Θ^{ch} holomorphic and bounded on $\{|a_K A| \le 1\}$, and $\mathsf{M}_\Theta = \max\{1, \sup|\Theta^{\mathrm{ch}}|\}$, the supremum taken there, an absolute number; the members of Θ^{ch} of the orders one, two and at least three in A give

$$\dot{A} = \Delta_a A + \mathcal{N}(A,A) + \mathcal{R}(A), \qquad \mathcal{N}(A,A') = \sum_j c_j \, (\tau_{h_j} \nabla_{\lambda_j} A_{\nu_j}) \wedge (\tau_{h'_j} A'_{\nu'_j}),$$

a finite sum, with absolute coefficients, over the directions and the fixed finite translation set used in the rough regularity quantity.

Fields in the chain gauge. Assume (E); let $d_0 \ge n_2$ be a top depth, $K > d_0$, $t = K - d_0$, $\mathbb{E}^{[d_0]}$ the normalised chain measure from the top depth d_0, and $V_k = \mathcal{M}^k U$. On a set of full measure: U is a gauge transform of U_0, where $\mathsf{U}_t = U_{\mathrm{top}}$ is a background over the top field V_t and $\mathsf{U}_k = e^{i a_K \mathbf{h}_{K-k}} \mathsf{U}_{k+1}$ ($k < t$) is a gauge transform of the background over V_k, in a gauge that is a function of V_{k+1} (the

gauge of the chain); the layer $\mathbf{h}_d$ of the depth $d = K - k$ is a function of the fluctuation B_k of the chart at V_{k+1}, with $|\mathbf{h}_d| \le c_{\mathsf{h}} L^d \delta(L^d \mathsf{b})$, c_{h} a constant of the construction; its linear part in the fluctuation, carried by $\mathrm{Ad}(\gamma)$ below, is $\mathsf{b}^{-1/2} L^{d/2} \mathsf{J}_d B_k$, with the layer map J_d, a function of V_{k+1}, of the norms $\|\mathsf{J}_d X\|_{L^2(T_\ell)} \le c_{\mathsf{J}} L^{-3d/2} \|X\|$ and $\|\mathsf{J}_d X\|_\infty \le c_{\mathsf{J}} \max |X|$, c_{J} a constant of the construction, and at every point of the box the derivative of $\mathbf{h}_d$ in B_k is $\mathsf{b}^{-1/2} L^{d/2}$ times a map with the same two bounds; in its regular gauge γ the top is $U^\gamma_{\mathrm{top}} = e^{i a_K \mathbf{A}^{\mathrm{top}}}$ with $|\mathbf{A}^{\mathrm{top}}| \le \mathsf{a}_{\mathrm{top}} \le C L^{10 d_0}$, $\mathsf{a}_{\mathrm{top}}$ a number that reads neither K nor the field; and U is a gauge transform of $e^{i a_K \mathbf{B}_K}$,

$$\mathbf{B}_K = \tfrac{1}{i a_K} \log(e^{i a_K \mathbf{h}'_K} \cdots e^{i a_K \mathbf{h}'_{d_0+1}} e^{i a_K \mathbf{A}^{\mathrm{top}}}), \qquad \mathbf{h}'_d = \mathrm{Ad}(\gamma) \mathbf{h}_d,$$

bond by bond, the logarithm being formed by its series if $\sum_d a_K |\mathbf{h}_d| \le \frac14$ and $a_K \mathsf{a}_{\mathrm{top}} \le \frac18$, and otherwise by a fixed Borel $\mathfrak{su}(2)$ logarithm of norm at most π, principal off $-\mathbf{1}$, on positive bonds and extended by reversal. Admissibility and the conclusion (Adm) have the meaning stated before Thm. AV; the layer fields and their conditional estimates are specified in the proof below.

(a) There are $c_1, c_2 \in (0,1]$, $c_{\mathrm{Lip}} \in (0, c_1]$ and c_3, c_4, c_5, depending only on κ, such that for all K, m and $\beta > 0$, every real 1-form $\mathbf{B}$ on $\Lambda_{K,\ell}$ and $R \ge \max\{1, \|\mathbf{B}\|_{\kappa;a}\}$, with $s_0(R) = (c_1 R^{-2})^{4/(1-2\kappa)}$, $s_{\mathrm{Lip}}(R) = (c_{\mathrm{Lip}} R^{-2})^{4/(1-2\kappa)}$ and $a_0(R) = (c_2 R^{-1})^{2/(1-2\kappa)}$, the following hold. If $a_K \le a_0(R)$, the gauge-fixed flow A_s from $A_0 = \mathbf{B}$ exists on the closed interval $[0, s_0(R)]$, $\Phi_s(e^{i a_K \mathbf{B}})$ is gauge equivalent to $e^{i a_K A_s}$, and for $0 < s \le s_0(R)$: $\|A_s - P_s \mathbf{B}\|_\infty \le c_3 R s^{-\kappa}$, $\|\nabla(A_s - P_s \mathbf{B})\|_\infty \le c_3 R s^{-\frac12 - \kappa}$, $\|A_s\|_\infty \le (1 + c_3) R s^{-\frac14 - \frac{\kappa}{2}}$, $\|\nabla A_s\|_\infty \le (2 + c_3) R s^{-\frac34 - \frac{\kappa}{2}}$.

If besides $\mathbf{B}'$ is a real 1-form with $\|\mathbf{B}'\|_{\kappa;a} \le R$, A'_s its gauge-fixed flow and D the rough distance of $\mathbf{B}$ and $\mathbf{B}'$, then for $0 < s \le s_{\mathrm{Lip}}(R)$: $\|(A_s - A'_s) - P_s(\mathbf{B} - \mathbf{B}')\|_\infty \le c_5 \mathsf{D} s^{-\kappa}$, $\|\nabla[(A_s - A'_s) - P_s(\mathbf{B} - \mathbf{B}')]\|_\infty \le c_5 \mathsf{D} s^{-\frac12 - \kappa}$, so $\|A_s - A'_s\|_\infty \le (1 + c_5) \mathsf{D} s^{-\frac14 - \frac{\kappa}{2}}$. For every a_K: $A^a(\Phi_s e^{i a_K \mathbf{B}}) \le c_4 \ell^3 (1 + R)^{24} s^{-\frac32 - \kappa}$ for $0 < s \le 1$. These statements, the short-time theory of the gauge-fixed flow, are deterministic and rest on its equation alone; they hold on every lattice $\Lambda_{K,\ell}$, uniformly in a_K.

Assume (E): for every admissible d_0, $K > d_0$ and $q < \infty$, $\mathbb{E}^{[d_0]}[\|\mathbf{B}_K\|^q_{\kappa;a}]^{1/q} \le C(q,\kappa) L^{20 d_0}$, with $C(q,\kappa)$ free of K and d_0; and for every top depth $d_1 \ge n_2$, $K > d_1$, invariant Borel $F \ge 0$ and $r > 1$,

$$\langle F \rangle_{\mu^K} \le \mathbb{E}^{[d_1]}[F] + \sum_{d_1 \le d \le K-2} \mathsf{R}_d^{1 - 1/r} \, \mathbb{E}^{[d+1]}[F^r]^{1/r} + \mathsf{R}_{K-1} \sup F,$$

where $\sum_d \mathsf{R}_d^{1/2} L^{pd}$ and $\sup_K \mathsf{R}_{K-1} L^{pK}$ are finite for every p. Under (E), the conclusion (Adm) proved below gives $\mathsf{M}_q = \mathsf{M}_q(\kappa) = \sup_K \langle \mathsf{N}'^q \rangle_{\mu^K} < \infty$ for every $q < \infty$, and: for every $\epsilon > 0$ there are $\bar s \in (0,1]$, $\bar C$, $\bar K$ with $\mu^K(\mathsf{O}_K(\bar C, \bar s)) \ge 1 - \epsilon$ for $K \ge \bar K$; $\sup_K \langle (\sup_{0 < s \le 1} s^{\frac32 + \kappa} A^a(U_s))^q \rangle_{\mu^K} < \infty$ for every q; for $0 < \sigma \le 1$ and $1 \le p < \infty$ there is C_3, bounded for σ in compact subsets of $(0,1]$, with $\mu^K(\max_p a_p(U_\sigma) > 10^{-4}) \le C_3 a_K^p$ and $\int [(a_K^{-4} \max_p a_p(U_\sigma))^p + (J,J)_a(U_\sigma)^p] \, \mathrm{d}\mu^K \le C_3$ for all K; and for $0 < \sigma \le 1$ the laws of Λ_σ under μ^K are tight.

(b) Under (E), the flow-approximation task seeks the following conclusion. For every closed path C of bonds of some $\Lambda_{d,\ell}$ and $s > 0$, the flowed Wilson loop $W_C \circ \Phi_s$ (W_C the half trace of the holonomy) converges strongly along the full sequence of cutoffs; there are an invariant Borel set of threads of full measure and a Borel map $\mathfrak{F}$ from it into flow lines, $\mathfrak{F}(Y)_s$ represented by a $C^{1,1}$ connection for $s > 0$, such that the limit is $W_C(\mathfrak{F}(Y)_s)$. For every flowed observable and $\sigma > 0$, with $\Xi^\sigma_{K,D}(V) = H^\sigma_K(U)$ for $V \in \{\varpi < \theta_D\} \subset \Omega_D$, U any element of the minimal orbit over V in Ω_K, and $\Xi^\sigma_{K,D} = 0$ elsewhere:

$$\lim_{D \to \infty} \sup_{K > D} \|H^\sigma_K - \Xi^\sigma_{K,D} \circ \pi_{K,D}\|_{L^2(\mu^K)} = 0, \qquad \lim_{D \to \infty} \limsup_{K \to \infty} \|\Xi^\sigma_{K,D}(Y_D) - \mathsf{G}_\sigma\|_{L^2(\mu)} = 0 :$$

the flowed observable at the background of the block field of a large depth.

(c) The same flow identification is intended to give the following joint convergence under (E). For $s > 0$ and Lipschitz f the joint laws of $\mathcal{O}(f) \circ \Phi_s$ and of the $W_C \circ \Phi_s$ under μ^K converge with all moments along the full sequence to the joint law of $\mathcal{O}^0(f)(\mathfrak{F}(Y)_s)$ and the $W_C(\mathfrak{F}(Y)_s)$ under μ. Separately, under (E) alone, for $\epsilon, \eta > 0$ there are Borel sets G_K with $\mu^K(\mathsf{G}_K) \geq 1 - \epsilon$, closed paths $C_1, \dots, C_q$ and a continuous Θ on $[-1,1]^q$ with $|\mathcal{O}(f)(U_s) - \Theta((W_{C_i}(U_s))_i)| \leq \eta$ on G_K for all large K.

■

Proof sketch. (a)

The gauge-fixed flow. Write $A_s = P_s\mathbf{B} + w_s$; the mild form $w_s = \int_0^s P_{s-r}[\mathcal{N}(A_r, A_r) + \mathcal{R}(A_r)]\,\mathrm{d}r$ has a fixed point in the norm $\sup_{0<s\leq s_0(R)} \max\{s^\kappa \|w_s\|_\infty, s^{\frac{1}{2}+\kappa}\|\nabla w_s\|_\infty\}$. The first iterate $\int_0^s P_{s-r}\mathcal{N}(P_r\mathbf{B}, P_r\mathbf{B})\,\mathrm{d}r$ is bounded through the quadratic member of the rough norm, by $\|P_{s-r}f\|_\infty \leq (s-r)^{-1/4}|f|_{-\frac{1}{2};a}$ and $\int_0^s (s-r)^{-\frac{1}{4}} r^{-\frac{3}{4}-\kappa} dr \leq c\, s^{-\kappa}$; the cross terms $\mathcal{N}(P_r\mathbf{B}, w_r)$, $\mathcal{N}(w_r, P_r\mathbf{B})$, $\mathcal{N}(w_r, w_r)$ give the unweighted factor $s^{\frac{1}{4}-\frac{3}{2}\kappa}$.

This is a positive power because $\kappa < \frac{1}{6}$; multiplication by the weight s^κ in the fixed-point norm gives the contraction condition $R^2 s^{\frac{1}{4}-\frac{\kappa}{2}} \leq c_1$; the members of order at least three are bounded in the chart, $|\mathcal{R}(A)| \leq 2 \sup|\Theta^{\mathrm{ch}}|\, \|A\|_\infty^3$ for $a_K\|A\|_\infty \leq \frac{1}{2}$ by Cauchy's estimate, and the flow does not leave the chart on $[0, s_0(R)]$ for $a_K \leq a_0(R)$, by the choice of c_2, as $a_K\|\mathbf{B}\|_\infty \leq e^{12} R a_K^{\frac{1}{2}-\kappa}$. The other two bounds follow with $s^{\frac{1}{4}+\frac{\kappa}{2}}\|P_s\mathbf{B}\|_\infty \leq R$, $s^{\frac{3}{4}+\frac{\kappa}{2}}\|\nabla P_s\mathbf{B}\|_\infty \leq 2R$.

The velocities of V_s and of $\Phi_s(V_0)$ differ by a gauge direction, so the two are gauge equivalent. For two data, $(A_s - A'_s) - P_s(\mathbf{B} - \mathbf{B}')$ solves the same kind of equation, into which the quadratic member enters through $\mathcal{N}(P_r\mathbf{B}, P_r\mathbf{B}) - \mathcal{N}(P_r\mathbf{B}', P_r\mathbf{B}')$, which the rough distance bounds; the fixed point closes on $(0, s_{\mathrm{Lip}}(R)]$. The action: for $s \leq s_0(R)$ and $a_K \leq a_0(R)$, $A^a \leq \frac{3}{2}\ell^3(a_K^{-2}\varpi)^2$ and $\varpi(e^{ia_K A}) \leq 2a_K^2(\|A\|_\infty^2 + \|\nabla A\|_\infty)$ (Lem. 16.2(c), (e)) give with the four bounds $A^a(\Phi_s e^{ia_K\mathbf{B}}) \leq 6((1+c_3)^2 + 2 + c_3)^2\ell^3 R^4 s^{-\frac{3}{2}-\kappa}$; for $s_0(R) < s \leq 1$ the action does not increase (Lem. 16.2(c)), and its value at $s_0(R)$ is at most $CR^4(R^2/c_1)^{4(\frac{3}{2}+\kappa)/(1-2\kappa)}$, a power of R below 24; for $a_K > a_0(R)$, $A^a \leq 6\ell^3 a_K^{-4} < 6\ell^3(R/c_2)^{8/(1-2\kappa)}$, a power of R below 12.

So c_4 depends only on κ.

The field of the gauge of the chain. Given $\mathcal{F}_{d-1}$, the σ-algebra of V_{K-d+1}, the law of B_{K-d} is p_W, $W = V_{K-d+1}$ (Lem. 15.3(ii)). The layers are made step by step, for $k = t-1, \dots, 0$, starting from $\mathsf{U}_t = U_{\mathrm{top}}$, a background over V_t, and ending at $k = 0$, where $\mathcal{M}^0$ is the identity, so that U is a gauge transform of U_0. At the step k, with $W = V_{k+1}$, let U_+ be a background over W whose block field $\mathcal{M}^k U_+$ is the base point $V_W(0)$ of the chart, a background of the level k as well, over $\mathcal{M}^k U_+$ (the background and chart identification in Prop. YM3_I:52.3), chosen as a Borel function of W, and let g_{k+1}, a function of V_{k+1}, be the gauge with $\mathsf{U}_{k+1}^{\mathsf{g}_{k+1}} = U_+$. The chart writes $V_k = V_W(B_k)$, and $\zeta = x_k^{-1/2} B_k$, on the bonds that are not pivot bonds, has $\sup|\zeta| < \delta(x_k) \leq \frac{1}{6}\varepsilon_{\mathrm{A}}$; by the global shift identification in Prop. YM3_I:51.5, and at $k = 0$ by the pivot chart in equation YM3_I:(800), the background over V_k is a gauge transform of $e^{iL^{-k}\mathbf{A}[U_+;\zeta]}U_+$. As $a_K L^d = L^{-k}$ and $(e^{iX}U)^g = e^{i\mathrm{Ad}(g)X}U^g$, the layer $\mathbf{h}_d = L^d \mathrm{Ad}(\mathsf{g}_{k+1}^{-1})\mathbf{A}[U_+;\zeta]$ makes $\mathsf{U}_k = e^{ia_K\mathbf{h}_d}\mathsf{U}_{k+1}$ the transform of $e^{iL^{-k}\mathbf{A}[U_+;\zeta]}U_+$ by g_{k+1}^{-1}, a function of V_{k+1}; U_k is a function of V_k, and so is the gauge g_k of the next step. The bound $|\mathbf{A}[U_+;\zeta]| \leq c_{\mathsf{h}} \sup|\zeta|$ (Prop. YM3_I:51.5 and the pivot estimate YM3_I:(800)) gives $|\mathbf{h}_d| \leq c_{\mathsf{h}} L^d \delta(L^d \mathsf{b})$. The product that defines $\mathbf{B}_K$ lies within $\frac{3}{8}$ of $\mathbf{1}$ when $\sum_d a_K|\mathbf{h}_d| \leq \frac{1}{4}$ and $a_K \mathsf{a}_{\mathrm{top}} \leq \frac{1}{8}$.

The regular gauge of the top. The regularity number of the level k of a configuration U' on the bare lattice is $L^{2k}\max_p|\partial U'(p) - \mathbf{1}|$, the maximum over the bare plaquettes p ([7, Def. P.3]); at the level 0 it is $\varpi(U')$, and at the level k it is L^{2k} times its value at the level 0 (by that definition

in the two units). For $k \geq 1$, Cut_k consists of the block fields of the level k whose backgrounds exist and have regularity numbers of the level k below $c_\rho\varepsilon(x_k)$ (the definition of the field cut). As $V_t \in \mathsf{I}_{d_0} \subset \mathsf{Cut}_t$ and $x_t = L^{d_0}\mathsf{b}$, every background U' over V_t has $L^{2t}\varpi(U') < c_\rho\varepsilon(L^{d_0}\mathsf{b})$, so $\varpi(U') < c_\rho\varepsilon(L^{d_0}\mathsf{b})L^{-2t} = c_\rho\varepsilon(L^{d_0}\mathsf{b})L^{2d_0}a_K^2$. Lem. 15.7(a) with $\mathfrak{n} = N$, at $\kappa' = \max\{c_\rho\varepsilon(L^{d_0}\mathsf{b})L^{2d_0}, \ell^{-2}\}a_K^2$, for which $N \geq \kappa'^{-1/2}$, gives a gauge γ, which a measurable selection makes depend on V_t alone in a Borel way, with $|U^\gamma_{\mathrm{top}}(b) - \mathbf{1}| \leq C_{\mathrm{BG}}\sqrt{\kappa'} \leq C_{\mathrm{BG}} \max\{(c_\rho\varepsilon(L^{d_0}\mathsf{b}))^{1/2}, \ell^{-1}\}L^{d_0}a_K \leq CL^{d_0}a_K$ at every bond; at the finitely many K with $\kappa' > \kappa_0$ or $N < \lambda_{\mathrm{BG},3}$ we take $\gamma = \mathbf{1}$, and there $\pi/a_K \leq CL^{d_0}$. So $U^\gamma_{\mathrm{top}} = e^{ia_K\mathbf{A}^{\mathrm{top}}}$, with $ia_K\mathbf{A}^{\mathrm{top}}$ a fixed Borel $\mathfrak{su}(2)$ logarithm of norm at most π, equal to the principal logarithm off $-\mathbf{1}$, at each positive bond, and extended by reversal, so that $\mathbf{A}^{\mathrm{top}}$ too depends on V_t alone in a Borel way, and $|\mathbf{A}^{\mathrm{top}}| \leq \frac{\pi}{2}a_K^{-1}|U^\gamma_{\mathrm{top}} - \mathbf{1}|$ gives $|\mathbf{A}^{\mathrm{top}}| \leq \mathsf{a}_{\mathrm{top}} = CL^{d_0} \leq CL^{10d_0}$. The bounds of J_d, and those of the derivative of the layer on the box, come from the locality of the shift. Let $\mathfrak{n}$ be the distance, in the maximum norm and in units of their side, of the cubes of π^M_{k+1} that hold β_- and b_-, for a bare bond β and a bond b of the level k, and $e^{\bar\kappa}$ the radius of the polydisc $\mathbb{D}(Y)$ of the parameters of the shift systems (Prop. YM3_I:51.5 and the hypotheses preceding it). For $\mathfrak{n} \geq 3$ the value of $\mathbf{A}[U_+;\zeta]$ at β differs from that of the shift of the cube of $(2\mathfrak{n}-1)^3$ cubes about the cube of β_-, which does not read $\zeta(b)$, by at most $2c_{\mathsf{h}} \sup|\zeta| e^{-\bar\kappa(\mathfrak{n}-2)}$: let every fine cube, that is every cube of π^M_k, in the shell of the cubes of this cube at the distance j from the cube of β_-, $2 \leq j \leq \mathfrak{n}-1$, one of its $\mathfrak{n}-2$ outer shells, carry a single variable s_j, $|s_j| < e^{\bar\kappa}$, and every fine cube outside these shells the value 1; by the holomorphy, support and restriction properties of Prop. YM3_I:51.5 the difference is then the value at $s_2 = \cdots = s_{\mathfrak{n}-1} = 1$ of a holomorphic function of these $\mathfrak{n}-2$ variables, of modulus at most $2c_{\mathsf{h}} \sup|\zeta|$ on their polydisc, and zero as soon as some s_j is zero, since that shell then separates the cubes nearer than it to the cube of β_-, among them those that hold the two ends of β, from the cubes beyond it; and Schwarz's lemma in the s_j gives the bound. So, by Cauchy's estimate, for $\sup|\zeta| \leq \frac{1}{2}\varepsilon_{\mathrm{A}}$ the derivative of $\mathbf{A}[U_+;\zeta]$ at β in the value of ζ at b is at most $4c_{\mathsf{h}}e^{-\bar\kappa(\mathfrak{n}-2)_+}$, and for $k = 0$ it vanishes unless $\mathfrak{n} \leq 1$ (the double-block dependence of the pivot chart YM3_I:(800)). As at most 125 cubes lie at a distance ≤ 2 from a given one and at most $24\mathfrak{n}^2 + 2$ at the distance $\mathfrak{n}$, Schur's test gives the two norms of J_d, and the same bounds for the derivative of the layer on the box, with $c_{\mathsf{J}} = \sqrt{27}(LM)^3 \cdot 4c_{\mathsf{h}}[125 + \sum_{\mathfrak{n}\geq 3}(24\mathfrak{n}^2+2)e^{-\bar\kappa(\mathfrak{n}-2)}]$, and $\|\mathsf{J}_d^* f\| \leq c_{\mathsf{J}} \min\{L^{-3d/2}\|f\|_2, \|f\|_1\}$.

Each layer splits as $\mathbf{h}'_d = \mathbf{n}_d + u_{(d)} + e_{(d)}$: $\mathbf{n}_d$ its conditional mean given $\mathcal{F}_{d-1}$; $u_{(d)} = \mathsf{b}^{-1/2}L^{d/2}\mathrm{Ad}(\gamma)\mathsf{J}_d\tilde B_d$, linear in the centred fluctuation $\tilde B_d = B_{K-d} - \int B\,\mathrm{d}p_W$ with a predictable coefficient; $e_{(d)}$ the centred rest, which carries besides the factor $\delta(L^d\mathsf{b})$. So $\mathbf{B}_K = u + \rho$, $u = \sum_d u_{(d)}$, ρ the sum of the $e_{(d)}$, of the $\mathbf{n}_d$, of $\mathbf{A}^{\mathrm{top}}$ and of the remainder of the ordered product beyond the sum of its factors.

The linear fluctuation. By Lem. 15.3(iv), given $\mathcal{F}_{d-1}$ every Lipschitz function of B_{K-d} has Gaussian tails, with the variance proxy $\frac{8}{\gamma_{\mathrm{r}}}$ times the square of its Lipschitz constant; for $\langle u_{(d)}, f\rangle$ this square is at most $c_{\mathsf{J}}^2\mathsf{b}^{-1}L^d \min\{\|f\|_1^2, L^{-3d}\|f\|_2^2\}$, so $(P_s u_{(d)})(x)$, $f = k_s(x,\cdot)$ the heat kernel, has the proxy $C\mathsf{b}^{-1}L^d\min\{1, L^{-3d}s^{-3/2}\}$, whose sum over d is $\leq C\mathsf{b}^{-1}s^{-1/2}$. Exponential supermartingales over the depths, with test functions that are functions of $\mathcal{F}_{d-1}$, and a union over dyadic grids of times and sites, through the grid lemma for $|\cdot|_{\eta;a}$ and the bounds of the heat kernel, give Gaussian tails of $|u|_{-\frac{1}{2}-\kappa;a}$.

The quadratic fluctuation and colour cancellation. Sort the pairs in $\mathcal{N}_s[u,u]$ by the finer index. A pair of different depths is linear in the finer layer with a predictable coefficient: a martingale difference, with Gaussian tails as before. The diagonal $\mathcal{N}_s[u_{(d)}, u_{(d)}]$, after P_r at a point, is a quadratic form $(\tilde B_d, \mathcal{Q}_d\tilde B_d)$ of the centred fluctuation, $\|\mathcal{Q}_d\|_{\mathrm{tr}} \leq C\mathsf{b}^{-1}L^d s^{-1/2}\min\{1, L^{-3d}s^{-3/2}\}$; its centred part has exponential moments by (Q), through the norm and the Hilbert–Schmidt norm of $\mathcal{Q}_d$. Its compensator, the conditional mean $\mathrm{tr}(\mathcal{Q}_d\mathrm{Cov}_W)$, would sum by the sizes to the order

$\mathsf{b}^{-1}s^{-1}$, beyond the weight $s^{\frac{3}{4}+\kappa}$; but it reads the conditional covariance of the layer on the bare lattice, in the gauge of the chain, only through its part that is antisymmetric in colour, as $X \wedge Y$ is antisymmetric.

By (Col) the scalar part drops out, and almost surely

$$|\mathrm{tr}(\mathcal{Q}_d \mathrm{Cov}_W)| \le C s^{-1/2} \min\{1, L^{-3d} s^{-3/2}\} [L^{d/2} (\log x)^{\mathfrak{p}_0+6} + \mathsf{a}_{\mathrm{top}}],$$

$x = L^d \mathsf{b}$, whose sum over d is $\le C(1+\mathsf{a}_{\mathrm{top}}) s^{-\frac{3}{4}-\frac{\kappa}{2}}$, which the weight $s^{\frac{3}{4}+\kappa}$ admits: the cancellation in colour.

Means and nonlinear remainders. The $e_{(d)}$ by the bounds of the linear part, with the factor $\delta(L^d\mathsf{b})$; the means by (M), summable in $|\cdot|_{-\kappa;a}$ through $\|\cdot\|_{L^p}$, $p = \lceil 3/\kappa \rceil$; $\mathbf{A}^{\mathrm{top}}$ by $\mathsf{a}_{\mathrm{top}}$; the ordered product, at the second order linear in the finer factor with a predictable coefficient, at the third order small; each with $|\cdot|_{-\kappa;a} \le C(1+\mathsf{a}_{\mathrm{top}})$ in every L^q. The cross members of u and ρ by the sizes, as $(-\frac{1}{2}-\kappa)+(-\kappa) \ge -\frac{1}{2}-2\kappa$. Together $\mathbb{E}^{[d_0]}[\|\mathbf{B}_K\|_{\kappa;a}^q]^{1/q} \le C(q,\kappa)(1+\mathsf{a}_{\mathrm{top}})^2$, every constant reading d_0 only through $\mathsf{a}_{\mathrm{top}}$; with $\mathsf{a}_{\mathrm{top}} \le C L^{10 d_0}$ this is the bound of the statement; for the finitely many K with $a_K \mathsf{a}_{\mathrm{top}} > \frac{1}{8}$ the principal logarithm has $\|\mathbf{B}_K\|_\infty < 8\pi \mathsf{a}_{\mathrm{top}}$.

The statements on one fluctuation integral that enter, besides Lem. 15.3(iv), the bounds of the response, the bounds of the heat kernel and the grid lemma, are, with C free of K and d_0: (M) $\|\mathbf{n}_d\|_\infty \le C\mathsf{b}^{-1}(\log L^d\mathsf{b})^{2\mathfrak{p}_0}$ and $\|\mathbf{n}_d\|_{L^2}^2 \le C\ell^3\mathsf{b}^{-3}L^{-d}(\log L^d\mathsf{b})^{8\mathfrak{p}_0+12}$; (Col) in the gauge of the chain the kernel of $\mathrm{Ad}(\gamma)\mathsf{J}_d\mathsf{A}_W^{-1}\mathsf{J}_d^*\mathrm{Ad}(\gamma)^{\mathsf{T}}$ at two bare bonds whose partition cubes are at distance N is a number times the identity in colour up to w_N, $\sum_N (24N^2+2)\mathsf{w}_N \le C[x^{-1/2}(\log x)^{\mathfrak{p}_0+6} + \mathsf{a}_{\mathrm{top}} L^{-d}]$, the number being formed at the trivial background, where the Gaussian form and the response commute with the adjoint action; and $\|\mathrm{Cov}_W - \mathsf{A}_W^{-1}\| \le C x^{-1/2}(\log x)^{\mathfrak{p}_0+3}$, A_W the Gaussian form of p_W.

(Q) There are universal constants $C_{\mathrm{HW}}, c_{\mathrm{HW}} > 0$ with the following property. Let the real symmetric $\mathcal{Q}_d$ be predictable, that is, functions of $\mathcal{F}_{d-1}$, and assume almost surely the deterministic bounds $\|\mathcal{Q}_d\| \le \mathsf{q}$ and $C_{\mathrm{HW}}\gamma_{\mathrm{r}}^{-2}\sum_d \|\mathcal{Q}_d\|_{\mathrm{HS}}^2 \le \Sigma^2$. For the actual conditional covariance set $\mathsf{X}_d = (\tilde{B}_d, \mathcal{Q}_d\tilde{B}_d) - \mathrm{tr}(\mathcal{Q}_d\mathrm{Cov}_W)$. Then $\mathbb{E}^{[d_0]} e^{\epsilon\sum_d \mathsf{X}_d} \le e^{\epsilon^2\Sigma^2}$ for $|\epsilon|\mathsf{q} \le c_{\mathrm{HW}}\gamma_{\mathrm{r}}$.

Indeed the conditional box law has Hessian at least $\gamma_{\mathrm{r}}/4$ and log-Sobolev constant $8/\gamma_{\mathrm{r}}$ (Lem. 15.3(iv)); the convex-domain criterion [4, Thm. 2.1] applies. For nonzero $\mathcal{Q}$, Adamczak's dependent-vector inequality [3, Thm. 2.3] gives the centred quadratic form Y the conditional tail $2\exp[-c\min\{\gamma_{\mathrm{r}}^2 t^2/\|\mathcal{Q}\|_{\mathrm{HS}}^2, \gamma_{\mathrm{r}} t/\|\mathcal{Q}\|\}]$. For $\epsilon > 0$, centering gives

$$\mathbb{E}e^{\epsilon Y} - 1 = \epsilon \int_0^\infty [(e^{\epsilon t}-1)\mathbb{P}(Y > t) + (1-e^{-\epsilon t})\mathbb{P}(Y < -t)]\,dt.$$

Integrating the Gaussian and exponential tails bounds the conditional logarithmic moment by $C_{\mathrm{HW}}\epsilon^2\gamma_{\mathrm{r}}^{-2}\|\mathcal{Q}\|_{\mathrm{HS}}^2$ for $|\epsilon|\|\mathcal{Q}\| \le c_{\mathrm{HW}}\gamma_{\mathrm{r}}$; apply the same argument to $-Y$. Consequently $\exp[\epsilon\sum_{j\le d}\mathsf{X}_j - C_{\mathrm{HW}}\epsilon^2\gamma_{\mathrm{r}}^{-2}\sum_{j\le d}\|\mathcal{Q}_j\|_{\mathrm{HS}}^2]$ is a conditional supermartingale. Iteration and the pathwise budget prove (Q); independence of the layers is unnecessary.

The proof of (Adm). Here $d_0 < d \le K$, $k = K-d$, $x = L^d\mathsf{b}$, $g = x^{-1/2}$, $N_k = (\ell L^d)^3$, $W \in \mathsf{Cut}_{k+1}$, U_+ is as above, $\bar{B}_W = \int B\, p_W(\,\mathrm{d}B)$, and v is the sum of the potentials, so that $\Psi_W = \frac{1}{2}(B, \mathsf{A}_W B) - v$ up to a constant (Prop. 15.4(b)). (1)

Conditions on numbers. Besides the chain, the bounds of (2) use only the following conditions on numbers. At the top depth there are six,

$$1 \le d_0 < K, \qquad L^{d_0}\mathsf{b} \ge \mathsf{t}_\natural,$$
$$\mathsf{R}_{d_0} \le \tfrac{1}{2}, \qquad C_{\mathrm{hs}}\,\delta(L^{d_0}\mathsf{b}) \le \tfrac{1}{4}\gamma_{\mathrm{r}},$$
$$4c_{\mathsf{h}}\delta(L^{d_0}\mathsf{b}) \le \tfrac{1}{4}, \qquad c_\rho \varepsilon(L^{d_0}\mathsf{b}) \le C_{\mathrm{BG}}^{-2},$$

with $C_{\rm hs} = C_{\rm E,1}[338K_P + \frac{1}{2}K_\Lambda(1+\beta_{\rm s})N_\star^{\rm s}/\alpha_0]$, formed from $C_{\rm E,1}$ and the numbers that make up C_V in the setting; the fifth then holds at every depth $d > d_0$ too, δ being decreasing from the reciprocal of the squared coupling floor on, a number that b does not fall below under (E) (the decreasing cutoff functions in §2.3.1). Together with them, (2) uses at the depths $d > d_0$ the bound $C_{\rm Hs}\delta(L^d\mathsf{b}) \le \frac{1}{4}\gamma_{\rm r}$, with the constant $C_{\rm Hs} = 144\,C_{\rm E,1}C_V$ of the setting, which exceeds $C_{\rm hs}$, as $3\cdot 144 > 338$ and $144 > \frac{1}{2}$, so that at d_0 this bound contains the fourth of the six; by the choice of n_2 it holds at every depth $d \ge n_2$, and $n_2 \le n_{\rm a} \le d_{\rm adm}$, so that it holds at d_0 and at every depth $d > d_0$ as soon as $d_0 \ge d_{\rm adm}$ and asks nothing of $d_{\rm adm}$ beyond its member $n_{\rm a}$. At each depth $d > d_0$, with $x = L^d\mathsf{b}$ as above, $\bar\kappa \ge 1$ (Def. YM3_I:47.2) and

$$\nu_0 = \Big\lceil \frac{\log x}{2\bar\kappa} \Big\rceil, \qquad \nu_1 = \Big\lceil \frac{\lceil \log x \rceil + 2}{L} \Big\rceil,$$
$$\nu_2 = \max\Big\{\nu_1, \Big\lceil \frac{3\log x}{2\bar\kappa} \Big\rceil + 1\Big\}, \qquad \nu_3 = \max\Big\{\nu_1, 2\Big\lceil \frac{2\log x}{\bar\kappa} \Big\rceil + 19\Big\},$$

there are three conditions on cubes, at each of the three inner radii $\nu = \nu_1, \nu_2, \nu_3$:

$$2M(2\nu + 2\nu_0 + 3)^3 + 4 < L^{m+d-1},$$
$$4M(2\nu + 2\nu_0 + 3)^3\,\theta_{d-1} \le \varepsilon_{\mathcal W}\, c_{*,\rm rg},$$
$$4B_*M(2\nu + 2\nu_0 + 3)^3\,\theta_{d-1} \le \tfrac{1}{2}\varepsilon_{\mathcal W}\, a^{\rm us}(\alpha_0, \alpha_1),$$

with $\theta_{d-1} = 2c_\rho\varepsilon(x/L)$, the absolute numbers $B_* \ge 1$ and $c_{*,\rm rg} \in (0,1]$ of the local regular-gauge Thm. YM3_I:F, and the usable radius $a^{\rm us}(\alpha_0,\alpha_1)$ in equation YM3_I:(933) at the fixed radii. $(2\nu + 2\nu_0 + 3)^3$ is the number of the cubes of π^M_{k+1} at distances up to $\nu + \nu_0 + 1$ from a given one, in units of their side; they form the cube Q_ν about it, of side $(2\nu + 2\nu_0 + 3)M$ in the units of the level $k+1$, whose cubes at distances up to ν form its *inner cube*, parted from the cubes outside Q_ν by its *collar*, the ν_0 shells of the cubes at the distances $\nu + 2, \dots, \nu + \nu_0 + 1$. The *trivial triple* on a polymer Y is $P[0] = (\mathbf{1}, 0, 0)$, the value at 0 of the chart $P[\omega] = (\mathbf{1}, A, \mathbf{J})$ at the trivial pair, $\omega = (A, \mathbf{J})$, whose pair is $\mathbf{p}(P[\omega]) = (e^{iL^{-k-1}A}, \mathbf{J})$ (the triple and pair conventions of Def. YM3_I:49.1); a triple P on Y is *within r of the trivial triple in a gauge* $\mathfrak{v}$ if $P^{\mathfrak{v}}$ has the pair of $P[\omega]$ for a real ω with $\|\omega\|_Y < r$, the triple norm, namely the largest of $|A|$, $|\nabla_{\mathbf{1}}A|$ and $|\mathbf{J}|$ on $\mathrm{Bd}(Y)$. Every term of the exponent of p_W reads a triple through its pair alone (proof of Prop. 15.4(c)), and so do the structure of the step and the shift (Thm. YM3_I:U, Prop. YM3_I:51.5, and the actual-core verification in the proof of Thm. YM3_I:W). Under the three conditions Q_ν is small against the side of the torus of the level, and Thm. YM3_I:F at the level $k+1$, on Q_ν with $\mathsf{l} = (2\nu + 2\nu_0 + 3)M$ and at $\alpha = 2\theta_{d-1}/\varepsilon_{\mathcal W}$ (the first two conditions give its conditions on the sides and on $\mathsf{l}\alpha$), gives a gauge $\mathfrak{v}$ and a real 1-form A' with $U_+^{\mathfrak{v}} = e^{iL^{-k-1}A'}$ on the bare bonds of the concentric cube of side $\mathsf{l}+4$ and $|A'|, |\nabla_{\mathbf{1}}A'| < B_*\mathsf{l}\alpha = 2B_*(2\nu + 2\nu_0 + 3)M\theta_{d-1}/\varepsilon_{\mathcal W}$ on $\mathrm{Bd}(Q_\nu)$ (the potential and first-difference estimates of that theorem). Its hypotheses on U_+, at the spacing $\eta = \eta_{k+1}$, are the bounds $|\partial U_+(p) - \mathbf{1}| < \alpha\eta^2$ at the bare plaquettes p with a corner in the concentric cube of side $\mathsf{l}+4$ and $|(D^{\mathsf T}_{U_+}\partial U_+)(b)| < \alpha\eta^2$ at the bare bonds b with an end point in that cube; here, for a configuration U, $D^{\mathsf T}_U\partial U$ is the 1-form whose value at $b = \langle x, x+\eta e_\mu\rangle$ is $\eta^{-1}\sum_{\nu\ne\mu}[X_{\mu\nu}(x) - \mathrm{Ad}(U(\langle x, x-\eta e_\nu\rangle))X_{\mu\nu}(x-\eta e_\nu)]$, with $X_{\mu\nu}(y)$ the value of ∂U at the plaquette of the directions $\mu < \nu$ based at y and $X_{\nu\mu} = -X_{\mu\nu}$ (the covariant-difference convention of Section YM3_I:22), that is, the transpose, at the spacing η and at U, of the covariant exterior derivative of 1-forms into 2-forms, applied to the matrix-valued 2-form $p \mapsto \partial U(p)$. Both bounds hold at all the bare plaquettes and bonds of the torus: U_+, a background over $W \in \mathsf{Cut}_{k+1}$, has regularity number of the level $k+1$ less than $c_\rho\varepsilon(x/L) = \frac{1}{2}\theta_{d-1}$ (the definition of the field cut), so $|\partial U_+ - \mathbf{1}| < \frac{1}{2}\theta_{d-1}\eta_{k+1}^2 < \alpha\eta_{k+1}^2$; its current $\mathbf{J}_+$ is below $\theta_{d-1}/\varepsilon_{\mathcal W}$ at every bare bond, as $\varpi(W) < \theta_{d-1}$ admits the plain-core verification in the proof of Prop. YM3_I:52.3 at the free radii $\theta_{d-1}/\varepsilon_{\mathcal W}$, α_1 (as in the proof of Prop. 15.4(a)), which puts P_W in the core with these radii, and

the condition (iii) of its simple set bounds the current by the first radius (Def. YM3_I:49.1); so $|D^{\mathsf T}_{U_+}\partial U_+| \le \eta_{k+1}^2|\mathbf J_+| + \frac14\eta_{k+1}^3\theta_{d-1}^2 < \alpha\eta_{k+1}^2$ (equation YM3_I:(579)). As $|\mathrm{Ad}(\mathfrak v)\mathbf J_+| = |\mathbf J_+|$, P_W is within $r_\nu := 4B_*M(2\nu{+}2\nu_0{+}3)^3\theta_{d-1}/\varepsilon_{\mathcal W}$ of the trivial triple in the gauge $\mathfrak v$, with $\omega = (A', \mathrm{Ad}(\mathfrak v)\mathbf J_+)$, on Q_ν and on every polymer in it; r_ν is below a multiple of $x^{-1/2}(\log x)^{\mathfrak p_0+3}$ and below half the usable radius, so that the chart carries the ball of radius $2r_\nu$ into the analyticity cores. The cubes Q_{ν_1}, Q_{ν_2}, Q_{ν_3} serve (i), (ii), (iii) of (2) in turn; in each the collar is the room in which $\mathsf A_W^{-1}$ on the inner cube forgets the cubes outside, up to a multiple of $x^{-1/2}$ (see (i)). The six conditions at the top depth and the conditions on cubes hold for every integer $d_0 \ge d_{\mathrm{num}}(\ell,\mathsf b)$, every $K > d_0$ and every depth $d > d_0$, where

$$
\begin{aligned}
d_{\mathrm{num}}(\ell,\mathsf b) := \max\Big\{1,\ &\Big\lceil\frac{\log(\mathsf t_{\mathrm{adm}}/\mathsf b)}{\log L}\Big\rceil,\\
&\Big\lceil\frac{\mathsf y_{\mathsf R}(\ell,\mathsf b)-\log\mathsf b}{\log L}\Big\rceil,\\
&\Big\lceil\frac{2}{\log L}\log\frac{12\cdot10^5\,LM(\log\mathsf b)^3}{\ell}\Big\rceil\Big\},
\end{aligned}
$$

$$
\begin{aligned}
\mathsf t_{\mathrm{adm}} = \max\Big\{&\mathsf t_\natural,\\
&\Big(\frac{4\mathfrak p_0}{e}\Big)^{4\mathfrak p_0}\max\Big\{\frac{4C_{\mathrm{hs}}A_1}{\gamma_{\mathrm r}},\,16c_{\mathsf h}A_1,\,C_{\mathrm{BG}}^2c_\rho A_0\Big\}^4,\\
&\Big(\frac{125000\,c_\rho A_0L^{1/2}M}{\varepsilon_{\mathcal W}}\Big)^4\Big(\frac{4\mathfrak p_0+12}{e}\Big)^{4\mathfrak p_0+12}\\
&\qquad\max\Big\{\frac{1}{c_{*,\mathrm{rg}}},\,\frac{2B_*}{a^{\mathrm{us}}(\alpha_0,\alpha_1)}\Big\}^4\Big\},
\end{aligned}
$$

$$
\mathsf y_{\mathsf R}(\ell,\mathsf b) = \max\Big\{\Big(\frac{6}{c'A_0^2}\Big)^{\frac{1}{2\mathfrak p_0-1}},\ \Big(\frac{2\,\mathsf E(\ell,\mathsf b)}{c'A_0^2}\Big)^{\frac{1}{2\mathfrak p_0}}\Big\},
$$

$$
\mathsf E(\ell,\mathsf b) = \log 4 + 3\log\frac{\ell}{M} + \max\{\mathsf k(\mathsf b),0\}\,\ell^3,
$$

with the rate $c'>0$ in the bound defining $\mathsf R_d$ and $\mathsf k(\mathsf b) = \mathsf C^{\Sigma}_{\mathrm{maj}}+2+C_{\mathrm{low}}(\mathsf b)$, the factor of ℓ^3 in the last exponent of the definition of $\mathsf R_d$; $\mathsf E(\ell,\mathsf b)>0$, as $\ell = 2L^m \ge 2MR(\mathsf b) \ge 2M$ under (E). $\mathsf t_{\mathrm{adm}}$ is formed from the constants alone, and $d_{\mathrm{num}}(\ell,\mathsf b)$ from ℓ, $\mathsf b$ and the constants of the construction; it reads neither K nor a level nor a field. For the proof: for $\mathsf t>1$ and $r>0$, $\mathsf t^{-1/2}(\log\mathsf t)^r \le (4r/e)^r\mathsf t^{-1/4}$, as $r\log z - \frac{z}{4}$ is largest at $z=4r$. Let $d_0 \ge d_{\mathrm{num}}(\ell,\mathsf b)$ and $K>d_0$, which is the first of the six. Then $L^{d_0}\mathsf b \ge \mathsf t_{\mathrm{adm}}$, which gives the second and, as $\delta(\mathsf t)$ and $\varepsilon(\mathsf t)$ are A_1 and A_0 times $\mathsf t^{-1/2}(\log\mathsf t)^{\mathfrak p_0}$, by that inequality at $r=\mathfrak p_0$ the last three; and $z_0 = \log L^{d_0}\mathsf b \ge \mathsf y_{\mathsf R}(\ell,\mathsf b)$, so that half of $c'A_0^2z_0^{2\mathfrak p_0}$ is at least $3z_0$ and at least $\mathsf E(\ell,\mathsf b)$, while the definition of $\mathsf R_{d_0}$, at the $(\ell L^{d_0})^3$ sites of the depth d_0 and with $3d_0\log L < 3z_0$, gives $\log 2\mathsf R_{d_0} = \log 4 + 3\log\frac{\ell}{M} + 3d_0\log L + \mathsf k(\mathsf b)\ell^3 - c'A_0^2z_0^{2\mathfrak p_0} < \mathsf E(\ell,\mathsf b) + 3z_0 - c'A_0^2z_0^{2\mathfrak p_0} \le 0$: the third. At a depth $d>d_0$, $x \ge L\mathsf t_{\mathrm{adm}}$, and $\log x > 3$, as $\log\mathsf b > 2$ (the label floor of §2.3.1); as $\lceil z\rceil < z+1$, $L\ge13$ and $\bar\kappa\ge1$, each of ν_1,ν_2,ν_3 is less than $4\log x + 21$ and $\nu_0 < \frac12\log x + 1$, so that $(2\nu+2\nu_0+3)^3 < (9\log x+47)^3 < 15625(\log x)^3$. With $\theta_{d-1} < 2c_\rho A_0L^{1/2}x^{-1/2}(\log x)^{\mathfrak p_0}$ and that inequality at $r=\mathfrak p_0+3$, the left member of the second condition on cubes is less than $125000\,c_\rho A_0L^{1/2}M(\frac{4\mathfrak p_0+12}{e})^{\mathfrak p_0+3}x^{-1/4}$ and that of the third less than B_* times this number; and $x\ge\mathsf t_{\mathrm{adm}}$ makes these two bounds at most $\varepsilon_{\mathcal W}c_{*,\mathrm{rg}}$ and $\frac12\varepsilon_{\mathcal W}a^{\mathrm{us}}(\alpha_0,\alpha_1)$. And as $\log x \le \log\mathsf b\,(1+d\log L)$ and $(1+z)^3 < 17.74\,e^{z/2}$ for $z\ge0$, the left member of the first is less than $31251M(\log x)^3 < 6\cdot10^5M(\log\mathsf b)^3L^{d/2}$, which is below L^{m+d-1}, as $L^{d/2} > L^{d_0/2} \ge 6\cdot10^5L^{1-m}M(\log\mathsf b)^3$ by the last member of $d_{\mathrm{num}}(\ell,\mathsf b)$, ℓ being $2L^m$.

A uniform second moment. For the fluctuation fields the bound holds under p_W at every W. Fix W and a bond b_0. At a bond b, the divergence theorem in the variable $B(b)$ on the ball $\{|B(b)| < \mathfrak{b}(x_k)\}$ of the box, applied to $B(b)e^{-\Psi_W}$, gives the flux $\mathfrak{b}(x_k)\int_{|B(b)|=\mathfrak{b}(x_k)} e^{-\Psi_W} \geq 0$, so $\int B(b)\cdot\nabla_b\Psi_W \,\mathrm{d}p_W \leq 3$. By Prop. 15.4(b), $\Psi_W = \frac{1}{2}(B, \mathsf{A}_W B) - v$ up to a constant, v the sum of the potentials; $\mathsf{A}_W \geq \frac{1}{2}\gamma_{\mathrm{r}}$, and Cauchy's estimate gives $|\nabla_b v| \leq C\delta(x_k)\mathfrak{b}(x_k) \leq \eta$ on the box, η a number free of x_k (the proof of Lem. 15.3(iv)). Take the weights $w_b = e^{-\mu\,\mathrm{dist}(b,b_0)}$, dist the distance in the units of the level k. The entries of A_W fall off exponentially with the distance of the bonds, as A_W is a sum of local terms with the bounds of the polymers (Prop. 15.4(b), Prop. YM3_I:51.4 and its proof), so Schur's test gives $\|w^{1/2}\mathsf{A}_W w^{-1/2} - \mathsf{A}_W\| \leq \frac{1}{4}\gamma_{\mathrm{r}}$ for a small $\mu > 0$ that reads neither K, d nor W. Then $\frac{1}{4}\gamma_{\mathrm{r}}\sum_b w_b \int |B(b)|^2 \,\mathrm{d}p_W \leq \sum_b w_b \int (3 + \eta|B(b)|)\,\mathrm{d}p_W$, and $\eta|B(b)| \leq 2\eta^2/\gamma_{\mathrm{r}} + \frac{1}{8}\gamma_{\mathrm{r}}|B(b)|^2$ gives $\int |B(b_0)|^2 \,\mathrm{d}p_W \leq m_2 = \frac{8}{\gamma_{\mathrm{r}}}(3 + \frac{2\eta^2}{\gamma_{\mathrm{r}}})\sum_b w_b$, which reads neither K, d, d_0 nor W, as $\sum_b w_b \leq C\mu^{-3}$. The log-Sobolev bound of Lem. 15.3(iv), together with this second moment, gives Gaussian tails about a uniformly bounded mean. Consequently the boundary layers used below have mass smaller than every power of x. (2)

The bounds on one integral, for $d_0 \geq d_{\mathrm{adm}}$; C stands for numbers formed from $\ell/\mathfrak{b}$ and the constants of the construction. (i)

The mean of the fluctuation. Integration by parts in each variable of the box, against a cut-off that vanishes on its boundary, gives $\mathsf{A}_W\bar{B}_W$ as the mean of ∇v up to terms carried by $\{|B(b)| \geq \frac{1}{2}\mathfrak{b}(x)\}$, whose mass under p_W is at most a multiple of $e^{-c\,\mathfrak{b}(x)^2}$, $c > 0$, and so smaller than every power of x; ∇v is at most a multiple of $\delta(x)\mathfrak{b}(x) = A_1^2 x^{-1/2}(\log x)^{2\mathfrak{p}_0}$ at each bond; and the kernel of A_W^{-1} decays at the rate $\bar{\kappa}$: $\|\mathsf{A}_W^{-1}(b,b')\| \leq c_{\mathbf{K}} e^{-\bar{\kappa}(\mathfrak{n}-3)_+}$, $\mathfrak{n}$ the distance of the cubes of π_{k+1}^M that hold b_- and b'_-, with $c_{\mathbf{K}} = \frac{2}{\gamma_{\mathrm{r}}} + \frac{4}{\gamma_{\mathrm{r}}^2}(c_{\mathrm{r}} + \gamma_{\mathrm{r}})^2$, formed from γ_{r} and c_{r} alone. Indeed A_W^{-1} is the whole covariance of the complete structure of the step at P_W, a decoupled Gaussian structure with the constants $(\bar{\kappa}; \gamma_{\mathrm{r}}, c_{\mathrm{r}}, \hat{\varepsilon})$, $\hat{\varepsilon} = \frac{1}{2}\mathsf{u}_{\mathrm{G}}(LM)^{-3} \leq \frac{1}{2}\gamma_{\mathrm{r}}$ (Def. YM3_I:47.2, Thm. YM3_I:U, and the actual-core verification in Thm. YM3_I:W); on the union of the two cubes it is $\mathbf{C}+\mathbf{C}\Gamma\Gamma^{\mathsf{T}}\mathbf{C}$ at the decoupling parameters 1 (the conditioning identity in Thm. YM3_I:U), where $\mathbf{C} = \mathbf{C}(Z_\circ, s)$ and $\Gamma = \mathbf{\Gamma}(Z_\circ, s)$ are the conditioned covariance and the connecting operator, the two families of operators that the structure assigns to a region $Z_\circ$, here the union of the two cubes, and to the decoupling parameters s (Def. YM3_I:47.2); at all parameters s of the polydisc of radius $e^{\bar{\kappa}}$ this is an operator of norm at most $c_{\mathbf{K}}$, as (St3) with $\hat{\varepsilon} \leq \frac{1}{2}\gamma_{\mathrm{r}}$ gives $\|\mathbf{C}\| \leq 2/\gamma_{\mathrm{r}}$ and $\|\Gamma^{\mathsf{T}}\| = \|\Gamma\| \leq c_{\mathrm{r}} + \hat{\varepsilon} \leq c_{\mathrm{r}} + \gamma_{\mathrm{r}}$; for $\mathfrak{n} \geq 4$ its block between the two cubes is holomorphic in $\mathfrak{n} - 3$ variables, each the parameter of all the cubes of one of the shells at the distances $2, \dots, \mathfrak{n} - 2$ from the cube of b_-, the other parameters being 1 ((St1)), and vanishes as soon as one of them is 0, since that shell then parts the two cubes into different wall-components ((St2)); Schwarz's lemma in these variables gives the factor $e^{-\bar{\kappa}(\mathfrak{n}-3)}$, and for $\mathfrak{n} \leq 3$ the bound is $\|\mathsf{A}_W^{-1}\| \leq \frac{2}{\gamma_{\mathrm{r}}}$. As at most $24\mathfrak{n}^2 + 2$ cubes lie at the distance $\mathfrak{n}$ from a given one and $\bar{\kappa} \geq 1$, the rows of A_W^{-1} have bounded sums: so $\sup|\bar{B}_W| \leq Cx^{-1/2}(\log x)^{2\mathfrak{p}_0}$. The same argument, with one variable for each of the ν_0 shells of the collar of a cube Q_ν of (1), compares the parameters 1 and 0 on the cubes outside Q_ν: as $\nu_0\bar{\kappa} \geq \frac{1}{2}\log x$, on the inner cube A_W^{-1} lies within $2c_{\mathbf{K}}x^{-1/2}$ of the operator at the parameters 1 on the collar and 0 outside Q_ν, which is that of the structure on Q_ν at the restriction of P_W (restriction consistency in Thm. YM3_I:U); carried by the gauge $\mathfrak{v}$ of Q_ν it is that of the structure at $P[\omega]$ (gauge covariance and dependence on the pair), which at $P[z\omega]$ is holomorphic in z for $|z| < a^{\mathrm{us}}(\alpha_0, \alpha_1)/r_\nu$, by joint holomorphy in $(s, A', \mathbf{J})$ at a fixed real factor, taken at the real factor $\mathbf{1}$, as the chart carries the ball of radius $a^{\mathrm{us}}(\alpha_0, \alpha_1)$ into the cores (equation YM3_I:(933) and its proof), and at most $c_{\mathbf{K}}$ there, by (St3) as above, since Thm. YM3_I:U gives at every triple of the cores a structure with the same constants; so Schwarz's lemma in z puts A_W^{-1} on the inner cube, carried by $\mathfrak{v}$, within $2c_{\mathbf{K}}(x^{-1/2} + r_\nu/a^{\mathrm{us}}(\alpha_0, \alpha_1)) \leq Cx^{-1/2}(\log x)^{\mathfrak{p}_0+3}$ of the operator formed at the trivial triple on Q_ν, which commutes with the constant gauge transformations, as these fix

$P[0]$ (gauge covariance). In the ℓ^2 norm a further factor $x^{-1/2}$ comes, up to powers of $\log x$: along the laws proportional to $e^{-\frac12(B,\mathsf{A}_W B)+\tau v}$ on the box, $0\le\tau\le1$, whose exponents have Hessians $\ge\frac14\gamma_{\rm r}$, the Poincaré inequality makes the means of ∇v at $\tau=0$ and at $\tau=1$ differ in ℓ^2 by at most $\frac{4}{\gamma_{\rm r}}C_{\rm Hs}\delta(x)\sup\|\nabla v\|$; and at $\tau=0$, the Gaussian law of covariance A_W^{-1} conditioned on the box, the gradient at a bond b of each potential v_Y with $b_-\in Y$ and $d(Y)<\lceil\log x\rceil$ has a mean smaller than its size by a factor $Cx^{-1/2}(\log x)^{\mathfrak{p}_0+3}$: such a Y lies within $(\lceil\log x\rceil+2)M\le\nu_1 LM$ of b_- in the units of the level k, so in the inner cube of Q_{ν_1} about the cube of b_-; in the gauge $\mathfrak{v}$ of Q_{ν_1} the triple is within r_{ν_1} of the trivial triple, at which the potentials are invariant under the constant gauge transformations; carried by $\mathfrak{v}$, the covariance lies on the inner cube within $Cx^{-1/2}(\log x)^{\mathfrak{p}_0+3}$ of an operator that commutes with these (above); and $\mathfrak{g}$ holds no non-zero vector fixed by the adjoint action. The potentials with $d(Y)\ge\lceil\log x\rceil$ add at most $C\delta(x)\mathfrak{b}(x)x^{-1}$ at b, by the sum over polymers [7, eq. 320] at $p=1$. So $\|\bar B_W\|\le CN_k^{1/2}x^{-1}(\log x)^{3\mathfrak{p}_0+3}$. (ii)

The mean of the shift. At a bare bond β the parts of $\mathbf{A}[U_+;gB](\beta)$ of degree two and of higher degree in B are at most $c_{\mathfrak{h}}\delta(x)^2/\varepsilon_{\rm A}$ and $2c_{\mathfrak{h}}\delta(x)^3/\varepsilon_{\rm A}^2$, by Cauchy's estimate; the mean of the part of degree two is its value at $g\bar B_W$, at most a multiple of $g^2\sup|\bar B_W|^2$, plus its contraction with $g^2{\rm Cov}_W$, and the latter is smaller than the size of that part by a factor $x^{-1/2}$, up to powers of $\log x$: by the locality of the shift at $\mathfrak{n}=\nu_2+1$ (above), as $\bar\kappa(\nu_2-1)\ge\frac32\log x$, it may be replaced, at a cost below $4c_{\mathfrak{h}}\delta(x)x^{-3/2}$, by the shift of the inner cube of Q_{ν_2} about the cube of β_-, of $(2\nu_2+1)^3$ cubes of π^M_{k+1}; by gauge covariance in Prop. YM3_I:51.5 it may be formed in the gauge $\mathfrak{v}$ of Q_{ν_2}, in which the triple is within $r_{\nu_2}\le Cx^{-1/2}(\log x)^{\mathfrak{p}_0+3}$ of the trivial one, and by pair dependence and slice holomorphy its part of degree two then differs from the one formed at the trivial triple by a multiple of r_{ν_2} times its size; at the trivial triple the part of degree two is covariant under the adjoint action of the constant gauge transformations, by covariance under constant gauges, so that its contraction with an operator that commutes with them is an element of $\mathfrak{g}^c=\mathfrak{g}\otimes\mathbb{C}$ fixed by the adjoint action, hence 0; and on the inner cube the covariance of p_W, carried by $\mathfrak{v}$, lies within $\epsilon_d^{\rm cv}+2c_{\mathbf{K}}(x^{-1/2}+r_{\nu_2}/a^{\rm us}(\alpha_0,\alpha_1))\le Cx^{-1/2}(\log x)^{\mathfrak{p}_0+3}$ of such an operator, by (i) and (iv). Hence $|\int\mathbf{A}[U_+;gB](\beta)\,p_W(\,{\rm d}B)-(D\mathbf{A}[U_+;0]g\bar B_W)(\beta)|\le\epsilon_d^{\rm s}\le Cx^{-3/2}(\log x)^{4\mathfrak{p}_0+6}$ at every bare bond; and as $|D\mathbf{A}[U_+;0]X|\le c_{\mathfrak{h}}\sup|X|$, and by the kernel bound of $D\mathbf{A}[U_+;0]$ above, which reads β only through its cube, with Schur's test, $\sup_\beta|\int\mathbf{A}[U_+;gB](\beta)\,p_W(\,{\rm d}B)|\le\epsilon_d^{\rm m}$ and $\sum_y\max_{\beta_-\in B^k(y)}|\int\mathbf{A}[U_+;gB](\beta)\,p_W(\,{\rm d}B)|^2\le\epsilon_d^{\rm q}$, the sum over the sites y of the level k, where $\epsilon_d^{\rm m}$ is $c_{\mathfrak{h}}g$ times the bound of $\sup|\bar B_W|$ in (i) plus $\epsilon_d^{\rm s}$, and $\epsilon_d^{\rm q}$ is $2c_{\mathfrak{J}}^2g^2$ times the square of the bound of $\|\bar B_W\|$ in (i) plus $2N_k(\epsilon_d^{\rm s})^2$. (iii)

The kernel. The kernel of $D\mathbf{A}[U_+;0]\mathsf{A}_W^{-1}D\mathbf{A}[U_+;0]^*$ at two bare bonds β,β' is

$$\sum_{b,b'}\partial_b\mathbf{A}[U_+;0](\beta)\,\mathsf{A}_W^{-1}(b,b')\,\partial_{b'}\mathbf{A}[U_+;0](\beta')^{\mathsf{T}},$$

with the kernel $\mathsf{A}_W^{-1}(b,b')\in{\rm End}\,\mathfrak{g}$ of A_W^{-1}, and its outer factors decay at the rate $\bar\kappa$ in the distances of the cubes (above), and so does its middle one, by (i); so, by the triangle inequality for these distances, it is at most $Ce^{-\bar\kappa\mathfrak{n}/2}$, $\mathfrak{n}$ the distance of the cubes of β_- and β'_-. Let $\mathfrak{n}\le n_\flat:=\lceil2\log x/\bar\kappa\rceil+9$, so that $\nu_3=\max\{\nu_1,2n_\flat+1\}$, and let $\mathfrak{v}$ be the gauge of Q_{ν_3} about the cube of β_-, in which P_W is within $r_{\nu_3}\le Cx^{-1/2}(\log x)^{\mathfrak{p}_0+3}$ of the trivial triple; the inner cube of Q_{ν_3} holds every cube within $n_\flat$ of the cubes of β_- and β'_-, and the bonds b, b' farther than $n_\flat$ from these add at most $Cx^{-2}(\log x)^2$, as $\bar\kappa(n_\flat-2)\ge2\log x$. There both factors lie within $Cx^{-1/2}(\log x)^{\mathfrak{p}_0+3}$ of the operators formed at the trivial triple, which commute with the adjoint action of the constant gauge transformations, and a linear map of $\mathfrak{g}$ that commutes with it is a multiple of $\mathbf{1}$; so, up to that error, the factors are numbers formed at the trivial triple times ${\rm Ad}(\mathfrak{v}(\beta_-)^{-1}\mathfrak{v}(b_-))$ and ${\rm Ad}(\mathfrak{v}(b_-)^{-1}\mathfrak{v}(b'_-))$, the inner factors cancel, and the kernel is $\mathsf{s}_0(\beta,\beta'){\rm Ad}(\mathfrak{v}(\beta_-)^{-1}\mathfrak{v}(\beta'_-))$ up to $Cx^{-1/2}(\log x)^{\mathfrak{p}_0+3}$, with a number $\mathsf{s}_0(\beta,\beta')$ formed at the trivial triple. In the gauge of the chain the kernel is multiplied

on the left by $\mathrm{Ad}((\gamma \mathsf{g}_{k+1}^{-1})(\beta_-))$ and on the right by the transpose of $\mathrm{Ad}((\gamma \mathsf{g}_{k+1}^{-1})(\beta'_-))$, which turns $\mathrm{Ad}(\mathfrak{v}(\beta_-)^{-1}\mathfrak{v}(\beta'_-))$ into $\mathrm{Ad}(\mathfrak{u}(\beta_-)^{-1}\mathfrak{u}(\beta'_-))$, $\mathfrak{u} = \mathfrak{v}\mathsf{g}_{k+1}\gamma^{-1}$. As $(\mathsf{U}_{k+1}^{\gamma})^{\mathfrak{u}} = U_+^{\mathfrak{v}} = e^{iL^{-k-1}A'}$ with $|A'| \le Cx^{-1/2}(\log x)^{\mathfrak{p}_0+3}$ on the cube, the ordered product of the values of $\mathsf{U}_{k+1}^{\gamma}$ along a path of bare bonds from β_- to β'_- of length at most $3(\mathfrak{n}+1)LML^{-d}$ lies within $3(\mathfrak{n}+1)M \sup|A'|$ of $\mathfrak{u}(\beta_-)^{-1}\mathfrak{u}(\beta'_-)$, and within $3(\mathfrak{n}+1)LM[c_{\mathsf{h}}\delta(x) + L^{-d}\mathsf{a}_{\mathrm{top}}]$ of $\mathbf{1}$, as $\mathsf{U}_{k+1}^{\gamma}$ is the ordered product of the layers of the depths below d and of the top field, which in its regular gauge has $|\mathbf{A}^{\mathrm{top}}| \le \mathsf{a}_{\mathrm{top}}$ at each bare bond. So in the gauge of the chain the kernel at β, β' is a number times $\mathbf{1}$ up to a $\mathsf{w}_{\mathfrak{n}}$ that is affine in $\mathsf{a}_{\mathrm{top}}L^{-d}$; for the pairs at distances $\mathfrak{n} > n_\flat$ the number is taken 0 and $\mathsf{w}_{\mathfrak{n}}$ is the bound $Ce^{-\bar\kappa \mathfrak{n}/2} \le Cx^{-1}e^{-\bar\kappa(\mathfrak{n}-n_\flat)/2}$ of the kernel; and $\sum_{\mathfrak{n}}(24\mathfrak{n}^2+2)\mathsf{w}_{\mathfrak{n}} \le \epsilon_d^{\Sigma}$, a number formed from x, the constants and $\mathsf{a}_{\mathrm{top}}L^{-d}$. For $k=0$, where no shift system enters, (ii) and (iii) use the local pivot construction and its holomorphy and gauge covariance in equations YM3_I:(800)–YM3_I:(803) in place of the corresponding shift properties: off the pivot bonds $\mathbf{A}[U_+;\zeta] = \zeta$ is linear in ζ and free of the triple; at the pivot bond $b_0(c)$ of a cube c the pivot value $\beta_c(P,\zeta)$ depends on the block field and on ζ through their values on P_c alone, so on the triple through its pair alone, the block field being formed from the configuration of the pair; for $\sup|\zeta| < r_P^\circ$ and a fixed real factor it is holomorphic in $(A', \mathbf{J}, \zeta)$ on the simple chart set, which contains the cores by the common-domain verification in the proof of Thm. YM3_I:W, with $|\beta_c| \le 2C_P \sup|\zeta|$; and $\beta_c(P^u, \mathrm{Ad}(u)\zeta) = \mathrm{Ad}(u(b_0(c)_-))\,\beta_c(P,\zeta)$ for every gauge transformation u of the bare sites, constant or not. Carried by the gauge $\mathfrak{v}$ of Q_ν, the pivot value is therefore the one formed at $P[\omega]$, and $z \mapsto \beta_c(P[z\omega],\zeta)$ is holomorphic and at most $2C_P\sup|\zeta|$ for $|z| < a^{\mathrm{us}}(\alpha_0,\alpha_1)/r_\nu$, where $P[z\omega]$ lies in the cores; Schwarz's lemma in z, with Cauchy's estimate in ζ, puts its parts of degree one and two in ζ within a multiple of r_ν times their size of the parts formed at the trivial triple, as pair dependence and slice holomorphy do for $k \ge 1$. (iv)

The covariance. On the convex box $\mathsf{A}_W - C_{\mathrm{Hs}}\delta(x) \le D^2\Psi_W \le \mathsf{A}_W + C_{\mathrm{Hs}}\delta(x)$, with $C_{\mathrm{Hs}}\delta(x) \le \frac{1}{4}\gamma_{\mathrm{r}}$ as $d > d_0 \ge n_2$ (the bound $C_{\mathrm{Hs}}\delta(x)$ of the Hessians of the potentials in the proof of Lem. 15.3(iv), which Cauchy's estimate draws from their bound on the whole product of the analyticity cores and the polydisc of radius $C_A\mathfrak{b}(x)$, and which so holds at the triple P_W of every $W \in \mathsf{Cut}_{k+1}$ and at every point of the box); the Brascamp–Lieb inequality gives $\mathrm{Cov}_W \le (\mathsf{A}_W - C_{\mathrm{Hs}}\delta(x))^{-1}$, and integration by parts and the Cauchy–Schwarz inequality against linear functions give $\mathrm{Cov}_W \ge (1-\iota_d)(\mathsf{A}_W + C_{\mathrm{Hs}}\delta(x))^{-1}$, with a share ι_d of the boundary of the box that is smaller than every power of x; so $\|\mathrm{Cov}_W - \mathsf{A}_W^{-1}\| \le \epsilon_d^{\mathrm{cv}} = \frac{8}{\gamma_{\mathrm{r}}^2}C_{\mathrm{Hs}}\delta(x) + \frac{2}{\gamma_{\mathrm{r}}}\iota_d$. (3)

One constant. The numbers of (2) are formed from $x = L^d\mathsf{b}$, ℓL^d and the constants of the construction, ϵ_d^{Σ} besides from $\mathsf{a}_{\mathrm{top}}L^{-d}$, in which it is affine; save for the factor $N_k = (\ell L^d)^3$ that multiplies both members of ϵ_d^{q} and stays in its bound below, $\ell L^d = (\ell/\mathsf{b})x$ enters only through counts of bonds that multiply masses of boundary layers of the box, which fall faster than every power of x. Collecting the multiples of (2) gives four numbers C_6, C_6', C_Σ, C_{cv}, formed from ℓ, b and the constants, which read neither K, d_0, d, W, a gauge nor $\mathsf{a}_{\mathrm{top}}$, with $\epsilon_d^{\mathrm{m}} \le C_6x^{-1}(\log x)^{2\mathfrak{p}_0}$, $\epsilon_d^{\mathrm{q}} \le C_6'N_kx^{-3}(\log x)^{8\mathfrak{p}_0+12}$, $\epsilon_d^{\Sigma} \le C_\Sigma[x^{-1/2}(\log x)^{\mathfrak{p}_0+6} + \mathsf{a}_{\mathrm{top}}L^{-d}]$ and $\epsilon_d^{\mathrm{cv}} \le C_{\mathrm{cv}}x^{-1/2}(\log x)^{\mathfrak{p}_0+3}$ at $x = L^d\mathsf{b}$. From these, (M) and (Col): given $\mathcal{F}_{d-1}$ the law of B_k is p_W, $W = V_{k+1}$, and γ and g_{k+1} are functions of V_{k+1} that act at each bond by isometries, so $|\mathbf{n}_d(\beta)| = L^d|\int \mathbf{A}[U_+;gB](\beta)\,p_W(\,\mathrm{d}B)|$; the factor L^d, with $L^dx^{-1} = \mathsf{b}^{-1}$, gives $C_6\mathsf{b}^{-1}(\log L^d\mathsf{b})^{2\mathfrak{p}_0}$ as a bound of $|\mathbf{n}_d(\beta)|$ for every bare bond β; three bare bonds start at every bare site, and the norm of (M) takes at a site the euclidean norm of the three values, so $\|\mathbf{n}_d\|_\infty \le \sqrt{3}\,C_6\mathsf{b}^{-1}(\log L^d\mathsf{b})^{2\mathfrak{p}_0}$; and as $\|\mathbf{n}_d\|_{L^2}^2 = a_K^3\sum_\beta|\mathbf{n}_d(\beta)|^2$ and $B^k(y)$ holds L^{3k} bare sites, each the initial point of three bare bonds, the sum over the blocks gives $\|\mathbf{n}_d\|_{L^2}^2 \le 3a_K^3L^{3k}L^{2d}\epsilon_d^{\mathrm{q}} \le 3C_6'\ell^3\mathsf{b}^{-3}L^{-d}(\log L^d\mathsf{b})^{8\mathfrak{p}_0+12}$, as $a_K^3L^{3k} = L^{-3d}$. The kernel of $\mathrm{Ad}(\gamma)\mathsf{J}_d\mathsf{A}_W^{-1}\mathsf{J}_d^*\mathrm{Ad}(\gamma)^{\mathsf{T}}$ is that of (iii) in the gauge of the chain, in which the top field is at most $\mathsf{a}_{\mathrm{top}}$ at each bare bond; so the first two clauses of (Col) hold with C_Σ, $\mathbb{E}^{[d_0]}$-almost surely and at all pairs of bare bonds, and the third holds with C_{cv} at every W. With the four explicit

conditions, of which $d_0 \ge n_{\mathrm{a}}$ and $d_0 \ge d_{\mathrm{num}}(\ell, \mathsf{b})$ follow from $d_0 \ge d_{\mathrm{adm}} = \max\{n_{\mathrm{a}}, d_{\mathrm{num}}(\ell, \mathsf{b})\}$ and the bounds of $c_\rho \varepsilon(L^{d_0}\mathsf{b})$ and of $4c_{\mathsf{h}}\delta(L^{d_0}\mathsf{b})$ from (1), every top depth $d_0 \ge d_{\mathrm{adm}}$ is admissible with $C_{\mathrm{adm}} = \max\{\sqrt{3}\, C_6, 3C_6', C_\Sigma, C_{\mathrm{cv}}\}$: this is (Adm).

Layer estimates. At an admissible d_0, with C free of K and d_0, the chain and these statements give: (L1) for $d_0 < d \le K$ the ordered product $e^{ia_K \mathbf{h}'_d} \cdots e^{ia_K \mathbf{h}'_{d_0+1}} e^{ia_K \mathbf{A}^{\mathrm{top}}}$ of the top field and of the layers of the depths $\le d$ is $\mathsf{U}^{\gamma}_{K-d}$, a gauge transform of a background over the block field V_{K-d} at depth d; (L2) $|\mathbf{h}_d| \le c_{\mathsf{h}} L^d \delta(L^d \mathsf{b})$ at every bond, and $\sum_d a_K |\mathbf{h}_d| \le \frac{1}{4}$; (L3) for a real bond function f, every component ξ of $\langle f, \mathbf{h}'_d - \mathbf{n}_d\rangle$ has $\mathbb{E}^{[d_0]}[e^{\lambda\xi}|\mathcal{F}_{d-1}] \le e^{4\lambda^2 G_d(f)^2/\gamma_{\mathrm{r}}}$ for real λ, $G_d(f)^2 = c_{\mathsf{J}}^2 \mathsf{b}^{-1} L^d \min\{\|f\|_1^2, L^{-3d}\|f\|_2^2\}$, by Lem. 15.3(iv), as the derivative of $\mathbf{h}'_d$ in the fluctuation obeys on the box the bounds of $\mathsf{b}^{-1/2} L^{d/2} \mathsf{J}_d$; (L4) $\sup|\mathbf{n}_d| \le C\mathsf{b}^{-1}(\log L^d \mathsf{b})^{2\mathfrak{p}_0}$ by (M), a bound whose sums over $d_0 < d' \le d$, times $L^{-d/2}$, tend to 0 as $d \to \infty$, and almost surely $\|\sum_{d_0 < d \le K} \mathbf{n}_d\|_{L^{12}(T_\ell)} \le \mathsf{c}_{12}$, a number free of K and d_0, as $\|\mathbf{n}_d\|_{L^{12}(T_\ell)} \le \|\mathbf{n}_d\|_\infty^{5/6} \|\mathbf{n}_d\|_{L^2}^{1/6}$ is summable over d by (M); (L5) the moments of the fluctuation fields and of the layers: at every bond b, $\mathbb{E}^{[d_0]}|B_{K-d}(b)|^2 \le m_2$, and for $q < \infty$, $\|a_K(\sum_d |\mathbf{h}_d(b)|)^2\|_{L^q(\mathbb{E}^{[d_0]})} \le \mathsf{c}_{(q)}\mathsf{b}^{-1}$, with m_2 and $\mathsf{c}_{(q)}$ free of K, d and d_0.

The fluctuation-field second moment was proved above before the mean and covariance estimates. For the layers: (L3) at $f = a_K^{-3}\delta_b$, for which $\langle f, \cdot\rangle$ is the value at b and $G_d(f)^2 = c_{\mathsf{J}}^2 \mathsf{b}^{-1} L^d$, bounds the centred layers at b in $L^{2q}(\mathbb{E}^{[d_0]})$ by $C\sqrt{q}\, \mathsf{b}^{-1/2} L^{d/2}$, whose sum over $d \le K$ is at most $C\sqrt{q}\, \mathsf{b}^{-1/2} a_K^{-1/2}$, and the means add at most $2a_K(\sum_{d \le K} C\mathsf{b}^{-1}(\log L^d \mathsf{b})^{2\mathfrak{p}_0})^2$, a number free of K times b^{-1}.

In its regular gauge the top field is bounded by $\mathsf{a}_{\mathrm{top}} \le CL^{10 d_0}$.

Passage from the chain measure to the Wilson measure. The inequality over the top depths: on invariant functions the chain measure from the top depth d has a density $\bar{\mathfrak{w}}_d \in [0,1]$ with respect to μ^K, with $\int(1 - \bar{\mathfrak{w}}_d)\, \mathrm{d}\mu^K \le \mathsf{R}_d$ (Lem. 15.3(i)); split $1 = \bar{\mathfrak{w}}_{d_1} + \sum_{d_1 < d \le K-1}(1 - \bar{\mathfrak{w}}_{d_1}) \cdots (1 - \bar{\mathfrak{w}}_{d-1})\bar{\mathfrak{w}}_d + (1 - \bar{\mathfrak{w}}_{d_1}) \cdots (1 - \bar{\mathfrak{w}}_{K-1})$, bound each product of the sum by its last two factors and the last product by its last factor, and use Hölder's inequality; $\sum_d \mathsf{R}_d^{1/2} L^{pd}$ and $\sup_K \mathsf{R}_{K-1} L^{pK}$ are finite, as R_d decays faster than every exponential in d.

By (Adm) let d_{adm} be a depth, free of K, such that every top depth $\ge d_{\mathrm{adm}}$ is admissible. At $d_1 = d_{\mathrm{adm}}$ the moment bound applies at every top depth $d+1$ of the inequality, and $\mathsf{N}' \le \|\mathbf{B}_K\|_{\kappa;a}$, $\mathsf{N}' \le 23 L^{3K}$: this gives $\mathsf{M}_q < \infty$. The bound of the action at $R = \max\{1, \mathsf{N}'\}$ and M_{24q} give the moments of $\sup_s s^{\frac{3}{2}+\kappa} A^a(U_s)$. The good sets: at an admissible d_0 with $\mathsf{R}_{d_0} \le \frac{1}{2}\epsilon$, on $\{\|\mathbf{B}_K\|_{\kappa;a} \le R\}$ and for $a_K \le a_0(R)$ the four bounds put U in $\mathsf{O}_K(\bar{C}, \bar{s})$, $\bar{C} = (2 + c_3)R$, $\bar{s} = s_0(R)$, and Chebyshev's inequality with Lem. 15.3(i) gives $\mu^K(\mathsf{O}_K(\bar{C}, \bar{s})^{\mathrm{c}}) \le \mathsf{R}_{d_0} + C(1,\kappa)L^{20 d_0}/R$.

With $E = A^a(U_{\sigma/2})$, Lem. 16.2(f) bounds $a_K^{-4} \max_p a_p(U_\sigma)$ and $(J,J)_a(U_\sigma)$ by polynomials in E if $a_K E \le e_0$, and otherwise by $2a_K^{-4}$ and $48\ell^3 a_K^{-6}$, against $\mu^K(a_K E > e_0) \le (a_K/e_0)^{6p}\langle E^{6p}\rangle_{\mu^K}$. Tightness of Λ_σ: on $\mathsf{O}_K(\bar{C}, \bar{s})$, which has mass $\ge 1 - \epsilon$, Λ_σ is bounded by a number free of K (Step 6).

(b) Choose D_* so that for $D \ge D_*$

$$L^{m+D} \ge 3R_1 \mathfrak{M}, \qquad L^{m+D} > 6\rho_*, \qquad 2\theta_D \le \bar{\varepsilon}, \qquad 2B_3\theta_D \le \alpha_*,$$
$$2B_3\theta_D L^{-4} < \min\{\omega_M, (\kappa_2^{\mathrm{SO}})^{-1}\},$$

where α_*, ρ_* are from Prop. 16.10. All these constants are fixed, so the conditions eventually hold because $\theta_D \to 0$. For $V \in \mathcal{K}_D$, set $\eta = L^{-(K-D)}$ and $N_D = (2L^{m+D})^3$. Take the minimal backgrounds $U^{\mathrm{A}} = U_{K,D}(V)$ and $U^{\mathrm{B}} = U_{K+1,D}(V)$. Their regularity on the larger open ball $\{\varpi < 2\theta_D\}$ and the proposition give

$$(U^{\mathrm{A}})^g = e^{i\eta A}\mathcal{M}(U^{\mathrm{B}}), \qquad \sup|A| \le 2CB_3\theta_D N_D^{1/5}\eta^{1/5}, \qquad \varpi(U^{\mathrm{B}}) \le 2B_3\theta_D(\eta/L)^2.$$

The one-step contour estimate, Lem. YM3_I:14.6(iii), gives $\mathcal{M}(U^{\mathrm{B}})(c) = e^{iY_c}U^{\mathrm{B}}[c]$, $|Y_c| \le \kappa_3\varpi(U^{\mathrm{B}})$, where $[c]$ subdivides c into its L fine bonds. A path C of physical length at most R uses at most RL^D/η bonds of the first lattice. Telescoping the products therefore gives

$$|W_C(U^{\mathrm{A}}) - W_C(U^{\mathrm{B}})| \le 2RL^D B_3\theta_D \left(CN_D^{1/5}\eta^{1/5} + \frac{\kappa_3}{L^2}\eta\right).$$

For each fixed D this is summable in K, uniformly in $V \in \mathcal{K}_D$ and in the closed paths of length at most R defined at that cutoff. Its constant may depend on D through N_D; no uniformity in D is used in this step. Thus the unflowed loops have limits $w_C^D(V)$. The positive-time assertion remains a sketch at (Flow approximation). Its proposed approximation step, together with compactness of $\mathcal{K}_D$, would give a convergent subsequence of any sequence of top data and background flows; its limit solves the specified equation with the specified initial holonomies. Uniqueness identifies the limiting orbit curve. Uniform convergence of the potential representatives gives convergence of flowed holonomies: the difference of their path products is bounded by the path length times their supremum discrepancy. The sequential assertion and compactness of $\mathcal{K}_D$ make this convergence uniform over V and bounded-length paths on compact positive time intervals. The cutoff-uniform comparison between the bare field and the background is proved separately below.

The bare field and the background over $\pi_{K,D}U$. By Lem. 15.3(i) at an admissible $d_0 < D$ with R_{d_0} small, which the proved conclusion (Adm) provides, the comparison is made under $\mathbb{E}^{[d_0]}$. There U is a gauge transform of $e^{ia_K\mathbf{B}_K}$; and $e^{ia_K\mathbf{B}_K^{\le D}}$, with $\mathbf{B}_K^{\le D}$ the logarithm of the ordered product of the top and of the layers of the depths $\le D$, the field cut at the depth D, a function of $\pi_{K,D}U$, is a gauge transform of U_{K-D}, that is of the background over $\pi_{K,D}U$. (i) $\mathbf{B}_K - \mathbf{B}_K^{\le D}$ is small in the rough distance, in every L^q uniformly in K, at the rate $L^{-\kappa D/8}$: its linear part is the sum of the martingale differences of the depths beyond D, its bilinear members are those of (a) with a factor of a depth beyond D, and its means and the remainders of the product are small in L^p.

(ii) The rough norm of the cut field has moments free of K and D: its martingale part $u_{\le D} = \sum_{d\le D} u_{(d)}$ is the conditional mean $\mathbb{E}[u|\mathcal{F}_D]$ of that of the whole field, so $|u_{\le D}|_{-\frac12-\kappa;a} \le \mathbb{E}[|u|_{-\frac12-\kappa;a}|\mathcal{F}_D]$; for a bilinear member Υ of the rough norm (one tuple of directions and translations, one colour, one time, one point) the pairs of different depths whose finer depth exceeds D have vanishing conditional means given $\mathcal{F}_D$, so $\mathbb{E}[\Upsilon(u,u)|\mathcal{F}_D] = \Upsilon(u_{\le D}, u_{\le D}) + \sum_{d>D}\mathbb{E}[\mathsf{c}_d|\mathcal{F}_D]$, $\mathsf{c}_d = \mathbb{E}[\Upsilon(u_{(d)},u_{(d)})|\mathcal{F}_{d-1}]$ the compensators of (a), whose sum is bounded almost surely by $C(1+\mathsf{a}_{\mathrm{top}})s^{-\frac34-\frac{\kappa}{2}}$; with the rest as in (a) and Jensen's inequality, $\mathbb{E}[\|\mathbf{B}_K^{\le D}\|_{\kappa;a}^q]^{1/q} \le C(q,\kappa)(1+\mathsf{a}_{\mathrm{top}})^2$.

So for a large R and a large D, off a probability that is small uniformly in K, the two fields have rough norms $\le R$ and the rough distance $\le \eta$: they are the two data of the Lipschitz bound of (a).

From the short time to the time s. Let $s_* = \min\{s_{\mathrm{Lip}}(R), \frac12 s\}$ and $a_K \le a_0(R)$. At s_* the two flowed fields are gauge transforms of e^{ia_KA}, $e^{ia_KA'}$ with the four bounds and $\|A - A'\|_\infty \le (1+c_5)\eta s_*^{-\frac14-\frac{\kappa}{2}}$; by Lem. 16.2(c), (e) the actions of the flowed fields at $\frac12 s_*$, and those of the fields $e^{ia_K((1-r)A+rA')}$, $0 \le r \le 1$, are at most $E = 6(2+c_3)^4\ell^3R^4(\frac12 s_*)^{-\frac32-\kappa}$. For $a_KE \le e_0$, Lem. 16.2(f), started at $\frac12 s_*$, bounds the curvature $a_K^{-2}\varpi$ of the flow of the bare field on $[s_*, s]$ by a number F_1, and, started at the time 0 of the fields of the segment, bounds the number Λ_{s-s_*} of their flows by a number Λ_1; both read E and s alone.

For a flowed observable H of the flow time s: by Lem. 16.2(b), (d), the square field of $H\circ\Phi_{s-s_*}$ is at most L^2, L free of K, at the fields of the segment, whose ℓ^2 length is at most $\sqrt{3}\ell^{3/2}a_K^{-1/2}\|A'-A\|_\infty$; so the two values of H differ by at most $\sqrt{3}\ell^{3/2}\mathsf{L}\|A'-A\|_\infty$. With (i), (ii) this gives the first display: the mean squared difference is at most $3\ell^3\mathsf{L}^2(1+c_5)^2\eta^2 s_*^{-\frac12-\kappa}$ plus $4\sup|\varphi|^2$ times a small probability, uniformly in K. For the loops, whose square field is of the order of the length times a_K^{-2}, the comparison is made in the supremum distance modulo gauge, in the units of a_K, which is

at most $\|A-A'\|_\infty$ at s_*: by Lem. 15.7(a) at $\kappa' = F_1 a_K^2$ the flowed bare field has at every time of $[s_*, s]$ a gauge with a potential bounded by a number $C_0(F_1, \ell)$; the short-time theory of (a), restarted from these bounded data in steps of a length $\sigma_1(F_1, \ell, \kappa)$, multiplies the distance by at most a factor C_1 in each step, and asks nothing of the second field; after $m = \lceil (s-s_*)/\sigma_1 \rceil$ steps the factor is C_1^m, free of K; and $|W_C(V) - W_C(V')|$ is at most the length of C times the distance.

Hence $\lim_D \limsup_K \|W_C \circ \Phi_s - \Psi_D \circ \pi_{K,D}\|_{L^1(\mu^K)} = 0$ with Ψ_D free of K, which is strong convergence. Two configurations with equal loops are gauge equivalent (characters of SU(2)), and the orbits with $\|\mathfrak{A}\|_\infty + \operatorname{Lip}\mathfrak{A} \le C$ form a compact set on which the loops give the topology: this gives $\mathfrak{F}$. The second display follows from the first, from (C3), and from total variation at the depth D.

(c) The good sets at the time s: by the moments of the action in (a) and Lem. 16.2(f), started at $\frac14 s$, for $\epsilon > 0$ there is F_1, free of K, such that the closed invariant set $\mathsf{G}_K = \{a_K^{-2}\varpi(\Phi_r U) \le F_1 \text{ for } \frac12 s \le r \le s\}$ has $\mu^K(\mathsf{G}_K) \ge 1-\epsilon$ for all large K. On G_K, Lem. 15.7(a) at $\kappa' = F_1 a_K^2$ gives at the time $s - \sigma_1$, $\sigma_1 = \min\{s_0''(R_1), \frac12 s\}$, where $s_0''(R) = (c_1'' R^{-2})^{4/(1-2\kappa)} \le s_0(R)$ and $c_1'' \in (0, c_1]$ depends only on κ, a gauge with a potential bounded by $C_0(F_1, \ell)$, of rough norm at most $R_1 = \max\{1, C_0 + C_0^2\}$, as $\|\mathbf{B}\|_{\kappa;a} \le \|\mathbf{B}\|_\infty + \|\mathbf{B}\|_\infty^2$; the short-time theory of (a) restarted from it gives at the time s a gauge transform $e^{ia_K A}$ of the flowed field with $\|A\|_\infty + \|\nabla A\|_\infty + \|\nabla\nabla A\|_\infty \le C$, a number $C \ge 1$ free of K: the first two members by the bounds of (a) at the time $\sigma_1 \le s_0(R_1)$ after the restart, the second differences by the same fixed point with one more derivative, which closes up to the time $s_0''(R_1)$ and gives $\|\nabla\nabla A_\sigma\|_\infty \le c_8 R_1 \sigma^{-\frac54 - \frac\kappa2}$ for $0 < \sigma \le s_0''(R_1)$, σ the time after the restart, c_8 depending only on κ; here $\sigma = \sigma_1 \le s_0''(R_1)$.

For $a_K C \le \frac{1}{24}$ and Lipschitz f, $|\mathcal{O}(f)(e^{ia_K A}) - \mathcal{O}^0(f)(\mathfrak{A})| \le c\,\ell^3 C^5 (\|f\|_\infty + \operatorname{Lip} f)\, a_K$ for the interpolated connection $\mathfrak{A}$; $\mathcal{O}^0(f)$ is continuous on the compact set of these orbits, so the theorems of Stone and Weierstrass and of Tietze give Θ; the moments come from (a). □

16.4. **Deterministic estimates along the Wilson flow.**

Lemma 16.2 (Deterministic bounds)**.** For all K, m, $\beta > 0$, every U and real X; $\partial U(p) = c_p \mathbf{1} + i\mathsf{s}_p$, $Y_1, \dots, Y_4$ the values of X at the four bonds of p carried to its base point by the parallel transport of U, with the signs $\epsilon = (+,+,-,-)$, and $Z_p(X) = \sum_i \epsilon_i Y_i$:

(a) $\langle X, \partial a_p\rangle = \langle Z_p(X), \mathsf{s}_p\rangle$, so $|\partial_b a_p| = |\mathsf{s}_p|$ for $b \in p$;

$$\nabla^2 a_p[X,X] = c_p |Z_p(X)|^2 + \sum_{i<j} \epsilon_i \epsilon_j \langle i[Y_i, Y_j], \mathsf{s}_p\rangle,$$

$$-3\varpi_p \sum_{b\in p} |X(b)|^2 \le \nabla^2 a_p[X,X] \le (4 + 3\varpi_p) \sum_{b\in p} |X(b)|^2,$$

and

$$-12\varpi(U)\|X\|^2 \le \nabla^2 A_K(U)[X,X] \le 40\|X\|^2$$

(second derivatives along $e^{isX}U$);

(b) with $w_s = D\Phi_s(U)X$: $\frac{d}{ds}\|w_s\|^2 = -2a_K^{-2}\nabla^2 A_K(U_s)[w_s, w_s]$; so $\|D\Phi_\sigma(U)\| \le e^{12\Lambda_\sigma(U)}$ and $\Gamma_K(G\circ\Phi_\sigma)(U) \le e^{24\Lambda_\sigma(U)}\Gamma_K(G)(U_\sigma)$ for $G \in C^1(\Omega_K)$; $D\Phi_s(U)\mathrm{d}_U\lambda = \mathrm{d}_{U_s}\lambda$ and $D\Phi_s(U)J(U) = J(U_s)$, and $D\Phi_\sigma(U)^{\mathsf{T}}$ maps horizontal vectors at U_σ to horizontal vectors at U;

(c) $\partial_\sigma A^a(U_\sigma) = -(J,J)_a(U_\sigma)$; $(J,J)_a(U_\sigma) \le e^{24(\Lambda_\sigma - \Lambda_s)(U)}(J,J)_a(U_s)$ for $s \le \sigma$; $A^a(V) \le \frac32 \ell^3 (a_K^{-2}\varpi(V))^2$; $A^a(U_{\sigma/2}) \le 24\ell^3 \Lambda_\sigma(U)^2 \sigma^{-2}$; and $(J,J)_a(U_\sigma) \le \frac{2}{\sigma} e^{24\Lambda_\sigma(U)} A^a(U_{\sigma/2})$;

(d) $\Gamma_K(\mathcal{O}(f)) \le 2\|f\|_\infty^2 (J,J)_a + 256\|\nabla f\|_\infty^2 A^a$ and $|v\,\mathcal{O}(f)| \le ((J,J)_a \Gamma_K(\mathcal{O}(f)))^{1/2}$;

(e) on $\mathsf{O}_K(\bar C, \bar s)$, $a_K^{-2}\varpi(U_s) \le 2\bar C s^{-\frac34 - \frac\kappa2} + 2\bar C^2 s^{-\frac12 - \kappa}$ for $0 < s \le \bar s$, and so $\Lambda_\sigma \le (8\bar C \sigma^{\frac14 - \frac\kappa2} + 4\bar C^2 \sigma^{\frac12 - \kappa})/(1-2\kappa)$ for $\sigma \le \bar s$;

(f) let $e_0 = 10^{-4}$, $c_{10} = 5.1 \cdot 10^4$, $x_0 = (1440 c_{10})^{-1}$, $E = A^a(U_{s_1})$ and $a_K E \le e_0$; at $E = 0$ set $E^{-2} = +\infty$, the flat field being stationary. Then for $s \ge s_1$, $a_p(U_s) \le e_0$, and $z_p = a_K^{-4} a_p(U_s)$ satisfies $\partial_s z_p \le \Delta_a z_p + c_{10} \bar z_p z_p^{1/2}$ (Δ_a the Laplacian on parallel plaquettes, $\bar z_p$ the largest z on a box of plaquettes about p); $a_K^{-4} \max_p a_p(U_s) \le \frac{5}{2} E \min\{s - s_1, x_0^4 E^{-2}, 1\}^{-3/2}$ for $s > s_1$; and at $s_1 = \frac{\sigma}{2} \le \frac{1}{2}$, $(J,J)_a(U_\sigma) \le e^{48}[\frac{4E}{\sigma} + 7E^{3/2}\sigma^{-3/4} + 3x_0^{-3}E^3]$;

(g) (Bochner.) For $H \in C^\infty(\Omega_K)$, with ∇^2 the Hessian of the bi-invariant metric of Ω_K, a product of unit spheres S^3 of Ricci curvature 2,

$$\|\mathcal{L}_K H\|^2_{L^2(\mu^K)} = a_K^{-2} \int \|\nabla^2 H\|^2_{\mathrm{HS}} \,\mathrm{d}\mu^K + 2a_K^{-1} \mathcal{E}_K(H) + a_K^{-2}\beta \int \nabla^2 A_K(\partial H, \partial H)\,\mathrm{d}\mu^K;$$

consequently $(\int |\partial H|^4 \,\mathrm{d}\mu^K)^{1/2} \le \sup|H|\,(3\mathsf{m} + 2(28\beta \sup|H|\,\mathsf{m})^{1/2})$, $\mathsf{m} = a_K \|\mathcal{L}_K H\|_{L^2(\mu^K)}$, and $H_K \in C^\infty(\Omega_K)$ with $\sup_K \|\mathcal{L}_K H_K\|_{L^2(\mu^K)} < \infty$ and $\liminf_K \mathcal{E}_K(H_K) > 0$ have $a_K^{-2}\beta \int \nabla^2 A_K(\partial H_K, \partial H_K)\,\mathrm{d}\mu^K \to -\infty$; and $(\mathcal{L}_K \cos G)^2 + (\mathcal{L}_K \sin G)^2 = (\mathcal{L}_K G)^2 + \Gamma_K(G)^2$ for real $G \in C^\infty(\Omega_K)$;

(h) (Weitzenböck and Kato.) $\nabla^2 A_K(U)[z,z] = \langle z, (\nabla_U^{\mathsf{T}} \nabla_U + R_U) z\rangle$ for horizontal z at U, where $(\nabla_U^{\mathsf{T}} \nabla_U z)(b) = \sum_{b' \sim b} (z(b) - \tau_{bb'} z(b'))$, the sum over the six parallel neighbours with the parallel transports $\tau_{bb'}$ of U, and $\|R_U(b,b')\| \le 4\varpi(U)\mathsf{n}(b,b')$, $\mathsf{n}(b,b') = 1$ if $b = b'$ or b, b' lie in a common plaquette and 0 otherwise. For invariant $G \in C^1(\Omega_K)$ and every bond b,

$$|\partial_b(G \circ \Phi_\sigma)(U)| \le \big[e^{4\Lambda_\sigma(U)\mathsf{n}}\, e^{-\sigma a_K^{-2}\Delta_0}\, |\partial G|(U_\sigma)\big](b),$$

Δ_0 the free Laplacian on the bonds (six parallel neighbours) and $|\partial G|(V)$ the function $b \mapsto |\partial_b G(V)|$; so $\|\partial(G \circ \Phi_\sigma)(U)\|_{\ell^q} \le e^{52\Lambda_\sigma(U)} \|\partial G(U_\sigma)\|_{\ell^q}$ for $1 \le q \le \infty$, and $\sup_b |J(U_\sigma)(b)| \le e^{52(\Lambda_\sigma - \Lambda_s)(U)} \sup_b |J(U_s)(b)|$ for $s \le \sigma$;

(i) (One-bond fibres.) For $N \ge 3$, $p \ni b_0$, $s \ge 0$, the map $g \mapsto \partial(\Phi_s U^{(g)})(p)$ of G to itself, $U^{(g)}$ the field U with the value g at b_0, is onto; so $\sup_U \varpi(U_s) = 2$, and $\sup_U \mathcal{O}(f)(U_s) \ge 2f(x(p))/a_K$ for $f \ge 0$, at every flow time. At a field with values in the centre $\{\pm\mathbf{1}\}$, $\partial A_K = 0$ and $\nabla^2 A_K[X,X] = \sum_p c_p |Z_p(X)|^2$, $c_p = \pm 1$; at the field that is $-\mathbf{1}$ at one bond and $\mathbf{1}$ elsewhere, $\|D\Phi_\sigma\| \ge e^{4\sigma/a_K^2}$.

■

Proof sketch. (a) Expand $\partial(e^{itX}U)(p)$ to second order in t, with the values of X carried to the base point of p; the bounds follow from $|\mathsf{s}_p| \le \varpi_p$, $|c_p| \le 1$, $|Z_p|^2 \le 4\sum_{b \in p} |X(b)|^2$, and four plaquettes at each bond. (b) $\frac{d}{ds}\|D\Phi_s X\|^2 = -2a_K^{-2}\nabla^2 A_K(U_s)[D\Phi_s X, D\Phi_s X]$ and Gronwall; the flow commutes with gauge transformations and moves along the current. (c) The current solves the linearised flow; A^a is non-increasing along the flow, ϖ bounds a_p, and averaging the bound of the current over $[\frac{\sigma}{2}, \sigma]$ gives the last inequality. (d) $\partial_b \mathcal{O}(f) = a_K^{-1} f(x_b)\partial_b A_K + a_K^{-1} \sum_{p \ni b} (f(x(p)) - f(x_b))\partial_b a_p$, x_b a point of b, so that $|x(p) - x_b| \le 2a_K$ for $p \ni b$, and $\sum_b (\sum_{p \ni b} \varpi_p)^2 \le 32 A_K$.

(e) $\varpi(e^{iaA}) \le 2a^2(\|A\|_\infty^2 + \|\nabla A\|_\infty)$, integrated in s. (f) The lattice Bianchi identity in a gauge near p gives the differential inequality where $a_K^4 \bar z_p \le e_0$; Duhamel's formula with $\|P_t z\|_\infty \le t^{-3/2}E$ bounds $\sup_t t^{3/2} \max_p z_p$ as long as $t^{1/4}E^{1/2} \le x_0$, and the action does not increase; then (c) on an interval before σ on which Λ grows by at most 1. (g) Bochner's formula with the drift; the limit follows from the formula, as its first member on the right is ≥ 0 and the second, $2a_K^{-1}\mathcal{E}_K(H_K)$, tends to ∞; the bound follows from $\int \langle \partial H, \partial F\rangle \,\mathrm{d}\mu^K = -a_K \int F\, \mathcal{L}_K H \,\mathrm{d}\mu^K$ at $F = H|\partial H|^2$ and using $\nabla^2 A_K \ge -28$; the identity is the chain rule $\mathcal{L}_K \phi(G) = \phi'(G)\mathcal{L}_K G + \phi''(G)\Gamma_K(G)$.

(h) Expanding $\sum_p |Z_p(X)|^2 + \sum_x |(\mathrm{d}_U^{\mathsf{T}} X)(x)|^2$, the products of parallel bonds give the covariant Laplacian and those of the corners of a plaquette cancel but for $2\sum_p \langle X_1, (1 - \mathrm{Ad}\,\partial U(p)) X_4\rangle$; with (a) this is the identity. The adjoint of the linearised flow keeps horizontal vectors horizontal, and $|z(b)|$ is then a subsolution of the parabolic system with the off-diagonal entries ≥ 0; the comparison principle and the commutation of Δ_0 with n (Δ_0 is the same scalar parallel-neighbour convolution

in every bond direction) give the bound; $e^{-t\Delta_0}$ is doubly stochastic and n has row sums 13. The current is a forward horizontal solution, whence its maximum principle.

(i) $s \mapsto (g \mapsto \partial(\Phi_s U^{(g)})(p))$ is a homotopy from a translation of S^3, or a translation composed with $g \mapsto g^{-1}$, and a map of S^3 to itself that misses a point is homotopic to a constant; at the centre fields $\mathsf{s}_p = 0$, and $X = e\,\delta_{b_0}$ gives $\langle X, \nabla^2 A_K X\rangle = -4|e|^2$ at the flipped field. □

Lemma 16.3 (Exact identities for the Laplacian along the flow)**.** For all K, m, $\beta > 0$:

(a) for $f \in C^\infty(\Omega_K)$, $[\Delta_K, v]f = -8a_K^{-1} v f - 2a_K^{-3}\langle \nabla^2 A_K, \nabla^2 f\rangle_{\mathrm{HS}}$, and the diagonal blocks of $\nabla^2 A_K$ are $\delta_{\alpha\beta}\sum_{p\ni b}(1 - a_p)$;

(b) Φ_σ is a diffeomorphism of Ω_K, and $\log\det D\Phi_\sigma(U) = -12 a_K^{-2}\int_0^\sigma (3N^3 - A_K(U_s))\,\mathrm{d}s$;

(c) at every U at which d_U is injective on the site functions (all U off a closed null set), with $\mathsf{G}_U = (\mathrm{d}_U^*\mathrm{d}_U)^{-1}$, $\mathsf{I}(U) = \frac{1}{2}\log\det \mathrm{d}_U^*\mathrm{d}_U$ and $\mathsf{j}^{\mathrm{gh}}_U(b) = i\sum_{\alpha,\gamma}\mathsf{G}_U((b_-,\alpha),(b_+,\gamma))[\sigma_\alpha, \mathrm{Ad}(U(b))\sigma_\gamma]$: $\mathsf{j}^{\mathrm{gh}}_U = \partial\mathsf{I}(U)$, and for invariant $f \in C^2(\Omega_K)$

$$\Delta_K f(U) = a_K^{-1}\sum_e \frac{d^2}{dt^2} f(e^{ite}U)\Big|_{t=0} + a_K^{-1}\sum_b \langle \mathsf{j}^{\mathrm{gh}}_U(b), \partial_b f(U)\rangle = \Delta_K^{\mathrm{hor}} f + \Gamma_K(\mathsf{I}, f),$$

(e) an orthonormal basis of the horizontal vectors at U; for a site function λ,

$$\frac{d^2}{dt^2} f(e^{it\,\mathrm{d}_U\lambda}U)|_0 = \sum_b \langle i[\lambda(b_-), \mathrm{Ad}(U(b))\lambda(b_+)], \partial_b f(U)\rangle;$$

and at $f = G\circ\Phi_\sigma$ the second member is $a_K^{-1}\langle D\Phi_\sigma(U)\mathsf{j}^{\mathrm{gh}}_U, \partial G(U_\sigma)\rangle$: the gauge term is evaluated once, at the bare field, and then carried by the linearised flow.

■

Proof sketch. (a) On a Riemannian manifold, $[\Delta, \langle\nabla\phi, \nabla\cdot\rangle]f = \langle\nabla\Delta\phi, \nabla f\rangle + 2\langle\nabla^2\phi, \nabla^2 f\rangle + 2\,\mathrm{Ric}(\nabla\phi, \nabla f)$; here $\mathrm{Ric} = 2$, and $a_K\Delta_K A_K = \sum_{b,\alpha}(\partial_b^\alpha)^2 A_K = 12(3N^3 - A_K)$, because each bond variable enters a plaquette through a matrix coefficient of the fundamental representation, an eigenfunction of the Laplacian of S^3 of eigenvalue -3, and four bonds enter each of the $3N^3$ plaquettes. (b) Liouville's formula $\frac{d}{ds}\log\det D\Phi_s = (\mathrm{div}\, v)\circ\Phi_s = -a_K^{-1}(\Delta_K A_K)\circ\Phi_s$. (c) The trace of the Hessian over an orthonormal basis splits into the horizontal part and the part along the gauge directions; along $\mathrm{d}_U\lambda$ the curve $e^{it\mathrm{d}_U\lambda}U$ differs from the gauge orbit at second order, which gives the formula for one site function by invariance of f; summing over a basis orthonormal for $\mathrm{d}_U^*\mathrm{d}_U$ gives j^{gh}, and differentiating $\log\det$ identifies it with $\partial\mathsf{I}$.

The last statement is the chain rule with (b) of Lem. 16.2. □

16.5. **The lattice defect in generator estimates.**

Lemma 16.4 (A lattice defect)**.** On $(\mathbb{Z}/N)^3$ let $\varsigma_p = -1$ on the four plaquettes that contain a bond b_0 and $+1$ elsewhere. For $N \ge N_0$ the operator $\mathrm{d}^{\mathsf{T}}\varsigma\,\mathrm{d}$ on real 1-forms, per colour the Hessian of A_K at the field that is $-\mathbf{1}$ on b_0 and $\mathbf{1}$ elsewhere, has exactly three negative eigenvalues, $-\lambda_1$ simple and $-\lambda_2$ double, $6.6 < \lambda_1 < 6.62$, $0.79 < \lambda_2 < 0.8$, with orthonormal eigenvectors e_1 (of $-\lambda_1$) and e_2, e_3 (of $-\lambda_2$) that decay exponentially away from b_0. Let $\hat e_1, \hat e_2$ be the lattice unit vectors of the two axes orthogonal to b_0, so that $b_0 + \hat e_1$, $b_0 + \hat e_2$ are the two translates of b_0 parallel to it that share a plaquette with it, and the three bonds b_0, $b_0 + \hat e_1$, $b_0 + \hat e_2$ lie in ten plaquettes; the matrix $[e_i(b_0 + \hat e_j)]_{i=1,2,3;\, j=0,1,2}$, $\hat e_0 := 0$, is upper triangular and invertible, with an inverse bounded by an absolute constant. And $\langle X, \mathrm{d}^{\mathsf{T}}\varsigma\,\mathrm{d}X\rangle \ge c_1|\mathrm{d}X|^2 - 7\sum_i\langle e_i, X\rangle^2$ with an absolute $c_1 > 0$. For $\varepsilon \in (0,1]$ and $\Lambda_0 \ge 1$ there are $N_*, T_* < \infty$ and $\delta_*, \rho_0, \theta, c_\sharp > 0$, $0 < \delta \le \frac{1}{10}$, $C_\sharp < \infty$, depending only on ε and Λ_0, such that for $N = \ell/a_K \ge N_*$, $T \ge T_*$, every bond b_0 and every U whose flow in lattice time, $U_t^\circ = \Phi_{a_K^2 t}U$, has $\sup_{t\le T}\varpi(U_t^\circ) \le \delta_*$, $\int_0^T \varpi(U_t^\circ)\,\mathrm{d}t \le \Lambda_0$, $T \le \Lambda_0 N^2$ and $A_K(U^\circ_{T/2}) \le \delta_*^2$, there is a Borel set $\mathcal{S}$ of $\zeta = (\zeta_0, \zeta_1, \zeta_2) \in \mathfrak{g}^3$, $|\zeta| \le \rho_0 \le 1$, of Lebesgue measure $\ge c_\sharp e^{-(3\lambda_1 + 6\lambda_2 + \varepsilon)T}$, such that

the field W_ζ, equal to U but for $W_\zeta(b_0) = -e^{i\zeta_0}U(b_0)$ and $W_\zeta(b_0+\hat e_j) = e^{i\zeta_j}U(b_0+\hat e_j)$ $(j=1,2)$, and the fields $\tilde V_t$, equal to $\Phi_{a_K^2 t}W_\zeta$ but for the sign of their value at b_0, have the properties (a)–(c) below. Here $a_p(\Phi_{a_K^2 t}W_\zeta) = \varsigma_p a_p(\tilde V_t) + 1 - \varsigma_p$ for every plaquette p, so that $A_K(\Phi_{a_K^2 t}W_\zeta) = 8 + \sum_p \varsigma_p a_p(\tilde V_t)$, and $t \mapsto \tilde V_t$ is the gradient flow, in lattice time, of $\tilde A = \sum_p \varsigma_p a_p + 8$. (a) $\varpi(\tilde V_t) \le \delta$ for $0 \le t \le T$: the four plaquettes of $\Phi_{a_K^2 t}W_\zeta$ at b_0 lie within δ of $-\mathbf{1}$ and all the others within δ of $\mathbf{1}$; (b) for every real function $\mathsf f$ of the plaquettes and real $\mathsf f_0$,

$$\big|\partial_{b_0}\big(\textstyle\sum_p \mathsf f_p\, a_p \circ \Phi_{a_K^2 T}\big)(W_\zeta)\big| \ge \Big[c_\sharp|\mathsf f_0| - C_\sharp\Big(\textstyle\sum_p |\mathsf f_p - \mathsf f_0|^2 e^{-2\theta|p|}\Big)^{1/2}\Big]e^{(\lambda_1-\varepsilon)T},$$

$|p|$ the distance of p from b_0; in particular $|\partial_{b_0}(A_K \circ \Phi_{a_K^2 T})(W_\zeta)| \ge c_\sharp e^{(\lambda_1-\varepsilon)T}$; (c) $\int_{T/2}^T \sum_p a_p(\tilde V_t)\,\mathrm dt \le \frac{3}{2}T\,A_K(U^\circ_{T/2}) + C_\sharp$.

(d) (Traversing the fibre of one bond.) Let $\lambda_1' = \lambda_1 - 0.027$ and $\lambda_2'' = \lambda_2 + \frac{1}{700}(\lambda_1 - \lambda_2) + 10^{-3}$; then $\lambda_1' - 6\lambda_2'' > \frac{2}{5}$. For $\Lambda_0 \ge 1$ there are $\epsilon_{\mathrm d}, \rho_2, c_{\mathrm d} > 0$ and $N_{\mathrm d} < \infty$, depending only on Λ_0, such that the following holds for $N \ge N_{\mathrm d}$, $T \ge N_{\mathrm d}$ with $N^3 \le c_{\mathrm d}T^2$, every bond b_0 and every U with $\int_0^T \varpi(U_t^\circ)\,\mathrm dt + T/N^2 \le \Lambda_0$ and $\sup_{t\le T}\varpi(U_t^\circ) \le \epsilon_{\mathrm d}$. There is a compact set $\mathsf Y \subset \mathfrak g^2$ of Lebesgue measure $\ge c_{\mathrm d}e^{-6\lambda_2'' T}$ such that for every $(y_1,y_2) \in \mathsf Y$, with W the field equal to U but for $W(b_0+\hat e_j) = e^{iy_j}U(b_0+\hat e_j)$ $(j=1,2)$ and $W^{(g)}$ the field W with the value g at b_0, and for every unit vector $e \in \mathfrak g$, there are $x_0 \in \mathfrak g$, with $|(x_0,y_1,y_2)| \le 30C_\triangle\rho_2 \le 1$, where $C_\triangle = 4c_\triangle$ and $c_\triangle$ is the absolute constant that bounds the inverse of the matrix $[e_i(b_0+\hat e_j)]$, and $r_* \in (0, \frac{4}{3}\rho_2 e^{-\lambda_1' T}]$ for which $\mathsf v(r) = A_K(\Phi_{a_K^2 T}W^{(g_r)})$, $g_r = -e^{ire}e^{ix_0}U(b_0)$, has $\mathsf v(0) \in [7,9]$ and $\mathsf v(0) - \mathsf v(r_*) \ge 2\rho_2^2$. If $f \in C^\infty(T_\ell)$ is real, $f(x_*) = \max|f| > 0$ and b_0 is based at a site nearest to x_*, the same holds, with constants that depend besides on f and for $a_K \le c_{\mathrm d}(f)$, for $\mathsf v_f(r) = \sum_p f(x(p))\,a_p(\Phi_{a_K^2 T}W^{(g_r)})$ in the place of $\mathsf v$, and with the single condition $\mathsf v_f(0) - \mathsf v_f(r_*) \ge f(x_*)\rho_2^2$ in the place of the two conditions on $\mathsf v$. ∎

Sketch. Flipping the sign of $U(b_0)$ conjugates the flow of A_K to that of $\tilde A = \sum_p \varsigma_p a_p + 8$, whose Hessian at small fields is close to $\mathrm d^{\mathsf T}\varsigma\,\mathrm d$. The number of negative eigenvalues is found exactly. For $E > 0$, with $S = (\mathrm d^{\mathsf T}\mathrm d + E)^{1/2}$ and Π_4 the projection onto the four plaquettes, $\mathrm d^{\mathsf T}\varsigma\,\mathrm d + E = S(1 - 2R^{\mathsf T}R)S$, $R = \Pi_4 \mathrm d S^{-1}$; by Sylvester's law of inertia the number of eigenvalues below $-E$ is the number of eigenvalues > 1 of $2RR^{\mathsf T} = K(E) = 2\Pi_4[\frac{-\Delta}{-\Delta+E} - \mathrm d_2^{\mathsf T}G_E\mathrm d_2]\Pi_4$, $\mathrm d_2$ the coboundary of 2-forms and $G_E = (-\Delta+E)^{-1}$, which decreases in E. On $\mathbb Z^3$, $K(E)$ is diagonal in three patterns of signs on the four plaquettes, taken relative to the signs of $\mathrm d\delta_{b_0}$: $(1,1,1,1)$, the doublet $(1,0,-1,0)$, $(0,1,0,-1)$, and $(1,-1,1,-1)$, with the entries

$$K_1 = 2 - \tfrac{E}{3}(1 - EG_E(0)), \qquad K_2 = 2 - F_2, \qquad F_2 = 2E\big(G_E(0) + G_E(e_1)\big) + 4\big(G_E(0) - G_E(e_1+e_2)\big),$$
$$K_3 = 2\big[1 - E\big(G_E(0) - G_E(e_1)\big) - 4G_E(0) + 8G_E(e_1) - 4G_E(e_1+e_2)\big],$$

where $G_E(x) = (2\pi)^{-3}\int_{[-\pi,\pi]^3} e^{ik\cdot x}\big(E + \sum_\mu 2(1-\cos k_\mu)\big)^{-1}dk$ is the Green function of $-\Delta + E$ on the sites of $\mathbb Z^3$. Its Neumann series $G_E(x) = \sum_{m\ge 0} c_m(x)(6+E)^{-m-1}$, with the number $c_m(x) = \sum_n \binom{m}{n}\binom{n}{(n+x_1+x_2)/2}\binom{n}{(n+x_1-x_2)/2}\binom{m-n}{(m-n+x_3)/2}$ of the walks of m steps from 0 to x (a binomial coefficient whose lower entry is not an integer between 0 and its upper entry read as 0), encloses $G_E(x)$ between its partial sum up to $m = M$ and that sum plus $E^{-1}(\frac{6}{6+E})^{M+1}$; at $M = 240$ the width is below 10^{-12} at the four energies 6.6, 6.62, 0.79, 0.8. This gives, to the digits shown, $E(1 - EG_E(0)) = 2.997958$ at $E = 6.6$ and 3.002351 at $E = 6.62$, and $F_2 = 0.99944880$ at $E = 0.79$ and 1.00144415 at $E = 0.8$; so $K_1(6.6) > 1 > K_1(6.62)$ and $K_2(0.79) > 1 > K_2(0.8)$. As $E \to 0$, $K_1 \to 2$, $K_2 \to 2 - 4(g_0 - g_2)$ and $K_3 \to 8(g_0 - g_2) - \frac{2}{3}$, $g_0 - g_2 = G_0(0) - G_0(e_1+e_2)$; since $0 \le 1 - \cos k_1\cos k_2 \le (1-\cos k_1) + (1-\cos k_2)$, the symmetry of the axes gives $0 \le g_0 - g_2 - [G_E(0) - G_E(e_1+e_2)] \le \frac{E}{3}G_E(0)$, and the series at $E = \frac{1}{10}$

with $M = 700$, of width $\le 9.3 \cdot 10^{-5}$, give $0.19195 < g_0 - g_2 < 0.19970$; so $K_2(0+) > 1.2$ and $K_3(0+) < 0.94$. As the entries decrease in E, the third pattern gives no negative eigenvalue, the first gives the simple $-\lambda_1 \in (-6.62, -6.6)$, the doublet gives the double $-\lambda_2 \in (-0.8, -0.79)$, and there is no other. On $(\mathbb{Z}/N)^3$ the Green function is $\sum_{k\in\mathbb{Z}^3} G_E(x+Nk)$, which differs from $G_E(x)$ by at most $E^{-1}\sum_{k\neq 0}(\frac{6}{6+E})^{|x+Nk|_1}$, uniformly small for $E \ge 0.7$ and $|x|_1 \le 2$, and the entries at $E \to 0$ are Riemann sums of those on $\mathbb{Z}^3$: the strict inequalities persist for $N \ge N_0$. The eigenvectors: $e_i \propto \mathrm{d}^{\mathsf{T}} G_{\lambda_i}\eta_i$, η_i the pattern of the mode on the four plaquettes, so they decay at the rate of $G_E(x) \le E^{-1}(\frac{6}{6+E})^{|x|_1}$: as $\frac{6}{6+\lambda_i} \le \frac{6}{6.79} < e^{-0.12}$, $|e_i(b)| \le C_e e^{-c_e|b|}$ with $c_e = 0.12$, an absolute number C_e and $|b|$ the distance of the bond b from b_0; $e_1(b_0)^2 \ge 2/\lambda_1$; the doublet patterns are odd under the reflections across the two planes through b_0 that contain $\hat e_2$, resp. $\hat e_1$, which fix b_0 and $b_0 + \hat e_2$, resp. b_0 and $b_0 + \hat e_1$, so $e_2(b_0) = e_2(b_0+\hat e_2) = 0 = e_3(b_0) = e_3(b_0+\hat e_1)$, while $|e_2(b_0+\hat e_1)| = |e_3(b_0+\hat e_2)| \ge 0.049$: the matrix is triangular. *The coercivity.* On the range of d the form $\langle \eta, \varsigma\eta'\rangle$ is that of $B = 1 - 2\mathsf{P}\Pi_4\mathsf{P}$, P the projection onto the range of d; the non-zero eigenvalues of $\mathsf{P}\Pi_4\mathsf{P}$ are those of $\frac12 K(0+)$, so the spectrum of B lies in $\{-1, 1-K_2(0+), 1-K_3(0+), 1\}$: $\|B\| \le 1$, $|B| \ge c_B = 0.06$ for $N \ge N_0$, and B has three negative eigenvalues. On the span of the $\mathrm{d}e_i$, three-dimensional, $B \le -\frac{\lambda_2}{12}$, as $\langle \mathrm{d}e_i, \varsigma\,\mathrm{d}e_j\rangle = -\lambda_j\delta_{ij}$ and $|\mathrm{d}Y|^2 \le 12|Y|^2$. A self-adjoint B with $\|B\| \le 1$, $|B| \ge c_B$ and an n-dimensional negative spectral subspace is $\ge c_- c_B$ on the B-orthogonal complement of every n-dimensional subspace on which $B \le -c_-$ (compare the two spectral parts in the norm of $|B|$). For $X = \sum_i a_i e_i + X'$, $a_i = \langle e_i, X\rangle$, one has $\langle \mathrm{d}X', \varsigma\,\mathrm{d}e_i\rangle = -\lambda_i\langle X', e_i\rangle = 0$, so $\langle X, \mathrm{d}^{\mathsf{T}}\varsigma\,\mathrm{d}X\rangle = -\sum_i \lambda_i a_i^2 + \langle \mathrm{d}X', \varsigma\,\mathrm{d}X'\rangle \ge -\lambda_1\sum_i a_i^2 + 0.0039(\frac12|\mathrm{d}X|^2 - 12\sum_i a_i^2)$: the bound with $c_1 = 0.0019$, as $\lambda_1 + 0.05 \le 7$. *The unstable coordinates.* Let $\tilde\Phi_t$ be the flow of $\tilde A$ in lattice time, $\tilde V_t = \tilde\Phi_t \tilde V_0$, $\tilde V_0$ the field W_ζ with the sign of its value at b_0 reversed. The nine coordinates $\xi_i(V) = \sum_p \omega_i(p)\,\mathrm{Ad}(T_p(V))\,\mathsf{s}_p(V) \in \mathfrak{g}$, $i = 1,2,3$, are the curvature paired with $\omega_i = -\lambda_i^{-1}\varsigma\,\mathrm{d}e_i$ (so $\mathrm{d}^{\mathsf{T}}\omega_i = e_i$, and $|\omega_i(p)| \le C_e e^{-c_e|p|}$ once C_e is enlarged by an absolute factor), cut off at the distance $\frac14 N$ from b_0 and carried to one root along a comb of paths: $T_x(V)$ is the ordered product of the values of V along the path of the comb from the root to the site x, and $T_p = T_x$, $T_b = T_x$ for a plaquette p and a bond b based at x; along a flow line of $\tilde A$, $\dot\xi_i = \lambda_i\xi_i + r_i$ with $|r_i| \le C(q(\tilde V_t) + N^2 e^{-c_e N/4})$, $q(V) = \sum_p e^{-c|p|}\varpi_p(V)^2$ with $c = \frac12 c_e$, a weighted sum of the plaquettes, at most $C_q\varpi(V)^2$ with an absolute C_q. The other directions are controlled by the coercivity bound in the rough field, through a lattice Weitzenböck identity and Kato's inequality, as long as the nine coordinates stay in the box $\mathfrak{B} = \{\max_i|\xi_i| \le \delta_u\}$, with the number $\delta_u > 0$ fixed below, say on $[0, t_1]$, $t_1 \le T$; the bound does not grow with T, as the unstable coordinates are integrated backwards in time. Let $D(t)$ be the ℓ^2 distance of the gauge orbits of $\tilde V_t$ and U_t°, an absolutely continuous function with $D(0) \le |\zeta| \le \rho_0$; at a minimising gauge $\tilde V_t = e^{iY_t}U_t^\circ$, Y_t horizontal at U_t°; let $\mathcal{D}_t = \sum_p |Z_p(Y_t)|^2$ be the energy of the difference and $\varpi_t^\circ = \varpi(U_t^\circ)$. (i) While $D \le \bar D$, a radius $\le \frac12$, half the derivative of D^2 is at most $-\sum_p \varsigma_p|Z_p(Y_t)|^2$ plus the source $2\sum_{p\ni b_0}\langle Z_p(Y_t), \mathsf{s}_p(U_t^\circ)\rangle$ of the four plaquettes at b_0, a member $28\varpi_t^\circ D^2$ and members of the orders three and four in Y_t; the coercivity bound applies to the first member, the three $\mathfrak{g}$-valued linear coordinates of Y_t that it leaves free equal the $\xi_i(\tilde V_t)$ up to errors bounded by $C\varpi_t^\circ$ and by $C(\varpi_t^\circ + D)(\mathcal{D}_t + (\varpi_t^\circ + N^{-2})D^2)^{1/2}$, and the Weitzenböck identity with Kato's inequality bounds $\sum_b |Y_t(b)|^4$ by $CD^2(\mathcal{D}_t + (\varpi_t^\circ + N^{-2})D^2)$; for $\bar D$ and δ_* small this gives, with absolute numbers $c' > 0$ and C, $\frac12\frac{d}{dt}D^2 \le -c'\mathcal{D}_t + C|\xi(\tilde V_t)|^2 + C(\varpi_t^\circ + N^{-2})D^2 + C(\varpi_t^\circ)^2$. (ii) Here the hypotheses $\int_0^T \varpi_t^\circ\,\mathrm{d}t \le \Lambda_0$ and $T \le \Lambda_0 N^2$ enter: they give $\int_0^T(\varpi_t^\circ + N^{-2})\,\mathrm{d}t \le 2\Lambda_0$, and with $\varpi_t^\circ \le \delta_*$ also $\int_0^T(\varpi_t^\circ)^2\,\mathrm{d}t \le \delta_*\Lambda_0$; so Gronwall's lemma gives, with $t_2 \le t_1$ the largest time with $D \le \bar D$ on $[0, t_2]$, $D(t)^2 + 2c'\int_0^t \mathcal{D}_s\,\mathrm{d}s \le e^{4C\Lambda_0}[\rho_0^2 + 2C\int_0^{t_2}|\xi(\tilde V_s)|^2 ds + 2C\delta_*\Lambda_0]$ for $t \le t_2$. (iii) The equation $\dot\xi_i = \lambda_i\xi_i + r_i$ is integrated backwards from t_2: $\xi_i(t) = e^{-\lambda_i(t_2-t)}\xi_i(t_2) - \int_t^{t_2} e^{-\lambda_i(u-t)}r_i(u)\,\mathrm{d}u$ with $|\xi_i(t_2)| \le \delta_u$, and Young's inequality gives $\int_0^{t_2}|\xi_i|^2 \le \delta_u^2/\lambda_i + 2\lambda_i^{-2}\int_0^{t_2}|r_i|^2$, a bound that does not grow with t_2. (iv) By the bound of r_i, $\int_0^{t_2}|r_i|^2 \le C[\sup_{[0,t_2]} q\int_0^{t_2} q + N^4 e^{-c_e N/2}T]$ with $q = q(\tilde V_t)$,

where $q(\tilde V_t)\le C(\mathcal D_t+(\varpi_t^\circ)^2+(\varpi_t^\circ+N^{-2})D^2)$, $\sup q\le C(\delta_*+\bar D)^2$ and $N^4e^{-c_eN/2}T\le\Lambda_0N^6e^{-c_eN/2}$. (v) Inserting (iii) and (iv) into (ii), $\mathcal J=\sup_{[0,t_2]}D^2+2c'\int_0^{t_2}\mathcal D_t\,\mathrm dt$ obeys $\mathcal J\le e^{C\Lambda_0}[\rho_0^2+\delta_u^2+\delta_*+N^6e^{-c_eN/2}]+e^{C\Lambda_0}(\delta_*+\bar D)^2\mathcal J$; for $\bar D$ and δ_* small with Λ_0, and N large with δ_*, this gives $\mathcal J\le\frac14\rho^2$ with $\rho^2=C_*(\rho_0^2+\delta_u^2+\delta_*)$, C_* depending only on Λ_0, and if $\rho\le\bar D$ the continuity of D gives $t_2=t_1$. So on $[0,t_1]$ the distance of the two flows modulo gauge stays below ρ, small with ρ_0, δ_u and δ_*, and the energy of the difference is integrable in time, $\int_0^{t_1}\mathcal D_t\,\mathrm dt\le\rho^2/c'$. *The numbers.* Given ε and Λ_0, the numbers are fixed in this order. First C_* and $\bar D$, which read Λ_0 alone, and $C_{**}=(C_*(64C_{\mathsf E}^2+2))^{1/2}$, with $C_{\mathsf E}$ the absolute bound of the inverse of the triangular matrix. Next θ and a ceiling $\bar\delta\le\frac{1}{10}$ for δ, small with ε and Λ_0, such that along flows with $\varpi(\tilde V_t)\le\bar\delta$, for N and T large, $\|\Lambda^9D\tilde\Phi_T\|\le e^{(3\lambda_1+6\lambda_2+\varepsilon)T}$, and the cone condition of (b) holds with the loss ε in the rate and leaves outside the strongest mode a part of w_T that is small against $|x_T|/(1+C_{**})$. Next δ_u, small with all these. Then $\delta_*=\delta_u^2$ and $\rho_0=8C_{\mathsf E}\delta_u$, so that the radius of (v) is $\rho=C_{**}\delta_u\le\bar D$ and $\delta=\delta_*+4\rho\le(1+4C_{**})\delta_u\le\bar\delta$. Last N_* and T_*, large with all these, so that for $N\ge N_*$ the number $N^2e^{-c_eN/4}$ is small against δ_u and $N^6e^{-c_eN/2}\le\delta_*$. These choices serve three steps below: on the faces of $\mathfrak B$, $|r_i|\le C(C_q\delta^2+N^2e^{-c_eN/4})\le\frac12\lambda_2\delta_u$, so that the exit is strict; at $t=0$ the map $\zeta\mapsto\xi(\tilde V_0)$ differs from its linear part, on the ball $|\zeta|\le\rho_0$, by a map with a Lipschitz constant at most $1/(2C_{\mathsf E})$, so that its image holds the ball of radius $\rho_0/(2C_{\mathsf E})=4\delta_u$ about $\xi(U)$, which holds $\mathfrak B$ as $|\xi(U)|\le C\delta_*\le\delta_u$; and at the end $|\partial\tilde A(\tilde V_T)|\le C(\delta_*+\rho)\le C(1+C_{**})\delta_u$, so that the errors in (b) are small multiples of $\delta_u|x_T|$. *The set.* Let $\mathcal K_T$ be the set of the ζ, $|\zeta|\le\rho_0=8C_{\mathsf E}\delta_u$, $C_{\mathsf E}$ the bound of the inverse of the triangular matrix, whose nine coordinates stay in $\mathfrak B$ up to the time T. At $t=0$ the map $\zeta\mapsto\xi(\tilde V_0)$ is close to the linear map $\zeta\mapsto(\sum_j e_i(b_0+\hat e_j)\mathrm{Ad}(T_{b_0+\hat e_j}(U))\zeta_j)_i$, invertible by the triangular matrix; on the faces of $\mathfrak B$ the exit is strict, as $\frac{d}{dt}\frac12|\xi_i|^2\ge\delta_u(\lambda_2\delta_u-|r_i|)>0$ there; so by a degree argument the end map $\Xi_T(\zeta)=\xi(\tilde V_T)$ carries $\mathcal K_T$ onto the interior of $\mathfrak B$. Put $\mathcal S=\mathcal K_T\cap\Xi_T^{-1}(\mathsf S)$, $\mathsf S=\{\frac12\delta_u<|\xi_1|<\delta_u,\ |\xi_2|,|\xi_3|<\delta_u\}$: its image under Ξ_T is the shell $\mathsf S$, of the volume $c\,\delta_u^9$, and the area formula with $\|\Lambda^9D\tilde\Phi_T\|\le e^{(3\lambda_1+6\lambda_2+\varepsilon)T}$ bounds the measure of $\mathcal S$ from below. (a) holds on $\mathcal K_T$, with $\delta=\delta_*+4\rho$. *(b) on $\mathcal S$.* With $w_T=D\tilde\Phi_T(\tilde V_0)[e\,\delta_{b_0}]$, $e\in\mathfrak g$,

$$\langle e,\partial_{b_0}(\textstyle\sum_p\mathsf f_pa_p\circ\Phi_{a_K^2T})(W_\zeta)\rangle=\mathsf f_0\langle\partial\tilde A(\tilde V_T),w_T\rangle+\sum_p(\mathsf f_p-\mathsf f_0)\varsigma_p\langle Z_p(w_T),\mathsf s_p(\tilde V_T)\rangle.$$

By a cone condition, also with the weight $e^{\theta|b|}$, w_T follows the strongest mode: $|w_T|^2\le3.4|x_T|^2$, x_T its coordinate along e_1 carried to the root, the rest of w_T small against $|x_T|$, and $|x_T|\ge\frac12|e|e^{(\lambda_1-\varepsilon)T}$. At the end the gradient is small, $|\partial\tilde A(\tilde V_T)|\le C(\delta_*+\rho)$, as $A_K(U^\circ_{T/2})\le\delta_*^2$, and, since $\varsigma\,\mathrm de_1=-\lambda_1\omega_1$, $\langle\partial\tilde A(\tilde V_T),w_T\rangle=-\lambda_1\langle\xi_1(\tilde V_T),x_T\rangle$ up to a small multiple of $\delta_u|x_T|$. The map $e\mapsto x_T(e)$ is onto; for e with $x_T(e)$ a positive multiple of $\xi_1(\tilde V_T)$, of modulus $>\frac12\delta_u$ on $\mathcal S$: $|\langle\partial\tilde A(\tilde V_T),w_T\rangle|\ge\frac14\lambda_1\delta_u|x_T|$, while the second sum is at most $21\,\delta(\sum_p|\mathsf f_p-\mathsf f_0|^2e^{-2\theta|p|})^{1/2}|x_T|$; divide by $|e|$. (c): on $\mathcal S$, up to a gauge, $\tilde V_t=e^{iY_t}U_t^\circ$ with Y_t horizontal at U_t° and small, and the energy $\mathcal D_t=\sum_p|Z_p(Y_t)|^2$ of the difference has $\int_0^T\mathcal D_t\,\mathrm dt$ bounded by a number that depends only on ε and Λ_0; as $\sum_pa_p(e^{iY}U)\le3A_K(U)+\frac32\sum_p|Z_p(Y)|^2+6\sum_p(\sum_{b\in p}|Y(b)|^2)^2$, with a quartic member that is integrable in t by the same bound and the inequality of Sobolev on the discrete torus, and as $A_K(U_t^\circ)$ does not increase, the integral over $[\frac T2,T]$ gives (c). (d): here the unstable part is read through nine coordinates of its own, taken from the velocity of the flow and cut off at a fixed radius. Let R be a radius that reads ε_c and Λ_0 alone, χ_R a cutoff of the bonds with values in $[0,1]$, equal to 1 within the distance $\frac R2$ of b_0 and to 0 beyond R, which varies by at most π/R across a plaquette, and $\mathsf t_x(V)$ the transport from the root to the site x along the comb of the coordinates ξ_i; with (σ_α) an orthonormal basis of $\mathfrak g$, $\lambda_3=\lambda_2$ and $i,\alpha=1,2,3$, let $e_{i\alpha}(V)(b)=\chi_R(b)\,e_i(b)\,\mathrm{Ad}(\mathsf t_{b_-}(V))^{-1}\sigma_\alpha$ and $u_{i\alpha}(V)=\lambda_i^{-1}\langle e_{i\alpha}(V),-\partial\tilde A(V)\rangle$. These nine numbers read the velocity of the flow along the cut eigenvectors, and they are the nine unstable coordinates of (d): $u_1=(u_{1\alpha})_\alpha\in\mathbb R^3$ is the coordinate in the strongest mode, and $(u_2,u_3)\in\mathbb R^6$ are the six slow coordinates. Along a flow line of $\tilde A$ the velocity solves the linearised

flow exactly, as the current does in Lem. 16.2(c), so that $\frac{d}{ds}u_{i\alpha}(\tilde V_s) = \lambda_i u_{i\alpha}(\tilde V_s) + \mathrm{err}_{i\alpha}(\tilde V_s)$ with $|\mathrm{err}_{i\alpha}(V)| \le C_R q_R(V) + 4\lambda_2^{-1}\epsilon_R q_R(V)^{1/2}$; here $q_R(V)$ is the sum of $\varpi_p(V)^2$ over the plaquettes within the distance $R+1$ of b_0, $\epsilon_R = \max_i \|(\mathrm{d}^\mathsf{T}\varsigma\,\mathrm{d} + \lambda_i)\chi_R e_i\| + \max_i \|(1-\chi_R)e_i\|_{\ell^{6/5}} \le CR^{5/2}(\frac{6}{6+\lambda_2})^{R/2}$ is the defect of the cut eigenvectors, and the numbers C_R read R and Λ_0 alone. For $V' = e^{i\eta}V$ with $\varpi(V)$ and $\|\eta\|$ small with R, $|\mathrm{err}_{i\alpha}(V') - \mathrm{err}_{i\alpha}(V)| \le [C_R(\varpi(V) + \|\eta\|) + 44\lambda_2^{-1}\epsilon_R]\|\eta\|$, and $|u_{i\alpha}(V') - u_{i\alpha}(V) - \langle e_{i\alpha}(V), \eta\rangle| \le \frac{\varepsilon_c}{100}\|\eta\|$ if $\varpi(V), \|\eta\| \le \epsilon_\sharp$, a number that reads R and ε_c. Let $\tilde V_s$ and $\tilde V'_s = e^{i\eta_s}\tilde V_s$ be two flow lines of $\tilde A$ in lattice time along which $\varpi \le \epsilon_\sharp$ and $n_s = \|\eta_s\| \le \epsilon_\sharp$, and $p_s = |u_1(\tilde V'_s) - u_1(\tilde V_s)|$ the difference of their coordinates in the strongest mode. They obey the cone inequalities $\frac{d}{ds}\frac12 n^2 \le \lambda_2 n^2 + (\lambda_1 - \lambda_2)p^2 + \varepsilon_c n^2$ and $\frac{d}{ds}\frac12 p^2 \ge \lambda_1 p^2 - \varepsilon_c pn$, $\varepsilon_c \le 10^{-3}$; so where $n^2 \le 700p^2$ the difference p grows at the rate $\ge \lambda_1 - \varepsilon_c\sqrt{700} \ge \lambda'_1$, and this cone is never left, while where $n^2 > 700p^2$ or $p = 0$ the distance n grows at the rate $\le \lambda_2 + \frac{1}{700}(\lambda_1 - \lambda_2) + \varepsilon_c \le \lambda''_2$. As long as the nine unstable coordinates stay in the ball $\{|u| \le \rho_1\}$, the flow of $\tilde A$ from the three changed bonds stays near the flow of A_K from U, modulo gauge and in ℓ^2 of all the bonds: by an energy inequality, which holds while the distance $D(s)$ of the two flows modulo gauge is at most a number D_0, $D(s) \le C'(\Lambda_0)\rho_1 + C_R\epsilon_{\mathrm d}^{1/2}$, with $C'(\Lambda_0)$ free of ε_c and of the radii; so $\varpi(\tilde V_s) \le \epsilon_{\mathrm d} + 4D(s) \le 5C'(\Lambda_0)\rho_1 \le \epsilon_\sharp$ and $|\mathrm{err}(\tilde V_s)| \le \frac{1}{12}\lambda_2\rho_1$ in $\mathbb{R}^9$ there. The values $(x, y_1, y_2) \in \mathfrak{g}^3$ at b_0 and at the two neighbouring bonds parametrise the nine unstable coordinates at the time 0 by a map close to an invertible linear one, triangular through the matrix $[e_i(b_0 + \hat e_j)]$, whose inverse has the norm at most $c_\triangle$. The data are taken in the ball $\{|(x, y_1, y_2)| \le 30C_\triangle\rho_2\}$, of radius $C_\triangle\rho_1 = 4c_\triangle\rho_1$: there ϖ and the changes stay below $\epsilon_\sharp$, and the bound on $u_{i\alpha}(V') - u_{i\alpha}(V)$ above, with the change of the $e_{i\alpha}(V)$ over the ball, small with ρ_1, leaves to the difference of the map and its linearisation at the centre a Lipschitz constant at most $\frac{1}{2c_\triangle}$ on the ball; so the map carries the ball homeomorphically onto its image, the images of the points of the boundary sphere lie at the distance $\ge \frac{C_\triangle\rho_1}{2c_\triangle} = 2\rho_1$ from $u(U)$, the image of the centre, and as $|u(U)| \le C_R\epsilon_{\mathrm d} \le \rho_1$ the image holds the ball $\{|u| \le \rho_1\}$, $\rho_1 = 30\rho_2$; on the sphere $\{|u| = \rho_1\}$ the exit is strict, as $\frac{d}{ds}\frac12|u|^2 \ge \lambda_2\rho_1^2 - \frac{1}{12}\lambda_2\rho_1^2 > 0$ there; so by a shooting argument with Brouwer's theorem, for every $\mathsf w \in \mathbb{R}^6$ with $|\mathsf w| \le \frac14\rho_1$ there are data in the ball $\{|(x, y_1, y_2)| \le 30C_\triangle\rho_2\}$ whose nine coordinates stay in the ball $\{|u| \le \rho_1\}$ up to T, whose coordinate in the strongest mode vanishes at T, and whose six slow coordinates at T equal $\mathsf w$. Here $e_1(b_0)^2 \ge 4/\lambda_1 > 0.604$: as $e_2(b_0) = e_3(b_0) = 0$, $\delta_{b_0} = e_1(b_0)e_1 + z$ with z in the orthogonal complement of e_1, e_2, e_3, on which the form of $\mathrm{d}^\mathsf{T}\varsigma\,\mathrm{d}$ is ≥ 0, as it has no other negative eigenvalue, and the cross term vanishes, e_1 being an eigenvector; so $-4 = \langle\delta_{b_0}, \mathrm{d}^\mathsf{T}\varsigma\,\mathrm{d}\,\delta_{b_0}\rangle \ge -\lambda_1 e_1(b_0)^2$, the value -4 coming from the four plaquettes at b_0; thus $|e_1(b_0)| > 0.777$. For two such data let $s_1 \le T$ be the last time up to which $n \le \epsilon_\sharp$. On $[0, s_1]$ the pair stays out of the cone: inside it p would grow from then on, against $p_T = 0$ if $s_1 = T$, as the coordinates in the strongest mode both vanish at T, and against $p_{s_1} \ge \epsilon_\sharp/\sqrt{700} > 2\rho_1 \ge p_{s_1}$ if $s_1 < T$. Hence $n_{s_1} \le e^{\lambda''_2 s_1}n_0$, and the slow coordinates at T differ by at most $3e^{\lambda''_2 T}n_0$, by $\sqrt6 \cdot 1.01\,n_T$ if $s_1 = T$ and by $2\rho_1 < \epsilon_\sharp = n_{s_1}$ if $s_1 < T$. At the time 0 the difference η_0 lives on the three changed bonds; $p_0 \le n_0/\sqrt{700}$, $n_0 \le |\eta_0(b_0)| + |\eta_0(b_0+\hat e_1)| + |\eta_0(b_0+\hat e_2)|$ and $p_0 \ge |e_1(b_0)||\eta_0(b_0)| - |\eta_0(b_0+\hat e_1)| - |\eta_0(b_0+\hat e_2)| - \varepsilon_c n_0$, so $0.73|\eta_0(b_0)| \le 1.04(|\eta_0(b_0+\hat e_1)| + |\eta_0(b_0+\hat e_2)|)$ and $n_0 \le 2.5(|\eta_0(b_0+\hat e_1)| + |\eta_0(b_0+\hat e_2)|) \le 4|y - y'|$, y, y' the values of the data at the two neighbouring bonds: the slow coordinates at T differ by at most $12e^{\lambda''_2 T}|y - y'|$; a Lipschitz map changes Lebesgue measure at most by the sixth power of its constant, which gives the measure of $\mathsf Y$. On the fibre of b_0 through such a datum, along the geodesic g_r, the two flows differ at the time 0 by $\eta_0 = re\,\delta_{b_0}$, so $p_0 \ge (|e_1(b_0)| - \varepsilon_c)r \ge 0.75\,r \ge 0.75\,n_0$: the difference in the strongest mode starts at $\ge 0.75\,r$, inside the cone, and grows at the rate $\ge \lambda'_1$, with $n_s \le 1.35p_s$ by the same two inequalities; let r_* be the first r at which it reaches ρ_2 at the time T (meanwhile $n_s \le 1.35\rho_2 < \epsilon_\sharp$, and $|u| \le \frac13\rho_1 + 3.1n_s < \frac12\rho_1$, where $\frac13\rho_1$ bounds $|u|$ along the datum by the equation of u read backwards from T), so that $r_* \le \frac43\rho_2 e^{-\lambda'_1 T}$. At the time T

the velocity of the flow is orthogonal to the strongest mode, $\langle \partial\tilde{A}(\tilde{V}_T), e_{1\alpha}(\tilde{V}_T)\rangle = -\lambda_1 u_{1\alpha}(\tilde{V}_T) = 0$, as the coordinate of the datum in that mode vanishes there, and it is of the order ρ_2 with a constant of Λ_0 alone: $\|\partial\tilde{A}(\tilde{V}_T)\| \le \|\partial A_K(U_T^\circ)\| + 16\epsilon_{\mathrm{d}} + 44D(T)$, $D(T)$ the distance of the two flows modulo gauge, where $\|\partial A_K(U_T^\circ)\|^2 \le 48e^{24\Lambda_0}\Lambda_0^2 N^3 T^{-3}$ (the condition $N^3 \le c_{\mathrm{d}} T^2$ enters here as well) and $D(T) \le C'(\Lambda_0)\rho_1 + C_R\epsilon_{\mathrm{d}}^{1/2}$; so $\|\partial\tilde{A}(\tilde{V}_T)\| \le 45\,C'(\Lambda_0)\rho_1 = 1350\,C'(\Lambda_0)\rho_2$, $\rho_2 = \rho_1/30$, by the last condition on ϵ_{d} below. The numbers are chosen in this order: $C'(\Lambda_0)$; $\varepsilon_c = \min\{10^{-3}, (4000\,C'(\Lambda_0))^{-2}\}$; the radius R, which reads ε_c and Λ_0; $\epsilon_\sharp$, which bounds ϖ and n wherever the cone inequalities are used, and D_0, the distance modulo gauge up to which the energy inequality holds, both depending on R and ε_c; ρ_1, small with R, with $5C'(\Lambda_0)\rho_1 \le \epsilon_\sharp$, $2\sqrt{700}\,\rho_1 < \epsilon_\sharp$ and $5C_\triangle\rho_1 \le \min\{1, \epsilon_\sharp\}$, so that $30C_\triangle\rho_2 \le 1$ and, as $\epsilon_{\mathrm{d}} \le \frac{1}{5}\epsilon_\sharp$ by the conditions on ϵ_{d} below, ϖ and the changes stay below $\epsilon_\sharp$ on the ball of the data; $\rho_2 = \rho_1/30$; and last ϵ_{d}, N_{d} and c_{d}, which depend on everything chosen before: small enough for the energy inequality, and such that $C_R\epsilon_{\mathrm{d}} \le \rho_1$, $\epsilon_{\mathrm{d}} + 4C_R\epsilon_{\mathrm{d}}^{1/2} \le C'(\Lambda_0)\rho_1$, $\|\partial A_K(U_T^\circ)\| + 16\epsilon_{\mathrm{d}} + 44C_R\epsilon_{\mathrm{d}}^{1/2} \le C'(\Lambda_0)\rho_1$ and $0.8\,e^{-11N_{\mathrm{d}}} \le \varepsilon_c$. At $r = r_*$ the ratio $\theta_s = n_s^2/p_s^2$ is defined on $[0,T]$, as $p_s > 0$ there, and the cone inequalities give $\theta' \le -2(\lambda_1 - \lambda_2)(\theta - 1) + 2\varepsilon_c\theta(1 + \theta^{1/2})$; as $\lambda_1 - \lambda_2 > 5.8$, the right side is at most $-11(\theta - 1) + 10\varepsilon_c$ where $1 \le \theta \le 2$, and $\theta_0 \le 0.75^{-2} < 1.8$, so that θ stays below 1.8 and $\theta_T - 1 \le 0.8\,e^{-11T} + \varepsilon_c \le 2\varepsilon_c$, by $T \ge N_{\mathrm{d}}$ and the last condition on N_{d}. Let $\hat{e}_\alpha = e_{1\alpha}(\tilde{V}_T)/\|\chi_R e_1\|$, three orthonormal vectors, and $\eta_T = \sum_\alpha c_\alpha\hat{e}_\alpha + \eta_T^\perp$ with $\eta_T^\perp$ orthogonal to them; the bound on $u_{i\alpha}(V') - u_{i\alpha}(V) - \langle e_{i\alpha}(V), \eta\rangle$ above, with $\|\chi_R e_1\| \le 1$, gives $|c| \ge p_T - \varepsilon_c n_T$, whence $\|\eta_T^\perp\|^2 = n_T^2 - |c|^2 \le (\theta_T - 1)p_T^2 + 2\varepsilon_c p_T n_T \le 5\varepsilon_c p_T^2$, and $p_T = \rho_2$ at $r = r_*$. Thus the displacement η_T at $r = r_*$ lies along the strongest mode up to the part $\eta_T^\perp$, of norm $\le (5\varepsilon_c)^{1/2}\rho_2$, so the member of the first order of the Taylor expansion of $\tilde{A}$ at the end of the flow, which sees only that part, is at most $1350\,C'(\Lambda_0)\rho_2\,(5\varepsilon_c)^{1/2}\rho_2 \le 0.76\,\rho_2^2$; the Hessian of $\tilde{A}$ on that mode is close to $-\lambda_1$, and its quadratic form at every X is at most $40\|X\|^2$, by the bounds of Lem. 16.2(a) at each plaquette, so that $\eta_T^\perp$ moves the member of the second order by at most $0.2\,\rho_2^2$, and this member is at most $-3\rho_2^2$: the fall is at least $2\rho_2^2$. The value $\mathsf{v}(0)$ is 8 plus a number of modulus at most $\sum_p a_p$ of the field with the reversed sign at b_0, and this sum is ≤ 1 by the bound used for (c), as the two flows are close and $A_K(U_T^\circ) \le 24\Lambda_0^2 N^3 T^{-2} \le 24\Lambda_0^2 c_{\mathrm{d}}$, the action being non-increasing and $\varpi(U_s^\circ) \le 4\Lambda_0/T$ at some $s \in [\frac{T}{4}, \frac{T}{2}]$. For a general f the same steps hold, with 8000 in the place of 4000 in ε_c: the plaquettes within a fixed distance of b_0 change $f(x(p))$ by $O(a_K)$ from $f(x_*)$, and the others see only the part of η_T outside the strongest mode, which is small; this gives the fall $f(x_*)\rho_2^2$. □

16.6. **Regularisation cutoffs and lower energy bounds.**

Lemma 16.5 (Condenser cutoffs)**.** For all K, m, $\beta > 0$, let $\eta \in D(\mathcal{E}_K)$, $0 \le \eta \le 1$, $\mathsf{A} = \{\eta = 0\}$, $\mathsf{B} = \{\eta = 1\}$, $D_0 = \{\phi \in D(\mathcal{E}_K) : \phi = 0 \text{ a.e. on } \mathsf{A}\cup\mathsf{B}\}$, $\mathcal{E}_{K,1}(\phi,\psi) = \mathcal{E}_K(\phi,\psi) + \int \phi\psi\,\mathrm{d}\mu^K$. There is exactly one $w \in \eta + D_0$ with $\mathcal{E}_{K,1}(w,\phi) = 0$ for $\phi \in D_0$; with $\varkappa = \mathcal{E}_{K,1}(w,w)$: $0 \le w \le 1$, $w = 0$ on A, $w = 1$ on B, $\Gamma_K(w,\cdot) = 0$ a.e. on $\mathsf{A}\cup\mathsf{B}$, $\varkappa \le \mathcal{E}_{K,1}(\eta,\eta)$, $\int w\,\mathrm{d}\mu^K \le \varkappa$, and $|\mathcal{E}_K(w,\phi)| \le \operatorname{osc}(\phi)\,\varkappa$ for every $\phi \in D(\mathcal{E}_K) \cap L^\infty$; w is invariant if η is.

With $q(u) = 3u^2 - 2u^3$: $q(w) \in D(\mathcal{E}_K)$, $q(w) = 0$ on A, $q(w) = 1$ on B, $\mathcal{E}_K(q(w)) \le \frac{9}{4}\varkappa$, and $\mathcal{E}_K(q(w),\phi) = -\int \mathsf{I}\,\phi\,\mathrm{d}\mu^K$ for $\phi \in D(\mathcal{E}_K) \cap L^\infty$, with $\mathsf{I} = 6w^2(1-w) + 6(1-2w)\Gamma_K(w)$, $\|\mathsf{I}\|_{L^1(\mu^K)} \le \frac{15}{2}\varkappa$. ■

Proof sketch. w is the element of least norm of the closed convex set $\{v \ge 1 \text{ on } \mathsf{B},\ v \le 0 \text{ on } \mathsf{A}\}$; truncation does not raise the norm, so $0 \le w \le 1$. The variational inequality $\mathcal{E}_{K,1}(w,\psi) \ge 0$ for $\psi \ge 0$ on B, $\psi \le 0$ on A, applied to $\psi = \phi\eta$, $(M-\phi)\eta$, $-\phi(1-\eta)$, $-(M-\phi)(1-\eta)$ for $0 \le \phi \le M = \operatorname{osc}\phi$, gives

$$M\Big(\int w\,\mathrm{d}\mu^K - \varkappa\Big) \le \mathcal{E}_{K,1}(w,\phi) \le M\varkappa, \qquad 0 \le \int w\phi\,\mathrm{d}\mu^K \le M\int w\,\mathrm{d}\mu^K,$$

so $|\mathcal{E}_K(w,\phi)| = |\mathcal{E}_{K,1}(w,\phi) - \int w\phi \,\mathrm{d}\mu^K| \le M\varkappa$. Gauge transformations preserve μ^K, Γ_K and the constraint set, so uniqueness gives invariance. For $q(w)$, apply the equation of w to $6w(1-w)\phi \in D_0$: $\Gamma_K(w, 6w(1-w)\phi) = 6w(1-w)\Gamma_K(w,\phi) + 6(1-2w)\phi\,\Gamma_K(w)$; and $6w^2(1-w) \le \frac{3}{2}w$, $\int \Gamma_K(w) \le \varkappa$. □

Proposition 16.6 (Gauge-invariant regularisation cutoffs)**.** For all K, m: let $0 < \bar{s} \le 1$ with $64\bar{s}^{7/16} \le \ell$ and $4a_K^2 \le \bar{s}$. For a site y and an integer $k \ge 1$ let $Q(y,k)$ be the cube of k^3 sites with lowest corner y, $\mathsf{e}_k(V;y) = a_K^{-1}\sum \varpi_p(V)^2$ over the plaquettes with base point in $Q(y,k)$, $\mathsf{e}_k^* = \max_y \mathsf{e}_k(\cdot;y)$, $\mathfrak{s}_k(V) = a_K^3 \sum_y \mathsf{e}_k(V;y)^8$, $k(r) = \lceil r^{7/16}/a_K \rceil$, and

$$\mathscr{N}(U) = \int_{a_K^2}^{\bar{s}} r\,\mathfrak{s}_{16k(r)}(U_r)\,\mathrm{d}r,$$

a smooth invariant function defined without choosing a gauge.

(a) If $\varpi(V) \le 1$, $1 \le k < N$ and X is horizontal at V, then $\nabla^2 A_K(V)[X,X] \ge \frac{1}{4}\sum_p |Z_p(X)|^2 - \hat{\lambda}_k(V)\|X\|^2$ with $a_K^{-2}\hat{\lambda}_k(V) = 16[16(144C)^3 \mathsf{e}_{k+1}^*(V)^2 + (144(a_K k)^2)^{-1}]$, C absolute: on horizontal vectors the Hessian is bounded below by the square of the energy in windows of side $a_K k$, with no supremum of the curvature and no gauge.

(b) Let $R > 0$ and K so large that $e^{64}(C_1 R)^{1/8} a_K^{11/64} \le \frac{1}{144}$, C_1 absolute. If $\mathscr{N}(U) \le R$, then $\varpi(U_r) \le \frac{1}{12}$ for $0 \le r \le \bar{s}$; $\Gamma_K(G\circ\Phi_\sigma)(U) \le e^{2\bar{\Lambda}(R)}\Gamma_K(G)(U_\sigma)$ for invariant $G \in C^2(\Omega_K)$ and $\sigma \le \bar{s}$, $\bar{\Lambda}(R) = 1 + c_\Lambda \bar{s}^{11/64} R^{1/4} + \bar{s}^{1/8}$, c_Λ absolute; for $4a_K^2 \le \sigma \le \bar{s}$, $A^a(U_{\sigma/2}) \le \mathsf{A}_\sigma(R) = \frac{1}{2}\cdot 16^{-3}(\frac{\sigma}{4})^{-25/16}\ell^{21/8}R^{1/8}$ and $(J,J)_a(U_\sigma) \le \frac{2}{\sigma}e^{2\bar{\Lambda}(R)}\mathsf{A}_\sigma(R)$; so $\Gamma_K(H_K^\sigma)$ and $|\partial_\sigma H_K^\sigma|$ are bounded on $\{\mathscr{N} \le R\}$ by numbers free of K for $4a_K^2 \le \sigma \le \bar{s}$, and for all $\sigma \ge 4a_K^2$ if besides $2a_K \mathsf{A}_{\bar{s}}(R) \le 1$; and $\Gamma_K(\mathscr{N}) \le 512\, e^{2\bar{\Lambda}(R)}\mathscr{N}'$, with $\mathscr{N}'$ the same integral with $r^2(16\rho(r))^3 a_K^3 \sum_y \mathsf{e}_{16k(r)}(\cdot;y)^{15}$, $\rho(r) = a_K k(r)$, in the place of $r\,\mathfrak{s}_{16k(r)}$.

(c) Assume (E), so that (Adm) of Prop. 16.1 holds, and let $\epsilon > 0$. There are $\bar{s}$, R, $\bar{K}$, and for every $q < \infty$ a number C_q, none depending on K, such that for $K \ge \bar{K}$ the function $\chi_K = \chi_0(\mathscr{N}/R)$, $\chi_0 \in C^\infty(\mathbb{R})$, $\chi_0 = 1$ on $(-\infty, \frac{1}{2}]$, $\chi_0 = 0$ on $[1,\infty)$, $0 \le -\chi_0' \le 3$, is smooth and invariant, vanishes off $\{\mathscr{N} \le R\}$, is 1 on the set $\mathsf{O}_K(\bar{C},\bar{s})$ of Prop. 16.1(a) formed at $\kappa = \frac{1}{64}$ and this ϵ, so that $\mu^K(\chi_K < 1) \le \epsilon$, and has $\int \Gamma_K(\chi_K)^q d\mu^K \le \mathsf{C}_q$.

(d) In the setting of (c), for a flowed observable $H = H_K^\sigma$ at a fixed $\sigma > 0$, $K \ge \bar{K}(\epsilon,\sigma)$, $\mathsf{h} = \sup_{\{\mathscr{N}\le R\}}\Gamma_K(H)$, $\mathsf{h}' = \sup_{\{\mathscr{N}\le R\}}|\partial_\sigma H|$: the vector field $\zeta_K = \chi_K \partial H$ has $a_K^{-1}\int |\zeta_K|^2\,\mathrm{d}\mu^K \le \mathsf{h}$ and the divergence $\chi_K\Delta_K H + \mathsf{b}\,\chi_K\partial_\sigma H + \Gamma_K(\chi_K, H)$; for $F \in C^1(\Omega_K)$, $|\int F\chi_K\Delta_K H\,\mathrm{d}\mu^K| \le \mathsf{h}^{1/2}\mathcal{E}_K(F)^{1/2} + [(\mathsf{hC}_1)^{1/2} + \mathsf{bh}']\|F\|_{L^2(\mu^K)}$; $|\int \chi_K(\mathsf{b}\partial_\sigma + \Delta_K)H\,\mathrm{d}\mu^K| \le (\mathsf{hC}_1)^{1/2}\mu^K(\frac{1}{2}R < \mathscr{N} < R)^{1/2}$; every sequence of cutoffs has a subsequence along which $\chi_K H_K^\sigma$ converges strongly to an element of $D(\mathcal{E}_\star)$; and $\|\chi_K H - g_n^K \circ \pi_{K,n}\|^2_{L^2(\mu^K)} \le C\,a_n^2/\mathsf{b} + 4\sup|\varphi|^2\mathsf{R}_n$ uniformly in K, g_n^K the top of the tower of $\chi_K H$, $n \ge n''$.

■

Proof sketch. (a) As $\varpi(V) \le 1$, $c_p \ge \frac{1}{2}$, and Lem. 16.2(a) gives $\nabla^2 A_K(V)[X,X] \ge \frac{1}{2}\sum_p |Z_p(X)|^2 - 3\sum_p |\mathsf{s}_p|\,|X|_p^2$, $|X|_p^2 = \sum_{b\in p}|X(b)|^2$. With $f(x) = (\sum_\mu |X_\mu(x)|^2)^{1/2}$ and $t(x)$ the largest $|\mathsf{s}_p|$ over the plaquettes with the corner x, the remainder is at most $12\mathcal{T}$, $\mathcal{T} = \sum_x t(x)f(x)^2$. For horizontal X the lattice Weitzenböck identity of Lem. 16.2(h), its remainder bounded plaquette by plaquette, gives $\|\nabla X\|^2 \le 2\sum_p |Z_p(X)|^2 + 36\mathcal{T}$, and Kato's inequality $\|\nabla f\| \le \|\nabla X\|$. In the windows $Q(y,k)$, each site lying in k^3 of them: Hölder's inequality and the inequality of Sobolev on a discrete cube, $\|f\|^2_{\ell^6(Q)} \le C_{\mathrm{Sob}}(\|\nabla f\|^2_{2,Q} + k^{-2}\|f\|^2_{\ell^2(Q)})$, $C_{\mathrm{Sob}} = 864$, give $\mathcal{T} \le E^{1/2}\|f\|^{1/2}\mathsf{B}^{3/4}$, $E = 4a_K\mathsf{e}_{k+1}^*(V)$, $\mathsf{B} = C_{\mathrm{Sob}}(\|\nabla f\|^2 + k^{-2}\|f\|^2) \le C_{\mathrm{Sob}}(2\sum_p |Z_p(X)|^2 + 36\mathcal{T} + k^{-2}\|X\|^2)$; the ℓ^2 member of the inequality of Sobolev is the source of the term $(144(a_K k)^2)^{-1}$.

Young's inequality at $(144C_{\mathrm{Sob}})^{-1}$ gives $\frac34\mathcal{T}\le\frac{1}{72}\sum_p|Z_p(X)|^2+[(144C_{\mathrm{Sob}})^3E^2+(144k^2)^{-1}]\|X\|^2$, so $12\mathcal{T}\le\frac29\sum_p|Z_p(X)|^2+\hat\lambda_k(V)\|X\|^2$, and $\frac12-\frac29>\frac14$; the absolute C of the statement is C_{Sob}. (b) Let $\mathscr{N}(U)\le R$.

The window energies. At every field a local energy inequality holds, from the derivative of a cut-off window energy along the flow: $\mathsf{e}_k(U_r;y)\le\mathsf{e}_{k+2h}(U_{r'};y-h)+64(a_Kh)^{-2}\int_{r'}^r\mathsf{e}_{k+2h}(U_\tau;y-h)\,\mathrm{d}\tau$ for $r'<r$, $y-h=y-a_Kh(1,1,1)$. At $k=16k(r)$, $h=8k(r)$, the mean over $r'\in[\frac r2,r]$, the bound $\mathsf{e}_k^*(V)^8\le8^9(a_Kk)^{-3}\mathfrak{s}_k(V)$ (a window is covered by eight windows of the same side whose corners run over a sub-cube of side $\frac k2$) and Jensen's inequality give $\mathsf{e}^*_{16k(r)}(U_r)^8\le C_1r^{-53/16}R$ for $2a_K^2\le r\le\bar s$, directly from $\mathscr{N}(U)\le R$. So $\varpi(U_r)^2\le a_K\mathsf{e}^*_{16k(r)}(U_r)\le(C_1R)^{1/8}a_K^{11/64}$ there, and for $r<2a_K^2$, as $\varpi(U_{r'})^2\le e^{32|r-r'|/a_K^2}\varpi(U_r)^2$, with the factor e^{64}: by the condition on K, $\varpi(U_r)\le\frac{1}{12}$ on $[0,\bar s]$.

The floor in two regimes. For horizontal X at U_r, $a_K^{-2}\nabla^2A_K(U_r)[X,X]\ge\frac14a_K^{-2}\sum_p|Z_p(X)|^2-\lambda(r)\|X\|^2$ with $\lambda(r)=12a_K^{-2}\varpi(U_r)$ for $r<a_K^2$ (Lem. 16.2(a), with $c_p\ge\frac12$; its integral over $[0,a_K^2]$ is ≤1) and $\lambda(r)=a_K^{-2}\hat\lambda_{k(r)}(U_r)$ for $a_K^2\le r\le\bar s$ (by (a)); Hölder's inequality with the exponents $4,\frac43$ in r bounds the integral of the window member by $c_\Lambda\bar s^{11/64}R^{1/4}$, and the member $(144\rho(r)^2)^{-1}$, $\rho(r)=a_Kk(r)\ge r^{7/16}$, contributes $\frac89\bar s^{1/8}\le\bar s^{1/8}$: $\int_0^{\bar s}\lambda(r)\,\mathrm{d}r\le\bar\Lambda(R)$.

The bounds. Lem. 16.2(b)–(d), read on horizontal vectors with λ in the place of $12a_K^{-2}\varpi$, give the bounds of $\Gamma_K(G\circ\Phi_\sigma)$, of $(J,J)_a$, of $\Gamma_K(H_K^\sigma)$ and of $\partial_\sigma H_K^\sigma$; $\mathsf{A}_\sigma(R)$ comes from $A^a=\frac12(16\rho)^{-3}a_K^3\sum_y\mathsf{e}_{16k}$, Hölder's inequality in y, the monotonicity of A^a along the flow and the mean over $[\frac\sigma4,\frac\sigma2]$. For continuation at later fixed times use the action-only horizontal Hessian bound from Prop. 16.6(a), as in Step 6 below: $a_KE\le\frac12$ gives $\max_p a_p\le\frac12$ along the later flow and a lower Hessian bound by $-c(E^2+\ell^{-2})$. This is distinct from the smaller small-action threshold in Lem. 16.2(f).

The square field of $\mathscr{N}$. With $M_r(V)=\int_r^{\bar s}w(\tau)\,\mathfrak{s}_{16k(\tau)}(\Phi_{\tau-r}V)\,\mathrm{d}\tau$, $w(\tau)=\tau\mathbf{1}_{\tau\ge a_K^2}$, and $\Xi_r=\partial M_r(U_r)$, horizontal, $\Xi_0=\partial\mathscr{N}(U)$, $\Xi_{\bar s}=0$: $\frac{d}{dr}\|\Xi_r\|^2=2a_K^{-2}\nabla^2A_K(U_r)[\Xi_r,\Xi_r]-2w(r)\langle\Xi_r,\partial\mathfrak{s}_{16k(r)}(U_r)\rangle$. The gradient of $\mathfrak{s}_{\hat k}$ is a covariant divergence: $\langle\Xi,\partial\mathfrak{s}_{\hat k}\rangle=\sum_p\mathsf{f}(p)\langle\mathfrak{s}_p,Z_p(\Xi)\rangle\le\mathsf{G}_r^{1/2}(\sum_p|Z_p(\Xi)|^2)^{1/2}$, $\mathsf{f}(p)=16a_K^2\sum_{y:\,\mathrm{base}(p)\in Q(y,\hat k)}\mathsf{e}_{\hat k}(\cdot;y)^7$, $\mathsf{G}_r\le256a_K^{-1}(16\rho)^3a_K^3\sum_y\mathsf{e}_{\hat k}(\cdot;y)^{15}$; it is paid by the member $\frac14\sum_p|Z_p|^2$ of the floor, $2w\mathsf{G}_r^{1/2}(\sum_p|Z_p(\Xi)|^2)^{1/2}\le\frac12a_K^{-2}\sum_p|Z_p(\Xi)|^2+2a_K^2w^2\mathsf{G}_r$.

Gronwall's inequality from $\bar s$ back to 0 gives

$$\|\partial\mathscr{N}(U)\|^2\le2a_K^2e^{2\bar\Lambda(R)}\int w^2\mathsf{G}_r\,\mathrm{d}r\le512\,a_Ke^{2\bar\Lambda(R)}\mathscr{N}'(U),$$

that is $\Gamma_K(\mathscr{N})\le512\,e^{2\bar\Lambda(R)}\mathscr{N}'$. (c) $\{\mathscr{N}\le R\}$ contains O_K, on which the window energies are bounded by the sizes of the flowed field; at each dyadic scale r_j the square field of the functional is bounded on the good set of that scale by a positive power of r_j, and off it crudely by r_j^{-1} (by a_K^{-12} at the last scale) against a mass $\le t_Mr_j^M$ for every M, from the moments of Prop. 16.1(a) and the super-exponential decay of R_d; the sum over the dyadic scales converges in every L^q.

(d) The divergence is the product rule and (C1) of Thm. AV; the bounds are Cauchy–Schwarz; the convergence is (A2) applied to χ_KH, whose energy is at most $2\mathsf{h}+2\sup|\varphi|^2\mathsf{C}_1$; the last bound is the key estimate of Step 1 applied to χ_KH. □

Lemma 16.7 (Lower bounds for the flowed action)**.** Assume (E), so that (Adm) of Prop. 16.1 holds; let $\mathbb{P}_K$ be the normalised chain measure from the least top depth d_0 with $\mathsf{R}_{d_0}\le\frac12$ such that every top depth $\ge d_0$ is admissible, $\mathbb{E}_K$ its mean, $\mathbf{B}_K$ and $\mathsf{a}_{\mathrm{top}}$ as in Prop. 16.1, and $\|\cdot\|_2$ the norm of $L^2(T_\ell)$. For a real function f and a $\mathfrak{g}$-valued function h of the bare bonds, $\langle f,h\rangle=a_K^3\sum_b f(b)h(b)\in\mathfrak{g}$, $\|f\|_1=a_K^3\sum_b|f(b)|$, $\|f\|_2^2=a_K^3\sum_b|f(b)|^2$. For a function Y of the bonds, $\mathrm{d}Y(p)=Y(b_1)+Y(b_2)-Y(b_3)-Y(b_4)$ is the plain sum around the plaquette p, with the signs $(+,+,-,-)$ of Lem. 16.2 and no factor of length; for a 1-form A per unit length, $\mathrm{d}_aA=a_K^{-2}\mathrm{d}(a_KA)$

is its lattice curl, $(\mathrm{d}_a A)(p) = \partial^+_\mu A_\nu(x) - \partial^+_\nu A_\mu(x)$ for $p = (x;\mu\nu)$, $\|\mathrm{d}_a A\|_2^2 = a_K^3 \sum_p |\mathrm{d}_a A(p)|^2$, and $\mathrm{d}_a P_s = P_s \mathrm{d}_a$.

(a) There are $k_4 = 100\, c_{\mathsf{J}} \gamma_{\mathrm{r}}^{-1/2}$, c_{J} the constant of the layer maps of Prop. 16.1, $p_* = 9/(64\Lambda_+^2 k_4^4)$, $k_\Sigma = 15/(16\Lambda_+)$ and $\theta_0 > 0$, depending only on the constants of the construction, and d_L, free of K, with $\mathbb{P}_K(\|\mathrm{d}_a P_s \mathbf{B}_K\|_2^2 \geq k_\Sigma \mathsf{b}^{-1} \ell^3 a_d^{-3}) \geq p_*$ for $d \geq d_L$, $K > d$ with $a_K \mathsf{a}_{\mathrm{top}} \leq \frac{1}{8}$, and $0 \leq s \leq \theta_0 a_d^2$. Let $k = K - d$ and $\mathbf{Q}_k^{\mathrm{lin}}$ the k-fold linear average at $\mathbf{1}$ from the bare bonds to the bonds of depth d, the composite of k maps YM3_I:(42).

For a plaquette P of depth d none of whose bonds is a pivot bond,

$$X_P = \mathrm{d}\, \mathbf{Q}_k^{\mathrm{lin}}(a_K P_s \mathbf{B}_K)(P) \in \mathfrak{g},$$

the plain sum around P, with no factor of length, is $\langle P_s f_P, \mathbf{B}_K \rangle$ with the real bond function $f_P = a_K^{-2} q_P$, q_P the kernel of $\mathsf{a} \mapsto \mathrm{d}\, \mathbf{Q}_k^{\mathrm{lin}} \mathsf{a}(P)$, $\|f_P\|_1 \leq 4a_d$, $\|f_P\|_2^2 \leq 4a_d^{-1}$; and X_P has $\mathbb{E}_K |X_P|^2 \geq \frac{3}{4\Lambda_+} x_k^{-1}$ and $\mathbb{E}_K |X_P|^4 \leq k_4^4 x_k^{-2}$; at least $3(1 - 4L^{-3})(\ell L^d)^3$ of the plaquettes of depth d have no pivot bond. Conditionally: let d_0 be any admissible top depth with $\mathsf{R}_{d_0} \leq \frac{1}{2}$, $\mathbb{P}_K$ the normalised chain measure from d_0, and $\mathbb{E}_K^0$ its conditional mean given the block field at depth d_0.

There are $d_L' > d_0$ and numbers $\tau_d \to 0$, which depend on d_0 but not on K, such that for $d \geq d_L'$, $K > d$ with $a_K \mathsf{a}_{\mathrm{top}} \leq \frac{1}{8}$ and $0 \leq s \leq \theta_0 a_d^2$ there is an event $\mathcal{T}_d$ of the block field at depth d_0 with $\mathbb{P}_K(\mathcal{T}_d) \geq 1 - \tau_d$ on which, N_P being the number of the plaquettes of depth d without pivot bond and the sums running over them, $N_P^{-1} \sum_P \mathbb{E}_K^0 |X_P|^2 \geq \frac{3}{4\Lambda_+} x_k^{-1}$, $N_P^{-1} \sum_P \mathbb{E}_K^0 |X_P|^4 \leq k_4^4 x_k^{-2}$, and

$$\mathbb{P}_K(\|\mathrm{d}_a P_s \mathbf{B}_K\|_2^2 \geq k_\Sigma \mathsf{b}^{-1} \ell^3 a_d^{-3} \,|\, \text{block field at depth } d_0) \geq p_*.$$

(b) Let $R \geq \max\{1, \|\mathbf{B}\|_{\kappa;a}\}$, $\bar{C} = (2 + c_3)R$, $a_K \leq a_1(R)$, $0 < \sigma \leq \min\{s_0(R), s_{\mathrm{lin}}(R)\}$, $a_K \bar{C} \sigma^{-1/4-\kappa/2} \leq \frac{1}{8}$, $P_\sigma \mathbf{B} \neq 0$, $U = e^{i a_K \mathbf{B}}$, where c_3, $s_0(R)$, $a_0(R)$ are those of Prop. 16.1(a) and M_Θ that of its head, $s_{\mathrm{lin}}(R) = (2c_6 \bar{C}^2)^{-4/(1-2\kappa)}$, $a_1(R) = \min\{a_0(R), (8e^{12}R)^{-2/(1-2\kappa)}, (8(4\mathsf{M}_\Theta)^{1/6} c_3 R)^{-1/(1-2\kappa)}\}$, a positive number that reads R and κ alone, and c_6, c', c_g are absolute. For $a_K \leq a_1(R)$ the gauge-fixed flow A_s from $\mathbf{B}$ has $a_K \|A_s\|_\infty \leq \frac{1}{2}$ on the closed interval $[0, s_0(R)]$, so that the chart and the linearised equation serve from $s = 0$ on: $a_K \|\mathbf{B}\|_\infty \leq e^{12} R a_K^{\frac{1}{2}-\kappa} \leq \frac{1}{8}$, as $\|f\|_\infty \leq e^{12} \|P_{a_K^2} f\|_\infty$; on $[0, a_K^2/(4\mathsf{M}_\Theta)]$ the chart variable $a_K A_s$ moves at most at the speed $a_K^{-2} \mathsf{M}_\Theta$, so that $a_K \|A_s\|_\infty \leq \frac{1}{8} + \frac{1}{4}$ there; later the first bound of Prop. 16.1(a) gives

$$\begin{aligned} a_K \|A_s - P_s \mathbf{B}\|_\infty &\leq (4\mathsf{M}_\Theta)^\kappa c_3 R a_K^{1-2\kappa} \\ &\leq (4\mathsf{M}_\Theta)^{1/6} c_3 R a_K^{1-2\kappa} \leq \frac{1}{8}. \end{aligned}$$

, and with $\|P_s \mathbf{B}\|_\infty \leq \|\mathbf{B}\|_\infty$, $a_K \|A_s\|_\infty \leq \frac{1}{4}$.

Then

$$\Gamma_K(A^a \circ \Phi_\sigma)(U)^{1/2} \geq \frac{\|\mathrm{d}_a P_{3\sigma/2} \mathbf{B}\|_2^2}{\|P_\sigma \mathbf{B}\|_2} - c' \ell^{3/2} \bar{C}^3 \sigma^{-1-\kappa}, \qquad \|P_\sigma \mathbf{B}\|_2 \leq \sqrt{3} \ell^{3/2} R \sigma^{-\frac{1}{4}-\frac{\kappa}{2}},$$

$\Gamma_K(A^a \circ \Phi_\sigma)(U)^{1/2} \leq \frac{8}{7}[\sqrt{2} c_g \sigma^{-1/2} \|\mathrm{d}_a P_\sigma \mathbf{B}\|_2 + c' \ell^{3/2} \bar{C}^3 \sigma^{-1-\kappa}]$, and $|(2A^a(U_\sigma))^{1/2} - \|\mathrm{d}_a P_\sigma \mathbf{B}\|_2| \leq \sqrt{3} \ell^{3/2} (2\bar{C} + 8\bar{C}^2) \sigma^{-\frac{1}{2}-\kappa}$.

(c) Let $\mathfrak{b} > 0$, $p = e^{-\Psi} dB$ a probability measure on the box $\prod_b \{|B(b)| < \mathfrak{b}\}$ with $\Psi \in C^2$ and $0 \leq D^2\Psi \leq \Lambda_+$, ∂_e the derivative in one coordinate of one bond b_1, and $\eta(B) = \phi(|B(b_1)|/\mathfrak{b})$ with $\phi = 1$ on $[0, \frac{1}{2}]$ and $\phi(t) = \cos^2(\pi(t - \frac{1}{2}))$ on $[\frac{1}{2}, 1]$. Then $\mathrm{Var}_p(u) \geq (\int \eta\, \partial_e u\, \mathrm{d}p)^2 / (\Lambda_+ + \pi^2 \mathfrak{b}^{-2})$ for every bounded Lipschitz u.

■

Proof sketch. (c) Integrate by parts against η, which vanishes at the faces in the direction e, so there is no boundary term: $\int \eta\, \partial_e u \,\mathrm{d}p = \int (u-\bar u)(\eta \partial_e \Psi - \partial_e \eta)\,\mathrm{d}p$, and bound the second factor in $L^2(p)$ through $D^2\Psi \le \Lambda_+$ (one more integration by parts) and $|\partial_e \eta| \le \pi/\mathfrak{b}$. (a)

Signs. They are those of the head of the lemma: the pairing $\langle f, \mathbf{h}\rangle = a_K^3 \sum_b f(b)\mathbf{h}(b) \in \mathfrak{g}$ of a real bond function f with a $\mathfrak{g}$-valued $\mathbf{h}$ on $\Lambda_{K,\ell}$; for a bond function Y without a factor of length, its plain sum $\mathrm{d}Y(p) = Y(b_1)+Y(b_2)-Y(b_3)-Y(b_4)$ around the plaquette p, with the signs of Lem. 16.2, the bonds taken as in $\partial U(p)$; and for a 1-form A per unit length, its lattice curl $\mathrm{d}_a A = a_K^{-2}\mathrm{d}(a_K A)$, with $\|\mathrm{d}_a A\|_2^2 = a_K^3 \sum_p |\mathrm{d}_a A(p)|^2$ and $\mathrm{d}_a P_s = P_s \mathrm{d}_a$. Besides, $\|f\|_r^r = a_K^3 \sum_b |f(b)|^r$ for $1 \le r < \infty$.

The roman d and d_a are these sums around a plaquette, and the italic d is a depth. For a plaquette P of depth d, $k = K-d$, X_P and f_P are those of (a): $X_P = \mathrm{d}\,\mathbf{Q}_k^{\mathrm{lin}}(a_K P_s \mathbf{B}_K)(P)$ is the sum around P of the k-fold linear average $\mathbf{Q}_k^{\mathrm{lin}}$ of equation YM3_I:(42), YM3_I:(42), applied to the bond function $a_K P_s \mathbf{B}_K$, which carries no factor of length, and f_P is the real bond function with $\mathrm{d}\,\mathbf{Q}_k^{\mathrm{lin}}(a_K A)(P) = \langle f_P, A\rangle$ for every 1-form A, so that $X_P = \langle P_s f_P, \mathbf{B}_K\rangle$. The numbers k_4, p_*, k_Σ of (a), and the number θ_0 with k_1, are

$$\theta_0 = \frac{3}{16\Lambda_+ k_1^2}, \qquad k_4 = 100\, c_{\mathsf{J}} \gamma_{\mathrm{r}}^{-1/2}, \qquad p_* = \frac{9}{64\Lambda_+^2 k_4^4}, \qquad k_\Sigma = \frac{15}{16\Lambda_+}, \qquad k_1 = 48\, c_\nabla \Big(\frac{24\, c_{\mathsf{J}}^2}{\gamma_{\mathrm{r}}(1-L^{-1})}\Big)^{1/2},$$

with the absolute constant c_∇ of the bound $\|\partial_\nu^{\pm} P_s f\|_1 \le c_\nabla s^{-1/2}\|f\|_1$ of the heat kernel and the constant c_{J} of the layer maps of Prop. 16.1, which enters (L3) below; they depend only on the constants of the construction.

The linear average. In the sum of YM3_I:(42) around a coarse plaquette the contour averages cancel, so one step maps the curl of a fine bond function to L^{-3} times the sum, over the sites x of the block, of its curls at the L^2 fine plaquettes of the square of side L with the corner x: a matrix with the row sums L^2 and the column sums L^{-1}. By the Cauchy–Schwarz inequality, $\sum_P |\mathrm{d}\,\mathbf{Q}_k^{\mathrm{lin}}\mathsf{a}(P)|^2 \le L^k \sum_p |\mathrm{d}\mathsf{a}(p)|^2$, and at $\mathsf{a} = a_K A$, as $\sum_p |\mathrm{d}(a_K A)(p)|^2 = a_K \|\mathrm{d}_a A\|_2^2$ and $L^k a_K = a_d$,

$$\|\mathrm{d}_a A\|_2^2 \;\ge\; a_d^{-1} \sum_P \big|\mathrm{d}\,\mathbf{Q}_k^{\mathrm{lin}}(a_K A)(P)\big|^2 : \tag{16.i}$$

the curls of the linear averages bound $\|\mathrm{d}_a A\|_2^2$ from below, with the constant one between the L^2 norms at the spacings a_K and a_d. Iterated, $\mathrm{d}\,\mathbf{Q}_k^{\mathrm{lin}}\mathsf{a}(P)$ is the mean, over the L^{3k} sites u of the block, of the sum of a around the square of side L^k with the corner u; so $a_K^2 f_P$ is a signed sum of four tents along the bonds of P, $\|f_P\|_1 \le 4a_d$, $\|f_P\|_2^2 \le 4a_d^{-1}$ and $\sum_\nu \|\partial_\nu^+ f_P\|_1 \le 24$; these norms do not grow under P_s, $\|P_s f_P\|_{12/11} \le \|P_s f_P\|_1^{5/6}\|P_s f_P\|_2^{1/6} \le 4a_d^{3/4}$, and $(1-P_s)f = -\int_0^s \sum_\nu \partial_\nu^- P_r \partial_\nu^+ f\,\mathrm{d}r$ gives $\|(1-P_s)f_P\|_1 \le 48 c_\nabla \sqrt{s}$.

The plaquette of a block field, by (c). Let $W \in \mathsf{Cut}_{k+1}$ and let P be a plaquette of level k none of whose bonds is a pivot bond, b_1 its first bond. Its four bonds are coordinates of the chart, so $\partial V_W(B)(P) = e^{i x_k^{-1/2} B(b_1)} C$ with C free of $B(b_1)$. Write $\partial V_W(B)(P) = c_P \mathbf{1} + i\mathsf{s}_P$, let (t_α) be an orthonormal basis of $\mathfrak{g}$ and $\mathsf{E} = \int_0^1 \mathrm{Ad}(e^{iu x_k^{-1/2} B(b_1)})\,\mathrm{d}u$; the derivative of $\langle \mathsf{s}_P, t_\alpha\rangle$ in the coordinate $B(b_1)^\alpha$ is $x_k^{-1/2}\,\frac12 \mathrm{tr}(t_\alpha(\mathsf{E}t_\alpha)\,\partial V_W(B)(P)) \ge x_k^{-1/2}(1-\delta(x_k)-\theta_d)$, as $|\mathsf{E}t_\alpha - t_\alpha| \le x_k^{-1/2}|B(b_1)| < \delta(x_k)$ on the box and $|\partial V_W(B)(P) - \mathbf{1}| < \theta_d$ on the cut: in the variable B of p_W, in which $D^2\Psi_W \le \Lambda_+$ holds (Lem. 15.3(iv)), the derivative carries the chart factor $x_k^{-1/2}$.

(c) at $p = p_W$, $\mathfrak{b} = \mathfrak{b}(x_k)$ and $u = \langle \mathsf{s}_P, t_\alpha\rangle$, with $\int \eta\,\mathrm{d}p_W \ge p_W(|B(b_1)| \le \frac12 \mathfrak{b}(x_k))$, summed over the three α and integrated over $W = V_{k+1}$ with Jensen's inequality, and $\mathbb{P}_K(|B_k(b_1)| > \frac12 \mathfrak{b}(x_k)) \le 4m_2 \mathfrak{b}(x_k)^{-2}$, m_2 the bound, free of k, of the second moments of one fluctuation field in (L5) of the proof of Prop. 16.1(a), recalled below (its Gaussian tails, Lem. 15.3(iv)), give, as

$|\partial V_W(B)(P)-\mathbf{1}|\ge|\mathsf{s}_P|$,

$$\mathbb{E}_K|\partial V_k(P)-\mathbf{1}|^2\ \ge\ \frac{3x_k^{-1}\varsigma_d^2}{\Lambda_+ +\pi^2\mathfrak{b}(x_k)^{-2}},\qquad \varsigma_d=\big(1-\delta(x_k)-\theta_d\big)_+\big(1-4m_2\mathfrak{b}(x_k)^{-2}\big)_+\to 1\quad(d\to\infty):$$

the upper bound of the Hessian of one fluctuation integral keeps the plaquette of the block field from degenerating. There are $3(\ell L^{d-1})^3$ pivot bonds, each in four plaquettes, so at most $12L^{-3}(\ell L^d)^3$ of the $3(\ell L^d)^3$ plaquettes of level k hold one.

The layers. By (Adm), as in the head of the lemma, d_0 and every larger top depth are admissible. By Prop. 16.1 (the field of the gauge of the chain), $\mathbf{B}_K=\frac{1}{ia_K}\log(e^{ia_K\mathbf{h}'_K}\cdots e^{ia_K\mathbf{h}'_{d_0+1}}e^{ia_K\mathbf{A}^{\mathrm{top}}})$, the logarithm of the ordered product of the top field, $|\mathbf{A}^{\mathrm{top}}|\le\mathsf{a}_{\mathrm{top}}$, and of the layers $\mathbf{h}'_{d'}=\mathrm{Ad}(\gamma)\mathbf{h}_{d'}$, $d_0<d'\le K$, with the regular gauge γ of the top, a function of the top field, and $\mathbf{h}_{d'}$ a function of the block field at depth d'; let $\mathcal{F}_{d'}$ be the σ-algebra of that block field, which increases with d', and $\mathbf{n}_{d'}=\mathbb{E}_K[\mathbf{h}'_{d'}|\mathcal{F}_{d'-1}]$.

The facts on the layers in the proof of Prop. 16.1(a) give: (L1) the ordered product of the top field and of the layers of the depths $\le d$ is a gauge transform of a background over the block field V_k, $k=K-d$, and the k-fold average of a background over V_k is V_k (Defs. YM3_I:9.3 and YM3_I:9.5, §5.2); (L2) $|\mathbf{h}'_{d'}|=|\mathbf{h}_{d'}|\le c_{\mathsf{h}}L^{d'}\delta(L^{d'}\mathsf{b})$ at every bond, with the constant c_{h} of the construction, and $\sum_{d'}a_K|\mathbf{h}'_{d'}|\le\frac14$, as d_0 is admissible; (L3) for a real bond function f, every component ξ of $\langle f,\mathbf{h}'_{d'}-\mathbf{n}_{d'}\rangle$ has $\mathbb{E}_K[e^{\lambda\xi}|\mathcal{F}_{d'-1}]\le e^{4\lambda^2G_{d'}(f)^2/\gamma_{\mathrm{r}}}$ for real λ, $G_{d'}(f)^2=c_{\mathsf{J}}^2\mathsf{b}^{-1}L^{d'}\min\{\|f\|_1^2,L^{-3d'}\|f\|_2^2\}$: the log-Sobolev inequality of one fluctuation integral (Lem. 15.3(iv)), carried by the map from the fluctuation field to the layer, whose derivative on the box obeys the bounds of the layer map with the constant c_{J}; on $L^2(T_\ell)$ this is the covariance bound $Ca_{d'}^2/\mathsf{b}$; (L4) $\sup|\mathbf{n}_{d'}|\le\mathsf{n}_{d'}=C\mathsf{b}^{-1}(\log L^{d'}\mathsf{b})^{2\mathfrak{p}_0}$, by the bounds (M), with $L^{-d/2}\mathsf{s}_d\to0$, $\mathsf{s}_d=\sum_{d_0<d'\le d}\mathsf{n}_{d'}$, and almost surely $\|\sum_{d_0<d'\le K}\mathbf{n}_{d'}\|_{L^{12}(T_\ell)}\le\mathsf{c}_{12}$, a number free of K; (L5) at every bond b, $\mathbb{E}_K|B_{K-d'}(b)|^2\le m_2$, and the moments of the layers are bounded so that $\|a_K(\sum_{d'}|\mathbf{h}'_{d'}(b)|)^2\|_{L^q(\mathbb{P}_K)}\le\mathsf{c}_{(q)}\mathsf{b}^{-1}$, with m_2 and $\mathsf{c}_{(q)}$ free of K and d'.

For $\sum_j|x_j|=s'\le\frac12$ the product $e^{ix_1}\cdots e^{ix_m}$ of exponentials of elements of $\mathfrak{g}$ is e^{iy} with $|y|\le1.02s'$ and $|y-\sum_j x_j|\le1.02s'^2$; so, with $a_K\mathsf{a}_{\mathrm{top}}\le\frac18$ and (L2), the remainder of the product formula, $\mathbf{B}_K-\mathbf{A}^{\mathrm{top}}-\sum_{d'}\mathbf{h}'_{d'}$, is at most $10a_K[(\sum_{d'}|\mathbf{h}'_{d'}|)^2+\mathsf{a}_{\mathrm{top}}^2]$ at every bond.

The field of the depths $\le d$. Let $d>d_0$ and let $\mathbf{B}^{\le d}$ be the logarithm of the ordered product of the top field and of the layers of the depths $\le d$. At every bare bond the arguments of the exponentials have the sum $s'=a_K\mathsf{a}_{\mathrm{top}}+a_K\sum_{d'\le d}|\mathbf{h}'_{d'}|\le L^{-k}\bar y_d$, $\bar y_d=a_d\mathsf{a}_{\mathrm{top}}+c_{\mathsf{h}}\delta(L^d\mathsf{b})/(1-L^{-1/2})$, by (L2) and $L^{d'}\delta(L^{d'}\mathsf{b})\le L^{-(d-d')/2}L^d\delta(L^d\mathsf{b})$ for $d'\le d$; so $y=L^k\sup a_K|\mathbf{B}^{\le d}|\le2\bar y_d$, and the remainder $\mathbf{B}^{\le d}-\mathbf{A}^{\mathrm{top}}-\sum_{d'\le d}\mathbf{h}'_{d'}$ is at most $1.02a_d^{-1}\bar y_d^2$ at every bond.

By (L1) and the gauge covariance of the average, $\mathcal{M}(U^u)=\mathcal{M}(U)^{u|}$ (Prop. YM3_I:4.5, YM3_I:(40)), $\mathcal{M}^k(e^{ia_K\mathbf{B}^{\le d}})$ is a gauge transform of V_k, so its plaquette at P has the modulus $|\partial V_k(P)-\mathbf{1}|$. For $L^k\sup|\mathsf{a}|<c_Q$, $\mathcal{M}^k(e^{i\mathsf{a}})(c)=e^{iQ_k(\mathsf{a})(c)}$ with $|Q_k(\mathsf{a})(c)|\le C_QL^k\sup|\mathsf{a}|$ and $|Q_k(\mathsf{a})(c)-\mathbf{Q}_k^{\mathrm{lin}}\mathsf{a}(c)|\le4C_Qc_Q^{-1}(L^k\sup|\mathsf{a}|)^2$: the linearisation of the average, with constants that depend on L alone (Prop. YM3_I:51.1 and Lem. 5.30).

At $\mathsf{a}=a_K\mathbf{B}^{\le d}$, with the product formula for the four factors around P,

$$\big|\langle f_P,\mathbf{B}^{\le d}\rangle\big|\ \ge\ |\partial V_k(P)-\mathbf{1}|-4\big(8C_Q^2+16C_Qc_Q^{-1}\big)\,\bar y_d^2,$$

a bound of the modulus alone, which is the quantity that (c) bounds from below; the error tends to 0 as $d\to\infty$, uniformly in K.

The heat kernel at the depths $\le d$. Let $f^1=(1-P_s)f_P$, $\|f^1\|_1\le48c_\nabla\sqrt{s}$. Of the parts of $\mathbf{B}^{\le d}$, the top field, the conditional means $\mathbf{n}_{d'}$ and the remainder are bounded at every bond by $\mathsf{a}_{\mathrm{top}}$, s_d and $1.02a_d^{-1}\bar y_d^2$, and pair with f^1 to at most $\|f^1\|_1$ times these; the centred layers $\mathbf{h}'_{d'}-\mathbf{n}_{d'}$, $d'\le d$, are martingale differences for $(\mathcal{F}_{d'})$, so orthogonal, and by (L3) at $f=f^1$ their pairing with f^1 has

the $L^2(\mathbb{P}_K)$ norm at most

$$\Big(3\cdot\frac{8}{\gamma_{\mathrm{r}}}\,c_{\mathsf{J}}^2\mathsf{b}^{-1}\|f^1\|_1^2\sum_{d'\le d}L^{d'}\Big)^{1/2}\;\le\;k_1\Big(\frac{s}{a_d^2}\Big)^{1/2}x_k^{-1/2},$$

as $\mathsf{b}^{-1}L^d=x_k^{-1}a_d^{-2}$. This cost does not vanish as d grows: at $s=\theta_0a_d^2$ it is $\frac14(3/\Lambda_+)^{1/2}x_k^{-1/2}$, and the range $0\le s\le\theta_0a_d^2$ of (a) is made for it.

The finer layers. Let $f=P_sf_P$. $\mathbf{B}_K-\mathbf{B}^{\le d}$ is the sum of the centred layers of the depths $d'>d$, of their conditional means and of the difference of the two remainders. $Y_P=\langle f,\mathbf{B}^{\le d}\rangle$ is $\mathcal{F}_d$-measurable and bounded, and $Z_P=\sum_{d'>d}\langle f,\mathbf{h}'_{d'}-\mathbf{n}_{d'}\rangle$ has $\mathbb{E}_K[Z_P|\mathcal{F}_d]=0$, so $\mathbb{E}_K|Y_P+Z_P|^2=\mathbb{E}_K|Y_P|^2+\mathbb{E}_K|Z_P|^2\ge\mathbb{E}_K|Y_P|^2$: in the second moment the finer layers can only add. The conditional means of the depths $>d$ pair with f to at most $4a_d^{3/4}\mathsf{c}_{12}+4a_d\mathsf{s}_d$, by Hölder's inequality and (L4), and the remainders, by (L5), to at most $40a_d(\mathsf{c}_{(4)}\mathsf{b}^{-1}+a_K\mathsf{a}_{\mathrm{top}}^2)+4.08\,\bar y_d^2$ in $L^4(\mathbb{P}_K)$.

(a). All these errors are at most $\mathsf{z}_dx_k^{-1/2}$ with $\mathsf{z}_d\to0$ uniformly in K, as $\bar y_d^2x_k^{1/2}\to0$, $a_dx_k^{1/2}=(a_d\mathsf{b})^{1/2}\to0$, $a_dx_k^{1/2}\mathsf{s}_d\to0$ by (L4) and $a_d^{3/4}x_k^{1/2}\to0$; d_L is the least depth $>d_0$ beyond which $\bar y_d\le\min\{\frac12,\frac{1}{16C_Q},\frac14c_Q\}$, $[3/(\Lambda_++\pi^2\mathsf{b}(L^d\mathsf{b})^{-2})]^{1/2}\varsigma_d-\mathsf{z}_d\ge\frac34(3/\Lambda_+)^{1/2}$ and $\mathsf{z}_d\le15c_{\mathsf{J}}\gamma_{\mathrm{r}}^{-1/2}$, a depth free of K. For P without pivot bond, $d\ge d_L$ and $0\le s\le\theta_0a_d^2$,

$$\|X_P\|_{L^2(\mathbb{P}_K)}\;\ge\;x_k^{-1/2}\Big\{\Big[\frac{3}{\Lambda_++\pi^2\mathfrak{b}(x_k)^{-2}}\Big]^{1/2}\varsigma_d-k_1\Big(\frac{s}{a_d^2}\Big)^{1/2}-\mathsf{z}_d\Big\}\;\ge\;x_k^{-1/2}\Big(\frac34-\frac14\Big)\Big(\frac{3}{\Lambda_+}\Big)^{1/2},$$

as $k_1\theta_0^{1/2}=\frac14(3/\Lambda_+)^{1/2}$; since $(\frac34-\frac14)^2\frac{3}{\Lambda_+}=\frac{3}{4\Lambda_+}$, this is $\mathbb{E}_K|X_P|^2\ge\frac{3}{4\Lambda_+}x_k^{-1}$. For the fourth moment, the centred layers of all depths form a martingale whose components have, by (L3) at $f=P_sf_P$ with $\|f\|_1\le4a_d$ at the depths $d'\le d$ and $\|f\|_2^2\le4a_{d'}^{-1}$ at the depths $d'>d$, the total variance proxy $v=\frac{4}{\gamma_{\mathrm{r}}}\sum_{d'}G_{d'}(f)^2\le\frac{4c_{\mathsf{J}}^2}{\gamma_{\mathrm{r}}}x_k^{-1}[\frac{16}{1-L^{-1}}+\frac{4}{L^2-1}]\le\frac{100\,c_{\mathsf{J}}^2}{\gamma_{\mathrm{r}}}x_k^{-1}$ $(L\ge3)$; so each component M^α has $\mathbb{E}_K|M^\alpha|^4\le64v^2$, the martingale has the $L^4(\mathbb{P}_K)$ norm at most $60\sqrt2\,c_{\mathsf{J}}\gamma_{\mathrm{r}}^{-1/2}x_k^{-1/2}$, the other members at most $\mathsf{z}_dx_k^{-1/2}$, and $\mathbb{E}_K|X_P|^4\le k_4^4x_k^{-2}$, for every P.

Paley and Zygmund. Let $\Sigma=a_d^{-1}\sum_P|X_P|^2$, the sum over the $N_P\ge3(1-4L^{-3})(\ell L^d)^3\ge\frac52(\ell/a_d)^3$ plaquettes without pivot bond $(L\ge3)$. By eq. 16.i at $A=P_s\mathbf{B}_K$, $\|\mathrm{d}_aP_s\mathbf{B}_K\|_2^2\ge\Sigma$; $\mathbb{E}_K\Sigma\ge a_d^{-1}N_P\frac{3}{4\Lambda_+}x_k^{-1}$, and by the Cauchy–Schwarz inequality for the pairs of plaquettes $\mathbb{E}_K\Sigma^2\le a_d^{-2}N_P^2k_4^4x_k^{-2}$; so $\mathbb{P}_K(\Sigma\ge\frac12\mathbb{E}_K\Sigma)\ge(\mathbb{E}_K\Sigma)^2/4\mathbb{E}_K\Sigma^2\ge\frac14(3/4\Lambda_+k_4^2)^2=p_*$, and $\frac12\mathbb{E}_K\Sigma\ge\frac12\cdot\frac52\cdot\frac{3}{4\Lambda_+}\,\ell^3a_d^{-3}\,x_k^{-1}a_d^{-1}=k_\Sigma\mathsf{b}^{-1}\ell^3a_d^{-3}$, as $x_k^{-1}a_d^{-1}=\mathsf{b}^{-1}$.

This is (a). The conditional form: the bounds that hold at every bond or almost surely, and the martingale structure of the layers, are conditional on the block field at depth d_0 as they stand, so the argument goes through under $\mathbb{E}_K^0$ with the norms $(N_P^{-1}\sum_P\mathbb{E}_K^0|\cdot|^r)^{1/r}$, $r=2,4$, but for the two places where unconditional moments of one layer enter, the bound m_2 and (L5); there Markov's inequality bounds their conditional means off an event of probability at most τ_d, and the inequality of Paley and Zygmund under $\mathbb{E}_K^0$, with the Cauchy–Schwarz inequality over P for the second moment of Σ, gives the last bound.

(b) By Duhamel's formula the linearised gauge-fixed flow from an L^2 datum z is a small perturbation of the heat flow, $\|w_s-P_sz\|_2\le2\epsilon(s)\|z\|_2$ with $\epsilon(s)=c_6\bar C^2s^{\frac14-\frac\kappa2}$; differentiating $A^a\circ\Phi_\sigma$ along $e^{ia_K(\mathbf{B}+rz)}$ at $z=P_\sigma\mathbf{B}$ and using $(\mathrm{d}_aP_\sigma\mathbf{B},\mathrm{d}_aP_\sigma P_\sigma\mathbf{B})=\|\mathrm{d}_aP_{3\sigma/2}\mathbf{B}\|_2^2$ and $\Gamma_K(F)^{1/2}\ge|\frac{d}{dr}F(e^{ia_K(\mathbf{B}+rz)})|/\|z\|_2$ gives the lower bound; the upper bound and the last inequality are the same expansion. □

The lower bound requires control of the frequency of the terminal gradient. For the action this frequency is controlled by the change in the squared current between two positive times. The following estimate makes this control uniform in the lattice spacing.

Lemma 16.8 (Lower propagation on a regular time interval)**.** Let $a = a_K$, $U_s = \Phi_s U$, $0 < \kappa < \frac{1}{2}$, and put

$$r = \frac{3}{4} + \frac{\kappa}{2}, \qquad \theta = 1 - r = \frac{1}{4} - \frac{\kappa}{2}.$$

Suppose that $C_0 > 0$, $\sigma > 0$, and

$$a^{-2}\varpi(U_s) \le C_0 s^{-r} \quad (0 < s \le \sigma). \tag{16.ii}$$

Write $g(s) = (J, J)_a(U_s)$ and assume $g(\sigma) > 0$. For $x \ge 0$ and $j > 0$ define

$$c_\kappa(x, j) = e^{8x/\theta}\left(\log \max\{1, j\} + \frac{64x}{\theta} + 1072x^2\right) + 32x,$$

$$E_\kappa(x, c) = 64x(1 + \theta^{-1}) + 16e^{8x/\theta}x(c + 32x) + \frac{672}{\theta}x^2 + \frac{7168}{\theta}e^{8x/\theta}x^3.$$

If $j \ge g(\sigma/2)/g(\sigma)$ and $x = C_0\sigma^\theta$, then

$$\Gamma_K(A^a \circ \Phi_\sigma)(U) \ge g(\sigma) \exp\left[-2c_\kappa(x, j) - E_\kappa(x, c_\kappa(x, j))\right]. \tag{16.iii}$$

The factor on the right is positive, decreases as j increases, and depends only on κ, C_0, σ, j. ∎

Proof. All the operators in the proof act on the finite periodic lattice at spacing a. Use the scalar product $(X, Y)_a = a^3 \sum_b \langle X(b), Y(b)\rangle$ on positive bonds. Thus $\mathrm{grad}_a F = a^{-2}\partial F$ satisfies $\|\mathrm{grad}_a F\|_a^2 = \Gamma_K(F)$, and $\mathrm{grad}_a A^a = J$. At a field V, let

$$T_\mu h(x) = \mathrm{Ad}(V_\mu(x))h(x + ae_\mu), \qquad \nabla_\mu = a^{-1}(T_\mu - 1), \qquad (DX)_{\mu\nu} = \nabla_\mu X_\nu - \nabla_\nu X_\mu.$$

Here $D_0^* X = \sum_\mu \nabla_\mu^* X_\mu$. In this proof $\Delta = \sum_\mu \nabla_\mu^* \nabla_\mu$ is positive and acts on each component; it is distinct from the heat operator Δ_a of Prop. 16.1. Set

$$\phi(V) = a^{-2}\varpi(V), \qquad \mathcal{F}(p) = \frac{\partial V(p) - \partial V(p)^*}{2ia^2}.$$

The orientations and base points of $\mathcal{F}$ agree with those of D. Differentiating the plaquette action gives $J = D^*\mathcal{F}$ and $a|J|_\infty \le 4\phi$.

We first compare the action Hessian with Δ. Define the symmetric operator H by

$$(X, HX)_a = \frac{d^2}{dt^2} A^a(e^{iatX}V)\Big|_{t=0}.$$

For $p = p_{\mu\nu}(x)$ put $\alpha_p = (-T_\nu X_\mu, -X_\nu, X_\mu, T_\mu X_\nu)$ and $S_p = \sum_{i<j}[\alpha_{p,i}, \alpha_{p,j}]$. The four perturbed factors in the plaquette can be transported to x; their sum is $a(DX)(p)$. For $P_t = \prod_i e^{it\alpha_{p,i}}$,

$$P_0'' = -\left(\sum_i \alpha_{p,i}\right)^2 - S_p.$$

Every traceless Hermitian 2×2 matrix Z has $Z^2 = |Z|^2\mathbf{1}$. Since $\partial V(p) = (1 - a_p)\mathbf{1} + ia^2\mathcal{F}(p)$, taking the trace in the second variation gives

$$(X, HX)_a = a^3 \sum_p \left((1 - a_p)|(DX)(p)|^2 + \langle \mathcal{F}(p), iS_p\rangle\right). \tag{16.iv}$$

Expansion of the squares also gives

$$\|DX\|_a^2 + \|D_0^* X\|_a^2 = (X, \Delta X)_a + w(X), \qquad w(X) = \sum_{\mu\neq\nu} (X_\mu, [\nabla_\mu, \nabla_\nu^*]X_\nu)_a.$$

The commutator compares the two transports around a plaquette, so its norm is at most 2ϕ. Thus $|w(X)| \le 4\phi\|X\|_a^2$. Moreover, $|\mathcal{F}(p)| \le \phi$, $|S_p| \le 3\sum_i |\alpha_{p,i}|^2$, and every bond belongs to four

plaquettes. Consequently the second term of eq. 16.iv has modulus at most $12\phi\|X\|_a^2$. Finally, $a_p = \frac{1}{2}|\partial V(p) - \mathbf{1}|^2$, $|\partial V(p) - \mathbf{1}| \le 2$, and $|DX(p)|^2 \le 4a^{-2}\sum_{b\in p}|X(b)|^2$ imply

$$0 \le a^3 \sum_p a_p |DX(p)|^2 \le 16\phi \|X\|_a^2.$$

It follows that

$$H + D_0 D_0^* = \Delta + \mathcal{U}, \qquad \|\mathcal{U}\| \le 32\phi. \tag{16.v}$$

In particular, $H = \Delta + \mathcal{U}$ on divergence-free forms.

We next record the two evolution equations used below. If Y_s is the differential of the flow, in the frame $\delta U_s = iaY_s U_s$, then

$$Y_s' = -H_s Y_s - \frac{a}{2} i[J_s, Y_s].$$

Indeed, for $R_X F = \frac{d}{dt} F(e^{iatX} V)\Big|_0$ one has $[R_X, R_Y] = -aR_{i[X,Y]}$. The symmetric part of the second derivative is the Hessian, and hence $R_Y J = HY - \frac{a}{2} i[J, Y]$. Differentiating $\dot U = -iaJU$ now gives the displayed equation. For a smooth gauge-invariant G with $\Gamma_K(G)(U_\sigma) > 0$, put

$$\xi(s) = \operatorname{grad}_a (G \circ \Phi_{\sigma - s})(U_s), \qquad n(s) = \frac{(\xi, H_s \xi)_a}{\|\xi\|_a^2}.$$

The pairing $(\xi(s), Y_s)_a$ is constant. Since $D\Phi_s$ is invertible, this gives

$$\xi' = H_s \xi - \frac{a}{2} i[J_s, \xi], \qquad \Gamma_K(G \circ \Phi_\sigma)(U) = \Gamma_K(G)(U_\sigma) \exp\left[-2\int_0^\sigma n(s)\,\mathrm{d}s\right]. \tag{16.vi}$$

Here ξ never vanishes, and $D_0^* \xi = 0$ because the flow is gauge covariant. The current $j(s) = J(U_s)$ satisfies

$$j' = -H_s j, \qquad D_0^* j = 0, \qquad g' = -2(j, H_s j)_a. \tag{16.vii}$$

Thus g vanishes at one time only if it vanishes identically.

The derivative of Δ has a useful cancellation. For fixed X, $\partial_s \nabla_\mu X_\nu = -i[J_\mu, T_\mu X_\nu]$. The part containing $a\nabla_\mu X_\nu$ pairs to zero with $\nabla_\mu X_\nu$. Setting $Y_\mu = \sum_\nu i[X_\nu, \nabla_\mu X_\nu]$ therefore gives

$$(X, \Delta' X)_a = -2(J, Y)_a = -2(\mathcal{F}, DY)_a.$$

The discrete product rule $\nabla_\mu [A, B] = [\nabla_\mu A, T_\mu B] + [A, \nabla_\mu B]$ shows explicitly that

$$(DY)_{\mu\lambda} = \sum_\nu i([\nabla_\mu X_\nu, T_\mu \nabla_\lambda X_\nu] - [\nabla_\lambda X_\nu, T_\lambda \nabla_\mu X_\nu] + [X_\nu, [\nabla_\mu, \nabla_\lambda] X_\nu]).$$

The first two terms contain only first differences. Translations are isometries, $|[A, B]| \le 2|A||B|$, and the sum over the three pairs of directions is bounded by twice the sum of the squared first differences. The last term uses $\|[\nabla_\mu, \nabla_\lambda]\| \le 2\phi$. Multiplying by $2|\mathcal{F}|$ and summing gives

$$|(X, \Delta' X)_a| \le 8\phi (X, \Delta X)_a + 24\phi^2 \|X\|_a^2. \tag{16.viii}$$

For either $\eta = \xi$ or $\eta = j$, let $h = (\eta, \Delta\eta)_a / \|\eta\|_a^2$ and $n_\eta = (\eta, H\eta)_a / \|\eta\|_a^2$. Then $h \ge 0$, $|h - n_\eta| \le 32\phi$, and

$$h' = \frac{2(\eta', (\Delta - h)\eta)_a + (\eta, \Delta'\eta)_a}{\|\eta\|_a^2}.$$

For ξ the leading term is $2\|(\Delta - h)\xi\|_a^2$, whereas for j it is its negative. The remaining part of ξ' has norm at most $(32 + 4)\phi\|\xi\|_a$, using $a|J|_\infty \le 4\phi$. The inequality $72\phi\|\xi\|_a \|(\Delta - h)\xi\|_a \le 2\|(\Delta - h)\xi\|_a^2 + 648\phi^2\|\xi\|_a^2$ and the corresponding bound with 64 and 512 for j give

$$h' \ge -8\phi h - 672\phi^2 \quad (\eta = \xi), \qquad h' \le 8\phi h + 536\phi^2 \quad (\eta = j). \tag{16.ix}$$

Put $\psi(s) = C_0 s^{-r}$, so $\int_0^\sigma \psi = x/\theta$. For the adjoint solution, backward integration of eq. 16.ix yields

$$h(s) \le e^{8x/\theta}\left(h(\sigma) + 1344 C_0^2 s^{1-2r}\right), \qquad 0 < s \le \sigma,$$

because $2r - 1 \ge \frac{1}{2}$. The square ψ^2 is not integrable at zero, but $s\psi^2$ is. Accordingly, integrate against the weight s:

$$\int_0^\sigma h = \sigma h(\sigma) - \int_0^\sigma s h' \le \sigma h(\sigma) + \int_0^\sigma s(8\psi h + 672\psi^2).$$

There is no boundary term at zero: for each fixed lattice the flow, the operators, and the nonzero adjoint solution are smooth there. If $c \ge \sigma n(\sigma)$, then $0 \le \sigma h(\sigma) \le c + 32x$. Combining the last two inequalities with $|n - h| \le 32\psi$ gives

$$2\int_0^\sigma n(s)\,\mathrm{d}s \le 2\sigma n(\sigma) + E_\kappa(x, c). \tag{16.x}$$

Indeed, the endpoint and $n - h$ terms give $64x(1 + \theta^{-1})$; the constant part of $h(s)$ gives at most $16e^{8x/\theta}x(c + 32x)$, since $2 - r \ge 1$; and the two remaining integrals are

$$1344 C_0^2 \int_0^\sigma s^{1-2r}\,\mathrm{d}s = \frac{672}{\theta}x^2, \qquad 21504 C_0^3 e^{8x/\theta}\int_0^\sigma s^{2-3r}\,\mathrm{d}s = \frac{7168}{\theta}e^{8x/\theta}x^3.$$

It remains to control the terminal frequency when $G = A^a$. Apply the second inequality of eq. 16.ix to the current and integrate forward from s to σ. Averaging over $\sigma/2 \le s \le \sigma$ gives

$$\frac{\sigma}{2}h(\sigma) \le e^{8x/\theta}\left(\int_{\sigma/2}^\sigma h(s)\,\mathrm{d}s + \frac{\sigma}{2}536\int_{\sigma/2}^\sigma \psi(s)^2\,\mathrm{d}s\right).$$

By eq. 16.vii,

$$\int_{\sigma/2}^\sigma h(s)\,\mathrm{d}s \le \frac{1}{2}\log\frac{g(\sigma/2)}{g(\sigma)} + 32\int_{\sigma/2}^\sigma \psi(s)\,\mathrm{d}s, \qquad \int_{\sigma/2}^\sigma \psi(s)^2\,\mathrm{d}s \le 2C_0^2\sigma^{1-2r}.$$

Consequently

$$0 \le \sigma h(\sigma) \le e^{8x/\theta}\left(\log\frac{g(\sigma/2)}{g(\sigma)} + \frac{64x}{\theta} + 1072x^2\right).$$

The terminal adjoint vector for $G = A^a$ is the current itself. Thus its terminal frequency satisfies $\sigma n(\sigma) \le c_\kappa(x, j)$. Equations eq. 16.vi and eq. 16.x prove eq. 16.iii. □

Proposition 16.9 (Lower energy bounds for flowed observables)**.** For all K, m, $\beta > 0$, with $a = a_K$, the lattice time $\tau = s/a^2$, $T = \sigma/a^2$ and $U_\tau^\circ = \Phi_{a^2\tau}U$: let $|X|_\infty = \max_b |X(b)|$, let $\nabla^3 A_K(V)$ be the symmetric trilinear form with $\nabla^3 A_K(V)[X, X, X] = \frac{d^3}{dt^3}A_K(e^{itX}V)|_{t=0}$, and let Z_p be as in Lem. 16.2, taken at the field V.

(a) (Log-convexity with a defect.) Let $G \in C^2(\Omega_K)$ be invariant, $U \in \Omega_K$, $\sigma > 0$ and $\partial G(U_\sigma) \ne 0$; for $0 \le \tau \le T$ let $\gamma_\tau = \partial(G \circ \Phi_{a^2(T-\tau)})(U_\tau^\circ)$, $N(\tau) = \|\gamma_\tau\|^2$, $\mathsf{q}(\tau) = \nabla^2 A_K(U_\tau^\circ)[\gamma_\tau, \gamma_\tau]$ and $\mathsf{f}_\tau = \partial A_K(U_\tau^\circ)$. Then $N > 0$, $\dot N = 2\mathsf{q}$, $(\log N)'' \ge -\eta$ with $\eta(\tau) = 2|\nabla^3 A_K(U_\tau^\circ)[\mathsf{f}_\tau, \gamma_\tau, \gamma_\tau]|/N(\tau)$, and

$$\Gamma_K(G \circ \Phi_\sigma)(U) \;\ge\; \Gamma_K(G)(U_\sigma)\,\exp\Big[-2\nu - \int_0^T \tau\,\eta(\tau)\,\mathrm{d}\tau\Big], \qquad \nu = \frac{T\,\mathsf{q}(T)}{N(T)},$$

ν the frequency of the gradient at the flowed field. For all V and real β, γ:

$$|\nabla^3 A_K(V)[\beta, \gamma, \gamma]| \le |\gamma|_\infty\{4(\sum_p |Z_p(\beta)|^2)^{1/2}\|\gamma\| + 8(\sum_p |Z_p(\gamma)|^2)^{1/2}\|\beta\| + 64\,\varpi(V)\|\gamma\|\|\beta\|\}.$$

(b) (An effective form.) In (a) let besides $0<\kappa<\frac16$, $C_F\ge1$, $0<\sigma\le1$ and $a^{-2}\varpi(U_s)\le C_F s^{-\frac34-\frac{\kappa}{2}}$ for $0<s\le\sigma$. Put $\lambda_F=6C_F$, $\mathsf{e}_1=(1+56\lambda_F e^{56\lambda_F})^{1/2}$,

$$\mathsf{r}=(\ell/a)^{3/2}|\partial G(U_\sigma)|_\infty/\|\partial G(U_\sigma)\|,$$

$\nu_+=\max\{\nu,0\}$, $\varrho_F=3.5\,e^{80\lambda_F}C_F\mathsf{r}$ and

$$\mathsf{k}_A=28e^{56\lambda_F}C_F+e^{28\lambda_F}\varrho_F(7\mathsf{e}_1+256C_F).$$

If

$$4e^{2\nu_++1}\varrho_F\big[48\mathsf{e}_1+384C_F+12\nu_+^{1/2}+32\mathsf{k}_A^{1/2}+768\varrho_F\big]\sigma^{\frac14-\frac{\kappa}{2}}\le1,$$

then $\Gamma_K(G\circ\Phi_\sigma)(U)\ge e^{-2\nu_+-\frac12}\,\Gamma_K(G)(U_\sigma)$. At $G=A^a$: $\Gamma_K(A^a)(U_\sigma)=(J,J)_a(U_\sigma)$ and $\nu=-\frac{\sigma}{2}\,\partial_\sigma\log(J,J)_a(U_\sigma)$.

(c) (Lower propagation at every time.) Let $0<\kappa<\frac16$, $\mathbf{B}$ a real 1-form, $R\ge\max\{1,\|\mathbf{B}\|_{\kappa;a}\}$ and

$$a\le a_2(R):=\min\big\{a_1(R),\ (16e^{12}R)^{-2/(1-2\kappa)},\ \big(8(16\mathsf{M}_\Theta)^{1/6}c_3R\big)^{-1/(1-2\kappa)}\big\},$$

a positive number that depends on R and κ alone, with c_3, $a_0(R)$, $s_0(R)$ of the short-time theory of the gauge-fixed flow in Prop. 16.1(a), $a_1(R)\le a_0(R)$ the threshold of Lem. 16.7(b), and $\mathsf{M}_\Theta=\max\{1,\sup|\Theta^{\mathrm{ch}}|\}$, the absolute number of the head of Prop. 16.1, with the chart equation $\frac{d}{ds}A_s(b)=a^{-3}\Theta^{\mathrm{ch}}(aA_s|_{\mathsf{N}(b)})$ of the gauge-fixed flow set there and the supremum taken where $|aA|\le1$; so $a_2(R)\le a_1(R)$, and $a\le a_2(R)$ serves both Lem. 16.7(b) and (c); let U be gauge equivalent to $e^{ia\mathbf{B}}$, $\sigma>0$, $\sigma'=\min\{\sigma,s_0(R)\}$, $(J,J)_a(U_\sigma)>0$, $\Lambda\ge\Lambda_\sigma(U)$ and $\mathsf{j}_\sharp\ge(J,J)_a(U_{\sigma'/2})/(J,J)_a(U_\sigma)$.

Put $r=3/4+\kappa/2$, $\theta=1-r$ and $C_0(R)=4((2+c_3)R)^2$. For $0<\sigma\le s_0(R)$ set $x=C_0(R)\sigma^\theta$ and

$$\mathsf{G}_\kappa^{\mathrm{short}}(R,\sigma,\mathsf{j}_\sharp)=\exp\left[-2c_\kappa(x,\mathsf{j}_\sharp)-E_\kappa(x,c_\kappa(x,\mathsf{j}_\sharp))\right],$$

where c_κ and E_κ are defined in Lem. 16.8. Then

$$\Gamma_K(A^a\circ\Phi_\sigma)(U)\ge(J,J)_a(U_\sigma)\mathsf{G}_\kappa^{\mathrm{short}}(R,\sigma,\mathsf{j}_\sharp).\tag{16.xi}$$

This positive factor is independent of K, ℓ, and b, and decreases as $\mathsf{j}_\sharp$ increases.

For arbitrary $\sigma>0$, set $s=\sigma'$, $\tau=s/2$, $x=C_0(R)s^\theta$ and $j=\max\{1,\mathsf{j}_\sharp\}$. Put

$$c_*=e^{8x/\theta}(\log j+64\Lambda+672x^2)+32x,\qquad \mathsf{G}_\kappa(R,s,\Lambda,\mathsf{j}_\sharp)=j^{-1}\exp[-2c_*-E_\kappa(x,c_*)].$$

Then

$$\Gamma_K(A^a\circ\Phi_\sigma)(U)\ge(J,J)_a(U_\sigma)\mathsf{G}_\kappa(R,\sigma',\Lambda,\mathsf{j}_\sharp).$$

The factor is positive, decreases in Λ and $\mathsf{j}_\sharp$, and is independent of the cutoff, side and label.

The upper bound $\Gamma_K(A^a\circ\Phi_\sigma)(U)\le e^{24\Lambda_\sigma(U)}(J,J)_a(U_\sigma)$ holds as well.

(d) (Positive energies and the flowed current.) Assume (E) and $0<\kappa<\frac16$; let $\varphi\in C^\infty(\mathbb{R})$ be bounded with all its derivatives and $\varphi'>0$ on $[0,\infty)$, and $0<\sigma<\sigma_2\le1$. Say that (J_σ) holds if there is $c_0>0$ with $p_0:=\liminf_K\mu^K((J,J)_a(U_\sigma)\ge c_0)>0$, and $(\mathrm{D}_{\sigma,\sigma_2})$ if $\liminf_K\langle A^a\circ\Phi_\sigma-A^a\circ\Phi_{\sigma_2}\rangle_{\mu^K}>0$. In (i) and in the first two implications of (iii) use the consequence (Adm) of (E) proved in Prop. 16.1. (i) If (J_σ) holds, then $\liminf_K\int\chi_K\Gamma_K(\varphi(A^a)\circ\Phi_\sigma)\,\mathrm{d}\mu^K>0$ for every sequence of Borel $\chi_K:\Omega_K\to[0,1]$ with $\limsup_K\mu^K(\chi_K<1)<p_0$.

(ii) Conversely, if $R\ge R_\flat$, the Borel $\chi_K:\Omega_K\to[0,1]$ vanish off $\mathsf{O}_K(8R,\bar s(R))$ and $\liminf_K\int\chi_K\Gamma_K(\varphi(A^a)\circ\Phi_\sigma)\,\mathrm{d}\mu^K>0$, then (J_σ) holds. (iii) (J_{σ_2}) implies $(\mathrm{D}_{\sigma,\sigma_2})$, and $(\mathrm{D}_{\sigma,\sigma_2})$ implies (J_σ); and $\liminf_K\langle A^a\circ\Phi_\sigma\rangle_{\mu^K}>\sup_K\langle A^a\circ\Phi_1\rangle_{\mu^K}$ implies $(\mathrm{D}_{\sigma,1})$.

■

Proof sketch. (a) $N > 0$ because $D\Phi$ is invertible. As G is invariant, γ_τ is horizontal, and along the trajectory $\dot\gamma_\tau = \nabla^2 A_K(U_\tau^\circ)\gamma_\tau - \frac{i}{2}[\mathsf{f}_\tau, \gamma_\tau]$, the Hessian taken as a symmetric operator; since $\langle \gamma, i[\mathsf{f},\gamma]\rangle = 0$, $\dot N = 2\mathsf{q}$, and a second differentiation, in which the Hessians cancel, gives $\dot{\mathsf{q}} = 2\|\nabla^2 A_K\gamma\|^2 - \nabla^3 A_K[\mathsf{f},\gamma,\gamma] + \frac{1}{6}\|i[\mathsf{f},\gamma]\|^2$. So

$$(\log N)'' = \frac{4}{N^2}(\|\nabla^2 A_K\gamma\|^2\|\gamma\|^2 - \langle\gamma, \nabla^2 A_K\gamma\rangle^2) + \frac{2}{N}\Big(\tfrac{1}{6}\|i[\mathsf{f},\gamma]\|^2 - \nabla^3 A_K[\mathsf{f},\gamma,\gamma]\Big) \geq -\eta$$

by the Cauchy–Schwarz inequality. Integrating twice from T, $\log\frac{N(\tau_1)}{N(T)} \geq -2(T-\tau_1)\frac{\mathsf{q}(T)}{N(T)} - \int_{\tau_1}^T(\tau - \tau_1)\eta(\tau)\,\mathrm{d}\tau$ for $0 \leq \tau_1 \leq T$; at $\tau_1 = 0$, $N(0) = a\Gamma_K(G\circ\Phi_\sigma)(U)$ and $N(T) = a\Gamma_K(G)(U_\sigma)$. The weight τ under the integral keeps the defect finite at rough fields. The bound of $\nabla^3 A_K$ follows from the expansion of the third derivative plaquette by plaquette, as in Lem. 16.2(a), and the Cauchy–Schwarz inequality over the plaquettes. (b) The hypothesis gives $\Lambda_\sigma \leq \lambda_F$; by Lem. 16.2(b), (c), (h), $(J,J)_a(U_s)^{1/2} \leq \mathsf{j}_0 s^{-\frac{5}{4}-\frac{\kappa}{2}}$ with $\mathsf{j}_0 = 3.5\,e^{28\lambda_F}\ell^{3/2}C_F$, and $|\gamma_\tau|_\infty \leq e^{52\lambda_F}|\gamma_T|_\infty$, $\|\gamma_\tau\| \leq e^{28\lambda_F}\|\gamma_T\|$.

Lem. 16.2(a) gives $\sum_p |Z_p(X)|^2 \leq \nabla^2 A_K(V)[X,X] + 28\varpi(V)\|X\|^2$ for all X; with $\frac{d}{d\tau}\|\mathsf{f}_\tau\|^2 = -2\nabla^2 A_K[\mathsf{f}_\tau,\mathsf{f}_\tau]$ this makes the energy $\sum_p|Z_p(\mathsf{f}_\tau)|^2$ of the current integrable in τ, $\int_0^T\sum_p|Z_p(\mathsf{f}_\tau)|^2 d\tau \leq \frac{1}{2}\|\mathsf{f}_0\|^2 + 28\lambda_F\sup_\tau\|\mathsf{f}_\tau\|^2$, and with (a) it bounds the energy $\sum_p|Z_p(\gamma_\tau)|^2 \leq \mathsf{q} + 28\varpi N$ of γ_τ. On the largest interval $[\tau_1, T]$ on which $N \geq e^{-2\nu_+ - 1}N(T)$, the bound of $\nabla^3 A_K$ and these estimates give $\int_{\tau_1}^T(\tau-\tau_1)\eta \leq \frac{1}{2}$ under the displayed condition; so (a) gives $\log\frac{N(\tau_1)}{N(T)} \geq -2\nu_+ - \frac{1}{2}$, and $\tau_1 = 0$ by continuity.

At $G = A^a$, $\partial A^a = a^{-1}\mathsf{f}$, and $\frac{d}{d\tau}\|\mathsf{f}_\tau\|^2 = -2\langle\mathsf{f}_\tau, \nabla^2 A_K\mathsf{f}_\tau\rangle$. (c) First suppose $0 < \sigma \leq s_0(R)$. Since $a \leq a_2(R) \leq a_0(R)$, Prop. 16.1(a) gives a gauge-fixed representative e^{iaA_s} of U_s for $0 \leq s \leq s_0(R)$, with

$$\|A_s\|_\infty \leq (1+c_3)Rs^{-1/4-\kappa/2}, \qquad \|\nabla A_s\|_\infty \leq (2+c_3)Rs^{-3/4-\kappa/2}.$$

For a plaquette, split its holonomy into the two differences between opposite bond potentials and the group commutator of two adjacent bond exponentials. The exponential is Lipschitz on Hermitian matrices, and the group commutator has norm at most twice the product of their norms. This gives

$$a^{-2}\varpi(e^{iaA_s}) \leq 2\left(\|\nabla A_s\|_\infty + \|A_s\|_\infty^2\right),$$

as in Lem. 16.2(e). Put $\bar C = (2+c_3)R \geq 1$. Since $s_0(R) \leq 1$ and $1/2 + \kappa \leq 3/4 + \kappa/2$, the right side is at most $4\bar C^2 s^{-3/4-\kappa/2}$. Curvature is gauge invariant, so eq. 16.ii holds with $C_0 = C_0(R)$. Here $\sigma' = \sigma$, and the ratio assumed in (c) is exactly the ratio required in Lem. 16.8. That lemma proves eq. 16.xi. Now let $\sigma > 0$ be arbitrary and put $s = \sigma'$ and $\tau = s/2$. Use the adjoint solution

$$\xi(t) = \mathrm{grad}_a(A^a \circ \Phi_{\sigma-t})(U_t), \qquad g(t) = \|J(U_t)\|_a^2.$$

It is horizontal and nonzero. The adjoint identity eq. 16.vi and $D\Phi_{\sigma-t}(U_t)J(U_t) = J(U_\sigma)$ give

$$(\xi(t), J(U_t))_a = g(\sigma), \qquad \|\xi(\tau)\|_a^2 \geq \frac{g(\sigma)^2}{g(\tau)} \geq \frac{g(\sigma)}{j}.$$

For this same solution put $h(t) = (\xi(t), \Delta_t\xi(t))_a/\|\xi(t)\|_a^2$ and $n(t) = (\xi(t), H_t\xi(t))_a/\|\xi(t)\|_a^2$. Since $h \geq 0$ and $|n-h| \leq 32\phi$,

$$\|\xi(s)\|_a^2 \leq e^{64(\Lambda_\sigma - \Lambda_s)}g(\sigma), \qquad \int_\tau^s h(t)\,\mathrm{d}t \leq \tfrac{1}{2}\log j + 32\Lambda.$$

Indeed, integrate $(\log\|\xi\|_a^2)' = 2n$ first on $[s,\sigma]$ and then on $[\tau,s]$, using the lower bound at τ. Only the integrated curvature up to σ enters here.

On $[0,s]$ the short-time estimate gives $\phi(t) \leq C_0(R)t^{-r}$. Backward integration of the adjoint inequality in eq. 16.ix, followed by averaging over $[\tau,s]$, yields

$$\tau h(\tau) \leq e^{8x/\theta}\left(\int_\tau^s h(t)\,\mathrm{d}t + 672\tau\int_\tau^s C_0(R)^2 t^{-2r}\,\mathrm{d}t\right).$$

Here $\tau\int_\tau^{2\tau} C_0(R)^2 t^{-2r}\,\mathrm{d}t \le C_0(R)^2\tau^{2\theta} \le x^2$. Consequently $\tau n(\tau) \le c_*$. The integrated adjoint estimate eq. 16.x, applied on $[0,\tau]$ to this solution, gives

$$2\int_0^\tau n(t)\,\mathrm{d}t \le 2c_* + E_\kappa(x, c_*).$$

We used $C_0(R)\tau^\theta \le x$ and monotonicity of E_κ in its nonnegative arguments. Finally,

$$\Gamma_K(A^a\circ\Phi_\sigma)(U) = \|\xi(0)\|_a^2 = \|\xi(\tau)\|_a^2 \exp\left[-2\int_0^\tau n(t)\,\mathrm{d}t\right] \ge \frac{g(\sigma)}{j} e^{-2c_* - E_\kappa(x,c_*)}.$$

Thus no power-law curvature estimate beyond the regular interval $[0,s]$ is needed. The upper bound follows as before from Lem. 16.2(b).

(d) (i): fix R, R' and s_+ such that, for large K, the regularised fields $\mathsf{O}_K(8R,\bar s(R))$ of (C4), the event that the gauge orbit holds a 1-form of rough norm $\le R'$ at the index $\frac{\kappa}{2}$, and $\{A^a\circ\Phi_\sigma \le s_+\}$ all have probability close to 1 (Prop. 16.1(a), by (Adm), and Chebyshev's inequality).

On their intersection with $\{\chi_K = 1\}\cap\{(J,J)_a(U_\sigma)\ge c_0\}$, whose mass has lower limit > 0, the proof of (C2)(a) bounds Λ_σ and $(J,J)_a(U_{\sigma'/2})$, the proved lower bound of (c) at the index $\frac{\kappa}{2}$, for $a_K \le a_2(R')$, bounds $\Gamma_K(A^a\circ\Phi_\sigma)$ below by c_0 times a positive number free of K, and $\Gamma_K(\varphi(A^a)\circ\Phi_\sigma) = \varphi'(A^a\circ\Phi_\sigma)^2\Gamma_K(A^a\circ\Phi_\sigma) \ge (\min_{[0,s_+]}\varphi')^2\Gamma_K(A^a\circ\Phi_\sigma)$. (ii): on $\mathsf{O}_K(8R,\bar s(R))$, for large K, the proof of (C2)(a) gives $\Lambda_\sigma \le \lambda_\sharp$ and $(J,J)_a(U_\sigma)\le \mathsf{j}_\sigma$ with numbers free of K, and $\Gamma_K(\varphi(A^a)\circ\Phi_\sigma) \le \|\varphi'\|_\infty^2 e^{24\lambda_\sharp}(J,J)_a(U_\sigma)$ (Lem. 16.2(b)); splitting at a level c, $\int\chi_K\Gamma_K \le \|\varphi'\|_\infty^2 e^{24\lambda_\sharp}[c + \mathsf{j}_\sigma\mu^K((J,J)_a(U_\sigma)\ge c)]$.

(iii): $A^a\circ\Phi_\sigma - A^a\circ\Phi_{\sigma_2} = \int_\sigma^{\sigma_2}(J,J)_a(U_s)\,\mathrm{d}s$ (Lem. 16.2(c)), and $(J,J)_a(U_{s'}) \le e^{24\lambda_\sharp}(J,J)_a(U_s)$ for $s\le s'$ on $\mathsf{O}_K(8R,\bar s(R))$, whose complement has small mass for large R by (Adm) (Step 8); in the second arrow the moments of $(J,J)_a(U_s)$ of Prop. 16.1(a), under (Adm), and the Cauchy–Schwarz inequality give $\langle(J,J)_a(U_\sigma)\rangle_{\mu^K} \ge 2c_0$, hence $\mu^K((J,J)_a(U_\sigma)\ge c_0) \ge c_0^2/\sup_K\langle(J,J)_a(U_\sigma)^2\rangle_{\mu^K}$. The last arrow is $\liminf_K(x_K - y_K) \ge \liminf_K x_K - \limsup_K y_K$. □

16.7. **Comparison of backgrounds at consecutive cutoffs.** The convergence argument in Prop. 16.1(b) first fixes the coarse lattice and then removes the ultraviolet cutoff. For this step a comparison whose constant depends on the coarse volume is sufficient. We prove that comparison here. The separate estimate between a rough field and its background, used subsequently in that proposition, remains uniform in the cutoff.

We measure lengths in units of the fixed coarse lattice. Its torus has side $2L^n$ and volume $N = (2L^n)^3$. Let $\eta = L^{-j}$, $j\ge 1$, and $\eta' = \eta/L$. The two fine lattices have spacings η and η'. For a configuration U at spacing ξ, write

$$F_U(p) = \frac{\partial U(p) - \partial U(p)^{-1}}{2i\xi^2}, \qquad J(U) = D_U^* F_U, \qquad \mathcal{A}^\xi(U) = \xi^{-1}\sum_p \left(1 - \tfrac{1}{2}\operatorname{tr}\partial U(p)\right).$$

For quaternionic matrices we use the real scalar product $\operatorname{Re}\operatorname{tr}(X^*Y)/2$; its norm agrees with the operator norm and restricts to the stated norm on the Lie algebra. All sums over bonds and plaquettes below use positive orientations. The pairing and L^p norms at spacing ξ include the measure ξ^3. The notation D_U denotes the covariant exterior derivative. We call U regular at radius α when

$$\sup_p|\partial U(p) - 1| < \alpha\xi^2, \qquad \sup_b|J(U)(b)| < \alpha.$$

These are the constant-scale regularity conditions used in the background construction. All constants in this subsection depend only on the fixed blocking factor L. We use the Hölder exponent $1/2$.

Proposition 16.10 (Comparison at a fixed coarse lattice)**.** There are $\alpha_* > 0$, $C < \infty$ and a fixed room radius ρ_* with the following property. Suppose the constant-torus background setting is admitted, $L^n \ge 3L\mathfrak{M}$ and $L^n > 6\rho_*$, where $\mathfrak{M}$ is its fixed analytic block size. Suppose that U^A and

U^B are regular real critical backgrounds, at radius $0 < \alpha \le \alpha_*$, on the lattices of spacings η and η', respectively, and that they have the same coarse datum:

$$\mathcal{M}^j U^A = \mathcal{M}^{j+1} U^B.$$

Then there are a gauge transformation g and a real one-form A on the η lattice such that

$$(U^A)^g = e^{i\eta A}\mathcal{M}(U^B), \qquad \|A\|_\infty \le C\alpha N^{1/5}\eta^{1/5}. \tag{16.xii}$$

The constants α_*, C, ρ_* are independent of n, j and of the coarse datum. The factor $N^{1/5}$ is retained in the estimate. ■

The proof compares the first variations of the actions on the two lattices. The average of a critical field need not be critical. Its failure of criticality is small when tested against a coarse variation and a compatible fine variation. We first quantify this fact.

Lemma 16.11 (One-step regularity and the action defect)**.** Suppose U is regular at radius α on the η' lattice, with α sufficiently small and $L^n > 6\rho_*$. Put $W = \mathcal{M}(U)$. Then W is regular at radius $C\alpha$ on the η lattice. For every real fine one-form h,

$$\left| \frac{d}{dt}\Big[\mathcal{A}^{\eta'}(e^{it\eta' h}U) - \mathcal{A}^{\eta}(\mathcal{M}(e^{it\eta' h}U))\Big]_{t=0} \right| \le C\alpha\eta^{1/2}\|D_U h\|_1 + C\alpha\eta'\|h\|_{1,\sup}. \tag{16.xiii}$$

Here $\|h\|_{1,\sup} = \eta^3 \sum_{p'} \sup_{b\subset Y(p')} |h(b)|$; $Y(p')$ is the union of the four fine blocks at the corners of the coarse plaquette p'. ■

Proof. Choose ρ_* to be a fixed multiple of the radius in [7, Thm. BT], large enough to contain each averaging stencil with a collar of width four. At every coarse site that theorem, with constant level, supplies a gauge $U = e^{i\eta' P}$ on this cube with

$$|P| + |\nabla P| + |D_1^* D_1 P| + |\Delta P| \le C\alpha, \qquad |\nabla P(x) - \nabla P(y)| \le C\alpha |x-y|^{1/2}. \tag{16.xiv}$$

The enlarged cube fits in the torus by $L^n > 6\rho_*$. Its radius is fixed, so the smallness condition in that theorem follows by reducing α_*. In particular the constants in eq. 16.xiv do not depend on η or N.

We give the local estimates needed for averaging. On the fixed stencil of a coarse bond write

$$\Phi(a) = \frac{1}{i}\log \mathcal{M}(e^{ia}), \qquad \Phi(0) = 0, \qquad \Phi'(0) = Q_{\mathrm{lin}}.$$

The exponential and logarithmic formulas defining $\mathcal{M}$ are holomorphic on a ball of fixed radius, since each path contains a bounded number, depending only on L, of factors. Their first three derivatives are bounded on a smaller fixed ball by Cauchy's formula. The differential Q_{lin} is the block mean of straight path sums. Summing the oriented boundary of each path surface gives

$$dQ_{\mathrm{lin}} a(p') = L^2 \langle da\rangle_{p'}, \qquad \langle f\rangle_{p'} = L^{-5} \sum_{x\in B(p'_-)} \sum_{p\in S_x(p')} f(p). \tag{16.xv}$$

The interior edges cancel in each surface $S_x(p')$; its remaining four edges are precisely the four path sums in the left member. The mean has total mass one. Every fine plaquette occurs with total weight L^{-3} when p' is summed, and hence

$$\eta^3 \sum_{p'} \langle f\rangle_{p'} = \eta'^3 \sum_{p} f(p). \tag{16.xvi}$$

For later use we also record the elementary nonlinear curvature bounds. If $V = e^{i\xi P}$, $|P| \le a \le 1$ and $\xi \le 1/13$, then

$$F_V = D_1 P + \mathsf{n}(P), \qquad |\mathsf{n}(P)| \le 16 a^2.$$

Indeed, expand the four exponentials of a plaquette to first order; the remainder is at most the square of the sum of the four angles. Cauchy's formula applied to this quadratic remainder gives

$|\mathsf{n}(P') - \mathsf{n}(P)| \le 64a \max|P' - P|$. In the current, the difference of the two nonlinear remainders on neighboring plaquettes contributes at most $64a^2$ per direction when $|\nabla P| \le a$. The parallel transport in the other term differs from the identity by at most $2\xi a$, while $|F_V| \le 2a + 16a^2$. Thus

$$|J(V) - D_1^* D_1 P| \le 200a^2. \tag{16.xvii}$$

Let $P^M = \eta^{-1}\Phi(\eta' P)$ in the gauge eq. 16.xiv. The derivative bounds for Φ give $|P^M| + |\nabla P^M| \le C\alpha$. Split it into $P^{\mathrm{lin}} = \eta^{-1} Q_{\mathrm{lin}}(\eta' P)$ and P^{nl}. For $\Psi = \Phi - \Phi'(0)$, $|\Psi'(a)| \le C|a|$ and $|\Psi''(a)| \le C$. A mixed second difference of $\Psi(\eta' P)$ is the double integral of

$$\Psi''(a(s,t))[\partial_s a, \partial_t a] + \Psi'(a(s,t))\partial_s\partial_t a,$$

where $a(s,t)$ is the bilinear interpolation of its four translated arguments. Here $|a| \le C\eta'\alpha$, $|\partial_s a| + |\partial_t a| \le C\eta'\eta\alpha$ and $|\partial_s\partial_t a| \le C\eta'\eta\alpha$. After division by η^3 this bounds $|D_1^* D_1 P^{\mathrm{nl}}|$ by $C\alpha^2$.

There is a cancellation needed to bound the linear term without a negative power of η. Fix distinct directions μ, ν, λ, put $F_{\mu\sigma} = \nabla'_\mu P_\sigma - \nabla'_\sigma P_\mu$ and $g_\sigma(w) = \eta'^{-1}[F_{\mu\sigma}(w) - F_{\mu\sigma}(w - \eta' e_\sigma)]$. The linear Stokes identity gives

$$(D_1^* D_1 P^{\mathrm{lin}})_\mu = K_\lambda(g_\nu + g_\lambda) + (K_\nu - K_\lambda) g_\nu,$$

where

$$K_\sigma f(x) = L^{-6} \sum_{z \in B(x)} \sum_{0 \le i, i', r < L} f(z + i\eta' e_\mu + (i' - r)\eta' e_\sigma).$$

The first term is bounded by $C\alpha$ because $g_\nu + g_\lambda = (D_1^* D_1 P)_\mu$ and K_λ is a positive average. To treat the second, multiply P by a product cutoff equal to one on the stencil, supported in its collar, whose first and second differences are bounded independently of the mesh, and extend it by zero. The product rule and eq. 16.xiv give $\|\Delta' \widetilde{P}\|_\infty \le C\alpha$. The two Fourier multipliers expressing $(K_\nu - K_\lambda) g_\nu$ in terms of $\Delta' \widetilde{P}_\nu, \Delta' \widetilde{P}_\mu$ are

$$-\kappa(\vartheta) \frac{\overline{d}_\nu d_\mu}{\sum_\sigma |d_\sigma|^2}, \qquad \kappa(\vartheta) \frac{|d_\nu|^2}{\sum_\sigma |d_\sigma|^2},$$

where $d_\sigma = e^{i\vartheta_\sigma} - 1$, $s(t) = L^{-1} \sum_{|q| \le (L-1)/2} e^{iqt}$, $e_L(t) = L^{-1} \sum_{q=0}^{L-1} e^{iqt}$ and

$$\kappa = s(\vartheta_\mu) e_L(\vartheta_\mu) s(\vartheta_\nu) s(\vartheta_\lambda) [s(\vartheta_\nu)^2 - s(\vartheta_\lambda)^2].$$

The last bracket vanishes to second order at zero. Each multiplier m therefore satisfies $|\partial^r m(\vartheta)| \le C|\vartheta|^{2-r}$ for $0 \le r \le 2$ near zero and is smooth elsewhere. These are also weak derivative bounds: the boundary terms on a sphere of radius δ tend to zero as $\delta \downarrow 0$. Twice integrating by parts and using Bessel's inequality gives

$$\sum_{z \in \mathbb{Z}^3} |\widehat{m}(z)| \le \left(\sum_z (1 + |z|^2)^{-2} \right)^{1/2} \|m - \Delta_\vartheta m\|_{L^2} < \infty.$$

The convolution kernels thus have ℓ^1 norms bounded only in terms of L. This bounds the second term by $C\alpha$. Together with eq. 16.xvii it proves $|J(W)| \le C\alpha$.

We prove the curvature bound and the action estimate together. In $Y(p')$ take the comb gauge based at p'_-. A comb comparison with the elementary plaquettes gives $U = e^{ia_0}$ in this gauge with $|a_0| \le C\varpi$, where $\varpi = \sup|\partial U - 1| \le \alpha\eta'^2$. Put $G_p = \partial U(p) - 1$ in this fixed gauge and $\bar{G} = \langle G \rangle_{p'}$. By eq. 16.xv, the map

$$R(a) = \partial \mathcal{M}(e^{ia})(p') - 1 - L^2 \langle \partial e^{ia} - 1 \rangle_{p'}$$

has zero value and derivative at zero. The holomorphic bounds above imply

$$|R(a_0)| \le C\varpi^2, \qquad |R'(a_0)\dot{a}| \le C\varpi|\dot{a}|. \tag{16.xviii}$$

Consequently $|\partial W - 1| \le L^2\varpi + C\varpi^2 \le C\alpha\eta^2$, completing regularity of W.

In the gauge eq. 16.xiv, curvature varies across a stencil by $C\alpha\eta^{1/2}$ in physical units: this follows from the Hölder bound for ∇P and the Lipschitz bound for $\mathsf{n}(P)$. Passing to the comb gauge changes the comparison of two curvatures by at most $C\alpha^2\eta$ in those units. Indeed, each comb has length $C\eta$, each bond angle is at most $C\eta'\alpha$, and $|\operatorname{Ad}(v)-1| \le 2|v-1|$. Thus

$$|G_p - \bar G| \le C\alpha\eta'^2\eta^{1/2}. \tag{16.xix}$$

The identity $1-\frac{1}{2}\operatorname{tr} v = |v-1|^2/2$ for $v \in SU(2)$ and eq. 16.xvi give the exact defect

$$\mathcal{A}^{\eta'}(U) - \mathcal{A}^{\eta}(W) = \eta^{-1}\sum_{p'} \delta_{p'}, \qquad \delta_{p'} = \frac{L^4}{2}\langle |G-\bar G|^2\rangle - L^2(\bar G, R) - \frac{1}{2}|R|^2.$$

Differentiate at $U_t = e^{it\eta' h}U$, retaining the comb gauge chosen at $t=0$. If $s_{p'} = \sup_{Y(p')}|h|$, the product rule on a plaquette and the derivative of the local logarithm give

$$|\dot G_p| \le \eta'^2 |D_U h(p)| + 4\varpi\eta' s_{p'}, \qquad |\dot a| \le 2\eta' s_{p'}, \qquad |\dot R| \le C\varpi\eta' s_{p'}.$$

In the derivative of the variance, the term containing $\dot{\bar G}$ vanishes because $\langle G - \bar G\rangle = 0$. Therefore

$$\dot\delta_{p'} = L^4\langle (G-\bar G, \dot G)\rangle - L^2(\bar G, \dot R) - L^2(R, \langle \dot G\rangle) - (R, \dot R).$$

Substitute eq. 16.xviii and eq. 16.xix, use $\eta' = \eta/L$ and $\alpha \le 1$, and divide by η. This yields

$$\eta^{-1}|\dot\delta_{p'}| \le C\eta^3\big[\alpha\eta^{1/2}\langle |D_U h|\rangle_{p'} + \alpha\eta' s_{p'}\big].$$

Summing and applying eq. 16.xvi proves eq. 16.xiii. □

Proof of Prop. 16.10. Put $W = \mathcal{M}(U^B)$. The preceding lemma makes W regular at radius $C\alpha$, and it has the same reduced datum as U^A. We check the setting of the companion's chart results. Take $\mathfrak{M} = M_{\mathrm{ps}}$ and choose once and for all $R_1 \ge L$ with $R_1\mathfrak{M} \ge w_S$. At depths $0,\dots,j$ let every region be the whole torus, and let the next region be empty. The required $\mathfrak{M}$-cubes exist since $\mathfrak{M} \le L^n$; the separation distances are infinite. Every layer below j is empty, so both its full and its reduced determining rows are empty. At level j all bonds are interior determining rows. There are no inward or outward rows, seam nodes or edge rows. Thus both determining maps are exactly $\mathcal{M}^j$, and both row spaces are the one-forms on the unit lattice. The local scale is one everywhere, so every weighted norm is the corresponding supremum. The same conclusions hold at the finer cutoff with depth $j+1$.

The companion's chart uses the matrix divergence regularity class. The current bound implies that class with radius $r + r^2$ when the curvature radius is r: the traceless part of the plaquette divergence is the current, and its scalar part is a difference of $1-\frac{1}{2}|\partial U - 1|^2$ at neighboring plaquettes. This is the real $SU(2)$ conversion of Lem. YM3_I:9.4. It costs only a fixed multiple of α here. The terminal torus has $2L^n \ge 6LM_{\mathrm{ps}}$ sites per direction. All geometric and regularity hypotheses of the common-chart argument in the proof of Prop. YM3_I:37.3 are therefore satisfied. After reducing α_* it gives

$$(U^A)^g = e^{i\eta A}W, \qquad RD_W^* A = 0, \qquad q(A) = 0, \qquad \|A\|_X \le C\alpha, \tag{16.xx}$$

where $\|A\|_X$ controls both $\sup|A|$ and the covariant first differences. Write $F(a) = \mathcal{A}^{\eta}(e^{i\eta a}W)$ and $F_B(b) = \mathcal{A}^{\eta'}(e^{i\eta' b}U^B)$. Here q is the nonlinear coarse constraint and $K_a = q'(a)$. Lem. YM3_I:30.8 converts criticality on the actual averaging fiber into the multiplier equation. Applying Prop. YM3_I:30.9 with zero constraint value bounds its multiplier, first at the reference W and then at U^B. Its gradient at the reference is bounded by $C\alpha$ in both cases, by regularity. It follows that on the common unit lattice

$$F'(A) = K_A^*\omega_A, \qquad F_B'(0) = (K_0^B)^*\omega_B, \qquad \|\omega_A\|_\infty + \|\omega_B\|_\infty \le C\alpha. \tag{16.xxi}$$

Here the two constraints are the actual relative constraints, before any orthogonal change of row frame. Their averaging recursions agree after the first fine step, as checked below.

We recall the precise linear and nonlinear estimates needed from Props. YM3_I:29.1 and YM3_I:30.5. With Q the corner linear constraint and T its fixed orthogonal row transport, $T^{-1}q(a) = Qa + \mathsf{C}(a)$, $\mathsf{C}(0) = 0$, and

$$\|\mathsf{C}'(a)\|_{2\to 2} \le C\alpha, \|\mathsf{C}'(a)^*\chi - \mathsf{C}'(b)^*\chi\|_\infty \le C\|a-b\|_\infty\|\chi\|_\infty \tag{16.xxii}$$

on a fixed multiple of the α ball containing the segment $[0, A]$. The first inequality follows also directly from the stated row and column bounds by applying them to the symmetric operator $\mathsf{C}'(a)^*\mathsf{C}'(a)$. Put $\Delta_R = F''(0) + D_W R D_W^*$ and $G = (\Delta_R + Q^*Q)^{-1}$, $M = QGQ^*$, $H = GQ^*M^{-1}$ and $\widetilde{G} = G - GQ^*M^{-1}QG$. These operators obey

$$QH = 1, \quad \widetilde{G}\Delta_R = 1 - HQ, \quad \Delta_R H = Q^*(M^{-1} - 1), \quad \|H\|_{2\to 2} + \|\widetilde{G}\|_{2\to 2} \le C. \tag{16.xxiii}$$

Thm. YM3_I:I gives positivity of G in this real regular setting, and Prop. YM3_I:29.1 gives positivity of M. Thus both M and G are positive definite, and $\widetilde{G}$ is positive semidefinite. To see the last assertion directly, complete the square in $(f, Gf) - (QGf, M^{-1}QGf)$. The L^2 bounds follow from the sup bounds for $\widetilde{G}, H, H^*$: a symmetric operator bounded by C in any norm has all eigenvalues bounded by C, and one applies this also to H^*H.

These facts imply coercivity along A. For $Qa = 0$, $a = \widetilde{G}\Delta_R a$ and the spectral inequality $\widetilde{G}^2 \le C\widetilde{G}$ give $(a, \Delta_R a) \ge c\|a\|_2^2$. For arbitrary h, set $a = h - HQh$. The mixed term in its quadratic form vanishes by eq. 16.xxiii, whereas the remaining HQh term is at least $-\|Qh\|_2^2$. Hence, whenever $C\|Qh\|_2 \le \|h\|_2$,

$$(h, \Delta_R h) \ge c(\|h\|_2 - C\|Qh\|_2)^2 - \|Qh\|_2^2.$$

The constraint in eq. 16.xx gives $QA = -\mathsf{C}(A)$, so eq. 16.xxii implies $\|QA\|_2 \le C\alpha\|A\|_2$. The gauge term vanishes on A. Finally $F''(0) = D_W^* D_W + \Delta'$ with $\|\Delta'\|_{2\to 2} \le C\alpha$ by the real Hessian bounds in the same results. Combining the two resulting lower bounds and reducing α_* proves

$$F''(0)[A, A] \ge cx^2, \qquad x^2 = \|D_W A\|_2^2 + \|A\|_2^2. \tag{16.xxiv}$$

We next construct the fine test form used to compare first variations. For a fine bond $b = (x, \mu)$, let Y be the lower corner of its coarse cell and $r = x - Y$, $0 \le r_\sigma < \eta$. Interpolate the four parallel coarse bonds c_s, with initial points $Y + \eta s$, $s \in \{0,1\}^{\{\nu:\nu\neq\mu\}}$, using the weights

$$w_s(x) = \prod_{\nu\neq\mu} (r_\nu/\eta)^{s_\nu}(1 - r_\nu/\eta)^{1-s_\nu}.$$

Transport their values to x along the coordinate combs in U^B and denote their weighted mean by $h^\uparrow(b)$. There is no length factor in this definition: h and $h^\uparrow$ are potentials in the same physical units. Define $h^\downarrow$ by

$$\eta h^\downarrow = Q_{\mathsf{m}}(U^B)(\eta' h^\uparrow),$$

where Q_{m} is the derivative of the relative one-step average, including its straightening gauge. The elementary interpolation estimates are

$$\begin{aligned} \|D_{U^B} h^\uparrow\|_1 &\le \|D_W h\|_1 + C\alpha\eta N\|h\|_\infty, \\ \|h^\uparrow\|_{1,\sup} &\le 3N\|h\|_\infty, \\ \|h - h^\downarrow\|_1 &\le C\eta N\|h\|_X. \end{aligned} \tag{16.xxv}$$

Here is the finite-stencil verification. In a comb gauge on the box containing the relevant neighboring cells, the fine angles are bounded by $C\alpha\eta\eta'$. Every comparison path has at most CL bonds. Replacing its transport by the identity changes the interpolated value by $C\alpha\eta^2 \sup|h|$ and its covariant curl by $C\alpha\eta \sup|h|$. Without these errors, subtracting the two linear interpolants in a plaquette gives the linear interpolation of $D_W h$ in the one transverse direction. Its weights are nonnegative; their total fine mass over each coarse plaquette is L^3. This proves the first estimate on multiplying by η'^3 and summing. The second follows from $|h^\uparrow| \le \|h\|_\infty$ and the $3N\eta^{-3}$ coarse

plaquettes. At zero fine angles the relative derivative Q_{m} is straight path averaging; applied to the interpolant it is a convex mean of neighboring values of h. At the given fine angles its derivative differs by $C\alpha\eta^2 \sup|h|$ after division by η, by the bounded second derivative of the relative average. Neighbors differ by at most $3\eta\|\nabla^W h\|_\infty + C\alpha\eta^2\|h\|_\infty$. This gives the last inequality in eq. 16.xxv.

Let $a_1(t)$ be the relative logarithm of the first average of $e^{it\eta' h^\uparrow}U^B$. It has derivative $a_1'(0) = \eta h^\downarrow$. After this first step, the reference configurations in the two averaging recursions are identical. Their subsequent relative logarithms therefore agree, by induction over the remaining j steps. Differentiating the final constraint gives

$$K_0^B h^\uparrow = K_0 h^\downarrow. \tag{16.xxvi}$$

Gauge invariance of the action removes the straightening gauge at the first step. Thus its action defect is exactly

$$E(h) = F_B'(0)[h^\uparrow] - F'(0)[h^\downarrow].$$

Put $d = F'(0) - K_0^*\omega_B$. From eq. 16.xxi and eq. 16.xxvi, $(d, h^\downarrow) = -E(h)$, and hence

$$F'(0)[h] = (d, h - h^\downarrow) + (\omega_B, K_0 h) - E(h). \tag{16.xxvii}$$

Regularity of W, the transpose row bound for K_0 and eq. 16.xxi give $\|d\|_\infty \le C\alpha$. Apply eq. 16.xiii and eq. 16.xxv with $h = A$, using $\|A\|_X \le C\alpha$ and $\|D_W A\|_1 \le (3N)^{1/2}\|D_W A\|_2$. We obtain

$$|(d, A - A^\downarrow)| + |E(A)| \le C\alpha\eta^{1/2}N^{1/2}x + C\alpha^2\eta N. \tag{16.xxviii}$$

The constraint multipliers contribute only quadratically. To see this, for fixed χ the Hessian of $(\chi, \mathsf{C}(a))$ is symmetric and has sup operator norm at most $C\|\chi\|_\infty$ by eq. 16.xxii. Its L^2 operator norm has the same bound, by the eigenvalue argument used above. Since $q(0) = q(A) = 0$, the fundamental theorem of calculus gives

$$T^{-1}K_0 A = -\int_0^1 [\mathsf{C}'(tA) - \mathsf{C}'(0)]A\,\mathrm{d}t, \qquad T^{-1}K_A A = \int_0^1 [\mathsf{C}'(A) - \mathsf{C}'(tA)]A\,\mathrm{d}t.$$

Pair these identities with $T^{-1}\omega_B$ and $T^{-1}\omega_A$, whose suprema are unchanged by the orthogonal row transport. Consequently

$$|(\omega_B, K_0 A)| + |F'(A)[A]| \le C\alpha\|A\|_2^2. \tag{16.xxix}$$

For completeness we bound the third derivative of the action; a bound on four separate bond derivatives would lose powers of the mesh. Put $\phi(t) = F(tA)$. At a plaquette with reference holonomy P, its varied holonomy has the form $Z(t) = e^{tX_1}\cdots e^{tX_4}P$, where $|X_i| \le \eta a_p$ and a_p is the largest $|A|$ on its boundary. Write $Z' = \Sigma Z$. Differentiating the four partial products gives

$$|\Sigma'| \le 12\eta^2 a_p^2, \qquad |\Sigma''| \le 72\eta^3 a_p^3, \qquad \Sigma(0) = i\eta^2 Y, \quad |Y - D_W A(p)| \le C\alpha\eta a_p.$$

The last error is precisely the transport around P rather than the identity; $|P - 1| \le C\alpha\eta^2$. Set $y_p = |Y| + 12a_p^2$. Then $|\Sigma(t)| \le \eta^2 y_p$ and $|Z(t) - 1| \le \eta^2(C\alpha + y_p)$ for $0 \le t \le 1$. Both Σ'' and Σ^3 are traceless, while $\Sigma^2 = -|\Sigma|^2 1$, since $\Sigma \in \mathfrak{su}(2)$. In

$$\frac{d^3}{dt^3}\left(1 - \tfrac{1}{2}\operatorname{tr} Z\right) = -\tfrac{1}{2}\operatorname{tr}[\Sigma'' + 2\Sigma'\Sigma + \Sigma\Sigma' + \Sigma^3]Z$$

one may therefore replace Z by $Z - 1$ in the first and last terms. With $a_p + |D_W A(p)| \le C\alpha$ the displayed bounds give

$$\left|\frac{d^3}{dt^3}\left(1 - \tfrac{1}{2}\operatorname{tr} Z\right)\right| \le C\eta^4\alpha(|D_W A(p)|^2 + a_p^2).$$

Each bond belongs to four plaquettes. Multiplication by η^{-1} and summation therefore prove $|\phi'''(t)| \le C\alpha x^2$.

Taylor's formula, eq. 16.xxiv, eq. 16.xxvii, eq. 16.xxviii and eq. 16.xxix now imply

$$cx^2 \le C\alpha x^2 + C\alpha\eta^{1/2}N^{1/2}x + C\alpha^2\eta N.$$

Choose α_* so the first term is absorbed. Young's inequality absorbs half of the remaining left member and yields

$$\|A\|_2^2 + \|D_W A\|_2^2 \le C\alpha^2 \eta N. \tag{16.xxx}$$

Finally let $s = \|A\|_\infty$ and choose $r = C\alpha$ so that $\|A\|_X \le r/2$. The norm of a component changes by at most $\eta r/2$ across an adjacent site: parallel transport is an isometry. If $s/r < \eta$, then $s < r\eta$. Otherwise set $R = \lfloor s/(r\eta) \rfloor \ge s/(2r\eta)$. On the lattice ℓ^1 ball of radius R about a maximizing site, that component has norm at least $s/2$. Its physical radius is at most $1/2$, so its sites are distinct on the torus. This ball has more than $4R^3/3$ sites, giving

$$\|A\|_2^2 \ge \eta^3 \frac{4R^3}{3} \frac{s^2}{4} \ge \frac{s^5}{24 r^3}.$$

Thus $s \le \max\{r\eta, (24r^3\|A\|_2^2)^{1/5}\}$. Substitution of eq. 16.xxx, with $N \ge 1$ and $\eta \le 1$, proves eq. 16.xii. □

17. A finite-size criterion for confinement

Let $Z(n_1, n_2, n_3; \beta)$ be the Wilson partition function on the periodic lattice box with the indicated integer sides. Its free-energy defect on a half box is

$$\mathsf{e}(l) = 2 \log Z(l, 2l, 2l; \beta) - \log Z(2l, 2l, 2l; \beta).$$

The following implication isolates the remaining quantitative confinement problem in one explicit inequality.

Sketch P (A uniform finite-size criterion). Suppose that for some physical side ℓ_0, label $b > 0$, and $0 < x_0 \le 0.069$,

$$\mathsf{e}(l_K) \le x_0, \qquad l_K = \frac{\ell_0}{2a_K}, \qquad \beta = L^K b,$$

at every sufficiently large cutoff, with $l_K \in \mathbb{N}$. Put $\theta = 12.9 x_0 < 1$. Then the periodic Gibbs limit and its continuum limit are unique, the vacuum is unique, and the reconstructed scalar action-density theory has a positive gap. The characteristic gap and area constants are

$$m_* = \frac{2\log(1/\theta)}{\ell_0}, \qquad \sigma_* = \frac{2\log(1/\theta) + 1.67}{\ell_0^2},$$

and the cutoff states obey $0 \le \nu_K(W_{R\times T}) \le e^{-\sigma_* RT}$ for coordinate rectangles with physical sides $R, T \in a_K\mathbb{N}$. The gap and continuum assertions are understood in the construction's admitted label range. ■□

Sketch of the argument. On one lattice, trace the transfer operator over a fixed section. The defect compares the trace at one height with the trace at twice that height, and hence controls the thermal eigenvalue weights above the largest eigenvalue. Change the time direction to compare the necessary transverse sections. Successive doublings give

$$\mathsf{e}(2l) \le 12.9\, \mathsf{e}(l)^2, \qquad \mathsf{e}(2^j l) \le \theta^{2^j}/12.9.$$

This is the decisive improvement: a small initial defect becomes exponentially small in the linear size.

The cube defect is a positive sum of the three one-direction defects. Summing its dyadic relation to the pressure bounds the finite-volume excess. The resulting thermal estimates give a simple transfer top and a lower bound for the excited levels. They also compare the local laws on successive tori in total variation, producing the unique periodic Gibbs state and exponential clustering.

The vortex-tube recursion bounds the small electric-flux weight. Resolving the transfer space into flux sectors then gives an asymptotically linear lower bound for electric-flux energies. Cutting a rectangular Wilson loop into layers and applying the transfer estimates gives the area bound. The constants survive the physical scaling because the assumed criterion uses one ℓ_0 and one x_0 for

every large cutoff. Uniform volume comparison then permits the cutoff and volume limits to be interchanged, and carries the gap to the reconstructed continuum theory.

This is a genuine conditional implication. Establishing the displayed box inequality at a physical scale is still the outstanding confinement step. The source's small-torus estimates can exclude it at small sides. A test at one cutoff proves a uniform criterion only when its explicit cutoff-comparison error fits inside the margin below the threshold. Finally, the cutoff area law does not construct a nonzero random sharp-loop observable in the continuum; that is distinct from the expectation renormalization in the Wilson-loop sketch.

17.1. **A finite-lattice criterion for confinement.** Let $Z(n_1, n_2, n_3)$ be the periodic partition function at coupling $\beta > 0$. We prove a fixed-lattice implication from

$$2 \log Z(l, 2l, 2l) - \log Z(2l, 2l, 2l) \le x_0, \qquad l \ge 3, \qquad 0 < x_0 \le 0.069.$$

The transfer spectrum controls the doubling defect. A vortex-tube recursion then gives area laws, electric-flux bounds and a gap over finite sections; excess-pressure estimates give uniqueness among periodic Gibbs states and convergence of periodic boxes of arbitrary shape.

All constants and parity restrictions are stated below. Physical conclusions uniform in the cutoff require the criterion at a fixed physical side for every sufficiently large cutoff. The stronger numerical tension and the simpler tension of Prop. 17.8 are kept separate.

17.1.1. *Notation and hypotheses. Boxes.* A *box* is a triple $\underline{n} = (n_1, n_2, n_3)$ of integers ≥ 3, $|\underline{n}| = n_1 n_2 n_3$; the cube of side n is (n, n, n). On the periodic lattice of a box a configuration U gives each bond b an element $U(b)$ of $G = \mathrm{SU}(2)$, with $U(\bar{b}) = U(b)^{-1}$ for the reversed bond; dU is the product of the Haar probability measures, $\partial U(p)$ the ordered product along the boundary of a plaquette p, and $a_p = 1 - \frac{1}{2} \operatorname{tr} \partial U(p) \in [0, 2]$. For $\beta \ge 0$ we write $Z(\underline{n}) = \int e^{-\beta \sum_p a_p} \, \mathrm{d}U$, which lies in $[e^{-6\beta|\underline{n}|}, 1]$, $F = \log Z$, and $\langle \cdot \rangle_{\underline{n}}$ for the mean under $Z^{-1} e^{-\beta \sum_p a_p} dU$. The pressure is $P = \inf_{\underline{n}} F(\underline{n})/|\underline{n}| \in [-6\beta, 0]$, the *excess* of a box is $\delta(\underline{n}) = F(\underline{n}) - |\underline{n}| P \ge 0$, and $\delta_{\mathrm{c}}(n) = \delta(n, n, n)$. For integers $l \ge 3$ we put

$$\mathsf{e}(l) = 2F(l, 2l, 2l) - F(2l, 2l, 2l), \quad d_1(l) = 2F(l, l, l) - F(2l, l, l), \quad d_2(l) = 2F(l, 2l, l) - F(2l, 2l, l), \quad X(l) = 8F(l, l, l) - F(2l, 2l, 2l).$$

Twists. For directions $\mu < \nu$ with third direction ρ, a *column* of the pair $\mu\nu$ is a set of plaquettes $p_{\mu\nu}(x + j e_\rho)$, $0 \le j < n_\rho$. For $z = (z_{12}, z_{13}, z_{23}) \in \{0, 1\}^3$, Z^z is Z with the sign of $\operatorname{tr} \partial U(p)$ reversed on one column of each pair $\mu\nu$ with $z_{\mu\nu} = 1$, and the *twisted box law* is the probability measure with this weight; $|z| = z_{12} + z_{13} + z_{23}$. We write $Z^- = Z^{(0,1,0)}$, $a_\pm = \frac{1}{2}(Z \pm Z^-)$ and $z_\pm = a_\pm / Z$; z_- is the *flux weight*. On a cube, r_j is the quotient Z^z/Z for $|z| = j$ (Lem. 17.10(c)); so $r_1 = Z^-/Z = 1 - 2z_-$.

Sections. A direction μ_0 taken as time has the side $t = n_{\mu_0}$; the two others, $\nu_1 < \nu_2$, span the *section* $\Sigma = (N_1, N_2) = (n_{\nu_1}, n_{\nu_2})$, and $Z(t, \Sigma) = Z(t, N_1, N_2)$, likewise for F and δ; by Lem. 17.10 these numbers do not depend on the order of the sides. Over Σ, $\lambda_0 > \lambda_1 \ge \lambda_2 \ge \cdots \ge 0$ are the eigenvalues of the transfer operator of Lem. 17.10, with $Z(t, \Sigma) = \sum_i \lambda_i^t$ for $t \ge 3$; λ_0 is the *vacuum*, $\mathsf{r}_t(\Sigma) = \sum_{i \ge 1} (\lambda_i/\lambda_0)^t$ the *thermal number*, and a number $y > 0$ *lies at least η above the vacuum* if $\log(\lambda_0/y) \ge \eta$.

The eigenvalues fall into four flux sectors $e \in \{0, 1\}^2$ (Lem. 17.10); Λ_e is the top of the sector e, the sectors $e \ne 00$ are those with electric flux, and over a section (N_2, N), with the first direction as time, so that $\nu_1 = 2$ runs along the side N_2 and $\nu_2 = 3$ along the side N, $\Lambda_- = \max\{\Lambda_{01}, \Lambda_{11}\}$ is the top of the sectors that are odd under J_2, that is, under the *flip of the side N*: the map ι_2 of Lem. 17.10, which changes the sign of the bonds along the side N that start on one line along the side N_2. These sectors carry an electric flux along the side N, a flux line of the length N. With $\lambda_-(t, N) = \lim_s Z^-(t, s, N)^{1/s}$, the number $\varepsilon_{\mathrm{v}}(t, N) = \log(\lambda_0(t, N)/\lambda_-(t, N)) \ge 0$ is the free

energy per unit length of the vortex line that runs along the second direction of the box (t, s, N); $\varepsilon_{\mathrm{v}}(l') = \varepsilon_{\mathrm{v}}(l', l')$.

Functions. $\varpi(x) = x/(1+\sqrt{1-x^2})$ on $[0,1]$; $\omega(D) = 2\varpi(e^D-1)^2$ on $[0, \log 2]$; $Y_1(x) = 2x + \omega(x)$, $Y_2 = 2Y_1 + \omega(Y_1)$, $\Psi = \omega(Y_2)$ and $\Phi = 4\omega + 2\omega(Y_1) + \omega(Y_2)$, where the arguments lie in $[0, \log 2]$.

Cells. For an integer $h \geq 2$, $N = 2h$, and a point $c \in (\mathbb{Z} + \frac{1}{2})^3$, the *cell* $\mathsf{C}_h(c)$ is the cube of the h layers below c, formed by the sites x with $c_\mu - h < x_\mu < c_\mu$, $\mu = 1, 2, 3$; a *function of the cell* is a bounded Borel function of the bonds with both ends in it. On the cube (N, N, N) the reflections $\mathfrak{r}_\mu \colon x_\mu \mapsto 2c_\mu - x_\mu$ map the lattice onto itself, each fixing the two planes $x_\mu = c_\mu$ and $x_\mu = c_\mu + h$; for a function A of the cell and $\epsilon \in \{0,1\}^3$ the *copy* $A^{[\epsilon]}$ is A composed with the $\mathfrak{r}_\mu$ with $\epsilon_\mu = 1$ and complex conjugated if $|\epsilon|$ is odd, and

$$\|\!| A |\!\|_N = \Big\langle \prod_{\epsilon \in \{0,1\}^3} A^{[\epsilon]} \Big\rangle^{1/8}_{(N,N,N)},$$

the mean being a number ≥ 0 by Lem. 17.21.

The infinite lattice. $\Omega = G^{\mathbb{B}}$, $\mathbb{B}$ the bonds of $\mathbb{Z}^3$; local functions are bounded Borel functions of finitely many bonds. A *cube of h layers* is $Q(x_0, h) = \{x_0 + (j_1, j_2, j_3) : 1 \leq j_\mu \leq h\}$; a function of the bonds with both ends in Q is a function on the configurations of every box with sides $\geq h+1$ through the projection of $\mathbb{Z}^3$, which is injective on Q, and box laws are laws on Ω by periodic extension. A *Gibbs state* is a probability measure on Ω under which, given the bonds outside a finite set $\mathfrak{A}$, those of $\mathfrak{A}$ have the density proportional to $e^{-\beta \sum a_p}$, the sum over the plaquettes that meet $\mathfrak{A}$; it is *periodic* if it is invariant under a subgroup of finite index of the translations.

The *symmetries of the lattice* are the maps $x \mapsto Ox + v$, O a signed permutation and $v \in \mathbb{Z}^3$. A *lattice mirror* is a plane $\{x_\mu = c\}$, $c \in \frac{1}{2}\mathbb{Z}$ (a *site mirror* if $c \in \mathbb{Z}$, a *bond mirror* otherwise), or a plane $\{x_\mu \pm x_\nu = c\}$, $\mu \neq \nu$, $c \in \mathbb{Z}$ (a *diagonal mirror*); a state s is *reflection positive* at it if $s(\overline{\mathfrak{r}A}\, A) \geq 0$ for every local A of the bonds in one closed side of the mirror, $\mathfrak{r}$ the reflection in it (bonds in a site mirror or a diagonal mirror are admitted; at a bond mirror the bonds that cross it lie in neither closed side). For a closed lattice path C, $W_C = \frac{1}{2}\operatorname{tr} U(C)$, $U(C)$ the ordered product along C; W_C is real, $|W_C| \leq 1$, and it depends neither on the starting point nor on the sense; $|C|$ is the number of bonds of C, and W_p is the loop along the boundary of one plaquette.

A *rectangle* $R \times T$ of the (μ, ν)-plane runs R steps along e_μ, T along e_ν, and back.

Hypotheses. The *criterion at the side l* is the condition: $\beta > 0$, $l \geq 3$ is an integer of either parity, $0 < x_0 \leq 0.069$, and $\mathsf{e}(l) \leq x_0$. Its numbers are $\theta = 12.9\, x_0 \leq 0.8901$, $\Lambda = \log \frac{1}{\theta} \geq 0.11642$, $\varsigma = 2\Lambda + 1.67$, $\mathsf{q} = \varsigma/2\Lambda = 1 + 0.835/\Lambda \leq 8.173$, and

$$\sigma^+ = \frac{\varsigma}{4l^2} = \frac{\log \frac{1}{\theta} + 0.835}{2l^2}, \qquad \sigma^+ l^2 = \frac{1}{2} \log \frac{e^{0.835}}{12.9 x_0} \geq 0.4757. \tag{17.i}$$

The *cube condition at the side l_0* is $X(l_0) \leq \frac{1}{20}$. For the physical scale let $L \geq 2$ be an integer and $a_K = L^{-K}$. The *uniform criterion* holds if there are $\mathsf{b} > 0$, $l_* > 0$, an integer K_0 and one $x_0 \in (0, 0.069]$ such that for every integer $K \geq K_0$ the number $l_* L^K$ is an integer ≥ 3 and the criterion holds at the side $l_* L^K$ and the coupling $L^K \mathsf{b}$ with this x_0. In the frame L is the blocking factor, a_K the spacing at the cutoff K, $\mathsf{b} = a_K \beta$ the label, and $\ell_0 = 2l_*$ the physical side of the cube of the criterion. In Lem. 17.24, in the steps of proofs that use it, and in Prop. 17.6, L is the blocking factor, an odd integer ≥ 13; the L of the uniform criterion is any integer ≥ 2, independent of it. We write x/yz for $x/(yz)$.

17.1.2. *Why doubling turns the box test into a gap.* The box defect cancels the extensive part of the free energy. At a fixed transverse section Σ, write the transfer eigenvalues as $\lambda_0 \geq \lambda_1 \geq \cdots \geq 0$ and put

$$\mathsf{r}_l(\Sigma) = \sum_{i \geq 1} (\lambda_i/\lambda_0)^l.$$

This includes any multiplicity of the largest eigenvalue. Factoring λ_0^l out of the transfer trace isolates the thermal correction $1 + \mathsf{r}_l$. The difference between twice the log trace at height l and the log trace at height $2l$ is therefore a function of these corrections, with the largest-eigenvalue contribution cancelled. The source proves

$$\log(1 + \mathsf{r}_l(\Sigma)) \le d(l; \Sigma) \le 2\log(1 + \mathsf{r}_l(\Sigma)).$$

Small defect thus controls all excited thermal weights together, including a possible top multiplicity. This is stronger than a comparison with one selected eigenvector.

Doubling time squares the eigenvalue ratios. To make a statement about larger cubes, however, one must also enlarge the transverse section. The proof alternates these two operations and changes which coordinate is time. It tracks the three sections (l, l), $(2l, l)$ and $(2l, 2l)$, bounds the intermediate defects, and closes the resulting scalar recursion. The numerical estimate that matters is

$$\mathsf{e}(2l) \le 12.9\mathsf{e}(l)^2, \qquad 0 < \mathsf{e}(l) \le 0.069.$$

With $\theta = 12.9x_0 < 1$, induction gives $\mathsf{e}(2^j l) \le \theta^{2^j}/12.9$. The exponent grows like the linear size, although the argument proceeds by doubling. This is the source of the exponential estimates that follow from a successful finite-size test.

The cube excess is controlled by a dyadic identity. If $\delta_{\mathrm{c}}(n)$ is the cube's finite-volume excess and $X(n)$ is the sum of its three directional defects, the source has

$$X(n) = 8\delta_{\mathrm{c}}(n) - \delta_{\mathrm{c}}(2n), \qquad \delta_{\mathrm{c}}(n) = \sum_{j\ge 0} 8^{-j-1} X(2^j n).$$

The pressure term cancels in the first relation. The recursion makes the second sum converge with an exponential bound. The transfer comparison then yields a simple top, a quantitative gap and rapid comparison of local laws on successive tori. In parallel, the vortex-tube estimates control the small electric-flux weight. Resolving the transfer trace into its flux sectors turns that bound into an energy cost for a long flux tube and then a rectangular Wilson area law. The gap and area law are two uses of the same small finite-volume excess, through different transfer observables.

For the continuum implication the tested box must have one physical side. Its lattice side is $l_K = \ell_0/(2a_K)$ and its coupling is $\beta_K = L^K b$. The criterion requires the same bound with the same x_0 for every sufficiently large K. This is exactly the uniformity that carries the finite-lattice gap and area constants to physical units. It also makes the volume-comparison error uniform while the cutoff is removed, allowing the volume and cutoff limits in this implication to commute.

Testing one cutoff is a different assertion. It can certify the uniform criterion only if the proved cutoff-comparison error fits within the available margin below the threshold. The detailed statement retains that margin. Likewise, small-torus asymptotics can rule out the criterion at small physical sides; a statement that the test is finite does not verify it at some larger side. Establishing it on the desired continuum trajectory remains the uncompleted confinement step.

17.1.3. *Principal results.*

Theorem AW (One lattice)**.** Assume the criterion at the side l.

(a) *The state.* For $n = 2^k l$, $k \ge 1$, and every bounded Borel function A of the bonds of a cube of h layers, $1 \le h \le n/2 - 1$,

$$|\langle A\rangle_{(n,n,n)} - \langle A\rangle_{(2n,2n,2n)}| \le 1.29 \sup|A|\, \theta^{n/2l}.$$

So $\nu(A) = \lim_k \langle A\rangle_{(2^k l, 2^k l, 2^k l)}$ exists for every local A; and for $n = 2^k l$, $k \ge 1$, and A a bounded Borel function of the bonds of a cube of h layers with $1 \le h \le n/2 - 1$, the n and A of the first bound, $|\nu(A) - \langle A\rangle_{(n,n,n)}| \le 1.29 \sup|A|\, \theta^{n/2l}/(1 - \theta^{n/2l})$. The measure ν is a Gibbs state, invariant under the symmetries of the lattice, reflection positive at every lattice mirror and ergodic under the translations; it is the only periodic Gibbs state, and the laws of the periodic boxes converge weakly to ν as their three sides tend to infinity.

(b) *Rectangles.* For integers $R, T \geq 1$ and every rectangle C of sides $R \times T$ in a coordinate plane,

$$0 \leq \nu(W_C) \leq e^{-\sigma^+ RT}; \tag{17.ii}$$

on the cube of every integer side $n \geq 2\mathsf{q}l$, for $R \leq n/2$ and $T \leq n-1$,

$$0 \leq \langle W_C \rangle_{(n,n,n)} \leq 8^{T/n} \exp\Big[-\sigma^+ RT + \frac{2\mathsf{q}RT}{ln}\Big(\frac{1}{2}\log\frac{2\mathsf{q}n}{l} + 0.86\Big)\Big];$$

and, if $R, T \geq 2$, also $\nu(W_C) \leq (\frac{8\beta}{8\beta+3})^{2(R+T)-4}$, a bound that holds in every Gibbs state.

(c) *Closed paths.* Let C be a closed lattice path that runs a bond of the direction ρ, and $\rho' \neq \rho$; let $\mathsf{A}_{\rho'\rho}(C)$ be the number of unit squares of the (ρ', ρ)-plane about which the projection of C to that plane winds an odd number of times, and $c_\rho(C)$ the number of changes from "down" to "up" in the cyclic sequence of the senses in which C runs its bonds of the direction ρ. Then $|\nu(W_C)| \leq 2^{c_\rho(C)-1} e^{-\sigma^+ \mathsf{A}_{\rho'\rho}(C)}$. More generally, for a gap τ between neighbouring layers of the direction ρ let $\ell^{\rho'}_\tau(C)$ be the number of integers α such that an odd number of the bonds of the direction ρ that C runs in that gap have the ρ'-coordinate $\leq \alpha$, so that $\sum_\tau \ell^{\rho'}_\tau(C) = \mathsf{A}_{\rho'\rho}(C)$; let $\mathsf{A}_\rho(C) = \sum_\tau \max_{\rho' \neq \rho} \ell^{\rho'}_\tau(C)$, the larger of the two counts taken gap by gap, so that $\mathsf{A}_\rho(C) \geq \max_{\rho' \neq \rho} \mathsf{A}_{\rho'\rho}(C)$; and let D be a continuous unitary representation of G on $\mathbb{C}^d$ with $D(-\mathbf{1}) = -\mathbf{1}$, and $W^D_C = d^{-1} \operatorname{tr} D(U(C))$.

Then $|\nu(W^D_C)| \leq d^{c_\rho(C)-1} e^{-\sigma^+ \mathsf{A}_\rho(C)}$; at $D(g) = g$, $d = 2$, this is the first bound with $\mathsf{A}_\rho(C)$ in the place of $\mathsf{A}_{\rho'\rho}(C)$. In particular a closed path that in each such gap runs either no bond of the direction ρ or exactly two, at the sites o_τ, r_τ of the section, has $c_\rho(C) = 1$, as its height in the direction ρ has one maximum, and $|\nu(W_C)| \leq \exp[-\sigma^+ \sum_\tau |r_\tau - o_\tau|_\infty]$. So $|\nu(W_C)| \leq e^{-\sigma^+ \sum_j |r_j - o_j|}$ for every *staircase loop*, a closed path in a coordinate plane that, in each gap between neighbouring layers of one direction ρ of the plane, runs either no bond of the direction ρ or exactly two, in opposite senses, at the ρ'-coordinates o_j and r_j; for a simple loop $\sum_j |r_j - o_j|$ is the enclosed area.

On every periodic box, of any parity, the staircase loops of a $(1,3)$-plane described in Lem. 17.12(f), with $\rho = 3$ and their bonds of the direction 3 in at most $n_3 - 2$ gaps, have the bound stated there. For a simple closed curve in a coordinate plane $4c_\rho(C) \leq |C|$, so that $|\nu(W_C)| \leq \frac{1}{2}\exp[-\sigma^+ \mathsf{A}_{\rho'\rho}(C) + 0.174\,|C|]$.

(d) *Twisted cubes.* For every integer $n \geq 2\mathsf{q}l$ and the seven $z \neq 0$, on (n, n, n):

$$z_- \leq \exp\Big[-\sigma^+ n^2 + \frac{2\mathsf{q}n}{l}\Big(\frac{1}{2}\log\frac{2\mathsf{q}n}{l} + 0.86\Big)\Big], \qquad 0 \leq 1 - Z^z/Z \leq 6z_-.$$

(e) *Flux sectors and static charges.* For the same n, over the section $n \times n$ the top of every sector with electric flux lies at least $\sigma^+ n - \frac{2\mathsf{q}}{l}(\frac{1}{2}\log\frac{2\mathsf{q}n}{l} + 0.86) - \frac{g_n}{n}$ above the vacuum, with $g_n = 0.0017$ if $n = 2^i l$ and $g_n = 0.105$ for every other n; and for $1 \leq R \leq n/2$ the transfer operator $\hat{\mathsf{T}}_{o,r}$ with two static charges at the distance R (Lem. 17.12) has

$$\|\hat{\mathsf{T}}_{o,r}\| \leq \lambda_0(n,n)\,\big[8(1 + \mathsf{r}_n(n,n))\big]^{1/n} \exp\Big[-\sigma^+ R + \frac{2\mathsf{q}R}{ln}\Big(\frac{1}{2}\log\frac{2\mathsf{q}n}{l} + 0.86\Big)\Big].$$

The same bound holds for any two sites $o \neq r$ of the section, with $R = |r - o|_\infty$, the larger of the distances of their two coordinates on the circle of n sites (Lem. 17.12(e)). So for any two sites $o \neq r$ of $\mathbb{Z}^2$, projected to the sections $n \times n$: $\limsup_{i \to \infty} \|\hat{\mathsf{T}}_{o,r}\|/\lambda_0(n,n) \leq e^{-\sigma^+ |r-o|_\infty}$ along $n = 2^i l$.

(f) *Gap.* Over every section $N_1 \times N_2$ with integers $N_1, N_2 \geq 2l$ the vacuum is simple, its eigenvector lies in the sector 00, and every λ_i, $i \geq 1$, the tops of the sectors with electric flux among them, lies at least $\frac{\Lambda}{l} + \frac{1.021}{n_0}$ above it, n_0 the largest $2^k l \leq \min\{N_1, N_2\}$. Over $N_2 \times n$, $n = 2^k l$, $k \geq 1$, and every integer $N_2 \geq n$ the same holds with $\frac{\Lambda}{l} + \frac{3.5065}{n}$, and over

$n\times n$ with $\frac{\Lambda}{l}+\frac{6.42}{n}$. For integers $N_1,N_2\ge n=2^k l$, $k\ge 1$, with $0.01655\,N_1N_2\,\theta^{n/l}\le n^2\log 2$: $\mathsf{r}_n(N_1,N_2)\le \frac{N_1N_2}{41.8\,n^2}\,\theta^{n/l}$.

(g) *Flux lines.* Over a section (N_2,N), with the first direction as time, $\Lambda_-=\max\{\Lambda_{01},\Lambda_{11}\}$ is the top of the sectors that are odd under the flip of the side N, the map ι_2 of Lem. 17.10 that changes the sign of the bonds along the side N starting on one line along the side N_2; these sectors carry a flux line of the length N.

(g1) Let $N=2wl$, $w\ge 2$ a power of two, $\chi=0.03\,\theta^{N/l}$, $\hat N=\lceil 2\sigma^+N^2/\log\frac{1}{\chi}\rceil$ and $\mathsf{g}=\log\hat N+1.71$. Then $4\le\hat N\le \mathsf{q}N/l+1$. For every integer $N_2\ge \hat N N/2$, Λ_- lies at least $\sigma^+N-(2\log\hat N+1.86)/N$ above the vacuum, and the lower limit of $\log(\lambda_0/\Lambda_-)$ as $N_2\to\infty$ is at least $\sigma^+N-\mathsf{g}/N$. For every integer N_2 with $\max\{N,2\mathsf{q}l\}\le N_2<\hat N N/2$, Λ_- lies at least $\sigma^+N-\frac{2\mathsf{q}}{l}(\log\frac{2\mathsf{q}N}{l}+0.86)$ above the vacuum. For integers N' with $N\le N'<2N$ and $N_2\ge\hat N N/2$, over (N_2,N') the top of the sectors odd under the flip of the side N' lies at least $\sigma^+N'-(4\log\hat N+3.72)/N-(2N_2/N^2)\,0.01655\,\theta^{N/l}$ above the vacuum; the last term is at most $0.0331/N$ for $N_2\le N\theta^{-N/l}$.

(g2) Let $N\ge 4l$ be any integer, $n_0=2wl$, $w\ge 2$ a power of two, the dyadic multiple of l with $n_0\le N<2n_0$, $\kappa=N/n_0$, $\tau=\theta^{n_0/l}$, $\chi_0=0.3602\,\tau$, $\mathsf{E}_0=\sigma^+n_0N$, $\hat N'$ the least even integer $\ge 2\mathsf{E}_0/\log\frac{1}{\chi_0}$, $\mathsf{h}=0.01655\,\hat N'\tau$ and $\mathsf{g}'=\kappa\log\hat N'+3.22+(3\kappa+0.68)\mathsf{h}$. Then $6\le\hat N'<\mathsf{q}N/l+2$; for every integer $N_2\ge\hat N'n_0/2$, Λ_- lies at least $\sigma^+N-\kappa\mathsf{g}'/N\ge\sigma^+N-(4\log\hat N'+18.2)/N$ above the vacuum; and for every integer N_2 with $\max\{N,2\mathsf{q}l\}\le N_2<\hat N'n_0/2$ it lies at least $\sigma^+N-\frac{2\mathsf{q}}{l}(\log\frac{2\mathsf{q}N}{l}+0.86)$ above it. Consequently, with

$$C(\theta)=\max\Big\{\frac{\mathsf{q}}{2}+\frac{1}{4},\ 4\log\Big(\mathsf{q}+\frac{1}{2}\Big)+18.2\Big\}\le 27,$$

for all integers $N\ge 4l$ and $N_2\ge C(\theta)N^2/l$ the top Λ_- lies at least $\sigma^+N-C(\theta)(1+\log\frac{N}{l})/N$ above the vacuum. If moreover $w\ge 4$ and $\tau\le 0.087$, then with $\hat N''=\lceil 4\sigma^+n_0^2/\log\frac{1}{0.2728\,\tau}\rceil$ it lies at least $\sigma^+N-(4\log\hat N''+10.8)/n_0$ above the vacuum for every integer $N_2\ge\hat N''n_0/2$.

(g3) For every integer $N\ge 4l$, every integer $N_2\ge\hat N'n_0/2$ and every rectangle C of sides $R\times T$, $T\le N-1$, in a plane spanned by the first and the third direction,

$$\limsup_{L_1\to\infty}\langle W_C\rangle_{(L_1,N_2,N)}\le 4^{T/N}\exp\Big[-\sigma^+RT+\frac{RT}{N^2}(4\log\hat N'+18.2)\Big].$$

(h) *Boxes of every shape, cylinders and twisted boxes.* For every even $N\ge 2l$, every bounded real Borel function A of the bonds of a cube of $N/2$ layers and every periodic box with all sides $\ge N$,

$$|\langle A\rangle_{\underline{n}}-\nu(A)|\le 5.42\,\theta^{N/4l}\sup|A|;$$

for the events of these bonds the bound is $2.71\,\theta^{N/4l}$. In particular $|\langle A\rangle_{(m,m,m)}-\nu(A)|\le 5.42\,\theta^{\lfloor m/2\rfloor/2l}\sup|A|$ for every integer $m\ge 2l$ and every such A of a cube of $\lfloor m/2\rfloor$ layers. Over every section Σ the laws of the boxes (t,Σ) converge as $t\to\infty$, on the bounded Borel functions of finitely many layers, to the law of the infinite cylinder over Σ; for every even $N'\ge 2l$, every $N\ge N'$ and every A as above of a cube of $N'/2$ layers, the cylinder over $N\times N$ satisfies the bound with N', so that these laws converge to ν as $N\to\infty$; and on the cylinder over $n\times n$, $n\ge 2l$, every rectangle with the side $R\le n/2$ across the axis and the side T along it has a mean of modulus at most $[(1+\mathsf{r}_n(n,n))\,z_-(n,n,n)]^{RT/n^2}$. The laws of twisted boxes whose three sides tend to infinity and whose columns recede, in the sense that every finite set of plaquettes meets the columns of only finitely many members, converge weakly to ν, and so do the mixtures, with any weights, of finitely many such sequences; if a

column passes through a fixed plaquette p_0 in every member, the law of $\operatorname{tr}\partial U(p_0)$ tends to the law of $-\operatorname{tr}\partial U(p_0)$ under ν, which is another law.

(i) *The pressure.* There is an open interval U containing β on which the criterion holds at the side $2l$ with $x_0 = 0.069$; on U the pressure is continuously differentiable, with $P'(\beta') = -3\,\nu_{\beta'}(a_p)$, $\nu_{\beta'}$ the state of (a) at β'; and $\nu(a_p) \le 1 - \frac{\beta}{4(\beta+1)} < 1$.

All constants are absolute; σ^+ and q depend only on l and x_0. Decimal upper and lower bounds used in the finite recursions are understood in their indicated directions. The interval bounds recorded in the proof of Prop. 17.17 verify the numerical part of the stronger tension estimate. ■□

Theorem AX (The criterion and the cube condition)**.** Let $\beta \ge 0$ and let $l \ge 3$ be an integer of either parity.

(a) $X(l) = \mathsf{e}(l) + 2d_2(l) + 4d_1(l)$, and $\log(1+\mathsf{r}_l(\Sigma_j)) \le d_j(l) \le 2\log(1+\mathsf{r}_l(\Sigma_j))$ for $\Sigma_1 = (l,l)$, $\Sigma_2 = (2l,l)$, $\Sigma_3 = (2l,2l)$ and $(d_1,d_2,d_3) = (d_1,d_2,\mathsf{e})$; so the three terms are ≥ 0, and the cube condition at l_0 is the inequality $\mathsf{e}(l_0) \le \frac{1}{20} - 2d_2(l_0) - 4d_1(l_0)$. If $\beta > 0$, the cube condition at l_0 gives the criterion at the same side l_0, with every $x_0 \in [X(l_0), \frac{1}{20}]$, $x_0 > 0$, and $\theta \le 0.645$.

(b) Under the criterion at l: $\mathsf{e}(2^k l) \le \theta^{2^k}/12.9$ for all $k \ge 0$, and $X(2^k l) \le 0.10643\,\theta^{2^k} \le 0.04194$ for all $k \ge 3$; so the cube condition holds at every side $2^k l$ with $k \ge 3$, with $k \ge 2$ if $x_0 \le 0.0650$, and with $k \ge 1$ if $x_0 \le 0.0527$.

(c) Under the criterion at l, along $n = 2^k l$ the rates $\mathsf{m}_\mathsf{e}(n) = \frac{1}{n}\log\frac{1}{12.9\,\mathsf{e}(n)}$ $(k \ge 0)$ and $\mathsf{m}_X(n) = \frac{1}{n}\log\frac{1}{18X(n)}$ $(k \ge 3)$, with values in $(0,\infty]$, do not decrease; for $k \ge 3$, $\mathsf{m}_\mathsf{e}(n/2) - \frac{0.6501}{n} \le \mathsf{m}_X(n) \le \mathsf{m}_\mathsf{e}(n) - \frac{0.3331}{n}$; so the two rates have one common limit, which is $\ge \Lambda/l$.

(d) If $0 < \mathsf{e}(l) < 0.0740538$ at one l, then $\mathsf{e}(2^k l) \to 0$ and $X(2^k l) \to 0$ as $k \to \infty$; so the criterion holds at some side $2^k l$ with any prescribed $x_0 \in (0, 0.069]$, and the cube condition holds at $2^k l$ for all large k.

■□

Theorem AY (Physical scale, uniformly in the cutoff)**.** Assume the uniform criterion, and let ν_K be the state of Thm. AW at $l = l_* L^K$ and $\beta = L^K \mathsf{b}$, read on the lattice $a_K\mathbb{Z}^3$. Put

$$\sigma_* = \frac{\frac{1}{2}\log\frac{1}{\theta} + 0.4175}{l_*^2} \;\ge\; \frac{0.4757}{l_*^2} = \frac{1.9028}{\ell_0^2}.$$

For every $K \ge K_0$, with physical lengths in $a_K\mathbb{N}$ and energies $a_K^{-1}\log(\lambda_0/\cdot)$:

(a) $0 \le \nu_K(W_C) \le e^{-\sigma_* RT}$ for every rectangle C of a coordinate plane with physical sides R, T; and if $R, T \ge 2a_K$, also $\nu_K(W_C) \le \exp[-\frac{1}{2}\sigma_* RT - \frac{3(|C|_{\mathrm{ph}} - 4a_K)}{2(8\mathsf{b}+3a_K)}]$, $|C|_{\mathrm{ph}} = 2(R+T)$ the physical length of C;

(b) on the periodic cube of physical side $\ell \ge 2\mathsf{q}l_*$, $0 \le 1 - Z^z/Z \le 6\exp[-\sigma_*\ell^2 + \frac{2\mathsf{q}\ell}{l_*}(\frac{1}{2}\log\frac{2\mathsf{q}\ell}{l_*} + 0.86)]$ for the seven $z \ne 0$, and every sector with electric flux over $\ell \times \ell$ lies at least $\sigma_*\ell - \frac{2\mathsf{q}}{l_*}(\frac{1}{2}\log\frac{2\mathsf{q}\ell}{l_*} + 0.86) - \frac{0.105}{\ell}$ above the vacuum, with $\frac{0.0017}{\ell}$ in the place of $\frac{0.105}{\ell}$ if ℓ/l_* is a power of two;

(c) over every section whose physical sides are $\ge 2l_*$ the gap is at least $\frac{1}{l_*}\log\frac{1}{\theta}$;

(d) a flux line of physical length $\ell = 2wl_*$, $w \ge 2$ a power of two, over a section $\ell_2 \times \ell$ with $\ell_2 \ge \hat{N}\ell/2$ lies at least $\sigma_*\ell - (2\log(\mathsf{q}\ell/l_* + 1) + 1.86)/\ell$ above the vacuum, and one of every physical length $\ell \ge 4l_*$ over $\ell_2 \times \ell$ with $\ell_2 \ge C(\theta)\ell^2/l_*$ lies at least $\sigma_*\ell - C(\theta)(1 + \log\frac{\ell}{l_*})/\ell$ above it, with the number $C(\theta) \le 27$ of Thm. AW(g2);

(e) the cube condition holds at the side $8l_*L^K$, with $18X \le 1.9157\,\theta^8$; conversely, the cube condition at the sides l_*L^K, $\beta = L^K\mathsf{b}$, for all $K \ge K_0$ gives the uniform criterion with $x_0 = \frac{1}{20}$.

The numbers σ_*, q and $C(\theta)$ depend on l_* and x_0 alone, and $\hat N$ on ℓ/l_* and x_0 alone; none of them depends on K, b or L. In the first bound of (a) the factor in front of $e^{-\sigma_* RT}$ is 1 and there is no perimeter term. ■□

Corollary 17.1 (The static potential)**.** Assume the criterion at the side l, and let $m(R,T)=\nu(W_C)$ for a rectangle C of sides $R\times T$ in a coordinate plane, $m(R,0)=1$.

(a) $m(R,T)$ depends on R,T alone, $m(R,T)=m(T,R)$, and $w^{RT}\le m(R,T)\le e^{-\sigma^+ RT}$, where $w=m(1,1)\ge \beta/(4(\beta+1))$.

(b) For every $n\ge 0$ and real $c_0,\dots,c_n$: $\sum_{h,h'} c_h c_{h'}\, m(R,h+h')\ge 0$ and $\sum_{h,h'} c_h c_{h'}\, m(R,h+h'+1)\ge 0$. Hence $m(R,T)^2\le m(R,T-1)\,m(R,T+1)$ for $T\ge 1$, and there is exactly one Borel probability measure ϱ_R on $[0,1]$ with $m(R,T)=\int \lambda^T\,\mathrm{d}\varrho_R(\lambda)$ for all $T\ge 0$.

(c) $d(R;T)=\log(m(R,T)/m(R,T+1))$ does not increase in $T\ge 0$; the static potential

$$V(R)=\lim_{T\to\infty} d(R;T)=\inf_T d(R;T)=\lim_{T\to\infty} -\frac{1}{T}\log m(R,T)$$

exists in $[0,\infty)$, $\max\operatorname{supp}\varrho_R=e^{-V(R)}$, and $m(R,T_2)\le e^{-(T_2-T_1)V(R)}m(R,T_1)$ for $T_2\ge T_1\ge 0$, $V(R)$ being the largest number with this property. For $0\le T_1<T_2\le T_3<T_4$: $[m(R,T_2)/m(R,T_1)]^{1/(T_2-T_1)}\le[m(R,T_4)/m(R,T_3)]^{1/(T_4-T_3)}$. For $T\ge 1$, $m(R,T)=\int_{[V(R),\infty)} e^{-TE}d\mu_R(E)$, μ_R the image of the restriction of ϱ_R to $(0,1]$ under $\lambda\mapsto-\log\lambda$, a measure of mass in $(0,1]$.

(d) $\sigma^+R\le V(R)\le R\log\frac{4(\beta+1)}{\beta}$.

(e) Under the uniform criterion, for every $K\ge K_0$ and physical R: the potential $V_K(R)=a_K^{-1}V(R/a_K)$ of ν_K in physical units is $\ge\sigma_*R$; and for every real κ the means $M_K(R,T)=e^{\kappa|C|_{\mathrm{ph}}}\nu_K(W_C)$ satisfy $M_K(R,T_2)\le e^{-(T_2-T_1)(V_K(R)-2\kappa)}M_K(R,T_1)$ for physical $T_2\ge T_1\ge 0$, while in the inequality of four rectangles of (c) the factors $e^{\kappa|C|_{\mathrm{ph}}}$ cancel. Along a sequence of cutoffs: let $K_j\to\infty$, $K_j\ge K_0$, let κ_j be arbitrary real numbers, and for R,T in $\bigcup_i a_i\mathbb{N}$ and all large j put $M_j(R,T)=e^{2\kappa_j(R+T)}\nu_{K_j}(W_C)$, C a rectangle of the physical sides $R\times T$, and $v_j(R)=V_{K_j}(R)-2\kappa_j$, so that $v_j(R)\ge\sigma_*R-2\kappa_j$. If $T_1<T_2$ and $M_j(R,T_i)\to\mathsf{W}(R,T_i)\in(0,\infty)$ for $i=1,2$, then

$$\limsup_j v_j(R)\le\frac{1}{T_2-T_1}\log\frac{\mathsf{W}(R,T_1)}{\mathsf{W}(R,T_2)},$$

and if $v_j(R)\ge B$ for infinitely many j, then $\mathsf{W}(R,T_2)\le e^{-(T_2-T_1)B}\,\mathsf{W}(R,T_1)$; in particular $\mathsf{W}(R,T_2)\le e^{-(T_2-T_1)(\sigma_*R-2\kappa)}\,\mathsf{W}(R,T_1)$ if $\kappa_j\le\kappa$ for infinitely many j. If $M_j(R,T)\to\mathsf{W}(R,T)\in(0,\infty)$ for all R,T, then $\mathsf{W}(R,T)=\mathsf{W}(T,R)$, $\mathsf{W}(R,T_2)^{T_3-T_1}\le\mathsf{W}(R,T_1)^{T_3-T_2}\,\mathsf{W}(R,T_3)^{T_2-T_1}$ for $T_1<T_2<T_3$, the inequality of four rectangles of (c) holds for W, and $V_{\mathrm{ren}}(R)=\lim_{T\to\infty}-\frac{1}{T}\log\mathsf{W}(R,T)$ exists in $[-\infty,\infty)$ and is the infimum of the quotients $\frac{1}{T_2-T_1}\log\frac{\mathsf{W}(R,T_1)}{\mathsf{W}(R,T_2)}$, $T_1<T_2$. The convergence of the M_j is a hypothesis of these statements.

(f) With no criterion, in every periodic Gibbs state s at $\beta\ge 0$, $s(W_C)\ge 0$ for every rectangle C of a coordinate plane with one even side and for every square.

(g) With no criterion: let $\beta>0$ and let s be a periodic Gibbs state that is positive at the bond mirrors on the functions of two crossing bonds, in the sense that

$$s\Big(\sum_{i,j,k,l}\overline{(\mathsf{z}_p)_{ji}}\,(\mathsf{z}_q)_{kl}\,F_{jk}\,\overline{\mathfrak{r}F_{il}}\Big)\ge 0$$

for every bond mirror, $\mathfrak{r}$ the reflection in it, z_p, z_q the values of U at the bonds from two sites x_p, x_q next to the mirror, equal or not, to their mirror images, and all continuous functions F_{jk}, $j,k\in\{1,2\}$, of finitely many bonds with both ends in the closed side of the mirror that

holds x_p and x_q, as in Lem. 17.21(c). Let $m_s(R,T) = s(W_C)$ for a rectangle C of sides $R \times T$ in a coordinate plane, $m_s(R,0) = 1$. Then $m_s(R,T)$ depends on R,T alone, $m_s(R,T) = m_s(T,R)$, and $[\beta/4(\beta+1)]^{RT} \le m_s(R,T) \le 1$ for all integers $R,T \ge 1$; both Hankel forms of (b) hold for m_s, and with them the log-convexity in T and exactly one Borel probability measure ϱ^s_R on $[0,1]$ with $m_s(R,T) = \int \lambda^T \, \mathrm{d}\varrho^s_R(\lambda)$; $d_s(R;T) = \log(m_s(R,T)/m_s(R,T+1))$ does not increase in T, $V_s(R) = \lim_T d_s(R;T) = \inf_T d_s(R;T) = \lim_T -\frac{1}{T}\log m_s(R,T)$ exists in $[0, R\log\frac{4(\beta+1)}{\beta}]$, $\max\operatorname{supp}\varrho^s_R = e^{-V_s(R)}$, $m_s(R,T_2) \le e^{-(T_2-T_1)V_s(R)} m_s(R,T_1)$ for $T_2 \ge T_1 \ge 0$, and the inequality of four rectangles of (c) holds for m_s.

On site mirrors alone, with no hypothesis at the bond mirrors: if $m_s(R,2) > 0$, as is the case for every even R, then $V_{\mathrm{e}}(R) = \lim_h -\frac{1}{2h}\log m_s(R,2h)$ exists in $[0,\infty)$, and $m_s(R,T_2) \le e^{-(T_2-T_1)V_{\mathrm{e}}(R)} m_s(R,T_1)$ for every even $T_1 \ge 0$ and every integer $T_2 \ge T_1$. The state ν of Thm. AW is positive at the bond mirrors in this sense, by Lem. 17.21(c) on the cubes of sides $2^k l$ and Thm. AW(a), as in Step 2 of the proof; so (g) contains (a)–(c), the upper bound $e^{-\sigma^+ RT}$ excepted. ■□

Proposition 17.2 (Where the criterion holds)**.**

(a) For every integer $l \ge 3$, every $x_0 \in (0, 0.069]$ and every $\beta \in (0, x_0/48l^3]$ the criterion holds at l.

(b) Let $0 < \beta < \frac{1}{6}$ and $l \ge 4$ even. Then $X(l) \le 64\, l^2 (6\beta)^{l/2}$; so the cube condition, and the criterion with every $x_0 \in [X(l), \frac{1}{20}]$, $x_0 > 0$, hold at every even l with $64\, l^2(6\beta)^{l/2} \le \frac{1}{20}$, for instance at all even $l \ge 10$ if $\beta = 0.01$, $l \ge 24$ if $\beta = 0.05$, $l \ge 62$ if $\beta = 0.1$, $l \ge 360$ if $\beta = 0.15$.

(c) For $0 < \beta < \frac{1}{6}$ there is exactly one Gibbs state, periodic or not.

■

Proposition 17.3 (What the criterion says)**.** Let $\beta \ge 0$ and let $l \ge 3$ be an integer.

(a) With $\mathsf{r} = \mathsf{r}_l(2l, 2l)$: $\log(1+\mathsf{r}) \le \mathsf{e}(l) \le 2\log(1+\mathsf{r})$. So $\mathsf{r} \le 0.0351$ implies $\mathsf{e}(l) \le 0.069$, and $\mathsf{e}(l) \le 0.069$ implies $\mathsf{r} \le 0.0357638$, in which case every λ_i, $i \ge 1$, over $2l \times 2l$, the tops of the sectors with electric flux among them, lies at least $3.33/l$ above the vacuum. Besides, $\mathsf{e}(l) \ge \log(1 + \sum_{e \ne 00}(\Lambda_e/\lambda_0)^l)$ over $2l \times 2l$; so $\mathsf{e}(l) \ge \log(1+3y)$ if each of the three sectors with electric flux has $(\Lambda_e/\lambda_0)^l \ge y$, a floor that tends to $\log 4$ as the three sectors approach the vacuum.

(b) Let $r_1 = Z^-/Z$ on the cube of side $2l$, and $\mathsf{f}(x) = 2\log(\sqrt{1-x} + \sqrt{2x})$ for $0 \le x \le 1$. Then $0 < r_1 \le 1$ and $\mathsf{e}(l) \ge \mathsf{f}(1-r_1)$. Hence $\mathsf{e}(l) \le 0.069$ forces $r_1 > 0.99937$, and $1 - r_1 \ge 0.0039$ forces $\mathsf{e}(l) \ge 0.16567$.

(c) If $\mathsf{e}(l) \le 0.069$, then $1 - Z^-/Z < 0.0039$ on the cube of every integer side $N \ge 16l$, and $\mathsf{e}(l') < 0.0234$ for every integer $l' \ge 16l$. So, if $1 - Z^-/Z \ge 0.0039$ on one cube of side N, then $\mathsf{e}(l) > 0.069$ for every integer $3 \le l \le N/16$.

(d) Under the criterion at l, the loop along the boundary of one plaquette has $l^2 \ge 0.4757/\log(1/\nu(W_p))$.

■

Remark 17.4 (Free fields). For comparison, take three free massless fields in three dimensions, one for each generator of SU(2), each with one mode for every momentum, and leave out the modes of momentum 0. Over the section $2l \times 2l$ the energies of the modes are $\pi|m|/l$, $m \in \mathbb{Z}^2 \setminus \{0\}$, so that the excited levels at the time l give

$$1 + \mathsf{r} = \prod_{m \in \mathbb{Z}^2 \setminus \{0\}} (1 - e^{-\pi|m|})^{-3}, \qquad 1 + \mathsf{r}' = \prod_{m \in \mathbb{Z}^2 \setminus \{0\}} (1 - e^{-2\pi|m|})^{-3},$$

numbers free of l, with $\log(1+\mathsf{r}) \approx 0.7201$ and $\log(1+\mathsf{r}') \approx 0.0242$. The number $D = 2\log(1+\mathsf{r}) - \log(1+\mathsf{r}')$ that stands for $\mathsf{e}(l)$ (Lem. 17.11) is approximately 1.416 at every scale, more than twenty times the bound 0.069 of the criterion. This is a computation for free fields.

Lemma 17.5 (A lower bound sufficient for anticoncentration)**.** There are absolute constants $c_{\mathrm{T}} > 0$ and $C_{\mathrm{T}} \geq 1$ such that, for every integer $N \geq 4$ and $\beta \geq C_{\mathrm{T}} N^{68}$,

$$Z(N,N,N) \geq c_{\mathrm{T}} \frac{N^2}{\beta^2} \mathsf{X}_0, \qquad \mathsf{X}_0 = (2\pi^2)^{-(2N^3-2)} \left(\frac{2\pi}{\beta}\right)^{3N^3-3} (\mathfrak{d}')^{-3/2}, \qquad \mathfrak{d}' = e^{S_0(N^{\times 3})}.$$

■

Proof. The periodic estimate in Prop. 17.47(a), proved in Prop. 11.14(g), gives

$$Z(N,N,N) \geq \tfrac{1}{2} \beta^{1-3N^3} \mathcal{Z}(N)$$

whenever $c_3 (N^{66} \log\beta/\beta)^{1/8} \leq x_*$, where $x_* \leq 1/2$. This condition holds throughout $\beta \geq C_{\mathrm{T}} N^{68}$ after an absolute increase of C_{T}. Indeed, $\log\beta/\beta$ decreases for $\beta \geq e$, and, for $C \geq e$ and $N \geq 4$,

$$\frac{N^{66}\log\beta}{\beta} \leq \frac{\log C + 68\log N}{CN^2} \leq \frac{\log C + 68\log 4}{16C} \quad (\beta \geq CN^{68}).$$

The last member tends to zero as $C \to \infty$.

Put

$$\mathcal{J}_N = N^{-2} \int_{[-\pi,\pi]^3} e^{S_0(N^{\times 3}) - S_\varphi(N^{\times 3})} \, \mathrm{d}\varphi.$$

The determinant estimate of Prop. 11.14(c) gives

$$\mathcal{J}_N \geq J_- := e^{-2C_S} \int_{[-\pi,\pi]^3} |\varphi|^{-2} d\varphi > 0.$$

The singularity is integrable in three dimensions. The normalizations in Prop. 17.47(a) give exactly

$$\frac{\beta^{1-3N^3} \mathcal{Z}(N)}{N^2 \beta^{-2} \mathsf{X}_0} = \frac{\mathcal{J}_N}{4\pi^3}.$$

Thus $c_{\mathrm{T}} = J_-/(8\pi^3)$ suffices. □

Proposition 17.6 (Failure on small tori, given Thm. YM3_I:D)**.** Given Thm. YM3_I:D among the construction results, there are numbers $t_\circ > 0$ and $\beta_\flat$, formed from the constants of Thm. YM3_I:D and of the setting (Th) of §2.3.1, the fixed calibration $\mathsf{t}_\natural$ of §2.3.1 among them, with the following properties.

(a) For every real $t \in (0, t_\circ]$ there is an integer m_t such that on the periodic cube of $2\mathsf{L}^j$ sites a side at the coupling $\beta = 2\mathsf{L}^j/t$ the quotient $r_1 = Z^-/Z$ satisfies $r_1 \leq 0.99596$ for every $j \geq m_t + 2$; there $\mathsf{e}(\mathsf{L}^j) \geq 0.16567$, and $\mathsf{e}(l) > 0.069$ for every integer $3 \leq l \leq \mathsf{L}^j/8$.

(b) For the sides $\ell_0 = 2\mathsf{L}^{m_0}$ and every label b with $\ell_0/\mathsf{b} \leq t_\circ$: $\mathsf{e}(\ell_0/2a_K) \geq 0.16567$ at $\beta = \mathsf{L}^K \mathsf{b}$ for all large K; so the uniform criterion with $L = \mathsf{L}$ and $l_* = \mathsf{L}^{m_0}$ fails at this label for every K_0 and every $x_0 \leq 0.069$.

(c) For every real $\beta \geq \beta_\flat$: $\mathsf{e}(l) > 0.069$ for every integer $3 \leq l \leq \frac{t_\circ}{16\mathsf{L}}\beta$. So the uniform criterion, for any integer $L \geq 2$, at any label $\mathsf{b} > 0$, forces $\ell_0 > \frac{t_\circ}{8\mathsf{L}}\mathsf{b}$.

(d) There are numbers $t_\star > 0$, $\mathsf{C}_\star \geq 1$ and $\mathsf{p}_\star \geq 1$, formed from the same constants, such that for every integer $j \geq 1$ and every real $\beta > 0$ with $t = 2\mathsf{L}^j/\beta \leq t_\star$, on the periodic cube of $N = 2\mathsf{L}^j$ sites a side,

$$\frac{\lambda_0(N,N)^N}{Z(N,N,N)} \leq \mathsf{C}_\star \, t \left(\log\frac{1}{t}\right)^{\mathsf{p}_\star}, \qquad \text{so that} \qquad \mathsf{e}(\mathsf{L}^j) \geq \log\frac{1}{t} - \mathsf{p}_\star \log\log\frac{1}{t} - \log\mathsf{C}_\star.$$

For a side $\ell_0 = 2\mathsf{L}^{m_0}$, $m_0 \in \mathbb{Z}$, and a label b with $\ell_0/\mathsf{b} \leq t_\star$ this says that at every cutoff $K \geq 1 - m_0$, at $\beta = \mathsf{L}^K \mathsf{b}$ and with $a_K = \mathsf{L}^{-K}$, $\mathsf{e}(\ell_0/2a_K) \geq \log\frac{\mathsf{b}}{\ell_0} - \mathsf{p}_\star \log\log\frac{\mathsf{b}}{\ell_0} - \log\mathsf{C}_\star$.

The proof of (d) rests, besides Thm. YM3_I:D, on Lem. 17.5 and the two estimates eq. 17.iii and eq. 17.iv stated in the next subsection: through them it bounds $\lambda_0(N,N)^N/Z(N,N,N)$ from above, and the floor of e follows from that ceiling.

■

Proposition 17.7 (What the trace identities alone allow)**.** Call *data* every family of numbers $Z^z(\underline{n})>0$, $\underline{n}$ a box and $z\in\{0,1\}^3$, with the properties (iii), (T1), (T2) and (T3) of Lem. 17.10, and form F, e, d_1, d_2, X, z_- from them as above. Let $0<q<1$.

(a) $Z^z(\underline{n})=\prod_\mu(1+q^{n_\mu})^{|\underline{n}|/n_\mu}$, the same for all z, are data, with $d_1:d_2:\mathsf{e}=1:2:4$ and $X=3\mathsf{e}$ at every l.

(b) The *sheets* $Z^z(\underline{n})=\prod_\rho(1+(-1)^{z_{\mu\nu}}q^{n_\mu n_\nu})^{n_\rho}$, $\{\mu,\nu,\rho\}=\{1,2,3\}$, are data. With $s=q^{l^2}$ they have $\mathsf{e}(l)=8l\log(1+s^2)-4l\log(1+s^4)$ and $X(l)=24l\log(1+s)-6l\log(1+s^4)$, and $X/\mathsf{e}\geq 2.95/s$ for $s\leq 0.03$ (at $l=4$, $s=0.03$: $\mathsf{e}\approx 0.028774$, $X\approx 2.837626$). So for no number C does $X(l)\leq C\,\mathsf{e}(l)$ hold for all data: at the same side, the criterion does not give the cube condition from these properties alone.

(c) For the sheets, $\frac12 q^{n^2}\leq z_-(n,n,n)\leq nq^{n^2}$, so that $-n^{-2}\log z_-(n,n,n)\to\log\frac1q$, while with $x_0=\mathsf{e}(l)$ one has $\sigma^+=\log\frac1q-\frac{\log(103.2e^{-0.835}l)}{2l^2}+o(1)$ and $\Lambda=2l^2\log\frac1q-\log(103.2\,l)+o(1)$ as $q\to0$. So an inequality $\liminf_n -n^{-2}\log z_-(n,n,n)\geq c'\Lambda/l^2-C(l)$ with a number $c'>\frac12$ is false for the sheets: from the properties of data alone, the coefficient $\frac12$ of Λ/l^2 in σ^+ cannot be raised.

■

Proposition 17.8 (An area law without the numerical recursion of Prop. 17.17)**.** Assume the criterion at the side l, and let $l_j=2^{j+1}l$. Then $l_j\varepsilon_{\mathrm{v}}(l_j)\leq 0.0165\cdot 2^{-j}\,\theta^{2\cdot4^j}$ for all $j\geq0$; $z_-(n,n,n)\leq\max\{1,0.0096\sqrt{n/l}\}^{n/l}\,\theta^{n^2/2l^2}$ for $n=2^il$, $i\geq1$; and $0\leq\nu(W_C)\leq\theta^{RT/2l^2}$ for every rectangle C of sides $R\times T$ in a coordinate plane. The tension $\log\frac1\theta/2l^2\geq0.0582/l^2$ so obtained does not use the values $E_1,\dots,E_6$ of Prop. 17.17; the difference $0.4175/l^2$ to σ^+ is what that proposition adds. ■

17.1.4. *Construction estimates.* We use: (i) the periodic lattice of a box, its configurations and Haar measures, and Wilson's action; (ii) for a direction taken as time, the layers and the functions A_Σ and $A_\times(V,S,V')=\sum_b(1-\frac12\operatorname{tr}V_bS_{b_+}V_b'^{-1}S_{b_-}^{-1})$ (Section YM3_I:85, Section YM3_I:85): the map $U\mapsto(V^\tau,S^\tau)_{\tau\in\mathbb{Z}/t}$, with $V^\tau\in\mathcal{X}=G^{\mathsf{B}(\Sigma)}$ the bonds of the layer τ and $S^\tau\in G^\Sigma$ those from the layer τ to the layer $\tau+1$, carries dU to the product of the Haar measures, $\sum_p a_p=\sum_\tau[A_\Sigma(V^\tau)+A_\times(V^\tau,S^\tau,V^{\tau+1})]$ with continuous summands in $[0,2]$, and $A_\times(V',S^{-1},V)=A_\times(V,S,V')$; (iii) the crossing operator (Section YM3_I:85, Prop. YM3_I:85.1): the kernel $\mathsf{K}(V',V)=\int e^{-\beta A_\times(V,S,V')}\,\mathrm{d}S$ is symmetric, continuous and has values in $(0,1]$, and the operator K on $\mathcal{H}=L^2(\mathcal{X},dV)$ is Hilbert–Schmidt and ≥0; (iv) for Lem. 17.13 and Prop. 17.6, Hölder's inequality for traces of products of an even number of Hilbert–Schmidt operators (Lem. YM3_I:86.1) and $\|A\|_{2k}^{2k}=\operatorname{Tr}(A^*A)^k$ (the norm definition preceding Lem. YM3_I:86.1); (v) for the uniqueness, the boxes and the pressure, the chessboard estimate on the tori of side $2\mathsf{L}^j$ (Prop. YM3_I:86.3), its form for two tori on one board (Cor. YM3_I:86.6), and its bound on every periodic Gibbs state (Prop. YM3_I:91.9); (vi) the invariance of every periodic Gibbs state under the symmetries of the lattice, its reflection positivity at every lattice mirror, the diagonal ones included, and its decomposition into ergodic components that are periodic Gibbs states (Props. YM3_I:103.4 and YM3_I:106.2); (vii) for Prop. 17.6 alone, Thm. YM3_I:D and the statements among the construction results through which it bears on the block fields at a fixed depth on tori that are small against the label: the setting (Th), what it gives, the weight of the small-field chain and the chain with its bound on the discarded mass (§2.3.1, Lem. YM3_I:118.1, the definitions preceding Prop. YM3_I:115.2, Prop. YM3_I:115.2); the fixed calibration and the inner balls (Defs. 2.3.1, YM3_I:113.10, Lem. YM3_I:113.10); the setting (E)

and the laws of the block fields (Defs. 2.3.1, 2.7); the numerical conditions at the chosen constants (Prop. YM3_I:69.1); the constants and the thresholds (§2.3.1 and equation YM3_I:(634)); the block-averaging map with its locality, gauge covariance, measurability and covariance under the lattice isometries (equations YM3_I:(37)–(39), Prop. YM3_I:4.5, YM3_I:4.5, [7, Lem. G.1]); the bound on the holonomies along rearranged paths (Lem. YM3_I:4.1); and, for its clause (d), the averages on small fields (Lem. YM3_I:4.4, Lem. 5.28) and the comparison of two consecutive cutoffs (§2.3.1, Section YM3_I:116, Props. YM3_I:116.1 and YM3_I:116.3 with their proofs, and §2.3.2), on which Cor. 2.8 rests, read from one cutoff to the next and summed into the estimate eq. 17.iv below, together with the signs of the constants of that comparison, among them $\kappa_{\mathrm{b}} > 0$ and $c' > 0$ (Prop. YM3_I:114.4 and §2.3.1).

From the textbooks we use the spectral theorem for compact self-adjoint operators, Jentzsch's theorem, the min–max principle, the inequalities of Young, Jensen and Schwarz, Kolmogorov's extension theorem, the weak compactness of the probability measures on a compact metrisable space, and Hausdorff's moment theorem.

Clause (d) of Prop. 17.6 rests, besides Thm. YM3_I:D and the statements of (vii), on two estimates of its own, which its proof takes up (Lem. 17.26, and Lem. 17.27 with the chain from a low cutoff). For a configuration U of a periodic cube and a site ξ let $\mathsf{a}(U,\xi) \in \mathbb{R}^5$ be the five invariants of the pair of the holonomies of U along the closed straight paths through ξ in the directions 1 and 2: their two class angles, $\alpha(g) = \arccos \frac{1}{2} \operatorname{tr} g \in [0,\pi]$ for $g \in G$, and the three entries of $(\mathsf{v}^{\mathrm{t}}\mathsf{v})^{1/2}$, v the real 3×2 matrix of their vector parts, numbers that do not change under a simultaneous conjugation of the pair. The first estimate concerns one lattice: there are absolute numbers C_{a}, c_{a}, c'_{a}, c''_{a} such that on the periodic cube of $N' \geq 4$ sites a side, at every coupling $\beta' \geq c'_{\mathrm{a}} N'^{c''_{\mathrm{a}}}$, for every closed cube Q of the half side w' in $\mathbb{R}^5$ with $(N'^5/2\beta')^{1/2} \leq w' \leq \frac{1}{20}$ and every $\omega > 0$ below a fixed negative power of N',

$$\mu_{N',\beta'}\Big(\mathsf{a}(U,\xi) \in \mathsf{Q},\ \max_p |\partial U(p) - \mathbf{1}| < \omega\Big) \leq C_{\mathrm{a}} N'^{c_{\mathrm{a}}}\, \beta'^{1/2}\, w'^3, \tag{17.iii}$$

$\mu_{N',\beta'}$ the law of that cube. Its proof combines a Gaussian ceiling with Lem. 17.5 for Wilson's measure on the cube in the gauge of a maximal tree, with constants that are powers of N': the ceiling integrates out all bonds but the three winding holonomies of ξ against the weight $\exp[-2\beta' N'^{-5} \sum_{\mu<\nu} |\mathsf{v}_\mu \times \mathsf{v}_\nu|^2]$, v_μ the vector part of the holonomy in the direction μ; the proved lower bound for $Z(N',N',N')$ supplies the normalizing denominator; and the integral of that weight over the triples whose first two members have their five invariants in Q is at most a fixed multiple of $w'^3(\beta'/N'^5)^{-3/2}$. The second estimate is drawn from the lemmas of (vii) on two consecutive cutoffs.

Let $\mathsf{b} = \ell/t$ be a label and $\ell = 2\mathsf{L}^m$ a side at which the setting (E) of §2.3.1 holds for it, with ℓ at most a fixed power of $y = \log \mathsf{b}$, and let μ_0^K be the law of the block field of the depth 0 at the cutoff K (Def. 2.7(2)). For every y above a number formed from the constants there is a lowest cutoff K_1, at which the bare cube has at most a fixed power of y sites a side, such that for every integer $K \geq K_1$ and every Borel set E of block fields of the depth 0 that the gauge transformations carry onto itself (Def. 2.7(3))

$$\mu_0^K(E) \leq e^{\mathsf{S}}(\mu_0^{K_1}(E) + \mathsf{R}_\Sigma), \tag{17.iv}$$

with S below an absolute number and $\mathsf{R}_\Sigma \leq e^{-3y^2}$. Cor. 2.8 compares two consecutive cutoffs additively; here, from one cutoff to the next, the errors ϵ_{fg} and ϵ_{cl} of §2.3.1, which decrease like powers of L^{-K} up to factors that are powers of K, y and ℓ, enter as factors on the probability, and the errors ϵ_{sh} and R_{t_1}, which are exponentially small in a power of y, as additive remainders and through the factors $(1-\epsilon_{\mathrm{sh}})^{-1}$ and $(1-\mathsf{R}_{t_1})^{-1}$ (Lem. 17.27); S and R_Σ are the sums over $K \geq K_1$ of the exponents of the factors and of the remainders. So an event whose probability at the cutoff K_1 is at most of the order of t has a probability at most of the order of t at every cutoff $K \geq K_1$, since $y \geq \log \frac{1}{t}$ makes e^{-3y^2} small against t.

Beyond (i)–(vii) the proofs use no result among the construction results; no statement of the section other than Prop. 17.6 rests on Thm. YM3_I:D or on Thms. YM3_I:AJ and YM3_I:AD. Its clause (d) combines the finite-cube lower bound of Lem. 17.5 with the estimates eq. 17.iii and eq. 17.iv.

17.1.5. *Estimates and proofs.*

Lemma 17.9 (Two identities for the crossing weight)**.** Let $(V^S)_b = S_{b_-} V_b S_{b_+}^{-1}$ and $\mathsf{K}_0(V,W) = \prod_b \exp[-\beta(1 - \frac{1}{2}\operatorname{tr} V_b W_b^{-1})]$ for $V, W \in \mathcal{X}$ and $S \in G^\Sigma$.

(a) $e^{-\beta A_\times(V,S,V')} = \mathsf{K}_0(V^g, V'^{gS})$ for all $V, V' \in \mathcal{X}$ and all $g, S \in G^\Sigma$.

(b) $\mathsf{K}_0(V,W) = \sum_\alpha c_\alpha \phi_\alpha(V)\overline{\phi_\alpha(W)}$, with α in a countable set, $c_\alpha \geq 0$, $\sum_\alpha c_\alpha = e^{\beta \#\mathsf{B}(\Sigma)}$, ϕ_α continuous and $|\phi_\alpha| \leq 1$.

■

Proof sketch. (a) $V_b((V'^S)_b)^{-1} = V_b S_{b_+} V_b'^{-1} S_{b_-}^{-1}$ is the matrix of the bond b in $A_\times$, so $e^{-\beta A_\times(V,S,V')} = \mathsf{K}_0(V, V'^S)$; $(V^g)_b((W^g)_b)^{-1} = g_{b_-} V_b W_b^{-1} g_{b_-}^{-1}$, so $\mathsf{K}_0(V^g, W^g) = \mathsf{K}_0(V,W)$ by the cyclicity of the trace; and $(V^S)^g = V^{gS}$. (b) For $v, w \in G$, $\operatorname{tr}(vw^{-1}) = \sum_{i,j} v_{ij}\overline{w_{ij}}$, with entries of modulus ≤ 1. Expand each factor $e^{-\beta}\exp[\frac{\beta}{2}\operatorname{tr} V_b W_b^{-1}]$ in powers, write out the powers and multiply over b: $\alpha = (n_b, (i_{b,s}, j_{b,s})_{s \leq n_b})_b$, $c_\alpha = e^{-\beta \#\mathsf{B}(\Sigma)} \prod_b (\beta/2)^{n_b}/n_b!$, $\phi_\alpha(V) = \prod_b \prod_s (V_b)_{i_{b,s} j_{b,s}}$; as there are 4^{n_b} index families for each n_b, $\sum_\alpha c_\alpha = e^{-\beta \#\mathsf{B}(\Sigma)} \prod_b e^{2\beta}$, and this bounds the series of the moduli. □

Lemma 17.10 (Transfer operator and flux sectors)**.** Fix a box and a direction μ_0 as time, with the other directions $\nu_1 < \nu_2$. For $i = 1, 2$ let ι_i change the sign of V_b on the bonds of the direction ν_i that start on a fixed line of the other direction of the section, $(\mathsf{J}_i \psi)(V) = \psi(\iota_i V)$, $\mathsf{J}^k = \mathsf{J}_1^{k_1}\mathsf{J}_2^{k_2}$ and $P_e = \frac{1}{4}(1 + (-1)^{e_1}\mathsf{J}_1)(1 + (-1)^{e_2}\mathsf{J}_2)$ for $e \in \{0,1\}^2$. Let $A_\Sigma^{(0)} = A_\Sigma$, let $A_\Sigma^{(1)}$ be A_Σ with the summand $1 - \frac{1}{2}\operatorname{tr}$ of one plaquette replaced by $1 + \frac{1}{2}\operatorname{tr}$, M_m the multiplication by $e^{-\beta A_\Sigma^{(m)}/2}$ and $\mathsf{T}_m = M_m \mathsf{K} M_m$; write $z \leftrightarrow (k; m)$ with $k = (z_{\mu_0\nu_1}, z_{\mu_0\nu_2})$, each pair written in its order, and $m = z_{\nu_1\nu_2}$. Then:

(i) $\mathsf{T}_m \geq 0$ is Hilbert–Schmidt with a continuous symmetric kernel with values in $(0,1]$; J_1, J_2 are commuting self-adjoint unitary involutions that commute with T_m; none of these objects depends on t.

(ii) $Z^{(k;m)} = \operatorname{Tr}(\mathsf{T}_m^t \mathsf{J}^k)$ for every integer $t \geq 3$.

(iii) Z^z does not depend on the places of the columns, and $Z^{\pi z}(\pi \underline{n}) = Z^z(\underline{n})$ for every permutation π of the directions.

Consequently: (T1) Z is symmetric in the sides; (T2) $Z(t, \Sigma) = \sum_i \lambda_i^t$ for $t \geq 3$, the $\lambda_i \geq 0$ being the eigenvalues of T_0, $\sum_i \lambda_i^2 < \infty$, and λ_0 is simple with an eigenvector in the range of P_{00}; (T3) with the eigenvalues $\mu_i(e,m) \geq 0$ of T_m on the range of P_e, which depend neither on t nor on k,

$$Z^{(k;m)} = \sum_e (-1)^{e\cdot k} W_t(e;m), \qquad W_t(e;m) = \operatorname{Tr}(\mathsf{T}_m^t P_e) = \sum_i \mu_i(e,m)^t \geq 0;$$

$\Lambda_e = \sup_i \mu_i(e,0)$ is the top of the sector e. Moreover:

(a) $W_t(00;0) \geq \lambda_0^t$;

(b) $0 \leq 1 - Z^{(k;0)}/Z \leq 2\mathsf{r}_t/(1+\mathsf{r}_t)$, so that $z_- \leq \mathsf{r}_t/(1+\mathsf{r}_t)$ when the time is the direction 1 or 3;

(c) on a cube Z^z/Z depends on $|z|$ alone, $r_2 \leq r_1$, $r_3 \leq r_1$, and $0 \leq 1 - Z^z/Z \leq |z|(1-r_1) \leq 6z_-$; with any direction as time, the weights $w_e = W_n(e;0)/Z$ of the cube of side n are $w_{00} = \frac{1}{4}(1 + 2r_1 + r_2)$, $w_{10} = w_{01} = \frac{1}{4}(1 - r_2)$, $w_{11} = \frac{1}{4}(1 - 2r_1 + r_2)$, and $1 - r_1 \leq \frac{4}{3}\mathsf{r}_n(n,n)/(1+\mathsf{r}_n(n,n))$;

(d) $a_- = \sum_i \mu_i^{n_1} = \sum_i \mu_i'^{n_3}$ and $Z^- = \sum_i \mu_i''^{n_2}$, with numbers $\mu_i, \mu_i', \mu_i'' \geq 0$ that do not depend on n_1, on n_3 and on n_2 respectively, and $\sup_i \mu_i'' = \lambda_-(n_1, n_3) \leq \lambda_0(n_1, n_3)$.

■

Proof sketch. Every plaquette is a summand of $A_\Sigma(V^\tau)$ or of $A_\times(V^\tau, S^\tau, V^{\tau+1})$. Put the columns at the origin of the section and at the crossing $\tau = 0$. A column of the pair $\nu_1\nu_2$ is one summand in every layer and turns A_Σ into $A_\Sigma^{(1)}$; a column of the pair $\mu_0\nu_i$ is the set of the crossing summands of the bonds flipped by ι_i, and since $1 - \frac{1}{2}\operatorname{tr}(-g) = 1 + \frac{1}{2}\operatorname{tr} g$, reversing it is the substitution $V^0 \mapsto \iota_i V^0$ in that crossing. Integrating each S^τ first,

$$Z^{(k;m)} = \int \mathsf{T}_m(V^1, \iota^k V^0) \prod_{\tau\neq 0} \mathsf{T}_m(V^{\tau+1}, V^\tau) \prod_\tau \mathrm{d}V^\tau = \operatorname{Tr}(\mathsf{T}_m^t \mathsf{J}^k),$$

the trace of a chain of $t \geq 3$ bounded kernels: (ii). The functions $A_\Sigma^{(m)}$ and $A_\times$ are invariant under ι_i, because the two bonds of the direction ν_i of a plaquette are both flipped or neither and $-\mathbf{1}$ is central; so J_i commutes with T_m, and $\langle\psi, \mathsf{T}_m\psi\rangle = \langle M_m\psi, \mathsf{K}M_m\psi\rangle \geq 0$: (i). (iii): changing $U(b)$ into $-U(b)$ on a set $\mathfrak{F}$ of bonds keeps dU and multiplies $\operatorname{tr}\partial U(p)$ by $(-1)^{\#(p\cap\mathfrak{F})}$; for $\mathfrak{F}$ the bonds of the direction ν that start on the line of the direction ρ through $x + e_\mu$ this sign is -1 exactly on the columns of the pair $\mu\nu$ through x and through $x + e_\mu$ (the sides are ≥ 3), so a column moves by one step; a permutation of the axes renumbers the bonds and carries a plaquette to a plaquette or to its inverse, which has the same trace. (T2) and (T3) follow from the spectral theorem on the four ranges of the P_e.

By Jentzsch's theorem λ_0 is simple with an eigenvector $\varphi > 0$, and $P_{00}\varphi = \frac{1}{4}\sum_k \varphi\circ\iota^k \geq \frac{1}{4}\varphi \neq 0$: (a). (b): $Z - Z^{(k;0)} = 2\sum_{e\cdot k \text{ odd}} W_t(e;0) \leq 2(Z - W_t(00;0)) \leq 2(Z - \lambda_0^t)$. (c): on a cube (iii) makes Z^z depend on $|z|$. With two twisted pairs, the time a direction of one of them but not of the other, $(k;m) = (10;1)$ and $r_2 = \sum_e (-1)^{e_1} W(e;1)/Z \leq \sum_e W(e;1)/Z = r^{(00;1)} = r_1$; in the same way $r_3 = r^{(11;1)} \leq r_1$; and with the time common to two twisted pairs, $(k;m) = (11;0)$, $4W(11;0)/Z = 1 - 2r_1 + r_2 \geq 0$ gives $1 - r_2 \leq 2(1 - r_1)$. With all three pairs twisted, $(k;m) = (11;1)$ for every direction as time, and (T3) at $m = 1$ gives $4W(11;1)/Z = r^{(00;1)} - r^{(10;1)} - r^{(01;1)} + r^{(11;1)} = r_1 - 2r_2 + r_3 \geq 0$; added to $1 - 2r_1 + r_2 \geq 0$ this is $1 - r_3 \leq (1 - r_1) + (1 - r_2)$, and so $1 - r_3 \leq 3(1 - r_1)$.

The formula for the weights is obtained by inverting $Z^{(k;0)}/Z = \sum_e (-1)^{e\cdot k} w_e$, with $Z^{(10;0)} = Z^{(01;0)} = Zr_1$ and $Z^{(11;0)} = Zr_2$. With $u = w_{10}$, $v = w_{11}$ one has $1 - r_1 = 2(u + v)$, $1 - r_2 = 4u$, so $r_2 \leq r_1$ means $v \leq u$, which gives $2(u + v) \leq \frac{4}{3}(2u + v) = \frac{4}{3}(1 - w_{00})$, and $1 - w_{00} \leq 1 - \lambda_0^n/Z = \mathsf{r}_n/(1 + \mathsf{r}_n)$ by (a). (d) is (T3) with the time 1, 3 and 2; $\lambda_- \leq \lambda_0$ because $Z^- \leq Z$. □

Lemma 17.11 (One positive sequence; the excess)**.** Let $\kappa_0 \geq \kappa_1 \geq \cdots \geq 0$, $\kappa_0 > 0$, $\sum \kappa_i < \infty$; $p_M = \sum_i \kappa_i^M$ for real $M \geq 1$, $\eta = \sum_{i\geq 1} \kappa_i/\kappa_0$, $s_M = \sum_{i\geq 1} (\kappa_i/\kappa_0)^M$, $D = 2\log p_1 - \log p_2$ and $D' = 2\log p_2 - \log p_4$. Then

(a) $\log(1 + \eta) \leq D \leq 2\log(1 + \eta)$;
(b) if $D < \log 2$, then $\eta \leq \varpi(e^D - 1)$;
(c) $D' \leq 2\log(1 + s_2) \leq 2s_2 \leq 2\eta^2$, so that $D' \leq \omega(D)$ if $D < \log 2$;
(d) $p_{MM'} \leq p_M^{M'}$ for real $M, M' \geq 1$;
(e) $s_M \leq s_{M_0}^{M/M_0}$ for real $M \geq M_0 \geq 1$.

For $\kappa_i = \lambda_i(\Sigma)^t$ one has $p_M = Z(Mt, \Sigma)$, $\eta = \mathsf{r}_t(\Sigma)$ and $s_M = \mathsf{r}_{Mt}(\Sigma)$ whenever Mt is an integer ≥ 3. Hence:

(f) $F/|\underline{n}|$ and $\delta/|\underline{n}|$ do not increase in any side, $F/|\underline{n}| \to P$ as the three sides tend to infinity, $\log(1 + \mathsf{r}_t(\Sigma)) \leq \delta(t, \Sigma)$, and $\log\lambda_0(N_1, N_2) \geq N_1 N_2 P$;
(g) for integers $A, B \geq N \geq 3$, the Casimir term $\varepsilon(A, B) = \log\lambda_0(A, B) - ABP$ satisfies $0 \leq \varepsilon(A, B) \leq (A + B)\,\delta_{\mathrm{c}}(N)/N^2$.

■

Proof sketch. $p_1 = \kappa_0(1 + \eta)$ and $p_M = \kappa_0^M(1 + s_M)$, so $e^D = (1 + \eta)^2/(1 + s_2)$. For $u_i \geq 0$ with the sum u and real $m \geq 1$, $\sum u_i^m \leq u^m$; this gives (d), (e) and $s_2 \leq \min\{\eta, \eta^2\}$, hence (a), and (c) as $D' = 2\log(1 + s_2) - \log(1 + s_4)$. (b): $e^D \geq (1 + \eta)^2/(1 + \eta^2) = 1 + 2\eta/(1 + \eta^2)$, and $\eta < 1$ by (a);

solve the quadratic inequality. (f): (d) with $M' = t'/t$ gives $Z(t', \Sigma) \le Z(t, \Sigma)^{t'/t}$ for $t' \ge t$, and (T1) the other sides; $\delta(t, \Sigma) = t(\log \lambda_0 - N_1 N_2 P) + \log(1 + \mathsf{r}_t)$, with $\log \lambda_0 = \lim_s F(s, \Sigma)/s \ge N_1 N_2 P$. (g): put $\varkappa = \delta_{\mathrm{c}}(N)/N^2$. For $t, A, B \ge N$, $\delta(t, A, B) = t\,\varepsilon(A, B) + \log(1 + \mathsf{r}_t(A, B))$, and by (f) $\log(1 + \mathsf{r}_t(A, B)) \le \log(1 + \mathsf{r}_N(A, B)) \le \delta(N, A, B) \le AB\varkappa$. By (f), $\varepsilon(A, B)/AB \le \delta(t, A, B)/tAB$ for every t, so $\varepsilon(A, B)/AB$ does not increase in A or B and tends to 0 as $A, B \to \infty$. With B as time, $\delta(s, t, B) = B\varepsilon(s, t) + \log(1 + \mathsf{r}_B(s, t)) \le B\varepsilon(s, t) + st\varkappa$; divide by s and let $s \to \infty$: $\varepsilon(t, B)/t \le B \lim_s \varepsilon(s, t)/st + \varkappa$, and as $t \to \infty$, $\limsup_t \varepsilon(t, B)/t \le \varkappa$. With A as time, $\delta(t, A, B) \le A\varepsilon(t, B) + tB\varkappa$; divide by t and let $t \to \infty$: $\varepsilon(A, B) \le (A + B)\varkappa$. □

Lemma 17.12 (Two static charges)**.** Fix a box, the time 3, $o \in \Sigma$ and $r = o + Re_1$, $1 \le R \le n_1 - 1$; let γ be the straight path of R bonds from o to r in the section, $V(\gamma)$ its ordered product, $\hat{\mathcal{H}} = L^2(\mathcal{X}; M_2(\mathbb{C}))$ with $\langle \Psi, \Psi' \rangle = \int \operatorname{tr}(\Psi^* \Psi')\, \mathrm{d}V$,

$$(\hat{\mathsf{K}}\Psi)(V) = \iint e^{-\beta A_\times(V,S,V')}\, S_o \Psi(V') S_r^{-1}\, \mathrm{d}S\, \mathrm{d}V', \qquad \hat{\mathsf{T}} = M_0 \hat{\mathsf{K}} M_0, \qquad (\Upsilon\psi)(V) = \psi(V) V(\gamma);$$

let C be the rectangle $R \times T$, $1 \le T \le n_3 - 1$, with the side γ in the layer 0, and p_ξ half the trace of the line that winds once along the direction 3 through $\xi \in \Sigma$. Then:

(a) $\hat{\mathsf{K}}$ and $\hat{\mathsf{T}}$ are Hilbert–Schmidt, self-adjoint and ≥ 0, they do not depend on n_3, and $\Upsilon^* \Upsilon = 2$;
(b) $2Z\langle W_C \rangle = \operatorname{Tr}(\mathsf{T}_0^{n_3 - T} \Upsilon^* \hat{\mathsf{T}}^T \Upsilon)$ and $\operatorname{Tr} \hat{\mathsf{T}}^{n_3} = 4Z\langle p_o p_r \rangle$;
(c) $0 \le \langle W_C \rangle \le (4\langle p_o p_r \rangle)^{T/n_3}$ and $\|\hat{\mathsf{T}}\|^{n_3} \le 4Z\langle p_o p_r \rangle$;
(d) $0 \le \langle p_o p_r \rangle \le z_+^{1-\varrho} z_-^{\varrho} + z_-^{1-\varrho} z_+^{\varrho}$, $\varrho = R/n_1$, and for $R \le n_1/2$

$$0 \le \langle W_C \rangle \le \big(8 z_-^{R/n_1}\big)^{T/n_3}, \qquad \frac{\|\hat{\mathsf{T}}\|}{\lambda_0(n_1, n_2)} \le \Big[8\big(1 + \mathsf{r}_{n_3}(n_1, n_2)\big) z_-^{R/n_1}\Big]^{1/n_3}.$$

(e) *Any two sites.* For any two sites $o \ne r$ of the section, $\hat{\mathsf{K}}$ and $\hat{\mathsf{T}} = \hat{\mathsf{T}}_{o,r}$ are defined by the same formulas, (a) holds, $\operatorname{Tr} \hat{\mathsf{T}}^{n_3} = 4Z\langle p_o p_r \rangle$ and $\|\hat{\mathsf{T}}\|^{n_3} \le 4Z\langle p_o p_r \rangle$, and (d) holds for $\langle p_o p_r \rangle$ and $\|\hat{\mathsf{T}}\|$ with R the distance of o_1 and r_1 on the circle of n_1 sites, $1 \le R \le n_1/2$; on a cube it holds as well with R the distance of o_2 and r_2 on the circle of n_2 sites.
(f) *Staircase loops in a box.* Let $1 \le T \le n_3 - 2$, let o be a site of the section, α_j and $R_j \ge 1$ integers, $o_j = o + \alpha_j e_1$ and $r_j = o + (\alpha_j + R_j) e_1$ for $0 \le j < T$, and let C be the closed path that runs in the layer 0 straight along the line of the direction 1 from o_0 to r_0, then for $j = 0, \dots, T - 1$ in turn one bond up at r_j and, if $j < T - 1$, straight from r_j to r_{j+1}, in the layer T straight from r_{T-1} to o_{T-1}, and then for $j = T - 1, \dots, 0$ in turn one bond down at o_j and, if $j > 0$, straight from o_j to o_{j-1} ("straight" from the site of the integer α to that of α': along the $|\alpha' - \alpha|$ bonds between them). Then, with the thermal numbers over (n_1, n_2),

$$|\langle W_C \rangle| \le \big(1 + \mathsf{r}_{n_3 - T}\big) \prod_{j=0}^{T-1} \frac{\|\hat{\mathsf{T}}_{o_j, r_j}\|}{\lambda_0(n_1, n_2)}, \qquad \text{and if all } R_j \le \tfrac{n_1}{2}: \quad |\langle W_C \rangle| \le \big(1 + \mathsf{r}_{n_3 - T}\big)\big[8(1 + \mathsf{r}_{n_3})\big]^{T/n_3} z_-^{\sum_j R_j/n_1 n_3}.$$

No side need be even. ■

Proof sketch. The inequality that replaces Hölder's inequality for traces is this. Let $A, C \ge 0$ be compact with eigenvalues a_i, c_j and eigenbases (e_i), (f_j), B bounded, $p, q \ge 1$ integers and $s = p + q$. Then $\langle e_i, A^p B^* C^q B e_i \rangle = \sum_j w_{ij} a_i^p c_j^q$ with $w_{ij} = |\langle f_j, B e_i \rangle|^2 \ge 0$, the row and column sums of (w_{ij}) are $\le \|B\|^2$. A zero normaliser below makes the weighted product sum vanish, and infinite normalisers give only a trivial bound. For finite positive normalisers, Young's inequality $a^p c^q \le \mathsf{a}^{p/s} \mathsf{c}^{q/s} \big(\frac{p}{s} \frac{a^s}{\mathsf{a}} + \frac{q}{s} \frac{c^s}{\mathsf{c}}\big)$, with $\mathsf{a} = \sum w_{ij} a_i^s$ and $\mathsf{c} = \sum w_{ij} c_j^s$, gives

$$0 \le \sum_i \langle e_i, A^p B^* C^q B e_i \rangle \le \|B\|^2 \Big(\sum_i a_i^s\Big)^{p/s} \Big(\sum_j c_j^s\Big)^{q/s}. \tag{17.v}$$

Its form with two operators bounds the modulus. For bounded B_1, B_2 the same eigenbases give $\sum_i \langle e_i, A^p B_1^* C^q B_2 e_i\rangle = \sum_{i,j} a_i^p c_j^q \overline{\langle f_j, B_1 e_i\rangle}\,\langle f_j, B_2 e_i\rangle$, and $\sum_{i,j} a_i^p c_j^q |\langle f_j, B_k e_i\rangle|^2 = \sum_i \langle e_i, A^p B_k^* C^q B_k e_i\rangle$ for $k = 1, 2$; Schwarz's inequality with the weights $a_i^p c_j^q \ge 0$, and then eq. 17.v for B_1 and for B_2, give

$$\Big|\sum_i \langle e_i, A^p B_1^* C^q B_2 e_i\rangle\Big| \le \|B_1\|\,\|B_2\| \Big(\sum_i a_i^s\Big)^{p/s} \Big(\sum_j c_j^s\Big)^{q/s}. \tag{17.vi}$$

(a) Symmetry follows from $A_\times(V', S^{-1}, V) = A_\times(V, S, V')$. For positivity substitute $S = g^{-1}Q$, use $e^{-\beta A_\times(V, g^{-1}Q, V')} = \mathsf{K}_0(V^g, V'^Q)$ (Lem. 17.9(a)), average over $g \in G^\Sigma$ (the left side does not depend on g) and expand K_0 by Lem. 17.9(b):

$$\langle \Psi, \hat{\mathsf{K}}\Psi\rangle = \sum_\alpha c_\alpha \operatorname{tr}(X_\alpha^* X_\alpha) \ge 0, \qquad X_\alpha = \iint \overline{\phi_\alpha(V^Q)}\, Q_o \Psi(V) Q_r^{-1}\, \mathrm{d}Q\, \mathrm{d}V.$$

The argument uses of o and r only that they are sites of the section. (b) With $\mathsf{P}_\xi = S_\xi^0 \cdots S_\xi^{T-1}$, $2W_C = \operatorname{tr}[V^0(\gamma)^* \mathsf{P}_o V^T(\gamma) \mathsf{P}_r^{-1}]$, and the layers write $2Z\langle W_C\rangle$ as the trace of the chain of the kernels of $\mathsf{T}_0^{n_3-T}$ and of $\Upsilon^* \hat{\mathsf{T}}^T \Upsilon$; at $T = n_3$ the trace over $M_2(\mathbb{C})$ gives $\operatorname{tr} \mathsf{P}_o \operatorname{tr} \mathsf{P}_r^{-1}$. (c) is eq. 17.v with $A = \mathsf{T}_0$, $C = \hat{\mathsf{T}}$, $B = \Upsilon$, $p = n_3 - T$, $q = T$ and $\sum a_i^{n_3} = Z$. (d) Take the time 1. The two winding lines become the multiplication $\hat{\mathsf{p}}$ by one function of the variables of a layer, taken at the layers o_1 and $o_1 + R$; each line holds exactly one bond flipped by ι_2, so $\hat{\mathsf{p}}$ exchanges the ranges of $P_\pm = \frac{1}{2}(1 \pm \mathsf{J}_2)$, on which $\operatorname{Tr}(\mathsf{T}_0^{n_1} P_\pm) = a_\pm$ by Lem. 17.10(ii). By eq. 17.v

$$Z\langle p_o p_r\rangle = \sum_\pm \operatorname{Tr}\big[(\mathsf{T}_0 P_\pm)^{n_1 - R}\, \hat{\mathsf{p}}\, (\mathsf{T}_0 P_\mp)^R\, \hat{\mathsf{p}}\big] \le \sum_\pm a_\pm^{1-\varrho} a_\mp^\varrho,$$

and for $\varrho \le \frac{1}{2}$ and $z_- \le z_+ \le 1$ each term is $\le z_-^\varrho Z$; finally $Z = \lambda_0^{n_3}(1 + \mathsf{r}_{n_3})$. (e) The proof of (a) uses of o and r only that they are sites of the section, and the trace of $\hat{\mathsf{T}}^{n_3}$ in (b) and the second bound of (c) do not use γ; so, for any two sites, $\hat{\mathsf{T}} \ge 0$, $4Z\langle p_o p_r\rangle = \operatorname{Tr} \hat{\mathsf{T}}^{n_3} \ge 0$ and $\|\hat{\mathsf{T}}\|^{n_3} \le 4Z\langle p_o p_r\rangle$: the sign of $\langle p_o p_r\rangle$ comes from the time 3. For the bound take the time 1. The line through $\xi \in \Sigma$ is then the function p_{ξ_2} of the variables V^{ξ_1} of the layer ξ_1, half the trace of the ordered product of the variables of its bonds of the direction 3 with the second coordinate ξ_2; whatever ξ_2 is, the line holds exactly one bond flipped by ι_2. So the multiplications $\hat{\mathsf{p}}_{o_2}$ and $\hat{\mathsf{p}}_{r_2}$, two different ones if $o_2 \ne r_2$, are real, of norm ≤ 1, and each exchanges the ranges of $P_\pm$. With R the distance of o_1 and r_1 on the circle, $1 \le R \le n_1/2$, the trace has $n_1 - R$ factors T_0 on one side of the two lines and R on the other, and, as in (d), with the names o and r exchanged if need be,

$$Z\langle p_o p_r\rangle = \sum_\pm \operatorname{Tr}\big[(\mathsf{T}_0 P_\pm)^{n_1 - R}\, \hat{\mathsf{p}}_{o_2}\, (\mathsf{T}_0 P_\mp)^R\, \hat{\mathsf{p}}_{r_2}\big];$$

eq. 17.vi with $A = \mathsf{T}_0 P_\pm$, $C = \mathsf{T}_0 P_\mp$, $B_1 = \hat{\mathsf{p}}_{o_2}$, $B_2 = \hat{\mathsf{p}}_{r_2}$, $p = n_1 - R$, $q = R$ bounds the term of $\pm$ in modulus by $a_\pm^{1-\varrho} a_\mp^\varrho$. With the sign from the time 3 this is the first statement of (d) for o and r, at every parity of n_1, n_2, n_3, and the bound of $\|\hat{\mathsf{T}}\|$ in (d) follows from it and from $\|\hat{\mathsf{T}}\|^{n_3} \le 4Z\langle p_o p_r\rangle$ as there. On a cube the time 2 does the same with o_2, r_2, the twisted pair being then the pair 23, whose quotient on the cube equals that of the pair 13 (Lem. 17.10(c)). (f) With the time 3, $U(C)$ is the ordered product of the runs in the layers and of the bonds $S_{r_j}^j$, $(S_{o_j}^j)^{-1}$; put $(\Upsilon_j \psi)(V) = \psi(V) V(o_j \to r_j)$, with $V(o_j \to r_j)$ the product along the straight run from o_j to r_j, $(\Gamma_j \Psi)(V) = V(o_{j-1} \to o_j) \Psi(V) V(r_j \to r_{j-1})$, a unitary operator of $\hat{\mathcal{H}}$ that commutes with M_0, and $\mathsf{Q} = \Upsilon_0^* \hat{\mathsf{T}}_{o_0, r_0} \Gamma_1 \hat{\mathsf{T}}_{o_1, r_1} \Gamma_2 \cdots \Gamma_{T-1} \hat{\mathsf{T}}_{o_{T-1}, r_{T-1}} \Upsilon_{T-1}$.

As in (b), $2Z\langle W_C\rangle = \operatorname{Tr}(\mathsf{T}_0^{n_3-T} \mathsf{Q})$; in an eigenbasis of T_0, $|\operatorname{Tr}(\mathsf{T}_0^{n_3-T}\mathsf{Q})| \le \|\mathsf{Q}\| \lambda_0^{n_3-T}(1 + \mathsf{r}_{n_3-T})$, with $n_3 - T \ge 2$; and $\|\mathsf{Q}\| \le 2 \prod_j \|\hat{\mathsf{T}}_{o_j, r_j}\|$, $Z \ge \lambda_0^{n_3}$. The second bound is (d) for each j. □

Lemma 17.13 (Several static charges; closed paths)**.** Fix a box with n_1 even and the time 3.

(a) For a finite set $\mathcal{V}$ of runs, each with a site $u(v) \in \Sigma$ and a sense, let $D(S) = \bigotimes_v D_v(S_{u(v)})$ on $(\mathbb{C}^2)^{\otimes\mathcal{V}}$, with $D_v(g) = g$ for the sense up and $D_v(g) = \bar g$ for the sense down, $(\hat{\mathsf{K}}_D\Psi)(V) = \iint e^{-\beta A_\times(V,S,V')} D(S)\Psi(V')\,\mathrm{d}S\,\mathrm{d}V'$ and $\hat{\mathsf{T}}_D = M_0\hat{\mathsf{K}}_D M_0$. Then $\hat{\mathsf{T}}_D \ge 0$ and $\|\hat{\mathsf{T}}_D\|^{n_3} \le \operatorname{Tr}\hat{\mathsf{T}}_D^{n_3} = 2^{\#\mathcal{V}} Z\langle\prod_v p_{u(v)}\rangle$.

(b) Let $\#\mathcal{V}$ be even, m_τ the number of the $u(v)$ in the layer $\tau \in \{1,\dots,n_1\}$ of the direction 1, and ℓ the number of τ with $m_1 + \dots + m_\tau$ odd. If $\ell \le n_1/2$, then $|\langle\prod_v p_{u(v)}\rangle| \le 2z_-^{\ell/n_1}$.

(c) Let C be a closed lattice path in a cube of h layers, $h+1 \le \min_\mu n_\mu$, that runs bonds of the direction 3 in T gaps, $n_3 \ge T+2$; in the gap j let v_j be the number of these bonds and ℓ_j the number of integers α such that an odd number of them have first coordinate $\le \alpha$, and assume $\ell_j \le n_1/2$. Then, with r taken over the section (n_1, n_2),

$$|\langle W_C\rangle| \le 2^{c_3(C)-1}(1+\mathsf{r}_{n_3-T}) \prod_j \Big[2^{\mathsf{v}_j+1}(1+\mathsf{r}_{n_3}) z_-^{\ell_j/n_1}\Big]^{1/n_3}.$$

(d) Let D be a continuous unitary representation of G on $\mathbb{C}^d$ with $D(-\mathbf{1}) = -\mathbf{1}$, and $W_C^D = d^{-1}\operatorname{tr} D(U(C))$. Then (a)–(c) hold with $D_v = D$ for the sense up and $D_v = \overline{D}$ for the sense down, with $d^{-1}\operatorname{tr} D$ of the line that winds once along the direction 3 through ξ in the place of p_ξ, $d^{\#\mathcal{V}}$ in the place of $2^{\#\mathcal{V}}$, and W_C^D, $d^{c_3(C)-1}$, $2d^{\mathsf{v}_j}$ in the place of W_C, $2^{c_3(C)-1}$, 2^{v_j+1}. ■

Proof sketch. (a) As in Lem. 17.12(a), with $D(g^{-1}Q) = D(g)^{-1}D(Q)$; the trace of the n_3-th power gives $\prod_v \operatorname{tr} D_v(\mathsf{P}_{u(v)})$, and $\operatorname{tr}\bar g = \operatorname{tr} g$. (b) With the time 1, $Z\langle\prod_v p_{u(v)}\rangle = \operatorname{Tr}\prod_\tau(\mathsf{T}_0\hat{\mathsf{p}}^\tau)$, $\hat{\mathsf{p}}^\tau$ the multiplication by the product of the line functions of the layer τ. Each line holds one flipped bond, so $\hat{\mathsf{p}}^\tau$ maps the range of P_s to that of $P_{s(-1)^{m_\tau}}$; splitting the trace over $s = \pm$, the factor of the layer τ becomes $\mathsf{T}_0 P_{s_\tau}\hat{\mathsf{p}}^\tau$, with $s_\tau = -s$ for exactly ℓ layers. Hölder's inequality for the trace of the n_1 factors (Lem. YM3_I:86.1), and $\|\mathsf{C}\hat{\mathsf{p}}\|_{n_1}^{n_1} = \operatorname{Tr}(\mathsf{C}\hat{\mathsf{p}}^2\mathsf{C})^{n_1/2} \le \operatorname{Tr}\mathsf{C}^{n_1} = a_\pm$ for $\mathsf{C} = \mathsf{T}_0 P_\pm$ (by min–max, as $0 \le \mathsf{C}\hat{\mathsf{p}}^2\mathsf{C} \le \mathsf{C}^2$), bound the term of s by $a_s^{1-\ell/n_1} a_{-s}^{\ell/n_1}$. (c) Cut C into pieces in single layers and runs of the direction 3: $2W_C = \Gamma_0 D_0(S^0)\Gamma_1 \cdots D_{T-1}(S^{T-1})\Gamma_T$, where D_j is the tensor product over the runs of the gap j and Γ_j, a function of V^j, is the tensor product over the pieces of the layer j of unitary 2×2 matrices, of cups (linear forms on $\mathbb{C}^2\otimes\mathbb{C}^2$ with the entries of a matrix of G) and of caps (vectors with such entries).

Cups and caps have the norm $\sqrt{2}$ and occur at the changes from down to up and from up to down, $c_3(C)$ times each, so $\prod_j \sup_V \|\Gamma_j(V)\| = 2^{c_3(C)}$. As in Lem. 17.12(b), $2Z\langle W_C\rangle = \operatorname{Tr}(\mathsf{T}_0^{n_3-T}\mathsf{Q})$ with $\mathsf{Q} = \Gamma_0\hat{\mathsf{T}}_{D_0}\Gamma_1\cdots\hat{\mathsf{T}}_{D_{T-1}}\Gamma_T$; in an eigenbasis of T_0, $|\operatorname{Tr}(\mathsf{T}_0^{n_3-T}\mathsf{Q})| \le \|\mathsf{Q}\|\lambda_0^{n_3-T}(1+\mathsf{r}_{n_3-T})$, and (a), (b) bound $\|\hat{\mathsf{T}}_{D_j}\|^{n_3} \le 2^{\mathsf{v}_j+1}\lambda_0^{n_3}(1+\mathsf{r}_{n_3})z_-^{\ell_j/n_1}$; numbering the layers so that C lies in the layers $1,\dots,h$ makes the ℓ of (b) equal to ℓ_j. (d) The proofs of (a)–(c) use of the representation $g \mapsto g$ only that it is continuous and unitary, which gives the positivity of $\hat{\mathsf{T}}_D$ and pointwise unitary pieces in the layers, and that it changes its sign at $-\mathbf{1}$, which makes every line function odd under the flip and of modulus ≤ 1; a cup or a cap of D, a linear form or a vector with the entries of a matrix of $D(G)$, has the norm $\sqrt{d}$, and the trace over $(\mathbb{C}^d)^{\otimes\mathcal{V}}$ of the n_3-th power gives $d^{\#\mathcal{V}}Z$ times the mean of the product of the line functions. □

Lemma 17.14 (A floor at one length bounds the loops)**.** Let $\beta \ge 0$ and $L_1, L_2, L_3 \ge 3$ integers; take the first direction as time over the section (L_2, L_3), with its λ_0, r_s and Λ_-, and put $\phi = \log(\lambda_0/\Lambda_-) \in [0,\infty]$. Let C be a rectangle $R\times T$ of a $(1,3)$-plane with $1 \le R \le L_1 - 3$ and $1 \le T \le L_3 - 1$, and p_o, p_r the lines of Lem. 17.12 through its corners in one layer of the direction 3. Then

$$0 \le \langle W_C\rangle \le (4\langle p_o p_r\rangle)^{T/L_3}, \qquad \langle p_o p_r\rangle \le \frac{1+\mathsf{r}_{L_1-R}}{1+\mathsf{r}_{L_1}}\Big[e^{-\phi R} + z_-(L_1-R, L_2, L_3)\Big];$$

so, if $\phi > 0$, $\limsup_{L_1\to\infty} \langle W_C\rangle_{(L_1,L_2,L_3)} \le 4^{T/L_3} e^{-\phi R T/L_3}$. ■

Proof sketch. The first inequality is Lem. 17.12(c). With the time 1, as in the proof of Lem. 17.12(d), $Z\langle p_o p_r\rangle = \sum_\pm \operatorname{Tr}(\mathsf{T}_\pm^{L_1-R}\hat{\mathsf{p}}\,\mathsf{T}_\mp^{R}\hat{\mathsf{p}})$ with $\mathsf{T}_\pm = \mathsf{T}_0 P_\pm$ and $\|\hat{\mathsf{p}}\| \le 1$. The nonzero eigenvalues of T_- are those of the sectors 01 and 11, so $\|\mathsf{T}_-\| = \Lambda_-$, and $\|\mathsf{T}_+\| \le \lambda_0$. For $B \ge 0$ of trace class and $Y \ge 0$ bounded, $\operatorname{Tr}(BY) = \operatorname{Tr}(B^{1/2} Y B^{1/2}) \le \|Y\| \operatorname{Tr} B$. With $B = \mathsf{T}_+^{L_1-R}$ and $Y = \hat{\mathsf{p}}\mathsf{T}_-^R\hat{\mathsf{p}}$ the term of $+$ is $\le \Lambda_-^R \lambda_0^{L_1-R}(1+\mathsf{r}_{L_1-R})$; with $B = \mathsf{T}_-^{L_1-R}$ and $Y = \hat{\mathsf{p}}\mathsf{T}_+^R\hat{\mathsf{p}}$ the term of $-$ is $\le \lambda_0^R \operatorname{Tr}\mathsf{T}_-^{L_1-R} = \lambda_0^R a_-(L_1-R,L_2,L_3) = \lambda_0^{L_1}(1+\mathsf{r}_{L_1-R}) z_-(L_1-R,L_2,L_3)$. Divide by $Z = \lambda_0^{L_1}(1+\mathsf{r}_{L_1})$. As $L_1\to\infty$ the quotient of the thermal factors tends to 1, λ_0 being simple, and $z_-(s,L_2,L_3) \le \lambda_0^{-s}\operatorname{Tr}\mathsf{T}_-^s \le (\Lambda_-/\lambda_0)^{s-2}\lambda_0^{-2}\operatorname{Tr}\mathsf{T}_-^2 \to 0$ if $\phi > 0$. □

Lemma 17.15 (Flux weight and vortex line)**.**

(a) For integers $n_1' \ge n_1 \ge 3$, $n_3' \ge n_3 \ge 3$, $n_2 \ge 3$ and $\kappa = n_1' n_3'/(n_1 n_3)$, a real number,

$$a_-(n_1',n_2,n_3') \le a_-(\underline{n})^\kappa, \qquad z_-(n_1',n_2,n_3') \le \big[z_-(\underline{n})\, e^{\delta(\underline{n})}\big]^\kappa.$$

(b) For all integers $t, N, n \ge 3$, with $\varepsilon = \varepsilon_{\mathrm{v}}(t,N)$,

$$2z_-(t,n,N) \le n\varepsilon + \mathsf{r}_n(t,N), \qquad n\varepsilon \le \mathsf{r}_n(t,N) - \log(1-2z_-(t,2n,N)).$$

■

Proof sketch. (a) By Lem. 17.10(d), $a_- = \sum_i \mu_i^{n_1}$ with μ_i independent of n_1, so Lem. 17.11(d) with a real exponent gives $a_-(n_1',n_2,n_3) \le a_-(n_1,n_2,n_3)^{n_1'/n_1}$; the same holds in the third side, and $Z(n_1',n_2,n_3') \ge e^{\kappa|\underline{n}|P}$. (b) With the time along the column, $Z(t,n,N) = \lambda_0^n(1+\mathsf{r}_n)$ and $Z^-(t,n,N) = \sum_i \mu_i''^{\,n}$ with $\sup_i \mu_i'' = \lambda_0 e^{-\varepsilon}$; so $1-2z_- = Z^-/Z \ge e^{-n\varepsilon}/(1+\mathsf{r}_n) \ge 1-n\varepsilon-\mathsf{r}_n$. For the second inequality, which uses that the twisted spectrum is ≥ 0, $\sum \mu_i''^{\,2n} \le \lambda_-^n \sum \mu_i''^{\,n}$ gives $\lambda_-^n \ge Z^-(t,2n,N)/Z^-(t,n,N) \ge (1-2z_-(t,2n,N))\lambda_0^{2n}/(\lambda_0^n(1+\mathsf{r}_n))$. □

Proposition 17.16 (Recursion; excess and thermal numbers)**.** Let $\beta \ge 0$ and let $l \ge 3$ be an integer.

(a) With $\Sigma_1 = (l,l)$, $\Sigma_2 = (2l,l)$, $\Sigma_3 = (2l,2l)$ and $(d_1,d_2,d_3) = (d_1,d_2,\mathsf{e})$: $\log(1+\mathsf{r}_l(\Sigma_j)) \le d_j(l) \le 2\log(1+\mathsf{r}_l(\Sigma_j))$, and $X = 4d_1 + 2d_2 + \mathsf{e}$.

(b) If $Y_2(\mathsf{e}(l)) < \log 2$, then $d_1(2l) \le \omega(\mathsf{e}(l))$, $d_2(2l) \le \omega(Y_1(\mathsf{e}(l)))$, $\mathsf{e}(2l) \le \Psi(\mathsf{e}(l))$ and $X(2l) \le \Phi(\mathsf{e}(l))$.

(c) $\Psi(x) \le 12.9x^2$ and $\Phi(x) \le 19.89x^2$ for $0 < x \le 0.069$; $\Psi(x) \le 11.03x^2$ and $\Phi(x) \le 17.71x^2$ for $0 < x \le \frac{1}{20}$; $\Psi(x) < x$ for $0 < x < 0.0740538$.

(d) Under the criterion, for $n = 2^k l$, $k \ge 1$, and $x = \mathsf{e}(n/2) \le \theta^{n/2l}/12.9$:

$$\mathsf{r}_{n/2}(n,n) \le 0.51832\,x, \quad \mathsf{r}_{n/2}(2n,n) \le 1.09990\,x, \quad \mathsf{r}_{n/2}(2n,2n) \le 2.53905\,x;$$

$$\mathsf{r}_n(n,n) \le 0.001615\,\theta^{n/l},$$

$$\mathsf{r}_n(2n,n) \le 0.00732\,\theta^{n/l},$$

$\delta_{\mathrm{c}}(n) \le 0.01655\,\theta^{n/l}$, $d_1(n) \le 0.0032288\,\theta^{n/l}$, $d_2(n) \le 0.0145396\,\theta^{n/l}$; and $\delta_{\mathrm{c}}(N) < 0.1324\,\theta^{N/2l}$ for every integer $N \ge 2l$.

(e) Let $\tilde\omega(D) = 2\log(1+\varpi(e^D-1)^2) \le \omega(D)$, and let $\tilde Y_1, \tilde Y_2, \tilde\Psi, \tilde\Phi$ be Y_1, Y_2, Ψ, Φ with $\tilde\omega$ in the place of ω. Then (b) holds with $\tilde\omega$, $\tilde Y_1$, $\tilde\Psi$, $\tilde\Phi$. With $X_0 = 0.069$ and $X_{k+1} = \tilde\Psi(X_k)$, the iterates $X_1,\dots,X_5$ are 0.0604446, 0.0432902, 0.0195722, 0.0034309, 0.0000960 to the digits shown, the X_k decrease to 0, and $\bar\delta_j = \sum_{s\ge 0} 8^{-s-1}\tilde\Phi(X_{j-1+s})$ decreases in j, with $\bar\delta_4 \le 0.00073051$. If $\mathsf{e}(l) \le 0.069$, with no other hypothesis, then $\mathsf{e}(2^k l) \le X_k$ and $\delta_{\mathrm{c}}(2^j l) \le \bar\delta_j$ for all $k \ge 0$ and $j \ge 1$.

■

Proof sketch. (a) Double the sides of the cube one after the other: by (T1), X is the sum of $4[2F(l,l,l)-F(2l,l,l)]$, $2[2F(2l,l,l)-F(2l,2l,l)]$ and $2F(2l,2l,l)-F(2l,2l,2l)$, and each bracket is the D of Lem. 17.11 for $\kappa_i=\lambda_i(\Sigma_j)^l$. (b) Over $(2l,2l)$ with $\kappa_i=\lambda_i^l$: $D=\mathsf{e}(l)$ and $D'=d_1(2l)\le\omega(\mathsf{e})$. Doubling one side of the section at most doubles a defect, up to defects already bounded: by Lem. 17.11(d) with the time in the second direction, $F(l,4l,2l)\le 2F(l,2l,2l)$, so

$$2F(l,4l,2l)-F(2l,4l,2l)\le 2\mathsf{e}(l)+d_1(2l)\le Y_1(\mathsf{e}(l)),$$

and Lem. 17.11(c) over $(4l,2l)$ gives $d_2(2l)\le\omega(Y_1)$; in the same way $2F(l,4l,4l)-F(2l,4l,4l)\le Y_2$, and over $(4l,4l)$ one doubling of the time squares the defect: $\mathsf{e}(2l)\le\omega(Y_2)=\Psi$. Then (a) at $2l$. (c) $\varpi(x)/x$ and $(e^D-1)/D$ increase, so $\Psi(x)/x^2$ and $\Phi(x)/x^2$ do not decrease on $(0,0.0741]$; their values at 0.069 are 12.893528 and 19.881793, at $\frac{1}{20}$ they are 11.029686 and 17.709146, to the digits shown; $\Psi(x)/x$ increases; its positive fixed point lies in $(0.0740538,0.0740539)$, and $Y_2<\log 2$ throughout this interval. (d) Induction with (b), (c): $12.9\,\mathsf{e}(2^{k+1}l)\le(12.9\,\mathsf{e}(2^kl))^2$, so $\mathsf{e}(2^kl)\le\theta^{2^k}/12.9\le 0.069$ and $X(2^{k+1}l)\le\frac{19.89}{166.41}\theta^{2^{k+1}}\le 0.11953\,\theta^{2^{k+1}}$. At $l'=n/2$ the sections $(2l',2l')$, $(4l',2l')$, $(4l',4l')$ have at the time l' the defects x, $\le Y_1(x)$, $\le Y_2(x)$, and Lem. 17.11(b) with the monotone quotients at 0.069 gives the first three bounds. Then $\mathsf{r}_n=s_2\le\eta^2$ gives $\mathsf{r}_n(n,n)\le\frac12\omega(x)\le 0.268652\,x^2\le 0.001615\,\theta^{n/l}$, likewise $\mathsf{r}_n(2n,n)$, and $d_1(n)\le\omega(x)$, $d_2(n)\le\omega(Y_1(x))$. For the excess, the pressure cancels in $X(n)=8\delta_{\mathrm{c}}(n)-\delta_{\mathrm{c}}(2n)$, and Lem. 17.11(f) gives

$$\delta_{\mathrm{c}}(n)=\sum_{i\ge0}8^{-i-1}X(2^in)\le 0.11953\,\theta^{n/l}\sum_{i\ge0}8^{-i-1}\theta^{(2^i-1)n/l}\le 0.01655\,\theta^{n/l}.$$

For an integer N with $2^jl\le N<2^{j+1}l$, $j\ge1$, Lem. 17.11(f) gives $\delta_{\mathrm{c}}(N)\le 8\,\delta_{\mathrm{c}}(2^jl)\le 0.1324\,\theta^{2^j}<0.1324\,\theta^{N/2l}$. (e) By Lem. 17.11(b), (c), $D'\le 2\log(1+\eta^2)\le\tilde\omega(D)$, and the proof of (b) uses of ω only that it increases. The numbers $X_1,\dots,X_5$ and $\bar\delta_4$ are the values of the recursion $X_{k+1}=\tilde\Psi(X_k)$ from $X_0=0.069$ and of the series that defines $\bar\delta_4$; the X_k decrease to 0 because $\tilde\Psi\le\Psi$ and $\Psi(x)<x$ on $(0,0.0740538)$, and $\bar\delta_j$ decreases because $\tilde\Phi$ increases. If $\mathsf{e}(l)\le X_0$, the induction gives $\mathsf{e}(2^kl)\le X_k$, and $\delta_{\mathrm{c}}(2^jl)=\sum_{s\ge0}8^{-s-1}X(2^{j+s}l)\le\sum_s 8^{-s-1}\tilde\Phi(X_{j+s-1})=\bar\delta_j$. □

Proposition 17.17 (Vortex lines on tubes)**.** Assume the criterion; put $l_j=2^{j+1}l$ and $w=2^j$.

(a) $l'\varepsilon_{\mathrm{v}}(l')\le 0.01643\,\theta^{l'/l}$ for $l'=2^kl$, $k\ge1$.

(b) Let $l'=2^kl$, $k\ge1$, $l'\varepsilon_{\mathrm{v}}(l')\le e^{-E}$, $\hat N\ge2$ an integer, $y=\hat Ne^{-E}$, $\mathsf{t}=\theta^{l'/l}$, $\bar z_1=y+\frac12(0.001615\,\mathsf{t})^{2\hat N}$ and $\bar z_2=\bar z_1^4e^{8\hat N\cdot0.01655\,\mathsf{t}}<\frac12$. Then

$$2l'\varepsilon_{\mathrm{v}}(2l')\le\frac{2}{\hat N}\Big\{(0.001615\,\mathsf{t}^2)^{\hat N/2}+\frac{2\bar z_2}{1-2\bar z_2}\Big\}.$$

(c) $l_j\varepsilon_{\mathrm{v}}(l_j)\le e^{-\sigma^+l_j^2}$ for all $j\ge0$.

(d) For real $M\ge w\mathsf{q}$ with Ml_j an integer: $z_-(l_j,Ml_j,l_j)\le Me^{-\varsigma w^2}$ and $\delta(l_j,Ml_j,l_j)\le 0.01655\,M\theta^{l_j/l}$.

(e) Let $l'=l_j$ and let $N_3\ge l'$ and $n_2\ge\mathsf{q}l'^2/2l$ be integers. Over the section (n_2,N_3), with the first direction as time, the top Λ_- of the sectors odd under the flip of the side N_3 lies at least $\sigma^+N_3-\frac{N_3}{l'^2}\big[\log\frac{n_2}{l'}+0.01655\,\frac{n_2}{l'}\theta^{l'/l}\big]$ above the vacuum.

■

Proof sketch. (a) The second inequality of Lem. 17.15(b) at $t=N=n=l'$, with $z_-(l',2l',l')\le\mathsf{r}_{l'}(2l',l')\le 0.00732\,\mathsf{t}$ (Lem. 17.10(b), the time 1) and $\mathsf{r}_{l'}(l',l')\le 0.001615\,\mathsf{t}$ (Prop. 17.16(d)). (b) Three moves. The first inequality of Lem. 17.15(b) on the box $(l',2\hat Nl',l')$, with $\mathsf{r}_{2\hat Nl'}(l',l')\le\mathsf{r}_{l'}(l',l')^{2\hat N}$ (Lem. 17.11(e)), gives $z_-\le\bar z_1$. Lem. 17.15(a) with $\kappa=4$ passes to the box $(2l',2\hat Nl',2l')$, a fourth power at the cost $e^{4\delta}$, with $\delta(l',2\hat Nl',l')\le 2\hat N\delta_{\mathrm{c}}(l')$. The second inequality of Lem. 17.15(b) for the tube $2l'\times2l'$ at the time $\hat Nl'$, with $\mathsf{r}_{\hat Nl'}(2l',2l')\le\mathsf{r}_{2l'}(2l',2l')^{\hat N/2}$, returns to the vortex line.

(c) Take $\hat N_j = 4\cdot 3^j$ and write $\mathsf{t}_j = \theta^{l_j/l} = e^{-2^j\cdot 2\Lambda}$. For real E and $\mathsf{t}\in(0,1]$ let $\mathsf{H}_j(E,\mathsf{t})$ be the right member of (b) at $l'=l_j$ and $\hat N=\hat N_j$, formed with $y=\hat N_j e^{-E}$, $\bar z_1 = y+\frac12(0.001615\,\mathsf{t})^{2\hat N_j}$ and $\bar z_2 = \bar z_1^4 e^{8\hat N_j\cdot 0.01655\,\mathsf{t}}$, where $\bar z_2<\frac12$. Identically $\mathsf{H}_j = \frac{2}{\hat N_j}y^4 e^{8\hat N_j\cdot 0.01655\,\mathsf{t}}\mathsf{B}_j$, that is,

$$-\log\mathsf{H}_j(E,\mathsf{t}) = 4E - 3\log\hat N_j - 8\hat N_j\cdot 0.01655\,\mathsf{t} - \log(2\mathsf{B}_j),\qquad \mathsf{B}_j = \frac{(0.001615\,\mathsf{t}^2)^{\hat N_j/2}}{y^4\,e^{8\hat N_j\cdot 0.01655\,\mathsf{t}}} + \frac{2(\bar z_1/y)^4}{1-2\bar z_2}.$$

One value of θ serves all. Let $\Delta = 2\Lambda - 0.23284$, which is ≥ 0 as $2\log\frac{1}{0.8901}\ge 0.23284$, and $\mathring{\mathsf{t}}_j = e^{-2^j\cdot 0.23284}$, the value of t_j at $2\Lambda=0.23284$, so that $\mathsf{t}_j = \mathring{\mathsf{t}}_j e^{-2^j\Delta}$. Let $E_0=4.3414$, and let $E_1,E_2,\dots$ be numbers with $E_{j+1}\le -\log\mathsf{H}_j(E_j,\mathring{\mathsf{t}}_j)$, $\bar z_2<\frac12$ at $(E_j,\mathring{\mathsf{t}}_j)$. Then $l_j\varepsilon_{\rm v}(l_j)\le e^{-E_j-4^j\Delta}$ for all $j\ge 0$ and every $\theta\le 0.8901$. At $j=0$ this is (a) at $l'=2l$, as $0.01643\,\theta^2 = \exp[-\log\frac{1}{0.01643}-0.23284-\Delta]$ and $\log\frac{1}{0.01643}+0.23284\ge 4.3414$. From j to $j+1$ apply (b) with $E=E_j+4^j\Delta$ and $\mathsf{t}=\mathsf{t}_j$, and compare each number with its value at $(E_j,\mathring{\mathsf{t}}_j)$: y is that value times $e^{-4^j\Delta}$; $(0.001615\,\mathsf{t})^{2\hat N_j}$ is it times $e^{-2^{j+1}\hat N_j\Delta}\le e^{-4^j\Delta}$; so $\bar z_1$ is at most it times $e^{-4^j\Delta}$; as $e^{8\hat N_j\cdot 0.01655\,\mathsf{t}}$ increases in t, $\bar z_2$ is at most it times $e^{-4^{j+1}\Delta}$, and so is $\frac{2\bar z_2}{1-2\bar z_2}$, as $u\mapsto 1/(1-2u)$ increases; and $(0.001615\,\mathsf{t}^2)^{\hat N_j/2}$ is it times $e^{-2^j\hat N_j\Delta}\le e^{-4^{j+1}\Delta}$, as $\hat N_j\ge 4\cdot 2^j$.

So each of the two terms of the right member of (b) is at most its value at $(E_j,\mathring{\mathsf{t}}_j)$ times $e^{-4^{j+1}\Delta}$, and

$$l_{j+1}\varepsilon_{\rm v}(l_{j+1})\le \mathsf{H}_j(E_j+4^j\Delta,\mathsf{t}_j)\le \mathsf{H}_j(E_j,\mathring{\mathsf{t}}_j)\,e^{-4^{j+1}\Delta}\le e^{-E_{j+1}-4^{j+1}\Delta}.$$

Numerical bounds. The functions in the recursion are explicit and monotone on the intervals used above. Outward interval evaluation gives

x	$\Psi(x)/x^2$	$\Phi(x)/x^2$
0.069	$(12.89352831, 12.89352832)$	$(19.88179283, 19.88179284)$
0.05	$(11.02968611, 11.02968612)$	$(17.70914610, 17.70914611)$.

For the successive lower bounds $E_0,\dots,E_6$ below, the differences $-\log\mathsf{H}_j(E_j,\mathring{\mathsf{t}}_j)-E_{j+1}$ exceed, respectively,

$$9.7\cdot 10^{-5},\quad 5.1\cdot 10^{-5},\quad 3.8\cdot 10^{-6},\quad 6.4\cdot 10^{-5},\quad 3.5\cdot 10^{-5},\quad 7.6\cdot 10^{-5}.$$

In each evaluation $\bar z_2<1/2$. The geometric tail displayed below gives a lower bound greater than 1.909266. The sharper defect recursion likewise gives $\bar\delta_4<0.00073051$. These bounds justify the directions of rounding used in the argument.

The numbers at $2\Lambda=0.23284$. There B_j is a number close to 2, and one may take $E_1,\dots,E_6 =$ 11.3558, 35.5848, 128.3243, 495.6445, 1962.8155, 7829.1628, each at most $-\log\mathsf{H}_j(E_j,\mathring{\mathsf{t}}_j)$ at the one before. For $j\ge 6$ the losses $3\log\hat N_j + 8\hat N_j\cdot 0.01655\,\mathring{\mathsf{t}}_j + \log(2\mathsf{B}_j)$ are at most $3\log\hat N_j+\log 4.01$, so that $E_{j+1}=4E_j-3\log\hat N_j-\log 4.01$ may be taken, and this closed recursion gives

$$4^{-j}E_j\ge 4^{-6}E_6 - \sum_{i\ge 6}4^{-i-1}(3\log 4+3i\log 3+\log 4.01)\ge 1.909266.$$

So $4^{-j}E_j\ge 0.23284+1.67$ for every $j\ge 0$, hence $4^{-j}(E_j+4^j\Delta)\ge 2\Lambda+1.67$, and $l_j\varepsilon_{\rm v}(l_j)\le\exp[-4^j(2\Lambda+1.67)] = e^{-\sigma^+ l_j^2}$. (d) The first inequality of Lem. 17.15(b) with $n=Ml_j$, (c), and $\mathsf{r}_{Ml_j}(l_j,l_j)\le(0.001615\,\theta^{2w})^M\le e^{-2\Lambda Mw}\le e^{-\varsigma w^2}$, as $2\Lambda\mathsf{q}=\varsigma$, give $2z_-\le(M+1)e^{-\varsigma w^2}$; Lem. 17.11(f) gives $\delta\le M\delta_{\rm c}(l_j)$. (e) For every integer $t\ge l'$, by Lem. 17.10(d) and Lem. 17.15(a) at the real factors t/l' and N_3/l', $\Lambda_-^t\le a_-(t,n_2,N_3)\le a_-(l',n_2,l')^{tN_3/l'^2}$; so $\log\Lambda_-\le\frac{N_3}{l'^2}[F+\log z_-](l',n_2,l')$, while $\log\lambda_0(n_2,N_3)\ge n_2N_3P$. Subtracting, $\log(\lambda_0/\Lambda_-)\ge\frac{N_3}{l'^2}[-\log z_--\delta](l',n_2,l')$, and (d) with $M=n_2/l'\ge w\mathsf{q}$ bounds the bracket below by $\varsigma w^2-\log M-0.01655\,M\theta^{l'/l}$, with $\varsigma w^2=\sigma^+l'^2$. □

Proposition 17.18 (Slabs)**.** For integers $N, a, b, t \geq 3$ let $\mathsf{e}^{(N)}(t) = 2F(t, 2t, N) - F(2t, 2t, N)$, $d^{(N)}(t) = 2F(t, t, N) - F(2t, t, N)$, $P^{(N)} = \inf_{a,b} F(a, b, N)/ab$ and $\delta^{(N)}(a, b) = F(a, b, N) - abP^{(N)}$.

(a) For every $\beta \geq 0$: $0 \leq \delta^{(N)}(a, b) \leq \delta(a, b, N)$, $\delta^{(N)}(a, b)/ab$ does not increase in a or b, $\delta^{(N)}(a, a)/a^2 \to 0$, and $\log(1 + \mathsf{r}_a(b, N)) \leq \delta^{(N)}(a, b)$; if $Y_1(\mathsf{e}^{(N)}(t)) < \log 2$, then $d^{(N)}(2t) \leq \omega(\mathsf{e}^{(N)}(t))$ and $\mathsf{e}^{(N)}(2t) \leq \omega(Y_1(\mathsf{e}^{(N)}(t)))$.

Assume now the criterion.

(b) Let $n_0 = 2^k l \leq N < 2n_0$, $k \geq 1$, $\chi' = 0.3602\,\theta^{n_0/l}$ and $t_i = 2^i n_0$. Then $2.72\,\mathsf{e}^{(N)}(t_i) \leq \chi'^{2^i}$, $\delta^{(N)}(t_i, t_i) \leq 0.133\,\chi'^{2^i}$, $\delta^{(N)}(n_0, n_0) \leq 0.01655\,\frac{N}{n_0}\theta^{n_0/l}$, $\mathsf{r}_{t_i}(t_i, N) \leq 0.14\,\chi'^{2^i}$, and $\mathsf{r}_{t_i}(b, N) \leq 0.28\,\chi'^{2^i}$ for $t_i \leq b < 2t_i$.

(c) Let $N = 2^k l$, $k \geq 1$, $\chi = 0.03\,\theta^{N/l}$ and $t_i = 2^i N$. Then $2.06\,\mathsf{e}^{(N)}(t_i) \leq \chi^{2^i}$, $d^{(N)}(t_i) \leq 0.1202\,\chi^{2^i}$, $\mathsf{r}_{t_i}(t_i, N) \leq 0.0613\,\chi^{2^i}$, $\mathsf{r}_{t_i}(2t_i, N) \leq \chi^{2^i}$ and $\delta^{(N)}(t_i, t_i) \leq 0.1826\,\chi^{2^i}$. If moreover $N = 2wl$ with $w \geq 2$, and $\hat{N}$, g are as in Thm. AW(g1), then $t_i\varepsilon_{\mathrm{v}}(t_i, N) \leq \exp[-2^i(\sigma^+ N^2 - \mathsf{g})]$ for all $i \geq 0$.

■

Proof sketch. (a) The slab of thickness N is a datum in two dimensions: $(a, b) \mapsto F(a, b, N)$ is symmetric and has the trace formula in each variable, so Lem. 17.11 and the proof of Prop. 17.16(b) go over with one side of the section frozen; $P^{(N)} \geq NP$ gives $\delta^{(N)} \leq \delta(\cdot, \cdot, N)$. (b) The pressures cancel at the start: $\mathsf{e}^{(N)}(n_0) \leq 2\delta(n_0, 2n_0, N) \leq 4\frac{N}{n_0}\delta_{\mathrm{c}}(n_0) < 0.1324\,\theta^{n_0/l} \leq 0.1049$. On $(0, 0.1049]$, $\omega(Y_1(x)) \leq 2.72x^2$ and $\omega(x) \leq 0.5593x^2$, so $2.72\,\mathsf{e}^{(N)}(t_{i+1}) \leq (2.72\,\mathsf{e}^{(N)}(t_i))^2$ and $2.72\,\mathsf{e}^{(N)}(n_0) \leq \chi'$. Since $4F(t, t, N) - F(2t, 2t, N) = 2d^{(N)}(t) + \mathsf{e}^{(N)}(t) = 4\delta^{(N)}(t, t) - \delta^{(N)}(2t, 2t)$, and for $i \geq 1$, $2d^{(N)}(t_i) + \mathsf{e}^{(N)}(t_i) \leq (2 \cdot 0.5593 + 2.72)\,\mathsf{e}^{(N)}(t_{i-1})^2 \leq 0.518842\,\chi'^{2^i}$, summation with the weights 4^{-j-1} gives $\delta^{(N)}(t_i, t_i) \leq 0.13236\,\chi'^{2^i}$ for $i \geq 1$; at $i = 0$, $\delta^{(N)}(n_0, n_0) \leq \delta(n_0, n_0, N) \leq \frac{N}{n_0}\delta_{\mathrm{c}}(n_0) < 0.0331\,\theta^{n_0/l} \leq 0.0919\,\chi'$.

Finally $\log(1 + \mathsf{r}_{t_i}(b, N)) \leq \delta^{(N)}(t_i, b) \leq \frac{b}{t_i}\delta^{(N)}(t_i, t_i)$. (c) At the dyadic thickness the start is $\mathsf{e}^{(N)}(N) = d_2(N) \leq 0.0145396\,\theta^{N/l} \leq 0.01152$ and $d^{(N)}(N) = d_1(N) \leq 0.0032288\,\theta^{N/l}$; for $x \leq 0.0116$, $\omega(Y_1(x)) \leq 2.06x^2$, $\omega(x) \leq 0.5059x^2$ and $e^x - 1 \leq 1.006x$, and for $D \leq 0.00286$, $\varpi(e^D - 1) \leq 0.5008D$. So $2.06\,\mathsf{e}^{(N)}(t_i) \leq \chi^{2^i}$; $d^{(N)}(t_i) \leq 0.5059\,\mathsf{e}^{(N)}(t_{i-1})^2 \leq 0.1202\,\chi^{2^i}$; Lem. 17.11(b) over (t_i, N) gives $\mathsf{r}_{t_i}(t_i, N) \leq 0.5008\,d^{(N)}(t_i)$, and Lem. 17.11(a) over $(2t_i, N)$ gives $\mathsf{r}_{t_i}(2t_i, N) \leq e^{\mathsf{e}^{(N)}(t_i)} - 1$; the sum with the weights 4^{-j-1} gives the bound on $\delta^{(N)}$.

In the flat tubes $t \times N$ one doubling of the width squares $t\varepsilon_{\mathrm{v}}(t, N)$. At $i = 0$, Prop. 17.17(c) gives $N\varepsilon_{\mathrm{v}}(N, N) \leq e^{-E_0}$, $E_0 = \sigma^+ N^2$. From $t = t_i$ to $2t$, with $n = \hat{N}t$ and $y_i = \hat{N}e^{-E_i}$: the first inequality of Lem. 17.15(b) at the time $2n$, with $\mathsf{r}_{2n}(t, N) \leq \mathsf{r}_t(t, N)^{2\hat{N}} \leq e^{-4E_i}$, gives $z_-(t, 2n, N) \leq (1 + 10^{-7})y_i$; Lem. 17.15(a) with $\kappa = 2$ in the width squares it at the cost $2F(t, 2n, N) - F(2t, 2n, N) \leq 4\hat{N}\delta^{(N)}(t, t) \leq \xi_i := 0.732\,\hat{N}\chi^{2^i}$; and the second inequality of Lem. 17.15(b) for the tube $2t \times N$ at the time n gives

$$2t\,\varepsilon_{\mathrm{v}}(2t, N) \leq 4.01\,\hat{N}e^{-2E_i + \xi_i} =: e^{-E_{i+1}}.$$

As $\hat{N}\chi \leq 0.7936$, $\sum_i 2^{-i-1}\xi_i \leq 0.294$, and $2^{-i}E_i \geq \sigma^+ N^2 - \log\hat{N} - \log 4.01 - 0.294 \geq \sigma^+ N^2 - \mathsf{g}$. □

Proposition 17.19 (Flux lines of every integer length)**.** Assume the criterion; let $N \geq 4l$ be an integer, n_0, κ, τ, χ_0, E_0, $\hat{N}'$, h and g' as in Thm. AW(g2), $\mathsf{E} = \sigma^+ n_0^2$ and $t_i = 2^i n_0$. Then $6 \leq \hat{N}' < \mathsf{q}N/l + 2$, $\hat{N}' \leq 1.3453\,\mathsf{E}_0 + 2$, $\mathsf{h} \leq 0.4274\,\kappa + 0.0208$, and $t_i\varepsilon_{\mathrm{v}}(t_i, N) \leq \exp[-2^i(\mathsf{E}_0 - \mathsf{g}')]$ for all $i \geq 0$. ■

Proof sketch. Numbers. $\log\frac{1}{\chi_0} = -\log 0.3602 + 2w\Lambda \geq 1.48677$; the bounds on $\hat{N}'$ follow from $2\mathsf{E}_0/\log\frac{1}{\chi_0} < 2\kappa\mathsf{q}w = \mathsf{q}N/l$, from the fact that the quotient exceeds 4 at $w = 2$ and increases in w, and from $2/1.48677 \leq 1.3453$. As $\tau = \theta^{2w} \leq 0.8901^4$, the number $c_\tau = 0.01655\,\tau$ satisfies

$c_\tau \le 0.010389$, and $\mathsf{h} = c_\tau \hat N'$. With $4\mathsf{q}\cdot 0.01655\, we^{-2w\Lambda} \le 0.8548$ (as in the proof of Thm. AW(d)), $\mathsf{h} < 0.01655(2\kappa\mathsf{q}w+2)\tau \le 0.4274\kappa + 0.0208$. By the choice of $\hat N'$, $\chi_0^{\hat N'/2} \le e^{-\mathsf{E}_0}$. Here $\mathsf{E}_0 = \kappa\mathsf{E}$ and $\mathsf{E} = \sigma^+ n_0^2 = \varsigma w^2 \ge 4\cdot 1.90284 = 7.61136$. For $\kappa\in[1,2]$, $E \ge 7.6$ and $0 < c \le 0.052$ the functions

$$\gamma_c(\kappa, E) = \kappa\big[\log(1.3453\,\kappa E + 2) + c\,(1.3453\,\kappa E + 2) - E\big]$$

increase in c, are convex in κ and decrease in E. As $\log\hat N' \le \log(1.3453\,\kappa\mathsf{E}+2)$ and $\mathsf{h} \le c_\tau(1.3453\,\kappa\mathsf{E}+2)$,

$$\log\big[\hat N' e^{-\mathsf{E}+\mathsf{h}}\big]^\kappa \le \gamma_{0.010389}(\kappa,\mathsf{E}), \qquad \log\big[\hat N' e^{-\mathsf{E}+5\mathsf{h}}\big]^\kappa \le \gamma_{0.051945}(\kappa,\mathsf{E}), \qquad 0.051945 = 5\cdot 0.010389.$$

On $\kappa\in[1,2]$, $E \ge 7.61136$ each of the two functions is largest at $E = 7.61136$ and at $\kappa = 1$ or $\kappa = 2$, where $\gamma_{0.010389}$ has approximate values -4.97953 and -8.53047, and $\gamma_{0.051945}$ approximate values -4.47090 and -6.66219. So $\bar z_0 := [\hat N' e^{-\mathsf{E}+\mathsf{h}}]^\kappa \le e^{-4.97952} \le 0.00688$ and $6.1\,[\hat N' e^{-\mathsf{E}+5\mathsf{h}}]^\kappa \le 6.1\,e^{-4.47090} \le 0.0698$. *The start.* The first inequality of Lem. 17.15(b) for the tube $n_0\times n_0$ at the time $\hat N' n_0$, Prop. 17.17(c) and $\mathsf{r}_{\hat N' n_0}(n_0,n_0) \le \chi_0^{\hat N'} \le e^{-\mathsf{E}}$ give $z_-(n_0, \hat N' n_0, n_0) \le \hat N' e^{-\mathsf{E}}$; with $\delta(n_0, \hat N' n_0, n_0) \le \hat N' \delta_{\mathrm c}(n_0) \le \mathsf{h}$, Lem. 17.15(a) at the real factor κ in the third side gives $z_-(n_0, \hat N' n_0, N) \le \bar z_0$; and the second inequality of Lem. 17.15(b) for the tube $n_0\times N$ at the time $\hat N' n_0/2$, with $\mathsf{r}_{\hat N' n_0/2}(n_0,N) \le (0.14\chi_0)^{\hat N'/2} \le \bar z_0$ (Prop. 17.18(b)), gives $n_0\varepsilon_{\mathrm v}(n_0,N) \le \frac{6.1}{\hat N'}\bar z_0 = e^{-E_0'}$, $E_0' = \mathsf{E}_0 - (\kappa-1)\log\hat N' - \kappa\mathsf{h} - \log 6.1$.

The recursion. As in Prop. 17.18(c), with the slab of the integer thickness N of Prop. 17.18(b), one doubling of the width squares $t\varepsilon_{\mathrm v}(t,N)$: $E_{i+1}' = 2E_i' - \log\hat N' - \log 4.1 - \xi_i'$, with $\xi_0' = 4\kappa\mathsf{h}$ and $\xi_i' = 0.532\,\hat N'\chi_0^{2^i}$ for $i\ge 1$; the bound $6.1[\hat N' e^{-\mathsf{E}+5\mathsf{h}}]^\kappa \le 0.0698$ keeps $y_i = \hat N' e^{-E_i'}$ so small that the step closes at every i, and $\sum_i 2^{-i-1}\xi_i' \le (2\kappa+0.68)\mathsf{h}$. Hence $2^{-i}E_i' \ge E_0' - \log\hat N' - \log 4.1 - (2\kappa + 0.68)\mathsf{h} \ge \mathsf{E}_0 - \mathsf{g}'$, as $\log 6.1 + \log 4.1 \le 3.22$. More precisely, the unused positive tail of this same recurrence gives

$$\begin{aligned} E_i' &= 2^i\Big(E_0' - \log\hat N' - \log 4.1 - \sum_{j\ge 0} 2^{-j-1}\xi_j'\Big) + \log\hat N' + \log 4.1 + \sum_{j\ge i} 2^{i-j-1}\xi_j' \\ &\ge 2^i(\mathsf{E}_0 - \mathsf{g}') + \log\hat N' + \log 4.1 + \tfrac12\xi_i'. \end{aligned}$$

Thus subtracting $\log\hat N' + \xi_i'/4$ still leaves the floor $2^i(\mathsf{E}_0 - \mathsf{g}')$, also at $i = 0$. □

Proposition 17.20 (A local function in a long box)**.** Let $\beta \ge 0$, Σ a section, $h \ge 1$ and $t \ge h+2$ integers, $s = t - h - 1$, and A a bounded Borel function of the variables of h consecutive layers and of the crossings between them. There is a number $a_\infty(\Sigma, A)$, $|a_\infty| \le \sup|A|$, that depends neither on t nor on the position of the layers, with $|\langle A\rangle_{(t,\Sigma)} - a_\infty(\Sigma,A)| \le 2\sup|A|\,\mathsf{r}_s(\Sigma)$. As $\mathsf{r}_s(\Sigma)\to 0$, the laws of the boxes (t,Σ) converge, as $t\to\infty$, on the bounded Borel functions of finitely many layers: to the law of the infinite cylinder over Σ. ■

Proof sketch. Integrating the weights of the h layers and the $h+1$ crossings with A inserted gives a kernel $\hat A$ dominated by $\sup|A|$ times that of T_0^{h+1}, so $\|\hat A\| \le \sup|A|\lambda_0^{h+1}$, and $Z\langle A\rangle = \mathrm{Tr}(\mathsf{T}_0^s\hat A) = \sum_i \lambda_i^s\langle e_i, \hat A e_i\rangle$ in an eigenbasis. With $a_\infty = \lambda_0^{-h-1}\langle e_0, \hat A e_0\rangle$, which is determined because λ_0 is simple: $Z\langle A\rangle = \lambda_0^t(a_\infty + \vartheta\sup|A|\,\mathsf{r}_s)$ with $|\vartheta|\le 1$, and $Z = \lambda_0^t(1+\mathsf{r}_t)$. The thermal number tends to 0 by dominated convergence, λ_0 being simple and $\sum\lambda_i^2 < \infty$; the limits are consistent, and Kolmogorov's theorem gives the law of the cylinder. □

Lemma 17.21 (Mirrors of a box)**.** Fix a box and a direction μ as time; let $c\in\mathbb{Z}+\frac12$, and $\mathfrak{r}$ the reflection $x_\mu \mapsto 2c - x_\mu$ of the box.

(a) Let $\vartheta \ge 2$ be an integer with $2\vartheta \le n_\mu$, and A a bounded Borel function of the bonds with both ends in the ϑ layers below c. Then $\langle A\,\overline{\mathfrak{r}A}\rangle \ge 0$ and $|\langle A\rangle|^2 \le \langle A\,\overline{\mathfrak{r}A}\rangle$; n_μ may be odd. If $n_\mu = 2\vartheta$ and A' is another such function, $|\langle A\,\mathfrak{r}\overline{A'}\rangle|^2 \le \langle A\,\overline{\mathfrak{r}A}\rangle\langle A'\,\overline{\mathfrak{r}A'}\rangle$.

(b) Let $N = 2h \geq 4$, $c \in (\mathbb{Z} + \frac{1}{2})^3$, and H_ϵ, $\epsilon \in \{0,1\}^3$, functions of the cell $\mathsf{C}_h(c)$. On the cube (N,N,N), $|\langle \prod_\epsilon H_\epsilon^{[\epsilon]} \rangle| \leq \prod_\epsilon |||H_\epsilon|||_N$; so $|||\cdot|||_N$ is a seminorm, $|\langle A \rangle_{(N,N,N)}| \leq |||A|||_N \leq \sup |A|$, the mean of the product of the copies of A over $\epsilon \in E$ is at most $|||A|||_N^{\#E}$ in modulus for every $E \subset \{0,1\}^3$, and $|||A|||_N^8 \leq \langle |A|^8 \rangle_{(N,N,N)}$.

(c) Let n_μ be even and the mirror lie between the layers 0 and 1 of the direction μ; let x_p, x_q be two sites of the layer 1, equal or not, z_p, z_q the values of U at the bonds from x_p to $\mathfrak{r} x_p$ and from x_q to $\mathfrak{r} x_q$, and A_{jk}, $j,k \in \{1,2\}$, bounded Borel functions of the bonds with both ends in the layers $1, \dots, n_\mu/2$. Then

$$\Big\langle \sum_{i,j,k,l} \overline{(\mathsf{z}_p)_{ji}}\, (\mathsf{z}_q)_{kl}\, A_{jk}\, \overline{\mathfrak{r} A_{il}} \Big\rangle \geq 0.$$

■

Proof sketch. Let $\mathsf{T}' = \mathsf{K}^{1/2} M_0^2 \mathsf{K}^{1/2}$, which has the eigenvalues of T_0. For a function A of ϑ consecutive layers and the $\vartheta - 1$ crossings between them let S_A be the operator whose kernel, a function of the first and the last of these layers, is the integral of A against the weights of the ϑ layers and of the inner crossings over all other variables, and $\mathsf{Y}_A = \mathsf{K}^{1/2} \mathsf{S}_A \mathsf{K}^{1/2}$, a Hilbert–Schmidt operator; $\mathsf{Y}_1 = \mathsf{T}'^{\vartheta}$. Integrating the layers in turn as in Lem. 17.10, the mean of a product of such functions on consecutive slabs, with free layers between them, is the trace of the product of their operators and of powers of T'; and, the weights being invariant under $\mathfrak{r}$ because $A_\times(V', S^{-1}, V) = A_\times(V, S, V')$, the operator of $\overline{\mathfrak{r} A}$ on the mirrored slab is Y_A^*.

So $Z\langle A \rangle = \operatorname{Tr}(\mathsf{T}'^{n_\mu - \vartheta} \mathsf{Y}_A)$ and $Z\langle A\, \overline{\mathfrak{r} A'} \rangle = \operatorname{Tr}(\mathsf{T}'^{n_\mu - 2\vartheta} \mathsf{Y}_{A'}^* \mathsf{Y}_A)$. (a) Write $\operatorname{Tr}(\mathsf{T}'^{n_\mu - \vartheta} \mathsf{Y}_A) = \operatorname{Tr}(\mathsf{T}'^{n_\mu/2} \cdot \mathsf{Y}_A \mathsf{T}'^{v})$ with $v = \frac{1}{2}(n_\mu - 2\vartheta) \geq 0$, a real power of a positive operator; Schwarz's inequality for Hilbert–Schmidt operators gives $|Z\langle A \rangle|^2 \leq \operatorname{Tr} \mathsf{T}'^{n_\mu} \, \|\mathsf{Y}_A \mathsf{T}'^{v}\|_2^2 = Z \cdot Z\langle A\, \overline{\mathfrak{r} A} \rangle$, with $\|\mathsf{Y}_A \mathsf{T}'^{v}\|_2^2 = \operatorname{Tr}(\mathsf{T}'^{2v} \mathsf{Y}_A^* \mathsf{Y}_A) \geq 0$; at $n_\mu = 2\vartheta$, Schwarz's inequality for $\operatorname{Tr}(\mathsf{Y}_{A'}^* \mathsf{Y}_A)$ is the second statement. (b) The reflection $\mathfrak{r}_\mu$ of the cube fixes the two bond mirrors $x_\mu = c_\mu$ and $x_\mu = c_\mu + h$, and the half below the first of them is the slab of the cell.

Group the eight factors into the four with $\epsilon_\mu = 0$ and their four mirror images: the second statement of (a) gives $|\Lambda(H)|^2 \leq \Lambda(H^0)\Lambda(H^1)$ for $\Lambda(H) = \langle \prod_\epsilon H_\epsilon^{[\epsilon]} \rangle$, where H^j is the family with the index ϵ_μ set to j; three directions give the first bound, and the mean of the eight copies of one function is ≥ 0. Expanding $|||H + H'|||_N^8$ into 2^8 such means gives the triangle inequality. With $H_\epsilon = A$ for $\epsilon \in E$ and $H_\epsilon = 1$ otherwise one gets the bound for subsets; Hölder's inequality with eight exponents 8 and the invariance of the cube law under the $\mathfrak{r}_\mu$ give the last bound. (c) Take μ as time; the mirror also fixes the plane between the layers $n_\mu/2$ and $n_\mu/2 + 1$.

Integrating in each half the layers and crossings between its two outer layers produces from A_{jk} a function Φ_{jk} of $(V^1, V^{n_\mu/2})$, and from $\overline{\mathfrak{r} A_{il}}$, by the symmetry of the weights, $\overline{\Phi_{il}}$ at the mirrored layers 0 and $n_\mu/2 + 1$. The crossing between the layers 0 and 1 carries the two cut bonds, which enter exactly as in the operator $\hat{\mathsf{K}}$ of Lem. 17.12 with the charges at the sites of x_p and x_q: the function under the mean is $\operatorname{tr}[(\overline{\mathfrak{r} A})^{\mathrm{t}} \mathsf{z}_p^{-1} A\, \mathsf{z}_q]$ for the matrices $A = (A_{jk})$, $\overline{\mathfrak{r} A} = (\overline{\mathfrak{r} A_{il}})$. The crossing between $n_\mu/2$ and $n_\mu/2 + 1$ enters through K.

Hence Z times the mean equals $\langle \Phi, (\hat{\mathsf{K}} \otimes \mathsf{K}) \Phi \rangle \geq 0$, by Lem. 17.12(a), (e), whose proof uses of the two sites only that they lie in the section, and the positivity of K. □

Lemma 17.22 (Comparison of a functional with its dominant state)**.** Let $\mathcal{A}$ be a real vector space with an element 1, N a seminorm and g a linear functional on $\mathcal{A}$ such that, with $\mathsf{R}(x) = (1+x)^8 - 1 - 8x$,

$$\mathsf{N}(1 + tG)^8 \leq 1 + 8t\, \mathsf{g}(G) + \mathsf{R}(|t| \mathsf{N}(G)) \quad \text{and} \quad |\mathsf{g}(G)| \leq \mathsf{N}(G)$$

for all $G \in \mathcal{A}$ and all real t. Let $\mathfrak{s}$ be a linear functional with $\mathfrak{s}(1) = 1$ and $\mathfrak{s}(G) \leq Q\, \mathsf{N}(G)$ for all G, where $1 \leq Q \leq 1.11$. Then $|\mathfrak{s}(G) - \mathsf{g}(G)| \leq 4.84\, [Q(Q-1)]^{1/2}\, \mathsf{N}(G)$. ■

Proof sketch. From $(1+u)^{1/8} \le 1+u/8$, $\mathsf{N}(1+tG) \le 1 + t\mathsf{g}(G) + \frac{1}{8}\mathsf{R}(x)$, $x = |t|\mathsf{N}(G)$; so $1+t\mathfrak{s}(G) = \mathfrak{s}(1+tG) \le Q[1+t\mathsf{g}(G)+\frac{1}{8}\mathsf{R}(x)]$, that is $t[\mathfrak{s}(G)-\mathsf{g}(G)] \le (Q-1)(1+x) + \frac{Q}{8}\mathsf{R}(x)$. If $\mathsf{N}(G)=0$ the bracket vanishes, and if $Q=1$ it vanishes as $x\to 0$. Otherwise take $t = \pm x/\mathsf{N}(G)$ with $x^2 = (Q-1)/4.74Q$, so that $x \le 0.1446$ and $\mathsf{R}(x)/8x^2 \le 4.718$:

$$|\mathfrak{s}(G)-\mathsf{g}(G)| \le \mathsf{N}(G)\Big[\frac{Q-1}{x} + (Q-1) + 4.74\,Qx\Big] = \mathsf{N}(G)\,[Q(Q-1)]^{1/2}\Big[2\sqrt{4.74} + \Big(\frac{Q-1}{Q}\Big)^{1/2}\Big],$$

and $2\sqrt{4.74} + (0.11/1.11)^{1/2} \le 4.67$. □

Proposition 17.23 (Boxes of every shape)**.** Let $\beta \ge 0$, $N = 2h \ge 4$, $c \in (\mathbb{Z}+\frac{1}{2})^3$, $\underline{n}$ a box with all sides $\ge N$, and $M_\mu = 2^{j_\mu}h$, $j_\mu \ge 1$, the largest number of this form that is $\le n_\mu$. For every function A of the cell $\mathsf{C}_h(c)$:

(a) $|\langle A\rangle_{\underline{n}}| \le \exp\{\frac{\delta_{\mathrm{c}}(N)}{8}[1+3N(\frac{1}{M_1}+\frac{1}{M_2}+\frac{1}{M_3})]\}\,\|A\|_N \le e^{\frac{5}{4}\delta_{\mathrm{c}}(N)}\,\|A\|_N$;

(b) if A is real and $\delta_{\mathrm{c}}(N) \le \frac{1}{2}$, then $|\langle A\rangle_{\underline{n}} - \langle A\rangle_{(N,N,N)}| \le 7.31\,\delta_{\mathrm{c}}(N)^{1/2}\,\|A\|_N$.

Under the criterion at the side l and for $N \ge 2l$: $|\langle A\rangle_{\underline{n}} - \langle A\rangle_{(N,N,N)}| \le 2.71\,\theta^{N/4l}\,\|A\|_N$ for every real A. None of these constants depends on the shape of the box. ■

Proof sketch. (a) *Doubling on the one box.* In the direction 1 put $A_0 = A$ and $A_{i+1} = A_i\,\overline{\mathfrak{r}^{(i)}A_i}$, $\mathfrak{r}^{(i)}$ the reflection in the plane $x_1 = c_1 + (2^i-1)h$; A_i is a function of the bonds of the 2^ih layers below that plane, and by Lem. 17.21(a), as long as $2^{i+1}h \le n_1$, $|\langle A_i\rangle_{\underline{n}}|^2 \le \langle A_{i+1}\rangle_{\underline{n}}$. So $|\langle A\rangle_{\underline{n}}| \le \langle A_{j_1}\rangle_{\underline{n}}^{2^{-j_1}}$, and A_{j_1} is a row of M_1/h alternating copies of the cell, which lies below c_2 and c_3; the same in the directions 2 and 3, always on the box $\underline{n}$, gives $|\langle A\rangle_{\underline{n}}| \le \langle \mathfrak{B}A\rangle_{\underline{n}}^{1/\mathcal{N}}$, with $\mathfrak{B}A$ an array of $\mathcal{N} = M_1M_2M_3/h^3$ copies on a block of $M_1\times M_2\times M_3$ sites.

The free layers cost their top eigenvalue. With the time 1 and Y the operator of the slab of h layers of $\mathfrak{B}_3\mathfrak{B}_2A$, $Z(\underline{m})\langle\mathfrak{B}A\rangle_{\underline{m}} = \mathrm{Tr}(\mathsf{T}'^{m_1-M_1}(\mathsf{Y}^*\mathsf{Y})^{M_1/N})$, and $\mathrm{Tr}(\mathsf{O}_1\mathsf{O}_2) \le \|\mathsf{O}_1\|\,\mathrm{Tr}\,\mathsf{O}_2$ for positive operators $\mathsf{O}_1,\mathsf{O}_2$, the second of trace class, gives

$$Z(\underline{m})\langle\mathfrak{B}A\rangle_{\underline{m}} \le \lambda_0(m_2,m_3)^{m_1-M_1}\,Z(M_1,m_2,m_3)\langle\mathfrak{B}A\rangle_{(M_1,m_2,m_3)};$$

with the times 2 and 3 in turn,

$$Z(\underline{n})\langle\mathfrak{B}A\rangle_{\underline{n}} \le \lambda_0(n_2,n_3)^{n_1-M_1}\lambda_0(M_1,n_3)^{n_2-M_2}\lambda_0(M_1,M_2)^{n_3-M_3}\,Z(\underline{M})\langle\mathfrak{B}A\rangle_{\underline{M}}.$$

On the block $\underline{M}$, whose sides lie in $N\mathbb{N}$, $\mathrm{Tr}(\mathsf{Y}^*\mathsf{Y})^q \le (\mathrm{Tr}\,\mathsf{Y}^*\mathsf{Y})^q$ in each direction gives $Z(\underline{M})\langle\mathfrak{B}A\rangle_{\underline{M}} \le [Z(N,N,N)\,\|A\|_N^8]^{M_1M_2M_3/N^3}$. *The count.* Write $\log\lambda_0(\mathsf{A},\mathsf{B}) = \mathsf{AB}P + \varepsilon(\mathsf{A},\mathsf{B})$ and $F = |\cdot|P+\delta$: the terms with P cancel, and with $\varepsilon(\mathsf{A},\mathsf{B}) \le (\mathsf{A}+\mathsf{B})\delta_{\mathrm{c}}(N)/N^2$ (Lem. 17.11(g)), $n_\mu < 2M_\mu$ and $\delta(\underline{n}) \ge 0$, the remaining exponent is at most $3\frac{\delta_{\mathrm{c}}(N)}{N^2}(M_1M_2+M_1M_3+M_2M_3) + \frac{M_1M_2M_3}{N^3}\delta_{\mathrm{c}}(N)$; divided by $\mathcal{N} = 8M_1M_2M_3/N^3$ this is the exponent of (a), and $M_\mu \ge N$. (b) Lem. 17.22 with $\mathcal{A}$ the real functions of the cell, $\mathsf{N} = \|\cdot\|_N$, $\mathsf{g} = \langle\cdot\rangle_{(N,N,N)}$, $\mathfrak{s} = \langle\cdot\rangle_{\underline{n}}$ and $Q = e^{\frac{5}{4}\delta_{\mathrm{c}}(N)}$: by Lem. 17.21(b), $\|1+tG\|_N^8$ is the sum over $E\subset\{0,1\}^3$ of $t^{\#E}$ times the mean of the copies of G over E, which is 1 for $E=\emptyset$, $\langle G\rangle_{(N,N,N)}$ for the eight one-element E, and at most $\|G\|_N^{\#E}$ else; and $\mathfrak{s}(G) \le Q\mathsf{N}(G)$ is (a).

So, if $\delta_{\mathrm{c}}(N) \le 0.08348$, which suffices for $Q \le 1.11$, the difference is at most $4.84\,[Q(Q-1)]^{1/2}\|A\|_N \le 5.86\,\delta_{\mathrm{c}}(N)^{1/2}\|A\|_N$. For $0.08348 < \delta_{\mathrm{c}}(N) \le \frac{1}{2}$, (a) and Lem. 17.21(b) bound it by $(e^{\frac{5}{4}\delta_{\mathrm{c}}(N)}+1)\|A\|_N$, and $(e^{5\delta/4}+1)\delta^{-1/2}$ decreases on $(0,\frac{1}{2}]$ and is ≤ 7.31 at 0.08348. Under the criterion $\delta_{\mathrm{c}}(N) < 0.1324\,\theta^{N/2l} \le \frac{1}{2}$ (Prop. 17.16(d)), and $7.31\cdot 0.1324^{1/2} \le 2.71$. □

Lemma 17.24 (The excess controls every periodic state)**.** Let $\beta > 0$, let $N = 2\mathsf{L}^j$ be the side of one of the tori of the chessboard estimate (Prop. YM3_I:86.3), and let B be the set of the bonds with both ends in one of the eight closed cubes of side $N/2$, with faces through midpoints of bonds,

that tile the torus (N,N,N). For every periodic Gibbs state s the law of s on B has a density h with respect to the law of the torus (N,N,N) on B, with $\int h^{8/7} \le e^{\delta_c(N)/7}$, and

$$\sup_A |s(A) - \langle \mathbf{1}_A \rangle_{(N,N,N)}| \le \tfrac{1}{2}(\sqrt{\epsilon} + \epsilon), \qquad \epsilon = 22.3\big(e^{\delta_c(N)/7} - 1\big),$$

the supremum over the Borel events A of the bonds of B. ■

Proof sketch. One board for all sides. Let $N' = 2\mathsf{L}^{j'}$, $j' \ge j$, be a side of the family. The cubes of side $N/2$, with faces through midpoints of bonds, that are the translates of the closed cube of B by the multiples of $N/2$ in the three directions tile the torus (N',N',N'), $2N'/N$ of them a side, an even number, as N'/N is a power of the odd number L: these *cells* form one board, and the planes between neighbouring cells are its walls. Number the cells by $k \in \{0,\dots,2N'/N-1\}^3$, the cell of B being $k=0$. For a bounded Borel function f of the bonds of B the copy $f^{[k]}$ is f composed, for $\mu = 1,2,3$, with the reflections in the k_μ walls of the direction μ that carry the cell 0 step by step to the cell k, and complex conjugated if $k_1+k_2+k_3$ is odd; it is a function of the bonds of the cell k. Put

$$\|f\|_{N'} = \Big\langle \prod_k f^{[k]} \Big\rangle_{(N',N',N')}^{(N/2N')^3},$$

the product over all $(2N'/N)^3$ cells. At $N' = N$ the cell of B is a cell $\mathsf{C}_{N/2}(c)$, the eight copies are the copies $f^{[\epsilon]}$, $\epsilon \in \{0,1\}^3$, by which $|\!|\!| \cdot |\!|\!|_N$ is defined, and $\|f\|_N = |\!|\!| f |\!|\!|_N$. With the time 1 the torus (N',N',N') is cut by the walls into $2N'/N$ slabs of $N/2$ layers, and, as in the proof of Prop. 17.23, with Y the operator of the first slab of the product (that of the cells with $k_1 = 0$), the operator of the next slab is Y^*, and $Z(N',N',N')\langle \prod_k f^{[k]} \rangle_{(N',N',N')} = \mathrm{Tr}(\mathsf{Y}^*\mathsf{Y})^{N'/N}$, a number ≥ 0 and at most $(\mathrm{Tr}\,\mathsf{Y}^*\mathsf{Y})^{N'/N}$, where $\mathrm{Tr}\,\mathsf{Y}^*\mathsf{Y}$ is the same expression on the box (N,N',N'). The same in the directions 2 and 3 gives the chessboard estimate on this board for the two tori (Prop. YM3_I:86.3, in the form of Cor. YM3_I:86.6):

$$Z(N',N',N')\,\|f\|_{N'}^{(2N'/N)^3} \le \big[Z(N,N,N)\,\|f\|_N^8\big]^{(N'/N)^3}.$$

Its root of the order $(2N'/N)^3$ is

$$\|f\|_{N'} \le \|f\|_N \exp\Big[\Big(\frac{N}{2}\Big)^3\Big(\frac{F(N,N,N)}{N^3} - \frac{F(N',N',N')}{N'^3}\Big)\Big] \le \|f\|_N\, e^{\delta_c(N)/8},$$

since $(N/2N')^3(N'/N)^3 = \frac{1}{8}$, $(N/2N')^3 = (N/2)^3/N'^3$ and $F(N',N',N')/N'^3 \ge P$. The domination of periodic Gibbs states (Prop. YM3_I:91.9), read with this same board and seminorm, gives $|s(f)| \le \liminf_{j'\to\infty} \|f\|_{N'}$. Hölder's inequality and the reflection invariance of the torus law give $\|f\|_N \le \langle |f|^8 \rangle_{(N,N,N)}^{1/8}$. So $|s(f)| \le e^{\delta_c(N)/8}\|f\|_{L^8}$, and by duality s has on B a density h with $\|h\|_{8/7} \le e^{\delta_c(N)/8}$. With $p = \frac{8}{7}$, $u^p - 1 - p(u-1) \ge 0.045\min\{(u-1)^2, |u-1|^p\}$ for $u \ge 0$; integrating at $u = h$ gives $\int_{|h-1|\le 1}(h-1)^2 + \int_{|h-1|>1}|h-1|^p \le \epsilon$, so $\int |h-1| \le \sqrt{\epsilon} + \epsilon$, and the total variation is half of it. □

Lemma 17.25 (One bond; a bound by the perimeter)**.**

(a) For $\kappa \ge 0$ let $\langle\cdot\rangle_\kappa$ be the probability measure on G with the density proportional to $e^{\kappa x_0(V)}$ with respect to the Haar measure, $x_0(V) = \frac{1}{2}\,\mathrm{tr}\,V$, and $r(\kappa) = \langle x_0 \rangle_\kappa$. Then $\langle V \rangle_\kappa = r(\kappa)\mathbf{1}$ and

$$\frac{\kappa}{\kappa+4} \le r(\kappa) \le \frac{2\kappa}{2\kappa+3}.$$

(b) Let $\beta \ge 0$, s a Gibbs state, periodic or not, C a closed lattice path and B a finite set of bonds which C runs through exactly once each, as b or as $\bar{b}$, no plaquette containing two of them. Then $|s(W_C)| \le (\frac{8\beta}{8\beta+3})^{\#B} \le \exp[-3\#B/(8\beta+3)]$. For a rectangle with $R,T \ge 2$ one may take $\#B = 2(R+T)-4$.

■

Proof sketch. (a) Conjugation invariance makes $\langle V\rangle_\kappa$ commute with G, so it is a multiple of $\mathbf{1}$, and the multiple is $r(\kappa)$. Write $V = x_0\mathbf{1} + \mathrm{i}\sum_j x_j\sigma_j$ with $\sum_{j=0}^3 x_j^2 = 1$. By the invariance of the Haar measure under $V \mapsto e^{\mathrm{i}s\sigma_j}V$, the derivatives at $s=0$ of $\int (x_j e^{\kappa x_0})(e^{\mathrm{i}s\sigma_j}V)\,\mathrm{d}V$ and of $\int (x_j x_0 e^{\kappa x_0})(e^{\mathrm{i}s\sigma_j}V)\,\mathrm{d}V$ vanish; as the derivatives of x_0 and x_j along this path are $-x_j$ and x_0, summing over $j=1,2,3$ gives

$$3\,r(\kappa) = \kappa\langle 1-x_0^2\rangle_\kappa, \qquad 4\langle x_0^2\rangle_\kappa - 1 = \kappa\langle x_0(1-x_0^2)\rangle_\kappa.$$

The right member of the second identity lies in $[0, \kappa\langle 1-x_0^2\rangle_\kappa]$, its sign by $V\mapsto -V$; so $(4+\kappa)\langle x_0^2\rangle_\kappa \le 1+\kappa$, and the first identity gives $r(\kappa) \ge \frac{\kappa}{3}\cdot\frac{3}{4+\kappa}$; and $3r = \kappa\langle (1-x_0)(1+x_0)\rangle \le 2\kappa(1-r)$. (b) As no plaquette contains two bonds of B, given the other bonds the weight of the bonds of B factorises: $V(b)$ has the density proportional to $\exp[\beta \mathsf{k}_b \frac12 \operatorname{tr}(V(b)g_b)]$, where the sum of the four staples of b is $\mathsf{k}_b g_b$, $g_b\in G$, $0\le \mathsf{k}_b \le 4$, as a sum of four unit quaternions. W_C is a sum of products that hold one entry of each $V(b)^{\pm1}$, $b\in B$; by (a) and $V\mapsto Vg^{-1}$ the conditional mean of $V(b)^{\pm1}$ is $r(\beta\mathsf{k}_b)g_b^{\mp1}$, so the conditional mean of W_C is $\prod_b r(\beta\mathsf{k}_b)$ times half the trace of an element of G, of modulus at most $(\frac{8\beta}{8\beta+3})^{\#B}$; the Gibbs equation of B carries this to s. In a rectangle with $R,T\ge 2$ take all bonds but the first of each side: two bonds of one plaquette are either perpendicular with a common end, which happens on C only at a corner, or parallel at the distance 1, which does not happen for opposite sides at the distance ≥ 2. Finally $\log(1+y)\ge y/(1+y)$ at $y = 3/8\beta$. □

17.1.6. *Proofs of the results.*

Proof of Thm. AW. Step 1 (the cubes converge). In a box (t,Σ) with $t\in\{n,2n\}$ a function A of a cube of $h\le n/2-1$ layers is a function of h consecutive layers, and Prop. 17.20 applies with $s\ge n/2$ and one and the same $a_\infty(\Sigma,A)$ for both t. Doubling the three sides one after the other, each time with the time in the doubled direction,

$$|\langle A\rangle_{(n,n,n)} - \langle A\rangle_{(2n,2n,2n)}| \le 4\sup|A|\,[\mathsf{r}_{n/2}(n,n) + \mathsf{r}_{n/2}(2n,n) + \mathsf{r}_{n/2}(2n,2n)] \le 4 \cdot 4.15727\sup|A|\,\mathsf{e}(n/2) \le 1.29\sup|A|\,\theta^{n/2l}$$

by Prop. 17.16(d). Summing over $n, 2n, 4n, \dots$ with $\theta^{2^i n/2l} \le \theta^{n/2l}\theta^{in/2l}$ gives the second bound of (a). On each finite set of bonds the cube laws form a Cauchy sequence in total variation; the limits are consistent, and Kolmogorov's theorem gives ν.

Step 2 (Gibbs state, symmetries). For a finite set $\mathfrak{A}$ of bonds and a large cube, the action is the sum over the plaquettes that meet $\mathfrak{A}$ plus a remainder that does not depend on the bonds of $\mathfrak{A}$; integrating these bonds first shows that every cube law, and so ν, satisfies the Gibbs equation of $\mathfrak{A}$. A symmetry of the lattice induces a map of each cube of side $2^k l$ that keeps dU and the action, hence each cube law and ν.

Step 3 (one periodic state). By Step 2, ν is periodic. Let s be a periodic Gibbs state and A a local event. For all large $N = 2\mathsf{L}^j$, A is an event of the bonds inside one cell of side $N/2$, and Lem. 17.24, applied to s and to ν, gives $|s(A)-\nu(A)| \le \sqrt{\epsilon}+\epsilon$ with $\epsilon = 22.3(e^{\delta_{\mathrm{c}}(N)/7}-1)$, which tends to 0 as $\delta_{\mathrm{c}}(N) < 0.1324\,\theta^{N/2l}$ (Prop. 17.16(d)). So $s=\nu$. Prop. YM3_I:103.4 gives reflection positivity at every lattice mirror. By Prop. YM3_I:106.2, the ergodic components of ν are periodic Gibbs states; uniqueness makes them equal to ν, so ν is ergodic.

Step 4 (all boxes). Let the three sides of boxes $\underline{n}^{(k)}$ tend to infinity. Their laws on Ω lie in a compact set; every limit point satisfies the Gibbs equations, as in Step 2, and is invariant under all translations, as every box law is; so it is a periodic Gibbs state, that is ν, and the whole sequence converges.

Step 5 (twisted cubes). Let $n\ge 2\mathsf{q}l$ be an integer, j the largest integer with $4^j \le n/2\mathsf{q}l$, $w = 2^j$, $l' = l_j = 2wl$ and $M = n/l'$, a real number with $w\mathsf{q} \le M < 4w\mathsf{q}$ and $M^2 < 2\mathsf{q}n/l$. Lem. 17.15(a)

from the box (l',n,l') to the cube, with $\kappa = M^2$, and Prop. 17.17(d) give

$$z_-(n,n,n) \le \big[M e^{-\varsigma w^2 + 0.01655\, M\theta^{2w}}\big]^{M^2}, \qquad M^2\varsigma w^2 = \sigma^+ n^2,$$

and $0.01655\, M\theta^{2w} < 4\mathsf{q}\cdot 0.01655\, we^{-2w\Lambda} \le \frac{4\cdot 0.01655}{e}(\frac{1}{2\Lambda} + \frac{1.67}{4\Lambda^2}) \le 0.855$ and $\log M < \frac12 \log\frac{2\mathsf{q}n}{l}$: the first bound of (d). The bound for all seven $z \ne 0$ then follows from Lem. 17.10(c).

Step 6 (flux sectors, static charges). With the time 1 over $n\times n$ and $e \in \{01, 11\}$, $\Lambda_e^n \le W_n(01;0)+W_n(11;0) = a_-(n,n,n) = z_-\lambda_0^n(1+\mathsf{r}_n(n,n))$; for $e = 10$ the same holds because $Z^{(1,0,0)} = Z^{(0,1,0)}$ on the cube (Lem. 17.10(iii)). So $\log(\lambda_0/\Lambda_e) \ge \frac1n[-\log z_-(n,n,n) - \log(1+\mathsf{r}_n(n,n))]$, and $\log(1+\mathsf{r}_n(n,n))$ is at most $\mathsf{r}_n(n,n) \le 0.001615\,\theta^{n/l} \le 0.0017$ at $n = 2^i l$, and in general, with $n_0 = 2^i l \le n < 2n_0$, at most $\delta_{\mathrm{c}}(n) \le 8\delta_{\mathrm{c}}(n_0) \le 8\cdot 0.01655\cdot 0.792279 \le 0.105$ (Lem. 17.11(f), Prop. 17.16(d)). With Step 5 this is the first statement of (e); the second is Lem. 17.12(d) on the cube with $n_3 = n$, and Step 5. For any two sites $o \ne r$ it is Lem. 17.12(e) on the cube, with the time 1 or 2, whichever direction of the section realises $|r-o|_\infty$, and Step 5; for fixed o, r and $n = 2^i l$, $i\to\infty$, the factor $[8(1+\mathsf{r}_n(n,n))]^{1/n}$ tends to 1 and the correction $\frac{2\mathsf{q}R}{ln}(\frac12\log\frac{2\mathsf{q}n}{l} + 0.86)$ to 0, while $R = |r-o|_\infty$ once $n > 2|r-o|_\infty$.

Step 7 (rectangles). By the symmetries of ν and of the cube laws it suffices to take C in a $(1,3)$-plane with the side R along e_1. On the cube of an integer side $n \ge 2\mathsf{q}l$ with $R \le n/2$, $T \le n-1$, Lem. 17.12(d) gives $0 \le \langle W_C\rangle_{(n,n,n)} \le 8^{T/n} z_-(n,n,n)^{RT/n^2}$, and Step 5 gives the bound on the cube. Along $n = 2^k l$ the left side tends to $\nu(W_C)$ by Step 1, while $8^{T/n}\to 1$ and $(\log n)/n \to 0$: this is eq. 17.ii. The bound by the perimeter is Lem. 17.25(b).

Step 8 (closed paths). Let C and (ρ',ρ) be as in (c); by symmetry $(\rho',\rho) = (1,3)$. On the cubes $n = 2^k l$, which are even, Lem. 17.13(c) applies for large k: the sum of the ℓ_j is $\mathsf{A}_{13}(C)$, $\sum_j(\mathsf{v}_j+1)/n \to 0$, the thermal factors tend to 1, and $z_-(n,n,n)^{\mathsf{A}/n^2} \le e^{-\sigma^+\mathsf{A}+o(1)}$ by Step 5. For the general form let $\rho = 3$: by the symmetry of the cube that exchanges the directions 1 and 2, Lem. 17.13(b) holds with the direction 2 in the place of the direction 1, the two twisted pairs having one quotient on the cube (Lem. 17.10(c)), and in the proof of Lem. 17.13(c) each norm $\|\hat{\mathsf{T}}_{D_j}\|$ is bounded by (b) on its own; so in each gap ℓ_j may be taken as the larger of the two counts $\ell_j^1(C)$, $\ell_j^2(C)$, whose sum over the gaps is $\mathsf{A}_3(C)$, and Lem. 17.13(d) carries all this to D, with $d^{c_3(C)-1}$ in front and $\sum_j \mathsf{v}_j \log d/n \to 0$.

For a staircase loop the senses of its runs of the direction ρ are up in T consecutive gaps and then down in the same gaps, so $c_\rho = 1$, and $\mathsf{A}_{\rho'\rho}(C) = \sum_j |r_j - o_j|$. For a simple closed curve of a coordinate plane each passage from down to up and each from up to down holds at least one bond of the other direction, and between two such passages there is at least one bond of the direction ρ; so $|C| \ge 4c_\rho(C)$ and $2^{c_\rho-1} \le \frac12 e^{(\log 2/4)|C|}$.

Step 9 (gap). Let $N = N_1 \le N_2$ (by (T1)), $n_0 \le N < 2n_0$, and i with $t_i \le N_2 < 2t_i$ in Prop. 17.18(b): for $a \ge 1$, $(\lambda_a/\lambda_0)^{t_i} \le \mathsf{r}_{t_i}(N_2,N) \le 0.28\,\chi'^{2^i} \le \chi'^{t_i/n_0}$, so $\log(\lambda_0/\lambda_a) \ge \frac{1}{n_0}\log\frac{1}{\chi'} \ge \frac{\Lambda}{l} + \frac{1.021}{n_0}$. The vacuum is simple and its eigenvector lies in the sector 00 (Lem. 17.10); the tops of the sectors with electric flux are among the λ_a, $a \ge 1$. For $N = n = 2^k l$ and an integer $N_2 \ge n$, with $t_i = 2^i n \le N_2 < 2t_i$, Prop. 17.18(a), (c) give $\log(1+\mathsf{r}_{t_i}(N_2,n)) \le \frac{N_2}{t_i}\delta^{(n)}(t_i,t_i) \le 0.3652\,\chi^{2^i}$, so $(\lambda_a/\lambda_0)^{t_i} \le 0.367\,\chi^{2^i} \le \chi^{t_i/n}$ and $\log(\lambda_0/\lambda_a) \ge \frac1n\log\frac1\chi \ge \frac{\Lambda}{l} + \frac{3.5065}{n}$. Over $n\times n$, $(\lambda_a/\lambda_0)^n \le \mathsf{r}_n(n,n) \le 0.001615\,\theta^{n/l}$, and $\log\frac{1}{0.001615} \ge 6.42$. Finally, for $N_1, N_2 \ge n$, Lem. 17.11(f) gives $\log(1+\mathsf{r}_n(N_1,N_2)) \le \delta(n,N_1,N_2) \le \frac{N_1N_2}{n^2}\delta_{\mathrm{c}}(n) \le 0.01655\,\frac{N_1N_2}{n^2}\theta^{n/l} =: u \le \log 2$, and $e^u - 1 \le u/\log 2$, $\log 2/0.01655 \ge 41.8$.

Step 10 (flux lines of dyadic length). $2\sigma^+N^2/\log\frac1\chi = 2\varsigma w^2/(-\log 0.03 + 2w\Lambda)$ increases in w, is ≥ 3.83 at $w = 2$ and $< 2\mathsf{q}w = \mathsf{q}N/l$: the bounds on $\hat N$. With the time 1 over (N_2,N), Lem. 17.10(d) gives $\Lambda_-^t \le a_-(t,N_2,N) = z_-(t,N_2,N)\lambda_0^t(1+\mathsf{r}_t(N_2,N))$ for every t. Take $t = t_i = 2^i N$ with i the largest integer such that $\mathsf{M} = N_2/t \ge \hat N/2$, so that $\hat N/2 \le \mathsf{M} < \hat N$. The first

inequality of Lem. 17.15(b) at the time N_2, Prop. 17.18(c), $\mathsf{r}_{N_2}(t,N) \le \mathsf{r}_t(t,N)^{\mathsf{M}} \le e^{-2^i\sigma^+N^2}$ and $\log(1+\mathsf{r}_t(N_2,N)) \le \mathsf{M}\delta^{(N)}(t,t) \le 0.1826\,\mathsf{M}\chi^{2^i}$ give

$$\log\frac{\lambda_0}{\Lambda_-} \ge \frac{1}{t}\Big[2^i(\sigma^+N^2-\mathsf{g}) - \log\mathsf{M} - 0.1826\,\mathsf{M}\chi^{2^i}\Big] \ge \sigma^+N - \frac{2\log\hat N + 1.86}{N},$$

as $\hat N\chi \le 0.7936$. As $N_2\to\infty$, $i\to\infty$, and the bounded last two terms are divided by 2^iN: the lower limit. For $\max\{N,2\mathsf{q}l\} \le N_2 < \hat N N/2$ let w' be the largest power of two with $w' \le w$ and $2\mathsf{q}w'^2l \le N_2$, and $l' = 2w'l \le N$, so that $N_2 \ge \mathsf{q}l'^2/2l$; Prop. 17.17(e) with $N_3 = N$, $n_2 = N_2$ and $M = N_2/l'$ applies. If $w' < w$, then $8\mathsf{q}w'^2l > N_2$, so $N/l'^2 < 2\mathsf{q}N/N_2l \le 2\mathsf{q}/l$ and $M < 4\mathsf{q}w'$; if $w' = w$, then $N/l'^2 = 1/N \le 2\mathsf{q}/l$ and $M < \hat N/2 < 4\mathsf{q}w$. In both cases $\log M < \log\frac{2\mathsf{q}N}{l}$ and $0.01655\,M\theta^{l'/l} < 4\mathsf{q}\cdot 0.01655\,w'e^{-2w'\Lambda} \le 0.855$, which gives the second floor. For $N \le N' < 2N$: with the time 1, Lem. 17.15(a) in the third side gives $\Lambda_-(N_2,N')^t \le a_-(t,N_2,N') \le a_-(t,N_2,N)^{N'/N}$, and $a_-(t,N_2,N)^{1/t} \to \Lambda_-(N_2,N)$; so $\log\Lambda_-(N_2,N') \le \frac{N'}{N}\log\Lambda_-(N_2,N)$. Also $\log\lambda_0(N_2,N') \ge N_2N'P$ and $\log\lambda_0(N_2,N) \le \frac{1}{N}F(N,N_2,N) \le N_2NP + \frac{N_2}{N^2}\delta_{\mathrm{c}}(N)$ (Lem. 17.11(f)). Hence $\log\frac{\lambda_0}{\Lambda_-}(N_2,N') \ge \frac{N'}{N}[\log\frac{\lambda_0}{\Lambda_-}(N_2,N) - \frac{N_2}{N^2}\delta_{\mathrm{c}}(N)]$, and the first floor, $\delta_{\mathrm{c}}(N) \le 0.01655\,\theta^{N/l}$ and $N'/N < 2$ give the third.

Step 11 (flux lines of every integer length; the slabs). Let $N \ge 4l$. Prop. 17.19 gives $t_i\varepsilon_{\mathrm{v}}(t_i,N) \le e^{-E_i'}$ with $2^{-i}E_i' \ge \mathsf{E}_0 - \mathsf{g}'$. As in Step 10, with i the largest integer such that $\mathsf{M} = N_2/t_i \ge \hat N'/2$, the first inequality of Lem. 17.15(b) at the time N_2, with $\mathsf{r}_{N_2}(t_i,N) \le \chi_0^{2^i\hat N'/2} \le e^{-2^i\mathsf{E}_0}$, gives $z_-(t_i,N_2,N) \le \hat N'e^{-E_i'}$, and Prop. 17.18(b) gives $\log(1+\mathsf{r}_{t_i}(N_2,N)) \le \mathsf{M}\delta^{(N)}(t_i,t_i) \le \frac14\xi_i'$; so $\log(\lambda_0/\Lambda_-) \ge \frac{1}{t_i}[E_i' - \log\hat N' - \frac14\xi_i'] \ge (\mathsf{E}_0-\mathsf{g}')/n_0 = \sigma^+N - \kappa\mathsf{g}'/N$. The penultimate inequality uses the unused positive recurrence tail displayed in the proof of Prop. 17.19.

As $\kappa\mathsf{g}' = \kappa^2\log\hat N' + \kappa[3.22 + (3\kappa+0.68)\mathsf{h}]$ and $\mathsf{h} \le 0.4274\kappa + 0.0208$, $\kappa\mathsf{g}'$ increases in κ, and at $\kappa = 2$ it is at most $4\log\hat N' + 18.14$. The window $\max\{N,2\mathsf{q}l\} \le N_2 < \hat N'n_0/2$ is treated as in Step 10, with $l' = 2w'l \le n_0$. For the closed form, $\hat N'n_0/2 < (\mathsf{q}N/l+2)N/2 \le (\frac{\mathsf{q}}{2}+\frac14)N^2/l$ and $\hat N' < (\mathsf{q}+\frac12)N/l$, as $N \ge 4l$; so $C(\theta)N^2/l \ge \hat N'n_0/2$ and $4\log\hat N' + 18.2 \le 4\log\frac{N}{l} + 4\log(\mathsf{q}+\frac12) + 18.2 \le C(\theta)(1+\log\frac{N}{l})$; and $4\log 8.673 + 18.2 \le 26.85$ at $\mathsf{q} \le 8.173$. If $w \ge 4$ and $\tau \le 0.087$, the same road, with the slab of the thickness N started from $\mathsf{e}^{(N)}(n_0) \le 2\delta(n_0,2n_0,N) \le 8\cdot 0.01655\,\tau \le 0.0116$, where $\omega(Y_1(x)) \le 2.06x^2$, so that $2.06\,\mathsf{e}^{(N)}(t_i) \le (0.2728\,\tau)^{2^i}$, and with the start from Prop. 17.17(d) at $l' = n_0$ carried to the thickness N by Lem. 17.15(a) at the real factor N/n_0, gives the floor with $\hat N''$ and 10.8. (g3) is Lem. 17.14 with $L_2 = N_2$, $L_3 = N$ and $\phi \ge \sigma^+N - (4\log\hat N' + 18.2)/N$.

Step 12 (boxes of every shape, cylinders, twisted boxes). A cube of $N/2$ layers is a cell $\mathsf{C}_{N/2}(c)$, and all box laws are invariant under translations. For a real function A of the cell, Prop. 17.23 gives $|\langle A\rangle_{\underline{n}} - \langle A\rangle_{(N,N,N)}| \le 2.71\,\theta^{N/4l}\|\!|A|\!\|_N$ for every box with sides $\ge N$. If A is continuous, let the sides of a second box tend to infinity: by Step 4, $|\nu(A) - \langle A\rangle_{(N,N,N)}| \le 2.71\,\theta^{N/4l}\|\!|A|\!\|_N$, so $|\langle A\rangle_{\underline{n}} - \nu(A)| \le 5.42\,\theta^{N/4l}\|\!|A|\!\|_N \le 5.42\,\theta^{N/4l}\sup|A|$ by Lem. 17.21(b). For a bounded real Borel A, take continuous A_k with $|A_k| \le \sup|A|$ and $A_k \to A$ in L^8 of the mean of the images of $\langle\cdot\rangle_{(N,N,N)}$, $\langle\cdot\rangle_{\underline{n}}$ and ν on the bonds of the cell; both sides of the bound pass to the limit, the seminorm because $\|\!|A_k - A|\!\|_N \le \langle|A_k-A|^8\rangle^{1/8}_{(N,N,N)}$ (Lem. 17.21(b)).

For an event take $A = \mathbf{1}_E - \frac12$. For the cube of side $m \ge 2l$ take $N = 2\lfloor m/2\rfloor \le m$. The cylinder over Σ is the limit of Prop. 17.20, and the bound passes to it; with $\Sigma = (n,n)$ and $M\to\infty$, Lem. 17.15(a) in the third side gives $z_-(n,n,Mn) \le [z_-(n,n,n)\lambda_0^n(1+\mathsf{r}_n(n,n))]^M\lambda_0^{-Mn} = [(1+\mathsf{r}_n(n,n))z_-(n,n,n)]^M$, and Lem. 17.12(d) on the box (n,n,Mn) gives the area bound on the cylinder, as $8^{T/Mn}\to 1$. Twisted boxes: let the columns recede. For a finite set $\mathfrak{A}$ of bonds the plaquettes that meet $\mathfrak{A}$ carry no twist in all but finitely many members, so every limit point satisfies the Gibbs equations, as in Step 2.

The translate by e_μ of a twisted box law is the twisted law with the columns moved by one step, and by the proof of Lem. 17.10(iii) this is the image of the law under $U(b) \mapsto -U(b)$ on a set of bonds next to the columns; a fixed local function depends on none of these bonds in all but finitely many members. So every limit point is a periodic Gibbs state, that is ν; the two equations are linear in the law, so they hold for finite mixtures. If a column passes through p_0 for all members, one flip moves it off p_0 and changes the sign of $\operatorname{tr} \partial U(p_0)$, and the column so moved recedes after further flips that do not touch p_0; so the law of $\operatorname{tr} \partial U(p_0)$ tends to that of $-\operatorname{tr} \partial U(p_0)$ under ν, which differs because $\nu(\operatorname{tr} \partial U(p_0)) = 2\nu(W_p) > 0$ by Cor. 17.1(a).

Step 13 (the pressure). By Thm. AX(b), $\mathsf{e}(2l) \le \theta^2/12.9 \le 0.0615 < 0.069$ at β. Each $F(\underline{n}; \beta')$ is a smooth function of β', so there is an open interval U containing β on which $\mathsf{e}(2l; \beta') \le 0.069$: the criterion holds on U at the side $2l$ with $x_0 = 0.069$, so that what is proved above holds at each $\beta' \in \mathsf{U}$, with the state $\nu_{\beta'}$ and with $\delta_{\mathrm{c}}(N; \beta') < 0.1324 \cdot 0.8901^{N/4l}$ for $N \ge 4l$, uniformly on U. Let $N_m = 2\mathsf{L}^m$ and $f_m(\beta') = F(N_m, N_m, N_m; \beta')/N_m^3$: these are smooth, tend to P pointwise, and $f_m'(\beta') = -3\langle a_p \rangle_{(N_m,N_m,N_m)}$ by the invariance of the cube under translations and permutations of the axes. As $0 \le a_p \le 2$ is a function of one cell of the torus, Lem. 17.24 gives $|\langle a_p \rangle_{(N_m,N_m,N_m)} - \nu_{\beta'}(a_p)| \le \sqrt{\epsilon_m} + \epsilon_m$, with $\epsilon_m \to 0$ uniformly on U. So the f_m' converge uniformly on U to $-3\nu_{\beta'}(a_p)$, which is therefore continuous, and $P(\beta_2) - P(\beta_1) = \lim_m \int_{\beta_1}^{\beta_2} f_m' = \int_{\beta_1}^{\beta_2} (-3\nu_{\beta'}(a_p))\, \mathrm{d}\beta'$. Finally $\nu(a_p) = 1 - \nu(W_p) \le 1 - \frac{\beta}{4(\beta+1)}$ by Cor. 17.1(a). □

Proof of Thm. AX. (a) is Prop. 17.16(a); if $\beta > 0$, $\mathsf{e}(l_0) \le X(l_0) \le \frac{1}{20} \le 0.069$, and $12.9/20 = 0.645$. (b) By Prop. 17.16(b), (c), $12.9\,\mathsf{e}(2n) \le (12.9\,\mathsf{e}(n))^2$ as long as $\mathsf{e}(n) \le 0.069$, and the induction keeps $\mathsf{e}(2^k l) \le \theta^{2^k}/12.9$. For $k \ge 3$, $\mathsf{e}(2^{k-1} l) \le \theta^4/12.9 \le 0.04866 \le \frac{1}{20}$, so

$$X(2^k l) \le 17.71\,\mathsf{e}(2^{k-1} l)^2 \le \frac{17.71}{166.41}\,\theta^{2^k} \le 0.10643\,\theta^{2^k} \le 0.10643 \cdot 0.394013.$$

$\Phi(\Psi(x)) = \frac{1}{20}$ at $x = 0.065062\ldots$ and $\Phi(x) = \frac{1}{20}$ at $x = 0.052727\ldots$; both functions increase, $X(2l) \le \Phi(x_0)$, $X(4l) \le \Phi(\Psi(x_0))$, and $\Phi(\Psi^k(x_0))$ does not increase in k because $\Psi(x) \le x$ there. (c) $\mathsf{m}_\mathsf{e}(2n) \ge \mathsf{m}_\mathsf{e}(n)$ is $12.9\,\mathsf{e}(2n) \le (12.9\,\mathsf{e}(n))^2$. For $k \ge 3$, $\mathsf{e}(n) \le X(n) \le \frac{1}{20}$ gives $X(2n) \le 17.71\,\mathsf{e}(n)^2 \le 18X(n)^2$, so m_X does not decrease. $X(n) \ge \mathsf{e}(n)$ gives $\mathsf{m}_X(n) \le \mathsf{m}_\mathsf{e}(n) - \frac{1}{n}\log\frac{18}{12.9}$, with $\log\frac{18}{12.9} \ge 0.3331$; $X(n) \le 17.71\,\mathsf{e}(n/2)^2$ gives $18X(n) \le 1.91563\,(12.9\,\mathsf{e}(n/2))^2$, so $\mathsf{m}_X(n) \ge \mathsf{m}_\mathsf{e}(n/2) - \frac{1}{n}\log 1.91563$, with $\log 1.91563 \le 0.6501$. Both rates being monotone and squeezed, they have one limit, and $\mathsf{m}_\mathsf{e}(l) \ge \Lambda/l$ is the criterion. (d) $Y_2(0.0740538) \approx 0.3156 < \log 2$; for $0 < x < 0.0740538$ the iterates $\Psi^k(x)$ decrease, and to 0, since a positive limit would be a fixed point below 0.0740538; so $\mathsf{e}(2^k l) \le \Psi^k(\mathsf{e}(l)) \to 0$ and $X(2^{k+1} l) \le \Phi(\Psi^k(\mathsf{e}(l))) \to 0$; and $\beta > 0$ because $\mathsf{e}(l) > 0$. □

Proof of Thm. AY. Apply Thms. AW and AX at $l = l_* L^K$ and $\beta = L^K \mathsf{b}$, with the one x_0 of the uniform criterion; then θ, Λ, q, $C(\theta)$ do not depend on K. By eq. 17.i, $\sigma^+ l^2 = \frac{1}{2}\log\frac{1}{\theta} + 0.4175$ depends on θ alone. A rectangle of physical sides R, T has $RL^K \times TL^K$ bonds, and $\sigma^+ \cdot RL^K \cdot TL^K = \frac{\sigma^+ l^2}{l_*^2} RT = \sigma_* RT$. For $R, T \ge 2a_K$, Lem. 17.25(b) with $\#B = 2(R+T)/a_K - 4$ and $\beta = \mathsf{b}/a_K$ gives $\nu_K(W_C) \le \exp[-3(|C|_{\mathrm{ph}} - 4a_K)/(8\mathsf{b} + 3a_K)]$, and the smaller of two bounds is at most their geometric mean: (a). For $n = \ell/a_K$: $\sigma^+ n^2 = \sigma_* \ell^2$, $n/l = \ell/l_*$, $a_K^{-1}\sigma^+ n = \sigma_* \ell$, $a_K^{-1}/l = 1/l_*$ and $a_K^{-1}/n = 1/\ell$; this turns Thm. AW(d), (e), (f) and (g) into (b), (c) and (d), with $\hat{N} \le \mathsf{q}\ell/l_* + 1$ and $C(\theta) \le 27$. (e) is Thm. AX(b) at $k = 3$, with $18 \cdot \frac{17.71}{166.41} \le 1.91563 \le 1.9157$, and Thm. AX(a); finally $4 \cdot 0.4757 = 1.9028$. □

Proof of Cor. 17.1. Step 1 (symmetry). The symmetries of ν carry a rectangle to every other one with the same sides, or with the sides exchanged and run backwards.

Step 2 (two Hankel forms). At a site mirror orthogonal to e_μ that contains a site x, let γ_h run from x up h steps along e_μ, R steps along e_λ, and down to $x + Re_\lambda$ (γ_0 lies in the mirror).

Then $\sum_{j,k} U(\gamma_{h'})_{jk}\overline{U(\mathfrak{r}\gamma_h)_{jk}} = 2W$ of a rectangle $R \times (h+h')$, and the reflection positivity of ν at this mirror (Prop. YM3_I:103.4; ν is a periodic Gibbs state), applied to $A_{jk} = \sum_h c_h U(\gamma_h)_{jk}$, gives $\sum_{h,h'} c_h c_{h'} m(R, h+h') \geq 0$. At a bond mirror, take the paths γ'_h from a site x_p next to the mirror up h steps, along R steps and down to $x_q = x_p + Re_\lambda$; then $\sum \overline{(\mathsf{z}_p)_{ji}}(\mathsf{z}_q)_{kl} A_{jk}\overline{\mathfrak{r}A_{il}}$ with $A_{jk} = \sum_h c_h U(\gamma'_h)_{jk}$ is $2\sum_{h,h'} c_h c_{h'} W$ of rectangles $R \times (h+h'+1)$. Its mean is ≥ 0 on the cubes of sides $2^k l$, which are even, by Lem. 17.21(c) after a translation that brings the mirror between the layers 0 and 1, and it passes to ν by Thm. AW(a): $\sum_{h,h'} c_h c_{h'} m(R, h+h'+1) \geq 0$.

Step 3 (log-convexity). The 2×2 minors of the two forms give $m(R,T)^2 \leq m(R,T-1)m(R,T+1)$ for all $T \geq 1$.

Step 4 ($w > 0$). Given the other bonds, $U(b)$ has the density proportional to $\exp[\beta \mathsf{k} \frac{1}{2}\operatorname{tr}(U(b)g)]$, $\mathsf{k}g$ the sum of the four staples of b, $0 \leq \mathsf{k} \leq 4$; so, by the Gibbs equation and Lem. 17.25(a), $4w = \nu(\mathsf{k}\, r(\beta\mathsf{k})) \geq \nu(\beta\mathsf{k}^2/(\beta\mathsf{k}+4)) \geq \beta\nu(\mathsf{k}^2)/(4\beta+4)$. Two different staples of b are mirror images at a mirror that contains b, a site mirror if they lie in one plane and a diagonal mirror if they lie in two; so $\nu(\mathsf{k}^2) = 4 + \sum_{s \neq s'} \nu(\frac{1}{2}\operatorname{tr} s s'^{-1}) \geq 4$ by reflection positivity, and $w \geq \beta/(4(\beta+1))$.

Step 5 (positivity). From $m(1,0) = 1$, $m(1,1) = w > 0$ and Step 3, $m(1,T) > 0$ for all T; then $m(R,1) = m(1,R) > 0$ and $m(R,T) > 0$. As $T \mapsto -\log m(R,T)$ is concave and vanishes at 0, $m(R,T) \geq m(R,1)^T \geq w^{RT}$; the upper bound of (a) is Thm. AW(b).

Step 6 (the potential). The increments of a concave sequence do not increase; their limit is their infimum and also the limit of $-\frac{1}{T}\log m(R,T)$; summing the increments from T_1 to T_2 gives $m(R,T_2) \leq e^{-(T_2-T_1)V(R)} m(R,T_1)$; if a number v has this property for infinitely many T_2, dividing by T_2 gives $v \leq V(R)$; and the increasing slopes of the convex function $\log m(R,\cdot)$ give the inequality of four rectangles.

Step 7 (the measure). Let Λ be the linear functional on the real polynomials with $\Lambda(\lambda^T) = m(R,T)$. Step 2 says $\Lambda(p^2) \geq 0$ and $\Lambda(\lambda p^2) \geq 0$. With $s_r = \Lambda(\lambda^r p^2) \geq 0$, Schwarz's inequality gives $s_r^2 \leq s_0 s_{2r}$, and, $|m| \leq 1$ bounding every s_r, iteration gives $s_2 \leq s_0$ and then $s_1 \leq s_0$: $\Lambda((1-\lambda)p^2) \geq 0$; and $\Lambda(\lambda(1-\lambda)p^2) = \Lambda((1-\lambda)(\lambda p)^2) + \Lambda(\lambda((1-\lambda)p)^2) \geq 0$. Writing $\lambda^T(1-\lambda)^k$ as $\lambda^\epsilon(1-\lambda)^{\epsilon'} q^2$ with $\epsilon, \epsilon' \in \{0,1\}$ gives $\sum_i (-1)^i \binom{k}{i} m(R, T+i) \geq 0$ for all $k, T \geq 0$: the sequence is completely monotone, and Hausdorff's moment theorem gives the one measure ϱ_R. Since $m(R,T) \leq (\max\operatorname{supp}\varrho_R)^T$ and $m(R,T) \geq (\lambda_* - \varepsilon)^T \varrho_R([\lambda_* - \varepsilon, 1])$ for $\lambda_* = \max\operatorname{supp}\varrho_R > 0$, $m(R,T)^{1/T} \to \lambda_*$, so $\lambda_* = e^{-V(R)}$; for $T \geq 1$ the point 0 gives nothing, and the substitution $\lambda = e^{-E}$ gives the representation, with the mass $\varrho_R((0,1]) \geq m(R,1) > 0$.

Step 8. (d) follows from (a): $V(R) \geq \sigma^+ R$ from the upper bound, $V(R) \leq -R\log w$ from the lower one. (e) At $l = l_* L^K$, $\sigma^+ = \sigma_* a_K^2$, so $a_K^{-1} V(R/a_K) \geq a_K^{-1}\sigma^+ R/a_K = \sigma_* R$; and $M_K(R,T_2)/M_K(R,T_1) = e^{2\kappa(T_2-T_1)} m(R/a_K, T_2/a_K)/m(R/a_K, T_1/a_K)$, to which (c) applies; in the inequality of four rectangles the factors $e^{2\kappa(T'-T)}$, raised to the powers $1/(T'-T)$, give one and the same factor on both sides. Along a sequence, $M_j(R,T_2) \leq e^{-(T_2-T_1)v_j(R)} M_j(R,T_1)$ at each j, as just shown with $\kappa = \kappa_j$; so $v_j(R) \leq \frac{1}{T_2-T_1}\log\frac{M_j(R,T_1)}{M_j(R,T_2)}$, whose right member converges, and $v_j(R) \geq B$ for infinitely many j gives $B \leq \limsup_j v_j(R)$; and $V_{K_j}(R) \geq \sigma_* R$ gives $v_j(R) \geq \sigma_* R - 2\kappa_j$.

The symmetry, the inequality of three rectangles, which is the log-convexity of (b) in the form of chords, and that of four rectangles hold for each M_j, the factors $e^{2\kappa_j(R+T)}$ cancelling in both inequalities, and pass to the limits. With $\mathsf{F}(T) = -\log \mathsf{W}(R,T)$, the inequality of three rectangles says that the slope $(\mathsf{F}(T') - \mathsf{F}(T_1))/(T'-T_1)$ does not increase in $T' > T_1$; so it has a limit in $[-\infty, \infty)$ as $T' \to \infty$, $\mathsf{F}(T')/T'$ tends to this limit for every T_1, and $V_{\mathrm{ren}}(R)$ is the infimum of all these slopes. (f) A rectangle with an even side $2h$ is $\gamma(\mathfrak{r}\gamma)^{-1}$ for the site mirror halfway along that side, and a square is $\gamma(\mathfrak{r}\gamma)^{-1}$ for a diagonal mirror through two of its corners; as in Step 2, Prop. YM3_I:103.4 gives $s(W_C) \geq 0$.

Step 9 (no criterion). (g) Steps 1–7 use of ν only that it is a periodic Gibbs state, invariant under the symmetries of the lattice and reflection positive at every lattice mirror (Prop. YM3_I:103.4), and,

for the second Hankel form, that it is positive at the bond mirrors on the functions of two crossing bonds. With the hypothesis of (g) they hold for s, but for the upper bound of (a), in whose place $m_s \le 1$ stands; and $m_s(R,T) \ge w^{RT}$ gives $V_s(R) \le -R \log w \le R \log \frac{4(\beta+1)}{\beta}$. On site mirrors alone the first Hankel form, with all c but two equal to 0, gives $m_s(R, h+h')^2 \le m_s(R,2h)\, m_s(R,2h')$. At $(h-1, h+1)$, and from $m_s(R,0) = 1$ and $m_s(R,2) > 0$, induction gives $m_s(R,2h) > 0$ for all h, and $\psi(h) = -\log m_s(R,2h)$ is concave, ≥ 0, with $\psi(0) = 0$; so $\psi(h)/h$ tends to $2V_{\mathrm{e}}(R) = \inf_h(\psi(h+1) - \psi(h)) \ge 0$, and $\psi(h_2) - \psi(h_1) \ge 2(h_2 - h_1)V_{\mathrm{e}}(R)$ is the inequality for even T_1 and T_2. For an odd $T_2 = 2h_2+1 > T_1$, the pair (h_2, h_2+1) gives $m_s(R, 2h_2+1) \le [m_s(R,2h_2)\, m_s(R,2h_2+2)]^{1/2}$, and the even case twice gives the inequality. For even R: the pair $(0,1)$ gives $m_s(1,2) \ge m_s(1,1)^2 = w^2 > 0$, by Step 4, and $m_s(2,2) \ge m_s(2,1)^2 = m_s(1,2)^2 > 0$; the induction at the width 2 gives $m_s(2,2h) > 0$, and $m_s(2h,2) = m_s(2,2h)$. □

Proof of Prop. 17.2. (a) $F(l,2l,2l) \le 0$ and $-F(2l,2l,2l) \le 6\beta(2l)^3$, so $0 \le \mathsf{e}(l) \le 48\beta l^3 \le x_0$. (b) For $\mathsf{k} \in \mathbb{R}^4$ let π_{k} be the law on $G \simeq S^3$ with the density proportional to $e^{\langle U, \mathsf{k}\rangle}$. Every linear component $\langle U, \mathsf{e}\rangle$, $|\mathsf{e}| = 1$, has variance $\le \frac14$ under π_{k}: with $\kappa = |\mathsf{k}|$ and $x = \langle U, \mathsf{k}\rangle/\kappa$, the mean $A(\kappa)$ of x satisfies $A' = \operatorname{Var} x$ and, by an integration by parts, $\langle 1 - x^2\rangle = 3A/\kappa$, so $A' = 1 - 3A/\kappa - A^2$; a comparison with the positive root $\mathsf{B}(\kappa)$ of $1 - 3a/\kappa - a^2 = \frac14$, whose derivative is $< \frac14$, together with $A \le \kappa/4$, which follows from $3A = \kappa\langle 1 - x^2\rangle$ and $\langle x^2\rangle \ge \frac14$, the two identities in the proof of Lem. 17.25(a), x having the law of x_0 under $\langle\cdot\rangle_\kappa$, gives $A \ge \mathsf{B}$ and so $\operatorname{Var} x \le \frac14$, while the components orthogonal to k have the variance $A/\kappa \le \frac14$.

So two such laws differ in total variation by at most $\frac14|\mathsf{k} - \mathsf{k}'|$. The conditional law of one bond given the others is $\pi_{\beta\mathsf{h}}$, h the sum of the four unit vectors of its staples, and changing one of its 12 neighbouring bonds moves h by at most 2. For a bounded Borel function f of finitely many bonds let $\delta_b(f)$ be its oscillation in the single bond b, and $\Delta(f) = \sum_b \delta_b(f)$. The conditional expectation E_b in the bond b gives $\delta_b(E_b f) = 0$, $\delta_{b'}(E_b f) \le \delta_{b'}(f) + \frac{\beta}{2}\delta_b(f)$ for the 12 neighbours b' of b, and $\delta_{b'}(E_b f) \le \delta_{b'}(f)$ for the other bonds; so after any sequence of updates the oscillations are, bond by bond, at most the entries of the vector v that arises from $(\delta_b(f))_b$ by the linear rules $v_b \to 0$, $v_{b'} \to v_{b'} + \frac{\beta}{2}v_b$ for the neighbours b' of the updated bond b.

Let S_0 be the set of the bonds of f, $S \supset S_0$ a finite set of bonds, $d(b)$ the distance of b from S_0 in the graph of neighbouring bonds, $r = \min\{d(b) : b \notin S\}$, $\alpha = 6\beta < 1$, and f_k the function that k sweeps of updates through S, each bond of S once in a sweep, make of f. An update at b takes $v_b\alpha^{-\min\{d(b),r\}}$ from the weighted sum $\sum_{b'} v_{b'}\alpha^{-\min\{d(b'),r\}}$ and adds at most $\frac{\beta}{2}\cdot 12\, v_b\, \alpha^{-\min\{d(b),r\}-1} = v_b\, \alpha^{-\min\{d(b),r\}}$, as $d(b') \le d(b)+1$; so the weighted sum does not increase, it equals $\Delta(f)$ at the start, when v lives on S_0, where $d = 0$, and the bonds outside S carry the weight α^{-r}: $\sum_{b\notin S} v_b \le \alpha^r \Delta(f)$ at all times.

An update at $b \in S$ changes $\sum_{b'\in S} v_{b'}$ by at most $-(1-\alpha)v_b$, and in a sweep the entries of the bonds of S at the times of their updates add up to at least the sum at the beginning of the sweep; so $\sum_{b\in S} v_b \le \alpha^k \Delta(f)$ after k sweeps. As f_k is a function of the bonds of S and of their neighbours, and two configurations of these bonds are joined by changing one bond at a time, $\sup f_k - \inf f_k \le (\alpha^k + \alpha^r)\Delta(f)$. Now let $l \ge 4$ be even, S the set of the bonds with both ends in the cube $\{0, \dots, l-3\}^3$, and $f = a_p$ for the plaquette $p = p_{12}(c_0, c_0, c_0)$, $c_0 = \frac{l-4}{2}$. The ends of the bonds at distance $\le j$ from the four bonds of p lie in $[c_0 - j, c_0 + 1 + j]^3$, which lies in $[0, l-3]^3$ for $j \le c_0$; so every bond not in S has $d \ge c_0 + 1 = \frac{l}{2} - 1$, that is $r \ge \frac{l}{2} - 1$; and $\Delta(a_p) \le 8$.

The bonds of S and their neighbours lie in a cube of l sites a side, which the projections modulo l and modulo $2l$ map one to one into the tori of sides l and $2l$, the plaquettes at the bonds of S onto plaquettes; so on both tori the conditional laws of the bonds of S are those used above, the functions f_k are the same, and both means of a_p equal means of f_k, which lie in $[\inf f_k, \sup f_k]$. Letting $k \to \infty$: $|\bar a_{2l}(b) - \bar a_l(b)| \le 8(6b)^{l/2-1}$, $\bar a_n(b)$ the mean of a_p on the cube of side n at the

coupling b. As $\frac{d}{d\beta}F(n,n,n) = -3n^3\bar{a}_n(\beta)$ and $X = 0$ at $\beta = 0$,

$$X(l) = 24\,l^3 \int_0^\beta (\bar{a}_{2l}(b) - \bar{a}_l(b))\,\mathrm{d}b \le 64\,l^2(6\beta)^{l/2}.$$

At each of the four pairs of a coupling and a side named in (b), $64\,l^2(6\beta)^{l/2} \le \frac{1}{20}$; and $l \mapsto l^2(6\beta)^{l/2}$ decreases for $l > 4/\log\frac{1}{6\beta}$, which lies below each of the four sides. Thm. AX(a) gives the criterion. (c) The same contraction, with S the set of the bonds at the distance $< r$ from the bonds of f, applied to two Gibbs states s, s', for which $s(E_b F) = s(F)$ by the Gibbs equation of one bond, gives $|s(f) - s'(f)| = |s(f_k) - s'(f_k)| \le (\alpha^k + \alpha^r)\Delta(f)$; let $k \to \infty$ and then $r \to \infty$. □

Proof of Prop. 17.3. (a) Lem. 17.11(a), (b) with $\kappa_i = \lambda_i(2l, 2l)^l$, so that $D = \mathsf{e}(l)$ and $\eta = \mathsf{r}$: $2\log 1.0351 \le 0.069$, and $\varpi(e^{0.069} - 1) \le 0.0357638$. The tops Λ_e, $e \ne 00$, are among the λ_i, $i \ge 1$ (Lem. 17.10), so $(\Lambda_e/\lambda_0)^l \le \mathsf{r}$, and $\log\frac{1}{0.0357638} \ge 3.33$; and $\mathsf{r} \ge \sum_{e\ne 00}(\Lambda_e/\lambda_0)^l$ gives the last statement. (b) With the time 1 over $(2l, 2l)$ let $w_f = W_{2l}(f; 0)/Z$ on the cube of side $2l$. Inside each sector $(\sum_i \mu_i^l)^2 \ge \sum_i \mu_i^{2l}$, so

$$\mathsf{e}(l) = 2\log Z(l, 2l, 2l) - \log Z(2l, 2l, 2l) \ge 2\log \sum_f \sqrt{w_f}.$$

By Lem. 17.10(c), $w_{00} = \frac{1}{4}(1 + 2r_1 + r_2)$, $w_{10} = w_{01} = \frac{1}{4}(1 - r_2)$, $w_{11} = \frac{1}{4}(1 - 2r_1 + r_2)$, and $w_{11} \ge 0$, $r_2 \le r_1$ give $2r_1 - 1 \le r_2 \le r_1$; here $r_1 \le 1$ by Lem. 17.10(c), and $r_1 > 0$ because, with the time the direction that is not in the twisted pair, $Z^- = \operatorname{Tr}\mathsf{T}_1^{2l}$ is the integral over the diagonal of a kernel with values > 0 (Lem. 17.10(i), (ii)). The sum of the square roots is concave in r_2, so on this interval it is at least the smaller of its two end values; with $x = 1 - r_1 \in [0, 1)$ these are $\sqrt{1-x} + \sqrt{2x}$ at $r_2 = 2r_1 - 1$ and $\frac{1}{2}\sqrt{4 - 3x} + \frac{3}{2}\sqrt{x}$ at $r_2 = r_1$, and the first is the smaller, as $\frac{1}{4}(4 - 3x) \ge 1 - x$ and $\frac{3}{2} > \sqrt{2}$.

The function f increases on $[0, \frac{2}{3}]$ and decreases on $[\frac{2}{3}, 1]$, with $\mathsf{f}(1) = \log 2$; and $\mathsf{f}(0.00063) > 0.069$, $\mathsf{f}(0.0039) \ge 0.16567$. (c) Let $\mathsf{e}(l) \le 0.069$ and $N \ge 16l$, and let $\bar{N} = 2^j l \le N < 2\bar{N}$, $j \ge 4$. By Lem. 17.11(f), $\log(1 + \mathsf{r}_N(N, N)) \le \log(1 + \mathsf{r}_{\bar{N}}(N, N)) \le \delta(\bar{N}, N, N) \le 4\delta_{\mathrm{c}}(\bar{N}) \le 4\bar{\delta}_4 \le 0.0029221$ (Prop. 17.16(e)); and by Lem. 17.10(c), $1 - Z^-/Z = 1 - r_1 \le \frac{4}{3}(1 - e^{-0.0029221}) \le 0.0038905 < 0.0039$. For an integer $l' \ge 16l$ with $\bar{N} = 2^j l \le l' < 2\bar{N}$, $j \ge 4$: $\mathsf{e}(l') \le 2\log(1 + \mathsf{r}_{l'}(2l', 2l')) \le 2\delta(\bar{N}, 2l', 2l') \le 2(2l'/\bar{N})^2\delta_{\mathrm{c}}(\bar{N}) < 32\,\bar{\delta}_4 \le 0.02338$.

The last sentence is the contrapositive of the first, at $l \le N/16$. (d) Thm. AW(b) at $R = T = 1$ gives $\sigma^+ \le \log(1/\nu(W_p))$, and $\sigma^+ l^2 \ge 0.4757$. □

The numbers of Remark 17.4. Over a torus of side $2l$ the momenta are $\frac{\pi}{l}m$, $m \in \mathbb{Z}^2$; three fields with one mode for each momentum $m \ne 0$ give the Fock levels whose sum at the time l is the product displayed, and the same at the time $2l$; in the notation of Lem. 17.11, $1 + \eta = 1 + \mathsf{r}$ and $1 + s_2 = 1 + \mathsf{r}'$. Summing $-3\log(1 - e^{-\pi|m|})$ over the shells $|m|^2 = 1, 2, 4, 5, 8, 9, 10, 13, \dots$ gives approximately 0.72010, and $-3\log(1 - e^{-2\pi|m|})$ gives approximately 0.02415; so $D \approx 1.416$.

Proof of Prop. 17.6. The exact part. On every periodic cube of an even side N, at every β, take as time a direction of the twisted pair, let ν be the other direction of that pair, let $\vartheta \ge 2$ and $k \ge 2$ be integers with $2k\vartheta = N$, and cut the cube into $2k$ slabs of ϑ layers. The transfer operator over ϑ layers, $\mathsf{Y} = \mathsf{T}'^{\vartheta}$ in the notation of the proof of Lem. 17.21, commutes with the flip J of one layer, the operator J_i of Lem. 17.10 that changes the sign of the bonds of the direction ν that cross one plane $\{x_\nu = c\}$, $c \in \mathbb{Z} + \frac{1}{2}$; $Z = \operatorname{Tr}\mathsf{Y}^{2k}$ and $Z^- = \operatorname{Tr}(\mathsf{J}\mathsf{Y}^{2k})$ (Lem. 17.10(ii), (c)); so $P_\pm = \frac{1}{2}(1 \pm \mathsf{J})$ have $\operatorname{Tr}(P_\pm \mathsf{Y}^{2k}) = \frac{1}{2}(1 \pm r_1)Z$: the weights of the two sectors of the flip are $\frac{1}{2}(1 \pm r_1)$.

A function of the bonds is *odd under the flip* if it changes its sign when the bonds of the direction ν that cross the plane $\{x_\nu = c\}$ change their signs in all layers. Let D, D' be bounded Borel functions of the bonds with both ends in one and the same slab of ϑ layers, D' odd under the

flip; of D no parity is asked. For $1 \le j \le 2k-1$ let t_j be the translation by $j\vartheta$ along the time, and $\mathsf{t} = \mathsf{t}_k$ the translation by $N/2$. With the slab operators Y_D, $\mathsf{Y}_{D'}$ of the proof of Lem. 17.21, $Z\langle D \cdot \mathsf{t}_j D'\rangle = \mathrm{Tr}(\mathsf{Y}^{2k-j-1}\mathsf{Y}_{D'}\mathsf{Y}^{j-1}\mathsf{Y}_D)$.

The weights of the layers and of the crossings do not change when all layers are flipped (proof of Lem. 17.10); so $P_\sigma \mathsf{Y} = \mathsf{Y} P_\sigma$ and, D' being odd, $\mathsf{J}\mathsf{Y}_{D'}\mathsf{J} = -\mathsf{Y}_{D'}$, that is $P_\sigma \mathsf{Y}_{D'} = \mathsf{Y}_{D'} P_{-\sigma}$. Splitting the trace over the two sectors, the sector changes between D and D':

$$Z\langle D \cdot \mathsf{t}_j D'\rangle = \sum_{\sigma=\pm 1} \mathrm{Tr}\big((P_\sigma \mathsf{Y})^{2k-j-1}\, \mathsf{Y}_{D'}\, (P_{-\sigma}\mathsf{Y})^{j-1}\, \mathsf{Y}_D\big),$$

a trace of $2k$ Hilbert–Schmidt operators. Hölder's inequality for it (Lem. YM3_I:86.1), with $\|P_\sigma \mathsf{Y}\|_{2k}^{2k} = \mathrm{Tr}(P_\sigma \mathsf{Y}^{2k})$ and $\|\mathsf{Y}_D\|_{2k}^{2k} = \mathrm{Tr}(\mathsf{Y}_D^* \mathsf{Y}_D)^k \le \sup|D|^{2k} Z$ (the norm definition preceding Lem. YM3_I:86.1; the trace is Z times the mean of a product of $2k$ copies of D and of its mirror image), gives

$$|\langle D \cdot \mathsf{t}_j D'\rangle| \le \sup|D|\, \sup|D'| \sum_{\sigma=\pm 1} \Big(\frac{1+\sigma r_1}{2}\Big)^{\frac{2k-j-1}{2k}} \Big(\frac{1-\sigma r_1}{2}\Big)^{\frac{j-1}{2k}}$$

(with $0^0 = 1$), and at $j = k$, where the two summands are equal,

$$|\langle D \cdot \mathsf{t} D'\rangle| \le 2 \sup|D|\, \sup|D'| \Big[\frac{1-r_1^2}{4}\Big]^{\frac12 - \frac{1}{2k}}.$$

The calibrated construction. The constants are chosen as in (Th1), (Th2) of §2.3.1, the setting in which Thm. YM3_I:D is used; among them are the cube size $M = \mathsf{L}^{m'}$, with $m' \ge 5$ (the setting preceding equation YM3_I:(634) has $M \ge L^4 M_1$ and $M_1 \ge L$), the exponents $r \ge 2$ and $\mathfrak{p}_0$, the constant $A_0 > 0$ of the thresholds (equation YM3_I:(634)), $\gamma \in (0, \frac12]$, and the numbers $c' > 0$, $\mathsf{C}^\Sigma_{\mathrm{maj}}$, c_{low} of (Th2). The setting (E) of §2.3.1 gives besides a label $\mathsf{b} \ge \gamma^{-2}$ and an integer $m \ge 1$ with $M\, R(\mathsf{b}) \le \mathsf{L}^m$, $R(x) = \min\{\mathsf{L}^i : i \in \mathbb{Z}_{\ge 0},\ \mathsf{L}^i \ge (\log x)^r\} \ge 1$, and $\ell = 2\mathsf{L}^m$. At the cutoff K the coupling is $\beta = \mathsf{L}^K \mathsf{b}$, and the cube has $N = \ell \mathsf{L}^K = 2\mathsf{L}^{m+K}$ sites a side. For an integer $0 \le d < K$ the block field of the depth d is $V = \mathcal{M}^{K-d} U$, $\mathcal{M}$ the block-averaging map of equations YM3_I:(37)–(39); it lives on a periodic cube of

$$\mathsf{n} = \ell \mathsf{L}^d = 2\mathsf{L}^{m+d}$$

blocks a side, each block having $\vartheta = \mathsf{L}^{K-d}$ sites a side, so that $N = \mathsf{n}\vartheta$; its law is μ_d^K (Def. 2.7). As $R \ge 1$, $\mathsf{L}^m \ge M = \mathsf{L}^{m'} \ge \mathsf{L}^5$, and with $d \ge 1$ and $\mathsf{L} \ge 13$:

$$\mathsf{n} \ge 2 \cdot 13^6.$$

Let $|\partial V - \mathbf{1}|_\infty$ be the largest $|\partial V(p) - \mathbf{1}|$ over the plaquettes of the block lattice, $x_d = \mathsf{L}^d \mathsf{b}$, $\varepsilon = A_0 x_d^{-1/2} (\log x_d)^{\mathfrak{p}_0}$ the threshold of equation YM3_I:(634) at the coupling x_d, $\delta = \exp[7\mathsf{n}^2\varepsilon] - 1$, and R_d the number R_t of the definitions preceding Prop. YM3_I:115.2 at the level $t = K - d$; as there $N_t = \ell^3 \mathsf{L}^{3d}$ and $x_t = x_d$, R_d does not depend on K. The hypotheses (H) are: $d \ge 1$; $x_d \ge \mathsf{t}_\natural$, the fixed calibration of §2.3.1; $\delta \le \frac{1}{20}$; and $\mathsf{R}_d \le \frac{1}{40}$.

Step 1: under (H) the block field is nearly flat with probability at least $\frac{39}{40}$, at every cutoff $K > d$. The data K, b, m and the top level $t = K - d$ make the setting (Th) of §2.3.1, as $1 \le t \le K-1$ and $x_t = x_d \ge \mathsf{t}_\natural$; Lem. YM3_I:118.1 gives what Prop. YM3_I:115.2 asks of it. Let $F = \mathbf{1}_{\mathsf{In}_t}(\mathcal{M}^t U)$, $\mathsf{In}_t = \{|\partial V - \mathbf{1}|_\infty < 2\varepsilon\}$ the inner ball of the definitions preceding Lem. YM3_I:113.10: a bounded gauge-invariant Borel function, by Prop. YM3_I:4.5, YM3_I:4.5 and YM3_I:113.10. The weight $\mathfrak{w}_t^{\mathrm{sf}}$ of the definitions preceding Prop. YM3_I:115.2 carries the factor $\mathbf{1}_{\mathsf{In}_t}(\mathcal{M}^t U)$, so $\mathfrak{w}_t^{\mathrm{sf}} = F\, \mathfrak{w}_t^{\mathrm{sf}}$, and the right inequality of Prop. YM3_I:115.2, domination and lost-mass bounds gives $Z - \int F \rho_0\, \mathrm{d}U \le Z - \int \mathfrak{w}_t^{\mathrm{sf}} \rho_0\, \mathrm{d}U \le \mathsf{R}_d Z$, $\rho_0 = e^{-\beta A}$ at $\beta = \mathsf{L}^K \mathsf{b}$. That is

$$\mu_d^K\big(|\partial V - \mathbf{1}|_\infty \ge 2\varepsilon\big) \le \mathsf{R}_d \le \tfrac{1}{40}.$$

Thm. YM3_I:D enters here, through Prop. YM3_I:115.2.

Step 2: a winding loop of the block field, its parity and its translate. Fix a block site ξ; let Γ_1, Γ_2, Γ_{12} be the closed paths of the block lattice from ξ with the step sequences (e_1^{n}), (e_2^{n}), $(e_1^{\mathsf{n}}, e_2^{\mathsf{n}})$, $P_i = V(\Gamma_i)$, and for each of them $D_\Gamma(U) = W_\Gamma(V) = \frac{1}{2}\operatorname{tr} V(\Gamma)$. Take the direction 3 as time, the flip of the bonds of the direction 1 that cross the plane $\{x_1 = c\}$, $c = \xi_1 + \frac{1}{2}\vartheta$, and t the translation by $N/2$ along e_3. Then: (i) D_Γ is a real Borel function with $|D_\Gamma| \le 1$ (Prop. YM3_I:4.5); (ii) it is a function of the bonds with both ends in one slab of ϑ layers, the slab of the blocks of the sites of Γ, by the locality of $\mathcal{M}^{K-d}$ (Prop. YM3_I:4.5); (iii) D_{Γ_1} and $D_{\Gamma_{12}}$ are odd under the flip: as L is odd, the plane $\{x_1 = c\}$ separates blocks at every level from d to K, on each double block the flip is the gauge transformation that is $-\mathbf{1}$ on one side of the plane, so by Prop. YM3_I:4.5 the averaging map carries the flip of the bonds that cross the plane at one level to the flip at the next level; and Γ_1, Γ_{12} run exactly one block bond that crosses the plane, Γ_2 none; (iv) $\mathsf{t}D_\Gamma = W_{\mathsf{t}\Gamma}(V)$, as t is the translation by $\frac{\mathsf{n}}{2}$ blocks and $\mathcal{M}^{K-d}$ commutes with it ([7, Lem. G.1]).

And $\mathsf{m}_\Gamma := \langle D_\Gamma \cdot \mathsf{t}D_\Gamma\rangle = \int W_\Gamma W_{\mathsf{t}\Gamma}\, \mathrm{d}\mu_d^K$ has $\mathsf{m}_{\Gamma_2} = \mathsf{m}_{\Gamma_1}$: the exchange of e_1 and e_2 about ξ keeps the law of the cube, commutes with t and with $\mathcal{M}^{K-d}$ ([7, Lem. G.1]), and carries Γ_1 to Γ_2.

On $\mathsf{B} = \{|\partial V - \mathbf{1}|_\infty < 2\varepsilon\}$ the two winding holonomies nearly commute: the step sequences $(e_1^{\mathsf{n}}, e_2^{\mathsf{n}})$ and $(e_2^{\mathsf{n}}, e_1^{\mathsf{n}})$ are rearrangements of one another, of the length $2\mathsf{n}$, and Lem. YM3_I:4.1 gives

$$|P_2P_1P_2^{-1}P_1^{-1} - \mathbf{1}| \le (1 + |\partial V - \mathbf{1}|_\infty)^{\mathsf{n}(2\mathsf{n}-1)} - 1 < e^{4\mathsf{n}^2\varepsilon} - 1 \le \delta.$$

As $|g - \mathbf{1}|^2 = 2 - \operatorname{tr} g$ for $g \in G$, $\frac{1}{2}\operatorname{tr}(P_2P_1P_2^{-1}P_1^{-1}) \ge 1 - \frac{1}{2}\delta^2$. The trace identity of Fricke, $\operatorname{tr}(ABA^{-1}B^{-1}) = (\operatorname{tr} A)^2 + (\operatorname{tr} B)^2 + (\operatorname{tr} AB)^2 - \operatorname{tr} A \operatorname{tr} B \operatorname{tr} AB - 2$ for $A, B \in \mathrm{SL}(2,\mathbb{C})$, at $A = P_2$, $B = P_1$, gives with $x_i = W_{\Gamma_i}(V)$ and $x_{12} = W_{\Gamma_{12}}(V)$

$$x_1^2 + x_2^2 + x_{12}^2 - 2x_1x_2x_{12} \ge 1 - \tfrac{1}{4}\delta^2, \qquad \text{so} \qquad 2x_1^2 + 2x_2^2 + x_{12}^2 \ge 1 - \tfrac{1}{4}\delta^2,$$

as $|x_{12}| \le 1$ gives $-2x_1x_2x_{12} \le x_1^2 + x_2^2$. On B a winding loop is also close to its translate: with ϵ the path of $\frac{\mathsf{n}}{2}$ steps e_3 from ξ to $\mathsf{t}\xi$ and $g = V(\epsilon)$, the step sequences of Γ followed by ϵ and of ϵ followed by $\mathsf{t}\Gamma$ are rearrangements of one another, of the length $\le \frac{5}{2}\mathsf{n}$, and Lem. YM3_I:4.1 gives $|gV(\mathsf{t}\Gamma)g^{-1} - V(\Gamma)| \le (1 + |\partial V - \mathbf{1}|_\infty)^{\frac{25}{8}\mathsf{n}^2} - 1 < e^{6.25\,\mathsf{n}^2\varepsilon} - 1 \le \delta$, so that $|W_{\mathsf{t}\Gamma} - W_\Gamma| \le \delta$. As $W_\Gamma W_{\mathsf{t}\Gamma} \ge W_\Gamma^2 - |W_\Gamma - W_{\mathsf{t}\Gamma}|$ and $|W_\Gamma - W_{\mathsf{t}\Gamma}| \le 2$ off B, Step 1 gives $\mathsf{m}_\Gamma \ge \mu_d^K(W_\Gamma^2) - \delta - 2\mathsf{R}_d$; with the weights $2, 2, 1$ and $\mathsf{m}_{\Gamma_2} = \mathsf{m}_{\Gamma_1}$,

$$4\mathsf{m}_{\Gamma_1} + \mathsf{m}_{\Gamma_{12}} \ge (1 - \tfrac{1}{4}\delta^2)(1 - \mathsf{R}_d) - 5(\delta + 2\mathsf{R}_d) \ge \tfrac{1599}{1600} \cdot \tfrac{39}{40} - \tfrac{1}{2} = \tfrac{30361}{64000} > 0.4743,$$

the middle member decreasing in δ and in R_d on $[0,1]$. So $\mathsf{m}_\Gamma \ge 0.0948 > 0.09$ for $\Gamma = \Gamma_1$ or for $\Gamma = \Gamma_{12}$; both loops give real functions of modulus ≤ 1 that are odd under the flip and are functions of one slab of the thickness of a block.

Step 3: the ceiling. The exact part at $D = D' = D_\Gamma$, with that Γ, $\vartheta = \mathsf{L}^{K-d} \ge 2$ and $2k = \mathsf{n}$, so that $2k\vartheta = N$ and $\frac{1}{2k} = \frac{1}{\mathsf{n}} = \frac{a_d}{\ell}$, the ratio of the block size $a_d = \mathsf{L}^{-d}$ at the depth d to the side ℓ, gives $0.09 \le \mathsf{m}_\Gamma \le 2[\frac{1}{4}(1 - r_1^2)]^{\frac{1}{2}(1-2/\mathsf{n})}$, that is, with $\mathsf{x} = (0.045)^{2/(1-2/\mathsf{n})}$,

$$r_1^2 \le 1 - 4\mathsf{x}, \qquad r_1 \le 1 - 2\mathsf{x} = 1 - 2\,(0.045)^{2/(1-2/\mathsf{n})},$$

the second because $(1 - 2\mathsf{x})^2 \ge 1 - 4\mathsf{x}$, $1 - 2\mathsf{x} > 0$ and $r_1 > 0$. The right member of the second inequality decreases in n to the limit 0.99595, and it is ≤ 0.99596 if and only if $\mathsf{n} \ge 5020$; at $\mathsf{n} = 2 \cdot 13^6$ it is ≤ 0.99595002. So under (H), $r_1 \le 0.99596$ on the cube of $2\mathsf{L}^{m+K}$ sites a side at $\beta = \mathsf{L}^K\mathsf{b}$, for every $K > d$.

Step 4: the range of (H); the numbers $t_\circ$ and m_t; (a). Let $\mathsf{C}_+ = \max\{0, \mathsf{C}_{\mathrm{maj}}^\Sigma + 2 + c_{\mathrm{low}}\}$, and let $y_\circ \ge 2$ be the least real number such that for all $y \ge y_\circ$

$$28\mathsf{L}^{11/2}M^2A_0\, y^{2r}(y + \log \mathsf{L})^{\mathfrak{p}_0} e^{-y/2} \le \log\tfrac{21}{20}, \qquad \log(16\mathsf{L}^9 y^{3r}) + (\mathsf{C}_+ + \tfrac{9}{2}y)(2\mathsf{L}^2M)^3 y^{3r} - c' A_0^2 y^{2\mathfrak{p}_0} \le -\log 40;$$

it exists, as the first left member tends to 0 and, in the second, $2\mathfrak{p}_0 > 3r+2$ at the chosen constants (the choice $\mathfrak{p}_0 \ge 6r+1$ in Prop. YM3_I:69.1) and $c'A_0^2 > 0$. Put $t_\circ = \min\{2\mathsf{L}\gamma^2,\ 2\mathsf{L}^2/\mathsf{t}_\natural,\ 2\mathsf{L}e^{-y_\circ}\}$. For $0 < t \le t_\circ$ let m_t be the least integer $m \ge 1$ with $M\,R(2\mathsf{L}^m/t) \le \mathsf{L}^m$, which exists as $R(x) \le \max\{1, \mathsf{L}(\log x)^r\}$; let $\ell_t = 2\mathsf{L}^{m_t}$, $\mathsf{b}_t = \ell_t/t$ and $y = \log \mathsf{b}_t$. Then (E) holds for the label b_t and the integer m_t, as $\mathsf{b}_t \ge 2\mathsf{L}/t_\circ \ge \gamma^{-2}$; $y \ge \log(2\mathsf{L}/t_\circ) \ge y_\circ$; and $\ell_t \le 2\mathsf{L}^2 M y^r$, as $m_t = 1$ or $\mathsf{L}^{m_t-1} < M\,R(\mathsf{b}_t/\mathsf{L}) \le M\mathsf{L}y^r$.

At $d=1$: $x_1 = \mathsf{L}\mathsf{b}_t \ge 2\mathsf{L}^2/t_\circ \ge \mathsf{t}_\natural$; $\mathsf{n} = \mathsf{L}\ell_t \le 2\mathsf{L}^3 M y^r$ and $x_1 = \mathsf{L}e^y$ give $7\mathsf{n}^2\varepsilon \le \log\frac{21}{20}$ by the first inequality, that is $\delta \le \frac{1}{20}$; and $\log x_1 \ge y$, $2M^{-3}\ell_t^3\mathsf{L}^3 \le 16\mathsf{L}^9 y^{3r}$ and $\ell_t^3 \le (2\mathsf{L}^2M)^3 y^{3r}$ give $\mathsf{R}_1 \le \frac{1}{40}$ by the second. So (H) holds for (b_t, m_t) at $d=1$. At the cutoff $K \ge 2$ the cube has $N = 2\mathsf{L}^{m_t+K}$ sites a side and $\beta = \mathsf{L}^K\mathsf{b}_t = N/t$; by Step 3, $r_1 \le 0.99596$ on the cube of $2\mathsf{L}^j$ sites a side at $\beta = 2\mathsf{L}^j/t$, for every $j \ge m_t+2$. Then $1-r_1 \ge 0.00404 \ge 0.0039$, and Prop. 17.3(b), (c) give $\mathsf{e}(\mathsf{L}^j) \ge \mathsf{f}(0.0039) \ge 0.16567$ and $\mathsf{e}(l) > 0.069$ for $3 \le l \le \mathsf{L}^j/8$: (a).

(b) The cube of the criterion at the cutoff K has $2l_*\mathsf{L}^K = 2\mathsf{L}^{m_0+K}$ sites a side and $\beta = \mathsf{L}^K\mathsf{b} = 2\mathsf{L}^{m_0+K}/t$ with $t = \ell_0/\mathsf{b} \le t_\circ$; apply (a) with $j = m_0+K$, for all $K \ge m_t+2-m_0$. (c) Let $\bar y = \frac{4}{3}[\log(2\mathsf{L}^2M) + r\log(4r) + \log(\mathsf{L}/t_\circ)]$ and $\beta_\flat = 2\mathsf{L}^5 M\bar y^r/t_\circ$. For $\beta \ge \beta_\flat$ choose j with $2\mathsf{L}^j \le t_\circ\beta < 2\mathsf{L}^{j+1}$ and put $t = 2\mathsf{L}^j/\beta \in (t_\circ/\mathsf{L}, t_\circ]$. With the numbers of Step 4 at this t, $y \le \log\ell_t + \log(\mathsf{L}/t_\circ)$ and $\ell_t \le 2\mathsf{L}^2 M y^r$; with $r\log y \le \frac{y}{4} + r\log(4r) - r$ this gives $y \le \bar y$, so $2\mathsf{L}^{m_t+2} = \mathsf{L}^2\ell_t \le 2\mathsf{L}^4 M\bar y^r = \frac{t_\circ}{\mathsf{L}}\beta_\flat \le \frac{t_\circ}{\mathsf{L}}\beta < t\beta = 2\mathsf{L}^j$, that is $j \ge m_t+2$; and (a) gives $\mathsf{e}(l) > 0.069$ for $3 \le l \le \mathsf{L}^j/8$, while $\mathsf{L}^j/8 > t_\circ\beta/16\mathsf{L}$. For the uniform criterion at a label b take $K \ge K_0$ with $L^K\mathsf{b} \ge \beta_\flat$: the criterion at $l = l_*L^K$ forces $l_*L^K > \frac{t_\circ}{16\mathsf{L}}L^K\mathsf{b}$.

(d) Put $\mathsf{G}_N = \log(Z(N,N,N)/\lambda_0(N,N)^N) = \log(1+\mathsf{r}_N(N,N))$. By Prop. 17.3(a), and as r_t decreases in t, $\mathsf{e}(N/2) \ge \log(1+\mathsf{r}_{N/2}(N,N)) \ge \mathsf{G}_N$; and for every integer $q \ge 2$, $Z(N,N,qN) \ge \lambda_0(N,N)^{qN}$ gives $\mathsf{G}_N \ge \mathsf{d}_q(N)/q$, $\mathsf{d}_q(N) = qF(N,N,N) - F(N,N,qN)$. Three facts bound $e^{-\mathsf{d}_q(N)} = Z(N,N,qN)/Z(N,N,N)^q$ from above: a swap on the long box, which bounds it through the largest probability on the cube (N,N,N) that the five invariants of the pair of the winding holonomies at one site lie in one small cell; a bound of that probability on a bare cube (Lem. 17.26); and a comparison of the laws of the block fields at two consecutive cutoffs in which the errors that bound quotients of densities enter as factors (Lem. 17.27), which carries the bound from one low cutoff K_1 to all higher ones. Besides Thm. YM3_I:D and the statements among the construction results named for this clause, (d) rests on Lem. 17.5 and these two lemmas, which give the estimate eq. 17.iii and, with the chain from a low cutoff, the estimate eq. 17.iv, and are stated below with their proofs.

A swap on the long box. For $g \in G$ write $g = \cos\alpha(g)\,\mathbf{1} + i\sum_j \mathsf{v}_j(g)\tau_j$, τ_1,τ_2,τ_3 the Pauli matrices, with the class angle $\alpha(g) = \arccos\frac{1}{2}\operatorname{tr} g \in [0,\pi]$ and the vector part $\mathsf{v}(g) \in \mathbb{R}^3$, $|\mathsf{v}(g)| = \sin\alpha(g)$. To a pair (g_1,g_2) of elements of G attach the real 3×2 matrix v with the columns $\mathsf{v}(g_1)$, $\mathsf{v}(g_2)$, the matrix $\mathsf{s} = (\mathsf{v}^{\mathsf{t}}\mathsf{v})^{1/2}$, and the five invariants $\mathsf{a}(g_1,g_2) = (\alpha(g_1),\alpha(g_2),\mathsf{s}_{11},\mathsf{s}_{12},\mathsf{s}_{22}) \in \mathbb{R}^5$: they do not change under a simultaneous conjugation of the pair, and if $|g_i - hg_i'h^{-1}| \le \delta \le 1$ for $i=1,2$ and one $h \in G$, then $|\mathsf{a}(g_1,g_2) - \mathsf{a}(g_1',g_2')|_\infty \le 2\delta$, as $|\mathsf{v}(g) - \mathsf{v}(g')| \le |g-g'|$, $|g-g'| \ge 2\sin\frac{1}{2}|\alpha(g)-\alpha(g')|$, and $\mathsf{v} \mapsto (\mathsf{v}^{\mathsf{t}}\mathsf{v})^{1/2}$ has the Lipschitz constant $\sqrt{2}$ for the Hilbert–Schmidt norm.

Let V be either the bare field, with $\mathsf{n} = N$, or the block field of the depth 0, on a lattice of $\mathsf{n} = \ell$ sites a side in the section, formed on the long box by the same local formulas; let $\omega \in (0,1]$ with $\delta_q = (1+\omega)^{q\mathsf{n}^2} - 1 \le 1$. To a site ξ attach $\mathsf{a}(V,\xi) = \mathsf{a}(P_1(\xi),P_2(\xi))$, the five invariants of the pair of its winding holonomies in the directions 1 and 2. Cut $\mathbb{R}^5$ into half-open cubes c_k of a side $w \ge 2\delta_q$, and let c_k^+ be the half-open cube of side $3w$ with the same centre and the same choice of included faces, so that every point lies in exactly 3^5 of them. On the event that every plaquette of V on the box (N,N,qN) is within ω of $\mathbf{1}$, let $\xi' = \xi + ke_3$, $0 \le k < q\mathsf{n}$, and t the transport of V from ξ to ξ' along the line of the direction 3: for $\mu = 1,2$ the transports $\mathsf{t}P_\mu(\xi')$ and $P_\mu(\xi)\mathsf{t}$ run along the step sequences $(e_3^k, e_\mu^{\mathsf{n}})$ and $(e_\mu^{\mathsf{n}}, e_3^k)$, which pass into one another by $k\mathsf{n} \le q\mathsf{n}^2$ exchanges of adjacent

entries, and each exchange multiplies the transport by a conjugate of $\partial V(p)^{\pm 1}$ for one plaquette p, as in the proof of Lem. YM3_I:4.1; so $|\mathsf{t} P_\mu(\xi')\mathsf{t}^{-1} - P_\mu(\xi)| \le \delta_q$, with one t for both μ, and $\mathsf{a}(V,\cdot)$ stays within $2\delta_q$ of $\mathsf{a}(V,\xi)$ at all the sites of that line.

Cut the box by walls into q slabs of N layers, with the mirrors θ_i in their middle planes and sites ξ_i^- of one line of the direction 3 just below these planes; then the indicator of the event is at most $\sum_k \prod_i g_i^k\,(g_i^k \circ \theta_i)$, $g_i^k = \mathbf{1}\{\mathsf{a}(V,\xi_i^-) \in c_k^+\}$, a function of the bonds of the lower half of the slab i (for the block field, whose walls of the blocks contain these planes, as $N = \mathsf{n}\mathsf{L}^K$ with n even, by the locality of the averaging map and its covariance under the lattice isometries, Prop. YM3_I:4.5, [7, Lem. G.1]). The slab operator of $g_i^k\,(g_i^k \circ \theta_i)$ is $\mathsf{Y}^*\mathsf{Y} \ge 0$, Y that of the lower half with g_i^k, as in the proof of Lem. 17.21; and $|\operatorname{Tr}(\mathsf{O}_1 \cdots \mathsf{O}_q)| \le \prod_i \operatorname{Tr}\mathsf{O}_i$ for positive operators of trace class.

So every slab closes into a cube: $Z(N,N,qN)\langle \prod_i g_i^k\,(g_i^k \circ \theta_i)\rangle \le \prod_i Z(N,N,N)\langle g_i^k\rangle_{(N,N,N)}$. The plaquettes off the event are bounded by Schwarz's inequality at a mirror (Lem. 17.21(a)) and the same closing. With P_* the largest probability on the cube (N,N,N) that $\mathsf{a}(V,\xi)$ lies in one c_k^+, and ρ the largest probability on it that one plaquette of V is not within ω of $\mathbf{1}$, this gives $x \le 243\,P_*^{q-1} + 3q\mathsf{n}^3(\rho x)^{1/2}$ for $x = e^{-\mathsf{d}_q(N)}$, whence

$$e^{-\mathsf{d}_q(N)} \le 486\,P_*^{q-1} + 9q^2\mathsf{n}^6\rho.$$

Anticoncentration on a bare cube. For an integer $N' \ge 4$ and a coupling $\beta' > 0$ let $\mu_{N',\beta'}$ be the law $Z^{-1}e^{-\beta'\sum_p a_p}dU$ of the periodic cube of N' sites a side, $Z = Z(N',N',N')$, and $|\partial U - \mathbf{1}|_\infty$ the largest $|\partial U(p) - \mathbf{1}|$ over its plaquettes.

Lemma 17.26 (Anticoncentration on a bare cube)**.** Let $N' \ge 4$ be an integer, $\beta' \ge C_{\mathrm{T}} N'^{68}$ and $0 < \omega \le 1.6 \cdot 10^{-7} N'^{-13}$, with C_{T} of Lem. 17.5, enlarged to be at least $1.1 \cdot 10^9$. For every site ξ, every w' with $(N'^5/2\beta')^{1/2} \le w' \le \frac{1}{20}$ and every closed cube Q of the half side w' in $\mathbb{R}^5$,

$$\mu_{N',\beta'}\big(\mathsf{a}(U,\xi) \in \mathsf{Q},\ \max_p |\partial U(p) - \mathbf{1}| < \omega\big) \le C_{\mathrm{a}}\,N'^{29.5}\,\beta'^{1/2}\,w'^3. \qquad \blacksquare$$

Here C_{a} is an absolute constant. Thus the bound $C_{\mathrm{a}}N'^{c_{\mathrm{a}}}\beta'^{1/2}w'^3$ of eq. 17.iii holds with $c_{\mathrm{a}} = 29.5$, $c'_{\mathrm{a}} = C_{\mathrm{T}}$ and $c''_{\mathrm{a}} = 68$.

Sketch of the proof of Lem. 17.26. Write $\mathsf{V} = N'^3$, take $\xi = 0$, let $|\cdot|_2$ be the Euclidean norm of a family indexed by bonds or by plaquettes, and $Z[F] = \int F\,e^{-\beta'\sum_p a_p}\,\mathrm{d}U$. (i) *The gauge of a maximal tree.* Let T be the tree of the bonds $\langle x, x+e_3\rangle$ with $x_3 \le N'-2$, $\langle x, x+e_2\rangle$ with $x_3 = 0$, $x_2 \le N'-2$, and $\langle x, x+e_1\rangle$ with $x_2 = x_3 = 0$, $x_1 \le N'-2$. A bounded Borel function that the gauge transformations do not change has the same integral against dU as against the Haar measures of the bonds off T at $U = \mathbf{1}$ on T; there the winding holonomies at 0 are the values P_μ of U at the three bonds b_μ^* from $(N'-1)e_\mu$ to 0.

On the set S_0 of the other $2\mathsf{V}-2$ bonds write $U(b) = V^P(b)\,e^{a(b)}$, $a(b) \in \mathfrak{su}(2)$, $|a(b)| < \pi$, where $V^P(b) = P_\mu$ if b is a bond of the direction μ that starts in the plane $x_\mu = N'-1$, and $V^P(b) = \mathbf{1}$ else; in $a(b)$ the Haar measure has a density at most $(2\pi^2)^{-1}$. (ii) *Small fields.* Along the tree every bond $b \in S_0$ is rebuilt from at most $2N'^2$ plaquettes, none of them a corner plaquette $p_{\mu\nu}(x)$ with $x_\mu = x_\nu = N'-1$: $|U(b) - V^P(b)| \le 2N'^2|\partial U - \mathbf{1}|_\infty$. The same rebuilding, linearised, gives $|a|_2 \le 2N'^{5/2}|d_P a|_2$ for every triple P and every family a on S_0, d_P being the linearisation at V^P of $a \mapsto (\partial U(p))_p$, a covariant lattice curl.

So on $\{|\partial U - \mathbf{1}|_\infty < \omega\}$ every $|a(b)| \le \pi N'^2\omega$, and at the corner plaquettes, where $\partial U(p)$ is the commutator $C_{\mu\nu} = P_\mu P_\nu P_\mu^{-1}P_\nu^{-1}$ up to four factors $e^{\pm a(b)}$, $|C_{\mu\nu} - \mathbf{1}| \le 9N'^2\omega$. Stokes' formula over the N'^2 plaquettes $p_{\mu\nu}(x)$ with $x_\rho = 0$, ρ the third direction, gives $2|\mathsf{v}(P_\mu) \times \mathsf{v}(P_\nu)| = |C_{\mu\nu} - \mathbf{1}| \le \sum_p |\partial U(p) - \mathbf{1}|$, and Schwarz's inequality then $\sum_p a_p \ge 2N'^{-2}\sum_{\mu<\nu}|\mathsf{v}(P_\mu) \times \mathsf{v}(P_\nu)|^2$. (iii) *The ceiling at a fixed triple.* Let κ be the family of the logarithms of the $C_{\mu\nu}$, placed at the corner plaquettes of the planes $\mu\nu$.

With $|e^{X_1}\cdots e^{X_k}-e^{X_1+\cdots+X_k}|\le\frac12(\sum_i|X_i|)^2$ for $X_i\in\mathfrak{su}(2)$, (ii) gives on $\{|\partial U-\mathbf{1}|_\infty<\omega\}$ that $2\sum_p a_p\ge(1-\eta_1)^2|d_Pa+\kappa|_2^2$, $\eta_1=39\pi N'^{9/2}\omega\le 2\cdot10^{-5}N'^{-8.5}$; keeping the fraction N'^{-3} of the action for the holonomies,

$$\sum_p a_p\ge\tfrac12(1-N'^{-3})(1-\eta_1)^2|d_Pa+\kappa|_2^2+\frac{2}{N'^5}\sum_{\mu<\nu}|\mathsf v(P_\mu)\times\mathsf v(P_\nu)|^2.$$

At a fixed triple the Gaussian integral over the $6\mathsf V-6$ real coordinates of a is at most $(2\pi/\beta'')^{3\mathsf V-3}D(P)^{-1/2}$, with $\beta''=\beta'(1-N'^{-3})(1-\eta_1)^2$ and $D(P)$ the determinant of $d_P^*d_P$ on the families on S_0, the shift by κ only lowering it; and $(\beta'/\beta'')^{3\mathsf V-3}\le(1-N'^{-3})^{-3N'^3}(1-\eta_1)^{-6N'^3}\le e^{3.2}$. (iv) *Determinants.* A triple whose three commutators lie within ϵ of $\mathbf1$ lies within $(2\epsilon)^{1/2}$ of a triple P' with one axis, $P'_\mu=\cos\alpha_\mu\,\mathbf1+i\sin\alpha_\mu\sum_j\hat n_j\tau_j$, and $D(P)\ge(1-48N'^{5/2}\delta)^{12\mathsf V}D(P')$ for $\max_\mu|P_\mu-P'_\mu|\le\delta$; at $\epsilon=9N'^2\omega$, $\omega\le1.6\cdot10^{-7}N'^{-13}$, this gives $D(P)^{-1/2}\le e^{0.5}D(P')^{-1/2}$.

At P' the families split into the colour along $\hat n$, on which $d_{P'}$ is the lattice curl, and the complex colour across $\hat n$, on which it is the curl twisted by the phases $e^{-2i\alpha_\mu}$ at the bonds where $V^{P'}=P'_\mu$: $D(P')=D(0)\,D(2\alpha)^2$, where $D(\theta)$, for $\theta\in\mathbb R^3$, is the determinant of the curl twisted by the phases $e^{i\theta_\mu}$ on the families on S_0, an even function, 2π-periodic in each θ_μ. For every momentum $k\in(\frac{2\pi}{N'}\mathbb Z/2\pi\mathbb Z)^3$ the twisted curl has the eigenvalue $\sum_\mu(2-2\cos(k_\mu+\theta_\mu/N'))$ twice, besides the gradients; let $\mathfrak d(\theta)$ be the product of these numbers over all k, and $\mathfrak d'$ their product over $k\ne0$ at $\theta=0$.

Then $D(0)=\mathfrak d'$, the change of coordinates along the tree being unimodular, and $D(\theta)\ge\mathfrak d'/(1.5\cdot10^{10}N'^{24})$ for every θ: by 2π-periodicity first take $\theta\in[-\pi,\pi]^3$; for $|\theta|_\infty\le\frac{1}{64}N'^{-11/2}$ by a comparison of the forms that uses the bound $|a|_2\le2N'^{5/2}|d_Pa|_2$ of (ii), and else by Cauchy's interlacing from the determinant $|1-e^{i\theta_\mu}|^2\mathfrak d(\theta)$ on the families that vanish on T and at b_μ^* alone, $|\theta_\mu|=|\theta|_\infty$, together with $\mathfrak d(\theta)\ge4\sin^2\frac{\theta_\mu}{2}\,N'^{-2}\mathfrak d'$, which the product formula $\prod_{j=0}^{n-1}(2\cosh m-2\cos\frac{2\pi j+\vartheta}{n})=2\cosh nm-2\cos\vartheta$ gives. This covers the triples near $P_\mu=\pm\mathbf1$ as well.

Together, for every Borel set $\mathcal P$ of triples that the simultaneous conjugations carry onto itself,

$$Z\big[\mathbf1_{\mathcal P}(P)\,\mathbf1\{|\partial U-\mathbf1|_\infty<\omega\}\big]\le10^{12}N'^{24}\,\mathsf X_0\int_{\mathcal P}\exp\Big[-\frac{2\beta'}{N'^5}\sum_{\mu<\nu}|\mathsf v(P_\mu)\times\mathsf v(P_\nu)|^2\Big]dP,\qquad \mathsf X_0=(2\pi^2)^{-(2\mathsf V-2)}\Big(\frac{2\pi}{\beta'}\Big)^{3\mathsf V-3}\mathfrak d'^{-3/2},$$

dP is Haar measure on G^3. (v) *The normalizing lower bound.* Lem. 17.5 gives $Z\ge c_{\mathrm T}N'^2\mathsf X_0/\beta'^2$ throughout the stated range, by the quantitative periodic Laplace estimate.

(vi) *The toron integral.* Let $B=2\beta'/N'^5$, so that $B^{-1/2}\le w'\le\frac{1}{20}$. The triples with $\mathsf a(P_1,P_2)\in\mathsf Q$ carry at most $261\,w'^3B^{-3/2}$ of the integral of $\exp[-B\sum_{\mu<\nu}|\mathsf v(P_\mu)\times\mathsf v(P_\nu)|^2]$. Integrate P_3 first: with the singular values $s_1\ge s_2$ of the matrix $\mathsf v$ of the pair (P_1,P_2), in the eigenframe of $\mathsf v\mathsf v^{\mathrm t}$, $\sum_{i=1,2}|\mathsf v(P_i)\times\mathsf v(P_3)|^2\ge s_2^2x_1^2+s_1^2(x_2^2+x_3^2)$, $x=\mathsf v(P_3)$, so that this integral is at most $\min\{1,(Bs_1^2)^{-1}\}\min\{1,3.81(Bs_2^2)^{-1/2}\}$; and $|\mathsf v(P_1)\times\mathsf v(P_2)|=s_1s_2$.

If $|\mathsf v(P_i)|<0.9$ for $i=1,2$ on the whole set, the Haar measure of the pair has in the coordinates $\mathsf s$ a density at most $\frac{2}{0.19\pi^2}\,4s_1s_2(s_1+s_2)$, the product of $4s_1s_2(s_1+s_2)$ with the bound of the P_3-integral is at most $30.5\,B^{-3/2}$, and the projection of $\mathsf Q$ onto the three coordinates $(\mathsf s_{11},\mathsf s_{12},\mathsf s_{22})$ has volume $8w'^3$. Else $\sin\alpha(P_i)\ge0.8$ on the set for one i, and the two angles alone give $12.6\,w'^2B^{-2}\le12.6\,w'^3B^{-3/2}$: in general the triples whose two angles lie in a set $\mathcal A$ carry at most $2.52\,B^{-2}\int_{\mathcal A}\min\{B^{1/2}\mathsf m,\mathsf m^{-1}\}\,\mathrm d\alpha_1d\alpha_2$, $\mathsf m=\max_i\sin\alpha(P_i)$. (vii) By (iv) and Lem. 17.5,

$$\mu_{N',\beta'}(P\in\mathcal P,\ |\partial U-\mathbf1|_\infty<\omega)\le\frac{10^{12}}{c_{\mathrm T}}N'^{22}\beta'^2\int_{\mathcal P}e^{-B\sum_{\mu<\nu}|\mathsf v(P_\mu)\times\mathsf v(P_\nu)|^2}\,\mathrm dP.$$

Using (vi) proves the lemma with any absolute $C_{\mathrm a}\ge261\cdot10^{12}/(2^{3/2}c_{\mathrm T})$, enlarged to be at least 1.

From one cutoff to the next. In the setting (E) of §2.3.1, with the side $\ell = 2\mathsf{L}^m$ and the label b, take at every integer $K \geq 2$ the numbers of §2.3.1 at $n = 1$: the depths $u = \lceil K/1260 \rceil$ and $s = \lceil K/9 \rceil$, $t_1 = K - u$, and, with $N_{[d]} = \ell^3 \mathsf{L}^{3d}$ and $\mathsf{y}_{[d]} = (\log(\mathsf{L}^d \mathsf{b}))^{2\mathfrak{p}_0}$, the four errors

$$\epsilon_{\mathrm{fg}} = C_{\mathrm{old}} C_{\mathrm{fg}} \tfrac{\mathsf{L}}{\mathsf{L}-1} E_0 \, N_{[u]} \mathsf{L}^{-(s-u)},$$
$$\epsilon_{\mathrm{cl}} = 4\mathsf{K}_{\mathrm{w}} c_\rho^2 A_0^2 (1 + \tfrac{2\varrho}{\mathsf{b}}) \mathsf{y}_{[K]} N_{[s]} K^{1/2} \mathsf{L}^{-(K-s)/2},$$
$$\epsilon_{\mathrm{sh}} = 3(s-u) N_{[s]} \exp\bigl[C_{\mathrm{sh}} - \tfrac{1}{4}\kappa_{\mathrm{b}} A_1^2 \mathsf{y}_{[u]}\bigr],$$
$$\mathsf{R}_{t_1} = 2M^{-3} N_{[u]} \exp\bigl[-c' A_0^2 \mathsf{y}_{[u]} + (\mathsf{C}^{\Sigma}_{\mathrm{maj}} + 2 + c_{\mathrm{low}} + \tfrac{9}{2}\log \mathsf{b}) \ell^3\bigr],$$

with the constants named there. The first two fall like powers of L^{-K}, up to factors that are powers of K, ℓ and $\log \mathsf{b}$; the last two are exponentially small in $(\log(\mathsf{L}^u \mathsf{b}))^{2\mathfrak{p}_0}$. Cor. 2.8(a) bounds the difference of the means of a gauge-invariant function at two consecutive cutoffs by $\mathfrak{o}_K = 8\mathsf{R}_{t_1} + 12\epsilon_{\mathrm{fg}} + 8\epsilon_{\mathrm{cl}} + 12\epsilon_{\mathrm{sh}}$ times its supremum; here its comparison is run with the errors that bound quotients of densities kept as factors, so that an event of a probability of the order of t keeps this order.

Lemma 17.27 (One step between consecutive cutoffs)**.** In the setting (E), let $K \geq 2$ be an integer at which the conditions (k0)–(k4) and (k6) of §2.3.1 hold at $n = 1$, and put

$$\epsilon_K^{\times} = 4\epsilon_{\mathrm{fg}} + 2\epsilon_{\mathrm{cl}} + 1.06\,\epsilon_{\mathrm{sh}} + 1.39\,\mathsf{R}_{t_1}, \qquad \epsilon_K^{+} = \mathsf{R}_{t_1} + 1.29\,\epsilon_{\mathrm{sh}}.$$

Then for every integer $0 \leq d \leq u$ and every Borel set E of block fields of the depth d that the gauge transformations carry onto itself (Def. 2.7(3)),

$$\mu_d^{K+1}(E) \leq e^{\epsilon_K^{\times}} \mu_d^K(E) + \epsilon_K^{+}, \qquad \mu_d^K(E) \leq e^{\epsilon_K^{\times}} \mu_d^{K+1}(E) + \epsilon_K^{+}.$$

■

Sketch of the proof of Lem. 17.27. It suffices to take $d = u$: for $d < u$ replace E by its preimage under the averaging maps from the depth u to the depth d, which are the same on the two lattices (Section YM3_I:116) and keep it Borel and gauge invariant (Prop. YM3_I:4.5, YM3_I:4.5). Take the lattices $\mathrm{A} = (\ell, K)$ and $\mathrm{B} = (\ell, K+1)$ with the top levels t_1 and $t_1 + 1$, the objects of Section YM3_I:116 with $n = 1$ and $f_1 = 1$, at the source 0, and $\mathsf{R} = \mathsf{R}_{t_1}$, which is the same number on both (§2.3.1). For a bounded Borel function $F \geq 0$ of the block fields of the depth u that the gauge transformations do not change, and for each lattice $\bullet$, of the cutoff $K^\bullet$ and the top level $t^\bullet$, let $\langle F \rangle_0^\bullet$ be the mean of $F(\mathcal{M}^{K^\bullet - u} U)$ at the cutoff $K^\bullet$, $\langle F \rangle_1^\bullet$ the normalised integral of F against the measure of the chain on the inner ball $\mathsf{In}_{[u]}$, and $\langle F \rangle_2^\bullet$ that against the classical start on it. (1) *The discarded mass, a remainder.* The measure of the chain is the image of $\mathfrak{w}_{t^\bullet}^{\mathrm{sf}} \rho_0 \, \mathrm{d}U$ under the averaging, with $\mathfrak{w}_{t^\bullet}^{\mathrm{sf}} \geq 0$ (Prop. YM3_I:115.2, image identity); the right inequality of Prop. YM3_I:115.2, domination bound, for F and for $\sup F - F$ composed with the averaging, which are gauge invariant (Prop. YM3_I:4.5), and Prop. YM3_I:115.2, lost-mass bound, by which the discarded mass is at most R times the partition function, give $\langle F \rangle_0^\bullet \leq \langle F \rangle_1^\bullet + \mathsf{R} \sup F$ and $\langle F \rangle_1^\bullet \leq (1 - \mathsf{R})^{-1} \langle F \rangle_0^\bullet$. (2) *Forgetting, a factor.* On $\mathsf{In}_{[u]}$ the two measures have the densities $e^{c^\bullet(0)} e^{\Phi^{\bullet,0}}$ and $e^{\hat{c}^\bullet(0)} e^{\hat{\Phi}^{\bullet,0}}$ (the proof of Prop. YM3_I:116.1), both positive almost everywhere, as both measures are positive (Prop. YM3_I:115.2, image identity, Prop. YM3_I:116.1); and the real parts of the two exponents differ by at most ϵ_{fg} at every point of $\mathsf{In}_{[u]}$ (equation YM3_I:(2180)).

So the quotient of the densities lies in $\varkappa^\bullet [e^{-\epsilon_{\mathrm{fg}}}, e^{\epsilon_{\mathrm{fg}}}]$ with one number $\varkappa^\bullet > 0$, which cancels between the numerator and the normaliser: $e^{-2\epsilon_{\mathrm{fg}}} \langle F \rangle_2^\bullet \leq \langle F \rangle_1^\bullet \leq e^{2\epsilon_{\mathrm{fg}}} \langle F \rangle_2^\bullet$. (3) *The two classical starts, a factor and a remainder.* Both are integrals of $F(\mathcal{M}^{s-u} W)$ over the block fields W of the depth s against $\mathsf{k}^\bullet(W) \hat{F}_0^\bullet(W) \, \mathrm{d}W$, $\mathsf{k}^\bullet \geq 0$ (§2.3.2); where k^{A} and k^{B} are both positive they are equal and $\hat{F}_0^{\mathrm{B}} = \hat{F}_0^{\mathrm{A}} e^{\mathsf{q}_0}$, $|\mathsf{q}_0| \leq \epsilon_{\mathrm{cl}}$ (Prop. YM3_I:116.1, density comparison in the proof); and for $(\bullet, \circ) = (\mathrm{A}, \mathrm{B})$ and (B, A) the fields at which k° vanishes carry at most the fraction ϵ_{sh} of the mass of the start of $\bullet$ (§2.3.2).

So $\langle F\rangle_2^\bullet \le e^{2\epsilon_{\rm cl}}(1-\epsilon_{\rm sh})^{-1}\langle F\rangle_2^\circ + \epsilon_{\rm sh}\sup F$. Chaining (1), (2), (3), (2), (1),

$$\langle F\rangle_0^\bullet \le \frac{e^{4\epsilon_{\rm fg}+2\epsilon_{\rm cl}}}{(1-\epsilon_{\rm sh})(1-\mathsf{R})}\langle F\rangle_0^\circ + \big(\mathsf{R}+e^{2\epsilon_{\rm fg}}\epsilon_{\rm sh}\big)\sup F,$$

and (k2), (k3), (k6) give $e^{2\epsilon_{\rm fg}} \le e^{1/4} < 1.29$, $-\log(1-\epsilon_{\rm sh}) \le 1.06\,\epsilon_{\rm sh}$ and $-\log(1-\mathsf{R}) \le 2\log 2\cdot\mathsf{R} \le 1.39\,\mathsf{R}$. At $F=\mathbf{1}_E$ this is the lemma.

The chain from a low cutoff. Let $p = 2\mathfrak{p}_0$, $y = \log \mathsf{b}$, $\bar\ell(y) = 2\mathsf{L}^2 M y^r$, $\mathsf{k}(y) = \bar\ell(y)^3 y^{p+1}$, $\mathsf{N}(y) = \mathsf{L}\,\bar\ell(y)^{31} y^{10(p+1)}$, and let $K_1 = K_1(y)$ be the least integer $K \ge 10$ with $\mathsf{L}^{K/10} \ge \mathsf{k}(y)$. With the constants of §2.3.1 and the number C_+ of Step 4 put

$$a_1 = C_{\rm old}C_{\rm fg}\tfrac{\mathsf{L}}{\mathsf{L}-1}E_0\mathsf{L}^4,$$
$$a_2 = 4\mathsf{K}_{\rm w}c_\rho^2 A_0^2(1+2\varrho)\mathsf{L}^{7/2},$$
$$a_3 = \sum_{K\ge1}(1+K\log\mathsf{L})^p K^{1/2}\mathsf{L}^{-K/90},$$
$$a_4 = 529200\,e^{C_{\rm sh}}\sum_{i\ge1} i\,\mathsf{L}^{420i}e^{-\frac14\kappa_{\rm b}A_1^2(i\log\mathsf{L})^p},$$
$$a_5 = 2520\,M^{-3}\sum_{i\ge1}\mathsf{L}^{3i}e^{-c'A_0^2(i\log\mathsf{L})^p},$$
$$a_6 = \frac{256\,\mathsf{K}_u c_\rho^2 A_0^2\mathsf{L}^4}{\gamma_{\rm r}A_1^2},$$
$$\mathsf{R}(y) = 1.29\,\bar\ell(y)^3\Big[a_4 e^{-\frac14\kappa_{\rm b}A_1^2 y^p} + a_5 e^{(\mathsf{C}_+ + \frac92 y)\bar\ell(y)^3 - c'A_0^2 y^p}\Big];$$

these are formed from the constants alone, and the three series converge, as $p\ge2$, $\kappa_{\rm b}, c' > 0$ (Prop. YM3_I:114.4 and §2.3.1) and $A_0, A_1 > 0$ (equation YM3_I:(634)). Suppose that (E) holds, $\ell \le \bar\ell(y)$, $\mathsf{b} \ge \Theta_\star$, the number of the condition (k1), $y \ge \max\{8a_1, 20a_2a_3, a_6\}$ and $\mathsf{R}(y) \le e^{-3y^2}$. Then (k0)–(k4) and (k6) hold at every integer $K \ge K_1$, and the sums $\mathsf{S} = \sum_{K\ge K_1}\epsilon_K^\times$ and $\mathsf{R}_\Sigma = \sum_{K\ge K_1}\epsilon_K^+$ satisfy

$$\mathsf{S} \le \frac{17.72\,a_1 + 2a_2a_3}{y} + 1.39\,\mathsf{R}(y), \qquad \mathsf{R}_\Sigma \le \mathsf{R}(y) \le e^{-3y^2}, \qquad \ell\mathsf{L}^{K_1} < \mathsf{N}(y).$$

Indeed, for $K \ge K_1$ one has $u < \frac{K}{1260}+1$ and $\frac{K}{9} \le s < \frac{K}{9}+1$, so $4u - s \le 4 - \frac{K}{10}$, $\frac{7s-K}{2} \le \frac72 - \frac{K}{10} - \frac{K}{90}$ and $4s - K \le 4 - \frac{K}{10}$; and $\ell^3\mathsf{L}^{-K/10} \le \ell^3/\mathsf{k}(y) \le y^{-p-1}$. Hence $\epsilon_{\rm fg} \le a_1\ell^3\mathsf{L}^{-K/10}$, $\epsilon_{\rm cl} \le a_2 y^p(1+K\log\mathsf{L})^p K^{1/2}\mathsf{L}^{-K/90}\ell^3\mathsf{L}^{-K/10}$, and the left member of (k4) is at most $a_6\gamma_{\rm r}A_1^2\ell^3\mathsf{L}^{-K/10}$. Summed over $K \ge K_1$, as $(1-13^{-1/10})^{-1} \le 4.43$, these give $\sum_{K\ge K_1}(4\epsilon_{\rm fg}+2\epsilon_{\rm cl}) \le (17.72\,a_1 + 2a_2a_3)/y$. (k0) holds as $K \ge 10$, (k1) as $\mathsf{b} \ge \Theta_\star$, and $y \ge \max\{8a_1, 20a_2a_3, a_6\}$ gives (k2), (k4) and the first half of (k3).

For the two other errors, $(y+u\log\mathsf{L})^p \ge y^p + (u\log\mathsf{L})^p$, $s-u \le 140u$, $N_{[s]} \le \ell^3\mathsf{L}^{420u}$, and at most 1260 cutoffs have the same u: $\sum_{K\ge K_1}\epsilon_{\rm sh} \le a_4\ell^3 e^{-\frac14\kappa_{\rm b}A_1^2y^p}$ and $\sum_{K\ge K_1}\mathsf{R}_{t_1} \le a_5\ell^3 e^{(\mathsf{C}_+ + \frac92 y)\ell^3 - c'A_0^2y^p}$. So $\mathsf{R}_\Sigma \le \mathsf{R}(y)$, which gives the second half of (k3) and (k6), and $\sum_{K\ge K_1}(1.06\,\epsilon_{\rm sh} + 1.39\,\mathsf{R}_{t_1}) \le 1.39\,\mathsf{R}(y)$. Finally $\mathsf{k}(y) > \mathsf{L}$, so $K_1 \ge 11$ and $\mathsf{L}^{(K_1-1)/10} < \mathsf{k}(y)$: $\mathsf{L}^{K_1} < \mathsf{L}\,\mathsf{k}(y)^{10}$ and $\ell\mathsf{L}^{K_1} < \mathsf{N}(y)$. Lem. 17.27 at $d=0$ and an induction on K give $\mu_0^K(E) \le e^{\mathsf{S}_K}(\mu_0^{K_1}(E) + \mathsf{R}_K)$, with S_K and R_K the sums of the $\epsilon_{K'}^\times$ and of the $\epsilon_{K'}^+$ over $K_1 \le K' < K$; so eq. 17.iv holds: for every integer $K \ge K_1$ and every Borel set E of block fields of the depth 0 that the gauge transformations carry onto itself

$$\mu_0^K(E) \le e^{\mathsf{S}}(\mu_0^{K_1}(E) + \mathsf{R}_\Sigma).$$

The parameters. Let $\mathsf{y}_\star$ be the least real number

$$y_0 \ge \max\{122, \log\gamma^{-2}, \log\Theta_\star, 8a_1, 20a_2a_3, a_6, 2(17.72\,a_1 + 2a_2a_3)\}$$

such that at every $y \ge y_0$

$$\text{(y1)}\ \ \mathsf{R}(y) \le e^{-3y^2};$$
$$\text{(y2)}\ \ \log\mathsf{L} + 30\log\bar\ell(y) + 10(p+1)\log y \le y;$$
$$\text{(y3)}\ \ 26\,\mathsf{N}(y)^{28}\,\bar\ell(y)\,y^4e^{-y} \le 2.5\cdot10^{-14};$$
$$\text{(y4)}\ \ 26\,\mathsf{N}(y)^{6}\,\bar\ell(y)\,y^2e^{-y} \le (c_2^{\rm SO})^2;$$
$$\text{(y5)}\ \ 31\,(y+1)^2\,\mathsf{N}(y)^6 \le e^{2y^2};$$
$$\text{(y6)}\ \ C_{\rm T}\bar\ell(y)\mathsf{N}(y)^{67}e^{-y} \le 1,$$

$c_2^{\rm SO}$ the number of Lem. 5.28. It exists: $\bar\ell$ and N are powers of y, so the left member of (y2) is $O(\log y)$, those of (y3) and (y4) tend to 0, that of (y5) is a power of y, that of (y6) tends to zero, and the two exponents in $\mathsf{R}(y)e^{3y^2}$ are $-\frac14\kappa_{\rm b}A_1^2y^p + 3y^2 + O(\log y)$ and

$$-c'A_0^2y^p + (\mathsf{C}_+ + \frac92 y)\bar\ell(y)^3 + 3y^2 + O(\log y),$$

which tend to $-\infty$, as $p = 2\mathfrak{p}_0 > 3r+2$ at the chosen constants (the choice $\mathfrak{p}_0 \ge 6r+1$ in Prop. YM3_I:69.1) and $r \ge 2$; each condition compares continuous functions, so the numbers y_0 at which all hold on $[y_0,\infty)$ form a closed half line. Put $t_\star = 2\mathsf{L}e^{-\mathsf{y}_\star}$. Now let $j \ge 1$ and $\beta > 0$ with $t = 2\mathsf{L}^j/\beta \le t_\star$, and $N = 2\mathsf{L}^j$; let m be the least integer $m \ge 1$ with $M\,R(2\mathsf{L}^m/t) \le \mathsf{L}^m$, $\ell = 2\mathsf{L}^m$, $\mathsf{b} = \ell/t$, $y = \log\mathsf{b}$ and $K = j - m \in \mathbb{Z}$, so that $N = \ell\mathsf{L}^K$ and $\beta = \mathsf{L}^K\mathsf{b}$; and $q = \lceil y\rceil$.

Then $y \ge \log(2\mathsf{L}/t) \ge \mathsf{y}_\star$; as in Step 4, (E) holds for (b, m) and $\ell \le \bar\ell(y)$; by (y2), $\log\ell \le y/30$, so that $\log\frac1t \le y \le 2\log\frac1t$, and $y \le q < y+1$, $q \ge 122$; $e^{-y} \le t = \ell e^{-y} \le \bar\ell(y)e^{-y}$; the chain holds from $K_1 = K_1(y)$, with $e^{\mathsf{S}} \le e^{0.53} < 1.7$ and $\mathsf{R}_\Sigma \le e^{-3y^2}$; and $\beta_1 = \mathsf{L}^{K_1}\mathsf{b} = \ell\mathsf{L}^{K_1}/t$ has $\log\beta_1 = y + K_1\log\mathsf{L} < y + \log\mathsf{L} + 10\log\mathsf{k}(y) \le 2y$, by (y2).

Bare cubes. Let (N',β') be $(\ell\mathsf{L}^{K_1},\beta_1)$, the bare cube at the cutoff K_1, or (N,β) if $N < \ell\mathsf{L}^{K_1}$. Then N' is of the form $2\mathsf{L}^{j'}$, $j' \ge 1$, so even, $26 \le N' < \mathsf{N}(y)$, $\beta' = N'/t$, and $\log\beta' \le 2y$. Put $\Lambda' = 3N'^3\log(8e^8\beta'^{3/2}) + \log(3N'^3) + 3y^2$, $\omega_0 = (2\Lambda'/\beta')^{1/2}$ and $\upsilon = N'^2\omega_0$. The Haar measure of $\{g \in G : |g-\mathbf{1}| \le \beta'^{-1/2}\}$ is at least $\frac18\beta'^{-3/2}$, and $\beta'\sum_p a_p \le 24N'^3$ when all bonds lie in this set; so $Z(N',N',N') \ge (8e^8\beta'^{3/2})^{-3N'^3}$, every plaquette has $|\partial U(p) - \mathbf{1}| \ge \omega_0$ with a probability at most $Z^{-1}e^{-\beta'\omega_0^2/2}$, and

$$\mu_{N',\beta'}(|\partial U - \mathbf{1}|_\infty \ge \omega_0) \le 3N'^3\,(8e^8\beta'^{3/2})^{3N'^3}e^{-\Lambda'} = e^{-3y^2}.$$

Further $N'^3 \le \Lambda' \le 13N'^3y^2$, so $\upsilon^2 \le 26N'^6y^2t \le 26\,\mathsf{N}(y)^6\bar\ell(y)y^2e^{-y}$; and (y3), (y4) give $\omega_0 \le 1.6\cdot10^{-7}N'^{-13}$, $q\upsilon \le 1.7\cdot10^{-7}$, $\beta' \ge N'e^y/\bar\ell(y) \ge C_{\rm T}N'^{68}$ by (y6) and $\upsilon \le c_2^{\rm SO}$. With $w = 2.03\,q\upsilon$ and $w' = \frac32 w + 1.03\,\upsilon$ one has $(N'^5/2\beta')^{1/2} \le 3\upsilon \le w' \le 4q\upsilon \le \frac{1}{20}$, and Lem. 17.26 at $\omega = \omega_0$ gives, for every site ξ' and every closed cube Q of the half side w',

$$\mu_{N',\beta'}(\mathsf{a}(U,\xi') \in \mathsf{Q},\ |\partial U - \mathbf{1}|_\infty < \omega_0) \le C_{\rm a}N'^{29.5}\beta'^{1/2}(4q\upsilon)^3 \le P^{\rm b} := C_{\rm b}\,q^3y^3\,\mathsf{N}(y)^{39}\,t,$$

where $C_{\rm b} = 8500C_{\rm a}$, since $(2\Lambda')^{3/2} \le 132.6\,N'^{4.5}y^3$ and $N'^{40}/\beta' = N'^{39}t$.

The cutoffs $K \ge K_1$. Take the swap with the block field $V = \mathcal{M}^K U$ of the depth 0, $\mathsf{n} = \ell$, and $\omega = \theta_{\mathsf{L}}\mathsf{L}^{2K_1}\omega_0$, with ω_0 and υ those of $(\ell\mathsf{L}^{K_1},\beta_1)$ and the number $\theta_{\mathsf{L}} = \exp\frac{1}{2(\mathsf{L}^2-1)} < 1.003$ of Lem. 5.28: $\mathsf{n}^2\omega = \theta_{\mathsf{L}}\upsilon$, so $\delta_q \le e^{q\theta_{\mathsf{L}}\upsilon} - 1 \le 1.015\,q\upsilon$, $2\delta_q \le w$ and $\omega \le 1$. The sets $E_p = \{|\partial V(p) - \mathbf{1}| \ge \omega\}$ and $\{\mathsf{a}(V,\xi') \in c_k^+\}$ are Borel sets of block fields of the depth 0 that the gauge transformations carry onto themselves, as these conjugate $\partial V(p)$, and the two winding holonomies at ξ' by one element; so the chain gives $\mu_0^K(E) \le 1.7\,(\mu_0^{K_1}(E) + e^{-3y^2})$ for each of them.

At the cutoff K_1, where $|\partial U-\mathbf{1}|_\infty<\omega_0$ on the bare cube: $\mathsf{L}^{2K_1}\omega_0\le \upsilon\le c_2^{\rm SO}$, so $|\partial V-\mathbf{1}|_\infty<\omega$, and the winding holonomies of $V=\mathcal{M}^{K_1}U$ differ from the bare ones along the same lines by at most $13.3\,\ell\mathsf{L}^{2K_1}\omega_0=13.3\,\upsilon/\ell\le 0.512\,\upsilon$ (Lem. 5.29), so that $|\mathsf{a}(V,\xi')-\mathsf{a}(U,\xi')|_\infty\le 1.03\,\upsilon$; and c_k^+ lies in a closed cube of the half side $\frac32 w$. With the bounds on the bare cube $(\ell\mathsf{L}^{K_1},\beta_1)$ this gives, for the swap, $\rho\le 3.4\,e^{-3y^2}$ and

$$P_*\le 1.7(P^{\rm b}+2e^{-3y^2})\le \Pi(y)\,t,\qquad \Pi(y)=C_\Pi y^6\,\mathsf{N}(y)^{39},\qquad C_\Pi=2C_{\rm b}+4,$$

as $q^3\le 1.025\,y^3$ and $e^{-3y^2}\le e^{-y}\le t$; and $9q^2\mathsf{n}^6\rho\le 30.6\,(y+1)^2\bar\ell(y)^6e^{-3y^2}\le e^{-y^2}$, by (y5).

The cutoffs $K<K_1$, among them those with $K\le 0$. Take the swap with the bare field on the cube of $N<\ell\mathsf{L}^{K_1}$ sites a side, $\mathsf{n}=N$, and $\omega=\omega_0$ of (N,β): $\mathsf{n}^2\omega=\upsilon$, $2\delta_q\le w$, $\rho\le e^{-3y^2}$, $P_*\le P^{\rm b}+e^{-3y^2}$, and $9q^2N^6\rho\le e^{-y^2}$; the same bounds hold.

The end. As $\Pi(y)\ge 1$, $t\ge e^{-y}$ and $q-1\le y$, $(\Pi(y)t)^{q-1}\ge e^{-y^2}$; so in all cases $e^{-\mathsf{d}_q(N)}\le 486\,P_*^{q-1}+9q^2\mathsf{n}^6\rho\le 487\,(\Pi(y)t)^{q-1}$. Where $\Pi(y)t<1$, as $q\ge y\ge\log\frac1t$ and $q\ge 122$,

$$\mathsf{G}_N\ge\frac{\mathsf{d}_q(N)}{q}\ge\frac{q-1}{q}\Big(\log\frac1t-\log\Pi(y)\Big)-\frac{\log 487}{q}\ge\log\frac1t-\log\Pi(y)-1.06,$$

that is $\lambda_0(N,N)^N/Z(N,N,N)=e^{-\mathsf{G}_N}\le 2.89\,\Pi(y)\,t$. As $\mathsf{N}(y)^{39}=\mathsf{L}^{39}(2\mathsf{L}^2M)^{1209}y^{1209r+390p+390}$ and $y\le 2\log\frac1t$, this is the bound of (d), with

$$\mathsf{p}_\star=1209\,r+780\,\mathfrak{p}_0+396,\qquad \mathsf{C}_\star=3C_\Pi\,\mathsf{L}^{39}(2\mathsf{L}^2M)^{1209}\,2^{\mathsf{p}_\star},$$

formed from L, M, r and $\mathfrak{p}_0$, and with $t_\star$, formed from the constants. Where $\Pi(y)t\ge 1$ the bound holds as well, as $\mathsf{C}_\star(\log\frac1t)^{\mathsf{p}_\star}\ge 2.89\,\Pi(y)$ and $\lambda_0(N,N)^N\le Z(N,N,N)$. With $\mathsf{e}(\mathsf{L}^j)=\mathsf{e}(N/2)\ge\mathsf{G}_N$ this gives the floor of e. The last sentence of (d) is the first at $j=m_0+K$, as $N=\ell_0/a_K$ and $t=\ell_0/\mathsf{b}$. □

Proof of Prop. 17.7. (a) With the time μ_0 and the section (N_1,N_2), $Z(t,N_1,N_2)=(1+q^t)^{N_1N_2}\mathsf{c}^t=\sum_k\binom{N_1N_2}{k}(\mathsf{c}q^k)^t$, $\mathsf{c}=(1+q^{N_1})^{N_2}(1+q^{N_2})^{N_1}$: (T2), and (T3) with all the weight in the sector 00; (T1) and (iii) are read off. With $f(u)=\log(1+q^u)$: $F(l,l,l)=3l^2f(l)$, $F(2l,l,l)=l^2f(2l)+4l^2f(l)$, $F(2l,2l,l)=4l^2f(2l)+4l^2f(l)$, $F(2l,2l,2l)=12l^2f(2l)$; so $d_1=l^2\mathsf{h}$, $d_2=2l^2\mathsf{h}$, $\mathsf{e}=4l^2\mathsf{h}$ with $\mathsf{h}=2f(l)-f(2l)$, and $X=12l^2\mathsf{h}$. (b) Expanding the two products of the factors that contain t,

$$Z^{(k;m)}=\sum_{\mathsf{a},\mathsf{b}}\binom{N_2}{\mathsf{a}}\binom{N_1}{\mathsf{b}}(-1)^{k_1\mathsf{a}+k_2\mathsf{b}}\big(\mathsf{c}_mq^{N_1\mathsf{a}+N_2\mathsf{b}}\big)^t,\qquad \mathsf{c}_m=1+(-1)^mq^{N_1N_2}>0,$$

which is (T3) with the sector $e=(\mathsf{a},\mathsf{b})$ mod 2; (T1), (T2) and (iii) are read off. The formulas for e and X follow from $F(\underline n)=\sum_\rho n_\rho\log(1+q^{n_\mu n_\nu})$; then $X\ge 24l(s-\frac{s^2}{2})-6ls^4$ and $\mathsf{e}\le 8ls^2$, so $X/\mathsf{e}\ge\frac3s(1-\frac s2-\frac{s^3}{4})\ge 2.95/s$ for $s\le 0.03$. (c) On the cube, $Z^-/Z=(\frac{1-u}{1+u})^n$ with $u=q^{n^2}$, so $\frac12q^{n^2}\le z_-(n,n,n)\le nq^{n^2}$, while $x_0=\mathsf{e}(l)=8ls^2(1+O(s^2))$ gives $\theta=103.2\,ls^2(1+O(s^2))$ and the two expansions, with the exact constant $103.2e^{-0.835}$. If the inequality held with $c'>\frac12$, then $\log\frac1q\ge 2c'\log\frac1q-C'(l)$ for all small q, which is false. □

Proof of Prop. 17.8. Put $u_j=l_j\varepsilon_{\rm v}(l_j)$ and $\mathsf{t}_j=\theta^{2^{j+1}}$. By Prop. 17.17(a), $u_0\le 0.01643\,\theta^2$. Let $u_j\le 0.0165\cdot 2^{-j}\theta^{2\cdot4^j}$, and take $\hat N=2^{j+2}$ in Prop. 17.17(b). Then the powers of θ match: $y=\hat Nu_j\le 0.066\,\theta^{2\cdot4^j}$, $(0.001615\,\mathsf{t}_j)^{2\hat N}\le 0.001615^8\theta^{2\cdot4^j}$, $(0.001615\,\mathsf{t}_j^2)^{\hat N/2}\le 0.001615^2\theta^{2\cdot4^{j+1}}$, and $8\hat N\cdot0.01655\,\mathsf{t}_j=16\cdot0.01655\,\mathsf{m}\theta^{\mathsf{m}}\le 0.835$ with $\mathsf{m}=2^{j+1}$, as $\mathsf{m}\,0.8901^{\mathsf{m}}\le 3.153$. So $\bar z_1\le 0.06601\,\theta^{2\cdot4^j}$, $\bar z_2\le 4.38\cdot10^{-5}\theta^{2\cdot4^{j+1}}$, and $u_{j+1}\le 2^{-j-1}\cdot 9.04\cdot10^{-5}\,\theta^{2\cdot4^{j+1}}$, which closes the induction.

For the cubes let $n=2^il$, $i\ge1$, $j=\lfloor(i-1)/2\rfloor$ and $M=n/l_j$, an integer with $M^2\le n/l\le 2M^2$. The first inequality of Lem. 17.15(b) on (l_j,n,l_j), with $\mathsf{r}_n(l_j,l_j)\le(0.001615\,\mathsf{t}_j)^M\le 0.001615\,\theta_j^{l_j^2/2l^2}$, gives $z_-(l_j,n,l_j)\le 0.00906\,M\theta_j^{l_j^2/2l^2}$; and $\delta(l_j,n,l_j)\le M\delta_{\rm c}(l_j)\le 0.01655\,M\theta^{l_j/l}\le$

$0.01655\, y\theta^y \le 0.0523$ with $y = \sqrt{n/l}$, as $M \le y \le l_j/l$ and $y\theta^y \le 1/e\Lambda$. Lem. 17.15(a) with $\kappa = M^2$ gives $z_-(n,n,n) \le (0.0096\, M)^{M^2\theta^{n^2/2l^2}}$, and the first factor is ≤ 1 if $0.0096\, M \le 1$, else at most $(0.0096\sqrt{n/l})^{n/l}$. Lem. 17.12(d) on these cubes and Thm. AW(a) give the area law, as $\frac{RT}{nl}\log(0.0096\sqrt{n/l}) \to 0$; and $\log \frac{1}{0.8901}/2 \ge 0.0582$. □

17.1.7. *Scope of the conclusions.* Every statement that uses the criterion is an implication from it on one lattice. Prop. 17.2 gives the criterion in two ranges of the coupling, both below $\frac16$: at every integer side $l \ge 3$ for $0 < \beta \le x_0/48l^3$ (its clause (a)), and at the even sides $l \ge 4$ with $64\, l^2(6\beta)^{l/2} \le \frac{1}{20}$ for $0 < \beta < \frac16$ (its clause (b)); from a side at which it holds the criterion passes, at the same coupling, to every integer side $l' \ge 16l$, of either parity (Prop. 17.3(c)), and to the sides $2^k l$ (Thm. AX(b)). With $\beta = L^K\mathsf{b}$ and $l = l_*L^K$ the first range reads $\mathsf{b} \le x_0 L^{-4K}/48l_*^3$ and the second $\mathsf{b} < L^{-K}/6$, while $L^K\mathsf{b} \ge \frac16$ for all large K at every label $\mathsf{b} > 0$; so neither range, nor its passage to other sides at the same coupling, contains a fixed label as K grows.

The uniform criterion is the hypothesis of Thm. AY and of Cor. 17.1(e). The number σ_* is a floor valid at each cutoff. On the centre twists of the cubes of sides $n \ge 2\mathsf{q}l$ the floors are those of the pure numbers $-\log z_-(n,n,n) \ge \sigma^+ n^2 - \frac{2\mathsf{q}n}{l}(\frac12 \log\frac{2\mathsf{q}n}{l} + 0.86)$ and $-\log(1 - Z^z/Z)$, which is at least the same number less $\log 6$ (Thm. AW(d); in physical units Thm. AY(b)); the free energy $-\log(Z^z/Z)$ of a twist itself lies between 0 and $-\log(1 - 6z_-)$ where $6z_- < 1$, and the free energy ε_{v} of a vortex line per unit length is bounded from above (Props. 17.17, 17.8).

The energies of the flux sectors and of the flux lines, $\log(\lambda_0/\Lambda_e)$ and $\log(\lambda_0/\Lambda_-)$, have the floors of Thm. AW(e), (g) and Thm. AY(b), (d), and the gap has those of Thm. AW(f) and Thm. AY(c). For a path that lies in no coordinate plane, the area in Thm. AW(c) is that of a projection counted modulo 2, which may vanish although every surface bounded by the path is large: two parallel squares of side s at the distance s, joined by a segment run twice, form a path C with $\mathsf{A}_{\rho'\rho}(C) = 0$ for all $\rho' \ne \rho$, while every set of plaquettes whose boundary modulo 2 is the path has at least $2s^2$ members.

For two static charges at sites o, r of a section that lie on no common line of the lattice, Thm. AW(e) gives the floor $\sigma^+|r-o|_\infty$, with the maximum norm, which is at least $2^{-1/2}\sigma^+|r-o|$.

17.2. **From the box criterion to continuum confinement.** A box inequality at one fixed physical size, imposed at every sufficiently large cutoff along a fixed label, controls the infrared limit. Under this hypothesis the periodic Gibbs state is unique at each such cutoff, the continuum limit is independent of the order of the volume and cutoff limits, and its reconstructed vacuum is unique with a positive Hamiltonian gap.

The fixed-lattice transfer and tube arguments are given in §17.1. Here we prove the comparisons uniform in the cutoff, passage of the gap to the continuum, transport between labels and the single-cutoff certificates with their required thresholds and margins. The box inequality itself is an additional input; the elementary small-coupling examples do not verify it along a fixed positive label.

17.2.1. *Notation and hypotheses. One lattice.* Let $\beta \ge 0$. A *box* $\underline{n} = (n_1, n_2, n_3)$ is the periodic lattice with $n_\mu \ge 3$ sites along the side μ; $|\underline{n}| = n_1n_2n_3$, and N^3 is the cube. Every bond carries the Haar probability measure of SU(2), every plaquette p the variable $a_p = 1 - \frac12 \operatorname{tr}\partial U(p) \in [0,2]$, and the box the weight $\exp(-\beta\sum_p a_p)$; $Z(\underline{n})$ is its partition function, $\mu_{\underline{n}}$ its measure, $F = \log Z \le 0$, $P = \inf_{\underline{n}} F(\underline{n})/|\underline{n}| \in [-6\beta, 0]$, and

$$\delta(\underline{n}) = F(\underline{n}) - |\underline{n}|P \ge 0, \qquad D_q(N) = q^3F(N^3) - F((qN)^3), \qquad \mathsf{e}(l) = 2F(l,2l,2l) - F((2l)^3).$$

We call δ the *excess*. For a box and a direction i, $\mathsf{d}^{(i)}(\underline{n}) = 2F(\underline{n}) - F(\underline{n}')$, $\underline{n}'$ the box with the side i doubled. None of these numbers depends on a spacing. For a bounded function Ψ of the bonds of $a\mathbb{Z}^3$ and a box, $\langle\Psi\rangle_{\underline{n}}$ is the mean of Ψ under the measure of the box extended periodically to $a\mathbb{Z}^3$.

The transfer operator. Fix a coordinate direction μ_0 as the time, the two others $\nu_1 < \nu_2$, and a cross-section $\Sigma = (A, B)$, $A, B \ge 3$; the box with t layers over Σ is (t, Σ). By Section YM3_I:85 and Prop. YM3_I:85.1 the configurations of a layer form a compact space $\mathcal{X}$ with the product Haar measure dV, a layer carries the continuous action $A_\Sigma \ge 0$ of its plaquettes, and the crossing operator K, which integrates the time-like bonds between two layers, is positive, Hilbert–Schmidt, with a continuous kernel with values in $(0, 1]$. With $\mathsf{m} = e^{-\beta A_\Sigma}$ (the function, and the operator of multiplication) we put

$$\mathsf{D} = \mathsf{D}_\Sigma = \mathsf{m}^{1/2}\mathsf{K}\,\mathsf{m}^{1/2}, \qquad \mathsf{T}' = \mathsf{K}^{1/2}\mathsf{m}\,\mathsf{K}^{1/2}.$$

D is positive, Hilbert–Schmidt, with the continuous kernel $\mathsf{m}(V')^{1/2}\mathsf{K}(V', V)\mathsf{m}(V)^{1/2} \in (0, 1]$, and $Z(t, \Sigma) = \operatorname{Tr} \mathsf{D}^t$ for every integer $t \ge 3$ (Prop. 17.31). T', which stands with the slab operators of the trace formula (M1) below, is positive as well, as $\mathsf{K} \ge 0$ and $\mathsf{m} > 0$, and has the same nonzero eigenvalues $\lambda_0 \ge \lambda_1 \ge \cdots > 0$, with multiplicities; they depend neither on t nor on the spacing. We put $\mathsf{r}_t(\Sigma) = \sum_{i \ge 1} (\lambda_i/\lambda_0)^t$, $\mathsf{G}(t; \Sigma) = \log(1 + \mathsf{r}_t)$, $\mathsf{R}_t = 1 - e^{-\mathsf{G}}$, $\varepsilon(\Sigma) = \log \lambda_0 - |\Sigma| P$, and on a cube $\mathsf{r}_N = \mathsf{r}_N(N, N)$, $\mathsf{G}_N = \mathsf{G}(N; N, N)$.

The transfer operator of a state. A periodic Gibbs state ν on $a\mathbb{Z}^3$ is invariant under $a\mathbb{Z}^3$ and reflection positive at every lattice mirror (M3), so it has the picture $(e_{\mu_0}, \frac{a}{2}, a e_{\mu_0})$ defined immediately before Lem. YM3_I:99.1. On the bounded Borel functions F, F' of finitely many bonds with both end points in $\{x_{\mu_0} \ge a\}$ the form $\langle F, F' \rangle = \nu(\overline{\Theta_0 F}\, F')$, Θ_0 the reflection $x_{\mu_0} \mapsto a - x_{\mu_0}$, is positive; $\mathcal{H}(\nu)$ is the completion of its quotient by the null vectors, $[F]$ the class of F, $\Omega = [1]$, and P_Ω the projection on $\mathbb{C}\Omega$. The *transfer operator* of ν is the operator T_ν with $T_\nu[F] = [\tau_{a e_{\mu_0}} F]$ of Lem. YM3_I:99.1: it is self-adjoint, $0 \le T_\nu \le 1$, $T_\nu \Omega = \Omega$, and $\langle [F], T_\nu^n [F'] \rangle = \nu(\overline{\Theta_0 F}\, \tau_{n a e_{\mu_0}} F')$, which is the number $h_{AB}(n)$ of Prop. 17.38(f) for $A = \overline{\Theta_0 F'}$ and $B = \overline{\Theta_0 F}$, functions of the slab of the layers $x_{\mu_0} \in \{(1 - \vartheta)a, \dots, 0\}$ when F, F' are functions of the bonds with both end points in the layers $a, \dots, \vartheta a$, $\vartheta \ge 2$; there $h_{AA}(0) - |\nu(A)|^2 = \|(1 - P_\Omega)[F']\|^2$.

So, if $\mathsf{e}(l) \le x_0 \le 0.069$ at β for one $l \ge 3$, that proposition at $n_0 = 0$ gives, with $\theta = 12.9\, x_0$, $|\langle [F], T_\nu^n [F'] \rangle - \langle [F], \Omega \rangle \langle \Omega, [F'] \rangle| \le \theta^{n/l} \|(1 - P_\Omega)[F]\|\, \|(1 - P_\Omega)[F']\|$; the classes being dense, $\|T_\nu^n - P_\Omega\| \le \theta^{n/l}$, hence $\ker(1 - T_\nu) = \mathbb{C}\Omega$ and $\|T_\nu\| \le \theta^{1/l}$ on $\Omega^\perp$.

Twists and loops. A *column* of the plane $(\mu\nu)$ is the set of the plaquettes $p_{\mu\nu}(y + j e_\rho)$, $j \in \mathbb{Z}$, ρ the third direction, in lattice units. For $z \in \{0, 1\}^3$, indexed by the three coordinate planes, Z^z is the partition function with the sign of $\operatorname{tr} \partial U(p)$ reversed on one column of each plane with $z = 1$; it does not depend on the columns chosen, and $r^z = Z^z / Z$. On a cube r^z depends only on the number of twisted planes; we write r_1, r_2, r_3. With a time μ_0 we write $z = (k; m)$, $k \in \{0, 1\}^2$ recording the planes $(\mu_0 \nu_i)$ and m the plane $(\nu_1 \nu_2)$; the weights of the four classes of electric flux $f \in \{0, 1\}^2$ are $w_f = \frac{1}{4} \sum_k (-1)^{f \cdot k} r^{(k;0)}$.

For a closed lattice path C, $W_C = \frac{1}{2} \operatorname{tr} U(C)$, $U(C)$ the ordered product along C; $W_{R \times T}$ is the loop of a coordinate rectangle with $R \times T$ bonds. A *staircase loop* of the plane (13) is given by a site o, an integer $T \ge 1$ and integers α_j and $R_j \ge 1$, $0 \le j < T$: with $o_j^k = o + \alpha_j a e_1 + k a e_3$ and $r_j^k = o + (\alpha_j + R_j) a e_1 + k a e_3$, the loop $C(\alpha, R)$ runs from o_0^0 straight to r_0^0; then, for $j = 0, \dots, T - 1$ in turn, one step up from r_j^j to r_j^{j+1} and, if $j < T - 1$, straight to r_{j+1}^{j+1}; from r_{T-1}^T straight to o_{T-1}^T; and then, for $j = T - 1, \dots, 0$ in turn, one step down from o_j^{j+1} to o_j^j and, if $j > 0$, straight to o_{j-1}^j.

It has one upward and one downward bond between any two consecutive layers that it meets, and consecutive rows need not overlap. Its physical area is $|A(C)| = a^2 \sum_j R_j$; it has $c_3(C) = 1$ and $A_{13}(C) = |A(C)|$ for the numbers c_3, A_{13} of Thm. AZ(4). A staircase loop of another coordinate plane is the image of such a loop under a lattice isometry.

The construction. Fix the common construction constants of Prop. YM3_I:69.1 and Section YM3_I:82, and use the first-passage data of Prop. YM3_I:74.1. The notation is: the blocking factor L, an odd integer; the spacings $a_K = L^{-K}$; the sides $\mathfrak{L} = \{2L^m : m \in \mathbb{N}\}$; the constants γ and $\gamma_W \le \gamma$; the reference couplings $0 < g \le \gamma$, the sides $\mathfrak{L}(g) \subset \mathfrak{L}$ at g and the first-passage data

at g; the numbers M, $r \ge 2$, $\mathfrak{p}_0 \in \mathbb{N}$ of the constants, $R(x) = \min\{L^j : j \in \mathbb{Z}_{\ge 0},\ L^j \ge (\log x)^r\}$, and $\mathsf{p}_0 = 2\mathfrak{p}_0 + \frac{1}{2}$; and the interval of labels $I(g) = (g^{-2} + \bar{B},\ X_g - \bar{B}]$, $X_g = L(g^{-2} + B_\sharp)$, $\bar{B} = LB_\sharp/(L-1)$, such that all pairs $(K, L^K \mathsf{b})$ with $\mathsf{b} \in I(g)$ are first-passage data at g.

A *label* is a real $\mathsf{b} > 0$; its *line* is the set of the pairs $(K, L^K \mathsf{b})$. The torus ℓ at the cutoff K is the cube $(\ell/a_K)^3$ of $a_K\mathbb{Z}^3$; $\mu_{K,\ell,\beta}$ is its measure, extended periodically to $a_K\mathbb{Z}^3$, and $Z_{K,\ell}$ its partition function at $\beta = L^K\mathsf{b}$; $p_\ell = \ell^{-3} \log Z_{K,\ell}$, $p_\infty = a_K^{-3} P$. On $a\mathbb{Z}^3$, for a real Lipschitz function f of compact support, $\mathcal{O}_a(f) = a^{-1} \sum_p f(x(p)) a_p$, $x(p)$ the centre of p, and $\mathcal{O}_K = \mathcal{O}_{a_K}$. For $X = (f_1, \dots, f_n)$, $S^T_{n;K,\ell,\beta}(X)$ and $S^T_n[\nu](X)$ are the joint cumulants of $\mathcal{O}_K(f_1), \dots, \mathcal{O}_K(f_n)$ on the torus and in a state ν, and a periodic Gibbs state is a Gibbs state on $a_K\mathbb{Z}^3$ invariant under a subgroup of finite index of $a_K\mathbb{Z}^3$ (equation YM3_I:(1819) and the definition following it).

Gibbs sequences and their components are defined in Def. 13.4(2), and the volume-first and growing-torus limits in [7, §K.1]. We write S^σ_n for the connected functions, as in Prop. YM3_I:92.1, and use the reflection-space notation $\mathcal{H}(\sigma)$, H_σ, Ω_σ introduced before and in Prop. YM3_I:94.1. On the line of a label we write, at $\beta = L^K\mathsf{b}$,

$$\delta_K(\ell) = \delta((\ell/a_K)^3), \qquad \mathsf{e}_K(\mathsf{b}, \ell_0) = \mathsf{e}(\ell_0/2a_K), \qquad X_K(\mathsf{b}, \ell) = D_L(\ell/La_K) = L^3 \log Z_{K,\ell/L} - \log Z_{K,\ell},$$

and call X_K the *cube number*; then $\ell^3(p_\ell - p_\infty) = \delta_K(\ell)$. A *number of the constants* is formed from this fixed common construction assignment; it depends on no cutoff, coupling, side or label.

The chessboard seminorm. For $\ell'' \in \mathfrak{L}$ with $\ell''/2a_K \ge 3$, a *board* of side $\ell''/2$ is a board of [7, §B.1] of the side $\ell''/2$ with an offset $o \in a_K\mathbb{Z}^3$: its cells are the closed cubes $o + \frac{\ell''}{2}\mathbf{k} + [-\frac{\ell''}{4}, \frac{\ell''}{4}]^3$, $\mathbf{k} \in \mathbb{Z}^3$, and its walls are the planes $x_\mu = o_\mu + \frac{\ell''}{2}(t + \frac{1}{2})$, $t \in \mathbb{Z}$, which lie in $a_K(\mathbb{Z} + \frac{1}{2})$, as $\ell''/2a_K$ is odd: planes of lattice reflections through no site. The *bonds of a cell*, its *inner bonds*, are the bonds with both end points in the closed cube. A function F of the bonds of one cell has a mirror copy in every cell, its image under the reflections in the walls between, complex conjugated at an odd number of reflections.

For a torus $\ell \in \mathfrak{L}$ with $\ell \ge \ell''$, a period of the torus holds $(2\ell/\ell'')^3$ cells of the board, and $\|F\|^{(\ell'')}_{\mathrm{ch},\ell}$, the upper index in parentheses naming the side ℓ'' of the board, is the root of order $(2\ell/\ell'')^3$ of the mean under $\mu_{K,\ell,\beta}$ of the product of the copies of F in these cells: the chessboard seminorm of F at the side ℓ, for the board of F ([7, §B.1], equation [7, eq. 189]). At $\ell = \ell''$ there are eight copies, and we write $\|F\|_{\mathrm{ch},\ell''} = \|F\|^{(\ell'')}_{\mathrm{ch},\ell''}$. So $\|\cdot\|_{\mathrm{ch},\ell}$ without an upper index is always the seminorm of eight copies on the torus ℓ, for a board of side $\ell/2$: a function of a cell of side $\ell''/2 < \ell/2$ is then read under the board of side $\ell/2$ whose cell is centred at its own cell, whose walls are walls of its own board, ℓ/ℓ'' being a power of the odd L, and whose eight copies of it are among its copies under its own board; (M2) below compares the two seminorms.

On a box: for an even $N = 2h$ and a corner $c \in a(\mathbb{Z} + \frac{1}{2})^3$, the cell below c is $\mathsf{C}_h(c) = c + [-ha, 0]^3$, and for a function Ψ of its bonds $\|\Psi\|_{\mathrm{ch};N}$ is the eighth root of the mean on the cube N^3 of the product of its eight mirror copies in the planes $x_\mu = c_\mu$; at $N = \ell/a_K$, $\ell \in \mathfrak{L}$, it is the seminorm $\|\Psi\|_{\mathrm{ch},\ell}$ of the torus ℓ for the board of side $\ell/2$ whose cell is $\mathsf{C}_h(c)$.

Throughout, $0 < x_0 \le 0.069$, $\theta = 12.9\, x_0 \le 0.8901$ and $\Lambda = \log(1/\theta) \ge 0.1164$.

Definition 17.28 (The box inequality BI)**.** Let $\mathsf{b} > 0$ be real, $\ell_0 = 2cL^{m_0}$ with $c \in \mathbb{N}$ and $m_0 \in \mathbb{Z}$, and let K_0 be an integer with $m_0 + K_0 \ge 0$ and $cL^{m_0+K_0} \ge 3$. BI$[\mathsf{b}, \ell_0, K_0; x_0]$ is the statement: for every integer $K \ge K_0$, with $l_K = \ell_0/2a_K = cL^{m_0+K}$,

$$\mathsf{e}(l_K) = 2 \log Z(l_K, 2l_K, 2l_K) - \log Z(2l_K, 2l_K, 2l_K) \le x_0 \qquad \text{at } \beta = L^K\mathsf{b}.$$

Its rate is $m_* = 2\log(1/\theta)/\ell_0 \ge 0.2328/\ell_0$; then $\theta^{1/l_K} = e^{-m_* a_K}$. For every $z \in \mathbb{Z}$ the data $(L^z\mathsf{b}, L^z\ell_0, K_0 - z)$ give the same statement, since they name the same pairs $(l_K, L^K\mathsf{b})$. In the proofs below only the fact that l_K is an integer ≥ 3 is used, except where $\ell_0 = 2L^{m_0}$ is assumed. ■

Remark 17.29 (The spectral reading). At every $\beta \ge 0$ and $l \ge 3$, with $\mathsf{r} = \mathsf{r}_l(2l, 2l)$ over the cross-section $(2l, 2l)$ of the box $(l, 2l, 2l)$, Prop. 17.31(b) at $q = 2$ gives $\log(1+\mathsf{r}) \le \mathsf{e}(l) \le 2\log(1+\mathsf{r})$; so $\mathsf{r} \le e^{x_0/2} - 1$ implies $\mathsf{e}(l) \le x_0$, and $\mathsf{e}(l) \le x_0$ implies $\mathsf{r} \le \varphi(e^{x_0} - 1)$, $\varphi(\eta) = \eta/(1 + \sqrt{1-\eta^2})$; at $x_0 = 0.069$ the two numbers are 0.035102 and 0.035764. Up to these constants BI is the statement that the thermal sum $\sum_{i\ge1}(\lambda_i/\lambda_0)^{l_K}$ over the cross-section $\ell_0 \times \ell_0$ is small uniformly in the cutoff.

17.2.2. *Principal results.*

Theorem AZ (One label: one state, one limit, one vacuum, a gap, an area law)**.** Assume BI$[\mathsf{b}, \ell_0, K_0; x_0]$. Let $K \ge \max\{K_0, 1\}$ and $\beta = L^K\mathsf{b}$. No constant below depends on K.

(1) *The lattice.* (a) For every $\ell \ge \ell_0$ with $\ell/a_K \in \mathbb{N}$: $\delta_K(\ell) < 0.13736\,\theta^{\ell/\ell_0}$ and $\delta_K(\ell) \le 0.5892\,\theta^{2\ell/\ell_0} = 0.5892\,e^{-m_*\ell}$; at $\ell = 2^j\ell_0$, $\delta_K(\ell) \le 0.017170\,\theta^{2\ell/\ell_0}$; if ℓ/a_K is even, $\mathsf{e}(\ell/2a_K) < 0.3702\,\theta^{\ell/2\ell_0}$ and $\mathsf{e}(\ell/2a_K) \le 0.4079\,\theta^{\ell/\ell_0}$; on the cube ℓ/a_K all seven $0 \le 1 - r^z < 0.4121\,\theta^{\ell/\ell_0}$; for $\ell \ge L\ell_0$, $X_K(\mathsf{b}, \ell) < 0.13736\,L^3\theta^{\ell/L\ell_0}$; and for every box whose least physical side ℓ_- is $\ge \ell_0$, $\delta(\underline{n}) < |\underline{n}|(a_K/\ell_-)^3\,0.13736\,\theta^{\ell_-/\ell_0}$. (b) The lattice $a_K\mathbb{Z}^3$ has exactly one periodic Gibbs state ν_K at β; it is ergodic under $a_K\mathbb{Z}^3$, invariant under all isometries of the lattice, reflection positive at every lattice mirror, trivial on the tail σ-field, and an extreme point of the set of all Gibbs states at β; the pressure P is continuously differentiable near β with $P'(\beta) = -3\nu_K(a_p)$. (c) For every $\ell \in \mathfrak{L}$ with $\ell \ge \ell_0$, on the inner bonds of a cell of side $\ell/2$ the laws of ν_K and of the torus ℓ are at a distance in total variation at most $0.541\,\theta^{\ell/2\ell_0}$, and $|\nu_K(G) - \mu_{K,\ell',\beta}(G)| \le 0.66\,\theta^{\ell'/2\ell_0}\|G\|_{\mathrm{ch},\ell''}$ for real G of a cell of side $\ell''/2$, $\ell_0 \le \ell'' \le \ell'$ in $\mathfrak{L}$. For every $\ell \ge \ell_0$ with $N = \ell/a_K$ even, every corner c, every bounded real Borel function Ψ of the bonds of $\mathsf{C}_{N/2}(c)$ and every periodic box $\underline{n}$ with $n_1, n_2, n_3 \ge N$, of any integer sides:

$$|\langle\Psi\rangle_{\underline{n}} - \langle\Psi\rangle_{N^3}| \le 2.71\,\theta^{\ell/2\ell_0}\|\Psi\|_{\mathrm{ch};N}, \qquad |\langle\Psi\rangle_{\underline{n}} - \nu_K(\Psi)| \le 5.42\,\theta^{\ell/2\ell_0}\|\Psi\|_{\mathrm{ch};N},$$

and $|\mu_{\underline{n}}(A) - \nu_K(A)| \le 2.71\,\theta^{\ell/2\ell_0}$ for the events A of the bonds of the cell. Periodic boxes with receding closed twist sets, and finite mixtures of them, converge to ν_K as in Prop. 17.38(d). (d) Over every cross-section with physical sides $l_1 \le l_2$, $l_1 \ge \ell_0$, $l_i/a_K \in \mathbb{N}$, the top eigenvalue is simple, its eigenvector carries no electric flux, and

$$-a_K^{-1}\log(\lambda_1/\lambda_0) > m_* + \frac{1.42}{l_1}; \qquad \text{at } l_1 = \ell_0: \ -a_K^{-1}\log(\lambda_1/\lambda_0) > \frac{2}{\ell_0}\log\frac{1}{2.42\,x_0} \ge \frac{3.579}{\ell_0};$$

so the lowest level carrying electric flux, around either side, lies more than $m_* + 1.42/l_1$ above the top; by contraposition, if over a cross-section with $l_1 = l_h$ that level lies at most $1.42/l_h$ above the top at infinitely many cutoffs, then BI fails at the label b at every side $\ell_0 \le l_h$. Over the square (ℓ, ℓ), $\ell \ge \ell_0$, with $s_\ell = 2^{k-1}\ell_0 \in (\frac{\ell}{4}, \frac{\ell}{2}]$, $2^k\ell_0 \le \ell < 2^{k+1}\ell_0$, the levels $E_i = -a_K^{-1}\log(\lambda_i/\lambda_0)$ satisfy $E_1 \ge m_* + 1.5898/s_\ell \ge m_* + 3.179/\ell$ and $\#\{i \ge 1 : E_i \le E\} \le 0.20395\,e^{(E-m_*)s_\ell}$. For $t \in a_K\mathbb{N}$:

$$\sum_{i\ge1}(\lambda_i/\lambda_0)^{t/a_K} \le \exp\Big[0.37\Big\langle\frac{l_1}{2t}\Big\rangle\Big\langle\frac{l_2}{2t}\Big\rangle e^{-m_*t}\Big] - 1 \le \exp\Big[\frac{0.37}{\ell_0^2}\,l_1l_2\,e^{-m_*t}\Big] - 1 \quad (t \ge \tfrac{\ell_0}{2}),$$

$\langle u\rangle = \max\{1, u\}$, and the sum is $\le 0.2314\,e^{-m_*t}$ for $t \ge l_2/2$. No such floor holds for magnetic flux: over the square (ℓ, ℓ) the free energy of a vortex line, $E_{\mathrm{v}} = a_K^{-1}\log(\lambda_0/\|\mathsf{T}^{(1)}\|)$, satisfies $\ell E_{\mathrm{v}} < 1.0292\,\theta^{\ell/\ell_0}$, and $\ell E_{\mathrm{v}} \le e^{-\sigma_*\ell^2}$ at $\ell = 2^j\ell_0$, $j \ge 0$, σ_* as in (4). (e) If A, B are bounded functions of the bonds of two slabs between which lie n layers, then $|\nu_K(AB) - \nu_K(A)\nu_K(B)| \le e^{-m_*na_K}\sup|A|\sup|B|$. For F, G of two cells of side $\ell''/2$ of a board, $\ell'' \in \mathfrak{L}$, $\ell'' \ge \ell_0$, $j \ge 1$ cells apart in a direction, $|\nu_K(FG) - \nu_K(F)\nu_K(G)| \le 1.0350\,e^{-m_*(j-1)\ell''/2}\|F\|_{\mathrm{ch},\ell''}\|G\|_{\mathrm{ch},\ell''}$. For every n with $na_K \ge \frac{3}{4}\ell_0 + \frac{1}{2}a_K$ and every bounded real Borel F of the bonds of the cube $\{|x|_\infty \le \lfloor n/3\rfloor a_K\}$, $\|\nu_K(F\,|\,\text{bonds outside } \{|x_3| < na_K\}) - \nu_K(F)\|_{L^2(\nu_K)} \le 0.7057\,\mathrm{osc}(F)\,e^{-m_*(2n-1)a_K/6}$. (f) For $\ell \in \mathfrak{L}$, $\ell \ge \ell_0$, the free energy

$E = \log Z_\Gamma - \log Z_{K,\ell}$ of the torus sheared by half a period, $\Gamma = \mathbb{Z}\ell(e_1-e_3)\oplus\mathbb{Z}\ell e_2\oplus\mathbb{Z}\frac{\ell}{2}(e_1+e_3)$, satisfies $0 \le E \le \mathsf{e}(\ell/2a_K) \le 0.4079\,\theta^{\ell/\ell_0}$; the gap alone gives $E \ge -1.24\,\theta^{\ell/\ell_0}$.

(2) *One limit.* Let $n \ge 1$, and let $X = (f_1, \dots, f_n)$ consist of real Lipschitz functions, none 0, with pairwise disjoint compact supports in $(-\ell''/4, \ell''/4)^3$ for some $\ell'' \in \mathfrak{L}$, $\ell'' \ge \ell_0$. Put $X_i = \mathcal{O}_K(f_i) - \mu_{K,\ell'',\beta}(\mathcal{O}_K(f_i))$, $G_I = \prod_{i\in I} X_i$, $M_I = \sup_{K\in\mathbb{N}} \|G_I\|_{\mathrm{ch},\ell''}$. Then $M_I < \infty$, and with $C(X) = 0.65\sum_\pi |\pi|! \prod_{I\in\pi}(1.0174\,M_I)$, π running over the partitions of $\{1,\dots,n\}$, for every $\ell' \in \mathfrak{L}$ with $\ell' \ge \ell''$:

$$\sup_{K\ge\max\{K_0,1\}} |S_n^T[\nu_K](X) - S^T_{n;K,\ell',L^K\mathsf{b}}(X)| \le C(X)\,\theta^{\ell'/2\ell_0},$$

$$\sup_{K\ge\max\{K_0,1\}} \sup_{\min_\mu n_\mu \ge \ell'/a_K} |S_n^T[\mu_{\underline{n}}](X) - S_n^T[\nu_K](X)| \le C^{\mathrm{sh}}(X)\,\theta^{\ell'/2\ell_0},$$

$C^{\mathrm{sh}}(X) = 3.36\sum_\pi |\pi|! \prod_{I\in\pi}(1.208\,M_I)$. Hence, for $n \ge 2$, $S_n(X) = \lim_K S_n^T[\nu_K](X)$ exists along all cutoffs; it equals $\lim_{\ell'}\lim_K S^T_{n;K,\ell',L^K\mathsf{b}}(X)$, the limit of the torus cumulants along every pair of sequences $K_j \to \infty$, $\mathfrak{L} \ni \ell_j \to \infty$, and the limit of the cumulants of periodic boxes $\underline{n}^{(j)}$ at cutoffs $K_j \to \infty$ along every sequence of boxes of any shape with $a_{K_j}\min_\mu n_\mu^{(j)} \to \infty$. If $\mathsf{b} \in I(g)$, $0 < g \le \gamma$, and the supports lie in one closed ball of radius R at pairwise distances $\ge \mathsf{d}$, then for $\ell' \in \mathfrak{L}(g)$, $\ell' \ge \max\{\ell_0, 128n(R+1)\}$, the first bound holds with $C^\flat(X) = 0.65\sum_\pi |\pi|! \prod_{I\in\pi}(1.0174\,\mathsf{m}_I^\flat)$, $\mathsf{m}_I^\flat = \mathsf{M}_{8|I|}(\mathsf{d}';R)^{1/8}\prod_{i\in I}\|f_i\|_\sharp$, which depends only on g, the constants (with the radius fixed in Thms. YM3_I:AJ and YM3_I:AD), n, $\mathsf{d}' = \min\{\mathsf{d},1\}$, R and the norms $\|f_i\|_\sharp = \max\{\sup|f_i|, 40M\,\mathrm{Lip}\,f_i\}$; here $\mathsf{M}_k(\mathsf{d}';R)$ is the sum over the partitions of $\{1,\dots,k\}$ into blocks of at least two members of the products of the bounds $C^g_{|B|}(\mathsf{d}';R)$ of (M5).

With the same $C^\flat(X)$ it holds with a torus $\ell^+ \ge \ell'$ of $\mathfrak{L}$ in the place of ν_K, and with 0.69 for 0.65 with the box (q_1N, q_2N, q_3N), $N = \ell'/a_K$, $q \in \mathbb{N}^3$, in the place of ν_K; for every periodic box $\underline{n}$ with all sides $\ge \ell'/a_K$, $|S_n^T[\mu_{\underline{n}}](X) - S_n^T[\nu_K](X)| \le 3.36\sum_\pi |\pi|! \prod_{I\in\pi}(1.19\,\mathsf{m}_I^\flat)\,\theta^{\ell'/2\ell_0}$; and for $n \ge 2$, $|S_n(X)| \le C_n^g(\mathsf{d};R)\prod_i \|f_i\|_\sharp$.

(2′) *One far field.* Let $n \ge 1$, $\rho \ge 1$, $\mathsf{d} > 0$, $f_1, \dots, f_n$ real Lipschitz, none 0, with supports in $[-\rho,\rho]^3$ at pairwise sup distances $\ge \mathsf{d}$, h real Lipschitz, not 0, with support in $v + [-\rho,\rho]^3$, $\Pi = \|h\|_\sharp \prod_i \|f_i\|_\sharp$, and $\ell'' \in \mathfrak{L}$, $\ell'' \ge \max\{\ell_0, 8\rho+4\mathsf{d}+4\}$. There is C_n, formed from $n, \rho, \mathsf{d}, \ell''$, b and the constants, such that $|S^T_{n+1}[\nu_K](h,X)| \le \mathsf{C}_n\, e^{-m_*(|v|_\infty - \ell'')}\,\Pi$ for $|v|_\infty \ge \ell''$, and on every torus $\ell \ge L\ell''$ of $\mathfrak{L}$, $|S^T_{n+1;K,\ell,\beta}(h,X)| \le 3.44\,\mathsf{C}_n\, e^{-m_*(|v|_{\infty,\ell} - \ell'')}\,\Pi$ for $|v|_{\infty,\ell} \ge \frac{3}{2}\ell''$, $|\cdot|_{\infty,\ell}$ the distance on the torus.

The same two bounds hold with h replaced by any bounded real Borel function W of the bonds with both ends in $v + [-\rho,\rho]^3$, with $\mathsf{C}'_n \sup_\square \|W - \mu_{K,\ell'',\beta}(W)\|_{\mathrm{ch},\ell''}\prod_i\|f_i\|_\sharp$ in the place of $\mathsf{C}_n\Pi$, C'_n formed as C_n and the supremum over the cells $\square$ of side $\ell''/2$ of all boards that hold $v + [-\rho,\rho]^3$ at a sup distance $\ge \mathsf{d}/2$ from their walls; so the decay of a far insertion W_K, for instance a local density smeared at a fixed scale, is uniform in the cutoff whenever these seminorms are bounded in K. In particular, let u be a real bounded Lipschitz function on $\mathbb{R}^3$ that vanishes on $\{|x|_\infty < r\}$, $r \ge \ell'' + \rho$, and let $\chi_v = \chi_0(\cdot - v)$, $v \in \rho\mathbb{Z}^3$, be the translates of the tent $\chi_0(x) = \prod_\mu \max\{0, 1 - |x_\mu|/\rho\}$: a partition of unity by Lipschitz functions with $\operatorname{supp}\chi_v \subset v + [-\rho,\rho]^3$.

Then $u\chi_v = 0$ unless $|v|_\infty \ge r - \rho \ge \ell''$, $\sup_v \|u\chi_v\|_\sharp < \infty$, and for every $K \ge \max\{K_0, 1\}$

$$\sum_{v\in\rho\mathbb{Z}^3} |S^T_{n+1}[\nu_K](u\chi_v, X)| \le \mathsf{C}_n \sup_v \|u\chi_v\|_\sharp \prod_i \|f_i\|_\sharp \sum_{v\in\rho\mathbb{Z}^3,\ |v|_\infty \ge r-\rho} e^{-m_*(|v|_\infty - \ell'')},$$

a bound free of K that tends to 0 as $r \to \infty$ at a fixed bound of $\sup_v \|u\chi_v\|_\sharp$. Thus the connected-cumulant series $\sum_v S^T_{n+1}[\nu_K](u\chi_v, X)$ converges absolutely, uniformly in K, and is the limit of the cumulants with finite partial sums of the far insertions; it tends

to 0 as the region of u recedes. On a torus $\ell \geq L\ell''$ of $\mathfrak{L}$, take $\rho = \ell/q \geq 1$ with q an integer and $\ell'' \geq 8\rho + 4\mathsf{d} + 4$, use the finite set $V = \rho\mathbb{Z}^3/\ell\mathbb{Z}^3$ and the periodized tents $\chi_v^{\mathrm{per}}(x) = \sum_{k\in\mathbb{Z}^3} \chi_0(x - v + \ell k)$. For a bounded Lipschitz function u on the torus vanishing where $|x|_{\infty,\ell} < r$, $r \geq \frac{3}{2}\ell'' + \rho$, the finite sum over V has the same bound with $3.44\,\mathsf{C}_n$ and periodic distances. Its tail is bounded by the displayed infinite-grid tail, so the decay as $r \to \infty$ is uniform over these tori.

(3) *One vacuum, the Euclidean group, a gap.* Let $\mathsf{b} \in I(g)$, $0 < g \leq \gamma$. Every Gibbs sequence at g with the pairs $(K_j, L^{K_j}\mathsf{b})$, $K_j \geq \max\{K_0, 1\}$, is $(\nu_{K_j})_j$; the full sequence (ν_K) converges to a functional σ with $S_n^\sigma = S_n$ for $n \geq 2$; σ is its own and only ergodic component, invariant under $\mathbb{R}^3 \rtimes \mathrm{O}(3)$ and reflection positive at every plane; $\ker H_\sigma = \mathbb{C}\Omega_\sigma$, and $\mathcal{H}(\sigma)$ carries a Wightman theory of one Hermitian scalar field on $\mathbb{R}^{1+2}$ with a unique vacuum and the Schwinger functions of σ. Every continuum limit at g along growing tori with the pairs $(K_j, L^{K_j}\mathsf{b})$, those with the spacing removed first among them, is $(S_n^\sigma)_{n\geq 2}$. Finally $\operatorname{spec} H_\sigma \subset \{0\} \cup [m_*, \infty)$: for all F, F' in the span of separated tuples in the upper half space and $t \geq 0$,

$$\left|\sigma(\Theta\bar{F} \cdot \tau_{te_3}F') - \overline{\sigma(F)}\,\sigma(F')\right| \leq e^{-m_* t}\,\|(1 - P_\Omega)[F]\|\,\|(1 - P_\Omega)[F']\|,$$

Θ the reflection $x_3 \mapsto -x_3$, P_Ω the projection on Ω_σ; the same in every direction.

(4) *An area law.* For every rectangle C of a coordinate plane with physical sides $R, T \in a_K\mathbb{N}$,

$$0 \leq \nu_K(W_C) \leq e^{-\sigma_* RT}, \qquad \sigma_* = \frac{2\log(1/\theta) + 1.67}{\ell_0^2} \geq \frac{1.9028}{\ell_0^2};$$

$|\nu_K(W_C)| \leq e^{-\sigma_*|A(C)|}$ for every staircase loop of a coordinate plane, $|A(C)|$ its physical area; and for every closed lattice path C with a bond of the direction 3, $|\nu_K(W_C)| \leq 2^{c_3(C)-1}e^{-\sigma_* A_{13}(C)}$, where $A_{13}(C)$ is the area of the union of the squares of side a_K of the plane (13) about which the projection of C winds an odd number of times, and $c_3(C)$ is the number of passages from a downward to an upward bond of the direction 3 in the cyclic order along C; the same holds for every ordered pair of directions. The static potential $V_K(r) = \lim_{T\to\infty} -T^{-1}\log \nu_K(W_{r\times T})$, $r, T \in a_K\mathbb{N}$ physical, exists in $[0, \infty]$, is concave and non-decreasing, and $V_K(r) \geq V_K(r') + \sigma_*(r - r')$ for $r \geq r'$. For every ℓ with ℓ/a_K an integer ≥ 3: $0 \leq 1 - r_1(\ell/a_K) \leq 2\exp(-8.8 \cdot 10^{-4}\,\ell^2/\ell_0^2)$, with $5.707 \cdot 10^{-3}$ in the place of $8.8 \cdot 10^{-4}$ for $\ell \geq 128\,\ell_0$; and with $\mathsf{q} = 1 + 0.835/\log(1/\theta)$, for $\ell \geq \mathsf{q}\ell_0$: $\frac{1}{2}(1 - r_1(\ell/a_K)) \leq \exp[-\sigma_*\ell^2 + (4\mathsf{q}\ell/\ell_0)(\frac{1}{2}\log(4\mathsf{q}\ell/\ell_0) + 0.86)]$. None of these bounds depends on K, b or L.

■□

Theorem BA (All labels: one theory up to dilation)**.** Assume $\mathrm{BI}[\mathsf{b}, \ell_0, K_0; x_0]$ at one label $\mathsf{b} > 0$.

(1) There are $\beta_0 > 0$ and $C_+ < \infty$ such that for every real $\beta \geq \beta_0$ there is an integer l with $3 \leq l \leq C_+\beta$ and $\delta(l^3) \leq 0.008625$, hence $\mathsf{e}(l) \leq 0.069$, at β.

(2) For every $\mathsf{b}' > 0$ and every integer m with $L^m \geq 48C_+\mathsf{b}'$, $\mathrm{BI}[\mathsf{b}', 2L^m, K_0'; 0.069]$ holds, K_0' the least integer with $L^{K_0'}\mathsf{b}' \geq \beta_0$ and $K_0' + m \geq 1$; on every compact set of labels one side and one threshold serve. Hence Thm. AZ holds at every label, its clause (3) at every label of every $I(g)$; and the set of the labels at which BI holds at some side and some threshold is empty or all of $(0, \infty)$.

(2′) *Near the label, with the same threshold.* Let $\ell_1 \in \mathfrak{L}$, $\ell_1 \geq 75\,\ell_0$, $f(x) = \max\{0, 1 - 8|x|_\infty/\ell_1\}$ and $\mathsf{M} = \sup\{\|\mathcal{O}_K(f) - \mu_{K,\ell_1,L^K\mathsf{b}'}(\mathcal{O}_K(f))\|_{\mathrm{ch},\ell_1} : K \in \mathbb{N},\ \mathsf{b}' \in [\frac{\mathsf{b}}{2}, 2\mathsf{b}]\}$. Then $\mathsf{M} < \infty$, and with $\eta = \min\{\frac{\mathsf{b}}{2},\ 0.0086/(0.17516 \cdot 1458\,\mathsf{M})\}$: for all $|\mathsf{b}' - \mathsf{b}| \leq \eta$ and all $K \geq \max\{K_0, 1\}$, $\delta_K(\ell_1) \leq 0.008625$ at $\beta = L^K\mathsf{b}'$, so $\mathrm{BI}[\mathsf{b}', 2\ell_1, \max\{K_0, 1\}; 0.069]$; $\mathsf{b}' \mapsto S_n^{(\mathsf{b}')}(X)$ is continuous there, and $S_n^T[\nu_{K_j,\mathsf{b}_j}](X) \to S_n^{(\mathsf{b})}(X)$ along all $K_j \to \infty$, $\mathsf{b}_j \to \mathsf{b}$.

Moreover, let $\ell_0 = 2L^{m_0}$, let $\ell_1^\flat = 2L^{m_1}$ with m_1 the least $m \in \mathbb{N}$ with $2L^m \geq \ell_0$ and $L^3\, 0.13736\, \theta^{2L^m/\ell_0} \leq 0.03$, let $f^\flat$, $\mathsf{M}^\flat$ be formed as f, M with $\ell_1^\flat$ for ℓ_1, and $r = \min\{\frac{\mathsf{b}}{2}, 0.03/(16L^3\mathsf{M}^\flat)\}$. Then for all $|\mathsf{b}' - \mathsf{b}| \leq r$ and all $K \geq \max\{K_0, 1\}$, at $\beta = L^K\mathsf{b}'$: $0 \leq D_L(\ell_1^\flat/a_K) < 0.06$, $\mathsf{e}(\ell_1^\flat/a_K) < 2.2 \cdot 10^{-4}$, and $\mathrm{BI}[\mathsf{b}', L^z\ell_1^\flat, \max\{K_0, 1\}; 6.1 \cdot 10^{-9}]$ for every integer $z \geq 1$. Neither η nor r depends on the cutoff.

(3) For every $K \in \mathbb{N}$ and every real $\beta \geq \beta_0$ the lattice $a_K\mathbb{Z}^3$ has exactly one periodic Gibbs state ν at β, which is also the limit of the tori; $|\nu(AB) - \nu(A)\nu(B)| \leq e^{-c_* n/\beta} \sup|A| \sup|B|$ for bounded functions of two slabs between which lie n layers, $c_* = \log(1/0.8901)/C_+$: the correlation length in lattice units is at most β/c_*, and the transfer operator T_ν of ν (the setting, in every coordinate direction as the time) has $\ker(1 - T_\nu) = \mathbb{C}\Omega$ and norm at most $e^{-c_*/\beta}$ on $\Omega^\perp$. Moreover $0 \leq 1 - r_1(N) \leq 2\exp[-2.2 \cdot 10^{-4}(N/C_+\beta)^2]$ at β for every integer $N \geq 3$, and in lattice units every slope of the static potential of ν is at least $0.4757\,(C_+\beta)^{-2}$. For $0 < g \leq \gamma$ and $\ell_0^g = 2C_+L(g^{-2} + \bar{B})$: $|S^T_{n;K,\ell,\beta}(X) - S^T_{n;K,\ell',\beta}(X)| \leq C_g(X)\, 0.8901^{\ell/2\ell_0^g}$ for $\ell \leq \ell'$ in $\mathfrak{L}(g)$ above a side that X fixes, uniformly in the first-passage data (K, β) at g with $\beta \geq \beta_0$, with $C_g(X)$ free of K, β, ℓ, ℓ'; the same with ν for the torus ℓ'.

(4) For every $c > 0$ there is one functional σ_c such that for all real spacings $a_j \to 0$ and couplings β_j with $a_j\beta_j \to c$, and every choice of periodic Gibbs states on $a_j\mathbb{Z}^3$ at β_j, the sequence converges to σ_c. With $X_s = (f_1(\cdot/s), \ldots, f_n(\cdot/s))$,

$$S_n^{\sigma_{tc}}(X) = t^{-n}\, S_n^{\sigma_c}(X_{1/t}) \qquad (t, c > 0,\ n \geq 2),$$

and σ_c is the same for every admitted blocking factor. In lattice units, $H_n(Y) = \lim_{\beta\to\infty} \kappa_n(\beta; Y)$ exists along the real β, where $\kappa_n(\beta; Y)$ is the joint cumulant of the $\mathcal{O}_{1/\beta}(y_i)$ in the periodic Gibbs state on $\beta^{-1}\mathbb{Z}^3$ at β, and $S_n^{\sigma_c}(X) = c^{-n}H_n(X_{1/c})$. For $c \in I(g)$, σ_c is the functional σ of Thm. AZ(3) at the label c; its gap scales like $1/c$.

(5) *The rotational Ward identity.* Let $\mathsf{b} \in I(g)$, $0 < g \leq \gamma$, $n \geq 2$, $\mathsf{d} > 0$, $R \geq 1$, $0 < \kappa < \frac{1}{16}$. There are C and p', depending on g, n, d, R, κ, the data of BI and the constants, but on no side and on no place, such that for all large K, all $\ell \in \mathfrak{L}(g)$ and $v \in a_K\mathbb{Z}^3$ with $|v|_\infty + R + 2 \leq \ell/4$, all $\omega \in \mathfrak{so}(3)$, and all real $\chi, f_1, \ldots, f_n \in C^{1,1}(\mathbb{R}^3)$ that vanish outside $v + [-R, R]^3$, the f_i at pairwise sup distances $\geq \mathsf{d}$, χ equal to 1 at the centres and at the bonds of the plaquettes at which some f_i does not vanish, the bulk term $\mathfrak{U}^{(K,\ell,L^K\mathsf{b})}_{\omega,\chi}(f_1, \ldots, f_n)$ of the rotational Ward identity YM3_I:(2108) on the torus ℓ (the joint cumulant of the $\mathcal{O}_K(f_i)$ with one insertion of the local density of the rotation ω weighted by χ, together with the like cumulants in which the insertion is weighted by one f_i) satisfies

$$|\mathfrak{U}^{(K,\ell,L^K\mathsf{b})}_{\omega,\chi}(f_1, \ldots, f_n)| \leq (1 + 2|v|_1)\, C\, |\omega|\, K^{p'} L^{-\kappa K}\, \|\chi\|_{C^{1,1}} \prod_i \|f_i\|_{C^{1,1}},$$

and the same bound holds for the bulk term in the state ν_K. The bound rests, besides on BI and on Thm. YM3_I:D, on the bound (M7) of the member $\mathfrak{U}^\sharp_{\omega,\chi}$ of the bulk term of third order in the curvature.

■□

Theorem BB (One cutoff decides; a form without cutoff)**.** Let $\mathsf{b} > 0$ and $\ell = 2L^m \in 2L^{\mathbb{Z}}$. Let $\varsigma = \sum_{j\geq 0}(1 + j)^{\mathsf{p}_0} L^{-j/10}$, and let $z = z(\mathsf{b}, \ell)$ be the least integer $z \geq 0$ with $L^z\mathsf{b} \geq \gamma^{-2}$, $m + z \geq 2$ and $LM\, R(L^z\mathsf{b}) \leq L^{m+z-1}$.

(1) There are an explicit threshold $K_\star(\mathsf{b}, \ell)$, formed from the constants, b and ℓ, with $K_\star(\mathsf{b}, \ell) - z$ at or above the thresholds K_{bf} of (M4)(ii) on the two tori $L^{z-1}\ell$ and $L^z\ell$ at the label $L^z\mathsf{b}$, to which the dilation of Prop. 17.44(c) by $z = z(\mathsf{b}, \ell)$ brings the data, and the margins

$$\eta_K(\mathsf{b}, \ell) = 3\varsigma\, \mathsf{C}_{\mathrm{bf}}(L^z\mathsf{b})\, (L^z\ell)^3 (K - z)^{\mathsf{p}_0} L^{-(K-z)/10} \leq C(\mathsf{b}, \ell)\, K^{\mathsf{p}_0} L^{-K/10},$$

$\mathsf{C}_{\rm bf}(\mathsf{b}')$ a number of the constants and of b', bounded on compact sets of labels, so that $C(\cdot,\ell)$ is bounded on compact sets of labels (where $z=0$, $\eta_K = 3\varsigma\,\mathsf{C}_{\rm bf}(\mathsf{b})\,\ell^3 K^{\mathsf{p}_0} L^{-K/10}$), such that $|X_{K'}(\mathsf{b},\ell) - X_K(\mathsf{b},\ell)| \le \eta_K(\mathsf{b},\ell)$ for $K' \ge K \ge K_\star(\mathsf{b},\ell)$. So $X_\infty(\mathsf{b},\ell) = \lim_K X_K(\mathsf{b},\ell) \ge 0$ exists, uniformly on compact sets of labels. For every $y>0$: if $X_{K_1} \le y - \eta_{K_1}$ at one $K_1 \ge K_\star$, then $X_K \le y$ for all $K \ge K_1$; $X_\infty < y$ if and only if $X_{K_1} < y - \eta_{K_1}$ at one $K_1 \ge K_\star$; $X_\infty > y$ if and only if $X_{K_1} > y + \eta_{K_1}$ at one $K_1 \ge K_\star$, and then $X_K > y$ for all $K \ge K_1$.

(2) *The cube criterion.* Let $2^k \le L/2 < 2^{k+1}$, $\Phi_L(x) = (L/2^{k+1})^3[\epsilon + \frac{20}{7}\epsilon^2]$ with $\epsilon = (12.9x)^{2^k}/12.9$, $x_1 = \max\{x \in (0, 0.069] : \Phi_L(x) \le x_0\}$, $y_1 = \frac{L^3}{8}x_1$ and $y_0 = \frac{L^3}{8}x_0$. If $X_{K_1}(\mathsf{b},\ell) \le y_1 - \eta_{K_1}(\mathsf{b},\ell)$ at one $K_1 \ge K_\star(\mathsf{b},\ell)$, then $\mathrm{BI}[\mathsf{b},\ell,K_1;x_0]$ holds; if $X_{K_1}(\mathsf{b},\ell) \le y_0 - \eta_{K_1}(\mathsf{b},\ell)$ there, then $\mathrm{BI}[\mathsf{b},2\ell/L,K_1;x_0]$ holds (with $c=2$), and with it Thm. AZ at the side $2\ell/L$, of rate $L\log(1/\theta)/\ell$. Conversely $\mathrm{BI}[\mathsf{b},2\ell/L,K_0;x_0]$ gives $X_K(\mathsf{b},L\ell) \le L^3\bar\delta_{x_0}(L)$ for all $K \ge K_0$, with $\bar\delta_x(\nu) = (\nu/2^j)^3\mathsf{c}_0(12.9x)^{2^j}$ for $2^j \le \nu < 2^{j+1}$, $\mathsf{c}_0 = \frac{20}{7\cdot 12.9^2} \approx 0.0171693$; so $X_{K_1}(\mathsf{b},L\ell) > L^3\bar\delta_{x_0}(L) + \eta_{K_1}(\mathsf{b},L\ell)$ at one $K_1 \ge K_\star(\mathsf{b},L\ell)$ refutes it. For every K with $K+m \ge 2$,

$$\min\Big\{0.069,\ \Big[\tfrac{7}{20}L^{-3}X_K(\mathsf{b},L\ell)\Big]^{1/2}\Big\} \le \mathsf{e}_K(\mathsf{b},\ell) \le \min\Big\{\tfrac{2}{L}X_K(\mathsf{b},\ell),\ \Phi_L\big(\tfrac{8}{L^3}X_K(\mathsf{b},\ell)\big)\Big\},$$

the last member where its argument is ≤ 0.069; so $\mathrm{BI}[\mathsf{b},\ell,K_0;x_0]$ gives $X_K(\mathsf{b},L\ell) \le \frac{20}{7}L^3x_0^2$ for $K \ge K_0$, and if $X_{K_1}(\mathsf{b},L\ell) > \frac{20}{7}L^3x_0^2 + \eta_{K_1}(\mathsf{b},L\ell)$ at one $K_1 \ge K_\star(\mathsf{b},L\ell)$, then $\mathsf{e}_K(\mathsf{b},\ell) > x_0$ for every $K \ge K_1$. The same bracket holds for $\liminf_K \mathsf{e}_K$ and $\limsup_K \mathsf{e}_K$ with X_∞.

(3) *The inequality of BI itself at one cutoff.* Let $\ell_0 = 2L^{m_0}$, $i \ge 1$ and $\ell = L^i\ell_0$. If $\mathsf{e}_{K_1}(\mathsf{b},\ell_0) \le x_0$ at one $K_1 \ge K_\star(\mathsf{b},\ell)$ with $L^3\bar\delta_{x_0}(2L^{i-1}) + \eta_{K_1}(\mathsf{b},\ell) \le y_1$, then $X_K(\mathsf{b},\ell) \le y_1$ for all $K \ge K_1$ and $\mathrm{BI}[\mathsf{b},L^i\ell_0,K_1;x_0]$ holds, of rate $L^{-i}m_*$; with y_0 in the place of y_1 it gives $\mathrm{BI}[\mathsf{b},2L^{i-1}\ell_0,K_1;x_0]$. The margin condition reads $\bar\delta_{x_0}(2L^{i-1}) \le \frac{1}{8}x_1 - L^{-3}\eta_{K_1}$; at $x_0 = 0.069$ and $i=2$ it holds, once η_{K_1} is small enough, at every odd $L \ge 33$, so at every admitted blocking factor, and at $i=1$ it never holds. So, wherever $L^3\bar\delta_{x_0}(2L) < y_1$, with $K_{\star\star}(\mathsf{b},\ell_0)$ the least $K \ge K_\star(\mathsf{b},L^2\ell_0)$ such that $\eta_{K'}(\mathsf{b},L^2\ell_0) \le y_1 - L^3\bar\delta_{x_0}(2L)$ for all $K' \ge K$, an explicit expression, the inequality $\mathsf{e}_{K_1}(\mathsf{b},\ell_0) \le x_0$ at one $K_1 \ge K_{\star\star}(\mathsf{b},\ell_0)$ gives $\mathrm{BI}[\mathsf{b},L^2\ell_0,K_1;x_0]$.

(4) The following are equivalent: (i) $\mathrm{BI}[\mathsf{b},\ell_0,K_0;x_0]$ holds for some $\ell_0 \in 2\mathbb{N}L^{\mathbb{Z}}$ and some K_0; (ii) $X_\infty(\mathsf{b},\ell) < y_0$ for some $\ell \in 2L^{\mathbb{Z}}$; (iii) $X_{K_1}(\mathsf{b},\ell) < y_0 - \eta_{K_1}(\mathsf{b},\ell)$ for some $\ell \in 2L^{\mathbb{Z}}$ and one $K_1 \ge K_\star(\mathsf{b},\ell)$; (iv) $X_\infty(\mathsf{b},\ell) \to 0$ as $2L^{\mathbb{Z}} \ni \ell \to \infty$. More precisely, (iii) at (ℓ,K_1) gives (i) at $\ell_0 = 2\ell/L$, $K_0 = K_1$; and (i) at (ℓ_0,K_0) gives $X_K(\mathsf{b},\ell) \le L^3\bar\delta_{x_0}(2\ell/L\ell_0) < 0.13736\,L^3\theta^{\ell/L\ell_0}$ for all $K \ge K_0$ and all $\ell \in 2L^{\mathbb{Z}}$ with $\ell \ge L\ell_0$.

(5) *The half box.* This branch remains a sketch at the full interacting Hessian estimate described with Thm. AH. Let $\ell_0 = 2L^{m_0}$ and J a compact set of labels. There are C and K_* with $|\mathsf{e}_{K+1}(\mathsf{b}',\ell_0) - \mathsf{e}_K(\mathsf{b}',\ell_0)| \le C\,L^{-K/10}$ for $\mathsf{b}' \in J$ and $K \ge K_*$. Hence $\mathsf{e}_\infty = \lim_K \mathsf{e}_K(\mathsf{b},\ell_0)$ exists, and with $Q_K = \sum_{k\ge K} CL^{-k/10}$, $|\mathsf{e}_K - \mathsf{e}_\infty| \le Q_K$: if $\mathsf{e}_{K_1}(\mathsf{b},\ell_0) \le x_0 - Q_{K_1}$ at one $K_1 \ge K_*$, then $\mathrm{BI}[\mathsf{b},\ell_0,K_1;x_0]$ holds, at the same side; $\mathrm{BI}[\mathsf{b},\ell_0,K_0;x_0']$ with $x_0' < x_0$ gives $\mathsf{e}_{K_1} \le x_0 - Q_{K_1}$ at every large K_1; $\mathsf{e}_\infty < x_0$ if and only if $\mathsf{e}_{K_1}(\mathsf{b},\ell_0) < x_0 - Q_{K_1}$ at one $K_1 \ge K_*$; and if $\mathsf{e}_{K_1}(\mathsf{b},\ell_0) > x_0 + Q_{K_1}$ at one $K_1 \ge K_*$, then $\mathrm{BI}[\mathsf{b},\ell_0,K_0;x_0]$ fails for every K_0.

(6) *A form without cutoff.* For every $\ell \in 2L^{\mathbb{Z}}$ and every real $\mathsf{b} > 0$ the limit $D_\infty(t) = \lim_K D_L(\ell/a_K)|_{\beta=L^K\mathsf{b}}$ exists, depends on $t = \ell/\mathsf{b}$ alone, is continuous on $(0,\infty)$, and $\sup_{t\in[t_-,t_+]}|D_L(\ell/a_K) - D_\infty(t)| \le C\,K^{\mathsf{p}_0}L^{-K/10}$ on compact sets for large K. The function $\delta_\infty(t) = \sum_{r\ge 0} L^{-3(r+1)}D_\infty(L^r t)$ is continuous and positive, $\delta_K(\ell) \to \delta_\infty(\ell/\mathsf{b})$ uniformly on compact sets of labels, $D_\infty(t) = L^3\delta_\infty(t) - \delta_\infty(Lt)$ and $(L-1)^3\delta_\infty \le D_\infty \le L^3\delta_\infty$. δ_∞ is the same for every admitted blocking factor; $t^{-3}\delta_\infty(t)$ does not increase, and $\delta_\infty(\rho t) \le$

$(3\rho^2-3\rho+1)\delta_\infty(t)\le 3\rho^2\delta_\infty(t)$ for real $\rho\ge 1$. Either $\delta_\infty\ge 0.008625$ on $(0,\infty)$, or $\delta_\infty(t_0)<0.008625$ at some t_0, and then $\delta_\infty(t)\le 0.1374\,(103.2\,\delta_\infty(t_0))^{t/2t_0}$ for all real $t\ge 2t_0$. The second case holds if and only if BI holds at some side of some label, and it holds as soon as $D_{q,\infty}(t_0)<(q-1)^3x_0/8$ at one t_0 for one admitted q, $D_{q,\infty}(t)=q^3\delta_\infty(t)-\delta_\infty(qt)$.

(7) *The size of the thresholds.* Read as inequalities, the conditions of the list of constants admit exactly the odd blocking factors $L>48\cdot 4096^2e^{24}=2.1331\ldots\cdot 10^{19}$, the least being 21 331 858 733 892 671 083. At every admissible choice of the constants, at a cutoff $K_1\ge K_\star(\mathsf{b},\ell)$ the small cube of the criterion of (2) has more than $10^{1.42\cdot 10^{11}}$ sites a side, and at a cutoff at which the criterion of (3) with $i=2$ applies each of the two boxes of BI has more than $10^{1.89\cdot 10^{12}}$ sites. Independently of all constants but L, $\eta_K(\mathsf{b},\ell)\ge 48\,\varsigma_\flat(L)\,L^6(K-z)^{5/2}L^{-(K-z)/10}$ with $\varsigma_\flat(L)=\sum_{j\ge 0}(1+j)^{5/2}L^{-j/10}$; as $X_K\ge 0$, the margin alone forces more than 10^{152} sites a side on the small cube of any inequality $X_{K_1}\le y-\eta_{K_1}$ with $y\le y_0$.

■□

To complete clause (5)(c) below and the indicated part of Prop. 17.48(b), one still needs a uniform probability estimate for tubes around holonomy classes. We call the target the **Heaviest-class estimate**. For every integer $j\ge 1$, put $N=2L^j$. For every $\beta>0$ and every $\omega>0$ with $\Xi=\beta\omega^2\ge 1$ and $\Xi N^5\omega\le 3/640$, every w with $N^2\omega\le w\le 1/10$, every pair $Q\in G^2$ and every site y of the periodic cube, its Wilson probability law satisfies

$$\mu_{N^3,\beta}\big(\mathsf{D}((P_1,P_2)(y),Q)\le w,\ \max_p|\partial U(p)-\mathbf{1}|<\omega\big)\le 10^{11}\Xi^5N^7w^6\omega^{-4}.$$

Here P_1,P_2 are the straight winding holonomies through y, and $\mathsf{D}(P,Q)=\min_{g\in G}\max_{i=1,2}|P_i-gQ_ig^{-1}|$ is the distance of pairs modulo simultaneous conjugation. This probability bound remains to be proved. The pointwise density maximum below does not control the tube measure, especially near degenerate or central pairs.

Theorem BC (Small tori without the box criterion)**.**

(1) *Comparison.* For $\beta\ge 0$, let $N,l\ge 3$ and $q\ge 2$ be integers. Then $0\le\mathsf{e}(N)\le D_2(N)\le 8\delta(N^3)$ and $D_q(N)\ge\frac{q^3}{8}D_2(N)$; $(q-1)^3\delta(N^3)\le D_q(N)\le q^3\delta(N^3)$, and $\delta(M^3)\le(3\rho^2-3\rho+1)\delta(N^3)\le 3\rho^2\delta(N^3)$ for $M=\rho N\ge N$; $\delta(\underline{n})\le(n_1n_2+n_1n_3+n_2n_3)N^{-2}\delta(N^3)$ for every box with all sides $\ge N$, so $\mathsf{e}(l)\le 16(l/N)^2\delta(N^3)$ for $l\ge N$; $\mathsf{e}(l)\le\frac{2}{L}D_L(2l/L)$ if $l=Ll'$, $l'\ge 2$; if $D_L(N)\le\frac{L^3}{8}x$ with $x\le 0.069$, then $D_2(N)\le x$, $\mathsf{e}(N)\le x$ and, for even N, $\mathsf{e}(LN/2)\le\Phi_L(x)$; $\mathsf{e}(l)\le x\le 0.069$ gives $D_L(2l)\le\frac{20}{7}L^3x^2$; $\delta(N^3)\le x_0/8$ gives $\mathsf{e}(N)\le x_0$; $\mathsf{e}(l)\le x_0$ gives $\delta(N'^3)\le 0.008625$ for $N'\ge 48l$; and $\mathsf{e}(l)\le 24\beta l^3$, so $\mathsf{e}(l)\le x_0$ whenever $0<\beta\le x_0/24l^3$.

(2) *Dichotomy.* At each $\beta\ge 0$ the following are equivalent: $\mathsf{e}(l)\le 0.069$ for one $l\ge 3$; $\mathsf{e}(l)\to 0$; $\delta(N^3)\to 0$; $\delta(N^3)\le 0.008625$ for one $N\ge 3$. Then $\delta((2N)^3)\le 183\,\delta(N^3)^2$ and $\delta(N'^3)\le 0.13736\,(103.2\,\delta(N^3))^{N'/2N}$ for $N'\ge 2N$. The numbers 0.07489 and 0.00936 may stand for 0.069 and 0.008625 in this equivalence. In the other case $\mathsf{e}(l)>0.069$ and $\delta(N^3)>0.008625$ for all $l,N\ge 3$, and indeed $\mathsf{e}(2^kl)>0.07489$ for all l and k; this case holds whenever there are two periodic Gibbs states at β, and then $\liminf_m\delta((2L^{m+K})^3)\ge\log 2$. Every phase is reached by the tori: if $\liminf_m\delta((2L^{m+K})^3)\le D<\infty$ at β, there are sides along which the tori converge to a periodic Gibbs state μ^* with $\mu^*\ge e^{-D}\nu$ for every ergodic periodic Gibbs state ν, and every periodic Gibbs state is at most $e^D\mu^*$.

(3) *A floor.* Let $0<g\le\gamma_W$, let $\mathsf{d}_0(g,\vartheta)$ and $s(\mathsf{d})$ be the threshold and the depth of the law of the two-point function at short distance of (M5), $\vartheta_1=3\sqrt{3}/8$, let $0<\mathsf{d}_\circ\le\mathsf{d}_0(g,\vartheta_1)$, let (K,β) be a first-passage datum at g with $K>s(\mathsf{d}_\circ)$, and $N_4=\lfloor 4\mathsf{d}_\circ/a_K\rfloor$. Then for every

integer $l \ge 3$

$$\mathsf{e}(l) \ge \min\{0.069,\ \tfrac{1}{12.9}\,31250^{-l/N_4}\} \ge \min\{0.069,\ 0.0775\,e^{-2.6\,l a_K/\mathsf{d}_\circ}\};$$

in particular $\mathsf{e}(\ell/2a_K) \ge \min\{0.069, 0.0775\,e^{-1.3\ell/\mathsf{d}_0(g,\vartheta_1)}\}$ whenever $\ell/a_K \ge 6$ is even, the same floor lies under D_2, 8δ and $\frac{8}{L^3}D_L$ at the side $\ell/2$, and $\mathsf{e}(l) \le 0.069$ fails for every l with $la_K < 0.0449\,\mathsf{d}_\circ$.

(4) *What BI forces.* If $\mathrm{BI}[\mathsf{b}, \ell_0, K_0; x_0]$ holds at some label, then $\ell_0 \ge 0.0895\,(\mathsf{d}_0^\star/\kappa_0)\,\mathsf{b}$, where $\kappa_0 = L(\gamma_W^{-2} + \bar{B})$, $\mathsf{d}_0^\star = \min\{\mathsf{d}_0(g_1,\vartheta_1), \mathsf{d}_0(g_2,\vartheta_1)\}$, $g_1 = \gamma_W$ and $g_2 = (\gamma_W^{-2} + 2\bar{B}/L)^{-1/2}$. On one lattice, every box and $t \ge 3$: $2F(t,\Sigma) - F(2t,\Sigma) \ge 2\log\sum_f w_f^{1/2}$, the w_f the weights of the box $(2t,\Sigma)$; on the cube $2l$, $\mathsf{e}(l) \ge 2\log(\sqrt{1-x} + \sqrt{2x})$ for $x = 1 - r_1 \le \frac{1}{2}$, $\mathsf{e}(l) \ge \log\frac{4}{1+3r_1}$, and $\mathsf{e}(l) < \log 2$ forces $r_1 > \frac{8}{9}$; so $\mathsf{e}(l) \le 0.069$ forces $r_1 \ge 0.99937$. At every $K \ge K_0$, on the cube $2l_K$ at $\beta = L^K\mathsf{b}$, with $x_0 = 0.069$: $1 - r_1 \le 0.000628$, $1 - r_2 \le 0.001255$, $1 - r_3 \le 0.001882$, $\delta_K(\ell_0) \le 0.013603$, and $\sum_{i\ge1}(\lambda_i/\lambda_0)^{l_K} \le 0.035764$ over the cross-section $(2l_K, 2l_K)$, so $\log(\lambda_0/\lambda_i) \ge 6.66\,a_K/\ell_0$ for $i \ge 1$. $\mathrm{BI}[\cdot; 0.069]$ is false as soon as $r_1 < 0.99937$ on the cube ℓ_0 at one cutoff $K \ge K_0$.

(5) *Small tori.* Let $\vartheta(\tau,u) = \sum_{k\in\mathbb{Z}} e^{-\tau(2\pi k+u)^2}$,

$$C_\Delta = \int_0^\infty \Big[1 - e^{-\tau} - \vartheta(\tau,0)^3 + (4\pi\tau)^{-3/2}\Big]\frac{d\tau}{\tau},$$

$$\Upsilon(u) = \exp\int_0^\infty \Big[e^{-\tau} - 1 + \vartheta(\tau,0)^3 - \prod_{\mu=1}^3 \vartheta(\tau,u_\mu)\Big]\frac{d\tau}{\tau},$$

$\mathscr{J} = \int_{[-\pi,\pi]^3} du/\Upsilon(u)$ and $c' = \log(\mathscr{J}/8\pi^2) - \frac{3}{2}C_\Delta$; numerically $c' \approx 2.0919$. There are numbers $t_1, t_2, t_4 > 0$ and $C, C_\flat$ of the constants, with $\mathsf{a} = \frac{9}{2}r + \mathfrak{p}_0$ and $\mathsf{q} = 1116\,r + 720\,\mathfrak{p}_0 + 366$, such that for every side $\ell = 2L^m$, $m \in \mathbb{Z}$, every real $\mathsf{b} > 0$ and every cutoff $K \ge 1 - m$, with $t = \ell/\mathsf{b}$: (a) if $t \le t_1$, then $|\delta_K(\ell) - \log(\mathsf{b}/\ell) - c'| \le C\,t^{1/1500} + C\,(a_K/\ell)^2\log^{5/2}(\ell/a_K)$, and $|\delta_\infty(t) - \log(1/t) - c'| \le C\,t^{1/1500}$; (b) if $t \le t_2$, then $r_1(\ell/a_K) \le \eta_\flat(t) = C_\flat(\log\frac{1}{t})^{\mathsf{a}}t^{1/500}$ at $\beta = L^K\mathsf{b}$, the weights of the four classes of electric flux on the cube satisfy $|w_f - \frac{1}{4}| \le \frac{3}{4}\eta_\flat(t)$, and $\delta_K(\ell)$ and $\mathsf{e}(\ell/2a_K)$ are both $\ge \mathsf{G}_{\ell/a_K} \ge \log\frac{4}{1+3\eta_\flat(t)}$; so $\liminf_{t\to0}\inf_K \delta_K(\ell) \ge \log 4$, the same for $\mathsf{e}(\ell/2a_K)$, and if $\ell_0 = 2L^{m_0}$, $\ell_0/\mathsf{b} \le t_2$ and $\eta_\flat(\ell_0/\mathsf{b}) < 0.99937$, the inequality of $\mathrm{BI}[\mathsf{b}, \ell_0, K_0; x_0]$ fails at every cutoff $K \ge 1 - m_0$, for every $x_0 \le 0.069$; (c) The sharper remainder in this clause requires the **(Heaviest class)** probability estimate: if $t \le t_4$, the intended conclusion is that over the cross-section $(\ell/a_K, \ell/a_K)$ the top level carries the weight $\lambda_0^N/Z(N^3) \le C\,t\,(\log\frac{1}{t})^{\mathsf{q}}$, $N = \ell/a_K$; so $\delta_K(\ell)$ and $\mathsf{e}(\ell/2a_K)$ are $\ge \mathsf{G}_N \ge \log\frac{\mathsf{b}}{\ell} - \mathsf{q}\log\log\frac{\mathsf{b}}{\ell} - \log C$: the coefficient of $\log(\mathsf{b}/\ell)$ in (a) is reached from below at every cutoff; (d) there are numbers $t_\circ > 0$ and $\beta_\flat$ of the constants such that for every real $\beta \ge \beta_\flat$: $\mathsf{e}(l) > 0.069$ for every integer $3 \le l \le t_\circ\beta$; so BI at any label b forces $\ell_0 > 2t_\circ\mathsf{b}$. On one lattice, exactly: if $1 - r_1(N) \ge 0.0039$ on one cube, then $\mathsf{e}(l) > 0.069$ for every integer $3 \le l \le N/16$; and $\mathsf{e}(l) \le 0.069$ gives $\mathsf{e}(l') < 0.0234$ for every integer $l' \ge 16l$; (e) the pressure of one lattice: $|P(\beta) + 3\log\beta - z_\infty| \le C\beta^{-1/1500}$ for all large real β, with $z_\infty = \log\frac{2}{\pi} - \frac{3}{16\pi^3}\int_{[-\pi,\pi]^3}\log\sum_\mu(2 - 2\cos p_\mu)\,dp \approx -2.96167$.

The leading logarithmic lower bound in clause (c) also holds without (Heaviest class), with q replaced by $\mathsf{p}_\star = 1209r + 780\mathfrak{p}_0 + 396$ and with possibly different constants t_4, C. Indeed, Prop. 17.6(d), with $\mathsf{L} = L$ and $j = m + K \ge 1$, applies to $N = \ell/a_K = 2L^j$ and $\beta = N/t$ and gives $\lambda_0(N,N)^N/Z(N,N,N) \le \mathsf{C}_\star t(\log(1/t))^{\mathsf{p}_\star}$. Since $\mathsf{G}_N \le \delta_K(\ell)$ and $\mathsf{G}_N \le \mathsf{e}(N/2)$, taking negative logarithms gives both lower bounds.

(6) *An area from above.* If $t = \ell/\mathsf{b} > t_1$ and j is the least integer with $L^{-j}t \le t_1$, then $\delta_K(\ell) \le 3L^2(t/t_1)^2\,[\log(L/t_1) + c' + C]$ at every cutoff $K \ge 1 - m + j$. So $\sup_K \delta_K(\ell) < \infty$ on every torus at every label, and $\mathsf{e}(\ell/a_K)$, $D_L(\ell/a_K)$ are $O((\ell/\mathsf{b})^2)$ uniformly in the cutoff. ■□

17.2.3. *Construction estimates.* Beyond the exact statements on one lattice proved below, the argument uses the following statements, (M1)–(M6) and (M8), one statement of the section on plane fields and the stress tensor (§5.1), Lem. 5.16, entered as (M7), and the comparison of two cutoffs with a factor, Prop. 17.30, which is derived at the end of this subsection from the statements named in (M4)(ii).

(M1) The layers and the crossing operator of one direction (Section YM3_I:85, Prop. YM3_I:85.1): the objects of the setting; a bounded Borel function of the bonds with both ends in a slab of $\vartheta \ge 2$ consecutive layers is Φ_{τ_0} for a bounded Borel function Φ on the configurations $\mathcal{Y}_\vartheta$ of a slab, which depends neither on the length of the box in the time nor on τ_0; its slab operator $\mathfrak{Y}[\Phi] = \mathsf{K}^{1/2}\mathfrak{S}[\Phi]\mathsf{K}^{1/2}$ is Hilbert–Schmidt, $\mathfrak{Y}[1_\vartheta] = \mathsf{T}'^{\vartheta}$, $\mathfrak{Y}[\Phi^{+j}] = \mathsf{T}'^{j}\mathfrak{Y}[\Phi]$ for Φ regarded as a function Φ^{+j} on a slab thicker by j layers, read on the first ϑ of its $\vartheta + j$ layers, so that the j added layers lie above, towards the later slabs of the trace formula below, in which the first slab stands rightmost, and $\mathfrak{Y}[\Phi^\Theta] = \mathfrak{Y}[\Phi]^*$ for the conjugated mirror image Φ^Θ, which, placed on a slab of ϑ layers, is the complex conjugate of Φ read on the configuration mirrored in the plane midway between that slab and the slab of Φ; and the trace formula (Prop. YM3_I:85.2): for $r \ge 2$ functions $\Phi^1, \dots, \Phi^r$ on consecutive slabs filling the box (t, Σ),

$$Z(t,\Sigma)\,\mu_{(t,\Sigma)}\Big(\prod_s (\Phi^s)_{\tau_{s-1}}\Big) = \mathrm{Tr}(\mathfrak{Y}[\Phi^r]\cdots\mathfrak{Y}[\Phi^1]).$$

(M2) The chessboard estimate on the tori $\ell \in \mathfrak{L}$, for one board of side $\ell''/2$, $\ell'' \le \ell$ in $\mathfrak{L}$ ([7, Cor. B.1](b), (c), (e)): $\|\cdot\|^{(\ell'')}_{\mathrm{ch},\ell}$ is a seminorm on the functions of one cell, with $\|1\|^{(\ell'')}_{\mathrm{ch},\ell} = 1$ and $|\mu_{K,\ell,\beta}(F)| \le \|F\|^{(\ell'')}_{\mathrm{ch},\ell} \le \sup|F|$; in its array form $|\mu_{K,\ell,\beta}(\prod_{\mathbf{k}} F_{\mathbf{k}})| \le \prod_{\mathbf{k}} \|F_{\mathbf{k}}\|^{(\ell'')}_{\mathrm{ch},\ell}$ for functions $F_{\mathbf{k}}$ of distinct cells $\mathbf{k}$ of one period, given in all of them or in some of them only; in its fine-to-coarse form, for one board and two tori, $\|F\|^{(\ell'')}_{\mathrm{ch},\ell} \le \|F\|^{(\ell'')}_{\mathrm{ch},\ell'} \exp[(\ell''/2)^3(p_{\ell'} - p_\ell)]$ and $p_\ell \le p_{\ell'}$ for $\ell'' \le \ell' \le \ell$ in $\mathfrak{L}$; and the domination $|\nu(\prod_{\mathbf{k}} F_{\mathbf{k}})| \le \liminf_{\ell\to\infty} \prod_{\mathbf{k}} \|F_{\mathbf{k}}\|^{(\ell'')}_{\mathrm{ch},\ell}$ of a periodic Gibbs state ν, for one board and growing tori (Prop. YM3_I:91.9). Together: for a function F of a cell of side $\ell''/2$ and its board of side $\ell/2$ of the setting, whose eight copies of F are copies of F under the board of F, the array form with the constant 1 in the other cells and the fine-to-coarse form give

$$\|F\|_{\mathrm{ch},\ell} \le \|F\|^{(\ell'')}_{\mathrm{ch},\ell} \le e^{(\ell''/2)^3(p_{\ell''} - p_\ell)}\,\|F\|_{\mathrm{ch},\ell''} :$$

the seminorm of eight copies on the torus ℓ is at most that of all the copies under the board of F, which is at most a factor times the seminorm on the torus ℓ''. (M3) Limits of tori and of periodic boxes are periodic Gibbs states (Prop. YM3_I:91.2); the DLR equations and the conditional laws of a Gibbs state; all periodic Gibbs states are invariant under the isometries of the lattice and reflection positive (Prop. YM3_I:103.4); a periodic Gibbs state is an integral of ergodic ones, and it is extreme if and only if it is ergodic (Prop. YM3_I:106.1). (M4) Thm. YM3_I:D, through the following consequences: (i) on one fixed torus the cumulants $S^T_{n;K,\ell,L^K\mathsf{b}}$ of fields with disjoint supports are bounded at all cutoffs, with bounds that depend on the side, a compact set of labels and the distance of the supports (Thm. YM3_I:AS, uniform bound YM3_I:(2214)), and converge as $K \to \infty$, uniformly on compact sets of labels, to continuous functions of the label (Thm. YM3_I:AS, limit and local uniformity), with the Cauchy rate proved in Thm. YM3_I:AS: on a compact interval of labels, $|S^T_{n;K',\ell,L^{K'}\mathsf{b}}(X) - S^T_{n;K,\ell,L^K\mathsf{b}}(X)| \le C\,K^{2\mathfrak{p}_0(n+1)+\frac12}L^{-K/10}\prod_i \|f_i\|_\sharp$ for all $K' \ge K$ above a threshold, where C and the threshold depend on n, the side, the interval and a floor of the spread of the supports, and on the functions only through their norms, at every admitted choice of

the constants; (ii) for a label $\mathsf{b} \geq \gamma^{-2}$, a side $\ell = 2L^m$ with $MR(\mathsf{b}) \leq L^m$ and a cutoff $K \geq 2$ at which the conditions (k0)–(k4) and (k6) of §2.3.1 hold, with its numbers at $n = 1$ ($u = \lceil K/1260 \rceil$, $s = \lceil K/9 \rceil$, $t_1 = K - u$),

$$\left|\log Z_{K+1,\ell} - \log Z_{K,\ell} - \mathfrak{K}_K(\ell, \mathsf{b})\right| \leq 4\mathsf{R}_{t_1} + 8\epsilon_{\mathrm{fg}} + 8\epsilon_{\mathrm{cl}} + 12\epsilon_{\mathrm{sh}} \leq \mathfrak{o}_K(\ell, \mathsf{b}),$$

where $\mathfrak{K}_K$, a sum of local vacuum constants, is the real part of the difference at the source 0 of the contact terms of the two cutoffs, and $\mathfrak{o}_K = 8\mathsf{R}_{t_1} + 12\epsilon_{\mathrm{fg}} + 8\epsilon_{\mathrm{cl}} + 12\epsilon_{\mathrm{sh}}$ is the number (2.xiv). This is the comparison of the two cutoffs at the source 0: the discarding of Prop. YM3_I:115.2, lost-mass bound gives $(1 - \mathsf{R}_{t_1})Z \leq T(0) \leq Z$ at each cutoff for the chain T of the top level; the form $T = e^c I$ and the contact terms of the proof of Prop. YM3_I:116.1 and Section YM3_I:116, with the proof of Prop. YM3_I:116.3, give $|I(0)/\hat{I}(0) - 1| \leq 2\epsilon_{\mathrm{fg}}$ against the classical start $\hat{T} = e^{\hat{c}}\hat{I}$; §2.3.2 and Prop. YM3_I:116.1 give $|\hat{T}_{K+1}(0)/\hat{T}_K(0) - 1| \leq 4\epsilon_{\mathrm{cl}} + 6\epsilon_{\mathrm{sh}}$ for the classical starts of the two cutoffs; and $|\log(1+\zeta)| \leq 2|\zeta|$ for $|\zeta| \leq \frac{1}{2}$.

The rate $\mathfrak{o}_K \leq \mathsf{C}_{\mathrm{bf}}\, \ell^3 K^{\mathfrak{p}_0} L^{-K/10}$ holds at every $K \geq K_{\mathrm{bf}}$, for the labels of an interval J of labels admitted on the side ℓ (Def. 2.7(5), Cor. 2.8(a), (2.xvii)): $K_{\mathrm{bf}} = K_{\mathrm{bf}}(\ell, J)$ is the least cutoff from which the conditions hold and the two fast members, the large fields $8\bar{\mathsf{R}}$ and the shell term $12\bar{\epsilon}_{\mathrm{sh}}$, majorised on J, lie below half of $\mathsf{C}_{\mathrm{bf}}\ell^3 K^{\mathfrak{p}_0} L^{-K/10}$; it depends on the side, through the factor $\exp[(\mathsf{C}^{\Sigma}_{\mathrm{maj}} + 2 + C_{\mathrm{low}}(\mathsf{b}))\ell^3]$ of R_{t_1}, and on J, while C_{bf} depends on J and on no side.

The same explicit numbers give a form uniform in the label, derived in the proof of Prop. 17.47(b): there are numbers $y_\sharp$, $C_{\mathfrak{o}}$ of the constants such that at every label with $\log \mathsf{b} \geq y_\sharp$, on every side $2MR(\mathsf{b}) \leq \ell \leq 2L^3 M(\log \mathsf{b})^r$ and at every $K \geq 11(5 + 2\mathfrak{p}_0/r)\log_L \ell$, the conditions (k0)–(k4), (k6) hold and $\mathfrak{o}_K(\ell) \leq C_{\mathfrak{o}}\, \ell^{3+2\mathfrak{p}_0/r} L^{-K/11}$, by the condition $2\mathfrak{p}_0 > 3r + 2$ of the list; (iii) for a label $\mathsf{b} \geq \gamma^{-2}$, a side $\ell = 2L^m$ with $MR(\mathsf{b}) \leq L^m$, a cutoff $K \geq 2$ at which the conditions (k0)–(k4) and (k6) of §2.3.1 hold, every integer $0 \leq d \leq \lceil K/1260 \rceil$ and every bounded Borel function F, invariant under the gauge transformations, of the configurations of the lattice of the spacing L^{-d} on the torus ℓ:

$$\left|\langle F(\mathcal{M}^{K+1-d}U)\rangle_{K+1} - \langle F(\mathcal{M}^{K-d}U)\rangle_K\right| \leq \mathfrak{o}_K \sup|F|$$

(Cor. 2.8(a)), with the number $\mathfrak{o}_K$ of (ii), which is $\leq \mathsf{C}_{\mathrm{bf}}\, \ell^3 K^{\mathfrak{p}_0} L^{-K/10}$ at every $K \geq K_{\mathrm{bf}}$ by (2.xvii), so that $\sum_K \mathfrak{o}_K < \infty$; the block field is taken at the fixed spacing L^{-d}, and the number $K - d$ of its averaging steps grows with the cutoff; (iv) the exact identification of the zero-source contact terms with the vacuum constants of their first $K^\bullet - u$ steps, proved in Step 3(iv) of Prop. 13.7, and their two-torus comparison in Lem. 13.31; Prop. 17.44 below retains this stopping depth and proves the resulting summable estimate; (v) the averaging map among the construction results (equations YM3_I:(37)–(39)), which sends a configuration U to its block field $V_j = \mathcal{M}^j U$ of level j, whose blocks are the cubes of L^j bare sites a side centred at the block sites, the points of $L^j\mathbb{Z}^3$, so that the block layer 1 of level q lies in the bare layers $\frac{1}{2}(L^q + 1), \ldots, \frac{1}{2}(3L^q - 1)$: it is Borel (Prop. YM3_I:4.5) and gauge covariant, it commutes with the isometries of the lattice that carry $L^j\mathbb{Z}^3$ onto itself ([7, Lem. G.1]), a block variable depends only on the bare bonds of the two blocks that its bond joins (Prop. YM3_I:4.5), on bare fields whose plaquettes all lie within ω_0 of $\mathbf{1}$, with $L^{2q}\omega_0$ below a number of L, the block plaquettes of level $j \leq q$ lie within $1.003\, L^{2j}\omega_0$ of $\mathbf{1}$ (Lem. 5.28), and on a cube of N bare sites a side the holonomies of the straight closed lines of the block field of level q lie within $13.3\, NL^q\omega_0$ of those of the bare field (Lem. 5.29); (vi) the comparison of (iii) with a factor, Prop. 17.30 at the end of this subsection, derived there from Prop. YM3_I:115.2 and the statements of Section YM3_I:116 named in (ii): in the setting of (iii), for every gauge-invariant Borel function F with values in $[0,1]$ of the block field of depth d, $\mu_d^{K+1}(F) \leq e^{m_K}\mu_d^K(F) + \epsilon_K$ and $\mu_d^K(F) \leq e^{m_K}\mu_d^{K+1}(F) + \epsilon_K$, where μ_d^K is the law of $\mathcal{M}^{K-d}U$ at the cutoff K and m_K, ϵ_K are the explicit numbers of that proposition: the errors that are powers of L^{-K} stand in the exponent m_K, the large fields of Prop. YM3_I:115.2, lost-mass bound and the shell term in the additive remainder ϵ_K.

Multiplying these inequalities: there are a number $y_\times$ and a power $\mathsf{N}(y)$ of y, both formed from the constants, such that at every label with $y = \log \mathsf{b} \ge y_\times$, on the least side ℓ at which (iii) applies, and from the cutoff $K_1 = K_1(y)$ of that proposition, at which the bare cube has $\ell L^{K_1} < \mathsf{N}(y)$ sites a side,

$$\mu_0^K(E) \le 1.7\,(\mu_0^{K_1}(E) + e^{-4y}) \qquad \text{for every } K \ge K_1$$

and every Borel set E of block fields of depth 0 that the gauge transformations carry onto itself: a factor and a remainder, not an additive error summed over the cutoffs. (M5) Thms. YM3_I:AJ and YM3_I:AD, through: (i) the bounds on cumulants at first-passage data: for $k \ge 2$, every first-passage datum (K,β) at g, every torus $\ell \in \mathfrak{L}(g)$ with $\ell \ge 16k(R+1)$ and k real Lipschitz functions with supports in $(-\frac{\ell}{4},\frac{\ell}{4})^3$, each in a ball of radius R, any two at a distance $\ge \mathsf{d}$, $|S^T_{k;K,\ell,\beta}(X)| \le C^g_k(\mathsf{d};R)\prod_i \|f_i\|_\sharp$; the bounds depend, besides on g, k, d, R and the constants, only on a radius fixed once in that theorem, and neither on the cutoff nor on the coupling nor on the side nor on the place; (ii) the theorem on the canonical decomposition, the ergodic components and the reconstruction, Thm. YM3_I:AP: each ergodic component of the limit of a convergent Gibbs sequence is a limit along the same pairs whose approximants are periodic Gibbs states at these pairs (Thm. YM3_I:AP, component approximation in the proof), and it carries the reconstruction of Thm. YM3_I:AP; (iii) the law of the two-point function at short distance (Thm. YM3_I:AV, short-distance bound and torus version), with the ceiling γ_W and the threshold $\mathsf{d}_0(g,\vartheta)$ of equations YM3_I:(2346) and YM3_I:(2349)–(2350), the depth $s(\mathsf{d})$ of equation YM3_I:(2321) and the test pairs of [7, Lem. I.1]: for $0 < g \le \gamma_W$, $\mathsf{d} \le \mathsf{d}_0(g,\vartheta)$, every first-passage datum (K,β) at g with $K > s(\mathsf{d})$, every periodic Gibbs state ν at it and every test pair (f,h) at $(\mathsf{d};\vartheta)$,

$$S_2^T[\nu](f,h) = \mathsf{c}_2\,\mathsf{b}^{-2}\,\mathcal{K}_{\text{free}}(f,h)\,(1+\varepsilon'), \qquad |\varepsilon'| \le \tfrac{1}{3}, \qquad \mathcal{K}_{\text{free}}(f,h) = \iint f(x)h(y)|x-y|^{-6}\,\mathrm{d}x\,\mathrm{d}y,$$

with $\mathsf{c}_2 > 0$ and $\mathcal{K}_{\text{free}}$ as in Thm. YM3_I:C and equation YM3_I:(2240); for $f(y) = (1 - |y - 2\mathsf{d}e_3|^2/\mathsf{d}^2)_+^2$ and $0 \le t \le 4\mathsf{d}$ the pair $(f(\Theta\,\cdot), f(\cdot - te_3))$, Θ the reflection $x_3 \mapsto -x_3$, is a test pair at $(\mathsf{d};\vartheta_1)$ ([7, Lem. I.1](c)), and $K > s(\mathsf{d})$ gives $\mathsf{d}/a_K > 4L^3M$. Besides, $I(g) = (u, Lu - 2\bar B]$ with $u = g^{-2} + \bar B$, $\bar B \ge 0$, and $g^{-2} > L^2\bar B$ for $g \le \gamma_W$ (the coupling ceiling preceding Thm. YM3_I:AV, equation YM3_I:(2346)). The labels $a_K\beta$ of all first-passage data (K,β) at g lie in the window $(g^{-2} + B_\sharp\frac{L-2}{L-1},\, L(g^{-2}+\bar B)]$ (YM3_I:(1531) in Prop. YM3_I:74.1), whose upper end is $X_g + \bar B$. (M6) The ordered choice leaves the blocking factor free upward (the choice of L in the proof of Prop. YM3_I:69.1, followed by the source choices in Section YM3_I:82): every odd integer above a threshold is the value of L at an admissible choice, and Thms. YM3_I:D, YM3_I:AJ and YM3_I:AD hold at every admissible choice. (M7) The member of third order in the curvature of the bulk term of the rotational Ward identity, on growing tori: Lem. 5.16 of the section on plane fields and the stress tensor (§5.1), which carries the bound of the cubic remainder on one torus, Lem. 5.11, to all sides at once. For $\chi, f_1, \dots, f_n$ and ω as in Thm. BA(5), this member of the bulk term of YM3_I:(2108) on the torus ℓ is

$$\mathfrak{U}^\sharp_{\omega,\chi}(f_1,\dots,f_n) = -\beta\langle \mathfrak{K}^\sharp[\chi\,\omega y]; \mathcal{O}_K(f_1); \dots; \mathcal{O}_K(f_n)\rangle_{K,\ell,\beta} + a_K^{-1}\sum_{i=1}^n \langle \mathfrak{K}^\sharp[f_i\,\omega y]; (\mathcal{O}_K(f_l))_{l\ne i}\rangle_{K,\ell,\beta},$$

the joint cumulants in which the local density of the rotation is replaced by its part $\mathfrak{K}^\sharp[\boldsymbol{\eta}] = \sum_p a_K^{-1}\boldsymbol{\eta}^{\rho(p)}(x(p))\,\mathfrak{r}^\sharp_{\rho(p)}(p)$ of [7, §H], built with the remainders $\mathfrak{r}^\sharp$ of third order in the field strength ([7, Prop. H.1]), with the weight β/a_K; here $\boldsymbol{\eta}^\rho(y) = \chi(y)(\omega y)_\rho$, resp. $f_i(y)(\omega y)_\rho$. As $\mathfrak{K}^\sharp$ depends on its field only through its values at the centres of the plaquettes, at which $\chi f_i = f_i$, this member is the cubic remainder $\mathfrak{U}^\sharp[\chi\,\omega y](f_1,\dots,f_n)$ of that section at the field $\chi\,\omega y$. Let $0 < g \le \gamma$, $n \ge 2$, $\mathsf{d} > 0$ and $\rho_0 \ge 1$. There are $C_\sharp$ and $K_\sharp$, formed from g, n, d, ρ_0 and the constants and depending on no side, such that for all first-passage data (K,β) at g with $K \ge K_\sharp$, all sides $\ell \ge 2\rho_0 + 2$ of $\mathfrak{L}(g)$, all $\omega \in \mathfrak{so}(3)$ and all real $\chi, f_1, \dots, f_n \in C^{1,1}(\mathbb{R}^3)$ that vanish outside $[-\rho_0,\rho_0]^3$, χ equal to 1

at the centres and at the bonds of the plaquettes at which some f_i does not vanish, the supports of the f_i at pairwise sup distances $\geq \mathsf{d}$,

$$|\mathfrak{U}^\sharp_{\omega,\chi}(f_1,\dots,f_n)| \leq C_\sharp\,|\omega|\,L^{-K/16}\,\|\chi\|_{C^{1,1}}\prod_i \|f_i\|_{C^{1,1}}.$$

This is Lem. 5.16 at the field $\chi\,\omega y$, for the cube of side $\frac14\ell'$ centred at 0, ℓ' the least side of $\mathfrak{L}(g)$ with $\ell' \geq 8\rho_0$, and for a rate $c \in (\frac{1}{16},\frac{1}{14})$: there $K^pL^{-cK} \leq C'L^{-K/16}$, the norm $\mathsf{H}(\eta) = \max_\rho \sup|\eta^\rho| + \max_\rho \mathrm{Lip}_\infty \eta^\rho$ of a field in that lemma, Lip_∞ the Lipschitz constant for the sup distance, is $\mathsf{H}(\chi\,\omega y) \leq C\rho_0|\omega|\,\|\chi\|_{C^{1,1}}$ with a pure number C, and $\|f_i\|_\sharp \leq \max\{1,40M\}\|f_i\|_{C^{1,1}}$. On each of the finitely many sides of $\mathfrak{L}(g)$ from $2\rho_0+2$ to ℓ' it is Lem. 5.11, whose setting holds on that side at all first-passage data at g: their labels lie in $[\gamma^{-2}, X_g+\bar B]$, and $MR(X_g+\bar B) \leq L^m$ on every side $2L^m$ of $\mathfrak{L}(g)$. The idea of the proof. On one torus the entry $\mathfrak{K}^\sharp$ enters each renormalisation step as one more local term, inside the radii of the one-step theorem with sources (Prop. 6.32). Each of its terms vanishes to the third order at $\mathbf{1}$ and is odd under the mirror of its plaquette, while the steps are covariant under the mirrors of their lattices; so only the differences of the backgrounds and their mirror images and the asymmetric part of the weight feed the families that the steps make of it, and what is fed at a level is forgotten at the rate $L^{-1/2}$ a level. The conditional mean of the entry, given the block field at a level $K - \lceil \phi K\rceil$ with a small $\phi > 0$, is therefore a number up to an oscillation of the order K^pL^{-cK} (Lem. 5.10(b)), the large fields costing less than every power of L^{-K}; the cumulant is a mixed derivative at 0 of the logarithm of the partition function with the entry and the sources, and Cauchy's estimates in the sources give $|\mathfrak{U}^\sharp| \leq CK^pL^{-cK}$ times the norms at every rate $c < \frac{1}{14}$ (Lem. 5.11), which is at most $C'L^{-K/16}$.

On all the sides at once, the carrier is put in one cell of a board of side $\frac12\ell'$, ℓ' a fixed side of $\mathfrak{L}(g)$; the chessboard estimate controls the generating function of the entry and of the sources on the torus ℓ by the mean of their mirror copies on the torus ℓ', with a factor $e^{(\ell'/2)^3(p_{\ell'}-p_\ell)}$ that is bounded uniformly at the first-passage data at g; on the torus ℓ' the bound of one torus applies to the copies, and the part of the cumulants that links different copies carries the oscillation of the conditional mean of Lem. 5.10(b); the finitely many sides between $2\rho_0+2$ and ℓ' are tori of their own. (M8) The rotational Ward identity among the construction results, which Thm. BA(5) and its proof read: the bulk term YM3_I:(2107) of equation YM3_I:(2107) and the source calculation in the proof of Prop. YM3_I:112.1, the bulk insertion of equation YM3_I:(2107) and the source calculation in the proof of Prop. YM3_I:112.1, whose mean at the sources gives the bulk term by differentiation in the source variables, and the identity YM3_I:(2108); the orbital potential $\Psi_\rho(p)$ and the remainder of third order $\mathfrak{r}^\sharp_\rho(p)$ of [7, §H], with the bound of third order [7, eq. 295], and the summation by parts of [7, Prop. H.1], which writes the bulk insertion as the sum of the spin part, a member of order 2 in the curvature and the cubic member [7, eq. 297]; the small-field density with a local insertion and its derivative as a conditional mean in Lem. 5.10(a), under its stated entry and cutoff hypotheses, obtained from the holomorphic fourth-family map of Prop. 6.32; and the two-constants inequality of Lem. YM3_I:116.2, used in the source variables.

Proposition 17.30 (Two cutoffs with a factor)**.** Let $\mathsf{b} \geq \gamma^{-2}$ and $\ell = 2L^m$ with $MR(\mathsf{b}) \leq L^m$; for a cutoff $K \geq 2$ take the numbers $\epsilon_{\mathrm{fg}}, \epsilon_{\mathrm{cl}}, \epsilon_{\mathrm{sh}}, \mathsf{R}_{t_1}$ of §2.3.1 at $n=1$ ($u = \lceil K/1260\rceil$, $s = \lceil K/9\rceil$, $t_1 = K-u$), and put

$$m_K = 4\epsilon_{\mathrm{fg}} + 2\epsilon_{\mathrm{cl}} + 1.06\,\epsilon_{\mathrm{sh}} + 1.39\,\mathsf{R}_{t_1}, \qquad \epsilon_K = \mathsf{R}_{t_1} + 1.29\,\epsilon_{\mathrm{sh}}.$$

(a) At every cutoff $K \geq 2$ at which (k0)–(k4) and (k6) hold, for every integer $0 \leq d \leq \lceil K/1260\rceil$ and every Borel function F with values in $[0,1]$ of the block fields of depth d on the torus ℓ that is invariant under their gauge transformations,

$$\mu_d^{K+1}(F) \leq e^{m_K}\mu_d^K(F) + \epsilon_K, \qquad \mu_d^K(F) \leq e^{m_K}\mu_d^{K+1}(F) + \epsilon_K,$$

μ_d^K the law of $\mathcal{M}^{K-d}U$ at the cutoff K (Def. 2.7(2)).

(b) If the conditions hold at every $K \ge K_1$, and $\mathsf{S} = \sum_{K\ge K_1} m_K$ and $\mathsf{R}_\Sigma = \sum_{K\ge K_1} \epsilon_K$ are finite, then $e^{-\mathsf{S}}\mu_d^{K_1}(E) - \mathsf{R}_\Sigma \le \mu_d^K(E) \le e^{\mathsf{S}}(\mu_d^{K_1}(E) + \mathsf{R}_\Sigma)$ for every $K \ge K_1$, every integer $0 \le d \le \lceil K_1/1260\rceil$ and every Borel set E of block fields of depth d that the gauge transformations carry onto itself.

(c) Let $p = 2\mathfrak{p}_0$, $y = \log \mathsf{b}$, $\bar\ell(y) = 2L^2 M y^r$, $\mathsf{N}(y) = L\,\bar\ell(y)^{31} y^{10(p+1)}$, and let $K_1(y)$ be the least integer $K \ge 10$ with $L^{K/10} \ge \bar\ell(y)^3 y^{p+1}$. There is a number $y_\times$, formed from the constants, such that at every label b with $y \ge y_\times$, on the least side $\ell = 2L^m$, $m \in \mathbb{N}$, with $MR(\mathsf{b}) \le L^m$, which has $\ell \le \bar\ell(y)$: the conditions (k0)–(k4) and (k6) hold at every $K \ge K_1(y)$; $\ell L^{K_1(y)} < \mathsf{N}(y)$; from $K_1 = K_1(y)$, $\mathsf{S} \le 0.53$ and $\mathsf{R}_\Sigma \le e^{-4y}$; and so $\mu_0^K(E) \le 1.7\,(\mu_0^{K_1(y)}(E) + e^{-4y})$ for every $K \ge K_1(y)$ and every Borel set E of block fields of depth 0 that the gauge transformations carry onto itself.

■

Proof. (a) Write $\langle F\rangle_K = \mu_d^K(F)$. The averaging maps from the depth u to the depth d are the same at the two cutoffs and commute with the gauge transformations, so $F \circ \mathcal{M}^{u-d}$ is a function of the same kind at the depth u, and we may take $d = u$. Three comparisons follow, each with the same function $F \ge 0$ inserted on both sides. (1) *Large fields.* By Prop. YM3_I:115.2, image identity and domination bound, the chain of the top level t_1, on its inner ball, is the image under $\mathcal{M}^{t_1}$ of $\mathfrak{w}_{t_1}^{\mathrm{sf}}\rho_0\,\mathrm{d}U$, and its integral of a gauge-invariant $F \ge 0$ is at most $\int F(\mathcal{M}^{t_1}U)\rho_0\,\mathrm{d}U$; by the discarding of Prop. YM3_I:115.2, lost-mass bound its mass is at least $(1 - \mathsf{R}_{t_1})Z$.

So the mean of F under the law exceeds its mean under the normalised chain by at most $\mathsf{R}_{t_1}\sup F$, and the latter is at most $(1-\mathsf{R}_{t_1})^{-1}$ times the former. (2) *The chain and the classical start.* On the small-field set the two measures have the positive densities $e^{c(0)}e^{\Phi}$ and $e^{\hat c(0)}e^{\hat\Phi}$ (the proof of Prop. YM3_I:116.1), and $|\Phi - \hat\Phi| \le \epsilon_{\mathrm{fg}}$ at every block field of the cut set (equation YM3_I:(2180)); so the two normalised means of F differ by a factor within $e^{\pm 2\epsilon_{\mathrm{fg}}}$, the constants $e^{c(0)}$, $e^{\hat c(0)}$ cancelling against $F = 1$. (3) *The classical starts of the two cutoffs.* They are integrals over the configurations of the depth s of the weights $\mathsf{k} \ge 0$ of the two cutoffs times densities that are positive on the cut sets (Section YM3_I:116, §2.3.2); where both weights are positive they are equal and the densities differ by a factor within $e^{\pm\epsilon_{\mathrm{cl}}}$ (Prop. YM3_I:116.1, density comparison in the proof), and where one of them vanishes the other carries at most the part ϵ_{sh} of the mass (§2.3.2); so the normalised mean at one cutoff is at most $e^{2\epsilon_{\mathrm{cl}}}(1-\epsilon_{\mathrm{sh}})^{-1}$ times that at the other plus $\epsilon_{\mathrm{sh}}\sup F$.

Chaining (1), (2), (3), (2), (1), for $\{K', K''\} = \{K, K+1\}$,

$$\langle F\rangle_{K'} \le \frac{e^{4\epsilon_{\mathrm{fg}}+2\epsilon_{\mathrm{cl}}}}{(1-\epsilon_{\mathrm{sh}})(1-\mathsf{R}_{t_1})}\,\langle F\rangle_{K''} + \big(\mathsf{R}_{t_1} + e^{2\epsilon_{\mathrm{fg}}}\epsilon_{\mathrm{sh}}\big)\sup F,$$

and (k2), (k3), (k6) give $e^{2\epsilon_{\mathrm{fg}}} \le e^{1/4} < 1.29$ and, by convexity, $-\log(1-\epsilon_{\mathrm{sh}}) \le 1.06\,\epsilon_{\mathrm{sh}}$ and $-\log(1-\mathsf{R}_{t_1}) \le 1.39\,\mathsf{R}_{t_1}$. (b) By induction on K, $\langle F\rangle_K \le e^{\sigma_K}(\langle F\rangle_{K_1} + \rho_K)$ and $\langle F\rangle_K \ge e^{-\sigma_K}\langle F\rangle_{K_1} - \rho_K$, with the sums σ_K, ρ_K of the m_k, ϵ_k over $K_1 \le k < K$; $d \le \lceil K_1/1260\rceil \le \lceil K/1260\rceil$ keeps every step in the range of (a), and $F = \mathbf{1}_E$. (c) Let $K \ge K_1(y)$; with $\ell \le \bar\ell(y)$, $\ell^3 L^{-K/10} \le y^{-p-1}$. As $u \ge K/1260$ and $s \ge K/9$, the definitions YM3_I:(2192) and YM3_I:(2193) give $\epsilon_{\mathrm{fg}} \le a_1 \ell^3 L^{-K/10}$ and $\epsilon_{\mathrm{cl}} \le a_2\, y^p(1 + K\log L)^p K^{1/2} L^{-K/90}\,\ell^3 L^{-K/10}$ with numbers a_1, a_2 of the constants, so that $\sum_{K\ge K_1(y)}(4\epsilon_{\mathrm{fg}} + 2\epsilon_{\mathrm{cl}}) \le (17.72\,a_1 + 2a_2a_3)/y$, $a_3 = \sum_{K\ge1}(1+K\log L)^p K^{1/2}L^{-K/90}$.

The numbers R_{t_1} and ϵ_{sh} carry the factors $e^{-c'A_0^2 \mathsf{y}_{[u]}}$ and $e^{-\frac14\kappa_{\mathrm{b}}A_1^2\mathsf{y}_{[u]}}$, where $\mathsf{y}_{[u]} = (y + u\log L)^p \ge y^p + (u\log L)^p$, against the growth $\ell^3 L^{3s}$, $s \le 140u$, and, in R_{t_1}, $\exp[(\mathsf{C}^{\Sigma}_{\mathrm{maj}} + 2 + c_{\mathrm{low}} + \frac92 y)\ell^3] \le \exp[O(y^{3r+1})]$; summed over the cutoffs, of which at most 1260 have the same u, and by $2\mathfrak{p}_0 > 3r+2$, which follows from the exponent choice $r \ge 2$, $\mathfrak{p}_0 \ge 6r+1$ in Thm. YM3_I:D, $\sum_{K\ge K_1(y)}(\mathsf{R}_{t_1} + 1.29\,\epsilon_{\mathrm{sh}}) \le e^{-4y}$ for all large y. The same bounds give (k0)–(k4) and (k6) at every $K \ge K_1(y)$

for large y; $y_\times$ is the least number from which all this holds, with $\mathsf{b} \ge \max\{\gamma^{-2}, \Theta_\star\}$ and $y \ge 2(17.72\,a_1 + 2a_2a_3)$.

Then $\mathsf{S} \le \frac12 + 1.39\,e^{-4y} \le 0.53$, $e^{0.53} < 1.7$, and (b) at $d = 0$ gives the last inequality. As $L < \bar\ell(y)^3$, $K_1(y) \ge 11$, and by its least choice $L^{(K_1(y)-1)/10} < \bar\ell(y)^3 y^{p+1}$, so $\ell L^{K_1(y)} < \bar\ell(y)\,L\,(\bar\ell(y)^3 y^{p+1})^{10} = \mathsf{N}(y)$. □

17.2.4. *Statements on one lattice.*

Proposition 17.31 (The exact layer)**.** Let $\beta \ge 0$, $\Sigma = (A, B)$ a cross-section with sides ≥ 3, any coordinate direction being the time, and $t, t' \ge 3$, $q \ge 2$ integers.

(a) $Z(t,\Sigma) = \operatorname{Tr} \mathsf{D}^t = \sum_i \lambda_i^t$, $\sum_i \lambda_i^2 < \infty$, λ_0 is simple and its eigenvector can be taken strictly positive. F is symmetric in the three sides, and $e^{-6\beta|\underline{n}|} \le Z \le 1$.

(b) $\mathsf{r}_{t+1} \le \mathsf{r}_t$, $\mathsf{r}_{t+t'} \le \mathsf{r}_t\mathsf{r}_{t'}$, $\mathsf{r}_{qt} \le \mathsf{r}_t^q$; $F(t,\Sigma) = t\log\lambda_0 + \mathsf{G}(t;\Sigma)$, $F(t,\Sigma)/t$ does not increase, and $F(t+t',\Sigma) \le F(t,\Sigma) + F(t',\Sigma)$;

$$qF(t,\Sigma) - F(qt,\Sigma) = q\,\mathsf{G}(t;\Sigma) - \mathsf{G}(qt;\Sigma) \in \big[(q-1)\mathsf{G}(t;\Sigma),\, q\,\mathsf{G}(t;\Sigma)\big].$$

If $2F(t,\Sigma) - F(2t,\Sigma) \le x < \log 2$, then $\mathsf{r}_t \le \varphi(e^x - 1)$ and $2F(2t,\Sigma) - F(4t,\Sigma) \le \psi(x)$, where $\varphi(\eta) = \eta/(1+\sqrt{1-\eta^2})$ and $\psi(x) = 2\log(1+\varphi(e^x-1)^2)$.

(c) $F/|\underline{n}|$ and $\delta/|\underline{n}|$ do not increase when one side grows, and $F/|\underline{n}| \to P$ along boxes all of whose sides tend to ∞; $\delta(t,\Sigma) = t\,\varepsilon(\Sigma) + \mathsf{G}(t;\Sigma)$ with $\varepsilon(\Sigma) \ge 0$, so $\mathsf{G} \le \delta$; $D_q(N) = q^3\delta(N^3) - \delta((qN)^3) \in [0, q^3\delta(N^3)]$, $\mathsf{e}(l) = 2\delta(l,2l,2l) - \delta((2l)^3)$, and

$$\delta(N^3) = \sum_{s\ge0} q^{-3(s+1)} D_q(q^s N).$$

From above, an area: for all integers $A, B, t \ge N$, $\mathsf{G}(t;A,B) \le AB\varkappa$ and $\varepsilon(A,B) \le (A+B-N)\varkappa$, $\varkappa = \delta(N^3)/N^2$; so for every box with all sides $\ge N$

$$\delta(\underline{n}) \le \big[n_1n_2 + n_1n_3 + n_2n_3 - N(n_1+n_2+n_3) + N^2\big]\varkappa \le (n_1n_2 + n_1n_3 + n_2n_3)\,\varkappa,$$

and $\delta(M^3) \le (3\rho^2 - 3\rho + 1)\,\delta(N^3)$ for $M = \rho N \ge N$.

(d) There are commuting unitary involutions $\mathsf{J}_1, \mathsf{J}_2$ that commute with $\mathsf{T}^{(0)} = \mathsf{D}$ and with an operator $\mathsf{T}^{(1)} \ge 0$ of the same kind (D with the sign of one plaquette of the cross-section reversed in m), such that, with $\mathsf{J}^k = \mathsf{J}_1^{k_1}\mathsf{J}_2^{k_2}$ and $\mathsf{P}_e = \frac14\sum_k (-1)^{e\cdot k}\mathsf{J}^k$,

$$Z^{(k;m)}(t,\Sigma) = \operatorname{Tr}\big(\mathsf{J}^k(\mathsf{T}^{(m)})^t\big) = \sum_e (-1)^{e\cdot k} W(e;m), \qquad W(e;m) = \operatorname{Tr}\big(\mathsf{P}_e(\mathsf{T}^{(m)})^t\big) \ge 0.$$

Each $W(e;m)$ is a sum of t-th powers of positive numbers that do not depend on t; the eigenvector of λ_0 lies in the range of P_{00}, and $W(00;0) \ge \lambda_0^t$; $Z^z > 0$ does not depend on the columns chosen. On every box: $0 \le 1 - r^{(k;0)} \le 2\mathsf{R}_t$, $0 < r^{(k;m)} \le r^{(00;m)}$, $r^{(11;m)} \ge r^{(10;m)} + r^{(01;m)} - r^{(00;m)}$, and

$$\delta(t,\Sigma) = t\,\varepsilon(\Sigma) + \log(1+\mathsf{r}_t^{00}) + \log(1/\zeta_t), \qquad \zeta_t = \frac{W(00;0)}{Z} = \frac14\Big(1 + \sum_{k\ne0} r^{(k;0)}\Big),$$

three terms ≥ 0, $1 + \mathsf{r}_t^{00} = W(00;0)/\lambda_0^t$. On a cube N^3: $r_2, r_3 \le r_1 \le 1$; all seven $0 \le 1 - r^z \le 3\mathsf{R}_N \le 3\min\{\mathsf{G}_N, \mathsf{r}_N\}$; $1 - r_1 \le \frac43\mathsf{R}_N$, $1 - r_2 \le 2(1-r_1)$ and $1 - r_3 \le 3(1-r_1)$.

(e) $2F(t,\Sigma) - F(2t,\Sigma) \ge 2\log\sum_{f\in\{0,1\}^2} w_f^{1/2}$, the w_f being the weights of the box $(2t,\Sigma)$; and $\mathsf{G}(t;\Sigma) \ge \log(1/w_{00})$, w_{00} the weight of the box (t,Σ).

(f) Let $\Gamma = \mathbb{Z}n_1e_1 \oplus \mathbb{Z}n_2e_2 \oplus \mathbb{Z}(v_1e_1 + v_2e_2 + n_3e_3)$, $v \in \mathbb{Z}^2$, and Z_Γ the partition function of the lattice with these periods. With the time e_3 over $\Sigma = (n_1, n_2)$: $Z_\Gamma = \operatorname{Tr}(\mathsf{U}_v\mathsf{D}^{n_3})$, U_v the unitary of the translation of a layer by v, and $\lambda_0^{n_3}(1 - \mathsf{r}_{n_3}) \le Z_\Gamma \le Z(\underline{n})$.

(g) *The sheared cube.* Let $N = 2h$ be even, $\Gamma^\diamond = \mathbb{Z}N(e_1 - e_3) \oplus \mathbb{Z}Ne_2 \oplus \mathbb{Z}h(e_1 + e_3)$, a period lattice of the index N^3, and $Z_{\Gamma^\diamond}$ the partition function of the lattice with these periods. Then $Z(N^3) \le Z_{\Gamma^\diamond}$.

■

Proof. (a) The weight of the box is a product over the layers of $\mathsf{m}(V^\tau)$ and of the crossing weights $e^{-\beta A_\times(V^\tau, S^\tau, V^{\tau+1})}$, each S^τ standing in one factor; integrating the S^τ (Fubini) gives

$$\int \prod_\tau \mathrm{d}V^\tau \prod_\tau \mathsf{m}(V^\tau)\mathsf{K}(V^{\tau+1}, V^\tau),$$

and splitting each $\mathsf{m}(V^\tau)$ into two roots this is

$$\int \prod_\tau \mathsf{D}(V^{\tau+1}, V^\tau)\, \mathrm{d}V^\tau = \operatorname{Tr} \mathsf{D}^t,$$

for every $t \ge 3$. With $\mathsf{A} = \mathsf{K}^{1/2}\mathsf{m}^{1/2}$, $\mathsf{A}^*\mathsf{A} = \mathsf{D}$ and $\mathsf{A}\mathsf{A}^* = \mathsf{T}'$ have the same nonzero spectrum. The kernel of D is continuous and bounded below by a positive number on a compact space; if e_0 is a real eigenvector of λ_0, so is $|e_0|$, hence so are $|e_0| \pm e_0 \ge 0$, one of which vanishes on a set of positive measure and is therefore 0.

So every eigenvector of λ_0 has a strict sign, and no two of them are orthogonal. A permutation of the coordinates carries plaquettes to plaquettes and $\operatorname{tr} \partial U$ to itself; $0 \le a_p \le 2$ on $3|\underline{n}|$ plaquettes. (b) With $x_i = (\lambda_i/\lambda_0)^t \le 1$: $x_i^{(t+1)/t} \le x_i$, $\sum x_i^q \le (\sum x_i)^q$, $\sum \lambda_i^{t+t'} \le \sum \lambda_i^t \sum \lambda_i^{t'}$; $F(t)/t = \log \|\lambda\|_{\ell^t}$ does not increase in t. The identity and its range follow from $F(t) = t \log \lambda_0 + \mathsf{G}(t)$ and $0 \le \mathsf{G}(qt) \le \mathsf{G}(t)$. If $2F(t) - F(2t) \le x$, then $\mathsf{G}(t) \le x < \log 2$, $\mathsf{r}_t < 1$, and

$$e^x \ge \frac{(1+\mathsf{r}_t)^2}{1+\mathsf{r}_{2t}} \ge \frac{(1+\mathsf{r}_t)^2}{1+\mathsf{r}_t^2} = 1 + \frac{2\mathsf{r}_t}{1+\mathsf{r}_t^2},$$

which gives $\mathsf{r}_t \le \varphi(e^x - 1)$ as $r \mapsto 2r/(1+r^2)$ increases on $[0,1]$; and $2F(2t) - F(4t) \le 2\mathsf{G}(2t) \le 2\log(1+\mathsf{r}_t^2)$. (c) is (b) in each direction, divided by the product of the two other sides; $\varepsilon(\Sigma) = \lim_t (F(t,\Sigma) - t|\Sigma|P)/t \ge 0$; in D_q and e the terms with P cancel, and the series telescopes, its tail $q^{-3S}\delta((q^S N)^3) = N^3[F/|\cdot| - P]$ tending to 0. For the area: $\mathsf{G}(t; A, B) \le \mathsf{G}(N; A, B) \le \delta(N, A, B) \le AB\varkappa$ by the monotonicity of G and of $\delta/|\underline{n}|$. With the side B as the time, $\delta(s,t,B) = B\varepsilon(s,t) + \mathsf{G}(B; s, t) \le B\varepsilon(s,t) + st\varkappa$; dividing by s and then t and letting both tend to ∞, as $\varepsilon(s,t)/st \to 0$, gives $\limsup_t \varepsilon(t,B)/t \le \varkappa$.

Since $\delta(a,b,c) = a\varepsilon(b,c) + \mathsf{G}(a; b, c) \le \delta(N,b,c) + (a-N)\varepsilon(b,c)$ for $a \ge N$, $\varepsilon(A,B) \le \varepsilon(N,B) + (A-N)\varkappa \le \varepsilon(N,N) + (A+B-2N)\varkappa$, and $\varepsilon(N,N) \le \delta(N^3)/N = N\varkappa$. The same inequality in the three sides in turn gives $\delta(\underline{n}) \le \delta(N^3) + (n_3 - N)\varepsilon(N,N) + (n_2 - N)\varepsilon(N, n_3) + (n_1 - N)\varepsilon(n_2, n_3)$, which is the bracket. (d) As -1 is central in SU(2), reversing the sign of $\operatorname{tr} \partial U(p)$ on a column that crosses one time step is the same as multiplying by -1 the bonds of one line of the next layer; this change of variables keeps the Haar measure, appears under the trace as an involution J_i of the layer variables, and commutes with m and K.

A reversed column in the plane of the cross-section lies in every layer and replaces m by m_1. Moving a column by one step is a further flip of the bonds of a line, which does not change the trace. The P_e are the isotypical projections of $k \mapsto \mathsf{J}^k$; they commute with $\mathsf{T}^{(m)}$, so every eigenspace splits along them and $W(e;m) = \sum_{i: f(i) = e} \mu_i^t$. The positive top eigenvector is invariant under the flips, so it lies in the range of P_{00}. Then $Z - Z^{(k;0)} = 2\sum_{e \cdot k \text{ odd}} W(e;0) \le 2(Z - \lambda_0^t)$, $Z^{(k;m)} \le \sum_e W(e;m) = Z^{(00;m)}$, $4W(11;m) = Z^{(00;m)} - Z^{(10;m)} - Z^{(01;m)} + Z^{(11;m)} \ge 0$, and $\delta = t\varepsilon + \log(Z/\lambda_0^t)$ with $Z/\lambda_0^t = [W(00;0)/\lambda_0^t][Z/W(00;0)]$.

On a cube, by the cubic symmetry, with $u = w_{10} = w_{01}$, $v = w_{11}$: $1 - r_1 = 2(u+v)$, $1 - r_2 = 4u$, $2u + v = 1 - w_{00} \le \mathsf{R}_N$; two planes with a common direction as the time give $r_2 \le r^{(00;1)} = r_1$, hence $v \le u$ and $1 - r_1 \le \frac{4}{3}\mathsf{R}_N$; three planes give $r_3 \ge 2r_2 - r_1$; and the sum over e with a vanishing

component gives $1-r^{(1,1,1)} \le (1-r^{z'})+(1-r^{z''})$, so $1-r_3 \le (1-r_1)+(1-r_2) \le 3(1-r_1)$. (e) Within each class $(\sum \mu_i^t)^2 \ge \sum \mu_i^{2t}$, so $Z(t,\Sigma) \ge \sum_f W_{2t}(f)^{1/2} = Z(2t,\Sigma)^{1/2} \sum_f w_f^{1/2}$; and $Z/\lambda_0^t \ge Z/W(00;0)$. (f) Integrating the layers as in (a), the last layer is glued to the first through the translation by v, which produces U_v under the trace; U_v commutes with D, so in a common eigenbasis $Z_\Gamma = \sum_i \omega_i \lambda_i^{n_3}$ with $|\omega_i| = 1$ and $\omega_0 = 1$ (the top eigenvector is positive and unique, hence invariant); Z_Γ is real and positive, so $\lambda_0^{n_3} - \sum_{i\ge 1} \lambda_i^{n_3} \le Z_\Gamma \le \sum_i \lambda_i^{n_3}$. (g) Put $\mathsf{v} = e_1 + e_3$.

The planes $x_1 + x_3 = c$, $c \in \mathbb{Z}$, are planes of lattice reflections, $(x_1, x_2, x_3) \mapsto (c - x_3, x_2, c - x_1)$; $N\mathbb{Z}^3 = \Gamma_\perp \oplus \mathbb{Z}Ne_1$ and $\Gamma^\diamond = \Gamma_\perp \oplus \mathbb{Z}h\mathsf{v}$ with $\Gamma_\perp = \mathbb{Z}N(e_1 - e_3) \oplus \mathbb{Z}Ne_2 \perp \mathsf{v}$, and both generators Ne_1, $h\mathsf{v}$ raise the height $x_1 + x_3$ by N. Cut either torus along the planes of the heights $-h, 0, h$ into two slabs of the height h, periodic under $\Gamma_\perp$. No plaquette lies in such a plane, and no bond crosses it; a plaquette cut by it has two corners z, z' in it and one on either side, and its weight is $e^{-\beta(1-\frac{1}{2}\operatorname{tr} vv'^{-1})}$, v and v' the transports from z to z' through the upper and through the lower corner.

As $\frac{1}{2}\operatorname{tr} vv'^{-1} = \frac{1}{2}\sum_{i,j} v_{ij}\overline{v'_{ij}}$ on SU(2), the exponential series writes this weight as a sum with coefficients ≥ 0 of products $\Phi(v)\overline{\Phi(v')}$ of matrix elements. With the data ϕ of a plane, its bonds and the indices of this expansion at its cut plaquettes, under the product of Haar measure and of the coefficients, the partition function of the torus $\Gamma_\perp \oplus \mathbb{Z}g$ is $\iint k_2(\tau_g\phi_0, \phi_1)\, k_1(\phi_1, \phi_0)$, where k_1, k_2 are the kernels of the two slabs between the data of their walls and τ_g, the translation by g, carries the data of the height $-h$ to those of the height h and keeps their measure.

Schwarz' inequality gives $Z(N^3)^2 \le \|k_1\|_2^2\, \|k_2\|_2^2$. The reflection in the middle plane exchanges the two sides of the cut plaquettes, conjugates the kernels, and acts on the plane of the height $-h$ as the translation by $h\mathsf{v}$; so $\|k_i\|_2^2$ is the partition function of the slab i glued to its mirror image, that is of the torus $\Gamma_\perp \oplus \mathbb{Z}h\mathsf{v} = \Gamma^\diamond$, and $Z(N^3) \le Z_{\Gamma^\diamond}$. □

Proposition 17.32 (The recursion under doubling)**.** Let $\beta \ge 0$, $l \ge 3$, and $\mathsf{e}(l) \le x_0$ at β; put $\mathsf{s}(x) = \psi(4x + 2\psi(x) + \psi(2x + \psi(x)))$.

(a) $\mathsf{e}(2l) \le \mathsf{s}(\mathsf{e}(l)) \le 12.696\, \mathsf{e}(l)^2 \le 12.9\, \mathsf{e}(l)^2$, $D_2(2l) \le 19.669\, \mathsf{e}(l)^2 \le 20\, \mathsf{e}(l)^2$, and $\mathsf{e}(2^k l) \le \theta^{2^k}/12.9$ for $k \ge 0$.

(b) $\delta((2^j l)^3) \le \frac{20}{7}\mathsf{e}(2^{j-1}l)^2 \le 0.017170\, \theta^{2^j}$, and $\delta((2^j l)^3) \le 0.01655\, \theta^{2^j}$, for $j \ge 1$; $\delta(l, 2l, 2l) \le \frac{x_0}{2} + \frac{10}{7}x_0^2$; for every integer $N \ge 2l$: $\delta(N^3) < 0.13736\, \theta^{N/2l}$; $\mathsf{e}(l') < 0.3702\, \theta^{l'/2l}$ for every integer $l' \ge l$; $D_q(N) < q^3\, 0.13736\, \theta^{N/2l}$; and $\delta(\underline{n}) < |\underline{n}|\, n_-^{-3}\, 0.13736\, \theta^{n_-/2l}$ for every box whose least side n_- is $\ge 2l$.

(c) On every cube N^3 with $N \ge 2l$: $\mathsf{G}_N \le 0.06754\, \theta^{N/2l}$, $\mathsf{r}_N < 0.1473\, \theta^{N/2l}$, all seven $0 \le 1 - r^z < 0.4121\, \theta^{N/2l}$, and $1 - r_1 \le 0.09006\, \theta^{N/2l}$, $1 - r_2 \le 0.1351\, \theta^{N/2l}$, $1 - r_3 \le 0.2252\, \theta^{N/2l}$; at $N = 2^{k+1}l$, $\mathsf{r}_N \le 0.001615\, \theta^{N/l}$.

(d) If $x_0 = 0.069$: $1 - r_1(N) < 0.0039$ for every $N \ge 16l$, and $\mathsf{e}(l') < 0.0234$ for every integer $l' \ge 16l$.

(e) *The threshold.* $\mathsf{s}(x) \le 0.99991\, x$ on $(0, 0.07489]$, and the least positive fixed point of s lies in $(0.0748941, 0.0748942)$. No threshold $\ge \log 2$ can serve for a recursion of this kind: the function $F = |\underline{n}|P + \log 2$, which in each direction is a sum of t-th powers of two equal positive numbers and has the properties of Prop. 17.31(b), (c), has $\mathsf{e} \equiv \delta \equiv \log 2$; and the positivity of the λ_i is used: $e^F = e^{|\underline{n}|P} \prod_i (1 + (-1)^{n_i})$ has $\mathsf{e} = -\infty$ at odd l and $\mathsf{e} = \log 8$ at even l.

■

Proof. Let $\epsilon = \mathsf{e}(l)$, the doubling defect at $t = l$ over the cross-section $(2l, 2l)$. We double the three sides in turn, each time with Prop. 17.31(b). First $\mathsf{a} = 2F((2l)^3) - F(2l, 2l, 4l) \le \psi(\epsilon)$. Next $y_1 = 2F(l, 2l, 4l) - F(2l, 2l, 4l) \le 2\epsilon + \mathsf{a}$, because $F(l, 2l, 4l) \le 2F(l, 2l, 2l)$, and

$\mathsf{b}' = 2F(2l,2l,4l) - F(2l,4l,4l) \le \psi(y_1)$. Last $2F(l,4l,4l) - F(2l,4l,4l) \le 4\epsilon + 2\mathsf{a} + \mathsf{b}'$, and

$$\mathsf{e}(2l) \le \psi(4\epsilon + 2\mathsf{a} + \mathsf{b}') \le \mathsf{s}(\epsilon), \qquad D_2(2l) = 4\mathsf{a} + 2\mathsf{b}' + \mathsf{e}(2l).$$

ψ and s are power series in x with coefficients ≥ 0 beginning with $\frac{1}{2}x^2$ and $8x^2$, so $\mathsf{s}(x)/x^2$ does not decrease, and $\epsilon \le 0.069$ gives $12.696\,\epsilon^2$ and $19.669\,\epsilon^2$, all arguments staying below $0.293 < \log 2$; since $12.9 \cdot 0.069 < 1$, iteration gives (a). Put $\mathsf{s}_D(x) = 4\psi(x) + 2\psi(2x + \psi(x)) + \mathsf{s}(x)$, a power series in x with coefficients ≥ 0 beginning with $14x^2$, so that $\mathsf{s}_D(x)/x^2$ does not decrease; it is at most 19.668665 at 0.069, the number 19.669 of (a) before rounding. As ψ increases, the display, with $m = 2^i l$, $i \ge 0$, in the place of l, gives $D_2(2m) \le \mathsf{s}_D(\mathsf{e}(m)) \le 19.668665\,\mathsf{e}(m)^2$, as $\mathsf{e}(m) \le \theta^{2^i}/12.9 \le 0.069$ by (a); and e decreases along these sides, as $\mathsf{e}(2m) \le 12.9 \cdot 0.069\,\mathsf{e}(m)$. The series of Prop. 17.31(c) at $q = 2$, $\delta((2^j l)^3) = \sum_{s\ge 0} 8^{-(s+1)} D_2(2^{s+j} l)$, gives for $j \ge 1$

$$\delta((2^j l)^3) \le 19.668665 \sum_{s\ge 0} 8^{-(s+1)} \mathsf{e}(2^{s+j-1} l)^2 \le \frac{19.668665}{7}\,\mathsf{e}(2^{j-1} l)^2 \le \frac{19.668665}{7 \cdot 12.9^2}\,\theta^{2^j} \le 0.016885\,\theta^{2^j},$$

which is below $\frac{20}{7}\mathsf{e}(2^{j-1}l)^2$ and below $\mathsf{c}_0\theta^{2^j}$, $\mathsf{c}_0 = \frac{20}{7\cdot 12.9^2} \approx 0.0171693$. Keeping the decay of the terms, $\mathsf{e}(2^{s+j-1}l)^2 \le \theta^{2^{s+j}}/12.9^2$, the same series gives $\delta((2^j l)^3) \le \frac{19.668665}{8\cdot 12.9^2}\,\theta^{2^j} \sum_{s\ge 0} 8^{-s}\theta^{2^{s+j}-2^j} \le 0.0147743 \cdot 1.10720\,\theta^{2^j} \le 0.01636\,\theta^{2^j}$, the sum being greatest at $j = 1$ and $\theta = 0.8901$. For $2^j l \le N < 2^{j+1} l$, $j \ge 1$, the monotone density gives $\delta(N^3) \le (N/2^j l)^3 \delta((2^j l)^3) < 8 \cdot 0.016885\,\theta^{2^j} < 0.13736\,\theta^{N/2l}$, and boxes follow against the cube of their least side. For $\mathsf{e}(l')$ with $n = 2^k l \le l' < 2n$, $k \ge 0$: $\mathsf{e}(l') \le 2\mathsf{G}(n; 2l', 2l') \le 2\delta(n, 2l', 2l') \le 2(l'/n)^2 \delta(n, 2n, 2n) < 8\delta(n, 2n, 2n) = 4[\mathsf{e}(n) + \delta((2n)^3)]$ by Prop. 17.31(b), (c), and the bound above at $j = k + 1$ with $\mathsf{e}(n) \le 0.069$ gives $4[\mathsf{e}(n) + \delta((2n)^3)] \le 4\,\mathsf{e}(n)(1 + \frac{19.668665}{7}\,0.069) \le \frac{4\cdot 1.193877}{12.9}\,\theta^{2^k} \le 0.370195\,\theta^{2^k} < 0.3702\,\theta^{l'/2l}$. (c) For $\bar{N} = 2^j l \le N < 2\bar{N}$, $j \ge 1$, with the dyadic side as the time, $\mathsf{G}_N \le \mathsf{G}(\bar{N}; N, N) \le \delta(\bar{N}, N, N) \le 4\delta(\bar{N}^3) \le \frac{4\cdot 19.668665}{7\cdot 12.9^2}\,\theta^{2^j} \le 0.0675395\,\theta^{2^j} \le 0.06754\,\theta^{N/2l}$.

Then Prop. 17.31(d): in the signs of its proof, $1 - r_1 \le \frac{4}{3}\mathsf{R}_N$, $1 - r_2 = 4u \le 2(2u + v) \le 2\mathsf{R}_N$ and $1 - r_3 \le (1 - r_1) + (1 - r_2) \le \frac{10}{3}\mathsf{R}_N$, with $\mathsf{R}_N \le \mathsf{G}_N$, which give 0.0900527, 0.135079, 0.225132 times $\theta^{N/2l}$; all seven $1 - r^z \le 3\mathsf{R}_N \le 3\mathsf{G}_N$, and $\mathsf{r}_N = e^{\mathsf{G}_N} - 1$ with $e^{cy} - 1 \le y(e^c - 1)$ for $y \le 1$. At $N = 2^{k+1} l$, with $n = 2^k l$: $\mathsf{r}_N \le \mathsf{r}_n(N, N)^2 \le \varphi(e^{\mathsf{e}(n)} - 1)^2$ by Prop. 17.31(b), and $\varphi(e^x - 1)^2/x^2$ does not decrease and is at most 0.268652 at 0.069; with $\mathsf{e}(n) \le \theta^{2^k}/12.9$ this is $\mathsf{r}_N \le 0.268652\,\theta^{N/l}/12.9^2 \le 0.001615\,\theta^{N/l}$. (d) Iterate $X_{k+1} = \mathsf{s}(X_k)$ from $X_0 = 0.069$: $X_1, \dots, X_5 \approx 0.0604446,\ 0.0432902,\ 0.0195722,\ 0.0034309,\ 0.0000960$; the series gives $\delta((2^j l)^3) \le \sum_s 8^{-(s+1)} \mathsf{s}_D(X_{j-1+s})$, with the s_D above and $\mathsf{e}(2^i l) \le X_i$ (s increases), which at $j = 4$ is at most 0.00073051; so for $N \ge 16l$, $\mathsf{G}_N \le 0.0029221$ and $1 - r_1 \le \frac{4}{3}(1 - e^{-\mathsf{G}_N}) < 0.0039$, and $\mathsf{e}(l') \le 32 \cdot 0.00073051 < 0.0234$. (e) s is star-shaped, and $\mathsf{s}(x)/x$ is approximately 0.99999914 at 0.0748941 and approximately 1.00000135 at 0.0748942.

The two examples satisfy the identities of Prop. 17.31(a)–(c) with two equal top eigenvalues, respectively with an eigenvalue $-\lambda$. □

Proposition 17.33 (Every cross-section: a simple top, a gap, the thermal law)**.** Let $\beta \ge 0$, $l \ge 3$, $\mathsf{e}(l) \le x_0$ at β, and $A \le B$ integers $\ge 2l$; let $n = 2^k l$ with $2n \le A < 4n$ and $n' = 2^j n$ with $2n' \le B < 4n'$; any direction is the time.

(a) Over (A, B) the top λ_0 is simple and lies in the sector 00, and $\lambda_1/\lambda_0 < 0.49067^{1/n}\theta^{1/l} < \theta^{1/l}$; so every level of a sector $e \ne 00$, and every level of the sector 00 but the top, lies more than $\frac{1}{l}\log\frac{1}{\theta} + \frac{1.4239}{A}$ a step above the top. If $A = 2l$: $\lambda_1/\lambda_0 < (2.42\,x_0)^{1/l}$, and $\mathsf{r}_t(2l, B) \le 0.5628\,(2.42\,x_0)^{t/l}$ for $t \ge n'$. Over the square (N, N), $N \ge 2l$, with $2n \le N < 4n$: $\mathsf{r}_n(N, N) \le 0.20395\,\theta^{n/l}$.

(b) $\mathsf{r}_t(A,B) \le 0.2314\,\theta^{t/l}$ for every integer $t \ge n'$, and for every integer $t \ge l$

$$\mathsf{G}(t;A,B) \le 0.37\,\Big\langle\frac{A}{2t}\Big\rangle\Big\langle\frac{B}{2t}\Big\rangle\,\theta^{t/l} \le 0.37\,\frac{AB}{4l^2}\,\theta^{t/l},$$

in detail $0.27263\frac{AB}{4t^2}\theta^{t/l}$ if $2t \le A$, $0.3695\frac{B}{2t}\theta^{t/l}$ if $A < 2t \le B$, and $\mathsf{r}_t(A,B) \le 0.26261\,\theta^{t/l}$ if $B < 2t$.

(c) $\mathsf{e}(l') \le 0.4079\,\theta^{l'/l}$ for every integer $l' \ge l$; $\delta(l',2l',2l') \le 0.27263\,\theta^{l'/l}$; $D_2(N) \le 4.1242\,\theta^{N/l}$ and $\delta(N^3) \le 0.5892\,\theta^{N/l}$ for every integer $N \ge 2l$.

(d) Over the square (N,N), $N \ge 2l$, the vortex free energy $\varepsilon_{\mathrm{v}}(N,N) = \log(\lambda_0/\|\mathsf{T}^{(1)}\|)$ satisfies $N\varepsilon_{\mathrm{v}}(N,N) < 1.0292\,\theta^{N/2l}$.

(e) Conversely, on one lattice: if $\lambda_1/\lambda_0 \le e^{-\mu}$ over the single cross-section $(2l,2l)$ and $3 \le s < l$, then $\mathsf{e}(l) \le 2e^{-\mu(l-s)}(e^{\delta(s,2l,2l)} - 1)$; so $\mathsf{e}(l) \le x_0$ as soon as $\mu(l-s) \ge \delta(s,2l,2l) + \log(2/x_0)$. ■

Proof. Freeze the side A: the boxes (A,s,u) form a two-dimensional system, with the pressure $P_A = \inf F(A,s,u)/su$ and the excess $\delta_A(s,u) = F(A,s,u) - suP_A$, which has the properties of Prop. 17.31(b), (c) in its two sides and bounds $\mathsf{G}(t;A,u)$ from above. The first two doublings of the proof of Prop. 17.32, carried out in this system, give for its doubling defect $\eta = \mathsf{e}_A(m) = 2F(A,m,2m) - F(A,2m,2m)$: if $\eta \le 0.16521$, then $\mathsf{e}_A(2m) \le 3.4522\,\eta^2 \le 0.5704\,\eta$ and $\delta_A(2m,2m) \le 1.5514\,\eta^2$; the map $x \mapsto \psi(2x + \psi(x))$ has its least positive fixed point in $(0.2236, 0.2237)$, the threshold of this recursion. It is started at the scale $A/2$: by Prop. 17.31(b), (c) and Prop. 17.32(b),

$$\mathsf{e}_A(n) \le 2\,\mathsf{G}(n;A,2n) \le 2\,\delta(n,A,2n) \le \frac{A}{n}\,\delta(n,2n,2n) \le 0.185604\,\theta^{n/l} \le 0.16521.$$

Iterating, $\mathsf{e}_A(2^j n) \le 0.185604\,\theta^{2^j n/l}$ and $\delta_A(m,2m) \le 0.11659\,\theta^{m/l}$ along $m = 2^j n$; so for $2m \le u < 4m$, $\mathsf{r}_m(A,u) \le (e^{0.23318} - 1)\theta^{m/l} \le 0.26261\,\theta^{m/l} < 1$. At $m = n'$, $u = B$ this excludes a double top and gives $(\lambda_1/\lambda_0)^{n'} < \theta^{n'/l}$, and $\mathsf{r}_t \le \mathsf{r}_{n'}^{t/n'}$; the sector is that of Prop. 17.31(d), as $\mathsf{A} = \mathsf{K}^{1/2}\mathsf{m}^{1/2}$ commutes with the flips and carries the eigenvectors of D to those of T'. *The constants of (a) and (b).* They come from a finer start and from finer constants of the same recursion. For $2n \le A < 4n$, Prop. 17.31(b) in the thickness and at the time n over the cross-section $(2n,2n)$ give $\mathsf{e}_A(n) \le \frac{A}{2n}[\mathsf{e}(n) + \frac{1}{2}\psi(\mathsf{e}(n))] \le 2.037074\,\mathsf{e}(n) \le 0.140559$, as $\mathsf{e}(n) \le 0.069$.

On $[0, 0.140559]$ the doubling of the slab, $\mathsf{e}_A(2m) \le \psi(2\mathsf{e}_A(m) + \psi(\mathsf{e}_A(m)))$, gives $\mathsf{e}_A(2m) \le 3.1072\,\mathsf{e}_A(m)^2$, the inner argument staying below $0.292638 < \log 2$. So from a start $\mathsf{e}_A(n) \le x_1 \le 0.140559$, with $\vartheta = 3.1072\,x_1 < 1$: $\mathsf{e}_A(2^j n) \le \vartheta^{2^j}/3.1072$; the series of the slab gives $\delta_A(n',2n') \le 0.19314\,\vartheta^{n'/n}$; over (B,A) with $2n' \le B < 4n'$, $\mathsf{r}_{n'} \le e^{2\delta_A(n',2n')} - 1 \le 0.4715\,\vartheta^{n'/n} < 1$, so the top is simple, $\lambda_1/\lambda_0 < \vartheta^{1/n}$, and $\mathsf{r}_t \le 0.4715\,\vartheta^{t/n}$ for every $t \ge n'$. With $x_1 = 2.037074\,\mathsf{e}(n)$ and $\mathsf{e}(n) \le \theta^{n/l}/12.9$ (Prop. 17.32(a)), $\vartheta \le 6.3296\,\mathsf{e}(n) \le 0.49067\,\theta^{n/l}$: this is the factor $0.49067^{1/n}$ in (a), and $0.4715 \cdot 0.49067 \le 0.2314$ is the number of (b).

At $A = 2l$ the hypothesis itself is the start, $\mathsf{e}_{2l}(l) = \mathsf{e}(l) \le x_0 \le 0.069$, and on $[0, 0.069]$ the same recursion has the constants 2.42 and 0.5628 in the places of 3.1072 and 0.4715: $\vartheta = 2.42\,x_0$, $\lambda_1/\lambda_0 < (2.42\,x_0)^{1/l}$ and $\mathsf{r}_t(2l,B) \le 0.5628\,(2.42\,x_0)^{t/l}$ for $t \ge n'$. For the square, the box (n,N,N) with the short side as the time gives $\mathsf{G}(n;N,N) \le \frac{N^2}{4n^2}\delta(n,2n,2n) \le 0.185604\,\theta^{n/l}$ and $\mathsf{r}_n \le (e^{0.185604} - 1)\theta^{n/l}$. The thermal law (b) follows in three zones: if $2t \le A$, over the cross-section $(2t,2t)$, with $m = 2^i l \le t < 2m$, $\mathsf{r}_t(2t,2t) \le 0.20395\,\theta^{t/l}$, so $\mathsf{e}(t) \le 2\mathsf{r}_t \le 0.4079\,\theta^{t/l}$ and $\delta(t,2t,2t) = \frac{1}{2}[\mathsf{e}(t) + \delta((2t)^3)] \le 0.27263\,\theta^{t/l}$, and the monotone density carries this to (A,B); if $A < 2t \le B$, the slab gives $\delta_A(t,2t) \le 0.3695\,\theta^{t/l}$ and $\mathsf{G}(t;A,B) \le \frac{B}{2t}\delta_A(t,2t)$; if $B < 2t$, $\mathsf{r}_t \le 0.26261\,\theta^{t/l}$.

This also gives the first two clauses of (c); since $D_2(N) \le 8\mathsf{G}(N;N,N) + 4\mathsf{G}(N;N,2N) + 2\mathsf{G}(N;2N,2N)$, the three zones and the series give the last two. (d) In the box (N,N,h) with one twisted column of length h and $\zeta = \frac12(1-r)$, the second inequality of step (ii) in the proof of Prop. 17.39 below, which is exact on one lattice, gives $N\varepsilon_{\mathrm{v}}(N,N) \le \mathsf{d}(N;N,N) - \log(1 - 2\zeta(N,N;2N))$, with $\zeta(N,N;2N) \le \delta(N,N,2N) \le 2\delta(N^3)$ and $\mathsf{d}(N;N,N) \le 2\mathsf{G}_N \le 2\delta(N^3)$; with $\delta(N^3) < 0.13736\,y$, $y = \theta^{N/2l} \le 0.8901$, the first term is $< 0.27472\,y$, and $-\log(1-0.54944\,y) \le 0.75442\,y$ by convexity on $[0, 0.8901]$; the sum is $< 1.0292\,y$. (e) $\mathsf{e}(l) \le 2\mathsf{r}_l(2l,2l) \le 2e^{-\mu(l-s)}\mathsf{r}_s(2l,2l)$ and $\mathsf{r}_s = e^{\mathsf{G}(s)} - 1 \le e^{\delta(s,2l,2l)} - 1$. □

The two seminorms on a larger torus. By the setting and (M2), for $K \in \mathbb{N}$, $\beta > 0$, $\ell'' \le \ell$ in $\mathfrak{L}$ and a function F of the bonds of a cell of side $\ell''/2$, with the seminorm $\|F\|_{\mathrm{ch},\ell}^{(\ell'')}$ of all its mirror copies on the torus ℓ under its own board and the seminorm $\|F\|_{\mathrm{ch},\ell}$ of its eight mirror copies under the board of side $\ell/2$ whose cell is centred at the cell of F, the cells of the two boards being nested (Lem. YM3_I:104.1):

$$\|F\|_{\mathrm{ch},\ell} \le \|F\|_{\mathrm{ch},\ell}^{(\ell'')} \le e^{(\ell''/2)^3(p_{\ell''}-p_\ell)}\,\|F\|_{\mathrm{ch},\ell''} \le e^{\delta((\ell''/a_K)^3)/8}\,\|F\|_{\mathrm{ch},\ell''},$$

the last as $p_\ell \ge p_\infty = a_K^{-3}P$ and $(\ell''/2)^3(p_{\ell''} - p_\infty) = \frac18\delta((\ell''/a_K)^3)$.

Proposition 17.34 (The excess bounds the distance from the torus)**.** Let $K \in \mathbb{N}$, $\beta > 0$, $\ell' \in \mathfrak{L}$, $\delta' = \delta((\ell'/a_K)^3)$ and $Q' = e^{\delta'/8}$; let $\mathfrak{s}$ be either $\mu_{K,\ell,\beta}$ with $\ell \in \mathfrak{L}$, $\ell \ge \ell'$, or a periodic Gibbs state at β.

(a) For every bounded function F of the bonds of a cell of side $\ell'/2$, of any board: $|\mathfrak{s}(F)| \le Q'\|F\|_{\mathrm{ch},\ell'} \le Q'\,\mu_{K,\ell',\beta}(|F|^8)^{1/8}$.

(b) On the inner bonds of such a cell $\mathfrak{s} = h\,\mu_{K,\ell',\beta}$ with $\int h^{8/7} d\mu_{K,\ell',\beta} \le e^{\delta'/7}$, and the two laws are at a distance in total variation at most $\frac12(\sqrt{\varepsilon} + \varepsilon)$, $\varepsilon = 22.3\,(e^{\delta'/7} - 1)$.

(c) If $Q' \le 1.11$, then for every real G of the bonds of a cell of side $\ell''/2$, $\ell'' \in \mathfrak{L}$, $\ell'' \le \ell'$:

$$|\mathfrak{s}(G) - \mu_{K,\ell',\beta}(G)| \le 4.84\,[Q'(Q'-1)]^{1/2}\,\|G\|_{\mathrm{ch},\ell'} \le 4.84\,[Q'(Q'-1)]^{1/2}\,Q''\,\|G\|_{\mathrm{ch},\ell''},$$
$$Q'' = e^{\delta((\ell''/a_K)^3)/8};$$

for complex G the same with twice the constant.

(d) Let $\ell_i \in \mathfrak{L}$, $\ell_i \to \infty$, with $\limsup_i \delta((\ell_i/a_K)^3) \le D < \infty$ and $\mu_{K,\ell_i,\beta} \to \mu^*$ weakly. Then μ^* is a periodic Gibbs state; $\mu^*(A) \ge e^{-D}\mathfrak{s}(A)^8$ for every periodic Gibbs state $\mathfrak{s}$ and every Borel set A; $\mu^* \ge e^{-D}\nu$ for every ergodic periodic Gibbs state ν; and $\mathfrak{s} \le e^D\mu^*$ for every periodic Gibbs state $\mathfrak{s}$. So n mutually singular periodic Gibbs states at β have $n \le e^D$, and indeed $n \le \liminf_{\mathfrak{L}\ni\ell'\to\infty} e^{\delta'}$.

■

Proof. (a) By the fine-to-coarse form of (M2), $\|F\|_{\mathrm{ch},\ell}^{(\ell')} \le \|F\|_{\mathrm{ch},\ell'}\exp[(\ell'/2)^3(p_{\ell'} - p_\ell)]$, and since $p_\ell \ge p_\infty = a_K^{-3}P$ the exponent is at most $\frac18\ell'^3(p_{\ell'} - p_\infty) = \delta'/8$. For a torus $|\mu_{K,\ell,\beta}(F)| \le \|F\|_{\mathrm{ch},\ell}^{(\ell')}$, by the array form with the factor 1 in the other cells; and a periodic Gibbs state is dominated by the tori, $|\mathfrak{s}(F)| \le \liminf_\ell \|F\|_{\mathrm{ch},\ell}^{(\ell')}$ (M2). The second inequality is Hölder's in the array form, the mirror copies of $|F|^8$ all having the same mean. (b) By (a), $\mathfrak{s}(A) \le Q'\mu_{K,\ell',\beta}(A)^{1/8}$ on the inner bonds, so $\mathfrak{s} = h\mu$ with $\|h\|_{8/7} \le Q'$ by duality. With $p = \frac87$ and $\phi(u) = u^p - 1 - p(u-1) \ge 0.045\min\{(u-1)^2, |u-1|^p\}$ for $u \ge 0$ (Taylor's formula on $[0,2]$, monotonicity of $\phi(1+v)/v^p$ beyond), $\int\phi(h) \le e^{\delta'/7} - 1$ gives $\int|h-1| \le \sqrt{\varepsilon} + \varepsilon$. (c) We use an abstract form. Let $\mathcal{A}$ be a real vector space with a member 1, N a seminorm and g a linear functional on it with

$$\mathsf{N}(1+tG)^8 \le 1 + 8t\,\mathsf{g}(G) + R(|t|\mathsf{N}(G)), \quad R(x) = (1+x)^8 - 1 - 8x, \qquad |\mathsf{g}(G)| \le \mathsf{N}(G),$$

and $\mathfrak{s}$ a linear functional with $\mathfrak{s}(1)=1$, $\mathfrak{s}(F)\le Q\,\mathsf{N}(F)$, $1\le Q\le 1.11$. As $(1+u)^{1/8}\le 1+\frac{u}{8}$: $t[\mathfrak{s}(G)-\mathsf{g}(G)]\le (Q-1)(1+x)+\frac{Q}{8}R(x)$, $x=|t|\mathsf{N}(G)$; if $\mathsf{N}(G)=0$, the bounds on g and the linearity bound on $\mathfrak{s}$, applied to G and $-G$, give $\mathfrak{s}(G)=\mathsf{g}(G)=0$. For $Q=1$, use both signs of t and let $t\to 0$, since $R(x)=O(x^2)$. For $\mathsf{N}(G)>0$ and $Q>1$, take $t=\pm x/\mathsf{N}(G)$, $x^2=\frac{Q-1}{4.74Q}$ (so $x\le 0.1446$, where $R(x)\le 8\cdot 4.7179\,x^2$); this gives $|\mathfrak{s}(G)-\mathsf{g}(G)|\le [2\sqrt{4.74}+(0.11/1.11)^{1/2}][Q(Q-1)]^{1/2}\mathsf{N}(G)\le 4.6692\,[Q(Q-1)]^{1/2}\mathsf{N}(G)$. For $\mathsf{N}=\|\cdot\|_{\mathrm{ch},\ell'}$, $\mathsf{g}=\mu_{K,\ell',\beta}$ on the real functions of a cell, the product of the eight copies of $1+tG$ expands into $\sum_S t^{|S|}m_S$ with $m_\emptyset=1$, $m_{\{\mathbf{k}\}}=\mu(G)$ for the eight one-point sets and $|m_S|\le \mathsf{N}(G)^{|S|}$ for $|S|\ge 2$ (array form), and $Q=Q'$ by (a).

A function of a cell of side $\ell''/2$ is a function of the cell of side $\ell'/2$ of the setting, centred at its cell, as ℓ'/ℓ'' is a power of the odd L, and $\|G\|_{\mathrm{ch},\ell'}\le \|G\|_{\mathrm{ch},\ell'}^{(\ell'')}\le Q''\|G\|_{\mathrm{ch},\ell''}$ by the chain of (M2) recalled before the Proposition. (d) μ^* is a periodic Gibbs state invariant under $a_K\mathbb{Z}^3$ by (M3). For an event A of the bonds of a cell of side $\ell_i/2$, (a) with $F=1_A$ gives $\mathfrak{s}(A)\le e^{\delta_i/8}\mu_{K,\ell_i,\beta}(A)^{1/8}$, $\delta_i=\delta((\ell_i/a_K)^3)$; the conditional laws of finitely many bonds given the others are continuous in the others, so the means of local events converge, and $\mu^*(A)\ge e^{-D}\mathfrak{s}(A)^8$ for local events, hence for all Borel sets by approximation.

An ergodic ν has an invariant carrier N_ν that carries no other ergodic state, so the ergodic decomposition of μ^* gives $\mu^*(\cdot\cap N_\nu)=\mu^*(N_\nu)\,\nu$ with $\mu^*(N_\nu)\ge e^{-D}\nu(N_\nu)^8=e^{-D}$; every $\mathfrak{s}$ is an integral of ergodic states (M3), each $\le e^D\mu^*$. Mutually singular states have disjoint carriers C_j, and $\sum_j \mu^*(C_j)\le 1$ with $\mu^*(C_j)\ge e^{-D}$; for the last clause take sides along which δ' tends to its lower limit and a weakly convergent subsequence. □

Lemma 17.35 (Boxes of every shape)**.** Let $\beta\ge 0$, $N=2h\ge 4$, c a corner, $\mathsf{C}=\mathsf{C}_h(c)$, and $\underline{n}$ any box with $n_\mu\ge N$; let $M_\mu=2^{j_\mu}h$, $j_\mu\in\mathbb{N}$, be the greatest number of this form that is $\le n_\mu$, so $N\le M_\mu\le n_\mu<2M_\mu$.

(a) Let μ be a direction, $\vartheta\ge 2$ with $2\vartheta\le n_\mu$, θ' the reflection in a plane $x_\mu=c'_\mu$, $c'_\mu\in a(\mathbb{Z}+\frac12)$, and F a bounded Borel function of the bonds with both ends in the ϑ layers below that plane. Then $\langle F\,\overline{\theta' F}\rangle_{\underline{n}}\ge 0$ and $|\langle F\rangle_{\underline{n}}|^2\le\langle F\,\overline{\theta' F}\rangle_{\underline{n}}$, also for odd n_μ. On the cube N^3, $|\langle\prod_\epsilon H_\epsilon^{[\epsilon]}\rangle_{N^3}|\le\prod_\epsilon\|H_\epsilon\|_{\mathrm{ch};N}$, so $\|\cdot\|_{\mathrm{ch};N}$ is a seminorm, and $|\langle\Psi\rangle_{N^3}|\le\|\Psi\|_{\mathrm{ch};N}\le\sup|\Psi|$.

(b) For every function Ψ of the bonds of C:

$$|\langle\Psi\rangle_{\underline{n}}|\le\exp\Big\{\frac{\delta(N^3)}{8}\Big[1+3N\Big(\frac{1}{M_1}+\frac{1}{M_2}+\frac{1}{M_3}\Big)\Big]\Big\}\|\Psi\|_{\mathrm{ch};N}\le e^{\frac54\delta(N^3)}\|\Psi\|_{\mathrm{ch};N};$$

and for the boxes (q_1N,q_2N,q_3N), $q\in\mathbb{N}^3$: $|\langle\Psi\rangle_{\underline{n}}|\le\exp[\frac18(\delta(N^3)-\delta(\underline{n})/q_1q_2q_3)]\|\Psi\|_{\mathrm{ch};N}$.

(c) Let Ψ be real. If $\delta(N^3)\le\frac12$, then $|\langle\Psi\rangle_{\underline{n}}-\langle\Psi\rangle_{N^3}|\le 7.31\,\delta(N^3)^{1/2}\|\Psi\|_{\mathrm{ch};N}$; for the boxes $(q_\mu N)$ and $\delta(N^3)\le 0.8348$, the same with 1.86 for 7.31.

None of these constants depends on the shape of the box. ■

Proof. (a) With the time μ and $\mathsf{v}=\frac12(n_\mu-2\vartheta)$, the trace formula (M1) and the cyclicity of the trace give $Z(\underline{n})\langle F\rangle_{\underline{n}}=\mathrm{Tr}(\mathsf{T}'^{\,n_\mu/2}\cdot Y\mathsf{T}'^{\mathsf{v}})$, $Y=\mathfrak{Y}[\Phi_F]$; Schwarz's inequality in the Hilbert–Schmidt class, with $\|\mathsf{T}'^{\,n_\mu/2}\|_2^2=Z(\underline{n})$ and $\|Y\mathsf{T}'^{\mathsf{v}}\|_2^2=\mathrm{Tr}(\mathsf{T}'^{\,2\mathsf{v}}Y^*Y)=Z(\underline{n})\langle F\overline{\theta' F}\rangle_{\underline{n}}$, is the claim. Applied in the three directions in turn this is the array bound on the cube. (b) *Doubling on a single box.* In the direction 1 let $F_0=\Psi$ and $F_{i+1}=F_i\,\overline{\theta^{(i)}F_i}$, $\theta^{(i)}$ the reflection in $x_1=c_1+(2^i-1)ha$; by (a), as long as the doubled function fits, $|\langle F_i\rangle_{\underline{n}}|^2\le\langle F_{i+1}\rangle_{\underline{n}}$; so $|\langle\Psi\rangle_{\underline{n}}|\le\langle\mathfrak{B}_1\Psi\rangle_{\underline{n}}^{2^{-j_1}}$ with an array $\mathfrak{B}_1\Psi$ of M_1/h cells.

The same in the directions 2, 3 gives an array $\mathfrak{B}\Psi$ on a block of $M_1\times M_2\times M_3$ sites with $\mathcal{N}=M_1M_2M_3/h^3$ cells and $|\langle\Psi\rangle_{\underline{n}}|\le\langle\mathfrak{B}\Psi\rangle_{\underline{n}}^{1/\mathcal{N}}$. *The free layers cost their top eigenvalue.* Let $\underline{m}$ be a box with $m_\mu\ge M_\mu$ for each μ; it is taken in turn as $\underline{n}$, (M_1,n_2,n_3) and (M_1,M_2,n_3). Take the direction 1 as the time over the cross-section (m_2,m_3), let Φ be the slab function of the cell

of $\mathfrak{B}_3\mathfrak{B}_2\Psi$, which lies in the h layers below c_1, and let $Y_\Phi = \mathfrak{Y}[\Phi]$ be its slab operator (another operator than the Y of (a)).

In the direction 1 the array $\mathfrak{B}\Psi$ is the row $\Phi, \Phi^\Theta, \Phi, \Phi^\Theta, \dots$ on M_1/h consecutive slabs of h layers, so its operators close up into $(Y_\Phi^* Y_\Phi)^{M_1/N}$, $M_1/N = 2^{j_1-1}$, and the $m_1 - M_1$ free layers carry $\mathsf{T}'^{m_1-M_1}$ (a free rest of one layer is $\mathfrak{Y}[\Phi^{+1}] = \mathsf{T}'\mathfrak{Y}[\Phi]$ of (M1)). By the trace formula (M1), $Z(\underline{m})\langle\mathfrak{B}\Psi\rangle_{\underline{m}} = \mathrm{Tr}(\mathsf{T}'^{m_1-M_1}(Y_\Phi^* Y_\Phi)^{M_1/N})$ and $\mathrm{Tr}(Y_\Phi^* Y_\Phi)^{M_1/N} = Z(M_1, m_2, m_3)\langle\mathfrak{B}\Psi\rangle_{(M_1,m_2,m_3)}$; and $\mathrm{Tr}(\mathsf{AB}) = \mathrm{Tr}(\mathsf{B}^{1/2}\mathsf{AB}^{1/2}) \le \|\mathsf{A}\|\,\mathrm{Tr}\,\mathsf{B}$ for $\mathsf{A}, \mathsf{B} \ge 0$, B of trace class.

So, direction by direction,

$$Z(\underline{n})\langle\mathfrak{B}\Psi\rangle_{\underline{n}} \le \lambda_0(n_2, n_3)^{n_1-M_1}\lambda_0(M_1, n_3)^{n_2-M_2}\lambda_0(M_1, M_2)^{n_3-M_3}\, Z(M_1, M_2, M_3)\langle\mathfrak{B}\Psi\rangle_{(M_1,M_2,M_3)}.$$

On the box (M_1, M_2, M_3), whose sides lie in $N\mathbb{N}$, the inequality $\mathrm{Tr}(Y_\Phi^* Y_\Phi)^p \le (\mathrm{Tr}\, Y_\Phi^* Y_\Phi)^p$ at $p = M_\mu/N$ in each direction μ, with Φ the cell of the array in that direction, reduces the array to the cube N^3, where it is the product of the eight mirror copies of Ψ: $Z(M_1, M_2, M_3)\langle\mathfrak{B}\Psi\rangle_{(M_1,M_2,M_3)} \le [Z(N^3)\|\Psi\|_{\mathrm{ch};N}^8]^{M_1M_2M_3/N^3}$. *The count.* With $\log\lambda_0(\Sigma) = |\Sigma|P + \varepsilon(\Sigma)$ and $F = |\cdot|P + \delta$ the terms with P cancel, and by the area bound $\varepsilon(\mathsf{A}, \mathsf{B}) \le (\mathsf{A} + \mathsf{B})\delta(N^3)/N^2$ for $\mathsf{A}, \mathsf{B} \ge N$ (Prop. 17.31(c)) the free layers cost at most $3\delta(N^3)N^{-2}(M_1M_2 + M_1M_3 + M_2M_3)$ against $\mathcal{N} = 8M_1M_2M_3/N^3$ cells: an area against a volume.

The boxes (q_1N, q_2N, q_3N), $q \in \mathbb{N}^3$, are treated with another array, which fills the box for every q, so that no layer is free. For a function F of the bonds in the h layers below c_μ, with slab function Φ_F and slab operator $Y_F = \mathfrak{Y}[\Phi_F]$, the time being μ, let $\mathfrak{A}_\mu F$ be the product of the $2q_\mu$ cells $\Phi_F, \Phi_F^\Theta, \Phi_F, \Phi_F^\Theta, \dots$ on all the $2q_\mu$ consecutive slabs of h layers: F and its successive conjugated mirror images in the planes $x_\mu = c_\mu + jha$. By (M1), $Z(\underline{n})\langle\mathfrak{A}_\mu F\rangle_{\underline{n}} = \mathrm{Tr}(Y_F^* Y_F)^{q_\mu} = \|Y_F\|_{2q_\mu}^{2q_\mu}$ and $Z(\underline{n}) = \mathrm{Tr}\,\mathsf{T}'^{2q_\mu h} = \|\mathsf{T}'^h\|_{2q_\mu}^{2q_\mu}$, the norms of the Schatten class of order $2q_\mu$; Hölder's inequality in these classes for the $2q_\mu$ factors $Y_F, \mathsf{T}'^h, \dots, \mathsf{T}'^h$ gives $Z(\underline{n})|\langle F\rangle_{\underline{n}}| = |\,\mathrm{Tr}(\mathsf{T}'^h \cdots \mathsf{T}'^h Y_F)| \le \|Y_F\|_{2q_\mu}\|\mathsf{T}'^h\|_{2q_\mu}^{2q_\mu-1}$, that is $|\langle F\rangle_{\underline{n}}| \le \langle\mathfrak{A}_\mu F\rangle_{\underline{n}}^{1/2q_\mu}$.

Applied on the one box $\underline{n}$ to Ψ in the direction 1, to $\mathfrak{A}_1\Psi$, which lies below c_2, in the direction 2, and to $\mathfrak{A}_2\mathfrak{A}_1\Psi$, which lies below c_3, in the direction 3, this gives $|\langle\Psi\rangle_{\underline{n}}| \le \langle\mathfrak{A}\Psi\rangle_{\underline{n}}^{1/8q_1q_2q_3}$, $\mathfrak{A} = \mathfrak{A}_3\mathfrak{A}_2\mathfrak{A}_1$, the product of the mirror copies of Ψ in all the $8q_1q_2q_3$ cells of the box. Only then is the box made smaller: with the time 1 and $F = \mathfrak{A}_3\mathfrak{A}_2\Psi$, $Z(q_1N, n_2, n_3)\langle\mathfrak{A}\Psi\rangle_{(q_1N,n_2,n_3)} = \mathrm{Tr}(Y_F^* Y_F)^{q_1} \le (\mathrm{Tr}\, Y_F^* Y_F)^{q_1} = [Z(N, n_2, n_3)\langle\mathfrak{A}\Psi\rangle_{(N,n_2,n_3)}]^{q_1}$, and the same in the directions 2, 3 give $Z(\underline{n})\langle\mathfrak{A}\Psi\rangle_{\underline{n}} \le [Z(N^3)\|\Psi\|_{\mathrm{ch};N}^8]^{q_1q_2q_3}$.

As $|\underline{n}| = q_1q_2q_3N^3$, the terms with P cancel, and $|\langle\Psi\rangle_{\underline{n}}| \le \exp\frac{1}{8}[\delta(N^3) - \delta(\underline{n})/q_1q_2q_3]\,\|\Psi\|_{\mathrm{ch};N}$.
(c) The abstract form of the proof of Prop. 17.34(c) with $\mathsf{N} = \|\cdot\|_{\mathrm{ch};N}$, $\mathfrak{g} = \langle\cdot\rangle_{N^3}$, $\mathfrak{s} = \langle\cdot\rangle_{\underline{n}}$: its first condition is the array bound of (a) on the cube, the eight one-point terms being $\langle G\rangle_{N^3}$ by the mirror symmetry of the cube, and $\mathfrak{s}(F) \le Q\,\mathsf{N}(F)$ is (b). With $Q = e^{\frac{5}{4}\delta(N^3)}$, which is ≤ 1.11 if and only if $\delta(N^3) \le \frac{4}{5}\log 1.11$: as $u \mapsto e^u(e^u - 1)/u$ increases, $Q(Q-1) \le \frac{1.11\cdot 0.11}{\log 1.11}\cdot\frac{5}{4}\delta(N^3)$, so $[Q(Q-1)]^{1/2} \le 1.209334\,\delta(N^3)^{1/2}$ and $4.84 \cdot 1.209334 \le 5.86 \le 7.31$; for $\frac{4}{5}\log 1.11 < \delta(N^3) \le \frac{1}{2}$, by (b) and (a) directly, $|\langle\Psi\rangle_{\underline{n}} - \langle\Psi\rangle_{N^3}| \le (e^{\frac{5}{4}\delta} + 1)\|\Psi\|_{\mathrm{ch};N}$, and $(e^{\frac{5}{4}\delta} + 1)\delta^{-1/2}$ decreases on $(0, \frac{1}{2}]$, so it is at most its value at $\frac{4}{5}\log 1.11$, which is less than 7.3025.

For the boxes $(q_\mu N)$ take $Q = e^{\delta(N^3)/8}$ by the second clause of (b): $Q \le 1.11$ if and only if $\delta(N^3) \le 8\log 1.11$, and then $Q(Q-1) \le \frac{1.11\cdot 0.11}{\log 1.11}\cdot\frac{\delta(N^3)}{8}$, $[Q(Q-1)]^{1/2} \le 0.382425\,\delta(N^3)^{1/2}$, and $4.84 \cdot 0.382425 \le 1.85094 \le 1.86$. □

Lemma 17.36 (Cells are nearly independent under every boundary condition)**.**

(a) Let $\mathcal{K}$ be a finite set, Y_κ measurable maps of a probability space (E, R) to spaces E_κ with probability measures ref_κ, and $\Delta \ge 0$ with $\log R(\exp\sum_\kappa \Phi_\kappa \circ Y_\kappa) \le \sum_\kappa[\Delta + \frac{1}{8}\log\mathrm{ref}_\kappa(e^{8\Phi_\kappa})]$

for all bounded real Φ_κ. Then for every probability $P = fR$ with $\mathsf{H} = R(f\log f) < \infty$ and all bounded real non-constant G_κ: $\sum_\kappa [P(G_\kappa \circ Y_\kappa) - \mathrm{ref}_\kappa(G_\kappa)]^2 / \operatorname{osc}(G_\kappa)^2 \le 4\mathsf{H} + 4\#\mathcal{K}\,\Delta$.

(b) Let $K \in \mathbb{N}$, $\beta > 0$, $\ell' \le \ell$ in $\mathfrak{L}$, $n = \ell/a_K$, $n' = \ell'/a_K$ and $\delta' = \delta(n'^3)$; these letters serve in (c) as well. The $(2n/n')^3 = 8(n/n')^3$ cells of a period box of a board of side $\ell'/2$ on the torus ℓ satisfy the hypothesis of (a) with $R = \mu_{K,\ell,\beta}$, ref the law of a cell under $\mu_{K,\ell',\beta}$ and $\Delta = \delta'/8$.

(c) Let D be the set of the sites of a period cube of the torus ℓ, n sites a side, that is the union of the $8(n/n')^3$ cells of side $\ell'/2$ of a period box of the board, let W be the set of the positively oriented bonds with initial point in D, which holds the bonds of these cells, and let $\gamma_W(\cdot)(\xi)$ be the law of the bonds of W given the configuration ξ outside W. For every ξ and all real non-constant $G_{\mathbf{k}}$ of the cells:

$$\sum_{\mathbf{k}} \frac{[(\gamma_W G_{\mathbf{k}})(\xi) - \mu_{K,\ell',\beta}(G_{\mathbf{k}})]^2}{\operatorname{osc}(G_{\mathbf{k}})^2} \le 144\,\beta n^2 + 4\,(n/n')^3\delta';$$

the mean over the cells is at most $\varepsilon_1(n,n')^2 = 18\beta n'^3/n + \delta'/2$.

(d) Let $\lambda \ge 3$ be odd and $\underline{n} = (\mathsf{n}_1\lambda, \mathsf{n}_2\lambda, \mathsf{n}_3\lambda)$ a periodic box with even $\mathsf{n}_1, \mathsf{n}_2, \mathsf{n}_3$, cut into the $\mathsf{n}_1\mathsf{n}_2\mathsf{n}_3$ cells of λ sites a side of a board whose walls are planes of lattice reflections. The cells satisfy the hypothesis of (a) with $R = \mu_{\underline{n}}$, ref the law of one cell of the cube $(2\lambda)^3$ under $\mu_{(2\lambda)^3}$, and $\Delta = \delta((2\lambda)^3)/8$.

■

Proof. (a) By the variational formula for the relative entropy, $P(\Phi) \le \mathsf{H} + \log R(e^\Phi)$ for $\Phi = \sum_\kappa t_\kappa(G_\kappa - \mathrm{ref}_\kappa G_\kappa)\circ Y_\kappa$; the hypothesis and Hoeffding's bound $\log \mathrm{ref}(e^{s(G-\mathrm{ref}G)}) \le s^2 \operatorname{osc}(G)^2/8$ give $\sum_\kappa t_\kappa[P(G_\kappa) - \mathrm{ref}_\kappa(G_\kappa)] \le \mathsf{H} + \#\mathcal{K}\Delta + \sum_\kappa t_\kappa^2 \operatorname{osc}(G_\kappa)^2$; optimise over the t_κ. (b) This is the proof of Prop. 17.34(a) for a product: for real $\Phi_{\mathbf{k}}$ of the cells, the array form of the chessboard estimate on the torus ℓ for the board of side $\ell'/2$ gives $\mu_{K,\ell,\beta}(\prod_{\mathbf{k}} e^{\Phi_{\mathbf{k}}}) \le \prod_{\mathbf{k}} \|e^{\Phi_{\mathbf{k}}}\|^{(\ell')}_{\mathrm{ch},\ell}$, the fine-to-coarse form gives the factor $e^{\delta'/8}$ per cell, $\|e^\Phi\|^{(\ell')}_{\mathrm{ch},\ell} \le e^{\delta'/8}\|e^\Phi\|_{\mathrm{ch},\ell'}$, and Hölder's inequality on the torus ℓ' gives $\|e^\Phi\|_{\mathrm{ch},\ell'} \le \mu_{K,\ell',\beta}(e^{8\Phi})^{1/8}$, the eight mirror copies of $e^{8\Phi}$ having the mean of one cell. (c) Identify the bonds of W with those of the torus ℓ.

The conditional law of the bonds of W and the law of the torus have the same weight but on the plaquettes that are not common to both, the $12n^2 - 3n$ plaquettes with a bond in W that meet the outside and the $6n^2 - 3n$ plaquettes of the torus that wrap around it, a surface; as $a_p \in [0,2]$, the density of the first law against the second lies between $e^{-36\beta n^2}$ and $e^{36\beta n^2}$ (Prop. YM3_I:91.8), so their relative entropy is $\mathsf{H} \le 36\beta n^2$. Then (a), with (b) on the torus ℓ and $\#\mathcal{K} = 8(n/n')^3$: the sum is at most $4\cdot 36\beta n^2 + 4\cdot 8(n/n')^3\delta'/8$, and the mean over the cells is at most $18\beta n'^3/n + \delta'/2$. (d) The reflection in a wall plane is a symmetry of the box, as every n_μ is even, and it is reflection positive by the trace formula (Lem. 17.35(a)).

So the chessboard estimate on the box holds (Prop. YM3_I:86.3): the mean of a product over the cells is at most the product of the seminorms of the factors on the box (equation YM3_I:(1756)), the $(\mathsf{n}_1\mathsf{n}_2\mathsf{n}_3)$-th roots of the means of the products of all their mirror copies. By the subadditivity in the volume (Prop. YM3_I:86.5) at $\underline{\mathsf{n}}' = (2,2,2)$, which is the trace formula with the inequality $\mathrm{Tr}(Y^*Y)^q \le (\mathrm{Tr}\,Y^*Y)^q$ in each direction as in the proof of Lem. 17.35(b), the seminorm on the box of a function G of one cell is at most $\|G\|_{\mathrm{ch};2\lambda}$ times $\exp[\frac{1}{8}F((2\lambda)^3) - \lambda^3 F(\underline{n})/|\underline{n}|] \le e^{\delta((2\lambda)^3)/8}$, as $F(\underline{n})/|\underline{n}| \ge P$.

For $G = e^\Phi$, Hölder's inequality on the cube gives $\|e^\Phi\|_{\mathrm{ch};2\lambda} \le \langle e^{8\Phi}\rangle^{1/8}_{(2\lambda)^3}$, the eight mirror copies of $e^{8\Phi}$ having the mean of one cell; this is the hypothesis of (a) with the reference law of one cell of the cube $(2\lambda)^3$. □

Lemma 17.37 (The operator of a cell; two cells cluster)**.** Let $K \in \mathbb{N}$, $\beta > 0$, a board of side $\mathsf{l} = \ell''/2$, $\ell'' \in \mathfrak{L}$, and $\ell \in \mathfrak{L}$, $\ell \geq 2\mathsf{l}$, $N = \ell/a_K = \mathsf{nw}$ with $\mathsf{w} = \mathsf{l}/a_K$ layers to a cell; a direction as the time of the cube N^3, $\Sigma = (N, N)$.

(a) For F of the bonds of one cell, Φ_F its slab function: $\|\mathfrak{Y}[\Phi_F]\| \leq Z(N^3)^{1/\mathsf{n}} \|F\|_{\mathrm{ch},\ell}^{(\ell'')} = \lambda_0^{\mathsf{w}}(1+\mathsf{r}_N)^{1/\mathsf{n}} \|F\|_{\mathrm{ch},\ell}^{(\ell'')}$, with the seminorm of the setting of all the mirror copies of F on the torus ℓ under the board of side l.

(b) Let $\mathsf{e}(l) \leq x_0$ at β, ν the periodic Gibbs state of Prop. 17.38, $\ell'' \geq 2la_K$, $Q'' = e^{\delta((\ell''/a_K)^3)/8}$. For F, G of the bonds of two cells whose indices differ by $j \geq 1$ in a direction: $|\nu(FG) - \nu(F)\nu(G)| \leq \theta^{(j-1)\mathsf{w}/l} Q''^2 \|F\|_{\mathrm{ch},\ell''} \|G\|_{\mathrm{ch},\ell''}$; on every torus $\ell \geq L\ell''$ of $\mathfrak{L}$, with $j_- = \min\{j, \mathsf{n} - j\} \geq 2$ the distance on the torus, the same with $3.44\,\theta^{(j_- - 1)\mathsf{w}/l}$.

The constants do not depend on K, β, ℓ, F, G or the offset. ■

Proof. (a) The operator norm is at most the Schatten norm of order n ($\mathsf{n} = 2\ell/\ell''$ is even), and by the trace formula (M1) $\mathrm{Tr}(\mathfrak{Y}^*\mathfrak{Y})^{\mathsf{n}/2}$ is $Z(N^3)$ times the mean of the product of the n mirror copies of F along the time, cells of the board of side l in one column, which the array form of (M2) bounds by $(\|F\|_{\mathrm{ch},\ell}^{(\ell'')})^{\mathsf{n}}$. (b) *The state ν.* A function of the second cell is the reflected image of a function of the slab of the first, so the spectral formula of Prop. 17.38(f) applies; over the cross-section (N', N'), $N' = \ell'/a_K$, $\ell' \geq \ell''$ in $\mathfrak{L}$, the covariance is $\lambda_0^{-2\mathsf{w}} \sum_{i \geq 1} (\lambda_i/\lambda_0)^n \overline{\mathsf{a}_i} \mathsf{b}_i$ with $n = (j-1)\mathsf{w}$ layers between the cells (at $n = 0$ with the term of the kernel of T' besides), and Schwarz's inequality bounds it by $(\lambda_1/\lambda_0)^n \lambda_0^{-2\mathsf{w}} \|\mathfrak{Y}[\Phi_F]\| \|\mathfrak{Y}[\Phi_G]\|$.

The gap of Prop. 17.33(a) gives $(\lambda_1/\lambda_0)^n < \theta^{n/l}$, and (a) on the cube N' with the fine-to-coarse form of (M2), $\|F\|_{\mathrm{ch},\ell'}^{(\ell'')} \leq Q'' \|F\|_{\mathrm{ch},\ell''}$ as $p_{\ell'} \geq p_\infty$, gives $\lambda_0^{-\mathsf{w}} \|\mathfrak{Y}[\Phi_F]\| \leq (1+\mathsf{r}_{N'})^{1/\mathsf{n}'} Q'' \|F\|_{\mathrm{ch},\ell''}$, $\mathsf{n}' = N'/\mathsf{w}$, the same for G; as $\ell' \to \infty$ the first factor tends to 1, and the formula passes to ν (Prop. 17.38(c), (f)). *The torus $\ell \geq L\ell''$.* The cube $N = \mathsf{nw}$ has $\mathsf{n} = 2\ell/\ell'' \geq 2L \geq 6$ cells along the time, an even number. The two cells, $j_- \geq 2$ apart the shorter way, leave $n = (j_- - 1)\mathsf{w}$ layers between them on one side and $E = (\mathsf{n} - j_- - 1)\mathsf{w} \geq n$ on the other, and $n, E \geq \mathsf{w} \geq l$, as $\ell'' \geq 2la_K$.

Write, as in the proof of Lem. 17.40, $\mathsf{T}'^m = \lambda_0^m(|e_0\rangle\langle e_0| + \mathsf{R}_{(m)})$, e_0 the top eigenvector, so that $\|\mathsf{R}_{(m)}\| = \varrho^m$, $\varrho = \lambda_1/\lambda_0$, and $\mathrm{Tr}\,|\mathsf{R}_{(m)}| = \mathsf{r}_m$, and put $y_C = \lambda_0^{-\mathsf{w}} \mathfrak{Y}[\Phi_C]$. The trace formula (M1) with the four slabs F, 1_n, G, 1_E gives $\mu_{K,\ell,\beta}(FG) = \mathrm{Tr}\big((|e_0\rangle\langle e_0| + \mathsf{R}_{(E)})\, y_G\, (|e_0\rangle\langle e_0| + \mathsf{R}_{(n)})\, y_F\big)/(1+\mathsf{r}_N)$, and the mean of one cell is $\mu_{K,\ell,\beta}(C) = [\langle e_0, y_C e_0\rangle + \mathrm{Tr}(\mathsf{R}_{(N-\mathsf{w})} y_C)]/(1+\mathsf{r}_N)$.

Expanding, the products $\langle e_0, y_F e_0\rangle\langle e_0, y_G e_0\rangle$ cancel up to the factors $(1+\mathsf{r}_N)^{-1}$ and $(1+\mathsf{r}_N)^{-2}$, and $|\mathrm{Tr}(XW)| \leq \|X\| \,\mathrm{Tr}\,|W|$ bounds the rest:

$$|\mu_{K,\ell,\beta}(FG) - \mu_{K,\ell,\beta}(F)\mu_{K,\ell,\beta}(G)| \leq \|y_F\|\,\|y_G\|\,[\mathsf{r}_N + \varrho^n + \varrho^E + \varrho^n \mathsf{r}_E + 2\mathsf{r}_{N-\mathsf{w}} + \mathsf{r}_{N-\mathsf{w}}^2],$$

the fourth term as the trace norm of $\mathsf{R}_{(E)}$ times the operator norm of $y_G \mathsf{R}_{(n)} y_F$. By (a) and the fine-to-coarse form, $\|y_C\| \leq (1+\mathsf{r}_N)^{1/\mathsf{n}} Q'' \|C\|_{\mathrm{ch},\ell''}$. Prop. 17.33 bounds $\varrho^n, \varrho^E \leq \theta^{n/l}$ by (a), and r_N, $\mathsf{r}_{N-\mathsf{w}}$, r_E by its thermal law (b) over the cross-section (N, N): in the zone $2t > N$ by $0.26261\,\theta^{t/l}$, which serves for $t = N$, $N - \mathsf{w}$ and for $t = E$ if $2E > N$, and in the zone $2t \leq N$ by $\mathsf{r}_t \leq e^{0.27263(N/2t)^2\theta^{t/l}} - 1$. Every term of the bracket, divided by $\theta^{n/l}$, and the factor $(1+\mathsf{r}_N)^{2/\mathsf{n}}$ increase with $u = \theta^{\mathsf{w}/l} \leq \theta \leq 0.8901$; at $u = 0.8901$ the product of the bracket over $\theta^{n/l}$ and $(1+\mathsf{r}_N)^{2/\mathsf{n}}$ is at most 3.33 for all even $\mathsf{n} \geq 6$ and $2 \leq j_- \leq \mathsf{n}/2$, the greatest value 3.3214 being taken at $\mathsf{n} = 6$, $j_- = 3$, where $n = E = 2\mathsf{w}$ and the bracket is at most $u^2[2 + 0.6258 + 0.26261(u^4 + 2u^3 + 0.26261\,u^8)]$. This is the torus clause, with $3.33 \leq 3.44$. □

Proposition 17.38 (One lattice under the inequality)**.** Let $K \in \mathbb{N}$, $\beta > 0$, and $\mathsf{e}(l) \leq x_0$ at β for one integer $l \geq 3$.

(a) The lattice $a_K\mathbb{Z}^3$ has exactly one periodic Gibbs state ν at β; it is ergodic under $a_K\mathbb{Z}^3$, invariant under all isometries of the lattice and reflection positive at every lattice mirror;

there is an open interval $U \ni \beta$ on which $\mathsf{e}(2l) \le 0.069$, P is continuously differentiable on U, and $P'(\beta') = -3\nu_{\beta'}(a_p)$.

(b) For every bounded real Borel F of the bonds of the cell $[-\frac{\ell'}{4}, \frac{\ell'}{4}]^3$, $\ell' \in \mathfrak{L}$, $n' = \ell'/a_K \ge 2l$, every $n = \ell/a_K$ with $\ell \ge \ell'$ in $\mathfrak{L}$, and the cube $Q_M = \{|x|_\infty \le Ma_K\}$, $M \ge n$, of $2M+1$ sites a side: $\|\nu(F \,|\, \text{bonds outside } Q_M) - \nu(F)\|_{L^2(\nu)} \le \operatorname{osc}(F)\, \varepsilon_1(n, n')$, $\varepsilon_1(n,n')^2 < 18\beta n'^3/n + 0.06868\, \theta^{n'/2l}$. So ν is trivial on the tail σ-field and an extreme point of the set of all Gibbs states at β, and for ν-almost every boundary datum the laws in growing cubes converge weakly to ν. The generator $\mathsf{H}_\nu \ge 0$ of the reversible lattice Langevin dynamics of ν, the operator of the closure of the form $\sum_b \nu(\nabla_b F \cdot \nabla_b F')$ on smooth local functions, has $\ker \mathsf{H}_\nu = \mathbb{C}1$, and $e^{-s\mathsf{H}_\nu} F \to \nu(F)$ in $L^2(\nu)$ for every F.

(b′) With a coordinate direction as the time, $\lambda = 2h+1 \ge \max\{3, l\}$, $t \ge \max\{2, l\}$ and $n = h+t$: for every bounded real Borel F of the bonds of $\{|x|_\infty \le ha_K\}$, $\|\nu(F \,|\, \text{bonds outside } \{|x_3| < na_K\}) - \nu(F)\|_{L^2(\nu)} \le \operatorname{osc}(F)\, \varepsilon_2$, $\varepsilon_2^2 = 0.27472\, (\lambda/t)^2 \theta^{t/l} + 0.06868\, \theta^{\lambda/l}$; with $h = \lfloor n/3 \rfloor$, for $n \ge \frac{1}{2}(3l+1)$, this is $\le 0.7057 \operatorname{osc}(F)\, \theta^{(2n-1)/6l}$.

(c) For every even $N \ge 2l$, every corner c, every bounded real Borel Ψ of the bonds of $\mathsf{C}_{N/2}(c)$ and every periodic box $\underline{n}$ with all sides $\ge N$: $|\langle\Psi\rangle_{\underline{n}} - \langle\Psi\rangle_{N^3}| \le 2.71\, \theta^{N/4l} \|\Psi\|_{\mathrm{ch};N}$ and $|\langle\Psi\rangle_{\underline{n}} - \nu(\Psi)| \le 5.42\, \theta^{N/4l} \|\Psi\|_{\mathrm{ch};N} \le 5.42\, \theta^{N/4l} \sup|\Psi|$; for events A of the cell, $|\mu_{\underline{n}}(A) - \nu(A)| \le 2.71\, \theta^{N/4l}$. As all sides tend to ∞ the measures of the periodic boxes converge weakly to ν, uniformly over all boxes with three sides $\ge N_0$.

(d) *Twisted boxes.* Let Γ_n be period lattices and T_n closed twist sets (Γ_n-periodic sets of plaquettes such that every elementary cube has an even number of faces in T_n; the sign of $\operatorname{tr} \partial U$ is reversed on them) that recede: the shortest period of Γ_n tends to ∞ and every finite set of plaquettes misses T_n at last. Then the laws $\mu^{\mathsf{T}_n}_{\Gamma_n,\beta}$, and all mixtures of a bounded number of such sequences with any weights (the sum over the eight sectors among them), converge weakly to ν; and if $\mathsf{T}'_n = \mathsf{T}_n \triangle d\mathsf{S}_n$ with any Γ_n-periodic set S_n of bonds, $d\mathsf{S}$ the plaquettes with an odd number of bonds in S, the joint cumulants of the twisted fields $\mathcal{O}_a^{\mathsf{T}'_n}(f) = a^{-1} \sum_p f(x(p)) a_p^{\mathsf{T}'_n}$, $a_p^{\mathsf{T}} = 1 + \frac{1}{2} \operatorname{tr} \partial U(p)$ on T and a_p elsewhere, under $\mu^{\mathsf{T}'_n}_{\Gamma_n,\beta}$ converge to those of the $\mathcal{O}_a(f)$ under ν. The word *recede* cannot be dropped: $P(\beta) > -3\beta$, so $\nu(\operatorname{tr} \partial U(p)) > 0$, and a column through a fixed plaquette p_0 gives in the limit the law of $-\operatorname{tr} \partial U(p_0)$ under ν, which differs from that of $\operatorname{tr} \partial U(p_0)$.

(e) Over every cross-section (A, B) with $A, B \ge 2l$ the top is simple, lies in the range of P_{00}, and $\lambda_1/\lambda_0 < \theta^{1/l}$ (Prop. 17.33(a)).

(f) Let Sl be a slab of $\vartheta \ge 2$ consecutive layers, Θ_n the lattice reflection such that exactly n layers lie between Sl and Θ_nSl, $A, B \in L^2(\nu)$ functions of the bonds of Sl, and $h_{AB}(n) = \nu(\overline{\Theta_n A}\, B)$. For all integers $n \ge n_0 \ge 0$,

$$|h_{AB}(n) - \overline{\nu(A)}\, \nu(B)| \le \theta^{(n-n_0)/l} \prod_{C=A,B} [h_{CC}(n_0) - |\nu(C)|^2]^{1/2},$$

and $n \mapsto h_{CC}(n) - |\nu(C)|^2$ is nonnegative and log-convex; at $n_0 = 0$, for bounded A, B: $|\operatorname{Cov}_\nu| \le \theta^{n/l} \sup|A| \sup|B|$.

■

Proof. (a) A periodic Gibbs state exists as a limit of tori (M3). If ρ and ν are two of them and A is a local event inside the central cell of side $\ell'/2$, Prop. 17.34(b) gives $|\rho(A) - \nu(A)| \le \sqrt{\varepsilon} + \varepsilon$, $\varepsilon = 22.3(e^{\delta'/7} - 1)$, and $\delta' \to 0$ as $\ell' \to \infty$ by Prop. 17.32(b); so $\rho = \nu$. The symmetries and reflection positivity come from (M3); almost every ergodic component of ν is a periodic Gibbs state, hence ν. By Prop. 17.32(a), $\mathsf{e}(2l) \le 12.9 \cdot 0.069^2 < 0.069$ at β, hence on an open interval U, the box functions being smooth in β; there the derivatives $-3\mu_{K,\ell_m,\beta'}(a_p)$ of $(\ell_m/a_K)^{-3} \log Z_{K,\ell_m}$, $\ell_m = 2L^m$, converge uniformly to $-3\nu_{\beta'}(a_p)$ by Prop. 17.34(b) and the bound of Prop. 17.32(b),

uniform on U. (b) Let D be the cube of n sites a side that is the union of the $8(n/n')^3$ cells $\square_{\mathbf{k}} = [-\frac{\ell'}{4}, \frac{\ell'}{4}]^3 + \frac{\ell'}{2}\mathbf{k}$, $0 \le \mathbf{k}_\mu < 2n/n'$, W the set of the positively oriented bonds with initial point in D, as in Lem. 17.36(c), and $F_{\mathbf{k}}$ the translate of F into $\square_{\mathbf{k}}$.

Lem. 17.36(c), integrated against ν, bounds the mean over $\mathbf{k}$ of $\|\nu(F_{\mathbf{k}}|\text{outside } W) - \mu_{K,\ell',\beta}(F)\|^2$ by $\operatorname{osc}(F)^2\varepsilon_1^2$, and by translation invariance the term of $\mathbf{k}$ is $\|\nu(F|\text{outside } W - \frac{\ell'}{2}\mathbf{k}) - \mu_{K,\ell',\beta}(F)\|^2$. Each translate $W - \frac{\ell'}{2}\mathbf{k}$ holds the cell and has its bonds in $[-\ell,\ell]^3$, so it lies in $Q_n \subset Q_M$; conditioning on the smaller field outside Q_M is a contraction in $L^2(\nu)$, so every term, and with them the mean, is at least $\|\nu(F|\text{outside } Q_M) - \mu_{K,\ell',\beta}(F)\|^2$; $\delta' < 0.13736\,\theta^{n'/2l}$; and the mean $\nu(F)$ is the best constant in $L^2(\nu)$.

Let $n \to \infty$, then $n' \to \infty$: tail triviality, which for a Gibbs state is extremality. A function in $\ker \mathsf{H}_\nu$ has $\nabla_b F = 0$ for every bond, so (the conditional law of one bond being equivalent to Haar measure) it is measurable outside every finite set of bonds, hence constant; the rest is the spectral theorem. (b′) Over the cross-section the top eigenvector e_0 is positive and continuous; in the cylinder the joint density of the two layers $V^{\pm n}$ against $\pi \otimes \pi$, $\pi = e_0^2 dV$, is $p_{2n} = \mathsf{D}^{2n}/\lambda_0^{2n} e_0 \otimes e_0$, and given them the slab of the cells has the density $\varrho = p_t(x,y)p_t(y',x')/p_{2n}(x,x')$ against its own law; the mean relative entropy is $2\iint p_t \log p_t - \iint p_{2n}\log p_{2n} \le 2\log \iint p_t^2 = 2\log(1+\mathsf{r}_{2t})$.

Lem. 17.36(a) for the n_2^2 cells of the slab (their sub-independence is Lem. 17.36(d) on the boxes $(\lambda\mathsf{n}_1, N', N')$, $N' = \lambda\mathsf{n}_2$, with even $\mathsf{n}_1, \mathsf{n}_2$, in the limit $\mathsf{n}_1 \to \infty$, the reference law being that of one cell of the cube $(2\lambda)^3$ and $\Delta = \delta((2\lambda)^3)/8)$, translation invariance of the cylinder along the cross-section, and $\log(1+\mathsf{r}_{2t}) \le \delta(2t, N', N') < 0.13736(N'/2t)^2\theta^{t/l}$ give the bound; then $N' \to \infty$ by (c) and density. (c) Lem. 17.35(c) with $\delta(N^3) < 0.13736\,\theta^{N/2l} \le \frac{1}{2}$, $7.31 \cdot 0.370622 \le 2.71$; a second box with all sides tending to ∞ gives ν; from continuous to Borel Ψ by approximation in L^8 of a measure that dominates both laws; for events take $\Psi = 1_A - \frac{1}{2}$.

The uniform convergence: limit points of box measures are periodic Gibbs states (M3), hence ν. (d) For a Γ-periodic set S of bonds, the flip $U_b \mapsto -U_b$, $b \in \mathsf{S}$, keeps the Haar measure and carries μ_Γ^{T} to $\mu_\Gamma^{\mathsf{T}\triangle d\mathsf{S}}$ (-1 is central). For a closed T and the bonds S whose plaquette swept in the direction μ lies in T, $\tau_{-ae_\mu}\mathsf{T} = \mathsf{T}\triangle d\mathsf{S}$, by counting the faces of the elementary cubes; so the translates of the twisted measure are flips of it, which a local function does not see once T_n has receded. Limit points satisfy the DLR equations away from the twist sets, hence are periodic Gibbs states, hence ν; both identities are linear in the law.

For $P(\beta) > -3\beta$: on the $\prod\lfloor n_i/3\rfloor \ge |\underline{n}|/125$ plaquettes $p_{12}(x)$, $x_i \in 3\mathbb{Z}$, which share no site, Jensen's inequality gives $F/|\underline{n}| \ge -3\beta + \frac{1}{125}\log\int e^{\frac{\beta}{2}\operatorname{tr} u}\,\mathrm{d}u > -3\beta$; P is convex, $P(0) = 0$, so $P'(\beta) > -3$, and $P' = -3\nu(a_p)$ by (a). (e) is Prop. 17.33(a). (f) In the box with $N = 2\vartheta + n + E'$ layers over (N', N'), (M1) gives $\mu(\overline{\Theta_n A}\,B) = \operatorname{Tr}(\mathsf{T}'^{E'}Y_A^*\mathsf{T}'^m Y_B)/\operatorname{Tr}\mathsf{T}'^N$, $Y_C = \mathfrak{Y}[\Phi_C]$ the slab operator of the function C. As $E' \to \infty$ the gap projects onto the top eigenvector e_0:

$$\mu(\overline{\Theta_n A}\,B) \longrightarrow \lambda_0^{-2\vartheta}\Big[\sum_{j\ge 0}(\lambda_j/\lambda_0)^n\,\overline{\mathsf{a}_j}\,\mathsf{b}_j + 1_{n=0}\,\langle \mathsf{P}_{\ker}Y_A e_0, \mathsf{P}_{\ker}Y_B e_0\rangle\Big], \qquad \mathsf{a}_j = \langle e_j, Y_A e_0\rangle,\ \mathsf{b}_j = \langle e_j, Y_B e_0\rangle,$$

e_j the eigenvectors of the nonzero eigenvalues and $\mathsf{P}_{\ker}$ the projection on $\ker\mathsf{T}'$; the term $j = 0$ is the product of the means, which converge in the same way. Schwarz's inequality and $\lambda_j/\lambda_0 < \theta^{1/l}$ for $j \ge 1$ give the bound (for $n > n_0$ the kernel term is absent on the left); $N' \to \infty$ by (c), and $L^2(\nu)$ by density. The same formula exhibits $h_{CC}(n) - |\nu(C)|^2$ as a limit of $\int x^n \varrho(\mathrm{d}x)$, $0^0 = 1$, with $\varrho = \lambda_0^{-2\vartheta}[\sum_{j\ge1}|\mathsf{c}_j|^2\delta_{\lambda_j/\lambda_0} + \|\mathsf{P}_{\ker}Y_C e_0\|^2\delta_0] \ge 0$ on $[0,1)$; such functions are ≥ 0 and log-convex. □

Proposition 17.39 (Flux and loops on one lattice)**.** Let $\beta > 0$, $l \ge 3$, $\mathsf{e}(l) \le x_0$, $\Lambda = \log(1/\theta)$, $\sigma^+ = (2\Lambda + 1.67)/4l^2$, so that $\sigma^+ l^2 = \frac{\Lambda}{2} + 0.4175 \ge 0.4757$, $\mathsf{q} = 1 + 0.835/\Lambda$, and ν as in Prop. 17.38; lengths in lattice units.

(a) $0 \le 1 - r_1(N) \le 2\exp[-2.2 \cdot 10^{-4}(N/l)^2]$ for every integer $N \ge 3$; $\frac{1}{2}(1 - r_1(N)) \le \exp[-1.4268 \cdot 10^{-3}(N/l)^2]$ for $N \ge 256\, l$; and for every $N \ge 2\mathsf{q}l$,

$$\tfrac{1}{2}(1 - r_1(N)) \le \exp\Big[-\sigma^+ N^2 + \frac{2\mathsf{q}N}{l}\Big(\frac{1}{2}\log\frac{2\mathsf{q}N}{l} + 0.86\Big)\Big];$$

and $1 - r^z \le 6\exp[\cdots]$ for all seven classes $z \ne 0$, with each of the three exponents.

(b) $0 \le \nu(W_C) \le e^{-\sigma^+ RT}$ for every rectangle C with $R \times T$ bonds in a coordinate plane; in the finite cube, for $n \ge 2\mathsf{q}l$, $1 \le R \le n/2$, $1 \le T \le n - 1$: $0 \le \langle W_C\rangle_{(n,n,n)} \le 8^{T/n}\exp[-\sigma^+ RT + \frac{2\mathsf{q}RT}{ln}(\frac{1}{2}\log\frac{2\mathsf{q}n}{l} + 0.86)]$; and on every box, with no hypothesis on e, for $1 \le R \le n_1/2$, $1 \le T \le n_3 - 1$ and $\zeta = \frac{1}{2}(1 - r)$ of one twisted column of the plane (13): $0 \le \langle W_{R\times T}\rangle_{\underline{n}} \le (8\,\zeta^{R/n_1})^{T/n_3}$.

(c) $|\nu(W_C)| \le e^{-\sigma^+|A(C)|}$ for every staircase loop of a coordinate plane, and $|\nu(W_C)| \le 2^{c_3(C)-1}e^{-\sigma^+ A_{13}(C)}$ for every closed lattice path C with a bond of the direction 3, with A_{13}, c_3 as in Thm. AZ(4) at unit spacing; the same for every ordered pair of directions. Here a *staircase loop* $C = C(\alpha, R)$ of the plane (13) is the staircase loop of the setting (Twists and loops), read in lattice units: it is given by a site o, an integer $T \ge 1$ and integers α_j and $R_j \ge 1$, $0 \le j < T$; with $o_j = o + \alpha_j e_1$ and $r_j = o + (\alpha_j + R_j)e_1$ it runs at the height 0 (in the direction 3, counted from o) straight from o_0 to r_0; then, for $j = 0, \dots, T - 1$ in turn, one step up above r_j from the height j to the height $j + 1$ and, if $j < T - 1$, straight at that height to above r_{j+1}; at the height T straight back from above r_{T-1} to above o_{T-1}; then, for $j = T - 1, \dots, 0$ in turn, one step down above o_j from the height $j + 1$ to the height j and, if $j > 0$, straight at that height to above o_{j-1}.

Its area is $|A(C)| = \sum_j R_j$; the staircase loops of the other coordinate planes are the images of these under the symmetries of the lattice. A staircase loop winds once about each unit square of the rows $[\alpha_j, \alpha_j + R_j] \times [j, j + 1]$ and about no other, and its steps in the direction 3 are T upward ones followed by T downward ones; so $A_{13}(C) = |A(C)|$, $c_3(C) = 1$, and the first clause is a case of the second.

(d) For every weak limit point of the measures of periodic boxes all of whose sides tend to infinity, the limits of the tori among them, $w(R, T) = \langle W_{R\times T}\rangle$ satisfies $0 \le w \le 1$, $w(R, T)^2 \le w(R, T - 1)w(R, T + 1)$ and $w(R, T)^2 \le w(R - 1, T)w(R + 1, T)$, with $w(R, 0) = w(0, T) = 1$; so $V(R) = \lim_T -T^{-1}\log w(R, T) = \inf_T(-T^{-1}\log w(R, T))$ exists in $[0, \infty]$, $V(0) = 0$, V is either finite everywhere or infinite at every $R \ge 1$, V is concave and non-decreasing, and, in the finite case, $V(R) \ge V(R') + \sigma_\infty(R - R')$ for $R \ge R' \ge 0$, with $\sigma_\infty = \lim_R V(R)/R \in [0, \infty)$. Under the hypothesis, in ν, which is the limit of the tori: $V(R) \ge V(R') + \sigma^+(R - R')$ for $R \ge R' \ge 0$.

(e) With $\varepsilon_{\mathrm{v}}(l') = \varepsilon_{\mathrm{v}}(l', l')$ the vortex free energy of Prop. 17.33(d) and $l_j = 2^{j+1}l$: $l_j\varepsilon_{\mathrm{v}}(l_j) \le e^{-(2\Lambda+1.67)4^j} = e^{-\sigma^+ l_j^2}$ for all $j \ge 0$.

■

Proof. For a box $(t, u; h)$, t, u the sides of a plane with one twisted column of length h, let $\zeta(t, u; h) = \frac{1}{2}(1 - r)$ for that twist, and let $\varepsilon_{\mathrm{v}}(t, u) = \log(\|\mathsf{T}^{(0)}\|/\|\mathsf{T}^{(1)}\|)$ over the cross-section (t, u), the time along h. (i) By Prop. 17.31(d), in each of the two sides t, u of the twisted plane ζZ and $(1 - \zeta)Z$ are sums of powers of numbers ≥ 0; Hölder's inequality for such sums and the excess give

$$\zeta(t', u'; h) \le [\zeta(t, u; h)\, e^{\delta(t,u,h)}]^{t'u'/tu} e^{-\delta(t',u',h)} \qquad (t' \ge t,\ u' \ge u),$$

and $\zeta(t, u; h) \le 1 - e^{-\mathsf{G}(t;u,h)} \le \delta(t, u, h)$. (ii) In the third side $Z = \mathrm{Tr}(\mathsf{T}^{(0)})^h = \lambda_0^h(1 + \mathsf{r}_h(t, u))$ and $Z^z = \mathrm{Tr}(\mathsf{T}^{(1)})^h$, with $\mathsf{T}^{(1)} \ge 0$ of norm $\lambda_0 e^{-\varepsilon_{\mathrm{v}}(t,u)}$. So $1 - 2\zeta(t, u; h) = Z^z/Z \ge e^{-h\varepsilon_{\mathrm{v}}}/(1 + \mathsf{r}_h) \ge 1 - h\varepsilon_{\mathrm{v}} - \mathsf{r}_h$ if $h\varepsilon_{\mathrm{v}} \le 1$, and $2\zeta < 1$ always: the first inequality below. Further $\mathrm{Tr}(\mathsf{T}^{(1)})^{2h} \le \|\mathsf{T}^{(1)}\|^h \mathrm{Tr}(\mathsf{T}^{(1)})^h$, $Z^z \le Z$ at the length h (Prop. 17.31(d), with the time in the twisted plane) and

$Z \ge \lambda_0^{2h}$ at the length $2h$ give $e^{-h\varepsilon_{\rm v}} \ge (1-2\zeta(t,u;2h))/(1+\mathsf{r}_h)$, the second:

$$2\zeta(t,u;h) \le h\,\varepsilon_{\rm v}(t,u) + \mathsf{r}_h(t,u), \qquad h\,\varepsilon_{\rm v}(t,u) \le \mathsf{G}(h;t,u) - \log(1-2\zeta(t,u;2h)) \le \mathsf{d}(h;t,u) - \log(1-2\zeta(t,u;2h)),$$

where $\mathsf{d}(h;t,u) = 2F(t,u,h) - F(t,u,2h) = 2\mathsf{G}(h;t,u) - \mathsf{G}(2h;t,u) \ge \mathsf{G}(h;t,u)$ is taken for the box (t,u,h) with the side h doubled; also $\mathsf{r}_h(t,u) \ge \mathsf{G}(h;t,u)$ may stand for $\mathsf{G}(h;t,u)$. (iii) By (i) and (ii), doubling the cross-section of a long tube, on which the thermal sums lie below the fourth power, replaces $l'\varepsilon_{\rm v}(l')$ by a bounded multiple of its fourth power. Let $\mathsf{t} = \theta^{l'/l}$ at the dyadic sides $l' = l_j = 2^{j+1}l$, and let $\mathsf{k}_1 = 0.01655$ and $\mathsf{k}_2 = 0.001615$ be the constants of the bounds $\delta((2^jl)^3) \le \mathsf{k}_1\theta^{2^j}$, $j \ge 1$, of Prop. 17.32(b) and $\mathsf{r}_N \le \mathsf{k}_2\theta^{N/l}$ at $N = 2^{k+1}l$ of Prop. 17.32(c); so $\delta(l',l',h) \le (h/l')\,\mathsf{k}_1\mathsf{t}$ by the monotone density, $\mathsf{r}_{l'}(l',l') \le \mathsf{k}_2\mathsf{t}$ and $\mathsf{r}_{2l'}(2l',2l') \le \mathsf{k}_2\mathsf{t}^2$; and $\mathsf{r}_{Ms} \le \mathsf{r}_s^M$ for real $M \ge 1$ with Ms an integer, as in Prop. 17.31(b).

The start. The second inequality of (ii) at $t = u = h = 2l$, with $\mathsf{G} \le \mathsf{r}$ and $\zeta(2l,2l;4l) \le \mathsf{R}_{2l}(2l,4l) = \mathsf{r}/(1+\mathsf{r})$, $\mathsf{r} = \mathsf{r}_{2l}(2l,4l)$, by (i), gives $l_0\varepsilon_{\rm v}(l_0) \le \mathsf{r}_{2l}(2l,2l) + \log\frac{1+\mathsf{r}}{1-\mathsf{r}}$. By Prop. 17.31(b), $\mathsf{r}_{2l}(2l,2l) \le \mathsf{r}_l(2l,2l)^2 \le \varphi(e^{x_0}-1)^2$, and $\mathsf{r} \le \mathsf{r}_l(2l,4l)^2 \le \varphi(e^{y_1}-1)^2$ with $y_1 = 2F(l,2l,4l) - F(2l,2l,4l) \le 2x_0 + \psi(x_0)$, as in the proof of Prop. 17.32. Both bounds are power series in x_0 with coefficients ≥ 0 beginning with multiples of x_0^2, and so is $\log\frac{1+\mathsf{r}}{1-\mathsf{r}}$ in r; the sum divided by $\theta^2 = (12.9\,x_0)^2$ is therefore greatest at $x_0 = 0.069$, where it is at most 0.016154: $l_0\varepsilon_{\rm v}(l_0) \le 0.016154\,\theta^2$.

One doubling. Let $\hat N \ge 2$ be an integer. The first inequality of (ii) in the tube $(l',l';2\hat N l')$, with $\mathsf{r}_{2\hat N l'}(l',l') \le (\mathsf{k}_2\mathsf{t})^{2\hat N}$, gives $\zeta(l',l';2\hat N l') \le \bar z_1 = \hat N l'\varepsilon_{\rm v}(l') + \frac12(\mathsf{k}_2\mathsf{t})^{2\hat N}$; (i) with $t'u'/tu = 4$ and $\delta(l',l',2\hat N l') \le 2\hat N\mathsf{k}_1\mathsf{t}$ gives $\zeta(2l',2l';2\hat N l') \le \bar z_2 = \bar z_1^4 e^{8\hat N\mathsf{k}_1\mathsf{t}}$; and the second inequality of (ii) in the tube $(2l',2l';\hat N l')$, with $\mathsf{G} \le \mathsf{r}$, $\mathsf{r}_{\hat N l'}(2l',2l') \le \mathsf{r}_{2l'}(2l',2l')^{\hat N/2} \le (\mathsf{k}_2\mathsf{t}^2)^{\hat N/2}$ and $-\log(1-x) \le x/(1-x)$, gives, if $\bar z_2 < \frac12$,

$$2l'\varepsilon_{\rm v}(2l') \le \mathsf{U} := \frac{2}{\hat N}\Big\{(\mathsf{k}_2\mathsf{t}^2)^{\hat N/2} + \frac{2\bar z_2}{1-2\bar z_2}\Big\}.$$

The iteration. Take $\hat N_j = 4\cdot 3^j$ at $l' = l_j$, and let $\Lambda_0 = \log(1/0.8901) \approx 0.116421$. As $\hat N_j \ge 4\cdot 2^j$ and $2^{j+1}\hat N_j \ge 4^j$, every term of U at $\theta \le 0.8901$ is at most the term at $\theta = 0.8901$ times $e^{-4^{j+1}(2\Lambda-2\Lambda_0)}$, if the bounds on $l_j\varepsilon_{\rm v}(l_j)$ carry the factor $e^{-4^j(2\Lambda-2\Lambda_0)}$; so it is enough to follow numbers E_j with $l_j\varepsilon_{\rm v}(l_j) \le e^{-E_j-4^j(2\Lambda-2\Lambda_0)}$ at $\theta = 0.8901$, $\mathsf{t}_j = 0.8901^{2^{j+1}}$. The start gives such an $E_0 \ge \log(1/0.016154) + 2\Lambda_0 \ge 4.3414$; as U increases with $l'\varepsilon_{\rm v}(l')$, it is enough to take $E_0 = 4.3414$ and $E_{j+1} = -\log\mathsf{U}$ at E_j.

With $y_j = \hat N_j e^{-E_j}$ this reads $E_{j+1} = 4E_j - 3\log\hat N_j - 8\hat N_j\mathsf{k}_1\mathsf{t}_j - \log(2B_j)$, where $B_j = (\mathsf{k}_2\mathsf{t}_j^2)^{\hat N_j/2}y_j^{-4}e^{-8\hat N_j\mathsf{k}_1\mathsf{t}_j} + 2(\bar z_1/y_j)^4/(1-2\bar z_2)$; along this sequence B_j is at most 2.0920 at $j=0$, 2.0001 at $j=1$ and 2 to ten digits from $j=2$ on, $\bar z_2 \le 1.2\cdot 10^{-5}$, and the numbers $4^{-j}E_j - 2\Lambda_0$ are at least 4.1085, 2.6061, 1.9912, 1.7722, 1.7032, 1.6839, 1.6785 at $j = 0,\dots,6$. For $j \ge 6$, $8\hat N_j\mathsf{k}_1\mathsf{t}_j \le 0.000131$ and $B_j \le 2.0023$, so $E_{j+1} \ge 4E_j - 3\log\hat N_j - \log 4.01$, and summing the losses, $4^{-j}E_j \ge 4^{-6}E_6 - \sum_{i\ge6} 4^{-i-1}(3\log 4 + 3i\log 3 + \log 4.01) \ge 1.909266$; so $4^{-j}E_j - 2\Lambda_0 \ge 1.6764 \ge 1.67$ for every $j \ge 0$, and $l_j\varepsilon_{\rm v}(l_j) \le e^{-(2\Lambda_0+1.67)4^j - (2\Lambda-2\Lambda_0)4^j} = e^{-(2\Lambda+1.67)4^j}$: this is (e).

From the tubes to the cubes. For an integer $N \ge 2\mathsf{q}l$ let $j \ge 0$ be the largest integer with $4^j \le N/2\mathsf{q}l$, $w = 2^j$, $l' = l_j = 2wl$ and $M = N/l'$, a real number; then $w\mathsf{q} \le M < 4w\mathsf{q}$ and $M^2 < 2\mathsf{q}N/l$. The first inequality of (ii) in the tube $(l',l';N)$, with $N\varepsilon_{\rm v}(l') = M\,l'\varepsilon_{\rm v}(l') \le Me^{-(2\Lambda+1.67)w^2}$ by (e) and $\mathsf{r}_N(l',l') \le (\mathsf{k}_2\mathsf{t})^M \le e^{-2\Lambda wM} \le e^{-(2\Lambda+1.67)w^2}$, as $M \ge w\mathsf{q}$ and $2\Lambda\mathsf{q} = 2\Lambda + 1.67$, gives $\zeta(l',l';N) \le \frac{M+1}{2}e^{-(2\Lambda+1.67)w^2} \le Me^{-(2\Lambda+1.67)w^2}$; and (i) with $t'u'/tu = M^2$ and $\delta(l',l',N) \le M\mathsf{k}_1\mathsf{t}$ gives $\frac12(1-r_1(N)) = \zeta(N,N;N) \le [Me^{-(2\Lambda+1.67)w^2}e^{M\mathsf{k}_1\mathsf{t}}]^{M^2}$.

Here $M^2(2\Lambda+1.67)w^2 = \sigma^+N^2$; $M\mathsf{k}_1\mathsf{t} < 4\mathsf{q}\mathsf{k}_1we^{-2\Lambda w} \le \frac{4\mathsf{q}\mathsf{k}_1}{2e\Lambda} = \frac{4\mathsf{k}_1}{e}\big(\frac{1}{2\Lambda} + \frac{1.67}{4\Lambda^2}\big) \le 0.85478$, the last member decreasing in $\Lambda \ge \Lambda_0$; and $0 \le \log M < \frac12\log\frac{2\mathsf{q}N}{l}$. So the exponent is at most

$-\sigma^+N^2 + M^2(\log M + 0.855) \le -\sigma^+N^2 + \frac{2\mathsf{q}N}{l}(\frac12 \log\frac{2\mathsf{q}N}{l} + 0.86)$: (a), the last display; the other two bounds of (a) are the same argument with the start at $64\,l$, where $D_2(64l) \le 0.12019\,\theta^{64}$, and Prop. 17.31(d) gives the seven classes. (iv) Two static charges at the ends o, r of the side of length R evolve by an operator $\hat{\mathsf{T}} = \mathsf{m}^{1/2}\hat{\mathsf{K}}\mathsf{m}^{1/2}$ on functions with values in the 2×2 matrices, $(\hat{\mathsf{K}}\Psi)(V) = \iint e^{-\beta A_\times(V,S,V')} S_o \Psi(V') S_r^{-1}\,\mathrm{d}S\,\mathrm{d}V'$, and

$$2Z[W_C] = \mathrm{Tr}\big((\mathsf{T}^{(0)})^{n_3-T}\Phi_\gamma^*\hat{\mathsf{T}}^T\Phi_\gamma\big) \ge 0, \qquad \mathrm{Tr}\,\hat{\mathsf{T}}^{n_3} = 4Z[\mathsf{PP}'],$$

$\Phi_\gamma\psi = \psi\,V(\gamma)$, P, P' the half-traces of the two straight closed lines through o and r; the positivity of $\hat{\mathsf{T}}$ is proved as that of K, with the crossing variables acting by a unitary representation. Hölder's inequality for the two spectral families gives $0 \le \langle W_C\rangle \le (4\langle \mathsf{PP}'\rangle)^{T/n_3}$, and (i) in the direction of R gives $\langle \mathsf{PP}'\rangle \le (1-\zeta)^{1-\varrho}\zeta^\varrho + \zeta^{1-\varrho}(1-\zeta)^\varrho \le 2\zeta^\varrho$, $\varrho = R/n_1 \le \frac12$: the bound on every box. With (a) on the cube n and $n\to\infty$ (Prop. 17.38(c)) the factor in front tends to 1: (b). (c) is the same argument with one charge for each passage of C through a layer: several static charges evolve by a positive operator on functions with values in $(\mathbb{C}^2)^{\otimes\mathcal{V}}$, and the projection of the path to the plane (13) is counted modulo 2 layer by layer; a passage from a downward to an upward bond costs a factor 2. (d) On every periodic box, for $T \le n_3 - 2$ and $R \le n_1 - 2$, the positivity of the operators of (iv) gives the log-convexity in T, and the reflection positivity across a plane between two charges gives it in R; these are inequalities between finitely many means of continuous local functions, and they pass to every weak limit of box measures whose sides tend to infinity.

If $w(R,T_0) = 0$ for one $T_0 \ge 1$, the log-convexity gives $w(R,T) = 0$ for all $T \ge 1$, and $V(R) = \infty$; else $T \mapsto w(R,T)^{1/T}$ does not decrease, and V is the infimum. The concavity holds in $[0,\infty]$, so V is infinite at every $R \ge 1$ as soon as it is infinite at one; with $V(R)/R \to \sigma_\infty$ it gives the slopes. Under the hypothesis $w(R,T) \le e^{-\sigma^+RT}$, so $V(R) \ge \sigma^+R$ for all R, and a concave function with $V(R) \ge \sigma^+R$ has every slope $\ge \sigma^+$. □

Lemma 17.40 (One step on a periodic box)**.** Let $\beta \ge 0$, Σ a cross-section, integers $\vartheta \ge 2$, $d \ge 3$, $N = 2d\vartheta$, $n = (d-1)\vartheta$, $\tau_s = \tau_0 + s\vartheta$; let Φ be a bounded complex Borel function on $\mathcal{Y}_\vartheta$, $\Phi^s = \Phi$ for odd s and $\Phi^s = \Phi^\Theta$ for even s, μ_N the measure of the box (N,Σ), $\varrho = \lambda_1/\lambda_0$, and

$$\mathsf{p} = \mu_N\Big(\prod_{s=1}^{2d}(\Phi^s)_{\tau_{s-1}}\Big), \qquad \mathsf{y} = [(1+\mathsf{r}_N)\,\mathsf{p}]^{1/2d}.$$

Then $\mathsf{p} \ge 0$, $\lambda_0^{-\vartheta}\|\mathfrak{Y}[\Phi]\| \le \mathsf{y}$, and

$$0 \le \mu_N(\Phi_{\tau_0}(\Phi^\Theta)_{\tau_d}) \le \big[(1+\mathsf{r}_N)\,|\mu_N(\Phi_{\tau_0})| + \mathsf{r}_{N-\vartheta}\,\mathsf{y}\big]^2 + (2+\mathsf{r}_n)\,\varrho^n\,\mathsf{y}^2.$$

■

Proof. By (M1), $Z(N,\Sigma)\mathsf{p} = \mathrm{Tr}(Y^*Y)^d = \|Y\|_{2d}^{2d} \ge \|Y\|^{2d}$, $Y = \mathfrak{Y}[\Phi]$, and $Z(N,\Sigma) = \lambda_0^N(1+\mathsf{r}_N)$: the bound on $\|Y\|$. The mean of the two cells d slabs apart is $\mathrm{Tr}(\mathsf{T}'^nY^*\mathsf{T}'^nY)/Z$; write $\mathsf{T}'^n = \lambda_0^n(|e_0\rangle\langle e_0| + \mathsf{R}_{(n)})$ with $\|\mathsf{R}_{(n)}\| = \varrho^n$ and $\mathrm{Tr}\,|\mathsf{R}_{(n)}| = \mathsf{r}_n$, and $\lambda_0^{-\vartheta}\langle e_0, Ye_0\rangle = (1+\mathsf{r}_N)\mu_N(\Phi_{\tau_0}) - \lambda_0^{-\vartheta}\,\mathrm{Tr}(\mathsf{R}_{(N-\vartheta)}Y)$ from the mean of one cell; expanding, the term $|\langle e_0, Ye_0\rangle|^2$ gives the square, and the others are bounded by ϱ^n and r_n times $\|Y\|^2$. □

Lemma 17.41 (Contraction of the seminorm)**.** Let $L' \ge 13$ be an odd integer, $k \in \mathbb{N}$, $\beta > 0$, on the lattice $L'^{-k}\mathbb{Z}^3$ with the sides $2L'^{\mathbb{N}}$. Let $\ell_f, \ell_c \in 2L'^{\mathbb{N}}$ with $d = \ell_c/\ell_f \ge L'$, $\mathsf{h} = \ell_f L'^k/2$ the number of layers of a fine cell, and let $\mathsf{e}(l) \le x_0$ at β with $24l \le \mathsf{h}$. Then for every real G of the bonds of the cell $[-\ell_f/4, \ell_f/4]^3$,

$$\big\|G - \mu_{k,\ell_c,\beta}(G)\big\|_{\mathrm{ch},\ell_c} \le 1.343\,\theta^{(d-1)\mathsf{h}/8l}\,\big\|G - \mu_{k,\ell_f,\beta}(G)\big\|_{\mathrm{ch},\ell_f},$$

where on the left G is regarded as a function of the cell $[-\ell_c/4, \ell_c/4]^3$, which holds the cell $[-\ell_f/4, \ell_f/4]^3$. The constant 1.343 depends on none of G, k, β, d, L'. ■

Proof. Let $X = G - \mu_{k,\ell_c,\beta}(G)$, $B_1 = X$, $B_{2p} = B_p \cdot \Theta_i B_p$ with Θ_i the reflection in the coarse wall plane of the direction i, and $m_p = \mu_{k,\ell_c,\beta}(B_p)$; then $m_1 = 0$, and m_8 is the eighth power of the left member. In the direction i, B_p is a function of one slab of h layers, and $\Theta_i B_p$ lies $n = (d-1)\mathsf{h}$ layers further, on the cube of $N = 2d\mathsf{h}$ sites. Lem. 17.40 with Y the slab operator of B_p, the gap of Prop. 17.33(a), and $\lambda_0^{-\mathsf{h}}\|Y\| \le \mathsf{c}\,u^p$, $\mathsf{c} = (1+\mathsf{r}_N)^{1/2d}$, $u = \|X\|_{\mathrm{ch},\ell_c}^{(\ell_f)}$, the seminorm of X on the torus ℓ_c for the fine board, of side $\ell_f/2$ (the array form of this board, applied to the pile of $2d$ copies of B_p, which are $2dp$ copies of X in cells of the fine board, with the factor 1 in the other cells, bounds the p of Lem. 17.40 by u^{2dp}), give

$$0 \le m_{2p} \le \left[(1+\mathsf{r}_N)\, m_p + \mathsf{r}_{N-\mathsf{h}}\, \mathsf{c}\, u^p\right]^2 + (2+\mathsf{r}_n)\, \theta^{n/l}\, \mathsf{c}^2 u^{2p}.$$

With the bounds of Prop. 17.32 on the r, three steps give $m_8 \le 3.68\, u^8 \theta^{n/l}$. To bound u, put $G_0 = G - \mu_{k,\ell_f,\beta}(G)$ and $Q_f = e^{\delta((2\mathsf{h})^3)/8}$, $2\mathsf{h} = \ell_f L'^k$ being the number of sites of a side of the torus ℓ_f. The fine-to-coarse form gives $u \le Q_f \|X\|_{\mathrm{ch},\ell_f}$; as $X = G_0 - \mu_{k,\ell_c,\beta}(G_0)$ and $\|1\|_{\mathrm{ch},\ell_f} = 1$, $\|X\|_{\mathrm{ch},\ell_f} \le \|G_0\|_{\mathrm{ch},\ell_f} + |\mu_{k,\ell_c,\beta}(G_0)|$; and Prop. 17.34(c) at $\ell' = \ell'' = \ell_f$, with $\mathfrak{s} = \mu_{k,\ell_c,\beta}$ and $\mu_{k,\ell_f,\beta}(G_0) = 0$, gives $|\mu_{k,\ell_c,\beta}(G_0)| \le 4.84\,[Q_f(Q_f-1)]^{1/2}\|G_0\|_{\mathrm{ch},\ell_f}$. By Prop. 17.33(c) at $N = 2\mathsf{h} \ge 48l$, $\delta((2\mathsf{h})^3) \le 0.5892\,\theta^{48} \le 0.0022046$, so $Q_f \le 1.000276$ and $u \le Q_f(1 + 4.84\,[Q_f(Q_f-1)]^{1/2})\|G_0\|_{\mathrm{ch},\ell_f} \le 1.0807\,\|G_0\|_{\mathrm{ch},\ell_f}$. As m_8 is the eighth power of the left member and $n = (d-1)\mathsf{h}$, the left member is at most $3.68^{1/8} \cdot 1.0807\,\theta^{(d-1)\mathsf{h}/8l}\|G_0\|_{\mathrm{ch},\ell_f} \le 1.272\,\theta^{(d-1)\mathsf{h}/8l}\|G_0\|_{\mathrm{ch},\ell_f}$, and $1.272 \le 1.343$. □

Lemma 17.42 (A second blocking factor; dense labels)**.** (a) For every integer $L \ge 2$ there are an odd prime $L' \ge 13$ with $L' > L$ and an admissible choice of the constants with the blocking factor L'; for such L', $\alpha = \log L / \log L' \in (0,1)$ is irrational. (b) Let $\mathsf{b} > 0$, $k(K) = \lfloor K\alpha \rfloor$ and $\mathsf{c}_K = \mathsf{b}\, L'^{K\alpha - k(K)} \in [\mathsf{b}, L'\mathsf{b})$. Then $L^K \mathsf{b} = L'^{k(K)} \mathsf{c}_K$; k does not decrease, $k(K+1) - k(K) \in \{0,1\}$ and $k(K) \to \infty$, so every integer $k > k(K_1)$ is $k(K)$ for some $K > K_1$; and $\{\mathsf{c}_K : K \ge K_1\}$ is dense in $[\mathsf{b}, L'\mathsf{b}]$ for every K_1. ■

Proof. (a) is (M6) with the infinitude of primes; $L^q = L'^p$ is impossible for a prime $L' \nmid L$. (b) is arithmetic; the density is that of $\{K\alpha\}$ modulo 1 for irrational α. □

17.2.5. *Statements uniform in the cutoff.*

Proposition 17.43 (Products of action densities on one torus)**.** Let $\ell'' \in \mathfrak{L}$, $J' = [\mathsf{b}_-, \mathsf{b}_+] \subset (0,\infty)$ and $\mathsf{d} > 0$. For $k \ge 2$ let $\mathsf{B}_k(\ell'', J', \mathsf{d})$ be the bound of (M4)(i) on the k-th cumulants on the torus ℓ'' at the labels of J', for supports at pairwise sup distances $\ge \mathsf{d}$ and unit norms, and $\mathsf{M}_k = \sum_\pi \prod_{B\in\pi} \mathsf{B}_{|B|}$ over the partitions of $\{1,\dots,k\}$ into blocks of at least two members. Let $K \in \mathbb{N}$, $\mathsf{b}' \in J'$, $\beta = L^K \mathsf{b}'$, □ a cell of a board of side $\ell''/2$ with any offset, $f_1, \dots, f_r$ real Lipschitz functions, none 0, with supports in □ at sup distance $\ge \mathsf{d}/2$ from its walls and pairwise $\ge \mathsf{d}$ apart, and $X_i = \mathcal{O}_K(f_i) - \mu_{K,\ell'',\beta}(\mathcal{O}_K(f_i))$. Then for $\emptyset \ne I \subset \{1,\dots,r\}$, $X_I = \prod_{i\in I} X_i$ is a real function of the bonds of □ and

$$\|X_I\|_{\mathrm{ch},\ell''} \le \mathsf{M}_{8|I|}^{1/8} \prod_{i\in I} \|f_i\|_\sharp.$$

■

Proof. The walls of the board lie between lattice sites. Thus a plaquette whose centre lies strictly inside □ has all its corners in the closed cell; the supports lie inside the cell, so X_I depends only on its bonds. By the array form of (M2), $\|X_I\|_{\mathrm{ch},\ell''}^8$ is the mean on the torus ℓ'' of the product of the eight mirror copies of X_I; the mirror copy of $\mathcal{O}_K(f)$ is $\mathcal{O}_K(f \circ \hat{\mathsf{r}})$ for the mirror map $\hat{\mathsf{r}}$, with the same mean. So it is a moment of $8|I|$ centred fields with pairwise disjoint supports on one torus, at distances $\ge \mathsf{d}$ inside one cell and across the walls, hence a sum over the partitions into blocks of at least two members of products of joint cumulants; (M4)(i) bounds each by $\mathsf{B}_{|B|}$ times the norms. □

Proposition 17.44 (The vacuum constants of two tori; the cube number)**.**

(a) Let $\mathsf{b} \geq \gamma^{-2}$, $\ell \in \mathfrak{L}$ with $LM\,R(\mathsf{b}) \leq \ell/2$, and $K \geq 10$ with $L^{\lceil K/1260\rceil}\mathsf{b} \geq \Theta_\star$, a number of the constants, so that the auxiliary sequence of renormalization steps is formed. At every level of the auxiliary sequence started from zero families with zero counterterms, the local numbers on the tori ℓ and $L\ell$ are equal on every region that lifts to both. Consequently

$$|L^3\mathfrak{K}_K(\ell,\mathsf{b}) - \mathfrak{K}_K(L\ell,\mathsf{b})| \leq C\,(L\ell)^3 \exp\bigl[-\kappa_1 \ell L^{\lceil K/1260\rceil}\bigr]$$

with numbers $C, \kappa_1 > 0$ of the constants.

(b) For every $\mathsf{b} > 0$, $\ell \in 2L^{\mathbb{Z}}$ and $K \geq K_\star(\mathsf{b},\ell)$: $|X_{K+1}(\mathsf{b},\ell) - X_K(\mathsf{b},\ell)| \leq L^3\mathfrak{o}_K(\ell/L) + \mathfrak{o}_K(\ell) + C\ell^3 e^{-\kappa_1 \ell L^{\lceil K/1260\rceil - 1}}$ after the dilation of (c), and $\sum_{k\geq K}|X_{k+1} - X_k| \leq \eta_K(\mathsf{b},\ell)$, with the η_K of Thm. BB(1).

(c) The dilation $(\mathsf{b},\ell,K) \mapsto (L^z\mathsf{b}, L^z\ell, K - z)$ names the same lattices at the same couplings, so $X_K(\mathsf{b},\ell) = X_{K-z}(L^z\mathsf{b}, L^z\ell)$, and it brings every label and side into the range of (a).

■

Proof. (a) The zero initial families and zero counterterms have a common infinite-lattice companion and satisfy the input size bound. The induction in Thm. YM3_I:Y, applied to the auxiliary sequence, therefore constructs the same family on each common lift. Its proof transports the entire local input through the chart at every step, including its domain. The floor $L^{\lceil K/1260\rceil}\mathsf{b} \geq \Theta_\star$ is the coupling floor needed for this auxiliary run. The Gaussian numbers also agree under these charts, by the local normalizer construction in equations YM3_I:(1399)–(1400).

To estimate the contact terms, we keep the stopping depth in their vacuum constants. Put $u = \lceil K/1260\rceil$, and mark the cutoffs $K^{\mathrm{A}} = K$ and $K^{\mathrm{B}} = K + 1$. The zero-source chain identity proved in Step 3(iv) of Prop. 13.7, or directly eq. 13.ii summed over its steps, gives

$$c_\lambda^\bullet(0) = E^\bullet_{(K^\bullet - u)}[T_\lambda], \qquad \mathfrak{K}_K(\lambda,\mathsf{b}) = c_\lambda^{\mathrm{B}}(0) - c_\lambda^{\mathrm{A}}(0).$$

Here $E^\bullet$ is built from the line $x_j^\bullet = L^{K^\bullet - j}\mathsf{b}$ on T_λ. These constants are real. The identity uses the actual chains; it does not require the error estimate comparing their masses. The line has zero increments and is admissible by the proof of Lem. YM3_I:118.1. The assumed room $LMR(\mathsf{b}) \leq \ell/2$ admits it on both tori. It also implies the side floor $\ell \geq C_L M$: the retained room condition [7, eq. 215] gives $C_L \leq 2LR(\gamma^{-2}) \leq 2LR(\mathsf{b})$.

Apply eq. 13.iii at each cutoff, with $n = K^\bullet - u$, to the sides ℓ and $L\ell$. Both lower summation indices are u. The triangle inequality gives

$$\left|\frac{\mathfrak{K}_K(\ell,\mathsf{b})}{\ell^3} - \frac{\mathfrak{K}_K(L\ell,\mathsf{b})}{(L\ell)^3}\right| \leq 4C_{\mathrm{tt}}E_0 \sum_{s\geq u} L^{3s} e^{-\kappa\ell L^s/(5M)} \leq 8C_{\mathrm{tt}}E_0 L^{3u} e^{-\kappa\ell L^u/(5M)}.$$

The last inequality is eq. 13.iv. Since $\kappa\ell/(5M) > 4$, the factor L^{3u} can be absorbed in half the exponent. For example, take

$$\kappa_1 = \frac{\kappa}{10M}, \qquad C = 8C_{\mathrm{tt}}E_0 \sup_{v\geq 1} v^3 e^{-2v} < \infty.$$

Multiplication by $(L\ell)^3$ proves (a).

(b) The zero-source read-out (M4)(ii) gives

$$|\log Z_{K+1,\ell} - \log Z_{K,\ell} - \mathfrak{K}_K(\ell,\mathsf{b})| \leq \mathfrak{o}_K(\ell,\mathsf{b}).$$

Apply this on the sides ℓ/L and ℓ, multiply the first identity by L^3, and subtract. The contact terms cancel up to the error in (a), giving the asserted bound for $X_{K+1} - X_K$. The two read-out errors have the summable rate in (M4)(ii). With $\varsigma = \sum_{j\geq 0}(1+j)^{\mathsf{p}_0}L^{-j/10}$, their tail is bounded by η_K once K is above the thresholds where (M4)(ii) and (a) apply and where the exponential term is at most $\mathsf{C}_{\mathrm{bf}}\ell^3 K^{\mathsf{p}_0}L^{-K/10}$. After the dilation in (c), these requirements are imposed on $K - z$ and

on both sides $L^{z-1}\ell$, $L^z\ell$, at label $L^z\mathsf{b}$, including their thresholds $K_{\rm bf}$. This is the choice of $K_\star$ in Thm. BB(1).

(c) The dilation leaves both the bare lattice and its coupling unchanged. This proves the identity for X_K and permits the fixed dilation used in (b). □

Proposition 17.45 (Two cutoffs on boxes with mixed sides)**.** The following comparison remains a sketch at the full interacting Hessian estimate described with Thm. AH. Its intended conclusion is: for every compact set J of labels, every class $z \in \{0,1\}^3$, all sides $s_\mu \in L^{\mathbb{Z}} \cup 2L^{\mathbb{Z}}$ such that each twisted plane has a side in $L^{\mathbb{Z}}$, and every $l \in 2L^{\mathbb{Z}}$, there are C and K_* such that

$$\Big|\log Z^z_{K+1}(\underline{s};\mathsf{b}) - \log Z^z_K(\underline{s};\mathsf{b}) - \frac{|\underline{s}|}{l^3}(\log Z_{K+1,l} - \log Z_{K,l})\Big| \le C\,L^{-K/10} \qquad (\mathsf{b}\in J,\ K \ge K_*),$$

where $Z^z_K(\underline{s};\mathsf{b})$ is the partition function of the box with s_μ/a_K sites along the side μ and the twist z, at $\beta = L^K\mathsf{b}$, and $|\underline{s}| = s_1s_2s_3$. ■

Proof. The box $\underline{s}$ is mixed for z in the sense of Lem. 11.9. After a common dilation by a power of L, choose a periodic covering cube $\ell_C \in 2L^{\mathbb{Z}}$ for both this box and the reference cube l^3. The latter is mixed for class 0. Thm. AH(4), under **(Hess)**, gives the two comparisons with the same extensive term $\Delta_K[\ell_C,\mathsf{b}]$.

Subtract $|\underline{s}|/l^3$ times the comparison for l^3 from the comparison for $(\underline{s},z)$. The coefficients $|\underline{s}|/\ell_C^3$ of Δ_K cancel. The two errors are bounded by $C_1L^{-K/10}$ and $(|\underline{s}|/l^3)C_2L^{-K/10}$, uniformly on J, after increasing the common threshold. This proves the stated estimate.

The errors are $\le CL^{-K/10}$. □

Lemma 17.46 (A summable bound closes the criterion)**.** Let $(\mathsf{e}_K)_{K\ge K_*}$ be real numbers with $|\mathsf{e}_{K+1}-\mathsf{e}_K| \le \mathsf{q}_K$ and $Q_K = \sum_{k\ge K}\mathsf{q}_k < \infty$. Then (a) $\mathsf{e}_\infty = \lim \mathsf{e}_K$ exists and $|\mathsf{e}_K - \mathsf{e}_\infty| \le Q_K$; (b) if $\mathsf{e}_{K_1} \le x_0 - Q_{K_1}$ at one $K_1 \ge K_*$, then $\mathsf{e}_K \le x_0$ for all $K \ge K_1$; (c) if $\mathsf{e}_K \le x_0' < x_0$ for all large K, then $\mathsf{e}_{K_1} \le x_0 - Q_{K_1}$ at every large K_1; (d) $\mathsf{e}_\infty < x_0$ if and only if $\mathsf{e}_{K_1} < x_0 - Q_{K_1}$ at one K_1, and $\mathsf{e}_\infty > x_0$ if and only if $\mathsf{e}_{K_1} > x_0 + Q_{K_1}$ at one K_1, and then $\mathsf{e}_K > x_0$ for all $K \ge K_1$; at $\mathsf{e}_\infty = x_0$ neither holds. ■

Proof. $|\mathsf{e}_K - \mathsf{e}_{K'}| \le Q_K$ for $K' \ge K$, and $Q_K \to 0$. □

Proposition 17.47 (Small tori)**.** For an integer $N \ge 3$ let $\lambda_N(k,u) = \sum_\mu(2-2\cos\frac{2\pi k_\mu+u_\mu}{N})$ for $k \in (\mathbb{Z}/N)^3$, $u \in \mathbb{R}^3$; $\mathrm{Det}'(N) = \prod_{k\ne0}\lambda_N(k,0)$; $\Upsilon_N(u) = N^2\prod_k \lambda_N(k,u)/\mathrm{Det}'(N)$; $\mathscr{J}_N = \int_{[-\pi,\pi]^3} du/\Upsilon_N(u)$; $f_\Delta = (2\pi)^{-3}\int_{[-\pi,\pi]^3}\log\sum_\mu(2-2\cos p_\mu)\,\mathrm{d}p$; and $c'_N = \log(\mathscr{J}_N/8\pi^2) - \frac{3}{2}(\log\mathrm{Det}'(N) - f_\Delta N^3 - 2\log N)$.

(a) There are absolute numbers $c_3 > 0$ and $x_* \in (0,\frac12]$ with the following two properties. On one bare periodic cube: for every integer $N \ge 3$ and every real $\beta \ge 3$ with $\mathsf{er}_N = c_3(N^{66}\log\beta/\beta)^{1/8} \le x_*$,

$$\Big|\frac{\beta^{3N^3-1}Z(N^3)}{\mathcal{Z}(N)} - 1\Big| \le \mathsf{er}_N, \qquad \mathcal{Z}(N) = \frac{(2\pi)^{3N^3-1}\,N^2\,\mathscr{J}_N}{(2\pi^2)^{2N^3}\,4\pi\,\mathrm{Det}'(N)^{3/2}}, \qquad \log\mathcal{Z}(N) = N^3z_\infty - \log N + c'_N,$$

$Z(N^3)$ the partition function at β and z_∞ the number of (d). Consequently, for all integers $N \ge 3$, $q \ge 2$ and real $\beta \ge 3$ with $\mathsf{er} = c_3((qN)^{66}\log\beta/\beta)^{1/8} \le x_*$:

$$|D_q(N) - (q^3-1)\log(\beta/N) - q^3c'_N + c'_{qN} - \log q| \le 2(q^3+1)\,\mathsf{er}.$$

(b) There are numbers $t_0 \in (0,\frac{1}{36}]$ and C_B of the constants such that for every integer $i \ge 1$, $N = 2L^i$, and every real β with $t = N/\beta \le t_0$: $|D_L(N) - (L^3-1)\log(1/t) - L^3c'_N + c'_{LN} - \log L| \le C_Bt^{1/1500}$.

(c) With $t_1 = t_0^2$ and a number C_C of the constants: $|\delta(N^3) - \log(1/t) - c'_N| \le C_C t^{1/1500}$ for $N = 2L^i$, $t = N/\beta \le t_1$. Moreover $c'_N \to c'$ and $|c'_N - c'| \le C_A N^{-2} (\log N)^{5/2}$ with an absolute C_A.

(d) With $z_\infty = \log \frac{2}{\pi} - \frac{3}{2} f_\Delta \approx -2.96167$ there are numbers β_1, C of the constants with $|P(\beta) + 3\log\beta - z_\infty| \le C\beta^{-1/1500}$ for every real $\beta \ge \beta_1$.

■

Proof. (a) *The value.* As $\beta \to \infty$ the measure of the bare cube concentrates on the flat fields. Up to gauge these are the commuting triples of holonomies; at a generic triple the orbit is that of a maximal torus, parametrised by $u \in [0, 2\pi]^3$, and the transverse fluctuations are Gaussian with the quadratic form of the lattice Laplacian twisted by u, whose lowest eigenvalue $\lambda_N(0,u)$ is of the order $|u|^2/N^2$. The volume of the orbits, the Haar measure near them and the Gaussian integral give $\beta^{3N^3-1} Z(N^3) \to \mathcal{Z}(N)$ and $\log \mathcal{Z}(N) = (3N^3-1)\log 2\pi - 2N^3 \log 2\pi^2 - \log 4\pi - \frac{3}{2}\log \mathrm{Det}'(N) + 2\log N + \log \mathscr{J}_N = N^3 z_\infty - \log N + c'_N$, by $3\log 2\pi - 2\log 2\pi^2 = \log \frac{2}{\pi}$, $\log 2\pi + \log 4\pi = \log 8\pi^2$ and the definition of c'_N.

The error, with all its powers of N. (1) *The cut.* In the gauge in which the bonds of a maximal tree carry $\mathbf{1}$, which removes the gauge group exactly, with no gauge copies and no Jacobian, the field is bounded by its curvature: with $A = \sum_p a_p$, every other bond variable lies within $2NA^{1/2}$ of $\mathbf{1}$ or, at a bond that crosses the boundary of the fundamental cube, of the holonomy h_μ of its direction, and $|h_\mu h_\nu h_\mu^{-1} h_\nu^{-1} - \mathbf{1}| \le N(2A)^{1/2}$. The fields with $A \ge \alpha_\beta = 4N^3 \log\beta/\beta$ carry at most β^{-4N^3}, which is $(C/\beta)^{N^3}$ times $\mathcal{Z}(N)\beta^{1-3N^3}$: this cut is the source of $\log\beta$. (2) *The generic stratum.* On $\{A < \alpha_\beta\}$ the integral is squeezed between two Gaussian integrals: in $d_\nu \le 6N^3$ transverse variables ν with $\|\nu\|^2 \le \Lambda_\delta A$ and $|A - Q| \le 43N^2 |\nu|_\infty \|\nu\|^2$, Q the quadratic part of A and $43N^2$ the bound of the cubic remainder of the plaquettes, the integral of $e^{-\beta A}$ over $\{A < \alpha_\beta\}$ lies between $e^{\mp(d_\nu \epsilon + C\rho_\delta)}$ times the Gaussian integral of Q, $\rho_\delta = (\Lambda_\delta \alpha_\beta)^{1/2}$, $\epsilon = 43N^2 \Lambda_\delta \rho_\delta$; the Gaussian determinants need not be computed, as the limit $\mathcal{Z}(N)$ names them.

Where some holonomy has $1 - (\frac{1}{2}\operatorname{tr} h_\mu)^2 \ge \delta^2$, the base is the commuting triple, there are $6N^3 - 2$ transverse variables, and the softest of them costs of the order δ^2/N^2: $\Lambda_\delta \le 2\pi^2 N^5 + \frac{\pi^2}{4} N^2 \delta^{-2}$. (3) *The eight points* at which every h_μ is $\pm\mathbf{1}$: there the stabiliser is all of SU(2), the nine constant modes $c = (c_\mu)$ have the quartic action $N^3 |c \wedge c|^2$, $|c \wedge c|^2 = \sum_{\mu<\nu} |c_\mu \times c_\nu|^2$, and the Gaussian picture fails for $|u| \lesssim (N/\beta)^{1/4}$. An upper bound suffices there. In Landau's gauge the action is at least $\frac{1}{2}(1-\epsilon')\|d_1 Y'\|^2 + N^3 |c \wedge c|^2$ for the non-constant part Y' of the field, d_1 the lattice exterior derivative of 1-forms, the Jacobian is at most $(\mathrm{Det}'(N)/N^3)^{3/2}(1 + 7N^2 r_\delta)^{3N^3}$ (Kirchhoff's theorem), $r_\delta^2 = 6\pi^2 N^2 \delta^2 + 24 N^5 \alpha_\beta$, and the integral over the constant modes is at most $48\pi^3 r_\delta N^{-3/2} (\beta N^3)^{-2}$; measured against $\mathcal{Z}(N)\beta^{1-3N^3}$, in which $\mathscr{J}_N = \int \mathrm{d}u / \Upsilon_N(u)$ lies between two absolute numbers, as $c_0 |u|^2 \le \Upsilon_N(u) \le c_0^{-1}$ with an absolute $c_0 > 0$ and $|u|^{-2}$ is integrable in three dimensions, the weight of these fields is at most $C\, N^{-1/2} r_\delta\, e^{C N^5 r_\delta}$. (4) *The balance.* The error of (2) is of the order of $N^8 \delta^{-3} (N^3 \log\beta/\beta)^{1/2}$, that of (3) of the order of $N^{1/2}\delta$; with $\delta = N^{-6}(N^{66} \log\beta/\beta)^{1/8}$ both, and the cut, are at most an absolute multiple of $(N^{66}\log\beta/\beta)^{1/8}$: the exponent $\frac{1}{8}$ is the balance of $\delta^{-3}\beta^{-1/2}$ against δ, all the dependence on N sits in N^{66}, and c_3, x_* are absolute.

The consequence. With $|\log(1+x)| \le 2|x|$ for $|x| \le \frac{1}{2}$, the bound at N and at qN (the condition at qN implies the one at N) gives $\log Z(N^3) = -(3N^3-1)\log\beta + \log \mathcal{Z}(N) + O(2\mathsf{er})$ and the same at qN; as $-q^3(3N^3-1) + (3q^3N^3 - 1) = q^3 - 1$, the terms in N^3 cancel in $D_q(N) = q^3 F(N^3) - F((qN)^3)$, and $-q^3 \log N + \log(qN) = -(q^3-1)\log N + \log q$. (b) Let $t \le t_0$. *The frame.* Let j_t be the least integer $j \ge 1$ with $L^j \ge LM\, R(2L^j/t)$. As $R(Lx) \le L\, R(x)$ once $\log x \ge r$, the exponent of $R(2L^j/t)$ grows by at most one when j grows by one, so equality holds at j_t: $\ell' = 2L^{j_t}$, $\mathsf{b}' = \ell'/t$ and $y' = \log \mathsf{b}'$ satisfy $\ell' = 2LM\, R(\mathsf{b}')$, which is L times the least side at which (M4)(ii), (iii) apply at the label b', and $2LMy'^r \le \ell' < 2L^2 M y'^r$, $\log\frac{1}{t} < y' < 2\log\frac{1}{t}$ for small t.

Every pair (N,β) with $N=2L^i$ and $N/\beta=t$ is the pair $(\ell'/a_K, L^K\mathsf{b}')$ of the cutoff $K=i-j_t$, of either sign, on the line of b' (Def. 17.28). Write $D_K=D_L(\ell'/a_K)$ at $\beta=L^K\mathsf{b}'$, which is $X_K(\mathsf{b}',L\ell')$. *Two cutoffs, uniformly in the label.* The number $\mathfrak{o}_K$ of (M4)(ii) is $8\mathsf{R}_{t_1}+12\epsilon_{\rm fg}+8\epsilon_{\rm cl}+12\epsilon_{\rm sh}$ ((2.xiv)), with the explicit numbers of §2.3.1 at $n=1$: with $y=\log\mathsf{b}$, $u=\lceil K/1260\rceil$, $s=\lceil K/9\rceil$ and the top level $t_1=K-u$ of (M4)(ii) in R_{t_1}, $\epsilon_{\rm fg}\le c\,\ell^3L^{-34K/315}$ and $\epsilon_{\rm cl}\le c\,(y+K\log L)^{2\mathfrak{p}_0}K^{1/2}\ell^3L^{-K/9}$, while R_{t_1} carries the factor $e^{-c'A_0^2(y+u\log L)^{2\mathfrak{p}_0}}$ against $e^{(C+\frac92 y)\ell^3}$, and $\epsilon_{\rm sh}$ the factor $e^{-\frac14\kappa_{\mathfrak{b}}A_1^2(y+u\log L)^{2\mathfrak{p}_0}}$.

On the sides $2MR(\mathsf{b})\le\ell\le 2L^3My^r$ one has $\ell^3\le 8L^9M^3y^{3r}$ and $y^{2\mathfrak{p}_0}\le(\ell/2M)^{2\mathfrak{p}_0/r}$, and the condition $2\mathfrak{p}_0>3r+2$ of the list of (H3) lets $e^{-c'A_0^2y^{2\mathfrak{p}_0}}$ beat $e^{(C+\frac92 y)\ell^3}$. So there are numbers $y_\sharp$, $C_{\mathfrak{o}}$ of the constants such that at every label with $y\ge y_\sharp$, on every such side ℓ and at every $K\ge 11(5+2\mathfrak{p}_0/r)\log_L\ell$, the conditions (k0)–(k4), (k6) of §2.3.1 hold and $\mathfrak{o}_K(\ell)\le C_{\mathfrak{o}}\,\ell^{3+2\mathfrak{p}_0/r}L^{-K/11}$, uniformly in the label; this is the form needed here, as $\mathsf{b}'\to\infty$ when $t\to0$, while the constant $\mathsf{C}_{\rm bf}$ of (M4)(ii) is formed on bounded sets of labels.

The two sides ℓ', $L\ell'$ lie in this range, and (M4)(ii) on both, with Prop. 17.44(a), gives

$$|D_{K+1}-D_K|\le L^3\mathfrak{o}_K(\ell')+\mathfrak{o}_K(L\ell')+C\ell'^3L^{-K}\le C\,\ell'^{3+2\mathfrak{p}_0/r}L^{-K/11}\qquad(K\ge 11(5+2\mathfrak{p}_0/r)\log_L(L\ell')).$$

The base. Let $K_\circ$ be the least integer with $N_\circ=\ell'L^{K_\circ}\ge t^{-1/130}$, so that $t^{-1/130}\le N_\circ<Lt^{-1/130}$; as ℓ' is a power of $\log\frac1t$, $K_\circ$ lies above the bound on K of the display once $t\le t_0$. Let $N=2L^i$ and $\beta=N/t$. (1) If $N\le N_\circ$, then $(LN)^{66}\log\beta/\beta=L^{66}N^{65}t\log\frac Nt\le L^{131}t^{1/2}(\log L+\frac{131}{130}\log\frac1t)$, and (a) at $q=L$ applies for $t\le t_0$ and bounds the left member of (b) by $2(L^3+1)c_3L^{17}t^{1/16}(\log L+\frac{131}{130}\log\frac1t)^{1/8}$. (2) If $N>N_\circ$, then $K=i-j_t>K_\circ$, and with $\mathsf{k}_M=L^3c'_M-c'_{LM}+\log L$,

$$D_L(N)-(L^3-1)\log\tfrac1t-\mathsf{k}_N=[D_K-D_{K_\circ}]+[D_L(N_\circ)-(L^3-1)\log\tfrac1t-\mathsf{k}_{N_\circ}]+[\mathsf{k}_{N_\circ}-\mathsf{k}_N]:$$

the first bracket is at most $C(1-L^{-1/11})^{-1}\ell'^{3+2\mathfrak{p}_0/r+1/11}t^{1/1430}$ by the display, as $L^{-K_\circ}\le\ell't^{1/130}$; the second is case (1) at $N_\circ$; the third is at most $2(L^3+1)C_AN_\circ^{-2}(\log N_\circ)^{5/2}\le 2(L^3+1)C_At^{1/65}(\log L+\frac{1}{130}\log\frac1t)^{5/2}$ by the rate of (c), whose proof does not use (b). The exponents $\frac{1}{16}$, $\frac{1}{1430}$, $\frac{1}{65}$ exceed $\frac{1}{1500}$, and the powers of $\log\frac1t$ are absorbed into C_B; t_0 is fixed by $\log\frac{1}{t_0}\ge y_\sharp$, by the place of $K_\circ$ and by the condition of (a) in case (1). (c) By Prop. 17.31(c) and $\delta(M^3)\le 3(M/N)^2\delta(N^3)$, $(1-3L^{-S-1})\delta(N^3)\le\sum_{s=0}^S L^{-3(s+1)}D_L(L^sN)\le\delta(N^3)$, and with $Y(M)=\log(\beta/M)+c'_M$ the main terms of (b) telescope:

$$\sum_{s=0}^{S}L^{-3(s+1)}\big[(L^3-1)\log(\beta/L^sN)+L^3c'_{L^sN}-c'_{L^{s+1}N}+\log L\big]=Y(N)-L^{-3(S+1)}Y(L^{S+1}N);$$

choosing L^S of the order $t^{-1/2}$ gives (c). The limit c' and the rate are those of Riemann sums for lattice determinants, written through heat kernels: $\log\mathrm{Det}'(N)$, $f_\Delta N^3$ and $2\log N$ are integrals $\int_0^\infty[\cdots]\frac{d\tau}{\tau}$ of one-dimensional heat kernels, which converge to those of $\vartheta(\tau,\cdot)$. (d) At $N=2L$ and $\beta\ge 2L/t_1$, (a) gives $F(N^3)=-(3N^3-1)\log\beta+N^3z_\infty-\log N+c'_N+O((\log\beta/\beta)^{1/8})$ and (c) gives $\delta(N^3)=\log(\beta/N)+c'_N+O(\beta^{-1/1500})$; the difference is $N^3P=-3N^3\log\beta+N^3z_\infty+O(\beta^{-1/1500})$. □

Proposition 17.48 (The flux quotient of small cubes at every cutoff)**.**

(a) There are numbers $t_2>0$, $C_{\mathfrak{b}}$ of the constants, and $\mathsf{a}=\frac92 r+\mathfrak{p}_0$, such that for every integer $j\ge1$ and every real β with $t=2L^j/\beta\le t_2$: $r_1(2L^j)\le\eta_{\mathfrak{b}}(t)=C_{\mathfrak{b}}(\log\frac1t)^{\mathsf{a}}t^{1/500}$ at β. Equivalently: for every side $\ell=2L^m$, $m\in\mathbb{Z}$, every real label b with $\ell/\mathsf{b}\le t_2$, and every cutoff $K\ge1-m$.

(b) To obtain the following sharper bound, one must establish the **(Heaviest class)** probability estimate. The intended conclusion is that there are numbers $t_4>0$, C of the constants, and $\mathsf{q}=1116\,r+720\,\mathfrak{p}_0+366$, such that for every integer $j\ge1$ and every real β with $t=2L^j/\beta\le t_4$, over the cross-section (N,N), $N=2L^j$: $\lambda_0^N/Z(N^3)\le C\,t\,(\log\frac1t)^{\mathsf{q}}$. ■

Proof. (a) *On one lattice.* (i) Multiplying by -1 the bonds of a sheet (a half-plane of bonds bounded by two columns) moves a twisted column; so Z^z does not depend on the place of its column, and reflection positivity across two layers with the column in the cut gives $0 \le Z^z[\chi\chi'] \le Z[\chi]^{1/2} Z[\chi']^{1/2}$ for $[0,1]$-valued χ, χ' of the two halves. On configurations whose block plaquettes at a level q (or the bare plaquettes) lie within ω of $\mathbf{1}$, a twist forces the holonomies P_1, P_3 of the block field around the twisted plane to anticommute up to $\varpi = (1+\omega)^{\mathsf{n}^2} - 1 \le \frac{1}{20}$, n the number of block sites a side; hence $|W_1| \le \varpi/2$ and $W_2^2 \ge 1 - 0.52\,\varpi^2$ for the half-traces $W_\mu(y)$ of the straight closed lines of the block field through a site, and

$$r_1(N) \le \max_y \mu_{N^3}(\mathcal{C}^\varpi(y)) + 3\mathsf{n}^3 \mathsf{R}^{1/2}, \qquad \mathcal{C}^\varpi(y) = \{|W_1(y)| \le \tfrac{\varpi}{2},\ W_2(y)^2 \ge 1 - 0.52\,\varpi^2\},$$

R the largest probability of a rough block plaquette. (ii) On a bare cube small against β: with $\Xi_\beta = 3N^3 \log(8e^8\beta^{3/2}) + 2\log(3N^3\beta)$ and $\omega_0 = (2\Xi_\beta/\beta)^{1/2}$, $Z \ge (8e^8\beta^{3/2})^{-3N^3}$, so a plaquette is rough, $|\partial U(p) - \mathbf{1}| \ge \omega_0$, with probability $\le (3N^3\beta)^{-2}$. Then the angle of P_1 is shifted along the seam: with $P_1(y_0) = e^{i\alpha_0 \hat n \cdot \sigma}$ at the site y_0 of the event, $0 < \alpha_0 < \pi$, σ the Pauli matrices, $\mathsf{N}(x)$ the matrix $\hat n \cdot \sigma$ carried to x along a path from y_0 that uses no seam bond $\langle x, x + e_k\rangle$, $x_k = 0$, $k = 1, 2$, and $V(b) \mapsto V(b) e^{-i s_k \mathsf{N}(b_+)}$ on the seam bonds of the direction k, this map, defined for $|s|_\infty \le \frac{1}{10}$, commutes with gauge transformations, turns $P_1(y_0)$ by $-s_1$ and $P_2(y_0)$ by $-s_2$ about $\hat n$, changes the action by at most $131N^4\beta\omega_0^2|s|_\infty$ on the fields whose plaquettes all lie within ω_0 of $\mathbf{1}$, and changes the Haar measure by the factor $\sin^2(\alpha_0 - s_1)/\sin^2\alpha_0$.

Let D be the event $\mathcal{C}^\varpi(y_0)$ intersected with that of no rough plaquette. For $s \in 2\varpi\mathbb{Z}^2$ with $|s|_\infty \le s_0 = (131N^4\beta\omega_0^2)^{-1}$, which is $< \frac{1}{10}$, the action rises by at most 1, so the image of D has the measure $\ge e^{-1}\cos^2(0.13)\,\mu_{N^3}(D) \ge 0.36\,\mu_{N^3}(D)$; on D the angles of $P_1(y_0)$ and $P_2(y_0)$ about $\hat n$ fix s up to 1.02ϖ and 1.46ϖ, so the images of distinct points of the grid are disjoint, and there are at least $(s_0/2\varpi)^2$ of them. Hence $\mu_{N^3}(D) \le \frac{4}{0.36}\varpi^2(131N^4\beta\omega_0^2)^2$, which is of the order $N^{12}\Xi_\beta^3/\beta$, as $\beta\omega_0^2 = 2\Xi_\beta$ and $\varpi \le 2.1N^2\omega_0$; with the rough plaquettes, $\mu_{N^3}(\mathcal{C}^\varpi) \le [6.8 \cdot 10^6 N^{12}\Xi_\beta^3 + 1]\beta^{-1}$, for the block field of a bare lattice that is not too large against β as well, by (M4)(v). (iii) *Through the cutoffs.* Write $2L^j = \ell L^K$ with $\ell = 2L^m$ the least side at which (M4) applies at the label $\mathsf{b} = \ell/t$, $y = \log \mathsf{b}$, $\ell \le 2L^2 M y^r$.

At the cutoffs $K \le K_1 = \lceil y/(25\log L)\rceil$ the lattice is a bare cube small against β, and (ii) applies. Above K_1 we use (M4)(iii) at the depth $d = 0$: at the cutoff K the block field is $\mathcal{M}^K U$, on the unit lattice of the torus ℓ, with ℓ sites a side at every cutoff; the roughness ω and the width ϖ are chosen at K_1 and do not depend on K. The events $\mathcal{C}^\varpi(y')$ of this block field and the events that one of its plaquettes lies at a distance $\ge \omega$ from $\mathbf{1}$ are closed and gauge invariant, and the same at all cutoffs; so by (M4)(iii) their probabilities move by at most $\mathfrak{o}_K$ from one cutoff to the next, and $\sum_{K \ge K_1} \mathfrak{o}_K \le Q(y) e^{-y/250}$, Q a polynomial formed from the constants; (i) at the cutoff K then gives $r_1 \le 4\bar\ell(y)^3 Q(y)^{1/2} e^{-y/500}$, $\bar\ell(y) = 2L^2 M y^r$, which is $\le C_\flat (\log \frac{1}{t})^{\mathfrak{a}} t^{1/500}$. (b) The proposed sharper estimate still requires **(Heaviest class)**; the remaining reduction uses another label of a layer and a multiplicative comparison through the cutoffs.

The top level against the heaviest cell of a label. Let Φ map the configurations on the bonds of the directions $1, 2$ of one layer into a set X, let $\mathcal{P}$ be a finite partition of X into cells whose preimages are Borel, $\sim$ a reflexive and symmetric relation on $\mathcal{P}$ with at most ν neighbours to each cell, and ι_k the cell of the label of the layer $x_3 = k$. Then over the cross-section (N, N) of an even cube $N = 2h$

$$e^{-\mathsf{G}_N} = \frac{\lambda_0^N}{Z(N^3)} \le \big(\nu^2\sqrt{p^\sharp} + \nu\sqrt{\varepsilon_2} + \sqrt{\varepsilon_1}\big)^2 \le 2\nu^4(p^\sharp + \max\{\varepsilon_1, \varepsilon_2\}) \quad (\nu \ge 2),$$

$p^\sharp = \max_I \mu_{N^3}(\iota_1 = I)$, $\varepsilon_1 = \mu_{N^3}(\iota_1 \not\sim \iota_{\mathsf{n}/2-1})$, $\varepsilon_2 = \mu_{N^3}(\iota_{\mathsf{n}/2-1} \not\sim \iota_{\mathsf{n}/2+1})$, in two cases: ($\alpha$) the label of a bare layer, $\mathsf{n} = N$; (β) the label of a layer of the block field $\mathcal{M}^q U$ of the bare cube, with $\mathsf{n} \ge 8$ block sites a side. Indeed, let the first slab be the bare layers $1, \dots, s$, with $s = 2$ in (α), $s = \frac{1}{2}(3L^q - 1)$ in (β), and $2s < h$, which in (β) follows from $\mathsf{n} \ge 8$. The label ι_1 is a Borel function

of the bonds with both ends in the first slab: in (α) it reads the layer $x_3 = 1$ alone, and in (β), by the locality of the averaging map (Prop. YM3_I:4.5), only the bare bonds of the blocks of the block layer 1, which, centred at the height L^q, lie in the bare layers $\frac{1}{2}(L^q+1), \dots, \frac{1}{2}(3L^q-1)$.

The reflection in the plane $x_3 = \frac{h}{2}$, the translation by he_3 and the reflection in the plane $x_3 = 0$ carry the first slab onto the layers $h-s, \dots, h-1$, the layers $h+1, \dots, h+s$ and the layers $N-s, \dots, N-1$; the layers 0 and h lie in none of these four slabs. The three maps are isometries of the bare lattice and, in (β), as $h = \frac{\mathsf{n}}{2}L^q$ is a multiple of L^q, of the block lattices, and the averaging maps commute with them ([7, Lem. G.1]); so the label ι_1, read on the configuration moved by them, is $\iota_{\mathsf{n}/2-1}$, $\iota_{\mathsf{n}/2+1}$, $\iota_{\mathsf{n}-1}$ respectively. Let χ_I be the indicator function of the label I on the first slab, $\mathfrak{Y}[\chi_I]$ its slab operator of (M1), and $Y_I = \mathfrak{Y}[\chi_I]\,\mathsf{T}'^{1/2}$, the operator $\mathsf{T}' = \mathsf{K}^{1/2}\mathsf{m}\,\mathsf{K}^{1/2}$ being ≥ 0; then $\sum_I Y_I = \mathsf{T}'^{s+1/2}$, $R = \mathsf{T}'^{h-2s-1} \geq 0$ and $\mathsf{T}'^{h} = \sum_{I,J} Y_J^* R Y_I$.

The conjugated mirror image χ_J^Θ of (M1), placed on the layers $h-s, \dots, h-1$, is χ_J read on the configuration mirrored in the plane $x_3 = \frac{h}{2}$ midway between these layers and the first slab, that is the indicator of $\iota_{\mathsf{n}/2-1} = J$; placed on the layers $N-s, \dots, N-1$, it is χ_J read on the configuration mirrored in the plane $x_3 = h$, which on the box of $N = 2h$ layers is the reflection in the plane $x_3 = 0$, that is the indicator of $\iota_{\mathsf{n}-1} = J$; and $\chi_{I'}$ placed on the layers $h+1, \dots, h+s$ is the indicator of $\iota_{\mathsf{n}/2+1} = I'$. So the trace formula of (M1), in which the first slab stands rightmost, for the functions $\chi_I^{+(h-2s-1)}$, $(\chi_J^\Theta)^{+1}$, $\chi_{I'}^{+(h-2s-1)}$, $(\chi_{J'}^\Theta)^{+1}$ on the consecutive slabs of $h-s-1$, $s+1$, $h-s-1$, $s+1$ layers from the layer 1 on, with $\mathfrak{Y}[\Phi^{+j}] = \mathsf{T}'^j\mathfrak{Y}[\Phi]$ (the j added layers lying above those of Φ), $\mathfrak{Y}[\chi_J^\Theta] = \mathfrak{Y}[\chi_J]^*$ and the cyclicity of the trace, gives $\mathrm{Tr}(Y_{J'}^* R Y_{I'}\, Y_J^* R Y_I) = \mathrm{Tr}(\mathsf{T}'\mathfrak{Y}[\chi_{J'}^\Theta]\, R\, \mathfrak{Y}[\chi_{I'}]\, \mathsf{T}'\mathfrak{Y}[\chi_J^\Theta]\, R\, \mathfrak{Y}[\chi_I]) = Z(N^3)\,\mu_{N^3}(\iota_1 = I, \iota_{\mathsf{n}/2-1} = J, \iota_{\mathsf{n}/2+1} = I', \iota_{\mathsf{n}-1} = J')$: probabilities of one and the same cube.

The terms with $I \not\sim J$ have together the Hilbert–Schmidt norm $\leq (Z\varepsilon_1)^{1/2}$. The terms with $I \sim J$ are $\mathbb{X}\mathbb{B}\mathbb{X}^*$, with $\mathbb{X}(\varphi_I)_I = \sum_I Y_I^* R^{1/2}\varphi_I$ and the neighbour matrix $\mathbb{B}$ of norm $\leq \nu$ (Schur's test); $\mathbb{X}^*\mathbb{X}$ is the matrix of the $R^{1/2}Y_I Y_J^* R^{1/2}$, whose Hilbert–Schmidt norms are $(Z\mu_{N^3}(\iota_1 = \iota_{\mathsf{n}/2-1} = I,\ \iota_{\mathsf{n}/2+1} = \iota_{\mathsf{n}-1} = J))^{1/2}$, so Schur's test on the neighbouring entries and the Hilbert–Schmidt norm of the others give $\|\mathbb{X}^*\mathbb{X}\| \leq \nu(Zp^\sharp)^{1/2} + (Z\varepsilon_2)^{1/2}$. Hence $\lambda_0^h = \|\mathsf{T}'^h\| \leq Z^{1/2}[\nu(\nu\sqrt{p^\sharp} + \sqrt{\varepsilon_2}) + \sqrt{\varepsilon_1}]$.

The label: the class of the pair of holonomies. For pairs $P = (P_1, P_2)$, Q in $\mathrm{SU}(2)^2$ let $\mathsf{D}(P,Q) = \min_g \max_{i=1,2} |P_i - gQ_ig^{-1}|$, a distance between the classes under simultaneous conjugation, $\mathcal{O}_w(Q) = \{P : \mathsf{D}(P,Q) \leq w\}$, and $\mathsf{v}(r)$ the volume of a ball of radius r of SU(2). For $0 < w \leq \frac{2}{35}$ a maximal family of points $Q^{(k)}$ at mutual class distances $> w$ gives a finite partition of $\mathrm{SU}(2)^2$ into Borel cells $I_k \subset \mathcal{O}_w(Q^{(k)})$, invariant under simultaneous conjugation; with $I_k \sim I_l$ if and only if $\mathsf{D}(Q^{(k)}, Q^{(l)}) \leq 3w$, two pairs at the class distance $\leq w$ lie in neighbouring cells, and $\nu \leq \mathsf{v}(\frac{7}{2}w)^2/\mathsf{v}(\frac{1}{2}w)^2 \leq 1.21 \cdot 10^5$.

The label of a layer is the cell of the pair (P_1, P_2) of the holonomies of its straight closed lines through one site, and E_k is the event that this pair lies in I_k. The label is rigid across the layers: if all plaquettes of a lattice of n sites a side lie within ϖ of $\mathbf{1}$, one conjugation carries both holonomies at $a + ke_3$ to within $k\mathsf{n}\varpi$ of those at a, so that $\mathsf{D} \leq k\mathsf{n}\varpi$. *No class is heavy on a bare even cube.* Let ρ_ω be the density, against the Haar measure of $\mathrm{SU}(2)^3$, of the three holonomies (P_1, P_2, P_3) through a site on the event that all plaquettes lie within ω of $\mathbf{1}$; it vanishes unless $|[P_\mu, P_\nu]| \leq N^2\omega$, and it is greatest at $(\mathbf{1}, \mathbf{1}, \mathbf{1})$, as follows. Choose a pointwise version of ρ_ω by solving one seam bond on each of the three straight winding loops for P_μ and integrating over all other bonds. The loops have no bonds in common, and these substitutions preserve product Haar measure. Fix μ and cut at the site planes $x_\mu = 0, h$, where $N = 2h$. Let B be their tangential bonds and X_+, X_- the private bonds in the two halves, identified by reflection. No plaquette crosses a site plane, so the cut Wilson weight factors as $w_0(B)w_+(B, X_+)w_+(B, X_-)$, with $dU = dB\,\mathrm{d}X_+\,\mathrm{d}X_-$ and nonnegative factors bounded by one. The plaquette indicators factor in the same way. At fixed B, solve one private bond on the half-path from 0 to he_μ for its transport. Its weighted Haar density k_B is bounded and hence belongs

to $L^2(G)$. If the two half-transports from 0 to he_μ are g_+, g_-, then $P_\mu = g_+ g_-^{-1}$ and its unnormalised conditional density is $q_B(p) = \int k_B(pg)k_B(g)\,\mathrm{d}g \le \int k_B(g)^2\,\mathrm{d}g = q_B(\mathbf{1})$ by Schwarz and Haar invariance. The other two loops lie in $x_\mu = 0$, so their holonomies and their solved seam bonds are functions of B. Integrating this inequality over the remaining boundary bonds with those holonomies fixed proves $\rho_\omega(P) \le \rho_\omega(P_1, \dots, P_{\mu-1}, \mathbf{1}, P_{\mu+1}, \dots, P_3)$ in the pointwise version just specified. Apply it for $\mu = 1, 2, 3$. The factorization is also the site-mirror case of the proof of Prop. YM3_I:91.4, which keeps the boundary bonds as separate Haar variables. To complete **(Heaviest class)**, one must establish the following quantitative class-probability estimate; the subsequent calculation describes its use. Take $\Xi' = \beta\omega_0^2 \ge 1$ with $\Xi' N^5 \omega_0 \le 3/640$ and $N^2\omega_0 \le w \le 1/10$.

The estimate required here is, for every pair Q and every site y,

$$\mu_{N^3}(\mathsf{D}((P_1, P_2)(y), Q) \le w, \text{ no rough plaquette}) \le 10^{11}\,\Xi'^5 N^7 w^6 \omega_0^{-4}.$$

Given this estimate, on the bare cube with ω_0 and Ξ_β of (a)(ii), $\Xi' = 2\Xi_\beta$ and $w = N^2\omega_0$, as long as $2\Xi_\beta N^5\omega_0 \le \frac{3}{640}$: where no plaquette is rough, D between the labels of the layers 1 and $\frac{N}{2} - 1$ is $< w$ and between those of the layers $\frac{N}{2} \mp 1$ it is $\le 2N\omega_0 \le w$, so $\varepsilon_1, \varepsilon_2$ are at most the probability, $\le \beta^{-2}$, that some plaquette is rough, $p^\sharp \le 10^{11}(2\Xi_\beta)^6 N^{19}\beta^{-1} + \beta^{-2}$, and, the bound increasing with ν, with $1.21 \cdot 10^5$ for ν and $2\nu^4 = 4.2872 \cdot 10^{20}$: $e^{-\mathsf{G}_N} \le 2.8 \cdot 10^{33}\Xi_\beta^6 N^{18}\, t$ on the bare cube. *Through the cutoffs.* With $y = \log \mathsf{b}$ and the side ℓ of (a)(iii), this serves at the cutoffs below the cutoff $K_1(y)$ of (M4)(vi), Prop. 17.30(c), which in (b) we call K_1: the least $K \ge 10$ with $L^{K/10} \ge \bar\ell(y)^3 y^{2\mathfrak{p}_0+1}$, so that $L^{K_1} < L\,(\bar\ell(y)^3 y^{2\mathfrak{p}_0+1})^{10}$ and the bare cube has $N_1 = \ell L^{K_1} < \mathsf{N}(y) = L\,\bar\ell(y)^{31} y^{10(2\mathfrak{p}_0+1)}$ sites a side, a fixed power of y; at the bare cubes $N \le N_1$ one has $\Xi_\beta \le 10N^3 y$, and the condition $2\Xi_\beta N^5\omega_0 \le \frac{3}{640}$ holds once y is large, as $t \le \bar\ell(y)e^{-y}$.

Above K_1 the additive transport of (a) does not serve: even summed from the higher cutoff $\lceil y/(25\log L)\rceil$ of (a)(iii), its error $\sum_K \mathfrak{o}_K \le Q(y)e^{-y/250}$ is of the order of $t^{1/250}$ up to powers of $\log\frac{1}{t}$, and swamps the first power of t; summed from K_1 it is only of the order $1/y$. We use instead the comparison with a factor of (M4)(vi), Prop. 17.30(c): from K_1 on its factors multiply to at most $e^{0.53} < 1.7$, the sums from K_1 on of the numbers of §2.3.1 being of the order $1/y$, and its remainders add up to at most e^{-4y}. At K_1 take ω_0 of (a)(ii) at $\beta_1 = N_1/t$, $w = N_1^2\omega_0$, the cells I_k of this w, and for the block field the roughness $\omega = 1.003\,L^{2K_1}\omega_0$; none of these reads K.

By (M4)(v), where all bare plaquettes lie within ω_0 of $\mathbf{1}$ (for large y, $L^{2K_1}\omega_0 \le N_1^2\omega_0$ lies below the number of L that (M4)(v) asks), the block plaquettes of $\mathcal{M}^{K_1}U$ lie within ω of $\mathbf{1}$ and its holonomies within $13.3\,N_1L^{K_1}\omega_0 \le 0.512\,w$ of the bare ones; so the bound on the classes of the bare cube N_1, taken at $1.52\,w$, gives for each of the finitely many events E_k of the block field of depth 0 at a site $\mu_0^{K_1}(E_k) \le \beta_1^{-2} + 7.9 \cdot 10^{19} N_1^{36} y^6\, t$, and for the event $\mathfrak{R}$ of a rough block plaquette $\mu_0^{K_1}(\mathfrak{R}) \le \beta_1^{-2}$. By (M4)(vi), in the form of Prop. 17.30(c), used for each of these gauge-invariant events apart (nothing is summed over the cells: $p^\sharp$ is a maximum), at every $K \ge K_1$ each has a probability at most 1.7 times its probability at K_1 plus $1.7\,e^{-4y}$.

At the cutoff K the label (β) of the block field of depth 0, on the torus ℓ with $\mathsf{n} = \ell$ block sites a side, is rigid with ω for ϖ: between the layers 1 and $\frac{\mathsf{n}}{2} - 1$ the class distance is $< \frac{1}{2}\ell^2\omega < w$, between $\frac{\mathsf{n}}{2} \mp 1$ it is $\le 2\ell\omega < w$, so $\varepsilon_1, \varepsilon_2 \le \mu_0^K(\mathfrak{R})$. As $\beta_1^{-2} \le t$ and $e^{-4y} \le t$, the bound of the top level by the heaviest cell gives $e^{-\mathsf{G}_N} \le 6 \cdot 10^{40} N_1^{36} y^6\, t$, which is at most $C\,t\,(\log\frac{1}{t})^{\mathfrak{q}}$, as $N_1 < \mathsf{N}(y)$ and $\log\frac{1}{t} \le y \le 2\log\frac{1}{t}$; the bound $2.8 \cdot 10^{33}\Xi_\beta^6 N^{18} t \le 2.8 \cdot 10^{39}\mathsf{N}(y)^{36} y^6 t$ of the bare cubes below K_1 is smaller. □

17.2.6. *Proof of the theorems. Step 1: Thm. AZ(1).* BI is the hypothesis $\mathsf{e}(l) \le x_0$ of Props. 17.32, 17.33, 17.38, 17.39 and of Lem. 17.37 at $l = l_K$, $\beta = L^K\mathsf{b}$, with one x_0 for all K. For $N = \ell/a_K$,

$$\frac{N}{2l_K} = \frac{\ell}{\ell_0}, \qquad \theta^{1/l_K} = e^{-m_* a_K}, \qquad N \ge 2l_K \iff \ell \ge \ell_0.$$

(a) is Prop. 17.32(b), (c) and Prop. 17.33(c), with $X_K(\mathsf{b},\ell) \le L^3\delta((\ell/La_K)^3)$ (Prop. 17.31(c)). (b) is Prop. 17.38(a), (b). (c): with $y = \theta^{\ell/\ell_0} \le 0.8901$, convexity gives $\varepsilon \le 22.3(e^{0.13736/7}-1)\,y \le 0.441912\,y$ in Prop. 17.34(b), and $\frac12(\sqrt{\varepsilon}+\varepsilon) \le 0.540843\sqrt{y}$; with $y = \theta^{\ell'/\ell_0}$, $Q', Q'' \le e^{0.13736/8} \le 1.017319$ and $Q'(Q'-1) \le 0.0176182\,y$, so Prop. 17.34(c) gives $4.84\cdot 0.132734\cdot 1.017319 \le 0.66$. The boxes are Prop. 17.38(c) with $\theta^{N/4l_K} = \theta^{\ell/2\ell_0}$, the twisted boxes Prop. 17.38(d). (d) is Prop. 17.33(a), (b) with $A = l_1/a_K$, $B = l_2/a_K$ and t/a_K for t: $\frac{A}{2(t/a_K)} = \frac{l_1}{2t}$, $\frac{AB}{4l_K^2} = \frac{l_1 l_2}{\ell_0^2}$, $\theta^{(t/a_K)/l_K} = e^{-m_* t}$, $\frac{1.4239}{A} = a_K\frac{1.4239}{l_1}$, and $\sum_{i\ge1}(\lambda_i/\lambda_0)^{t/a_K} = e^{\mathsf{G}} - 1$; at $l_1 = \ell_0$, $(2.42x_0)^{1/l_K} = \exp[-a_K\frac{2}{\ell_0}\log\frac{1}{2.42x_0}]$, and $2\log(1/(2.42\cdot 0.069)) \approx 3.5798$.

For the square, $n = s_\ell/a_K = 2^k l_K$ and $(\lambda_i/\lambda_0)^n \le i^{-1}\mathsf{r}_n \le 0.20395\,i^{-1}\theta^{n/l_K}$, which is the bound on E_i and on their number; $\log(1/0.20395) \ge 1.5898$. The vortex line: Prop. 17.33(d) and Prop. 17.39(e), at $\ell = 2^j\ell_0$, $\ell/a_K = 2^{j+1}l_K$, $\sigma^+ a_K^{-2} = \sigma_*$. (e) is Prop. 17.38(f) at $n_0 = 0$; Lem. 17.37(b) with $\theta^{(j-1)\mathsf{w}/l_K} = e^{-m_*(j-1)\ell''/2}$ and $Q''^2 \le 1.0350$; and Prop. 17.38(b′), $\theta^{(2n-1)/6l_K} = e^{-m_*(2n-1)a_K/6}$, $n \ge \frac12(3l_K+1)$ being $na_K \ge \frac34\ell_0 + \frac12 a_K$. (f) With the time e_3, the lattice Γ is the box $(\ell/2a_K; 2\ell/a_K, \ell/a_K)$ closed up with the translation by $\frac{\ell}{2}e_1$ (its periods are also $2\ell e_1$, ℓe_2, $\frac{\ell}{2}(e_1+e_3)$).

With $l' = \ell/2a_K$, Prop. 17.31(f) over the cross-section $(4l', 2l')$ gives $\lambda_0^{l'}(1-\mathsf{r}) \le Z_\Gamma \le Z(l', 4l', 2l')$, $\mathsf{r} = \mathsf{r}_{l'}(2l', 4l')$. Above: $F(l', 4l', 2l') \le 2F(l', 2l', 2l')$ (Prop. 17.31(b)), so $E \le \mathsf{e}(l')$. Below: $\log\lambda_0 = 8l'^2P + \varepsilon(2l', 4l') \ge 8l'^2P$ and $\log Z((2l')^3) = 8l'^3P + \delta_K(\ell)$, so $E \ge \log(1-\mathsf{r}) - \delta_K(\ell)$; by Prop. 17.33(b) in the zone $2t \le A$, $\mathsf{r} \le e^c - 1$ with $c = 0.54526\,y$, $y = \theta^{\ell/\ell_0}$; as $c \mapsto \log(2-e^c)$ is concave and vanishes at 0, $\log(2-e^c) \ge -2.0196\,c$ for $0 \le c \le 0.54526\cdot 0.8901$; with $\delta_K(\ell) < 0.13736\,y$ the sum is $\ge -1.24\,y$.

The sign $E \ge 0$ is Prop. 17.31(g) at $N = \ell/a_K$, the lattice Γ being $\Gamma^\diamond$ in physical units.

Step 2: Thm. AZ(2). Fix $K \ge \max\{K_0, 1\}$, $\beta = L^K\mathsf{b}$, boards of the offset 0. The X_i and G_I are real functions of the cell $[-\ell''/4, \ell''/4]^3$ at every cutoff. *(i)* $M_I < \infty$: $\|G_I\|^8_{\mathrm{ch},\ell''}$ is a moment of $8|I|$ centred fields with disjoint supports on the single torus ℓ'', a polynomial in their cumulants of orders ≥ 2, which converge as $K \to \infty$ by (M4)(i) (Prop. 17.43). *(ii) One factor.* With $y = \theta^{\ell'/\ell_0}$, by Step 1, $[Q'(Q'-1)]^{1/2} \le 0.132734\sqrt{y}$ and $Q'' \le 1.0174$; Prop. 17.34(a), (c) give $|\nu_K(G_I)|, |\mu_{K,\ell',\beta}(G_I)| \le 1.0174\,M_I$ and $|\nu_K(G_I) - \mu_{K,\ell',\beta}(G_I)| \le 0.65\sqrt{y}\,1.0174\,M_I$. *(iii) The cumulant.* A cumulant of order $n \ge 2$ does not see the centring, nor, for $n = 1$, does the difference of the means of two laws; and

$$\kappa_n = \sum_\pi (-1)^{|\pi|-1}(|\pi|-1)!\prod_{I\in\pi}\mathbb{E}\Big(\prod_{i\in I}X_i\Big), \qquad \Big|\prod_I \mathsf{a}_I - \prod_I \mathsf{b}_I\Big| \le \sum_I |\mathsf{a}_I - \mathsf{b}_I| \prod_{I'\ne I}\max\{|\mathsf{a}_{I'}|, |\mathsf{b}_{I'}|\},$$

so the bound holds with $C(X)$. *(iv) Limits.* As the torus cumulants converge at each fixed ℓ' by (M4)(i), $\limsup_K - \liminf_K$ of $S_n^T[\nu_K](X)$ is at most $2C(X)\theta^{\ell'/2\ell_0}$ for every ℓ', hence 0; letting $K \to \infty$ in the bound gives the iterated limit, and along (K_j, ℓ_j) the difference is $\le C(X)\theta^{\ell_j/2\ell_0}$. *(v) Every shape.* At $N' = \ell'/a_K$ the corner $c' = \frac{\ell'}{4}(1,1,1)$ has $\mathsf{C}_{N'/2}(c') = [-\frac{\ell'}{4}, \frac{\ell'}{4}]^3$, and $\|\cdot\|_{\mathrm{ch};N'} = \|\cdot\|_{\mathrm{ch},\ell'}$ there; by Lem. 17.35(b), (c) with $\delta(N'^3) < 0.13736\,y \le \frac12$: $|\langle G_I\rangle_{\underline{n}}| \le e^{0.1717}\,1.0174\,M_I \le 1.208\,M_I$ and $|\langle G_I\rangle_{\underline{n}} - \mu_{K,\ell',\beta}(G_I)| \le 7.31\cdot 0.370622\sqrt{y}\,1.0174\,M_I$; with (ii), $|\langle G_I\rangle_{\underline{n}} - \nu_K(G_I)| \le 3.36\sqrt{y}\,1.0174\,M_I$, and (iii) gives $C^{\mathrm{sh}}(X)$.

For boxes of any shape with $a_{K_j}\min_\mu n_\mu^{(j)} \to \infty$ take ℓ_j the greatest member of $\mathfrak{L}$ below that length. *(vi) A constant free of label, side and place.* For $\mathsf{b} \in I(g)$ the pairs are first-passage data at g. Put the supports by a corner of the cell, $c_\mu = \lfloor z_\mu + R\rfloor + \frac32$, z the centre of the ball: one corner serves all cutoffs because L is odd; the cell $c + [-\frac{\ell'}{2}, 0]^3$ holds the supports farther than $\frac12$ from its walls; and the $8n$ mirror images of the f_i in the planes $x_\mu = c_\mu$ have the same norms, supports each in a ball of radius R, pairwise $\ge \mathsf{d}'$ apart (two images on the two sides of a wall are more than 1 apart), all in $c + [-(2R+\frac32), 2R+\frac32]^3$, which a lattice translation carries into $(-\frac{\ell'}{4}, \frac{\ell'}{4})^3$.

So the expansion of Prop. 17.43, with the bounds (M5)(i) in the place of (M4)(i), gives $\|G_I\|_{\mathrm{ch},\ell'} \le \mathsf{m}_I^\flat$, centring by the mean on the torus ℓ' itself (the floor $128n(R+1)$ is the floor $16k(R+1)$ of (M5)(i) at $k=8n$). Then (ii), (iii) at $\ell''=\ell'$, where $4.84\cdot 0.132734 = 0.642433$ and no factor Q'' enters, give $C^\flat(X)$, for ν_K and for a torus $\ell^+ \ge \ell'$ alike, both being states $\mathfrak{s}$ of Prop. 17.34; for the boxes $(q_\mu N)$, Lem. 17.35(b), (c) give $1.86\cdot 0.370622 = 0.68936$ in the place of 0.642433; for every box, Lem. 17.35(b), (c) and (ii) give $|\langle G_I\rangle_{\underline{n}} - \nu_K(G_I)| \le (2.70925 + 0.642433)\sqrt{y}\,\mathsf{m}_I^\flat$ and $e^{\frac{5}{4}\cdot 0.13736} \le 1.19$.

For $n\ge 2$, $|S_n(X)| \le C_n^g \prod \|f_i\|_\sharp$ is the limit of (M5)(i) along the tori.

Step 3: Thm. AZ(2′). For bounded $Y, X_1, \dots, X_n$, with $X_I = \prod_{i\in I} X_i$,

$$\kappa(Y, X_1, \dots, X_n) = \sum_{\emptyset\ne I\subset[n]} \mathrm{Cov}(Y, X_I)\,\mathsf{c}([n]\setminus I), \qquad \mathsf{c}(S) = \sum_{\pi\in \mathrm{Part}(S)} (-1)^{|\pi|}|\pi|! \prod_{B\in\pi} \mathbb{E}[X_B],$$
$$\mathsf{c}(\emptyset) = 1.$$

Take a board of side $\ell''/2$ with a cell $\square$ about the tuple and a cell $\square'$ about the support of h; centring by the means on the torus ℓ'', Prop. 17.43 bounds the seminorms of the products of the centred fields in each cell, Prop. 17.34(a) bounds the means $\mathbb{E}[X_B]$, and Lem. 17.37(b) bounds $\mathrm{Cov}(Y, X_I)$ by $e^{-m_*(j-1)\ell''/2}$ times the seminorms, the indices of $\square$, $\square'$ differing by $j \ge 2(|v|_\infty - \ell'')/\ell'' + 1$ in some direction; on a torus both ways around enter. The constants depend on the label only through the compact set of Prop. 17.43. For W in the place of h the same argument runs with $Y = W$, whose centred seminorm in its cell takes the place of the bound of Prop. 17.43 for h. Summing over the plaquettes of a far region with a partition of unity by translates of a fixed bump gives the absolute convergence and the decay: with the tents χ_v, $v\in\rho\mathbb{Z}^3$, of (2′) and u vanishing on $\{|x|_\infty < r\}$, $r \ge \ell'' + \rho$, the bound for $h = u\chi_v$, which vanishes unless $|v|_\infty \ge r - \rho \ge \ell''$, summed over v, gives

$$\sum_{v\in\rho\mathbb{Z}^3} |S^T_{n+1}[\nu_K](u\chi_v, X)| \le \mathsf{C}_n \sup_v \|u\chi_v\|_\sharp \prod_i \|f_i\|_\sharp \sum_{v\in\rho\mathbb{Z}^3,\ |v|_\infty \ge r-\rho} e^{-m_*(|v|_\infty - \ell'')},$$

On the torus use V and χ_v^{per} from (2′). The conditions $\ell \ge L\ell''$ and $\ell'' \ge 8\rho + 4\mathsf{d} + 4$ give $2\rho < \ell$, so each periodized tent is a single local tent in its torus chart and $\sum_{v\in V}\chi_v^{\mathrm{per}} = 1$. Choose half-open representatives of V with minimal sup norm. They form a subset of $\rho\mathbb{Z}^3$ and satisfy $|v|_\infty = |v|_{\infty,\ell}$. The one-far-insertion bound summed over this finite set, for $r \ge \frac{3}{2}\ell'' + \rho$, is therefore bounded by the displayed infinite-grid tail, with the same $3.44\,\mathsf{C}_n$.

Step 4: Thm. AZ(3), the vacuum and the Euclidean group. For $\mathsf{b}\in I(g)$ all pairs $(K, L^K\mathsf{b})$ are first-passage data, and by Step 1 the only periodic Gibbs state there is ν_K. The moments of the centred fields under ν_K are polynomials in the cumulants $S^T_{|I|}[\nu_K]$, $|I|\ge 2$, which converge by Step 2; so (ν_K) converges, and $S_n^\sigma = S_n$. By (M5)(ii) each ergodic component of σ is a limit along the full sequence whose approximants are periodic Gibbs states at the same pairs, that is ν_K; so σ is its own component, and Thm. YM3_I:AP gives the invariance under $\mathbb{R}^3 \rtimes \mathrm{O}(3)$, reflection positivity at every plane, $\ker H_\sigma = \mathbb{C}\Omega_\sigma$ and the Wightman theory with a unique vacuum. Step 2 (iv), with its iterated limit and its bound along every pair of sequences $K_j\to\infty$, $\mathfrak{L}\ni\ell_j\to\infty$, identifies every continuum limit at g along growing tori with the pairs $(K_j, L^{K_j}\mathsf{b})$ with $(S_n^\sigma)_{n\ge 2}$, the limits with the spacing removed first among them; so each of them has the Schwinger functions of σ, and with them its invariance under $\mathbb{R}^3\rtimes\mathrm{O}(3)$, its reflection positivity, its reconstruction with a unique vacuum and, by Step 5, its gap.

Step 5: Thm. AZ(3), the gap. Let $F = \sum_i \zeta_i X_i$ be a finite combination of separated tuples of fields supported in $\{\epsilon < x_3 < \mathsf{h}'\}$, Θ the reflection $x_3\mapsto -x_3$, τ_t the translation by te_3 with $t\in a_K\mathbb{N}$, and F_K the same combination of the lattice fields $\mathcal{O}_K(f)$, centred by their means in ν_K, a function of one slab above the plane $x_3 = 0$, with n_0 layers between the slab and its mirror image. By translation invariance $\nu_K(\overline{\Theta F_K}\,\tau_t F_K)$ is $h_{F_KF_K}(n_0 + t/a_K)$, the mirror being moved to the plane

$x_3 = -t/2$; so Prop. 17.38(f) with $A = B = F_K$, $n = n_0 + t/a_K$ and $l = l_K$ gives

$$\nu_K(\overline{\Theta F_K}\,\tau_t F_K) - |\nu_K(F_K)|^2 \le e^{-m_* t}[\nu_K(\overline{\Theta F_K}\,F_K) - |\nu_K(F_K)|^2],$$

with a rate that does not depend on K and a prefactor that converges. Fix $t \in a_{K_*}\mathbb{N}$. It belongs to every finer time grid, so we may let $K \to \infty$ at this fixed time. With $\psi = (1 - P_\Omega)[F] \in \mathcal{H}(\sigma)$: $\langle \psi, e^{-tH_\sigma}\psi\rangle \le e^{-m_* t}\|\psi\|^2$ on the unbounded set $\bigcup_K a_K\mathbb{N}$ of times; so the spectral measure of ψ gives mass at most $e^{-\eta t}\|\psi\|^2$ to $[0, m_* - \eta]$ for arbitrarily large t, that is none, and such ψ are dense in $\Omega_\sigma^\perp$. The bilinear bound is the spectral theorem on $\Omega_\sigma^\perp$; the other directions follow by the invariance.

Step 6: Thm. AZ(4). Prop. 17.39 at $l = l_K$, lengths in units of a_K: $\sigma^+ a_K^{-2} = (2\Lambda + 1.67)/4l_K^2 a_K^2 = \sigma_*$, and $\sigma_* \ge (2 \cdot 0.116421 + 1.67)/\ell_0^2$ at $\theta \le 0.8901$; the areas become physical areas. For the flux quotient, $N/l_K = 2\ell/\ell_0$ turns $2.2 \cdot 10^{-4}(N/l_K)^2$ into $8.8 \cdot 10^{-4}\ell^2/\ell_0^2$, $1.4268 \cdot 10^{-3}$ into $5.707 \cdot 10^{-3}$ at $N \ge 256 l_K$, that is $\ell \ge 128\ell_0$, and $2\mathsf{q}N/l_K$ into $4\mathsf{q}\ell/\ell_0$. For the potential, $V_K(r) = a_K^{-1}V(r/a_K)$ with the V of Prop. 17.39(d), whose slopes are $\ge \sigma^+ = \sigma_* a_K^2$.

Step 7: Thm. BA(1). By Lem. 17.42 take a second admitted blocking factor $L' > L$, a prime with $\alpha = \log L/\log L'$ irrational; Thm. YM3_I:D holds in the scheme of L' (spacings L'^{-k}, sides $2L'^{\mathbb{N}}$), and $L^K\mathsf{b} = L'^{k(K)}\mathsf{c}_K$ with the labels $\mathsf{c}_K \in [\mathsf{b}, L'\mathsf{b})$ dense. In the scheme of L' let $\ell_f \in 2L'^{\mathbb{N}}$, $\ell_f \ge 24L'\ell_0$, $w_0 = \ell_f/2L'$, $f(x) = \prod_i \max\{0, 1 - |x_i|/w_0\}$ (the tent), and $\mathsf{M}_k(\ell; c) = \|\mathcal{O}_k(f) - \mu_{k,\ell,L'^k c}(\mathcal{O}_k(f))\|_{\mathrm{ch},\ell}$. (i) $\mathsf{M}_k(\ell;\cdot)^8$ is a polynomial in torus cumulants of orders 2 to 8, which by (M4)(i) at L' converges, uniformly on $[\mathsf{b}, L'\mathsf{b}]$, to a continuous function; so $\mathsf{M}_f = \sup_{k\in\mathbb{N},\ c\in[\mathsf{b},L'\mathsf{b}]} \mathsf{M}_k(\ell_f; c) < \infty$. (ii) At the labels c_K, BI gives $\mathsf{e}(l_K) \le x_0$ at $\beta = L'^{k(K)}\mathsf{c}_K$ with $l_K = \ell_0 L^K/2$, and a fine cell has $\mathsf{h} = \ell_f L'^{k(K)}/2$ layers with $\mathsf{h}/l_K \ge \ell_f/L'\ell_0 \ge 24$; Lem. 17.41 with $\ell_c = d\ell_f$, $d = L'^j$, gives

$$\mathsf{M}_{k(K)}(\ell_c; \mathsf{c}_K) \le \epsilon_\star(d) = 1.343\,\mathsf{M}_f\,\theta^{(d-1)\ell_f/8L'\ell_0}.$$

(iii) Let $x(c) = L'^{-3}D_{L'}(N)$, $N = \ell_c L'^k$, at $\beta = L'^k c$. Since $\partial_\beta \log Z(n^3)$ is $-3n^3$ times the mean plaquette variable, which is the same for all plaquettes of a torus, and the translates of the tent are a partition of unity,

$$x'(c) = -(\ell_c/w_0)^3[\mu_{k,\ell_c,\beta}(\mathcal{O}_k(f)) - \mu_{k,L'\ell_c,\beta}(\mathcal{O}_k(f))], \qquad \ell_c/w_0 = 2L'd.$$

Wherever $x \le 0.0064$, $\delta(N^3) \le (L'/(L'-1))^3 x \le 0.0086$ by Thm. BC(1), and Prop. 17.34(c) bounds the bracket by $0.16\,\mathsf{M}_k(\ell_c; c)$; so $|x'| \le 0.0058/((L'-1)\mathsf{b})$ there as soon as $\mathsf{M}_k(\ell_c; c) \le \epsilon_{\mathrm{req}}(d) = 0.036/[(2L'd)^3(L'-1)\mathsf{b}]$. The required smallness shrinks with d like d^{-3}, while $\epsilon_\star(d)/\epsilon_{\mathrm{req}}(d)$ is a constant times $d^3\theta^{(d-1)\ell_f/8L'\ell_0}$, which tends to 0 as $d = L'^j \to \infty$: fix d with $\epsilon_\star(d) \le \frac{1}{2}\epsilon_{\mathrm{req}}(d)$. By (i) the functions $\mathsf{M}_k(\ell_c;\cdot)$ converge uniformly on $[\mathsf{b}, L'\mathsf{b}]$ to a continuous limit; at the labels c_K, $K \ge K_0$, which are dense, with $k(K) \to \infty$, they are at most $\frac{1}{2}\epsilon_{\mathrm{req}}(d)$; so the limit is at most $\frac{1}{2}\epsilon_{\mathrm{req}}(d)$ everywhere, and there is $\bar{k} \ge k(K_0)$ with $\mathsf{M}_k(\ell_c; c) \le \epsilon_{\mathrm{req}}(d)$ for all $c \in [\mathsf{b}, L'\mathsf{b}]$ and all $k \ge \bar{k}$. (iv) At each $k \ge \bar{k}$ there is K with $k(K) = k$, and Prop. 17.32(b) gives $x(\mathsf{c}_K) < 0.000514$; by the mean value theorem the component of $\{x \le 0.0064\}$ that contains c_K carries $x \le 0.006314$, so it is all of $[\mathsf{b}, L'\mathsf{b}]$.

Hence $\delta(N^3) \le 0.008625$ for all c and all $k \ge \bar{k}$, and every $\beta \ge \beta_0 = L'^{\bar{k}}\mathsf{b}$ equals $L'^k c$ with $l = N \le C_+\beta$, $C_+ = \ell_c/\mathsf{b}$. Of L the argument uses only that $\log L/\log L'$ is irrational.

Step 8: Thm. BA(2), (2′), (3). At $\beta = L^K\mathsf{b}' \ge \beta_0$ there is $l \le C_+\beta$ with $\mathsf{e}(l) \le 0.069$, and Prop. 17.32(b) gives $\delta(N'^3) \le 0.13736 \cdot 0.8901^{24} < 0.008625$, so $\mathsf{e}(N') \le 0.069$, for all $N' \ge 48l$. Since $L^{m+K} \ge 48C_+\mathsf{b}'L^K \ge 48l$, the cube of BI at the side $2L^m$ qualifies; on $[\mathsf{b}_-, \mathsf{b}_+]$ take $L^m \ge 48C_+\mathsf{b}_+$ and the threshold of b_-. If BI holds at one label, it holds at all: the set of the labels is empty or $(0, \infty)$. (2′): $\mathsf{M} < \infty$ by Prop. 17.43 and (M4)(i). Fix $K \ge \max\{K_0, 1\}$ and write $\delta(s)$ for $\delta_K(\ell_1)$ at the label s. The derivative of the excess in the label is L^K times the sum over the plaquettes of the torus of the differences between the mean plaquette variable on the torus and in the periodic Gibbs state at the label s (Prop. 17.38(a)); all plaquettes having one such difference, by the symmetries of

both laws, it is a multiple R_K, $|R_K| \le 1458$, of the difference of the means of $\mathcal{O}_K(f)$ on the torus and in that state:

$$\delta(\mathsf{b}_2) - \delta(\mathsf{b}_1) = -\int_{\mathsf{b}_1}^{\mathsf{b}_2} R_K \left[\mu_{K,\ell_1,L^K s}(\mathcal{O}_K(f)) - \nu_{K,s}(\mathcal{O}_K(f))\right] \mathrm{d}s.$$

While $\delta(s) \le 0.008625$, Prop. 17.34(c) bounds the bracket by $0.17516\,\mathsf{M}$, with constants free of K. At $s = \mathsf{b}$, $\delta(\mathsf{b}) < 0.13736\,\theta^{\ell_1/\ell_0} \le 0.13736 \cdot 0.8901^{75} < 0.000025$; so δ cannot reach 0.008625 within η of b, at every cutoff. The continuity and the limits along labels follow from Step 2, made with a supremum over the labels of $[\mathsf{b} - \eta, \mathsf{b} + \eta]$ (one side $2\ell_1$, one threshold, $\theta = 0.8901$), and the continuity in the label of (M4)(i). For the second form put $N = \ell_1^\flat / a_K$. At b, Prop. 17.32(b) gives $D_L(N) < L^3\, 0.13736\, \theta^{\ell_1^\flat/\ell_0} \le 0.03$.

At a label s with $D_L(N) \le 0.069$, $\delta(N^3) \le 0.069/(L-1)^3$ by Thm. BC(1), and the derivative of $D_L(N) = L^3 F(N^3) - F((LN)^3)$ in the label is $-L^3 R_K$ times the difference of the means of $\mathcal{O}_K(f^\flat)$ on the tori $\ell_1^\flat$ and $L\ell_1^\flat$, with the number R_K of the first form, now for $f^\flat$, $|R_K| \le 1458$; Prop. 17.34(c) bounds that difference by $4.84\,[Q'(Q'-1)]^{1/2}\mathsf{M}^\flat \le 0.010813\,\mathsf{M}^\flat$. So $D_L(N)$ rises by at most 0.03 within r of b and stays below 0.06; then $\mathsf{e}(N) \le \frac{8}{L^3} D_L(N) < 0.48/L^3 < 2.2 \cdot 10^{-4}$ by Thm. BC(1), and Prop. 17.32(b) at $l = N$, with $\theta_1 = 12.9 \cdot 2.2 \cdot 10^{-4}$, gives $\mathsf{e}(L^z N/2) < 0.3702\,\theta_1^{L^z/4} \le 6.1 \cdot 10^{-9}$, which is the inequality of BI at the side $L^z \ell_1^\flat$. (3): Prop. 17.38 at this l: the rate $\theta^{n/l}$ with $\theta = 0.8901$ and $l \le C_+\beta$ is at most $e^{-c_* n/\beta}$; the transfer operator is the operator T_ν of the setting, whose matrix elements are the numbers $h_{AB}(n)$ of Prop. 17.38(f): at $\theta = 0.8901$ and $l \le C_+\beta$, $\ker(1 - T_\nu) = \mathbb{C}\Omega$ and $\|T_\nu\| \le 0.8901^{1/l} \le e^{-c_*/\beta}$ on $\Omega^\perp$; and Prop. 17.39(a) with $l \le C_+\beta$ gives the flux quotient; Prop. 17.39(d) gives the slopes, $\sigma^+ l^2 = \frac{1}{2}\log\frac{1}{0.8901} + 0.4175 \ge 0.4757$.

The labels $a_K\beta$ of all first-passage data (K, β) at g lie in the window $(g^{-2} + B_\sharp \frac{L-2}{L-1},\ L(g^{-2} + \bar{B})]$ (YM3_I:(1531) in Prop. YM3_I:74.1), so at a datum with $\beta \ge \beta_0$ the cube of the side $2la_K \le 2C_+\beta a_K \le 2C_+ L(g^{-2} + \bar{B}) = \ell_0^g$ satisfies the inequality: Thm. AZ(1) and Step 2 hold there with ℓ_0^g for ℓ_0 and 0.8901 for θ, uniformly over these data, the seminorms of Prop. 17.43 being bounded over the window.

Step 9: Thm. BA(4). On $a\mathbb{Z}^3$ the weight does not contain a, and $\mathcal{O}_a(f(\cdot/t)) = t^{-1}\mathcal{O}_{a/t}(f)$; hence

$$\kappa_n(\beta, a/t; X) = t^n\, \kappa_n(\beta, a; X_t)$$

for the cumulants in the unique periodic Gibbs state at $\beta \ge \beta_0$. By Step 8, BI holds near every label with one side and one threshold, and Step 2, taken with a supremum over a compact set of labels and of dilations of X, gives one limit along every sequence of pairs whose labels converge, continuous in the label and the dilation. The supremum over the dilations is had through the rate of (M4)(i): by the Cauchy estimate in the proof of Thm. YM3_I:AS in Thm. YM3_I:AS, on one torus and on a compact interval of labels, $|S^T_{n;K',\ell,L^{K'}\mathsf{b}}(Y) - S^T_{n;K,\ell,L^K\mathsf{b}}(Y)| \le C\, K^{2\mathfrak{p}_0(n+1)+\frac{1}{2}} L^{-K/10} \prod_i \|y_i\|_\sharp$ for all $K' \ge K$ above a threshold, where C and the threshold depend on n, the side, the interval and a floor of the spread of the supports of $Y = (y_1, \dots, y_n)$, and on Y otherwise only through these norms.

For $Y = X_{t'}$ with t' in a compact interval $J'' \subset (0, \infty)$,

$$\|f_i(\cdot/t')\|_\sharp \le \max\{\sup|f_i|, 40M(\inf J'')^{-1} \operatorname{Lip} f_i\}$$

and the spread is at least $\inf J''$ times that of X; so the torus cumulants converge uniformly in the label and the dilation. At a fixed cutoff they are continuous in both, being finite sums with the coefficients $f_i(x(p)/t')$; so the limit is jointly continuous, and the seminorms M_I of Step 2 are bounded uniformly in $t' \in J''$ in the same way. In the scheme of L' the lattices of the line of b are the lattices of the dense labels c_K, dilated by b/c_K; so the limit of the scheme of L' at every label c is the dilation of σ_b by c/b.

The dilations by L and by L', which are trivial in their own schemes, generate a dense subgroup of $(0,\infty)$, and by continuity σ_{tc} is the dilation of σ_c for every real t; the identity above reduces arbitrary real spacings to the scheme of L, and H_n is $S_n^{\sigma_1}$, so $S_n^{\sigma_c}(X) = c^{-n} H_n(X_{1/c})$. At $c \in I(g)$, Step 4 identifies σ_c; the gap of σ_{tc} is that of σ_c divided by t.

Step 10: Thm. BA(5). For φ vanishing outside $v + [-\rho,\rho]^3$ the insertion of the rotation weighted by φ depends only on the bonds within $2a_K$ of that cube and is bounded by $3402(1 + |v|_\infty)\rho^4 L^{4K} |\omega| \sup|\varphi|$, and $\sup|\mathcal{O}_K(f)| \le 162\rho^3 L^{4K} \sup|f|$ for f vanishing outside $v + [-\rho,\rho]^3$; so in every state, for χ vanishing outside $v + [-\rho,\rho]^3$ and the f_i outside $v + [-R,R]^3$, $|\mathfrak{U}| \le \mathsf{c}_n(1+\mathsf{b})(1+|v|_\infty)\rho^4 R^{3n} L^{(4n+5)K} |\omega| \sup|\chi| \prod \sup|f_i|$, with $\mathsf{c}_n = 3402 \cdot 162^n \, \mathsf{F}_{n+1}$ and $\mathsf{F}_m = \sum_\pi |\pi|!$, the sum over the partitions π of $\{1,\dots,m\}$; in the theorem $\rho = R$. (i) *Torus against state.* The bulk term is a polynomial in means of products of local functions of one cube; by Thm. AZ(1)(c), in total variation, on all tori $\ell \ge \ell''$ and in ν_K it agrees up to $2.164\,\mathsf{c}_n(1+\mathsf{b})(1+|v|_\infty)\rho^4 R^{3n} L^{(4n+5)K} e^{-m_* \ell''/4}$ times the sup norms, which is $\le$ the same expression with L^{-K} for $L^{(4n+5)K} e^{-m_* \ell''/4}$ once $\ell'' \ge \Lambda_0 K$, $\Lambda_0 = 4(4n+6)\log L/m_*$. (ii) *The place.* For a state invariant under $a_K\mathbb{Z}^3$, translating the tuple by v adds the translation $\omega v = \sum_\lambda v_\lambda \, \omega e_\lambda$ to the generator of the insertion, and the insertion of the translation by the vector ωe_λ is the difference of the bulk terms of the tuple moved by e_λ and of the tuple, both carried by $[-R-1, R+1]^3$; summed over the components of v this gives the factor $1 + 2|v|_1$. (iii) So it suffices to bound the bulk term of tuples carried by a fixed cube about the origin on one side in $[\Lambda_0 K, L\Lambda_0 K]$ of $\mathfrak{L}(g)$, which by (i) serves for ν_K and all larger sides, and on the sides $\ell \le L\Lambda_0 K$ a power of ℓ is a power of K.

There $\mathfrak{U}$ of YM3_I:(2108) is split as follows, with the objects of (M8). A joint cumulant with the $\mathcal{O}_K(f_i)$ is the derivative in the z_i of a mean at the source $w_z = \sum_i z_i f_i$, so the bulk term of YM3_I:(2107) in equation YM3_I:(2107) and the source calculation in the proof of Prop. YM3_I:112.1 is $-\partial_z^n$ at $z = 0$ of the mean at the source w_z of the bulk insertion $(\mathfrak{S}_{\omega,\chi} + \mathfrak{K}_{\omega,\chi})[\beta - a_K^{-1} w_z]$ (equation YM3_I:(2107) and the source calculation in the proof of Prop. YM3_I:112.1). With the orbital potential $\Psi_\rho(p)$ and the remainder of third order $\mathfrak{r}_\rho^\sharp(p) = \mathfrak{r}_\rho(p) - \Psi_\rho(p) + \Psi_\rho(p_-)$ of [7, §H], the summation by parts of [7, Prop. H.1] writes this insertion as the sum of three members, affine in z: the spin part $\mathfrak{S}_\omega[\phi_z]$, $\phi_z = \beta\chi - a_K^{-1} w_z$; the member $-\sum_p \sum_\rho \tilde{c}_\rho(x(p)) \Psi_\rho(p)$, $\tilde{c}_\rho(y) = (\omega y)_\rho \, a_K^{-1}[\phi_z(y + a_K e_\rho) - \phi_z(y)]$, of order 2 in the curvature with the weight β; and the cubic member $\mathfrak{K}^\sharp[\phi_z \, \omega y]$ of [7, eq. 297], of order 3 with the weight $\beta/a_K = \mathsf{b} L^{2K}$.

Accordingly $\mathfrak{U} = \mathfrak{U}^\sharp + (\mathfrak{U} - \mathfrak{U}^\sharp)$ with

$$\mathfrak{U}^\sharp_{\omega,\chi}(f) = -\beta \langle \mathfrak{K}^\sharp[\chi \, \omega y]; \mathcal{O}_K(f_1); \dots; \mathcal{O}_K(f_n) \rangle + a_K^{-1} \sum_{i=1}^n \langle \mathfrak{K}^\sharp[f_i \, \omega y]; (\mathcal{O}_K(f_l))_{l \ne i} \rangle .$$

All but the cubic member. Use the exact lattice decomposition in Lem. 5.9(b,c,e,f), as in Step 8 of the proof of Thm. O. For $\varphi \in \{\chi, f_1, \dots, f_n\}$, antisymmetry of ω rewrites $a_K^{-1}\mathfrak{S}_\omega[\varphi]$ as an order-2 entry with weights $k^\varphi_{b,p,\rho} = a_K^{-1}(\varphi(x(p)) - \varphi(x_b)) \sum_\lambda \omega_{\rho\lambda} a_K^{-1}(x_b - x(p))_\lambda$, bounded by $|\omega| \operatorname{Lip}\varphi/4$. The bilinear orbital entry has the vector weight $\mathsf{k}^{\varphi,\rho}(y) = (\omega y)_\rho \nabla^+_\rho \varphi(y)$, whose Lipschitz norm is at most $5|\omega|(1+\ell)\|\varphi\|_{C^{1,1}}$. Lem. 5.9(c) gives $\mathfrak{J}^{\mathrm{bl}}[\mathsf{k}] = -\mathfrak{N}[\mathsf{k}]/4 + \sum_{v=1}^5 \mathfrak{J}_v[\mathsf{k}] - c_{\mathsf{k}}/4$: the five entries have orders $2,2,2,3,4$, and only the last two carry β. The null term is evaluated by the contact formulas (N3), (N4) of that lemma. They leave numbers, action densities and order-2 entries without a factor β; the numbers drop from the cumulants. Terms absorbing all the sources vanish, and merging the absorbed supports leaves spread at least $\mathsf{d} - 3a_K \ge \mathsf{d}/2$. Thus Lem. 5.15, applied in its spread form, bounds every remaining term.

For the dependence on ℓ, use the proof of Lem. 5.15 before its constants are collected. A local insertion of order m is carried to depth $u = \lceil \phi K \rceil$, $\phi = (\frac{1}{2} - c_1)/(3n+4)$. The entry contributes at most a factor ℓ^3 from its sites, and the source radii are $(2nN_t \mathsf{C}_t \|f_q\|_\sharp)^{-1}$, with $N_t = \ell^3 L^{3u}$ and $\mathsf{C}_t \le C K^{2\mathfrak{p}_0}$. Their Cauchy cost contributes ℓ^{3n}; the powers of L are $L^{(3n+4)u - \mathsf{g}_m K}$, where $\mathsf{g}_2 = 1$ and $\mathsf{g}_m = 3/2$ for $m \ge 3$. The factors β in the orders $3,4$ leave the decay $L^{-c_1 K}$. The discarded mass of

the definitions preceding Prop. YM3_I:115.2 has the form $2M^{-3}\ell^{3}L^{3u}\exp[-c\{\log(L^{u}\mathfrak{b}')\}^{2\mathfrak{p}_0}+C_g\ell^{3}]$. Since $\mathfrak{p}_0\ge 13$, the labels lie in a compact first-passage interval and $\ell\le L\Lambda_0 K$, this mass and its fixed-order moment factors decay faster than every L^{-AK}, uniformly in these sides. The locality remainders have the same property. Collecting the weights from the preceding paragraph gives, times the norms, $|\mathfrak{U}-\mathfrak{U}^{\sharp}|\le C\ell^{3(n+1)}(R+5)|\omega|K^{(2n+1)\mathfrak{p}_0}L^{-c_1K}$ for every $0<c_1<\frac12$ on $2R+4\le\ell\le L\Lambda_0K$, uniformly over first-passage data at g, with C free of the side.

The cubic member obeys an estimate on all the sides at once, the bound (M7), which is Lem. 5.16 of the section on plane fields and the stress tensor (§5.1) taken at the field $\chi\,\omega y$: for $n\ge 2$, $\mathsf{d}>0$ and $\rho_0\ge 1$ there are $C_\sharp$ and a threshold, depending on g, n, d, ρ_0 and the constants and on no side, such that for all real $\chi, f_i\in C^{1,1}$ vanishing outside $[-\rho_0,\rho_0]^3$, χ covering the f_i, the supports of the f_i at pairwise sup distances $\ge\mathsf{d}$,

$$|\mathfrak{U}^{\sharp}_{\omega,\chi}(f)|\le C_\sharp|\omega|L^{-K/16}\|\chi\|_{C^{1,1}}\prod_i\|f_i\|_{C^{1,1}}$$

at all first-passage data at g with K above the threshold, on all sides $\ell\ge 2\rho_0+2$ of $\mathfrak{L}(g)$. It is proved in three steps, as in the idea of the proof of (M7). On one torus, the conditional mean of the cubic insertion given the block field of the level $K-\lceil\phi' K\rceil$, $\phi'>0$ small, under the small-field chain of Thm. YM3_I:D, differs from a constant by at most a power of K times L^{-cK}, for every $c<\frac{1}{14}$ (Lem. 5.10(b)): the density is of third order, $|\mathfrak{r}^{\sharp}_{\rho(p)}(p)|\le C_{\mathrm{orb}}\,\bar a(p)^{3/2}$ ([7, eq. 295]), plain power counting against the weight $\mathfrak{b}L^{2K}$ falls short by the factor $a_K^{-1/2}$, and the parity under the lattice mirrors, the covariance of the renormalisation step and a Hölder bound of a background against its mirror image win this factor back.

The joint cumulants of the insertion with the sources then obey the same bound, for every $c<\frac{1}{14}$, by the two-constants inequality of Lem. YM3_I:116.2 in the source variables; this is the bound of one torus, Lem. 5.11. Last, the bound on one torus serves on the finitely many sides of $\mathfrak{L}(g)$ from $2\rho_0+2$ up to a fixed side ℓ', where (M7) is Lem. 5.11, and it passes from ℓ' to every larger side (Lem. 5.16) by the chessboard estimate (M2) on the board of side $\ell'/2$, which writes the difference of a cumulant on the torus ℓ and on the torus ℓ' through cumulants on the torus ℓ' of mirror copies of the entries that join cells, every block that holds a copy of the cubic insertion carrying the small factor, at the cost of the factor $e^{(\ell'/2)^3(p_{\ell'}-p_\ell)}\le e^{\delta_K(\ell')/8}$ of the fine-to-coarse form, which Thm. BC(5)(a), (6) bound at the fixed side ℓ', uniformly in the cutoff and over the window of the labels at g.

With $\rho_0=R+1$ the sum is $\le C|\omega|K^{p'}L^{-\kappa K}$ times the norms for every $\kappa<\frac{1}{16}$. The condition $|v|_\infty+R+2\le\ell/4$ keeps the moved tuples and the insertion away from the seam of the torus, the planes $x_\mu=\pm\ell/2$ along which the cube $[-\ell/2,\ell/2]^3$ is closed up.

Step 11: Thm. BB(1)–(4). (1) is Prop. 17.44(b), (c); the statements on a height y follow from $|X_{K'}-X_K|\le\eta_K$ for $K'\ge K\ge K_\star$ and $\eta_K\to 0$. (2) If $D_L(N)\le\frac{L^3}{8}x$ with $x\le 0.069$, then $\mathsf{e}(N)\le D_2(N)\le\frac{8}{L^3}D_L(N)\le x$, because $D_L(N)-\frac{L^3}{8}D_2(N)=\frac{L^3}{8}\delta((2N)^3)-\delta((LN)^3)\ge 0$ by the monotone density; Prop. 17.32 at $l=N$ gives $\mathsf{e}(2^kN)\le(12.9x)^{2^k}/12.9=\epsilon$, and since $2^kN\le LN/2<2^{k+1}N$,

$$\mathsf{e}(LN/2)\le 2\,\delta(\tfrac{LN}{2},LN,LN)\le\Big(\frac{L}{2^{k+1}}\Big)^3[\epsilon+\tfrac{20}{7}\epsilon^2]=\Phi_L(x),\qquad D_L(2l)\le L^3\delta((2l)^3)\le\tfrac{20}{7}L^3\mathsf{e}(l)^2.$$

At $N=\ell/La_K$ we have $LN/2=\ell/2a_K$, and by (1) $X_K\le X_{K_1}+\eta_{K_1}\le y_1$ for all $K\ge K_1$; as $\Phi_L(x_1)\le x_0$, this gives BI[$\mathfrak{b},\ell,K_1;x_0$]; with y_0, $\mathsf{e}(\ell/La_K)\le x_0$, which is BI at the side $2\ell/L$ with $c=2$, and Thm. AZ holds at every side $2cL^{m_0}$. Conversely BI at the side $2\ell/L$ has $l_K=\ell/La_K$, and $X_K(\mathfrak{b},L\ell)=D_L(Ll_K)\le L^3\delta((Ll_K)^3)\le L^3\bar\delta_{x_0}(L)$ by the proof of Prop. 17.32(b). The bracket: the upper members are Thm. BC(1) at $l=\ell/2a_K$, $2l/L=\ell/La_K$; the lower member is the contrapositive of $D_L(2l)\le\frac{20}{7}L^3\mathsf{e}(l)^2$ for $\mathsf{e}(l)\le 0.069$, at $2l=\ell/a_K$.

So BI gives $X_K(\mathsf{b}, L\ell) \le \frac{20}{7} L^3 x_0^2$, and (1) excludes this for $K \ge K_1$ when $X_{K_1}(\mathsf{b}, L\ell) > \frac{20}{7} L^3 x_0^2 + \eta_{K_1}$; the members of the bracket are continuous and monotone in X_K, and $X_K \to X_\infty$, which gives the bracket for $\liminf_K \mathsf{e}_K$ and $\limsup_K \mathsf{e}_K$. (3) At the single cutoff K_1, Prop. 17.32(b) at $l = l_{K_1}$ gives $X_{K_1}(\mathsf{b}, L^i\ell_0) = D_L(2L^{i-1} l_{K_1}) \le L^3 \bar\delta_{x_0}(2L^{i-1})$; with the margin, (1) gives $X_K \le y_1$ for all $K \ge K_1$, and (2) applies. At $i = 2$, $\bar\delta_{x_0}(2L) < 8\mathsf{c}_0 \theta^L \le 0.13736\,\theta^L$, which at $x_0 = 0.069$ and $L \ge 33$ is at most 0.0030; and for $L \ge 33$, $2^k \ge 16$, so $\Phi_L(0.05) \le 8 \cdot 1.01\,(0.645)^{16}/12.9 < 0.069$ and $x_1 \ge 0.05$, $\frac{1}{8} x_1 \ge 0.00625$: the margin condition holds once $\eta_{K_1} \le 0.003 L^3$.

At $i = 1$ it never holds at $x_0 = 0.069$, as $\bar\delta_{0.069}(2) = \frac{20}{7} \cdot 0.069^2 > \frac{1}{8} \cdot 0.069$. (4) (iii) and (1) give $X_K \le y_0$ for $K \ge K_1$, hence $\mathsf{e}(\ell/La_K) \le x_0$, which is BI at the side $\ell_0 = 2\ell/L \in 2 \cdot 2L^{\mathbb{Z}}$; (i) gives $X_K(\mathsf{b}, \ell) \le L^3 \bar\delta_{x_0}(2\ell/L\ell_0) < 0.13736 L^3 \theta^{\ell/L\ell_0}$ for $\ell \ge L\ell_0$ by Prop. 17.32(b), hence (iv); (iv) implies (ii); and (ii) implies (iii) by (1).

Step 12: Thm. BB(5), (6). (5): the reduction, requiring the interacting Hessian estimate, applies Prop. 17.45 at $\underline{s} = (\ell_0/2, \ell_0, \ell_0)$, $z = 0$, $l = \ell_0$, where $|\underline{s}|/l^3 = \frac{1}{2}$, gives

$$\mathsf{e}_{K+1} - \mathsf{e}_K = 2\Big[\log Z_{K+1}(\underline{s}; \mathsf{b}) - \log Z_K(\underline{s}; \mathsf{b}) - \tfrac{1}{2}\big(\log Z_{K+1,\ell_0} - \log Z_{K,\ell_0}\big)\Big],$$

so $|\mathsf{e}_{K+1} - \mathsf{e}_K| \le 2CL^{-K/10}$ uniformly on J, and Lem. 17.46 gives the rest. (6): $D_L(\ell/a_K) = X_K(\mathsf{b}, L\ell)$ converges uniformly on compact sets of labels by (1), with the rate of Prop. 17.44(b); the pairs $(\ell/a_K, L^K\mathsf{b})$ depend on (ℓ, b) only through ℓ/b up to the index of the cutoff, so D_∞ is a continuous function of t. The series for $\delta_K(\ell)$ of Prop. 17.31(c) at $q = L$, $\delta_K(\ell) = \sum_s L^{-3(s+1)} D_L(L^s\ell/a_K)$, has the majorant $L^{-3(s+1)} D_L(L^s\ell/a_K) \le L^{-3s}\delta_K(L^s\ell) \le C(t)\,L^{-s}$, uniformly in the large K, with $C(t)$ bounded on compact sets, by the bounds at a fixed cutoff of Thm. BC(5)(a) and (6), whose proofs do not use the present clause; dominated convergence gives $\delta_K(\ell) \to \delta_\infty(t)$, locally uniformly, and $L^3\delta_\infty(t) - \delta_\infty(Lt) = D_\infty(t)$; the inequalities $(L-1)^3\delta \le D_L \le L^3\delta$ of Thm. BC(1) pass to the limit.

One function for all block sizes. Let L' be a second admitted blocking factor with $\log L/\log L'$ irrational. At one coupling, cubes $N \le M$ with $M/N \le \rho$ have $\delta(M^3) \le \rho^3\delta(N^3)$ by the monotone density. Pass to a subsequence of the lattices (N, β) that converge to $\delta_\infty^{(L)}(t)$, and choose along it cubes $M = 2L'^j$ with $M/N \to \rho_0 \in [1, \rho]$, which the density of $\{L'^j/L^i\}$ allows; the lattices (M, β) are lattices of the scheme of L' whose labels converge, so $\delta_\infty^{(L')}(\rho_0 t) \le \rho^3 \delta_\infty^{(L)}(t)$. Letting $\rho \to 1$ and using the continuity, $\delta_\infty^{(L')} \le \delta_\infty^{(L)}$, and by symmetry the two are equal. The same comparison of nearly equal cubes, now with the lattice inequalities of Prop. 17.31(c) and Thm. BC(1),(2) between the cubes N and $M = \lfloor \rho N \rfloor$, gives that $t^{-3}\delta_\infty(t)$ does not increase, the bound $(3\rho^2 - 3\rho + 1)$ for real ρ, and the alternative for real $t \ge 2t_0$.

Positivity. $\delta_\infty \ge 0$ from the series. At every t, choosing by a dilation a label in $I(g_1) \cup I(g_2) = (\gamma_W^{-2} + \bar B, \kappa_0]$ and the side $\ell = t\mathsf{b}$, Thm. BC(1), (3) give $8\delta_K(\ell) \ge \mathsf{e}(\ell/a_K) \ge \min\{0.069, 0.0775\, e^{-2.6\ell/\mathsf{d}_\circ}\}$ at all large K, so $\delta_\infty(t) > 0$; this uses Thms. YM3_I:AJ and YM3_I:AD through (M5)(iii). *The case of BI.* If BI holds at (b, ℓ_0), then $\delta_\infty(\ell/\mathsf{b}) \le 0.13736\,\theta^{\ell/\ell_0}$ for large ℓ by Thm. AZ(1); conversely, $\delta_\infty(t_0) < 0.008625$ and $\ell/\mathsf{b} = t_0$ give $\delta_K(\ell) \le 0.008625$ for all large K, hence $\mathsf{e}(\ell/a_K) \le 0.069$: BI at the side 2ℓ.

If $D_{q,\infty}(t_0) < (q-1)^3 x_0/8$ for an admitted q, then $\delta_\infty(t_0) < x_0/8 \le 0.008625$, as $(q-1)^3\delta_\infty \le D_{q,\infty}$. In the second case $\delta_\infty < 0.008625$ on a whole interval $[t_2', Lt_2']$, and the uniform convergence on the compact set of labels $[\mathsf{b}, L\mathsf{b}]$ at one side gives, for every real β above a threshold, a cube $N \le C\beta$ with $\delta(N^3) \le 0.008625$: Thm. BA(1) a second time.

Step 13: Thm. BB(7). The conditions of the list (H3), each attached to its constant, are inequalities between numbers; the one that binds L alone, with its constant 4096, is $L > 48 \cdot 4096^2 e^{24} \approx 21\,331\,858\,733\,892\,671\,081.7\ldots$, and every odd integer above it is admitted. The list also forces floors on the other constants that enter $K_\star$ and η_K: $r \ge 2$, $\mathfrak{p}_0 \ge 13$ (so $\mathsf{p}_0 \ge 26.5$), $M \ge L^7$, $\gamma^{-2} \ge (2L)^{1602}$, $A_0 \ge 256 L^5 A_1$, and three constants: $\mathsf{c}_\mathrm{a} = \mathsf{C}_\mathrm{ex}$, the expansion constant preceding equation YM3_I:

(705), with $\mathsf{C}_{\rm ex} \geq 8e^8$; $\mathsf{c}_{\rm c} = \kappa > 0$, the constant of the list that enters $\lambda_{\rm out} = (1+4\beta_{\rm s})\kappa$ of YM3_I:(889); and $\mathsf{c}_{\rm b} = \lambda_1 = \lambda_{\rm out} + 2^3 c_{\rm w} + \mathsf{C}_{\rm ex} + \lambda_{\rm s}$ of YM3_I:(705), where $c_{\rm w} = c_{\rm v} + 3$, $c_{\rm v} = 2 + \mathsf{C}_{\rm ex} c_G (LM)^3 u$ with $c_G = \mathsf{C}_{\rm ex}(\gamma_{\rm r}^{-1} + c_{\rm r}^2) \geq 2\mathsf{C}_{\rm ex}$ and $(LM)^3 u = \mathsf{u}_{\rm S}^* = 3 + c_{\rm r} + \mathsf{u}_{\rm G} > 4$ by YM3_I:(889), so that $\mathsf{c}_{\rm b} > \mathsf{c}_{\rm c} + 64\,\mathsf{c}_{\rm a}^2 \geq \mathsf{c}_{\rm c} + 4096\,e^{16}$, where $4096\,e^{16} \approx 3.6397 \cdot 10^{10}$.

The threshold $K_\star$ is the least cutoff at which the conditions of (M4)(ii)–(iii) on the construction constants hold for both tori, with the thresholds $K_{\rm bf}$ of (M4)(ii) on both after the dilation by z, as in Thm. BB(1), together with the comparison of Prop. 17.44. At their floors these conditions give $(K-z)\log L > 9\,\mathsf{c}_{\rm b} > 36864\,e^{16} \approx 3.2757 \cdot 10^{11}$ for every $K \geq K_\star(\mathsf{b}, \ell)$. Indeed, at the dilated data $(L^z\mathsf{b}, L^z\ell)$, whose cutoff $K - z$ is written K in this sentence: the number $\kappa_{\rm b}$ of YM3_I:(2181) is $w_\star e^{-\lambda_V}/32M^3$ with $w_\star = w_0 e^{-4\lambda_1}/(\mathsf{C}_{\rm ex}(L+2)^3 S_{\rm s})$, $w_0 = (4S_{\rm s}L^3)^{-1}$ (equation YM3_I:(705)), so $\kappa_{\rm b}^{-1} > L^6 M^3 e^{4\mathsf{c}_{\rm b}}$; the condition $\epsilon_{\rm sh} \leq \frac{1}{20}$ of (k3) forces $\exp[\frac{1}{4}\kappa_{\rm b} A_1^2 \mathsf{y}_{[u]}] \geq 60$, so $A_1^2 \mathsf{y}_{[u]} > 16/\kappa_{\rm b}$ and $A_0^2 \mathsf{y}_{[u]} > L^{16} M^3 e^{4\mathsf{c}_{\rm b}}$; then $\epsilon_{\rm cl} \leq \frac{1}{20}$ forces $L^{(K-s)/2} \geq 80 A_0^2 \mathsf{y}_{[u]} > e^{4\mathsf{c}_{\rm b}}$, $s = \lceil K/9 \rceil$, and as $K - s \leq \frac{8}{9}K$, $K \log L > 9\,\mathsf{c}_{\rm b}$.

So the small cube of the criterion of (2) has $2L^{m+K-1} \geq 2L^{K-z+1}$ sites a side, as $m + z \geq 2$, hence more than $e^{9\mathsf{c}_{\rm b}} > 10^{1.4226 \cdot 10^{11}}$. Where the margin η_K is below the height of the criterion, as the margin condition of (3) requires of $\eta_{K_1}(\mathsf{b}, L^2\ell_0)$, the same conditions, with $\mathsf{C}_{\rm bf} \geq 64 \mathsf{K}_{\rm w} c_\rho^2 A_0^2 \mathsf{y}_{[1]} L^{7/2}$ from (2.xv) and $\mathsf{y}_{[1]}(K-z)^{2\mathsf{p}_0} \geq \mathsf{y}_{[u]}$, make $L^{(K-z)/10}$ larger than $e^{4\mathsf{c}_{\rm b}} \geq e^{4(\mathsf{c}_{\rm b} - \mathsf{c}_{\rm c})}$ times a power of L and of M, and $e^{4(\mathsf{c}_{\rm b} - \mathsf{c}_{\rm c})} > e^{256\,\mathsf{c}_{\rm a}^2} \geq e^{16384\,e^{16}}$; so $(K-z)\log L > 163840\,e^{16} \approx 1.4559 \cdot 10^{12}$.

At the cutoff of (3) the half side $l = \ell_0/2a_{K_1}$ of the cube of BI is then $> 10^{6.3228 \cdot 10^{11}}$, and the two boxes $(l, 2l, 2l)$ and $(2l)^3$, of $4l^3$ and $8l^3$ sites, have more than $10^{1.8968 \cdot 10^{12}}$ sites each. For the floor of η_K: $\varsigma \geq \varsigma_\flat(L)$ as $\mathsf{p}_0 \geq \frac{5}{2}$, $\mathsf{C}_{\rm bf} \geq 2$, and $L^z\ell = 2L^{m+z} \geq 2L^2$ as $m + z \geq 2$; so $\eta_K \geq 3\varsigma_\flat \cdot 2 \cdot 8L^6 (K-z)^{5/2} L^{-(K-z)/10}$. An inequality $X_{K_1} \leq y - \eta_{K_1}$ with $X_{K_1} \geq 0$, $y \leq y_0$, forces $\eta_{K_1} \leq \frac{L^3}{8}\,0.069$, hence $K_1 - z \geq K_\flat(L)$ for the least $K_\flat$ at which the floor drops below that number, and the small cube then has $2L^{m+K_1-1} \geq 2L^{1+K_\flat}$ sites a side; at every admitted L this exceeds 10^{152} (already at $L = 13$, $2L^{1+K_\flat} > 10^{152.9}$).

Step 14: Thm. BC(1)–(4). (1): $D_2(N) = 4\mathsf{d}^{(3)}(N,N,N) + 2\mathsf{d}^{(2)}(N,N,2N) + \mathsf{d}^{(1)}(N,2N,2N)$, three doubling defects ≥ 0 by Prop. 17.31(b), the last being $\mathsf{e}(N)$; $D_2 \leq 8\delta$ and $D_q - \frac{q^3}{8} D_2 = \frac{q^3}{8}\delta((2N)^3) - \delta((qN)^3) \geq 0$ by Prop. 17.31(c); $(q-1)^3 \delta(N^3) \leq D_q(N)$ is $\delta((qN)^3) \leq (3q^2 - 3q + 1)\delta(N^3)$, and the box and cube bounds are Prop. 17.31(c); $\mathsf{e}(l) \leq 2\delta(l, 2l, 2l) \leq 16(l/N)^2 \delta(N^3)$. For $l = Ll'$, $n = 2l/L$, with the time along a side of the cube $Ln = 2l$: $D_L(n) \geq LF(n, 2l, 2l) - F(2l, 2l, 2l) = L\mathsf{G}(n) - \mathsf{G}(2l) \geq L\mathsf{G}(l) - \mathsf{G}(2l) \geq \frac{L}{2}\mathsf{e}(l)$ over the cross-section $(2l, 2l)$, as $n \leq l$ and $\mathsf{e}(l) = 2\mathsf{G}(l) - \mathsf{G}(2l)$, the two other thermal defects of D_L being ≥ 0; this is $\mathsf{e}(l) \leq \frac{2}{L} D_L(2l/L)$.

The cube criterion and $D_L(2l) \leq \frac{20}{7} L^3 x^2$ are Step 11; $\mathsf{e} \leq 8\delta$; and $0.13736 \cdot 0.8901^{24} \leq 0.0084022$. Finally $F \leq 0$ and, by Jensen's inequality for the Haar measure, $F \geq -3\beta|\underline{n}|$, so $\mathsf{e}(l) \leq -F((2l)^3) \leq 24\beta l^3$. (2) is Prop. 17.32 with $x_0 = 8\delta(N^3)$, $\theta = 103.2\,\delta(N^3)$ (if $\delta(N^3) = 0$ the excess vanishes for all larger cubes), $\delta((2N)^3) \leq \frac{20}{7}\mathsf{e}(N)^2 \leq \frac{20}{7} \cdot 64\,\delta(N^3)^2$, and Prop. 17.32(e) for 0.07489. Two periodic Gibbs states exclude the first alternative by Prop. 17.38(a); by (M3) each is an integral of ergodic ones, two different ergodic ones are mutually singular, and Prop. 17.34(d) gives $2 \leq \liminf e^\delta$.

The last sentence is Prop. 17.34(d), the sides being chosen with δ tending to its lower limit and the laws converging, by the compactness of the laws on a compact space. (3): if $\mathsf{e}(l) > 0.069$ there is nothing to prove; otherwise let ν be the state of Prop. 17.38, f the bump of (M5)(iii) at radius $\mathsf{d}_\circ$ and height $2\mathsf{d}_\circ$, and $C(n)$ the covariance under ν of $\mathcal{O}_K(f \circ \Theta)$ and $\mathcal{O}_K(f(\cdot - na_K e_3))$, Θ the reflection $x_3 \mapsto -x_3$. For $0 \leq n \leq N_4$ the pair is a test pair at $(\mathsf{d}_\circ; \vartheta_1)$ and two points of the supports are between $2\mathsf{d}_\circ + na_K$ and $6\mathsf{d}_\circ + na_K$ apart; so (M5)(iii) gives $C(n) > 0$ and

$$\frac{C(0)}{C(N_4)} \leq \frac{4/3}{2/3}\Big(\frac{10\,\mathsf{d}_\circ}{2\,\mathsf{d}_\circ}\Big)^6 = 2 \cdot 5^6 = 31250.$$

By translation invariance $C(n)$ is the covariance of $\mathcal{O}_K(f)$ and its mirror image in the plane $x_3 = -\frac{n}{2}a_K$, so Prop. 17.38(f) with $A = B = \mathcal{O}_K(f)$, as in Step 5, gives $C(n) \le \theta^{n/l}C(0)$, $C \ge 0$, and C is log-convex, so all $C(n) > 0$ and the quotients $C(n+1)/C(n)$ do not decrease: $C(n) \ge C(N_4)\,31250^{-(n-N_4)/N_4}$. Taking n-th roots, $\theta^{1/l} \ge 31250^{-1/N_4}$, with $\theta = 12.9\,\mathsf{e}(l)$; and $\log 31250 < 10.35 \le 2.6\,N_4 a_K/\mathsf{d}_\circ$ as $a_K/\mathsf{d}_\circ \le \frac{1}{52}$. With $\mathsf{d}_\circ = \mathsf{d}_0$, $l = \ell/2a_K$; (1) carries the floor to D_2, 8δ, $\frac{8}{L^3}D_L$; and $\frac{1}{12.9}31250^{-l/N_4} > 0.069$ if and only if $l/N_4 < \log(1/(12.9\cdot 0.069))/\log 31250$, which $la_K < 0.0449\,\mathsf{d}_\circ$ gives, as $N_4 > 4\mathsf{d}_\circ/a_K - 1$ and $\mathsf{d}_\circ/a_K > 4L^3M$. (4): dilating by L^z, the label enters the period $(\gamma_W^{-2} + \bar{B}, \kappa_0]$, which $I(g_1)\cup I(g_2)$ covers by the form of the intervals $I(g)$ in (M5); there (3) at the side $L^z\ell_0$ gives $L^{-z}m_* \le 2.6/\mathsf{d}_0^\star$, and $L^z \le \kappa_0/\mathsf{b}$, so $m_* \le 2.6\kappa_0/(\mathsf{d}_0^\star\mathsf{b})$, which is the bound since $\log(1/\theta)/1.3 \ge 0.0895$.

On the cube $(2l)^3$, Prop. 17.31(d), (e) give $w_{00} = \frac{1}{4}(1+2r_1+r_2)$, $w_{10} = w_{01} = \frac{1}{4}(1-r_2) = u$, $w_{11} = \frac{1}{4}(1-2r_1+r_2) = v$; with $x = 1-r_1 = 2(u+v)$ the sum $\sum_f w_f^{1/2} = (1-u-\frac{x}{2})^{1/2} + 2u^{1/2} + (\frac{x}{2}-u)^{1/2}$ is concave in u on the range $\frac{x}{4} \le u \le \frac{x}{2}$ allowed by $r_2 \le r_1$ and $v \ge 0$, so least at an end of it, which is $u = \frac{x}{2}$ for $x \le \frac{1}{2}$; this gives $\mathsf{e}(l) \ge 2\log(\sqrt{1-x} + \sqrt{2x})$, which exceeds 0.069 for $x > 0.00062714$ and is at least $\log 2$ for $\frac{1}{9} \le x \le \frac{1}{2}$; for $x > \frac{1}{2}$ both end values of the sum are at least $\sqrt{2}$, so $\mathsf{e}(l) \ge \log 2$ there; and $\mathsf{e}(l) \ge \mathsf{G}_{2l} \ge \log\frac{1}{w_{00}} \ge \log\frac{4}{1+3r_1}$.

Then Prop. 17.31(d) for r_2, r_3, Prop. 17.32(b) for $\delta((2l)^3) \le \frac{20}{7}\cdot 0.069^2 \le 0.0136029$, and Prop. 17.31(b) for $\mathsf{r}_l(2l, 2l) \le \varphi(e^{0.069} - 1) \le 0.0357638$, $\log(1/0.0357638) \approx 3.33082$, $l_K a_K = \ell_0/2$.

Step 15: Thm. BC(5), (6). (5)(a) is Prop. 17.47(c) at $N = \ell/a_K$, $\beta = L^K\mathsf{b}$, where $N/\beta = \ell/\mathsf{b}$, with $|c'_N - c'| \le C_A N^{-2}(\log N)^{5/2}$; and the limit $K \to \infty$ by Thm. BB(6). (b) Prop. 17.48(a) and Prop. 17.31(d), (e): as $r_2 \le r_1$, w_{00} is the largest weight and $\frac{1}{4} \le w_{00} \le \frac{1}{4}(1+3r_1)$, $|4w_f - 1| \le 3r_1$; $\delta_K(\ell) \ge \mathsf{G}_N \ge \log\frac{1}{w_{00}} \ge \log\frac{4}{1+3r_1}$, and $\mathsf{e}(\ell/2a_K) \ge \mathsf{G}(\ell/2a_K; N, N) \ge \mathsf{G}_N$; the last assertion follows from (4): $\eta_\flat(\ell_0/\mathsf{b}) < 0.99937$ gives $1 - r_1 > 0.000628$ on the cube ℓ_0/a_K at every cutoff. (c) Completing the **(Heaviest class)** probability estimate would make the reduction in Prop. 17.48(b) give $\mathsf{G}_N = \log(Z/\lambda_0^N) \ge \log\frac{1}{t} - \mathsf{q}\log\log\frac{1}{t} - \log C$. (d) Choose $t_\circ > 0$ with $16Lt_\circ \le t_2$ and $\eta_\flat(t) \le 0.9961$ for $t \le 16Lt_\circ$, and $\beta_\flat = 1/8t_\circ$.

For $\beta \ge \beta_\flat$ there is $j \ge 1$ with $2L^j/\beta \in (16t_\circ, 16Lt_\circ]$; then $1 - r_1(2L^j) \ge 0.0039$ by (b), and Prop. 17.32(d), by contraposition, excludes $\mathsf{e}(l) \le 0.069$ for every $l \le 2L^j/16$, which includes $3 \le l \le t_\circ\beta$. For BI take K with $L^K\mathsf{b} \ge \beta_\flat$: $l_K = \ell_0 L^K/2 > t_\circ L^K\mathsf{b}$. The exact statement on one lattice is the contrapositive of Prop. 17.32(d). (e) is Prop. 17.47(d). (6): with $N = L^{-j}\ell/a_K$ at the same β, (a) at the side $L^{-j}\ell$, where $t_1/L < L^{-j}t \le t_1$, bounds $\delta(N^3)$ by $\log(L/t_1) + c' + C$, and Thm. BC(1) gives $\delta_K(\ell) \le 3L^{2j}\delta(N^3)$ with $L^j \le Lt/t_1$; then $\mathsf{e} \le 8\delta$ and $D_L \le L^3\delta$.

Remark 17.49 (Scope). The theorems do not assert BI at any label; by Thm. BC it fails at every cutoff on cubes small against the label, and it can hold only on cubes of side at least $0.0895\,(\mathsf{d}_0^\star/\kappa_0)\,\mathsf{b}$. Thm. BA(2) carries BI from one label to every other label at other sides and from other thresholds, with numbers β_0, C_+ that are not explicit. Uniqueness in Thm. AZ(1)(b) is uniqueness among the periodic Gibbs states; of Gibbs states without periods it says only that ν_K is extreme and trivial at infinity. The area law bounds the loops in the state ν_K at each cutoff. Thm. BA(5) concerns tuples carried by a cube that keeps away from the seam of the torus.

18. Further propositions

This section develops further results for the lattice construction and its continuum limits. The proofs give the additional arguments for the stated hypotheses, including the box inequality, the lattice gap and the uniform Poincaré bound where they are used. The effective-density and source-stability inputs are Theorems YM3_I:D and YM3_I:AD.

18.1. **Further lattice statements: the adjoint weight of $SU(2)$, plaquette weights that coincide with Wilson's at small plaquette action, and lattices of all dimensions.** This

subsection states, and sketches the proofs of, statements on one lattice that complement Section 9.1 in three directions. For $SO(3) = SU(2)/\{\pm\mathbf{1}\}$ in its defining representation, whose Wilson weight is the adjoint weight of $SU(2)$, it expands the partition function on a single periodic lattice with even sides into a sum, over sign fields on the plaquettes, of integrals with a weight of one well, and shows that at a cube where the cube number of Proposition 9.33 (i) is $-\mathbf{1}$, a $\mathbb{Z}_2$ monopole, the rotation angles of the six faces add up to at least 2π, a bound that some configuration attains, so that one of the faces turns by at least $\frac{\pi}{3}$ (Proposition 18.3); that on such a lattice r prescribed cubes are monopoles together with probability at most $(C_0\beta^{9/2}e^{-\beta/6})^r$ whenever $\beta \geq 1$, where $\beta := \frac{8}{3}\beta_A$, β_A being the coupling at which this Wilson weight is taken, and where C_0 does not depend on the lattice size, so that on the tori of the main text the expected number of monopoles per unit volume vanishes in the limit of infinite cutoff at a fixed label, whereas no bound of this shape can hold uniformly as $\beta \to 0$ (Proposition 18.4); and that, conditioned on the absence of monopoles, the adjoint measure is a mixture of eight measures of one well, indexed by the fluxes through the three coordinate planes, so that replacing the adjoint measure by this mixture changes the joint cumulants of the plaquette observables by an explicit amount that vanishes in the limit (Proposition 18.5). For a probability measure on the configurations over $a\mathbb{Z}^d$, $d \geq 2$, with values in a compact metrisable group, which is invariant under the translations of $a\mathbb{Z}^d$ and need not be a Gibbs state, it derives $(1-T)^2 \geq T(2 - V(ae_\lambda) - V(ae_\lambda)^*)$ from reflection positivity at two coordinate mirrors and at one diagonal mirror (Proposition 18.6), and, for $d \geq 3$ and positivity at the coordinate mirrors and at one diagonal mirror across each coordinate plane, the decay of the correlations of arbitrary bounded Borel functions of finitely many bonds like $(ra/|x|)^{d-2}$, with constants determined by d only; if the group contains an element of order two, no larger exponent is possible (Proposition 18.7). For the Wilson weight of any closed subgroup of $U(N)$, at any $\beta \geq 0$ and in any dimension $d \geq 2$, all periodic Gibbs states are reflection positive at all lattice mirrors; in four dimensions their correlations therefore decay like $(ra/|x|)^2$, the three-dimensional rate $ra/|x|$ squared (Proposition 18.8). For $SU(2)$ on one torus of the main text, the probability of $a_p \geq \delta$ is bounded for a whole class of plaquette weights, so that two weights of the class that agree below a cut of the plaquette action give measures whose distance these bounds control (Proposition 18.9); a weight that equals Wilson's weight $e^{-\beta s}$ at plaquette actions $s < s_0$, for a fixed $s_0 > 0$, and does not exceed $(2+\beta)^{C_v}e^{-c_0\beta s}$ at $s \geq s_0$, for fixed $c_0 \in (0,1]$ and $C_v \geq 0$, has, at an unchanged label, the limits of Wilson's action, at Wilson's rate, as the cutoff runs through all integers and along arbitrary sequences of bare couplings, and its block fields have, in total variation, the limiting laws of Wilson's action (Proposition 18.10). Proposition 18.1 contains the facts on the planes through the sites that these arguments share.

Planes through the sites. The first proposition is stated for $SU(2)$ and for every bounded plaquette weight that is a Borel function of the trace; its clauses (a) and (b) serve the adjoint weight and the class of weights of Proposition 18.9, its clause (c) every comparison of two measures below.

Proposition 18.1 (Planes through the sites; cumulants under a change of measure)**.** Let $G = SU(2)$ with its Haar probability measure, let $\kappa\colon [-1,1] \to [0,\infty)$ be a bounded Borel function, and put $k(g) := \kappa(\frac{1}{2}\operatorname{tr} g)$ for $g \in G$; since the trace on $SU(2)$ is real and is unchanged by conjugation and by inversion, $k(hgh^{-1}) = k(g) = k(g^{-1})$ for all $g, h \in G$. Let $\Lambda = \prod_{\mu\leq 3}\mathbb{Z}/N_\mu$ with even $N_\mu \geq 2$, and let $|\Lambda|$ be its number of sites. Bonds and plaquettes are the labels (x,μ) and $(x;\mu\nu)$, $\mu < \nu$: the bond (x,μ) runs from x to $x+e_\mu$, a configuration U takes the value $U(x,\mu) \in G$ on it and $U(x,\mu)^{-1}$ on its reverse, and $\partial U(x;\mu\nu) = U(x,\mu)U(x+e_\mu,\nu)U(x+e_\nu,\mu)^{-1}U(x,\nu)^{-1}$. The cube c_x has the sites $x+\{0,1\}^3$, the twelve bonds (y,μ) with $y \in x+\{0,1\}^3$ and $y_\mu = x_\mu$, and the six faces $(x;\mu\nu)$ and $(x+e_\lambda;\mu\nu)$, $\{\lambda,\mu,\nu\} = \{1,2,3\}$. For a bounded Borel function F of the configuration put $\langle F\rangle := \int F \prod_p k(\partial U(p))\, dU$, and suppose $\langle 1\rangle > 0$.

(a) Let μ be a direction, $t \in \mathbb{Z}/N_\mu$, and θ the reflection in the plane $x_\mu = t$: on sites $x_\mu \mapsto 2t - x_\mu$; on bonds $(x,\nu) \mapsto (\theta x,\nu)$ for $\nu \neq \mu$, while (x,μ) goes to the reverse of $(\theta x - e_\mu,\mu)$; on

configurations $(\theta^*U)(b) = U(\theta b)$; and $(\theta F)(U) = F(\theta^*U)$. Let B_+ consist of the bonds (x,μ) with $x_\mu \in \{t, \dots, t + \frac{1}{2}N_\mu - 1\}$ and the bonds (x,ν), $\nu \neq \mu$, with $x_\mu \in \{t, \dots, t + \frac{1}{2}N_\mu\}$. Then $\langle \theta F\rangle = \langle F\rangle$ for every F, and $\langle \overline{\theta F}\, F\rangle \geq 0$ whenever F depends on the values of the configuration on B_+ alone.

(b) For $x \in \Lambda$ let ρ_x map the sites of c_x onto the model cube $\{0,1\}^3$ by $(\rho_x y)_\mu = y_\mu - x_\mu$ for even x_μ and $(\rho_x y)_\mu = x_\mu + 1 - y_\mu$ for odd x_μ; for a function F of the configurations of the model cube let $F^{(x)}(U)$ be the value of F at the configuration that takes on the bond of the model cube from y_0 to y_0' the value of U on the bond of c_x from $\rho_x^{-1}y_0$ to $\rho_x^{-1}y_0'$. Then, whenever each F_x, $x \in \Lambda$, is a non-negative bounded Borel function on the configurations of the model cube,

$$\Big\langle \prod_x F_x^{(x)} \Big\rangle \leq \prod_x \Big\langle \prod_{x'} F_x^{(x')} \Big\rangle^{1/|\Lambda|}.$$

(c) Let two probability measures μ_1, μ_2 on a common measurable space satisfy $|\mu_1(E) - \mu_2(E)| \leq \varepsilon$ at every event E, and let $X_1, \dots, X_n$ be real measurable functions bounded by $|X_i| \leq \mathsf{s}_i$. Then the joint cumulant of $X_1, \dots, X_n$ under μ_1 lies within $2\,\mathsf{fu}_n\,\varepsilon \prod_i \mathsf{s}_i$ of that under μ_2, where $\mathsf{fu}_n := \sum_\pi |\pi|!$, the sum running over the partitions π of $\{1, \dots, n\}$ and $|\pi|$ being the number of blocks; $\mathsf{fu}_1, \dots, \mathsf{fu}_5$ are $1, 3, 13, 75, 541$.

■

Sketch of proof. Step 1: clause (a). The map θ is an involution of the labels that respects the incidences of sites, bonds, plaquettes and cubes and fixes the two planes $x_\mu = t$ and $x_\mu = t + \frac{1}{2}N_\mu$ pointwise. It permutes the factors of the product Haar measure and inverts some of them, so dU is preserved; and $\partial(\theta^*U)(p)$ is $\partial U(\theta p)$ read from another corner or in the other sense, so that $k(\partial(\theta^*U)(p)) = k(\partial U(\theta p))$ by the two invariances of k. Hence $\langle \theta F\rangle = \langle F\rangle$. Put $\mathsf{B}_- := \theta\mathsf{B}_+$ and $\mathsf{B}_0 := \mathsf{B}_+ \cap \mathsf{B}_-$; the latter consists of the bonds of the directions $\nu \neq \mu$ in the two fixed planes, each mapped to itself with its orientation, and $\mathsf{B}_+ \cup \mathsf{B}_-$ is the set of all bonds. Each plaquette has its four bonds in B_+ or in B_-, and in both precisely when it lies in a fixed plane. Accordingly $\prod_p k(\partial U(p)) = W_0\, W_+\, \theta W_+$, where $W_0 \geq 0$, the product over the plaquettes of the fixed planes, is a function of $U_0 := U|_{\mathsf{B}_0}$, and W_+ is the product over the remaining plaquettes with bonds in B_+. Splitting $U = (U_0, U_1, U_2)$ with U_1 on $\mathsf{B}_+ \setminus \mathsf{B}_0$ and U_2 on $\mathsf{B}_- \setminus \mathsf{B}_0$, and using that $U_2 \mapsto (\theta^*U)|_{\mathsf{B}_+ \setminus \mathsf{B}_0}$ carries dU_2 to the Haar measure of U_1, Fubini's theorem gives

$$\langle \overline{\theta F}\, F\rangle = \int dU_0\, W_0(U_0) \Big| \int (F W_+)(U_0, U_1)\, dU_1 \Big|^2 \geq 0.$$

Only the two invariances of k enter, so the weight is subject to no condition at a site mirror; Proposition 9.27 (i) notes the same of every real even class weight.

Step 2: a lemma on families indexed by a cycle. Let $m \geq 1$, X a set and $\Phi\colon \mathsf{X}^{\mathbb{Z}/2m} \to [0,\infty)$ with $\mathsf{N}(e) := \Phi(e, \dots, e)^{1/2m} > 0$ for every $e \in \mathsf{X}$. For $t \in \mathbb{Z}/2m$ let $R_t^+ g$ be the family that agrees with g at the $j \in \{t, \dots, t + m - 1\}$ and takes the value g_j at $2t - 1 - j$, and $R_t^- g$ the family that agrees with g at the $j \in \{t - m, \dots, t - 1\}$ and takes the value g_j at $2t - 1 - j$. If $\Phi(g)^2 \leq \Phi(R_t^+ g)\, \Phi(R_t^- g)$ for all t and g, then $\Phi(g) \leq \prod_j \mathsf{N}(g_j)$. Indeed $Q(g) := \Phi(g) / \prod_j \mathsf{N}(g_j)$ satisfies the same inequality, as $\prod_j \mathsf{N}((R_t^+ g)_j)\, \mathsf{N}((R_t^- g)_j) = \prod_j \mathsf{N}(g_j)^2$. Among the finitely many families whose entries are taken from those of a given g, choose one, g^*, at which Q is maximal and which, among the maximisers, has a run of equal neighbouring entries of the greatest length r, beginning at t. If $r < 2m$, then $Q(g^*)^2 \leq Q(R_t^+ g^*)\, Q(R_t^- g^*) \leq Q(R_t^+ g^*)\, Q(g^*)$ shows that $R_t^+ g^*$ is a maximiser too, while its run has the length $\min\{2r, 2m\} > r$; hence g^* is constant and $Q \leq 1$. The chessboard estimate of Proposition YM3_I:86.3 rests on the same principle, applied there at walls between the sites.

Step 3: clause (b). Assume first that all $F_x \geq \epsilon > 0$, and put $\Phi((F_x)_x) := \langle \prod_x F_x^{(x)} \rangle > 0$. Fix μ, put $m = \frac{1}{2}N_\mu$, and regard the family as $(g_j)_{j \in \mathbb{Z}/N_\mu}$ with $g_j := (F_x)_{x_\mu = j}$. The reflection in $x_\mu = t$ maps c_x onto $\mathsf{c}_{x'}$, where $x'_\mu = 2t - 1 - x_\mu$ and the other coordinates are kept, and $\rho_{x'} \circ \theta = \rho_x$ on

c_x for either parity of x_μ, whence $\theta F^{(x)} = F^{(x')}$; moreover the cubes with $x_\mu \in \{t, \dots, t+m-1\}$ have all their bonds in B_+. Thus $\prod_x F_x^{(x)} = A \cdot \theta A'$ with real functions A, A' of the configuration on B_+; by (a), $(F, F') \mapsto \langle \theta F \cdot F' \rangle$ is a symmetric positive form on such real functions, and Schwarz's inequality yields $\Phi(g)^2 \le \langle \theta A \cdot A \rangle \langle \theta A' \cdot A' \rangle = \Phi(R_t^+ g)\, \Phi(R_t^- g)$. Applying Step 2 successively in the directions 1, 2, 3 gives (b) for these F_x, and letting $\epsilon \to 0$, both sides being continuous in ϵ, gives it for $F_x \ge 0$.

Step 4: clause (c). The joint cumulant equals $\sum_\pi (-1)^{|\pi|-1}(|\pi|-1)! \prod_{B\in\pi} \mu(\prod_{i\in B} X_i)$ by expanding the logarithm of the moment generating function in squarefree coefficients. For bounded real Y, $|\mu_1(Y) - \mu_2(Y)| \le (\sup Y - \inf Y)\, \varepsilon \le 2\varepsilon \sup |Y|$. In the term of π each of the $|\pi|$ factors is bounded by $\prod_{i\in B} \mathsf{s}_i$ and changes by at most $2\varepsilon \prod_{i\in B} \mathsf{s}_i$, so the product changes by at most $2|\pi|\, \varepsilon \prod_i \mathsf{s}_i$; the weights $(|\pi|-1)!\, |\pi|$ add up to fu_n. □

The adjoint weight of $SU(2)$ on a single periodic lattice. In Propositions 18.3 to 18.5, $G = SO(3) \subset U(3)$ is the image of $SU(2)$ under the adjoint representation Ad; it is the quotient $\tilde{G}/\mathsf{Z}$ of Proposition 9.33 with $\tilde{G} = SU(2)$ and $\mathsf{Z} = \{\pm \mathbf{1}\}$, and for the pair $SO(3)$ on $\mathbb{C}^3$ of Definition 9.1 the action density is $a(X) = 1 - \frac{1}{3}\mathrm{tr}\, X$. For $u \in SU(2)$ let $w(u) := \frac{1}{2}\mathrm{tr}\, u \in [-1, 1]$ and let $\alpha(u) \in [0, \pi]$ be the angle with $\cos \alpha(u) = w(u)$, so that u has the eigenvalues $e^{\pm i\alpha(u)}$; then $\mathrm{tr}\, \mathrm{Ad}(u) = 4w(u)^2 - 1$, $a(\mathrm{Ad}(u)) = \frac{4}{3}(1 - w(u)^2)$, and $\mathrm{Ad}(u)$ is the rotation by the angle $2\min\{\alpha(u), \pi - \alpha(u)\} \in [0, \pi]$, which belongs to G^π if and only if $w(u) \ne 0$. The lattice $\Lambda = \prod_{\mu\le 3} \mathbb{Z}/N_\mu$ has even sides $N_\mu \ge 2$, as in Proposition 18.1, and its bonds, plaquettes and cubes carry the labels of that proposition; we write Bd for its set of bonds, which has $3|\Lambda|$ members (x, μ), and Pl for its set of plaquettes $(x; \mu\nu)$, which has $3|\Lambda|$ members as well; there are $|\Lambda|$ cubes, each plaquette borders two cubes and each bond belongs to four plaquettes. A configuration U of $SO(3)$ is $\mathrm{Ad} \circ \tilde{U}$ for a configuration $\tilde{U}$ of $SU(2)$, its lift, determined by U up to a sign field $\gamma \in \{\pm 1\}^{\mathrm{Bd}}$ acting by $(\gamma\tilde{U})(b) = \gamma_b \tilde{U}(b)$; the map $\tilde{U} \mapsto \mathrm{Ad} \circ \tilde{U}$ carries the Haar measure $d\tilde{U}$ to the Haar measure dU, and composition with it makes the bounded Borel functions of U correspond to the bounded Borel functions of $\tilde{U}$ that are *centre-blind*, $F(\gamma\tilde{U}) = F(\tilde{U})$ for every γ. Write $w_p := w(\partial\tilde{U}(p))$, $\alpha_p := \alpha(\partial\tilde{U}(p))$ and $\tilde{\alpha}_p := \min\{\alpha_p, \pi - \alpha_p\}$, so that $\vartheta_p := 2\tilde{\alpha}_p$ is the rotation angle of $\partial U(p) = \mathrm{Ad}(\partial\tilde{U}(p))$, and $(\delta\gamma)_p := \prod_{b\in p} \gamma_b$, so that $w_p(\gamma\tilde{U}) = (\delta\gamma)_p w_p(\tilde{U})$. Let $\beta_A > 0$ be the coupling of the Wilson weight $\prod_p e^{-\beta_A a_p}$ of the pair, and $\beta := \frac{8}{3}\beta_A$; then $\beta_A a_p = \frac{\beta}{2}(1 - w_p^2) = \frac{\beta}{2}\sin^2 \alpha_p$, whose quadratic part at $\alpha_p = 0$, $\frac{\beta}{2}\alpha_p^2$, is that of the exponent $\beta(1 - w_p)$ of the Wilson weight of $SU(2)$ at the same β. Let $W^A(\tilde{U}) := \prod_p e^{-\frac{\beta}{2}(1-w_p^2)}$, $Z^A := \int W^A\, d\tilde{U} = \int \prod_p e^{-\beta_A a_p}\, dU$ and $\mu^A := (Z^A)^{-1} W^A\, d\tilde{U}$, whose image under $\tilde{U} \mapsto \mathrm{Ad} \circ \tilde{U}$ is the Wilson measure of $SO(3)$ at β_A; the *one-well weight* is $\mathsf{h}_\beta(s) := \mathbf{1}\{s > 0\}\, e^{-\frac{\beta}{2}(1-s^2)}$, $s \in [-1, 1]$.

Lemma 18.2 (Site-mirror dissemination for closed blocks)**.** Let every period of a rectangular torus be an even multiple of an integer $R \ge 1$, and use the closed blocks of side R with boundaries on site planes. For a bounded nonnegative function F of the bonds of one block, its normalized mean is at most the normalized mean of the product of its alternating reflected copies, raised to the reciprocal of the number of blocks. ■

Proof. In one direction let there be $2k$ slabs. For a function F_j in slab j, integrate over the bonds off its two end slices against the plaquette weights, using half the exponent for the plaquettes in an end slice. This gives an integral kernel A_j on the end-slice variables. The integral of the product is $\mathrm{Tr}(A_1 \cdots A_{2k})$. Reflection reverses the order of the two slice variables, so the partition integral for the alternating reflected copies of one slab function is $\mathrm{Tr}[(A_j A_j^*)^k]$. The kernels are bounded on probability spaces, hence Hilbert–Schmidt. Schatten Hölder bounds the product trace by $\prod_j (\mathrm{Tr}[(A_j A_j^*)^k])^{1/(2k)}$. The term for an unmarked slab is the partition function raised to $1/(2k)$. Divide by the partition function and apply this estimate successively in the three directions. The plaquette weights that lie on a wall are counted exactly once because the adjacent slab kernels each include half of their exponent. □

Proposition 18.3 (The adjoint weight: sign fields and the cube number)**.** In this setting let $\mathcal{N} := \bigcup_p \{w_p = 0\}$, let $\eta_p(\tilde U) := \operatorname{sign} w_p(\tilde U)$ off $\mathcal{N}$, and for a cube c with the set $\partial\mathsf{c}$ of its six faces let E_c be the event that $w_p \neq 0$ at each of the six faces $p \in \partial\mathsf{c}$ and $\prod_{p\in\partial\mathsf{c}} \operatorname{sign} w_p = -1$.

(a) $\mathcal{N}$ is a null set, and off $\mathcal{N}$

$$W^A(\tilde U) = \sum_{\eta\in\{\pm1\}^{\mathrm{Pl}}} \prod_p \mathsf{h}_\beta(\eta_p w_p(\tilde U)),$$

with $\eta = \eta(\tilde U)$ as the single non-vanishing term; hence $Z^A = \sum_\eta \int \prod_p \mathsf{h}_\beta(\eta_p w_p)\, d\tilde U$.

(b) Off $\mathcal{N}$ every plaquette variable $\partial U(p)$ of $U = \mathrm{Ad}\circ\tilde U$ lies in G^π, the central part of Proposition 9.33 is $z_{\tilde U}(p) = \eta_p(\tilde U)\,\mathbf{1}$, and the cube number of U at c is $\prod_{p\in\partial\mathsf{c}} \eta_p(\tilde U)\,\mathbf{1}$. So, off $\mathcal{N}$, E_c is the event that the cube number at c equals $-\mathbf{1}$, a $\mathbb{Z}_2$ monopole at c; it is an open centre-blind event determined by the configuration on c.

(c) If $\tilde U \notin \mathcal{N}$ and c is a monopole, then $\sum_{p\in\partial\mathsf{c}} \vartheta_p \geq 2\pi$; consequently one of its faces has $\vartheta_p \geq \frac{\pi}{3}$, which means $1 - w_p^2 \geq \frac14$ and $a_p = 1 - \frac13 \operatorname{tr} \partial U(p) \geq \frac13$; and $\sum_{p\in\partial\mathsf{c}}(1-w_p^2) \geq \frac23$. Put otherwise, if all six faces of a cube have rotation angles below $\frac{\pi}{3}$, its cube number is $\mathbf{1}$: for $SO(3)$ the threshold $\frac{\pi}{6}$ of the eigen-angles in Proposition 9.33 (i) may be raised to $\frac{\pi}{3}$. Some configuration of the twelve bonds of c makes c a monopole at which $\sum_{p\in\partial\mathsf{c}} \vartheta_p = 2\pi$; so the bound on the sum of the six angles is attained.

■

Sketch of proof. Step 1: clause (a). Freeze all bonds but one bond b of a plaquette p: then $\partial\tilde U(p)$ or its inverse is a conjugate of $\tilde U(b)g$ for a frozen g, hence Haar distributed. Under the Haar measure of $SU(2)$ the angle α has the law $\frac{2}{\pi}\sin^2\alpha\, d\alpha$ on $[0,\pi]$, by Lemma 9.14 (i) applied to the pair $SU(2)$ on $\mathbb{C}^2$, where $j_{SU(2)}(Y) = (\frac{\sin|Y|}{|Y|})^2$ and $\mathrm{vol}(SU(2)) = 2\pi^2$, as recorded after Definition 9.1; so $\{w_p = 0\} = \{\alpha_p = \frac{\pi}{2}\}$ is null, and by Fubini's theorem so is $\mathcal{N}$. For $s \neq 0$ the sum $\mathsf{h}_\beta(s) + \mathsf{h}_\beta(-s)$ has a single non-zero term, equal to $e^{-\frac{\beta}{2}(1-s^2)}$; so off $\mathcal{N}$, $W^A = \prod_p(\mathsf{h}_\beta(w_p) + \mathsf{h}_\beta(-w_p)) = \sum_\eta \prod_p \mathsf{h}_\beta(\eta_p w_p)$, and a term can be non-zero only if $\eta_p w_p > 0$ for all p, that is, only for $\eta = \eta(\tilde U)$.

Step 2: clause (b). Let $g = e^{iY} \in G^\pi$ with $Y \in \mathfrak{g}$ and $|Y| = \vartheta < \pi$, the rotation angle of g. The element $\tilde Y$ of the Lie algebra of $SU(2)$ that the covering sends to Y has the eigenvalues $\pm\frac{\vartheta}{2}$, since $\operatorname{ad}\tilde Y$, which the covering identifies with Y, has the eigenvalues $0, \pm\vartheta$; therefore the canonical lift $s(g) = e^{i\tilde Y}$ of Proposition 9.33 is the lift of g with $w(s(g)) = \cos\frac{\vartheta}{2} > 0$. As g has the two lifts $\pm s(g)$, $s(\mathrm{Ad}(u)) = \operatorname{sign}(w(u))\, u$ whenever $w(u) \neq 0$; hence $z_{\tilde U}(p) = \partial\tilde U(p)\, s(\partial U(p))^{-1} = \eta_p \mathbf{1}$, and the cube number, the product of the $z_{\tilde U}(p)^{\pm1}$ over the six faces, is $\prod_{p\in\partial\mathsf{c}} \eta_p\, \mathbf{1}$, the signs being central and their own inverses. Each of the twelve bonds of c borders exactly two faces of c, so a sign field multiplies the product by a square and E_c is centre-blind; and E_c is the union of the open sets $\{\epsilon_p w_p > 0 \text{ for } p \in \partial\mathsf{c}\}$ over the sign patterns (ϵ_p) with $\prod_p \epsilon_p = -1$.

Step 3: the identity of the six faces. Label the sites of c by $(i,j,k) \in \{0,1\}^3$ and let t be the tree of seven bonds of c consisting of the four bonds of the direction 3, the two bonds of the direction 2 at $k = 0$ and the bond of the direction 1 at $j = k = 0$. A gauge transformation of the eight sites, $\mathbf{1}$ at $(0,0,0)$ and propagated along t, makes $\tilde U$ trivial on t; it conjugates every face variable and so changes no w_p, α_p or η_p. Call A_i the new value on the bond from $(i,0,1)$ to $(i,1,1)$, $i = 0, 1$, and B, C, D the new values on the bonds of the direction 1 that start at $(0,1,0)$, $(0,0,1)$ and $(0,1,1)$. Taken from a suitable corner and in a suitable sense, the faces $x_1 = 0$, $x_1 = 1$, $x_3 = 0$, $x_2 = 0$, $x_2 = 1$ and $x_3 = 1$ carry

$$h_1 = A_0, \quad h_2 = A_1, \quad h_3 = B, \quad h_4 = C, \quad h_5 = BD^{-1}, \quad h_6 = CA_1D^{-1}A_0^{-1},$$

and eliminating D yields $h_6 = h_4 h_2 h_3^{-1} h_5 h_1^{-1}$: the lattice form of the identity of Bianchi that underlies the cube number of Proposition 9.33 (i).

Step 4: clause (c). Let $\tilde U \notin \mathcal{N}$, $\eta_i := \operatorname{sign} w(h_i)$ and $v_i := \eta_i h_i$; then $w(v_i) = |w(h_i)| > 0$, $\alpha(v_i) = \tilde\alpha_i < \frac{\pi}{2}$, and the identity of Step 3 turns into $v_6^{-1}v_4v_2v_3^{-1}v_5v_1^{-1} = \prod_i \eta_i\,\mathbf{1}$, which equals $-\mathbf{1}$ at a monopole. Viewing $SU(2)$ as the unit quaternions in $\mathbb{R}^4$, $w(u^{-1}u')$ is the Euclidean inner product of u and u', so $\alpha(u^{-1}u')$ is their spherical distance, which left and right multiplications preserve; hence $\alpha(u_1\cdots u_6) \le \sum_i \alpha(u_i)$ and $\alpha(u^{-1}) = \alpha(u)$, and at a monopole $\pi = \alpha(-\mathbf{1}) \le \sum_i \tilde\alpha_i$, that is, $\sum_p \vartheta_p \ge 2\pi$. Then some $\tilde\alpha_i \ge \frac{\pi}{6}$, where $1 - w^2 = \sin^2\tilde\alpha_i \ge \frac14$ and $a_p = \frac43(1-w_p^2) \ge \frac13$; and $\sin x \ge \frac{2}{\pi}x$ on $[0,\frac{\pi}{2}]$ with the inequality of Cauchy and Schwarz gives $\sum_i \sin^2\tilde\alpha_i \ge \frac{4}{\pi^2}\sum_i \tilde\alpha_i^2 \ge \frac{4}{\pi^2}\cdot\frac16\big(\sum_i\tilde\alpha_i\big)^2 \ge \frac23$. To attain the bound on the sum take $u := \operatorname{diag}(e^{i\pi/6}, e^{-i\pi/6})$, $\mathbf{1}$ on t, $A_0 = B = u^{-1}$, $A_1 = C = u$ and $D = u^{-2}$: then $h_1 = h_3 = u^{-1}$, $h_2 = h_4 = h_5 = u$ and $h_6 = u^5$, so $w(h_i) = \cos\frac{\pi}{6} > 0$ for $i \le 5$ and $w(h_6) = \cos\frac{5\pi}{6} < 0$; the cube is a monopole, $\tilde\alpha_i = \frac{\pi}{6}$ at each face, and $\sum_p \vartheta_p = 2\pi$. □

Proposition 18.4 (The rarity of $\mathbb{Z}_2$ monopoles)**.** In the setting of Proposition 18.3, with its even sides, put $C_0 := (\frac{3\pi^3}{8})^3 e^{24}$.

(a) For $\beta \ge 1$ and any $r \ge 1$ distinct cubes $\mathsf{c}_1, \dots, \mathsf{c}_r$,

$$\mu^A(\mathsf{E}_{\mathsf{c}_1} \cap \cdots \cap \mathsf{E}_{\mathsf{c}_r}) \le \big(C_0\,\beta^{9/2}e^{-\beta/6}\big)^r,$$

and $(Z^A)^{1/|\Lambda|} \ge C_0^{-1}\beta^{-9/2}$; neither C_0 nor the exponents $\frac92$ and $\frac16$ involve the sides. In the coupling of the pair, $\beta^{9/2}e^{-\beta/6} = (\frac83)^{9/2}\beta_A^{9/2}e^{-4\beta_A/9}$. At every $\beta > 0$, $\mu^A(\mathsf{E}_{\mathsf{c}}) \le C_0 \max\{1,\beta\}^{9/2}e^{-\beta/6}$.

(b) For a given lattice, $\mu^A(\mathsf{E}_{\mathsf{c}}) \to \int \mathbf{1}_{\mathsf{E}_{\mathsf{c}}}\,d\tilde U > 0$ as $\beta \to 0$; hence no choice of C, c and $q > 0$ makes $\mu^A(\mathsf{E}_{\mathsf{c}}) \le C\beta^q e^{-c\beta}$ valid at all $\beta > 0$.

(c) On the lattice $\Lambda_{K,\ell}$ of the main text, $\ell = 2L^m$, which has $\ell/a_K = 2L^{m+K}$ sites on an axis, at $\beta = L^K\mathsf{b} \ge 1$, $\mathsf{b} > 0$, the expected number of monopoles, and with it the probability that at least one cube carries an odd number of faces on which $\operatorname{tr}\partial\tilde U(p)$ is negative, is at most

$$(\ell/a_K)^3\,C_0\,\beta^{9/2}\,e^{-\mathsf{b}/(6a_K)},$$

and the expected number per unit volume of T_ℓ is at most $a_K^{-3}C_0\beta^{9/2}e^{-\mathsf{b}/(6a_K)}$; as $K \to \infty$ both vanish, uniformly on compact sets of labels in $(0,\infty)$, the first also uniformly on bounded sets of sides.

■

Sketch of proof. Step 1: a lower bound for Z^A. Let $\beta \ge 1$ and $\varrho := \beta^{-1/2} \le 1$. If $\alpha(\tilde U(b)) \le \varrho$ at every bond, the subadditivity of the spherical distance (Step 4 of the proof of Proposition 18.3) gives $\alpha_p \le 4\varrho$, so $1 - w_p^2 = \sin^2\alpha_p \le 16\varrho^2$ and $W^A \ge e^{-\frac{\beta}{2}\cdot 16\varrho^2\cdot 3|\Lambda|} = e^{-24|\Lambda|}$; since α has the density $\frac{2}{\pi}\sin^2\alpha$ and $\sin\alpha \ge \frac{2}{\pi}\alpha$ on $[0,\frac{\pi}{2}]$, the Haar measure of $\{\alpha(u) \le \varrho\}$ is at least $\frac{8}{3\pi^3}\varrho^3$. Therefore $(Z^A)^{1/|\Lambda|} \ge (\frac{8}{3\pi^3})^3 e^{-24}\beta^{-9/2} = C_0^{-1}\beta^{-9/2}$.

Step 2: all cubes at once. Let F_{M} be the function on the configurations of the model cube that is 1 when w vanishes at none of its six faces and the signs of w at these faces multiply to -1, and 0 otherwise. The maps ρ_x of Proposition 18.1 (b) send faces to faces, and the value of w at a face does not depend on the corner or the sense in which the face is read; so $F_{\mathsf{M}}^{(x)} = \mathbf{1}_{\mathsf{E}_{\mathsf{c}_x}}$. On the event that all cubes are monopoles, Proposition 18.3 (c) at each cube, together with the two cubes bordering each plaquette, gives $2\sum_p(1-w_p^2) \ge \frac23|\Lambda|$, hence $W^A \le e^{-\beta|\Lambda|/6}$; thus $\langle\prod_x F_{\mathsf{M}}^{(x)}\rangle \le e^{-\beta|\Lambda|/6}$, where the bracket of Proposition 18.1 is formed with $\kappa(s) := e^{-\frac{\beta}{2}(1-s^2)}$, that is, with $k(u) = e^{-\frac{\beta}{2}(1-w(u)^2)}$, so that $\langle 1\rangle = Z^A$.

Step 3: clause (a). Proposition 18.1 (b) with $F_x := F_{\mathsf{M}}$ at the r cubes and $F_x := 1$ at all others yields $Z^A\mu^A(\bigcap_i \mathsf{E}_{\mathsf{c}_i}) \le e^{-\beta r/6}(Z^A)^{1-r/|\Lambda|}$, that is, $\mu^A(\bigcap_i \mathsf{E}_{\mathsf{c}_i}) \le (e^{-\beta/6}(Z^A)^{-1/|\Lambda|})^r \le (C_0\beta^{9/2}e^{-\beta/6})^r$ by Step 1. For $\beta < 1$ simply $\mu^A(\mathsf{E}_{\mathsf{c}}) \le 1 \le C_0e^{-1/6}$. The formula in β_A is the substitution $\beta = \frac83\beta_A$.

Step 4: clause (b). Since $e^{-\frac{3}{2}|\Lambda|\beta} \le W^A \le 1$, the quotient W^A/Z^A stays between $e^{-\frac{3}{2}|\Lambda|\beta}$ and $e^{\frac{3}{2}|\Lambda|\beta}$, so $|\mu^A(E) - \int \mathbf{1}_E \, d\tilde{U}| \le e^{\frac{3}{2}|\Lambda|\beta} - 1$ for every event E, which tends to 0 with β. The set $\mathsf{E_c}$ is open (Proposition 18.3 (b)) and contains the configuration of Proposition 18.3 (c), so it has positive Haar measure, whereas $C\beta^q e^{-c\beta} \to 0$.

Step 5: clause (c). The sides $2L^{m+K}$ of $\Lambda_{K,\ell}$ are even and $|\Lambda| = (\ell/a_K)^3$. By (a) the expected number of monopoles, $\sum_{\mathsf{c}} \mu^A(\mathsf{E_c})$, is at most $|\Lambda| \, C_0 \beta^{9/2} e^{-\beta/6}$, and by Markov's inequality it dominates the probability of at least one monopole; moreover $\beta/6 = \mathsf{b}/(6a_K)$, and off the null set $\mathcal{N}$ a cube is a monopole exactly when its faces with negative trace are odd in number. Division by the volume ℓ^3 of T_ℓ gives the density, and $L^{3K}(L^K\mathsf{b})^{9/2} e^{-L^K\mathsf{b}/6} \to 0$ uniformly on compact sets of labels. □

Proposition 18.5 (Sectors of the adjoint measure)**.** In the setting of Proposition 18.3, with its even sides $N_\mu \ge 2$, call a sign field $\eta \in \{\pm 1\}^{\mathrm{Pl}}$ closed if $\prod_{p \in \partial \mathsf{c}} \eta_p = 1$ at every cube, let $\mathcal{Z}^2$ be the set of closed sign fields and $\mathcal{B}^2 := \{\delta\gamma\}$. For $\{\lambda,\mu,\nu\} = \{1,2,3\}$, $\mu < \nu$, let the column C_λ be the set of the plaquettes $(x;\mu\nu)$ with $x_\mu = x_\nu = 0$, and let $\Phi_{\lambda,t}(\eta) := \prod_{x:\, x_\lambda = t} \eta_{(x;\mu\nu)}$ be the flux of η through the plane $x_\lambda = t$; for $n \in \{0,1\}^3$ let $\sigma^n_p := -1$ if $p \in \mathsf{C}_\lambda$ for a λ with $n_\lambda = 1$, and $\sigma^n_p := 1$ otherwise. For bounded Borel F put $Z^+[n;F] := \int F \prod_p \mathsf{h}_\beta(\sigma^n_p w_p)\, d\tilde{U}$ and $Z^+[n] := Z^+[n;1]$, and let $\mu^+_{\mathrm{mix}} := \sum_n Z^+[n;\,\cdot\,]/\sum_n Z^+[n]$, the mixture of the eight one-well weights $Z^+[n;\,\cdot\,]$.

(a) $|\mathcal{Z}^2| = 2^{2|\Lambda|+1}$ and $|\mathcal{B}^2| = 2^{2|\Lambda|-2}$. On closed fields the flux through $x_\lambda = t$ is the same for all t; call it $\Phi_\lambda(\eta)$. The map $\eta \mapsto (\Phi_\lambda(\eta))_{\lambda \le 3}$ from $\mathcal{Z}^2$ to $\{\pm1\}^3$ is a surjective homomorphism whose kernel is exactly $\mathcal{B}^2$, and $\Phi_\lambda(\sigma^n) = (-1)^{n_\lambda}$; hence each closed field lies in exactly one of the eight cosets $\sigma^n\mathcal{B}^2$.

(b) Let F be bounded, Borel and centre-blind. Then

$$\int F\, \mathbf{1}\{\text{no cube is a monopole}\}\, W^A \, d\tilde{U} = 2^{2|\Lambda|-2} \sum_{n \in \{0,1\}^3} Z^+[n;F],$$

and replacing σ^n in $Z^+[n;F]$ by another closed sign field with the same three fluxes leaves $Z^+[n;F]$ unchanged. Off $\mathcal{N}$ and where no cube is a monopole, let the sector $\mathsf{n}(\tilde{U}) \in \{0,1\}^3$ be given by $(-1)^{\mathsf{n}_\lambda} = \Phi_\lambda(\eta(\tilde{U}))$; it is a centre-blind function, and under μ^A the event that no cube is a monopole and $\mathsf{n} = n$ has the probability $2^{2|\Lambda|-2} Z^+[n]/Z^A$.

(c) Let $\varepsilon = \mu^A(\text{some cube is a monopole})$. For centre-blind F,

$$\mu^+_{\mathrm{mix}}(F) = \mu^A(F \mid \text{no monopole}), \qquad |\mu^A(F) - \mu^+_{\mathrm{mix}}(F)| \le 2\varepsilon \|F\|_\infty.$$

Centre-blind variables $X_1, \dots, X_r$ bounded by M have joint cumulants under μ^A and under μ^+_{mix} within $2\,\mathsf{fu}_r\, \mathsf{M}^r \varepsilon$ of each other. The mixture gives the class n the weight $\mu^A(\mathsf{n} = n \mid \text{no monopole}) = Z^+[n]/\sum_{n'} Z^+[n']$.

(d) On $\Lambda_{K,\ell}$ at $\beta = L^K\mathsf{b} \ge 1$, for real bounded $f_1, \dots, f_r$ on T_ℓ and the observables $\mathcal{O}^A(f_i) := a_K^{-1} \sum_p f_i(x(p)) \, \frac{3}{8} a_p = a_K^{-1} \sum_p f_i(x(p)) \, \frac{1}{2}(1 - w_p^2)$, $x(p)$ the barycentre of p, the joint cumulants of $\mathcal{O}^A(f_1), \dots, \mathcal{O}^A(f_r)$ under μ^A and under μ^+_{mix} differ by at most

$$2\,\mathsf{fu}_r \Big(\tfrac{3}{2} \ell^3 L^{4K} \max_i \|f_i\|_\infty \Big)^r \ell^3 L^{3K} \, C_0 \, (L^K\mathsf{b})^{9/2} \, e^{-L^K\mathsf{b}/6},$$

a quantity that tends to 0 as $K \to \infty$ with no condition on the supports of the f_i, uniformly on compact sets of labels and on bounded sets of sides.

■

Sketch of proof. Step 1: the complex of sign fields. Sign fields on the sites, bonds, plaquettes and cubes, written multiplicatively, are vector spaces $\mathcal{C}^0, \dots, \mathcal{C}^3$ over the field of two elements, of the dimensions $|\Lambda|$, $3|\Lambda|$, $3|\Lambda|$ and $|\Lambda|$, linked by $(\delta_0 g)_{(x,\mu)} := g_x g_{x+e_\mu}$, $\delta_1 := \delta$ and $(\delta_2\eta)_{\mathsf{c}} := \prod_{p \in \partial \mathsf{c}} \eta_p$. One has $\delta_1\delta_0 = 1$, and $\delta_2\delta_1 = 1$ since each edge of a cube borders exactly two of its faces; so $\mathcal{B}^2 \subset \mathcal{Z}^2 = \ker \delta_2$. The sign field equal to -1 at a single plaquette is sent by δ_2 to the field equal

to -1 at the two cubes bordering it; the cubes form a connected graph when cubes sharing a face are joined, and $\prod_{\mathsf{c}}(\delta_2\eta)_{\mathsf{c}} = \prod_p \eta_p^2 = 1$. Hence the range of δ_2 is the set of cube fields with an even number of entries -1, its rank is $|\Lambda|-1$, and $\dim \mathcal{Z}^2 = 2|\Lambda|+1$.

Step 2: the fluxes. Multiply the conditions $(\delta_2\eta)_{\mathsf{c}} = 1$ of a closed η over the layer of cubes c_x with $x_\lambda = t$: the faces containing the direction λ are each shared by two cubes of the layer, while every face lying in one of the planes $x_\lambda = t$ and $x_\lambda = t+1$, distinct since $N_\lambda \geq 2$, belongs to one cube of the layer; so $\Phi_{\lambda,t}(\eta) = \Phi_{\lambda,t+1}(\eta)$. Each bond of the plane $x_\lambda = t$ is shared by exactly two plaquettes of that plane, so $\Phi_\lambda(\delta\gamma) = 1$; every cube meets the column C_λ in none or two of its faces, so σ^n is closed; and the plane $x_\lambda = t$ contains one plaquette of C_λ and no plaquette of the two other columns, so $\Phi_\lambda(\sigma^n) = (-1)^{n_\lambda}$. Therefore Φ is onto $\{\pm1\}^3$, $\mathcal{B}^2 \subset \ker\Phi$, and $\dim\ker\Phi = 2|\Lambda| - 2$. For $\eta = \eta(\tilde U)$, $\Phi_{\lambda,t}$ is the product of the central parts over the plane $x_\lambda = t$, which is the plane number of Proposition 9.33 (i), and $\Phi(\eta(\tilde U))$ is the sector of Lemma 9.11 (iv), here available at every configuration without monopoles.

Step 3: $\mathcal{B}^2 = \ker\Phi$. With coordinates in $\{0,\dots,N_\mu - 1\}$ let t consist of the bonds $(x,1)$ with $x_1 \leq N_1 - 2$, the bonds $(x,2)$ with $x_1 = 0$ and $x_2 \leq N_2 - 2$, and the bonds $(x,3)$ with $x_1 = x_2 = 0$ and $x_3 \leq N_3 - 2$, a spanning tree with $|\Lambda| - 1$ bonds. Any $\gamma \in \ker\delta_1$ is the product of some $\delta_0 g$, g constructed along t, with an element of $\ker\delta_1$ that is 1 on t; and $\dim\operatorname{im}\delta_0 = |\Lambda| - 1$ as the lattice is connected. An element γ of $\ker\delta_1$ that equals 1 on t is determined by three signs: the plaquettes $(x;1\nu)$ with $x_1 \leq N_1 - 2$ force γ on the bonds of the directions 2 and 3 to be independent of x_1; the plaquettes $(x;23)$ with $x_2 \leq N_2 - 2$ then force γ on the bonds of the direction 3 to depend on x_3 only, hence to be 1 except at $x_3 = N_3 - 1$, where it takes one value ϵ_3; the plaquettes $(x;23)$ with $x_2 = N_2 - 1$ force γ to take one value ϵ_2 on the bonds $(x,2)$ with $x_2 = N_2 - 1$, it being 1 on the other bonds of the direction 2; and the plaquettes $(x;1\nu)$ with $x_1 = N_1 - 1$ force one value ϵ_1 on the bonds $(x,1)$ with $x_1 = N_1 - 1$. So $\dim\ker\delta_1 \leq |\Lambda| + 2$, $\dim\mathcal{B}^2 = 3|\Lambda| - \dim\ker\delta_1 \geq 2|\Lambda| - 2 = \dim\ker\Phi$, and $\mathcal{B}^2 = \ker\Phi$, which is (a).

Step 4: clause (b). By Proposition 18.3 (a), off $\mathcal{N}$ the product $\mathbf{1}\{\text{no monopole}\}\,W^A$ equals $\sum_{\eta\in\mathcal{Z}^2}\prod_p \mathsf{h}_\beta(\eta_p w_p)$, the only possibly non-zero term being that of $\eta(\tilde U)$, and $\eta(\tilde U)$ is closed exactly when there is no monopole. By (a), each $\eta \in \mathcal{Z}^2$ has the form $\sigma^n\delta\gamma$ for a unique n, and each coset has $2^{2|\Lambda|-2}$ elements. For $\eta = \sigma^n\delta\gamma$ the change of variables $\tilde U = \gamma\tilde U'$ preserves $d\tilde U$ and the centre-blind F and turns $\eta_p w_p(\tilde U)$ into $\sigma_p^n w_p(\tilde U')$; hence $\int F\prod_p \mathsf{h}_\beta(\eta_p w_p)\,d\tilde U = Z^+[n;F]$, and the sum over η gives the identity. The same change of variables shows that only the coset of σ^n matters. Since $\Phi_\lambda(\eta(\gamma\tilde U)) = \Phi_\lambda(\eta(\tilde U))\Phi_\lambda(\delta\gamma) = \Phi_\lambda(\eta(\tilde U))$, the sector is centre-blind; where $\prod_p \mathsf{h}_\beta(\sigma_p^{n'} w_p) \neq 0$ one has $\eta(\tilde U) = \sigma^{n'}$ and $\mathsf{n} = n'$; and the identity applied to $F = \mathbf{1}\{\mathsf{n} = n\}$, extended by 0 at monopoles and on $\mathcal{N}$, gives the probability of the sector.

Step 5: clause (c). Let A be the event without monopoles. Applying (b) to F and to 1 gives $\mu^A(F\mathbf{1}_{\mathsf{A}})/\mu^A(\mathsf{A}) = \mu^+_{\mathrm{mix}}(F)$. From $\mu^A(F) - \mu^A(F \mid \mathsf{A}) = \mu^A(F\mathbf{1}_{\mathsf{A}^c}) - \mu^A(F\mathbf{1}_{\mathsf{A}})\,\mu^A(\mathsf{A}^c)/\mu^A(\mathsf{A})$ the difference is at most $2\varepsilon\|F\|_\infty$, and similarly $|\mu^A(E) - \mu^A(E \mid \mathsf{A})| \leq \varepsilon$ at every event E; so Proposition 18.1 (c), on the centre-blind functions, bounds the change of the cumulants. The weights of the classes follow from (b) at $F = \mathbf{1}\{\mathsf{n} = n\}$.

Step 6: clause (d). Each $\mathcal{O}^A(f)$ is centre-blind, and $|\mathcal{O}^A(f)| \leq a_K^{-1}\cdot 3|\Lambda|\cdot\frac12\|f\|_\infty = \frac32\ell^3L^{4K}\|f\|_\infty$ because $|\Lambda| = \ell^3L^{3K}$; inserting $\varepsilon \leq \ell^3L^{3K}C_0(L^K\mathsf{b})^{9/2}e^{-L^K\mathsf{b}/6}$ from Proposition 18.4 (c) into (c) gives the display, whose exponential factor beats every power of L^K uniformly on compact sets of labels. For the joint cumulants of the plaquette observables the Wilson measure of $SO(3)$ may thus be traded for the mixture of the eight one-well weights, at this rate. □

Lattices of every dimension. In Propositions 18.6 to 18.8 let $d \geq 2$, $a > 0$, and let G be a compact metrisable group. Let $\mathcal{V} = \mathcal{V}(a\mathbb{Z}^d)$ be the product space of G-valued configurations on nearest-neighbour bonds. A reversed bond carries the inverse of its forward value. An isometry r acts by $(\mathsf{r}U)_b = U_{\mathsf{r}^{-1}b}$ and $(\mathsf{r}F)(U) = F(\mathsf{r}^{-1}U)$. In particular τ_x is translation by $x \in a\mathbb{Z}^d$. For

$S \subset \mathbb{R}^d$, $\mathsf{Bd}(S)$ consists of the bonds contained in S as closed segments. Write $\mathcal{F}(B)$ for the bounded complex Borel functions depending only on bonds in the finite set B, and $\mathcal{F}_{\mathrm{loc}}(S)$ for their union over finite $B \subset \mathsf{Bd}(S)$.

A mirror $\Sigma(n,c)$ is the plane $n \cdot x = c$ with reflection $\theta_{n,c}$ and closed sides $\mathbb{H}^{\pm}(n,c) = \{x : \pm(n \cdot x - c) \geq 0\}$. The normals are $\pm e_\mu$ with $c \in (a/2)\mathbb{Z}$, and $\pm(e_\mu + s e_\lambda)$ with $s = \pm 1$, $\mu \neq \lambda$ and $c \in a\mathbb{Z}$. For a coordinate normal the mirror is a site mirror when $c \in a\mathbb{Z}$, and a bond mirror otherwise. There are d coordinate and $d(d-1)$ diagonal families. Reflection positivity at a mirror means $\nu(\overline{\theta_{n,c}F}F) \geq 0$ for $F \in \mathcal{F}_{\mathrm{loc}}(\mathbb{H}^+(n,c))$.

For an inward normal lattice vector v assume invariance under translations and the mirror reflection, and reflection positivity both at $\Sigma(n,c)$ and at $\Sigma(n, c+n\cdot v/2)$. The picture (n,c,v) is the Hilbert space obtained by completing $\mathcal{F}_{\mathrm{loc}}(\mathbb{H}^+(n,c))$ for the inner product $\langle [F],[G]\rangle = \nu(\overline{\theta_{n,c}F}G)$ after quotienting the null space. Reflection invariance gives $\|[F]\|^2 \leq \nu(|F|^2)$. On local functions the normal translation is symmetric for this form and reflection Cauchy–Schwarz gives $\|[\tau_v F]\|^2 \leq \|[F]\|\|[\tau_{2v}F]\|$. Iterating this inequality and using $\|[\tau_{2^k v}F]\| \leq \|F\|_\infty$ shows $\|[\tau_v F]\| \leq \|[F]\|$, including when $[F]=0$. Thus $T[F] = [\tau_v F]$ is a self-adjoint contraction. Reflection positivity at the half-way mirror applied to $\tau_v F$ gives $\langle [F], T[F]\rangle \geq 0$. Tangential lattice translations w preserve the form, and their inverses are also tangential. They give commuting unitaries $V(w)[F] = [\tau_w F]$ that commute with T. In a coordinate picture $(e_\mu, c, a e_\mu)$ the joint spectral theorem gives the projection-valued measure $\mathfrak{E}$ on $[0,1]\times(S^1)^{d-1}$ with

$$T^k V(ay) = \int t^k \zeta^y \, d\mathfrak{E}(t,\zeta) \qquad (k \geq 0,\ y \in \mathbb{Z}^d,\ y_\mu = 0).$$

The diagonal picture uses $v = an$. The same construction in dimension three is Lemma YM3_I: 99.1. W_d is the group of signed permutations of the coordinates, $\theta_\mu := \theta_{e_\mu,0}$, S_{2^d} the group of the 2^d sign changes, generated by the θ_μ, $\mathcal{I}$ the σ-field of the translation invariant Borel sets, and $Q_r := [-ra, ra]^d$ for an integer $r \geq 1$. In the site picture $(e_\mu, 0)$ the cube vector of $A \in \mathcal{F}(\mathsf{Bd}(Q_r))$ is $\psi^\mu_A := [\tau_{rae_\mu} A]$; its reflection size is $\mathcal{N}_r(A) := \max_\mu \max_{\mathsf{g}\in\mathsf{S}_{2^d}} \|\psi^\mu_{\mathsf{g}A}\|^2$, and $\mathcal{N}'_{r,\mu}(A) := \max\{\|\psi^\mu_A\|^2, \max_{\lambda\neq\mu}\|\psi^\lambda_{\theta_\lambda A}\|\,\|\psi^\lambda_A\|\}$, as in YM3_I:(1959). A Borel probability measure ν on $\mathcal{V}$ is said to have (Mir) if it is invariant under the translations of $a\mathbb{Z}^d$, reflection positive at $\Sigma(e_\mu,0)$ and at $\Sigma(e_\mu, \frac{a}{2})$ for $\mu = 1,\dots,d$, and, for every pair $\mu<\lambda$, reflection positive at some mirror $\Sigma(e_\mu + s e_\lambda, c)$ with $s=\pm1$ and $c \in a\mathbb{Z}$.

Proposition 18.6 (A spectral inequality from one diagonal mirror, in dimension d)**.** Fix two different directions μ,λ and a sign $\epsilon \in \{1,-1\}$, and let ν be a Borel probability measure on $\mathcal{V}(a\mathbb{Z}^d)$, Gibbs state or not, which does not change under the translations of $a\mathbb{Z}^d$ and is reflection positive at $\Sigma(e_\mu,0)$, at $\Sigma(e_\mu,\frac{a}{2})$ and at the diagonal mirror $\Sigma(e_\mu + \epsilon e_\lambda, 0)$.

(a) The reflection positivity of ν extends to all mirrors $\Sigma(e_\mu, c)$, $c \in \frac{a}{2}\mathbb{Z}$, and $\Sigma(e_\mu + \epsilon e_\lambda, c)$, $c \in a\mathbb{Z}$, and ν is invariant under their reflections; in particular ν has the picture (e_μ, c_0) for each $c_0 \in \frac{a}{2}\mathbb{Z}$.

(b) In each picture (e_μ, c_0),

$$(1-T)^2 \ \geq\ T\big(2 - V(ae_\lambda) - V(ae_\lambda)^*\big),$$

which is the same as $\mathfrak{E}(\{(t,\zeta) : (1-t)^2 < t\,|1-\zeta_\lambda|^2\}) = 0$; moreover $\mathfrak{E}$ lives on $\{t \leq e^{-\mathsf{k}|\arg\zeta_\lambda|}\}$, $\mathsf{k} = \frac{2}{\pi}\log(1+\sqrt{2})$.

(c) If ν has (Mir), its symmetry group contains $a\mathbb{Z}^d \rtimes W_d$ and it is reflection positive at all lattice mirrors of $a\mathbb{Z}^d$, for the functions of the bonds of either closed side; in each picture (e_μ, c_0) the inequality of (b) holds for all $d-1$ directions $\lambda \neq \mu$, and $\mathfrak{E}$ lives on

$$\mathcal{C}_\mu := \{(t,\zeta) : (1-t)^2 \geq t\,|1-\zeta_\lambda|^2 \text{ for all } \lambda \neq \mu\};$$

writing a point of $\mathcal{C}_\mu$ with $t > 0$ as $t = e^{-x}$ and $\zeta_\lambda = e^{i\alpha_\lambda}$, $\alpha_\lambda \in (-\pi, \pi]$, one has $x \geq \mathsf{k} \max_{\lambda \neq \mu} |\alpha_\lambda|$.

■

Sketch of proof. Step 1: invariance from positivity. (i) Reflection positivity at $\Sigma(n, c)$ and at $\Sigma(-n, -c)$ are the same condition: for F on $\mathbb{H}^-(n, c)$, $G := \theta_{n,c} F$ is a function on $\mathbb{H}^+(n, c)$, and $\nu(\overline{\theta F} F) = \overline{\nu(\overline{\theta G} G)}$. (ii) If ν is reflection positive at a lattice mirror and invariant under an isometry r, then it is reflection positive at the image of the mirror under r. (iii) At a site or diagonal mirror, $n \cdot x$ moves by 0 or $\pm a$ along a bond and lies in $c + a\mathbb{Z}$ at sites, so each bond is contained in one of the two closed sides; since a positive form is Hermitian, $\nu(\theta F \cdot F') = \nu(\theta F' \cdot F)$ for F, F' on $\mathbb{H}^+$, that is, $\nu(\theta\Phi) = \nu(\Phi)$ for $\Phi = F' \cdot \theta F$; the indicator of a set given by conditions $U_b \in E_b$ at finitely many bonds b, with Borel sets $E_b \subset G$, is of this form: the conditions at the bonds contained in $\mathbb{H}^+$ give F', and those at the other bonds, which are contained in $\mathbb{H}^- = \theta_{n,c}\mathbb{H}^+$, give θF; these sets form a family stable under intersections that generates the Borel σ-field, as the coordinates generate it, so ν and its image under $\theta_{n,c}$, which agree on this family, agree on all Borel sets, and ν is invariant under $\theta_{n,c}$. Applying (ii) to the translations τ_{kae_μ} moves $\Sigma(e_\mu, 0)$ and $\Sigma(e_\mu, \frac{a}{2})$ onto all $\Sigma(e_\mu, c)$, $c \in \frac{a}{2}\mathbb{Z}$, and $\Sigma(e_\mu + \epsilon e_\lambda, 0)$ onto all $\Sigma(e_\mu + \epsilon e_\lambda, c)$, $c \in a\mathbb{Z}$, since $(e_\mu + \epsilon e_\lambda) \cdot kae_\mu = ka$; the reflection in a bond mirror is a site reflection composed with a translation. The half-way mirror of the picture $(e_\mu, c_0) = (e_\mu, c_0, ae_\mu)$ is $\Sigma(e_\mu, c_0 + \frac{a}{2})$. This gives (a).

Step 2: two pictures. By the construction of the pictures, the transfer operator of the picture (e_μ, c_0) satisfies $0 \leq T \leq 1$, the tangential unitaries $V(v')$, $v' \perp e_\mu$, commute with T, and $\langle [F], T^k V(v')[F'] \rangle = \nu(\overline{\theta_{e_\mu,c_0} F}\, \tau_{kae_\mu + v'} F')$. The same is true in the picture (n, c, an) of the diagonal normal $n := e_\mu + \epsilon e_\lambda$, $c \in a\mathbb{Z}$, whose half-way mirror is $\Sigma(n, c + a)$; call its transfer operator $T_\parallel$, and $V_\perp := V(an^\perp)$ with $n^\perp := e_\mu - \epsilon e_\lambda$.

Step 3: a lemma on the closed disc. Suppose η is a finite positive Borel measure on $\overline{\mathbb{D}}$ and $C \geq 0$ is such that $\int |\mathfrak{q}|^2 d\eta \leq C \sup_{[0,1]} |\mathfrak{q}| \sup_{S^1} |\mathfrak{q}|$ for all polynomials $\mathfrak{q}$. Then η gives no mass to $\{z : |1+z|^2 < 4|z|\}$ (Lemma YM3_I:99.2). To see this, write $\operatorname{Log} \frac{1+\zeta}{1-\zeta} = x + i\phi$ with $|\phi| < \frac{\pi}{2}$ for $\zeta \in \mathbb{D}$, an odd function of ζ. For $s > 0$ the function $\cos(s(x + i\phi))$ is even in ζ, so $\cos(s(x + i\phi)) = g_s(\zeta^2)$ for a function g_s holomorphic in the disc, and $|g_s|^2 = \cos^2(sx) + \sinh^2(s\phi)$ is at least $\sinh^2(s\phi)$ and at most $\cosh^2 \frac{s\pi}{2}$, while $|g_s| \leq 1$ on $[0, 1)$, where ζ is real. In the variable $z = \zeta^2$, $\tan|\phi| = \sqrt{2(|z| - \operatorname{Re} z)}/(1 - |z|)$, so $|\phi| > \frac{\pi}{4}$ if and only if $|1 + z|^2 < 4|z|$. The assumed inequality extends to $z \mapsto g_s(\varrho z)$, $0 < \varrho < 1$, which are uniform limits of polynomials on $\overline{\mathbb{D}}$, and Fatou's lemma as $\varrho \to 1$ yields $\int \sinh^2(s\Phi)\, d\eta \leq C \cosh \frac{s\pi}{2}$, with Φ the function $|\phi|$ of z, set equal to $\frac{\pi}{2}$ on $S^1 \setminus \{1\}$ and to 0 at 1, so that $\liminf_{\varrho \to 1} |g_s(\varrho z)|^2 \geq \sinh^2(s\Phi(z))$ at every $z \in \overline{\mathbb{D}}$, Φ being continuous on $\overline{\mathbb{D}} \setminus \{1\}$. Hence $\eta(\{\Phi \geq \frac{\pi}{4} + \delta\}) \leq C \cosh(\frac{s\pi}{2}) / \sinh^2(s(\frac{\pi}{4} + \delta))$, which vanishes as $s \to \infty$, for each $\delta > 0$; as $\delta \downarrow 0$ these sets exhaust $\{z : |1 + z|^2 < 4|z|\}$, which contains $S^1 \setminus \{1\}$, where $\Phi = \frac{\pi}{2}$, but not the point 1.

Step 4: the inequality. Let $F \in \mathcal{F}_{\mathrm{loc}}(\mathbb{H}^+(e_\mu, c_0))$, $\psi := [F]$, $\varrho_\psi := \langle \psi, \mathfrak{E}(\cdot)\psi \rangle$, and let ς be the image of ϱ_ψ under $(t, \zeta) \mapsto z := t\zeta_\lambda^\epsilon \in \overline{\mathbb{D}}$. For a polynomial $\mathfrak{q} = \sum_k \mathfrak{q}_k z^k$ and an integer $N \geq 0$, as $(k + l)e_\mu + \epsilon(k - l)e_\lambda = kn + ln^\perp$,

$$\int |\mathfrak{q}|^2 |z|^{2N}\, d\varsigma = \sum_{k,l} \mathfrak{q}_k \bar{\mathfrak{q}}_l\, \nu\big(\overline{\theta_{e_\mu,c_0} F}\, \tau_{kan + lan^\perp} \tau_{2Nae_\mu} F\big).$$

Choose $c \in a\mathbb{Z}$ with $c \geq n \cdot x$ at the end points of the bonds on which $\theta_{e_\mu,c_0} F$ depends, and N with $n \cdot x + 2Na \geq c$ at those of the bonds on which F depends; then $F_1 := \theta_{n,c}\theta_{e_\mu,c_0} F$ and $F_2 := \tau_{2Nae_\mu} F$ are functions of the bonds of $\mathbb{H}^+(n, c)$, $\theta_{n,c} F_1 = \theta_{e_\mu,c_0} F$, and in the picture (n, c, an) the sum equals

$\langle [F_1], \mathfrak{q}(T_\parallel)\bar{\mathfrak{q}}(V_\perp)[F_2]\rangle$, $\bar{\mathfrak{q}}(z) := \sum_l \bar{\mathfrak{q}}_l z^l$. As $0 \le T_\parallel \le 1$, $V_\perp$ is unitary and $\|[F_i]\| \le \|F\|_\infty$,

$$\int |\mathfrak{q}|^2 |z|^{2N}\, d\varsigma \le \|F\|_\infty^2 \sup_{[0,1]} |\mathfrak{q}| \sup_{S^1} |\mathfrak{q}|.$$

Step 3 applied to $|z|^{2N}\varsigma$ gives $\varsigma(\{|1+z|^2 < 4|z|\}) = 0$, the point 0 not lying in this set. At $z = t\zeta_\lambda^\epsilon$, $|1+z|^2 - 4|z| = (1-t)^2 - t|1-\zeta_\lambda|^2$; so $\mathfrak{E}(\{(1-t)^2 < t|1-\zeta_\lambda|^2\})[F] = 0$ for all F, the classes $[F]$ are dense, and the spectral form of the inequality follows; it is equivalent to $(1-T)^2 - T(2 - V(ae_\lambda) - V(ae_\lambda)^*) \ge 0$. For $d = 3$ this is Proposition YM3_I:99.3, whose argument involves two directions only.

Step 5: the chord. Put $t = e^{-x}$ and $\zeta_\lambda = e^{i\alpha_\lambda}$; then $(1-t)^2 - t|1-\zeta_\lambda|^2 = 4t(\sinh^2\frac{x}{2} - \sin^2\frac{\alpha_\lambda}{2})$. The function $\alpha \mapsto 2\,\mathrm{arsinh}(\sin\frac{\alpha}{2})$ is zero at 0 and concave on $[0,\pi]$, since its derivative $\cos\frac{\alpha}{2}\,(1 + \sin^2\frac{\alpha}{2})^{-1/2}$ decreases; a concave function vanishing at 0 dominates its secant through the point at π, here $\alpha \mapsto \frac{\alpha}{\pi}\,2\,\mathrm{arsinh}\,1 = \mathsf{k}\alpha$. So $x \ge \mathsf{k}|\alpha_\lambda|$ wherever $\mathfrak{E}$ lives.

Step 6: clause (c). Under (Mir), Step 1 gives positivity at all coordinate mirrors and invariance under the θ_μ, which generate S_{2^d}. In each plane (μ,λ) translations move the given diagonal mirror $\Sigma(e_\mu + se_\lambda, c)$ to all offsets in $a\mathbb{Z}$; the reflection $\theta_{e_\mu + se_\lambda, 0}$, which leaves ν invariant, sends e_μ to $-se_\lambda$ and e_λ to $-se_\mu$, and composed with sign changes it exchanges the two coordinates; transpositions and sign changes generate W_d. Since θ_λ maps $\Sigma(e_\mu + se_\lambda, c)$ onto $\Sigma(e_\mu - se_\lambda, c)$, (ii) yields the other diagonal family of the plane, and (i) both closed sides. Now (b) applies to every ordered pair of directions, and finitely many null sets have a null union. □

Proposition 18.7 (Decay of correlations like $(ra/|x|)^{d-2}$, and optimality of the exponent)**.** Let $d \ge 3$, and let ν be a Borel probability measure on $\mathcal{V}(a\mathbb{Z}^d)$ with (Mir); ν need not be a Gibbs state. Use the function Ψ constructed in Step 2 below, $\mathsf{c}_\Psi = (1+\sqrt{2})^{-1}$ and the chord constant k of Proposition 18.6. Put $D := d-1$,

$$\mathsf{s}_d := \int_0^\infty (5+9s)^D e^{-s}\, ds = \sum_{i=0}^{D} \binom{D}{i} 5^{D-i} 9^i\, i!, \qquad \mathsf{C}'_d := \frac{8\,D\,(D-2)!\,\mathsf{s}_d}{\mathsf{c}_\Psi\,(\pi\mathsf{k})^D}, \qquad C_d := 2D\,\mathsf{C}'_d\,(2\sqrt{d})^{d-2}, \qquad c_d := 4\sqrt{d},$$

numbers determined by d. Let $r \ge 1$ be an integer and $A, B \in \mathcal{F}(\mathsf{Bd}(Q_r))$, bounded Borel functions of finitely many bonds of Q_r.

(a) In the site picture of e_μ let Π_0 denote the projection on $\ker(1-T)$. Then $\Pi_0 = \mathfrak{E}(\{(1,1,\dots,1)\})$, $T^n \to \Pi_0$ strongly, $V(v')\Pi_0 = \Pi_0$, the vectors of $L^2(\nu)$ fixed by τ_{ae_μ} are those fixed by all of $a\mathbb{Z}^d$, $\langle \psi^\mu_{\theta_\mu A}, \Pi_0 \psi^\mu_B\rangle = \nu(\overline{E_\nu[A|\mathcal{I}]}\,E_\nu[B|\mathcal{I}])$, and, for every integer $n \ge 1$,

$$0 \le \langle \psi^\mu_A, T^n(1-\Pi_0)\psi^\mu_A\rangle \le 4\,\mathsf{C}'_d\,\mathcal{N}_r(A)\Big[(D-1)\Big(\frac{r}{n}\Big)^D + \frac{1}{2}\Big(\frac{r}{n}\Big)^{D-1}\Big].$$

(b) Let $x \in a\mathbb{Z}^d$ and $n := |x|_\infty/a - 2r \ge 1$. Then

$$\Big|\nu(\bar{A}\,\tau_x B) - \nu(\overline{E_\nu[A|\mathcal{I}]}\,E_\nu[B|\mathcal{I}])\Big| \le 4\,\mathsf{C}'_d\,\mathcal{N}_r(A)^{1/2}\mathcal{N}_r(B)^{1/2}\Big[(D-1)\Big(\frac{r}{n}\Big)^D + \frac{1}{2}\Big(\frac{r}{n}\Big)^{D-1}\Big].$$

(c) If $x \in a\mathbb{Z}^d$ and $|x| \ge c_d\, ra$, then

$$\Big|\nu(\bar{A}\,\tau_x B) - \nu(\overline{E_\nu[A|\mathcal{I}]}\,E_\nu[B|\mathcal{I}])\Big| \le C_d\,\mathcal{N}_r(A)^{1/2}\mathcal{N}_r(B)^{1/2}\Big(\frac{ra}{|x|}\Big)^{d-2} \le C_d\,\|A\|_\infty\|B\|_\infty\Big(\frac{ra}{|x|}\Big)^{d-2},$$

and for ergodic ν the subtracted term equals $\nu(\bar{A})\nu(B)$. The numbers c_d and C_d are independent of G, a, ν, r, A, B and x; $C_3 < 4.8\cdot 10^4$ and $C_4 < 7.8\cdot 10^6$.

(d) If some $g_0 \ne e$ in G satisfies $g_0^2 = e$, no larger exponent is possible: for some ergodic ν on $\mathcal{V}(\mathbb{Z}^d)$ with (Mir), some real $A \in \mathcal{F}(\mathsf{Bd}(Q_1))$ with $|A| \le 1$ and $\nu(A) = 0$, and some number $c > 0$, one has $\nu(A\,\tau_x A) \ge c\,|x|^{-(d-2)}$ at every $x \in \mathbb{Z}^d \setminus \{0\}$.

■

Sketch of proof. Step 1: the projection at $t=1$. By Proposition 18.6 (c), ν is invariant under $a\mathbb{Z}^d \rtimes W_d$ and reflection positive at all lattice mirrors, and in the site picture of e_μ, $\mathfrak{E}$ lives on $\mathcal{C}_\mu$, where $t=1$ only if every $\zeta_\lambda = 1$; hence $\Pi_0 = \mathfrak{E}(\{t=1\})$ equals $\mathfrak{E}(\{(1,1,\dots,1)\})$, $V(v')\Pi_0 = \Pi_0$, and $T^n \to \Pi_0$ strongly by bounded convergence of t^n to $\mathbf{1}\{t=1\}$. For $A, B \in \mathcal{F}(\mathsf{Bd}(Q_r))$ and $x \in a\mathbb{Z}^d$ with $x_\mu \geq 2ra$, $x_\perp := x - x_\mu e_\mu$, the definitions of the picture and the invariance under τ_x give (translate both entries by rae_μ in the reflection form)

$$\nu(\bar{A}\,\tau_x B) = \langle \psi^\mu_{\theta_\mu A}, T^{x_\mu/a-2r} V(x_\perp)\psi^\mu_B\rangle, \qquad \nu(\bar{A}\,\tau_{-x}B) = \overline{\nu(\bar{B}\,\tau_x A)}.$$

Letting $k\to\infty$, $\nu(\bar{A}\,\tau_{kae_\mu+y}B) \to \langle \psi^\mu_{\theta_\mu A}, \Pi_0 \psi^\mu_B\rangle$ for $y \perp e_\mu$; the ergodic theorem of von Neumann for the unitary τ_{ae_μ} of $L^2(\nu)$ identifies the limit of the Cesàro means with $\langle A, \mathsf{P}_\mu \tau_y B\rangle_{L^2(\nu)}$, P_μ projecting on the vectors fixed by τ_{ae_μ}. The limit is the same for every $y \perp e_\mu$, and local functions are dense in $L^2(\nu)$; so $\mathsf{P}_\mu \tau_y = \mathsf{P}_\mu$, the range of P_μ is invariant under all of $a\mathbb{Z}^d$, P_μ coincides with the projection $E_\nu[\,\cdot\,|\mathcal{I}]$ on the invariant vectors by the characterization of conditional expectation as an orthogonal projection, and $\langle \psi^\mu_{\theta_\mu A}, \Pi_0 \psi^\mu_B\rangle = \nu(\overline{E_\nu[A|\mathcal{I}]}\, E_\nu[B|\mathcal{I}])$.

Step 2: the kernel. Let z_m be the even period-four sequence with $(z_0,z_1,z_2,z_3) = (0,1,-2,1)$. Let $\mathsf{B}(u) = 8(1-4|u|/\pi)_+$ on $[-\pi,\pi]$, extended periodically. Its Fourier coefficients are $b_0 = 1$ and $b_m = (8\sin(m\pi/8)/(m\pi))^2$ for $m \neq 0$. Put $\gamma_m = z_m b_m$ and

$$\mathsf{B}_z(u) = \sum_{m\in\mathbb{Z}} \gamma_m e^{imu}, \qquad \Psi(t) = 2\sum_{m\geq 1} \gamma_m t^{m-1} \quad (0 \leq t \leq 1),$$

with $0^0 = 1$. The series converge absolutely. The scalar calculation of Lemma YM3_I:99.5 gives $\gamma_0 = 0$, $\gamma_{-m} = \gamma_m$, $\sum_m |\gamma_m| = 8$, $|\mathsf{B}_z| \leq 8$, $\mathsf{B}_z(u) = 0$ whenever $\operatorname{dist}(u, 2\pi\mathbb{Z}) \leq \frac{\pi}{4}$, $0 \leq \Psi \leq 8$, $\Psi(1) = 0$, and $\Psi(e^{-x}) \geq \mathsf{c}_\Psi \tanh x$ for $x > 0$.

Step 3: the density in the momenta. Let $\Phi \in \mathcal{F}(\mathsf{Bd}(Q_r))$ satisfy $\theta_\mu \Phi = \epsilon\Phi$, $\epsilon = \pm 1$; put $\psi := \psi^\mu_\Phi$ and $\varrho_\psi := \langle \psi, \mathfrak{E}(\cdot)\psi\rangle$ in the site picture of e_μ, and let $\tilde{\varrho}_\psi$ be the image on $(S^1)^D$, under $(t,\zeta) \mapsto \zeta$, of $\Psi(t^{2r})\, d\varrho_\psi$, a positive measure of total mass at most $8\|\psi\|^2$. With $K(x) := \nu(\bar{\Phi}\,\tau_x\Phi)$ one has $K(-x) = \overline{K(x)}$, K is unchanged under the linear part of θ_μ because ν is invariant under θ_μ and $\theta_\mu\Phi = \epsilon\Phi$, and Step 1 gives $\langle \psi, T^j V(ay)\psi\rangle = \epsilon K((j+2r)ae_\mu + ay)$ for $j \geq 0$ and $y \in \mathbb{Z}^d$ with $y_\mu = 0$. Hence $\tilde{\varrho}_\psi$ has the Fourier coefficients

$$c(y) = 2\sum_{m\geq 1} \gamma_m \langle \psi, T^{2r(m-1)} V(ay)\psi\rangle = \epsilon \sum_{m\in\mathbb{Z}} \gamma_m K(2rmae_\mu + ay),$$

and $|c(y)| \leq 8\|\psi\|^2$. For $|y|_\infty \geq 2r$ pick λ with $|y_\lambda| = |y|_\infty$; as $c(-y) = \overline{c(y)}$ we may let $y_\lambda > 0$. With $y' := y - y_\lambda e_\lambda$ and $n' := y_\lambda - 2r$, Step 1 in the site picture of e_λ gives

$$\epsilon\, c(y) = \Big\langle \psi^\lambda_{\theta_\lambda\Phi}, \int t^{n'} \zeta^{y'}\, \mathsf{B}_z(2ru)\, d\mathfrak{E}_\lambda(t,\zeta)\, \psi^\lambda_\Phi \Big\rangle, \qquad \zeta_\mu = e^{iu},\ u \in (-\pi,\pi].$$

If $\mathsf{B}_z(2ru) \neq 0$, then $|u| > \frac{\pi}{8r}$ and $|\sin\frac{u}{2}| \geq \frac{1}{8r}$; on $\mathcal{C}_\lambda$ (Proposition 18.6 (c)) every point with $t > 0$ then has $t = e^{-x}$ and $\sinh\frac{x}{2} \geq \frac{1}{8r}$, whence $x \geq 2\log(1+\frac{1}{8r}) \geq \frac{2}{9r}$ and $|c(y)| \leq 8e^{-2(|y|_\infty - 2r)/(9r)} \|\psi^\lambda_{\theta_\lambda\Phi}\|\, \|\psi^\lambda_\Phi\|$. Summation by parts over the shells $|y|_\infty = k$ of $\mathbb{Z}^D$, together with $4r + 2i + 1 \leq r(5 + 9\beta' i)$ for $\beta' := \frac{2}{9r}$, gives $\sum_y |c(y)| \leq 8\,\mathsf{s}_d\, r^D\, \mathcal{N}'_{r,\mu}(\Phi)$. Therefore $\tilde{\varrho}_\psi = h\, d\alpha$ on $(-\pi,\pi]^D$ with the continuous function $h(e^{i\alpha}) = (2\pi)^{-D}\sum_y c(y) e^{-iy\cdot\alpha}$, and

$$0 \leq h \leq 8\,(2\pi)^{-D}\, \mathsf{s}_d\, r^D\, \mathcal{N}'_{r,\mu}(\Phi).$$

For $d = 3$ this is Lemma YM3_I:99.6; only the number of circles and the sums over the shells are different.

Step 4: clause (a). For A with $\theta_\mu A = \epsilon A$ and $\psi = \psi^\mu_A$, $\mathfrak{q}_n(\psi) := \langle \psi, T^n(1-\Pi_0)\psi\rangle = \int_{\{0<t<1\}} t^n\, d\varrho_\psi$. On $\mathcal{C}_\mu \cap \{0 < t < 1\}$, with $t = e^{-x}$ and $|\alpha|_\infty := \max_{\lambda\neq\mu} |\alpha_\lambda|$, $x \geq \mathsf{k}|\alpha|_\infty$ (Proposition 18.6 (c))

and, by Step 2, $t^n = e^{-nx}\Psi(t^{2r})/\Psi(e^{-2rx}) \le g(\alpha)\Psi(t^{2r})$ with $g(\alpha) := e^{-n\mathsf{k}|\alpha|_\infty}/(\mathsf{c}_\Psi \tanh(2r\mathsf{k}|\alpha|_\infty))$, because $e^{-nx}/\tanh(2rx)$ is a decreasing function of x. By Step 3 and $\coth s \le 1 + \frac{1}{s}$,

$$\mathsf{q}_n(\psi) \le \int g\, h\, d\alpha \le \|h\|_\infty\, \mathsf{c}_\Psi^{-1} \int_{\mathbb{R}^D} e^{-n\mathsf{k}|\alpha|_\infty}\Big(1 + \frac{1}{2r\mathsf{k}|\alpha|_\infty}\Big) d\alpha = \|h\|_\infty\, \mathsf{c}_\Psi^{-1} D 2^D \Big[\frac{(D-1)!}{(n\mathsf{k})^D} + \frac{(D-2)!}{2r\mathsf{k}\,(n\mathsf{k})^{D-1}}\Big],$$

where the integral converges only because $D \ge 2$, that is, $d \ge 3$; with the bound on h this is $\mathsf{C}'_d \mathcal{N}'_{r,\mu}(A)\,[(D-1)(\frac{r}{n})^D + \frac{1}{2}(\frac{r}{n})^{D-1}]$. A general A splits as $A_+ + A_-$ with $A_\pm := \frac{1}{2}(A \pm \theta_\mu A)$ and $\theta_\mu A_\pm = \pm A_\pm$; q_n being a positive quadratic form, $\mathsf{q}_n(\psi^\mu_A) \le 2\mathsf{q}_n(\psi^\mu_{A_+}) + 2\mathsf{q}_n(\psi^\mu_{A_-})$, and $\mathcal{N}'_{r,\mu}(A_\pm) \le \mathcal{N}_r(A)$ since $\|\psi^\kappa_{\mathsf{g}A_\pm}\| \le \frac{1}{2}(\|\psi^\kappa_{\mathsf{g}A}\| + \|\psi^\kappa_{\mathsf{g}\theta_\mu A}\|)$ and $\mathsf{g}\theta_\mu \in \mathsf{S}_{2^d}$.

Step 5: clauses (b) and (c). Choose μ with $|x_\mu| = |x|_\infty$. If $x_\mu > 0$, Step 1 and $V\Pi_0 = \Pi_0$ express the difference as $\langle \psi^\mu_{\theta_\mu A}, T^n V(x_\perp)(1-\Pi_0)\psi^\mu_B\rangle$; since $V(x_\perp)$ is unitary and commutes with $T^{n/2}(1-\Pi_0)$, its modulus is at most $\mathsf{q}_n(\psi^\mu_{\theta_\mu A})^{1/2}\mathsf{q}_n(\psi^\mu_B)^{1/2}$, and $\mathcal{N}_r(\theta_\mu A) = \mathcal{N}_r(A)$; if $x_\mu < 0$, the second identity of Step 1 exchanges A and B. For (c): $|x|_\infty \ge |x|/\sqrt{d} \ge 4ra$, hence $n \ge \frac{|x|_\infty}{2a}$ and $\frac{r}{n} \le \frac{2ra}{|x|_\infty} \le \frac{1}{2}$, and $(D-1)(\frac{r}{n})^D + \frac{1}{2}(\frac{r}{n})^{D-1} \le \frac{D}{2}(\frac{r}{n})^{D-1} \le \frac{D}{2}(2\sqrt{d})^{d-2}(\frac{ra}{|x|})^{d-2}$; moreover $\mathcal{N}_r(A) \le \nu(|A|^2) \le \|A\|^2_\infty$, as $\|\psi^\mu_A\|^2 = \nu(\overline{\theta_\mu A}\,\tau_{2rae_\mu}A)$. For ergodic ν, $E_\nu[A|\mathcal{I}] = \nu(A)$. Numerically $\mathsf{s}_3 = 277$, $\mathsf{s}_4 = 7604$, $\mathsf{C}'_3 < 3444$, $\mathsf{C}'_4 < 80438$, $C_3 < 4.8\cdot 10^4$ and $C_4 < 7.73\cdot 10^6$. For $d = 3$, (b) and (c) are the bounds of Proposition YM3_I:99.7, with different constants.

Step 6: clause (d). Let ϕ be the centred Gaussian field on $\mathbb{Z}^d$, $d \ge 3$, with covariance $\mathsf{C} = (-\Delta)^{-1}$, Δ the lattice Laplacian. $\mathsf{C}(x,y) = \mathsf{C}(x-y)$ is $(2d)^{-1}$ times the mean number of visits to x of the simple random walk started at y; thus $\mathsf{C} > 0$, $\mathsf{C}(x) \to 0$, and the classical asymptotics of the lattice Green function give $|x|^{d-2}\mathsf{C}(x) \to \mathsf{c}_d > 0$. The law of ϕ is reflection positive at all lattice mirrors. At a site or diagonal mirror $\Sigma(n,c)$, $n\cdot x$ moves by 0 or ± 1 along a bond, so a walk going from one closed side to the other visits $\mathsf{M} := \mathbb{Z}^d \cap \Sigma(n,c)$; the strong Markov property yields $\mathsf{C}(x,y) = \sum_{z\in\mathsf{M}} g_y(z)\,\mathsf{C}(x,z)$ for $n\cdot x \le c \le n\cdot y$, g_y being the law of the first point of M that the walk from y visits; since $\theta_{n,c}$ fixes every point of M and preserves C, for real f of finite support in $\mathbb{H}^+(n,c)$ and $\hat f := \sum_y f(y)g_y$ one obtains $\langle \theta f, \mathsf{C}f\rangle = \langle \hat f, \mathsf{C}\hat f\rangle \ge 0$. At a bond mirror $\Sigma(e_\mu, \frac{1}{2})$,

$$\mathsf{C}(x) = \int_{[-\pi,\pi]^{d-1}} e^{i\alpha\cdot x_\perp}\, \frac{e^{-\varepsilon(\alpha)|x_\mu|}}{2\sinh\varepsilon(\alpha)}\, \frac{d\alpha}{(2\pi)^{d-1}}, \qquad \cosh\varepsilon(\alpha) = 1 + \sum_{\lambda\ne\mu}(1-\cos\alpha_\lambda),$$

and $|(\theta x)_\mu - y_\mu| = x_\mu + y_\mu - 1$ when x, y lie in the closed side, so that $\langle \theta f, \mathsf{C}f\rangle = \int |\sum_x f(x) e^{i\alpha\cdot x_\perp} e^{-\varepsilon(\alpha)(x_\mu - \frac{1}{2})}|^2 \frac{d\alpha}{(2\pi)^{d-1}2\sinh\varepsilon(\alpha)} \ge 0$. In either case $(\langle \theta f_k, \mathsf{C}f_l\rangle)_{k,l}$ is a positive semidefinite Gram matrix; by Schur's product theorem so is its entrywise exponential, and for $F = \sum_k c_k e^{i\phi(f_k)}$ one gets $\mathbb{E}[\overline{\theta F} F] = \sum_{k,l} \bar c'_k c'_l \exp\langle \theta f_k, \mathsf{C}f_l\rangle \ge 0$ with $c'_k := c_k e^{-\frac{1}{2}\langle f_k, \mathsf{C}f_k\rangle}$; such F are dense. Now set $U_{(x,y)} := g_0$ when $\phi_x + \phi_y < 0$ and $U_{(x,y)} := e$ otherwise, so that $U_{\bar b} = U_b = U_b^{-1}$, and let ν be the law of U. Since U is built bond by bond from the values of ϕ at the two ends, every isometry of $\mathbb{Z}^d$ acts compatibly on ϕ and on U, and every function of the bonds in a closed side depends only on the values of ϕ at the sites of that side; hence ν is translation invariant and reflection positive at all lattice mirrors, and it inherits from ϕ, whose covariance vanishes at infinity, the mixing property, so it is ergodic. Take $b = (0, e_1)$ and $A := \mathbf{1}\{U_b = e\} - \mathbf{1}\{U_b = g_0\}$, which is almost surely $\operatorname{sign}(\phi_0 + \phi_{e_1})$; then $|A| \le 1$ and $\nu(A) = 0$ by the symmetry $\phi \mapsto -\phi$. For two centred jointly Gaussian variables of correlation $\varrho \ge 0$ the expectation of the product of their signs is $\frac{2}{\pi}\arcsin\varrho \ge \frac{2}{\pi}\varrho$; here

$$\varrho_x = \frac{2\mathsf{C}(x) + \mathsf{C}(x+e_1) + \mathsf{C}(x-e_1)}{2\mathsf{C}(0) + 2\mathsf{C}(e_1)}$$

is positive for every x and $|x|^{d-2}\varrho_x$ has a positive limit, so $\inf_{x\ne 0}|x|^{d-2}\varrho_x > 0$. Combined with (c), this shows that when G has an element of order two the exponent $d-2$ cannot be raised in a bound valid for all measures with (Mir). □

Gibbs states in dimension d. Let dg be the Haar probability measure of G and $w\colon G \to (0,\infty)$ a continuous class function with $w(g^{-1}) = w(g)$. For a plaquette p of $a\mathbb{Z}^d$ write U_p for the holonomy of U around the boundary of p, taken from any of its corners and in either orientation; by the two invariances of w, $w(U_p)$ is the same for all these choices. For a finite set B of bonds let γ_B be the kernel that resamples the bonds of B with the normalised density $\prod_p w(U_p)$, over the plaquettes p containing a bond of B, against $\prod_{b\in B} dU_b$. A Gibbs state of w is a Borel probability measure ν on $\mathcal{V}$ that every γ_B leaves unchanged, $\nu(\gamma_B F) = \nu(F)$ for continuous local F; it is periodic if some sublattice of $a\mathbb{Z}^d$ of finite index leaves it invariant. Here $\mathcal{G}(w)$ denotes the set of Gibbs states and $\mathcal{G}^{\mathrm{per}}(w)$ that of the periodic ones. For an irreducible unitary representation π of G, of dimension d_π and character χ_π, let $c_\pi(w) := d_\pi^{-1} \int w\chi_\pi \, dg$, a real number, as in Proposition 9.27 (i).

Proposition 18.8 (Wilson weights in d dimensions: the mirrors of their periodic Gibbs states)**.** Let $d \geq 2$, $a > 0$, G a compact metrisable group and w as above.

(a) Let N be a closed normal subgroup of G such that w is constant on the cosets $g\mathsf{N}$ and $c_\pi(w) > 0$ for each irreducible π whose kernel contains N. Then $\mathcal{G}^{\mathrm{per}}(w) \neq \emptyset$, and each of its members is reflection positive at all lattice mirrors $\Sigma(n,c)$ of the three kinds, for the functions in $\mathcal{F}_{\mathrm{loc}}(\mathbb{H}^+(n,c))$ and in $\mathcal{F}_{\mathrm{loc}}(\mathbb{H}^-(n,c))$, and is invariant under $a\mathbb{Z}^d \rtimes W_d$; in particular it has (Mir). Moreover $\mathcal{G}^{\mathrm{per}}(w)$ coincides with the set of translation invariant Gibbs states of w, a convex weakly compact set.

(b) For every finite-dimensional unitary representation ρ of G and every $\beta \geq 0$, the weight $w = e^{\beta \operatorname{Re}\chi_\rho}$ meets the hypothesis of (a), with $\mathsf{N} = \ker\rho$ if $\beta > 0$ and $\mathsf{N} = G$ if $\beta = 0$. This covers, for every $\beta \geq 0$, the Wilson weight $e^{-\beta a(U_p)}$ of every closed subgroup G of $U(N)$, which for $G \neq \{\mathbf{1}\}$ is the weight of the Wilson measure of Proposition 9.27; here $a(X) = 1 - \frac{1}{N}\operatorname{Re}\operatorname{tr} X$, the formula that Definition 9.1 uses for the action density. In particular the Wilson weight of $SU(2)$ is covered.

(c) Let $d = 4$ and ν a periodic Gibbs state of a weight as in (a), for instance of the Wilson weight of $SU(2)$ at some $\beta \geq 0$. In every picture (e_μ, c_0) of ν, $c_0 \in \frac{a}{2}\mathbb{Z}$, and for each of the three directions $\lambda \neq \mu$,

$$(1-T)^2 \geq T\big(2 - V(ae_\lambda) - V(ae_\lambda)^*\big);$$

and for every integer $r \geq 1$, all $A, B \in \mathcal{F}(\mathsf{Bd}([-ra, ra]^4))$ and every $x \in a\mathbb{Z}^4$ with $|x| \geq 8ra$,

$$\Big|\nu(\bar{A}\,\tau_x B) - \nu\big(\overline{E_\nu[A|\mathcal{I}]}\,E_\nu[B|\mathcal{I}]\big)\Big| \leq C_4\,\|A\|_\infty\|B\|_\infty\Big(\frac{ra}{|x|}\Big)^2, \qquad C_4 < 7.8\cdot 10^6,$$

which is the three-dimensional rate $ra/|x|$ of Proposition YM3_I:99.7 squared; if ν is ergodic, the subtracted term is $\nu(\bar{A})\nu(B)$.

■

Sketch of proof. Step 1: Gibbs states. For a family $\Psi = (A_i)$ of real continuous local functions such that each bond lies in finitely many of their supports, the kernel of (w,Ψ) carries in addition the factor $\exp\sum A_i$ over the A_i whose support meets B, and Gibbs states of (w,Ψ) are defined in the same manner. The kernels map continuous local functions to continuous local functions. For a finite set B consider two kernels of this kind, the density of each being the product of the factors of the plaquettes that contain a bond of B with its exponential factor. If at every configuration the quotient of the two densities lies between $e^{-\epsilon}$ and e^{ϵ}, then after normalisation it lies between $e^{-2\epsilon}$ and $e^{2\epsilon}$, and the values of the two kernels at F differ by no more than $(e^{2\epsilon}-1)\|F\|_\infty$. For B fixed, $w_k \to w$ uniformly and $t_k \to 0$, the quotient for the kernels of B of $(w_k, t_k\Psi)$ and of w tends to 1 uniformly, since w is continuous and positive on a compact group and only finitely many of the bounded A_i meet B; consequently any weak limit of Gibbs states of $(w_k, t_k\Psi)$ is a Gibbs state of w, and $\mathcal{G}(w)$ is convex and weakly compact. Isometries map Gibbs states to Gibbs states. A gauge transformation at a single site leaves the kernel of the bonds at that site unchanged, and so

does $U_b \mapsto U_b z$, $z \in \mathsf{N}$, at a single bond b when w is constant on the cosets of the closed normal subgroup N; hence every Gibbs state is invariant under both.

Step 2: kernels of positive type. Schur's lemma gives $\int w(vg^{-1})\pi_{ij}(v)\,dv = c_\pi(w)\,\pi_{ij}(g)$ for the matrix coefficients of π. If every $c_\pi(w) \ge 0$, the integral operator with the kernel $w(gh^{-1})$ is positive on $L^2(G)$ by the theorem of Peter and Weyl, and, w being continuous, $(g,h) \mapsto w(gh^{-1})$ is a positive definite kernel. A product of positive definite kernels, or the composition of one with a map, is again positive definite, and a bounded positive definite kernel $\mathcal{K}$ on a probability space satisfies $\iint \overline{\Phi(y)}\,\mathcal{K}(y,y')\,\Phi(y') \ge 0$ for bounded Borel Φ, as sampling independent points shows. The hypothesis of (a) entails $c_\pi(w) \ge 0$ for every π: when π is not trivial on N, $\int_{\mathsf{N}} \pi(z)\,dz = 0$ and hence $c_\pi(w) = 0$.

Step 3: tori adapted to one normal. Put $a = 1$, let $\nu \in \mathcal{G}^{\mathrm{per}}(w)$ be invariant under $\mathsf{N}_0\mathbb{Z}^d$, let $n = e_\mu$ or $n = e_\mu + se_\lambda$, $c \in \mathrm{Adm}(n)$, and let h be a multiple of $|n|^2\mathsf{N}_0$, $M \ge 2$, and $\ell \ge 3$ a multiple of N_0. With $\mathsf{p}(x) := n \cdot x$, $z_0 := \frac{h}{|n|^2} n$ and the rank-$(d-1)$ lattice $\Gamma_\perp := \{z \in \mathbb{Z}^d : n \cdot z = 0\}$, set $\Gamma := 2Mz_0\mathbb{Z} + \ell\Gamma_\perp$ and $\mathbb{T} := \mathbb{Z}^d/\Gamma$. Every non-zero vector of Γ has a coordinate of modulus at least 3, so near each site the quotient map is injective on bonds and plaquettes. The reflections $\theta_k := \theta_{n,c+kh}$ and the translation τ_{z_0} act on $\mathbb{T}$; each of them moves points along n by an amount that depends on $\mathsf{p}(x)$ alone, so it is determined by what it does to p, which gives $\theta_k\theta_l = \tau_{2(k-l)z_0}$ and $\theta_{k+M} = \theta_k$ on $\mathbb{T}$. The slabs $\mathsf{Sl}_j := \{c + (j-1)h \le \mathsf{p} \le c + jh\}$, $j \in \mathbb{Z}/2M$, the halves $\mathbb{T}^+_k := \mathsf{Sl}_{k+1} \cup \cdots \cup \mathsf{Sl}_{k+M}$ and the maps $\mathsf{s}_j := \theta_{j-1}\cdots\theta_1$ satisfy $\mathsf{s}_j\mathsf{Sl}_1 = \mathsf{Sl}_j$ and $\theta_k\mathsf{s}_j = \mathsf{s}_{2k+1-j}$. Let $\mu_0 \propto \prod_p w(U_p)\prod_b dU_b$ on the configurations of $\mathbb{T}$, a measure invariant under the isometries of $\mathbb{T}$. For $d = 3$ these are the adapted tori of Proposition YM3_I:91.4.

Step 4: positivity on these tori. Suppose every $c_\pi(w) \ge 0$. Let $k \in \mathbb{Z}/2M$, S_+ a real bounded Borel function of the bonds of $\mathbb{T}^+_k$, and $\mu \propto e^{\mathsf{S}_+ + \theta_k\mathsf{S}_+}\mu_0$. Then μ is invariant under θ_k, and $\mu(\overline{\theta_k F}F) \ge 0$ for every bounded Borel F of the bonds of $\mathbb{T}^+_k$. At a site mirror no plaquette crosses the mirror, and Fubini's theorem writes the integral as $\int dU_0\, W_0\,|\int F\,W_+ e^{\mathsf{S}_+}\,dU_+|^2$, by the computation of Step 1 of the proof of Proposition 18.1, which uses of the group its Haar measure alone and of the weight only its two invariances. At a diagonal mirror the plaquettes of the plane (μ,λ) whose corners have $\mathsf{p} = m-1, m, m, m+1$ with $m \in \{c+kh, c+(k+M)h\}$ cross the mirror: two corners y, y' lie on it, one corner x^+ in $\mathbb{T}^+_k$, the last is $\theta_k x^+$; with $\mathsf{A}_p(U) := U(\langle y, x^+\rangle)U(\langle x^+, y'\rangle)$ their variables are $\mathsf{A}_p(U_+)\mathsf{A}_p(U'_+)^{-1}$, U'_+ denoting the reflected half, and the product of $w(\mathsf{A}_p(U_+)\mathsf{A}_p(U'_+)^{-1})$ over these plaquettes is a positive definite kernel in (U_+, U'_+) by Step 2. At a bond mirror the bonds crossing the two mirrors act on the reflected half as a gauge transformation V of its two outermost layers, and the crossing plaquettes contribute $w(U_+(b)\,[(V\cdot U'_+)(b)]^{-1})$; since W_+ and this kernel are gauge invariant, one may replace $F\,W_+e^{\mathsf{S}_+}$ by its average Φ^{av} over these gauge transformations, and the integral becomes $\iint \Phi^{\mathrm{av}}(U_+)\,\overline{\Phi^{\mathrm{av}}(U'_+)}\prod_b w(U_+(b)U'_+(b)^{-1}) \ge 0$. For $d = 3$ this is Proposition YM3_I:91.4 and Corollary YM3_I:85.3.

Step 5: a chessboard estimate along the normal. Under the same assumption, for bounded Borel functions $\mathsf{g}_j > 0$ of the bonds of Sl_1, $j \in \mathbb{Z}/2M$, the quantity $\Phi(\mathsf{g}_1, \dots, \mathsf{g}_{2M}) := \mu_0(\prod_j \mathsf{s}_j\mathsf{g}_j)$ satisfies $\Phi \le \prod_j \Phi(\mathsf{g}_j, \dots, \mathsf{g}_j)^{1/2M}$: by $\theta_k\mathsf{s}_j = \mathsf{s}_{2k+1-j}$ and Step 4 with $\mathsf{S}_+ = 0$, Schwarz's inequality supplies the hypothesis of Step 2 of the proof of Proposition 18.1. For $d = 3$ this is Proposition YM3_I:91.7.

Step 6: a reflection positive state above ν. Suppose every $c_\pi(w) \ge 0$, let $\nu \in \mathcal{G}^{\mathrm{per}}(w)$, $\Sigma(n,c)$ a lattice mirror and A a real continuous local function of the bonds of $\mathbb{H}^+(n,c)$ that depends on bonds in $[-r,r]^d$ only. We find $\nu_0 \in \mathcal{G}^{\mathrm{per}}(w)$, reflection positive at $\Sigma(n,c)$, with $\nu_0(A) \ge \nu(A)$, and see that $\mathcal{G}^{\mathrm{per}}(w) \ne \emptyset$. Take h large enough that A depends on bonds of Sl_1 only, let A_1 be the sum of the translates of A by $\mathsf{N}_0\Gamma_\perp/\ell\Gamma_\perp$, $\mathsf{S}_{\mathrm{alt}} := \sum_j \mathsf{s}_j A_1$ and $\mathsf{S}_{\mathrm{uni}} := \sum_j \tau_{(j-1)z_0}A_1$, each a sum of $\mathcal{N} := 2M(\ell/\mathsf{N}_0)^{d-1}$ copies of A, and $\mu_t \propto e^{t\mathsf{S}_{\mathrm{alt}}}\mu_0$ for $t > 0$. Since $t\mathsf{S}_{\mathrm{alt}} = \mathsf{S}_+ + \theta_k\mathsf{S}_+$ with $\mathsf{S}_+ := t\sum_{i=1}^M \mathsf{s}_{k+i}A_1$, Step 4 shows that μ_t is invariant under all θ_k and reflection positive at them; it is also invariant under $\mathsf{N}_0\Gamma_\perp$, whence $\mu_t(\mathsf{S}_{\mathrm{alt}}) = \mathcal{N}\mu_t(A)$. (i) With $\mathsf{m} := \theta_1\tau_{z_0}$, which mirrors

Sl_1 onto itself, $e^{t\mathsf{S}_{\rm uni}} = \prod_{j\ \rm odd} \mathsf{s}_j e^{tA_1} \prod_{j\ \rm even} \mathsf{s}_j \mathsf{m} e^{tA_1}$, and Step 5 together with the invariance of μ_0 under m gives $\mu_0(e^{t\mathsf{S}_{\rm uni}}) \le \mu_0(e^{t\mathsf{S}_{\rm alt}})$. (ii) A fundamental domain D of Γ has $2Mh\ell^{d-1} = h\mathsf{N}_0^{d-1}\mathcal{N}$ sites. Let B_D be the set of the bonds based in D; a plaquette of the infinite lattice whose four bonds all belong to B_D is said to be inner. Put $n_\partial := n_1 + n_2$, where n_1 counts the plaquettes having some, but not all, of their four bonds in B_D, and n_2 counts the plaquettes of $\mathbb{T}$ that do not arise from an inner plaquette under the quotient map. Both n_∂ and the number $\mathcal{N}_\partial$ of copies of A whose lifts are not inside D at distance $r+1$ from the boundary are at most $(\frac{1}{M} + \frac{1}{\ell})\mathcal{N}$ times a number depending on d, r, h and N_0. The densities of the kernel $\gamma_{B_D}(\cdot\,|U)$ and of μ_0 have in common the product over the inner plaquettes; besides it the first has n_1 factors and the second n_2, each between $\min w$ and $\max w$, so that the normalised density of μ_0 is at least $\kappa_w^{-n_\partial}$ times that of the kernel, $\kappa_w := \max w / \min w$. As ν is a Gibbs state, $\mu_0(f) \ge \kappa_w^{-n_\partial}\nu(f \circ \mathrm{res})$ for $f \ge 0$, where res is the restriction to B_D (Proposition YM3_I:91.8 for $d = 3$); with Jensen's inequality and the periodicity of ν, $\mu_0(e^{t\mathsf{S}_{\rm uni}}) \ge \exp(-n_\partial \log \kappa_w + t\mathcal{N}\nu(A) - 2t\mathcal{N}_\partial \|A\|_\infty)$. (iii) The function $\psi(t) := \log \mu_0(e^{t\mathsf{S}_{\rm alt}})$ is convex, $\psi(0) = 0$ and $\psi'(t) = \mathcal{N}\mu_t(A)$, so $\psi(t) \le t\mathcal{N}\mu_t(A)$, and together with (i) and (ii)

$$\mu_t(A) \ge \nu(A) - 2\frac{\mathcal{N}_\partial}{\mathcal{N}}\|A\|_\infty - \frac{n_\partial \log \kappa_w}{t\mathcal{N}}.$$

(iv) Let $M = \ell \to \infty$ along multiples of N_0 with t fixed, and let ν_t be a weak limit point of the periodic extensions of μ_t: it is invariant under the finite-index lattice $2z_0\mathbb{Z} + \mathsf{N}_0\Gamma_\perp$, reflection positive at $\Sigma(n,c)$, a Gibbs state of $(w, t\Psi)$ for the family Ψ of the copies $\tau_z \mathsf{s}_j A$, and $\nu_t(A) \ge \nu(A)$. (v) Any weak limit point ν_0 of $(\nu_{1/k})_k$ is by Step 1 a periodic Gibbs state of w, reflection positive at $\Sigma(n,c)$, with $\nu_0(A) \ge \nu(A)$. At $A = 0$, (iv) and (v) do not use ν, so $\mathcal{G}^{\rm per}(w) \ne \emptyset$. For $d = 3$ this is Proposition YM3_I:102.5.

Step 7: agreement on a half space forces equality. Under the hypothesis of (a), if two Gibbs states ν_1, ν_2 of w agree on all functions of the bonds of a half space $\{\epsilon x_\mu \ge c\}$, $c \in \mathbb{Z}$, then $\nu_1 = \nu_2$. Suppose first $\mathsf{N} = \{e\}$, $\mu = d$ and $\epsilon = 1$. For a bond b of $\mathbb{Z}^{d-1}$ let $v(b)$ and $u'(b)$ be the values of U on its copies at the heights c and $c-1$, D_y the value on the vertical bond from (y,c) to $(y,c-1)$, and $\mathsf{g}(b) := \mathsf{D}_{b_-} u'(b) \mathsf{D}_{b_+}^{-1}$, so that the plaquette below b has the variable $v(b)\mathsf{g}(b)^{-1}$. For a finite set S of bonds of $\mathbb{Z}^{d-1}$ with copies B at the height c, and continuous local $\psi(\mathsf{g}|_S, v, \mathsf{r})$, r standing for the values on the remaining bonds of $\{x_d \ge c\}$, the invariance under γ_B reads

$$\nu_i(\psi) = \int \varsigma_i(d\mathsf{g}, d\mathsf{r}) \int \prod_{b \in S} w(v(b)\mathsf{g}(b)^{-1})\, \Phi(v;\mathsf{r})\, \psi(\mathsf{g}, v, \mathsf{r})\, dv,$$

with a continuous $\Phi > 0$ and finite positive measures ς_i. Choose $\psi = \phi(v)\chi(\mathsf{r})/\Phi(v;\mathsf{r})$ with ϕ a product of matrix coefficients $\pi^b_{ij}(v(b))$: then ψ depends on bonds of the half space only, so $\nu_1(\psi) = \nu_2(\psi)$, while by Step 2 the right side is $\prod_b c_{\pi^b}(w) \int \phi(\mathsf{g})\chi(\mathsf{r})\, d\varsigma_i$. As all $c_\pi(w) > 0$, the theorems of Peter and Weyl and of Stone and Weierstrass give $\varsigma_1 = \varsigma_2$, so ν_1 and ν_2 agree on the functions of g, v and r. A gauge transformation at the sites $(y, c-1)$ fixes these variables and moves the D_y; averaging over it (Step 1), the two states agree on the functions of the bonds of $\{x_d \ge c-1\}$, because $u'(b) = \mathsf{D}_{b_-}^{-1}\mathsf{g}(b)\mathsf{D}_{b_+}$; induction downwards gives $\nu_1 = \nu_2$. For general N pass to G/N, whose irreducible representations are the π trivial on N, using the invariance under N from Step 1. For $d = 3$ and $\beta > 0$ the main text has the analogue for the chain of layers: by Lemma YM3_I:103.2, among the Borel probability measures that solve the equations of that chain, each is singled out by its image on any two adjacent layers; a half space needs neither layers nor a gauge fixing.

Step 8: clause (a). Let $\nu \in \mathcal{G}^{\rm per}(w)$. For a site or bond mirror $\Sigma = \Sigma(e_\mu, c)$ let $\mathcal{R}$ be the set of the $\nu' \in \mathcal{G}(w)$ that are reflection positive at Σ on the continuous local functions of $\mathbb{H}^+$; by Step 1 it is convex and weakly compact, and by Step 6 it is not empty. Its restrictions to the bonds of $\mathbb{H}^+$ form a convex weak-$*$ compact set; were the restriction of ν outside it, a continuous function of these bonds, approximated by a real continuous local A, would separate them, $\nu(A) \ge \nu'(A) + \delta$ for all $\nu' \in \mathcal{R}$ and some $\delta > 0$, contradicting Step 6. Hence some $\nu' \in \mathcal{R}$ has the restriction of ν to the

bonds of $\mathbb{H}^+$, which are the bonds of $\{x_\mu \geq \lceil c \rceil\}$, and Step 7 gives $\nu' = \nu$; L^2 approximation with respect to $\nu + \theta_*\nu$ carries positivity from continuous to bounded Borel local functions. Hermiticity of the positive form gives $\nu(\theta F) = \nu(F)$ for local functions on the positive side. For any local A, choose a sufficiently large lattice period v along the positive normal so that $\tau_v A$ lies on this side. Since $\theta\tau_v = \tau_{-v}\theta$, periodicity gives $\nu(\theta A) = \nu(\theta\tau_v A) = \nu(\tau_v A) = \nu(A)$. This gives full reflection invariance and positivity on the other closed side. From $\theta_{e_\mu,1/2}\theta_{e_\mu,0} = \tau_{e_\mu}$, every periodic Gibbs state is translation invariant, and $\mathcal{G}^{\mathrm{per}}(w)$ is the convex, weakly compact set of translation invariant Gibbs states. For a diagonal mirror let $\mathcal{R}$ be the set of members of $\mathcal{G}^{\mathrm{per}}(w)$ that are reflection positive at it; it contains the ν_0 of Step 6, and the same separation yields $\nu' \in \mathcal{R}$ with the restriction of ν to the bonds of $\mathbb{H}^+(n,c)$; both being translation invariant, $\nu'(f) = \nu'(\tau_{ke_\mu} f) = \nu(\tau_{ke_\mu} f) = \nu(f)$ for every continuous local f and large k, so $\nu = \nu'$. The reflections in $\Sigma(e_\mu, 0)$ and $\Sigma(e_\mu - e_\lambda, 0)$ generate W_d. In three dimensions see Proposition YM3_I:103.4 for the $SU(2)$ Wilson weight and Proposition 9.27 (iii) for closed $G \subset U(N)$.

Step 9: clause (b). Since $\operatorname{Re}\chi_\rho = \frac{1}{2}\chi_{\rho\oplus\bar\rho}$, $e^{\beta \operatorname{Re}\chi_\rho} = \sum_m \frac{(\beta/2)^m}{m!}\chi_{(\rho\oplus\bar\rho)^{\otimes m}}$ and $d_\pi c_\pi(w) = \sum_m \frac{(\beta/2)^m}{m!}\operatorname{mult}(\pi, (\rho\oplus\bar\rho)^{\otimes m}) \geq 0$; also w is constant on the cosets of $\ker\rho$. The functions $g \mapsto \langle u, (\rho\oplus\bar\rho)^{\otimes m}(g)u'\rangle$, $m \geq 0$, together form a subalgebra of the continuous functions on $G/\ker\rho$ that contains the constants, is stable under complex conjugation and separates points, ρ being faithful on $G/\ker\rho$; it is dense by the theorem of Stone and Weierstrass, while an irreducible π trivial on $\ker\rho$ that occurred in no tensor power would have a matrix coefficient orthogonal to it, by the relations of Schur. Hence $c_\pi(w) > 0$ for $\beta > 0$; at $\beta = 0$, $w = 1$ and $\mathsf{N} = G$. The Wilson weight of a closed $G \subset U(N)$ equals $e^{-\beta} e^{\frac{\beta}{N}\operatorname{Re}\chi_\rho}$ with ρ the inclusion, and a constant factor does not affect Gibbs states; Proposition 9.27 (i) records the same positivity of its coefficients.

Step 10: clause (c). By (a) and (b), ν has (Mir); Proposition 18.6 (c) gives the inequality for the three directions $\lambda \neq \mu$ in every picture (e_μ, c_0), and Proposition 18.7 (c) for $d = 4$, with $c_4 = 8$, $(ra/|x|)^{d-2} = (ra/|x|)^2$ and $C_4 < 7.8 \cdot 10^6$, gives the decay of correlations. □

Other plaquette weights of $SU(2)$. In Propositions 18.9 and 18.10, $G = SU(2)$; a torus Λ with even sides $N_\mu \geq 4$, its configurations, the Haar measure dU and $a_p = 1 - \frac{1}{2}\operatorname{tr}\partial U(p) \in [0,2]$ are as in Proposition 18.1; on the torus $\Lambda_{K,\ell}$, $\ell = 2L^m$, with $2L^{K+m}$ sites on an axis, $\mathcal{O}(f) = a_K^{-1}\sum_p f(x(p))\, a_p$, and $S^T_{n;K,\ell,\beta}$ is the joint cumulant of $\mathcal{O}(f_1), \dots, \mathcal{O}(f_n)$ under the Wilson measure; for a function f on T_ℓ, $\|f\|_\sharp$ denotes $\max\{\sup|f|, 40M \operatorname{Lip} f\}$ with the construction constant M. A *plaquette weight* is a bounded Borel function $\mathsf{w}\colon [0,2] \to [0,\infty)$ with $Z_\Lambda[\mathsf{w}] := \int \prod_p \mathsf{w}(a_p)\, dU > 0$; $\mu^{\mathsf{w}}_\Lambda := Z_\Lambda[\mathsf{w}]^{-1}\prod_p \mathsf{w}(a_p)\, dU$, and $S^{\mathsf{w}}_{n;K,\ell}$ is the joint cumulant of $\mathcal{O}(f_1), \dots, \mathcal{O}(f_n)$ under $\mu^{\mathsf{w}}_{\Lambda_{K,\ell}}$; Wilson's weight at β is $e^{-\beta s}$. Given $c_0 \in (0,1]$, $C_v \geq 0$, $r_0 \in (0,2]$, $M_\phi > 0$ and $\beta > 0$, the class $\mathcal{W}_\beta(c_0, C_v, r_0, M_\phi)$ is formed by the weights $\mathsf{w} = e^{-\beta\phi - v_\beta}$ in which $\phi\colon [0,2] \to [0,\infty)$ and $v_\beta\colon [0,2] \to (-\infty, \infty]$ are Borel, $\phi(s) \geq c_0 s$, $v_\beta \geq -C_v \log(2+\beta)$, and, on $[0, r_0)$, ϕ and v_β extend holomorphically to the disc $D(0, r_0)$ with $\phi(0) = 0$, $|\phi| \leq M_\phi$ and $|v_\beta| \leq C_v$ on it; ϕ and v_β may vary with β. Put $\mathsf{a} := 3C_v + \frac{9}{2}$, $\mathsf{A} := 9C_v + 24M_\phi/r_0 + 3\log 7$ and $q_\beta(\delta) := \exp[-\frac{1}{2}c_0\beta\delta + \mathsf{a}\log(2+\beta) + \mathsf{A}]$.

Proposition 18.9 (Tails of the plaquette action; weights that differ only at large plaquette action)**.** Let Λ be a torus with even sides $N_\mu \geq 4$.

(a) Let $\mathsf{w} \in \mathcal{W}_\beta(c_0, C_v, r_0, M_\phi)$ with $\beta \geq 1$ and $\beta > 8/r_0$, and $0 < \delta \leq 2$. Then, for each plaquette p and each family Y of plaquettes,

$$\mu^{\mathsf{w}}_\Lambda(a_p \geq \delta) \leq q_\beta(\delta), \qquad \mu^{\mathsf{w}}_\Lambda(a_p \geq \delta \text{ for all } p \in Y) \leq q_\beta(\delta)^{|Y|/3} \text{ if } q_\beta(\delta) \leq 1, \qquad \mu^{\mathsf{w}}_\Lambda(\max_p a_p \geq \delta) \leq 3|\Lambda|\, q_\beta(\delta).$$

Neither the sides nor any feature of w other than the four numbers of its class enters $q_\beta(\delta)$.

(b) Let $\mathsf{w}_i \in \mathcal{W}_\beta(c_0^{(i)}, C_v^{(i)}, r_0^{(i)}, M_\phi^{(i)})$, $i = 1, 2$, with $\beta \geq 1$ and $\beta > 8/r_0^{(i)}$, let $0 < \delta \leq 2$ and $\epsilon \geq 0$, and assume that both weights are positive on $[0, \delta)$ and that some real c satisfies

$|\log \mathsf{w}_1(s) - \log \mathsf{w}_2(s) - c| \le \epsilon$ for $0 \le s < \delta$. Then for every event E

$$|\mu_\Lambda^{\mathsf{w}_1}(E) - \mu_\Lambda^{\mathsf{w}_2}(E)| \le \mathsf{E} := 3|\Lambda|(q_\beta^{(1)}(\delta) + q_\beta^{(2)}(\delta)) + e^{6|\Lambda|\epsilon} - 1,$$

where $q_\beta^{(i)}$ is q_β formed with the numbers of w_i. Consequently any Borel function of U, in particular the block field of any depth, has laws under the two measures whose distance on events is at most E; and on $\Lambda_{K,\ell}$, for $n \ge 1$ and real bounded $f_1, \dots, f_n$ with no condition on their supports, $|S^{\mathsf{w}_1}_{n;K,\ell}(\underline{f}) - S^{\mathsf{w}_2}_{n;K,\ell}(\underline{f})| \le 2\,\mathsf{fu}_n\,\mathsf{E}\prod_i 6\ell^3 L^{4K}\|f_i\|_\infty$. The same holds for $\mathsf{w}_2 := \mathbf{1}_{[0,\delta)}\mathsf{w}_1$, without the term $q_\beta^{(2)}(\delta)$.

■

Sketch of proof. Step 1: the planes through the sites. On $SU(2)$, $k(g) := \mathsf{w}(1 - \frac{1}{2}\mathrm{tr}\, g)$ equals $\kappa(\frac{1}{2}\mathrm{tr}\, g)$ for the bounded Borel function $\kappa(t) := \mathsf{w}(1-t)$ on $[-1,1]$, so Proposition 18.1 (a), (b) applies to μ_Λ^{w}, which is moreover invariant under the translations of Λ.

Step 2: bounds on the weight. Multiplying w by the constant $e^{v_\beta(0)}$ leaves μ_Λ^{w} as it is; afterwards $\mathsf{w} = e^{-g}$ with $g = \beta\phi + v_\beta - v_\beta(0)$. From $\phi \ge c_0 s$, $-v_\beta \le C_v \log(2+\beta)$ and $v_\beta(0) \le C_v$, $\mathsf{w}(s) \le \mathsf{B}e^{-c_0\beta s}$ with $\mathsf{B} := e^{C_v(1+\log(2+\beta))} \ge 1$. In the other direction, if $|U(b) - \mathbf{1}| \le \beta^{-1/2}$ at all bonds, then $|\partial U(p) - \mathbf{1}| \le 4\beta^{-1/2}$ and $a_p = \frac{1}{2}|\partial U(p) - \mathbf{1}|^2 \le 8/\beta < r_0$, so the lemma of Schwarz gives $\beta\phi(a_p) \le \beta M_\phi a_p / r_0 \le 8M_\phi/r_0$, while $|v_\beta(a_p) - v_\beta(0)| \le 2C_v$. The set $\{|g - \mathbf{1}| \le \varrho\}$, $\varrho \le 1$, has Haar measure at least $\frac{2}{\pi}\int_0^\varrho \sin^2\vartheta\, d\vartheta \ge \frac{1}{7}\varrho^3$, because $|g - \mathbf{1}| = 2\sin\frac{\vartheta}{2} \le \vartheta$ for the class angle ϑ of g and $\sin\vartheta \ge \frac{5}{6}\vartheta$ on $[0,1]$. Hence

$$Z_\Lambda[\mathsf{w}] \ge \left(\tfrac{1}{7}\beta^{-3/2}\right)^{3|\Lambda|} e^{-(8M_\phi/r_0 + 2C_v)\,3|\Lambda|}.$$

Step 3: one plaquette. Let the plaquette p span the directions μ, ν and let λ be the remaining direction; by translation invariance we may take for p the face p_0 of c_0 in the plane $x_\lambda = 0$. Proposition 18.1 (b) with $F_0 := \mathbf{1}[a_{p_0} \ge \delta]$, read on the model cube, and $F_x := 1$ for $x \ne 0$, gives $\mu_\Lambda^{\mathsf{w}}(a_{p_0} \ge \delta) \le \mathfrak{z}(F_0)^{1/|\Lambda|}$ with $\mathfrak{z}(F) := \langle \prod_x F^{(x)} \rangle / \langle 1 \rangle$. As x runs over Λ, the faces $\rho_x^{-1} p_0$ run over the plaquettes spanned by μ, ν that lie in planes with even x_λ, $\frac{1}{2}|\Lambda|$ of them; bounding w by B on all plaquettes and by $\mathsf{B}e^{-c_0\beta\delta}$ on those,

$$\begin{aligned}\mathfrak{z}(F_0)^{1/|\Lambda|} &\le \mathsf{B}^3 e^{-\frac{1}{2}c_0\beta\delta} Z_\Lambda[\mathsf{w}]^{-1/|\Lambda|} \\ &\le \exp\big[-\tfrac{1}{2}c_0\beta\delta + 3C_v(1+\log(2+\beta)) \\ &\qquad + \tfrac{9}{2}\log\beta + 3\log 7 + 24M_\phi/r_0 + 6C_v\big] \le q_\beta(\delta).\end{aligned}$$

A face of the model cube at $x_\lambda = 1$ leads in the same way to the planes with odd x_λ, with as many plaquettes.

Step 4: families of plaquettes. Assign the plaquette $(x; \mu\nu)$ to the cube c_x, so that each cube is assigned three plaquettes, and let F_x, read on the model cube, be the product of the indicators $\mathbf{1}[a_p \ge \delta]$ over the plaquettes of Y assigned to c_x. Whenever $F_x \ne 1$ it is dominated by the indicator for one face of the model cube, so $\mathfrak{z}(F_x)^{1/|\Lambda|} \le q_\beta(\delta)$ by Step 3 and the monotonicity of $\mathfrak{z}$; at least $|Y|/3$ cubes are assigned a plaquette of Y, and Proposition 18.1 (b) gives the second bound. Summing the first bound over the $3|\Lambda|$ plaquettes gives the third.

Step 5: clause (b). Let $\mathsf{G} := \{a_p < \delta \text{ for all } p\}$ and $\mu_i := \mu_\Lambda^{\mathsf{w}_i}$. By (a), $\mu_i(\mathsf{G}^c) \le 3|\Lambda| q_\beta^{(i)}(\delta)$, and $|\mu_i(E) - \mu_i(E \mid \mathsf{G})| \le \mu_i(\mathsf{G}^c)$ for every event E. On G the two products $\prod_p \mathsf{w}_i(a_p)$ differ by a factor in $e^{3|\Lambda|c}[e^{-3|\Lambda|\epsilon}, e^{3|\Lambda|\epsilon}]$; after normalisation the conditioned measures $\mu_1(\cdot \mid \mathsf{G})$ and $\mu_2(\cdot \mid \mathsf{G})$ have a mutual density between $e^{-6|\Lambda|\epsilon}$ and $e^{6|\Lambda|\epsilon}$, so they differ by at most $e^{6|\Lambda|\epsilon} - 1$ on every event. Passing to the law of a Borel function of U can only diminish this distance, and block fields are Borel functions of U (Definition 2.7). For the cumulants use Proposition 18.1 (c) with $|\mathcal{O}(f_i)| \le a_K^{-1} \cdot 3|\Lambda| \cdot 2\|f_i\|_\infty = 6\ell^3 L^{4K}\|f_i\|_\infty$. If $\mathsf{w}_2 = \mathbf{1}_{[0,\delta)}\mathsf{w}_1$, then $\mu_2 = \mu_1(\cdot \mid \mathsf{G})$. □

Proposition 18.10 (The limits of Wilson's action for weights that coincide with Wilson's at small plaquette action)**.** Let $0 < s_0 \le 2$, $c_0 \in (0,1]$ and $C_v \ge 0$, and call a plaquette weight w *admitted at* β if $\mathsf{w}(s) = e^{-\beta s}$ for $0 \le s < s_0$ and $0 \le \mathsf{w}(s) \le (2+\beta)^{C_v} e^{-c_0 \beta s}$ for $s_0 \le s \le 2$; the values of w on $[s_0, 2]$ may vary with β arbitrarily. Such a weight belongs to $\mathcal{W}_\beta(c_0, C_v, s_0, s_0)$; let q_β be formed with these numbers, so that $\mathsf{A} = 9C_v + 24 + 3\log 7$, and q_β^W with the numbers $(1, 0, s_0, s_0)$ of Wilson's weight. Let $m \in \mathbb{N}$, $\ell = 2L^m$, $n \ge 2$, and let $f_1, \dots, f_n$ be real Lipschitz functions on T_ℓ, each different from 0, such that no point of T_ℓ lies in all of their supports; let $S_n^{(\ell,\mathsf{b})}(\underline{f})$ be the limit of the cutoffs of Wilson's action at the label $\mathsf{b} > 0$ on the side ℓ and $\mathsf{d}_\infty(\underline{f}) > 0$ the spread of the tuple, $\min_{x_i \in \operatorname{supp} f_i} \max_{i,j} \operatorname{dist}_\infty(x_i, x_j)$ on T_ℓ, with the constants of the main text as in Theorem YM3_I:AS.

(a) For each $\mathsf{b} > 0$ and any choice of weights w_K admitted at $\beta = L^K \mathsf{b}$, $S^{\mathsf{w}_K}_{n;K,\ell}(\underline{f}) \to S_n^{(\ell,\mathsf{b})}(\underline{f})$ as $K \to \infty$ through all integers, uniformly on compact sets of labels in $(0,\infty)$. Quantitatively, let $0 < \mathsf{b}_- \le \mathsf{b}_+$, let $s \ge 0$ be an integer with YM3_I:(2210), $0 < \mathsf{d} \le \mathsf{d}_\infty(\underline{f})$, and let C'_s, $K_{1,s}(\mathsf{d})$ and p_n be the numbers of Theorem YM3_I:AS. For $\mathsf{b} \in [\mathsf{b}_-, \mathsf{b}_+]$, integers $K \ge K_{1,s}(\mathsf{d}) + s$ with $\beta = L^K \mathsf{b} > 8/s_0$, and every weight w admitted at β,

$$|S^{\mathsf{w}}_{n;K,\ell}(\underline{f}) - S_n^{(\ell,\mathsf{b})}(\underline{f})| \le L^{sn} \mathsf{C}'_s (L^s \ell)^{3(n+1)} (K-s)^{\mathsf{p}_n} L^{-(K-s)/10} \prod_i \|f_i\|_\sharp$$
$$+ 6\,\mathsf{fu}_n\, \ell^3 L^{3K} \big(q_\beta(s_0) + q_\beta^W(s_0)\big) \prod_i 6\ell^3 L^{4K} \|f_i\|_\infty;$$

here the first term is the bound that Theorem YM3_I:AS gives for Wilson's action, and the second is at most $e^{-\frac{1}{2}c_0 s_0 \mathsf{b}_- L^K} \prod_i \|f_i\|_\infty$ times a sum of two powers of L^K whose exponents and coefficients are formed from n, ℓ, b_+ and C_v.

(b) Let $\mathsf{b} > 0$, let $K_j \to \infty$ and $a_{K_j}\beta_j \to \mathsf{b}$, where the K_j need not increase and may repeat, and let w_j be admitted at β_j. Then $S^{\mathsf{w}_j}_{n;K_j,\ell}(\underline{f}) \to S_n^{(\ell,\mathsf{b})}(\underline{f})$ along the given sequence itself.

(c) For each $\mathsf{b} > 0$ and each integer $d \ge 0$ there is a Borel probability measure $\mu_d[\ell, \mathsf{b}]$ on $\mathcal{V}(\Lambda_{d,\ell})$, the limiting law of the block field of depth d of Wilson's action, such that, for weights w_K admitted at $\beta = L^K \mathsf{b}$, the law of $\mathcal{M}^{K-d}U$ under $\mu^{\mathsf{w}_K}_{\Lambda_{K,\ell}}$ converges to $\mu_d[\ell, \mathsf{b}]$ in total variation as $K \to \infty$. In the setting (E) at the label b (§2.3.1) the distance is at most $\sum_{K' \ge K} \mathfrak{o}_{K'} + 3\ell^3 L^{3K}(q_\beta(s_0) + q_\beta^W(s_0))$ for every integer $K \ge K_{\mathrm{bf}}$ with $\lceil K/1260 \rceil \ge d$ and $\beta > 8/s_0$, where $\mathfrak{o}_{K'}$ is the number of Definition 2.7 (4) and K_{bf} the threshold of Definition 2.7 (5) formed for the interval $\mathsf{J} = \{\mathsf{b}\}$ that consists of the label alone, as in Corollary 2.8 (b).

■

Sketch of proof. Step 1: admitted weights belong to the class. Set $\phi(s) := s$ and $v_\beta(s) := 0$ on $[0, s_0)$, $\phi(s) := c_0 s$ and $v_\beta(s) := -\log(\mathsf{w}(s) e^{c_0 \beta s}) \in (-\infty, \infty]$ on $[s_0, 2]$. Then $\phi \ge c_0 s$ because $c_0 \le 1$, $v_\beta \ge -C_v \log(2+\beta)$, and on $[0, s_0)$ the two functions are the restrictions of the entire functions s and 0, with $|\phi| \le s_0$ and $v_\beta = 0$ on $D(0, s_0)$; thus $\mathsf{w} \in \mathcal{W}_\beta(c_0, C_v, s_0, s_0)$, and Wilson's weight lies in $\mathcal{W}_\beta(1, 0, s_0, s_0)$. Since $s_0 \le 2$, the condition $\beta > 8/s_0$ implies $\beta > 1$.

Step 2: a single cutoff. Proposition 18.9 (b) with $\mathsf{w}_1 = \mathsf{w}$, $\mathsf{w}_2 = e^{-\beta s}$, $\delta = s_0$ and $c = \epsilon = 0$ gives, on $\Lambda_{K,\ell}$, where $|\Lambda_{K,\ell}| = \ell^3 L^{3K}$, and for every event E,

$$|\mu^{\mathsf{w}}_{\Lambda_{K,\ell}}(E) - \mu^{e^{-\beta s}}_{\Lambda_{K,\ell}}(E)| \le 3\ell^3 L^{3K} \big(q_\beta(s_0) + q_\beta^W(s_0)\big),$$
$$|S^{\mathsf{w}}_{n;K,\ell}(\underline{f}) - S^T_{n;K,\ell,\beta}(\underline{f})| \le 6\,\mathsf{fu}_n\, \ell^3 L^{3K} \big(q_\beta(s_0) + q_\beta^W(s_0)\big)$$
$$\times \prod_i 6\ell^3 L^{4K} \|f_i\|_\infty.$$

For $\beta = L^K\mathsf{b}$ with $\mathsf{b} \in [\mathsf{b}_-, \mathsf{b}_+]$, using $c_0 \le 1$ and $2+\beta \le L^K(2+\mathsf{b}_+)$, $q_\beta(s_0) + q^W_\beta(s_0) \le e^{-\frac12 c_0 s_0 \mathsf{b}_- L^K}((L^K(2+\mathsf{b}_+))^{\mathsf{a}} e^{\mathsf{A}} + 7^3 e^{24}(L^K(2+\mathsf{b}_+))^{9/2})$.

Step 3: clauses (a) and (b). For Wilson's action, YM3_I:(2213) bounds $|S^T_{n;K,\ell,L^K\mathsf{b}}(\underline{f}) - S^{(\ell,\mathsf{b})}_n(\underline{f})|$ by the first term of the display, for $\mathsf{b} \in [\mathsf{b}_-, \mathsf{b}_+]$ and $K \ge K_{1,s}(\mathsf{d}) + s$, with s chosen as in YM3_I:(2210). Step 2 contributes the second term, which decays faster than any power of L^{-K}. Theorem YM3_I:AS provides the limit at each $\mathsf{b} > 0$ and the uniformity on compact sets. For (b), YM3_I:(2216) gives $S^T_{n;K_j,\ell,\beta_j}(\underline{f}) \to S^{(\ell,\mathsf{b})}_n(\underline{f})$ along the given sequence, and Step 2 at (K_j, β_j), where $\beta_j = L^{K_j} a_{K_j} \beta_j \ge \frac12 \mathsf{b} L^{K_j}$ for large j, contributes a term tending to 0.

Step 4: clause (c) in the setting (E). Corollary 2.8 (b), (c) supplies Borel probability measures μ_d on $\mathcal{V}(\Lambda_{d,\ell})$ with $|\mu^K_d(E) - \mu_d(E)| \le \sum_{K' \ge K} \mathfrak{o}_{K'}$ for every Borel E and every cutoff K named in (c), where μ^K_d is the law of $\mathcal{M}^{K-d}U$ under Wilson's measure at $\beta = L^K\mathsf{b}$ (Definition 2.7); by Step 2 the law of the same Borel function of U under $\mu^{\mathsf{w}}_{\Lambda_{K,\ell}}$ is within $3\ell^3 L^{3K}(q_\beta(s_0) + q^W_\beta(s_0))$ of μ^K_d.

Step 5: clause (c) at every label on every side. Fix $\mathsf{b} > 0$ and $d \ge 0$, and choose $s \ge d$ so large that $L^s\mathsf{b} \ge \gamma^{-2}$ and $MR(L^s\mathsf{b}) \le L^{m+s}$. Such s exists because $R(x)$ grows at most as a power of $\log x$. Then the setting (E) of §2.3.1 holds at label $L^s\mathsf{b}$ and side $L^s\ell$. The torus $\Lambda_{K,\ell}$ at $\beta = L^K\mathsf{b}$ and the torus $\Lambda_{K-s,L^s\ell}$ at $\beta = L^{K-s}(L^s\mathsf{b})$ are the same lattice with the same Haar measure and the same weight, the dilation δ_s carrying $\Lambda_{j,\ell}$ onto $\Lambda_{j-s,L^s\ell}$; at a bond c of the coarse lattice the averaging map of Section YM3_I:4 is formed from the configuration on the bonds of the double block of c, through plaquette variables, products along contours and their group means, and from the constants ω_M and L^{-3}, while the cutoff and the level occur in that definition only as names of the fine and the coarse lattice. Hence the map $\mathcal{V}(\Lambda_{j,\ell}) \to \mathcal{V}(\Lambda_{j-1,\ell})$ is the same for every cutoff $K \ge j$; and as δ_s maps each block and each contour of $\Lambda_{j,\ell}$, with its sites, bonds and plaquettes, onto the corresponding block or contour of $\Lambda_{j-s,L^s\ell}$, the averaging maps of the two sides are conjugate under δ_s. So the block field of depth s on the side ℓ is the block field of depth 0 on the side $L^s\ell$. By Step 4 on the side $L^s\ell$, its law converges in total variation as $K - s \to \infty$; the block field of depth d is its image under $\mathcal{M}^{s-d}$, and pushing measures forward by a Borel map does not enlarge their distance. So the laws at depth d converge in total variation to the image $\mu_d[\ell, \mathsf{b}]$ of the depth-0 limit on the side $L^s\ell$, which does not depend on s since limits are unique; Step 2 transfers the convergence to admitted weights. □

18.2. **The action density on $\mathbb{R}^3$: its two-point function at and near the diagonal.** For the scalar part of every limit through periodic Gibbs states at a coupling $0 < g \le \gamma_W$ (with γ_W the ceiling of the two-point estimate), that is at the entries $f\mathbf{1}$ of the tensor field $\mathcal{O}^T_K$, $\mathbf{1}$ being the unit 3×3 matrix, where $\mathcal{O}^T_K(f\mathbf{1}) = \mathcal{O}_K(f)$ is the action density, for each component of each canonical decomposition (Theorem YM3_I:AL) of this scalar part, and more generally for every limit in the frame of a Gibbs sequence at g, we describe the connected two-point function S_2 of the action density at and near the diagonal. Away from the diagonal it is the integral of a radial kernel $\mathbf{s}$, the Laplace transform of a positive measure, and $r^6\mathbf{s}(r) \to \frac{9}{16\pi^2\mathsf{b}^2}$ as $r \downarrow 0$, b being the label (Proposition 18.11). For every small $\mathsf{d} > 0$ there is a separated pair of functions with values in $[0,1]$, both supported in one fixed ball and with supports d apart, at which S_2 is at least a fixed positive multiple of d^{-3}; hence S_2 is represented neither by a complex Radon measure nor by a locally integrable function on $\mathbb{R}^3 \times \mathbb{R}^3$. The two functions of such a pair have the norm $\|\cdot\|_\sharp$ equal to one fixed multiple of d^{-1}, so that the product of their norms is a fixed multiple of d^{-2}; since d^{-3} outgrows d^{-2}, no continuation of S_2 is continuous for the norm $\|\cdot\|_\sharp$ (Proposition 18.12). The kernel becomes a distribution in the difference of the two points once two local terms are removed whose coefficients grow like $\frac{3}{4\pi\mathsf{b}^2}\epsilon^{-3}$ and $\frac{3}{8\pi\mathsf{b}^2}\epsilon^{-1}$ as the excised radius ϵ tends to 0, and every extension of it has order at least 4 at the origin and scaling degree at least 6 (Proposition 18.13). Those extensions which are invariant under O(3) and of scaling degree at most 6 form precisely the plane of the distributions $\mathbf{T}_0 + c_0\delta + c_1\Delta\delta$, $(c_0, c_1) \in \mathbb{R}^2$; they share one scaling limit, $\frac{9}{16\pi^2\mathsf{b}^2}$ times the

finite part of $|u|^{-6}$, which acts as $\varphi \mapsto \frac{3}{64\mathsf{b}^2}\int |p|^3\hat{\varphi}(p)\frac{dp}{(2\pi)^3}$ (Proposition 18.14). The points (c_0, c_1) at which the bilinear form of $\mathbf{T}_0 + c_0\delta + c_1\Delta\delta$ is of positive type, that is non-negative at every pair (f, f) with f real, smooth and compactly supported, form a convex set which is not empty and contains with each of its points a whole quadrant; so asking for positive type leaves admissible extensions and does not pick out one of them (Proposition 18.15). The torus, with all six entries, is treated in (F8) of Theorem D; here the space is $\mathbb{R}^3$ and the entries are scalar.

Setting. Fix the construction of the scalar field in YM3 I and the ceiling γ_W of its two-point estimate (Section YM3_I:127). Write $\bar{\varepsilon}_{g,\vartheta}(\mathsf{d})$ and $\mathsf{d}_0(g,\vartheta)$ for its tolerance and threshold, as in YM3_I:(2349)–(2350). The tolerance depends only on the construction, g, ϑ and d. It is non-decreasing in d, tends to zero as $\mathsf{d} \downarrow 0$ and is at most $1/3$ for $0 < \mathsf{d} \le \mathsf{d}_0(g,\vartheta)$, where $0 < \mathsf{d}_0(g,\vartheta) \le 1/200$.

The test pairs $\mathfrak{T}_{\mathbb{R}^3}(\mathsf{d};\vartheta)$ in the estimate consist of two nonzero nonnegative Lipschitz functions f, h supported in closed balls of radius d. The distance of their supports lies in $[\mathsf{d}, 6\mathsf{d}]$. Each entry u satisfies $\int u \ge \vartheta\|u\|_\infty \mathsf{d}^3$ and $\operatorname{Lip} u \le \|u\|_\infty/(\vartheta\mathsf{d})$. The superscript 1 requires both functions to be of class C^1. Set $\mathcal{K}_{\mathrm{free}}(f,h) = \iint f(x)h(y)|x-y|^{-6}\,dx\,dy$ and $\mathsf{c}_2 = 9/(16\pi^2)$.

Write $\mathfrak{S}_n(\mathbb{R}^3)$ for the tuples of n real C^1 functions with compact pairwise disjoint supports, including zero entries. Let $\mathfrak{S} = \bigcup_{n\ge 0}\mathfrak{S}_n(\mathbb{R}^3)$, where $\mathfrak{S}_0(\mathbb{R}^3) = \{\varnothing\}$. Put $\|f\|_\sharp = \max\{\sup|f|, 40M\operatorname{Lip} f\}$, let $\mathbb{B}(x,R)$ be the closed ball and B_R the open ball about zero, and write $(f\otimes h)(x,y) = f(x)h(y)$. For $\eta > 0$, let $\mathsf{v}^\eta_z(y) = (1-|y-z|^2/\eta^2)^2$ on $\mathbb{B}(z,\eta)$ and zero outside. The derivative vanishes on the boundary, so this function is C^1. It has supremum 1, integral $4\pi\eta^3(1/3-2/5+1/7) = 32\pi\eta^3/105$, and Lipschitz constant $8/(3\sqrt{3}\,\eta)$, the maximum of $4s(1-s^2)/\eta$ for $0 \le s \le 1$.

Limits in a frame. For $0 < g \le \gamma_W$, a Gibbs sequence consists of first-passage data $(K_j, \beta_j) \in \mathfrak{d}(g)$ with $K_j \to \infty$ and periodic Gibbs laws $\nu_j \in \mathcal{G}^{\mathrm{per}}_{K_j,\beta_j}$. For $X = (f_1,\dots,f_n) \in \mathfrak{S}$ and a law ν at cutoff K, set $X^\nu_K = \prod_i(\mathcal{O}_K(f_i) - \nu(\mathcal{O}_K(f_i)))$, with the empty product equal to one. A limit in the frame is a functional $\sigma'(X) = \lim_j \rho_j(X^{\nu_j}_{K_j})$ on all these tuples for some approximants $\rho_j \in \mathcal{G}^{\mathrm{per}}_{K_j,\beta_j}$. We use three cases.

(i) If the Gibbs sequence converges in its own centering, its limit is obtained with $\rho_j = \nu_j$.

(ii) Theorem YM3_I:AP, with Theorem YM3_I:AL, supplies canonical decompositions of such a limit and obtains every component along the same sequence of cutoffs. Its approximants are $\rho_j = \nu_j(\,\cdot\,|E_j)$ for Borel events E_j invariant under all lattice translations and of positive ν_j-probability. These conditional laws are periodic Gibbs laws.

(iii) For a tensor limit through periodic Gibbs states, the scalar part is $X \mapsto \sigma^T(f_1\mathbf{1},\dots,f_n\mathbf{1})$. At cutoff K_j the scalar tensor insertion $\mathcal{O}^T_{K_j}(f\mathbf{1})$ is exactly $\mathcal{O}_{K_j}(f)$. The moment–cumulant formula for products centered in ν_j sums only over partitions with no singleton blocks. Thus convergence of the tensor cumulants on separated tuples gives convergence of the scalar centered products, with limit equal to the scalar part. This is case (i). The approximants in (F6) of Theorem D likewise give the scalar parts of its components as limits in the same frame when $g \le \min\{\gamma_2, \gamma_W\}$, where γ_2 is the ceiling in (F6).

The two-point function and the label. For a limit in the frame define $S_2(f,h) = \sigma'(f,h) - \sigma'((f))\sigma'((h))$ on separated pairs. Subtracting the product of the two one-point expectations cancels the choice of centering at each cutoff. Hence $S_2(f,h) = \lim_j\langle\mathcal{O}_{K_j}(f);\mathcal{O}_{K_j}(h)\rangle_{\rho_j}$. Proposition YM3_I:127.8 gives convergence of the physical labels $a_{K_j}\beta_j$ to a single b and the leading law for S_2. The limit label depends only on the sequence of data and lies in $I^{\mathrm{lab}}(g) \subset [g^{-2}, \bar{x}_0]$, where $\bar{x}_0 = L(g^{-2}+\bar{B})$, by the first-passage bounds (Corollary YM3_I:74.2). In case (iii), $S_2(f,h) = S_2^T(f\mathbf{1}, h\mathbf{1})$ and b is the label of the tensor limit in (F1) of Theorem D. Write $\mathfrak{a} = \mathsf{c}_2\mathsf{b}^{-2}$.

Words of analysis. $\mathcal{D} = C^\infty_c(\mathbb{R}^3)$ is the space of the real smooth functions with compact support, $\mathcal{S}(\mathbb{R}^3)$ the Schwartz space, and $\hat{\psi}(p) = \int e^{-ip\cdot u}\psi(u)\,du$; $\|\varphi\|_{C^N} = \max_{|\alpha|\le N}\sup|\partial^\alpha\varphi|$, $\|\varphi\|_{(4)} = \sup_u \max_{|e|=1}|(e\cdot\nabla)^4\varphi(u)|$, and $\mathrm{av}_r\varphi = \frac{1}{4\pi}\int_{\mathbb{S}^2}\varphi(r\mathbf{n})\,d\mathbf{n}$ is the mean of φ over the sphere of

radius r about 0. By a distribution we mean a real linear functional $\mathbf{T}$ on $\mathcal{D}$ which, on the functions supported in each ball $\mathbb{B}(0,R)$, obeys a bound $|\mathbf{T}(\varphi)| \le C\|\varphi\|_{C^N}$ with some N and C depending on R; when such a bound is available at $R=1$ with a given N, $\mathbf{T}$ has order $\le N$ near 0. δ and $\Delta\delta$ denote $\varphi \mapsto \varphi(0)$ and $\varphi \mapsto \Delta\varphi(0)$. A distribution $\mathbf{T}$ extends a continuous function $\mathbf{s}$ on $(0,\infty)$, and is then called an extension of $\mathbf{s}$, when $\mathbf{T}(\varphi) = \int \mathbf{s}(|u|)\varphi(u)\,du$ for all $\varphi \in \mathcal{D}$ whose support avoids some ball about 0. For $f, h \in \mathcal{D}$ let $\varphi_{f,h}(u) = \int f(y+u)h(y)\,dy$; then $\varphi_{f,h} \in \mathcal{D}$, $\varphi_{f,h}(0) = \int fh$, $\Delta\varphi_{f,h}(0) = \int(\Delta f)h = -\int \nabla f \cdot \nabla h$, $\int \varphi_{f,f} = (\int f)^2$, $\hat\varphi_{f,f} = |\hat f|^2$, a common translation of f and h leaves $\varphi_{f,h}$ unchanged, and $\varphi_{f\circ\mathsf{R}^{-1},h\circ\mathsf{R}^{-1}} = \varphi_{f,h}\circ\mathsf{R}^{-1}$ for $\mathsf{R} \in \mathrm{O}(3)$; the letter R always denotes an element of $\mathrm{O}(3)$, while R and R_0 are radii. To a distribution $\mathbf{T}$ we attach the bilinear form $\mathbf{S}_{\mathbf{T}}(f,h) = \mathbf{T}(\varphi_{f,h})$ on $\mathcal{D}$. When $\operatorname{dist}(\operatorname{supp} f, \operatorname{supp} h) = d' > 0$, $\varphi_{f,h}$ vanishes on $B_{d'}$, and if $\mathbf{T}$ extends $\mathbf{s}$, Fubini's theorem turns $\mathbf{S}_{\mathbf{T}}(f,h)$ into $\iint f(x)h(y)\mathbf{s}(|x-y|)\,dx\,dy$.

For $t>0$ the dilate of a distribution $\mathbf{T}$ is $\mathbf{T}_t(\varphi) = t^{-3}\mathbf{T}(\varphi(\cdot/t))$; we say that $\mathbf{T}$ is of scaling degree at most ω_0 when $t^\omega \mathbf{T}_t(\varphi) \to 0$ as $t \downarrow 0$ for all $\omega > \omega_0$ and all $\varphi \in \mathcal{D}$. We call an extension admissible when $\mathbf{T}(\varphi\circ\mathsf{R}) = \mathbf{T}(\varphi)$ for every $\mathsf{R} \in \mathrm{O}(3)$ and its scaling degree is at most 6; and $\mathbf{T}$ has positive type when $\mathbf{S}_{\mathbf{T}}(f,f) \ge 0$ for all $f \in \mathcal{D}$. Hadamard's finite part of $|u|^{-6}$ is the functional $\mathrm{Pf}|u|^{-6}(\varphi) = 4\pi\int_0^\infty r^{-4}[\mathrm{av}_r\varphi - \varphi(0) - \frac{r^2}{6}\Delta\varphi(0)]\,dr$, defined for φ in $\mathcal{D}$ and in $\mathcal{S}(\mathbb{R}^3)$; the integral converges absolutely, by Step 6 of the proof of Proposition 18.14. By a complex Radon measure on $\mathbb{R}^3\times\mathbb{R}^3$ we mean a $\mathbb{C}$-linear functional m on the continuous compactly supported complex functions on $\mathbb{R}^6$ that is bounded on each compact set: for each compact $\mathcal{C} \subset \mathbb{R}^6$ there is $C_{\mathcal{C}}$ with $|m(F)| \le C_{\mathcal{C}} \sup|F|$ whenever $\operatorname{supp} F \subset \mathcal{C}$. Integration against a locally integrable function k is one, with $C_{\mathcal{C}} = \int_{\mathcal{C}} |k|$. e is Euler's number.

Proposition 18.11 (The radial kernel at short distance)**.** Let $0 < g \le \gamma_W$, and let σ' be a limit in the frame of a Gibbs sequence at g, with S_2, $\mathfrak{b}$ and $\mathfrak{a}$ as above.

(a) A unique continuous function $\mathbf{s}:(0,\infty)\to\mathbb{R}$ represents S_2 on separated pairs: $S_2(f,h) = \iint f(x)h(y)\,\mathbf{s}(|x-y|)\,dx\,dy$ for every $(f,h) \in \mathfrak{S}_2(\mathbb{R}^3)$. This $\mathbf{s}$ is real-analytic on $(0,\infty)$ and satisfies $0 \le \mathbf{s}(r) \le C_{\mathrm{pt}}^2 \min\{1, \frac{r}{2}\}^{-8}$, where $C_{\mathrm{pt}} = C_{\mathrm{pt}}^g$ is the point-field constant in YM3_I:(2097).

(b) The kernel is a Laplace transform: $\mathbf{s}(r) = \int_{[0,\infty)} e^{-rE}\,d\mathfrak{m}(E)$ for all $r>0$, with a positive Borel measure $\mathfrak{m}$ on $[0,\infty)$ that gives finite mass to bounded sets; hence $\mathbf{s}$ is non-increasing.

(c) As $r \downarrow 0$, $r^6\mathbf{s}(r) \to \mathfrak{a} = \frac{9}{16\pi^2\mathfrak{b}^2}$. Quantitatively: given $\zeta \in (0, \frac12]$, with $\vartheta_\zeta := \frac{32\pi}{105}\zeta^3$ and $q_\zeta := \frac{3+2\zeta}{3-2\zeta}$, every $r \in (0, (3-2\zeta)\,\mathfrak{d}_0(g,\vartheta_\zeta)]$ satisfies $q_\zeta^{-6}(1-\bar\varepsilon_{g,\vartheta_\zeta}(\frac{r}{3-2\zeta})) \le r^6\mathbf{s}(r)/\mathfrak{a} \le q_\zeta^6(1+\bar\varepsilon_{g,\vartheta_\zeta}(\frac{r}{3+2\zeta}))$. For $\zeta = \frac12$ this yields $\frac{1}{96}\mathfrak{a}\,r^{-6} \le \mathbf{s}(r) \le \frac{256}{3}\mathfrak{a}\,r^{-6}$ on $0 < r \le \mathbf{r}_g := 2\,\mathfrak{d}_0(g, \frac{4\pi}{105})$, and $\mathbf{c}_6 := \sup_{0<r\le1} r^6\mathbf{s}(r) \le \frac{256}{3}\mathfrak{a}\,\mathbf{r}_g^{-6}$. The numbers ϑ_ζ, q_ζ, $\bar\varepsilon_{g,\vartheta}$, $\mathfrak{d}_0(g,\vartheta)$ and $\mathbf{r}_g$ are common to all Gibbs sequences at g and all limits in their frames.

■

Proof. Step 1: the centred functional. For each f, the difference $\rho_j(\mathcal{O}_{K_j}(f)) - \nu_j(\mathcal{O}_{K_j}(f))$ converges to $\sigma'((f))$. Expanding a product centered in ρ_j in terms of products centered in ν_j thus gives a limit on every separated tuple. Denote it by σ'^{c}. It has zero one-point function and equals S_2 on separated pairs. Proposition YM3_I:111.1 gives scalar reconstruction data for this centred limit, with exponent $\wp = 4$ and point-field constant C_{pt}^g. Its proof gives the centred insertion estimates required for the point fields of Proposition YM3_I:97.2.

Step 2: the radial kernel. Proposition YM3_I:108.3 represents σ'^{c} on separated pairs by a unique real-analytic kernel on $\{(x,y) : x \ne y\}$. Euclidean invariance makes this kernel $\mathbf{s}(|x-y|)$. To identify its value, let $u = (r/2)e_3$ and let h_δ be nonnegative smooth approximate identities at u of integral 1, carried by $0 < x_3 < r$. Write $\theta x = (x_1, x_2, -x_3)$. In the reflection Hilbert space of σ'^{c}, $\|[(h_\delta)]\|^2 = \sigma'^{\mathrm{c}}(\theta h_\delta, h_\delta) \to \mathbf{s}(r)$. The slab insertion formula and the point-field integral of

Proposition YM3_I:97.2 give

$$[(h_\delta)] = \int h_\delta(x)\, B(x, re_3 - x)\Omega_{\mathrm{c}}\, dx \;\longrightarrow\; B(u,u)\Omega_{\mathrm{c}}.$$

Here the slab operator for h_δ applied to the vacuum is its one-entry vector, and the convergence follows from norm continuity of B near (u,u). The shift law and $\mathsf{T}_3(te_3)\Omega_{\mathrm{c}} = \Omega_{\mathrm{c}}$ allow the second argument to be changed from u to e_3. Apply the shift law from the smaller to the larger height in the second slot. Therefore $\mathbf{s}(r) = \|B((r/2)e_3, e_3)\Omega_{\mathrm{c}}\|^2$, and the norm estimate in Proposition YM3_I:97.2 gives (a).

Step 3: the semigroup and its spectral measure. Put $\psi_\epsilon = B(\epsilon e_3, e_3)\Omega_{\mathrm{c}}$ for $\epsilon > 0$. The shift law in the first slot gives $\psi_\epsilon = e^{-(\epsilon-\epsilon')H}\psi_{\epsilon'}$ for $0 < \epsilon' \le \epsilon$. Here $H \ge 0$ is the generator of the reflection semigroup. Let $\mathfrak{m}_\epsilon$ be the positive spectral measure of H in ψ_ϵ. By the spectral theorem,

$$d\mathfrak{m}_\epsilon(E) = e^{-2(\epsilon-\epsilon')E}\, d\mathfrak{m}_{\epsilon'}(E).$$

Thus $e^{2\epsilon E} d\mathfrak{m}_\epsilon(E)$ is independent of ϵ. Denote this positive measure by $d\mathfrak{m}(E)$. Its mass on $[0,N]$ is at most $e^{2\epsilon N}\|\psi_\epsilon\|^2$ for every $\epsilon > 0$. Since $\mathbf{s}(r) = \|\psi_{r/2}\|^2$, we obtain $\mathbf{s}(r) = \int e^{-rE} d\mathfrak{m}(E)$ and monotonicity, proving (b).

Step 4: the two-point law, at each shape. Proposition YM3_I:127.8 applies to the converging covariances in the frame. It gives, at each $0 < \mathsf{d} \le \mathsf{d}_0(g,\vartheta)$ and each $(f,h) \in \mathfrak{T}^1_{\mathbb{R}^3}(\mathsf{d};\vartheta)$,

$$|S_2(f,h) - \mathfrak{a}\,\mathcal{K}_{\mathrm{free}}(f,h)| \le \bar\varepsilon_{g,\vartheta}(\mathsf{d})\,\mathfrak{a}\,\mathcal{K}_{\mathrm{free}}(f,h). \tag{L}$$

Step 5: a test pair made of two small bumps. Let $0 < \zeta \le \frac12$ and $\mathsf{d} > 0$, and consider $f = \mathsf{v}_0^{\zeta\mathsf{d}}$ and $h = \mathsf{v}_{3\mathsf{d}e_3}^{\zeta\mathsf{d}}$, with the centres 0 and $3\mathsf{d}e_3$. The test-pair conditions at $(\mathsf{d};\vartheta_\zeta)$ hold. Each support is the ball of radius $\zeta\mathsf{d} \le \mathsf{d}$ about its centre, and the two supports are $(3-2\zeta)\mathsf{d} \in [2\mathsf{d}, 3\mathsf{d})$ apart; $\int f = \int h = \vartheta_\zeta \mathsf{d}^3$, both suprema being 1; and $\operatorname{Lip} f = \operatorname{Lip} h = \frac{8}{3\sqrt3\,\zeta\mathsf{d}}$ does not exceed $\frac{1}{\vartheta_\zeta\mathsf{d}}$, because $\vartheta_\zeta \le \frac{32\pi}{420}\zeta < 0.24\,\zeta < \frac{3\sqrt3}{8}\zeta$. Also $0 < \vartheta_\zeta \le \frac{4\pi}{105} < 1$, and f, h are non-negative, non-zero and of class C^1. Thus $(f,h) \in \mathfrak{T}^1_{\mathbb{R}^3}(\mathsf{d};\vartheta_\zeta)$.

Step 6: (c). Take $0 < \mathsf{d} \le \mathsf{d}_0(g,\vartheta_\zeta)$, the pair (f,h) of Step 5, and $m = \int f \int h > 0$. For $x \in \operatorname{supp} f$ and $y \in \operatorname{supp} h$ the distance $|x-y|$ lies between $(3-2\zeta)\mathsf{d}$ and $(3+2\zeta)\mathsf{d}$. Since f and h are non-negative and $\mathbf{s}$ is non-increasing, the representation (a) squeezes $S_2(f,h)$ between $\mathbf{s}((3+2\zeta)\mathsf{d})\,m$ and $\mathbf{s}((3-2\zeta)\mathsf{d})\,m$, and the same reasoning squeezes $\mathcal{K}_{\mathrm{free}}(f,h)$ between $((3+2\zeta)\mathsf{d})^{-6}m$ and $((3-2\zeta)\mathsf{d})^{-6}m$. Combined with (L) at $\vartheta = \vartheta_\zeta$ this gives $\mathbf{s}((3+2\zeta)\mathsf{d}) \le (1+\bar\varepsilon_{g,\vartheta_\zeta}(\mathsf{d}))\,\mathfrak{a}\,((3-2\zeta)\mathsf{d})^{-6}$ and $\mathbf{s}((3-2\zeta)\mathsf{d}) \ge (1-\bar\varepsilon_{g,\vartheta_\zeta}(\mathsf{d}))\,\mathfrak{a}\,((3+2\zeta)\mathsf{d})^{-6}$. Writing r for $(3+2\zeta)\mathsf{d}$ in the upper and for $(3-2\zeta)\mathsf{d}$ in the lower inequality turns them into the two bounds of (c), valid for r up to $(3+2\zeta)\mathsf{d}_0(g,\vartheta_\zeta)$ and up to $(3-2\zeta)\mathsf{d}_0(g,\vartheta_\zeta)$ respectively. Letting $r \downarrow 0$ at fixed ζ, where $\bar\varepsilon_{g,\vartheta_\zeta} \to 0$, shows that the lower and upper limits of $r^6\mathbf{s}(r)/\mathfrak{a}$ lie in $[q_\zeta^{-6}, q_\zeta^6]$; as $q_\zeta \to 1$ for $\zeta \to 0$, the limit is 1. For $\zeta = \frac12$ one has $q_\zeta = 2$, $\vartheta_\zeta = \frac{4\pi}{105}$ and $3-2\zeta = 2$, and the tolerances at $\frac{r}{2}$ and $\frac{r}{4}$, both $\le \mathsf{d}_0(g, \frac{4\pi}{105})$ when $r \le \mathbf{r}_g$, are at most $\frac13$; this gives the factors $\frac23 \cdot 2^{-6} = \frac{1}{96}$ and $\frac43 \cdot 2^6 = \frac{256}{3}$. Finally $\mathbf{r}_g \le \frac{1}{100}$: on $[\mathbf{r}_g, 1]$, $r^6\mathbf{s}(r) \le \mathbf{s}(r) \le \mathbf{s}(\mathbf{r}_g) \le \frac{256}{3}\mathfrak{a}\,\mathbf{r}_g^{-6}$ by (b), and on $(0, \mathbf{r}_g]$, $r^6\mathbf{s}(r) \le \frac{256}{3}\mathfrak{a} \le \frac{256}{3}\mathfrak{a}\,\mathbf{r}_g^{-6}$. The numbers listed at the end of (c) are built from the radius, the constants and g (YM3_I:(2349)), not from the sequence or the limit. □

Proposition 18.12 (Neither a measure nor a locally integrable function; no continuous continuation)**.** Let g, σ', S_2, $\mathfrak{a}$ and $\mathbf{s}$ be as in Proposition 18.11. For $0 < \mathsf{d} \le \mathsf{d}_0(g, \frac12)$ let $N = \lfloor \frac{1}{8\mathsf{d}} \rfloor$, $\mathsf{G} = \{0, \dots, N-1\}^3$, $x_\iota = 8\mathsf{d}\,\iota$ and $y_\iota = x_\iota + 3\mathsf{d}\,e_3$ for $\iota \in \mathsf{G}$, and $f_\mathsf{d} = \sum_{\iota\in\mathsf{G}} \mathsf{v}^{\mathsf{d}}_{x_\iota}$, $h_\mathsf{d} = \sum_{\iota\in\mathsf{G}} \mathsf{v}^{\mathsf{d}}_{y_\iota}$.

(a) The pair $(f_\mathsf{d}, h_\mathsf{d})$ is separated; both functions take values in $[0,1]$, both supports lie in $\mathbb{B}(0,3)$ and are d apart, $\operatorname{Lip} f_\mathsf{d} = \operatorname{Lip} h_\mathsf{d} = \frac{8}{3\sqrt3}\mathsf{d}^{-1}$, $\|f_\mathsf{d}\|_\sharp = \|h_\mathsf{d}\|_\sharp = \frac{320M}{3\sqrt3\,\mathsf{d}}$, and $S_2(f_\mathsf{d}, h_\mathsf{d}) \ge \frac{10^{-6}}{24576}\,\mathfrak{a}\,\mathsf{d}^{-3}$.

(b) There is no complex Radon measure m on $\mathbb{R}^3\times\mathbb{R}^3$ such that $S_2(f,h)=m(f\otimes h)$ on all of $\mathfrak{S}_2(\mathbb{R}^3)$; a fortiori no locally integrable k on $\mathbb{R}^3\times\mathbb{R}^3$ gives $S_2(f,h)=\iint f(x)h(y)k(x,y)\,dx\,dy$ on these pairs. Moreover $\int_{B_R}\mathfrak{s}(|u|)\,du=\infty$ for every $R>0$.

(c) Fix $0<\mathsf{d}\le\mathsf{d}_0(g,\frac12)$. A number C with $|S_2(f,h)|\le C\|f\|_\sharp\|h\|_\sharp$ for every $(f,h)\in\mathfrak{S}_2(\mathbb{R}^3)$ whose supports lie in $\mathbb{B}(0,3)$ and are at least d apart must be $\ge 10^{-14}\mathfrak{a}\,M^{-2}\mathsf{d}^{-1}$; with $\sup|f|\sup|h|$ instead of $\|f\|_\sharp\|h\|_\sharp$ it must be $\ge\frac{10^{-6}}{24576}\mathfrak{a}\,\mathsf{d}^{-3}$. Hence $|S_2(f,h)|$ admits no bound $C\|f\|_\sharp\|h\|_\sharp$ with one C for all separated pairs supported in $\mathbb{B}(0,3)$. In particular, for $R\ge3$, the number $C_2^g(\mathsf{d};R)$ of the uniform cumulant bound in Proposition YM3_I: 91.13 is at least $10^{-14}\mathsf{c}_2\,\bar{x}_0^{-2}M^{-2}\mathsf{d}^{-1}$; and when moreover $g\le\gamma_1$, γ_1 being the bound on the coupling in (C1) and (D1) of Theorem C, the number $C_2^T(\mathsf{d};R)$ of (C1) obeys $3600\,C_2^T(\mathsf{d};R)\ge\frac{10^{-6}}{24576}\mathsf{c}_2\,\mathsf{d}^{-3}$.

(d) The separated pairs $(\mathsf{d}^{5/4}f_\mathsf{d},\mathsf{d}^{5/4}h_\mathsf{d})$ satisfy $\|\mathsf{d}^{5/4}f_\mathsf{d}\|_\sharp=\|\mathsf{d}^{5/4}h_\mathsf{d}\|_\sharp=\frac{320M}{3\sqrt3}\mathsf{d}^{1/4}\to0$ and $S_2(\mathsf{d}^{5/4}f_\mathsf{d},\mathsf{d}^{5/4}h_\mathsf{d})\ge\frac{10^{-6}}{24576}\mathfrak{a}\,\mathsf{d}^{-1/2}\to\infty$ as $\mathsf{d}\to0$. Consequently, if Φ is any map into $\mathbb{R}$ defined on a set of pairs of real C^1 functions supported in $\mathbb{B}(0,3)$ which contains $(0,0)$ and all these pairs, and $\Phi=S_2$ at the separated pairs of that set, then Φ is discontinuous at $(0,0)$ for the distance $\max\{\|f-f'\|_\sharp,\|h-h'\|_\sharp\}$. In particular no bilinear form on these functions, continuous for $\|\cdot\|_\sharp$, coincides with S_2 on separated pairs, and S_2 itself is discontinuous at the separated pair $(0,0)$, where its value is 0.

■

Proof. Step 1: geometry of the bumps. From $\mathsf{d}\le\mathsf{d}_0(g,\frac12)\le\frac{1}{200}$ we get $N\ge\frac{1}{8\mathsf{d}}-1\ge\frac{1}{16\mathsf{d}}\ge1$. Every coordinate of an x_ι is in $[0,1)$, and $y_\iota\in[0,1)^2\times[3\mathsf{d},1+3\mathsf{d})$, so each ball $\mathbb{B}(x_\iota,\mathsf{d})$ or $\mathbb{B}(y_\iota,\mathsf{d})$ is contained in $\mathbb{B}(0,\sqrt3+4\mathsf{d})\subset\mathbb{B}(0,3)$. Two different x_ι, or two different y_ι, are at least $8\mathsf{d}$ apart. The vector $x_\iota-y_{\iota'}=8\mathsf{d}(\iota-\iota')-3\mathsf{d}e_3$ has length $3\mathsf{d}$ for $\iota=\iota'$; for $\iota\ne\iota'$ its length is $\ge5\mathsf{d}$: if ι and ι' differ in the third entry only, $|8n-3|\ge5$ for each integer $n\ne0$, and otherwise some other coordinate has modulus $\ge8\mathsf{d}$. Hence no two of the $2N^3$ balls meet; the support of f_d is the union of the balls centred at the x_ι, that of h_d the union of those centred at the y_ι, both compact, and their distance is $3\mathsf{d}-2\mathsf{d}=\mathsf{d}$; so $(f_\mathsf{d},h_\mathsf{d})\in\mathfrak{S}_2(\mathbb{R}^3)$. As the bumps of one sum have disjoint supports, f_d and h_d are C^1 with values in $[0,1]$ and gradients of supremum $\frac{8}{3\sqrt3}\mathsf{d}^{-1}$, which is their Lipschitz constant by the mean-value formula. The definition of $\|\cdot\|_\sharp$ gives $\|f_\mathsf{d}\|_\sharp=\|h_\mathsf{d}\|_\sharp=\max\{1,\frac{320M}{3\sqrt3\,\mathsf{d}}\}=\frac{320M}{3\sqrt3\,\mathsf{d}}$.

Step 2: (a). Expanding f_d and h_d in the representation of Proposition 18.11(a) writes $S_2(f_\mathsf{d},h_\mathsf{d})$ as $\sum_{\iota,\iota'\in\mathsf{G}}\iint\mathsf{v}^\mathsf{d}_{x_\iota}(x)\mathsf{v}^\mathsf{d}_{y_{\iota'}}(y)\,\mathfrak{s}(|x-y|)\,dx\,dy$, a sum of non-negative numbers since $\mathfrak{s}\ge0$; the summand with $\iota=\iota'$ equals $S_2(\mathsf{v}^\mathsf{d}_{x_\iota},\mathsf{v}^\mathsf{d}_{y_\iota})$. Each diagonal pair is in $\mathfrak{T}^1_{\mathbb{R}^3}(\mathsf{d};\frac12)$. Its two supports have radius d and distance d, and the bump integral and Lipschitz bounds above give the remaining two conditions. Inequality (L) of the proof of Proposition 18.11 at $\vartheta=\frac12$, with $\bar\varepsilon_{g,1/2}(\mathsf{d})\le\frac13$, together with $\mathcal{K}_{\text{free}}\ge(32\pi/105)^2 5^{-6}>10^{-6}/4$ (all point distances are at most $5\mathsf{d}$), bounds each diagonal summand below by $\frac23\,\mathfrak{a}\,10^{-6}\cdot\frac14=\frac{10^{-6}}{6}\mathfrak{a}$. There are $N^3\ge(16\mathsf{d})^{-3}$ of them, so $S_2(f_\mathsf{d},h_\mathsf{d})\ge\frac{10^{-6}}{6\cdot16^3}\mathfrak{a}\,\mathsf{d}^{-3}$, where $6\cdot16^3=24576$.

Step 3: (b). Suppose m were a measure as excluded in (b), and let $\mathcal{C}=\mathbb{B}(0,3)\times\mathbb{B}(0,3)$. The continuous function $f_\mathsf{d}\otimes h_\mathsf{d}$ is supported in $\mathcal{C}$ and bounded by 1, so $S_2(f_\mathsf{d},h_\mathsf{d})=m(f_\mathsf{d}\otimes h_\mathsf{d})$ would stay below $C_\mathcal{C}$ for all $\mathsf{d}\in(0,\mathsf{d}_0(g,\frac12)]$, which (a) contradicts for small d. Since integration against a locally integrable k is such a measure, no such k exists either. For the last claim, Proposition 18.11(c) gives $\mathfrak{s}(r)\ge\frac{1}{96}\mathfrak{a}\,r^{-6}$ on $(0,\mathbf{r}_g]$, hence $\int_{B_R}\mathfrak{s}(|u|)\,du\ge\frac{4\pi}{96}\mathfrak{a}\int_0^{\min\{R,\mathbf{r}_g\}}r^{-4}dr=\infty$.

Step 4: (c). The hypothesis on C covers the pair $(f_\mathsf{d},h_\mathsf{d})$. From (a) and Step 1, $C\ge\frac{10^{-6}}{24576}\mathfrak{a}\,\mathsf{d}^{-3}\cdot\frac{27\,\mathsf{d}^2}{320^2M^2}=\frac{27\cdot10^{-6}}{24576\cdot102400}\mathfrak{a}\,M^{-2}\mathsf{d}^{-1}\ge10^{-14}\mathfrak{a}\,M^{-2}\mathsf{d}^{-1}$, since $\frac{27}{24576\cdot102400}>1.07\cdot10^{-8}$; with suprema, both equal to 1, one gets $C\ge\frac{10^{-6}}{24576}\mathfrak{a}\,\mathsf{d}^{-3}$. A single C valid for all separated pairs in $\mathbb{B}(0,3)$ would be admissible at every d, which these lower bounds forbid as $\mathsf{d}\to0$. For the two consequences,

note that the pair has supports in $\mathbb{B}(0,R)$ at distance d for every $R \geq 3$. Choose a Gibbs sequence at a fixed label in $I(g)$ and extract a convergent subsequence by Theorem YM3_I:AP. Its limit has $\mathfrak{a} \geq \mathsf{c}_2 \bar{x}_0^{-2}$. The uniform bound of Proposition YM3_I:91.13 passes to its S_2 with the same $C_2^g(\mathsf{d};R)$, so the first estimate of (c) applies with $C = C_2^g(\mathsf{d};R)$. If also $g \leq \gamma_1$, (D1) of Theorem C extracts from any Gibbs sequence at g a subsequence along which all $S_n^T[\nu_j](X)$ converge, that is a limit through periodic Gibbs states, whose scalar part is a case (iii) of σ'; by (D1) its S_2^T obeys the bound of (C1) with b for b, so that $S_2(f_\mathsf{d}, h_\mathsf{d}) = S_2^T(f_\mathsf{d}\mathbf{1}, h_\mathsf{d}\mathbf{1}) \leq 3600\, C_2^T(\mathsf{d};R)\, \mathsf{b}^{-2}$, the suprema of the entries $f_\mathsf{d}\mathbf{1}$, $h_\mathsf{d}\mathbf{1}$ being 1; comparing with (a) gives the last bound.

Step 5: (d). Multiplying both entries by $c > 0$ keeps the pair separated, with the supports of Step 1, and multiplies S_2 by c^2 (Proposition 18.11(a)); for $c = \mathsf{d}^{5/4}$, Step 1 and (a) give the norms and the lower bound stated. Were Φ continuous at $(0,0)$, its values at these pairs would converge to the real number $\Phi(0,0)$ as $\mathsf{d} \to 0$; but they are values of S_2 and diverge. A bilinear form continuous for $\|\cdot\|_\sharp$ would be such a Φ. Finally, $(0,0)$ is a separated pair, the zero function being an allowed entry, and $S_2(0,0) = 0$ by Proposition 18.11(a). □

Proposition 18.13 (Two divergent local counterterms; an admissible extension)**.** Let g, σ', S_2, $\mathfrak{a}$, $\mathbf{s}$ and $\mathbf{c}_6$ be as in Proposition 18.11.

(a) Let f and h be real, continuous and compactly supported on $\mathbb{R}^3$. Then, as $\epsilon \downarrow 0$,
$$\epsilon^3 \iint_{|x-y|>\epsilon} f(x)h(y)\,\mathbf{s}(|x-y|)\,dx\,dy \longrightarrow \frac{4\pi}{3}\mathfrak{a}\int fh = \frac{3}{4\pi\mathsf{b}^2}\int fh.$$

(b) Set $\mathbf{A}_0(\epsilon) = 4\pi\int_\epsilon^1 \mathbf{s}(r)r^2\,dr$ and $\mathbf{A}_2(\epsilon) = \frac{2\pi}{3}\int_\epsilon^1 \mathbf{s}(r)r^4\,dr$ for $0 < \epsilon < 1$. As $\epsilon \downarrow 0$, $\epsilon^3\mathbf{A}_0(\epsilon) \to \frac{3}{4\pi\mathsf{b}^2}$ and $\epsilon\,\mathbf{A}_2(\epsilon) \to \frac{3}{8\pi\mathsf{b}^2}$.

(c) For each $\varphi \in \mathcal{D}$ the limit $\mathbf{T}_0(\varphi) := \lim_{\epsilon\downarrow 0}[\int_{|u|>\epsilon} \mathbf{s}(|u|)\varphi(u)\,du - \mathbf{A}_0(\epsilon)\varphi(0) - \mathbf{A}_2(\epsilon)\Delta\varphi(0)]$ is finite, equal to $4\pi\int_0^1 \mathbf{s}(r)r^2[\mathrm{av}_r\varphi - \varphi(0) - \frac{r^2}{6}\Delta\varphi(0)]\,dr + \int_{|u|\geq 1}\mathbf{s}(|u|)\varphi(u)\,du$, and bounded by $\frac{\pi}{6}\mathbf{c}_6\|\varphi\|_{(4)} + \mathbf{s}(1)\|\varphi\|_{L^1}$. $\mathbf{T}_0$ is a distribution, of order at most 4, which extends $\mathbf{s}$, satisfies $\mathbf{T}_0(\varphi\circ\mathsf{R}) = \mathbf{T}_0(\varphi)$ for $\mathsf{R} \in \mathrm{O}(3)$, and has $t^6(\mathbf{T}_0)_t(\varphi) \to \mathfrak{a}\,\mathrm{Pf}|u|^{-6}(\varphi)$ as $t \downarrow 0$ for each $\varphi \in \mathcal{D}$; hence its scaling degree is at most 6 and $\mathbf{T}_0$ is admissible. Its form is $\mathbf{S}_{\mathbf{T}_0}(f,h) = \lim_{\epsilon\downarrow 0}[\iint_{|x-y|>\epsilon} f(x)h(y)\,\mathbf{s}(|x-y|)\,dx\,dy - \mathbf{A}_0(\epsilon)\int fh + \mathbf{A}_2(\epsilon)\int \nabla f\cdot\nabla h]$ for $f, h \in \mathcal{D}$, and it agrees with $S_2(f,h)$ when $\mathrm{supp}\, f \cap \mathrm{supp}\, h = \emptyset$.

(d) Every extension of $\mathbf{s}$, admissible or not, has order larger than 3 near 0: there is no bound $|\mathbf{T}(\varphi)| \leq C\|\varphi\|_{C^N}$ with $N \leq 3$ for the φ supported in $\mathbb{B}(0,1)$; in particular no bound by $C\sup|\varphi|$, so no extension is given by a measure.

(e) For every extension $\mathbf{T}$ of $\mathbf{s}$ and every $\omega_0 < 6$, the scaling degree of $\mathbf{T}$ is not at most ω_0. ■

Proof. Step 1: Taylor's formula on spheres. Fix $\varphi \in C^4(\mathbb{R}^3)$, $r > 0$ and $\mathbf{n} \in \mathbb{S}^2$. Expanding $t \mapsto \varphi(t\mathbf{n})$ on $[0,r]$ to third order with Lagrange's remainder gives $\varphi(r\mathbf{n}) = \sum_{k=0}^3 \frac{r^k}{k!}(\mathbf{n}\cdot\nabla)^k\varphi(0) + \frac{r^4}{24}(\mathbf{n}\cdot\nabla)^4\varphi(t_0 r\mathbf{n})$ with some $t_0 \in (0,1)$. Averaging over $\mathbb{S}^2$ kills the terms with $k = 1$ and $k = 3$, which are odd in $\mathbf{n}$, and turns the term $k = 2$ into $\frac{r^2}{6}\Delta\varphi(0)$, by $\frac{1}{4\pi}\int_{\mathbb{S}^2} n_i n_j\,d\mathbf{n} = \frac{1}{3}\delta_{ij}$; the remainder depends continuously on $\mathbf{n}$, as a difference of continuous functions, and has modulus $\leq \frac{r^4}{24}\|\varphi\|_{(4)}$. So $|\mathrm{av}_r\varphi - \varphi(0) - \frac{r^2}{6}\Delta\varphi(0)| \leq \frac{r^4}{24}\|\varphi\|_{(4)}$.

Step 2: (a). The function $\varphi = \varphi_{f,h}$, given by the same formula, is continuous with compact support, and $\varphi(0) = \int fh$. On $[\epsilon,\infty)$ the kernel is bounded (Proposition 18.11(b)); so, changing variables to $x = y + u$ and using Fubini's theorem, $I_\epsilon := \iint_{|x-y|>\epsilon} f(x)h(y)\,\mathbf{s}(|x-y|)\,dx\,dy$ equals $\int_{|u|>\epsilon}\varphi(u)\,\mathbf{s}(|u|)\,du$. Given $\eta > 0$, pick $r_0 \in (0,1]$ so small that $|r^6\mathbf{s}(r) - \mathfrak{a}| \leq \eta$ on $(0,r_0]$ (Proposition 18.11(c)) and $|\varphi(u) - \varphi(0)| \leq \eta$ on $|u| \leq r_0$. As $\int_{\epsilon<|u|\leq r_0}|u|^{-6}du = \frac{4\pi}{3}(\epsilon^{-3} - r_0^{-3})$, for $\epsilon < r_0$ we may write $I_\epsilon - \frac{4\pi}{3}\mathfrak{a}\,\varphi(0)\,\epsilon^{-3} = \int_{\epsilon<|u|\leq r_0}[\varphi(u)|u|^6\mathbf{s}(|u|) - \mathfrak{a}\,\varphi(0)]\,|u|^{-6}du - \frac{4\pi}{3}\mathfrak{a}\,\varphi(0)\,r_0^{-3} +$

$\int_{|u|>r_0}\varphi(u)\,\mathbf{s}(|u|)\,du$. In modulus the expression in brackets does not exceed $|\varphi(u)-\varphi(0)|\,|u|^6\mathbf{s}(|u|)+|\varphi(0)|\,\big||u|^6\mathbf{s}(|u|)-\mathfrak{a}\big|\le\eta(\mathfrak{a}+\eta+|\varphi(0)|)$, and beyond r_0 the kernel is at most $\mathbf{s}(r_0)$. Therefore $|\epsilon^3 I_\epsilon-\frac{4\pi}{3}\mathfrak{a}\,\varphi(0)|\le\frac{4\pi}{3}\eta(\mathfrak{a}+\eta+|\varphi(0)|)+\epsilon^3[\frac{4\pi}{3}\mathfrak{a}\,|\varphi(0)|\,r_0^{-3}+\mathbf{s}(r_0)\|\varphi\|_{L^1}]$. First ϵ and then η go to 0; and $\frac{4\pi}{3}\mathsf{c}_2=\frac{3}{4\pi}$.

Step 3: (b). Given $\eta>0$, take $r_0\in(0,1]$ with $|r^6\mathbf{s}(r)-\mathfrak{a}|\le\eta$ on $(0,r_0]$, let $\epsilon<r_0$, and split $\mathbf{A}_0(\epsilon)$ and $\mathbf{A}_2(\epsilon)$ at r_0. Since $\mathbf{s}(r)r^2=r^{-4}\cdot r^6\mathbf{s}(r)$, $|\mathbf{A}_0(\epsilon)-4\pi\mathfrak{a}\int_\epsilon^{r_0}r^{-4}dr|\le 4\pi\eta\int_\epsilon^{r_0}r^{-4}dr+4\pi\int_{r_0}^1\mathbf{s}(r)r^2dr$, and $\int_\epsilon^{r_0}r^{-4}dr=\frac13(\epsilon^{-3}-r_0^{-3})$; multiplying by ϵ^3 and letting $\epsilon\to0$ leaves $\limsup|\epsilon^3\mathbf{A}_0(\epsilon)-\frac{4\pi}{3}\mathfrak{a}|\le\frac{4\pi}{3}\eta$. In the same way, from $\mathbf{s}(r)r^4=r^{-2}\cdot r^6\mathbf{s}(r)$ and $\int_\epsilon^{r_0}r^{-2}dr=\epsilon^{-1}-r_0^{-1}$, $\limsup|\epsilon\,\mathbf{A}_2(\epsilon)-\frac{2\pi}{3}\mathfrak{a}|\le\frac{2\pi}{3}\eta$. The constants are $\frac{4\pi}{3}\mathsf{c}_2=\frac{3}{4\pi}$ and $\frac{2\pi}{3}\mathsf{c}_2=\frac{3}{8\pi}$.

Step 4: (c). Passing to polar coordinates one finds, for $0<\epsilon<1$, $\int_{|u|>\epsilon}\mathbf{s}(|u|)\varphi(u)\,du-\mathbf{A}_0(\epsilon)\varphi(0)-\mathbf{A}_2(\epsilon)\Delta\varphi(0)=4\pi\int_\epsilon^1\mathbf{s}(r)r^2[\mathrm{av}_r\varphi-\varphi(0)-\frac{r^2}{6}\Delta\varphi(0)]\,dr+\int_{|u|\ge1}\mathbf{s}(|u|)\varphi(u)\,du$, because $\frac{2\pi}{3}r^4=4\pi r^2\cdot\frac{r^2}{6}$. Step 1 and $\mathbf{s}(r)\le\mathbf{c}_6r^{-6}$ on $(0,1]$ bound the function integrated in the first term by $\frac{\mathsf{c}_6}{24}\|\varphi\|_{(4)}$. So the limit exists and has the stated form, the expression at ϵ is within $\frac{\pi}{6}\mathbf{c}_6\|\varphi\|_{(4)}\,\epsilon$ of it, and $|\mathbf{T}_0(\varphi)|$ obeys the stated bound. Each of the 81 terms of $(e\cdot\nabla)^4=\sum e_ie_je_ke_l\partial_i\partial_j\partial_k\partial_l$ carries a coefficient of modulus ≤1, whence $\|\varphi\|_{(4)}\le81\|\varphi\|_{C^4}$; with $\|\varphi\|_{L^1}\le\frac{4\pi}{3}R^3\sup|\varphi|$ for $\mathrm{supp}\,\varphi\subset\mathbb{B}(0,R)$, $\mathbf{T}_0$ is a distribution of order at most 4. For φ vanishing near 0 both $\varphi(0)$ and $\Delta\varphi(0)$ are 0, and for small ϵ the expression reduces to $\int\mathbf{s}(|u|)\varphi(u)\,du$; thus $\mathbf{T}_0$ extends $\mathbf{s}$. Rotations and reflections $\mathsf{R}\in\mathrm{O}(3)$ preserve $|u|$, the spherical means and the value of the Laplacian at 0, so $\mathbf{T}_0(\varphi\circ\mathsf{R})=\mathbf{T}_0(\varphi)$. For the scaling, take φ supported in B_{R_0}, $0<t<1/R_0$ and $\varphi^t=\varphi(\cdot/t)$: then $\mathrm{av}_r\varphi^t=\mathrm{av}_{r/t}\varphi$, $\varphi^t(0)=\varphi(0)$ and $\Delta\varphi^t(0)=t^{-2}\Delta\varphi(0)$, while φ^t vanishes for $|u|\ge1$. Substituting $r=tv$ in the formula of (c), $t^6(\mathbf{T}_0)_t(\varphi)=t^3\mathbf{T}_0(\varphi^t)=4\pi\int_0^{1/t}(tv)^6\mathbf{s}(tv)\,v^{-4}[\mathrm{av}_v\varphi-\varphi(0)-\frac{v^2}{6}\Delta\varphi(0)]\,dv$. The factor $(tv)^6\mathbf{s}(tv)$ stays below $\mathbf{c}_6$ and converges to $\mathfrak{a}$ for each fixed v as $t\downarrow0$ (Proposition 18.11(c)); the remaining factor is bounded in modulus both by $\frac{1}{24}\|\varphi\|_{(4)}$ (Step 1) and by $2\sup|\varphi|\,v^{-4}+\frac16|\Delta\varphi(0)|\,v^{-2}$, so it has an integrable majorant on $(0,\infty)$. Dominated convergence gives $t^6(\mathbf{T}_0)_t(\varphi)\to\mathfrak{a}\,\mathrm{Pf}|u|^{-6}(\varphi)$; hence $t^\omega(\mathbf{T}_0)_t(\varphi)=t^{\omega-6}\cdot t^6(\mathbf{T}_0)_t(\varphi)\to0$ when $\omega>6$. The formula for $\mathbf{S}_{\mathbf{T}_0}$ is the definition of $\mathbf{T}_0$ evaluated at $\varphi_{f,h}$, using $\varphi_{f,h}(0)=\int fh$, $\Delta\varphi_{f,h}(0)=-\int\nabla f\cdot\nabla h$ and the identity of Step 2 between the two cut-off integrals; for disjoint supports, $\mathbf{T}_0$ being an extension, $\mathbf{S}_{\mathbf{T}_0}(f,h)=\iint f(x)h(y)\,\mathbf{s}(|x-y|)\,dx\,dy$, which is $S_2(f,h)$ by Proposition 18.11(a).

Step 5: (d). Let $\chi\in\mathcal{D}$ be non-negative with $\int\chi=1$ and support in $\mathbb{B}(e_3,\frac14)$, and $\chi_j(u)=2^{-3j}\chi(2^ju)$, $j\ge1$. Then χ_j is supported in the shell $\frac34 2^{-j}\le|u|\le\frac54 2^{-j}$; no two of these shells meet, since $\frac54 2^{-j-1}<\frac34 2^{-j}$, and all of them lie in $\mathbb{B}(0,1)\setminus\{0\}$. For $|\alpha|\le3$, $\sup|\partial^\alpha\chi_j|=2^{(|\alpha|-3)j}\sup|\partial^\alpha\chi|\le\sup|\partial^\alpha\chi|$; so each $\Phi_J=\sum_{j=1}^J\chi_j$ lies in $\mathcal{D}$, vanishes near 0, has support in $\mathbb{B}(0,1)$, and $\|\Phi_J\|_{C^3}\le\|\chi\|_{C^3}$. If $\mathbf{T}$ extends $\mathbf{s}$, then, $\mathbf{s}$ being non-increasing and $\int\chi_j=2^{-6j}$, $\mathbf{T}(\Phi_J)=\sum_{j\le J}\int\mathbf{s}(|u|)\chi_j(u)\,du\ge\sum_{j\le J}\mathbf{s}(\frac54 2^{-j})\,2^{-6j}$, and the summands $\mathbf{s}(\frac54 2^{-j})\,2^{-6j}=(\frac45)^6(\frac54 2^{-j})^6\mathbf{s}(\frac54 2^{-j})$ converge to $(\frac45)^6\mathfrak{a}>0$. Hence $\mathbf{T}(\Phi_J)$ is unbounded in J, although $\|\Phi_J\|_{C^N}\le\|\chi\|_{C^3}$ for $N\le3$, which excludes the bounds named in (d); $N=0$ is the bound by $C\sup|\varphi|$ that a measure would satisfy.

Step 6: (e). Take $\varphi\in\mathcal{D}$ non-negative, not identically 0, with support in the shell $\{\eta\le|u|\le R_0\}$, $0<\eta<R_0$. When $0<t<1/R_0$, $\varphi(\cdot/t)$ vanishes near 0, so for any extension $\mathbf{T}$ one has $t^6\mathbf{T}_t(\varphi)=t^3\int\mathbf{s}(|u|)\varphi(u/t)\,du=\int(t|v|)^6\mathbf{s}(t|v|)\,|v|^{-6}\varphi(v)\,dv$; dominated convergence, with the majorant $\mathbf{c}_6\eta^{-6}\varphi$, shows that this tends to $\mathfrak{a}\int|v|^{-6}\varphi(v)\,dv>0$. Hence $t^\omega\mathbf{T}_t(\varphi)$ has no limit 0 when $\omega\le6$; but scaling degree at most ω_0 for some $\omega_0<6$ would require that limit at $\omega=6$. □

Proposition 18.14 (The plane of admissible extensions and their common scaling limit)**.** Let g, σ', $\mathfrak{a}$, $\mathbf{s}$ and $\mathbf{T}_0$ be as in Propositions 18.11 and 18.13.

(a) Any two admissible extensions $\mathbf{T}$, $\mathbf{T}'$ of $\mathbf{s}$ satisfy $\mathbf{T}-\mathbf{T}'=c_0\delta+c_1\Delta\delta$ for some $(c_0,c_1)\in\mathbb{R}^2$, that is $\mathbf{S}_{\mathbf{T}}(f,h)-\mathbf{S}_{\mathbf{T}'}(f,h)=c_0\int fh-c_1\int\nabla f\cdot\nabla h$ for $f,h\in\mathcal{D}$; and for each $(c_0,c_1)\in\mathbb{R}^2$ the distribution $\mathbf{T}_0+c_0\delta+c_1\Delta\delta$ is an admissible extension. Thus the admissible extensions

of $\mathbf{s}$ form the plane $\{\mathbf{T}_0 + c_0\delta + c_1\Delta\delta : (c_0, c_1) \in \mathbb{R}^2\}$, and each of them has order exactly 4 near 0. Asking only for invariance under W_3, the group of the signed permutation matrices, yields this very plane; asking for no symmetry at all, two extensions of scaling degree at most 6 still differ only by a distribution $\varphi \mapsto \sum_{|\alpha|\le 3} c_\alpha \partial^\alpha\varphi(0)$, $c_\alpha \in \mathbb{R}$.

(b) Whichever admissible extension $\mathbf{T}$ is taken, $t^6\mathbf{T}_t(\varphi)$ tends to $\mathfrak{a}\,\mathrm{Pf}|u|^{-6}(\varphi)$ as $t \downarrow 0$, for each $\varphi \in \mathcal{D}$; the coordinates c_0, c_1 leave no trace in this limit. $\mathrm{Pf}|u|^{-6}$ is a distribution of order at most 4, homogeneous in the sense that $(\mathrm{Pf}|u|^{-6})_t = t^{-6}\,\mathrm{Pf}|u|^{-6}$ for $t > 0$, and $\mathrm{Pf}|u|^{-6}(\varphi) = \lim_{\epsilon\downarrow 0}[\int_{|u|>\epsilon} |u|^{-6}\varphi(u)\,du - \frac{4\pi}{3}\epsilon^{-3}\varphi(0) - \frac{2\pi}{3}\epsilon^{-1}\Delta\varphi(0)]$, the subtraction involving pure powers of ϵ only. On $\mathcal{S}(\mathbb{R}^3)$, $\mathrm{Pf}|u|^{-6}(\varphi) = \frac{\pi^2}{12}\int |p|^3\hat\varphi(p)\frac{dp}{(2\pi)^3}$; consequently $\mathfrak{a}\,\mathrm{Pf}|u|^{-6}(\varphi_{f,f}) = \frac{3}{64\,\mathfrak{b}^2}\int |p|^3|\hat f(p)|^2\frac{dp}{(2\pi)^3} \ge 0$ for every $f \in \mathcal{D}$.

■

Proof. Step 1: distributions carried by the origin. Let $\mathbf{D}$ be a distribution that annihilates every $\varphi \in \mathcal{D}$ vanishing near 0. Fix N and C such that $|\mathbf{D}(\varphi)| \le C\|\varphi\|_{C^N}$ whenever $\operatorname{supp}\varphi \subset \mathbb{B}(0,1)$, and a radial $\chi \in \mathcal{D}$, $0 \le \chi \le 1$, equal to 1 on $B_{1/2}$ and supported in B_1; for $0 < \eta \le 1$ write $\chi_\eta = \chi(\cdot/\eta)$, so that $\chi_1 = \chi$. Because $(1-\chi_\eta)\varphi$ vanishes near 0, $\mathbf{D}(\varphi) = \mathbf{D}(\chi_\eta\varphi)$ for all $\varphi \in \mathcal{D}$. Now let ψ be smooth with all derivatives of order $\le N$ vanishing at 0. Taylor's formula gives $|\partial^\beta\psi(u)| \le C_\psi|u|^{N+1-|\beta|}$ for $|u| \le 1$ and $|\beta| \le N$; with Leibniz' rule, for $|\alpha| \le N$ and $|u| \le \eta$, $|\partial^\alpha(\chi_\eta\psi)(u)| \le \sum_{\beta\le\alpha}\binom{\alpha}{\beta}\eta^{-|\alpha-\beta|}\sup|\partial^{\alpha-\beta}\chi|\,C_\psi\eta^{N+1-|\beta|} \le C'\eta$, and $\chi_\eta\psi$ is 0 for $|u| \ge \eta$. For $\eta \le \frac12$ the support of χ_η lies in $B_{1/2}$, where $\chi = 1$, so $\chi_\eta\chi = \chi_\eta$; applying the identity above to $\chi\psi \in \mathcal{D}$ we get $|\mathbf{D}(\chi\psi)| = |\mathbf{D}(\chi_\eta\chi\psi)| = |\mathbf{D}(\chi_\eta\psi)| \le CC'\eta$ for all $\eta \in (0, \frac12]$, so $\mathbf{D}(\chi\psi) = 0$. Applying this to $\psi = \varphi - P$, with $P(u) = \sum_{|\alpha|\le N}\frac{u^\alpha}{\alpha!}\partial^\alpha\varphi(0)$ the Taylor polynomial of $\varphi \in \mathcal{D}$, gives $\mathbf{D}(\varphi) = \mathbf{D}(\chi\varphi) = \mathbf{D}(\chi P) = \sum_{|\alpha|\le N} c_\alpha\partial^\alpha\varphi(0)$, where $c_\alpha = \mathbf{D}(\chi u^\alpha)/\alpha!$ is real.

Step 2: homogeneity bounds the order of the derivatives. Let $\mathbf{T}$, $\mathbf{T}'$ be extensions of $\mathbf{s}$ of scaling degree at most 6 and $\mathbf{D} = \mathbf{T} - \mathbf{T}'$. Then $\mathbf{D}$ annihilates the functions vanishing near 0, and $t^\omega\mathbf{D}_t(\varphi) \to 0$ for $\omega > 6$. By Step 1, $\mathbf{D}(\varphi) = \sum_{|\alpha|\le N} c_\alpha\partial^\alpha\varphi(0)$, and since $\partial^\alpha[\varphi(\cdot/t)](0) = t^{-|\alpha|}\partial^\alpha\varphi(0)$, $\mathbf{D}_t(\varphi) = \sum_\alpha c_\alpha t^{-3-|\alpha|}\partial^\alpha\varphi(0)$. Test with $\varphi = \chi u^{\alpha_0}$, χ as in Step 1: its derivative ∂^α at 0 is $\alpha_0!$ for $\alpha = \alpha_0$ and 0 for every other α, so $t^{13/2}\mathbf{D}_t(\varphi) = c_{\alpha_0}\alpha_0!\,t^{7/2-|\alpha_0|}$, which can tend to 0 for $|\alpha_0| \ge 4$ only if $c_{\alpha_0} = 0$. This is the statement of (a) without symmetry.

Step 3: symmetry leaves two coefficients. Suppose in addition that $\mathbf{T}$ and $\mathbf{T}'$ are invariant under W_3, as admissible extensions are. Then so is $\mathbf{D}$, and $c_\alpha = \mathbf{D}(\chi u^\alpha)/\alpha!$ with χ radial. The point reflection $u \mapsto -u$ sends χu^α to $(-1)^{|\alpha|}\chi u^\alpha$, which removes the coefficients with $|\alpha| = 1$ and $|\alpha| = 3$. For $i \ne j$, reversing the sign of the coordinate u_i reverses the sign of χu_iu_j, which removes the mixed second derivatives; interchanging u_i and u_j maps χu_i^2 to χu_j^2, which equalises the coefficients of ∂_i^2 and ∂_j^2. All three maps belong to $W_3 \subset \mathrm{O}(3)$. What is left is $\mathbf{D} = c_0\delta + c_1\Delta\delta$ with real $c_0 = \mathbf{D}(\chi)$ and $c_1 = \frac12\mathbf{D}(\chi u_1^2)$, and $\mathbf{D}(\varphi_{f,h}) = c_0\int fh - c_1\int\nabla f\cdot\nabla h$.

Step 4: every point of the plane occurs; orders. The distributions δ and $\Delta\delta$ annihilate the functions vanishing near 0, are invariant under $\mathrm{O}(3)$, and satisfy $\delta_t(\varphi) = t^{-3}\varphi(0)$ and $(\Delta\delta)_t(\varphi) = t^{-5}\Delta\varphi(0)$, which tend to 0 after multiplication by t^ω, $\omega > 6$. Together with Proposition 18.13(c), this makes each $\mathbf{T}_0 + c_0\delta + c_1\Delta\delta$ an admissible extension, and by Steps 2 and 3 there are no others, also when only invariance under W_3 is asked. As to the orders: $\mathbf{T}_0$ has order at most 4 near 0, δ and $\Delta\delta$ have orders 0 and 2, and by Proposition 18.13(d) no extension has order ≤ 3; so the order is exactly 4.

Step 5: the limit in (b). For $\mathbf{T}_0$ it was obtained in Step 4 of the proof of Proposition 18.13. For $\mathbf{T} = \mathbf{T}_0 + c_0\delta + c_1\Delta\delta$ the two extra terms contribute $t^6\delta_t(\varphi) = t^3\varphi(0)$ and $t^6(\Delta\delta)_t(\varphi) = t\,\Delta\varphi(0)$, both tending to 0.

Step 6: properties of $\mathrm{Pf}|u|^{-6}$. For φ in $\mathcal{D}$ or in $\mathcal{S}(\mathbb{R}^3)$, Step 1 of the proof of Proposition 18.13 bounds $r^{-4}|\mathrm{av}_r\varphi - \varphi(0) - \frac{r^2}{6}\Delta\varphi(0)|$ by $\frac{1}{24}\|\varphi\|_{(4)}$, and the same quantity is also at most $2\sup|\varphi|\,r^{-4} + \frac16|\Delta\varphi(0)|\,r^{-2}$; using the first bound on $(0,1]$ and the second on $[1,\infty)$, the integral converges

absolutely and $|\mathrm{Pf}|u|^{-6}(\varphi)| \le 4\pi[\frac{1}{24}\|\varphi\|_{(4)} + \frac{2}{3}\sup|\varphi| + \frac{1}{6}|\Delta\varphi(0)|]$, so $\mathrm{Pf}|u|^{-6}$ is a distribution of order at most 4. Substituting $r = tv$ shows $\mathrm{Pf}|u|^{-6}(\varphi(\cdot/t)) = t^{-3}\,\mathrm{Pf}|u|^{-6}(\varphi)$, i.e. $(\mathrm{Pf}|u|^{-6})_t = t^{-6}\,\mathrm{Pf}|u|^{-6}$. Polar coordinates give $\int_{|u|>\epsilon}|u|^{-6}\varphi(u)\,du = 4\pi\int_\epsilon^\infty r^{-4}[\mathrm{av}_r\varphi - \varphi(0) - \frac{r^2}{6}\Delta\varphi(0)]\,dr + \frac{4\pi}{3}\epsilon^{-3}\varphi(0) + \frac{2\pi}{3}\epsilon^{-1}\Delta\varphi(0)$, and letting $\epsilon\to 0$ gives the subtracted formula of (b).

Step 7: $\mathrm{Pf}|u|^{-6}$ *in Fourier variables.* Let $\varphi\in\mathcal{S}(\mathbb{R}^3)$ and $k = |p|$. Fourier inversion, $\varphi(u) = \int e^{ip\cdot u}\hat\varphi(p)\frac{dp}{(2\pi)^3}$, gives $\varphi(0) = \int\hat\varphi\frac{dp}{(2\pi)^3}$ and $\Delta\varphi(0) = -\int|p|^2\hat\varphi\frac{dp}{(2\pi)^3}$, and, with $\frac{1}{4\pi}\int_{\mathbb{S}^2}e^{irp\cdot\mathbf{n}}d\mathbf{n} = \frac{\sin kr}{kr}$ and Fubini's theorem, $\mathrm{av}_r\varphi - \varphi(0) - \frac{r^2}{6}\Delta\varphi(0) = \int[\frac{\sin kr}{kr} - 1 + \frac{k^2r^2}{6}]\hat\varphi(p)\frac{dp}{(2\pi)^3}$. For $x\ge 0$ one has $x - \frac{x^3}{6} \le \sin x \le \min\{x, x - \frac{x^3}{6} + \frac{x^5}{120}\}$, so $0 \le \frac{\sin x}{x} - 1 + \frac{x^2}{6} \le \min\{\frac{x^4}{120}, \frac{x^2}{6}\}$; the substitution $x = kr$ then gives $\int_0^\infty r^{-4}[\frac{\sin kr}{kr} - 1 + \frac{k^2r^2}{6}]\,dr = k^3 I$ with $I = \int_0^\infty x^{-5}(\sin x - x + \frac{x^3}{6})\,dx \in (0,\infty)$, and since $\int k^3|\hat\varphi|\,dp < \infty$ the order of integration may be exchanged: $\mathrm{Pf}|u|^{-6}(\varphi) = 4\pi I\int|p|^3\hat\varphi(p)\frac{dp}{(2\pi)^3}$. To evaluate I, put $F = \sin x - x + \frac{x^3}{6}$, so that $F' = \cos x - 1 + \frac{x^2}{2}$, $F'' = x - \sin x$ and $F''' = 1 - \cos x$; these behave like x^5, x^4, x^3, x^2 near 0 and like $x^3, x^2, x, 1$ near ∞, so integrating by parts three times, the boundary terms vanishing at both ends, yields $I = \frac{1}{4}\int_0^\infty F'x^{-4}dx = \frac{1}{12}\int_0^\infty F''x^{-3}dx = \frac{1}{24}\int_0^\infty(1-\cos x)\,x^{-2}dx = \frac{1}{24}\cdot\frac{\pi}{2} = \frac{\pi}{48}$, hence $4\pi I = \frac{\pi^2}{12}$. For $f\in\mathcal{D}$, $\hat\varphi_{f,f} = |\hat f|^2$, and $\frac{\pi^2}{12}\mathsf{c}_2 = \frac{3}{64}$. □

Remark. By Lemma 2.26(c), the contact terms of Lemma 2.26(b) at $n = 2$ on $\mathbb{R}^3$ that are invariant under translations, real, symmetric under the exchange of the two entries and covariant under O(3) make up a six-dimensional space, which the lemma spans by six forms written in Fourier space. Evaluated at the entries $f\mathbf{1}$ and $h\mathbf{1}$, these six forms become multiples of $\int fh$ (the two without p, as $\mathrm{tr}\,\mathbf{1} = 3$) and of $\int\nabla f\cdot\nabla h$ (the four with two factors p, as $p\cdot\mathbf{1}p = |p|^2$); so on the scalar entries the contact terms with these four properties are exactly the forms $c_0\int fh - c_1\int\nabla f\cdot\nabla h$ of Proposition 18.14(a).

Proposition 18.15 (Positive type does not select a point of the plane)**.** Let g, σ', $\mathbf{s}$, $\mathfrak{m}$, $\mathbf{c}_6$ and $\mathbf{T}_0$ be as in Propositions 18.11 and 18.13. For $p\in\mathbb{R}^3$ and $\psi\in\mathcal{D}$ let

$$\mathbf{w}(p) = \int_{(0,1]}\frac{8\pi E}{(E^2+|p|^2)^2}\,d\mathfrak{m}(E) + \int_{(1,\infty)}\frac{8\pi|p|^4(3E^2+2|p|^2)}{E^5(E^2+|p|^2)^2}\,d\mathfrak{m}(E) \in [0,\infty],$$

$$\mathbf{T}_+(\psi) = \mathfrak{m}(\{0\})\int\psi(u)\,du + \int\mathbf{w}(p)\hat\psi(p)\frac{dp}{(2\pi)^3}.$$

(a) The integral over $(0,1]$ in $\mathbf{w}$ is an integrable function of p, with integral $(2\pi)^3\mathfrak{m}((0,1])$ and finite values at all $p\ne 0$; the integral over $(1,\infty)$ is bounded by $264\pi\,\mathrm{e}\,\mathbf{c}_6|p|^3$.

(b) $\mathbf{T}_+$ takes real values on $\mathcal{D}$, is a distribution, extends $\mathbf{s}$, is admissible, and has positive type: $\mathbf{S}_{\mathbf{T}_+}(f,f) = \mathfrak{m}(\{0\})(\int f)^2 + \int\mathbf{w}(p)|\hat f(p)|^2\frac{dp}{(2\pi)^3} \ge 0$ for $f\in\mathcal{D}$.

(c) Let $\mathbf{Q}\subset\mathbb{R}^2$ consist of the (c_0,c_1) such that $\mathbf{T}_0 + c_0\delta + c_1\Delta\delta$ has positive type. Then $\mathbf{Q}\ne\emptyset$, $\mathbf{Q}$ is convex, and with (c_0,c_1) it contains $(c_0+\eta_0, c_1-\eta_1)$ for all $\eta_0,\eta_1\ge 0$. So the demand that the form be of positive type leaves admissible extensions and does not determine (c_0,c_1). ■

Proof. Step 1: Fourier transform of the exponential. For $E>0$ and $k = |p| > 0$, in polar coordinates $\int e^{-E|u|}e^{-ip\cdot u}\,du = 4\pi\int_0^\infty e^{-Er}\frac{\sin kr}{kr}r^2dr = \frac{4\pi}{k}\,\mathrm{Im}\int_0^\infty re^{-(E-ik)r}dr = \frac{4\pi}{k}\,\mathrm{Im}\frac{(E+ik)^2}{(E^2+k^2)^2} = \Upsilon_E(k) := \frac{8\pi E}{(E^2+k^2)^2}$, while for $p=0$ the integral is $8\pi E^{-3} = \Upsilon_E(0)$; replacing $e^{-ip\cdot u}$ by $e^{ip\cdot u}$ does not change the value. With $k = E\tan s$, $\int\Upsilon_E(|p|)\,dp = 32\pi^2E\int_0^\infty\frac{k^2\,dk}{(E^2+k^2)^2} = 32\pi^2\int_0^{\pi/2}\sin^2 s\,ds = 8\pi^3 = (2\pi)^3$.

Step 2: removing two Taylor terms. Put $x = k^2/E^2$. Then $\Upsilon_E(k) - \frac{8\pi}{E^3} + \frac{16\pi k^2}{E^5} = \frac{8\pi}{E^3}[(1+x)^{-2} - 1 + 2x] = \frac{8\pi}{E^3}\cdot\frac{x^2(3+2x)}{(1+x)^2} = \frac{8\pi k^4(3E^2+2k^2)}{E^5(E^2+k^2)^2}$, by the identity $1-(1-2x)(1+x)^2 = 3x^2+2x^3$. Hence the subtracted kernel is non-negative and bounded both by $\frac{8\pi}{E^3}3x^2 = 24\pi k^4 E^{-7}$, because $3+2x \le 3(1+x)^2$, and by $\frac{8\pi}{E^3}2x = 16\pi k^2 E^{-5}$, because $(1+x)^{-2} \le 1$.

Step 3: how fast $\mathfrak{m}$ grows. Write $\mathfrak{M}(v) = \mathfrak{m}([0,v])$. For $v \ge 1$ the weight $e^{-E/v}$ is at least e^{-1} on $[0,v]$, so $\mathfrak{M}(v) \le \mathrm{e}\int e^{-E/v}d\mathfrak{m}(E) = \mathrm{e}\,\mathbf{s}(1/v) \le \mathrm{e}\,\mathbf{c}_6 v^6$ (Proposition 18.11(b), (c)). For $k \ge 1$, writing $E^{-7} = \int_E^\infty 7v^{-8}dv$ and using Tonelli's theorem, $\int_{(k,\infty)} E^{-7}d\mathfrak{m}(E) = \int_k^\infty 7v^{-8}\mathfrak{m}((k,v])\,dv \le 7\mathrm{e}\,\mathbf{c}_6 k^{-1}$; writing $E^{-5} = k^{-5} + \int_E^k 5v^{-6}dv$ for $E \le k$, $\int_{(1,k]} E^{-5}d\mathfrak{m}(E) \le k^{-5}\mathfrak{M}(k) + \int_1^k 5v^{-6}\mathfrak{M}(v)\,dv \le 6\mathrm{e}\,\mathbf{c}_6 k$.

Step 4: (a), and the bound that makes $\mathbf{T}_+$ a distribution. Let $\mathbf{w}_1$ and $\mathbf{w}_2$ be the integrals over $(0,1]$ and over $(1,\infty)$ in $\mathbf{w}$. Tonelli's theorem and Step 1 give $\mathbf{w}_1 \ge 0$ and $\int \mathbf{w}_1 dp = (2\pi)^3\mathfrak{m}((0,1])$, and $\Upsilon_E(k) \le 8\pi k^{-4}$ for $E \le 1$ gives $\mathbf{w}_1(p) \le 8\pi|p|^{-4}\mathfrak{m}((0,1])$ for $p \ne 0$. By Step 2, $\mathbf{w}_2 \ge 0$; for $k = |p| \ge 1$, using the bound $24\pi k^4 E^{-7}$ on $E > k$ and the bound $16\pi k^2 E^{-5}$ on $1 < E \le k$, Step 3 gives $\mathbf{w}_2(p) \le 24\pi k^4\cdot 7\mathrm{e}\,\mathbf{c}_6 k^{-1} + 16\pi k^2\cdot 6\mathrm{e}\,\mathbf{c}_6 k = 264\pi\mathrm{e}\,\mathbf{c}_6 k^3$, and for $k < 1$ the first bound alone gives $\mathbf{w}_2(p) \le 24\pi k^4\cdot 7\mathrm{e}\,\mathbf{c}_6 \le 168\pi\mathrm{e}\,\mathbf{c}_6 k^3$. For $\psi \in \mathcal{D}$ supported in $\mathbb{B}(0,R)$, $(1+|p|^2)^4|\hat\psi(p)| = |\widehat{(1-\Delta)^4\psi}(p)| \le \|(1-\Delta)^4\psi\|_{L^1} \le C_R\|\psi\|_{C^8}$; therefore $|\mathbf{T}_+(\psi)| \le \mathfrak{m}(\{0\})\|\psi\|_{L^1} + (2\pi)^{-3}[\|\mathbf{w}_1\|_{L^1} + 264\pi\mathrm{e}\,\mathbf{c}_6\int|p|^3(1+|p|^2)^{-4}dp]\,C_R\|\psi\|_{C^8}$, and $\mathbf{T}_+$ is a linear functional with this bound.

Step 5: $\mathbf{T}_+$ agrees with $\mathbf{s}$ off the origin. Take $\psi \in \mathcal{D}$ with support avoiding a ball about 0. Inserting $\mathbf{s}(|u|) = \int e^{-E|u|}d\mathfrak{m}(E)$ (Proposition 18.11(b)) and interchanging the integrals, which is legitimate since $\iint e^{-E|u|}|\psi(u)|\,du\,d\mathfrak{m}(E) = \int \mathbf{s}(|u|)|\psi(u)|\,du < \infty$, one gets $\int \mathbf{s}(|u|)\psi(u)\,du = \int_{[0,\infty)}(\int e^{-E|u|}\psi(u)\,du)\,d\mathfrak{m}(E)$. The mass of $\mathfrak{m}$ at $E = 0$ contributes $\mathfrak{m}(\{0\})\int\psi$. For $E > 0$, by Fourier inversion, Fubini's theorem and Step 1, $\int e^{-E|u|}\psi(u)\,du = \int \Upsilon_E(|p|)\hat\psi(p)\frac{dp}{(2\pi)^3}$. Here $\int\hat\psi\,dp = (2\pi)^3\psi(0) = 0$ and $\int|p|^2\hat\psi\,dp = -(2\pi)^3\Delta\psi(0) = 0$, so that for $E > 1$ one may subtract $\frac{8\pi}{E^3} - \frac{16\pi|p|^2}{E^5}$ from Υ_E without changing the integral, which brings in the kernel of Step 2. Interchanging next the integrations over E and over p, as permitted by Tonelli's and Fubini's theorems because the kernels are non-negative and $\int(\mathbf{w}_1 + |p|^3)|\hat\psi|\,dp < \infty$ (Step 4), one arrives at $\mathbf{T}_+(\psi)$.

Step 6: (b). For real ψ, $\hat\psi(-p) = \overline{\hat\psi(p)}$, and $\mathbf{w}$ is a real function of $|p|$; the substitution $p \mapsto -p$ shows that $\int \mathbf{w}\hat\psi\,dp$ is real, so $\mathbf{T}_+$ is real-valued and, by Step 4, a distribution. For $\mathsf{R} \in \mathrm{O}(3)$, $\widehat{\psi\circ\mathsf{R}} = \hat\psi\circ\mathsf{R}$, and $\mathbf{w}$ depends on $|p|$ only, so $\mathbf{T}_+(\psi\circ\mathsf{R}) = \mathbf{T}_+(\psi)$. Since $\widehat{\psi(\cdot/t)}(p) = t^3\hat\psi(tp)$, $(\mathbf{T}_+)_t(\psi) = \mathfrak{m}(\{0\})\int\psi + \int\mathbf{w}(p)\hat\psi(tp)\frac{dp}{(2\pi)^3}$; here $|\int\mathbf{w}_1(p)\hat\psi(tp)\,dp| \le \|\mathbf{w}_1\|_{L^1}\sup|\hat\psi|$ and, by (a), $|\int\mathbf{w}_2(p)\hat\psi(tp)\,dp| \le 264\pi\mathrm{e}\,\mathbf{c}_6 t^{-6}\int|q|^3|\hat\psi(q)|\,dq$. Thus $t^\omega(\mathbf{T}_+)_t(\psi) \to 0$ for $\omega > 6$, and with Step 5, $\mathbf{T}_+$ is an admissible extension of $\mathbf{s}$. Lastly $\mathbf{T}_+(\varphi_{f,f}) = \mathfrak{m}(\{0\})\int\varphi_{f,f} + \int\mathbf{w}\,\hat\varphi_{f,f}\frac{dp}{(2\pi)^3} = \mathfrak{m}(\{0\})(\int f)^2 + \int\mathbf{w}|\hat f|^2\frac{dp}{(2\pi)^3}$, which is non-negative.

Step 7: (c). Proposition 18.14(a) writes $\mathbf{T}_+ = \mathbf{T}_0 + c_0^+\delta + c_1^+\Delta\delta$ with real (c_0^+, c_1^+), and by (b) this point lies in $\mathbf{Q}$. For each $f \in \mathcal{D}$ the value $(\mathbf{T}_0 + c_0\delta + c_1\Delta\delta)(\varphi_{f,f})$ is an affine function of (c_0, c_1); $\mathbf{Q}$ is the intersection, over $f \in \mathcal{D}$, of the sets where these affine functions are ≥ 0, hence convex. Finally $(\eta_0\delta - \eta_1\Delta\delta)(\varphi_{f,f}) = \eta_0\int f^2 + \eta_1\int|\nabla f|^2 \ge 0$ for $\eta_0, \eta_1 \ge 0$, so passing from $(c_0, c_1) \in \mathbf{Q}$ to $(c_0 + \eta_0, c_1 - \eta_1)$ keeps the form of positive type. □

18.3. **Energy and momentum as charges of a conserved tensor with a flux law; for data.** Over scalar reconstruction data with a unique vacuum we consider a tensor of the frame (M1)–(M5) of Definition 3.1 (3) and impose one further condition on its Euclidean kernels, the flux law of the translations (Definition 18.16): the inward flux of the kernel $\mathsf{s}_{ij}(\cdot\,; y)$ across the boundary of a region holding the first scalar point y_1 but none of the other scalar points is the derivative of the scalar kernel s_m in the j-th coordinate of y_1. At one scalar point every tensor of the frame satisfies this law, and the law for the full kernels is the same condition as the law for the connected

kernels (Lemma 18.20). The results are the following. Let α be a smooth function of the time with compact support and integral 1, g a smooth function of the space variables with compact support, equal to 1 on the disc of radius r_g about 0, and $g_R = g(\cdot/R)$. If the law holds in the direction e_3, respectively e_j with $j \in \{1,2\}$, then for every test function f of compact support the commutator of $T_{00}(\alpha \otimes g_R)$, respectively $T_{0j}(\alpha \otimes g_R)$, with $\phi(f)$ is $i\phi(\partial_\mu f)$, the commutator of the generator P_μ of the translations with $\phi(f)$, where $\mu = 0$, respectively $\mu = j$; this holds on the whole of $\mathcal{D}_\infty$ as soon as $R \geq R_0 = (r_O + t_* + 1)/r_g$, where I_α is a bounded interval containing the support of α, I_f a bounded open interval and $r_O > 0$ are such that $I_f \times \{|\vec{x}| < r_O\}$ contains the support of f, and $t_* = \sup\{|t - t'| : t \in I_\alpha,\ t' \in I_f\}$; conversely, these commutators for one μ imply the law in the corresponding direction (Proposition 18.23). If the tensor is besides covariant on the vacuum, (M5), the vacuum vectors $T_{00}(\alpha \otimes g_R)\Omega$ are of the order $R^{-1/2}$ in every graph norm of H; for every χ in the span $\mathcal{D}_0$ of the polynomial vectors, $T_{00}(\alpha \otimes g_R)\chi$ converges to $H\chi$ in the norm of $\mathcal{H}$ with the rate $R^{-1/2}$; the squared norms of $T_{0j}(\alpha \otimes g_R)\Omega$ converge to an explicit limit; and $T_{0j}(\alpha \otimes g_R)\chi$ converges weakly in $\mathcal{H}$ to $P_j\chi$ (Proposition 18.28). The commutators come from analytic continuation in two Euclidean heights. Between damping factors, the product of a smeared component and the scalar field in either order is the boundary value of a holomorphic function of the two heights, one function for each order. Conservation turns the derivative of each of them in the height of the component into the like function formed with the components whose first index is a direction of the plane, smeared with the derivative of the cut-off in that direction; these derivatives of the cut-off vanish near the support of the scalar test function, locality relative to ϕ gives the two functions so formed equal boundary values at nearby times, and so they are one holomorphic function on a domain that crosses the equal heights (Lemma 18.21). Hence each of the two original functions is a common primitive of that function, in the height of the component, plus a term that depends on the scalar height only; the difference of these two terms is, for real heights, the Euclidean kernel integrated against the gradient of a cut-off equal to 1 about the scalar point, and its values at imaginary heights give the commutator (Lemma 18.22). The limits rest on the form of the two-point measure of a conserved tensor covariant on the vacuum: two mass measures, finite near the mass 0, and a massless scalar part of weight $c_\partial \geq 0$ (Lemma 18.26). No decay at large distances is used. By Theorem F (d) and the uniqueness clause of Proposition YM3_I:110.1, the frame holds over every component of a tensor limit at a reference coupling not larger than γ_W^{sd}, and the results apply to it whenever its kernels obey the flux law (Remark 18.29).

Data. Throughout, σ' satisfies (Rec) of Section YM3_I:107 with numbers $C_\wp, \wp, c_0$ and unique vacuum, $\ker H = \mathbb{C}\Omega$. We call the latter property (Rec5). Let $\mathbb{E}_+ = \{x_3 > 0\}$. A tuple over $\mathbb{E}_+$ is a finite family of real $C^1_c(\mathbb{E}_+)$ functions with pairwise disjoint supports, including the empty tuple $\varnothing$. A tuple is smooth when all its entries are smooth. The classes $[F]$ of complex linear combinations of tuples form a dense subspace of the reflection Hilbert space $\mathcal{H} = \mathcal{H}(\sigma')$. Write $\Omega = [\varnothing]$ and take $H, \mathcal{U}(v)$ and the joint spectral measure $\mathcal{P}$ of the reflection translations as in YM3_I:(2031). Scalar products are antilinear in the first entry. Apart from the real tuple entries, the test-function spaces in this subsection are complex unless specified otherwise. For $X = (h_1, \dots, h_n)$ and $z = (z_1, \dots, z_n)$ put $X(z) = \prod_b h_b(z_b)$. Write $z + ae_3$ and $\theta(z + ae_3)$ for the tuples of translated and reflected points, where $\theta x = (\vec{x}, -x_3)$.

Let $R_H = (1 + H)^{-1}$ and $\mathcal{D}_\infty = \bigcap_{n \geq 1} \operatorname{dom} H^n$. The spectral projections onto $\{E \leq N\}$ have range in $\mathcal{D}_\infty$ and increase strongly to the identity, so $\mathcal{D}_\infty$ is dense. Spectral calculus gives $(1+H)^n\mathcal{D}_\infty = \mathcal{D}_\infty$, with inverse R_H^n on that subspace. For $\operatorname{Re}\tau > 0$, the operator $e^{-\tau H} = \int e^{-\tau E} d\mathcal{P}$ is norm holomorphic and maps $\mathcal{H}$ into $\mathcal{D}_\infty$, since $E^n e^{-\tau E}$ is bounded for every n. We fix an integer $l > \wp$. The scalar point insertion $\mathsf{B}_0(s, s')$ is the operator of Proposition YM3_I:107.2. Write A_l for the bounded operator in YM3_I:(2044), and q_0 for the form YM3_I:(2047) restricted to $\mathcal{D}_\infty$. Lemma YM3_I:107.3 gives

$$\mathsf{B}_0(s, s') = e^{-sH}(1 + H)^l \mathsf{A}_l (1 + H)^l e^{-s'H}, \qquad \mathsf{B}_0(s, s')^* = \mathsf{B}_0(s', s),$$

using YM3_I:(2046) for the first equality and the point-insertion adjoint relation for the second. The kernels s_n of σ' (Proposition YM3_I:108.3) are continuous and real-analytic on $(\mathbb{R}^3)^n_{\neq}$, symmetric and Euclidean invariant. In particular s_1 is constant and $\mathsf{s}_0 = 1$.

Real time and spectrum. Points of $\mathbb{R}^{1+2}$ are written $x = (t, \vec{x}) = (x^0, x^1, x^2)$; $x \cdot p = x^0 p^0 - \vec{x} \cdot \vec{p}$ and $\mathsf{g} = \mathrm{diag}(1, -1, -1)$, $p_\mu = \mathsf{g}_{\mu\mu} p^\mu$, $\partial_\mu = \partial/\partial x^\mu$ and $\partial_t = \partial_0$. Minkowski indices μ, ν, μ', ν' run through $\{0, 1, 2\}$ and are summed when repeated; Euclidean indices i, i', j, j' run through $\{1, 2, 3\}$, and k, k' through $\{1, 2\}$. $U(x) = e^{itH}\mathcal{U}(\vec{x})$; $P_0 = H$, and P_1, P_2 are the joint self-adjoint generators of $\mathcal{U}$, so that $U(x) = \exp i(x^0 P_0 + x^1 P_1 + x^2 P_2)$, and $\|P_k \psi\| \leq \|H\psi\|$ on $\mathrm{dom}\, H$ by (Rec3). E is the image of $\mathcal{P}$ under $(E, q) \mapsto (E, -q)$, so that $U(x) = \int e^{ix\cdot p}\, d\mathsf{E}(p)$. $V_+ = \{p^0 > |\vec{p}|\}$, and $\partial V_+ = \{p^0 = |\vec{p}| > 0\}$ is the cone without its apex. By the cone condition (Rec3), E is carried by $V_+ \cup \partial V_+ \cup \{0\}$, and as the range of $\mathsf{E}(\{0\})$ lies in $\ker H$ and contains the invariant vector Ω, (Rec5) gives $\mathsf{E}(\{0\}) = \langle \Omega, \cdot \rangle \Omega$. For $\psi \in \mathcal{H}$, $m_\psi = \langle \psi, \mathsf{E}(\cdot)\psi \rangle$, and for a bounded Borel function f, $\mathsf{f}(P) = \int \mathsf{f}\, d\mathsf{E}$. $\mathcal{S} = \mathcal{S}(\mathbb{R}^{1+2}; \mathbb{C})$; $\phi(f)$, $f \in \mathcal{S}$, is the field of Proposition YM3_I:109.1 on $\mathcal{D}_\infty$, and $\mathcal{D}_0$ is the span of Ω and of the vectors $\phi(f_1) \cdots \phi(f_n)\Omega$ (Theorem YM3_I:AO); $U(a, \Lambda) = U(a)U(\Lambda)$, $a \in \mathbb{R}^{1+2}$, $\Lambda \in \mathcal{L}$, is the representation of Proposition YM3_I:110.1, and $B(s)$ is the boost $B(s)(p^0, p^1, p^2) = (p^0 \cosh s + p^1 \sinh s, p^0 \sinh s + p^1 \cosh s, p^2)$. A *polynomial* is a finite linear combination Q of 1 and of products $\phi(h_1) \cdots \phi(h_n)$, $h_b \in \mathcal{S}$, acting on $\mathcal{D}_\infty$; its degree is the largest n that occurs, and $\mathsf{Q}^\dagger$ arises by reversing the order of the factors and conjugating the coefficients and the test functions, so that $\langle \psi', \mathsf{Q}\psi \rangle = \langle \mathsf{Q}^\dagger \psi', \psi \rangle$ on $\mathcal{D}_\infty$ by YM3_I:(2076); every vector of $\mathcal{D}_0$ is $\mathsf{Q}\Omega$ for a polynomial Q. For $F \in \mathcal{S}$, $\hat{F}(p) = \int F(x) e^{ix\cdot p}\, dx$; then $\int F(x)U(x)\, dx = \hat{F}(P)$ weakly, $\widehat{\partial_\mu F} = -ip_\mu \hat{F}$, $\widehat{F(\cdot - a)} = e^{ia\cdot p}\hat{F}$, and $\widehat{F \circ \Lambda^{-1}} = \hat{F} \circ \Lambda^{-1}$ for $\Lambda \in \mathcal{L}$. For $\alpha \in \mathcal{S}(\mathbb{R})$ and $g \in \mathcal{S}(\mathbb{R}^2)$ we write $(\alpha \otimes g)(t, \vec{x}) = \alpha(t) g(\vec{x})$, $\hat{\alpha}(E) = \int \alpha(t) e^{iEt}\, dt$ and $\hat{g}(\vec{k}) = \int g(\vec{x}) e^{-i\vec{k}\cdot\vec{x}}\, d^2x$, so that $\widehat{\alpha \otimes g}(p) = \hat{\alpha}(p^0)\hat{g}(\vec{p})$; for $R > 0$, $g_R = g(\cdot/R)$ has the transform $R^2 \hat{g}(R\,\cdot)$.

The frame. We use the frame (M1)–(M5) of Definition 3.1 (3) over σ', with its signs $\vartheta_1 = \vartheta_2 = 1$, $\vartheta_3 = -1$, its forms $\mathfrak{q}_{ij} = \mathfrak{q}_{ji}$ and its kernels s_{ij}, and with the real-time components $T_{00} = -T^E_{33}$, $T_{0k} = T_{k0} = -iT^E_{3k}$ and $T_{kk'} = T^E_{kk'}$; we read it as follows. The integer in the bound of (M1) is l, a bound with one integer giving the bound with every larger one, and $T^E_{ij}(F)$, $F \in \mathcal{S}$, is the operator $\mathfrak{q}_{ij}(F)$ of Lemma 18.17 (c). We put $\bar{c} = \max\{\|\mathsf{A}_l\|, c_T\}$ and $\mathsf{c}_N(f) = \bar{c} \sum_{n \leq N} \binom{N}{n} \|\partial_t^n f\|_{L^1}$ for $f \in \mathcal{S}$; this number is at least the number YM3_I:(2071), so that YM3_I:(2075) holds with it. In (M2), for every $m \geq 1$, $\mathsf{s}_{ij}(x; y_1, \ldots, y_m)$ is continuous on $(\mathbb{R}^3)^{1+m}_{\neq}$ and symmetric in $y_1, \ldots, y_m$, and $\mathsf{s}_{ij}(\mathsf{R}x + b; \mathsf{R}y_1 + b, \ldots, \mathsf{R}y_m + b) = \mathsf{R}_{ii'}\mathsf{R}_{jj'}\mathsf{s}_{i'j'}(x; y)$ for $\mathsf{R} \in \mathrm{O}(3)$ and $b \in \mathbb{R}^3$. For tuples X, X' over $\mathbb{E}_+$ with smooth entries, the empty tuple admitted, for $a > 0$ and for $x \neq y$ in $\{|x_3| < a\}$ we put $\mathsf{s}_{ij}^{X',X}(x, y) = \int X'(z') X(z)\, \mathsf{s}_{ij}(x; y, \theta(z' + ae_3), z + ae_3)\, dz'\, dz$, and the relation of the kernels to the forms is the product formula

$$\mathsf{s}_{ij}^{X',X}(x, y) = \begin{cases} \big\langle [X'], \mathcal{U}(\vec{x})\, \mathsf{B}_{ij}\big(a + x_3, \frac{y_3 - x_3}{2}\big)\, \mathcal{U}(\vec{y} - \vec{x})\, \mathsf{B}_0\big(\frac{y_3 - x_3}{2}, a - y_3\big)\, \mathcal{U}(-\vec{y})[X] \big\rangle & (x_3 < y_3), \\ \big\langle [X'], \mathcal{U}(\vec{y})\, \mathsf{B}_0\big(a + y_3, \frac{x_3 - y_3}{2}\big)\, \mathcal{U}(\vec{x} - \vec{y})\, \mathsf{B}_{ij}\big(\frac{x_3 - y_3}{2}, a - x_3\big)\, \mathcal{U}(-\vec{x})[X] \big\rangle & (y_3 < x_3), \end{cases} \tag{18.i}$$

where $\mathsf{B}_{ij} = \mathsf{B}_{\mathfrak{q}_{ij}}$ is the operator of Lemma 18.17 (b): the factor of the lower height stands on the left. The two lines are the relation $\mathsf{s}_{ij}^{F',G}(x, y) = \mathfrak{q}_{ij}(\psi'_x, \psi_x)$ to which (M2) of Definition 3.1 (3) refers, with the vectors ψ'_x, ψ_x that the proof of Lemma 3.5 (c) writes out for $x_3 < y_3$ and for $y_3 < x_3$, taken at the tuples $F' = X'$ and $G = X$, whose entries are real, so that the smeared kernel there is $\mathsf{s}_{ij}^{X',X}$, and at the intermediate height $h_* = \frac{1}{2}(x_3 + y_3)$: since the unitary group $\mathcal{U}$ commutes with the functions of H, the formula for the matrix elements in Lemma 18.17 (b) carries $\mathfrak{q}_{ij}(\psi'_x, \psi_x)$ into the first line when $x_3 < y_3$, and, with $\mathsf{B}_0(s, s')^* = \mathsf{B}_0(s', s)$, into the second line when $y_3 < x_3$. In (M3) the test functions F_1 and f lie in $C^\infty_c(\mathbb{R}^{1+2}; \mathbb{C})$ and their supports are spacelike separated: $|\vec{x} - \vec{x}'| > |t - t'|$ for $(t, \vec{x})$ in the one and $(t', \vec{x}')$ in the other. (M4) (a) reads $T_{0\nu}(\partial_0 F)\psi = \sum_{k=1,2} T_{k\nu}(\partial_k F)\psi$ for $F \in \mathcal{S}$, $\psi \in \mathcal{D}_0$ and $\nu = 0, 1, 2$. Let T be the space of the

symmetric complex tensors $t = (t^{\mu\nu})$, $T[t] = t^{\mu\nu}T_{\mu\nu}$, $e \vee e' = \frac{1}{2}(e \otimes e' + e' \otimes e)$ and $ee = e \vee e$ for $e, e' \in \mathbb{C}^3$, and $(\Lambda t)^{\mu\nu} = \Lambda^{\mu}{}_{\mu'}\Lambda^{\nu}{}_{\nu'}t^{\mu'\nu'}$; then $T[e \vee e'] = e^{\mu}e'^{\nu}T_{\mu\nu} = T[e, e']$, and (M5), extended by linearity, reads $U(\Lambda)T[t](F)\Omega = T[\Lambda t](F \circ \Lambda^{-1})\Omega$ for $\Lambda \in \mathcal{L}$, $F \in \mathcal{S}$ and $t \in \mathsf{T}$. We write $\mathfrak{M}^4$ for (M1), (M2), (M3) and (M4) (a), and $\mathfrak{M}^5$ for $\mathfrak{M}^4$ together with (M5).

Definition 18.16 (Flux laws of the translations)**.** Let kernels s_{ij} as in (M2) be given over σ'. A function $\eta \in C_c^\infty(\mathbb{R}^3)$ is *admitted at* $y \in (\mathbb{R}^3)^m_{\neq}$ if $\eta = 1$ on a neighbourhood of y_1 and $y_r \notin \operatorname{supp}\eta$ for $2 \le r \le m$. For $j \in \{1, 2, 3\}$ the *flux law* (FL_j) is the condition

$$\sum_{i=1}^{3} \int \partial_i\eta(x)\, \mathsf{s}_{ij}(x; y)\, dx = \partial_{y_1^j}\mathsf{s}_m(y) \qquad \text{for all } m \ge 1,\ y \in (\mathbb{R}^3)^m_{\neq} \text{ and } \eta \text{ admitted at } y, \tag{18.ii}$$

where $\partial_{y_1^j}$ is the derivative in the j-th coordinate of y_1. The *connected kernels* are given by recursion on m: $\mathsf{s}^{\mathrm{c}}_m$, $m \ge 1$, by $\mathsf{s}_m(y) = \sum_{A \ni 1} \mathsf{s}^{\mathrm{c}}_{|A|}(y_A)\, \mathsf{s}_{m-|A|}(y_{A^c})$, and $\mathsf{s}^{\mathrm{c}}_{ij}(x; y_1, \dots, y_m)$ by $\mathsf{s}_{ij}(x; y_1, \dots, y_m) = \sum_{A \neq \varnothing} \mathsf{s}^{\mathrm{c}}_{ij}(x; y_A)\, \mathsf{s}_{m-|A|}(y_{A^c})$; here A runs through the subsets of $\{1, \dots, m\}$, $y_A = (y_r)_{r \in A}$, A^c is the complement of A, and in each recursion the term $A = \{1, \dots, m\}$ carries the new function with the factor $\mathsf{s}_0 = 1$. The condition ($\mathrm{FL}^{\mathrm{c}}_j$) is eq. 18.ii for all $m \ge 2$, with $\mathsf{s}^{\mathrm{c}}_{ij}$ and $\mathsf{s}^{\mathrm{c}}_m$ in the places of s_{ij} and s_m. ■

The connected kernels are polynomials in the kernels s_n and s_{ij}; the scalar ones are real-analytic, so the right members are defined. Under (M1), (M2) and (M4) (a) the left member of eq. 18.ii is the same for all functions admitted at y (Lemma 18.20 (a)): the law fixes the flux of $\mathsf{s}_{ij}(\cdot\,; y)$ through the boundary of a neighbourhood of y_1 that avoids the other points, and with it the normalisation of the tensor relative to the generators of the translations of σ'. The real part of a function η admitted at y is again admitted at y, and $\eta - \operatorname{Re}\eta = i \operatorname{Im}\eta$ lies in $C_c^\infty(\mathbb{R}^3 \setminus \{y_1, \dots, y_m\})$, as $\operatorname{Im}\eta$ vanishes near y_1; so, under the same conditions, Lemma 18.18 (d) carries eq. 18.ii from the real functions admitted at y to all of them.

Lemma 18.17 (Forms, operators and boundary values)**.** Let $\mathfrak{q}$ be a sesquilinear form on $\mathcal{D}_\infty \times \mathcal{D}_\infty$, antilinear in the first entry, with $|\mathfrak{q}(\psi', \psi)| \le c\, \|(1 + H)^l\psi'\|\, \|(1 + H)^l\psi\|$.

(a) There is exactly one bounded operator $\mathsf{A}_\mathfrak{q}$ with $\mathfrak{q}(\psi', \psi) = \langle (1 + H)^l\psi', \mathsf{A}_\mathfrak{q}(1 + H)^l\psi\rangle$ for $\psi, \psi' \in \mathcal{D}_\infty$, and $\|\mathsf{A}_\mathfrak{q}\| \le c$. For $g \in L^1(\mathbb{R}^2)$, $\mathfrak{q}[g](\psi', \psi) = \int g(v)\, \mathfrak{q}(\mathcal{U}(-v)\psi', \mathcal{U}(-v)\psi)\, dv$ is a form of the same kind, with $c\|g\|_{L^1}$ in the place of c, and $\mathsf{A}_{\mathfrak{q}[g]} = \int g(v)\, \mathcal{U}(v)\mathsf{A}_\mathfrak{q}\mathcal{U}(-v)\, dv$ weakly.

(b) For $\operatorname{Re}\tau, \operatorname{Re}\tau' > 0$ the operator $\mathsf{B}_\mathfrak{q}(\tau, \tau') = e^{-\tau H}(1 + H)^l\mathsf{A}_\mathfrak{q}(1 + H)^l e^{-\tau' H}$ is bounded and holomorphic in norm; $\langle \xi', \mathsf{B}_\mathfrak{q}(\tau, \tau')\xi\rangle = \mathfrak{q}(e^{-\bar\tau H}\xi', e^{-\tau' H}\xi)$ for $\xi, \xi' \in \mathcal{H}$; $\|\mathsf{B}_\mathfrak{q}(\tau, \tau')\| \le c\, b_l(\operatorname{Re}\tau)\, b_l(\operatorname{Re}\tau')$ with $b_l(r) = \sup_{E \ge 0}(1 + E)^l e^{-rE} \le e^r(l/er)^l$; $e^{-zH}\mathsf{B}_\mathfrak{q}(\tau, \tau')e^{-z' H} = \mathsf{B}_\mathfrak{q}(\tau + z, \tau' + z')$ for $\operatorname{Re} z, \operatorname{Re} z' \ge 0$; and $\mathsf{B}_{\mathfrak{q}[g]} = \int g(v)\, \mathcal{U}(v)\mathsf{B}_\mathfrak{q}\mathcal{U}(-v)\, dv$. For $\mathfrak{q} = q_0$, $\mathsf{A}_{q_0} = \mathsf{A}_l$ and $\mathsf{B}_{q_0}(s, s') = \mathsf{B}_0(s, s')$.

(c) For $F \in \mathcal{S}$ there is exactly one operator $\mathfrak{q}(F) : \mathcal{D}_\infty \to \mathcal{D}_\infty$ with $\langle \psi', \mathfrak{q}(F)\psi\rangle = \int F(x)\, \mathfrak{q}(U(x)^{-1}\psi', U(x)^{-1}\psi)\, dx$ for $\psi, \psi' \in \mathcal{D}_\infty$. It satisfies $\|(1 + H)^n\mathfrak{q}(F)\psi\| \le c \sum_{n' \le l+n} \binom{l+n}{n'} \|\partial_t^{n'}F\|_{L^1}\, \|(1 + H)^{2l+n}\psi\|$, $[H, \mathfrak{q}(F)] = i\,\mathfrak{q}(\partial_t F)$ on $\mathcal{D}_\infty$, $U(a)\mathfrak{q}(F)U(a)^{-1} = \mathfrak{q}(F(\cdot - a))$, $\langle \psi', \mathfrak{q}(\alpha \otimes g)\psi\rangle = \int \alpha(t)\, \mathfrak{q}[g](e^{-itH}\psi', e^{-itH}\psi)\, dt$, $\mathfrak{q}(F)\Omega = (1 + H)^l\hat F(P)\mathsf{A}_\mathfrak{q}\Omega$ and $(1 + H)^N\mathfrak{q}(F)\Omega = \mathfrak{q}((1 + i\partial_t)^N F)\Omega$. If $\overline{\mathfrak{q}(\psi', \psi)} = \vartheta\, \mathfrak{q}(\psi, \psi')$ with $\vartheta = \pm 1$, then $\langle \psi', \mathfrak{q}(F)\psi\rangle = \vartheta\langle \mathfrak{q}(\bar F)\psi', \psi\rangle$; so $\langle \psi', T_{\mu\nu}(F)\psi\rangle = \langle T_{\mu\nu}(\bar F)\psi', \psi\rangle$. For $\mathfrak{q} = q_0$, $\mathfrak{q}(f) = \phi(f)$.

(d) Let $\mathfrak{q}_1, \mathfrak{q}_2$ be two such forms with the numbers c_1, c_2, let $g_1, g_2 \in C_c^\infty(\mathbb{R}^2)$, $\chi, \chi' \in \mathcal{H}$, $a > 0$, and $\mathcal{G}_< = \{(s, u) \in \mathbb{C}^2 : -a < \operatorname{Re} s < \operatorname{Re} u < a\}$. The function $\Gamma_<(s, u) = \langle \chi', \mathsf{B}_{\mathfrak{q}_1[g_1]}(a + s, \frac{u-s}{2})\, \mathsf{B}_{\mathfrak{q}_2[g_2]}(\frac{u-s}{2}, a - u)\chi\rangle$ is holomorphic on $\mathcal{G}_<$ and bounded by $c_1c_2\|g_1\|_{L^1}\|g_2\|_{L^1}\|\chi\|\|\chi'\|\, b_l(a + \operatorname{Re} s)\, b_l(\frac{1}{2}\operatorname{Re}(u - s))^2\, b_l(a - \operatorname{Re} u)$; and for

$r\in(-a,a)$, $\psi'=e^{-(a+r)H}\chi'$, $\psi=e^{-(a-r)H}\chi$ and $\alpha_1,\alpha_2\in C_c^\infty(\mathbb{R})$,

$$\lim_{\epsilon\downarrow0}\iint\alpha_1(t)\alpha_2(t')\,\Gamma_<(r-it,r+\epsilon-it')\,dt\,dt'=\langle\psi',\mathfrak{q}_1(\alpha_1\otimes g_1)\,\mathfrak{q}_2(\alpha_2\otimes g_2)\psi\rangle.$$

■

Proof. (a) $(1+H)^l$ maps $\mathcal{D}_\infty$ onto the dense subspace $\mathcal{D}_\infty$, so $(\xi',\xi)\mapsto\mathfrak{q}(R_H^l\xi',R_H^l\xi)$ is a sesquilinear form on a dense subspace bounded by $c\|\xi'\|\|\xi\|$, and the theorem of Riesz gives $\mathsf{A}_\mathfrak{q}$. $\mathcal{U}(-v)$ is unitary and commutes with $(1+H)^l$ on $\mathcal{D}_\infty$ by joint spectral calculus; hence $\mathfrak{q}(\mathcal{U}(-v)\psi',\mathcal{U}(-v)\psi)=\langle(1+H)^l\psi',\mathcal{U}(v)\mathsf{A}_\mathfrak{q}\mathcal{U}(-v)(1+H)^l\psi\rangle$ is continuous in v and bounded by $c\|(1+H)^l\psi'\|\|(1+H)^l\psi\|$, and its integral against g has the stated properties. (b) On the carrier of $\mathcal{P}$, $|e^{-\tau E}(1+E)^l|\le b_l(\operatorname{Re}\tau)$, and $(1+E)^le^{-rE}=e^ry^le^{-ry}$ at $y=1+E$, where $\sup_{y>0}y^le^{-ry}=(l/er)^l$; holomorphy in norm, the semigroup law and $(e^{-\tau H})^*=e^{-\bar\tau H}$ belong to the calculus of $\mathcal{P}$, and the matrix element is (a) at the vectors $e^{-\bar\tau H}\xi',e^{-\tau'H}\xi$ of $\mathcal{D}_\infty$. For q_0, YM3_I:(2047) gives $\mathsf{A}_{q_0}=\mathsf{A}_l$, and the point-insertion identity above gives $\mathsf{B}_{q_0}=\mathsf{B}_0$. (c) Let $\mathfrak{q}_F(\psi',\psi)$ denote the integral of the statement; as in (a) its integrand is continuous and bounded, and $|\mathfrak{q}_F(\psi',\psi)|\le c\|F\|_{L^1}\|(1+H)^l\psi'\|\|(1+H)^l\psi\|$. For $\psi\in\mathcal{D}_\infty$ the map $t\mapsto e^{-itH}(1+H)^l\psi$ is continuously differentiable in norm with the derivative $-iHe^{-itH}(1+H)^l\psi$ by the spectral theorem; as the form is antilinear in its first entry, $\partial_t\,\mathfrak{q}(U(x)^{-1}\psi',U(x)^{-1}\psi)=i\,\mathfrak{q}(U(x)^{-1}H\psi',U(x)^{-1}\psi)-i\,\mathfrak{q}(U(x)^{-1}\psi',U(x)^{-1}H\psi)$, and an integration by parts in t gives

$$\mathfrak{q}_F(H\psi',\psi)-\mathfrak{q}_F(\psi',H\psi)=i\,\mathfrak{q}_{\partial_tF}(\psi',\psi).$$

Hence $\mathfrak{q}_F((1+H)\chi',\psi)=\mathfrak{q}_F(\chi',(1+H)\psi)+i\,\mathfrak{q}_{\partial_tF}(\chi',\psi)$; the two operations commute, so $\mathfrak{q}_F((1+H)^N\chi',\psi)=\sum_{n'}\binom{N}{n'}i^{n'}\mathfrak{q}_{\partial_t^{n'}F}(\chi',(1+H)^{N-n'}\psi)$, and at $\chi'=R_H^l\psi'$, $N=l+n$, one obtains $|\mathfrak{q}_F((1+H)^n\psi',\psi)|\le c\sum_{n'}\binom{l+n}{n'}\|\partial_t^{n'}F\|_{L^1}\|\psi'\|\,\|(1+H)^{2l+n}\psi\|$. At $n=0$ the theorem of Riesz gives the vector $\mathfrak{q}(F)\psi$; for $n\ge1$, the graph-domain criterion YM3_I:(2073) puts it into $\operatorname{dom}H^n$ with the bound of the statement, so that $\mathfrak{q}(F)\mathcal{D}_\infty\subset\mathcal{D}_\infty$, and the last display is $[H,\mathfrak{q}(F)]=i\,\mathfrak{q}(\partial_tF)$. The covariance under $U(a)$ is a change of variables, and the formula for $\alpha\otimes g$ is the theorem of Fubini with $U(x)^{-1}=e^{-itH}\mathcal{U}(-\vec x)$. As $U(x)^{-1}\Omega=\Omega=(1+H)^l\Omega$, $\langle\psi',\mathfrak{q}(F)\Omega\rangle=\int F(x)\langle(1+H)^l\psi',U(x)\mathsf{A}_\mathfrak{q}\Omega\rangle\,dx=\langle(1+H)^l\psi',\hat F(P)\mathsf{A}_\mathfrak{q}\Omega\rangle$, and $(1+p^0)^l\hat F(p)$ is bounded; this is the formula on Ω, and the next one follows from $(1+p^0)^N\hat F=\widehat{(1+i\partial_t)^NF}$. The relation of adjoints follows by complex conjugation under the integral; for T_{00} and $T_{kk'}$ the sign is $\vartheta=1$, and for $T_{0k}=-iT^E_{3k}$ it is $\vartheta_3\vartheta_k=-1$, which the factor $-i$ compensates. For q_0 the defining relation is YM3_I:(2074). (d) On $\mathcal{G}_<$ all arguments of the factors have positive real parts, and (b) gives holomorphy and the bound. Let $0<\epsilon<a-r$, $s=r-it$, $u=r+\epsilon-it'$, $\psi_\epsilon=e^{-(a-r-\epsilon)H}\chi$ and $\mathsf{A}_b=\mathsf{A}_{\mathfrak{q}_b[g_b]}$. As $e^{-(a+s)H}=e^{-(a+r)H}e^{itH}$, $e^{-(u-s)H}=e^{-\epsilon H}e^{-itH}e^{it'H}$ and $e^{-(a-u)H}=e^{-it'H}e^{-(a-r-\epsilon)H}$,

$$\Gamma_<(s,u)=\langle\chi',\mathsf{Y}_t\,e^{-\frac{\epsilon}{2}H}e^{it'H}(1+H)^l\mathsf{A}_2(1+H)^le^{-it'H}\psi_\epsilon\rangle,\qquad\mathsf{Y}_t=e^{-(a+r)H}e^{itH}(1+H)^l\mathsf{A}_1(1+H)^le^{-itH}e^{-\frac{\epsilon}{2}H},$$

with Y_t bounded; the integrand of the double integral is continuous with compact support. For $\xi\in\mathcal{H}$, the bracket with ξ in the place of $\mathsf{Y}_t^*\chi'$ equals $\mathfrak{q}_2[g_2](e^{-it'H}e^{-\frac{\epsilon}{2}H}\xi,e^{-it'H}\psi_\epsilon)$, and its integral against $\alpha_2(t')$ is $\langle\xi,e^{-\frac{\epsilon}{2}H}\mathfrak{q}_2(\alpha_2\otimes g_2)\psi_\epsilon\rangle$ by (c); with $\xi=\mathsf{Y}_t^*\chi'$ and (c) for $\mathfrak{q}_1$ the double integral becomes $\langle\psi',\mathfrak{q}_1(\alpha_1\otimes g_1)\zeta_\epsilon\rangle$, $\zeta_\epsilon=e^{-\epsilon H}\mathfrak{q}_2(\alpha_2\otimes g_2)\psi_\epsilon\in\mathcal{D}_\infty$. Since $\|(1+H)^N(\psi_\epsilon-\psi)\|\le\sup_{E\ge0}(1+E)^Ne^{-(a-r)E}(e^{\epsilon E}-1)\,\|\chi\|\to0$, (c) gives $(1+H)^{2l}\zeta_\epsilon\to(1+H)^{2l}\mathfrak{q}_2(\alpha_2\otimes g_2)\psi$ as $\epsilon\downarrow0$, and (c) once more gives the limit. □

Lemma 18.18 (Generators, a core, local conservation)**.** (a) For $f\in\mathcal{S}$, $\psi\in\mathcal{D}_\infty$ and $\mu=0,1,2$, $[P_\mu,\phi(f)]\psi=i\,\phi(\partial_\mu f)\psi$; moreover $P_\mu\Omega=0$ and $P_\mu\phi(f_1)\cdots\phi(f_n)\Omega=i\sum_{b=1}^n\phi(f_1)\cdots\phi(\partial_\mu f_b)\cdots\phi(f_n)\Omega$.

(b) $\mathcal{D}_0$ is a core of $(1+H)^n$ for every integer $n\ge0$.

(c) Let (M1) and (M4) (a) hold. Then (M4) (a) holds for all $\psi \in \mathcal{D}_\infty$, and for $\psi', \psi \in \mathcal{D}_\infty$, $j = 1, 2, 3$ and $g' \in C_c^\infty(\mathbb{R}^2)$

$$\begin{aligned} \mathfrak{q}_{3j}(H\psi', \psi) - \mathfrak{q}_{3j}(\psi', H\psi) &= i \sum_{k\le 2} \big[\mathfrak{q}_{kj}(P_k\psi', \psi) - \mathfrak{q}_{kj}(\psi', P_k\psi)\big], \\ \mathfrak{q}_{3j}[g'](H\psi', \psi) - \mathfrak{q}_{3j}[g'](\psi', H\psi) &= -\sum_{k\le 2} \mathfrak{q}_{kj}[\partial_k g'](\psi', \psi). \end{aligned} \tag{18.iii}$$

If (M3) holds as well, it holds for all $\psi \in \mathcal{D}_\infty$.

(d) Let (M1), (M2) and (M4) (a) hold. For $m \ge 1$, $y \in (\mathbb{R}^3)^m_{\neq}$, $\zeta \in C_c^\infty(\mathbb{R}^3 \setminus \{y_1, \dots, y_m\})$ and $j = 1, 2, 3$, $\sum_i \int \partial_i \zeta(x)\, \mathsf{s}_{ij}(x; y)\, dx = 0$; the same holds for the connected kernels $\mathsf{s}^{\mathrm{c}}_{ij}$ of Definition 18.16.

■

Proof. (a) For $\mu = 0$ this is the last relation of YM3_I:(2076). For $\mu = k \le 2$, $U(se_k) = e^{isP_k}$, and the second relation of YM3_I:(2076) gives $\langle e^{isP_k}\psi', \phi(f) e^{isP_k}\psi\rangle = \langle \psi', \phi(f(\cdot + se_k))\psi\rangle$ for $\psi, \psi' \in \mathcal{D}_\infty$. Both members are differentiable at $s = 0$: on the left, $(1 + H)^{2l}\psi$ lies in $\operatorname{dom} P_k$, so $s \mapsto e^{isP_k}\psi$ is differentiable for the norm $\|(1 + H)^{2l} \cdot \|$, and YM3_I:(2075) applies; on the right, $s^{-1}(f(\cdot + se_k) - f) \to \partial_k f$ for every c_N. The derivatives give $-i\langle \psi', [P_k, \phi(f)]\psi\rangle = \langle \psi', \phi(\partial_k f)\psi\rangle$, as $\phi(f)\psi \in \mathcal{D}_\infty \subset \operatorname{dom} P_k$. $U(x)\Omega = \Omega$ gives $P_\mu \Omega = 0$, and moving P_μ to the right through the factors gives the last formula. (b) Let $\xi \in \operatorname{dom} H^n$ with $\langle (1 + H)^n \xi, (1 + H)^n \psi\rangle = 0$ for all $\psi \in \mathcal{D}_0$, and let $\zeta = (1 + H)^n \xi$. As $e^{itH} = U((t, \vec{0}))$ maps $\mathcal{D}_0$ onto itself (Theorem YM3_I: AO), $\int (1 + E)^n e^{itE}\, d\mu_\psi(E) = 0$ for all real t, where $\mu_\psi = \langle \zeta, \mathcal{P}_H(\cdot)\psi\rangle$ and $\mathcal{P}_H$ is the spectral measure of H; $(1 + E)^n \mu_\psi$ is a finite complex measure with vanishing Fourier transform, so it vanishes, and $\langle \zeta, \psi\rangle = \mu_\psi(\mathbb{R}) = 0$; $\mathcal{D}_0$ being dense, $\zeta = 0$ and $\xi = 0$. (c) By Lemma 18.17 (c) and YM3_I:(2075), both members of (M4) (a), and of (M3), are bounded by $C\|(1 + H)^{4l}\psi\|$; by (b) they extend from $\mathcal{D}_0$ to $\operatorname{dom} H^{4l} \supset \mathcal{D}_\infty$. In the Euclidean components, (M4) (a) at $\nu = k$ reads $-iT^E_{3k}(\partial_t F) = \sum_{k' \le 2} T^E_{k'k}(\partial_{k'} F)$, and at $\nu = 0$, multiplied by i and with $\mathfrak{q}_{k'3} = \mathfrak{q}_{3k'}$, it reads $-iT^E_{33}(\partial_t F) = \sum_{k' \le 2} T^E_{k'3}(\partial_{k'} F)$; so $-iT^E_{3j}(\partial_t F) - \sum_{k \le 2} T^E_{kj}(\partial_k F) = 0$ on $\mathcal{D}_\infty$ for $j = 1, 2, 3$. For $\psi', \psi \in \mathcal{D}_\infty$ the functions $\mathfrak{t}_{ij}(x) = \mathfrak{q}_{ij}(U(x)^{-1}\psi', U(x)^{-1}\psi)$ are continuously differentiable and bounded with their first derivatives, and an integration by parts against every $F \in \mathcal{S}$ gives $i\,\partial_t \mathfrak{t}_{3j} + \sum_{k \le 2} \partial_k \mathfrak{t}_{kj} = 0$. At $x = 0$, $\partial_t \mathfrak{t} = i\,\mathfrak{q}(H\psi', \psi) - i\,\mathfrak{q}(\psi', H\psi)$ by the proof of Lemma 18.17 (c), and $\partial_k \mathfrak{t} = i\,\mathfrak{q}(P_k\psi', \psi) - i\,\mathfrak{q}(\psi', P_k\psi)$, as $\mathcal{U}(-\vec{x}) = e^{-i\vec{x}\cdot P}$; this is the first identity of eq. 18.iii. As $\mathcal{U}(-v)$ commutes with H on $\mathcal{D}_\infty$, the first identity at the vectors $\mathcal{U}(-v)\psi', \mathcal{U}(-v)\psi$ has the right member $\sum_{k \le 2} \partial_{v_k} \mathfrak{q}_{kj}(\mathcal{U}(-v)\psi', \mathcal{U}(-v)\psi)$; multiplication by $g'(v)$ and an integration by parts in v give the second identity. (d) This is the last part of Lemma 3.5 (c). Its argument runs as follows. At a configuration with pairwise different heights in which a scalar point y_{r_0} is the lowest point above x, an open slab contains x and y_{r_0}, and its closure contains no other point; after a translation, under which (M2) and Proposition YM3_I:108.3 keep the kernels, it is $\{|x_3| < a\}$, and the other points, smeared by bumps, form a tuple X above the slab and, through their mirror images, a tuple X' below it. By the first line of eq. 18.i, the symmetry in the scalar points and the semigroup law of Lemma 18.17 (b), for x near the given point and $x_3 < h_* < y_{r_0,3}$, $\mathsf{s}^{X',X}_{ij}(x, y_{r_0}) = \mathfrak{q}_{ij}(\psi'_x, \psi_x)$ with $\psi'_x = \mathcal{U}(-\vec{x}) e^{-(a+x_3)H}[X']$ and $\psi_x = \mathcal{U}(-\vec{x}) e^{-(h_* - x_3)H} \mathcal{U}(\vec{y}_{r_0}) \mathsf{B}_0(y_{r_0,3} - h_*, a - y_{r_0,3}) \mathcal{U}(-\vec{y}_{r_0})[X]$. As $\partial_{x_3}\psi'_x = -H\psi'_x$, $\partial_{x_3}\psi_x = H\psi_x$, $\partial_{x_k}\psi'_x = -iP_k\psi'_x$ and $\partial_{x_k}\psi_x = -iP_k\psi_x$, the divergence $\sum_i \partial_{x_i} \mathsf{s}^{X',X}_{ij}(x, y_{r_0})$ is the difference of the two members of the first identity of eq. 18.iii with the sign changed, hence 0, and approximate units for the bumps give the identity for ζ carried near x. A rotation of SO(3), under which (M2) moves the indices as a tensor, brings every configuration of distinct points to pairwise different heights with y_1 above x, and a partition of unity gives every ζ. For the connected kernels, by induction on m: in the recursion of Definition 18.16 the terms with $A \neq \{1, \dots, m\}$ are connected kernels of fewer points times factors free of x, and ζ vanishes near the points of every y_A. □

Lemma 18.19 (One scalar insertion; smooth tuples; products). (a) For tuples X, X' over $\mathbb{E}_+$ with n and n' entries, $a > 0$ and $y \in \mathbb{R}^3$ with $|y_3| < a$,

$$\int X'(z')X(z)\,\mathsf{s}_{1+n'+n}(y, \theta(z' + ae_3), z + ae_3)\,dz'\,dz = \mathsf{n}^{X',X}(y),$$

$$\mathsf{n}^{X',X}(y) = \langle [X'], \mathcal{U}(\vec{y})\,\mathsf{B}_0(a + y_3, a - y_3)\,\mathcal{U}(-\vec{y})[X]\rangle.$$

For $\chi, \chi' \in \mathcal{H}$ the function $\mathsf{n}(y) = \langle \chi', \mathcal{U}(\vec{y})\mathsf{B}_0(a+y_3, a-y_3)\mathcal{U}(-\vec{y})\chi\rangle$, with $\mathsf{B}_0(a+y_3, a-y_3) = e^{-(a+y_3)H}(1+H)^l \mathsf{A}_l (1+H)^l e^{-(a-y_3)H}$, is defined for $\vec{y} \in \mathbb{R}^2$ and complex y_3 with $|\operatorname{Re} y_3| < a$; it is smooth there and holomorphic in y_3, and $\mathsf{n}(\vec{y}, -it') = q_0(U(y)^{-1}e^{-aH}\chi', U(y)^{-1}e^{-aH}\chi)$ at $y = (t', \vec{y}) \in \mathbb{R}^{1+2}$.

(b) The classes of the tuples over $\mathbb{E}_+$ with smooth entries span a dense subspace of $\mathcal{H}$.

(c) Let $I \subset \mathbb{R}^{n_1}$ and $B \subset \mathbb{R}^{n_2}$ be bounded open sets and $f \in C_c^\infty(I \times B; \mathbb{C})$. There are finite sums $f_M = \sum_b \beta_b \otimes \varkappa_b$ with $\beta_b \in C_c^\infty(I; \mathbb{C})$ and $\varkappa_b \in C_c^\infty(B; \mathbb{C})$, all carried by one compact subset of $I \times B$, such that $f_M \to f$ in every C^N.

■

Proof. (a) The right member is continuous in y by Proposition YM3_I:107.2 and the strong continuity of $\mathcal{U}$, the left member by the continuity of $\mathsf{s}_{1+n'+n}$ on the compact set of the points involved; so it suffices to integrate both against every $h \in C_c^1(\mathbb{R}^3)$ carried by $\{|y_3 - u_0| < c_0\eta\}$, where $|u_0| < a$ and $0 < \eta \le \min\{1, a - |u_0|\}$. By YM3_I:(2065) the integral of the left member against h is $\sigma'(\theta\tau_{ae_3}X' \cdot (h) \cdot \tau_{ae_3}X)$, and by (Rec1) for the translation by $-u_0e_3$, which turns $\theta\tau_{ae_3}$ into $\theta\tau_{(a+u_0)e_3}$ and τ_{ae_3} into $\tau_{(a-u_0)e_3}$, it equals $\sigma'(\theta\tau_{\eta e_3}Y' \cdot (h_0) \cdot \tau_{\eta e_3}Y)$ with $h_0 = h(\cdot + u_0e_3) \in C_c^1(\mathsf{Sl}_\eta)$ and the tuples $Y = \tau_{(a-u_0-\eta)e_3}X$, $Y' = \tau_{(a+u_0-\eta)e_3}X'$ over $\mathbb{E}_+$, the two shifts being ≥ 0. By YM3_I:(2037) and YM3_I:(2039) this is $\int h_0(x)\langle [Y'], \mathcal{U}(\vec{x})\mathsf{B}_0(\eta + x_3, \eta - x_3)\mathcal{U}(-\vec{x})[Y]\rangle\,dx$; $[Y] = e^{-(a-u_0-\eta)H}[X]$ and $[Y'] = e^{-(a+u_0-\eta)H}[X']$ by (Rec2), and the shift law YM3_I:(2038) turns the bracket into $\langle [X'], \mathcal{U}(\vec{x})\mathsf{B}_0(a + u_0 + x_3, a - u_0 - x_3)\mathcal{U}(-\vec{x})[X]\rangle$; the substitution $y = x + u_0e_3$ gives the right member integrated against h. Smoothness in $\vec{y}$ follows from $\mathcal{U}(\vec{y}) = e^{i\vec{y}\cdot P}$ and the boundedness of $P_k e^{-sH}$ for $s > 0$, holomorphy in y_3 from Lemma 18.17 (b); at $y_3 = -it'$, $e^{-(a\mp it')H} = e^{-aH}e^{\pm it'H}$ and $U(y)^{-1} = e^{-it'H}\mathcal{U}(-\vec{y})$ give the last formula by YM3_I:(2047). (b) The classes of the tuples are dense. For a tuple X with n entries let X^ε be the tuple of their mollifications, whose supports lie, for small ε, in fixed pairwise disjoint compact subsets of $\mathbb{E}_+$; by YM3_I:(2065), $\|[X] - [X^\varepsilon]\|^2 = \int \mathsf{s}_{2n}(\theta z', z)\,(X(z') - X^\varepsilon(z'))\,(X(z) - X^\varepsilon(z))\,dz'\,dz$, where the kernel is bounded on the compact set of the points involved and $X^\varepsilon \to X$ uniformly. (c) Choose $\chi_1 \in C_c^\infty(I)$ and $\chi_2 \in C_c^\infty(B)$ equal to 1 on the projections of $\operatorname{supp} f$, place $I \times B$ in an open cube, continue f periodically from it and expand it in its Fourier series, which converges in every C^N; the partial sums, multiplied by $\chi_1 \otimes \chi_2$, are the f_M. □

Lemma 18.20 (The flux law at one scalar point and for connected kernels). Let (M1), (M2) and (M4) (a) hold, and let $j \in \{1, 2, 3\}$.

(a) For $y \in (\mathbb{R}^3)^m_{\neq}$ the left member of eq. 18.ii is the same for all functions η admitted at y.

(b) For $y_1 \in \mathbb{R}^3$ and every η admitted at (y_1), $\sum_i \int \partial_i\eta(x)\,\mathsf{s}_{ij}(x; y_1)\,dx = 0 = \partial_{y_1^j}\mathsf{s}_1$; so eq. 18.ii holds at $m = 1$.

(c) (FL_j) holds if and only if $(\mathrm{FL}_j^{\mathrm{c}})$ holds.

■

Proof. (a) Two functions admitted at y differ by a function of $C_c^\infty(\mathbb{R}^3 \backslash \{y_1, \dots, y_m\})$; Lemma 18.18 (d). (b) By (M2) for the translations, $\mathsf{s}_{ij}(x; y_1) = \mathsf{t}_{ij}(x - y_1)$ with $\mathsf{t}_{ij}(\xi) = \mathsf{s}_{ij}(\xi; 0)$, and $\mathsf{t}_{ij}(\mathsf{R}\xi) = \mathsf{R}_{ii'}\mathsf{R}_{jj'}\mathsf{t}_{i'j'}(\xi)$ for $\mathsf{R} \in \mathrm{SO}(3)$. By (a) we may take $\eta(x) = \eta_0(|x - y_1|)$ with $\eta_0 \in C_c^\infty([0, \infty))$ equal to 1 near 0; the left member is then $\mathsf{V}_j = \sum_i \int \eta_0'(|\xi|)\,\xi_i|\xi|^{-1}\mathsf{t}_{ij}(\xi)\,d\xi$, and the substitution $\xi \mapsto \mathsf{R}\xi$ gives $\mathsf{V}_j = \mathsf{R}_{jj'}\mathsf{V}_{j'}$ for every $\mathsf{R} \in \mathrm{SO}(3)$, so $\mathsf{V} = 0$; s_1 is a constant. (c) Fix $y \in (\mathbb{R}^3)^m_{\neq}$ and η admitted

at y, and put $\mathsf{F}(A) = \sum_i \int \partial_i \eta(x)\, \mathsf{s}^{\mathrm{c}}_{ij}(x; y_A)\, dx$ for $\varnothing \neq A \subset \{1, \dots, m\}$. By the recursion, the left member of eq. 18.ii is $\sum_{A \neq \varnothing} \mathsf{F}(A)\, \mathsf{s}_{m-|A|}(y_{A^c})$. If $1 \notin A$, η vanishes near the points of y_A, and $\mathsf{F}(A) = 0$ by Lemma 18.18 (d) for the connected kernels; $\mathsf{s}^{\mathrm{c}}_{ij}(x; y_1) = \mathsf{s}_{ij}(x; y_1)$, so $\mathsf{F}(\{1\}) = 0$ by (b); if $1 \in A$ and $|A| \geq 2$, η is admitted at y_A. The right member is $\sum_{A \ni 1} \partial_{y_1^j} \mathsf{s}^{\mathrm{c}}_{|A|}(y_A)\, \mathsf{s}_{m-|A|}(y_{A^c})$, whose term $A = \{1\}$ vanishes, $\mathsf{s}^{\mathrm{c}}_1 = \mathsf{s}_1$ being a constant. Hence $(\mathrm{FL}^{\mathrm{c}}_j)$ gives (FL_j) term by term; conversely, by induction on m, (FL_j) at m together with $(\mathrm{FL}^{\mathrm{c}}_j)$ at the orders $2, \dots, m-1$ leaves $\mathsf{F}(\{1, \dots, m\}) = \partial_{y_1^j} \mathsf{s}^{\mathrm{c}}_m(y)$. □

Lemma 18.21 (An edge of the wedge in one complex variable, with a parameter)**.** Let $a, d_* > 0$, let $\mathcal{G}_<$ be as in Lemma 18.17 (d), $\mathcal{G}_> = \{(s,u) \in \mathbb{C}^2 : -a < \operatorname{Re} u < \operatorname{Re} s < a\}$ and $\mathcal{G}_{d_*} = \{(s,u) \in \mathbb{C}^2 : |\operatorname{Re} s| < a,\ |\operatorname{Re} u| < a,\ |\operatorname{Im}(s-u)| < d_*\}$. Let $\Gamma_<$ and $\Gamma_>$ be holomorphic on $\mathcal{G}_<$ and $\mathcal{G}_>$, and let there be an integer $p \geq 1$ and for every $\delta > 0$ a number C_δ with $|\Gamma_{\lessgtr}(s,u)| \leq C_\delta |\operatorname{Re}(u-s)|^{-p}$ wherever $|\operatorname{Re} s|, |\operatorname{Re} u| \leq a - \delta$. Suppose that for every $r \in (-a, a)$ and all $\alpha_1, \alpha_2 \in C^\infty_c(\mathbb{R})$ with $|t - t'| < d_*$ on $\operatorname{supp}\alpha_1 \times \operatorname{supp}\alpha_2$

$$\lim_{\epsilon \downarrow 0} \iint \alpha_1(t)\alpha_2(t') \big[\Gamma_<(r - it, r + \epsilon - it') - \Gamma_>(r + \epsilon - it, r - it')\big]\, dt\, dt' = 0.$$

Then there is exactly one holomorphic function Γ on $\mathcal{G}_{d_*}$ equal to $\Gamma_<$ on $\mathcal{G}_< \cap \mathcal{G}_{d_*}$ and to $\Gamma_>$ on $\mathcal{G}_> \cap \mathcal{G}_{d_*}$. ■

Proof. (1) Write $c = \frac{1}{2}(s+u)$ and $\varsigma = u - s$; then $\max\{|\operatorname{Re} s|, |\operatorname{Re} u|\} = |\operatorname{Re} c| + \frac{1}{2}|\operatorname{Re}\varsigma|$. Fix $0 < \delta < a/3$, put $\mathcal{C} = \{|\operatorname{Re} c| < a - 2\delta\}$ and $\mathcal{W}_\pm = \{0 < \pm \operatorname{Re}\varsigma < 2\delta\}$, and let $\Gamma_+(c,\varsigma)$ and $\Gamma_-(c,\varsigma)$ be $\Gamma_<$ and $\Gamma_>$ at $(s,u) = (c - \frac{\varsigma}{2}, c + \frac{\varsigma}{2})$: they are holomorphic on $\mathcal{C} \times \mathcal{W}_\pm$ and bounded there by $C_\delta |\operatorname{Re}\varsigma|^{-p}$. (2) Put $\Phi^0_\pm = \Gamma_\pm$ and $\Phi^{k+1}_\pm(c,\varsigma) = \int_{\pm\delta}^{\varsigma} \Phi^k_\pm(c,\varsigma')\, d\varsigma'$, the integral being taken in the convex strip $\mathcal{W}_\pm$; these functions are holomorphic in (c,ς). Integrating first along the segment from $\pm\delta$ to $\pm\delta + i\operatorname{Im}\varsigma$ and then parallel to the real axis, one finds on $\{|\operatorname{Im}\varsigma| \leq d_*\}$, with constants that do not depend on $c \in \mathcal{C}$, $|\Phi^k_\pm| \leq C_k |\operatorname{Re}\varsigma|^{k-p}$ for $k < p$, $|\Phi^p_\pm| \leq C_p(1 + |\log|\operatorname{Re}\varsigma||)$ and $|\Phi^{p+1}_\pm| \leq C_{p+1}$. So $\Phi_\pm = \Phi^{p+2}_\pm$ is Lipschitz in ς, uniformly in c; it extends continuously to $\operatorname{Re}\varsigma = 0$, its boundary values $\Psi_\pm(c,\tau) = \lim_{\epsilon\downarrow 0} \Phi_\pm(c, \pm\epsilon - i\tau)$ are uniform limits and therefore holomorphic in c (theorem of Weierstrass), and $\Gamma_\pm(c, \pm\epsilon - i\tau) = (i\partial_\tau)^{p+2}[\Phi_\pm(c, \pm\epsilon - i\tau)]$. (3) Put $N = p + 2$, $t_+ = \frac{1}{2}(t + t')$ and $t_- = t' - t$, a change of variables with Jacobian 1; the points $(r - it, r + \epsilon - it')$ and $(r + \epsilon - it, r - it')$ are $(c,\varsigma) = (r + \frac{\epsilon}{2} - it_+, \pm\epsilon - it_-)$. For $|r| < a - 2\delta$ and $\Upsilon \in C^\infty_c(\mathbb{R}^2)$, a function of (t_+, t_-) carried by $\{|t_-| < d_*\}$, N integrations by parts in t_- and the uniform convergence give $\iint \Upsilon\, \Gamma_\pm(r + \frac{\epsilon}{2} - it_+, \pm\epsilon - it_-)\, dt_+\, dt_- \to \iint ((-i\partial_{t_-})^N \Upsilon)\, \Psi_\pm(r - it_+, t_-)\, dt_+\, dt_-$. For $\Upsilon = \alpha_1(t)\alpha_2(t')$ the hypothesis says that the two limits are equal. With a partition of unity, a function Υ carried by $\{|t_-| < d_*\}$ is written as a sum of finitely many functions, each carried by a product of intervals in the variables (t,t') that lies inside $\{|t - t'| < d_*\}$; Lemma 18.19 (c) then shows that each of them is a limit in C^N of sums of such products, all carried by a single compact set. Hence the continuous function $\mathsf{d}(t_+, t_-) = (\Psi_+ - \Psi_-)(r - it_+, t_-)$ satisfies $\partial^N_{t_-} \mathsf{d} = 0$ as a distribution on $\mathbb{R} \times (-d_*, d_*)$; after a mollification it is a polynomial in t_- of degree $< N$, and so is d, with coefficients depending on t_+. Thus $(\Psi_+ - \Psi_-)(c,\tau) = \sum_{n<N} a_n(c)\tau^n$ for $c \in \mathcal{C}$ and $|\tau| < d_*$, and the values at N fixed points τ, through a Vandermonde system, show that the a_n are holomorphic on $\mathcal{C}$. (4) With $\mathsf{p}(c,\varsigma) = \sum_{n<N} a_n(c)(i\varsigma)^n$ put $\Phi = \Phi_+ - \mathsf{p}$ where $\operatorname{Re}\varsigma \geq 0$ and $\Phi = \Phi_-$ where $\operatorname{Re}\varsigma \leq 0$, on $\mathcal{C} \times \{|\operatorname{Re}\varsigma| < 2\delta,\ |\operatorname{Im}\varsigma| < d_*\}$. At $\operatorname{Re}\varsigma = 0$ the two definitions agree, as $\mathsf{p}(c, -i\tau) = \sum_n a_n(c)\tau^n$; Φ is continuous, holomorphic in ς for fixed c by the theorem of Morera, and holomorphic in c for fixed ς; by the theorem of Osgood it is holomorphic. $\Gamma = \partial^N_\varsigma \Phi$ equals $\Gamma_\pm$ on the two sides, p having degree $< N$ in ς. As $\delta \downarrow 0$, the sets so obtained, together with $\mathcal{G}_< \cap \mathcal{G}_{d_*}$ and $\mathcal{G}_> \cap \mathcal{G}_{d_*}$, cover $\mathcal{G}_{d_*}$; on overlaps the functions agree off $\{\operatorname{Re}(u-s) = 0\}$, hence everywhere by continuity. $\mathcal{G}_{d_*}$ is convex, so the identity theorem gives uniqueness. □

Lemma 18.22 (The commutator of a smeared component with the scalar field)**.** Let $\mathfrak{M}^4$ hold and $j \in \{1,2,3\}$. Put $\mathcal{J}_j(\alpha, g) = T^E_{3j}(\alpha \otimes g)$, $\mathfrak{L}_j f = \partial_j f$ for $j \le 2$ and $\mathfrak{L}_3 f = i\partial_t f$. Let $\alpha \in C^\infty_c(\mathbb{R})$, let $\varkappa, g \in C^\infty_c(\mathbb{R}^2)$ with $g = 1$ near $\operatorname{supp}\varkappa$, and let $0 < d_* < \operatorname{dist}(\operatorname{supp}\varkappa, \operatorname{supp}\nabla g)$.

(a) For tuples X, X' over $\mathbb{E}_+$ with smooth entries and $a > 0$ there is exactly one holomorphic function D on $\{u \in \mathbb{C} : |\operatorname{Re} u| < a\}$ such that $\sum_i \iint \partial_i\eta(x)\, h(y)\, \mathsf{s}^{X',X}_{ij}(x,y)\, dx\, dy = \int \beta_1(u)\, \mathsf{D}(u)\, du$ whenever $\beta_1, \chi_1 \in C^\infty_c((-a,a))$, $\chi_1 = 1$ near $\operatorname{supp}\beta_1$, $\eta(x) = \chi_1(x_3) g(\vec{x})$ and $h(y) = \beta_1(y_3)\varkappa(\vec{y})$.

(b) For $\beta \in C^\infty_c(\mathbb{R})$ with $|t - t'| < d_*$ on $\operatorname{supp}\alpha \times \operatorname{supp}\beta$, $\psi = e^{-aH}[X]$ and $\psi' = e^{-aH}[X']$: $\langle \psi', [\mathcal{J}_j(\alpha, g), \phi(\beta \otimes \varkappa)]\psi\rangle = (\int \alpha) \int \beta(t')\, \mathsf{D}(-it')\, dt'$.

(c) The function $\mathsf{N}(u) = \int \varkappa(\vec{y})\, (\partial_j \mathsf{n}^{X',X})(\vec{y}, u)\, d\vec{y}$, with $\mathsf{n}^{X',X}$ of Lemma 18.19 (a) and ∂_3 the derivative in y_3, is holomorphic on $\{|\operatorname{Re} u| < a\}$, and $\int \beta(t')\, \mathsf{N}(-it')\, dt' = -\langle \psi', \phi(\mathfrak{L}_j(\beta \otimes \varkappa))\psi\rangle$ for every $\beta \in C^\infty_c(\mathbb{R})$.

(d) If (FL_j) holds, then $\mathsf{D} = \mathsf{N}$. If moreover $f \in C^\infty_c(\mathbb{R}^{1+2}; \mathbb{C})$ is carried by $I_f \times B_f$, where I_f is a bounded open interval and B_f an open disc, $\operatorname{supp}\alpha$ lies in a bounded interval I_α, $t_* = \sup\{|t - t'| : t \in I_\alpha, t' \in I_f\}$, $g = 1$ near $\overline{B_f}$ and $\operatorname{dist}(B_f, \operatorname{supp}\nabla g) > t_*$, then $[\mathcal{J}_j(\alpha, g), \phi(f)]\psi = -(\int \alpha)\phi(\mathfrak{L}_j f)\psi$ for all $\psi \in \mathcal{D}_\infty$.

■

Proof. Fix $\chi, \chi' \in \mathcal{H}$ and $a > 0$, and let $\mathfrak{x} = \mathfrak{q}_{3j}[g]$, $\mathfrak{z} = \sum_{k\le 2} \mathfrak{q}_{kj}[\partial_k g]$ and $\mathfrak{y} = q_0[\varkappa]$ (Lemma 18.17 (a)); by the second identity of eq. 18.iii at $g' = g$, $\mathfrak{x}(H\psi', \psi) - \mathfrak{x}(\psi', H\psi) = -\mathfrak{z}(\psi', \psi)$ on $\mathcal{D}_\infty \times \mathcal{D}_\infty$. (1) On $\mathcal{G}_<$ and $\mathcal{G}_>$ (Lemma 18.21) put

$$G_<(s,u) = \langle \chi', \mathsf{B}_\mathfrak{x}(a+s, \tfrac{u-s}{2})\mathsf{B}_\mathfrak{y}(\tfrac{u-s}{2}, a-u)\chi\rangle, \qquad G_>(s,u) = \langle \chi', \mathsf{B}_\mathfrak{y}(a+u, \tfrac{s-u}{2})\mathsf{B}_\mathfrak{x}(\tfrac{s-u}{2}, a-s)\chi\rangle,$$

and let $Z_<, Z_>$ be the same with $\mathfrak{z}$ in the place of $\mathfrak{x}$; they are holomorphic by Lemma 18.17 (b). We have $\partial_s G_\lessgtr = Z_\lessgtr$. Indeed, by Lemma 18.17 (b), $G_<(s,u) = \mathfrak{x}(e^{-(a+\bar{s})H}\chi', \xi_s)$ with $\xi_s = \mathsf{B}_\mathfrak{y}(u - s, a - u)\chi \in \mathcal{D}_\infty$; as s moves, the factor $e^{-(a+s)H}(1+H)^l$ to the left of $\mathsf{A}_\mathfrak{x}$ acquires the derivative $-He^{-(a+s)H}(1+H)^l$ and $(1+H)^l\xi_s$ the derivative $(1+H)^l H\xi_s$, so that $\partial_s G_< = -\mathfrak{x}(He^{-(a+\bar{s})H}\chi', \xi_s) + \mathfrak{x}(e^{-(a+\bar{s})H}\chi', H\xi_s) = \mathfrak{z}(e^{-(a+\bar{s})H}\chi', \xi_s) = Z_<$; in the same way $G_>(s,u) = \mathfrak{x}(\mathsf{B}_\mathfrak{y}(a+u, s-u)^*\chi', e^{-(a-s)H}\chi)$, where again the factor to the left of $\mathsf{A}_\mathfrak{x}$ acquires $-H$ and the one to the right $+H$. (2) $Z_<$ and $Z_>$ satisfy the hypotheses of Lemma 18.21 with $p = 2l$. The bound follows from Lemma 18.17 (d), (b) and $b_l(r) \le e^{2a}(l/e)^l r^{-l}$ for $0 < r \le 2a$. By Lemma 18.17 (d), applied as it stands and with the two forms exchanged, the two smeared boundary values at $r \in (-a, a)$ are $\langle \psi'_r, \mathfrak{z}(\alpha_1)\phi(\alpha_2 \otimes \varkappa)\psi_r\rangle$ and $\langle \psi'_r, \phi(\alpha_2 \otimes \varkappa)\mathfrak{z}(\alpha_1)\psi_r\rangle$, where $\mathfrak{z}(\alpha_1) = \sum_{k\le 2} T^E_{kj}(\alpha_1 \otimes \partial_k g)$, the vectors $\psi'_r = e^{-(a+r)H}\chi'$ and $\psi_r = e^{-(a-r)H}\chi$ lie in $\mathcal{D}_\infty$, and $q_0(\alpha_2 \otimes \varkappa) = \phi(\alpha_2 \otimes \varkappa)$ by Lemma 18.17 (c). If $|t - t'| < d_*$ on $\operatorname{supp}\alpha_1 \times \operatorname{supp}\alpha_2$, the supports of $\alpha_1 \otimes \partial_k g$ and of $\alpha_2 \otimes \varkappa$ are spacelike separated, since $|\vec{x} - \vec{y}| \ge \operatorname{dist}(\operatorname{supp}\nabla g, \operatorname{supp}\varkappa) > d_* > |t - t'|$ on them, and (M3) on $\mathcal{D}_\infty$ (Lemma 18.18 (c)) makes the two boundary values equal. So there is one holomorphic function Z on $\mathcal{G}_{d_*}$ equal to $Z_\lessgtr$ on $\mathcal{G}_\lessgtr \cap \mathcal{G}_{d_*}$. (3) $\mathcal{Z}(s,u) = \int_u^s Z(w,u)\, dw$, along the segment from u to s, which keeps the point in $\mathcal{G}_{d_*}$, is holomorphic on $\mathcal{G}_{d_*}$, and $\partial_s\mathcal{Z} = Z$. On $\mathcal{G}_< \cap \mathcal{G}_{d_*}$, $\partial_s(G_< - \mathcal{Z}) = 0$, and for fixed u the set of the s with (s,u) in it is convex; so $G_< = \mathcal{Z} + c_<(u)$ there, and likewise $G_> = \mathcal{Z} + c_>(u)$ on $\mathcal{G}_> \cap \mathcal{G}_{d_*}$, with $c_<, c_>$ holomorphic on $\{|\operatorname{Re} u| < a\}$ (near u_0, $c_<(u) = (G_< - \mathcal{Z})(u - \epsilon_0, u)$ for a fixed small $\epsilon_0 > 0$). Put $\mathsf{D} = c_< - c_>$. (4) Let $\chi = [X]$ and $\chi' = [X']$. Integrating eq. 18.i against $g(\vec{x})\varkappa(\vec{y})$, respectively against $\partial_k g(\vec{x})\varkappa(\vec{y})$ summed over $k \le 2$, and using the last formula of Lemma 18.17 (b), one finds for real $s \ne u$ in $(-a, a)$

$$\iint g(\vec{x})\varkappa(\vec{y})\, \mathsf{s}^{X',X}_{3j}((\vec{x}, s), (\vec{y}, u))\, d\vec{x}\, d\vec{y} = G_\lessgtr(s,u),$$

$$\sum_{k\le 2} \iint \partial_k g(\vec{x})\varkappa(\vec{y})\, \mathsf{s}^{X',X}_{kj}((\vec{x}, s), (\vec{y}, u))\, d\vec{x}\, d\vec{y} = Z_\lessgtr(s,u) \quad (s \lessgtr u).$$

The second integrand stays continuous at $s=u$, as $\operatorname{supp}\nabla g$ and $\operatorname{supp}\varkappa$ are disjoint and the kernels are continuous; so the second integral equals Z on the whole of $(-a,a)^2$. With $\partial_3\eta=\chi_1'(x_3)g(\vec{x})$ and $\partial_k\eta=\chi_1(x_3)\partial_k g(\vec{x})$, the integrand of (a) is continuous with compact support, and the left member of (a) is $\int\beta_1(u)\big[\int\chi_1'(s)G(s,u)\,ds+\int\chi_1(s)Z(s,u)\,ds\big]du$, with $G=G_<$ for $s<u$ and $G=G_>$ for $s>u$. By (3), $\int\chi_1'(s)G(s,u)\,ds=-\int\chi_1(s)Z(s,u)\,ds+(c_<(u)-c_>(u))\chi_1(u)$, and $\chi_1=1$ on $\operatorname{supp}\beta_1$: this is (a), and, β_1 being arbitrary, D is fixed on $(-a,a)$, hence on the strip. (5) At $r=0$, Lemma 18.17 (d) gives $\iint\alpha(t)\beta(t')G_<(-it,\epsilon-it')\,dt\,dt'\to\langle\psi',\mathcal{J}_j(\alpha,g)\phi(\beta\otimes\varkappa)\psi\rangle$ and $\iint\alpha(t)\beta(t')G_>(\epsilon-it,-it')\,dt\,dt'\to\langle\psi',\phi(\beta\otimes\varkappa)\mathcal{J}_j(\alpha,g)\psi\rangle$ as $\epsilon\downarrow 0$. On the supports the points lie in $\mathcal{G}_{d_*}$, and by (3) the difference of the two integrals is $\iint\alpha\beta\,[\mathcal{Z}(-it,\epsilon-it')-\mathcal{Z}(\epsilon-it,-it')]+\iint\alpha\beta\,[c_<(\epsilon-it')-c_>(-it')]$, which tends to $(\int\alpha)\int\beta(t')\,\mathsf{D}(-it')\,dt'$ by the continuity of $\mathcal{Z}$ and of $c_<$: this is (b). (6) By Lemma 18.19 (a), $\mathsf{n}^{X',X}$ is holomorphic in y_3, and at $y_3=-it'$ it equals $\mathsf{w}(y)=q_0(U(y)^{-1}\psi',U(y)^{-1}\psi)$, $y=(t',\vec{y})$, with $(\partial_k\mathsf{n}^{X',X})(\vec{y},-it')=\partial_{y^k}\mathsf{w}(y)$ and $(\partial_3\mathsf{n}^{X',X})(\vec{y},-it')=i\partial_{t'}\mathsf{w}(y)$. So $\int\beta(t')\mathsf{N}(-it')\,dt'$ is $\int(\beta\otimes\varkappa)\,\partial_j\mathsf{w}\,dy$ for $j\le 2$ and $\int(\beta\otimes\varkappa)\,i\partial_{t'}\mathsf{w}\,dy$ for $j=3$, and an integration by parts and YM3_I:(2074) give (c). (7) If (FL_j) holds, multiply eq. 18.ii at the points $(y_1;\theta(z'+ae_3),z+ae_3)$ by $h(y_1)X'(z')X(z)$ and integrate: $\eta=\chi_1(x_3)g(\vec{x})$ is admitted at these points, the left member gives $\int\beta_1\mathsf{D}$ by (a), and the right member gives $\int h\,\partial_j\mathsf{n}^{X',X}=\int\beta_1\mathsf{N}$ by Lemma 18.19 (a); so $\mathsf{D}=\mathsf{N}$ on $(-a,a)$, and on the strip. With (b) and (c), $\langle\psi',[\mathcal{J}_j(\alpha,g),\phi(\beta\otimes\varkappa)]\psi\rangle=-(\int\alpha)\langle\psi',\phi(\mathfrak{L}_j(\beta\otimes\varkappa))\psi\rangle$. For fixed a both members are bounded sesquilinear forms of $([X'],[X])$, by Lemma 18.17 (c), YM3_I:(2075) and the boundedness of $(1+H)^Ne^{-aH}$; by Lemma 18.19 (b) the identity holds at $\psi'=e^{-aH}\chi'$, $\psi=e^{-aH}\chi$ for all $\chi,\chi'\in\mathcal{H}$, and as $e^{-aH}\psi\to\psi$ for $a\downarrow 0$ in every norm $\|(1+H)^N\cdot\|$, $\psi\in\mathcal{D}_\infty$, it is an identity of vectors on $\mathcal{D}_\infty$. Finally, by Lemma 18.19 (c), f is a limit in every C^N of sums of products $\beta_b\otimes\varkappa_b$ with $\operatorname{supp}\beta_b\subset I_f$ and $\operatorname{supp}\varkappa_b\subset B_f$, all carried by one compact set; each product satisfies the hypotheses above with any d_* strictly between t_* and $\operatorname{dist}(B_f,\operatorname{supp}\nabla g)$, and $\mathsf{c}_N(f-f_M)+\mathsf{c}_N(\mathfrak{L}_j(f-f_M))\to 0$; Lemma 18.17 (c) and YM3_I:(2075) carry the identity to f. □

Proposition 18.23 (The commutators of $T_{0\mu}(\alpha\otimes g_R)$ with the scalar field)**.** Let $\mathfrak{M}^4$ hold. Let $\alpha\in C_c^\infty(\mathbb{R})$ with $\int\alpha=1$ and $\operatorname{supp}\alpha$ in a bounded interval I_α, and let $g\in C_c^\infty(\mathbb{R}^2)$ with $g=1$ on $\{|\vec{x}|\le r_g\}$, $r_g>0$; α and g may be complex. For $f\in C_c^\infty(\mathbb{R}^{1+2};\mathbb{C})$ carried by $I_f\times\{|\vec{x}|<r_O\}$, where I_f is a bounded open interval and $r_O>0$, put $t_*=\sup\{|t-t'|:t\in I_\alpha,\,t'\in I_f\}$ and $R_0=(r_O+t_*+1)/r_g$. Let $j_0=3$, $j_1=1$ and $j_2=2$.

(a) Let $\mu\in\{0,1,2\}$ and let (FL_{j_μ}) hold. Then for every such f, every $R\ge R_0$ and every $\psi\in\mathcal{D}_\infty$

$$[T_{0\mu}(\alpha\otimes g_R),\phi(f)]\psi=i\,\phi(\partial_\mu f)\psi=[P_\mu,\phi(f)]\psi;\tag{18.iv}$$

in particular the commutator does not depend on α, on g and on $R\ge R_0$.

(b) Conversely, let $\mu\in\{0,1,2\}$, and let the first equality of eq. 18.iv hold, for the given α and g, for every f as above, every $R\ge R_0$ and every $\psi\in\mathcal{D}_0$. Then (FL_{j_μ}) holds.

■

Proof. (a) By the components of (M1), $\mathcal{J}_3(\alpha,g_R)=T_{33}^E(\alpha\otimes g_R)=-T_{00}(\alpha\otimes g_R)$ with $\mathfrak{L}_3=i\partial_t$, and $\mathcal{J}_k(\alpha,g_R)=T_{3k}^E(\alpha\otimes g_R)=iT_{0k}(\alpha\otimes g_R)$ with $\mathfrak{L}_k=\partial_k$. As $g_R=1$ on $\{|\vec{x}|\le Rr_g\}\supset\{|\vec{x}|\le r_O+t_*+1\}$ and $\operatorname{supp}\nabla g_R\subset\{|\vec{x}|\ge Rr_g\}$ for $R\ge R_0$, the disc $B_f=\{|\vec{x}|<r_O\}$ has $g_R=1$ near $\overline{B_f}$ and $\operatorname{dist}(B_f,\operatorname{supp}\nabla g_R)\ge t_*+1$. Lemma 18.22 (d) with $\int\alpha=1$ gives $[T_{00}(\alpha\otimes g_R),\phi(f)]\psi=i\phi(\partial_t f)\psi$ and $[T_{0k}(\alpha\otimes g_R),\phi(f)]\psi=i\phi(\partial_k f)\psi$ for $\psi\in\mathcal{D}_\infty$; Lemma 18.18 (a) gives the second equality. (b) Let $j=j_\mu$. Both members of the first equality are bounded by $C\|(1+H)^{4l}\psi\|$, so by Lemma 18.18 (b) the hypothesis holds on $\mathcal{D}_\infty$, and by the components listed in (a) it says $[\mathcal{J}_j(\alpha,g_R),\phi(f)]\psi=-\phi(\mathfrak{L}_j f)\psi$. Fix $\varkappa\in C_c^\infty(\mathbb{R}^2)$ carried by $\{|\vec{x}|<r_O\}$, tuples X,X' over $\mathbb{E}_+$ with smooth entries, $a>0$, and $R\ge R_0$ for the given I_f and r_O; let D and N be the functions of Lemma 18.22 (a), (c) for g_R and $d_*=t_*+\frac12$, which is admissible as $\operatorname{dist}(\operatorname{supp}\varkappa,\operatorname{supp}\nabla g_R)\ge t_*+1$. For $\beta\in C_c^\infty(I_f)$, Lemma 18.22 (b), (c) and the hypothesis at $f=\beta\otimes\varkappa$ give $\int\beta(t')(\mathsf{D}-\mathsf{N})(-it')\,dt'=0$; so $\mathsf{D}-\mathsf{N}$

vanishes on the segment $\{-it' : t' \in I_f\}$, hence on the strip (identity theorem), and at real u: by Lemma 18.22 (a) and Lemma 18.19 (a), $\sum_i \iint \partial_i\eta(x)\,h(y)\,\mathsf{s}_{ij}^{X',X}(x,y)\,dx\,dy = \int h\,\partial_j \mathsf{n}^{X',X}\,dy$ for $\eta = \chi_1(x_3)g_R(\vec{x})$ and $h = \beta_1 \otimes \varkappa$. Now let $y \in (\mathbb{R}^3)^m_{\neq}$ with $m \geq 2$ and $y_{r,3} \neq y_{1,3}$ for $r \geq 2$. The case $m = 1$ is Lemma 18.20 (b). Translate all points by $-y_1$, carrying the admitted functions by the same translation. Both members of eq. 18.ii are invariant under this operation by (M2) and Proposition YM3_I:108.3. In these coordinates $y_1 = 0$. Take $0 < a < \min_{r\geq 2}|y_{r,3}|$ and choose $\chi_1 = 1$ near 0. Let β_1, $\varkappa$ and the entries of X and X' run through real approximate units about 0, about $\vec{y}_1$, about the points $y_r - ae_3$ with $y_{r,3} > a$, and about the points $\theta y_r - ae_3$ with $y_{r,3} < -a$. By the continuity of the kernels and of $\partial_{y_1}\mathsf{s}_m$, the identity becomes eq. 18.ii at y for the function $\chi_1(x_3)g_R(\vec{x})$, which is admitted at y; by Lemma 18.20 (a) it holds for every function admitted at y. Such y are dense in $(\mathbb{R}^3)^m_{\neq}$, and for a fixed η both members of eq. 18.ii are continuous at every point at which η is admitted; so (FL_j) holds. □

Lemma 18.24 (Products of fields; from compact supports to Schwartz functions)**.** Let (M1) hold.

(a) For $f_1, \dots, f_n \in \mathcal{S}$, $\xi \in \mathcal{D}_\infty$ and $N \geq 0$,

$$\|(1+H)^N \phi(f_1)\cdots\phi(f_n)\xi\| \leq \prod_{b=1}^{n} \mathsf{c}_{(2b-1)l+N}(f_b)\,\|(1+H)^{2nl+N}\xi\|.$$

(b) Let $\kappa \in C_c^\infty(\mathbb{R}^{1+2})$ with $0 \leq \kappa \leq 1$, $\kappa = 1$ on $\{|x| \leq 1\}$ and $\kappa = 0$ off $\{|x| < 2\}$, $|x|$ the Euclidean norm of $x \in \mathbb{R}^{1+2}$; for $r \geq 1$ and $f \in \mathcal{S}$ put $f^{(r)} = \kappa(\cdot/r)f$, and put $\|f\|_{(N)} = \sum_{n\leq N}\int(1+|x|)^N|\partial_t^n f|\,dx$. There are numbers $C_{N,M}$, formed from N, M, κ and $\bar{c}$, with $\mathsf{c}_N(f) + \mathsf{c}_N(f^{(r)}) \leq C_{N,0}\|f\|_{(N)}$ and $\mathsf{c}_N(f - f^{(r)}) \leq C_{N,M}\,r^{-M}\|f\|_{(N+M)}$ for all $r \geq 1$.

(c) For $R \geq 1$ let $\mathsf{K}_R = \sum_a c_a\,\mathfrak{q}_a(F_{a,R})$ be a finite sum, with $c_a \in \mathbb{C}$, each $\mathfrak{q}_a$ one of the forms $\mathfrak{q}_{ij}$, and $F_{a,R} \in \mathcal{S}$ with $\mathsf{c}_N(F_{a,R}) \leq C_N R^3$ for every N. Let $\mathfrak{D}$ be a differential operator of the first order with polynomial coefficients and $c_1, r_1 > 0$, and suppose that $[\mathsf{K}_R, \phi(f)]\psi = \phi(\mathfrak{D}f)\psi$ for all $\psi \in \mathcal{D}_\infty$ whenever $f \in C_c^\infty(\mathbb{R}^{1+2};\mathbb{C})$ is carried by $\{|x| < r\}$ and $R \geq c_1(r + r_1)$. Then for $f_1, \dots, f_n \in \mathcal{S}$ and integers $N, M \geq 0$ there is a number C, formed from N, M, n, l, $\bar{c}$, c_1, r_1, κ, the c_a, the numbers $C_{N'}$ and the f_b, with

$$\Big\|(1+H)^N\Big[\mathsf{K}_R\phi(f_1)\cdots\phi(f_n)\Omega - \phi(f_1)\cdots\phi(f_n)\mathsf{K}_R\Omega - \sum_{b=1}^{n}\phi(f_1)\cdots\phi(\mathfrak{D}f_b)\cdots\phi(f_n)\Omega\Big]\Big\| \leq C\,R^{-M} \qquad (R \geq 1).$$

■

Proof. (a) Induction on n with YM3_I:(2075). (b) $\partial_t^n(\kappa(\cdot/r)f)$ is the sum of the terms $\binom{n}{n'}r^{-n'}(\partial_t^{n'}\kappa)(\cdot/r)\,\partial_t^{n-n'}f$, bounded by multiples of $|\partial_t^{n-n'}f|$ for $r \geq 1$; for $f - f^{(r)}$ the same terms, with $1-\kappa$ in the place of κ, vanish on $\{|x| < r\}$, and $\int_{|x|\geq r}|\partial_t^{n'}f| \leq r^{-M}\int|x|^M|\partial_t^{n'}f|$. (c) By Lemma 18.17 (c), $\|(1+H)^{N'}\mathsf{K}_R\xi\| \leq CR^3\|(1+H)^{2l+N'}\xi\|$ for $\xi \in \mathcal{D}_\infty$, in particular $\|(1+H)^{N'}\mathsf{K}_R\Omega\| \leq CR^3$. For $1 \leq R \leq R_1 = c_1(3+2r_1)$ the left member is bounded by these bounds and (a). For $R \geq R_1$ put $r = \frac{1}{3}(R/c_1 - r_1)$; then $r \geq 1$ and $r \geq R/(6c_1)$. The functions $f_b^{(r)}$ are carried by $\{|x| < 3r\}$ and $R = c_1(3r + r_1)$; moving K_R to the right through the factors, one commutator at a time on vectors of $\mathcal{D}_\infty$, gives the identity with the remainder 0 for the $f_b^{(r)}$ in the places of the f_b. Each of the three expressions formed with the f_b differs from the one formed with the $f_b^{(r)}$ by a sum of terms with one entry $f_b - f_b^{(r)}$ or $\mathfrak{D}(f_b - f_b^{(r)})$; the latter is $(1-\kappa(\cdot/r))\mathfrak{D}f_b - [\mathfrak{D},\kappa(\cdot/r)]f_b$, a Schwartz function times smooth factors bounded with their derivatives uniformly in $r \geq 1$ and vanishing on $\{|x| < r\}$, so that its numbers $\mathsf{c}_{N'}$ are at most $C_{M'}r^{-M'}$, as in (b). With (a) every term is at most $CR^3r^{-M'}$, and the choice $M' = M + 3$ gives the claim. □

Lemma 18.25 (The spectrum near zero momentum)**.** (a) $U(\Lambda)\mathsf{E}(\Delta)U(\Lambda)^{-1} = \mathsf{E}(\Lambda\Delta)$ for $\Lambda \in \mathcal{L}$ and Borel sets $\Delta \subset \mathbb{R}^3$, and $\mathsf{E}(\{p : \vec{p} = 0\}) = \mathsf{E}(\{0\})$.

(b) Let ξ_R, $R \geq 1$, be vectors orthogonal to Ω with $\sup_R \|\xi_R\| < \infty$ and $\|\mathsf{E}(\{|\vec{p}| > \delta\})\xi_R\| \to 0$ as $R \to \infty$ for every $\delta > 0$. Then $\xi_R \to 0$ weakly in $\mathcal{H}$.

■

Proof. (a) By YM3_I:(2087) and the invariance of the Minkowski product, $U(\Lambda)U(a)U(\Lambda)^{-1} = U(\Lambda a) = \int e^{ia\cdot\Lambda^{-1}p}\, d\mathsf{E}(p)$; the projection-valued measures $\Delta \mapsto U(\Lambda)\mathsf{E}(\Delta)U(\Lambda)^{-1}$ and $\Delta \mapsto \mathsf{E}(\Lambda\Delta)$ both have this Fourier transform, and such a measure is unique. Let $\ell = \{(E,0,0) : E > 0\}$. The rays $B(s)\ell = \{E(\cosh s, \sinh s, 0) : E > 0\}$, $s \in \mathbb{R}$, are pairwise disjoint, so the projections $\mathsf{E}(B(s)\ell) = U(B(s))\mathsf{E}(\ell)U(B(s))^{-1}$ are pairwise orthogonal and all unitarily equivalent to $\mathsf{E}(\ell)$; as $\mathcal{H}$ is separable, $\mathsf{E}(\ell) = 0$. The set $\{\vec{p} = 0\}$ meets $V_+ \cup \partial V_+ \cup \{0\}$, which carries E, in $\ell \cup \{0\}$. (b) Let $\psi \in \mathcal{H}$, $\delta > 0$ and $S_\delta = \{p \neq 0 : |\vec{p}| \leq \delta\}$. As $\mathsf{E}(\{0\})\xi_R = \langle\Omega, \xi_R\rangle\Omega = 0$, $|\langle\psi, \xi_R\rangle| \leq \|\mathsf{E}(S_\delta)\psi\| \sup_{R'} \|\xi_{R'}\| + \|\psi\|\, \|\mathsf{E}(\{|\vec{p}| > \delta\})\xi_R\|$; as $\delta \downarrow 0$, $\|\mathsf{E}(S_\delta)\psi\|$ tends to $\|\mathsf{E}(\{p \neq 0 : \vec{p} = 0\})\psi\|$, which is 0 by (a). □

Lemma 18.26 (The two-point measure of the tensor)**.** Let (M1), (M4) (a) and (M5) hold.

(a) For $t, t' \in \mathsf{T}$ there is exactly one complex Radon measure $\mathcal{M}_{t,t'}$ on $\mathbb{R}^3$ with $\langle T[t](F')\Omega, \mathsf{E}(\Delta)T[t'](F)\Omega\rangle = \int_\Delta \overline{\hat{F}'}\hat{F}\, d\mathcal{M}_{t,t'}$ for all $F, F' \in \mathcal{S}$ and Borel sets Δ. This measure depends antilinearly on t and linearly on t'; moreover $\mathcal{M}_{t,t} \geq 0$, it is concentrated on $V_+ \cup \partial V_+$, $\int (1+p^0)^{-2l}\, d|\mathcal{M}_{t,t'}| < \infty$, and $\mathcal{M}_{\Lambda t,\Lambda t'}(\Lambda\Delta) = \mathcal{M}_{t,t'}(\Delta)$ holds for each $\Lambda \in \mathcal{L}$.

(b) There are positive Radon measures ϱ_0, ϱ_2 on $(0,\infty)$ with $\int (3+2m)^{-2l}(\varrho_0 + \varrho_2)(dm) < \infty$, so finite on $(0,1]$ and polynomially bounded, and a number $c_\partial \geq 0$ such that, with $\omega_m(\vec{p}) = (m^2 + |\vec{p}|^2)^{1/2}$, the measures $\mathsf{v}_i = \varrho_i(dm)\, d^2p/\omega_m(\vec{p})$ on V_+, written in the coordinates $(m, \vec{p}) \mapsto (\omega_m(\vec{p}), \vec{p})$ $(i = 0, 2)$, the measure $\mathsf{v}_\partial = c_\partial\, d^2p/|\vec{p}|$ on ∂V_+, $\Pi_{\mu\nu} = p_\mu p_\nu - \mathsf{g}_{\mu\nu}\, p\cdot p$, $t\cdot\Pi = t^{\mu\nu}\Pi_{\mu\nu}$, $\langle t\Pi t\Pi\rangle = t^{\mu\nu}\Pi_{\nu\mu'}t^{\mu'\nu'}\Pi_{\nu'\mu}$ and $t\cdot pp = t^{\mu\nu}p_\mu p_\nu$,

$$\mathcal{M}_{t,t} = (t\cdot\Pi)^2\,\mathsf{v}_0 + \big[2\langle t\Pi t\Pi\rangle - (t\cdot\Pi)^2\big]\mathsf{v}_2 \ \text{ on } V_+ \text{ for real } t, \qquad \mathcal{M}_{t,t'} = \overline{(t\cdot pp)}\,(t'\cdot pp)\,\mathsf{v}_\partial \ \text{ on } \partial V_+.$$

In particular, with $\varrho = \varrho_0 + \varrho_2$, $T[e_0e_0] = T_{00}$, $T[e_0 \vee e_k] = T_{0k}$ and $n = (0,\vec{n})$, $\vec{n} \in \mathbb{R}^2$: $\mathcal{M}_{e_0e_0,e_0e_0} = |\vec{p}|^4\big[\varrho(dm)\, d^2p/\omega_m(\vec{p}) + c_\partial\, d^2p/|\vec{p}|\big]$ and $\mathcal{M}_{e_0\vee n,e_0\vee n} = (p^0)^2(\vec{n}\cdot\vec{p})^2(\mathsf{v}_0 + \mathsf{v}_\partial) + |\vec{p}|^2\big((\vec{n}\cdot\vec{p})^2 + |\vec{n}|^2\, p\cdot p\big)\mathsf{v}_2$.

■

Proof. (1) Put $\mathsf{A}_{00} = -\mathsf{A}_{\mathsf{q}_{33}}$, $\mathsf{A}_{0k} = \mathsf{A}_{k0} = -i\mathsf{A}_{\mathsf{q}_{3k}}$, $\mathsf{A}_{kk'} = \mathsf{A}_{\mathsf{q}_{kk'}}$ and $\mathsf{A}_t = t^{\mu\nu}\mathsf{A}_{\mu\nu}$; by Lemma 18.17 (c), $T[t](F)\Omega = (1+H)^l\hat{F}(P)\mathsf{A}_t\Omega$, and $\mathcal{M}_{t,t'}(\Delta) = \int_\Delta (1+p^0)^{2l}\, d\langle\mathsf{A}_t\Omega, \mathsf{E}(p)\mathsf{A}_{t'}\Omega\rangle$ for bounded Borel sets Δ has the property of (a) by the calculus of E. Every $\chi \in C_c^\infty(\mathbb{R}^3)$ is a product $\overline{\hat{F}'}\hat{F}$, with $\hat{F} = \chi$ and $\hat{F}' \in C_c^\infty(\mathbb{R}^3)$ equal to 1 on $\operatorname{supp}\chi$: this gives uniqueness. Sesquilinearity, positivity and the bound of the variation of $(1+p^0)^{-2l}\mathcal{M}_{t,t'}$ by $\|\mathsf{A}_t\Omega\|\|\mathsf{A}_{t'}\Omega\|$ can be read off; $\mathcal{M}_{t,t'}(\{0\}) = \overline{\langle\Omega, \mathsf{A}_t\Omega\rangle}\langle\Omega, \mathsf{A}_{t'}\Omega\rangle = 0$, as $\langle\Omega, \mathsf{A}_{\mathsf{q}_{ij}}\Omega\rangle = \mathsf{q}_{ij}(\Omega,\Omega) = 0$. (M5), the unitarity of $U(\Lambda)$, Lemma 18.25 (a) and $\widehat{F\circ\Lambda^{-1}} = \hat{F}\circ\Lambda^{-1}$ give

$$\int \chi(\Lambda^{-1}p)\, \mathcal{M}_{\Lambda t,\Lambda t'}(dp) = \int \chi\, d\mathcal{M}_{t,t'} \qquad (\chi \in C_c^\infty(\mathbb{R}^3),\ \Lambda \in \mathcal{L}). \tag{18.v}$$

(2) For every finite family $t_1, t_2, \ldots$ and Borel Δ the matrix $(\mathcal{M}_{t_a,t_b}(\Delta))$ is positive semi-definite, hence bounded by its trace; so every $\mathcal{M}_{t,t'}$ is absolutely continuous with respect to the positive Radon measure $\bar{\mathcal{M}} = \sum_{\mu\le\nu}\mathcal{M}_{e_\mu\vee e_\nu, e_\mu\vee e_\nu}$, and the theorem of Radon and Nikodym gives $\mathcal{M}_{t,t'} = \mathsf{M}_p(t,t')\,\bar{\mathcal{M}}(dp)$ with Borel densities; outside a null set, M_p is a positive semi-definite Hermitian form on T, and for such a form $\mathsf{M}_p(t,t) = 0$ implies $\mathsf{M}_p(t,\cdot) = 0$. (3) For real $\mathsf{u} \in \mathbb{R}^3$, (M4) (a) at $\psi = \Omega$, contracted with u^ν, is $\sum_{\mu,\nu}\mathsf{g}^{\mu\mu}\mathsf{u}^\nu T_{\mu\nu}(\partial_\mu F)\Omega = 0$, that is $-i(1+H)^l\sum_{\mu,\nu}(p^\mu\mathsf{u}^\nu\hat{F})(P)\,\mathsf{A}_{\mu\nu}\Omega = 0$; the square of its norm is $\int |\hat{F}|^2\,\mathsf{M}_p(p\vee\mathsf{u}, p\vee\mathsf{u})\, d\bar{\mathcal{M}}$, and a choice of F with $\hat{F} > 0$ gives, outside a null set and then for all $\mathsf{u} \in \mathbb{C}^3$,

$$\mathsf{M}_p(p\vee\mathsf{u}, \cdot) = 0. \tag{18.vi}$$

(4) Let J, N_1, N_2 be the matrices with $Jp = (0, -p^2, p^1)$, $N_1 p = (p^1, p^0, 0)$ and $N_2 p = (p^2, 0, p^0)$; $e^{sJ} = 1 \oplus \mathsf{R}_s$ is the rotation by the angle s, $e^{sN_1} = B(s)$, and e^{sN_2} is the boost in the plane of (x^0, x^2), the conjugate of $B(s)$ by a rotation, all in $\mathcal{L}$ by the displayed boost and rotation formulas. For a real matrix G put $(\mathsf{G}t)^{\mu\nu} = \mathsf{G}^{\mu}{}_{\mu'} t^{\mu'\nu} + \mathsf{G}^{\nu}{}_{\mu'} t^{\mu\mu'}$. Differentiating eq. 18.v at $\Lambda = e^{s\mathsf{G}}$, $s = 0$, for $\mathsf{G} \in \{J, N_1, N_2\}$ (the measure $\mathcal{M}_{\Lambda t, \Lambda t'}$ is a polynomial in the entries of Λ with fixed measures as coefficients) gives $-\int (\mathsf{G}p) \cdot \nabla\chi \, d\mathcal{M}_{t,t'} + \int \chi \, d(\mathcal{M}_{\mathsf{G}t,t'} + \mathcal{M}_{t,\mathsf{G}t'}) = 0$. With $\Xi_p = p^0 J + p^2 N_1 - p^1 N_2$ one has $\Xi_p p = 0$; adding these identities for $(J, p^0\chi)$, $(N_1, p^2\chi)$ and $(N_2, -p^1\chi)$, the terms with derivatives of χ cancel, because $\Xi_p p = 0$ and $(Jp)^0 = (N_1 p)^2 = (N_2 p)^1 = 0$, and what remains is $\int \chi(p) \, [\mathsf{M}_p(\Xi_p t, t') + \mathsf{M}_p(t, \Xi_p t')] \, \bar{\mathcal{M}}(dp) = 0$. So, outside a null set,

$$\mathsf{M}_p(e^{s\Xi_p} t, e^{s\Xi_p} t') = \mathsf{M}_p(t, t') \qquad (s \in \mathbb{R}, \ t, t' \in \mathsf{T}); \tag{18.vii}$$

Ξ_p is real, antisymmetric for the Minkowski product, like J, N_1, N_2, and not 0 for $p \neq 0$. (5) Let $p \in V_+$ lie outside the null sets. $W = \{w \in \mathbb{R}^3 : w \cdot p = 0\}$ is spacelike, so $\langle w, w' \rangle = -w \cdot w'$ is a scalar product on it; $\mathbb{C}^3 = W_{\mathbb{C}} \oplus \mathbb{C}p$, and by eq. 18.vi, $\mathsf{M}_p(t, t') = \mathsf{M}_p(t_\perp, t'_\perp)$, $t_\perp$ the image of t under $n \mapsto n - \frac{n \cdot p}{p \cdot p} p$ in both slots. Ξ_p maps W into itself, is antisymmetric for $\langle \cdot, \cdot \rangle$ and not 0 on W; in an orthonormal basis u_1, u_2 of W, $\Xi_p u_1 = \varpi u_2$ and $\Xi_p u_2 = -\varpi u_1$ with $\varpi \neq 0$, and $\mathsf{z}_\pm = u_1 \pm i u_2$ satisfy $e^{s\Xi_p} \mathsf{z}_\pm = e^{\mp i\varpi s} \mathsf{z}_\pm$. By eq. 18.vii the tensors $\mathsf{I} = u_1 u_1 + u_2 u_2 = \mathsf{z}_+ \vee \mathsf{z}_-$, $\mathsf{z}_+ \mathsf{z}_+$ and $\mathsf{z}_- \mathsf{z}_-$, which are multiplied by the different characters 1, $e^{-2i\varpi s}$, $e^{2i\varpi s}$, are pairwise orthogonal for M_p. With $\beta_0 = \frac{1}{4} \mathsf{M}_p(\mathsf{I}, \mathsf{I})$ and $\beta_2 = \frac{1}{16} [\mathsf{M}_p(\mathsf{z}_+ \mathsf{z}_+, \mathsf{z}_+ \mathsf{z}_+) + \mathsf{M}_p(\mathsf{z}_- \mathsf{z}_-, \mathsf{z}_- \mathsf{z}_-)]$, both ≥ 0, a real $\mathsf{s} \in \mathrm{Sym}^2(W)$ with the matrix (s_{ab}) in the basis is $\frac{1}{2}(\mathrm{tr}\, \mathsf{s})\mathsf{I} + \frac{1}{2}(c_3 - ic_1)\mathsf{z}_+ \mathsf{z}_+ + \frac{1}{2}(c_3 + ic_1)\mathsf{z}_- \mathsf{z}_-$, where $c_3 = \frac{1}{2}(\mathsf{s}_{11} - \mathsf{s}_{22})$ and $c_1 = \mathsf{s}_{12}$, and $\mathsf{M}_p(\mathsf{s}, \mathsf{s}) = (\mathrm{tr}\, \mathsf{s})^2 \beta_0 + 4(c_3^2 + c_1^2)\beta_2 = (\mathrm{tr}\, \mathsf{s})^2 \beta_0 + [2\, \mathrm{tr}\, \mathsf{s}^2 - (\mathrm{tr}\, \mathsf{s})^2]\beta_2$. For real t the matrix of $t_\perp$ is $\mathsf{s}_{ab} = t^{\mu\nu} (u_a)_\mu (u_b)_\nu$, and $\sum_a (u_a)_\mu (u_a)_\nu = \Pi_{\mu\nu}/(p \cdot p)$; so $\mathrm{tr}\, t_\perp = t \cdot \Pi/(p \cdot p)$ and $\mathrm{tr}\, t_\perp^2 = \langle t \Pi t \Pi \rangle / (p \cdot p)^2$, and, for all real t at once,

$$(p \cdot p)^2 \, \mathsf{M}_p(t, t) = (t \cdot \Pi)^2 \beta_0 + [2\langle t \Pi t \Pi \rangle - (t \cdot \Pi)^2] \beta_2. \tag{18.viii}$$

At $t = \mathsf{g}^{-1}$ the bracket is 0 and $t \cdot \Pi = -2\, p \cdot p$; at $t = e_1 e_1$ the bracket is $(t \cdot \Pi)^2 = ((p^1)^2 + p \cdot p)^2$; so β_0 and $\beta_0 + \beta_2$ are Borel functions. (6) Let $p \in \partial V_+$ lie outside the null sets, and choose real $e_\perp, n_*$ with $e_\perp \cdot e_\perp = -1$, $e_\perp \cdot p = 0$, $n_* \cdot n_* = 0$, $n_* \cdot p = 1$ and $n_* \cdot e_\perp = 0$; $(p, e_\perp, n_*)$ is a basis. The antisymmetry and $\Xi_p p = 0$ give $\Xi_p e_\perp = c_p p$ and $\Xi_p n_* = c_p e_\perp$ with a real c_p, which is $\neq 0$ as $\Xi_p \neq 0$; so $e^{s\Xi_p} n_* = n_* + s c_p e_\perp + \frac{1}{2} s^2 c_p^2 p$ and, modulo the tensors $p \vee \mathsf{u}$, $e^{s\Xi_p}(n_* n_*) \equiv n_* n_* + 2 s c_p \, e_\perp \vee n_* + s^2 c_p^2 \, e_\perp e_\perp$. By eq. 18.vii and eq. 18.vi, $\mathsf{M}_p(e^{s\Xi_p} n_* n_*, e^{s\Xi_p} n_* n_*)$ is a polynomial in s that does not depend on s: its coefficient of s^4, $c_p^4 \mathsf{M}_p(e_\perp e_\perp, e_\perp e_\perp)$, vanishes, and then its coefficient of s^2, $4 c_p^2 \mathsf{M}_p(e_\perp \vee n_*, e_\perp \vee n_*)$. So $e_\perp e_\perp$ and $e_\perp \vee n_*$ lie in the radical, with the $p \vee \mathsf{u}$; as $t \mapsto t \cdot pp$ is 1 at $n_* n_*$ and 0 at the other five basis tensors, every t equals $(t \cdot pp)\, n_* n_*$ modulo the radical, and $\mathsf{M}_p(t, t') = \overline{(t \cdot pp)} (t' \cdot pp) \beta_\partial(p)$ with $\beta_\partial = \mathsf{M}_p(n_* n_*, n_* n_*) = (p^0)^{-4} \mathsf{M}_p(e_0 e_0, e_0 e_0)$. (7) Put $\mathsf{v}_i = (p \cdot p)^{-2} \beta_i \bar{\mathcal{M}}$ on V_+ $(i = 0, 2)$ and $\mathsf{v}_\partial = \beta_\partial \bar{\mathcal{M}}$ on ∂V_+; these are Radon measures, as $\mathsf{v}_0 \leq \mathsf{v}_0 + \mathsf{v}_2 = ((p^1)^2 + p \cdot p)^{-2} \mathcal{M}_{e_1 e_1, e_1 e_1}$ on V_+ and $\mathsf{v}_\partial = (p^0)^{-4} \mathcal{M}_{e_0 e_0, e_0 e_0}$ on ∂V_+, and eq. 18.viii and (6) are the formulas of (b). They are invariant under $\mathcal{L}$, which keeps V_+ and ∂V_+: $\mathsf{v}_0 = \frac{1}{4}(p \cdot p)^{-2} \mathcal{M}_{\mathsf{g}^{-1}, \mathsf{g}^{-1}}$ with $\Lambda \mathsf{g}^{-1} = \mathsf{g}^{-1}$, so eq. 18.v applies; for real $w \in \mathbb{R}^3$, $\mathcal{M}_{ww,ww} = [(w \cdot p)^2 - (w \cdot w)(p \cdot p)]^2 (\mathsf{v}_0 + \mathsf{v}_2)$ on V_+ and $(w \cdot p)^4 \mathsf{v}_\partial$ on ∂V_+, the weights being unchanged under $(w, p) \mapsto (\Lambda w, \Lambda p)$, so eq. 18.v at $w = e_1$, with the weight $((p^1)^2 + p \cdot p)^2 > 0$ on V_+, gives the invariance of $\mathsf{v}_0 + \mathsf{v}_2$, hence of v_2, and at $w = e_0$, with the weight $(p^0)^4 > 0$ on ∂V_+, that of v_∂. (8) $(m, \vec{p}) \mapsto (\omega_m(\vec{p}), \vec{p})$ is a diffeomorphism of $(0, \infty) \times \mathbb{R}^2$ onto V_+, in which e^{sN_k} keeps m and moves $\vec{p}$ with the velocity $\omega_m(\vec{p}) e_k$ at $s = 0$. For an invariant Radon measure $\tilde{\mathsf{v}}$ the derivative at $s = 0$ gives $\int \omega_m(\vec{p}) \, \partial_{p_k} \chi \, d\tilde{\mathsf{v}} = 0$; so, for $0 \leq \chi_1 \in C_c^\infty((0, \infty))$, the positive distribution $\chi_2 \mapsto (\omega_m \tilde{\mathsf{v}})(\chi_1 \otimes \chi_2)$ on $\mathbb{R}^2$ has vanishing first derivatives and is a multiple $c(\chi_1)$ of the Lebesgue integral, $c(\chi_1) = \int \chi_1 \, d\varrho$ by the theorem of Riesz, and with Lemma 18.19 (c), $\tilde{\mathsf{v}} = \varrho(dm) \, d^2p / \omega_m(\vec{p})$. On ∂V_+, in the coordinate $\vec{p} \in \mathbb{R}^2 \setminus \{0\}$ moved with the velocity $|\vec{p}| e_k$, the measure $|\vec{p}| \tilde{\mathsf{v}}$ has vanishing first derivatives on a connected open set, so $\tilde{\mathsf{v}} = c_\partial \, d^2p / |\vec{p}|$. (9) By (a), $\infty > \int (1 + p^0)^{-2l} \, d\mathcal{M}_{e_0 e_0, e_0 e_0} \geq \int |\vec{p}|^4 (1 + p^0)^{-2l} \, \varrho(dm) \, d^2p / \omega_m(\vec{p})$; with $d^2p / \omega_m = 2\pi \, d\omega$ on $\omega \geq m$,

$|\vec{p}|^2 = \omega^2 - m^2$, and with $\omega^2 - m^2 \geq 1$ and $1 + \omega \leq 3 + 2m$ for $\omega \in [2m+1, 2m+2]$, the right member is at least $2\pi \int (3+2m)^{-2l} \varrho(dm)$. The two special formulas are (b) at $t = e_0 e_0$, where $t \cdot \Pi = |\vec{p}|^2$, $t \cdot pp = (p^0)^2$, which is $|\vec{p}|^2$ on ∂V_+, and the bracket is $|\vec{p}|^4$, and at $t = e_0 \vee n$, where $t \cdot \Pi = t \cdot pp = -p^0(\vec{n} \cdot \vec{p})$ and the bracket is $\Pi(e_0, e_0)\,\Pi(n,n) = |\vec{p}|^2((\vec{n} \cdot \vec{p})^2 + |\vec{n}|^2 p \cdot p)$. □

Lemma 18.27 (Norms of the vacuum vectors)**.** Let (M1), (M4) (a) and (M5) hold, and let $\varrho_0, \varrho_2, \varrho, c_\partial$ be as in Lemma 18.26. For $\alpha \in \mathcal{S}(\mathbb{R})$ put $\hat{\alpha}^*(m) = \sup_{E \geq m} |\hat{\alpha}(E)|$ and $\mathsf{a}(\alpha) = \int \hat{\alpha}^*(m)^2 \, \varrho(dm) + c_\partial \sup |\hat{\alpha}|^2$, which is finite. Let $g \in \mathcal{S}(\mathbb{R}^2)$ and $R \geq 1$.

(a) The vacuum norm satisfies

$$\|T_{00}(\alpha \otimes g_R)\Omega\|^2 \leq \mathsf{a}(\alpha)\, R^{-1} \int |\vec{k}|^3 |\hat{g}(\vec{k})|^2 \, d^2k,$$

and, as $R \to \infty$,

$$R\, \|T_{00}(\alpha \otimes g_R)\Omega\|^2 \to c_\partial\, |\hat{\alpha}(0)|^2 \int |\vec{k}|^3 |\hat{g}(\vec{k})|^2 \, d^2k.$$

(b) For $j \in \{1,2\}$, $\|T_{0j}(\alpha \otimes g_R)\Omega\|^2$ is bounded in $R \geq 1$ and tends, as $R \to \infty$, to

$$\mathsf{L}_j(\alpha, g) = \int m |\hat{\alpha}(m)|^2 \, \varrho_0(dm) \int k_j^2 |\hat{g}(\vec{k})|^2 \, d^2k + \int m |\hat{\alpha}(m)|^2 \, \varrho_2(dm) \int |\vec{k}|^2 |\hat{g}(\vec{k})|^2 \, d^2k.$$

(c) For every $\delta > 0$ and all μ, ν, $\|\mathsf{E}(\{|\vec{p}| > \delta\})\, T_{\mu\nu}(\alpha \otimes g_R)\Omega\| \to 0$ as $R \to \infty$.

■

Proof. By Lemma 18.26 (a) at $F' = F$ and $\Delta = \mathbb{R}^3$, $\|T[t](F)\Omega\|^2 = \int |\hat{F}|^2 \, d\mathcal{M}_{t,t}$, with $\hat{F}(p) = \hat{\alpha}(p^0) R^2 \hat{g}(R\vec{p})$ for $F = \alpha \otimes g_R$. We insert Lemma 18.26 (b) and substitute $\vec{p} = \vec{k}/R$; with $\omega = \omega_m(\vec{k}/R)$ one has $R\omega = (R^2 m^2 + |\vec{k}|^2)^{1/2} \geq |\vec{k}|$, $\omega \geq m$, and $\omega \to m$ as $R \to \infty$. $\mathsf{a}(\alpha)$ is finite because $\hat{\alpha}^*$ decreases faster than every power and $\int (3+2m)^{-2l} \varrho(dm) < \infty$. (a) At $t = e_0 e_0$ the powers of R give

$$\|T_{00}(\alpha \otimes g_R)\Omega\|^2 = \frac{1}{R} \int \varrho(dm) \int d^2k \, \frac{|\vec{k}|^4 |\hat{g}(\vec{k})|^2 |\hat{\alpha}(\omega)|^2}{R\omega} + \frac{c_\partial}{R} \int d^2k \, |\vec{k}|^3 |\hat{g}(\vec{k})|^2 |\hat{\alpha}(|\vec{k}|/R)|^2;$$

$R\omega \geq |\vec{k}|$ and $|\hat{\alpha}(\omega)| \leq \hat{\alpha}^*(m)$ give the bound. In the first integral the integrand is dominated by $|\vec{k}|^3 |\hat{g}|^2 \hat{\alpha}^*(m)^2$ and tends to 0 for every $m > 0$; in the second, $\hat{\alpha}(|\vec{k}|/R) \to \hat{\alpha}(0)$ boundedly. (b) At $t = e_0 \vee e_j$, that is $\vec{n} = e_j$ in Lemma 18.26 (b),

$$\begin{aligned} \|T_{0j}(\alpha \otimes g_R)\Omega\|^2 = {} & \int \varrho_0(dm) \int d^2k \, k_j^2 |\hat{g}|^2 \, \omega |\hat{\alpha}(\omega)|^2 \\ & + \int \varrho_2(dm) \int d^2k \, |\vec{k}|^2 |\hat{g}|^2 \, \frac{m^2 + k_j^2/R^2}{\omega} |\hat{\alpha}(\omega)|^2 \\ & + \frac{c_\partial}{R} \int d^2k \, |\vec{k}| \, k_j^2 |\hat{g}|^2 |\hat{\alpha}(|\vec{k}|/R)|^2; \end{aligned}$$

as $m^2 + k_j^2/R^2 \leq \omega^2$, both integrands over $m > 0$ are at most $|\vec{k}|^2 |\hat{g}(\vec{k})|^2 \sup_{E \geq m} E |\hat{\alpha}(E)|^2$, which is integrable for $\varrho(dm)\, d^2k$; this gives the bound, and dominated convergence with $\omega \to m$ gives the limit, the last term being $O(R^{-1})$. (c) The square is at most $R^4 \sup_{|\vec{k}| > R\delta} |\hat{g}(\vec{k})|^2 \int |\hat{\alpha}(p^0)|^2 \, d\mathcal{M}_{t,t}$ with $t = e_\mu \vee e_\nu$; the integral is finite by Lemma 18.26 (a), as $|\hat{\alpha}(p^0)|^2 \leq C(1+p^0)^{-2l}$, and $\sup_{|\vec{k}| > R\delta} |\hat{g}(\vec{k})|^2 \leq C(R\delta)^{-6}$. □

Proposition 18.28 (Energy and momentum as limits of smeared components)**.** Let (M1), (M4) (a) and (M5) hold; let α, I_α, g and r_g be as in Proposition 18.23, let ϱ and c_∂ be as in Lemma 18.26, a and L_j as in Lemma 18.27, and put $\alpha_N = (1 + i\partial_t)^N \alpha$.

(a) For all integers $N \geq 0$ and all $R \geq 1$,

$$\|(1+H)^N T_{00}(\alpha \otimes g_R)\Omega\| \leq \Big(\mathsf{a}(\alpha_N) \int |\vec{k}|^3 |\hat{g}(\vec{k})|^2 \, d^2k\Big)^{1/2} R^{-1/2},$$

$$R\, \|T_{00}(\alpha \otimes g_R)\Omega\|^2 \to c_\partial \int |\vec{k}|^3 |\hat{g}(\vec{k})|^2 \, d^2k \quad (R \to \infty).$$

If besides (M2), (M3) and (FL_3) hold, then for every $\chi \in \mathcal{D}_0$ there is a number C, formed from l, $\bar{c}$, ϱ, c_∂, α, I_α, g, r_g and a polynomial Q with $\chi = \mathsf{Q}\Omega$, such that

$$\|T_{00}(\alpha \otimes g_R)\chi - H\chi\| \leq C\, R^{-1/2} \qquad (R \geq 1);$$

in particular $\langle \psi, T_{00}(\alpha \otimes g_R)\chi\rangle \to \langle \psi, H\chi\rangle$ for every $\psi \in \mathcal{H}$.

(b) For $j \in \{1,2\}$, $\|T_{0j}(\alpha \otimes g_R)\Omega\|^2 \to \mathsf{L}_j(\alpha, g)$ as $R \to \infty$. If besides (M2), (M3) and (FL_j) hold, then $T_{0j}(\alpha \otimes g_R)\chi \to P_j\chi$ weakly in $\mathcal{H}$ as $R \to \infty$, for every $\chi \in \mathcal{D}_0$.

■

Proof. (1) Let $\mu \in \{0,1,2\}$, let $\mathfrak{M}^4$ and (FL_{j_μ}) hold, and let $t_\alpha = \sup_{t \in I_\alpha} |t|$. The operators $\mathsf{K}_R = T_{0\mu}(\alpha \otimes g_R)$ satisfy the hypotheses of Lemma 18.24 (c) with $\mathfrak{D} = i\partial_\mu$, $c_1 = 2/r_g$ and $r_1 = (t_\alpha + 1)/2$: $\mathsf{c}_N(\alpha \otimes g_R) = \bar{c} R^2 \|g\|_{L^1} \sum_n \binom{N}{n} \|\partial_t^n \alpha\|_{L^1}$, which is at most $C_N R^3$ for $R \geq 1$; and for f carried by $\{|x| < r\}$ one may take $I_f = (-r, r)$ and $r_O = r$ in Proposition 18.23, so that $t_* \leq r + t_\alpha$ and $R_0 \leq c_1(r + r_1)$, and Proposition 18.23 (a) gives the commutator. With Lemma 18.18 (a), Lemma 18.24 (c) gives, for a polynomial Q, $\chi = \mathsf{Q}\Omega$ and all N, M, a number C with

$$\|(1+H)^N [T_{0\mu}(\alpha \otimes g_R)\chi - P_\mu \chi - \mathsf{Q}\, T_{0\mu}(\alpha \otimes g_R)\Omega]\| \leq C\, R^{-M} \qquad (R \geq 1). \tag{18.ix}$$

(2) By Lemma 18.17 (c), $(1+H)^N T_{0\mu}(\alpha \otimes g_R)\Omega = T_{0\mu}(\alpha_N \otimes g_R)\Omega$, with $\alpha_N \in C_c^\infty(\mathbb{R})$. (a) The bound is Lemma 18.27 (a) for α_N, and the limit is Lemma 18.27 (a) for α, as $\hat{\alpha}(0) = \int \alpha = 1$. If Q has the degree n, Lemma 18.24 (a) and the bound at $2nl$ in the place of N give $\|\mathsf{Q}\, T_{00}(\alpha \otimes g_R)\Omega\| \leq C' \|(1+H)^{2nl} T_{00}(\alpha \otimes g_R)\Omega\| \leq C'' R^{-1/2}$, and eq. 18.ix at $\mu = 0$, $N = 0$ and $M = 1$ gives the claim, as $P_0 = H$. (b) The limit is Lemma 18.27 (b). The vector $\xi_R = T_{0j}(\alpha \otimes g_R)\Omega$ is orthogonal to Ω, since $\langle \Omega, T_{0j}(F)\Omega\rangle = -i \int F(x)\, \mathfrak{q}_{3j}(\Omega, \Omega)\, dx = 0$; it is bounded in R, and $\|\mathsf{E}(\{|\vec{p}| > \delta\})\xi_R\| \to 0$ (Lemma 18.27 (b), (c)); so $\xi_R \to 0$ weakly (Lemma 18.25 (b)). For a polynomial Q of degree n, $\|\mathsf{Q}\,\xi_R\| \leq C\|(1+H)^{2nl}\xi_R\| = C\|T_{0j}(\alpha_{2nl} \otimes g_R)\Omega\|$ is bounded in R (Lemma 18.24 (a), Lemma 18.27 (b)), and $\langle \psi, \mathsf{Q}\,\xi_R\rangle = \langle \mathsf{Q}^\dagger \psi, \xi_R\rangle \to 0$ for $\psi \in \mathcal{D}_\infty$; a bounded family that tends to 0 against the dense set $\mathcal{D}_\infty$ tends to 0 weakly, and eq. 18.ix at $\mu = j$ gives the claim. □

Remark 18.29 (Over the components of the tensor limits). Let σ_ξ^c be the recentred component of a tensor limit at a reference coupling not larger than γ_W^{sd}, as in Theorem F (b), (c). The functional $\sigma'(f_1, \dots, f_n) = \sigma_\xi^c(f_1\mathbf{1}, \dots, f_n\mathbf{1})$ on real tuples is scalar data (Rec) with (Rec5): by Theorem F (c), $(\mathrm{Rec1})_V$–$(\mathrm{Rec4})_V$ restrict to the entries in V_0, with the numbers $\sqrt{3}\, C_V$, p_V, $\frac{1}{2}$, as $|\mathbf{1}| = \sqrt{3}$ in the invariant norm; its space is the scalar subspace of $\mathcal{H}(\sigma_\xi^c)$, which reduces H, so that the kernel of H there is $\mathbb{C}\Omega$; and its field is the restriction of $T(f\mathbf{1})$. By Theorem F (d) the frame (M1)–(M5) holds over σ', with the forms $q_0(\cdot, \cdot; \mathsf{m}_{ij})$ at the label matrices m_{ij} of the stress tensor and the kernels $\mathsf{s}_{ij}(x; y) = \mathsf{s}_{1+m}(x, y; \mathsf{m}_{ij}, \mathbf{1}, \dots, \mathbf{1})$; on the vectors of $\mathcal{D}_\infty$ in the scalar subspace, the operators that these forms define are the compressions $T_{ij}^E(F) = \Pi_0 T(F\mathsf{m}_{ij})\Pi_0$, where Π_0 is the orthogonal projection of $\mathcal{H}(\sigma_\xi^c)$ onto the scalar subspace (Step 7 of the proof of Theorem F). In Theorem F (d), (M5) is formed with the restriction to the scalar subspace of the representation $U(a, \Lambda)$ of Theorem E (d) (ii) for σ_ξ^c. For $\Lambda \in \mathcal{L}$ the restricted operator $U(0, \Lambda)$ fixes Ω and satisfies $U(0,\Lambda)\phi(f)U(0,\Lambda)^{-1}\psi = \phi(f \circ \Lambda^{-1})\psi$ for $f \in \mathcal{S}$ and ψ in the span $\mathcal{D}_0$ of Ω and of the vectors $\phi(f_1) \cdots \phi(f_n)\Omega$ (Theorem F (d)); used on the factors one at a time, from the right, this gives $U(0,\Lambda)\phi(f_1) \cdots \phi(f_n)\Omega = \phi(f_1 \circ \Lambda^{-1}) \cdots \phi(f_n \circ \Lambda^{-1})\Omega$, so that $U(0, \Lambda)$ and

its inverse $U(0,\Lambda^{-1})$ map $\mathcal{D}_0$ into itself, and $U(0,\Lambda)\mathcal{D}_0 = \mathcal{D}_0$. Proposition YM3_I:110.1 allows only one unitary operator with these three properties, so the restricted $U(0,\Lambda)$ coincides with the $U(\Lambda)$ of σ' that enters (M5) in this subsection. Hence Lemmas 18.26 and 18.27 and the first statements of Proposition 18.28 (a), (b) hold for this tensor; Proposition 18.23 (a) and the second statements of Proposition 18.28 (a), (b) hold for it in every direction in which its kernels satisfy the flux law eq. 18.ii; and, by Proposition 18.23 (b), the commutator identity eq. 18.iv in one direction is equivalent to the flux law in that direction.

18.4. **The three-point function against the two-point functions: free rings, the value $\frac{4}{9}$, and the free template that the functions of two and three points single out.** For a triple $f = (f_1, f_2, f_3)$ of test functions of one scale with separated supports, the free ratio $\mathcal{R}(f) = \mathcal{K}^{(3)}_{\text{free}}(f)^2 / \prod_{i<j} \mathcal{K}^{(2)}_{\text{free}}(f_i, f_j)$, the square of the ring integral of three dipole kernels $\mathbb{K}_3$ divided by the three pair integrals, depends on the shapes and the positions of the three functions alone; for the free template with n photon fields and the label b the ratio $S_3^2/(S_2S_2S_2)$ equals $\mathcal{R}(f)/\mathsf{n}$, in which the label b does not enter. We show that at three distinct points x_1, x_2, x_3 the kernels satisfy $k_3(x)^2 / \prod_{i<j} k_2(x_i, x_j) = \frac{1}{3}(1 + 3\mathsf{q}_\triangle)^2$, where $\mathsf{q}_\triangle$ is the product of the three cosines of the angles of the triangle that the points span, so that this kernel quotient takes the value $\frac{4}{3}$ when the points lie on one line; that $0 \le \mathcal{R} < \frac{4}{3}$ on all test triples; that $\mathcal{R} \ge \frac{4}{3}(1 - 412w^2)$ for bumps of the radius $w\mathsf{d}$ about three points of a line, d being the scale, and $\mathcal{R} \to \frac{1}{3}(1 + 3\mathsf{q}_\triangle)^2$ for a family of bumps that shrink to the corners of a triangle whose three sides have lengths in $[\frac{9}{8}, 6]$; and that over the test triples of a fixed shape parameter ϑ the supremum of $\mathcal{R}$ lies below $\frac{4}{3}$ and tends to $\frac{4}{3}$ as $\vartheta \to 0$ (Propositions 18.32 to 18.37). For the limits $S_n^{(\ell,\mathsf{b})}$ of the connected functions of the action density on a torus, as the cutoff is removed at a fixed label, the law of two points $S_2^{(\ell,\mathsf{b})} = 3\mathsf{b}^{-2}\mathcal{K}^{(2)}_{\text{free}}\,(1 + o(1))$ at short distance is valid on every side $\ell = 2L^m$ and at every real label $\mathsf{b} > 0$ (Proposition 18.38). Let g be a reference coupling with $g \le \gamma_W$, the ceiling YM3_I:(2346) of the law of two points of the main text, and let T_ℓ be the torus of a side ℓ at g. If on T_ℓ the law of three points $S_3^T \approx 3\mathsf{b}^{-3}\mathcal{K}^{(3)}_{\text{free}}$ holds in the form of Definition 18.31, whose numbers are the same at all cutoffs, then the ratio $Q = S_3^2/(S_2S_2S_2)$ obeys $|3Q - \mathcal{R}| \le \Theta_{\text{Q}}$ at every large cutoff, where the bound sees the cutoff only through the quotient $a_K/(\vartheta\mathsf{d})$ and is uniform in β, in the label, in the triple and in its position, and Q obeys the same inequality in the limit, where the bound is free of the cutoff and tends to 0 with the scale; so $Q \to \frac{1}{3}\mathcal{R}$ along the dilates of one shape, $Q \to \frac{4}{9} = \frac{4}{3\dim \mathfrak{su}(2)}$ as the scale and then the width of bumps about three points of a line tend to 0, and the supremum of Q over the collinear triples of a shape parameter ϑ tends to a number below $\frac{4}{9}$ which increases to $\frac{4}{9}$ as $\vartheta \to 0$; besides, at every scale below a threshold that depends on ϑ, the function of three points is positive at the collinear triples of the shape parameter ϑ, and along the periodised dilates of the bumps of Proposition 18.36 about the corners of a triangle whose three sides have lengths in $[\frac{9}{8}, 6]$, $Q \to \frac{1}{9}(1 + 3\mathsf{q}_\triangle)^2$ as first the scale and then the width tend to 0, while in the case $1 + 3\mathsf{q}_\triangle \ne 0$ the function of three points at these dilates is not 0 and has the same sign as $-(1 + 3\mathsf{q}_\triangle)$ whenever the width is small enough and, for that width, the scale is small enough (Propositions 18.39 to 18.41). If the hypothesis is met on all sides at g for every small g, the conclusions on the limits are valid on all tori and at all real labels, and the ratios at the cutoffs converge to the ratio of the limits (Proposition 18.42). The function of two points of a free template depends on its data (n, b) through $\mathsf{n}\mathsf{b}^{-2}$ alone, so that, for instance, one photon field at the label $\mathsf{b}/\sqrt{3}$ and three at the label b share it; when the hypothesis holds on every side at g for all small g, the functions of two and of three points of $S^{(\ell,\mathsf{b})}$, divided by the free functionals along the periodised dilates of a collinear shape, tend at short distance to $3\mathsf{b}^{-2}$ and $3\mathsf{b}^{-3}$, and of all free templates only the one with three photon fields and the label b shows these two limits (Proposition 18.43). Conversely, given the law of two points, $3Q/\mathcal{R} \to 1$ is equivalent to the law of three points in modulus, and $Q \to \frac{4}{9}$ in the form with a width is equivalent to the law of three points in modulus in the limit of narrow bumps; an explicit

pair of functions of two and of three points has the law of two points and $Q \to \frac{4}{9}$, while along the dilates of a collinear shape its function of three points exceeds the free one by one fixed factor at every scale (Propositions 18.44 and 18.45). Under the ceiling (Γ) of Section 4.1, the uniform law follows by comparing each state with the cubic source jet on the least torus and estimating that jet (Remark 18.46). We use the setting and the signs of Section 4.1. For g under the ceiling (Γ) of that Section, its Theorem G bounds the function of three points at two scales against the lattice ring on the least torus (its clause (A.1)) and, through its clause (A.4)(i), for each state at a first-passage datum, the periodic extensions of the torus measures on all sides at g among them; taken at one scale and joined with Lemma 4.22(a), which passes from the lattice ring to the free ring, these bounds give the hypothesis on every side at g, for every such g (Remark 18.47).

Setting. We use the construction and the notation of Section 4.1 and its hypotheses where its estimates are invoked. From Section 4.1 we take the constants, among them the odd blocking factor $L \geq 13$, $M = L^{m'}$, $a_j = L^{-j}$, $B_\sharp \geq 0$ and $\bar B = B_\sharp L/(L-1)$, so that $LB_\sharp = (L-1)\bar B$, and $\gamma \in (0, \frac{1}{2}]$; the reference couplings $0 < g \leq \gamma$; the first-passage data (K, β) at g and their labels $\mathsf{b} = a_K\beta$, which satisfy $g^{-2} < \mathsf{b} \leq \bar{\mathsf{b}} = L(g^{-2} + \bar B)$ (Lemma 4.3(a)), so that every limit of such labels is a number $\geq g^{-2} > 0$; the interval $I(g) = (g^{-2} + \bar B, X_g - \bar B]$, $X_g = L(g^{-2} + B_\sharp)$, YM3_I: (1527), for whose points b every pair $(K, L^K\mathsf{b})$, $K \in \mathbb{N}$, is a first-passage datum at g with the label b; the tori $T_\ell = \mathbb{R}^3/\ell\mathbb{Z}^3$ of the sides $\ell = 2L^m$, $m \in \mathbb{N}$, with the lattices $a_K\mathbb{Z}^3/\ell\mathbb{Z}^3$; the sides at g, $\mathfrak{L}(g) = \{2L^m : m \in \mathbb{N},\ m \geq m_{\mathfrak{L}}(g)\}$ with $L^{m_{\mathfrak{L}}(g)} = LM \max\{R(L(g^{-2} + \bar B)), L^{\lceil \log_L C_{\mathrm{s}} \rceil}\}$, where for $x > 1$ the number $R(x)$ is the least power L^{u}, $\mathsf{u} \geq 0$ an integer, with $L^{\mathsf{u}} \geq (\log x)^r$, and $r \in \mathbb{N}$, $C_{\mathrm{s}} \geq 2$ are constants, and the least side ℓ_g; the torus measures $\mu_{K,\ell,\beta}$ and their periodic extensions $\tilde\mu_{K,\ell,\beta}$; the observable $\mathcal{O}(f) = a_K^{-1} \sum_p f(x(p))\, a_p$; the connected functions $S^T_{n;K,\ell,\beta}(f_1, \dots, f_n)$; the witness depth $s(\varrho)$ at g, an integer formed from the constants, g and the length ϱ, in which no cutoff enters; the periodisation $f^{\langle \ell \rangle}$ of a function f on $\mathbb{R}^3$; and the kernels $\mathbb{K}_3$ and k_3.

Free kernels and test classes. For a unit vector e put $D(e) = 3ee^{\mathsf{T}} - 1$, so that $\mathbb{K}_3(u) = (4\pi)^{-1}|u|^{-3}D(\hat u)$. For distinct $x_1, x_2, x_3 \in \mathbb{R}^3$, let a, b, c be the unit directions of $x_1 - x_2$, $x_2 - x_3$ and $x_3 - x_1$, and let $r_{ij} = |x_i - x_j|$. The product of the three angle cosines is $\mathsf{q}_\triangle = -(a \cdot b)(b \cdot c)(c \cdot a)$. Since the three vectors are coplanar, their Gram determinant gives

$$(a \cdot b)^2 + (b \cdot c)^2 + (c \cdot a)^2 = 1 - 2\mathsf{q}_\triangle .$$

Expanding the product of the three matrices then gives $\tau_\triangle = \operatorname{tr}(D(a)D(b)D(c)) = -3 - 9\mathsf{q}_\triangle$. On a line $\mathsf{q}_\triangle = -1$ and at an equilateral triangle $\mathsf{q}_\triangle = 1/8$. The two cyclic orders in the three-point ring have equal trace because D is even and symmetric. Thus

$$k_3(x) = \tau_\triangle(x)(4\pi)^{-3}(r_{12}r_{13}r_{23})^{-3}, \qquad k_2(x_1, x_2) = \tfrac{1}{2} \operatorname{tr} \mathbb{K}_3(x_1 - x_2)^2 = \tfrac{3}{16\pi^2} r_{12}^{-6}.$$

For bounded Borel functions with compact pairwise separated supports define $\mathcal{K}^{(n)}_{\mathrm{free}}(f) = \int k_n(x) \prod_i f_i(x_i)\, dx$ for $n = 2, 3$, taking the integrand to be zero where the product of the tests vanishes. These forms are symmetric and multilinear. The pair form is $\mathcal{K}^{(2)}_{\mathrm{free}} = \frac{3}{16\pi^2}\mathcal{K}_6$, where $\mathcal{K}_6(f, h) = \iint f(x)h(y)|x - y|^{-6}\, dx\, dy$. In the scalar two-point law, $\mathcal{K}_{\mathrm{free}} = \mathcal{K}_6$ and $\mathsf{c}_2 = 9/(16\pi^2)$, so $3\mathcal{K}^{(2)}_{\mathrm{free}} = \mathsf{c}_2\mathcal{K}_{\mathrm{free}}$.

The class $\mathfrak{T}_{\mathbb{E}}(\mathsf{d}; \vartheta)$, where $\mathbb{E} = \mathbb{R}^3$ or T_ℓ with $\ell \geq 20\mathsf{d}$, consists of pairs of nonzero nonnegative Lipschitz functions supported in closed balls of radius d, with support distance in $[\mathsf{d}, 6\mathsf{d}]$. Each entry u satisfies $\int u \geq \vartheta \|u\|_\infty \mathsf{d}^3$ and $\operatorname{Lip} u \leq \|u\|_\infty/(\vartheta\mathsf{d})$. The centers of any two such supports are at most $8\mathsf{d}$ apart by the triangle inequality. Write $\mathfrak{Q}^{(3)}_{\mathbb{E}}(\mathsf{d}; \vartheta)$ for the triples whose pairs all belong to this class, imposing $36\mathsf{d} < \ell$ on a torus. Let $\mathfrak{T}^{(3),\mathrm{col}}_{\mathbb{E}}(\mathsf{d}; \vartheta)$ be its subclass whose supports on $\mathbb{R}^3$ lie in balls of radius $\mathsf{d}/16$ with collinear centers, or on a torus have a common lift with that property.

To lift a triple from T_ℓ, fix a lift of the first center, and choose the lifts of the other centers within $8\mathsf{d}$ of it. The three lifted balls lie within $9\mathsf{d} < \ell/4$ of the first center. The covering map is an isometry on such a ball because every point difference there has length less than $\ell/2$. Each test is

pulled back there and extended by zero, preserving its supremum, integral and Lipschitz constant. For the latter, if one point lies outside the lifted ball, the test vanishes on the boundary and the segment to that point bounds the difference. The triple remains in its test class. Conversely a triple in the Euclidean test class periodises to a torus triple when $36\mathsf{d} < \ell$, by the same isometry. For a pair the midpoint of the two lifted centers gives a ball of radius $5\mathsf{d}$ containing both supports, and their mutual point distances are at most $10\mathsf{d} \le \ell/2$. The same conclusions hold for pairs. Any common lift of supports in a ball of radius less than $\ell/4$ gives the same relative positions, as every point difference is the unique shortest lift of that difference on the torus. All such lifts differ by a common period. Define the torus free forms at these lifts. They are independent of the lift by translation invariance. We write $\mathsf{N}(f) = \prod_a \|f_a\|_\infty$.

For $r > 0$ put $f_r(x) = f(x/r)$ and write Ψ_r for dilation of every entry. This preserves the test and width classes with d changed to $r\mathsf{d}$. Since k_n is homogeneous of degree $-3n$, $\mathcal{K}^{(n)}_{\mathrm{free}}(\Psi_r) = \mathcal{K}^{(n)}_{\mathrm{free}}(\Psi)$. Let $\psi_{i,r}^{\langle\ell\rangle} = ((\psi_i)_r)^{\langle\ell\rangle}$ and $\Psi_r^{\langle\ell\rangle} = (\Psi_r)^{\langle\ell\rangle}$ denote the periodised dilates, where $f^{\langle\ell\rangle}(x) = \sum_{z\in\ell\mathbb{Z}^3} f(x+z)$. For a periodically extended configuration the bare plaquette action is periodic. Grouping plaquettes by their images on T_ℓ therefore identifies the observable tested at f on the extended lattice with the observable tested at $f^{\langle\ell\rangle}$ on the torus, and identifies their joint cumulants. On $T_{L^s\ell}$ put $h^{(s)}(x) = h(L^{-s}x)$. The free template is $S_n^{\mathrm{fr}}[\mathsf{n},\mathsf{b}] = \mathsf{n}\mathsf{b}^{-n}\mathcal{K}^{(n)}_{\mathrm{free}}$ for n photon fields and label b.

On the test pairs, point distances lie in $[\mathsf{d}, 10\mathsf{d}]$, and the test-integral and ball-volume bounds give

$$10^{-6}\vartheta^2\|f\|_\infty\|h\|_\infty \le \mathcal{K}_6(f,h) \le (4\pi/3)^2\|f\|_\infty\|h\|_\infty.$$

On the test triples, $|\tau_\triangle| \le 6$ by Proposition 18.32. The three support volumes then give $|\mathcal{K}^{(3)}_{\mathrm{free}}(f)| \le (2/9)\mathsf{N}(f) \le (4/9)\mathsf{N}(f)$. For collinear supports of radius at most $\mathsf{d}/16$, orient each of a, b, c toward the common line direction. Each perpendicular component has length at most $1/8$, so the mutual dot products are at least $1 - 2(1/8)^2$. Reversing a direction leaves both D and the product of the three dot products invariant. The formula for $\tau_\triangle$ then gives $\tau_\triangle \ge 6 - 216(1/16)^2 > 3.78$. The integral lower bounds and $r_{ij} \le 10\mathsf{d}$ give $\mathcal{K}^{(3)}_{\mathrm{free}}(f) \ge c_3^{\mathrm{col}}\vartheta^3\mathsf{N}(f)$ with $c_3^{\mathrm{col}} = 3.78(4\pi)^{-3}10^{-9}$.

Limits on a torus; the law of two points. Let $\ell = 2L^m$, $m \in \mathbb{N}$, let $\mathsf{b} > 0$ be real, and let $f_1, \dots, f_n$ be real Lipschitz functions on T_ℓ, none of them 0, whose supports have no common point. Then the limit $S_n^{(\ell,\mathsf{b})}(f_1,\dots,f_n) = \lim_{K\to\infty} S^T_{n;K,\ell,L^K\mathsf{b}}(f_1,\dots,f_n)$ exists (Theorem YM3_I:AS); for all pairs $(K_j,\beta_j) \in \mathbb{N}\times(0,\infty)$ with $K_j \to \infty$ and $a_{K_j}\beta_j \to \mathsf{b}$ the numbers $S^T_{n;K_j,\ell,\beta_j}(f_1,\dots,f_n)$ tend to it along the whole sequence (YM3_I:(2216)), so that the limit along first-passage data of Section 4.1 is this limit; and $S_n^{(\ell,\mathsf{b})}(f_1,\dots,f_n) = L^{sn}S_n^{(L^s\ell,L^s\mathsf{b})}(f_1^{(s)},\dots,f_n^{(s)})$ for every integer $s \ge 0$, YM3_I:(2215). $\gamma_W \in (0,\gamma]$ is the ceiling YM3_I:(2346) of the law of two points of the main text; for $0 < g \le \gamma_W$ and $\vartheta \in (0,1)$, $\mathsf{d}_0(g,\vartheta) \in (0, \frac{1}{200}]$ is its threshold YM3_I:(2350), and $\bar\varepsilon_{g,\vartheta}$ its tolerance YM3_I:(2349), a function formed from the setting, g and ϑ. By Theorem YM3_I:AV: if $0 < g \le \gamma_W$, (K,β) is a first-passage datum at g with the label b, $0 < \mathsf{d} \le \mathsf{d}_0(g,\vartheta)$ and $s(\mathsf{d}) < K$, then for every $\ell \in \mathfrak{L}(g)$ and every $(f,h) \in \mathfrak{T}_{T_\ell}(\mathsf{d};\vartheta)$

$$S^T_{2;K,\ell,\beta}(f,h) = \mathsf{c}_2\,\mathsf{b}^{-2}\,\mathcal{K}_{\mathrm{free}}(f,h)\,(1+\varepsilon'), \qquad |\varepsilon'| \le \bar\varepsilon_{g,\vartheta}(\mathsf{d}),$$

where the same function $\bar\varepsilon_{g,\vartheta}$ serves every first-passage datum at g, every pair and every position; it is non-decreasing, $\bar\varepsilon_{g,\vartheta}(\mathsf{d}) \to 0$ as $\mathsf{d} \to 0$, and $\bar\varepsilon_{g,\vartheta} \le \frac{1}{3}$ on $(0,\mathsf{d}_0(g,\vartheta)]$.

What is used where. Propositions 18.32 to 18.37, parts (a) and (b) of Proposition 18.43, part (a) of Proposition 18.44 and parts (b) and (c) of Proposition 18.45 concern the free kernels alone. Proposition 18.38, parts (b) and (c) of Proposition 18.44 and part (a) of Proposition 18.45 rest besides on the limits on a torus and on the law of two points recalled above; though the last three speak of the function of three points and of the ratio Q, they ask no bound on that function. Propositions 18.39 to 18.42 and part (c) of Proposition 18.43 rest, beyond these, on the hypothesis of Definition 18.31.

Definition 18.30 (Width classes; the two ratios; four numbers)**.** (1) *Width classes.* Let $0 < w \le \frac{1}{16}$, $\mathsf{d} > 0$ and $\vartheta \in (0,1)$. $\mathfrak{T}^{\mathrm{col}}_{\mathbb{R}^3}(\mathsf{d};\vartheta;w)$ is the set of the triples $f \in \mathfrak{Q}^{(3)}_{\mathbb{R}^3}(\mathsf{d};\vartheta)$ for which there are points c_1, c_2, c_3 on one line with $\operatorname{supp} f_i \subset \mathbb{B}(c_i, w\mathsf{d})$ for $i = 1,2,3$; for a side ℓ with $36\mathsf{d} < \ell$, $\mathfrak{T}^{\mathrm{col}}_{T_\ell}(\mathsf{d};\vartheta;w)$ is the set of the triples of $\mathfrak{Q}^{(3)}_{T_\ell}(\mathsf{d};\vartheta)$ one lift of which lies in $\mathfrak{T}^{\mathrm{col}}_{\mathbb{R}^3}(\mathsf{d};\vartheta;w)$; then all lifts lie in it, since any two lifts are translates of each other and a translation takes a line to a line and a ball to a ball of equal radius. The width classes increase with w, and at $w = \frac{1}{16}$ they are the classes $\mathfrak{T}^{(3),\mathrm{col}}_{\mathbb{E}}(\mathsf{d};\vartheta)$.

(2) *The free ratio.* For bounded Borel functions $f_1, f_2, f_3 \ge 0$ on $\mathbb{R}^3$ with compact supports at pairwise positive distance and with $\mathcal{K}^{(2)}_{\mathrm{free}}(f_i, f_j) > 0$ for $i < j$,

$$\mathcal{R}(f) = \frac{\mathcal{K}^{(3)}_{\mathrm{free}}(f_1,f_2,f_3)^2}{\mathcal{K}^{(2)}_{\mathrm{free}}(f_1,f_2)\,\mathcal{K}^{(2)}_{\mathrm{free}}(f_1,f_3)\,\mathcal{K}^{(2)}_{\mathrm{free}}(f_2,f_3)};$$

for a triple of $\mathfrak{Q}^{(3)}_{T_\ell}(\mathsf{d};\vartheta)$, $\mathcal{R}$ is formed at a lift, at which the four functionals have the values that they have on the torus. On the test classes the denominator is positive, by the lower bound of $\mathcal{K}^{(2)}_{\mathrm{free}}$ recalled above. For $0 < \vartheta \le \frac{\pi}{13440}$ we put $\bar{\mathcal{R}}(\vartheta) = \sup\{\mathcal{R}(f) : f \in \mathfrak{Q}^{(3)}_{\mathbb{R}^3}(1;\vartheta)\}$, $\bar{\mathcal{R}}^{\mathrm{col}}(\vartheta) = \sup\{\mathcal{R}(f) : f \in \mathfrak{T}^{(3),\mathrm{col}}_{\mathbb{R}^3}(1;\vartheta)\}$ and $w_\vartheta = (\frac{105\,\vartheta}{32\pi})^{1/3}$, so that $w_\vartheta \le \frac{1}{16}$; both classes have elements (Proposition 18.35(b)).

(3) *The ratio of the connected functions.* For a triple f of real Lipschitz functions on T_ℓ, none of them 0, with pairwise disjoint supports,

$$Q_{K,\ell,\beta}(f) = \frac{S^T_{3;K,\ell,\beta}(f_1,f_2,f_3)^2}{\prod_{i<j} S^T_{2;K,\ell,\beta}(f_i,f_j)}, \qquad Q^{(\ell,\mathsf{b})}(f) = \frac{S^{(\ell,\mathsf{b})}_3(f_1,f_2,f_3)^2}{\prod_{i<j} S^{(\ell,\mathsf{b})}_2(f_i,f_j)},$$

each where the three factors below the line are positive. For the free template with the data (n,b) the same quotient is $(\mathsf{n}\mathsf{b}^{-3})^2\mathcal{K}^{(3)}_{\mathrm{free}}(f)^2/((\mathsf{n}\mathsf{b}^{-2})^3\prod_{i<j}\mathcal{K}^{(2)}_{\mathrm{free}}(f_i,f_j)) = \mathcal{R}(f)/\mathsf{n}$.

(4) *Numbers.* $\mathsf{C}_{\mathrm{P}} = (\frac{16\pi^2}{3})^3\,10^{18}$; for $\vartheta \in (0,1)$ and real $t, e \ge 0$,

$$\Theta_{\mathrm{Q}}(\vartheta;t,e) = \tfrac{27}{8}\,\mathsf{C}_{\mathrm{P}}\,\vartheta^{-6}(\tfrac{8}{9}\,t + t^2) + \tfrac{37}{2}\,e;$$

and for $0 < w \le \frac{1}{16}$, $\mathsf{t}_w = 1 - 90w^2 - 72w^3$ and $\varkappa_w = \log\frac{1+2w}{1-2w}$. ■

Definition 18.31 (The uniform law of three points)**.** Let $0 < g \le \gamma$ and $\ell \in \mathfrak{L}(g)$. We say that the uniform law of three points holds at (g,ℓ), and write $\mathrm{UL}_3(g,\ell)$, if to each $\vartheta \in (0,1)$ one can find $\mathsf{d}_{\mathrm{III}} \in (0, \frac{1}{200}]$, $C_{\mathrm{III}} \ge 0$, integers $K_{\mathrm{III}}(\mathsf{d})$ for $0 < \mathsf{d} \le \mathsf{d}_{\mathrm{III}}$, and a function ϵ_{III} on $(0,\mathsf{d}_{\mathrm{III}}]$ with values in $[0,\infty)$ and the limit 0 at 0, depending on the setting, on g, ℓ, ϑ and, for K_{III} and ϵ_{III}, on d, but on none of K, β, the triple and the place of the triple, with the following property: whenever (K,β) is a first-passage datum at g, $\mathsf{b} = a_K\beta$ its label, $0 < \mathsf{d} \le \mathsf{d}_{\mathrm{III}}$, $K \ge K_{\mathrm{III}}(\mathsf{d})$ and $f \in \mathfrak{Q}^{(3)}_{T_\ell}(\mathsf{d};\vartheta)$, then

$$\left|S^T_{3;K,\ell,\beta}(f_1,f_2,f_3) - 3\,\mathsf{b}^{-3}\,\mathcal{K}^{(3)}_{\mathrm{free}}(f_1,f_2,f_3)\right| \le \left[\epsilon_{\mathrm{III}}(\mathsf{d}) + C_{\mathrm{III}}\,\frac{a_K}{\vartheta\,\mathsf{d}}\right]\mathsf{b}^{-3}\,\mathsf{N}(f).$$

By UL^*_3 we mean: for some $\gamma_{\mathrm{III}} \in (0,\gamma]$, $\mathrm{UL}_3(g,\ell)$ is true whenever $0 < g \le \gamma_{\mathrm{III}}$ and $\ell \in \mathfrak{L}(g)$. ■

The free ratio.

Proposition 18.32 (Three points)**.** Let $x_1, x_2, x_3 \in \mathbb{R}^3$ be pairwise different.

(a) $-1 \le \mathsf{q}_\triangle(x) \le \frac{1}{8}$ and $-\frac{33}{8} \le \tau_\triangle(x) \le 6$; $\mathsf{q}_\triangle(x) = -1$, and equally $\tau_\triangle(x) = 6$, if and only if the three points lie on one line.

(b) $k_3(x) = \frac{\tau_\triangle(x)}{3\sqrt{3}}\prod_{i<j} k_2(x_i,x_j)^{1/2}$, and

$$\frac{k_3(x)^2}{k_2(x_1,x_2)\,k_2(x_1,x_3)\,k_2(x_2,x_3)} = \frac{\tau_\triangle(x)^2}{27} = \frac{(1+3\,\mathsf{q}_\triangle(x))^2}{3}.$$

(c) For the template with the data (n,b) the quotient of its kernels is

$$(\mathsf{n}\mathsf{b}^{-3}k_3(x))^2/\prod_{i<j}\mathsf{n}\mathsf{b}^{-2}k_2(x_i,x_j)=(1+3\mathsf{q}_\triangle(x))^2/(3\mathsf{n}).$$

It depends neither on b nor on the size of the triangle. At three points of a line it is $\frac{4}{3\mathsf{n}}$ for all distances between the points, so $\frac{4}{9}$ for $\mathsf{n}=3=\dim\mathfrak{su}(2)$; at the corners of an equilateral triangle it is $\frac{121}{192\,\mathsf{n}}$, and there $k_3<0$. At $(x_1,x_2,x_3)=(0,e_3,2e_3)$, $(4\pi)^3k_3=6\cdot 2^{-3}=\frac{3}{4}$. ■

Proposition 18.33 (The free ratio stays below $\frac{4}{3}$)**.** (a) For bounded Borel functions $f_1,f_2,f_3\ge 0$ on $\mathbb{R}^3$ whose supports are compact and at pairwise positive distance,

$$\mathcal{K}^{(3)}_{\mathrm{free}}(f_1,f_2,f_3)^2\le\tfrac{4}{3}\,\mathcal{K}^{(2)}_{\mathrm{free}}(f_1,f_2)\,\mathcal{K}^{(2)}_{\mathrm{free}}(f_1,f_3)\,\mathcal{K}^{(2)}_{\mathrm{free}}(f_2,f_3).$$

(b) If moreover each f_i is continuous and not the zero function, then $\mathcal{K}^{(2)}_{\mathrm{free}}(f_i,f_j)>0$ for $i<j$, and strict inequality is valid in (a): $0\le\mathcal{R}(f)<\frac{4}{3}$.

(c) For every $\mathsf{d}>0$, $\vartheta\in(0,1)$ and $f\in\mathfrak{Q}^{(3)}_{\mathbb{E}}(\mathsf{d};\vartheta)$, on $\mathbb{E}=\mathbb{R}^3$ and on $\mathbb{E}=T_\ell$ with $36\mathsf{d}<\ell$, $0\le\mathcal{R}(f)<\frac{4}{3}$; so the quotient $\mathcal{R}(f)/\mathsf{n}$ of the template with the data (n,b) lies in $[0,\frac{4}{3\mathsf{n}})$. ■

Proposition 18.34 (Narrow bumps about three points of a line)**.** Let $\mathbb{E}=\mathbb{R}^3$, or $\mathbb{E}=T_\ell$ and $36\mathsf{d}<\ell$; let $\mathsf{d}>0$, $\vartheta\in(0,1)$, $0<w\le\frac{1}{16}$ and $f\in\mathfrak{T}^{\mathrm{col}}_{\mathbb{E}}(\mathsf{d};\vartheta;w)$. Then $\mathsf{t}_w>0$, $\mathcal{K}^{(3)}_{\mathrm{free}}(f)>0$, and

$$\tfrac{4}{3}\,(1-412\,w^2)\le\tfrac{4}{3}\,\mathsf{t}_w^2\,e^{-\frac{27}{2}\varkappa_w^2}\le\mathcal{R}(f)<\tfrac{4}{3}.$$

So $0<\frac{4}{3}-\mathcal{R}(f)\le\frac{1648}{3}\,w^2$, a bound quadratic in w that is uniform in the scale d and in the shape parameter ϑ. ■

Proposition 18.35 (The collinear classes and the suprema of the free ratio)**.** Let $\mathbb{E}$, d, ϑ and w be as in Proposition 18.34.

(a) A width class $\mathfrak{T}^{\mathrm{col}}_{\mathbb{E}}(\mathsf{d};\vartheta;w)$ that has an element has $\vartheta\le\frac{4\pi}{3}w^3$.

(b) For $\vartheta\le\frac{32\pi}{105}w^3$ the width class $\mathfrak{T}^{\mathrm{col}}_{\mathbb{E}}(\mathsf{d};\vartheta;w)$ has elements: for a unit vector e and $c_j=2(j-1)\mathsf{d}\,e$, the functions $u_j(x)=(1-|x-c_j|^2/(w\mathsf{d})^2)^2$ for $|x-c_j|\le w\mathsf{d}$ and $u_j(x)=0$ otherwise, $j=1,2,3$, form a triple of it on $\mathbb{R}^3$, and their periodisations one on T_ℓ.

(c) Let $0<\vartheta\le\frac{\pi}{13440}$. For every d and on both spaces, $\sup\{\mathcal{R}(f):f\in\mathfrak{Q}^{(3)}_{\mathbb{E}}(\mathsf{d};\vartheta)\}=\bar{\mathcal{R}}(\vartheta)$ and $\sup\{\mathcal{R}(f):f\in\mathfrak{T}^{(3),\mathrm{col}}_{\mathbb{E}}(\mathsf{d};\vartheta)\}=\bar{\mathcal{R}}^{\mathrm{col}}(\vartheta)$; both are non-increasing functions of ϑ; and

$$\tfrac{4}{3}\,(1-412\,w_\vartheta^2)\le\bar{\mathcal{R}}^{\mathrm{col}}(\vartheta)\le\bar{\mathcal{R}}(\vartheta)\le\tfrac{4}{3},$$

so that both tend to $\frac{4}{3}$ as $\vartheta\to 0$. ■

Proposition 18.36 (Bumps at the corners of a triangle)**.** Let $c_1,c_2,c_3\in\mathbb{R}^3$ with $\frac{9}{8}\le|c_i-c_j|\le 6$ for $i\ne j$, let $0<w\le\frac{1}{16}$, let $u_j(x)=(1-|x-c_j|^2/w^2)^2$ for $|x-c_j|\le w$ and $u_j(x)=0$ otherwise, and $\Psi^{c,w}=(u_1,u_2,u_3)$. Then $\Psi^{c,w}\in\mathfrak{Q}^{(3)}_{\mathbb{R}^3}(1;\vartheta)$ for $0<\vartheta\le\frac{32\pi}{105}w^3$; as $w\to 0$, $\mathcal{K}^{(3)}_{\mathrm{free}}(\Psi^{c,w})/\prod_j\int u_j\to k_3(c)$ and

$$\mathcal{R}(\Psi^{c,w})\to\tfrac{1}{3}\,(1+3\,\mathsf{q}_\triangle(c))^2.$$

■

Proposition 18.37 (A fixed shape parameter keeps the free ratio away from $\frac{4}{3}$)**.** For every $0<\vartheta\le\frac{\pi}{13440}$, $\bar{\mathcal{R}}^{\mathrm{col}}(\vartheta)\le\bar{\mathcal{R}}(\vartheta)<\frac{4}{3}$. Consequently, if for $k\in\mathbb{N}$ the space $\mathbb{E}_k$ is $\mathbb{R}^3$ or a torus, $\mathsf{d}_k>0$, $\vartheta_k\in(0,1)$, $f^{(k)}\in\mathfrak{Q}^{(3)}_{\mathbb{E}_k}(\mathsf{d}_k;\vartheta_k)$, and $\mathcal{R}(f^{(k)})\to\frac{4}{3}$, then $\vartheta_k\to 0$. ■

The law of two points for the limits on all tori.

Proposition 18.38 (The law of two points on all sides and at all labels)**.** (a) Let $0 < g \le \gamma_W$, $\ell \in \mathfrak{L}(g)$ and $\vartheta \in (0,1)$, and let b be the limit of the labels $a_{K_j}\beta_j$ of first-passage data (K_j,β_j) at g with $K_j \to \infty$, for instance any $\mathsf{b} \in I(g)$. Then for every $0 < \mathsf{d} \le \mathsf{d}_0(g,\vartheta)$ and every $(f,h) \in \mathfrak{T}_{T_\ell}(\mathsf{d};\vartheta)$

$$S_2^{(\ell,\mathsf{b})}(f,h) = 3\,\mathsf{b}^{-2}\,\mathcal{K}^{(2)}_{\mathrm{free}}(f,h)\,(1+\eta), \qquad |\eta| \le \bar\varepsilon_{g,\vartheta}(\mathsf{d}).$$

(b) Let $\ell = 2L^m$, $m \in \mathbb{N}$, be any side, $\mathsf{b} > 0$ any real number and $\vartheta \in (0,1)$. One can choose a number $\mathsf{d}_{\mathrm{II}} = \mathsf{d}_{\mathrm{II}}(\ell,\mathsf{b},\vartheta) \in (0, \frac{1}{200}]$ and a non-decreasing function ϵ_{II} from $(0,\mathsf{d}_{\mathrm{II}}]$ to $[0,\frac13]$ with $\epsilon_{\mathrm{II}}(\mathsf{d}) \to 0$ as $\mathsf{d} \to 0$, both depending on the setting, ℓ, b and ϑ only, such that for every $0 < \mathsf{d} \le \mathsf{d}_{\mathrm{II}}$ and every $(f,h) \in \mathfrak{T}_{T_\ell}(\mathsf{d};\vartheta)$ the display of (a) holds with $\epsilon_{\mathrm{II}}(\mathsf{d})$ in the place of $\bar\varepsilon_{g,\vartheta}(\mathsf{d})$.

(c) In (a) and in (b), $S_2^{(\ell,\mathsf{b})}(f,h) \ge 2\,\mathsf{b}^{-2}\mathcal{K}^{(2)}_{\mathrm{free}}(f,h) > 0$.

■

Three points, under the uniform law.

Proposition 18.39 (The ratio at a cutoff and in the limit)**.** Let $0 < g \le \gamma_W$ and $\ell \in \mathfrak{L}(g)$, and suppose $\mathrm{UL}_3(g,\ell)$. Let $\vartheta \in (0,1)$, let $\mathsf{d}_{\mathrm{III}}$, C_{III}, ϵ_{III}, K_{III} be the numbers of Definition 18.31 at ϑ, and $0 < \mathsf{d} \le \min\{\mathsf{d}_0(g,\vartheta), \mathsf{d}_{\mathrm{III}}\}$.

(a) For every first-passage datum (K,β) at g with $K > s(\mathsf{d})$ and $K \ge K_{\mathrm{III}}(\mathsf{d})$ and every $f \in \mathfrak{Q}^{(3)}_{T_\ell}(\mathsf{d};\vartheta)$, the three numbers $S^T_{2;K,\ell,\beta}(f_i,f_j)$ are positive, and

$$|3\,Q_{K,\ell,\beta}(f) - \mathcal{R}(f)| \le \Theta_{\mathrm{Q}}\Big(\vartheta;\ \tfrac13\Big[\epsilon_{\mathrm{III}}(\mathsf{d}) + C_{\mathrm{III}}\frac{a_K}{\vartheta\,\mathsf{d}}\Big],\ \bar\varepsilon_{g,\vartheta}(\mathsf{d})\Big);$$

this bound involves only the setting, g, ℓ, ϑ, d and a_K.

(b) Let (K_j,β_j) be first-passage data at g with $K_j \to \infty$ and $a_{K_j}\beta_j \to \mathsf{b}$. Then for every $f \in \mathfrak{Q}^{(3)}_{T_\ell}(\mathsf{d};\vartheta)$, $Q_{K_j,\ell,\beta_j}(f) \to Q^{(\ell,\mathsf{b})}(f)$ along the whole sequence, and

$$|S_3^{(\ell,\mathsf{b})}(f) - 3\,\mathsf{b}^{-3}\mathcal{K}^{(3)}_{\mathrm{free}}(f)| \le \epsilon_{\mathrm{III}}(\mathsf{d})\,\mathsf{b}^{-3}\,\mathsf{N}(f), \qquad |3\,Q^{(\ell,\mathsf{b})}(f) - \mathcal{R}(f)| \le \Theta_{\mathrm{Q}}(\vartheta;\tfrac13\epsilon_{\mathrm{III}}(\mathsf{d}),\bar\varepsilon_{g,\vartheta}(\mathsf{d}));$$

the last right member tends to 0 as $\mathsf{d} \to 0$.

■

Proposition 18.40 (Short distance: along one shape, at a small width, and for the supremum)**.** Let $0 < g \le \gamma_W$ and $\ell \in \mathfrak{L}(g)$, suppose $\mathrm{UL}_3(g,\ell)$, and let (K_j,β_j) be first-passage data at g with $K_j \to \infty$ and $a_{K_j}\beta_j \to \mathsf{b}$. For $\vartheta \in (0,1)$ the numbers $\mathsf{d}_{\mathrm{III}}$, ϵ_{III} are those of Definition 18.31 at ϑ.

(a) *Along one shape.* For every $\vartheta \in (0,1)$ and every $\Psi \in \mathfrak{Q}^{(3)}_{\mathbb{R}^3}(1;\vartheta)$, $\lim_{r\to0} Q^{(\ell,\mathsf{b})}(\Psi_r^{\langle\ell\rangle}) = \frac13\mathcal{R}(\Psi)$, which is smaller than $\frac49$.

(b) *At a small width.* Let $\vartheta \in (0,1)$ and $0 < w \le \frac{1}{16}$. For $0 < \mathsf{d} \le \min\{\mathsf{d}_0(g,\vartheta), \mathsf{d}_{\mathrm{III}}\}$ and $f \in \mathfrak{T}^{\mathrm{col}}_{T_\ell}(\mathsf{d};\vartheta;w)$,

$$|Q^{(\ell,\mathsf{b})}(f) - \tfrac49| \le 184\,w^2 + \tfrac13\,\Theta_{\mathrm{Q}}(\vartheta;\tfrac13\epsilon_{\mathrm{III}}(\mathsf{d}),\bar\varepsilon_{g,\vartheta}(\mathsf{d})).$$

Consequently, for every function $w \mapsto \vartheta(w)$ on $(0,\frac{1}{16}]$ with $0 < \vartheta(w) \le \frac{32\pi}{105}w^3$,

$$\lim_{w\to0}\ \limsup_{\mathsf{d}\to0}\ \sup_{f\in\mathfrak{T}^{\mathrm{col}}_{T_\ell}(\mathsf{d};\vartheta(w);w)}\ \limsup_{j\to\infty}\ |Q_{K_j,\ell,\beta_j}(f) - \tfrac49| = 0;$$

the classes that occur are not empty, and $\vartheta(w) \to 0$ as $w \to 0$.

(c) *For the supremum.* For $0<\vartheta\le\frac{\pi}{13440}$,

$$\lim_{\mathsf{d}\to0}\sup_{f\in\mathfrak{T}^{(3),\mathrm{col}}_{T_\ell}(\mathsf{d};\vartheta)}Q^{(\ell,\mathsf{b})}(f)=\tfrac{1}{3}\,\bar{\mathcal{R}}^{\mathrm{col}}(\vartheta),\qquad \lim_{\mathsf{d}\to0}\sup_{f\in\mathfrak{Q}^{(3)}_{T_\ell}(\mathsf{d};\vartheta)}Q^{(\ell,\mathsf{b})}(f)=\tfrac{1}{3}\,\bar{\mathcal{R}}(\vartheta);$$

both limits lie in $[\frac{4}{9}(1-412\,w_\vartheta^2),\frac{4}{9})$, both are non-increasing in ϑ, and both tend to $\frac{4}{9}$ as $\vartheta\to0$.

■

Proposition 18.41 (Positivity at the collinear triples; the corners of a triangle)**.** Let g, ℓ, (K_j,β_j) and b be as in Proposition 18.40, under $\mathrm{UL}_3(g,\ell)$.

(a) Let $\vartheta\in(0,1)$, $0<\mathsf{d}\le\min\{\mathsf{d}_0(g,\vartheta),\mathsf{d}_{\mathrm{III}}\}$ and $\epsilon_{\mathrm{III}}(\mathsf{d})<3c_3^{\mathrm{col}}\vartheta^3$. Then for every $f\in\mathfrak{T}^{(3),\mathrm{col}}_{T_\ell}(\mathsf{d};\vartheta)$, $S_3^{(\ell,\mathsf{b})}(f)\ge(3c_3^{\mathrm{col}}\vartheta^3-\epsilon_{\mathrm{III}}(\mathsf{d}))\,\mathsf{b}^{-3}\mathsf{N}(f)>0$.

(b) For c and $\Psi^{c,w}$ as in Proposition 18.36,

$$\lim_{w\to0}\lim_{r\to0}Q^{(\ell,\mathsf{b})}((\Psi^{c,w})_r^{\langle\ell\rangle})=\tfrac{1}{9}\,(1+3\,\mathsf{q}_\triangle(c))^2.$$

If $1+3\mathsf{q}_\triangle(c)\neq0$, there is $w_0>0$ such that for every $0<w\le w_0$ there is $r_0(w)>0$ with the following property: for $0<r\le r_0(w)$ the number $S_3^{(\ell,\mathsf{b})}((\Psi^{c,w})_r^{\langle\ell\rangle})$ is not 0, and its sign is that of $-(1+3\mathsf{q}_\triangle(c))$. When the c_i lie on one line this sign is $+$; for an equilateral triangle it is $-$, and the iterated limit equals $\frac{121}{576}$.

■

Proposition 18.42 (All sides and all real labels)**.** Suppose UL_3^*. Let $\ell=2L^m$, $m\in\mathbb{N}$, be any side, $\mathsf{b}>0$ any real number and $\vartheta\in(0,1)$. There are a number $\mathsf{d}_\sharp>0$ and functions $\epsilon_\sharp,e_\sharp$ on $(0,\mathsf{d}_\sharp]$, formed from the setting, ℓ, b, ϑ and the numbers of UL_3^*, with $\epsilon_\sharp(\mathsf{d})\to0$, $0\le e_\sharp\le\frac{1}{3}$ and $e_\sharp(\mathsf{d})\to0$ as $\mathsf{d}\to0$, such that for $0<\mathsf{d}\le\mathsf{d}_\sharp$, $f\in\mathfrak{Q}^{(3)}_{T_\ell}(\mathsf{d};\vartheta)$ and $i<j$

$$|S_3^{(\ell,\mathsf{b})}(f)-3\,\mathsf{b}^{-3}\mathcal{K}^{(3)}_{\mathrm{free}}(f)|\le\epsilon_\sharp(\mathsf{d})\,\mathsf{b}^{-3}\mathsf{N}(f),\qquad |S_2^{(\ell,\mathsf{b})}(f_i,f_j)-3\,\mathsf{b}^{-2}\mathcal{K}^{(2)}_{\mathrm{free}}(f_i,f_j)|\le e_\sharp(\mathsf{d})\,3\,\mathsf{b}^{-2}\mathcal{K}^{(2)}_{\mathrm{free}}(f_i,f_j),$$

and $|3Q^{(\ell,\mathsf{b})}(f)-\mathcal{R}(f)|\le\Theta_\sharp(\mathsf{d}):=\Theta_{\mathrm{Q}}(\vartheta;\frac{1}{3}\epsilon_\sharp(\mathsf{d}),e_\sharp(\mathsf{d}))$, which tends to 0 as $\mathsf{d}\to0$. The assertions of Propositions 18.40 and 18.41 hold on T_ℓ at the label b, with $\mathsf{d}_\sharp$, $\epsilon_\sharp$ and $\Theta_\sharp$ in the places of $\min\{\mathsf{d}_0(g,\vartheta),\mathsf{d}_{\mathrm{III}}\}$, ϵ_{III} and $\Theta_{\mathrm{Q}}(\vartheta;\frac{1}{3}\epsilon_{\mathrm{III}}(\cdot),\bar\varepsilon_{g,\vartheta}(\cdot))$, all at the shape parameter at hand, and, in the display of Proposition 18.40(b), with any pairs $(K_j,\beta_j)\in\mathbb{N}\times(0,\infty)$ such that $K_j\to\infty$ and $a_{K_j}\beta_j\to\mathsf{b}$ in the place of the first-passage data; for such pairs and every f as above, $Q_{K_j,\ell,\beta_j}(f)$ is formed for all large j and tends to $Q^{(\ell,\mathsf{b})}(f)$. ■

The number of photon fields and the label.

Proposition 18.43. *Within the free templates, the function of two points fixes one number, and the functions of two and three points fix both.*

(a) Let $\mathsf{n},\mathsf{n}'\in\mathbb{N}$ and $\mathsf{b},\mathsf{b}'>0$. The functions $S_2^{\mathrm{fr}}[\mathsf{n},\mathsf{b}]$ and $S_2^{\mathrm{fr}}[\mathsf{n}',\mathsf{b}']$ agree at one test pair if and only if $\mathsf{n}'\mathsf{b}'^{-2}=\mathsf{n}\,\mathsf{b}^{-2}$, and then they agree at every pair at which they are defined. The templates whose function of two points is $3\mathsf{b}^{-2}\mathcal{K}^{(2)}_{\mathrm{free}}$, the leading term of Proposition 18.38, are exactly the $S^{\mathrm{fr}}[\mathsf{n}',\mathsf{b}(\mathsf{n}'/3)^{1/2}]$, $\mathsf{n}'\in\mathbb{N}$; among them are $S^{\mathrm{fr}}[1,\mathsf{b}/\sqrt{3}]$, $S^{\mathrm{fr}}[3,\mathsf{b}]$ and $S^{\mathrm{fr}}[12,2\mathsf{b}]$.

(b) The map $(\mathsf{n},\mathsf{b})\mapsto(A_2,A_3)=(\mathsf{n}\,\mathsf{b}^{-2},\mathsf{n}\,\mathsf{b}^{-3})$ of $(0,\infty)^2$ into itself is injective, with the inverse $(A_2,A_3)\mapsto(A_2^3/A_3^2,\ A_2/A_3)$.

(c) Suppose UL_3^*. Let $\ell=2L^m$, $m\in\mathbb{N}$, $\mathsf{b}>0$, $\vartheta\in(0,1)$ and $\Psi=(\psi_1,\psi_2,\psi_3)\in\mathfrak{T}^{(3),\mathrm{col}}_{\mathbb{R}^3}(1;\vartheta)$. Then the limits

$$A_2=\lim_{r\to0}\frac{S_2^{(\ell,\mathsf{b})}(\psi_{i,r}^{\langle\ell\rangle},\psi_{j,r}^{\langle\ell\rangle})}{\mathcal{K}^{(2)}_{\mathrm{free}}(\psi_i,\psi_j)}\quad(i<j),\qquad A_3=\lim_{r\to0}\frac{S_3^{(\ell,\mathsf{b})}(\Psi_r^{\langle\ell\rangle})}{\mathcal{K}^{(3)}_{\mathrm{free}}(\Psi)}$$

exist, with $A_2 = 3\,\mathsf{b}^{-2}$ for each of the three pairs and $A_3 = 3\,\mathsf{b}^{-3}$; the first of them exists, with this value, without UL_3^*. Hence $\lim_{r\to 0}\mathcal{R}(\Psi)/Q^{(\ell,\mathsf{b})}(\Psi_r^{\langle\ell\rangle}) = A_2^3/A_3^2 = 3$ and $A_2/A_3 = \mathsf{b}$; and the only template $S^{\mathrm{fr}}[\mathsf{n}',\mathsf{b}']$, $\mathsf{n}' \in \mathbb{N}$, $\mathsf{b}' > 0$, with these two limits at the periodised dilates of Ψ is $S^{\mathrm{fr}}[3,\mathsf{b}]$, where $3 = \dim \mathfrak{su}(2)$.

■

How the ratio stands to the law of three points.

Proposition 18.44 (The ratio and the law of three points in modulus)**.** (a) Let $\mathbb{E} = \mathbb{R}^3$, or $\mathbb{E} = T_\ell$ with $36\mathsf{d} < \ell$, $f \in \mathfrak{Q}^{(3)}_{\mathbb{E}}(\mathsf{d};\vartheta)$ with $\mathcal{K}^{(3)}_{\mathrm{free}}(f) \neq 0$, $\mathsf{b} > 0$, $y_3 \in \mathbb{R}$, and $y_{ij} = 3\mathsf{b}^{-2}\mathcal{K}^{(2)}_{\mathrm{free}}(f_i,f_j)(1+e_{ij})$ with $|e_{ij}| < 1$ for $i < j$. Then

$$\Big(\frac{y_3}{3\,\mathsf{b}^{-3}\mathcal{K}^{(3)}_{\mathrm{free}}(f)}\Big)^2 = \frac{3\,y_3^2/(y_{12}y_{13}y_{23})}{\mathcal{R}(f)}\ \prod_{i<j}(1+e_{ij}).$$

(b) Let $\ell = 2L^m$, $m \in \mathbb{N}$, $\mathsf{b} > 0$, $\vartheta \in (0,1)$, and for $k \in \mathbb{N}$ let $f^{(k)} \in \mathfrak{Q}^{(3)}_{T_\ell}(\mathsf{d}_k;\vartheta)$ with $\mathcal{K}^{(3)}_{\mathrm{free}}(f^{(k)}) \neq 0$ and $\mathsf{d}_k \to 0$. Then $3Q^{(\ell,\mathsf{b})}(f^{(k)})/\mathcal{R}(f^{(k)}) \to 1$ if and only if $|S_3^{(\ell,\mathsf{b})}(f^{(k)})|/(3\mathsf{b}^{-3}|\mathcal{K}^{(3)}_{\mathrm{free}}(f^{(k)})|) \to 1$.

(c) On the collinear classes $27\,(c_3^{\mathrm{col}})^2\vartheta^6 \le \mathcal{R}(f) < \frac{4}{3}$. For every side $\ell = 2L^m$, $m \in \mathbb{N}$, every $\mathsf{b} > 0$ and every $\vartheta \in (0,1)$, the supremum over the empty set being taken as 0,

$$\lim_{\mathsf{d}\to 0}\ \sup_{f\in\mathfrak{T}^{(3),\mathrm{col}}_{T_\ell}(\mathsf{d};\vartheta)} \big|3\,Q^{(\ell,\mathsf{b})}(f) - \mathcal{R}(f)\big| = 0 \quad\Longleftrightarrow\quad \lim_{\mathsf{d}\to 0}\ \sup_{f\in\mathfrak{T}^{(3),\mathrm{col}}_{T_\ell}(\mathsf{d};\vartheta)} \Big|\frac{|S_3^{(\ell,\mathsf{b})}(f)|}{3\,\mathsf{b}^{-3}\,\mathcal{K}^{(3)}_{\mathrm{free}}(f)} - 1\Big| = 0.$$

■

Proposition 18.45. *The value $\frac{4}{9}$ and the law in modulus for narrow bumps; an explicit pair.*

(a) Let $\ell = 2L^m$, $m \in \mathbb{N}$, $\mathsf{b} > 0$, $0 < w \le \frac{1}{16}$, $\vartheta \in (0,1)$, $0 < \mathsf{d} \le \mathsf{d}_{\mathrm{II}}(\ell,\mathsf{b},\vartheta)$ with ϵ_{II} as in Proposition 18.38(b), $f \in \mathfrak{T}^{\mathrm{col}}_{T_\ell}(\mathsf{d};\vartheta;w)$, and $\mathsf{v}(f) = |S_3^{(\ell,\mathsf{b})}(f)|/(3\,\mathsf{b}^{-3}\mathcal{K}^{(3)}_{\mathrm{free}}(f))$, the denominator being positive. Then $\frac{9}{4}\,Q^{(\ell,\mathsf{b})}(f) = \mathsf{A}(f)\,\mathsf{v}(f)^2$, where the number $\mathsf{A}(f)$ satisfies

$$\frac{1-412\,w^2}{(1+\epsilon_{\mathrm{II}}(\mathsf{d}))^3} \le \mathsf{A}(f) \le \frac{1}{(1-\epsilon_{\mathrm{II}}(\mathsf{d}))^3}.$$

Consequently, for every function $w \mapsto \vartheta(w)$ on $(0,\frac{1}{16}]$ with $0 < \vartheta(w) \le \frac{32\pi}{105}w^3$ and every sequence of pairs $(K_j,\beta_j) \in \mathbb{N}\times(0,\infty)$ with $K_j \to \infty$ and $a_{K_j}\beta_j \to \mathsf{b}$, the display of Proposition 18.40(b) holds if and only if

$$\lim_{w\to 0}\ \limsup_{\mathsf{d}\to 0}\ \sup_{f\in\mathfrak{T}^{\mathrm{col}}_{T_\ell}(\mathsf{d};\vartheta(w);w)} \big|\mathsf{v}(f) - 1\big| = 0.$$

(b) For bounded Borel functions $f_1,f_2,f_3 \ge 0$ on $\mathbb{R}^3$ with compact supports at pairwise positive distance put $\hat{S}_2 = S_2^{\mathrm{fr}}[3,\mathsf{b}]$ and

$$\hat{S}_3(f) = 18\,\mathsf{b}^{-3}(4\pi)^{-3}\int \prod_i f_i(x_i)\,(r_{12}r_{13}r_{23})^{-3}\,dx,$$

and, if each f_i is positive on a set of positive measure, $\hat{Q}(f) = \hat{S}_3(f)^2/\prod_{i<j}\hat{S}_2(f_i,f_j)$; on a torus both are formed at a lift, which is possible as $\hat{S}_3$ does not change under a common translation of the entries. Let Ψ° be the triple of Proposition 18.35(b) at $\mathsf{d} = 1$, $w = \frac{1}{16}$ and $e = e_3$. Then: (i) $\hat{S}_2$ satisfies the display of Proposition 18.38(a) with $\eta = 0$. (ii) $\hat{Q}(f) \le \frac{4}{9}$; for $f \in \mathfrak{T}^{\mathrm{col}}_{\mathbb{E}}(\mathsf{d};\vartheta;w)$, at every d and ϑ, $\hat{Q}(f) \ge \frac{4}{9}\,e^{-\frac{27}{2}\varkappa_w^2} \ge \frac{4}{9}\,(1-223\,w^2)$, so that $|\hat{Q}(f) - \frac{4}{9}| \le 100\,w^2$; and for $0 < \vartheta \le \frac{\pi}{13440}$ the supremum of $\hat{Q}$ over $\mathfrak{T}^{(3),\mathrm{col}}_{\mathbb{E}}(\mathsf{d};\vartheta)$ lies

in $[\frac{4}{9}(1-223\,w_\vartheta^2),\frac{4}{9}]$ at every d, and so tends to $\frac{4}{9}$ as $\vartheta\to 0$. (iii) For $f\in\mathfrak{T}^{\mathrm{col}}_{\mathbb{E}}(\mathsf{d};\vartheta;w)$, $1<\hat S_3(f)/(3\,\mathsf{b}^{-3}\mathcal{K}^{(3)}_{\mathrm{free}}(f))\le \mathsf{t}_w^{-1}$; for $\Psi\in\mathfrak{T}^{(3),\mathrm{col}}_{\mathbb{R}^3}(1;\vartheta)$ the quotient takes the same value $1+\hat\delta(\Psi)$, with $\hat\delta(\Psi)>0$, at all the dilates Ψ_r, $r>0$, and $3\hat Q(\Psi_r)=\mathcal{R}(\Psi)\,(1+\hat\delta(\Psi))^2>\mathcal{R}(\Psi)$; for every $0<\vartheta\le\frac{\pi}{13440}$ and every $\mathsf{d}>0$ (on T_ℓ, $36\mathsf{d}<\ell$)

$$\sup_{f\in\mathfrak{T}^{(3),\mathrm{col}}_{\mathbb{E}}(\mathsf{d};\vartheta)}\Big(\frac{\hat S_3(f)}{3\,\mathsf{b}^{-3}\mathcal{K}^{(3)}_{\mathrm{free}}(f)}-1\Big)\ge\hat\delta(\Psi^\circ)>0,$$

and $|\hat S_3(\Psi_r^\circ)-3\,\mathsf{b}^{-3}\mathcal{K}^{(3)}_{\mathrm{free}}(\Psi_r^\circ)|=3\,\mathsf{b}^{-3}\mathcal{K}^{(3)}_{\mathrm{free}}(\Psi^\circ)\,\hat\delta(\Psi^\circ)$ for every $r>0$, while $\mathsf{N}(\Psi_r^\circ)=1$.

(c) So the pair $(\hat S_2,\hat S_3)$ has the law of two points with $\eta=0$, the bound $|\hat Q-\frac{4}{9}|\le 100\,w^2$ on the width classes and suprema over the collinear classes that tend to $\frac{4}{9}$ as $\vartheta\to0$, while at the collinear triples of every shape parameter $\vartheta\le\frac{\pi}{13440}$ its function of three points exceeds $3\mathsf{b}^{-3}\mathcal{K}^{(3)}_{\mathrm{free}}$ by the factor $1+\hat\delta(\Psi^\circ)$ along the dilates of Ψ°, at every scale.

■

Remark 18.46 (Comparison to the cubic source jet). Let g satisfy (Γ) of Definition 4.1(1). Take $\mathsf{d}_{\mathrm{III}}=\min\{\mathsf{d}_4(g),1/200\}$ and let $K_{\mathrm{III}}(\mathsf{d})$ be the least integer $K\ge\max\{2,s(\mathsf{d})\}$ with $4a_K\le\mathsf{d}$ and $\sqrt{3}\,a_K<2\vartheta\mathsf{d}$. For $\ell\in\mathfrak{L}(g)$, lift $f\in\mathfrak{Q}^{(3)}_{T_\ell}(\mathsf{d};\vartheta)$ to a triple $\hat f$ in the same class on $\mathbb{R}^3$. Its three supports lie in a ball of radius $9\mathsf{d}<\ell_g/8-1$, so periodisation on the least torus preserves the suprema and gives weights $w_a=\hat f_a^{\langle\ell_g\rangle}\circ x$ placed at $(\mathsf{d},\mathsf{d},\mathsf{d})$. Each $\hat f_a$ is positive on the open ball of radius $\vartheta\mathsf{d}$ about a point at which it attains its maximum. The barycentres of plaquettes of a fixed orientation form a translate of $a_K\mathbb{Z}^3$, which meets every closed ball of radius $\sqrt{3}\,a_K/2$. Thus the sampled weights on T_{ℓ_g} are nonzero.

Let J_3 be the cubic jet on T_{ℓ_g} at these weights used in Proposition 4.19(a). By the finite-cutoff identity for periodic extensions, $S^T_{3;K,\ell,\beta}(f)=S^T_3[\tilde\mu_{K,\ell,\beta}](\hat f)$. Theorem G(A.4)(i) compares this cumulant to the one on T_{ℓ_g}, and Lemma 4.21 compares the least-torus cumulant to J_3. Here $s(\mathsf{d})\le K$ and $\mathsf{d}\le\mathsf{d}_4(g)\le r_1(\bar{\mathsf{b}})\le r_1(\mathsf{b})$. Adding the estimates and using $\mathsf{b}\le\bar{\mathsf{b}}$ yields

$$|S^T_{3;K,\ell,\beta}(f)-\mathsf{J}_3|\le[C_3^{\mathrm{tr3}}(g)+13\Lambda_3^3(\bar{\mathsf{M}}^{(24)})^3]\bar{\mathsf{b}}^3\mathsf{d}^{3/2}\mathsf{b}^{-3}\mathsf{N}(f).$$

Proposition 4.19(a) compares J_3 to the lattice ring. Lemma 4.22(a), applied at $\rho=r=\mathsf{d}$ and $\mathsf{d}^+=18\mathsf{d}$ with $4a_K\le\mathsf{d}$ and $72\mathsf{d}<\ell_g$, compares the lattice ring to the free integral. Together they give

$$|\mathsf{J}_3-3\mathsf{b}^{-3}\mathcal{K}^{(3)}_{\mathrm{free}}(f)|\le\left[\epsilon^{\mathrm{nar}}(g,\mathsf{d})+3C^{(3)}\left(\frac{a_K}{\vartheta\mathsf{d}}+\left(\frac{18\mathsf{d}}{\ell_g}\right)^3\right)\right]\mathsf{b}^{-3}\mathsf{N}(f).$$

The triangle inequality proves Definition 18.31 with $C_{\mathrm{III}}=3C^{(3)}$. The remaining error tends to zero uniformly in the cutoff, label, triple and position. Thus UL^*_3 holds with γ_{III} the least of γ and the coupling bounds in (Γ). The conclusions of Propositions 18.39 to 18.43 then apply for $0<g\le\min\{\gamma_W,\gamma_{\mathrm{III}}\}$ and, by the change of units in Proposition 18.42, on all tori at all labels.

Remark 18.47 (The hypothesis from the law of three points of Section 4.1). Let g be a reference coupling under the ceiling (Γ) of Definition 4.1(1); let $r_1(\cdot)$ and $\mathsf{d}_4(g)$ be the function and the number of Definition 4.1(2), $\bar{\mathsf{M}}^{(24)}$, Λ_3, $C^{(3)}$ and $C_3^{\mathrm{tr3}}(g)$ the numbers of its item (3), and $\epsilon^{\mathrm{nar}}(g,\cdot)$ the function of its item (4). Then $\mathrm{UL}_3(g,\ell)$ holds for every $\ell\in\mathfrak{L}(g)$: at every $\vartheta\in(0,1)$ one may take $C_{\mathrm{III}}=3C^{(3)}$, $\mathsf{d}_{\mathrm{III}}=\min\{\mathsf{d}_4(g),\frac{1}{200}\}$, $\epsilon_{\mathrm{III}}(\mathsf{d})=\epsilon^{\mathrm{nar}}(g,\mathsf{d})+(13\Lambda_3^3(\bar{\mathsf{M}}^{(24)})^3+C_3^{\mathrm{tr3}}(g))\bar{\mathsf{b}}^3\mathsf{d}^{3/2}+3C^{(3)}(18\mathsf{d}/\ell_g)^3$, and for $K_{\mathrm{III}}(\mathsf{d})$ the least integer $K\ge\max\{1,s(\mathsf{d})\}$ with $4a_K\le\mathsf{d}$. Indeed, let (K,β) be a first-passage datum at g with the label b, $0<\mathsf{d}\le\mathsf{d}_{\mathrm{III}}$, $K\ge K_{\mathrm{III}}(\mathsf{d})$ and $f\in\mathfrak{Q}^{(3)}_{T_\ell}(\mathsf{d};\vartheta)$. By the lifting argument above, f has a lift $\hat f$ whose entries form test pairs on $\mathbb{R}^3$, with $\operatorname{supp}\hat f_a\subset\mathbb{B}(\hat x_a,\mathsf{d})$, $|\hat x_a-\hat x_1|\le8\mathsf{d}$ and all three supports inside $\mathbb{B}(\hat x_1,9\mathsf{d})$; by the periodic extension identity $S^T_{3;K,\ell,\beta}(f)=S^T_3[\tilde\mu_{K,\ell,\beta}](\hat f)$,

and $\tilde{\mu}_{K,\ell,\beta}$ is a state at (K,β) in the sense of Section 4.1. We read Theorem G at $\rho = r = \mathsf{d}$, where the natural size of Section 4.1 is $\mathrm{nat}_3 = \mathsf{b}^{-3}\mathsf{N}(f)$. Its hypotheses hold: $s(\mathsf{d}) \le K$; $\mathsf{d} \le \mathsf{d}_4(g) \le r_1(\bar{\mathsf{b}}) \le r_1(\mathsf{b})$, as $\mathsf{b} \le \bar{\mathsf{b}}$ and r_1 does not increase; $9\mathsf{d} < \frac{1}{8}\ell_g - 1$, as $\ell_g \ge 2LM \ge 26$; and on T_{ℓ_g}, where the covering map is isometric on every ball of a radius $< \ell_g/4$, the weights $w_a = \hat{f}_a^{\langle \ell_g \rangle} \circ x$ are placed at $(\mathsf{d},\mathsf{d},\mathsf{d})$, in the sense of Section 4.1, about the images of the points $\hat{x}_a$. Clause (A.4)(i) bounds $|S_3^T[\tilde{\mu}_{K,\ell,\beta}](\hat{f}) - S_{3;K,\ell_g,\beta}^T(w_1,w_2,w_3)|$ by $C_3^{\mathrm{tr}3}(g)\mathsf{b}^3\mathsf{d}^{3/2}\,\mathsf{b}^{-3}\mathsf{N}(f)$; clause (A.1), whose number M' is at most $\bar{\mathsf{M}}^{(24)}$, bounds the distance of $S_{3;K,\ell_g,\beta}^T(w_1,w_2,w_3)$ from the lattice ring $3\mathsf{b}^{-3}\,\mathrm{Tr}\prod_a(w_a C_K^{\mathrm{free}})$ by $[\epsilon^{\mathrm{nar}}(g,\mathsf{d}) + 13\Lambda_3^3(\bar{\mathsf{M}}^{(24)})^3\mathsf{b}^3\mathsf{d}^{3/2}]\,\mathsf{b}^{-3}\mathsf{N}(f)$; and Lemma 4.22(a) on T_{ℓ_g}, at $\rho = r = \mathsf{d}$ and $\mathsf{d}^+ = 18\mathsf{d}$, applicable as $4a_K \le \mathsf{d}$, $72\mathsf{d} < \ell_g$ and the periodisations $\hat{f}_a^{\langle \ell_g \rangle}$ form test pairs on T_{ℓ_g} with the sup norms of the f_a, bounds the distance of that trace from $\int k_3 \prod_a \hat{f}_a = \mathcal{K}_{\mathrm{free}}^{(3)}(f)$ by $C^{(3)}[a_K/(\vartheta\mathsf{d}) + (18\mathsf{d}/\ell_g)^3]\,\mathsf{N}(f)$, the least lifts of the differences of points of one ball of a radius $< \ell_g/4$ being the differences of their lifts in one ball. The three bounds together, with $\mathsf{b} \le \bar{\mathsf{b}}$, give the inequality of Definition 18.31 with the numbers above, none of which depends on K, β, the triple or its place, and $\epsilon_{\mathrm{III}}(\mathsf{d}) \to 0$ as $\mathsf{d} \to 0$. Each member of (Γ) bounds g from above by a number formed from the constants (Definition 4.1(1)); with γ_{III} the least of these numbers and of γ, UL_3^* therefore holds, and the conclusions of Propositions 18.39 to 18.43 hold for $0 < g \le \min\{\gamma_W, \gamma_{\mathrm{III}}\}$, and on all tori at all labels.

The argument.

Lemma 18.48 (A Cauchy inequality for three kernels)**.** Let μ_1,μ_2,μ_3 be finite Borel measures on $\mathbb{R}^3$ and $F_{12},F_{13},F_{23} \ge 0$ bounded Borel functions on $\mathbb{R}^3 \times \mathbb{R}^3$. Then

$$\int F_{12}(x_1,x_2)F_{13}(x_1,x_3)F_{23}(x_2,x_3)\,d\mu_1(x_1)\,d\mu_2(x_2)\,d\mu_3(x_3) \le \prod_{i<j}\Big(\int F_{ij}^2\,d\mu_i\,d\mu_j\Big)^{1/2}.$$

■

Proof. Integrate first in x_3: the function $G(x_1,x_2) = \int F_{13}(x_1,x_3)F_{23}(x_2,x_3)\,d\mu_3(x_3)$ is bounded and Borel, and by Tonelli's theorem the integral to be bounded equals $\int F_{12}\,G\,d\mu_1 d\mu_2$. The Cauchy–Schwarz inequality for $\mu_1 \otimes \mu_2$ bounds this by $(\int F_{12}^2\,d\mu_1 d\mu_2)^{1/2}(\int G^2\,d\mu_1 d\mu_2)^{1/2}$. For fixed x_1,x_2 the same inequality for μ_3 gives $G(x_1,x_2)^2 \le (\int F_{13}(x_1,\cdot)^2\,d\mu_3)(\int F_{23}(x_2,\cdot)^2\,d\mu_3)$; integrating in x_1 and x_2 separates the two factors, and $\int G^2\,d\mu_1 d\mu_2 \le \int F_{13}^2\,d\mu_1 d\mu_3 \cdot \int F_{23}^2\,d\mu_2 d\mu_3$. □

Lemma 18.49 (Jensen's inequality and a bound for $\langle e^{2Y}\rangle$)**.** Let ν be a probability measure, $\langle\cdot\rangle$ the mean for ν, and Y a bounded real measurable function. (J) $\langle e^Y\rangle \ge e^{\langle Y\rangle}$. (V) If the values of Y lie in an interval of the length ω, then $\langle e^{2Y}\rangle \le e^{2\langle Y\rangle + \omega^2/2}$. ■

Proof. (J) Since $e^u \ge 1 + u$ for real u, averaging at $u = Y - \langle Y\rangle$ gives $\langle e^Y\rangle\, e^{-\langle Y\rangle} \ge 1$. (V) Let $\psi(\sigma) = \log\langle e^{\sigma Y}\rangle$ for real σ. As Y is bounded, one may differentiate under the average; ψ is twice continuously differentiable, $\psi(0) = 0$ and $\psi'(0) = \langle Y\rangle$. With $\langle\cdot\rangle_\sigma$ the average for the probability measure $e^{\sigma Y}\nu/\langle e^{\sigma Y}\rangle$, $\psi''(\sigma) = \langle Y^2\rangle_\sigma - \langle Y\rangle_\sigma^2 = \min_{c\in\mathbb{R}}\langle (Y-c)^2\rangle_\sigma$; choosing for c the centre of an interval of the length ω that holds the values of Y gives $\psi''(\sigma) \le \frac{\omega^2}{4}$. Taylor's formula at 0 with the remainder in Lagrange's form then yields $\psi(2) \le 2\psi'(0) + 2\sup\psi'' \le 2\langle Y\rangle + \frac{\omega^2}{2}$. □

Proof of Proposition 18.32. (a) The three numbers $u = (a\cdot b)^2$, $v = (b\cdot c)^2$, $z = (c\cdot a)^2$ lie in $[0,1]$, and $\mathsf{q}_\triangle^2 = uvz \le 1$. The value $\mathsf{q}_\triangle = -1$ forces $uvz = 1$, hence $u = v = z = 1$; then $b = \pm a$, so $x_1 - x_2$ is a multiple of $x_2 - x_3$, and x_1,x_2,x_3 are collinear. Collinear points give $\mathsf{q}_\triangle = -1$. Suppose $\mathsf{q} = \mathsf{q}_\triangle > 0$. From $u + v + z = 1 - 2\mathsf{q} \ge 0$ we get $\mathsf{q} \le \frac{1}{2}$, and as a geometric mean does not exceed the arithmetic mean of the same numbers, $\mathsf{q}^2 = uvz \le (\frac{u+v+z}{3})^3 = (\frac{1-2\mathsf{q}}{3})^3$, i.e. $27\mathsf{q}^2 \le (1-2\mathsf{q})^3$. The function $\mathsf{q} \mapsto 27\mathsf{q}^2 - (1-2\mathsf{q})^3$ is increasing on $(0,\frac{1}{2}]$ and vanishes at $\mathsf{q} = \frac{1}{8}$,

where $27\mathsf{q}^2 = (1-2\mathsf{q})^3 = \frac{27}{64}$; hence $\mathsf{q} \le \frac{1}{8}$. The bounds for $\tau_\triangle$ follow, $\tau_\triangle = -3 - 9\mathsf{q}_\triangle$ being a decreasing affine function of $\mathsf{q}_\triangle$ with the values 6 and $-\frac{33}{8}$ at $\mathsf{q}_\triangle = -1$ and at $\mathsf{q}_\triangle = \frac{1}{8}$.

(b) By the formula for k_2, $k_2(x_i,x_j)^{1/2} = \frac{\sqrt{3}}{4\pi} r_{ij}^{-3}$, so $\prod_{i<j} k_2(x_i,x_j)^{1/2} = 3\sqrt{3}\,(4\pi)^{-3}(r_{12}r_{13}r_{23})^{-3}$, while $k_3(x) = \tau_\triangle(x)(4\pi)^{-3}(r_{12}r_{13}r_{23})^{-3}$; the quotient is $\tau_\triangle(x)/(3\sqrt{3})$. Its square is $\tau_\triangle^2/27 = 9(1+3\mathsf{q}_\triangle)^2/27$.

(c) In the quotient of the kernels of the template the factors n and b give $\mathsf{n}^2\mathsf{b}^{-6}/(\mathsf{n}^3\mathsf{b}^{-6}) = \mathsf{n}^{-1}$; the rest is (b), with $(1-3)^2/3 = \frac{4}{3}$ at $\mathsf{q}_\triangle = -1$, and with $(\frac{11}{8})^2/3 = \frac{121}{192}$ and $\tau_\triangle = -\frac{33}{8} < 0$ at $\mathsf{q}_\triangle = \frac{1}{8}$. At $(0, e_3, 2e_3)$ the points lie on a line, so $\tau_\triangle = 6$, and $r_{12}r_{13}r_{23} = 1\cdot 2\cdot 1$. □

Proof of Proposition 18.33. (a) Put $\mu_i = f_i\,dx$, finite Borel measures, and define $F_{ij}(x,y) = k_2(x,y)^{1/2}$ when $x \in \operatorname{supp} f_i$ and $y \in \operatorname{supp} f_j$, and $F_{ij}(x,y) = 0$ otherwise; F_{ij} is bounded and Borel, since $|x-y|$ is at least the positive distance of the two supports wherever $F_{ij} \ne 0$, and $\int F_{ij}^2\,d\mu_i d\mu_j = \mathcal{K}^{(2)}_{\text{free}}(f_i,f_j)$. On $\prod_i \operatorname{supp} f_i$, Proposition 18.32(a), (b) give $|k_3| = \frac{|\tau_\triangle|}{3\sqrt{3}} \prod_{i<j} F_{ij}$ with $|\tau_\triangle| \le 6$, so that $|k_3| \le \frac{2}{\sqrt{3}} \prod_{i<j} F_{ij} = \sqrt{\frac{4}{3}} \prod_{i<j} F_{ij}$. Writing I for $\int F_{12}F_{13}F_{23}\,d\mu_1 d\mu_2 d\mu_3$ and using Lemma 18.48,

$$|\mathcal{K}^{(3)}_{\text{free}}(f)| \le \int |k_3|\,d\mu_1 d\mu_2 d\mu_3 \le \sqrt{\tfrac{4}{3}}\,\mathsf{I} \le \sqrt{\tfrac{4}{3}} \prod_{i<j} \Big(\int F_{ij}^2\,d\mu_i d\mu_j \Big)^{1/2} = \sqrt{\tfrac{4}{3}} \prod_{i<j} \mathcal{K}^{(2)}_{\text{free}}(f_i,f_j)^{1/2}.$$

(b) Let $G_i = \{f_i > 0\}$, an open set that is not empty. The integrand $k_2(x,y)f_i(x)f_j(y)$ of $\mathcal{K}^{(2)}_{\text{free}}(f_i,f_j)$ is non-negative, and continuous and positive on $G_i \times G_j$; so $\mathcal{K}^{(2)}_{\text{free}}(f_i,f_j) > 0$. Pick $x_1 \in G_1$ and $x_2 \in G_2$, and, as a non-empty open subset of $\mathbb{R}^3$ is contained in no line, a point $x_3 \in G_3$ outside the line through x_1 and x_2. These points are distinct, and Proposition 18.32(a) gives $-1 < \mathsf{q}_\triangle(x) \le \frac{1}{8}$, hence $|\tau_\triangle(x)| < 6$. By the continuity of $\mathsf{q}_\triangle$ on the distinct triples there is an open set $U \ni x$ inside $G_1 \times G_2 \times G_3$ on which $|\tau_\triangle| < 6$. The function $\sqrt{4/3}\prod_{i<j} F_{ij} - |k_3| = \frac{6-|\tau_\triangle|}{3\sqrt{3}} \prod_{i<j} F_{ij}$, non-negative on $\prod_i \operatorname{supp} f_i$, is positive on U, and U carries positive mass for $\mu_1 \otimes \mu_2 \otimes \mu_3$, the density $\prod_i f_i(x_i)$ being positive there; so $\int |k_3|\,d\mu_1 d\mu_2 d\mu_3 < \sqrt{4/3}\,\mathsf{I}$, and the inequality of (a) is strict.

(c) A triple of $\mathfrak{Q}^{(3)}_{\mathbb{R}^3}(\mathsf{d};\vartheta)$ consists of Lipschitz functions ≥ 0, none of them zero, whose supports are compact and at least d apart; on T_ℓ one passes to a lift in $\mathfrak{Q}^{(3)}_{\mathbb{R}^3}(\mathsf{d};\vartheta)$, at which $\mathcal{R}$ is formed. So (b) applies. □

Proof of Proposition 18.34. Since $\mathcal{K}^{(3)}_{\text{free}}$ and $\mathcal{R}$ of a triple on T_ℓ are their values at a lift, and a lift of a triple of the width class on T_ℓ lies in the width class on $\mathbb{R}^3$ (Definition 18.30(1), (2)), we may take $\mathbb{E} = \mathbb{R}^3$. Fix points c_i on a line $y + \mathbb{R}e$, $|e| = 1$, with $\operatorname{supp} f_i \subset \mathbb{B}(c_i, w\mathsf{d})$, put $m_i = \int f_i > 0$ and $\nu_i = m_i^{-1} f_i\,dx$, and write $\langle\cdot\rangle$ for averages with respect to $\nu_1 \otimes \nu_2 \otimes \nu_3$.

Step 1: distances. For $x_i \in \operatorname{supp} f_i$ and $x_j \in \operatorname{supp} f_j$, $r_{ij} = |x_i - x_j| \ge \mathsf{d}$, the supports being at least d apart. The difference $x_i - x_j$ is $c_i - c_j$ plus the vector $u_{ij} = (x_i - c_i) - (x_j - c_j)$, whose length is at most $2w\mathsf{d} \le 2w\,r_{ij}$; hence $|c_i - c_j|$ lies between $(1-2w)r_{ij}$ and $(1+2w)r_{ij}$, so that $|c_i - c_j| > 0$, and $\log r_{ij}$ ranges in the interval from $\log|c_i - c_j| - \log(1+2w)$ to $\log|c_i - c_j| - \log(1-2w)$, of the length $\varkappa_w$.

Step 2: directions. The vector $c_i - c_j$ is a multiple of e, so the part of $x_i - x_j$ perpendicular to e is the part of u_{ij} perpendicular to e, of a length at most $2w\,r_{ij}$. Dividing by r_{ij}: the unit vector along $x_i - x_j$ makes with $\mathbb{R}e$ an angle whose sine is at most $2w \le \frac{1}{8}$. Orient the three unit vectors toward e. Their projections on e are at least $\sqrt{1-4w^2}$ and their perpendicular components have norm at most $2w$. Thus each mutual dot product is at least $1 - 8w^2$. The trace identity gives $\tau_\triangle \ge -3 + 9(1-8w^2)^3 \ge 6 - 216w^2 \ge 6\,\mathsf{t}_w$. Also $\mathsf{t}_w \ge \mathsf{t}_{1/16} = 1 - \frac{90}{256} - \frac{72}{4096} > 0$, t_w decreasing in w.

Step 3: the bound. On $\operatorname{supp} f_i \times \operatorname{supp} f_j$ put $\lambda_{ij} = \log k_2(x_i,x_j)^{1/2} = \log\frac{\sqrt3}{4\pi} - 3\log r_{ij}$; by Step 1 it takes its values in an interval of the length $3\varkappa_w$. Proposition 18.32(b) writes $k_3 = \frac{\tau_\triangle}{3\sqrt3}\exp(\lambda_{12}+\lambda_{13}+\lambda_{23})$ on the product of the three supports; so by Step 2 and Lemma 18.49(J)

$$\mathcal{K}^{(3)}_{\text{free}}(f) = m_1m_2m_3\,\langle k_3\rangle \ge m_1m_2m_3\,\frac{6\,\mathsf{t}_w}{3\sqrt3}\,e^{\langle\lambda_{12}\rangle+\langle\lambda_{13}\rangle+\langle\lambda_{23}\rangle} > 0,$$

while Lemma 18.49(V), for $Y=\lambda_{ij}$ and $\omega = 3\varkappa_w$, gives

$$\mathcal{K}^{(2)}_{\text{free}}(f_i,f_j) = m_im_j\,\langle e^{2\lambda_{ij}}\rangle \le m_im_j\,e^{2\langle\lambda_{ij}\rangle+\frac92\varkappa_w^2}.$$

The numbers m_i enter $\mathcal{K}^{(3)}_{\text{free}}(f)^2$ and $\prod_{i<j}\mathcal{K}^{(2)}_{\text{free}}(f_i,f_j)$ through one and the same factor $(m_1m_2m_3)^2$, and the averages $\langle\lambda_{ij}\rangle$ through one and the same factor $\exp(2\sum_{i<j}\langle\lambda_{ij}\rangle)$; dividing, $\mathcal{R}(f) \ge \frac{36}{27}\,\mathsf{t}_w^2\exp(-3\cdot\frac92\varkappa_w^2) = \frac43\,\mathsf{t}_w^2\,e^{-\frac{27}{2}\varkappa_w^2}$.

Step 4: a polynomial lower bound. For $0\le x<1$ the series $\log\frac{1+x}{1-x} = 2\sum_{k\ge0}\frac{x^{2k+1}}{2k+1}$ is at most $2x\sum_{k\ge0}x^{2k} = \frac{2x}{1-x^2}$; with $x = 2w \le \frac18$ this gives $\varkappa_w \le \frac{4w}{1-1/64} = \frac{256}{63}w$, and $\frac{27}{2}(\frac{256}{63})^2 = \frac{884736}{3969} < 223$. Hence $e^{-\frac{27}{2}\varkappa_w^2} \ge e^{-223w^2} \ge 1-223w^2$, a positive number as $w^2 \le \frac{1}{256}$; also $\mathsf{t}_w^2 \ge 1-2(90w^2+72w^3) \ge 1-189w^2 > 0$, using $144w \le 9$. The product of these two positive lower bounds is at least $1-412w^2$. The upper bound $\mathcal{R}(f) < \frac43$ is Proposition 18.33(c), and $\frac43\cdot412 = \frac{1648}{3}$. □

Proof of Proposition 18.35. (a) Passing to a lift changes neither $\|f_1\|_\infty$ nor $\int f_1$. As f_1 vanishes off a ball of the radius $w\mathsf{d}$, $\int f_1 \le \frac{4\pi}{3}(w\mathsf{d})^3\|f_1\|_\infty$; together with $\int f_1 \ge \vartheta\|f_1\|_\infty\mathsf{d}^3$ and $\|f_1\|_\infty > 0$ this gives $\vartheta \le \frac{4\pi}{3}w^3$.

(b) The bump formula in the subsection on the scalar two-point function gives $u_j \in C^1$, $\operatorname{supp}u_j = \mathbb{B}(c_j,w\mathsf{d}) \subset \mathbb{B}(c_j,\mathsf{d})$, $\|u_j\|_\infty = 1$, $\int u_j = \frac{32\pi}{105}w^3\mathsf{d}^3 \ge \vartheta\mathsf{d}^3$ and $\operatorname{Lip}u_j = \frac{8}{3\sqrt3\,w\mathsf{d}}$; the last number is at most $\frac{1}{\vartheta\mathsf{d}}$, as $\vartheta \le \frac{32\pi}{105}w^3 \le \frac{32\pi}{105\cdot256}w < \frac{3\sqrt3}{8}w$. If the centres of two balls of the radius $w\mathsf{d}$ are $D > 2w\mathsf{d}$ apart, the balls themselves are $D-2w\mathsf{d}$ apart; here $D = 2\mathsf{d}$ or $D = 4\mathsf{d}$, so the supports are between $\frac{15}{8}\mathsf{d}$ and $4\mathsf{d}$ apart, inside $[\mathsf{d},6\mathsf{d}]$. Hence each pair (u_i,u_j) is a test pair with the centres c_i,c_j, the triple belongs to $\mathfrak{Q}^{(3)}_{\mathbb{R}^3}(\mathsf{d};\vartheta)$, and, the centres being on $\mathbb{R}e$, to $\mathfrak{T}^{\text{col}}_{\mathbb{R}^3}(\mathsf{d};\vartheta;w)$. On T_ℓ, by the lifting argument above, the periodisations form a triple of $\mathfrak{Q}^{(3)}_{T_\ell}(\mathsf{d};\vartheta)$ having (u_1,u_2,u_3) as a lift.

(c) The dilations $\Psi\mapsto\Psi_{\mathsf{d}}$ and $\Psi\mapsto\Psi_{1/\mathsf{d}}$ are mutually inverse maps between the classes on $\mathbb{R}^3$ at the scale 1 and at the scale d, both for $\mathfrak{Q}^{(3)}$ and for $\mathfrak{T}^{(3),\text{col}}$, and they leave $\mathcal{K}^{(3)}_{\text{free}}$ and $\mathcal{K}^{(2)}_{\text{free}}$, hence $\mathcal{R}$, unchanged. On T_ℓ with $36\mathsf{d} < \ell$, every triple of a torus class has a lift in the matching class on $\mathbb{R}^3$ with the same value of $\mathcal{R}$; conversely, the periodisations of a triple of a class on $\mathbb{R}^3$ form a triple of the matching torus class that has the given triple as a lift. So on both spaces and at every scale the suprema equal those on $\mathbb{R}^3$ at the scale 1. When ϑ decreases the classes can only become larger by the test inequalities, the requirement $\operatorname{supp}f_i \subset \mathbb{B}(c_i,\mathsf{d}/16)$ being free of ϑ; so the suprema are non-increasing in ϑ. Proposition 18.33(c) gives the upper bound $\frac43$. For the lower bound note that $\frac{\pi}{13440} = \frac{32\pi}{105}16^{-3}$; so $w_\vartheta \le \frac{1}{16}$ and $\vartheta = \frac{32\pi}{105}w_\vartheta^3$, by (b) the width class $\mathfrak{T}^{\text{col}}_{\mathbb{R}^3}(1;\vartheta;w_\vartheta)$ has elements, it is contained in $\mathfrak{T}^{(3),\text{col}}_{\mathbb{R}^3}(1;\vartheta) \subset \mathfrak{Q}^{(3)}_{\mathbb{R}^3}(1;\vartheta)$, and on it $\mathcal{R} \ge \frac43(1-412w_\vartheta^2)$ by Proposition 18.34. Since $w_\vartheta \to 0$ as $\vartheta\to0$, both suprema tend to $\frac43$. □

Proof of Proposition 18.36. The argument of Proposition 18.35(b) at the scale 1, with the centres c_j, gives the membership: $\operatorname{supp}u_j = \mathbb{B}(c_j,w) \subset \mathbb{B}(c_j,1)$, and two supports are $|c_i-c_j|-2w$ apart, a number in $[1,6]$ as $\frac98-\frac18 = 1$. After normalisation, $\nu_j^w = u_j\,dx/\int u_j$ is a probability measure carried by $\mathbb{B}(c_j,w)$, and $\mathcal{K}^{(3)}_{\text{free}}(\Psi^{c,w})/\prod_j\int u_j = \int k_3\,d\nu_1^w d\nu_2^w d\nu_3^w$. Its distance from $k_3(c)$ is bounded by the oscillation $\sup\{|k_3(x)-k_3(c)| : |x_j-c_j| \le w \text{ for } j=1,2,3\}$, which goes to 0 with w because k_3 is continuous at the triple c of distinct points; likewise $\mathcal{K}^{(2)}_{\text{free}}(u_i,u_j)/(\int u_i\int u_j) \to k_2(c_i,c_j) > 0$.

In $\mathcal{R}(\Psi^{c,w})$ the factors $\int u_j$ drop out, so it tends to $k_3(c)^2/\prod_{i<j}k_2(c_i,c_j)$, which is $\frac{1}{3}(1+3\mathsf{q}_\triangle(c))^2$ by Proposition 18.32(b). □

Proof of Proposition 18.37. Fix $\vartheta\in(0,\frac{\pi}{13440}]$.

Step 1: an auxiliary family. Let $\mathcal{C}_\vartheta$ consist of the triples $h=(h_1,h_2,h_3)$ of functions from $\mathbb{R}^3$ to $[0,1]$ for which one can find $x_1=0$ and points $x_2,x_3\in\mathbb{B}(0,8)$ with: (i) $\sup h_i=1$; (ii) $|h_i(x)-h_i(y)|\le|x-y|/\vartheta$ for all x,y; (iii) $h_i(x)=0$ whenever $|x-x_i|>1$; (iv) for $i\ne j$, $h_i(x)h_j(y)>0$ implies $|x-y|\ge1$. Every such h_i is continuous, non-negative, not the zero function and zero outside $\mathbb{B}(0,9)$, and by (iv) two of the supports, the closures of the sets where the functions are positive, are at least 1 apart; so $\mathcal{R}(h)$ is formed and $\mathcal{R}(h)<\frac{4}{3}$ (Proposition 18.33(b)).

Step 2: reduction to the family. Take $f\in\mathfrak{Q}^{(3)}_{\mathbb{R}^3}(1;\vartheta)$ and points x_i with $\operatorname{supp}f_i\subset\mathbb{B}(x_i,1)$. Any two of these points serve as centres of a test pair, so $|x_i-x_j|\le8$. Multiplying each f_i by some $c_i>0$ multiplies both $\mathcal{K}^{(3)}_{\mathrm{free}}(f)^2$ and $\prod_{i<j}\mathcal{K}^{(2)}_{\mathrm{free}}(f_i,f_j)$ by $(c_1c_2c_3)^2$, and a common translation of the three functions changes neither; so $\mathcal{R}$ takes the same value at f and at $h=(f_i(\cdot+x_1)/\|f_i\|_\infty)_{i=1,2,3}$. This h belongs to $\mathcal{C}_\vartheta$ with the points x_i-x_1: (i) and (iii) are immediate, (ii) is the Lipschitz condition of a test pair at the scale 1, on which the normalisation has no effect, and (iv) restates that the supports of f_i and f_j are at least 1 apart. Hence $\bar{\mathcal{R}}(\vartheta)\le R^*:=\sup_{\mathcal{C}_\vartheta}\mathcal{R}$.

Step 3: a maximising triple. Choose $h^{(k)}\in\mathcal{C}_\vartheta$, with points $x_i^{(k)}$, such that $\mathcal{R}(h^{(k)})\to R^*$. On the compact ball $\mathbb{B}(0,9)$ the functions $h_i^{(k)}$ are bounded by 1 and ϑ^{-1}-Lipschitz, so by the Arzelà–Ascoli theorem a subsequence converges uniformly on that ball, hence on $\mathbb{R}^3$, as all of them vanish outside it; after a further subsequence, $x_i^{(k)}\to x_i$, where $x_1=0$ and $|x_i|\le8$. Call the limit h. It lies in $\mathcal{C}_\vartheta$: its values are in $[0,1]$; $\sup h_i=\lim_k\sup h_i^{(k)}=1$ by the uniform convergence; the bound (ii) passes to pointwise limits; a point y with $|y-x_i|>1$ has $|y-x_i^{(k)}|>1$ for all large k, so $h_i(y)=0$; and $h_i(x)h_j(y)>0$ with $i\ne j$ gives $h_i^{(k)}(x)h_j^{(k)}(y)>0$ for all large k, hence $|x-y|\ge1$.

Step 4: passage to the limit in $\mathcal{R}$. Define $\tilde k_3(x)=k_3(x)$ if $r_{12},r_{13},r_{23}\ge1$ and $\tilde k_3(x)=0$ otherwise, and $\tilde k_2(x,y)=k_2(x,y)$ if $|x-y|\ge1$ and $\tilde k_2(x,y)=0$ otherwise; these Borel functions satisfy $|\tilde k_3|\le6(4\pi)^{-3}$, by Proposition 18.32(a), (b), and $0\le\tilde k_2\le\frac{3}{16\pi^2}$. For $h\in\mathcal{C}_\vartheta$ property (iv) allows one to replace k_3 and k_2 by $\tilde k_3$ and $\tilde k_2$ in $\mathcal{K}^{(3)}_{\mathrm{free}}(h)$ and $\mathcal{K}^{(2)}_{\mathrm{free}}(h_i,h_j)$, and to integrate over $\mathbb{B}(0,9)^3$ and $\mathbb{B}(0,9)^2$ only. As $|\alpha_1\alpha_2\alpha_3-\alpha_1'\alpha_2'\alpha_3'|\le\sum_i|\alpha_i-\alpha_i'|$ for numbers in $[0,1]$, we get, with $V=\frac{4\pi}{3}9^3$, $|\mathcal{K}^{(3)}_{\mathrm{free}}(h^{(k)})-\mathcal{K}^{(3)}_{\mathrm{free}}(h)|\le6(4\pi)^{-3}V^3\sum_i\|h_i^{(k)}-h_i\|_\infty\to0$ and $|\mathcal{K}^{(2)}_{\mathrm{free}}(h_i^{(k)},h_j^{(k)})-\mathcal{K}^{(2)}_{\mathrm{free}}(h_i,h_j)|\le\frac{3}{16\pi^2}V^2(\|h_i^{(k)}-h_i\|_\infty+\|h_j^{(k)}-h_j\|_\infty)\to0$. The numbers $\mathcal{K}^{(2)}_{\mathrm{free}}(h_i,h_j)$ are positive (Proposition 18.33(b)); hence $\mathcal{R}(h^{(k)})\to\mathcal{R}(h)$, so $R^*=\mathcal{R}(h)<\frac{4}{3}$ by Step 1, and with Step 2, $\bar{\mathcal{R}}^{\mathrm{col}}(\vartheta)\le\bar{\mathcal{R}}(\vartheta)<\frac{4}{3}$.

The consequence. Suppose that ϑ_k does not tend to 0. Then some $\vartheta_0\in(0,\frac{\pi}{13440}]$ satisfies $\vartheta_k\ge\vartheta_0$ for infinitely many k. For each such k, by the test inequalities, $f^{(k)}$ lies in $\mathfrak{Q}^{(3)}_{\mathbb{E}_k}(\mathsf{d}_k;\vartheta_0)$ as well, and Proposition 18.35(c) gives $\mathcal{R}(f^{(k)})\le\bar{\mathcal{R}}(\vartheta_0)$, a number below $\frac{4}{3}$; so $\mathcal{R}(f^{(k)})$ does not tend to $\frac{4}{3}$. □

Lemma 18.50 (The ratio near the free one)**.** Let $\mathbb{E}=\mathbb{R}^3$, or $\mathbb{E}=T_\ell$ with $36\mathsf{d}<\ell$, let $f\in\mathfrak{Q}^{(3)}_{\mathbb{E}}(\mathsf{d};\vartheta)$, let $\mathsf{n}>0$, $\mathsf{b}>0$, $t\ge0$ and $0\le e\le\frac{1}{3}$ be real numbers, and let y_3 and y_{ij}, $i<j$, be real numbers with

$$|y_3-\mathsf{n}\,\mathsf{b}^{-3}\mathcal{K}^{(3)}_{\mathrm{free}}(f)|\le t\,\mathsf{n}\,\mathsf{b}^{-3}\,\mathsf{N}(f),\qquad |y_{ij}-\mathsf{n}\,\mathsf{b}^{-2}\mathcal{K}^{(2)}_{\mathrm{free}}(f_i,f_j)|\le e\,\mathsf{n}\,\mathsf{b}^{-2}\mathcal{K}^{(2)}_{\mathrm{free}}(f_i,f_j).$$

Then $y_{ij}>0$, and $\bigl|\mathsf{n}\,y_3^2/(y_{12}y_{13}y_{23})-\mathcal{R}(f)\bigr|\le\Theta_{\mathrm{Q}}(\vartheta;t,e)$. ■

Proof. Set $\kappa=\mathcal{K}^{(3)}_{\mathrm{free}}(f)/\mathsf{N}(f)$ and $P=\prod_{i<j}\mathcal{K}^{(2)}_{\mathrm{free}}(f_i,f_j)/\mathsf{N}(f)^2$; each $\|f_a\|_\infty$ occurs in two of the three pairs, so $\mathcal{R}(f)=\kappa^2/P$. The hypotheses say that $y_3=\mathsf{n}\mathsf{b}^{-3}\mathsf{N}(f)(\kappa+\delta)$ with $|\delta|\le t$ and $y_{ij}=\mathsf{n}\mathsf{b}^{-2}\mathcal{K}^{(2)}_{\mathrm{free}}(f_i,f_j)(1+e_{ij})$ with $|e_{ij}|\le e<1$; so $y_{ij}>0$, as $\mathcal{K}^{(2)}_{\mathrm{free}}(f_i,f_j)>0$. With

$\Pi = \prod_{i<j}(1+e_{ij})$,

$$\mathsf{n}\,\frac{y_3^2}{y_{12}y_{13}y_{23}} = \frac{\mathsf{n}^3\mathsf{b}^{-6}\mathsf{N}(f)^2(\kappa+\delta)^2}{\mathsf{n}^3\mathsf{b}^{-6}\mathsf{N}(f)^2 P\,\Pi} = \frac{(\kappa+\delta)^2}{P\,\Pi}, \qquad \frac{(\kappa+\delta)^2}{P\,\Pi} - \frac{\kappa^2}{P} = \frac{\delta\,(2\kappa+\delta)}{P\,\Pi} + \mathcal{R}(f)\,\frac{1-\Pi}{\Pi}.$$

We bound the pieces. $|\kappa| \le \frac{4}{9}$ gives $|\delta(2\kappa+\delta)| \le \frac{8}{9}t + t^2$; $\Pi \ge (1-e)^3 \ge \frac{8}{27}$; the lower bound for $\mathcal{K}^{(2)}_{\mathrm{free}}$ at the test pairs gives $P \ge (\frac{3}{16\pi^2}10^{-6}\vartheta^2)^3 = \vartheta^6/\mathsf{C}_{\mathrm{P}}$; Π lies between $(1-e)^3$ and $(1+e)^3$, and $(1+e)^3 + (1-e)^3 = 2+6e^2 \ge 2$ shows $1-(1-e)^3 \le (1+e)^3 - 1$, so that $|1-\Pi| \le (1+e)^3 - 1 = e(3+3e+e^2) \le \frac{37}{9}e$ for $e \le \frac{1}{3}$; finally $\mathcal{R}(f) \le \frac{4}{3}$ by Proposition 18.33(c). Adding up, $|\mathsf{n}\,y_3^2/(y_{12}y_{13}y_{23}) - \mathcal{R}(f)| \le \frac{27}{8}\mathsf{C}_{\mathrm{P}}\vartheta^{-6}(\frac{8}{9}t+t^2) + \frac{4}{3}\cdot\frac{27}{8}\cdot\frac{37}{9}\,e$, and the last coefficient equals $\frac{37}{2}$. □

Lemma 18.51 (From any side and any label to a reference coupling)**.** Let $m \in \mathbb{N}$, $\mathsf{b} > 0$ and $\gamma_* \in (0,\gamma]$, and for an integer $s \ge 0$ put $g_s = (\frac{1}{2}L^s\mathsf{b})^{-1/2}$. There is an integer $s_0 \ge 0$ such that for every integer $s \ge s_0$: $g_s \le \gamma_*$, $L^s\mathsf{b} \in I(g_s)$ and $2L^{m+s} \in \mathfrak{L}(g_s)$; and for such s the pairs $(K, L^{K+s}\mathsf{b})$, $K \in \mathbb{N}$, are first-passage data at g_s with the label $L^s\mathsf{b}$. ■

Proof. Since $g_s^{-2} = \frac{1}{2}L^s\mathsf{b}$ grows without bound, both $g_s \le \gamma_*$ and $g_s^{-2} > \bar{B}$ are true for all large s. Write $\mathsf{b}' = L^s\mathsf{b}$, so that $\mathsf{b}' = 2g_s^{-2}$. *Membership in $I(g_s)$.* On the left, $g_s^{-2} + \bar{B} < 2g_s^{-2} = \mathsf{b}'$ because $g_s^{-2} > \bar{B}$. On the right, $X_{g_s} - \bar{B} = Lg_s^{-2} + LB_\sharp - \bar{B} = Lg_s^{-2} + (L-2)\bar{B}$ by $LB_\sharp = (L-1)\bar{B}$, and this equals $\frac{L}{2}\mathsf{b}' + (L-2)\bar{B} \ge \mathsf{b}'$. *The side.* $\mathfrak{L}(g_s)$ is defined, as $g_s \le \gamma \le \frac{1}{2}$. Put $x_s = L(g_s^{-2} + \bar{B})$; then $x_s \ge 4L > e$, so $(\log x_s)^r > 1$, and $R(x_s) = L^{\mathsf{u}}$ with an integer $\mathsf{u} \ge 1$ for which $L^{\mathsf{u}-1} < (\log x_s)^r$; thus $R(x_s) < L(\log x_s)^r$. From $x_s \le L^{s+1}(\mathsf{b}+\bar{B})$ we get $\log x_s \le (s+1)\log L + |\log(\mathsf{b}+\bar{B})|$, whence $L^{m_{\mathfrak{L}}(g_s)} = LM\max\{R(x_s), L^{\lceil \log_L C_{\mathrm{s}}\rceil}\} \le L^2M\,[(s+1)\log L + |\log(\mathsf{b}+\bar{B})|]^r + LM\,L^{\lceil\log_L C_{\mathrm{s}}\rceil}$. This bound grows polynomially in s and L^{m+s} exponentially, so $m+s \ge m_{\mathfrak{L}}(g_s)$, that is $2L^{m+s} \in \mathfrak{L}(g_s)$, for all large s. The statement on the pairs $(K, L^K\mathsf{b}')$ is the property of the points of $I(g_s)$ recalled in the setting, at the coupling $g_s \le \gamma$. □

Lemma 18.52 (Dilating a triple of a torus by L^s)**.** Let $\ell = 2L^m$, $m \in \mathbb{N}$, $s \ge 0$ an integer, $\mathsf{d} > 0$ with $36\mathsf{d} < \ell$, $\vartheta \in (0,1)$ and $f \in \mathfrak{Q}^{(3)}_{T_\ell}(\mathsf{d};\vartheta)$, and let $f^{(s)} = (f_1^{(s)}, f_2^{(s)}, f_3^{(s)})$. Then $f^{(s)} \in \mathfrak{Q}^{(3)}_{T_{L^s\ell}}(L^s\mathsf{d};\vartheta)$; $\|f_a^{(s)}\|_\infty = \|f_a\|_\infty$; $\mathcal{K}^{(3)}_{\mathrm{free}}(f^{(s)}) = \mathcal{K}^{(3)}_{\mathrm{free}}(f)$, $\mathcal{K}^{(2)}_{\mathrm{free}}(f_i^{(s)}, f_j^{(s)}) = \mathcal{K}^{(2)}_{\mathrm{free}}(f_i, f_j)$ and $\mathcal{R}(f^{(s)}) = \mathcal{R}(f)$; if $f \in \mathfrak{T}^{\mathrm{col}}_{T_\ell}(\mathsf{d};\vartheta;w)$, then $f^{(s)} \in \mathfrak{T}^{\mathrm{col}}_{T_{L^s\ell}}(L^s\mathsf{d};\vartheta;w)$; and for every $\mathsf{b} > 0$, $S_n^{(\ell,\mathsf{b})} = L^{sn}S_n^{(L^s\ell, L^s\mathsf{b})}$ at the triple and at its pairs and their dilates ($n = 3, 2$), so that $Q^{(\ell,\mathsf{b})}(f) = Q^{(L^s\ell, L^s\mathsf{b})}(f^{(s)})$ whenever one of the two is formed. ■

Proof. The map $x \mapsto L^s x$ from T_ℓ onto $T_{L^s\ell}$ multiplies distances by L^s. It preserves the supremum of every transformed test $f^{(s)}$, multiplies its integral by L^{3s} and divides its Lipschitz constant by L^s. Therefore the image triple is in the test class at scale $L^s\mathsf{d}$. If $\hat{f}$ is a common lift inside a ball of radius $\rho < \ell/4$, the lifted transformed triple in the dilated ball is $(\hat{f})_{L^s}$, since the covering maps commute with $x \mapsto L^s x$. Thus both free forms are invariant by homogeneity. The centers of narrow collinear supports are carried to collinear centers, and their widths multiply by L^s just as the scale does. The identity for the limits is YM3_I:(2215), and in the quotient Q the powers of L cancel, as $(L^{3s})^2 = (L^{2s})^3$. □

Proof of Proposition 18.38. (a) $\mathsf{b} \ge g^{-2} > 0$, as recalled in the setting. Let $0 < \mathsf{d} \le \mathsf{d}_0(g,\vartheta)$ and $(f,h) \in \mathfrak{T}_{T_\ell}(\mathsf{d};\vartheta)$. Both supports lie in one ball of the radius $5\mathsf{d} \le \frac{1}{40} < \frac{\ell}{4}$; so $\mathcal{K}^{(2)}_{\mathrm{free}}(f,h)$ is formed on T_ℓ, and $\mathsf{c}_2\mathcal{K}_{\mathrm{free}}(f,h) = 3\mathcal{K}^{(2)}_{\mathrm{free}}(f,h)$. The witness depth $s(\mathsf{d})$ is one integer, in which no cutoff enters, so $s(\mathsf{d}) < K_j$ once j is large. For such j, Theorem YM3_I:AV gives $|S^T_{2;K_j,\ell,\beta_j}(f,h) - 3\mathsf{b}_j^{-2}\mathcal{K}^{(2)}_{\mathrm{free}}(f,h)| \le \bar{\varepsilon}_{g,\vartheta}(\mathsf{d})\,3\mathsf{b}_j^{-2}\mathcal{K}^{(2)}_{\mathrm{free}}(f,h)$, where $\mathsf{b}_j = a_{K_j}\beta_j$. As $j \to \infty$, YM3_I:(2216), applicable as f, h are real, Lipschitz, not 0 and have disjoint supports, gives $S^T_{2;K_j,\ell,\beta_j}(f,h) \to S_2^{(\ell,\mathsf{b})}(f,h)$, and $\mathsf{b}_j^{-2} \to \mathsf{b}^{-2}$; a

non-strict inequality between convergent sequences persists for the limits. For $\mathsf{b} \in I(g)$ take $K_j = j$ and $\beta_j = L^j\mathsf{b}$.

(b) Take for s the smallest s_0 allowed by Lemma 18.51 at $(m, \mathsf{b}, \gamma_* = \gamma_W)$; then $g' = g_s$, $\ell' = L^s\ell \in \mathfrak{L}(g')$ and $\mathsf{b}' = L^s\mathsf{b} \in I(g')$. Put $\mathsf{d}_{\mathrm{II}} = L^{-s}\mathsf{d}_0(g', \vartheta)$ and $\epsilon_{\mathrm{II}}(\mathsf{d}) = \bar\varepsilon_{g',\vartheta}(L^s\mathsf{d})$; these have the properties stated, by YM3_I:(2349). For $0 < \mathsf{d} \le \mathsf{d}_{\mathrm{II}}$ and $(f, h) \in \mathfrak{T}_{T_\ell}(\mathsf{d}; \vartheta)$, the same dilation gives $(f^{(s)}, h^{(s)}) \in \mathfrak{T}_{T_{\ell'}}(L^s\mathsf{d}; \vartheta)$ with $\mathcal{K}^{(2)}_{\mathrm{free}}(f^{(s)}, h^{(s)}) = \mathcal{K}^{(2)}_{\mathrm{free}}(f, h)$. By YM3_I:(2215) and by (a) on $T_{\ell'}$ at the label b', which is the label of the first-passage data $(K, L^K\mathsf{b}')$ at g',

$$S_2^{(\ell,\mathsf{b})}(f, h) = L^{2s} S_2^{(\ell',\mathsf{b}')}(f^{(s)}, h^{(s)}) = L^{2s}\, 3\,(L^s\mathsf{b})^{-2}\,\mathcal{K}^{(2)}_{\mathrm{free}}(f, h)\,(1 + \eta) = 3\,\mathsf{b}^{-2}\,\mathcal{K}^{(2)}_{\mathrm{free}}(f, h)\,(1 + \eta), \qquad |\eta| \le \bar\varepsilon_{g',\vartheta}(L^s\mathsf{d}).$$

(c) $|\eta| \le \frac{1}{3}$, and $\mathcal{K}^{(2)}_{\mathrm{free}}(f, h) > 0$ at a test pair. □

Proof of Proposition 18.39. (a) $36\mathsf{d} \le \frac{36}{200} < \ell$, so the class is formed, and every pair (f_i, f_j) of the triple lies in $\mathfrak{T}_{T_\ell}(\mathsf{d}; \vartheta)$. The hypotheses of Theorem YM3_I:AV hold ($g \le \gamma_W$, $\mathsf{d} \le \mathsf{d}_0(g, \vartheta)$, $s(\mathsf{d}) < K$); with $\mathsf{c}_2\mathcal{K}_{\mathrm{free}} = 3\mathcal{K}^{(2)}_{\mathrm{free}}$, its torus estimate says that $y_{ij} = S^T_{2;K,\ell,\beta}(f_i, f_j)$ satisfies the second hypothesis of Lemma 18.50 with $\mathsf{n} = 3$ and $e = \bar\varepsilon_{g,\vartheta}(\mathsf{d}) \le \frac{1}{3}$. By $\mathrm{UL}_3(g, \ell)$ ($\mathsf{d} \le \mathsf{d}_{\mathrm{III}}$, $K \ge K_{\mathrm{III}}(\mathsf{d})$), $y_3 = S^T_{3;K,\ell,\beta}(f)$ satisfies the first with $\mathsf{n} = 3$ and $t = \frac{1}{3}[\epsilon_{\mathrm{III}}(\mathsf{d}) + C_{\mathrm{III}}\frac{a_K}{\vartheta\mathsf{d}}]$, since the bound in Definition 18.31 equals $t \cdot 3\,\mathsf{b}^{-3}\mathsf{N}(f)$. Lemma 18.50 now gives the claim. The bound contains ϵ_{III} and C_{III}, which depend on the setting, g, ℓ and ϑ, the tolerance $\bar\varepsilon_{g,\vartheta}$, and d and a_K; nothing else enters.

(b) For $j \ge j_1$, $K_j > s(\mathsf{d})$ and $K_j \ge K_{\mathrm{III}}(\mathsf{d})$. By YM3_I:(2216) at $n = 3$ and at $n = 2$, the entries of f being real Lipschitz functions, none of them 0, with pairwise disjoint supports, $S^T_{3;K_j,\ell,\beta_j}(f) \to S_3^{(\ell,\mathsf{b})}(f)$ and $S^T_{2;K_j,\ell,\beta_j}(f_i, f_j) \to S_2^{(\ell,\mathsf{b})}(f_i, f_j)$, and the latter limits are positive by Proposition 18.38(a), (c); hence $Q_{K_j,\ell,\beta_j}(f) \to Q^{(\ell,\mathsf{b})}(f)$. In the inequality of Definition 18.31 at (K_j, β_j) let $j \to \infty$: $a_{K_j} \to 0$ and $a_{K_j}\beta_j \to \mathsf{b}$, which gives the first display. With it and Proposition 18.38(a), Lemma 18.50 at $\mathsf{n} = 3$, $t = \frac{1}{3}\epsilon_{\mathrm{III}}(\mathsf{d})$, $e = \bar\varepsilon_{g,\vartheta}(\mathsf{d})$, $y_3 = S_3^{(\ell,\mathsf{b})}(f)$ and $y_{ij} = S_2^{(\ell,\mathsf{b})}(f_i, f_j)$ gives the second. For fixed ϑ the function $\Theta_{\mathrm{Q}}(\vartheta; t, e)$ is continuous in (t, e) and vanishes at $(0, 0)$. □

Proof of Proposition 18.40. (a) For $36r < \ell$, $\Psi_r \in \mathfrak{Q}^{(3)}_{\mathbb{R}^3}(r; \vartheta)$, and $\Psi_r^{\langle\ell\rangle} \in \mathfrak{Q}^{(3)}_{T_\ell}(r; \vartheta)$ has the lift Ψ_r; so $\mathcal{R}(\Psi_r^{\langle\ell\rangle}) = \mathcal{R}(\Psi_r) = \mathcal{R}(\Psi)$. Proposition 18.39(b) at $\mathsf{d} = r$ gives $|3Q^{(\ell,\mathsf{b})}(\Psi_r^{\langle\ell\rangle}) - \mathcal{R}(\Psi)| \le \Theta_{\mathrm{Q}}(\vartheta; \frac{1}{3}\epsilon_{\mathrm{III}}(r), \bar\varepsilon_{g,\vartheta}(r)) \to 0$ as $r \to 0$; and $\mathcal{R}(\Psi) < \frac{4}{3}$ by Proposition 18.33(c).

(b) $|Q - \frac{4}{9}| \le \frac{1}{3}|3Q - \mathcal{R}(f)| + \frac{1}{3}|\mathcal{R}(f) - \frac{4}{3}|$; the first term is bounded by Proposition 18.39(b), and by Proposition 18.34 the second is at most $\frac{1}{3} \cdot \frac{1648}{3}w^2 = \frac{1648}{9}w^2 < 184\,w^2$. For the display fix w and $\vartheta = \vartheta(w)$; by Proposition 18.35(b) the classes $\mathfrak{T}^{\mathrm{col}}_{T_\ell}(\mathsf{d}; \vartheta(w); w)$ with $36\mathsf{d} < \ell$ are not empty. For $\mathsf{d} \le \min\{\mathsf{d}_0(g, \vartheta(w)), \mathsf{d}_{\mathrm{III}}\}$, with $\mathsf{d}_{\mathrm{III}}$ at $\vartheta(w)$, Proposition 18.39(b) gives $\limsup_j |Q_{K_j,\ell,\beta_j}(f) - \frac{4}{9}| = |Q^{(\ell,\mathsf{b})}(f) - \frac{4}{9}|$ for every f of the class; so the supremum over the class is at most $184w^2 + \frac{1}{3}\Theta_{\mathrm{Q}}(\vartheta(w); \frac{1}{3}\epsilon_{\mathrm{III}}(\mathsf{d}), \bar\varepsilon_{g,\vartheta(w)}(\mathsf{d}))$, its upper limit as $\mathsf{d} \to 0$ is at most $184w^2$, and $184w^2 \to 0$ as $w \to 0$. Finally $\vartheta(w) \le \frac{32\pi}{105}w^3 \to 0$.

(c) For real-valued bounded F, G defined on the same non-empty set, $|\sup F - \sup G| \le \sup|F - G|$. On the class, which is not empty (Proposition 18.35(b)), take $F = Q^{(\ell,\mathsf{b})}$ and $G = \frac{1}{3}\mathcal{R}$, both bounded by Propositions 18.33(c) and 18.39(b); so the difference of the two suprema is at most $\frac{1}{3}\Theta_{\mathrm{Q}}(\vartheta; \frac{1}{3}\epsilon_{\mathrm{III}}(\mathsf{d}), \bar\varepsilon_{g,\vartheta}(\mathsf{d}))$, a number that goes to 0 as $\mathsf{d} \to 0$. The suprema of $\mathcal{R}$ over the two torus classes are $\bar{\mathcal{R}}^{\mathrm{col}}(\vartheta)$ and $\bar{\mathcal{R}}(\vartheta)$ at every d, with the lower bound, the monotonicity and the limit of Proposition 18.35(c); and they are $< \frac{4}{3}$ by Proposition 18.37. □

Proof of Proposition 18.41. (a) By the first display of Proposition 18.39(b) and the lower bound $\mathcal{K}^{(3)}_{\mathrm{free}}(f) \ge c_3^{\mathrm{col}}\vartheta^3\mathsf{N}(f)$ on the collinear class, $S_3^{(\ell,\mathsf{b})}(f) \ge 3\mathsf{b}^{-3}\mathcal{K}^{(3)}_{\mathrm{free}}(f) - \epsilon_{\mathrm{III}}(\mathsf{d})\mathsf{b}^{-3}\mathsf{N}(f) \ge (3c_3^{\mathrm{col}}\vartheta^3 - \epsilon_{\mathrm{III}}(\mathsf{d}))\mathsf{b}^{-3}\mathsf{N}(f)$.

(b) Fix $w \in (0, \frac{1}{16}]$ and $\vartheta_w = \frac{32\pi}{105} w^3$. By Proposition 18.36, $\Psi^{c,w} \in \mathfrak{Q}^{(3)}_{\mathbb{R}^3}(1; \vartheta_w)$, and Proposition 18.40(a) at ϑ_w gives $\lim_{r\to 0} Q^{(\ell,\mathsf{b})}((\Psi^{c,w})^{\langle\ell\rangle}_r) = \frac{1}{3}\mathcal{R}(\Psi^{c,w})$; as $w \to 0$ this tends to $\frac{1}{9}(1+3\mathsf{q}_\triangle(c))^2$ by Proposition 18.36. *The sign.* By Proposition 18.36, $\mathcal{K}^{(3)}_{\mathrm{free}}(\Psi^{c,w})/\prod_j \int u_j \to k_3(c)$, whose sign, by Proposition 18.32(b), is that of $\tau_\triangle(c) = -3(1+3\mathsf{q}_\triangle(c)) \neq 0$; so for all w up to some $w_0 > 0$ the number $\mathcal{K}^{(3)}_{\mathrm{free}}(\Psi^{c,w})$ is non-zero with that sign. By homogeneity and the lift, it is the value of $\mathcal{K}^{(3)}_{\mathrm{free}}$ at $(\Psi^{c,w})^{\langle\ell\rangle}_r$, where $\mathsf{N} = 1$, the u_j having the sup norm 1. By Proposition 18.39(b) at ϑ_w and $\mathsf{d} = r$, $|S^{(\ell,\mathsf{b})}_3((\Psi^{c,w})^{\langle\ell\rangle}_r) - 3\mathsf{b}^{-3}\mathcal{K}^{(3)}_{\mathrm{free}}(\Psi^{c,w})| \le \epsilon_{\mathrm{III}}(r)\mathsf{b}^{-3}$, which is $< 3\mathsf{b}^{-3}|\mathcal{K}^{(3)}_{\mathrm{free}}(\Psi^{c,w})|$ for all small r. On a line $\mathsf{q}_\triangle = -1$ and $-(1+3\mathsf{q}_\triangle) = 2 > 0$; at the corners of an equilateral triangle with a side in $[\frac{9}{8}, 6]$, $\mathsf{q}_\triangle = \frac{1}{8}$, $-(1+3\mathsf{q}_\triangle) < 0$, and $\frac{1}{9}(\frac{11}{8})^2 = \frac{121}{576}$. □

Proof of Proposition 18.42. Take for s the smallest s_0 allowed by Lemma 18.51 at $(m, \mathsf{b}, \gamma_* = \min\{\gamma_W, \gamma_{\mathrm{III}}\})$, and let $g' = g_s$, $\ell' = L^s\ell \in \mathfrak{L}(g')$ and $\mathsf{b}' = L^s\mathsf{b} \in I(g')$. By UL^*_3, $\mathrm{UL}_3(g', \ell')$ holds; let $\mathsf{d}'_{\mathrm{III}}$, ϵ'_{III} be its numbers at ϑ. The pairs $(K, L^K\mathsf{b}')$, $K \in \mathbb{N}$, are first-passage data at g' with the label b'. Put $\mathsf{d}_\sharp = L^{-s}\min\{\mathsf{d}_0(g', \vartheta), \mathsf{d}'_{\mathrm{III}}\}$, $\epsilon_\sharp(\mathsf{d}) = \epsilon'_{\mathrm{III}}(L^s\mathsf{d})$ and $e_\sharp(\mathsf{d}) = \bar\varepsilon_{g',\vartheta}(L^s\mathsf{d})$. For $0 < \mathsf{d} \le \mathsf{d}_\sharp$ and $f \in \mathfrak{Q}^{(3)}_{T_\ell}(\mathsf{d}; \vartheta)$, Lemma 18.52 gives $f^{(s)} \in \mathfrak{Q}^{(3)}_{T_{\ell'}}(L^s\mathsf{d}; \vartheta)$ with the same sup norms, the same values of $\mathcal{K}^{(3)}_{\mathrm{free}}$, $\mathcal{K}^{(2)}_{\mathrm{free}}$ and $\mathcal{R}$, and $S^{(\ell,\mathsf{b})}_n = L^{sn}S^{(\ell',\mathsf{b}')}_n$ at the triple and its pairs. The first display of Proposition 18.39(b) on $T_{\ell'}$ at g', b' and $L^s\mathsf{d}$, multiplied by L^{3s}, with $L^{3s}(L^s\mathsf{b})^{-3} = \mathsf{b}^{-3}$, gives the first bound; Proposition 18.38(a) on $T_{\ell'}$, multiplied by L^{2s}, gives the second, with $e_\sharp \le \frac{1}{3}$ by YM3_I:(2349); and Lemma 18.50 with $\mathsf{n} = 3$, $t = \frac{1}{3}\epsilon_\sharp(\mathsf{d})$, $e = e_\sharp(\mathsf{d})$ gives the third. In the proofs of Propositions 18.40 and 18.41, Proposition 18.39 enters only through the two inequalities of its part (b), used at all small scales and uniformly on the classes, and through the convergence of the ratios at the cutoffs; all three are now available on T_ℓ at the label b. For the pairs (K_j, β_j) with $K_j \to \infty$ and $a_{K_j}\beta_j \to \mathsf{b}$, YM3_I:(2216) at $n = 3$ and $n = 2$, which needs no further property of the pairs, gives the convergence of the connected functions at the triple and its pairs; by the second bound the limits at the pairs are at least $2\mathsf{b}^{-2}\mathcal{K}^{(2)}_{\mathrm{free}}(f_i, f_j) > 0$; so $Q_{K_j,\ell,\beta_j}(f)$ is formed for all large j and tends to $Q^{(\ell,\mathsf{b})}(f)$. □

Proof of Proposition 18.43. (a) $S^{\mathrm{fr}}_2[\mathsf{n}, \mathsf{b}] = \mathsf{n}\mathsf{b}^{-2}\mathcal{K}^{(2)}_{\mathrm{free}}$, and $\mathcal{K}^{(2)}_{\mathrm{free}} > 0$ at a test pair; so agreement at one test pair forces $\mathsf{n}'\mathsf{b}'^{-2} = \mathsf{n}\mathsf{b}^{-2}$, which in turn gives agreement wherever both are defined. $\mathsf{n}'\mathsf{b}'^{-2} = 3\mathsf{b}^{-2}$ holds exactly for $\mathsf{b}' = \mathsf{b}(\mathsf{n}'/3)^{1/2}$: at $\mathsf{n}' = 1$, $(\mathsf{b}/\sqrt{3})^{-2} = 3\mathsf{b}^{-2}$, and at $\mathsf{n}' = 12$, $12(2\mathsf{b})^{-2} = 3\mathsf{b}^{-2}$.

(b) $A_2/A_3 = \mathsf{b}$ and $A_2^3/A_3^2 = \mathsf{n}^3\mathsf{b}^{-6}/(\mathsf{n}^2\mathsf{b}^{-6}) = \mathsf{n}$.

(c) For $36r < \ell$, $(\psi^{\langle\ell\rangle}_{i,r}, \psi^{\langle\ell\rangle}_{j,r}) \in \mathfrak{T}_{T_\ell}(r; \vartheta)$, with the lift $((\psi_i)_r, (\psi_j)_r)$ and the value $\mathcal{K}^{(2)}_{\mathrm{free}}(\psi_i, \psi_j)$ of $\mathcal{K}^{(2)}_{\mathrm{free}}$; so by Proposition 18.38(b) the first quotient is $3\mathsf{b}^{-2}(1+\eta)$ with $|\eta| \le \epsilon_{\mathrm{II}}(r) \to 0$, for $r \le \mathsf{d}_{\mathrm{II}}(\ell, \mathsf{b}, \vartheta)$. The first bound of Proposition 18.42 at $\mathsf{d} = r$, at the triple $\Psi^{\langle\ell\rangle}_r$, whose lift is Ψ_r and whose value of $\mathcal{K}^{(3)}_{\mathrm{free}}$ and of N are those of Ψ, divided by $\mathcal{K}^{(3)}_{\mathrm{free}}(\Psi) \ge c_3^{\mathrm{col}}\vartheta^3\mathsf{N}(\Psi) > 0$, gives $|S^{(\ell,\mathsf{b})}_3(\Psi^{\langle\ell\rangle}_r)/\mathcal{K}^{(3)}_{\mathrm{free}}(\Psi) - 3\mathsf{b}^{-3}| \le \epsilon_\sharp(r)\mathsf{b}^{-3}/(c_3^{\mathrm{col}}\vartheta^3) \to 0$. So, for small r, $Q^{(\ell,\mathsf{b})}(\Psi^{\langle\ell\rangle}_r)$ is formed, and

$$\frac{\mathcal{R}(\Psi)}{Q^{(\ell,\mathsf{b})}(\Psi^{\langle\ell\rangle}_r)} = \frac{\prod_{i<j} S^{(\ell,\mathsf{b})}_2(\psi^{\langle\ell\rangle}_{i,r}, \psi^{\langle\ell\rangle}_{j,r})/\mathcal{K}^{(2)}_{\mathrm{free}}(\psi_i, \psi_j)}{\big(S^{(\ell,\mathsf{b})}_3(\Psi^{\langle\ell\rangle}_r)/\mathcal{K}^{(3)}_{\mathrm{free}}(\Psi)\big)^2} \to \frac{A_2^3}{A_3^2} = \frac{27\,\mathsf{b}^{-6}}{9\,\mathsf{b}^{-6}} = 3.$$

For a template $S^{\mathrm{fr}}[\mathsf{n}', \mathsf{b}']$ the two quotients, formed at the periodised dilates of Ψ and evaluated at the lift Ψ_r, are $\mathsf{n}'\mathsf{b}'^{-2}$ and $\mathsf{n}'\mathsf{b}'^{-3}$ at every r; by (b) they are $(3\mathsf{b}^{-2}, 3\mathsf{b}^{-3})$ exactly when $(\mathsf{n}', \mathsf{b}') = (3, \mathsf{b})$. □

Proof of Proposition 18.44. (a) Inserting the y_{ij}, $3y_3^2/(y_{12}y_{13}y_{23}) = 3y_3^2/(27\mathsf{b}^{-6}\prod_{i<j}\mathcal{K}^{(2)}_{\mathrm{free}}(f_i, f_j)\prod_{i<j}(1+e_{ij}))$. Dividing by $\mathcal{R}(f)$, which is positive as $\mathcal{K}^{(3)}_{\mathrm{free}}(f) \neq 0$, puts $\mathcal{K}^{(3)}_{\mathrm{free}}(f)^2$ in the place of $\prod_{i<j}\mathcal{K}^{(2)}_{\mathrm{free}}(f_i, f_j)$, and multiplying by $\prod_{i<j}(1+e_{ij})$ leaves $y_3^2/(9\mathsf{b}^{-6}\mathcal{K}^{(3)}_{\mathrm{free}}(f)^2)$.

(b) For $\mathsf{d}_k \le \mathsf{d}_{\mathrm{II}}(\ell,\mathsf{b},\vartheta)$, Proposition 18.38(b), (c) shows that the numbers $y_{ij} = S_2^{(\ell,\mathsf{b})}(f_i^{(k)}, f_j^{(k)})$ are positive and of the form of (a) with $|e_{ij}| \le \epsilon_{\mathrm{II}}(\mathsf{d}_k) \to 0$; so $Q^{(\ell,\mathsf{b})}(f^{(k)})$ is formed, and (a) at $y_3 = S_3^{(\ell,\mathsf{b})}(f^{(k)})$ says $\mathsf{v}_k^2 = (3Q^{(\ell,\mathsf{b})}(f^{(k)})/\mathcal{R}(f^{(k)}))\,\Pi_k$, with $\mathsf{v}_k = |S_3^{(\ell,\mathsf{b})}(f^{(k)})|/(3\mathsf{b}^{-3}|\mathcal{K}^{(3)}_{\mathrm{free}}(f^{(k)})|) \ge 0$ and $\Pi_k = \prod_{i<j}(1+e_{ij}) \to 1$. So $3Q/\mathcal{R} \to 1$ if and only if $\mathsf{v}_k^2 \to 1$, that is $\mathsf{v}_k \to 1$.

(c) By the bounds recalled in the setting, $\mathcal{K}^{(3)}_{\mathrm{free}}(f) \ge c_3^{\mathrm{col}}\vartheta^3\mathsf{N}(f)$ and $\prod_{i<j}\mathcal{K}^{(2)}_{\mathrm{free}}(f_i,f_j) \le \frac{1}{27}\mathsf{N}(f)^2$; so $\mathcal{R}(f) \ge 27(c_3^{\mathrm{col}})^2\vartheta^6 =: \mathcal{R}_\vartheta > 0$, and $\mathcal{R}(f) < \frac{4}{3}$ by Proposition 18.33(c). Let $\mathsf{d} \le \mathsf{d}_{\mathrm{II}}(\ell,\mathsf{b},\vartheta)$ and f in the collinear class; with v the quotient of the right display, $\Pi = \prod_{i<j}(1+e_{ij})$ formed as in (b), $\pi(\mathsf{d}) = (1+\epsilon_{\mathrm{II}}(\mathsf{d}))^3 - 1$, which bounds $|\Pi - 1|$ as in the proof of Lemma 18.50 and tends to 0, and with $\mathsf{X}(\mathsf{d})$, $\mathsf{Y}(\mathsf{d})$ the two suprema of the display at d, (a) says $\mathsf{v}^2 = (3Q/\mathcal{R})\Pi$. ($\Rightarrow$) $\mathsf{v}^2 - 1 = \frac{3Q-\mathcal{R}}{\mathcal{R}}\Pi + (\Pi - 1)$, so $|\mathsf{v}^2-1| \le \frac{\mathsf{X}}{\mathcal{R}_\vartheta}(1+\pi) + \pi$, and $|\mathsf{v}-1| = |\mathsf{v}^2-1|/(\mathsf{v}+1) \le |\mathsf{v}^2-1|$. ($\Leftarrow$) $3Q - \mathcal{R} = \mathcal{R}\,\frac{(\mathsf{v}^2-1)+(1-\Pi)}{\Pi}$, $|\mathsf{v}^2-1| \le \mathsf{Y}(2+\mathsf{Y})$ and $\Pi \ge \frac{8}{27}$, so $\mathsf{X} \le \frac{4}{3}\cdot\frac{27}{8}\,[\mathsf{Y}(2+\mathsf{Y}) + \pi]$. Where the class is empty both suprema are 0. □

Proof of Proposition 18.45. (a) f lies in $\mathfrak{T}^{(3),\mathrm{col}}_{T_\ell}(\mathsf{d};\vartheta)$, as $w \le \frac{1}{16}$; so $\mathcal{K}^{(3)}_{\mathrm{free}}(f) \ge c_3^{\mathrm{col}}\vartheta^3\mathsf{N}(f) > 0$. By Proposition 18.38(b) the numbers $y_{ij} = S_2^{(\ell,\mathsf{b})}(f_i,f_j)$ are as in Proposition 18.44(a), with $|e_{ij}| \le \epsilon_{\mathrm{II}}(\mathsf{d}) \le \frac{1}{3}$, and Proposition 18.44(a) at $y_3 = S_3^{(\ell,\mathsf{b})}(f)$ gives $\mathsf{v}^2 = (3Q/\mathcal{R})\,\Pi$, $\Pi = \prod_{i<j}(1+e_{ij})$. So $\frac{9}{4}Q = \mathsf{A}\mathsf{v}^2$ with $\mathsf{A} = \frac{3}{4}\mathcal{R}(f)/\Pi$, and $1 - 412w^2 \le \frac{3}{4}\mathcal{R}(f) < 1$ (Proposition 18.34), $(1-\epsilon_{\mathrm{II}}(\mathsf{d}))^3 \le \Pi \le (1+\epsilon_{\mathrm{II}}(\mathsf{d}))^3$. *The equivalence.* Fix the function $\vartheta(\cdot)$; for each w, d_{II} and ϵ_{II} are taken at $(\ell,\mathsf{b},\vartheta(w))$. For $\mathsf{d} \le \mathsf{d}_{\mathrm{II}}$ let $\mathsf{X}(w,\mathsf{d}) = \sup_f |\frac{9}{4}Q^{(\ell,\mathsf{b})}(f) - 1|$ and $\mathsf{Y}(w,\mathsf{d}) = \sup_f |\mathsf{v}(f) - 1|$, the suprema over $\mathfrak{T}^{\mathrm{col}}_{T_\ell}(\mathsf{d};\vartheta(w);w)$, which is not empty (Proposition 18.35(b)), and $\alpha(w,\mathsf{d}) = \max\{1 - (1-412w^2)(1+\epsilon_{\mathrm{II}}(\mathsf{d}))^{-3},\ (1-\epsilon_{\mathrm{II}}(\mathsf{d}))^{-3} - 1\}$, which bounds $|\mathsf{A}(f) - 1|$ on the class; as $\epsilon_{\mathrm{II}}(\mathsf{d}) \to 0$, $\alpha(w,\mathsf{d}) \to 412w^2$. By YM3_I:(2216) at $n=3$, $n=2$ and Proposition 18.38(c), $Q_{K_j,\ell,\beta_j}(f) \to Q^{(\ell,\mathsf{b})}(f)$; so the display of Proposition 18.40(b) says $\lim_{w\to 0}\limsup_{\mathsf{d}\to 0}\frac{4}{9}\mathsf{X}(w,\mathsf{d}) = 0$. From $\frac{9}{4}Q - 1 = \mathsf{A}(\mathsf{v}^2-1) + (\mathsf{A}-1)$ and $|\mathsf{v}^2-1| \le |\mathsf{v}-1|(2+|\mathsf{v}-1|)$, $\mathsf{X} \le (1+\alpha)\mathsf{Y}(2+\mathsf{Y}) + \alpha$. From $\mathsf{v}^2 - 1 = [(\frac{9}{4}Q - 1) - (\mathsf{A}-1)]/\mathsf{A}$, $\mathsf{A} \ge 1-\alpha$ and $|\mathsf{v}-1| \le |\mathsf{v}^2-1|$, $\mathsf{Y} \le (\mathsf{X}+\alpha)/(1-\alpha)$ wherever $\alpha < 1$, that is for $412w^2 < 1$ and small d. Taking the upper limit as $\mathsf{d} \to 0$ in each inequality, the right members being increasing continuous functions of Y, α, respectively X, α, and then letting $w \to 0$, each of the two iterated limits is 0 if the other one is.

(b) Let $\mu_i = f_i\,dx$, F_{ij} and I be as in the proof of Proposition 18.33. As $\prod_{i<j}F_{ij} = 3\sqrt{3}\,(4\pi)^{-3}(r_{12}r_{13}r_{23})^{-3}$ on $\prod_i \operatorname{supp} f_i$, $\hat{S}_3(f) = \frac{18}{3\sqrt{3}}\mathsf{b}^{-3}\mathsf{I} = 2\sqrt{3}\,\mathsf{b}^{-3}\mathsf{I}$. (i) $\hat{S}_2 = 3\mathsf{b}^{-2}\mathcal{K}^{(2)}_{\mathrm{free}}$. (ii) $\hat{Q}(f) = \frac{12\,\mathsf{b}^{-6}\mathsf{I}^2}{27\,\mathsf{b}^{-6}\prod_{i<j}\mathcal{K}^{(2)}_{\mathrm{free}}(f_i,f_j)} = \frac{4}{9}\,\mathsf{I}^2/\prod_{i<j}\int F_{ij}^2\,d\mu_i d\mu_j \le \frac{4}{9}$ by Lemma 18.48. On a width class, at a lift on $\mathbb{R}^3$ and in the signs of the proof of Proposition 18.34: by Lemma 18.49(J), $\mathsf{I} = m_1m_2m_3\langle e^{\lambda_{12}+\lambda_{13}+\lambda_{23}}\rangle \ge m_1m_2m_3\,e^{\sum_{i<j}\langle\lambda_{ij}\rangle}$, and by Lemma 18.49(V), $\int F_{ij}^2\,d\mu_i d\mu_j = m_im_j\langle e^{2\lambda_{ij}}\rangle \le m_im_j\,e^{2\langle\lambda_{ij}\rangle + \frac{9}{2}\varkappa_w^2}$; so $\mathsf{I}^2/\prod_{i<j}\int F_{ij}^2 \ge e^{-\frac{27}{2}\varkappa_w^2} \ge 1 - 223w^2$ (Step 4 of that proof), and $\frac{4}{9}\cdot 223 < 100$. For $0 < \vartheta \le \frac{\pi}{13440}$ the width class at w_ϑ is not empty and lies in $\mathfrak{T}^{(3),\mathrm{col}}_{\mathbb{E}}(\mathsf{d};\vartheta)$ (Proposition 18.35(b)), which gives the bounds of the supremum. (iii) By Proposition 18.32(b), $3\mathsf{b}^{-3}\mathcal{K}^{(3)}_{\mathrm{free}}(f) = \frac{\mathsf{b}^{-3}}{\sqrt{3}}\int \tau_\triangle \prod_{i<j}F_{ij}\,d\mu_1d\mu_2d\mu_3$, while $\hat{S}_3(f) = \frac{\mathsf{b}^{-3}}{\sqrt{3}}\,6\,\mathsf{I}$; so

$$\frac{\hat{S}_3(f)}{3\,\mathsf{b}^{-3}\mathcal{K}^{(3)}_{\mathrm{free}}(f)} = \frac{\int 6\prod_{i<j}F_{ij}\,d\mu_1d\mu_2d\mu_3}{\int \tau_\triangle\prod_{i<j}F_{ij}\,d\mu_1d\mu_2d\mu_3}.$$

On a width class $6\mathsf{t}_w \le \tau_\triangle \le 6$ on $\prod_i \operatorname{supp} f_i$ (Step 2 of the proof of Proposition 18.34, and Proposition 18.32(a)), which gives the upper bound t_w^{-1}; and $\tau_\triangle < 6$ on an open set on which $\prod_i f_i \prod_{i<j}F_{ij} > 0$ (proof of Proposition 18.33(b)), which makes the quotient larger than 1. For the dilates Ψ_r the substitution $x_i = rx_i'$ multiplies dx by r^9 and $\prod_{i<j}F_{ij}$ by r^{-9}, and does not change

$\tau_\triangle$, which depends only on the directions of x_1-x_2, x_2-x_3, x_3-x_1; so neither integral depends on r, and the quotient takes one value, $1+\hat\delta(\Psi)$, for all r; $\hat\delta(\Psi)>0$ by what was just shown, as Ψ_r lies in $\mathfrak{T}^{(3),\mathrm{col}}_{\mathbb{R}^3}(r;\vartheta)$, the width class at $w=\frac{1}{16}$. Proposition 18.44(a), with $y_3=\hat S_3(\Psi_r)$ and $e_{ij}=0$, gives $(1+\hat\delta(\Psi))^2=3\hat Q(\Psi_r)/\mathcal{R}(\Psi_r)$, and $\mathcal{R}(\Psi_r)=\mathcal{R}(\Psi)>0$. For $0<\vartheta\le\frac{\pi}{13440}$, $\Psi^\circ\in\mathfrak{T}^{(3),\mathrm{col}}_{\mathbb{R}^3}(1;\vartheta)$ by Proposition 18.35(b); so Ψ°_{d} lies in $\mathfrak{T}^{(3),\mathrm{col}}_{\mathbb{R}^3}(\mathsf{d};\vartheta)$, and its periodisation in $\mathfrak{T}^{(3),\mathrm{col}}_{T_\ell}(\mathsf{d};\vartheta)$ with the lift Ψ°_{d}; the quotient there is $1+\hat\delta(\Psi^\circ)$, which gives the display. As $\mathcal{K}^{(3)}_{\mathrm{free}}(\Psi^\circ_r)=\mathcal{K}^{(3)}_{\mathrm{free}}(\Psi^\circ)$, the last identity is the same fact, and $\mathsf{N}(\Psi^\circ_r)=1$ as the entries of Ψ° have the sup norm 1.

(c) This collects (i)–(iii). □

18.5. **The dilation bulk term with one constant in every periodic Gibbs state and on all tori above one side; the flux alone in the limits.** This subsection bounds the bulk term of the dilation Ward identity, whose cubic member is bounded on one torus in Theorem O(d), by CK^pL^{-cK} for every $c<\frac{1}{14}$, with one constant C in every periodic Gibbs state and on all the tori whose sides are at least one fixed side, and it takes the identity, with the flux alone beside the term of the label, to every volume-first limit at g and, for $0<g\le\gamma^{\mathrm{ref}}_W$, to the components of the limits of Gibbs sequences at g; for $\gamma^{\mathrm{ref}}_W<g\le\gamma$ it reaches these components for the flux and the term of the label taken together, and for the flux alone along the subsequences on which the term of the label converges. The bulk term is a sum of hard combinations of local insertions with the sources (Proposition 18.53); a hard combination is an algebraic function of means of functions of one cell (Lemma 18.54); the chessboard bound of Lemma 5.8 carries such a mean, in every law at hand, to the eighth root of the mean on one torus of the product of its eight mirror copies (Lemma 18.55); on that torus the eight copies enter the exponent of the small-field chain, where copies in different cells and sources far apart decouple, and each copy is read off at one source at a time (Lemmas 18.56, 18.57 and Proposition 18.58). Theorem BD is the resulting transfer, Theorem BE the bound of the bulk term, and Corollary 18.60 the identity in the limits.

Setting. We keep the setting and the signs of Section 5.1, with $0<g\le\gamma$ throughout: the lattice $\mathbb{L}_K=a\mathbb{Z}^3$, $a=a_K=L^{-K}$, the tori T_ℓ, $\ell=2L^m$, the label $\mathsf{b}=a\beta$, the set $\mathfrak{d}(g)$ of the first-passage data at g (the set $\mathfrak{D}(g)$ of Section 5.1), the set $\mathfrak{L}(g)$ of the sides at g, the number G_g, $\mathsf{b}_+=X_g+\bar B$, and the lattice calculus with $\mathfrak{d}_{b,p,\rho}$, $\mathfrak{s}_{l\rho}$, Ψ_ρ, $\mathfrak{r}^\sharp_\rho$, $G_\rho(b)$, Θ_v, $\mathfrak{K}^{\mathrm{v}}$, $\mathfrak{K}^\sharp$, div^s, e^η, ∇^+_ρ, ∇^c_ρ, the derivations v_η and the null terms $N_\beta[\eta]$. Every label of $\mathfrak{d}(g)$ lies in $[\gamma^{-2},\mathsf{b}_+]$, and $MR(\mathsf{b}_+)\le L^m$ on every side $2L^m\in\mathfrak{L}(g)$ ([7, Lemma 3.12](a)). A *law* at (K,β) is a Borel probability measure on the configurations of $\mathbb{L}_K$; $\mathcal{G}^{\mathrm{per}}_{K,\beta}$ is the set of the periodic Gibbs states at (a_K,β), and $\mathcal{G}^{\mathrm{tor}}_{K,\beta}$ the set of the weak limit points, as $m\to\infty$, of the laws $\tilde\mu_{K,2L^m,\beta}$ of the tori, a subset of $\mathcal{G}^{\mathrm{per}}_{K,\beta}$ by Proposition YM3_I:91.2; the law $\tilde\mu_{K,\ell,\beta}$ is the measure of T_ℓ at (K,β) carried to $\mathbb{L}_K$ by the periodic extension of the configurations, so that the mean under it of a bounded Borel function of the bonds inside the chart $(-\frac{\ell}{2},\frac{\ell}{2})^3$ equals the mean on T_ℓ of that function read in the chart, and joint cumulants of such functions agree in the same way; for action densities $\mathcal{O}(f)$ with supports anywhere, the periodised weights give the same observables by grouping bare plaquettes in each period class, and hence the same cumulants. For a law ν, $\langle Y_1;\dots;Y_q\rangle_\nu$ is the joint cumulant, $\mathcal{O}(h)=a^{-1}\sum_p h(p)a_p$ for a weight h of the plaquettes or a function h read at the barycentres, and $S^T_q[\nu](h_1,\dots,h_q)=\langle\mathcal{O}(h_1);\dots;\mathcal{O}(h_q)\rangle_\nu$. As in Section 5.1, a distance is Euclidean unless it is called a sup distance, Lip_∞ is the Lipschitz constant for the sup distance and Lip the Euclidean one, and $\|h\|_{C^{1,1}}=\max\{\sup|h|,\sup|\nabla h|,\mathrm{Lip}\,\nabla h\}$; $\mathsf{A}_r=\{y\in\mathbb{R}^3:|y|_\infty\le r\}$, and $\|h\|_\sharp=\max\{\sup|h|,40M\,\mathrm{Lip}\,h\}$. Since $\mathrm{Lip}\,h\le\sup|\nabla h|$ and $|y|\le\sqrt{3}|y|_\infty$, one has $\max_\rho\sup|\partial_\rho h|\le\|h\|_{C^{1,1}}$, $\max_\rho\mathrm{Lip}_\infty\partial_\rho h\le\sqrt{3}\|h\|_{C^{1,1}}$, $\mathrm{Lip}_\infty h\le\sqrt{3}\|h\|_{C^{1,1}}$ and $\|h\|_\sharp\le40M\|h\|_{C^{1,1}}$. For a function h of the class C^3, $\|h\|_{C^3}$ is the sum of the suprema of $|h|$, $|\nabla h|$, $|\nabla^2h|$ and $|\nabla^3h|$. The numbers $\mathsf{op}_q=\sum_\pi(\#\pi-1)!$ and Bl_q, the number of the partitions π of q points, are those of [7, § 3]; by the moment–cumulant formula a joint cumulant of q bounded functions is at most op_q times the product of their suprema.

The dilation objects. For a real function φ on $\mathbb{R}^3$, and on a torus for a function in the chart $(-\frac{\ell}{2},\frac{\ell}{2})^3$ that vanishes within the sup distance $2a$ of its boundary, put $\nabla^s_{\mathrm{dil}}\varphi(y)=\sum_\rho y_\rho(2a)^{-1}[\varphi(y+ae_\rho)-\varphi(y-ae_\rho)]$ and $\delta^2_\rho\varphi(y)=\varphi(y+ae_\rho)+\varphi(y-ae_\rho)-2\varphi(y)$, and for a plaquette p

$$(\mathsf{L}\varphi)(p)=-\tfrac14\sum_{\nu\in P(p)}\delta^2_\nu\varphi(x(p))-\tfrac12\delta^2_{\lambda(p)}\varphi(x(p)),\qquad \varphi^{\square}(p)=\tfrac14\sum_{b\subset\partial p}\varphi(x_b),\qquad v^{\mathrm{dil}}_\varphi(p)=\sum_{\rho\in P(p)}x(p)_\rho\,\nabla^c_\rho\varphi(x(p)),$$

$$h^{\mathrm{dil}}_\varphi(p)=a^{-1}\sum_{b\subset\partial p}\sum_\rho[\varphi(x_b)-\varphi(x(p))](x_b)_\rho\,\mathfrak{d}_{b,p,\rho},\qquad \psi(p)=\tfrac12\sum_{\nu\in P(p)}\big[\Psi_\nu(p)+\Psi_\nu(p-ae_\nu)\big],$$

the last one the in-plane orbital potential of p. With the identity field $\mathsf{y}(y)=y$ and real functions χ and f, the *dilation flux insertion* and the two *bulk insertions* are

$$\mathfrak{B}^{\mathrm{dil}}_\chi=\mathsf{b}\,\mathcal{O}(\nabla^s_{\mathrm{dil}}\chi)-\beta\sum_p h^{\mathrm{dil}}_\chi(p)-\tfrac32 a\,\mathcal{O}(v^{\mathrm{dil}}_\chi),\qquad \mathfrak{X}^{(0)}_\chi=\mathfrak{K}^{\mathrm{v}}[\beta\chi\mathsf{y}]+\beta\sum_p\chi(x(p))(\psi(p)-a_p^2)+\mathsf{b}\,\mathcal{O}(\mathsf{L}\chi)+3a\,\mathcal{O}(\chi^{\square}),$$

$$\mathfrak{X}^{(f)}=-a^{-1}\mathfrak{K}^{\mathrm{v}}[f\mathsf{y}]-a^{-1}\sum_p f(x(p))(\psi(p)-a_p^2)-\mathcal{O}(\mathsf{L}f),$$

and for a law ν and $\underline{f}=(f_1,\dots,f_n)$ the *flux* and the *bulk term* are

$$\mathsf{Fl}_K[\nu](\underline{f})[\chi]=-\langle\mathfrak{B}^{\mathrm{dil}}_\chi;\mathcal{O}(f_1);\dots;\mathcal{O}(f_n)\rangle_\nu,\qquad \mathfrak{U}^{\mathrm{dil}}_K[\nu](\underline{f})[\chi]=\langle\mathfrak{X}^{(0)}_\chi;\mathcal{O}(f_1);\dots;\mathcal{O}(f_n)\rangle_\nu+\sum_{i=1}^n\langle\mathfrak{X}^{(f_i)};(\mathcal{O}(f_l))_{l\neq i}\rangle_\nu.$$

The flux insertion depends on the configuration only near the closed set $\mathrm{Var}(\chi)$ of the points near which χ is not constant, and $\nabla^s_{\mathrm{dil}}\varphi$ tends to $y\cdot\nabla\varphi=\sum_\rho y_\rho\partial_\rho\varphi$ as $a\to0$.

Insertion tuples and hard combinations. An *insertion tuple* over real functions $g_1,\dots,g_n$, the *sources*, is a tuple $\mathbb{J}=(\mathfrak{J}_0,\mathfrak{J}_1,\dots,\mathfrak{J}_n)$ of bounded functions of finitely many bonds; for a law ν and $B\subset\{1,\dots,n\}$ its *hard combination* is

$$\mathsf{T}_\nu[\mathbb{J}](B)=\big\langle\mathfrak{J}_0;(\mathcal{O}(g_i))_{i\in B}\big\rangle_\nu+\sum_{i\in B}\big\langle\mathfrak{J}_i;(\mathcal{O}(g_l))_{l\in B\setminus\{i\}}\big\rangle_\nu,\qquad \mathsf{T}_\nu[\mathbb{J}](\emptyset)=\nu(\mathfrak{J}_0).$$

Thus $\mathfrak{U}^{\mathrm{dil}}_K[\nu](\underline{f})[\chi]=\mathsf{T}_\nu[\mathbb{X}](\{1,\dots,n\})$ for $\mathbb{X}=(\mathfrak{X}^{(0)}_\chi,\mathfrak{X}^{(f_1)},\dots,\mathfrak{X}^{(f_n)})$ over the sources f_i. A *local observable* is one of [7, Definition 5.19], with its order $m_{\mathfrak{J}}$, its type $(m_{\mathfrak{J}},c,\hat c,N)$, its weights $\varkappa$, the number $|\varkappa|_\infty$, its boxes and its carrier; an insertion in the sense of [7, Definition 3.2](1), and an entry in the sense of Lemma 5.10, of a given type is a local observable of that type, as the three definitions ask the same of the boxes, of the densities with their heights and orders, and of the number of the boxes that have one lowest site; an insertion tuple is *local* if its members are local observables of orders ≥2 with real weights and with densities that are real on the configurations with values in SU(2).

Data of the transfer. Let $n\geq1$, $R\geq1$ and $0<\mathsf{d}\leq1$. Let m_1 be the least integer with $2L^{m_1}\in\mathfrak{L}(g)$ and $L^{m_1}\geq32n(R+1)$, $\mathsf{l}=L^{m_1}$, and $\ell'=\ell'(g,n,R)=2\mathsf{l}$. The board of cubic cells of side l whose cell $\square_0=[-\frac{\mathsf{l}}{2},\frac{\mathsf{l}}{2}]^3$ is centred at 0 lies on $\mathbb{L}_K$, on $T_{\ell'}$ and on every T_ℓ with $\ell\geq\ell'$ in $\mathfrak{L}(g)$, as l divides $\frac{\ell'}{2}$; $\square_0$ holds A_{R+2} at sup distance $\geq\frac{\mathsf{l}}{2}-R-2\geq16$ from its walls. $\mathsf{Z}=(\mathbb{Z}/2)^3$ numbers the eight cells of $T_{\ell'}$, and r_κ is the diagonal isometry of the lattice of $T_{\ell'}$ that carries $\square_0$ onto the cell $\kappa\in\mathsf{Z}$ by reflections in walls; for a real bounded Borel function F of the bonds inside $\square_0$ its mirror copy $F^{[\kappa]}=F\circ\mathsf{r}_\kappa^{-1}$ is a function of the cell κ, and $F\mapsto F^{[\kappa]}$ is real linear, multiplicative and the identity on constants by its definition as composition. The measure of $T_{\ell'}$ at (K,β), whose means and cumulants we write $\langle\cdot\rangle'$, is invariant under every r_κ by relabelling plaquettes and Haar variables. We put $\mathsf{Q}=e^{G_g\mathsf{l}^3}$, and the *laws of the transfer* at $(K,\beta)\in\mathfrak{d}(g)$ are those of $\mathfrak{S}_{K,\beta}=\mathcal{G}^{\mathrm{per}}_{K,\beta}\cup\{\tilde\mu_{K,\ell,\beta}:\ell\in\mathfrak{L}(g),\ \ell\geq\ell'\}$. The *sources of the transfer* are real Lipschitz functions $g_1,\dots,g_n\neq0$ on $\mathbb{R}^3$ that vanish outside A_R, with supports at pairwise sup distances $\geq\mathsf{d}$; $O_i=\mathcal{O}(g_i)$, and on $T_{\ell'}$, $g_i^\kappa=g_i\circ\mathsf{r}_\kappa^{-1}$, so that $O_i^{[\kappa]}=\mathcal{O}(g_i^\kappa)$ and $\|g_i^\kappa\|_\sharp=\|g_i\|_\sharp$ by relabelling the plaquettes. *The chain on $T_{\ell'}$.* For $0<\phi\leq\frac{1}{84}$, $K\geq\max\{K_\phi,5\}$ with the number K_ϕ of [7, Lemma 3.12](b), $u=\lceil\phi K\rceil$ and $t=K-u$, the setting (Th) of §2.3.1 holds on $T_{\ell'}$ with (K,b,m_1,t)

([7, Lemma 3.12](b)); we use its cuts Cut_k, its inner ball $\mathsf{In}_t \subset \mathsf{Cut}_t$, the backgrounds U over $V \in \mathsf{Cut}_k$, the weights p_k and $\mathfrak{w}_t^{\mathrm{sf}}$ of [7, Definition 3.2](3), the discarded-mass bound R_t of Proposition YM3_I:115.2, the source class $\mathsf{Src} = \mathsf{Src}_{\mathsf{Bc}}(2\varrho, 2\varrho)$ of the calibrated construction, and $\rho_0^w = \rho_0 e^{\mathcal{O}(w)}$, the numbers $N_k = (\ell' L^{K-k})^3$, $\mathsf{m}_t^S = C_{\mathrm{old}} L E_0^{\mathrm{s}} N_t/(L-1)$ and $\mathsf{M}_t^{\mathrm{rd}} = \max\{\varrho^{-1}, \frac{3}{2}(\mathsf{e}_t^{\mathrm{c}})^2 L^{K-t} N_t, \varrho^{-1}\mathsf{m}_t^S\}$, the source-action bound and read-out radius of Section YM3_I:114 and Lemma YM3_I:114.5, so that $\mathsf{M}_t^{\mathrm{rd}} \geq \varrho^{-1}\mathsf{m}_t^S$, $\mathsf{M} = \max\{1, 8\mathsf{M}_t^{\mathrm{rd}}\}$, the small-field functional $\Lambda'(F) = \int F\, \mathfrak{w}_t^{\mathrm{sf}} \rho_0\, dU$, $Z = Z_{K,\ell',\beta}$, and for the sources the radii $r_i = (8\mathsf{M}_t^{\mathrm{rd}} \|g_i\|_\sharp)^{-1}$; by Lemma 5.10(b), $\mathsf{M} \leq C K^{2\mathfrak{p}_0} \ell'^3 L^{3u}$ with C a function of the constants and b_+. *The numbers of an observable.* [7, Lemma 5.20] applies on $T_{\ell'}$ with $m = m_1$, $\mathsf{b}_- = \gamma^{-2}$ and $\mathsf{b}' = \mathsf{b}$ ([7, Lemma 3.12](a)), as $\phi < \frac{1}{42}$ and as every $K \geq K_\phi$ meets the two conditions with which that Lemma fixes its own threshold, both of them among the conditions on K_ϕ in [7, Lemma 3.12](b). For a local observable $\mathfrak{J}$ of an order ≥ 2 on $T_{\ell'}$, m_j, $\mathsf{n}_{\mathfrak{J}}^{\mathrm{f}}$, d_i, n_j and $r_{\mathfrak{J}}$ are the numbers of [7, Lemma 5.20](a), and $\mathsf{o}_t(\mathfrak{J})$ is the number of [7, Lemma 5.20](b); in the bounds below, C is a function of the constants, m_1, b_+ and the type of $\mathfrak{J}$, and does not read ϕ. [7, Lemma 5.20](a) bounds $\mathsf{o}_t(\mathfrak{J})$ by $CK^{2\mathfrak{p}_0} L^{4u-K} |\varkappa|_\infty$ at the order 2 and by $CK^{3\mathfrak{p}_0} L^{4u-\frac{3}{2}K} |\varkappa|_\infty$ at the orders ≥ 3, and [7, Lemma 5.20](c) gives $1/r_{\mathfrak{J}} \leq C|\varkappa|_\infty$ and $\sup|\mathfrak{J}| \leq N\hat{c}\,\ell'^3 L^{3K} |\varkappa|_\infty$ at real configurations. The numbers m_j, n_j and d_i are given by the formulas of [7, Proposition 3.13], except $\mathsf{d}_1 = K^{\mathrm{d}} \mathsf{n}_{\mathfrak{J}}^{\mathrm{f}}$, with $K^{\mathrm{d}} = (351/2) L^3 K_\Pi$, as in the construction preceding Proposition YM3_I:114.3. As [7, Lemma 5.20](a) gives $\mathsf{n}_{\mathfrak{J}}^{\mathrm{f}} \leq C_{\mathrm{hz}} N \hat{c}\, x_0^{-1/2} (\log x_0)^{\mathfrak{p}_0} |\varkappa|_\infty$, where C_{hz} is formed from the constants alone and $x_0 = \beta = L^K \mathsf{b} \in [\gamma^{-2} L^K, \mathsf{b}_+ L^K]$, one has $\mathsf{d}_1 \leq C K^{\mathfrak{p}_0} L^{-K/2} |\varkappa|_\infty$. The proof of [7, Lemma 3.14](c) uses d_1 only through a bound of this form, with constants that do not read ϕ when $\phi \leq \frac{1}{84}$ and $K \geq \max\{K_\phi, 5\}$; carried out for the present numbers, it gives $\max_i \mathsf{d}_i \leq C|\varkappa|_\infty$ and, at the order 2, $\sum_{i \leq k} L^i \mathsf{d}_i \leq C K^{\mathfrak{p}_0} L^{-(K-k)/2} |\varkappa|_\infty$ for $1 \leq k \leq t$.

Proposition 18.53 (The dilation identity; the bulk term as a sum of hard combinations)**.** Let $K \in \mathbb{N}$, $\beta > 0$, $n \geq 1$, and let $\chi, f_1, \dots, f_n$ be bounded real functions on $\mathbb{R}^3$ of bounded supports such that χ *covers* the f_i: $\chi = 1$ at $x(p)$ and at the four x_b, $b \subset \partial p$, wherever $f_i(x(p)) \neq 0$ for some i. Let ν be a Gibbs state at (a_K, β), or the measure of a torus T_ℓ when all the supports lie in $(-\frac{\ell}{2} + 2a, \frac{\ell}{2} - 2a)^3$.

(a) *(The identity.)* $\sum_{i=1}^n S_n^T[\nu](f_1, \dots, f_i - \nabla_{\mathrm{dil}}^s f_i, \dots, f_n) = \mathsf{b}\, S_{n+1}^T[\nu](\chi, f_1, \dots, f_n) + \mathsf{Fl}_K[\nu](\underline{f})[\chi] + \mathfrak{U}_K^{\mathrm{dil}}[\nu](\underline{f})[\chi]$.

(b) *(The pieces.)* For complex vector fields k and complex functions f, f_l, w on $\mathbb{R}^3$ that vanish outside bounded sets, read on $\mathbb{L}_K$, the relations (N1) and (N4) of Lemma 5.9(f) hold as they are written there, (N2) holds with the mean under ν in the place of the integral against $e^{-\beta A} dU$, and (N3) holds with the joint cumulants under ν in the place of $\langle\cdot\rangle_{K,\ell,\beta}$; for the measure of a torus and functions read in its chart these are the relations of that Lemma. Let besides $n \geq 2$, let the supports of the f_i lie at pairwise sup distances $> 4a$, and let $\chi = 1$ at every point within sup distance $2a$ of $\bigcup_i \operatorname{supp} f_i$. For a real function φ let k^φ be the vector field with $\mathsf{k}^{\varphi,\rho}(z) = \nabla_\rho^+(\varphi\, y_\rho)(z) = \varphi(z + a e_\rho) + z_\rho \nabla_\rho^+ \varphi(z)$; let $\mathfrak{J}_v[\mathsf{k}]$, $1 \leq v \leq 5$, be the five sums of Lemma 5.9(c); and, with the constant field $\mathbf{1} = (1,1,1)$ and the notation of Lemma 5.9(f), let $\mathfrak{R}^\circ[f] = \mathfrak{R}[\mathbf{1}; f] - \mathsf{b}^{-1} c_{f\mathbf{1}} = -3\mathsf{b}^{-1} \sum_p a_p\, \mathsf{e}^{f\mathbf{1}}(p) + \mathfrak{C}'[\mathbf{1}; f]$. Then $\mathfrak{U}_K^{\mathrm{dil}}[\nu](\underline{f})[\chi] = \sum_\pi \mathsf{T}_\nu[\mathbb{J}^\pi](\{1, \dots, n\})$, over the sources $f_1, \dots, f_n$ and the ten insertion tuples

$$\mathbb{J}^\sharp = (\beta \mathfrak{K}^\sharp[\chi \mathsf{y}],\ (-a^{-1} \mathfrak{K}^\sharp[f_i \mathsf{y}])_i),$$

$$\mathbb{J}^\psi = \Big(\beta \sum_p \chi(x(p)) \psi(p),\ (-a^{-1} \sum_p f_i(x(p)) \psi(p))_i\Big),$$

$$\mathbb{J}^{a^2} = \Big(-\beta \sum_p \chi(x(p)) a_p^2,\ (a^{-1} \sum_p f_i(x(p)) a_p^2)_i\Big),$$

$$\mathbb{J}^v = \big(-\mathfrak{J}_v[\mathsf{k}^\chi],\, (\mathsf{b}^{-1}\mathfrak{J}_v[\mathsf{k}^{f_i}])_i\big)\ (1 \le v \le 5),$$
$$\mathbb{J}^{\mathfrak{R}} = \big(0,\, (\tfrac{1}{4}\mathfrak{R}^\circ[f_i])_i\big),$$
$$\mathbb{J}^{\mathrm{s}} = \big(\mathsf{b}\,\mathcal{O}(\mathsf{L}\chi) + 3a\,\mathcal{O}(\chi^\square),\, (-\mathcal{O}(\mathsf{L}f_i))_i\big).$$

(c) *(Carriers, orders, weights.)* In each of these tuples $\mathfrak{J}_0$ is a sum over plaquettes and boxes within sup distance $4a$ of $\operatorname{supp}\chi$, and $\mathfrak{J}_i$ a sum over plaquettes and boxes within the sup distance $4a$ of $\operatorname{supp} f_i$. Let besides $(K,\beta) \in \mathfrak{d}(g)$ and $a \le \frac{1}{4}$, and let χ and the f_i belong to $C^{1,1}(\mathbb{R}^3)$ and vanish outside A_R; write $\|\cdot\| = \|\cdot\|_{C^{1,1}}$, and let C_R be a function of R. All the members are local observables of types formed with pure numbers. In $\mathbb{J}^\psi$ and $\mathbb{J}^{a^2}$ the densities are the in-plane orbital potentials of $\psi(p)$, of the order 2, each on the box of a planar 1×2 rectangle, and a_p^2, of the order 4, with weights at most $\beta \sup|\chi|$ in $\mathfrak{J}_0$ and $a^{-1}\sup|f_i|$ in $\mathfrak{J}_i$. In $\mathbb{J}^v$ the orders are $2,2,2,3,4$ for $v = 1,\dots,5$, and the weights are at most $C_R(1+\mathsf{b})\|\chi\|$ and $C_R\|f_i\|$ for $v \le 3$, and $C_R\beta\|\chi\|$ and $C_R a^{-1}\|f_i\|_\sharp$ for $v = 4,5$. In $\mathbb{J}^{\mathfrak{R}}$ the densities a_p and $\langle G_\rho(b), \mathbf{w}(W_{b,p})\rangle$ have the order 2 and weights at most $\|f_i\|_\sharp$. In $\mathbb{J}^{\mathrm{s}}$ the density a_p has weights at most $3\sup|\chi| + 2\mathsf{b}a\|\chi\|$ in $\mathfrak{J}_0$ and $2a\|f_i\|$ in $\mathfrak{J}_i$. And $\mathbb{J}^\sharp = (\beta\mathfrak{K}^\sharp[\Xi], (-a^{-1}\mathfrak{K}^\sharp[f_i\Xi])_i)$ for the field $\Xi = \chi\mathsf{y}$, as $f_i\chi = f_i$, with $\mathsf{H}(\Xi) \le C_R\|\chi\|$: its members are weighted sums of the cubic density $\mathfrak{r}^\sharp_{\lambda(p)}(p)$, with weights at most $L^{2K}\mathsf{b}_+\mathsf{H}(\Xi)$ and $L^{2K}\sup|f_i|\,\mathsf{H}(\Xi)$, and in the sense of [7, Definition 5.19] this density has the order 3: the paragraph that follows that Definition presents it on a box of $2\times2\times3$ sites with the type $(3, 2^{36}\hat{c}_\sharp, \hat{c}_\sharp, N)$, $\hat{c}_\sharp$ and N pure numbers. Here the order measures a bound by a constant times δ^3 where the plaquette variables lie within δ of $\mathbf{1}$; it is not the least order 2 of the monomials with which Lemma 5.9(b) writes $\mathfrak{r}^\sharp$ as a block form.

■

Sketch. Null terms with sources. Let $\mathbf{X}$ be a polynomial field of directions that vanishes off a finite set Λ of bonds, as in Lemma 5.9(a), and w a complex weight of finite support. In a Gibbs state the conditional law of the bonds of Λ, given the other bonds, is $e^{-\beta\sum_{p\cap\Lambda\neq\emptyset} a_p}$ times the Haar measure of these bonds, normalised by a function of the other bonds; the integration by parts of Lemma 5.9(a) in the bonds of Λ, applied to $e^{\mathcal{O}(w)}$, gives $\nu((\beta v_{\mathbf{X}}A - \operatorname{div} v_{\mathbf{X}})e^{\mathcal{O}(w)}) = \nu((v_{\mathbf{X}}\mathcal{O}(w))e^{\mathcal{O}(w)})$, and on a torus this is the integration by parts against its measure. At $w = \sum_i z_i f_i$ we divide by $\nu(e^{\mathcal{O}(w)})$, which is 1 at $z=0$, and differentiate once in each z_i at $z=0$: the means under the measure with the source become joint cumulants, and $v_{\mathbf{X}}\mathcal{O}(w) = \sum_i z_i v_{\mathbf{X}}\mathcal{O}(f_i)$ is linear in z, so that

$$\langle \beta v_{\mathbf{X}}A - \operatorname{div} v_{\mathbf{X}}; \mathcal{O}(f_1); \dots; \mathcal{O}(f_n)\rangle_\nu = \sum_i \langle v_{\mathbf{X}}\mathcal{O}(f_i); (\mathcal{O}(f_l))_{l\neq i}\rangle_\nu.$$

For the field $\mathbf{X}^{\mathsf{k}}$ of Lemma 5.9(c), k of bounded support, the identity before the derivatives is (N2) in ν; (N1) and (N4) are identities in U whose terms read k, f and the configuration near the supports, so they hold on $\mathbb{L}_K$ as on T_ℓ; and (N1), inserted into the right member of the display, gives (N3) in ν by induction on the number of sources.

The identity (a). The field $\eta = \chi\mathsf{y}$ has bounded support, and for the field of directions of v_η the left member of the display is $\langle N_\beta[\eta]; \mathcal{O}(f_1); \dots; \mathcal{O}(f_n)\rangle_\nu$, while $v_\eta\mathcal{O}(f) = -\mathrm{Ct}[f;\eta]$ (Lemma 5.9(a) and the identities recalled in Section 5.1). Write $\eta^\rho(x_b) - \eta^\rho(x(p)) = [\chi(x_b) - \chi(x(p))](x_b)_\rho + \chi(x(p))(x_b - x(p))_\rho$; then $\Theta_v[\chi\mathsf{y}] = \sum_p v(p)h^{\mathrm{dil}}_\chi(p) + \sum_p v(p)\chi(x(p))\operatorname{tr}\mathfrak{s}(p)$ with $\operatorname{tr}\mathfrak{s} = \sum_\rho \mathfrak{s}_{\rho\rho}$, and the formulas for $\mathfrak{s}$ in Lemma 5.9(e) ($\mathfrak{s}_{\lambda\rho}(p) = 0$ for $\lambda = \lambda(p)$, and the value of $\mathfrak{s}_{\nu\nu}(p)$ for $\nu \in P(p)$) give $\operatorname{tr}\mathfrak{s}(p) = 4a_p - a_p^2 + \frac{1}{4}\sum_{\nu\in P(p)}(a_{p+ae_\nu} + a_{p-ae_\nu} - 2a_p) + \psi(p)$. Moved onto the weight, the shifted plaquette terms give $\sum_p \varphi(x(p))(a_{p+ae_\nu} + a_{p-ae_\nu} - 2a_p) = \sum_p a_p\,\delta^2_\nu\varphi(x(p))$, both sums over the plaquettes with $\nu \in P(p)$. Moreover $\operatorname{div}^s(\chi\mathsf{y}) = \nabla^s_{\mathrm{dil}}\chi + 3\chi + \frac{1}{2}\sum_\rho \delta^2_\rho\chi$, and $e^{\chi\mathsf{y}}(p) = v^{\mathrm{dil}}_\chi(p) + 2\chi^\square(p)$, since the two bonds of p normal to $\rho \in P(p)$ have their midpoints at $x(p) \pm \frac{a}{2}e_\rho$. Inserted at $v \equiv \beta$ into $N_\beta[\eta] = \Theta_\beta[\eta] + \mathfrak{K}^{\mathrm{v}}[\beta\eta] - \mathsf{b}\,\mathcal{O}(\operatorname{div}^s\eta) + \frac{3}{2}a\,\mathcal{O}(e^\eta)$, these formulas give $N_\beta[\chi\mathsf{y}] = -\mathfrak{B}^{\mathrm{dil}}_\chi + \mathsf{b}\,\mathcal{O}(\chi) + \mathfrak{X}^{(0)}_\chi$.

At $v = f$: as χ covers f, the terms of a plaquette with $f(x(p)) \neq 0$ see $\chi = 1$ at all their points, and $f\chi = f$ at every barycentre; so $\Theta_f[\chi\mathsf{y}] = \sum_p f(x(p)) \operatorname{tr} \mathfrak{s}(p)$, $\operatorname{div}^s(f\chi\mathsf{y}) = \operatorname{div}^s(f\mathsf{y})$ at the barycentres, $\mathfrak{K}^{\mathrm{v}}[f\chi\mathsf{y}] = \mathfrak{K}^{\mathrm{v}}[f\mathsf{y}]$, and $\mathrm{Ct}[f;\chi\mathsf{y}] = \mathcal{O}(\operatorname{div}^s(f\mathsf{y})) - a^{-1}\Theta_f[\chi\mathsf{y}] - a^{-1}\mathfrak{K}^{\mathrm{v}}[f\mathsf{y}] = -\mathcal{O}(f - \nabla^s_{\mathrm{dil}} f) + \mathfrak{X}^{(f)}$. The display, with these two expansions and the multilinearity of the cumulants, is (a).

The pieces (b). Lemma 5.9(b) at the vector weights $\beta\chi\mathsf{y}$ and $f\mathsf{y}$, with $\nabla^+_\rho(\varphi y_\rho) = \mathsf{k}^{\varphi,\rho}$, gives

$$\mathfrak{K}^{\mathrm{v}}[\beta\chi\mathsf{y}] = -\mathfrak{J}^{\mathrm{II}}[\mathsf{k}^\chi] + \beta\mathfrak{K}^\sharp[\chi\mathsf{y}], \qquad -a^{-1}\mathfrak{K}^{\mathrm{v}}[f\mathsf{y}] = \mathsf{b}^{-1}\mathfrak{J}^{\mathrm{II}}[\mathsf{k}^f] - a^{-1}\mathfrak{K}^\sharp[f\mathsf{y}], \qquad \mathfrak{J}^{\mathrm{II}}[\mathsf{k}] = \beta\sum_{p,\rho} \mathsf{k}^\rho(x(p))\Psi_\rho(p),$$

and Lemma 5.9(c) gives $\mathfrak{J}^{\mathrm{II}}[\mathsf{k}] = -\frac{1}{4}\mathfrak{N}[\mathsf{k}] + \sum_v \mathfrak{J}_v[\mathsf{k}] - \frac{1}{4}c_{\mathsf{k}}$. Numbers drop from joint cumulants of order ≥ 2, and as $n \geq 2$ every cumulant of the bulk term has at least two entries. The field of directions of $\mathfrak{N}[\mathsf{k}^{f_i}]$ vanishes off the bonds whose midpoints lie within the sup distance a of $\operatorname{supp} f_i$, and no $\mathcal{O}(f_l)$, $l \neq i$, reads these bonds, the supports lying at sup distances $> 4a$; so $\langle \mathfrak{N}[\mathsf{k}^{f_i}]; (\mathcal{O}(f_l))_{l\neq i}\rangle_\nu = 0$ (Lemma 5.9(a)). In (N3) for $\langle \mathfrak{N}[\mathsf{k}^\chi]; \mathcal{O}(f_1); \dots; \mathcal{O}(f_n)\rangle_\nu$ every term with $j \geq 2$ holds $\mathfrak{R}[f_{q_1}\cdots f_{q_{j-1}}\mathsf{k}^\chi; f_{q_j}]$ with $q_1 \neq q_j$, whose weights and whose number vanish by (N4), as the supports of f_{q_1} and f_{q_j} lie at a sup distance $> 2a$; the terms with $j = 1$ are $\langle \mathfrak{R}[\mathsf{k}^\chi; f_q]; (\mathcal{O}(f_l))_{l\neq q}\rangle_\nu$, and $\mathfrak{R}[\mathsf{k}^\chi; f_q]$ reads k^χ at bond midpoints within the sup distance $\frac{a}{2}$ of $\operatorname{supp} f_q$, where $\mathsf{k}^\chi = \mathbf{1}$ because $\chi = 1$ within the sup distance $2a$ of that support; so it is $\mathfrak{R}[\mathbf{1}; f_q]$, whose number drops. Grouping the members of $\mathfrak{X}^{(0)}_\chi$ and of the $\mathfrak{X}^{(f_i)}$ by their kind gives the ten tuples.

Carriers and weights (c). The weights of the $\mathfrak{J}_v[\mathsf{k}]$ and of $\mathfrak{R}[\mathbf{1}; f]$ are those listed in Lemma 5.9(c) and in (N4), the latter multiplied by $\frac{1}{4}$; with $|\mathsf{k}^\varphi|_\infty \leq \sup|\varphi| + (R+1)\max_\rho \sup|\partial_\rho\varphi|$ and $\mathrm{Lip}_\infty \mathsf{k}^\varphi \leq 2\,\mathrm{Lip}_\infty\varphi + (R+1)\max_\rho \mathrm{Lip}_\infty \partial_\rho\varphi$ for φ that vanish outside A_R, with $|\mathsf{L}\varphi| \leq a^2 \max_\rho \mathrm{Lip}_\infty \partial_\rho\varphi$, $|\chi^\square| \leq \sup|\chi|$, with the comparisons of the Setting between these numbers and $\|\varphi\|$, and with $4 \leq \mathsf{b} \leq \mathsf{b}_+$ at first-passage data ([7, Lemma 3.12](a)), they give the bounds. The orders are those of Lemma 5.9(c) for $\Psi_\rho(p)$, a_p, $\langle G_\rho(b), \mathbf{w}(W_{b,p})\rangle$, $\mathfrak{e}_{\lambda(p)}(p)$ and a_p^2, and the order and the type of the cubic density $\mathfrak{r}^\sharp_{\lambda(p)}(p)$ are those of the paragraph that follows [7, Definition 5.19]. Every density sits on a box within the sup distance $2a$ of its plaquette, and k^{f_i} vanishes beyond the sup distance a of $\operatorname{supp} f_i$, which gives the carriers. □

Lemma 18.54 (Moments for hard combinations)**.** Let ν be a probability measure, $\mathfrak{J}_0, \dots, \mathfrak{J}_n, O_1, \dots, O_n$ bounded real measurable functions, and $\bar o_1, \dots, \bar o_n, \alpha_\emptyset, \alpha_1, \dots, \alpha_n$ real numbers; put $\hat O_i = O_i - \bar o_i$, $\alpha_{\{i\}} = \alpha_i$, $\alpha_A = 0$ for $\#A \geq 2$, $\mathsf{T}_\nu(A) = \langle \mathfrak{J}_0; (O_i)_{i\in A}\rangle_\nu + \sum_{i\in A}\langle \mathfrak{J}_i; (O_l)_{l\in A\setminus\{i\}}\rangle_\nu$ with $\mathsf{T}_\nu(\emptyset) = \nu(\mathfrak{J}_0)$, $P_\nu(C) = \nu(\prod_{i\in C}\hat O_i)$ with $P_\nu(\emptyset) = 1$, and

$$H_B = (\mathfrak{J}_0 - \alpha_\emptyset)\prod_{i\in B}\hat O_i + \sum_{i\in B}(\mathfrak{J}_i - \alpha_i)\prod_{l\in B\setminus\{i\}}\hat O_l.$$

Then $\nu(H_B) = \sum_{A\subset B}(\mathsf{T}_\nu(A) - \alpha_A)P_\nu(B\setminus A)$ for every $B \subset \{1,\dots,n\}$. Consequently, if $|\nu(H_A)| \leq \eta\prod_{i\in A}\gamma_i$ for all $A \subset B$ and $|P_\nu(C)| \leq \bar P\prod_{i\in C}\gamma_i$ for all non-empty $C \subset B$, with numbers $\eta, \bar P \geq 0$ and $\gamma_i > 0$, then $|\mathsf{T}_\nu(B) - \alpha_B| \leq \mathsf{c}_{\#B}\,\eta\prod_{i\in B}\gamma_i$, where $\mathsf{c}_0 = 1$ and $\mathsf{c}_k = 1 + \bar P\sum_{j<k}\binom{k}{j}\mathsf{c}_j$. ■

Sketch. For $z \in \mathbb{C}^n$ put $Y_z = \mathfrak{J}_0 + \sum_i z_i\mathfrak{J}_i$, $\mathsf{a}_\circ(z) = \alpha_\emptyset + \sum_i \alpha_i z_i$, $\Gamma(z) = \nu(e^{\sum_i z_i\hat O_i})$ and $\mathcal{N}(z) = \nu(Y_z e^{\sum_i z_i\hat O_i})$, entire functions with $\Gamma(0) = 1$, and near 0 let $\mathcal{X} = \mathcal{N}/\Gamma$, the mean of Y_z under the measure with the source $\sum_i z_i O_i$, from which the numbers $\bar o_i$ drop. Let ∂_A be the derivative once in each z_i, $i \in A$, at $z = 0$. The derivatives of a mean under a source are joint cumulants, and Y_z is affine in z; so $\partial_A\mathcal{X} = \mathsf{T}_\nu(A)$, while $\partial_A\mathsf{a}_\circ = \alpha_A$ and $\partial_C\Gamma = P_\nu(C)$. As a function of the configuration, by the rule of Leibniz for a derivative in which each variable stands once and as $Y_z - \mathsf{a}_\circ(z)$ is affine, $\partial_B[(Y_z - \mathsf{a}_\circ(z))e^{\sum_i z_i\hat O_i}] = H_B$. The functions are bounded, so ν and ∂_B commute, and $\nu(H_B) = \partial_B[(\mathcal{X} - \mathsf{a}_\circ)\Gamma] = \sum_{A\subset B}(\mathsf{T}_\nu(A) - \alpha_A)P_\nu(B\setminus A)$. For the bound let $\delta_A = \mathsf{T}_\nu(A) - \alpha_A$; then $\delta_B = \nu(H_B) - \sum_{A\subsetneq B}\delta_A P_\nu(B\setminus A)$, and induction on $\#B$ gives $|\delta_B| \leq \eta\prod_B\gamma_i + \bar P\sum_{j<\#B}\binom{\#B}{j}\mathsf{c}_j\,\eta\prod_B\gamma_i$. □

Lemma 18.55 (Domination at one side; moments of the sources)**.** In the data of the transfer let $(K,\beta)\in\mathfrak{d}(g)$.

(a) For every $\rho\in\mathfrak{S}_{K,\beta}$ and every bounded real Borel function F of the bonds inside $\square_0$, $|\rho(F)|\le \mathsf{Q}\,|\langle\prod_{\kappa\in\mathsf{Z}}F^{[\kappa]}\rangle'|^{1/8}$.

(b) Let $\bar o_i=\langle O_i\rangle'$. There is a number $\bar P$, a function of g, n, R and d, such that $|\rho(\prod_{i\in C}(O_i-\bar o_i))|\le \bar P\prod_{i\in C}\|g_i\|_\sharp$ for every $\rho\in\mathfrak{S}_{K,\beta}$ and every non-empty $C\subset\{1,\dots,n\}$.

■

Sketch. (a) is Lemma 5.8 for the board of the side $\mathsf{l}'=\mathsf{l}$, which divides $\frac{\ell'}{2}$, with $N'=8$: the law ρ is a periodic Gibbs state at (K,β) or the measure of a torus T_ℓ, $\ell\ge\ell'$ in $\mathfrak{L}(g)$, lifted to $\mathbb{L}_K$, and F is a function of the bonds inside one cell. (b) A moment of the centred functions $O_i-\bar o_i$ is the sum, over the partitions of C, of the products of the joint cumulants under ρ of the blocks. A block $\{i\}$ gives $\rho(O_i)-\langle O_i\rangle'$, the difference of the means of $\mathcal{O}(g_i)$ under two laws of $\mathfrak{S}_{K,\beta}$; the support of g_i lies in the Euclidean ball of radius $2R$ about 0, every pair of $\mathfrak{d}(g)$ has the property (Nr1) on $\mathfrak{L}(g)$ (Theorem YM3_I:91.12), and the one-point comparison in Proposition YM3_I:91.13 bounds the difference by $C_1(2R)\|g_i\|_\sharp$. A block of $N\ge2$ sources, whose supports lie at pairwise Euclidean distances $\ge\mathsf{d}$, has a joint cumulant at most $C_N^{\mathrm{per}}(\mathsf{d};2R)\prod\|g_i\|_\sharp$ in a periodic Gibbs state (Proposition YM3_I:91.13) and at most $C_N^{g}(\mathsf{d};2R)\prod\|g_i\|_\sharp$ on a torus T_ℓ, $\ell\ge\ell'\ge16N(2R+1)$, the supports lying in $(-\frac{\ell}{4},\frac{\ell}{4})^3$ (Proposition YM3_I:91.13). So $\bar P=\mathsf{Bl}_n\max\{1,C_1(2R),\max_{2\le N\le n}C_N^{\mathrm{per}}(\mathsf{d};2R),\max_{2\le N\le n}C_N^{g}(\mathsf{d};2R)\}^n$ serves. □

Lemma 18.56 (The chain on $T_{\ell'}$ with observables in the exponent; read-outs)**.** In the data of the transfer let $(K,\beta)\in\mathfrak{d}(g)$, $0<\phi\le\frac{1}{84}$ and $K\ge\max\{K_\phi,5\}$, and let $\mathfrak{J}_\iota$, $\iota\in\mathcal{A}_0$, finitely many, be local observables of orders ≥2 on $T_{\ell'}$ with complex weights $\varkappa_\iota$ and with the radii $r_{\mathfrak{J}_\iota}$ of [7, Lemma 5.20](a), so that $1/r_{\mathfrak{J}_\iota}\le C_r|\varkappa_\iota|_\infty$ with C_r a function of ℓ', b_+ and the type ([7, Lemma 5.20](c)).

(a) For $w\in\mathsf{Src}$ and $\mathbf{s}\in\mathbb{C}^{\mathcal{A}_0}$ with $\sum_\iota|s_\iota|/r_{\mathfrak{J}_\iota}<1$ let $\mathcal{L}(\mathbf{s},w;V)=\sum_{i\le t}\mathsf{c}_i(w,\mathbf{s})+\Phi_t^{w,\mathbf{s}}(V)$, $V\in\mathsf{Cut}_t$, with the numbers and the exponent of [7, Lemma 5.20](a) formed for $\sum_\iota s_\iota\mathfrak{J}_\iota$. Then

$$(\mathcal{M}^t)_*(\mathsf{p}_t e^{\sum_\iota s_\iota\mathfrak{J}_\iota}\rho_0^w dU)=\mathbf{1}_{\mathsf{Cut}_t}e^{\mathcal{L}(\mathbf{s},w;\cdot)}dV,\qquad (\mathcal{M}^t)_*(\mathfrak{w}_t^{\mathrm{sf}}e^{\sum_\iota s_\iota\mathfrak{J}_\iota}\rho_0^w dU)=\mathbf{1}_{\mathsf{In}_t}e^{\mathcal{L}(\mathbf{s},w;\cdot)}dV;$$

$\mathcal{L}(\mathbf{s},w;V)$ is holomorphic in $(\mathbf{s},w)$, and for every background U over V

$$\mathcal{L}(\mathbf{s},w;V)=\mathsf{c}_\circ+\sum_{i\le t}\sum_{X,z}E^{(i)}[w,\mathbf{s}](X,U,z)-A^{\eta_t}(x_t^w,U)+\sum_\iota s_\iota\mathfrak{J}_\iota(U),$$

with the families $E^{(i)}[w,\mathbf{s}]$ of that Lemma and a number $\mathsf{c}_\circ$ that reads neither $\mathbf{s}$ nor w nor V.

(b) *(Mixed derivatives across a gap.)* Let $\Theta\subset\mathbb{C}^{\mathcal{V}}$ be a polydisc about 0 with the radii ρ_v, and $\theta\mapsto(\mathbf{s}(\theta),w(\theta))$ a polynomial map of Θ into the domain of (a) with $w(\theta)$ linear in θ; for each variable v let $S(v)\subset T_{\ell'}$ be a closed set such that $w(\theta)(p)$ reads θ_v only if $x(p)\in S(v)$, and $s_\iota(\theta)$ reads θ_v only if the boxes of $\mathfrak{J}_\iota$ lie in $S(v)$. Let $b\subset\mathcal{V}$, $\#b\ge2$, hold two variables v,v' whose sets $S(v)$, $S(v')$ lie at sup distance $\ge\mathsf{d}_*>2a$. Then at every $V\in\mathsf{Cut}_t$, with a background U over V,

$$\Big|\prod_{v\in b}\partial_{\theta_v}\Big[\mathcal{L}(\mathbf{s}(\theta),w(\theta);V)-\sum_\iota s_\iota(\theta)\mathfrak{J}_\iota(U)\Big]_{\theta=0}\Big|\le\tau_t(\mathsf{d}_*)\prod_{v\in b}\rho_v^{-1},\qquad \tau_t(\mathsf{d}_*)=e^2E_0C_{\mathrm{E},0}\sum_{i\le t}N_ie^{-(\mathsf{d}_*-2a)L^{K-i}/M},$$

with E_0 of (Th1) and $C_{\mathrm{E},0}$ of Proposition YM3_I:45.1(a4).

(c) *(Read-outs.)* For one local observable $\mathfrak{J}$ of an order ≥2 and $w\in\mathsf{Src}$ put $\dot F[\mathfrak{J};w](V)=\partial_s\mathcal{L}(s,w;V)|_{s=0}$, $V\in\mathsf{Cut}_t$. Then $(\mathcal{M}^t)_*(\mathsf{p}_t\mathfrak{J}\rho_0^w dU)=\dot F[\mathfrak{J};w]\,\mathbf{1}_{\mathsf{Cut}_t}e^{\mathcal{L}(0,w;\cdot)}dV$. At a real source w the measure $\nu_t^w=(\mathcal{M}^t)_*(\mathsf{p}_t\rho_0^w dU)=\mathbf{1}_{\mathsf{Cut}_t}e^{\mathcal{L}(0,w;\cdot)}dV$ is positive and has the null sets of the Haar measure on Cut_t, and $\dot F[\mathfrak{J};w]$ is, almost everywhere on Cut_t, the

conditional mean $\mathsf{Q}^w_{t\leftarrow 0}\mathfrak{J}$ of $\mathfrak{J}$ given the block field of the level t under $\mathsf{p}_t\rho_0^w dU$, that is, the density of $(\mathcal{M}^t)_*(\mathsf{p}_t\mathfrak{J}\rho_0^w dU)$ against ν_t^w. For two observables, $\dot F[\mathfrak{J}_1+\zeta\mathfrak{J}_2;w]=[\partial_{s_1}+\zeta\partial_{s_2}]\mathcal{L}((s_1,s_2),w;\cdot)|_{s=0}$; $\dot F[\mathfrak{J};w](V)$ is holomorphic in w; and at every $V\in\mathsf{Cut}_t$ and every $w\in\mathsf{Src}$, $|\dot F[\mathfrak{J};w](V)|\le C_B L^{3K}|\varkappa|_\infty$, C_B a function of ℓ', b_+ and the type of $\mathfrak{J}$.

(d) *(One source.)* For a real Lipschitz function $h\neq 0$ on $T_{\ell'}$ and $|\zeta|<(8\mathsf{M}_t^{\mathrm{rd}}\|h\|_\sharp)^{-1}$, $\zeta h\in\mathsf{Src}$ and $\mathcal{L}(0,\zeta h;V)-\mathcal{L}(0,0;V)=\mathsf{k}(\zeta)+\Delta\Phi(\zeta,V)$, where k is holomorphic with $\mathsf{k}(0)=0$ and reads no V, and $\Delta\Phi=\Phi_t^{\zeta h}-\Phi_t^0$, formed with the field-dependent exponents of the chain at the two sources, is holomorphic in ζ and at most $\frac14$ in modulus on Cut_t.

■

Sketch. (a) [7, Lemma 5.20](a), for several observables and at every source, complex sources included, gives at the level t the display $(\mathcal{M}^t)_*(\mathsf{p}_t e^{\sum_\iota s_\iota\mathfrak{J}_\iota}\rho_0^w dU)=\exp[\sum_{i\le t}\mathsf{c}_i(w,\mathbf{s})]\mathbf{1}_{\mathsf{Cut}_t}e^{\Phi_t^{w,\mathbf{s}}}dV$, with families $E^{(i)}[w,\mathbf{s}]$ that are holomorphic in $(w,\mathbf{s})$ and in the fields, local in w and in the observables, and of norms below E_0 on the domain; multiplied by $\mathbf{1}_{\mathsf{In}_t}\circ\mathcal{M}^t$ it gives the second display, as $\mathfrak{w}_t^{\mathrm{sf}}=(\mathbf{1}_{\mathsf{In}_t}\circ\mathcal{M}^t)\mathsf{p}_t$ and $\mathsf{In}_t\subset\mathsf{Cut}_t$. The number $\mathsf{c}_i(w,\mathbf{s})$ is $\log(\mathsf{pf}_{i-1}(x_{i-1})\mathsf{Z}^{(i-1)}[(\mathbf{1},0,0)])$, which reads neither w nor $\mathbf{s}$, plus the sum of the members of $E^{(i)}[w,\mathbf{s}]$ at the trivial configuration (Proposition YM3_I:114.2 and [7, Lemma 5.20(a)]), while $\Phi_t^{w,\mathbf{s}}(V)$ is $-A^{\eta_t}(x_t^w,U)$ plus the sums of the same members at U less their values at $\mathbf{1}$, plus $\sum_\iota s_\iota\mathfrak{J}_\iota(U)$; adding them gives the formula, $\mathsf{c}_\circ$ being the sum of the logarithms.

(b) In the bracket the term $\mathsf{c}_\circ$ reads no variable, and $-A^{\eta_t}(x_t^{w(\theta)},U)$ is affine in w, so affine in θ; the derivative in the $\#b\ge 2$ variables of b kills both. The member of $E^{(i)}$ at (X,z), taken at U, reads w on the plaquettes of X alone and s_ι only if $\mathfrak{J}_\iota$ meets X ([7, Lemma 5.20](a), whose families are local in w and in the observables); so it reads θ_v only if X meets $S(v)$, and its derivative in the variables of b vanishes unless X meets both $S(v)$ and $S(v')$. Such a domain of the level i holds two points at sup distance $\ge\mathsf{d}_*-2a$, so that $d_i(X)\ge(\mathsf{d}_*-2a)L^{K-i}/M-2$ (Proposition YM3_I:45.1(b3)). On Θ the member is at most $E_0e^{-(\kappa+1)d_i(X)}$, where κ is the rate of decay in the norms of the families of the main text, and Cauchy's estimate bounds its mixed derivative by this number times $\prod_{v\in b}\rho_v^{-1}$; the sum over the domains through each of the N_i sites of the level i, with Proposition YM3_I:45.1(a4) as in the proof of [7, Proposition 3.15](b), and over $i\le t$ gives $\tau_t(\mathsf{d}_*)$.

(c) The densities in (a) are bounded on a disc of s uniformly in V, the members having norms below E_0; so (a), integrated against a bounded Borel function of V, may be differentiated in s at 0 under the integrals, which gives the display, and at a real source the measures are images of positive measures, the density $e^{\mathcal{L}(0,w;\cdot)}$ has no zero on Cut_t, and the conditional mean is the quotient of the two densities. The rule for two observables holds because the recursion of [7, Lemma 5.20](a) reads the observables through $\sum_\iota s_\iota\mathfrak{J}_\iota$ alone, and holomorphy in w comes from (a). For the bound, by (a), $\dot F[\mathfrak{J};w](V)=\sum_{i\le t}\mathrm{Sum}[\mathsf{g}_i](U)+\mathfrak{J}(U)$ with the pointed families $\mathsf{g}_i=\partial_s E^{(i)}[w,s]|_{s=0}$, of norms at most the numbers d_i of the Lipschitz bound of [7, Lemma 5.20](a), which holds at every source; Proposition YM3_I:45.1(a4) at $D=0$ bounds $|\mathrm{Sum}[\mathsf{g}_i](U)|$ by $C_{\mathrm{E},0}N_i\mathsf{d}_i$, $\max_i\mathsf{d}_i\le C|\varkappa|_\infty$ (the numbers of an observable, in the data of the transfer), $\sum_{i\ge1}N_i\le 2\ell'^3L^{3K}$, and $|\mathfrak{J}(U)|\le N\hat c\,\ell'^3L^{3K}|\varkappa|_\infty$ at the real configuration U ([7, Lemma 5.20](c)).

(d) This is Lemma YM3_I:114.5 for the single function h, with $\|h\|_{\flat;T_{\ell'}}\le\|h\|_\sharp$: below the radius $(8\mathsf{M}_t^{\mathrm{rd}}\|h\|_\sharp)^{-1}$ the source ζh lies in Src and $|\Phi_t^{\zeta h}-\Phi_t^0|\le\frac14$ on Cut_t. At $s=0$ the families of [7, Lemma 5.20](a) are those of the chain at the source, so $\mathcal{L}(0,\zeta h;V)=\sum_i\mathsf{c}_i(\zeta h)+\Phi_t^{\zeta h}(V)$, and $\mathsf{k}(\zeta)=\sum_i(\mathsf{c}_i(\zeta h)-\mathsf{c}_i(0))$ is holomorphic by Proposition YM3_I:114.2. □

Lemma 18.57 (Oscillations of read-outs)**.** In the setting of Lemma 18.56:

(a) *(Power counting, at every source.)* For a local observable $\mathfrak{J}$ of the order $m_{\mathfrak{J}}\ge2$ and every $w\in\mathsf{Src}$, real or complex, $\dot F[\mathfrak{J};w]$ differs at every point of Cut_t from one number by at

most $\mathsf{o}_t(\mathfrak{J})$, the number of [7, Lemma 5.20](b); and $\mathsf{o}_t(\mathfrak{J}) \le CK^{2\mathfrak{p}_0}L^{4u-K}|\varkappa|_\infty$ if $m_{\mathfrak{J}} = 2$, $\mathsf{o}_t(\mathfrak{J}) \le CK^{3\mathfrak{p}_0}L^{4u-\frac{3}{2}K}|\varkappa|_\infty$ if $m_{\mathfrak{J}} \ge 3$, C a function of ℓ', b_+ and the type.

(b) *(In-plane orbital potentials, at a real source.)* Let $0 < \kappa_{\mathrm{H}} < 1$, $\phi < \psi_{\mathrm{c}} < 1$ and $n_0 = K - \lceil \psi_{\mathrm{c}} K \rceil$. There are K_ψ and C_ψ, functions of ℓ', b_+, κ_{H}, ψ_{c} and ϕ, such that for $K \ge K_\psi$, every real weight $\varkappa$ of the plaquettes and every real source $w \in \mathsf{Src}$ there is a number a with $|\dot F[\mathfrak{J}_\varkappa; w] - \mathsf{a}| \le \epsilon_\psi |\varkappa|_\infty$ almost everywhere on Cut_t, where $\mathfrak{J}_\varkappa = \sum_p \varkappa(p)\psi(p)$ and

$$\epsilon_\psi = C_\psi K^{3\mathfrak{p}_0} L^{-K} \big[L^{-(2\kappa_{\mathrm{H}} - (3+2\kappa_{\mathrm{H}})\psi_{\mathrm{c}})K} + L^{3u - \frac{1}{2}\psi_{\mathrm{c}} K} \big].$$

(c) *(The cubic insertion, at a real source.)* Let $0 < \tau \le \frac{1}{4}$, $\phi < \frac{1}{42}$, $c_0' = \min\{\frac{1}{14} - \frac{\tau}{7}, \frac{1}{14} - 3\phi\}$, and let K_2, C_2, p_2 be the numbers of Lemma 5.10(b) on $T_{\ell'}$ for $\mathsf{b}_- = \gamma^{-2}$ and b_+. For $K \ge K_2$, every field Ξ on $T_{\ell'}$, every real Lipschitz $h \ne 0$ on $T_{\ell'}$ and every real x with $|x| < (8\mathsf{M}_t^{\mathrm{rd}} \|h\|_\sharp)^{-1}$ there is a number $\mathsf{a}(x)$ with $|\dot F[\mathfrak{K}^\sharp[(\beta - a^{-1}xh)\Xi]; xh] - \mathsf{a}(x)| \le C_2 \mathsf{H}(\Xi) K^{p_2} L^{-c_0' K}$ almost everywhere on Cut_t.

■

Sketch. (a) At a background U over $V \in \mathsf{Cut}_t$ the formula of Lemma 18.56(a) gives $\Phi_t^{w,s}(V) - \Phi_t^{w,s'}(V) = (s - s')\mathfrak{J}(U) + \sum_{i \le t}(\mathrm{Sum}[\delta E_i](U) - \mathrm{Sum}[\delta E_i](\mathbf{1}))$ with $\delta E_i = E^{(i)}[w, s] - E^{(i)}[w, s']$, of norms at most $|s - s'|\mathsf{d}_i$ at every source ([7, Lemma 5.20](a)). The sum is the action at U of a bare array with vanishing counterterms and an empty slot, at most $C_{\mathrm{old}} N_t \sum_i L^{-(t-i)}|s - s'|\mathsf{d}_i$ by Proposition YM3_I:113.13, which reads no source, and $|\mathfrak{J}(U)| \le N_t N c(\mathsf{e}_t^{\mathrm{c}})^{m_{\mathfrak{J}}} |\varkappa|_\infty L^{-(2m_{\mathfrak{J}}-3)t}$, the plaquette variables of U lying within $\mathsf{e}_t^{\mathrm{c}} L^{-2t}$ of $\mathbf{1}$; this is the computation of the proof of [7, Lemma 5.20](a), and it gives $|\partial_s \Phi_t^{w,s}|_{s=0}| \le \mathsf{o}_t(\mathfrak{J})$ at every point of Cut_t. The difference $\dot F[\mathfrak{J}; w] - \partial_s \Phi_t^{w,s}|_{s=0} = \sum_i \partial_s \mathsf{c}_i(w, s)|_{s=0}$ reads no V, and the bounds of o_t are those of [7, Lemma 5.20](a).

(b) Let w be a real source and $\mathfrak{J} = \mathfrak{J}_\varkappa$, an observable of the order 2. We follow the proof of [7, Lemma 3.18](a) at the source w. The chain measures $\nu_k^w = (\mathcal{M}^k)_*(\mathsf{p}_k \rho_0^w dU)$ are positive and obey $\nu_{k+1}^w = \mathbf{1}_{\mathsf{Cut}_{k+1}} \mathbf{T}(\mathfrak{G}_k[x_k] \chi_{x_k}^{\mathrm{sf}} \nu_k^w)$, by the recursion of the weights p_k ([7, Definition 3.2](3)); so the conditional mean $\mathsf{Q}_k^w h$, the density of $\mathbf{1}_{\mathsf{Cut}_{k+1}} \mathbf{T}(\mathfrak{G}_k[x_k] \chi_{x_k}^{\mathrm{sf}} h\, \nu_k^w)$ against ν_{k+1}^w, averages h against a positive kernel, $|\mathsf{Q}_k^w h - \mathsf{a}_0| \le \|h - \mathsf{a}_0\|_{L^\infty(\nu_k^w)}$ for real h and numbers a_0, and $\mathsf{Q}_{t \leftarrow 0}^w = \mathsf{Q}_{t \leftarrow n_0}^w \mathsf{Q}_{n_0 \leftarrow 0}^w$ with $\mathsf{Q}_{t \leftarrow n_0}^w = \mathsf{Q}_{t-1}^w \cdots \mathsf{Q}_{n_0}^w$. The display of [7, Lemma 5.20](a) at the level n_0, differentiated in s as in Lemma 18.56(c), gives $\mathsf{Q}_{n_0 \leftarrow 0}^w \mathfrak{J} = \dot c + \mathfrak{J}(U_{n_0}) + \sum_{i \le n_0}(\mathrm{Sum}[\mathsf{g}_i](U_{n_0}) - \mathrm{Sum}[\mathsf{g}_i](\mathbf{1}))$ almost everywhere on Cut_{n_0}, with a number $\dot c$ and the pointed families $\mathsf{g}_i = \partial_s E^{(i)}[w, s]|_{s=0}$, of norms $\le \mathsf{d}_i$. A pointed family g of a level $i' \le k$ is carried by one conditional mean as $\mathsf{Q}_k^w \Sigma_k[\mathsf{g}] = \Sigma_{k+1}[\mathsf{g} + \mathsf{S}_k^w \mathsf{g}]$, $\Sigma_k[\mathsf{g}] = \mathrm{Sum}[\mathsf{g}] \circ U_k$, with $\|\mathsf{S}_k^w \mathsf{g}\| \le \theta_k L^{-(k-i')}\|\mathsf{g}\|$: the derivation of [7, Lemma 3.14](a), which perturbs an early member of the run by $\mu\mathsf{g}$ and differentiates the identity of the step in μ, rests on the step identity of Proposition YM3_I:114.2 and on the loss of memory of Proposition YM3_I:114.3, both of them stated at one source, and so holds at w. Hence, as in the proof of [7, Lemma 3.18](a), the families carried from the level n_0 to the level t differ at every point of Cut_t from a number by at most $e\mathsf{B}_0' L^{3u-K} \sum_{i \le n_0} L^i \mathsf{d}_i \le CK^{3\mathfrak{p}_0} L^{3u-K-(K-n_0)/2}|\varkappa|_\infty$ ([7, Lemma 3.16](b), and the bound of $\sum_{i \le k} L^i \mathsf{d}_i$ at the order 2 and $k = n_0$ among the numbers of an observable), while $|\mathsf{Q}_{t \leftarrow n_0}^w(\mathfrak{J} \circ U_{n_0})| \le \epsilon_X$, the supremum of $|\mathfrak{J}|$ over the backgrounds over Cut_{n_0}. To bound ϵ_X we use Lemma 5.9(e), which writes at every real configuration

$$\psi(p) = -\tfrac{1}{8} \sum_{b \subset \partial p} |G_{\nu(p,b)}(b)|^2 - \tfrac{1}{8} \sum_{\nu \in P(p)} (2a_p^2 + a_{p+ae_\nu}^2 + a_{p-ae_\nu}^2),$$

$\nu(p, b)$ the direction of $P(p)$ normal to b; and $G_\rho(b)$ is the vector $\mathbf{y}_{b,\rho}$ of [7, § 3]. A background U over $V \in \mathsf{Cut}_{n_0}$ is regular of level n_0 at the radius ε_{n_0}, with $\varepsilon_{n_0}^2 \le \bar{\mathsf{c}}_\varepsilon (K \log L + \log \mathsf{b}_+)^{2\mathfrak{p}_0} L^{-(K-n_0)}$ ([7, Lemma 3.16](a)); so $a_q(U) = \frac{1}{2}|\partial U(q) - \mathbf{1}|^2 < \frac{1}{2}\varepsilon_{n_0}^2 L^{-4n_0}$, and for $K \ge K_\psi$, which brings the data of [7, Lemma 3.16](c) at $n = n_0$ and $\varepsilon = \varepsilon_{n_0}$ together with the thresholds of [7, Lemma 3.18], $|\mathbf{y}_{b,\rho}(U)| \le 48\, \mathsf{c}_{\mathrm{g}}(\kappa_{\mathrm{H}}) \rho_{\mathrm{H}} \varepsilon_{n_0} L^{-(2+\kappa_{\mathrm{H}})n_0}$. With $3\ell'^3 L^{3K}$ plaquettes this gives $\epsilon_X \le CK^{2\mathfrak{p}_0} L^{2K-(3+2\kappa_{\mathrm{H}})n_0}|\varkappa|_\infty$. By

Lemma 18.56(c) the read-out is $\mathsf{Q}^{w}_{t\leftarrow 0}\mathfrak{J}$ almost everywhere on Cut_t; with $n_0 \ge (1-\psi_{\rm c})K-1$ and $K-n_0 \ge \psi_{\rm c}K$ the two bounds give ϵ_ψ.

(c) Lemma 5.10(b), on $T_{\ell'}$ with $\mathsf{b}_- = \gamma^{-2}$ and b_+ ([7, Lemma 3.12](a)), with $n=1$ and the function h, gives at real x in its polydisc, whose radius is $(8\mathsf{M}_t^{\rm rd}\|h\|_\sharp)^{-1}$, a number $\mathsf{a}(x)$ with $|\mathsf{Q}^{xh}_{t\leftarrow 0}\mathfrak{K}^\sharp[(\beta-a^{-1}xh)\Xi]-\mathsf{a}(x)| \le C_2\mathsf{H}(\Xi)K^{p_2}L^{-c_0'K}$ almost everywhere for ν_t^{xh} on Cut_t; by Lemma 18.56(c) the read-out is that conditional mean almost everywhere, and ν_t^{xh} has the null sets of the Haar measure on Cut_t. □

Proposition 18.58 (The mean of the eight mirror copies)**.** In the data of the transfer, with $\|g_i\|_\sharp = 1$ for every i, let $\mathbb{J} = (\mathfrak{J}_0,\dots,\mathfrak{J}_n)$ be a local insertion tuple over the sources, of types with constants at most $(c,\hat c,N)$, with $\mathfrak{J}_0$ carried by A_{R+1} and $\mathfrak{J}_i$ within sup distance $4a$ of $\operatorname{supp} g_i$, and with $\varkappa^* = |\varkappa_0|_\infty + \max_i|\varkappa_i|_\infty = 1$, $\varkappa_i$ the weights of $\mathfrak{J}_i$, $0\le i\le n$. Let $(K,\beta)\in\mathfrak{d}(g)$, $0<\phi\le\frac{1}{84}$, $K\ge\max\{K_\phi,5\}$ and $16a\le\mathsf{d}$. Put $\alpha_\emptyset = \langle\mathfrak{J}_0\rangle'$, $\alpha_i = \langle\mathfrak{J}_0;O_i\rangle' + \langle\mathfrak{J}_i\rangle'$, $\bar o_i = \langle O_i\rangle'$, let H_B be the functions of Lemma 18.54, and for $\kappa\in\mathsf{Z}$, $1\le i\le n$ and $|\zeta|<r_i$ let $\dot F_i^\kappa(\zeta,\cdot) = \dot F[\mathfrak{J}_0^{[\kappa]} + \zeta\mathfrak{J}_i^{[\kappa]};\zeta g_i^\kappa]$, the read-out of Lemma 18.56(c), which applies as each of the mirror copies $\mathfrak{J}_0^{[\kappa]},\dots,\mathfrak{J}_n^{[\kappa]}$ is a local observable on $T_{\ell'}$ of the order of the member that it copies, of a type formed from the type of that member alone, and with weights of the same moduli ([7, Lemma 5.20](d)). Suppose that for numbers $0<\epsilon_1\le B_1$ and all κ and i:

(OI-C) $|\dot F_i^\kappa(\zeta,V)|\le B_1$ for $|\zeta|<r_i$ and $V\in\mathsf{Cut}_t$;

(OI-R) for every real $x\in(-r_i,r_i)$ there is a number a with $|\dot F_i^\kappa(x,V)-\mathsf{a}|\le\epsilon_1$ for almost every $V\in\mathsf{In}_t$.

Then there are $K_\star$, a function of g, n, R, d, ϕ and $(c,\hat c,N)$, an integer q_n and a number C of the same data, such that for $K\ge K_\star$, every $B\subset\{1,\dots,n\}$, $k=\#B$, and every $0<\sigma\le 1$

$$\Big|\Big\langle\prod_{\kappa\in\mathsf{Z}}H_B^{[\kappa]}\Big\rangle'\Big|^{1/8} \le [(1+k)\epsilon_2+\vartheta_K]\mathsf{M}^k, \qquad \epsilon_2 = \frac{4}{\pi\sigma}(2\epsilon_1)^{1-\sigma}(2B_1)^\sigma + 2\epsilon_1, \qquad \vartheta_K \le (CL^K)^{q_n}[\tau_t(\tfrac12\mathsf{d})^{1/8} + \mathsf{R}_t^{1/8}],$$

with τ_t of Lemma 18.56(b). For $B=\emptyset$ only the read-outs at $\zeta=0$ enter. ■

Sketch. 1. The identity. Let $\mathcal{A} = \{\mu_\kappa : \kappa\in\mathsf{Z}\}\cup\{z_i^\kappa : \kappa\in\mathsf{Z},\ i\in B\}$, $8(1+k)$ variables, and let Θ be the polydisc $|\mu_\kappa|<\mu^\sharp$, $|z_i^\kappa|<\zeta_\circ$, with $\zeta_\circ = \min\{1,\varrho/(8n)\}$ and $\mu^\sharp = [16(1+n)C_r]^{-1}$, C_r the number of Lemma 18.56 for the types of the copies. On Θ put $w(\theta) = \sum_{\kappa,i} z_i^\kappa g_i^\kappa$, which lies in Src by Lemma YM3_I:114.5(b), as $\sum_{\kappa,i}|z_i^\kappa|\,\|g_i^\kappa\|_{\flat;T_{\ell'}} < 8k\zeta_\circ \le \varrho$, and give the observables $\mathfrak{J}_0^{[\kappa]}$ and $\mathfrak{J}_i^{[\kappa]}$ the coefficients μ_κ and $\mu_\kappa z_i^\kappa$, which keeps $\sum_\iota |s_\iota|/r_{\mathfrak{J}_\iota}<1$. Then $\sum_\iota s_\iota\mathfrak{J}_\iota + \mathcal{O}(w) = \sum_\kappa[\mu_\kappa Y^\kappa_{z^\kappa} + \sum_i z_i^\kappa O_i^\kappa]$ with $Y_z^\kappa = \mathfrak{J}_0^{[\kappa]} + \sum_{i\in B} z_i\mathfrak{J}_i^{[\kappa]}$ and $O_i^\kappa = O_i^{[\kappa]}$, and Lemma 18.56(a) gives $\Lambda'(\exp\sum_\kappa[\mu_\kappa Y^\kappa_{z^\kappa} + \sum_i z_i^\kappa O_i^\kappa]) = \int_{\mathsf{In}_t} e^{\mathcal{L}(\theta;V)}dV$, $\mathcal{L}(\theta;V) = \mathcal{L}(\mathbf{s}(\theta),w(\theta);V)$. We multiply by $\exp[-\sum_\kappa\mu_\kappa\mathsf{a}_\circ(z^\kappa) - \sum_{\kappa,i}\bar o_i z_i^\kappa]$, $\mathsf{a}_\circ(z) = \alpha_\emptyset + \sum_i\alpha_i z_i$, and differentiate once in each variable of $\mathcal{A}$ at $\theta=0$, under the integrals, the integrands being holomorphic on Θ and bounded there uniformly in the point. As the copies are multiplicative and fix constants, the left member becomes $\Lambda'(\prod_\kappa H_B^{[\kappa]})$ by the rule of Leibniz of the proof of Lemma 18.54, and the right member becomes $\int D\,d\tilde\nu$, with the positive measure $\tilde\nu = \mathbf{1}_{\mathsf{In}_t}e^{\mathcal{L}(0;\cdot)}dV = (\mathcal{M}^t)_*(\mathfrak{w}_t^{\rm sf}\rho_0 dU)$ and

$$D(V) = \prod_{v\in\mathcal{A}}\partial_v\, e^{\bar{\mathcal{E}}(\theta;V)}\Big|_{\theta=0} = \sum_\pi\prod_{b\in\pi}\partial^b\bar{\mathcal{E}}(0;V), \qquad \bar{\mathcal{E}}(\theta;V) = \mathcal{L}(\theta;V) - \mathcal{L}(0;V) - \sum_\kappa\mu_\kappa\mathsf{a}_\circ(z^\kappa) - \sum_{\kappa,i}\bar o_i z_i^\kappa,$$

the sum over the partitions π of $\mathcal{A}$, ∂^b the derivative once in each variable of b.

2. Discarding. For a bounded real gauge invariant Borel function F, Proposition YM3_I:115.2, applied to its positive and to its negative part, give $|Z\langle F\rangle' - \Lambda'(F)| \le \sup|F|\,\mathsf{R}_tZ$, and $0\le Z-\Lambda'(1)\le\mathsf{R}_tZ$. With $\tilde\nu(\mathsf{In}_t) = \Lambda'(1)\le Z$ this gives $|\langle\prod_\kappa H_B^{[\kappa]}\rangle'| \le \operatorname{ess\,sup}_{\mathsf{In}_t}|D| + \mathsf{R}_tS_H^8$, $S_H\ge\sup|H_B|$; and, with $\langle F\rangle_{\Lambda'} = \Lambda'(F)/\Lambda'(1)$, $|\langle F\rangle' - \langle F\rangle_{\Lambda'}| \le 2\mathsf{R}_t\sup|F|$ once $\mathsf{R}_t\le\frac12$.

3. Blocks across a gap. Let $S(\mu_\kappa)$ be the union of the boxes of $\mathfrak{J}_0^{[\kappa]},\dots,\mathfrak{J}_n^{[\kappa]}$, and $S(z_i^\kappa)$ the union of $\{x(p): g_i^\kappa(x(p))\neq 0\}$ with the boxes of $\mathfrak{J}_i^{[\kappa]}$. They are as Lemma 18.56(b) asks. The sets of one cell lie in the copy of A_{R+2}, at sup distance ≥ 16 from the walls, so sets of two different cells lie at sup distance ≥ 32; and $S(z_i^\kappa)$, $S(z_l^\kappa)$, $i\neq l$, lie at sup distance $\geq \mathsf{d}-8a\geq \frac12\mathsf{d}$. So every block b with $\#b\geq 2$ that is not of the form $\{\mu_\kappa,z_i^\kappa\}$ holds two variables whose sets lie at sup distance $\geq \frac12\mathsf{d}>2a$. The subtracted terms of $\bar{\mathcal{E}}$ and the sum $\sum_\iota s_\iota(\theta)\mathfrak{J}_\iota(U)$ are sums of the monomials μ_κ, $\mu_\kappa z_i^\kappa$, z_i^κ, which ∂^b kills; Lemma 18.56(b) gives $|\partial^b\bar{\mathcal{E}}(0;V)|\leq \tau_t(\frac12\mathsf{d})\prod_{v\in b}\rho_v^{-1}$ at every $V\in\mathsf{Cut}_t$.

4. Read-outs. For $K\geq K_\star$, $r_i\leq\zeta_\circ$, and $\dot F_i^\kappa(\zeta,V)$ is $\partial_{\mu_\kappa}\mathcal{L}(\theta;V)$ at $\mu=0$, $z_i^\kappa=\zeta$ and the other z equal to 0 (the rule for two observables of Lemma 18.56(c)); it is holomorphic in ζ at every V. Let $\mathsf{In}^\circ\subset\mathsf{In}_t$ be the set of the V at which (OI-R) holds for all κ, i and all x of a countable dense subset of $(-r_i,r_i)$; its complement in In_t is a Haar null set, and by continuity in x, $|\dot F_i^\kappa(x,V)-\dot F_i^\kappa(x,V')|\leq 2\epsilon_1$ for all real $x\in(-r_i,r_i)$ and $V,V'\in\mathsf{In}^\circ$. The one-variable estimate in the proof of Lemma YM3_I: 116.2, applied to $\zeta\mapsto \dot F_i^\kappa(\zeta,V)-\dot F_i^\kappa(\zeta,V')$, bounded by $2B_1$ on the disc and by $2\epsilon_1$ on the real interval, gives $|\partial_\zeta\dot F_i^\kappa(0,V)-\partial_\zeta\dot F_i^\kappa(0,V')|\leq\omega/r_i$ with $\omega=\frac{4}{\pi\sigma}(2\epsilon_1)^{1-\sigma}(2B_1)^\sigma$.

5. The blocks $\{\mu_\kappa\}$, $\{z_i^\kappa\}$, $\{\mu_\kappa,z_i^\kappa\}$, at $V\in\mathsf{In}^\circ$, with the probability measure $\hat\nu=\tilde\nu/\tilde\nu(\mathsf{In}_t)$, which charges no Haar null set, and with the invariance of $\langle\cdot\rangle'$ under the r_κ. $\{\mu_\kappa\}$: $\partial_{\mu_\kappa}\bar{\mathcal{E}}(0;V)=\dot F_i^\kappa(0,V)-\alpha_\emptyset$; the identity of Step 1 differentiated in μ_κ alone gives $\langle\mathfrak{J}_0^{[\kappa]}\rangle_{\Lambda'}=\int\dot F_i^\kappa(0,\cdot)\,d\hat\nu$, while $\alpha_\emptyset=\langle\mathfrak{J}_0^{[\kappa]}\rangle'$, and by (OI-R) at $x=0$, $|\dot F_i^\kappa(0,V)-\int\dot F_i^\kappa(0,\cdot)\,d\hat\nu|\leq 2\epsilon_1$; so $|\partial_{\mu_\kappa}\bar{\mathcal{E}}(0;V)|\leq 2\epsilon_1+2\mathsf{R}_tS_0$, $S_0\geq\sup|\mathfrak{J}_0|$. $\{z_i^\kappa\}$: by Lemma 18.56(d) for $h=g_i^\kappa$, $\partial_{z_i^\kappa}\bar{\mathcal{E}}(0;V)=\mathsf{k}'(0)+\varphi(V)-\bar o_i$ with $\varphi=\partial_\zeta\Delta\Phi(0,\cdot)$, $|\varphi|\leq\frac{1}{4r_i}$ by Cauchy's estimate; $T(\zeta)=\Lambda'(e^{\zeta O_i^\kappa})=e^{\mathsf{k}(\zeta)}\int e^{\Delta\Phi(\zeta,\cdot)}d\tilde\nu$ gives $\mathsf{k}'(0)+\int\varphi\,d\hat\nu=\langle O_i^\kappa\rangle_{\Lambda'}$, while $\bar o_i=\langle O_i^\kappa\rangle'$; so $|\partial_{z_i^\kappa}\bar{\mathcal{E}}(0;V)|\leq\frac{1}{2r_i}+2\mathsf{R}_tS_O$, $S_O\geq\max_i\sup|O_i|$. $\{\mu_\kappa,z_i^\kappa\}$: $\partial_{\mu_\kappa}\partial_{z_i^\kappa}\bar{\mathcal{E}}(0;V)=\partial_\zeta\dot F_i^\kappa(0,V)-\alpha_i$. The function $N(\zeta)=\Lambda'(Y_\zeta^\kappa e^{\zeta O_i^\kappa})$, $Y_\zeta^\kappa=\mathfrak{J}_0^{[\kappa]}+\zeta\mathfrak{J}_i^{[\kappa]}$, is $e^{\mathsf{k}(\zeta)}\int\dot F_i^\kappa(\zeta,\cdot)e^{\Delta\Phi(\zeta,\cdot)}d\tilde\nu$, by Step 1 differentiated in μ_κ alone at the source ζg_i^κ; so $(N/T)'(0)=\int\partial_\zeta\dot F_i^\kappa(0,\cdot)\,d\hat\nu+\int(\dot F_i^\kappa(0,\cdot)-\mathsf{a})(\varphi-\int\varphi\,d\hat\nu)\,d\hat\nu$, a the number of (OI-R) at $x=0$, and the second integral has modulus at most $\frac{\epsilon_1}{2r_i}$. On the other hand $(N/T)'(0)=\langle\mathfrak{J}_0^{[\kappa]}O_i^\kappa\rangle_{\Lambda'}-\langle\mathfrak{J}_0^{[\kappa]}\rangle_{\Lambda'}\langle O_i^\kappa\rangle_{\Lambda'}+\langle\mathfrak{J}_i^{[\kappa]}\rangle_{\Lambda'}$, and α_i is the same expression under $\langle\cdot\rangle'$; by Step 2 they differ by at most $6\mathsf{R}_tS_0S_O+2\mathsf{R}_tS_*$, $S_*\geq\max_i\sup|\mathfrak{J}_i|$. With Step 4: $|\partial_{\mu_\kappa}\partial_{z_i^\kappa}\bar{\mathcal{E}}(0;V)|\leq(\omega+\frac12\epsilon_1)r_i^{-1}+6\mathsf{R}_tS_0S_O+2\mathsf{R}_tS_*$.

6. The sum over the partitions. Let $K_\star$ be such that for $K\geq K_\star$: $\mathsf{R}_t\leq\frac12$, $2\mathsf{R}_tS_O\leq\frac{1}{2r_i}$, $r_i\leq\zeta_\circ$; this is possible, as R_t lies below every power of L^{-K} ([7, Lemma 3.14](d)), $S_O\leq 6\ell'^3L^{4K}$ and $r_i^{-1}=8\mathsf{M}_t^{\mathrm{rd}}\geq 8\varrho^{-1}\mathsf{m}_t^S$. Then $|\partial_{z_i^\kappa}\bar{\mathcal{E}}(0;V)|\leq r_i^{-1}$. Call a partition good if its blocks are of the forms $\{\mu_\kappa\}$, $\{z_i^\kappa\}$, $\{\mu_\kappa,z_i^\kappa\}$: in a cell κ it has either $\{\mu_\kappa\}$ and all the $\{z_i^\kappa\}$, or one $\{\mu_\kappa,z_{i_0}^\kappa\}$ and the $\{z_i^\kappa\}$, $i\neq i_0$; so the good partitions add up to the product over κ of $\partial_{\mu_\kappa}\bar{\mathcal{E}}\prod_i\partial_{z_i^\kappa}\bar{\mathcal{E}}+\sum_{i_0}\partial_{\mu_\kappa}\partial_{z_{i_0}^\kappa}\bar{\mathcal{E}}\prod_{i\neq i_0}\partial_{z_i^\kappa}\bar{\mathcal{E}}$, each factor of modulus at most $[(1+k)\epsilon_2+\delta_{\mathsf{R}}]\prod_{i\in B}r_i^{-1}$, with $\delta_{\mathsf{R}}=2\mathsf{R}_tS_0+k(6\mathsf{R}_tS_0S_O+2\mathsf{R}_tS_*)$, as $r_i\leq 1$ and $2\epsilon_1+k(\omega+\frac12\epsilon_1)\leq(1+k)\epsilon_2$. Every other partition holds a block across a gap (Step 3), and every block obeys $|\partial^b\bar{\mathcal{E}}(0;V)|\leq G_*\prod_{v\in b}\rho_v^{-1}$ with $G_*=\sup_{\Theta\times\mathsf{Cut}_t}|\bar{\mathcal{E}}|$, by Cauchy's estimate; there are $\mathsf{Bl}_{8(1+k)}$ partitions. So on In°

$$|D|\leq\Big([(1+k)\epsilon_2+\delta_{\mathsf{R}}]\prod_{i\in B}r_i^{-1}\Big)^8+\mathsf{Bl}_{8(1+k)}\,\tau_t(\tfrac12\mathsf{d})\max\{1,G_*\}^{8(1+k)}(\mu^\sharp)^{-8}\zeta_\circ^{-8k},$$

and Step 2, $(x^8+y+y')^{1/8}\leq x+y^{1/8}+y'^{1/8}$ and $r_i^{-1}\leq\mathsf{M}$ give the claim with $\vartheta_K=\delta_{\mathsf{R}}+\mathsf{Bl}_{8(1+n)}^{1/8}\tau_t(\frac12\mathsf{d})^{1/8}\max\{1,G_*\}^{1+n}(\mu^\sharp)^{-1}\zeta_\circ^{-n}+\mathsf{R}_t^{1/8}S_H$. As $\varkappa^*=1$, the suprema S_0, S_* are at most $N\hat c\,\ell'^3L^{3K}$ ([7, Lemma 5.20](c)); with $|\mathfrak{J}_0-\alpha_\emptyset|\leq 2S_0$, $|\mathfrak{J}_i-\alpha_i|\leq 2S_*+2S_0S_O$ and $|O_i-\bar o_i|\leq 2S_O$ one may take $S_H=(n+1)(2+2S_O)^n(2S_0+2S_*+2S_0S_O)$, a power of L^K times a number; and G_*, which holds the members of the families (of norms below E_0, summed with Proposition YM3_I: 45.1(a4)), the source term A^{η_t} and the polynomial terms on Θ, is a power of L^K times a number; this gives the bound of ϑ_K. □

Theorem BD (Hard combinations in every law of the transfer)**.** In the data of the transfer let $\mathbb{J}$ be a local insertion tuple over the sources g_i as in Proposition 18.58, but with sources of any norms and with $\varkappa^* = |\varkappa_0|_\infty + \max_i |\varkappa_i|_\infty / \|g_i\|_\sharp$, and let its read-outs satisfy (OI-C) and (OI-R) of that Proposition, with the radii $r_i = (8\mathsf{M}_t^{\mathrm{rd}}\|g_i\|_\sharp)^{-1}$ and numbers $0 < \epsilon_1 \le B_1$. Then for $(K,\beta) \in \mathfrak{d}(g)$ with $K \ge K_\star$ and $16a \le \mathsf{d}$, every law $\rho \in \mathfrak{S}_{K,\beta}$, every $B \subset \{1,\dots,n\}$ with $\#B \ge 2$ and every $0 < \sigma \le 1$,

$$|\mathsf{T}_\rho[\mathbb{J}](B)| \le \mathsf{c}_{\#B}\,\mathsf{Q}\,[(1+n)\epsilon_2 + \varkappa^*\vartheta_K]\,\mathsf{M}^{\#B}\prod_{i\in B}\|g_i\|_\sharp,$$

with ϵ_2, ϑ_K and $K_\star$ of Proposition 18.58 and the numbers c_k of Lemma 18.54 at the number $\bar{P}$ of Lemma 18.55(b). The numbers c_k and Q read g, n, R and d, and no K, no β, no side and no law. ■□

Sketch. Normalisation. Replacing $(g_i, \mathfrak{J}_i)$ by $(g_i/\|g_i\|_\sharp, \mathfrak{J}_i/\|g_i\|_\sharp)$ divides $\mathsf{T}_\rho[\mathbb{J}](B)$ by $\prod_{i\in B}\|g_i\|_\sharp$ and leaves $\varkappa^*$, (OI-C) and (OI-R) as they are: the new read-out at ζ is the old one at $\zeta/\|g_i\|_\sharp$, and the radius r_i is multiplied by $\|g_i\|_\sharp$. Dividing the tuple by $\varkappa^*$ divides $\mathsf{T}_\rho[\mathbb{J}](B)$, ϵ_1, B_1 and ϵ_2 by $\varkappa^*$. So we may take $\|g_i\|_\sharp = 1$ and $\varkappa^* = 1$.

The moments. Let $\alpha_\emptyset$, α_i, $\bar{o}_i$ and H_A be those of Proposition 18.58. Every H_A, $A \subset B$, is a bounded real Borel function of the bonds inside $\square_0$, all the carriers lying in A_{R+2}; Lemma 18.55(a) and Proposition 18.58 give $|\rho(H_A)| \le \eta\prod_{i\in A}\gamma_i$ with $\eta = \mathsf{Q}[(1+n)\epsilon_2 + \vartheta_K]$ and $\gamma_i = \mathsf{M}$, and Lemma 18.55(b) gives $|\rho(\prod_{i\in C}(O_i - \bar{o}_i))| \le \bar{P} \le \bar{P}\prod_{i\in C}\gamma_i$, as $\mathsf{M} \ge 1$. Lemma 18.54, applied in the law ρ with these numbers, gives $|\mathsf{T}_\rho[\mathbb{J}](B) - \alpha_B| \le \mathsf{c}_{\#B}\,\eta\,\mathsf{M}^{\#B}$, and $\alpha_B = 0$ for $\#B \ge 2$. □

Theorem BE (The dilation bulk term with one constant in every periodic Gibbs state and on all tori above one side)**.** Let $0 < g \le \gamma$, $n \ge 2$, $R \ge 1$, $0 < \mathsf{d} \le 1$, $\delta_1 > 0$ and $c < \frac{1}{14}$, and let $\ell' = \ell'(g,n,R)$ be the side of the data of the transfer. There are p, C and K_1, functions of g, n, R, d, δ_1 and c, such that for every first-passage datum $(K,\beta) \in \mathfrak{d}(g)$ with $K \ge K_1$, all real $\chi, f_1, \dots, f_n \in C^{1,1}(\mathbb{R}^3)$ that vanish outside A_R, with the supports of the f_i at pairwise sup distances $\ge \mathsf{d}$ and $\chi = 1$ on the closed sup δ_1-neighbourhood of $\bigcup_i \operatorname{supp} f_i$, and every law ν that is a periodic Gibbs state at (a_K, β) or the measure of a torus T_ℓ with $\ell \in \mathfrak{L}(g)$, $\ell \ge \ell'$ (the functions read in its chart):

$$|\mathfrak{U}_K^{\mathrm{dil}}[\nu](\underline{f})[\chi]| \le CK^pL^{-cK}\|\chi\|_{C^{1,1}}\prod_{i=1}^n\|f_i\|_{C^{1,1}},$$

$$\begin{aligned}&\Big|\mathsf{FI}_K[\nu](\underline{f})[\chi]\\ &\quad - \sum_{i=1}^n S_n^T[\nu](f_1,\dots,f_i - \nabla_{\mathrm{dil}}^s f_i,\dots,f_n)\\ &\quad + \mathsf{b}\,S_{n+1}^T[\nu](\chi,\underline{f})\Big|\\ &\quad \le CK^pL^{-cK}\|\chi\|_{C^{1,1}}\prod_{i=1}^n\|f_i\|_{C^{1,1}}.\end{aligned}$$

On these tori the part of the bulk term that comes from the tuple $\mathbb{J}^{\mathrm{s}}$ of Proposition 18.53(b), the soft combination $\mathsf{T}_\nu[\mathbb{J}^{\mathrm{s}}](\{1,\dots,n\})$, is moreover at most $Ca_K\|\chi\|_{C^{1,1}}\prod_{i=1}^n\|f_i\|_{C^{1,1}}$. With a larger C the first bound holds at every $(K,\beta) \in \mathfrak{d}(g)$, and the second at every one with $a_K \le \delta_1$. One C serves for all these tori and all these states at once. ■□

Sketch. Vanishing sources. Let first $f_{i_0} = 0$ for some i_0. Then $\mathcal{O}(f_{i_0}) = 0$, and $\mathfrak{X}^{(f_{i_0})}$, $\mathcal{O}(\mathsf{L}f_{i_0})$ and $\mathcal{O}(f_{i_0} - \nabla_{\mathrm{dil}}^s f_{i_0})$ are 0 too, all three being linear in f_{i_0}. Every joint cumulant in the bulk term, in the flux, in the sums S_n^T and S_{n+1}^T of the second display and in the soft combination has one of these four as an entry: $\mathcal{O}(f_{i_0})$ wherever it stands, and in the remaining terms, those of the index i_0, one of the other three. So all these cumulants vanish, and with them both members of each

bound of the statement, the right ones through the factor $\|f_{i_0}\|_{C^{1,1}} = 0$. In what follows every f_i is non-zero; then $\bigcup_i \operatorname{supp} f_i$ is not empty, $\chi = 1$ on it, and $\|\chi\|_{C^{1,1}} \geq \sup|\chi| \geq 1$.

Choices. We may take $0 < c < \frac{1}{14}$. Let $\delta = \frac{1}{14} - c$, $\tau = \delta$, $\phi = \delta/(30(n+1))$, $\sigma = \delta/30$, $\kappa_{\mathrm{H}} = \frac{3}{4}$ and $\psi_{\mathrm{c}} = \frac{1}{4}$, and let $K_1 \geq \max\{K_\phi, 5\}$ be so large that $16a_{K_1} \leq \min\{\mathsf{d}, \delta_1\}$, $8\mathsf{M}_t^{\mathrm{rd}} \geq 1$, and that the thresholds $K_\star$ of Theorem BD for the ten tuples below and K_ψ, K_2 of Lemma 18.57 are reached. Let $K \geq K_1$; the sources of the transfer are the f_i.

The tuples. As $2a \leq \delta_1$ and $4a < \mathsf{d}$, Proposition 18.53(b) writes the bulk term in ν as the sum of the ten hard combinations $\mathsf{T}_\nu[\mathbb{J}^\pi](\{1,\dots,n\})$, and the laws ν of the statement are laws of the transfer. By Proposition 18.53(c) every tuple is local, with $\mathfrak{J}_0$ carried by A_{R+1} and $\mathfrak{J}_i$ within the sup distance $4a$ of $\operatorname{supp} f_i$; with $\|\cdot\| = \|\cdot\|_{C^{1,1}}$, $\mathsf{N} = \|\chi\| + \max_i \|f_i\|/\|f_i\|_\sharp$, and C_R now a function of R and g (it absorbs $\mathsf{b} \leq \mathsf{b}_+$), its number $\varkappa^*$ is at most $C_R\mathsf{N}$ for $\mathbb{J}^1$, $\mathbb{J}^2$, $\mathbb{J}^3$, $\mathbb{J}^{\mathfrak{R}}$, $\mathbb{J}^{\mathrm{s}}$, at most $C_R L^K \mathsf{N}$ for $\mathbb{J}^\psi$, $\mathbb{J}^{a^2}$, $\mathbb{J}^4$, $\mathbb{J}^5$, and at most $C_R L^{2K}\mathsf{N}$ for $\mathbb{J}^\sharp$.

The read-outs. The read-out of the cell κ and the source i is that of $\mathfrak{J}_0^{[\kappa]} + \zeta\mathfrak{J}_i^{[\kappa]}$ at the source ζf_i^κ. The copies of a local observable are local observables of a type formed from its type alone, with weights of the same moduli ([7, Lemma 5.20](d)). The copy of $\sum_p \varkappa(p)\psi(p)$ is $\sum_p \varkappa(\mathsf{r}_\kappa^{-1}p)\psi(p)$: a diagonal isometry r maps bonds to bonds with $(\mathsf{r}U)(\mathsf{r}b) = U(b)$, so the holonomy of $\mathsf{r}U$ along the image of a closed path is that of U along the path; around the image of a plaquette or of a planar 1×2 rectangle this image path may start at another corner and run in the other sense, which changes the holonomy by a conjugation or an inversion and leaves its trace, and so the plaquette or rectangle term, as it is; hence r carries the plaquette terms and the terms of the planar 1×2 rectangles to those of the image plaquettes and rectangles, and $\psi(p)$ is symmetric in the two pairs of neighbours of p in each direction of its plane. The copies of $\beta\mathfrak{K}^\sharp[\Xi]$ and of $a^{-1}\mathfrak{K}^\sharp[f_i\Xi]$ are $\beta\mathfrak{K}^\sharp[\Xi^\kappa]$ and $a^{-1}\mathfrak{K}^\sharp[f_i^\kappa\Xi^\kappa]$, $\Xi^\kappa_\rho = s_\rho\,\Xi_\rho \circ \mathsf{r}_\kappa^{-1}$ with the signs s_ρ of r_κ, and $\mathsf{H}(\Xi^\kappa) = \mathsf{H}(\Xi)$, as $\mathfrak{r}^\sharp$ transforms as a vector under the coordinate mirrors (Lemma 5.9(b)); so the cubic read-out is that of $\mathfrak{K}^\sharp[(\beta - a^{-1}\zeta f_i^\kappa)\Xi^\kappa]$ at the source ζf_i^κ. For $|\zeta| < r_i$ the weights of $\mathfrak{J}_0^{[\kappa]} + \zeta\mathfrak{J}_i^{[\kappa]}$ are at most $\varkappa^*$. Hence (OI-C) holds with $B_1 = C_B L^{3K}\varkappa^*$ (Lemma 18.56(c)), and (OI-R) holds with $\epsilon_1 = \mathsf{o}_t$ for $\mathbb{J}^{a^2}$, $\mathbb{J}^v$, $\mathbb{J}^{\mathfrak{R}}$, $\mathbb{J}^{\mathrm{s}}$ (Lemma 18.57(a)), with $\epsilon_1 = \epsilon_\psi\varkappa^*$ for $\mathbb{J}^\psi$ (Lemma 18.57(b)), and with $\epsilon_1 = C_2\mathsf{H}(\Xi)K^{p_2}L^{-c_0'K}$, $c_0' = \frac{1}{14} - \frac{\delta}{7}$, for $\mathbb{J}^\sharp$ (Lemma 18.57(c)), where $3\phi < \frac{\delta}{7}$. A tuple with $\varkappa^* = 0$ has only vanishing members, so that its hard combinations vanish; we leave it out. For the others $B_1 > 0$, and (OI-C) gives (OI-R) with the number $\mathsf{a} = 0$ and with B_1 in the place of ϵ_1; as (OI-R) persists when its number grows, it holds with every number of $(0, B_1]$ that is not below $\min\{\epsilon_1, B_1\}$. Theorem BD applies with each of these numbers in the place of ϵ_1; since ϵ_2 is continuous and increasing in ϵ_1, its bound then holds with ϵ_2 formed from the numbers ϵ_1 listed above.

The rates. With $\epsilon_2 \leq \frac{C}{\sigma}\epsilon_1^{1-\sigma}B_1^\sigma + 2\epsilon_1$, $u \leq \phi K + 1$, $\mathsf{M}^n \leq CK^{2n\mathfrak{p}_0}L^{3n\phi K}$, and the bounds of o_t and of ϵ_ψ, in which $2\kappa_{\mathrm{H}} - (3+2\kappa_{\mathrm{H}})\psi_{\mathrm{c}} = \frac{3}{8}$ and $\frac{1}{2}\psi_{\mathrm{c}} = \frac{1}{8}$, the number $\epsilon_2\mathsf{M}^n$ is at most $CK^p\mathsf{N}$ times L^{-rK} with $r = (1-\sigma)(1-4\phi) - 3\sigma - 3n\phi$ for $\mathbb{J}^1$, $\mathbb{J}^2$, $\mathbb{J}^3$, $\mathbb{J}^{\mathfrak{R}}$, $\mathbb{J}^{\mathrm{s}}$; $r = (1-\sigma)(\frac{1}{2} - 4\phi) - 4\sigma - 3n\phi$ for $\mathbb{J}^{a^2}$, $\mathbb{J}^4$, $\mathbb{J}^5$; $r = (1-\sigma)(\frac{1}{8} - 3\phi) - 4\sigma - 3n\phi$ for $\mathbb{J}^\psi$; and $r = (1-\sigma)c_0' - 5\sigma - 3n\phi$ for $\mathbb{J}^\sharp$. As $3n\phi \leq \frac{\delta}{10}$ and $\sigma = \frac{\delta}{30}$, the last is at least $\frac{1}{14} - \frac{\delta}{7} - \frac{71}{14}\sigma - \frac{\delta}{10} = \frac{1}{14} - \frac{173}{420}\delta > c$, and the other three exceed $\frac{1}{14}$, as $\delta < \frac{1}{14}$. The term $\varkappa^*\vartheta_K\mathsf{M}^n$ lies below every power of L^{-K} times N, as $\mathsf{R}_t \leq C_{\mathrm{R}}L^{3u}e^{-c'A_0^2(\phi K\log L)^{2\mathfrak{p}_0}}$ ([7, Lemma 3.14](d)) and $\tau_t(\frac{1}{2}\mathsf{d}) \leq e^2E_0C_{\mathrm{E},0}tN_1e^{-(\frac{1}{2}\mathsf{d}-2a)L^u/M}$. Theorem BD at $B = \{1,\dots,n\}$ bounds each of the ten hard combinations by $CK^pL^{-cK}\mathsf{N}\prod_i\|f_i\|_\sharp$, in every law of the transfer, and $\mathsf{N}\prod_i\|f_i\|_\sharp \leq (n+1)(40M)^n\|\chi\|\prod_i\|f_i\|$, because $\|\chi\| \geq \sup|\chi| \geq 1$.

The soft combination on the tori. Let ν be the measure of T_ℓ, $\ell \in \mathfrak{L}(g)$, $\ell \geq \ell'$, so that $\ell \geq 128(R+1)$; then $\mathsf{T}_\nu[\mathbb{J}^{\mathrm{s}}](\{1,\dots,n\})$ equals $\mathsf{b}\langle\mathcal{O}(\mathsf{L}\chi);\mathcal{O}(f_1);\dots;\mathcal{O}(f_n)\rangle_\nu + 3a\langle\mathcal{O}(\chi^\square);\mathcal{O}(f_1);\dots;\mathcal{O}(f_n)\rangle_\nu - \sum_i\langle\mathcal{O}(\mathsf{L}f_i);(\mathcal{O}(f_l))_{l\neq i}\rangle_\nu$, a sum of joint cumulants on T_ℓ of action densities of weights of the plaquettes. To use Lemma 18.59, write each weight as a Lipschitz function on T_ℓ. Two distinct barycentres of plaquettes are at least $a/\sqrt{2}$ apart, since two of their coordinates differ by at least $\frac{a}{2}$ when their planes differ; so for a real weight v with $|v| \leq s_0$ and $\Lambda = 2\sqrt{2}s_0/a$ the function

$\tilde v = \max_p[v^+(p) - \Lambda\,\mathrm{dist}(\cdot, x(p))]_+ - \max_p[v^-(p) - \Lambda\,\mathrm{dist}(\cdot, x(p))]_+$, with the distance of T_ℓ, takes the value $v(q)$ at every $x(q)$, as every other term there is at most $s_0 - \Lambda a/\sqrt{2} = -s_0$; each of its two maxima is Λ-Lipschitz with values in $[0, s_0]$, so that $\|\tilde v\|_\sharp \le \max\{s_0, 80M\Lambda\} \le 227 M s_0/a$; $\tilde v$ vanishes beyond the distance $\frac{a}{2}$ of the barycentres at which $v \ne 0$; and $\mathcal{O}(v) = \mathcal{O}(\tilde v)$. Second differences give $|\mathsf{L}\varphi| \le a^2 \max_\rho \mathrm{Lip}_\infty \partial_\rho\varphi \le \sqrt{3}a^2\|\varphi\|$ and, the midpoints of the four bonds of p being $x(p) \pm \frac{a}{2}e_\lambda$, $\lambda \in P(p)$, $\chi^\square(p) = \chi(x(p)) + r(p)$ with $|r(p)| \le \frac{1}{8}a^2 \mathrm{Lip}\,\nabla\chi$. So the entries other than the sources are $\mathcal{O}(\widetilde{\mathsf{L}\chi})$, $\mathcal{O}(\chi) + \mathcal{O}(\tilde r)$ and $\mathcal{O}(\widetilde{\mathsf{L}f_i})$, with $\|\widetilde{\mathsf{L}\chi}\|_\sharp \le 400Ma\|\chi\|$, $\|\tilde r\|_\sharp \le 29Ma\|\chi\|$, $\|\chi\|_\sharp \le 40M\|\chi\|$ and $\|\widetilde{\mathsf{L}f_i}\|_\sharp \le 400Ma\|f_i\|$; all these functions, and the f_l, vanish outside A_{R+1}, on which the chart of T_ℓ keeps the distances, the suprema and the Lipschitz constants of $\mathbb{R}^3$. In each of these cumulants some two of the supports lie at a distance $\ge \frac{1}{2}\mathsf{d}$ from each other: those of two of the sources among its entries, or, for $\langle \mathcal{O}(\widetilde{\mathsf{L}f_i}); \mathcal{O}(f_l)\rangle_\nu$ at $n = 2$, the support of $\widetilde{\mathsf{L}f_i}$, which lies within $2a$ of that of f_i, and that of f_l, as $4a \le \mathsf{d}$. Lemma 18.59 bounds each cumulant by a number formed from g, n, R and d times the product of the norms $\|\cdot\|_\sharp$ of its entries; with $\mathsf{b} \le \mathsf{b}_+$, $\|f_l\|_\sharp \le 40M\|f_l\|$ and $\|\chi\| \ge 1$ this gives $|\mathsf{T}_\nu[\mathbb{J}^{\mathrm{s}}](\{1,\dots,n\})| \le C a_K \|\chi\| \prod_i \|f_i\|$.

Small K and the flux. For the finitely many $K < K_1$ the moment–cumulant formula bounds each cumulant of the bulk term, in every law, by op_{n+1} times the product of the suprema of its entries. Each of these suprema is bounded by the norm of the function it is built on, χ or one of the f_i, times a number formed from K_1, R, b_+ and the constants: the label enters $\mathfrak{X}^{(0)}_\chi$ through $\beta = L^K\mathsf{b} \le L^{K_1}\mathsf{b}_+$ and through $\mathsf{b} \le \mathsf{b}_+$ ([7, Lemma 3.12](a)). With $\|\chi\| \ge 1$ and $K^p L^{-cK} \ge L^{-cK_1}$ there, the first bound follows with a C that is still a function of g, n, R, d, δ_1 and c, since $\mathsf{b}_+ = X_g + \bar B$ is formed from g and the constants. The second display is Proposition 18.53(a), χ covering the f_i once $a_K \le \delta_1$. □

Lemma 18.59 (A cumulant bound with one separated pair)**.** Fix g, an integer $q \ge 2$, and $R, \delta > 0$. On every sufficiently large torus in $\mathfrak{L}(g)$, a tuple of real Lipschitz functions $v_1, \dots, v_q$ supported in A_R with one pair of supports at Euclidean distance at least δ satisfies

$$|\langle \mathcal{O}(v_1); \dots; \mathcal{O}(v_q)\rangle_{K,\ell,\beta}| \le C(g, q, R, \delta) \prod_i \|v_i\|_\sharp$$

uniformly over first-passage data at g. The same bound holds for the torus laws in the dilation bulk estimate. ■

Proof. Place all supports in the central cell of a board of fixed side $h = L^m$, with positive wall margin. Take $2h \in \mathfrak{L}(g)$ and $\ell \ge 2h$, so the reference torus is admissible and the central cell has a distinct image on T_ℓ. One may use the board chosen for the transfer in the bulk estimate when $q \le n + 1$ and the supports are in A_{R+1}. It contains this ball with the required margin. The fixed-board source normalizer of Proposition YM3_I:89.2, with the first-passage bounds used in the proof of Proposition YM3_I:91.12, gives a holomorphic $E(z)$ with $E(0) = 0$. Its normalized exponential mean has one bound on the source polydisc independent of the data and side. The full mixed derivative of E has the bound YM3_I:(1793), after numbering the separated pair first. That bound uses only the separation of the first two supports, as its preceding cell estimate shows. The cumulant of the normalized exponential mean is bounded by Proposition YM3_I:91.11. Adding the bound for the mixed derivative of E gives the stated estimate. All board, wall and radius constants are fixed by g, q, R, δ before the cutoff and torus are chosen. □

Corollary 18.60 (The local dilation identity with the flux alone in the limits)**.** Let $0 < g \le \gamma$, let $(K_j, \beta_j, \nu_j)_{j\in\mathbb{N}}$ be a Gibbs sequence at g, in the sense of the subsection on the scalar two-point function, $\mathsf{b}_j = a_{K_j}\beta_j$, let σ' be a limit in its frame with approximants $\rho_j \in \mathcal{G}^{\mathrm{per}}_{K_j,\beta_j}$, and put $S'_q = S^{\sigma'}_q$. Let $n \ge 2$, $R, \mathsf{d}, \delta_1 > 0$, and let $f_1, \dots, f_n \in C^3(\mathbb{R}^3)$ and $\chi \in C^{1,1}(\mathbb{R}^3)$ be real functions that vanish outside A_R, with the supports of the f_i at pairwise sup distances $\ge \mathsf{d}$ and $\chi = 1$ on the closed sup δ_1-neighbourhood of $\bigcup_i \mathrm{supp}\, f_i$; $(y \cdot \nabla f)(y) = \sum_\rho y_\rho \partial_\rho f(y)$.

(a) Along the full sequence $\mathfrak{U}^{\mathrm{dil}}_{K_j}[\rho_j](\underline{f})[\chi]\to 0$, and

$$\mathsf{b}_j\, S^T_{n+1}[\rho_j](\chi,\underline{f})+\mathsf{Fl}_{K_j}[\rho_j](\underline{f})[\chi]\ \longrightarrow\ nS'_n(\underline{f})-\sum_{i=1}^n S'_n(f_1,\dots,y\cdot\nabla f_i,\dots,f_n).$$

(b) Along every subsequence on which b_j and $S^T_{n+1}[\rho_j](\chi,\underline{f})$ converge, to b and to $S^+_{n+1}(\chi;\underline{f})$ say, the flux $\mathsf{Fl}_{K_j}[\rho_j](\underline{f})[\chi]$ converges, to $\mathsf{Fl}'(\underline{f})[\chi]$ say, and

$$nS'_n(\underline{f})-\sum_{i=1}^n S'_n(f_1,\dots,y\cdot\nabla f_i,\dots,f_n)=\mathsf{b}\,S^+_{n+1}(\chi;\underline{f})+\mathsf{Fl}'(\underline{f})[\chi]:$$

beside the term of the label, the flux alone carries the rest.

(c) The numbers $S^T_{n+1}[\rho_j](\chi,\underline{f})$ are bounded uniformly in j, so that the subsequences of (b) exist, (i) if $\rho_j\in\mathcal{G}^{\mathrm{tor}}_{K_j,\beta_j}$ for every j, and (ii) for every choice of the approximants $\rho_j\in\mathcal{G}^{\mathrm{per}}_{K_j,\beta_j}$ if $0<g\le\gamma^{\mathrm{ref}}_W$.

(d) (a)–(c) hold for every volume-first limit S at g, with $\sigma'=\sigma_S$, $S'_q=S_q$ for $q\ge 2$ and $\rho_j=\nu_j$ a defining sequence of states of S, whose states lie in $\mathcal{G}^{\mathrm{tor}}$, so that (c)(i) applies. For a convergent Gibbs sequence at g, (a) and (b) hold for every component σ_ξ of every canonical decomposition of its limit, an ergodic component with $\ker H_{\sigma_\xi}=\mathbb{C}\,\Omega_{\sigma_\xi}$, with the approximants $\rho_j=(\nu_j)_{1_{E_j}}\in\mathcal{G}^{\mathrm{per}}_{K_j,\beta_j}$ of Theorem YM3_I:AP; when $0<g\le\gamma^{\mathrm{ref}}_W$, (c)(ii) applies to these approximants, so that the subsequences of (b) exist, while for $\gamma^{\mathrm{ref}}_W<g\le\gamma$ the components carry what (a) states, and the flux alone enters along every subsequence on which b_j and $S^T_{n+1}[\rho_j](\chi,\underline{f})$ converge.

■□

Sketch. (a) We may take $R\ge 1$ and $\mathsf{d}\le 1$. For the j with $a_{K_j}\le\delta_1$, χ covers the f_i, and Proposition 18.53(a) at $\nu=\rho_j$ gives $\mathsf{b}_jS^T_{n+1}[\rho_j](\chi,\underline{f})+\mathsf{Fl}_{K_j}[\rho_j](\underline{f})[\chi]=\sum_i S^T_n[\rho_j](f_1,\dots,f_i-\nabla^s_{\mathrm{dil}}f_i,\dots,f_n)-\mathfrak{U}^{\mathrm{dil}}_{K_j}[\rho_j](\underline{f})[\chi]$. The bulk term tends to 0 by Theorem BE, as $K_j\to\infty$. Write $\nabla^s_{\mathrm{dil}}f_i=y\cdot\nabla f_i+e_{i,j}$: the function $e_{i,j}$ vanishes beyond sup distance a_{K_j} of $\operatorname{supp}f_i$, and Taylor's formula to the third order gives $\|e_{i,j}\|_\sharp\le C_R a_{K_j}\|f_i\|_{C^3}$. For large j the tuples with the entry $e_{i,j}$ have supports at pairwise Euclidean distances $\ge\frac12\mathsf{d}$ in balls of radius $2R+2$, and Proposition YM3_I:91.13 bounds their cumulants by $C^{\mathrm{per}}_n(\frac12\mathsf{d};2R+2)\|e_{i,j}\|_\sharp\prod_{l\ne i}\|f_l\|_\sharp\to 0$. The tuples $\underline{f}$ and $(f_1,\dots,y\cdot\nabla f_i,\dots,f_n)$ are separated tuples of C^1_c functions, and on them $S^T_n[\rho_j]\to S'_n$ by Proposition YM3_I:101.3, applied to the Gibbs frame classes of Proposition YM3_I:106.3.

(b) follows from (a): along such a subsequence $\mathsf{Fl}_{K_j}[\rho_j](\underline{f})[\chi]$ is the difference of two convergent sequences.

(c)(i) Each $\rho_j\in\mathcal{G}^{\mathrm{tor}}_{K_j,\beta_j}$ is a weak limit of the laws $\tilde\mu_{K_j,\ell_k,\beta_j}$ along sides $\ell_k=2L^{m_k}\to\infty$, which lie in $\mathfrak{L}(g)$ and exceed $16(n+1)(R+1)$ for large k. The joint cumulant of the continuous functions $\mathcal{O}(\chi),\mathcal{O}(f_1),\dots,\mathcal{O}(f_n)$ of finitely many bonds passes to the weak limit, and under $\tilde\mu_{K_j,\ell_k,\beta_j}$ it is the cumulant on T_{ℓ_k} of the periodised functions, at most $C(g,n+1,R,\mathsf{d})\|\chi\|_\sharp\prod_i\|f_i\|_\sharp$ by Lemma 18.59: the supports lie in sup-balls of radius R, those of f_1 and f_2 are at Euclidean distance $\ge\mathsf{d}$, and the support of χ may lie anywhere; the number reads no K, no β and no side.

(c)(ii) Let $0<g\le\gamma^{\mathrm{ref}}_W$. The flux insertion is a rough entry of Lemma 5.7 carried within $2a$ of $\mathrm{Var}(\chi)$: the petal form of $\mathfrak{d}_{b,p,\rho}$ that Lemma 5.7 uses for $h_{\omega,\chi}$, with $(x_b)_\rho$ in the place of $(\omega x_b)_\rho$, gives $-\beta\sum_p h^{\mathrm{dil}}_\chi(p)=\mathsf{b}\,\mathrm{Pf}(k^{\mathrm{dil}})$ with $|k^{\mathrm{dil}}|_1\le 12\,\mathrm{Lip}_\infty(\chi)(3R+1)$, and $\mathfrak{B}^{\mathrm{dil}}_\chi-\mathsf{b}\,\mathrm{Pf}(k^{\mathrm{dil}})=\mathcal{O}(v)$ with $v(p)=\mathsf{b}\nabla^s_{\mathrm{dil}}\chi(x(p))-\frac32 a\,v^{\mathrm{dil}}_\chi(p)$ and $|v|_\infty\le(\mathsf{b}+2a)(3R+1)\mathrm{Lip}_\infty\chi$, each of these vanishing at a plaquette unless χ varies on the points $x(p)\pm ae_\rho$, $x(p)\pm\frac a2 e_\rho$, x_b that it reads. As $\chi=1$ on the closed sup δ_1-neighbourhood of the supports, $\mathrm{Var}(\chi)$ lies at sup distance $\ge\delta_1$ from every $\operatorname{supp}f_i$. The rule of Theorem N(a)(ii), for a rough entry carried within $2a$ of a closed set at distance $\ge\frac12\min\{\delta_1,1\}$ from each support, bounds $|\mathsf{Fl}_{K_j}[\rho_j](\underline{f})[\chi]|$ by a number free of j, of the side and

of the state once K_j is above a threshold. With (a), the bound of $\mathfrak{U}^{\mathrm{dil}}$ in Theorem BE and the bound of Proposition YM3_I:91.13 for the separated tuples $(f_1, \dots, f_i - \nabla^s_{\mathrm{dil}} f_i, \dots, f_n)$, this bounds $\mathsf{b}_j S^T_{n+1}[\rho_j](\chi, \underline{f})$, and $\mathsf{b}_j \geq \gamma^{-2}$; the finitely many other j give finitely many numbers.

(d) In a defining sequence of a volume-first limit, each ν_j is a limit of periodic torus laws at (K_j, β_j) and hence is periodic Gibbs. The convergent cumulants determine all limits of the self-centered products by the moment–cumulant formula. Thus the sequence converges to σ_S with $S_q^{\sigma_S} = S_q$ for $q \geq 2$. This is a limit in its own frame with $\rho_j = \nu_j$. For every convergent Gibbs sequence, Theorem YM3_I:AP obtains each component in the frame of the full sequence by conditioning on invariant events of positive probability, and proves $\ker H_{\sigma_\xi} = \mathbb{C}\Omega_{\sigma_\xi}$. □

18.6. **A logarithmic rate in the leading law of two points.** For $0 < g \leq \gamma_W$ the main text gives the leading law $c_2 b^{-2} \mathcal{K}_6$ of the two-point function with a tolerance $\eta_{g,\vartheta}(d)$ that is one function of the scale d for every state at g at a first-passage datum with $s(d) < K$, for the torus of every side at g, for the continuum limits at g on a torus, along growing tori and volume first, for the limits through periodic Gibbs states and for every component of a canonical decomposition, and that tends to 0 with d (Theorems YM3_I:AV and YM3_I:AW, Proposition YM3_I:127.8, Corollary YM3_I:127.11). Here this tolerance is bounded by an explicit function of d. The couplings $0 < g \leq \gamma_W$ whose carried coupling $\check{g}$ does not exceed γ_W^{sd} form an interval $(0, \gamma_{\mathrm{rt}}]$; it contains $(0, \gamma_{\mathrm{rt}}^\flat]$, where $\gamma_{\mathrm{rt}}^\flat > 0$ is a number formed at the values of the constants in which no least tolerance of a two-point function occurs, and it is all of $(0, \gamma_W]$ as soon as $\gamma_W^{\mathrm{sd}} \geq \gamma^2$ (Proposition 18.61). For these g, every $\vartheta \in (0,1)$ and every d in a set $\mathsf{Z}(g)$ of scales that does not depend on ϑ and contains all small scales, $\eta_{g,\vartheta}(d) \leq C_{\mathrm{un}} \vartheta^{-3} (\log \frac{L}{g^2 d})^{-2p_0} + \vartheta^{-4} \mathsf{C}^\circ_{\mathrm{law}}(g)\, d$, where $p_0 \geq 13$, the number C_{un} depends on the constants alone, and $\mathsf{C}^\circ_{\mathrm{law}}(g)$ is an explicit expression in the radius, the constants and g (Proposition 18.63); below the scale $L^2 \mathsf{d}_0^{\mathrm{ref}}(\check{g}, \vartheta)$ the same holds with a constant $C^{\mathrm{ref}}(\check{g}, \vartheta)$ in the place of $C_{\mathrm{un}} \vartheta^{-3}$. Consequently each of the laws listed above holds with these relative errors at the scales $d \leq \mathsf{d}_0(g, \vartheta)$ of the respective ranges (Proposition 18.64).

We use the constants and notation of Section 8.1, with the scalar two-point law of Theorems YM3_I:AV and YM3_I:AW. The tolerance $\eta_{g,\vartheta}(d)$ and the threshold $\mathsf{d}_0(g, \vartheta)$ are those in YM3_I:(2349)–(2350). The cutoff spacing is $a_K = L^{-K}$, the physical label is $b = a_K \beta$, the first-passage endpoint is $X_g = L(g^{-2} + B_\sharp)$, and $\mathfrak{d}(g)$ and $\mathfrak{L}(g)$ are the sets of first-passage data and admissible sides. We use $r \geq 2$ and $p_0 = \mathfrak{p}_0 \geq 6r + 1$, hence $p_0 \geq 13$, as in the construction constants of Theorem YM3_I:D. The classes $\mathfrak{T}_{\mathbb{E}}(d; \vartheta)$ are the scalar test pairs defined above, and $\mathfrak{T}^1_{\mathbb{E}}(d; \vartheta)$ restricts their entries to C^1. Write $c_2 = 9/(16\pi^2)$.

A state at a datum at g means a periodic Gibbs state on the cutoff lattice or the periodic extension of the Wilson law on a torus of side in $\mathfrak{L}(g)$, as in Section YM3_I:127. Let $\mathcal{G}^{\mathrm{per}}_{K,\beta}$ be the periodic Gibbs laws, $\mathcal{G}^{\mathrm{inv}}_{K,\beta}$ its translation-invariant members and $\mathcal{G}^{\mathrm{tor}}_{K,\beta}$ the weak limit points of the periodic extensions of the cubic Wilson tori at fixed (K, β). A separated tuple is a family of real C^1 functions with compact pairwise disjoint supports, on the relevant torus or on $\mathbb{R}^3$. We use the frame and the centered products X^ν_K defined in the subsection on the scalar two-point function. Proposition YM3_I:106.3 supplies the common bound system for these Gibbs classes.

A fixed-torus limit at g is a limit on all separated tuples at first-passage data at g, on one torus of side in $\mathfrak{L}(g)$ as $K_j \to \infty$. A growing-torus limit at g is formed at (K_j, ℓ_j, β_j) with $\ell_j \in \mathfrak{L}(g)$ tending to infinity, by evaluating at the periodisations of each fixed separated tuple over $\mathbb{R}^3$. A volume-first limit at g is the limit of cumulants of such tuples under $\nu_j \in \mathcal{G}^{\mathrm{tor}}_{K_j,\beta_j}$ at first-passage data with $K_j \to \infty$. A sequence giving one of these limits is a defining sequence. A label of S is a limit of the physical labels along a defining sequence, or a subsequence with converging labels. We write $\mathsf{b}(S)$ when there is just one such value. Theorem YM3_I:AW in fact proves convergence of the labels along every defining sequence in the range of interest.

For a reference coupling $g' \le 1$, an integer $s \ge 0$ and $d > 0$ let $\bar{x}_s[g'] = L^{s+1}(g'^{-2} + \bar{B})$, $Y_s[g'] = \log \bar{x}_s[g']$, $C_W(s)[g'] = \mathsf{c}_W\, Y_s[g']^{2p_0}$ and $s(d)[g'] = \min\{s \in \mathbb{Z} :\ s \ge 0,\ L^s d \ge C_W(s)[g']\}$, where $\mathsf{c}_W = C_{W,0}A_0^2$ and $C_{W,0} = \max\{1, 4L^2M/A_0^2\}$ is the room constant YM3_I:(2320), so that $\mathsf{c}_W \ge 4L^2M$: these are the envelope numbers, the witness rooms and the witness depth. Without a bracket they are taken at $g' = g$; in this subsection $s(d)$ is the witness depth at g, the depth in the hypotheses of Theorem YM3_I:AV. As $s(d)[g']$ is the least element of the set in its definition, $L^{s(d)[g']}d \ge C_W(s(d)[g'])[g']$. We write $R(x)$, for real $x > 1$, for the least power L^u with $u \in \mathbb{Z}$, $u \ge 0$ and $L^u \ge (\log x)^r$.

Let $0 < g' \le \gamma$, $\vartheta \in (0,1)$, $0 < d \le \frac{1}{100}$ and $\ell' = 2M\,R(X_{g'})$. A number $\epsilon \in [0,\infty)$ *serves* at (g', ϑ, d) if $|S^T_{2;K,\ell',\beta}(\Psi) - c_2(a_K\beta)^{-2}\mathcal{K}_6(\Psi)| \le \epsilon\, c_2(a_K\beta)^{-2}\mathcal{K}_6(\Psi)$ for every $(K,\beta) \in \mathfrak{d}(g')$ with $s(d)[g'] \le K$ and every $\Psi \in \mathfrak{T}_{T_{\ell'}}(d;\vartheta)$. The least tolerance $\mathsf{e}_{g',\vartheta}(d) \in [0,\infty]$ is the infimum of these numbers, and when it is finite it serves since the inequalities defining it are closed in the tolerance. The set $\mathfrak{G}$ consists of the $\gamma' \in (0,\gamma]$ such that $\mathsf{e}_{g',\vartheta}(d) \to 0$ as $d \to 0$ for all $g' \in (0,\gamma']$ and all $\vartheta \in (0,1)$, and $\hat{\gamma}_W = \frac{1}{2}\sup\mathfrak{G} \in (0, \frac{1}{2}\gamma]$ is the first ceiling YM3_I:(2345). With $\mathsf{a}_1 := \max\{\hat{\gamma}_W^{-2}, \gamma^{-4}, e^{C_{\mathrm{s}}}\}$ we have $\gamma_W = (L^2(\bar{B} + \mathsf{a}_1))^{-1/2}$, and for $0 < g \le \gamma_W$ the carried coupling $\check{g} = (L^{-2}g^{-2} - \bar{B})^{-1/2}$ is a reference coupling with $\check{g}^{-2} + \bar{B} = L^{-2}g^{-2}$, $\check{g} \le \hat{\gamma}_W$, $\check{g}^{-2} \ge \gamma^{-4}$ and $g^{-2} \ge L^2\check{g}^{-2}$ by YM3_I:(2346)–(2347).

For $0 < g \le \gamma_W$ and $\vartheta \in (0,1)$ use the two-point comparison constant of [1, *Transfer of the two-point law*],

$$C_2^{\mathrm{tr}} = 2\log 2\left(\frac{50}{3}\right)^3 \frac{[160M(\mathsf{Q}_1+1)]^2}{\varrho^2\vartheta^2}, \qquad \mathsf{Q}_1 = \exp[(E_+ + C_{\mathrm{low}} + C_{\mathrm{gap}})L^3 + \Phi_1(2,1;L)],$$

with the constants of that construction at g. Then $\mathsf{C}_{\mathrm{law}}(g,\vartheta) = 10^6 C_2^{\mathrm{tr}}\,\bar{x}_0^2/(c_2\vartheta^2)$ with $\bar{x}_0 = L(g^{-2} + \bar{B})$; for $0 < d \le \frac{1}{100}$, $\eta_{g,\vartheta}(d) = \sup_{0<d'\le L^{-2}d} \mathsf{e}_{\check{g},\vartheta}(d') + \mathsf{C}_{\mathrm{law}}(g,\vartheta)\,d$; and $\mathsf{d}_0(g,\vartheta)$ is one half of the supremum of the $d \in (0, \frac{1}{100}]$ with $\eta_{g,\vartheta}(d) \le \frac{1}{3}$.

For $0 < g' \le \gamma_W^{\mathrm{ref}}$ use the small-torus error functions $\varrho_{g',\vartheta}(s)$ and $\bar{\varepsilon}_{\mathrm{ref}}(d;g',\vartheta)$ and the threshold $\mathsf{d}_0^{\mathrm{ref}}(g',\vartheta) = L^{-s_0(g',\vartheta)-1}$ of YM3_I:(2334)–(2335), with $s_0(g',\vartheta) \ge 3$. Set $\varepsilon^{\mathrm{ref}}(d;g',\vartheta) = \bar{\varepsilon}_{\mathrm{ref}}(d;g',\vartheta)$ and $C^{\mathrm{ref}}(g',\vartheta) = \sup_{s\ge s_0(g',\vartheta)} Y_s[g']^{2p_0}\varrho_{g',\vartheta}(s)$. The error formulas there give a finite supremum. For $0 < d \le \mathsf{d}_0^{\mathrm{ref}}$, the error is non-decreasing in d, takes values in $[0,1/2]$ and is at most $C^{\mathrm{ref}}(g',\vartheta)Y_{s(d)[g']}[g']^{-2p_0}$ since Y_s increases with s. Use γ_W^{ref} and $\gamma_W^{\mathrm{sd}} = \min\{\gamma_W^{\mathrm{ref}}, e^{-Y_{\mathrm{sd}}/2}\}$ from Section YM3_I:126.2 and YM3_I:(2327), with $\gamma_W^{\mathrm{sd}} \le \gamma_W^{\mathrm{ref}} \le \gamma$. Let $Y_{\mathrm{u}} \ge 2\log(2L^3)$ and C_{un} be the constants of Lemma 18.62.

We put $\mathsf{a}_2 := (\gamma_W^{\mathrm{sd}})^{-2}$,

$$\gamma_{\mathrm{rt}} := \big(L^2(\bar{B} + \max\{\mathsf{a}_1, \mathsf{a}_2\})\big)^{-1/2}, \qquad \gamma_{\mathrm{rt}}^{\flat} := \big(L^2(\bar{B} + \max\{4\mathsf{a}_2, \gamma^{-4}, e^{C_{\mathrm{s}}}\})\big)^{-1/2}, \tag{18.x}$$

and, for $0 < g \le \gamma_W$, $d > 0$ and $\vartheta \in (0,1)$, with Q_1 taken at g,

$$\begin{aligned} \mathsf{y}_g(d) &:= \log\frac{L}{g^2 d}, \qquad \mathsf{C}^{\circ}_{\mathrm{law}}(g) := \frac{2\cdot 10^6 \log 2\,(\frac{50}{3})^3\,(160M(\mathsf{Q}_1+1))^2\,L^2(g^{-2}+\bar{B})^2}{c_2\,\varrho^2}, \\ \mathsf{F}_g(d,\vartheta) &:= C_{\mathrm{un}}\,\vartheta^{-3}\,\mathsf{y}_g(d)^{-2p_0} + \vartheta^{-4}\,\mathsf{C}^{\circ}_{\mathrm{law}}(g)\,d \quad \text{where } \mathsf{y}_g(d) > 0, \\ \mathsf{Z}(g) &:= \{d > 0 :\ \mathsf{y}_g(d) \ge Y_{\mathrm{u}},\ 8L^3 d\,(2\,\mathsf{y}_g(d))^{3p_0+1} \le 1\}. \end{aligned} \tag{18.xi}$$

Proposition 18.61 (The ceiling of the carried coupling)**.** (a) $0 < \gamma_{\mathrm{rt}}^{\flat} \le \gamma_{\mathrm{rt}} \le \gamma_W \le \gamma$, and $\gamma_{\mathrm{rt}} = \min\{\gamma_W, (L^2(\bar{B} + (\gamma_W^{\mathrm{sd}})^{-2}))^{-1/2}\}$.

(b) Let $0 < g \le \gamma_W$. Then $\check{g} \le \gamma_W^{\mathrm{sd}}$ if and only if $g \le \gamma_{\mathrm{rt}}$, and for these g we have $\check{g} \le \gamma_W^{\mathrm{sd}} \le \gamma_W^{\mathrm{ref}} \le \gamma$.

(c) $\gamma_{\mathrm{rt}} = \gamma_W$ if and only if $\gamma_W^{\mathrm{sd}} \ge \min\{\hat{\gamma}_W, \gamma^2, e^{-C_{\mathrm{s}}/2}\}$; in particular $\gamma_{\mathrm{rt}} = \gamma_W$ whenever $\gamma_W^{\mathrm{sd}} \ge \gamma^2$.

(d) $\gamma_{\mathrm{rt}}^{\flat}$ is a function of L, $\bar{B}$, γ, C_{s} and γ_W^{sd}; and γ_W^{sd} is the minimum of the explicit construction threshold γ_W^{ref} and $e^{-Y_{\mathrm{sd}}/2}$ in YM3_I:(2327). Neither the least tolerances $\mathsf{e}_{g',\vartheta}$ nor the set $\mathfrak{G}$ occur in the definition of $\gamma_{\mathrm{rt}}^{\flat}$. Every $0 < g \le \gamma_{\mathrm{rt}}^{\flat}$ has $\breve{g} \le \gamma_W^{\mathrm{sd}}$.

■

Sketch of proof. The map $\mathsf{a} \mapsto (L^2(\bar{B}+\mathsf{a}))^{-1/2}$ is positive and strictly decreasing on $(0,\infty)$, since $\bar{B} \ge 0$; so its value at the larger of two numbers is the smaller of its two values, and $\gamma_{\mathrm{rt}} = \min\{\gamma_W, (L^2(\bar{B}+\mathsf{a}_2))^{-1/2}\}$ by YM3_I:(2346). Corollary YM3_I:126.5 gives a reference-torus tolerance tending to zero for every $g' \le \gamma_W^{\mathrm{sd}}$ and every ϑ. Thus $\gamma_W^{\mathrm{sd}} \in \mathfrak{G}$. Hence $\sup\mathfrak{G} \ge \gamma_W^{\mathrm{sd}}$, and the definition of $\hat{\gamma}_W$ gives $\frac{1}{2}\gamma_W^{\mathrm{sd}} \le \hat{\gamma}_W \le \frac{1}{2}\gamma$, that is, $\hat{\gamma}_W^{-2} \le 4\mathsf{a}_2$. Therefore $\max\{\mathsf{a}_1,\mathsf{a}_2\} \le \max\{4\mathsf{a}_2, \gamma^{-4}, e^{C_{\mathrm{s}}}\} < \infty$, which yields $0 < \gamma_{\mathrm{rt}}^{\flat} \le \gamma_{\mathrm{rt}}$; moreover $\gamma_{\mathrm{rt}} \le \gamma_W$, and $\gamma_W \le \gamma$ follows from its defining formula. This is (a).

For (b), let $0 < g \le \gamma_W$; by YM3_I:(2346), $\breve{g}^{-2} = L^{-2}g^{-2} - \bar{B} > 0$. So $\breve{g} \le \gamma_W^{\mathrm{sd}}$ means $L^{-2}g^{-2} - \bar{B} \ge \mathsf{a}_2$, that is, $g \le (L^2(\bar{B}+\mathsf{a}_2))^{-1/2}$; together with $g \le \gamma_W$ this is $g \le \gamma_{\mathrm{rt}}$, by (a). The other two inequalities are part of the construction ceilings.

For (c), the map of the first paragraph is injective; so $\gamma_{\mathrm{rt}} = \gamma_W$ means $\max\{\mathsf{a}_1,\mathsf{a}_2\} = \mathsf{a}_1$, that is, $\mathsf{a}_2 \le \mathsf{a}_1$. As $\mathsf{a} \mapsto \mathsf{a}^{-1/2}$ decreases on $(0,\infty)$, this is $\gamma_W^{\mathrm{sd}} \ge \mathsf{a}_1^{-1/2} = \min\{\hat{\gamma}_W, \gamma^2, e^{-C_{\mathrm{s}}/2}\}$, a minimum that does not exceed γ^2.

For (d), the formula for $\gamma_{\mathrm{rt}}^{\flat}$ shows its listed dependence. The reference ceiling $\gamma_W^{\mathrm{ref}} = x_W^{-1/2}$ comes from the pointwise construction tests YM3_I:(2326) and the accompanying inequalities for $x \ge x_W$. The second ceiling in YM3_I:(2327) uses Y_{sd} large enough that

$$C_{\mathrm{sd}} Y\,[MR(e^Y)]^3 \le \tfrac{1}{2}c' A_0^2 Y^{2p_0} \quad (Y \ge Y_{\mathrm{sd}}).$$

Here C_{sd} is the constant in the small-torus criterion of Section YM3_I:126.2. This choice exists since $R(e^Y) < LY^r$ for large Y and $2p_0 > 3r+2$. Both ceilings are therefore formed entirely from construction constants. The least tolerances and $\mathfrak{G}$ are subsequently defined using the reference-torus estimates. The last assertion follows from (a) and (b). □

Lemma 18.62 (A relative error uniform at the reference torus)**.** There are constants Y_{u} and $C_{\mathrm{un}} > 0$, depending only on the construction constants, with the following property. Let $0 < u \le \gamma_W^{\mathrm{sd}}$, $\bar{x}_s = L^{s+1}(u^{-2}+\bar{B})$, $Y_s = \log \bar{x}_s$ and $s = s(d)[u]$. Let $0 < d < 1$ and $w = \log(\bar{x}_0/d)$. If

$$w \ge Y_{\mathrm{u}}, \qquad 8L^5 d(2w)^{3p_0+1} \le 1,$$

then on the torus of side $2MR(X_u)$, for all first-passage data with $K \ge s$ and all pairs of type $(d;\vartheta)$, the relative error in the short-distance two-point estimate is at most

$$C_{\mathrm{un}}\,\vartheta^{-3} Y_s^{-2p_0} \le C_{\mathrm{un}}\,\vartheta^{-3} w^{-2p_0}.$$

One can take $Y_{\mathrm{u}} \ge 2\log(2L^3)$ and $Y_{\mathrm{u}} > 8p_0$. ■

Proof. Use the witness and the error functions from the short-distance estimate of [1, *Errors depending only on the physical separation*], with $p = p_0$. Write $Y = Y_s$, $\mathsf{L}_s = L^2 C_W(s) Y^{3p}$, and $\mathsf{c}_W = C_{W,0}A_0^2$. By the minimality of the witness and the increase of $C_W(s)$,

$$C_W(s) \le L^s d < L C_W(s), \qquad Y = w + \log(L^s d) \ge w.$$

Increase Y_{u}, using only the construction constants, so that $\log(L\mathsf{c}_W) + 2p\log Y \le Y/2$ when $Y \ge Y_{\mathrm{u}}$. Then $Y < 2w$. Consequently

$$L^s \ge \mathsf{c}_W\, Y^{2p}/d \ge 8L^5 \mathsf{c}_W\, Y^{5p+1}, \qquad \mathsf{L}_s \le L^{s-3}.$$

The same estimate gives $LC_W(s)L^{-s} \le (8L^4 Y^{3p+1})^{-1}$.

Here are the bounds on the small factors in the proof of the short-distance estimate. The normalization gives $\bar{B}u^2 \le 13/24$, so $L^{s+1}u^{-2} \le e^Y \le (37/24)L^{s+1}u^{-2}$ and $s\log L \le Y$. In particular

$$x_s^- = \frac{4L^{s-2}u^{-2}}{5\mathsf{L}_s} \;\ge\; \frac{96\,e^Y}{185L^3\mathsf{L}_s}, \qquad x_s^- \le e^Y.$$

By increasing Y_{u}, we have $x_s^- \ge e^{2p}$ and $\log x_s^- \ge Y/2$. Both entries in the minimum defining $d_s^- + 1$ are at least $2Y+1$ for sufficiently large Y. Indeed the first, $c_A\alpha_0\sqrt{x_s^-}/(2MB_3A_0Y^p)$, grows exponentially divided by a power of Y, and the second satisfies

$$L^{s-1}/(2\mathsf{L}_s) - 2/M \ge 4L^2Y - 2/M.$$

Thus $\bar{x}_s e^{-d_s^-} \le e^{-Y}$. The two entries $(x_s^-)^{-1/2}$ and $\delta(x_s^-) + e^{\mathrm{co}}(x_s^-)$ of $\sigma_u(s)$ decay faster than any inverse power of Y, since the functions $\delta(t)$ and $e^{\mathrm{co}}(t)$ are bounded by constants times $t^{-1/2}(\log t)^p + \exp[-c(\log t)^{2p}]$ for $t \ge e^{2p}$. The other two entries are bounded by $\exp[Y + 3 - 4Y^{2p}/\sqrt{3}]$ and

$$M^{-3}c'A_0^2Y^{2p}\exp[5Y - \tfrac{1}{2}c'A_0^2(Y/2)^{2p}].$$

Here $L^{3s}\bar{x}_s^2 \le e^{5Y}$. Therefore $\sigma_u(s)$ decays faster than any inverse power of Y uniformly in u. Enlarge Y_{u} so that $\sigma_u(s) \le 1$. Since $Y - \log(2L^3) \ge Y/2$, the six terms of $\epsilon_u(s)$ are bounded by constant multiples of

$$Y^{-2p}, \quad Y^{1-5p}, \quad Y^{-6p}, \quad Y^{-3p}, \quad Y^{-14p}, \quad Y^{27p+4}\sigma_u(s),$$

respectively. This gives $\epsilon_u(s) \le CY^{-2p}$ with a constant independent of u.

The reference side is small over the run when $u \le \gamma_W^{\mathrm{sd}}$. The proof of the small-torus read-out bound uses, at the selected witness, $\mathsf{L}_s \le L^{s-3}$, $x_s^- \ge e^{2p}$, $Y - \log(2L^3) > 0$ and $\sigma_u(s) \le 1$. We have checked all four inequalities. Its bound and the other two estimates in the proof of the small-torus two-point law give the relative error $\varrho_{u,\vartheta}(s)$ at this witness. Its lattice-kernel terms are bounded by constants times

$$\vartheta^{-2}\left[\frac{1}{\vartheta\,\mathsf{c}_W Y^{2p}} + (8L^4Y^{3p+1})^{-3} + \sup_{t \ge L^{-2}\mathsf{c}_W Y^{2p}} t^3 e^{-\delta_{\mathrm{tp}}Lt/\sqrt{3}}\right].$$

The supremum is bounded by CY^{-2p}, as $t^3e^{-\delta_{\mathrm{tp}}Lt/\sqrt{3}}$ decreases once $t \ge 3\sqrt{3}/(\delta_{\mathrm{tp}}L)$ and decays exponentially there. The running-coupling term is at most $3\bar{B}u^2L^{2-s} \le (37/8)\bar{B}L^3e^{-Y}$, and the read-out term is at most a constant times $\vartheta^{-2}\epsilon_u(s)$. Summing yields $\varrho_{u,\vartheta}(s) \le C_{\mathrm{un}}\vartheta^{-3}Y^{-2p}$. The last inequality follows from $Y \ge w$. All increases in Y_{u} and all majorants used here involve only the construction constants. □

Proposition 18.63 (The rate of the tolerance)**.** Let $0 < g \le \gamma_{\mathrm{rt}}$ and $\vartheta \in (0,1)$; thus $\check{g} \le \gamma_W^{\mathrm{sd}}$ by Proposition 18.61 (b).

(a) $0 \le C^{\mathrm{ref}}(\check{g},\vartheta) < \infty$ and $L^2\mathsf{d}_0^{\mathrm{ref}}(\check{g},\vartheta) \le L^{-2} < \frac{1}{100}$, and for $0 < d \le L^2\mathsf{d}_0^{\mathrm{ref}}(\check{g},\vartheta)$

$$\eta_{g,\vartheta}(d) \le \varepsilon^{\mathrm{ref}}(L^{-2}d;\check{g},\vartheta) + \mathsf{C}_{\mathrm{law}}(g,\vartheta)\,d \le C^{\mathrm{ref}}(\check{g},\vartheta)\,\mathsf{y}_g(d)^{-2p_0} + \mathsf{C}_{\mathrm{law}}(g,\vartheta)\,d \le C^{\mathrm{ref}}(\check{g},\vartheta)\Big(\log\frac{L^2}{d}\Big)^{-2p_0} + \mathsf{C}_{\mathrm{law}}(g,\vartheta)\,d.$$

(b) $\mathsf{C}_{\mathrm{law}}(g,\vartheta) = \vartheta^{-4}\,\mathsf{C}^\circ_{\mathrm{law}}(g)$, where $\mathsf{C}^\circ_{\mathrm{law}}(g) > 0$ depends on the radius, the constants and g, and not on ϑ; moreover $C_{\mathrm{un}} \ge 0$ and $Y_{\mathrm{u}} > 8p_0$. The set $\mathsf{Z}(g)$ does not depend on ϑ; it is not empty, it lies in $(0,(8L^3)^{-1}]$, and with every number it contains every smaller positive number. For every $d \in \mathsf{Z}(g)$ and every $\vartheta \in (0,1)$ we have $\mathsf{y}_g(d) \ge Y_{\mathrm{u}}$ and

$$\eta_{g,\vartheta}(d) \le \mathsf{F}_g(d,\vartheta) = C_{\mathrm{un}}\,\vartheta^{-3}\Big(\log\frac{L}{g^2d}\Big)^{-2p_0} + \vartheta^{-4}\,\mathsf{C}^\circ_{\mathrm{law}}(g)\,d; \tag{18.xii}$$

the right side is non-decreasing in $d \in \mathsf{Z}(g)$ and tends to 0 as $d \to 0$.

(c) The number $d_\sharp(g,\vartheta) := \min\{L^2\mathsf{d}_0^{\mathrm{ref}}(\check g,\vartheta),\ L^2\exp[-(6\,C^{\mathrm{ref}}(\check g,\vartheta))^{1/(2p_0)}],\ (6\,\mathsf{C}_{\mathrm{law}}(g,\vartheta))^{-1}\}$ is positive, and $\mathsf{d}_0(g,\vartheta) \ge \frac12 d_\sharp(g,\vartheta)$. If $d \in \mathsf{Z}(g)$ and $\mathsf{F}_g(d,\vartheta) \le \frac13$, then $\mathsf{d}_0(g,\vartheta) \ge \frac12 d$ and $(0,\frac12 d] \subset \mathsf{Z}(g) \cap (0,\mathsf{d}_0(g,\vartheta)]$; such numbers d exist.

None of $C^{\mathrm{ref}}(\check g,\vartheta)$, $\mathsf{C}_{\mathrm{law}}(g,\vartheta)$, $\mathsf{C}^\circ_{\mathrm{law}}(g)$, $d_\sharp(g,\vartheta)$, C_{un}, Y_{u} and $\mathsf{Z}(g)$ depends on a cutoff, on β, on a side, a state, a pair or the position of a pair; C_{un} and Y_{u} depend on the constants alone. ■

Sketch of proof. (1) *The carried coupling.* By Proposition 18.61 (a), (b), $g \le \gamma_W$ and $\check g \le \gamma_W^{\mathrm{sd}} \le \gamma_W^{\mathrm{ref}} \le \gamma$. So the tolerance, its threshold and its transfer constant are defined. The small-torus estimates of Theorem YM3_I:AU and Corollary YM3_I:126.5 apply at $\check g$. By the carried-coupling formula, $\bar x_s[\check g] = L^{s+1}L^{-2}g^{-2} = L^{s-1}g^{-2}$ for $s \ge 0$, and since $\check g^{-2} \ge \gamma^{-4}$ and $\gamma \le \frac12$, $g^{-2} \ge L^2\check g^{-2} \ge L^2\gamma^{-4} \ge 16L^2$. Hence $Y_s[\check g] \ge \log(L^{-1}g^{-2}) \ge \log(16L) > 1$ for all $s \ge 0$, and for $0 < d < 1$ we get $\mathsf{y}_g(d) = \log(Lg^{-2}/d) \ge \log(16L) + \log\frac{L^2}{d} > \log\frac{L^2}{d} > 0$.

(2) *What serves at the carried coupling.* Let $\ell' = 2M\,R(X_{\check g})$ and $0 < d' \le 1/100$. The cumulant in YM3_I:(2337) is $S^T_{2;K',\ell',\beta}(f,h)$. Thus every uniform relative tolerance for that law on the reference torus at $\check g$ and d' bounds $\mathsf{e}_{\check g,\vartheta}(d')$ from above by its definition.

(3) *Clause (a).* As $s_0(\check g,\vartheta) \ge 3$, we have $\mathsf{d}_0^{\mathrm{ref}}(\check g,\vartheta) \le L^{-4}$ and $L^2\mathsf{d}_0^{\mathrm{ref}}(\check g,\vartheta) \le L^{-2} \le \frac{1}{169}$. At the scale $\mathsf{d}_0^{\mathrm{ref}}(\check g,\vartheta)$, Theorem YM3_I:AU and the bound on $\varepsilon^{\mathrm{ref}}$ above at $\check g$ give $0 \le \varepsilon^{\mathrm{ref}} \le C^{\mathrm{ref}}(\check g,\vartheta)\,Y^{-2p_0}$ with a number $Y = Y_s[\check g] > 1$ (by (1)); so $C^{\mathrm{ref}}(\check g,\vartheta) \ge 0$. Now let $0 < d \le L^2\mathsf{d}_0^{\mathrm{ref}}(\check g,\vartheta)$, so that $d \le \frac{1}{100}$, and let $0 < d' \le L^{-2}d$. Corollary YM3_I:126.5 at $\check g$ gives YM3_I:(2337) on the side ℓ', at

by (2) and the monotonicity of $\varepsilon^{\mathrm{ref}}$, $\mathsf{e}_{\check g,\vartheta}(d') \le \varepsilon^{\mathrm{ref}}(d';\check g,\vartheta) \le \varepsilon^{\mathrm{ref}}(L^{-2}d;\check g,\vartheta)$. The supremum over d' and YM3_I:(2349) give the first inequality. Put $\check s := s(L^{-2}d)[\check g]$; by the bound on $\varepsilon^{\mathrm{ref}}$ above at $\check g$, $\varepsilon^{\mathrm{ref}}(L^{-2}d;\check g,\vartheta) \le C^{\mathrm{ref}}(\check g,\vartheta)\,Y_{\check s}[\check g]^{-2p_0}$. By (1), $C_W(\check s)[\check g] = \mathsf{c}_W\,Y_{\check s}[\check g]^{2p_0} > 1$, so $L^{\check s}L^{-2}d > 1$ and $e^{Y_{\check s}[\check g]} = L^{\check s-1}g^{-2} > Lg^{-2}/d$; that is, $Y_{\check s}[\check g] > \mathsf{y}_g(d) > \log\frac{L^2}{d} > 0$. As $t \mapsto t^{-2p_0}$ decreases on $(0,\infty)$ and $C^{\mathrm{ref}}(\check g,\vartheta) \ge 0$, the second and the third inequality follow.

(4) *The constants of (b).* The displayed formula for C_2^{tr} and $\bar x_0 = L(g^{-2} + \bar B)$ give $\mathsf{C}_{\mathrm{law}}(g,\vartheta) = \vartheta^{-4}\mathsf{C}^\circ_{\mathrm{law}}(g)$. This number is positive, and Q_1 depends on g and the construction constants. By Lemma 18.62, $Y_{\mathrm{u}} > 8p_0 > 3p_0 + 1$.

(5) *The set* $\mathsf{Z}(g)$. Put $n_0 := 3p_0 + 1$ and $\phi(u) := u\,(2\,\mathsf{y}_g(u))^{n_0}$ for $0 < u < Lg^{-2}$, where $\mathsf{y}_g(u) > 0$; the second condition of $\mathsf{Z}(g)$ is $8L^3\phi(u) \le 1$, and, as the derivative of y_g at u is $-1/u$, $\phi'(u) = 2\,(2\,\mathsf{y}_g(u))^{n_0-1}(\mathsf{y}_g(u) - n_0)$. Let $d \in \mathsf{Z}(g)$. Then $2\,\mathsf{y}_g(d) \ge 2Y_{\mathrm{u}} > 1$, so the second condition gives $8L^3d \le 1$; thus $d \le (8L^3)^{-1} < 1 < Lg^{-2}$. For $0 < u \le d$, $\mathsf{y}_g(u) \ge \mathsf{y}_g(d) \ge Y_{\mathrm{u}} > n_0$ by (4); so the first condition holds at u, ϕ is non-decreasing on $(0,d]$, and $8L^3\phi(u) \le 8L^3\phi(d) \le 1$. Hence $(0,d] \subset \mathsf{Z}(g)$. As $u \to 0$, $\mathsf{y}_g(u) \to \infty$ and $\phi(u) = Lg^{-2}e^{-\mathsf{y}_g(u)}(2\,\mathsf{y}_g(u))^{n_0} \to 0$; so all small $u > 0$ lie in $\mathsf{Z}(g)$. No ϑ occurs in the two conditions.

(6), (7) *The uniform law at the carried coupling.* Let $d'' \in \mathsf{Z}(g)$ and $d' = L^{-2}d''$. At the reference coupling $\check g$ the logarithm $\log(\bar x_0[\check g]/d')$ equals $\mathsf{y}_g(d'')$. The two conditions of Lemma 18.62 become exactly the two inequalities defining $\mathsf{Z}(g)$ at d''. Apply that lemma at $\check g$ on $T_{\ell'}$. For every admitted first-passage datum (K',β) with $s(d')[\check g] \le K'$, the relative error is at most $C_{\mathrm{un}}\vartheta^{-3}Y^{-2p_0}$, where $Y = Y_{s(d')[\check g]}[\check g] \ge \mathsf{y}_g(d'')$. The constant C_{un} is positive by its construction.

(8) *Clause (b).* By (6), (7) and $Y \ge \mathsf{y}_g(d'') > 0$, the number $C_{\mathrm{un}}\vartheta^{-3}\mathsf{y}_g(d'')^{-2p_0} \in [0,\infty)$ bounds the relative error in the reference-torus law for all these data and pairs; by (2) it serves at $(\check g,\vartheta,L^{-2}d'')$, and $\mathsf{e}_{\check g,\vartheta}(L^{-2}d'') \le C_{\mathrm{un}}\vartheta^{-3}\mathsf{y}_g(d'')^{-2p_0}$ for every $d'' \in \mathsf{Z}(g)$. Let $d \in \mathsf{Z}(g)$; then $d \le \frac{1}{100}$ by (5). Each $d' \in (0,L^{-2}d]$ is $L^{-2}d''$ with $d'' \in (0,d] \subset \mathsf{Z}(g)$, and $\mathsf{y}_g(d'') \ge \mathsf{y}_g(d) > 0$; as $C_{\mathrm{un}} \ge 0$, $\mathsf{e}_{\check g,\vartheta}(d') \le C_{\mathrm{un}}\vartheta^{-3}\mathsf{y}_g(d)^{-2p_0}$. The supremum over d', YM3_I:(2349) and (4) give eq. 18.xii. Its right side is non-decreasing in d, because $C_{\mathrm{un}} \ge 0$, $\mathsf{C}^\circ_{\mathrm{law}}(g) > 0$ and y_g decreases, and it tends to 0 with d, because $\mathsf{y}_g(d) \to \infty$.

(9) *Clause (c).* The three members of the minimum are positive, as $\mathsf{C}_{\mathrm{law}}(g,\vartheta) > 0$ by (4). At $d := d_\sharp(g,\vartheta) \le L^2\mathsf{d}_0^{\mathrm{ref}}(\check g,\vartheta) \le L^{-2}$ clause (a) applies: $\log\frac{L^2}{d} \ge (6\,C^{\mathrm{ref}}(\check g,\vartheta))^{1/(2p_0)}$ gives $C^{\mathrm{ref}}(\check g,\vartheta)(\log\frac{L^2}{d})^{-2p_0} \le \frac16$, this term being 0 when $C^{\mathrm{ref}}(\check g,\vartheta) = 0$, and $\mathsf{C}_{\mathrm{law}}(g,\vartheta)\,d \le \frac16$. So $\eta_{g,\vartheta}(d) \le \frac13$, and $d \in (0, \frac{1}{100}]$ lies in the set of YM3_I:(2350), whose supremum is therefore at least d. For $d \in \mathsf{Z}(g)$ with $\mathsf{F}_g(d,\vartheta) \le \frac13$ the same argument, with (b) in the place of (a), gives $\mathsf{d}_0(g,\vartheta) \ge \frac12 d$, and $(0,\frac12 d] \subset \mathsf{Z}(g)$ by (5). Such d exist, since all small scales lie in $\mathsf{Z}(g)$ and $\mathsf{F}_g(d,\vartheta) \to 0$ as $d \to 0$. The dependence of the constants follows from Lemma 18.62, the definition of $\mathsf{C}_{\mathrm{law}}$ and the reference-torus error formulas of [1, *Errors depending only on the physical separation*]. □

Proposition 18.64 (The leading law with the logarithmic rate)**.** Let $0 < g \le \gamma_{\mathrm{rt}}$ and $\vartheta \in (0,1)$, and let $0 < d \le \mathsf{d}_0(g,\vartheta)$ satisfy $d \in \mathsf{Z}(g)$ or $d \le L^2\mathsf{d}_0^{\mathrm{ref}}(\check g,\vartheta)$. In the first case let $\varepsilon_{\mathrm{rt}} := \mathsf{F}_g(d,\vartheta)$, in the second $\varepsilon_{\mathrm{rt}} := C^{\mathrm{ref}}(\check g,\vartheta)\,\mathsf{y}_g(d)^{-2p_0} + \mathsf{C}_{\mathrm{law}}(g,\vartheta)\,d$; when both hold, either number may be taken. Then $\eta_{g,\vartheta}(d) \le \varepsilon_{\mathrm{rt}}$, and the following hold.

(1) *States at a finite cutoff.* For every $(K,\beta) \in \mathfrak{d}(g)$ with $s(d) < K$ and $b = a_K\beta$, every state $\mathfrak{s}$ at g at (K,β), among them every $\nu \in \mathcal{G}^{\mathrm{per}}_{K,\beta}$, and every $\Psi \in \mathfrak{T}_{\mathbb{R}^3}(d;\vartheta)$: $S_2^T[\mathfrak{s}](\Psi) = c_2 b^{-2}\mathcal{K}_6(\Psi)(1+\eta)$ with $|\eta| \le \varepsilon_{\mathrm{rt}}$.

(2) *Tori at a finite cutoff.* For the same (K,β), every $\ell \in \mathfrak{L}(g)$ and every $\Psi \in \mathfrak{T}_{T_\ell}(d;\vartheta)$: $S^T_{2;K,\ell,\beta}(\Psi) = c_2 b^{-2}\mathcal{K}_6(\Psi)(1+\eta)$ with $|\eta| \le \varepsilon_{\mathrm{rt}}$.

(3) *Continuum limits.* Let S be a continuum limit at g on a torus T_ℓ with $\ell \in \mathfrak{L}(g)$, and $\mathbb{E} := T_\ell$, or a continuum limit at g along growing tori or a volume-first limit at g, and $\mathbb{E} := \mathbb{R}^3$. Then S has exactly one label $\mathsf{b}(S)$ at g, and $S_2(\Psi) = c_2\mathsf{b}(S)^{-2}\mathcal{K}_6(\Psi)(1+\eta)$ with $|\eta| \le \varepsilon_{\mathrm{rt}}$ for every $\Psi \in \mathfrak{T}^1_{\mathbb{E}}(d;\vartheta)$.

(4) *Limits through periodic Gibbs states; frames and components.* Let $(K_j,\beta_j) \in \mathfrak{d}(g)$ for $j \in \mathbb{N}$, with $K_j \to \infty$. (i) If $\rho_j \in \mathcal{G}^{\mathrm{per}}_{K_j,\beta_j}$ and the limit $S_2(\Phi) := \lim_j S_2^T[\rho_j](\Phi)$ exists for every separated pair Φ, then $a_{K_j}\beta_j$ converges to a number b, and $S_2(\Psi) = c_2 b^{-2}\mathcal{K}_6(\Psi)(1+\eta)$ with $|\eta| \le \varepsilon_{\mathrm{rt}}$ for every $\Psi \in \mathfrak{T}^1(d;\vartheta)$. (ii) If $\nu_j \in \mathcal{G}^{\mathrm{per}}_{K_j,\beta_j}$ and the frame $\mathfrak{j} = (K_j,\nu_j)_j$ converges, then $a_{K_j}\beta_j$ converges to a number b, and for every limit σ' in $\mathfrak{j}$, every canonical decomposition $(\mathsf{X},\lambda,(\sigma_\xi))$ of $\sigma_{\mathfrak{j}}$, every $\xi \in \mathsf{X}$ and every $\Psi \in \mathfrak{T}^1(d;\vartheta)$: $S_2^{\sigma'}(\Psi) = c_2 b^{-2}\mathcal{K}_6(\Psi)(1+\eta')$ and $S_2^{\sigma_\xi}(\Psi) = c_2 b^{-2}\mathcal{K}_6(\Psi)(1+\eta_\xi)$ with $|\eta'| \le \varepsilon_{\mathrm{rt}}$ and $|\eta_\xi| \le \varepsilon_{\mathrm{rt}}$.

(5) *Volume-first limits and their components.* For every volume-first limit S at g, every canonical decomposition $(\mathsf{X},\lambda,(\sigma_\xi))$ of σ_S, for $\mathsf{S} = S_2$ and $\mathsf{S} = S_2^\xi$, $\xi \in \mathsf{X}$, and for every $\Psi \in \mathfrak{T}^1(d;\vartheta)$: $|\mathsf{S}(\Psi) - c_2\mathsf{b}(S)^{-2}\mathcal{K}_6(\Psi)| \le \varepsilon_{\mathrm{rt}}\,c_2\mathsf{b}(S)^{-2}\mathcal{K}_6(\Psi)$.

At fixed g and ϑ, $\varepsilon_{\mathrm{rt}}$ is a function of d alone: one bound serves all the data, states, sides, pairs and positions of pairs, limits, frames, functionals and components of (1)–(5). For each of the two choices of $\varepsilon_{\mathrm{rt}}$ the admitted scales d include all sufficiently small positive numbers (Proposition 18.63 (c)). ■

Sketch of proof. (1) *The bound.* By Proposition 18.61 (a), $g \le \gamma_W$. If $d \in \mathsf{Z}(g)$, Proposition 18.63 (b) gives $\eta_{g,\vartheta}(d) \le \mathsf{F}_g(d,\vartheta)$; if $d \le L^2\mathsf{d}_0^{\mathrm{ref}}(\check g,\vartheta)$, its clause (a) gives $\eta_{g,\vartheta}(d) \le C^{\mathrm{ref}}(\check g,\vartheta)\,\mathsf{y}_g(d)^{-2p_0} + \mathsf{C}_{\mathrm{law}}(g,\vartheta)\,d$. So $\eta_{g,\vartheta}(d) \le \varepsilon_{\mathrm{rt}}$ in each case.

(2) *A larger tolerance.* Since $\mathcal{K}_6(\Psi) > 0$ at every test pair, the relative form and the absolute difference form of the estimate are equivalent. Increasing the tolerance from $\eta_{g,\vartheta}(d)$ to $\varepsilon_{\mathrm{rt}}$ preserves both forms.

(3) *The statements.* Theorem YM3_I:AV gives (1) and (2) at finite cutoff. Theorem YM3_I:AW gives (3), including convergence of the labels along every defining sequence. Proposition YM3_I:127.8 gives (4)(i). For (4)(ii), Corollary YM3_I:127.11 gives the same label and estimate for the limit in the frame and all components. The component approximation is along the full chosen sequence by Theorem YM3_I:AP. A volume-first defining sequence has converging self-centered moments by the moment–cumulant formula, with the functional σ_S. Applying (3) and (4)(ii) to this sequence gives (5). In every case use (2) to substitute the larger tolerance.

(4) *One bound.* The number $\varepsilon_{\mathrm{rt}}$ is formed from g, ϑ, d and the numbers C_{un}, $\mathsf{C}^{\circ}_{\mathrm{law}}(g)$, $C^{\mathrm{ref}}(\check{g},\vartheta)$ and $\mathsf{C}_{\mathrm{law}}(g,\vartheta)$, which depend on no datum, state, side, pair or position (Proposition 18.63).

(5) *The objects exist.* Choose a label $b' \in I(g)$ and take first-passage data $(K, L^K b')$ from Corollary YM3_I:74.2, with $K > s(d)$. The admissible sides are all powers of the stated form beyond a fixed size, hence are infinite in number. Proposition YM3_I:91.2 gives nonempty $\mathcal{G}^{\mathrm{tor}}_{K,\beta} \subset \mathcal{G}^{\mathrm{inv}}_{K,\beta} \subset \mathcal{G}^{\mathrm{per}}_{K,\beta}$. Lemma YM3_I:127.5 gives smooth Euclidean test pairs and periodisation preserves the test-pair conditions by Lemma YM3_I:127.2. Fixed-torus limits exist by Theorem YM3_I:A, and growing-torus and volume-first limits exist by the compactness of Proposition YM3_I:92.1 applied to sequences of the respective laws. To get a convergent frame choose periodic Gibbs states along first-passage data with $K_j \to \infty$ and extract a subsequence by Theorem YM3_I:AP. It also gives the canonical decomposition. By Proposition 18.63 (c) the admitted scales include, for the first choice of $\varepsilon_{\mathrm{rt}}$, the interval $(0, \frac{1}{2}d_1]$ for every $d_1 \in \mathsf{Z}(g)$ with $\mathsf{F}_g(d_1,\vartheta) \le \frac{1}{3}$, and, for the second, the interval $(0, \frac{1}{2}d_\sharp(g,\vartheta)]$, as $\frac{1}{2}d_\sharp(g,\vartheta) \le \min\{\mathsf{d}_0(g,\vartheta), L^2\mathsf{d}_0^{\mathrm{ref}}(\check{g},\vartheta)\}$. □

18.7. **Three complements: the lattice gradient flow and the continuum Yang–Mills flow; the limit form of the block observables, with nothing cut; a torus with one unit of centre flux.** This subsection adds three groups of results to Section 15.1, to which the words *the section* refer in what follows, and it uses the signs of that Section wherever it does not set them anew. The first compares the gradient flow Φ_s of Wilson's action, started at the lattice field $U_K[Q]$ of a 1-form Q of class C^3 on T_ℓ, with the Yang–Mills flow line q of Q: up to a lattice gauge transformation, bond by bond and for $0 \le s \le s_1$, the two configurations lie within $c\,e^{cs_1}a_K^3(1+\|Q\|_{C^3})^{3p}\exp(120\int_0^{s_1}\sup|F_{q_r}|\,dr)$ of each other, with absolute numbers c and p, for every side ℓ and every cutoff K (Proposition 18.66); behind this stand the existence and the uniqueness of the flow line and bounds of its curvature and of three covariant derivatives of the curvature that read neither the side nor the cutoff (Proposition 18.65). The second treats the block observables read through a smoothed block map $\tilde{\mathcal{M}}$, with no cutoff in the functions and none in the measures: at every depth d the lattice Dirichlet forms converge along the full sequence of cutoffs to a form $\mathcal{E}_\infty^{\tilde{\mathcal{M}}}$ given by a matrix field $\mathsf{Mat}_d^{\tilde{\mathcal{M}}}$ of the threads, which reads neither the depth at which an observable is read nor more than its class, and which lies between $\frac{1}{6}\Gamma_d$ and $\frac{7}{6}\Gamma_d$ at every thread Y such that $\varpi(Y_j) < \theta_j$ for all large j, a property of $\mu_{\ell,\mathsf{b}}$-almost every thread, and such that $\varpi(Y_j) \le \varepsilon_{\mathrm{tr}}$ and $\varpi(Y_j) < \omega_{\tilde{\mathcal{M}}}$ for all $j > d$ and $\sum_{j>d}\varpi(Y_j) \le (36C_{\mathrm{tr}})^{-1}$, where $\varepsilon_{\mathrm{tr}}$, C_{tr} are numbers of L and $\omega_{\tilde{\mathcal{M}}}$ is a number of the map; once d is at least a depth $d_{\tilde{\mathcal{M}}}$ of the map, these threads include all threads with $\varpi(Y_j) < \theta_j$ for every $j > d$ (Propositions 18.67, 18.68); for the smoothed map $\mathcal{M}^\chi$ of Proposition 18.67(d), the form of every single depth is closable (Proposition 18.69); and when the forms of the single depths are closable and a contraction of the energy holds at infinitely many depths, the form on all depths is closable, its closure is a symmetric Dirichlet form for which $\mu_{\ell,\mathsf{b}}$ is invariant and reversible, and every observable of positive energy moves under its semigroup (Proposition 18.70). The third concerns a cube of side l with twisted boundary conditions that carry one unit of centre flux: modulo gauge the action has exactly two minimisers, exchanged by a flip, with the floor $4l^{-2}$ of the Hessian across the gauge directions, and when ϵ obeys the explicit bounds (h1), (h2) in terms of the size of the lattice, the fields with all plaquettes within ϵ of $\mathbf{1}$ form, modulo gauge, two pieces, one around each minimiser, which the flip interchanges; each is a copy of one subset of the Landau slice, and a radial reparametrisation of the slice makes this subset convex and open (Proposition 18.71); on the cube with $N = lL^n$ sites along each axis, a measure there whose density is the exponential of minus the multiple $x = L^n\mathsf{b}$ of an analytic action with that floor, plus an analytic term, obeys a log-Sobolev inequality of the order l^2/b on the gauge-invariant even functions once x is large and $N \le C_N(\log x)^{r_N}$ for fixed numbers $C_N > 0$ and $r_N \ge 0$ with $5r_N < 2\mathfrak{p}_0$, $\mathfrak{p}_0$ the power of the logarithm in the thresholds θ_d, and the gauge invariance cannot be dropped (Proposition 18.72); an entropy tower adds the fibres of a chain with log-concave fibres (Proposition 18.73); and if, with x and N as just said, the small-field chain of the twisted cube is

given as a measure below the Wilson measure, with log-concave fibres, an analytic top and one-step bounds of its tower (the data (D1), (D2) below), then, for gauge-invariant even functions (real and smooth ones in the inequalities of the two measures, bounded Borel ones in those of the dynamics), the chain measure obeys a log-Sobolev inequality of the order l^2/b uniformly in the cutoff, the Wilson measure a weak Poincaré inequality with the defect $\mathsf{R}_{\rm tw}\,\mathrm{osc}^2$, the Langevin dynamics relaxes exponentially down to the level $\mathsf{R}_{\rm tw}\,\mathrm{osc}^2$, and, if $\mathsf{R}_{\rm tw} < 0.19$, along every sequence of cutoffs on which, under the stationary dynamics, the joint laws at two times of all bounded Borel functions of the top field V_n converge, some $\{0,1\}$-valued function of V_n changes its value with positive probability over every positive time in the limit (Proposition 18.74).

I. The lattice gradient flow and the continuum flow. The statements of this part hold for every side $\ell = 2L^m$ and every cutoff K; the setting (E) does not enter. In this part a flow line, with or without the words Yang–Mills, is always a solution in the classical sense, started at a 1-form, as set out below: a map into the 1-forms on T_ℓ, and never the line of gauge orbits of the setting of the section. As in the setting of the section, Φ_s is the flow of $\frac{d}{ds}U(b) = -ia_K J(U)(b)U(b)$ on Ω_K, with the current $J(U) = D_U^{\mathsf{T}} F_U$ of YM3_I:(69) in physical units.

Continuum fields and their lattice fields. A 1-form of class C^k on T_ℓ is $Q = (Q_1, Q_2, Q_3)$ with $Q_\mu \in C^k(T_\ell; \mathfrak{g})$, and $\|Q\|_{C^k} = \max_\mu \max_{|\alpha|\le k} \sup |\partial^\alpha Q_\mu|$, α a multi-index; we write $N_Q := \|Q\|_{C^3}$. For a piecewise C^1 path $\gamma : [t_0, t_1] \to T_\ell$ the holonomy $\mathrm{hol}_Q(\gamma) := H(t_1)$ is the solution of $H'(t) = H(t)\, iQ_\lambda(\gamma(t))\dot\gamma^\lambda(t)$, $H(t_0) = \mathbf{1}$, earlier points standing on the left; it does not read the parametrisation, $\mathrm{hol}_Q(\gamma_1\gamma_2) = \mathrm{hol}_Q(\gamma_1)\mathrm{hol}_Q(\gamma_2)$, and the lattice field $U_K[Q] \in \Omega_K$ takes at a bond b the holonomy of the straight segment from b_- to b_+, so that $U_K[Q](\Gamma) = \mathrm{hol}_Q(\Gamma)$ for every path Γ of $\Lambda_{K,\ell}$. With $F_{\mu\nu} = \partial_\mu Q_\nu - \partial_\nu Q_\mu + i[Q_\mu, Q_\nu]$, $D_\lambda\Psi = \partial_\lambda\Psi + i[Q_\lambda, \Psi]$, $(d_Q\vartheta)_\mu = D_\mu\vartheta$, $(d_Q\eta)_{\mu\nu} = D_\mu\eta_\nu - D_\nu\eta_\mu$ and $(d_Q^* F)_\mu = \sum_\nu D_\nu F_{\mu\nu}$ one has $[D_\mu, D_\nu]\Psi = i[F_{\mu\nu}, \Psi]$, the Bianchi identity $D_\lambda F_{\mu\nu} + D_\mu F_{\nu\lambda} + D_\nu F_{\lambda\mu} = 0$, $\partial_\lambda\langle\Psi, \Psi'\rangle = \langle D_\lambda\Psi, \Psi'\rangle + \langle\Psi, D_\lambda\Psi'\rangle$, and $\int_{T_\ell}\sum_\mu\langle\eta_\mu, (d_Q^* F)_\mu\rangle = \int_{T_\ell}\sum_{\mu<\nu}\langle(d_Q\eta)_{\mu\nu}, F_{\mu\nu}\rangle$; so the flow $\partial_s q = -d_q^* F_q$ is the gradient flow of $\frac12\int_{T_\ell}\sum_{\mu<\nu}|F_{\mu\nu}|^2$. For $g \in C^1(T_\ell; G)$ put $Q_\mu^g = gQ_\mu g^{-1} + i(\partial_\mu g)g^{-1}$; then $\mathrm{hol}_{Q^g}(\gamma) = g(\gamma(t_0))\,\mathrm{hol}_Q(\gamma)\,g(\gamma(t_1))^{-1}$, so that $U_K[Q^g] = U_K[Q]^{g|}$ exactly, $g|$ the restriction of g to the sites of $\Lambda_{K,\ell}$, and for $g \in C^2$ one has $F^{Q^g} = \mathrm{Ad}(g)F^Q$ and $D^{Q^g}(\mathrm{Ad}(g)\Psi) = \mathrm{Ad}(g)D^Q\Psi$. The numbers $\Phi_k(Q) := \max\sup_x |D_{\lambda_1}\cdots D_{\lambda_k}F_{\mu\nu}(x)|$, the maximum over all indices, are gauge invariant, and by the rule of Leibniz and $|[X,Y]| \le 2|X||Y|$, $\Phi_k(Q) \le 2^{k^2+k+1}(1 + \|Q\|_{C^{k+1}})^{k+2}$; in particular $\Phi_0 \le 2(1+N_Q)^2$, $\Phi_1 \le 8(1+N_Q)^3$, $\Phi_2 \le 128(1+N_Q)^4$, and $\Phi_3 \le 2\|Q\|_{C^4} + 2^{13}(1+N_Q)^5$. For a 1-form η and $b = \langle x, x + a_K e_\mu\rangle$ the bond average is $\bar\eta(b) := a_K^{-1}\int_0^{a_K}\mathrm{Ad}(H_b(t))\,\eta_\mu(x + te_\mu)\,dt$, $H_b(t)$ the holonomy of Q along the segment from x to $x + te_\mu$.

Flow lines and gauged flows. A Yang–Mills flow line from Q on $[0, s_1]$ is a map $q : [0, s_1] \times T_\ell \to \mathfrak{g}^3$ such that q, $\partial_s q$ and $\partial_x^\alpha q$, $|\alpha| \le 2$, are continuous, $\partial_s q = -d_q^* F_q$ and $q_0 = Q$; it is a flow line on $[0, \infty)$ if this holds on every $[0, s_1]$. A regular gauged flow on $[s', s'']$ is a pair of maps $q : [s', s''] \times T_\ell \to \mathfrak{g}^3$, $\vartheta : [s', s''] \times T_\ell \to \mathfrak{g}$ such that q, $\partial_s q$, $\partial_x q$, $\partial_x^2 q$, ϑ and $\partial_x\vartheta$ are continuous, $\partial_s q = -d_q^* F_q - d_q\vartheta$, $s \mapsto q_s$ is continuous from $(s', s'']$ into C^4, and $\mathfrak{t}(s) := 2[\frac{1}{12}\Phi_3 + \frac{37}{3}\Phi_0\Phi_1 + \frac13\Phi_0^3 + 12\Phi_0^2\Phi_1](q_s)$ is integrable on $(s', s'']$. A piecewise regular gauged flow from Q on $[0, s_1]$ consists of points $0 = \sigma_0 < \cdots < \sigma_k = s_1$ and regular gauged flows $(q^{(j)}, \vartheta^{(j)})$ on $[\sigma_{j-1}, \sigma_j]$ with $q_0^{(1)} = Q$ and $q_{\sigma_j}^{(j+1)} = (q_{\sigma_j}^{(j)})^{g_j}$, $g_j \in C^1(T_\ell; G)$; its numbers are $\mathfrak{T} := \sum_j\int_{\sigma_{j-1}}^{\sigma_j}\mathfrak{t}\,ds$ and $\mathfrak{E} := \exp(10s_1 + 40\int_0^{s_1}\Phi_0(q_s)\,ds)$, q_s the field of a piece that holds s (the Φ_k do not see the joints, being gauge invariant). For $Q \in C^3$ with its flow line q (Proposition 18.65(a)) and a cutoff K put $\mathrm{err}_K(Q; s) := \min_u \max_b \|\Phi_s(U_K[Q])^u(b) - U_K[q_s](b)\|$, the minimum over the gauge transformations u of $\Lambda_{K,\ell}$, $\|\cdot\|$ the operator norm.

Lattice notation. On $\Lambda_{K,\ell}$, with $a = a_K$: for a bond $b = \langle x, x + ae_\mu\rangle$ and $\alpha \in \{\pm1, \pm2, \pm3\}$ with $|\alpha| \ne \mu$, $e_{-\nu} := -e_\nu$, $W_x(\mu, \alpha)$ is the value of a configuration along the closed path $x \to x + ae_\mu \to x + ae_\mu + ae_\alpha \to x + ae_\alpha \to x$. For a 2×2 matrix V, $\mathbf{w}(V) := \frac{1}{2i}(V - V^*)$ and $\mathbf{c}(V) := \frac12\mathrm{Re}\,\mathrm{tr}\,V$; for $V \in G$, $V = \mathbf{c}(V)\mathbf{1} + i\mathbf{w}(V)$, $\|V - \mathbf{1}\|^2 = 2(1 - \mathbf{c}(V))$ and $|\mathbf{w}(V)|^2 = (1 - \mathbf{c}(V))(1 + \mathbf{c}(V))$.

$\Phi_0^{\mathrm{lat}}(V) := a^{-2}\max\|W_x(\mu,\alpha)(V)-\mathbf{1}\|$, the maximum over x,μ,α. For 1-forms, D_V is the covariant exterior derivative and D_V^{T} its transpose in physical units at spacing a, as in YM3_I:(67)–(68). On site functions D_V is the covariant gradient and D_V^* is its adjoint on 1-forms, with the formulas below. With $V(x,\nu)$ the value at $\langle x, x+ae_\nu\rangle$, put $(\nabla_\nu^V f)(x) := a^{-1}[\mathrm{Ad}(V(x,\nu))f(x+ae_\nu)-f(x)]$ and $(\nabla_\nu^{V*}f)(x) := a^{-1}[\mathrm{Ad}(V(x-ae_\nu,\nu)^{-1})f(x-ae_\nu)-f(x)]$ on site functions, so that $(D_V\lambda)(\langle x, x+ae_\mu\rangle) = (\nabla_\mu^V\lambda)(x)$ and $D_V^*E = \sum_\nu \nabla_\nu^{V*}E_\nu$; $\Delta^V := -\sum_\nu \nabla_\nu^{V*}\nabla_\nu^V$ acts on each component of a 1-form, and $|E|_\infty := \max_b |E(b)|$.

Proposition 18.65 (The Yang–Mills flow from a field of class C^3; curvature bounds free of the side)**.** Let $Q\in C^3(T_\ell;\mathfrak{g}^3)$ and $N_Q = \|Q\|_{C^3}$.

(a) There is exactly one Yang–Mills flow line q from Q on $[0,\infty)$. With $c_* := (1152\pi)^{-2}$, $s_0 := c_*(1+N_Q)^{-2}$, Δ the Laplacian of T_ℓ acting on each component and $\mathcal{N}(q)_\mu := \sum_\nu\{2i[q_\nu,\partial_\nu q_\mu]-i[q_\nu,\partial_\mu q_\nu]-[q_\nu,[q_\nu,q_\mu]]\}$, there is exactly one $\hat q\in C([0,s_0];C^3)$ with $\hat q_s = e^{s\Delta}Q+\int_0^s e^{(s-r)\Delta}\mathcal{N}(\hat q_r)\,dr$; it has $\|\hat q_s\|_{C^3}\le 2N_Q$ and $\|\hat q_s\|_{C^4}\le 2N_Q s^{-1/2}$, and with $\hat\vartheta := -\sum_\nu \partial_\nu\hat q_\nu$ the pair $(\hat q,\hat\vartheta)$ is a regular gauged flow on $[0,s_0]$; if h solves $\partial_s h = -ih\hat\vartheta$, $h_0=\mathbf{1}$, then $q_s = \hat q_s^{h_s}$ for $s\le s_0$, so that $U_K[q_s] = U_K[\hat q_s]^{h_s|}$. For every $s_1>0$ there is a piecewise regular gauged flow from Q on $[0,s_1]$ whose first piece is $(\hat q,\hat\vartheta)$ on $[0,\min\{s_0,s_1\}]$; if $s_1\le s_0$ this is its only piece, and if $s_1>s_0$ it has exactly one piece more, defined on $[s_0,s_1]$, of class C^∞ and with vanishing ϑ; its numbers Φ_k are those of q.

(b) Let $s_1>0$, let $e_k := \sum_{\lambda_1,\dots,\lambda_k}\sum_{\mu<\nu}|D_{\lambda_1}\cdots D_{\lambda_k}F_{\mu\nu}|^2$, formed with q_s, $y_k(s) := \max_x e_k(s,x)$, $\mathsf{E}_{\mathrm{loc}}(s) := \sup_x\int_{x+[0,1]^3} e_0(s,z)\,dz$ with e_0 read as a periodic function on $\mathbb{R}^3$, $\bar{\mathsf{E}} := 27\mathsf{E}_{\mathrm{loc}}(0)e^{324s_1}$, $\tau_* := \min\{1,155^{-4}\bar{\mathsf{E}}^{-2}\}$ if $\bar{\mathsf{E}}>0$ and $\tau_*:=1$ if $\bar{\mathsf{E}}=0$, and let $\mathsf{u}_1,\mathsf{u}_2,\mathsf{u}_3$ be the absolute numbers of the equations of the curvature set out in the proof. Then for $0\le s\le s_1$, with no number that reads ℓ or K: (i) $\mathsf{E}_{\mathrm{loc}}(s)\le 27\mathsf{E}_{\mathrm{loc}}(0)e^{324s}\le\bar{\mathsf{E}}$; (ii) $y_0(s)\le \mathsf{Y}_0 := 2\max\{y_0(0),[2(4\pi\tau_*)^{-1/2}+1]^3\bar{\mathsf{E}}\}$, and we put $\mathsf{h}_0 := \mathsf{Y}_0^{1/2}$; (iii) $y_1(s)\le\mathsf{Y}_1 := y_1(0)+2\mathsf{u}_1\mathsf{h}_0y_0(0)+10\mathsf{u}_1\mathsf{h}_0^4 s_1$ and $y_2(s)\le\mathsf{Y}_2 := y_2(0)+\mathsf{q}_2y_1(0)+(\mathsf{u}_2\mathsf{Y}_1^2+4\mathsf{u}_1\mathsf{q}_2\mathsf{h}_0\mathsf{Y}_1)s_1$, $\mathsf{q}_2 := \frac12\mathsf{u}_2(4\mathsf{h}_0+1)$; (iv) for $0<s\le s_1$: $\min\{s,1\}\,y_3(s)\le\mathsf{Y}_3 := \mathsf{q}_3\mathsf{Y}_2+2\mathsf{u}_3\mathsf{Y}_1\mathsf{Y}_2+\mathsf{q}_3[\mathsf{u}_2(4\mathsf{h}_0+1)\mathsf{Y}_2+\mathsf{u}_2\mathsf{Y}_1^2]$, $\mathsf{q}_3 := \frac12[1+\mathsf{u}_3(4\mathsf{h}_0+2)]$.

(c) There are absolute numbers c and p such that for every side ℓ, every $Q\in C^3(T_\ell;\mathfrak{g}^3)$ and every $s>0$: $\Phi_0(q_s)\le ce^{cs}(1+N_Q)^8$, $\Phi_1(q_s)\le ce^{cs}(1+N_Q)^p$, $\Phi_2(q_s)\le ce^{cs}(1+N_Q)^p$ and $\Phi_3(q_s)\le ce^{cs}(1+N_Q)^p(1+s^{-1/2})$; and $\Phi_0(q_s)\le 8(1+N_Q)^2$ for $s\le s_0$.

■

Sketch. (a) *The local fixed point.* The heat semigroup $e^{s\Delta}$ commutes with derivatives and contracts every $C^k(T_\ell)$, and the heat kernel p_s of $\mathbb{R}^3$ has $\|\partial_\lambda p_s\|_{L^1} = (\pi s)^{-1/2}$, so $\|e^{s\Delta}f\|_{C^{k+1}}\le s^{-1/2}\|f\|_{C^k}$ for $0<s\le 1$. $\mathcal{N}=\mathcal{N}_2+\mathcal{N}_3$ with $\mathcal{N}_2$ bilinear in $(q,\partial q)$ and $\mathcal{N}_3$ trilinear; by the rule of Leibniz and $|[X,Y]|\le 2|X||Y|$, $\|\mathcal{N}_2(q)-\mathcal{N}_2(q')\|_{C^k}\le 18\cdot 2^k(\|q-q'\|_{C^k}\|q\|_{C^{k+1}}+\|q'\|_{C^k}\|q-q'\|_{C^{k+1}})$ and $\|\mathcal{N}_3(q)-\mathcal{N}_3(q')\|_{C^k}\le 12\cdot 3^k(\|q\|_{C^k}^2+\|q\|_{C^k}\|q'\|_{C^k}+\|q'\|_{C^k}^2)\|q-q'\|_{C^k}$. On the ball of radius $2N_Q$ of the Banach space of the $q\in C([0,s_0];C^3)\cap C((0,s_0];C^4)$ with the norm $\max\{\sup_s\|q_s\|_{C^3},\sup_s s^{1/2}\|q_s\|_{C^4}\}$, the map $q\mapsto e^{s\Delta}Q+\int_0^s e^{(s-r)\Delta}\mathcal{N}(q_r)\,dr$ moves $e^{\cdot\Delta}Q$ (of norm at most N_Q) by at most $576\pi N_Q^2 s_0^{1/2}+5184N_Q^3 s_0\le N_Q$ and contracts with the factor $576\pi N_Q s_0^{1/2}+7776N_Q^2 s_0\le\frac34$, by the choice of c_*; here $\int_0^s(s-r)^{-1/2}dr = 2s^{1/2}$ and $\int_0^s(s-r)^{-1/2}r^{-1/2}dr=\pi$ enter. This gives $\hat q$ with its two bounds. Two mild solutions in $C([0,s];C^1)$ from one datum agree: the function $s\mapsto\|q_s-q'_s\|_{C^1}$ is at most $C\int_0^s\max\{1,(s-r)^{-1/2}\}\|q_r-q'_r\|_{C^1}\,dr$, and one iteration and Gronwall's lemma make it vanish. The same fixed point with (C^k,C^{k+1}) in the place of (C^3,C^4) gives, for every $k\ge1$, solutions on intervals of the length $c_{*,k}(1+\|\cdot\|_{C^k})^{-4}$ from data of class C^k, with absolute numbers $c_{*,k}$; restarted at small positive times, by the semigroup law, it shows that a mild solution is C^∞ on $(0,s]\times T_\ell$ and solves $\partial_s\hat q=\Delta\hat q+\mathcal{N}(\hat q)$ there, and for data of class C^3 the equation holds on $[0,s_0]$, all its terms being

continuous. The algebra $\sum_\nu[D_\nu F_{\nu\mu}+D_\mu(\partial_\nu q_\nu)]=\Delta q_\mu+\mathcal{N}(q)_\mu$, in which the terms $\pm i[\partial_\nu q_\nu,q_\mu]$ cancel, shows that $(\hat q,\hat\vartheta)$ is a regular gauged flow, with $\Phi_3(\hat q_s)\le 4N_Q s^{-1/2}+2^{13}(1+2N_Q)^5$, an integrable function.

Back to the flow line; uniqueness. $\hat\vartheta\in C([0,s_0];C^2)$ and $\|\hat\vartheta_s\|_{C^3}\le 6N_Q s^{-1/2}$, so the linear equation $\partial_s h=-ih\hat\vartheta$, $h_0=\mathbf{1}$, has a solution with $\partial_x^\alpha h$, $|\alpha|\le 3$, and $\partial_s\partial_x h$ continuous (differentiation in the parameter x under an integrable majorant). With $\partial_s h=i\eta h$, $\eta=-\mathrm{Ad}(h)\hat\vartheta$, one has $\partial_s(\hat q^h)_\mu=\mathrm{Ad}(h)\partial_s\hat q_\mu-D^{\hat q^h}_\mu\eta$, and with the covariance of F and of D this is $-(d^*_{\hat q^h}F_{\hat q^h})_\mu$: $q:=\hat q^h$ is a flow line on $[0,s_0]$, and $U_K[q_s]=U_K[\hat q_s]^{h_s|}$. If q,q' are two flow lines from Q and $\eta:=q-q'$, then $F_q=F_{q'}+d_{q'}\eta+\mathsf{c}$ and $d_q\eta=d_{q'}\eta+2\mathsf{c}$ with $\mathsf{c}_{\mu\nu}=i[\eta_\mu,\eta_\nu]$, and in $L^2(T_\ell)$: $\frac12\frac{d}{ds}\|\eta\|^2=-(d_q\eta,F_q)+(d_{q'}\eta,F_{q'})=-\|d_{q'}\eta\|^2-3(d_{q'}\eta,\mathsf{c})-2\|\mathsf{c}\|^2-2(\mathsf{c},F_{q'})\le\frac14\|\mathsf{c}\|^2+2\|\mathsf{c}\|_{L^1}\|F_{q'}\|_\infty$ (as $-x^2+3xy-2y^2\le\frac14y^2$); with $|\mathsf{c}|\le 2|\eta|^2$ this gives $\frac{d}{ds}\|\eta\|^2\le C(\|\eta\|^2_\infty+\|F_{q'}\|_\infty)\|\eta\|^2$ and $\eta_0=0$, so $\eta=0$.

All times. From a datum $p\in C^\infty$ the fixed point at the level $k=1$, the regularity and the gauge transformation give a C^∞ flow line on $[0,c_{*,1}(1+\|p\|_{C^1})^{-4}]$. Let a C^∞ flow line exist on $[t_0,T)$, $T<\infty$. For $t_0<s<T$ the bounds of (b), whose proof reads only the equations of the curvature, hold on $[t_0,s]$ with $s-t_0$ in the place of s_1 and the values at t_0 in the place of those at 0; the numbers $\mathsf{Y}_0,\mathsf{Y}_1,\mathsf{Y}_2$ increase with s_1, so, for every $s<T$, y_0,y_1,y_2 are bounded on $[t_0,s]$ by the numbers that $s_1:=T-t_0$ gives, finite numbers that do not depend on s; as $|\partial_s q_\mu|\le 2y_1^{1/2}$ and $|\partial_\rho\partial_s q_\mu|\le|D_\rho\partial_s q_\mu|+2|q_\rho||\partial_s q_\mu|\le 2y_2^{1/2}+4\|q\|_\infty y_1^{1/2}$, the norm $\|q_s\|_{C^1}$ stays bounded on $[t_0,T)$, so the length of the local interval of existence from q_t stays bounded below, and the local flow line from q_t, t close to T, continues the flow line beyond T; by the uniqueness the local pieces fit together, and the flow line exists on $[0,\infty)$. For $Q\in C^3$ and $s_1>s_0$ take $(\hat q,\hat\vartheta)$ on $[0,s_0]$ and, behind it, the C^∞ flow line from the C^∞ field $\hat q_{s_0}$ on $[s_0,s_1]$ (with $g_1=\mathbf{1}$): a piecewise regular gauged flow with two pieces; for $s_1\le s_0$ the restriction of $(\hat q,\hat\vartheta)$ to $[0,s_1]$, again a regular gauged flow, is the only piece; and $\hat q^h$ on $[0,s_0]$, followed by the flow line from $\hat q_{s_0}$ transformed by h_{s_0}, held fixed in time, is the flow line from Q.

(b) *The equations of the curvature.* On a gauged flow write $\partial_s q=\mathsf{j}+d_q\vartheta_c$, $\mathsf{j}_\nu=\sum_\mu D_\mu F_{\mu\nu}$, $\vartheta_c=-\vartheta$. From $\partial_s F_{\mu\nu}=D_\mu\partial_s q_\nu-D_\nu\partial_s q_\mu$, the commutation rule and the Bianchi identity: $\partial_s F_{\mu\nu}=\sum_\lambda D_\lambda D_\lambda F_{\mu\nu}+2i\sum_\lambda[F_{\mu\lambda},F_{\lambda\nu}]+i[F_{\mu\nu},\vartheta_c]$. By induction every component T of D^kF obeys $\partial_s T=\Delta_D T+R_T+i[T,\vartheta_c]$, $\Delta_D=\sum_\lambda D_\lambda D_\lambda$, where R_T is a sum, with absolute coefficients, of commutators of a component of D^iF with one of D^jF, $i+j=k$ (one more derivative: $\partial_s(D_\kappa T)=\Delta_D(D_\kappa T)+2i\sum_\lambda[F_{\kappa\lambda},D_\lambda T]+D_\kappa R_T+i[D_\kappa T,\vartheta_c]$, the two terms with $\partial_s q_\kappa$ cancelling). As $\langle T,i[T,\vartheta_c]\rangle=0$ and $\Delta|T|^2=2\langle T,\Delta_D T\rangle+2\sum_\lambda|D_\lambda T|^2$: $\partial_s e_k=\Delta e_k-2e_{k+1}+\varrho_k$ with $|\varrho_k|\le 2\mathsf{u}_k\sqrt{e_k}\sum_{i+j=k}\sqrt{e_ie_j}$, absolute numbers u_k, and $|\varrho_0|\le 5e_0^{3/2}$: for $k=0$ only the third index contributes to each pair $\mu<\nu$, each of the twelve terms is at most $2|F_{12}||F_{13}||F_{23}|\le 2(e_0/3)^{3/2}$, and $24\cdot 3^{-3/2}<5$. These identities hold on each piece of the flow of (a); e_0,e_1,e_2 are continuous on $[0,s_1]\times T_\ell$ and e_3 on $(0,s_1]\times T_\ell$. *The maximum principle used:* if u,h are bounded, continuous and periodic on $[t_a,t_b]\times\mathbb{R}^3$, $\partial_s u,\partial_x u,\partial_x^2 u$ are continuous on each of finitely many intervals $(t,t']$ that make up $(t_a,t_b]$, and $\partial_s u-\Delta u\le h$ there, then $u_s\le e^{(s-t_a)\Delta}u_{t_a}+\int_{t_a}^s e^{(s-t)\Delta}h_t\,dt$ (differentiate $t\mapsto(e^{(s-t)\Delta}u_t)(x)$; the heat kernel is positive); in particular $\max u_s\le\max u_{t_a}+h_0(s-t_a)$ if $h\le h_0$, a constant.

(i) Let ϕ_1 be 1 on $[0,1]$, 0 off $[-1,2]$ and linear in between, and $\phi_x(z):=\prod_i\phi_1(z_i-x_i)$: $|\nabla\phi_x|^2\le 3$ almost everywhere and the support lies in 27 unit cubes. From $\partial_s e_0=2\sum_{\mu,\nu}\langle F_{\mu\nu},D_\mu\mathsf{j}_\nu\rangle$ (the gauge term drops, being orthogonal to F) and an integration by parts on $\mathbb{R}^3$: $\frac{d}{ds}\int\phi_x^2e_0=-2\int\phi_x^2\sum_\nu|\mathsf{j}_\nu|^2-4\int\phi_x\sum_\nu\langle G_\nu,\mathsf{j}_\nu\rangle\le 4\int|\nabla\phi_x|^2e_0\le 324\,\mathsf{E}_{\mathrm{loc}}(s)$, $G_\nu:=\sum_\mu\partial_\mu\phi_x F_{\mu\nu}$, $\sum_\nu|G_\nu|^2\le 2|\nabla\phi_x|^2e_0$; so $\mathsf{E}_{\mathrm{loc}}(s)\le 27\mathsf{E}_{\mathrm{loc}}(0)+324\int_0^s\mathsf{E}_{\mathrm{loc}}$, $\mathsf{E}_{\mathrm{loc}}$ is continuous, and Gronwall's lemma gives (i).

(ii) Cut $\mathbb{R}^3$ into unit cubes, eight of which have a corner at the point at which $e^{\tau\Delta}u$ is read; with the one-dimensional heat kernel, $\sup|e^{\tau\Delta}u| \le [2(4\pi\tau)^{-1/2}+1]^3 \sup_x \int_{x+[0,1]^3}|u|$ and, by Cauchy–Schwarz on each cube, $\sup|e^{\tau\Delta}u| \le [2(8\pi\tau)^{-1/4}+1]^3 \sup_x(\int_{x+[0,1]^3} u^2)^{1/2}$. The maximum principle with $u = e_0$, $h = 5e_0^{3/2}$, started at $\max\{0, s-\tau_*\}$, and $\int_{x+[0,1]^3} e_0(r)^3 \le y_0(r)^2\bar{\mathsf{E}}$ by (i), give $y_0(s) \le \max\{y_0(0), [2(4\pi\tau_*)^{-1/2}+1]^3\bar{\mathsf{E}}\} + 77.1\,\bar{\mathsf{E}}^{1/2}\tau_*^{1/4}\sup_{[0,s_1]} y_0$, using $(x+y)^3 \le 4(x^3+y^3)$, $\tau_* \le 1$ and $20[32(8\pi)^{-3/4}+1] < 77.1$; and $77.1\,\bar{\mathsf{E}}^{1/2}\tau_*^{1/4} \le \frac12$ by the choice of τ_*. The supremum is finite, and (ii) follows.

(iii) With $|\varrho_1| \le 4\mathsf{u}_1\sqrt{e_0}\,e_1$, $u = e_1 + 2\mathsf{u}_1\mathsf{h}_0 e_0$ has $\partial_s u - \Delta u \le (4\mathsf{u}_1\sqrt{e_0} - 4\mathsf{u}_1\mathsf{h}_0)e_1 + 10\mathsf{u}_1\mathsf{h}_0 e_0^{3/2} \le 10\mathsf{u}_1\mathsf{h}_0^4$; with $|\varrho_2| \le \mathsf{u}_2(4\mathsf{h}_0+1)e_2 + \mathsf{u}_2 e_1^2$, $u = e_2 + \mathsf{q}_2 e_1$ has $\partial_s u - \Delta u \le \mathsf{u}_2\mathsf{Y}_1^2 + 4\mathsf{u}_1\mathsf{q}_2\mathsf{h}_0\mathsf{Y}_1$; the maximum principle on $[0,s]$.

(iv) Let $0 < t < s \le s_1$, $s-t \le 1$. On $[t,s]$ the function $u(r) = (r-t)e_3 + \mathsf{q}_3 e_2$ has, by $|\varrho_3| \le \mathsf{u}_3(4\mathsf{h}_0+2)e_3 + 2\mathsf{u}_3 e_1 e_2$, $\partial_r u - \Delta u \le [1+\mathsf{u}_3(4\mathsf{h}_0+2) - 2\mathsf{q}_3]e_3 + \mathsf{h}_3$ with $\mathsf{h}_3 := 2\mathsf{u}_3\mathsf{Y}_1\mathsf{Y}_2 + \mathsf{q}_3[\mathsf{u}_2(4\mathsf{h}_0+1)\mathsf{Y}_2 + \mathsf{u}_2\mathsf{Y}_1^2]$, and the bracket vanishes; so $(s-t)y_3(s) \le \max u(s) \le \mathsf{q}_3 y_2(t) + \mathsf{h}_3(s-t) \le \mathsf{Y}_3$. Let $t \downarrow 0$ if $s \le 1$, and take $t = s-1$ if $s > 1$.

(c) $\mathsf{E}_{\mathrm{loc}}(0) \le y_0(0) \le 3\Phi_0(Q)^2$, $y_1(0) \le 9\Phi_1(Q)^2$, $y_2(0) \le 27\Phi_2(Q)^2$, and the bounds of the $\Phi_k(Q)$ by N_Q give $\bar{\mathsf{E}} \le 324(1+N_Q)^4 e^{324 s_1}$, $\tau_*^{-1/2} \le 1 + 155^2\bar{\mathsf{E}}$, $\mathsf{Y}_0 \le c e^{1296 s_1}(1+N_Q)^{16}$, $\mathsf{h}_0 \le c e^{648 s_1}(1+N_Q)^8$; $\mathsf{Y}_1, \mathsf{Y}_2, \mathsf{Y}_3$ are polynomials with absolute coefficients in $\mathsf{h}_0, y_0(0), y_1(0), y_2(0), s_1$. Apply (b) with $s_1 := s$; $\Phi_k(q_s)^2 \le y_k(s)$, $s \le e^s$, $\min\{s,1\}^{-1/2} \le 1 + s^{-1/2}$. No step reads ℓ. For $s \le s_0$: $\Phi_0(q_s) = \Phi_0(\hat{q}_s) \le 2(1+\|\hat{q}_s\|_{C^1})^2 \le 2(1+2N_Q)^2 \le 8(1+N_Q)^2$. □

Proposition 18.66 (The lattice gradient flow against the Yang–Mills flow)**.** Let K be a cutoff, $a := a_K$, and let all lattice objects be those of $\Lambda_{K,\ell}$.

(a) (Exact identities.) For $U \in \Omega_K$ and $b = \langle x, x+ae_\mu\rangle$: $J(U)(b) = a^{-3}\sum_{|\alpha|\ne\mu} \mathbf{w}(W_x(\mu,\alpha))$. $\Phi_s(U^u) = \Phi_s(U)^u$ for every gauge transformation u. If $s \mapsto \lambda_s$ is a continuous family of site functions on $[s', s'']$ and $\hat{U}$ solves $\partial_s\hat{U}(b) = -ia[J(\hat{U}) + D_{\hat{U}}\lambda_s](b)\hat{U}(b)$, then $\hat{U}_s = (\Phi_{s-s'}\hat{U}_{s'})^{u_s}$ with $\partial_s u_s(x) = i\lambda_s(x)u_s(x)$, $u_{s'} = \mathbf{1}$. For every 1-form E: $(D_V^{\mathsf{T}}D_V E + D_V D_V^* E)_\mu = -(\Delta^V E)_\mu + \sum_{\nu\ne\mu}[\nabla_\mu^V, \nabla_\nu^{V*}]E_\nu$, and $|\sum_{\nu\ne\mu}[\nabla_\mu^V, \nabla_\nu^{V*}]E_\nu| \le 4\Phi_0^{\mathrm{lat}}(V)|E|_\infty$.

(b) (The current near a configuration.) Let $V, \hat{U} \in \Omega_K$, $R(b) := \hat{U}(b)V(b)^{-1}$, $\eta := \max_b\|R(b) - \mathbf{1}\| \le \frac14$, $y := \eta/a$ and $\tilde{E}(b) := a^{-1}\mathbf{w}(R(b))$. Then $|\tilde{E}|_\infty \le y$ and $|J(\hat{U}) - J(V) - D_V^{\mathsf{T}}D_V\tilde{E}|_\infty \le 36\,\Phi_0^{\mathrm{lat}}(V)\,y + 40\,a^{-1}y^2$.

(c) (Consistency.) For $q \in C^4$ and $V = U_K[q]$: $\Phi_0^{\mathrm{lat}}(V) \le \Phi_0(q)$, and, the bond average being formed with q, $|J(V)(b) - \overline{d_q^* F_q}(b)| \le 2a^2[\frac{1}{12}\Phi_3 + \frac{37}{3}\Phi_0\Phi_1 + \frac13 a\Phi_0^3 + 12a^2\Phi_0^2\Phi_1](q)$, which is at most $a^2\mathfrak{t}$ if $a \le 1$.

(d) (Comparison along a regular gauged flow.) Let (q,ϑ) be a regular gauged flow on $[s', s'']$, $U' \in \Omega_K$, $y' := a^{-1}\max_b\|U'(b) - U_K[q_{s'}](b)\|$ and $\mathfrak{Y} := (y' + a^2\int_{s'}^{s''}\mathfrak{t}\,ds)\exp(10(s''-s') + 40\int_{s'}^{s''}\Phi_0(q_s)\,ds)$. If $a \le 1$ and $\mathfrak{Y} < a/4$, there are gauge transformations u_s of $\Lambda_{K,\ell}$, C^1 in s, $u_{s'} = \mathbf{1}$, with $\max_b\|(\Phi_{s-s'}U')^{u_s}(b) - U_K[q_s](b)\| \le a\,\mathfrak{Y}$ for $s' \le s \le s''$.

(e) (Every cutoff.) For a piecewise regular gauged flow from Q on $[0, s_1]$ with the numbers $\mathfrak{T}, \mathfrak{E}$ and every $s \in [\sigma_{j-1}, \sigma_j]$ there is a gauge transformation v_s with $\max_b\|\Phi_s(U_K[Q])^{v_s}(b) - U_K[q_s^{(j)}](b)\| \le a^3\max\{\mathfrak{T}\mathfrak{E}, 128\,\mathfrak{T}^3\mathfrak{E}^3\}$, and with the bound $a^3\mathfrak{T}\mathfrak{E}$ when $4a\mathfrak{T}\mathfrak{E} < 1$.

(f) There are absolute numbers c and p such that for every side ℓ, every cutoff K, every $s_1 > 0$, every $Q \in C^3(T_\ell;\mathfrak{g}^3)$ with its Yang–Mills flow line q and every $0 \le s \le s_1$: $\mathrm{err}_K(Q;s) \le c\,e^{cs_1}a_K^3(1+N_Q)^{3p}\exp(120\int_0^{s_1}\Phi_0(q_r)\,dr)$, where $\Phi_0(q_r) = \sup|F_{q_r}|$. The exponential factor is at most e^{960c_*} if $s_1 \le c_*(1+N_Q)^{-2}$, and at most $\exp(ce^{cs_1}(1+N_Q)^8)$ always.

■

Sketch. Facts on G used throughout: for $X \in \mathfrak{g}$, $\|e^{iX} - \mathbf{1}\| = 2|\sin\frac{|X|}{2}| \le |X|$ and $|\mathbf{w}(e^{iX}) - X| \le \frac{1}{6}|X|^3$; for $\Sigma \in \mathfrak{g}$ and $W \in G$, $\mathbf{c}(i\Sigma W) = -\langle \Sigma, \mathbf{w}(W)\rangle$ and $\langle \mathrm{Ad}(W)Z, \mathbf{w}(W)\rangle = \langle Z, \mathbf{w}(W)\rangle$; $\|\mathrm{Ad}(V) - \mathrm{Ad}(V')\| \le 2\|V - V'\|$; and if $V' = i\Xi_1 V$, $V_0' = i\Xi_0 V_0$ on $[0,r]$ with $V(0) = V_0(0)$, then $\|V(r) - V_0(r)\| \le \int_0^r |\Xi_1 - \Xi_0|$.

(a) By the curvature formula YM3_I:(69), $F_V(p) = a^{-2}\mathbf{w}(\partial V(p))$, and for $\mu \ne \nu$ the component $F_{\mu\nu}(x)$ is $a^{-2}\mathbf{w}(W_x(\mu,\nu))$, as $W_x(\nu,\mu) = W_x(\mu,\nu)^{-1}$ and $\mathbf{w}(V^{-1}) = -\mathbf{w}(V)$; the term $-\mathrm{Ad}(V(x - ae_\nu,\nu)^{-1})F_{\mu\nu}(x - ae_\nu)$ of YM3_I:(68) is $a^{-2}\mathbf{w}(W_x(\mu,-\nu))$, the transported loop reducing to $W_x(\mu,-\nu)^{-1}$; this is the plaquette form of $J = D_V^{\mathsf{T}} F_V$. The covariance of Φ_s follows from $J(U^u) = \mathrm{Ad}(u)J(U)$ since each based holonomy transforms by conjugation at its base and the uniqueness of solutions, the flow existing for all s as its field is tangent to the compact manifold Ω_K. For the gauged flow put $\tilde U_s := (\Phi_{s-s'}\hat U_{s'})^{u_s}$; with $\partial_s(u^{-1}) = -u^{-1}i\lambda$ and the covariance of J, $\partial_s\tilde U(b) = [i\lambda(b_-) - iaJ(\tilde U)(b) - i\mathrm{Ad}(\tilde U(b))\lambda(b_+)]\tilde U(b) = -ia[J(\tilde U) + D_{\tilde U}\lambda](b)\tilde U(b)$ by the covariant-gradient formula: the equation of $\hat U$ with the same initial value. For the identity of Weitzenböck type: by YM3_I:(67), YM3_I:(68), $(D_V^{\mathsf{T}} D_V E)_\mu = \sum_{\nu\ne\mu}\nabla_\nu^{V*}\nabla_\nu^V E_\mu - \sum_{\nu\ne\mu}\nabla_\nu^{V*}\nabla_\mu^V E_\nu$ and $(D_V D_V^* E)_\mu = \sum_\nu \nabla_\mu^V\nabla_\nu^{V*}E_\nu$; $\nabla_\nu^{V*}\nabla_\nu^V = \nabla_\nu^V\nabla_\nu^{V*}$, and the sum of these over ν is $-\Delta^V$; for $\nu \ne \mu$ the expressions $\nabla_\mu^V\nabla_\nu^{V*}f$ and $\nabla_\nu^{V*}\nabla_\mu^V f$ agree but for the terms at $z = x + ae_\mu - ae_\nu$, and $V(x,\mu)V(z,\nu)^{-1} = W_x(\mu,-\nu)V(x - ae_\nu,\nu)^{-1}V(x - ae_\nu,\mu)$ gives $([\nabla_\mu^V, \nabla_\nu^{V*}]f)(x) = a^{-2}[\mathrm{Ad}(W_x(\mu,-\nu)) - 1]\mathrm{Ad}(V(x - ae_\nu,\nu)^{-1}V(x - ae_\nu,\mu))f(z)$, of norm at most $2\Phi_0^{\mathrm{lat}}(V)|f|_\infty$.

(b) $|\tilde E| \le a^{-1}\|R - \mathbf{1}\|$. Fix b and α, let $\beta_1,\dots,\beta_4$ be the positively oriented bonds of the loop of $W_x(\mu,\alpha)$, run in the orientations $\epsilon_j = \pm 1$, and P_j the product of the values of V on the bonds that the loop runs from x until it reaches $(\beta_j)_-$, the initial point of the positively oriented bond β_j; the loop reaches this point before it runs β_j if $\epsilon_j = 1$, and after it if $\epsilon_j = -1$. With $\mathsf{e} := \mathbf{w}(R) = a\tilde E$: $\hat W = \tilde R_1\tilde R_2\tilde R_3\tilde R_4 W$, $\tilde R_j := \mathrm{Ad}(P_j)R(\beta_j)^{\epsilon_j} = \mathbf{c}_j + i\tilde{\mathsf{e}}_j$, $\tilde{\mathsf{e}}_j = \epsilon_j\mathrm{Ad}(P_j)\mathsf{e}(\beta_j)$; $\|\tilde R_j - \mathbf{1}\| \le \eta$ and $0 \le 1 - \mathbf{c}_j \le \eta^2/2$, so $\prod_j \tilde R_j = \mathbf{1} + i\tilde\Sigma + Z$, $\tilde\Sigma := \sum_j \tilde{\mathsf{e}}_j$, $\|Z\| \le 10\eta^2$ for $\eta \le \frac14$, and $\mathbf{w}(\hat W) - \mathbf{w}(W) = \tilde\Sigma + r_1$, $|r_1| \le 4\eta a^2\Phi_0^{\mathrm{lat}}(V) + 10\eta^2$. For $\alpha = \nu > 0$, $\tilde\Sigma$ differs from $a(D_V\mathsf{e})_{\mu\nu}(x)$ (YM3_I:(67)) by two factors $\mathrm{Ad}(W)$, that is by at most $4a^2\Phi_0^{\mathrm{lat}}\eta$; for $\alpha = -\nu$ it differs from $-a\,\mathrm{Ad}(V(x - ae_\nu,\nu)^{-1})(D_V\mathsf{e})_{\mu\nu}(x - ae_\nu)$ by three such factors, by at most $6a^2\Phi_0^{\mathrm{lat}}\eta$. Summing over the four α with (a) and YM3_I:(68): $a^3(J(\hat U) - J(V))(b) = a^2(D_V^{\mathsf{T}} D_V\mathsf{e})(b) + r_2$, $|r_2| \le 36a^2\Phi_0^{\mathrm{lat}}\eta + 40\eta^2$; divide by a^3 and put $\eta = ay$.

(c) For the rectangle $x \to x + a_1e_\mu \to x + a_1e_\mu + re_\nu \to x + re_\nu \to x$ the holonomy $W(r)$ has $W'(r)W(r)^{-1} = i\Xi(r)$, $\Xi(r) = \int_0^{a_1}\mathrm{Ad}(T(r,t))F_{\mu\nu}(x + te_\mu + re_\nu)\,dt$, $T(r,t)$ the holonomy along the boundary up to the point $x + te_\mu + re_\nu$; this is the variation formula $W'W^{-1} = i\int \mathrm{Ad}(H)F_{\alpha\beta}\partial_r\sigma^\alpha\partial_t\sigma^\beta dt$ of the holonomy of a family $\sigma(r,\cdot)$ of closed paths with fixed ends. So $|\Xi| \le a_1\Phi_0$ and $\|W(a_2) - \mathbf{1}\| \le |a_1a_2|\Phi_0$: $\Phi_0^{\mathrm{lat}}(V) \le \Phi_0(q)$. Fix $\nu \ne \mu$ and let $W(r)$ be as above with $a_1 = a$, $|r| \le a$, so that $W(\pm a) = W_x(\mu,\pm\nu)$. Iterating $W(r) = \mathbf{1} + i\int_0^r \Xi W$ twice and using $[\Xi(r),\Xi(r')] = [\Xi(r) - \Xi(r'),\Xi(r')]$ with $|\Xi(r) - \Xi(r')| \le a|r - r'|\Phi_1 + 12a^4\Phi_0\Phi_1$: $\mathbf{w}(W(a)) = \int_0^a \Xi(r)\,dr + \varepsilon_+$, $|\varepsilon_+| \le \frac16 a^5\Phi_0\Phi_1 + 6a^7\Phi_0^2\Phi_1 + \frac16 a^6\Phi_0^3$, and likewise $\mathbf{w}(W(-a)) = -\int_0^a \Xi(-r)\,dr + \varepsilon_-$. With $H(t)$ the holonomy from x to $z := x + te_\mu$, $S_\nu(r)$ that from z to $z + re_\nu$, and Rect that of the small rectangle $z \to z + (a-t)e_\mu \to z + (a-t)e_\mu + re_\nu \to z + re_\nu \to z$: $T(r,t) = H(t)\,\mathrm{Rect}\,S_\nu(r)$; the matrix $\exp(i(a-t)rF_{\mu\nu}(z))$ commutes with $F_{\mu\nu}(z)$ and lies within $\frac23|(a-t)r|(2(a-t) + |r|)\Phi_1 \le 2a^3\Phi_1$ of Rect, so $\mathrm{Ad}(T(r,t))F_{\mu\nu}(z + re_\nu) = \mathrm{Ad}(H(t))[\varphi_t(r) + \delta]$ with $\varphi_t(r) := \mathrm{Ad}(S_\nu(r))F_{\mu\nu}(z + re_\nu)$ and $|\delta| \le 6a^3\Phi_0\Phi_1$. By the covariant Taylor formula, $\varphi_t^{(j)}(r) = \mathrm{Ad}(S_\nu(r))(D_\nu^j F_{\mu\nu})(z + re_\nu)$, so $|\varphi_t(r) - \varphi_t(-r) - 2rD_\nu F_{\mu\nu}(z)| \le \frac13 r^3\Phi_3$, the even orders cancelling. Integrating in t and r: $\mathbf{w}(W_x(\mu,\nu)) + \mathbf{w}(W_x(\mu,-\nu)) = a^2\int_0^a \mathrm{Ad}(H(t))(D_\nu F_{\mu\nu})(x + te_\mu)\,dt + \varepsilon_\nu$, $|\varepsilon_\nu| \le a^5[\frac{1}{12}\Phi_3 + \frac{37}{3}\Phi_0\Phi_1 + \frac13 a\Phi_0^3 + 12a^2\Phi_0^2\Phi_1]$. Sum over the two $\nu \ne \mu$, divide by a^3, and use (a) and $(d_q^*F)_\mu = \sum_\nu D_\nu F_{\mu\nu}$.

(d) *The equation of* $V_s := U_K[q_s]$. The variation of a holonomy under a variation of the field is $\partial_s \mathrm{hol}\cdot \mathrm{hol}^{-1} = i\int \mathrm{Ad}(H)\,\partial_s q_\lambda \dot\gamma^\lambda dt$, and the transport rule $\frac{d}{dt}\mathrm{Ad}(H(t))\Psi(x+te_\mu) = \mathrm{Ad}(H(t))(D_\mu\Psi)(x+te_\mu)$ gives $\int_0^a \mathrm{Ad}(H(t))(D_\mu\vartheta)(x+te_\mu)\,dt = \mathrm{Ad}(V(b))\vartheta(x+ae_\mu) - \vartheta(x)$; with (c): $\partial_s V(b)V(b)^{-1} = -ia[J(V) - \tau_s + D_V\vartheta_s](b)$, $|\tau_s|_\infty \le a^2\mathfrak{t}(s)$ for $s > s'$. *The gauged lattice flow, with the continuum flow as the reference.* Let $\hat U$ solve $\partial_s\hat U(b) = -ia[J(\hat U) + D_{\hat U}\lambda](b)\hat U(b)$, $\hat U_{s'} = U'$, with $\lambda := \vartheta_s| + D^*_{V_s}\tilde E$ and $\tilde E(b) := a^{-1}\mathbf{w}(\hat U(b)V_s(b)^{-1})$; the field is continuous in s and tangent to the compact manifold, so $\hat U$ exists on $[s', s'']$, and $\hat U_s = (\Phi_{s-s'}U')^{u_s}$ by (a). *A scalar identity.* With $R := \hat U V^{-1}$, $\mathsf{e} := \mathbf{w}(R) = a\tilde E$ and $n_b := 1 - \mathbf{c}(R(b)) = \frac12\|R(b) - \mathbf{1}\|^2$: $\partial_s R\, R^{-1} = ia\mathsf{y}$, $\mathsf{y} = -J(\hat U) - D_{\hat U}\lambda + \mathrm{Ad}(R)[J(V) - \tau + D_V\vartheta]$, and $n_b' = a\langle \mathsf{y}, \mathsf{e}\rangle(b)$. As $\langle \mathrm{Ad}(R)Z, \mathsf{e}\rangle = \langle Z, \mathsf{e}\rangle$, also $\langle D_{\hat U}\lambda, \mathsf{e}\rangle(b) = \langle D_V\lambda, \mathsf{e}\rangle(b)$; so ϑ drops out, and with (a): $n_b' = a\langle \Delta^V\tilde E - \sum_{\nu\neq\mu}[\nabla^V_\mu, \nabla^{V*}_\nu]\tilde E_\nu - (J(\hat U) - J(V) - D_V^{\mathsf{T}}D_V\tilde E) - \tau, \mathsf{e}\rangle(b)$. *The maximum principle.* Let $y(s) := a^{-1}\max_b \|R_s(b) - \mathbf{1}\|$ and $n := \max_b n_b = \frac12 a^2y^2$; n is Lipschitz, and where it is differentiable $n' = n_b'$ at every b where the maximum is attained; there $|\mathsf{e}(b')| \le |\mathsf{e}(b)|$ for all b', as $|\mathsf{e}|^2 = n_b(2 - n_b)$ increases with $n_b \le 1$, so that $\langle(\Delta^V\tilde E)_\mu(x), \mathsf{e}(b)\rangle \le 0$. By (a), (b), $\Phi_0^{\mathrm{lat}}(V_s) \le \Phi_0(q_s)$ and $40a^{-1}y^2 \le 10y$, as long as $y \le a/4$: $n' \le a\sqrt{2n}\,[|\tau_s|_\infty + (40\Phi_0(q_s) + 10)y]$ almost everywhere; so $(\sqrt{n+\delta})' \le \frac{a}{\sqrt2}[\cdots]$, and letting $\delta \downarrow 0$: $y(s) \le y' + \int_{s'}^s [a^2\mathfrak{t} + (40\Phi_0 + 10)y]$. Gronwall's lemma gives $y \le \mathfrak{Y} < a/4$ wherever $y \le a/4$ held before, and y is continuous: the bound holds on all of $[s', s'']$, and $\|\hat U_s(b) - V_s(b)\| = \|R_s(b) - \mathbf{1}\| \le a\mathfrak{Y}$.

(e) Apply (d) on each piece with $U' := \Phi_{\sigma_{j-1}}(U_K[Q])^v$, v the gauge transformation reached at σ_{j-1}, and $\Phi_{s-\sigma_{j-1}}(U') = \Phi_s(U_K[Q])^v$ by (a); at σ_j pass from v to $(g_j|)v$, as $U_K[q^{(j+1)}_{\sigma_j}] = U_K[q^{(j)}_{\sigma_j}]^{g_j|}$ and a common gauge transformation keeps the distances. Starting from $y' = 0$, the bounds compose to $y \le a^2\mathfrak{T}\mathfrak{E}$ whenever $a^2\mathfrak{T}\mathfrak{E} < a/4$, that is, whenever $4a\mathfrak{T}\mathfrak{E} < 1$; this is the bound $a^3\mathfrak{T}\mathfrak{E}$. If $4a\mathfrak{T}\mathfrak{E} \ge 1$, then $\|\cdot\| \le 2 \le 2(4a\mathfrak{T}\mathfrak{E})^3 = 128a^3\mathfrak{T}^3\mathfrak{E}^3$.

(f) By Proposition 18.65(a) the flow line q from Q is, up to the gauge transformations h_s for $s \le s_0$ and h_{s_0} for $s > s_0$, the field of a piecewise regular gauged flow from Q, with one piece if $s_1 \le s_0$ and with two if $s_1 > s_0$, whose numbers $\mathfrak{T}, \mathfrak{E}$ are formed with the $\Phi_k(q_s)$; a gauge transformation of the continuum field changes $U_K[q_s]$ by a lattice gauge transformation, which the minimum in err_K absorbs. By Proposition 18.65(c) and $\int_0^{s_1}(1 + s^{-1/2})\,ds = s_1 + 2s_1^{1/2}$: $\mathfrak{T} \le ce^{cs_1}(1 + N_Q)^p$; and $\max\{\mathfrak{T}\mathfrak{E}, 128\mathfrak{T}^3\mathfrak{E}^3\} \le 128(1 + \mathfrak{T})^3\mathfrak{E}^3$, $\mathfrak{E}^3 = e^{30s_1}\exp(120\int_0^{s_1}\Phi_0(q_r)\,dr)$; with (e) this is the bound, with new numbers c, p. For $s_1 \le c_*(1 + N_Q)^{-2} = s_0$: $120\int_0^{s_1}\Phi_0 \le 120s_1 \cdot 8(1 + N_Q)^2 \le 960c_*$; always $120\int_0^{s_1}\Phi_0 \le 120s_1ce^{cs_1}(1 + N_Q)^8$. □

II. The limit form of the block observables. Throughout this part the setting (E) of §2.3.1 holds, and $\Lambda_{d,\ell}$, a_d, Ω_d, μ^K, the averaging map $\mathcal{M}$, ϖ, θ_d, I_d, R_n, o_n, $\mathsf{t}_\natural$, n_2, ω_L, the threads, $\mu = \mu_{\ell,\mathsf{b}}$, μ_d, μ_d^K, $\pi_{K,d}$, Γ_d, $\mathcal{E}_K$, L^2_{inv}, the strong convergence and the limits, together with the tangent vectors X at V, with their curves $s \mapsto e^{isX}V$ and their norm $\|X\|^2 = \sum_b |X(b)|^2$, the gradient ∂g, the gauge directions $d_V\lambda$, $(d_V\lambda)(b) = \lambda(b_-) - \mathrm{Ad}(V(b))\lambda(b_+)$ for $\mathfrak{g}$-valued functions λ of the sites, and the horizontal vectors, which are orthogonal to the gauge directions, are those of the setting of Section 15.1; $\omega_M = (200L^2)^{-1}$ is the averaging radius of YM3_I:(30). For each coarse bond c write $\mathcal{S}_c$ for the configurations with $|\partial U(p) - \mathbf{1}| < \omega_M$ at every plaquette of its double block. The block map on $\mathcal{S}_c$ is the analytic formula YM3_I:(37), with $U[c]$ the straight transport along c and W_x the relative contour transports in that formula.

Horizontal vectors, the forms t*, the transport constants.* For $V \in \Omega_d$, Π_V is the orthogonal projection, for the norm $\|B\|^2 = \sum_b |B(b)|^2$ of the 1-forms of $\Lambda_{d,\ell}$, onto the horizontal vectors at V; $C^1_{\mathrm{inv}}(\Omega_d)$ is the space of the invariant C^1 functions on Ω_d, and the gradient $\partial g(V)$ of such a function is horizontal. For a map F from $\Omega_{j'}$ to Ω_j, $j \le j'$, that is differentiable at W, the tangent vector $DF(W)X$ at $F(W)$ is given by $\frac{d}{ds}F(e^{isX}W)(b)|_{s=0} = i\,(DF(W)X)(b)\,F(W)(b)$ at every bond b of $\Lambda_{j,\ell}$, and $DF(W)^*$ is the transpose of $DF(W)$ for the scalar products $\sum_b\langle X(b), Y(b)\rangle$ of the two lattices. A covariant map carries gauge directions to gauge directions: if $F(W^u) = F(W)^{u|}$

for all gauge transformations u and F is differentiable at W, then $DF(W)\,d_W\varphi = d_{F(W)}(\varphi|)$ (differentiate at $u = e^{it\varphi}$), so that $DF(W)^*$ maps the horizontal vectors at $F(W)$ to horizontal vectors at W, and $g \circ F$ is invariant with $\partial(g\circ F) = DF^*(\partial g \circ F)$ for invariant g. The forms $\mathsf{t}[\alpha,\beta;V]$ are those of Proposition 15.2(b); for every $\varrho \in [0,1]$ the form $\mathsf{t}[(\frac23,\frac16)+\varrho(\frac13,-\frac16);V]$ lies between $\frac13\|\cdot\|^2$ and $\|\cdot\|^2$, and $\mathsf{t}[2,-1;V](B) = \sum_c |B(c) - \mathrm{Ad}(V(c))B(c^+)|^2 \le 4\|B\|^2$. A_L is the linear map $(\alpha,\beta)\mapsto(\frac{(2L^2+1)\alpha+4(L^2-1)\beta}{3L^2}, \frac{(L^2-1)\alpha+2(L^2+2)\beta}{6L^2})$, so that $A_L(\frac23,\frac16) = (\frac23,\frac16)$ and $A_L(\frac13,-\frac16) = L^{-2}(\frac13,-\frac16)$. The transport constants are two numbers $\varepsilon_{\rm tr} \in (0,(48000L^4)^{-1}]$ and $C_{\rm tr} \ge 1$, functions of L alone, such that (Tr) holds: for every $d \ge 0$, every $U \in \Omega_{d+1}$ with $\varpi(U) \le \epsilon \le \varepsilon_{\rm tr}$, $V := \mathcal{M}U$, every horizontal B at V and all real α,β, the map $\mathcal{M}$ is real-analytic near U and $|L\,\mathsf{t}[\alpha,\beta;U](D\mathcal{M}(U)^*B) - \mathsf{t}[A_L(\alpha,\beta);V](B)| \le C_{\rm tr}(|\alpha|+|\beta|)\,\epsilon\,\|B\|^2$; such numbers exist by the transport of the forms t under the transpose of the differential of $\mathcal{M}$ in the proof of Proposition 15.2(b), taken at one step, whose linear part is computed in the proof of Proposition 18.67.

Depths and events. n_θ is the least integer $n \ge 1$ with $L^n\mathsf{b} \ge \mathsf{t}_\natural$, and $d_\diamond$ the least integer $d \ge \max\{n_\theta, n_2\}$ with $\theta_j \le \varepsilon_{\rm tr}$ for $j > d$ and $36C_{\rm tr}\sum_{j>d}\theta_j \le 1$, which exists as θ_j decreases to 0 and $\sum_j \theta_j < \infty$. For a thread Y, $\mathfrak{d}(Y)$ is the least $n \ge n_\theta$ with $\varpi(Y_j) < \theta_j$ for all $j \ge n$, and ∞ if there is none; by Lemma 15.1(c), $\mu(\mathfrak{d} > n) \le \mathsf{R}_n$, so that $\mathrm{Thr}^\circ := \{Y : \mathfrak{d}(Y) < \infty\}$ has full measure. At a cutoff K and for $n_\theta \le n \le K$, $\mathsf{A}_n^K := \{U \in \Omega_K : \varpi(\pi_{K,j}U) < \theta_j \text{ for } n \le j \le K\}$, an invariant Borel set ($\mathcal{M}$ is Borel, Proposition YM3_I:4.5) with $\mu^K(\Omega_K \setminus \mathsf{A}_n^K) \le \mathsf{R}_n$ for $n < K$ (Lemma 15.1(c)); $\mathsf{A}_{K+1}^K := \Omega_K$; the sets A_n^K increase with n. For $n \ge n_2$ the set A_n^K is open, and $\pi_{K,n}$ is continuous on it, by induction on j from K down to n: $\mathsf{A}_K^K = \{\varpi < \theta_K\}$ is open; for $n \le j < K$, on A_{j+1}^K the map $\pi_{K,j+1}$ is continuous and $\varpi(\pi_{K,j+1}U) < \theta_{j+1} \le \omega_L < \omega_M$ (by the choice of n_2, $\theta_i \le \omega_L$ for $i \ge n_2$), and $\mathcal{M}$ is real-analytic on the open set $\{\varpi < \omega_M\}$, which lies in every set $\mathcal{S}_c$ (Proposition YM3_I:4.5); so $\pi_{K,j} = \mathcal{M}\circ\pi_{K,j+1}$ is continuous on A_{j+1}^K, and A_j^K is open. $\mathsf{tv}_j^K := \sup_E |\mu_j^K(E) - \mu_j(E)|$, which tends to 0 as $K\to\infty$ (Corollary 2.8). A gauge transformation v of $\bigcup_d \Lambda_{d,\ell}$ acts on the threads by $(v\cdot Y)_d := Y_d^{v|}$, $v|$ the restriction to $\Lambda_{d,\ell}$; μ is invariant under these maps, as μ^K is invariant under the bare gauge transformations and $\pi_{K,d}(U^u) = (\pi_{K,d}U)^{u|}$ YM3_I:(40).

Smoothed block maps. A family of maps $\tilde{\mathcal{M}} : \Omega_{j+1} \to \Omega_j$, $j \ge 0$, is admissible if there are numbers $\omega_{\tilde{\mathcal{M}}} \in (0,\omega_M]$ and $c_{\tilde{\mathcal{M}}} \ge 1$, which read neither j nor m, such that: (N1) $\tilde{\mathcal{M}}$ is C^1; (N2) $\tilde{\mathcal{M}}(U^u) = \tilde{\mathcal{M}}(U)^{u|}$ for every gauge transformation u of $\Lambda_{j+1,\ell}$; (N3) $\tilde{\mathcal{M}}(U) = \mathcal{M}(U)$ whenever $\varpi(U) < \omega_{\tilde{\mathcal{M}}}$; (N4) $\|D\tilde{\mathcal{M}}(U)\| \le c_{\tilde{\mathcal{M}}}$ for all U and j, in the operator norm for $\|\cdot\|$. Put $\mathsf{k}_{\tilde{\mathcal{M}}} := Lc_{\tilde{\mathcal{M}}}^2$, $d_{\tilde{\mathcal{M}}}$ the least $d \ge d_\diamond$ with $\theta_j < \omega_{\tilde{\mathcal{M}}}$ for $j > d$, and $\tilde{\mathcal{M}}^n$ the n-fold composite. For $Y \in \mathrm{Thr}^\circ$ and $d \ge 0$ let $\mathsf{j}_0(Y) := \max\{d, d_{\tilde{\mathcal{M}}}, \mathfrak{d}(Y)-1\}$ and $Y_d^{\tilde{\mathcal{M}}} := \tilde{\mathcal{M}}^{J-d}Y_J$ for $J \ge \mathsf{j}_0(Y)$, which does not read J (Proposition 18.67(c)); $Y_d^{\tilde{\mathcal{M}}} := \mathbf{1}$ off Thr°. For $g \in C^1_{\rm inv}(\Omega_d)$ and $J \ge d$ put $g^{(J)} := g\circ\tilde{\mathcal{M}}^{J-d}$, and at a cutoff $K \ge d$, $f_g^K := g\circ\tilde{\mathcal{M}}^{K-d}$ on Ω_K; $d^\vee := \max\{d, d_{\tilde{\mathcal{M}}}\}$ and $\mathsf{z}_J(g) := a_d^{-1/2}\mathsf{k}_{\tilde{\mathcal{M}}}^{(J-d)/2}\|\partial g\|_\infty$. $\mathcal{C}_d^{\tilde{\mathcal{M}}} := \{g(Y_d^{\tilde{\mathcal{M}}}) : g \in C^1_{\rm inv}(\Omega_d)\}$ and $\mathrm{Cyl}^{\tilde{\mathcal{M}}} := \bigcup_d \mathcal{C}_d^{\tilde{\mathcal{M}}}$.

Proposition 18.67 (The matrix of the threads; smoothed block maps)**.** (a) (Many steps.) Let $0 \le d < j$, $W \in \Omega_j$, $V_i := \mathcal{M}^{j-i}W$ for $d \le i \le j$, with $\varpi(V_i) \le \varepsilon_{\rm tr}$ for $d < i \le j$ and $\sigma := \sum_{d<i\le j}\varpi(V_i) \le (6C_{\rm tr})^{-1}$. Then $\mathcal{M}^{j-d}$ is real-analytic near W, and for horizontal λ at V_d, with $q_i(\lambda) := a_i^{-1}|D\mathcal{M}^{i-d}(V_i)^*\lambda|^2$: $|a_dq_j(\lambda) - \mathsf{t}[(\frac23,\frac16)+L^{-2(j-d)}(\frac13,-\frac16);V_d](\lambda)| \le 6C_{\rm tr}\sigma\|\lambda\|^2$, so that $(\frac13 - 6C_{\rm tr}\sigma)\|\lambda\|^2 \le a_dq_j(\lambda) \le (1+6C_{\rm tr}\sigma)\|\lambda\|^2$; and for $d \le J \le j$: $a_d|q_j(\lambda) - q_J(\lambda)| \le \rho_{d,J}\|\lambda\|^2$ with $\rho_{d,J} := \frac23 L^{-2(J-d)} + 12C_{\rm tr}(\sum_{d<i\le J}L^{-2(J-i)}\varpi(V_i) + \sum_{J<i\le j}\varpi(V_i))$; the same holds for the polarised forms, with $\|\lambda\|\|\lambda'\|$ in the place of $\|\lambda\|^2$.

(b) (The matrix.) For $Y \in \mathrm{Thr}^\circ$ and $d \ge \max\{d_\diamond, \mathfrak{d}(Y)-1\}$ the limit $\mathsf{Mat}_d(Y) := \lim_j a_j^{-1}\Pi_{Y_d}D\mathcal{M}^{j-d}(Y_j)D\mathcal{M}^{j-d}(Y_j)^*\Pi_{Y_d}$ exists; $|a_d\langle\lambda, \mathsf{Mat}_d(Y)\lambda\rangle - \mathsf{t}[\frac23,\frac16;Y_d](\lambda)| \le 6C_{\rm tr}\sum_{j>d}\varpi(Y_j)\,\|\lambda\|^2$ for horizontal λ, so that $\frac16\Gamma_d(g)(Y_d) \le \langle\partial g(Y_d), \mathsf{Mat}_d(Y)\partial g(Y_d)\rangle \le$

$\frac{7}{6}\Gamma_d(g)(Y_d)$ for $g \in C^1_{\mathrm{inv}}(\Omega_d)$; $\mathsf{Mat}_d = \Pi_{Y_d} D\mathcal{M}(Y_{d+1})\mathsf{Mat}_{d+1} D\mathcal{M}(Y_{d+1})^* \Pi_{Y_d}$; $Y \mapsto \mathsf{Mat}_d(Y)$ is Borel on $\{\mathfrak{d} \le d+1\}$, and $\mathsf{Mat}_d(v\cdot Y) = \mathrm{Ad}_v \mathsf{Mat}_d(Y)\mathrm{Ad}_v^{-1}$, $(\mathrm{Ad}_v X)(b) := \mathrm{Ad}(v(b_-))X(b)$.

(c) (Admissible maps.) Let $\tilde{\mathcal{M}}$ be admissible, $Y \in \mathrm{Thr}^\circ$ and $d \ge 0$. Then $Y_d^{\tilde{\mathcal{M}}}$ does not read $J \ge \mathsf{j}_0(Y)$, $Y_d^{\tilde{\mathcal{M}}} = \tilde{\mathcal{M}}(Y_{d+1}^{\tilde{\mathcal{M}}})$, and $Y_j^{\tilde{\mathcal{M}}} = Y_j$ for $j \ge \max\{d_{\tilde{\mathcal{M}}}, \mathfrak{d}(Y)-1\}$. The limit $\mathsf{Mat}_d^{\tilde{\mathcal{M}}}(Y) := \lim_j a_j^{-1} \Pi_{Y_d^{\tilde{\mathcal{M}}}} D\tilde{\mathcal{M}}^{j-d}(Y_j) D\tilde{\mathcal{M}}^{j-d}(Y_j)^* \Pi_{Y_d^{\tilde{\mathcal{M}}}}$ exists and equals $\Pi_{Y_d^{\tilde{\mathcal{M}}}} D\tilde{\mathcal{M}}^{J-d}(Y_J)\mathsf{Mat}_J(Y) D\tilde{\mathcal{M}}^{J-d}(Y_J)^* \Pi_{Y_d^{\tilde{\mathcal{M}}}}$ for every $J \ge \mathsf{j}_0(Y)$; with the value 0 off Thr° it is a Borel map into the non-negative symmetric operators, $\mathsf{Mat}_d^{\tilde{\mathcal{M}}}(v\cdot Y) = \mathrm{Ad}_v \mathsf{Mat}_d^{\tilde{\mathcal{M}}}(Y)\mathrm{Ad}_v^{-1}$, $\mathsf{Mat}_d^{\tilde{\mathcal{M}}} = \Pi_{Y_d^{\tilde{\mathcal{M}}}} D\tilde{\mathcal{M}}(Y_{d+1}^{\tilde{\mathcal{M}}}) \mathsf{Mat}_{d+1}^{\tilde{\mathcal{M}}} D\tilde{\mathcal{M}}(Y_{d+1}^{\tilde{\mathcal{M}}})^* \Pi_{Y_d^{\tilde{\mathcal{M}}}}$, $\|a_d \mathsf{Mat}_d^{\tilde{\mathcal{M}}}(Y)\| \le \frac{7}{6}\mathsf{k}_{\tilde{\mathcal{M}}}^{\mathsf{j}_0(Y)-d}$, and $\int \|a_d \mathsf{Mat}_d^{\tilde{\mathcal{M}}}\|^p d\mu \le (\frac{7}{6})^p [\mathsf{k}_{\tilde{\mathcal{M}}}^{p(d^\vee - d)} + \sum_{J > d^\vee} \mathsf{k}_{\tilde{\mathcal{M}}}^{p(J-d)} \mathsf{R}_J] < \infty$ for $1 \le p < \infty$. If $d \ge d_{\tilde{\mathcal{M}}}$ and $\mathfrak{d}(Y) \le d+1$, then $Y_d^{\tilde{\mathcal{M}}} = Y_d$ and $\mathsf{Mat}_d^{\tilde{\mathcal{M}}}(Y) = \mathsf{Mat}_d(Y)$.

(d) (Existence.) Let $\chi_0 \in C^\infty(\mathbb{R}; [0,1])$ be 1 on $(-\infty, \frac{1}{4}]$ and 0 on $[\frac{1}{2}, \infty)$; for a positively oriented bond c of $\Lambda_{j,\ell}$ and $U \in \Omega_{j+1}$ let $\vartheta_c(U) := \prod_p \chi_0(|\partial U(p) - \mathbf{1}|^2/\omega_M^2)$, the product over the plaquettes of the double block of c, and $\mathcal{M}^\chi(U)(c) := \exp(\vartheta_c(U) L^{-3} \sum_{x \in B(c_-)} \log W_x)\, U[c]$ for $U \in \mathcal{S}_c$, $\mathcal{M}^\chi(U)(c) := U[c]$ where $\vartheta_c(U) = 0$, with W_x and $U[c]$ from the analytic block formula above. Then $\mathcal{M}^\chi$ is C^∞ and admissible, with $\omega_{\mathcal{M}^\chi} = \frac{1}{2}\omega_M$, $d_{\mathcal{M}^\chi} = d_\diamond$ and $c_{\mathcal{M}^\chi} = \max\{1, \sqrt{6}\, r_\chi\}$, where $r_\chi < \infty$ is a number of L and χ_0 alone.

■

Sketch. (a) $\mathcal{M}$ is real-analytic near every V_i, as $\varpi(V_i) \le \varepsilon_{\mathrm{tr}} < \omega_M$ (Proposition YM3_I:4.5). Put $\lambda_d := \lambda$, $\lambda_i := D\mathcal{M}(V_i)^* \lambda_{i-1}$, horizontal at V_i, so that $\lambda_j = D\mathcal{M}^{j-d}(W)^* \lambda$, and $\tau_i(\alpha,\beta) := L^{i-d}\, \mathsf{t}[\alpha,\beta; V_i](\lambda_i)$, so that $a_d q_i(\lambda) = \tau_i(1,0)$. (Tr) at the step from V_{i-1} to V_i, multiplied by L^{i-1-d}, reads $\tau_i(\alpha,\beta) = \tau_{i-1}(A_L(\alpha,\beta)) + e_i$ with $|e_i| \le C_{\mathrm{tr}}(|\alpha|+|\beta|)\varpi(V_i)\tau_{i-1}(1,0)$. The iterates $A_L^k(1,0) = (\frac{2}{3}, \frac{1}{6}) + L^{-2k}(\frac{1}{3}, -\frac{1}{6})$ have $|\alpha|+|\beta| \le \frac{7}{6}$; by induction on i, $\tau_i(1,0) \le (1 + \frac{7}{3} C_{\mathrm{tr}}\sigma)\|\lambda\|^2 \le 2\|\lambda\|^2$, and $|\tau_j(1,0) - \mathsf{t}[A_L^{j-d}(1,0); V_d](\lambda)| \le \frac{7}{3} C_{\mathrm{tr}}\sigma \|\lambda\|^2$. For the differences: $\tau_j(1,0) - \tau_J(1,0) = -\frac{1}{6}(1 - L^{-2(j-J)})\tau_J(2,-1) + \sum_{J<i\le j} e_i$, as $A_L^{j-J}(1,0) - (1,0) = -\frac{1}{6}(1 - L^{-2(j-J)})(2,-1)$, and $(2,-1)$ is an eigenvector of A_L with the eigenvalue L^{-2}, so $\tau_J(2,-1) = L^{-2(J-d)}\mathsf{t}[2,-1; V_d](\lambda) + \sum_{d<i\le J} L^{-2(J-i)} e_i'$ with $|e_i'| \le 6 C_{\mathrm{tr}}\varpi(V_i)\|\lambda\|^2$; with $\mathsf{t}[2,-1] \le 4\|\cdot\|^2$ this gives $\rho_{d,J}$. The polarised forms follow by polarisation. *The linear part of (Tr).* At the configuration $\mathbf{1}$ the differential of $\mathcal{M}$ is the linear average $\mathbf{Q}^{\mathrm{lin}}$ of YM3_I:(42) (Lemma YM3_I:4.6); as the contour paths of $B(c_+)$ are the translates of those of $B(c_-)$, $(\mathbf{Q}^{\mathrm{lin}} A)(c) = (SA)(c) + \Lambda_A(c_-) - \Lambda_A(c_+)$ with the straight average $(SA)(c) := L^{-3}\sum_{x \in B(c_-)} A([x,x'])$ and $\Lambda_A(y) := L^{-3}\sum_{x \in B(y)} \bar{A}(y,x)$; the transpose of the second part is the transpose of a coarse gradient, which kills the horizontal vectors, so $\mathbf{Q}^{\mathrm{lin}*} B = S^* B$ for horizontal B. Along the axis of a coarse bond c of the direction μ, S^*B takes at the fine bonds of the direction μ that start in $B(c_-)$, at the m-th position along the axis ($1 \le m \le L$), the value $L^{-3}[mB(c) + (L-m)B(c^-)]$, c^- the coarse bond before c on its axis; at a fine bond of another direction, S^*B takes the value given by the same rule from the coarse bonds of that direction, so that $B(c)$ enters only the values at fine bonds of the direction μ. With $\sum_m [m^2 + (L-m)^2] = \frac{1}{3}L(2L^2+1)$, $\sum_m 2m(L-m) = \frac{1}{3}L(L^2-1)$ for the norm, and the analogous sums $\frac{2}{3}(L^2-1)L$, $\frac{1}{3}L(L^2+2)$ for the pairing of neighbouring fine bonds along the axis: $L\,\mathsf{t}[\alpha,\beta;\mathbf{1}](S^*B) = \mathsf{t}[A_L(\alpha,\beta);\mathbf{1}](B)$. At a field U with $\varpi(U) \le \epsilon$ one goes over, for each coarse bond, to the gauge of its triple block in which the bond variables lie within $15L^2\epsilon$ of $\mathbf{1}$ (Lemma YM3_I:4.1); by the gauge covariance YM3_I:(40) and the real-analyticity of $\mathcal{M}$ on $\mathcal{S}_c$ (Proposition YM3_I:4.5), the differential differs there from the linear average by at most a number of L times ϵ, the supremum of a second differential on a compact ball of configurations of one double block, and the transports along the axes enter the forms t through $\mathrm{Ad}(V(c))$; this is the one-step transport of the proof of Proposition 15.2(b).

(b) For $d \geq \max\{d_\diamond, \mathfrak{d}(Y)-1\}$ and $j > d$, (a) applies to $W = Y_j$ with $\sigma \leq \sum_{i>d}\theta_i \leq (36C_{\rm tr})^{-1}$. The polarised forms $a_j^{-1}\langle D\mathcal{M}^{j-d}(Y_j)^*\lambda, D\mathcal{M}^{j-d}(Y_j)^*\lambda'\rangle$ are Cauchy in j, uniformly in unit λ,λ', as $\rho_{d,J} \to 0$ when $J \to \infty$ (split the middle sum at $i = \frac12(d+J)$ and use $\theta_i \to 0$); the limit, extended by Π_{Y_d}, is $\mathsf{Mat}_d(Y)$, and the bound follows from (a) with $6C_{\rm tr}\sigma \leq \frac16$. Consistency follows from $D\mathcal{M}^{j-d}(Y_j) = D\mathcal{M}(Y_{d+1})D\mathcal{M}^{j-d-1}(Y_j)$ and the horizontality of the values of $D\mathcal{M}(Y_{d+1})^*\Pi_{Y_d}$. Borel: on $\{\mathfrak{d} = D\}$ the matrix is a pointwise limit of continuous functions of finitely many coordinates, and $W \mapsto \Pi_W = \lim_{t\to\infty}\exp(-t\,d_W d_W^*)$ is a pointwise limit of continuous maps. Covariance: $\mathcal{M}^n(W^u) = \mathcal{M}^n(W)^{u|}$ gives $D\mathcal{M}^n(W^u) = \mathrm{Ad}_{u|}D\mathcal{M}^n(W)\mathrm{Ad}_u^{-1}$, and $\Pi_{W^u} = \mathrm{Ad}_u\Pi_W\mathrm{Ad}_u^{-1}$.

(c) For $J \geq \mathsf{j}_0(Y)$, $\varpi(Y_{J+1}) < \theta_{J+1} < \omega_{\tilde{\mathcal{M}}}$, so $\tilde{\mathcal{M}}(Y_{J+1}) = \mathcal{M}(Y_{J+1}) = Y_J$ by (N3), and $\tilde{\mathcal{M}}^{J+1-d}Y_{J+1} = \tilde{\mathcal{M}}^{J-d}Y_J$; the other identities follow at one J that serves all the depths named. For $j \geq J \geq \mathsf{j}_0(Y)$ the open set $\{W \in \Omega_j : \varpi(\mathcal{M}^{j-i}W) < \theta_i,\ J < i \leq j\}$ holds Y_j, and on it $\tilde{\mathcal{M}}^{j-J} = \mathcal{M}^{j-J}$, by downward induction with (N3); so $D\tilde{\mathcal{M}}^{j-d}(Y_j) = D\tilde{\mathcal{M}}^{J-d}(Y_J)D\mathcal{M}^{j-J}(Y_j)$; for horizontal λ at $Y_d^{\tilde{\mathcal{M}}}$ the vector $\eta := D\tilde{\mathcal{M}}^{J-d}(Y_J)^*\lambda$ is horizontal at Y_J, by (N2) and the remark on covariant maps, and $a_j^{-1}\langle D\mathcal{M}^{j-J}(Y_j)^*\eta, D\mathcal{M}^{j-J}(Y_j)^*\eta'\rangle \to \langle\eta, \mathsf{Mat}_J(Y)\eta'\rangle$ by (b). By (N4), $|\eta| \leq c_{\tilde{\mathcal{M}}}^{J-d}|\lambda|$, and by (b) $a_d\langle\eta, \mathsf{Mat}_J\eta\rangle \leq \frac76 a_d a_J^{-1}|\eta|^2 \leq \frac76 \mathsf{k}_{\tilde{\mathcal{M}}}^{J-d}|\lambda|^2$ at $J = \mathsf{j}_0(Y)$; $\{\mathsf{j}_0 = J\} \subset \{\mathfrak{d} > J\}$ for $J > d^\vee$ gives the moments. Borel, covariance and consistency follow as in (b), with (N2) for the covariance of $\tilde{\mathcal{M}}^n$. If $d \geq d_{\tilde{\mathcal{M}}}$ and $\mathfrak{d}(Y) \leq d+1$, then $\varpi(Y_j) < \theta_j < \omega_{\tilde{\mathcal{M}}}$ for $j > d$, so $\tilde{\mathcal{M}}^{J-d}Y_J = \mathcal{M}^{J-d}Y_J = Y_d$ and $D\tilde{\mathcal{M}}^{J-d}(Y_J) = D\mathcal{M}^{J-d}(Y_J)$, whence $\mathsf{Mat}_d^{\tilde{\mathcal{M}}} = \mathsf{Mat}_d$ by the consistency of (b).

(d) The set F_c of the U with $|\partial U(p) - \mathbf{1}|^2 \leq \frac12\omega_M^2$ at the plaquettes of the double block of c is closed and lies in $\mathcal{S}_c$, and $\vartheta_c = 0$ off F_c; so the two clauses define $\mathcal{M}^\chi(U)(c)$ on the two open sets $\mathcal{S}_c$ and $\Omega_{j+1}\setminus F_c$, which cover Ω_{j+1}, and they agree where both apply ($\exp(0)U[c] = U[c]$). On $\mathcal{S}_c$ the W_x are real-analytic with $|W_x - \mathbf{1}| < 1$ (Proposition YM3_I:4.5), where the logarithm is analytic, and ϑ_c is χ_0 composed with polynomials of the entries: $\mathcal{M}^\chi$ is C^∞ and reads U on the double block of c alone. (N2): plaquette variables are conjugated by gauge transformations, $W_x(U^u) = u(c_-)W_x(U)u(c_-)^{-1}$ and $U^u[c] = u(c_-)U[c]u(c_+)^{-1}$ (the intermediate factors of a path product cancel under a gauge transformation), and log and exp are covariant under conjugation. (N3): if $\varpi(U) < \frac12\omega_M$, every argument of χ_0 is below $\frac14$, $\vartheta_c = 1$, and $\mathcal{M}^\chi(U)(c)$ is the analytic branch of $\mathcal{M}$ (Proposition YM3_I:4.5); as $\theta_j \leq \varepsilon_{\rm tr} < \frac12\omega_M$ for $j > d_\diamond$, $d_{\mathcal{M}^\chi} = d_\diamond$. (N4): in the coordinates of a double block, $\mathcal{M}^\chi(U)(c)$ is one C^∞ function F_ν of the values on the double block, the same at every depth, for every side and at every bond of one direction ν, since the shift between two double blocks of one direction keeps the coordinates, the plaquettes, the contour families and the straight paths; $r_\chi := \max_\nu \sup\|DF_\nu\|$ is finite on the compact space of these values; a fine bond lies in at most six double blocks, so $\|D\mathcal{M}^\chi(U)\xi\|^2 \leq 6r_\chi^2\|\xi\|^2$. □

Proposition 18.68 (The limit of the lattice forms on block observables, with nothing cut)**.** Assume (E); let $\tilde{\mathcal{M}}$ be admissible, $d \geq 0$ and $g, h \in C^1_{\rm inv}(\Omega_d)$.

(a) For every $K \geq d$, f_g^K is C^1 and invariant on Ω_K, so it lies in $D(\mathcal{E}_K)$; for $\tilde{\mathcal{M}} = \mathcal{M}^\chi$ and $g \in C^\infty$ it is C^∞. For every $1 \leq p < \infty$: $\sup_{K>d^\vee}\int\Gamma_K(f_g^K)^p d\mu^K \leq (\frac76)^p[\mathsf{z}_{d^\vee}(g)^{2p} + \sum_{J>d^\vee}\mathsf{z}_J(g)^{2p}\mathsf{R}_{J-1}] < \infty$.

(b) Along the full sequence of cutoffs, $\lim_K \mathcal{E}_K(f_g^K, f_h^K) = \mathcal{E}_\infty^{\tilde{\mathcal{M}}}(g,h) := \int\langle\partial g(Y_d^{\tilde{\mathcal{M}}}), \mathsf{Mat}_d^{\tilde{\mathcal{M}}}\,\partial h(Y_d^{\tilde{\mathcal{M}}})\rangle\,d\mu$. For $d^\vee \leq J < J'$ and $J'+1 < K$ the difference is at most $a_d^{-1}\|\partial g\|_\infty\|\partial h\|_\infty\{\mathsf{k}_{\tilde{\mathcal{M}}}^{J-d}[2\bar\rho_{J,J'} + \frac73\mathsf{R}_{J'+1} + \frac73\mathsf{tv}_{J'}^K] + \frac76\sum_{i>J}\mathsf{k}_{\tilde{\mathcal{M}}}^{i-d}(\mathsf{R}_{i-1}+\mathsf{R}_i)\}$, with $\bar\rho_{J,J'} := \frac23 L^{-2(J'-J)} + 12C_{\rm tr}(\sum_{J<i\leq J'}L^{-2(J'-i)}\theta_i + \sum_{i>J'}\theta_i)$; it tends to 0 as first K, then J', then J tends to ∞.

(c) The integrand $\Gamma_\infty^{\tilde{\mathcal{M}}}(g,h) := \langle\partial g(Y_d^{\tilde{\mathcal{M}}}), \mathsf{Mat}_d^{\tilde{\mathcal{M}}}\,\partial h(Y_d^{\tilde{\mathcal{M}}})\rangle$ is the same function for (d,g,h) and for $(d+1, g\circ\tilde{\mathcal{M}}, h\circ\tilde{\mathcal{M}})$; if $g(Y_d^{\tilde{\mathcal{M}}}) = \tilde g(Y_{\tilde d}^{\tilde{\mathcal{M}}})$ μ-almost everywhere, the integrands of (g,h)

and of $(\tilde g, h)$ agree μ-almost everywhere. So $\mathcal{E}^{\tilde{\mathcal{M}}}_\infty$ and $\Gamma^{\tilde{\mathcal{M}}}_\infty$ are defined on the classes of $\mathrm{Cyl}^{\tilde{\mathcal{M}}}$; $\Gamma^{\tilde{\mathcal{M}}}_\infty(\varphi\circ u, w) = \varphi'(u)\Gamma^{\tilde{\mathcal{M}}}_\infty(u,w)$ for $\varphi\in C^1(\mathbb{R})$; $\Gamma^{\tilde{\mathcal{M}}}_\infty(u,u)\ge 0$ lies in every $L^p(\mu)$, $p<\infty$.

(d) If $Y\in \mathrm{Thr}^\circ$ has $\varpi(Y_j)\le \varepsilon_{\mathrm{tr}}$ and $\varpi(Y_j)<\omega_{\tilde{\mathcal{M}}}$ for all $j>d$, and $\sum_{j>d}\varpi(Y_j)\le (36C_{\mathrm{tr}})^{-1}$, then $Y^{\tilde{\mathcal{M}}}_d = Y_d$ and $\frac16\Gamma_d(g)(Y_d)\le \Gamma^{\tilde{\mathcal{M}}}_\infty(g,g)(Y)\le \frac76\Gamma_d(g)(Y_d)$. If $d\ge d_{\tilde{\mathcal{M}}}$, every thread of the event $\{\mathfrak{d}\le d+1\}$, whose μ-mass is at least $1-\mathsf{R}_{d+1}$, is such a thread; so, for $d\ge d_{\tilde{\mathcal{M}}}$, $\mathcal{E}^{\tilde{\mathcal{M}}}_\infty(g,g)\ge \frac16\int_{\{\mathfrak{d}\le d+1\}}\Gamma_d(g)(Y_d)\,d\mu$, $\mu(Y^{\tilde{\mathcal{M}}}_d\ne Y_d)\le \mathsf{R}_{d+1}$, and $\mu^K(\tilde{\mathcal{M}}^{K-d}U\ne \pi_{K,d}U)\le \mathsf{R}_{d+1}$ for $K>d+1$.

(e) $f^K_g\to g(Y^{\tilde{\mathcal{M}}}_d)$ strongly along the full sequence of cutoffs; so $\mathcal{E}_\star(g(Y^{\tilde{\mathcal{M}}}_d))\le \mathcal{E}^{\tilde{\mathcal{M}}}_\infty(g,g)$ for every limit. $\mathrm{Cyl}^{\tilde{\mathcal{M}}}$ is an algebra of bounded Borel invariant functions on the threads, nested in d, with $1\in\mathcal{C}^{\tilde{\mathcal{M}}}_0$ and $\varphi\circ u\in\mathrm{Cyl}^{\tilde{\mathcal{M}}}$ for $u\in\mathrm{Cyl}^{\tilde{\mathcal{M}}}$, $\varphi\in C^1(\mathbb{R})$; its classes are dense in L^2_{inv}. ∎

Sketch. (a) $\tilde{\mathcal{M}}^{K-d}$ is C^1 and covariant, so $f^K_g\in C^1_{\mathrm{inv}}(\Omega_K)$ and $|\partial f^K_g|\le c^{K-d}_{\tilde{\mathcal{M}}}\|\partial g\|_\infty$ by (N4). For $U\in\Omega_K$ let $\mathsf{j}_*(U)$ be the least $J\ge d^\vee$ with $U\in\mathsf{A}^K_{J+1}$. For $J>d^\vee$, $\{\mathsf{j}_*=J\}\subset\Omega_K\setminus\mathsf{A}^K_J\subset\Omega_K\setminus\mathsf{A}^K_{J-1}$, so $\mu^K(\mathsf{j}_*=J)\le\mathsf{R}_{J-1}$. On the open set A^K_{J+1}, $\tilde{\mathcal{M}}^{K-J}=\mathcal{M}^{K-J}$ (downward induction with (N3), $\theta_i<\omega_{\tilde{\mathcal{M}}}$ for $i>J\ge d_{\tilde{\mathcal{M}}}$), so $f^K_g = g^{(J)}\circ\mathcal{M}^{K-J}$ near U; $\partial g^{(J)}(V_J)$ is horizontal at $V_J:=\pi_{K,J}U$, of norm at most $c^{J-d}_{\tilde{\mathcal{M}}}\|\partial g\|_\infty$, and by Proposition 18.67(a), with $\sigma\le\sum_{i>J}\theta_i\le(36C_{\mathrm{tr}})^{-1}$: $\frac16 a_J^{-1}|\partial g^{(J)}(V_J)|^2\le\Gamma_K(f^K_g)(U)\le\frac76 a_J^{-1}|\partial g^{(J)}(V_J)|^2\le\frac76\mathsf{z}_J(g)^2$. Summing over the values of j_* gives the moment bound; the series converges, as R_J decays faster than every exponential in J.

(b) Fix $d^\vee\le J<J'$ with $J'+1<K$. Then $\mathcal{E}_K(f^K_g,f^K_h)=\int_{\mathsf{A}^K_{J+1}}\Gamma_K(g^{(J)}\circ\mathcal{M}^{K-J}, h^{(J)}\circ\mathcal{M}^{K-J})\,d\mu^K + \int_{\{\mathsf{j}_*>J\}}\Gamma_K(f^K_g,f^K_h)\,d\mu^K$, and by (a) the second term is at most $\frac76\sum_{i>J}\mathsf{z}_i(g)\mathsf{z}_i(h)\mathsf{R}_{i-1}$. In the first, with $\lambda:=\partial g^{(J)}(V_J)$, $\lambda':=\partial h^{(J)}(V_J)$, the integrand is $a_K^{-1}\langle D\mathcal{M}^{K-J}(U)^*\lambda, D\mathcal{M}^{K-J}(U)^*\lambda'\rangle$; by Proposition 18.67(a) it lies within $a_J^{-1}|\lambda||\lambda'|\bar\rho_{J,J'}$ of $\phi(V_{J'}):=a_{J'}^{-1}\langle D\mathcal{M}^{J'-J}(V_{J'})^*\lambda, D\mathcal{M}^{J'-J}(V_{J'})^*\lambda'\rangle$, a function of $V_{J'}=\pi_{K,J'}U$ with $|\phi|\le\frac76 a_J^{-1}|\lambda||\lambda'|$. Replacing A^K_{J+1} by the event of $V_{J'}$ that the fields of the depths $J+1,\dots,J'$ are small costs at most $\mathsf{R}_{J'+1}$ in mass; passing from $\mu^K_{J'}$ to $\mu_{J'}$ costs at most $2\mathsf{tv}^K_{J'}$ in total variation; returning to the event $\{\mathfrak{d}\le J+1\}$ costs $\mathsf{R}_{J'+1}$ again; and on $\{\mathfrak{d}\le J+1\}$, $\phi(Y_{J'})$ lies within $a_J^{-1}|\lambda||\lambda'|\bar\rho_{J,J'}$ of $\langle\lambda,\mathsf{Mat}_J(Y)\lambda'\rangle=\Gamma^{\tilde{\mathcal{M}}}_\infty(g,h)(Y)$, by Proposition 18.67(b), (c). Finally $|\int_{\{\mathfrak{d}>J+1\}}\Gamma^{\tilde{\mathcal{M}}}_\infty(g,h)\,d\mu|\le\frac76\sum_{i>J}\mathsf{z}_i(g)\mathsf{z}_i(h)\mathsf{R}_i$, as $\{\mathsf{j}_0=i\}\subset\{\mathfrak{d}=i+1\}$. With $a_J^{-1}|\lambda||\lambda'|\le a_d^{-1}\mathsf{k}^{J-d}_{\tilde{\mathcal{M}}}\|\partial g\|_\infty\|\partial h\|_\infty$ this is the error bound. It tends to 0 as $K\to\infty$ (Corollary 2.8), then $J'\to\infty$ ($\bar\rho_{J,J'}\to0$, $\mathsf{R}_{J'+1}\to0$), then $J\to\infty$ (the decay of R_i beats the growth of $\mathsf{k}^i_{\tilde{\mathcal{M}}}$).

(c) $g\circ\tilde{\mathcal{M}}\in C^1_{\mathrm{inv}}(\Omega_{d+1})$ and $(g\circ\tilde{\mathcal{M}})^{(J)}=g^{(J)}$ for $J\ge d+1$, so both integrands are $\langle\partial g^{(J)}(Y_J),\mathsf{Mat}_J\partial h^{(J)}(Y_J)\rangle$ at $J\ge\mathsf{j}_0(Y)$. For the classes, let $g(Y^{\tilde{\mathcal{M}}}_d)=0$ on a Borel set A of threads; for $J\ge d^\vee$ the set $Z_J:=\{W: g^{(J)}(W)=0,\ \partial g^{(J)}(W)\ne0\}$ is a C^1 hypersurface, hence Haar-null, and the image of $\mathbf{1}_{\{\mathfrak{d}\le J\}}\mu$ under Y_J is absolutely continuous (Proposition 18.69(a)); on $A\cap\{\mathfrak{d}\le J\}$, where $\mathsf{j}_0\le J$, $g^{(J)}(Y_J)=g(Y^{\tilde{\mathcal{M}}}_d)=0$, so $\partial g^{(J)}(Y_J)=0$ and $\Gamma^{\tilde{\mathcal{M}}}_\infty(g,g)=\langle\partial g^{(J)}(Y_J),\mathsf{Mat}_J\partial g^{(J)}(Y_J)\rangle=0$ μ-almost everywhere there; the union over J of these sets is $A\cap\mathrm{Thr}^\circ$; by Cauchy–Schwarz for the non-negative form $\Gamma^{\tilde{\mathcal{M}}}_\infty$, the integrand of (g,h) reads the class of $g(Y^{\tilde{\mathcal{M}}}_d)$ alone. The chain rule is that of ∂ at the point Y_J; the L^p bounds follow from Proposition 18.67(c).

(d) For $j>d$ the field Y_j lies in the open set $\{\varpi<\omega_{\tilde{\mathcal{M}}}\}$, on which $\tilde{\mathcal{M}}=\mathcal{M}$ by (N3); so $\tilde{\mathcal{M}}Y_j=Y_{j-1}$ and $D\tilde{\mathcal{M}}(Y_j)=D\mathcal{M}(Y_j)$, whence $Y^{\tilde{\mathcal{M}}}_d=Y_d$ and, by the chain rule, $D\tilde{\mathcal{M}}^{j-d}(Y_j)=D\mathcal{M}^{j-d}(Y_j)$. For horizontal λ at Y_d, Proposition 18.67(c) gives $\langle\lambda,\mathsf{Mat}^{\tilde{\mathcal{M}}}_d(Y)\lambda\rangle=\lim_j a_j^{-1}|D\mathcal{M}^{j-d}(Y_j)^*\lambda|^2$, and Proposition 18.67(a) at $W=Y_j$, where $\sigma\le(36C_{\mathrm{tr}})^{-1}$, puts a_d times each term of this sequence

between $\frac{1}{6}\|\lambda\|^2$ and $\frac{7}{6}\|\lambda\|^2$; take $\lambda = \partial g(Y_d)$. If $d \geq d_{\tilde{\mathcal{M}}}$ and $\mathfrak{d}(Y) \leq d+1$, then $\varpi(Y_j) < \theta_j$ for $j > d$, while $\theta_j \leq \varepsilon_{\rm tr}$, $\theta_j < \omega_{\tilde{\mathcal{M}}}$ for $j > d$ and $\sum_{j>d}\theta_j \leq (36C_{\rm tr})^{-1}$, by the choice of $d_\diamond$ and of $d_{\tilde{\mathcal{M}}} \geq d_\diamond$; and $\mu(\mathfrak{d} > d+1) \leq \mathsf{R}_{d+1}$. On A^K_{d+1}, $\tilde{\mathcal{M}}^{K-d}U = \pi_{K,d}U$, by the induction of (a).

(e) For $n \geq d^\vee$, $\Psi_n := g^{(n)}(Y_n)$ is a bounded invariant cylinder function; $\Psi_n = g(Y_d^{\tilde{\mathcal{M}}})$ on $\{\mathfrak{d} \leq n+1\}$, so $\|\Psi_n - g(Y_d^{\tilde{\mathcal{M}}})\|^2_{L^2(\mu)} \leq 4\|g\|^2_\infty \mathsf{R}_{n+1}$; and $\Psi_{n,K} = g^{(n)} \circ \pi_{K,n} = f_g^K$ on A^K_{n+1}, so $\|\Psi_{n,K} - f_g^K\|^2_{L^2(\mu^K)} \leq 4\|g\|^2_\infty \mathsf{R}_{n+1}$ for $K > n+1$: this is the strong convergence of the setting, and Lemma 15.1(b) gives $\mathcal{E}_\star(g(Y_d^{\tilde{\mathcal{M}}})) \leq \liminf_j \mathcal{E}_{K_j}(f_g^{K_j}) = \mathcal{E}^{\tilde{\mathcal{M}}}_\infty(g,g)$ for every limit (K_j). Two members of $\mathrm{Cyl}^{\tilde{\mathcal{M}}}$ are read at one common depth, by (c), where sums, products and $\varphi \circ \cdot$ stay in $C^1_{\rm inv}$; $1 = g(Y_0^{\tilde{\mathcal{M}}})$ at $g = 1$. Density: let $u \in L^2_{\rm inv}$ and $\eta > 0$. By the definition of $L^2_{\rm inv}$ there are a depth J_1 and a bounded invariant Borel function Ψ on Ω_{J_1} with $\|\Psi(Y_{J_1}) - u\|_{L^2(\mu)} \leq \eta$; put $B := \sup|\Psi|$ and choose $J \geq \max\{J_1, d_{\tilde{\mathcal{M}}}\}$ so large that $4(B+1)^2\mathsf{R}_{J+1} \leq \eta^2$. Then $F := \Psi \circ \mathcal{M}^{J-J_1}$ is a bounded invariant Borel function on Ω_J with $F(Y_J) = \Psi(Y_{J_1})$ on the threads. There is a polynomial g_0 of the matrix entries with $\|g_0 - F\|_{L^2(\mu_J)} \leq \eta$ (Stone–Weierstrass); take a smooth ψ with $\psi(t) = t$ for $|t| \leq B$, $|\psi'| \leq 1$ and $|\psi| \leq B+1$, so that $|\psi(g_0) - F| = |\psi(g_0) - \psi(F)| \leq |g_0 - F|$; the average g of $\psi(g_0)$ over the gauge group of $\Lambda_{J,\ell}$ is smooth and invariant, has $|g| \leq B+1$, a bound that reads neither g_0 nor J, and has $\|g - F\|_{L^2(\mu_J)} \leq \eta$ (Minkowski's inequality and the invariance of μ_J and of F), that is, $\|g(Y_J) - \Psi(Y_{J_1})\|_{L^2(\mu)} \leq \eta$, as μ_J is the law of Y_J. Last, $g(Y_J^{\tilde{\mathcal{M}}}) = g(Y_J)$ on $\{\mathfrak{d} \leq J+1\}$, by Proposition 18.67(c), and the complement has mass at most R_{J+1}; so $\|g(Y_J^{\tilde{\mathcal{M}}}) - g(Y_J)\|^2_{L^2(\mu)} \leq 4(B+1)^2\mathsf{R}_{J+1} \leq \eta^2$, and $\|g(Y_J^{\tilde{\mathcal{M}}}) - u\|_{L^2(\mu)} \leq 3\eta$. □

Proposition 18.69 (The laws of one depth; the form of one depth is closable)**.** Assume (E).

(a) For $J \geq d_\diamond$ and $K > J$: $\mu^K(\pi_{K,J}U \notin \mathsf{I}_J) \leq \mathsf{R}_J$ and $\mu(Y_J \notin \mathsf{I}_J) \leq \mathsf{R}_J$. For every Borel set $E \subset \mathsf{I}_J$: $\mu(Y_J \in E,\ \mathfrak{d} \leq J) \geq (1 - \mathsf{R}_J)e^{-\mathsf{o}_J}\,\mathrm{Haar}(E)$. The image of $\mathbf{1}_{\{\mathfrak{d}\leq J\}}\mu$ under $Y \mapsto Y_J$ is absolutely continuous with respect to the Haar measure of Ω_J.

(b) Let $\tilde{\mathcal{M}}$ be the map $\mathcal{M}^\chi$ of Proposition 18.67(d). For every $d \geq 0$ the form $\mathcal{E}^{\tilde{\mathcal{M}}}_\infty$ on $\mathcal{C}_d^{\tilde{\mathcal{M}}}$ is closable in $L^2_{\rm inv}$; equivalently, $g \mapsto \mathcal{E}^{\tilde{\mathcal{M}}}_\infty(g,g)$ on $C^1_{\rm inv}(\Omega_d)$ is closable in $L^2(\mu_d^{\tilde{\mathcal{M}}})$, $\mu_d^{\tilde{\mathcal{M}}}$ the law of $Y_d^{\tilde{\mathcal{M}}}$ under μ.

■

Sketch. (a) Proposition 15.2(c) at the depth J, at the cutoff K and with no cutoff, applied to the invariant function $f = \mathbf{1}_{\Omega_J \setminus \mathsf{I}_J}$, whose integral over I_J vanishes, gives the first two bounds. For invariant E the lower bound is Proposition 15.2(c) at the depth J and $f = \mathbf{1}_E$, as $|\mathsf{I}_J| \leq 1$ and $\{\varpi(Y_e) < \theta_e \text{ for all } e \geq J\} = \{\mathfrak{d} \leq J\}$. Absolute continuity: let $E \subset \Omega_J$ be Haar-null and $D > J$; the set $E_D := \{W \in \Omega_D : \mathcal{M}^{D-J}W \in E,\ \varpi(\mathcal{M}^{D-e}W) < \theta_e \text{ for } J < e \leq D\}$ is Haar-null, because $\mathcal{M}^{D-J}$ is a real-analytic submersion on the open set of the second condition (there $\theta_e \leq \omega_L$, and the derivative of $\mathcal{M}$ in the values at the pivot bonds is invertible: proof of Lemma 15.3(iv)); in the same way, for $K > D+1$, the set of the $U \in \mathsf{A}^K_{D+1}$ with $\pi_{K,D}U \in E_D$ is Haar-null, and μ^K has a density; so $\mu^K_D(E_D) \leq \mathsf{R}_{D+1}$, $\mu_D(E_D) \leq \mathsf{R}_{D+1}$ by total variation, and $\mu(Y_J \in E,\ \mathfrak{d} \leq J) \leq \mu(Y_D \in E_D) \leq \mathsf{R}_{D+1} \to 0$. General $E \subset \mathsf{I}_J$: the measure $m_J(E) := \mu(Y_J \in E,\ \mathfrak{d} \leq J)$ is invariant under the gauge transformations of $\Lambda_{J,\ell}$ (the invariance of μ under the maps $Y \mapsto v \cdot Y$; $\mathfrak{d}$ and I_J are invariant) and has a density r; its average $\bar{r}$ over the gauge group is again a density, invariant, and the invariant set $\{\bar{r} < (1 - \mathsf{R}_J)e^{-\mathsf{o}_J}\} \cap \mathsf{I}_J$ carries a non-negative integral of a negative function, by the bound for invariant sets: it is Haar-null.

(b) Here J runs over the depths $J \geq d^\vee$ at which $\mathsf{R}_J < 1$; every large J is one of them, as $\mathsf{R}_J \to 0$. Put $B_J := \{\mathfrak{d} \leq J\} \cap \{Y_J \in \mathsf{I}_J\}$. On B_J, $J \geq \mathfrak{j}_0(Y)$, so $u := g(Y_d^{\tilde{\mathcal{M}}}) = g^{(J)}(Y_J)$ and, by Proposition 18.67(b), (c), $\frac{1}{6}a_J^{-1}|\partial g^{(J)}(Y_J)|^2 \leq \Gamma^{\tilde{\mathcal{M}}}_\infty(g,g) \leq \frac{7}{6}a_J^{-1}|\partial g^{(J)}(Y_J)|^2$. By (a), $\mu(B_J) \geq 1 - 2\mathsf{R}_J \to 1$, and the image m_J of $\mathbf{1}_{B_J}\mu$ under Y_J is absolutely continuous and at least

$c_J\,dW$ on I_J, with the positive number $c_J := (1-\mathsf{R}_J)e^{-\mathsf{o}_J}$. Let $u_n = g_n(Y_d^{\tilde{\mathcal{M}}}) \in \mathcal{C}_d^{\tilde{\mathcal{M}}}$ with $\|u_n\|_{L^2} \to 0$ and $\mathcal{E}_\infty^{\tilde{\mathcal{M}}}(u_n - u_m) \to 0$. Fix J and put $G_n := g_n^{(J)}$, a C^1 function on Ω_J (the map $g \mapsto g^{(J)}$ is linear, and on I_J it reads the class of $g(Y_d^{\tilde{\mathcal{M}}})$ alone, as $m_J \geq c_J\,dW$ there and $g^{(J)}$ is continuous): $\int_{\mathsf{I}_J} G_n^2 dm_J = \int_{B_J} u_n^2 d\mu \to 0$ and $\int_{\mathsf{I}_J} |\partial G_n - \partial G_m|^2 dm_J \leq 6a_J \int_{B_J} \Gamma_\infty^{\tilde{\mathcal{M}}}(u_n - u_m)\,d\mu \to 0$; so $\partial G_n \to \xi$ in $L^2(\mathsf{I}_J, m_J)$, and, as $m_J \geq c_J\,dW$ on I_J, $G_n \to 0$ and $\partial G_n \to \xi$ in $L^2(\mathsf{I}_J, dW)$. For a smooth vector field ϕ with compact support in I_J, $\int \langle \xi, \phi\rangle dW = \lim \int \langle \partial G_n, \phi\rangle dW = -\lim \int G_n \operatorname{div}\phi\, dW = 0$, the derivatives along $e^{itX}V$ being skew for the Haar measure; so $\xi = 0$ dW-almost everywhere on I_J, hence m_J-almost everywhere, and $\int_{B_J} \Gamma_\infty^{\tilde{\mathcal{M}}}(u_n)\,d\mu \leq \frac{7}{6} a_J^{-1} \int |\partial G_n|^2 dm_J \to 0$. Now let $\epsilon > 0$ and n_0 with $\mathcal{E}_\infty^{\tilde{\mathcal{M}}}(u_n - u_{n_0}) \leq \epsilon$ for $n \geq n_0$; as $\Gamma_\infty^{\tilde{\mathcal{M}}}(u_{n_0}) \in L^1(\mu)$ and $\mu(B_J) \to 1$, there is J with $\int_{B_J^c} \Gamma_\infty^{\tilde{\mathcal{M}}}(u_{n_0}) \leq \epsilon$; and $\Gamma_\infty^{\tilde{\mathcal{M}}}(u_n) \leq 2\Gamma_\infty^{\tilde{\mathcal{M}}}(u_{n_0}) + 2\Gamma_\infty^{\tilde{\mathcal{M}}}(u_n - u_{n_0})$ pointwise; so $\limsup_n \mathcal{E}_\infty^{\tilde{\mathcal{M}}}(u_n) \leq 4\epsilon$. Last, $g \mapsto g(Y_d^{\tilde{\mathcal{M}}})$ maps $L^2(\mu_d^{\tilde{\mathcal{M}}})$ isometrically into $L^2(\mu)$; it carries $C^1_{\mathrm{inv}}(\Omega_d)$ onto $\mathcal{C}_d^{\tilde{\mathcal{M}}}$, which lies in the closed subspace L^2_{inv} (Proposition 18.68(e)), and the form on the one onto the form on the other. A form whose domain lies in a closed subspace is closable in that subspace exactly when it is closable in the whole space; hence the two wordings of (b) agree. □

Proposition 18.70. *Closability of the limit form from a contraction of the energy; the Dirichlet form, its semigroup, motion.*

(a) Let $(\mathfrak{X}, \mathfrak{B}, \nu)$ be a probability space, $\mathsf{H} := L^2(\nu)$, $\mathcal{C}_0 \subset \mathcal{C}_1 \subset \cdots$ linear spaces of bounded measurable functions with $1 \in \mathcal{C}_0$ and $\mathcal{C} := \bigcup_d \mathcal{C}_d$ dense in H, and $\mathcal{E}$ a symmetric non-negative bilinear form on $\mathcal{C}$ with $\mathcal{E}(1) = 0$ and a square field: a symmetric bilinear $\Gamma : \mathcal{C} \times \mathcal{C} \to L^1(\nu)$ with $\Gamma(u,u) \geq 0$, $\mathcal{E}(u,w) = \int \Gamma(u,w)\,d\nu$, and $\varphi \circ u \in \mathcal{C}$, $\Gamma(\varphi \circ u, w) = \varphi'(u)\Gamma(u,w)$ for $\varphi \in C^1(\mathbb{R})$ with bounded φ'. Let H_d be the closure of $\mathcal{C}_d$ and $\mathcal{P}_d$ the orthogonal projection onto it. Suppose (Cl) for every d the form $(\mathcal{E}, \mathcal{C}_d)$ is closable in H, with the closure $(\bar{\mathcal{E}}_d, \mathrm{Dom}_d)$; and (CE) there are numbers $C > 0$ and $\lambda_0 \geq 0$ and an infinite set $\mathbb{I}$ of depths with $\mathcal{P}_d u \in \mathrm{Dom}_d$ and $\bar{\mathcal{E}}_d(\mathcal{P}_d u) \leq C(\mathcal{E}(u) + \lambda_0 \|u\|^2)$ for all $u \in \mathcal{C}$, $d \in \mathbb{I}$. Then: (1) $(\mathcal{E}, \mathcal{C})$ is closable in H; (2) its closure $(\bar{\mathcal{E}}, \mathrm{Dom})$ is a symmetric Dirichlet form with $1 \in \mathrm{Dom}$, $\bar{\mathcal{E}}(1) = 0$, and $\bar{\mathcal{E}}(u,w) = \int \bar{\Gamma}(u,w)\,d\nu$ with the continuous extension $\bar{\Gamma}$ of Γ, which keeps the chain rule; (3) its semigroup $P_t = e^{-tA}$ is a strongly continuous semigroup of self-adjoint contractions, $0 \leq f \leq 1$ implies $0 \leq P_t f \leq 1$, $P_t 1 = 1$, $\int P_t f\,d\nu = \int f\,d\nu$ and $\int g\,P_t f\,d\nu = \int f\,P_t g\,d\nu$: ν is invariant and reversible; $t^{-1}(u, (1-P_t)u)$ is non-increasing in t and tends to $\bar{\mathcal{E}}(u) \in [0,\infty]$ as $t \downarrow 0$, so that $(u,(1-P_t)u) > 0$ for every $t > 0$ whenever $\bar{\mathcal{E}}(u) > 0$, the value $+\infty$ included; (4) $\mathcal{P}_d \mathrm{Dom} \subset \mathrm{Dom}_d$ and $\bar{\mathcal{E}}_d(\mathcal{P}_d u) \leq C(\bar{\mathcal{E}}(u) + \lambda_0\|u\|^2)$ for $d \in \mathbb{I}$, and $\mathcal{P}_d u \to u$ in the norm $(\bar{\mathcal{E}} + \|\cdot\|^2)^{1/2}$ along $\mathbb{I}$, for $u \in \mathrm{Dom}$; (5) $e^{-tA_d}\mathcal{P}_d f \to P_t f$ in H, A_d the operator of $\bar{\mathcal{E}}_d$ in H_d; (6) if $\lambda_0 = 0$: $(u,(1-P_t)u) \geq \mathfrak{J}^{\bar{\mathcal{E}}_d}_{t/C}(\mathcal{P}_d u) \geq \frac{1}{2}(\mathcal{P}_d u, (1 - e^{-(t/C)A_d})\mathcal{P}_d u)$ for $u \in \mathsf{H}$, $t > 0$, $d \in \mathbb{I}$, where $\mathfrak{J}^{\mathcal{Q}}_s(v) := \inf_w \{s\mathcal{Q}(w) + \|v - w\|^2\}$.

(b) (The block observables.) Assume (E), let $\tilde{\mathcal{M}}$ be admissible, ν the restriction of μ to the invariant σ-algebra (so that $\mathsf{H} = L^2_{\mathrm{inv}}$), $\mathcal{C}_d := \mathcal{C}_d^{\tilde{\mathcal{M}}}$, $\mathcal{E} := \mathcal{E}_\infty^{\tilde{\mathcal{M}}}$ and $\Gamma := \Gamma_\infty^{\tilde{\mathcal{M}}}$. Then every datum of (a) other than (Cl) and (CE) holds, and (Cl) holds when $\tilde{\mathcal{M}} = \mathcal{M}^\chi$. So (1)–(6) hold for $(\mathcal{E}_\infty^{\tilde{\mathcal{M}}}, \mathrm{Cyl}^{\tilde{\mathcal{M}}})$ whenever (Cl) and (CE) hold for these data; for $\tilde{\mathcal{M}} = \mathcal{M}^\chi$ it suffices that (CE) holds.

■

Sketch. (a) (1) For $u \in \mathsf{H}$ put $\mathfrak{N}(u) := \sup_{d \in \mathbb{I}} \bar{\mathcal{E}}_d(\mathcal{P}_d u) \in [0,\infty]$; each $u \mapsto \bar{\mathcal{E}}_d(\mathcal{P}_d u)$ is lower semicontinuous on H (a closed form is lower semicontinuous, $\mathcal{P}_d$ is continuous), and so is $\mathfrak{N}$. On $\mathcal{C}$: $\mathfrak{N} \leq C(\mathcal{E} + \lambda_0\|\cdot\|^2)$ by (CE), and $\mathcal{E}(u) = \bar{\mathcal{E}}_d(\mathcal{P}_d u) \leq \mathfrak{N}(u)$ for $u \in \mathcal{C}_e$ and $d \in \mathbb{I}$, $d \geq e$. If $u_n \in \mathcal{C}$, $u_n \to 0$ and $\mathcal{E}(u_n - u_m) \to 0$, then for a fixed m, $u_m - u_n \to u_m$ as $n \to \infty$, and $\mathcal{E}(u_m) \leq \mathfrak{N}(u_m) \leq \liminf_n \mathfrak{N}(u_m - u_n) \leq C \sup_{n \geq m}(\mathcal{E} + \lambda_0\|\cdot\|^2)(u_m - u_n) \to 0$. (4) By the lower semicontinuity (CE) passes to the closure; so the $\mathcal{P}_d$, $d \in \mathbb{I}$, are uniformly bounded operators of

the Hilbert space $(\mathrm{Dom}, \bar{\mathcal{E}} + \|\cdot\|^2)$, equal to the identity on $\mathcal{C}_e$ for $d \geq e$, and $\mathcal{P}_d u \to u$ there along $\mathbb{I}$. (2) $|\Gamma(u,w)| \leq \Gamma(u,u)^{1/2}\Gamma(w,w)^{1/2}$ almost everywhere, so Γ extends continuously to $\bar{\Gamma} : \mathrm{Dom} \times \mathrm{Dom} \to L^1(\nu)$; if $u_n \in \mathcal{C}$ tends to u in the norm of the closure and almost everywhere, then $\Gamma(\varphi \circ u_n - \varphi \circ u_m) = \varphi'(u_n)^2\Gamma(u_n) - 2\varphi'(u_n)\varphi'(u_m)\Gamma(u_n,u_m) + \varphi'(u_m)^2\Gamma(u_m) \to 0$ in L^1, so $\varphi \circ u \in \mathrm{Dom}$ and $\bar{\Gamma}(\varphi \circ u, w) = \varphi'(u)\bar{\Gamma}(u,w)$; with the near-unit contractions φ_ϵ ($0 \leq \varphi_\epsilon' \leq 1$, $\varphi_\epsilon(t) = t$ on $[0,1]$, $|\varphi_\epsilon(t) - (0 \vee t) \wedge 1| \leq \epsilon/2$) and the lower semicontinuity, $\bar{\mathcal{E}}((0 \vee u) \wedge 1) \leq \bar{\mathcal{E}}(u)$. (3) By the spectral theorem for A, with the spectral measure m_u of u: P_t is a strongly continuous semigroup of self-adjoint contractions; for $0 \leq f \leq 1$ and $\alpha > 0$, $\alpha(\alpha + A)^{-1} f$ is the minimiser of $v \mapsto \bar{\mathcal{E}}(v) + \alpha\|v - f\|^2$, which the truncation $(0 \vee v) \wedge 1$ does not increase, so $0 \leq \alpha(\alpha + A)^{-1} f \leq 1$, and $((n/t)(n/t + A)^{-1})^n f \to P_t f$; $\bar{\mathcal{E}}(1) = 0$ puts m_1 at 0, so $P_t 1 = 1$, and with the self-adjointness $\int P_t f = (P_t f, 1) = (f, 1)$; $(u, (1 - P_t)u) = \int (1 - e^{-t\lambda})\, m_u(d\lambda)$, and $(1 - e^{-t\lambda})/t$ decreases in t to λ as $t \downarrow 0$; if $\bar{\mathcal{E}}(u) = \int \lambda\, dm_u > 0$, then $m_u((0,\infty)) > 0$. (5) $(\alpha + A_d)^{-1}\mathcal{P}_d f$ is the orthogonal projection, in $(\mathrm{Dom}, \bar{\mathcal{E}} + \alpha\|\cdot\|^2)$, of $(\alpha + A)^{-1} f$ onto Dom_d, and these closed subspaces increase with a dense union; so the resolvents converge strongly, and with them every continuous function of the resolvents, $e^{-tA_d}\mathcal{P}_d$ among them. (6) $(u, (1 - P_t)u) \geq \mathfrak{J}_t^{\bar{\mathcal{E}}}(u) = \int \frac{t\lambda}{1+t\lambda} dm_u$, as $\frac{x}{1+x} \leq 1 - e^{-x}$; for $w \in \mathrm{Dom}$, $t\bar{\mathcal{E}}(w) \geq \frac{t}{C}\bar{\mathcal{E}}_d(\mathcal{P}_d w)$ and $\|u - w\| \geq \|\mathcal{P}_d u - \mathcal{P}_d w\|$; and $1 - e^{-x} \leq \frac{2x}{1+x}$ gives the last inequality.

(b) $\mathcal{C}_d^{\tilde{\mathcal{M}}} \subset \mathcal{C}_{d+1}^{\tilde{\mathcal{M}}}$, $1 \in \mathcal{C}_0^{\tilde{\mathcal{M}}}$, density in L^2_{inv}, the definition on classes, $\mathcal{E}_\infty^{\tilde{\mathcal{M}}}(1) = 0$ and the square field with the chain rule and values in L^1: Proposition 18.68(c), (e). The values of $\Gamma_\infty^{\tilde{\mathcal{M}}}$ are Borel functions on the threads that the maps $Y \mapsto v \cdot Y$ keep, by the covariance in Proposition 18.67(c); such a function agrees μ-almost everywhere with a function measurable for the invariant σ-algebra, as its conditional means given Y_d, averaged over the gauge group of $\Lambda_{d,\ell}$, are invariant cylinder functions that converge to it as $d \to \infty$ (the martingale convergence theorem for bounded truncations). (Cl) for $\tilde{\mathcal{M}} = \mathcal{M}^\chi$: Proposition 18.69(b). □

III. A torus with one unit of centre flux. This part works on a twisted torus; its lattices, configurations and gauge groups, and the maps and measures on them, are all introduced below. From outside it takes G, $\mathfrak{g}$, $\langle\cdot,\cdot\rangle$, the blocking factor L and the spacings $a_d = L^{-d}$ of the setting of the section ($|\cdot|$ is the operator norm of 2×2 matrices, which on $\mathfrak{g}$ is the norm of the setting, and of linear maps of $\mathfrak{g}$, and the Euclidean norm on coordinate spaces), the constants A_0, $\mathfrak{p}_0$ and the threshold function $\varepsilon(t) = A_0 t^{-1/2}(\log t)^{\mathfrak{p}_0}$ for $t > 1$, the Pauli matrices

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \qquad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},$$

and the map $\mathbf{w}$ of Part I. The setting (E) is not assumed in this part: here $\mathsf{b} > 0$ is a free number, and the conditions on it are stated where they are used.

A twisted lattice. Let $N_1, N_2, N_3 \geq 3$ be integers, $\mathsf{v} := N_1N_2N_3$, $N := \max_\mu N_\mu$, $\mathsf{B}_{\mathrm{tw}} := \prod_\mu \{0, \dots, N_\mu - 1\}$; we work in lattice units on $\mathbb{Z}^3$, with bonds (x,μ) from x to $x + e_\mu$, plaquettes $p_{\mu\nu}(x)$, $\mu < \nu$, and $\partial U(p_{\mu\nu}(x)) = U(x,\mu)U(x+e_\mu,\nu)U(x+e_\nu,\mu)^{-1}U(x,\nu)^{-1}$. Put $\mathsf{i}_a := i\sigma_a$ and $\mathsf{z}_1 := \mathsf{i}_1$, $\mathsf{z}_2 := \mathsf{i}_2$, $\mathsf{z}_3 := \mathbf{1}$, so that $\mathsf{z}_1\mathsf{z}_2 = -\mathsf{z}_2\mathsf{z}_1$ while the rotations $\mathrm{Ad}(\mathsf{z}_\mu)$ commute. $\boldsymbol{\Omega} = \boldsymbol{\Omega}(N_1,N_2,N_3)$ is the set of the maps U from the bonds of $\mathbb{Z}^3$ to G with $U(b + N_\mu e_\mu) = \mathsf{z}_\mu U(b)\mathsf{z}_\mu^{-1}$, and $\mathcal{G}^{\mathrm{tw}}$ the group of the maps $g : \mathbb{Z}^3 \to G$ with $g(x + N_\mu e_\mu) = \mathsf{z}_\mu g(x)\mathsf{z}_\mu^{-1}$, acting by $U^g(b) = g(b_-)U(b)g(b_+)^{-1}$; the values at the bonds and at the sites based in B_{tw} are free coordinates, and dU, dg are the products of the Haar probability measures over them. $A(U) := \sum_p (1 - \frac{1}{2}\mathrm{tr}\,\partial U(p))$ and $\varpi(U) := \max_p |\partial U(p) - \mathbf{1}|$, over the plaquettes based in B_{tw}; as the $\mathrm{Ad}(\mathsf{z}_\mu)$ are commuting isometries, sums over B_{tw} of invariant expressions do not read the fundamental domain. For $\varsigma = \pm$, $R_\varsigma \in \boldsymbol{\Omega}$ is $\varsigma\mathbf{1}$ at the bonds $(x,3)$ with $x_3 \equiv N_3 - 1 \bmod N_3$ and $\mathbf{1}$ at the other bonds; the flip $U \mapsto R_- U$ keeps dU, every $\partial U(p)$, A and ϖ, and a function is even if the flip keeps it. The line $\mathcal{P}(U)(x) := \prod_{j=0}^{N_3-1} U(x + je_3, 3)$ has $\mathcal{P}(U^g)(x) = g(x)\mathcal{P}(U)(x)g(x)^{-1}$ and $\mathcal{P}(R_- U) = -\mathcal{P}(U)$.

Forms, the slice, the sets. $\mathscr{F}^0$ and $\mathscr{F}^1$ are the spaces of the $\mathfrak{g}$-valued functions f on the sites, resp. the bonds, of $\mathbb{Z}^3$ with $f(\cdot+N_\mu e_\mu)=\mathrm{Ad}(\mathsf{z}_\mu)f(\cdot)$; $\|f\|^2:=\sum_{\mathsf{B}_{\rm tw}}|f|^2$, $|f|_\infty:=\max|f|$; $(\nabla\lambda)(b):=\lambda(b_+)-\lambda(b_-)$ and ∇^* is its adjoint; $dX(p):=X(b_1)+X(b_2)-X(b_3)-X(b_4)$ for the bonds $b_1,\dots,b_4$ of p in the order of $\partial U(p)$; $\mathscr{H}:=\ker\nabla^*\subset\mathscr{F}^1$, the Landau slice, with the Euclidean structure of $\|\cdot\|$ and its Lebesgue measure dX; $(D_U\lambda)(b):=\mathrm{Ad}(U(b))\lambda(b_+)-\lambda(b_-)$. A tangent vector at U is an $X\in\mathscr{F}^1$, through $s\mapsto e^{isX}U$, and the gradient $\partial h(U)\in\mathscr{F}^1$ of a C^1 function is given by $\frac{d}{ds}h(e^{isX}U)|_{s=0}=\langle X,\partial h(U)\rangle$; for $X\in\mathfrak{g}$, $\mathsf{T}_X:=\int_0^1\mathrm{Ad}(e^{i\tau X})\,d\tau$, so that $\frac{d}{ds}e^{i(X+sY)}|_{s=0}=i(\mathsf{T}_XY)e^{iX}$, applied bond by bond. Put $\lambda_*:=\min\{4\sin^2\frac{\pi}{2N_1},4\sin^2\frac{\pi}{2N_2}\}\in[4N^{-2},1]$, $\Xi_N:=\frac{N}{2}\sqrt{3\mathsf{v}}$ and $r_1:=(16\Xi_N)^{-1}$. For $\epsilon>0$: $\mathrm{In}(\epsilon):=\{U\in\mathbf{\Omega}:\varpi(U)<\epsilon\}$, $\mathrm{In}^\varsigma(\epsilon):=\{U\in\mathrm{In}(\epsilon):\varsigma\,\mathrm{tr}\,\mathcal{P}(U)(0)>0\}$, $\mathcal{D}(\epsilon):=\{X\in\mathscr{H}:|X|_\infty\le r_1,\ \varpi(e^{iX})<\epsilon\}$, $f_p(X):=1-\frac12\mathrm{tr}\,\partial e^{iX}(p)=\frac12|\partial e^{iX}(p)-\mathbf{1}|^2$ and $f:=\sum_p f_p=A(e^{i\cdot})$; $\varkappa_\epsilon:=6/\epsilon$ and $\Sigma_\epsilon(Z):=(1+\varkappa_\epsilon\|Z\|^2)Z$ on $\mathscr{H}$, a radial real-analytic diffeomorphism of $\mathscr{H}$ with the inverse Θ_ϵ; $\Upsilon_\varsigma(g,X):=(R_\varsigma e^{iX})^g$; $\mathbb{A}(X):\mathscr{F}^0\oplus\mathscr{H}\to\mathscr{F}^1$, $\mathbb{A}(X)(\lambda,Y):=-D_{e^{iX}}\lambda+\mathsf{T}_XY$, and $\mathcal{J}(X):=|\det\mathbb{A}(X)|$ in orthonormal bases. The conditions on ϵ are (h1) $36\pi\mathsf{v}N^3\epsilon\le1$ and (h2) $1800N^2\mathsf{v}\epsilon\le1$.

The cube, the units, the measure. Let $l\ge L$ be a power of L; at the depth d put $N_1=N_2=N_3=lL^d$, $\mathbf{\Omega}_d:=\mathbf{\Omega}(lL^d,lL^d,lL^d)$, $\mathcal{G}_d^{\rm tw}$ its gauge group and $x_d:=L^d\mathsf{b}$. Physical fields are X/a_d, with the inner product $a_d^3\sum_b$, and $\Gamma_d(h):=a_d^{-1}\|\partial h\|^2$; so a lower bound $\varrho\|Y\|^2$ of a Hessian in lattice units is the bound ϱ/a_d in the physical inner product, and $\mathrm{Ent}\le c\int\|\partial h\|^2$ reads $\mathrm{Ent}\le c\,a_d\int\Gamma_d(h)$. At the cutoff K: $\mu_K^{\rm tw}:=Z^{-1}e^{-\beta A}dU$ on $\mathbf{\Omega}_K$, $\beta=L^K\mathsf{b}$; $\mathcal{L}_K^{\rm tw}:=a_K^{-1}[\sum_b\partial_b^2-\beta\langle\partial A,\partial\cdot\rangle]$, $\mathcal{E}_K^{\rm tw}(F):=\int\Gamma_K(F)\,d\mu_K^{\rm tw}$ and $P_u^{\rm tw}:=e^{u\mathcal{L}_K^{\rm tw}}$, $u\ge0$: the Langevin dynamics, a reversible elliptic diffusion with smooth coefficients on a compact manifold; $P_u^{\rm tw}$ commutes with $\mathcal{G}_K^{\rm tw}$ and with the flip, maps bounded Borel functions to smooth ones for $u>0$, and $\mathrm{osc}(P_u^{\rm tw}F)\le\mathrm{osc}(F)$. For a finite positive measure m and a bounded Borel $F\ge0$: $\Xi(F,c):=F\log\frac{F}{c}-F+c\ge0$ for $c>0$ (with $0\log0=0$), $\mathrm{Ent}_m(F):=\inf_{c>0}\int\Xi(F,c)\,dm=\int F\log F\,dm-m(F)\log\frac{m(F)}{m(1)}$, the infimum being taken at $c=m(F)/m(1)$ when $m(F)>0$, and $\mathrm{Var}_m(G):=\min_c\int(G-c)^2dm$.

Two data. Fix a top depth $n\ge1$ with $L^n\mathsf{b}>1$ and put $N:=lL^n$, $x:=L^n\mathsf{b}$ and $\epsilon:=2\varepsilon(x)$, with the function $\varepsilon(\mathsf{t})=A_0\mathsf{t}^{-1/2}(\log\mathsf{t})^{\mathsf{p}_0}$, $\mathsf{t}>1$, the one that enters the thresholds θ_d of the setting of the section. All the numbers named in the data are free of K. (D1) For every $K>n$ there are: (D1.1) Borel maps $\mathcal{M}^{\rm tw}:\mathbf{\Omega}_d\to\mathbf{\Omega}_{d-1}$, $n<d\le K$, that carry gauge equivalent fields to gauge equivalent fields and R_-U to a field gauge equivalent to $R_-\mathcal{M}^{\rm tw}(U)$, so that a gauge-invariant even function of $V_d:=(\mathcal{M}^{\rm tw})^{K-d}U$ is a gauge-invariant even function of U, and open sets $\mathrm{Cut}_d^{\rm tw}\subset\mathbf{\Omega}_d$, $n\le d\le K$, carried onto themselves by $\mathcal{G}_d^{\rm tw}$ and by the flip, with $\mathrm{In}(\epsilon)\subset\mathrm{Cut}_n^{\rm tw}$; (D1.2) a positive Borel measure $\tilde\mu_K$ on $\mathbf{\Omega}_K$ and a number $\mathsf{R}_{\rm tw}\in[0,1)$ with $\int F\,d\tilde\mu_K\le\int F\,d\mu_K^{\rm tw}\le\int F\,d\tilde\mu_K+\mathsf{R}_{\rm tw}\sup F$ for every bounded Borel gauge-invariant $F\ge0$; the image $\tilde\mu_{K,[d]}$ of $\tilde\mu_K$ under $U\mapsto V_d$ sits on $\mathrm{Cut}_d^{\rm tw}$, and $\tilde{\mathcal{E}}_K(G):=\int\Gamma_K(G)\,d\tilde\mu_K$; (D1.3) for $n<d\le K$ and $W\in\mathrm{Cut}_{d-1}^{\rm tw}$: an open bounded convex set $\mathrm{Rg}_d\subset\mathbb{R}^{\dim_d}$, a C^1 map $\tilde V_W:\mathrm{Rg}_d\to\mathrm{Cut}_d^{\rm tw}$ with $\|D\tilde V_W(B)\zeta\|^2\le C_{\rm ch}^{\rm tw}x_d^{-1}|\zeta|^2$, and a probability measure $\tilde p_W=e^{-\tilde\Psi_W}dB$ on Rg_d with $\tilde\Psi_W\in C^2$, $D^2\tilde\Psi_W\ge\gamma_\flat>0$, such that $\int h(V_d)\phi(V_{d-1})\,d\tilde\mu_K=\int\phi(W)[\int h(\tilde V_W(B))\tilde p_W(dB)]\,\tilde\mu_{K,[d-1]}(dW)$ for bounded Borel h,ϕ, the inner integral being Borel in W; (D1.4) if h on $\mathrm{Cut}_d^{\rm tw}$ is gauge invariant and even, so is $W\mapsto\int h(\tilde V_W(B))\tilde p_W(dB)$; (D1.5) $\tilde\mu_{K,[n]}=\mathbf{1}_{\mathrm{In}(\epsilon)}\mathsf{g}^{\rm tw}dV$ with $\mathsf{g}^{\rm tw}=\mathsf{c}_K^{\rm tw}\exp[-x\mathcal{A}^{\rm tw}+\mathcal{S}^{\rm tw}]$, a number $\mathsf{c}_K^{\rm tw}>0$ and real gauge-invariant even functions $\mathcal{A}^{\rm tw},\mathcal{S}^{\rm tw}$ on $\mathrm{In}(\epsilon)$ such that $\mathsf{a}(X):=\mathcal{A}^{\rm tw}(e^{iX})$ and $\mathsf{s}(X):=\mathcal{S}^{\rm tw}(e^{iX})$ satisfy (a1)–(a3) and (s) of Proposition 18.72(b) with $M_\mathsf{a}=C_\mathsf{a}N^3\rho_\mathsf{a}^2$, $M_\mathsf{s}=E_\mathsf{s}N^3$ and $\rho_\mathsf{s}=\vartheta_\mathsf{s}\varepsilon(x)$. (D2) For $n\le d\le K$ and $V\in\mathrm{Cut}_d^{\rm tw}$ there are seminorms $\mathsf{n}_d(V;\cdot)$ on the tangent vectors at V, with $\mathsf{n}_K(U;\xi)^2=a_K\min_\lambda\|\xi-D_U\lambda\|^2$ and (N) $\mathsf{n}_d(V;\xi)^2\le c_\flat^{-1}a_d\|\xi\|^2$; put $\hat\Gamma_d(g)(V):=\sup\{\langle\xi,\partial g(V)\rangle^2:\mathsf{n}_d(V;\xi)\le1\}$, a Borel function of V for $g\in C^1$. For every real gauge-invariant $f_0\in C^\infty(\mathbf{\Omega}_K)$ with its tower $f_{k+1}(W):=\int f_k(\tilde V_W(B))\tilde p_W(dB)$, $W\in\mathrm{Cut}_{K-k-1}^{\rm tw}$, every $n<d\le K$, $k=K-d$, every $W\in\mathrm{Cut}_{d-1}^{\rm tw}$ and every tangent vector ξ at W: (S0) $f_{k+1}\in$

$C^1(\mathrm{Cut}^{\mathrm{tw}}_{d-1})$, and there are a Borel field $B \mapsto \dot V_B$ of tangent vectors at $\tilde V_W(B)$ and a $\psi_\xi \in C^1(\mathrm{Rg}_d)$, neither reading f_0, with $\langle \xi, \partial f_{k+1}(W)\rangle = \int \langle \dot V_B, \partial f_k(\tilde V_W(B))\rangle\, \tilde p_W(dB) + \mathrm{Cov}_{\tilde p_W}(f_k \circ \tilde V_W, \psi_\xi)$; (S1a) $\mathsf{n}_d(\tilde V_W(B); \dot V_B) \le (1+\eta'_d)\,\mathsf{n}_{d-1}(W;\xi)$ for all B; (S1b) $|\nabla_B \psi_\xi| \le \iota_d\, \mathsf{n}_{d-1}(W;\xi)$ on Rg_d; (Σ) with $\eta_d := \eta'_d + \gamma_\flat^{-1}\iota_d (C^{\mathrm{tw}}_{\mathrm{ch}}/c_\flat)^{1/2} a_d \mathsf{b}^{-1/2}$: $\prod_{d>n}(1+\eta_d)^2 \le \Pi_{\mathrm{tw}} < \infty$.

Proposition 18.71 (The fields with small plaquettes on a twisted lattice)**.** Let $N_1, N_2, N_3 \ge 3$.

(o) (The sign picture.) The map J that keeps the value at every bond (x,μ), $x \in \mathsf{B}_{\mathrm{tw}}$, $x_\mu < N_\mu - 1$, and multiplies the value at the seam bonds (x,μ), $x_\mu = N_\mu - 1$, on the right by z_μ, is a bijection of $\mathbf{\Omega}$ onto the configurations of the periodic lattice $\prod_\mu \mathbb{Z}/N_\mu\mathbb{Z}$ that carries dU to the Haar measure and U^g to $(\mathsf{J}U)^{g|_{\mathsf{B}_{\mathrm{tw}}}}$, and $\partial(\mathsf{J}U)(p) = \eta(p)\,\partial U(p)$ with $\eta(p) = -1$ at the plaquettes $p_{12}(x)$, $x_1 = N_1 - 1$, $x_2 = N_2 - 1$, and $\eta(p) = 1$ elsewhere: A is the Wilson action with one column of plaquettes of the planes $(1,2)$ taken with the sign -1, one unit of centre flux through these planes.

(a) $A(U) = 0$ if and only if $U = R_\varsigma^g$ for some $g \in \mathcal{G}^{\mathrm{tw}}$ and $\varsigma = \pm$; the two orbits are disjoint ($\frac12 \mathrm{tr}\,\mathcal{P} = \varsigma$ on the orbit of R_ς) and exchanged by the flip, and the stabiliser of R_ς is $\{\pm\mathbf{1}\}$.

(b) $\frac{d^2}{ds^2} A(e^{isX} R_\varsigma)|_{s=0} = \|dX\|^2$; the tangent space of the orbit at R_ς is $\nabla\mathscr{F}^0$, with the orthogonal complement $\mathscr{H}$; $\|dX\|^2 + \|\nabla^* X\|^2 \ge \lambda_* \|X\|^2$ on $\mathscr{F}^1$ and $\|\nabla\lambda\|^2 \ge \lambda_*\|\lambda\|^2$ on $\mathscr{F}^0$, with equality for some $X \in \mathscr{H}$ and some λ; so $\dim \mathscr{H} = 6\mathsf{v}$ and $\|dX\|^2 \ge \lambda_* \|X\|^2 \ge 4N^{-2}\|X\|^2$ on $\mathscr{H}$. On the cube of side l at the depth d, the Hessian at R_ς of $a_d^{-1}A$, on the orthogonal complement of the gauge directions and in the inner product $a_d^3 \sum_b$ of the physical fields, is at least $a_d^{-2} 4 \sin^2 \frac{\pi a_d}{2l} \ge 4l^{-2}$, and this floor tends to $\pi^2 l^{-2}$ as $d \to \infty$.

(c) If $\varpi(U) \le \epsilon$ and $4N^2\epsilon \le 1$, there are $g \in \mathcal{G}^{\mathrm{tw}}$ and ς with $\max_b |U^g(b) - R_\varsigma(b)| \le 3N^2\epsilon$ and $\varsigma\, \frac12 \mathrm{tr}\,\mathcal{P}(U)(x) \ge \frac{23}{32}$ for all x.

(d) Under (h1): $\mathrm{In}(\epsilon)$ is the disjoint union of $\mathrm{In}^+(\epsilon)$ and $\mathrm{In}^-(\epsilon)$, both open, carried onto themselves by $\mathcal{G}^{\mathrm{tw}}$ and exchanged by the flip; $\mathcal{D}(\epsilon)$ is open in $\mathscr{H}$, and $\max_p |dX(p)| < 2\epsilon$ and $|X|_\infty \le \|X\| < 2\Xi_N \epsilon < r_1$ on it; $\Upsilon_\varsigma : \mathcal{G}^{\mathrm{tw}} \times \mathcal{D}(\epsilon) \to \mathrm{In}^\varsigma(\epsilon)$ is real-analytic, onto, a local diffeomorphism, and $\Upsilon_\varsigma(g,X) = \Upsilon_\varsigma(g',X')$ if and only if $X' = X$ and $g' = \pm g$; for every bounded Borel F on $\mathrm{In}^\varsigma(\epsilon)$: $\int_{\mathrm{In}^\varsigma(\epsilon)} F\, dU = \frac12 (2\pi^2)^{-2\mathsf{v}} \int_{\mathcal{G}^{\mathrm{tw}}} dg \int_{\mathcal{D}(\epsilon)} dX\, \mathcal{J}(X) F(\Upsilon_\varsigma(g,X))$.

(e) Let (h1) and (h2) hold. Then $\Theta_\epsilon(\mathcal{D}(\epsilon))$ is a convex open subset of $\mathscr{H}$ containing 0 and contained in $\mathcal{B}_\epsilon := \{\|Z\|^2 < \epsilon/600\}$, and throughout $\mathcal{B}_\epsilon$ one has $\varkappa_\epsilon \|Z\|^2 < \frac{1}{100}$ and $\|Y\| \le \|\Sigma'_\epsilon(Z)Y\| \le 1.03\|Y\|$. So $\mathcal{D}(\epsilon)$ is star-shaped, $\mathrm{In}^\pm(\epsilon)$ are connected, and modulo gauge $\mathrm{In}(\epsilon)$ consists of two pieces, $\mathrm{In}^+(\epsilon)$ and $\mathrm{In}^-(\epsilon)$, exchanged by the flip, each of which is parametrised by $\mathcal{D}(\epsilon)$ through (d), and the coordinate $Z = \Theta_\epsilon(X)$ of the slice maps $\mathcal{D}(\epsilon)$ to a convex open subset of the $6\mathsf{v}$-dimensional space $\mathscr{H}$.

(f) For $X, Y \in \mathscr{F}^1$ with $r := |X|_\infty$ and $\varpi := \varpi(e^{iX}) \le 1$: $(1 - \frac12\varpi^2)(\|dY\| - 36r\|Y\|)_+^2 - 16\varpi\|Y\|^2 \le D^2 f(X)(Y,Y) \le (\|dY\| + 36r\|Y\|)^2 + 16\varpi\|Y\|^2$ and $|Df(X)Y| \le (\|dY\| + 36r\|Y\|)\sqrt{3\mathsf{v}}\,\varpi$. Under (h1), (h2), for $X \in \mathcal{D}(\epsilon)$ and $Y \in \mathscr{H}$: $D^2 f(X)(Y,Y) \ge 0.99\|dY\|^2 \ge 0.99\lambda_*\|Y\|^2$ and $|Df(X)Y| \le 4.01\sqrt{3\mathsf{v}}\,\epsilon\|Y\|$.

(g) On $\{X \in \mathscr{F}^1 \otimes \mathbb{C} : |X|_\infty < (5N)^{-1}\}$ the function $\mathfrak{l}(X) := \log\det(\mathbb{A}(0)^{-1}\mathbb{A}(X))$ is holomorphic with $|\mathfrak{l}| \le 9\mathsf{v}\log 2$; for real X, $\mathcal{J}(X) = \mathcal{J}(0) e^{\mathfrak{l}(X)} > 0$, $\mathcal{J}(0) = \det(\nabla^*\nabla)^{1/2}$; and for real X with $|X|_\infty \le (10N)^{-1}$ and real Y: $|D\log\mathcal{J}(X)Y| \le 63\mathsf{v}N|Y|_\infty$ and $|D^2 \log\mathcal{J}(X)(Y,Y)| \le 1250\mathsf{v}N^2|Y|_\infty^2$.

■

Sketch. Facts on G: for $u \in G$, $u = \mathbf{c} + i\mathbf{v}$ with $\mathbf{c} = \frac12\mathrm{tr}\,u$, $\mathbf{v} = \mathbf{w}(u)$, $\mathbf{c}^2 + |\mathbf{v}|^2 = 1$ and $1 - \frac12\mathrm{tr}\,u = \frac12|u - \mathbf{1}|^2$; $|e^{iX} - \mathbf{1}| \le |X|$, $|e^{iX} - \mathbf{1} - iX| \le \frac12|X|^2$; for $u \ne -\mathbf{1}$, $\mathsf{d}(u) := |\log u|$, the geodesic distance from u to $\mathbf{1}$ on the unit sphere G, is at most $\frac{\pi}{2}|u - \mathbf{1}|$; $|\mathrm{Ad}(e^{iX}) - 1| \le 2|X|$; $|\mathsf{T}_X Y| \le |Y|$, $|\mathsf{T}_X Y - Y| \le |X||Y|$, and on the plane orthogonal to X, $\langle \mathsf{T}_X^{-1} Y, Y\rangle \ge \theta\cot\theta\,|Y|^2$,

$\theta = |X| < \frac{\pi}{2}$; for unitaries $|u_1 \cdots u_k - \mathbf{1} - \sum_j (u_j - \mathbf{1})| \le \sum_{i<j} |u_i - \mathbf{1}||u_j - \mathbf{1}|$; on the ball $|\lambda| \le \Lambda_0 < \frac{\pi}{2}$ of $\mathfrak{g}$, $|e^{i\lambda} - e^{i\lambda'}| \le |\lambda - \lambda'| \le (\cos \Lambda_0)^{-1} |e^{i\lambda} - e^{i\lambda'}|$.

(o) At a seam bond, with $x' = x + e_\mu - N_\mu e_\mu$: $g(x)U(x,\mu)g(x+e_\mu)^{-1}\mathsf{z}_\mu = g(x)U(x,\mu)\mathsf{z}_\mu g(x')^{-1}$, by the twisted periodicity of g. In a plaquette the seam factors cancel or commute, except at the corner of the planes $(1,2)$, where they meet as $\mathsf{z}_2\mathsf{z}_1^{-1} = -\mathsf{z}_1^{-1}\mathsf{z}_2$.

(a) $\partial R_\varsigma(p) = \mathbf{1}$, as a plaquette holds none or two of the bonds with the value $\varsigma\mathbf{1}$, in opposite orientations; and $\partial U^g(p) = g(x)\partial U(p) g(x)^{-1}$. Conversely, if all $\partial U(p) = \mathbf{1}$, the product $h(x)$ of U along a path from x to 0 does not depend on the path, and $U(b) = h(b_-)h(b_+)^{-1}$; the twisted periodicity gives $h(x + N_\mu e_\mu) = \mathsf{z}_\mu h(x) C_\mu$ with constants $C_\mu \in G$, and adding two periods in the two orders, $C_1C_2 = -C_2C_1$ and C_3 commutes with C_1, C_2. The algebraic fact below at $\delta = 0$, applied to $(-C_1, -C_2, C_3)$, gives c and ς with $c(-C_j)c^{-1} = \mathsf{i}_j$, $C_3 = \varsigma\mathbf{1}$; then $h' := hc^{-1}$ has $h'(x + N_j e_j) = \mathsf{z}_j h'(x) \mathsf{z}_j^{-1}$ $(j = 1,2)$ and $h'(x + N_3 e_3) = \varsigma h'(x)$, and with $s(x) := \varsigma^{\lfloor x_3/N_3 \rfloor}$, $g := sh' \in \mathcal{G}^{\mathrm{tw}}$ and $U = R_\varsigma^g$. $\mathcal{P}(R_\varsigma) = \varsigma\mathbf{1}$, and $\operatorname{tr}\mathcal{P}$ is gauge invariant; $R_\varsigma^g = R_\varsigma$ forces g constant and commuting with $\mathsf{i}_1, \mathsf{i}_2$: $g = \pm\mathbf{1}$.

The algebraic fact. If $C_1, C_2, C_3 \in G$, $0 \le \delta \le \frac{1}{4}$, $|C_1C_2 + C_2C_1| \le \delta$ and $|C_3C_j - C_jC_3| \le \delta$ $(j = 1,2)$, there are $c \in G$ and $\varsigma = \pm 1$ with $|cC_1c^{-1} - \mathsf{i}_1| \le 0.71\delta$, $|cC_2c^{-1} - \mathsf{i}_2| \le 0.72\delta$ and $|C_3 - \varsigma\mathbf{1}| \le 0.77\delta$. Write $C_j = \cos\vartheta_j + i \sin\vartheta_j\, n_j \cdot \sigma$, $\sin\vartheta_j \ge 0$, $\nu := n_1 \cdot n_2$: $\frac{1}{4}|C_1C_2 + C_2C_1|^2 = \cos^2\vartheta_1 + \sin^2\vartheta_1(\cos^2\vartheta_2 + \sin^2\vartheta_2\nu^2)$ and the same with $1 \leftrightarrow 2$; so $|\cos\vartheta_j| \le \frac{\delta}{2}$ and $|\nu| \le 0.508\delta$; turning n_1 to e_1 and n_2 into the plane of e_1, e_2 gives the first two bounds; $|C_3C_j - C_jC_3| = 2\sin\vartheta_3 \sin\vartheta_j |n_3 \times n_j|$ and $\sum_j |n_3 \times n_j|^2 \ge 1 - |\nu| \ge 0.873$ give $\sin^2\vartheta_3 \le 0.582\delta^2$, and $|C_3 - \varsigma\mathbf{1}|^2 = 2 - 2|\cos\vartheta_3|$ with ς the sign of $\cos\vartheta_3$.

(b) $\partial(e^{isX}R_\varsigma)(p) = e^{isX_1}e^{isX_2}e^{-isX_3}e^{-isX_4}$, the central factors multiplying to $\mathbf{1}$, and $1 - \frac{1}{2}\operatorname{tr} u = \frac{1}{2}|u - \mathbf{1}|^2 = \frac{1}{2}s^2|dX(p)|^2 + O(s^3)$; $R_\varsigma^{e^{is\lambda}}$ has the velocity $-\nabla\lambda$. As $\operatorname{Ad}(\mathsf{i}_a)\sigma_c = \pm\sigma_c$, with $+$ if and only if $c = a$, the colours $X = \sum_a X^a\sigma_a$ decouple, $X^a(\cdot + N_\mu e_\mu) = (-1)^{\theta^a_\mu} X^a$ with $\theta^1 = (0,1,0)$, $\theta^2 = (1,0,0)$, $\theta^3 = (1,1,0)$, and Fourier series in the momenta $p_\mu \in \frac{2\pi}{N_\mu}(\mathbb{Z} + \frac{1}{2}\theta^a_\mu)$ with $k_\mu = e^{ip_\mu} - 1$ and the identity of Lagrange $\sum_{\mu<\nu} |k_\mu \hat{X}_\nu - k_\nu \hat{X}_\mu|^2 + |\sum_\mu \bar{k}_\mu \hat{X}_\mu|^2 = |k|^2|\hat{X}|^2$ give the inequalities: every colour has a direction $\mu \in \{1,2\}$ with $\theta^a_\mu = 1$, and there $|k_\mu|^2 \ge 4\sin^2\frac{\pi}{2N_\mu}$. Equality: if $N_2 \ge N_1$, $X = \cos(\pi x_2/N_2)\sigma_1$ at the bonds $(x,1)$ and 0 elsewhere. The physical floor: $a_d^{-1}\|dX\|^2 \ge a_d^{-2}\lambda_* \cdot a_d\|X\|^2$ with $N = l/a_d$, and $4\sin^2 y \ge 4(2y/\pi)^2$.

(c) *The comb.* For $x \in \mathsf{B}_{\mathrm{tw}}$ let $h(x)$ be the product of U along the path from 0 by x_1 steps e_1, then x_2 steps e_2, then x_3 steps e_3, and $U'(b) := h(b_-)U(b)h(b_+)^{-1}$ at the bonds with both ends in B_{tw}. As the value along the boundary of a ladder of k plaquettes is within $k\epsilon$ of $\mathbf{1}$: $U'(x,3) = \mathbf{1}$, $|U'(x,2) - \mathbf{1}| \le x_3\epsilon$, $|U'(x,1) - \mathbf{1}| \le (x_2 + x_3)\epsilon$. *The seams.* For $x \in \mathsf{B}_{\mathrm{tw}}$ with $x_\mu = N_\mu - 1$ and $x' = x + e_\mu - N_\mu e_\mu$ put $C_\mu(x) := h(x)U(x,\mu)\mathsf{z}_\mu h(x')^{-1}$; a plaquette through the seam bonds (x,μ), $(x + e_\nu, \mu)$ gives $|C_\mu(x) - C_\mu(x + e_\nu)| \le \epsilon + |U'(x',\nu) - \mathbf{1}| + |U'(x,\nu) - \mathbf{1}|$; moving along the seams in a suitable order, $C_\mu(x)$ lies within $2N(N-1)\epsilon$ of $C_1 := C_1(N_1 - 1, 0, 0)$, $C_2 := C_2(0, N_2 - 1, 0)$, $C_3 := C_3(0, 0, N_3 - 1)$ respectively. *The corners.* At $x = (N_1 - 1, N_2 - 1, 0)$ the plaquette of the planes $(1,2)$, read in the sign picture (o), gives $|C_1(x)C_2C_1^{-1}C_2(x)^{-1} + \mathbf{1}| \le \epsilon$; along the seams at $x_3 = 0$ the comb gives $|C_1(x) - C_1| \le (N_2 - 1)\epsilon$ and $|C_2(x) - C_2| \le (N_1 - 1)N_2\epsilon$, so $|C_1C_2 + C_2C_1| = |C_1C_2C_1^{-1}C_2^{-1} + \mathbf{1}| \le N_1N_2\epsilon$; in the same way the corners $(N_1 - 1, 0, N_3 - 1)$ and $(0, N_2 - 1, N_3 - 1)$ give $|C_1C_3 - C_3C_1| \le N_1N_3\epsilon$ and $|C_2C_3 - C_3C_2| \le N_2N_3\epsilon$. The algebraic fact at $\delta = N^2\epsilon \le \frac{1}{4}$ gives c, ς; with $g \in \mathcal{G}^{\mathrm{tw}}$ extending $(ch)|_{\mathsf{B}_{\mathrm{tw}}}$: at the inner bonds $U^g = cU'c^{-1}$ lies within $2(N-1)\epsilon$ of $\mathbf{1}$; at a seam bond $U^g(x,\mu) = cC_\mu(x)c^{-1}\mathsf{z}_\mu^{-1}$ lies within $2N(N-1)\epsilon + 0.77N^2\epsilon \le 3N^2\epsilon$ of $R_\varsigma(x,\mu)$. $\mathcal{P}(U^g)(x_1, x_2, 0) = cC_3(x_1, x_2, N_3 - 1)c^{-1}$ lies within $3N^2\epsilon \le \frac{3}{4}$ of $\varsigma\mathbf{1}$, and $\varsigma\frac{1}{2}\operatorname{tr} u = 1 - \frac{1}{2}|u - \varsigma\mathbf{1}|^2 \ge 1 - \frac{9}{32}$; the trace of $\mathcal{P}(U)(x)$ reads no x beyond (x_1, x_2) in the box.

(d) *The slice.* For $X \in \mathscr{H}$ with $|X|_\infty \le r_1$ and $\varpi(e^{iX}) < \epsilon$, put $D_X := \max_p |dX(p)|$; by (b), $|X|_\infty \le \|X\| \le \lambda_*^{-1/2}\|dX\| \le \Xi_N D_X$, and by (f) below, $|\partial e^{iX}(p) - \mathbf{1} - i\,dX(p)| \le 8|X|_\infty^2$,

so $D_X < \epsilon + 8r_1\Xi_N D_X = \epsilon + \frac12 D_X$: $D_X < 2\epsilon$, $\|X\| < 2\Xi_N\epsilon$, and $2\Xi_N\epsilon < r_1$ by (h1): the constraint $|X|_\infty \le r_1$ is not active, and $\mathcal{D}(\epsilon)$ is open. *Onto.* Let $\varpi(U) < \epsilon$; (c) gives g_0, ς with $U_1 := R_\varsigma U^{g_0}$ within $3N^2\epsilon$ of $\mathbf{1}$ at every bond. Let g_1 minimise $\Phi(g) := \sum_b \mathsf{d}(U_1^g(b))^2$ on the compact group $\mathcal{G}^{\mathrm{tw}}$; $\Phi(g_1) \le \Phi(\mathbf{1}) \le 3\mathsf{v}(\frac{\pi}{2}\,3N^2\epsilon)^2$, so $|X|_\infty \le r_1$ for $X := \log U_1^{g_1} \in \mathscr{F}^1$, by (h1). As $\frac{d}{ds}\frac12\mathsf{d}(e^{is\lambda}e^{iX})^2|_{s=0} = \langle\lambda, X\rangle$ and $\frac{d}{ds}\frac12\mathsf{d}(e^{iX}e^{-is\lambda})^2|_{s=0} = -\langle\lambda, X\rangle$, the minimality gives $\langle X, \nabla\lambda\rangle = 0$ for all λ: $X \in \mathscr{H}$, so $X \in \mathcal{D}(\epsilon)$, and as R_ς is central with $R_\varsigma^2 = \mathbf{1}$, $U = \Upsilon_\varsigma((g_1g_0)^{-1}, X)$. *The two pieces.* For $U = \Upsilon_\varsigma(g, X)$, $\operatorname{tr}\mathcal{P}(U)(0) = \varsigma \operatorname{tr}\mathcal{P}(e^{iX})(0)$ and $|\mathcal{P}(e^{iX}) - \mathbf{1}| \le N_3 r_1 \le \frac{1}{72}$: $\mathrm{In}(\epsilon)$ is the disjoint union of the two open invariant sets $\mathrm{In}^\pm(\epsilon)$, exchanged by the flip as $\mathcal{P}(R_- U) = -\mathcal{P}(U)$. *One to one.* Let $R_\varsigma e^{iX'} = (R_\varsigma e^{iX})^g$ with $X, X' \in \mathcal{D}(\epsilon)$. By (h1), $|X|_\infty, |X'|_\infty < 2\Xi_N\epsilon \le \frac{2}{3\pi N} r_1 < 0.071 r_1$, and $e^{iX'(b)} = g(b_-)e^{iX(b)}g(b_+)^{-1}$ gives $|g(b_-) - g(b_+)| < 0.142 r_1$ at every bond; following the periods of the torus in the directions 1 and 2 gives $|[\mathsf{i}_j, g(0)]| < 0.142 N r_1$ for $j = 1, 2$, which puts $g(0)$ within $0.15 N r_1$ of $\pm\mathbf{1}$; as $N r_1 \le \frac{1}{72}$, after the choice of the sign $g = e^{i\lambda}$ with $\lambda \in \mathscr{F}^0$, $|\lambda|_\infty \le 0.02$ and $|\nabla\lambda|_\infty \le 0.15 r_1$. Along $U_\tau := (e^{iX})^{e^{i\tau\lambda}}$, $0 \le \tau \le 1$, with $X_\tau := \log U_\tau$, of $|X_\tau|_\infty \le 0.23 r_1$, the function $\varphi(\tau) := \frac12\|X_\tau\|^2$ has $\varphi'(\tau) = -\langle X_\tau, \nabla\lambda\rangle$, which vanishes at $\tau = 0$ and at $\tau = 1$, as $X_0 = X$ and $X_1 = X'$ lie in $\mathscr{H}$; and $\varphi'' \ge (1 - \frac12|X_\tau|_\infty^2)\|\nabla\lambda\|^2 - 2.1\sqrt{3}|X_\tau|_\infty\|\lambda\|\|\nabla\lambda\| \ge 0.99\|\nabla\lambda\|^2$, by $\langle \mathsf{T}_Z^{-1}Y, Y\rangle \ge \min\{1, \theta\cot\theta\}|Y|^2 \ge (1 - \frac12\theta^2)|Y|^2$ and $\|\mathsf{T}_Z^{-1}\| \le \theta/\sin\theta$ for $\theta = |Z|$, $|\mathrm{Ad}(e^{iZ}) - 1| \le 2|Z|$ and $\|\lambda\| \le \frac{N}{2}\|\nabla\lambda\|$; so $\nabla\lambda = 0$, and λ, a constant fixed by $\mathrm{Ad}(\mathsf{i}_1)$ and by $\mathrm{Ad}(\mathsf{i}_2)$, is 0: $g = \pm\mathbf{1}$ and $X' = X$. *The integral.* The velocity of $s \mapsto \Upsilon_\varsigma(e^{is\lambda'}g, X)$ is $-D_V\lambda'$, $V = \Upsilon_\varsigma(g, X)$, that of $s \mapsto \Upsilon_\varsigma(g, X + sY)$ is $\mathrm{Ad}_g(\mathsf{T}_X Y)$, $(\mathrm{Ad}_g\xi)(b) := \mathrm{Ad}(g(b_-))\xi(b)$, and $D_V(\mathrm{Ad}(g)\lambda) = \mathrm{Ad}_g(D_{e^{iX}}\lambda)$, R_ς being central; so $D\Upsilon_\varsigma(g, X)(\lambda', Y) = \mathrm{Ad}_g\mathbb{A}(X)(\mathrm{Ad}(g)^{-1}\lambda', Y)$, of the determinant $\pm\det\mathbb{A}(X) \ne 0$ by (g) (on $\mathcal{D}(\epsilon)$, $|X|_\infty < 2\Xi_N\epsilon \le (10N)^{-1}$ by (h1)). The metrics are bi-invariant, each factor G is the unit sphere S^3 of the volume $2\pi^2$, and Υ_ς is two-to-one.

(e) *Step 1.* $\Theta_\epsilon(\mathcal{D}(\epsilon)) = \{Z \in \mathcal{B}_\epsilon : \tilde f_p(Z) < \frac12\epsilon^2 \text{ for all } p\}$, $\tilde f_p := f_p \circ \Sigma_\epsilon$: on one side $\|Z\| \le \|X\| < 2\Xi_N\epsilon$ and $4\Xi_N^2\epsilon^2 \le \epsilon/600$ by (h2); on the other $|X|_\infty \le (\epsilon/400)^{1/2} \le r_1$, as $256\Xi_N^2\epsilon \le 400$. On $\mathcal{B}_\epsilon$: $\theta := \varkappa_\epsilon\|Z\|^2 < \frac{1}{100}$, $q := 1 + \theta$, $\Sigma_\epsilon'(Z)Y = qY + 2\varkappa_\epsilon\langle Z, Y\rangle Z$ with the eigenvalues q and $1 + 3\theta$, $\Sigma_\epsilon''(Z)(Y, Y) = 2\varkappa_\epsilon\|Y\|^2 Z + 4\varkappa_\epsilon\langle Z, Y\rangle Y$; for $X = \Sigma_\epsilon(Z) = qZ$, $\omega := |X|_\infty^2 \le \epsilon/400$ and $|u_p - \mathbf{1}| \le 2\|X\| \le \frac12$. *Step 2.* At every $Z \in \mathcal{B}_\epsilon$ with $|u_p - \mathbf{1}| \ge \frac{\epsilon}{2}$: $D^2\tilde f_p(Z)(Y, Y) \ge 0$ for all Y. Indeed, with the notation of (f) at X, $Y' := \Sigma_\epsilon'(Z)Y$, $\mathbf{v}_p = \mathbf{w}(u_p)$, $c_p \ge \frac78$, $\alpha := \varkappa_\epsilon\langle Z, Y\rangle$ and $P := \langle\zeta_p(Z), \mathbf{v}_p\rangle$: $D^2\tilde f_p(Y, Y) = c_p|\zeta_p(Y')|^2 + \langle\dot\zeta_p(Y', Y'), \mathbf{v}_p\rangle + 2\varkappa_\epsilon\|Y\|^2 P + 4\alpha\langle\zeta_p(Y), \mathbf{v}_p\rangle$, $\zeta_p(Y') = q\zeta_p(Y) + 2\alpha\zeta_p(Z)$, $\alpha^2 \le \varkappa_\epsilon\theta\|Y\|^2$, $|\dot\zeta_p(Y', Y')| \le 4(1 + 3\theta)^2\|Y\|^2$, $|\zeta_p(X) - \mathbf{v}_p| \le 20\omega$, so $P \ge q^{-1}|\mathbf{v}_p|(|\mathbf{v}_p| - 20\omega) > 0$ as $|\mathbf{v}_p| \ge |u_p - \mathbf{1}|(1 - \frac{1}{16})^{1/2} \ge 0.484\epsilon$; completing the square in $|\zeta_p(Y')|$: $D^2\tilde f_p(Y, Y) \ge \|Y\|^2|\mathbf{v}_p|\{1.855\varkappa_\epsilon|\mathbf{v}_p| - 40\varkappa_\epsilon\omega - 4.244\}$, and $1.855 \cdot 6 \cdot 0.484 - 40 \cdot \frac{6}{400} - 4.244 > 0.54$. *Step 3.* On a segment of $\mathcal{B}_\epsilon$, a C^2 function that is convex wherever it exceeds the level $c_0 := \frac18\epsilon^2$ has the maximum with c_0 convex: a component of the set where it exceeds c_0 with both ends inside the segment would carry a convex function equal to c_0 at both ends and larger between. *Step 4.* $\Theta_\epsilon(\mathcal{D}(\epsilon)) = \bigcap_p\{Z \in \mathcal{B}_\epsilon : \max\{\tilde f_p, \frac18\epsilon^2\} < \frac12\epsilon^2\}$ is an intersection of open convex sets that holds 0; Σ_ϵ is radial, so $\mathcal{D}(\epsilon)$ is star-shaped, and $\mathrm{In}^\varsigma(\epsilon) = \Upsilon_\varsigma(\mathcal{G}^{\mathrm{tw}} \times \mathcal{D}(\epsilon))$ is connected, $\mathcal{G}^{\mathrm{tw}}$ being connected.

(f) Let $X_s = X + sY$, $U_j = e^{iX_j(s)}$ at the bonds $1, \dots, 4$ of p in the order of $\partial U(p)$; then $\dot U_j = i\xi_j U_j$, $\xi_j = \mathsf{T}_{X_j + sY_j}Y_j$, and $\dot u_p = i\zeta_p u_p$ with $\zeta_p(Y) := \xi_1 + \mathrm{Ad}(U_1)\xi_2 - \mathrm{Ad}(U_1U_2U_3^{-1})\xi_3 - \mathrm{Ad}(u_p)\xi_4$. So $Df_p(X)Y = \langle\zeta_p(Y), \mathbf{v}_p\rangle$ and $D^2f_p(X)(Y, Y) = c_p|\zeta_p(Y)|^2 + \langle\dot\zeta_p, \mathbf{v}_p\rangle$, $c_p = \frac12\mathrm{tr}\,u_p$, with $|\dot\zeta_p| \le (\sum_{b\in p}|Y_b|)^2$, $|\zeta_p(Y) - dY(p)| \le 9r\sum_{b\in p}|Y_b|$, $|\zeta_p(X) - dX(p)| \le 12r^2$, and $|u_p - \mathbf{1} - i\,dX(p)| \le \frac12(\sum_{b\in p}|X_b|)^2 \le 8r^2$. Summing over the plaquettes, each bond lying in four: $\|\zeta(Y) - dY\| \le 36r\|Y\|$, $\sum_p|\dot\zeta_p| \le 16\|Y\|^2$, $c_p \in [1 - \frac12\varpi^2, 1]$, $|\mathbf{v}_p| \le \varpi$ and $\sum_p|\mathbf{v}_p|^2 \le 3\mathsf{v}\varpi^2$. On $\mathcal{D}(\epsilon)$: $r < 2\Xi_N\epsilon$, $\varpi < \epsilon$, $\|Y\| \le \frac{N}{2}\|dY\|$ on $\mathscr{H}$, and by (h2) $36N\Xi_N\epsilon \le 0.0034$, $4N^2\epsilon \le 10^{-4}$, $72\Xi_N\epsilon \le 0.01$.

(g) On $\mathfrak{g} \otimes \mathbb{C}$ with the Hermitian norm, $\|\mathrm{ad}X\| \le 2|X|$; for $|X| \le \frac14$, $\|e^{i\,\mathrm{ad}X} - 1\| \le 2.6|X|$ and $\|\mathsf{T}_X - 1\| \le 1.19|X|$; so $\|\mathbb{A}(X) - \mathbb{A}(0)\| \le 4.66|X|_\infty$, while $\mathbb{A}(0)(\lambda, Y) = -\nabla\lambda + Y$ is an

orthogonal sum with $\|\mathbb{A}(0)^{-1}\| \le \lambda_*^{-1/2} \le \frac{N}{2}$ and $|\det \mathbb{A}(0)| = \det(\nabla^*\nabla)^{1/2}$. For $|X|_\infty < (5N)^{-1}$, $\mathfrak{e} := \mathbb{A}(0)^{-1}\mathbb{A}(X) - 1$ has $\|\mathfrak{e}\| \le 2.33N|X|_\infty < \frac12$, and $\mathfrak{l} = \operatorname{tr}\log(1+\mathfrak{e})$ is holomorphic with $|\mathfrak{l}| \le 9\mathsf{v}\log 2$, $9\mathsf{v}$ being the dimension; for real X the determinant is real, non-zero and 1 at 0, on a convex set, hence positive. Cauchy's estimate on the disc $|z| < (10N|Y|_\infty)^{-1}$ for $z \mapsto \mathfrak{l}(X+zY)$ gives the two bounds. □

Proposition 18.72 (A log-Sobolev inequality on the small fields of the twisted lattice)**.** Let (h1), (h2) hold.

(a) Let $\nu = \mathsf{g}\,dU$ be a finite measure on $\mathrm{In}(\epsilon)$ with $\mathsf{g} > 0$ Borel, $\mathsf{g}(U^g) = \mathsf{g}(U) = \mathsf{g}(R_- U)$, and $\mathsf{g}(e^{iX}) = \exp[-w(X) - w_0(X)]$ on $\mathcal{D}(\epsilon)$, where (i) $w \in C^2(\mathcal{D}(\epsilon))$, $D^2w(X)(Y,Y) \ge \varrho_0\|Y\|^2$ and $|Dw(X)Y| \le \mathsf{L}_0\|Y\|$ for $X \in \mathcal{D}(\epsilon)$, $Y \in \mathscr{H}$; (ii) w_0 is Borel with $\operatorname{osc} w_0 \le \mathsf{e}_{\mathrm{osc}}$; (iii) $\varrho := \varrho_0 - 1250\mathsf{v}N^2 - 72\Xi_N(\mathsf{L}_0 + 63\mathsf{v}N) - 84\mathsf{v}/\epsilon > 0$. Then for gauge-invariant bounded Borel F: $\int_{\mathrm{In}^\varsigma(\epsilon)} F\,d\nu = \mathsf{k}\int_{\mathcal{D}(\epsilon)} F(R_\varsigma e^{iX})e^{-W(X)-w_0(X)}dX$ with a number $\mathsf{k} > 0$, $W := w - \log\mathcal{J}$, $D^2W \ge \varrho_0 - 1250\mathsf{v}N^2$ and $\|DW\| \le \mathsf{L} := \mathsf{L}_0 + 63\mathsf{v}N$; in the coordinate $Z = \Theta_\epsilon(X)$ the density on the convex set $\Theta_\epsilon(\mathcal{D}(\epsilon))$ is $e^{-\tilde W - w_0\circ\Sigma_\epsilon}$ with $D^2\tilde W \ge \varrho$; and for every bounded, gauge-invariant, even $h \in C^1(\mathrm{In}(\epsilon))$: $\mathrm{Ent}_\nu(h^2) \le \frac{2.13\,e^{\mathsf{e}\mathrm{osc}}}{\varrho}\int\|\partial h\|^2 d\nu$.

(b) Let $x > 0$, $w = x\mathsf{a} - \mathsf{s}$ and $w_0 = 0$, with real functions on $\mathcal{D}(\epsilon)$ such that (a1) a extends holomorphically to $\{X \in \mathscr{H}\otimes\mathbb{C} : |X|_\infty < \rho_\mathsf{a}\}$, with $|\mathsf{a}| \le M_\mathsf{a}$ there, and $4\Xi_N\epsilon \le \rho_\mathsf{a}$; (a2) $D\mathsf{a}(0) = 0$; (a3) $D^2\mathsf{a}(0)(Y,Y) \ge c_\mathsf{a}\|dY\|^2$ for real Y; (s) $\mathsf{s} \in C^2(\mathcal{D}(\epsilon))$, and for real $X \in \mathcal{D}(\epsilon)$ and $Y \in \mathscr{H}$ the function $z \mapsto \mathsf{s}(X+zY)$ extends holomorphically to $\{|z||Y|_\infty < \rho_\mathsf{s}\}$, with modulus at most M_s. Then (a)(i) holds with $\varrho_0 = x[c_\mathsf{a}\lambda_* - 216M_\mathsf{a}\Xi_N\epsilon\rho_\mathsf{a}^{-3}] - 2M_\mathsf{s}\rho_\mathsf{s}^{-2}$ and $\mathsf{L}_0 = 32xM_\mathsf{a}\Xi_N\epsilon\rho_\mathsf{a}^{-2} + M_\mathsf{s}\rho_\mathsf{s}^{-1}$.

(c) (The top of the cube.) In the setting of (a) and (b), let $N_1 = N_2 = N_3 = N = lL^n$, $x = L^n\mathsf{b} > 1$, $\epsilon = 2\varepsilon(x)$, and suppose $N \le C_N(\log x)^{r_N}$ with numbers $C_N > 0$ and $r_N \ge 0$, $2\mathfrak{p}_0 > 5r_N$, $M_\mathsf{a} = C_\mathsf{a}N^3\rho_\mathsf{a}^2$, $M_\mathsf{s} = E_\mathsf{s}N^3$ and $\rho_\mathsf{s} = \vartheta_\mathsf{s}\varepsilon(x)$, with fixed $\rho_\mathsf{a}, C_\mathsf{a}, c_\mathsf{a}, E_\mathsf{s}, \vartheta_\mathsf{s}$. There is $x_\sharp$, formed from $C_N, r_N, A_0, \mathfrak{p}_0, \rho_\mathsf{a}, C_\mathsf{a}, c_\mathsf{a}, E_\mathsf{s}, \vartheta_\mathsf{s}$ alone, such that for $x \ge x_\sharp$: (h1), (h2), $4\Xi_N\epsilon \le \rho_\mathsf{a}$ and $\varrho \ge \frac12 c_\mathsf{a}x\lambda_* \ge 2c_\mathsf{a}x/N^2$ hold; in physical units the Hessian of W is at least $2c_\mathsf{a}\mathsf{b}\,l^{-2}$, and $\mathrm{Ent}_\nu(h^2) \le \frac{1.07}{c_\mathsf{a}}\frac{l^2}{\mathsf{b}}\int\Gamma_n(h)\,d\nu$ for every bounded, gauge-invariant, even $h \in C^1(\mathrm{In}(\epsilon))$.

(d) (The gauge invariance cannot be dropped.) In the setting of (a), for a site x_0 let $h(\Upsilon_\varsigma(g,X)) := \langle\sigma_3, \mathrm{Ad}(g(x_0))\sigma_3\rangle$; it is well defined, real-analytic on $\mathrm{In}(\epsilon)$, even and not gauge invariant, and $\mathrm{Var}_\nu(h) \ge (8N^2)^{-1}\int\|\partial h\|^2 d\nu$. So on the even functions the Poincaré constant of ν is at least $a_n/(8N^2)$ in physical units, which exceeds $C\,l^2/\mathsf{b} = CN^2a_n/x$ as soon as $x > 8CN^4$.

■

Sketch. Three facts on measures are used. (LC) A probability measure $e^{-V}dz$ on an open convex set with $D^2V \ge \varrho > 0$ obeys $\mathrm{Ent}(\phi^2) \le \frac{2}{\varrho}\int|\nabla\phi|^2$ for bounded $\phi \in C^1$ (the fact used in the proof of Lemma 15.3(iv)). (BP) If $\mathrm{Ent}_p(\phi^2) \le c\int|\nabla\phi|^2dp$ and $dp' = Z'^{-1}e^{-\psi}dp$ with $\operatorname{osc}\psi \le e_1$, then $\mathrm{Ent}_{p'}(\phi^2) \le e^{e_1}c\int|\nabla\phi|^2dp'$: with $c' = p(\phi^2)$, $\mathrm{Ent}_{p'}(\phi^2) \le \int\Xi(\phi^2,c')\,dp' \le Z'^{-1}e^{-\inf\psi}\mathrm{Ent}_p(\phi^2)$, and $dp \le Z'e^{\sup\psi}dp'$. (Cov) If a probability p on an open bounded convex set obeys $\mathrm{Ent}_p(\phi^2) \le \frac{2}{\gamma}\int|\nabla\phi|^2dp$, $|\nabla\psi| \le \ell_\psi$, and $F \in C^1$ is bounded with $\inf F > 0$, then $|\mathrm{Cov}_p(F,\psi)| \le \frac{\ell_\psi}{\gamma}(\int F\,dp\int|\nabla F|^2F^{-1}dp)^{1/2}$: the inequality at $\phi = e^{s\psi/2}$ gives $\frac{d}{ds}(s^{-1}\log\int e^{s\psi}dp) \le \frac{\ell_\psi^2}{2\gamma}$, so $\log\int e^{s(\psi-\bar\psi)}dp \le \frac{s^2\ell_\psi^2}{2\gamma}$; the entropy inequality $s\int\varphi(\psi-\bar\psi)\,dp \le \mathrm{Ent}_p(\varphi) + \log\int e^{s(\psi-\bar\psi)}dp$ at $\varphi = F/\int F$, optimised in s, and $\mathrm{Ent}_p(\varphi) \le \frac{1}{2\gamma}\int|\nabla\varphi|^2\varphi^{-1}dp$.

(a) Proposition 18.71(d) at $F\mathsf{g}$, both gauge invariant, and $\mathsf{g}(R_\varsigma e^{iX}) = \mathsf{g}(e^{iX})$ (evenness) give the integral; on $\mathcal{D}(\epsilon)$, $|X|_\infty < (10N)^{-1}$, so Proposition 18.71(g) with $|Y|_\infty \le \|Y\|$ bounds the derivatives of $\log\mathcal{J}$. In the coordinate Z: $dX = \det\Sigma_\epsilon'(Z)\,dZ$, $\det\Sigma_\epsilon'(Z) = (1+\varkappa_\epsilon\|Z\|^2)^{6\mathsf{v}-1}(1+3\varkappa_\epsilon\|Z\|^2)$,

so $\tilde W(Z)=W(\Sigma_\epsilon Z)-(6\mathsf v-1)\log(1+\varkappa_\epsilon\|Z\|^2)-\log(1+3\varkappa_\epsilon\|Z\|^2)$; $D^2(W\circ\Sigma_\epsilon)(Y,Y)\ge(\varrho_0-1250\mathsf v N^2)\|\Sigma'_\epsilon(Z)Y\|^2-\mathsf L\|\Sigma''_\epsilon(Z)(Y,Y)\|$ with $\|\Sigma'_\epsilon(Z)Y\|\ge\|Y\|$ and $\|\Sigma''_\epsilon(Z)(Y,Y)\|\le 6\varkappa_\epsilon\|Z\|\|Y\|^2\le 72\Xi_N\|Y\|^2$; and $D^2\log(1+c\|Z\|^2)\le 2c$, so the two logarithms take away at most $\frac{12(6\mathsf v+2)}{\epsilon}\le\frac{84\mathsf v}{\epsilon}$. For h as stated, $\hat h(Z):=h(e^{i\Sigma_\epsilon(Z)})$: as h^2, $\|\partial h\|^2$ and $\mathsf g$ are gauge invariant and even and the flip exchanges $\mathrm{In}^\pm(\epsilon)$, $\mathrm{Ent}_\nu(h^2)=\nu(1)\mathrm{Ent}_{\hat p}(\hat h^2)$ with the normalised measure $\hat p$ on $\Theta_\epsilon(\mathcal D(\epsilon))$; (LC) for $e^{-\tilde W}$ and (BP) for $w_0\circ\Sigma_\epsilon$ give $\mathrm{Ent}_{\hat p}(\hat h^2)\le e^{\mathsf e_{\mathrm{osc}}}\frac{2}{\varrho}\int|\nabla\hat h|^2d\hat p$; and $D\hat h(Z)Y=\langle\partial h(e^{iX}),\mathsf T_X(\Sigma'_\epsilon(Z)Y)\rangle$, so $|\nabla\hat h|^2\le 1.0609\|\partial h\|^2$.

(b) For real $X\in\mathcal D(\epsilon)$, $|X|_\infty<2\Xi_N\epsilon\le\frac12\rho_{\mathsf a}$; for $0\le\tau\le1$ the function $(z,z')\mapsto\mathsf a(\tau X+zX+z'Y)$ is holomorphic and bounded by $M_{\mathsf a}$ on $|z||X|_\infty<\frac16\rho_{\mathsf a}$, $|z'||Y|_\infty<\frac13\rho_{\mathsf a}$, so by Cauchy's estimate $|D^3\mathsf a(\tau X)(X,Y,Y)|\le 108M_{\mathsf a}\rho_{\mathsf a}^{-3}|X|_\infty|Y|_\infty^2$, and integrating in τ: $|D^2\mathsf a(X)(Y,Y)-D^2\mathsf a(0)(Y,Y)|\le 216M_{\mathsf a}\Xi_N\epsilon\rho_{\mathsf a}^{-3}\|Y\|^2$; on the polydisc of the radii $\frac14\rho_{\mathsf a}$, $|D^2\mathsf a(\tau X)(X,Y)|\le 16M_{\mathsf a}\rho_{\mathsf a}^{-2}|X|_\infty|Y|_\infty$, and with (a2) $|D\mathsf a(X)Y|\le 32M_{\mathsf a}\Xi_N\epsilon\rho_{\mathsf a}^{-2}\|Y\|$. By (s), $|D\mathsf s(X)Y|\le M_{\mathsf s}\rho_{\mathsf s}^{-1}|Y|_\infty$ and $|D^2\mathsf s(X)(Y,Y)|\le 2M_{\mathsf s}\rho_{\mathsf s}^{-2}|Y|_\infty^2$. With (a3) and $\|dY\|^2\ge\lambda_*\|Y\|^2$ on $\mathscr H$ this is (a)(i).

(c) With $\mathsf v=N^3$, $\Xi_N=\frac{\sqrt3}{2}N^{5/2}$, $\lambda_*\ge 4N^{-2}$ and $x\varepsilon(x)^2=A_0^2(\log x)^{2\mathfrak p_0}$, it suffices that: (1) $36\pi N^6\epsilon\le1$; (2) $1800N^5\epsilon\le1$; (3) $4\Xi_N\epsilon\le\rho_{\mathsf a}$; (4) $216C_{\mathsf a}N^3\Xi_N\epsilon\rho_{\mathsf a}^{-1}\le\frac12 c_{\mathsf a}N^{-2}$; (5) $2E_{\mathsf s}N^3\vartheta_{\mathsf s}^{-2}\varepsilon(x)^{-2}\le\frac12 c_{\mathsf a}xN^{-2}$; (6) $1250N^5+72\Xi_N(32xC_{\mathsf a}N^3\Xi_N\epsilon+E_{\mathsf s}N^3\vartheta_{\mathsf s}^{-1}\varepsilon(x)^{-1}+63N^4)+84N^3\epsilon^{-1}\le c_{\mathsf a}xN^{-2}$; for then $\varrho\ge xc_{\mathsf a}\lambda_*(1-\frac18-\frac18-\frac14)$. Divided by x where x stands on the right, the left members of (1)–(4) and (6) are at most a number times $N^{10}x^{-1/2}(\log x)^{\mathfrak p_0}$, which tends to 0 as $N\le C_N(\log x)^{r_N}$; (5) reads $4E_{\mathsf s}N^5\le c_{\mathsf a}\vartheta_{\mathsf s}^2A_0^2(\log x)^{2\mathfrak p_0}$, which holds for large x as $5r_N<2\mathfrak p_0$. Units: $\varrho/a_n\ge 2c_{\mathsf a}x/(N^2a_n)=2c_{\mathsf a}\mathsf b/l^2$, and $\frac{2.13}{\varrho}a_n\le\frac{1.07}{c_{\mathsf a}}\frac{l^2}{\mathsf b}$.

(d) Since ν is gauge invariant, Proposition 18.71(d) shows that, with respect to $\nu/\nu(1)$, the variable g is independent of X and, modulo $\pm$, Haar distributed; the function $\phi(g_0):=\langle\sigma_3,\mathrm{Ad}(g_0)\sigma_3\rangle$ on $G=S^3$ is a matrix coefficient of the spin-one representation, and it has mean 0, $\int\phi^2=\frac13$ and $-\Delta\phi=8\phi$ (the Laplacian of S^3 has the eigenvalues $j(j+2)$), whence $\int|\nabla\phi|^2=\frac83$. Every tangent vector ξ at $V=\Upsilon_\varsigma(g,X)$ has the form $\xi=\mathrm{Ad}_g\mathbb A(X)(\mathrm{Ad}(g)^{-1}\lambda',Y)$, where $\|(\lambda',Y)\|\le\|(1+\mathfrak e)^{-1}\|\|\mathbb A(0)^{-1}\|\|\xi\|\le N\|\xi\|$, and $\langle\xi,\partial h(V)\rangle$ equals the derivative of ϕ at the point $g(x_0)$ in the direction $\lambda'(x_0)$; it follows that $\int\|\partial h\|^2d\nu\le N^2\frac83\nu(1)=8N^2\mathrm{Var}_\nu(h)$. As the flip takes $\Upsilon_\varsigma(g,X)$ to $\Upsilon_{-\varsigma}(g,X)$, the function h is even; besides, $\mathrm{Ad}(-g)=\mathrm{Ad}(g)$. □

Proposition 18.73 (An entropy tower)**.** Let $t\ge1$. For $0\le k\le t$ let $\mathcal U_k$ be an open subset of a product of copies of G indexed by the bonds of a lattice, with tangent vectors and gradients as above, m_k a finite positive Borel measure on $\mathcal U_k$, $\mathsf n_k(V;\cdot)$ seminorms on the tangent vectors at V, and $\hat\Gamma_k(g)(V):=\sup\{\langle\xi,\partial g(V)\rangle^2:\mathsf n_k(V;\xi)\le1\}$, Borel in V. For each $k<t$ and each $W\in\mathcal U_{k+1}$, let an open, bounded, convex set $\mathrm{Rg}_k\subset\mathbb R^{\dim_k}$, a map $\tilde V_W:\mathrm{Rg}_k\to\mathcal U_k$ of class C^1 and a probability measure $\tilde p_W$ on Rg_k be given, subject to: (T1) $\int h\,dm_k=\int[\int h(\tilde V_W(B))\tilde p_W(dB)]\,m_{k+1}(dW)$ for every Borel $h\ge0$; (T2) $\mathrm{Ent}_{\tilde p_W}(\phi^2)\le\frac{2}{\kappa_k}\int|\nabla\phi|^2d\tilde p_W$ for every bounded $\phi\in C^1(\mathrm{Rg}_k)$; (T3) $\mathsf n_k(\tilde V_W(B);D\tilde V_W(B)\zeta)\le\mathsf l_k|\zeta|$. Fix a bounded $f_0\in C^1(\mathcal U_0)$ with $\inf f_0>0$, and let its tower $f_{k+1}(W):=\int f_k(\tilde V_W(B))\,\tilde p_W(dB)$ meet the following conditions for all $k<t$, all $W\in\mathcal U_{k+1}$ and every tangent vector ξ at W: (S0) $f_{k+1}\in C^1(\mathcal U_{k+1})$, and for some Borel field $B\mapsto\dot V_B$ of tangent vectors at $\tilde V_W(B)$ and some $\psi_\xi\in C^1(\mathrm{Rg}_k)$ one has $\langle\xi,\partial f_{k+1}(W)\rangle=\int\langle\dot V_B,\partial f_k(\tilde V_W(B))\rangle\,\tilde p_W(dB)+\mathrm{Cov}_{\tilde p_W}(f_k\circ\tilde V_W,\psi_\xi)$; (S1a) $\mathsf n_k(\tilde V_W(B);\dot V_B)\le(1+\eta'_k)\,\mathsf n_{k+1}(W;\xi)$; (S1b) $|\nabla_B\psi_\xi|\le\iota_k\,\mathsf n_{k+1}(W;\xi)$. Put $\eta_k:=\eta'_k+\iota_k\mathsf l_k/\kappa_k$ and $\mathcal I_k:=\int\hat\Gamma_k(f_k)f_k^{-1}dm_k$. Then $\mathcal I_{k+1}\le(1+\eta_k)^2\mathcal I_k$, and $\mathrm{Ent}_{m_0}(f_0)=\sum_{k<t}\int\mathrm{Ent}_{\tilde p_W}(f_k\circ\tilde V_W)\,m_{k+1}(dW)+\mathrm{Ent}_{m_t}(f_t)\le\sum_{k<t}\frac{\mathsf l_k^2}{2\kappa_k}\mathcal I_k+\mathrm{Ent}_{m_t}(f_t)$. ■

Sketch. The identity. By (T1) at $h=f_k\log f_k$: $\int f_k\log f_k\,dm_k=\int[\mathrm{Ent}_{\tilde p_W}(f_k\circ\tilde V_W)+f_{k+1}(W)\log f_{k+1}(W)]\,m_{k+1}(dW)$; sum over $k<t$, and use $m_k(f_k)=m_0(f_0)$ and $m_k(1)=m_0(1)$, by (T1) at $h=f_k$ and $h=1$. *One fibre.* (T2) at $\phi=(f_k\circ\tilde V_W)^{1/2}$: $\mathrm{Ent}_{\tilde p_W}(f_k\circ\tilde V_W)\le$

$\frac{1}{2\kappa_k}\int|\nabla_B(f_k\circ\tilde V_W)|^2(f_k\circ\tilde V_W)^{-1}d\tilde p_W$, and $|\nabla_B(f_k\circ\tilde V_W)|\le \mathsf{l}_k\hat\Gamma_k(f_k)^{1/2}\circ\tilde V_W$ by (T3) and the definition of $\hat\Gamma_k$ ($|\langle\eta,\partial g(V)\rangle|\le \mathsf{n}_k(V;\eta)\hat\Gamma_k(g)(V)^{1/2}$); integrate with m_{k+1} and use (T1). *The Fisher information contracts.* Let $\mathcal{I}_k<\infty$; for m_{k+1}-almost every W, $\mathsf{I}(W):=\int\hat\Gamma_k(f_k)f_k^{-1}(\tilde V_W(B))\,\tilde p_W(dB)<\infty$. Fix such a W, a tangent vector ξ and $\mathsf{n}:=\mathsf{n}_{k+1}(W;\xi)$. In (S0) the first term is at most $(1+\eta_k')\mathsf{n}\int\hat\Gamma_k(f_k)^{1/2}d\tilde p_W\le(1+\eta_k')\mathsf{n}\,\mathsf{I}(W)^{1/2}f_{k+1}(W)^{1/2}$, by (S1a) and Cauchy–Schwarz; the second, by (Cov) of the proof of Proposition 18.72 with $\gamma=\kappa_k$, $\ell_\psi=\iota_k\mathsf{n}$ by (S1b), $F=f_k\circ\tilde V_W$ and the bound of $|\nabla_BF|$ above, is at most $\frac{\iota_k\mathsf{n}}{\kappa_k}\mathsf{l}_k(f_{k+1}(W)\mathsf{I}(W))^{1/2}$. So $\langle\xi,\partial f_{k+1}(W)\rangle^2\le(1+\eta_k)^2\mathsf{n}^2f_{k+1}(W)\mathsf{I}(W)$, that is $\hat\Gamma_{k+1}(f_{k+1})/f_{k+1}\le(1+\eta_k)^2\mathsf{I}$ at W; integrate with m_{k+1} and use (T1). □

Proposition 18.74 (The twisted cube: the chain measure, the Wilson measure, the dynamics)**.** Let $N=lL^n$, $x=L^n\mathsf{b}$, $\epsilon=2\varepsilon(x)$; assume (D1), (D2), and let the hypotheses of Proposition 18.72(c) hold with $x\ge x_\sharp$. Put $C_{\mathrm{LS}}:=\frac{\Pi_{\mathrm{tw}}}{c_\flat}\big[\frac{2C_{\mathrm{ch}}^{\mathrm{tw}}}{\gamma_\flat(L^2-1)N^2}+\frac{1.07}{c_{\mathsf{a}}}\big]$ and $C_{\mathrm{W}}:=\frac12C_{\mathrm{LS}}\,l^2/\mathsf{b}$, numbers free of K.

(a) For all $K>n$ and all real gauge-invariant even $G\in C^\infty(\mathbf{\Omega}_K)$, with $\mathsf{f}_{\mathrm{top}}$ the top of the tower of G^2: $\mathrm{Ent}_{\tilde\mu_K}(G^2)\le\frac{2\Pi_{\mathrm{tw}}C_{\mathrm{ch}}^{\mathrm{tw}}}{c_\flat\gamma_\flat(L^2-1)}\frac{a_n^2}{\mathsf{b}}\tilde{\mathcal{E}}_K(G)+\mathrm{Ent}_{\tilde\mu_{K,[n]}}(\mathsf{f}_{\mathrm{top}})\le C_{\mathrm{LS}}\frac{l^2}{\mathsf{b}}\tilde{\mathcal{E}}_K(G)\le C_{\mathrm{LS}}\frac{l^2}{\mathsf{b}}\mathcal{E}_K^{\mathrm{tw}}(G)$.

(b) For the same G: $\mathrm{Var}_{\mu_K^{\mathrm{tw}}}(G)\le C_{\mathrm{W}}\,\mathcal{E}_K^{\mathrm{tw}}(G)+\mathsf{R}_{\mathrm{tw}}\,\mathrm{osc}(G)^2$, where $\mathsf{S}_G=\sup G^2$. If $c=\tilde\mu_K(G^2)/\tilde\mu_K(1)>0$, then $\mathrm{Ent}_{\mu_K^{\mathrm{tw}}}(G^2)\le C_{\mathrm{LS}}\frac{l^2}{\mathsf{b}}\mathcal{E}_K^{\mathrm{tw}}(G)+\mathsf{R}_{\mathrm{tw}}\mathsf{S}_G(1+\log\frac{\mathsf{S}_G}{c})$. If $c=0$, write $m_G=\mu_K^{\mathrm{tw}}(G^2)\le\mathsf{R}_{\mathrm{tw}}\mathsf{S}_G$. The entropy is at most $m_G\log(\mathsf{S}_G/m_G)$ when $m_G>0$ and is zero when $m_G=0$.

(c) For all $K>n$, $u\ge0$ and all bounded real Borel gauge-invariant even H on $\mathbf{\Omega}_K$: $(H,(1-P_u^{\mathrm{tw}})H)_{L^2(\mu_K^{\mathrm{tw}})}\ge(1-e^{-u/C_{\mathrm{W}}})[\mathrm{Var}_{\mu_K^{\mathrm{tw}}}(H)-\mathsf{R}_{\mathrm{tw}}\,\mathrm{osc}(H)^2]$ and $\|P_u^{\mathrm{tw}}H-\int H\,d\mu_K^{\mathrm{tw}}\|^2_{L^2(\mu_K^{\mathrm{tw}})}\le e^{-2u/C_{\mathrm{W}}}\mathrm{Var}_{\mu_K^{\mathrm{tw}}}(H)+\mathsf{R}_{\mathrm{tw}}\,\mathrm{osc}(H)^2$.

(d) A finite set $\mathcal{F}_{\mathrm{tw}}$ of even, gauge-invariant Borel functions $\phi:\mathbf{\Omega}_n\to\{0,1\}$ can be chosen independently of K so that each $K>n$ admits a $\phi\in\mathcal{F}_{\mathrm{tw}}$ with $\mathrm{Var}_{\mu_K^{\mathrm{tw}}}(\phi(V_n))\ge0.24(1-\mathsf{R}_{\mathrm{tw}})$. Consequently, when $\mathsf{R}_{\mathrm{tw}}<0.19$, along every sequence $K_j\to\infty$ one can pass to a subsequence and pick one $\phi\in\mathcal{F}_{\mathrm{tw}}$ such that the stationary Langevin process (U_u) at the cutoff K_j satisfies $\liminf_j\frac12\mathbb{E}|\phi(V_n(U_u))-\phi(V_n(U_0))|^2\ge(1-e^{-u/C_{\mathrm{W}}})(0.24-1.24\mathsf{R}_{\mathrm{tw}})>0$ for every $u>0$. In particular, if along a sequence $K_j\to\infty$ the laws of the pairs $(F(V_n(U_0)),F(V_n(U_u)))$ converge for all bounded Borel F on $\mathbf{\Omega}_n$ and all $u\ge0$, then for the ϕ found for that sequence the limit law of $(\phi(V_n(U_0)),\phi(V_n(U_u)))$ gives positive mass to $\{(a,b):a\ne b\}$, for every $u>0$.

■

Sketch. (a) Proposition 18.73 with $t=K-n$, $\mathcal{U}_k=\mathrm{Cut}_{K-k}^{\mathrm{tw}}$, $m_k=\tilde\mu_{K,[K-k]}$ and the seminorms of (D2): (T1) is (D1.3) at $\phi=1$ with monotone limits; (T2) is (LC) of the proof of Proposition 18.72 with $\kappa_k=\gamma_\flat$; (T3) follows from (N) and (D1.3): $\mathsf{n}_d(\tilde V_W(B);D\tilde V_W(B)\zeta)^2\le c_\flat^{-1}a_dC_{\mathrm{ch}}^{\mathrm{tw}}x_d^{-1}|\zeta|^2=\frac{C_{\mathrm{ch}}^{\mathrm{tw}}}{c_\flat}\frac{a_d^2}{\mathsf{b}}|\zeta|^2=:\mathsf{l}_d^2|\zeta|^2$, so that the η_d of (Σ) are those of Proposition 18.73. Let $\delta>0$ and $f_0=G^2+\delta$; its tower is that of G^2 plus δ, and $\hat\Gamma_K=\Gamma_K$ on gauge-invariant functions; $\mathcal{I}_0=\int4G^2\Gamma_K(G)(G^2+\delta)^{-1}d\tilde\mu_K\le4\tilde{\mathcal{E}}_K(G)$, so $\mathcal{I}_k\le4\Pi_{\mathrm{tw}}\tilde{\mathcal{E}}_K(G)$ for all k. The fibres contribute at most $\sum_{d>n}\frac{\mathsf{l}_d^2}{2\gamma_\flat}4\Pi_{\mathrm{tw}}\tilde{\mathcal{E}}_K(G)$ with $\sum_{d>n}a_d^2=\frac{a_n^2}{L^2-1}$. The top $f_t=\mathsf{f}_{\mathrm{top}}+\delta$: $f_t^{1/2}$ is bounded, C^1 on $\mathrm{Cut}_n^{\mathrm{tw}}\supset\mathrm{In}(\epsilon)$ by (S0), as $f_t\ge\delta$, gauge invariant and even by (D1.4), and $\tilde\mu_{K,[n]}$ is a measure ν as in Proposition 18.72(c) by (D1.5), the number $\mathsf{c}_K^{\mathrm{tw}}$ dropping out of both members; so $\mathrm{Ent}_{\tilde\mu_{K,[n]}}(f_t)\le\frac{1.07}{c_{\mathsf{a}}}\frac{l^2}{\mathsf{b}}\int\Gamma_n(f_t^{1/2})\,d\tilde\mu_{K,[n]}$, and $\Gamma_n(f_t^{1/2})=\Gamma_n(f_t)/(4f_t)\le\hat\Gamma_n(f_t)/(4c_\flat f_t)$ by (N), of the integral at most $\mathcal{I}_t/(4c_\flat)\le\Pi_{\mathrm{tw}}\tilde{\mathcal{E}}_K(G)/c_\flat$. With $a_n=l/N$ this gives the first two inequalities for $G^2+\delta$ and its top f_t; let $\delta\downarrow0$, the entropies of bounded functions being continuous in δ. The last one is (D1.2) at the gauge-invariant $\Gamma_K(G)$.

(b) (a) at $1+sG$: $\mathrm{Ent}_{\tilde\mu_K}((1+sG)^2) = 2s^2\int (G-c)^2 d\tilde\mu_K + O(s^3)$ with $c = \tilde\mu_K(G)/\tilde\mu_K(1)$, and $\tilde{\mathcal{E}}_K(1+sG) = s^2\tilde{\mathcal{E}}_K(G)$; so $\int (G-c)^2 d\tilde\mu_K \le C_{\mathrm{W}}\tilde{\mathcal{E}}_K(G)$, and (D1.2) at $F=(G-c)^2$, with $\inf G \le c \le \sup G$, gives the first inequality. For the entropy bound when $c = \tilde\mu_K(G^2)/\tilde\mu_K(1) > 0$, $\mathrm{Ent}_{\mu_K^{\mathrm{tw}}}(G^2) \le \int \Xi(G^2,c)\, d\mu_K^{\mathrm{tw}} \le \mathrm{Ent}_{\tilde\mu_K}(G^2) + \mathsf{R}_{\mathrm{tw}} \sup \Xi(G^2,c)$ by (D1.2), and $\Xi(\cdot,c)$ is convex with $0 \le G^2 \le \mathsf{S}_G$, $c \le \mathsf{S}_G$, so $\sup \Xi(G^2,c) \le \mathsf{S}_G(1+\log \frac{\mathsf{S}_G}{c})$. If $c=0$, (D1.2) at $F=G^2$ gives $m_G \le \mathsf{R}_{\mathrm{tw}}\mathsf{S}_G$. When $m_G > 0$, $\int G^2 \log G^2\, d\mu_K^{\mathrm{tw}} \le m_G \log \mathsf{S}_G$, and subtracting $m_G \log m_G$ gives the stated bound. When $m_G = 0$ the entropy is zero.

(c) For $s>0$, $H_s := P_s^{\mathrm{tw}} H$ is smooth, gauge invariant and even, with $\mathrm{osc}(H_s) \le o := \mathrm{osc}(H)$. $y(s) := \mathrm{Var}(H_s) - \mathsf{R}_{\mathrm{tw}} o^2$ has $y' = -2\mathcal{E}_K^{\mathrm{tw}}(H_s) \le -\frac{2}{C_{\mathrm{W}}} y$ by (b); so $y(s) \le e^{-2(s-s_0)/C_{\mathrm{W}}} y(s_0)$ whatever the sign, and $s_0 \downarrow 0$ gives $\mathrm{Var}(P_u^{\mathrm{tw}} H) \le e^{-2u/C_{\mathrm{W}}}\mathrm{Var}(H) + (1-e^{-2u/C_{\mathrm{W}}})\mathsf{R}_{\mathrm{tw}} o^2$, the second inequality, since $\int P_u^{\mathrm{tw}} H\, d\mu_K^{\mathrm{tw}} = \int H\, d\mu_K^{\mathrm{tw}}$, the probability measure μ_K^{tw} being invariant for the dynamics; and $(H, (1-P_u^{\mathrm{tw}})H) = \mathrm{Var}(H) - \mathrm{Var}(P_{u/2}^{\mathrm{tw}} H)$ gives the first.

(d) Fix a unit vector $e \in \mathscr{H}$ of the top lattice. By Proposition 18.71(d), $V = \Upsilon_\varsigma(g,X) \mapsto X =: X(V)$ is a continuous, gauge-invariant, even map $\mathrm{In}(\epsilon) \to \mathcal{D}(\epsilon)$; put $\phi_\upsilon := \mathbf{1}\{V \in \mathrm{In}(\epsilon),\ \langle e, X(V)\rangle \le \upsilon\}$ for real υ. With $\hat\mu := \tilde\mu_{K,[n]}/\tilde\mu_K(1)$ and Proposition 18.72(a) (here $w_0 = 0$): $\varphi_K(\upsilon) := \hat\mu(\phi_\upsilon) = \int_{\mathcal{D}(\epsilon)\cap\{\langle e,X\rangle \le \upsilon\}} e^{-W} dX / \int_{\mathcal{D}(\epsilon)} e^{-W} dX$. $\mathcal{D}(\epsilon)$ is star-shaped, $\|X\| < 2\Xi_N\epsilon$ and $\|DW\| \le \mathsf{L}$ there, so $\mathrm{osc}\, W \le 4\Xi_N\epsilon\mathsf{L} =: \mathsf{O}$, a number free of K, the normalised density is at most $e^{\mathsf{O}}|\mathcal{D}(\epsilon)|^{-1}$, and $0 \le \varphi_K(\upsilon') - \varphi_K(\upsilon) \le \mathsf{L}'(\upsilon'-\upsilon)$ with $\mathsf{L}' := e^{\mathsf{O}}|\mathcal{D}(\epsilon)|^{-1} \sup_\upsilon \mathrm{vol}_{6\mathsf{v}-1}(\mathcal{D}(\epsilon) \cap \{\langle e, \cdot\rangle = \upsilon\}) < \infty$, free of K; $\varphi_K(-2\Xi_N\epsilon) = 0$ and $\varphi_K(2\Xi_N\epsilon) = 1$. On the grid $\upsilon_j = -2\Xi_N\epsilon + 0.1j/\mathsf{L}'$ there is a j with $\varphi_K(\upsilon_j) \le \frac12 < \varphi_K(\upsilon_{j+1})$, so $\varphi_K(\upsilon_j) \in [0.4, 0.5]$ and $\mathrm{Var}_{\hat\mu}(\phi_{\upsilon_j}) \ge 0.24$; $\mathcal{F}_{\mathrm{tw}}$ is the set of the ϕ_{υ_j} with $|\upsilon_j| \le 2\Xi_N\epsilon$. By (D1.1), $\phi(V_n)$ is gauge invariant and even on $\mathbf{\Omega}_K$; by (D1.2), $\mathrm{Var}_{\mu_K^{\mathrm{tw}}}(\phi(V_n)) = \min_c \int (\phi(V_n)-c)^2 d\mu_K^{\mathrm{tw}} \ge \min_c \int (\phi - c)^2 d\tilde\mu_{K,[n]} = \tilde\mu_K(1)\mathrm{Var}_{\hat\mu}(\phi)$, and $\tilde\mu_K(1) \ge 1 - \mathsf{R}_{\mathrm{tw}}$. Along a sequence one $\phi \in \mathcal{F}_{\mathrm{tw}}$ serves infinitely often; (c) with osc $= 1$, and $\frac12 \mathbb{E}|\phi(V_n(U_u)) - \phi(V_n(U_0))|^2 = (H, (1-P_u^{\mathrm{tw}})H)$ for $H = \phi(V_n)$ and the stationary process. As ϕ takes the values 0 and 1, this mean is half the mass that the law of the pair gives to $\{a \ne b\}$, and a law on the four points of $\{0,1\}^2$ converges point by point; so, where the laws of the pairs converge along the whole sequence, the limit of this mass is its limit along the subsequence. □

18.8. **The measure on the threads of one torus: the one invariant measure with the limits of the lattice means of the block algebra, the continuous translations on the gauge-invariant functions, and the local semigroup at a pair of mirrors.** This subsection adds three groups of statements to Proposition 10.2(a). Proposition 18.75 holds on every side $\ell = 2L^m$ and at every label $\mathsf{b} > 0$: the laws of the block fields converge in total variation; their images under $V \mapsto (\mathcal{M}^D V, \dots, \mathcal{M}V, V)$ are the finite pieces of exactly one Borel probability measure $\mu_{\ell,\mathsf{b}}$ on $\mathcal{X}$, which is carried by the threads; the joint laws of the block fields of the depths $0, \dots, D$ converge to its images in total variation; it is invariant under every θ_v; the lattice means of every element of the block algebra converge to its integral against $\mu_{\ell,\mathsf{b}}$; and among the Borel probability measures on $\mathcal{X}$ that are invariant under all the θ_v, these integrals single out $\mu_{\ell,\mathsf{b}}$. Proposition 18.76 is set in the one-torus setting. In its part (A), shifting the threads through vectors of $\mathbb{Q}_L^3$, $\mathbb{Q}_L = \bigcup_s a_s\mathbb{Z}$, gives Borel bijections that preserve μ; restricted to $\mathcal{K}^{\mathrm{inv}}$, the closed linear hull in $L^2(\mu)$ of the gauge-invariant functions of block fields, these shifts continue in a unique way to a strongly continuous unitary representation of $\mathbb{R}^3/\ell\mathbb{Z}^3$, and $\mathcal{K}^{\mathrm{inv}}$ splits orthogonally into the joint eigenspaces of the three momenta. In its part (B) we fix a coordinate mirror of T_ℓ, i.e. the two planes $x_\mu = c$ and $x_\mu = c + \frac{\ell}{2}$; on the functions of the block variables strictly between them the reflection form is non-negative; shifts normal to the planes act continuously for this form, with explicit moduli when the functions stay away from both planes; a shifted class vanishes only when the class itself vanishes; the classes of functions that stay away from the planes are dense; and a single self-adjoint operator $\mathfrak{h}_c$ has a semigroup that reproduces the shifts on these classes, so that the shifts make up a local semigroup with one generator. The continuity in (A) rests on Lemma 18.93. Together with a depth D_1 formed

from the constants and b, set below, it supplies numbers s_1 and C_A, formed from the constants, ℓ and b, with this property: for $s \geq s_1$, when the field of the depth s is shifted by one spacing a_s and transported back along the shifted bonds, the block field of a depth d with $D_1 \leq d < \lfloor s/5 \rfloor$ moves at a given bond by more than ρ, for any $\rho \in [L^{-s/6}, 2]$, on the threads whose plaquettes are small from the depth d to the depth s, with probability at most $\mathsf{C}_A \rho^{-1} L^{-s/21}$. There, the displacement of a block bond at the depth $j = \lfloor s/5 \rfloor + \lfloor \frac{1}{2} \lfloor s/5 \rfloor \rfloor$ is controlled by the plaquette moduli; the displacement at the depth $\lfloor s/5 \rfloor$ is, up to small errors, an average over columns of such displacements; and the average over one column is smaller by a power of the spacing, which comes from the reflection positivity of the Wilson measure at mirrors of both types, used for quantities with one adjoint charge carried along straight bare segments. Iterating the block averaging map is Lipschitz near small fields, and this lets one compose the shifts of all the finer spacings.

Further signs. We keep the setting and the labels of Section 10.1, and its signs wherever they are not set anew; the one-torus setting is the one that Section sets, with its constants. Several of its letters are set anew here for other objects, each at the place where it is used: among them are k, κ, N, γ and r, the letter m apart from the side $\ell = 2L^m$, and c', which also names bonds. Where one letter carries two meanings, its arguments tell them apart: with a depth alone as index, π_d is the coordinate of $\mathcal{X}$, whereas $\pi_i(b)$, with a bond as argument, is the imaginary part of the defect set below; with two sites as arguments, $\mathcal{G}(y,x)$ is a contour family, whereas $\mathcal{G}$ and $\mathcal{G}_i$, without arguments, are events, set in Lemmas 18.91 and 18.92 and in the proof of the latter. $\mathbb{Q}_L = \bigcup_{s \geq 0} a_s \mathbb{Z}$; as L is odd, $\frac{1}{2} \notin \mathbb{Q}_L$ and $\frac{1}{2}\mathbb{Q}_L \setminus \mathbb{Q}_L = \bigcup_s a_s(\mathbb{Z} + \frac{1}{2})$. For $y \in \Lambda_j$ and $n \geq 0$, the block of order n is

$$B^n(y) = \left\{ y + a_{j+n} \sum_{\mu=1}^{3} r_\mu e_\mu : r_\mu \in \mathbb{Z}, \quad |r_\mu| \leq (L^n - 1)/2 \right\}, \qquad B(y) = B^1(y).$$

The blocks partition the fine lattice and satisfy $B^{n+n'}(y) = \bigsqcup_{x \in B^n(y)} B^{n'}(x)$. Both facts follow by successive integer division with the centered remainder. A fine bond either stays in one such block or crosses to the adjacent block in its own direction. In the latter case it lies over the coarse bond joining the two block centers. Let c be a bond of Λ_j. We write P_c for the set of those bonds of Λ_{j+1} whose two end points lie in $B(c_-) \cup B(c_+)$, $\mathcal{G}(y,x)$ for the paths from $y \in \{c_-, c_+\}$ to $x \in B(y)$ that traverse each nonzero coordinate displacement in one straight segment, with all orders of those segments allowed, $[x, x']$ for the straight path leading from $x \in B(c_-)$ to $x' = x + c_+ - c_-$, $V[c]$ for the transport of a field V of Λ_{j+1} along the straight path leading from c_- to c_+, and $V(y,x)$, where it is defined, for the unique logarithmic group mean of $\{V(\Gamma) : \Gamma \in \mathcal{G}(y,x)\}$ in YM3_I:(31). The averaging radius is $\omega_M = 1/(200L^2)$ by YM3_I:(30). $\mathfrak{g}$ denotes the hermitian traceless 2×2 matrices, a real vector space carrying $\langle X, Y \rangle = \frac{1}{2} \operatorname{tr} XY$, so that $\langle X, X \rangle = |X|^2$; ${}^w X = wXw^{-1}$ for $w \in G$. For a complex 2×2 matrix Y put $\operatorname{Im} Y = \frac{1}{2i}(Y - Y^*)$, and for $w \in G$ put $\operatorname{Re} w = \frac{1}{2} \operatorname{tr} w \in [-1, 1]$; then $w = (\operatorname{Re} w)\mathbf{1} + i \operatorname{Im} w$, $\operatorname{Im} w \in \mathfrak{g}$ and $|w - \mathbf{1}|^2 = 2 - 2 \operatorname{Re} w$. By a 1-form we mean a function A of the bonds of a lattice, real or $\mathfrak{g}$-valued, with $A(\bar{b}) = -A(b)$; $A(\Gamma)$ adds its values along a path Γ, $\sup |A| = \max_b |A(b)|$, $(\mathrm{d}\lambda)(b) = \lambda(b_+) - \lambda(b_-)$ for a function λ on the sites, and $\mathrm{d}A(p) = A(\partial p)$ for a plaquette p; e^{iA} is the field $b \mapsto e^{iA(b)}$. For two fields W, W' of one lattice, $\operatorname{dist}_{\mathrm{orb}}(W', W) = \min_v \max_b |W'(b) - W^v(b)|$, the minimum over the gauge transformations v; it is symmetric, and the triangle inequality holds for it since multiplying by elements of G does not change the norm.

Block observables and symmetries. The block algebra consists of bounded continuous functions of finitely many bond variables from depths $0, \dots, s$ which are invariant under the joint action of any gauge transformation of Λ_s. At each smaller depth the gauge transformation is restricted to that depth. For such a function F, the notation $F \circ \mathcal{M}^{K-s}$ means evaluation of F on all the block fields it uses at the cutoff K.

Let Iso_i be the group of affine isometries of the torus that preserve Λ_i and its nearest-neighbor bonds. Their linear parts are signed coordinate permutations, and their translation vectors lie in $a_i\mathbb{Z}^3$ modulo the period. Thus $\mathsf{Iso}_i \subset \mathsf{Iso}_{i'}$ for $i \le i'$. We use the action $(rV)(b) = V(r^{-1}b)$ on oriented bonds. Isometries of the coarse lattice carry its centered blocks, contour families and straight paths onto the corresponding objects of the image bond. In the three real rules preceding Proposition YM3_I:4.5 a positively oriented image preserves the formula by relabeling these objects. If the image bond is reversed, set $P = V[c]$ and write the transport from c_- to c_+ through the two blocks as T, so that the relative loop is $W = TP^{-1}$. This applies to both averaged-contour and individual-contour transports. Reversal replaces T, P by T^{-1}, P^{-1} and W by $T^{-1}P = P^{-1}W^{-1}P$. It preserves the tests for each rule, since the operator norm is unitarily invariant and the exceptional point $-\mathbf{1}$ is central. Both logarithms send the new loop to $-P^{-1}(\log W)P$. Therefore the exponential of the mean followed by P^{-1} is the inverse of the old output. The rule with just P also transforms by inversion. This proves $\mathcal{M}(rV) = r\mathcal{M}(V)$ for a coarse-lattice isometry, and iteration gives the same identity for any compatible depths. The Wilson law at a finite cutoff is invariant under its lattice isometries. Indeed, the action is preserved by relabeling plaquettes, conjugating their holonomies or replacing them by their inverses, and the product Haar measure is preserved by relabeling or reversing bonds.

Numbers. $\varepsilon_{[i]} = \varepsilon(x_{[i]}) = A_0(L^i\mathsf{b})^{-1/2}(\log L^i\mathsf{b})^{\mathfrak{p}_0}$. Whenever $L^i\mathsf{b} > 1$, both $\varepsilon_{[i]}$ and $\varepsilon_{[i+1]}$ are defined, $0 < \log L^i\mathsf{b} < \log L^{i+1}\mathsf{b}$, and $\varepsilon_{[i]} \le L^{1/2}\varepsilon_{[i+1]}$ because $(\log L^i\mathsf{b})^{\mathfrak{p}_0} \le (\log L^{i+1}\mathsf{b})^{\mathfrak{p}_0}$; and if $\log(L^i\mathsf{b}) \ge 8\mathfrak{p}_0$, then $\varepsilon_{[i+1]}/\varepsilon_{[i]} = L^{-1/2}(1 + \frac{\log L}{\log L^i\mathsf{b}})^{\mathfrak{p}_0} \le L^{-1/2}e^{(\log L)/8} = L^{-3/8}$, so that $\sum_{i'>i}\varepsilon_{[i']} \le 0.62\,\varepsilon_{[i]}$ for $L \ge 13$. In the one-torus setting, $R_d \le \mathsf{A}_R L^{3d} e^{-a_R d^2}$ with $\mathsf{A}_R = 2\ell^3 M^{-3}\exp[(\mathsf{C}^{\Sigma}_{\mathrm{maj}} + 2 + c_{\mathrm{low}} + \frac{9}{2}\log\mathsf{b})\ell^3]$ and $a_R = c'A_0^2(\log L)^2$, because $(\log x_{[d]})^{2\mathfrak{p}_0} \ge (d\log L)^2$; d_Σ is the least integer $d \ge d_\natural$ with $c'A_0^2\log x_{[d]} \ge 2$, so that $R_{d+1} \le L^{-1}R_d$ and $\sum_{d'\ge d} R_{d'} \le \frac{L}{L-1}R_d$ for $d \ge d_\Sigma$ (Proposition 10.2(b) and its proof). For integers $D \le s$, $\mathsf{Gd}_{D,s}$ is the set of the $V \in \mathcal{V}_s$ with $\mathcal{M}^{s-i}V \in I_i$ for $D \le i \le s$, a Borel set, and Gd_D is the set of the threads $V_\bullet$ with $V_i \in I_i$ for all $i \ge D$.

Translations and reflections of fields and of threads. For $h \in a_s\mathbb{Z}^3$ and a field V of a depth $j \ge s$ put $(\mathrm{t}_h V)(b) = V(b-h)$; the translation by h lies in Iso_s. An element of Iso_j is called diagonal when its linear part is a diagonal matrix. The diagonal elements of Iso_j form a group; each of them is the composite of a translation through a vector of $a_j\mathbb{Z}^3$ and of a change of sign of some of the coordinates, and among them are the translations through vectors of $a_j\mathbb{Z}^3$ and the coordinate mirrors $x \mapsto x - 2(x_\mu - c)e_\mu$ with $c \in \frac{a_j}{2}\mathbb{Z}$. For $h \in \mathbb{Q}^3_L$, s the smallest non-negative integer with $h \in a_s\mathbb{Z}^3$, and $V_\bullet \in \mathcal{X}$ put $(\mathrm{T}_h V_\bullet)_j = \mathrm{t}_h V_j$ for $j \ge s$ and $(\mathrm{T}_h V_\bullet)_j = \mathcal{M}^{s-j}(\mathrm{t}_h V_s)$ for $j < s$; for a function F on $\mathcal{X}$, or on Thr, put $\mathcal{U}_h F = F \circ \mathrm{T}_{-h}$, which on Thr is defined because T_{-h} maps Thr into itself (Lemma 18.77(b)). In the same way, for a diagonal element r of $\bigcup_{s\ge 0}\mathsf{Iso}_s$, with s now the smallest non-negative integer such that $\mathsf{r} \in \mathsf{Iso}_s$, put $(\mathsf{r}^\bullet V_\bullet)_j = \mathsf{r}V_j$ for $j \ge s$ and $(\mathsf{r}^\bullet V_\bullet)_j = \mathcal{M}^{s-j}(\mathsf{r}V_s)$ for $j < s$. Since $\mathsf{Iso}_s \subset \mathsf{Iso}_{s'}$ for $s \le s'$, these formulas make sense with any integer $s' \ge s$ in the place of s; on threads the outcome is then unchanged (Lemma 18.77(b)), while elsewhere the choice just made fixes $\mathsf{r}^\bullet$. $\mathcal{K}^{\mathrm{inv}}$ is the closed span in $L^2(\mu)$ of the functions $F \circ \pi_d$, $d \ge 0$, F a bounded gauge-invariant Borel function on $\mathcal{V}_d$.

The covariant shift. Let $\nu \in \{1,2,3\}$, $s \ge 1$, $h = a_s e_\nu$ or $h = -a_s e_\nu$, and $V \in \mathcal{V}_s$; put $V_i = \mathcal{M}^{s-i}V$ and $V'_i = \mathcal{M}^{s-i}(\mathrm{t}_h V)$ for $i \le s$. With $g(x) = V(\langle x-h, x\rangle)$, $x \in \Lambda_s$, the covariantly shifted field is $\check V(b) = g(b_-)^{-1}(\mathrm{t}_h V)(b)\,g(b_+)$; we put $\check V_i = \mathcal{M}^{s-i}\check V$, and for the bonds b of Λ_i, $i \le s$, the defect $\mathsf{P}_i(b) = \check V_i(b)V_i(b)^{-1}$ and $\pi_i(b) = \operatorname{Im}\mathsf{P}_i(b) \in \mathfrak{g}$. Where a thread $V_\bullet$ is given, these objects are formed from its entry V_s, and $V'_{d,s} = \mathcal{M}^{s-d}(\mathrm{t}_h V_s) = (\mathrm{T}_h V_\bullet)_d$ for threads.

Bare lines. At a cutoff $K > s$ the bare field is $U \in \mathcal{V}_K$, $V_i = \mathcal{M}^{K-i}U$, and then $V'_i = \mathcal{M}^{K-i}(\mathrm{t}_h U)$ by block equivariance; $\mathbb{E}$ is the Wilson mean $\langle\cdot\rangle_K$ at any $\beta > 0$, and at $\beta = L^K\mathsf{b}$ when the one-torus setting is assumed; $\|F\|_2 = (\mathbb{E}|F|^2)^{1/2}$ and $\mathbb{E}[F;\mathcal{E}] = \mathbb{E}[F\mathbf{1}_\mathcal{E}]$. The height of a point

is its coordinate x_ν, and a column is the set of the sites of Λ_K with two given coordinates x_κ, $\kappa \neq \nu$. When x, y are sites on a common column with $|x_\nu - y_\nu| < \frac{\ell}{2}$, we write $\mathsf{L}(x,y) = U([x,y])$ for the bare transport along the straight segment from x to y: $\mathsf{L}(x,x) = \mathbf{1}$, $\mathsf{L}(y,x) = \mathsf{L}(x,y)^{-1}$, $\mathsf{L}(x,y)\mathsf{L}(y,w) = \mathsf{L}(x,w)$ for three sites of one column in a window of heights shorter than $\frac{\ell}{2}$, and a gauge transformation v of Λ_K carries $\mathsf{L}(x,y)$ to $v(x)\mathsf{L}(x,y)v(y)^{-1}$. A function S of the bare field with values in G or in $\mathfrak{g}$ is covariant at a site x if $S(U^v) = v(x)S(U)v(x)^{-1}$ for all v. For a positively oriented bond $b \parallel e_\nu$ of Λ_i, $i \le K$, a site x of Λ_i and $n \ge 0$ with $na_i < \frac{\ell}{4}$ put $z_i(b) = V_i(b)\mathsf{L}(b_+, b_-)$, $Z_i(x;n) = V_i([x, x+na_ie_\nu])\,\mathsf{L}(x+na_ie_\nu, x)$ and $\sigma_i(n) = \|Z_i(x;n) - \mathbf{1}\|_2$. $z_i(b)$ and $Z_i(x;n)$ are covariant at b_- and at x; $\sigma_i(n)$ is the same for all x, since translating Λ_K leaves the Wilson measure unchanged and $\mathcal{M}^{K-i}$ commutes with the translations by $a_i\mathbb{Z}^3$ by block equivariance; $\sigma_i(0) = 0$, $\sigma_K \equiv 0$, and $\sigma_i(n)^2 = 2 - 2\,\mathbb{E}\,\mathrm{Re}\,Z_i(x;n)$.

Constants of the shift. $\rho_L = \frac{1}{1600L^2}$, $C_g = 50L^2$, $\mathsf{k}_1 = 600L^3$, $c_{\mathrm{d}} = 2(\sqrt{2}+1)$; C_N is the number of Lemma 18.80(a), $\mathsf{k}_2 = 8C_NL^{-1}(3.9\,C_g\mathsf{k}_1 + 2.8\,\mathsf{k}_1^2)$ and $c_* = \min\{\frac{L(L-1)}{32C_N}, \frac{\rho_L}{4}\}$; all these are formed from L alone. D_g is the least integer $j \ge 0$ with $\log(L^j\mathsf{b}) \ge 8\mathfrak{p}_0$, $C_g\varepsilon_{[j]} \le \frac{\rho_L}{2}$, $\mathsf{k}_1\varepsilon_{[j]} \le \frac{\rho_L}{4}$ and $53\,C_NC_g\varepsilon_{[j]} \le L$; it is formed from the constants and b, and $17000L^3\varepsilon_{[j]} \le 1$ for $j \ge D_g$. $D_1 = \max\{D_g, d_\Sigma, d_\Theta\}$, where d_Θ is the least integer $d \ge 1$ with $x_{[d]} = L^d\mathsf{b} \ge \Theta_\star$, and $\Theta_\star > 1$ is chosen from the fixed constants so that $2c_\rho\varepsilon(x) < \omega_M$ for all $x \ge \Theta_\star$, possible since $x^{-1/2}(\log x)^{\mathfrak{p}_0} \to 0$; so $2c_\rho\varepsilon_{[i]} < \omega_M$ for every $i \ge D_1$. For integers $j \le s$ put $\mathsf{E}_s = L^s\varepsilon_{[s]}^2 = A_0^2\mathsf{b}^{-1}(\log L^s\mathsf{b})^{2\mathfrak{p}_0}$ and $\bar\eta_j = \mathsf{k}_1(2+s-j)\varepsilon_{[s]} + \mathsf{k}_2\mathsf{E}_sL^{-j}$; and, for the integer $j < s$ that the context fixes, $\mathsf{H}_s = \mathsf{k}_2\mathsf{E}_s + \mathsf{k}_1(2+s)L^j\varepsilon_{[s]}$. The diagonal of an integer s: $D = \lfloor s/5 \rfloor$, $\mathfrak{m} = \lfloor D/2 \rfloor$, $j = D + \mathfrak{m}$, $N = L^{\mathfrak{m}}$, $K = 1260s$, $\mathfrak{t} = 31000$; $\mathsf{e} = \mathsf{e}(s) = \max\{1, A_0\mathsf{b}^{-1/2}(\log L^{1260s}\mathsf{b})^{\mathfrak{p}_0}\}$, so that $L^{i/2}\varepsilon_{[i]} \le \mathsf{e}$ for $i \le K$, $\mathsf{E}_s \le \mathsf{e}^2$ and $\mathsf{e} \le C_{\mathsf{e}}s^{\mathfrak{p}_0}$ with C_{e} formed from A_0, $\mathfrak{p}_0$, L and b; on the diagonal, H_s takes the j of the diagonal. We say that a statement holds for large s when it is true for every integer s beyond some threshold formed from the constants, ℓ and b.

Mirrors acting on threads. Let $\mu \in \{1,2,3\}$, $d_0 \ge 0$ an integer and $c \in \frac{a_{d_0}}{2}\mathbb{Z}$; θ is the coordinate mirror $x \mapsto x - 2(x_\mu - c)e_\mu$, a diagonal element of Iso_{d_0}; it fixes the two planes $x_\mu = c$ and $x_\mu = c + \frac{\ell}{2}$, and $\mathbb{E}_+ = \{c < x_\mu < c + \frac{\ell}{2}\}$. It is of the site type if $c \in \mathbb{Q}_L$, and then $c \in a_j\mathbb{Z}$ for every $j \ge d_0$; it is of the bond type otherwise, and then $c \in a_j(\mathbb{Z} + \frac{1}{2})$ for every $j \ge d_0$, as L^{j-d_0} is odd. For $j \ge d_0$ and real $\alpha, \alpha' \ge 0$, $\mathsf{Bd}_j(\alpha,\alpha')$ collects the bonds of Λ_j whose two end points obey $c + \alpha + \frac{1}{2}a_j \le x_\mu \le c + \frac{\ell}{2} - \alpha' - \frac{1}{2}a_j$; $\mathcal{A}_j(\alpha,\alpha')$ consists of the functions $V_\bullet \mapsto \mathsf{f}(V_j|_{\mathsf{Bd}_j(\alpha,\alpha')})$ on Thr, f bounded, complex and Borel, and $\mathcal{A}(\alpha,\alpha') = \bigcup_{j \ge d_0}\mathcal{A}_j(\alpha,\alpha')$. As $\mathcal{M}$ is Borel, Thr is a Borel subset of $\mathcal{X}$, and $\mu(\mathsf{Thr}) = 1$ (Proposition 18.75(2)); so these functions are elements of $L^2(\mu)$, and the maps T_h and $\mathsf{r}^\bullet$, r diagonal, which send Thr into itself (Lemma 18.77(b)), act on them by composition. For functions F on Thr, $\Theta F = F \circ \theta^\bullet$; $\langle F, F'\rangle_c = \int \overline{\Theta F}\,F'\,d\mu$ and $\|F\|_c = \langle F, F\rangle_c^{1/2}$ for $F, F' \in \mathcal{A}(0,0)$; $\mathcal{K}_c$ is the completion of $\mathcal{A}(0,0)$ modulo the vectors F with $\|F\|_c = 0$, and $[F]$ the class of F. For $t \in \mathbb{Q}_L$, $\mathcal{U}_t = \mathcal{U}_{te_\mu}$, and $\varphi_F(t) = \int \overline{\Theta F}\,\mathcal{U}_tF\,d\mu$. The objects of a second mirror (μ, c^*) carry the index c^*: Θ_{c^*}, $\langle\cdot,\cdot\rangle_{c^*}$, $\mathcal{A}^{c^*}(\alpha,\alpha')$. ϱ is the reflection at the plane $x_\mu = c + \frac{\ell}{4}$, and $RF = F \circ \varrho^\bullet$.

Proposition 18.75 (One measure on the threads at every label, and the integrals of the block algebra)**.** Let $m \in \mathbb{N}$, $\ell = 2L^m$ and let $\mathsf{b} > 0$ be real; let $\langle\cdot\rangle_K$ be the Wilson measure on $\mathcal{V}_K$ with the density $Z^{-1}e^{-L^K\mathsf{b}\sum_p a_p}$ for dU, and for $K \ge \max\{d,1\}$ let μ_d^K be the law of $\mathcal{M}^{K-d}U$ under it. Put $\iota_D(V) = (\mathcal{M}^DV, \dots, \mathcal{M}V, V)$ for $V \in \mathcal{V}_D$ and $\pi_{\le D} = (\pi_0, \dots, \pi_D)$.

(1) For every $d \ge 0$ the laws μ_d^K converge in total variation, as $K \to \infty$ along all the integers, to a gauge-invariant Borel probability measure μ_d on $\mathcal{V}_d$, and $\mathcal{M}_*\mu_{d+1} = \mu_d$.

(2) Exactly one Borel probability measure $\mu_{\ell,\mathsf{b}}$ on $\mathcal{X}$ satisfies $(\pi_{\le D})_*\mu_{\ell,\mathsf{b}} = (\iota_D)_*\mu_D$ for all $D \ge 0$; moreover $(\pi_D)_*\mu_{\ell,\mathsf{b}} = \mu_D$ and $\mu_{\ell,\mathsf{b}}(\mathsf{Thr}) = 1$. For every $D \ge 0$ the joint law ν_D^K of

$(\mathcal{M}^K U, \mathcal{M}^{K-1}U, \dots, \mathcal{M}^{K-D}U)$ under $\langle\cdot\rangle_K$ converges to the image of $\mu_{\ell,\mathsf{b}}$ under $\pi_{\le D}$ in total variation as $K\to\infty$. In the one-torus setting $\mu_{\ell,\mathsf{b}}$ is the measure μ of Proposition 10.2(a).

(3) For every map $v\colon \bigcup_d \Lambda_d \to G$, θ_v is a homeomorphism of $\mathcal{X}$ that carries Thr onto itself and $\mu_{\ell,\mathsf{b}}$ to itself.

(4) Let F belong to the block algebra, with s the largest depth occurring in it, and regard F as a function on $\mathcal{V}_0\times\cdots\times\mathcal{V}_s$ of the variables $V_{s'}(b)$, $s'\le s$. Then $\lim_{K\to\infty}\langle F\circ\mathcal{M}^{K-s}\rangle_K = \int F\circ\pi_{\le s}\,d\mu_{\ell,\mathsf{b}}$. If μ' is a Borel probability measure on $\mathcal{X}$ that is invariant under every θ_v and satisfies $\int F\circ\pi_{\le s}\,d\mu' = \lim_K\langle F\circ\mathcal{M}^{K-s}\rangle_K$ for every element F of the block algebra, s its largest depth, then $\mu'=\mu_{\ell,\mathsf{b}}$.

■

Sketch of proof. (1), (2). Choose an integer $s\ge 0$ so that $\mathsf{b}' = L^s\mathsf{b}\ge\gamma^{-2}$ and $MQ(\mathsf{b}')\le L^{m+s}$. Such an s exists because Q grows at most as a power of the logarithm. At depth $d\ge s$, the dilation $D_s : x\mapsto L^s x$ identifies the lattice of T_ℓ with the depth-$d-s$ lattice of $T_{L^s\ell}$, preserving bond orientation and the relative positions of the blocks. The block map commutes with this identification. For $K\ge d$ the Wilson couplings agree, $L^{K-s}\mathsf{b}' = L^K\mathsf{b}$, and the Haar measures correspond bond by bond. Thus

$$(D_s)_*\mu_d^K[\ell,\mathsf{b}] = \mu_{d-s}^{K-s}[L^s\ell,\mathsf{b}'].$$

Proposition 10.2(a) gives total variation convergence on the admitted torus on the right. Pulling back gives convergence for $d\ge s$. For $d<s$ the identity $\mu_d^K = (\mathcal{M}^{s-d})_*\mu_s^K$ and contraction of total variation under a push-forward give convergence. Gauge invariance follows from gauge covariance of the block map and gauge invariance of the finite-cutoff measures. The finite-cutoff relation $\mathcal{M}_*\mu_{d+1}^K = \mu_d^K$ passes to the limit by the same contraction. Consequently the laws $(\iota_D)_*\mu_D$ on the finite products are consistent. Kolmogorov extension gives a unique measure $\mu_{\ell,\mathsf{b}}$ with these finite pieces and marginals μ_D. It gives each relation $\mathcal{M}V_{d+1}=V_d$ probability one, hence gives the threads full measure by countable intersection. In the one-torus setting it agrees with the measure of Proposition 10.2(a) under every $\pi_{\le D}$, hence the two measures coincide. For $d\le D\le K$ one has $\mathcal{M}^{K-d} = \mathcal{M}^{D-d}\circ\mathcal{M}^{K-D}$, so $\nu_D^K = (\iota_D)_*\mu_D^K$, and ι_D is Borel because $\mathcal{M}$ is (Proposition YM3_I: 4.5). Pushing two measures forward by one measurable map cannot enlarge their distance in total variation; thus $\sup_B|\nu_D^K(B)-(\iota_D)_*\mu_D(B)|\le\sup_E|\mu_D^K(E)-\mu_D(E)|$, which tends to 0, while $(\iota_D)_*\mu_D = (\pi_{\le D})_*\mu_{\ell,\mathsf{b}}$.

(3). At a bond b of Λ_d the entry of $\theta_v V_\bullet$ equals $v(b_-)V_d(b)v(b_+)^{-1}$, which depends continuously on $V_\bullet$; hence θ_v is continuous, the gauge rule $V^v(b) = v(b_-)V(b)v(b_+)^{-1}$ gives $\theta_{v'}\theta_v=\theta_{v'v}$, and $\theta_{v^{-1}}$ undoes θ_v. Averaging commutes with gauge transformations by YM3_I:(40). Iterating this identity restricts the gauge transformation to the coarsest lattice. In particular, $\mathcal{M}(V_{d+1}^{v|\Lambda_{d+1}}) = \mathcal{M}(V_{d+1})^{v|\Lambda_d}$. Therefore θ_v and $\theta_{v^{-1}}$ send threads to threads. On $\mathcal{X}_D=\mathcal{V}_0\times\cdots\times\mathcal{V}_D$ let a gauge transformation w of Λ_D act entrywise, $\theta_w^{(D)}\colon V_d\mapsto V_d^{w|\Lambda_d}$. Then $\pi_{\le D}\circ\theta_v = \theta^{(D)}_{v|\Lambda_D}\circ\pi_{\le D}$, and $D-d$ applications of the covariance give $\iota_D(V^w) = \theta_w^{(D)}\iota_D(V)$. Consequently $(\pi_{\le D})_*(\theta_v)_*\mu_{\ell,\mathsf{b}}$ is the push-forward of μ_D by $V\mapsto\iota_D(V^{v|\Lambda_D})$; as μ_D is gauge invariant, this is $(\iota_D)_*\mu_D$, and the uniqueness part of (2) yields $(\theta_v)_*\mu_{\ell,\mathsf{b}}=\mu_{\ell,\mathsf{b}}$.

(4), the limit. Unwinding the definitions, $\langle F\circ\mathcal{M}^{K-s}\rangle_K = \int F\circ\iota_s\,d\mu_s^K$, with a bounded Borel integrand. If ν,ν' are probability measures on a common σ-algebra and f is bounded, complex and measurable, then $|\int f\,d\nu-\int f\,d\nu'|\le 4\sup|f|\sup_E|\nu(E)-\nu'(E)|$, as one sees by writing f as a signed and complex combination of four non-negative functions, each bounded by $\sup|f|$. By (1), therefore, the means converge to $\int F\circ\iota_s\,d\mu_s$, and (2) identifies this number with $\int F\circ\pi_{\le s}\,d\mu_{\ell,\mathsf{b}}$.

(4), the uniqueness. Fix D. The gauge transformations of Λ_D form the compact group G^{Λ_D}; write dw for its normalized Haar measure. A continuous f on the compact metrisable space $\mathcal{X}_D$ is averaged over the gauge orbits: $\bar f(X) = \int f(\theta_w^{(D)}X)\,dw$. The integrand is jointly continuous in

(w, X) and bounded, so dominated convergence makes $\bar f$ continuous; and $\bar f \circ \theta^{(D)}_{w'} = \bar f$ for all w', since $\theta^{(D)}_w \theta^{(D)}_{w'} = \theta^{(D)}_{ww'}$ and right translation by w' preserves dw. Thus $\bar f$ is a bounded continuous function of the finitely many variables $V_d(b)$, $d \le D$, unchanged when one gauge transformation of Λ_D acts on all of them together; that is, $\bar f$ belongs to the block algebra, with largest depth D. Write $\nu'_D = (\pi_{\le D})_* \mu'$ and $\nu_D = (\pi_{\le D})_* \mu_{\ell,\mathsf{b}}$. Extending w by $\mathbf{1}$ off Λ_D gives a map v with $\pi_{\le D} \circ \theta_v = \theta^{(D)}_w \circ \pi_{\le D}$, so the hypothesis on μ' and part (3) make ν'_D and ν_D invariant under all the $\theta^{(D)}_w$. Averaging the identity $\int f \, d\nu'_D = \int f \circ \theta^{(D)}_w \, d\nu'_D$ in w and exchanging the integrals (Fubini's theorem) yields $\int f \, d\nu'_D = \int \bar f \, d\nu'_D$; likewise $\int f \, d\nu_D = \int \bar f \, d\nu_D$. Applying the hypothesis to $F = \bar f$ and comparing with the limit above, $\int \bar f \, d\nu'_D = \int \bar f \, d\nu_D$. So ν'_D and ν_D integrate every continuous function alike; as finite Borel measures on compact metrisable spaces are determined by their integrals of continuous functions, $\nu'_D = \nu_D$. As D was arbitrary, the uniqueness part of (2) gives $\mu' = \mu_{\ell,\mathsf{b}}$. □

Proposition 18.76 (Translations of the threads; the local semigroup at a pair of mirrors)**.** Assume the one-torus setting.

(A) *The translations.*

(A1) Let $h, h' \in \mathbb{Q}^3_L$. Then $\mathrm{T}_h \colon \mathcal{X} \to \mathcal{X}$ is Borel, restricts to a bijection of Thr, and satisfies $(\mathrm{T}_h)_* \mu = \mu$. On Thr, moreover, T_h is unchanged when any integer $s' \ge s$ takes the place of s in its definition, $\mathrm{T}_h \mathrm{T}_{h'} = \mathrm{T}_{h+h'}$, and T_h is the identity when $h \in \ell\mathbb{Z}^3$. Hence $h \mapsto \mathcal{U}_h$ represents the group $\mathbb{Q}^3_L/\ell\mathbb{Z}^3$ by unitary operators on $L^2(\mu)$, each mapping $\mathcal{K}^{\mathrm{inv}}$ onto itself. In the same way, for each diagonal element r of $\bigcup_s \mathsf{Iso}_s$, $\mathsf{r}^\bullet$ is Borel, permutes Thr and preserves μ.

(A2) For every $d \ge 0$ and every bounded gauge-invariant Borel function F on $\mathcal{V}_d$, $\|\mathcal{U}_v(F \circ \pi_d) - F \circ \pi_d\|_{L^2(\mu)} \to 0$ as $v \to 0$ in $\mathbb{Q}^3_L$.

(A3) A unique strongly continuous unitary representation $x \mapsto \mathcal{U}(x)$ of $\mathbb{R}^3/\ell\mathbb{Z}^3$ on $\mathcal{K}^{\mathrm{inv}}$ satisfies $\mathcal{U}(v) = \mathcal{U}_v|_{\mathcal{K}^{\mathrm{inv}}}$ for all $v \in \mathbb{Q}^3_L$; and $\mathcal{K}^{\mathrm{inv}}$ decomposes as the orthogonal direct sum of the spaces $\mathcal{K}_k = \{\psi \in \mathcal{K}^{\mathrm{inv}} : \mathcal{U}(x)\psi = e^{ik\cdot x}\psi \text{ for all } x\}$, $k \in \frac{2\pi}{\ell}\mathbb{Z}^3$.

(B) *A pair of mirrors.* Fix a coordinate mirror (μ, c), $c \in \frac{a_{d_0}}{2}\mathbb{Z}$, of either type, together with the objects that the notation attaches to it.

(B1) On $\mathcal{A}(0,0)$ the form $\langle \cdot, \cdot \rangle_c$ is hermitian, $\langle F, F\rangle_c \ge 0$, and $\|F\|_c \le \|F\|_{L^2(\mu)}$.

(B2) For every $F \in \mathcal{A}(0,0)$, $\int \overline{\Theta F}\, \mathcal{U}_t F \, d\mu \to \int \overline{\Theta F}\, F \, d\mu$ as $t \to 0$ in $\mathbb{Q}_L$, from either side. For $\alpha' > 0$ and $F \in \mathcal{A}(0, \alpha')$, $\|\mathcal{U}_t F - F\|_c \to 0$ as $t \downarrow 0$ in $\mathbb{Q}_L$; for $\alpha > 0$ and $F \in \mathcal{A}(\alpha, 0)$, $\|\mathcal{U}_{-t} F - F\|_c \to 0$ as $t \downarrow 0$ in $\mathbb{Q}_L$. For $\alpha, \alpha' > 0$, $F \in \mathcal{A}(\alpha, \alpha')$ and $t \in \mathbb{Q}_L$: $\|\mathcal{U}_t F - F\|_c^2 \le t(\frac{1}{2\alpha} + \frac{1}{\alpha'})\|F\|^2_{L^2(\mu)}$ if $0 \le t \le \frac{\alpha'}{2}$, and $\|\mathcal{U}_{-t} F - F\|_c^2 \le t(\frac{1}{\alpha} + \frac{1}{2\alpha'})\|F\|^2_{L^2(\mu)}$ if $0 \le t < \frac{\alpha}{2}$.

(B3) For $\alpha' > 0$, $F \in \mathcal{A}(0, \alpha')$ and $t \in \mathbb{Q}_L$ with $0 < t < \alpha'$: $\|\mathcal{U}_t F\|_c = 0$ if and only if $\|F\|_c = 0$.

(B4) $\bigcup_{\alpha, \alpha' > 0} \mathcal{A}(\alpha, \alpha')$ is dense in $\mathcal{A}(0,0)$ for the norm of $L^2(\mu)$, with approximants of a function F bounded by $\sup |F|$; so the classes of its elements are dense in $\mathcal{K}_c$.

(B5) On $\mathcal{K}_c$ a unique self-adjoint operator $\mathfrak{h}_c$ has the following property: whenever $\alpha' > 0$, $F \in \mathcal{A}(0, \alpha')$ and $t \in \mathbb{Q}_L \cap [0, \alpha')$, the class $[F]$ belongs to the domain of $e^{-t\mathfrak{h}_c}$ and $e^{-t\mathfrak{h}_c}[F] = [\mathcal{U}_t F]$. Moreover $\mathfrak{h}_c[1] = 0$; the identity holds also at $t = \alpha'$ if $\alpha' \in \mathbb{Q}_L$; and for $\alpha, \alpha' > 0$, $F \in \mathcal{A}(\alpha, \alpha')$ and $t \in \mathbb{Q}_L \cap (0, \alpha]$, $[F]$ lies in the domain of $e^{t\mathfrak{h}_c}$ and $e^{t\mathfrak{h}_c}[F] = [\mathcal{U}_{-t} F]$.

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Lemma 18.77 (Translations and reflections of the threads; the form at a mirror)**.** Assume the one-torus setting.

(a) For every diagonal element r of Iso_j: $\mathsf{r}_* \mu_j^K = \mu_j^K$ for $K \ge \max\{j, 1\}$, and $\mathsf{r}_* \mu_j = \mu_j$.

(b) Let $s \ge 0$ be an integer and $h, h' \in a_s \mathbb{Z}^3$. For a thread $V_\bullet$ and every integer $s' \ge s$, the sequence that the definition of $\mathrm{T}_h V_\bullet$ gives with s' in the place of the smallest non-negative

integer used there is $\mathrm{T}_h V_\bullet$. The map $\mathrm{T}_h\colon \mathcal{X}\to\mathcal{X}$ is Borel and sends Thr into Thr; on Thr, $\mathrm{T}_h\mathrm{T}_{h'}=\mathrm{T}_{h+h'}$, and T_h is the identity when $h\in\ell\mathbb{Z}^3$; and $(\mathrm{T}_h)_*\mu=\mu$. Any Borel probability measure on $\mathcal{X}$ that gives Thr full measure and is mapped by π_j to μ_j for every $j\ge s$ equals μ. The same holds for $\mathsf{r}^\bullet$, r a diagonal element of Iso_s; and for two diagonal elements $\mathsf{r}_1,\mathsf{r}_2$ of Iso_s, whose product is again one, $(\mathsf{r}_1\mathsf{r}_2)^\bullet=\mathsf{r}_1^\bullet\mathsf{r}_2^\bullet$ on Thr.

(c) Let (μ,c) be a mirror with $c\in\frac{a_{d_0}}{2}\mathbb{Z}$ and let $j\ge d_0$; all functions are understood as functions on Thr, where $\mathcal{U}_t$, Θ, Θ_{c^*} and R act by means of (b). Then $\mathcal{A}_j(\alpha,\alpha')\subset\mathcal{A}_{j+1}(\alpha,\alpha')$, hence $\mathcal{A}(\alpha,\alpha')$ is an algebra that is stable under conjugation; $\mathsf{Bd}_j(0,0)$ consists of those bonds of Λ_j whose two end points lie in $\mathbb{E}_+$; if the mirror is of the site type, then $\mathcal{A}_j(0,0)=\mathcal{A}_j(\frac{a_j}{2},\frac{a_j}{2})$; for $t\in\mathbb{Q}_L$ with $-\alpha\le t\le\alpha'$ one has $\mathcal{U}_t\mathcal{A}(\alpha,\alpha')\subset\mathcal{A}(\alpha+t,\alpha'-t)$; $\Theta\mathcal{U}_t=\mathcal{U}_{-t}\Theta$ and $\Theta^2=1$, while $\Theta_{c^*}\mathcal{U}_t=\Theta_c$ when $c^*=c+\frac{t}{2}$; further, ϱ is one of the diagonal elements of Iso_{d_0}, and $R\mathcal{A}(\alpha,\alpha')=\mathcal{A}(\alpha',\alpha)$, $R\Theta=\Theta R$, $R\mathcal{U}_tR=\mathcal{U}_{-t}$, $R^2=1$.

(d) For mirrors of both types, the form $\langle\cdot,\cdot\rangle_c$ on $\mathcal{A}(0,0)$ is hermitian and non-negative, and $\|F\|_c\le\|F\|_{L^2(\mu)}$.

■

Sketch of proof. (a) Take a Borel set $E\subset\mathcal{V}_j$. Since r is a diagonal isometry of each Λ_i with $i\ge j$, block equivariance lets it pass through all the averaging steps from Λ_K down to Λ_j: $\mu_j^K(\mathsf{r}^{-1}E)=\langle\mathbf{1}_E(\mathsf{r}\mathcal{M}^{K-j}U)\rangle_K=\langle\mathbf{1}_E(\mathcal{M}^{K-j}(\mathsf{r}U))\rangle_K$. Diagonal isometries of Λ_K do not change the Wilson measure, so the last mean is $\mu_j^K(E)$; letting $K\to\infty$ in total variation (Corollary 2.8(c)) gives $\mathsf{r}_*\mu_j=\mu_j$. (b) Let $V_\bullet$ be a thread, $h\in a_s\mathbb{Z}^3$ and $s'>s$. For $s\le j<s'$, block equivariance yields $\mathcal{M}^{s'-j}(\mathrm{t}_hV_{s'})=\mathrm{t}_h\mathcal{M}^{s'-j}V_{s'}=\mathrm{t}_hV_j$, and for $j<s$ one has $\mathcal{M}^{s'-j}(\mathrm{t}_hV_{s'})=\mathcal{M}^{s-j}(\mathcal{M}^{s'-s}(\mathrm{t}_hV_{s'}))=\mathcal{M}^{s-j}(\mathrm{t}_hV_s)$. Thus s and s' produce one and the same sequence, and the same computation shows $\mathcal{M}((\mathrm{T}_hV_\bullet)_{j+1})=(\mathrm{T}_hV_\bullet)_j$, so that $\mathrm{T}_hV_\bullet$ is again a thread. Using one s for both h and h', the entries of $\mathrm{T}_h\mathrm{T}_{h'}V_\bullet$ are $\mathrm{t}_{h+h'}V_j$ for $j\ge s$ and $\mathcal{M}^{s-j}(\mathrm{t}_{h+h'}V_s)$ for $j<s$, i.e. those of $\mathrm{T}_{h+h'}V_\bullet$; and t_h is the identity for $h\in\ell\mathbb{Z}^3$. Hence T_h permutes Thr, with the inverse T_{-h}, and its entries are Borel because $\mathcal{M}$ is (Proposition YM3_I:4.5). The push-forward $(\mathrm{T}_h)_*\mu$ gives Thr full measure, and for $j\ge s$ its image under π_j is $(\mathrm{t}_h)_*\mu_j=\mu_j$ by (a). Now let μ' be any Borel probability measure on $\mathcal{X}$ with $\mu'(\mathsf{Thr})=1$ and $(\pi_j)_*\mu'=\mu_j$ for $j\ge s$. On Thr the identity $\pi_{\le j}=\iota_j\circ\pi_j$ holds, so $(\pi_{\le j})_*\mu'=(\iota_j)_*\mu_j$ for $j\ge s$, and the smaller j follow by projecting; the uniqueness in Proposition 10.2(a) then gives $\mu'=\mu$, and in particular $(\mathrm{T}_h)_*\mu=\mu$. For $\mathsf{r}^\bullet$ the argument is identical, and the rule for products comes from $(\mathsf{r}_1(\mathsf{r}_2V))(b)=V(\mathsf{r}_2^{-1}\mathsf{r}_1^{-1}b)$. (c) *Nesting.* On a thread $V_j(b)=\mathcal{M}(V_{j+1})(b)$, and by Proposition YM3_I:4.5 this value involves only those bonds of Λ_{j+1} whose two end points lie in $B(b_-)\cup B(b_+)$, and the coordinate x_μ of such a point is within $\frac12(a_j-a_{j+1})$ of that of b_- or of b_+. So if both end points of b obey $c+\alpha+\frac12a_j\le x_\mu\le c+\frac{\ell}{2}-\alpha'-\frac12a_j$, these points obey the same bounds with a_{j+1} in place of a_j. *The case $\alpha=\alpha'=0$.* For $x\in\Lambda_j\cap\mathbb{E}_+$, the number $x_\mu-c$ is positive and lies in $a_j(\mathbb{Z}+\frac12)$ for the bond type and in $a_j\mathbb{Z}$ for the site type; so it is at least $\frac12a_j$, resp. at least a_j. The same is true of $c+\frac{\ell}{2}-x_\mu$, because $\frac{\ell}{2}=L^m\in a_j\mathbb{Z}$. *Shifts.* For $t\in a_j\mathbb{Z}$ one has $(\mathrm{T}_{-te_\mu}V_\bullet)_j(b)=V_j(b+te_\mu)$; hence $\mathcal{U}_tF$ depends on V_j only through the bonds of $\mathsf{Bd}_j(\alpha,\alpha')+te_\mu=\mathsf{Bd}_j(\alpha+t,\alpha'-t)$. *Reflections.* As maps of points, θ followed by the translation through $-te_\mu$ is the translation through te_μ followed by θ; and the reflection at $c^*=c+\frac{t}{2}$ followed by the translation through $-te_\mu$ is θ. Part (b) carries these relations over to threads, which gives $\Theta\mathcal{U}_t=\mathcal{U}_{-t}\Theta$ and $\Theta_{c^*}\mathcal{U}_t=\Theta_c$; and $\Theta^2=1$ because θ^2 is the identity. For $j\ge d_0$ the height $c+\frac{\ell}{4}$ lies in $\frac{a_j}{2}\mathbb{Z}$, as $\frac{\ell}{4}=\frac{a_j}{2}L^{m+j}$; so ϱ is a diagonal element of Iso_{d_0}. It preserves $\mathbb{E}_+$ and swaps the two boundary planes, whence $\varrho\,\mathsf{Bd}_j(\alpha,\alpha')=\mathsf{Bd}_j(\alpha',\alpha)$; the products $\varrho\theta$ and $\theta\varrho$ are the translations through $\pm\frac{\ell}{2}e_\mu$, which are the same map of T_ℓ; and conjugation by ϱ turns the translation through $-te_\mu$ into the one through te_μ. (d) The form is hermitian, since $\overline{\langle F,F'\rangle_c}=\int\Theta F\,\overline{F'}\,d\mu=\int F\,\overline{\Theta F'}\,d\mu$ by (b) and $\Theta^2=1$. For its sign write $F=\mathsf{f}(V_j|_{\mathsf{Bd}_j(0,0)})$ with $j\ge d_0$, and let $K\ge j$. Applying

the nesting $K-j$ times, $\tilde F(U)=\mathsf f((\mathcal M^{K-j}U)|_{\mathsf{Bd}_j(0,0)})$ is a bounded Borel function of those bonds of Λ_K whose end points both lie in $\mathbb E_+$, so all these bonds belong to the positive half of the bare lattice; moreover $\mathsf f((\theta\mathcal M^{K-j}U)|_{\mathsf{Bd}_j(0,0)})=\tilde F(\theta U)$ by block equivariance. Since $c\in\frac{a_K}{2}\mathbb Z$, θ is a lattice mirror at cutoff K. Apply Proposition 18.1(a) for a site mirror and Corollary YM3_I:85.3 for a bond mirror. They give $\int\overline{\mathsf f((\theta V)|_{\mathsf{Bd}_j(0,0)})}\,\mathsf f(V|_{\mathsf{Bd}_j(0,0)})\,d\mu_j^K(V)=\langle\overline{\tilde F(\theta U)}\tilde F(U)\rangle_K\ge 0$. Let $K\to\infty$ in total variation, recalling $(\theta^\bullet V_\bullet)_j=\theta V_j$. Finally $\|\tilde F\|_c\le\|F\|_{L^2(\mu)}$ follows from the Schwarz inequality in $L^2(\mu)$, because $\|\Theta F\|_{L^2(\mu)}=\|F\|_{L^2(\mu)}$. □

Lemma 18.78 (Elementary facts on SU(2) and on one averaging step at small fields)**.** (a) For $a,b,w\in G$: $|awb-\mathbf 1|\le|w-\mathbf 1|+|ab-\mathbf 1|$. For $w,w'\in G$ and $X,Y\in\mathfrak g$: $|{}^wX-{}^{w'}X|\le 2|w-w'||X|$ and $\langle{}^wX,{}^wY\rangle=\langle X,Y\rangle$. For complex 2×2 matrices, $|\operatorname{Im}Y-\operatorname{Im}Y'|\le|Y-Y'|$; for $w,w'\in G$, $|\operatorname{Im}w|\le|w-\mathbf 1|$, $\operatorname{Im}w^{-1}=-\operatorname{Im}w$ and $\operatorname{Im}({}^{w'}w)={}^{w'}\operatorname{Im}w$.

(b) For hermitian X,Y: $|e^{iX}e^{iY}-e^{i(X+Y)}|\le\frac12|[X,Y]|\le|X||Y|$ and $|e^{iX}-e^{iY}|\le|X-Y|$. For $X,Y\in\mathfrak g$: $|e^{iX}-\mathbf 1|=2\sin\frac{|X|}{2}\ge\frac{2}{\pi}|X|$ if $|X|\le\pi$; $|X-Y|\le(1-\rho)^{-1}|e^{iX}-e^{iY}|$ if $|X|,|Y|\le\rho<1$; and $|X-Y|\le2|e^{iX}-e^{iY}|$ if $|X|,|Y|\le1.1$.

(c) Let U be a field of a lattice Λ_i, and suppose that a path γ' is obtained from a path γ with the same two ends by k swaps of two consecutive steps, each swap crossing one plaquette, and by deleting steps that are immediately retraced. Then $|U(\gamma)U(\gamma')^{-1}-\mathbf 1|\le k\varpi(U)$, and $|A(\gamma)-A(\gamma')|\le k\max_p|\mathrm dA(p)|$ for a 1-form A. Any two members of a contour family differ by fewer than $\frac34L^2$ swaps; so if $\varpi(U)<\omega_M$, then $|U(\Gamma)U(y,x)^{-1}-\mathbf 1|\le3L^2\varpi(U)$ for every $\Gamma\in\mathcal G(y,x)$.

(d) If $V\in\mathcal V_i$ and $\varpi(V)<\omega_M$, then $|\mathcal M(V)(c)V[c]^{-1}-\mathbf 1|\le8L^2\varpi(V)$ for every bond c of Λ_{i-1}. ■

Sketch of proof. (a) Multiplication by unitaries leaves the norm unchanged; so $|awb-\mathbf 1|=|w-a^{-1}b^{-1}|\le|w-\mathbf 1|+|\mathbf 1-a^{-1}b^{-1}|$, and $|\mathbf 1-a^{-1}b^{-1}|=|ba-\mathbf 1|=|ab-\mathbf 1|$ after conjugating by a. The identity ${}^wX-{}^{w'}X=(w-w')Xw^{-1}+w'X(w^{-1}-w'^{-1})$ gives the bound for conjugates; Im is a contraction with $\operatorname{Im}\mathbf 1=0$, and $w^{-1}=w^*$ for $w\in G$. (b) Differentiating $t\mapsto e^{itX}e^{itY}e^{i(1-t)(X+Y)}$ produces $e^{itX}[iX,e^{itY}]e^{i(1-t)(X+Y)}$, whose norm is at most $t|[X,Y]|$, since $|[X,e^{itY}]|\le t|[X,Y]|$; integrating over $[0,1]$ gives the first bound, and $|[X,Y]|\le2|X||Y|$ because only the traceless parts enter and $[x\cdot\sigma,y\cdot\sigma]=2i(x\times y)\cdot\sigma$ for $x,y\in\mathbb R^3$, σ the triple of the Pauli matrices. Next, $e^{iX}-e^{iY}=\int_0^1e^{iuX}\,i(X-Y)\,e^{i(1-u)Y}du$; hence $|e^{iX}-e^{iY}|\le|X-Y|$, and as the integrand differs from $i(X-Y)$ by at most $(u|X|+(1-u)|Y|)\,|X-Y|$, also $|e^{iX}-e^{iY}-i(X-Y)|\le\rho|X-Y|$ when $|X|,|Y|\le\rho$, which gives the bound with the factor $(1-\rho)^{-1}$. Identify G with the unit sphere of the quaternions, the norm becoming the euclidean norm of $\mathbb R^4$; then $X\mapsto e^{iX}$ is the exponential map at $\mathbf 1$, and $|X|$ is the spherical distance of e^{iX} from $\mathbf 1$, which gives the formula $2\sin\frac{|X|}{2}$. If $|X|,|Y|\le1.1<\frac\pi2$, both points lie in a geodesically convex ball, where the logarithm is Lipschitz with the constant $1.1/\sin1.1$ for the spherical distance, and there the spherical distance is at most $1.1/\sin1.1$ times the chord, being at most 2.2; since $(1.1/\sin1.1)^2<2$, the last bound follows. (c) A swap across the plaquette p changes the transport, on one side, by a factor conjugate to $\partial U(p)^{\pm1}$, and (a) bounds each such change. Two shortest paths from y to x inside a box with $|n_\mu|\le\frac12(L-1)$ are turned into each other by at most $|n_1n_2|+|n_1n_3|+|n_2n_3|<\frac34L^2$ swaps, by sorting the three coordinate directions through adjacent exchanges. Thus the transports $\{U(\Gamma)\}_{\Gamma\in\mathcal G(y,x)}$ form a family of diameter $d=\max_{\Gamma,\Gamma'}|U(\Gamma)-U(\Gamma')|\le\frac34L^2\varpi(U)<\frac{1}{40}$, and Proposition YM3_I: 4.3 gives $|U(\Gamma)U(y,x)^{-1}-\mathbf 1|\le4d$. (d) When $\varpi<\omega_M$, the small-curvature formula YM3_I: (37) applies, and $\mathcal M(V)(c)V[c]^{-1}=\exp(L^{-3}\sum_x\log W_x)$, $W_x=V(c_-,x)V([x,x'])V(c_+,x')^{-1}V[c]^{-1}$. Pick $\Gamma\in\mathcal G(c_-,x)$ and let Γ'' be Γ moved by c_+-c_-; it belongs to $\mathcal G(c_+,x')$ by the contour definition. Then W_x factors into three pieces: $V(c_-,x)V(\Gamma)^{-1}$; the transport $V(\Gamma)V([x,x'])V(\Gamma'')^{-1}V[c]^{-1}$

around the strip of at most $\frac{3}{2}(L-1)L$ plaquettes that Γ covers while sliding along c; and a conjugate of $V(\Gamma'')V(c_+,x')^{-1}$. By (a) and (c), $|W_x-\mathbf{1}|\le(3+\frac{3}{2}+3)L^2\varpi\le 0.0375$, hence $|\log W_x|\le|W_x-\mathbf{1}|/(1-|W_x-\mathbf{1}|)\le 7.8L^2\varpi$. The exponent $L^{-3}\sum_x\log W_x$ lies in $i\mathfrak{g}$ and has norm at most $7.8L^2\varpi$, and (b) concludes. □

Lemma 18.79 (Linearizing the averaging map)**.** Let $\mathbf{Q}$ be the linear map of YM3_I:(42), $(\mathbf{Q}A)(c)=L^{-3}\sum_{x\in B(c_-)}[\bar A(c_-,x)+A([x,x'])-\bar A(c_+,x')]$, $\bar A(y,x)$ the mean of A over the members of $\mathcal{G}(y,x)$; put $(\mathbf{S}A)(c)=L^{-3}\sum_{x\in B(c_-)}A([x,x'])$ and $\mathsf{c}_A(y)=L^{-3}\sum_{x\in B(y)}\bar A(y,x)$ for the sites y of the coarser lattice. For 1-forms A with values in $\mathbb{R}$ or in $\mathfrak{g}$:

(a) $\mathbf{Q}A=\mathbf{S}A-\mathrm{d}\mathsf{c}_A$, and $\mathbf{Q}(\mathrm{d}\lambda)=\mathrm{d}(\lambda|_{\mathrm{coarse}})$.

(b) $\mathbf{Q}^nA=\mathbf{S}^nA+\mathrm{d}\zeta_n(A)$ with $\zeta_n(A)=-\sum_{i<n}\mathsf{c}_{\mathbf{S}^iA}$, restricted to the coarsest lattice, and $(\mathbf{S}^nA)(c)=L^{-3n}\sum_{x\in B^n(c_-)}A([x,x+c_+-c_-])$, which averages over the block the sums of A along straight paths of L^n bonds.

(c) $\sup|\mathbf{S}^nA|\le L^n\sup|A|$, $|\mathsf{c}_A|\le\frac{3}{2}(L-1)\sup|A|$, $|\zeta_n(A)|\le\frac{3}{2}(L^n-1)\sup|A|$, $|\mathrm{d}\zeta_n(A)|\le 3L^n\sup|A|$, and $(\mathbf{Q}^nA)(c)$ is a fixed linear combination of the values $A(b)$ whose coefficients have the total modulus at most $4L^n$; in particular $\sup|\mathbf{Q}^nA|\le 4L^n\sup|A|$, and for random forms ϕ and an event $\mathcal{E}$, $\mathbb{E}[|(\mathbf{Q}^n\phi)(c)|;\mathcal{E}]\le 4L^n\max_b\mathbb{E}[|\phi(b)|;\mathcal{E}]$.

(d) For a path γ of the coarser lattice with $|\gamma|$ bonds, $|(\mathbf{Q}^nA)(\gamma)|\le(|\gamma|L^n+2\mathsf{m}_1\frac{L^n-1}{L-1})\sup|A|$, $\mathsf{m}_1=\frac{3(L^2-1)}{4L}$; so $\sup|\mathbf{Q}^nA|\le c_{\mathbf{Q}}L^n\sup|A|$ with $c_{\mathbf{Q}}=1+\frac{3(L+1)}{2L}\le 2.62$.

(e) If A is a form on Λ_s and $h=\pm a_se_\nu$, then $\sup|\mathbf{S}^k(\mathrm{t}_hA-A)|\le 2\sup|A|$ for every $k\ge 0$, where $(\mathrm{t}_hA)(b)=A(b-h)$.

(f) $(\mathbf{Q}^nA)(c)$ reads A at the bonds with both end points in $B^n(c_-)\cup B^n(c_+)$ alone, and it is the derivative at $\mathbf{1}$ of $\mathcal{M}^n$ as in Lemma YM3_I:4.6.

■

Sketch of proof. (a) Since $x\mapsto x'$ maps $B(c_-)$ bijectively onto $B(c_+)$, the last sum equals $-\mathsf{c}_A(c_+)$. For $A=\mathrm{d}\lambda$ the three contributions are $\lambda(x)-\lambda(c_-)$, $\lambda(x')-\lambda(x)$ and $\lambda(c_+)-\lambda(x')$, adding up to $\lambda(c_+)-\lambda(c_-)$. (b) Induction on n with (a): $\mathbf{Q}(\mathbf{S}^nA+\mathrm{d}\zeta_n)=\mathbf{S}^{n+1}A-\mathrm{d}\mathsf{c}_{\mathbf{S}^nA}+\mathrm{d}(\zeta_n|)$. Concatenating straight paths gives the formula for $\mathbf{S}^n$, because $B^n(c_-)$ splits into the pairwise disjoint blocks $B^{n-1}(z)$, $z\in B(c_-)$. (c) Straight paths have L^n bonds and contours at most $\frac{3}{2}(L-1)$; the bound on ζ_n sums the geometric series $\sum_{i<n}L^i$; the coefficients of $\mathbf{S}^nA$ have total modulus L^n, and those of $\mathrm{d}\zeta_n$ at most $3L^n$. (d) Let γ run in the coarser lattice from Y to Y'. Adding (a) over the bonds of γ, the contour means at the intermediate sites cancel, and $(\mathbf{Q}A)(\gamma)=L^{-3}\sum_o[\bar A(Y,Y+o)+A(\gamma+o)-\bar A(Y',Y'+o)]$, where o ranges over the L^3 offsets inside a block and $\gamma+o$ is the concatenation of the corresponding straight paths of the finer lattice. Every summand averages A along paths of $L|\gamma|+2|o|_1$ bonds, $|o|_1$ being measured in finer spacings, and the average of $|o|_1$ over the offsets is $3\cdot\frac{1}{L}\sum_{|k|\le(L-1)/2}|k|=\mathsf{m}_1$. Inducting on n via $\mathbf{Q}^nA=\mathbf{Q}(\mathbf{Q}^{n-1}A)$ gives $(|\gamma|L+2\mathsf{m}_1)L^{n-1}+2\mathsf{m}_1\frac{L^{n-1}-1}{L-1}=|\gamma|L^n+2\mathsf{m}_1\frac{L^n-1}{L-1}$. (e) If $c\perp e_\nu$, the shifted and the unshifted families of L^{3k} straight paths have all but $2L^{2k}$ of their paths in common, each of L^k bonds; if $c\parallel e_\nu$, shifting a straight path removes one bond and adds one. (f) The locality of the one-step paths and nesting of the centered blocks give the support assertion by induction. The derivative formula follows from Lemma YM3_I:4.6 by the chain rule. □

Lemma 18.80 (One averaging step in coordinates)**.** (a) There is a number C_N, formed from L alone, with the following property. Let c be a bond of a lattice Λ_{i-1}, $i\ge 1$, and let A,A' be $\mathfrak{g}$-valued forms on P_c with $\sup|A|,\sup|A'|\le\rho_L$. Then e^{iA} satisfies the small-field condition of c at the radius ω_M, $\mathcal{M}(e^{iA})(c)=\exp i[(\mathbf{Q}A)(c)+q_c(A)]$ with $q_c(0)=0$, and $|q_c(A)-q_c(A')|\le C_N\max\{\sup|A|,\sup|A'|\}\sup|A-A'|$.

(b) For every $\kappa > 0$ there is $\rho_{\mathrm{Lip}}(\kappa) \in (0, \frac{1}{5}\omega_M]$, formed from κ and L, such that for every bond c as in (a) the map $\psi_c(A) = -i \log \mathcal{M}(e^{iA})(c)$ is real-analytic on $\{\sup|A| < \frac{1}{4}\omega_M\}$ and $|D\psi_c(A)[Y] - (\mathbf{Q}Y)(c)| \le \frac{\kappa}{c_{\mathbf{Q}}} \sup|Y|$ whenever $\sup|A| \le \rho_{\mathrm{Lip}}(\kappa)$.

■

Sketch of proof. Take $\sup|A| < 2\rho_L = \frac{1}{4}\omega_M$. Every plaquette then has $|\partial e^{iA}(p) - \mathbf{1}| \le 4\sup|A| < \omega_M$, so the small-curvature branch YM3_I:(37) applies, and $A \mapsto \mathcal{M}(e^{iA})(c)$ is real-analytic by Proposition YM3_I:4.5. Lemma 18.78(b), (d) gives $|\mathcal{M}(e^{iA})(c) - \mathbf{1}| \le 8L^2\omega_M + \frac{L}{4}\omega_M < 1$; hence ψ_c is real-analytic, with $\psi_c(0) = 0$ and $D\psi_c(0) = (\mathbf{Q}\cdot)(c)$ by Lemma YM3_I:4.6. Next we put integer coordinates on the double block. Let c be positively oriented, along e_ν; the sites of its double block are the points $c_- + a_i n$, n running through the box N_ν of the $n \in \mathbb{Z}^3$ with $-\frac{L-1}{2} \le n_\nu \le \frac{3L-1}{2}$ and $|n_\kappa| \le \frac{L-1}{2}$ for $\kappa \ne \nu$, which has $2L$ points along e_ν and L along each other axis. As $\ell = 2L^m$ with $m \ge 1$, each line of Λ_i, $i \ge 1$, parallel to an axis is a circle of $2L^{m+i} \ge 2L^2$ sites, so more than $2L$; hence $n \mapsto c_- + a_i n$ is one-to-one on N_ν, two points of N_ν are neighbours in $\mathbb{Z}^3$ exactly when their images are neighbours in Λ_i, and so this map matches the unit squares, the straight segments and the contour paths of N_ν with the plaquettes, the straight paths and the members of the contour families inside the double block. The three real block rules form $\mathcal{M}(V)(c)$ out of these objects, the number L and the radius ω_M alone, and it puts $\mathcal{M}(V)(\bar{c}) = \mathcal{M}(V)(c)^{-1}$. Read in the coordinates n, the map $V|_{\mathsf{P}_c} \mapsto \mathcal{M}(V)(c)$ is therefore one and the same for all positively oriented bonds along e_ν, independently of the depth, of the side and of the cutoff; moreover $\psi_{\bar{c}} = -\psi_c$ and $(\mathbf{Q}A)(\bar{c}) = -(\mathbf{Q}A)(c)$. So $q_c = \psi_c - (\mathbf{Q}\cdot)(c)$ is, up to its sign, one of three real-analytic functions, each vanishing at 0 together with its derivative. Define C_N as the largest supremum of $\|D^2 q_c\|$ over the closed ball of radius ρ_L, the operator norm being taken from the sup norm on forms to $|\cdot|$. Then $\|Dq_c(A)\| \le C_N \sup|A|$ on that ball, and the mean value theorem along segments of this convex set proves (a). Part (b) is obtained likewise from the continuity of $D\psi_c$ at 0, there being, up to sign, only three functions to consider. □

Lemma 18.81. *The hierarchical axial gauge; a Lipschitz bound for iterates of the averaging map.*

(a) Let $D' > d' \ge 0$, $V_{D'} \in \mathcal{V}_{D'}$, $V_i = \mathcal{M}^{D'-i}V_{D'}$, and $\varpi_i = \varpi(V_i) < \omega_M$ for $d' < i \le D'$. Given a gauge transformation v of $\Lambda_{d'}$, extend it step by step to $\Lambda_{D'}$ through $v(x) = v(y)V_i(y,x)$ for $d' < i \le D'$, $y \in \Lambda_{i-1}$, $x \in B(y) \setminus \{y\}$, and set $U_i = V_i^{v|\Lambda_i}$. Then $U_{i-1} = \mathcal{M}(U_i)$ and $U_i(y,x) = \mathbf{1}$ for all $y \in \Lambda_{i-1}$, $x \in B(y)$. For a bond b of Λ_i, $d' < i \le D'$: if both end points of b lie in one block $B(y)$, then $|U_i(b) - \mathbf{1}| \le 9.2L^2\varpi_i$; if b lies over the bond b' of Λ_{i-1}, then $|U_i(b) - U_{i-1}(b')| \le 31L^2\varpi_i$.

(b) Let $\kappa > 0$, $\rho = \rho_{\mathrm{Lip}}(\kappa)$, $0 \le d' < D'$, c a bond of $\Lambda_{d'}$, and $\mathcal{R}_i$, $d' \le i \le D'$, the set of the bonds of Λ_i with both end points in $B^{i-d'}(c_-) \cup B^{i-d'}(c_+)$. Let $V_{D'}, \tilde{V}_{D'} \in \mathcal{V}_{D'}$, $V_i = \mathcal{M}^{D'-i}V_{D'}$, $\varpi(V_i) < \omega_M$ for $d' < i \le D'$, $49L^2 \sum_{i=d'+1}^{D'} \varpi(V_i) \le \frac{\rho}{2}$, $|\tilde{V}_{D'}(b) - V_{D'}(b)| \le \eta$ for $b \in \mathcal{R}_{D'}$, and $3(L+\kappa)^{D'-d'}\eta \le \frac{\rho}{2}$. Then $|\mathcal{M}^{D'-d'}(\tilde{V}_{D'})(c) - V_{d'}(c)| \le 3(L+\kappa)^{D'-d'}\eta$.

■

Sketch of proof. (a) Proposition YM3_I:4.5, the gauge covariance of the averaging map and of the averaged contour variables, gives $U_{i-1} = \mathcal{M}(U_i)$ and $U_i(y,x) = v(y)V_i(y,x)v(x)^{-1} = \mathbf{1}$. Consider a bond $b = \langle x, x^+\rangle$, $x^+ = x + a_i e_\kappa$, inside one block $B(y)$. Suppose first that the κ-component of $x - y$ is ≥ 0; then the contour of $\mathcal{G}(y, x^+)$ that moves in the direction κ last has b as its final step, and dropping that step leaves a contour Γ of $\mathcal{G}(y,x)$. Otherwise the roles of x and x^+ are interchanged. Thus $V_i(y,x)V_i(b)V_i(y,x^+)^{-1}$ factors as $V_i(y,\cdot)V_i(\Gamma)^{-1}$ times $V_i(\Gamma^+)V_i(y,\cdot)^{-1}$, both within $3L^2\varpi_i$ of $\mathbf{1}$ by Lemma 18.78(c); conjugating by $v(y)$ gives $U_i(b)$. Next let $b = \langle x, x^+\rangle$ lie over $b' = \langle y, y'\rangle$ with $x \in B(y)$, $x^+ \in B(y')$, and put $x' = x + y' - y$. Then $|U_i(b) - U_{i-1}(b')| = |E\,\mathcal{M}(V_i)(b')^{-1} - \mathbf{1}|$, where $E = V_i(y,x)V_i(b)V_i(y',x^+)^{-1}$. Because $V_i([x,x']) = V_i(b)V_i([x^+,x'])$, we may write $E = W_x V_i[b']Z$ with W_x as in Lemma 18.78(d) and $Z = V_i(y',x')V_i([x^+,x'])^{-1}V_i(y',x^+)^{-1}$. Continuing the contour

of $\mathcal{G}(y',x^+)$ that ends in the direction κ by the segment $[x^+,x']$ backtracks over its final steps and leaves the contour of $\mathcal{G}(y',x')$ that ends in the direction κ; therefore $|Z-\mathbf{1}|\le 6L^2\varpi_i$. Together with $|W_x-\mathbf{1}|\le 7.5L^2\varpi_i$ and $|V_i[b']\mathcal{M}(V_i)(b')^{-1}-\mathbf{1}|\le 7.8L^2\varpi_i$ (Lemma 18.78(d) and its proof) this gives $|E\,\mathcal{M}(V_i)(b')^{-1}-\mathbf{1}|\le(7.5+6+7.8)L^2\varpi_i\le 31L^2\varpi_i$. (b) *Choice of gauge.* Neither the claim nor its hypotheses change if one gauge transformation of $\Lambda_{D'}$ acts on both $V_{D'}$ and $\tilde V_{D'}$ (Proposition YM3_I:4.5). We use the extension, as in (a), of the gauge transformation of $\Lambda_{d'}$ that equals $V_{d'}(c)$ at c_+ and $\mathbf{1}$ at every other site; then $U_{d'}(c)=\mathbf{1}$. Every bond $b\in\mathcal{R}_i$ lies over a bond of Λ_{i-1}, which lies over a bond of Λ_{i-2}, and so on, until either the bond reached at some depth $k>d'$ sits inside a block, or the depth d' is reached, where the bond is c or $\bar c$. Adding the bounds of (a) along this chain, $|U_i(b)-\mathbf{1}|\le 31L^2\sum_{k=d'+1}^{i}\varpi(V_k)$; so $U_i=e^{i\Phi_i}$ on $\mathcal{R}_i$ with $|\Phi_i|\le\frac{\pi}{2}\cdot 31L^2\sum_k\varpi(V_k)\le\frac{\rho}{2}$ (Lemma 18.78(b)). Set $\tilde U_{D'}=\tilde V^v_{D'}$ and $\tilde U_{i-1}=\mathcal{M}(\tilde U_i)$ on $\mathcal{R}_{i-1}$, which involves $\tilde U_i$ on $\mathcal{R}_i$ only. *An adapted norm.* For forms δ on $\mathcal{R}_i$ let $\|\delta\|_{*,i}=\max_{0\le n\le i-d'}L^{-n}\sup_{\mathcal{R}_{i-n}}|\mathbf{Q}^n\delta|$. Lemma 18.79(d) gives $\sup|\delta|\le\|\delta\|_{*,i}\le c_{\mathbf{Q}}\sup|\delta|$, and the definition gives $\|\mathbf{Q}\delta\|_{*,i-1}\le L\|\delta\|_{*,i}$ at once. *Descending induction.* We show that $\tilde U_i=e^{i\tilde\Phi_i}$ on $\mathcal{R}_i$ and that $\delta_i=\tilde\Phi_i-\Phi_i$ satisfies $\|\delta_i\|_{*,i}\le c_{\mathbf{Q}}(1-\rho)^{-1}(L+\kappa)^{D'-i}\eta$. For $i=D'$ we have $|\tilde U_{D'}-U_{D'}|\le\eta\le\frac{\rho}{6}$. The norm $|X|$ of $X\in\mathfrak{g}$ is the spherical distance from $\mathbf{1}$ to e^{iX} on G, and an arc no longer than π is at most $\frac{\pi}{2}$ times its chord; so $|\tilde\Phi_{D'}|\le|\Phi_{D'}|+\frac{\pi}{2}\eta\le\frac{\rho}{2}+\frac{\pi}{12}\rho\le\rho$, and Lemma 18.78(b) gives $\sup|\delta_{D'}|\le(1-\rho)^{-1}\eta$. Passing from i to $i-1$: $\sup|\delta_i|\le\|\delta_i\|_{*,i}\le 2.62\cdot 1.07\,(L+\kappa)^{D'-d'}\eta\le\frac{\rho}{2}$, so the whole segment from Φ_i to $\tilde\Phi_i$ stays in $\{\sup|\cdot|\le\rho\}$, and at each $c'\in\mathcal{R}_{i-1}$, $\delta_{i-1}(c')=\int_0^1 D\psi_{c'}(\Phi_i+\lambda\delta_i)[\delta_i]\,d\lambda=(\mathbf{Q}\delta_i)(c')+\epsilon_i(c')$, where $|\epsilon_i(c')|\le\frac{\kappa}{c_{\mathbf{Q}}}\sup|\delta_i|$ by Lemma 18.80(b). Hence $\|\delta_{i-1}\|_{*,i-1}\le L\|\delta_i\|_{*,i}+c_{\mathbf{Q}}\frac{\kappa}{c_{\mathbf{Q}}}\|\delta_i\|_{*,i}=(L+\kappa)\|\delta_i\|_{*,i}$. At $i=d'$ this yields $|\tilde U_{d'}(c)-U_{d'}(c)|\le|\delta_{d'}(c)|\le 2.79(L+\kappa)^{D'-d'}\eta$; and $\tilde U_{d'}(c)$, $U_{d'}(c)$ come from $\mathcal{M}^{D'-d'}(\tilde V_{D'})(c)$ and $V_{d'}(c)$ by the same multiplications, by $v(c_-)$ on the left and by $v(c_+)^{-1}$ on the right. □

Lemma 18.82 (Gauge fixing on a box, with small bond variables at all depths)**.** Let $j_0\ge 0$ satisfy $\log(L^{j_0}\mathsf{b})\ge 8\mathfrak{p}_0$ and $17000L^3\varepsilon_{[j_0]}\le 1$, let $s>j_0$, and let $V_i\in\mathcal{V}_i$, $j_0\le i\le s$, satisfy $V_{i-1}=\mathcal{M}(V_i)$ and $\varpi(V_i)<2\varepsilon_{[i]}$. For a bond c_0 of Λ_{j_0} let $\mathsf{Box}(c_0)$ be the smallest closed coordinate box containing the four points $c_{0\pm}\pm a_{j_0}(1,1,1)$; for every depth $i\ge j_0$ it contains the double block $B^{i-j_0}(c_{0-})\cup B^{i-j_0}(c_{0+})$, at a distance at least $\frac12 a_{j_0}$ from its boundary. Then one gauge transformation u of Λ_s makes all the bonds in the box small at once: for $j_0\le i\le s$ and every bond b of Λ_i whose end points lie in $\mathsf{Box}(c_0)$, $V_i^{\mathsf{u}|\Lambda_i}(b)=e^{i\Phi_i(b)}$ with $\Phi_i(b)\in\mathfrak{g}$ and $|\Phi_i(b)|\le C_g\varepsilon_{[i]}$. On the Borel set of the V_s satisfying these hypotheses, u and the forms Φ_i may be taken to be Borel functions of V_s. ■

Sketch of proof. Within each step, lengths are expressed in units of the spacing of Λ_i. *Interpolating a form.* Let A be a $\mathfrak{g}$-valued form on the bonds of Λ_{i-1} inside the box and $b=\langle x,x+e_\mu\rangle$ a bond of Λ_i inside the box. Choose a cell of Λ_{i-1} inside the box that contains b (a cube of side L with its corners in Λ_{i-1}), let y^0 be its corner with the smallest coordinates, write $t_\kappa=(x_\kappa-y^0_\kappa)/L\in[0,1]$ for the two directions $\kappa\ne\mu$, and set $(\mathsf{H}A)(b)=L^{-1}\sum_{\epsilon\in\{0,1\}^2}\prod_\kappa t_\kappa^{\epsilon_\kappa}(1-t_\kappa)^{1-\epsilon_\kappa}A(\langle y^\epsilon,y^\epsilon+Le_\mu\rangle)$, $y^\epsilon=y^0+L\sum_\kappa\epsilon_\kappa e_\kappa$. Adjacent cells agree on their common face, so $\mathsf{H}A$ is well defined. It satisfies $|\mathsf{H}A|\le L^{-1}\sup|A|$; its sum over the L bonds beneath a bond $c=\langle y,y'\rangle$ of Λ_{i-1} is $A(c)$; and $\max_p|\mathrm{d}\mathsf{H}A(p)|\le L^{-2}\max_P|\mathrm{d}A(P)|$, since the μ-component of $\mathsf{H}A$ is affine in each t_κ separately. *The first depth.* Let $\mathsf{u}_{j_0}(y)=V_{j_0}(\gamma_y)$, where γ_y runs from the corner of the box with the smallest coordinates to y, first along e_1, then along e_2, then along e_3. For a bond b in the box, $V^{\mathsf{u}}_{j_0}(b)$ equals the transport of V_{j_0} along the closed path $\gamma_{b_-}b\gamma_{b_+}^{-1}$, which encloses at most 5 plaquettes of the box; so $|V^{\mathsf{u}}_{j_0}(b)-\mathbf{1}|<10\varepsilon_{[j_0]}$ and $|\Phi_{j_0}|\le 1.01\cdot 10\varepsilon_{[j_0]}$. *From $i-1$ to i.* Write $\alpha=\sup|\Phi_{i-1}|\le C_g\varepsilon_{[i-1]}$. For $y\in\Lambda_{i-1}$ and $x\in B(y)$ inside the box let $\mathsf{u}_i(x)=e^{-i\lambda(x)}\mathsf{u}_{i-1}(y)V_i(y,x)$, where $\lambda(x)=(\mathsf{H}\Phi_{i-1})(\Gamma^1_{y,x})$ and $\Gamma^1_{y,x}$ is the contour of $\mathcal{G}(y,x)$ that moves along e_1, e_2, e_3 in this order. Then u_i agrees with u_{i-1} on Λ_{i-1}, and $|\lambda|\le\frac32\alpha$. Abbreviate $\varpi=\varpi(V_i)$.

If $b=\langle x,x''\rangle$ has both end points in one block $B(y)$, then $V_i^{\mathsf{u}_i}(b)=e^{-i\lambda(x)\,\mathsf{u}_{i-1}(y)}E\,e^{i\lambda(x'')}$ with $E=V_i(y,x)V_i(b)V_i(y,x'')^{-1}$; Lemma 18.78(c) gives $|E-\mathbf{1}|\le(\frac34+6)L^2\varpi$ and, applied to the form $\mathsf{H}\Phi_{i-1}$, also $|\lambda(x'')-\lambda(x)-(\mathsf{H}\Phi_{i-1})(b)|\le\frac34\max_P|\mathrm{d}\Phi_{i-1}(P)|$. If b leads from $B(y)$ to $B(y'')$ and lies over $c=\langle y,y''\rangle$, put $T_b=V_i(y,x)V_i(b)V_i(y'',x'')^{-1}$; two contour means and at most L^2 plaquettes stand between the path that crosses over, runs along and comes back, and the segment $[y,y'']$, so $|T_bV_i[c]^{-1}-\mathbf{1}|\le7L^2\varpi$. Writing $V_{i-1}(c)=e^YV_i[c]$ with $|e^Y-\mathbf{1}|\le8L^2\varpi$ (Lemma 18.78(d)), we get $T_b=E_bV_{i-1}(c)$ with $|E_b-\mathbf{1}|\le15L^2\varpi$, and $V_i^{\mathsf{u}_i}(b)=e^{-i\lambda(x)\,\mathsf{u}_{i-1}(y)}E_b\,e^{i\Phi_{i-1}(c)}e^{i\lambda(x'')}$. Around the loop $[y,y'']\,\Gamma^1_{y'',x''}\,\bar b\,(\Gamma^1_{y,x})^{-1}$, which encloses at most $2.5L^2$ plaquettes, the sum of $\mathsf{H}\Phi_{i-1}$ equals $\Phi_{i-1}(c)+\lambda(x'')-(\mathsf{H}\Phi_{i-1})(b)-\lambda(x)$ and has modulus at most $2.5\max_P|\mathrm{d}\Phi_{i-1}(P)|$. *Curvature.* Three uses of Lemma 18.78(b) give $|\partial V_{i-1}^{\mathsf{u}}(P)-e^{i\,\mathrm{d}\Phi_{i-1}(P)}|\le6\alpha^2$, hence $|\mathrm{d}\Phi_{i-1}(P)|\le\frac\pi2(\varpi(V_{i-1})+6\alpha^2)$. *Collecting.* Two more uses of Lemma 18.78(b), costing at most $3(\frac32\alpha)^2$, and $|\mathsf{H}\Phi_{i-1}|\le\frac\alpha L$ yield $|V_i^{\mathsf{u}_i}(b)-\mathbf{1}|\le\frac\alpha L+15L^2\varpi(V_i)+4\varpi(V_{i-1})+31\alpha^2$. Insert $\varpi(V_i)<2\varepsilon_{[i]}$, $\varepsilon_{[i-1]}\le L^{1/2}\varepsilon_{[i]}$, $\alpha\le50L^{5/2}\varepsilon_{[i]}$ and $|\Phi|\le1.01|e^{i\Phi}-\mathbf{1}|$ in this range: $|\Phi_i|\le1.01(50L^{3/2}+31L^2+77500L^5\varepsilon_{[i]})\varepsilon_{[i]}\le50L^2\varepsilon_{[i]}$, because $50L^{-1/2}\le13.9$, $1.01(13.9+31)\le45.35$, and $1.01\cdot77500L^3\varepsilon_{[i]}\le4.605$ by $17000L^3\varepsilon_{[j_0]}\le1$ and $\varepsilon_{[i]}\le\varepsilon_{[j_0]}$. Finally let $\mathsf{u}=\mathsf{u}_s$ inside the box and $\mathbf{1}$ outside it; then $\mathsf{u}_s|\Lambda_i=\mathsf{u}_i$. *Measurability.* Only transports, averaged contour variables, which are real-analytic on the open small-curvature set by Lemma YM3_I:4.4, and the logarithm near $\mathbf{1}$ enter the construction. □

Lemma 18.83 (Identities for the covariant shift, and a bound on its defect through the plaquette moduli)**.** Let $s\ge1$, $h=\pm a_se_\nu$ and $V\in\mathcal{V}_s$, with the objects of the covariant shift.

(a) $\mathrm{t}_hV=\check V^{\,g}$ and $V_i'=\check V_i^{\,g|\Lambda_i}$ for $i\le s$; so $\mathrm{dist}_{\mathrm{orb}}(V_i',V_i)\le\max_b|\mathsf{P}_i(b)-\mathbf{1}|$. At every bond b of Λ_s, $|\mathsf{P}_s(b)-\mathbf{1}|\le\varpi(V)$.

(b) For a gauge transformation w of Λ_s, the covariant shift of V^w is $\check V^w$, and the defect formed from V^w at a bond b of Λ_i is $w(b_-)\mathsf{P}_i(b)w(b_-)^{-1}$.

(c) Let $s>j\ge D_g$, $\varpi(\mathcal{M}^{s-i}V)<2\varepsilon_{[i]}$ for $j\le i\le s$, and $\mathsf{k}_2L^{s-j}\varepsilon_{[s]}^2\le c_*$. Then $|\mathsf{P}_j(c)-\mathbf{1}|\le\bar\eta_j$ for every bond c of Λ_j, and so $\mathrm{dist}_{\mathrm{orb}}(V_j',V_j)\le\bar\eta_j$.

■

Sketch of proof. (a) The gauge rule gives $\check V=(\mathrm{t}_hV)^{g^{-1}}$; since the iterates of $\mathcal{M}$ commute with gauge transformations (Proposition YM3_I:4.5), $\check V_i=V_i'^{\,g^{-1}|\Lambda_i}$. Taking $g|\Lambda_i$ as the gauge transformation in the definition of $\mathrm{dist}_{\mathrm{orb}}$, $|V_i'(b)-V_i^g(b)|=|g(b_-)^{-1}V_i'(b)g(b_+)-V_i(b)|=|\mathsf{P}_i(b)-\mathbf{1}|$. At the depth s there are two cases. If $b\perp e_\nu$ and p_b is the plaquette spanned by b and $b-h$, then $\check V(b)V(b)^{-1}=g(b_-)^{-1}\partial V(p_b)^{\pm1}g(b_-)$. If $b\parallel e_\nu$, then $\check V(b)=V(b)$, both factors g being transports along bonds of the line through b. (b) Formed from V^w, the transports become $w(x-h)g(x)w(x)^{-1}$, and $\mathrm{t}_h(V^w)=(\mathrm{t}_hV)^{w(\cdot-h)}$; substituting, the covariant shift of V^w at b is $w(b_-)\check V(b)w(b_+)^{-1}$, and the covariance of $\mathcal{M}^{s-i}$ does the rest. (c) Fix a bond c_0 of Λ_j. As $j\ge D_g$, Lemma 18.82 with $j_0=j$ provides u and the forms Φ_i on the bonds inside $\mathsf{Box}(c_0)$. Let $\mathcal{R}_i$ consist of the bonds of Λ_i whose end points lie in $B^{i-j}(c_{0-})\cup B^{i-j}(c_{0+})$; then $\mathsf{P}_c\subset\mathcal{R}_i$ whenever $c\in\mathcal{R}_{i-1}$. On $\mathcal{R}_s$ let $\Phi_s'=\mathrm{t}_h\Phi_s$, which uses Φ_s only inside the box, and let $\mathsf{u}'=\mathsf{u}(\cdot-h)$; then $e^{i\Phi_s'}=(\mathrm{t}_hV)^{\mathsf{u}'}$ on $\mathcal{R}_s$. With $m_i=s-i$ set $X_i=\mathsf{k}_1(1+m_i)\varepsilon_{[s]}+\mathsf{k}_2L^{m_i}\varepsilon_{[s]}^2$. Since $(1+m)\varepsilon_{[s]}\le(1+m)L^{-3m/8}\varepsilon_{[s-m]}\le\varepsilon_{[s-m]}$, we get $X_i\le\mathsf{k}_1\varepsilon_{[i]}+c_*\le\frac{\rho_L}{2}$. *Claim:* for $j\le i\le s$ the forms $\Phi_i'=-i\log\mathcal{M}^{s-i}(e^{i\Phi_s'})$ on $\mathcal{R}_i$ exist, and $x_i=\sup_{\mathcal{R}_i}|\Phi_i'-\Phi_i|\le X_i$. For $i=s$, $x_s\le2C_g\varepsilon_{[s]}\le X_s$. Assume the claim for all $i>j'$. Then $\sup|\Phi_i'|\le\rho_L$, Lemma 18.80(a) is available at every $c\in\mathcal{R}_{i-1}$, and the covariance gives $\mathcal{M}(e^{i\Phi_i})=e^{i\Phi_{i-1}}$ there; hence $\Phi_{i-1}'-\Phi_{i-1}=\mathbf{Q}(\Phi_i'-\Phi_i)+r_i$ with $|r_i|\le C_N(C_g\varepsilon_{[i]}+X_i)X_i$. Iterating down to j' and writing $m=m_{j'}$: $\Phi_{j'}'-\Phi_{j'}=\mathbf{Q}^m(\mathrm{t}_h\Phi_s-\Phi_s)+\sum_{i=j'+1}^s\mathbf{Q}^{i-1-j'}r_i$. Lemma 18.79(b), (c), (e) bounds the first term by $2C_g\varepsilon_{[s]}(1+3(L-1)m)\le\frac12\mathsf{k}_1(1+m)\varepsilon_{[s]}$. By Lemma 18.79(c) the sum is at most $\mathsf{S}=4C_N\sum_{\mathsf{m}=0}^{m-1}L^{m-\mathsf{m}-1}(2C_g\varepsilon_{[s-\mathsf{m}]}+X)X$, where $X=\mathsf{k}_1(1+\mathsf{m})\varepsilon_{[s]}+\mathsf{k}_2L^{\mathsf{m}}\varepsilon_{[s]}^2$. Using $\varepsilon_{[s-\mathsf{m}]}\le L^{\mathsf{m}/2}\varepsilon_{[s]}$, $X^2\le2\mathsf{k}_1^2(1+\mathsf{m})^2\varepsilon_{[s]}^2+2\mathsf{k}_2^2L^{2\mathsf{m}}\varepsilon_{[s]}^4$, $\sum_{\mathsf{m}\ge0}(1+$

$\mathsf{m})L^{-\mathsf{m}/2} \le 1.92$, $\sum_{\mathsf{m}\ge 0}(1+\mathsf{m})^2 L^{-\mathsf{m}} \le 1.4$, $\sum_{i>j'} \varepsilon_{[i]} \le 1.62\,\varepsilon_{[j']}$ and $\mathsf{k}_2 L^m \varepsilon_{[s]}^2 \le c_*$, one finds $\mathsf{S} \le 4C_N L^{m-1}\varepsilon_{[s]}^2[3.9C_g\mathsf{k}_1 + 2.8\mathsf{k}_1^2 + \mathsf{k}_2(3.3C_g\varepsilon_{[j']} + \frac{2c_*}{L-1})] \le \frac{1}{2}\mathsf{k}_2 L^m\varepsilon_{[s]}^2 + \frac{1}{2}\mathsf{k}_2 L^m \varepsilon_{[s]}^2$; here the first half is the definition of k_2, and the second holds because $4C_N L^{-1}\cdot 3.3 C_g \varepsilon_{[j']} \le \frac{1}{4}$, from $53 C_N C_g \varepsilon_{[j]} \le L$, and $4C_N L^{-1}\frac{2c_*}{L-1} \le \frac{1}{4}$, from $c_* \le \frac{L(L-1)}{32 C_N}$. Therefore $x_{j'} \le X_{j'}$. *Conclusion.* By the covariance, $e^{i\Phi'_j(c_0)} = \mathsf{u}'(c_{0-})V'_j(c_0)\mathsf{u}'(c_{0+})^{-1}$ and $e^{i\Phi_j(c_0)} = \mathsf{u}(c_{0-})V_j(c_0)\mathsf{u}(c_{0+})^{-1}$; so, with $\mathsf{w} = \mathsf{u}'^{-1}\mathsf{u}$, Lemma 18.78(b) gives $|V'_j(c_0) - V_j^{\mathsf{w}}(c_0)| \le x_j$. Moreover $g(y) = \mathsf{u}'(y)^{-1} e^{i\Phi_s(\langle y-h, y\rangle)}\mathsf{u}(y)$, whence $|g(y) - \mathsf{w}(y)| \le C_g\varepsilon_{[s]}$. Altogether $|\mathsf{P}_j(c_0) - \mathbf{1}| = |V'_j(c_0) - V_j^g(c_0)| \le X_j + 2C_g\varepsilon_{[s]} \le \bar\eta_j$, as $2C_g \le \mathsf{k}_1$. □

Lemma 18.84 (One event for all the levels of a cutoff; regular fields and threads)**.** Assume the one-torus setting.

(a) Let $d_\natural \le D < K$, and let G_D^K be the set of the $U \in \mathcal{V}_K$ with $\varpi(\mathcal{M}^{K-D}U) < 2\varepsilon_{[D]}$, $\varpi(\mathcal{M}^{K-i}U) < 2c_\rho\varepsilon_{[i]}$ for $D < i < K$, and $\varpi(U) < c_\rho\varepsilon_{[K]}$. Then $\langle \mathbf{1} - \mathbf{1}_{\mathsf{G}_D^K}\rangle_K \le R_D$.

(b) For $d_\Sigma \le j \le s < K$, $\langle \mathbf{1}\{\mathcal{M}^{K-s}U \notin \mathsf{Gd}_{j,s}\}\rangle_K \le \frac{L}{L-1}R_j$ and $\mu_s(\mathcal{V}_s \setminus \mathsf{Gd}_{j,s}) \le \frac{L}{L-1}R_j$; for $D \ge d_\Sigma$, $\mu(\mathsf{Thr} \setminus \mathsf{Gd}_D) \le \frac{L}{L-1}R_D$.

■

Sketch of proof. (a) The proof of Proposition 10.2(b) uses Theorem F on the line of couplings $x_{[d']} = L^{d'}\mathsf{b}$, with the run in which every u_t vanishes; for $d_\natural \le D < K$ the small-field chain of that run goes up to the level $t = K - D$, at which the lattice is Λ_D. This chain carries the bounded nonnegative Borel weight $\mathfrak{w}_t^{\mathrm{sf}}$ defined immediately before Proposition YM3_I:115.2, containing the factors $\mathbf{1}_{\mathsf{In}_t}(\mathcal{M}^t U)$ and $\mathbf{1}_{\mathsf{Cut}_k}(\mathcal{M}^k U)$, $0 \le k < t$; on this line $\mathsf{In}_t = I_D$, $\mathsf{Cut}_0 = \{\varpi < c_\rho\varepsilon(x_0)\}$ with $x_0 = x_{[K]}$, and $\mathsf{Cut}_k \subset \{\varpi < 2c_\rho\varepsilon(x_k)\}$, $x_k = x_{[K-k]}$, for $1 \le k < t$ by the definitions preceding Lemma YM3_I:113.10 and its cut bound YM3_I:(2159). Consequently $\mathfrak{w}_t^{\mathrm{sf}}$ is zero outside G_D^K, which is Borel and gauge invariant (Proposition YM3_I:4.5). Write $\rho_0 = e^{-\beta\sum_p a_p}$ for the bare density at $\beta = x_0 = L^K\mathsf{b}$, so that $Z = \int \rho_0\, dU$ is the normalisation of $\langle\cdot\rangle_K$. Taking $F = \mathbf{1}_{\mathsf{G}_D^K}$ in YM3_I:(2189) and using the discarded-mass bound in Proposition YM3_I:115.2, $Z\langle \mathbf{1}_{\mathsf{G}_D^K}\rangle_K \ge \int \mathbf{1}_{\mathsf{G}_D^K}\mathfrak{w}_t^{\mathrm{sf}}\rho_0\, dU = \int \mathfrak{w}_t^{\mathrm{sf}}\rho_0\, dU \ge (1 - \mathsf{R}_t)Z$, and on this line the number R_t of YM3_I:(2187) equals R_D. (b) Proposition 10.2(b) gives $\mu_i^K(\mathcal{V}_i \setminus I_i) \le R_i$ and $\mu_i(\mathcal{V}_i \setminus I_i) \le R_i$ for $d_\natural \le i < K$; add these over $j \le i \le s$ and use $\sum_{i\ge j} R_i \le \frac{L}{L-1}R_j$, valid for $j \ge d_\Sigma$. For threads, $\mathsf{Thr} \setminus \mathsf{Gd}_D = \mathsf{Thr} \cap \bigcup_{i\ge D}\pi_i^{-1}(\mathcal{V}_i \setminus I_i)$. □

Lemma 18.85 (Reflection positivity for one adjoint charge)**.** Let $K \ge 1$ and $\beta > 0$. Let $\Sigma = (\nu, c)$, $c \in \frac{a_K}{2}\mathbb{Z}$, be a lattice mirror of T_ℓ at the cutoff K, θ its reflection and $\mathbb{E}_+ = \{c < x_\nu < c + \frac{\ell}{2}\}$, and fix a column. For a bond mirror, $x_\pm$ denote the sites of the column at the heights $c \pm \frac{1}{2}a_K$, $b_x = \langle x_-, x_+\rangle$, and $\mathsf{B}(\Sigma)$ is the set of bonds of Λ_K whose end points both lie in $\mathbb{E}_+$. For a site mirror, $x_+ = x_-$ denotes the site of the column at the height c, $U(b_x) := \mathbf{1}$, and $\mathsf{B}(\Sigma)$ is the set of bonds whose end points both lie in $\{c \le x_\nu \le c + \frac{\ell}{2}\}$. Let S be a bounded Borel $\mathfrak{g}$-valued function of the bare field that depends only on the bonds of $\mathsf{B}(\Sigma)$ and, for a bond mirror, is covariant at x_+. Then $\langle S, S\rangle_\Sigma := \big\langle \langle S(\theta U), {}^{U(b_x)}S(U)\rangle\big\rangle_K \ge 0$. ■

Sketch of proof. Since S takes hermitian values, $\langle S(\theta U), S(U)\rangle = \frac{1}{2}\sum_{i,k}\overline{S_{ki}(\theta U)}\, S_{ki}(U)$. *Site mirror.* Every summand has a non-negative mean by Proposition 18.1(a) applied to the bounded Borel function $F = S_{ki}$ on the closed positive-side bonds; the covariance plays no role here. *Bond mirror.* First, $H(U) = \langle S(\theta U), {}^{U(b_x)}S(U)\rangle$ does not change under gauge transformations: from $\theta(U^v) = (\theta U)^{v\circ\theta}$ one gets $S(\theta(U^v)) = v(x_-)S(\theta U)v(x_-)^{-1}$, while $U^v(b_x) = v(x_-)U(b_x)v(x_+)^{-1}$. Let $\mathcal{C}$ denote the bonds that cross the plane $x_\nu = c$ or the plane $x_\nu = c + \frac{\ell}{2}$, each oriented so that it ends in $\mathbb{E}_+$; as $\frac{\ell}{2} \ge 2a_K$, no site is an end point of two of them. Given U, let v_U equal $U(b)$ at the end point in $\mathbb{E}_+$ of each $b \in \mathcal{C}$, and $\mathbf{1}$ at all remaining sites; then U^{v_U} is $\mathbf{1}$ on $\mathcal{C}$. For fixed

values on $\mathcal{C}$, the substitution $U|_{\mathcal{C}^c} \mapsto U^{v_U}|_{\mathcal{C}^c}$ multiplies every bond variable on both sides by group elements that do not depend on $U|_{\mathcal{C}^c}$, and therefore preserves the Haar measure. Since $H e^{-\beta\sum_p a_p}$ is gauge invariant, it follows that $Z\langle H\rangle_K = \int [H e^{-\beta\sum_p a_p}](\mathbf{1}_{\mathcal{C}}, U_+, U_-)\, dU_+ dU_-$, where $U_\pm$ is the restriction of U to the bonds with both end points in $\mathbb{E}_\pm = \{\pm(x_\nu - c) > 0\}$, heights being read on the circle. For bounded Borel functions Γ, Γ' of U_+, $\Theta\Gamma = \Gamma(\theta\,\cdot)$ depends on U_- only, and $\int \overline{\Theta\Gamma}\,\Gamma'\, dU_+ dU_- = \overline{\int \Gamma\, dU_+}\int \Gamma'\, dU_+$: indeed θ relabels the bonds and inverts the variables on the bonds along e_ν, and both operations preserve the Haar measure. When $U|_{\mathcal{C}} = \mathbf{1}$, every plaquette has either all its corners in $\mathbb{E}_+$, or all in $\mathbb{E}_-$, or it is made of a bond b_p of $\mathbb{E}_+$ parallel to the planes, the bond θb_p and two bonds of $\mathcal{C}$; in the last case $\operatorname{tr}\partial U(p) = \sum_{i,k} C_{p,ik}\overline{\Theta C_{p,ik}}$ with $C_{p,ik} = U(b_p)_{ik}$. Hence, writing A_+ for the action of the plaquettes inside $\mathbb{E}_+$ and $N_\times$ for the number of crossing plaquettes, $e^{-\beta\sum_p a_p} = e^{-\beta A_+}\,\Theta(e^{-\beta A_+})\, e^{-\beta N_\times}\sum_{n\ge 0}\frac{(\beta/2)^n}{n!}\sum_{(p_l,i_l,k_l)_{l\le n}}\prod_l C_{p_l,i_l k_l}\overline{\Theta\prod_l C_{p_l,i_l k_l}}$, with uniform convergence; and at $U(b_x) = \mathbf{1}$ one has $H = \frac12\sum_{i,k}\overline{\Theta S_{ki}}\, S_{ki}$. Inserting both expansions, $Z\langle H\rangle_K = e^{-\beta N_\times}\sum_n \frac{(\beta/2)^n}{n!}\sum_{(p_l,i_l,k_l)}\frac12\sum_{i,k}\bigl|\int e^{-\beta A_+}\prod_l C_{p_l,i_l k_l}\, S_{ki}\, dU_+\bigr|^2 \ge 0$. □

Lemma 18.86 (A bound on second differences for kernels of positive type)**.** Let $n \ge 1$ be an integer, and let $e\colon \{-n,\dots,n\} \to \mathbb{R}$ satisfy $\sum_{\alpha,\beta}\xi_\alpha\xi_\beta\, e(\alpha+\beta) \ge 0$ whenever ξ is a finitely supported real family whose support is contained both in $[-\frac{n}{2}, \frac{n}{2}]$ and in a single one of the classes $\mathbb{Z}$, $\mathbb{Z}+\frac12$. Then $0 \le e(1) + e(-1) - 2e(0) \le n^{-2}[e(n) + e(-n) - 2e(0)]$. ■

Sketch of proof. Introduce a formal variable x and the linear functional $\mathcal{L}$ on the span of the x^k, $|k| \le n$, given by $\mathcal{L}(x^k) = e(k)$. In this language the hypothesis reads $\mathcal{L}(p^2) \ge 0$ for every $p = \sum_\alpha \xi_\alpha x^\alpha$ with ξ as above. Write $\mathsf{q} = x^{1/2} - x^{-1/2}$, so that $\mathsf{q}^2 = x - 2 + x^{-1}$; let $r(\gamma) = \mathcal{L}(\mathsf{q}^2 x^\gamma) = e(\gamma+1) + e(\gamma-1) - 2e(\gamma)$ for $|\gamma| \le n-1$, and $\Delta_k = e(k) + e(-k) - 2e(0) = \mathcal{L}((x^{k/2} - x^{-k/2})^2)$. First, $\Delta_1 = \mathcal{L}(\mathsf{q}^2) \ge 0$. Next, for $0 \le \gamma \le n-1$ the product $\mathsf{q}(x^{\gamma/2} - x^{-\gamma/2})$ has four exponents, all in $[-\frac{n}{2}, \frac{n}{2}]$ and all in the same class; so $0 \le \mathcal{L}(\mathsf{q}^2(x^\gamma - 2 + x^{-\gamma})) = r(\gamma) + r(-\gamma) - 2r(0)$. Finally $x^{n/2} - x^{-n/2} = \mathsf{q}\sum_{a\in I_n} x^a$, where $I_n = \{\frac{n-1}{2}, \frac{n-3}{2}, \dots, -\frac{n-1}{2}\}$ is symmetric about 0; therefore $\Delta_n = \sum_{a,b\in I_n} r(a+b) = \sum_{a,b\in I_n}\frac12[r(a+b) + r(-a-b)] \ge n^2 r(0) = n^2\Delta_1$. □

Lemma 18.87 (Block bonds compared with straight bare segments)**.** (a) Let $K \ge 1$, $\beta > 0$, $0 \le i \le K$ and $n \ge 0$. If $(n+1)a_i < \frac{\ell}{4}$, then $\sigma_i(n+1) \le \sigma_i(n) + \sigma_i(1)$; if $(2n+2)a_i < \frac{\ell}{4}$, then $\sigma_i(2n+1) \le \sqrt{2}\,\sigma_i(n) + \sigma_i(1)$. Consequently $\sigma_i(n) \le c_{\mathrm{d}}(\sqrt{n+1} - 1)\sigma_i(1) \le c_{\mathrm{d}}\sqrt{n}\,\sigma_i(1)$ whenever $(n+1)a_i < \frac{\ell}{4}$.

(b) Assume the one-torus setting, and let $\mathfrak{t} \ge 1$ and $D_1 \le i_* < K$ be integers. Then

$$\sigma_{i_*}(1) \le 0.27\, c_{\mathrm{d}}^{\lceil (K-i_*)/\mathfrak{t}\rceil}\Bigl[23\, c_\rho L^{\mathfrak{t}/2+1}\max_{i_*\le i'\le K} L^{i'/2}\varepsilon_{[i']} + 2L^{K/2}R_{i_*}^{1/2}\Bigr]L^{-i_*/2}.$$

(c) On the diagonal, for large s and every i with $\lfloor s/5\rfloor \le i \le s$: $\sigma_i(1) \le \mathsf{C}_3\,\mathsf{e}\, L^{s/40 - i/2}$ with $\mathsf{C}_3 = 32(c_\rho + 1)L^{15501}$.

■

Sketch of proof. (a) For $i = K$ every σ_K vanishes, so let $i < K$; write $x_k = x + k a_i e_\nu$ and $b_k = \langle x_{k-1}, x_k\rangle$. The rule $\mathsf{L}(p,q)\mathsf{L}(q,w) = \mathsf{L}(p,w)$ gives $Z_i(x; n+1) = V_i([x, x_n])\, z_i(b_{n+1})\,\mathsf{L}(x_n, x)$, from which the first inequality follows by Lemma 18.78(a); it also gives $Z_i(x; 2n+1) = Z^{(1)} \cdot {}^{\mathsf{L}(x,x_n)}z_i(b_{n+1}) \cdot {}^{\mathsf{L}(x,x_{n+1})}Z^{(2)}$, where $Z^{(1)} = Z_i(x; n)$ and $Z^{(2)} = Z_i(x_{n+1}; n)$. By Lemma 18.78(a), $\sigma_i(2n+1) \le \sigma_i(1) + \|Y - \mathbf{1}\|_2$ for $Y = Z^{(1)} \cdot {}^{\mathsf{L}(x,x_{n+1})}Z^{(2)}$, and $\|Y - \mathbf{1}\|_2^2 = 2 - 2\,\mathbb{E}\operatorname{Re} Y$. *The mirror.* We reflect at the midpoint $c = (x_n)_\nu + \frac{a_i}{2}$ of b_{n+1}. Since $a_i/a_K = L^{K-i}$ is odd, $c \in a_K(\mathbb{Z} + \frac12)$; so this is a bond mirror of the cutoff K, and its reflection θ belongs to Iso_i. Let $x_\pm$ be the sites of the column of x next to the mirror, as in Lemma 18.85, and put $\mathcal{Y} = {}^{\mathsf{L}(x_+,x_{n+1})}Z^{(2)}$; it is G-valued and covariant at x_+. Only bonds with both end points in $\mathbb{E}_+$ enter $\mathcal{Y}$. The one-step locality in Proposition YM3_I:4.5 iterates: each finer block about an endpoint in a given block is contained in that block. Consequently

$V_i(b_k)$ for $k \ge n+2$ is a function of U on the bonds with end points in $B^{K-i}(x_{k-1}) \cup B^{K-i}(x_k)$, and the lowest of these end points have the height $(x_{n+1})_\nu - \frac12(a_i - a_K) = c + \frac12 a_K$; the bare transports involved join x_+ to x_{2n+1}; and every height stays below $c + (n+2)a_i < c + \frac{\ell}{2}$. *The reflected quantity.* We have $\theta x_{n+1+k} = x_{n-k}$, and block equivariance gives $V_i[\theta U](\langle x_{n+k}, x_{n+k+1}\rangle) = V_i(\langle x_{n+1-k}, x_{n-k}\rangle)$, whence $V_i[\theta U]([x_{n+1}, x_{2n+1}]) = V_i([x, x_n])^{-1}$. Combined with $\mathsf{L}[\theta U](p,q) = \mathsf{L}(\theta p, \theta q)$, this yields $Z^{(2)}(\theta U) = ({}^{\mathsf{L}(x_n,x)}Z^{(1)})^{-1}$ and $\mathcal{Y}(\theta U) = ({}^{\mathsf{L}(x_-,x)}Z^{(1)})^{-1}$. As Re is unchanged by conjugation and $\mathsf{L}(x_-, x_{n+1}) = U(b_x)\mathsf{L}(x_+, x_{n+1})$, we obtain $\operatorname{Re} Y = \operatorname{Re}[\mathcal{Y}(\theta U)^{-1} \cdot {}^{U(b_x)}\mathcal{Y}(U)]$. Split $\mathcal{Y} = \mathsf{c} + i\mathsf{S}$, where $\mathsf{c} = \operatorname{Re}\mathcal{Y}$ is real and gauge invariant and $\mathsf{S} = \operatorname{Im}\mathcal{Y}$; then $\mathcal{Y}^{-1} = \mathsf{c} - i\mathsf{S}$, and since S and its conjugates are traceless, $\operatorname{Re} Y = \mathsf{c}(\theta U)\mathsf{c}(U) + \langle \mathsf{S}(\theta U), {}^{U(b_x)}\mathsf{S}(U)\rangle$. Cauchy–Schwarz in the scalar reflection form, applied to 1 and c and using reflection invariance, gives $\mathbb{E}[\mathsf{c}(\theta U)\mathsf{c}(U)] \ge (\mathbb{E}\mathsf{c})^2$, and by Lemma 18.85 the second term has a non-negative mean. Put $w = \mathbb{E}\mathsf{c} = \mathbb{E}\operatorname{Re} Z^{(2)} = 1 - \frac12\sigma_i(n)^2 \in [-1,1]$; then $\|Y - \mathbf{1}\|_2^2 \le 2 - 2w^2 \le 4(1-w) = 2\sigma_i(n)^2$. *Induction.* If $\sigma_i(1) = 0$, the two inequalities give $\sigma_i(n) = 0$ for all admissible n; otherwise divide by $\sigma_i(1)$ and assume $\sigma_i(1) = 1$. The bound holds for $n = 0, 1, 2$, because $c_{\mathrm{d}}(\sqrt{2} - 1) = 2$ and $c_{\mathrm{d}}(\sqrt{3} - 1) \ge 2$. For $n \ge 3$ with $(n+1)a_i < \frac{\ell}{4}$ let $n' = \lfloor \frac{n-1}{2} \rfloor$; as $2n' + 2 \le n + 1$, both inequalities may be used, and $\sigma_i(n) \le \sqrt{2}\sigma_i(n') + 2 \le c_{\mathrm{d}}\sqrt{2n'+2} - \sqrt{2}c_{\mathrm{d}} + 2 = c_{\mathrm{d}}\sqrt{2n'+2} - c_{\mathrm{d}} \le c_{\mathrm{d}}(\sqrt{n+1} - 1)$. (b) On $\mathsf{G}_{i_*}^K$ (Lemma 18.84(a)) we have $\varpi(V_{i'}) < 2c_\rho\varepsilon_{[i']} \le \omega_M$ for $i_* < i' \le K$, the last bound by the choice of d_Θ, as $i' > i_* \ge D_1 \ge d_\Theta$; so Lemma 18.78(d) gives $|V_{i'-1}(c')V_{i'}[c']^{-1} - \mathbf{1}| \le 16c_\rho L^2\varepsilon_{[i']}$ for these i' and every bond c' of $\Lambda_{i'-1}$. *One step.* Let $i_* \le i - \mathfrak{t}' < i \le K$ with $1 \le \mathfrak{t}' \le \mathfrak{t}$, let $c \parallel e_\nu$ be a bond of $\Lambda_{i-\mathfrak{t}'}$, and let $V_{i'}[[c]]$ be the transport of $V_{i'}$ along the straight path of $L^{i'-i+\mathfrak{t}'}$ bonds of $\Lambda_{i'}$ from c_- to c_+. For $i - \mathfrak{t}' < i' \le i$, $V_{i'-1}[[c]]V_{i'}[[c]]^{-1}$ is a product of $L^{i'-1-i+\mathfrak{t}'}$ conjugates of factors $V_{i'-1}(c')V_{i'}[c']^{-1}$, hence within $16c_\rho L^2 L^{i'-1-i+\mathfrak{t}'}\varepsilon_{[i']}$ of $\mathbf{1}$. Summing over i', with $\varepsilon_{[i']} \le L^{(i-i')/2}\varepsilon_{[i]}$ and $\sum_{q\ge0} L^{-q/2} \le 1.384$, gives $|V_{i-\mathfrak{t}'}(c)V_i[[c]]^{-1} - \mathbf{1}| \le 23c_\rho L^{\mathfrak{t}'+1}\varepsilon_{[i]}$ on $\mathsf{G}_{i_*}^K$; off this event the left-hand side cannot exceed 2, and the event fails with probability at most R_{i_*}. Since $z_{i-\mathfrak{t}'}(c) = (V_{i-\mathfrak{t}'}(c)V_i[[c]]^{-1})\, Z_i(c_-; L^{\mathfrak{t}'})$ and $(L^{\mathfrak{t}'} + 1)a_i \le 2a_{i-\mathfrak{t}'} \le 2 < \frac{\ell}{4}$, part (a) gives $\sigma_{i-\mathfrak{t}'}(1) \le c_{\mathrm{d}}L^{\mathfrak{t}'/2}\sigma_i(1) + 23c_\rho L^{\mathfrak{t}'+1}\varepsilon_{[i]} + 2R_{i_*}^{1/2}$. In terms of $y_i = L^{i/2}\sigma_i(1)$ this reads $y_{i-\mathfrak{t}'} \le c_{\mathrm{d}}y_i + \mathsf{f}$, with f the bracket in the claim. *Chaining.* Descend from K, where $y_K = 0$, to i_* in $q = \lceil (K - i_*)/\mathfrak{t} \rceil$ steps, each of length at most $\mathfrak{t}$: $y_{i_*} \le \mathsf{f}(1 + c_{\mathrm{d}} + \cdots + c_{\mathrm{d}}^{q-1}) \le \frac{c_{\mathrm{d}}^q}{c_{\mathrm{d}}-1}\mathsf{f} \le 0.27\, c_{\mathrm{d}}^q \mathsf{f}$, the same event $\mathsf{G}_{i_*}^K$ being used at every step. (c) Here $\mathfrak{t} = 31000$, $q \le \frac{K}{\mathfrak{t}} + 1$ and $c_{\mathrm{d}}^{K/\mathfrak{t}} = L^{s\cdot\frac{1260}{31000}\log_L c_{\mathrm{d}}} \le L^{0.02496\,s} \le L^{s/40}$, as $L \ge 13$; $L^{i'/2}\varepsilon_{[i']} \le \mathsf{e}$; $L^{K/2}R_i^{1/2} \le \mathsf{A}_R^{1/2}L^{630s+3i/2}e^{-a_R i^2/2}$ with $i \ge \frac{s}{5} - 1$, which is at most $\frac12 L^{\mathfrak{t}/2+1}$ for large s; and $0.27\, c_{\mathrm{d}}(23c_\rho + 1) \le 32(c_\rho + 1)$. For large s also $\lfloor s/5 \rfloor \ge D_1$, so (b) applies with $i_* = i$. □

Lemma 18.88 (Defects stacked in a column transverse to the shift add up at most like independent ones)**.** Let $K > s > j \ge 0$, $\beta > 0$, $h = a_s e_\nu$, and $n \ge 1$ with $(n+2)a_j < \frac{\ell}{4}$. Take a bond $b_1 \perp e_\nu$ of Λ_j, its translates $b_k = b_1 + (k-1)a_j e_\nu$ with the lower ends $y_k = (b_k)_-$, a number $\eta > 0$, and the truncated defects $\hat\pi_k = \pi_j(b_k)\mathbf{1}\{|\pi_j(b_k)| \le \eta\}$, formed from $V_s = \mathcal{M}^{K-s}U$. Then $\|\sum_{k=1}^n {}^{\mathsf{L}(y_1,y_k)}\hat\pi_k\|_2 \le \eta\,(\sqrt{3n} + n\sqrt{2\sigma_s(1)})$. ■

Sketch of proof. Since conjugation is an isometry and $\mathsf{L}(y_1,y_k)^{-1}\mathsf{L}(y_1,y_l) = \mathsf{L}(y_k,y_l)$, we have $|\sum_k {}^{\mathsf{L}(y_1,y_k)}\hat\pi_k|^2 = \sum_{k,l}\langle \hat\pi_k, {}^{\mathsf{L}(y_k,y_l)}\hat\pi_l\rangle$. There are at most $3n$ pairs with $|k-l| \le 1$, each contributing at most η^2. Now let $l \ge k+2$ (the case $l \le k-2$ is symmetric), and let t_k denote the height of b_k. *The mirror.* We place it at the height $c = \frac12(t_k + t_l - a_s) \in \frac{a_K}{2}\mathbb{Z}$. Because $(l-k)a_j - a_s = a_s((l-k)L^{s-j} - 1)$ with L^{s-j} odd, the mirror is of the site type when $l-k$ is odd and of the bond type when $l-k$ is even. Its reflection is $\vartheta = \mathsf{r} \circ \mathrm{t}_h$, where r is the reflection at the height $\frac12(t_k + t_l) \in \frac{a_j}{2}\mathbb{Z}$; thus $\mathsf{r} \in \mathsf{Iso}_j$, $\vartheta \in \mathsf{Iso}_s$, $\mathsf{r}b_l = b_k$ and $\vartheta y_l = y_k - h$. Block equivariance gives $V_j[\vartheta U] = \mathsf{r}V_j'$, $V_j'[\vartheta U] = \mathsf{r}V_j$ and $V_s[\vartheta U] = \vartheta V_s$, so that $g[\vartheta U](x) = g(\mathsf{r}x)^{-1}$. Therefore $\mathsf{P}_j(b_l)[\vartheta U] = g(y_k)V_j(b_k)g(b_{k+})^{-1}V_j'(b_k)^{-1} = {}^{g(y_k)}(\mathsf{P}_j(b_k)^{-1})$, and Lemma 18.78(a) gives $\hat\pi_l \circ \vartheta = -{}^{g(y_k)}\hat\pi_k$, the truncation depending on the norm

only. Consequently $\langle\hat\pi_k, {}^{\mathsf{L}(y_k,y_l)}\hat\pi_l\rangle = -\langle\hat\pi_l\circ\vartheta, {}^{g(y_k)\mathsf{L}(y_k,y_l)}\hat\pi_l\rangle$. *Replacing g by a bare transport.* Since $g(y_k)\mathsf{L}(y_k,y_k-h) = z_s(\langle y_k-h,y_k\rangle)$, we get $|g(y_k)\mathsf{L}(y_k,y_l)-\mathsf{L}(y_k-h,y_l)| = |z_s(\langle y_k-h,y_k\rangle)-\mathbf{1}|$, a quantity of mean at most $\sigma_s(1)$; by Lemma 18.78(a), making this replacement changes the mean of the pairing by at most $2\sigma_s(1)\eta^2$. *Positivity.* Let $x_\pm$ be the sites of the column of the y_k next to the mirror, and $S = {}^{\mathsf{L}(x_+,y_l)}\hat\pi_l$, which is covariant at x_+ by Lemma 18.83(b). By the locality of the averaging, $\mathsf{P}_j(b_l)$ depends on U only through the bonds with end points in $B^{K-j}((b_l)_\pm)$, in the translates of these blocks by $-h$, and in $B^{K-s}((b_l)_\pm)$, $B^{K-s}((b_l)_\pm-h)$. All these end points have heights at least $t_l-a_s-\frac12(a_j-a_K)$, which is $\ge c+\frac12 a_K$ because $t_l-t_k\ge 2a_j\ge a_j+a_s$, and at most $t_l+\frac12 a_j < c+\frac{\ell}{2}$. Hence S meets the hypotheses of Lemma 18.85 for either type. Moreover $S(\vartheta U) = {}^{\mathsf{L}(x_-,y_k-h)}(\hat\pi_l\circ\vartheta)$ and $\mathsf{L}(y_k-h,x_-)U(b_x)\mathsf{L}(x_+,y_l) = \mathsf{L}(y_k-h,y_l)$, so $\langle S,S\rangle_\Sigma = \mathbb{E}\langle\hat\pi_l\circ\vartheta, {}^{\mathsf{L}(y_k-h,y_l)}\hat\pi_l\rangle \ge 0$. *Collecting.* For $|k-l|\ge 2$ this gives $\mathbb{E}\langle\hat\pi_k, {}^{\mathsf{L}(y_k,y_l)}\hat\pi_l\rangle \le 2\sigma_s(1)\eta^2$, and therefore $\|\sum_k {}^{\mathsf{L}(y_1,y_k)}\hat\pi_k\|_2^2 \le 3n\eta^2+2n^2\sigma_s(1)\eta^2$. □

Lemma 18.89 (Defects in a column along the shift, controlled through increments of a correlation of positive type)**.** Let $K>s>j\ge 0$, $\beta>0$, $h=a_se_\nu$, and $n\ge 1$ with $2(n+2)a_j<\frac{\ell}{4}$. Take a positively oriented bond $b_1\parallel e_\nu$ of Λ_j, $b_k=b_1+(k-1)a_je_\nu$ and $y_k=(b_k)_-$, and set $\mathfrak{a}_k=\operatorname{Im} z_j(b_k)$, $\mathfrak{a}_k^\flat(U)=\mathfrak{a}_k(\mathrm{t}_hU)$, $\tilde{\mathfrak{a}}_k = {}^{\mathsf{L}(y_k,y_k-h)}\mathfrak{a}_k^\flat$ and $\mathsf{v}=\mathbb{E}|\tilde{\mathfrak{a}}_1-\mathfrak{a}_1|^2$. Then $\|\sum_{k=1}^n {}^{\mathsf{L}(y_1,y_k)}(\tilde{\mathfrak{a}}_k-\mathfrak{a}_k)\|_2^2 \le n\,[5\mathsf{v}+7L^{-2(s-j)}\sigma_j(1)^2]$. Moreover, writing $\mathsf{P}_j^{\mathsf{L}}(b)=\mathsf{L}(b_-,b_--h)\,V_j'(b)\,\mathsf{L}(b_+-h,b_+)\,V_j(b)^{-1}$, the identity $\tilde{\mathfrak{a}}_k-\mathfrak{a}_k=\operatorname{Im}\mathsf{P}_j^{\mathsf{L}}(b_k)+\operatorname{Im}[(\mathsf{P}_j^{\mathsf{L}}(b_k)-\mathbf{1})(z_j(b_k)-\mathbf{1})]$ holds exactly, $|\tilde{\mathfrak{a}}_k-\mathfrak{a}_k|\le|\mathsf{P}_j^{\mathsf{L}}(b_k)-\mathbf{1}|$, and $|\mathsf{P}_j^{\mathsf{L}}(b)-\mathsf{P}_j(b)|\le|z_s(\langle b_--h,b_-\rangle)-\mathbf{1}|+|z_s(\langle b_+-h,b_+\rangle)-\mathbf{1}|$ for every bond b of Λ_j. ■

Sketch of proof. The identity. Fix $b=b_k$ and write $z=z_j(b)$. Since $V_j[\mathrm{t}_hU]=V_j'$ and $\mathsf{L}[\mathrm{t}_hU](p,q)=\mathsf{L}(p-h,q-h)$, we get $z^\flat = z\circ\mathrm{t}_h = V_j'(b)\mathsf{L}(b_+-h,b_--h)$; let $\tilde z = {}^{\mathsf{L}(b_-,b_--h)}z^\flat$. Along one column $\mathsf{L}(b_+-h,b_--h)\mathsf{L}(b_--h,b_-)\mathsf{L}(b_-,b_+)=\mathsf{L}(b_+-h,b_+)$, so $\tilde z z^{-1}=\mathsf{P}_j^{\mathsf{L}}(b)=:P$. Then $\tilde{\mathfrak{a}}-\mathfrak{a}=\operatorname{Im}(Pz-z)$, where $Pz-z=(P-\mathbf{1})+(P-\mathbf{1})(z-\mathbf{1})$, and $|\operatorname{Im}(Pz-z)|\le|P-\mathbf{1}|$. In P_j, compared with $\mathsf{P}_j^{\mathsf{L}}$, the factor $g(b_-)^{-1}$ stands in place of $\mathsf{L}(b_-,b_--h)$ and $g(b_+)$ in place of $\mathsf{L}(b_+-h,b_+)$; and $|g(x)-\mathsf{L}(x-h,x)|=|z_s(\langle x-h,x\rangle)-\mathbf{1}|$. *Shifted copies.* For $y\in a_K\mathbb{Z}$ let y be the site of the column at the height y, and $\mathfrak{a}^{(y)}(U)=\mathfrak{a}_1(\mathrm{t}_{(t_1-y)e_\nu}U)$, t_k being the height of y_k. Block equivariance gives $\mathfrak{a}^{(t_k)}=\mathfrak{a}_k$ and $\mathfrak{a}^{(t_k-a_s)}=\mathfrak{a}_k^\flat$. Each $\mathfrak{a}^{(y)}$ satisfies $|\mathfrak{a}^{(y)}|\le 1$, is covariant at y, and depends on U only at heights in $[y-\frac12(a_j-a_K),\,y+a_j+\frac12(a_j-a_K)]$. *Behaviour under reflections.* For $c\in\frac{a_K}{2}\mathbb{Z}$ let θ_c be the reflection at the height c, and $y^*=2c-y-a_j$. Then $\mathfrak{a}^{(y)}\circ\theta_c = -{}^{\mathsf{L}(\mathsf{y}^*+a_je_\nu,\mathsf{y}^*)}\mathfrak{a}^{(y^*)}$. To see this, reflect at the midpoint $c_1=t_1+\frac{a_j}{2}$ of b_1: $z_j(b_1)[\theta_{c_1}U]=V_j(b_1)^{-1}\mathsf{L}(b_{1-},b_{1+})=({}^{\mathsf{L}(b_{1+},b_{1-})}z_j(b_1))^{-1}$; and on heights, θ_c followed by the translation through t_1-y equals the translation through t_1-y^* followed by θ_{c_1}. *A correlation function.* For $T\in a_K\mathbb{Z}$ with $0\le T<\frac{\ell}{2}$, $C(T)=\mathbb{E}\langle\mathfrak{a}^{(y)}, {}^{\mathsf{L}(\mathsf{y},\mathsf{y}+Te_\nu)}\mathfrak{a}^{(y+T)}\rangle$ is independent of y, and $|C(T)|\le\sigma_j(1)^2$; let $v=-C$. *Positive type.* Let $c\in\frac{a_K}{2}\mathbb{Z}$, let $y_\alpha\in a_K\mathbb{Z}$ be finitely many heights with $c+\frac{a_j}{2}\le y_\alpha<c+\frac{\ell}{4}-a_j$, and let ξ_α be real. Then $S=\sum_\alpha\xi_\alpha\,{}^{\mathsf{L}(x_+,\mathsf{y}_\alpha)}\mathfrak{a}^{(y_\alpha)}$ meets the hypotheses of Lemma 18.85 at mirrors of both types: each $\mathfrak{a}^{(y_\alpha)}$ depends on U only at heights $\ge c+\frac12 a_K$, the transport $\mathsf{L}(x_+,\mathsf{y}_\alpha)$ climbs the column starting at x_+, whose height is $c+\frac12 a_K$ at a bond mirror and c at a site mirror, and all these heights stay below $c+\frac{\ell}{2}$; so every bond that enters S lies in $\mathsf{B}(\Sigma)$. By the reflection rule, $S_\alpha(\theta U)=-{}^{\mathsf{L}(x_-,\mathsf{y}_\alpha^*)}\mathfrak{a}^{(y_\alpha^*)}$, so that $\mathbb{E}\langle S_\alpha(\theta U), {}^{U(b_x)}S_\beta(U)\rangle = v(y_\alpha+y_\beta+a_j-2c)$; hence $\sum_{\alpha,\beta}\xi_\alpha\xi_\beta\,v(y_\alpha+y_\beta+a_j-2c)=\langle S,S\rangle_\Sigma\ge 0$. *Pairs of distinct bonds.* For $l>k$ and $T=t_l-t_k$, the pairing $D_{kl}=\mathbb{E}\langle\tilde{\mathfrak{a}}_k-\mathfrak{a}_k, {}^{\mathsf{L}(y_k,y_l)}(\tilde{\mathfrak{a}}_l-\mathfrak{a}_l)\rangle$ expands into four correlations, with the arguments T, T, $T+a_s$ and $T-a_s$; thus $D_{kl}=v(T+a_s)+v(T-a_s)-2v(T)$. Suppose $l-k\ge 3$, put $n_{kl}=(l-k-2)L^{s-j}$, so that $n_{kl}a_s=T-2a_j$, and $e(\gamma)=v(T+\gamma a_s)$ for $|\gamma|\le n_{kl}$. Given ξ supported in $[-\frac{n_{kl}}{2},\frac{n_{kl}}{2}]$ and in one class, choose $y_\circ\in a_K\mathbb{Z}$ for the class $\mathbb{Z}$ and $y_\circ\in a_K(\mathbb{Z}+\frac12)$ for the class $\mathbb{Z}+\frac12$ (recall that a_s/a_K is odd), and set $y_\alpha=y_\circ+\alpha a_s$ and $c=y_\circ+\frac12(a_j-T)$. Then $y_\alpha+y_\beta+a_j-2c=T+(\alpha+\beta)a_s$, $y_\alpha\ge c+\frac{a_j}{2}$, and $y_\alpha-c\le T-\frac32 a_j<\frac{\ell}{4}-a_j$ because $2T<\frac{\ell}{4}$. So

the positivity just established is precisely the hypothesis of Lemma 18.86, with both classes and both types of mirrors occurring. That lemma, together with $e(0) = v(T) \geq 0$ (positivity for a single α), yields $0 \leq D_{kl} \leq n_{kl}^{-2}[e(n_{kl}) + e(-n_{kl}) - 2e(0)] \leq 2n_{kl}^{-2}\sigma_j(1)^2$. *Summation.* By the Schwarz inequality and translation invariance, each of the at most $5n$ pairs with $|k-l| \leq 2$ contributes at most v; all remaining pairs together contribute at most $n \cdot 2\sum_{q\geq 1} 2q^{-2}L^{-2(s-j)}\sigma_j(1)^2 \leq 7nL^{-2(s-j)}\sigma_j(1)^2$. □

Lemma 18.90 (At a coarse depth the defect is, after a small gauge change and up to small errors, an average of finer defects along straight paths)**.** Let $D_g \leq d < D < j < s$ be integers, $\mathfrak{m} = j - D$, $N = L^{\mathfrak{m}}$, with

$$(\mathrm{Z}^\sharp) \qquad \mathsf{k}_2\mathsf{E}_s L^{-D} \leq \min\{c_*, \tfrac{\rho_L}{200}\}, \qquad \mathsf{k}_1(2+s)^2\varepsilon_{[s]} \leq \tfrac{\rho_L}{200}, \qquad \mathsf{k}_1(2+s)L^j\varepsilon_{[s]} \leq \mathsf{k}_2\mathsf{E}_s,$$

and put $\mathsf{H}_s = \mathsf{k}_2\mathsf{E}_s + \mathsf{k}_1(2+s)L^j\varepsilon_{[s]}$ with the j of this lemma; then $\mathsf{H}_s \leq 2\mathsf{k}_2\mathsf{E}_s$, and $L^i\bar\eta_i \leq \mathsf{H}_s$ for $i \leq j$. Let $h = \pm a_s e_\nu$, $V \in \mathsf{Gd}_{d,s}$, c_0 a bond of Λ_d, u and Φ_i, $d \leq i \leq s$, the objects of Lemma 18.82 at $j_0 = d$, $\mathcal{R}_i$ the set of the bonds of Λ_i with both end points in $B^{i-d}(c_{0-}) \cup B^{i-d}(c_{0+})$, and $E_i'(b) = \mathsf{u}(b_- - h)V_i'(b)\mathsf{u}(b_+ - h)^{-1}$ for $b \in \mathcal{R}_i$. Let i_1 be the least integer $\geq D_g$ with $\mathsf{k}_2\mathsf{E}_s L^{-i_1} \leq \min\{c_*, \frac{\rho_L}{200}\}$; then $i_1 \leq D$ and $L^{i_1} \leq L\max\{L^{D_g}, \mathsf{k}_2\mathsf{E}_s/\min\{c_*, \frac{\rho_L}{200}\}\}$.

(a) $E_i' = \mathcal{M}^{s-i}(e^{i\,\mathrm{t}_h\Phi_s})$ on $\mathcal{R}_i$, and $||E_i'(b) - e^{i\Phi_i(b)}| - |\mathsf{P}_i(b) - \mathbf{1}|| \leq 2C_g\varepsilon_{[s]}$ for $d \leq i \leq s$ and $b \in \mathcal{R}_i$. For $\max\{d, i_1\} \leq i \leq s$: $|E_i'(b) - e^{i\Phi_i(b)}| \leq \bar\eta_i + 2C_g\varepsilon_{[s]}$, $E_i' = e^{i\Phi_i'}$ on $\mathcal{R}_i$, and $\delta_i = \Phi_i' - \Phi_i$ has $|\delta_i| \leq 3\bar\eta_i$, $|\Phi_i'| \leq \rho_L$.

(b) For $b \in \mathcal{R}_j$: $|\delta_j(b) - {}^{\mathsf{u}(b_-)}\pi_j(b)| \leq \varrho^{(3)} = 3C_g\varepsilon_{[j]}\bar\eta_j + 5\bar\eta_j^3 + 2C_g\varepsilon_{[s]}$.

(c) Let $\zeta = \zeta_{\mathfrak{m}}(\delta_j)$ (Lemma 18.79(b)), defined at the sites of Λ_D that lie in $B^{D-d}(c_{0-}) \cup B^{D-d}(c_{0+})$, and $\tilde E_D(c) = e^{i\zeta(c_-)}E_D'(c)e^{-i\zeta(c_+)}$. Then $|\zeta| \leq 4.5L^{-D}\mathsf{H}_s$; $\tilde E_D = e^{i\tilde\Phi_D}$ on $\mathcal{R}_D$; and $\tilde\delta_D = \tilde\Phi_D - \Phi_D$ satisfies $|\tilde\delta_D(c) - \mathsf{M}(c)| \leq \varrho$ for $c \in \mathcal{R}_D$, where $\mathsf{M}(c) = L^{-3\mathfrak{m}}\sum_{x\in B^{\mathfrak{m}}(c_-)}\sum_{b\in[x,x+c_+-c_-]} {}^{\mathsf{u}(b_-)}\pi_j(b)$, the inner sum over the N bonds of a straight path of Λ_j, and $L^D\varrho \leq \mathsf{H}_s[(C_N+21)C_g\varepsilon_{[D]} + (C_N+100)L^{-D}\mathsf{H}_s] + 2C_gL^j\varepsilon_{[s]}$.

The objects u, Φ_i, Φ_i', ζ and $\tilde\delta_D$ can be chosen as Borel functions of V on $\mathsf{Gd}_{d,s}$. ■

Sketch of proof. (a) Write $\mathsf{u}' = \mathsf{u}(\cdot - h)$. For $b \in \mathcal{R}_s$ the bond $b - h$ lies in the box, and $(\mathrm{t}_hV)^{\mathsf{u}'}(b) = V^{\mathsf{u}}(b-h) = e^{i\Phi_s(b-h)}$; covariance and locality of the averaging then give $\mathcal{M}^{s-i}(e^{i\,\mathrm{t}_h\Phi_s})(b) = \mathsf{u}'(b_-)V_i'(b)\mathsf{u}'(b_+)^{-1} = E_i'(b)$. Let $\varsigma(x) = \mathsf{u}(x-h)g(x)\mathsf{u}(x)^{-1} = V^{\mathsf{u}}(\langle x-h, x\rangle)$, so that $|\varsigma(x) - \mathbf{1}| \leq C_g\varepsilon_{[s]}$. Using $V_i'(b) = g(b_-)\check V_i(b)g(b_+)^{-1}$ (Lemma 18.83(a)) one finds

$$E_i'(b) = \varsigma(b_-)\,{}^{\mathsf{u}(b_-)}\mathsf{P}_i(b)\,e^{i\Phi_i(b)}\,\varsigma(b_+)^{-1},$$

and the first inequality follows. If $\max\{d, i_1\} \leq i < s$, then $i \geq D_g$ and $\mathsf{k}_2L^{s-i}\varepsilon_{[s]}^2 = \mathsf{k}_2\mathsf{E}_sL^{-i} \leq c_*$, so Lemma 18.83(c) gives $|\mathsf{P}_i(b) - \mathbf{1}| \leq \bar\eta_i$; for $i = s$, $|\mathsf{P}_s(b) - \mathbf{1}| \leq \varpi(V) < 2\varepsilon_{[s]} \leq \bar\eta_s$ (Lemma 18.83(a)). Condition $(\mathrm{Z}^\sharp)$ gives $\bar\eta_i \leq \frac{\rho_L}{100}$; hence $|E_i' - \mathbf{1}| < 1$, the logarithm $\Phi_i' = -i\log E_i'$ exists, Lemma 18.78(b) gives $|\delta_i| \leq 2(\bar\eta_i + 2C_g\varepsilon_{[s]}) \leq 3\bar\eta_i$, and $|\Phi_i'| \leq \frac{\rho_L}{2} + \frac{3\rho_L}{100}$. (b) From the display, $|E_j'e^{-i\Phi_j} - {}^{\mathsf{u}(b_-)}\mathsf{P}_j(b)| \leq 2C_g\varepsilon_{[s]}$. Lemma 18.78(b) gives $|e^{i\Phi_j'}e^{-i\Phi_j} - e^{i\delta_j}| \leq |[\delta_j, \Phi_j]|/2 \leq |\delta_j||\Phi_j|$. Further, $|\operatorname{Im} e^{i\delta} - \delta| = |\delta| - \sin|\delta| \leq \frac{1}{6}|\delta|^3$. Finally Im is 1-Lipschitz and commutes with ${}^{\mathsf{u}(b_-)}$. (c) For $D < i \leq j$ part (a) applies, because $i \geq D \geq i_1$, and gives $\sup_{\mathcal{R}_i}|\Phi_i|, \sup_{\mathcal{R}_i}|\Phi_i'| \leq \rho_L$; Lemma 18.80(a) on $\mathcal{R}_{i-1}$ then gives $\delta_{i-1} = \mathbf{Q}\delta_i + r_i$ with $|r_i| \leq C_N(C_g\varepsilon_{[i]} + 3\bar\eta_i)\,3\bar\eta_i$. Iterating, and using Lemma 18.79(b), (c): $\delta_D = \mathbf{Q}^{\mathfrak{m}}\delta_j + \sum_{i=D+1}^{j}\mathbf{Q}^{i-1-D}r_i = \mathbf{S}^{\mathfrak{m}}\delta_j + \mathrm{d}\zeta + \rho^{(1)}$, where $|\rho^{(1)}| \leq \sum_{i=D+1}^{j} 4L^{i-1-D}\,3C_N(C_g\varepsilon_{[i]} + 3\bar\eta_i)\bar\eta_i$. With $L^i\bar\eta_i \leq \mathsf{H}_s$, $\sum_{i>D}\varepsilon_{[i]} \leq 0.62\varepsilon_{[D]}$ and $\sum_{i>D}L^{-i} = \frac{L^{-D}}{L-1}$ this gives $L^D|\rho^{(1)}| \leq C_N\mathsf{H}_s[C_g\varepsilon_{[D]} + L^{-D}\mathsf{H}_s]$. The value $\zeta(y)$ depends only on δ_j inside $B^{\mathfrak{m}}(y)$, and $|\zeta| \leq \frac{3}{2}(L^{\mathfrak{m}} - 1)\sup|\delta_j| \leq 4.5L^{\mathfrak{m}}\bar\eta_j \leq 4.5L^{-D}\mathsf{H}_s$. *Removing* $\mathrm{d}\zeta$ *by a gauge change.* Two applications of Lemma 18.78(b) give $|e^{i\zeta(c_-)}e^{i\Phi_D'(c)}e^{-i\zeta(c_+)} - e^{i(\Phi_D'(c) - (\mathrm{d}\zeta)(c))}| \leq |\zeta(c_-)||\Phi_D'(c)| + |\zeta(c_+)|(|\Phi_D'(c)| + |\zeta(c_-)|)$. All exponents having norm below 1.1, it follows that $\tilde\Phi_D = \Phi_D' - \mathrm{d}\zeta + \rho^{(2)}$ with $|\rho^{(2)}| \leq$

$2\sup|\zeta|(2\sup|\Phi'_D|+\sup|\zeta|)$, and, as $|\Phi'_D|\le C_g\varepsilon_{[D]}+3L^{-D}\mathsf{H}_s$, $L^D|\rho^{(2)}|\le 9\mathsf{H}_s[2C_g\varepsilon_{[D]}+10.5L^{-D}\mathsf{H}_s]$. *Combining.* We have $\tilde\delta_D=\mathbf{S}^{\mathfrak m}\delta_j+\rho^{(1)}+\rho^{(2)}$. The value $(\mathbf{S}^{\mathfrak m}\delta_j)(c)$ is a sum of $L^{3\mathfrak m}\cdot L^{\mathfrak m}$ values $\delta_j(b)$, each with the weight $L^{-3\mathfrak m}$; so (b) gives $|(\mathbf{S}^{\mathfrak m}\delta_j)(c)-\mathsf{M}(c)|\le L^{\mathfrak m}\varrho^{(3)}$, and $L^{D+\mathfrak m}\varrho^{(3)}\le \mathsf{H}_s(3C_g\varepsilon_{[D]}+5L^{-D}\mathsf{H}_s)+2C_gL^j\varepsilon_{[s]}$. *Measurability* follows from Lemma 18.82 and from that of the logarithm near $\mathbf{1}$. □

Lemma 18.91 (The mean of the defect on the diagonal)**.** Assume the one-torus setting; let s be large, $D,\mathfrak m,j,N,K$ on the diagonal, $D_g\le d<D$, c_0 a bond of Λ_d, $h=a_se_\nu$, and $\mathcal G=\{U\in\mathcal V_K:\mathcal M^{K-s}U\in\mathsf{Gd}_{d,s}\}$, the objects of Lemma 18.90 being formed from $V=\mathcal M^{K-s}U$ on $\mathcal G$. Then ($\mathsf{Z}^\sharp$) holds, and $X_D=\max_{c\in\mathcal R_D}\mathbb E[|\tilde\delta_D(c)|;\mathcal G]\le \mathsf{C}_7\,\mathsf{e}^4L^{-D}L^{-s/20}$ with a number C_7 formed from L and c_ρ. ■

Sketch of proof. Step 0. On the diagonal $D\ge\frac{s}{5}-1$ and $j\le\frac32D\le0.3s$. The quantities $\mathsf{k}_2\mathsf{E}_sL^{-D}\le\mathsf{k}_2\mathsf{e}^2L^{1-s/5}$, $\mathsf{k}_1(2+s)^2\varepsilon_{[s]}\le\mathsf{k}_1(2+s)^2\mathsf{e}L^{-s/2}$ and $\mathsf{k}_1(2+s)L^j\varepsilon_{[s]}\le\mathsf{k}_1(2+s)\mathsf{e}L^{-s/5}$ tend to 0, whereas $\mathsf{k}_2\mathsf{E}_s\ge\mathsf{k}_2A_0^2\mathsf{b}^{-1}$; so ($\mathsf{Z}^\sharp$) holds once s is large, and then $\mathsf{H}_s\le2\mathsf{k}_2\mathsf{e}^2$. *Step 1: reduction to columns.* Fix $c\in\mathcal R_D$. If the direction κ of c differs from ν, fix in $\mathsf{M}(c)$ the position along the straight path together with the two coordinates of x other than the height (N^3 choices); what remains is a sum over the N heights of x, namely $\Sigma=\sum_{k=1}^N{}^{\mathsf u(y_k)}\pi_j(b_k)$ over a column of bonds $b_k=b_1+(k-1)a_je_\nu\perp e_\nu$ with $y_k=(b_k)_-$. If $c\parallel e_\nu$, fix instead $x\in B^{\mathfrak m}(c_-)$; the sum along the path is again of the form Σ, now with $b_k\parallel e_\nu$. As $L^{-3\mathfrak m}N^3=1$, we get $\mathbb E[|\mathsf M(c)|;\mathcal G]\le\max\mathbb E[|\Sigma|;\mathcal G]$, where the maximum runs over all columns of N bonds of $\mathcal R_j$ of these two kinds. *Step 2: change of frame.* Put $\mathsf L^{\mathsf u}(y_1,y_k)=\mathsf u(y_1)\mathsf L(y_1,y_k)\mathsf u(y_k)^{-1}$. Then ${}^{\mathsf u(y_k)}X={}^{\mathsf L^{\mathsf u}(y_1,y_k)^{-1}\mathsf u(y_1)\mathsf L(y_1,y_k)}X$, and Lemma 18.78(a) gives $|\Sigma|\le|\sum_k{}^{\mathsf L(y_1,y_k)}\pi_j(b_k)|+2\sum_k|\mathsf L^{\mathsf u}(y_1,y_k)-\mathbf1||\pi_j(b_k)|$. Because $V_j^{\mathsf u}([y_1,y_k])\,\mathsf L^{\mathsf u}(y_1,y_k)^{-1}={}^{\mathsf u(y_1)}Z_j(y_1;k-1)$, and because on $\mathcal G$ we have $|V_j^{\mathsf u}([y_1,y_k])-\mathbf1|\le(k-1)C_g\varepsilon_{[j]}$ (the bonds of the column lie inside the box), we get $|\mathsf L^{\mathsf u}(y_1,y_k)-\mathbf1|\le|Z_j(y_1;k-1)-\mathbf1|+NC_g\varepsilon_{[j]}$. On $\mathcal G$, $|\pi_j(b)|\le|\mathsf P_j(b)-\mathbf1|\le\bar\eta_j$ by Lemma 18.83(c), whose hypotheses at the depth j follow from ($\mathsf{Z}^\sharp$); and $\mathbb E|Z_j(y_1;k-1)-\mathbf1|\le\sigma_j(k-1)\le c_{\mathrm d}\sqrt N\sigma_j(1)$ by Lemma 18.87(a). Hence the last sum has, on $\mathcal G$, a mean of at most $2N\bar\eta_j[c_{\mathrm d}\sqrt N\sigma_j(1)+NC_g\varepsilon_{[j]}]$. *Step 3: columns of bonds transverse to the shift.* On $\mathcal G$ the truncation at $\eta=\bar\eta_j$ changes nothing, $\pi_j(b_k)=\hat\pi_k$, and Lemma 18.88, whose hypothesis $(N+2)a_j<\frac{\ell}{4}$ is satisfied, gives $\mathbb E[|\sum_k{}^{\mathsf L(y_1,y_k)}\pi_j(b_k)|;\mathcal G]\le\bar\eta_j(\sqrt{3N}+N\sqrt{2\sigma_s(1)})$. *Step 4: columns of bonds along the shift.* Let $\zeta'_k=|z_s(\langle y_k-h,y_k\rangle)-\mathbf1|+|z_s(\langle y_{k+1}-h,y_{k+1}\rangle)-\mathbf1|$; then $\mathbb E\zeta'_k\le2\sigma_s(1)$ and $\mathbb E\zeta'^2_k\le4\sigma_s(1)^2$. Lemma 18.89 gives, on $\mathcal G$, $|\pi_j(b_k)-(\tilde{\mathfrak a}_k-\mathfrak a_k)|\le|\mathsf P_j(b_k)-\mathsf P_j^{\mathsf L}(b_k)|+|\mathsf P_j^{\mathsf L}(b_k)-\mathbf1||z_j(b_k)-\mathbf1|\le3\zeta'_k+\bar\eta_j|z_j(b_k)-\mathbf1|$, whose mean is at most $6\sigma_s(1)+\bar\eta_j\sigma_j(1)$. Moreover $\mathsf v\le2\mathbb E|\mathsf P_j(b_1)-\mathbf1|^2+2\mathbb E\zeta'^2_1\le2\bar\eta_j^2+8\frac{L}{L-1}R_j+8\sigma_s(1)^2$, using Lemma 18.83(c) on $\{\mathcal M^{K-s}U\in\mathsf{Gd}_{j,s}\}$ and Lemma 18.84(b) for the complement. Lemma 18.89, whose hypothesis $2(N+2)a_j<\frac{\ell}{4}$ is satisfied, together with $\sigma_j(1)\le2$, then gives $\mathbb E[|\sum_k{}^{\mathsf L(y_1,y_k)}\pi_j(b_k)|;\mathcal G]\le\sqrt N[\sqrt{10}\,\bar\eta_j+7R_j^{1/2}+7\sigma_s(1)+6L^{-(s-j)}]+N[6\sigma_s(1)+\bar\eta_j\sigma_j(1)]$. *Step 5: bookkeeping.* Lemma 18.90(c) and Steps 1–4, with $L^DN=L^j$ and $L^j\bar\eta_j\le\mathsf H_s$, give

$$L^DX_D\le\mathsf H_s\Big[\sqrt{\tfrac{10}{N}}+\sqrt{2\sigma_s(1)}+(2c_{\mathrm d}\sqrt N+1)\sigma_j(1)+2NC_g\varepsilon_{[j]}\Big]+L^j\Big[\tfrac{7}{\sqrt N}(R_j^{1/2}+\sigma_s(1)+L^{-(s-j)})+6\sigma_s(1)\Big]+L^D\varrho.$$

Using $\frac{s}{10}-1\le\mathfrak m\le\frac{D}{2}$, $j\le0.3s$ and Lemma 18.87(c), we have: $N^{-1/2}\le L^{1/2}L^{-s/20}$; $\sqrt{2\sigma_s(1)}\le(2\mathsf C_3\mathsf e)^{1/2}L^{-19s/80}$; $\sqrt N\sigma_j(1)\le\mathsf C_3\mathsf eL^{\mathfrak m/2+s/40-j/2}=\mathsf C_3\mathsf eL^{s/40-D/2}\le\mathsf C_3\mathsf eL^{1/2}L^{-3s/40}$; $NC_g\varepsilon_{[j]}\le C_g\mathsf eL^{(\mathfrak m-D)/2}\le C_g\mathsf eL^{1/4}L^{-s/20}$; $L^j\sigma_s(1)\le\mathsf C_3\mathsf eL^{-7s/40}$; $L^jL^{-(s-j)}\le L^{-2s/5}$; $L^jR_j^{1/2}\le A_R^{1/2}L^{5j/2}e^{-a_Rj^2/2}\le L^{-s}$ for large s; $\varepsilon_{[D]}\le\mathsf eL^{1/2}L^{-s/10}$; $L^{-D}\mathsf H_s\le2\mathsf k_2\mathsf e^2L^{1-s/5}$; $L^j\varepsilon_{[s]}\le\mathsf eL^{-s/5}$. So each term is bounded by a number formed from L and c_ρ, times $\mathsf e^4L^{-s/20}$. □

Lemma 18.92 (Going down the depths, first moments grow by at most the factor L per depth)**.** Keep the situation of Lemma 18.90, let $\mathbb P$ be a Borel probability measure on $\mathcal V_s$, regard V as

the random field with the law $\mathbb{P}$, and put $\mathcal{G} = \{V \in \mathsf{Gd}_{d,s}\}$, $X_D = \max_{c\in\mathcal{R}_D} \mathbb{E}_{\mathbb{P}}[|\tilde\delta_D(c)|;\mathcal{G}]$, and $C_G = 4\exp(0.015\,C_N)$. Then for every $\rho' > 0$

$$\mathbb{P}(\mathcal{G} \cap \{|\mathsf{P}_d(c_0) - \mathbf{1}| > \rho' + 9L^{-D}\mathsf{H}_s + 2C_g\varepsilon_{[s]}\}) \le C_G L^D X_D\Big[\frac{L^{-d}}{\rho'} + 12A_0^{-1}\mathsf{b}^{1/2}L^{\frac52 i_1}\Big].$$

■

Sketch of proof. Following the descent. For $d \le i \le D$ and $b \in \mathcal{R}_i$ let $\tilde E_i = \mathcal{M}^{D-i}(\tilde E_D)$; by the covariance, $\tilde E_i(b) = e^{i\zeta(b_-)}E'_i(b)e^{-i\zeta(b_+)}$. Let $\mathsf{x}_i(b) = |\tilde E_i(b) - e^{i\Phi_i(b)}|$. It differs from $|E'_i(b) - e^{i\Phi_i(b)}|$ by at most $2\sup|\zeta|$, and Lemma 18.90(a), (c) yields $|\mathsf{P}_d(c_0) - \mathbf{1}| \le \mathsf{x}_d(c_0) + 9L^{-D}\mathsf{H}_s + 2C_g\varepsilon_{[s]}$. *Thresholds.* Let $\varkappa_i = 20\bar\eta_i$ for $i_1 \le i \le D$ and $\varkappa_i = \varepsilon_{[i]}$ for $d \le i < i_1$. For $i_1 \le i \le D$, everywhere on $\mathcal{G}$, $\mathsf{x}_i \le \bar\eta_i + 2C_g\varepsilon_{[s]} + 9L^{-D}\mathsf{H}_s \le 20\bar\eta_i$, because $\mathsf{H}_s \le 2\mathsf{k}_2\mathsf{E}_s$ and $\mathsf{k}_2\mathsf{E}_sL^{-D} \le \bar\eta_i$. Set $\mathcal{G}_{D+1} = \mathcal{G}$ and $\mathcal{G}_i = \mathcal{G} \cap \{\mathsf{x}_{i'}(b) \le \varkappa_{i'}$ for all $b \in \mathcal{R}_{i'}$, $i \le i' \le D\}$; then $\mathcal{G}_i = \mathcal{G}$ for $i \ge i_1$, and $\mathcal{G}_i \subset \mathcal{G}_{i+1}$. On $\mathcal{G}_i$ and at each depth $i' \ge i$ we have $\tilde E_{i'} = e^{i\tilde\Phi_{i'}}$, the difference $\tilde\delta_{i'} = \tilde\Phi_{i'} - \Phi_{i'}$ satisfies $|\tilde\delta_{i'}| \le 2\varkappa_{i'}$ (Lemma 18.78(b)), and $\mathsf{m}_{i'} = \max\{\sup_{\mathcal{R}_{i'}}|\Phi_{i'}|, \sup_{\mathcal{R}_{i'}}|\tilde\Phi_{i'}|\} \le C_g\varepsilon_{[i']} + 2\varkappa_{i'} \le \rho_L$. In fact $\sum_{i'>d}(C_g\varepsilon_{[i']} + 2\varkappa_{i'}) \le \rho_L$, by ($\mathrm{Z}^\sharp$), by $\sum_{i'\ge i_1}\bar\eta_{i'} \le \mathsf{k}_1(2+s)^2\varepsilon_{[s]} + \frac{L}{L-1}\mathsf{k}_2\mathsf{E}_sL^{-i_1}$, and by $\sum_{i'>d}(C_g + 2)\varepsilon_{[i']} \le 0.62(C_g+2)\varepsilon_{[d]}$. *Recursion.* On $\mathcal{G}_{i+1}$, $d \le i < D$, Lemma 18.80(a) applied at the depths $i' > i$ produces $\tilde\Phi_i$ on $\mathcal{R}_i$ with $\tilde E_i = e^{i\tilde\Phi_i}$ and $\tilde\delta_i = \mathbf{Q}^{D-i}\tilde\delta_D + \sum_{i'=i+1}^D \mathbf{Q}^{i'-1-i}(q(\tilde\Phi_{i'}) - q(\Phi_{i'}))$, where $|q_c(\tilde\Phi_{i'}) - q_c(\Phi_{i'})| \le C_N\mathsf{m}_{i'}\sum_{b\in\mathsf{P}_c^+}|\tilde\delta_{i'}(b)|$ and $\#\mathsf{P}_c^+ \le 6L^3$. Let $X_i = \max_{b\in\mathcal{R}_i}\mathbb{E}[|\tilde\delta_i(b)|;\mathcal{G}_{i+1}]$. Since $\mathcal{G}_{i+1} \subset \mathcal{G}_{i'+1}$ for $i' \ge i$ and all integrands are non-negative, Lemma 18.79(c) gives $X_i \le 4L^{D-i}X_D + \sum_{i'=i+1}^D 4L^{i'-1-i}\,6L^3C_N(C_g\varepsilon_{[i']} + 2\varkappa_{i'})X_{i'}$. In terms of $Y_i = L^iX_i$ and $\kappa_{i'} = 24L^2C_N(C_g\varepsilon_{[i']} + 2\varkappa_{i'})$ this reads $Y_i \le 4Y_D + \sum_{i'>i}\kappa_{i'}Y_{i'}$, and a descending induction gives $Y_i \le 4Y_D\prod_{i<i'\le D}(1+\kappa_{i'}) \le 4Y_De^{\sum\kappa}$, where $\sum\kappa_{i'} \le 24L^2C_N\rho_L = 0.015\,C_N$. Hence $X_i \le C_GL^{D-i}X_D$ for $d \le i \le D$. *Existence of the logarithms.* For $d < i < i_1$, as $\mathsf{x}_i \le |\tilde\delta_i|$ on $\mathcal{G}_{i+1}$, Chebyshev's inequality and a union bound over the at most $6L^{3(i-d)}$ bonds of $\mathcal{R}_i$ give $\mathbb{P}(\mathcal{G}_{i+1}\setminus\mathcal{G}_i) \le 6L^{3(i-d)}X_i/\varepsilon_{[i]}$, and $\varepsilon_{[i]}^{-1} \le A_0^{-1}\mathsf{b}^{1/2}L^{i/2}$; adding up, $\mathbb{P}(\mathcal{G}\setminus\mathcal{G}_{d+1}) \le 12\,C_GA_0^{-1}\mathsf{b}^{1/2}L^DX_DL^{\frac52 i_1}$. *Conclusion.* $\mathbb{P}(\mathcal{G}\cap\{\mathsf{x}_d(c_0) > \rho'\}) \le \mathbb{P}(\mathcal{G}\setminus\mathcal{G}_{d+1}) + X_d/\rho'$; combine this with the bound for $|\mathsf{P}_d(c_0) - \mathbf{1}|$ found at the start. □

Lemma 18.93 (A shift by one fine spacing moves coarser block fields only slightly, outside a rare event)**.** Assume the one-torus setting and let $\nu \in \{1,2,3\}$. One can find numbers s_1 and $\mathsf{C}_A \ge 1$, formed from the constants, ℓ and b, with the following property. Whenever $s \ge s_1$ and d are integers with $D_1 \le d < \lfloor s/5\rfloor$, c is a bond of Λ_d, $\rho \in [L^{-s/6}, 2]$ and $h = \pm a_se_\nu$,

$$\mu(V_s \in \mathsf{Gd}_{d,s} \text{ and } |\mathsf{P}_d(c) - \mathbf{1}| > \rho) \le \mathsf{C}_A\,\rho^{-1}L^{-s/21},$$

where the defect is formed from V_s; and the same bound holds when the law μ_s of V_s is replaced by μ_s^K with $K = 1260s$. Consequently, for every $d \ge D_1$ and every $\rho > 0$, $\mu(\mathsf{Gd}_d \cap \{\mathrm{dist}_{\mathrm{orb}}((\mathrm{T}_hV_\bullet)_d, V_d) > \rho\}) \le 3\ell^3L^{3d}\mathsf{C}_A\rho^{-1}L^{-s/21}$ for all $s \ge s_1$ with $d < \lfloor s/5\rfloor$, which tends to 0 as $s \to \infty$. ■

Sketch of proof. Let s be large, let $D, \mathfrak{m}, j, N, K$ be the numbers of the diagonal, $D_1 \le d < D$, and $h = a_se_\nu$. The event in question is $\{V_s \in E\}$ for a Borel set $E \subset \mathcal{V}_s$ (Proposition YM3_I:4.5). Take the cutoff $K = 1260s$ and let $\mathbb{P}$ be the law μ_s^K of $V_s = \mathcal{M}^{K-s}U$; Lemma 18.91 gives ($\mathrm{Z}^\sharp$) and $L^DX_D \le \mathsf{C}_7\mathsf{e}^4L^{-s/20}$. When $\rho \ge L^{-s/6}$ and s is large, $9L^{-D}\mathsf{H}_s + 2C_g\varepsilon_{[s]} \le 18\mathsf{k}_2\mathsf{e}^2L^{1-s/5} + 2C_g\mathsf{e}L^{-s/2} \le \frac12 L^{-s/6} \le \frac{\rho}{2}$, so Lemma 18.92 with $\rho' = \frac{\rho}{2}$ gives $\mu_s^K(E) \le C_G\mathsf{C}_7\mathsf{e}^4L^{-s/20}[2\rho^{-1}L^{-d} + 12A_0^{-1}\mathsf{b}^{1/2}L^{\frac52 i_1}]$. Here $L^{i_1} \le \mathsf{Ce}^2$ with C formed from the constants and b, $1 \le 2\rho^{-1}$, $\mathsf{e} \le C_\mathsf{e}s^{\mathfrak{p}_0}$, and $s^{9\mathfrak{p}_0}L^{-s/20} \le \mathsf{C}'L^{-s/21}$. To pass from the cutoff to the limit, Corollary 2.8(b), (c) and eq. 2.xvii give, at $K = 1260s$ (so that $\lceil K/1260\rceil = s$) and for $K \ge K_{\mathrm{bf}}$, $|\mu_s(E) - \mu_s^K(E)| \le \mathsf{C}_{\mathrm{bf}}\ell^3\sum_{K'\ge K}K'^{\mathfrak{p}}L^{-K'/10}$ with $\mathfrak{p} = 2\mathfrak{p}_0 + \frac12$, and this is at most $\rho^{-1}L^{-s/21}$ when s is large. *The sign* $-$. The reflection $x_\nu \mapsto -x_\nu$ is a diagonal element of Iso_0. It preserves the Wilson measure and μ_s (by isometry invariance and Lemma 18.77(a)), commutes with $\mathcal{M}$, preserves ϖ and $\mathsf{Gd}_{d,s}$, and turns the covariant shift

through $a_s e_\nu$ into the one through $-a_s e_\nu$; the defects of the reflected field are those of the field at the reflected bonds, possibly reversed, and a reversal does not change $|\mathsf{P}-\mathbf{1}|$. *The last assertion.* By Lemma 18.83(a), $\mathrm{dist}_{\mathrm{orb}}(V'_{d,s}, V_d) \le \max_b |\mathsf{P}_d(b)-\mathbf{1}| \le 2$. The lattice Λ_d has $3\ell^3 L^{3d}$ positively oriented bonds, and $\mathsf{Gd}_d \subset \{V_s \in \mathsf{Gd}_{d,s}\}$. For $L^{-s/6} \le \rho \le 2$ the bound therefore follows from the first assertion by a union over bonds; for $\rho > 2$ the event is empty; and for $\rho < L^{-s/6}$ the right side exceeds 1, as $\mathsf{C}_A \ge 1$. □

Lemma 18.94 (Uniform control of all shifts shorter than a fine spacing)**.** Assume the one-torus setting. There is an integer $D_3 \ge D_1$, formed from the constants and b, such that for every $d \ge D_3$, every $\nu \in \{1,2,3\}$ and every $\rho > 0$:

$$\lim_{s'\to\infty} \sup_{t\in\mathbb{Q}_L,\ 0\le t<a_{s'}} \mu\big(\mathsf{Gd}_d \cap \{\mathrm{dist}_{\mathrm{orb}}((\mathrm{T}_{te_\nu}V_\bullet)_d, V_d) > \rho\}\big) = 0.$$

■

Sketch of proof. Parameters. Let $\kappa = 1$, $\lambda = L+1 \le L^{1.03}$, $\rho_* = \rho_{\mathrm{Lip}}(1)$ (Lemma 18.80(b)), $\delta(i) = \lfloor i/200 \rfloor$ and $\rho_i = \lambda^{-\delta(i)} L^{-i/84}$. Let D_3 be the least $d \ge D_1$ with $3\varepsilon_{[d]} < \omega_M$, $49L^2 \cdot 3 \cdot 0.62\,\varepsilon_{[d]} \le \frac12 \rho_*$ and $L^{-d/2}\mathsf{b}^{1/2} \le A_0$; the last condition gives $\lambda^{-i'} \le L^{-i'} \le \varepsilon_{[i']}$ for $i' \ge d$. Fix $d \ge D_3$ and choose s' so large that $s' \ge \max\{s_1, 200(d+1)\}$, $12L\sum_{i>s'} L^{-i/84} \le 1$ and $3L^{-s'/84} \le \frac12\rho_*$. For $i > s'$ one then has $D_1 \le d \le \delta(i) < \lfloor i/5 \rfloor$ and $L^{-i/6} \le L^{-0.0171\,i} \le \rho_i \le 2$, so that Lemma 18.93 is available with $(s,d,\rho) = (i,\delta(i),\rho_i)$. *Decomposing the shift.* Expand $t = \sum_{i=s'+1}^{S} k_i a_i$ in base L, $0 \le k_i \le L-1$, and carry out the $\sum_i k_i$ elementary shifts in the order of increasing i, the n-th one being through $a_{i_n}e_\nu$: $W^{(0)} = V_\bullet$ and $W^{(n+1)} = \mathrm{T}_{a_{i_n}e_\nu} W^{(n)}$. By Lemma 18.77(b) the final thread is $\mathrm{T}_{te_\nu}V_\bullet$, and each $W^{(n)}$ has the law μ. Let $\mathcal{E}_n$ be the event that (e1) $W^{(n)} \in \mathsf{Gd}_{\delta(i_n)}$, (e2) $W^{(n+1)} \in \mathsf{Gd}_{\delta(i_n)}$, and (e3) $|\mathsf{P}_{\delta(i_n)}(c) - \mathbf{1}| \le \rho_{i_n}$ at every bond c of $\Lambda_{\delta(i_n)}$, for the defects of the shift through $a_{i_n}e_\nu$ formed from $W^{(n)}$. Lemmas 18.84(b) and 18.93 give $\mu(\mathcal{E}_n^c) \le q_{i_n}$, where $q_i = 2\frac{L}{L-1}R_{\delta(i)} + 3\ell^3 L^{3\delta(i)}\mathsf{C}_A\rho_i^{-1}L^{-i/21} \le 2\frac{L}{L-1}R_{\delta(i)} + 3\ell^3\mathsf{C}_A L^{-i/65}$, because $\frac{3}{200} + \frac{1.03}{200} + \frac{1}{84} - \frac{1}{21} < -\frac{1}{65}$; and $\sum_i q_i < \infty$. *Claim:* on $\mathsf{Gd}_d \cap \bigcap_n \mathcal{E}_n$, for every n, (R_n) $\varpi(W^{(n)}_{i'}) < 2\varepsilon_{[i']} + 12\lambda^{-i'}\sum_{n'<n} L^{-i_{n'}/84}$ for all $i' \ge d$, and consequently $\varpi(W^{(n)}_{i'}) < 3\varepsilon_{[i']}$; and ($\mathrm{M}_n$) $\mathrm{dist}_{\mathrm{orb}}(W^{(n+1)}_{i'}, W^{(n)}_{i'}) \le 3\lambda^{-i'}L^{-i_n/84}$ for $d \le i' \le \delta(i_n)$. For $n = 0$, (R_0) says that $V_\bullet \in \mathsf{Gd}_d$. Suppose ($\mathrm{R}_n$) holds; write $i = i_n$, $W = W^{(n)}$, and let $\hat W_{i'} = \mathcal{M}^{i-i'}(\check W_i)$ denote the block fields of the covariantly shifted field at the depth i; by Lemma 18.83(a), $W^{(n+1)}_{i'}$ and $\hat W_{i'}$ differ by a gauge transformation. For $d \le i' < \delta(i)$ we may use Lemma 18.81(b) with $\kappa = 1$, $D' = \delta(i)$, $d' = i'$, $V_{D'} = W_{\delta(i)}$, $\tilde V_{D'} = \hat W_{\delta(i)}$ and $\eta = \rho_i$: its hypotheses hold, since $\varpi(W_{i''}) < 3\varepsilon_{[i'']} < \omega_M$, $49L^2\sum_{i''>i'} 3\varepsilon_{[i'']} \le \frac12\rho_*$, $|\hat W_{\delta(i)}(c) - W_{\delta(i)}(c)| = |\mathsf{P}_{\delta(i)}(c) - \mathbf{1}| \le \rho_i$ by (e3), and $3\lambda^{\delta(i)-i'}\rho_i \le 3L^{-i/84} \le \frac12\rho_*$. It yields $|\hat W_{i'}(c) - W_{i'}(c)| \le 3\lambda^{-i'}L^{-i/84}$ on every bond of $\Lambda_{i'}$; at $i' = \delta(i)$ the same bound is (e3); together this is (M_n). Each plaquette has four sides, so $\varpi(W^{(n+1)}_{i'}) = \varpi(\hat W_{i'}) \le \varpi(W_{i'}) + 12\lambda^{-i'}L^{-i/84}$ for $i' < \delta(i)$, while for $i' \ge \delta(i)$ the statement (R_{n+1}) is just (e2). Since each i occurs at most $L-1$ times among the i_n, $12\lambda^{-i'}\sum_{n'} L^{-i_{n'}/84} \le \lambda^{-i'} \le \varepsilon_{[i']}$. *Conclusion.* On $\mathsf{Gd}_d \cap \bigcap_n \mathcal{E}_n$ the triangle inequality for $\mathrm{dist}_{\mathrm{orb}}$ gives $\mathrm{dist}_{\mathrm{orb}}((\mathrm{T}_{te_\nu}V_\bullet)_d, V_d) \le 3\lambda^{-d}(L-1)\sum_{i>s'} L^{-i/84}$, which tends to 0 as $s' \to \infty$; and $\mu(\bigcup_n \mathcal{E}_n^c) \le (L-1)\sum_{i>s'} q_i \to 0$. Neither bound depends on t. □

Lemma 18.95 (The reflected correlation at a mirror is log-convex and convex)**.** Assume the one-torus setting; let (μ, c) be a mirror as in the notation, $\alpha' > 0$, $F \in \mathcal{A}(0,\alpha')$, and $I = \mathbb{Q}_L \cap [0, 2\alpha']$.

(a) Let $u \in \mathbb{Q}_L$, $c^* = c + \frac{u}{2}$, and $x, y \in \mathbb{Q}_L$ with $\frac{u}{2} \le x, y \le \alpha' + \frac{u}{2}$. Then $\mathcal{U}_x F$ and $\mathcal{U}_y F$ lie in $\mathcal{A}^{c^*}(0,0)$, and $\langle \mathcal{U}_x F, \mathcal{U}_y F\rangle_{c^*} = \varphi_F(x+y-u)$. In particular $\langle \mathcal{U}_{t_1}F, \mathcal{U}_{t_2}F\rangle_c = \varphi_F(t_1+t_2)$ for $t_1, t_2 \in \mathbb{Q}_L \cap [0,\alpha']$, and $0 \le \varphi_F \le \|F\|^2_{L^2(\mu)}$ on I.

(b) $\varphi_F(t)^2 \le \varphi_F(t-h)\,\varphi_F(t+h)$ whenever $t, h \in \mathbb{Q}_L$, $0 \le h \le t$ and $t + h \le 2\alpha'$.

(c) φ_F is convex on I and has a continuous extension to $(0, 2\alpha')$; the limit $\varphi_F(0+)$ exists and does not exceed $\varphi_F(0)$; and $\|\mathcal{U}_t F - F\|_c^2 \to \varphi_F(0) - \varphi_F(0+)$ as $t \downarrow 0$ in $\mathbb{Q}_L$.
(d) For $\alpha > 0$ and $F \in \mathcal{A}(\alpha, \alpha')$, the two moduli stated in Proposition 18.76(B2) hold.
(e) If $\varphi_F(0) = 0$, then φ_F vanishes on $\mathbb{Q}_L \cap [0, 2\alpha')$. If $\varphi_F(t_0) = 0$ for some $t_0 \in I$ with $t_0 > 0$, then φ_F vanishes on $\mathbb{Q}_L \cap (0, t_0]$.

■

Sketch of proof. (a) Choose j so large that $x, y, u \in a_j\mathbb{Z}$. Then the bonds of Λ_j on which $\mathcal{U}_x F$ depends have end points with $c + x + \frac{1}{2}a_j \le x_\mu \le c + \frac{\ell}{2} - \alpha' + x - \frac{1}{2}a_j$ (Lemma 18.77(c)); as $c + x \ge c^*$ and $c + \frac{\ell}{2} - \alpha' + x \le c^* + \frac{\ell}{2}$, these bonds lie in the window of $\mathcal{A}^{c^*}(0, 0)$, and the same holds for $\mathcal{U}_y F$. Lemma 18.77(c) also gives $\Theta_{c^*} = \Theta_c \mathcal{U}_{-u}$ and $\Theta_c \mathcal{U}_{x-u} = \mathcal{U}_{u-x}\Theta_c$, and μ is translation invariant (Lemma 18.77(b)). Hence $\langle \mathcal{U}_x F, \mathcal{U}_y F\rangle_{c^*} = \int \overline{\mathcal{U}_{u-x}\Theta_c F}\, \mathcal{U}_y F \, d\mu = \varphi_F(x + y - u)$. Taking $u = 0$ gives the second statement. Taking $u = x = y = t \in I$ gives $\varphi_F(t) = \|\mathcal{U}_t F\|^2_{c+t/2}$, which is ≥ 0 by Lemma 18.77(d) applied to the mirror at $c + \frac{t}{2} \in \frac{1}{2}\mathbb{Q}_L$, whatever its type; we argue in this way because $\frac{t}{2}$ need not belong to $\mathbb{Q}_L$. Finally $\varphi_F(t) \le \|\Theta F\|_2 \|\mathcal{U}_t F\|_2 = \|F\|_2^2$. (b) Write $p = t - h$ and $q = t + h$, and use (a) with $u = x = p$ and $y = t$, which is admissible. Since $x + y - u = t$, $2x - u = p$ and $2y - u = q$, the Schwarz inequality for $\langle \cdot, \cdot \rangle_{c^*}$ gives (b). (c) From (b) and $\sqrt{ab} \le \frac{1}{2}(a + b)$, on every grid $a_s\mathbb{Z} \cap [0, 2\alpha']$ the sequence $k \mapsto \varphi_F(ka_s)$ has non-negative second differences; any three points of I lie on a common grid, so φ_F is convex on I. Being convex and bounded on I, which is dense in $[0, 2\alpha']$, φ_F is locally Lipschitz on $I \cap (0, 2\alpha')$ and therefore has a continuous extension to $(0, 2\alpha')$; as its difference quotients increase, the limit $\varphi_F(0+)$ exists and is at most $\varphi_F(0)$. By (a), $\|\mathcal{U}_t F - F\|_c^2 = \varphi_F(2t) - 2\varphi_F(t) + \varphi_F(0)$. (d) Let v be convex on $\mathbb{Q}_L \cap [0, b]$ with $0 \le v \le \mathsf{m}$. Increasing difference quotients give $-\frac{\mathsf{m}}{x} \le \frac{v(x) - v(0)}{x} \le \frac{v(y) - v(x)}{y - x} \le \frac{v(b') - v(y)}{b' - y} \le \frac{\mathsf{m}}{b' - y}$ for $0 < x < y < b' \le b$ in $\mathbb{Q}_L$; letting b' increase to b, one gets $v(x + 2t) - 2v(x + t) + v(x) \le t\mathsf{m}(\frac{1}{b - x - 2t} + \frac{1}{x})$ whenever $x > 0$ and $x + 2t < b$. Now let $\alpha'' \in \mathbb{Q}_L \cap (0, \alpha]$ and $G = \mathcal{U}_{-\alpha''} F \in \mathcal{A}(0, \alpha' + \alpha'')$, so that $\|G\|_2 = \|F\|_2$, and apply this to $v = \varphi_G$ and $b = 2(\alpha' + \alpha'')$. By (a), $\|\mathcal{U}_t F - F\|_c^2$ is the second difference of v with the step t at the point $x = 2\alpha''$, where $b - x - 2t = 2\alpha' - 2t \ge \alpha'$ if $t \le \frac{\alpha'}{2}$; and $\|\mathcal{U}_{-t} F - F\|_c^2$ equals the second difference taken at $x = 2\alpha'' - 2t \ge \alpha''$ (when $2t \le \alpha''$), where $b - x - 2t = 2\alpha'$. Then let α'' increase to α. (e) If $\varphi_F(0) = 0$, then (b) with $h = t$ gives $\varphi_F = 0$ on $\mathbb{Q}_L \cap [0, \alpha']$. Next, suppose $\varphi_F = 0$ on $\mathbb{Q}_L \cap [0, z)$ with $0 < z < 2\alpha'$, and let $x \in \mathbb{Q}_L$ with $z \le x < \frac{1}{2}(z + 2\alpha')$; choosing $h \in \mathbb{Q}_L \cap (x - z, 2\alpha' - x)$ with $h \le x$, (b) gives $\varphi_F(x)^2 \le \varphi_F(x - h)\varphi_F(x + h) = 0$. So the zero set grows from $[0, z_k)$ to $[0, z_{k+1})$ with $z_{k+1} = \frac{1}{2}(z_k + 2\alpha')$, and $z_k \to 2\alpha'$. If instead $\varphi_F(t_0) = 0$, then for $\frac{1}{2}t_0 \le x < t_0$ the choice $h = t_0 - x$ in (b) gives $\varphi_F(x)^2 \le \varphi_F(2x - t_0)\varphi_F(t_0) = 0$. Since $\frac{1}{2}t_0$ may fail to lie in $\mathbb{Q}_L$, pick $t_1 \in \mathbb{Q}_L \cap [\frac{1}{2}t_0, \frac{2}{3}t_0]$ and repeat with t_1 in place of t_0; the intervals $[\frac{1}{2}t_k, t_k]$ overlap, and $t_k \le (\frac{2}{3})^k t_0$, so together they cover $(0, t_0]$. □

Lemma 18.96 (Near a bond mirror, the averaging forgets the first layer of fine bonds)**.** Let (μ, c), $c \in \frac{a_{d_0}}{2}\mathbb{Z}$, be a mirror of the bond type. For $j \ge d_0$ let $\ell_1^{(j)}$ denote the sites of Λ_j at the height $x_\mu = c + \frac{1}{2}a_j$, the first layer above the mirror, and $\mathcal{L}_j$ the bonds of Λ_j that have both end points in $\mathbb{E}_+$ and touch $\ell_1^{(j)}$. Put $\lambda_{\mathrm{w}} = 1 + \frac{2}{L} + \frac{L^2 - 1}{2L^2} < 1.654$ and $\lambda_{\mathrm{l}} = 1.8$; then $\lambda_{\mathrm{l}}/\sqrt{L} < \frac{1}{2}$.

(a) Let $i > d_0$ and let c' be a bond of Λ_{i-1} with both end points in $\mathbb{E}_+$. The coefficients of $(\mathbf{Q}A)(c') = \sum_b \mathsf{w}_{c'}(b) A(b)$ satisfy $\sum_{b \in \mathcal{L}_i} |\mathsf{w}_{c'}(b)| \le \lambda_{\mathrm{w}}$; and $\mathsf{P}_{c'}$ meets $\mathcal{L}_i$ only if $c' \in \mathcal{L}_{i-1}$.
(b) There is a number $\rho_{\mathrm{l}} \in (0, \frac{1}{4}\omega_M)$, formed from L alone, with the following property. Let $i > d_0$, let c' be a bond of Λ_{i-1} with both end points in $\mathbb{E}_+$, let $V, \tilde{V} \in \mathcal{V}_i$ and $\varpi', \delta \ge 0$. Suppose that $\tilde{V} = V$ on $\mathsf{P}_{c'} \setminus \mathcal{L}_i$, that $|\partial V(p) - \mathbf{1}| \le \varpi'$ for the plaquettes p of the double block of c', that $|\tilde{V} - V| \le \delta$ on $\mathsf{P}_{c'} \cap \mathcal{L}_i$, and that $\frac{\pi}{2}(3L\varpi' + \delta) \le \rho_{\mathrm{l}}$. Then $|\mathcal{M}(\tilde{V})(c') - \mathcal{M}(V)(c')| \le \lambda_{\mathrm{l}}\delta$.

(c) For $j > d_0$ and $V \in \mathcal{V}_j$ define $V^\flat \in \mathcal{V}_j$ by: $V^\flat = \mathbf{1}$ on the bonds $\langle x, x + a_j e_\mu\rangle$ with $x \in \ell_1^{(j)}$; $V^\flat(b) = V(b + a_j e_\mu)$ on the bonds b with both end points in $\ell_1^{(j)}$; $V^\flat(b) = V(b)$ on every other bond; reversed bonds receive the inverse values. Let $d_0 \le D < j$, $V_i = \mathcal{M}^{j-i}V$ and $V_i^\flat = \mathcal{M}^{j-i}V^\flat$. For real $\alpha' \ge 0$, the restriction $V_D^\flat|_{\mathsf{Bd}_D(0,\alpha')}$ is a Borel function of $V|_{\mathsf{Bd}_j(a_j,\alpha')}$. If $\varpi_i \ge \varpi(V_i)$ and $\frac{\pi}{2}(3L\varpi_i + \lambda_1^{j-i}\varpi_j) \le \rho_1$ for $D < i \le j$, then at every bond c' of Λ_D with both end points in $\mathbb{E}_+$, $V_D^\flat(c') = V_D(c')$ when $c' \notin \mathcal{L}_D$, and $|V_D^\flat(c') - V_D(c')| \le \lambda_1^{j-D}\varpi_j$ when $c' \in \mathcal{L}_D$.

(d) In the one-torus setting let $C_\lambda = \sup_{n\ge 0}(\lambda_1/\sqrt{L})^n(1+n)^{\mathfrak{p}_0}$, and let D_* be the least integer $D \ge \max\{d_\natural, d_0\}$ with $\pi(3L + C_\lambda)\sup_{i\ge D}\varepsilon_{[i]} \le \rho_1$. On Gd_D with $D \ge D_*$, for all $j > D$, $|(\mathcal{M}^{j-D}V_j^\flat)(c') - V_D(c')| \le 2\varepsilon_{[D]}(\lambda_1/\sqrt{L})^{j-D}(1+j-D)^{\mathfrak{p}_0}$ at every bond c' of Λ_D whose end points both lie in $\mathbb{E}_+$.

(e) In the one-torus setting, each $F \in \mathcal{A}(0,\alpha')$, $\alpha' \ge 0$, is the limit in $L^2(\mu)$ of functions $F_n \in \mathcal{A}_{j_n}(a_{j_n},\alpha')$ with $\sup|F_n| \le \sup|F|$; and each $F \in \mathcal{A}(\alpha,0)$, $\alpha \ge 0$, is such a limit of functions $F_n \in \mathcal{A}_{j_n}(\alpha, a_{j_n})$, with the same bound.

■

Sketch of proof. (a) Work in units of Λ_i and put $\mathsf{h} = \frac{1}{2}(L-1)$. For a path Γ let $N(\Gamma)$ count those of its bonds that belong to $\mathcal{L}_i$, in either orientation. The formula for $\mathbf{Q}$ gives $\sum_{b\in\mathcal{L}_i}|\mathsf{w}_{c'}(b)| \le L^{-3}\sum_{x\in B(c'_-)}[\bar N(c'_-,x) + N([x,x']) + \bar N(c'_+,x')]$, with $\bar N$ the average over the contour family. For $y \in \Lambda_{i-1}\cap\mathbb{E}_+$, the block $B(y)$ contains sites of $\ell_1^{(i)}$ exactly when $y \in \ell_1^{(i-1)}$, and these are then its L^2 sites with $n_\mu = -\mathsf{h}$; this proves the second assertion. On a contour every coordinate moves monotonically from 0 to n_κ. Hence a contour reaches the layer only if $n_\mu = -\mathsf{h}$; in that case its last step in the direction μ ends in the layer, the earlier steps stay off it, and all later steps run inside it, so that $N(\Gamma_\sigma) = 1 + \sum\{|n_\kappa| : \kappa \text{ comes after } \mu \text{ in } \sigma\}$. Every $\kappa \ne \mu$ with $n_\kappa \ne 0$ comes after μ in half of the orders; therefore $\bar N(y,x) = 1 + \frac{1}{2}\sum_{\kappa\ne\mu}|n_\kappa|$, and since $\sum_{|k|\le\mathsf{h}}|k| = \frac{1}{4}(L^2-1)$, $L^{-3}\sum_{x\in B(y)}\bar N(y,x) = \frac{1}{L} + \frac{L^2-1}{4L^2}$. When c' is parallel to the mirror, $N([x,x'])$ equals L for the L^2 sites with $n_\mu = -\mathsf{h}$ and 0 otherwise, which contributes 1; adding the two blocks, the total is at most $1 + \frac{2}{L} + \frac{L^2-1}{2L^2}$. When c' is perpendicular to the mirror with $c'_- \in \ell_1^{(i-1)}$, the segment $[x,x']$ contains one bond of the layer if $n_\mu = -\mathsf{h}$ and none otherwise, which contributes $\frac{1}{L}$, and the block of c'_+ contains no site of the layer. (b) We use the integer coordinates n of the proof of Lemma 18.80, attached to whichever of c', $\bar c'$ is positively oriented. In them the map $V|_{\mathsf{P}_{c'}} \mapsto \mathcal{M}(V)(c')$ depends only on the direction and the orientation of c', at every depth and on every side. By the nesting in the proof of Lemma 18.77(c), the double block of c' lies in $\mathbb{E}_+$, so that $\mathsf{P}_{c'}\cap\mathcal{L}_i$ consists of the bonds of $\mathsf{P}_{c'}$ that touch $\ell_1^{(i)}$; and by the proof of (a), the sites of the double block in $\ell_1^{(i)}$ are, for each end point y of c' in $\ell_1^{(i-1)}$, the L^2 sites of $B(y)$ with the smallest coordinate x_μ, and there are no others. In the coordinates n the set $\mathsf{P}_{c'}\cap\mathcal{L}_i$ therefore depends only on the direction and the orientation of c', on μ, and on which end points of c' lie in $\ell_1^{(i-1)}$, and not on the depth, the side or the height c. So there are finitely many cases, and ρ_1 is the smallest of the numbers obtained below for the individual cases. In one case, let $\mathsf{P}^+ \subset \mathsf{P}_{c'}$ be the subset of positively oriented bonds, let $E = \mathfrak{g}^{\mathsf{P}^+}$ with $|A|_\infty = \max_b|A(b)|$, and let $E_\mathcal{L} \subset E$ be the forms vanishing off $\mathcal{L}_i$. On $|A|_\infty < \frac{1}{4}\omega_M$ the map $\Psi(A) = \mathcal{M}(e^{iA})(c')$ is continuously differentiable (Lemma 18.80(a) and its proof), and $D\Psi(0)\delta A = i\sum_b \mathsf{w}_{c'}(b)\delta A(b)$ (Lemma YM3_I:4.6). Let $\Lambda(\rho) = \sup\{|D\Psi(A)\delta A| : |A|_\infty \le \rho,\ \delta A \in E_\mathcal{L},\ |\delta A|_\infty \le 1\}$. By (a), $\Lambda(0) \le \lambda_\mathrm{w} < \lambda_1$, and by compactness Λ is continuous at 0; so ρ_1 can be chosen with $\Lambda(\rho_1) \le (1-\rho_1)\lambda_1$.

Gauge fixing on the double block. Let ν' be the direction of c' and κ, κ' the two other directions; let o be the vertex of the double block whose coordinates are smallest, and write every site of the double block as $o + n_1 e_{\nu'} + n_2 e_\kappa + n_3 e_{\kappa'}$ with $0 \le n_1 \le 2L-1$ and $0 \le n_2, n_3 \le L-1$. Let Γ_x go from o first n_3 steps along $e_{\kappa'}$, then n_2 steps along e_κ, then n_1 steps along $e_{\nu'}$, and set

$u(x) = V(\Gamma_x)$ on the double block and $\mathbf{1}$ elsewhere. Then $V^u = \mathbf{1}$ on all bonds along $e_{\nu'}$, on the bonds along e_κ with $n_1 = 0$, and on the bonds along $e_{\kappa'}$ with $n_1 = n_2 = 0$. If b and $b + e_{\nu'}$ are bonds of a direction other than ν', the plaquette they span has its two remaining sides equal to $\mathbf{1}$, so $|V^u(b + e_{\nu'}) - V^u(b)| \le \varpi'$; walking back to the trivial bonds gives $|V^u - \mathbf{1}| \le 3L\varpi'$ on $\mathsf{P}_{c'}$. By Lemma 18.78(b), $V^u = e^{iA}$ and $\tilde V^u = e^{i\tilde A}$ on P^+ with $|A|_\infty, |\tilde A|_\infty \le \rho_1$, $\tilde A - A \in E_{\mathcal{L}}$ and $|\tilde A - A|_\infty \le (1-\rho_1)^{-1}\delta$. As the ball is convex, $|\Psi(\tilde A) - \Psi(A)| \le \Lambda(\rho_1)(1-\rho_1)^{-1}\delta \le \lambda_1\delta$; and $\Psi(A)$, $\Psi(\tilde A)$ come from $\mathcal{M}(V)(c')$, $\mathcal{M}(\tilde V)(c')$ through the same multiplications, by $u(c'_-)$ on the left and by $u(c'_+)^{-1}$ on the right (Proposition YM3_I:4.5). (c) *The first claim.* By locality and nesting, for $c' \in \mathsf{Bd}_D(0,\alpha')$ the value $V^\flat_D(c')$ depends on $V^\flat$ only at bonds $b \in \mathsf{Bd}_j(0,\alpha')$ inside the blocks $B^{j-D}(c'_\pm)$. For such b: if neither end point of b is in $\ell_1^{(j)}$, then $V^\flat(b) = V(b)$ and $b \in \mathsf{Bd}_j(a_j,\alpha')$; if b is perpendicular to the mirror with an end point in $\ell_1^{(j)}$, then $V^\flat(b) = \mathbf{1}$; and if b is contained in the first layer, then $V^\flat(b) = V(b + a_j e_\mu)$, the bond $b + a_j e_\mu$ connecting two sites one layer higher and still inside the same blocks, as these have $L^{j-D} \ge 2$ layers. *A gauge change that the coarser depths ignore.* Let $g(x) = V(\langle x, x + a_j e_\mu\rangle)^{-1}$ for $x \in \ell_1^{(j)}$, and $g = \mathbf{1}$ at all other sites. The heights of the sites of Λ_{j-1} lie in $c + a_{j-1}(\mathbb{Z} + \frac12)$, which does not contain $c + \frac12 a_j$; so $g = \mathbf{1}$ on Λ_{j-1}, and $\mathcal{M}^{j-i}(V^g) = V_i$ for $i < j$ (Proposition YM3_I:4.5). Put $V_j^\sharp = V^g$ and $V_i^\sharp = V_i$ for $i < j$. *Claim* (I_i), $D \le i \le j$: at each bond b of Λ_i with both end points in $\mathbb{E}_+$, $V_i^\flat(b) = V_i^\sharp(b)$ when $b \notin \mathcal{L}_i$, and $|V_i^\flat(b) - V_i^\sharp(b)| \le \lambda_1^{j-i}\varpi_j$ when $b \in \mathcal{L}_i$. For $i = j$: away from the layer $V^g = V = V^\flat$; on a bond leaving the layer perpendicularly, $V^g = \mathbf{1} = V^\flat$; and on a bond $b = \langle x, x + \mathsf{f}\rangle$ inside the layer, $V^g(b)V^\flat(b)^{-1}$ is conjugate to the transport of V around the plaquette with the corners x, $x + \mathsf{f}$, $x + \mathsf{f} + a_j e_\mu$, $x + a_j e_\mu$, so $|V^g(b) - V^\flat(b)| \le \varpi_j$. From i to $i-1$: if $c' \notin \mathcal{L}_{i-1}$, then $\mathsf{P}_{c'} \cap \mathcal{L}_i = \emptyset$ by (a), so $V_i^\flat = V_i^\sharp$ on $\mathsf{P}_{c'}$, and the averages coincide (Proposition YM3_I:4.5); if $c' \in \mathcal{L}_{i-1}$, apply (b) with $V = V_i^\sharp$, $\tilde V = V_i^\flat$, $\varpi' = \varpi_i$ (the plaquette variables of $V_i^\sharp$ are conjugate to those of V_i) and $\delta = \lambda_1^{j-i}\varpi_j$. The case ($\mathrm{I}_D$) is the second claim. (d) On Gd_D take $\varpi_i = 2\varepsilon_{[i]}$. For $i \ge 1$, since $\mathsf{b} \ge 1$, $\log L^j\mathsf{b} \le (1 + j - i)\log L^i\mathsf{b}$; therefore $\lambda_1^{j-i}\varepsilon_{[j]} \le \varepsilon_{[i]}(\lambda_1/\sqrt{L})^{j-i}(1 + j - i)^{\mathfrak{p}_0} \le C_\lambda\varepsilon_{[i]}$, and $\frac{\pi}{2}(3L\varpi_i + \lambda_1^{j-i}\varpi_j) \le \pi(3L + C_\lambda)\varepsilon_{[i]} \le \rho_1$ for $i \ge D_*$. So (c) applies. (e) Let $\epsilon > 0$ and $F \in \mathcal{A}(0,\alpha')$. By the nesting, $F = \mathsf{f}(V_D|_{\mathsf{Bd}_D(0,\alpha')})$ for every large D; take $D \ge D_*$ so large that $\mu(\mathsf{Thr} \setminus \mathsf{Gd}_D) \le \epsilon^2$ (Lemma 18.84(b)). For every finite Borel measure ν on a compact metrisable space, continuous functions are dense in $L^2(\nu)$; composing them with the radial retraction of $\mathbb{C}$ onto the disc of radius $\sup|\mathsf{f}|$ makes their modulus at most $\sup|\mathsf{f}|$ without increasing their distance to f. Take such a continuous f_ϵ with $\|F - \mathsf{f}_\epsilon(V_D|_{\mathsf{Bd}_D(0,\alpha')})\|_{L^2(\mu)} \le \epsilon$. Then $F_{\epsilon,j} = \mathsf{f}_\epsilon((\mathcal{M}^{j-D}V_j^\flat)|_{\mathsf{Bd}_D(0,\alpha')})$ belongs to $\mathcal{A}_j(a_j,\alpha')$ by (c), and on Gd_D it converges to $\mathsf{f}_\epsilon(V_D|_{\mathsf{Bd}_D(0,\alpha')})$ uniformly as $j \to \infty$, by (d) and the uniform continuity of f_ϵ. Hence $\limsup_j \|F_{\epsilon,j} - F\|_{L^2(\mu)} \le \epsilon + 2\sup|F|\,\epsilon$. For the second assertion use R: it maps $\mathcal{A}(\alpha,0)$ onto $\mathcal{A}(0,\alpha)$ and $\mathcal{A}_j(a_j,\alpha)$ onto $\mathcal{A}_j(\alpha,a_j)$, and it preserves μ (Lemma 18.77(b), (c)). □

Sketch of proof of Proposition 18.76. (A1). The statements about T_h and $\mathsf{r}^\bullet$ are Lemma 18.77(b). Since T_h preserves μ and T_{-h} inverts it on Thr, $\mathcal{U}_h$ is unitary on $L^2(\mu)$ and $\mathcal{U}_h\mathcal{U}_{h'} = \mathcal{U}_{h+h'}$. Let $v \in a_s\mathbb{Z}^3$ and let F be a bounded gauge-invariant Borel function on $\mathcal{V}_d$. Then $\mathcal{U}_v(F \circ \pi_d)$ equals $(F \circ \mathrm{t}_{-v}) \circ \pi_d$ if $d \ge s$, and $(F \circ \mathcal{M}^{s-d} \circ \mathrm{t}_{-v}) \circ \pi_s$ if $d < s$; both are bounded, Borel and gauge invariant, because $\mathrm{t}_{-v}(V^w) = (\mathrm{t}_{-v}V)^{w(\cdot+v)}$ and by Proposition YM3_I:4.5. So $\mathcal{U}_v$ and $\mathcal{U}_{-v}$ send $\mathcal{K}^{\mathrm{inv}}$ into itself. *(A2).* Fix F, bounded, gauge invariant and Borel on $\mathcal{V}_d$, together with $\epsilon > 0$ and $\nu \in \{1,2,3\}$. Choose $d' \ge \max\{d, D_3, d_\Sigma\}$ with $\frac{L}{L-1}R_{d'} \le \epsilon^2$, so that $\mu(\mathsf{Thr} \setminus \mathsf{Gd}_{d'}) \le \epsilon^2$ (Lemma 18.84(b)). The function $F' = F \circ \mathcal{M}^{d'-d}$ on $\mathcal{V}_{d'}$ is bounded, Borel and gauge invariant, and along threads $F(V_d) = F'(V_{d'})$ and $F((\mathrm{T}_{te_\nu}V_\bullet)_d) = F'((\mathrm{T}_{te_\nu}V_\bullet)_{d'})$. Pick a continuous Φ_0 on $\mathcal{V}_{d'}$ with $|\Phi_0| \le \sup|F|$ and $\|F' - \Phi_0\|_{L^2(\mu_{d'})} \le \epsilon$, and average it over gauge transformations: $\Phi(W) = \int \Phi_0(W^w)\,dw$, dw being the normalized Haar measure of the group of gauge transformations

of $\Lambda_{d'}$. Then Φ is a continuous gauge-invariant function, and $\|F'-\Phi\|_{L^2(\mu_{d'})}\le\epsilon$ because both $\mu_{d'}$ and F' are gauge invariant. Write ω for the modulus of continuity of Φ with respect to $\max_b|W'(b)-W(b)|$; gauge invariance gives $|\Phi(W')-\Phi(W)|\le\omega(\mathrm{dist}_{\mathrm{orb}}(W',W))$. Since $(\mathrm{T}_{te_\nu}V_\bullet)_{d'}$ has the law $\mu_{d'}$ (Lemma 18.77(b)), we obtain for every $\rho>0$

$$\|F((\mathrm{T}_{te_\nu}V_\bullet)_d)-F(V_d)\|_{L^2(\mu)}\le 2\epsilon+\omega(\rho)+2\sup|F|\,[\mu(\mathsf{Thr}\setminus\mathsf{Gd}_{d'})+\mu(\mathsf{Gd}_{d'}\cap\{\mathrm{dist}_{\mathrm{orb}}((\mathrm{T}_{te_\nu}V_\bullet)_{d'},V_{d'})>\rho\})]^{1/2}.$$

By Lemma 18.94, the upper limit as $s'\to\infty$ of the supremum of the left side over $t\in\mathbb{Q}_L\cap[0,a_{s'})$ is at most $2\epsilon+\omega(\rho)+2\sup|F|\,\epsilon$; now let $\rho\to0$ and then $\epsilon\to0$. Finally, unitarity gives $\|\mathcal{U}_{-w}\psi-\psi\|=\|\mathcal{U}_w\psi-\psi\|$ and $\|\mathcal{U}_{w+w'}\psi-\psi\|\le\|\mathcal{U}_w\psi-\psi\|+\|\mathcal{U}_{w'}\psi-\psi\|$, which yields the limit along all $v\in\mathbb{Q}_L^3$ with $|v|_\infty<a_{s'}$, $s'\to\infty$. *(A3).* Since the $\mathcal{U}_v$ are unitary, the vectors $\psi\in\mathcal{K}^{\mathrm{inv}}$ with $\|\mathcal{U}_v\psi-\psi\|\to0$ as $v\to0$ in $\mathbb{Q}_L^3$ form a closed subspace; by (A2) it contains a family whose linear hull is dense, so it is all of $\mathcal{K}^{\mathrm{inv}}$. As $\|\mathcal{U}_v\psi-\mathcal{U}_{v'}\psi\|=\|\mathcal{U}_{v-v'}\psi-\psi\|$, the map $v\mapsto\mathcal{U}_v\psi$ is uniformly continuous on $\mathbb{Q}_L^3/\ell\mathbb{Z}^3$, a dense subgroup of the compact group $\mathbb{R}^3/\ell\mathbb{Z}^3$; it therefore has exactly one continuous extension, and the extension inherits the homomorphism property and unitarity. For $k\in\frac{2\pi}{\ell}\mathbb{Z}^3$ let $P_k\psi=\ell^{-3}\int_{T_\ell}e^{-ik\cdot x}\,\mathcal{U}(x)\psi\,dx$, an integral of a continuous function. Then $\mathcal{U}(y)P_k\psi=e^{ik\cdot y}P_k\psi$, P_k restricts to the identity on $\mathcal{K}_k$, and P_k is self-adjoint because $\mathcal{U}(x)^*=\mathcal{U}(-x)$; hence P_k is the orthogonal projection onto $\mathcal{K}_k$, and distinct $\mathcal{K}_k$ are orthogonal. If ψ is orthogonal to all the $\mathcal{K}_k$, then for each ϕ the continuous function $x\mapsto\langle\phi,\mathcal{U}(x)\psi\rangle$ has all its Fourier coefficients $\langle\phi,P_k\psi\rangle$ equal to 0 and so vanishes identically; at $x=0$ this gives $\langle\phi,\psi\rangle=0$ for all ϕ, hence $\psi=0$. *(B1)* is Lemma 18.77(d). *(B4).* For a site mirror, $\mathcal{A}_j(0,0)=\mathcal{A}_j(\frac{a_j}{2},\frac{a_j}{2})$ (Lemma 18.77(c)). For a bond mirror, Lemma 18.96(e) approximates $F\in\mathcal{A}(0,0)$ by some $F'\in\mathcal{A}(a_j,0)$, and then F' by some $F''\in\mathcal{A}(a_j,a_{j'})$, with the stated bounds. The density of the classes follows, since $\|\cdot\|_c\le\|\cdot\|_{L^2(\mu)}$. *(B2).* The two moduli are Lemma 18.95(d). Let $F\in\mathcal{A}(0,\alpha')$ with $\alpha'>0$, $\tilde F\in\mathcal{A}(\alpha,\alpha')$ with $\alpha>0$, and $0\le t\le\frac{\alpha'}{2}$. Then $\mathcal{U}_t(F-\tilde F)\in\mathcal{A}(0,0)$ and, by (B1), $\|\mathcal{U}_t(F-\tilde F)\|_c\le\|F-\tilde F\|_{L^2(\mu)}$; hence $\|\mathcal{U}_tF-F\|_c\le2\|F-\tilde F\|_{L^2(\mu)}+\|\mathcal{U}_t\tilde F-\tilde F\|_c$. Lemma 18.96(e) for a bond mirror, and the choice $\tilde F=F$ for a site mirror, make the first term arbitrarily small, and the moduli send the second term to 0 as $t\downarrow0$. For $F\in\mathcal{A}(\alpha,0)$ we have $\|\mathcal{U}_{-t}F-F\|_c=\|\mathcal{U}_tRF-RF\|_c$, because $\langle RF,RF'\rangle_c=\int\overline{R\Theta F}\,RF'\,d\mu=\langle F,F'\rangle_c$ (Lemma 18.77(b), (c)). For $F\in\mathcal{A}(0,0)$ and both signs of t: given $\epsilon>0$, choose by (B4) some $\tilde F\in\mathcal{A}(\alpha,\alpha')$, $\alpha,\alpha'>0$, with $\|F-\tilde F\|_{L^2(\mu)}\le\epsilon$. For $|t|\le\min\{\alpha,\alpha'\}$ the function $\mathcal{U}_t\tilde F$ lies in $\mathcal{A}(0,0)$, and $\int\overline{\Theta F}(\mathcal{U}_tF-F)\,d\mu=\int\overline{\Theta F}(\mathcal{U}_t(F-\tilde F)-(F-\tilde F))\,d\mu+\langle F,\mathcal{U}_t\tilde F-\tilde F\rangle_c$; its modulus is at most $2\|F\|_{L^2(\mu)}\epsilon+\|F\|_{L^2(\mu)}\|\mathcal{U}_t\tilde F-\tilde F\|_c$, and the last norm tends to 0 by the moduli. *(B3).* By Lemma 18.95(a), $\|\mathcal{U}_tF\|_c^2=\varphi_F(2t)$ for $0<2t<2\alpha'$. If $\|F\|_c^2=\varphi_F(0)=0$, Lemma 18.95(e) gives $\varphi_F(2t)=0$. Conversely, if $\varphi_F(2t)=0$, the same item gives $\varphi_F=0$ on $\mathbb{Q}_L\cap(0,2t]$, hence $\varphi_F(0+)=0$; and $\varphi_F(0)=\varphi_F(0+)$ by (B2) and Lemma 18.95(c). *(B5).* We rely on the theorem of A. Klein and L. J. Landau, *Construction of a unique self-adjoint generator for a symmetric local semigroup*, J. Funct. Anal. 44 (1981), 121–137, in the following form. Let $\mathcal{H}$ be a Hilbert space and $T>0$, and for each real $\alpha\in[0,T]$ let S_α be a symmetric linear operator with domain $\mathcal{D}_\alpha\subset\mathcal{H}$, such that (i) $\mathcal{D}_\alpha\supset\mathcal{D}_\beta$ for $\alpha\le\beta$, and $\bigcup_{0<\alpha\le T}\mathcal{D}_\alpha$ is dense; (ii) for $\gamma\in[0,T]$ and $\chi\in\mathcal{D}_\gamma$ the function $\alpha\mapsto\langle\chi,S_\alpha\chi\rangle$ is continuous on $[0,\gamma]$; (iii) S_0 is the identity, $S_\beta\mathcal{D}_\alpha\subset\mathcal{D}_{\alpha-\beta}$ for $0\le\beta\le\alpha$, and $S_\alpha S_\beta=S_{\alpha+\beta}$ on $\mathcal{D}_{\alpha+\beta}$ for $\alpha+\beta\le T$. Then a unique self-adjoint operator H on $\mathcal{H}$ satisfies $\mathcal{D}_\alpha\subset\mathrm{Dom}(e^{-\alpha H})$ and $e^{-\alpha H}|_{\mathcal{D}_\alpha}=S_\alpha$ for all $\alpha\in[0,T]$. We also use three consequences of the spectral theorem, for a self-adjoint H and a vector ϕ with spectral measure ρ_ϕ: $\phi\in\mathrm{Dom}(e^{-tH})$ if and only if $\int e^{-2t\lambda}d\rho_\phi(\lambda)<\infty$; if $\phi\in\mathrm{Dom}(e^{-t_0H})$ with $t_0>0$, then $t\mapsto e^{-tH}\phi$ is norm continuous on $[0,t_0]$; and e^{-tH} is injective with range $\mathrm{Dom}(e^{tH})$, and $e^{tH}e^{-tH}\phi=\phi$. *Step 1: the vectors $\phi(F,\gamma)$.* Let $F\in\mathcal{A}(0,\alpha'_F)$ with $\alpha'_F>0$. By Lemma 18.95(a), $\|\mathcal{U}_tF-\mathcal{U}_{t'}F\|_c^2=\varphi_F(2t)-2\varphi_F(t+t')+\varphi_F(2t')$ for $t,t'\in\mathbb{Q}_L\cap[0,\alpha'_F)$, and by Lemma 18.95(c) and (B2) the function φ_F extends continuously to $[0,2\alpha'_F)$. Hence, for real $\gamma\in[0,\alpha'_F)$, the limit $\phi(F,\gamma)=\lim_{\substack{t\to\gamma\\ t\in\mathbb{Q}_L\cap[0,\alpha'_F)}}[\mathcal{U}_tF]$ exists in $\mathcal{K}_c$; it equals $[\mathcal{U}_tF]$ when $\gamma=t\in\mathbb{Q}_L$, depends continuously on γ in norm, and is independent of the

choice of α'_F. For two such functions F, G and times $t_1 \in \mathbb{Q}_L \cap [0, \alpha'_F)$, $t_2 \in \mathbb{Q}_L \cap [0, \alpha'_G)$, the number $\langle \mathcal{U}_{t_1} F, \mathcal{U}_{t_2} G\rangle_c = \int \overline{\Theta F}\, \mathcal{U}_{t_1+t_2} G \, d\mu$ depends only on $t_1 + t_2$. Approximating real numbers with $\gamma_1 + \gamma'_1 = \gamma_2 + \gamma'_2$ by $t_1, t'_1, t_2 \in \mathbb{Q}_L$ and setting $t'_2 = t_1 + t'_1 - t_2$ (with $t_2 \le t_1 + t'_1$ when $\gamma'_2 = 0$), one obtains a continuous function $\mathsf{U}_{F,G}$ on $[0, \alpha'_F + \alpha'_G)$ with $\langle \phi(F,\gamma), \phi(G,\gamma')\rangle = \mathsf{U}_{F,G}(\gamma + \gamma')$. *Step 2: domains and operators.* For real $\alpha \in [0, \frac{\ell}{2}]$ let $\mathcal{D}_\alpha$ consist of the finite linear combinations of the $\phi(F,\gamma)$ with $\alpha'_F > \alpha + \gamma$, and for $0 \le \beta \le \alpha$ and $\psi = \sum_i z_i \phi(F_i, \gamma_i) \in \mathcal{D}_\alpha$ let $S_\beta \psi = \sum_i z_i \phi(F_i, \gamma_i + \beta)$. This does not depend on how ψ is written. Indeed, with $\psi_x = \sum_i z_i \phi(F_i, \gamma_i + x)$, $0 \le x \le \alpha$, Step 1 gives $\langle \psi_x, \psi_y\rangle = w(x+y)$ with w continuous on $[0, 2\alpha]$, $w(2x) = \|\psi_x\|^2$ and $|w(x+y)|^2 \le w(2x) w(2y)$. If $\psi_0 = 0$, then w vanishes on $[0, \alpha]$; and if w vanishes on $[0, z]$, then it vanishes on $[0, \frac{z}{2} + \alpha]$. Hence $w = 0$ on $[0, 2\alpha]$, and $\psi_x = 0$ for all $x \le \alpha$. *Step 3: checking the hypotheses, with $T = \frac{\ell}{2}$.* (i) The $\mathcal{D}_\alpha$ decrease, and their union contains $[F] = \phi(F, 0)$ for every $F \in \bigcup_{\alpha' > 0} \mathcal{A}(0, \alpha')$, a dense set by (B4). (ii) $\langle \psi, S_\beta \psi\rangle = w(\beta)$. (iii) S_0 is the identity, $S_\beta \mathcal{D}_\alpha \subset \mathcal{D}_{\alpha-\beta}$, the semigroup law is immediate from the definition, and $\langle S_\beta \psi, \psi'\rangle = \sum_{i,k} \bar z_i z'_k \mathsf{U}_{F_i, F'_k}(\gamma_i + \beta + \gamma'_k) = \langle \psi, S_\beta \psi'\rangle$. The theorem provides $\mathfrak{h}_c$ with $e^{-\alpha \mathfrak{h}_c}|_{\mathcal{D}_\alpha} = S_\alpha$. If $\alpha' \le \frac{\ell}{2}$, $F \in \mathcal{A}(0, \alpha')$ and $t \in \mathbb{Q}_L \cap [0, \alpha')$, then $[F] \in \mathcal{D}_t$ and $e^{-t\mathfrak{h}_c}[F] = \phi(F, t) = [\mathcal{U}_t F]$. The constant function 1 involves no bond, lies in every $\mathcal{A}(0, \alpha')$ with $\alpha' < \frac{\ell}{2}$, and satisfies $\mathcal{U}_t 1 = 1$; so $e^{-t\mathfrak{h}_c}[1] = [1]$ for $0 \le t < \frac{\ell}{2}$, the spectral measure of $[1]$ is concentrated at 0, and $\mathfrak{h}_c[1] = 0$. If $\alpha' > \frac{\ell}{2}$, no site of any Λ_j obeys the two bounds that define $\mathsf{Bd}_j(0, \alpha')$, so every $F \in \mathcal{A}(0, \alpha')$ is a constant multiple of 1; then $\mathcal{U}_t F = F$, and $e^{-t\mathfrak{h}_c}[F] = [F]$ for every real $t \ge 0$ because $\mathfrak{h}_c[1] = 0$, which gives the identities of (B5) for such α', the one at $t = \alpha'$ included. *Step 4: uniqueness.* Let H' be self-adjoint with the property stated in (B5). For real $\gamma < \alpha'_F$ choose $t \in \mathbb{Q}_L$ with $\gamma < t < \alpha'_F$; the map $x \mapsto e^{-xH'}[F]$ is norm continuous on $[0, t]$ and equals $[\mathcal{U}_x F]$ at the points $x \in \mathbb{Q}_L$, so $e^{-\gamma H'}[F] = \phi(F, \gamma)$, and $e^{-\beta H'} \phi(F, \gamma) = \phi(F, \gamma + \beta)$ whenever $\gamma + \beta < \alpha'_F$. Thus $e^{-\alpha H'}|_{\mathcal{D}_\alpha} = S_\alpha$, and the uniqueness part of the theorem gives $H' = \mathfrak{h}_c$. *Step 5: the end point.* Let $\alpha' \in \mathbb{Q}_L$, $F \in \mathcal{A}(0, \alpha')$, and $t_n \uparrow \alpha'$ in $\mathbb{Q}_L$. Then $\mathcal{U}_{\alpha'} F \in \mathcal{A}(\alpha', 0)$, and (B2) gives $\|\mathcal{U}_{t_n} F - \mathcal{U}_{\alpha'} F\|_c = \|\mathcal{U}_{-(\alpha' - t_n)}(\mathcal{U}_{\alpha'} F) - \mathcal{U}_{\alpha'} F\|_c \to 0$. Hence $\int e^{-2t_n \lambda} d\rho_{[F]} = \|\mathcal{U}_{t_n} F\|_c^2$ stays bounded, Fatou's lemma gives $\int e^{-2\alpha' \lambda} d\rho_{[F]} < \infty$, and $e^{-t_n \mathfrak{h}_c}[F] \to e^{-\alpha' \mathfrak{h}_c}[F]$; consequently $e^{-\alpha' \mathfrak{h}_c}[F] = [\mathcal{U}_{\alpha'} F]$. *Step 6: negative times.* Let $F \in \mathcal{A}(\alpha, \alpha')$ with $\alpha, \alpha' > 0$, and $t \in \mathbb{Q}_L \cap (0, \alpha]$. Then $G = \mathcal{U}_{-t} F \in \mathcal{A}(\alpha - t, \alpha' + t) \subset \mathcal{A}(0, \alpha' + t)$ and $t < \alpha' + t$, so $[G] \in \mathrm{Dom}(e^{-t\mathfrak{h}_c})$ and $e^{-t\mathfrak{h}_c}[G] = [\mathcal{U}_t G] = [F]$. As the range of $e^{-t\mathfrak{h}_c}$ is the domain of $e^{t\mathfrak{h}_c}$ and $e^{t\mathfrak{h}_c} e^{-t\mathfrak{h}_c}[G] = [G]$, the class $[F]$ lies in the domain of $e^{t\mathfrak{h}_c}$, and $e^{t\mathfrak{h}_c}[F] = [\mathcal{U}_{-t} F]$. □

18.9. **The two minima of one unit of centre flux, and the energy of a pair of $\mathbb{Z}_2$ monopoles.** This subsection sets out two classical layers under the partition functions of the section. The first is one unit of centre flux through the plane 12 of a box: the action of the twisted fields of Lemma 11.9 vanishes on exactly two gauge orbits, each with the stabiliser $\{\pm\mathbf{1}\}$, which the change of sign of the bonds of the direction 3 across one plane exchanges; at them, on a cube of side l whose spacing is a, the second variation in physical units has, on the orthogonal complement of the gauge directions, the floor $a^{-2}\, 4 \sin^2 \frac{\pi a}{2l}$, a number that is at least $4l^{-2}$ at every spacing and tends to $\pi^2 l^{-2}$ as $a \to 0$; and on a box with $|N|$ sites and largest side $\bar N$, when every plaquette variable lies at a distance less than ε from $\mathbf{1}$ and $36\pi |N| \bar N^3 \varepsilon \le 1$, a bound on ε that shrinks as the box grows, the field is, up to a gauge transformation fixed up to its sign, one of the two minima times the exponential of an element of the Landau slice, and the set of these elements lies between two convex sets of the slice (Propositions 18.97 and 18.98); this is the description of the minima that the sketch of Proposition 11.14(a) uses, for a class with one twisted plane. The second is the point operator of a periodic box, the sum of the Wilson integrals over the sign fields whose monopoles are two given cubes, divided by the same sum over the closed sign fields. The least action of one $\mathbb{Z}_2$ monopole on $\mathbb{Z}^3$ is a number c_{mon}, attained at an abelian field, with $0.9960 < c_{\mathrm{mon}} \le \frac{\pi^2}{2} \mathcal{G}(0)$, $\mathcal{G}$ the Green function of $\mathbb{Z}^3$ (Proposition 18.101); the least action of a pair at the lattice vector v is $2c_{\mathrm{mon}} - \pi/(4|v|) + O(|v|^{-2})$

over all SU(2) fields (Proposition 18.103); and on the lattices $\Lambda_{K,\ell}$ of the torus T_ℓ, for every side $\ell = 2L^m$ and every label, the point operator of two distinct points, both centres of cubes of a single lattice of this family, is at most $\exp[-2c_{\mathrm{mon}}\beta + (192L+69)\beta^{3/4}(\log\beta)^{1/4}]$ at all large cutoffs, while for two cubes that differ by a lattice vector v_K with $|v_K| \to \infty$ and $|v_K|^6 \log\beta = o(\beta)$ as the cutoff K tends to infinity, its logarithm is $-2c_{\mathrm{mon}}\beta + \frac{\pi}{4}\beta/|v_K|$ up to $o(\beta/|v_K|)$ (Proposition 18.105). Propositions 18.99, 18.100, 18.102 and 18.104 hold the tools: Stokes inequalities for sets of cubes, abelian fields made from flows and the asymptotics of $\mathcal{G}$, the potential of one monopole at large distances, and the exact relations of the point operator on one box.

Twisted fields. In Propositions 18.97 and 18.98, $N = (N_1, N_2, N_3)$ is a box with all $N_\mu \geq 3$, $|N| = N_1N_2N_3$ and $\bar{N} = \max_\mu N_\mu$; the class is the one whose one twisted plane is 12, with the twist triple $\Omega = (i\sigma_1, i\sigma_2, \mathbf{1})$ of Lemma 11.9, the other classes with one twisted plane being reached by a permutation of the axes. $\mathcal{V}_N$ is the set of the twisted fields of Lemma 11.9, the maps U from the bonds of $\mathbb{Z}^3$ to G with $U(b + N_\mu e_\mu) = \Omega_\mu U(b)\Omega_\mu^{-1}$; $\mathscr{G}_N$ is the group of the maps $g\colon \mathbb{Z}^3 \to G$ with $g(x + N_\mu e_\mu) = \Omega_\mu g(x)\Omega_\mu^{-1}$, which act by $U^g(b) = g(b_-)U(b)g(b_+)^{-1}$; $\mathsf{B}_N = \prod_\mu \{0, \dots, N_\mu - 1\}$, and dU, dg are the products of the Haar probability measures over the bonds and the sites that begin in B_N; (x, μ) is the bond from x to $x + e_\mu$; $\partial U(p)$ is the ordered product around the oriented boundary of p, taking the inverse on a bond traversed backwards, and $A(U) = \sum_p (1 - \frac{1}{2}\mathrm{tr}\,\partial U(p))$, the sum over the $3|N|$ plaquettes based in B_N, is the action A' of Lemma 11.9; $\|\partial U - \mathbf{1}\|_\infty$ is the largest $|\partial U(p) - \mathbf{1}|$ over all plaquettes of $\mathbb{Z}^3$, $|\cdot|$ the operator norm, and it is attained at a plaquette based in B_N, since a translation by a period conjugates the plaquette variable. The map of the sketch of Lemma 11.9, which we call I and which multiplies $U(b)$ on the right by Ω_μ at the bonds b of the direction μ that begin in B_N and end outside it, is a bijection of $\mathcal{V}_N$ onto the periodic fields of the box N that carries dU to Haar measure, with $\mathrm{tr}\,\partial(\mathsf{I}U)(p) = \eta^{n_{12}}(p)\,\mathrm{tr}\,\partial U(p)$ for the sign field $\eta^{n_{12}}$ whose reversed column lies over the point $(N_1 - 1, N_2 - 1)$ of the plane 12, and with $\mathsf{I}(U^g) = (\mathsf{I}U)^{g|_{\mathsf{B}_N}}$; so $A_{\eta^{n_{12}}} \circ \mathsf{I} = A$, $\int e^{-\beta A} dU = Z[n_{12}]$, and every statement below on $\mathcal{V}_N$ is a statement on the class n_{12}. For $\varsigma \in \{+, -\}$ the field $R_\varsigma \in \mathcal{V}_N$ is $\varsigma\mathbf{1}$ at the bonds $(x, 3)$ with $x_3 \equiv N_3 - 1$ modulo N_3 and $\mathbf{1}$ at all other bonds; the flip is $U \mapsto R_- U$, the product bond by bond, that is the change of sign of the bonds of the direction 3 that cross one plane normal to 3; it keeps dU and every plaquette variable. $\mathcal{P}_3(U)(x) = U(x,3)U(x + e_3, 3) \cdots U(x + (N_3 - 1)e_3, 3)$ is the holonomy of the closed line through x in the direction 3. *Forms.* C^0, C^1 and C^2 are the spaces of the $\mathfrak{g}$-valued functions f of the sites, of the bonds, respectively of the plaquettes of $\mathbb{Z}^3$ that obey $f(\cdot + N_\mu e_\mu) = \Omega_\mu f(\cdot)\Omega_\mu^{-1}$; in each of them $\|f\|^2 = \sum |f|^2$, the sum running over the sites of B_N, over the bonds that begin there, respectively over the plaquettes whose base point lies there, with $|X|^2 = \frac{1}{2}\mathrm{tr}\,X^2$ and $\langle X, Y\rangle = \frac{1}{2}\mathrm{tr}\,XY$ on $\mathfrak{g}$, and $|f|_\infty = \max|f|$. $(\nabla\lambda)(b) = \lambda(b_+) - \lambda(b_-)$, ∇^* is its adjoint, $(dX)(p) = X(b_1) + X(b_2) - X(b_3) - X(b_4)$ where b_1, b_2 are the positive bonds on the path from the base of p along its first and second directions, and b_3, b_4 are the positive bonds on the path in the reverse order, so that $X \mapsto dX$ maps C^1 to C^2, and $\mathcal{H}_N = \ker\nabla^* \subset C^1$ is the Landau slice; for $X \in C^1$, $e^{iX} \in \mathcal{V}_N$. We put

$$\lambda_* = \min\{4\sin^2\tfrac{\pi}{2N_1},\, 4\sin^2\tfrac{\pi}{2N_2}\}, \qquad \Xi = \tfrac{\bar{N}}{2}(3|N|)^{1/2}, \qquad r_\Xi = \frac{1}{16\,\Xi}.$$

Proposition 18.97 (One unit of centre flux: the two minima and the floor of the second variation)**.** Let N be a box with all $N_\mu \geq 3$.

(a) $A(U) = 0$ if and only if $U = R_\varsigma^g$ for some $g \in \mathscr{G}_N$ and some $\varsigma \in \{+, -\}$.

(b) $\frac{1}{2}\mathrm{tr}\,\mathcal{P}_3 = \varsigma$ on the orbit of R_ς, so the two orbits are disjoint; the flip maps each of them onto the other; and the stabiliser of R_ς in $\mathscr{G}_N$ is $\{\pm\mathbf{1}\}$.

(c) The holonomies of the periodic field $\mathsf{I}R_\varsigma$ along the straight closed lattice paths from the origin in the directions 1, 2, 3 are $i\sigma_1$, $i\sigma_2$ and $\varsigma\mathbf{1}$.

(d) For $X \in C^1$, $\frac{d^2}{ds^2}A(e^{isX}R_\varsigma)\big|_{s=0} = \|dX\|^2$; the tangent space at R_ς of its orbit is ∇C^0, and its orthogonal complement in C^1 is $\mathcal{H}_N$.

(e) $\|dX\|^2 + \|\nabla^* X\|^2 \geq \lambda_* \|X\|^2$ for $X \in C^1$ and $\|\nabla\lambda\|^2 \geq \lambda_* \|\lambda\|^2$ for $\lambda \in C^0$, with equality in each at some element $\neq 0$; so ∇ is injective, $\dim \mathcal{H}_N = 6|N|$, $\|dX\|^2 \geq \lambda_* \|X\|^2 \geq 4\bar{N}^{-2}\|X\|^2$ on $\mathcal{H}_N$, and $\lambda_* \leq 1$.

(f) On the cube with N sites a side, of spacing a and side $l = Na$, the second variation at R_ς of the action $a^{-1}A$ in physical units, taken in the physical field X/a with a^3 times the lattice-sum inner product, is at least $a^{-2}\, 4\sin^2 \frac{\pi a}{2l} \geq 4l^{-2}$ on the orthogonal complement of the gauge directions, and $a^{-2}\, 4\sin^2 \frac{\pi a}{2l} \to \pi^2 l^{-2}$ as $a \to 0$ at fixed l; for every $j > -m$ the lattice $\Lambda_{j,\ell}$ of the torus T_ℓ is the cube of side ℓ with $N = 2L^{m+j} \geq 2L$, and on it this floor is at least $4\ell^{-2}$.

■

Sketch. (a)–(c). At every plaquette $\partial R_\varsigma(p) = \mathbf{1}$, since a plaquette holds either none of the bonds at which $R_\varsigma = \varsigma\mathbf{1}$ or two of them, run in opposite senses, and $\partial U^g(p)$ is conjugate to $\partial U(p)$; so $A = 0$ on the two orbits. Conversely, if $A(U) = 0$, all plaquette variables of $\mathbb{Z}^3$ are $\mathbf{1}$; two lattice paths with the same ends differ by moves across plaquettes and by backtrackings, so the holonomy $h(x)$ of U along any path from x to 0 is well defined, and $U(b) = h(b_-)h(b_+)^{-1}$. By the twisted periodicity of U the function $x \mapsto \Omega_\mu^{-1} h(x + N_\mu e_\mu)$ has the same increments as h along every bond, so it is $h(x)C_\mu$ with a constant $C_\mu \in G$, that is $h(x + N_\mu e_\mu) = \Omega_\mu h(x) C_\mu$; computing $h(x + N_\mu e_\mu + N_\nu e_\nu)$ in the two orders, with $\Omega_1\Omega_2 = -\Omega_2\Omega_1$ and $\Omega_3 = \mathbf{1}$, gives $C_1C_2 = -C_2C_1$ and $C_3C_j = C_jC_3$ for $j = 1, 2$. Two anticommuting elements of G are imaginary unit quaternions with orthogonal axes; as $\mathrm{Ad}\colon G \to \mathrm{SO}(3)$ is onto, there is $c \in G$ with $cC_jc^{-1} = -i\sigma_j = \Omega_j^{-1}$ for $j = 1, 2$, and an element that commutes with both is $\pm\mathbf{1}$, so $C_3 = \varsigma\mathbf{1}$. Then $h' = hc^{-1}$ has $h'(x + N_j e_j) = \Omega_j h'(x) \Omega_j^{-1}$ for $j = 1, 2$ and $h'(x + N_3 e_3) = \varsigma h'(x)$; with $s(x) = \varsigma^{\lfloor x_3/N_3 \rfloor}$ the map $g = sh'$ lies in $\mathscr{G}_N$, and $U(b) = s(b_-)s(b_+)\, g(b_-)g(b_+)^{-1} = R_\varsigma^g(b)$, since $s(b_-)s(b_+) = \varsigma$ exactly at the bonds where $R_\varsigma = \varsigma\mathbf{1}$. As $\Omega_3 = \mathbf{1}$, $\mathcal{P}_3(U^g)(x) = g(x)\mathcal{P}_3(U)(x)g(x)^{-1}$, so $\mathrm{tr}\,\mathcal{P}_3$ is gauge invariant, and $\mathcal{P}_3(R_\varsigma) = \varsigma\mathbf{1}$, as the closed line meets exactly one bond where $R_\varsigma = \varsigma\mathbf{1}$. $R_- R_\varsigma = R_{-\varsigma}$ and $R_- U^g = (R_- U)^g$, R_- being central. $R_\varsigma^g = R_\varsigma$ says $g(b_-) = g(b_+)$ at every bond, so g is constant, and $g = \Omega_j g \Omega_j^{-1}$ for $j = 1, 2$ leaves $g = \pm\mathbf{1}$. Along the direction μ from the origin, $\mathsf{I}R_\varsigma$ is $\mathbf{1}$ at $N_\mu - 1$ bonds and $R_\varsigma((N_\mu - 1)e_\mu, \mu)\, \Omega_\mu$ at the last one, which gives (c).

(d)–(f). As R_ς is central, $\partial(e^{isX} R_\varsigma)(p) = e^{isX(b_1)} e^{isX(b_2)} e^{-isX(b_3)} e^{-isX(b_4)} = \mathbf{1} + is\, dX(p) + O(s^2)$, and $1 - \frac{1}{2}\mathrm{tr}\, u = \frac{1}{2}|u - \mathbf{1}|^2$ for $u \in G$; so $A(e^{isX} R_\varsigma) = \frac{1}{2} s^2 \|dX\|^2 + O(s^3)$. The curve $s \mapsto R_\varsigma^{e^{is\lambda}}$ has the velocity $-\nabla\lambda$ at $s = 0$, and $(\nabla C^0)^\perp = \ker \nabla^*$. For (e): $\mathrm{Ad}(i\sigma_a)\sigma_c$ is σ_c for $c = a$ and $-\sigma_c$ for $c \neq a$, so the colours of $X = \sum_a X^a \sigma_a$ decouple, with $X^a(\cdot + N_\mu e_\mu) = (-1)^{\theta_\mu^a} X^a$, $\theta^1 = (0, 1, 0)$, $\theta^2 = (1, 0, 0)$, $\theta^3 = (1, 1, 0)$: the colour phases of Proposition 11.14 for the class n_{12}, divided by π. Such functions have the orthogonal basis $e^{ip\cdot x}$, $p_\mu \in \frac{2\pi}{N_\mu}(\mathbb{Z} + \frac{1}{2}\theta_\mu^a)$; with $k_\mu = e^{ip_\mu} - 1$ one has $\widehat{dX}_{\mu\nu} = k_\mu \hat{X}_\nu - k_\nu \hat{X}_\mu$, $\widehat{\nabla^* X} = \sum_\mu \bar{k}_\mu \hat{X}_\mu$ and $\widehat{\nabla\lambda}_\mu = k_\mu \hat{\lambda}$, and Lagrange's identity reads $\sum_{\mu<\nu} |k_\mu \hat{X}_\nu - k_\nu \hat{X}_\mu|^2 + |\sum_\mu \bar{k}_\mu \hat{X}_\mu|^2 = |k|^2 |\hat{X}|^2$ with $|k|^2 = \sum_\mu 4\sin^2 \frac{p_\mu}{2}$. Every θ^a has a $\mu \in \{1, 2\}$ with $\theta_\mu^a = 1$, and there $\frac{p_\mu}{2}$ lies at the distance $\geq \frac{\pi}{2N_\mu}$ from $\pi\mathbb{Z}$; so $|k|^2 \geq \lambda_*$ on every mode, which gives both inequalities. $\sin y \geq \frac{2y}{\pi}$ on $[0, \frac{\pi}{2}]$ gives $\lambda_* \geq 4\bar{N}^{-2}$, and $N_\mu \geq 3$ gives $\lambda_* \leq 1$. If $N_2 \geq N_1$, the field $X = \cos(\pi x_2/N_2)\, \sigma_1$ at the bonds of the direction 1 and 0 at the others lies in $\mathcal{H}_N$ and has $\|dX\|^2 = \lambda_* \|X\|^2$, and $\lambda = \cos(\pi x_2/N_2)\, \sigma_1$ has $\|\nabla\lambda\|^2 = \lambda_* \|\lambda\|^2$; if $N_1 > N_2$ the colour 2 with the axes 1, 2 exchanged serves. ∇ is injective on C^0, of the dimension $3|N|$, so ∇^* maps C^1, of the dimension $9|N|$, onto C^0, and $\dim \mathcal{H}_N = 6|N|$. For (f): in $\mathbf{A} = X/a$ the second variation of $a^{-1}A$ is $a^{-1}\|d(a\mathbf{A})\|^2 = a\|d\mathbf{A}\|^2$, while $(\mathbf{A}, \mathbf{A})_a = a^3 \|\mathbf{A}\|^2$; on $\mathcal{H}_N$ their quotient is at least $a^{-2}\lambda_*$, and on the cube $\lambda_* = 4\sin^2 \frac{\pi a}{2l} \geq 4a^2 l^{-2}$, with $a^{-2}\, 4\sin^2 \frac{\pi a}{2l} \to \pi^2 l^{-2}$; on $\Lambda_{j,\ell}$, $j > -m$, the side is $l = \ell$ and $N = 2L^{m+j} \geq 3$. □

Proposition 18.98 (Small plaquettes: two pieces, each between two convex sets of the slice)**.** Let N be a box with all $N_\mu \geq 3$ and $\varepsilon > 0$.

(a) If $\|\partial U-\mathbf{1}\|_\infty \le \varepsilon$ and $4\bar N^2\varepsilon \le 1$, there are $g\in\mathscr{G}_N$ and $\varsigma\in\{+,-\}$ with $\max_b |U^g(b)-R_\varsigma(b)| \le 3\bar N^2\varepsilon$ and $\varsigma\,\frac12\operatorname{tr}\mathcal{P}_3(U)(x) \ge 1-\frac12(3\bar N^2\varepsilon)^2 \ge \frac{23}{32}$ at every site x.

Let now $36\pi|N|\bar N^3\varepsilon\le 1$, and put $\mathcal{V}_{N,\varepsilon}=\{U\in\mathcal{V}_N : \|\partial U-\mathbf{1}\|_\infty<\varepsilon\}$, $\mathcal{V}^\varsigma_{N,\varepsilon}=\{U\in\mathcal{V}_{N,\varepsilon} : \varsigma\operatorname{tr}\mathcal{P}_3(U)(0)>0\}$, $\mathcal{S}_\varepsilon=\{X\in\mathcal{H}_N : |X|_\infty\le r_\Xi,\ \|\partial e^{iX}-\mathbf{1}\|_\infty<\varepsilon\}$ and, for $\rho>0$, $\mathcal{K}_\rho=\{X\in\mathcal{H}_N : \max_p|dX(p)|<\rho\}$, a convex set symmetric about 0.

(b) $\mathcal{V}_{N,\varepsilon}$ is the disjoint union of $\mathcal{V}^+_{N,\varepsilon}$ and $\mathcal{V}^-_{N,\varepsilon}$; both are open and carried onto themselves by $\mathscr{G}_N$, and the flip exchanges them.

(c) $\mathcal{S}_\varepsilon$ is open in $\mathcal{H}_N$, and on it $\max_p|dX(p)|<2\varepsilon$ and $|X|_\infty\le\|X\|<2\Xi\varepsilon<r_\Xi$.

(d) The map $\Psi_\varsigma\colon\mathscr{G}_N\times\mathcal{S}_\varepsilon\to\mathcal{V}^\varsigma_{N,\varepsilon}$, $\Psi_\varsigma(g,X)=(R_\varsigma e^{iX})^g$, is real-analytic and onto, and $\Psi_\varsigma(g,X)=\Psi_\varsigma(g',X')$ if and only if $X'=X$ and $g'=\pm g$; so, modulo gauge transformations, each of the two pieces $\mathcal{V}^\pm_{N,\varepsilon}$ is a copy of $\mathcal{S}_\varepsilon$.

(e) $\mathcal{K}_{\varepsilon(1-8\Xi^2\varepsilon)}\subset\mathcal{S}_\varepsilon\subset\mathcal{K}_{\varepsilon(1+32\Xi^2\varepsilon)}$, and $8\Xi^2\varepsilon=6\bar N^2|N|\varepsilon\le 0.018$.

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Sketch. (a) *Two nearly anticommuting elements.* Let $C_1,C_2,C_3\in G$, $0\le\delta\le\frac14$, $|C_1C_2+C_2C_1|\le\delta$ and $|C_3C_j-C_jC_3|\le\delta$ for $j=1,2$. Write $C_j=\cos\vartheta_j+i\sin\vartheta_j\,n_j\cdot\sigma$ with $\sin\vartheta_j\ge0$, $|n_j|=1$, and $\nu=n_1\cdot n_2$. Expanding the scalar and vector parts of the anticommutator gives $\frac14|C_1C_2+C_2C_1|^2=\cos^2\vartheta_1+\sin^2\vartheta_1(\cos^2\vartheta_2+\sin^2\vartheta_2\,\nu^2)$, and the same holds with 1 and 2 exchanged; so $|\cos\vartheta_j|\le\frac{\delta}{2}$, $\sin^2\vartheta_j\ge\frac{63}{64}$ and $|\nu|\le0.508\delta$. A rotation $\mathrm{Ad}(c)$ that carries n_1 to e_1 and n_2 to $\nu e_1+(1-\nu^2)^{1/2}e_2$ gives $|cC_1c^{-1}-i\sigma_1|^2=2-2\sin\vartheta_1\le\frac{\delta^2}{2}$ and $|cC_2c^{-1}-i\sigma_2|^2=2-2\sin\vartheta_2(1-\nu^2)^{1/2}\le\frac{32}{63}\delta^2$. As $C_3C_j-C_jC_3=-2i\sin\vartheta_3\sin\vartheta_j\,(n_3\times n_j)\cdot\sigma$ and $\sum_j|n_3\times n_j|^2\ge1-|\nu|$, $\sin^2\vartheta_3\le0.582\delta^2$, and with ς the sign of $\cos\vartheta_3$, $|C_3-\varsigma\mathbf{1}|^2=2-2|\cos\vartheta_3|\le0.588\delta^2$. So $|cC_1c^{-1}-i\sigma_1|\le0.71\delta$, $|cC_2c^{-1}-i\sigma_2|\le0.72\delta$ and $|C_3-\varsigma\mathbf{1}|\le0.77\delta$. *The comb.* Let $\epsilon=\|\partial U-\mathbf{1}\|_\infty$. For $x\in\mathsf{B}_N$ let $h(x)$ be the holonomy of U along the path from 0 that makes x_1 steps e_1, then x_2 steps e_2, then x_3 steps e_3, and put $U'(b)=h(b_-)U(b)h(b_+)^{-1}$ at the bonds with both ends in B_N. Then $U'(x,3)=\mathbf{1}$, and $U'(x,2)$, $U'(x,1)$ are conjugate to the holonomies along the boundaries of ladders of x_3, respectively x_2+x_3, plaquettes; taking the plaquettes of a ladder away one at a time multiplies such a holonomy by a conjugate of a plaquette variable or of its inverse, so $|U'(x,2)-\mathbf{1}|\le x_3\epsilon$ and $|U'(x,1)-\mathbf{1}|\le(x_2+x_3)\epsilon$. *The seams.* For $x\in\mathsf{B}_N$ with $x_\mu=N_\mu-1$ and $x'=x+e_\mu-N_\mu e_\mu$ let $C_\mu(x)=h(x)U(x,\mu)\Omega_\mu h(x')^{-1}$, the value at the bond (x,μ) of $\mathsf{I}(U^{g_0})$, $g_0\in\mathscr{G}_N$ the extension of $h|_{\mathsf{B}_N}$; the plaquette variables of $\mathsf{I}(U^{g_0})$ lie within ϵ of $\eta^{n_{12}}(p)\mathbf{1}$. A plaquette of the periodic box through two neighbouring seam bonds (x,μ), $(x+e_\nu,\mu)$ with $x_\nu<N_\nu-1$ is not reversed, so $|C_\mu(x)-C_\mu(x+e_\nu)|\le\epsilon+|U'(x',\nu)-\mathbf{1}|+|U'(x,\nu)-\mathbf{1}|$. With $C_1=C_1(N_1-1,0,0)$, $C_2=C_2(0,N_2-1,0)$ and $C_3=C_3(0,0,N_3-1)$, moving on each seam first in the lower of its two free directions at the height 0 of the other and then in the other, the bounds of the comb give $|C_1(x)-C_1|\le(x_2+x_3)\epsilon$, $|C_2(x)-C_2|\le(x_1N_2+x_3)\epsilon$ and $|C_3(x)-C_3|\le(x_1+x_2)N_3\epsilon$, all at most $2\bar N(\bar N-1)\epsilon$. *The corners.* The reversed plaquette at $(N_1-1,N_2-1,0)$ gives $|C_1(x)C_2+C_2(x)C_1|\le\epsilon$ there, hence $|C_1C_2+C_2C_1|\le N_1N_2\epsilon$; the plaquettes of the planes 13 and 23 at the corners $(N_1-1,0,N_3-1)$ and $(0,N_2-1,N_3-1)$ give $|C_1C_3-C_3C_1|\le N_1N_3\epsilon$ and $|C_2C_3-C_3C_2|\le N_2N_3\epsilon$. With $\delta=\bar N^2\epsilon\le\frac14$ the first step gives c and ς, and the $g\in\mathscr{G}_N$ that extends $(ch)|_{\mathsf{B}_N}$ has $U^g=cU'c^{-1}$, within $2(\bar N-1)\epsilon$ of $\mathbf{1}$, at the inner bonds, and $U^g(x,\mu)=cC_\mu(x)c^{-1}\Omega_\mu^{-1}$, within $2\bar N(\bar N-1)\epsilon+0.77\bar N^2\epsilon\le3\bar N^2\epsilon$ of $R_\varsigma(x,\mu)$, at the seam bonds. The line through $(x_1,x_2,0)$ has $\mathcal{P}_3(U^g)=cC_3(x_1,x_2,N_3-1)c^{-1}$, within $3\bar N^2\epsilon\le\frac34$ of $\varsigma\mathbf{1}$, and $|u-\varsigma\mathbf{1}|^2=2-\varsigma\operatorname{tr}u$; $\operatorname{tr}\mathcal{P}_3$ is gauge invariant and reads x only through (x_1,x_2) modulo the periods.

(c), (e). Let $X\in\mathcal{H}_N$ with $|X|_\infty\le r_\Xi$ and $\|\partial e^{iX}-\mathbf{1}\|_\infty<\varepsilon$, and let $D=\max_p|dX(p)|$. By Proposition 18.97(e), as there are $3|N|$ plaquettes, $|X|_\infty\le\|X\|\le\lambda_*^{-1/2}\|dX\|\le\Xi D$. The plaquette variable u_p of e^{iX} is a product of four exponentials, and $|e^{iY}-\mathbf{1}-iY|\le\frac12|Y|^2$, $|e^{iY}-\mathbf{1}|\le|Y|$ and $|u_1\cdots u_4-\mathbf{1}-\sum_j(u_j-\mathbf{1})|\le\sum_{i<j}|u_i-\mathbf{1}||u_j-\mathbf{1}|$ for unitary u_j give $|u_p-\mathbf{1}-i\,dX(p)|\le$

$\frac12(\sum_{b\in p}|X(b)|)^2\le 8|X|_\infty^2$. So $D<\varepsilon+8r_\Xi\,\Xi D=\varepsilon+\frac12 D$, that is $D<2\varepsilon$ and $\|X\|<2\Xi\varepsilon$; and $32\Xi^2\varepsilon=24\bar N^2|N|\varepsilon\le\frac{24}{36\pi\bar N}<1$ gives $2\Xi\varepsilon<r_\Xi$, so the condition $|X|_\infty\le r_\Xi$ holds strictly on $\mathcal{S}_\varepsilon$, which is therefore open. The same bound gives $D<\varepsilon+8(2\Xi\varepsilon)^2$ on $\mathcal{S}_\varepsilon$; and for $X\in\mathcal{K}_{\rho'}$, $\rho'=\varepsilon(1-8\Xi^2\varepsilon)$, it gives $|X|_\infty\le\Xi\rho'<r_\Xi$ and $|u_p-\mathbf{1}|<\rho'(1+8\Xi^2\varepsilon)<\varepsilon$. Last, $8\Xi^2\varepsilon=6\bar N^2|N|\varepsilon\le\frac{6}{36\pi\bar N}\le 0.018$.

(b), (d). Let $\|\partial U-\mathbf{1}\|_\infty<\varepsilon$; as $4\bar N^2\varepsilon\le 1$, (a) gives g_0 and ς with $|U_1(b)-\mathbf{1}|\le 3\bar N^2\varepsilon$ for $U_1=R_\varsigma U^{g_0}$. Let $\mathsf{d}(u)\in[0,\pi]$ be the angle of $u\in G$, so that $u=e^{iY}$ with $|Y|=\mathsf{d}(u)$, and let g_1 minimise the continuous function $\Phi(g)=\sum_b\mathsf{d}(U_1^g(b))^2$, the sum over the bonds that begin in B_N, on the compact group $\mathscr{G}_N$. Every angle $\mathsf{d}(U_1^{g_1}(b))$ is at most $\Phi(g_1)^{1/2}\le\Phi(\mathbf{1})^{1/2}\le(3|N|)^{1/2}\frac{3\pi}{2}\bar N^2\varepsilon$, by $\mathsf{d}(u)\le\frac\pi2|u-\mathbf{1}|$, and this number is at most r_Ξ exactly when $36\pi|N|\bar N^3\varepsilon\le 1$, which we assume; so no $U_1^{g_1}(b)$ equals $-\mathbf{1}$, and at every bond we put $X=\frac1i\log U_1^{g_1}$, log being the branch on $G\setminus\{-\mathbf{1}\}$ with angle in $[0,\pi)$. Then $e^{iX}=U_1^{g_1}$, $|X(b)|=\mathsf{d}(U_1^{g_1}(b))$, $\|X\|^2=\Phi(g_1)$ and $|X|_\infty\le r_\Xi$; the intrinsic logarithm commutes with Ad, so $X\in C^1$. At $s=0$ the derivative of $\frac12\mathsf{d}(e^{is\lambda}e^{iY})^2$ is $\langle\lambda,Y\rangle$ and that of $\frac12\mathsf{d}(e^{iY}e^{-is\lambda})^2$ is $-\langle\lambda,Y\rangle$; so the stationarity of Φ at g_1 along $s\mapsto e^{is\lambda}g_1$ is $\langle\nabla^*X,\lambda\rangle=0$ for all $\lambda\in C^0$, that is $X\in\mathcal{H}_N$. Gauge transformations and the multiplication by R_ς keep $\|\partial U-\mathbf{1}\|_\infty$; so $X\in\mathcal{S}_\varepsilon$ and $U=(R_\varsigma e^{iX})^g$ with $g=(g_1g_0)^{-1}$: Ψ_ς is onto. For such U, $\frac12\mathrm{tr}\,\mathcal{P}_3(U)=\varsigma\frac12\mathrm{tr}\,\mathcal{P}_3(e^{iX})$ and $|\mathcal{P}_3(e^{iX})-\mathbf{1}|\le N_3r_\Xi\le\frac{1}{72}$, as $|N|\ge 27$; so $\mathrm{tr}\,\mathcal{P}_3(U)(0)$ does not vanish on $\mathcal{V}_{N,\varepsilon}$ and has the sign ς, which gives (b), the flip changing the sign of $\mathcal{P}_3$. *One to one.* Let $X,X'\in\mathcal{S}_\varepsilon$, $g\in\mathscr{G}_N$ and $e^{iX'(b)}=g(b_-)e^{iX(b)}g(b_+)^{-1}$ at all bonds. Then $|g(b_-)-g(b_+)|\le 2r_\Xi$. Writing $g(0)=\alpha+iw\cdot\sigma$, the bound $|\Omega_jg(0)\Omega_j^{-1}-g(0)|=|g(N_je_j)-g(0)|\le 2N_jr_\Xi$ reads $(w_2^2+w_3^2)^{1/2}\le\bar Nr_\Xi$ and $(w_1^2+w_3^2)^{1/2}\le\bar Nr_\Xi$, so $|w|\le\sqrt2\,\bar Nr_\Xi$; replacing g by $-g$, which acts in the same way, $\alpha\ge0$ and $|g(0)-\mathbf{1}|\le 2\bar Nr_\Xi$; every site is congruent modulo the periods to a site at most $\frac32\bar N$ steps from 0, so $|g-\mathbf{1}|_\infty\le 5\bar Nr_\Xi\le\frac{5}{72}$, and $\lambda=\frac1i\log g\in C^0$, taken at every site with the same intrinsic logarithm, has $|\lambda|_\infty\le 0.07$ and, by $|\lambda-\lambda'|\le|e^{i\lambda}-e^{i\lambda'}|/\cos 0.07$ on the ball $\{|\lambda|\le 0.07\}$, $|\nabla\lambda|_\infty\le 2.01r_\Xi$. Along $U_\tau=(e^{iX})^{e^{i\tau\lambda}}$, $0\le\tau\le1$, $|U_\tau(b)-\mathbf{1}|\le r_\Xi+2.01r_\Xi$, so $X_\tau=\frac1i\log U_\tau$ has $|X_\tau|_\infty\le 3.02r_\Xi$, $X_0=X$ and $X_1=X'$; and $\varphi(\tau)=\frac12\|X_\tau\|^2$ has $\varphi'(\tau)=-\langle\nabla^*X_\tau,\lambda\rangle$, which vanishes at $\tau=0$ and at $\tau=1$. With $T_Y=\int_0^1\mathrm{Ad}(e^{i\sigma Y})\,d\sigma$, the formulas $\frac{d}{d\tau}e^{iX_\tau}=i(T_{X_\tau}\dot X_\tau)e^{iX_\tau}$, $\dot U_\tau=i(\lambda(b_-)-\mathrm{Ad}(U_\tau)\lambda(b_+))U_\tau$ and $T_Y^{-1}(\mathrm{Ad}(e^{iY})-1)=i\,\mathrm{ad}\,Y$ give $\dot X_\tau=-T_{X_\tau}^{-1}\nabla\lambda-i[X_\tau,\lambda(b_+)]$, and

$$\varphi''(\tau)=\sum_b\big(\langle T_{X_\tau}^{-1}\nabla\lambda,\nabla\lambda\rangle+\langle i[X_\tau,\lambda(b_+)],\nabla\lambda\rangle\big)\ge(1-\tfrac12|X_\tau|_\infty^2)\|\nabla\lambda\|^2-2\sqrt3\,|X_\tau|_\infty\|\lambda\|\,\|\nabla\lambda\|\ge 0.92\,\|\nabla\lambda\|^2,$$

by $\langle T_Y^{-1}Z,Z\rangle\ge|Y|\cot|Y|\,|Z|^2$, $\|\lambda\|\le\frac{\bar N}{2}\|\nabla\lambda\|$ (Proposition 18.97(e)) and $\bar Nr_\Xi\le\frac{1}{72}$. As $\varphi'(0)=\varphi'(1)$, $\nabla\lambda=0$, so $\lambda=0$, $g=\pm\mathbf{1}$ and $X'=X$. The general case $\Psi_\varsigma(g,X)=\Psi_\varsigma(g',X')$ is this one with $g'^{-1}g$ in the place of g, as R_ς is central. □

Monopoles. From here on, on $\mathbb{Z}^3$, the lattice of Propositions 18.99–18.103, a unit cube is named by its centre $c\in\mathbb{Z}^3$, the closed cube being $c+[-\frac12,\frac12]^3$, so that the sites have half-integer coordinates. On a periodic box, the lattice of Propositions 18.104 and 18.105, the sites keep the integer coordinates of the periodic lattice of the box, each cube has the form $x+[0,1]^3$ with x a site, and for a cube c and $v\in\mathbb{Z}^3$ we write $c+v$ for the image of c under the translation by v; a set of cubes of a box is brought to $\mathbb{Z}^3$ by a translation that moves the midpoints of all its cubes to points of $\mathbb{Z}^3$, and brought back by the inverse translation. In both cases a plaquette is the common face of two neighbouring cubes, that is an edge $\{c,c'\}$ of the dual lattice; for $c\in\mathbb{Z}^3$, $|c|$ is the Euclidean length, $|c|_\infty$ and $|c|_1$ are the two other norms, and $\mathbb{B}_r=\{c\in\mathbb{Z}^3:|c|_\infty\le r\}$. For a finite set $\mathfrak{B}$ of cubes, $\partial\mathfrak{B}$ is the set of the plaquettes that are a face of exactly one cube of $\mathfrak{B}$, $\mathcal{P}^\circ(\mathfrak{B})$ the set of those that are a face of two cubes of $\mathfrak{B}$, and $\mathfrak{B}^\circ$ the set of the cubes of $\mathfrak{B}$ whose six neighbours lie in $\mathfrak{B}$; the closure of $\mathfrak{B}$ is the set of the bonds and the plaquettes of its cubes. For

a sign field η on the plaquettes, of $\mathbb{Z}^3$ or of a periodic box, $\delta\eta(c) = \prod_{p\subset\partial c}\eta(p)$ on the cubes, and $\mathfrak{m}(\eta) = \{c : \delta\eta(c) = -1\}$ is the set of the monopoles of η; the closed sign fields of the setting are those with $\mathfrak{m}(\eta) = \emptyset$. A configuration on a set P of plaquettes is a pair (U, η) of a field on the bonds of the plaquettes of P and a sign field on P; $\alpha_p \in [0, \pi]$ is given by $\cos\alpha_p = \frac{1}{2}\eta(p)t_p$, $t_p = \operatorname{tr}\partial U(p)$, so that $1 - \frac{1}{2}\eta(p)t_p = \mathsf{f}(\alpha_p)$ with $\mathsf{f}(t) = 1 - \cos t$; for a field U and a sign field η on the whole of $\mathbb{Z}^3$, $A_\eta(U) = \sum_p(1 - \frac{1}{2}\eta(p)t_p) \in [0, \infty]$, summed over every plaquette of $\mathbb{Z}^3$, is the action of eq. 11.i on that lattice; and $\alpha_p = d_G(\partial U(p), \eta(p)\mathbf{1})$, where $d_G(X, Y) = \arccos\frac{1}{2}\operatorname{tr}(X^*Y) \in [0, \pi]$ is the angle of X and Y as unit vectors of $\mathbb{R}^4 \supset G$, a metric that is invariant under left and right translations and under $X \mapsto X^{-1}$, with $d_G(\mathbf{1}, -\mathbf{1}) = \pi$. A flow is a real function j of the oriented edges of the dual lattice with $j(c' \to c) = -j(c \to c')$, and $\operatorname{div} j(c) = \sum_{c'\sim c} j(c \to c')$; for a function u on the cubes $\nabla u(c \to c') = u(c) - u(c')$, so that $\sum \nabla u\, j = \sum_c u \operatorname{div} j$, the first sum over the edges, when one of the two factors has a finite support or both decay fast enough; on a plaquette p a flow is read across p in the orientation fixed by the duality, and only $\cos j(p)$ and parities enter below. Last, $\mathsf{f}^*(y) = y\arcsin y - 1 + (1 - y^2)^{1/2}$ for $|y| \le 1$; so $\mathsf{f}(j) + \mathsf{f}^*(\sin j) = j\sin j$ for $|j| \le \frac{\pi}{2}$, and $(\mathsf{f}^*)' = \arcsin$.

Proposition 18.99 (Stokes inequalities for sets of cubes)**.** (a) A polygon is a closed lattice path; its sides are its steps, and its corners are the pairs of cyclically consecutive sides. Let finitely many polygons $w_1, \dots, w_F$ be given with a pairing of all their sides, two paired sides lying on one bond; let E be the number of pairs and V the number of the classes of corners for the equivalence generated by the rule that the corners next to two paired sides at the same end of their bond are equivalent. If the graph whose vertices are the polygons and whose edges are the pairs is connected and $V - E + F = 2$, then the polygons can be put in an order, and exponents $\epsilon_k = \pm 1$ chosen, independently of the field, so that every field U has $\prod_k g_k U(w_k)^{\epsilon_k} g_k^{-1} = \mathbf{1}$, the factors in that order, where $g_k = U(\zeta_k)$ is the product of U along a lattice path ζ_k; so for all signs s_k, $\sum_k d_G(U(w_k), s_k\mathbf{1}) \ge \pi$ if $\prod_k s_k = -1$, and $d_G(U(w_{k_0}), \prod_{k\ne k_0} s_k\mathbf{1}) \le \sum_{k\ne k_0} d_G(U(w_k), s_k\mathbf{1})$ for each k_0.

(b) A finite set R of cubes is good if $\sum_{p\in\partial R}\alpha_p \ge \pi$ for every configuration on ∂R with $\prod_{p\in\partial R}\eta(p) = -1$, that is with $\prod_{c\in R}\delta\eta(c) = -1$. We call a set Ω of cells of $\mathbb{Z}^2$ admissible when it is finite and non-empty, each of its rows is an interval, its non-empty rows occupy an interval, and each pair of consecutive non-empty rows shares a column; a set of cubes $\bigcup_{h_-\le h\le h_+}\Omega_h \times \{h\}$ for which every Ω_h and every $\Omega_h \cap \Omega_{h-1}$, $h_- < h \le h_+$, is admissible is called a generalised stack along the axis 3, and generalised stacks along the other axes are defined in the same way. Every generalised stack is good. Boxes and staircases are generalised stacks, a staircase being a finite set $R \ni 0$ that contains, along with any c in it, every c' each coordinate of which lies between 0 and the corresponding coordinate of c; and when $K \subset \mathbb{R}^3$ is compact, convex and non-empty, the set $\mathsf{R}(K)$ of all cubes meeting K is a generalised stack.

(c) Let (U, η) be given on the plaquettes of $\mathbb{Z}^3$, or on $\mathcal{P}^\circ(\mathfrak{B})$ for a finite set $\mathfrak{B}$ of cubes; let μ be a finite positive measure on a family of good sets R, contained in $\mathfrak{B}^\circ$ in the second case, each with $\prod_{c\in R}\delta\eta(c) = -1$, and let $y_p = \mu\{R : p \in \partial R\} \le \frac{1}{2}$ for all p. Then $\sum_p \mathsf{f}(\alpha_p) \ge \pi\,\mu(\text{all}) - \sum_p \mathsf{f}^*(y_p)$, the sums over all plaquettes, respectively over $\mathcal{P}^\circ(\mathfrak{B})$. ∎

Sketch. (a) If a side of w on the bond b is paired with a side of another polygon w', rotate and reverse the two so that $w = \mu b$ and $w' = b^{-1}\mu'$; the polygon $\mu\mu'$ has the holonomy $U(w)U(w')$, keeps the other pairs, and replaces the two pairs of equivalent corners at the ends of b by one corner at each end: F and E drop by one and V stays. Merging along a spanning tree of the graph of the polygons leaves one polygon $\hat{w}$ whose $2m$ sides are paired among themselves, $m = E - F + 1 = V - 1$, and whose holonomy is a product of conjugates of the $U(w_k)^{\pm 1}$, each k once; a rotation of a polygon

conjugates its holonomy by $U(\zeta)$ for a path ζ of some of its sides, and a reversal inverts it, so the order of the factors and their exponents come from the merging alone, and each conjugating element is $U(\zeta_k)$ for a path ζ_k made of sides of the polygons. In the graph $\hat\Gamma$ whose vertices are the classes of corners and whose edges are the pairs, a side joins the classes of the corners at its two ends, and $\hat w$ passes through all of them; so $\hat\Gamma$ is connected with V vertices and $V-1$ edges, a tree. At a leaf, every corner of that class is formed by the only pair $\{s,s'\}$ at it; hence s and s' follow one another in $\hat w$, lie on a single bond, and their meeting point is the site of that corner, so that $\hat w$ goes along a bond and straight back; erasing these two steps leaves the holonomy unchanged and takes the leaf away. By induction down to $m=1$, where $\hat w$ is one bond run there and back, $U(\hat w)=\mathbf{1}$. Solving the relation for one factor, the invariance of d_G under translations and $X\mapsto X^{-1}$, $d_G(XY,ss'\mathbf{1})\le d_G(X,s\mathbf{1})+d_G(Y,s'\mathbf{1})$ for signs s,s', and $d_G(\mathbf{1},-\mathbf{1})=\pi$ give the inequalities.

(b) *In a plane.* Let Ω be admissible. Because consecutive rows overlap, it never happens that two cells of Ω touch only at a corner while neither of the other two cells sharing that corner is present; the union of its closed cells, built up row by row with each new row meeting the union of the earlier ones in an interval of edges, has (vertices) − (edges) + (cells) = 1; and the edges lying on exactly one cell make up a single simple closed path $\partial\Omega$, which climbs along the right ends of the rows and descends along their left ends. Let $\Omega''\subset\Omega$ be admissible or empty, in a lattice plane of $\mathbb{Z}^3$. The polygons $\partial\Omega$, $\partial\Omega''$ and the boundaries of the plaquettes of $\Omega\setminus\Omega''$, with the pairing forced by the edges, form a connected complex in which the corners at one lattice vertex form one class whenever there are any; counting, $V-E+F=[V(\Omega)-E(\Omega)+|\Omega|]-[V_{\rm in}(\Omega'')-E_{\rm in}(\Omega'')+|\Omega''|]+2=1-1+2$ when $\Omega''\neq\emptyset$, the vertices and edges inner to Ω'' being those that carry no corner and no pair; when $\Omega''=\emptyset$ the complex consists of $\partial\Omega$ and the boundaries of the plaquettes of Ω, and $V-E+F=V(\Omega)-E(\Omega)+|\Omega|+1=2$. By (a), for every sign s, $d_G(U(\partial\Omega),s\varpi\mathbf{1})\le d_G(U(\partial\Omega''),s\mathbf{1})+\mathsf{s}$ and $d_G(U(\partial\Omega''),s\varpi\mathbf{1})\le d_G(U(\partial\Omega),s\mathbf{1})+\mathsf{s}$, with $\varpi=\prod_{\Omega\setminus\Omega''}\eta(p)$, $\mathsf{s}=\sum_{\Omega\setminus\Omega''}\alpha_p$ and $U(\partial\emptyset)=\mathbf{1}$. *A stack.* ∂R is made of the horizontal plaquettes over $\Omega_h\triangle\Omega_{h-1}$ between the layers $h-1$ and h, with $\Omega_{h_+ +1}=\Omega_{h_- -1}=\emptyset$, and of the bands of vertical plaquettes of each layer h over the edges of $\partial\Omega_h$. A band with the two copies of $\partial\Omega_h$ on its upper and lower planes is a complex as in (a) with $V-E+F=2n-3n+(n+2)=2$, n the length of $\partial\Omega_h$. Going down from the top faces, through each band, and between consecutive layers from $\partial\Omega_h$ to $\partial(\Omega_h\cap\Omega_{h-1})$ and on to $\partial\Omega_{h-1}$ by the inequalities in a plane, each plaquette of ∂R is used once, and the lower copy of $\partial\Omega_{h_-}$ comes within $\sum_{\partial R\setminus\text{bottom}}\alpha_p$ of one sign and within $\sum_{\text{bottom}}\alpha_p$ of the other, the product of the two signs being $\prod_{\partial R}\eta(p)=-1$; so $\sum_{\partial R}\alpha_p\ge\pi$. *Examples.* A staircase of $\mathbb{Z}^2$ is admissible, and two staircases meet in a staircase; so staircases of $\mathbb{Z}^3$, and boxes, are generalised stacks. For a compact convex $K'\subset\mathbb{R}^2$ the row y of $\mathsf{R}_2(K')$, the cells that meet K', is the set of the integers in $[\alpha-\frac12,\beta+\frac12]$, $[\alpha,\beta]$ the projection to the first axis of $K'\cap\{|x_2-y|\le\frac12\}$, and a segment between points of K' in two consecutive rows crosses the line between them in a point of K', which lies in a cell of each row; and $\mathsf{R}(K)$ has the layers $\Omega_h=\mathsf{R}_2(\pi_{12}(K\cap\{|x_3-h|\le\frac12\}))$, with $\Omega_h\cap\Omega_{h-1}=\mathsf{R}_2(\pi_{12}(K\cap\{x_3=h-\frac12\}))$ by the same crossing over one cell.

(c) For $0\le y\le\frac12$ and $0\le\alpha\le\pi$, $\mathsf{f}(\alpha)\ge y\alpha-\mathsf{f}^*(y)$: with $y=\sin j$, $0\le j\le\frac{\pi}{6}$, the difference $\mathsf{f}(\alpha)-\mathsf{f}(j)-(\alpha-j)\sin j$ vanishes at j, falls on $[0,j]$, rises on $[j,\pi-j]$, falls on $[\pi-j,\pi]$, and is $1+\cos j-(\pi-j)\sin j\ge0.55$ at π. Summing with $y=y_p$, and $\sum_p y_p\alpha_p=\int\mu(dR)\sum_{p\in\partial R}\alpha_p\ge\pi\mu(\text{all})$ by (b) and the definition of a good set, give the claim; for $R\subset\mathfrak{B}^\circ$, $\partial R\subset\mathcal{P}^\circ(\mathfrak{B})$. □

Proposition 18.100 (Abelian fields from flows; the Green function)**.** (a) Let j be a flow on $\mathbb{Z}^3$ and γ an integer flow with $\operatorname{div} j=\pi\operatorname{div}\gamma$. There is a real function ϕ of the bonds such that the field $U(b)=e^{i\phi(b)\sigma_3/2}$ and the sign field $\eta(p)=(-1)^{\gamma(p)}$ have $\frac12\eta(p)t_p=\cos j(p)$ at every plaquette; then $\mathfrak{m}(\eta)=\{c:\operatorname{div}\gamma(c)\text{ odd}\}$ and $A_\eta(U)=\sum_p\mathsf{f}(j(p))$. If $j-\pi\gamma$ vanishes off the plaquettes of a box of sites, ϕ may be taken 0 off the bonds of that box, so that $U=\mathbf{1}$ there.

(b) Every closed sign field on the plaquettes of $\mathbb{Z}^3$ is $d\gamma$ for a sign field γ on the bonds; so $\inf\{A_\eta(U) : \mathfrak{m}(\eta) = M\}$, over the fields on $\mathbb{Z}^3$, is the same for every sign field η with the monopoles M.

(c) Let $\lambda(k) = \sum_\mu (2 - 2\cos k_\mu)$ and $\mathcal{G}(c) = (2\pi)^{-3} \int_{[-\pi,\pi]^3} e^{ik\cdot c} \lambda(k)^{-1} dk$. $\mathcal{G}$ is the one function on $\mathbb{Z}^3$ with $\sum_{c' \sim c} (\mathcal{G}(c) - \mathcal{G}(c')) = \delta_{c,0}$ that tends to 0 at infinity. For every $n \geq 0$ there is C_n with $|\nabla_{e^{(1)}} \cdots \nabla_{e^{(n)}} (\mathcal{G} - \mathcal{G}_\mathrm{c})(c)| \leq C_n |c|^{-3-n}$ for every choice of $e^{(1)}, \dots, e^{(n)}$ among the six lattice unit vectors $\pm e_\mu$ and every $c \in \mathbb{Z}^3$ with $|c| \geq 2n + 2$, where $\mathcal{G}_\mathrm{c}(x) = \frac{1}{4\pi|x|}$ and $\nabla_e F(c) = F(c + e) - F(c)$; so there is an absolute $C_\mathcal{G}$ with $|\mathcal{G}(c) - \frac{1}{4\pi|c|}| \leq C_\mathcal{G} |c|^{-3}$, $|\nabla \mathcal{G}| \leq C_\mathcal{G} |c|^{-2}$ and $|\nabla \nabla \mathcal{G}| \leq C_\mathcal{G} |c|^{-3}$ at all $c \neq 0$; and $\sum (\nabla \mathcal{G})^2 = \mathcal{G}(0)$, the sum over the edges.

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Sketch. (a) Through the duality between the plaquettes of $\mathbb{Z}^3$ and the edges of the dual lattice, oriented so that the coboundary of a real function Θ of the plaquettes, read on a cube, is the divergence of the corresponding flow, put $\Theta = 2(j - \pi\gamma)$; then $d\Theta = 2(\operatorname{div} j - \pi \operatorname{div} \gamma) = 0$. A closed real function of the plaquettes of $\mathbb{Z}^3$ is $d\phi$ for a real function ϕ of the bonds: put $\phi = 0$ on the bonds of the direction 3, fix ϕ on the plane $x_3 = 0$ by summing Θ over the plaquettes of that plane along rows, and continue it upwards and downwards by summing Θ over the vertical plaquettes; $d\phi - \Theta$ is then closed and vanishes on all vertical plaquettes and on the plane $x_3 = 0$, so it vanishes. The abelian field $U = e^{i\phi\sigma_3/2}$ has $\partial U(p) = e^{i(d\phi)(p)\sigma_3/2}$, so $\frac{1}{2} t_p = \cos(j(p) - \pi\gamma(p)) = (-1)^{\gamma(p)} \cos j(p)$, and $\delta\eta(c) = (-1)^{\operatorname{div}\gamma(c)}$. If Θ vanishes off the plaquettes of a box of sites, sum from below instead: $\phi = 0$ on the bonds of the direction 3 and, on the bond (ξ, τ) of the direction $\nu \in \{1, 2\}$, minus the sum of Θ over the vertical plaquettes of the direction ν over ξ below the height τ; this vanishes below the box and off the columns over it, and at the height τ_+ of the highest layer of sites of the box and at every greater height it is one and the same closed function of the plane, which vanishes off the projection of the box and is therefore the gradient of a function $\varkappa$ of the plane that vanishes off that projection, whose complement is connected; subtracting the gradient of the function that is $\varkappa(\xi)$ at the sites (ξ, τ) with $\tau \geq \tau_+$ and 0 at the others leaves the same $d\phi$, makes ϕ vanish on the horizontal bonds at the heights $\geq \tau_+$, and alters it on the vertical bonds only between the heights $\tau_+ - 1$ and τ_+ over the projection, which are bonds of the box; so ϕ then vanishes off the bonds of the box. (b) The same construction with signs in the place of real numbers gives the first claim, and $U \mapsto \gamma U$ keeps the Haar measure, with $A_{\eta\, d\gamma}(U) = A_\eta(\gamma U)$.

(c) $\lambda \geq \frac{4}{\pi^2} |k|^2$ on the cube $[-\pi, \pi]^3$, so λ^{-1} is integrable, $\sum_{c' \sim c} (\mathcal{G}(c) - \mathcal{G}(c')) = \delta_{c,0}$, and $\mathcal{G} \to 0$ by the Riemann–Lebesgue lemma; the difference of two such functions is harmonic and tends to 0, so a positive maximum or a negative minimum would be attained and would spread along a line to infinity. For the expansion fix $\chi \in C_c^\infty(\mathbb{R}^3)$ with $\chi = 1$ on $\{|k| \leq 1\}$ and $\chi = 0$ on $\{|k| \geq 2\}$, and write $\mathcal{G} = \mathcal{G}_1 + \mathcal{G}_2 + \mathcal{G}_3$ with the integrands $\chi |k|^{-2}$, $\chi \mathsf{r}$ and $(1 - \chi)\lambda^{-1}$, where $\mathsf{r} = \lambda^{-1} - |k|^{-2} = \mathsf{q}/(\lambda |k|^2)$ and $\mathsf{q} = |k|^2 - \lambda$. For real m let $\mathcal{S}^m$ be the set of the smooth functions F on $\{0 < |k| < \pi\}$ with $|\partial^\alpha F| \leq A_\alpha |k|^{m - |\alpha|}$ for all α. Then $\mathcal{S}^m \mathcal{S}^{m'} \subset \mathcal{S}^{m+m'}$, by the Leibniz rule, and $\mathcal{S}^m \subset \mathcal{S}^{m'}$ for $m \geq m'$, as $|k| < \pi$; for an integer $m \geq 0$, every function smooth near the closed ball that vanishes at 0 to the order m lies in $\mathcal{S}^m$, by Taylor's formula; $|k|^{-2} \in \mathcal{S}^{-2}$ and $|k|^{-4} \in \mathcal{S}^{-4}$, by homogeneity; and $1/F \in \mathcal{S}^0$ for every $F \in \mathcal{S}^0$ bounded below by a positive constant, by induction on $|\alpha|$ in $\partial^\alpha (F \cdot F^{-1}) = 0$. As q vanishes at 0 to the fourth order, $\mathsf{q} \in \mathcal{S}^4$; so $\mathsf{q}|k|^{-4} \in \mathcal{S}^0$ and $\lambda/|k|^2 = 1 - \mathsf{q}|k|^{-2} \in \mathcal{S}^0$, and since $\lambda/|k|^2 \geq \frac{4}{\pi^2}$, also $|k|^2/\lambda \in \mathcal{S}^0$; hence $\mathsf{r} = (\mathsf{q}|k|^{-4})(|k|^2/\lambda) \in \mathcal{S}^0$. For $m > -3$, an integer $M > m + 3$ and F supported in $\{|k| \leq 2\}$ with $|\partial^\alpha F| \leq A |k|^{m - |\alpha|}$ for $|\alpha| \leq M$, one has $|\int F e^{ik\cdot x} dk| \leq C(m, M) A |x|^{-3-m}$: split F with $\chi(|x|k)$; the part near 0 is at most $A \int_{|k| \leq 2/|x|} |k|^m dk$, and in the rest integrate by parts M times in a coordinate with $|x_\mu| \geq |x|/\sqrt{3}$. Applied to $(ik)^\gamma \chi \mathsf{r}$ with $m = |\gamma|$, this gives $|\partial^\gamma \mathcal{G}_2(x)| \leq C_\gamma |x|^{-3-|\gamma|}$ on $\mathbb{R}^3 \setminus \{0\}$. $(1 - \chi)\lambda^{-1}$ continues to a smooth periodic function, so on $\mathbb{Z}^3$, where $e^{ik\cdot c}$ is periodic

in k, $\mathcal{G}_3$ and all its finite differences decay faster than every power, by integration by parts on the torus. And $\mathcal{G}_{\mathrm{c}}(x) = \lim_{\varepsilon\to 0}(2\pi)^{-3}\int |k|^{-2}e^{-\varepsilon|k|^2}e^{ik\cdot x}dk$, by $|k|^{-2}e^{-\varepsilon|k|^2} = \int_\varepsilon^\infty e^{-t|k|^2}dt$ and $\int_0^\infty (4\pi t)^{-3/2}e^{-|x|^2/4t}dt = \frac{1}{4\pi|x|}$; the part with $1-\chi$, whose symbol bounds hold uniformly in $\varepsilon\in(0,1]$, decays with all its derivatives faster than every power of $|x|$, and so does $\mathcal{G}_{\mathrm{c}}-\mathcal{G}_1$. Hence $\mathcal{G}-\mathcal{G}_{\mathrm{c}} = \mathcal{G}_2+\mathcal{G}_3-(\mathcal{G}_{\mathrm{c}}-\mathcal{G}_1)$ on $\mathbb{Z}^3\setminus\{0\}$, and a finite difference of order n at c is the mean of the corresponding derivative over a cube of points at the distance $\geq |c|/2$ from 0 when $|c|\geq 2n+2$. The bounds on $\mathcal{G}$ follow with $|\partial^\gamma \mathcal{G}_{\mathrm{c}}|\leq C_\gamma |x|^{-1-|\gamma|}$, and a larger constant covers the finitely many $c\neq 0$ with $|c|<8$. Last, summation by parts over growing boxes, with $\mathcal{G}\leq C|c|^{-1}$ and $|\nabla\mathcal{G}|\leq C|c|^{-2}$, gives $\sum(\nabla\mathcal{G})^2 = \sum_c \mathcal{G}(c)\delta_{c,0} = \mathcal{G}(0)$. □

Proposition 18.101 (One monopole)**.** For $r\geq 1$ let $c_{\mathrm{mon}}(r)$ be the least value of $\sum_{p\in\mathcal{P}^\circ(\mathbb{B}_r)}(1-\frac{1}{2}\eta(p)t_p)$ over the configurations on the plaquettes of the closure of $\mathbb{B}_r$ with $\delta\eta(0)=-1$ and $\delta\eta(c)=1$ for $c\in\mathbb{B}_{r-1}\setminus\{0\}$, and let $c_{\mathrm{mon}} = \inf A_\eta(U)$ over the fields U on $\mathbb{Z}^3$ and the sign fields η on its plaquettes with $\mathfrak{m}(\eta)=\{0\}$, A_η the sum over all plaquettes. Let Γ_r be the graph with the vertices $\mathbb{B}_r$ and the edges $\mathcal{P}^\circ(\mathbb{B}_r)$, $\mathcal{F}_r$ the set of the flows j on Γ_r with $\operatorname{div} j = \pi\delta_0$ on $\mathbb{B}_{r-1}$, and $F_r(j) = \sum \mathsf{f}(j(p))$.

(a) $c_{\mathrm{mon}}(r) = \min_{\mathcal{F}_r} F_r$, and $c_{\mathrm{mon}}(r)$ is attained at a field with values in $\{e^{i\varphi\sigma_3/2} : \varphi\in\mathbb{R}\}$.

(b) There are a minimiser j_r^* of F_r on $\mathcal{F}_r$ and a function $u_r\geq 0$ on $\mathbb{Z}^3$, equal to 0 off $\mathbb{B}_{r-1}$, with $j_r^* = \arcsin\nabla u_r$ and $\sum_{c'\sim c}\arcsin(u_r(c)-u_r(c')) = \pi\delta_{c,0}$ for $c\in\mathbb{B}_{r-1}$; u_r is invariant under the 48 isometries of $\mathbb{Z}^3$ that fix 0 and does not increase when one $|c_i|$ increases, so the sets $\{u_r>s\}$, $0\leq s<u_r(0)$, are staircases; and $|j_r^*|\leq\frac{\pi}{6}$, so that $|\nabla u_r|\leq\frac{1}{2}$, with the outward flow $\frac{\pi}{6}$ through each face of the cube 0.

(c) $c_{\mathrm{mon}}(r)$ increases to c_{mon}; c_{mon} is attained at an abelian field, and it is the same for every sign field with the one monopole 0.

(d) $c_{\mathrm{mon}}(1) = 6(1-\cos\frac{\pi}{6})$, and $0.9960 < \sum_{r\geq 0} n_r\,\mathsf{f}(\pi/n_r)\leq c_{\mathrm{mon}}\leq\frac{\pi^2}{2}\mathcal{G}(0)$, with $n_r = 6(2r+1)^2$ and $\mathcal{G}$ of Proposition 18.100(c).

(e) For $3\leq r<r'$, $0\leq c_{\mathrm{mon}}(r')-c_{\mathrm{mon}}(r)\leq 6.25/\lfloor r/3\rfloor\leq 32/r$; so $c_{\mathrm{mon}}-c_{\mathrm{mon}}(r)\leq 32/r$ for $r\geq 3$.

(f) u_r increases with r to a function u_∞ with the symmetry and the monotonicity of (b), $0\leq u_\infty\leq u_\infty(0)$ and $u_\infty(c)\leq 1.99/\lfloor|c|_\infty/3\rfloor$ for $|c|_\infty\geq 3$; $j_\infty = \arcsin\nabla u_\infty$ is a flow on $\mathbb{Z}^3$ with $|j_\infty|\leq\frac{\pi}{6}$, $\operatorname{div} j_\infty = \pi\delta_0$ and $\sum_p \mathsf{f}(j_\infty(p)) = c_{\mathrm{mon}}$.

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Sketch. (a), (b). *Flows give fields.* For $j\in\mathcal{F}_r$ lead the amount $-\operatorname{div} j(c)$ from each cube c of $\mathbb{B}_r\setminus\mathbb{B}_{r-1}$ straight outwards to infinity, along $\pm e_i$ for the first i with $c_i=\pm r$; these rays are disjoint and use no edge of $\mathcal{P}^\circ(\mathbb{B}_r)$, and the continued flow has the divergence $\pi\delta_0$ on $\mathbb{Z}^3$. Proposition 18.100(a) with γ the unit flow along $0, e_3, 2e_3, \dots$ gives an abelian field with the one monopole 0 whose restriction to the closure of $\mathbb{B}_r$ is admitted in $c_{\mathrm{mon}}(r)$ with the value $F_r(j)$; so $c_{\mathrm{mon}}(r)\leq\inf_{\mathcal{F}_r} F_r$. *A convex problem.* Let $\tilde{\mathsf{f}}$ be the even C^2 function equal to f on $[-1,1]$ and to $\mathsf{f}(1)+(|t|-1)\sin 1+\frac{1}{2}(|t|-1)^2\cos 1$ beyond; $\tilde{\mathsf{f}}''\geq\cos 1>0.54$ and $\tilde{\mathsf{f}}\geq\mathsf{f}$. On the non-empty affine space $\mathcal{F}_r$ the strictly convex and coercive $\tilde{F}_r = \sum\tilde{\mathsf{f}}(j(p))$ has one minimiser j_r^*, invariant under the 48 isometries. Its first variation vanishes on the flows without divergence on $\mathbb{B}_{r-1}$, which form the orthogonal complement of the gradients of the functions supported in $\mathbb{B}_{r-1}$; so $\tilde{\mathsf{f}}'(j_r^*) = \nabla u_r$ with u_r supported in $\mathbb{B}_{r-1}$, and $\sum_{c'\sim c} g(u_r(c)-u_r(c')) = \pi\delta_{c,0}$ on $\mathbb{B}_{r-1}$, $g = (\tilde{\mathsf{f}}')^{-1}$ strictly increasing. *Sign and moving planes.* At a negative minimum of u_r all terms of the equation are ≤ 0 while their sum is ≥ 0, so the neighbours share the value, and walking in one direction one leaves $\mathbb{B}_{r-1}$: so $u_r\geq 0$. Let σ be an isometry of $\mathbb{Z}^3$ and H a set of cubes with $0\notin H$, $H+e\subset H$ for a unit vector e, $\sigma(H\cap\mathbb{B}_{r-1})\subset\mathbb{B}_{r-1}$, and such that every neighbour $c'\notin H$ of a cube $c\in H$ has $w(c')=0$ or $w(c')=-w(c)$, where $w = u_r\circ\sigma - u_r$. Then $w\geq 0$ on H: off $\mathbb{B}_{r-1}$ this is $u_r\geq 0$; at a negative minimum of w on $H\cap\mathbb{B}_{r-1}$, the equation at σc has the right member ≥ 0, the one at c the right member 0, and the arguments of g differ

by $w(c)-w(c')\le 0$, strictly where $c'\notin H\cap\mathbb{B}_{r-1}$; so all differences vanish, and the minimum spreads in the direction e out of $\mathbb{B}_{r-1}$, which is absurd. For $\sigma(c)=(2k+1-c_1,c_2,c_3)$, $k\ge 0$, $H=\{c_1\ge k+1\}$, $e=e_1$, the neighbour of $c\in H$ outside H is σc; this gives $u_r(c-e_1)\ge u_r(c)$ for $c_1\ge 1$, and with the symmetries the monotonicity. *The size of the flow.* For an edge p from c to c' with $u_r(c)>u_r(c')$ and $s\in[u_r(c'),u_r(c))$, the set $R_s=\{u_r>s\}\subset\mathbb{B}_{r-1}$ holds 0, so the outflow $\pi=\sum_{\partial R_s}j_r^*$ is a sum of positive terms; R_s is invariant under the 48 isometries, which carry p to at least six plaquettes of ∂R_s with the same flow; so $6j_r^*(p)\le\pi$. Hence $\tilde{\mathsf{f}}=\mathsf{f}$ and $g=\arcsin$ at j_r^*, $|\nabla u_r|=\sin|j_r^*|\le\frac12$, and by symmetry each face of the cube 0 carries $\frac{\pi}{6}$. *Every field costs as much.* Let (U,η) be admitted in $c_{\mathrm{mon}}(r)$. By the tangent inequality in the sketch of Proposition 18.99(c), $\mathsf{f}(\alpha_p)\ge\mathsf{f}(|j_r^*(p)|)+(\alpha_p-|j_r^*(p)|)|\nabla u_r(p)|$; summing, $\sum|\nabla u_r||j_r^*|=\sum\nabla u_r\,j_r^*=\pi u_r(0)$, and by the coarea formula $\sum_p|\nabla u_r(p)|\alpha_p=\int_0^{u_r(0)}\sum_{p\in\partial R_s}\alpha_p\,ds\ge\pi u_r(0)$, since each R_s is a staircase in $\mathbb{B}_{r-1}$ that holds the one monopole 0, good by Proposition 18.99(b), with $\partial R_s\subset\mathcal{P}^\circ(\mathbb{B}_r)$. So the value of (U,η) is at least $F_r(j_r^*)$; with the first step, $c_{\mathrm{mon}}(r)=F_r(j_r^*)=\min_{\mathcal{F}_r}F_r$, attained at the abelian field of j_r^*.

(c), (f). A configuration admitted at $r+1$, or on $\mathbb{Z}^3$ with the one monopole 0, restricts to one admitted at r with no larger value: $c_{\mathrm{mon}}(r)\le c_{\mathrm{mon}}(r+1)\le c_{\mathrm{mon}}$. The comparison principle, that $u_1\ge u_2$ when $\sum_{c'\sim c}g(u_1(c)-u_1(c'))\ge\sum_{c'\sim c}g(u_2(c)-u_2(c'))$ on a finite set $\mathcal{D}$ and $u_1\ge u_2$ off $\mathcal{D}$, follows by the same spreading of a negative minimum of u_1-u_2; with $\mathcal{D}=\mathbb{B}_{r-1}$ it gives $u_r\le u_{r'}$ for $r<r'$. As $\mathsf{f}(j)\ge 0.488j^2$ on $[0,\frac{\pi}{6}]$, $\pi u_r(0)=\sum\nabla u_r\,j_r^*\le\sum(j_r^*)^2\le 2.1\,c_{\mathrm{mon}}$, so u_r increases to a bounded u_∞; the equation at each cube passes to the limit, so $j_\infty=\arcsin\nabla u_\infty$ has $\operatorname{div}j_\infty=\pi\delta_0$, and by Fatou's lemma $\sum\mathsf{f}(j_\infty)\le\lim c_{\mathrm{mon}}(r)$. Proposition 18.100(a) turns j_∞ into an abelian field with the one monopole 0 and the action $\sum\mathsf{f}(j_\infty)$, so $c_{\mathrm{mon}}\le\sum\mathsf{f}(j_\infty)\le\lim c_{\mathrm{mon}}(r)\le c_{\mathrm{mon}}$; by Proposition 18.100(b), c_{mon} does not depend on the sign field. The bound on u_∞ is that of the proof of (e) as $r'\to\infty$, with $\frac{100}{16\pi}<1.99$, together with the monotonicity.

(d) At $r=1$ the potential is $u_1=\frac12\delta_0$ and the flow is $\frac{\pi}{6}$ through each face of the cube 0. The surfaces $\partial\mathbb{B}_r$, $r\ge 0$, are disjoint, each with n_r plaquettes, and each $\mathbb{B}_r$ is good and holds the one monopole; with $y=\sin\frac{\pi}{n_r}\le\frac12$ the tangent inequality gives $\sum_{\partial\mathbb{B}_r}\mathsf{f}(\alpha_p)\ge\pi y-n_r\mathsf{f}^*(y)=n_r\mathsf{f}(\pi/n_r)$, and summing over r bounds every action with the one monopole 0 from below. Numerically $n_0\mathsf{f}(\frac{\pi}{6})=0.803848$, and for $r\ge 1$, $n_r\mathsf{f}(\pi/n_r)\ge\frac{\pi^2}{2n_r}(1-\frac{\pi^2}{12n_r^2})$, with $\sum_{r\ge1}\frac{\pi^2}{2n_r}=\frac{\pi^2}{12}(\frac{\pi^2}{8}-1)=0.192211$ and a correction below $3\cdot10^{-5}$. From above, the flow $\pi\nabla\mathcal{G}$ has the divergence $\pi\delta_0$ by Proposition 18.100(c), and $\mathsf{f}(t)\le\frac12t^2$ gives its abelian field the action $\le\frac{\pi^2}{2}\sum(\nabla\mathcal{G})^2=\frac{\pi^2}{2}\mathcal{G}(0)$.

(e) Let $3\le r<r'$, $u=u_{r'}$, $s=u(re_1)$ and $\rho'=\lfloor r/3\rfloor$. *A dual bound.* For v supported in $\mathbb{B}_{r-1}$ and $j\in\mathcal{F}_r$, Young's inequality with the even convex conjugate $\tilde{\mathsf{f}}^*$, which does not decrease in $|y|$, gives $\tilde F_r(j)\ge\sum[j\nabla v-\tilde{\mathsf{f}}^*(\nabla v)]=\pi v(0)-\sum\tilde{\mathsf{f}}^*(\nabla v)$; as $c_{\mathrm{mon}}(r)=\tilde F_r(j_r^*)=\min_{\mathcal{F}_r}\tilde F_r$, this bounds $c_{\mathrm{mon}}(r)$ from below, with equality at r' for $v=u$. With $v=(u-s)^+$, supported in $\mathbb{B}_{r-1}$ by the monotonicity, $|\nabla v|\le|\nabla u|$ and $v(0)=u(0)-s$: $c_{\mathrm{mon}}(r)\ge c_{\mathrm{mon}}(r')-\pi s$. *A floor near* 0. The reflections $\sigma(c)=(c_2+k,c_1-k,c_3)$, $k\ge0$, with $H=\{c_1-c_2\ge k+1\}$ and $e=e_1$ meet the conditions of the moving planes at r', the neighbours outside H lying on the plane fixed by σ; so $u(c_1-1,c_2+1,c_3)\ge u(c)$ when $c_1-c_2\ge1$. Stepping from $(r,0,0)$ to $(r-b,b,0)$ and then, with the axes 1 and 3, to $(r-b-q,b,q)$, for integers $b\ge q\ge0$ with $r-b-q\ge b$, the first coordinate exceeding the growing one before each step, and using the symmetries and the monotonicity, $u\ge s$ on $\{|c|_1\le r\}\supset\mathbb{B}_{\rho'}$. *A capacity bound.* Let $\bar u=\min(u,s)$; on each edge $\nabla\bar u$ has the sign of ∇u and $|\nabla\bar u|\le|\nabla u|\le|j_{r'}^*|$, so $\|\nabla\bar u\|^2\le\sum\nabla\bar u\,j_{r'}^*=\pi\bar u(0)=\pi s$; for every flow k whose divergence is ≥0, vanishes off $\mathbb{B}_{\rho'}$ and has the sum 1, $s=\sum\bar u\operatorname{div}\mathsf{k}=\sum\nabla\bar u\,\mathsf{k}\le(\pi s)^{1/2}\|\mathsf{k}\|$, so $s\le\pi\|\mathsf{k}\|^2$. Take for k the flux of the field $x/4\pi|x|^3$ through the plaquettes off $\mathcal{P}^\circ(\mathbb{B}_{\rho'})$, and 0 on $\mathcal{P}^\circ(\mathbb{B}_{\rho'})$; its divergence is 0 off $\mathbb{B}_{\rho'}$ and, on $\mathbb{B}_{\rho'}$, the outward flux through the outer faces, of the sum 1; the $72\varrho_n^2+48\varrho_n+12$ plaquettes of $\mathcal{P}^\circ(\mathbb{B}_{n+1})\setminus\mathcal{P}^\circ(\mathbb{B}_n)$ lie at the distance $\ge\varrho_n=n+\frac12$ from 0 and carry at most $(4\pi\varrho_n^2)^{-1}$

each, so, with $\sum_{n\ge\rho'}\varrho_n^{-m}\le\int_{\rho'}^\infty y^{-m}dy$, $\|\mathsf{k}\|^2\le\frac{1}{16\pi^2}(\frac{72}{\rho'}+\frac{24}{\rho'^2}+\frac{4}{\rho'^3})\le\frac{100}{16\pi^2\rho'}$ and $\pi s\le 6.25/\rho'$; and $\lfloor r/3\rfloor\ge r/5$ for $r\ge 3$. □

Proposition 18.102 (The potential of one monopole at large distances)**.** Let u_∞ and j_∞ be as in Proposition 18.101(f), and $\mathsf{h}=j_\infty-\nabla u_\infty$. There is an absolute constant C such that, for every $c\ne 0$ and every edge with an end at c:

(a) $|u_\infty(c)-\pi\mathcal{G}(c)|\le C|c|^{-3}$, and so $|u_\infty(c)-\frac{1}{4|c|}|\le C|c|^{-3}$;
(b) $|\nabla u_\infty|\le C(1+|c|)^{-2}$, at every edge;
(c) $|\mathsf{h}|\le 0.19\,|\nabla u_\infty|^3\le C(1+|c|)^{-6}$, at every edge;
(d) $c_{\rm mon}=\pi u_\infty(0)-\sum_p \mathsf{f}^*(\nabla u_\infty(p))$.

■

Sketch. (c) $\arcsin y-y\le 0.19y^3$ on $[0,\frac12]$, and $|\nabla u_\infty|\le\frac12$. (d) $\mathsf{f}(j)+\mathsf{f}^*(\sin j)=j\sin j$, and $\sum j_\infty\nabla u_\infty=\pi u_\infty(0)$ by summation by parts over $\mathbb{B}_r$, whose boundary term is $\le Cr^2\cdot r^{-1}\cdot r^{-2}$ by (b) and Proposition 18.101(f); and $\sum\mathsf{f}(j_\infty)=c_{\rm mon}$. *The decay of* h. For $\rho\ge 3$ every edge off $\mathcal{P}^\circ(\mathbb{B}_\rho)$ has both ends at $|\cdot|_\infty\ge\rho$, where $u_\infty\le 10/\rho$, so $|\nabla u_\infty|\le 10/\rho$ there; the restriction of j_∞ to Γ_ρ lies in $\mathcal{F}_\rho$, so $\sum_{\rm off}\mathsf{f}(j_\infty)\le c_{\rm mon}-c_{\rm mon}(\rho)\le 32/\rho$ by Proposition 18.101(a), (e), and $\sum_{\rm off}|\nabla u_\infty|^2\le 68/\rho$; so $\sum_{\rm off}|\mathsf{h}|\le 130\rho^{-2}$, and h is summable. *A representation.* Let $\psi(c)=\pi\mathcal{G}(c)-\sum_{\mathsf{e}}(\nabla\mathcal{G}(c-\cdot))_{\mathsf{e}}\,\mathsf{h}_{\mathsf{e}}$, the sum over the edges $\mathsf{e}=(c_1\to c_2)$ with $(\nabla\mathcal{G}(c-\cdot))_{\mathsf{e}}=\mathcal{G}(c-c_1)-\mathcal{G}(c-c_2)$; it is well defined, tends to 0 at infinity, and $\sum_{c'\sim c}(\psi(c)-\psi(c'))=\pi\delta_{c,0}-\operatorname{div}\mathsf{h}=\operatorname{div}\nabla u_\infty$; so $u_\infty-\psi$ is harmonic and tends to 0, and $u_\infty=\psi$. With $\rho=\lfloor|c|_\infty/2\rfloor$, by Proposition 18.100(c) the edges of $\mathcal{P}^\circ(\mathbb{B}_\rho)$ contribute at most $C|c|^{-2}\|\mathsf{h}\|_1$ to $\psi-\pi\mathcal{G}$ and $C|c|^{-3}\|\mathsf{h}\|_1$ to its increments, and the other edges at most $C\rho^{-2}$; so $|u_\infty-\pi\mathcal{G}|\le C|c|^{-2}$ and $|\nabla u_\infty|\le C|c|^{-2}$, which is (b), and then $|\mathsf{h}_{\mathsf{e}}|\le C(1+|c_1|)^{-6}$. *The next order.* u_∞ is even, so $\mathsf{h}_{-\mathsf{e}}=\mathsf{h}_{\mathsf{e}}$ for $-\mathsf{e}=(-c_1\to-c_2)$, and the sum in ψ is $\frac12\sum_{\mathsf{e}}\mathsf{h}_{\mathsf{e}}\mathsf{g}_{\mathsf{e}}$ with $\mathsf{g}_{\mathsf{e}}=\nabla_e\mathcal{G}(c-c_1-e)-\nabla_e\mathcal{G}(c+c_1)$, $c_2=c_1+e$: a difference of the values of $\nabla_e\mathcal{G}$ at two points joined by a monotone lattice path of at most $4|c_1|+1$ steps. For $|c|\ge 16$ and $|c_1|\le|c|/8$ that path stays at the distance $\ge|c|/2$ from 0, so $|\mathsf{g}_{\mathsf{e}}|\le C(4|c_1|+1)|c|^{-3}$, which is summable against $(1+|c_1|)^{-6}$; for $|c|/8<|c_1|\le 4|c|$, $|\mathsf{h}_{\mathsf{e}}|\le C|c|^{-6}$ against $\sum_{|z|\le 5|c|+1}(1+|z|)^{-2}\le C|c|$; and for $|c_1|>4|c|$ the terms are at most $C|c_1|^{-8}$. So the sum is at most $C|c|^{-3}$, and Proposition 18.100(c) gives (a); a larger constant covers $|c|<16$. □

Proposition 18.103 (The least action of a pair of monopoles)**.** For $v\in\mathbb{Z}^3\setminus\{0\}$ let $E_{\rm pair}(v)=\inf A_\eta(U)$ over the fields on $\mathbb{Z}^3$ and the sign fields with $\mathfrak{m}(\eta)=\{0,v\}$; let $E^{\rm ab}_{\rm pair}(v)=\inf\sum_p\mathsf{f}(J(p))$ over the flows J with $\operatorname{div}J=\pi(\delta_0-\delta_v)$; and, for a finite set $\mathfrak{B}$ of cubes with $0,v\in\mathfrak{B}^\circ$, let $E_{\rm pair}(v;\mathfrak{B})$ be the least value of $\sum_{p\in\mathcal{P}^\circ(\mathfrak{B})}(1-\frac12\eta(p)t_p)$ over the configurations on the plaquettes of the closure of $\mathfrak{B}$ with $\delta\eta=-1$ at 0 and at v and $\delta\eta=1$ at the other cubes of $\mathfrak{B}^\circ$. There are absolute constants C and t_0 such that:

(a) for all $v\ne 0$, $E_{\rm pair}(v)\le E^{\rm ab}_{\rm pair}(v)\le\sum_p\mathsf{f}(j_\infty(p)-j_\infty(p-v))\le 2c_{\rm mon}-\pi u_\infty(v)+C|v|^{-2}$, where $p-v$ is the plaquette p moved by $-v$;
(b) for $|v|\ge t_0$, $\varrho\ge 2|v|$ and every finite $\mathfrak{B}$ with $\mathfrak{B}^\circ\supset\mathbb{Z}^3\cap(\{|x|<\varrho\}\cup\{|x-v|<\varrho\})$, $E_{\rm pair}(v;\mathfrak{B})\ge 2c_{\rm mon}-\pi u_\infty(v)-C|v|^{-2}-\pi|v|/\varrho^2$; and $E_{\rm pair}(v)\ge 2c_{\rm mon}-\pi u_\infty(v)-C|v|^{-2}$ for $|v|\ge t_0$, the infimum being over all SU(2) fields;
(c) for all $v\ne 0$, $|E_{\rm pair}(v)-2c_{\rm mon}+\frac{\pi}{4|v|}|\le C|v|^{-2}$ and $0\le E^{\rm ab}_{\rm pair}(v)-E_{\rm pair}(v)\le C|v|^{-2}$;
(d) for $t=|v|$ and every integer $R\ge 4t+8$ there is a flow J'' that vanishes off $\mathcal{P}^\circ(\mathbb{B}_R)$, with $\operatorname{div}J''=\pi(\delta_0-\delta_v)$ and $\sum\mathsf{f}(J''(p))\le 2c_{\rm mon}-\pi u_\infty(v)+C(t^{-2}+R^{-1})$.

■

Sketch. Signs. $t=|v|$, $u_1=u_\infty$, $u_2=u_\infty(\cdot-v)$, $a=\nabla u_1$, $b=\nabla u_2$, $j_i=\arcsin\nabla u_i$, $\mathsf{h}_i=j_i-\nabla u_i$, $\mathsf{U}=u_1-u_2$, and $\iota(c)=v-c$, so that $u_1\circ\iota=u_2$ and $\mathsf{U}\circ\iota=-\mathsf{U}$, u_∞ being even; $\mathcal{N}_1$ is the set of the edges with both ends in $\{|c|\le|c-v|\}$ and $\mathcal{N}_2$ the set of the others, whose ends lie at the

distance $\geq \frac{t}{2}-1$ from 0. By Proposition 18.102, $|b| \leq C(1+t)^{-2}$ on $\mathcal{N}_1$ and $|a| \leq C(1+t)^{-2}$ on $\mathcal{N}_2$. *Two sums.* Summation by parts gives $\sum j_1 b = \pi u_2(0) = \pi u_\infty(v)$; and $\sum |\mathsf{h}_1||b| + \sum |\mathsf{h}_2||a| \leq Ct^{-2}$, by the summability of h on $\mathcal{N}_1$, by $|\mathsf{h}_1| \leq C(1+|c|)^{-6}$ on $\mathcal{N}_2$, and by ι.

(a) $J = j_1 - j_2$ has the divergence $\pi(\delta_0 - \delta_v)$, and $\mathsf{f}(x-x') = \mathsf{f}(x)+\mathsf{f}(x')-\mathsf{f}(x)\mathsf{f}(x') - \sin x \sin x' \leq \mathsf{f}(x) + \mathsf{f}(x') - \sin x \sin x'$; so $\sum \mathsf{f}(J) \leq 2c_{\mathrm{mon}} - \sum ab$, and $\sum ab = \sum (j_1 - \mathsf{h}_1)b \geq \pi u_\infty(v) - Ct^{-2}$. Proposition 18.100(a), with γ the unit flow along a dual path from 0 to v, gives $E_{\mathrm{pair}} \leq E^{\mathrm{ab}}_{\mathrm{pair}}$.

(b) *The dual value of* U. By Taylor's formula with $(\mathsf{f}^*)'' = (1-y^2)^{-1/2}$, $\mathsf{f}^*(x-x') \leq \mathsf{f}^*(x)+\mathsf{f}^*(x') - x' \arcsin x + C(x^2+x'^2)x'^2$ for $|x|, |x'| \leq \frac{1}{2}$; used with $(x,x') = (a,b)$ on $\mathcal{N}_1$ and $(x,x') = (b,a)$ on $\mathcal{N}_2$, with $j_2 a - j_1 b = \mathsf{h}_2 a - \mathsf{h}_1 b$, the two sums and Proposition 18.102(d), it gives

$$\mathcal{D}(\mathsf{U}) := \pi(\mathsf{U}(0) - \mathsf{U}(v)) - \sum_p \mathsf{f}^*(\nabla \mathsf{U}(p)) \geq 2c_{\mathrm{mon}} - \pi u_\infty(v) - Ct^{-2}.$$

Two families of good sets. Let $\rho = t^{2/3}$, $w_0 = \frac{1}{4t}$, $s_a = \frac{1}{4\rho}$, $s_b = \frac{1}{8\rho}$, and $\psi(x) = \frac{1}{4|x|} - \frac{1}{4|x-v|}$ on $\mathbb{R}^3$. For $s > 0$ the set $C_s = \{0\} \cup \{\psi \geq s\}$ is compact, convex and contained in $\{|x| < |x-v|\}$: if the gradient of ψ is orthogonal to a unit vector e, that is $(e \cdot x)/|x|^3 = (e \cdot (x-v))/|x-v|^3 = \lambda$, then $4e^{\mathsf{T}} \nabla^2 \psi\, e = -3\lambda^2(|x-v| - |x|) - (|x|^{-3} - |x-v|^{-3}) < 0$, so on a segment between two points of C_s the function ψ has no interior minimum below s; and for $\varrho \geq 2|v|$ and $s > |v|/2\varrho^2$, $C_s \subset \{|x| < \varrho\}$, as $\psi(x) \leq |v|/2|x|^2$ for $|x| \geq 2|v|$. Let $K_s = \{x : x + [-\frac{1}{2}, \frac{1}{2}]^3 \subset C_s\}$, compact and convex, $R_s = \mathsf{R}(K_s)$, a set of cubes inside C_s that decreases in s, and $u^\flat(c) = \sup\{s : c \in R_s\}$; then $R_s = \{u^\flat > s\}$ for all but countably many s, and $0 \leq u^\flat \leq \psi^+$. On the unit cube $\prod_i [c_i, c_i + \varsigma_i]$, ς_i the sign of $\partial_i \psi(c)$, the first order term of ψ about c is ≥ 0 and the second is $\geq -\frac{3}{2} \sup \|\nabla^2 \psi\| \geq -6(|c|^{-3} + |c-v|^{-3})$; so $\psi^+(c) - u^\flat(c) \leq 6(|c|^{-3} + |c-v|^{-3})$ for $|c|, |c-v| \geq 4$. For $0 < s \leq s_a$ and t large, $0 \in R_s$, $v \notin R_s$, and R_s is good by Proposition 18.99(b). Next, with $\mathsf{a} = (u_\infty - w_0)^+$ and $s_{\mathrm{top}} = u_\infty(0) - w_0$, the sets $T_s = \{\mathsf{a} > s\}$, $0 < s < s_{\mathrm{top}}$, are staircases by Proposition 18.101(f), so good, and by Proposition 18.102(a) $T_s \subset \{|c| < 3\rho\}$, so $v \notin T_s$, for $s \geq s_b$. *The measure.* Let $\theta(s) = \min\{1, \max\{0, (s-s_b)/(s_a - s_b)\}\}$, of the Lipschitz constant 8ρ, $\Theta(s) = \int_0^s \theta$ and $\bar{\Theta}(s) = s - \Theta(s)$; let $\mu^+ = \int_0^{s_{\mathrm{top}}} [\theta(s)\delta_{T_s} + (1-\theta(s))\delta_{R_s}]\, ds$ and $\mu = \mu^+ + \iota_* \mu^+$. Every set charged by μ is good and holds exactly one of 0 and v, so it has $\prod_{c \in R} \delta\eta(c) = -1$ whenever $\mathfrak{m}(\eta) = \{0, v\}$; $\mu(\mathrm{all}) = 2s_{\mathrm{top}}$; and by the coarea formula $y_p = \mu\{p \in \partial R\} = y_p^0 + y_{\iota p}^0$ with $y_p^0 = |\nabla \Theta(\mathsf{a})(p)| + |\nabla \bar{\Theta}(u^\flat)(p)|$. Put $\tilde{u}_0 = \Theta(\mathsf{a}) + \bar{\Theta}(u^\flat)$, $\tilde{u} = \tilde{u}_0 - \tilde{u}_0 \circ \iota$ and $\mathsf{r} = \tilde{u} - \mathsf{U}$. *Three zones.* On $\{|c|_\infty \leq \rho/4\}$, for large t, $u_\infty \geq 0.55/\rho$ and the unit cube at c lies in C_{s_a}, so $\mathsf{a} \geq s_a$ and $u^\flat \geq s_a$, and $\tilde{u}_0 = u_\infty - w_0$, as $\Theta(s) = s - \frac{1}{2}(s_a + s_b)$ and $\bar{\Theta}(s) = \frac{1}{2}(s_a + s_b)$ for $s \geq s_a$; on $\{|c| \geq 3\rho\}$, a and $u^\flat$ lie below s_b, so $\tilde{u}_0 = u^\flat$, which vanishes where $|c| \geq |c-v|$; hence $\tilde{u} = \tilde{u}_0$ on $\{|c| \leq |c-v|\}$ and $\tilde{u} = -\tilde{u}_0 \circ \iota$ on the other side. *The defect.* In the inner zone $\mathsf{r} = u_2 - w_0$ and $|\nabla \mathsf{r}| = |b| \leq Ct^{-2}$ on $C\rho^3$ edges; where $|c| \geq \rho/8$ and $|c| \leq |c-v|$, $|u^\flat - \mathsf{U}| \leq C|c|^{-3}$ by the bound on $\psi^+ - u^\flat$ and Proposition 18.102(a), which gives $\sum |\nabla \mathsf{r}|^2 \leq C\rho^{-3}$ over the outer zone; in the middle zone $\Theta(\mathsf{a}) - \Theta(u^\flat) = (\mathsf{a} - u^\flat)\mathsf{m}$ with $\mathsf{m} \in [0,1]$ of increments $\leq 8\rho \max(|\nabla \mathsf{a}|, |\nabla u^\flat|) \leq C\rho^{-1}$, $|\mathsf{a} - u^\flat| \leq C\rho^{-3} + C\rho t^{-2} \leq Ct^{-4/3}$ and $|\nabla(\mathsf{a} - u^\flat)| \leq Ct^{-2}$, so $|\nabla \mathsf{r}| \leq Ct^{-2}$ on $C\rho^3$ edges. As $\rho^3 = t^2$, $\|\nabla \mathsf{r}\|^2 \leq Ct^{-2}$. *From* y *to* $\nabla \tilde{u}$. $y_p = |\nabla \tilde{u}(p)|$ except on edges of the middle zone where $\nabla \Theta(\mathsf{a})$ and $\nabla \bar{\Theta}(u^\flat)$ have opposite signs, and there both are at most $|\nabla(\mathsf{a} - u^\flat)| \leq Ct^{-2}$; and $y_p \leq \frac{1}{2}$. So $\sum \mathsf{f}^*(y_p) \leq \sum \mathsf{f}^*(\nabla \tilde{u}) + Ct^{-2}$. *The end.* By Proposition 18.99(c), every (U, η) on $\mathbb{Z}^3$ with $\mathfrak{m}(\eta) = \{0, v\}$ has $A_\eta(U) \geq 2\pi s_{\mathrm{top}} - \sum \mathsf{f}^*(y_p)$. With $\mathsf{f}^*(x+x') \leq \mathsf{f}^*(x) + x' \arcsin x + x'^2$ for $|x|, |x+x'| \leq 0.6$, summation by parts, $\arcsin(a-b) = j_1 - j_2 - \kappa$ with $|\kappa| \leq C \max(a^2, b^2) \min(|a|, |b|)$ and so $\|\kappa\|_2 \leq Ct^{-2}$, and $\mathsf{r}(0) = -\mathsf{r}(v) = u_\infty(v) - w_0$, one gets $\sum \mathsf{f}^*(\nabla \tilde{u}) \leq \sum \mathsf{f}^*(\nabla \mathsf{U}) + 2\pi(u_\infty(v) - w_0) + Ct^{-2}$, hence $A_\eta(U) \geq \mathcal{D}(\mathsf{U}) - Ct^{-2}$. For the box keep in μ^+ the values $s > t/2\varrho^2$ alone: by the ball bound for C_s and $T_s \subset \{|c| < 3\rho\}$ all sets then lie in $\{|x| < \varrho\}$, those of $\iota_* \mu^+$ in $\{|x - v| < \varrho\}$, so in $\mathfrak{B}^\circ$; $\pi\mu(\mathrm{all})$ drops by $\pi t/\varrho^2$, while y_p decreases and f^* increases on $[0, \frac{1}{2}]$. A field on $\mathbb{Z}^3$ restricts to the closure of every box, which gives the bound on $E_{\mathrm{pair}}(v)$ as $\varrho \to \infty$.

(c) This is (a), (b) and Proposition 18.102(a) for $|v| \ge t_0$; for $|v| < t_0$, $0 \le E_{\text{pair}} \le E^{\text{ab}}_{\text{pair}} \le 2c_{\text{mon}} + C$.

(d) Let $\chi(c) = \min\{1, \max\{0, 2 - 2|c|_\infty/R\}\}$, equal to 1 on $\mathbb{B}_{R/2} \ni 0, v$ and to 0 off $\mathbb{B}_{R-1}$, let χ_p be its mean over the two ends of p, and $J^\chi = \chi_p J$ with $J = j_1 - j_2$; J^χ vanishes off $\mathcal{P}^\circ(\mathbb{B}_R)$. Its divergence is $\chi \operatorname{div} J + \omega'$ with $\omega'(c) = \frac12 \sum_{c' \sim c} (\chi(c') - \chi(c)) J(c \to c')$, carried by the shell $\{R/2 - 1 \le |c|_\infty \le R\}$, where $|c - v| \ge R/4 + 1$ and $|J| \le CR^{-2}$; so $|\omega'| \le CR^{-3}$, $\|\omega'\|_2^2 \le CR^{-3}$, $\|\omega'\|_1 \le C$, and $\sum \omega' = 0$. The flow k of least norm on Γ_R with $\operatorname{div} \mathsf{k} = -\omega'$ is a gradient $\nabla\varphi$, and $\|\mathsf{k}\|^2 = -\sum(\varphi - \bar\varphi)\omega'$; the Poincaré inequality $\|\varphi - \bar\varphi\| \le (3/2)^{1/2}(2R+1)\|\nabla\varphi\|$ on the box, by paths that adjust the coordinates in turn, each edge lying on at most $(2R+1)^4$ of them, gives $\|\mathsf{k}\|^2 \le C/R$. Then $J'' = J^\chi + \mathsf{k}$ has the divergence $\pi(\delta_0 - \delta_v)$, and by $\mathsf{f}(x + x') \le \mathsf{f}(x) + x' \sin x + \frac12 x'^2$ and $\mathsf{f}(\chi_p J) \le \mathsf{f}(J)$, $\sum \mathsf{f}(J'') \le \sum \mathsf{f}(J) + \sum \sin(\chi_p J)\, \mathsf{k} + \frac12 \|\mathsf{k}\|^2$; writing $\sin(\chi_p J) = \nabla \mathsf{U} + \mathsf{e}$, one has $\mathsf{e} = -a(1 - \cos j_2) + b(1 - \cos j_1)$ where $\chi_p = 1$ and $|\mathsf{e}| \le CR^{-2}$ elsewhere, so $\|\mathsf{e}\|_2 \le Ct^{-2} + CR^{-1/2}$, and $\sum \nabla \mathsf{U}\, \mathsf{k} = -\sum \mathsf{U} \omega'$ with $|\mathsf{U}| \le C/R$ on the shell; with (a) this is the claim. □

Proposition 18.104 (The point operator on one box)**.** Let N be a box with all N_μ even and ≥ 4, and $\beta \ge 0$. For a set M of cubes of the box let $\mathsf{Z}(M) = \sum_{\eta:\, \mathfrak{m}(\eta) = M} Z_\beta[\eta; 1]$, the sum over the sign fields on the plaquettes of the box, Z_β as in eq. 11.i, and $\mathsf{D}(M) = \mathsf{Z}(M)/\mathsf{Z}(\emptyset)$.

(a) $|\mathfrak{m}(\eta)|$ is even for every η, and every even set of cubes is an $\mathfrak{m}(\eta)$. For every sign field η, $\mathsf{Z}(\mathfrak{m}(\eta)) = 2^{2|N|-2} \sum_n Z_\beta[\eta\eta^n; 1]$, the sum over the eight classes; so $\mathsf{D}(\mathfrak{m}(\eta)) = \sum_n Z_\beta[\eta\eta^n; 1] / \sum_n Z[n]$, $\mathsf{D}(M) > 0$ exactly for $|M|$ even, D is invariant under the isometries of the box, and $Z_\beta[\eta; 1]/Z[0] \le 8\, \mathsf{D}(\mathfrak{m}(\eta))$.

(b) (Site mirrors.) Let θ be the reflection in the two planes $x_\mu = h$ and $x_\mu = h + \frac12 N_\mu$ through sites, and $M = M_+ \cup M_-$ with $M_\pm$ in the two closed halves that they bound. Then $\mathsf{Z}(M_+ \cup M_-)^2 \le \mathsf{Z}(M_+ \cup \theta M_+)\, \mathsf{Z}(M_- \cup \theta M_-)$.

(c) (Chessboard.) Let $N_\mu = 2n_\mu \mathsf{l}$ with integers $n_\mu, \mathsf{l} \ge 1$, let the blocks $\prod_\mu [k_\mu \mathsf{l}, (k_\mu + 1)\mathsf{l}]$ be named by $k \in \mathfrak{B} = \prod_\mu \mathbb{Z}/2n_\mu$, and let τ_k be the map of the block 0 onto the block k that in the direction μ is $x_\mu \mapsto k_\mu \mathsf{l} + x_\mu$ for even k_μ and $x_\mu \mapsto (k_\mu + 1)\mathsf{l} - x_\mu$ for odd k_μ. For a map $\mathcal{A}$ from $\mathfrak{B}$ to the sets of cubes of the block 0 put $M(\mathcal{A}) = \bigcup_k \tau_k \mathcal{A}(k)$, and for one such set P put $\|P\| = \mathsf{D}(M(\mathcal{A} \equiv P))^{1/|\mathfrak{B}|}$, $\mathcal{A} \equiv P$ the constant map. Then $\mathsf{D}(M(\mathcal{A})) \le \prod_k \|\mathcal{A}(k)\|$.

(d) For $\beta \ge 1$, $\log Z[0] \ge -|N|(\frac92 \log \beta + 29.4)$; and on a cube box every plaquette has the mean $\bar a = \langle 1 - \frac12 t_p \rangle \le (3 \log \beta + 19.6)/\beta$ under the periodic measure.

(e) (A triangular map.) On a cube box with N sites a side and $\beta \ge 0$, let Q be a box of sites with $H_\mu + 1 \le N - 1$ sites in the direction μ, V a field with $V(b) = \mathbf{1}$ unless b is a bond of Q that lies in no plaquette leaving Q, and η a sign field equal to 1 off the plaquettes of Q. Then, with $\bar a$ the mean plaquette and $|Q|$ the number of the sites of Q, $\log(Z_\beta[\eta; 1]/Z[0]) \ge -\beta A_\eta(V) - 48\, \beta \bar a\, |Q|\, (H_1 + H_2)^2$.

■

Sketch. (a) $\prod_c \delta\eta(c) = 1$, each plaquette being a face of two cubes; the sign field that is -1 on the plaquettes pierced by a dual path from c to c' has the monopoles c and c', and products of such fields give every even set. As $\mathfrak{m}(\eta\eta') = \mathfrak{m}(\eta) \triangle \mathfrak{m}(\eta')$, the sign fields with the monopoles $\mathfrak{m}(\eta)$ are the ηs with s closed; the closed sign fields are the disjoint union of the eight classes $[n]$ of the setting of Section 11.1, and $[n] \ni \eta^n$ consists of the $2^{2|N|-2}$ sign fields $\eta^n d\gamma$, γ a sign field on the bonds. Both facts are cellular cohomology of the torus with coefficients in $\mathbb{Z}_2$: every closed sign field of the class 0 has the form $d\gamma$, and every $d\gamma$ lies in the class 0, a bond lying in none or in two plaquettes of a lattice plane, so that $[n] = \eta^n[0]$, products over a plane being multiplicative; exactly $2^{|N|+2}$ sign fields γ have $d\gamma = 1$, since their products along three straight closed lines, one in each direction, take all eight values, and two of them with the same products differ by a field $\sigma(b_-)\sigma(b_+)$, σ one of the $2^{|N|}$ sign functions of the sites, with σ and $-\sigma$ giving the same field; dividing $2^{3|N|}$, the number of the sign fields on the $3|N|$ bonds, by $2^{|N|+2}$ gives the number of the distinct $d\gamma$. And $U \mapsto \gamma U$ gives

$Z_\beta[\eta\eta^n d\gamma;1]=Z_\beta[\eta\eta^n;1]$. An isometry of the box permutes the cells and keeps the Haar measure and the traces. By Theorem AG(1), $Z[n]\le Z[0]$, so $\sum_n Z[n]\le 8Z[0]$ and $\mathsf{D}(\mathfrak{m}(\eta))\ge Z_\beta[\eta;1]/8Z[0]$.

(b) Every cell lies in one of the two closed halves $\Lambda_\pm$, and the cells in both lie in the two planes, where θ fixes them with their orientations. With $s_c=-1$ for $c\in M$ and 1 else, $\mathsf{Z}(M)$ is the sum over the sign fields of the integral of $\prod_p e^{-\beta(1-\frac12\eta(p)t_p)}\prod_c \mathbf{1}[\delta\eta(c)=s_c]$; this weight is $W_0(\mathcal{X}_0)\,G_{M_+}(\mathcal{X}_+,\mathcal{X}_0)\,G_{\theta M_-}(\theta^*\mathcal{X}_-,\mathcal{X}_0)$, where $\mathcal{X}_0$ are the bond and plaquette variables in the planes, $\mathcal{X}_\pm$ those of the other cells of $\Lambda_\pm$, $W_0>0$ is the product over the plaquettes in the planes, and $G_{M'}$ is the product over the other plaquettes and over all cubes of Λ_+, with the signs of M'; θ^*, the pull back by θ, carries the Haar and counting measures of $\mathcal{X}_-$ to those of $\mathcal{X}_+$. Integrating out $\mathcal{X}_\pm$, $\mathsf{Z}(M_+\cup M_-)=\int W_0\,g_{M_+}g_{\theta M_-}\,d\mathcal{X}_0$ with bounded $g_{M'}\ge0$ and $\int W_0 g_{M'}^2\,d\mathcal{X}_0=\mathsf{Z}(M'\cup\theta M')$; the Schwarz inequality ends the argument.

(c) The planes $x_\mu=i\mathsf{l}$ and $x_\mu=(i+n_\mu)\mathsf{l}$ are a pair of site mirrors whose reflection θ maps the block k to the block θk with $(\theta k)_\mu=2i-1-k_\mu$ and has $\theta\circ\tau_k=\tau_{\theta k}$. Put $\phi(\mathcal{A})=\mathsf{Z}(M(\mathcal{A}))/\prod_k\|\mathcal{A}(k)\|$, and for such a mirror let $\mathcal{A}^\pm$ be $\mathcal{A}$ on the half $\pm$, continued by $\mathcal{A}^\pm(\theta k)=\mathcal{A}(k)$; by (b), $\phi(\mathcal{A})^2\le\phi(\mathcal{A}^+)\phi(\mathcal{A}^-)$, and $\phi(\mathcal{A}\equiv P)=\mathsf{Z}(\emptyset)$ for every P, $|\mathfrak{B}|$ being even, so that $\|P\|>0$. Among the finitely many maps with values in the range of a given $\mathcal{A}$ the maximisers of ϕ are closed under $\mathcal{A}\mapsto\mathcal{A}^\pm$; a maximiser with a longest run of equal slabs in the direction 1, if it were not constant in that direction, would be reflected at the mirror at one end of the run into a maximiser that is constant in that direction or has a longer run; the same in the directions 2 and 3, whose reflections keep the constancy already reached, gives a constant maximiser; so $\max\phi=\mathsf{Z}(\emptyset)$.

(d) Let every bond lie within the angle $\beta^{-1/2}$ of $\mathbf{1}$, a set of Haar measure $\frac{2}{\pi}\int_0^{\beta^{-1/2}}\sin^2\varphi\,d\varphi\ge\frac{8}{15\pi}\beta^{-3/2}$ for each bond; then $|\partial U(p)-\mathbf{1}|\le4\beta^{-1/2}$ and $1-\frac12 t_p\le8/\beta$, and the $3|N|$ bonds and $3|N|$ plaquettes give $\log Z[0]\ge-3|N|(\frac32\log\beta+\log\frac{15\pi}{8})-24|N|$, with $3\log\frac{15\pi}{8}+24\le29.4$. The derivative of $\log Z[0]$ in β is minus the mean action, which does not increase, $\log Z[0]$ being convex in β; so $\frac{\beta}{2}\langle A\rangle_\beta\le\log Z_{\beta/2}[0]-\log Z_\beta[0]\le|N|(\frac92\log\beta+29.4)$, as $Z\le1$; on a cube box the isometries make all $3|N|$ plaquettes have one mean.

(e) In Q take the tree T of the bonds of the direction 3 on the line through a corner x_0, the bonds of the direction 2 in the plane of the directions 2, 3 through x_0, and all bonds of the direction 1; let $P_x(U)$ be the holonomy along T from x_0 to x and $W(b)=P_{b_-}U(b)P_{b_+}^{-1}$, which is $\mathbf{1}$ on T. The map $\Phi(U)(b)=U(b)P_{b_+}^{-1}V(b)P_{b_+}$ at the bonds of Q, $\Phi(U)=U$ elsewhere, is at the bonds of T, ordered from x_0, a left translation of $U(b)$ by a function of earlier bonds, and at the other bonds a right translation by a function of the bonds of T; so it keeps the Haar measure, and $Z_\beta[\eta;1]=Z[0]\langle e^{-\beta(A_\eta\circ\Phi-A)}\rangle\ge Z[0]e^{-\beta\langle A_\eta\circ\Phi-A\rangle}$ by Jensen's inequality, $\langle\cdot\rangle$ the periodic mean. As $\Phi(U)$ is a gauge transform of the bondwise product WV on Q, a plaquette whose bonds all have $V=\mathbf{1}$ and whose sign is 1 keeps its term, and the at most $3|Q|$ other plaquettes have $\partial(WV)(p)$ equal to a product of eight unitary factors that differ from those of $\partial V(p)$ by $\omega_i=W(b)^{\pm1}-\mathbf{1}$ at four places. A global conjugation keeps the periodic measure and conjugates every $W(b)$, so $\langle W(b)^{\pm1}\rangle$ is a multiple of $\mathbf{1}$ and $\langle\omega_i\rangle=-\frac12\langle|\omega_i|^2\rangle\mathbf{1}$; expanding to the second order, the mean of each such term exceeds its value at V by at most $2\sum_{b\in p}\langle|W(b)-\mathbf{1}|^2\rangle$. $W(b)$ is the holonomy along the boundary of a ladder of at most H_1+H_2 plaquettes, so $|W(b)-\mathbf{1}|\le\sum_{\text{ladder}}|\partial U(p')-\mathbf{1}|$ and $\langle|W(b)-\mathbf{1}|^2\rangle\le2\bar a(H_1+H_2)^2$; dropping the terms $-(1-\frac12 t_p)\le0$ of $-A$ gives $\langle A_\eta\circ\Phi-A\rangle\le A_\eta(V)+3|Q|\cdot4\cdot2\cdot2\bar a(H_1+H_2)^2$. □

Proposition 18.105. *The point operator at large inverse coupling: the self-energy and the Coulomb term.*

(a) Let N be a cube box with $N=2n\mathsf{l}$ sites a side, $\mathsf{l}=2r+1$, $r\ge1$, $\beta\ge1$, and let $x\ne y$ be the central cubes of two blocks of side l. Then $\mathsf{D}(\{x,y\})\le8^{1/4}\exp[-2\beta c_{\text{mon}}(r)+\mathsf{l}^3(9\log\beta+58.8)]$.

(b) Let $\ell=2L^m$, $b>0$, $K_0\ge0$, and let $x\ne y$ be two points of T_ℓ that are centres of cubes of $\Lambda_{K_0,\ell}$; for $K\ge K_0$ let x_K,y_K be the cubes of $\Lambda_{K,\ell}$ with these centres, $\beta=L^Kb$, and

$D_K(x,y) = \mathsf{D}(\{x_K, y_K\})$ on $\Lambda_{K,\ell}$. There is K_1, formed from L, K_0 and b, such that for all $K \ge K_1$

$$\log D_K(x,y) \le -2c_{\mathrm{mon}}\beta + (192L+69)\,\beta^{3/4}(\log\beta)^{1/4};$$

so $\limsup_K \beta^{-1}\log D_K(x,y) \le -2c_{\mathrm{mon}}$; and the same holds for $Z_\beta[\eta;1]/Z[0]$, for every sign field η on $\Lambda_{K,\ell}$ with $\mathfrak{m}(\eta) = \{x_K, y_K\}$, with $\log 8$ added to the right member.

(c) Let N be a cube box with N sites a side, N even, $\beta \ge 1$, $v \in \mathbb{Z}^3 \setminus \{0\}$, $t = |v|$, c a cube, and C, t_0 the constants of Proposition 18.103. For every integer $R \ge 4t+8$ with $N \ge 2R+6$, the sign field η that is -1 exactly on the plaquettes pierced by a shortest dual path from c to $c+v$ has $\log(Z_\beta[\eta;1]/Z[0]) \ge -\beta[2c_{\mathrm{mon}} - \pi u_\infty(v) + C(t^{-2}+R^{-1})] - 48(2R+4)^3(4R+6)^2(3\log\beta + 19.6)$. For $t \ge t_0$, $\varrho \ge 2t$ and $N = 2n\mathsf{l}$ with $\mathsf{l} \ge t + 2\varrho + 5$, $\log \mathsf{D}(\{c, c+v\}) \le -\beta[2c_{\mathrm{mon}} - \pi u_\infty(v) - Ct^{-2} - \pi t/\varrho^2] + \mathsf{l}^3(\frac{9}{2}\log\beta + 29.4) + 1$.

(d) On $\Lambda_{K,\ell}$, for every $\ell = 2L^m$ and every $b > 0$, let $D_K(v) = \mathsf{D}(\{c, c+v\})$, which does not depend on the cube c. For every sequence $v_K \in \mathbb{Z}^3$ with $|v_K| \to \infty$ and $|v_K|^6 \log\beta/\beta \to 0$,

$$\log D_K(v_K) = -2c_{\mathrm{mon}}\beta + \frac{\pi}{4}\,\frac{\beta}{|v_K|} + o\Big(\frac{\beta}{|v_K|}\Big),$$

uniformly in the direction of v_K; in the physical distance $R_K = |v_K|a_K$ this reads $\log(e^{2c_{\mathrm{mon}}\beta} D_K(v_K)) = \frac{\pi}{4}\frac{b}{R_K}(1+o(1))$. The same holds for $Z_\beta[\eta;1]/Z[0]$ with η the sign field of a shortest dual path from c to $c+v_K$.

■

Sketch. (a) With $\mathcal{A}$ equal to the central cube at the blocks of x and of y and empty elsewhere, $M(\mathcal{A}) = \{x,y\}$, as τ_k maps the central cube of the block 0 to that of the block k, l being odd; Proposition 18.104(c) gives $\mathsf{D}(\{x,y\}) \le \|P\|^2 = \mathsf{D}(\mathfrak{C})^{2/|\mathfrak{B}|}$, P the central cube and $\mathfrak{C}$ the set of all central cubes. For $\mathfrak{m}(\eta) = \mathfrak{C}$ the restriction of (U,η) to each block, moved onto the closure of $\mathbb{B}_r$, is admitted in $c_{\mathrm{mon}}(r)$, and the sets $\mathcal{P}^\circ$ of different blocks are disjoint; so $Z_\beta[\eta;1] \le e^{-\beta|\mathfrak{B}|c_{\mathrm{mon}}(r)}$, and by Proposition 18.104(a), (d), $\mathsf{D}(\mathfrak{C}) \le 8e^{-\beta|\mathfrak{B}|c_{\mathrm{mon}}(r)}/Z[0] \le 8\exp[-\beta|\mathfrak{B}|c_{\mathrm{mon}}(r) + |\mathfrak{B}|\mathsf{l}^3(\frac{9}{2}\log\beta + 29.4)]$, with $|\mathfrak{B}| \ge 8$.

(b) The centre of a cube of $\Lambda_{K_0,\ell}$ is in lattice units $(k'+\frac12)L^{K-K_0} = (k+\frac12)L^j$ with $2k+1 = (2k'+1)L^{K-K_0-j}$; so for $1 \le j \le K-K_0$ the cubes x_K, y_K are central cubes of blocks of side L^j, and $N/L^j = 2L^{m+K-j}$ is even. Let $Y = (\beta/\log\beta)^{1/4}$ and $L^j \le Y < L^{j+1}$; for $K \ge K_1$, $\log\beta \ge 1$ and $1 \le j \le K-K_0$. Then $r = \frac{L^j-1}{2}$ satisfies $r \ge \frac{L^j}{3}$ because $L^j \ge 3$, and $r \ge 6$ because $L^j \ge L \ge 13$; and by Proposition 18.101(e) $2\beta(c_{\mathrm{mon}} - c_{\mathrm{mon}}(r)) \le 192\beta/L^j \le 192L\beta/Y$, while $L^{3j}(9\log\beta + 58.8) \le 68\,Y^3\log\beta$; and $\beta/Y = Y^3\log\beta = \beta^{3/4}(\log\beta)^{1/4}$, $\frac14\log 8 < 1$. The last sentence is Proposition 18.104(a).

(c) *Below.* Let J'' be the flow of Proposition 18.103(d) moved to c, and γ the unit flow along the shortest dual path, which stays in $c + \mathbb{B}_t$. By Proposition 18.100(a) there is an abelian field V on $\mathbb{Z}^3$, equal to $\mathbf{1}$ off the bonds of the box of the $(2R+2)^3$ sites of the cubes of $c+\mathbb{B}_R$, with $A_\eta(V) = \sum \mathsf{f}(J'')$; as $2R+2 < N$, it is placed on the periodic box without change of its action. Proposition 18.104(e), with Q that box enlarged by one layer, of $2R+4 \le N-1$ sites a side, and $H_1 = H_2 = 2R+3$, together with Proposition 18.104(d), gives the bound. *Above.* Let $\mathfrak{B}_v = \prod_i[\min(0,v_i) - \lceil\varrho\rceil - 1, \max(0,v_i) + \lceil\varrho\rceil + 1]$, with at most $t + 2\varrho + 5 \le \mathsf{l}$ cubes a side, placed with c so that $c+\mathfrak{B}_v$ lies in one block; $\mathfrak{B}_v^\circ$ holds the cubes within ϱ of 0 or of v. Proposition 18.104(c), with $\mathcal{A}$ equal to $\{c, c+v\}$ at that block and empty elsewhere, gives $\mathsf{D}(\{c,c+v\}) \le \|\{c,c+v\}\|$. For $\mathfrak{m}(\eta) = M(\mathcal{A} \equiv \{c,c+v\})$ the restriction of (U,η) to $\tau_k(c+\mathfrak{B}_v)$, carried back to $c+\mathfrak{B}_v$ by τ_k, which keeps the Haar measure and the traces, has among the cubes of $c+\mathfrak{B}_v^\circ$ the monopoles c and $c+v$ alone, so after the translation that takes c to the cube 0 of $\mathbb{Z}^3$ it is admitted in $E_{\mathrm{pair}}(v;\mathfrak{B}_v)$; the sets $\mathcal{P}^\circ(\tau_k(c+\mathfrak{B}_v))$ of different blocks are disjoint, so $Z_\beta[\eta;1] \le e^{-\beta|\mathfrak{B}|E_{\mathrm{pair}}(v;\mathfrak{B}_v)}$, and Proposition 18.103(b) and Proposition 18.104(a), (d) end as in (a), with $8^{1/|\mathfrak{B}|} < e$.

(d) Let $t = |v_K|$, $\varepsilon = t^6 \log\beta/\beta \to 0$, $\lambda = \varepsilon^{-1/6}$, $R = \lceil \lambda t \rceil$, $\varrho = \lambda t$, and $\mathsf{l} = L^j$ with $L^{j-1} < t + 2\varrho + 5 \le L^j$. Then $\beta t^{-2} = o(\beta/t)$, $\beta/R \le \varepsilon^{1/6}\beta/t$, $R^5 \log\beta \le 32\,\varepsilon^{1/6}\beta/t$, $\beta t/\varrho^2 = \varepsilon^{1/3}\beta/t$ and $\mathsf{l}^3 \log\beta \le (4L\lambda t)^3 \log\beta = 64L^3\varepsilon^{1/2}t^{-2}\beta/t$; for large K, $R \ge 4t + 8$, $N \ge 2R + 6$, $t \ge t_0$, and $j \le m + K$, so that N/l is even. By Proposition 18.102(a), $\pi u_\infty(v_K) = \frac{\pi}{4t} + O(t^{-3})$. The inequality $\mathsf{D} \ge Z_\beta[\eta; 1]/8Z[0]$ of Proposition 18.104(a) carries the lower bound of (c) from η to D, and the same inequality carries the upper bound of (c) from D to η; $\log 8 = o(\beta/t)$. All constants are absolute, so the error does not depend on the direction; and $\beta/|v_K| = b/R_K$. □

18.10. **The two-point function of a limit and of its components: the Källén–Lehmann measure, the law r^{-6} at a point and the growth M^5.** Fix a frame of Definition 13.1 and a reference coupling $0 < g \le \gamma$. Let (K_j, β_j, ν_j) be a converging Gibbs sequence, with limit σ and canonical decomposition $(\mathsf{X}, \lambda, (\sigma_\xi)_{\xi\in\mathsf{X}})$ in the sense of Definition 13.4(2). We use the scalar reconstruction, canonical decomposition and clustering of the components from Thms. YM3_I:AL and YM3_I:AP. Write $\bar c_g$ for the common bound on the one-point constants and $C^g_{\mathrm{rad}} = C^g_\wp + \bar c_g$ for the common point-kernel constant in that reconstruction. For $g \le \gamma_W$ we use the separated two-point law of Thm. YM3_I:C, with the common limiting label, thresholds and tolerances there. In particular that theorem gives the same estimates on σ and on every component. A volume-first limit has the centred functional of its defining Gibbs sequence and the same connected functions at orders $n \ge 2$. Proposition 13.25 writes the radial kernel of σ and of every σ_ξ as a constant plus a superposition of the Yukawa kernels $e^{-mr}/(4\pi r)$ with a positive measure ρ, written ρ_ξ for σ_ξ. We identify this constant, to which the kernel tends as $r \to \infty$ and which we call the constant at infinity: it is $\mathfrak{c}_\xi^2$ for σ_ξ and $\int \mathfrak{c}_\xi^2\, d\lambda$ for σ. In a component the kernel $\mathbf{s}^{\mathrm c}_\xi$ of the connected function of two points is therefore the superposition itself, completely monotone together with $r\,\mathbf{s}^{\mathrm c}_\xi(r)$, and from this form we obtain the convexity of $\log(r\,\mathbf{s}^{\mathrm c}_\xi(r))$ when $\rho_\xi \ne 0$, the bound $\mathbf{s}^{\mathrm c}_\xi(r) \le (r_0/r)\,\mathbf{s}^{\mathrm c}_\xi(r_0)$ for $0 < r_0 \le r$, the identity of $\rho_\xi(\{0\})/(4\pi)$ with the constant of the decay $|v|^{-1}$ at two points, and the bound $\rho_\xi([0,\mathsf{M}]) \le 1024\pi e\,(C^g_{\mathrm{rad}})^2\mathsf{M}^7$ for $\mathsf{M} \ge \frac12$, C^g_{rad} being the common reconstruction constant just specified; the connected functions of every order have real-analytic kernels at distinct points. Below the ceiling γ_W we show that r^6 times the connected kernel tends to $\mathsf{c}_2\mathsf{b}^{-2} = 9/(16\pi^2\mathsf{b}^2)$ as $r \to 0$, b the limit of the labels; that the kernel less its constant at infinity, which in a component is the connected kernel, once multiplied by $r^6\mathsf{b}^2/\mathsf{c}_2$, lies for small r between a lower and an upper member, neither of which depends on the sequence, the decomposition or the component; that $\rho([0,\mathsf{M}]) \sim \frac{3}{160\pi}\mathsf{b}^{-2}\mathsf{M}^5$ and $\mu^E([0,\mathsf{M}]) \sim \frac{\mathsf{c}_2}{720}\mathsf{b}^{-2}\mathsf{M}^6$ as $\mathsf{M} \to \infty$, uniformly in the sequence, the decomposition and the functional, limit or component, μ^E being the measure whose Laplace transform is the kernel less its constant at infinity; and that the Hamiltonian of the reconstruction is unbounded and its Hilbert space infinite-dimensional.

Definition 18.106. *The constant at infinity, the two parts of the kernel, the label, the energy of one field.*

(1) σ' denotes σ or a component σ_ξ, and $\mathfrak{c}'$ its one-point constant: $\mathfrak{c}' = 0$ for σ, because $S^\sigma_1 = 0$ (Thm. YM3_I:AP), and $\mathfrak{c}' = \mathfrak{c}_\xi$ for σ_ξ (Definition 13.4 (2)). $\mathbf{s}$ is the radial kernel of σ' (Definition 13.4 (3)), and $\kappa' = \lim_{r\to\infty}\mathbf{s}(r)$ and ρ are the number and the measure of Proposition 13.25 (a), (b), so that $\mathbf{s}(r) = \kappa' + \int e^{-mr}(4\pi r)^{-1}\, d\rho(m)$ for $r > 0$; we call κ' the *constant at infinity* and ρ the *Källén–Lehmann measure* of σ'; where the functional matters we write $\mathbf{s}_\sigma, \kappa'_\sigma, \rho_\sigma$ and $\mathbf{s}_\xi, \kappa'_\xi, \rho_\xi$, and we put the same subscripts on the objects that are formed from them below.

(2) The *Källén–Lehmann part* of the kernel is $\mathbf{s}^{\mathrm{KL}} = \mathbf{s} - \kappa'$, so that $4\pi r\,\mathbf{s}^{\mathrm{KL}}(r) = \int e^{-mr}\, d\rho(m)$, and the *connected kernel* is $\mathbf{s}^{\mathrm c} = \mathbf{s} - \mathfrak{c}'^2$. By the two-entry cumulant formula, $S^{\sigma'}_2(f,h) = \sigma'(f,h) - \sigma'((f))\,\sigma'((h))$, and $\sigma'((f)) = \mathfrak{c}'\int f$; so $S^{\sigma'}_2(f,h) = \iint f(x)h(y)\,\mathbf{s}^{\mathrm c}(|x-y|)\,dx\,dy$ for all $(f,h) \in \mathfrak{S}_2(\mathbb{R}^3)$. For the limit, $\mathbf{s}^{\mathrm c}_\sigma = \mathbf{s}_\sigma$.

(3) By Proposition 13.25 (a), $\mathbf{s}$ is completely monotone and non-increasing with the limit κ'; so $\mathbf{s}^{\mathrm{KL}} \geq 0$ is completely monotone, and by Bernstein's theorem there is exactly one positive Borel measure μ^E on $[0,\infty)$ with $\mathbf{s}^{\mathrm{KL}}(r) = \int e^{-rE}\, d\mu^E(E)$ for all $r > 0$, the *energy measure* of σ'.

(4) Let $0 < g \leq \gamma_W$, where γ_W is the ceiling in Thm. YM3_I:C. By that theorem the labels $a_{K_j}\beta_j$ converge to a positive number b, the *label* of the sequence. With the constant $\mathsf{c}_2 = \frac{9}{16\pi^2}$ of Thm. YM3_I:C we put $\mathsf{A}_{\mathsf{b}} = 4\pi\mathsf{c}_2\mathsf{b}^{-2} = \frac{9}{4\pi\mathsf{b}^2}$, and for $r > 0$ we put $y^{\mathrm{c}}(r) = r^6\, \mathbf{s}^{\mathrm{c}}(r)\, \mathsf{b}^2/\mathsf{c}_2$ and $y(r) = r^6\, \mathbf{s}^{\mathrm{KL}}(r)\, \mathsf{b}^2/\mathsf{c}_2$. For $0 < \delta \leq \frac{1}{2}$ let $\vartheta_\delta = \frac{32\pi}{105}\, \delta^3 \in (0, \frac{1}{8})$ and $a_\delta = \left(\frac{3+2\delta}{3-2\delta}\right)^6 \in (1, 64]$; with the threshold $\mathsf{d}_0(g,\vartheta)$ and the tolerance $\bar\varepsilon_{g,\vartheta}$ of Thm. YM3_I:C, let $r_\delta = (3-2\delta)\, \mathsf{d}_0(g,\vartheta_\delta)$, and $\varepsilon_\delta(r) = \bar\varepsilon_{g,\vartheta_\delta}(r/(3-2\delta))$ for $0 < r \leq r_\delta$. By Thm. YM3_I:C, ε_δ is non-decreasing, $\varepsilon_\delta(r) \leq \frac{1}{3}$, $\varepsilon_\delta(r) \to 0$ as $r \to 0$, and r_δ and ε_δ are formed from the frame, g and δ alone; they depend on no sequence, decomposition or component.

(5) For $m \geq 0$ and $\mathsf{q} \in \mathbb{R}^2$ let $\omega_m(\mathsf{q}) = (m^2 + |\mathsf{q}|^2)^{1/2}$, and for a real $h \in C^1_c(\mathbb{E}_+)$, writing $y = (y_\perp, y_3)$, let $\hat h_m(\mathsf{q}) = \int h(y)\, e^{-\omega_m(\mathsf{q})y_3 + i\mathsf{q}\cdot y_\perp}\, dy$. In $\mathcal{H}(\sigma', e_3)$ let $\Omega = [\varnothing]$, let P_0 be the orthogonal projection onto $\ker H$, and, with the vector ψ_h and the measure μ_h of Definition 13.4 (4), let $\mu_h^H(B) = \mu_h(B \times \mathbb{R}^2)$ for Borel sets $B \subset [0,\infty)$. Since $H = \int E\, d\mathcal{P}(E,\mathsf{q})$, μ_h^H is the spectral measure of H in the vector ψ_h: $\mu_h^H(B) = \|\mathbf{1}_B(H)\psi_h\|^2$, its mass is $\|\psi_h\|^2$, and $\int e^{-tE}\, d\mu_h^H(E) = \langle \psi_h, e^{-tH}\psi_h\rangle$ for $t \geq 0$ by the spectral theorem.

■

Lemma 18.107 (Laplace transforms of positive measures; Karamata's theorem)**.** Let $\mathfrak{m}$ be a positive Borel measure on $[0,\infty)$ with $\mathcal{L}_{\mathfrak{m}}(r) = \int e^{-mr}\, d\mathfrak{m}(m) < \infty$ for every $r > 0$.

(a) $\mathcal{L}_{\mathfrak{m}}$ is completely monotone on $(0,\infty)$; $\mathfrak{m}$ is the only positive Borel measure on $[0,\infty)$ whose Laplace transform is $\mathcal{L}_{\mathfrak{m}}$; $\mathcal{L}_{\mathfrak{m}}(r) \to \mathfrak{m}(\{0\})$ as $r \to \infty$; and $\mathfrak{m}([0,\mathsf{M}]) \leq e\, \mathcal{L}_{\mathfrak{m}}(1/\mathsf{M})$ for every $\mathsf{M} > 0$.

(b) Either $\mathfrak{m} = 0$, or $\mathcal{L}_{\mathfrak{m}} > 0$ on $(0,\infty)$ and $\log \mathcal{L}_{\mathfrak{m}}$ is convex and non-increasing on $(0,\infty)$.

(c) Let $\alpha > 0$, and let $\mathfrak{F}$ be a set of pairs $(\mathfrak{m}, A)$ of such measures and real numbers $A > 0$ such that $\sup_{(\mathfrak{m},A)\in\mathfrak{F}} |r^\alpha \mathcal{L}_{\mathfrak{m}}(r)/A - 1| \to 0$ as $r \downarrow 0$. Then $\sup_{(\mathfrak{m},A)\in\mathfrak{F}} |\Gamma(\alpha+1)\, \mathsf{M}^{-\alpha}\, \mathfrak{m}([0,\mathsf{M}])/A - 1| \to 0$ as $\mathsf{M} \to \infty$.

■

Sketch of proof. (a) For $r \geq r_0 > 0$ and an integer $k \geq 0$, $m^k e^{-mr} \leq (2k/(er_0))^k\, e^{-mr_0/2}$, a function that $\mathfrak{m}$ integrates; differentiation under the integral sign gives $(-1)^k \mathcal{L}_{\mathfrak{m}}^{(k)}(r) = \int m^k e^{-mr}\, d\mathfrak{m}(m) \geq 0$. The uniqueness is that of the measure in Bernstein's theorem. As $r \to \infty$, $e^{-mr} \to \mathbf{1}_{\{0\}}(m)$, dominated by e^{-m} once $r \geq 1$; and since $e^{-m/\mathsf{M}} \geq e^{-1}$ for $0 \leq m \leq \mathsf{M}$, $\mathcal{L}_{\mathfrak{m}}(1/\mathsf{M}) \geq e^{-1}\mathfrak{m}([0,\mathsf{M}])$.

(b) If $\mathfrak{m} \neq 0$, the positivity of e^{-mr} gives $\mathcal{L}_{\mathfrak{m}} > 0$. For $r_1, r_2 > 0$ and $0 < t < 1$, Hölder's inequality with the exponents $1/t$ and $1/(1-t)$, applied to e^{-tmr_1} and $e^{-(1-t)mr_2}$, gives $\mathcal{L}_{\mathfrak{m}}(tr_1 + (1-t)r_2) \leq \mathcal{L}_{\mathfrak{m}}(r_1)^t\, \mathcal{L}_{\mathfrak{m}}(r_2)^{1-t}$, which is the convexity of $\log \mathcal{L}_{\mathfrak{m}}$; by (a) it is non-increasing.

(c) *Step 1: two positive functionals.* For a bounded Borel function G on $[0,1]$, a pair $(\mathfrak{m}, A) \in \mathfrak{F}$ and $r > 0$ put

$$\Phi_r(G) = \frac{r^\alpha}{A} \int e^{-mr}\, G(e^{-mr})\, d\mathfrak{m}(m), \qquad \Phi_0(G) = \frac{1}{\Gamma(\alpha)} \int_0^\infty e^{-t}\, G(e^{-t})\, t^{\alpha-1}\, dt.$$

Both are linear and monotone in G, a pointwise inequality $G_1 \leq G_2$ giving $\Phi(G_1) \leq \Phi(G_2)$, and $\Phi_0(1) = 1$ by the integral of the Gamma function.

Step 2: polynomials. For an integer $k \geq 0$, $\Phi_r(x^k) = (k+1)^{-\alpha}\, \big((k+1)r\big)^\alpha \mathcal{L}_{\mathfrak{m}}\big((k+1)r\big)/A$, which tends to $(k+1)^{-\alpha}$ as $r \downarrow 0$, uniformly on $\mathfrak{F}$; and $\Phi_0(x^k) = \Gamma(\alpha)^{-1} \int_0^\infty e^{-(k+1)t} t^{\alpha-1}\, dt = (k+1)^{-\alpha}$. Hence $\Phi_r(P) \to \Phi_0(P)$ uniformly on $\mathfrak{F}$ for every polynomial P.

Step 3: an indicator between two continuous functions. Let $G_*(x) = x^{-1}$ for $e^{-1} \leq x \leq 1$ and $G_*(x) = 0$ for $0 \leq x < e^{-1}$. Then $e^{-t} G_*(e^{-t}) = \mathbf{1}_{[0,1]}(t)$, so that $\Phi_r(G_*) = r^\alpha\, \mathfrak{m}([0,1/r])/A$ and

$\Phi_0(G_*) = \Gamma(\alpha)^{-1}\int_0^1 t^{\alpha-1}\,dt = \Gamma(\alpha+1)^{-1}$. Let $0<\epsilon<1$. The function G_+ coincides with x^{-1} on $[e^{-1},1]$, vanishes on $[0,e^{-1-\epsilon}]$ and is affine on $[e^{-1-\epsilon},e^{-1}]$; the function G_- coincides with x^{-1} on $[e^{-1+\epsilon},1]$, vanishes on $[0,e^{-1}]$ and is affine on $[e^{-1},e^{-1+\epsilon}]$. Both are continuous and enclose G_*, $G_- \le G_* \le G_+$, because on $[e^{-1},e^{-1+\epsilon}]$ the values of G_- do not exceed $e^{1-\epsilon} \le x^{-1}$; and $xG_+(x) \le e^{-1}\cdot e = 1$ on $[e^{-1-\epsilon},e^{-1}]$. So $e^{-t}G_+(e^{-t}) \le \mathbf{1}_{[0,1+\epsilon]}(t)$ and $e^{-t}G_-(e^{-t}) \ge \mathbf{1}_{[0,1-\epsilon]}(t)$, whence $\Phi_0(G_+) \le (1+\epsilon)^\alpha/\Gamma(\alpha+1)$ and $\Phi_0(G_-) \ge (1-\epsilon)^\alpha/\Gamma(\alpha+1)$. By Weierstrass' approximation theorem there are polynomials $P_\pm$, depending on ϵ alone, with $G_+ \le P_+ \le G_+ + 2\epsilon$ and $G_- - 2\epsilon \le P_- \le G_-$ on $[0,1]$.

Step 4: the squeeze. By monotony $\Phi_r(P_-) \le r^\alpha\,\mathfrak{m}([0,1/r])/A \le \Phi_r(P_+)$, while $\Phi_0(P_+) \le (1+\epsilon)^\alpha/\Gamma(\alpha+1)+2\epsilon$ and $\Phi_0(P_-) \ge (1-\epsilon)^\alpha/\Gamma(\alpha+1)-2\epsilon$. Step 2 gives $r_\epsilon>0$ with $|\Phi_r(P_\pm)-\Phi_0(P_\pm)| \le \epsilon$ for $0<r\le r_\epsilon$ and every pair of $\mathfrak{F}$; with $\mathsf{M}=1/r$,

$$\frac{(1-\epsilon)^\alpha}{\Gamma(\alpha+1)} - 3\epsilon \le \frac{\mathfrak{m}([0,\mathsf{M}])}{A\,\mathsf{M}^\alpha} \le \frac{(1+\epsilon)^\alpha}{\Gamma(\alpha+1)} + 3\epsilon \qquad (\mathsf{M}\ge 1/r_\epsilon,\ (\mathfrak{m},A)\in\mathfrak{F}),$$

and ϵ is arbitrary. □

Lemma 18.108 (One field in the direction e_3)**.** Let σ' be σ or a component, and let $h \in C^1_c(\mathbb{E}_+)$ be real, with $h=0$ on $\{y_3<a\}$ for a number $a>0$.

(a) For every $t\ge 0$, all the integrals converging absolutely,

$$\langle\psi_h, e^{-tH}\psi_h\rangle = \kappa'\Big(\int h\Big)^2 + \int_{[0,\infty)} d\rho(m)\int_{\mathbb{R}^2} e^{-t\omega_m(\mathsf{q})}\,\frac{|\hat h_m(\mathsf{q})|^2}{2\,\omega_m(\mathsf{q})}\,\frac{d^2\mathsf{q}}{(2\pi)^2}.$$

(b) $\mu_h^H = \kappa'\big(\int h\big)^2\delta_0 + \mu_h^{\mathrm{sh}}$, where $\mu_h^{\mathrm{sh}}(B) = \int d\rho(m)\int \mathbf{1}_B(\omega_m(\mathsf{q}))\,|\hat h_m(\mathsf{q})|^2\,(2\omega_m(\mathsf{q}))^{-1}\,(2\pi)^{-2}\,d^2\mathsf{q}$ for Borel sets $B\subset[0,\infty)$.

(c) Let $h\ge 0$ and $h\ne 0$. For every $\mathsf{M}\ge 0$ with $\rho((\mathsf{M},\infty))>0$, $\mathbf{1}_{(\mathsf{M},\infty)}(H)\,\psi_h \ne 0$. If $\rho((\mathsf{M},\infty))>0$ for every $\mathsf{M}\ge 0$, then H is unbounded and $\mathcal{H}(\sigma',e_3)$ has infinite dimension.

■

Sketch of proof. Step 1: the Yukawa kernel as an integral over a plane. For $c,\omega>0$, $m\ge 0$, $t>0$ and $w\in\mathbb{R}^2$ we use the classical integrals

$$\int_0^\infty \frac{e^{-c^2/(4t)-\omega^2 t}}{(4\pi t)^{1/2}}\,dt = \frac{e^{-c\omega}}{2\omega}, \qquad \int_0^\infty \frac{e^{-c^2/(4t)-m^2 t}}{(4\pi t)^{3/2}}\,dt = \frac{e^{-mc}}{4\pi c}, \qquad \int_{\mathbb{R}^2} e^{i\mathsf{q}\cdot w - t|\mathsf{q}|^2}\,\frac{d^2\mathsf{q}}{(2\pi)^2} = \frac{e^{-|w|^2/(4t)}}{4\pi t}.$$

Let $\tau>0$ and $z=(w,\tau)$. Express $e^{-\tau\omega_m(\mathsf{q})}/(2\omega_m(\mathsf{q}))$ by the first integral at $c=\tau$, $\omega=\omega_m(\mathsf{q})$ (for $\mathsf{q}\ne 0$ if $m=0$, a null set being left out), integrate in q before t by the third, and apply the second at $c=|z|$; as $(4\pi t)^{-1/2}(4\pi t)^{-1} = (4\pi t)^{-3/2}$, this gives

$$\frac{e^{-m|z|}}{4\pi|z|} = \int_{\mathbb{R}^2} e^{i\mathsf{q}\cdot w}\,\frac{e^{-\tau\omega_m(\mathsf{q})}}{2\,\omega_m(\mathsf{q})}\,\frac{d^2\mathsf{q}}{(2\pi)^2}.$$

The exchange of the two integrations is Fubini's theorem: at $w=0$ every function in the computation is positive, so by Tonelli's theorem the integral of the modulus of the integrand is $e^{-m\tau}/(4\pi\tau)<\infty$.

Step 2: (a). By Definition 13.4 (4), $\langle\psi_h,e^{-tH}\psi_h\rangle = \langle[(h)],[(\tau_{te_3}h)]\rangle = \sigma'(\theta h,\tau_{te_3}h)$, where $\theta h = h\circ\theta$ is supported in $\{x_3\le -a\}$ and $\tau_{te_3}h$ in $\{x_3\ge a+t\}$. The radial kernel (Definition 13.4 (3)), the form of Proposition 13.25 (b) and the substitution $x\mapsto\theta x$, $y\mapsto y+te_3$ give

$$\sigma'(\theta h,\tau_{te_3}h) = \kappa'\Big(\int h\Big)^2 + \int d\rho(m)\iint h(x)h(y)\,\frac{e^{-m|z|}}{4\pi|z|}\,dx\,dy, \qquad z=(x_\perp-y_\perp,\ x_3+y_3+t),$$

the order of $d\rho$ and $dx\,dy$ being free by Fubini's theorem, as the modulus of the integrand is integrable: $|z|\ge 2a$ on the supports, and $\int e^{-2am}\,d\rho(m)<\infty$. Insert Step 1 at $w=x_\perp-y_\perp$ and $\tau=x_3+y_3+t\ge 2a$: the modulus of the integrand in (m,x,y,q) has the integral $\int d\rho(m)\iint|h(x)h(y)|\,e^{-m\tau}(4\pi\tau)^{-1}\,dx\,dy \le (8\pi a)^{-1}\|h\|_{L^1}^2\int e^{-2am}\,d\rho(m)$, so Fubini's

theorem lets us integrate in x and y first; h being real, $\int h(x)\, e^{i\mathsf{q}\cdot x_\perp - x_3\omega_m(\mathsf{q})}\, dx = \hat h_m(\mathsf{q})$ and $\int h(y)\, e^{-i\mathsf{q}\cdot y_\perp - y_3\omega_m(\mathsf{q})}\, dy = \overline{\hat h_m(\mathsf{q})}$, which gives (a).

Step 3: (b). The measure ρ is σ-finite. Assign the value 0 to the integrand defining μ_h^{sh} at $(m,\mathsf{q}) = (0,0)$. For every Borel set B the resulting integrand $\mathbf{1}_B(\omega_m(\mathsf{q}))\,|\hat h_m(\mathsf{q})|^2/(2\omega_m(\mathsf{q}))$ is nonnegative and Borel, so μ_h^{sh} is a positive Borel measure. By Tonelli's theorem the right member of (a) is $\int e^{-tE}\, d(\kappa'(\int h)^2\delta_0 + \mu_h^{\mathrm{sh}})(E)$, equal to $\|\psi_h\|^2 < \infty$ at $t = 0$. The left member is $\int e^{-tE}\, d\mu_h^H(E)$ (Definition 18.106 (5)). Two positive Borel measures on $[0,\infty)$ whose Laplace transforms are finite and equal coincide (Lemma 18.107 (a)).

Step 4: (c). Put $\varpi_h(m) = \int |\hat h_m(\mathsf{q})|^2\, (2\omega_m(\mathsf{q}))^{-1}(2\pi)^{-2}\, d^2\mathsf{q}$. For $m > \mathsf{M}$, $\omega_m(\mathsf{q}) \geq m > \mathsf{M}$ for every q; so by (b) $\|\mathbf{1}_{(\mathsf{M},\infty)}(H)\psi_h\|^2 = \mu_h^H((\mathsf{M},\infty)) \geq \int_{(\mathsf{M},\infty)} \varpi_h\, d\rho$. As $h \geq 0$ and $h \neq 0$, the function $\mathsf{q} \mapsto \hat h_m(\mathsf{q})$ is continuous with $\hat h_m(0) = \int h(y)\, e^{-my_3}\, dy > 0$; so $\varpi_h > 0$, and the last integral is positive when $\rho((\mathsf{M},\infty)) > 0$. If this is so for every M, the spectral projection $\mathbf{1}_{(\mathsf{M},\infty)}(H)$ is not 0 for any M, whereas $\mathbf{1}_{(\|H\|,\infty)}(H) = 0$ for a bounded H; so H is unbounded. Moreover $\mu_h^H((\mathsf{M}',\mathsf{M}''])$ increases to $\mu_h^H((\mathsf{M}',\infty)) > 0$ as $\mathsf{M}'' \to \infty$, so there are numbers $0 = \mathsf{M}_0 < \mathsf{M}_1 < \mathsf{M}_2 < \cdots$ with $\mu_h^H((\mathsf{M}_k,\mathsf{M}_{k+1}]) > 0$; the vectors $\mathbf{1}_{(\mathsf{M}_k,\mathsf{M}_{k+1}]}(H)\,\psi_h$, $k \geq 0$, are non-zero and pairwise orthogonal, and $\mathcal{H}(\sigma', e_3)$ has infinite dimension. □

Lemma 18.109 (The constants at infinity; the kernel of the limit)**.** (a) For every $\xi \in \mathsf{X}$ the constant at infinity of σ_ξ is $\kappa'_\xi = \mathfrak{c}_\xi^2$.

(b) The constant at infinity of σ is $\kappa'_\sigma = \int_{\mathsf{X}} \mathfrak{c}_\xi^2\, d\lambda(\xi)$, and $\kappa'_\sigma \leq \bar c_g^2$.

(c) At each $r > 0$, $\xi \mapsto \mathbf{s}_\xi^{\mathrm{c}}(r)$ is Borel on X, and its integral against λ is $\mathbf{s}_\sigma^{\mathrm{KL}}(r)$: $\mathbf{s}_\sigma^{\mathrm{KL}}(r) = \int_{\mathsf{X}} \mathbf{s}_\xi^{\mathrm{c}}(r)\, \lambda(d\xi)$.

■

Sketch of proof. Step 1: one field at large times. Let σ' be σ or σ_ξ, and let $h \in C_c^1(\mathbb{E}_+)$ be real with $h \geq 0$, $\int h > 0$ and $h = 0$ on $\{y_3 < a\}$, $a > 0$. In $\mathcal{H}(\sigma', e_3)$, $\langle \psi_h, e^{-tH}\psi_h\rangle = \sigma'(\theta h, \tau_{te_3} h)$ (Definition 13.4 (4)), and the radial kernel (Definition 13.4 (3)) with the substitution $x \mapsto \theta x$, $y \mapsto y + te_3$ turns this into $\iint h(x)h(y)\, \mathbf{s}(|z_t|)\, dx\, dy$, where $z_t = (x_\perp - y_\perp, x_3 + y_3 + t)$ and $|z_t| \geq 2a + t$ on the supports. By Proposition 13.25 (a), $\mathbf{s}$ is non-increasing with the limit κ'; dominated convergence gives $\langle \psi_h, e^{-tH}\psi_h\rangle \to \kappa'(\int h)^2$ as $t \to \infty$. By the spectral theorem $e^{-tH} \to P_0$ strongly as $t \to \infty$; hence $\kappa'(\int h)^2 = \langle \psi_h, P_0\psi_h\rangle = \|P_0\psi_h\|^2$.

Step 2: (a). In $\mathcal{H}(\sigma_\xi, e_3)$ the kernel of H is $\mathbb{C}\,\Omega$ (Thm. YM3_I:AP), $\|\Omega\|^2 = \sigma_\xi(\varnothing) = 1$, and $\langle \Omega, \psi_h\rangle = \sigma_\xi(\theta\varnothing \cdot (h)) = \sigma_\xi((h)) = \mathfrak{c}_\xi \int h$ (Definition 13.4 (2), (4)). So $\|P_0\psi_h\|^2 = |\langle \Omega, \psi_h\rangle|^2 = \mathfrak{c}_\xi^2(\int h)^2$, and Step 1 gives $\kappa'_\xi = \mathfrak{c}_\xi^2$.

Step 3: the kernel of the limit; (b) and (c). Let $r > 0$. Choose $x_0, y_0 \in \mathbb{R}^3$ at the distance r and, for each integer $n > 4/r$, non-negative C^1 functions f_n, h_n, neither of them 0, whose supports lie in the closed balls of radius $1/n$ centred at x_0 and at y_0; the pair (f_n, h_n) belongs to $\mathfrak{S}_2(\mathbb{R}^3)$, and the distance of a point of $\operatorname{supp} f_n$ from a point of $\operatorname{supp} h_n$ lies in $[r - \frac{2}{n}, r + \frac{2}{n}]$. For σ'' equal to σ or to a component let $F_n(\sigma'') = \sigma''(f_n, h_n)/(\int f_n \int h_n)$. By Definition 13.4 (3), $F_n(\sigma'')$ is an average, with non-negative weights, of the radial kernel $\mathbf{s}_{\sigma''}$ over $[r - \frac{2}{n}, r + \frac{2}{n}]$, and that kernel is continuous and non-increasing (Proposition 13.25 (a)); so $\mathbf{s}_{\sigma''}(r + \frac{2}{n}) \leq F_n(\sigma'') \leq \mathbf{s}_{\sigma''}(r - \frac{2}{n})$, and $F_n(\sigma'') \to \mathbf{s}_{\sigma''}(r)$ as $n \to \infty$. By continuity of evaluations in the canonical decomposition every $\xi \mapsto F_n(\sigma_\xi)$ is a continuous function on X, and their pointwise limit $\xi \mapsto \mathbf{s}_\xi(r)$ is therefore Borel; and the barycentre identity gives $F_n(\sigma) = \int F_n(\sigma_\xi)\, \lambda(d\xi)$. Hence $\int \mathbf{s}_\xi(r + \frac{2}{n})\, \lambda(d\xi) \leq F_n(\sigma) \leq \int \mathbf{s}_\xi(r - \frac{2}{n})\, \lambda(d\xi)$. As n grows, $\mathbf{s}_\xi(r + \frac{2}{n})$ increases to $\mathbf{s}_\xi(r)$, and monotone convergence gives $\int \mathbf{s}_\xi(r)\, \lambda(d\xi) \leq \mathbf{s}_\sigma(r)$, at every $r > 0$; in particular $\xi \mapsto \mathbf{s}_\xi(r/2)$ is λ-integrable. As n grows, $\mathbf{s}_\xi(r - \frac{2}{n})$ decreases to $\mathbf{s}_\xi(r)$ and stays below $\mathbf{s}_\xi(r/2)$, and dominated convergence gives the reverse inequality: $\mathbf{s}_\sigma(r) = \int_{\mathsf{X}} \mathbf{s}_\xi(r)\, \lambda(d\xi)$ for every $r > 0$. As $r \to \infty$, $\mathbf{s}_\xi(r)$ decreases to $\kappa'_\xi = \mathfrak{c}_\xi^2$ by (a), and for $r \geq 1$ it stays below the

integrable $\mathbf{s}_\xi(1)$; dominated convergence gives $\kappa'_\sigma = \int \mathfrak{c}_\xi^2\, d\lambda$. By Thm. YM3_I:AP, $|\mathfrak{c}_\xi| \le \bar{c}_g$ and $\xi \mapsto \mathfrak{c}_\xi$ is continuous; with $\lambda(\mathsf{X}) = 1$ the first gives the bound of (b). Finally $\xi \mapsto \mathbf{s}^{\mathrm{c}}_\xi(r) = \mathbf{s}_\xi(r) - \mathfrak{c}_\xi^2$ is a Borel function, and $\mathbf{s}^{\mathrm{KL}}_\sigma(r) = \mathbf{s}_\sigma(r) - \kappa'_\sigma = \int (\mathbf{s}_\xi(r) - \mathfrak{c}_\xi^2)\, \lambda(d\xi)$, which is (c). □

Proposition 18.110 (The connected kernel of a component: Källén–Lehmann form, log-convexity, the weight of the mass 0)**.** Under the statements granted at the head of this subsection, let $0 < g \le \gamma$ and $\xi \in \mathsf{X}$.

(a) The connected function of two points of σ_ξ is $S_2^{\sigma_\xi}(f,h) = \iint f(x)h(y)\, \mathbf{s}^{\mathrm{c}}_\xi(|x-y|)\, dx\, dy$ on $\mathfrak{S}_2(\mathbb{R}^3)$, with $\mathbf{s}^{\mathrm{c}}_\xi(r) = \int_{[0,\infty)} e^{-mr}(4\pi r)^{-1}\, d\rho_\xi(m)$ for all $r > 0$; ρ_ξ is the only positive Borel measure on $[0,\infty)$ for which this identity is true for all $r > 0$, and $\mathbf{s}^{\mathrm{c}}_\xi = \mathbf{s}^{\mathrm{KL}}_\xi$.

(b) $\mathbf{s}^{\mathrm{c}}_\xi$ and $r \mapsto r\, \mathbf{s}^{\mathrm{c}}_\xi(r)$ are completely monotone on $(0,\infty)$. Either $\rho_\xi = 0$ and $\mathbf{s}^{\mathrm{c}}_\xi = 0$, or $r\, \mathbf{s}^{\mathrm{c}}_\xi(r) > 0$ for every $r > 0$ and $r \mapsto \log(r\, \mathbf{s}^{\mathrm{c}}_\xi(r))$ is convex and non-increasing on $(0,\infty)$.

(c) $\mathbf{s}^{\mathrm{c}}_\xi(r) \le \frac{r_0}{r}\, \mathbf{s}^{\mathrm{c}}_\xi(r_0)$ for $0 < r_0 \le r$.

(d) As $r \to \infty$, $4\pi r\, \mathbf{s}^{\mathrm{c}}_\xi(r)$ tends to $\rho_\xi(\{0\})$.

■

Sketch of proof. Step 1: (a). The constant at infinity of σ_ξ is $\kappa'_\xi = \mathfrak{c}_\xi^2$ (Lemma 18.109 (a)); so $\mathbf{s}^{\mathrm{c}}_\xi = \mathbf{s}_\xi - \kappa'_\xi = \mathbf{s}^{\mathrm{KL}}_\xi$, and Definition 18.106 (2) with Proposition 13.25 (b) gives the formula and the uniqueness.

Step 2: (b)–(d). In the notation of Lemma 18.107, $4\pi r\, \mathbf{s}^{\mathrm{c}}_\xi(r) = \mathcal{L}_{\rho_\xi}(r)$; by clauses (a) and (b) of that lemma it is completely monotone, identically 0 if $\rho_\xi = 0$, and otherwise positive with a convex and non-increasing logarithm; $\mathbf{s}^{\mathrm{c}}_\xi = \mathbf{s}^{\mathrm{KL}}_\xi$ is completely monotone by Definition 18.106 (3). As $r\, \mathbf{s}^{\mathrm{c}}_\xi(r)$ is non-increasing, $r\, \mathbf{s}^{\mathrm{c}}_\xi(r) \le r_0\, \mathbf{s}^{\mathrm{c}}_\xi(r_0)$ for $r \ge r_0$, which is (c); and (d) is Lemma 18.107 (a), or Proposition 13.25 (c) with $\kappa'_\xi = \mathfrak{c}_\xi^2$. □

Proposition 18.111 (The law at a point; the growth of the Källén–Lehmann measure and of the energy measure)**.** Under the statements granted at the head of this subsection, let $0 < g \le \gamma$. For every $\xi \in \mathsf{X}$ the connected function of two points of σ_ξ has the kernel $\mathbf{s}^{\mathrm{c}}_\xi(|x-y|)$, where $\mathbf{s}^{\mathrm{c}}_\xi(r) = \int e^{-mr}(4\pi r)^{-1}\, d\rho_\xi(m)$ with exactly one positive Borel measure ρ_ξ on $[0,\infty)$, and both $\mathbf{s}^{\mathrm{c}}_\xi$ and $r\, \mathbf{s}^{\mathrm{c}}_\xi(r)$ are completely monotone. Let now $0 < g \le \gamma_W$, let b be the label of the sequence (Definition 18.106 (4)), and let σ' be σ or a component.

(a) *The law at a point.* For $0 < \delta \le \frac{1}{2}$ and $0 < r \le r_\delta$

$$a_\delta^{-1}(1 - \varepsilon_\delta(r)) \le y(r) \le a_\delta(1 + \varepsilon_\delta(r)), \qquad |y(r) - 1| \le a_\delta(1 + \varepsilon_\delta(r)) - 1;$$

for a component $y = y^{\mathrm{c}}$. For the limit and for every component, $r^6 \mathbf{s}^{\mathrm{c}}(r) \to \mathsf{c}_2 \mathsf{b}^{-2} = \frac{9}{16\pi^2 \mathsf{b}^2}$ as $r \to 0$.

(b) *Uniformity.* For every $\eta > 0$ there is $r(\eta) > 0$, formed from the frame, g and η alone, such that $|y^{\mathrm{c}}(r) - 1| \le \eta$ and $|y(r) - 1| \le \eta$ for $0 < r \le r(\eta)$, for every convergent Gibbs sequence at g, every canonical decomposition of its limit, every component and the limit itself.

(c) *The growth.* As $\mathsf{M} \to \infty$

$$\rho([0,\mathsf{M}]) \sim \frac{4\pi \mathsf{c}_2}{\Gamma(6)}\, \mathsf{b}^{-2}\mathsf{M}^5 = \frac{3}{160\pi}\, \mathsf{b}^{-2}\mathsf{M}^5, \qquad \mu^E([0,\mathsf{M}]) \sim \frac{\mathsf{c}_2}{\Gamma(7)}\, \mathsf{b}^{-2}\mathsf{M}^6 = \frac{\mathsf{c}_2}{720}\, \mathsf{b}^{-2}\mathsf{M}^6,$$

and uniformly: the suprema of $|\frac{160\pi}{3}\mathsf{b}^2\mathsf{M}^{-5}\rho([0,\mathsf{M}]) - 1|$ and of $|\frac{720}{\mathsf{c}_2}\mathsf{b}^2\mathsf{M}^{-6}\mu^E([0,\mathsf{M}]) - 1|$ over the convergent Gibbs sequences at g, the canonical decompositions of their limits, the components and the limits tend to 0 as $\mathsf{M} \to \infty$. Moreover $d\mu^E(E) = \frac{1}{4\pi}\rho([0,E])\, dE$.

(d) $\rho([0,\infty)) = \infty$; and for every $C > 1$ there is M_C, formed from the frame, g and C alone, with $\rho((\mathsf{M}, C\mathsf{M}]) > 0$ for all $\mathsf{M} \ge \mathsf{M}_C$ and all these σ'.

■

Sketch of proof. Step 0: the first sentence is Proposition 18.110 (a), (b).

Step 1: narrow pairs. Let $0 < \delta \le \frac12$, $\mathsf{d} > 0$, and $x_0, y_0 \in \mathbb{R}^3$ with $|x_0 - y_0| = 3\mathsf{d}$; let $\varphi(u) = (1-|u|^2)^2$ for $|u| \le 1$ and $\varphi(u) = 0$ for $|u| > 1$, and put $f(x) = \varphi((x-x_0)/(\delta\mathsf{d}))$ and $h(y) = \varphi((y-y_0)/(\delta\mathsf{d}))$. These are non-negative functions of class C^1 with $\|f\|_\infty = \|h\|_\infty = 1$; each is supported by the closed ball of radius $\delta\mathsf{d} \le \mathsf{d}$ about its centre, and the two balls lie at the distance $(3-2\delta)\mathsf{d}$, a number of $[2\mathsf{d}, 3\mathsf{d})$; $\int f = \int h = (\delta\mathsf{d})^3 \int \varphi = \vartheta_\delta\, \mathsf{d}^3$, as $\int \varphi = 4\pi \int_0^1 (1-s^2)^2 s^2\, ds = \frac{32\pi}{105}$; and $\operatorname{Lip} f = \operatorname{Lip} h = (\delta\mathsf{d})^{-1} \max_{0\le s\le 1} 4s(1-s^2) = \frac{8}{3\sqrt3\,\delta\mathsf{d}} \le \frac{1}{\vartheta_\delta \mathsf{d}}$, because $\vartheta_\delta \cdot \frac{8}{3\sqrt3\,\delta} = \frac{256\pi}{315\sqrt3}\,\delta^2 < \frac25$. So the test-pair conditions of Thm. YM3_I:C are satisfied with $\vartheta = \vartheta_\delta \in (0,1)$, and $(f,h) \in \mathfrak{T}^1_{\mathbb{R}^3}(\mathsf{d};\vartheta_\delta) \subset \mathfrak{S}_2(\mathbb{R}^3)$.

Step 2: the law on narrow pairs in a component. Let $\xi \in \mathsf{X}$. Thm. YM3_I:C at $\vartheta = \vartheta_\delta$ gives, for $0 < \mathsf{d} \le \mathsf{d}_0(g,\vartheta_\delta)$ and the pair of Step 1, $S_2^{\sigma_\xi}(f,h) = \mathsf{c}_2 \mathsf{b}^{-2} \mathcal{K}_{\text{free}}(f,h)(1+\eta_\xi)$ with $|\eta_\xi| \le \bar\varepsilon_{g,\vartheta_\delta}(\mathsf{d})$, where $\mathcal{K}_{\text{free}}(f,h) = \iint f(x)h(y)|x-y|^{-6}\,dx\,dy$ is the free kernel in Thm. YM3_I:C. For $x \in \operatorname{supp} f$ and $y \in \operatorname{supp} h$, $(3-2\delta)\mathsf{d} \le |x-y| \le (3+2\delta)\mathsf{d}$. Since f and h are non-negative and $\mathbf{s}^{\mathrm{c}}_\xi = \mathbf{s}_\xi - \mathfrak{c}_\xi^2$ is non-increasing (Proposition 13.25 (a)), Definition 18.106 (2) gives, with $I = \int f \int h > 0$,

$$\mathbf{s}^{\mathrm{c}}_\xi\big((3+2\delta)\mathsf{d}\big)\, I \le S_2^{\sigma_\xi}(f,h) \le \mathbf{s}^{\mathrm{c}}_\xi\big((3-2\delta)\mathsf{d}\big)\, I, \qquad \big((3+2\delta)\mathsf{d}\big)^{-6} I \le \mathcal{K}_{\text{free}}(f,h) \le \big((3-2\delta)\mathsf{d}\big)^{-6} I.$$

Hence $\mathbf{s}^{\mathrm{c}}_\xi((3+2\delta)\mathsf{d}) \le \mathsf{c}_2\mathsf{b}^{-2}((3-2\delta)\mathsf{d})^{-6}(1+\bar\varepsilon_{g,\vartheta_\delta}(\mathsf{d}))$ and $\mathbf{s}^{\mathrm{c}}_\xi((3-2\delta)\mathsf{d}) \ge \mathsf{c}_2\mathsf{b}^{-2}((3+2\delta)\mathsf{d})^{-6}(1-\bar\varepsilon_{g,\vartheta_\delta}(\mathsf{d}))$.

Step 3: (a). Let $0 < r \le r_\delta$. In a component σ_ξ the first inequality of Step 2 at $\mathsf{d} = r/(3+2\delta)$ and the second at $\mathsf{d} = r/(3-2\delta)$, both numbers being at most $\mathsf{d}_0(g,\vartheta_\delta)$, together with $\bar\varepsilon_{g,\vartheta_\delta}(r/(3+2\delta)) \le \varepsilon_\delta(r)$, give the two-sided bound for y^{c}_ξ, and $y^{\mathrm{c}}_\xi = y_\xi$ because $\mathbf{s}^{\mathrm{c}}_\xi = \mathbf{s}^{\mathrm{KL}}_\xi$ (Proposition 18.110 (a)). For the limit, Lemma 18.109 (c) gives $y_\sigma(r) = \int_{\mathsf{X}} y_\xi(r)\,\lambda(d\xi)$, the label being one number for the sequence and all the components; as neither member of the bound depends on ξ and $\lambda(\mathsf{X}) = 1$, $y_\sigma(r)$ lies between the same two members. With $\varepsilon = \varepsilon_\delta(r)$, $y - 1 \le a_\delta(1+\varepsilon) - 1$, and $1 - y \le 1 - a_\delta^{-1}(1-\varepsilon) \le a_\delta(1+\varepsilon) - 1$, because $a \mapsto a(1+\varepsilon) + a^{-1}(1-\varepsilon)$ takes the value 2 at $a = 1$ and has the derivative $(1+\varepsilon) - a^{-2}(1-\varepsilon) \ge 0$ for $a \ge 1$. As $r \to 0$, $\varepsilon_\delta(r) \to 0$, so that $a_\delta^{-1} \le \liminf y \le \limsup y \le a_\delta$; and $a_\delta \to 1$ as $\delta \to 0$; so $y(r) \to 1$. By Definition 18.106 (2), $\mathbf{s}^{\mathrm{c}} - \mathbf{s}^{\mathrm{KL}} = \kappa' - \mathfrak{c}'^2$, which is κ'_σ for the limit and 0 for a component (Lemma 18.109 (a)); so $r^6 \mathbf{s}^{\mathrm{c}}(r)\,\mathsf{b}^2/\mathsf{c}_2 = y(r) + r^6(\kappa' - \mathfrak{c}'^2)\,\mathsf{b}^2/\mathsf{c}_2 \to 1$.

Step 4: (b). Let $\eta > 0$. Choose δ with $a_\delta - 1 \le \eta/3$, and then $r' \in (0, r_\delta]$ with $64\,\varepsilon_\delta(r') \le \eta/3$; as $a_\delta \le 64$ and ε_δ is non-decreasing, (a) gives $|y(r) - 1| \le \frac23\eta$ for $0 < r \le r'$, for the limit and for every component. For a component $y^{\mathrm{c}} = y$. For the limit, $0 \le y^{\mathrm{c}}(r) - y(r) = r^6 \kappa'_\sigma \mathsf{b}^2/\mathsf{c}_2 \le r^6\, \bar c_g^2 L^2 (g^{-2} + \bar B)^2/\mathsf{c}_2$, by Lemma 18.109 (b) and since $\mathsf{b} \le L(g^{-2} + \bar B)$, the upper end of $I^{\mathrm{lab}}(g)$ (Definition 13.1 (2)). Let $r(\eta)$ be the largest number of $(0, r']$ with $r(\eta)^6\, \bar c_g^2 L^2 (g^{-2} + \bar B)^2 \le \eta\, \mathsf{c}_2/3$; it is formed from the frame, g and η alone and serves for all the functionals named in (b).

Step 5: (c). Over the convergent Gibbs sequences at g, the canonical decompositions of their limits, their components and their limits, the pairs $(\rho, \mathsf{A}_\mathsf{b})$ form a set $\mathfrak{F}_5$, and the pairs $(\mu^E, \mathsf{c}_2\mathsf{b}^{-2})$ a set $\mathfrak{F}_6$, of the kind of Lemma 18.107 (c) at $\alpha = 5$ and at $\alpha = 6$. Indeed, by Definition 18.106 (2), (3), $\mathcal{L}_\rho(r) = 4\pi r\, \mathbf{s}^{\mathrm{KL}}(r)$ and $\mathcal{L}_{\mu^E}(r) = \mathbf{s}^{\mathrm{KL}}(r)$, so that $r^5 \mathcal{L}_\rho(r)/\mathsf{A}_\mathsf{b} = r^6 \mathcal{L}_{\mu^E}(r)\,\mathsf{b}^2/\mathsf{c}_2 = y(r)$, which tends to 1 uniformly by (b). Lemma 18.107 (c) gives the two uniform laws, with $\Gamma(6) = 120$, $\Gamma(7) = 720$ and $\frac{4\pi\mathsf{c}_2}{120} = \frac{36}{1920\pi} = \frac{3}{160\pi}$. For the density of μ^E: $e^{-mr}/r = \int_m^\infty e^{-Er}\,dE$, and Tonelli's theorem gives $\mathbf{s}^{\mathrm{KL}}(r) = \int_0^\infty e^{-Er}\,\frac{\rho([0,E])}{4\pi}\,dE$; the uniqueness in Lemma 18.107 (a) identifies the two measures.

Step 6: (d). By (c), $\rho([0,\infty)) = \lim_{\mathsf{M}\to\infty} \rho([0,\mathsf{M}]) = \infty$. Let $C > 1$ and let $\epsilon_\rho(\mathsf{M})$ be the supremum in (c) for ρ; it tends to 0, so there is M_C, formed from the frame, g and C alone, with $\epsilon_\rho(\mathsf{M}') < (C^5-1)/(C^5+1)$ for all $\mathsf{M}' \ge \mathsf{M}_C$. For $\mathsf{M} \ge \mathsf{M}_C$

$$\mathsf{M}^{-5}\rho((\mathsf{M}, C\mathsf{M}]) = C^5(C\mathsf{M})^{-5}\rho([0, C\mathsf{M}]) - \mathsf{M}^{-5}\rho([0,\mathsf{M}]) \ge \frac{3}{160\pi}\,\mathsf{b}^{-2}\big[C^5 - 1 - (C^5+1)\max\{\epsilon_\rho(\mathsf{M}), \epsilon_\rho(C\mathsf{M})\}\big] > 0.$$

□

Proposition 18.112 (The two-point function of a component)**.** Under the statements granted at the head of this subsection, let $0 < g \le \gamma$ and $\xi \in \mathsf{X}$.

(a) There is exactly one positive Borel measure ρ_ξ on $[0,\infty)$ with $\int e^{-mr}\,d\rho_\xi(m) < \infty$ for all $r > 0$ such that $S_2^{\sigma_\xi}(f,h) = \iint f(x)h(y)\,G_\xi(|x-y|)\,dx\,dy$ for all $(f,h) \in \mathfrak{S}_2(\mathbb{R}^3)$, where $G_\xi(r) = (4\pi r)^{-1}\int e^{-mr}\,d\rho_\xi(m)$; and $G_\xi = \mathbf{s}^{\mathrm{c}}_\xi$. This measure has polynomial growth: $\rho_\xi([0,\mathsf{M}]) \le 1024\pi e\,(C^g_{\mathrm{rad}})^2 \max\{\mathsf{M}, \frac12\}^7$ for every $\mathsf{M} \ge 0$, with the common reconstruction constant C^g_{rad}.

(b) G_ξ is real-analytic, non-negative and non-increasing on $(0,\infty)$; $r \mapsto rG_\xi(r)$ is completely monotone and non-increasing with the limit $\rho_\xi(\{0\})/(4\pi)$ at infinity; and $G_\xi(r) \le \frac{r_0}{r}\,G_\xi(r_0)$ for $0 < r_0 \le r$.

(c) Let $0 < g \le \gamma_W$ and let b be the label. Then $r^6 G_\xi(r) \to \mathsf{c}_2\mathsf{b}^{-2}$ as $r \to 0$, and $r^6 G_\xi(r)\,\mathsf{b}^2/\mathsf{c}_2$ is held between a lower and an upper member: $a_\delta^{-1}(1-\varepsilon_\delta(r)) \le r^6 G_\xi(r)\,\mathsf{b}^2/\mathsf{c}_2 \le a_\delta(1+\varepsilon_\delta(r))$ for $0 < \delta \le \frac12$ and $0 < r \le r_\delta$, with members that are the same for all components of all canonical decompositions of the limits of all convergent Gibbs sequences at g. Further $\rho_\xi([0,\mathsf{M}]) \sim \frac{3}{160\pi}\,\mathsf{b}^{-2}\mathsf{M}^5$ as $\mathsf{M} \to \infty$, so that $\rho_\xi([0,\infty)) = \infty$; the Hilbert space $\mathcal{H}(\sigma_\xi, e_3)$ has infinite dimension; and its Hamiltonian H is unbounded.

■

Sketch of proof. Step 1: the measure. By Proposition 18.110 (a), ρ_ξ has the property, with $G_\xi = \mathbf{s}^{\mathrm{c}}_\xi$. Let ρ'' be a second measure with the property and G'' its function. Dominated convergence makes G'' continuous on $(0,\infty)$, and $\iint f(x)h(y)\,(G'' - \mathbf{s}^{\mathrm{c}}_\xi)(|x-y|)\,dx\,dy = 0$ on $\mathfrak{S}_2(\mathbb{R}^3)$; non-negative functions of class C^1 and of mass 1 whose supports shrink to two points at the distance r give $G''(r) = \mathbf{s}^{\mathrm{c}}_\xi(r)$. So $\mathcal{L}_{\rho''} = \mathcal{L}_{\rho_\xi}$, and $\rho'' = \rho_\xi$ by Lemma 18.107 (a). The same argument shows that a continuous radial kernel of σ_ξ on $\mathfrak{S}_2(\mathbb{R}^3)$ is unique.

Step 2: polynomial growth. By Thm. YM3_I:AP, the component is reconstruction data with exponent 4 and constant C^g_{rad}. Prop. YM3_I:108.3 bounds its two-point kernel at a separation re_3 by $(C^g_{\mathrm{rad}})^2 \min\{1, r/2\}^{-8}$, by taking the tube distance parameter $\eta = \min\{1, r/2\}$ in its point estimate. Rotation invariance gives the same bound at every point of separation r. This kernel is $\mathbf{s}_\xi$ by Definition 13.4(3). Since $0 \le G_\xi = \mathbf{s}_\xi - \mathfrak{c}_\xi^2 \le \mathbf{s}_\xi$, Lemma 18.107(a) gives for $\mathsf{M} \ge 1/2$

$$\rho_\xi([0,\mathsf{M}]) \le e\,\mathcal{L}_{\rho_\xi}(1/\mathsf{M}) = 4\pi e\,\mathsf{M}^{-1}G_\xi(1/\mathsf{M}) \le 4\pi e\,\mathsf{M}^{-1}(C^g_{\mathrm{rad}})^2(2\mathsf{M})^8 = 1024\pi e\,(C^g_{\mathrm{rad}})^2\mathsf{M}^7,$$

and $\rho_\xi([0,\mathsf{M}]) \le \rho_\xi([0,\frac12])$ for $0 \le \mathsf{M} < \frac12$.

Step 3: (b). $\mathbf{s}_\xi$ is real-analytic (Definition 13.4 (3)), and so is $G_\xi = \mathbf{s}_\xi - \mathfrak{c}_\xi^2$; the other assertions are Proposition 18.110 (b)–(d), a completely monotone function being non-negative and non-increasing.

Step 4: (c). The law, its enclosure between the two members and the growth are Proposition 18.111 (a), (c), (d) at $\sigma' = \sigma_\xi$, where $y^{\mathrm{c}}(r) = r^6 G_\xi(r)\,\mathsf{b}^2/\mathsf{c}_2$; the members a_δ, $\varepsilon_\delta(r)$ and the range r_δ are formed from the frame, g, δ and r (Definition 18.106 (4)). As $\rho_\xi([0,\mathsf{M}]) < \infty$ by (a), $\rho_\xi((\mathsf{M},\infty)) = \infty$ for every $\mathsf{M} \ge 0$, and Lemma 18.108 (c), at a real $h \in C^1_c(\mathbb{E}_+)$ with $h \ge 0$ and $h \ne 0$, gives the last two assertions. □

Proposition 18.113 (The connected functions of a limit and of its components)**.** Under the statements granted at the head of this subsection, let $0 < g \le \gamma$, and let σ' be the limit σ or a component σ_ξ; $(\mathbb{R}^3)^n_{\ne}$ is the open set of the n-tuples $x = (x_1,\dots,x_n)$ of pairwise distinct points of $\mathbb{R}^3$, and $x_J = (x_i)_{i\in J}$ for $J \subset \{1,\dots,n\}$.

(i) For every $n \ge 2$ there is exactly one continuous function k_n on $(\mathbb{R}^3)^n_{\ne}$ with $S_n^{\sigma'}(f_1,\dots,f_n) = \int k_n(x)\prod_{i=1}^n f_i(x_i)\,dx$ for all $(f_1,\dots,f_n) \in \mathfrak{S}_n(\mathbb{R}^3)$, and it is real-analytic; $k_2(x_1,x_2) = \mathbf{s}^{\mathrm{c}}(|x_1-x_2|)$.

(ii) For all $r > 0$, $\mathbf{s}^{\mathrm{c}}_\sigma(r) = \kappa'_\sigma + \int e^{-mr}(4\pi r)^{-1}\,d\rho_\sigma(m)$ with $\kappa'_\sigma = \int_{\mathsf{X}} \mathfrak{c}_\xi^2\,d\lambda(\xi)$, and $\mathbf{s}^{\mathrm{c}}_\xi(r) = \int e^{-mr}(4\pi r)^{-1}\,d\rho_\xi(m)$. In each of the two identities the pair formed by the constant term,

κ'_σ in the first and 0 in the second, and the positive Borel measure with finite Laplace transforms is unique.

(iii) Let $0 < g \le \gamma_W$ and let b be the label. Then $|x|^6 k_2(0,x) \to \mathsf{c}_2\mathsf{b}^{-2}$ as $x \to 0$.

(iv) In the situation of (iii), $\rho([0,\mathsf{M}]) \sim \frac{4\pi\mathsf{c}_2}{5!}\,\mathsf{b}^{-2}\mathsf{M}^5$ as $\mathsf{M} \to \infty$.

(v) In the situation of (iii), the Hamiltonian H of $\mathcal{H}(\sigma', e_3)$ is unbounded, and $\dim \mathcal{H}(\sigma', e_3) = \infty$.

(vi) In a component, $4\pi|x|\,k_2(0,x) \to \rho_\xi(\{0\})$ as $|x| \to \infty$, and $|v|\,S_2^{\sigma_\xi}(f, \tau_v h) \to \frac{1}{4\pi}\,\rho_\xi(\{0\}) \int f \int h$ as $|v| \to \infty$ for every $(f,h) \in \mathfrak{S}_2(\mathbb{R}^3)$. Hence $\rho_\xi(\{0\}) > 0$ if and only if there is a pair $(f,h) \in \mathfrak{S}_2(\mathbb{R}^3)$ with $\limsup_{|v|\to\infty} |v|\,|\sigma_\xi((f)\cdot\tau_v(h)) - \sigma_\xi((f))\,\sigma_\xi((h))| > 0$, that is, a pair at which the decay $|v|^{-1}$ of the cluster bound of Thm. YM3_I:AP is reached. ■

Sketch of proof. Step 1: (i). Thm. YM3_I:AP gives reconstruction data for the limit and each component. By Prop. YM3_I:108.3, for every $p \ge 1$ there is a real-analytic function s_p on $(\mathbb{R}^3)^p_{\neq}$ with $\sigma'(f_1,\dots,f_p) = \int \mathsf{s}_p(x) \prod_i f_i(x_i)\,dx$ on $\mathfrak{S}_p(\mathbb{R}^3)$. Write $X = (f_1,\dots,f_n) \in \mathfrak{S}_n(\mathbb{R}^3)$, and for a non-empty $J \subset \{1,\dots,n\}$ let X_J be the tuple of the f_i with $i \in J$, taken by increasing i. By the moment–cumulant formula, $S_n^{\sigma'}(X)$ is the sum over the partitions π of $\{1,\dots,n\}$ of $(-1)^{|\pi|-1}(|\pi|-1)!\prod_{J\in\pi}\sigma'(X_J)$; by Fubini's theorem each product is the integral of $\prod_{J\in\pi}\mathsf{s}_{|J|}(x_J)$ against $\prod_i f_i(x_i)$, and for $x \in (\mathbb{R}^3)^n_{\neq}$ every x_J consists of pairwise distinct points. So $k_n(x) = \sum_\pi (-1)^{|\pi|-1}(|\pi|-1)!\prod_{J\in\pi}\mathsf{s}_{|J|}(x_J)$ is real-analytic and represents $S_n^{\sigma'}$. A continuous function on $(\mathbb{R}^3)^n_{\neq}$ whose integrals against all these products vanish is 0: at a point x, take non-negative functions f_i of class C^1 and of mass 1 supported by pairwise disjoint balls about the points x_i, and let the balls shrink. This gives the uniqueness, and at $n = 2$ Definition 18.106 (2) gives k_2.

Step 2: (ii). By Lemma 18.109, $\kappa'_\sigma = \int \mathfrak{c}_\xi^2\,d\lambda$ and $\kappa'_\xi = \mathfrak{c}_\xi^2$; as $\mathfrak{c}' = 0$ for σ and $\mathfrak{c}' = \mathfrak{c}_\xi$ for σ_ξ, Proposition 13.25 (b) gives the two identities. If $\mathsf{s}^{\mathrm{c}}(r) = \varkappa + \int e^{-mr}(4\pi r)^{-1}\,d\rho''(m)$ for all $r > 0$, with a number $\varkappa$ and a positive Borel measure ρ'' with finite Laplace transforms, the integral is at most $\mathcal{L}_{\rho''}(1)/(4\pi r)$ for $r \ge 1$ and tends to 0; so $\varkappa = \lim_{r\to\infty}\mathsf{s}^{\mathrm{c}}(r)$, then $\mathcal{L}_{\rho''}(r) = 4\pi r(\mathsf{s}^{\mathrm{c}}(r) - \varkappa)$ is determined, and so is ρ'' (Lemma 18.107 (a)).

Step 3: (iii)–(v). By (i), $k_2(0,x) = \mathsf{s}^{\mathrm{c}}(|x|)$, so $|x|^6 k_2(0,x) = \mathsf{c}_2\mathsf{b}^{-2}y^{\mathrm{c}}(|x|)$, and (iii) is Proposition 18.111 (a); (iv) is Proposition 18.111 (c), as $5! = \Gamma(6)$. By Proposition 18.111 (d) and Lemma 18.107 (a), $\rho((\mathsf{M},\infty)) = \infty$ for every $\mathsf{M} \ge 0$; Lemma 18.108 (c), at a real $h \in C^1_c(\mathbb{E}_+)$ with $h \ge 0$ and $h \ne 0$, gives (v).

Step 4: (vi). In a component $k_2(0,x) = \mathsf{s}^{\mathrm{c}}_\xi(|x|)$, and Proposition 18.110 (d) gives the first limit. Let $(f,h) \in \mathfrak{S}_2(\mathbb{R}^3)$ have its supports in the ball of radius R about 0, and let $|v| > 2R$. Then $(f,\tau_v h) \in \mathfrak{S}_2(\mathbb{R}^3)$, and $S_2^{\sigma_\xi}(f,\tau_v h) = \iint f(x)h(y)\,\mathsf{s}^{\mathrm{c}}_\xi(|x-y-v|)\,dx\,dy$ by Definition 18.106 (2). On the supports $\big||x-y-v|-|v|\big| \le 2R$, so $|v|/|x-y-v| \to 1$ and, by Proposition 18.110 (d), $|x-y-v|\,\mathsf{s}^{\mathrm{c}}_\xi(|x-y-v|) \to \rho_\xi(\{0\})/(4\pi)$, both uniformly there as $|v| \to \infty$; this gives the second limit. If $\rho_\xi(\{0\}) > 0$, a pair with $f, h \ge 0$ and $\int f \int h > 0$ has $|v|\,S_2^{\sigma_\xi}(f,\tau_v h) \to \rho_\xi(\{0\}) \int f \int h/(4\pi) > 0$; if $\rho_\xi(\{0\}) = 0$, the limit is 0 at every pair. Finally, by the two-entry cumulant formula and $\sigma_\xi((\tau_v h)) = \mathfrak{c}_\xi \int h = \sigma_\xi((h))$ (Definition 13.4 (2)), $\sigma_\xi((f)\cdot\tau_v(h)) - \sigma_\xi((f))\,\sigma_\xi((h)) = S_2^{\sigma_\xi}(f,\tau_v h)$, the quantity that the cluster bound of Thm. YM3_I:AP controls at $X = (f)$, $Y = (h)$. □

When, beyond what is granted at the head of this subsection, a Setting is given whose first frame is the frame fixed there (Definition 13.6; there is one by Lemma 13.5), and Theorem F, Theorem S and the statements (M1)–(M12) are granted at both of its frames as in Theorem AL, the clauses (c) 1 and (c) 7 of that theorem, which by its clause (e) 4 apply at every reference coupling $g' \in (0,\gamma]$ to every convergent Gibbs sequence at g' and to every component of its limit, give $\mathfrak{c}_\xi = 0$ and $\rho_\xi(\{0\}) = 0$, the measure of its conclusion 3 being ρ_ξ once $\mathfrak{c}_\xi = 0$; the constant at infinity of σ is then 0 by Lemma 18.109 (b), and $|v|\,S_2^{\sigma_\xi}(f,\tau_v h) \to 0$ for every $(f,h) \in \mathfrak{S}_2(\mathbb{R}^3)$ by Proposition 18.113 (vi).

18.11. **Reflection positivity at the diagonal mirrors from a sheared torus; the pressure to two loops; loops of block fields under the box inequality.** Three further results hold in the setting of Section 17.2. First, with no hypothesis of the kind of BI: if along one defining sequence of tori of a continuum limit at a reference coupling the free energy E of the shear of the cube by half a period, the number of Theorem AZ(1)(f), has a finite lower limit, then the centred functional of the limit is reflection positive at every diagonal mirror of $\mathbb{R}^3$, and so, with the coordinate mirrors, at every mirror of the cubic lattice (Proposition 18.114); the lower limit is finite as soon as the doubling defect e of the half cube, or the cube number, stays bounded along the sequence, as $0 \le E \le \mathsf{e}$ and $E \le D_L$ hold exactly on one lattice. The cube is compared with the diamond, the torus of the same volume sheared back, by Schwarz's inequality across the shear; the diamond carries a chessboard estimate along three orthogonal normals that is dominated by the chessboard seminorms of the cubes, so that its cumulants obey bounds free of the cutoff and converge along a subsequence; and the operator that σ_S defines in the Hilbert space of the limit along the diamonds is positive, by the positivity of σ_S at a coordinate mirror, by bounded multiplications by columns, and because the half turn about the line in which the diagonal and the coordinate mirror meet acts trivially at infinity. Second, the pressure of one lattice to two loops, $P(\beta) = -3\log\beta + z_\infty + p_1/\beta + O((\log\beta)^2\beta^{-1.02})$ as $\beta \to \infty$ through the reals, with the constant z_∞ of Proposition 17.47(d) and $p_1 = 9G_0 - 36G_0^2 - 9J_{\mathrm{gh}} + 18J_3$, formed from the Green function G of the lattice Laplacian on $\mathbb{Z}^3$; p_1 is the limit per site of the vacuum diagrams with two loops of the Dirichlet boxes and of the periodic cubes (Proposition BF). The pressure is pinned between two periodic cubes of the side about $\beta^{0.34}$, where one chart about the flat field in Landau's gauge and three levels of large-field estimates give $\log Z$ to the order $1/\beta$, and the coefficient is computed on the Dirichlet boxes, the odd part of the cubes of twice their side, in the gauge of Feynman. Third, under the inequality BI of one box at one label, Wilson loops of the block fields of a fixed depth d obey an area law with the tension $0.66/\ell_0^2$, uniformly in the cutoff, on the tori, in the limit of the cutoffs and in the one periodic Gibbs state: the rectangle with four tapered links at all sizes, and the rectangle of block links itself with the constant 17 as long as its shorter side is at most a length ϱ_d, which grows without bound with the depth, besides a second term at all sizes; the two loops differ, uniformly in the side of the torus, by at most a number τ'_d that tends to 0 with the depth (Proposition 18.115). The flux ratio of the boxes obeys an area law under BI; loops that split at two planes are bounded through the flux of the transfer operator in the transverse direction; and the large fields of one cell of a block field are bounded by the discarding of the small-field chain, with no factor of the volume.

Setting. Use the constants and labels of §17.2. For a nonzero normal n write $\Theta_{n,c}x = x - 2(n\cdot x - c)n/|n|^2$. For the coordinate normals e_μ the lattice heights are $c \in (a/2)\mathbb{Z}$, and for $n = e_\mu \pm e_\nu$ they are $c \in a\mathbb{Z}$. We put $\mathsf{n}_\pm = e_1 \pm e_3$, $u_\diamond = \mathsf{n}_+/\sqrt{2}$, $\Sigma_\diamond = (u_\diamond, 0)$, $\Theta_\diamond = \Theta_{u_\diamond,0}$, $\Theta_2 = \Theta_{e_2,0}$, and $\mathsf{s}_y = \tau_{y\mathsf{n}_+/2}$. The last map raises the height $x_1 + x_3$ by y.

Write $\mathfrak{S}_n(O)$ for the n-tuples of real $C_c^1(O)$ tests with pairwise disjoint supports, and $\mathbb{C}\mathfrak{S}(O)$ for their finite complex linear combinations, including the empty tuple. Conjugate the coefficients and concatenate tuples when the combined supports are disjoint. Reflection positivity on an open half-space means $\sigma(\Theta\bar{F}\cdot F) \ge 0$ there. Put $\mathcal{D} = \mathbb{C}\mathfrak{S}(\{x_1 + x_3 > 0\})$. Use the cumulant and centred moment relations of Definition 18.205, and the norm $\|f\|_\sharp = \max(\sup|f|, 40M\,\mathrm{Lip}\,f)$. A tuple of type $(\mathsf{d}; R)$ has each support in a ball of radius R and distances at least d between distinct supports. A bound table bounds each n-linear form on such tuples by a constant $C_n(\mathsf{d}; R)$ times the product of their norms. For compactly supported Lipschitz tests and a law ν put $X_K^\nu = \prod_q[\mathcal{O}_K(f_q) - \nu(\mathcal{O}_K(f_q))]$. For a second law ν' set $\sigma_{\nu',\nu}(X) = \nu'(X_K^\nu)$. By expanding the centred product and using the moment–cumulant relations of ν',

$$\sigma_{\nu',\nu}(X) = \sum_{\pi\in\mathrm{Part}(n)} \prod_{I\in\pi,\ |I|=1} \delta_{\nu',\nu}(f_I) \prod_{I\in\pi,\ |I|\ge 2} S^T_{|I|}[\nu'](X_I), \qquad \delta_{\nu',\nu}(f) := \nu'(\mathcal{O}_K(f)) - \nu(\mathcal{O}_K(f)),$$

for $X = (f_1, \dots, f_n)$. A period lattice Γ is a finite-index subgroup of $a\mathbb{Z}^3$. Use the Wilson law $\mu_{\Gamma,\beta}$ on the quotient, periodically extended, and the unnormalized integral $\mathsf{I}_\Gamma(H) = Z_\Gamma[0]\mu_{\Gamma,\beta}(H)$. The cube is $\Gamma = \ell\mathbb{Z}^3$ and $p_\ell = \ell^{-3}\log Z_{K,\ell,\beta}$.

A board (i, o) of side $\mathsf{l} = a_i$, with $i < K$ and $o \in a_K\mathbb{Z}^3$, has cells of side l about $o + \mathsf{l}\mathbf{k}$, $\mathbf{k} \in \mathbb{Z}^3$. Use the bonds with both endpoints in a cell. The coordinate walls have heights $o_\mu + \mathsf{l}(\mathbb{Z} + 1/2)$. On a compatible cube of side ℓ' the chessboard seminorm $\|F\|_{\mathrm{ch},\ell'}$ is the $(\ell'/\mathsf{l})^3$-th root of the Wilson mean of the product of mirror copies in one period, conjugated for an odd number of reflections. These are the board and fold of Section YM3_I:86. Positivity, the product bound, and the comparison across volumes are Propositions YM3_I:86.3 and YM3_I:86.5. The triangle inequality follows by expanding the fold of a sum and using the product bound term by term. Each seminorm is at most the uniform norm, and $\|1\|_{\mathrm{ch}} = 1$. The eight-cell convention of §17.2 agrees with this one when $\ell' = 2\mathsf{l}$.

The diamond. For $K, m \in \mathbb{N}$, $\ell = 2L^m$ and $\beta \geq 0$ let

$$\Gamma^\diamond_\ell := \mathbb{Z}\,\ell\,\mathsf{n}_- \oplus \mathbb{Z}\,\ell e_2 \oplus \mathbb{Z}\,\tfrac{\ell}{2}\mathsf{n}_+ \subset a_K\mathbb{Z}^3, \qquad \mu^\diamond_{K,\ell,\beta} := \mu_{\Gamma^\diamond_\ell,\beta}, \qquad E(K,\ell,\beta) := \log Z_{\Gamma^\diamond_\ell}[0] - \log Z_{K,\ell,\beta}.$$

This is the lattice Γ of Theorem AZ(1)(f), and E is the free energy of the shear of that clause, here at every coupling: the cube $\ell\mathbb{Z}^3$ arises from $\Gamma^\diamond_\ell$ when its third generator is moved by $\frac{\ell}{2}\mathsf{n}_-$, half a period across the height. Use the fixed source radius ϱ and $C_{\mathrm{v}} = C_{\mathrm{S}} + C_g + C_{\mathrm{gap}}$ of [7, Lem. N.1]. Use $\mathsf{Q}_1(L), Q^\natural(n,R)$ and the functions $\Phi_1(R,D;\mathsf{l}), \mathsf{U}(R,\mathsf{d})$ of §5.2.1. Write $\mathsf{l}(n,R)$ for the least power of L at least $8n(R+1)$. Let $C^g_n(\mathsf{d};R)$ be the bound table for the cube cumulants of Proposition YM3_I:91.13.

Proposition 18.114 (A bounded free energy of the shear gives reflection positivity at the diagonal mirrors)**.** Let $0 < g \leq \gamma$, and let $S = (S_n)_{n\geq 2}$ be a continuum limit at g along growing tori in Section YM3_I:92 which has a defining sequence of tori $(K_j, \ell_j, \beta_j)_{j\in\mathbb{N}}$ with

$$\liminf_{j\to\infty} E(K_j, \ell_j, \beta_j) < \infty.$$

(a) For every $F \in \mathcal{D}$, $\sigma_S(\Theta_\diamond \bar{F} \cdot F) \geq 0$: the functional σ_S is reflection positive at the diagonal mirror $\Sigma_\diamond$.

(b) σ_S is reflection positive at every diagonal mirror $(2^{-1/2}(s\,e_\mu + s'e_\nu), c)$ of $\mathbb{R}^3$, $\mu \neq \nu$, $s, s' \in \{1, -1\}$, $c \in \mathbb{R}$; with Proposition YM3_I:92.4, it is reflection positive at every coordinate and at every diagonal mirror of $\mathbb{R}^3$.

(c) At all $K, m \in \mathbb{N}$, $\ell = 2L^m$ and $\beta \geq 0$: $0 \leq E(K,\ell,\beta) \leq \mathsf{e}(\ell/2a_K)$, and $E(K,\ell,\beta) \leq D_L(\ell/La_K)$ if $m \geq 2$. So the hypothesis holds whenever $\liminf_j \mathsf{e}(\ell_j/2a_{K_j}) < \infty$ or $\liminf_j D_L(\ell_j/La_{K_j}) < \infty$, each number taken at $\beta = \beta_j$.

■

Proof. At a fixed cutoff we write $a = a_K$. Steps 1–4 are exact statements on one lattice at every $\beta \geq 0$, and Steps 5 and 6, which work with the boards and chessboard seminorms above, are such statements at every $\beta > 0$; Steps 7–10 read the bounds at the first-passage data; Steps 11–18 work with the limit functionals.

Step 1: the diamond and its mirrors. The determinant of the three period generators has modulus ℓ^3, so the lattice index is $(\ell/a)^3$. A nonzero period has sup norm at least $\ell/2$: its first and third coordinates are integer multiples of $\ell/2$ and the second is a multiple of ℓ. Equality holds for $(\ell/2)\mathsf{n}_+$. The three generators are orthogonal. The period splits into a transverse sublattice and a pure normal period for $\mathsf{n}_+, \mathsf{n}_-, e_2$, with height periods $\ell, 2\ell, \ell$. These are the adapted-period conditions of YM3_I:(1820). Each corresponding lattice reflection flips just the normal generator in its linear part and preserves the quotient. It permutes the plaquette actions and product Haar factors, as do all lattice translations. The cube law is invariant under all signed coordinate permutations as well. With $\Gamma_\perp := \mathbb{Z}\ell\mathsf{n}_- \oplus \mathbb{Z}\ell e_2 \subset \mathsf{n}_+^\perp$ we have $\Gamma^\diamond_\ell = \Gamma_\perp \oplus \mathbb{Z}\frac{\ell}{2}\mathsf{n}_+$ and $\ell\mathbb{Z}^3 = \Gamma_\perp \oplus \mathbb{Z}\ell e_1$, as $\ell e_3 = \ell e_1 - \ell\mathsf{n}_-$; both $\frac{\ell}{2}\mathsf{n}_+$

and ℓe_1 raise the height by ℓ, and they differ by $\frac{\ell}{2}\mathsf{n}_-$. Moreover $\Gamma_\ell^\diamond = (\mathbb{Z}2\ell e_3 \oplus \mathbb{Z}\ell e_2) \oplus \mathbb{Z}\frac{\ell}{2}(e_1+e_3)$, as the vectors of $\Gamma_\ell^\diamond$ with first coordinate 0 are those of $\mathbb{Z}2\ell e_3 \oplus \mathbb{Z}\ell e_2$ and the first coordinates fill $\frac{\ell}{2}\mathbb{Z}$: with e_1 as the time, the diamond is the box of $\ell/2a$ layers over the cross-section of ℓ/a and $2\ell/a$ sites in the directions e_2 and e_3, closed up with the translation of a layer by $\frac{\ell}{2}e_3$.

Step 2: Schwarz's inequality across the shear. For all bounded Borel functions F, G of the bonds of the closed slab $\{0 \le x_1 + x_3 \le \frac{\ell}{2}\}$,

$$\left|\mathsf{I}_{\ell\mathbb{Z}^3}(\overline{\Theta_\diamond G}\, F)\right|^2 \le \mathsf{I}_{\Gamma_\ell^\diamond}(\overline{\Theta_\diamond G}\, G)\, \mathsf{I}_{\Gamma_\ell^\diamond}(\overline{\Theta_\diamond F}\, F),$$

both factors on the right being ≥ 0. At $F = G = 1$ this is Proposition 17.31(g), that is $E \ge 0$; at $G = F$, divided by $Z^2_{K,\ell,\beta}$, it gives $|\mu_{K,\ell,\beta}(\overline{\Theta_\diamond F} F)| \le e^{E(K,\ell,\beta)}\, \mu^\diamond_{K,\ell,\beta}(\overline{\Theta_\diamond F} F)$. The argument of Proposition 17.31(g) applies, with the two insertions carried along. The planes of the heights in $\frac{\ell}{2}\mathbb{Z}$ are diagonal lattice mirrors, as $\frac{\ell}{2} = L^{m+K}a \in a\mathbb{Z} = \mathrm{Adm}(\mathsf{n}_+)$; no bond crosses them because a bond changes the height by at most a. A plaquette meeting both open sides has two opposite corners on the plane and one corner at each of the heights $\pm a$ from it, by listing its four heights. Orient the two paths between the corners on the plane in the same direction and call their transports v, v'. The factor $e^{-\beta}\exp[(\beta/2)\,\mathrm{tr}(vv'^{-1})]$ has a positive kernel expansion: its m-th Taylor term is $e^{-\beta}(\beta/2)^m/m!$ times the trace of $v^{\otimes m}(v'^{\otimes m})^*$, the sum of products of corresponding matrix entries and their conjugates. The series is absolutely summable, since the sum of the absolute entry products at each m is at most 2^m by Hilbert–Schmidt Cauchy–Schwarz. Both lattices are $\Gamma_\perp \oplus \mathbb{Z}g$ with a g that raises the height by ℓ; cut at the heights $-\frac{\ell}{2}$, 0, $\frac{\ell}{2}$, the integral $\mathsf{I}_{\Gamma_\perp\oplus\mathbb{Z}g}(F_1F_2)$, F_1 a function of the lower and F_2 of the upper slab, is $\iint k_{F_2}(\tau_g\phi_0, \phi_1)\, k_{F_1}(\phi_1, \phi_0)$ over the data ϕ of the walls (their bonds modulo $\Gamma_\perp$ and the indices of the expansion at their cut plaquettes, under Haar measure and the weights of the expansion), with the kernels k_{F_i} of the two slabs, and τ_g carries the data of the height $-\frac{\ell}{2}$ to those of the height $\frac{\ell}{2}$ and keeps their measure. Schwarz's inequality in the wall data bounds $|\mathsf{I}_{\ell\mathbb{Z}^3}(F_1F_2)|^2$ by $\|k_{F_1}\|_2^2\|k_{F_2}\|_2^2$. The reflection $\Theta_{\mathsf{n}_+,0} = \Theta_\diamond$ exchanges the two slabs, exchanges the two corners of the cut plaquettes of the height 0, conjugates the kernels, and acts on the plane of the height $-\frac{\ell}{2}$ as the translation by $\frac{\ell}{2}\mathsf{n}_+$; so $\|k_{F_i}\|_2^2$ is the integral over the slab of F_i glued to its mirror image, that is over $\Gamma_\perp \oplus \mathbb{Z}\frac{\ell}{2}\mathsf{n}_+ = \Gamma_\ell^\diamond$, of F_i times its conjugated mirror image: the gluing in the proof of Proposition YM3_I:91.4, with nonnegativity from the squared norm. Take $F_1 = \overline{\Theta_\diamond G}$ and $F_2 = F$.

Step 3: the chessboard estimate along one normal, walls at all admissible heights. Let n be a normal, $c_0 \in \mathrm{Adm}(n)$, $h \in a\mathbb{N}$, $N \in \mathbb{N}$, Γ a period lattice adapted to n with the height period $2Nh$, and $\mu = \mu_{\Gamma,\beta}$. The heights $c_0 + jh$ lie in $\mathrm{Adm}(n)$; let $\Theta_j := \Theta_{n,c_0+jh}$ and $\mathbb{S}_j := \{c_0 + (j-1)h \le n\cdot x \le c_0 + jh\}$. For a function f of the bonds of $\mathbb{S}_1$ put $f^{[1]} := f$ and $f^{[j+1]} := \overline{\Theta_j f^{[j]}}$, a function of the bonds of $\mathbb{S}_{j+1}$; then $\overline{\Theta_i f^{[j]}} = f^{[2i+1-j]}$, $f^{[j+2]}$ is the translate of $f^{[j]}$ by $2h|n|^{-2}n \in a\mathbb{Z}^3$, and $f^{[j+2N]} = f^{[j]}$ on the Γ-periodic configurations. For a function F of the bonds of $\mathbb{S}_j$ let $\mathrm{Fold}\, F := \prod_{i=1}^{2N} f^{[i]}$, where $f^{[j]} = F$. Then $\mu(\mathrm{Fold}\, F) \ge 0$, and for functions F_j of the bonds of $\mathbb{S}_j$

$$\left|\mu\Big(\prod_{j=1}^{2N} F_j\Big)\right| \le \prod_{j=1}^{2N} \mu(\mathrm{Fold}\, F_j)^{1/2N}.$$

The index rule $\overline{\Theta_i f^{[j]}} = f^{[2i+1-j]}$ uses only the reflection maps and the lattice translation by $2hn/|n|^2$. In particular it is valid at odd diagonal spacings. For any division into two half-arrays at walls N slabs apart, Proposition YM3_I:91.4 and Cauchy–Schwarz bound the squared mean of the array by the product of the two means obtained by reflecting each half-array into the other half. Each reflected mean is nonnegative. For the constant array this also gives nonnegativity of the folded mean.

When all constant-array roots are positive, maximize the ratio of the modulus of the array mean to the product of its roots over arrays taking their entries in the finite set of the given model

functions. The denominator of the product of the two reflected ratios is the square of the original denominator. Maximality and Cauchy–Schwarz force both reflections of a maximizing array to maximize the ratio. Among maximizers choose one with a longest constant cyclic string. At its terminal wall the half-array retaining that string doubles its length if it is at most N and produces a constant array if it is longer than N. Thus a constant array maximizes the ratio, with value one. If a root vanishes, add a common real ϵ to the model functions. The folded means are nonnegative monic polynomials in ϵ of degree $2N$. Use values approaching zero outside their finitely many roots and pass to the limit. This proves the array estimate for all the admissible heights used here.

Step 4: three normals on the diamond. Let $h \in a\mathbb{N}$ with $M_h := \ell/h$ an even integer and $c_1, c_2, c_3 \in a\mathbb{Z}$. The walls are the lattice mirrors $\Sigma(\mathsf{n}_+, c_1 + h\mathbb{Z})$, $\Sigma(\mathsf{n}_-, c_2 + h\mathbb{Z})$, $\Sigma(e_2, c_3 + h\mathbb{Z})$; for each of the three normals the slabs of Step 3 are formed with these heights, with $2N = M_h$, $2M_h$, M_h respectively, and the cells are the intersections $\mathsf{C}_{\underline{j}}$ of three slabs, one of each normal; $\mathsf{C} := \mathsf{C}_{(1,1,1)}$, a box of the volume $\frac{1}{2}h^3$. The group W generated by the reflections in the walls acts simply transitively on the cells; its translations by $\Gamma_\ell^\diamond$ form a normal subgroup of the index $\mathsf{N} := 2M_h^3$. For a function G of the bonds of a cell put $\operatorname{FOLD} G := \prod_{[\mathsf{w}]\in \mathsf{W}/\Gamma_\ell^\diamond} (\mathsf{w}G)^{(\cdot)}$ on the $\Gamma_\ell^\diamond$-periodic configurations, the factor conjugated when w is a product of an odd number of reflections, and $q_\diamond(G) := \mu^\diamond_{K,\ell,\beta}(\operatorname{FOLD} G)^{1/\mathsf{N}}$. Then $\mu^\diamond(\operatorname{FOLD} G) \ge 0$; $|\mu^\diamond(\prod_{\mathsf{c}} G_{\mathsf{c}})| \le \prod_{\mathsf{c}} q_\diamond(G_{\mathsf{c}})$ for functions G_{c} of the bonds of cells c that are pairwise incongruent modulo $\Gamma_\ell^\diamond$; $q_\diamond$ is a seminorm on the functions of one cell, $q_\diamond(1) = 1$, and $q_\diamond(\mathsf{w}G) = q_\diamond(G)$ for $\mathsf{w} \in \mathsf{W}$; and for a function f of the bonds of C, with its copies $f^{[i]}$ along n_+ of Step 3, the column seminorm $\|f\|_{\mathrm{col},h} := \mu^\diamond(\prod_{i=1}^{M_h} f^{[i]})^{1/M_h}$ is at most $q_\diamond(f)$. Indeed, Step 3 along n_+, then along n_- for each slab of the first normal, then along e_2, gives the product bound, the reflections in walls of orthogonal normals commuting and those of one normal changing the height along that normal alone; the exponents multiply to $1/\mathsf{N}$, and the threefold fold is FOLD. Homogeneity holds as half of the factors are conjugated, the triangle inequality follows on expanding $\operatorname{FOLD}(G + G')$ and bounding each term by the product bound, and the column bound is Step 3 along n_- and then along e_2, with the factor 1 in all the other slabs.

Step 5: products over the cells of a board of $a\mathbb{Z}^3$, on the diamond. Let (i, o) be a board of $a\mathbb{Z}^3$ with the side $\mathsf{l} = a_i$ and $\ell' \in \mathfrak{L}$ with $2\mathsf{l} \le \ell' \le \ell/L$. For a set $\mathcal{K}^\diamond$ of indices that represents the classes of $\mathbb{Z}^3$ modulo $\mathsf{l}^{-1}\Gamma_\ell^\diamond$ and functions $H_{\mathbf{k}}$ of the inner bonds of the cells $\square_{\mathbf{k}}$,

$$\Big|\mathsf{I}_{\Gamma_\ell^\diamond}\Big(\prod_{\mathbf{k}\in\mathcal{K}^\diamond} H_{\mathbf{k}}\Big)\Big| \le Z_{K,\ell',\beta}^{(\ell/\ell')^3} \prod_{\mathbf{k}\in\mathcal{K}^\diamond} \|H_{\mathbf{k}}\|_{\mathrm{ch},\ell'}.$$

Here $\Gamma_\ell^\diamond \subset \mathsf{l}\mathbb{Z}^3$. A translation by this subgroup preserves the coordinate wall set and conjugates its reflection group to itself. It therefore just reindexes the mirror copies on the board. The folds and their means on the comparison cube are unchanged by reindexing and lattice translation, including if the parity convention conjugates the entire fold, since its mean is real. By Step 1 we may take $\mathcal{K}^\diamond = \{1 \le \mathbf{k}_1 \le q',\ 1 \le \mathbf{k}_2 \le \ell/\mathsf{l},\ 1 \le \mathbf{k}_3 \le 2\ell/\mathsf{l}\}$, where $q' := \ell/2\mathsf{l}$ equals L^{m+i} and is therefore odd, L being odd; we write $q' = 2r + 1$, and $\ell'/2\mathsf{l} \le q'/L \le r$. With e_1 as the time, $F_1 := \prod_{\mathbf{k}_1 \le r} H_{\mathbf{k}}$ and $F_2 := \prod_{\mathbf{k}_1 > r} H_{\mathbf{k}}$ are functions of the layers between the wall planes $c_0 = o_1 + \frac{\mathsf{l}}{2}$, $c_0 + r\mathsf{l}$ and $c_0 + (2r+1)\mathsf{l} = c_0 + \frac{\ell}{2}$; the trace formula of (M1), closed up through the unitary translation U_v of a layer by $v = \frac{\ell}{2}e_3$ as in the proof of Proposition 17.31(f), gives $\mathsf{I}_{\Gamma_\ell^\diamond}(F_1F_2) = \operatorname{Tr}(\mathsf{U}_v\mathfrak{Y}[\Phi_2]\mathfrak{Y}[\Phi_1])$. Schwarz's inequality in the Hilbert–Schmidt class and $\mathfrak{Y}[\Phi^\Theta] = \mathfrak{Y}[\Phi]^*$ bound its square by $\operatorname{Tr}(\mathfrak{Y}[\Phi_1]^*\mathfrak{Y}[\Phi_1])\operatorname{Tr}(\mathfrak{Y}[\Phi_2]^*\mathfrak{Y}[\Phi_2])$, the integrals over the coordinate lattices B_1, B_2 with $2r\mathsf{l}/a$ and $(2r+2)\mathsf{l}/a$ layers over the same cross-section of F_i times its conjugated mirror image in the wall $c_0 + r\mathsf{l}$. The mirror image of $H_{\mathbf{k}}$ is its mirror copy in the cell of the index $2r + 1 - \mathbf{k}_1$ in the direction 1, with the same seminorm; so each integrand is a product over all the cells of a period of a box with the even numbers $2r$ or $2r+2$, ℓ/l, $2\ell/\mathsf{l}$, all $\ge \ell'/\mathsf{l}$, of cells a side, and the chessboard estimate on a box (Proposition YM3_I:86.3) and the volume comparison

of Proposition YM3_I:86.5 down to the torus ℓ', bounds it by $Z^{\mathsf{N}_i/\mathsf{N}'}_{K,\ell',\beta}\prod\|H_{\mathbf{k}}\|^2_{\mathrm{ch},\ell'}$, the product over the indices of F_i, with $\mathsf{N}_1=2r\cdot 2(\ell/\mathsf{l})^2$, $\mathsf{N}_2=(2r+2)\cdot 2(\ell/\mathsf{l})^2$, $\mathsf{N}'=(\ell'/\mathsf{l})^3$; and $\frac12(\mathsf{N}_1+\mathsf{N}_2)=(\ell/\mathsf{l})^3$. With all $H_{\mathbf{k}}=1$, the board $(K-1,0)$ and $\ell'=\ell/L$, $m\ge 2$, this is $Z_{\Gamma^\diamond_\ell}[0]\le Z^{L^3}_{K,\ell/L,\beta}$, that is $E(K,\ell,\beta)\le D_L(\ell/La)$, at every $\beta>0$. At $\beta=0$ every weight is 1 and the bonds carry Haar probability measures, so all the partition functions equal 1, the numbers E, e and D_L vanish, and (c) is true there trivially. The other bound of (c), $E(K,\ell,\beta)\le\mathsf{e}(\ell/2a)$, is the upper bound in Step 1 of the proof of Theorem AZ(1)(f), which rests on Proposition 17.31(b), (f) alone and holds at every $\beta\ge 0$; and $E\ge 0$ is Step 2. This proves (c).

Step 6: the diamond is dominated by the cubes. A diagonal board $(h;c_1,c_2,c_3)$ of Step 4 is *compatible* with a board (i,o) of the side l if h is a power of L with $\mathsf{l}\le h\le\frac{\ell}{2}$ and $c_1\equiv o_1+o_3$, $c_2\equiv o_1-o_3$, $c_3\equiv o_2$ modulo l. Then: every $\mathsf{w}\in\mathsf{W}$ has $\mathsf{w}o-o\in\mathsf{l}\mathbb{Z}^3$, as $\Theta_{\mathsf{n}_+,c}o-o=(c-o_1-o_3)\mathsf{n}_+$, $\Theta_{\mathsf{n}_-,c}o-o=(c-o_1+o_3)\mathsf{n}_-$ and $\Theta_{e_2,c}o-o=2(c-o_2)e_2$, so that the offsets $\mathsf{w}o$ and o differ by a vector of $\mathsf{l}\mathbb{Z}^3$; the two offsets thus give the same walls and cells. Each generator of W permutes the coordinate directions up to sign and conjugates the coordinate-wall reflections to coordinate-wall reflections. It maps folds to folds, up to reindexing and possible overall conjugation, so preserves the seminorms by invariance of the cube law; a wall meets the interior of a cell $\square_{\mathbf{k}}$ only if it passes through its centre $o+\mathsf{l}\mathbf{k}$, and then its reflection maps $\square_{\mathbf{k}}$ onto itself; and the stabiliser $\mathsf{W}_{\mathbf{k}}$ of $\square_{\mathbf{k}}$ in W is generated by the reflections in the walls through its centre, $s_{\mathbf{k}}:=\#\mathsf{W}_{\mathbf{k}}\in\{1,2,4,8\}$. Let $\ell'\in\mathfrak{L}$, $2\mathsf{l}\le\ell'\le\ell/L$, let $\mathcal{K}_0$ be a finite set of indices of cells whose interiors meet that of C, let $F_{\mathbf{k}}$ be functions of the bonds of $\square_{\mathbf{k}}\cap\mathsf{C}$, and $H_{\mathbf{k}}:=\prod_{\mathsf{v}\in\mathsf{W}_{\mathbf{k}}}(\mathsf{v}F_{\mathbf{k}})^{(\cdot)}$, conjugated for odd v. Then

$$q_\diamond\Big(\prod_{\mathbf{k}\in\mathcal{K}_0}F_{\mathbf{k}}\Big)\le\exp\big[\tfrac12h^3(p_{\ell'}-p_\ell)\big]\prod_{\mathbf{k}\in\mathcal{K}_0}\|H_{\mathbf{k}}\|^{1/s_{\mathbf{k}}}_{\mathrm{ch},\ell'}.$$

Indeed, if $\gamma\mathsf{w}\square_{\mathbf{k}}=\mathsf{w}'\square_{\mathbf{k}'}$ with $\gamma\in\Gamma^\diamond_\ell$, $\mathsf{w},\mathsf{w}'\in\mathsf{W}$, $\mathbf{k},\mathbf{k}'\in\mathcal{K}_0$, then $\mathbf{k}=\mathbf{k}'$ and $\mathsf{w}'^{-1}\gamma\mathsf{w}\in\mathsf{W}_{\mathbf{k}}$, by the simple transitivity and the description of the stabilisers; so $\mathrm{FOLD}\prod_{\mathbf{k}}F_{\mathbf{k}}$ is a product of the functions $(\mathsf{w}H_{\mathbf{k}})^{(\cdot)}$ on cells of (i,o) pairwise incongruent modulo $\Gamma^\diamond_\ell$, $2M_h^3/s_{\mathbf{k}}$ of them for each $\mathbf{k}$, each of the seminorm $\|H_{\mathbf{k}}\|_{\mathrm{ch},\ell'}$; Step 5 with 1 in the other cells bounds $\mathsf{l}_{\Gamma^\diamond_\ell}(\mathrm{FOLD}\prod F_{\mathbf{k}})$ by $e^{\ell^3p_{\ell'}}\prod_{\mathbf{k}}\|H_{\mathbf{k}}\|^{2M_h^3/s_{\mathbf{k}}}_{\mathrm{ch},\ell'}$, Step 2 gives $Z_{\Gamma^\diamond_\ell}[0]\ge e^{\ell^3p_\ell}$, and $\ell^3/2M_h^3=\frac12h^3$.

Step 7. At a first-passage datum $(K,\beta)\in\mathfrak{d}(g)$ on any admitted side λ, Theorem YM3_I:AD and [7, Lem. N.1] give a vacuum energy E_λ with

$$E_\lambda-C_g\lambda^3\le\log Z_{K,\lambda,\beta}\le E_\lambda+C_{\mathrm{S}}\lambda^3,\qquad|E_{\lambda'}\lambda'^{-3}-E_\lambda\lambda^{-3}|\le C_{\mathrm{gap}}.$$

Thus $p_{\lambda'}-p_\lambda\le C_{\mathrm{v}}$. The cell-normalizer construction of §5.2.1 also applies to supports in balls of radius R with their centres at distance at least $R+D$ from the walls of a board of side l. If $\underline{f}$ has n such entries, use the polydisc $\mathbb{D}_\varrho(\underline{f})=\{z:\sum_q|z_q|\|f_q\|_\sharp<\varrho\}$ and write $\Pi_{\ell''}(z)$ for the product over occupied cells of the seminorms of $e^{\mathcal{O}_K(\sum z_qf_q)}$ in those cells, with each sum restricted to that cell. Equation eq. 5.xxviii gives a holomorphic $\mathcal{E}$ with $\mathcal{E}(0)=0$ and

$$|e^{-\mathcal{E}(z)}|\Pi_{\ell''}(z)\le\mathsf{Q}(\mathsf{l};R,D)^n,\qquad\mathsf{Q}(\mathsf{l};R,D)=\exp[C_{\mathrm{v}}\mathsf{l}^3+\Phi_1(R,D;\mathsf{l})]$$

on the stated polydisc and all admitted $\ell''\ge 2\mathsf{l}$. The bound is uniform in the cutoff, coupling datum, side, board offset and position.

Step 8: centred polynomials in one cell. Let $(K,\beta)\in\mathfrak{d}(g)$, (i,o) a board with the side l, $R,D,\mathsf{d}>0$, x_0 a point at the distance $\ge R+D$ from the walls and $\square$ the cell that holds it, $N\ge 1$, $X=(f_1,\dots,f_N)$ functions of Lip_c with supports in $\mathbb{B}(x_0,R)$ at pairwise distances $\ge\mathsf{d}$, and $\ell,\ell'\in\mathfrak{L}(g)$ with $\ell'\ge 2\mathsf{l}$ and $\ell\ge\max\{2\mathsf{l},16N(R+1),8R+8\}$. Then $X_K^{\mu_{K,\ell,\beta}}$ is a function of the inner bonds of $\square$, and

$$\big\|X_K^{\mu_{K,\ell,\beta}}\big\|_{\mathrm{ch},\ell'}\le\mathsf{P}_N\prod_{q=1}^N\|f_q\|_\sharp,\qquad\mathsf{P}_N:=\sum_{j=0}^N\binom{N}{j}\mathsf{k}_{N-j}\,\mathsf{g}_j,$$

where $\mathsf{g}_0 := 1$, $\mathsf{g}_j := \mathsf{Q}(\mathsf{l}; R, D)^N (N/\varrho)^j$ for $j \ge 1$, $\mathsf{t}_m := \sum_{\pi \in \mathrm{Part}_{\ge 2}(m)} \prod_{I \in \pi} C^g_{|I|}(\mathsf{d}; R)$, $\mathsf{k}_0 := 1$ and $\mathsf{k}_m := \mathsf{t}_m + \sum_{j=1}^m \binom{m}{j} \mathsf{k}_{m-j} \mathsf{g}_j$: a number that depends on none of K, β, ℓ, ℓ', the offset and the position. If some f_q is 0, then $\mathcal{O}_K(f_q) = 0$, the centred product $X_K^{\mu_{K,\ell,\beta}}$ is 0 and $\|f_q\|_\sharp = 0$, so that the two sides of the claim are 0; let now no f_q be 0. Let $\mathsf{X}_q := \mathcal{O}_K(f_q)$, real bounded continuous functions of the inner bonds of $\square$, $m_q := \mu_{K,\ell,\beta}(\mathsf{X}_q)$, $\mathcal{E}$ the function of Step 7 for X and $\mathsf{E}_z := \exp[\sum_q z_q \mathsf{X}_q - \mathcal{E}(z)]$; as one cell is occupied, homogeneity gives $\|\mathsf{E}_z\|_{\mathrm{ch},\ell''} \le \mathsf{Q}(\mathsf{l}; R, D)^N$ at $\ell'' = \ell$ and at $\ell'' = \ell'$. For $B \subset \{1, \dots, N\}$ let $\mathsf{G}_B := \partial_B \mathsf{E}_z|_{z=0}$, a polynomial in the X_q. Cauchy's formula on the polydisc of the radii $r_q < \varrho/(N\|f_q\|_\sharp)$, which lies in $\mathbb{D}_\varrho(X)$, writes G_B as a limit in the sup norm of Riemann sums of values of E_z; the seminorm being subadditive, homogeneous and below the sup norm, $\|\mathsf{G}_B\|_{\mathrm{ch},\ell''} \le \mathsf{g}_{|B|} \prod_{q \in B} \|f_q\|_\sharp$. From $e^{\sum z_q(\mathsf{X}_q - m_q)} = e^{\mathcal{E}(z) - \sum z_q m_q} \mathsf{E}_z$ and Leibniz's rule, $\prod_{q \in A}(\mathsf{X}_q - m_q) = \sum_{B \subset A} \kappa_{A \setminus B} \mathsf{G}_B$ with $\kappa_C := \partial_C (e^{\mathcal{E}(z) - \sum z_q m_q})(0)$. Taking the mean under $\mu_{K,\ell,\beta}$, the left side is $\sigma_{\mu_{K,\ell,\beta}}(X_A)$, a sum over $\mathrm{Part}_{\ge 2}(A)$ of products of torus cumulants; take $y \in a_K \mathbb{Z}^3$ with $|y - x_0|_\infty \le 1$: as $\mu_{K,\ell,\beta}$ is invariant under the translation by $-y$ (Step 1), each of these cumulants equals the cumulant of the functions translated by $-y$, which are supported in $[-R-1, R+1]^3$, a subset of $(-\frac{\ell}{4}, \frac{\ell}{4})^3$ as $\ell \ge 8R + 8$, and Proposition YM3_I:91.13 bounds it; so $|\sigma_{\mu_{K,\ell,\beta}}(X_A)| \le \mathsf{t}_{|A|} \prod_{q \in A} \|f_q\|_\sharp$, and induction on $|A|$ with $|\mu_{K,\ell,\beta}(\mathsf{G}_B)| \le \|\mathsf{G}_B\|_{\mathrm{ch},\ell}$ gives $|\kappa_A| \le \mathsf{k}_{|A|} \prod_{q \in A} \|f_q\|_\sharp$. The identity at $A = \{1, \dots, N\}$ and the bounds at $\ell'' = \ell'$ give the claim.

Step 9: a bound table on the diamonds. For $n \ge 2$ and $R > 0$ let $h(n, R)$ be the least power of L that is $\ge (n+1)\,\mathsf{l}(n, R)$, and put

$$Q^\diamond(n, R) := e^{\frac{n}{2} h(n,R)^3 C_{\mathrm{v}}} Q^\natural(n, R)^n, \qquad C^\diamond_n(\mathsf{d}; R) := \Big(\frac{n}{\varrho}\Big)^n \Big[\log 2 \,(2Q^\diamond(n, R) + 2)^n + \mathsf{U}(R, \mathsf{d})\Big],$$

$$Q^\diamond_1 := e^{\frac{1}{2} L^6 C_{\mathrm{v}}} \mathsf{Q}_1(L), \qquad C^\diamond_1(R) := (2R+3)^3 (1 + 40\sqrt{3}M)\, 2 \log 2\, (2Q^\diamond_1 + 2)/\varrho.$$

There are numbers $\ell_\diamond(n, R)$ such that for every $(K, \beta) \in \mathfrak{d}(g)$, every $\ell \in \mathfrak{L}$ with $\ell/L \in \mathfrak{L}(g)$ and $\ell \ge \ell_\diamond(n, R)$, and all functions $f_1, \dots, f_n$ of Lip_c of the type $(\mathsf{d}; R)$ with supports in $(-\frac{\ell}{8}, \frac{\ell}{8})^3$: $|S^T_n[\mu^\diamond_{K,\ell,\beta}](f_1, \dots, f_n)| \le C^\diamond_n(\mathsf{d}; R) \prod_q \|f_q\|_\sharp$ for $n \ge 2$, and $|\delta_{\mu^\diamond_{K,\ell,\beta}, \mu_{K,\ell,\beta}}(f_1)| \le C^\diamond_1(R) \|f_1\|_\sharp$. For $n \ge 2$, let $y_1, \dots, y_n$ be the centres of the balls and $\mathsf{l} = \mathsf{l}(n, R) \ge 8n(R+1)$. Fix a direction μ. The walls $x_\mu \in o_\mu + \mathsf{l}(\mathbb{Z} + \frac{1}{2})$ stay at the distance $\ge 2R+1$ from all the y_q unless o_μ falls, modulo l, within less than $2R+1$ of one of the n points $y_{q,\mu} - \frac{1}{2}$; on the circle $\mathbb{R}/\mathsf{l}\mathbb{Z}$ this excluded set is a union of n intervals of a total measure $\le n(4R+2) < \mathsf{l} - 4n$, so its complement consists of no more than n closed arcs whose measures add up to more than $4n$; one of these arcs is longer than 4, hence than a_K, and so meets $a_K\mathbb{Z}$. With o_μ taken there for each μ, every y_q lies at the distance $\ge 2R+1$ from the walls of the board (i, o) of the side l, and each support lies in the corresponding cell interior. Take besides a compatible diagonal board with $h = h(n, R)$ none of whose walls meets the interior of an occupied cell, which exists as $h/\mathsf{l} \ge n+1$ and each occupied cell excludes one class of c_r modulo h for each r; so each occupied cell lies in one diagonal cell, with $s_{\mathbf{k}} = 1$. For $\ell \ge 4h + 8\mathsf{l}$ no two diagonal cells that hold occupied cells are congruent modulo $\Gamma^\diamond_\ell$, as a diagonal cell has the sup-diameter h and $\mathrm{sys}(\Gamma^\diamond_\ell) = \frac{\ell}{2}$. With $w = \sum_q z_q f_q$, Step 4 over the at most n diagonal cells, Step 6 in each of them (after an element of W that carries it to C) and Step 7 at $\ell' = \ell/L \ge 2\mathsf{l}$ give $|\mu^\diamond_{K,\ell,\beta}(e^{\mathcal{O}_K(w)})| \le e^{\frac{n}{2} h^3 C_{\mathrm{v}}} \Pi_{\ell/L}(z)$; the normaliser of Proposition YM3_I:91.12(2) bounds $\Pi_{\ell/L}(z)$ by $|e^{\mathcal{E}(z)}| Q^\natural(n, R)^n$ on $\mathbb{D}_\varrho(\underline{f})$ and bounds $\partial_{z_1} \cdots \partial_{z_n} \mathcal{E}(0)$ by $(n/\varrho)^n \prod \|f_q\|_\sharp\, \mathsf{U}(R, \mathsf{d})$, as U does not increase in its second argument; and Proposition YM3_I:91.11 gives the bound, with $\ell_\diamond(n, R) := \max\{4h + 8\mathsf{l}, 2L\mathsf{l}\}$. For $n = 1$: write $f = \sum_{v \in \mathbb{Z}^3} f\chi(\cdot - v)$ with $\chi(x) = \prod_\mu \max\{0, 1 - |x_\mu|\}$, at most $(2R+3)^3$ terms, each of the norm $\le (1 + 40\sqrt{3}M)\|f\|_\sharp$; for each v take the board of the side L with the offset v, which has v as the centre of a cell at the distance $\frac{L}{2} \ge 3$ from its walls since $L \ge 13$, and the compatible diagonal board $h = L^2$, $c_r = \mathsf{n}_r \cdot v + L$ ($\mathsf{n}_1 = \mathsf{n}_+$, $\mathsf{n}_2 = \mathsf{n}_-$, $\mathsf{n}_3 = e_2$), none of whose walls passes through v; Proposition YM3_I:91.12(1) gives a normaliser with the bound $\mathsf{Q}_1(L)$ on all sides of

$\mathfrak{L}(g)$, Steps 4, 6, 7 give $|e^{-\mathcal{E}}\mu^\diamond(e^{\mathcal{O}_K(zf_v)})| \le Q_1^\diamond$ for $\ell \ge 2L^3$, the cube chessboard bound gives $|e^{-\mathcal{E}}\mu_{K,\ell,\beta}(e^{\mathcal{O}_K(zf_v)})| \le \mathsf{Q}_1(L)$, and the difference bound of Proposition YM3_I:91.11 at $n=1$ and the sum over v give the bound.

Step 10: columns. Let $F = \sum_i \zeta_i X_i \in \mathcal{D}$, $\rho(F)$ the larger of 1 and the largest $|x|$ on the supports of its entries, and l_F the least power of L with $\mathsf{l}_F \ge 2\rho(F)+2$; a power h of L is *admitted* for F if $h \ge L\mathsf{l}_F$. Then $F \in \mathbb{C}\mathfrak{S}(\{0 < x_1 + x_3 < h\})$. Let $(K_j, \ell_j, \beta_j)_j$ be a sequence of tori at g, $\mu_j := \mu_{K_j,\ell_j,\beta_j}$, $\mu_j^\diamond := \mu^\diamond_{K_j,\ell_j,\beta_j}$, and $F_j := \sum_i \zeta_i (X_i)^{\mu_j}_{K_j}$. There is a number $\mathsf{q}_h(F)$, which depends on F, h, g and the constants alone, such that for all large j, with the diagonal board $(h; 0, -\mathsf{l}_F, -\mathsf{l}_F)$ on $\Gamma^\diamond_{\ell_j}$, F_j is a function of the bonds of C and $\|F_j\|_{\mathrm{col},h} \le q_\diamond(F_j) \le \mathsf{q}_h(F)$. Indeed, this diagonal board is compatible with the board $(i,0)$ of the side l_F, and the one wall that meets the interior of the cell $[-\frac{\mathsf{l}_F}{2}, \frac{\mathsf{l}_F}{2}]^3$ is the plane $\{x_1 + x_3 = 0\}$, so that $\mathsf{W}_0 = \{1, \Theta_\diamond\}$; the plaquettes at which an entry does not vanish lie in that cell and on the upper side of the plane for all sufficiently small a_K, by compact separation from the plane, so F_j is a function of the bonds of $\square_0 \cap \mathsf{C}$. Step 4, Step 6 with $\mathcal{K}_0 = \{0\}$ and $\ell' = \ell_j/L$, and Step 7 give $q_\diamond(F_j) \le e^{\frac12 h^3 C_{\mathrm{v}}} \|H_j\|^{1/2}_{\mathrm{ch},\ell_j/L}$ with $H_j = F_j \overline{\Theta_\diamond F_j} = \sum_{i,k} \zeta_i \bar\zeta_k \, (X_i \cdot \Theta_\diamond X_k)^{\mu_j}_{K_j}$, by the invariance of μ_j under $\Theta_\diamond$; and Step 8 at $x_0 = 0$, $R = \rho(F)$, $D = 1$, d the least of 1 and of the separations of the tuples $X_i \cdot \Theta_\diamond X_k$, bounds each term in which $X_i \cdot \Theta_\diamond X_k$ has at least one entry; a term with $X_i = X_k = \varnothing$ is the constant 1, of seminorm 1. So $\mathsf{q}_h(F) := e^{\frac12 h^3 C_{\mathrm{v}}} [\sum_{i,k} |\zeta_i \zeta_k| \, \mathsf{P}_{N_{ik}} \|X_i\|_\sharp \|X_k\|_\sharp]^{1/2}$ serves, N_{ik} the length of $X_i \cdot \Theta_\diamond X_k$, $\mathsf{P}_0 := 1$, and $\|X\|_\sharp$ the product of the norms of the entries, 1 for the empty tuple. Moreover, in the setting of Step 3 at $(\mathsf{n}_+, 0, h)$ on $\Gamma^\diamond_\ell$, for a power p of L with $2N' := \ell/\mathsf{p}h$ an even integer, a function f of the bonds of $\mathbb{S}_1$, functions G, G' of the bonds of $\{0 \le x_1 + x_3 \le \mathsf{p}h\}$ and $k = \mathsf{p}k'$ with $2k' + 2 \le 2N'$,

$$\Big| \mu^\diamond \Big(\overline{\Theta_\diamond G'} \prod_{i=1}^{2k} f^{[i]} \, \tau_{kh\mathsf{n}_+} G \Big) \Big| \le \|G'\|_{\mathrm{col},\mathsf{p}h} \, \|G\|_{\mathrm{col},\mathsf{p}h} \, \|f\|^{2k}_{\mathrm{col},h},$$

the column seminorms taken with the walls at the heights $\mathsf{p}h\mathbb{Z}$ and $h\mathbb{Z}$: the block $\mathsf{b} := \prod_{i \le \mathsf{p}} f^{[i]}$ is a function of the first slab of the height $\mathsf{p}h$, its copies at the height $\mathsf{p}h$ are the products of p consecutive copies of f, so $\|\mathsf{b}\|_{\mathrm{col},\mathsf{p}h} = \|f\|^{\mathsf{p}}_{\mathrm{col},h}$, and Step 3 at the height $\mathsf{p}h$ with the factors $\mathsf{b}^{[i'']}$, $i'' \le 2k'$, $G^{[2k'+1]} = \tau_{kh\mathsf{n}_+} G$, $G'^{[2N']} = \overline{\Theta_\diamond G'}$, and 1 in the other slabs, gives the bound.

Step 11: limits along the diamonds. Let now $(K_j, \ell_j, \beta_j)_j$ be the defining sequence of the hypothesis. Pass to a subsequence along which $E_j := E(K_j, \ell_j, \beta_j) \to E_\infty := \liminf_j E(K_j, \ell_j, \beta_j) \in [0, \infty)$; it is again a defining sequence of tori of S. With $O_j := (-\frac{\ell_j}{8}, \frac{\ell_j}{8})^3$, the forms $\Phi_n^{(j)} := S_n^T[\mu_j^\diamond]$ on $\mathfrak{S}_n(O_j)$, $n \ge 2$, and $\Phi_1^{(j)} := \delta_{\mu_j^\diamond, \mu_j}$ make a symmetric sequence of forms with the table $\underline{C}^\diamond = (C_n^\diamond)_{n \ge 1}$ from 1 on, by Step 9, the indices being those from which $\ell_j \ge \ell_\diamond(n, R)$ and $\ell_j/L \in \mathfrak{L}(g)$; by Proposition YM3_I:90.1 a further subsequence converges to a family $S^\diamond$ with the same table $\underline{C}^\diamond$, and along it $\sigma'(X) := \lim_j \sigma_{\mu_j^\diamond, \mu_j}(X) = \sum_{\pi \in \mathrm{Part}(n)} \prod_{I \in \pi} S^\diamond_{|I|}(X_I)$ for every separated tuple X, by the formula of the setting. We fix this subsequence. Then: (P1) σ' and $\sigma := \sigma_S$ are symmetric functionals on separated tuples, real on tuples, invariant under all translations of $\mathbb{R}^3$, under $\Theta_\diamond$ and under Θ_2; for σ this is Proposition YM3_I:92.3, since $\Theta_\diamond, \Theta_2 \in W_3$. For σ' use the finite-lattice invariance under those reflections and lattice translations from Step 1. To pass a translation $v \in \mathbb{R}^3$ to the limit, approximate it by $v_j \in a_{K_j}\mathbb{Z}^3$. The translated tests differ by $o(1)$ in C^1, have supports in fixed compact sets, and remain separated. The table and multilinearity bound each resulting difference by $o(1)$. For $\eta \in C^1_c$ with $\int \eta = 1$, use both orders of the convolution $f * \eta = \eta * f$. The Riemann sums of translates converge in C^1 with supports in a fixed ball. Translation invariance and the first-order table then give $S_1^\diamond(f) = S_1^\diamond(\eta) \int f$. Its scalar coefficient is real. The inverse moment–cumulant formula identifies all the connected functions of σ' with $S_n^\diamond$. (P2) σ' is reflection positive at $\Sigma_\diamond$ and at $(e_2, 0)$, and σ at $(e_2, 0)$: for $F \in \mathcal{D}$ and large j, F_j is a function of the bonds of $\{0 \le x_1 + x_3 \le \frac{\ell_j}{2}\}$, $\overline{\Theta_\diamond F_j} F_j = (\Theta_\diamond \bar F \cdot F)^{\mu_j}_{K_j}$, and $\mu_j^\diamond(\overline{\Theta_\diamond F_j} F_j) \ge 0$ by Proposition YM3_I:91.4 at

the pair of the heights 0 and $\frac{\ell_j}{2}$; so $\sigma'(\Theta_\diamond \bar F\cdot F)\ge 0$, and the same at the site mirror $\Sigma(e_2,0)$ and the pair of $x_2=0$ and $x_2=\frac{\ell_j}{2}$; for σ it is Proposition YM3_I:92.4. (P3) $|\sigma(\Theta_\diamond\bar F\cdot F)|\le e^{E_\infty}\sigma'(\Theta_\diamond\bar F\cdot F)$ for $F\in\mathcal D$: Step 2 at $G=F=F_j$ gives $|\mu_j(\overline{\Theta_\diamond F_j}F_j)|\le e^{E_j}\mu_j^\diamond(\overline{\Theta_\diamond F_j}F_j)$; on the left, $\mu_j((\Theta_\diamond\bar F\cdot F)^{\mu_j}_{K_j})$ is the sum over $\mathrm{Part}_{\ge 2}$ of products of torus cumulants by the moment–cumulant formula, which tends to $\sigma_S(\Theta_\diamond\bar F\cdot F)$ along a defining sequence; the right side tends to $e^{E_\infty}\sigma'(\Theta_\diamond\bar F\cdot F)$. (P4) By (P1), (P2), Proposition YM3_I:94.1 applies to σ' at the unit vectors $u_\diamond$ and e_2: $\mathcal H:=\mathcal H(\sigma',u_\diamond)$, the completion of $\mathcal D$ modulo the null vectors for $\langle F,F'\rangle=\sigma'(\Theta_\diamond\bar F\cdot F')$, carries the semigroup $\mathcal T_t[F]=[\tau_{tu_\diamond}F]$ of self-adjoint operators, $0\le\mathcal T_t\le 1$, the unitary group $\mathcal U(v)[F]=[\tau_vF]$, $v\perp u_\diamond$, commuting with it and fixing $\Omega=[\varnothing]$, and $\mathcal T_t\to P_0$ strongly as $t\to\infty$, P_0 the projection onto $\ker H$; and $\mathcal H^{(2)}:=\mathcal H(\sigma',e_2)$ carries the like objects $\mathcal T_t^{(2)}$, $\mathcal U^{(2)}(v)$, $P_0^{(2)}$. Note $[\mathsf s_yF]=\mathcal T_{y/\sqrt2}[F]$ for $y\ge 0$, and the classes do not depend on the order of the entries, σ' being symmetric. (P5) For $X,G\in\mathcal D$, h admitted for X and a power $\mathsf p$ of L with $\mathsf ph$ admitted for G, and with the copies $X^{[1]}:=X$, $X^{[i+1]}:=\overline{\Theta_{(u_\diamond,ih/\sqrt2)}X^{[i]}}$ in the slabs of the height h: for all $k\in\mathsf p\mathbb N$,

$$\big|\sigma'(\Theta_\diamond\bar G\cdot X^{[1]}\cdots X^{[2k]}\cdot\mathsf s_{2kh}G)\big|\le \mathsf q_{\mathsf ph}(G)^2\,\mathsf q_h(X)^{2k}.$$

At large j the heights ih lie in $a_{K_j}\mathbb Z$, so the copies are lattice copies and $\mathsf s_{2kh}=\tau_{kh\mathsf n_+}$ is a lattice translation; the column bound of Step 10 and the bounds of Step 10 on the column seminorms give the inequality at j, and the left side converges.

Step 12: a bound on columns gives a bounded multiplication. Let $X\in\mathcal D$ have its entries supported in $\{0<x_1+x_3<h\}$, and put $X^\dagger:=\mathsf s_h\Theta_\diamond\bar X$, the conjugated mirror image of X in the plane of the height $\frac h2$, which lies in the same slab; then $X^{[2]}=\mathsf s_hX^\dagger$ and $X^{[i+2]}=\mathsf s_{2h}X^{[i]}$. For $\varsigma\in\{\sigma,\sigma'\}$ and $G,G'\in\mathcal D$, the invariance of ς under $\mathsf s_{-h}$, the relation $\Theta_\diamond\mathsf s_y=\mathsf s_{-y}\Theta_\diamond$ and the symmetry give

$$\varsigma(\Theta_\diamond\bar G'\cdot X\cdot\mathsf s_hG)=\varsigma(\Theta_\diamond\overline{(X^\dagger\cdot\mathsf s_hG')}\cdot G),$$

without any positivity. If besides h is admitted for X, then $G\mapsto X\cdot\mathsf s_hG$ and $G\mapsto X^\dagger\cdot\mathsf s_hG$ pass to the classes and extend to bounded operators $A_{X,h}$ and $A_{X^\dagger,h}=A_{X,h}^*$ of $\mathcal H$ of norms $\le\mathsf q_h(X)$. Indeed, with $\mathsf C_1G:=X^\dagger\cdot\mathsf s_hG$, $\mathsf C_2G:=X\cdot\mathsf s_hG$ and $\mathsf B:=\mathsf C_2\mathsf C_1$, the identity for σ' says $\langle G',\mathsf C_2G\rangle=\langle\mathsf C_1G',G\rangle$, so $\langle G',\mathsf BG\rangle=\langle\mathsf C_1G',\mathsf C_1G\rangle=\langle\mathsf BG',G\rangle$ and $\mathsf B^kG=X^{[1]}\cdots X^{[2k]}\cdot\mathsf s_{2kh}G$; the numbers $a_k:=\langle G,\mathsf B^kG\rangle$ satisfy $a_{2i}=\|\mathsf B^iG\|^2$, $a_{2i+1}=\|\mathsf C_1\mathsf B^iG\|^2$ and, by Schwarz's inequality, $a_k^2\le a_{k-1}a_{k+1}$; hence $a_1\le a_0^{1-1/k}a_k^{1/k}$, and (P5) along $k\in\mathsf p\mathbb N$ gives $\|\mathsf C_1G\|^2=a_1\le\mathsf q_h(X)^2\|G\|^2$; then $|\langle G',\mathsf C_2G\rangle|=|\langle\mathsf C_1G',G\rangle|\le\mathsf q_h(X)\|G'\|\|G\|$. Call a pair (B,y), $y>0$, $B\in\mathbb C\mathfrak S(\{0<x_1+x_3<y\})$, *bounded* if there is a bounded operator $A_{B,y}$ of $\mathcal H$ with $A_{B,y}[G]=[B\cdot\mathsf s_yG]$ for $G\in\mathcal D$; then $A_{B,y}\Omega=[B]$; $(B\cdot\mathsf s_yB',y+y')$ is bounded with the operator $A_{B,y}A_{B',y'}$; and for $v\perp u_\diamond$, (τ_vB,y) is bounded with the operator $\mathcal U(v)A_{B,y}\mathcal U(v)^*$.

Step 13: the operator $\mathsf R$. The form $\mathfrak b(G,F):=\sigma(\Theta_\diamond\bar G\cdot F)$ on $\mathcal D$ is Hermitian, by the reality, the invariance under $\Theta_\diamond$ and the symmetry of σ (P1), and $|\mathfrak b(F,F)|\le e^{E_\infty}\|F\|^2$ by (P3); polarisation of the Hermitian form gives $|\mathfrak b(G,F)|\le e^{E_\infty}\|G\|\|F\|$. So there is exactly one bounded self-adjoint operator $\mathsf R$ of $\mathcal H$ with $\sigma(\Theta_\diamond\bar G\cdot F)=\langle[G],\mathsf R[F]\rangle$, $\|\mathsf R\|\le e^{E_\infty}$, and in particular $\sigma(Z)=\langle\Omega,\mathsf R[Z]\rangle$ for $Z\in\mathcal D$. The invariance of σ under the translations, with $\tau_{-tu_\diamond}\Theta_\diamond=\Theta_\diamond\tau_{tu_\diamond}$ and $\tau_v\Theta_\diamond=\Theta_\diamond\tau_v$ for $v\perp u_\diamond$, shows that $\mathsf R$ commutes with every $\mathcal T_t$ and every $\mathcal U(v)$; and the identity of Step 12 for $\varsigma=\sigma$ shows $\langle[G'],\mathsf RA_{X,h}[G]\rangle=\langle A_{X^\dagger,h}[G'],\mathsf R[G]\rangle$, so that $\mathsf R$ commutes with $A_{X,h}$ and with $A_{X^\dagger,h}$ for every $X\in\mathcal D$ and every h admitted for X. It remains to show $\mathsf R\ge 0$.

Step 14: plateaux in the plane of $u_\diamond$ *and* e_2. Fix $t_1,s_1>0$; $\mathsf T:=\mathcal T_{t_1}$, $U:=\mathcal U(s_1e_2)$, $\mathsf E_U$ the projection onto $\ker(U-1)$, $P':=P_0\mathsf E_U$, and $x(k,m):=kt_1u_\diamond+ms_1e_2$. As P_0 is the strong limit of the $\mathcal T_t$, it commutes with U and $\mathsf R$; by von Neumann's ergodic theorem $\mathsf E_U$ commutes with P_0, the $\mathcal T_t$ and $\mathsf R$; so P' is a projection with $P'\mathcal T_t=P'=P'U$, $P'\Omega=\Omega$, $P'\mathsf R=\mathsf RP'$. *Claim:* for $A,B\in\mathbb C\mathfrak S(\mathbb R^3)$ the numbers $\sigma'(A\cdot\tau_{x(k,m)}B)$ converge as k and m tend to $\pm\infty$ together, with one and the same limit $c(A,B)$ in the four quadrants; $c(\Theta_\diamond\bar G,F)=\langle[G],P'[F]\rangle$ for $F,G\in\mathcal D$; $c(\tau_zA,\tau_{z'}B)=c(A,B)$

for $z, z' \in \mathbb{R}u_\diamond + s_1\mathbb{Z}e_2$; and $\mathcal{T}_t U^m \to P'$ weakly as $t \to \infty$ and $|m| \to \infty$ together. In the first picture, after a translation along $u_\diamond$, $\sigma'(A \cdot \tau_{x(k,m)}B) = \langle \chi, \mathsf{T}^{k-k_0}U^m\chi'\rangle$ for $k \geq k_0$ and all m, with classes $\chi, \chi' \in \mathcal{H}$, and it differs from $h_1(m) := \langle \chi, P_0U^m\chi'\rangle$ by at most $\epsilon_1(k-k_0) \to 0$, uniformly in m; in the second picture, in $\mathcal{H}^{(2)}$, it differs from a function $h_2(k)$ by at most $\epsilon_2(m-m_0) \to 0$, uniformly in k; the same for $k \to -\infty$ or $m \to -\infty$ with the roles of A and B exchanged, as $\sigma'(A \cdot \tau_x B) = \sigma'(B \cdot \tau_{-x}A)$. These approximations make $h_1(m)$ and $h_2(k)$ Cauchy sequences at $\pm\infty$ with equal limits in each quadrant, and the means $\frac{1}{N}\sum_{m\leq N} h_1(\pm m)$, which tend to $\langle \chi, P_0\mathsf{E}_U\chi'\rangle$ by von Neumann's theorem, together with the like means of h_2 and of the second functions, identify the four limits. At $A = \Theta_\diamond\bar{G}$, $B = F$ this gives $h_1(m) = \langle [G], P_0U^m[F]\rangle \to \langle [G], P'[F]\rangle$ as $|m| \to \infty$, hence the weak convergence. The invariance of c follows from $P'\mathcal{T}_t = P'$.

Step 15: the half turn acts trivially at infinity. Let $\rho := \Theta_\diamond\Theta_2$, the linear map with $\rho u_\diamond = -u_\diamond$, $\rho e_2 = -e_2$, $\rho\mathsf{n}_- = \mathsf{n}_-$; σ and σ' are invariant under it. For $Y \in \mathbb{C}\mathfrak{S}(\mathbb{R}^3)$ put $\pi(Y) := P'[\mathsf{s}_y Y]$ with y so large that $\mathsf{s}_y Y \in \mathcal{D}$; it does not depend on y, it is linear, $\pi(\tau_x Y) = \pi(Y)$ for $x \in \mathbb{R}u_\diamond + s_1\mathbb{Z}e_2$, the vectors $\pi(Y)$ are dense in $\mathcal{K} := P'\mathcal{H}$, and $\langle \pi(G), \pi(F)\rangle = c(\Theta_\diamond\bar{G}, F)$ by Step 14. *Claim:* $\pi(\rho Y) = \pi(Y)$. The invariance of σ' under ρ and $\rho\tau_x = \tau_{-x}\rho$ for x in that plane give $\langle \pi(\rho G), \pi(\rho F)\rangle = \langle \pi(G), \pi(F)\rangle$, the four limits being one; so $\mathsf{V}\pi(Y) := \pi(\rho Y)$ extends to a unitary operator of $\mathcal{K}$ with $\mathsf{V}^2 = 1$, $\mathsf{V} = \mathsf{V}^*$. Further $\langle \pi(Y), \mathsf{V}\pi(Y)\rangle = c(\Theta_\diamond\bar{Y}, \rho Y) = c(\Theta_2\bar{Y}, Y)$, by the invariance under $\Theta_\diamond$ and Θ_2. For large m, $V := \tau_{ms_1e_2}Y \in \mathbb{C}\mathfrak{S}(\{x_2 > 0\})$, and $\varphi_m(k) := \sigma'(\Theta_2\bar{V} \cdot \tau_{kt_1u_\diamond}V) = \sigma'(\Theta_2\bar{Y} \cdot \tau_{x(k,2m)}Y)$; as Θ_2 commutes with $\tau_{tu_\diamond}$, $\sum_{a,b\leq N}\varphi_m(b-a) = \sigma'(\Theta_2\bar{V}_N \cdot V_N) \geq 0$ with $V_N := \sum_{a\leq N}\tau_{at_1u_\diamond}V$ (P2); $\varphi_m(k)$ tends to $h_1(2m)$ and to $h_1^-(2m)$ as $k \to \pm\infty$, where h_1 is the function of the first picture of Step 14 for the pair $(A, B) = (\Theta_2\bar{Y}, Y)$ and $h_1^-(m) := \langle \chi^-, P_0U^{-m}\chi'^-\rangle$ is built, as h_1 is, from the classes χ^-, χ'^- that the first picture gives to the pair taken in the other order, $(Y, \Theta_2\bar{Y})$, as $\sigma'(A \cdot \tau_{x(k,m)}B) = \sigma'(B \cdot \tau_{x(-k,-m)}A)$; so the means give $\frac{1}{2}(h_1(2m) + h_1^-(2m)) \geq 0$, and as $m \to \infty$, both terms tending to $c(\Theta_2\bar{Y}, Y)$ by Step 14, as limits in the quadrants of $k, m \to +\infty$ and of $k \to -\infty$, $m \to +\infty$, $c(\Theta_2\bar{Y}, Y) \geq 0$. So $\mathsf{V} \geq 0$ on a dense subspace, hence $\mathsf{V} \geq 0$; a self-adjoint unitary $\mathsf{V} \geq 0$ is 1.

Step 16: at infinity a multiplication reads π alone. Let (B, y_B) be a bounded pair, $H \in \mathcal{D}$, $y > 0$ with $H^\rho := \mathsf{s}_y\rho H \in \mathcal{D}$ and h' admitted for H^ρ. Then $P'A_{B,y_B}\pi(H) = P'A_{H^\rho,h'}\pi(B)$. For $x = x(k,m)$, $k, m \to +\infty$: $[B \cdot \tau_x H] = A_{B,y_B}\mathcal{T}_{kt_1 - y_B/\sqrt{2}}U^m[H]$, which tends weakly to $A_{B,y_B}P'[H]$ by Step 14, so $P'[B \cdot \tau_x H] \to P'A_{B,y_B}\pi(H)$; on the other side, by Step 15 and the invariance of π under τ_x, $P'[B \cdot \tau_x H] = \pi(\rho B \cdot \tau_{-x}\rho H) = \pi(H^\rho \cdot \tau_x\mathsf{s}_y\rho B)$, and writing $\tau_x\mathsf{s}_y\rho B = \mathsf{s}_{h'}\tau_{t'u_\diamond}\tau_{ms_1e_2}B^\rho$ with $B^\rho := \mathsf{s}_{y'}\rho B \in \mathcal{D}$ and $t' \to \infty$, this is $P'A_{H^\rho,h'}\mathcal{T}_{t'}U^m[B^\rho] \to P'A_{H^\rho,h'}\pi(\rho B) = P'A_{H^\rho,h'}\pi(B)$. As the $\pi(H)$, $H \in \mathcal{D}$, are dense in $\mathcal{K}$, $P'A_{B,y_B}P'$ depends only on $\pi(B)$.

Step 17: $\mathsf{R} \geq 0$. Let $F \in \mathcal{D}$, h admitted for F, $\Phi := \Theta_\diamond\bar{F} \cdot F$; up to the order of the entries $\Theta_\diamond\bar{\Phi} = \Phi$, so $\Theta_2\bar{\Phi} = \rho\Phi$. Choose s_1 with $|x_2| < s_1$ on the supports of Φ. $\mathsf{A} := A^*_{F,h}A_{F,h} = A_{F^\dagger,h}A_{F,h} \geq 0$ is the operator of the bounded pair $(\mathsf{s}_h\Phi, 2h)$; with $F_2 := \Theta_2\bar{F} \in \mathcal{D}$, for which h is admitted, $\mathsf{A}^\rho := A^*_{F_2,h}A_{F_2,h} \geq 0$ is the operator of $(\mathsf{s}_h\rho\Phi, 2h)$. Put $\mathsf{M} := P'\mathsf{A}P' \geq 0$. (i) $P'\mathsf{A}^\rho P' = \mathsf{M}$, by Step 16, as $\pi(\mathsf{s}_h\rho\Phi) = \pi(\rho\Phi) = \pi(\Phi) = \pi(\mathsf{s}_h\Phi)$. (ii) R commutes with A, A^ρ, M (Steps 13, 14). (iii) For every real polynomial $p(\lambda) = \sum_{i\leq d} c_i\lambda^i$, $\langle p(\mathsf{M})\Omega, \mathsf{R}\,p(\mathsf{M})\Omega\rangle \geq 0$: let $\Phi_N := \frac{1}{N}\sum_{a\leq N}\tau_{as_1e_2}\mathsf{s}_h\Phi$, supported in $\{x_2 > 0\}$ at heights in $(0, 2h)$, $\mathsf{g} > 2h$, $W_0 := \varnothing$, $W_i := \mathsf{s}_\mathsf{g}\Phi_N \cdot \mathsf{s}_{2\mathsf{g}}\Phi_N \cdots \mathsf{s}_{i\mathsf{g}}\Phi_N$ for $i \geq 1$, $W := \sum_i c_i W_i \in \mathbb{C}\mathfrak{S}(\{x_2 > 0\})$ and $\varphi(k) := \sigma(\Theta_2\bar{W} \cdot \tau_{kt_1u_\diamond}W)$; by the positivity of σ at $(e_2, 0)$ (P2), $\sum_{a,b\leq N'}\varphi(b-a) \geq 0$ for all N', and $\varphi(-k) = \overline{\varphi(k)}$; so if $\varphi(k)$ converges as $k \to \infty$, the real part of the limit is ≥ 0. With $\mathsf{A}_N := \frac{1}{N}\sum_a U^a\mathsf{A}U^{-a}$, $\mathsf{A}^\rho_N := \frac{1}{N}\sum_a U^{-a}\mathsf{A}^\rho U^a$ and $\mathsf{T}_\circ := \mathcal{T}_{(\mathsf{g}-2h)/\sqrt{2}}$, Step 12 gives $(\mathsf{T}_\circ\mathsf{A}_N)^i[G] = [\mathsf{s}_{-2h}W_i \cdot \mathsf{s}_{i\mathsf{g}}G]$ and $(\mathsf{T}_\circ\mathsf{A}^\rho_N)^i[G] = [\mathsf{s}_{-2h}\Theta_2\bar{W}_i \cdot \mathsf{s}_{i\mathsf{g}}G]$, and so, with $t(k) := kt_1 - i\mathsf{g}/\sqrt{2} \geq 0$ and Step 13,

$$\sigma\big(\Theta_2\bar{W}_i \cdot \tau_{kt_1u_\diamond}W_{i'}\big) = \big\langle \Omega,\ \mathsf{R}\,(\mathsf{T}_\circ\mathsf{A}^\rho_N)^i\,\mathcal{T}_{t(k)}\,(\mathsf{T}_\circ\mathsf{A}_N)^{i'}\Omega\big\rangle.$$

As $k\to\infty$, $\mathcal{T}_{t(k)}\to P_0$ strongly, so $\varphi(k)$ converges; as $\mathsf{g}\to\infty$, $\mathsf{T}_\circ\to P_0$ strongly; as $N\to\infty$, for every vector y of $\mathcal{K}$, on which U acts as 1, $P_0\mathsf{A}_NP_0y\to\mathsf{E}_UP_0\mathsf{A}y=\mathsf{M}y$ and $P_0\mathsf{A}_N^\rho P_0y\to P'\mathsf{A}^\rho P'y=\mathsf{M}y$, by von Neumann's theorem and (i), and the operators being uniformly bounded, $(P_0\mathsf{A}_NP_0)^i\Omega\to\mathsf{M}^i\Omega$ and $(P_0\mathsf{A}_N^\rho P_0)^i\Omega\to\mathsf{M}^i\Omega$; all the operators are bounded, self-adjoint and commute with R, and $P_0\Omega=\Omega$; so $\langle p(\mathsf{M})\Omega,\mathsf{R}p(\mathsf{M})\Omega\rangle$ is a limit of numbers with real parts ≥0, and it is real. (iv) With real polynomials $p_r\to\sqrt{\lambda}$ uniformly on $[0,\|\mathsf{M}\|]$, (ii) and (iii) give $\langle\Omega,\mathsf{RM}\Omega\rangle=\lim_r\langle p_r(\mathsf{M})\Omega,\mathsf{R}p_r(\mathsf{M})\Omega\rangle\ge0$. Finally, by Step 12, (ii), $P'\Omega=\Omega$ and $P'\mathsf{R}=\mathsf{R}P'$,

$$\sigma_S(\Theta_\diamond\bar F\cdot F)=\langle[F],\mathsf{R}[F]\rangle=\langle A_{F,h}\Omega,\mathsf{R}A_{F,h}\Omega\rangle=\langle\Omega,\mathsf{RA}\Omega\rangle=\langle\Omega,\mathsf{RM}\Omega\rangle\ge0.$$

This is (a); as the classes $[F]$ are dense, also $0\le\mathsf{R}\le e^{E_\infty}$.

Step 18: every diagonal mirror. By (a), σ_S is reflection positive at the diagonal mirror $\Sigma_\diamond$; it is symmetric and invariant under all translations of $\mathbb{R}^3$ and under W_3 (P1); so translate the plane and apply a signed permutation of the coordinates to obtain (b). Proposition YM3_I:92.4 supplies the coordinate planes. □

18.11.1. *Two-loop pressure. Weak coupling: the group and the cube.* We identify SU(2) with the unit quaternions, $U=\operatorname{Sc}U+\mathrm{i}\operatorname{Vec}U\cdot\sigma$, σ the Pauli matrices, and put $a(U):=1-\frac12\operatorname{tr}U=\frac12|U-\mathbf{1}|^2$, so that $a_p=a(\partial U(p))$ and, as $|\operatorname{Vec}U|^2=a(2-a)$,

$$a(U)=\tfrac12|\operatorname{Vec}U|^2+\tfrac12a(U)^2.$$

The chart $q(u):=(1-|u|^2)^{1/2}+\mathrm{i}u\cdot\sigma$, $u\in\mathbb{R}^3$, $|u|<1$, covers $\{\operatorname{Sc}U>0\}$, where the Haar measure is $\sigma_{\mathrm{H}}(1-|u|^2)^{-1/2}d^3u$, $\sigma_{\mathrm{H}}=\frac{1}{2\pi^2}$; and $\mathsf{z}(\lambda):=\int e^{-\lambda a(U)}dU$ lies between $c\lambda^{-3/2}$ and $(2/\pi)^{1/2}\lambda^{-3/2}$ for $\lambda\ge1$. For an even $N\ge4$ let T_N be the periodic cube N^3, $Z_N:=Z(N^3)$, $P_N:=N^{-3}\log Z_N$ and $\delta_N:=\delta(N^3)$ (Section 17.2); $h_\mu(U)$ is the holonomy of the closed straight line through 0 in the direction μ, and $\mathsf{H}_r:=\{|h_\mu-\mathbf{1}|\le r,\ \mu=1,2,3\}$. On T_N let C^0,C^1,C^2 be the $\mathbb{R}^3$-valued functions on the sites, on the positively oriented bonds and on the plaquettes, with their ℓ^2 products, d and d^* the coboundary and its adjoint; $C^1=dC^0\oplus\mathsf{K}\oplus\mathsf{T}_0$ with the constant forms K, one vector $c_\mu\in\mathbb{R}^3$ for each direction, and $\mathsf{T}_0:=d^*C^2$; $\mathsf{T}:=\mathsf{K}\oplus\mathsf{T}_0=\ker d^*$. $Q:=d^*d$ on T_0 is $\ge4\sin^2\frac{\pi}{N}\ge16/N^2$; $G_N(x):=N^{-3}\sum_{k\ne0}e^{\mathrm{i}k\cdot x}/|\mathsf{k}|^2$, $|\mathsf{k}|^2:=\sum_\mu(2-2\cos k_\mu)$, $k\in\frac{2\pi}{N}(\mathbb{Z}/N)^3$; $H_N:=G_N*G_N$; $\mathsf{Cv}:=Q^{-1}$ on T_0 (and 0 on its complement), $\mathsf{Gr}:=(d^*d)^{-1}d^*$ with values in the functions of mean zero, and $\Pi:=dQ^{-1}d^*$, the orthogonal projection of C^2 onto dC^1. E_0, P_0 are the mean and the law of the centred Gaussian field on T_0 with the covariance $\beta^{-1}\mathsf{Cv}$, the density being proportional to $\exp[-\frac{\beta}{2}|du|_2^2]$; $L_\mu(u):=\sum_{x\in l_\mu}u_\mu(x)$ is the sum of a form along the axis l_μ; and

$$\mathsf{N}_0(N,\beta):=\sigma_{\mathrm{H}}^{2N^3+1}(2\pi/\beta)^{3(N^3-1)}\exp\bigl[-\tfrac32\Sigma_N\bigr],\qquad\Sigma_N:=\sum_{k\ne0}\log|\mathsf{k}|^2.$$

The chart and the two-loop number of the cube. For $\tilde u=c+u\in\mathsf{T}$ with $|\tilde u|_\infty<1$ let $q(\tilde u)$ be the configuration $b\mapsto q(\tilde u_b)$; $\mathsf{N}_p(\tilde u):=\operatorname{Vec}\partial q(\tilde u)(p)-(d\tilde u)_p$ is the nonlinear part of the plaquette, with its Taylor parts $\mathsf{N}^{(2)}$, $\mathsf{N}^{(3)}$ of the degrees 2 and 3; $\mathsf{B}(\tilde u):=Q^{-1}d^*\mathsf{N}(\tilde u)\in\mathsf{T}_0$; $\Phi_c(u):=u+\mathsf{B}(c+u)$; $\mathsf{X}:=D_u\mathsf{B}$ on T_0, with its parts $\mathsf{X}_1,\mathsf{X}_2$ of the degrees $1,2$ in $\tilde u$; $\mathsf{Y}_{\tilde u}:=\mathsf{Gr}(D_{\tilde u}-d)$ on the functions of mean zero, $(D_{\tilde u}\xi)_b:=c_b(d\xi)_b-\tilde u_b\times(\xi_x+\xi_y)$ at $b=(x,y)$, $c_b:=(1-|\tilde u_b|^2)^{1/2}$, with its linear part Y_1. With

$$F^{\mathrm{lead}}(\tilde u):=\tfrac{\beta}{2}|(1-\Pi)\mathsf{N}^{(2)}(\tilde u)|_2^2+\tfrac{\beta}{8}\sum_p|(d\tilde u)_p|^4+\operatorname{tr}\mathsf{X}_2-\tfrac12\operatorname{tr}\mathsf{X}_1^2+\tfrac12\operatorname{tr}\mathsf{Y}_1^2-\frac{|\tilde u|_2^2}{2N^3},$$

$m_1(N):=-\beta E_0F^{\mathrm{lead}}(u)$ is the *two-loop number* of the cube T_N: a number that depends on N alone, as the mean of a monomial of the degree $2j$ under E_0 is β^{-j} times a number of N, and $|m_1(N)|\le CN^3$ (Step 7). Gaussian integration by parts writes it as $m_1(N)/\beta=-E_0[\beta\mathsf{S}_4^{\mathrm{L}}+\Xi_2^{\mathrm{L}}]+\frac12E_0(\beta\mathsf{S}_3^{\mathrm{L}})^2$, with the

three forms of the Landau chart on the cube $\mathsf{S}_3^{\mathrm{L}} = \langle du, \mathsf{N}^{(2)}\rangle$, $\mathsf{S}_4^{\mathrm{L}} = \frac12|\mathsf{N}^{(2)}|_2^2 + \langle du, \mathsf{N}^{(3)}\rangle + \frac18\sum_p |du_p|^4$ and $\Xi_2^{\mathrm{L}} = \frac12 \operatorname{tr}\mathsf{Y}_1^2 - |u|_2^2/2N^3$: the vacuum diagrams with two loops of the cube in Landau's gauge.

The Dirichlet box and the lattice. $\Lambda_n = \{0,\dots,n\}^3$, its free bonds and forms, its partition function $Z_{n,\beta}$, the gauge parameter $\tau_{\mathrm{gf}} > 0$, the covariance $\mathsf{C}^{[n]}_{\tau_{\mathrm{gf}}}$, the kernels $\mathsf{G}, \mathsf{E}, \Pi, \mathsf{g}, \mathsf{D}$, the cubic and quartic vertices $\mathsf{S}_3 = \sum_p \mathsf{a}_p^{(3)}$ and V_2 (the quartic Wilson vertex, the Haar term, the ghost tadpole and the ghost loop) and the Gaussian family $\langle\cdot\rangle_0$ are those of Definitions 7.2, 7.3, 7.7 and 7.8; $d_1 := 3(n-1)^2(2n+1)$ is the number of the real coordinates of the forms in the axial gauge of Lemma 7.6, and

$$q_1^{[n]} := -\langle \mathsf{V}_2\rangle_0 + \tfrac12\langle \mathsf{S}_3; \mathsf{S}_3\rangle_0$$

at τ_{gf}. On $\mathbb{Z}^3$ let $G(y) := (2\pi)^{-3}\int_{[-\pi,\pi]^3} e^{ik\cdot y}\big(\sum_\mu(2-2\cos k_\mu)\big)^{-1}dk$ be the Green function of the lattice Laplacian, with the normalisation of the Fourier conventions of Definition 7.1(5), $G_0 := G(0)$, and

$$J_{\mathrm{gh}} := \sum_{y\in\mathbb{Z}^3}\big[G(y+e_1) - G(y-e_1)\big]^2 G(y),$$

$$J_3 := \sum_{y\in\mathbb{Z}^3}\big[G(y)^2 - G(y+e_2)G(y-e_2)\big]\big[2G(y) + G(y+e_1) + G(y-e_1)\big],$$

two absolutely convergent sums, their terms being $O(\langle y\rangle^{-5})$; we put $p_1 := 9G_0 - 36G_0^2 - 9J_{\mathrm{gh}} + 18J_3$, numerically $p_1 \approx 0.69788$. In this part C, c denote pure numbers, which may change from one occurrence to the next, and $\mathsf{y} := \log\beta$.

Detailed sketch BF (Two-loop pressure of the infinite lattice). Let $P(\beta)$ be the pressure of the lattice $\mathbb{Z}^3$ at the real coupling β, as in the setting of Section 17.2, and let p_1, $q_1^{[n]}$ and $m_1(N)$ be the numbers defined above.

(a) There are pure numbers C and β_* such that for every real $\beta \ge \beta_*$

$$\Big|P(\beta) + 3\log\beta - z_\infty - \frac{p_1}{\beta}\Big| \le C\,(\log\beta)^2\,\beta^{-1.02},$$

$z_\infty = \log\frac{2}{\pi} - \frac32 f_\Delta \approx -2.96167$ the number of Proposition 17.47(d); and $|P(\beta)+3\log\beta - z_\infty| \le C/\beta$ for $\beta \ge \beta_*$.

(b) For every $\tau_{\mathrm{gf}} > 0$, $\lim_{n\to\infty} n^{-3}q_1^{[n]} = p_1$; and $|N^{-3}m_1(N) - p_1| \le C\log N/N$ for every even $N \ge 4$.

■□

Proof sketch. The pressure is pinned between two periodic cubes of the side $N \approx \beta^{0.34}$; there N^3/β is a small positive power of β, so that factors β^C cost nothing against the term $m_1(N)/\beta$, while $N^2 \ll \beta$ allows one chart about the flat field. The window $\beta^{1/3} \ll N \ll \beta^{1/2}$ is not empty because the dimension is three.

Step 1: the Green function. There is a pure number C_{G} such that for all even N and $x \in T_N$, $|x|$ the distance on the torus and D, D' first differences, $|G_N(x)| \le C_{\mathrm{G}}/(1+|x|)$, $|DG_N(x)| \le C_{\mathrm{G}}/(1+|x|)^2$, $|DD'G_N(x)| \le C_{\mathrm{G}}/(1+|x|)^3$ and $|DD'H_N(x)| \le C_{\mathrm{G}}/(1+|x|)$. Indeed $G_N = \int_0^\infty (P_s - N^{-3})ds$ and $H_N = \int_0^\infty s(P_s - N^{-3})ds$, P_s the heat kernel of T_N, a product of three periodisations of the kernel $p_s(x) = \frac{1}{2\pi}\int e^{ikx - s(2-2\cos k)}dk$ on $\mathbb{Z}$; moving the contour to $k + i\eta$, $\eta = \operatorname{sgn}(x)\min\{1, |x|/2.2s\}$, gives $|D^j p_s(x)| \le C(1+s)^{-(1+j)/2}e^{-w(x,s)}$, $w = \min\{x^2/9s, |x|/4\}$, $j \le 2$; the periodisation keeps this for $|x| \le N/2$, $s \le N^2$, the nonzero modes decay like e^{-16s/N^2} beyond, and the integrals give the claims. Hence the kernels of Cv, $Q^{-1}d^*$, Gr and Π decay like $|x|^{-1}$, $|x|^{-2}$, $|x|^{-2}$ and $|x|^{-3}$; $|\mathsf{Gr}f|_\infty + |Q^{-1}d^*f|_\infty \le CN|f|_\infty$ and $|d\,\mathsf{Gr}f|_\infty \le C\log N|f|_\infty$; each component of $L_\mu(u)$ has under P_0 a variance $\le CN\log N/\beta$; and $g_0 := \sup_N G_N(0) < \infty$. The same bounds hold on $\mathbb{Z}^3$, and for $|x| \le N/2$ the differences of the kernels of T_N and of $\mathbb{Z}^3$ of the decay $|x|^{-e}$ are at most CN^{-e}.

Step 2: reflections. For a bounded Borel function F of the bonds of a periodic box Λ, $Z_\Lambda[F] := \int F \exp[-\beta \sum_p a_p]\, dU = Z(\Lambda)\langle F\rangle_\Lambda$, so that $Z_\Lambda[1] = Z(\Lambda)$ for the constant function 1, and $Z_\Lambda[E] := Z_\Lambda[1_E]$ for an event E; $Z_N[\cdot]$ is this bracket on T_N, and Z, $Z[\cdot]$ with no index refer to the box under consideration; inside the bracket the function multiplies the weight here, while in $Z_\Gamma[0]$ before Proposition 18.114 the bracket carries a source added to the exponent. (i) For every even $N \ge 4$ and $\beta \ge 0$, $8P_{2N} - 7P_N \le P \le P_{2N} \le P_N$ and $0 \le \delta_N \le 8 \log Z_N - \log Z_{2N}$: Theorem BC(1) at $q = 2$ and Proposition 17.31(c). (ii) Let $R \ge 2$ with N_1/R an even integer, let B be a *block of the side* R of T_{N_1}, that is, a translate of the closed cube $[0,R]^3$, which has $R+1$ sites along each edge, and let E be an event of the bonds of B; we move B to $[0,R]^3$ and let F_8 be the product of 1_E over the images of B under the eight reflections $x \mapsto (\pm x_1, \pm x_2, \pm x_3)$ of T_{2R}, which tile T_{2R}. Then $\langle 1_E\rangle_{N_1^3} \le (Z_{2R}[F_8]/Z_{2R})^{1/8} e^{\delta_{2R}/8}$. Lemma 18.2 gives $\langle 1_E\rangle_{N_1^3} \le \langle F_{\mathrm{all}}\rangle_{N_1^3}^{(R/N_1)^3}$, with F_{all} the product over all reflected blocks. With the time μ, for a bounded Borel function G of the bonds of the closed slab $\{0 \le x_\mu \le R\}$ let $\mathsf{A}_G(V, V')$, V and V' the configurations of its two end slices, be the integral over its other bonds of G times the weights $e^{-\beta a_p}$ of its plaquettes, taken with the power $\frac12$ at the plaquettes that lie in an end slice: a bounded kernel, which the reflection of the slab transposes. On a box whose side in the direction μ is $2kR$, the product of the functions $G, \Theta G, G, \Theta G, \dots$ on its $2k$ consecutive closed slabs, each the mirror image of the one before, has the integral $\mathrm{Tr}\big[(\mathsf{A}_G \mathsf{A}_G^{\mathrm{T}})^k\big] \le (\mathrm{Tr}(\mathsf{A}_G \mathsf{A}_G^{\mathrm{T}}))^k$, as $\mathsf{A}_G \mathsf{A}_G^{\mathrm{T}} \ge 0$, and $\mathrm{Tr}(\mathsf{A}_G \mathsf{A}_G^{\mathrm{T}})$ is the integral of the product of G and ΘG on the box with the side $2R$ in that direction; in the three directions this gives $Z_{N_1}[F_{\mathrm{all}}] \le Z_{2R}[F_8]^{(N_1/2R)^3}$. Finally $Z_{N_1}^{(R/N_1)^3} = e^{R^3 P_{N_1}} \ge e^{R^3 P} = (Z_{2R} e^{-\delta_{2R}})^{1/8}$ by (i). (iii) For $\beta \ge 1$, $\varepsilon > 0$, even $N_1 \ge 4$ and every plaquette p, $\langle 1[|\partial U(p) - \mathbf{1}| > \varepsilon]\rangle_{N_1^3} \le \exp[-\beta\varepsilon^2/4 + 14 + 3\mathsf{y}/N_1]$: The same slab dissemination bound at width one, in the three directions in turn, bounds the probability by the N_1^3-th root of the mean of the product of the indicators over the images of p, which are the $N_1^3/2$ plaquettes of the orientation of p in the layers at an even distance from that of p; on that event $\beta \sum a_p \ge \beta\varepsilon^2 N_1^3/4$ over these plaquettes; the plaquettes of the two other orientations integrate to at most the following quantity. Fix the bonds normal to the tested plaquette. For each of the two tangential directions, the bonds of that direction form N_1^2 chains along the normal axis, coupled consecutively by the retained plaquettes. Drop one interaction in each closed chain, which increases the integral. Integrate the open chain starting at an end. At each step Haar invariance gives exactly $\mathsf{z}(\beta)$, since the two normal bonds and the neighboring tangential bond are held fixed. The resulting bound is $\mathsf{z}(\beta)^{2N_1^2(N_1-1)} \le \beta^{-3N_1^2(N_1-1)}$; and $Z_{N_1} \ge e^{N_1^3 P}$ by (i), where $P \ge -3\mathsf{y} - 13.84$ for $\beta \ge \frac14$. For the last bound, a gauge transformation sets the $N^3 - 1$ bonds of a spanning tree of the cube T_N to $\mathbf{1}$ and leaves the remaining $2N^3 + 1$ bonds Haar distributed; where these lie within $s \le 1$ of $\mathbf{1}$, a set of the measure $\ge (s^3/2\pi)^{2N^3+1}$, as the Haar probability of $\{|U - \mathbf{1}| \le s\}$ is at least $\frac{2}{\pi}\int_0^s \sin^2 t\, dt \ge s^3/2\pi$, every plaquette has $|\partial U(p) - \mathbf{1}| \le 4s$ and $a_p \le 8s^2$; so $\log Z_N \ge (2N^3+1)\log(s^3/2\pi) - 24N^3\beta s^2$, and at $s = \frac12 \beta^{-1/2}$, as $N \to \infty$, $P \ge -3\mathsf{y} - 6\log 2 - 2\log 2\pi - 6 \ge -3\mathsf{y} - 13.84$.

Step 3: the law of a line of holonomies is of positive type. Let $\Lambda = (2R, N_2, N_3)$, $F \ge 0$ a bounded Borel function of the bonds of the slab of the first $R+1$ layers that is invariant under the gauge transformations of the slab, $F \otimes \Theta F$ the product of F and its mirror image on the second slab, h the holonomy of a closed line in the direction 1, and $w = \sum_j \hat w_j \chi_j$ a class function with $\hat w_j \ge 0$, $\sum \hat w_j d_j < \infty$. Then $Z_\Lambda[(F \otimes \Theta F)\, w(h)] \ge \hat w_0 Z_\Lambda[F \otimes \Theta F]$. In the gauge in which the bonds of the line are $\mathbf{1}$ but the last, which carries h, the integral is $\int w(h)\, \mathrm{Tr}(\mathsf{A}^\circ \mathsf{A}^{\circ\,\mathrm{T}} R(h)) dh$, A° the kernel A_F of Step 2 (ii), with the time 1, in which the bonds of the line are set to $\mathbf{1}$ and not integrated, and R the unitary representation of the gauge transformations at the one site of the line in a layer, which commutes with A°; by the Peter–Weyl theorem $\int w(h) R(h) dh = \sum_j (\hat w_j/d_j) P_j$ with orthogonal projections P_j commuting with A°, $\mathrm{Tr}(\mathsf{A}^\circ \mathsf{A}^{\circ\,\mathrm{T}} P_j) = \|P_j \mathsf{A}^\circ\|_2^2 \ge 0$, and the term $j = 0$ is $\hat w_0 Z_\Lambda[F \otimes \Theta F]$. For $w = e^{-\lambda a(\cdot)}$, an exponential series of a character with coefficients ≥ 0, $\hat w_0 = \mathsf{z}(\lambda)$. Apply the estimate in the three coordinate directions in turn. For the line gauge used in this trace,

successively transport the gauge along the line starting at the origin. At fixed original line bonds this is a Haar-preserving substitution on every other bond. The last line variable becomes h by Haar invariance. In the second slab a gauge at the last endpoint sets this last line bond to $\mathbf{1}$ and acts on the end-slice variable by $R(h)$, with the inverse convention giving the same integral because $w(h^{-1}) = w(h)$. Averaging the trace over h at $w = 1$ therefore gives the unpinned integral and inserts $P_0 = \int R(h)\, dh$. In the estimate for direction μ, each previously inserted pin lies entirely in the site wall $x_\mu = 0$ because the three loops go through the origin. Split the square root of that pin weight between the two reflected slabs. The product retains the required form $F_\mu \otimes \Theta_\mu F_\mu$. Thus for any nonnegative gauge-invariant Φ having this form in every direction, in particular $\Phi = 1$, insertion of the three pin weights costs at most the factor $\mathsf{z}(\lambda)^{-3}$. Since $\prod_\mu e^{-\lambda a(h_\mu)} \le 1_{\mathsf{H}_r} + e^{-\lambda r^2/2}$, this gives

$$Z[\Phi] \le \mathsf{z}(\lambda)^{-3}\{Z[\Phi 1_{\mathsf{H}_r}] + e^{-\lambda r^2/2} Z[\Phi]\}, \qquad Z\left(\mathsf{z}(\lambda)^3 - \langle 1_{A^c}\rangle - e^{-\lambda r^2/2}\right) \le Z[A \cap \mathsf{H}_r] \ \text{ for every event } A.$$

Step 4: gauges. All distances of group elements below are measured in the Euclidean norm of the unit quaternions. Products of unit quaternions satisfy $|u_1 \cdots u_j - v_1 \cdots v_j| \le \sum_i |u_i - v_i|$. In particular, commuting a lattice path across a plaquette changes its transport by at most the plaquette error. This is the estimate used in the following gauge constructions.

(i) A periodic gauge from a comb. Suppose the plaquettes on $(\mathbb{Z}/m)^3$ are within ε of the identity and the three coordinate holonomies through the origin are within r. On the lifted cube $[-m, m]^3$ use the gauge $g(x)$ given by the transport along the path from 0 to x which traverses the coordinates in order. For an edge (x, y) the two comb paths differ by sliding this edge past at most two coordinate segments, of total length at most $2m$. Thus $|g(x)T_{xy} - g(y)| \le 2m\varepsilon$. For lifts of the same site separated by one period in a direction, compare the comb paths by inserting the straight period loop at that site. Moving that loop to the origin uses at most Cm^2 plaquettes, and moving the inserted segment to its position in the comb uses at most Cm^2 more. It follows that the two values of g are within $r + Cm^2\varepsilon$. Lifts with positive weights below differ by at most one period per coordinate, so their mutual distances are at most $C(r + m^2\varepsilon)$.

For each lattice site x, average the values $g(x + mj)$ using weights $\prod_\mu (1 - |x_\mu + mj_\mu|/m)_+$. The weights sum to one, the average is periodic, and its norm is at least $1 - C(r + m^2\varepsilon)$. The sum of the absolute weight differences across an edge is $O(m^{-1})$. In the term containing these differences, subtract one fixed lift value at the second endpoint, since the differences sum to zero. The resulting error is $O((r + m^2\varepsilon)/m)$. The term with fixed weights is bounded by the comb estimate. Dividing the average by its norm therefore gives a periodic gauge in which all bonds are within $C(m\varepsilon + r/m)$ of the identity, provided $r + m^2\varepsilon$ is less than a sufficiently small pure constant.

(ii) Gluing local gauges. Let R be divisible by 4, N_1 be divisible by R, and $N_1 \ge 2R$. Write $h = R/2$. Suppose that on each block $B_j = jh + [0, 2h]^3$ there is a gauge g_j making all bonds within ρ_1 of the identity, and that the three coordinate holonomies on the N_1-torus are within r. The overlap transition $g_j(x)g_k(x)^{-1}$ changes by at most $2\rho_1$ across an edge of the overlap. Its oscillation on an overlap of neighboring blocks is therefore $O(R\rho_1)$. On the grid of block indices, define a nearest-neighbor bond by evaluating the transition at one point of the overlap and use its inverse on the reverse bond. Around a coarse plaquette all four blocks share a point. Comparing the four transitions with their values at that point shows that the coarse plaquette error is $O(R\rho_1)$. For a coarse coordinate loop, evaluate the transitions along the corresponding fine coordinate axis. Consecutive transition points can be joined in one block by a straight segment of length h. In the block gauge each segment transport is within $h\rho_1$ of the identity. Inserting the segments compares the coarse loop product to a conjugate of the original fine holonomy, with error $O(N_1\rho_1)$, including the errors from changing transition points.

The coarse torus has side N_1/h. Part (i) therefore gives constant changes of the block gauges making all transitions within $C(N_1\rho_1 + rR/N_1)$ of the identity, if $r + N_1^2\rho_1/R$ is sufficiently small. The same bound holds for any blocks which share a point. Indeed, one can change their indices one

coordinate step at a time through blocks that all contain the point, and the transition products telescope. Average the adjusted block gauges with the product tent functions of width h centered in the blocks. Their weights sum to one. For a fine lattice edge, every block with positive weight at an endpoint contains the edge, since the block boundaries have integer coordinates and the tents vanish there. The same normalized-average estimate as in (i), now with weight variation $O(R^{-1})$, gives a global gauge with bond error at most $C((N_1/R)\rho_1 + r/N_1)$.

(iii) The Landau chart. Put $\mathsf{D}_N(\rho) = \{\tilde{u} \in \mathsf{T} : |\tilde{u}|_\infty \le \rho\}$ and let GG consist of the gauges with $g_0 = \mathbf{1}$. Set $\Psi(g, \tilde{u}) = q(\tilde{u})^g$. If $W = q(w)$ has bond error at most ρ_a, its vector part after the gauge $e^{i\mu}$ is $w - d\mu + B(\mu, w)$. Direct multiplication of the three unit quaternions gives

$$|B(\mu, w)|_\infty \le C(\|\mu\|_\infty + \|w\|_\infty)(\|w\|_\infty + \|d\mu\|_\infty).$$

For the difference of two remainders the corresponding bound is

$$\begin{aligned}|B(\mu, w) - B(\nu, w)|_\infty \le{}& C(\|w\|_\infty + \|d\mu\|_\infty + \|d\nu\|_\infty)\|\mu - \nu\|_\infty \\ &+ C(\|w\|_\infty + \|\mu\|_\infty + \|\nu\|_\infty)\|d(\mu - \nu)\|_\infty.\end{aligned}$$

These follow by writing the gauge product as the conjugation of W at its initial site times $e^{i\mu_x} e^{-i\mu_y}$. The latter factor has vector part $-(d\mu)_{xy}$ plus a remainder of size $C\|\mu\|_\infty |(d\mu)_{xy}|$, with the same product estimate for differences. The equation $d^* \operatorname{Vec}(W^{e^{i\mu}}) = 0$ is $\mu = \mathsf{Gr}(w + B(\mu, w))$ on mean-zero site fields. The row bounds in Step 1 make this a contraction on a ball of radius $C\rho_a$ for the norm $\max\{\|\mu\|_\infty/N, \|d\mu\|_\infty/\log N\}$ if $N\rho_a \log N$ is sufficiently small. The resulting vector part has norm $O(\rho_a \log N)$. Its scalar parts are positive because the gauge and field are near the identity. A constant conjugation of the new field makes the inverse gauge based and preserves transversality.

For uniqueness, suppose $q(v) = q(u)^g$ with $u, v \in \mathsf{D}_N(\rho)$ and g based. The bond equation gives $|g_x - g_y| \le C\rho$ on every edge. Paths of length $O(N)$ give $\|g - \mathbf{1}\|_\infty \le CN\rho$. Multiply g on the left by a constant near the identity so that its logarithm μ has mean zero, conjugating $q(v)$ by that constant. This constant exists: the map sending the logarithm of the constant to the spatial mean of the new logarithm has derivative within $O(N\rho)$ of the identity on a ball of radius $CN\rho$. The product estimate for the remainder also gives

$$\|B(\mu, u)\|_2 \le C\rho\|\mu\|_2 + C(\|\mu\|_\infty + \rho)\|d\mu\|_2.$$

Pairing $v - u = -d\mu + B(\mu, u)$ with $d\mu$ and using $\|\mu\|_2 \le CN\|d\mu\|_2$ makes $d\mu = 0$ for small $N\rho$. The original based gauge is then the identity.

The differential of Ψ has tangent block equal to the identity on T and normal block given by the projection of $D_{\tilde{u}}$ onto dC^0. The derivative on a constant gauge parameter is an infinitesimal constant conjugation of $\tilde{u}$, hence lies in T. Thus the normal determinant can be computed on mean-zero parameters and multiplied by the determinant of the orthogonal projection from based parameters to mean-zero parameters. This projection has determinant $N^{-3/2}$ per color: it is the identity on the intersection of the two hyperplanes and has singular value $N^{-3/2}$ on the remaining direction. On the mean-zero space the determinant of d onto its range has magnitude $e^{\frac{3}{2}\Sigma_N}$, by the Fourier eigenvalues. The normal block is therefore $N^{-9/2} e^{\frac{3}{2}\Sigma_N} \det(1 + \mathsf{Y}_{\tilde{u}})$. The product of the bond Haar densities contributes $\sigma_{\mathrm{H}}^{3N^3} \prod_b (1 - |\tilde{u}_b|^2)^{-1/2}$. Writing the gauge measure as normalized Haar divides out $\sigma_{\mathrm{H}}^{N^3-1}$. Consequently, for sufficiently small $N\rho$,

$$\Psi^*\Big(\prod_b dU_b\Big) = \sigma_{\mathrm{H}}^{2N^3+1} N^{-9/2} e^{\frac{3}{2}\Sigma_N} \det(1 + \mathsf{Y}_{\tilde{u}}) \prod_b (1 - |\tilde{u}_b|^2)^{-1/2}\, d\tilde{u} \prod_{x \ne 0} dg_x, \qquad d\tilde{u} = N^{27/2} dc\, du,$$

with $\|\mathsf{Y}_{\tilde{u}}\| \le N\rho$, $\operatorname{tr} \mathsf{Y}_{\tilde{u}} = (1 - N^{-3}) \sum_b (c_b - 1)$ and $\|\mathsf{Y}_{\tilde{u}}\|_{\mathrm{HS}}^2 \le C|\tilde{u}|_2^2$: the Jacobian is triangular in the splitting into T and dC^0, and its block on the gauge directions is $\det(d^*d)^{1/2} \det(1 + \mathsf{Y}_{\tilde{u}})$ up to the Gram factor $N^{-9/2}$ of the condition $g_0 = \mathbf{1}$; ϵ_* is a small pure number. Indeed, $\|\mathsf{Gr}\| \le N/4$, and $\|D_{\tilde{u}} - d\| \le \sqrt{12}(\rho + 1 - \sqrt{1 - \rho^2}) \le 4\rho$ for $\rho \le 1/8$, by summing the squared endpoint terms over the six bonds at each site. This gives the operator norm of $\mathsf{Y}_{\tilde{u}}$. Each column of $\mathsf{Gr}(D_{\tilde{u}} - d)$ is

a finite sum of first-difference Green columns with coefficients $O(|\tilde{u}_b|)$ from bonds at the site. The columns of Gr have uniformly bounded ℓ^2 norm by Step 1. Squaring and summing therefore gives $\|\mathsf{Y}_{\tilde{u}}\|_{\mathrm{HS}}^2 \le C|\tilde{u}|_2^2$. The cross-product blocks have trace zero in color. In the remaining scalar term, cyclicity gives the trace of $(c_b - 1)d\mathsf{Gr}$ over bonds and colors. The diagonal of $d\mathsf{Gr}$ is $(1 - N^{-3})/3$ per color by cubic symmetry and the rank $N^3 - 1$ of the gradient projection per color. This proves the stated trace formula.

Step 5: the action in the chart. For $\tilde{u} = c + u \in \mathsf{T}$, $|\tilde{u}|_\infty \le \rho \le \frac{1}{8}$: by the identity for $a(U)$ and as $\Pi\mathsf{N} = d\mathsf{B}$,

$$\sum_p a_p(q(\tilde{u})) = \tfrac{1}{2}|d\Phi_c(u)|_2^2 + \tfrac{1}{2}|(1-\Pi)\mathsf{N}(\tilde{u})|_2^2 + \tfrac{1}{2}\sum_p a_p(q(\tilde{u}))^2;$$

$|\mathsf{N}_p| \le 14m_p^2$ and $a_p \le 9m_p^2$, m_p the largest $|\tilde{u}_b|$ at p; $\|D\mathsf{N}\| \le 29\rho$; $\|\mathsf{X}\| \le \frac{29}{4}N\rho$, $\|\mathsf{X}\|_{\mathrm{HS}}^2 + |\operatorname{tr}\mathsf{X}| \le C|\tilde{u}|_2^2$, $\operatorname{tr}\mathsf{X}_1 = 0$ by the colour trace, $\mathsf{B}(c) = 0$ and $|\mathsf{B}|_\infty \le CN|\tilde{u}|_\infty^2$; and if $N\rho \le \epsilon_*$, then Φ_c is one-to-one on $\mathsf{D}_c(\rho) := \{u \in \mathsf{T}_0 : |c+u|_\infty \le \rho\}$ with $|u - v|_2 \le 2|\Phi_c(u) - \Phi_c(v)|_2$, and for $|c| \le \rho/4$ its image contains $\mathsf{Gd} := \{u' \in \mathsf{T}_0 : |u'|_\infty \le \rho/2\}$.

Here are the derivative and image estimates used in this step. On a plaquette write its four oriented factors as $1 + E_i$, where $|E_i| \le 1.01m_p$. The vector part of the linear sum of the E_i is precisely $d\tilde{u}$, so N_p is the vector part of the terms with at least two factors. Expanding the four factors gives $|\mathsf{N}_p| \le 14m_p^2$, while product telescoping and $a = \frac{1}{2}|\partial q - \mathbf{1}|^2$ give $a_p \le 9m_p^2$. In the derivative with respect to one bond, subtract the derivative of its vector part alone. Moving the three other unitary factors to the identity costs at most 3.03ρ times the derivative norm, which is at most 1.01 for $\rho \le 1/8$. Thus each derivative block has norm at most 4.1ρ. Each plaquette has four bonds and each bond belongs to four plaquettes, so the operator norm is at most 16.4ρ by the row and column sum bounds. This implies the stated bound for $D\mathsf{N}$ and, since $\|Q^{-1}d^*\| \le N/4$, the one for X.

Write $\mathsf{R} = Q^{-1}d^*$ on plaquette cochains. Each column of R has bounded squared ℓ^2 norm by the first-difference bound in Step 1. A column of $\mathsf{R}D\mathsf{N}$ is a sum of at most four such columns, with matrix coefficients bounded by Cm_p at their plaquettes. Summing the squared column norms proves $\|\mathsf{X}\|_{\mathrm{HS}}^2 \le C|\tilde{u}|_2^2$. The trace on T_0 equals the trace of the same product on all bond cochains because the range of R lies in T_0. The color-independent kernel of R multiplies each local color derivative of $\mathsf{N}^{(2)}$, which is a cross-product matrix and has trace zero. The derivative of the remainder on a plaquette is $O(m_p^2)$ by the Taylor expansion of the four factors. Taking the diagonal and summing gives $|\operatorname{tr}\mathsf{X}| \le C|\tilde{u}|_2^2$. For a constant form c, $\mathsf{N}(c)$ is a constant plaquette cochain, so $d^*\mathsf{N}(c) = 0$. The ℓ^∞ row bound on R in Step 1 gives $|\mathsf{B}|_\infty \le CN\rho^2$. On the convex domain $\mathsf{D}_c(\rho)$ the Lipschitz bound $\|D\mathsf{B}\| \le 1/2$ in ℓ^2 gives the stated injectivity and inverse bound. If $|c| \le \rho/4$ and $|u'|_\infty \le \rho/2$, the map $u \mapsto u' - \mathsf{B}(c+u)$ maps $\mathsf{D}_c(\rho)$ into itself because $|\mathsf{B}|_\infty \le \rho/4$ when ϵ_* is small. It is the same ℓ^2 contraction, so its fixed point gives the image assertion.

Step 6: the Gaussian number. Let $p_s^{(3)}$ be the infinite-lattice heat kernel. Subtract the heat integral for $N^3 f_\Delta$ from the one for Σ_N, using the periodic heat kernel and omitting its zero mode. The difference is

$$\int_0^\infty \left[1 - e^{-s} - N^3 \sum_{j \in \mathbb{Z}^3 \setminus \{0\}} p_s^{(3)}(Nj)\right] \frac{ds}{s}.$$

For $0 < s \le 1$ the image sum is bounded by the probability that a rate-six Poisson variable of mean $6s$ is at least N. The integral of this bound times N^3 is at most $N^2 e^6 6^N/N!$, uniformly bounded in N. For $1 \le s \le N^2$, the heat-kernel bound of Step 1 and the product of the three one-dimensional kernels bound the image sum times N^3 by $CN^3 s^{-3/2}(e^{-cN^2/s} + e^{-cN})$. For this sum, one may group the images by $|j|_\infty$ and use $\min\{N^2|j|_\infty^2/s, N|j|_\infty\} \ge |j|_\infty \min\{N^2/s, N\}$. The integral of this image bound against ds/s is uniformly bounded, and the $1 - e^{-s}$ term contributes at most $2\log N$. For $s \ge N^2$, write the integrand before division by s as $-e^{-s} - \sum_{k \ne 0} e^{-s|\mathsf{k}|^2} + N^3 p_s^{(3)}(0)$. The Fourier

sum is bounded by Ce^{-cs/N^2} using $|\mathsf{k}|^2 \ge c|m|^2/N^2$ for $k = 2\pi m/N$ in the centered frequency cube. The last term is at most $CN^3 s^{-3/2}$. These bounds are integrable there against ds/s with a uniform integral. Hence $|\Sigma_N - N^3 f_\Delta| \le C(1+\log N)$, and $|\log \mathsf{N}_0(N,\beta) - N^3(-3\mathsf{y} + z_\infty)| \le C(\mathsf{y} + (\log N)^2)$ and $|8\log \mathsf{N}_0(N,\beta) - \log \mathsf{N}_0(2N,\beta)| \le C(\mathsf{y} + (\log N)^2)$, as $3\log 2\pi - 2\log 2\pi^2 = \log \frac{2}{\pi}$.

Step 7: the chart, from below and from above. Assume (S): $N \ge 16$ even, $\beta \ge e$, $0 < r \le \rho$, $N\rho \le \epsilon_*$, $N^2 \log N \le \epsilon_* \beta$, and put $\mathsf{W}_N(\rho) := \Psi(\mathsf{GG} \times \mathsf{D}_N(\rho)) \cap \mathsf{H}_r$. With $u' = \Phi_c(u)$, Steps 4, 5 and the area formula give, for Borel $D' \subset \mathsf{D}_N(\rho)$,

$$\int_{\mathsf{GG}\times D'} \Psi^*(e^{-\beta \sum a_p} dU) = N^9 \mathsf{N}_0(N,\beta) \int dc\, E_0\big[1\{c + \Phi_c^{-1} u' \in D'\}\, e^{-F_c(\Phi_c^{-1}u')}\big],$$

$$F_c(u) := \tfrac{\beta}{2} |(1-\Pi)\mathsf{N}(\tilde u)|_2^2 + \tfrac{\beta}{2} \sum_p a_p^2 + \log\det(1+\mathsf{X}) - \log\det(1+\mathsf{Y}_{\tilde u}) + \tfrac12 \sum_b \log(1 - |\tilde u_b|^2).$$

(i) *Crude bounds.* The quadratic remainders of the log determinants and the Haar term give $e^{-F_c(u)} \le e^{C|c+u|_2^2}$; the two terms multiplied by β in F_c have nonnegative sign. For clarity, the determinant bound uses $|\log\det(1+B) - \operatorname{tr} B| \le C\|B\|_{\mathrm{HS}}^2$ when $\|B\| \le 1/2$. The linear parts of both traces vanish, and the remaining parts are $O(|c+u|_2^2)$. For $u' = \Phi_c(u)$ the smallness of $N\rho$ also gives $|c+u|_2^2 \le C(|u'|_2^2 + N^3|c|^2)$, since $\|Q^{-1} d^*\| \le CN$ and $|\mathsf{N}(\tilde u)|_2 \le C\rho|\tilde u|_2$. The axis holonomy differs in vector part from $Nc_\mu + L_\mu(u)$ by at most $CN^2\rho^2$ by multiplying the N factors and bounding the terms beyond first order. Also $|L_\mu(u) - L_\mu(u')| \le CN^2\rho^2$. On H_r this gives $N^3|c|^2 \le C(N\sum_\mu |L_\mu(u')|^2 + N^5\rho^4 + Nr^2)$. Here and below $|c| = \max_\mu |c_\mu|$.

For a centered Gaussian vector of covariance C_0, the formula $\mathbb{E}\exp[t|Z|^2] = \det(1-2tC_0)^{-1/2}$ bounds this expectation by $\exp(Ct \operatorname{tr} C_0)$ when $2t\|C_0\| \le 1/2$. Apply it to u' and to the nine axis components, using Cauchy–Schwarz to combine the bounds. The covariance of u' has norm $O(N^2/\beta)$ and trace $O(N^3/\beta)$, and that of the axis vector has norm $O(N\log N/\beta)$. The smallness of $N^2\log N/\beta$ makes these estimates valid even for twice the constants in the exponent above. The c integration has bounded volume because $|c| \le \rho \le 1/8$, and $Nr^2 \le N\rho^2 \le 1$. This gives

$$Z_N[\mathsf{W}_N(\rho)] \le N^9 \mathsf{N}_0 \exp[C(1 + N^3/\beta + N^5\rho^4)].$$

For the shell $\mathsf{W}_N(\rho) \setminus \mathsf{W}_N(\rho/2)$, injectivity and $|u-u'|_\infty \le CN\rho^2$ imply that either $|u'|_\infty \ge c\rho$ or $|c| \ge c'\rho$. In the latter case the holonomy estimate forces $\max_\mu |L_\mu(u')| \ge c'' N\rho$ after decreasing the smallness constant and using $r \le \rho$, $N \ge 16$. The first event has Gaussian probability at most $CN^3 e^{-c\beta\rho^2}$ by the uniform diagonal covariance bound. The second has probability at most $Ce^{-c\beta N\rho^2/\log N}$ by the axis covariance bound. One more application of Cauchy–Schwarz gives the shell bound $N^{13}\mathsf{N}_0 \exp[C(1+N^3/\beta+N^5\rho^4) - \beta\rho^2/C]$, with constants adjusted.

(ii) *The mean behind the change of variables.* On $\mathsf{D}_c(\rho)$ the Taylor bound is $|F_c - F^{\mathrm{lead}}(c+\cdot)| \le C(\beta N^3\rho^5 + N^4\rho^3)$. Indeed, in each action term the first discarded power has degree at least five. In a log determinant with matrix B, the terms beyond the quadratic in B cost at most $C\|B\|\|B\|_{\mathrm{HS}}^2 = O(N^4\rho^3)$. The remaining trace remainders are sums of local coefficients against the diagonal of a bounded kernel and cost $O(N^3\rho^3)$.

Both functions are Lipschitz in $|\cdot|_2$ with constant $C(\beta\rho^3+\rho)N^{3/2}$. For the squared action terms this follows by bounding the nonlinear term in ℓ^2 by $C\rho^2 N^{3/2}$ and its derivative in operator norm by $C\rho$. For a determinant, write the differential as $\operatorname{tr}(DB) + \operatorname{tr}(((1+B)^{-1} - 1)DB)$. The first trace has zero constant term and is bounded by $C\rho N^{3/2}|\delta u|_2$ from its local coefficients. For the second use $\|B\|_{\mathrm{HS}} \le C\rho N^{3/2}$ and $\|DB[\delta u]\|_{\mathrm{HS}} \le C|\delta u|_2$. These Hilbert–Schmidt estimates follow by expanding the local multiplication and difference operators in B and using the square-summable first-difference Green kernels in Step 1. The Haar factor follows by differentiating its product logarithm.

For $|c| \le c_0 \le \rho/4$ the change from u to u' costs $C(\beta\rho^3+\rho)N^{3/2} \cdot N^{3/2} N\rho^2$ by the Lipschitz estimate. In $E_0[F^{\mathrm{lead}}(c+u')]$ the terms odd in u' vanish. The quadratic terms involving c are bounded by $CN^3 c_0^2$ by the trace and Hilbert–Schmidt estimates. In a quartic term the contribution containing two copies of c costs $C\beta N^3 c_0^2 \beta^{-1}$ by the local diagonal covariance bound

and the contraction property of $1-\Pi$. The term containing four copies costs $C\beta N^3 c_0^4$. Thus $|E_0[F^{\mathrm{lead}}(c+u')] - E_0[F^{\mathrm{lead}}(u')]| \le C(N^3c_0^2 + \beta N^3 c_0^4)$. If $N \le \beta$ and $\rho^2 \ge K_*^\flat \mathsf{y}^3/\beta$, the Gaussian union bound gives $P_0(\mathsf{Gd}^c) \le e^{-\mathsf{y}^3}$. The polynomial moments on that complement are $O(1)$ by Cauchy–Schwarz, since the moments on the whole Gaussian space are bounded by fixed powers of N and β. Together these estimates give the asserted mean bound $C(\beta N^4\rho^5 + N^4\rho^3 + N^3 c_0^2 + \beta N^3 c_0^4 + 1)$.
(iii) *Two loops in the chart.* With $\mathsf{EE} := \beta N^4\rho^5 + N^4\rho^3 + \beta N^5\rho^6 + N^5\rho^2/\beta + N^{5/2}/\beta + 1$, under (S), $N \le \beta$ and $\rho^2 \ge K_*^\flat \mathsf{y}^3/\beta$:

$$\log Z_N \ge \log \mathsf{N}_0 + \frac{m_1(N)}{\beta} - C(\mathsf{y} + \mathsf{EE}),$$

by restricting to $|c_\mu| \le \beta^{-10}$ and to Gd, where Ψ is one-to-one, and Jensen's inequality; and if besides $N^{5/4}\rho^2\beta^{1/2} \le \epsilon_*$, $r \le N^{3/4}\beta^{-1/2}/8$ and $N^{1/2}/\log N \ge K_*^\flat(1 + N^3/\beta + N^5\rho^4)$,

$$\log Z_N[\mathsf{W}_N(\rho)] \le \log \mathsf{N}_0 + \frac{m_1(N)}{\beta} + C(\log N + \mathsf{EE}),$$

as follows. For $|c| \le 2N^{-1/4}\beta^{-1/2}$ this bound on c is at most $\rho/4$ under the stated radius assumptions, after increasing $K_*^\flat$. The inverse map is 2-Lipschitz. Extend $F_c \circ \Phi_c^{-1}$ from the image to T_0 with the same Lipschitz constant, by the infimum formula for Lipschitz extensions. The extension agrees on Gd, and its mean outside Gd costs $O(1)$ by the Gaussian tail and its linear growth. Gaussian concentration with covariance norm at most $N^2/(16\beta)$ bounds the logarithm of the exponential moment by minus its mean plus $C(\beta N^5\rho^6 + N^5\rho^2/\beta)$. The mean estimate in (ii) adds at most $CN^{5/2}/\beta$ from the constants c. The integral of these estimates over c costs a bounded volume.

If $|c| > 2N^{-1/4}\beta^{-1/2}$, the holonomy estimate of (i) forces $\max_\mu |L_\mu(u')| \ge N^{3/4}\beta^{-1/2}$: the two errors $CN^2\rho^2$ and r are smaller than the margin by the extra radius hypotheses. The Gaussian tail of the axis vector is bounded by $C\exp[-cN^{1/2}/\log N]$. Combining it with the exponential weight bound in (i) by Cauchy–Schwarz shows that this part is at most $N^9\mathsf{N}_0\exp[-C_1N^3/\beta]$ for any fixed C_1 if $K_*^\flat$ is sufficiently large. For the comparison, $|m_1(N)| \le CN^3$ follows directly from the local Gaussian fourth moments of $\mathsf{N}^{(2)}$ and du, and the Hilbert–Schmidt and trace bounds for the quadratic terms of F^{lead}. The prefactor N^9 then costs only $O(\log N)$ in the upper bound.

For the Gaussian identity used to identify $m_1(N)$, write $\mathsf{B}_j = Q^{-1}d^*\mathsf{N}^{(j)}$ for $j = 2, 3$ at $c = 0$. For the Gaussian measure on T_0, integration by parts gives

$$E_0[\beta\langle u, QV(u)\rangle] = E_0[\operatorname{div} V(u)]$$

for each polynomial vector field V. Since $\operatorname{div}\mathsf{B}_2 = \operatorname{tr}\mathsf{X}_1 = 0$ and $W = \beta\mathsf{S}_3^{\mathrm{L}} = \beta\langle u, Q\mathsf{B}_2\rangle$, taking $V = W\mathsf{B}_2$ gives

$$E_0 W^2 = E_0\Big[\beta|\Pi\mathsf{N}^{(2)}|_2^2 + \beta\langle u, Q(D\mathsf{B}_2)\mathsf{B}_2\rangle\Big] = E_0\Big[\beta|\Pi\mathsf{N}^{(2)}|_2^2 + \operatorname{tr}\mathsf{X}_1^2\Big].$$

In the second equality use integration by parts again and $\operatorname{div}((D\mathsf{B}_2)\mathsf{B}_2) = \operatorname{tr}(D\mathsf{B}_2)^2 + D(\operatorname{div}\mathsf{B}_2)[\mathsf{B}_2]$. Taking $V = \mathsf{B}_3$ gives $E_0[\beta\langle du, \mathsf{N}^{(3)}\rangle] = E_0[\operatorname{tr}\mathsf{X}_2]$. These equalities give the expression in terms of $\mathsf{S}_3^{\mathrm{L}}, \mathsf{S}_4^{\mathrm{L}}, \Xi_2^{\mathrm{L}}$.

Step 8: three levels. Let $R_\flat := 4\lceil\beta^{0.22}\rceil$, $n_\flat := 2R_\flat$, $k_\flat := \lceil\beta^{0.12}\rceil$, $N := 2k_\flat R_\flat$, so that $8\beta^{0.34} \le N \le 32\beta^{0.34}$, and let $N' \in \{N, 2N\}$; $\varepsilon_0^2 := K_0^\flat\mathsf{y}^3/\beta$, $\rho_1^2 := K_1^\flat\mathsf{y}^3/\beta$, $\rho_2^2 := K_2^\flat N'^3/\beta^2$, $\lambda := \beta^4$, $r := \beta^{-1.9}$, with large pure numbers $K_0^\flat, K_1^\flat, K_2^\flat$. There is a pure β_* such that for $\beta \ge \beta_*$:

(a) $\delta_{n_\flat} \le C\mathsf{y}^2$. By Step 2 (iii) all plaquettes of $T_{n_\flat}$ lie within ε_0 of $\mathbf{1}$ outside an event of the probability $e^{-9.5\mathsf{y}^3}$; there Step 4 (i), (iii) put the field in $\mathsf{W}_{n_\flat}(C\beta^{-0.28}\mathsf{y}^{5/2})$, if it lies in H_r; Step 3 with $\mathsf{z}(\lambda)^3 \ge c\beta^{-18}$ and Step 7 (i) give $\log Z_{n_\flat} \le \log\mathsf{N}_0(n_\flat) + C\mathsf{y}$, Step 7 (iii) at ρ_1 gives $\log Z_{n'} \ge \log\mathsf{N}_0(n') - C\mathsf{y}$ at $n' = n_\flat, 2n_\flat$, and Step 2 (i) with Step 6 bounds $\delta_{n_\flat} \le 8\log Z_{n_\flat} - \log Z_{2n_\flat}$.

(b) For every block B of side $R_\flat$ on $T_{N'}$, let E_B be the event that some gauge on B puts its bonds within ρ_1 of $\mathbf{1}$. Let F_8 be the product of the eight reflected indicators of E_B^c on $T_{n_\flat}$. It has the factorization into reflected halves in each coordinate used in Step 3. With $Z = Z_{n_\flat}$, that step gives

$$(\mathsf{z}(\lambda)^3 - e^{-\lambda r^2/2}) Z[F_8] \le Z[F_8\, 1_{\mathsf{H}_r}].$$

The factor on the left is at least $c\beta^{-18}$ for large β. The part on the right where a plaquette exceeds ε_0 is at most $Ze^{-9.5\mathsf{y}^3}$ by (a). On the rest the field lies in the chart of radius $C\beta^{-0.28}\mathsf{y}^{5/2}$ by (a). If its chart radius is at most $\rho_1/1.01$, its gauge restricts to a gauge of bond error at most ρ_1 on B, so $F_8 = 0$. The intervening dyadic shells number $O(\mathsf{y})$ and have radii at least $\rho_1/1.01$. Uniformly on them, $n_\flat^3/\beta = o(1)$ and $n_\flat^5\rho^4 = o(1)$. The shell bound of Step 7(i), divided by the lower bound $Z \ge \mathsf{N}_0 e^{-C\mathsf{y}}$ from (a), is at most $\exp(-10\mathsf{y}^3)$ after increasing $K_1^\flat$. Thus $Z[F_8]/Z$ is at most $\exp(-9\mathsf{y}^3)$ for large β. The eighth root in Step 2(ii) and $\delta_{n_\flat} \le C\mathsf{y}^2$ give $\langle 1_{\mathsf{E}_B^c}\rangle \le e^{-\mathsf{y}^3}$.

(c) $Z_{N'} \le \beta^{19} Z_{N'}[\mathsf{W}_{N'}(\rho_2)]$. Let A be the intersection of the events E_B for the blocks on the grid of spacing $R_\flat/2$ used in Step 4(ii). There are $O(\beta^{0.36})$ such blocks, so $\langle 1_{A^c}\rangle \le \beta e^{-\mathsf{y}^3}$ by (b). The gluing condition holds since $N'^2\rho_1/R_\flat = O(\beta^{-0.04}\mathsf{y}^{3/2})$. On $A \cap \mathsf{H}_r$, the gauge from Step 4(ii) has bond error $O(\beta^{-0.38}\mathsf{y}^{3/2})$, and Step 4(iii) puts the field in $\mathsf{W}_{N'}(\rho_{\max})$ with $\rho_{\max} = C\beta^{-0.38}\mathsf{y}^{5/2}$. By Step 3,

$$Z_{N'}[\mathsf{W}_{N'}(\rho_{\max})] \ge (\mathsf{z}(\lambda)^3 - \beta e^{-\mathsf{y}^3} - e^{-\beta^{0.2}/2}) Z_{N'} \ge c\beta^{-18} Z_{N'}.$$

To compare with the smaller chart, cover the radii between ρ_2 and $\rho_{\max}$ by $O(\mathsf{y})$ dyadic shells. At outer radius ρ in this range, Step 7(i) divided by the lower bound of Step 7(iii) at ρ_2 bounds the relative mass of a shell by

$$N'^{13} \exp\left[C(\mathsf{y} + \mathsf{EE}(\rho_2) + N'^3/\beta) + CN'^5\rho^4 - c\beta\rho^2\right].$$

Here we used $|m_1(N')| \le CN'^3$ from Step 7(iii). Since $N'^5\rho_{\max}^2/\beta = O(\beta^{-0.06}\mathsf{y}^5)$, the quartic term is at most $c\beta\rho^2/2$ for large β. Also $\mathsf{EE}(\rho_2) \le 2$ for large β by the powers listed in (d). The choice of $K_2^\flat$ then makes the sum of the shell bounds at most β^{-30} because $\beta\rho_2^2 = K_2^\flat N'^3/\beta$ and $N'^3/\beta \gg \mathsf{y}^2$. Subtracting this shell mass gives (c).

(d) $|\log Z_{N'} - N'^3(-3\mathsf{y} + z_\infty) - m_1(N')/\beta| \le C\mathsf{y}^2$: from below by Step 7 (iii) at ρ_2, from above by (c) and Step 7 (iii), and by Step 6; at ρ_2 every member of EE except 1 is a negative power of β, the largest $\beta N'^4\rho_2^5 \approx \beta^{-0.09}$, so that $\mathsf{EE} \le 2$, and the hypotheses of Step 7 hold, as $N'\rho_2 \approx \beta^{-0.15}$, $N'^{1/2} \approx \beta^{0.17}$ and $N'^3/\beta \approx \beta^{0.02}$.

Step 9: the Dirichlet box. For every $n \ge 2$ and $\tau_{\mathrm{gf}} > 0$ there are $q_0^{[n]}$ and $C(n)$ with $|\log Z_{n,\beta} + \frac{d_1}{2}\log\beta - q_0^{[n]} - q_1^{[n]}/\beta| \le C(n)\beta^{-2}$ for $\beta \ge 1$; so $q_1^{[n]}$ does not depend on τ_{gf}. The reduction in the proof of Proposition 7.12, taken at the insertion 1 (the axial gauge on the tree of Lemma 7.6(b), the exponential chart of Definition 7.4 and Lemma 7.5, the discarding of the large fields, and the change of variables eq. 7.viii to the covariant gauge with the Gaussian weight of d^*A), writes $Z_{n,\beta}$, up to $O(e^{-c\beta})$ and an explicit factor, as an integral $\int_{\mathcal{W}} e^{-\beta\Phi_\tau - \Theta}$ with $\Phi_\tau = S(e^{\mathrm{i}A}) + \frac{1}{2\tau}\|d^*A\|^2$, whose Hessian at 0 is the positive form S_2 of Definition 7.7 (Lemma 7.6(c)), and Θ the Haar and ghost terms. Laplace's method to the second order (the Gaussian scaling $A = \beta^{-1/2}z$, the Taylor expansion of the exponent, and the moment–cumulant relation for the coefficient) gives the expansion, with the coefficient $-\langle W_2\rangle_0 + \frac{1}{2}\langle W_1; W_1\rangle_0$, $W_1 = \mathsf{S}_3$ and $W_2 = \mathsf{V}_2$; the left side does not involve τ_{gf}, so neither does the coefficient. In the Landau chart of the box, with the gauges of the box (a forest of lines that each end on the boundary) in the place of the periodic ones and no constant modes, Step 4 (iii), Step 5, the change of variables displayed at the head of Step 7, without c and without the factor N^9, and the first two bounds of Step 7 (ii) hold with the same proofs at fixed n, with numbers $C(n)$ in the place of C; so on the chart of the radius ρ, $|F - F^{\mathrm{lead}}| \le C(n)(\beta\rho^5 + \rho^3)$, both are Lipschitz

with the constant $C(n)(\beta\rho^3+\rho)$, and $|F|\le C(n)(\beta\rho^4+\rho^2)$ by the bounds of Steps 4 (iii) and 5. Take $\rho=\mathsf{y}\beta^{-1/2}$. The fields outside the chart of the radius ρ have an action $\ge\rho^2/C(n)$, so that their part of $Z_{n,\beta}$ is at most $e^{-\mathsf{y}^2/C(n)}$, smaller than $\mathsf{N}_0^{\mathrm{D}}(n,\beta)$, a number of the order $\beta^{-d_1/2}$, by a factor that decays faster than every power of β; in the chart the shift of u to u' costs the Lipschitz constant times $|\mathsf{B}|_2\le C(n)\rho^2$, the Gaussian tails of single bonds make the probability of Gd^c and the mean of $|F^{\mathrm{lead}}|$ on Gd^c smaller than every power of β, and the expansion $e^{-F}=1-F+O(F^2)$ applies, with $F^2=O(\mathsf{y}^8\beta^{-2})$; as $\beta\rho^5=\mathsf{y}^5\beta^{-3/2}$, this gives $\log Z_{n,\beta}=\log\mathsf{N}_0^{\mathrm{D}}(n,\beta)+m_1^{\mathrm{D}}(n)/\beta+O(\beta^{-3/2}\mathsf{y}^5)$, where $m_1^{\mathrm{D}}(n):=-\beta E_0^{\mathrm{D}}F^{\mathrm{lead}}$ is taken in the box: E_0^{D} is the Gaussian mean of the covariance $\beta^{-1}\mathsf{Cv}$ of the box, every kernel is that of the box, the constants c are absent, and the final term of F^{lead} takes the form $\sum_b|u_b|^2[\frac{3}{2}(d\,\mathsf{Gr})_{bb}-\frac{1}{2}]$, the quadratic part of $-\operatorname{tr}\mathsf{Y}_u+\frac{1}{2}\sum_b\log(1-|u_b|^2)$, which for the periodic cube, where $(d\,\mathsf{Gr})_{bb}=(1-N^{-3})/3$, reduces to $-|\tilde u|_2^2/2N^3$. Further $\log\mathsf{N}_0^{\mathrm{D}}=-\frac{d_1}{2}\log\beta+\mathrm{a}$ constant. The two expansions have one coefficient of β^{-1}: $m_1^{\mathrm{D}}(n)=q_1^{[n]}$.

Step 10: comparison of the box and the torus. Reflect the cochains of T_{2n} in the three coordinate walls through the origin. Odd extension in all three reflections identifies the Dirichlet cochains of Λ_n with the corresponding subspace of the torus cochains. The coboundary and its adjoint commute with the reflected cochain maps, including their orientation signs. Each box kernel is therefore the signed sum of the eight reflected torus kernels. If r is the distance between the two arguments in the box and D is the distance from the first argument to the boundary, each nonidentity image with decay exponent e has size at most $C(1+\max(r,D))^{-e}$. This follows coordinatewise: the distance from x to the image of y in a reflected coordinate is $\min(x+y,2n-x-y)$ for $0\le x,y\le n$. This is at least both $|x-y|$ and the distance of x to the two endpoints. Offsets from bond and plaquette orientations change these bounds by at most a constant.

To estimate the coefficient, use the expression before Step 1 with the local quartic term, the quadratic determinant terms and the two cubic vertices. The local quartic terms have errors $O((1+D)^{-1})$ under image replacement. Each nonlocal summand is a product of three kernels with decay powers $(3,1,1)$ or $(2,2,1)$. The first two patterns come from the two cubic vertices: the two curl legs are either paired together or paired with plain legs at the opposite vertices. The determinant loop has two first-difference Green kernels and a plain covariance, which gives $(2,2,1)$. For each changed line the sum over the second cell is bounded by

$$C\sum_{r\ge0}(1+r)^2(1+\max(r,D))^{-e}(1+r)^{-(5-e)}\le\frac{C'}{1+D}.$$

For $r\le D$ sum the power $(1+r)^{e-3}$ and for $r>D$ sum $(1+r)^{-3}$. The omitted second cells outside the box give the same bound from their distance at least D. Since the number of first cells at any given integer boundary depth is $O(n^2)$, the total error is $O(n^2\log n)$. Consequently $|m_1^{\mathrm{D}}(n)-m_1(2n)/8|\le Cn^2\log n$ and Step 9 gives $|(2n)^{-3}m_1(2n)-n^{-3}q_1^{[n]}|\le C\log n/n$.

Step 11: the value. Let $\tau_{\mathrm{gf}}>0$. By Lemma 7.9 each term of $q_1^{[n]}$ is a finite sum, with coefficients that do not depend on the place, of products of the kernels $\mathsf{C},\mathsf{E},\Pi,\mathsf{D},\mathsf{g}$ of the box; the pairings of $\langle\mathsf{S}_3;\mathsf{S}_3\rangle_0$ with a line from a cubic vertex to itself vanish by the colour sum, so each of the others has three lines between two plaquettes. In the standard writing of Definition 7.8(3) every monomial of $\mathsf{a}_p^{(3)}$ carries exactly one curl leg, its two other legs being plain, so these three lines are a line between the two curl legs and two lines between plain legs, or two lines from a curl leg to a plain leg and one line between plain legs, and the majorants eq. 7.xxxi give them the decay exponents $3,1,1$ or $2,2,1$, a decay $\langle y\rangle^{-5}$ in all. These majorants hold for the kernels of $\mathbb{Z}^3$ (Step 1 of the proof of Lemma 7.27); by the images of the proof of Proposition 7.39 and Lemma 7.38(a), (b), the kernels of the boxes are signed sums of eight periodised kernels, which obey the same majorants with a constant that does not depend on n and have the kernels of $\mathbb{Z}^3$ as their pointwise limits; at the places at the distance $\ge R$ from the boundary the image terms change the density of a site by CR^{-1}, the truncation of

the second cell to $|y|_\infty < R$ costs CR^{-2}, and the remaining finite sums converge to those of $\mathbb{Z}^3$. So $n^{-3}q_1^{[n]} \to -v_2 + \frac{1}{2}s_{33}$, the density of $\mathbb{Z}^3$ at the site 0:

$$v_2 := \sum_{x_p=0} \langle \mathsf{a}_p^{(4)} \rangle_0 + \sum_{b_-=0} (1 + 2\mathsf{g}(b))\mathsf{C}(b,b) - 12 \sum_{b_-=0} \sum_{b'} \mathsf{D}(b;b')\mathsf{D}(b';b)\mathsf{C}(b,b'), \qquad s_{33} := \sum_{x_p=0} \sum_{p'} \langle \mathsf{a}_p^{(3)}; \mathsf{a}_{p'}^{(3)} \rangle_0,$$

which does not depend on τ_{gf}, as $q_1^{[n]}$ does not. At $\tau_{\mathrm{gf}} = 1$, $\mathsf{C}((x,\mu),(y,\nu)) = \delta_{\mu\nu}G(x-y)$, $\mathsf{g}(b) = \frac{1}{3}$ and $G(e_\nu) = G_0 - \frac{1}{6}$. (i) Write the signed vectors around a plaquette as $w_1, \dots, w_4$, with the two pairs of opposite edges $(1,3)$ and $(2,4)$. Put $g = G_0 - 1/6$. The variance of each color component is G_0, the covariance across opposite edges is $-g$, and the two directions are independent. In the fourth Taylor coefficient of the scalar part of the plaquette product, the contributions from a single fourth-order exponential and from two second-order exponentials have means $5G_0^2/2$ and $27G_0^2/2 + 3g^2$, respectively. The terms with two vector factors have mean $-28G_0 g$: for each opposite pair the terms with a cubic vector factor give $-5G_0 g$, and the terms with a second-order scalar factor on another edge give $-9G_0 g$. For the term with four vector factors, the Pauli trace identity gives the three products of dot products, with signs $+,-,+$. Their expectations are $3g^2, 9g^2, 3g^2$, giving $-3g^2$ after the signs. Reversing sign to obtain the action, $\langle \mathsf{a}_p^{(4)} \rangle_0 = -16G_0^2 + 28G_0 g = 12G_0^2 - \frac{14}{3}G_0$. There are three plaquettes per site.

(ii) The Haar term and the ghost tadpole give $(1 + \frac{2}{3})G_0$ for each of the three bonds at a site, $5G_0$ in all. (iii) The ghost loop: for bonds of one direction, $\mathsf{D}(b;b') = \frac{1}{2}[G(y - e_\mu) - G(y + e_\mu)] = -\mathsf{D}(b';b)$, and bonds of two directions do not interact at $\tau_{\mathrm{gf}} = 1$; so $-12 \sum \mathsf{DDC} = 3 \sum_\mu \sum_y [G(y + e_\mu) - G(y - e_\mu)]^2 G(y) = 9J_{\mathrm{gh}}$. So $v_2 = 36G_0^2 - 9G_0 + 9J_{\mathrm{gh}}$. (iv) The cubic vertex: $\mathsf{a}_p^{(3)} = T^{\mu\nu}(x) + T^{\nu\mu}(x)$ for $p = p_{\mu\nu}(x)$, with $T^{\mu\nu}(x) := -(A_\mu(x) \times A_\mu(x + e_\nu)) \cdot (A_\nu(x) + A_\nu(x + e_\mu))$; at $\tau_{\mathrm{gf}} = 1$ lines join equal directions, so only the pairs $T^{\mu\nu}, T^{\mu\nu}$ interact; the line of the two legs of the direction μ to each other vanishes by the antisymmetry, and the two pairings of these legs with those at y give $6G(y)^2$ and $-6G(y + e_\nu)G(y - e_\nu)$ times the line $2G(y) + G(y + e_\mu) + G(y - e_\mu)$ of the third legs; the six ordered pairs of directions give $s_{33} = 36J_3$. Hence $-v_2 + \frac{1}{2}s_{33} = 9G_0 - 36G_0^2 - 9J_{\mathrm{gh}} + 18J_3 = p_1$. This is the first claim of (b).

Step 12: the rate and the conclusion. For an even $N_1 \geq 4$ let $\mu_T(N_1)$ be the density per site of $-m_1(N_1)$ on T_{N_1}, and μ_∞ the same expression with the kernels of $\mathbb{Z}^3$ and the sums over $\mathbb{Z}^3$; by the last bounds of Step 1, replacing a kernel of the decay e costs $C\langle y \rangle^{-(5-e)} N_1^{-e}$ for $|y| \leq N_1/2$ and the second cells beyond $N_1/2$ cost CN_1^{-2}, so $|\mu_T(N_1) - \mu_\infty| \leq C \log N_1 / N_1$; by Steps 10 and 11, $-\mu_T(2n) = (2n)^{-3} m_1(2n) \to p_1$, so $\mu_\infty = -p_1$, and $|N_1^{-3} m_1(N_1) - p_1| \leq C \log N_1 / N_1$: the second claim of (b). Finally let $\pi(\beta) := -3 \log \beta + z_\infty + p_1/\beta$ and $\mathrm{Er}_{N'} := P_{N'} - \pi(\beta)$. By Step 2 (i), $8\mathrm{Er}_{2N} - 7\mathrm{Er}_N \leq P - \pi \leq \mathrm{Er}_{2N}$, so $|P - \pi| \leq 15 \max\{|\mathrm{Er}_N|, |\mathrm{Er}_{2N}|\}$; by Step 8 (d), $|\mathrm{Er}_{N'}| \leq |N'^{-3} m_1(N') - p_1|/\beta + C\mathsf{y}^2 N'^{-3}$, and $N'^{-3} \leq \beta^{-1.02}/512$; with (b) this is (a). The same with $|m_1(N')| \leq CN'^3$ in the place of (b) gives $|P + 3 \log \beta - z_\infty| \leq C/\beta$. □

Block fields and tapered links. $\mathcal{M}$ is the double-block average of Proposition YM3_I:4.5; on the torus ℓ at the cutoff K, $V_j := \mathcal{M}^{K-j} U$ is the block field of the depth j, a configuration of $\Lambda_{j,\ell} := a_j \mathbb{Z}^3 / \ell \mathbb{Z}^3$, $0 \leq j \leq K$, $V_K = U$. Its value at a bond c reads only the bare bonds with both ends in the double block $Q(c)$, the union of the two blocks of L^{K-j} sites a side about c_- and c_+ (Proposition YM3_I:4.5); the map is Borel on every branch, gauge covariant and equivariant under lattice isometries by Lemma 18.211; so the same formulas, read on $a_K \mathbb{Z}^3$ or on a periodic box whose sides are multiples of L^{K-j}, define block fields $\widetilde{\mathcal{M}}^{K-j} U$ there. For a closed path C of $\Lambda_{d,\ell}$, $W_C(V)$ is $\frac{1}{2}$ tr of the ordered product of V along C. Fix a depth $d \geq 0$ and put $w := a_d$. For a positively oriented bond c of $\Lambda_{d,\ell}$ of the direction e and $0 \leq i \leq K - d$ let $y_i^\mp := c_\mp \pm \frac{1}{2}(a_d - a_{d+i})e$; the straight path $[y_{i-1}^-, y_i^-]$ consists of $\frac{1}{2}(L-1)$ bonds of $\Lambda_{d+i,\ell}$, and $y_{K-d}^+ = y_{K-d}^- + a_K e$. The *tapered*

link is

$$\widetilde V(c) := \prod_{i=1}^{K-d} V_{d+i}([y^-_{i-1}, y^-_i]) \cdot U(\langle y^-_{K-d}, y^+_{K-d}\rangle) \cdot \prod_{i=K-d}^{1} V_{d+i}([y^+_i, y^+_{i-1}]) :$$

links L times thinner at each stage, one bare bond across the middle plane of c, and back. Let C be the boundary of a rectangle of bonds of $\Lambda_{d,\ell}$ in a coordinate plane, with the two sides of the length T along a direction e_{μ_0} (the *legs*) and the two sides of the length R along e_ν, $R, T \in a_d\mathbb{N}$. $W_C^{\rm tap}$ is $W_C(V_d)$ with V_d replaced by $\widetilde V$ at the first and at the last link of each leg; for $d \le D \le K$, $W_C^{(D)}$ is $W_C(V_d)$ with these four links tapered down to the depth D only, the piece across the middle plane being the link $V_D(c''_D)$, $c''_D := \langle y^-_{D-d}, y^+_{D-d}\rangle$; so $W_C^{(d)} = W_C(V_d)$ and $W_C^{(K)} = W_C^{\rm tap}$.

The numbers of the block fields. Use the canonical constants of §17.2 and the calibrated choice (Th1), (Th2) of §2.3.1. Write $\varepsilon_{\rm th}(x) = A_0 x^{-1/2}(\log x)^{\mathfrak{p}_0}$ as in YM3_I:(634), with the cut constant c_ρ and threshold $\mathfrak{t}_\natural$ of that calibration. Use the history constants $c', C_{\rm maj}, C^\Sigma_{\rm maj}, c_{\rm low}$ from Section YM3_I:115, and set $C_{\rm low}(x) = c_{\rm low} + \frac{9}{2}\log x$. Use $C_{\rm tt}, C'_{\rm tt}, E_0$ in Proposition YM3_I:74.5. As in Proposition YM3_I:74.1, $M = L^{m'}$, $R(x) = \min\{L^j : j \ge 0,\ L^j \ge (\log x)^r\}$, and the room constant $C_{\rm s}$ satisfies $2LC_{\rm s} \ge C_L$. Put

$$\mathsf{m}(x) = 1 + m' + \max\{\log_L R(x + \bar B), \lceil \log_L C_{\rm s}\rceil\}.$$

If $m \ge \mathsf{m}(x)$, then $L^m \ge LMR(x)$ and $2L^m \ge C_L M$, as required in Proposition YM3_I:74.5. For a label $\mathsf{b} \ge \gamma^{-2}$ and integers $D > D' \ge 0$ let $\eta_D := \log(L^D\mathsf{b})$, $\omega_D := 2c_\rho\,\varepsilon_{\rm th}(L^{D+1}\mathsf{b})$, $\mu(D') := \mathsf{m}(L^{D'}\mathsf{b})$, $\mathsf{n}(D') := 2L^{\mu(D')}$, $\Delta(D') := \mathsf{C}_{\rm maj} + 2 + C_{\rm low}(L^{D'}\mathsf{b}) + 2C_{\rm tt}C'_{\rm tt}E_0$,

$$\mathsf{R}(D', D) := 2M^{-3}\big(\mathsf{n}(D')L^{D-D'}\big)^3 e^{-c'A_0^2\eta_D^{2\mathfrak{p}_0}} e^{(\mathsf{C}^\Sigma_{\rm maj}+2+C_{\rm low}(L^{D'}\mathsf{b}))\,\mathsf{n}(D')^3}, \qquad \varepsilon''(D', D) := \mathsf{R}(D', D)^{1/\mathsf{n}(D')^3} e^{\Delta(D')},$$

$\tau^{\rm ch}_d := \sum_{D\ge d}(13L^2\omega_D + 2\varepsilon''(d-1, D))$ for $d \ge 1$; $D_\flat$ the least integer ≥ 1 such that $L^D\mathsf{b} \ge \mathfrak{t}_\natural$, $\omega_D \le 1/200L^2$ and $D - 1 + m' + \log_L R(\mathsf{b}) \ge \mu(D-1)$ for all $D \ge D_\flat$; $\varepsilon'_D := 13L^2\omega_D + 2\varepsilon''(D-1, D)$ for $D \ge D_\flat$ and $\varepsilon'_D := 2$ for $D < D_\flat$; $\tau'_d := 4\sum_{D\ge d}\varepsilon'_D$. With the tension $\sigma_{\rm bl} := 0.66/\ell_0^2$: $\varrho_d := w + \log(1/4\tau^{\rm ch}_d)/(\sigma_{\rm bl}Lw)$ and $\kappa_d := \log(1/2\tau^{\rm ch}_d)/(Lw)$. None of these numbers depends on K, m or ℓ.

Proposition 18.115 (Wilson loops of block fields under the box inequality)**.** Let $\mathsf{b} \ge \gamma^{-2}$, $\ell_0 = 2L^{m_0}$ with $m_0 \in \mathbb{N}$, and assume BI$[\mathsf{b}, \ell_0, K_0; 0.069]$ (Definition 17.28) with $\ell_0/2a_K \ge 4$ for $K \ge K_0$. Let $\langle\cdot\rangle_K$ be the mean on the torus ℓ at $\beta = L^K\mathsf{b}$, ν_K the periodic Gibbs state of Theorem AZ(1)(b), $d \ge 0$, $w = a_d$, $\sigma_{\rm bl} = 0.66/\ell_0^2$, and C a rectangle as above.

(a) *Tapered loops on the tori.* For every $\ell = \ell_0 L^j$, $j \ge 1$, every $K \ge \max\{K_0, d+1\}$ and $2w \le R, T \le \ell/2$: $|\langle W_C^{\rm tap}\rangle_K| \le 1.3\, e^{-\sigma_{\rm bl}(R-w)(T-w)}$. The same bound holds when the four links are replaced by arbitrary products $X_- \cdot S \cdot X_+$ of the bare bond $S = U(\langle y^-_{K-d}, y^+_{K-d}\rangle)$ of the straight path $[c]$ that crosses the middle plane of c and of unitary Borel transporters X_- from c_- to y^-_{K-d} and X_+ from y^+_{K-d} to c_+ that read only bare bonds of the block about c_-, respectively c_+; here, for bare sites x, y, a *transporter* from x to y is a Borel map X on the bare configurations, with values in the 2×2 matrices, such that $X(U^u) = u(x)X(U)u(y)^{-1}$ for all gauge transformations u, where $U^u(b) := u(b_-)U(b)u(b_+)^{-1}$; the ordered product of U along a path from x to y is one, and $V_d(c)$ and each product $X_- \cdot S \cdot X_+$ are transporters from c_- to c_+.

(b) *Tapered loops in the Gibbs state.* For $K \ge \max\{K_0, d+1\}$ and all $R, T \in a_d\mathbb{N}$ with $R, T \ge 2w$: $|\nu_K(W_C^{\rm tap})| \le e^{-\sigma_{\rm bl}(R-w)(T-w)}$, the block fields being formed with $\widetilde{\mathcal{M}}$.

(c) *The telescope, without the side.* For every $\ell = 2L^m$ with $MR(\mathsf{b}) \le L^m$, every $K > d$ and $2w \le R, T < \ell$: $|\langle W_C(V_d)\rangle_K - \langle W_C^{\rm tap}\rangle_K| \le \tau'_d$; the limits $\mu_d(W_C) := \lim_K \langle W_C(V_d)\rangle_K$ and $\mathsf{w}_C := \lim_K \langle W_C^{\rm tap}\rangle_K$ exist, $|\mu_d(W_C) - \mathsf{w}_C| \le \tau'_d$, and τ'_d, $\tau^{\rm ch}_d$ tend to 0 as $d \to \infty$. These assertions do not use BI. Under BI and at $\ell = \ell_0 L^j$, $j \ge 1$, with $MR(\mathsf{b}) \le \ell/2$ and

$2w \le R,T \le \ell/2$: $|\mathsf{w}_C| \le 1.3\, e^{-\sigma_{\rm bl}(R-w)(T-w)}$ and $|\mu_d(W_C)| \le 1.3\, e^{-\sigma_{\rm bl}(R-w)(T-w)} + \tau'_d$, with the error τ'_d the same on all these tori.

(d) *Block loops: a range and a second term.* Let $d \ge 1$ with $L^d\mathsf{b} \ge \mathsf{t}_\natural$ and $\omega_D \le 1/200L^2$ for all $D \ge d$, $K \ge \max\{K_0, d+1\}$, $\ell = 2L^m \ge 13\ell_0$ with $m+d-2 \ge \mu(d-1)$, and $2w \le R,T \le \ell/2$. If $\min\{R,T\} \le \varrho_d$, then $|\langle W_C(V_d)\rangle_K| \le 17\, e^{-\sigma_{\rm bl}(R-w)(T-w)}$. If $2\tau_d^{\rm ch} \le 1$, then for all such R,T, $|\langle W_C(V_d)\rangle_K| \le 5(2+\max\{R,T\}/Lw)^2\big[e^{-\sigma_{\rm bl}(R-w)(T-w)} + e^{-\kappa_d(\max\{R,T\}-3Lw)}\big]$. The same two bounds hold for $\mu_d(W_C)$ if besides $MR(\mathsf{b}) \le L^m$, and for $\nu_K\big(W_C(\widetilde{\mathcal{M}}^{K-d}U)\big)$ for all $R,T \ge 2w$. As $\tau_d^{\rm ch} \to 0$ and $w = a_d \to 0$, for a rectangle of fixed physical sides the first bound holds at every large depth.

■

Proof. Lengths are in lattice units of a_K where nothing else is said; $N := \ell_0/2a_K$, so that BI says $\mathsf{e}(N) \le 0.069$ at $\beta = L^K\mathsf{b}$, $K \ge K_0$; $N_k := 2^k N$ and $\chi_k := \delta((2N_k)^3)$. For a box $\Lambda = (n_0, n_1, n_2)$ with the time 0 over the slice (n_1, n_2), D is the transfer operator of the setting of Section 17.2 over that slice and $\lambda_0 = \|\mathsf{D}\|$, while T keeps its meaning of the statement, the length of the legs; $S_\mu(x;\Sigma)$ is the number r_x of Proposition 17.31 for the time μ over the cross-section Σ, and $S_0(x) := S_0(x;(n_1,n_2))$; J_1 is the involution of Proposition 17.31(d) that reverses the sign of the bonds of a layer that cross one plane $x_1 = $ const between sites; and Φ^Θ is the conjugated mirror image of a slab function, as in (M1) of Section 17.2; for a function Φ with values in matrices, Φ^Θ is the adjoint matrix of its mirror image, $(\Phi^\Theta)_{ab} := (\Phi_{ba})^\Theta$. For a time direction over a cross-section Σ of a box, with the action A_Σ of a layer, the crossing operator K and the transfer operator $\mathsf{D} = \mathsf{D}_\Sigma$ of the setting of Section 17.2, and for a bounded Borel function Φ on $\mathcal{Y}_\vartheta$, $\vartheta \ge 2$, or one with values in matrices, taken entry by entry, put $\hat\Phi := e^{\beta A_\Sigma/2}\,\mathfrak{S}[\Phi]\,e^{\beta A_\Sigma/2}$, with the operator $\mathfrak{S}[\Phi]$ of (M1): its kernel is that of $\mathfrak{S}[\Phi]$ with the weight $e^{-\beta A_\Sigma}$ of each of the two end layers taken with the power $\frac12$. Then $\mathfrak{Y}[\Phi] = \mathsf{K}^{1/2}e^{-\beta A_\Sigma/2}\,\hat\Phi\,e^{-\beta A_\Sigma/2}\mathsf{K}^{1/2}$ and $e^{-\beta A_\Sigma/2}\mathsf{K}\,e^{-\beta A_\Sigma/2} = \mathsf{D}$, so the identities of (M1) and the cyclicity of the trace give $\hat{\mathsf{1}}_\vartheta = \mathsf{D}^{\vartheta-1}$ and the trace formula in the form $Z(\Lambda)\langle\prod_s(\Phi^s)_{\tau_{s-1}}\rangle_\Lambda = \mathrm{Tr}(\hat\Phi^r\mathsf{D}\,\hat\Phi^{r-1}\mathsf{D}\cdots\hat\Phi^1\mathsf{D})$ for consecutive slabs of $\vartheta_s \ge 2$ layers that fill the time, one factor D standing for the step from each slab to the next; the weights being positive, the kernels obey $|\hat\Phi(V',V)| \le \sup|\Phi|\,\mathsf{D}^{\vartheta-1}(V',V)$; and as the mirror reverses the order of the layers and keeps the weight, the kernel of $\widehat{\Phi^\Theta}$ at (V',V) is the complex conjugate of that of $\hat\Phi$ at (V,V'), and for matrix values its adjoint matrix, that is $\widehat{\Phi^\Theta} = \hat\Phi^*$, the adjoint being taken in the second case on the L^2 space of the vector-valued functions.

Step 1: flux on one lattice. Let $Z^\tau(\Lambda) := Z^{(10;0)}(n_0,(n_1,n_2))$ be the partition function with the sign of $\mathrm{tr}\,\partial U$ reversed on one column of plaquettes of the plane $(0,1)$, $Z_e := \frac12(Z - Z^\tau)$, $\mathfrak{f} := Z_e/Z$, $\hat{\mathfrak{f}} := \mathfrak{f}e^{\delta(\Lambda)} = Z_e e^{-P|\Lambda|}$, and $q := \|\mathsf{D}P_o\|/\lambda_0$ with $P_o := \frac12(1-\mathsf{J}_1)$. By Proposition 17.31(d), J_1 is a unitary involution commuting with D, $Z^\tau = \mathrm{Tr}(\mathsf{J}_1\mathsf{D}^{n_0}) > 0$, $Z_e = \mathrm{Tr}(P_o\mathsf{D}^{n_0}) \ge 0$, the top lies in the even sector, and $0 \le \mathfrak{f} < \frac12$. Then: (i) $\mathfrak{f} \le \hat{\mathfrak{f}} \le e^{\delta(\Lambda)} - 1$, as $Z_e \le \lambda_0^{n_0}S_0(n_0)$ and $\delta(\Lambda) = n_0\varepsilon(n_1,n_2) + \log(1+S_0(n_0))$ with $\varepsilon \ge 0$ (Proposition 17.31(c)), so that $\hat{\mathfrak{f}} \le S_0(n_0)e^{n_0\varepsilon}$; (ii) $\hat{\mathfrak{f}}(n'_0,n'_1,n_2) \le \hat{\mathfrak{f}}(n_0,n_1,n_2)^{n'_0n'_1/n_0n_1}$ for $n'_i \ge n_i$, as $x \mapsto (\mathrm{Tr}\,B^x)^{1/x}$ does not increase for $B = \mathsf{D}P_o \ge 0$, as in Proposition 17.31(b), in the direction 0 and, with the flips of the other plane of Proposition 17.31(d), in the direction 1, $e^{-P|\Lambda|}$ being an exact power; (iii) $q^{n_0} \le \hat{\mathfrak{f}}$, as $(\lambda_0 q)^{n_0} \le Z_e$ and $\lambda_0^{n_0} \ge e^{Pn_0|\Sigma|}$; (iv) for integers $4 \le m \le n'$, at a fixed slice (n_0,n_1), with the time 2 and $S_2(x) := S_2(x;(n_0,n_1))$, $\mathfrak{f}(n_0,n_1,n') \le (n'/m)(\mathfrak{f}(n_0,n_1,2m) + \frac12 S_2(m)) + \frac12 S_2(n')$: with the time 2 the reversed column lies in every layer, $Z^\tau = \mathrm{Tr}(\mathsf{T}^{(1)})^{n_2}$ with the operator $\mathsf{T}^{(1)} \ge 0$ of Proposition 17.31(d), whose top is at least $(Z^\tau(2m)/Z^\tau(m))^{1/m}$, and $Z^\tau(m) \le Z(m)$; so $1 - 2\mathfrak{f}(n') \ge \big[(1-2\mathfrak{f}(2m))/(1+S_2(m))\big]^{n'/m}/(1+S_2(n'))$, and Bernoulli's inequality gives the claim.

Step 2: the numbers under BI. One doubling of the scale, as in Proposition 17.32, with the three sides doubled in turn and each defect read through Proposition 17.31(b), gives $\mathsf{e}(2n) \le$

$\Psi(\mathsf{e}(n))$ for $\mathsf{e}(n) \le 0.069$, where $\Psi(\epsilon) = 2\log(1+s_3^2)$, $s_1 = \varphi(e^{\epsilon}-1)$, $s_2 = \varphi(e^{4\log(1+s_1)}-1)$, $s_3 = \varphi(e^{4\log(1+s_2)}-1)$, φ the function of Remark 17.29, the three doublings having the defects at most $2\log(1+s_i^2)$, $i=1,2,3$; the identity $\delta(\Lambda) = \frac12(\delta(\Lambda')+2F(\Lambda)-F(\Lambda'))$ for a box Λ' with one side of Λ doubled, along the three doublings from $(2N_k)^3$ to $(4N_k)^3$, gives $\chi_k \le c_k + \chi_{k+1}/8$ with $c_k := \log(1+s_1^2)+\frac12\log(1+s_2^2)+\frac14\log(1+s_3^2)$ at $\epsilon = \mathsf{e}(N_k)$, and $8^{-J}\chi_{k+J} \to 0$; this gives the upper bounds $\mathsf{e}(N_k) \le 0.069$, 0.060454, 0.043318, 0.019604, 0.0034431 for $k = 0,\dots,4$, $\mathsf{e}(N_{k+1}) \le 8.5\,\mathsf{e}(N_k)^2$ and $8.5\,\mathsf{e}(N_k) \le \exp(-0.2207\cdot 2^k)$ for $k \ge 4$; $\chi_0,\dots,\chi_4 \le 0.012838$, 0.0090476, 0.0040840, 0.00073315, 0.000021695, $\chi_k \le 1.83\,\mathsf{e}(N_k)^2$ for $k \ge 4$ and $\chi_k \le 0.004084$ for $k \ge 2$; $S_\mu(N;(2N,2N)) \le 0.035764$, $\delta(N,2N,2N) \le 0.04092$, $N\varepsilon(2N,2N) \le \chi_0/2 \le 0.00642$; and $S_\mu(t;(2N_k,2N_k)) \le (e^{\chi_k}-1)^{t/2N_k}$ for real $t \ge 2N_k$.

Step 3: the area law of the flux. (a) If $n_2 = 2N$, $n_0 \ge N$, $n_1 \ge 2N$, then $\mathfrak{f} \le \hat{\mathfrak{f}} \le \exp(-1.66\,n_0n_1/N^2)$: at $(N,2N,2N)$, by the bound of Step 1 (i) through S_0, $\hat{\mathfrak{f}} \le 0.035764\,e^{0.00642} \le e^{-3.3243}$, and Step 1 (ii). (b) If $k \ge 1$, $24\cdot 4^{k-1}N \le n_2 \le 96\cdot 4^{k-1}N$ and $n_0, n_1 \ge 2^{k+1}N$, then $\mathfrak{f} \le \hat{\mathfrak{f}} \le \exp(-0.165\,n_0n_1/N^2)$. Let $b_k := 2^{k+1}N = 2N_k$, $h_k := 24\cdot 4^kN$, and let $\bar u_k$ bound $\hat{\mathfrak{f}}(b_k,b_k,n_2)$ for $n_2 \in [h_{k-1},h_k]$ ($k \ge 1$), $\bar u_0$ at $n_2 = h_0$. Step 1 (i) and $\delta(2N,2N,24N) \le 12\chi_0$ give $\bar u_0 := e^{12\chi_0}-1 \le 0.16656$; Step 1 (ii) gives $\mathfrak{f}(b_{k+1},b_{k+1},h_k) \le \hat{\mathfrak{f}}(b_k,b_k,h_k)^4 \le \bar u_k^4$; Step 1 (iv) at $m = h_k/2$ and $n' \in [h_k,h_{k+1}]$, with the thermal bounds $S_2(m) \le \xi_{k+1}^{3\cdot 2^k}$, $S_2(n') \le \xi_{k+1}^{6\cdot 2^k}$ of Step 2 over the slice (b_{k+1},b_{k+1}), $\xi_{k+1} := e^{\chi_{k+1}}-1$, gives $\mathfrak{f}(b_{k+1},b_{k+1},n') \le 8\bar u_k^4 + 4.5\,\xi_{k+1}^{3\cdot 2^k}$, and the factor $e^{\delta(b_{k+1},b_{k+1},n')} \le \exp(24\cdot 2^k\chi_{k+1})$ turns this into the bound $\bar u_{k+1} := (8\bar u_k^4 + 4.5\,\xi_{k+1}^{3\cdot 2^k})\exp(24\cdot 2^k\chi_{k+1})$ on $\hat{\mathfrak{f}}$ over the whole window; the resulting numbers $\bar u_1 \le e^{-4.872}$, $\bar u_2 \le e^{-17.21}$, $\bar u_3 \le e^{-66.70}$ and, inductively, $\bar u_k \le \exp(-0.66\cdot 4^k - 0.7)$ for $k \ge 3$ satisfy $-\log \bar u_k \ge 0.165\,b_k^2/N^2$, and Step 1 (ii) carries the bound to all $n_0, n_1 \ge b_k$. (c) In these two cases $q \le \exp(-1.66\,n_1/N^2)$, respectively $q \le \exp(-0.165\,n_1/N^2)$, over the slice (n_1,n_2), by Step 1 (iii). Every $n_2 \ge 24N$ lies in one of the windows of (b).

Step 4: two tools of the transfer operator. (i) *Two odd slab functions.* Let Φ, Φ' be bounded Borel functions of slabs of $\vartheta \ge 2$ layers that change their sign when the reversal J_1 is applied to all their layers, placed g and g' layers apart, $2 \le g \le g'$, $2\vartheta + g + g' = n_0$. Then $|\langle \Phi\Phi'\rangle_\Lambda| \le 2\sup|\Phi|\sup|\Phi'|\,q^{g+1}(1+S_0(n_0-g-1))$, and $\langle \Phi\,\Phi^\Theta\rangle_\Lambda \ge 0$ when Φ^Θ stands on the slab g layers after that of Φ: the operators $\hat\Phi$, $\hat\Phi'$ anticommute with J_1, so they exchange the even and the odd sector; in the trace formula $Z\langle\Phi\Phi'\rangle_\Lambda = \mathrm{Tr}(\mathsf{D}^{g'+1}\hat\Phi'\mathsf{D}^{g+1}\hat\Phi)$ each of the two parts carries on the odd sector at least the power D^{g+1}, as $g \le g'$, and Schwarz's inequality in the Hilbert–Schmidt class with $\widehat{\Phi^\Theta} = \hat\Phi^*$ bounds the rest by $\mathrm{Tr}\,\mathsf{D}^{n_0-g-1}$. (ii) *The temporal gauge at single steps.* In the integral of a bounded gauge-invariant Borel function, the time-like bonds of steps no two of which are equal or neighbours may be set to $\mathbf{1}$; at such a step the factor D of the trace formula is replaced by $\mathsf{D}_{\mathbf{1}} \ge 0$, the operator D with the crossing bonds set to $\mathbf{1}$, whose kernel is of positive type by the exponential series of $\frac12\,\mathrm{tr}(VV'^{-1})$, and $\mathsf{D} = P_G\mathsf{D}_{\mathbf{1}} = \mathsf{D}_{\mathbf{1}}P_G$ with the projection P_G onto the gauge-invariant functions of a layer.

Step 5: loops that split at two planes. Let $\Lambda = (n_0,n_1,n_2)$; in the direction 0 let the consecutive slabs S_B, S_I, S_T have ϑ_c, ϑ_I, ϑ_c layers and a rest $a \ge 2$ layers, the last layers of S_B and S_I being t_1 and t_2; in the direction 1 let Str_L, Str_R be two slabs of ϑ layers, $g \le g'$ layers apart; and let $\mathbf{B},\mathbf{T},\mathbf{R},\mathbf{L}$ be Borel transporters with values in the 2×2 matrices of norm ≤ 1, $\mathbf{B}$ of the inner bonds of S_B from (t_1,ξ_L) to (t_1,ξ_R), $\mathbf{R}$ of the inner bonds of $S_I \cap \mathrm{Str}_R$ from (t_1+1,ξ_R) to (t_2,ξ_R), $\mathbf{T}$ of S_T from (t_2+1,ξ_R) to (t_2+1,ξ_L), $\mathbf{L}$ of $S_I \cap \mathrm{Str}_L$ from (t_2,ξ_L) to (t_1+1,ξ_L), and $f := \frac12\,\mathrm{tr}[\mathbf{B}S^{t_1}_{\xi_R}\mathbf{R}S^{t_2}_{\xi_R}\mathbf{T}(S^{t_2}_{\xi_L})^{-1}\mathbf{L}(S^{t_1}_{\xi_L})^{-1}]$, S^t_ξ the time-like bond at (t,ξ). If $q(n_0',n_2) \le e^{-\varsigma n_0'}$ for all large n_0', over the slices (n_0',n_2) of the direction 1, then for every integer $\mathsf{k} \ge 4$

$$|\langle f\rangle_\Lambda| \le \big(1+S_0(n_0-\vartheta_I;(n_1,n_2))\big)\exp\big((\vartheta_I/\mathsf{k})\,\delta(\mathsf{k},n_1-g-1,n_2)\big)\,e^{-\varsigma\vartheta_I(g+1)}.$$

Indeed, with the steps t_1, t_2 set to $\mathbf{1}$ (Step 4 (ii)), $Z[f] = \frac{1}{2}\operatorname{Tr}(\mathcal{Q}X')$ on $L^2 \otimes \mathbb{C}^4$ with $\mathcal{Q} := \mathsf{D}_{\mathbf{1}}^{1/2} \hat{\Phi}\, \mathsf{D}_{\mathbf{1}}^{1/2}$, $\hat{\Phi}$ the operator of the head of the proof for the matrix-valued function $\Phi = \mathbf{R}^{\mathrm{T}} \otimes \mathbf{L}$ of S_I, and X' built in the same way from the caps and from D^{a+1}; so $|Z[f]| \le \frac{1}{2}\|\mathcal{Q}\|\,\|X'\|_1$, Schwarz's inequality with kernels of positive type bounds $\|X'\|_1$ by $2\operatorname{Tr}\mathsf{D}^{n_0-\vartheta_I} = 2\lambda_0^{n_0-\vartheta_I}(1+S_0(n_0-\vartheta_I))$, and $Z \ge \lambda_0^{n_0}$. For $\|\mathcal{Q}\|$: on the box $(2k\vartheta_I, n_1, n_2)$ let Φ_R be the trace of the product, upward through the $2k$ slabs, of alternating copies of $\mathbf{R}$, $\mathbf{R}^\Theta$, each followed by the time-like bond $S^t_{\xi_R}$ of the step from its last layer t to the next slab, and Φ_L the trace of the like product, downward, of the copies of $\mathbf{L}$, $\mathbf{L}^\Theta$, each followed by the inverse $(S^t_{\xi_L})^{-1}$ of the bond of the step below it; here $\mathbf{R}^\Theta$ and $\mathbf{L}^\Theta$ are the adjoint matrices of the mirror images, as at the head of the proof, so that $\Phi^\Theta = (\mathbf{R}^\Theta)^{\mathrm{T}} \otimes \mathbf{L}^\Theta$, and as $(u(x)Xu(y)^{-1})^* = u(y)X^*u(x)^{-1}$, $\mathbf{R}^\Theta$ is a transporter from the bottom to the top of its slab and $\mathbf{L}^\Theta$ one from the top to the bottom, like $\mathbf{R}$ and $\mathbf{L}$. So $\Phi_R\Phi_L$ is gauge invariant, and with the steps between the copies set to $\mathbf{1}$ (Step 4 (ii)) it has the integral $\operatorname{Tr}(\mathcal{Q}^*\mathcal{Q})^k \ge \|\mathcal{Q}\|^{2k}$, as $\widehat{\Phi^\Theta} = \hat{\Phi}^*$ and $\operatorname{tr}_{\mathbb{C}^4}\prod_i(A_i \otimes B_i) = \operatorname{tr}\prod_i A_i \operatorname{tr}\prod_i B_i$; Φ_R and Φ_L change their sign under the reversal of the time-like bonds of one step, so as functions of the slabs Str_R, Str_L of the direction 1 they fall under Step 4 (i) in the direction 1, which bounds their mean by $8q(2k\vartheta_I, n_2)^{g+1} e^{\delta(2k\vartheta_I, n_1-g-1, n_2)}$, while $Z(2k\vartheta_I, n_1, n_2) = \lambda_0^{2k\vartheta_I}(1 + S_0(2k\vartheta_I))$; with $\delta(2k\vartheta_I, \cdot) \le (2k\vartheta_I/\mathsf{k})\delta(\mathsf{k}, \cdot)$ (Proposition 17.31(c)), the root of the order $2k$ and $k \to \infty$ give $\|\mathcal{Q}\| \le \lambda_0^{\vartheta_I} e^{-\varsigma\vartheta_I(g+1)} \exp((\vartheta_I/\mathsf{k})\delta(\mathsf{k}, n_1-g-1, n_2))$, and the claim follows.

Step 6: tube, sign fields, tapered links, telescope. (i) $W_C \circ \mathcal{M}^{K-d}$ is a real gauge-invariant Borel function, $|\cdot| \le 1$, of the bare bonds of the tube of C, the bonds with both ends in one double block $Q(c)$, $c \in C$; its bonds lie at the sup distance $\le \frac{1}{2}(w - a_K)$ from the sites of C. For a direction μ and an integer i, the *block layer* i of μ is the slab of the L^{K-d} bare layers with $|x_\mu - ia_d| \le \frac{1}{2}(w - a_K)$, which contains the blocks centred at those sites of $\Lambda_{d,\ell}$ that have $x_\mu = ia_d$; the block layers i and $i+1$ abut, and together all block layers fill the direction. So the double blocks of the bonds of C whose ends have $x_\mu \in \{j_- a_d, \dots, j_+ a_d\}$ lie in the $(j_+ - j_- + 1)L^{K-d}$ layers of the block layers $j_-, \dots, j_+$; the tube of a side of C that lies in a plane $x_\mu = ja_d$ lies in the block layer j of μ, a slab of exactly L^{K-d} layers; and a link of the direction μ from the height $j-1$ to the height j reads the block layers $j-1$ and j, which its middle plane parts. (ii) If a field $\gamma = \pm 1$ on the bare bonds has the product 1 over the four bonds of every plaquette with its corners in $Q(c)$, then $\mathcal{M}^{K-d}(\gamma U)(c) = \gamma([c])\mathcal{M}^{K-d}(U)(c)$: on $Q(c)$, γ is a gauge transformation by $\pm\mathbf{1}$ (Proposition YM3_I:4.5); so a flip across a plane between sites multiplies $W_C(V_d)$ by (-1) to the number of the bonds of C that cross it. (iii) $\widetilde{V}(c) = A^-\, U(\langle y^-_{K-d}, y^+_{K-d}\rangle)\, A^+$ with Borel transporters A^- from c_- to y^-_{K-d} and A^+ from y^+_{K-d} to c_+, in the sense of (a), that read only the block about c_-, respectively c_+, as the double blocks of the links of the paths $[y^\mp_{i-1}, y^\mp_i]$ lie in it; so W_C^{tap} is gauge invariant, $|W_C^{\mathrm{tap}}| \le 1$, reads the tube alone and obeys (ii). (iv) $W_C^{(D)}$ is a bounded gauge-invariant Borel function of V_D alone, and pointwise $|W_C^{(D)} - W_C^{(D+1)}| \le \sum_{\text{four links}} |V_D(c''_D) - V_{D+1}([c''_D])|$, as $V_{D+1}([c''_D])$ is the next stage of the taper and $\frac{1}{2}|\operatorname{tr}(A(X-Y)B)| \le |X-Y|$ for unitary A, B.

Step 7: (a). Let $n := \ell/a_K = 2NL^j \ge 26N$, since L is an odd integer at least 13; $T' = T/a_K$, $R' = R/a_K$, $\vartheta = L^{K-d}$, and the legs lie along the direction 0, the sides of the length R along the direction 1. Take for S_B, S_I, S_T the block layers of the lower side, the layers between, and the block layers of the upper side, so that $\vartheta_I = T' - \vartheta$, and for Str_L, Str_R the block layers of the two legs, $g = R' - \vartheta \le g'$; the four replaced links each carry one bare bond across the planes of t_1 or t_2, the parts on either side joining the caps or the legs. So W_C^{tap}, and every loop with the four links replaced by products $X_- \cdot S \cdot X_+$ as in (a), is an f of Step 5, with caps of norm 1. By Step 3 (c), case (b), with the directions 0 and 1 exchanged (Proposition 17.31(a)) and $n_2 = n \ge 24N$, $\varsigma = 0.165/N^2$. Let k be the largest index with $2N_k \le n/2$, so that $k \ge 2$ and $n < 8N_k$. As S_0 does not increase and $\delta/|\underline{n}|$ does not increase when a side grows, $1 + S_0(n - \vartheta_I; (n,n)) \le e^{\delta(n/2,n,n)} \le e^{32\chi_k} \le 1.1397$, and

the factor $\exp((\vartheta_I/\mathsf{k})\delta(\mathsf{k},n-g-1,n))$ with $\mathsf{k}=2N_k$ is $\le e^{32\chi_k}\le 1.1397$, by Step 2; $1.1397^2\le 1.3$; and $0.165\,\vartheta_I(g+1)/N^2\ge 0.165\,(T-w)(R-w)/(Na_K)^2=\sigma_{\rm bl}(R-w)(T-w)$, as $Na_K=\ell_0/2$.

Step 8: the large fields of a cell, without the volume. (i) *One averaging step.* If $\varpi_c(V)$, the largest $|\partial V(p)-\mathbf{1}|$ over the plaquettes of the double block of c, is $<\omega_M=1/200L^2$, then $|\mathcal{M}(V)(c)-V[c]|\le 12.2\,L^2\varpi_c(V)$, and always ≤ 2: use the small-curvature branch in Proposition YM3_I:4.5, where $\mathcal{M}(V)(c)=e^X V[c]$ with $X=L^{-3}\sum_x \log W_x$. In the proof of Lemma YM3_I:4.4, retain the curvature $\varpi_c(V)$ in the two rearrangement bounds. Writing $u=L^2\varpi_c(V)$, the two contour families each have diameter at most $e^{9u/8}-1$, and their transported loops differ from the straight loop by at most $e^{25u/8}-1$. Each contour mean differs from any member of its family by at most four times the diameter, by Proposition YM3_I:4.3. Use the product of the resulting three unitaries in the formula for W_x and telescope from the identity. This gives $|W_x-\mathbf{1}|\le 8(e^{9u/8}-1)+(e^{25u/8}-1)\le 12.175u$ for $0\le u<1/200$, since each quotient $(e^{au}-1)/u$ is increasing there. The principal SU(2) angle is $2\arcsin(|W_x-\mathbf{1}|/2)$ and hence at most $12.177u$ on this interval. Thus $|e^X-\mathbf{1}|\le|X|\le 12.177u$, which gives the stated bound.

(ii) *Discarding on one torus.* In the calibrated setting of §2.3.1, the small-field weight through level t is supported where the level $t-1$ field belongs to Cut_{t-1}, by its defining product in Section YM3_I:115. By YM3_I:(2159) this implies $\varpi(\mathcal{M}^{t-1}U)<2c_\rho\varepsilon_{\rm th}(x_{t-1})$. Proposition YM3_I:115.2 gives mass at least $(1-R_t)Z$ to this weight and domination by the unsourced Wilson integral on nonnegative gauge-invariant functions. Thus the Wilson probability of the complementary curvature event is at most R_t in YM3_I:(2187). (iii) *The cell bound.* For $K>D'\ge 0$ let a (D')-box be a periodic box of the spacing a_K with the sides $\lambda\mathsf{n}_\mu$, $\lambda=L^{K-D'}$, $\mathsf{n}_\mu\ge\mathsf{n}(D')$ even, at $\beta=L^K\mathsf{b}$; a cell is a cube of the side $a_{D'}$ of a board with its offset in $a_{D'+1}\mathbb{Z}^3$; and for $D'<D<K$ the event $E_D(\square)$ is that a plaquette of the block field of the depth $D+1$ with its corners in $\square$ has $|\partial V_{D+1}(p)-\mathbf{1}|\ge\omega_D$. Then for pairwise different cells of one board and $D'<D_s<K$ with $L^{D_s}\mathsf{b}\ge\mathsf{t}_\natural$,

$$\Big\langle\prod_s 1_{E_{D_s}(\square_s)}\Big\rangle\le\prod_s\varepsilon''(D',D_s),\qquad \|1_{E_D(\square)}\|_{\rm ch}\le\varepsilon''(D',D).$$

The chessboard estimate of the box (Proposition YM3_I:86.3) gives the product of the seminorms; the subadditivity in the volume (Proposition YM3_I:86.5) passes from the (D')-box Γ, with $\mathsf{N}:=\prod_\mu\mathsf{n}_\mu$ cells in all, to the cube $\Gamma^\flat$ with $\mathsf{n}(D')$ cells along each side, $\mathsf{N}^\flat:=\mathsf{n}(D')^3$ cells in all, at the price $\exp[F(\Gamma^\flat)/\mathsf{N}^\flat-F(\Gamma)/\mathsf{N}]$, $F=\log Z$, which is at most $e^{\Delta(D')}$ as follows. Enclose the side counts of both boxes by a cubic count $2L^j$, and compare the unnormalized partition functions by Proposition YM3_I:86.5 with insertion 1. The pressure per cell of this larger cube is a lower bound on that of Γ. Relabel the two cubes by D' to cutoff $K-D'$ and label $L^{D'}\mathsf{b}$, leaving the bond action and Haar measures unchanged. The room conditions just verified apply to both cubes. Use the upper and lower vacuum bounds of Proposition YM3_I:74.4 and the per-volume vacuum comparison of Proposition YM3_I:74.5 to bound the pressure gap by $\Delta(D')$; on that cube the copies of an indicator are indicators, so the seminorm is at most the probability of $E_D(\square)$ to the power $\mathsf{n}(D')^{-3}$; and the cube is the torus $2L^{\mu(D')}$ at the cutoff $K-D'$ and the label $L^{D'}\mathsf{b}$, where (ii) at $t=K-D$ bounds this probability by $\mathsf{R}(D',D)$. Moreover, as $R(x)\le L(\log x)^r$ and $\mathsf{n}(D')\le 2L^2M\max\{\eta_{D'+1}^r,C_{\rm s}\}$, $\log\varepsilon''(D',D)\le -c_\natural\eta_D^{2\mathfrak{p}_0}/\max\{\eta_{D'+1}^r,C_{\rm s}\}^3+\eta_D+9\eta_{D'}+C_\natural$ with $c_\natural:=c'A_0^2/8L^6M^3$ and a number $C_\natural$ of the constants; by the floor $2\mathfrak{p}_0>3r+2$ of the list of constants, $\varepsilon''(D',D)\le\exp(-c_\natural\eta_D^2+10\eta_D+C_\natural)$ for all $D>D'$ when $\eta_{D'+1}^r\ge C_{\rm s}$. So $\sum_D\varepsilon''(D-1,D)<\infty$, $\sum_{D>D'}\varepsilon''(D',D)<\infty$ for every D', and $\sum_{D\ge d}\varepsilon''(d-1,D)\to 0$ as $d\to\infty$; as $\omega_D=2c_\rho A_0(L^{D+1}\mathsf{b})^{-1/2}\eta_{D+1}^{\mathfrak{p}_0}$, also $\tau_d^{\rm ch}\to 0$ and $\tau_d'\to 0$ as $d\to\infty$. (iv) *One link.* For $D\ge D_\flat$, every torus $\ell=2L^m$ with $MR(\mathsf{b})\le L^m$, every $K>D$ and every bond c of $\Lambda_{D,\ell}$: $\langle|V_D(c)-V_{D+1}([c])|\rangle_K\le\varepsilon_D'$, by (i) off $E_D(\square)$ for the cell $\square$ about c_- of the side a_{D-1} of the board with the offset c_-, which holds $Q(c)$ as $L\ge 3$, by $|V_D(c)-V_{D+1}([c])|\le 2$ on $E_D(\square)$, and by (iii) at $D'=D-1$: the torus is a $(D-1)$-box, as $m+D-1\ge m'+\log_L R(\mathsf{b})+D-1\ge\mu(D-1)$.

Step 9: (c). By Step 6 (iv) and Step 8 (iv), $|\langle W_C^{(D)}\rangle_K - \langle W_C^{(D+1)}\rangle_K| \le 4\varepsilon'_D$ for $d \le D < K$, and the sum over D is $\le \tau'_d$, a number free of K, m, ℓ and C. As $W_C^{(D)}$ is a bounded gauge-invariant Borel function of V_D, $\lim_K \langle W_C^{(D)}\rangle_K$ exists by the convergence of the laws of the block fields (Corollary 2.8; (M4)(iii) of Section 17.2), and these limits form a Cauchy sequence in D, whose limit is w_C, with $|\mu_d(W_C) - \mathsf{w}_C| \le \tau'_d$. With (a) the last bound of (c) follows.

Step 10: (b). By Theorem AZ(1)(b), (c) and Proposition 17.38(c), the torus means of bounded Borel functions of finitely many bonds converge to ν_K as $\ell \to \infty$; on the tori $\ell > 2(R+T+2w)$, $W_C^{\mathrm{tap}} \circ \widetilde{\mathcal{M}}$ is $W_C^{\mathrm{tap}} \circ \mathcal{M}$. In Step 7 on the torus $\ell = \ell_0 L^j$, with $\mathsf{k} = 2N_k$, the two factors in front are at most $\exp \delta(n/2, n, n) \le e^{32\chi_k}$ and $\exp((\vartheta_I/\mathsf{k})\delta(\mathsf{k}, n-g-1, n)) \le \exp((\vartheta_I/2N_k)16\chi_k)$; at fixed R, T both tend to 1 as $j \to \infty$, since $k \to \infty$, $\chi_k \to 0$ and $\vartheta_I/N_k \to 0$, while $e^{-0.165\,\vartheta_I(g+1)/N^2} \le e^{-\sigma_{\mathrm{bl}}(R-w)(T-w)}$ does not move; so $|\nu_K(W_C^{\mathrm{tap}})| \le e^{-\sigma_{\mathrm{bl}}(R-w)(T-w)}$ for all $R, T \ge 2w$.

Step 11: (d). Let $J := T/w$ and $\vartheta = L^{K-d}$; the legs lie along the direction 0, their links c_j^{R}, c_j^{L} join the heights $j-1$ and j in units of w, $1 \le j \le J$, the link at the height j reads the block layers $j-1$ and j, and its middle plane parts them. (i) *One defect.* On a $(d-1)$-box let $x_c := |V_d(c) - \widetilde{V}(c)|$ for a link c of a leg; it is a gauge-invariant Borel function with $x_c \le 2$, as $V_d(c)$ and $\widetilde{V}(c)$ are transporters of one kind. By Step 6 (iv) along the taper of the one link and Step 8 (i), with $\omega_D \le 1/200L^2$ for $D \ge d$, $x_c \le \sum_{D=d}^{K-1}(13L^2\omega_D + 2 \cdot 1_{E_D(\square)})$ for the cell $\square$ of the side a_{d-1} that holds $Q(c)$. The seminorm is homogeneous and subadditive with $\|1\|_{\mathrm{ch}} = 1$ (Proposition YM3_I: 86.4), and it does not decrease on functions ≥ 0 of a cell, whose fold is the product of their copies; so Step 8 (iii) at $D' = d-1$ gives $\|x_c\|_{\mathrm{ch}} \le \tau_d^{\mathrm{ch}}$. (ii) *The expansion.* $W_C(V_d)$ is linear in each link. At a height j write the pair $(V_d(c_j^{\mathrm{R}}), V_d(c_j^{\mathrm{L}}))$ as the tapered pair $(\widetilde{V}(c_j^{\mathrm{R}}), \widetilde{V}(c_j^{\mathrm{L}}))$, plus a defect of the right leg, $(V_d(c_j^{\mathrm{R}}) - \widetilde{V}(c_j^{\mathrm{R}}), V_d(c_j^{\mathrm{L}}))$, plus a defect of the left leg, $(\widetilde{V}(c_j^{\mathrm{R}}), V_d(c_j^{\mathrm{L}}) - \widetilde{V}(c_j^{\mathrm{L}}))$. Expand from below at the heights $t_s := 1 + sL$, in the two defect terms again at the next height; in a term with the tapered pair at t_s expand from above at the heights $u_{s'} := J - s'L$; and stop at $s + s' = s_B := \lfloor (J-2)/L \rfloor$, which keeps $u_{s'} > t_s$. Then $W_C(V_d)$ is the sum of the main terms $f[\iota, s; \iota', s']$, $s + s' \le s_B$, $\iota \in \{\mathrm{R}, \mathrm{L}\}^s$, $\iota' \in \{\mathrm{R}, \mathrm{L}\}^{s'}$, with a defect of the leg ι_i at the height t_i for $i < s$, one of the leg ι'_i at the height u_i for $i < s'$, and tapered pairs at t_s and at $u_{s'}$, and of $(s_B+2)2^{s_B+1}$ remainders, each with exactly s_B+1 defects. (iii) *Main terms.* A main term is an f of Step 5 in which S_B consists of the block layers $0, \dots, t_s - 1$ and S_T of the block layers $u_{s'}, \dots, J$, of $\vartheta_B = (sL+1)\vartheta$ and $\vartheta_T = (s'L+1)\vartheta$ layers, S_I of the block layers between, $\vartheta_I = (J-1-(s+s')L)\vartheta$, $g = R' - \vartheta$, the rest $a = n - (J+1)\vartheta$, and the halves $A^{\mp}$ of the tapered links at t_s and $u_{s'}$ joining the caps or the legs as in Step 7; the legs $\mathbf{R}$, $\mathbf{L}$ are unitary, and $|\mathbf{B}| \le \beta_B$, $|\mathbf{T}| \le \beta_T$, the products of the defects x_c of the caps. In Step 5 the caps may be paid in the mean, with caps of different thicknesses: Schwarz's inequality with kernels of positive type bounds the trace norm of the operator of the caps through $\mathrm{Tr}(\hat\beta_B^* \mathsf{D}_{\mathbf{1}} \hat\beta_B \mathsf{D}^{a_B+1})$, which is $Z(\Lambda_B)\langle \beta_B^\Theta \beta_B\rangle_{\Lambda_B}$ by the trace formula of the head of the proof with the step between the two slabs set to $\mathbf{1}$ (Step 4 (ii)), and through the like number of the top, using $\mathsf{D}^{a+1} \le \lambda_0^{a-a_B}\mathsf{D}^{a_B+1}$; this gives

$$|\langle f\rangle_\Lambda| \le \big[(1 + S_0(2\vartheta_B + a_B))(1 + S_0(2\vartheta_T + a_T))\big]^{1/2} \big[\langle \beta_B^\Theta \beta_B\rangle_{\Lambda_B} \langle \beta_T^\Theta \beta_T\rangle_{\Lambda_T}\big]^{1/2} e^{(\vartheta_I/\mathsf{k})\delta(\mathsf{k}, n-g-1, n)} e^{-\varsigma\vartheta_I(g+1)}$$

on the shortened boxes $\Lambda_B = (2\vartheta_B + a_B, n, n)$ and $\Lambda_T = (2\vartheta_T + a_T, n, n)$, on which β_B stands on a slab of ϑ_B layers and β_B^Θ on the next one, for integers $2 \le a_B, a_T \le a$. Take for a_B the largest integer $\le a$ with $x_B := 2\vartheta_B + a_B \in 2L\vartheta\mathbb{Z}$; then $a_B \ge 2$, and as $T \le \ell/2$ and $n = 2L^{m+d-1} \cdot L\vartheta \ge 8L\vartheta$, $x_B > n/2 - 2L\vartheta \ge n/4$; so Λ_B has the sides $L\vartheta(\mathsf{n}_0, \mathsf{n}, \mathsf{n})$ with $\mathsf{n} = 2L^{m+d-1}$ and an even $\mathsf{n}_0 \ge \mathsf{n}/4 \ge 2L^{m+d-2} \ge \mathsf{n}(d-1)$: a $(d-1)$-box; the like for Λ_T. Number the block layers of Λ_B so that β_B stands on $0, \dots, t_s - 1$ and β_B^Θ on $t_s, \dots, 2t_s - 1$: the defect at t_i reads the block layers iL, $iL+1$, and its mirror image the block layers $(2s-i)L$, $(2s-i)L+1$. A board of the side Lw with its offset in $a_d\mathbb{Z}^3$ whose walls across the direction 0 part the block layers $\equiv 1$ and $\equiv 2$ modulo L cuts no block, puts each of the $2s$ double blocks into one cell and these cells into $2s$ different layers of cells; so the

chessboard estimate and (i) give $\langle \beta_B^{\Theta} \beta_B \rangle_{\Lambda_B} \le (\tau_d^{\mathrm{ch}})^{2s}$, and $\langle \beta_T^{\Theta} \beta_T \rangle_{\Lambda_T} \le (\tau_d^{\mathrm{ch}})^{2s'}$. Now $\varsigma = 0.165/N^2$, as $n \ge 26N$ by $\ell \ge 13\ell_0$; let $k' \ge 1$ be the largest index with $2N_{k'} \le n/4$, so that $n < 16N_{k'}$ and $N_{k'} \ge 8$: then $\log(1 + S_0(x_B)) \le \delta(\lceil n/4 \rceil, n, n) \le \frac{(n/4+1)n^2}{8N_{k'}^3} \chi_{k'} \le 132 \chi_{k'} \le 1.195$, the like for the top, the factor $e^{(\vartheta_I/\mathsf{k})\delta(\mathsf{k}, n-g-1, n)} \le e^{32\chi_k} \le e^{0.131}$ as in Step 7, and with $C_\sharp := e^{1.195+0.131} \le 3.77$ and $\varsigma \vartheta_I (g+1) \ge \sigma_{\mathrm{bl}}(R-w)(T-w-(s+s')Lw)$:

$$|\langle f[\iota, s; \iota', s'] \rangle| \le C_\sharp (\tau_d^{\mathrm{ch}})^{s+s'} e^{\sigma_{\mathrm{bl}}(R-w)Lw(s+s')} e^{-\sigma_{\mathrm{bl}}(R-w)(T-w)}.$$

(iv) *Remainders.* A remainder is at most the product of its $s_B + 1$ defects. The torus is a $(d-1)$-box, as $m + d - 1 \ge \mu(d-1)$; a board of the side Lw with its offset in $a_d \mathbb{Z}^3$ whose walls across the direction 0 part neither the block layers $\equiv 0, 1$ nor the block layers $\equiv J-1, J$ modulo L, which exists as $L \ge 3$, puts every double block of a defect into one cell, two defects from below, or two from above, into different cells, L block layers apart, and a defect from below at a height $t_i \le t_s - L$ and one from above at a height $u \ge t_s + 1$ into different layers of cells; so $|\langle \mathsf{r} \rangle| \le (\tau_d^{\mathrm{ch}})^{s_B+1}$. (v) *The sums.* Let $\upsilon := 2\tau_d^{\mathrm{ch}} e^{\sigma_{\mathrm{bl}}(R-w)Lw}$. The main terms sum to at most $C_\sharp e^{-\sigma_{\mathrm{bl}}(R-w)(T-w)} \sum_{s+s' \le s_B} \upsilon^{s+s'}$ and the remainders to at most $(s_B+2)(2\tau_d^{\mathrm{ch}})^{s_B+1}$, where $(s_B+1)Lw \ge T - w$ and $s_B Lw \le T - 2w$. If $\upsilon \le \frac{1}{2}$, the first sum is ≤ 4, and $(2\tau_d^{\mathrm{ch}})^{s_B+1} = \upsilon^{s_B+1} e^{-\sigma_{\mathrm{bl}}(R-w)Lw(s_B+1)} \le 2^{-(s_B+1)} e^{-\sigma_{\mathrm{bl}}(R-w)(T-w)}$ with $(s_B+2)2^{-(s_B+1)} \le 1$: the total is at most $(4C_\sharp + 1) e^{-\sigma_{\mathrm{bl}}(R-w)(T-w)} \le 17\, e^{-\sigma_{\mathrm{bl}}(R-w)(T-w)}$. The condition $\upsilon \le \frac{1}{2}$ is $R \le \varrho_d$; the lattice, the tori and the hypotheses are symmetric in the two sides, so $\min\{R, T\} \le \varrho_d$ suffices. If only $2\tau_d^{\mathrm{ch}} \le 1$: $\sum_{s+s' \le s_B} \upsilon^{s+s'} \le (s_B+1)^2 \max\{1, \upsilon^{s_B}\}$ and $\upsilon^{s_B} e^{-\sigma_{\mathrm{bl}}(R-w)(T-w)} \le (2\tau_d^{\mathrm{ch}})^{s_B}$, so the total is at most $(C_\sharp + 1)(s_B+2)^2 [e^{-\sigma_{\mathrm{bl}}(R-w)(T-w)} + (2\tau_d^{\mathrm{ch}})^{s_B}]$, with $C_\sharp + 1 \le 5$, $s_B + 2 \le 2 + T/Lw$, and $(2\tau_d^{\mathrm{ch}})^{s_B} = e^{-\kappa_d Lw s_B} \le e^{-\kappa_d(T-2w-Lw)} \le e^{-\kappa_d(T-3Lw)}$; with the legs along the longer side this is the second bound. The limit $K \to \infty$ follows by the convergence of the laws of the block fields, as in Step 9, and the bounds in ν_K by the convergence of Step 10, $W_C \circ \widetilde{\mathcal{M}}^{K-d}$ being a bounded Borel function of finitely many bonds that equals $W_C \circ \mathcal{M}^{K-d}$ on large tori, on which the hypotheses on ℓ hold. □

18.12. **Free laws of the action density at every order: the law on atoms in every state, and the tangent field of every limit and of every component, given by the free template.** This subsection treats the action density $\mathcal{O}$, obtained by summing the three plaquette orientations, or by using the scalar entry of the curvature tensor, at short distance and at every number n of entries. Building on Theorems F and S and on the results of the main text, we establish the following. Let $0 < g \le \gamma_W^{\mathrm{sd}}$ be a reference coupling, $n \ge 2$, $\Lambda \ge 1$ and $0 < \vartheta < 1$, and let $0 < \mathsf{d} \le \mathsf{d}_0$, where $\mathsf{d}_0 > 0$ is a number formed with n, Λ and g, into which ϑ does not enter. Let (K, β) be a first-passage datum at g with label b whose cutoff lies above the depth of the scale d, that is, $K \ge s_g(\mathsf{d}) + 1$, and let $\mathfrak{s}$ be any state at g at (K, β). For real Lipschitz functions $f_1, \dots, f_n$, none identically 0, each carried by a ball of radius d, any two of them with supports at distance at least d, all of them carried by one ball of radius $\Lambda\mathsf{d}$, and with $\mathrm{Lip}\, f_i \le \|f_i\|/(\vartheta \mathsf{d})$, the n-th joint cumulant of $\mathcal{O}$ under $\mathfrak{s}$ at $(f_1, \dots, f_n)$ differs from $3b^{-n} \mathcal{K}_{\mathrm{free}}^{(n)}(f_1, \dots, f_n)$, the sum of the rings of the dipole kernel $\mathbb{K}_3$ of three free photon fields, by at most $[\Phi_n(\mathsf{d}) + 3C_n^{\mathrm{rg}} a_K/(\vartheta\mathsf{d})]\, b^{-n} \prod_i \|f_i\|$; here $\Phi_n(\mathsf{d}) = \mathsf{C}_\sharp(n)(\log \frac{1}{\mathsf{d}})^{\mathsf{c}_\sharp(n)} \mathsf{d}^{13/256}$, the numbers $\mathsf{C}_\sharp(n)$ and $\mathsf{c}_\sharp(n)$ involve neither g nor Λ, and C_n^{rg} is defined below. If the entries are not required to be carried by balls of radius d, a bound of this form still holds when d is at most a number of the same kind, with the bracket $(6N_\Lambda)^n [\Phi_n(\frac{1}{2}\mathsf{d}) + 255\, C_n^{\mathrm{rg}} a_K/(\vartheta\mathsf{d})]$, $N_\Lambda = (32\Lambda + 3)^3$, and under $K \ge s_g(\frac{1}{2}\mathsf{d}) + 1$, in every state and on every torus of a side $\ell \in \mathfrak{L}(g)$ with $8\Lambda\mathsf{d} < \ell$; and it holds at every pair (K, β) with $\beta > X_g$ and $b = a_K \beta$, for the periodic Gibbs states and for the torus measures whose sides, multiplied by $L^{K-K'}$, lie in $\mathfrak{L}(g)$, with $\Phi_n(\frac{1}{2} L^{K-K'} \mathsf{d})$ in the bracket, the smallness condition imposed on $L^{K-K'}\mathsf{d}$ in the place of d, and the condition $K' \ge s_g(\frac{1}{2} L^{K-K'} \mathsf{d}) + 1$, where (K', β) is the first-passage datum at g with the same β. In the limit σ of every convergent sequence of periodic Gibbs states at any reference coupling $0 < g_0 \le \gamma$, in every component of its

canonical decomposition and in every limit in its frame, the label $b = \lim_j a_{K_j}\beta_j$ exists; for σ' any of these, every $n \geq 2$ and every tuple X of C^1 functions with compact, pairwise disjoint supports, with no bound on the ratio of its diameter to its least separation and none on the Lipschitz constants of its entries, $b^n S_n^{\sigma'}(X_r) \to 3\mathcal{K}_{\mathrm{free}}^{(n)}(X)$ as the real dilation factor r tends to 0, at the explicit rate $\mathsf{C}(\log\frac{1}{r})^{\mathsf{c}} r^{13/256}$, of the order $z^{\mathsf{c}} L^{-13z/256}$ along $r = L^{-z}$; and, as $r \to 0$, the functionals $X \mapsto \sigma'(X_r)$, for σ' the limit or one of its components, converge on separated tuples to the centred functional of the free template $S^{\mathrm{fr}}[3, b]$. So at every point y the tangent field of the action density, by which we mean the functional $X \mapsto \lim_{r\to 0} \sigma'((f_i(y + (\cdot - y)/r))_i)$, is this centred functional; its connected functions are the limits, at a fixed label, of the joint cumulants of the density $\frac{1}{2}\sum_c (\theta^c)^2$ formed with the curvatures θ^c of three free photon fields on lattice tori (Proposition 18.128(b)).

The law on atoms is obtained on one small torus. A dilation by L^{z}, with z fixed by d from both sides, carries a first-passage datum at g to a first-passage datum at a lowered coupling g', the length d to $\mathsf{d}' = L^{\mathsf{z}}\mathsf{d} \leq \mathsf{d}^{15/16}$, and every state at g to a state over the sides at g' from the least one, ℓ', on, whose size is a power of $\log\frac{1}{\mathsf{d}}$ (Lemma 18.119). On $T_{\ell'}$ the dilated entries are placed in one cell of a board with eight cells and copied into the seven other cells by the mirror maps. The renormalisation group of the main text, entered with these $8n$ sources, gives their jets at the top level as the Gaussian rings of the curvature slices up to $(\log\frac{1}{\mathsf{d}})^{\mathsf{c}}\mathsf{d}^{15/32}$ (Proposition 18.122); the rings give the free value (Lemma 18.123); the jets give the cumulants of the small torus (Lemma 18.121); and the chessboard estimate together with the pressure gap of Theorem S carries these cumulants to every state, at the price of an eighth root and of a factor $\mathsf{d}^{-1/128}$ (Lemma 18.120); indeed $\frac{1}{8}(\frac{15}{32} - \frac{1}{16}) = \frac{13}{256}$. A decomposition of the entries into atoms, the exact covariance of the lattice theory under the dilations by powers of L, and the passage to the limits give the rest.

Setting.

Definition 18.116 (Notation of this subsection)**.** (1) Use the canonical real and sourced construction of Theorems YM3_I:D, YM3_I:AJ and YM3_I:AD at the same fixed auxiliary data. Write $X_g = L(g^{-2} + B_\sharp)$, $\bar{B} = LB_\sharp/(L-1)$, and use the first-passage data $(K, \beta) \in \mathfrak{d}(g)$, running couplings $x_0 = \beta, \dots, x_K$, labels $b = a_K\beta$ and sides $\mathfrak{L}(g)$ of Proposition YM3_I:74.1. Here $R(x)$ is the least power of L at least $(\log x)^{\mathfrak{r}}$, with the room exponent $\mathfrak{r}$ of that construction. The ceilings $\gamma_W^{\mathrm{sd}} \leq \gamma_W^{\mathrm{ref}}$ are the small-side and reference thresholds in Section YM3_I:126. For a set $\mathfrak{l}$ of sides, a state over $\mathfrak{l}$ means a periodic Gibbs state in $\mathcal{G}_{K,\beta}$ or the periodic lift $\tilde{\mu}_{K,\ell,\beta}$ of the Wilson law for some $\ell \in \mathfrak{l}$. A state at g means a state over $\mathfrak{L}(g)$. Periodic Gibbs states are invariant under all lattice isometries by Proposition YM3_I:103.4. Set $g_\star = \gamma_W^{\mathrm{sd}}$, $\mathsf{c}_\star = L(g_\star^{-2} + \bar{B})$ and

$$Y_s^g = (s+1)\log L + \log(g^{-2} + \bar{B}), \quad C_W(s) = C_{W,0} A_0^2 (Y_s^g)^{2\mathfrak{p}_0}, \quad C_{W,0} = \max\{1, 4L^2 M/A_0^2\}.$$

The witness depth $s_g(\mathsf{d})$ is the least nonnegative integer s with $L^s\mathsf{d} \geq C_W(s)$. It exists by exponential growth of L^s, is nonincreasing in d, and tends to infinity as $\mathsf{d} \downarrow 0$. These are the room numbers of Section YM3_I:126.

(2) For a compactly supported real weight f put $\mathcal{O}_K(f) = \mathcal{O}_K^T(f\mathbf{1}) = a_K^{-1}\sum_p f(x(p))\, a_p$. For a plaquette array w use $\mathcal{O}(w) = a_K^{-1}\sum_p w(p)\, a_p$. Write $S_n^T[\nu]$ for their joint cumulants under ν and $X_K^\nu = \prod_i[\mathcal{O}_K(f_i) - \nu(\mathcal{O}_K(f_i))]$ for a tuple $X = (f_i)_i$. For the lifted torus law, collecting the plaquettes in each period class gives the torus cumulant with each f_i replaced by $f_i^{\langle\ell\rangle}(x) = \sum_{m\in\mathbb{Z}^3} f_i(x + \ell m)$. Indeed the plaquette action is constant on each period class. Write $\|f\| = \sup|f|$.

(3) For $u \neq 0$ let $\mathbb{K}_3(u) = (4\pi|u|^3)^{-1}(3\widehat{u}\widehat{u}^{\mathsf{T}} - \mathbf{1})$. Write $\mathrm{Cyc}(n)$ for the cyclic orders of $\{1, \dots, n\}$, represented by the permutations with first entry 1, and read $i+1$ modulo n. Put

$$k_n(x) = \frac{1}{2}\sum_{\varsigma\in\mathrm{Cyc}(n)} \mathrm{tr}\prod_{i=1}^n \mathbb{K}_3(x_{\varsigma(i)} - x_{\varsigma(i+1)}), \qquad \mathcal{K}_{\mathrm{free}}^{(n)}(f_1, \dots, f_n) = \int \prod_i f_i(x_i)\, k_n(x)\, dx$$

for bounded Borel functions with compact, positively separated supports. On a torus these integrals use compact lifts in a common ball of radius less than $\ell/4$. The functional of fixed dual directions $\iota \in \{1,2,3\}^n$ is the same integral with the trace product replaced by the product of entries $[\mathbb{K}_3]_{\iota_{\varsigma(i)},\iota_{\varsigma(i+1)}}$. Summing over the directions restores the trace. For the sum $\mathcal{O}$ over all three plaquette orientations the free template is $S_n^{\mathrm{fr}}[3,b] = 3b^{-n}\mathcal{K}_{\mathrm{free}}^{(n)}$. Proposition 18.152 derives this normalization, including its kernel at $n=2$.

On the torus at spacing a_K, let $C_K^{\mathrm{free}}(p,q)$ be the matrix of the orthogonal projection onto the exact two-forms, in the positive plaquette basis. Its Fourier formula and its comparison with $\mathbb{K}_3$ are in §YM3_I:121.1. Define the lattice ring

$$\mathcal{K}_K^{(n)}(f_1,\dots,f_n) = \frac{1}{2}\sum_{\varsigma\in\mathrm{Cyc}(n)}\sum_{p_1,\dots,p_n}\prod_i f_i(x(p_i))\prod_i C_K^{\mathrm{free}}(p_{\varsigma(i)},p_{\varsigma(i+1)}).$$

Let C_{lat} be a constant in the absolute error bound of equation YM3_I:(2245) and set $C_n^{\mathrm{rg}} = 22\,n!\,50^n(1+C_{\mathrm{lat}})^n$.

(4) Let $n\ge 2$, $\mathsf{d}>0$, $\vartheta\in(0,1)$ and $\Lambda\ge 1$. $P_n(\mathsf{d};\vartheta;\Lambda)$ consists of the tuples $X=(f_1,\dots,f_n)$ of real Lipschitz functions on $\mathbb{R}^3$, no one of which vanishes identically, subject to: (P1) all the supports lie in one closed ball $\bar{\mathbb{B}}(x_0,\Lambda\mathsf{d})$; (P2) two different supports are at distance at least d; (P3) $\mathrm{Lip}\,f_i \le \|f_i\|/(\vartheta\mathsf{d})$ for every i. The atoms form the subset $\mathfrak{A}_n(\mathsf{d};\vartheta;\Lambda)$ of the tuples that satisfy moreover (P4): every support lies in a closed ball $\bar{\mathbb{B}}(x_i,\mathsf{d})$ of radius d. For a torus T_ℓ with $8\Lambda\mathsf{d}<\ell$, $P_n^{T_\ell}(\mathsf{d};\vartheta;\Lambda)$ is formed in the same way from functions on T_ℓ and the distance of T_ℓ. We put $N_\Lambda := (32\Lambda+3)^3$, $\mathsf{d}_1 := \frac{1}{2}\mathsf{d}$, $\vartheta_1 := 2\vartheta/(1+84\vartheta)$ and $\Lambda_1 := 2\Lambda+1$.

(5) For $r>0$ set $f_r(x)=f(x/r)$ and $X_r=(f_{i,r})_i$. For integral z set $f^{[z]}=f_{L^z}$, and let $\nu^{(z)}$ be the image of ν by the lattice relabeling $x\mapsto L^z x$.

(6) Use the separated tuples and the Gibbs sequences of Definition 18.205. A limit in the frame of a convergent sequence $(K_j,\beta_j,\nu_j)_j$ means a functional σ' with $\sigma'(X)=\lim_j \rho_j(X_{K_j}^{\nu_j})$ at all separated tuples X, where each ρ_j is periodic Gibbs at the same (K_j,β_j). The centering in $X_{K_j}^{\nu_j}$ uses the means of ν_j. The canonical decomposition is the one in Theorems YM3_I:AL and YM3_I:AP.

(7) Let B_m count the partitions of an m-element set and put $\mathsf{a}_m=\sum_{\pi\in\mathrm{Part}(m)}|\pi|!$ and $\mathsf{a}'_m=\sum_{\pi\in\mathrm{Part}(m)}(|\pi|-1)!$. For $u_a>0$ write $u^\gamma=\prod_{a\in\gamma}u_a$. For a normalized linear map ω on a commutative algebra and $M_A=\omega(\prod_{a\in A}F_a)$, the cumulant is

$$\kappa_\gamma = \sum_{\pi\in\mathrm{Part}(\gamma)}(-1)^{|\pi|-1}(|\pi|-1)!\prod_{A\in\pi}M_A.$$

In the algebra $z_a^2=0$, this is the coefficient of z^γ in $\log(1+\sum_{A\neq\emptyset}z^A M_A)$. Exponentiating gives the inverse relation $M_\gamma=\sum_\pi\prod_{A\in\pi}\kappa_A$ and uniqueness. Adding constants c_a to the F_a multiplies the moment polynomial by $\exp(\sum_a z_a c_a)$, so only the singletons change. If $|M_A|\le\prod_{a\in A}o_a$ the cumulant has modulus at most $\mathsf{a}'_{|\gamma|}\prod_{a\in\gamma}o_a$. These identities also hold for a map into bounded measurable functions modulo null sets, since each polynomial is finite.

■

The free kernel and the atoms.

Lemma 18.117 (The free kernel; decomposition into atoms)**.** Let $n\ge 2$, $\mathsf{d}>0$, $\vartheta\in(0,1)$ and $\Lambda\ge 1$.

(a) If $x_1,\dots,x_n\in\mathbb{R}^3$ satisfy $|x_i-x_j|\ge\mathsf{d}$ for $i\neq j$, then $|k_n(x)|\le\frac{3}{2}(n-1)!\,(2\pi\mathsf{d}^3)^{-n}$. For $X\in P_n(\mathsf{d};\vartheta;\Lambda)$, $|\mathcal{K}_{\mathrm{free}}^{(n)}(X)|\le\frac{3}{2}(n-1)!\,(\frac{2}{3}\Lambda^3)^n\prod_i\|f_i\|$, and for $X\in\mathfrak{A}_n(\mathsf{d};\vartheta;\Lambda)$, $|\mathcal{K}_{\mathrm{free}}^{(n)}(X)|\le$

$\frac{3}{2}(n-1)!\,(\frac{2}{3})^n\prod_i\|f_i\|$. $\mathcal{K}^{(n)}_{\mathrm{free}}$ is linear in each entry, and it does not change under a common translation of the entries nor under $f_i\mapsto f_{i,r}$, $r>0$.

(b) For every real $r>0$ the map $f_i\mapsto f_{i,r}$ takes $P_n(\mathsf{d};\vartheta;\Lambda)$ into $P_n(r\mathsf{d};\vartheta;\Lambda)$ and $\mathfrak{A}_n(\mathsf{d};\vartheta;\Lambda)$ into $\mathfrak{A}_n(r\mathsf{d};\vartheta;\Lambda)$; a common translation of the entries keeps both classes; and both classes become larger when Λ increases or ϑ decreases.

(c) Every $X=(f_1,\dots,f_n)\in P_n(\mathsf{d};\vartheta;\Lambda)$ admits decompositions $f_i=\sum_{\mu\in\mathcal{M}_i}\epsilon_{i,\mu}a_{i,\mu}$ into at most $2N_\Lambda$ non-negative Lipschitz pieces $a_{i,\mu}$ with signs $\epsilon_{i,\mu}\in\{1,-1\}$, such that $\|f_i\|\le\|a_{i,\mu}\|\le 3\|f_i\|$, each piece is carried by the $\frac{\sqrt3}{8}\mathsf{d}$-neighbourhood of $\operatorname{supp}f_i$, and each choice of one piece for every entry gives a tuple $(a_{1,\mu_1},\dots,a_{n,\mu_n})$ in $\mathfrak{A}_n(\mathsf{d}_1;\vartheta_1;\Lambda_1)$. Hence, if T is defined on the n-tuples of bounded Borel functions whose compact supports are mutually at positive distance and is linear in each entry, a bound $|\mathsf{T}(Y)|\le\tau\prod_i\|a_i\|$ for all $Y=(a_1,\dots,a_n)\in\mathfrak{A}_n(\mathsf{d}_1;\vartheta_1;\Lambda_1)$ implies $|\mathsf{T}(X)|\le(6N_\Lambda)^n\tau\prod_i\|f_i\|$ for all $X\in P_n(\mathsf{d};\vartheta;\Lambda)$. ■

Sketch. (a) The matrix $\mathbb{K}_3(u)=(4\pi|u|^3)^{-1}(3\hat u\hat u^{\mathrm T}-1)$ has the eigenvalues $2(4\pi|u|^3)^{-1}$ and $-(4\pi|u|^3)^{-1}$ (twice), so its operator norm is $(2\pi|u|^3)^{-1}$; the trace of a product of 3×3 matrices is at most 3 times the product of their operator norms, and $\mathrm{Cyc}(n)$ has $(n-1)!$ members; this gives the bound on k_n. Integrating it against $\prod_i|f_i(x_i)|$ over supports of volume at most $\frac{4\pi}{3}(\Lambda\mathsf{d})^3$, respectively $\frac{4\pi}{3}\mathsf{d}^3$, gives the two bounds, as $\frac{4\pi}{3}/(2\pi)=\frac23$. Linearity follows from the integral. A common translation preserves the differences. Under a common dilation by r the r^{-3n} from the kernel cancels the Jacobian r^{3n}. (b) Under $f\mapsto f_r$ supports and their distances are multiplied by r, Lipschitz constants by r^{-1}, and suprema are kept. (c) Let $h:=\mathsf{d}/16$ and let $\chi_m(x)=\prod_\mu\max\{0,1-|x_\mu/h-m_\mu|\}$, $m\in\mathbb{Z}^3$, be the tents of mesh h: they form a partition of unity, χ_m is carried by $hm+[-h,h]^3\subset\bar{\mathbb{B}}(hm,\sqrt3h)$, and $\operatorname{Lip}\chi_m\le\sqrt3/h$: the three partial slopes each have modulus at most $1/h$. With $A_i:=\|f_i\|>0$ and M_i the set of the m whose cube meets $\operatorname{supp}f_i$, put $g_{i,m}:=(f_i+2A_i)\chi_m$ and $e_{i,m}:=2A_i\chi_m$; then $f_i=\sum_{m\in M_i}(g_{i,m}-e_{i,m})$, and the $a_{i,\mu}$ are the $g_{i,m}$, with the sign $+$, and the $e_{i,m}$, with the sign $-$. A member of M_i has $|hm-x_0|_\infty\le\Lambda\mathsf{d}+h$, so each coordinate of m ranges over at most $32\Lambda+3$ integers, and $\#M_i\le N_\Lambda$. Since $A_i\le f_i+2A_i\le3A_i$, $A_i\le\|g_{i,m}\|\le3A_i$ and $\|e_{i,m}\|=2A_i$; both functions are carried by $\bar{\mathbb{B}}(hm,\sqrt3h)\subset\bar{\mathbb{B}}(hm,\mathsf{d}_1)$, which is (P4) at d_1. Their Lipschitz constants are at most $A_i(\frac{1}{\vartheta\mathsf{d}}+\frac{48\sqrt3}{\mathsf{d}})\le\|g_{i,m}\|\frac{1/\vartheta+84}{\mathsf{d}}$, respectively $2A_i\frac{16\sqrt3}{\mathsf{d}}\le\|e_{i,m}\|\frac{1/\vartheta+84}{\mathsf{d}}$, and $\frac{1/\vartheta+84}{\mathsf{d}}=\frac{1}{\vartheta_1\mathsf{d}_1}$: this is (P3). A point of the support of a piece lies within $2\sqrt3h=\frac{\sqrt3}{8}\mathsf{d}$ of the support of its entry, so points of pieces of two different entries are at a distance at least $(1-\frac{\sqrt3}{4})\mathsf{d}>\mathsf{d}_1$, which is (P2), and every such point lies within $\Lambda\mathsf{d}+\frac{\sqrt3}{8}\mathsf{d}<(2\Lambda+\frac12)\mathsf{d}_1\le\Lambda_1\mathsf{d}_1$ of x_0, which is (P1). Expanding $\mathsf{T}(X)$ entry by entry, which is legitimate because every intermediate tuple has supports at positive distances, gives at most $(2N_\Lambda)^n$ terms, each at most $\tau\,3^n\prod_iA_i$. □

Lemma 18.118 (Lattice rings at atoms)**.** Let (ℓ,K) be a lattice, $n\ge2$, $\vartheta\in(0,1)$, $\Lambda\ge1$, and $\mathsf{d}>0$ with $4a_K\le\mathsf{d}$ and $4\Lambda\mathsf{d}<\ell$. For every $(\hat f_1,\dots,\hat f_n)\in\mathfrak{A}_n(\mathsf{d};\vartheta;\Lambda)$, with $f_i:=\hat f_i^{\langle\ell\rangle}$,

$$\Big|\mathcal{K}^{(n)}_K(f_1,\dots,f_n)-\mathcal{K}^{(n)}_{\mathrm{free}}(\hat f_1,\dots,\hat f_n)\Big|\le C^{\mathrm{rg}}_n\Big[\frac{a_K}{\vartheta\mathsf{d}}+\Big(\frac{\mathsf{d}}{\ell}\Big)^3\Big]\prod_i\|\hat f_i\|.$$

■

Proof. Set $a=a_K$, $d=\mathsf{d}$ and $A=\prod_i\|\hat f_i\|$. There is a unique compact lift of each participating barycentre in the common ball. Since $2\Lambda d<\ell/2$, every difference of two such lifts is a least lift. Two barycentres contributing to a ring lie in different supports, so their separation is at least d. By equation YM3_I:(2245), dividing each signed projection entry by a^3 gives the corresponding dipole entry with error at most

$$C_{\mathrm{lat}}d^{-3}[(a/d)^2+(d/\ell)^3].$$

The two orientation signs attached to a plaquette cancel in a cycle. The dipole entries are bounded by d^{-3} and the divided lattice entries by $(1+C_{\mathrm{lat}})d^{-3}$. Replacing the n factors one at a time costs at most $n(1+C_{\mathrm{lat}})^n d^{-3n}[(a/d)^2+(d/\ell)^3]$ for each cycle and each choice of barycentres. At a fixed orientation, the a-cubes centred at the contributing barycentres are disjoint and contained in a ball of radius $d+\sqrt{3}a/2$, of volume less than $8d^3$. Including the a^{3n} from the entries, the resulting error after summing over the 3^n choices of orientation and $(n-1)!$ cycles is at most $\frac{1}{2}n!\,24^n(1+C_{\mathrm{lat}})^n[(a/d)^2+(d/\ell)^3]A$.

For the Riemann sums, fix the orientations and one cycle. Let y_i be a grid barycentre and x_i a point of its cube. If either the integrand at x or the summand at y is nonzero, all n cubes meet the respective supports. Along the straight segment between x and y every separation between different entries is at least $d-\sqrt{3}a>0.56d$. The dipole entries there have modulus at most d^{-3} and directional derivative at most $8d^{-4}$ per unit displacement, by differentiating $\mathbb{K}_3$ and using $\|\partial_e\mathbb{K}_3(u)\|\le 9/(4\pi|u|^4)$ for $|e|=1$. Thus replacing one dipole factor by its value at the grid point costs at most $14ad^{-4}$. Each function value changes by at most $\|\hat f_i\|a/(\vartheta d)$, by the Lipschitz bound and $|x_i-y_i|\le\sqrt{3}a/2$. The difference of the two full integrands is at most $15n\,a(\vartheta d)^{-1}d^{-3n}A$ by telescoping. The union of cubes meeting the support of entry i lies in a ball of radius $d+\sqrt{3}a$, of volume less than $13d^3$. Integrating, and summing the orientations and cycles with the factor $1/2$, gives an error at most $\frac{15}{2}n!\,39^n a(\vartheta d)^{-1}A$. The integral sum over orientations is the trace defining the continuum ring. Since $(a/d)^2\le a/(\vartheta d)$, the two error bounds are below the stated one. □

One small torus at a lowered coupling.

Lemma 18.119 (One small torus at a lowered coupling)**.** Let $0<g\le\gamma_W^{\mathrm{sd}}$, let $C_{\mathrm{S}}\ge 1$ be the upper stability constant of Theorem YM3_I:AD enlarged if necessary, $\mathsf{C}_{\mathrm{Q}}:=32L^9M^3C_{\mathrm{S}}$ and $\mathsf{C}_{\mathrm{r}}:=16\cdot 2^{3\mathfrak{r}+1}\mathsf{C}_{\mathrm{Q}}$. For $0<\mathsf{d}<1$ with $16\mathsf{C}_{\mathrm{Q}}(Y_1^g)^{3\mathfrak{r}+1}\le\log\frac{1}{\mathsf{d}}$ put $\mathsf{z}(\mathsf{d}):=\max\{z\ge 1: 16\mathsf{C}_{\mathrm{Q}}(Y_z^g)^{3\mathfrak{r}+1}\le\log\frac{1}{\mathsf{d}}\}$; it is formed with the constants, g and d, and tends to ∞ as $\mathsf{d}\to 0$. For $(K,\beta)\in\mathfrak{d}(g)$ with run $(x_0,\dots,x_K)$ and label b, and for such d with $K-s_g(\mathsf{d})\ge 1$, the dilated data are: $\mathsf{z}:=\mathsf{z}(\mathsf{d})$; $k:=K-\mathsf{z}$; the coupling g' with $g'^{-2}=L^{-1}x_k-B_\sharp$; the least side $\ell'=2L^m$ at g'; $\mathsf{d}':=L^{\mathsf{z}}\mathsf{d}$; $b':=L^{\mathsf{z}}b$; $y_j:=\log x_j$; the depth $s':=s_{g'}(\mathsf{d}')$; the working level $\mathsf{k}:=\max\{k-s',2\}$; $N:=k-\mathsf{k}$; $Y:=Y_{s'}^{g'}$; $\hat{\mathsf{d}}_i:=\mathsf{d}'/a_{k-i}$ and $\hat{\mathsf{d}}:=\hat{\mathsf{d}}_{\mathsf{k}}=L^N\mathsf{d}'$; $\mathfrak{q}:=\max\{x_{\mathsf{k}}^{-1/2},x_kL^{-N}\}$; the set of sides $\mathfrak{l}':=\{\ell\in\mathfrak{L}(g'):\ell\ge\ell'\}$; and $\mathsf{Q}':=\exp[\frac{1}{8}\ell'^3(2C_{\mathrm{S}}+C_{g'})]$, where $C_{g'}=C_{\mathrm{st}}+\frac{9}{2}\log X_{g'}$ is the lower stability constant of [7, Section 2]. There is a number $\mathsf{d}_{\mathrm{sm}}(g)\in(0,1)$, formed with the constants and g, such that for every $(K,\beta)\in\mathfrak{d}(g)$ and every $0<\mathsf{d}\le\mathsf{d}_{\mathrm{sm}}(g)$ with $K-s_g(\mathsf{d})\ge 1$ the following holds.

(a) $\mathsf{z}\ge 1$, $L^{k-2}\ge 6M$ and $g'<g$; (k,β) is a first-passage datum at g', with the run $(x_0,\dots,x_k)$ and the label b'; for every state $\mathfrak{s}$ at g at (K,β) the image law $\mathfrak{s}^{(\mathsf{z})}$ is a state over $\mathfrak{l}'$ at (k,β), and $S_n^T[\mathfrak{s}](f_1,\dots,f_n)=L^{\mathsf{z}n}S_n^T[\mathfrak{s}^{(\mathsf{z})}](f_1^{[\mathsf{z}]},\dots,f_n^{[\mathsf{z}]})$ for all $n\ge 1$ and all bounded real functions f_i with compact supports; and $Y_{\mathsf{z}}^g-2\log L\le y_k\le Y_{\mathsf{z}}^g$.

(b) $x_k\le\mathsf{d}^{-1/16}$, $\mathsf{d}'\le\mathsf{d}^{15/16}$ and $\frac{3}{4}x_k\le b'\le\frac{5}{4}x_k$; $2My_k^{\mathfrak{r}}\le\ell'<2L^3My_k^{\mathfrak{r}}$ and $\ell'\ge 2LM$, and ℓ' is small over $(x_0,\dots,x_k)$ in the sense of Lemma YM3_I:125.3; for every $\ell\in\mathfrak{l}'$ the pressures at (k,β) satisfy $e^{(\ell'/2)^3(p_{\ell'}-p_\ell)}\le\mathsf{Q}'$; and $\mathsf{Q}'^8\le e^{\mathsf{C}_{\mathrm{Q}}y_k^{3\mathfrak{r}+1}}\le\mathsf{d}^{-1/16}$.

(c) $\log\frac{1}{\mathsf{d}}<\mathsf{C}_{\mathrm{r}}\,y_k^{3\mathfrak{r}+1}$.

(d) $2\le\mathsf{k}\le k-1$ and $N\ge 1$; $4MY^{2\mathfrak{p}_0}\le\hat{\mathsf{d}}<LC_{W,0}A_0^2Y^{2\mathfrak{p}_0}$; $a_k/\mathsf{d}'\le\frac{1}{4}$; $x_{\mathsf{k}}>L^Ng'^{-2}$; $Y-\log(2L^3)<y_{\mathsf{k}}\le Y$; and $\frac{15}{16}\log\frac{1}{\mathsf{d}}\le\log\frac{1}{\mathsf{d}'}\le Y\le 2\log\frac{1}{\mathsf{d}}$.

(e) $\mathfrak{q}\le\mathsf{d}^{15/32}$.

■

Proof. Write $s=s_g(\mathsf{d})$ and $\Xi=\log(1/\mathsf{d})$. By the reference threshold of Section YM3_I:126, $g^{-2}\ge 4\bar B$. The first-passage estimates in Proposition YM3_I:74.1 give $g^{-2}<x_K\le X_g$ and

$|x_j - L^{-(j-i)}x_i| \le \bar{B}$. In particular,

$$L^{\mathsf{z}}(g^{-2} - \bar{B}) < x_k \le L^{\mathsf{z}}(X_g + \bar{B}) = e^{Y_{\mathsf{z}}^g}, \qquad Y_{\mathsf{z}}^g - 2\log L \le y_k \le Y_{\mathsf{z}}^g.$$

The lower logarithmic bound follows from $g^{-2} - \bar{B} \ge L^{-1}(g^{-2} + \bar{B})$. Because $C_{\mathrm{S}} \ge 1$, $\mathsf{z}\log L \le Y_{\mathsf{z}}^g \le \Xi/16$. The witness room condition gives $s\log L \ge \Xi$. Thus $\mathsf{z} \le s/16$ and $k \ge 15s/16$, so $L^{k-2} \ge 6M$ for small d.

For (a), Corollary YM3_I:74.3 gives $g' < g$, $X_{g'} = x_k$, the first-passage datum with the shortened run, and $L^{\mathsf{z}}\ell \in \mathfrak{L}(g')$ whenever $\ell \in \mathfrak{L}(g)$. Relabeling the sites by $x \mapsto L^{\mathsf{z}}x$ preserves the bond Haar factors, plaquette products and coupling. It therefore takes the torus law to the law on the dilated torus. On the infinite lattice it takes the Gibbs kernels for finite bond sets to the corresponding kernels and preserves periodicity. Since $a_k = L^{\mathsf{z}}a_K$, the smeared action at the original spacing is L^{z} times the smeared action with the dilated weight. Multilinearity gives the asserted cumulant scaling.

For (b), $x_k, L^{\mathsf{z}} \le e^{Y_{\mathsf{z}}^g} \le \mathsf{d}^{-1/16}$ and $|b' - x_k| \le \bar{B}$. Lemma YM3_I:126.3 puts the least side at g' in the small-side family, with $MR(x_k) \le L^m \le L^2 MR(x_k)$ and $L^m \ge LM$. This gives the stated side bounds. At zero source, Theorem YM3_I:AD writes $p_\ell = \ell^{-3}E_\ell + \ell^{-3}\log \widetilde{Z}_\ell(0)$, where the vacuum densities differ by at most C_{S} and $-C_{g'} \le \ell^{-3}\log \widetilde{Z}_\ell(0) \le C_{\mathrm{S}}$. It follows that $p_{\ell'} - p_\ell \le 2C_{\mathrm{S}} + C_{g'}$. Using $C_{g'} \le C_{\mathrm{S}}(1 + \log g'^{-2}) \le C_{\mathrm{S}}(1 + y_k)$ and $y_k \ge 1$ gives

$$\log(\mathsf{Q}'^8) \le \ell'^3 C_{\mathrm{S}}(3 + y_k) \le \mathsf{C}_{\mathrm{Q}} y_k^{3\mathfrak{r}+1} \le \Xi/16.$$

For (c), the next integer fails the test defining z. Since $Y_{\mathsf{z}}^g + \log L \le 2y_k$ for $y_k \ge 3\log L$, the failure is exactly the required strict upper bound on Ξ.

For (d), the envelope at g' at depth $\tilde{s}$ is $L^{\tilde{s}}(x_k + \bar{B}) \le e^{Y^g_{\tilde{s}+\mathsf{z}+1}}$. At $\tilde{s} = s - \mathsf{z} + 1$ the available room $L^{\tilde{s}}\mathsf{d}' = L \cdot L^s \mathsf{d}$ is therefore at least the required one. Indeed, $C_W(s+2)/C_W(s) = (1 + 2\log L/Y_s^g)^{2\mathfrak{p}_0} \le L$ for $Y_s^g \ge 4\mathfrak{p}_0$. Thus $s' \le s - \mathsf{z} + 1 \le k$. Apply Lemma YM3_I:126.1 at the shortened run, with $d = \mathsf{d}'$. Its room and coupling bounds YM3_I:(2323)–(2324) give the assertions about $\mathsf{k}, N, \hat{\mathsf{d}}, a_k/\mathsf{d}'$ and y_{k}. The witness inequality gives $\log(1/\mathsf{d}') \le Y$. Minimality of s' gives

$$s'\log L < \log(1/\mathsf{d}') + 2\mathfrak{p}_0 \log Y + \log(LC_{W,0}A_0^2).$$

Since $Y = s'\log L + \log(x_k + \bar{B})$ and $y_k \le \Xi/16$, this implies $Y \le (17/16)\Xi + 2\mathfrak{p}_0 \log Y + C$ with a fixed constant C. As $Y \to \infty$ with Ξ, this gives $Y \le 2\Xi$ for small d. Part (b) supplies the lower bound $15\Xi/16$ for $\log(1/\mathsf{d}')$.

Finally, $x_{\mathsf{k}} > L^N = \hat{\mathsf{d}}/\mathsf{d}' \ge 1/\mathsf{d}' \ge \mathsf{d}^{-15/16}$. Also $x_k L^{-N} \le \mathsf{d}^{-1/16}\mathsf{d}^{15/16}$. Taking the maximum gives (e). All smallness conditions are uniform in the cutoff and the datum because they use the fixed constants, g, and quantities bounded above or below by functions of Ξ. In particular $y_k \to \infty$ uniformly as $\mathsf{d} \downarrow 0$ by the lower bound in (a). □

Transfer by mirrors.

Lemma 18.120 (Transfer by mirrors)**.** Fix $K \in \mathbb{N}$ and $\beta > 0$, a board of $\mathbb{L}_K$ with centers $o + a_i\mathbf{k}$, $\mathbf{k} \in \mathbb{Z}^3$, and cell side $a_i = \ell'/2$, an odd multiple of a_K, with $o \in a_K\mathbb{Z}^3$, an unbounded set $\mathfrak{l}$ of sides $\ell \ge \ell'$, and a number $\mathsf{Q}_0 \ge 1$ that bounds $e^{(\ell'/2)^3(p_{\ell'} - p_\ell)}$ for all $\ell \in \mathfrak{l}$, with pressures $p_\ell = \ell^{-3}\log Z_{K,\ell,\beta}$; let $\mathfrak{s}$ be a state over $\mathfrak{l}$ at (K,β), and $\mathcal{K} := \{0,1\}^3$. Let $n \ge 2$; let $f_1, \dots, f_n$ be bounded real functions on $\mathbb{R}^3$, each equal to 0 off the open cube interior to $\square_0$, with the mirror images $f_{i,\mathbf{k}'} := f_i \circ \hat{\mathsf{r}}^{-1}_{\mathbf{k}' \leftarrow 0}$; let $c_i \in \mathbb{R}$; and for $\emptyset \ne \gamma \subset I := \{1,\dots,n\} \times \mathcal{K}$ write κ_γ for the joint cumulant, under $\tilde{\mu} := \tilde{\mu}_{K,\ell',\beta}$, of $\mathcal{O}_K(f_{i,\mathbf{k}'}) - c_i$, $(i,\mathbf{k}') \in \gamma$. Suppose that, with numbers $u_i > 0$, $u_{(i,\mathbf{k}')} := u_i$, $C_1 \ge 1$ and $0 \le \eta_\times \le 1$, (i) $|\kappa_\gamma| \le C_1 u^\gamma$ when γ lies in one cell $\{1,\dots,n\} \times \{\mathbf{k}'\}$, and (ii) $|\kappa_\gamma| \le \eta_\times u^\gamma$ when γ meets two cells. If $\eta := 2\mathsf{Q}_0 B_{8n}^{1/8} C_1^n \eta_\times^{1/8} \le 1$, then

$$|S_n^T[\mathfrak{s}](f_1,\dots,f_n) - S_n^T[\tilde{\mu}](f_1,\dots,f_n)| \le \mathsf{a}_n(B_nC_1^n + 1)^{n-1}\,\eta \prod_i u_i.$$

■

Sketch. 1. One state against the small torus. For a function F of the cell $\square_0$, $|\mathfrak{s}(F)| \le \mathsf{Q}_0 \|F\|_{\mathrm{ch},\ell'}$. If $\mathfrak{s} = \tilde{\mu}_{K,\ell,\beta}$ with $\ell \in \mathfrak{l}$, this is $|\mathfrak{s}(F)| \le \|F\|_{\mathrm{ch},\ell}$ followed by the comparison of the two sides, $\|F\|_{\mathrm{ch},\ell} \le \|F\|_{\mathrm{ch},\ell'} e^{(\ell'/2)^3(p_{\ell'}-p_\ell)}$ (Proposition YM3_I:86.3 and Corollary YM3_I:86.6). If $\mathfrak{s} \in \mathcal{G}_{K,\beta}$, the domination of Proposition YM3_I:91.9 along a sequence of sides of $\mathfrak{l}$ that tends to ∞ gives $|\mathfrak{s}(F)| \le \liminf \|F\|_{\mathrm{ch},\ell}$, and the same comparison applies. *2. The seminorm of the small torus.* At the side $\ell' = 2a_i$ the period box is $\mathcal{K}$, and $\|F\|^8_{\mathrm{ch},\ell'} = \tilde{\mu}(\prod_{\mathbf{k}' \in \mathcal{K}} F^{[\mathbf{k}']})$ by the definition of the chessboard seminorm in Proposition YM3_I:86.3. *3. Moments.* For $\emptyset \ne J \subset \{1,\dots,n\}$ put $X_i := \mathcal{O}_K(f_i) - c_i$, a function of the inner bonds of $\square_0$: a plaquette centered in the open cell stays inside it, since the walls are midplanes and the barycenter is a half-lattice site in each plaquette direction, $m_J := \tilde{\mu}(\prod_{i\in J} X_i)$ and $G_J := \prod_{i\in J} X_i - m_J$. The mirror maps are isometries of $\mathbb{L}_K$, $\mathsf{r}\mathcal{O}_K(f) = \mathcal{O}_K(f \circ \mathsf{r}^{-1})$, and the copies of a real function are its images; so the copy of G_J in $\square_{\mathbf{k}'}$ is $P_{\mathbf{k}'} - m_J$ with $P_{\mathbf{k}'} := \prod_{i\in J}(\mathcal{O}_K(f_{i,\mathbf{k}'}) - c_i)$. Expand $\tilde{\mu}(\prod_{\mathbf{k}'}(P_{\mathbf{k}'} - m_J)) = \sum_{S\subset\mathcal{K}} (-m_J)^{8-|S|} \tilde{\mu}(\prod_{\mathbf{k}'\in S} P_{\mathbf{k}'})$ and write each moment as the sum over the partitions of $J \times S$ of the products of the κ_γ. The partitions without a block that meets two cells give $\prod_{\mathbf{k}'\in S} \tilde{\mu}(P_{\mathbf{k}'})$, and these terms add up to $\prod_{\mathbf{k}'\in\mathcal{K}}(\tilde{\mu}(P_{\mathbf{k}'}) - m_J) = 0$, the factor of $\mathbf{k}' = 0$ being 0. Every other term has a block that meets two cells; by (i), (ii), $|m_J| \le B_{|J|} C_1^{|J|} u^J$ and $B_{q_1}^{q_3} B_{q_1 q_2} \le B_{q_1(q_2+q_3)}$ for integers $q_1, q_2, q_3 \ge 1$ (distinct choices of partitions of the parts give distinct partitions of their union), they add up to at most $2^8 B_{8n} C_1^{8n} \eta_\times (u^J)^8$. So $\|G_J\|_{\mathrm{ch},\ell'} \le 2 B_{8n}^{1/8} C_1^n \eta_\times^{1/8} u^J$, and by step 1, $|\mathfrak{s}(\prod_{i\in J} X_i) - m_J| \le \eta u^J$. *4. Cumulants.* $S_n^T[\cdot](f_1,\dots,f_n)$ is the joint cumulant of the X_i, the constants dropping out at the order $n \ge 2$; in the cumulant formula in Definition 18.116(7), for a partition π, $|\prod_{J\in\pi}(m_J + \epsilon_J) - \prod_{J\in\pi} m_J| \le |\pi|\, \eta\, (B_n C_1^n + 1)^{|\pi|-1} \prod_i u_i$ when $|\epsilon_J| \le \eta u^J \le u^J$, and $\sum_\pi (|\pi|-1)!\,|\pi| = \mathsf{a}_n$. □

Jets and the cumulants of a torus.

Lemma 18.121 (Jets and cumulants of the history measures)**.** On a finite Wilson torus, use the canonical real progression and assume that the history split is available through level T_*, as in YM3_I:(2270). Let $m_t[F]$ be the unnormalized measure of the all-small-field history through level t with insertion F, and set $\mathsf{m}_t = m_t[1]$ and $\Phi^{(t)}[F] = dm_t[F]/d\mathsf{m}_t$. Let w_t be the normalized mass of histories that first leave the small-field branch at level t. For a finite set of real sources w_a, $a \in I$, put $F_a = \mathcal{O}(w_a)$ and let $J_S^{(t)}$ be their conditional cumulant with respect to $\Phi^{(t)}$ for $\emptyset \ne S \subset I$. In particular $J^{(0)}_{\{a\}} = F_a$ and $J_S^{(0)} = 0$ for $|S| \ge 2$. Write

$$\mathsf{s}_{t,S} = \operatorname*{ess\,sup}_{\mathsf{m}_t} |J_S^{(t)}|, \qquad \mathsf{o}_{t,S} = \inf_{c\in\mathbb{C}} \operatorname*{ess\,sup}_{\mathsf{m}_t} |J_S^{(t)} - c|.$$

When the source progression of Proposition YM3_I:122.1 is admitted through level t, let $D_t^\lambda(U)$ be the sourced exponent less the unsourced exponent, namely

$$D_t^\lambda(U) = \mathcal{O}\Big(\sum_a \lambda_a w_a\Big)(U) + \sum_{i\le t} \sum_{X,z} [E_\lambda^{(i)} - E_0^{(i)}](X,U,z).$$

Write $\mathcal{J}_S^{(t)}(U) = \partial_S D_t^\lambda(U)|_{\lambda=0}$. Then $J_S^{(t)}(W) = \mathcal{J}_S^{(t)}(U_t(W))$ almost everywhere on the passing set $\mathcal{V}_t^\chi$ of Proposition YM3_I:123.1, where $U_t(W)$ is the minimal background over the output W. On this set we take $\mathcal{J}_S^{(t)}(U_t(W))$ as the Borel representative whenever the source progression is admitted.

(a) The $J_S^{(t)}$ are bounded Borel functions, real m_t-almost everywhere, and

$$\Phi^{(t)}\Big[\prod_{a\in S} F_a\Big] = \sum_{\pi\in \mathrm{Part}(S)} \prod_{\gamma\in\pi} J_\gamma^{(t)}.$$

(b) For $1 \le j \le T_*$ and $|S| \ge 2$,

$$\mathsf{s}_{j,S} \le \mathsf{s}_{j-1,S} + \sum_{\pi \in \mathrm{Part}(S),\, |\pi| \ge 2} \mathsf{a}'_{|\pi|} \prod_{\gamma \in \pi} \mathsf{o}_{j-1,\gamma}.$$

For every constant c, the essential supremum of $|J^{(j)}_{\{a\}} - c|$ is nonincreasing in j.

(c) Let $1 \le T \le T_*$ and partition I into cells. Take $c_a \in \mathbb{R}$, set $X_a = F_a - c_a$, and write $\tilde{J}_{\{a\}} = J^{(T)}_{\{a\}} - c_a$ and $\tilde{J}_\gamma = J^{(T)}_\gamma$ for $|\gamma| \ge 2$. Let $u_a > 0$, $C_2 \ge 1$, and let $\bar{J}_\gamma$ be real and zero on sets γ meeting more than one cell. Suppose $\epsilon = \epsilon_{\mathrm{sc}} + \epsilon_{\mathrm{lk}} + \epsilon_{\mathrm{rare}} \le 1$, and, almost everywhere under m_T:

(i) $|\tilde{J}_\gamma - \bar{J}_\gamma| \le \epsilon_{\mathrm{sc}} u^\gamma$ and $|\bar{J}_\gamma| \le C_2 u^\gamma$ in a single cell.

(ii) $|\tilde{J}_\gamma| \le \epsilon_{\mathrm{lk}} u^\gamma$ if γ meets more than one cell.

(iii) For every nonempty $\gamma \subset I$,

$$\sum_{t=1}^{T} \mathsf{w}_t \left(1 + (u^\gamma)^{-1} \operatorname*{ess\,sup}_{\mathsf{m}_{t-1}} \left| \Phi^{(t-1)}\Big[\prod_{a \in \gamma} X_a \Big] \right| \right) \le \epsilon_{\mathrm{rare}}.$$

Then the Wilson cumulants κ_γ of the X_a satisfy $|\kappa_\gamma - \bar{J}_\gamma| \le \mathsf{C}_{|\gamma|} \epsilon u^\gamma$, where $\mathsf{C}_m = \mathsf{a}_m [6 m B_m (C_2 + 1)^m]^m$.

■

Proof. The history measure is positive, so $|m_t[F]| \le \|F\|_\infty \mathsf{m}_t$. This gives bounded Borel versions of the quotients, and real versions for real insertions. Use the algebra with variables z_a and relations $z_a^2 = 0$. If $z^S = \prod_{a \in S} z_a$, the conditional moment polynomial is $M_t = 1 + \sum_{S \ne \emptyset} z^S \Phi^{(t)}[\prod_{a \in S} F_a]$. The coefficient of z^S in $\log M_t$ is $J^{(t)}_S$ by the cumulant formula. Expanding the exponential recovers the moments and proves (a).

For the identification with the local exponent, the source weights in Proposition YM3_I:122.1 enter as a single linear array $\sum_a \lambda_a w_a$. The polydisc condition there is a bound on the sum of the norms of its source entries, so any finite array is allowed on a sufficiently small polydisc. In the proof of Proposition YM3_I:123.1 the exact sum rule is applied to that one array and then divided by the positive unsourced density. Thus the density of $m_t[\exp(\sum_a \lambda_a F_a)]$ relative to m_t is $\exp D^\lambda_t(U_t(W))$. The identity holds as an analytic germ. To differentiate, test against any bounded Borel function of W and expand the bounded insertions in the original finite-volume integral. On the other side the finite-lattice exponent is uniformly bounded on a smaller polydisc by the local Lipschitz estimates of Proposition YM3_I:122.1. Coefficient extraction shows that $\exp D^\lambda_t(U_t(W))$ and M_t agree modulo λ_a^2, almost everywhere. Taking the logarithm gives the stated identification.

For (b), let g_j be the nonnegative all-small-field step factor. Define $Q_j[G] = d\mathcal{M}_*(G g_j \mathsf{m}_{j-1})/d\mathsf{m}_j$. The pushforward identity YM3_I:(2269) gives $\mathsf{m}_j = \mathcal{M}_*(g_j \mathsf{m}_{j-1})$ and $\Phi^{(j)}[F] = Q_j[\Phi^{(j-1)}[F]]$. Thus $Q_j[1] = 1$ and Q_j is positive, with the essential supremum bound from level $j-1$ to level j. In the same finite algebra,

$$\log M_j = \log Q_j \left[\exp \left(\sum_{\gamma \ne \emptyset} z^\gamma J^{(j-1)}_\gamma \right) \right].$$

The coefficient of z^S in this logarithm is the sum, over partitions π of S, of the Q_j-cumulants with entries $J^{(j-1)}_\gamma$, $\gamma \in \pi$. Indeed products with overlapping subsets vanish, and for each partition the surviving terms are exactly its cumulant expansion. The one-block term is $Q_j[J^{(j-1)}_S]$. For the other terms subtract constants from the entries, apply the supremum bound to their products, and take the infimum over the constants. The cumulant formula gives the factor $\mathsf{a}'_{|\pi|}$. The singleton case gives $J^{(j)}_{\{a\}} - c = Q_j[J^{(j-1)}_{\{a\}} - c]$ directly.

For (c), let $\bar{M}_\gamma$ be the moment polynomial formed from $\bar{J}$. By (a), the analogous polynomial formed from $\tilde{J}$ is the conditional moment of the centered insertions at T. Telescoping each product of at most $m = |\gamma|$ factors gives

$$\Big|\Phi^{(T)}\Big[\prod_{a\in\gamma} X_a\Big] - \bar{M}_\gamma\Big| \le mB_m(C_2+1)^{m-1}(\epsilon_{\mathrm{sc}}+\epsilon_{\mathrm{lk}})u^\gamma, \qquad |\bar{M}_\gamma| \le B_m C_2^m u^\gamma.$$

Use the first-large-field split YM3_I:(2270). The all-small-field term has mass $1-\sum_{t=1}^{T}\mathsf{w}_t$. For a history first leaving at t, the nonnegative step factor has mass w_t after normalization, and multiplies $\Phi^{(t-1)}[\prod_{a\in\gamma}X_a]$. Thus (iii) controls the sum of the discarded contributions and the mass deficit of the all-small-field term. The full moment differs from $\bar{M}_\gamma$ by at most $3mB_m(C_2+1)^m\epsilon u^\gamma$. For a fixed γ of size m, divide the moments on each subset A by u^A. Both families of normalized moments are bounded by $R = 6mB_m(C_2+1)^m$, and their difference is at most $R\epsilon$. In a product indexed by a partition π, telescope again to bound the difference by $|\pi|R^{|\pi|}\epsilon$. The cumulant coefficients have moduli $(|\pi|-1)!$ and $\sum_\pi |\pi|! = \mathsf{a}_m$. Since the cumulants of the reference moments are $\bar{J}$, this proves (c). □

The jets of the mirrored sources on the small torus.

Proposition 18.122 (The jets of the mirrored sources)**.** Let $0 < g \le \gamma_W^{\mathrm{sd}}$ and $n \ge 2$. There are numbers $\mathsf{C}_{\mathrm{jt}}, \mathsf{c}_{\mathrm{jt}} \ge 0$, formed with the constants and n, and a number $\mathsf{d}_n^\circ(g) \in (0, \mathsf{d}_{\mathrm{sm}}(g)]$, formed with the constants, n and g, such that the following holds; $C_{\mathrm{tp}} \ge 1$ and $\delta_{\mathrm{tp}} > 0$ are the kernel constants of Proposition YM3_I:120.6, taking common constants for its three bounds, and $\mathsf{c}_{\mathrm{g}} := \max\{1, 1900\max\{1, C_{\mathrm{tp}}\}\delta_{\mathrm{tp}}^{-3}\}$. Let $(K,\beta) \in \mathfrak{d}(g)$ and $0 < \mathsf{d} \le \mathsf{d}_n^\circ(g)$ with $K - s_g(\mathsf{d}) \ge 1$, with the dilated data of Lemma 18.119, $\ell' = 2L^m$. Let $o \in a_k\mathbb{Z}^3$, let $(-m, o)$ be the board of $\mathbb{L}_k$ of side $\ell'/2$ with the cells $\square_{\mathbf{k}'} = o + \frac{\ell'}{2}\mathbf{k}' + [-\frac{\ell'}{4}, \frac{\ell'}{4}]^3$ and the mirror maps $\hat{\mathsf{r}}_{\mathbf{k}'\leftarrow 0}$, $\mathbf{k}' \in \{0,1\}^3$; let $I := \{1,\dots,n\}\times\{0,1\}^3$, whose eight cells are the sets $\{1,\dots,n\}\times\{\mathbf{k}'\}$; and let $F_1,\dots,F_n$ be real Lipschitz functions on $\mathbb{R}^3$ with:

(M1) $|x-o|_\infty \le \ell'/8$ for every i and every $x \in \operatorname{supp} F_i$;
(M2) for every i some closed ball of radius d' contains $\operatorname{supp} F_i$, and every weight w_a below is non-zero;
(M3) $|x-y|_\infty \ge \mathsf{d}'/\sqrt{3}$ whenever $x \in \operatorname{supp} F_i$, $y \in \operatorname{supp} F_j$ and $i \ne j$.

Put $f_{(i,\mathbf{k}')} := (F_i \circ \hat{\mathsf{r}}^{-1}_{\mathbf{k}'\leftarrow 0})^{\langle \ell'\rangle}$, $w_a := f_a \circ x$ on the bare plaquettes of $T_{\ell'}$, and $u_{(i,\mathbf{k}')} := b'^{-1}\|F_i\|$. The jets $J_S^{(t)}$ are those of Lemma 18.121 on (ℓ', k) through level k. Let $\upsilon_j := 1/(a_{k-j+1}x_{j-1})$, let $\mathcal{C}_{(j)}$ be the curvature slices of YM3_I:(2226) on (ℓ', k), $G_i := \sum_{\mathsf{k}<j\le i}\upsilon_j\mathcal{C}_{(j)}$, W_a the multiplication by w_a, and, for $\emptyset \ne S \subset I$ and $\mathsf{k} \le i \le k$,

$$n_S^{(i)} := \tfrac{3}{2}\sum_{\varsigma\in\mathrm{Cyc}(S)}\operatorname{Tr}\prod_{t=1}^{|S|}(G_iW_{\varsigma(t)}) \quad (|S|\ge 2), \qquad n_{\{a\}}^{(i)} := \tfrac{3}{2}\operatorname{Tr}(G_iW_a) + \mathcal{J}_{\{a\}}^{(\mathsf{k})}(\mathbf{1}),$$

where, for $|S| \ge 2$, $\mathrm{Cyc}(S)$ consists of the bijections ς from $\mathbb{Z}/|S|\mathbb{Z}$ onto S that send 1 to a member of S chosen once and for all; these cyclic orders of S number $(|S|-1)!$; the product is ordered, $G_iW_{\varsigma(1)}G_iW_{\varsigma(2)}\cdots G_iW_{\varsigma(|S|)}$, and as the trace is cyclic the sum does not depend on the chosen member. Let $\epsilon_{\mathrm{jt}} := \mathsf{C}_{\mathrm{jt}}(\log\frac{1}{\mathsf{d}})^{\mathsf{c}_{\mathrm{jt}}}\mathsf{d}^{15/32}$. There are numbers $0 \le \rho_{\mathrm{fo}} \le \epsilon_{\mathrm{rare}}$ that depend on the constants, n, the run $(x_0,\dots,x_k)$, ℓ' and d' only, and in particular not on o or on the F_i, such that:

(a) $|J_S^{(k)}(W) - n_S^{(k)}| \le \epsilon_{\mathrm{jt}}u^S$ for every $W \in \mathcal{V}_k^\chi$ and every $S \subset I$ with $1 \le |S| \le 8n$;
(b) $|n_S^{(k)}| \le \frac{3}{2}(n-1)!\,\mathsf{c}_{\mathrm{g}}^n u^S$ if S lies in one cell and $2 \le |S| \le n$, and $|n_S^{(k)}| \le 254\,(8n)!\,\mathsf{c}_{\mathrm{g}}^{8n}\mathsf{d}'^6u^S$ if S meets two cells and $|S| \le 8n$;

(c) the mean $c_a := \langle \mathcal{O}(w_a)\rangle_{k,\ell',\beta}$ is real, it is the same for the eight members $(i,\mathbf{k}')$ of a row, and $|n^{(k)}_{\{a\}} - c_a| \le (\epsilon_{\rm jt} + 2\rho_{\rm fo})u_a$;

(d) $\rho_{\rm fo} \le \epsilon_{\rm rare} \le \frac14$, $\mathsf{Q}'^8\epsilon_{\rm rare} \le 2\mathsf{d}^{15/16}$, and with $X_a := \mathcal{O}(w_a) - c_a$, for every $\emptyset \ne \gamma \subset I$,

$$\sum_{t=1}^{k} \mathsf{w}_t\Big(1 + (u^\gamma)^{-1}\operatorname*{ess\,sup}_{\mathsf{m}_{t-1}}\Big|\Phi^{(t-1)}\Big[\prod_{a\in\gamma} X_a\Big]\Big|\Big) \le \epsilon_{\rm rare},$$

with the notation of Lemma 18.121;

(e) $b'\upsilon_j \le \frac32$ for $\mathsf{k} < j \le k$.

■

Sketch. Put $\Xi = \log(1/\mathsf{d})$, $\eta_i = L^{-i}$, $g_i = x_i^{-1/2}$ and $\mathsf{ang}_p^{[i]}(A) = \eta_i(\mathrm{d}A)(p)$. Use the shift system of Section YM3_I:124 and the curvature kernels of Proposition YM3_I:120.6. A number formed with the constants, n and the dilated data is called tame if it is at most $\mathsf{C}Y^{\mathsf{c}}$ with C,c formed with the constants and n; as $Y \le 2\Xi$ and $\mathsf{q} \le \mathsf{d}^{15/32}$ (Lemma 18.119(d), (e)), a tame number times q is at most $\mathsf{C}(2\Xi)^{\mathsf{c}}\mathsf{d}^{15/32}$. Every condition laid on d below has the form $\Xi \ge \Xi_0$, with Ξ_0 formed with the constants, n and g: the conditions bear on $y_k \ge (\mathsf{z}(\mathsf{d}) - 1)\log L$, on $Y \in [\frac{15}{16}\Xi, 2\Xi]$, on $\mathsf{q} \le \mathsf{d}^{15/32}$ and on $\mathsf{d}' \le \mathsf{d}^{15/16}$, and a tame number times a positive power of d is at most 1 for large Ξ. To these we add, already here, the four inequalities that Lemma 18.123 and Proposition 18.124 use: $\delta_{\rm tp}L(L-1)\hat{\mathsf{d}} \ge \sqrt{3}(3\log L + \log 2)$, the hypothesis of the fluctuation-kernel bound of Proposition YM3_I:120.6 at the depth N; $162L^3C_{\rm tp}\hat{\mathsf{d}}^3e^{-\delta_{\rm tp}L\hat{\mathsf{d}}/\sqrt{3}} \le \mathsf{d}'$; $\mathsf{c}'_{\rm val}\mathsf{d}' \le 1$; and $\mathsf{C}'\Xi^{\mathsf{c}_{\rm jt}/8}\mathsf{d}^{13/256} \le 1$, with the numbers $\mathsf{c}'_{\rm val}$ of Lemma 18.123 and C' of Proposition 18.124, which are formed with the constants and n. They too have the form $\Xi \ge \Xi_0$: the first because $\hat{\mathsf{d}} \ge 4MY^{2\mathfrak{p}_0}$ grows with Ξ; the second because, besides, $2\mathfrak{p}_0 \ge 2$ puts the exponential below every power of d for large Ξ, while $\hat{\mathsf{d}}^3$ is tame and $\mathsf{d}' \ge \mathsf{d}$; the third because $\mathsf{d}' \le \mathsf{d}^{15/16}$; the fourth by its form. $\mathsf{d}^\circ_n(g)$ is the least of all the resulting thresholds and of $\mathsf{d}_{\rm sm}(g)$.

Step 1: geometry and settings. The maps between cells are compositions of reflections at their walls. The walls are lattice midplanes because $\ell'/(2a_k)$ is odd. Thus the maps preserve the lattice, Euclidean and sup distances, and send the first cell center to $o + (\ell'/2)\mathbf{k}'$. By (M1) each support lifts to its ball inside its cell and is at least $\ell'/8$ from its walls. The periodization preserves the sup norm. Distinct supports in one cell are at sup distance at least $\mathsf{d}'/\sqrt{3}$, and those in different cells at least $\ell'/4$ in the torus metric. Counting points on the three translated grids of plaquette centers gives $n_a := \#\operatorname{supp} w_a \le 81(\mathsf{d}'/a_k)^3$. A plaquette corner is at sup distance at most $a_k/2$ from its center. Hence the distance between distinct sets of corners, in units of the working-level spacing, is at least $\hat{\mathsf{d}}/\sqrt{3} - 1 \ge 3M$ by Lemma 18.119(d).

The shortened run is first passage at $g' \le \gamma_W^{\rm ref}$, $k \ge 3$ and $MR(x_k) \le L^m$. Thus Lemma YM3_I:126.2 gives the canonical progression, density and regularity conditions through level k. Proposition YM3_I:122.1 then admits the linear array of all the sources for parameters near zero at these finitely many levels. The source parameter radii may depend on the cutoff here. The derivative estimates below use the array norm to give bounds uniform in the cutoff. We take d small enough that also $g'^{-2} \ge x_{\rm J}$ of Lemma 4.6, which is possible since $x_k \to \infty$ and $X_{g'} = x_k$. Write $\delta_i = A_1x_i^{-1/2}y_i^{\mathfrak{p}_0}$ and $\varepsilon_i = A_0x_i^{-1/2}y_i^{\mathfrak{p}_0}$. They are nondecreasing in i in this run, since $y_i \ge 2\mathfrak{p}_0$ and the x_i decrease. The flow estimate in Proposition YM3_I:74.1 gives $\frac34L^{j-i}x_j \le x_i \le \frac54L^{j-i}x_j$ for $i \le j$ and $|a_{k-j}x_j - b'| \le \bar{B}a_{k-j}$. Since $x_j \ge 4\bar{B}$, $a_{k-j}x_j \ge (2/3)b'$. This last bound gives (e).

Step 2: the exact step for all the sources. Use the chart and lowering map of Proposition YM3_I:124.2. For a height i let ϱ satisfy eq. 2.viii, let $A \in \mathsf{C}\mathsf{c}_{i+1}(\varrho)$, and set $\widetilde{U} = e^{i\eta_{i+1}A}$, $v = v_A^{[i+1]}$, $\varpi = \nu_{i,v}$, and $\widetilde{U}^{\rm lw}(B) = e^{i\eta_iA^{\rm lw}(v,g_iB)}$. The chart conditions ensure that the array actions used below are defined at the upper and the lowered backgrounds. In the weighted progression use the single bare array $f_{\lambda,i}$ formed with the weight $\sum_a \lambda_a w_a$, as in Proposition YM3_I:122.1. For any finite

source family, the norm condition for that array holds on a polydisc about zero. The counterterms of $f_{\lambda,i}$ are the same as those of $f_{0,i}$. Its difference $g_\lambda = f_{\lambda,i} - f_{0,i}$ has bare-array action $D_i^\lambda(U) - D_i^\lambda(\mathbf{1})$ by the weight normalization YM3_I:(2256). Consequently the sum rule YM3_I:(2289) gives

$$(\mathbb{V}'[g_\lambda])^{\mathrm{tot}}(P[v];B) = D_i^\lambda(\widetilde{U}^{\mathrm{lw}}(B)) - D_i^\lambda(\widetilde{U}).$$

The whole-torus step integral for $f_{\lambda,i}$ is $\exp \mathsf{Sum}[\mathsf{St}_i(f_{\lambda,i})](P[v])$ by Theorem YM3_I:V, as in eq. 2.xii. Dividing by the integral at zero gives the ϖ-mean of $\exp(\mathbb{V}'[g_\lambda])^{\mathrm{tot}}$, since the step potentials are linear in the added array. By the update of the source families in Proposition YM3_I:122.1, the same quotient is $\exp[D_{i+1}^\lambda(\widetilde{U}) - D_i^\lambda(\widetilde{U})]$. Thus

$$e^{D_{i+1}^\lambda(\widetilde{U})} = \int e^{D_i^\lambda(\widetilde{U}^{\mathrm{lw}}(B))}\, d\varpi(B).$$

The holomorphic local families are uniformly bounded on the fluctuation box on a smaller source polydisc by the bounds of Proposition YM3_I:122.1, so Cauchy's formula permits coefficient extraction under the integral. Take the logarithm in the algebra $\lambda_a^2 = 0$ used in Lemma 18.121. The coefficient of $\prod_{a\in S}\lambda_a$ is

$$\mathcal{J}_S^{(i+1)}(\widetilde{U}) = \sum_{\pi\in\mathrm{Part}(S)} \kappa_\varpi\big(\mathcal{J}_\gamma^{(i)}(\widetilde{U}^{\mathrm{lw}}(\cdot)) : \gamma\in\pi\big).$$

Step 3: the Gaussian template. For a real symmetric $\mathcal{G}\ge 0$ and a real diagonal W with $\|\mathcal{G}^{1/2}W\mathcal{G}^{1/2}\| < 1$ put $\mathcal{D}[\mathcal{G}](F) := -\frac{3}{2}\operatorname{Tr}\log(1-\mathcal{G}W) + \frac{1}{2}(F, W(1-\mathcal{G}W)^{-1}F)$ for $\mathfrak{g}$-valued functions F of the bare plaquettes. Completing the square, $\int \exp\mathcal{D}[\mathcal{G}](F+\phi)\, d\gamma_\Gamma(\phi) = \exp\mathcal{D}[\mathcal{G}+\Gamma](F)$ for the centred Gaussian measure γ_Γ with the covariance $\Gamma\otimes 1_3$; indeed, $e^{\mathcal{D}[\mathcal{G}](F)}$ is the expectation of $e^{\frac{1}{2}(F+\zeta, W(F+\zeta))}$ over an independent Gaussian ζ with covariance $\mathcal{G}\otimes 1_3$. This formula follows by completing the square on the range of $\mathcal{G}$, so it holds for singular covariances as well. Integrating first in ζ and then in ϕ replaces that covariance by $(\mathcal{G}+\Gamma)\otimes 1_3$. These Gaussian integrals converge for source parameters near zero, which is the domain needed to extract the mixed derivatives. At $W = a_k^{-1}\sum_a \lambda_a W_a$, $\mathcal{G} = a_k G_i$ and $\Gamma = a_k \upsilon_{i+1}\mathcal{C}_{(i+1)}$, the mixed derivative $\partial_S \mathcal{D}[a_k G_i](F)|_0$ is $n_S^{(i)} + \mathsf{Q}_S^{(i)}(F)$, without the constant at $|S| = 1$, where $\mathsf{Q}_S^{(i)}(F) := \frac{1}{2}\sum_{p,p'} k_S^{(i)}(p,p')\langle F_p, F_{p'}\rangle$ and $k_S^{(i)} := a_k^{-1}\sum W_{a_1}G_i W_{a_2}\cdots G_i W_{a_q}$, the sum over the orders $(a_1,\dots,a_q)$ of S (expand $-\operatorname{Tr}\log(1-x)$; the orders of S fall into the cyclic classes). Hence, with $P_\gamma(\phi) := \mathsf{Q}_\gamma^{(i)}(F+\phi) - \mathsf{Q}_\gamma^{(i)}(F)$ and the Gaussian joint cumulants κ^{G} for the covariance $a_k\upsilon_{i+1}\mathcal{C}_{(i+1)}\otimes 1_3$,

$$n_S^{(i)} + \mathsf{Q}_S^{(i)}(F) + \sum_{\pi\in\mathrm{Part}(S)} \kappa^{\mathrm{G}}(P_\gamma : \gamma\in\pi) = n_S^{(i+1)} + \mathsf{Q}_S^{(i+1)}(F).$$

Step 4: the recursion down the chain classes. Let the radii be $\varrho_i := L^{-1}\varrho_{i+1} + \frac{11}{10}\mathsf{c}_{\mathfrak{h}}\delta_i$ below a top radius ϱ_k, with the constant $\mathsf{c}_{\mathfrak{h}}\ge 1$ of the shift system in Section YM3_I:124, and consider for $\mathsf{k}\le i\le k$ the bounds

$$(\mathrm{I}_i)\qquad |\mathcal{J}_S^{(i)}(e^{i\eta_i A}) - n_S^{(i)} - \mathsf{Q}_S^{(i)}(\mathsf{ang}^{[i]}(A))| \le e_{S,i}\qquad (A\in \mathsf{Cc}_i(\varrho_i),\ 1\le|S|\le 8n).$$

Let $\mathsf{k}\le i<k$, and suppose that $\varrho_{i+1}\le\frac{1}{2}\varrho_{\mathrm{ch}}$, $\delta_i\le\frac{1}{2}\varrho_{\mathrm{fl}}$ and $6L^{-2}\varrho_{i+1} + 8C_{\mathrm{db}}\delta_i < \varepsilon_{\mathcal{W}}\alpha_0$, which are the conditions of Step 2 at $\varrho = \varrho_{i+1}$ with room to spare, and that the number $\bar{e}_i$ formed below is at most 1. Then (I_i) implies (I_{i+1}), with the numbers $e_{S,i+1}$ that the display below forms from the $e_{\gamma,i}$, $\gamma\subset S$. Both suppositions hold at every such i for small d: by the recursion, ϱ_{i+1} is at most ϱ_k plus a number of the constants times $\delta(x_k)$, since $\delta(\cdot)$ does not increase and $x_i\ge x_k$; $\bar{e}_i\le \mathsf{c}_{\mathrm{e}}[\delta(x_k)+\varrho_k]$; and $\delta(x_k)$ and ϱ_k (Step 7) tend to 0 with d. The lowered form lies in $\mathsf{Cc}_i(\varrho_i)$ (Proposition YM3_I:124.2). The plaquette angles of the lowered form are those of the upper form plus the linear image $g_i\mathsf{T}_i^{\mathfrak{h}}B$ of the fluctuation field, with $\mathsf{T}_i^{\mathfrak{h}} = \pm\mathsf{T}_i^{\mathrm{cv}}\otimes 1$ of row sums at most $C_{\mathrm{tp}}^{\mathrm{row}}\eta_i^2$ ($C_{\mathrm{tp}}^{\mathrm{row}} > 0$ the row-sum constant in YM3_I:(2239), formed with the constants), up to a remainder

of the second order (Lemmas YM3_I:124.3 and YM3_I:124.4); so each $\mathcal{J}_\gamma^{(i)}$ at the lowered form is $n_\gamma^{(i)}$ plus a polynomial of degree two in the fluctuation field plus an error of the size $e_{\gamma,i}$ and of the third order in the fields. On the support of ϖ, where $|g_iB| < \delta_i$, the angles of the upper form plus the linear image are at most $2\eta_i^2 x_i^{-1/2}\mathsf{t}_i$ in modulus, $\mathsf{t}_i := x_i^{1/2}(L^{-2}\varrho_{i+1} + \mathsf{c}_\mathfrak{h}\delta_i)$, and Lemma YM3_I:124.4 bounds the remainder as well; so, with $\mathsf{m}_{\gamma,i} := x_i^{-1}a_k^{-1}\eta_i^4\mathsf{N}_\gamma^{(i)}$, $\mathsf{N}_\gamma^{(i)}$ being the sum defined at the end of this step, the polynomial is at most $\mathsf{p}_{\gamma,i} := 2\mathsf{t}_i^2\mathsf{m}_{\gamma,i}$ in modulus, the ℓ^1 norm of its coefficients of degrees one and two is at most $\mathsf{l}_{\gamma,i} := 4\max\{1, C_{\mathrm{tp}}^{\mathrm{row}}\}^2(1+\mathsf{t}_i)\mathsf{m}_{\gamma,i}$, and the error is at most $e'_{\gamma,i} := e_{\gamma,i} + \mathsf{c}_1 x_i^{-1/2}\mathsf{t}_i^3\mathsf{m}_{\gamma,i}$, with a number c_1 of the constants. For Step 8 put $p_i := \max_{1\le|\gamma|\le 8n}\mathsf{p}_{\gamma,i}/u^\gamma$, $l_i := \max_{1\le|\gamma|\le 8n}\mathsf{l}_{\gamma,i}/u^\gamma$ and $s_i := \max_{1\le|\gamma|\le 8n}(e'_{\gamma,i} - e_{\gamma,i})/u^\gamma$. For the conditional measure $\varpi = \nu_{i,v}$ in the admitted chart, Lemma 4.14(a) gives the component cumulant comparison at every order $q \le 16n$. To see the dependence on the order, place each tested component in its entry block and put all q insertions into a single quadratic-sort origin on the polydisc $|z_a| < w_\star/(q\mathsf{n}_\mathrm{f})$. The difference of connected terms between the perturbed step and the uncut Gaussian is bounded on that disc by $(K_\Pi\mathsf{n}_i^{\mathbb{V}} + K_P^\Pi\mathsf{p}_{\star,i})e^{-\lambda_{\mathrm{out}}d_{i+1}(X)}$ by YM3_I:(724). The mixed derivative vanishes unless X contains all entry blocks. Anchoring at one of these blocks and applying Cauchy's inequality bounds the sum of derivatives by

$$C_{\mathrm{E},0}(K_\Pi\mathsf{n}_i^{\mathbb{V}} + K_P^\Pi\mathsf{p}_{\star,i})(q\mathsf{n}_\mathrm{f}/w_\star)^q.$$

Here $\mathsf{n}_i^{\mathbb{V}} = O(\delta_i)$ and $\mathsf{p}_{\star,i} = O(\delta_i)$ by YM3_I:(2294): the latter decays exponentially in $y_i^{2\mathfrak{p}_0}$, whereas $\delta_i = A_1 y_i^{\mathfrak{p}_0}e^{-y_i/2}$ and $\mathfrak{p}_0 \ge 1$. Thus the error is $O_n(\delta_i)$ uniformly in the entry blocks and the admitted background. The Gaussian covariance obeys $|\mathbf{C}_v - \mathbf{C}_0| \le 2\mathsf{c}_\mathbf{C}\varrho_{i+1}/r_\alpha$, with the constants of Proposition YM3_I:124.5, and $\mathbf{C}_0 = (\mathsf{A}_i^{\mathrm{cv}})^{-1}\otimes 1$. The curvature covariance identity in YM3_I:(2297) gives $\eta_i\mathcal{C}_{(i+1)}$ after applying $\mathsf{T}_i^{\mathrm{cv}}$, with the identity on colors. The radius recursion bounds the combined error by $\bar e_i \le \mathsf{c}_\mathrm{e}[\delta_i + L^{-(k-i-1)}\varrho_k]$, with c_e depending only on the construction and n. For polynomials of degree two in a vector whose cumulants of odd orders and of orders at least 4 are at most $\bar e$ and whose covariance is within $\bar e$ of a covariance with entries at most $\mathsf{c}_\mathbf{C}$, the joint cumulants of ρ such polynomials differ from the Gaussian ones by at most $2\rho B_{2\rho}(\mathsf{c}_\mathbf{C}+1)^{2\rho-1}\bar e$ times the ℓ^1 norms of their coefficients (expand over the partitions of the factors into blocks; under the Gaussian measure only pairs survive); and Step 3 identifies the Gaussian value. The constants drop out of the cumulants of order at least 2. Expanding each cumulant of Step 2 multilinearly in the polynomials and the errors of its entries, bounding every term that holds an error by $\mathsf{a}'_{|\pi|}$ times the product of the bounds of its entries (Definition 18.116(7)), and comparing the term without an error with its Gaussian value as above, one obtains (I_{i+1}) with

$$e_{S,i+1} := \sum_{\pi\in\mathrm{Part}(S)}\Big\{\mathsf{a}'_{|\pi|}\Big[\prod_{\gamma\in\pi}(\mathsf{p}_{\gamma,i}+e'_{\gamma,i}) - \prod_{\gamma\in\pi}\mathsf{p}_{\gamma,i}\Big] + 2|\pi|B_{2|\pi|}(\mathsf{c}_\mathbf{C}+1)^{2|\pi|-1}\bar e_i\prod_{\gamma\in\pi}\mathsf{l}_{\gamma,i}\Big\}.$$

At the top $|\mathsf{ang}_p^{[k]}(A)| \le 2\eta_k^2\varrho_k$, so that $|\mathcal{J}_S^{(k)}(e^{i\eta_k A}) - n_S^{(k)}| \le \mathsf{o}_S := e_{S,k} + 2a_k^{-1}\eta_k^4\mathsf{N}_S^{(k)}\varrho_k^2$ on $\mathsf{Cc}_k(\varrho_k)$, where $\mathsf{N}_S^{(i)}$ is the sum over the orders of S and over the plaquettes of the products $|w_{a_1}||G_i|\cdots|G_i||w_{a_q}|$.

Step 5: sizes. By Proposition YM3_I:120.6, $|\mathcal{C}_{(j)}(p,p')| \le C_{\mathrm{tp}}(a_k/a_{k-j+1})^3 e^{-\delta_{\mathrm{tp}}\mathsf{R}/a_{k-j+1}}$ at the sup distance R of the barycentres. With $\mathsf{R} \ge \mathsf{d}'/\sqrt{3}$, $\upsilon_j|w_a|_\infty \le \frac{3}{2}u_a$ (Step 1), $n_a(a_k/a_{k-j+1})^3 \le 81\hat{\mathsf{d}}_{j-1}^3$, and $\sum_{l\in\mathbb{Z}}(t_0L^l)^3e^{-ct_0L^l} \le 3c^{-3}$ for $c, t_0 > 0$ (the greatest term is at most $\frac{27}{e^3}c^{-3}$ and the others at most $(\log L)^{-1}\int_0^\infty x^2e^{-cx}\,dx = \frac{2}{c^3\log L}$ together, and $L \ge 13$), every link that joins two different sources has $n_{a'}|w_{a'}|_\infty \sup|G_i(p,p')| \le \mathsf{c}_\mathrm{g}u_{a'}$, and $\le 13\mathsf{c}_\mathrm{g}(\mathsf{d}'/\ell')^3u_{a'}$ if the cells differ ($\mathsf{R} \ge \ell'/4$). In an order, and in a cyclic order, two neighbouring members are never the same source. Summing over the plaquettes one link at a time, each order $(a_1,\dots,a_q)$ of S, $q = |S|$, contributes to $\mathsf{N}_S^{(i)}$ at most $n_{a_1}|w_{a_1}|_\infty\,\mathsf{c}_\mathrm{g}^{q-1}\prod_{t\ge 2}u_{a_t}$; as $|w_a|_\infty \le \|f_a\| = b'u_a$ and $n_a \le 81(\mathsf{d}'/a_k)^3$ for every a (Step 1), this is at

most $81(\mathsf{d}'/a_k)^3\, b'\, \mathsf{c}_{\mathrm{g}}^{q-1} u^S$, and therefore $\mathsf{N}_S^{(i)} \le q!\, \mathsf{c}_{\mathrm{g}}^{q-1}\, 81(\mathsf{d}'/a_k)^3\, b'\, u^S$. Also $|n_S^{(i)}| \le \frac{3}{2}(q-1)!\, \mathsf{c}_{\mathrm{g}}^q u^S$ for $q \ge 2$; a cyclic order that passes through two cells crosses between cells at least twice, and $\frac{3}{2}\cdot 13^2 < 254$ gives the bound $254\,(q-1)!\, \mathsf{c}_{\mathrm{g}}^q (\mathsf{d}'/\ell')^6 u^S$. With $\ell' \ge 1$ this is (b).

Step 6: the start at the working level. We first bound the mixed families on every level $t \le \mathsf{k}$. Use the bare-array iteration and its norm in Prop. YM3_I:122.1, equations YM3_I:(2251) and YM3_I:(2253). Write θ_j for the step Lipschitz constant in the first equation, $N_{\star,j}$ for its array ceiling, and put

$$m_{j,a} = 12M^3\alpha_0^2 L^{k-j}|w_a|_\infty, \qquad v_{j,a} = \theta_j m_{j,a}.$$

Here θ_j is a fixed multiple of $\delta(x_j)$ and $\theta_j N_{\star,j}$ is fixed. By the running-coupling comparison and the first-derivative recurrence in equation YM3_I:(2258), $N_j(\partial_a \mathfrak{f}_{\lambda,j}|_0) \le 2m_{j,a}$ at $1 \le j < k$, and the first derivative family at level i has norm at most $Cv_{i-1,a}$ at output rate κ^+. Indeed the numerical condition for (2258) holds once $\theta(x_k)$ is small enough, since it is an upper bound for all the θ_j. The ordinary input is in the half-ball $N_j \le N_{\star,j}/2$ by the same weighted ceiling used in Prop. YM3_I:122.2. All these conditions hold for small d, since $x_k \to \infty$.

For $S \subset I$ of size $s \ge 2$, let $D_i(S)$ be the largest of the pairwise distances between the sets of plaquette corners, in units of the spacing at level i. Every member of $E_S^{(i)} = \partial_S E_\lambda^{(i)}|_0$ vanishes unless its domain meets all the carriers. Thus its domain size is at least $[D_i(S)/M - 2]_+$. The locality and diameter assertions are Lem. 4.6(a), (d). Here $D_i(S) = L^{t-i} D_t(S)$ for $i \le t$ and $D_t(S) \ge 3M$ for $t \le \mathsf{k}$ by Step 1.

We claim that for $1 \le i \le \mathsf{k}$ there is a constant A_s depending only on the construction constants and s such that

$$\|E_S^{(i)}\|_{i,\kappa^+} \le A_s \prod_{a\in S} v_{i-1,a}. \tag{E}$$

The first-step estimate in Lem. 4.6(e), (f) implies (E) at $i = 1$, since $v_{0,a}$ is a constant times $x_0^{-1/2} y_0^{\mathfrak{p}_0} L^k |w_a|_\infty$ and $y_0 \ge 1$.

To pass from j to $j+1$, first suppose (E) for S up to level j and for all smaller subsets. For $s \ge 2$ the derivative $\mathfrak{f}_{S,j}$ of the input array has no explicit weight member. Its norm at input rate κ is at most

$$\sum_{i=1}^{j} \omega_{j+1-i} e^{-4\beta_{\mathrm{s}}\kappa[D_i(S)/M-2]_+} A_s \prod_{a\in S} v_{i-1,a} \le H_s A_s \prod_{a\in S} v_{j,a}. \tag{L}$$

For this inequality use $v_{i-1,a}/v_{j,a} \le CL^{(j-i+1)/2}(j-i+2)^{\mathfrak{p}_0}$, which follows from the comparison of the x_i and their logarithms in Step 1. The resulting sum over the lag $h = j - i$ is finite uniformly in j, since $D_i(S)/M \ge 3L^h$ and the weights $\omega_{h+1} = (h+2)L^{-h}$. Call its bound H_s.

For a derivative of the step in q input directions $G_1, \dots, G_q$, the bound on the half-ball is

$$\big\|D^q \mathsf{St}_j[G_1,\dots,G_q]\big\|_{j+1,\kappa^+} \le C_q \theta_j N_{\star,j}^{1-q} \prod_b N_j(G_b). \tag{C}$$

For $q = 1$ this is the Lipschitz estimate. For $q \ge 2$ apply Cauchy's formula to the first derivative in direction G_1, in the other directions with radii $N_{\star,j}/[2(q-1)N_j(G_b)]$, omitting zero directions. The polydisc lies in the admitted array ball. This gives $C_q = [2(q-1)]^{q-1}$. Equivalently the prefactor in (C) is a constant times θ_j^q.

By the chain rule of Lem. 4.6(a) the output derivative is the sum over partitions of S. The partition with one block contributes at most $\theta_j H_s A_s \prod_{a\in S} v_{j,a}$ by (L). For a partition with at least two blocks use (L) for the nonsingleton blocks, which have smaller size, and $2m_{j,a} = 2v_{j,a}/\theta_j$ for a singleton. There are at most q singleton blocks, so the factor θ_j^q in (C) absorbs their denominators, as $\theta_j \le 1$. The resulting sum is at most $A_s^* \prod_{a\in S} v_{j,a}$, with A_s^* formed from the constants and the A_b for $b < s$. Choose A_s at least twice A_s^* and at least the first-step constant. Impose $\theta(x_k) \max_{2\le s\le 8n} H_s \le 1/2$. Then (E) follows by induction on the size and on the level. There are finitely many orders and the added condition holds for all sufficiently small d.

At any configuration whose lower-level pairs lie in the cores, sum (E) over the domains and marked points up to level $t \le \mathsf{k}$. In a contributing family fix a carrier $a \in S$ and anchor the domain in a cube containing one of its plaquette corners. At level i there are at most $C(1+\hat{\mathsf{d}}_t)^3 L^{3(t-i)}$ choices of anchor. A domain of size $d_i(X)$ has at most $8M^3(1+d_i(X))$ marked points. The domain count in Lem. YM3_I:88.1 bounds the number of size-$d_i(X)$ domains through the anchor by $2048\,9216^{d_i(X)}$. Choose $c>0$ small enough that $\kappa^+ - c > \log 9216$. The norm decay therefore bounds the anchored sum, including the marked points, by a fixed constant times $e^{-c[D_i(S)/M-2]_+}$. Using the scale comparison of v in (L), the lag sum is bounded by a convergent series with terms at most $C_s L^{(3+s/2)h}(h+2)^{s\mathfrak{p}_0} e^{-3c(L^h-1)}$, $h=t-i$. For $s\ge 2$ the explicit source in D_t^λ is linear and contributes no mixed derivative. We have proved

$$|\mathcal{J}_S^{(t)}(U)| \le C_s(1+\hat{\mathsf{d}}_t)^3 e^{-c[D_t(S)/M-2]_+} \prod_{a\in S} v_{t,a} \qquad (1\le t\le \mathsf{k}). \tag{J}$$

This holds on the real passing backgrounds of Prop. YM3_I:123.1 as well as on the chart at the working level. In particular, after dropping the exponential, it is at most $(C_n x_t^5)^s u^S$: $\hat{\mathsf{d}}_t \le x_t$ and $v_{t,a} \le C\sqrt{x_t} y_t^{\mathfrak{p}_0} u_a \le C' x_t^{3/4} u_a$ by the coupling comparison. At $t=\mathsf{k}$ we have $D_t(S)/M \ge \hat{\mathsf{d}}/(\sqrt{3}M) - 1/M$, $\hat{\mathsf{d}} \ge 4MY^{2\mathfrak{p}_0}$, and $y_{\mathsf{k}} \le Y$. Hence the exponential in (J) absorbs $x_{\mathsf{k}}^{s/2}$ and leaves a tame number times $e^{-Y} \le \mathsf{q}$. Since $G_{\mathsf{k}}=0$, both the ring and quadratic terms of order at least two vanish at this level. This proves the corresponding initial estimates of (I_{k}).

For a singleton, the recursion for ϱ_i gives $\varrho_{\mathsf{k}} \le \mathsf{y}\mathsf{q}$, where $\mathsf{y} = 1 + 2\mathsf{c}_{\mathfrak{h}} A_1 Y^{\mathfrak{p}_0}$. Put $\varrho' = 2\varrho_{\mathsf{k}}$. For small d, the chart is regular at radius $3\varrho'$ and its lower pairs lie in the cores by Lem. 2.3. Apply Lem. YM3_I:124.4 at this chart with zero fluctuation. The difference between its plaquette action and half the squared linear angle is at most $C\eta_{\mathsf{k}}^4 \varrho'^3$ per plaquette. After smearing, this costs at most $C\hat{\mathsf{d}}^{\,3} x_{\mathsf{k}} \varrho'^3 u_a$. For the first-order source families apply the regular background estimate YM3_I:(2264) with $T=\mathsf{k}$ and $\varepsilon = 3\varrho'$. Its bound on their variation from the identity is at most

$$C(1+\hat{\mathsf{d}})^3[\varrho'^2 + e^{-d_*}]\delta(x_{\mathsf{k}}) x_{\mathsf{k}} u_a, \qquad d_* = \min\left\{\frac{c_{\mathrm{A}}\alpha_0}{3M\varrho'}, \frac{L^{m+N}-4}{2M}\right\} - 1.$$

Both entries in the minimum exceed $Y+1$ for small d, since $\varrho' \le 2\mathsf{y}\mathsf{q}$, $L^m \ge LM$ and $L^N \ge 1/\mathsf{d}'$. Thus $e^{-d_*} \le x_{\mathsf{k}}^{-1}$. Also $x_{\mathsf{k}}\mathsf{q}^2 \le C$: the $x_{\mathsf{k}}^{-1/2}$ term gives 1, and the other term is bounded by $Cx_k^3 L^{-N}$ using $x_{\mathsf{k}} \le (5/4)L^N x_k$. Here $x_k \le \mathsf{d}^{-1/16}$ and $L^{-N} \le \mathsf{d}' \le \mathsf{d}^{15/16}$. Consequently each bound is at most a tame number times $\mathsf{q}u_a$, using $\delta(x_{\mathsf{k}}) = A_1 x_{\mathsf{k}}^{-1/2} y_{\mathsf{k}}^{\mathfrak{p}_0}$. The constant of the singleton at the working level is $\mathcal{J}_{\{a\}}^{(\mathsf{k})}(\mathbf{1})$, and its quadratic part is precisely half the smeared squared linear angle. This proves (I_{k}) with $e_{S,\mathsf{k}} \le \mathsf{E}\mathsf{q}u^S$, where E is tame.

Step 7: the top. Write $N_0 = \ell'$ for the number of sites along each axis of the level-k torus. By the passing-test argument in Prop. YM3_I:68.1, every $W \in \mathcal{V}_k^\chi$ has $\varpi(W) \le 2\varepsilon_k$, where $\varepsilon_k = \varepsilon(x_k)$. For small d this is below $\varepsilon_{\mathcal{W}}\alpha_0$.

The periodic gauge. Put $\epsilon = \varpi(W)$, and let H_μ be the holonomy of the full positive axis through 0 in direction μ. Work with representatives $x \in \{0,\dots,N_0-1\}^3$. Take $g(x)$ to be the transport from 0 to x along the path which traverses its coordinate segments in order. Swapping two consecutive steps in different directions changes the transport by at most ϵ in operator norm: the two products differ by multiplication by a plaquette holonomy, and the other factors are unitary. This holds also for a swap across a periodic seam.

If a positive bond $b = (x, x+e_\mu)$ does not cross a seam, moving its last step in the path to x followed by b to the place of direction μ changes at most $2N_0$ adjacent pairs. Hence $|W^g(b) - \mathbf{1}| \le 2N_0\epsilon$. For a seam bond first move the N_0-1 steps in direction μ in the path to x to the beginning. This makes at most $2N_0^2$ swaps. Then move the seam step past the remaining transverse segment, at a cost of at most $2N_0$ swaps. The resulting path is the full positive axis followed by the path to the

endpoint of the seam bond, by periodicity. Thus $|W^g(b) - H_\mu| \le CN_0^2\epsilon$. Similarly the two paths with N_0 steps in direction μ followed by N_0 in direction ν, and the reverse order, require N_0^2 swaps. Their transports are $H_\mu H_\nu$ and $H_\nu H_\mu$, so $|H_\mu H_\nu - H_\nu H_\mu| \le N_0^2\epsilon$.

Write $H_\mu = a_\mu \mathbf{1} + i\mathbf{v}_\mu \cdot \sigma$ in quaternion coordinates and $D = N_0^2\epsilon$. The operator norm of the commutator difference is $2|\mathbf{v}_\mu \times \mathbf{v}_\nu|$. If $\max_\mu |\mathbf{v}_\mu| \le \sqrt{D}$, replace each H_μ by its nearest central sign. The change has norm at most $\sqrt{2}\sqrt{D}$ since $2(1 - |a_\mu|) \le 2|\mathbf{v}_\mu|^2$. Otherwise choose the direction of a largest $\mathbf{v}_\mu$ as a common axis. The component of any $\mathbf{v}_\nu$ perpendicular to it has modulus at most $D/(2|\mathbf{v}_\mu|) \le \sqrt{D}/2$. Discard these perpendicular components and renormalize the remaining quaternions to unit norm. For $D < 1/4$ the renormalization changes the result by at most the modulus of the discarded component. In either case there are commuting approximants $\widetilde{H}_\mu = e^{iB_\mu}$ with commuting Hermitian logarithms B_μ of norms at most π, and $|H_\mu - \widetilde{H}_\mu| \le C\sqrt{D}$.

In this fundamental box make the additional gauge change $s(x) = \exp[-i\sum_\mu x_\mu B_\mu/N_0]$. For the bond field with interiors equal to $\mathbf{1}$ and seams equal to $\widetilde{H}_\mu$, this gauge produces e^{iB_μ/N_0} on every bond of direction μ. On an interior bond this is the quotient at consecutive representatives, and on the seam the extra factor $\widetilde{H}_\mu$ gives the same quotient. The errors in the preceding comparisons are unchanged under multiplication on the left and right by these unitary gauge factors. The intrinsic angle distance on SU(2) is at most $\pi/2$ times the operator norm of the difference, since the chord for angle $r \in [0, \pi]$ is $2\sin(r/2)$. Its triangle inequality now gives a real bond logarithm B of W^{sg} with

$$\sup|B| \le \mathsf{b}_k := \frac{\pi}{\ell'} + C\big(\ell'^2\varepsilon_k + \ell'\sqrt{\varepsilon_k}\big).$$

Here the coarse gauge is defined on the torus by the chosen representatives. Both $D < 1/4$ and the smallness of b_k hold uniformly in W for small d, since $\ell' \asymp y_k^{\mathfrak{r}}$ and $\varepsilon_k = A_0 y_k^{\mathfrak{p}_0} e^{-y_k/2}$.

The critical response. Use the constant whole-torus tower of depth k, in level-k units. Its lower layers are empty, its reduced datum consists of all top bonds, and its constraint map is exactly $\mathcal{M}^k$. The region-separation inequalities are automatic for whole-torus regions. The fixed powers used for the partition divide the side in the respective level units once ℓ' is large. The uniform global response hypotheses are those of the constant-tower case following equation YM3_I:(2132); its side condition, $2L^{m+1} \ge C_L M$, holds for small d since $\ell' \to \infty$. In particular the top lattice has the fixed number of sites required in Thm. YM3_I:E. Take the reference datum $\mathbf{1}$ and the reference critical configuration $\mathbf{1}$, which has zero action and is the axial representative of the regular orbit over this datum. Apply the theorem to the perturbation e^{iB}, with $\varepsilon_1 = \varepsilon_k$. The curvature hypothesis for the reference is automatic, and ε_k and b_k tend to zero and thus fit the two fixed radii.

The theorem and Lem. YM3_I:51.6 give a real potential $A = \mathcal{H}(B)$ on the bare bonds and an axial representative $\Phi_{\mathbf{1}}(B) = (e^{i\eta_k A})^{u_B}$ of the regular critical orbit over e^{iB}. The response is defined holomorphically on the admitted datum ball. In equation YM3_I:(838) the local pillar scales are 1 for this constant tower. Integrating its derivative bound along sB, $0 \le s \le 1$, gives

$$\max\{|A(b)|, |\nabla^{\eta_k} A(b)|, |D^{\mathsf{T}} DA(b)|\} \le B_5 \sum_{c>0} e^{-\delta_{\mathrm{tw}} d(y, y_c)} |B(c)|.$$

Here y is a pillar top at an endpoint of b and y_c is an endpoint of the top bond c. This is also the integrated first-difference estimate YM3_I:(997). The sum of exponential weights is uniformly bounded on the level-k torus, as in the constant-tower application in the proof of Lem. YM3_I:113.6. Indeed the top lattice has unit spacing and the number of bond endpoints in successive distance shells is bounded by a fixed polynomial in the shell index. The derivatives here are with the identity as reference.

For the current use the current estimate of Thm. YM3_I:E with $\mathsf{L}(b) = 1$. The current is $D_U^{\mathsf{T}} F_U$ at bare spacing η_k, the normalization of equation YM3_I:(69). Its norm is unchanged by removing the gauge u_B, by covariance of the plaquette and divergence. We obtain

$$\max\{\sup|A|, \sup|\nabla^{\eta_k} A|, \sup|J^{(k)}(e^{i\eta_k A})|\} \le \mathsf{c}_{\mathrm{top}}(\mathsf{b}_k + \varepsilon_k) =: \varrho_k,$$

with a constant formed from the fixed construction constants.

The passing-field background $U_k(W)$ of Prop. YM3_I:123.1 is in the whole-torus regular critical orbit given by Thm. YM3_I:E, as in the proof of Prop. YM3_I:68.1. Extend sg from the top sites to the bare sites. The constraint and the Wilson action are gauge covariant, so this gauge carries the regular critical orbit over W to the one over $W^{sg} = e^{iB}$. The regular-orbit uniqueness in Thm. YM3_I:E now identifies $e^{i\eta_k A}$ as a gauge transform of $U_k(W)$. Together with the current bound and the cutoff on $\varpi(W)$, this is exactly membership of A in $\mathsf{Cc}_k(\varrho_k)$.

Finally, the lower-level pairs of the passing background are admitted by Prop. YM3_I:123.1, and the same is true after a real gauge change. Use the admitted plain triples of the real backgrounds. The local norm inequalities in Def. YM3_I:49.1 are gauge invariant. For a plain triple, the complex and real data on each smaller tower coincide, so the relative logarithm is zero. The averaged real data transform by endpoint gauges, and their regular critical orbits transform by covariance of the constraint and uniqueness. The continued background at zero perturbation is the real background. Admissible positive continuation radii exist at both data by the same real-background hypotheses, whose bounds are gauge invariant. This preserves the core conditions. In particular the relative exponent of the two gauge-related backgrounds is the same: the local terms are gauge-invariant functions of the admitted pairs by Def. YM3_I:49.4 and the explicit plaquette source is gauge invariant. Differentiating gives $J_S^{(k)}(W) = \mathcal{J}_S^{(k)}(e^{i\eta_k A})$ for every S. The bound on b_k and $y_k \to \infty$ give $\varrho_k \to 0$, so the top radius and all the smallness conditions in Step 4 hold for small d.

Step 8: summing the error recursion. Write $h = i-\mathsf{k}$. The radius recursion and $\delta_{i+j} \le \sqrt{5/4}L^{j/2}\delta_i$ give

$$\varrho_{i+1} \le L^{-(k-i-1)}\varrho_k + C\delta_i.$$

Also $x_i \le (5/4)L^{k-i}x_k$ and $\sqrt{x_k}L^{-N/2} \le \sqrt{\mathsf{q}}$. Hence $\mathsf{t}_i \le C(1+Y^{\mathfrak{p}_0})L^{h/2}$. By Step 5 and $b'/a_{k-i} \le (3/2)x_i$,

$$\mathsf{m}_{\gamma,i} \le C\,(\mathsf{d}'/a_{k-i})^3 u^\gamma = C\hat{\mathsf{d}}^{\,3}L^{-3h}u^\gamma,$$

where C can depend on the maximal order $8n$. Together with $x_i^{-1/2} \le CL^{h/2}x_{\mathsf{k}}^{-1/2}$ these give

$$p_i \le \mathsf{p}L^{-2h}, \quad l_i \le \mathsf{l}L^{-5h/2}, \quad s_i \le \mathsf{u}L^{-h},$$

where p,l are tame and u is at most a tame multiple of q. In the covariance error, $\delta_i \le CL^{h/2}\delta_{\mathsf{k}}$, so

$$\bar{e}_i \le CL^h(\delta_{\mathsf{k}} + L^{1-N}\varrho_k).$$

Put

$$e_0 = \max_{1\le|S|\le 8n} e_{S,\mathsf{k}}/u^S + \mathsf{u} + \delta_{\mathsf{k}} + L^{1-N}\varrho_k.$$

Step 6 and $L^{-N} \le \mathsf{q}$ bound e_0 by a tame multiple of q. In particular $Ne_0 \le 1$ for small d, since $N \le Y/\log L$.

We prove by induction on q that $e_{S,i}/u^S \le \Gamma_q e_0$ for $|S| \le q$ and every $\mathsf{k} \le i \le k$, with tame Γ_q. In the recurrence of Step 4 the partition with one block contributes $e_{S,i}/u^S$, plus an increment bounded by $s_i + C\bar{e}_i l_i$. The latter two terms have total at most a tame multiple of e_0 by their geometric bounds. This proves the assertion for $q = 1$. For a partition with at least two blocks, every block is smaller than q. Its Gaussian comparison term is bounded by $C\bar{e}_i l_i^{|\pi|}$ and is summable by the same geometric bounds. For the other term expand the difference of products. Every term contains an error factor $e'_{\gamma,i}/u^\gamma \le (\Gamma_{q-1}+1)e_0$. If a polynomial factor also occurs, its factor L^{-2h} makes the sum of the term over i at most a tame multiple of e_0. If all factors are errors, there are at least two, and their total is at most a tame multiple of $Ne_0^2 \le e_0$. The number of partitions and terms depends only on q. Adding the initial error gives a tame Γ_q and proves the assertion through $q = 8n$.

At the top, Step 5 bounds the omitted quadratic term by $Cb'\mathsf{d}'^3\varrho_k^2 u^S$. Since $b' \le (5/4)x_k \le (5/4)\mathsf{d}^{-1/16}$, $\mathsf{d}' \le \mathsf{d}^{15/16}$ and $\varrho_k \le 1$, this is at most $C\mathsf{d}^{44/16}u^S$. Also $\mathsf{q} \ge x_{\mathsf{k}}^{-1/2} \ge e^{-Y/2} \ge \mathsf{d}$. The

top error is therefore bounded by a tame multiple of $\mathsf{q}\,u^S$. Step 7 carries this estimate to the passing backgrounds. Finally $Y \le 2\log(1/\mathsf{d})$ and $\mathsf{q} \le \mathsf{d}^{15/32}$ give (a).

Step 9: the first order and the means; (c). At $1 \le t \le k$, Prop. YM3_I:123.1 writes $J^{(t)}_{\{a\}} = \mathsf{c}^{(t)}_a + X_{t,a} + I_{t,a}$, where $\mathsf{c}^{(t)}_a$ is the sum of the first-derivative families at the identity, $X_{t,a} = \mathcal{O}(w_a)(U_t)$, and $I_{t,a}$ is the change of those families from the identity to the regular background U_t. Equations YM3_I:(2264) and YM3_I:(2268) give on the passing set

$$|X_{t,a}| + |I_{t,a}| \le P_t u_a, \qquad P_t = C(1+\hat{\mathsf{d}}_t)^3(y_t^{2\mathfrak{p}_0} + \sqrt{x_t}\,y_t^{\mathfrak{p}_0}). \tag{P}$$

For the first term, multiply the plaquette count of Step 1 by $2B_3^2\varepsilon(x_t)^2L^{-4t}a_k^{-1}|w_a|_\infty$. It is at most $C\hat{\mathsf{d}}_t^3\varepsilon(x_t)^2x_tu_a$ since $L^{k-t}b' \le 3x_t/2$. For the second term the number of anchor cubes is at most $C(1+\hat{\mathsf{d}}_t)^3$. In (2264), take $\varepsilon = 2B_3\varepsilon(x_t)$. The bracket is bounded by a fixed constant: $d_*(t;t,\varepsilon) \ge 3$ because $L^{m+k-t} \ge LM \ge 8M+4$ and $\varepsilon(x_t) \le \varepsilon(x_k) \to 0$. Now $\delta(x_{t-1}) \le \delta(x_t)$ and the same bound on $L^{k-t}b'$ gives (P). At $t=0$ take $\mathsf{c}^{(0)}_a = 0$ and $P_0 = C\hat{\mathsf{d}}_0^3x_0$ from $a_p \le 2$. In particular $P_t \le C'x_t^4$ for $t \le \mathsf{k}$, using $y_t^{\mathfrak{p}_0} \le Cx_t^{1/4}$ and $\hat{\mathsf{d}}_t \le x_t$. For $\mathsf{k} \le t \le k$, (P) gives $P_t \le C(1+\hat{\mathsf{d}})^3(Y^{2\mathfrak{p}_0} + \sqrt{x_{\mathsf{k}}}Y^{\mathfrak{p}_0}) \le x_{\mathsf{k}}$ for small d, since $y_{\mathsf{k}} > Y - \log(2L^3)$.

The history bound Lem. YM3_I:125.3 on the small torus gives $W_k := \sum_{t\le k}\mathsf{w}_t < 1$ for small d by the side and coupling bounds in Lem. 18.119(b). In particular m_k has positive mass. The one-entry tower identity of Lem. 18.121(b) shows $|J^{(k)}_{\{a\}} - \mathsf{c}^{(t)}_a| \le P_tu_a$ m_k-almost everywhere. Combine it with (a) on the passing set to get $|\mathsf{c}^{(t)}_a - n^{(k)}_{\{a\}}| \le (P_t + \epsilon_{\mathrm{jt}})u_a$. The Wilson measure is invariant under the mirror maps: they permute the Haar factors and the plaquette traces. They carry the sources in each row to one another, so c_a is real and is constant along that row. Apply the first-creation identity eq. 4.xxxi to $\mathcal{O}(w_a) - n^{(k)}_{\{a\}}$. The top quotient is bounded by $\epsilon_{\mathrm{jt}}u_a$ and the quotient at $t-1$ is bounded by $(2P_{t-1} + \epsilon_{\mathrm{jt}})u_a$. The masses in that identity sum to one, so (c) holds with $\rho_{\mathrm{fo}} := \sum_{t=1}^k \mathsf{w}_tP_{t-1}$.

Step 10: the weighted rare histories; (d). Put $m_* = 8n$ and $\mathsf{v} = 1 + \sum_{\pi\in\mathrm{Part}(m_*)}\mathsf{a}'_{|\pi|}$. Choose $C_n \ge 1$ such that $2P_t + 1 \le C_nx_t^5$ for $t \le \mathsf{k}$ and the bound (J), with its exponential removed, is at most $(C_nx_t^5)^{|S|}u^S$ in the same range. Define

$$B^*_t = \begin{cases} C_nx_t^5, & 0 \le t \le \mathsf{k}, \\ 3\mathsf{v}Nx_{\mathsf{k}}, & \mathsf{k} < t < k, \end{cases} \qquad \epsilon_{\mathrm{rare}} = \sum_{t=1}^k \mathsf{w}_t\big[1 + B_{m_*}(B^*_{t-1})^{m_*}\big]. \tag{R}$$

By (P), P_{t-1} is at most the bracket of its term, so $\rho_{\mathrm{fo}} \le \epsilon_{\mathrm{rare}}$.

We first bound (R) using only the numerical history bound. For $t \le \mathsf{k}+1$, $y_t \ge Y - \log(2L^3) - \log(5L/4) \ge Y/2$ for small d. The running comparison gives $x_{t-1} \le (5/4)Lx_t$ and $L^{k-t} \le x_t$. By Lem. YM3_I:125.3 the part of (R) through $\mathsf{k}+1$ is thus at most

$$C'_n\ell'^3 \sum_{t\le \mathsf{k}+1} \exp[(3+5m_*)y_t - cy_t^{2\mathfrak{p}_0}], \qquad c = \tfrac{1}{2}c'A_0^2 > 0.$$

For large Y each summand is at most $e^{-y_t}e^{-c(Y/2)^{2\mathfrak{p}_0}/2}$. The sum of e^{-y_t} over this range is bounded by a geometric sum, since $x_t \ge (3/4)L^{\mathsf{k}+1-t}x_{\mathsf{k}+1}$. Also $\ell'^3 \le Cy_k^{3\mathfrak{r}} \le CY^{3\mathfrak{r}}$. Since $Y \ge (15/16)\log(1/\mathsf{d})$ and $2\mathfrak{p}_0 > 1$, this part is at most d. Multiplying by $\mathsf{Q}'^8 \le \mathsf{d}^{-1/16}$ makes it at most $\mathsf{d}^{15/16}$.

For $t \ge \mathsf{k}+2$, the common bracket is at most $\exp(C''_nY)$ since $N \le Y/\log L$ and $y_{\mathsf{k}} \le Y$. Use $\sum_t \mathsf{w}_t \le 2M^{-3}\ell'^3e^{-cy_k^{2\mathfrak{p}_0}}$ from the same history bound at k. By Lem. 18.119(b), (c), (d), $Y \le 2\log(1/\mathsf{d}) < 2\mathsf{C}_{\mathrm{r}}y_k^{3\mathfrak{r}+1}$ and $\mathsf{Q}'^8 \le e^{\mathsf{C}_{\mathrm{Q}}y_k^{3\mathfrak{r}+1}}$. As $2\mathfrak{p}_0 > 3\mathfrak{r}+1$ in the construction, the negative exponent absorbs all these factors and $\log(1/\mathsf{d})$ for sufficiently large y_k. This part, multiplied by Q'^8, is therefore at most d. Altogether $\mathsf{Q}'^8\epsilon_{\mathrm{rare}} \le 2\mathsf{d}^{15/16}$ and $\epsilon_{\mathrm{rare}} \le 1/4$ after shrinking d.

It remains to verify that (R) bounds the actual inserted moments. Let $\tilde J^{(t)}_{\{a\}} = J^{(t)}_{\{a\}} - c_a$ and $\tilde J^{(t)}_S = J^{(t)}_S$ if $|S| \ge 2$. Step 9 and $\rho_{\mathrm{fo}} \le \epsilon_{\mathrm{rare}} \le 1/4$ give $\|\tilde J^{(t)}_{\{a\}}\|_{L^\infty(\mathsf m_t)} \le (2P_t + 2\epsilon_{\mathrm{jt}} + 2\rho_{\mathrm{fo}})u_a$. Take $\epsilon_{\mathrm{jt}} \le 1/4$. For $t \le \mathsf k$ this is bounded by $B_t^* u_a$, and (J) and the zero initial mixed jets give the corresponding bounds $(B_t^*)^{|S|}u^S$ at order $|S| \ge 2$. For $\mathsf k \le t < k$ the singleton bound is Bu_a with $B = 3x_{\mathsf k}$. At $t = \mathsf k$ the stronger bound (J) gives $\|\tilde J^{(\mathsf k)}_S\|_\infty \le u^S$ for $|S| \ge 2$ on the passing set, as it has a tame prefactor times $\mathsf q$. This uses (J) on actual passing backgrounds, independently of the chain chart.

For each size $s = |S| \ge 2$, divide the recurrence of Lem. 18.121(b) by u^S. Its partition with one block transports the preceding jet by a positive conditional quotient of norm one. Assuming the bound $(\mathsf v N)^{b-1}B^b$ at every smaller size b, each partition with $q \ge 2$ blocks contributes at most $\mathsf a'_q(\mathsf v N)^{s-q}B^s$ by the moment–cumulant bound. Their sum is at most $(\mathsf v - 1)(\mathsf v N)^{s-2}B^s$. The constant shifts of singletons have no effect in these cumulants. With at most N steps and initial bound 1, the total is at most $(\mathsf v N)^{s-1}B^s$. Induction on s proves the bound through m_*. In particular it is at most $(B_t^*)^{|S|}$ for $t > \mathsf k$.

By Lem. 18.121(a), $\Phi^{(t)}[\prod_{a\in\gamma} X_a]$ is the sum over partitions of γ of products of these centred jets. Its essential supremum is at most $B_{m_*}(B_t^*)^{m_*}u^\gamma$ for $t < k$, since $B_t^* \ge 1$ and $|\gamma| \le m_*$. Substitution in (d) gives precisely the bound (R). □

Lemma 18.123 (The value of the rings)**.** In the setting of Proposition 18.122, let besides $\vartheta \in (0,1)$, $\Lambda \ge 1$, $(F_1,\dots,F_n) \in \mathfrak A_n(\mathsf d';\vartheta;\Lambda)$ and $4\Lambda\mathsf d' < \ell'$. With $\mathsf c'_{\mathrm{val}} := 365\,C_{\mathrm{tp}}\delta_{\mathrm{tp}}^{-2} + 1 + 81\,C_{\mathrm{tp}}$ and $\mathsf c_{\mathrm{val}} := \frac32\, n!\,\mathsf c'_{\mathrm{val}}(\mathsf c_{\mathrm g}+1)^{n-1} + 3C_n^{\mathrm{rg}}$, numbers formed with the constants and n,

$$\Big|n^{(k)}_{\{1,\dots,n\}\times\{0\}} - 3b'^{-n}\mathcal K^{(n)}_{\mathrm{free}}(F_1,\dots,F_n)\Big| \le \Big[\mathsf c_{\mathrm{val}}\mathsf d' + \frac{3C_n^{\mathrm{rg}}a_k}{\vartheta\mathsf d'}\Big]\prod_i b'^{-1}\|F_i\|.$$

■

Sketch. By YM3_I:(2226), $C_k^{\mathrm{free}} = \sum_{1\le j\le k}\mathcal C_{(j)} + C_{k,0}^{\mathrm{bg}}$ and $\sum_{j\le\mathsf k}\mathcal C_{(j)} = C_{k,N}^{\mathrm{fl}}$; so $\Delta := G_k - b'^{-1}C_k^{\mathrm{free}} = \sum_{\mathsf k<j\le k}(\upsilon_j - b'^{-1})\mathcal C_{(j)} - b'^{-1}C_{k,N}^{\mathrm{fl}} - b'^{-1}C_{k,0}^{\mathrm{bg}}$. At plaquettes p,p' of two different sources of the cell 0, the sup distance $\mathsf R$ of the barycentres is at least $\mathsf d'/\sqrt3$. The run gives $|\upsilon_j - b'^{-1}| \le \bar B a_{k-j+1}\upsilon_j/b'$ with $a_{k-j+1} = \mathsf d'/\hat{\mathsf d}_{j-1}$ by the flow estimate of Proposition YM3_I:74.1; with Proposition YM3_I:120.6 and $\sum_l (t_0L^l)^2e^{-ct_0L^l} \le 3c^{-2}$ (as in Step 5 there), the first part, multiplied by $n_a|w_a|_\infty$, is at most $1094\,C_{\mathrm{tp}}\delta_{\mathrm{tp}}^{-2}\bar B\mathsf d'b'^{-1}u_a$. the fluctuation-kernel bound of Proposition YM3_I:120.6 at the depth N, whose condition is met, as $\mathsf R \ge \mathsf d'/\sqrt3$ and $\mathsf d^\circ_n(g)$ was taken small enough for it, bounds the second part by $162L^3C_{\mathrm{tp}}\hat{\mathsf d}^3e^{-\delta_{\mathrm{tp}}L\hat{\mathsf d}/\sqrt3}u_a$, which is at most $\mathsf d'u_a$ for the same reason; the background-kernel bound of Proposition YM3_I:120.6 bounds the third by $81C_{\mathrm{tp}}\mathsf d'^3u_a$. With $\bar B \le \frac13 b'$: $n_a|w_a|_\infty \sup|\Delta| \le \mathsf c'_{\mathrm{val}}\mathsf d'u_a \le u_a$, where $\mathsf c'_{\mathrm{val}}\mathsf d' \le 1$ is one of the inequalities that fix $\mathsf d^\circ_n(g)$. Telescoping every cyclic product link by link, each link with its weight and its count being at most $\mathsf c_{\mathrm g}u$, $(\mathsf c_{\mathrm g}+1)u$ or $\mathsf c'_{\mathrm{val}}\mathsf d'u$ (Step 5 of the proof of Proposition 18.122), the ring number differs from $\frac32\sum_\varsigma \mathrm{Tr}\prod(b'^{-1}C_k^{\mathrm{free}}W_a) = 3b'^{-n}\mathcal K_k^{(n)}(f_{(1,0)},\dots,f_{(n,0)})$ by the definition of $\mathcal K_k^{(n)}$ and cyclicity of trace by at most $\frac32(n-1)!\,n\,\mathsf c'_{\mathrm{val}}\mathsf d'(\mathsf c_{\mathrm g}+1)^{n-1}\prod_a u_a$. As $f_{(i,0)} = F_i^{\langle\ell'\rangle}$, $4a_k \le \mathsf d'$ and $4\Lambda\mathsf d' < \ell'$, Lemma 18.118 on (ℓ',k) gives the free rings with the error $3C_n^{\mathrm{rg}}[a_k/(\vartheta\mathsf d') + (\mathsf d'/\ell')^3]\prod_i b'^{-1}\|F_i\|$, and $(\mathsf d'/\ell')^3 \le \mathsf d'$. □

The law on atoms.

Proposition 18.124 (The law on atoms, at every order and in every state)**.** For every $n \ge 2$ let $C_2 := 1 + \frac32(n-1)!\,\mathsf c_{\mathrm g}^n$, $\mathsf C_m := \mathsf a_m(6mB_m(C_2+1)^m)^m$, $C_1 := C_2 + \mathsf C_n$, $\mathsf C'_{\mathrm{jt}} := 2\mathsf C_{\mathrm{jt}} + 254\,(8n)!\,\mathsf c_{\mathrm g}^{8n} + 2$,

$\mathsf{C}' := 2B_{8n}^{1/8} C_1^n (\mathsf{C}_{8n}\mathsf{C}'_{\mathrm{jt}})^{1/8}$, and

$$\mathsf{C}_\sharp(n) := \mathsf{a}_n(B_nC_1^n+1)^{n-1}\mathsf{C}' + \mathsf{C}_n\mathsf{C}'_{\mathrm{jt}} + \mathsf{c}_{\mathrm{val}}, \qquad \mathsf{c}_\sharp(n) := \mathsf{c}_{\mathrm{jt}}, \qquad \Phi_n(t) := \mathsf{C}_\sharp(n)(\log\tfrac{1}{t})^{\mathsf{c}_\sharp(n)} t^{13/256} \quad (0<t<1),$$

numbers formed with the constants and n (Proposition 18.122, Lemma 18.123). Let $0 < g \le \gamma_W^{\mathrm{sd}}$, $\Lambda \ge 1$ and $\vartheta \in (0,1)$ be arbitrary; let $(K,\beta) \in \mathfrak{d}(g)$ have the label b; let $\mathfrak{s}$ be any state at g at (K,β); let d satisfy

$$0 < \mathsf{d} \le \mathsf{d}_n^{\mathrm{at}}(g,\Lambda) := \min\Big\{\mathsf{d}_n^\circ(g), \Big(\frac{LM}{4(\Lambda+1)}\Big)^{16/15}\Big\} \qquad \text{and} \qquad K - s_g(\mathsf{d}) \ge 1;$$

and let $(f_1,\dots,f_n) \in \mathfrak{A}_n(\mathsf{d};\vartheta;\Lambda)$. Then

$$\big|S_n^T[\mathfrak{s}](f_1,\dots,f_n) - 3b^{-n}\mathcal{K}_{\mathrm{free}}^{(n)}(f_1,\dots,f_n)\big| \le \Big[\Phi_n(\mathsf{d}) + \frac{3C_n^{\mathrm{rg}}a_K}{\vartheta\mathsf{d}}\Big] b^{-n}\prod_i \|f_i\|.$$

Of these data, only the threshold depends on Λ, only the lattice term of the bracket on ϑ, and only $\mathsf{d}_n^\circ(g)$ on g; the bracket involves K only by way of a_K and involves neither β nor a side nor the state, and outside the bracket the datum (K,β) enters the right side only through the factor b^{-n}. ∎

Sketch. Let x_0, x_i be the points of (P1), (P4), and take the dilated data of Lemma 18.119. Put $F_i := f_i^{[\mathsf{z}]}$ and choose $o \in a_k\mathbb{Z}^3$ with $|o - L^{\mathsf{z}}x_0|_\infty \le a_k$. By Lemma 18.117(a), (b), $(F_i) \in \mathfrak{A}_n(\mathsf{d}';\vartheta;\Lambda)$, $\|F_i\| = \|f_i\|$ and $\mathcal{K}_{\mathrm{free}}^{(n)}(F) = \mathcal{K}_{\mathrm{free}}^{(n)}(f)$. A point of $\operatorname{supp} F_i$ lies within the sup distance $\Lambda\mathsf{d}' + a_k \le (\Lambda+1)\mathsf{d}^{15/16} \le \frac14 LM \le \frac18\ell'$ of o, by the threshold and $\ell' \ge 2LM$ (Lemma 18.119(b)): this is (M1); also $4\Lambda\mathsf{d}' < \ell'$; (P4) gives the first half of (M2), and (P2) with $|v|_\infty \ge |v|/\sqrt{3}$ gives (M3). *Step 0: an entry that vanishes at every barycentre.* The mirror maps carry barycentres to barycentres, so $w_{(i,\mathbf{k}')} = 0$ if and only if F_i vanishes at every barycentre of $\mathbb{L}_k$. In that case $\mathcal{O}_k(F_i) = 0$, the cumulant of order $n \ge 2$ vanishes, and so does $S_n^T[\mathfrak{s}](f)$ by Lemma 18.119(a); a barycentre lies within $\frac{\sqrt3}{2}a_k$ of a point where $|F_i| = \|F_i\|$, so (P3) gives $a_K/(\vartheta\mathsf{d}) = a_k/(\vartheta\mathsf{d}') \ge \frac{2}{\sqrt3}$, and the right side is at least $3C_n^{\mathrm{rg}}b^{-n}\prod_i\|f_i\|$, while $3b^{-n}|\mathcal{K}_{\mathrm{free}}^{(n)}(f)| \le \frac92(n-1)!(\frac23)^n b^{-n}\prod_i\|f_i\|$ (Lemma 18.117(a)) is smaller. From here on (M2) holds, and Proposition 18.122 and Lemma 18.123 apply. *Step 1: the cumulants of the small torus.* Apply Lemma 18.121(c) on (ℓ',k) at $T = k$, with the eight cells, the means c_a of Proposition 18.122(c), $u_a = b'^{-1}\|F_i\|$, C_2, $\bar J_\gamma := \operatorname{Re} n_\gamma^{(k)}$ for γ in one cell with $|\gamma| \ge 2$, $\bar J_{\{a\}} := \operatorname{Re} n_{\{a\}}^{(k)} - c_a$, $\epsilon_{\mathrm{sc}} := \epsilon_{\mathrm{jt}}$, $\epsilon_{\mathrm{lk}} := \epsilon_{\mathrm{jt}} + 254\,(8n)!\,\mathsf{c}_{\mathrm{g}}^{8n}\mathsf{d}'^6$ and the ϵ_{rare} of Proposition 18.122(d). Its hypotheses (i)–(iii) are (a)–(d) there, the jets being real almost everywhere (Lemma 18.121(a)), so that the real parts are at least as near, and $|\bar J_{\{a\}}| \le (\epsilon_{\mathrm{jt}} + 2\rho_{\mathrm{fo}})u_a \le u_a$. So the joint cumulants κ_γ of the X_a under $\mu_{k,\ell',\beta}$ obey $|\kappa_\gamma - \bar J_\gamma| \le \mathsf{C}_{|\gamma|}\epsilon u^\gamma$ with $\epsilon := \epsilon_{\mathrm{sc}} + \epsilon_{\mathrm{lk}} + \epsilon_{\mathrm{rare}}$. *Step 2: the numbers.* With $\mathsf{d}' \le \mathsf{d}^{15/16}$ and $\mathsf{Q}'^8 \le \mathsf{d}^{-1/16}$ (Lemma 18.119(b)): $\mathsf{Q}'^8\epsilon \le \mathsf{d}^{-1/16}(2\epsilon_{\mathrm{jt}} + 254\,(8n)!\,\mathsf{c}_{\mathrm{g}}^{8n}\mathsf{d}^{45/8}) + 2\mathsf{d}^{15/16} \le \mathsf{C}'_{\mathrm{jt}}(\log\frac{1}{\mathsf{d}})^{\mathsf{c}_{\mathrm{jt}}}\mathsf{d}^{13/32}$, as $\frac{15}{32} - \frac{1}{16} = \frac{13}{32}$. With $\eta_\times := \mathsf{C}_{8n}\epsilon$ and $\eta' := 2\mathsf{Q}'B_{8n}^{1/8}C_1^n\eta_\times^{1/8} = 2B_{8n}^{1/8}C_1^n(\mathsf{C}_{8n}\mathsf{Q}'^8\epsilon)^{1/8}$: $\eta' \le \mathsf{C}'(\log\frac{1}{\mathsf{d}})^{\mathsf{c}_{\mathrm{jt}}/8}\mathsf{d}^{13/256} \le 1$, the last inequality being one of those that fix $\mathsf{d}_n^\circ(g) \ge \mathsf{d}_n^{\mathrm{at}}(g,\Lambda)$ (sketch of Proposition 18.122); and $\epsilon \le \eta_\times \le 1$. *Step 3: every state.* Apply Lemma 18.120 at (k,β) to the board $(-m,o)$, the set $\mathfrak{l}'$ and the number Q' of Lemma 18.119(b), the state $\mathfrak{s}^{(\mathsf{z})}$ over $\mathfrak{l}'$ (Lemma 18.119(a)), the functions F_i and the constants $c_i := c_{(i,0)}$: its cumulants are the κ_γ of step 1, by periodization and the equality of the means along the rows; its (i) holds with C_1, since $|\kappa_\gamma| \le |\bar J_\gamma| + \mathsf{C}_{|\gamma|}\epsilon u^\gamma$ and $|\gamma| \le n$ in one cell; its (ii) holds with $\eta_\times$; and $\eta = \eta' \le 1$. So $|S_n^T[\mathfrak{s}^{(\mathsf{z})}](F) - S_n^T[\tilde\mu_{k,\ell',\beta}](F)| \le \mathsf{a}_n(B_nC_1^n+1)^{n-1}\eta'\prod_i u_i$. *Step 4: the value.* $S_n^T[\tilde\mu_{k,\ell',\beta}](F) = \kappa_{\{1,\dots,n\}\times\{0\}}$ by periodization; by step 1 it is within $\mathsf{C}_n\epsilon\prod_i u_i$ of $\operatorname{Re} n_{\{1,\dots,n\}\times\{0\}}^{(k)}$, which by Lemma 18.123 is within $[\mathsf{c}_{\mathrm{val}}\mathsf{d}' + 3C_n^{\mathrm{rg}}a_k/(\vartheta\mathsf{d}')]\prod_i u_i$ of the real number

$3b'^{-n}\mathcal{K}^{(n)}_{\mathrm{free}}(F)$. Adding,

$$|S^T_n[\mathfrak{s}^{(\mathsf{z})}](F) - 3b'^{-n}\mathcal{K}^{(n)}_{\mathrm{free}}(f)| \le \Big[\mathsf{a}_n(B_nC_1^n+1)^{n-1}\eta' + \mathsf{C}_n\epsilon + \mathsf{c}_{\mathrm{val}}\mathsf{d}' + \frac{3C_n^{\mathrm{rg}}a_K}{\vartheta\mathsf{d}}\Big]\prod_i b'^{-1}\|f_i\|,$$

and the first three members are together at most $\Phi_n(\mathsf{d})$, because $\epsilon \le \mathsf{Q}'^8\epsilon$, $\mathsf{d}' \le \mathsf{d}^{15/16}$ and $\log\frac{1}{\mathsf{d}} \ge 1$. Multiplying by $L^{\mathsf{z}n}$, with $S^T_n[\mathfrak{s}](f) = L^{\mathsf{z}n}S^T_n[\mathfrak{s}^{(\mathsf{z})}](F)$ and $L^{\mathsf{z}}b'^{-1} = b^{-1}$ (Lemma 18.119(a)), gives the claim. □

Separated tuples of a bounded aspect; every pair; the limits.

Proposition 18.125 (Separated tuples of a bounded aspect, in every state and on every torus)**.** Let $0 < g \le \gamma_W^{\mathrm{sd}}$, $n \ge 2$, $\vartheta \in (0,1)$ and $\Lambda \ge 1$, and put $\mathsf{d}^P_n(g,\Lambda) := \min\{2\mathsf{d}^{\mathrm{at}}_n(g,\Lambda_1), L^{-2\mathfrak{p}_0}\}$. If $(K,\beta) \in \mathfrak{d}(g)$ has the label b, $\mathfrak{s}$ is a state at g at (K,β), $0 < \mathsf{d} \le \mathsf{d}^P_n(g,\Lambda)$ and $K - s_g(\frac{1}{2}\mathsf{d}) \ge 1$ (which holds when $K - s_g(\mathsf{d}) \ge 2$), then for each $X = (f_1,\dots,f_n) \in P_n(\mathsf{d};\vartheta;\Lambda)$

$$\Big|S^T_n[\mathfrak{s}](X) - 3b^{-n}\mathcal{K}^{(n)}_{\mathrm{free}}(X)\Big| \le (6N_\Lambda)^n\Big[\Phi_n(\tfrac{1}{2}\mathsf{d}) + 255\,C_n^{\mathrm{rg}}\frac{a_K}{\vartheta\mathsf{d}}\Big]b^{-n}\prod_i\|f_i\|.$$

For a torus T_ℓ with $\ell \in \mathfrak{L}(g)$ and $8\Lambda\mathsf{d} < \ell$, the inequality holds for $S^T_{n;K,\ell,\beta}(X)$ at each $X \in P^{T_\ell}_n(\mathsf{d};\vartheta;\Lambda)$, regardless of where on the torus the supports are placed, with $\mathcal{K}^{(n)}_{\mathrm{free}}$ taken at a lift. The threshold and the first member of the bracket are free of ϑ. ■

Sketch. The depth at half the length. For $0 < \mathsf{d} \le L^{-2\mathfrak{p}_0}$, $s_g(\frac{1}{2}\mathsf{d}) \le s_g(\mathsf{d}) + 1$: at $s := s_g(\mathsf{d})$, $L^s\mathsf{d} \ge C_W(s) \ge 1$ and $\mathsf{d} \le L^{-2\mathfrak{p}_0}$ give $L^s \ge L^{2\mathfrak{p}_0}$, that is $s \ge 2\mathfrak{p}_0$, so, as $Y^g_s \ge (s+1)\log L$ and $L \ge 13$, $C_W(s+1)/C_W(s) = (1+\log L/Y^g_s)^{2\mathfrak{p}_0} \le e^{2\mathfrak{p}_0/(s+1)} < \frac{L}{2}$, and $L^{s+1}\frac{\mathsf{d}}{2} \ge \frac{L}{2}C_W(s) \ge C_W(s+1)$. *The bound.* The map $\mathsf{T} := S^T_n[\mathfrak{s}] - 3b^{-n}\mathcal{K}^{(n)}_{\mathrm{free}}$ is linear in each entry on the bounded functions with compact supports at positive mutual distances by the cumulant and integral formulas. As $\mathsf{d}_1 = \frac{1}{2}\mathsf{d} \le \mathsf{d}^{\mathrm{at}}_n(g,\Lambda_1)$ and $K - s_g(\mathsf{d}_1) \ge 1$, Proposition 18.124 at $(\mathsf{d}_1;\vartheta_1;\Lambda_1)$ bounds T on $\mathfrak{A}_n(\mathsf{d}_1;\vartheta_1;\Lambda_1)$ by $\tau := [\Phi_n(\mathsf{d}_1) + 3C_n^{\mathrm{rg}}a_K/(\vartheta_1\mathsf{d}_1)]b^{-n}$, and $\vartheta_1\mathsf{d}_1 = \vartheta\mathsf{d}/(1+84\vartheta) \ge \vartheta\mathsf{d}/85$; Lemma 18.117(c) carries the bound to $P_n(\mathsf{d};\vartheta;\Lambda)$. *Tori.* Let x_0 be the point of (P1) and $\tilde{x}_0$ a lift of it. As $\Lambda\mathsf{d} < \ell/8$, the covering map is isometric on $\bar{\mathbb{B}}(\tilde{x}_0,\Lambda\mathsf{d})$, and the lifts $\hat{f}_i$ of the entries at this ball keep suprema, Lipschitz constants and the distances of the supports, and have $\hat{f}_i^{\langle\ell\rangle} = f_i$ by extending by zero off the lifted ball; so $\hat{X} \in P_n(\mathsf{d};\vartheta;\Lambda)$, $S^T_{n;K,\ell,\beta}(X) = S^T_n[\tilde{\mu}_{K,\ell,\beta}](\hat{X})$ by periodization, and $\tilde{\mu}_{K,\ell,\beta}$ is a state at g. □

Proposition 18.126 (Every pair (K,β); one modulus in d/b)**.** Let $0 < g \le \gamma_W^{\mathrm{sd}}$, $n \ge 2$, $\vartheta \in (0,1)$, $\Lambda \ge 1$, $K \in \mathbb{N}$, $\beta > X_g$ and $b := a_K\beta$; let K' be the integer for which $(K',\beta) \in \mathfrak{d}(g)$, and $z := K - K' \in \mathbb{Z}$. Then $g^{-2} < L^z b \le L(g^{-2}+\bar{B})$. Let $\mathsf{d} > 0$ with $L^z\mathsf{d} \le \mathsf{d}^P_n(g,\Lambda)$ and $K' - s_g(\frac{1}{2}L^z\mathsf{d}) \ge 1$. Then for each law ν that is a member of $\mathcal{G}_{K,\beta}$ or a torus measure $\tilde{\mu}_{K,\ell,\beta}$ with $L^z\ell \in \mathfrak{L}(g)$, and for each $X = (f_1,\dots,f_n) \in P_n(\mathsf{d};\vartheta;\Lambda)$,

$$\Big|S^T_n[\nu](X) - 3b^{-n}\mathcal{K}^{(n)}_{\mathrm{free}}(X)\Big| \le (6N_\Lambda)^n\Big[\Phi_n(\tfrac{1}{2}L^z\mathsf{d}) + 255\,C_n^{\mathrm{rg}}\frac{a_K}{\vartheta\mathsf{d}}\Big]b^{-n}\prod_i\|f_i\|.$$

At $g = g_\star$ put $\mathsf{t}_n(\Lambda) := \min\{\mathsf{d}^P_n(g_\star,\Lambda), 2e^{-256\,\mathsf{c}_\sharp(n)/13}\}$, a number that does not depend on ϑ; if $\mathsf{c}_\star\mathsf{d}/b \le \mathsf{t}_n(\Lambda)$ and $K' - s_{g_\star}(\frac{1}{2}L^z\mathsf{d}) \ge 1$, the same bound holds with $\Phi_n(\mathsf{c}_\star\mathsf{d}/(2b))$ in the place of $\Phi_n(\frac{1}{2}L^z\mathsf{d})$. ■

Sketch. Proposition YM3_I:74.1 gives the unique first-passage cutoff K' at this β and bounds the label $a_{K'}\beta = L^z b$ in $(g^{-2}, L(g^{-2}+\bar{B})]$. The lattice relabeling by L^z preserves the Gibbs kernels and Wilson torus density as in Lemma 18.119(a), so $\nu^{(z)}$ is a state at g at (K',β). $X^{[z]} \in P_n(L^z\mathsf{d};\vartheta;\Lambda)$, $\mathcal{K}^{(n)}_{\mathrm{free}}(X^{[z]}) = \mathcal{K}^{(n)}_{\mathrm{free}}(X)$ and $\|f^{[z]}\| = \|f\|$ (Lemma 18.117(a), (b)). Proposition 18.125 at (K',β), $\nu^{(z)}$, $L^z\mathsf{d}$ and $X^{[z]}$, together with $S^T_n[\nu](X) = L^{zn}S^T_n[\nu^{(z)}](X^{[z]})$ by the scaling of the smeared actions,

$L^{zn}(L^z b)^{-n} = b^{-n}$ and $a_{K'}/(L^z\mathsf{d}) = a_K/\mathsf{d}$, gives the bound. At $g_\star$: $L^z\mathsf{d} = (L^z b)\mathsf{d}/b \le \mathsf{c}_\star\mathsf{d}/b \le \mathsf{t}_n(\Lambda) \le \mathsf{d}_n^P(g_\star,\Lambda)$; for $\mathsf{C},\mathsf{c} \ge 0$ and $\alpha > 0$ the function $t \mapsto \mathsf{C}(\log\frac{1}{t})^{\mathsf{c}}t^\alpha$ is continuous on $(0,1)$, has the limit 0 at 0, and is non-decreasing on $(0, e^{-\mathsf{c}/\alpha}]$, where its logarithmic derivative $t^{-1}[\alpha - \mathsf{c}/\log\frac{1}{t}]$ is ≥ 0; as $\frac{1}{2}L^z\mathsf{d} \le \frac{1}{2}\mathsf{c}_\star\mathsf{d}/b \le e^{-256\mathsf{c}_\sharp(n)/13}$, $\Phi_n(\frac{1}{2}L^z\mathsf{d}) \le \Phi_n(\mathsf{c}_\star\mathsf{d}/(2b))$. □

Proposition 18.127 (The limits: every limit, every component, every limit in the frame; the label)**.** Let $n \ge 2$, $\vartheta \in (0,1)$ and $\Lambda \ge 1$.

(i) (Tori.) Let $0 < g \le \gamma_W^{\mathrm{sd}}$, $\ell \in \mathfrak{L}(g)$, $(K_j,\beta_j) \in \mathfrak{d}(g)$ with $K_j \to \infty$ and $a_{K_j}\beta_j \to b$, $0 < \mathsf{d} \le \mathsf{d}_n^P(g,\Lambda)$ with $8\Lambda\mathsf{d} < \ell$, and $X = (f_1,\dots,f_n) \in P_n^{T_\ell}(\mathsf{d};\vartheta;\Lambda)$. Any accumulation point S of the sequence $(S^T_{n;K_j,\ell,\beta_j}(X))_j$ obeys $|S - 3b^{-n}\mathcal{K}^{(n)}_{\mathrm{free}}(X)| \le (6N_\Lambda)^n\Phi_n(\frac{1}{2}\mathsf{d})\,b^{-n}\prod_i\|f_i\|$.

(ii) ($\mathbb{R}^3$.) Let $0 < g_0 \le \gamma$. Let σ be the limit of a convergent Gibbs sequence $(K_j,\beta_j,\nu_j)_j$ at g_0, and let σ' denote σ itself, any component of any canonical decomposition of σ, or any limit in the frame of that sequence. The labels $a_{K_j}\beta_j$ converge to a number $b \in [g_0^{-2}, L(g_0^{-2}+\bar B)]$, and for $\mathsf{d} > 0$ with $\mathsf{c}_\star\mathsf{d}/b < \mathsf{t}_n(\Lambda)$ and $X = (f_1,\dots,f_n) \in P_n(\mathsf{d};\vartheta;\Lambda)\cap\mathfrak{S}_n(\mathbb{R}^3)$,

$$\Big|S_n^{\sigma'}(X) - 3b^{-n}\mathcal{K}^{(n)}_{\mathrm{free}}(X)\Big| \le (6N_\Lambda)^n\,\Phi_n\Big(\frac{\mathsf{c}_\star\mathsf{d}}{2b}\Big)\,b^{-n}\prod_i\|f_i\|.$$

Since ϑ appears neither in the condition on d nor on the right side, the inequality is valid for each $X \in \mathfrak{S}_n(\mathbb{R}^3)$ without identically vanishing entries that satisfies (P1) and (P2) at (d,Λ), with no condition on the Lipschitz constants of its entries.

■

Sketch. (i) $s_g(\frac{1}{2}\mathsf{d})$ does not depend on the cutoff, so $K_j - s_g(\frac{1}{2}\mathsf{d}) \ge 1$ for large j; Proposition 18.125 on T_ℓ at (K_j,β_j), with $a_{K_j}\beta_j \to b \ge g^{-2} > 0$ and $a_{K_j} \to 0$. (ii) *1.* By Proposition YM3_I:74.1, $b_j := a_{K_j}\beta_j \in [b_-,b_+] := [g_0^{-2}, L(g_0^{-2}+\bar B)]$; as $K_j \to \infty$, $\beta_j = L^{K_j}b_j \to \infty$, and $\beta_j > X_{g_\star}$ for large j. *2.* σ uses the approximants ν_j themselves. Every component has approximants $\rho_j = \nu_j(\,\cdot\mid E_j)$ at the same data by Theorem YM3_I:AL, equation YM3_I:(1988). The events E_j are translation invariant, so these approximants belong to $\mathcal{G}_{K_j,\beta_j}$ by Proposition YM3_I:106.3. *3.* For a limit in the frame with approximants ρ_j, $S_n^{\sigma'}(X) = \lim_j S_n^T[\rho_j](X)$ for $n \ge 2$ and $X \in \mathfrak{S}_n(\mathbb{R}^3)$: the connected functions of order at least 2 of $X \mapsto \rho_j(X^{\nu_j}_{K_j})$ are the cumulants $S_n^T[\rho_j]$ because subtracting the means of ν_j does not affect cumulants of order at least two. The cumulant formula is a fixed polynomial in the values at subtuples. *4.* Let b be a limit point of (b_j) along an infinite set $\mathcal{J}$ of indices, let $\mathsf{c}_\star\mathsf{d}/b < \mathsf{t}_n(\Lambda)$ and $X \in P_n(\mathsf{d};\vartheta;\Lambda)\cap\mathfrak{S}_n(\mathbb{R}^3)$. For large $j \in \mathcal{J}$, $\mathsf{c}_\star\mathsf{d}/b_j \le \mathsf{t}_n(\Lambda)$. With K_j' the integer for which $(K_j',\beta_j) \in \mathfrak{d}(g_\star)$ and $z_j := K_j - K_j'$, $g_\star^{-2} < L^{z_j}b_j \le \mathsf{c}_\star$ (Proposition 18.126); so $L^{z_j} \le \mathsf{c}_\star/b_-$ and $K_j' \to \infty$, while $L^{z_j}\mathsf{d} \ge g_\star^{-2}\mathsf{d}/b_+$, so that $s_{g_\star}(\frac{1}{2}L^{z_j}\mathsf{d}) \le s_{g_\star}(g_\star^{-2}\mathsf{d}/(2b_+))$ by monotonicity of the witness depth and $K_j' - s_{g_\star}(\frac{1}{2}L^{z_j}\mathsf{d}) \ge 1$ for large j. Proposition 18.126 at $g_\star$ and $\nu = \rho_j$ gives

$$\Big|S_n^T[\rho_j](X) - 3b_j^{-n}\mathcal{K}^{(n)}_{\mathrm{free}}(X)\Big| \le (6N_\Lambda)^n\Big[\Phi_n\Big(\frac{\mathsf{c}_\star\mathsf{d}}{2b_j}\Big) + 255\,C_n^{\mathrm{rg}}\frac{a_{K_j}}{\vartheta\mathsf{d}}\Big]b_j^{-n}\prod_i\|f_i\|;$$

letting $j \to \infty$ in $\mathcal{J}$, with step 3, $a_{K_j} \to 0$ and the continuity of Φ_n, gives the bound with this b. *5. One label.* Let $\psi_1(x) := (1-|x|^2)_+^2$ and $\psi_2 := \psi_1(\cdot - 3e_1)$: two C^1 functions with $\|\psi_i\| = 1$ and $\operatorname{Lip}\psi_i = \frac{8}{3\sqrt{3}} < 2$, supported in the unit balls centred at 0 and at $3e_1$, which are one unit apart and are contained in $\bar{\mathbb{B}}(\frac{3}{2}e_1,\frac{5}{2})$. So $\Psi_r := (\psi_{1,r},\psi_{2,r}) \in P_2(r;\frac{1}{2};\frac{5}{2})\cap\mathfrak{S}_2(\mathbb{R}^3)$ for every $r > 0$, and $\kappa := \mathcal{K}^{(2)}_{\mathrm{free}}(\Psi_r) = \frac{3}{16\pi^2}\iint\psi_1(x)\psi_2(y)|x-y|^{-6}\,dx\,dy > 0$ does not depend on r by homogeneity and $\operatorname{tr}[(3\hat u\hat u^{\mathsf T} - \mathbf{1})^2] = 6$. If b and $\tilde b$ are two limit points, step 4 at $n=2$ and $X = \Psi_r$, for small r and for both, gives $3\kappa|b^{-2} - \tilde b^{-2}| \le (6N_{5/2})^2[\Phi_2(\frac{\mathsf{c}_\star r}{2b})b^{-2} + \Phi_2(\frac{\mathsf{c}_\star r}{2\tilde b})\tilde b^{-2}]$, which tends to 0 with r; so $b = \tilde b$,

and the bounded sequence (b_j) converges. The last sentence: an entry of $X \in \mathfrak{S}_n(\mathbb{R}^3)$ that is not identically 0 has $0 < \operatorname{Lip} f_i < \infty$, so (P3) holds at $\vartheta := \min\{\frac{1}{2}, \min_i \|f_i\|/(\mathsf{d} \operatorname{Lip} f_i)\}$. □

The tangent field.

Proposition 18.128 (The tangent field of the action density is the centred functional of the free template)**.** Let $0 < g_0 \le \gamma$, let σ be the limit of a convergent Gibbs sequence $(K_j, \beta_j, \nu_j)_j$ at g_0, let σ' be σ itself, any component of any canonical decomposition of σ, or any limit in the frame of that sequence, and let $b = \lim_j a_{K_j}\beta_j$ be the label of Proposition 18.127(ii).

(a) Let $n \ge 2$ and let $X = (f_1, \dots, f_n) \in \mathfrak{S}_n(\mathbb{R}^3)$ have no identically vanishing entry. Let $\mathsf{d}_X > 0$ be the least distance between two of its supports, let $\mathcal{R} > 0$ be such that some closed ball of radius $\mathcal{R}$ contains every support, and put $\Lambda := \max\{1, \mathcal{R}/\mathsf{d}_X\}$. Then for each real $r > 0$ with $\mathsf{c}_\star r \mathsf{d}_X / b < \mathsf{t}_n(\Lambda)$

$$\left| S_n^{\sigma'}(X_r) - 3b^{-n}\mathcal{K}_{\mathrm{free}}^{(n)}(X) \right| \le (6N_\Lambda)^n \, \Phi_n\Big(\frac{\mathsf{c}_\star \mathsf{d}_X}{2b}\, r\Big)\, b^{-n} \prod_i \|f_i\|,$$

and the right side tends to 0 as $r \to 0$; so $b^n S_n^{\sigma'}(X_r) \to 3\mathcal{K}_{\mathrm{free}}^{(n)}(X)$. At $r = L^{-z}$, $z \in \mathbb{N}$, the right side is $(6N_\Lambda)^n \mathsf{C}_\sharp(n) (\frac{\mathsf{c}_\star \mathsf{d}_X}{2b})^{13/256} (\log \frac{2b}{\mathsf{c}_\star \mathsf{d}_X} + z \log L)^{\mathsf{c}_\sharp(n)} L^{-13z/256}\, b^{-n} \prod_i \|f_i\|$, of the order $z^{\mathsf{c}_\sharp(n)} L^{-13z/256}$; in terms of the dilates defined by $S_n^{[z]}(Y) = L^{-zn} S_n(Y^{[-z]})$, $(L^z b)^n S_n^{\sigma'[z]}(X) = b^n S_n^{\sigma'}(X^{[-z]}) \to 3\mathcal{K}_{\mathrm{free}}^{(n)}(X)$ as $z \to \infty$, where $\sigma'[z]$ is the dilate of σ' by L^z, the limit along the dilated sequence, whose label is $L^z b$ by lattice relabeling.

(b) Let σ' be σ or a component. For every $n \ge 1$ and every $X \in \mathfrak{S}_n(\mathbb{R}^3)$,

$$\lim_{r \to 0} \sigma'(X_r) = \sum_{\pi \in \mathrm{Part}_{\ge 2}(n)} \prod_{J \in \pi} 3b^{-|J|} \mathcal{K}_{\mathrm{free}}^{(|J|)}(X_J) = \sigma_{S^{\mathrm{fr}}[3,b]}(X),$$

the centred functional of the free template with the data $(3, b)$. The same limit is obtained for the dilations about any point y, that is, for the tuples $(f_i(y + (\cdot - y)/r))_i$. The tangent field at a point $y \in \mathbb{R}^3$, in σ', of the action density is by definition the functional $X \mapsto \lim_{r \to 0} \sigma'((f_i(y + (\cdot - y)/r))_i)$ on $\mathfrak{S}(\mathbb{R}^3)$. At every point, for the limit and for each of its components, the tangent field is thus $\sigma_{S^{\mathrm{fr}}[3,b]}$. Its connected functions $3b^{-n}\mathcal{K}_{\mathrm{free}}^{(n)}$, $n \ge 2$, are, at every tuple of $P_n(\mathsf{d}; \vartheta; \Lambda)$ for every $\mathsf{d} > 0$, $\vartheta \in (0, 1)$ and $\Lambda \ge 1$, the limits as $a_K \to 0$ and $\ell \to \infty$, at the fixed label b, of the joint cumulants at the periodised entries of the variables $\mathcal{O}^{\mathrm{G}}$ of the Gaussian lattice model with the data $(\ell, K, 3, b)$: $\mathcal{O}^{\mathrm{G}}(f) = a_K^{-1} \sum_p f(x(p)) \frac{1}{2} \sum_{c \le 3} \theta^c(p)^2$ smears with f the plaquette action $\frac{1}{2} \sum_{c \le 3} \theta^c(p)^2$ of three independent free photon fields, $\theta^1, \theta^2, \theta^3$ being their curvatures.

■

Sketch. (a) $\mathsf{d}_X > 0$, as the supports are finitely many compact and pairwise disjoint sets. (P1) holds at (d_X, Λ), since $\mathcal{R} \le \Lambda \mathsf{d}_X$; every entry is of class C^1 with compact support and not identically 0, so $0 < \operatorname{Lip} f_i < \infty$, and (P3) holds at $\vartheta := \min\{\frac{1}{2}, \min_i \|f_i\|/(\mathsf{d}_X \operatorname{Lip} f_i)\}$; so $X \in P_n(\mathsf{d}_X; \vartheta; \Lambda) \cap \mathfrak{S}_n(\mathbb{R}^3)$, and $X_r \in P_n(r\mathsf{d}_X; \vartheta; \Lambda) \cap \mathfrak{S}_n(\mathbb{R}^3)$ with $\mathcal{K}_{\mathrm{free}}^{(n)}(X_r) = \mathcal{K}_{\mathrm{free}}^{(n)}(X)$ and $\|f_{i,r}\| = \|f_i\|$ (Lemma 18.117(a), (b)). Proposition 18.127(ii) at $\mathsf{d} = r\mathsf{d}_X$ is the bound, and $\Phi_n(t) \to 0$ as $t \to 0$. The value at $r = L^{-z}$ is the formula of Φ_n; $X_{L^{-z}} = X^{[-z]}$, and the definition $S_n^{[z]}(Y) = L^{-zn} S_n(Y^{[-z]})$ gives the last form. (b) By moment–cumulant inversion, $\sigma'(X_r) = \sum_{\pi \in \mathrm{Part}(n)} \prod_{J \in \pi} S_{|J|}^{\sigma'}((X_J)_r)$. A block with $|J| \ge 2$ tends to $3b^{-|J|}\mathcal{K}_{\mathrm{free}}^{(|J|)}(X_J)$ by (a), both sides being 0 when an entry is 0 by multilinearity. A block $\{i\}$ gives $S_1^{\sigma'}(f_{i,r})$: it is 0 for σ by the centering of the sequence, and for a component it is $\mathfrak{c}_\xi \int f_{i,r} = \mathfrak{c}_\xi r^3 \int f_i \to 0$ by Theorem YM3_I:AP. So the partitions with a block of one element drop out in the limit, and the sum over $\mathrm{Part}_{\ge 2}(n)$ is $\sigma_{S^{\mathrm{fr}}[3,b]}(X)$. Dilations

about y: the tuple dilated about y is the translate by y of the tuple $(f_i(\cdot+y))_i$ dilated about 0; σ and its components are invariant under the translations by Theorem YM3_I:AP, and so is $\mathcal{K}^{(n)}_{\mathrm{free}}$ by translation of the integration variables. So the tangent field at every point is the functional of the display. For the Gaussian lattice model, take the three curvature fields to be independent centered Gaussians with covariance $a_K b^{-1} C_K^{\mathrm{free}}$ in each color. If W_i multiplies by the periodized weight at the plaquettes, the log moment generating function of the smeared quadratic actions is $-\frac{3}{2}\operatorname{Tr}\log(1-b^{-1}C_K^{\mathrm{free}}\sum_i z_i W_i)$ near zero, by the finite-dimensional Gaussian integral on the range of the covariance. In the term of degree n, each cyclic order occurs n times, which cancels the denominator n in the logarithm. The mixed derivative is exactly $3b^{-n}\mathcal{K}^{(n)}_K(f_1^{\langle\ell\rangle},\dots,f_n^{\langle\ell\rangle})$. For $X=(f_1,\dots,f_n)\in P_n(\mathsf{d};\vartheta;\Lambda)$, once $4a_K\le \mathsf{d}_1$ and $4\Lambda_1\mathsf{d}_1<\ell$, Lemma 18.118 at the atoms of $\mathfrak{A}_n(\mathsf{d}_1;\vartheta_1;\Lambda_1)$, carried to X by Lemma 18.117(c) with the map $Y\mapsto \mathcal{K}^{(n)}_K(Y^{\langle\ell\rangle})-\mathcal{K}^{(n)}_{\mathrm{free}}(Y)$, which is linear in each entry, gives $|\mathcal{K}^{(n)}_K(X^{\langle\ell\rangle})-\mathcal{K}^{(n)}_{\mathrm{free}}(X)|\le (6N_\Lambda)^n C_n^{\mathrm{rg}}[\frac{85a_K}{\vartheta\mathsf{d}}+(\frac{\mathsf{d}}{2\ell})^3]\prod_i\|f_i\|$, as $\vartheta_1\mathsf{d}_1\ge\vartheta\mathsf{d}/85$; the right side tends to 0. □

18.13. **Under the Wilson measure the squared plaquette density decouples from the action density, and the shifted density has the torus limits.** We show that under the Wilson measure the density of the squared plaquette action, $\mathcal{O}^{(2)}_K(f)=(2a_K)^{-1}\sum_p f(x(p))\,a_p^2$, drops out of the joint cumulants when the ultraviolet cutoff is removed, at every label $b>0$ and on the torus of every side $\ell=2L^m$. For $n\ge 2$ real Lipschitz functions whose supports have no common point, each joint cumulant whose entries are $\mathcal{O}_K(f_i)$ or $\mathcal{O}^{(2)}_K(f_i)$, with an entry of the second kind among them, is bounded by a constant times $K^{\mathfrak{p}^\natural_n}L^{-K/10}\prod_i\|f_i\|_\sharp$ for all cutoffs beyond a threshold, where $\mathfrak{p}^\natural_n:=\max\{2\mathfrak{p}_0(n+1)+\frac12,(2\mathfrak{p}_0+2)n\}$ with the calibrated exponent $\mathfrak{p}_0$; consequently the joint cumulants of the shifted density $\mathcal{O}'_K(f)=\mathcal{O}_K(f)-\mathcal{O}^{(2)}_K(f)$ approach, at the same rate, the torus limits $S_n^{(\ell,b)}$ of the action density (Proposition 18.129). Three facts carry the argument. An insertion of a_p^2 is a second derivative in a source that acts at the single plaquette p; the renormalisation step with a source admits sources w whose numbers $\tau_p:=L^K w(p)$, by which the bare coupling $\beta=L^K b$ at the single plaquette p is moved to $\beta-\tau_p$ ($\rho_0^w=\rho_0\exp[\sum_p\tau_p a_p]$), may vary from a plaquette to a nearby one by up to the order $\beta^{1/2}$, and on this larger class the flows of effective-action terms of the main text keep their properties (Lemma 18.131); so one insertion is paid by a number of the order $L^K/\beta=b^{-1}$, up to a small power of L^K, in the place of a power of the volume. The generating function of the small-field term is the exponential of a sum of local terms times an integral (Lemma 18.132), and along these rough directions the logarithm of that integral varies by a small amount only (Step 4 of the proof of Proposition 18.135); the local part splits along groups of entries whose supports lie apart and drops out of each cumulant that links two groups, while the sums over plaquettes that the insertions produce involve one plaquette at a time, and their size does not grow with the volume (Lemmas 18.133 and 18.134, Proposition 18.135). Finally, small pieces of the test functions turn a tuple without a common point of the supports into tuples with at least two groups (Lemma 18.137), and an exact dilation carries the bound from the labels of an interval under (Tor) to every label and every side (Proposition 18.138 and Lemma 18.139).
Setting. We use the fixed construction and source calibration of §2.3.1, and write $\mathfrak{D}$ for its choices preceding the finite-lattice data. In this subsection $\varrho_\varsigma=\varrho$, $K_{\mathrm{Q}}=K_{\mathrm{Q}}^\vee$ and $K_{\mathrm{L}}=K_{\mathrm{L}}^\vee$ are the source radius and the constants of the direct source estimate YM3_I:(2167). Use its $\mu_\star$, $\sigma_{\mathrm{src}}=\mu_\star/K_{\mathrm{Q}}$ and set $o_{\mathrm{src}}(x)=\mu_\star/(K_{\mathrm{L}}\sqrt{x})$. The constants c_ρ, E_0 and κ are those of the calibration. The old-action bound and forgetting constant are C_{old} and C_{fg} of Props. YM3_I:113.13 and YM3_I:114.3. Take $C_{\mathrm{E},0}=2048\sum_{j\ge0}e^{-j}$, the rooted count used in the proof of Lem. YM3_I:114.6.

For $\ell=2L^m$ let $T_\ell=\mathbb{R}^3/\ell\mathbb{Z}^3$, $\Lambda_{K,\ell}=a_K\mathbb{Z}^3/\ell\mathbb{Z}^3$ with $a_K=L^{-K}$, and dU the product of the bond Haar probability measures. For each positively oriented plaquette p write $x(p)$ for its barycentre

and $a_p(U) = 1 - \frac{1}{2}\operatorname{tr}\partial U(p) \in [0,2]$. Use $\rho_0 = e^{-\beta\sum_p a_p}$ and

$$\mathcal{O}_K(f) = a_K^{-1}\sum_p f(x(p))a_p, qquad \mathcal{O}_K^{(2)}(f) = (2a_K)^{-1}\sum_p f(x(p))a_p^2, \qquad \mathcal{O}'_K = \mathcal{O}_K - \mathcal{O}_K^{(2)}.$$

There are $3\ell^3 a_K^{-3}$ positive plaquettes, and $0 \le a_p^2/2 \le a_p \le 2$. Therefore both insertions have modulus at most $6\ell^3 L^{4K}\|f\|_\infty$. They are real, continuous and gauge invariant as functions of the fields, and real linear in f. Write $\langle X_1;\dots;X_n\rangle_{K,\ell,\beta}$ for the joint cumulant under the normalized Wilson measure, defined by the mixed derivative of the local logarithm of the joint moment-generating function at zero. Thus $S^T_{n;K,\ell,\beta}(\underline{f})$ is the cumulant of the $\mathcal{O}_K(f_i)$. For $n \ge 2$, the limit $S_n^{(\ell,b)}$ on tuples with no common support point is that of Thm. YM3_I:AS.

Write $\|f\|_\sharp = \max\{\|f\|_\infty, 40M\operatorname{Lip} f\}$ with the Euclidean Lipschitz constant. The spread $\mathsf{d}_\infty(\underline{f})$ is the minimum, over $x_i \in \operatorname{supp} f_i$, of $\max_{i,i'}|x_i - x_{i'}|_\infty$ on T_ℓ. Set $\mathsf{p}_n = 2\mathfrak{p}_0(n+1) + 1/2$ as in Thm. YM3_I:AR, and $\mathsf{p}_n^\natural = \max\{\mathsf{p}_n, (2\mathfrak{p}_0+2)n\}$.

Proposition 18.129 (Decoupling of the squared plaquette density under the Wilson measure; torus limits of the shifted density)**.** Let $m \in \mathbb{N}$, $\ell = 2L^m$, let $n \ge 2$ be an integer, and let b_-, b_+, d be real numbers with $0 < b_- \le b_+$ and $d > 0$. A number $C^\flat$ and an integer $K^\flat$ exist, obtained from $\mathfrak{D}$, n, ℓ, b_-, b_+ and d through the formulas in the proof that closes this subsection, such that the two bounds stated next hold whenever $b \in [b_-, b_+]$ is a label, $K \ge K^\flat$ is an integer, $\beta := L^K b$, and $\underline{f} = (f_1,\dots,f_n)$ is a tuple of real-valued Lipschitz functions on T_ℓ, each different from 0, with no point common to all of their supports and with a spread obeying $\mathsf{d}_\infty(\underline{f}) \ge d$.

(a) For every set Q with $\emptyset \ne Q \subset \{1,\dots,n\}$, with $X_i^Q := \mathcal{O}_K^{(2)}(f_i)$ for $i \in Q$ and $X_i^Q := \mathcal{O}_K(f_i)$ for $i \notin Q$: $|\langle X_1^Q;\dots;X_n^Q\rangle_{K,\ell,\beta}| \le C^\flat K^{\mathsf{p}_n^\natural} L^{-K/10}\prod_{i=1}^n \|f_i\|_\sharp$.

(b) $|\langle \mathcal{O}'_K(f_1);\dots;\mathcal{O}'_K(f_n)\rangle_{K,\ell,\beta} - S_n^{(\ell,b)}(\underline{f})| \le C^\flat K^{\mathsf{p}_n^\natural} L^{-K/10}\prod_{i=1}^n \|f_i\|_\sharp$.

Hence, as $K \to \infty$ at a fixed label $b > 0$, the cumulants in (a) converge to 0 and the cumulant of $\mathcal{O}'_K(f_1),\dots,\mathcal{O}'_K(f_n)$ converges to $S_n^{(\ell,b)}(\underline{f})$, uniformly in b on every compact subset of $(0,\infty)$ and in the tuples of spread at least d with $\prod_i \|f_i\|_\sharp \le 1$. ∎

The proof occupies the rest of the subsection; it is completed after Lemma 18.139.

The level-wise classes of sources. Use the line $x_j = L^{K-j}\mathsf{b}'$ in the setting (Th3) of §2.3.1, with its top level t_*. The field cuts Cut_t and inner balls In_t are those of Lem. YM3_I:113.10, with $\mathsf{e}_t^{\mathrm{c}} = c_\rho\varepsilon(x_t)$. Let Bc be the set of bare plaquette barycentres and identify weights of the plaquettes with functions on it. For such a source w, the relative weight is $v_j^w = -w/\mathsf{b}'$ and $x_j^w = x_j - L^{K-j}w$. For $0 \le k < K$ set

$$\|w\|_{\mathrm{lw}}^{(k)} := \max_{k\le j<K}\max\left\{\frac{|v_j^w|_\infty}{\sigma_{\mathrm{src}}}, \frac{\mathrm{sw}_j(v_j^w)}{o_{\mathrm{src}}(x_j)}\right\}, \qquad \mathsf{Src}_k^{\mathrm{lw}} := \{w : \|w\|_{\mathrm{lw}}^{(k)} < 1\}.$$

The swing is the supremum of $|v(p)-v(q)|$ over plaquette centres at sup distance at most Ma_{K-j}. The original smooth-source class $\mathsf{Src} = \{w : \|w\|_{\flat;\mathsf{Bc}} < 2\varrho\}$ uses $\|w\|_{\flat;\mathsf{Bc}} = \max\{|w|_\infty, (40M/\sqrt{3})\operatorname{Lip}_\infty w\}$. The same formula defines $\|\cdot\|_{\flat;T_\ell}$ for functions on the torus. Finally put $\beta = x_0$ and $r_1 = \mu_\star\sqrt{\beta}/(2K_{\mathrm{L}})$.

Lemma 18.130 (The level-wise norms; rough directions)**.** Use the calibrated line above and let $0 \le k < K$.

(a) $\|\cdot\|_{\mathrm{lw}}^{(k)}$ is a norm on $\mathbb{C}^{\mathsf{Bc}}$, and $\mathsf{Src}_k^{\mathrm{lw}}$ is open and convex and contains 0; for $0 \le k' \le k$ one has $\|\cdot\|_{\mathrm{lw}}^{(k)} \le \|\cdot\|_{\mathrm{lw}}^{(k')}$, so that $\mathsf{Src}_{k'}^{\mathrm{lw}} \subset \mathsf{Src}_k^{\mathrm{lw}}$.

(b) $\mathsf{Src} \subset \mathsf{Src}_0^{\mathrm{lw}}$, and $\|w\|_{\mathrm{lw}}^{(0)} \le \|w\|_{\flat;\mathsf{Bc}}/(2\varrho_\varsigma)$ for every source w.

(c) If $2K_{\mathrm{L}}\, x_k^{1/2} \ge K_{\mathrm{Q}}$, then every source w, written as $w(p) = L^{-K}\tau_p$, satisfies $\|w\|_{\mathrm{lw}}^{(k)} \le (|\tau|_\infty/r_1)\, L^{-k/2}$.

∎

Sketch of proof. (a) The map $w \mapsto v_j^w$ is linear and injective, $|\cdot|_\infty$ is a norm and sw_j a seminorm; a maximum over fewer heights is smaller. (b) For $\|w\|_{\flat;\mathsf{Bc}} < 2\varrho$, $|v_j^w|_\infty = |w|_\infty/\mathsf{b}'$ and $\mathrm{sw}_j(v_j^w) \le M\,\mathrm{Lip}_\infty(w)/x_j$. The source-room inequalities YM3_I:(2171), together with $x_j^{-1/2} \le \gamma$, give $|v_j^w|_\infty < \sigma_{\mathrm{src}}$ and $\mathrm{sw}_j(v_j^w) < o_{\mathrm{src}}(x_j)$. Thus the open $\|\cdot\|_\flat/(2\varrho)$ unit ball is contained in the open $\|\cdot\|_{\mathrm{lw}}^{(0)}$ unit ball. Homogeneity gives the stated norm inequality. (c) One has $|v_j^w|_\infty = |\tau|_\infty/\beta$ and $\mathrm{sw}_j(v_j^w) \le 2|v_j^w|_\infty$. As $o_{\mathrm{src}}(x_j) = x_j^{-1/2}\mu_\star/K_{\mathrm{L}}$ increases with j and $x_k = L^{-k}\beta$, for $j \ge k$ the second quotient is at most $2|\tau|_\infty K_{\mathrm{L}} x_k^{1/2}/(\mu_\star\beta) = (|\tau|_\infty/r_1)L^{-k/2}$; the first quotient is $|\tau|_\infty K_{\mathrm{Q}}/(\mu_\star\beta)$, which the hypothesis bounds by the same number. □

Lemma 18.131 (The flows on the level-wise classes)**.** Use the calibrated line above. Let $0 \le k < t_*$ and let $\mathcal{S}'$ be an open set containing 0 in $\mathsf{Src}_k^{\mathrm{lw}}$. At level k, an early system on $\mathcal{S}'$ is a sequence of pointed families $G^{(i)}[w]$, $1 \le i \le k$, on the fixed pair cores, bounded in the norm of rate $\kappa+1$ by E_0, jointly holomorphic in w and the open pair slices, and depending at each domain X only on the source on its plaquettes. Form the flow by the recursion YM3_I:(2172) starting with this sequence.

(a) The flow continues through level K and preserves the three conditions on the families. Its fourth potential has norm at most $7w_\star/32$. Its density and normalization increments on the cuts are those of Prop. YM3_I:114.2. The zero early sequence is admitted and gives the classical start. The empty sequence at $k = 0$ gives the chain on $\mathsf{Src}_0^{\mathrm{lw}}$. If $\mathcal{S}' \subset \mathsf{Src}_0^{\mathrm{lw}}$, the chain can be restarted at k with its own first k families.

(b) The difference estimate of Prop. YM3_I:114.3 applies to two early systems at the same source. In particular the exponent difference between the chain and the classical start at k is at most $\epsilon_{\mathrm{fg}}(t,k)$ on Cut_t whenever both flows are defined, with the constant in equation YM3_I:(2179).

(c) For $k < t \le t_*$ and $V \in \mathsf{Cut}_t$, choose a representative $U = U_t(V)$ of its regular critical orbit. The exponent is

$$\Phi_t^w(V) = -A^{\eta_t}(x_t, U) + L^{K-t}A^{\eta_t}(w, U) + \Psi_V(w), \qquad \Psi_V(w) = S[\mathfrak{f}_t[w]](U).$$

The function Ψ_V is holomorphic and $|\Psi_V| \le \mathsf{m}_t^S = C_{\mathrm{old}} L E_0 N_t/(L-1)$ on $\mathcal{S}'$, where $N_t = \ell^3 L^{3(K-t)}$. The same applies to the classical start.

(d) Every flat member $E^{(j)}[w](X, \mathbf{1}, z')$ is holomorphic in w, bounded by $E_0 e^{-(\kappa+1)d_j(X)}$, and depends only on $w|_{\mathcal{P}(X)}$, with $\mathcal{P}(X)$ the bare plaquettes whose corners lie in X.

The flows agree with the original smooth-source flows on their common source class. Restricting a flow to a smaller parameter set leaves it unchanged there. ■

Proof. The direct source lemma in YM3 I, Lem. YM3_I:114.1, applies to an arbitrary complex plaquette array v. It proves holomorphy, locality and the exact weighted action difference on the actual averaging fibre, as well as the bound

$$\mathsf{n}^{\mathrm{f}}(\mathbb{W}^{\mathrm{src}}[v]) \le 2K_{\mathrm{Q}}|v|_\infty + K_{\mathrm{L}}\sqrt{x_j}\,\mathrm{sw}_j(v)$$

at height j, including height zero. For $w \in \mathcal{S}'$ the right side at $v = v_j^w$ is at most $3\mu_\star$ at every $j \ge k$. The old array has size at most $LE_0/(L-1)$, hence its fourth-potential contribution is at most $w_\star/8$ by the calibration. The sum is at most $w_\star/8 + 3\mu_\star \le 7w_\star/32$. In the proof of Prop. YM3_I:114.2, the unperturbed interaction costs at most $E_0/4$, the normalization family at most $E_0/2$, and the change on adding the source at most $3K^{\mathrm{d}}\mu_\star \le 3E_0/16$. Consequently the new norm is at most $15E_0/16$. The localized output map is holomorphic in these parameters. Its locality and that of the input arrays give the required dependence only on plaquettes of X. These estimates inductively admit every step. The exact fibre identity of the direct source lemma and the actual cut-domain sum rule apply to this same fourth potential. They give the contact increment YM3_I:(2173) and the density YM3_I:(2175), with the fixed real Gaussian normalizations. This proves (a), including equality on restarting, which follows from the recursion.

For (b) the direct source and normalizer cancel between the two recursions. The old-array norm and the fourth-family difference estimate thus give the recurrence of Prop. YM3_I:114.3, independently of the source class. Apply the action estimate of Prop. YM3_I:113.13 to the difference and use equations YM3_I:(2179)–(2180). The same action estimate for one flow bounds its $S[\mathfrak{f}_t[w]](U)$. It is a finite sum of values of holomorphic local members at a background independent of w, so it is holomorphic. This proves (c). The induction proves (d), including evaluation at the trivial pair. Each flow is specified by the same deterministic step formula, including the same early families for the chain or the classical start. This proves agreement on the common domain. □

The small-field term at a rough source.

Lemma 18.132 (Wilson's generating function on the level-wise class)**.** Use the calibrated line with $\mathsf{b}' = b$ and choose $n \geq 2$, $u = \lceil K/[180(3n+4)] \rceil$, $s = \lceil K/9 \rceil$, $t = t_1 = K - u$, $t_0 = K - s$ with $1 \leq u < s < K$ and $t \leq t_*$. Write $x_{[d]} = L^d b$ for a height indexed by its remaining depth. Write $N_{[u]} = \ell^3 L^{3u} = N_t$ and assume $\epsilon_{\mathrm{fg}} = \epsilon_{\mathrm{fg}}(t, t_0) \leq 1/8$, $\mathsf{R}_t \leq 1/2$, where R_t is the discarded-mass bound YM3_I:(2187). Let $\mathfrak{w}_t^{\mathrm{sf}}$ be the fixed cutoff weight in Prop. YM3_I:115.2 and put

$$T(w) = \int \mathfrak{w}_t^{\mathrm{sf}} \rho_0 e^{\mathcal{O}(w)}\, dU, \qquad \mathcal{O}(w) = L^K \sum_p w(p) a_p.$$

Use the chain on $\mathsf{Src}_0^{\mathrm{lw}}$ and the classical start at t_0 on $\mathsf{Src}_{t_0}^{\mathrm{lw}}$ from Lem. 18.131. Let Φ_t^w and $\hat{\Phi}_t^w$ be their exponents on Cut_t, and $I(w)$ and $\hat{I}(w)$ their exponentials integrated over In_t against normalized Haar measure.

(a) T is entire on $\mathbb{C}^{\mathsf{Bc}}$ and $T(0) \geq \frac{1}{2} \int \rho_0\, dU > 0$.
(b) On $\mathsf{Src}_0^{\mathrm{lw}}$, $T(w) = e^{c(w)} I(w)$ with $c(w)$ the sum of the chain contact increments through t.
(c) There is a real constant $c_\star$ such that $c(w) = c_\star + \sum_{j \leq t, X, z'} e_{j,X,z'}(w)$ with $e_{j,X,z'}(w) = E^{(j)}[w](X, \mathbf{1}, z')$ holomorphic, of modulus at most $E_0 e^{-(\kappa+1) d_j(X)}$ and depending only on $w|_{\mathcal{P}(X)}$.
(d) On their respective source classes the two exponents have the form in Lem. 18.131(c), with remainders $\Psi_V(w)$ and $\hat{\Psi}_V(w)$ both bounded by m_t^S. The integrals I and $\hat{I}$ are holomorphic there. Moreover $|\Phi_t^w(V) - \hat{\Phi}_t^w(V)| \leq \epsilon_{\mathrm{fg}}$ for $w \in \mathsf{Src}_0^{\mathrm{lw}}$ and $V \in \mathsf{Cut}_t$.

■

Proof. The finite-lattice exponential has a locally uniform integrable bound since the plaquette variables and the fixed cutoff weight are bounded. This gives entireness, and equation YM3_I:(2190) at zero source gives (a). The contact increment YM3_I:(2173) and Lem. 18.131(d) give (c). Parts (b) and (c) of that lemma prove the exponent claims of (d). On any compact source set the exponential is uniformly bounded on In_t: use the cut action estimate YM3_I:(2161) for $x_t^w = x_t - L^{K-t} w$ and the bound m_t^S. Cauchy's formula and Fubini's theorem then give holomorphy of both integrals. For smooth sources the exact image identity YM3_I:(2188) and the chain density YM3_I:(2175) give (b). Both sides are holomorphic on the connected set $\mathsf{Src}_0^{\mathrm{lw}}$, and its subset Src is nonempty and open in the finite-dimensional source space. The identity theorem gives (b) throughout $\mathsf{Src}_0^{\mathrm{lw}}$. □

Two tools.

Lemma 18.133 (Sums over plaquettes of one-plaquette derivatives)**.** Let P be a nonempty finite set, $n \geq 1$, $\rho^{\mathrm{z}} \in (0, \infty)^n$ and $\rho^{\mathrm{p}} > 0$. For $0 < \theta \leq 1$ let $\mathbb{D}_\theta := \{(z, \tau) \in \mathbb{C}^n \times \mathbb{C}^{\mathsf{P}} : \ |z_i| < \theta \rho_i^{\mathrm{z}},\ |\tau_p| < \theta \rho^{\mathrm{p}}\}$, let $\partial_p := \partial/\partial \tau_p$, and for $q \in \mathsf{P}$ let $(z, \tau)^{(q)}$ be the point (z, τ) with τ_q replaced by 0. For a holomorphic function F on $\mathbb{D}_\theta$ put $\|F\|_\theta := \sup_{\mathbb{D}_\theta} |F|$, $\mathrm{osc}_\theta F := \sup\{|F(z,\tau) - F(z,\tau')| : \ (z,\tau), (z,\tau') \in \mathbb{D}_\theta\}$ and $[F]_\theta := \sup_{q \in \mathsf{P}} \sup_{\xi \in \mathbb{D}_\theta} |F(\xi) - F(\xi^{(q)})|$, the largest change of F when one plaquette variable is put to 0; these three numbers do not increase as θ decreases. Let $0 < \theta' < \theta \leq 1$, $\Delta' := (\theta - \theta') \rho^{\mathrm{p}}$,

$\xi \in \mathbb{D}_{\theta'}$ and $g \in \mathbb{C}^{\mathsf{P}}$; let $G_1, \dots, G_k$ ($k \geq 0$) and $F_1, \dots, F_l$ ($l \geq 1$) be holomorphic on $\mathbb{D}_\theta$, and let $o_1, \dots, o_k, o'_1, \dots, o'_l \geq 1$ be integers. Then $|\partial_p^o G(\xi)| \leq o!\,[G]_\theta \Delta'^{-o}$ for every holomorphic G on $\mathbb{D}_\theta$, every $p \in \mathsf{P}$ and every integer $o \geq 1$, and

$$\Big|\sum_{p\in\mathsf{P}} g(p) \prod_{j\leq k} \partial_p^{o_j} G_j(\xi) \prod_{j\leq l} \partial_p^{o'_j} F_j(\xi)\Big| \leq |g|_\infty \prod_{j\leq k} \frac{o_j!\,[G_j]_\theta}{\Delta'^{o_j}} \prod_{j\leq l} \frac{o'_j!\,\min\{\|F_j\|_\theta, \operatorname{osc}_\theta F_j\}}{\Delta'^{o'_j}};$$

the bound does not depend on how many elements P has. ■

Sketch of proof. Let e_p be the unit vector of τ_p. The function $\zeta \mapsto G(\xi+\zeta e_p) - G(\xi^{(p)})$ is holomorphic on $|\zeta| < \Delta'$, where $\xi + \zeta e_p \in \mathbb{D}_\theta$, and bounded by $[G]_\theta$, since $(\xi+\zeta e_p)^{(p)} = \xi^{(p)}$; its derivative of order $o \geq 1$ at 0 is $\partial_p^o G(\xi)$, and Cauchy's estimate gives the first bound. So $\tilde g(p) := g(p) \prod_j \partial_p^{o_j} G_j(\xi)$ satisfies $|\tilde g|_\infty \leq |g|_\infty \prod_j o_j! [G_j]_\theta \Delta'^{-o_j}$; if $\tilde g = 0$ the sum vanishes. Otherwise let $o := \sum_j o'_j$, let $g_0(p)$ be an o-th root of $\tilde g(p)$, let $(\sigma_p)_{p\in\mathsf{P}}$ be independent random variables, each uniform on the o-th roots of unity, and put $\partial_\sigma := \sum_p \sigma_p g_0(p) \partial_p$. Expanding $\prod_j \partial_\sigma^{o'_j} F_j(\xi)$ into a sum over families of o plaquettes, the mean of the product of the o factors σ equals 1 when the o plaquettes coincide and vanishes for every other family, because $\mathbb{E}\,\sigma_p^\nu$ equals 1 when o divides ν and vanishes otherwise; so $\mathbb{E}\prod_j \partial_\sigma^{o'_j} F_j(\xi)$ is the sum to be bounded. For each value of σ the function $\zeta \mapsto F_j(\xi + \zeta(\sigma_p g_0(p))_p)$ is holomorphic on $|\zeta| < \Delta'/|g_0|_\infty$, with values of F_j at points of $\mathbb{D}_\theta$ that have the coefficient tuple z of ξ; its derivative of order o'_j at 0 is $\partial_\sigma^{o'_j} F_j(\xi)$, which is also the derivative of this function minus $F_j(\xi)$. Cauchy's estimate gives $|\partial_\sigma^{o'_j} F_j(\xi)| \leq o'_j! \min\{\|F_j\|_\theta, \operatorname{osc}_\theta F_j\} (|g_0|_\infty/\Delta')^{o'_j}$, and $|g_0|_\infty^o = |\tilde g|_\infty$. □

Lemma 18.134 (Cumulants in their dependence on a source; entries with a squared plaquette action)**.** Let m be a non-zero finite measure on a measurable space, let P be a finite set, and let a_p ($p \in \mathsf{P}$) and $O_1, \dots, O_n$ be real measurable functions that are bounded; let $Q \subset \{1, \dots, n\}$, and for $i \in Q$ let $g_i \in \mathbb{R}^{\mathsf{P}}$. Put $X_i := \frac12 \sum_p g_i(p)\, a_p^2$ for $i \in Q$, $X_i := O_i$ for $i \notin Q$, and

$$\mathcal{G}(z,\tau,\lambda) := \int \exp\Big[\sum_i z_i O_i + \sum_p \tau_p a_p + \sum_i \lambda_i X_i\Big] d\mathsf{m}, \qquad T_{\mathsf{m}}(z,\tau) := \mathcal{G}(z,\tau,0),$$

an entire function on $\mathbb{C}^n \times \mathbb{C}^{\mathsf{P}} \times \mathbb{C}^n$. On the open set where $T_{\mathsf{m}} \neq 0$, for $\emptyset \neq S \subset \{1, \dots, n\}$ let $\Xi_S(z,\tau) := \partial_\lambda^S \mathcal{W}(z,\tau,0)$, $\partial_\lambda^S := \prod_{j\in S} \partial_{\lambda_j}$, where $\mathcal{W}$ is a holomorphic logarithm of $\mathcal{G}$ near $(z,\tau,0)$; $\Xi_\emptyset$ denotes any holomorphic logarithm of T_{m} near the point in question, and only its derivatives enter. Then Ξ_S is well defined and holomorphic, $\Xi_{\{1,\dots,n\}}(0,0)$ is the joint cumulant of $X_1, \dots, X_n$ under the probability measure $\mathsf{m}/\mathsf{m}(1)$, and for $S \subset \{1, \dots, n\}$ and $i \notin S$, with $\partial_p := \partial/\partial\tau_p$:

(i) if $i \notin Q$: $\Xi_{S\cup\{i\}} = \partial_{z_i} \Xi_S$;
(ii) if $i \in Q$: $\Xi_{S\cup\{i\}} = \frac12 \sum_p g_i(p) \big[\partial_p^2 \Xi_S + \sum_{S'\subset S} \partial_p \Xi_{S'}\, \partial_p \Xi_{S\setminus S'}\big]$.

■

Sketch of proof. The integrands are bounded and m is finite, so $\mathcal{G}$ is entire and may be differentiated under the integral. Where $T_{\mathsf{m}}(z,\tau) \neq 0$, $\mathcal{G}$ has no zero near $(z,\tau,0)$ and a holomorphic logarithm there; two logarithms differ by a constant, which ∂_λ^S annihilates for $S \neq \emptyset$. At $(z,\tau) = (0,0)$, $\lambda \mapsto \mathcal{G}(0,0,\lambda)$ is the generating function of $X_1, \dots, X_n$ under m, and division by $\mathsf{m}(1)$ changes the logarithm by a constant. (i) $\partial_{\lambda_i} \mathcal{G} = \partial_{z_i} \mathcal{G}$; divide by $\mathcal{G}$ and apply ∂_λ^S at $\lambda = 0$. (ii) $\partial_{\lambda_i} \mathcal{G} = \frac12 \sum_p g_i(p) \partial_p^2 \mathcal{G}$, hence $\partial_{\lambda_i} \mathcal{W} = \frac12 \sum_p g_i(p)\, e^{-\mathcal{W}} \partial_p^2 e^{\mathcal{W}} = \frac12 \sum_p g_i(p) \big[\partial_p^2 \mathcal{W} + (\partial_p \mathcal{W})^2\big]$; apply ∂_λ^S at $\lambda = 0$, and Leibniz's rule to the square, each ∂_{λ_j}, $j \in S$, falling on one of its two factors. □

Tuples in groups.

Proposition 18.135 (Cumulants with squared entries over tuples in at least two groups)**.** Use the calibrated line at b and the data and two smallness conditions of Lem. 18.132. Let $n \geq 2$ and let $f_1, \dots, f_n$ be nonzero real Lipschitz functions on T_ℓ. Put $\beta = L^K b$, and let T, c, I, Φ_t^w, $\hat\Phi_t^w$, Ψ_V,

$\hat\Psi_V$ be as in Lemma 18.132. Let $d'>0$, and let $\{1,\dots,n\}=\bigsqcup_{a\in\mathfrak{A}} C_a$ be a partition into $\#\mathfrak{A}\ge 2$ groups such that $\operatorname{dist}_\infty(\operatorname{supp} f_i,\operatorname{supp} f_{i'})\ge d'$ whenever i and i' lie in two different groups. Put

$$r_1:=\frac{\mu_\star\beta^{1/2}}{2K_{\mathrm L}},\qquad \bar{\mathsf m}:=\max\{1,\mathsf m_t^S\},$$
$$r_{\mathrm{eff}}:=\frac{r_1}{16n\bar{\mathsf m}},\qquad \Delta:=\frac{r_{\mathrm{eff}}}{4n},$$
$$C_{\mathrm{pt}}:=8e^2(3M+1)^3C_{\mathrm{E},0},\qquad \Gamma:=\max\{1,2E_0\,t\,C_{\mathrm{pt}}\},$$

$\epsilon_{\mathrm{lk}}:=4e^3E_0C_{\mathrm{E},0}N_{[u]}e^{-d'L^u/M}$, $\epsilon_{\mathrm{rd}}:=\frac32(\mathsf e_t^{\mathrm c})^2N_{[u]}L^{-t}r_{\mathrm{eff}}+\frac{3}{16n}L^{-t_0/2}$, $\epsilon_{\mathrm{osc}}:=2\epsilon_{\mathrm{lk}}+8\epsilon_{\mathrm{fg}}+4\epsilon_{\mathrm{rd}}$, with $N_{[u]}=\ell^3L^{3u}$, and assume

(Y1) $\epsilon_{\mathrm{osc}}\le\frac12$ and $\epsilon_{\mathrm{rd}}\le\frac18$;
(Y2) $\frac32(\mathsf e_t^{\mathrm c})^2N_{[u]}L^{-t}\le\bar{\mathsf m}/r_1$;
(Y3) $2K_{\mathrm L}(L^sb)^{1/2}\ge K_{\mathrm Q}$;
(Y4) $d'L^u\ge 6M$;
(Y5) $16n\,\Gamma\le r_{\mathrm{eff}}$.

Let $\emptyset\neq Q\subset\{1,\dots,n\}$, $X_i^Q:=\mathcal O_K^{(2)}(f_i)$ for $i\in Q$, $X_i^Q:=\mathcal O_K(f_i)$ for $i\notin Q$, and $\Lambda'''(F):=\int F\,\mathfrak w_t^{\mathrm{sf}}\rho_0\,dU$. Then

$$|\mathsf{cum}_{\Lambda'''}(X_1^Q,\dots,X_n^Q)|\le\epsilon_{\mathrm{osc}}\prod_{i=1}^n\mathsf A_i,$$
$$\mathsf A_i:=\frac{8n}{r_i(t)}\quad(i\notin Q),$$
$$\mathsf A_i:=\frac{2^{n+4}\Gamma^2L^K\|f_i\|_\infty}{\Delta^2}=\frac{2^{n+18}n^4K_{\mathrm L}^2\bar{\mathsf m}^2\Gamma^2\,\|f_i\|_\infty}{\mu_\star^2\,b}\quad(i\in Q),$$

with the radii of Lem. YM3_I:114.5,

$$r_i(t)=(8n\mathsf M_t^{\mathrm{rd}}\|f_i\|_{\flat;T_\ell})^{-1},\qquad \mathsf M_t^{\mathrm{rd}}=\max\{\varrho^{-1},\tfrac32(\mathsf e_t^{\mathrm c})^2N_tL^{K-t},\varrho^{-1}\mathsf m_t^S\}.$$

■

Sketch of proof. Step 1: the variables. Put $\mathsf P_i:=\{p:\ f_i(x(p))\neq0\}$, $\mathsf P^a:=\bigcup_{i\in C_a}\mathsf P_i$ and $\mathsf P:=\bigcup_a\mathsf P^a$. If some $\mathsf P_i$ is empty, the corresponding insertion vanishes identically, so the asserted bound holds. Otherwise, as $x(p)\in\operatorname{supp} f_i$ for $p\in\mathsf P_i$, the sets $\mathsf P^a$ are disjoint, and barycentres of plaquettes of two different sets $\mathsf P^a$ lie at sup distance $\ge d'$. For $(z,\tau)\in\mathbb C^n\times\mathbb C^{\mathsf P}$ let $w(z,\tau):=\sum_i z_i\,f_i\circ x+L^{-K}\tau$, the vector τ being continued by 0 off $\mathsf P$, and $w_a(z,\tau):=\sum_{i\in C_a}z_i\,f_i\circ x+L^{-K}\tau\,\mathbf 1_{\mathsf P^a}$; then $w=\sum_a w_a$ and $\mathcal O(w(z,\tau))=\sum_i z_i\mathcal O_K(f_i)+\sum_p\tau_pa_p$. On sources put $\|h\|_\star:=\max\{\frac32(\mathsf e_t^{\mathrm c})^2N_tL^{K-t}|h|_\infty,\ \bar{\mathsf m}\|h\|_{\mathrm{lw}}^{(0)}\}$, a norm, with $N_t=N_{[u]}$. By Lem. 18.130(b) and the displayed read-out radii, $\|f_i\circ x\|_\star\le\mathsf M_t^{\mathrm{rd}}\|f_i\|_{\flat;T_\ell}=(8n\,r_i(t))^{-1}$; by Lemma 18.130, (c), at $k=0$, whose hypothesis follows from (Y3) since $x_0\ge x_{t_0}=L^sb$, and by (Y2), $\|L^{-K}\tau\|_\star\le\bar{\mathsf m}|\tau|_\infty/r_1$. Take $\rho_i^{\mathrm z}:=\frac14r_i(t)$, $\rho^{\mathrm p}:=\frac12r_{\mathrm{eff}}$ and the polydiscs $\mathbb D_\theta$ of Lemma 18.133. Every $(z,\tau)\in\mathbb D_1$ then satisfies $\|w(z,\tau)\|_\star<\frac1{32}+\frac{1}{32n}<\frac1{16}$, and so does every sum of some of the $w_a(z,\tau)$.

Step 2: where the generating function does not vanish. Suppose $\|h\|_\star<1/16$. In particular h belongs to both source classes used at t. On Cut_t, equation YM3_I:(2161) gives $|L^{K-t}A^{\eta_t}(h,U)|<1/16$. The function $\hat\Psi_V$ is bounded by $\mathsf m_t^S$, so Schwarz's lemma on the line through $0,h$ gives $|\hat\Psi_V(h)-\hat\Psi_V(0)|\le2\mathsf m_t^S\|h\|_{\mathrm{lw}}^{(t_0)}<1/8$. Hence $|\hat\Phi_t^h-\hat\Phi_t^0|<3/16$. Also $|\Phi_t^h-\hat\Phi_t^h|\le\epsilon_{\mathrm{fg}}\le1/8$ by Lem. 18.132(d).

At zero source the classical start and the operators in its density recursion are positive. By YM3_I:(2175), $e^{\hat\Phi_t^0}$ is a fixed nonzero scalar times a positive density, as in the proof of Prop. YM3_I:

116.3. The inner ball is open and contains the identity, and the exponential is everywhere nonzero there. Hence $\hat I(0)\neq 0$ and $e^{\hat\Phi^0_t}/\hat I(0)$ is a probability density on the inner ball. Let $\alpha = 3/16$. Relative to this probability measure, $\hat I(h)/\hat I(0)$ is an average of numbers e^z with $|z|<\alpha$. Its argument has modulus at most α, and its real part is at least $e^{-\alpha}\cos\alpha$. Similarly

$$\left|\frac{I(h)}{\hat I(h)}-1\right| \le (e^{\epsilon_{\rm fg}}-1)\frac{e^{2\alpha}}{\cos\alpha} \le 2\epsilon_{\rm fg} \le \frac14.$$

The numerical bound uses $e^{1/8+2\alpha}/\cos\alpha < 2$. Therefore $I(h)$ and $T(h)$ are nonzero, in particular on the polydisc of Step 1.

Step 3: the local part. For a triple (j,X,z') of the sum in Lemma 18.132, (c), let $\mathsf I(X) := \{a : \mathcal P(X)\cap \mathsf P^a \neq \emptyset\}$. If $\mathsf I(X)=\emptyset$, the source $w(z,\tau)$ vanishes on $\mathcal P(X)$; if $\mathsf I(X)=\{a\}$, it agrees there with $w_a(z,\tau)$. The locality in Lemma 18.132, (c), therefore gives, at $w=w(z,\tau)$, $w_a=w_a(z,\tau)$, $(z,\tau)\in\mathbb D_1$,

$$\begin{aligned} c(w) &= c(0)+\sum_a c_a(w_a)+c_{\rm lk}(w),\\ c_a(w_a) &:= \sum_{\mathsf I(X)=\{a\}} [e_{j,X,z'}(w_a)-e_{j,X,z'}(0)],\\ c_{\rm lk}(w) &:= \sum_{\#\mathsf I(X)\ge 2} [e_{j,X,z'}(w)-e_{j,X,z'}(0)], \end{aligned}$$

and $c_a(w_a(z,\tau))$ is a function of $(z_i)_{i\in C_a}$ and $(\tau_p)_{p\in\mathsf P^a}$ only. If $\#\mathsf I(X)\ge 2$, then $\mathcal P(X)$ contains plaquettes of two different sets $\mathsf P^a$, and their barycentres are at sup distance at least $d'L^{K-j}$ in the units of level j; their corners lie in X, within $\frac12$ of the barycentres, and the diameter bound of Prop. YM3_I:45.1 gives $d_j(X)\ge D_j := d'L^{K-j}/M-3$. The rooted count in the proof of Lem. YM3_I:114.6, with rate $\kappa+1$ and size at least D_j, gives a level sum at most $2E_0C_{\mathrm{E},0}N_je^{-D_j}$. Moving one level finer multiplies this bound by $L^3\exp[-(L-1)d'L^{K-j}/M]\le 1/2$ by (Y4). Summing therefore gives $|c_{\rm lk}|\le 4e^3E_0C_{\mathrm{E},0}N_te^{-d'L^{K-t}/M}=\epsilon_{\rm lk}$. Let $c_{\rm cl}(z,\tau):=\sum_a c_a(w_a(z,\tau))$. For $q\in\mathsf P$, putting τ_q to 0 changes each source w_a at the single plaquette q; by the locality in Lemma 18.132, (c), a term $e_{j,X,z'}$ moves only when $q\in\mathcal P(X)$, and then by no more than $2E_0e^{-(\kappa+1)d_j(X)}$. If y is a corner of q, $q\in\mathcal P(X)$ and $z'\in X$, then $|y-z'|_\infty\le M(d_j(X)+2)$ in the units of level j. The same rooted count with $D=\max\{0,|y-z'|_\infty/M-2\}$ bounds the sum over these X by $C_{\mathrm{E},0}e^{2-|y-z'|_\infty/M}$, and, choosing for each site of $\mathbb T^{(j)}$ the lift nearest to a lift of y, $\sum_{z'\in\mathbb T^{(j)}}e^{-|y-z'|_\infty/M}\le\prod_{\mu\le3}\sum_{k\in\mathbb Z}e^{-|k-y_\mu|/3M}\le[2(3M+1)]^3$, by $|x|_\infty\ge\frac13\sum_\mu|x_\mu|$. Summing over the levels $j\le t$: $[c_{\rm cl}]_1\le 2E_0\,t\,C_{\rm pt}\le\Gamma$.

Step 4: the non-local part. For $\|h\|_\star<1/16$ put

$$\vartheta(h)=c_{\rm lk}(h)+\mathrm{Log}\,\frac{\hat I(h)}{\hat I(0)}+\mathrm{Log}\,\frac{I(h)}{\hat I(h)}-\mathrm{Log}\,\frac{I(0)}{\hat I(0)}.$$

All logarithms are principal and holomorphic by Step 2. On the source polydisc the grouping of Step 3 gives $T(w)=T(0)\exp[c_{\rm cl}+\vartheta(w)]$.

For two points $(z,\tau),(z,\tau')\in\mathbb D_1$ let $h=w(z,\tau)$ and $h'=w(z,\tau')$. Then $h'-h=L^{-K}(\tau'-\tau)$ with $|\tau'-\tau|_\infty<r_{\rm eff}$. The difference of the two classical-start exponents has a linear action term bounded by $\frac32(\mathsf e_t^{\rm c})^2N_tL^{-t}r_{\rm eff}$ by equation YM3_I:(2161). For the other term, Lem. 18.130(c) at $k=t_0$, using (Y3), gives

$$\|h'-h\|^{(t_0)}_{\rm lw}\le\frac{r_{\rm eff}}{r_1}L^{-t_0/2}=\frac{L^{-t_0/2}}{16n\bar{\mathfrak m}}.$$

If $h' \neq h$, restrict $\hat\Psi_V$ to $h+\zeta(h'-h)$ on the disc of radius $(1-\|h\|_{\mathrm{lw}}^{(t_0)})/\|h'-h\|_{\mathrm{lw}}^{(t_0)} > 1$. After subtracting its value at zero on this line, Schwarz's lemma and the bound $2\mathsf{m}_t^S$ give

$$|\hat\Psi_V(h') - \hat\Psi_V(h)| \le \frac{32}{15}\mathsf{m}_t^S \|h'-h\|_{\mathrm{lw}}^{(t_0)} \le \frac{3}{16n} L^{-t_0/2}.$$

The same holds when $h' = h$. This proves $|\hat\Phi_t^{h'} - \hat\Phi_t^h| \le \epsilon_{\mathrm{rd}}$. As in Step 2, with $\alpha = 3/16$,

$$\left|\frac{\hat I(h')}{\hat I(h)} - 1\right| \le (e^{\epsilon_{\mathrm{rd}}} - 1)\frac{e^{2\alpha}}{\cos\alpha} \le 2\epsilon_{\mathrm{rd}} \le \frac14.$$

The principal logarithm of this ratio is the difference of the two principal logarithms relative to $\hat I(0)$, since their arguments differ by at most $2\alpha < \pi$. Its modulus is at most $4\epsilon_{\mathrm{rd}}$ by the power series for the logarithm on $|z-1| < 1/2$. Each logarithm $\mathrm{Log}(I/\hat I)$ has modulus at most $4\epsilon_{\mathrm{fg}}$ by the same bound. The linked local term changes by at most $2\epsilon_{\mathrm{lk}}$. Thus $\mathrm{osc}_1(\vartheta\circ w) \le 2\epsilon_{\mathrm{lk}} + 4\epsilon_{\mathrm{rd}} + 8\epsilon_{\mathrm{fg}} = \epsilon_{\mathrm{osc}}$.

Step 5: splitting the recursion into a local and a non-local part. Apply Lemma 18.134 with $\mathsf{m} := \mathfrak{w}_t^{\mathrm{sf}}\rho_0\,dU$, the functions $O_i := \mathcal{O}_K(f_i)$ and a_p, $p \in \mathsf{P}$, the set Q and $g_i(p) := L^K f_i(x(p))$; its X_i are the X_i^Q, its function T_{m} is $T\circ w$, which has no zero on $\mathbb{D}_1$ (Step 2), and $\Xi_{\{1,\dots,n\}}(0,0) = \mathsf{cum}_{\Lambda''}(X_1^Q,\dots,X_n^Q)$. By Step 4 every derivative of $\Xi_\emptyset$ on $\mathbb{D}_1$ is the same derivative of $c_{\mathrm{cl}} + \vartheta\circ w$. Since the cumulant does not change under a permutation of its entries, we may and do assume $Q = \{n_0+1,\dots,n\}$. Define holomorphic functions Ξ_S^{c}, Ξ_S^ϑ on $\mathbb{D}_1$ by recursion on $\#S$: $\Xi_\emptyset^{\mathrm{c}} := c_{\mathrm{cl}}$ and $\Xi_\emptyset^\vartheta := \vartheta\circ w$; for $S \neq \emptyset$ let $i := \max S$ and $S' := S\setminus\{i\}$; if $i \notin Q$, $\Xi_S^{\mathrm{c}} := \partial_{z_i}\Xi_{S'}^{\mathrm{c}}$ and $\Xi_S^\vartheta := \partial_{z_i}\Xi_{S'}^\vartheta$; if $i \in Q$,

$$\Xi_S^{\mathrm{c}} := \frac{L^K}{2}\sum_p f_i(x(p))\Big[\partial_p^2\Xi_{S'}^{\mathrm{c}} + \sum_{S''\subset S'} \partial_p\Xi_{S''}^{\mathrm{c}}\,\partial_p\Xi_{S'\setminus S''}^{\mathrm{c}}\Big], \qquad \Xi_S^\vartheta := \Xi_S - \Xi_S^{\mathrm{c}}.$$

Induction on $\#S$, with Lemma 18.134, shows $\Xi_S = \Xi_S^{\mathrm{c}} + \Xi_S^\vartheta$ whenever $S \neq \emptyset$, and for $i \in Q$

$$\Xi_S^\vartheta = \frac{L^K}{2}\sum_p f_i(x(p))\Big[\partial_p^2\Xi_{S'}^\vartheta + \sum_{S''\subset S'} \big(\partial_p\Xi_{S''}^{\mathrm{c}}\,\partial_p\Xi_{S'\setminus S''}^\vartheta + \partial_p\Xi_{S''}^\vartheta\,\partial_p\Xi_{S'\setminus S''}^{\mathrm{c}} + \partial_p\Xi_{S''}^\vartheta\,\partial_p\Xi_{S'\setminus S''}^\vartheta\big)\Big].$$

Step 6: the local part links no two groups. Let $S \neq \emptyset$. If S is contained in no single group, Ξ_S^{c} vanishes; if $S \subset C_a$, it is a function of $(z_j)_{j\in C_a}$ and $(\tau_p)_{p\in\mathsf{P}^a}$ only. This follows by induction on $\#S$, with $i = \max S \in C_a$. If $i \notin Q$: $\partial_{z_i}c_{\mathrm{cl}} = \partial_{z_i}c_a(w_a)$ by Step 3, and $\partial_{z_i}\Xi_{S'}^{\mathrm{c}} = 0$ when $\emptyset \neq S' \subset C_{a'}$ with $a' \neq a$, or when S' lies in no group. If $i \in Q$: the sum runs over $p \in \mathsf{P}_i \subset \mathsf{P}^a$; a factor $\partial_p^o\Xi_\emptyset^{\mathrm{c}}$ is $\partial_p^o c_a(w_a)$, and a factor $\partial_p^o\Xi_{S''}^{\mathrm{c}}$ with $S'' \neq \emptyset$ vanishes unless $S'' \subset C_a$, since otherwise $\Xi_{S''}^{\mathrm{c}}$ is 0 or does not depend on τ_p; so a product that is not 0 has $S' \subset C_a$ and depends on the variables of C_a alone. As $\#\mathfrak{A} \ge 2$, $\Xi_{\{1,\dots,n\}}^{\mathrm{c}} = 0$, and $\mathsf{cum}_{\Lambda''}(X_1^Q,\dots,X_n^Q) = \Xi_{\{1,\dots,n\}}^\vartheta(0,0)$.

Step 7: estimates by induction. Let $\theta_m := 1 - m/2n$ for $0 \le m \le n$, so that $(\theta_{m-1}-\theta_m)\rho^{\mathrm{p}} = \Delta$ and $(\theta_{m-1}-\theta_m)\rho_i^{\mathrm{z}} = 1/\mathsf{A}_i$ for $i \notin Q$. We show for every S with $\#S = m$

$$[\Xi_S^{\mathrm{c}}]_{\theta_m} \le \Gamma\prod_{j\in S}\mathsf{A}_j, \qquad \mathrm{osc}_{\theta_m}\Xi_S^\vartheta \le 2\epsilon_{\mathrm{osc}}\prod_{j\in S}\mathsf{A}_j, \qquad \|\Xi_S^\vartheta\|_{\theta_m} \le \epsilon_{\mathrm{osc}}\prod_{j\in S}\mathsf{A}_j \ \text{ if } S\cap Q \neq \emptyset.$$

At $m = 0$ this is Steps 3 and 4. Let $m \ge 1$, $i = \max S$, $\theta := \theta_{m-1}$ and $\theta' := \theta_m$. Passing to a smaller polydisc can only lower the three sizes $\|\cdot\|$, osc and $[\cdot]$; so the estimates already obtained for the sets with fewer than m elements, on $\mathbb{D}_{\theta_{m''}}$ with $\theta_{m''} \ge \theta$, are valid on $\mathbb{D}_\theta$. If $i \notin Q$, then $S\cap Q = \emptyset$; for F holomorphic on $\mathbb{D}_\theta$, Cauchy's estimate in z_i on the disc of radius $1/\mathsf{A}_i$, applied to $\xi \mapsto F(\xi) - F(\xi^{(q)})$ and to $z \mapsto F(z,\tau) - F(z,\tau')$, gives $[\partial_{z_i}F]_{\theta'} \le \mathsf{A}_i[F]_\theta$ and $\mathrm{osc}_{\theta'}\partial_{z_i}F \le \mathsf{A}_i\,\mathrm{osc}_\theta F$. Let $i \in Q$ and $\Pi := \prod_{j\in S'}\mathsf{A}_j$. For Ξ_S^ϑ, Lemma 18.133 with $\Delta' = \Delta$ and $g = \frac{L^K}{2}f_i\circ x$ bounds the sum with $\partial_p^2\Xi_{S'}^\vartheta$ by $\frac{L^K}{2}\|f_i\|_\infty\cdot 2\cdot 2\epsilon_{\mathrm{osc}}\Pi\Delta^{-2}$, each of the $2\cdot 2^{m-1}$ mixed sums by $\frac{L^K}{2}\|f_i\|_\infty\Gamma\cdot 2\epsilon_{\mathrm{osc}}\Pi\Delta^{-2}$, and each

of the 2^{m-1} sums with two factors Ξ^{ϑ} by $\frac{L^K}{2}\|f_i\|_\infty(2\epsilon_{\rm osc})^2\Pi\Delta^{-2}$; as $2\epsilon_{\rm osc}\le 1\le\Gamma$ by (Y1), and as $m\le n$,

$$\|\Xi^{\vartheta}_S\|_{\theta'}\le L^K\|f_i\|_\infty\Delta^{-2}\Pi\,\epsilon_{\rm osc}(2+2^m\Gamma+2^{m-1})\le 2^{n+4}\Gamma^2L^K\|f_i\|_\infty\Delta^{-2}\,\epsilon_{\rm osc}\Pi=\epsilon_{\rm osc}\prod_{j\in S}\mathsf{A}_j,$$

and $\mathrm{osc}_{\theta'}\le 2\|\cdot\|_{\theta'}$. For $\Xi^{\rm c}_S$ let $q\in\mathsf{P}$ and $\xi\in\mathbb{D}_{\theta'}$. The summand $p=q$ of $\Xi^{\rm c}_S(\xi)-\Xi^{\rm c}_S(\xi^{(q)})$ is at most $\frac{L^K}{2}\|f_i\|_\infty\cdot 2(2\Gamma+2^{m-1}\Gamma^2)\Pi\Delta^{-2}$, by the first bound of Lemma 18.133 at ξ and at $\xi^{(q)}$. For $p\neq q$ the derivative ∂_p commutes with $\xi\mapsto\xi^{(q)}$; with $F^q_R:=\Xi^{\rm c}_R-\Xi^{\rm c}_R(\cdot^{(q)})$, holomorphic on $\mathbb{D}_\theta$ with $\|F^q_R\|_\theta\le[\Xi^{\rm c}_R]_\theta$, the difference of the terms with ∂_p^2 is $\partial_p^2F^q_{S'}(\xi)$, and the difference of the products is $\partial_p\Xi^{\rm c}_{S''}(\xi)\,\partial_pF^q_{S'\setminus S''}(\xi)+\partial_pF^q_{S''}(\xi)\,\partial_p\Xi^{\rm c}_{S'\setminus S''}(\xi^{(q)})$. Lemma 18.133 with one function F, the other factor being put into the numbers $g(p)$ (and $g(q):=0$), which its first bound controls, gives $\frac{L^K}{2}\|f_i\|_\infty 2\Gamma\Pi\Delta^{-2}$ for the first sum over $p\neq q$ and $\frac{L^K}{2}\|f_i\|_\infty 2\Gamma^2\Pi\Delta^{-2}$ for each of the 2^{m-1} ordered pairs $(S'',S'\setminus S'')$. Adding,

$$[\Xi^{\rm c}_S]_{\theta'}\le\frac{L^K}{2}\|f_i\|_\infty\Delta^{-2}\Pi(6\Gamma+2^{m+1}\Gamma^2)\le 2^{n+4}\Gamma^2L^K\|f_i\|_\infty\Delta^{-2}\Pi=\mathsf{A}_i\Pi\le\Gamma\prod_{j\in S}\mathsf{A}_j.$$

Step 8: conclusion. The point $(0,0)$ lies in $\mathbb{D}_{\theta_n}$, and $\{1,\dots,n\}$ meets Q; Steps 6 and 7 give $|\mathsf{cum}_{\Lambda''}(X^Q_1,\dots,X^Q_n)|=|\Xi^{\vartheta}_{\{1,\dots,n\}}(0,0)|\le\epsilon_{\rm osc}\prod_i\mathsf{A}_i$. The second expression of A_i, $i\in Q$, follows from $\Delta^{-2}=2^{12}n^4\bar{\mathsf{m}}^2r_1^{-2}$ and $L^Kr_1^{-2}=4K_{\rm L}^2/(\mu_\star^2b)$. □

The numbers on an interval under (Tor).

Lemma 18.136 (Arithmetic of the bound of Proposition 18.135)**.** Let $m\in\mathbb{N}$, $\ell=2L^m$, let $\mathsf{J}=[b_-,b_+]$ be an admissible label interval on T_ℓ in §2.3.1, $n\ge 2$ and $d'>0$; take $\bar{\mathsf{C}}_{\mathsf{M}}$, $\bar{\mathsf{y}}_{[K]}$ and K_0 from the proof of Thm. YM3_I:AR, with $\bar{\mathsf{y}}_{[K]}=(\log(L^Kb_+))^{2\mathfrak{p}_0}$. Use $u=\lceil K/[180(3n+4)]\rceil$ and $s=\lceil K/9\rceil$. Put $c_{\mathsf{m}}:=\max\{1,C_{\rm old}LE_0/(L-1)\}$, $c_\Gamma:=\max\{1,2E_0C_{\rm pt}\}$,

$$\begin{aligned}
C_{\rm ins}&:=\max\Big\{64n^2\bar{\mathsf{C}}_{\mathsf{M}}\big(\log(Lb_+)\big)^{2\mathfrak{p}_0},\ \frac{2^{n+18}n^4K_{\rm L}^2c_{\mathsf{m}}^2c_\Gamma^2}{\mu_\star^2\,b_-}\Big\},\\
C_{\rm sm}&:=1+8C_{\rm old}C_{\rm fg}\tfrac{L}{L-1}E_0L^{6n+4}\\
&\quad+3c_\rho^2A_0^2\frac{\mu_\star}{K_{\rm L}}\big(\log(Lb_+)\big)^{2\mathfrak{p}_0}b_-^{-1/2}L^{6n+3}+L^{6n+1},\\
\bar\epsilon_{\rm rd}&:=\tfrac34c_\rho^2A_0^2\frac{\mu_\star}{K_{\rm L}}\,\bar{\mathsf{y}}_{[K]}N_{[u]}(L^Kb_-)^{-1/2}+\tfrac{3}{16n}L^{-(K-s)/2},
\end{aligned}$$

with $\epsilon_{\rm lk}$ of Proposition 18.135 and $\epsilon_{\rm fg}$ of equation YM3_I:(2192), and let $K_{\rm gr}(d')$ be the least integer $k\ge K_0$ such that at every integer $K\ge k$

$$\bar\epsilon_{\rm rd}\le\tfrac18,\qquad 2\epsilon_{\rm lk}+8\epsilon_{\rm fg}+4\bar\epsilon_{\rm rd}\le\tfrac12,\qquad 2K_{\rm L}(L^sb_-)^{1/2}\ge K_{\rm Q},$$
$$d'L^u\ge 6M,\qquad 2^9n^2K_{\rm L}c_\Gamma c_{\mathsf{m}}\,K\,N_{[u]}\le\mu_\star(L^Kb_-)^{1/2},\qquad 2\epsilon_{\rm lk}N_{[u]}^{2n}\le\ell^{6n+3}L^{-K/10}.$$

Then $K_{\rm gr}(d')$ exists, and for every $b\in\mathsf{J}$, every integer $K\ge K_{\rm gr}(d')$, every tuple and every partition into groups as in Proposition 18.135, the calibrated conditions of Lem. 18.132 and (Y1)–(Y5) are satisfied, and for every $\emptyset\neq Q\subset\{1,\dots,n\}$

$$\epsilon_{\rm osc}\prod_{i=1}^n\mathsf{A}_i\le C_{\rm ins}^n\,C_{\rm sm}\,\ell^{6n+3}K^{(2\mathfrak{p}_0+2)n}L^{-K/10}\prod_{i=1}^n\|f_i\|_\sharp.$$

■

Sketch of proof. By the choice of K_0 in the proof of Thm. YM3_I:AR, the calibrated conditions, $\epsilon_{\rm fg} \le 1/8$ and $\mathsf{R}_t \le 1/2$ hold for every $b \in \mathsf{J}$ and $K \ge K_0$. Enlarge K_0 if needed so that $L^u b_- \ge \mathsf{t}_\natural$, and take $t_* = t$. Then $\frac{3}{2}(\mathsf{e}_t^{\rm c})^2 L^{-t} = \frac{3}{2} c_\rho^2 A_0^2 \mathsf{y}_{[u]}/\beta$, with the numbers $\mathsf{y}_{[d]} = (\log x_{[d]})^{2\mathfrak{p}_0}$, and $\mathsf{y}_{[u]} \le \mathsf{y}_{[K]} \le \bar{\mathsf{y}}_{[K]} \le (\log(Lb_+))^{2\mathfrak{p}_0} K^{2\mathfrak{p}_0}$; with $r_{\rm eff} \le r_1$ and $\beta \ge L^K b_-$ this gives $\epsilon_{\rm rd} \le \bar\epsilon_{\rm rd}$; and (Y2), which after multiplication by r_1 reads $\frac{3}{4} c_\rho^2 A_0^2 (\mu_\star/K_{\rm L}) \mathsf{y}_{[u]} N_{[u]} \beta^{-1/2} \le \bar{\mathsf{m}}$, follows from $\bar\epsilon_{\rm rd} \le \frac{1}{8}$ and $\bar{\mathsf{m}} \ge 1$. So the first four inequalities of the list give (Y1)–(Y4), with $L^s b \ge L^s b_-$ in (Y3). By the radius bound YM3_I:(2204), $8n/r_i(t_1) \le 64 n^2 \bar{\mathsf{C}}_{\mathsf{M}} (\log(Lb_+))^{2\mathfrak{p}_0} K^{2\mathfrak{p}_0} N_{[u]} \|f_i\|_\sharp$; moreover $\bar{\mathsf{m}} \le c_{\mathsf{m}} N_{[u]}$ by equation YM3_I:(2179), $\Gamma \le c_\Gamma K$ and $b \ge b_-$. As $r_{\rm eff} = \mu_\star \beta^{1/2}/(32 n K_{\rm L} \bar{\mathsf{m}})$, (Y5) is the inequality $2^9 n^2 K_{\rm L} \Gamma \bar{\mathsf{m}} \le \mu_\star \beta^{1/2}$, which the fifth inequality of the list gives, by the bounds of $\bar{\mathsf{m}}$ and of Γ just stated and by $\beta \ge L^K b_-$. Hence $\mathsf{A}_i \le C_{\rm ins} K^{2\mathfrak{p}_0} N_{[u]} \|f_i\|_\sharp$ for $i \notin Q$ and $\mathsf{A}_i \le C_{\rm ins} K^2 N_{[u]}^2 \|f_i\|_\infty$ for $i \in Q$. With $N_{[u]} \ge 1$ and $\|f_i\|_\infty \le \|f_i\|_\sharp$, each A_i is at most $C_{\rm ins} N_{[u]}^2 \|f_i\|_\sharp$ times $K^{2\mathfrak{p}_0}$ if $i \notin Q$ and times K^2 if $i \in Q$; as $K \ge 1$, $\mathfrak{p}_0 \ge 1$ and Q has at least one element, the product of these powers of K is at most $K^{2\mathfrak{p}_0(n-1)+2}$, so that $\prod_i \mathsf{A}_i \le C_{\rm ins}^n K^{2\mathfrak{p}_0(n-1)+2} N_{[u]}^{2n} \prod_i \|f_i\|_\sharp$. Write $\phi_1 = \frac{1}{180(3n+4)}$ and $\phi_2 = \frac{1}{9}$, so that $u = \lceil \phi_1 K \rceil$ and $s = \lceil \phi_2 K \rceil$. With $\phi_1 K \le u < \phi_1 K + 1$ and $\phi_2 K \le s < \phi_2 K + 1$ and $(6n+4)\phi_1 = \frac{3n+2}{90(3n+4)} \le \frac{1}{90}$ one has $u(6n+4) - s \le -\frac{K}{10} + 6n + 4$, $3u(2n+1) - \frac{K}{2} \le -\frac{K}{10} + 6n + 3$ and $6nu - \frac{1}{2}(K - s) \le -\frac{K}{10} + 6n + \frac{1}{2}$, as $\frac{K}{90} - \frac{K}{9} = -\frac{K}{10}$, $\frac{K}{90} - \frac{K}{2} < -\frac{K}{10}$ and $\frac{K}{90} - \frac{K}{2} + \frac{K}{18} = -\frac{13K}{30} \le -\frac{K}{10}$. Hence $8\epsilon_{\rm fg} N_{[u]}^{2n} = 8 C_{\rm old} C_{\rm fg} \frac{L}{L-1} E_0 \, \ell^{6n+3} L^{u(6n+4)-s}$, $4\bar\epsilon_{\rm rd} N_{[u]}^{2n}$ and, by the last inequality of the list, $2\epsilon_{\rm lk} N_{[u]}^{2n}$ are bounded by the parts of $C_{\rm sm} K^{2\mathfrak{p}_0} \ell^{6n+3} L^{-K/10}$, so $\epsilon_{\rm osc} N_{[u]}^{2n} \le C_{\rm sm} K^{2\mathfrak{p}_0} \ell^{6n+3} L^{-K/10}$. The product of the two bounds gives $\epsilon_{\rm osc} \prod_i \mathsf{A}_i \le C_{\rm ins}^n C_{\rm sm} \ell^{6n+3} K^{2\mathfrak{p}_0 n + 2} L^{-K/10} \prod_i \|f_i\|_\sharp$, and $K^{2\mathfrak{p}_0 n + 2} \le K^{(2\mathfrak{p}_0+2)n}$ because $n \ge 1$ and $K \ge 1$. Each inequality of the list is true at all large K: $\bar\epsilon_{\rm rd}$, $\epsilon_{\rm fg}$, $\epsilon_{\rm lk} N_{[u]}^{2n} L^{K/10}$ and $K N_{[u]} L^{-K/2}$ tend to 0, by these exponents, by $3\phi_1 < \frac{1}{2}$ and by $L^u \ge L^{\phi_1 K}$, while L^s and L^u tend to infinity; so $K_{\rm gr}(d')$ exists. □

Supports without a common point.

Lemma 18.137 (Cutting the test functions into small pieces)**.** Let $m \in \mathbb{N}$, $\ell = 2L^m$, $n \ge 2$, $d > 0$, and let $f_1, \dots, f_n$ be real Lipschitz functions on T_ℓ, none of them 0, with $\mathsf{d}_\infty(\underline{f}) \ge d$. Put $\mathsf{H} := \lceil 4n\ell/d \rceil$, $\mathsf{l} := \ell/\mathsf{H}$, $\psi_a^{(1)}(y) := \max\{0, 1 - \operatorname{dist}(y, a\mathsf{l} + \ell\mathbb{Z})/\mathsf{l}\}$ on $\mathbb{R}/\ell\mathbb{Z}$ for $a \in \mathbb{Z}/\mathsf{H}\mathbb{Z}$, and $\psi_\alpha(x) := \prod_{\mu \le 3} \psi_{\alpha_\mu}^{(1)}(x_\mu)$ for $\alpha \in (\mathbb{Z}/\mathsf{H}\mathbb{Z})^3$. Then $\sum_\alpha \psi_\alpha = 1$; each $\psi_\alpha f_i$ is a real Lipschitz function with $\|\psi_\alpha f_i\|_\infty \le \|f_i\|_\infty$ and $\|\psi_\alpha f_i\|_\sharp \le (1 + 600 n M/d) \|f_i\|_\sharp$; and for every $\underline{\alpha} = (\alpha_1, \dots, \alpha_n)$ with $g_i := \psi_{\alpha_i} f_i \ne 0$ for all i there is a partition of $\{1, \dots, n\}$ into at least two groups such that $\operatorname{dist}_\infty(\operatorname{supp} g_i, \operatorname{supp} g_{i'}) \ge d/2n$ whenever i and i' lie in two different groups. ■

Sketch of proof. Two points of T_ℓ lie at sup distance at most $\frac{1}{2}\ell$, so $d \le \mathsf{d}_\infty(\underline{f}) \le \frac{1}{2}\ell$; hence $\mathsf{H} \ge 8n$, $\mathsf{l} \le d/4n$ and $1/\mathsf{l} \le 4n/d + 1/\ell \le 5n/d$. Each $\psi_a^{(1)}$ is piecewise linear with values in $[0,1]$, slopes $\pm 1/\mathsf{l}$ and an arc of length $2\mathsf{l}$ as support, and these functions sum to 1 on the circle; hence ψ_α takes values in $[0,1]$, the ψ_α sum to 1, the support of ψ_α has sup diameter at most $2\mathsf{l} \le d/2n$, and, a product of numbers in $[0,1]$ moving by at most the sum of the moves of its factors, $|\psi_\alpha(x) - \psi_\alpha(y)| \le \mathsf{l}^{-1} \sum_\mu |x_\mu - y_\mu| \le 3\mathsf{l}^{-1} |x - y|_\infty \le 3\mathsf{l}^{-1} |x - y|$, the coordinate differences being taken on the circle $\mathbb{R}/\ell\mathbb{Z}$, and the sup distance being at most the Euclidean one. So $\operatorname{Lip}(\psi_\alpha f) \le \operatorname{Lip} f + 3\mathsf{l}^{-1} \|f\|_\infty$ and $\|\psi_\alpha f\|_\sharp \le (1 + 120 M/\mathsf{l}) \|f\|_\sharp \le (1 + 600 n M/d) \|f\|_\sharp$. For the groups, call i and i' neighbours when $\operatorname{dist}_\infty(\operatorname{supp} g_i, \operatorname{supp} g_{i'}) < d/2n$, and let the groups be the classes of the equivalence relation that this relation generates; supports of indices in two different classes are then at sup distance at least $d/2n$. Suppose that all indices formed one class, and pick $x_i \in \operatorname{supp} g_i \subset \operatorname{supp} f_i$. Between two indices there is a path of neighbours through at most n indices; along it each support adds at most its diameter $d/2n$ and each passage to a neighbour a distance smaller than $d/2n$, so that $|x_i - x_{i'}|_\infty < n \cdot \frac{d}{2n} + (n-1)\frac{d}{2n} < d$ for all i and i'. The tuple $(x_1, \dots, x_n)$ would then have sup diameter smaller than d, which the hypothesis $\mathsf{d}_\infty(\underline{f}) \ge d$ excludes. □

Proposition 18.138 (The bound at the labels of an interval under (Tor))**.** Let $m \in \mathbb{N}$, $\ell = 2L^m$, let $\mathsf{J} = [b_-, b_+]$ be an admissible label interval on T_ℓ, $n \geq 2$ and $d > 0$; let $\mathsf{C} = C_n(\mathsf{J})$, $\mathsf{C}' = C'_n(\mathsf{J})$ and $K_1(d) = K_1(\ell, n, \mathsf{J}, d)$ be the numbers of Thm. YM3_I:AR and Cor. YM3_I:117.1, and C_{ins}, C_{sm}, $K_{\mathrm{gr}}(\cdot)$ those of Lemma 18.136. Put

$$C^\flat_{\mathsf{J}} := (2^n - 1)\Big[\mathsf{H}^{3n}\Big(1 + \frac{600nM}{d}\Big)^n C^n_{\mathrm{ins}} C_{\mathrm{sm}}\, \ell^{6n+3} + \tfrac{1}{4}\mathsf{C}\,\ell^{3(n+1)}\Big] + \mathsf{C}'\ell^{3(n+1)}, \qquad K^\flat_{\mathsf{J}} := \max\{K_1(d), K_{\mathrm{gr}}(d/2n)\},$$

with $\mathsf{H} = \lceil 4n\ell/d \rceil$. Then the bounds (a) and (b) of Proposition 18.129 are true with $C^\flat_{\mathsf{J}}$ and $K^\flat_{\mathsf{J}}$ in the place of $C^\flat$ and $K^\flat$, for every $b \in \mathsf{J}$, every integer $K \geq K^\flat_{\mathsf{J}}$ and every tuple as there. ■

Sketch of proof. Let $b \in \mathsf{J}$, $K \geq K^\flat_{\mathsf{J}}$, $\beta = L^K b$, and let the tuple be as there. The functionals $\Lambda(F) = \int F\rho_0\, dU$ and $\Lambda'''(F) = \int F\mathfrak{w}^{\mathrm{sf}}_t\rho_0\, dU$ are positive on the gauge-invariant algebra, and $\Lambda - \Lambda'''$ is positive there by equation YM3_I:(2189). The loss in total mass is at most $\mathsf{R}_t\Lambda(1)$ by equation YM3_I:(2190). Write $\mathsf{cum}_{\Lambda'''}$ for the cumulant of the normalized second functional. After normalizing both functionals, their means of any bounded gauge-invariant function F differ by at most $2\mathsf{R}_t\|F\|_\infty$: subtract the lost integral and then account for the difference of the normalizers. In the moment formula for the joint cumulant, replace the block moments of each partition one at a time. Summing the bounds with the cumulant coefficients gives at most $2\mathsf{op}_n\mathsf{R}_t\prod_i\|X_i^Q\|_\infty$, where $\mathsf{op}_n = \sum_\pi |\pi|!$. Using the common bound on the entries gives, in particular,

$$\begin{aligned}|\langle X_1^Q;\dots;X_n^Q\rangle_{K,\ell,\beta} - \mathsf{cum}_{\Lambda'''}(X_1^Q,\dots,X_n^Q)| &\leq 4\mathsf{op}_n\mathsf{R}_t\prod_i 6\ell^3L^{4K}\|f_i\|_\infty\\ &\leq \tfrac{1}{4}\mathsf{C}\ell^{3(n+1)}K^{\mathsf{p}_n}L^{-K/10}\prod_i\|f_i\|_\sharp.\end{aligned}$$

The last inequality is the discarded-mass estimate in the proof of Thm. YM3_I:AR, using its larger prefactor $8\mathsf{op}_n\bar{\mathsf{R}}(6\ell^3L^{4(K+1)})^n$ and its choice of K_0. The entries are real linear in the f_i, and the cumulant depends real-linearly on each of its entries; with the functions ψ_α of Lemma 18.137, $\mathsf{cum}_{\Lambda'''}(X_1^Q,\dots,X_n^Q)$ is the sum over the H^{3n} tuples $\underline{\alpha}$ of the cumulants formed with the pieces $g_i = \psi_{\alpha_i}f_i$, and the terms in which some piece vanishes are 0. Each remaining choice of pieces falls into at least two groups at sup distance $\geq d/2n$, and $K \geq K_{\mathrm{gr}}(d/2n)$; Proposition 18.135 and Lemma 18.136 bound its cumulant by $C^n_{\mathrm{ins}}C_{\mathrm{sm}}\ell^{6n+3}K^{(2\mathfrak{p}_0+2)n}L^{-K/10}\prod_i\|g_i\|_\sharp$, and $\|g_i\|_\sharp \leq (1 + 600nM/d)\|f_i\|_\sharp$. As $(2\mathfrak{p}_0 + 2)n \leq \mathfrak{p}^\natural_n$, $\mathfrak{p}_n \leq \mathfrak{p}^\natural_n$ and $K \geq 1$, the cumulant in (a) is at most the bracket in the formula for $C^\flat_{\mathsf{J}}$ times $K^{\mathfrak{p}^\natural_n}L^{-K/10}\prod_i\|f_i\|_\sharp$, which gives (a). For (b), $\mathcal{O}'_K = \mathcal{O}_K - \mathcal{O}^{(2)}_K$ and the multilinearity give $\langle\mathcal{O}'_K(f_1);\dots;\mathcal{O}'_K(f_n)\rangle_{K,\ell,\beta} = \sum_{Q\subset\{1,\dots,n\}}(-1)^{\#Q}\langle X_1^Q;\dots;X_n^Q\rangle_{K,\ell,\beta}$, whose term $Q = \emptyset$ is $S^T_{n;K,\ell,\beta}(\underline{f})$; and $|S^T_{n;K,\ell,\beta}(\underline{f}) - S^{(\ell,b)}_n(\underline{f})| \leq \mathsf{C}'\ell^{3(n+1)}K^{\mathsf{p}_n}L^{-K/10}\prod_i\|f_i\|_\sharp$ for $K \geq K_1(d)$, by Cor. YM3_I:117.1 in the limit $K' \to \infty$, with the limit identified in Thm. YM3_I:AS. Adding the bounds of (a) for the $2^n - 1$ sets $Q \neq \emptyset$ and this bound, in which $K^{\mathsf{p}_n} \leq K^{\mathfrak{p}^\natural_n}$, gives (b) with the constant $C^\flat_{\mathsf{J}}$. □

Every label, every side.

Lemma 18.139 (Dilation of the cumulants of the plaquette densities)**.** Let $K, m \in \mathbb{N}$, $\ell = 2L^m$, let s be an integer with $0 \leq s < K$, let $\beta > 0$ be real, and let $f_1,\dots,f_n$ be bounded real functions on T_ℓ; for each i let $Y_{i,K}$ be one of $\mathcal{O}_K$, $\mathcal{O}^{(2)}_K$ and $\mathcal{O}'_K$, and $Y_{i,K-s}$ the density of the same kind on $\Lambda_{K-s,L^s\ell}$. With $f^{(s)}(x) = f(L^{-s}x)$,

$$\langle Y_{1,K}(f_1);\dots;Y_{n,K}(f_n)\rangle_{K,\ell,\beta} = L^{sn}\,\langle Y_{1,K-s}(f_1^{(s)});\dots;Y_{n,K-s}(f_n^{(s)})\rangle_{K-s,L^s\ell,\beta}.$$

■

Sketch of proof. The map $U \mapsto U^{(s)}$ that relabels every bond and plaquette by dilation is a bijection of $\mathcal{V}(\Lambda_{K,\ell})$ onto $\mathcal{V}(\Lambda_{K-s,L^s\ell})$ that carries the Haar measure to the Haar measure, with $a_{L^sp}(U^{(s)}) =$

$a_p(U)$ and $x(L^s p) = L^s x(p)$; it therefore carries the density $e^{-\beta \sum_p a_p}$ of the first lattice to that of the second. As $a_K^{-1} = L^s a_{K-s}^{-1}$ and $f(x(p)) = f^{(s)}(x(L^s p))$, one has $Y_K(f)(U) = L^s Y_{K-s}(f^{(s)})(U^{(s)})$ for $\mathcal{O}$ and for $\mathcal{O}^{(2)}$, and so also for their difference $\mathcal{O}'$; the cumulant is multilinear, so multiplying each of its n entries by L^s multiplies it by L^{sn}. □

Proof of Proposition 18.129. Let $s_\star$ be the least integer ≥ 0 that satisfies YM3_I:(2210) for the pair (b_-, b_+) on the side ℓ; it exists by Lem. YM3_I:118.2, and $\mathsf{J}_\star := [L^{s_\star} b_-, L^{s_\star} b_+]$ is an admissible interval on $T_{L^{s_\star}\ell}$ by equation YM3_I:(2201). Let $C^\flat_{\mathsf{J}_\star}$ and $K^\flat_{\mathsf{J}_\star}$ be the numbers of Proposition 18.138 at the side $L^{s_\star}\ell$, the interval $\mathsf{J}_\star$, the integer n and the number $L^{s_\star} d$, and put

$$C^\flat := L^{s_\star(n+\frac{1}{10})}\, C^\flat_{\mathsf{J}_\star}, \qquad K^\flat := s_\star + K^\flat_{\mathsf{J}_\star};$$

they are formed from $\mathfrak{D}$, n, ℓ, $b_\pm$ and d. Let $b \in [b_-, b_+]$, $K \geq K^\flat$, and let $\underline{f}$ be as in the statement. Then $K - s_\star \geq K^\flat_{\mathsf{J}_\star} \geq 2$; the functions $f_i^{(s_\star)}$ are real Lipschitz functions on $T_{L^{s_\star}\ell}$, none of them 0, with $\|f_i^{(s_\star)}\|_\infty = \|f_i\|_\infty$ and $\|f_i^{(s_\star)}\|_\sharp \leq \|f_i\|_\sharp$, their supports have no common point, and their spread is $L^{s_\star}\mathsf{d}_\infty(\underline{f}) \geq L^{s_\star} d$ since dilation multiplies sup distances by $L^{s_\star}$; and $L^K b = L^{K-s_\star}(L^{s_\star} b)$ with $L^{s_\star} b \in \mathsf{J}_\star$. By Lemma 18.139, each cumulant of (a) and the cumulant of (b) at $(K, \ell, L^K b, \underline{f})$ is $L^{s_\star n}$ times the corresponding cumulant at $(K - s_\star, L^{s_\star}\ell, L^{K-s_\star}(L^{s_\star} b), \underline{f}^{(s_\star)})$, and $S_n^{(\ell,b)}(\underline{f}) = L^{s_\star n} S_n^{(L^{s_\star}\ell, L^{s_\star} b)}(\underline{f}^{(s_\star)})$ by display YM3_I:(2215). Proposition 18.138 at the dilated data bounds the dilated quantities by $C^\flat_{\mathsf{J}_\star}(K - s_\star)^{\mathsf{p}^\natural_n} L^{-(K-s_\star)/10} \prod_i \|f_i\|_\sharp$, and $L^{s_\star n}(K - s_\star)^{\mathsf{p}^\natural_n} L^{-(K-s_\star)/10} \leq L^{s_\star(n+\frac{1}{10})} K^{\mathsf{p}^\natural_n} L^{-K/10}$. This gives (a) and (b); the last sentence of the Proposition follows, as $K^{\mathsf{p}^\natural_n} L^{-K/10} \to 0$ and $C^\flat$, $K^\flat$ do not depend on $b \in [b_-, b_+]$ or on the tuple. □

18.14. **One torus: the label as a variable, the laws and kernels of the block fields, the action density at coinciding points, and the classical action.** We study the label dependence and the block fields on a fixed torus T_ℓ, $\ell = 2L^m$, together with the lattice fields of smooth one-forms. Proposition 18.141 says that a torus limit $S_n^{\ell,\mathsf{b}}(X)$, at a tuple in which some two entries have disjoint supports, is real-analytic in the label on $(0, \infty)$, on every side; hence it is either identically zero or has only finitely many zeros on each compact set of labels. It also says that for test pairs (f, h) at a small scale the integrated three-point function $S_3^{\ell,\mathsf{b}}(1, f, h) = -\partial_{\mathsf{b}} S_2^{\ell,\mathsf{b}}(f, h)$ takes a positive value somewhere between two labels of a window as soon as the ratio of their squares exceeds 2, with relative bounds that are the same for all the pairs of the class. Proposition 18.142 says that at every label one of the connected functions of order three or more is non-zero at a tuple $(1, \dots, 1, f, h)$, and that a law of three points on a torus, uniform in the cutoff, makes the limit of the third connected function positive at collinear triples of bumps. According to Proposition 18.144, the laws of the block fields of a fixed depth are, on every side, real-analytic in the label, their derivatives being given by the correlations with the total action, and the mixed cumulants of the action density with a bounded Borel function of a block field obey bounds that are uniform in the cutoff, for arbitrary supports. Proposition 18.145 treats a label $\mathsf{b} \geq \gamma^{-2}$ on a side $\ell = 2L^m$ with $M\, R(\mathsf{b}) \leq L^m$; write ς_K for the square root of the specific heat at the cutoff K. Once centred, multiplied by $a_K^{3/2}$ and divided by ς_K, the smeared action density has a limit in law as K runs through all the integers, a white noise whose covariance at two test functions f, h is $\int fh$, and in this limit it is independent of the block field V_d of any fixed depth d. Without the division by ς_K, a limit in law of the same kind, with the covariance $\varsigma_0^2 \int fh$ and again independent of V_d, exists along any infinite sequence of cutoffs for which ς_K^2 tends to a number ς_0^2. Finally, $\liminf_K \varsigma_K^2 \geq \frac{3}{4}\mathsf{b}^{-2}$ holds for every real label $\mathsf{b} > 0$ and every side. Proposition 18.147 says that the conditional laws of the block field of depth $d+1$ given the block field of depth d exist at $K = \infty$, are the limits of those at the cutoffs, and turn the sequence of the block fields into a Markov chain indexed by the depth. Proposition 18.148 says that the covariance of a bounded Borel function of the block field of depth d with the smeared action

density tends, as the cutoff runs through all the integers and with an error that decays geometrically, to the integral of that function against a density $X_d(f)$ that lies in L^p for every $p<\infty$; that these densities form a martingale in the depth; and that, for f supported in one half of the torus cut by a coordinate mirror with the reflection θ, their conditional expectations with respect to the block variables on the opposite side constitute a martingale converging almost surely and in L^2, with L^2 norms at most $S_2^{\ell,\mathsf{b}}(f\circ\theta,f)^{1/2}$. Propositions 18.150 to 18.152 contain those entries of the dictionary with the classical theory that involve no evolution in time, for the action and its density, the coupling, the shell flux and the free kernels: on the lattice field of a smooth one-form the Wilson action, the action density and the shell flux are the Yang–Mills action, its density and the flux of its stress tensor, up to errors of the order of a power of the spacing; $\mathsf{b}g_{\mathrm{std}}^2=4$; and, for each $n\ge 2$, the n-th connected function of $\frac12{:}|B|^2{:}$, where B is the curvature of three free photon fields at the label b, equals $3\mathsf{b}^{-n}\mathcal{K}_{\mathrm{free}}^{(n)}$, the function of order n of the free template of Definition 18.116(3).

The scalar torus limits and the law of two points are Theorems YM3_I:AS and YM3_I:AV. Proposition 18.161 supplies holomorphic convergence and the constant-source derivatives. The block-field laws use the calibrated comparison of §2.3.1 and Corollary 2.8. Lemma 18.212 constructs the compatible thread law. The final three propositions compute the classical and Gaussian expressions directly.

Definition 18.140 (Signs on one torus)**.** Use the sourced setting $\mathfrak{s}$, the cutoff cumulants $S^T_{n;K,\ell,\beta}$ and the limits $S_n^{\ell,\mathsf{b}}$ of Definition 18.159. On $\ell=2L^m$ let $a_j=L^{-j}$ and $\Lambda_{j,\ell}=a_j\mathbb{Z}^3/\ell\mathbb{Z}^3$ for $j\ge -m$. For $j>-m$, $\mathcal{V}(\Lambda_{j,\ell})$ consists of bond fields with values in $\mathrm{SU}(2)$ and $U(\bar b)=U(b)^{-1}$. Write $\mathsf{V}_d=\mathcal{V}(\Lambda_{d,\ell})$ for $d\ge 0$. At cutoff K write $\mathbb{E}_{K,s}$ for expectation at label s and $\beta=L^K s$ and $V_d=\mathcal{M}^{K-d}U$ for $d\le K$. Its law is $\mu_d^K[s]$. Write $\mu_d[s]=\mu_d[\ell,s]$ for the limit law and $\mu_{\ell,s}$ for the thread law of Lemmas 18.212 and 18.230. The observable is $\mathcal{O}(f)=a_K^{-1}\sum_p f(x(p))a_p$, with $a_p=1-\frac12\operatorname{tr}U(p)$, and the source norm $\|f\|=\|f\|_\flat$ is specified in Definition 18.143(1). Let $\mathfrak{S}_n(T_\ell)$ be the tuples of n real C^1 functions with pairwise disjoint supports.

(1) The *setting (E)* uses a fixed choice ς of the canonical constants and auxiliary data in (Th1), (Th2) of §2.3.1, and a label $\mathsf{b}\ge\gamma^{-2}$ with $MR(\mathsf{b})\le L^m$. Denote by ϱ_ς the source radius in that calibration. The real restriction of the sourced setting uses the same canonical construction of Theorems YM3_I:D and YM3_I:AD.

(2) Use the dilation $f^{(s)}$, the tuple $X^{(s)}$, the constant insertions $(1^\nu,X)$, and tuples with a separated pair as in Definition 18.159(2), (3).

(3) Use the class $\mathfrak{T}_{T_\ell}(\mathsf{d};\vartheta)$ and the pair kernel $\mathcal{K}_6$ of §18.4, where $\vartheta\in(0,1)$ and $20\mathsf{d}\le\ell$. In particular
$$10^{-6}\vartheta^2\|f\|_\infty\|h\|_\infty\le\mathcal{K}_6(f,h)\le(4\pi/3)^2\|f\|_\infty\|h\|_\infty,\qquad \mathsf{c}_2=\frac{9}{16\pi^2}.$$
The inequalities follow from the pair-distance bounds $[\mathsf{d},10\mathsf{d}]$, the test-integral lower bound and the support-volume upper bound.

(4) Let γ_W be the ceiling YM3_I:(2346), $\bar\varepsilon_{g,\vartheta}$ the tolerance YM3_I:(2349), and $\mathsf{d}_0(g,\vartheta)\in(0,1/200]$ the threshold YM3_I:(2350). For $0<g\le\gamma_W$ the tolerance is nondecreasing, tends to zero at zero, and stays at most $1/3$ through the threshold. Use the witness depth of YM3_I:(2321) with $C_{W,0}=\max\{1,4L^2M/A_0^2\}$ as in Theorem YM3_I:AV. This depth is formed before fixing the cutoff, label and side.

(5) For $36\mathsf{d}<\ell$, put $X_{\mathsf{d}}^\ell=(\beta_0^{\langle\ell\rangle},\beta_2^{\langle\ell\rangle},\beta_4^{\langle\ell\rangle})$, where $\beta_j(y)=(1-256|y-j\mathsf{d}e_1|^2/\mathsf{d}^2)_+^2$. These C^1 functions have supremum 1 and radius $\mathsf{d}/16$. The lifted support distances range from $15\mathsf{d}/8$ to $31\mathsf{d}/8$. Their integrals are
$$4\pi(\mathsf{d}/16)^3(1/3-2/5+1/7)=\pi\mathsf{d}^3/13440$$

and their Lipschitz constants are at most $128/(3\sqrt{3}\,\mathsf{d})$, by maximizing the radial derivative. Hence for $\vartheta \le \pi/13440$ the triple lies in $\mathfrak{T}^{(3),\mathrm{col}}_{T_\ell}(\mathsf{d};\vartheta)\cap\mathfrak{S}_3(T_\ell)$. The collinear class is contained in $\mathfrak{Q}^{(3)}_{T_\ell}(\mathsf{d};\vartheta)$, whose triples have all pairs in $\mathfrak{T}_{T_\ell}(\mathsf{d};\vartheta)$. The lifted kernel and the collinear bound of §18.4 give $\mathcal{K}^{(3)}_{\mathrm{free}} \ge c_3^{\mathrm{col}}\vartheta^3\prod_i\|f_i\|_\infty$ on that class, where $c_3^{\mathrm{col}} = 3.78(4\pi)^{-3}10^{-9}$. ∎

Proposition 18.141 (The torus limits as functions of the label; the sum rule; the integrated three-point function)**.** Work in a setting $\mathfrak{s}$. Let $m\in\mathbb{N}$ and $\ell = 2L^m$.

(1) *(Analytic on all of $(0,\infty)$.)* Let X be a tuple with a separated pair on T_ℓ, with $n\ge 2$ entries, and put $\varphi(\mathsf{b}) = S_n^{\ell,\mathsf{b}}(X)$, $\mathsf{b}>0$. Given $0<\mathsf{b}_-<\mathsf{b}_+$, choose an integer $s\ge 0$ with $L^s\mathsf{b}_-\ge 2\gamma^{-2}$ and $2L^{m+s}\in\mathfrak{L}\big((L^s\mathsf{b}_+/2)^{-1/2}\big)$; every sufficiently large integer qualifies. With $\upsilon_s = L^{-s}\upsilon([L^s\mathsf{b}_-,L^s\mathsf{b}_+],L^s\ell)>0$, where the width $\upsilon(\cdot,\cdot)$ of Definition 18.159(4) involves neither n nor X, the restriction of φ to $[\mathsf{b}_-,\mathsf{b}_+]$ extends to a holomorphic function on the set of the complex numbers at a distance less than υ_s from $[\mathsf{b}_-,\mathsf{b}_+]$. Hence φ is real-analytic on the whole half line $(0,\infty)$.

(2) *(Zeros; their order.)* In the setting of (1), φ is either the zero function or its zero set is discrete in $(0,\infty)$, so that each compact interval of labels contains only finitely many zeros. When φ is not constant, each label $\mathsf{b}>0$ has some $\nu\ge 1$ for which $S^{\ell,\mathsf{b}}_{n+\nu}(1^\nu,X) = (-1)^\nu\partial_{\mathsf{b}}^\nu\varphi(\mathsf{b})\ne 0$.

(3) *(The sum rule at a window.)* Let $0<g\le\gamma_W$, $\ell\in\mathfrak{L}(g)$, $\vartheta\in(0,1)$, $0<\mathsf{d}\le\mathsf{d}_0(g,\vartheta)$ and $(f,h)\in\mathfrak{T}_{T_\ell}(\mathsf{d};\vartheta)$. Put $\epsilon = \bar\varepsilon_{g,\vartheta}(\mathsf{d})\in[0,\frac13]$, $\mathsf{k} = \mathsf{c}_2\mathcal{K}_6(f,h)>0$, $\phi(\mathsf{b}) = S_2^{\ell,\mathsf{b}}(f,h)$ and $\psi(\mathsf{b}) = S_3^{\ell,\mathsf{b}}(1,f,h)$, and let $\mathsf{b}_{\mathrm{top}} = L(g^{-2}+\bar B)-2\bar B$ be the right end of $I(g)$. Then ϕ and ψ are real-analytic on $(0,\infty)$, $\phi' = -\psi$, and the following hold.

(a1) $|\phi(\mathsf{b}) - \mathsf{k}\,\mathsf{b}^{-2}|\le\epsilon\,\mathsf{k}\,\mathsf{b}^{-2}$ for every $\mathsf{b}\in I(g)$.

(a2) For all $\mathsf{b}_1<\mathsf{b}_2$ in $I(g)$: $\big|\int_{\mathsf{b}_1}^{\mathsf{b}_2}(\psi - 2\mathsf{k}\,\mathsf{b}^{-3})\,d\mathsf{b}\big|\le\epsilon\,\mathsf{k}\,(\mathsf{b}_1^{-2}+\mathsf{b}_2^{-2})$ and $\big|\int_{\mathsf{b}_1}^{\mathsf{b}_2}\mathsf{b}\,(\mathsf{b}\psi-2\phi)\,d\mathsf{b}\big|\le 2\epsilon\,\mathsf{k}$.

(a3) Between any two labels $\mathsf{b}_1<\mathsf{b}_2$ of $I(g)$ lies a label b'' with

$$\mathsf{k}\,\frac{(1-\epsilon)\mathsf{b}_1^{-2}-(1+\epsilon)\mathsf{b}_2^{-2}}{\mathsf{b}_2-\mathsf{b}_1}\ \le\ \psi(\mathsf{b}'')\ \le\ \mathsf{k}\,\frac{(1+\epsilon)\mathsf{b}_1^{-2}-(1-\epsilon)\mathsf{b}_2^{-2}}{\mathsf{b}_2-\mathsf{b}_1};$$

the left member is positive as soon as $\mathsf{b}_2^2>2\mathsf{b}_1^2$. The labels $\mathsf{b}_2=\mathsf{b}_{\mathrm{top}}$ and $\mathsf{b}_1=\frac12\mathsf{b}_{\mathrm{top}}$ lie in $I(g)$, and for them $\psi(\mathsf{b}'')\ge\frac83\mathsf{k}\,\mathsf{b}_{\mathrm{top}}^{-3}\ge\frac83\,10^{-6}\vartheta^2\mathsf{c}_2\,\mathsf{b}_{\mathrm{top}}^{-3}\|f\|_\infty\|h\|_\infty>0$.

(a4) At each fixed cutoff $K>s(\mathsf{d})$ the statements (a1)–(a3) remain true, with the same ϵ and k, for the functions $\mathsf{b}\mapsto S^T_{2;K,\ell,L^K\mathsf{b}}(f,h)$ and $\mathsf{b}\mapsto S^T_{3;K,\ell,L^K\mathsf{b}}(1,f,h)$ instead of ϕ and ψ.

(b) ψ does not vanish identically, its zero set is discrete in $(0,\infty)$, and ψ takes positive values inside every interval $(\mathsf{b}_1,\mathsf{b}_2)$ whose ends lie in $I(g)$ and satisfy $\mathsf{b}_2^2>2\mathsf{b}_1^2$.

The number ϵ is formed from g, ϑ, d, the radius and the choice alone, so it is one and the same at all the labels of $I(g)$; and every bound in (a1)–(a4) that carries the factor k is k times an explicit expression in ϵ and in the labels that appear in it (b, or b_1 and b_2, or $\mathsf{b}_{\mathrm{top}}$), an expression into which neither the side $\ell\in\mathfrak{L}(g)$, nor the pair in $\mathfrak{T}_{T_\ell}(\mathsf{d};\vartheta)$ and its position, nor the cutoff $K>s(\mathsf{d})$ enters.

(4) *(Every side.)* Now let $m\in\mathbb{N}$ be arbitrary, $\ell = 2L^m$, $0<g\le\gamma_W$, $\vartheta\in(0,1)$, s an integer with $s\ge 0$ and $s\ge m_{\mathfrak{L}}(g)-m$, and $(f,h)\in\mathfrak{T}_{T_\ell}(\mathsf{d};\vartheta)$ with $L^s\mathsf{d}\le\mathsf{d}_0(g,\vartheta)$. Then (a1)–(a3) and (b) are true on the side ℓ after three replacements: the window $I(g)$ by the dilated window $L^{-s}I(g) = \{L^{-s}\lambda:\lambda\in I(g)\}$, the number ϵ by $\bar\varepsilon_{g,\vartheta}(L^s\mathsf{d})$, and $\mathsf{b}_{\mathrm{top}}$ by $L^{-s}\mathsf{b}_{\mathrm{top}}$. ∎

Proof. Step 1. For a tuple Y with empty intersection of the supports, Theorem YM3_I:AS gives the real limits, including along moving labels by YM3_I:(2216). Proposition 18.161 gives

$\partial_{\mathsf{b}}^{\nu} S_n^{\ell,\mathsf{b}}(Y) = (-1)^{\nu} S_{n+\nu}^{\ell,\mathsf{b}}(1^{\nu}, Y)$ on every side. At a single cutoff the same identity follows by differentiating the logarithmic source generating function with weight $e^{-\mathsf{b}\mathcal{O}(1)}$. In particular $\phi' = -\psi$.

Step 2. For $g_s = (L^s\mathsf{b}_+/2)^{-1/2}$ the least side of $\mathfrak{L}(g_s)$ grows at most polynomially in s, by the formula for $m_{\mathfrak{L}}$ in Proposition YM3_I:74.1 and $R(x) \le L\max\{1, (\log x)^r\}$. Thus for all sufficiently large s the side $\ell_s = L^s\ell$ belongs to $\mathfrak{L}(g_s)$ and $L^s\mathsf{b}_- \ge 2\gamma^{-2}$. The bijective dilation preserves the separated pair, so $X^{(s)}$ is a tuple on ℓ_s.

Step 3. Proposition 18.161 gives holomorphic convergence of the cutoff cumulants of $X^{(s)}$ on the strip of width $\upsilon([L^s\mathsf{b}_-, L^s\mathsf{b}_+], \ell_s)$, with the torus limit $S_n^{\ell_s,x}(X^{(s)})$ on its real interval. The width is independent of the tuple and of n.

Step 4: back to the side ℓ; clause (1). The map $\mathsf{b} \mapsto L^s\mathsf{b}$ carries the strip of width υ_s about $[\mathsf{b}_-, \mathsf{b}_+]$ onto $\Omega_\upsilon(\mathsf{J})$, so $\Phi(\mathsf{b}) = L^{sn}F_\infty(L^s\mathsf{b})$ is holomorphic on that strip, and for $\mathsf{b} \in [\mathsf{b}_-, \mathsf{b}_+]$ the scaling law YM3_I:(2215) gives $\Phi(\mathsf{b}) = L^{sn}S_n^{L^s\ell, L^s\mathsf{b}}(X^{(s)}) = S_n^{\ell,\mathsf{b}}(X) = \varphi(\mathsf{b})$. Every $a > 0$ is an inner point of $[\frac{1}{2}a, 2a]$; at $\mathsf{b}_\pm = 2^{\pm 1}a$ the function Φ is holomorphic on a disc about a on whose real points it is φ, so φ is the sum of a convergent power series near a.

Step 5: clause (2). If the zeros of a real-analytic function on an interval accumulate at a point of the interval, the function is 0 near that point (identity theorem), and then on the whole interval, which is connected; and a compact interval cannot contain infinitely many zeros without containing one of their accumulation points (Bolzano–Weierstrass). If all the derivatives $\varphi^{(\nu)}(\mathsf{b})$, $\nu \ge 1$, vanished at one $\mathsf{b} > 0$, the power series of φ at b would be constant, $\varphi - \varphi(\mathsf{b})$ would vanish near b, and so on all of $(0, \infty)$. Step 1 identifies $\varphi^{(\nu)}(\mathsf{b})$.

Step 6: clause (3), (a1). Let $\mathsf{b} \in I(g)$. For every $K \in \mathbb{N}$ the pair $(K, L^K\mathsf{b})$ is a first-passage datum at g with the label b by Corollary YM3_I:74.2. The law of two points on the torus of every side at g, Theorem YM3_I:AV, read at this datum for every $K > s(\mathsf{d})$, gives $S^T_{2;K,\ell,L^K\mathsf{b}}(f,h) = \mathsf{c}_2\mathsf{b}^{-2}\mathcal{K}_6(f,h)(1+\varepsilon')$ with $|\varepsilon'| \le \epsilon \le \frac{1}{3}$. The witness depth involves no cutoff, so this holds for all large K; by Step 1 the left member tends to $\phi(\mathsf{b})$, and non-strict inequalities survive the limit. Analyticity of ϕ and ψ is clause (1) for the tuples (f,h) and $(1,f,h)$, which have a separated pair.

Step 7: (a2), (a3). Integrating $\phi' = -\psi$ from b_1 to b_2 and subtracting $\int_{\mathsf{b}_1}^{\mathsf{b}_2} 2\mathsf{k}\mathsf{b}^{-3} = \mathsf{k}(\mathsf{b}_1^{-2} - \mathsf{b}_2^{-2})$ shows that the first quantity in (a2) equals $[\phi(\mathsf{b}_1) - \mathsf{k}\mathsf{b}_1^{-2}] - [\phi(\mathsf{b}_2) - \mathsf{k}\mathsf{b}_2^{-2}]$; since $\frac{d}{d\mathsf{b}}(\mathsf{b}^2\phi) = -\mathsf{b}(\mathsf{b}\psi - 2\phi)$, the second equals $[\mathsf{b}_1^2\phi(\mathsf{b}_1) - \mathsf{k}] - [\mathsf{b}_2^2\phi(\mathsf{b}_2) - \mathsf{k}]$; and (a1) controls each bracket. By the mean value theorem $(\phi(\mathsf{b}_1) - \phi(\mathsf{b}_2))/(\mathsf{b}_2 - \mathsf{b}_1)$ is a value $\psi(\mathsf{b}'')$ with b'' between the two labels, and (a1) at b_1 and b_2 gives the two bounds of (a3); the lower bound is > 0 exactly when $(1-\epsilon)\mathsf{b}_2^2 > (1+\epsilon)\mathsf{b}_1^2$, which follows from $\mathsf{b}_2^2 > 2\mathsf{b}_1^2$, as $\frac{1+\epsilon}{1-\epsilon} \le 2$ for $\epsilon \le \frac{1}{3}$. The right end of $I(g)$ divided by the left end equals $L - 2\bar{B}/(g^{-2} + \bar{B}) > L - 2 \ge 11$; hence $\frac{1}{2}\mathsf{b}_{\mathrm{top}}$ exceeds $g^{-2} + \bar{B}$ and belongs to $I(g)$. For $\mathsf{b}_2 = \mathsf{b}_{\mathrm{top}} = 2\mathsf{b}_1$ the lower bound is $2(3 - 5\epsilon)\mathsf{k}\mathsf{b}_{\mathrm{top}}^{-3} \ge \frac{8}{3}\mathsf{k}\mathsf{b}_{\mathrm{top}}^{-3}$, and the lower bound for $\mathcal{K}_6$ in Definition 18.140(3) completes (a3).

Step 8: (a4), (b). At one cutoff $K > s(\mathsf{d})$ both ingredients of Step 7 are available for all $\mathsf{b} \in I(g)$, with the same ϵ: the derivative identity of Step 1 and the law of Step 6 at the datum $(K, L^K\mathsf{b})$; Step 7 uses nothing beyond them. (b): ψ is real-analytic by clause (1) and positive somewhere by (a3), so clause (2) applies, and the last assertion is a restatement of (a3).

Step 9: clause (4). As $m + s \ge m_{\mathfrak{L}}(g)$, the side $L^s\ell = 2L^{m+s}$ belongs to $\mathfrak{L}(g)$ by Proposition YM3_I:74.1. The dilation maps the test pair to scale $L^s\mathsf{d}$ and preserves $\mathcal{K}_6$ by homogeneity, as shown in §18.4. Also $1^{(s)} = 1$. Clause (3) therefore applies on the side $L^s\ell$ to the dilated pair at the scale $L^s\mathsf{d}$; call its two functions ϕ_s, ψ_s. The scaling law YM3_I:(2215) gives $\phi(\mathsf{b}) = L^{2s}\phi_s(L^s\mathsf{b})$ and $\psi(\mathsf{b}) = L^{3s}\psi_s(L^s\mathsf{b})$; as $L^{2s}(L^s\mathsf{b})^{-2} = \mathsf{b}^{-2}$, multiplying (a1) for ϕ_s at the label $L^s\mathsf{b} \in I(g)$ by L^{2s} yields (a1) for ϕ at $\mathsf{b} \in L^{-s}I(g)$. The arguments of Steps 7, 8 use only (a1), the derivative identity and the ratio of the two ends of the window, and this ratio is the same for $L^{-s}I(g)$ as for $I(g)$. □

Proposition 18.142 (At every label a higher connected function does not vanish; a law of three points gives a positive third connected function)**.** Work in a setting $\mathfrak{s}$. Let $m \in \mathbb{N}$, $\ell = 2L^m$ and $\vartheta \in (0,1)$.

(1) Given a real label $\mathsf{b}_1 > 0$, there is $\mathsf{d}_1 > 0$, depending on the setting, ℓ, b_1 and ϑ only, with the following property: whenever $0 < \mathsf{d} \le \mathsf{d}_1$ and $(f,h) \in \mathfrak{T}_{T_\ell}(\mathsf{d};\vartheta)$,

$$S_2^{\ell,\mathsf{b}_1}(f,h) \;\ge\; \tfrac{2}{3}\,\mathsf{c}_2\mathsf{b}_1^{-2}\mathcal{K}_6(f,h) \;>\; \tfrac{1}{3}\,\mathsf{c}_2\mathsf{b}_1^{-2}\mathcal{K}_6(f,h) \;\ge\; S_2^{\ell,2\mathsf{b}_1}(f,h) \;>\; 0.$$

For $\vartheta \le \pi/13440$ and $36\,\mathsf{d} < \ell$ the class $\mathfrak{T}_{T_\ell}(\mathsf{d};\vartheta)$ contains pairs.

(2) For a pair (f,h) as in (1), every label $\mathsf{b} > 0$ admits some $\nu \ge 1$ with $S_{2+\nu}^{\ell,\mathsf{b}}(1^\nu,f,h) \neq 0$. Moreover $\mathsf{b} \mapsto S_3^{\ell,\mathsf{b}}(1,f,h)$ is real-analytic on $(0,\infty)$, takes a positive value somewhere in $(\mathsf{b}_1, 2\mathsf{b}_1)$, and has a discrete set of zeros in $(0,\infty)$.

(3) *(From a law of three points.)* Let $0 < g \le \gamma$, $\ell' \in \mathfrak{L}(g)$ and $\vartheta = \pi/13440$. Suppose that there are a number $\mathsf{d}_3 > 0$ with $36\,\mathsf{d}_3 < \ell'$, a function $\epsilon_3\colon (0,\mathsf{d}_3] \to [0,\infty)$ with $\epsilon_3(\mathsf{d}) \to 0$ as $\mathsf{d} \to 0$, a number $C_3 \ge 0$ and integers $s_3(\mathsf{d})$, formed with no cutoff, such that for every first-passage datum $(K,\beta) \in \mathfrak{d}(g)$, with the label $\mathsf{b}' = a_K\beta$, every $\mathsf{d} \in (0,\mathsf{d}_3]$ with $K \ge s_3(\mathsf{d})$ and every $(f_1,f_2,f_3) \in \mathfrak{Q}^{(3)}_{T_{\ell'}}(\mathsf{d};\vartheta)$

$$\left|S^T_{3;K,\ell',\beta}(f_1,f_2,f_3) - 3\,\mathsf{b}'^{-3}\mathcal{K}^{(3)}_{\mathrm{free}}(f_1,f_2,f_3)\right| \;\le\; \left[\epsilon_3(\mathsf{d}) + C_3\frac{a_K}{\vartheta\,\mathsf{d}}\right] 3\,\mathsf{b}'^{-3}\prod_{i=1}^{3}\|f_i\|_\infty.$$

Let $\mathsf{d}_* \in (0,\mathsf{d}_3]$ be such that $\epsilon_3(\mathsf{d}) < c_3^{\mathrm{col}}\vartheta^3$ for $0 < \mathsf{d} \le \mathsf{d}_*$. Then for every sequence $(K_j,\beta_j)_j$ of first-passage data at g with $K_j \to \infty$ and $a_{K_j}\beta_j \to \mathsf{b}' \in (0,\infty)$, and for every $\mathsf{d}' \in (0,\mathsf{d}_*]$,

$$\lim_{j\to\infty} S^T_{3;K_j,\ell',\beta_j}(X^{\ell'}_{\mathsf{d}'}) = S_3^{\ell',\mathsf{b}'}(X^{\ell'}_{\mathsf{d}'}) \;\ge\; (c_3^{\mathrm{col}}\vartheta^3 - \epsilon_3(\mathsf{d}'))\,3\,\mathsf{b}'^{-3} \;>\; 0.$$

■

Proof. Clause (1). Put $\mathsf{b}_2 = 2\mathsf{b}_1$, and take an integer $s \ge 0$ with $L^s\mathsf{b}_1 \ge 2\gamma_W^{-2}$ and $\ell_s = 2L^{m+s} \in \mathfrak{L}(g'')$, $g'' = (L^s\mathsf{b}_2/2)^{-1/2}$; all large s are such, as in Step 2 of the proof of Proposition 18.141. Put $g' = (L^s\mathsf{b}_1/2)^{-1/2}$; then $g'' \le g' \le \gamma_W$, and $\ell_s \in \mathfrak{L}(g'') \subset \mathfrak{L}(g')$ by monotonicity of the least side in Proposition YM3_I:74.1. The labels $x_i = L^s\mathsf{b}_i$ satisfy $x_i \ge 2\gamma^{-2} > 2\bar{B}$. With $g_i^{-2} = x_i/2$, this gives $x_i > g_i^{-2} + \bar{B}$ and $x_i \le L(g_i^{-2} + \bar{B}) - 2\bar{B}$, hence $x_i \in I(g_i)$. Corollary YM3_I:74.2 gives the first-passage data $(K, L^K x_i)$ at $g_1 = g'$ and $g_2 = g''$ for every $K \in \mathbb{N}$. Put $\mathsf{d}_1 = L^{-s}\min\{\mathsf{d}_0(g',\vartheta), \mathsf{d}_0(g'',\vartheta)\}$, let $0 < \mathsf{d} \le \mathsf{d}_1$ and $(f,h) \in \mathfrak{T}_{T_\ell}(\mathsf{d};\vartheta)$. By dilation, $(f^{(s)},h^{(s)}) \in \mathfrak{T}_{T_{\ell_s}}(L^s\mathsf{d};\vartheta)$ with the same $\mathcal{K}_6 =: \mathcal{K} > 0$. Theorem YM3_I:AV, at g' and at g'', for all K above the two witness depths, gives $S^T_{2;K,\ell_s,L^K x_1}(f^{(s)},h^{(s)}) \ge \frac{2}{3}\mathsf{c}_2 x_1^{-2}\mathcal{K}$ and $\frac{2}{3}\mathsf{c}_2 x_2^{-2}\mathcal{K} \le S^T_{2;K,\ell_s,L^K x_2}(f^{(s)},h^{(s)}) \le \frac{4}{3}\mathsf{c}_2 x_2^{-2}\mathcal{K} = \frac{1}{3}\mathsf{c}_2 x_1^{-2}\mathcal{K}$. Let $K \to \infty$ (Theorem YM3_I:AS) and return to the side ℓ by YM3_I:(2215), with $L^{2s}x_i^{-2} = \mathsf{b}_i^{-2}$. Not empty: the first two entries of X^ℓ_{d} form a pair of $\mathfrak{T}_{T_\ell}(\mathsf{d};\vartheta)$ (Definition 18.140(5)).

Clause (2). The pairs (f,h) and the triples $(1,f,h)$ are tuples with a separated pair; by Proposition 18.141(1) the functions $\phi = S_2^\cdot(f,h)$ and $\psi = S_3^\cdot(1,f,h) = -\phi'$ are real-analytic on $(0,\infty)$. By (1), $\phi(\mathsf{b}_1) > \phi(2\mathsf{b}_1)$: ϕ is not constant, and Proposition 18.141(2) gives the integer ν. The mean value theorem gives $\mathsf{b} \in (\mathsf{b}_1, 2\mathsf{b}_1)$ with $\psi(\mathsf{b}) = (\phi(\mathsf{b}_1) - \phi(2\mathsf{b}_1))/\mathsf{b}_1 > 0$; so ψ is not identically 0, and Proposition 18.141(2) gives the last assertion.

Clause (3). Let $0 < \mathsf{d}' \le \mathsf{d}_*$ and $X = X^{\ell'}_{\mathsf{d}'}$; by Definition 18.140(5) it lies in $\mathfrak{T}^{(3),\mathrm{col}}_{T_{\ell'}}(\mathsf{d}';\vartheta) \subset \mathfrak{Q}^{(3)}_{T_{\ell'}}(\mathsf{d}';\vartheta)$, the product of the sup norms of its entries is 1, and $\mathcal{K}^{(3)}_{\mathrm{free}}(X) \ge c_3^{\mathrm{col}}\vartheta^3$. For all j with $K_j \ge s_3(\mathsf{d}')$ the hypothesis, read at the datum (K_j,β_j) with the label $\mathsf{b}_j = a_{K_j}\beta_j$, gives $S^T_{3;K_j,\ell',\beta_j}(X) \ge 3\mathsf{b}_j^{-3}[c_3^{\mathrm{col}}\vartheta^3 - \epsilon_3(\mathsf{d}') - C_3 a_{K_j}/(\vartheta\mathsf{d}')]$. The entries of X are real Lipschitz functions,

none of them 0, with pairwise disjoint supports, and $\beta_j > 0$; by YM3_I:(2216) the left member tends to $S_3^{\ell',\mathsf{b}'}(X)$, while $\mathsf{b}_j \to \mathsf{b}' > 0$ and $a_{K_j} \to 0$. □

Definition 18.143 (Sources, sourced means of a block field, mixed cumulants)**.** Let $K, m \in \mathbb{N}$, $\ell = 2L^m$, $\beta > 0$ and $\mathsf{b} = a_K\beta$.

(1) $\mathsf{Lp}(T_\ell)$ is the space of the complex Lipschitz functions w on T_ℓ, with the norm $\|w\|_\flat = \max\{\sup|w|, \frac{40}{\sqrt{3}} M \operatorname{Lip}_\infty w\}$, $\operatorname{Lip}_\infty$ being the Lipschitz number for the sup distance; and $\|1\|_\flat = 1$. For $r > 0$, $\mathsf{Src}(r) = \{w \in \mathsf{Lp}(T_\ell) : \|w\|_\flat < r\}$; a function on it is *holomorphic on the class* if $z \mapsto \Phi(w_0 + \sum_{i\le n} z_i f_i)$ is holomorphic on the open set where its argument lies in the class, for all n and all $w_0, f_i \in \mathsf{Lp}(T_\ell)$. For $w \in \mathsf{Lp}(T_\ell)$ we put $\mathcal{O}(w) = a_K^{-1}\sum_p w(x(p))\,a_p$.

(2) For an integer $0 \le d \le K$, a bounded complex Borel function F on V_d and $w \in \mathsf{Lp}(T_\ell)$ we put

$$N_K[F;w] = \int F(\mathcal{M}^{K-d}U)\, e^{-\beta\sum_p a_p(U) + \mathcal{O}(w)(U)}\, dU, \qquad Z_K[w] = N_K[1;w], \qquad \langle F\rangle_{K,w} = \frac{N_K[F;w]}{Z_K[w]} \quad (Z_K[w] \ne 0);$$

so $\langle F\rangle_{K,0} = \mu_d^K(F) = \int F\, d\mu_d^K$, and for a real number $\zeta < \mathsf{b}$, as $\mathcal{O}(\zeta) = \zeta L^K \sum_p a_p$, $\langle F\rangle_{K,\zeta} = \mu_d^K[\mathsf{b}-\zeta](F)$: a constant source shifts the label. Where the side and the label must be shown we write $\langle F\rangle_{K,w}^{\ell,\mathsf{b}}$, $N_K^{\ell,\mathsf{b}}$, $Z_K^{\ell,\mathsf{b}}$ and $\mu_d^K[\ell,\mathsf{b}]$, $\mu_d[\ell,\mathsf{b}]$.

(3) For complex Lipschitz $f_1, \dots, f_n$ the *mixed cumulant* $\langle F; \mathcal{O}(f_1); \dots; \mathcal{O}(f_n)\rangle_K$ is the cumulant $\mathsf{cum}(\mathcal{O}(f_1), \dots, \mathcal{O}(f_n), F\circ\mathcal{M}^{K-d})$ under the probability measure $Z_K[0]^{-1}e^{-\beta\sum_p a_p}\,dU$. Here the joint cumulant is the mixed derivative at zero of the local logarithm of the joint moment generating function.

(4) For real $\mathsf{b} > 0$ and integers $d \ge 0$ we put

$$Q_d(\ell,\mathsf{b}) = \exp\Big[(\mathsf{C}^{\mathrm{S}}_{\mathrm{maj}} + 2 + c_{\mathrm{low}} + \tfrac{9}{2}\log(L^d\mathsf{b}))(L^d\ell)^3\Big],$$

where $\mathsf{C}^{\mathrm{S}}_{\mathrm{maj}}$ is the constant in the history estimate of Prop. YM3_I:83.2, and c_{low} is the constant in the real vacuum floor of Prop. YM3_I:74.4. The source energy constants E_0 and κ below can be fixed before the source radius and the coupling ceiling by Thm. YM3_I:AD. $Q_d(\ell,\mathsf{b})$ reads (d,ℓ,b) through $L^d\ell$ and $L^d\mathsf{b}$ alone: $Q_{d-s}(L^s\ell, L^s\mathsf{b}) = Q_d(\ell,\mathsf{b})$.

■

Proposition 18.144 (The laws of the block fields are real-analytic in the label; mixed cumulants with the action density, uniformly in the cutoff)**.** Work in a setting $\mathfrak{s}$, with its radius ϱ. Let $m \in \mathbb{N}$, $\ell = 2L^m$, let $d \ge 0$ be an integer, and let F be a bounded complex Borel function on V_d, gauge invariant or not.

(1) Let $\mathsf{b} \ge \gamma^{-2}$ with $M\,R(\mathsf{b}) \le L^m$, let $K \ge d+1$, $Q_d = Q_d(\ell,\mathsf{b})$ and $r_1 = 2\varrho/(Q_d+1)$. Then $Z_K[w]$ has no zero on $\|w\|_\flat < r_1$, the sourced mean $w \mapsto \langle F\rangle_{K,w}$ is a holomorphic function on the class $\mathsf{Src}(r_1)$, and

$$|\langle F\rangle_{K,w}| \le \frac{Q_d\sup|F|}{1-\|w\|_\flat/r_1} \quad (\|w\|_\flat < r_1), \qquad |\langle F\rangle_{K,w}| \le 2Q_d\sup|F| \quad \Big(\|w\|_\flat \le \frac{\varrho}{Q_d+1}\Big).$$

(2) Under the hypotheses of (1), for each $n \ge 1$ and any complex Lipschitz $f_1, \dots, f_n$ on T_ℓ, with no condition on their supports,

$$\big|\langle F; \mathcal{O}(f_1); \dots; \mathcal{O}(f_n)\rangle_K\big| \le 2Q_d\sup|F|\Big(\frac{n(Q_d+1)}{\varrho}\Big)^n \prod_{i=1}^n \|f_i\|_\flat.$$

(3) *(Every label, every side.)* Let $\mathsf{b} > 0$, and let $s \ge 0$ be an integer with $L^s\mathsf{b} \ge \gamma^{-2}$ and $M\,R(L^s\mathsf{b}) \le L^{m+s}$; every sufficiently large integer qualifies. With $d' = \max\{d,s\}$ and $Q_{d'} = Q_{d'}(\ell,\mathsf{b})$, the statements (1) and (2) are true at (ℓ,b) for every $K \ge d'+1$ once Q_d is replaced by $Q_{d'}$ and ϱ by $L^{-s}\varrho$.

(4) *(The laws at $K=\infty$ are real-analytic in the label.)* For every $\mathsf{b}>0$, $\mu_d^K[\ell,\mathsf{b}](F)\to\mu_d[\ell,\mathsf{b}](F)$ as $K\to\infty$ along the full sequence, $\mu_d[\ell,\mathsf{b}]$ being the limit law obtained from Cor. 2.8(c) and dilation. The function $\mathsf{b}\mapsto\mu_d[\ell,\mathsf{b}](F)$ is real-analytic on $(0,\infty)$. Quantitatively, fix $\mathsf{b}_0>0$, take s, d' as in (3) at b_0, and put $\mathsf{r}=L^{-s}\varrho/(Q_{d'}(\ell,\mathsf{b}_0)+1)$. Then $\mathsf{r}\le\frac{1}{8}\mathsf{b}_0$, and there is a holomorphic function Φ on $D(\mathsf{b}_0,\mathsf{r})$ with $|\Phi|\le 2Q_{d'}(\ell,\mathsf{b}_0)\sup|F|$ that agrees with $\mathsf{b}\mapsto\mu_d[\ell,\mathsf{b}](F)$ on the real segment $(\mathsf{b}_0-\mathsf{r},\mathsf{b}_0+\mathsf{r})$; Φ is the locally uniform limit, as $K\to\infty$, of $\mathsf{b}\mapsto\langle F\rangle^{\ell,\mathsf{b}_0}_{K,\mathsf{b}_0-\mathsf{b}}$, the sourced mean at the constant complex source $\mathsf{b}_0-\mathsf{b}$.

(5) *(The derivatives.)* At each label $\mathsf{b}>0$ and for each $\nu\ge1$, the limit below exists and

$$\partial_{\mathsf{b}}^{\nu}\,\mu_d[\ell,\mathsf{b}](F)=(-1)^{\nu}\lim_{K\to\infty}\langle F;\underbrace{\mathcal{O}(1);\dots;\mathcal{O}(1)}_{\nu}\rangle_K,\qquad |\partial_{\mathsf{b}}^{\nu}\,\mu_d[\ell,\mathsf{b}](F)|\le 2Q_{d'}\sup|F|\Big(\frac{\nu L^{s}(Q_{d'}+1)}{\varrho}\Big)^{\nu},$$

the cumulants being taken at $(K,\ell,L^K\mathsf{b})$, and s, d', $Q_{d'}=Q_{d'}(\ell,\mathsf{b})$ as in (3).

(6) *(A common disc for every depth.)* Under the hypotheses of (1), let $Q_0=Q_0(\ell,\mathsf{b})$. Then $Z_K[w]$ does not vanish on $\|w\|_\flat<2\varrho/(Q_0+1)$, and on the smaller ball $\|w\|_\flat\le\varrho/(Q_0+1)$ one has $|\langle F\rangle_{K,w}|\le Q_d^3\sup|F|$. Consequently the disc $D(\mathsf{b},\varrho/(Q_0+1))$, whose radius does not involve d, carries a holomorphic extension of $\mathsf{b}'\mapsto\mu_d[\ell,\mathsf{b}'](F)$ for every depth d and every F, bounded there by $Q_d^3\sup|F|$.

(7) *(A rate.)* Let J be an interval of labels under (Tor) on the side ℓ in §2.3.1, let $\mathsf{b}\in J$ with $[\mathsf{b}-\mathsf{r},\mathsf{b}+\mathsf{r}]\subset J$, $\mathsf{r}=\varrho/(Q_d(\ell,\mathsf{b})+1)$, let C_{bf}, K_{bf} and $\mathsf{p}=2\mathfrak{p}_0+\frac12$ be the numbers of Def. 2.7(5) at J, for the cutoff comparison of §2.3.1, and let $\Sigma_K=\sum_{K'\ge K}K'^{\mathsf{p}}L^{-K'/10}$. Let $\nu\ge0$ and $0<\eta\le1$. For every integer $K\ge\max\{K_{\mathrm{bf}},1260\,d,d+1\}$ with $\mathsf{C}_{\mathrm{bf}}\ell^3\Sigma_K\le 2Q_d(\ell,\mathsf{b})$,

$$\Big|(-1)^{\nu}\langle F;\underbrace{\mathcal{O}(1);\dots;\mathcal{O}(1)}_{\nu}\rangle_K-\partial_{\mathsf{b}}^{\nu}\,\mu_d[\ell,\mathsf{b}](F)\Big|\le\big(2\mathsf{C}_{\mathrm{bf}}\ell^3\Sigma_K\big)^{1-\eta}\big(4Q_d(\ell,\mathsf{b})\big)^{\eta}\Big(\frac{4\nu^2}{\pi\eta\,\mathsf{r}}\Big)^{\nu}\sup|F|,$$

the cumulants at $(K,\ell,L^K\mathsf{b})$; at $\nu=0$ the left member is $|\mu_d^K(F)-\mu_d(F)|$.

■

Sketch. Step 1: the line of a label. At $\beta=L^K\mathsf{b}$ the prescribed sequence $u_j=L^{K-j}\mathsf{b}$ has $u_{j+1}=u_j/L$ and $u_K=\mathsf{b}\ge\gamma^{-2}$. The assumption $MR(\mathsf{b})\le L^m$ supplies the room on the torus for this line, in the setting (E) of §2.3.1. The fixed source radius and coupling ceiling are those of the sourced construction. Apply Thm. YM3_I:AJ to the prescribed sequence at $k=K-d\ge1$. Its active lattice is $\Lambda_{d,\ell}$, with $u_k=L^d\mathsf{b}$ and $\mathsf{N}=(L^d\ell)^3$ sites.

Step 2: the numerator. The sum of the sourced history measures is the pushforward of $e^{-\beta\sum_p a_p+\mathcal{O}(w)}\,dU$ by $\mathcal{M}^k$. Hence $N_K[F;w]$ is the sum of their integrals against F. Pulling F back to the free history coordinates preserves its supremum bound. Prop. YM3_I:83.2 bounds the sum of the total variations there after division by $|e^{E^w_{(k)}}|$. In particular

$$|e^{-E^w_{(k)}}N_K[F;w]|\le\sup|F|\,e^{(\mathsf{C}^{\mathrm{S}}_{\mathrm{maj}}+2)\mathsf{N}}\qquad(w\in\mathsf{Src}(2\varrho)).$$

On every finite-dimensional slice the bare integral for $N_K[F;w]$ is entire, since the integrand is entire in the source coordinates and uniformly bounded in U on compact sets of those coordinates.

Step 3: the denominator. By Prop. YM3_I:83.2, $\widetilde{Z}_k(w)=e^{-E^w_{(k)}}Z_K[w]$ is holomorphic on the class with $|\widetilde{Z}_k|\le e^{(\mathsf{C}^{\mathrm{S}}_{\mathrm{maj}}+2)\mathsf{N}}$, and by the lower bound of Prop. YM3_I:74.4, $\widetilde{Z}_k(0)\ge e^{-C_{\mathrm{low}}(L^d\mathsf{b})\mathsf{N}}>0$; so $e^{(\mathsf{C}^{\mathrm{S}}_{\mathrm{maj}}+2)\mathsf{N}}/\widetilde{Z}_k(0)\le Q_d$, and $Q_d\ge1$. The function $\widetilde{Z}_k/\widetilde{Z}_k(0)-1$ is holomorphic on the class, vanishes at 0 and has modulus at most Q_d+1; the Schwarz lemma on the disc along the complex line through 0 and w gives $|\widetilde{Z}_k(w)|\ge\widetilde{Z}_k(0)(1-\|w\|_\flat/r_1)>0$ for $\|w\|_\flat<r_1$. As $e^{-E^w_{(k)}}\ne0$, $Z_K[w]\ne0$ there.

Step 4: clause (1). For $\|w\|_\flat < r_1$, $\langle F\rangle_{K,w} = e^{-E^w_{(k)}} N_K[F;w]/\widetilde{Z}_k(w)$, and Steps 2, 3 give the first bound; at $\|w\|_\flat \le \frac12 r_1$ it is at most $2Q_d \sup|F|$. On every slice numerator and denominator are holomorphic and the denominator has no zero, so the quotient is holomorphic.

Step 5: a mixed cumulant is a derivative of a sourced mean. Let $Y_i = \mathcal{O}(f_i)$, $i \le n$, $Y_{n+1} = F \circ \mathcal{M}^{K-d}$ and $\mathsf{Z}(z) = \int \exp(\sum_{i\le n+1} z_i Y_i)\, e^{-\beta\sum_p a_p}\, dU$. By definition the mixed cumulant is $\partial_1\cdots\partial_{n+1}\operatorname{Log}(\mathsf{Z}/\mathsf{Z}(0))$ at 0, taking the derivative with the largest index first. By the chain rule $\partial_{n+1}\operatorname{Log}(\mathsf{Z}/\mathsf{Z}(0)) = \partial_{n+1}\mathsf{Z}/\mathsf{Z}$ near 0, and at the points $(z',0)$, by differentiation under the integral, $\partial_{n+1}\mathsf{Z}(z',0) = N_K[F;\sum_{i\le n} z_i f_i]$ and $\mathsf{Z}(z',0) = Z_K[\sum_{i\le n} z_i f_i]$. The derivatives $\partial_1,\dots,\partial_n$ at 0 read the slice $z_{n+1}=0$ alone. So

$$\langle F;\mathcal{O}(f_1);\dots;\mathcal{O}(f_n)\rangle_K = \partial_1\cdots\partial_n\Big[z \mapsto \langle F\rangle_{K,\sum_i z_i f_i}\Big](0).$$

Step 6: clause (2). If some f_i is 0, the function does not depend on z_i, and the derivative vanishes. Otherwise put $r_i = \varrho/((Q_d+1)\,n\,\|f_i\|_\flat)$; on the polydisc $|z_i| < r_i$ the source $\sum_i z_i f_i$ has $\|\cdot\|_\flat < \varrho/(Q_d+1)$, so by clause (1) the sourced mean is holomorphic there with modulus at most $2Q_d\sup|F|$, and Cauchy's estimate with Step 5 gives (2).

Step 7: clause (3), a relabelling. All large s have the two properties, because $R(x) \le L\max\{1,(\log x)^r\}$ (Thm. YM3_I:AD) grows at most like a power of s at $x = L^s\mathsf{b}$, while L^{m+s} grows like L^s. Let D_* be the relabelling for the dilation by L^s: it renames the factors of a finite power of SU(2), so it carries the product of the Haar measures to the product of the Haar measures; it carries the plaquettes p onto plaquettes $\mathsf{D}p$ with $a_{\mathsf{D}p}(\mathsf{D}_*U) = a_p(U)$ and $x(\mathsf{D}p) = L^s x(p)$; and it commutes with the averaging map. With $w' = L^s w^{(s)}$ the observable of w' on the lattice $(L^s\ell, K-s)$ at D_*U is $L^{K-s}\sum_p L^s w(x(p))a_p(U) = \mathcal{O}(w)(U)$, and $\beta = L^K\mathsf{b} = L^{K-s}(L^s\mathsf{b})$. So, with $F' = F\circ\mathcal{M}^{d'-d}\circ\mathsf{D}_*^{-1}$ on $\mathsf{V}_{d'-s}$ of $T_{L^s\ell}$, a bounded Borel function with $\sup|F'| \le \sup|F|$,

$$N_K^{\ell,\mathsf{b}}[F;w] = N_{K-s}^{L^s\ell,L^s\mathsf{b}}[F';L^s w^{(s)}], \qquad Z_K^{\ell,\mathsf{b}}[w] = Z_{K-s}^{L^s\ell,L^s\mathsf{b}}[L^s w^{(s)}],$$

the right members at the depth $d'-s$; and $\|L^s w^{(s)}\|_\flat = \max\{L^s\sup|w|, \frac{40}{\sqrt3} M\operatorname{Lip}_\infty w\} \le L^s\|w\|_\flat$. At $(L^s\ell, L^s\mathsf{b})$ clause (1) holds with $Q_{d'-s}(L^s\ell, L^s\mathsf{b}) = Q_{d'}(\ell,\mathsf{b})$; read through these identities it is clause (1) at (ℓ,b) for $\|w\|_\flat < L^{-s}\cdot 2\varrho/(Q_{d'}+1)$, the map $w\mapsto L^s w^{(s)}$ being linear. Step 5 at (ℓ,b) and Step 6 with the radii $L^{-s}\varrho/((Q_{d'}+1)n\|f_i\|_\flat)$ give clause (2) there.

Step 8: clause (4). Cor. 2.8(c) gives total variation convergence at admitted labels and sides. For an arbitrary positive label and side, use Step 7 at $w=0$: the deeper law on the dilated torus converges by that corollary, and total variation cannot increase under the fixed block projection and inverse dilation. for probability measures μ',μ'' and a bounded complex Borel F, $|\mu'(F)-\mu''(F)| \le 2\sup|F|\sup_E|\mu'(E)-\mu''(E)|$, first for simple F and then by uniform approximation. The source room condition in Lem. YM3_I:82.5, with $x_K = \mathsf{b}$, gives $2\varrho \le \frac12\mathsf{b}$ by taking constant sources with modulus tending to 2ϱ. Let first $\mathsf{b} \ge \gamma^{-2}$ with $M\,R(\mathsf{b}) \le L^m$, and $\mathsf{r} = \varrho/(Q_d+1) \le \frac12\varrho \le \frac18\mathsf{b}$. The functions $\phi_K(\zeta) = \langle F\rangle^{\ell,\mathsf{b}}_{K,\zeta}$, $K \ge d+1$, are holomorphic on $D(0,\mathsf{r})$ with $|\phi_K| \le 2Q_d\sup|F|$ (clause (1) at the source $\zeta\cdot 1$, $\|\zeta\cdot 1\|_\flat = |\zeta|$), and at real $\zeta\in(-\mathsf{r},\mathsf{r})$ they converge to $\mu_d[\ell,\mathsf{b}-\zeta](F)$, by Definition 18.143(2) and the first sentence. By Montel's theorem and uniqueness of analytic continuation the whole sequence converges uniformly on the compact subsets of $D(0,\mathsf{r})$ to a holomorphic ϕ with $|\phi| \le 2Q_d\sup|F|$, and $\mu_d[\ell,\mathsf{b}'](F) = \phi(\mathsf{b}-\mathsf{b}')$ for real $|\mathsf{b}'-\mathsf{b}| < \mathsf{r}$. At a general label b_0, the identities of Step 7 at $w=0$, at every $K \ge d'+1$, give $\mu_d^K[\ell,\mathsf{b}](F) = \mu_{d'-s}^{K-s}[L^s\ell, L^s\mathsf{b}](F')$ for all $\mathsf{b}>0$, so $\mu_d[\ell,\mathsf{b}](F) = \mu_{d'-s}[L^s\ell,L^s\mathsf{b}](F')$; at complex constant sources they give $\langle F\rangle^{\ell,\mathsf{b}_0}_{K,\zeta} = \langle F'\rangle^{L^s\ell,L^s\mathsf{b}_0}_{K-s,L^s\zeta}$ for $|\zeta|<\mathsf{r}$. The first case, at $(L^s\ell,L^s\mathsf{b}_0)$, the depth $d'-s$ and F', gives the clause; and $\mathsf{r} \le L^{-s}\frac12\varrho \le \frac18\mathsf{b}_0$, because $2\varrho \le \frac12 L^s\mathsf{b}_0$.

Step 9: clause (5). With ϕ_K and ϕ on $D(0,\mathsf{r})$ as in Step 8 at the label b, let $0<\mathsf{r}'<\mathsf{r}$ and $\epsilon_K = \sup_{|\zeta|\le\mathsf{r}'}|\phi_K-\phi| \to 0$. The function $(\phi_K-\phi)(z_1+\cdots+z_\nu)$ on the polydisc of the

radii r'/ν has the mixed derivative $(\phi_K-\phi)^{(\nu)}(0)$ at 0, and Cauchy's estimate bounds it by $\epsilon_K(\nu/\mathsf{r}')^\nu\to 0$. By Step 5 at $f_1=\dots=f_\nu=1$ and the chain rule, $\phi_K^{(\nu)}(0)=\langle F;\mathcal{O}(1);\dots;\mathcal{O}(1)\rangle_K$; and $\partial_{\mathsf{b}'}^\nu\phi(\mathsf{b}-\mathsf{b}')|_{\mathsf{b}'=\mathsf{b}}=(-1)^\nu\phi^{(\nu)}(0)$. The bound is clause (3) at $f_i=1$, in the limit.

Step 10: clause (6). Two facts of complex analysis are used: a holomorphic function without a zero on a disc is e^{-g} with g holomorphic there, and Harnack's inequality, $u(\zeta)\le u(0)\frac{\mathsf{r}_\circ+|\zeta|}{\mathsf{r}_\circ-|\zeta|}$ for a harmonic function $u\ge 0$ on $D(0,\mathsf{r}_\circ)$. Applied at the level K, where the depth is 0, Step 3 yields $Z_K[w]\neq 0$ for $\|w\|_\flat<\mathsf{r}_\circ=2\varrho/(Q_0+1)$; consequently $\widetilde{Z}_k$, taken at the level $k=K-d$, is free of zeros on the same set. For $0<\|w\|_\flat\le\frac12\mathsf{r}_\circ$ put $\hat w=w/\|w\|_\flat$ and $\vartheta(\zeta)=\widetilde{Z}_k(\zeta\hat w)\,e^{-(\mathsf{C}^{\mathrm{S}}_{\mathrm{maj}}+2)\mathsf{N}}$ on $D(0,\mathsf{r}_\circ)$: it is holomorphic, has no zero, $|\vartheta|\le 1$, and $Q_d^{-1}\le\vartheta(0)\le 1$ (Step 3). With $\vartheta=e^{-g}$, Harnack's inequality for $u=\operatorname{Re}g=-\log|\vartheta|\ge 0$ at $|\zeta|=\|w\|_\flat\le\frac12\mathsf{r}_\circ$ gives $|\vartheta(\zeta)|\ge\vartheta(0)^3\ge Q_d^{-3}$, and with Step 2, $|\langle F\rangle_{K,w}|\le Q_d^3\sup|F|$; at $w=0$ the bound is $\sup|F|$. For the last sentence, the functions $\phi_K(\zeta)=\langle F\rangle_{K,\zeta}^{\ell,\mathsf{b}}$ are holomorphic on $|\zeta|<\mathsf{r}_\circ$, bounded by $Q_d^3\sup|F|$ on $D(0,\frac12\mathsf{r}_\circ)$, and converge at its real points, where $\mathsf{b}-\zeta>0$ by the convergence proved in Step 8. Montel's theorem and uniqueness give the continuation, as in Step 8.

Step 11: clause (7). By Cor. 2.8 and Step 8, for K as in the clause and all real $|\zeta|<\mathsf{r}$, $|\phi_K(\zeta)-\phi(\zeta)|=|\mu_d^K[\mathsf{b}-\zeta](F)-\mu_d[\mathsf{b}-\zeta](F)|\le\epsilon_K:=2\sup|F|\,\mathsf{C}_{\mathrm{bf}}\ell^3\Sigma_K$, since $\mathsf{b}-\zeta\in J$ and $\lceil K/1260\rceil\ge d$; and $|\phi_K-\phi|\le B:=4Q_d\sup|F|$ on $D(0,\mathsf{r})$. If $F=0$ there is nothing to show; otherwise $0<\epsilon_K\le B$. The two-constants bound, Lem. YM3_I:116.2, for the function $(\phi_K-\phi)(z_1+\dots+z_\nu)$ on the polydisc of the radii r/ν, whose real points are carried into $(-\mathsf{r},\mathsf{r})$, gives $|\phi_K^{(\nu)}(0)-\phi^{(\nu)}(0)|\le\epsilon_K^{1-\eta}B^\eta(\frac{4\nu}{\pi\eta})^\nu(\frac{\nu}{\mathsf{r}})^\nu$; with Step 9 this is the clause. At $\nu=0$, $|\phi_K(0)-\phi(0)|\le\epsilon_K\le\epsilon_K^{1-\eta}B^\eta$. □

Proposition 18.145 (What diverges in the action density at coinciding points is white noise, independent of the block fields)**.** Work in a setting $\mathfrak{s}$, with its radius ϱ. Let $\kappa_{\mathcal{D}}=\log 9216$ and $S(q)=2^{12}(1-e^{\kappa_{\mathcal{D}}-q})^{-1}$ for $q>\kappa_{\mathcal{D}}$. Use the constants E_0 and $\kappa>\kappa_{\mathcal{D}}+2$ of the source energy estimate in Thm. YM3_I:AD. Put

$$C_e=\frac{2E_0\,S(\kappa)}{L^3-1},\qquad C_\Gamma=\frac{1536\,M^2E_0\,S(\kappa-1)}{L^3-1},$$

numbers formed from the constants before γ alone. Let $\mathsf{b}\ge\gamma^{-2}$ and $m\in\mathbb{N}$ with $M\,R(\mathsf{b})\le L^m$, $\ell=2L^m$, $Q_d=Q_d(\ell,\mathsf{b})$ and $r_d=\varrho/(2Q_d)$. At the cutoff K the measure is the Wilson measure at $\beta=L^K\mathsf{b}$, with the mean $\mathbb{E}_K=\mathbb{E}_{K,\mathsf{b}}$; put

$$Y_K(f)=a_K^{3/2}(\mathcal{O}(f)-\mathbb{E}_K\mathcal{O}(f)),\qquad \varsigma_K^2=a_K^3\ell^{-3}S^T_{2;K,\ell,L^K\mathsf{b}}(1,1)=a_K\ell^{-3}\operatorname{Var}_K\Big(\sum_p a_p\Big)=a_K\ell^{-3}\,\partial_\beta^2\log Z_K\ \ge 0,$$

the specific heat in physical units, $Z_K=Z_K[0]$ being read as a function of β. Test functions are real Lipschitz functions on T_ℓ, with the norm $\|f\|=\|f\|_\flat$ of Definition 18.143(1); nothing is asked of their supports.

(a) *Every cumulant grows at most like a_K^{-3}.* For all $K\ge 1$, $n\ge 1$ and all $f_1,\dots,f_n$,

$$|S^T_{n;K,\ell,L^K\mathsf{b}}(f_1,\dots,f_n)|\le n^n\Big[C_e\,\ell^3a_K^{-3}(2\varrho)^{-n}+\log 2\;r_0^{-n}\Big]\prod_{i=1}^n\|f_i\|;$$

so $a_K^{3n/2}S^T_{n;K,\ell,L^K\mathsf{b}}\to 0$ for $n\ge 3$, and $\varsigma_K^2\le\frac{C_e}{2\varrho^2}+\frac{2\log 2}{\ell^3r_0^2}\,a_K^3$.

(b) *Small sources change the law of a block field only a little.* Let $d\ge 0$, $K\ge d+1$, F a bounded complex Borel function on V_d and w a complex Lipschitz function with $\|w\|_\flat<r_d$. Then $Z_K[w]\neq 0$, $|\langle F\rangle_{K,w}|\le 2Q_d\sup|F|$ and $|\langle F\rangle_{K,w}-\mu_d^K(F)|\le\frac{4Q_d}{r_d}\sup|F|\,\|w\|_\flat$.

(c) *Gaussian behaviour at a finite cutoff, with an error term.* Let $d\ge 0$, $K\ge d+1$, F as in (b), $f_1,\dots,f_J$ test functions, $t\in\mathbb{C}^J$, $w_t=\sum_i t_if_i$ and $\Sigma_K(t)=a_K^3\sum_{i,j}t_it_jS^T_{2;K,\ell,L^K\mathsf{b}}(f_i,f_j)$. If

$a_K^{3/2}\|w_t\|_\flat \le \frac12 r_d$, then

$$\Big|\mathbb{E}_K\Big[F(V_d)\,e^{\sum_i t_i Y_K(f_i)}\Big] - \mu_d^K(F)\,e^{\frac12\Sigma_K(t)}\Big| \le \sup|F|\,e^{\frac12\operatorname{Re}\Sigma_K(t)+\epsilon_K}\Big[\frac{4Q_d}{r_d}\,a_K^{3/2}\|w_t\|_\flat + \epsilon_K\Big],$$

with $\epsilon_K = 2C_e\ell^3 a_K^{3/2}(\frac{\|w_t\|_\flat}{2\varrho})^3 + 2\log 2\; a_K^{9/2}(\frac{\|w_t\|_\flat}{r_d})^3$. This error is $O(a_K^{3/2})$ for $\sup|F|\le 1$ and t in a fixed compact subset of $\mathbb{C}^J$, with a constant independent of F and t.

(d) *A white-noise covariance; the same factor ς_K^2 serves every test function.* For all test functions f, h, $a_K^3 S^T_{2;K,\ell,L^K\mathsf{b}}(f,h) - \varsigma_K^2\int_{T_\ell} fh \to 0$ as K runs through all the integers. Quantitatively: given $\varrho'>0$, one finds an integer $i=i(\varrho')\ge 0$ and a number $r'=r'(\varrho')>0$, depending only on the constants before γ, on b, ℓ and ϱ', with

$$\Big|a_K^3 S^T_{2;K,\ell,L^K\mathsf{b}}(f,f) - \varsigma_K^2\int f^2\Big| \le C_\Gamma\,\ell^3\frac{(\operatorname{Lip}_\infty f)^2}{\varrho'^2} + \frac{4\log 2\,L^{2i}}{r'^2}(\|f^2\|+\|f\|^2)a_K^3 + \tfrac12\,\ell^3\varsigma_K^2\operatorname{Lip}_\infty(f^2)\,a_K$$

for every $K > i$.

(e) *A lower bound for the specific heat.* At each real label $\mathsf{b}>0$ and on each side $\ell = 2L^m$, $m\in\mathbb{N}$, every test function f has $\liminf_{K\to\infty} a_K^3\operatorname{Var}_K\mathcal{O}(f) \ge \frac34\mathsf{b}^{-2}\int_{T_\ell} f^2$; taking $f=1$, $\liminf_K \varsigma_K^2 \ge \frac34\mathsf{b}^{-2}$.

(f) *The limit laws.* Let $G_{ij} = \int_{T_\ell} f_i f_j$. (f1) From any infinite set of cutoffs one can extract an infinite subset $\mathcal{K}$ on which ς_K^2 has a limit $\varsigma_0^2 \in [\frac34\mathsf{b}^{-2}, \frac{C_e}{2\varrho^2}]$. On such a $\mathcal{K}$, simultaneously for all $d\ge 0$, all J and all test functions $f_1,\dots,f_J$, $\mathbb{E}_K[F(V_d)\,\phi(Y_K(f_1),\dots,Y_K(f_J))] \to \mu_d(F)\int\phi\,dN(0,\varsigma_0^2 G)$ for all bounded Borel F on V_d and all ϕ that are either bounded continuous functions or polynomials; in particular $(V_d,(Y_K(f_i))_{i\le J})$ converges in distribution to $\mu_d\otimes N(0,\varsigma_0^2 G)$. (f2) For K beyond some $K_\circ$ one has $\varsigma_K>0$, and then, without passing to a subsequence, $(V_d,(Y_K(f_i)/\varsigma_K)_i)$ converges in distribution to $\mu_d\otimes N(0,G)$, together with the moments as in (f1): after the normalisation by ς_K, the centred action density converges to a standard white noise, and in the limit this noise is independent of V_d for each d.

(g) *Shifts by constants do not make $\mathcal{O}(f)$ tight.* Let $f\neq 0$ be a test function and (c_K) any sequence of real numbers; then on no infinite set of cutoffs is the family of the laws of $\mathcal{O}(f)-c_K$ under $\mathbb{E}_K$ tight.

Here $N(0,C)$ is the centred normal law with the covariance matrix C, and $\int e^{t\cdot y}\,dN(0,C)(y) = e^{\frac12 t\cdot Ct}$ for $t\in\mathbb{C}^J$. ■

Sketch. Step 1: the source energies. Throughout this sketch, $S^T_{n;K}$ abbreviates $S^T_{n;K,\ell,L^K\mathsf{b}}$. Let $d\ge 0$, $K\ge d+1$ and $k=K-d$. Use the same prescribed line as in Step 1 of the proof of Proposition 18.144. Prop. YM3_I:83.2 gives $E^w_{(k)} = E_{(k)} + \mathcal{E}_k(w)$ with $\mathcal{E}_k(w) = \sum_{j=1}^k\sum_{X\in\mathcal{D}_j} e_j(X;w)$, where each $e_j(X;\cdot)$ is holomorphic on the class $\mathsf{Src}(2\varrho)$, vanishes at 0, is real at real sources, depends on the source through its values on $X^{\sim 1}$ alone, and obeys $|e_j(X;w)| \le 2M^3E_0\frac{\|w\|_\flat}{2\varrho}e^{-\kappa d_j(X)}$; the number $E_{(k)}$ does not depend on the source. The polymers of $\mathcal{D}_j$ are unions of cubes of the partition π_j, which has $(L^{K-j}\ell/M)^3$ cubes, and the entropy estimate in the proof of Lem. YM3_I:88.1 bounds the number of size-n domains containing a given cube by $2048\,9216^n$. The geometric sum is at most $S(\kappa')$ for $\kappa'>\kappa_{\mathcal{D}}$. So

$$\sum_{j=1}^K\sum_{X\in\mathcal{D}_j} e^{-\kappa' d_j(X)} \le \frac{S(\kappa')}{M^3}\,\ell^3\sum_{j=1}^K L^{3(K-j)} \le \frac{S(\kappa')\,\ell^3 a_K^{-3}}{M^3(L^3-1)},$$

and $|\mathcal{E}_k(w)| \le C_e\ell^3 a_K^{-3}\|w\|_\flat/(2\varrho) \le C_e\ell^3 a_K^{-3}$ on the class. As in Steps 2, 3 of the proof of Proposition 18.144, $|e^{-E^w_{(k)}}N_K[F;w]| \le Q_d\sup|F|\,\widetilde{Z}_k(0)$ on the class, and $\widetilde{Z}_k = e^{-E^w_{(k)}}Z_K$ is holomorphic

there, with $|\widetilde{Z}_k| \le Q_d \widetilde{Z}_k(0)$. As $\mathcal{E}_k(0) = 0$,

$$\frac{Z_K[w]}{Z_K[0]} = e^{\mathcal{E}_k(w)}\,\frac{\widetilde{Z}_k(w)}{\widetilde{Z}_k(0)}.$$

Step 2: the small ball. Apply the Schwarz lemma to $(\widetilde{Z}_k/\widetilde{Z}_k(0) - 1)/(Q_d + 1)$ on the disc through 0 and w. Since $Q_d \ge 1$, it gives $|\widetilde{Z}_k(w)/\widetilde{Z}_k(0) - 1| \le Q_d \|w\|_\flat/\varrho \le \frac12$ for $\|w\|_\flat \le r_d$. So $\mathsf{L}_k(w) = \mathrm{Log}(\widetilde{Z}_k(w)/\widetilde{Z}_k(0))$ is formed there, $|\mathsf{L}_k| \le \sum_{n\ge1} 2^{-n}/n = \log 2$, it is holomorphic on every slice of the ball, and it is real at real sources, since $\widetilde{Z}_k(\bar w) = \overline{\widetilde{Z}_k(w)}$ by Prop. YM3_I:83.2. With $\Xi = \mathcal{E}_k + \mathsf{L}_k$: $e^{\Xi(w)} = Z_K[w]/Z_K[0] = \mathbb{E}_K e^{\mathcal{O}(w)}$ and $\Xi(0) = 0$ for $\|w\|_\flat \le r_d$; in particular $Z_K[w] \ne 0$ there.

Step 3: clause (a). Take $d = 0$ and $k = K$. If an entry is 0, the cumulant is 0: the generating function does not depend on the corresponding variable. Otherwise $S^T_{n;K,\ell,L^K\mathsf{b}}(f_1,\dots,f_n) = \partial_z^n \log Z(0)$ for $Z(z) = Z_K[\sum_i z_i f_i]$, and any holomorphic g with $e^g = Z$ on a polydisc about 0 may be used because two local logarithms differ by a constant; by Step 2, $g(z) = \log Z(0) + \Xi(\sum_i z_i f_i)$ is one on the polydisc $|z_i| < r_0/(n\|f_i\|)$. Cauchy's estimate for $\mathcal{E}_K(\sum_i z_i f_i)$ on the polydisc of the radii $2\varrho/(n\|f_i\|)$, where it is bounded by $C_e \ell^3 a_K^{-3}$, and for $\mathsf{L}_K(\sum_i z_i f_i)$ on the radii $r_0/(n\|f_i\|)$, where it is bounded by $\log 2$, gives (a). For ς_K^2: the Taylor coefficients c_n of $\zeta \mapsto \Xi(\zeta\cdot 1)$ at 0 obey $|c_n| \le C_e \ell^3 a_K^{-3}(2\varrho)^{-n} + \log 2\, r_0^{-n}$ by Cauchy's inequalities on the discs $|\zeta| < 2\varrho$ and $|\zeta| < r_0$, as $\|\zeta\cdot 1\|_\flat = |\zeta|$; and $S^T_{2;K}(1,1) = 2c_2$ is a variance, hence ≥ 0.

Step 4: clause (b). For $\|w\|_\flat < r_d$, $\langle F\rangle_{K,w} = e^{-E^w_{(k)}} N_K[F;w]/\widetilde{Z}_k(w)$, with $|\widetilde{Z}_k(w)| \ge \frac12 \widetilde{Z}_k(0)$ by Step 2; Step 1 gives $|\langle F\rangle_{K,w}| \le 2Q_d \sup|F|$. The function $w \mapsto \langle F\rangle_{K,w} = N_K[F;w]/Z_K[w]$ is holomorphic on the class $\mathsf{Src}(r_d)$, as a quotient of two functions entire on every slice whose denominator has no zero, and its value at 0 is $\mu_d^K(F)$. Applying the Schwarz lemma to the difference from its value at zero on the complex line through w, with bound $4Q_d \sup|F|$, gives the second bound.

Step 5: clause (c). Let $w_t \ne 0$, and put $\phi(\zeta) = \Xi(\zeta w_t)$ for $|\zeta|\,\|w_t\|_\flat \le r_d$. Then $e^{\phi(\zeta)} = \mathbb{E}_K e^{\zeta\mathcal{O}(w_t)}$, so $\phi'(0) = \mathbb{E}_K\mathcal{O}(w_t)$ and $\phi''(0) = \sum_{i,j} t_i t_j S^T_{2;K}(f_i,f_j)$. By Cauchy's inequalities on the discs $|\zeta| < 2\varrho/\|w_t\|_\flat$ and $|\zeta| < r_d/\|w_t\|_\flat$ the Taylor coefficients of ϕ obey $|c_n| \le C_e\ell^3 a_K^{-3}(\|w_t\|_\flat/2\varrho)^n + \log 2\,(\|w_t\|_\flat/r_d)^n$. For $|\zeta|\,\|w_t\|_\flat \le \frac12 r_d$, where $\frac12 r_d \le \frac14\varrho$, and with $\sum_{n\ge3} x^n \le 2x^3$ for $0 \le x \le \frac12$, $\phi(\zeta) = \zeta\phi'(0) + \frac12\zeta^2\phi''(0) + \mathsf{R}_3(\zeta)$ with

$$|\mathsf{R}_3(\zeta)| \le 2C_e\ell^3 a_K^{-3}\Big(\frac{|\zeta|\,\|w_t\|_\flat}{2\varrho}\Big)^3 + 2\log 2\,\Big(\frac{|\zeta|\,\|w_t\|_\flat}{r_d}\Big)^3.$$

At $\zeta = a_K^{3/2}$ and $w = a_K^{3/2} w_t$ this gives $|\mathsf{R}_3| \le \epsilon_K$, and, $\mathcal{O}$ being linear,

$$\mathbb{E}_K\Big[F(V_d)\,e^{\sum_i t_i Y_K(f_i)}\Big] = \langle F\rangle_{K,w}\,\frac{Z_K[w]}{Z_K[0]}\,e^{-\mathbb{E}_K\mathcal{O}(w)} = \langle F\rangle_{K,w}\,e^{\frac12\Sigma_K(t)+\mathsf{R}_3}.$$

Subtracting $\mu_d^K(F)e^{\frac12\Sigma_K(t)}$ and using (b), $|e^{\mathsf{R}_3}| \le e^{\epsilon_K}$, $|e^{\mathsf{R}_3} - 1| \le \epsilon_K e^{\epsilon_K}$ and $|\mu_d^K(F)| \le \sup|F|$ gives (c); by (a), $\Sigma_K(t)$ is bounded uniformly in K for t in a compact set.

Step 6: clause (d). Both members are symmetric bilinear forms in (f,h), so $f = h$ suffices. (i) *Algebra.* Given an algebra of real-valued functions containing the constants and a symmetric real bilinear form q on it, set $\Gamma_q(u) = q(u^2,1) - q(u,u)$; a direct expansion gives $\Gamma_q(u-c) = \Gamma_q(u)$ for all constants c. With $\Gamma_K = \Gamma_{S^T_{2;K}}$, $S^T_{2;K}(f,f) = S^T_{2;K}(f^2,1) - \Gamma_K(f)$. (ii) *A larger radius on a dilated torus.* Let $\varrho' > 0$. Thm. YM3_I:AD permits the source radius ϱ' after reducing the coupling ceiling to some $\gamma' > 0$, while keeping the constants preceding the radius, in particular E_0, κ, L, M, r, fixed. Use its source assignment and the all-prefix form of the energy estimates in Prop. YM3_I:83.2. Let $i \ge 0$ be the least integer with $L^i\mathsf{b} \ge \gamma'^{-2}$ and $M\,R(L^i\mathsf{b}) \le L^{m+i}$; all large integers are such, since $R(x) < L(\log x)^r$ for $x > e$ (Thm. YM3_I:AD). For $K > i$ put $K' = K - i$, $\ell' = L^i\ell$

and $\mathsf{b}' = L^i\mathsf{b}$, so that $L^K\mathsf{b} = L^{K'}\mathsf{b}'$. Steps 1, 2 hold on the lattice (ℓ', K') at the label b' in the new setting, at the depth 0, with functions $e'_j(X;W)$, $|e'_j(X;W)| \le 2M^3E_0e^{-\kappa d_j(X)}$ for $\|W\|'_\flat < 2\varrho'$, and L' with $|\mathsf{L}'| \le \log 2$ on $\|W\|'_\flat \le r' := \varrho'/(2Q_0(\ell', \mathsf{b}'))$. By the exact dilation identity of Lem. YM3_I: 118.2, $S^T_{2;K,\ell,L^K\mathsf{b}}(u,v) = L^{2i}S^T_{2;K',\ell',L^{K'}\mathsf{b}'}(u^{(i)}, v^{(i)})$ for real Lipschitz u, v, and therefore, by Step 3 on (ℓ', K'),

$$S^T_{2;K,\ell,L^K\mathsf{b}}(u,v) = L^{2i}\Big[\sum_{j=1}^{K'}\sum_{X\in\mathcal{D}'_j} q_X(u^{(i)}, v^{(i)}) + q_\infty(u^{(i)}, v^{(i)})\Big],$$

with $q_X(u',v') = \partial_1\partial_2\big[e'_j(X; z_1u' + z_2v')\big](0)$ and $q_\infty(u',v') = \partial_1\partial_2\big[\mathsf{L}'(z_1u' + z_2v')\big](0)$. (iii) *Three properties.* Each q_X and q_∞ is a symmetric bilinear form, real at real arguments: mixed derivatives of a holomorphic function commute, the chain rule along $z(c_1u_1 + c_2u_2) + z_3v'$ gives linearity, and the functions are real at real sources. If $u' = u''$ on $X^{\sim 1}$, then $q_X(u',v') = q_X(u'',v')$, by the locality of $e'_j(X;\cdot)$. Cauchy's estimate on the polydiscs of the radii $\varrho'/\|u'\|'_\flat$, $\varrho'/\|v'\|'_\flat$, respectively $r'/(2\|u'\|'_\flat)$, $r'/(2\|v'\|'_\flat)$, gives $|q_X(u',v')| \le 2M^3E_0e^{-\kappa d_j(X)}\|u'\|'_\flat\|v'\|'_\flat/\varrho'^2$ and $|q_\infty(u',v')| \le 4\log 2\,\|u'\|'_\flat\|v'\|'_\flat/r'^2$. (iv) *One polymer.* Let $\mathsf{f} = f^{(i)}$, $\lambda = \mathrm{Lip}_\infty \mathsf{f} = L^{-i}\,\mathrm{Lip}_\infty f$, $X \in \mathcal{D}'_j$ and $\mathsf{d} = d_j(X)$. A connected set of d edges of the partition meeting all cubes of X has diameter at most d times the cube side. Every point of X is within one cube side of that set, and every point of $X^{\sim 1}$ is within one cube side of X. The cube side is at most M, so $\mathrm{diam}_\infty X^{\sim 1} \le D_X := M(\mathsf{d}+4)$. Fix $y_X \in X$, put $\mathsf{g} = \mathsf{f} - \mathsf{f}(y_X)$, $\delta = \lambda D_X$ and $\tilde{\mathsf{g}} = \max\{-\delta, \min\{\delta, \mathsf{g}\}\}$. Then $\tilde{\mathsf{g}} = \mathsf{g}$ on $X^{\sim 1}$, $\|\tilde{\mathsf{g}}\|'_\flat \le 6\lambda D_X$ and $\|\tilde{\mathsf{g}}^2\|'_\flat \le 12\lambda^2 D_X^2$, as $D_X \ge 4M$; and by (i), (iii)

$$|\Gamma_{q_X}(\mathsf{f})| = |q_X(\tilde{\mathsf{g}}^2, 1) - q_X(\tilde{\mathsf{g}}, \tilde{\mathsf{g}})| \le 2M^3E_0e^{-\kappa\mathsf{d}}\,\frac{48\,\lambda^2M^2(\mathsf{d}+4)^2}{\varrho'^2} \le 1536\,M^5E_0\,\frac{\lambda^2}{\varrho'^2}\,e^{-(\kappa-1)\mathsf{d}},$$

because $(\mathsf{d}+4)^2e^{-\mathsf{d}} \le 16$ for $\mathsf{d} \ge 0$. (v) *The sum.* The count of Step 1 on (ℓ', K') at $\kappa' = \kappa - 1 > \kappa_{\mathcal{D}}$, whose right member is the same number since $\ell'^3a_{K'}^{-3} = \ell^3a_K^{-3}$, together with $L^{2i}\lambda^2 = (\mathrm{Lip}_\infty f)^2$, the bound of q_∞, $\|\mathsf{f}\|'_\flat \le \|f\|$ and $\|\mathsf{f}^2\|'_\flat \le \|f^2\|$, gives

$$a_K^3|\Gamma_K(f)| \le C_\Gamma\,\ell^3\frac{(\mathrm{Lip}_\infty f)^2}{\varrho'^2} + \frac{4\log 2\,L^{2i}}{r'^2}(\|f^2\| + \|f\|^2)a_K^3 \qquad (K > i).$$

(vi) *The lattice symmetries.* Translations by lattice points and coordinate permutations relabel the plaquettes and bonds, preserving the Wilson measure and $\mathcal{O}(1)$. Hence $S^T_{2;K}(u \circ \mathsf{r}^{-1}, 1) = S^T_{2;K}(u, 1)$ for these isometries r. They form a finite group that acts transitively on the plaquettes and carries barycentres to barycentres, so the mean of $u \circ \mathsf{r}^{-1}$ over the group takes at every barycentre the value $\bar{u}^K = (3N_K)^{-1}\sum_p u(x(p))$, $N_K = \ell^3a_K^{-3}$; and $\mathcal{O}$ reads a function at the barycentres alone. So $a_K^3S^T_{2;K}(u,1) = \varsigma_K^2\,\ell^3\,\bar{u}^K$. (vii) *A Riemann sum.* For each orientation the barycentres form a translate of $\Lambda_{K,\ell}$, and the cubes of side a_K about them tile T_ℓ; so $|\ell^3\bar{u}^K - \int u| \le \frac{1}{2}\ell^3a_K\,\mathrm{Lip}_\infty u$. At $u = f^2$, with (i), (v) and (vi), this is the display of (d); ς_K^2 is bounded by (a). As ϱ' is arbitrary, the limit is 0.

Step 7: clause (e), at every label and on every side. Let K be so large that $\beta = L^K\mathsf{b} \ge 1$. Write $a = a_K$, $A = \sum_p a_p$, and use $\|P\|_2 = (\mathrm{tr}\,P^*P)^{1/2}$; for $V \in \mathrm{SU}(2)$, $1 - \frac{1}{2}\,\mathrm{tr}\,V = \frac{1}{4}\|V - \mathbf{1}\|_2^2$, and $a_p = 1 - \frac{1}{2}\,\mathrm{tr}\,\partial U(p)$. (i) *Half of the vertical bonds.* Let $\mathcal{B}$ be the set of the bonds $\langle x, x + ae_3\rangle$ with $(x_1 + x_2)/a$ even, a parity that is well defined because ℓ/a is even. A plaquette contains either no bond in the direction e_3 or two of them, based at sites of opposite parity, so no plaquette contains two bonds of $\mathcal{B}$; every bond lies in four plaquettes; and for $b \in \mathcal{B}$ and $p \ni b$, the trace being invariant under cyclic permutations and under inversion on $\mathrm{SU}(2)$, $\mathrm{tr}\,\partial U(p) = \mathrm{tr}(U(b)S_{p,b})$, where $S_{p,b} \in \mathrm{SU}(2)$ is a product of values of U at bonds that are not in $\mathcal{B}$. With U' the variables of the bonds outside $\mathcal{B}$, put $\mathsf{S}_b = \sum_{p\ni b} S_{p,b} = \lambda_bW_b$, with $\lambda_b \in [0,4]$ and $W_b \in \mathrm{SU}(2)$ Borel in U' ($\mathrm{SU}(2)$ is the unit sphere of the quaternions, and a sum of unit quaternions is a non-negative

multiple of one), $\mathsf{S}_b^f = \sum_{p\ni b} f(x(p))S_{p,b}$, $X_b = \frac12\operatorname{tr}(U(b)\mathsf{S}_b)$ and $\alpha_b = \sum_{p\ni b} a_p = 4 - X_b$. Then $\sum_{b\in\mathcal{B}}\alpha_b \le A$, $e^{-\beta A} = c_1(U')\prod_{b\in\mathcal{B}} e^{\beta X_b}$ and $\mathcal{O}(f) = c_2(U') - a^{-1}\sum_{b\in\mathcal{B}}\frac12\operatorname{tr}(U(b)\mathsf{S}_b^f)$. Conditionally on U' the $U(b)$, $b\in\mathcal{B}$, are independent, and the formula for the variance through conditional variances gives $\operatorname{Var}_K\mathcal{O}(f) \ge a^{-2}\sum_{b\in\mathcal{B}}\mathbb{E}_K\operatorname{Var}(\frac12\operatorname{tr}(U(b)\mathsf{S}_b^f)\mid U')$. (ii) *A single bond.* Given U', $V = U(b)W_b$ has the law $\propto e^{t\cos\theta}\,dV$, with $t = \beta\lambda_b$ and $\frac12\operatorname{tr}V = \cos\theta$, $\theta\in[0,\pi]$; under the Haar measure the angle has the density $\frac{2}{\pi}\sin^2\theta$ (Weyl's formula for SU(2)), so here it has the density $\propto \sin^2\theta\, e^{t\cos\theta}$. Let $v(t)$ be the variance of $\mathsf{X} = t(1-\cos\theta)$ under this law; X has the density $\propto [\mathsf{x}(2-\mathsf{x}/t)]^{1/2}e^{-\mathsf{x}}$ on $[0,2t]$, and dominated convergence gives $v(t)\to\frac{15}{4}-\frac94 = \frac32$, the variance of the law $\propto \mathsf{x}^{1/2}e^{-\mathsf{x}}$, as $t\to\infty$. Then $\operatorname{Var}(X_b\mid U') = \lambda_b^2\operatorname{Var}(\cos\theta\mid U') = \beta^{-2}v(\beta\lambda_b)$. As $\lambda_b\le 4$, $\alpha_b = 4-\lambda_b\cos\theta_b$ is at least $4(1-\cos\theta_b)$ where $\cos\theta_b\ge 0$ and at least 4 where $\cos\theta_b<0$; so $1-\cos\theta_b\le\alpha_b$. If $\lambda_b<1$, then $X_b<1$ and $\alpha_b>3$; so $\mathbb{P}_K(\lambda_b<1)\le\frac13\mathbb{E}_K\alpha_b$. (iii) *The variation of f inside a plaquette.* With x_b the midpoint of b and $f_b = f(x_b)$: $\frac12\operatorname{tr}(U(b)\mathsf{S}_b^f) = f_bX_b + Y_b$, $Y_b = \frac12\operatorname{tr}(U(b)R_b)$, $R_b = \sum_{p\ni b}(f(x(p))-f_b)S_{p,b}$, $\|R_b\|_2\le 2\sqrt2\,a\operatorname{Lip}_\infty f$, since $|x(p)-x_b|_\infty = \frac{a}{2}$. The conditional variance is at most the conditional mean square distance from the function $\frac12\operatorname{tr}(W_b^{-1}R_b)$ of U'; as $U(b)-W_b^{-1} = (V-\mathbf{1})W_b^{-1}$ and $|\frac12\operatorname{tr}(PQ)|\le\frac12\|P\|_2\|Q\|_2$, $\operatorname{Var}(Y_b\mid U')\le 8a^2(\operatorname{Lip}_\infty f)^2\,\mathbb{E}(1-\cos\theta_b\mid U')$. With $\operatorname{Var}(A'+B')\ge(1-\epsilon)\operatorname{Var}A' - \epsilon^{-1}\operatorname{Var}B'$ for $0<\epsilon<1$: $\operatorname{Var}(f_bX_b+Y_b\mid U')\ge(1-\epsilon)f_b^2\beta^{-2}v(\beta\lambda_b) - 8\epsilon^{-1}a^2(\operatorname{Lip}_\infty f)^2\,\mathbb{E}(1-\cos\theta_b\mid U')$. (iv) *A bound on the mean action.* The function $\beta'\mapsto\log\int e^{-\beta' A}\,dU$ is convex, vanishes at 0 and has the derivative $-\langle A\rangle_{\beta'}$, so $\beta\langle A\rangle_\beta\le -\log\int e^{-\beta A}\,dU$. On the set where $\|U(b)-\mathbf{1}\|_2\le\beta^{-1/2}$ for all $3N_K$ bonds, every $a_p\le 4/\beta$, so $A\le 12N_K/\beta$; and by Weyl's formula, $\sin\theta\ge 2\theta/\pi$ and $1-\cos\theta\le\theta^2/2$, each factor of this set has the Haar measure at least $\frac{4}{3\sqrt2\pi^3}\beta^{-3/2}\ge e^{-3.5}\beta^{-3/2}$. Hence $\beta\langle A\rangle_\beta\le N_K(22.5+\frac92\log\beta)$, and $\eta_K = a^3\mathbb{E}_KA\le\ell^3\frac{a}{\mathsf{b}}(22.5+\frac92\log(L^K\mathsf{b}))\to 0$. (v) *Conclusion.* Given ϵ, let t_ϵ be such that $v\ge\frac32-\epsilon$ on $[t_\epsilon,\infty)$. For $\beta\ge t_\epsilon$, using $v\ge 0$, $a\beta^{-2} = a^3\mathsf{b}^{-2}$ and (ii)–(iv),

$$a^3\operatorname{Var}_K\mathcal{O}(f)\ge(1-\epsilon)(\tfrac32-\epsilon)\mathsf{b}^{-2}\Big[a^3\sum_{b\in\mathcal{B}}f_b^2 - \tfrac13\sup f^2\,\eta_K\Big] - 8\epsilon^{-1}(\operatorname{Lip}_\infty f)^2\,\eta_K,$$

and $|2a^3\sum_{b\in\mathcal{B}}f_b^2 - \int f^2|\le\ell^3a\operatorname{Lip}_\infty(f^2)$, the cells made of the two cubes of side a about x_b and x_b+ae_1 tiling T_ℓ. Let $K\to\infty$ and then $\epsilon\to 0$. At $f=1$, $a^3\operatorname{Var}_K\mathcal{O}(1) = \ell^3\varsigma_K^2$.

Step 8: clause (f). By (a) the numbers ς_K^2 are bounded, so every infinite set of cutoffs holds a $\mathcal{K}$ along which they converge; the limit lies in the stated interval by (a) and (e). Along $\mathcal{K}$, by (d), $\Sigma_K(t)\to\varsigma_0^2\,t\cdot Gt$ uniformly for t in compact sets; by Cor. 2.8(c), $\mu_d^K(F)\to\mu_d(F)$; with (c), $\Psi_K(t) = \mathbb{E}_K[F(V_d)e^{t\cdot Y_K}]\to\mu_d(F)e^{\frac12\varsigma_0^2t\cdot Gt}$ uniformly on the compact subsets of $\mathbb{C}^J$. The functions Ψ_K are entire, the variables being bounded at each cutoff, and Cauchy's estimate, applied to Ψ_K minus its limit on a polydisc, carries this to all the mixed moments $\mathbb{E}_K[F(V_d)\prod_i Y_K(f_i)^{\nu_i}]$; this is the case of polynomials. For $\psi\in C_c^\infty(\mathbb{R}^J)$, Fourier inversion and Fubini's theorem give $\mathbb{E}_K[F(V_d)\psi(Y_K)] = \int\hat\psi(\tau)\Psi_K(i\tau)\,d\tau$, with $\hat\psi$ integrable and the normalisation of the Fourier transform for which this is the inversion formula, and dominated convergence, with $|\Psi_K(i\tau)|\le\sup|F|$, gives the limit $\mu_d(F)\int\psi\,dN(0,\varsigma_0^2G)$. For a bounded continuous ϕ, Chebyshev's inequality with the second moments bounded by (a) cuts off the large values, and on a ball ϕ is a uniform limit of such ψ; for a bounded continuous function of the pair, its uniform continuity on the compact set $\mathsf{V}_d\times\{|y|\le\mathsf{R}\}$ reduces it to finite sums of products of indicators of Borel sets of V_d and continuous functions of y. (f2): by (e) there is $K_\circ$ with $\varsigma_K^2\ge\frac12\mathsf{b}^{-2}$ for $K\ge K_\circ$; (c) read at t/ς_K and (d) divided by ς_K^2 give the same convergence of the generating functions along the full sequence, with 1 in the place of ς_0^2, and the rest of the argument is unchanged.

Step 9: clause (g). Suppose that the laws of $\mathcal{O}(f)-c_K$ are tight along an infinite set of cutoffs; pass to a subset $\mathcal{K}$ as in (f1), and put $\sigma^2 = \varsigma_0^2\int f^2$, which is positive by (e) since f is continuous and not 0. Then $Y_K(f) = a_K^{3/2}(\mathcal{O}(f)-c_K) + s_K$ with real numbers s_K, and the first summand

tends to 0 in probability; so $|\mathbb{E}_K e^{i\tau Y_K(f)}| \to 1$, while by (f1) at $d=0$, $F=1$, the mean tends to $e^{-\frac{1}{2}\sigma^2\tau^2} < 1$ for $\tau \neq 0$. □

Definition 18.146 (Kernels; the laws of two neighbouring depths; composed kernels; the two symmetries)**.** Let W and Y be compact metrisable spaces; finite and countable products of copies of SU(2), including $\mathsf{V}_d = \mathcal{V}(\Lambda_{d,\ell})$ and $\mathsf{V} = \prod_{d\geq 0} \mathsf{V}_d$, are such spaces.

(1) A *kernel* from W to Y is a function $\kappa(W, E)$ of the points $W \in \mathsf{W}$ and of the Borel sets $E \subset \mathsf{Y}$ such that $\kappa(W, \cdot)$ is a Borel probability measure for every W and $\kappa(\cdot, E)$ is a Borel function for every E. For a bounded Borel function F on $\mathsf{W}\times\mathsf{Y}$ we put $(\kappa F)(W) = \int F(W,V)\,\kappa(W, dV)$; for a Borel probability measure μ on W, $\mu\otimes\kappa$ is the measure $B \mapsto \int \kappa(W, B_W)\,\mu(dW)$ on $\mathsf{W}\times\mathsf{Y}$, where $B_W = \{V : (W,V) \in B\}$. If P is a Borel probability measure on $\mathsf{W}\times\mathsf{Y}$ with the first marginal μ, a *kernel of* P is a kernel κ with $P = \mu\otimes\kappa$: it gives the distribution of the second coordinate when the first one is known. For two Borel probability measures on one space, $\|\lambda - \lambda'\| = \sup_E |\lambda(E) - \lambda'(E)|$, the supremum over the Borel sets.

(2) Let $m \in \mathbb{N}$, $\ell = 2L^m$, $\mathsf{b} > 0$ and $d \geq 0$. The averaging map $\mathcal{M}\colon \mathsf{V}_{d+1} \to \mathsf{V}_d$ is the Borel block map of Prop. YM3_I:4.5; the cutoff and the level enter its definition only through the two lattices $\mathbb{T}^{(k)} = \Lambda_{d+1,\ell}$ and $\mathbb{T}^{(k+1)} = \Lambda_{d,\ell}$, so it is one map at all the cutoffs $K \geq d+1$, and $\mathcal{M}^{K-d} = \mathcal{M}\circ\mathcal{M}^{K-d-1}$. Put $\mathrm{gr}_d(V) = (\mathcal{M}V, V)$; for $K \geq d+1$ let $P_d^K = (\mathrm{gr}_d)_*\mu_{d+1}^K$, the law under $\mathbb{E}_{K,\mathsf{b}}$ of the pair (V_d, V_{d+1}) of the block fields, and let $P_d = (\mathrm{gr}_d)_*\mu_{d+1}$, with the limit laws of Proposition 18.144(4). Their first marginals are $\mathcal{M}_*\mu_{d+1}^K = \mu_d^K$ and $\mathcal{M}_*\mu_{d+1} = \mu_d$. Where the side and the label must be shown we write $\mu_d[\ell, \mathsf{b}]$, $P_d[\ell, \mathsf{b}]$ and $\kappa_d[\ell, \mathsf{b}]$.

(3) For $0 \leq d < D$ and kernels κ_j from V_j to V_{j+1}, $d \leq j < D$, the *composed kernel* is

$$\kappa_{d\to D}(W, E) = \int \kappa_d(W, dV_{d+1}) \int \kappa_{d+1}(V_{d+1}, dV_{d+2}) \cdots \int \kappa_{D-2}(V_{D-2}, dV_{D-1})\,\kappa_{D-1}(V_{D-1}, E).$$

On V, $\mathcal{F}_d$ is the σ-algebra generated by $(V_0, \dots, V_d)$, and $\iota_{d,D}(V) = (\mathcal{M}^{D-d}V, \dots, \mathcal{M}V, V)$ for $V \in \mathsf{V}_D$.

(4) A gauge transformation v of $\Lambda_{d+1,\ell}$, a map from its sites to SU(2), acts on V_{d+1} by $V \mapsto V^v$, $V^v(b) = v(b_-)V(b)v(b_+)^{-1}$, and its restriction $v|_d$ to the sites of $\Lambda_{d,\ell}$ acts on V_d; the set $\mathcal{G}_{d+1}$ of all these maps is the compact group of these transformations, with its Haar probability measure dv, and $(V^u)^v = V^{vu}$. Iso_d is the finite group induced on T_ℓ by translations through $a_d\mathbb{Z}^3$ and signed coordinate permutations, acting on fields by $(\mathsf{r}V)(b) = V(\mathsf{r}^{-1}b)$ and on gauge transformations by $(\mathsf{r}u)(x) = u(\mathsf{r}^{-1}x)$; $\mathsf{Iso}_d \subset \mathsf{Iso}_{d+1}$ since $a_d\mathbb{Z}^3 \subset a_{d+1}\mathbb{Z}^3$. For a set E of fields, E^v and $\mathsf{r}E$ are its images.

■

Proposition 18.147 (The kernels from one depth to the next at $K=\infty$; the sequence of the block fields as a Markov chain indexed by the depth)**.** Use the block laws of Proposition 18.144(4). Let $m \in \mathbb{N}$, $\ell = 2L^m$, $\mathsf{b} > 0$, and $d \geq 0$ be an integer. The laws μ_d satisfy $\mathcal{M}_*\mu_{d+1} = \mu_d$ since the identity holds at every cutoff and the limits are in total variation. Define the thread measure $\mu_{\ell,\mathsf{b}}$ on V by its finite laws $(\iota_{0,D})_*\mu_D$. These laws are consistent by the preceding identity. Kolmogorov's extension theorem gives a unique Borel probability measure with those finite laws, and the countably many identities $V_j = \mathcal{M}V_{j+1}$ then hold simultaneously almost surely.

(a) *Existence and uniqueness.* (a1) At each cutoff $K \geq d+1$ the pair law P_d^K has a kernel κ_d^K, so that $\mathbb{E}_{K,\mathsf{b}}F(V_d, V_{d+1}) = \int \mu_d^K(dW) \int \kappa_d^K(W, dV)\,F(W,V)$ whenever F is bounded and Borel on $\mathsf{V}_d \times \mathsf{V}_{d+1}$. The limit pair law P_d, which is the joint law of V_d and V_{d+1} under $\mu_{\ell,\mathsf{b}}$, has a kernel κ_d. A kernel of P_d^K is determined as a measure-valued function outside a Borel μ_d^K-null set, and a kernel of P_d outside a Borel μ_d-null set. (a2) The measure $\kappa_d(W, \cdot)$ is carried by the fibre $\mathcal{M}^{-1}\{W\}$ for all W outside a μ_d-null set, and likewise at every cutoff.

(b) *Convergence, for any kernels.* Whatever kernels κ_d^K ($K \ge d+1$) and κ_d as in (a1) are chosen, the function $\Delta_K(W) = \|\kappa_d^K(W,\cdot) - \kappa_d(W,\cdot)\|$ is Borel on V_d, takes values in $[0,1]$, and satisfies

$$\int \Delta_K \, d\mu_d = \|\mu_d \otimes \kappa_d^K - \mu_d \otimes \kappa_d\| \le \|\mu_d - \mu_d^K\| + \|\mu_{d+1}^K - \mu_{d+1}\| \le 2\,\|\mu_{d+1}^K - \mu_{d+1}\| \xrightarrow[K\to\infty]{} 0$$

as K runs through all the integers. Replacing μ_d by μ_d^K, both as the measure against which Δ_K is integrated and as the first factor of the two products, keeps the chain valid. In the other direction, $\|\mu_{d+1}^K - \mu_{d+1}\| \le \|\mu_d^K - \mu_d\| + \int \Delta_K \, d\mu_d$. When $\mathsf{b} \ge \gamma^{-2}$ and $M\,R(\mathsf{b}) \le L^m$, every $K \ge K_{\rm bf}$ with $\lceil K/1260 \rceil \ge d+1$ has

$$2\,\|\mu_{d+1}^K - \mu_{d+1}\| \le 2\,\mathsf{C}_{\rm bf}\,\ell^3 \sum_{K' \ge K} K'^{\mathsf{p}} L^{-K'/10} \le 2\,\mathsf{C}_{\rm bf}\,\Sigma_0\,\ell^3 K^{\mathsf{p}} L^{-K/10},$$

where $\mathsf{C}_{\rm bf}$, $K_{\rm bf}$ and $\mathsf{p} = 2\mathfrak{p}_0 + \frac12$ are the numbers of Def. 2.7(5) at the interval $\{\mathsf{b}\}$ and $\Sigma_0 = \sum_{j\ge0}(1+j)^{\mathsf{p}} L^{-j/10}$. (b2) Outside some Borel μ_d-null set $\mathsf{N} \subset \mathsf{V}_d$ the series $\sum_K L^{\theta K}\Delta_K(W)$ converges for every real $\theta < \frac{1}{10}$; thus at every $W \notin \mathsf{N}$, $\Delta_K(W) = o(L^{-\theta K})$ for each such θ, with no subsequence taken. (b3) For $D > d$ the composed kernels give the conditional law of V_D given V_d: $\kappa_{d\to D}^K$, built from the κ_j^K, at each cutoff $K \ge D$, and $\kappa_{d\to D}$, built from the κ_j, under $\mu_{\ell,\mathsf{b}}$; and $\int \|\kappa_{d\to D}^K(W,\cdot) - \kappa_{d\to D}(W,\cdot)\|\,\mu_d(dW) \le \|\mu_d - \mu_d^K\| + \|\mu_D^K - \mu_D\| \le 2\|\mu_D^K - \mu_D\|$.

(c) *The Markov chain.* Let κ_j be a kernel of P_j for every $j \ge 0$. (c1) Whenever $0 \le d \le D$ and F is a bounded Borel function of $(V_d,\dots,V_D)$,

$$\int F(V_d,\dots,V_D)\,d\mu_{\ell,\mathsf{b}} = \int \mu_d(dV_d) \int \kappa_d(V_d,dV_{d+1}) \cdots \int \kappa_{D-1}(V_{D-1},dV_D)\,F(V_d,\dots,V_D).$$

(c2) For $D > d$ and a bounded Borel function H on $\mathsf{V}_{d+1}\times\cdots\times\mathsf{V}_D$, the function $h(W) = \int \kappa_d(W,dV_{d+1}) \cdots \int \kappa_{D-1}(V_{D-1},dV_D)\,H$ is a bounded Borel function on V_d, and $h(V_d)$, which reads V_d alone, serves under $\mu_{\ell,\mathsf{b}}$ as the conditional expectation of $H(V_{d+1},\dots,V_D)$ with respect to $\mathcal{F}_d$, the σ-algebra of all the depths up to d. (c3) $\mu_{\ell,\mathsf{b}}$ is the only Borel probability measure on V for which (c1) holds at $d=0$ for every D: it is the Markov chain that starts with the law μ_0 and moves from the depth j to the depth $j+1$ by κ_j. (c4) Likewise at a fixed cutoff K: under $\mathbb{E}_{K,\mathsf{b}}$ the sequence $(V_0,\dots,V_K) = (\mathcal{M}^K U,\dots,\mathcal{M}U,U)$ satisfies (c1)–(c3), with the laws μ_d^K and kernels κ_d^K of P_d^K, $0 \le d \le D \le K$. Read towards smaller depths, each chain has no randomness: $V_d = \mathcal{M}V_{d+1}$ almost surely.

(d) *Covariance.* (d1) Let κ_d be a kernel of P_d, v a gauge transformation of $\Lambda_{d+1,\ell}$ and $\mathsf{r} \in \mathsf{Iso}_d$. Outside a Borel μ_d-null set of W, the identities $\kappa_d(W^{v|_d}, E^v) = \kappa_d(W,E)$ and $\kappa_d(\mathsf{r}W, \mathsf{r}E) = \kappa_d(W,E)$ are true for every Borel set $E \subset \mathsf{V}_{d+1}$. (d2) Some kernel of P_d satisfies these identities with no exceptional set: for every $W \in \mathsf{V}_d$, every v, every $\mathsf{r} \in \mathsf{Iso}_d$ and every E. Both statements carry over to the cutoffs $K \ge d+1$, with P_d^K, μ_d^K and κ_d^K.

(e) *Every label, every side; scaling.* (e1) Let $s \ge 0$ be an integer with $L^s\mathsf{b} \ge \gamma^{-2}$ and $M\,R(L^s\mathsf{b}) \le L^{m+s}$; every sufficiently large integer qualifies. Then each $K \ge d+1$ with $K - s \ge K'_{\rm bf}$ and $\lceil (K-s)/1260 \rceil \ge \max\{d+1-s, 0\}$ has

$$\int \Delta_K \, d\mu_d \le 2\,\mathsf{C}'_{\rm bf}\,\Sigma_0\,(L^s\ell)^3 (K-s)^{\mathsf{p}} L^{-(K-s)/10},$$

with $\mathsf{C}'_{\rm bf}$ and $K'_{\rm bf}$ taken from Def. 2.7(5) on the side $L^s\ell$ at the interval $\{L^s\mathsf{b}\}$; in particular (b2) is true at every label and on every side. (e2) Let $0 \le s \le d$ be integers and the kernels arbitrary, and let D_* be the relabelling of Proposition 18.144, Step 7, for the dilation by L^s. Outside a $\mu_d[\ell,\mathsf{b}]$-null set of W,

$$\kappa_d[\ell,\mathsf{b}](W,E) = \kappa_{d-s}[L^s\ell, L^s\mathsf{b}](\mathsf{D}_*W, \mathsf{D}_*E) \qquad \text{for every Borel set } E \subset \mathsf{V}_{d+1}.$$

For $s = d$ this says that each step of the chain, from the depth d to $d+1$, is, once relabelled, the first step, from the depth 0 to 1, of the chain on the side $L^d\ell$ at the label $L^d\mathsf{b}$.

■

Sketch. Step 1: kernels exist and are unique. Let P be a Borel probability measure on $\mathsf{W}\times\mathsf{Y}$ with the first marginal μ. The space $C_{\mathbb{R}}(\mathsf{Y})$ is separable; let $\mathcal{Q}$ be a countable dense subspace over $\mathbb{Q}$ that holds 1. For $g\in\mathcal{Q}$, the real measure $A\mapsto\int \mathbf{1}_A(W)g(V)\,dP$ on W is bounded by $\|g\|_\infty\mu$; let h_g be a real Radon–Nikodym density of it with respect to μ. Each of the countably many relations $h_{qg+q'g'}=qh_g+q'h_{g'}$, $|h_g|\le\|g\|_\infty$, $h_1=1$ ($q,q'\in\mathbb{Q}$) says of an integrable function that its integral over every Borel set is 0, respectively ≥ 0, so it holds off a Borel μ-null set; let N be the union of these sets. For $W\notin\mathsf{N}$ the map $g\mapsto h_g(W)$ is linear over $\mathbb{Q}$, bounded by the sup norm and equal to 1 at 1; it extends to a positive linear form on $C_{\mathbb{R}}(\mathsf{Y})$, the integral of a Borel probability measure $\kappa(W,\cdot)$ by the theorem of Riesz; for $W\in\mathsf{N}$ let $\kappa(W,\cdot)$ be the second marginal of P. The bounded Borel functions g on Y for which $W\mapsto\int g\,d\kappa(W,\cdot)$ is a Borel function and $\int\mathbf{1}_A(W)g(V)\,dP=\int_A(\int g\,d\kappa(W,\cdot))\mu(dW)$ for all Borel A form a vector space that holds $\mathcal{Q}$ and is closed under bounded pointwise limits, so it holds all of them; at $g=\mathbf{1}_E$ this is the measurability in W and $P(A\times E)=(\mu\otimes\kappa)(A\times E)$, and the products of Borel sets generate. The same argument shows that κF is a bounded Borel function for every bounded Borel F on $\mathsf{W}\times\mathsf{Y}$ and that $\int F\,d(\mu\otimes\kappa)=\int\kappa F\,d\mu$. Two kernels of P give the same integral to each $g\in\mathcal{Q}$ off a null set, hence, $\mathcal{Q}$ being dense, the same measure there.

Step 2: the distance of two kernels. For a Borel function h with values in $[0,1]$, $|\int h\,d\lambda-\int h\,d\lambda'|\le\|\lambda-\lambda'\|$, by the step functions $\frac1n\sum_{i\le n}\mathbf{1}_{\{h>i/n\}}$, which lie between $h-\frac1n$ and h. Let $g_1=0,g_2,\dots$ number the functions $\min\{\max\{g,0\},1\}$, $g\in\mathcal{Q}$. Then $\|\lambda-\lambda'\|=\sup_i(\int g_i\,d\lambda-\int g_i\,d\lambda')$ for all Borel probability measures λ,λ' on Y: for a Borel set E, approximate $\mathbf{1}_E$ in $L^1(\lambda+\lambda')$ by a continuous function, cut it off at 0 and 1, and approximate that uniformly by some g_i. Hence $\Delta(W)=\|\kappa(W,\cdot)-\kappa'(W,\cdot)\|=\sup_i(\kappa-\kappa')(1\otimes g_i)(W)$ is a Borel function. Let $A_{n,i}$, $i\le n$, be the Borel sets on which i is the first index $\le n$ at which $(\kappa-\kappa')(1\otimes g_i)$ is largest, and $F_n=\sum_{i\le n}\mathbf{1}_{A_{n,i}}\otimes g_i$, with values in $[0,1]$. By Step 1 and the first remark on $\mathsf{W}\times\mathsf{Y}$, $\int\max_{i\le n}(\kappa-\kappa')(1\otimes g_i)\,d\mu=\int F_n\,d(\mu\otimes\kappa)-\int F_n\,d(\mu\otimes\kappa')\le\|\mu\otimes\kappa-\mu\otimes\kappa'\|$, and monotone convergence gives $\int\Delta\,d\mu\le\|\mu\otimes\kappa-\mu\otimes\kappa'\|$; the reverse inequality is $|\int[\kappa(W,B_W)-\kappa'(W,B_W)]\,\mu(dW)|\le\int\Delta\,d\mu$.

Step 3: clause (b). Images under Borel maps do not raise $\|\cdot\|$, and $P_d^K(\mathsf{V}_d\times E)-P_d(\mathsf{V}_d\times E)=\mu_{d+1}^K(E)-\mu_{d+1}(E)$; so $\|P_d^K-P_d\|=\|\mu_{d+1}^K-\mu_{d+1}\|=:\epsilon_K$, and $\|\mu_d^K-\mu_d\|\le\epsilon_K$. For a Borel set $B\subset\mathsf{V}_d\times\mathsf{V}_{d+1}$ put $h=\kappa_d^K\mathbf{1}_B$, with values in $[0,1]$; as $\mu_d^K\otimes\kappa_d^K=P_d^K$ and $\mu_d\otimes\kappa_d=P_d$,

$$(\mu_d\otimes\kappa_d^K)(B)-(\mu_d\otimes\kappa_d)(B)=\Big[\int h\,d\mu_d-\int h\,d\mu_d^K\Big]+[P_d^K(B)-P_d(B)],$$

of modulus at most $\|\mu_d-\mu_d^K\|+\epsilon_K$; this uses no property of κ_d^K besides $\mu_d^K\otimes\kappa_d^K=P_d^K$, so its values on a μ_d^K-null set do not matter. With Step 2 this is the chain of (b). With $h'=\kappa_d\mathbf{1}_B$ the same computation gives the variant with μ_d^K, and $P_d^K(B)-P_d(B)=[\int h\,d\mu_d^K-\int h\,d\mu_d]+[(\mu_d\otimes\kappa_d^K)(B)-(\mu_d\otimes\kappa_d)(B)]$ gives the converse. The convergence $\epsilon_K\to0$ is Proposition 18.144(4). For the rate, $\{\mathsf{b}\}$ is an interval of labels under (Tor) on the side ℓ in §2.3.1, and Cor. 2.8 at the depth $d+1$ gives $\epsilon_K\le\mathsf{C}_{\mathrm{bf}}\ell^3\sum_{K'\ge K}K'^{\mathsf{p}}L^{-K'/10}$; as $K+j\le K(1+j)$, the sum is at most $K^{\mathsf{p}}L^{-K/10}\Sigma_0$, and $\Sigma_0<\infty$ because $(1+j)^{\mathsf{p}}L^{-j/20}$ is bounded.

Step 4: (b2). Let $\theta_n=\frac{1}{10}-\frac1n$, $n>10$. By the rate, $\sum_K L^{\theta_nK}\int\Delta_K\,d\mu_d\le2\sum_K L^{\theta_nK}\epsilon_K<\infty$, the finitely many K below the range adding finite terms; by monotone convergence $\int\sum_K L^{\theta_nK}\Delta_K\,d\mu_d<\infty$, so the series converges off a Borel null set N_n. Put $\mathsf{N}=\bigcup_n\mathsf{N}_n$; for $\theta<\frac{1}{10}$ take $\theta_n\ge\theta$. At a general label and side the same argument runs with (e1).

Step 5: clause (a). Step 1 gives the kernels and their uniqueness. Under $\mu_{\ell,\mathsf{b}}$ the law of $(V_0,\dots,V_{d+1})$ is the image of μ_{d+1} under $\iota_{0,d+1}$, so that of (V_d,V_{d+1}) is the image of μ_{d+1} under gr_d, which is P_d. (a2): the set $\mathsf{G}=\{(W,V):W=\mathcal{M}V\}$ is Borel, $\mathcal{M}$ being Borel and the diagonal closed;

its sections are the fibres $\mathcal{M}^{-1}\{W\}$, and $\mathrm{gr}_d^{-1}\mathsf{G} = \mathsf{V}_{d+1}$. So $1 = P_d(\mathsf{G}) = \int \kappa_d(W, \mathcal{M}^{-1}\{W\})\,\mu_d(dW)$, and the Borel function $1 - \kappa_d(W, \mathcal{M}^{-1}\{W\}) \geq 0$, whose integral is 0, vanishes almost everywhere; the same at the cutoff.

Step 6: clause (c) and (b3). As in Step 5, the law of $(V_d, \dots, V_D)$ under $\mu_{\ell,\mathsf{b}}$ is $(\iota_{d,D})_*\mu_D$. We show $\int F \circ \iota_{d,D}\, d\mu_D = \int \mu_d(dV_d) \int \kappa_d(V_d, dV_{d+1}) \cdots F$ by induction on D. As $\iota_{d,D+1}(V) = (\iota_{d,D}(\mathcal{M}V), V)$ and $P_D = \mu_D \otimes \kappa_D$, $\int F \circ \iota_{d,D+1}\, d\mu_{D+1} = \int \mu_D(dW) \int \kappa_D(W, dV)\, F(\iota_{d,D}(W), V)$; the inner integral is a bounded Borel function of $\iota_{d,D}(W)$, whose last entry is W, and the hypothesis of the induction applies to it. This is (c1). (c2): for $F = \Phi(V_0, \dots, V_d)H(V_{d+1}, \dots, V_D)$ the factor Φ is not integrated in the inner integrals of (c1) from the depth 0, so the iterated integral is that of $\Phi(V_0, \dots, V_d)h(V_d)$, and (c1) on the depths $0, \dots, d$ gives $\int \Phi H\, d\mu_{\ell,\mathsf{b}} = \int \Phi\, h(V_d)\, d\mu_{\ell,\mathsf{b}}$ for all bounded Borel Φ, in particular for the indicators of the sets of $\mathcal{F}_d$. (c3): a measure for which (c1) holds at $d = 0$ for all D has the images $(\iota_{0,D})_*\mu_D$ under $V \mapsto (V_0, \dots, V_D)$, so it is $\mu_{\ell,\mathsf{b}}$ since its finite laws determine it on the product Borel σ-algebra. (c4): the same induction at the cutoff K, with μ_K^K the Wilson measure and $\mathcal{M}_*\mu_{j+1}^K = \mu_j^K$. The last sentence: the threads have full measure, and at a cutoff it is the definition of the block fields. (b3): (c1) for $F = \mathbf{1}_A(V_d)\mathbf{1}_E(V_D)$ gives $\mu_D(\{V \in E : \mathcal{M}^{D-d}V \in A\}) = \int_A \kappa_{d\to D}(W, E)\,\mu_d(dW)$; $\kappa_{d\to D}$ is a kernel, measurable by Step 1 and σ-additive by monotone convergence; the same at the cutoff, and Step 3 with $\mathcal{M}^{D-d}$ in the place of $\mathcal{M}$ gives the bound.

Step 7: clause (d). (d1), gauge: $\mathcal{M}(V^v) = \mathcal{M}(V)^{v|_d}$ (Prop. YM3_I:4.5); $V \mapsto V^v$ and $W \mapsto W^{v|_d}$ are homeomorphisms, and μ_{d+1} is gauge invariant by covariance and invariance of the cutoff laws, followed by total variation convergence, so $\mu_d = \mathcal{M}_*\mu_{d+1}$ is invariant under $W \mapsto W^{v|_d}$. Put $\kappa^v(W, E) = \kappa_d(W^{v|_d}, E^v)$, a kernel (its measurability by Step 1, the map $W \mapsto W^{v|_d}$ being continuous). For Borel sets A, E,

$$\begin{aligned}(\mu_d \otimes \kappa^v)(A \times E) &= \int \mathbf{1}_{A^{v|_d}}(W')\,\kappa_d(W', E^v)\,\mu_d(dW') \\ &= \mu_{d+1}(\{V \in E^v : \mathcal{M}V \in A^{v|_d}\}) \\ &= \mu_{d+1}(\{V \in E : \mathcal{M}V \in A\}^v) = P_d(A \times E),\end{aligned}$$

so κ^v is a kernel of P_d, and Step 1 gives (d1). Isometries: $\mathsf{r} \in \mathsf{Iso}_d \subset \mathsf{Iso}_{d+1}$ acts on V_d and V_{d+1} by homeomorphisms, and $\mathcal{M}(\mathsf{r}V) = \mathsf{r}\mathcal{M}(V)$ ([7, Lem. G.1]). The Wilson measure at the cutoff K is invariant under r, which is induced by an isometry of $a_K\mathbb{Z}^3$; with $\mathcal{M}^{K-d-1}(\mathsf{r}U) = \mathsf{r}\mathcal{M}^{K-d-1}(U)$, $\mathsf{r}_*\mu_{d+1}^K = \mu_{d+1}^K$, and in the limit in total variation $\mathsf{r}_*\mu_{d+1} = \mu_{d+1}$. The computation above, with r in the place of v, gives the second equality. (d2): put $\bar\kappa(W, E) = \int_{\mathcal{G}_{d+1}} \kappa_d(W^{v|_d}, E^v)\,dv$. The integrand is $\int \kappa_d(W^{v|_d}, dV)\,\mathbf{1}_E(V^{v^{-1}})$, a bounded Borel function of (v, W) by Step 1, the maps $(v, W) \mapsto W^{v|_d}$ and $(v, V) \mapsto V^{v^{-1}}$ being continuous; so $\bar\kappa(\cdot, E)$ is Borel by Fubini's theorem, and $\bar\kappa(W, \cdot)$ is a probability measure by monotone convergence. By Fubini's theorem and (d1), $\mu_d \otimes \bar\kappa = \int dv\,(\mu_d \otimes \kappa^v) = P_d$. For every gauge transformation u and every W, $\bar\kappa(W^{u|_d}, E^u) = \int \kappa_d(W^{(vu)|_d}, E^{vu})\,dv = \bar\kappa(W, E)$, dv being invariant under $v \mapsto vu$. Next put $\bar{\bar\kappa}(W, E) = |\mathsf{Iso}_d|^{-1} \sum_{\mathsf{r}\in\mathsf{Iso}_d} \bar\kappa(\mathsf{r}W, \mathsf{r}E)$; each term is a kernel of P_d by the isometry part of (d1), so the mean is one; it is invariant under Iso_d at every W, the group acting by $\mathsf{r}(\mathsf{r}'V) = (\mathsf{r}\mathsf{r}')V$; and it keeps the gauge covariance at every W, because $\mathsf{r}(V^u) = (\mathsf{r}V)^{\mathsf{r}u}$ by the formula for the gauge action on an oriented bond and $(\mathsf{r}u)|_d = \mathsf{r}(u|_d)$, r carrying $\Lambda_{d,\ell}$ onto itself. At the cutoff K the same argument runs with μ_{d+1}^K, which is gauge invariant as the image of the gauge-invariant Wilson measure under the gauge-covariant map $\mathcal{M}^{K-d-1}$.

Step 8: clause (e). The relabelling D_* of Proposition 18.144, Step 7, for the dilation by L^s renames the factors of a finite power of SU(2), so it carries the Haar measure to the Haar measure; $a_{\mathsf{D}p}(\mathsf{D}_*U) = a_p(U)$; it commutes with $\mathcal{M}$; and $L^K\mathsf{b} = L^{K-s}(L^s\mathsf{b})$. Hence $(\mathsf{D}_*)_*\mu_j^K[\ell, \mathsf{b}] = \mu_{j-s}^{K-s}[L^s\ell, L^s\mathsf{b}]$ for $s \leq j \leq K$ and $K > s$, and D_*, a bijection, keeps $\|\cdot\|$. (e1): if $d + 1 \geq s$, $\|\mu_{d+1}^K[\ell, \mathsf{b}] - \mu_{d+1}[\ell, \mathsf{b}]\| = \|\mu_{d+1-s}^{K-s}[L^s\ell, L^s\mathsf{b}] - \mu_{d+1-s}[L^s\ell, L^s\mathsf{b}]\|$, and Cor. 2.8 on the side $L^s\ell$ at the

interval $\{L^s\mathsf{b}\}$, with the tail sum of Step 3, bounds it; if $d+1<s$, $\mu^K_{d+1}=(\mathcal{M}^{s-d-1})_*\mu^K_s$, the same holds at $K=\infty$ by consistency, and the bound at the depth s applies. With Step 3 this is (e1); that all large integers s have the two properties follows from $R(x)<L(\log x)^r$ for $x>e$ (Thm. YM3_I:AD). (e2): passing to the limit in the identity of the laws, $(\mathsf{D}_*)_*\mu_j[\ell,\mathsf{b}]=\mu_{j-s}[L^s\ell,L^s\mathsf{b}]$ for $j\ge s$. Let κ' be a kernel of $P_{d-s}[L^s\ell,L^s\mathsf{b}]$ and $\kappa''(W,E)=\kappa'(\mathsf{D}_*W,\mathsf{D}_*E)$, a kernel from V_d to V_{d+1} on T_ℓ. For Borel sets A, E, by $\mathcal{M}\mathsf{D}_*=\mathsf{D}_*\mathcal{M}$,

$$\begin{aligned}&(\mu_d[\ell,\mathsf{b}]\otimes\kappa'')(A\times E)\\&\quad=\int_{\mathsf{D}_*A}\kappa'(W',\mathsf{D}_*E)\,\mu_{d-s}[L^s\ell,L^s\mathsf{b}](dW')\\&\quad=\mu_{d+1-s}[L^s\ell,L^s\mathsf{b}](\{V'\in\mathsf{D}_*E:\mathcal{M}V'\in\mathsf{D}_*A\})\\&\quad=\mu_{d+1}[\ell,\mathsf{b}](\{V\in E:\mathcal{M}V\in A\}),\end{aligned}$$

so κ'' is a kernel of $P_d[\ell,\mathsf{b}]$, and Step 1 gives (e2). □

Proposition 18.148 (The covariance of a block function with the action density; a density in every L^p; a martingale in the depth; one half of the torus across a mirror)**.** Work in setting (E) of Definition 18.140(1) at label b and side ℓ. Use the calibration radius ϱ_ς, denoted by ϱ in the proof. Write $\mu_d=\mu_d[\ell,\mathsf{b}]$ and let $\mu_{\ell,\mathsf{b}}$ be the thread law. Use the filtration $\mathcal{F}_d$ of Definition 18.146(3) and the torus limit $S_2^{\ell,\mathsf{b}}$ of Theorem YM3_I:AS for separated pairs. For real Lipschitz f on T_ℓ, put $\|f\|_\sharp=\max\{\sup|f|,40M\operatorname{Lip}f\}$. For $K\ge\max\{d,2\}$ and bounded complex Borel F on V_d define

$$\mathsf{c}_K(F;f)=\mathbb{E}_{K,\mathsf{b}}\Big[F(V_d)\big(\mathcal{O}(f)-\mathbb{E}_{K,\mathsf{b}}\mathcal{O}(f)\big)\Big].$$

For an integer $n\ge1$ let $\alpha_n=\frac{1}{60(3n+4)}$, $\gamma_n=\frac{1}{20}-\alpha_n$, $\theta_n=1-20\alpha_n$ and $u_n(K)=\lceil K/(180(3n+4))\rceil$; so $\gamma_1=\frac{1}{21}$ and $\theta_1=\frac{20}{21}$. Let $c_0=3\mathfrak{p}_0+\frac14$ and $c_1=4\mathfrak{p}_0+\frac12$. Let $\bar K_n=\max\{K_{\rm bf},K_0\}$, where $K_{\rm bf}$ is the integer of Definition 2.7(5) and K_0 the uniform smallness threshold in Theorem YM3_I:AR at n, both at the interval $\{\mathsf{b}\}$, and let $K_*(d,n)$ be the least integer $K\ge\bar K_n$ with $u_n(K)\ge d$; $K_*(d)=K_*(d,1)$.

(a1) *The limit.* At each depth $d\ge0$ a unique finite signed Borel measure $\xi_{d,f}$ on V_d satisfies

$$\Big|\mathsf{c}_K(F;f)-\int F\,d\xi_{d,f}\Big|\le\mathsf{C}_1\,K^{c_0}L^{-K/21}\sup|F|\,\|f\|_\sharp$$

for all integers $K\ge K_*(d)$ and all bounded complex Borel F on V_d, whether gauge invariant or not; here C_1 depends on the choice, ℓ and b and on nothing else, in particular not on d, K, F or f. Moreover $\xi_{d,f}(\mathsf{V}_d)=0$, $\xi_{d,f}$ is gauge invariant and depends real linearly on f, and $\mathcal{M}_*\xi_{d+1,f}=\xi_{d,f}$.

(a2) *The density.* Given $d\ge0$, an integer $n\ge1$ and a real $\eta\in(0,\theta_n)$, one has $|\xi_{d,f}(E)|\le\mathsf{C}\|f\|_\sharp\,\mu_d(E)^{\theta_n-\eta}$ for all Borel sets $E\subset\mathsf{V}_d$, with a constant $\mathsf{C}=\mathsf{C}(d,n,\eta)$ depending only on the choice, ℓ, b, d, n and η. Hence $\xi_{d,f}$ has a density $X_d(f)$ with respect to μ_d, real and determined up to a μ_d-null set, with $\int X_d(f)\,d\mu_d=0$ and $X_d(f)\in L^p(\mu_d)$ for every $p<\infty$, $\|X_d(f)\|_{L^p(\mu_d)}\le\mathsf{C}'(d,p)\|f\|_\sharp$; and $\mathsf{c}_K(F;f)\to\int F\,X_d(f)\,d\mu_d$ for every bounded Borel F.

(a3) *The martingale.* Under $\mu_{\ell,\mathsf{b}}$ the sequence $(X_d(f)(V_d))_{d\ge0}$ is a martingale with respect to $(\mathcal{F}_d)_{d\ge0}$; its mean is 0, and each of its terms lies in $L^p(\mu_{\ell,\mathsf{b}})$ for all $p<\infty$.

(a4) *The growth in the depth.* Let $n\ge1$ be an integer and $1\le p<3(3n+4)$. Then $\|X_d(f)\|_{L^p(\mu_d)}\le\mathsf{C}_4\,(1+d)^{c_1}L^{3d}\,\|f\|_\sharp$ at every depth $d\ge0$, where C_4 depends on the choice, ℓ, b, n and p, but not on d.

(b) *Across a mirror.* Let $d_0\ge0$ be an integer, $\mu\in\{1,2,3\}$, $c\in\frac{a_{d_0}}{2}\mathbb{Z}$, θ the map of T_ℓ induced by $x\mapsto x-2(x_\mu-c)e_\mu$, $\mathbb{E}_+=\{c<x_\mu<c+\frac{\ell}{2}\}$ and $\mathbb{E}_-=\theta\mathbb{E}_+$, and let $f\ne0$ with $\operatorname{supp}f\subset\mathbb{E}_+$. For $D\ge d_0$ let $\mathcal{G}_D$ be the σ-algebra on V generated by the block variables $V_d(c')$, $d_0\le d\le D$, c' a bond of $\Lambda_{d,\ell}$ with both end points in $\mathbb{E}_-$; $\mathcal{G}_\infty=\sigma(\bigcup_D\mathcal{G}_D)$; and

$\Psi_D = \mathbb{E}_{\mu_{\ell,\mathsf{b}}}[X_D(f)(V_D) \mid \mathcal{G}_D]$. A bounded $\mathcal{G}_D$-measurable function is $H = h\big((V_d(c'))_{(d,c')}\big)$ with a bounded Borel function h; put $H^\theta = h\big((V_d(\theta c'))_{(d,c')}\big)$.

(b1) The Ψ_D, $D \ge d_0$, have mean 0 and form a martingale with respect to $(\mathcal{G}_D)$; and $\lim_K \mathbb{E}_{K,\mathsf{b}}\big[h\big(((\mathcal{M}^{K-d}U)(c'))_{(d,c')}\big)\big(\mathcal{O}(f) - \mathbb{E}_{K,\mathsf{b}}\mathcal{O}(f)\big)\big] = \int H\,\Psi_D\,d\mu_{\ell,\mathsf{b}}$.

(b2) We have $S_2^{\ell,\mathsf{b}}(f\circ\theta, f) \ge 0$ and

$$\Big|\int H\Psi_D\,d\mu_{\ell,\mathsf{b}}\Big|^2 \le \Big(\int H\,\overline{H^\theta}\,d\mu_{\ell,\mathsf{b}}\Big)\,S_2^{\ell,\mathsf{b}}(f\circ\theta, f) \le \|H\|^2_{L^2(\mu_{\ell,\mathsf{b}})}\,S_2^{\ell,\mathsf{b}}(f\circ\theta, f),$$

where the first factor in the middle is a real number ≥ 0. In particular the L^2 norms of all the Ψ_D are bounded by $S_2^{\ell,\mathsf{b}}(f\circ\theta, f)^{1/2}$.

(b3) There is a $\mathcal{G}_\infty$-measurable $\Psi_\infty(f)$, of mean 0 and with $\|\Psi_\infty(f)\|_{L^2} \le S_2^{\ell,\mathsf{b}}(f\circ\theta, f)^{1/2}$, such that $\Psi_D \to \Psi_\infty(f)$ both $\mu_{\ell,\mathsf{b}}$-almost surely and in L^2, and $\Psi_D = \mathbb{E}[\Psi_\infty(f) \mid \mathcal{G}_D]$. Among the elements of $L^2(\mathcal{G}_\infty)$ it is characterised by its integrals against the bounded $\mathcal{G}_D$-measurable H, $D \ge d_0$, which are the limits in (b1); and it depends real linearly on f within the Lipschitz functions supported in $\mathbb{E}_+$.

■

Proof. Step 1: a gauge-invariant function at a larger depth. Let $\bar F(V) = \int F(V^v)\,dv$, the mean over the gauge transformations of $\Lambda_{d,\ell}$; it is bounded, Borel and gauge invariant. For a gauge transformation v of $\Lambda_{d,\ell}$ let $\tilde v$ be the bare gauge transformation equal to v at the sites of $\Lambda_{d,\ell}$ and to $\mathbf{1}$ elsewhere; $\mathcal{M}^{K-d}(U^{\tilde v}) = (\mathcal{M}^{K-d}U)^v$ (Proposition YM3_I:4.5), the Wilson density and $\mathcal{O}(f)$ are gauge invariant, and dU is invariant under $U \mapsto U^{\tilde v}$; so $\mathsf{c}_K(F; f) = \mathsf{c}_K(F(\cdot^v); f)$, and, integrating over v, $\mathsf{c}_K(F; f) = \mathsf{c}_K(\bar F; f)$. For $d \le u \le K$, $\mathsf{c}_K(F; f)$ is also the covariance at the depth u of the function $F \circ \mathcal{M}^{u-d}$, gauge invariant when F is (Proposition YM3_I:4.5).

Step 2. Take $f \ne 0$, $n \ge 1$ and $K \ge K_0$. Use $u = u_n(K)$, $s = \lceil K/9\rceil$, and the two lattices $\mathrm{A} = (\ell, K)$, $\mathrm{B} = (\ell, K+1)$ at the common label. Use the hypotheses of the adjacent-cutoff comparison as in Theorem YM3_I:AR. Its extra spread condition concerns contact derivatives. The quotient below cancels the contact terms identically. For a mark $\bullet$ put $t^\bullet = K^\bullet - u$, and use the common source radius $r = r_1(t^{\mathrm{A}}) = (8nM^{\mathrm{rd}}_{t^{\mathrm{A}}}\|f\|_\flat)^{-1}$ from the polydisc of Lemma YM3_I:114.5, taking all n test functions equal to f and using just the first source coordinate. Use the common depth-u space and inner set $\mathsf{In}_{[u]}$ of Section YM3_I:116. Let F now be a bounded gauge-invariant function on that space, $Y_0 = F(\mathcal{M}^{t^\bullet}U)$, $Y_1 = \mathcal{O}(f)$, and $s_1^\bullet = 6\ell^3 L^{4K^\bullet}\|f\|_\infty$. The last number bounds $|Y_1|$ since $0 \le a_p \le 2$ and there are $3(\ell L^{K^\bullet})^3$ plaquettes.

For this lattice write $\Lambda_0(G) = \int G\rho_0\,dU$ and $\Lambda_1(G) = \int G\mathfrak{w}^{\mathrm{sf}}_{t^\bullet}\rho_0\,dU$. By YM3_I:(2189)–(2190), Λ_1 and $\Lambda_0 - \Lambda_1$ are positive on bounded gauge-invariant functions and the latter has mass at most $R_{t_1}\Lambda_0(1)$. In particular $\Lambda_1(1) \ge \Lambda_0(1)/2$. Write the normalized Λ_0 as the convex combination of the normalized Λ_1 and the normalized remainder, with remainder weight at most R_{t_1}. The normalized moments of any bounded G in the two expectations differ by at most $2R_{t_1}\|G\|_\infty$. Apply this to Y_0Y_1, Y_0 and Y_1, and telescope the product of the latter two means. The covariance error is at most $12R_{t_1}\sup|F|\,s_1^\bullet$.

Step 3. Put $q^\bullet(\zeta) = \Lambda_1(Y_0e^{\zeta Y_1})/\Lambda_1(e^{\zeta Y_1})$ near zero. Its derivative is the covariance for the normalized Λ_1. The exact pushforward YM3_I:(2188) and the density formula YM3_I:(2175) give

$$\Lambda_1(h(\mathcal{M}^{t^\bullet}U)e^{\zeta Y_1}) = e^{c^\bullet(\zeta)}\int_{\mathsf{In}_{[u]}} h(V)e^{\Phi^{\bullet,\zeta f}(V)}\,dV \qquad (|\zeta| < r)$$

for bounded Borel h. In the ratio $q^\bullet$ the prefactor, including the contact energies, cancels.

Step 4. The density $e^{c^\bullet(0)+\Phi^{\bullet,0}}$ is nonnegative by the pushforward identity. Put $\delta(V) = \Phi^{\bullet,\zeta f}(V) - \Phi^{\bullet,0}(V)$. By YM3_I:(2184), $|\delta| \le 1/4$ on the disc. The denominator, after the scalar prefactor is removed, differs from its zero-source value by at most $e^{1/4} - 1$ times the modulus of that value.

Thus it has no zeros. Bounding the numerator by $e^{1/4}$ times its zero-source integral with $|Y_0|$ gives

$$|q^\bullet(\zeta)| \le \frac{e^{1/4}}{2-e^{1/4}}\frac{\Lambda_1(|Y_0|)}{\Lambda_1(1)} < 1.794\frac{\Lambda_1(|Y_0|)}{\Lambda_1(1)}.$$

Step 5. For real $x \in (-r, r)$ the chain and the classical start define positive measures on $\mathsf{In}_{[u]}$. Their field-dependent exponents differ by at most ϵ_{fg} by YM3_I:(2180) and YM3_I:(2192). Up to a positive scalar, their densities differ by at most $(e^{\epsilon_{\mathrm{fg}}}-1)$ times the classical-start density. If positive measures P, Q satisfy $|(P-Q)(h)| \le \delta Q(1)\|h\|_\infty$, their normalized means differ by at most $2\delta\|h\|_\infty$, by adding and subtracting $P(h)/Q(1)$. Thus the chain and classical-start normalized values differ by at most $5\epsilon_{\mathrm{fg}} \sup|F|$.

For the two classical starts use the common space at depth s and its common map $\mathcal{M}^{s-u}$ to depth u. In the proof of Proposition YM3_I:116.1 each positive density gives mass at most ϵ_{sh} times its own mass to the set where the other density vanishes, by YM3_I:(2198). On their common positive set the gauge factors coincide and the log ratio of the starts is bounded by ϵ_{cl}. In particular the mass for B is at most twice the mass for A. Splitting the two integrals over the common set and the two exceptional sets bounds their difference, also with a multiplier $F(\mathcal{M}^{s-u}V)$, by $(2\epsilon_{\mathrm{cl}} + 3\epsilon_{\mathrm{sh}})\sup|F|$ times the mass for A. The same normalization argument gives

$$|q^{\mathrm{B}}(x) - q^{\mathrm{A}}(x)| \le \mathfrak{o}'_K \sup|F|, \qquad \mathfrak{o}'_K = 10\epsilon_{\mathrm{fg}} + 8\epsilon_{\mathrm{cl}} + 12\epsilon_{\mathrm{sh}}.$$

Step 6. The holomorphic difference on the disc is bounded by $4\sup|F|$ there and by $\mathfrak{o}'_K \sup|F|$ on the diameter. Apply Lemma YM3_I:116.2 with one variable and $\sigma = 1/2$. Use the discarding error on each lattice and Step 1 to pass to any $d \le u_n(K)$. This proves

$$\text{(P1)} \qquad |\mathsf{c}_{K+1}(F;f) - \mathsf{c}_K(F;f)| \le \sup|F|\Big[\frac{16}{\pi}(\mathfrak{o}'_K)^{1/2}r^{-1} + 144\,\ell^3 L^{4K+4}\,R_{t_1}\|f\|_\infty\Big].$$

For a Borel set $E \subset \mathsf{V}_d$, apply Step 4 to $F = \bar{\mathbf{1}}_E \circ \mathcal{M}^{u-d}$ on A. Cauchy's inequality and $\Lambda_1 \le \Lambda_0$ for nonnegative gauge-invariant functions give $|(q^{\mathrm{A}})'(0)| \le 3.6 r^{-1}\mu_d^K(E)$, using gauge invariance of μ_d^K to integrate $\bar{\mathbf{1}}_E$. The single-lattice discarding error then gives

$$\text{(P2)} \qquad |\mathsf{c}_K(\mathbf{1}_E;f)| \le 3.6\, r^{-1}\mu_d^K(E) + 72\,\ell^3 L^{4K}\,R_{t_1}\|f\|_\infty.$$

By YM3_I:(2204) and $N_{[u]} \le \ell^3 L^3 L^{\alpha_n K}$, $r^{-1} \le \mathsf{A}_1 K^{2\mathfrak{p}_0} L^{\alpha_n K}\|f\|_\sharp$. Here A_1 is formed from ℓ, b, n and the fixed constants. The bounds $4u - s \le -K/10 + 4$ and $(7s-K)/2 \le -K/9 + 7/2$ of YM3_I:(2205) in YM3_I:(2192)–(2194) imply that ϵ_{fg} and ϵ_{cl} are at most constants times $K^{\mathsf{p}}L^{-K/10}$, with $\mathsf{p} = 2\mathfrak{p}_0 + 1/2$. The majorants of ϵ_{sh} and R_{t_1} in the proof of Theorem YM3_I:AR decay faster than any exponential at fixed side and label. Hence choose $\mathsf{A}_2, \mathsf{A}_3$ independent of K, d, F, f with $\mathfrak{o}'_K \le \mathsf{A}_2 K^{\mathsf{p}} L^{-K/10}$ and $144\ell^3 L^{4K+4} R_{t_1} \le \mathsf{A}_3 L^{-K/10}$. Since $\mathsf{p}/2 + 2\mathfrak{p}_0 = c_0$, the bracket in (P1) is at most $\mathsf{A}_4 K^{c_0} L^{-\gamma_n K}\|f\|_\sharp$ with $\mathsf{A}_4 = (16/\pi)\mathsf{A}_2^{1/2}\mathsf{A}_1 + \mathsf{A}_3$. For summation use $\sum_{k\ge K} k^c L^{-\gamma k} \le K^c L^{-\gamma K}\mathsf{S}(c,\gamma)$ with $\mathsf{S}(c,\gamma) = \sum_{j\ge 0}(1+j)^c L^{-\gamma j}$, from $K + j \le K(1+j)$ for $K \ge 1$.

Step 7: clause (a1). At $f = 0$ everything vanishes. Let $f \ne 0$, $n = 1$, $K \ge K_*(d)$; every $k \ge K$ has $k \ge K_0$ and $u_1(k) \ge d$. Summing (P1) over $K \le k < K'$: $|\mathsf{c}_{K'}(F;f) - \mathsf{c}_K(F;f)| \le \mathsf{C}_1 K^{c_0} L^{-K/21}\sup|F|\|f\|_\sharp$ with $\mathsf{C}_1 = \mathsf{A}_4\mathsf{S}(c_0, \frac{1}{21})$ at $n = 1$. So $\mathsf{c}(F;f) = \lim_K \mathsf{c}_K(F;f)$ exists with the bound of (a1). Each $\xi^K(E) = \mathsf{c}_K(\mathbf{1}_E;f)$ is a finite signed measure, and $\xi_{d,f}(E) = \mathsf{c}(\mathbf{1}_E;f)$ is finitely additive and, as the limit is uniform in E, σ-additive: for Borel sets $R_N \downarrow \emptyset$ and $\eta' > 0$ take K with $\sup_E|\xi_{d,f}(E) - \xi^K(E)| \le \eta'$, and use $\xi^K(R_N) \to 0$. It is bounded, and $\int F\, d\xi_{d,f} = \mathsf{c}(F;f)$ by uniform approximation of F by simple functions. Uniqueness holds because a signed measure is determined by the integrals of indicators. The total mass vanishes because $\mathsf{c}_K(1;f) = 0$; gauge invariance comes from Step 1; linearity in f from that of $\mathcal{O}(f)$; and as $\mathcal{M}^{K-d} = \mathcal{M} \circ \mathcal{M}^{K-d-1}$, the covariance of $\mathbf{1}_{\mathcal{M}^{-1}E}$ computed at the depth $d+1$ equals that of $\mathbf{1}_E$ computed at the depth d, which gives $\mathcal{M}_*\xi_{d+1,f} = \xi_{d,f}$ in the limit.

Step 8: clause (a2). Let $f \neq 0$, $0 < \eta < \theta_n$, $K \geq K_*(d,n)$, E a Borel set and $\mu = \mu_d(E)$. As in Step 7 at n, $|\xi_{d,f}(E) - \mathsf{c}_K(\mathbf{1}_E; f)| \leq \mathsf{A}_4\mathsf{S}(c_0,\gamma_n)K^{c_0}L^{-\gamma_n K}\|f\|_\sharp$. By (P2), the bound of r^{-1} and $72\ell^3 L^{4K}R_{t_1} \leq \frac{1}{2}\mathsf{A}_3 L^{-K/10}$: $|\mathsf{c}_K(\mathbf{1}_E; f)| \leq 3.6\mathsf{A}_1 K^{2\mathfrak{p}_0}L^{\alpha_n K}\|f\|_\sharp \mu_d^K(E) + \frac{1}{2}\mathsf{A}_3 L^{-K/10}\|f\|_\sharp$. By Corollary 2.8(b), (c), eq. 2.xvii and the tail sum, as $\lceil K/1260\rceil = u_1(K) \geq u_n(K) \geq d$: $\mu_d^K(E) \leq \mu + \mathsf{C}_{\mathrm{bf}}\ell^3\mathsf{S}(\mathsf{p}, \frac{1}{10})K^{\mathsf{p}}L^{-K/10}$. With $2\mathfrak{p}_0 + \mathsf{p} = c_1 \geq c_0$ and $\alpha_n - \frac{1}{10} \leq -\gamma_n$:

$$|\xi_{d,f}(E)| \leq \mathsf{A}_5\|f\|_\sharp K^{c_1}\big[L^{\alpha_n K}\mu + L^{-\gamma_n K}\big] \qquad (K \geq K_*(d,n)),$$

$\mathsf{A}_5 = 3.6\mathsf{A}_1(1 + \mathsf{C}_{\mathrm{bf}}\ell^3\mathsf{S}(\mathsf{p}, \frac{1}{10})) + \frac{1}{2}\mathsf{A}_3 + \mathsf{A}_4\mathsf{S}(c_0,\gamma_n)$, formed without d. If $\mu = 0$, let $K \to \infty$. Let $0 < \mu \leq 1$ and $\Lambda = \log_L \frac{1}{\mu}$; the two terms are of equal size at $K = 20\Lambda$, because $\alpha_n + \gamma_n = \frac{1}{20}$. If $\lceil 20\Lambda\rceil \geq K_*$, take $K = \lceil 20\Lambda\rceil$: the bracket is at most $2L\mu^{\theta_n}$ and $K^{c_1} \leq \mathsf{a}_\eta\mu^{-\eta}$ with $\mathsf{a}_\eta = \sup_{\Lambda\geq 0}(1+20\Lambda)^{c_1}L^{-\eta\Lambda}$. Otherwise $1 \leq (L^{K_*/20}\mu)^{\theta_n}$; take $K = K_*$. So (a2) holds with $\mathsf{C} = 2\mathsf{A}_5\max\{L\mathsf{a}_\eta, K_*^{c_1}L^{K_*/20}\}$, $K_* = K_*(d,n)$, using $\alpha_n + \frac{\theta_n}{20} = \frac{1}{20}$. The density: since $\xi_{d,f}(E) = 0$ whenever $\mu_d(E) = 0$, the Radon–Nikodym theorem gives $X = X_d(f) \in L^1(\mu_d)$. With $\vartheta = \theta_n - \eta$, $q = \frac{1}{1-\vartheta}$ and $\mathsf{c} = \mathsf{C}\|f\|_\sharp$: $\lambda\mu_d(X > \lambda) \leq \xi_{d,f}(\{X > \lambda\}) \leq \mathsf{c}\,\mu_d(X > \lambda)^\vartheta$, so $\mu_d(X > \lambda) \leq (\mathsf{c}/\lambda)^q$, the same for $-X$, and $\int |X|^p\,d\mu_d \leq \mathsf{c}^p(1 + \frac{2p}{q-p})$ for $p < q$, by $\int |X|^p = p\int_0^\infty \lambda^{p-1}\mu_d(|X| > \lambda)\,d\lambda$. As $\frac{1}{1-\theta_n} = 3(3n+4) \to \infty$, every $p < \infty$ is reached.

Step 9: clause (a4). Let $f \neq 0$, $1 \leq p < 3(3n+4)$, $a = \mathsf{A}_5\|f\|_\sharp$, $K_* = K_*(d,n)$ and $\lambda_K = 2aK^{c_1}L^{\alpha_n K}$. For $K \geq K_*$ and $\lambda \geq \lambda_K$, the bound of Step 8 at $E = \{X > \lambda\}$ gives $\lambda\mu_d(E) \leq \frac{1}{2}\lambda_K\mu_d(E) + aK^{c_1}L^{-\gamma_n K}$, so $\mu_d(|X| > \lambda) \leq 2L^{-K/20}$. Splitting $[0,\infty)$ at the λ_K, $K \geq K_*$, and using $K + 1 \leq 2K$ and $\delta = \frac{1}{20} - p\alpha_n > 0$: $\int |X|^p\,d\mu_d \leq \lambda_{K_*}^p[1 + 2^{1+pc_1}L^{p\alpha_n}\mathsf{S}(pc_1,\delta)]$. As $K_* = \max\{\bar K_n, 180(3n+4)(d-1)+1\}$ and $\alpha_n\cdot 180(3n+4) = 3$: $K_* \leq (\bar K_n + 180(3n+4))(1+d)$ and $\alpha_n K_* \leq \alpha_n\bar K_n + 3d$, which gives (a4) with $\mathsf{C}_4 = 2\mathsf{A}_5(\bar K_n + 180(3n+4))^{c_1}L^{\alpha_n\bar K_n}[1 + 2^{1+pc_1}L^{p\alpha_n}\mathsf{S}(pc_1,\delta)]^{1/p}$.

Step 10: clause (a3). Under $\mu_{\ell,\mathsf{b}}$ the law of $(V_0,\dots,V_{d+1})$ is the image of μ_{d+1} under $\iota_{0,d+1}$, so for a bounded Borel Φ on $\mathsf{V}_0 \times \cdots \times \mathsf{V}_d$, by the consistency in (a1),

$$\int \Phi(V_0,\dots,V_d)\,X_{d+1}(f)(V_{d+1})\,d\mu_{\ell,\mathsf{b}} = \xi_{d+1,f}(\Phi\circ\iota_{0,d}\circ\mathcal{M}) = \xi_{d,f}(\Phi\circ\iota_{0,d}) = \int \Phi(V_0,\dots,V_d)\,X_d(f)(V_d)\,d\mu_{\ell,\mathsf{b}};$$

so $\mathbb{E}[X_{d+1}(f)(V_{d+1}) \mid \mathcal{F}_d] = X_d(f)(V_d)$; the integrability is (a2), and the mean is 0.

Step 11. For $K \geq d_0$, $c \in \frac{a_K}{2}\mathbb{Z}$ gives a coordinate lattice mirror at depth K. The same holds at each depth $d \geq d_0$. The reflection exchanges the two halves and commutes with blocking by Proposition YM3_I:4.5. The double-block dependence in that proposition confines a blocked bond to the bare bonds in the two endpoint blocks. Each endpoint block at depth d has coordinate half-width $(a_d - a_K)/2$, by summing the half-widths of its nested blocks. If an endpoint lies strictly in the positive half, its two distances to the boundary are positive multiples of $a_d/2$. Thus a blocked bond with both endpoints there reads only bonds in that half. For the plaquettes with $f(x(p)) \neq 0$, the positive distance of the compact support to the boundary ensures the same claim for every sufficiently large K. The Wilson law is reflection invariant by permutation of the bonds and their orientations. The plaquette barycentres transform with the plaquettes and their actions are unchanged, so reflection takes $\mathcal{O}(f)$ to $\mathcal{O}(f\circ\theta)$. Commutation with blocking and total variation convergence give the reflection invariance of μ_D^K and μ_D. (iv) *Schwarz at a cutoff.* Let $D \geq d_0$, h bounded Borel, $\pi(V) = ((\mathcal{M}^{D-d}V)(c'))_{(d,c')}$ on V_D, the pairs (d,c') as in the definition of $\mathcal{G}_D$, $H_D = h\circ\pi$, and take $K \geq \max\{D,2\}$ sufficiently large for the support condition above. Put $F_1(U) = \overline{H_D(\theta\mathcal{M}^{K-D}U)}$ and $F_2 = \mathcal{O}(f) - \mathbb{E}_{K,\mathsf{b}}\mathcal{O}(f)$; both read only bonds of the positive half, $\overline{F_1(\theta U)} = H_D(\mathcal{M}^{K-D}U)$, and $F_2(\theta U)$ is the centred $\mathcal{O}(f\circ\theta)$. The reflection-positive form from Proposition YM3_I:91.4 is positive semidefinite on these functions, so Cauchy–Schwarz gives

$$|\mathsf{c}_K(H_D; f)|^2 \leq \int H_D\,\overline{H_D\circ\theta}\,d\mu_D^K \cdot S^T_{2;K,\ell,L^K\mathsf{b}}(f\circ\theta, f).$$

(v) *Three limits.* $\mathsf{c}_K(H_D;f)\to\int H_D X_D(f)\,d\mu_D$ by (a1), (a2); the middle factor converges, the laws converging in total variation; and $S^T_{2;K,\ell,L^K\mathsf{b}}(f\circ\theta,f)\to S_2^{\ell,\mathsf{b}}(f\circ\theta,f)$ by Theorem YM3_I: AS, as $f\circ\theta$ and f are real Lipschitz, not 0, with supports in the disjoint sets $\mathbb{E}_-$ and $\mathbb{E}_+$. So both factors of the limit are ≥ 0, and the Cauchy–Schwarz inequality in $L^2(\mu_D)$ with $\theta_*\mu_D=\mu_D$ gives the second inequality. (vi) *The martingale and its bound.* On the threads, which have full measure, $V_d=\mathcal{M}^{D-d}V_D$, so $H=H_D(V_D)$ almost surely; $\mathcal{G}_D\subset\mathcal{G}_{D+1}$ and $\mathcal{G}_D\subset\mathcal{F}_D$; by the tower property and (a3), $\mathbb{E}[\Psi_{D+1}\mid\mathcal{G}_D]=\mathbb{E}[\mathbb{E}[X_{D+1}(f)(V_{D+1})\mid\mathcal{F}_D]\mid\mathcal{G}_D]=\Psi_D$, and $\int H\Psi_D=\int H_D X_D(f)\,d\mu_D=\lim_K\mathsf{c}_K(H_D;f)$, which is (b1). The first inequality of (b2) is (iv)–(v) read on the threads; it extends to all bounded $\mathcal{G}_D$-measurable H by uniform approximation. At $H_N=\Psi_D\mathbf{1}\{|\Psi_D|\le N\}$ it gives $\|H_N\|_2^2\le\|H_N\|_2 S_2^{\ell,\mathsf{b}}(f\circ\theta,f)^{1/2}$, and monotone convergence gives $\|\Psi_D\|_2\le S_2^{\ell,\mathsf{b}}(f\circ\theta,f)^{1/2}$. (vii) *Clause (b3).* Doob's theorem for martingales bounded in L^2 gives $\Psi_\infty(f)$, the convergence almost surely and in L^2, and $\Psi_D=\mathbb{E}[\Psi_\infty(f)\mid\mathcal{G}_D]$; the norm and the mean pass to the limit. If $\Phi\in L^2(\mathcal{G}_\infty)$ has $\int H\Phi=0$ for all bounded $\mathcal{G}_D$-measurable H and all D, the sets $G\in\mathcal{G}_\infty$ with $\int_G\Phi=0$ form a Dynkin system that holds the π-system $\bigcup_D\mathcal{G}_D$, so $\Phi=0$. Linearity: $\xi_{D,f}$ is linear in f, so are $X_D(f)$ up to null sets, Ψ_D and the limit. □

Definition 18.149 (Smooth one-forms, their holonomies and their lattice fields)**.** Let $K,m\in\mathbb{N}$, $\ell=2L^m$, $a=a_K$ and $\Lambda=\Lambda_{K,\ell}$; so $\ell\ge 2$, $0<a\le 1$ and $\ell/a\in\mathbb{N}$. $\mathfrak{g}$ is the space of the traceless hermitian 2×2 matrices with $\langle X,Y\rangle=\frac12\operatorname{tr}(XY)$ and $|X|=\langle X,X\rangle^{1/2}$; for $X\in\mathfrak{g}$, $X^2=|X|^2\mathbf{1}$, so $|X|$ is the operator norm $\|X\|$. For a 2×2 matrix M put $\mathbf{w}(M)=\frac{1}{2i}(M-M^*)$ and $\mathbf{c}(M)=\frac12\operatorname{Re}\operatorname{tr}M$; on SU(2), $\mathbf{w}$ is the vector part of YM3_I:(2103). In fact $V=\mathbf{c}(V)\mathbf{1}+i\mathbf{w}(V)$ and $\mathbf{c}(V)^2+|\mathbf{w}(V)|^2=1$, and $a_p=1-\mathbf{c}(\partial U(p))$.

(1) By a *one-form of class* C^k we mean a triple $Q=(Q_1,Q_2,Q_3)$ of functions $Q_\mu\in C^k(T_\ell;\mathfrak{g})$; $\|Q\|_{C^k}=\max_\mu\max_{|\kappa|\le k}\sup|\partial^\kappa Q_\mu|$, and $\|\chi\|_{C^k}$ is defined in the same way for a function χ.

(2) Let $\gamma\colon[t_0,t_1]\to T_\ell$ be continuous and of class C^1 on finitely many pieces, and let H_γ be the solution of $H'(t)=H(t)\,iQ_\lambda(\gamma(t))\,\gamma'^\lambda(t)$, $H(t_0)=\mathbf{1}$; the *holonomy* is $\operatorname{hol}_Q(\gamma)=H_\gamma(t_1)\in\mathrm{SU}(2)$. It does not depend on the parametrisation, $\operatorname{hol}_Q(\gamma_1\gamma_2)=\operatorname{hol}_Q(\gamma_1)\operatorname{hol}_Q(\gamma_2)$ for γ_1 followed by γ_2, and the reversed path has the inverse holonomy. The *lattice field* $U[Q]\in\mathcal{V}(\Lambda)$ is $U[Q](\langle x,x'\rangle)=\operatorname{hol}_Q$ of the straight segment of length a from x to x'; thus the ordered product of its bonds along any lattice path Γ is $\operatorname{hol}_Q(\Gamma)$. For a signed direction α write $e_\alpha=\operatorname{sgn}(\alpha)e_{|\alpha|}$. When $|\alpha|\ne|\alpha'|$, let $W_x(\alpha,\alpha')$ be the product along the loop from x with steps $ae_\alpha,ae_{\alpha'},-ae_\alpha,-ae_{\alpha'}$. Its underlying plaquette is $p(x;\alpha,\alpha')$. In particular $\partial U[Q](p_{\mu\nu}(x))=W_x(\mu,\nu)(U[Q])$.

(3) $F_{\mu\nu}=\partial_\mu Q_\nu-\partial_\nu Q_\mu+i[Q_\mu,Q_\nu]$, $D_\lambda\Psi=\partial_\lambda\Psi+i[Q_\lambda,\Psi]$ for $\mathfrak{g}$-valued Ψ, $(d^*_Q F)_\mu=\sum_\nu D_\nu F_{\mu\nu}$; $\mathrm{YM}(f)=\frac12\int_{T_\ell}f\sum_{\mu<\nu}|F_{\mu\nu}|^2$ for a bounded Borel function f; $B_k=\frac12\sum_{i,j}\epsilon_{kij}F_{ij}$; and the stress tensor is $T_{\alpha\rho}=\sum_\mu\langle F_{\alpha\mu},F_{\rho\mu}\rangle-\delta_{\alpha\rho}\frac12\sum_{\mu<\nu}|F_{\mu\nu}|^2$.

(4) For $Q\in C^{k+1}$, $\Phi_k=\Phi_k(Q)=\max\sup_x|D_{\lambda_1}\cdots D_{\lambda_k}F_{\mu\nu}(x)|$, maximised over all choices of the indices.

(5) For $g\in C^1(T_\ell;\mathrm{SU}(2))$ and a one-form Q of class C^0 put $Q^g_\mu=gQ_\mu g^{-1}-ig\,\partial_\mu g^{-1}$, a one-form of class C^0. Then $\operatorname{hol}_{Q^g}(\gamma)=g(\gamma(t_0))\operatorname{hol}_Q(\gamma)g(\gamma(t_1))^{-1}$, as the two sides, read as functions of the end point, satisfy one differential equation with one initial value; so $U[Q^g]=U[Q]^{g|_\Lambda}$. If moreover $g\in C^2$ and $Q\in C^1$, then $Q^g\in C^1$, $F^{Q^g}=\mathrm{Ad}(g)F^Q$, and $D^{Q^g}_\lambda(\mathrm{Ad}(g)\Psi)=\mathrm{Ad}(g)D^Q_\lambda\Psi$ for $\Psi\in C^1(T_\ell;\mathfrak{g})$, where D^Q is the D of (3) formed with Q; if $g\in C^{k+2}$ and $Q\in C^{k+1}$, then $Q^g\in C^{k+1}$, and the first identity, the second applied k times and $|\mathrm{Ad}(g)X|=|X|$ give $\Phi_k(Q^g)=\Phi_k(Q)$.

For a positively oriented bond $b=\langle x,x+ae_\mu\rangle$ and a plaquette p containing b, there is a unique α with $|\alpha|\ne\mu$ and $p=p(x;\mu,\alpha)$. Put $W_{b,p}=W_x(\mu,\alpha)$ and $x_b=x+\frac{a}{2}e_\mu$. The displacement and

elementary density of YM3_I:(2103), YM3_I:(2104) are

$$\mathbf{u}_{b,\rho}=-\tfrac12\{\mathbf{w}(W_x(\mu,\rho))-\mathbf{w}(W_x(\mu,-\rho))\}\quad(\rho\neq\mu),\qquad \mathbf{u}_{b,\mu}=0,\qquad \mathfrak{d}_{b,p,\rho}=\langle\mathbf{u}_{b,\rho},\mathbf{w}(W_{b,p})\rangle.$$

Here ρ ranges over the positive coordinate indices. Use central lifts $y\in[-\ell/2,\ell/2)^3$ for positions paired with ωy. A function has margin $2a$ if its support is contained in $(-\ell/2+2a,\ell/2-2a)^3$. The differences and the three sums of the shell insertion YM3_I:(2106) are

$$\begin{aligned}
\nabla^{\mathrm c}_\mu\chi(y)&=\frac{\chi(y+ae_\mu/2)-\chi(y-ae_\mu/2)}{a},\\
\nabla^{\mathrm s}_\omega\chi(y)&=\sum_\rho(\omega y)_\rho\frac{\chi(y+ae_\rho)-\chi(y-ae_\rho)}{2a},\\
S_1&=\beta\sum_p a_p\nabla^{\mathrm s}_\omega\chi(x(p)),\\
S_2&=\frac{\beta}{a}\sum_{p,\,b\subset\partial p,\,\rho}[\chi(x_b)-\chi(x(p))](\omega x_b)_\rho\,\mathfrak{d}_{b,p,\rho},\\
S_3&=\tfrac32\sum_{p=p_{\mu\nu},\,\mu<\nu}a_p[(\omega x(p))_\mu\nabla^{\mathrm c}_\mu\chi(x(p))+(\omega x(p))_\nu\nabla^{\mathrm c}_\nu\chi(x(p))],\\
\mathfrak{B}_{\omega,\chi}&=S_1-S_2-S_3.
\end{aligned}$$

Every bond in S_2 has the positive orientation. All positions in a local displacement use a common lift.

■

Proposition 18.150 (Tools: the numbers Φ_k, transport, planar loops, a rectangle, a square seen from its centre, Riemann sums)**.** In the setting of Definition 18.149:

(a) For $Q\in C^{k+1}$ and $N=\|Q\|_{C^{k+1}}$, $\Phi_k\le 2^{k^2+k+1}(1+N)^{k+2}$; in particular $\Phi_0\le 2(1+N)^2$, $\Phi_1\le 8(1+N)^3$, $\Phi_2\le 128(1+N)^4$.

(b) *Transport.* Let $Q\in C^1$, γ as in Definition 18.149(2) and $\Psi\in C^1(T_\ell;\mathfrak g)$; inside each piece the transported value $\mathrm{Ad}(H_\gamma(t))\Psi(\gamma(t))$ has the t-derivative $\mathrm{Ad}(H_\gamma(t))(\gamma'^\lambda D_\lambda\Psi)(\gamma(t))$. For $Q\in C^k$, $\Psi\in C^k$, $c\in T_\ell$, $z\in\mathbb R^3$ and $R(t)$ the holonomy of $s\mapsto c+sz$, $0\le s\le t$, the function $\varphi(t)=\mathrm{Ad}(R(t))\Psi(c+tz)$ has $\varphi^{(j)}(t)=\mathrm{Ad}(R(t))((z\cdot D)^j\Psi)(c+tz)$ for $j\le k$.

(c) *Planar loops.* Take $Q\in C^1$, a point $c\in T_\ell$, two directions $\mu\neq\nu$ and a loop $y=y^\mu e_\mu+y^\nu e_\nu\colon[0,1]\to\mathbb R^3$, continuous and C^1 on finitely many pieces, with $y(0)=y(1)=0$; put $m=y^\mu y'^\nu-y^\nu y'^\mu$ and $|m|_1=\int_0^1|m|$. For $0\le r,t\le1$ write $H(r,t)$ for the holonomy along $s\mapsto c+ry(s)$, $s\in[0,t]$, and $V(r)=H(r,1)$. Then $V(0)=\mathbf1$, $V'(r)V(r)^{-1}=ir\int_0^1\mathrm{Ad}(H(r,t))F_{\mu\nu}(c+ry(t))\,m(t)\,dt$, and $\|V(r)-\mathbf1\|\le\frac12r^2|m|_1\Phi_0$.

(d) *A rectangle seen from a corner.* For $x\in T_\ell$, $\mu\neq\nu$, real a_1,a_2 and Rect the holonomy of $x\to x+a_1e_\mu\to x+a_1e_\mu+a_2e_\nu\to x+a_2e_\nu\to x$: $\|\mathrm{Rect}-\mathbf1\|\le|a_1a_2|\Phi_0$ for $Q\in C^1$, and $\|\mathrm{Rect}-\exp(ia_1a_2F_{\mu\nu}(x))\|\le\frac23|a_1a_2|(2|a_1|+|a_2|)\Phi_1$ for $Q\in C^2$.

(e) *A square seen from its centre.* For $Q\in C^3$, $\mu\neq\nu$, $x\in T_\ell$, $c=x+\frac a2(e_\mu+e_\nu)$, T the holonomy of the segment from c to x and W the holonomy of $x\to x+ae_\mu\to x+ae_\mu+ae_\nu\to x+ae_\nu\to x$: $\|TWT^{-1}-\exp(ia^2F_{\mu\nu}(c))\|\le\frac14a^4(\Phi_2+4\Phi_0^2)$.

(f) *Riemann sums.* For $c_0\in\mathbb R^3$: $|\int_{T_\ell}\psi-a^3\sum_{c\in\Lambda+c_0}\psi(c)|\le\frac{5}{16}\ell^3a^2\max_{\kappa,\lambda}\sup|\partial_\kappa\partial_\lambda\psi|$ for $\psi\in C^2(T_\ell)$, and $\le\frac{\sqrt3}{2}\ell^3a\,\mathrm{Lip}\,\psi$ for Lipschitz ψ, Lip for the euclidean distance.

■

Proof. Step 0: matrix facts. For $V\in\mathrm{SU}(2)$: $(V-\mathbf1)^*(V-\mathbf1)=2(1-\mathbf c(V))\mathbf1$, so $\|V-\mathbf1\|^2=2(1-\mathbf c(V))$; $|\mathbf w(V)|^2=(1-\mathbf c(V))(1+\mathbf c(V))\le\|V-\mathbf1\|^2$; for $X\in\mathfrak g$, $e^{iX}=\cos|X|\,\mathbf1+i\frac{\sin|X|}{|X|}X$, so $\|e^{iX}-\mathbf1\|\le|X|$ and $|\mathbf w(e^{iX})-X|=||X|-\sin|X||\le\frac16|X|^3$; $\|\mathbf w(M)\|\le\|M\|$; $|[X,Y]|\le2|X||Y|$;

$\|\operatorname{Ad}(V)-\operatorname{Ad}(V')\|\le 2\|V-V'\|$; $\partial_\lambda\langle\Psi,\Psi'\rangle=\langle D_\lambda\Psi,\Psi'\rangle+\langle\Psi,D_\lambda\Psi'\rangle$. *Comparison:* let Ξ,Ξ_0 be continuous $\mathfrak{g}$-valued functions on $[0,r]$ and V,V_0 solutions of $V'=i\Xi V$, $V_0'=i\Xi_0V_0$ with $V(0)=V_0(0)\in\mathrm{SU}(2)$; then both take values in SU(2), $(V_0^{-1}V)'=V_0^{-1}i(\Xi-\Xi_0)V$, and so $\|V(r)-V_0(r)\|\le\int_0^r|\Xi-\Xi_0|$.

Step 1: (a). By Leibniz' rule and $|[X,Y]|\le 2|X||Y|$, $\|F\|_{C^k}\le 2N+2^{k+1}N^2\le 2^{k+1}(1+N)^2$, and $\|D_\lambda\Psi\|_{C^{j-1}}\le(1+2^jN)\|\Psi\|_{C^j}$; so $\Phi_k\le\prod_{j=1}^k(1+2^jN)\cdot 2^{k+1}(1+N)^2\le 2^{k^2}(1+N)^k\,2^{k+1}(1+N)^2$.

Step 2: (b) and a variation formula. If $\partial_tH=H\,iA(t;r)$, $H(t_0)=\mathbf{1}$, with A and ∂_rA continuous on pieces, then $X=\partial_rH\,H^{-1}$ has $X(t_0)=0$ and $\partial_tX=\operatorname{Ad}(H)\,i\partial_rA$, so $\partial_rH(t)H(t)^{-1}=i\int_{t_0}^t\operatorname{Ad}(H)\partial_rA$. At $A=Q_\lambda(\gamma)\gamma'^\lambda$, $\frac{d}{dt}(H\Psi H^{-1})=H(iA\Psi+\gamma'^\lambda\partial_\lambda\Psi-\Psi iA)H^{-1}$, which is the transport; the formula for $\varphi^{(j)}$ follows by induction, $z\cdot D\Psi$ being of class C^{k-1}. For a family of paths $\sigma(r,t)$ with fixed end points and $A=Q_\beta(\sigma)\partial_t\sigma^\beta$, the same computation shows that ∂_rA differs from $\operatorname{Ad}(H)^{-1}\partial_t[\operatorname{Ad}(H)(Q_\alpha\partial_r\sigma^\alpha)]$ by $F_{\alpha\beta}\partial_r\sigma^\alpha\partial_t\sigma^\beta$, and the total derivative integrates to 0, as $\partial_r\sigma=0$ at the ends; so the derivative of the holonomy of the loop in r is $i\int\operatorname{Ad}(H)F_{\alpha\beta}\partial_r\sigma^\alpha\partial_t\sigma^\beta$, times the holonomy.

Step 3: (c). At $\sigma(r,t)=c+ry(t)$, $\partial_r\sigma=y$ and $\partial_t\sigma=ry'$, and $F_{\alpha\beta}y^\alpha ry'^\beta=rF_{\mu\nu}m$; this is the formula. The bound is the comparison with $V_0=\mathbf{1}$: $\|V(r)-\mathbf{1}\|\le\int_0^r r'|m|_1\Phi_0\,dr'$.

Step 4: (d). Parametrise each of the four sides by $u\in[0,1]$: $y=ua_1e_\mu$, $a_1e_\mu+ua_2e_\nu$, $(1-u)a_1e_\mu+a_2e_\nu$, $(1-u)a_2e_\nu$, with $m=0,a_1a_2,a_1a_2,0$; so $|m|_1=2|a_1a_2|$, and (c) gives the first bound. A point of the second or of the third side is reached after a path length of at most $r(2|a_1|+|a_2|)$, so by (b) $|\operatorname{Ad}(H(r,t))F_{\mu\nu}(x+ry(t))-F_{\mu\nu}(x)|\le r(2|a_1|+|a_2|)\Phi_1$ where $m\neq 0$. So $V'V^{-1}=i[2ra_1a_2F_{\mu\nu}(x)+e(r)]$ with $|e(r)|\le 2r^2|a_1a_2|(2|a_1|+|a_2|)\Phi_1$, while $V_0(r)=\exp(ir^2a_1a_2F_{\mu\nu}(x))$ has $V_0'V_0^{-1}=i\,2ra_1a_2F_{\mu\nu}(x)$; the comparison and $\int_0^1 2r^2\,dr=\frac{2}{3}$ give the second bound.

Step 5: (e). Let y go straight from 0 to the corner $y_0=-\frac{a}{2}(e_\mu+e_\nu)$, then once around the square $[-\frac{a}{2},\frac{a}{2}]^2$ of the (μ,ν)-plane, starting in the direction e_μ and parametrised by arc length, and then straight back to 0; then $V(1)=TWT^{-1}$. On the segments $m=0$, on the boundary $m=\frac{a}{2}$; so $\int m=|m|_1=2a^2$, and $\int y^\lambda m=0$ by the symmetry $y\mapsto -y$ of the boundary. For a point $z=ry(t)$ of the boundary let $R(z)$ be the holonomy of the segment from c to $c+z$; $H(r,t)R(z)^{-1}$ is the holonomy of a planar loop with $|m|_1\le 2a^2$, so $\|H(r,t)R(z)^{-1}-\mathbf{1}\|\le r^2a^2\Phi_0$ by (c), and $|\operatorname{Ad}(H(r,t))F_{\mu\nu}(c+z)-\operatorname{Ad}(R(z))F_{\mu\nu}(c+z)|\le 2r^2a^2\Phi_0^2$. By (b) and Taylor's formula, $\operatorname{Ad}(R(z))F_{\mu\nu}(c+z)=F_{\mu\nu}(c)+z^\lambda D_\lambda F_{\mu\nu}(c)+\rho(z)$ with $|\rho(z)|\le\frac{1}{2}(|z^\mu|+|z^\nu|)^2\Phi_2\le\frac{1}{2}r^2a^2\Phi_2$. Into (c): $V'V^{-1}=ir[2a^2F_{\mu\nu}(c)+e(r)]$, the term with $D_\lambda F$ vanishing, with $|e(r)|\le r^2a^4(\Phi_2+4\Phi_0^2)$; the comparison with $\exp(ir^2a^2F_{\mu\nu}(c))$ and $\int_0^1 r^3\,dr=\frac{1}{4}$ give (e).

Step 6: (f). T_ℓ is the disjoint union of the $(\ell/a)^3$ cubes $c+[-\frac{a}{2},\frac{a}{2})^3$, $c\in\Lambda+c_0$. For $\psi\in C^2$ the first-order term of Taylor's formula integrates to 0 over a cube, and $\int z_\kappa^2\,dz=\frac{a^5}{12}$, $\int|z_\kappa||z_\lambda|\,dz=\frac{a^5}{16}$ for $\kappa\neq\lambda$, with $\frac{1}{2}(\frac{3}{12}+\frac{6}{16})=\frac{5}{16}$; for Lipschitz ψ, $|\psi(c+z)-\psi(c)|\le\frac{\sqrt{3}}{2}a\operatorname{Lip}\psi$. □

Proposition 18.151 (On the lattice field of a smooth one-form: the Wilson action, the action density, the elementary densities and the shell flux)**.** In the setting of Definition 18.149, let Q be a one-form, $N=\|Q\|_{C^3}$, and put $\mathsf{K}_1=\frac{1}{4}\Phi_0\Phi_2+\Phi_0^3+\frac{7}{24}a^2\Phi_0^4$ and $\mathsf{K}_2=5\Phi_0\Phi_1+\frac{1}{3}a^3\Phi_0^4$.

(1) *A single plaquette.* For $Q\in C^3$ and $p=p_{\mu\nu}(x)$: $0\le a_p(U[Q])\le\frac{1}{2}a^4\Phi_0^2$ and $|a_p(U[Q])-\frac{1}{2}a^4|F_{\mu\nu}(x(p))|^2|\le a^6\mathsf{K}_1$, $\mathsf{K}_1\le 77(1+N)^8$.

(2) *The action.* For $Q\in C^3$, $\mathsf{b}>0$ and $\beta=\mathsf{b}/a$: $|\beta A(U[Q])-\mathsf{b}\,\mathrm{YM}(1)|\le 3\mathsf{b}\ell^3a^2[\mathsf{K}_1+\frac{5}{16}(\Phi_0\Phi_2+\Phi_1^2)]\le 531\,\mathsf{b}\,\ell^3a^2(1+N)^8$, where $A=\sum_p a_p$.

(3) *The action density.* For $Q\in C^3$ and a complex Lipschitz function f on T_ℓ: $|\mathcal{O}(f)(U[Q])-\mathrm{YM}(f)|\le 3\ell^3[a^2\mathsf{K}_1\|f\|_\infty+\frac{\sqrt{3}}{2}a(\frac{1}{2}\Phi_0^2\operatorname{Lip}f+\sqrt{3}\Phi_0\Phi_1\|f\|_\infty)]\le(303+\frac{1}{7M})\,\ell^3a\,\|f\|_\sharp\,(1+N)^8$.

(4) *The coupling.* The Pauli matrices $\sigma_1,\sigma_2,\sigma_3$ form an orthonormal basis of $\mathfrak{g}$. Requiring $\mathsf{b}\cdot\frac12|F|^2=\frac{1}{2g_{\rm std}^2}\sum_c(F^c)^2$ for all $F=\sum_c F^c\sigma_c/2$ with real F^c fixes $g_{\rm std}>0$ as $g_{\rm std}=2\mathsf{b}^{-1/2}$; thus $\mathsf{b}\,g_{\rm std}^2=4$.

(5) *The elementary densities.* Let $Q\in C^2$ and $x\in\Lambda$. (a) For $\mu,\rho\in\mathbb{D}$ with $|\mu|\neq|\rho|$ and $\mathbb{D}=\{\pm1,\pm2,\pm3\}$: $|\mathbf{w}(W_x(\mu,\rho)(U[Q]))|\le a^2\Phi_0$ and $|\mathbf{w}(W_x(\mu,\rho)(U[Q]))-\operatorname{sgn}(\mu)\operatorname{sgn}(\rho)a^2F_{|\mu||\rho|}(x)|\le 2a^3\Phi_1+\frac16a^6\Phi_0^3$. (b) For $b=\langle x,x+ae_\mu\rangle$, $1\le\mu\le3$, $\alpha\in\mathbb{D}$ with $|\alpha|\neq\mu$, $p=p(x;\mu,\alpha)$ the underlying plaquette and $1\le\rho\le3$: $|\mathfrak{d}_{b,p,\rho}(U[Q])|\le a^4\Phi_0^2$ and

$$|a^{-4}\mathfrak{d}_{b,p,\rho}(U[Q])+\operatorname{sgn}(\alpha)\langle F_{\mu\rho},F_{\mu|\alpha|}\rangle(x_b)|\le a\,\mathsf{K}_2,\qquad \mathsf{K}_2\le 86(1+\|Q\|_{C^2})^8,$$

Here $\mathfrak{d}_{b,p,\rho}$ and x_b are as in Definition 18.149.

(6) *The shell flux.* Let $Q\in C^3$, $\mathsf{b}>0$, $\beta=\mathsf{b}/a$ and $\omega\in\mathfrak{so}(3)$, write $\|\omega\|_{\rm op}=\max\{|\omega v|:v\in\mathbb{R}^3,\ |v|=1\}$ for the operator norm of ω on euclidean $\mathbb{R}^3$, and let $\chi\in C^2(\mathbb{R}^3)$ be real with margin $2a$ as in Definition 18.149. For the shell flux $\mathfrak{B}_{\omega,\chi}$, use the integral over $[-\frac{\ell}{2},\frac{\ell}{2})^3$ and $\mathsf{K}_3=\frac92a\mathsf{K}_1+9\mathsf{K}_2+\frac{63}{2}\Phi_0^2+\frac{135}{4}\Phi_0\Phi_1$:

$$\begin{aligned}&\Big|\mathfrak{B}_{\omega,\chi}(U[Q])+\mathsf{b}\int\sum_{\alpha,\rho}\partial_\alpha\chi(y)\,(\omega y)_\rho\,T_{\alpha\rho}(y)\,dy\Big|\\&\quad\le\ell^4a\|\omega\|_{\rm op}\|\chi\|_{C^2}\big[\tfrac{27}{8}\Phi_0^2+\mathsf{b}\,\mathsf{K}_3\big]\\&\quad\le 3000\,(1+\mathsf{b})\,\ell^4a\|\omega\|_{\rm op}\|\chi\|_{C^2}(1+N)^8;\end{aligned}$$

the sum S_3 has modulus at most $\frac{27}{8}\ell^4a\|\omega\|_{\rm op}\|\chi\|_{C^1}\Phi_0^2$.

(7) *The tensor.* For $Q\in C^2$: (a) $T_{\alpha\rho}=T_{\rho\alpha}=\frac12\delta_{\alpha\rho}|B|^2-\langle B_\alpha,B_\rho\rangle$ with $|B|^2=\sum_k|B_k|^2=\sum_{\mu<\nu}|F_{\mu\nu}|^2$; (b) $\sum_\alpha\partial_\alpha T_{\alpha\rho}=-\sum_\mu\langle(d_Q^*F)_\mu,F_{\rho\mu}\rangle$; (c) for $\chi\in C^1$ vanishing outside a compact subset of $(-\frac{\ell}{2},\frac{\ell}{2})^3$, $-\int\sum\partial_\alpha\chi(\omega y)_\rho T_{\alpha\rho}=-\int\chi\sum_{\rho,\mu}(\omega y)_\rho\langle(d_Q^*F)_\mu,F_{\rho\mu}\rangle=-\frac{d}{dt}\big|_{t=0}\mathrm{YM}(1)(Q+t\,\delta Q)$ with $\delta Q_\mu=\chi\sum_\rho(\omega y)_\rho F_{\rho\mu}$. Thus, for $Q\in C^3$, $\omega\in\mathfrak{so}(3)$, $\mathsf{b}>0$, $\beta=\mathsf{b}/a$ and a real $\chi\in C^2(\mathbb{R}^3)$ whose support is a compact subset of $(-\frac{\ell}{2},\frac{\ell}{2})^3$, so that χ has the margin $2a$ once K is large, the shell flux at $U[Q]$ tends, as $K\to\infty$ with ℓ held fixed, to the negative of the derivative of $\mathsf{b}\,\mathrm{YM}(1)$ along the rotation field cut off by χ, a number that vanishes where $d_Q^*F=0$.

■

Proof. (1). $\partial U[Q](p)=W$ of Proposition 18.150(e) and $n:=a_p=1-\mathbf{c}(W)=1-\mathbf{c}(V)$ with $V=TWT^{-1}$. By Step 0 of the proof of Proposition 18.150 and its clause (d), $n=\frac12\|W-\mathbf{1}\|^2\le\frac12a^4\Phi_0^2$. By its clause (e) and Step 0, $|\mathbf{w}(V)-a^2F_{\mu\nu}(c)|\le\frac14a^4(\Phi_2+4\Phi_0^2)+\frac16a^6\Phi_0^3$, with $c=x(p)$; as $|\mathbf{w}(V)|\le\|V-\mathbf{1}\|\le a^2\Phi_0$, $||\mathbf{w}(V)|^2-a^4|F_{\mu\nu}(c)|^2|\le2a^2\Phi_0$ times that; and $|\mathbf{w}(V)|^2=n(2-n)$ gives $n-\frac12|\mathbf{w}(V)|^2=\frac12n^2\le\frac18a^8\Phi_0^4$. Together, $|n-\frac12a^4|F_{\mu\nu}(c)|^2|\le a^6\mathsf{K}_1$; Proposition 18.150(a) and $a\le1$ give $\mathsf{K}_1\le(64+8+\frac{14}{3})(1+N)^8$.

(2). $\beta A=\frac{\mathsf{b}}{a}\sum_p a_p$, and there are $3(\ell/a)^3$ plaquettes; by (1), putting $\frac12a^4|F_{\mu\nu}(x(p))|^2$ in the place of every a_p changes βA by at most $3\mathsf{b}\ell^3a^2\mathsf{K}_1$. The barycentres of the plaquettes with the directions $\mu<\nu$ are the points of $\Lambda+\frac{a}{2}(e_\mu+e_\nu)$, and Proposition 18.150(f) applies to $\psi=\frac12|F_{\mu\nu}|^2$, with $\partial_\kappa\partial_\lambda\psi=\langle D_\kappa D_\lambda F_{\mu\nu},F_{\mu\nu}\rangle+\langle D_\lambda F_{\mu\nu},D_\kappa F_{\mu\nu}\rangle$, of modulus at most $\Phi_0\Phi_2+\Phi_1^2$. Numbers: $3[77+\frac{5}{16}(256+64)]=531$.

(3). $\mathcal{O}(f)=a^{-1}\sum_p f(x(p))a_p$ (Definition 18.143(1)); the same substitution changes it by at most $3\ell^3a^2\mathsf{K}_1\|f\|_\infty$, and the Riemann sums of $\psi=f\cdot\frac12|F_{\mu\nu}|^2$, with $\operatorname{Lip}\psi\le\frac12\Phi_0^2\operatorname{Lip}f+\sqrt3\Phi_0\Phi_1\|f\|_\infty$, as $|\partial_\lambda\frac12|F|^2|=|\langle D_\lambda F,F\rangle|\le\Phi_0\Phi_1$, cost the rest. With $\operatorname{Lip}f\le\|f\|_\sharp/(40M)$, $a\le1$, $\Phi_0\Phi_1\le16(1+N)^5$ and $\Phi_0^2\le4(1+N)^4$: $3[77+24+\frac{\sqrt3}{40M}]\le303+\frac{1}{7M}$.

(4). $\sigma_c\sigma_e+\sigma_e\sigma_c=2\delta_{ce}\mathbf{1}$, so $\langle\sigma_c,\sigma_e\rangle=\delta_{ce}$ and $|\sum_c F^c\sigma_c/2|^2=\frac14\sum_c(F^c)^2$; the identity reads $\frac{\mathsf b}{8}\sum(F^c)^2=\frac{1}{2g_{\rm std}^2}\sum(F^c)^2$.

(5). (a) $W_x(\mu,\rho)(U[Q])$ is the Rect of Proposition 18.150(d) with the directions $|\mu|,|\rho|$, $a_1=\operatorname{sgn}(\mu)a$ and $a_2=\operatorname{sgn}(\rho)a$; Step 0 of its proof gives both bounds. (b) If $\rho=\mu$, the two sides are 0: $\mathbf{u}_{b,\mu}=0$ by definition, and $F_{\mu\mu}=0$. Let $\rho\neq\mu$ and $\eta_0=2a^3\Phi_1+\frac16 a^6\Phi_0^3$; by (a), $\mathbf{w}(W_x(\mu,\pm\rho))=\pm a^2F_{\mu\rho}(x)+r_\pm$ and, with the based holonomy $W_{b,p}$, $\mathbf{w}(W_{b,p})=\mathbf{w}(W_x(\mu,\alpha))=\operatorname{sgn}(\alpha)a^2F_{\mu|\alpha|}(x)+r_\alpha$, all $|r|\le\eta_0$. Substituting in the density formula gives

$$\mathfrak{d}_{b,p,\rho}=-\operatorname{sgn}(\alpha)a^4\langle F_{\mu\rho},F_{\mu|\alpha|}\rangle(x)-a^2\langle F_{\mu\rho}(x),r_\alpha\rangle-\tfrac12\langle r_+-r_-,\mathbf{w}(W_{b,p})\rangle,$$

the last two terms of modulus at most $2a^2\Phi_0\eta_0$; moving the point from x to $x_b=x+\frac a2e_\mu$ costs $a\Phi_0\Phi_1$, as $|\partial_\mu\langle F_{\mu\rho},F_{\mu|\alpha|}\rangle|\le2\Phi_0\Phi_1$. Sum: $a^{-4}\cdot2a^2\Phi_0\eta_0+a\Phi_0\Phi_1=a\mathsf K_2$. And $|\mathfrak{d}_{b,p,\rho}|\le\frac12(|\mathbf{w}(W_x(\mu,\rho))|+|\mathbf{w}(W_x(\mu,-\rho))|)\,|\mathbf{w}(W_{b,p})|\le a^4\Phi_0^2$.

(6). Read all points at their central lifts $y\in[-\frac\ell2,\frac\ell2)^3$; there $|y|\le\frac{\sqrt3}{2}\ell$, hence $\sum_\rho|(\omega y)_\rho|\le\sqrt3\,|\omega y|\le\frac32\ell\|\omega\|_{\rm op}$, and near the faces of the cube, within the sup distance $2a$, the function χ and all its derivatives are 0, so that the shifted values of χ in $\nabla^{\rm s}_\omega$, $\nabla^{\rm c}_\mu$ and S_2 are those of χ on $\mathbb{R}^3$. Use the three sums $\mathfrak{B}_{\omega,\chi}=S_1-S_2-S_3$ and $-\mathsf b\int\sum\partial_\alpha\chi(\omega y)_\rho T_{\alpha\rho}=I_1-I_2$, with $I_1=\mathsf b\int\frac12\sum_{\mu<\nu}|F_{\mu\nu}|^2\,(\omega y\cdot\nabla\chi)$ and $I_2=\mathsf b\int\sum_{\alpha,\rho}\partial_\alpha\chi(\omega y)_\rho\sum_\mu\langle F_{\alpha\mu},F_{\rho\mu}\rangle$. *$S_3$:* as $|\nabla^{\rm c}_\mu\chi|\le\|\chi\|_{C^1}$ and $a_p\le\frac12a^4\Phi_0^2$, $|S_3|\le\frac32\cdot3(\ell/a)^3\cdot\frac12a^4\Phi_0^2\cdot\frac32\ell\|\omega\|_{\rm op}\|\chi\|_{C^1}=\frac{27}{8}\ell^4a\|\omega\|_{\rm op}\|\chi\|_{C^1}\Phi_0^2$, which is the last assertion. *S_1-I_1:* as $|\nabla^{\rm s}_\omega\chi|\le\frac32\ell\|\omega\|_{\rm op}\|\chi\|_{C^1}$, (1) shows that putting $\frac12a^4|F_{\mu\nu}(x(p))|^2$ in the place of a_p costs at most $\frac92\mathsf b\ell^4a^2\|\omega\|_{\rm op}\|\chi\|_{C^1}\mathsf K_1$; next, since $|(2a)^{-1}[\chi(y+ae_\rho)-\chi(y-ae_\rho)]-\partial_\rho\chi(y)|\le\frac a2\|\chi\|_{C^2}$, putting $\omega y\cdot\nabla\chi$ in the place of $\nabla^{\rm s}_\omega\chi$ costs at most $\frac98\mathsf b\ell^4a\|\omega\|_{\rm op}\|\chi\|_{C^2}\Phi_0^2$; finally, for $\psi_{\mu\nu}=\frac12|F_{\mu\nu}|^2(\omega y\cdot\nabla\chi)$ one has $|\partial_\lambda(\omega y\cdot\nabla\chi)|\le\sqrt3\|\omega\|_{\rm op}\|\chi\|_{C^1}+\frac32\ell\|\omega\|_{\rm op}\|\chi\|_{C^2}\le\frac52\ell\|\omega\|_{\rm op}\|\chi\|_{C^2}$, because $\ell\ge2$, so $|\partial_\lambda\psi_{\mu\nu}|\le\ell\|\omega\|_{\rm op}\|\chi\|_{C^2}[\frac54\Phi_0^2+\frac32\Phi_0\Phi_1]$, the euclidean Lipschitz number of $\psi_{\mu\nu}$ is at most $\sqrt3$ times this, and Proposition 18.150(f) bounds the three Riemann sums by $\frac92\mathsf b\ell^4a\|\omega\|_{\rm op}\|\chi\|_{C^2}[\frac54\Phi_0^2+\frac32\Phi_0\Phi_1]$. *$S_2-I_2$:* a plaquette together with one of its four bonds corresponds one to one, through $p=p(x;\mu,\alpha)$, to a pair (b,α) of a positively oriented bond $b=\langle x,x+ae_\mu\rangle$ and an $\alpha\in\mathbb D$ with $|\alpha|\neq\mu$; there are $12(\ell/a)^3$ such pairs, and $x(p)=x_b+\frac a2e_\alpha$, $W_{b,p}=W_x(\mu,\alpha)$; so $S_2=\frac{\mathsf b}{a^2}\sum_{(b,\alpha)}\sum_\rho[\chi(x_b)-\chi(x_b+\frac a2e_\alpha)](\omega x_b)_\rho\,\mathfrak{d}_{b,p,\rho}$. Here $\chi(x_b)-\chi(x_b+\frac a2e_\alpha)=-\frac a2\operatorname{sgn}(\alpha)\partial_{|\alpha|}\chi(x_b)+r_\chi$, $|r_\chi|\le\frac{a^2}{8}\|\chi\|_{C^2}$, and by (5) $\mathfrak{d}_{b,p,\rho}=-\operatorname{sgn}(\alpha)a^4\langle F_{\mu\rho},F_{\mu|\alpha|}\rangle(x_b)+a^4r_{\mathfrak d}$, $|r_{\mathfrak d}|\le a\mathsf K_2$; the product of the main terms is $\frac{a^5}{2}\partial_{|\alpha|}\chi(x_b)\langle F_{\mu\rho},F_{\mu|\alpha|}\rangle(x_b)$, since $\operatorname{sgn}(\alpha)^2=1$, the remaining part of the product is at most $a^6[\frac12\|\chi\|_{C^1}\mathsf K_2+\frac18\|\chi\|_{C^2}\Phi_0^2]$ in modulus, and with $\sum_\rho|(\omega x_b)_\rho|\le\frac32\ell\|\omega\|_{\rm op}$ these remainders cost at most $18\mathsf b\ell^4a\|\omega\|_{\rm op}[\frac12\|\chi\|_{C^1}\mathsf K_2+\frac18\|\chi\|_{C^2}\Phi_0^2]$. The main term is $\mathsf b\sum_\mu a^3\sum_{x_b}\psi_\mu(x_b)$, a Riemann sum over $\Lambda+\frac a2e_\mu$ of $\psi_\mu=\sum_{\nu\neq\mu}\sum_\rho(\omega y)_\rho\partial_\nu\chi\langle F_{\mu\rho},F_{\mu\nu}\rangle$, the two $\alpha=\pm\nu$ giving the same term; the three ways in which ∂_λ can fall on ψ_μ give $|\partial_\lambda\psi_\mu|\le2[\sqrt3\|\chi\|_{C^1}\Phi_0^2+\frac32\ell\|\chi\|_{C^2}\Phi_0^2+3\ell\|\chi\|_{C^1}\Phi_0\Phi_1]\|\omega\|_{\rm op}\le2\ell\|\omega\|_{\rm op}\|\chi\|_{C^2}[\frac52\Phi_0^2+3\Phi_0\Phi_1]$, so the Riemann sums cost at most $9\mathsf b\ell^4a\|\omega\|_{\rm op}\|\chi\|_{C^2}[\frac52\Phi_0^2+3\Phi_0\Phi_1]$; and $\mathsf b\sum_\mu\int\psi_\mu=I_2$, as $F_{\mu\mu}$ vanishes and $\langle F_{\mu\rho},F_{\mu\nu}\rangle=\langle F_{\nu\mu},F_{\rho\mu}\rangle$. *Sum:* with $\|\chi\|_{C^1}\le\|\chi\|_{C^2}$, the costs that carry the factor $\mathsf b$ add up to $\mathsf b\ell^4a\|\omega\|_{\rm op}\|\chi\|_{C^2}$ times $\frac92a\mathsf K_1+9\mathsf K_2+(\frac98+\frac{45}{8}+\frac94+\frac{45}{2})\Phi_0^2+(\frac{27}{4}+27)\Phi_0\Phi_1=\mathsf K_3$, and S_3 adds $\frac{27}{8}\Phi_0^2$; with the bounds of (1), (5), with Proposition 18.150(a) and $a\le1$, $\mathsf K_3\le(\frac{693}{2}+774+126+540)(1+N)^8<1787(1+N)^8$ and $\frac{27}{8}\Phi_0^2\le\frac{27}{2}(1+N)^8$, which gives the second inequality.

(7). (a) $F_{\alpha\mu}=\sum_k\epsilon_{\alpha\mu k}B_k$ and $\sum_\mu\epsilon_{\alpha\mu k}\epsilon_{\rho\mu l}=\delta_{\alpha\rho}\delta_{kl}-\delta_{\alpha l}\delta_{k\rho}$ give $\sum_\mu\langle F_{\alpha\mu},F_{\rho\mu}\rangle=\delta_{\alpha\rho}|B|^2-\langle B_\rho,B_\alpha\rangle$. (b) $\partial_\alpha\sum_\mu\langle F_{\alpha\mu},F_{\rho\mu}\rangle=\langle D_\alpha F_{\alpha\mu},F_{\rho\mu}\rangle+\langle F_{\alpha\mu},D_\alpha F_{\rho\mu}\rangle$, summed over repeated indices, and the identity $D_\lambda F_{\mu\nu}+D_\mu F_{\nu\lambda}+D_\nu F_{\lambda\mu}=0$ follows by the Jacobi identity for $\partial_\lambda+iQ_\lambda$, whose commutator with $\partial_\mu+iQ_\mu$ is $iF_{\lambda\mu}$. Contracting with $F_{\alpha\mu}$ gives $\partial_\rho\frac14\sum_{\mu,\nu}|F_{\mu\nu}|^2=\langle F_{\alpha\mu},D_\alpha F_{\rho\mu}\rangle$; so $\partial_\alpha T_{\alpha\rho}=\langle D_\alpha F_{\alpha\mu},F_{\rho\mu}\rangle=-\langle(d_Q^*F)_\mu,F_{\rho\mu}\rangle$. (c) Integration by parts gives $\int\chi[\omega_{\rho\alpha}T_{\alpha\rho}+(\omega y)_\rho\partial_\alpha T_{\alpha\rho}]$, and the first term vanishes, ω being antisymmetric and T symmetric; and $\frac{d}{dt}|_0\mathrm{YM}(1)(Q+t\delta Q)=$

$\int\sum_{\mu,\nu}\langle F_{\mu\nu}, D_\mu\delta Q_\nu\rangle = \int\sum_\nu\langle(d^*_Q F)_\nu, \delta Q_\nu\rangle$. With (6): for $Q\in C^3$, $\omega\in\mathfrak{so}(3)$, $\mathsf{b}>0$ and a real $\chi\in C^2(\mathbb{R}^3)$ that vanishes outside a compact subset of $(-\frac{\ell}{2},\frac{\ell}{2})^3$, and so has the margin $2a_K$ for all large K, the shell flux at $U[Q]$ with $\beta=\mathsf{b}/a_K$ differs from $-\mathsf{b}\int\sum\partial_\alpha\chi(\omega y)_\rho T_{\alpha\rho}$ by at most $3000\,(1+\mathsf{b})\,\ell^4 a_K\|\omega\|_{\mathrm{op}}\|\chi\|_{C^2}(1+N)^8$, and so tends to it as $K\to\infty$, ℓ being fixed. □

Proposition 18.152 (Three free photons: the connected functions of half the Wick square of their curvature form the free template)**.** Let $\mathsf{b}>0$, Π the orthogonal projection of $L^2(\mathbb{R}^3;\mathbb{R}^3)$ with the Fourier multiplier $\delta_{ik}-k_ik_k/|k|^2$, $\mathsf{g}(x)=(4\pi|x|)^{-1}$, and $G(x)=\mathbb{K}_3(x)$.

(1) *Gaussian quadratic forms.* Fix $n\geq 2$, real symmetric $m\times m$ matrices $A_1,\dots,A_n$, and a centred Gaussian vector ξ with values in $\mathbb{R}^m$ and covariance Σ, which is allowed to be degenerate. The joint cumulant of $\frac12\xi^{\mathsf{T}}A_1\xi,\dots,\frac12\xi^{\mathsf{T}}A_n\xi$ is $\frac12\sum_\pi\operatorname{tr}(\Sigma A_1\Sigma A_{\pi2}\cdots\Sigma A_{\pi n})$, the sum over the $(n-1)!$ orders π of $2,\dots,n$; adding constants to the quadratic forms leaves it unchanged.

(2) *The field and the photons.* (a) A centred Gaussian process $B(\varphi)$, depending linearly on $\varphi=(\varphi^c)_{c=1,2,3}\in L^2(\mathbb{R}^3;\mathbb{R}^3)^3$, exists with the covariance $\mathbb{E}\,B(\varphi)B(\psi)=\mathsf{b}^{-1}\sum_c\langle\varphi^c,\Pi\psi^c\rangle_{L^2}$. (b) For bounded Borel $\varphi,\psi\in L^2(\mathbb{R}^3;\mathbb{R}^3)$ whose compact supports are at a positive distance, $\langle\varphi,\Pi\psi\rangle=\iint\varphi_i(x)G_{ik}(x-y)\psi_k(y)\,dx\,dy$; the entries $G_{ik}=\partial_i\partial_k\mathsf{g}$ are harmonic off the origin, and $G(-x)=G(x)=G(x)^{\mathsf{T}}$. (c) For $\zeta\geq0$, $\mathcal{A}$ denotes a centred Gaussian process indexed by the triples $\varphi=(\varphi^c)_{c=1,2,3}\in\mathcal{S}(\mathbb{R}^3;\mathbb{R}^3)^3$ of Schwartz vector fields, whose covariance is $\mathbb{E}\,\mathcal{A}(\varphi)\mathcal{A}(\psi)=\mathsf{b}^{-1}\sum_c\int\overline{\hat\varphi^c_i(k)}\,|k|^{-2}[\delta_{ik}-(1-\zeta)k_ik_k/|k|^2]\,\hat\psi^c_k(k)\frac{dk}{(2\pi)^3}$; for $\zeta>0$ this is the inverse Hessian of the quadratic action $\frac{\mathsf{b}}{2}\int|\operatorname{curl}\mathcal{A}|^2+\frac{\mathsf{b}}{2\zeta}\int(\operatorname{div}\mathcal{A})^2$. For every ζ, the process $\varphi\mapsto\mathcal{A}(\operatorname{curl}\varphi)$, $\varphi\in\mathcal{S}(\mathbb{R}^3;\mathbb{R}^3)^3$, with the curl applied colour by colour, and the restriction of B of (a) to these Schwartz triples have one and the same covariance; in this sense $B=\operatorname{curl}\mathcal{A}$ in law. (d) For $Q=\sum_c\mathcal{A}^c\sigma_c$ with real one-forms $\mathcal{A}^c\in C^1$, the terms of second order in $\mathcal{A}$ of $\mathrm{YM}(f)(Q)$ sum to $\frac12\int f\sum_c|\operatorname{curl}\mathcal{A}^c|^2$; so $\mathsf{b}\,\mathrm{YM}(1)$ has at second order the action $\mathsf{b}\cdot\frac12\int|\operatorname{curl}\mathcal{A}^c|^2$ of three photons, and $\mathrm{YM}(f)$ has at second order $\frac12\int f|B|^2$, $B^c=\operatorname{curl}\mathcal{A}^c$.

(3) *The connected functions of* $\frac12{:}|B|^2{:}$. Let $\varrho\in C^\infty_c(\mathbb{R}^3)$ be ≥0, of integral 1 and supported in $\{|x|\leq1\}$, $\varrho_\varepsilon(x)=\varepsilon^{-3}\varrho(x/\varepsilon)$, $B^c_{\varepsilon,k}(x)=B(\varrho_\varepsilon(x-\cdot)e_k\otimes e^c)$, $C^\varepsilon_{ik}(x-y)=\langle\varrho_\varepsilon(x-\cdot)e_i,\Pi\varrho_\varepsilon(y-\cdot)e_k\rangle$, and for a bounded Borel f with compact support $\mathcal{O}_\varepsilon(f)=\frac12\int f(x)[|B_\varepsilon(x)|^2-\mathbb{E}|B_\varepsilon(x)|^2]\,dx$, a Bochner integral in L^p for every $p<\infty$. Let $n\geq2$ and $f_1,\dots,f_n$ bounded Borel functions with compact supports at pairwise distances $\geq\delta_0>0$. For random variables $Y_1,\dots,Y_n$ with moments of every order, $\kappa_n(Y_1,\dots,Y_n)$ denotes their joint cumulant; the $\mathcal{O}_\varepsilon(f_i)$ are such variables. (a) For every $\varepsilon>0$, $\kappa_n(\mathcal{O}_\varepsilon(f_1),\dots,\mathcal{O}_\varepsilon(f_n))=\frac32\mathsf{b}^{-n}\sum_\pi\int\prod_i f_i(x_i)\operatorname{tr}[C^\varepsilon(x_1-x_{\pi2})C^\varepsilon(x_{\pi2}-x_{\pi3})\cdots C^\varepsilon(x_{\pi n}-x_1)]\,dx$. (b) For radial ϱ and $\varepsilon<\delta_0/2$ one has the identity $\kappa_n(\mathcal{O}_\varepsilon(f_1),\dots,\mathcal{O}_\varepsilon(f_n))=3\,\mathsf{b}^{-n}\mathcal{K}^{(n)}_{\mathrm{free}}(f_1,\dots,f_n)$, with the functional of Definition 18.116(3). (c) For any ϱ as above and $\varepsilon\leq\delta_0/4$, the modulus of the difference of the two members in (b) is at most $104\,n!\,(4/\pi)^{n-1}\mathsf{b}^{-n}\varepsilon\,\delta_0^{-3n-1}\prod_i\|f_i\|_{L^1}$; so the limit $\varepsilon\to0$ exists and depends on b and G alone, and this limit is what we call the connected function of $\frac12{:}|B|^2{:}$ at $(f_1,\dots,f_n)$. (d) For a continuous real $f\neq0$ with compact support, $\varepsilon^3\operatorname{Var}\mathcal{O}_\varepsilon(f)$ tends to a positive number, so $\operatorname{Var}\mathcal{O}_\varepsilon(f)\to\infty$: it is the set of the connected functions evaluated at tuples with separated supports that has a limit, not the variable $\mathcal{O}_\varepsilon(f)$.

(4) *The dictionary.* The connected functions of (3) form the free template $S^{\mathrm{fr}}[3,\mathsf{b}]$ of Definition 18.116(3). For $n=2$ one gets $\mathsf{c}_2\mathsf{b}^{-2}\mathcal{K}_6(f,h)$, $\mathsf{c}_2=\frac{9}{16\pi^2}$; for $n=3$ one gets $3\mathsf{b}^{-3}\mathcal{K}^{(3)}_{\mathrm{free}}$, where the kernel k_3 of $\mathcal{K}^{(3)}_{\mathrm{free}}$ is $k_3(x_1,x_2,x_3)=-\dfrac{3+9\,\mathsf{q}_\triangle(x_1,x_2,x_3)}{(4\pi)^3|x_1-x_2|^3|x_2-x_3|^3|x_3-x_1|^3}$, $\mathsf{q}_\triangle$ being the product of the cosines of the angles of the triangle. So, for every order $n\geq2$, the function $S^{\mathrm{fr}}_n[3,\mathsf{b}]=3\mathsf{b}^{-n}\mathcal{K}^{(n)}_{\mathrm{free}}$ of the free template is the connected function of order n

of $\frac12{:}|B|^2{:}$, where $B^c=\operatorname{curl}\mathcal{A}^c$ for three independent free photon fields $\mathcal{A}^c$ on $\mathbb{R}^3$ with the action $\mathsf{b}\cdot\frac12\int|\operatorname{curl}\mathcal{A}|^2$, at the label b. By (2)(d), point by point, the density $\mathsf{b}\cdot\frac12\sum_c|\operatorname{curl}\mathcal{A}^c|^2$ of their action and the density $\frac12 f|B|^2$ are the parts of second order in $\mathcal{A}$ of the integrands $\frac{\mathsf{b}}{2}\sum_{\mu<\nu}|F_{\mu\nu}|^2$ of $\mathsf{b}\,\mathrm{YM}(1)$ and $\frac12 f\sum_{\mu<\nu}|F_{\mu\nu}|^2$ of $\mathrm{YM}(f)$, formed at $Q=\sum_c\mathcal{A}^c\sigma_c$, while on the lattice of a torus T_ℓ the functions βA and $\mathcal{O}(f)$ stand for $\mathsf{b}\,\mathrm{YM}(1)$ and $\mathrm{YM}(f)$ (Proposition 18.151(2), (3)).

■

Proof. (1). Write $\xi=\Sigma^{1/2}\zeta$ with ζ standard Gaussian and $A(t)=\sum_j t_jA_j$; for small t the symmetric matrix $M=\Sigma^{1/2}A(t)\Sigma^{1/2}$ has norm <1, and diagonalising it gives $\mathbb{E}\exp(\frac12\zeta^{\mathsf{T}}M\zeta)=\prod_i(1-m_i)^{-1/2}$. The logarithm of the generating function is therefore $-\frac12\log\det(\mathbf{1}-M)=\frac12\sum_{k\ge1}\frac1k\operatorname{tr}\big((\Sigma A(t))^k\big)$, analytic near 0, and the joint cumulant is the coefficient of $t_1\cdots t_n$, carried by the term $k=n$ alone: $\frac{1}{2n}\sum_{\varsigma\in S_n}\operatorname{tr}(\Sigma A_{\varsigma1}\cdots\Sigma A_{\varsigma n})$. The cyclic rotations split S_n into classes of size n, each of which contains a single permutation fixing 1. Adding constants to the forms only adds a linear function of t to the logarithm.

(2). (a) Since Π is an orthogonal projection, the covariance form is symmetric and ≥0, and a Gaussian process with it exists. (b) For $\varphi,\psi\in C_c^\infty$ Plancherel's theorem gives $\langle\varphi,\Pi\psi\rangle=\int\varphi\cdot\psi-\iint(\operatorname{div}\varphi)(x)\mathsf{g}(x-y)(\operatorname{div}\psi)(y)\,dx\,dy$, as g has the Fourier transform $|k|^{-2}$. When the supports are disjoint, $\int\varphi\cdot\psi=0$, and integrating by parts in x and in y yields the kernel $\partial_i\partial_k\mathsf{g}(x-y)=G_{ik}(x-y)$, with $\partial_i\partial_k(4\pi|x|)^{-1}=(4\pi)^{-1}(3x_ix_k|x|^{-5}-\delta_{ik}|x|^{-3})$. For bounded Borel φ,ψ one approximates by mollified functions whose supports keep a distance $\ge\frac{\delta}{2}$; both sides pass to the limit, the left one in L^2 and the right one in L^1. Off the origin $\Delta G_{ik}=\partial_i\partial_k\Delta\mathsf{g}=0$. (c) The matrix $|k|^{-2}[\Pi(k)+\zeta kk^{\mathsf{T}}/|k|^2]$ is positive semidefinite and locally integrable in k; since $(\operatorname{curl}\varphi)^\wedge=ik\times\hat\varphi$ is orthogonal to k, the part with kk^{T} does not contribute, and $\overline{(k\times\hat\varphi)}\cdot(k\times\hat\psi)=|k|^2\,\overline{\hat\varphi}\cdot\Pi(k)\hat\psi$. (d) $F_{\mu\nu}=\sum_c(\partial_\mu\mathcal{A}^c_\nu-\partial_\nu\mathcal{A}^c_\mu)\sigma_c+i[Q_\mu,Q_\nu]$, and the commutator is of second order in $\mathcal{A}$; as $\langle\sigma_c,\sigma_e\rangle=\delta_{ce}$, the second-order part of $|F_{\mu\nu}|^2$ is $\sum_c(\partial_\mu\mathcal{A}^c_\nu-\partial_\nu\mathcal{A}^c_\mu)^2$, whose sum over $\mu<\nu$ is $\sum_c|\operatorname{curl}\mathcal{A}^c|^2$.

(3). Translation is continuous in L^2, so $B^c_{\varepsilon,k}(x)$ depends continuously on x in L^2 of the probability space, and, Gaussian variables having all their moments bounded by powers of the second one, also in every L^p. So the Bochner integral defining $\mathcal{O}_\varepsilon(f)$ makes sense, and the joint cumulant, continuous and multilinear on L^n, may be taken under the integral sign: $\kappa_n(\mathcal{O}_\varepsilon(f_1),\dots)=\int\prod_i f_i(x_i)\,\kappa_n(\frac12|B_\varepsilon(x_1)|^2,\dots,\frac12|B_\varepsilon(x_n)|^2)\,dx$. (a) Use (1) for the Gaussian vector ξ of the $9n$ numbers $B^c_{\varepsilon,k}(x_j)$, with A_j selecting the nine coordinates that belong to x_j; the covariance of ξ is made of the blocks $\mathsf{b}^{-1}\mathbf{1}_3\otimes C^\varepsilon(x_j-x_{j'})$, and the trace over the colours contributes the factor 3. No distance between the points x_j or between the supports enters this computation, coinciding points giving a degenerate Σ, which (1) admits; so the formula of (a) holds at all bounded Borel $f_1,\dots,f_n$ with compact supports. (b) For $|x-y|>2\varepsilon$, (2)(b) gives $C^\varepsilon(x-y)=\int\eta_\varepsilon(z)G(x-y-z)\,dz$ with $\eta_\varepsilon(z)=\int\varrho_\varepsilon(u)\varrho_\varepsilon(u-z)\,du$, a probability density supported in $|z|\le2\varepsilon$, radial when ϱ is. Averaging the harmonic function G_{ik} over spheres centred at $x-y$ returns its value there, so $C^\varepsilon(x-y)=G(x-y)$. Where $\prod_i f_i(x_i)\ne0$ all the distances are $\ge\delta_0>2\varepsilon$; the orders π of $2,\dots,n$ correspond to the cyclic orders ς with $\varsigma(1)=1$, and $\frac32\sum_\pi\operatorname{tr}[G\cdots G]=3k_n$ by the definition of k_n. The integral is therefore $3\mathsf{b}^{-n}\mathcal{K}^{(n)}_{\text{free}}$. (c) For a unit vector e, $\|\partial_eG(u)\|\le\frac{9}{4\pi|u|^4}$, and $\|G(u)\|=\frac{2}{4\pi|u|^3}$; so for $|x|\ge\delta_0\ge4\varepsilon$ and $|z|\le2\varepsilon$, $\|C^\varepsilon(x)\|,\|G(x)\|\le\frac{4}{\pi\delta_0^3}$ and $\|C^\varepsilon(x)-G(x)\|\le\frac{72\varepsilon}{\pi\delta_0^4}$. The trace of a 3×3 matrix is at most 3 times its norm; expanding the difference of the two products of n factors as a telescoping sum and multiplying by $\frac32\mathsf{b}^{-n}(n-1)!\prod\|f_i\|_{L^1}$ gives the bound, as $\frac92\cdot\frac{72}{\pi}<104$. (d) At $n=2$, $f_1=f_2=f$, the formula of (a), valid at coinciding supports as noted at the end of the proof of (a), reads $\operatorname{Var}\mathcal{O}_\varepsilon(f)=\frac32\mathsf{b}^{-2}\iint f(x)f(y)\operatorname{tr}C^\varepsilon(x-y)^2\,dx\,dy$, as C^ε is even; $C^\varepsilon(x)=\varepsilon^{-3}C^1(x/\varepsilon)$, so $\varepsilon^3\operatorname{Var}\mathcal{O}_\varepsilon(f)\to\frac32\mathsf{b}^{-2}\|\operatorname{tr}(C^1)^2\|_{L^1}\int f^2>0$, by Plancherel's theorem and $\operatorname{tr}\Pi(k)^2=2$.

(4). The template identity is (3)(b), (c) with Definition 18.116(3). For a unit vector $e \in \mathbb{R}^3$ let $D(e) = 3ee^{\mathsf{T}} - \mathbf{1}$, and for $u \neq 0$ let $\hat{u} = u/|u|$; then $G(u) = (4\pi)^{-1}|u|^{-3}D(\hat{u})$, and $D(-e) = D(e)$. For $n = 2$, $\mathrm{Cyc}(2)$ consists of a single cyclic order, and at $u = x_1 - x_2$ the matrix $D(\hat{u})^2 = 3\hat{u}\hat{u}^{\mathsf{T}} + \mathbf{1}$ has the trace 6, so $k_2 = \frac{1}{2} \cdot \frac{6}{(4\pi)^2}|u|^{-6}$ and $3k_2 = \mathsf{c}_2|u|^{-6}$. For $n = 3$ the two cyclic orders yield equal traces, as G is symmetric and even, so $k_3 = \operatorname{tr} G(x_1 - x_2)G(x_2 - x_3)G(x_3 - x_1)$; with the unit vectors $\mathsf{e}_1, \mathsf{e}_2, \mathsf{e}_3$ of $x_1 - x_2$, $x_2 - x_3$, $x_3 - x_1$, $\operatorname{tr} D(\mathsf{e}_1)D(\mathsf{e}_2)D(\mathsf{e}_3) = 27(\mathsf{e}_1 \cdot \mathsf{e}_2)(\mathsf{e}_2 \cdot \mathsf{e}_3)(\mathsf{e}_3 \cdot \mathsf{e}_1) - 9[(\mathsf{e}_1 \cdot \mathsf{e}_2)^2 + (\mathsf{e}_2 \cdot \mathsf{e}_3)^2 + (\mathsf{e}_3 \cdot \mathsf{e}_1)^2] + 6$, and the scalar products are minus the cosines of the angles of the triangle; with $\cos^2 A + \cos^2 B + \cos^2 C = 1 - 2\cos A \cos B \cos C$ this is $-(3 + 9\mathsf{q}_{\triangle})$. The last sentence is (2)(c), (d) with Proposition 18.151(2), (3). □

18.15. **On one torus: the torus limits in the label, the statistic on one thread, and other plaquette weights.** We use the block fields and the Wilson measure of Definition 18.140 on a fixed torus T_ℓ. Proposition 18.154 takes the torus limits as functions of the label: they are real-analytic on $(0, \infty)$ and extend to a complex strip whose width is fixed by the side and the interval of labels, their derivatives in the label are insertions of the constant function 1, and the law of two points turns into a sum rule for the three-point function with one entry 1. Proposition 18.155 takes the background action statistic T_d^K at $K = K(d)$, the cutoff that Corollary 18.213 attaches to the depth d, and compares $T_d = T_d^{K(d)}$ with the number $m_d(\mathsf{b}) = \int T_d \, d\mu_d^{K(d)}[\mathsf{b}]$, an integral against the law of the block field at that same cutoff: on almost every thread the two differ by at most a constant times $(dN_d)^{1/2}$ once d exceeds a depth that depends on the thread; hence $m_d(\mathsf{b})$ is of the form $N_d[\varkappa_0\mathsf{b}^{-1} + \varkappa_1 L^{-d}\mathsf{b}^{-2} + o(L^{-d})]$ precisely when $L^d[N_d^{-1}T_d - \varkappa_0\mathsf{b}^{-1}]$ tends to $\varkappa_1\mathsf{b}^{-2}$ on almost every thread, and when $m_d(\mathsf{b})$ is of this form its coefficient $\varkappa_1$ can be found from one thread alone; it shows besides that the singularity of the thread measure against the laws of Gaussian lattice fields, and the smallness of their affinity, hold for every gauge-invariant law in its place. Proposition 18.158, with Lemmas 18.156 and 18.157 on a real source in the chain of the main text, carries the laws of the block fields from Wilson's plaquette weight to the weights $e^{-\beta s}\hat{m}(s)e^{-h(s)}$, in which $\hat{m}(s) = \int e^{-\tau s} m(d\tau)$ for a probability measure m that lives on a short interval, and $0 \leq h(s) \leq M_h s^4$; they include the mixed action whose adjoint coupling is positive and small. The label stays the same, and at the cutoff K' the error is at most a constant times $L^{-K'/11}$.

Definition 18.153 (Signs)**.** Use the fixed constants ς of §2.3.1 and write ϱ_ς for its source radius. Let $\ell = 2L^m$, $m \geq 1$, and use V_d, $\mathcal{M}$, $\mu_d^K[s]$, $\mu_d[s]$ and $\mu_{\ell,s}$ of Definition 18.140, with $a_d = L^{-d}$ and $N_d = \ell^3 L^{3d}$. For an admitted interval $J = [\mathsf{b}_-, \mathsf{b}_+]$ on this side put $d_\natural = \min\{d \geq 1 : L^d\mathsf{b}_- \geq \mathsf{t}_\natural\}$. For $K > d \geq d_\natural$ and $t = K - d$, let $\mathsf{Cut}_t[s]$ and $\mathsf{In}_t[s]$ be the cut and inner ball of Lem. YM3_I:113.10 on the line $x_j = L^{K-j}s$. If $U_t(V)$ is the critical background over V, define

$$T_d^K(V) = \begin{cases} L^K \sum_p a_p(U_t(V)), & V \in \mathsf{Cut}_t[\mathsf{b}_-], \\ 0, & V \notin \mathsf{Cut}_t[\mathsf{b}_-]. \end{cases}$$

The critical orbit is independent of the label and the action is constant on it. The cut and the action are Borel and gauge invariant by Lem. YM3_I:113.10 and Prop. YM3_I:114.2. Use the constants c_{A}, C_{old} and $C_{\mathrm{E},0}$ of YM3_I:(2163) and put

$$\mathsf{Q}' = \max\left\{\varrho_\varsigma^{-1}, \tfrac{3}{2}c_\rho^2\ell^3 \sup_{u \geq 1} \varepsilon(L^u\mathsf{b}_-)^2, \varrho_\varsigma^{-1}C_{\mathrm{old}}\tfrac{L}{L-1}E_0\ell^3\right\},$$

$$\Xi_u[s] = \frac{c_{\mathrm{A}}\alpha_0\varepsilon_{\mathcal{W}}}{2Mc_\rho\varepsilon(L^u s)}, \qquad \theta_0 = c'A_0^2(\log L)^{2\mathfrak{p}_0},$$

$$\rho[s](u) = 2M^{-3}\ell^3 L^{3u}\exp[(C_{\mathrm{maj}}^{\Sigma} + 2 + C_{\mathrm{low}}(s))\ell^3 - \theta_0 u^2].$$

The constants of the cut are those of §2.3.1. By the action estimate YM3_I:(2161), $0 \leq T_d^K \leq \mathsf{Q}'L^{4d}$. Take $u_\flat$ to be the least integer $u \geq d_\natural$ with $\theta_0(2u+1) \geq 3\log L$ and $\rho[\mathsf{b}_+](u) \leq 1/4$. The Gaussian decay in u ensures that this integer exists and that $\rho[s](u) \leq 1/4$ for every $s \in J$ and $u \geq u_\flat$.

(1) *Tuples and limits.* Use the norms $\|f\|_\sharp$ and $\|f\| = \|f\|_\flat$ of Definitions 18.159(3) and 18.143(1). A tuple has $n \geq 2$ real Lipschitz entries with empty intersection of their supports. For nonzero entries, its spread $\mathsf{d}_\infty(\underline{f})$ is the minimum of the maximum pairwise sup distance over one point from each support, as in Definition 18.159(2). It is positive by compactness. Write $\underline{f}^{+\nu} = (1^\nu, \underline{f})$ and $\underline{f}^{(s)}$ for the dilated tuple on $T_{L^s\ell}$. The scalar cutoff cumulants and their torus limits $S_n^{\ell,\mathsf{b}}(\underline{f})$ are those of Thm. YM3_I:AS. For compact $J \subset (0,\infty)$ and $r > 0$ set $\Omega_r(J) = \{\zeta : \operatorname{dist}(\zeta, J) < r\}$.

(2) *Distances of laws.* Let P, Q be probability measures on one measurable space, with densities p, q against a σ-finite measure λ. Their total variation is $\|P - Q\| = \sup_E |P(E) - Q(E)| = \frac{1}{2}\int |p - q|\, d\lambda$, and their affinity is $\operatorname{aff}(P, Q) = \int (pq)^{1/2} d\lambda$; neither of the two depends on λ.

(3) *Gauge.* $\mathcal{G}_d$ is the compact group of the gauge transformations of the lattice of depth d, acting by $V^u(x,y) = u(x)V(x,y)u(y)^{-1}$ on the bond from x to y, and du is its Haar probability measure. A Borel probability measure P on V_d is *gauge invariant* if it is its own image under $V \mapsto V^u$ for every $u \in \mathcal{G}_d$; $\mathscr{I}_d$ is the σ-algebra of the Borel sets of V_d that every $u \in \mathcal{G}_d$ maps onto themselves; and the *gauge average* of a Borel probability measure ν on V_d is $\bar\nu = \int \nu^u\, du$, where ν^u is the image of ν under $V \mapsto V^u$. A site $a_d k$, $k \in \mathbb{Z}^3$, is *even* if $k_1 + k_2 + k_3$ is even; as ℓL^d is even this is a property of the site of T_ℓ, there are $\frac{1}{2}N_d$ even sites, and the two ends of a bond have different parities. For an even site y we write c_y for the bond from y to $y + a_d e_1$, and we put

$$E_d = \{V \in \mathsf{V}_d :\ |V(c_y) - \mathbf{1}|_{\mathrm{op}} \leq 1 \text{ for every even site } y\}, \qquad h_0 = \tfrac{1}{3} - \tfrac{\sqrt{3}}{4\pi} = 0.1955\ldots,$$

$|\cdot|_{\mathrm{op}}$ being the operator norm of 2×2 matrices.

(4) *Gaussian lattice fields.* Let $c > 0$. Let $A = (A^1, A^2, A^3)$ be three independent centred Gaussian fields indexed by the vector fields $\varphi \in L^2(T_\ell; \mathbb{R}^3)$, with

$$\mathbb{E}\, A^a(\varphi) A^{a'}(\psi) = \delta_{aa'}\, c^{-1}\ell^{-3} \sum_{k \in (2\pi/\ell)\mathbb{Z}^3 \setminus \{0\}} |k|^{-2}\, \overline{\hat\varphi(k)} \cdot \Pi(k) \hat\psi(k), \qquad \hat\varphi(k) = \int_{T_\ell} \varphi(x) e^{-ik\cdot x} dx, \qquad \Pi(k) = 1 - \frac{kk^{\mathsf{T}}}{|k|^2},$$

the field of the covariance $c^{-1}(d^*d)^{-1}$ on the orthogonal complement of the closed one-forms. For the bond b from x to $x + a_d e_\mu$ let $q_b(z) = a_d^{-3} \big|\{\tau \in [0, a_d] :\ z - x - \tau e_\mu \in [-\frac{1}{2}a_d, \frac{1}{2}a_d)^3\}\big|\, e_\mu$, so that $A^a(q_b)$ is the mean, over the shifts by the points of a cube of side a_d, of the line integral of the component μ of A^a along the shifted bond; put $B^a(b) = A^a(q_b)$, $B(b) = (B^a(b))_a$, and $V_d^{\mathrm{qf}}(b) = \exp(i \sum_a B^a(b)\sigma_a)$, σ_a the Pauli matrices, with $V_d^{\mathrm{qf}}(b^{-1}) = V_d^{\mathrm{qf}}(b)^{-1}$. The *quasi-free law* $\nu_d[c]$ is the law of V_d^{qf} on V_d, and $\nu[c]$ is the law on $\mathsf{V} = \prod_{d \geq 0} \mathsf{V}_d$ of the family $(V_d^{\mathrm{qf}})_{d \geq 0}$, all of whose members are built from one field A.

■

Proposition 18.154 (The torus limits as functions of the label)**.** Let $\ell = 2L^m$ be a side and $\underline{f} = (f_1, \ldots, f_n)$ a tuple on T_ℓ none of whose entries is 0.

(1) *(Each cutoff.)* Use the real construction of Definition 18.159(1). For every $K \in \mathbb{N}$ the function $\mathsf{b} \mapsto S^T_{n;K,\ell,L^K\mathsf{b}}(\underline{f})$ is real-analytic on $(0,\infty)$, and $\partial_{\mathsf{b}}^\nu S^T_{n;K,\ell,L^K\mathsf{b}}(\underline{f}) = (-1)^\nu S^T_{n+\nu;K,\ell,L^K\mathsf{b}}(\underline{f}^{+\nu})$ for every $\nu \geq 0$ and $\mathsf{b} > 0$.

(2) *(The limit is smooth.)* Under the same constants the function $\mathsf{b} \mapsto S_n^{\ell,\mathsf{b}}(\underline{f})$ is infinitely differentiable on $(0,\infty)$, with

$$\partial_{\mathsf{b}}^\nu S_n^{\ell,\mathsf{b}}(\underline{f}) = (-1)^\nu S_{n+\nu}^{\ell,\mathsf{b}}(\underline{f}^{+\nu}) \qquad (\nu \geq 0,\ \mathsf{b} > 0),$$

and for every ν the derivative of order ν at the cutoff K tends to the derivative of order ν of the limit as $K \to \infty$, uniformly on the compact subsets of $(0,\infty)$. With a rate: let $0 < \mathsf{b}_- \leq \mathsf{b}_+$, let $s \geq 0$ be an integer with $L^s\mathsf{b}_- \geq \gamma^{-2}$ and $M\, R(L^s\mathsf{b}_+) \leq L^{m+s}$ (all large s are such by Lem. YM3_I:118.2), let $0 < \mathsf{d} \leq \mathsf{d}_\infty(\underline{f})$, and let C'_s, $K_{1,s}(\mathsf{d})$ be the numbers of

Theorem YM3_I:AS formed for $n+\nu$ entries, with $\mathsf{p}_{n+\nu} = 2\mathfrak{p}_0(n+\nu+1)+\frac{1}{2}$; then for all $\mathsf{b} \in [\mathsf{b}_-, \mathsf{b}_+]$ and all integers $K \ge K_{1,s}(\mathsf{d}) + s$

$$|\partial_{\mathsf{b}}^{\nu} S^{T}_{n;K,\ell,L^K\mathsf{b}}(\underline{f}) - \partial_{\mathsf{b}}^{\nu} S^{\ell,\mathsf{b}}_{n}(\underline{f})| \le L^{s(n+\nu)}\, \mathsf{C}'_s\, (L^s\ell)^{3(n+\nu+1)} (K-s)^{\mathsf{p}_{n+\nu}} L^{-(K-s)/10} \prod_{i=1}^{n} \|f_i\|_\sharp,$$

and the two numbers C'_s, $K_{1,s}(\mathsf{d})$ serve all tuples of n entries with spread at least d and all labels of the interval.

(3) *(Lipschitz in the label.)* Under the same constants and with the same $\mathsf{b}_\pm$, s, d, put $r = n+1$, $Y = \underline{f}^{+1}$ and let $k_* = K_{1,s}(\mathsf{d})$ and C'_s here be the constants for r entries. Write $\mathsf{op}_r = \sum_\pi |\pi|!$ over partitions of $\{1,\dots,r\}$ and set

$$\mathsf{L} = \mathsf{op}_r \prod_{i=1}^{r} (6\ell^3 L^{4(k_*+s)} \|Y_i\|_\infty) + L^{sr}\mathsf{C}'_s (L^s\ell)^{3(r+1)} k_*^{\mathsf{p}_r} L^{-k_*/10} \prod_{i=1}^{r} \|Y_i\|_\sharp.$$

Then then $|S^{T}_{n;K,\ell,L^K\mathsf{b}}(\underline{f}) - S^{T}_{n;K,\ell,L^K\mathsf{b}'}(\underline{f})| \le \mathsf{L}|\mathsf{b}-\mathsf{b}'|$ for all $K \in \mathbb{N}$ and $\mathsf{b}, \mathsf{b}' \in [\mathsf{b}_-, \mathsf{b}_+]$, and the same bound holds for the limits.

(4) *(A strip.)* Let the constants be a sourced setting $\mathfrak{s}$ with radius ϱ from Definition 18.159(1). Let $0 < \mathsf{b}_- < \mathsf{b}_+$, and let $s \ge 0$ be an integer with

$$L^s\mathsf{b}_- \ge 2\gamma^{-2}, \qquad 2L^{m+s} \in \mathfrak{L}\big((\tfrac{1}{2}L^s\mathsf{b}_+)^{-1/2}\big);$$

all large s are such. Put

$$\upsilon_s = \tfrac{1}{4} L^{-s} \varrho \exp\Big[-\Big(C_{\mathrm{S}} + C_{\mathrm{st}} + \tfrac{9}{2}\log\big(L(\tfrac{1}{2}L^s\mathsf{b}_+ + B_\sharp)\big)\Big)(L^s\ell)^3\Big],$$

with the source stability constants C_{S} and C_{st} of Definition 18.159(4). There is exactly one holomorphic function $\Phi[\underline{f}]$ on $\Omega_{\upsilon_s}([\mathsf{b}_-, \mathsf{b}_+])$ that equals $S^{\ell,\mathsf{b}}_n(\underline{f})$ at the points $\mathsf{b} \in [\mathsf{b}_-, \mathsf{b}_+]$; it is bounded, and it equals $S^{\ell,\mathsf{b}}_n(\underline{f})$ at every real point b of its domain. Consequently $\mathsf{b} \mapsto S^{\ell,\mathsf{b}}_n(\underline{f})$ is real-analytic on $(0,\infty)$; for $\mathsf{b} \in [\mathsf{b}_-, \mathsf{b}_+]$ and real t with $|t| < \upsilon_s$

$$S^{\ell,\mathsf{b}+t}_n(\underline{f}) = \sum_{\nu \ge 0} \frac{(-t)^\nu}{\nu!}\, S^{\ell,\mathsf{b}}_{n+\nu}(\underline{f}^{+\nu}), \qquad |S^{\ell,\mathsf{b}}_{n+\nu}(\underline{f}^{+\nu})| \le \nu!\, \upsilon_s^{-\nu} \sup|\Phi[\underline{f}]|;$$

and for every $\mathsf{b} \ge \gamma^{-2}$ the interval $(\mathsf{b}-2\varrho, \mathsf{b}+2\varrho)$ lies in $(0,\infty)$, so the limit is real-analytic on it.

(5) *(A sum rule with one insertion of* 1*.)* Let the constants be a setting $\mathfrak{s}$, let $0 < g \le \gamma_W$ with the ceiling γ_W of YM3_I:(2346), $\ell \in \mathfrak{L}(g)$, $\vartheta \in (0,1)$, $0 < \mathsf{d} \le \mathsf{d}_0(g,\vartheta)$ with d_0 of YM3_I:(2350), and let $(f,h) \in \mathfrak{T}_{T_\ell}(\mathsf{d};\vartheta)$ be a test pair of Definition 18.159(5). Let $\mathsf{b}_1 > 0$ with $\mathsf{b}_1 \in I(g)$ and $2\mathsf{b}_1 \in I(g)$. Then $(1,f,h)$ is a tuple on T_ℓ none of whose entries is 0, and

$$\int_{\mathsf{b}_1}^{2\mathsf{b}_1} S^{\ell,\mathsf{b}}_3(1,f,h)\, d\mathsf{b} = S^{\ell,\mathsf{b}_1}_2(f,h) - S^{\ell,2\mathsf{b}_1}_2(f,h) \ge \tfrac{1}{3}\,\mathsf{c}_2\mathsf{b}_1^{-2}\mathcal{K}_{\mathrm{free}}(f,h) \ge \tfrac{1}{3}\,10^{-6}\vartheta^2\,\mathsf{c}_2\mathsf{b}_1^{-2}\|f\|_\infty\|h\|_\infty > 0,$$

with the constant c_2 and the pair kernel $\mathcal{K}_{\mathrm{free}} = \mathcal{K}_6$ of Definition 18.159(5). Hence $\mathsf{b} \mapsto S^{\ell,\mathsf{b}}_3(1,f,h)$ is not identically 0 on $(0,\infty)$, and its zeros have no limit point in $(0,\infty)$.

■

Proof. The exact finite-cutoff derivative and its convergence follow from Prop. 18.160. The constant insertions do not alter the spread and have $\|1\|_\sharp = 1$. For the explicit rate, apply Cor. YM3_I:117.1 on the dilated torus to $(\underline{f}^{+\nu})^{(s)}$, whose spread is at least $L^s\mathsf{d}$. One can use the weaker threshold formed at spread d because $s \ge 0$. The dilation identity YM3_I:(2209) supplies the factor $L^{s(n+\nu)}$, and the dilated entries have no larger norms. This proves clauses (1) and (2).

For clause (3), the cumulant partition formula and the bound $|\mathcal{O}(f)| \le 6\ell^3 L^{4K}\|f\|_\infty$ give the first term of L when $K \le k_* + s$, including at $K = k_* + s$. For $K \ge k_* + s$, apply YM3_I:(2207) at

the base cutoff k_* on the dilated torus, comparing with cutoff $K-s$, and use the same dilation as above. This costs at most the second term of L. It follows that the absolute value of the first derivative in the label is at most L at every cutoff. Integration gives the Lipschitz inequality, which survives the limit in clause (2).

For clause (4), Prop. 18.161 applies to every empty-intersection tuple. Its proof uses a finite partition of unity into small cubes, so each nonzero piece has a separated pair, and takes the same holomorphic width for all pieces. The width in the present statement is exactly the width there, since $X_{g(L^s\mathsf{b}_+)} = L(L^s\mathsf{b}_+/2 + B_\sharp)$. The same proposition gives the bounds and the absolutely convergent series with constant entries. The canonical source room gives $2\varrho \le \gamma^{-2} - \bar{B} < \gamma^{-2}$, so the indicated real interval about every $\mathsf{b} \ge \gamma^{-2}$ is positive.

Finally integrate $\partial_{\mathsf{b}} S_2^{\ell,\mathsf{b}}(f,h) = -S_3^{\ell,\mathsf{b}}(1,f,h)$ on $[\mathsf{b}_1, 2\mathsf{b}_1]$. The two-point estimates of Prop. 18.141(3)(a1) at the endpoints, with relative error at most $1/3$, give

$$S_2^{\ell,\mathsf{b}_1}(f,h) - S_2^{\ell,2\mathsf{b}_1}(f,h) \ge \mathsf{c}_2 \mathsf{b}_1^{-2} \mathcal{K}_6(f,h) \left[(1-\tfrac{1}{3}) - \tfrac{1}{4}(1+\tfrac{1}{3})\right] = \tfrac{1}{3}\mathsf{c}_2 \mathsf{b}_1^{-2} \mathcal{K}_6(f,h).$$

The lower bound for the kernel is in Definition 18.159(5). The integral is positive, so the analytic integrand has only isolated zeros in $(0,\infty)$. □

Proposition 18.155 (The statistic on one thread; what gauge invariance alone gives)**.** Use Definition 18.153 with an admitted interval $J = [\mathsf{b}_-, \mathsf{b}_+]$ on the side ℓ. For $d \ge d_\natural$ let $K(d)$ be the cutoff of Corollary 18.213, and put $T_d = T_d^{K(d)}$, $m_d^K(s) = \int T_d^K \, d\mu_d^K[s]$ and $m_d(s) = m_d^{K(d)}(s)$ for $s \in J$. Set

$$\mathsf{v}_d = 2E_0 C_{\mathrm{E},0} L^{-3} N_d, \qquad \mathsf{A} = 8E_0 C_{\mathrm{E},0} \varrho_\varsigma^{-2} L^{-3} \log L.$$

(1) *(Tails at a cutoff.)* Let $\mathsf{b} \in J$ and let $u_\flat \le d' < d < K$ be integers with $\Xi_d[\mathsf{b}_-] \ge 2$. For every real λ with $0 \le \lambda \le \mathsf{v}_d/(8\varrho_\varsigma^2 \mathsf{Q}' L^{4d'})$,

$$\mu_d^K[\mathsf{b}]\Big(\big|T_d^K - m_d^K(\mathsf{b})\big| \ge \lambda + 2\rho[\mathsf{b}](d')\,\mathsf{Q}' L^{4d}\Big) \le 4\exp\Big[-\frac{\varrho_\varsigma^2 \lambda^2}{2\mathsf{v}_d}\Big] + 2\rho[\mathsf{b}](d').$$

(2) *(On one thread.)* For every $\mathsf{b} \in J$ and $\mu_{\ell,\mathsf{b}}$-almost every thread V there is an integer d_V with

$$\big|T_d(V_d) - m_d(\mathsf{b})\big| \le (\mathsf{A}\, d\, N_d)^{1/2} + 1 \qquad \text{for all } d \ge d_V.$$

(3) *(The mean and one thread.)* Let $\mathsf{b} \in J$ and $\varkappa_0 \in \mathbb{R}$, and put $\xi_d(V) = L^d[N_d^{-1} T_d(V_d) - \varkappa_0 \mathsf{b}^{-1}]$ and $x_d = L^d[N_d^{-1} m_d(\mathsf{b}) - \varkappa_0 \mathsf{b}^{-1}]$. For $\mu_{\ell,\mathsf{b}}$-almost every V, $|\xi_d(V) - x_d| \le \mathsf{A}^{1/2} d^{1/2} \ell^{-3/2} L^{-d/2} + \ell^{-3} L^{-2d}$ for all large d; so for every real y the event $\{\xi_d \to y\}$ has the $\mu_{\ell,\mathsf{b}}$-probability 1 if $x_d \to y$ and 0 otherwise. In particular, for real $\varkappa_1$ the expansion

$$m_d(\mathsf{b}) = N_d[\varkappa_0 \mathsf{b}^{-1} + \varkappa_1 L^{-d} \mathsf{b}^{-2} + o(L^{-d})] \qquad (d \to \infty)$$

holds if and only if $\xi_d \to \varkappa_1 \mathsf{b}^{-2}$ $\mu_{\ell,\mathsf{b}}$-almost surely; and if it holds with $\varkappa_0 > 0$, then $\mu_{\ell,\mathsf{b}}$-almost surely

$$L^d\Big(\frac{\varkappa_0 N_d}{T_d(V_d)} - \mathsf{b}\Big) \to -\frac{\varkappa_1}{\varkappa_0}, \qquad L^d\Big[\frac{T_d(V_d)}{N_d} - \frac{T_{d+1}(V_{d+1})}{N_{d+1}}\Big] \to \frac{\varkappa_1}{\mathsf{b}^2}\Big(1 - \frac{1}{L}\Big).$$

(4) *(Gauge invariance alone.)* Let P be a gauge-invariant Borel probability measure on V_d. Under P the variables $V(c_y)$, y an even site, are independent, and each has the Haar law of SU(2); hence $P(E_d) = h_0^{N_d/2}$.

(5) *(Against the quasi-free laws.)* For every $c > 0$ and every $d \ge 0$, $\nu_d[c](\mathsf{V}_d \setminus E_d) \le 3N_d e^{-c/(12a_d)}$, and for every gauge-invariant Borel probability measure P on V_d

$$\mathrm{aff}(P, \nu_d[c]) \le h_0^{N_d/4} + (3N_d)^{1/2} e^{-c/(24a_d)} \le e^{-c/(48a_d)},$$

the second inequality from an integer d_c on, which depends on c and ℓ alone; for any $\mathsf{b} > 0$ the last member is $\exp[-k\ell^3/(\mathsf{b}^2 a_d)]$ with $k = c\mathsf{b}^2/(48\ell^3)$. The Borel set $\mathbf{E} = \{V \in \mathsf{V} : V_d \in$

E_d for all large $d\}$ has $\nu[c](\mathbf{E}) = 1$ for every $c > 0$, and $\mathbf{P}(\mathbf{E}) = 0$ for every Borel probability measure $\mathbf{P}$ on V whose images under the maps $V \mapsto V_d$ are gauge invariant. In particular $\mu_{\ell,\mathsf{b}}$ and $\nu[c]$ are mutually singular for all real $\mathsf{b} > 0$ and $c > 0$. What (4) and (5) assert on V_d uses of P only that it is gauge invariant; it applies to $P = \mu_d[\mathsf{b}]$ and, unchanged, to $P = \bar{\nu}_d[c']$, $c' > 0$, the average of a quasi-free law over the gauge group.

(6) *(On the invariant sets.)* Let P be a gauge-invariant and ν any Borel probability measure on V_d. Then ν and $\bar{\nu}$ agree on $\mathscr{I}_d$, and

$$\mathrm{aff}(P, \bar{\nu}) = \mathrm{aff}\big(P|_{\mathscr{I}_d}, \nu|_{\mathscr{I}_d}\big) \geq \mathrm{aff}(P, \nu).$$

■

Sketch. Step 1: the generating function of the statistic. Set $t = K - d$ and $t' = K - d'$ and take $t_* = t'$ in the calibration. Let $\tilde{\nu}$ and $\hat{\nu}$ be the laws of V_d and $V_{d'}$ under the normalized weight $\mathfrak{w}^{\mathrm{sf}}_{t'}\rho_0\, dU$ of Prop. YM3_I:115.2. The discarded fraction is at most $\mathsf{R}_{t'} \leq \rho[\mathsf{b}](d') \leq 1/4$ by YM3_I:(2187), since $\log L^{d'}\mathsf{b} \geq d' \log L$ and $2\mathfrak{p}_0 \geq 2$. The weight includes the cut at level $t < t'$. The cut criterion in Section YM3_I:113.4 and the decreasing ε give $\mathsf{Cut}_t[\mathsf{b}] \subset \mathsf{Cut}_t[\mathsf{b}_-]$. Thus $T_d^K = \mathcal{O}(1)(U_t(V))$ on the support of the cutoff density at level t.

Start the recursion YM3_I:(2172) at level t with the unsourced chain's first t families, held constant as the source z varies on $|z| < 2\varrho_\varsigma$. These satisfy all the early-family hypotheses of Prop. YM3_I:114.2. Write $\Phi_j^{[z]}$ for the exponents of this run. At level t the old array is independent of z, so the linear weighted-action term gives $\Phi_t^{[z]} - \Phi_t^{[0]} = zT_d^K$ on the cut. At $z = 0$ the run is the original chain from t on by the restart identity. Factor $\mathfrak{w}^{\mathrm{sf}}_{t'}$ at level t: the weight through the cut at t produces the chain's density at t, and the remaining factors depend on V_d and its later averages. Apply the exact density recursion YM3_I:(2175) to the frozen early system from t to t' and restrict to $\mathsf{In}_{t'}[\mathsf{b}]$. Divide by its value at $z = 0$. The contact increments YM3_I:(2173) have source-independent normalization terms, so this yields

$$\tilde{\nu}(e^{zT_d^K}) = e^{\mathsf{c}(z)-\mathsf{c}(0)}\hat{J}_0(z), \qquad \hat{J}_0(z) = \int e^{\Phi_{t'}^{[z]} - \Phi_{t'}^{[0]}}\, d\hat{\nu},$$

where c is the sum of the flat members of levels $t+1$ through t'. The rooted bound of Lem. YM3_I:114.6 gives $|\mathsf{c}| \leq 2E_0C_{\mathrm{E},0}L^{-3}N_d = \mathsf{v}_d$ on $|z| < 2\varrho_\varsigma$. Apply Lem. YM3_I:114.5 to this early system at $f = 1$ and level t'. It gives $|\hat{J}_0 - 1| \leq e^{1/4} - 1 < 0.2841$ on $|z| < r' = (8\mathsf{M}^{\mathrm{rd}}_{t'})^{-1}$. Its readout constant satisfies $\varrho_\varsigma^{-1} \leq \mathsf{M}^{\mathrm{rd}}_{t'} \leq \mathsf{Q}'L^{4d'}$ by the definition of Q' and $\varepsilon(L^{d'}\mathsf{b}) \leq \varepsilon(L^{d'}\mathsf{b}_-)$. The upper bound on $|\hat{J}_0 - 1|$ holds at $|z| = r'$ by continuity. The argument proves this identity and the stated tail bound for every $u_\flat \leq d' < d < K$.

Step 2: an exponential moment and a tail bound under $\tilde{\nu}$. Let $\tilde{m} = \tilde{\nu}(T_d^K)$; it is the real part of $(\mathsf{c})'(0) + \hat{J}_0'(0)$. For real z with $|z| \leq r'$ the left member of the identity of Step 1 is positive, and Cauchy's bounds of the Taylor coefficients, on the radius $2\varrho_\varsigma$ for c and on the radius r' for $\hat{J}_0 - 1$, give

$$\log \tilde{\nu}\big(e^{z(T_d^K - \tilde{m})}\big) \leq \mathsf{v}_d\, \frac{(z/2\varrho_\varsigma)^2}{1 - |z|/2\varrho_\varsigma} + \log 1.2841 + 0.2841 \leq \frac{\mathsf{v}_d z^2}{2\varrho_\varsigma^2} + 0.54.$$

For $0 \leq \lambda \leq \mathsf{v}_d r'/\varrho_\varsigma^2$, Markov's inequality at $z = \pm\lambda\varrho_\varsigma^2/\mathsf{v}_d$ gives $\tilde{\nu}(|T_d^K - \tilde{m}| \geq \lambda) \leq 2e^{0.54}\exp[-\varrho_\varsigma^2\lambda^2/(2\mathsf{v}_d)]$, and $2e^{0.54} < 4$.

Step 3: back to the Wilson measure; clause (1). On the gauge-invariant algebra, the difference between the Wilson probability and its unnormalized restriction to the cutoff weight is positive by YM3_I:(2189) and has mass at most $\rho[\mathsf{b}](d')$. For a bounded function F on this algebra, first remove that difference and then rescale the remaining mass to one. The change in its expectation is at most $2\rho[\mathsf{b}](d')\|F\|_\infty$. Apply this to the statistic with $0 \leq T_d^K \leq \mathsf{Q}'L^{4d}$ and to the indicator of $\{|T_d^K - \tilde{m}| \geq \lambda\}$. The first use bounds $|m_d^K(\mathsf{b}) - \tilde{m}|$ by $2\rho[\mathsf{b}](d')\mathsf{Q}'L^{4d}$, and the second transfers the tail. The permitted range of λ follows from $r' \geq (8\mathsf{Q}'L^{4d'})^{-1}$.

Step 4: clause (2). Take $d' = \lfloor d/5 \rfloor$, $K = K(d)$ and $\lambda_d = (\mathsf{A}\, dN_d)^{1/2}$. For all large d: $d' \ge u_\flat$ and $\Xi_d[\mathsf{b}_-] \ge 2$; λ_d, of the order $d^{1/2}L^{3d/2}$, lies below $\mathsf{v}_d/(8\varrho_\varsigma^2 \mathsf{Q}' L^{4d'})$, of the order $L^{11d/5}$; and $2\rho[\mathsf{b}](d')\mathsf{Q}'L^{4d} \le 1$, because $\theta_0 d'^2$ beats every multiple of d. As $\varrho_\varsigma^2\lambda_d^2/(2\mathsf{v}_d) = 2d\log L$, clause (1) and Corollary 18.213 give

$$\mu_d[\mathsf{b}](|T_d - m_d(\mathsf{b})| > \lambda_d + 1) \le 4L^{-2d} + 2\rho[\mathsf{b}](\lfloor d/5 \rfloor) + L^{-d},$$

a summable sequence. The map $V \mapsto V_d$ carries $\mu_{\ell,\mathsf{b}}$ to $\mu_d[\mathsf{b}]$ (Lemma 18.212(b)), and the lemma of Borel and Cantelli gives clause (2).

Step 5: clause (3). $|\xi_d - x_d| = L^d N_d^{-1}|T_d - m_d(\mathsf{b})|$, and clause (2) with $N_d = \ell^3 L^{3d}$ gives the bound, which tends to 0; so ξ_d converges exactly when the number x_d does, and to the same limit. The expansion says $x_d \to \varkappa_1 \mathsf{b}^{-2}$. If it holds with $\varkappa_0 > 0$, then almost surely $T_d/N_d = \varkappa_0 \mathsf{b}^{-1} + L^{-d}\xi_d$ with $\xi_d \to \varkappa_1 \mathsf{b}^{-2}$; so $T_d > 0$ for large d, $\varkappa_0 N_d/T_d = \mathsf{b}(1 + \mathsf{b}\varkappa_0^{-1}L^{-d}\xi_d)^{-1} = \mathsf{b} - \mathsf{b}^2\varkappa_0^{-1}L^{-d}\xi_d + O(L^{-2d})$, and $L^d[T_d/N_d - T_{d+1}/N_{d+1}] = \xi_d - L^{-1}\xi_{d+1}$.

Step 6: clause (4). Let $u \in \mathcal{G}_d$ be random, with independent Haar values at the even sites and the value $\mathbf{1}$ at the odd sites, independent of V, which has the law P. As P is invariant under every fixed gauge transformation, V^u has the law P. The bond c_y goes from the even site y to an odd site, so $V^u(c_y) = u(y)V(c_y)$; given V these are independent and Haar distributed, by the invariance of the Haar measure under left translations, the bonds c_y being distinct. An element of SU(2) with the eigenvalues $e^{\pm i\theta}$, $0 \le \theta \le \pi$, has $|g - \mathbf{1}|_{\mathrm{op}} = 2\sin(\theta/2)$, and the class angle has the Haar density $\frac{2}{\pi}\sin^2\theta$; so the Haar measure of $\{|g - \mathbf{1}|_{\mathrm{op}} \le 1\} = \{\theta \le \pi/3\}$ is $\frac{2}{\pi}\int_0^{\pi/3}\sin^2\theta\, d\theta = h_0$.

Step 7: one bond of the quasi-free field. For the bond b from x to $x + a_d e_\mu$, $|\hat{q}_b(k)| = a_d|\operatorname{sinc}(k_\mu a_d/2)|\prod_{\nu=1}^3 |\operatorname{sinc}(k_\nu a_d/2)| \le a_d \prod_\nu \min\{1, 2/(a_d|k_\nu|)\}$, with $\operatorname{sinc} v = v^{-1}\sin v$. With $p = a_d k \in (2\pi/n_s)\mathbb{Z}^3$, $n_s = \ell L^d$, and $\Pi_{\mu\mu} \le 1$,

$$\operatorname{Var} B^a(b) \le \frac{a_d}{c}\, n_s^{-3} \sum_{p \ne 0} |p|^{-2} \prod_{\nu=1}^3 \min\{1, 4p_\nu^{-2}\}.$$

Write $p = \kappa + 2\pi j$ with $\kappa \in (-\pi, \pi]^3$, $j \in \mathbb{Z}^3$. For the terms with $j = 0$ write $\kappa = 2\pi n/n_s$: the shell $|n|_\infty = r$ holds $24r^2 + 2$ points, all with $|n| \ge r$, so these terms add up to at most $(n_s/2\pi)^2(12n_s + \pi^2/3) \le 13n_s^3/(4\pi^2)$, as $n_s \ge 4$. The terms with $j \ne 0$ have $|p| \ge \pi$, and $\sum_{j_\nu \in \mathbb{Z}} \min\{1, 4(\kappa_\nu + 2\pi j_\nu)^{-2}\} \le 1 + \frac{8}{\pi^2}\sum_{m \ge 1,\ m \text{ odd}} m^{-2} = 2$ for $|\kappa_\nu| \le \pi$; so they add up to at most $8n_s^3/\pi^2$. Hence $\operatorname{Var} B^a(b) \le \frac{45}{4\pi^2}\frac{a_d}{c} \le \frac{2a_d}{c}$, and, the three colours being independent, $\mathbb{P}(|B(b)| > 1) \le 3 \cdot 2\exp[-\frac{1}{3}/(2 \cdot 2a_d/c)] = 6e^{-c/(12a_d)}$. As $i\sum_a B^a\sigma_a$ has the eigenvalues $\pm i|B|$, $|V_d^{\mathrm{qf}}(b) - \mathbf{1}|_{\mathrm{op}} = 2|\sin(|B(b)|/2)| \le |B(b)|$; a union over the $\frac{1}{2}N_d$ even sites gives $\nu_d[c](\mathsf{V}_d \setminus E_d) \le 3N_d e^{-c/(12a_d)}$.

Step 8: clause (5). Splitting the affinity over E_d and its complement, the inequality of Cauchy and Schwarz on each part gives $\operatorname{aff}(P, \nu_d[c]) \le P(E_d)^{1/2} + \nu_d[c](\mathsf{V}_d \setminus E_d)^{1/2}$, and Steps 6 and 7 give the first bound. As $N_d = \ell^3 a_d^{-3}$, the first term is $\exp[-\frac{1}{4}\ell^3 a_d^{-3}\log h_0^{-1}]$ and the second is a power of a_d^{-1} times $e^{-c/(24a_d)}$; both are below $\frac{1}{2}e^{-c/(48a_d)}$ from a threshold on that depends on c and ℓ alone. The set $\mathbf{E}$ is a lim inf of cylinder sets, hence Borel. As $\sum_d \nu_d[c](\mathsf{V}_d \setminus E_d) < \infty$, $\nu[c](\mathbf{E}) = 1$ by Borel and Cantelli; if the images $\mathbf{P}_d$ of $\mathbf{P}$ are gauge invariant, $\sum_d \mathbf{P}_d(E_d) = \sum_d h_0^{N_d/2} < \infty$, so $\mathbf{P}$-almost surely $V_d \notin E_d$ for all large d, and $\mathbf{P}(\mathbf{E}) = 0$. The images of $\mu_{\ell,\mathsf{b}}$ are the gauge-invariant laws $\mu_d[\mathsf{b}]$, for every $\mathsf{b} > 0$ (Lemma 18.230). Steps 6–8 use of the law only its gauge invariance, and a gauge average is gauge invariant, since $(\nu^u)^{u'} = \nu^{u'u}$ and the Haar measure is invariant.

Step 9: clause (6). For $E \in \mathscr{I}_d$ and $u \in \mathcal{G}_d$, $\nu^u(E) = \nu(\{V : V^u \in E\}) = \nu(E)$; so $\bar{\nu} = \nu$ on $\mathscr{I}_d$. Let $\lambda = \frac{1}{2}(P + \bar{\nu})$, a gauge-invariant law. If p is a density of a gauge-invariant law against λ, so is $V \mapsto p(V^u)$ for every u, and so is the average $\int p(V^u)\, du$, which is gauge invariant and hence $\mathscr{I}_d$-measurable; with such densities for P and $\bar{\nu}$, $\operatorname{aff}(P, \bar{\nu})$ is computed on $\mathscr{I}_d$, where it equals $\operatorname{aff}(P|_{\mathscr{I}_d}, \nu|_{\mathscr{I}_d})$. To compare with $\operatorname{aff}(P, \nu)$, let $\lambda_\nu = \frac{1}{2}(P + \nu)$, a probability measure

against which P and ν have densities p' and q' on all Borel sets. For a sub-σ-algebra $\mathscr{A}$ the conditional means $\mathbb{E}_{\lambda_\nu}[p'|\mathscr{A}]$ and $\mathbb{E}_{\lambda_\nu}[q'|\mathscr{A}]$ exist, λ_ν being finite, and they serve as densities of the restrictions $P|_{\mathscr{A}}$ and $\nu|_{\mathscr{A}}$ with respect to $\lambda_\nu|_{\mathscr{A}}$; the conditional inequality of Cauchy and Schwarz gives $\mathbb{E}_{\lambda_\nu}[(p'q')^{1/2}|\mathscr{A}] \le (\mathbb{E}_{\lambda_\nu}[p'|\mathscr{A}]\,\mathbb{E}_{\lambda_\nu}[q'|\mathscr{A}])^{1/2}$ λ_ν-almost everywhere, and integrating both members against λ_ν gives $\mathrm{aff}(P,\nu) \le \mathrm{aff}(P|_{\mathscr{A}}, \nu|_{\mathscr{A}})$. At $\mathscr{A} = \mathscr{I}_d$ this is the inequality of clause (6). □

Lemma 18.156 (A rough real source lies in the class; the chain forgets a source in the sup norm)**.** Use the calibrated line $x_j = L^{K'-j}\mathsf{b}'$ and top level t_* of §2.3.1 on (ℓ, K'). Use the cuts of Lem. YM3_I:113.10 with $\mathsf{e}^{\mathrm{c}}_j = c_\rho \varepsilon(x_j)$, and the barycentre source class $\mathsf{Src} = \mathsf{Src}_{\mathsf{Bc}}(2\varrho_\varsigma, 2\varrho_\varsigma)$ as in YM3_I:(1497). Here Bc is the set of bare plaquette barycentres, $|w|_\infty$ is the maximum modulus, and Lip_∞ uses the torus sup distance. Let $E^{(i)}[w]$, $\mathsf{c}_i(w)$, Φ^w_t be the chain families, contact increments and exponents in YM3_I:(2172)–(2175). Write $v^w_j = -w/\mathsf{b}'$, $\mathsf{s}^{\mathrm{abs}}_j = (1+\beta_{\mathrm{s}})\delta(x_j)/\alpha_0$, $\Delta_i(w) = \|E^{(i)}[w] - E^{(i)}[0]\|_{i,\kappa+1}$ and $n_t(w) = \sum_{i=1}^{t} L^{-(t-i)}\Delta_i(w)$. Put

$$C_n = 2C_{\mathrm{fg}}K^{\mathrm{d}}\big(\tfrac{13}{12}K^\vee_{\mathrm{Q}} + \tfrac{3}{2}K^\vee_{\mathrm{L}}\big), \qquad C_{\mathrm{src}} = \tfrac{3}{8} + C_{\mathrm{old}}C_n, \qquad \epsilon_{\mathrm{src}}(D;\vartheta) = C_{\mathrm{src}}\,\ell^3 L^{4D}\vartheta,$$

with K^{d} of Section YM3_I:114.1, C_{fg} of Prop. YM3_I:114.3 and C_{old} of YM3_I:(2163); C_n and C_{src} are formed from the constants alone.

(1) Let $\tau \ge 0$, and let w be a complex weight of the bare plaquettes with $|w(p) - w(p')| \le 2\tau L^{-K'}$ for all bare plaquettes p, p'. Then $\mathrm{Lip}_\infty(w; \mathsf{Bc}) \le 4\tau$; hence $w \in \mathsf{Src}_{\mathsf{Bc}}(2\varrho_0, 2\varrho_0)$ whenever $\varrho_0 > 0$, $|w|_\infty < 2\varrho_0$ and $\tau < \sqrt{3}\varrho_0/(80M)$.

(2) For $w \in \mathsf{Src}$ and $0 \le j < t_*$,

$$\Delta_{j+1}(w) \le K^{\mathrm{d}}K_\Lambda \mathsf{s}^{\mathrm{abs}}_j\, n_j(w) + K^{\mathrm{d}}\big(2K^\vee_{\mathrm{Q}} + 2K^\vee_{\mathrm{L}}x_j^{1/2}\big)\,|w|_\infty/\mathsf{b}'.$$

(3) For $w \in \mathsf{Src}$, $1 \le t \le t_*$ and $D = K' - t$: $n_t(w) \le C_n|w|_\infty L^{(D+1)/2}$, and $|\Phi^w_t(V) - \Phi^0_t(V)| \le \epsilon_{\mathrm{src}}(D; |w|_\infty)$ at every $V \in \mathsf{Cut}_t$.

■

Proof. Distinct plaquette barycentres in the periodic lattice differ in some coordinate by at least $a_{K'}/2$. This proves the Lipschitz estimate and the source admission in (1).

At level j, keep the early array fixed and remove the direct source potential. The fourth-family Lipschitz estimate YM3_I:(1603), in the range verified in the proof of Prop. YM3_I:114.2, bounds the change in the next family by $K^{\mathrm{d}}\mathsf{n}^{\mathrm{f}}(\mathbb{W}^{\mathrm{src}}[v^w_j])$. By YM3_I:(2167) and $\mathrm{sw}_j(v) \le 2|v|_\infty$, this is at most $K^{\mathrm{d}}(2K^\vee_{\mathrm{Q}} + 2K^\vee_{\mathrm{L}}\sqrt{x_j})|w|_\infty/\mathsf{b}'$. With the source removed, the change from the old array at w to that at zero is at most $K^{\mathrm{d}}K_\Lambda \mathsf{s}^{\mathrm{abs}}_j n_j(w)$ by the one-step bound in the proof of Prop. YM3_I:114.3. The old array is empty if $j = 0$. The triangle inequality proves (2).

For (3), put $q_j = K^{\mathrm{d}}K_\Lambda \mathsf{s}^{\mathrm{abs}}_j$ and let σ_j be the second summand of (2). Then $n_{j+1} \le (L^{-1} + q_j)n_j + \sigma_j$, with $n_0 = 0$. By YM3_I:(2176)–(2177), the product from $i = j+1$ to $t-1$ of $L^{-1} + q_i$ is at most $C_{\mathrm{fg}}L^{-(t-1-j)}$. Summing the two geometric series, with $D = K' - t$, gives

$$n_t \le 2C_{\mathrm{fg}}K^{\mathrm{d}}|w|_\infty \left(\frac{K^\vee_{\mathrm{Q}}}{\mathsf{b}'}\frac{L}{L-1} + \frac{K^\vee_{\mathrm{L}}}{\sqrt{\mathsf{b}'}}\frac{L^{(D+1)/2}}{1 - L^{-1/2}}\right) \le C_n|w|_\infty L^{(D+1)/2},$$

using $L \ge 13$ and $\mathsf{b}' \ge 4$. The action term in the exponent changes by $L^D A^{\eta_t}(w, U_t(V))$ and has modulus at most $\frac{3}{8}N_D L^D|w|_\infty$, by YM3_I:(2161) and $2\mathsf{e}^{\mathrm{c}}_t \le 1$. The remaining change is at most $C_{\mathrm{old}}N_D n_t(w)$ by YM3_I:(2164). Since $D \ge 1$ and $N_D = \ell^3 L^{3D}$, these bounds give $\epsilon_{\mathrm{src}}(D; |w|_\infty)$. □

Lemma 18.157 (A real source: floor, tails and discarding; the laws of the block fields)**.** Use the fixed choices of Definition 18.153 and the sourced construction of Definition 18.159(1) with radius ϱ, with common real data. Choose the calibration to satisfy Lem. YM3_I:82.5 as well as Section YM3_I:115, using the same $\eth \in (0,1)$. The history summability thresholds are imposed

before the torus, cutoff and label. For an admitted interval $J = [\mathsf{b}_-, \mathsf{b}_+]$ put $\varrho_\wedge = \min\{\varrho, \varrho_\varsigma\}$ and $\mathsf{Src}_\wedge = \mathsf{Src}_{\mathsf{Bc}}(2\varrho_\wedge, 2\varrho_\wedge)$. For $K' > d_\natural$ and $\mathsf{b}' \in J$ take $t_* = K' - d_\natural$. At depth $D = K' - t$ the mass bound YM3_I:(2187) is

$$\mathsf{R}(D;\mathsf{b}') = 2M^{-3}\ell^3 L^{3D}\exp\bigl[-c' A_0^2(\log L^D \mathsf{b}')^{2\mathfrak{p}_0}\bigr]\exp\bigl[(\mathsf{C}^\Sigma_{\mathrm{maj}} + 2 + C_{\mathrm{low}}(\mathsf{b}'))\ell^3\bigr],$$

with C_{low} of YM3_I:(1538). Let $D_1 \ge d_\natural$ be the least integer from which $\mathsf{R}(D;\mathsf{b}') \le 1/2$ holds for all $\mathsf{b}' \in J$. The exponential decay in D makes this threshold finite. Use the source history constants $c^{\mathrm{S}}_\star$ and $\mathsf{C}^{\mathrm{S}}_{\mathrm{maj}}$ of YM3_I:(1725)–(1726), and put $\lambda_L = (6L^3+1)\log 2$, $c'_{\mathrm{S}} = (1-\eth)c^{\mathrm{S}}_\star$ and $\mathsf{e}_1 = 4E_0 C_{\mathrm{E},0}\ell^3 L^{3(D_1-1)}$. For $\vartheta \ge 0$ set

$$\mathsf{R}^{\mathrm{src}}(D;\mathsf{b}',\vartheta) = 4M^{-3}\ell^3 L^{3D}\exp\bigl[-c'_{\mathrm{S}} A_0^2(\log L^D \mathsf{b}')^{2\mathfrak{p}_0}\bigr]\exp\bigl[(\mathsf{C}^{\mathrm{S}}_{\mathrm{maj}} + \lambda_L + 2 + C_{\mathrm{low}}(\mathsf{b}'))\ell^3 + \mathsf{e}_1 + \epsilon_{\mathrm{src}}(D_1;\vartheta)\bigr],$$

with ϵ_{src} of Lem. 18.156. Let $K' > D_1$, $\mathsf{b}' \in J$ and $w \in \mathsf{Src}_\wedge$ be real with $|w|_\infty \le \vartheta$. Put $\rho_0^w = \exp[-\sum_p (L^{K'}\mathsf{b}' - L^{K'}w(p))a_p]$, $Z[w] = \int \rho_0^w\, dU$, $P^w = Z[w]^{-1}\rho_0^w\, dU$ and $Z = Z[0]$. Use $\mathfrak{w}^{\mathrm{sf}}_t$ of Prop. YM3_I:115.2, the cut Cut_0 of Section YM3_I:113.4 and $a_t(w) = \sum_{0<i\le t}(\mathsf{c}_i(w) - \mathsf{c}_i(0))$.

(1) *(Floor.)* Every $a_t(w)$ is real, and with $t_\flat = K' - D_1$, $Z[w] \ge \frac{1}{2}e^{a_{t_\flat}(w) - \epsilon_{\mathrm{src}}(D_1;\vartheta)}Z$.

(2) *(Discarding.)* For $1 \le t \le t_*$ and $D = K' - t$, $0 \le Z[w] - \int \mathfrak{w}^{\mathrm{sf}}_t \rho_0^w\, dU \le \mathsf{R}^{\mathrm{src}}(D;\mathsf{b}',\vartheta)\, Z[w]$; and $P^w(\mathsf{V}_{K'} \setminus \mathsf{Cut}_0) \le \mathsf{R}^{\mathrm{src}}(K'-1;\mathsf{b}',\vartheta)$.

(3) *(Block laws.)* For $d_\natural \le D < K'$ and $0 \le d \le D$,

$$\bigl\|(\mathcal{M}^{K'-d})_* P^w - \mu_d^{K'}[\mathsf{b}']\bigr\| \le \eta(D;\mathsf{b}',\vartheta) := \mathsf{R}^{\mathrm{src}}(D;\mathsf{b}',\vartheta) + \mathsf{R}(D;\mathsf{b}') + 2\epsilon_{\mathrm{src}}(D;\vartheta).$$

(4) *(Mixtures; a thin factor.)* Take a Borel set $\mathcal{S} \subset \mathsf{Src}_\wedge$ whose members are real sources with $|w|_\infty \le \vartheta$, a positive Borel measure $\pi_0 \ne 0$ of finite mass on $\mathcal{S}$, and a bounded, gauge-invariant Borel function $H \ge 0$ on $\mathsf{V}_{K'}$ such that $H \le \delta_0$ on Cut_0; put $\Psi = \int \rho_0^w\, \pi_0(dw)$. Then $P = \Psi e^{-H}dU / \int \Psi e^{-H} dU$ is a gauge-invariant law, and for $d_\natural \le D < K'$ and $0 \le d \le D$

$$\bigl\|(\mathcal{M}^{K'-d})_* P - \mu_d^{K'}[\mathsf{b}']\bigr\| \le \delta_0 + \mathsf{R}^{\mathrm{src}}(K'-1;\mathsf{b}',\vartheta) + \eta(D;\mathsf{b}',\vartheta).$$

■

Proof. Step 1: the common regular families. The constant-coupling line has zero increments and meets the sequence and room requirements of YM3_I:(1524). A real barycentre source extends to a real function on T_ℓ with the same Lipschitz bound and supremum bound. For example, take the infimum of the functions $w(p) + \Lambda|x - x(p)|_\infty$ for $\Lambda = \mathrm{Lip}_\infty w$ and clip to the range of w. The sourced history construction of Thm. YM3_I:AJ therefore applies. It uses the same real data as the small-field chain. Its ordinary regular recursion YM3_I:(1683) uses the one-lattice companion and the real localized normalizer. For $j \ge 1$ compare the companion in YM3_I:(1635) with YM3_I:(2165). The weighted Wilson family in both formulas uses the same vertex brackets of the actual shift and the same reference plaquette for each nonpivot bond. The last paragraph of p. 799 identifies these weighted members on the one-lattice cores. The formula at $j = 0$ uses the same pivot completion by YM3_I:(1652). The companion is $-\mathbb{W}^{\mathrm{src}}[v_j^w]$ in both cases. The pointed term with this companion is the same local integral by Props. YM3_I:77.3 and YM3_I:77.4, with the normalization terms independent of w. The counterterms are zero on this line. Thus induction on j identifies the regular families in (1683) and (2172). The source contribution to the vacuum through level t in YM3_I:(1736)–(1737) equals $a_t(w)$ by YM3_I:(2173), so it is real at real sources.

Step 2: the floor. At depth $D = K' - t$ the image of $\mathfrak{w}^{\mathrm{sf}}_t \rho_0^w\, dU$ is a positive measure by YM3_I:(2188). On the inner ball its density is that of YM3_I:(2175). Taking the ratio of the densities at w and 0, and using Lemma 18.156(3), bounds it between $e^{a_t(w) - \epsilon_{\mathrm{src}}(D;\vartheta)}$ and $e^{a_t(w) + \epsilon_{\mathrm{src}}(D;\vartheta)}$. The same bounds hold if the exponents are read through their real parts, since the density ratio is positive. At $t_\flat = K' - D_1$ the zero-source restricted mass is at least $Z/2$ by YM3_I:(2190). The sourced restricted mass is at most $Z[w]$ by YM3_I:(2189), tested with the positive gauge-invariant function ρ_0^w/ρ_0. This proves (1).

Step 3: the vacuum comparison. Let $E^w_{(K')}$ be the sourced vacuum of YM3_I:(1737). Its difference from $E_{(K')}$ is $a_{t_\flat}(w)$ plus the flat-family differences for $j > t_\flat$. At level j the norm bound in YM3_I:(1729) and the rooted sum used in Prop. YM3_I:83.2 bound those flat differences by $2E_0C_{\mathrm{E},0}N_{K'-j}$. Their sum is at most e_1, since $\sum_{u=0}^{D_1-1} N_u \le 2\ell^3 L^{3(D_1-1)}$. The zero-source floor YM3_I:(1538) and clause (1) give

$$e^{E^w_{(K')}} \le 2Z[w]\exp[\mathsf{e}_1 + C_{\mathrm{low}}(\mathsf{b}')\ell^3 + \epsilon_{\mathrm{src}}(D_1;\vartheta)].$$

Step 4: the sourced rare histories. The sourced histories are positive at real w, since their measures are images of ρ_0^w multiplied by the fixed real nonnegative history weights of Prop. YM3_I:73.5. For each $h \in \mathcal{H}_{K'}$ denote the mass by $m_h(w)$ and its history label by $S(h)$. The creation estimate YM3_I:(1734) and deficit estimate YM3_I:(1735) give the hypotheses of YM3_I:(1452), applied to $f(h) = e^{\mathcal{E}^w_{K',S(h)}} m_h(w)$ and $e(S) = -\sum_n \mathfrak{E}'^w_n(S)$, with the sourced constants. Equation YM3_I:(1735) identifies $f(h)e^{e(S(h))}$ with $e^{-E^w_{(K')}} m_h(w)$. All vacuum exponents here are real by conjugation. In the proof of Prop. YM3_I:67.5, for fixed creation sets W_j the multiplicity bound YM3_I:(1447) cancels the binary factors, leaving at most $e^{\mathsf{C}^{\mathrm{S}}_{\mathrm{maj}}\ell^3}$ times the product of $e^{-c'_{\mathrm{S}}p_0(g_j)^2}$ over the created cells. If the bad region at level t is nonempty, at least one cell was created by level t. Mark one such cell and sum over it and then over the unrestricted creation sets. Exactly as in Lem. YM3_I:115.1, the resulting bound is

$$\sum_{h:Z_t(h)\neq\varnothing} m_h(w) \le e^{E^w_{(K')}} \Pi_{K'}(\mathsf{C}^{\mathrm{S}}_{\mathrm{maj}}, c'_{\mathrm{S}}) \sum_{j\le t} \#\mathcal{P}_j e^{-c'_{\mathrm{S}}p_0(g_j)^2}.$$

Here $\mathcal{P}_j$ is the level-j partition into candidate cells. The threshold in Lem. YM3_I:82.5 admits Prop. YM3_I:67.6 at the sourced constants, including $Y^{\mathrm{S}}(c'_{\mathrm{S}})$. Thus the stopped sum is at most $2M^{-3}N_D e^{-c'_{\mathrm{S}}A_0^2(\log x_t)^{2\mathfrak{p}_0}}$ and $\Pi_{K'}(\mathsf{C}^{\mathrm{S}}_{\mathrm{maj}}, c'_{\mathrm{S}})$ is at most $e^{(\mathsf{C}^{\mathrm{S}}_{\mathrm{maj}}+2)\ell^3}$, by YM3_I:(1451). Enlarging its first argument by λ_L keeps both bounds valid and gives the stated constant.

Step 5: discarding. The inequalities YM3_I:(2189), with $F = \rho_0^w/\rho_0$, compare the sourced partition function and the empty-history and cutoff masses. The fixed history weights have the prefix refinement used in the proof of Prop. YM3_I:115.2. The descendants of the empty prefix through level t are exactly the histories with $Z_t = \varnothing$. This property concerns their weights and holds also after multiplication by ρ_0^w. Thus the lost mass is nonnegative and at most the rare-history sum of Step 4. Step 3 yields $\mathsf{R}^{\mathrm{src}}(D;\mathsf{b}',\vartheta)Z[w]$. The cutoff weight at $t=1$ contains $\mathbf{1}_{\mathsf{Cut}_0}$; apply YM3_I:(2189) with $F = \mathbf{1}_{\mathsf{Cut}_0}\rho_0^w/\rho_0$ to bound its mass by the sourced mass of Cut_0. This proves (2).

Step 6: the block laws. Test the cutoff and full image laws at level t against an invariant function G with $0 \le G \le 1$. The positive difference in YM3_I:(2189) at $F = G\circ\mathcal{M}^t$ times ρ_0^w/ρ_0 shows that the full normalized law is a convex combination of the normalized restricted law and the law of the removed mass on this algebra. Its difference from the restricted law on G is at most the discarded fraction from clause (2). At zero source use YM3_I:(2190) instead. The ratio of the two normalized restricted densities is between $e^{-2\epsilon_{\mathrm{src}}(D;\vartheta)}$ and $e^{2\epsilon_{\mathrm{src}}(D;\vartheta)}$ by Step 2, so their variation distance is at most $1 - e^{-2\epsilon_{\mathrm{src}}(D;\vartheta)} \le 2\epsilon_{\mathrm{src}}(D;\vartheta)$. The full laws are gauge invariant, and gauge averaging an indicator reduces the supremum to such invariant tests by eq. 2.xx. This proves (3) for $d = D$. Taking the image under $\mathcal{M}^{D-d}$ cannot increase total variation.

Step 7: clause (4). The map $(w,U) \mapsto \rho_0^w(U)$ is continuous and positive, and Fubini's theorem gives $\Psi dU/\int \Psi dU = \int P^w\,\pi(dw)$ with $\pi \propto Z[w]\pi_0(dw)$. For $w \in \mathcal{S}$ let $P^w_H = e^{-H}P^w/P^w(e^{-H})$; its density ϕ against P^w is at least $e^{-H} \ge e^{-\delta_0}$ on Cut_0, so $\|P^w_H - P^w\| = \int(1-\phi)_+ dP^w \le (1-e^{-\delta_0}) + P^w(\mathsf{V}_{K'} \setminus \mathsf{Cut}_0) \le \delta_0 + \mathsf{R}^{\mathrm{src}}(K'-1;\mathsf{b}',\vartheta)$. In the same way $P = \int P^w_H\,\pi'(dw)$ with $\pi' \propto Z[w]P^w(e^{-H})\pi_0(dw)$; a mixture of gauge-invariant laws is gauge invariant, the total variation between a mixture and a fixed law is at most the supremum of the total variations of the mixed laws, and clause (3) ends the argument. □

Proposition 18.158 (The block fields under the plaquette weights $e^{-\beta s}\hat m(s)e^{-h(s)}$)**.** Let the constants be a setting $\mathfrak s$ with the radius ϱ, let $J=[\mathsf b_-,\mathsf b_+]$ be an interval of labels on the side ℓ, take the numbers of Lemmas 18.156 and 18.157, and put $\tau_\star=\sqrt3\,\varrho_\wedge/(80M)$. Let $\lambda\in\mathbb R$, $0\le\tau_0<\tau_\star$, $M_h\ge0$ and $\tau_1=|\lambda|+\tau_0$. For $K'\in\mathbb N$, $\mathsf b\in J$ and $\beta=L^{K'}\mathsf b$ let

$$\Psi_{K'}(U)=\prod_p e^{-\beta s_p}\,\hat m_p(s_p)\,e^{-h(s_p)},\qquad s_p=a_p(U),\qquad \hat m_p(s)=\int e^{-\tau s}m_p(d\tau),$$

the product over the bare plaquettes, where each m_p is a Borel probability measure on $[\lambda-\tau_0,\lambda+\tau_0]$ (one measure m for all p keeps the invariance under the lattice isometries) and h is a Borel function on $[0,2]$ with $0\le h(s)\le M_hs^4$; let $P_{\Psi_{K'}}=\Psi_{K'}dU/\int\Psi_{K'}dU$. Let K_0 be the least integer with $K_0\ge8D_1+8$, $\tau_1L^{-K_0}<2\varrho_\wedge$ and $\epsilon_{\rm src}(D_1;\tau_1L^{-K_0})\le1$, and put

$$\mathsf C_R=\sup_{D\ge1,\ \mathsf b'\in J}L^{8D}\mathsf R(D;\mathsf b'),\qquad \mathsf C'_R=\sup\bigl\{L^{8D}\mathsf R^{\rm src}(D;\mathsf b',\vartheta):\ D\ge1,\ \mathsf b'\in J,\ \epsilon_{\rm src}(D_1;\vartheta)\le1\bigr\},$$

$$\eta_\Psi(K')=(\mathsf C_R+2\mathsf C'_R)L^{-K'}+2C_{\rm src}\ell^3\tau_1L^{4-K'/2}+\tfrac{3}{16}M_h(c_\rho A_0)^8\ell^3\mathsf b_-^{-4}(\log L^{K'}\mathsf b_+)^{8\mathfrak p_0}L^{-K'};$$

both suprema are finite.

(1) For $K'\ge K_0$ and $K=\lceil K'/8\rceil$, $\bigl\|(\mathcal M^{K'-K})_*P_{\Psi_{K'}}-\mu_K^{K'}[\mathsf b]\bigr\|\le\eta_\Psi(K')$.

(2) *(Every block field, at the same label.)* For every integer $d\ge0$ and every integer $K'\ge\max\{K_0,8d,K_{\rm bf},1260d\}$, with $K_{\rm bf}$ and τ_J of Corollary 18.213,

$$\bigl\|(\mathcal M^{K'-d})_*P_{\Psi_{K'}}-\mu_d[\mathsf b]\bigr\|\le\eta_\Psi(K')+\tau_J(K')\le C_\Psi L^{-K'/11};$$

both laws are gauge invariant, so this total variation over all Borel sets of $\mathsf V_d$ equals the one over the sets of $\mathscr J_d$; C_Ψ depends on ς, ϱ, ℓ, J, τ_1 and M_h, and on no label of J.

(3) *(Functions of one block field.)* For bounded Borel functions $f_1,\dots,f_n$ on $\mathsf V_d$ and K' as in (2), the joint cumulants of $f_1(V_d),\dots,f_n(V_d)$ under $P_{\Psi_{K'}}$ and under $\mu_d[\mathsf b]$ differ by at most $2\bigl(\sum_\pi|\pi|!\bigr)\prod_i\sup|f_i|\,\bigl(\eta_\Psi(K')+\tau_J(K')\bigr)$, the sum over the partitions π of $\{1,\dots,n\}$.

(4) *(The mixed action.)* Let $0<\beta_A<\frac{3}{32}\min\{\tau_\star,\frac12\}^2$ and $a=(32\beta_A/3)^{1/2}$. The plaquette weight $\exp[-\beta(1-\frac12\operatorname{tr}\partial U(p))-\beta_A(1-\frac13\operatorname{tr}_{\rm ad}\partial U(p))]$ equals $e^{-\beta s_p}\hat m(s_p)e^{-h(s_p)}$ with $m=\frac34\delta_\lambda+\frac18(\delta_{\lambda+a}+\delta_{\lambda-a})$, $\lambda=\frac83\beta_A$, $\tau_0=a$, and $0\le h(s)\le\frac{1}{96}a^4\cosh(1)\,s^4$; so (1)–(3) hold for it.

(5) *(Four numbers that suffice.)* Let $u\in C^4([0,2])$ be real with $u(0)=0$, $u_k=u^{(k)}(0)/k!$ and $M_4=\frac{1}{24}\sup|u''''|$, and let a be real with $0<a\le\frac12$, $2a<\tau_\star$ and

$$0<32u_2\le a^2,\qquad 256|u_3|\le a^3,\qquad 2u_2a^2-6.4|u_3|a\ge80M_4.$$

Then $e^{-\beta s+u(s)}=e^{-\beta s}\hat m(s)e^{-h(s)}$ on $[0,2]$, with a probability measure m of at most six atoms in $[\lambda-2a,\lambda+2a]$, $|\lambda|\le|u_1|+a/32$, and $0\le h(s)\le(a^4/100+M_4)s^4$; so (1)–(3) hold for the weight $e^{-\beta s+u(s)}$, with $\tau_0=2a$.

■

Sketch. Step 1: the weight as a mixture of sources. With $w(\tau)(p)=-L^{-K'}\tau_p$ for $\tau=(\tau_p)_p$, Fubini's theorem gives

$$\prod_p e^{-\beta s_p}\hat m_p(s_p)=\int\exp\Bigl[-\sum_p(\beta+\tau_p)\,a_p(U)\Bigr]\bigotimes_p m_p(d\tau_p)=\int\rho_0^{w(\tau)}(U)\bigotimes_p m_p(d\tau_p),$$

ρ_0^w taken at the label $\mathsf b$; the constant λ stays in the source, and the label does not move. Here $|w(\tau)|_\infty\le\tau_1L^{-K'}=:\vartheta<2\varrho_\wedge$ and $|w(\tau)(p)-w(\tau)(p')|\le2\tau_0L^{-K'}$, so $w(\tau)\in\mathsf{Src}_\wedge$ by Lemma 18.156(1), as $\tau_0<\tau_\star$; the map $\tau\mapsto w(\tau)$ is continuous, so the image π_0 of $\bigotimes_p m_p$ is a probability measure on a compact set of real sources of $\mathsf{Src}_\wedge$. The factor $H=\sum_p h(s_p)$ is bounded, Borel and gauge invariant. On Cut_0, $\varpi(U)<c_\rho\varepsilon(x_0)$ with $x_0=L^{K'}\mathsf b$; since $\det\partial U(p)=1$,

$a_p = \frac{1}{2}\det(\partial U(p) - \mathbf{1}) \le \frac{1}{2}|\partial U(p) - \mathbf{1}|^2$, by the 2×2 determinant identity $\det(g-1) = 2 - \operatorname{tr} g$; with the $3\ell^3 L^{3K'}$ bare plaquettes this gives $H \le \delta_0$ on Cut_0, with

$$\delta_0 = 3\ell^3 L^{3K'} M_h \tfrac{1}{16} c_\rho^8 \varepsilon(x_0)^8 \le \tfrac{3}{16} M_h (c_\rho A_0)^8 \ell^3 \mathsf{b}_-^{-4} (\log L^{K'} \mathsf{b}_+)^{8\mathfrak{p}_0} L^{-K'}.$$

Step 2: clause (1). For $K' \ge K_0$, $K = \lceil K'/8 \rceil$ satisfies $D_1 + 1 \le K \le \frac{1}{8}K' + 1 < K'$, so Lemma 18.157(4) applies at $D = d = K$ and bounds the distance by $\delta_0 + \mathsf{R}^{\mathrm{src}}(K'-1;\mathsf{b},\vartheta) + \mathsf{R}^{\mathrm{src}}(K;\mathsf{b},\vartheta) + \mathsf{R}(K;\mathsf{b}) + 2\epsilon_{\mathrm{src}}(K;\vartheta)$. As $\epsilon_{\mathrm{src}}(D_1;\vartheta) \le 1$ and $8K \ge K'$, the three middle terms are at most $(\mathsf{C}_R + 2\mathsf{C}'_R)L^{-K'}$, and $4K \le \frac{1}{2}K' + 4$ gives $2\epsilon_{\mathrm{src}}(K;\vartheta) = 2C_{\mathrm{src}}\ell^3 L^{4K}\tau_1 L^{-K'} \le 2C_{\mathrm{src}}\ell^3\tau_1 L^{4-K'/2}$. The two suprema are finite, because $(\log L^D \mathsf{b}')^{2\mathfrak{p}_0} \ge (D\log L)^{2\mathfrak{p}_0}$, $2\mathfrak{p}_0 \ge 2$, beats every multiple of D.

Step 3: clauses (2) and (3). For $K' \ge 8d$, $K \ge d$; the image under $\mathcal{M}^{K-d}$ does not raise the total variation and carries $\mu_K^{K'}[\mathsf{b}]$ to $\mu_d^{K'}[\mathsf{b}]$, and Corollary 18.213 adds $\tau_J(K')$ for $K' \ge K_{\mathrm{bf}}$ and $\lceil K'/1260 \rceil \ge d$. Every term of $\eta_\Psi(K') + \tau_J(K')$ is at most a constant times $K'^{\mathfrak{p}} L^{-K'/10}$, which gives $C_\Psi L^{-K'/11}$. Both laws are gauge invariant. By Step 1, $P_{\Psi_{K'}}$ is the law P of Lemma 18.157(4), which is gauge invariant on $\mathsf{V}_{K'}$; the averaging map is gauge covariant by YM3_I:(37)–(39), so the image of $P_{\Psi_{K'}}$ under $\mathcal{M}^{K'-d}$ is gauge invariant on V_d; and $\mu_d[\mathsf{b}]$ is gauge invariant by Lemma 18.212(b), whose hypotheses $\mathsf{b} \ge \gamma^{-2}$ and $M\,R(\mathsf{b}) \le M\,R(\mathsf{b}_+) \le L^m$ hold on J. For gauge-invariant laws the gauge average of indicators eq. 2.xx makes the total variation over $\mathscr{I}_d$ equal to the one over all Borel sets. For (3): the means of a product of bounded functions under two laws differ by at most twice the total variation times the product of the suprema, and the joint cumulant is $\sum_\pi (-1)^{|\pi|-1}(|\pi|-1)! \prod_{B\in\pi} m_B$ in the mixed moments m_B; replacing the $|\pi|$ factors one at a time gives the bound.

Step 4: clause (4). For $g \in \mathrm{SU}(2)$, $\operatorname{tr}_{\mathrm{ad}} g = (\operatorname{tr} g)^2 - 1$; with $s = 1 - \frac{1}{2}\operatorname{tr} g$ this is $4(1-s)^2 - 1$, so $1 - \frac{1}{3}\operatorname{tr}_{\mathrm{ad}} g = \frac{1}{3}(8s - 4s^2)$, and the weight is $e^{-(\beta+\lambda)s} e^{\theta s^2}$ with $\lambda = \frac{8}{3}\beta_A$ and $\theta = \frac{4}{3}\beta_A$. Let $q = \frac{1}{4}$, so that $\frac{1}{2}qa^2 = \theta$, and $g_0(y) = \log(1 + q(\cosh y - 1)) - \frac{1}{2}qy^2$; then $e^{\theta s^2} = (1 - q + q\cosh(as))e^{-g_0(as)}$, and $1 - q + q\cosh(as) = \int e^{-\tau s} m^\circ(d\tau)$ with $m^\circ = \frac{3}{4}\delta_0 + \frac{1}{8}(\delta_a + \delta_{-a})$. Let $0 \le y = as \le 2a \le 1$. From $\log(1+x) \le x$ and $\cosh y - 1 - \frac{1}{2}y^2 = \sum_{k\ge 2} y^{2k}/(2k)! \le \frac{1}{24}y^4 \cosh y$, as $(2k)! \ge 24\,(2k-4)!$, we get $g_0(y) \le \frac{q}{24}y^4 \cosh 1$. From $\log(1+x) \ge x - \frac{1}{2}x^2$, $\cosh y - 1 - \frac{1}{2}y^2 \ge \frac{1}{24}y^4$ and $\cosh y - 1 \le \frac{1}{2}y^2 e^{y^2/12}$, which follows from $2\cdot 12^{k-1}(k-1)! \le (2k)!$ coefficient by coefficient, we get $g_0(y) \ge \frac{q}{24}y^4(1 - 3qe^{y^2/6}) \ge \frac{q}{24}y^4(1 - \frac{3}{4}e^{1/6}) \ge 0$. So $h(s) = g_0(as)$ lies between 0 and $\frac{1}{96}a^4\cosh(1)s^4$, and $e^{-\lambda s}\int e^{-\tau s} m^\circ(d\tau) = \hat{m}(s)$ for the translate m of m° by λ, which lies in $[\lambda - a, \lambda + a]$; and $a < \min\{\tau_\star, \frac{1}{2}\}$.

Step 5: clause (5). Let σ be the sign of u_3 and $y_3 = 6|u_3|/a^3 \le \frac{6}{256}$. The function $\varphi(q') = q'(1-q')(1-2q')$ increases on $[0, \frac{1}{32}]$ from 0 to $\frac{930}{32768} > 0.0283 > y_3$, so there is $q' \in [0, \frac{1}{32}]$ with $\varphi(q') = y_3$, and $q'(1-q') = y_3/(1-2q') \le \frac{16}{15}y_3$. Put $q = 2u_2/a^2 - q'(1-q')$; then $q \le \frac{1}{16}$ and $qa^4 \ge 2u_2a^2 - 6.4|u_3|a \ge 80M_4 \ge 0$. Let $X_1 \in \{0, \pm a\}$ with the weights $1-q, \frac{q}{2}, \frac{q}{2}$ and $X_2 \in \{0, \sigma a\}$ with the weights $1-q', q'$ be independent, let m° be the law of $-(X_1 + X_2)$, which has at most six atoms in $[-2a, 2a]$, and $\psi(s) = \log\int e^{-\tau s} m^\circ(d\tau) = \log \mathbb{E} e^{s(X_1+X_2)}$. At $s = 0$, $\psi'(0) = \sigma a q'$, $\frac{1}{2}\psi''(0) = \frac{1}{2}a^2(q + q'(1-q')) = u_2$, and $\frac{1}{6}\psi'''(0) = \frac{1}{6}\sigma a^3\varphi(q') = u_3$. For $s \in [0,2]$, $\psi''''(s)$ is the fourth cumulant of $X_1 + X_2$ under the law tilted by $e^{s(X_1+X_2)}$, the sum of the fourth cumulants of the two tilted variables. The tilted X_1 has the moments $\mu_1 = a\delta$, $\mu_2 = a^2 Q$, $\mu_3 = a^3\delta$, $\mu_4 = a^4 Q$, with $|\delta| \le Q = q\cosh(as)/(1 + q(\cosh(as) - 1)) \in [q, q\cosh 1] \subset [q, 0.0965]$; by $\kappa_4 = \mu_4 - 4\mu_3\mu_1 - 3\mu_2^2 + 12\mu_2\mu_1^2 - 6\mu_1^4$ its fourth cumulant lies in $[a^4Q(1-7Q-6Q^3), a^4Q(1+12Q^2)] \subset [0.3qa^4, 1.72qa^4]$. The tilted X_2 is σa times a Bernoulli variable whose parameter p is at most $eq' \le 0.085$, and its fourth cumulant $a^4 p(1-p)(1-6p(1-p))$ lies in $[0, 2.72q'a^4]$. Put $\lambda = \psi'(0) - u_1$ and $h(s) = \psi(s) - \psi'(0)s - u(s) + u_1 s$. Then h and its first three derivatives vanish at 0, and $h'''' = \psi'''' - u''''$ lies in $[0.3qa^4 - 24M_4, (1.72q + 2.72q')a^4 + 24M_4]$, whose lower end is ≥ 0; Taylor's

formula with the integral remainder gives $0 \le h(s) \le [\frac{1}{24}(\frac{1.72}{16} + \frac{2.72}{32})a^4 + M_4]s^4 \le (\frac{a^4}{100} + M_4)s^4$. Finally $e^{u(s)} = e^{-\lambda s}e^{\psi(s)}e^{-h(s)}$, where $e^{-\lambda s}e^{\psi(s)} = \hat{m}(s)$ for the translate m of m° by λ, and $|\lambda| \le |u_1| + aq' \le |u_1| + a/32$. □

18.16. **One torus: derivatives in the label as entries 1, analyticity at every label, and the sign of $S_3(1,f,h)$ between two labels.** The statements of this subsection concern the torus limits $S_n^{\ell,\mathsf{b}}$ of one side $\ell = 2L^m$, taken as functions of the label b on the whole half-line $(0,\infty)$. Proposition 18.160 says that for every tuple X on T_ℓ, at each cutoff and in the limit, a derivative in the label puts the constant function 1 in front of X as one more entry and changes the sign, $\partial_{\mathsf{b}}^{\nu} S_n^{\ell,\mathsf{b}}(X) = (-1)^{\nu} S_{n+\nu}^{\ell,\mathsf{b}}(1^{\nu}, X)$; that, for each order ν, the ν-th derivative at the cutoff K converges as $K \to \infty$ to the ν-th derivative of the limit, uniformly on every compact set of labels; and that $\int_{\mathsf{b}'}^{\mathsf{b}} S_3^{\ell,t}(1,f,h)\,dt = S_2^{\ell,\mathsf{b}'}(f,h) - S_2^{\ell,\mathsf{b}}(f,h)$ for two functions f,h with disjoint supports. Proposition 18.161 says that in a setting $\mathsf{b} \mapsto S_n^{\ell,\mathsf{b}}(X)$ is a real-analytic function of the label on the whole half-line, whatever the tuple X, whether two of its supports are disjoint or only the intersection of all of them is empty: about each compact interval of positive labels it extends to a holomorphic function on a strip whose width depends neither on X nor on n, and within that width it is the sum of its power series in the shift t of the label, in which the coefficient of t^{ν} is the torus limit of the tuple with ν entries 1 put in front, multiplied by $(-1)^{\nu}/\nu!$. Proposition 18.162 works at a reference coupling $0 < g \le \gamma_W$, on a side of $\mathfrak{L}(g)$, with a test pair (f,h) at a scale d and the tolerance $\epsilon = \bar{\varepsilon}_{g,\vartheta}(\mathsf{d})$ of the main text's law of two points: for any two labels $\mathsf{b}_1 < \mathsf{b}_2$ of the window $I(g)$ with $(\mathsf{b}_2/\mathsf{b}_1)^2 > (1+\epsilon)/(1-\epsilon)$ the function $t \mapsto S_3^{\ell,t}(1,f,h)$ takes a positive value between them, bounded below by a number given in closed form; since ϵ tends to 0 with d, any two labels of the window have this property at all the pairs of all the scales below a threshold formed from g, ϑ and $\mathsf{b}_2/\mathsf{b}_1$, and the classes of these pairs are not empty; and the zeros of $t \mapsto S_3^{\ell,t}(1,f,h)$ do not accumulate at any point of $(0,\infty)$. Proposition 18.163 admits any two positive labels $\mathsf{b}_1 < \mathsf{b}_2$ and treats all the sides $2L^m$, $m \in \mathbb{N}$, together: at every scale up to a first threshold the class of test pairs is not empty, and for each of its pairs the proposition gives the law of two points at b_1 and at b_2, and a label $t \in (\mathsf{b}_1, \mathsf{b}_2)$ at which $S_3^{\ell,t}(1,f,h)$ equals $(S_2^{\ell,\mathsf{b}_1}(f,h) - S_2^{\ell,\mathsf{b}_2}(f,h))/(\mathsf{b}_2 - \mathsf{b}_1)$, the difference quotient of $\mathsf{b} \mapsto S_2^{\ell,\mathsf{b}}(f,h)$ over $[\mathsf{b}_1, \mathsf{b}_2]$ with its sign reversed, and is bounded below by an expression in the two tolerances; for the scales up to a second, smaller threshold this lower bound is in turn at least a positive number given in closed form; the two thresholds and the two tolerances serve every m.

The cutoff differentiation uses the exact Wilson weight, and the passage to the limit uses locally uniform convergence from Theorem YM3_I:AS. The source factorization of Theorem YM3_I:AD gives a common strip. The two-point law in Theorem YM3_I:AV, together with the integral identity in Proposition 18.160, gives the sign conclusions. The Taylor coefficients are the cumulants with constant entries.

Definition 18.159 (Label, tuples, and the width of a strip)**.** (1) Use the canonical construction of Theorem YM3_I:D and the scalar torus cumulants $S^T_{n;K,\ell,\beta}$ for the Wilson action. In a sourced setting $\mathfrak{s}$ also use the fixed radius ϱ and the common constants of Theorem YM3_I:AD, with the canonical source room of [7, Lem. N.1]. This choice has $2\varrho \le \gamma^{-2} - \bar{B}$ by the source-radius inequality of the canonical construction. Write $S_n^{\ell,\mathsf{b}}(X)$ for the limit of the torus cumulants at $\beta = L^K\mathsf{b}$ through all cutoffs, as in Theorem YM3_I:AS. Use $X_g = L(g^{-2} + B_\sharp)$, $\bar{B} = LB_\sharp/(L-1)$, $I(g) = (g^{-2} + \bar{B}, X_g - \bar{B}]$, and the first-passage data and sides $\mathfrak{L}(g)$ of Proposition YM3_I:74.1.

(2) A tuple on T_ℓ has $n \ge 2$ real Lipschitz entries whose supports have empty intersection. For $\nu \ge 0$, $(1^\nu, X)$ prepends ν copies of the constant function 1. For a tuple with no identically zero entry put $\mathsf{d}_\infty(X) = \min_{x_i \in \operatorname{supp} f_i} \max_{i,j} |x_i - x_j|_\infty$ with torus sup distance. This minimum is positive by compactness and empty intersection. Such a tuple has a separated

pair if two supports are disjoint. In that case put $\hat{\mathsf{d}}(X) = \max_{i<j} \operatorname{dist}_\infty(\operatorname{supp} f_i, \operatorname{supp} f_j)$, with distance between sets measured in the torus sup metric.

(3) Put $f^{(s)}(x) = f(L^{-s}x)$ on $T_{L^s\ell}$ and $X^{(s)} = (f_i^{(s)})_i$. This dilation multiplies distances by L^s, preserves the supremum and decreases the Lipschitz constant. Thus $\|f^{(s)}\|_\sharp \le \|f\|_\sharp$, where $\|f\|_\sharp = \max\{\|f\|_\infty, 40M \operatorname{Lip} f\}$. It takes supports to the corresponding dilated supports, so it preserves the empty-intersection condition and multiplies $\hat{\mathsf{d}}(X)$ by L^s.

(4) An interval above the floor is $\mathsf{J} = [x_-, x_+]$ with $x_- \ge 2\gamma^{-2}$, and its sides are $\mathfrak{L}(g(x_+))$ with $g(x) = (x/2)^{-1/2}$. On a side λ set

$$Q = \exp[(C_{\mathrm{S}} + C_{g(x_+)})\lambda^3], \qquad \upsilon(\mathsf{J},\lambda) = \frac{\varrho}{4Q}, \qquad C_{g(x_+)} = C_{\mathrm{st}} + \frac{9}{2}\log X_{g(x_+)}.$$

Here C_{S} and C_g are the canonical source stability constants of [7, Lem. N.1]. Put

$$\mathsf{M}_n(d;\mathsf{J},\lambda) = \left(\frac{n}{\varrho}\right)^n 27 C_{\mathrm{S}} S(\kappa) C'_\Sigma(\kappa - \log 9216, 4)\left(\frac{\lambda}{d}\right)^3 + \log 2\left(\frac{n}{\upsilon(\mathsf{J},\lambda)}\right)^n,$$

where $S(\kappa)$ is the counting constant of Lemma YM3_I:88.1 and C'_Σ is from YM3_I:(1776). For a real interval J write $\Omega_r(J) = \{\zeta : \operatorname{dist}(\zeta, J) < r\}$.

(5) For $0 < \vartheta < 1$ and $20d \le \ell$, $\mathfrak{T}_{T_\ell}(d;\vartheta)$ consists of nonnegative Lipschitz pairs (f,h), neither zero, each supported in a closed ball of radius d, at mutual support distance between d and $6d$. Each entry satisfies $\int f \ge \vartheta \|f\|_\infty d^3$ and $\operatorname{Lip} f \le \|f\|_\infty/(\vartheta d)$. Put $\mathsf{c}_2 = 9/(16\pi^2)$ and $\mathcal{K}_6(f,h) = \iint f(x)h(y)|x-y|^{-6}\,dx\,dy$, using the torus distance. All cross distances lie in $[d, 10d]$. The integral bounds and the volumes of the two balls give

$$10^{-6}\vartheta^2 \|f\|_\infty \|h\|_\infty \le \mathcal{K}_6(f,h) \le (4\pi/3)^2 \|f\|_\infty \|h\|_\infty.$$

The class is nonempty: take the translates of $(1 - |x|/d)_+$ centered at 0 and $3de_1$, periodized to T_ℓ. Each has mass $\pi d^3/3 > d^3$ and Lipschitz constant $1/d$. Dilation by L^s maps a test pair to the class at $L^s d$ with the same ϑ and the same $\mathcal{K}_6$, by change of variables. Use the ceiling γ_W, the witness depth $s(d)[g]$, the tolerance $\bar{\varepsilon}_{g,\vartheta}(d)$ and the threshold $d_0(g,\vartheta)$ of Theorem YM3_I:AV and YM3_I:(2349). The tolerance is nondecreasing, tends to zero at zero and is at most $1/3$ on the indicated range.

■

Proposition 18.160 (Derivatives in the label at a cutoff and in the limit; their integrals over labels)**.** Use the real construction of Definition 18.159(1). Let $m \in \mathbb{N}$, $\ell = 2L^m$, $n \ge 2$, and let $X = (f_1, \dots, f_n)$ be a tuple on T_ℓ.

(1) *(One cutoff.)* For every $K \in \mathbb{N}$ the function $\mathsf{b} \mapsto S^T_{n;K,\ell,L^K\mathsf{b}}(X)$ is real-analytic on $(0,\infty)$, and

$$\partial_{\mathsf{b}}^\nu S^T_{n;K,\ell,L^K\mathsf{b}}(X) = (-1)^\nu S^T_{n+\nu;K,\ell,L^K\mathsf{b}}(1^\nu, X) \qquad (\nu \ge 0,\ \mathsf{b} > 0).$$

(2) *(The limit.)* The function $\mathsf{b} \mapsto S^{\ell,\mathsf{b}}_n(X)$ is of class C^∞ on $(0,\infty)$, and

$$\partial_{\mathsf{b}}^\nu S^{\ell,\mathsf{b}}_n(X) = (-1)^\nu S^{\ell,\mathsf{b}}_{n+\nu}(1^\nu, X) \qquad (\nu \ge 0,\ \mathsf{b} > 0).$$

For every $\nu \ge 0$ and every compact set $\mathsf{J}' \subset (0,\infty)$, $\sup_{\mathsf{b}\in\mathsf{J}'} \left|\partial_{\mathsf{b}}^\nu S^T_{n;K,\ell,L^K\mathsf{b}}(X) - \partial_{\mathsf{b}}^\nu S^{\ell,\mathsf{b}}_n(X)\right| \to 0$ as $K \to \infty$.

(3) *(Over an interval of labels.)* For all $0 < \mathsf{b}' < \mathsf{b}$,

$$\int_{\mathsf{b}'}^{\mathsf{b}} S^{\ell,t}_{n+1}(1,X)\,dt = S^{\ell,\mathsf{b}'}_n(X) - S^{\ell,\mathsf{b}}_n(X).$$

If f, h are real Lipschitz functions on T_ℓ with disjoint supports, then for all $0 < \mathsf{b}' < \mathsf{b}$

$$\int_{\mathsf{b}'}^{\mathsf{b}} S^{\ell,t}_3(1,f,h)\,dt = S^{\ell,\mathsf{b}'}_2(f,h) - S^{\ell,\mathsf{b}}_2(f,h), \qquad \int_{\mathsf{b}'}^{\mathsf{b}} t\,[t\,S^{\ell,t}_3(1,f,h) - 2S^{\ell,t}_2(f,h)]\,dt = \mathsf{b}'^2 S^{\ell,\mathsf{b}'}_2(f,h) - \mathsf{b}^2 S^{\ell,\mathsf{b}}_2(f,h).$$

The three identities hold as well at every cutoff K, each torus limit $S_k^{\ell,u}(Y)$ being replaced by $S^T_{k;K,\ell,L^K u}(Y)$.

■

Sketch. Step 1: an entry 0. If an entry of X is 0, so is an entry of each tuple $(1^\nu, X)$; the cumulants of such a tuple then vanish for all K and all $\beta > 0$, hence so do its torus limits at all labels (Theorem YM3_I:AS), and every member of (1)–(3) is 0. From here on no entry of X is 0, and then no entry of any $(1^\nu, X)$ is 0.

Step 2: the label as a constant source. For fixed K set $\mathcal{O}(f) = a_K^{-1}\sum_p f(x(p))a_p$ and

$$\mathcal{Z}_K(\zeta, z) = \int e^{-\zeta\mathcal{O}(1)+\sum_i z_i\mathcal{O}(f_i)}\, dU.$$

This is entire because the action and all insertions are bounded on the finite lattice. It is positive for real ζ, z. At any real $\mathsf{b} > 0$, the mixed derivative of $\log\mathcal{Z}_K(\mathsf{b}, z)$ in the z_i at zero is $S^T_{n;K,\ell,L^K\mathsf{b}}(X)$, since dividing by $\mathcal{Z}_K(\mathsf{b}, 0)$ adds a constant to the logarithm. The same derivative defines a holomorphic function of ζ where $\mathcal{Z}_K(\zeta, 0) \neq 0$. A local logarithm near each $(\zeta, 0)$ suffices, since the derivative is independent of its branch. A source z_0 for the entry 1 replaces ζ by $\zeta - z_0$. Differentiating once and then repeating yields the identity of (1) for every ν.

Step 3: the limit; clause (2). Put $\phi_{\nu,K}(\mathsf{b}) = (-1)^\nu S^T_{n+\nu;K,\ell,L^K\mathsf{b}}(1^\nu, X)$ and $\phi_\nu(\mathsf{b}) = (-1)^\nu S^{\ell,\mathsf{b}}_{n+\nu}(1^\nu, X)$. The tuple $(1^\nu, X)$ has $n + \nu \geq 2$ entries, none of them 0, and the intersection of its supports is empty; so by Theorem YM3_I:AS its torus limit exists at every label and is continuous in the label, and $\phi_{\nu,K} \to \phi_\nu$ uniformly on each compact subset of $(0, \infty)$. By (1), $\phi_{\nu,K}$ is continuously differentiable and $\phi'_{\nu,K} = \phi_{\nu+1,K}$. A sequence of C^1 functions on a compact interval that converges at every point, while the derivatives converge uniformly, has a C^1 limit, and differentiation commutes with the passage to the limit; so $\phi'_\nu = \phi_{\nu+1}$ in the interior of every compact interval of $(0, \infty)$, hence on $(0, \infty)$. By induction on ν, ϕ_0 has derivatives of every order and $\phi_0^{(\nu)} = \phi_\nu$. The uniform convergence of the derivatives is the convergence $\phi_{\nu,K} \to \phi_\nu$ just used.

Step 4: clause (3). By (2) at $\nu = 1$, ϕ_0 is continuously differentiable on $[\mathsf{b}', \mathsf{b}]$ with the derivative $t \mapsto -S^{\ell,t}_{n+1}(1, X)$, a continuous function, and integrating this derivative over $[\mathsf{b}', \mathsf{b}]$ gives the first identity. Two real Lipschitz functions with disjoint supports form a tuple (f, h) on T_ℓ, and $(1, f, h) = (1^1, (f, h))$; this gives the second identity. For the third, with $\phi(t) = S_2^{\ell,t}(f, h)$ and $\psi(t) = S_3^{\ell,t}(1, f, h)$, $\frac{d}{dt}[t^2\phi(t)] = 2t\phi(t) + t^2\phi'(t) = -t[t\psi(t) - 2\phi(t)]$, a continuous function, which is integrated in the same way. At a cutoff, (1) takes the place of (2) in these three arguments. □

Proposition 18.161 (Analyticity in the label on the whole half-line; the series of the entries 1)**.** Work in a setting $\mathfrak{s}$. Let $m \in \mathbb{N}$, $\ell = 2L^m$, $n \geq 2$, and let $X = (f_1, \dots, f_n)$ be a tuple on T_ℓ.

(1) *(A strip about a compact interval.)* Let $0 < \mathsf{b}_- < \mathsf{b}_+$, and fix an integer $s \geq 0$ such that

$$(\dagger) \qquad L^s\mathsf{b}_- \geq 2\gamma^{-2}, \qquad 2L^{m+s} \in \mathfrak{L}\big((L^s\mathsf{b}_+/2)^{-1/2}\big).$$

By the first inequality of (†), $(L^s\mathsf{b}_+/2)^{-1/2} < (L^s\mathsf{b}_-/2)^{-1/2} \leq \gamma$, so that the set of sides in the second condition is defined. If an integer $s \geq 0$ satisfies $L^s \geq \max\{2\gamma^{-2}/\mathsf{b}_-,\ L(\frac{1}{2}\mathsf{b}_+ + \bar{B})\}$, $L^{m+s/2} \geq L^2M\,(4r/e)^r$ and $L^{m+s} \geq L^2MC_{\mathrm{s}}$, then it satisfies (†); and these three inequalities hold for every s beyond some integer. Put $\mathsf{J}_s = [L^s\mathsf{b}_-, L^s\mathsf{b}_+]$ and $\upsilon_s = L^{-s}\upsilon(\mathsf{J}_s, L^s\ell)$. Then the real points of $\Omega_{\upsilon_s}([\mathsf{b}_-, \mathsf{b}_+])$, the numbers of $(\mathsf{b}_- - \upsilon_s, \mathsf{b}_+ + \upsilon_s)$, are positive, and a unique holomorphic function $\Phi[X]$ on $\Omega_{\upsilon_s}([\mathsf{b}_-, \mathsf{b}_+])$ takes the value $S_n^{\ell,\mathsf{b}}(X)$ at each real point b of this set; $\Phi[X]$ is bounded. The width υ_s is formed from the values of the constants before γ, from ϱ, s, b_+ and ℓ, and depends neither on n, nor on X, nor on a cutoff. If X has a separated pair, then $|\Phi[X]| \leq L^{sn}\,\mathsf{M}_n\big(L^s\hat{\mathsf{d}}(X); \mathsf{J}_s, L^s\ell\big)\prod_{i=1}^n \|f_i\|_\sharp$ on $\Omega_{\upsilon_s}([\mathsf{b}_-, \mathsf{b}_+])$.

(2) *(Every label.)* The function $\mathsf{b} \mapsto S_n^{\ell,\mathsf{b}}(X)$ is real-analytic at every point of $(0, \infty)$.

(3) *(The series of the entries 1.)* In the situation of (1) let $\mathsf{b} \in [\mathsf{b}_-, \mathsf{b}_+]$. Then $|S^{\ell,\mathsf{b}}_{n+\nu}(1^\nu, X)| \le \nu!\, \upsilon_s^{-\nu} \sup|\Phi[X]|$ for every $\nu \ge 0$, the supremum being taken over $\Omega_{\upsilon_s}([\mathsf{b}_-, \mathsf{b}_+])$, and whenever t is real and $|t| < \upsilon_s$,

$$S^{\ell,\mathsf{b}+t}_n(X) = \sum_{\nu=0}^{\infty} \frac{(-t)^\nu}{\nu!}\, S^{\ell,\mathsf{b}}_{n+\nu}(1^\nu, X),$$

with an absolutely convergent series.

■

Sketch. Step 1: the integer s. Let s satisfy the three inequalities; put $c = \frac{1}{2}\mathsf{b}_+$ and $g_s = (L^s c)^{-1/2}$. From the first, $L^s \mathsf{b}_- \ge 2\gamma^{-2}$; then $L^s c \ge \frac{1}{2} L^s \mathsf{b}_- \ge \gamma^{-2} \ge 4$, so $g_s \le \frac{1}{2}$ and $\mathfrak{L}(g_s)$ is defined. With $Y = L(g_s^{-2} + \bar{B}) = L(L^s c + \bar{B})$ we have $Y \ge 4L > e$, and, because $1 \le L^s$ and $L(c + \bar{B}) \le L^s$, also $Y \le L^s\, L(c + \bar{B}) \le L^{2s}$; so $0 < \log Y \le 4y$ with $y = \frac{1}{2} s \log L$. Raising $y/r \le e^{y/r - 1}$, a case of $u \le e^{u-1}$ for real u, to the power r gives $y^r \le (r/e)^r e^y = (r/e)^r L^{s/2}$, hence $(\log Y)^r \le (4r/e)^r L^{s/2}$, and the second and third inequalities yield $L^2 M \max\{(\log Y)^r, C_{\mathrm{S}}\} \le L^{m+s}$. The room formula in Proposition YM3_I:74.1 bounds the smallest side of $\mathfrak{L}(g_s)$, the number $2L^{m_{\mathfrak{L}}(g_s)}$, strictly by $2L^2 M \max\{(\log Y)^r, C_{\mathrm{S}}\}$; therefore $m_{\mathfrak{L}}(g_s) < m + s$, and $2L^{m+s} \in \mathfrak{L}(g_s)$ by the room condition. Since L^s, $L^{m+s/2}$ and L^{m+s} tend to infinity with s while the numbers they are compared with are fixed, the three inequalities hold for all large s.

Step 2: a uniform complex neighborhood on T_{ℓ_s}. By (†) the interval J_s is above the floor and $\ell_s = L^s \ell$ is a side of it. Use Q and υ from Definition 18.159(4). Because $2\upsilon \le \varrho/2$ and the lower endpoint of J_s is at least $2\gamma^{-2}$, every point at distance at most 2υ from J_s has positive real part. Fix $x \in \mathsf{J}_s$. For $g(x)^{-2} = x/2$ the label x lies in $I(g(x))$: the left endpoint is $x/2 + \bar{B} < x$, and the right endpoint is at least $Lx/2$. Thus $(K, L^K x)$ is first passage for each K by Corollary YM3_I:74.2. The side is admissible there by monotonicity of the room condition in Proposition YM3_I:74.1. For a tuple $Y = (h_i)$ on T_{ℓ_s} with nonzero entries, write $\mathcal{Z}_K$ for the finite-lattice integral in the proof of Proposition 18.160, with side ℓ_s and entries h_i. Use the source $w = \xi + \sum_i z_i h_i$ on the ball $|\xi| + \sum_i |z_i| \|h_i\|_\sharp < 2\varrho$. The factorization of Theorem YM3_I:AD at $(K, L^K x)$ gives

$$\frac{\mathcal{Z}_K(x - \xi, z)}{\mathcal{Z}_K(x, 0)} = e^{\mathcal{E}(w)} H(\xi, z), \qquad \mathcal{E}(w) = \sum_{j,X} e_j(X; w), \qquad H(\xi, z) = \frac{\widetilde{Z}(w)}{\widetilde{Z}(0)}.$$

Here $H(0,0) = 1$ and $|H| \le Q$, since $C_{g(x)} \le C_{g(\sup \mathsf{J}_s)}$. Schwarz's lemma on each radial complex line, applied to $(H - 1)/(Q + 1)$, gives

$$|H(\xi, z) - 1| \le \frac{Q + 1}{2\varrho} \left(|\xi| + \sum_i |z_i| \|h_i\|_\sharp \right).$$

It follows that H is nonzero when the parentheses are less than 4υ, and that $|\log H| \le \log 2$ when they are at most 2υ, by the series for $\log(1 + (H - 1))$.

Step 3: the uniform bound for a separated pair. Fix $|\xi| < \upsilon$. Cauchy's inequality on $|z_i| < \upsilon/(n\|h_i\|_\sharp)$ bounds the mixed derivative of $\log H$ at zero by $\log 2\, (n/\upsilon)^n \prod_i \|h_i\|_\sharp$. For the source energies use instead the radii $\varrho/(n\|h_i\|_\sharp)$, which still lie in the source ball. Each term contributes at most $(n/\varrho)^n C_{\mathrm{S}} e^{-\kappa d_j(X)} \prod_i \|h_i\|_\sharp$. It contributes zero unless the one-layer enlargement of X meets both supports in a separated pair, by source locality. Choose a pair with sup separation $d = \hat{\mathsf{d}}(Y)$. At level $j = K - r$ Lemma YM3_I:88.1 bounds the sum of the exponential factors of contributing domains by

$$27 S(\kappa) (\ell_s L^r / M)^3 e^{-(\kappa - \log 9216)[L^r d/M - 4]_+}.$$

Summing this over r by YM3_I:(1777) and adding the logarithm estimate gives $\mathsf{M}_n(d; \mathsf{J}_s, \ell_s) \prod_i \|h_i\|_\sharp$. The mixed log derivative of $\mathcal{Z}_K$ does not depend on the local branch, so these functions agree on overlaps of the discs $|x - \zeta| < \upsilon$ and define a bounded holomorphic function on $\Omega_\upsilon(\mathsf{J}_s)$ for each K.

On the real points it is the torus cumulant of Y. Theorem YM3_I:AS gives convergence at every such real point. Montel's theorem gives locally uniform subsequential limits. The identity theorem makes them equal, since they agree on the interval J_s, and hence the whole sequence converges locally uniformly to F_∞. It obeys the same bound. Apply this argument to $Y = X^{(s)}$.

Step 4: from T_{ℓ_s} to T_ℓ. Put $\Phi[X](\zeta) = L^{sn}F_\infty(L^s\zeta)$ for $\zeta \in \Omega_{\upsilon_s}([\mathsf{b}_-, \mathsf{b}_+])$, holomorphic by Step 2. At a real point b of this set, Step 3 and the scaling law YM3_I:(2215) give $\Phi[X](\mathsf{b}) = L^{sn}S_n^{L^s\ell, L^s\mathsf{b}}(X^{(s)}) = S_n^{\ell,\mathsf{b}}(X)$. The bound of (1) follows from that of Step 3, since $\hat{\mathsf{d}}(X^{(s)}) = L^s\hat{\mathsf{d}}(X)$ and $\|f_i^{(s)}\|_\sharp \le \|f_i\|_\sharp$. For uniqueness: being convex, the set $\Omega_{\upsilon_s}([\mathsf{b}_-, \mathsf{b}_+])$ is connected, and it contains a real interval; a holomorphic function on it vanishing on that interval vanishes identically, by the identity theorem. If an entry of X is 0, $\Phi[X] = 0$ has the required properties (Theorem YM3_I:AS).

Step 5: every tuple. Let no entry of X be 0, and put $\delta_X = \mathsf{d}_\infty(X) > 0$. Choose an integer $N \ge 3$ for which the mesh $\mathsf{w} = \ell/N$ satisfies $4\mathsf{w} < \delta_X$, and let $\mathrm{tent}(u) = \max\{0, 1 - |u|\}$; the translates $\mathrm{tent}(u - j)$, $j \in \mathbb{Z}$, add up to 1 at every real u. For $\alpha \in \{0, \dots, N-1\}^3$ define χ_α at a point $x \in T_\ell$ with a lift $\tilde{x}$ by $\chi_\alpha(x) = \prod_{\mu=1}^3 \sum_{j \equiv \alpha_\mu \pmod N} \mathrm{tent}(\tilde{x}_\mu/\mathsf{w} - j)$. A change of $\tilde{x}_\mu$ by $\ell = N\mathsf{w}$ shifts the index j by N and leaves the inner sum unchanged, so χ_α is well defined on T_ℓ. In an inner sum the terms that are not 0 live on disjoint intervals and vary at a rate at most $1/\mathsf{w}$; therefore χ_α takes its values in $[0,1]$ and is Lipschitz. The sum over α is $\prod_\mu \sum_{j\in\mathbb{Z}} \mathrm{tent}(\tilde{x}_\mu/\mathsf{w} - j) = 1$. If $\chi_\alpha(x) \ne 0$, each $\tilde{x}_\mu$ lies within w of a number $j\mathsf{w}$ with $j \equiv \alpha_\mu$, so x is at a sup distance less than w from the point p_α with the lift $\mathsf{w}\alpha$; consequently $\operatorname{supp}\chi_\alpha$ has sup diameter at most $2\mathsf{w}$. Each $f_i\chi_\alpha$ is a real Lipschitz function supported in $A_i = \operatorname{supp} f_i$, the closed sets $A_1, \dots, A_n$ have an empty intersection, and $f_i = \sum_\alpha f_i\chi_\alpha$. Since $S_n^{\ell,\mathsf{b}}$ is multilinear on the product of the spaces of real Lipschitz functions supported in $A_1, \dots, A_n$ (Proposition YM3_I:118.3),

$$S_n^{\ell,\mathsf{b}}(X) = \sum_{\alpha_1,\dots,\alpha_n} S_n^{\ell,\mathsf{b}}(f_1\chi_{\alpha_1}, \dots, f_n\chi_{\alpha_n}) \qquad (\mathsf{b} > 0).$$

Let a term have no entry 0, and suppose that every two of its supports intersect. Pick x_i in the support of its i-th entry for each i, and, for $i \ne i'$, a point y common to the supports of the i-th and the i'-th entries; then $|x_i - x_{i'}|_\infty \le |x_i - y|_\infty + |y - x_{i'}|_\infty \le 4\mathsf{w} < \delta_X$ for all i, i', although $x_i \in \operatorname{supp} f_i$ for every i, which contradicts the definition of δ_X as a least value. So each term either has an entry 0 and vanishes at every label, or has a separated pair and falls under Steps 3 and 4 with the same s and the same strip, which depend neither on the tuple nor on n. The sum of the functions Φ of the terms of the second kind is holomorphic and bounded on $\Omega_{\upsilon_s}([\mathsf{b}_-, \mathsf{b}_+])$ and equals $S_n^{\ell,\mathsf{b}}(X)$ at its real points; its uniqueness follows as in Step 4.

Step 6: clauses (2) and (3). Let $\mathsf{b}_0 > 0$. Clause (1) with $\mathsf{b}_\pm = 2^{\pm1}\mathsf{b}_0$ and an integer s satisfying (†) provides a holomorphic function on a disc about b_0 equal to $S_n^{\ell,\cdot}(X)$ at the real points of the disc; such a function is the sum of its Taylor series on a smaller disc, which is (2). For (3) let $\mathsf{c}_\Phi = \sup|\Phi[X]| < \infty$. As $D(\mathsf{b}, \upsilon_s) \subset \Omega_{\upsilon_s}([\mathsf{b}_-, \mathsf{b}_+])$, $\Phi[X](\zeta) = \sum_{\nu\ge0} c_\nu(\zeta - \mathsf{b})^\nu$ on this disc, and Cauchy's inequalities give $|c_\nu| \le \mathsf{c}_\Phi\upsilon_s^{-\nu}$. Along the real axis the derivatives of $\Phi[X]$ at b are those of $S_n^{\ell,\cdot}(X)$, so $\nu!\,c_\nu = \partial_\mathsf{b}^\nu S_n^{\ell,\mathsf{b}}(X) = (-1)^\nu S_{n+\nu}^{\ell,\mathsf{b}}(1^\nu, X)$ by Proposition 18.160(2). Setting $\zeta = \mathsf{b} + t$ with real t, $|t| < \upsilon_s$, gives (3), the terms being at most $\mathsf{c}_\Phi(|t|/\upsilon_s)^\nu$ in modulus. □

Proposition 18.162 (The three-point function with an entry 1 between two labels of a window)**.** Work in a setting $\mathfrak{s}$. Let $0 < g \le \gamma_W$, $\ell \in \mathfrak{L}(g)$, $\vartheta \in (0,1)$, $0 < \mathsf{d} \le \mathsf{d}_0(g,\vartheta)$ and $(f,h) \in \mathfrak{T}_{T_\ell}(\mathsf{d};\vartheta)$. Put $\epsilon = \bar{\varepsilon}_{g,\vartheta}(\mathsf{d})$, so that $0 \le \epsilon \le \frac13$, $\mathsf{k} = \mathsf{c}_2\mathcal{K}_6(f,h)$, $\phi(\mathsf{b}) = S_2^{\ell,\mathsf{b}}(f,h)$ and $\psi(\mathsf{b}) = S_3^{\ell,\mathsf{b}}(1,f,h)$ for $\mathsf{b} > 0$, and let $\mathsf{b}_{\mathrm{top}} = L(g^{-2} + \bar{B}) - 2\bar{B}$ be the right end of $I(g)$.

(1) (f,h) and $(1,f,h)$ are tuples on T_ℓ with a separated pair, and $\mathsf{k} \ge 10^{-6}\vartheta^2\mathsf{c}_2\|f\|_\infty\|h\|_\infty > 0$. Each of ϕ and ψ is a real-analytic function of the label on $(0,\infty)$, $\phi' = -\psi$, and $\int_{\mathsf{b}'}^{\mathsf{b}} \psi(t)\,dt = \phi(\mathsf{b}') - \phi(\mathsf{b})$ for $0 < \mathsf{b}' < \mathsf{b}$.

(2) *(The law of two points on the window.)* $|\phi(\mathsf{b}) - \mathsf{k}\,\mathsf{b}^{-2}| \le \epsilon\,\mathsf{k}\,\mathsf{b}^{-2}$ for every $\mathsf{b} \in I(g)$.

(3) *(Integrals.)* Whenever $\mathsf{b}_1 < \mathsf{b}_2$ lie in $I(g)$, $\left|\int_{\mathsf{b}_1}^{\mathsf{b}_2}(\psi(\mathsf{b}) - 2\mathsf{k}\,\mathsf{b}^{-3})\,d\mathsf{b}\right| \le \epsilon\,\mathsf{k}\,(\mathsf{b}_1^{-2} + \mathsf{b}_2^{-2})$ and $\left|\int_{\mathsf{b}_1}^{\mathsf{b}_2}\mathsf{b}\,(\mathsf{b}\,\psi(\mathsf{b}) - 2\phi(\mathsf{b}))\,d\mathsf{b}\right| \le 2\epsilon\,\mathsf{k}$.

(4) *(A positive value between two labels.)* Whenever $\mathsf{b}_1 < \mathsf{b}_2$ lie in $I(g)$, some $t \in (\mathsf{b}_1, \mathsf{b}_2)$ satisfies

$$\mathsf{k}\,\frac{(1-\epsilon)\mathsf{b}_1^{-2} - (1+\epsilon)\mathsf{b}_2^{-2}}{\mathsf{b}_2 - \mathsf{b}_1} \le \psi(t) \le \mathsf{k}\,\frac{(1+\epsilon)\mathsf{b}_1^{-2} - (1-\epsilon)\mathsf{b}_2^{-2}}{\mathsf{b}_2 - \mathsf{b}_1}.$$

If $(\mathsf{b}_2/\mathsf{b}_1)^2 > (1+\epsilon)/(1-\epsilon)$, the lower bound is a positive number, so that $\psi(t) > 0$ and $\int_{\mathsf{b}_1}^{\mathsf{b}_2}\psi > 0$. As $(1+\epsilon)/(1-\epsilon) \le 2$, this happens whenever $\mathsf{b}_2^2 > 2\mathsf{b}_1^2$; for instance $\mathsf{b}_1 = \frac{1}{2}\mathsf{b}_{\mathrm{top}}$ and $\mathsf{b}_2 = \mathsf{b}_{\mathrm{top}}$ are two labels of $I(g)$, and for this choice $\psi(t) \ge 2(3-5\epsilon)\mathsf{k}\,\mathsf{b}_{\mathrm{top}}^{-3} \ge \frac{8}{3}\mathsf{k}\,\mathsf{b}_{\mathrm{top}}^{-3}$.

(5) *(Any two labels of the window, at small scales.)* Let $\mathsf{b}_1 < \mathsf{b}_2$ lie in $I(g)$, and put $\rho = \mathsf{b}_2/\mathsf{b}_1$, $\eta_\rho = \frac{\rho^2-1}{2(\rho^2+1)}$ and

$$\mathsf{d}_\bullet(g,\vartheta,\rho) = \tfrac{1}{2}\sup\{\mathsf{d}' \in (0, \mathsf{d}_0(g,\vartheta)] :\ \bar{\varepsilon}_{g,\vartheta}(\mathsf{d}') \le \eta_\rho\}.$$

Then $0 < \mathsf{d}_\bullet(g,\vartheta,\rho) \le \frac{1}{2}\mathsf{d}_0(g,\vartheta)$; for every $\mathsf{d}' \in (0, \mathsf{d}_\bullet(g,\vartheta,\rho)]$ the class $\mathfrak{T}_{T_\ell}(\mathsf{d}';\vartheta)$ is not empty, and every pair (f',h') of it has a label $t \in (\mathsf{b}_1,\mathsf{b}_2)$ with

$$S_3^{\ell,t}(1,f',h') \ge \tfrac{1}{2}\,\mathsf{c}_2\,\mathcal{K}_6(f',h')\,\frac{\mathsf{b}_1+\mathsf{b}_2}{\mathsf{b}_1^2\mathsf{b}_2^2} \ge \tfrac{1}{2}\,10^{-6}\vartheta^2\,\mathsf{c}_2\,\frac{\mathsf{b}_1+\mathsf{b}_2}{\mathsf{b}_1^2\mathsf{b}_2^2}\,\|f'\|_\infty\|h'\|_\infty > 0.$$

Every open interval that lies in $I(g)$ and is not empty contains two labels $\mathsf{b}_1 < \mathsf{b}_2$; so at all the pairs of all the scales up to $\mathsf{d}_\bullet(g,\vartheta,\mathsf{b}_2/\mathsf{b}_1)$ the function $t \mapsto S_3^{\ell,t}(1,f',h')$ takes a positive value in that interval.

(6) *(Zeros.)* ψ is not identically 0, and its zeros do not accumulate at any point of $(0,\infty)$, so that every compact interval of positive labels holds finitely many of them; the set $\{\psi > 0\}$ is open, and it contains a point of each interval $(\mathsf{b}_1,\mathsf{b}_2)$ whose ends lie in $I(g)$ and satisfy $(\mathsf{b}_2/\mathsf{b}_1)^2 > (1+\epsilon)/(1-\epsilon)$.

(7) *(One cutoff.)* Let $K > s(\mathsf{d})[g]$. Then (2)–(4) remain true when ϕ and ψ are replaced by $\phi_K(\mathsf{b}) = S^T_{2;K,\ell,L^K\mathsf{b}}(f,h)$ and $\psi_K(\mathsf{b}) = S^T_{3;K,\ell,L^K\mathsf{b}}(1,f,h)$, ϵ and k being unchanged; and $\phi_K' = -\psi_K$ on $(0,\infty)$.

The numbers ϵ, $\mathsf{d}_0(g,\vartheta)$ and $\mathsf{d}_\bullet(g,\vartheta,\rho)$ are formed from the setting, g, ϑ, d and ρ alone. So one and the same number ϵ enters the bounds of (2)–(4) and (7) for all the sides $\ell \in \mathfrak{L}(g)$, all the pairs of $\mathfrak{T}_{T_\ell}(\mathsf{d};\vartheta)$ wherever they lie on the torus, all the labels of $I(g)$ and all the cutoffs $K > s(\mathsf{d})[g]$. ■

Sketch. Step 1: clause (1). The two functions of a test pair are non-negative and Lipschitz, neither vanishes identically, and their supports are at a distance at least $\mathsf{d} > 0$; so f and h are real Lipschitz functions, not 0, with disjoint supports, and the constant function 1 is real, Lipschitz and not 0. Hence (f,h) and $(1,f,h)$ are tuples on T_ℓ with a separated pair. The lower bound for k is Definition 18.159(5). Proposition 18.161(2) for the two tuples gives the analyticity, and Proposition 18.160(2), (3) the identity $\phi' = -\psi$ and the integral.

Step 2: clause (2). Let $\mathsf{b} \in I(g)$. As $g \le \gamma_W \le \gamma$, each pair $(K, L^K\mathsf{b})$, $K \in \mathbb{N}$, is a first-passage datum at g, and its label is b (Corollary YM3_I:74.2). For $K > s(\mathsf{d})[g]$, Theorem YM3_I:AV, the law of two points on the tori of the sides at g, applied at this datum, at the side $\ell \in \mathfrak{L}(g)$, at the scale $\mathsf{d} \in (0, \mathsf{d}_0(g,\vartheta)]$ and at the pair (f,h), gives $S^T_{2;K,\ell,L^K\mathsf{b}}(f,h) = \mathsf{c}_2\mathsf{b}^{-2}\mathcal{K}_6(f,h)(1+\varepsilon')$ with $|\varepsilon'| \le \epsilon$, that is, $|S^T_{2;K,\ell,L^K\mathsf{b}}(f,h) - \mathsf{k}\,\mathsf{b}^{-2}| \le \epsilon\,\mathsf{k}\,\mathsf{b}^{-2}$. The witness depth does not depend on K or on b, so this holds for all large K; as $K \to \infty$ the cumulant converges, by Theorem YM3_I:AS applied to the tuple (f,h) of clause (1), none of whose entries is 0, and its limit is $\phi(\mathsf{b})$ (YM3_I:(2212)); the inequality, whose right side is free of K, persists.

Step 3: clause (3). Integrating ψ and $2\mathsf{k}\,\mathsf{b}^{-3}$ over $[\mathsf{b}_1,\mathsf{b}_2]$ with the help of (1), one finds that the quantity inside the first modulus of (3) equals $[\phi(\mathsf{b}_1) - \mathsf{k}\mathsf{b}_1^{-2}] - [\phi(\mathsf{b}_2) - \mathsf{k}\mathsf{b}_2^{-2}]$, a difference of two numbers that (2) bounds. By the last identity of Proposition 18.160(3) the quantity inside

the second modulus equals $[\mathsf{b}_1^2\phi(\mathsf{b}_1)-\mathsf{k}]-[\mathsf{b}_2^2\phi(\mathsf{b}_2)-\mathsf{k}]$, and (2) times b^2 bounds each of the two brackets by $\epsilon\mathsf{k}$.

Step 4: clause (4). Applied to ϕ on the interval $[\mathsf{b}_1,\mathsf{b}_2]$, the mean value theorem and $\phi'=-\psi$ give a $t\in(\mathsf{b}_1,\mathsf{b}_2)$ with $\psi(t)=(\phi(\mathsf{b}_1)-\phi(\mathsf{b}_2))/(\mathsf{b}_2-\mathsf{b}_1)$; inserting $(1-\epsilon)\mathsf{k}\mathsf{b}_i^{-2}\le\phi(\mathsf{b}_i)\le(1+\epsilon)\mathsf{k}\mathsf{b}_i^{-2}$, $i=1,2$, from (2) gives the lower and the upper bound. As $\epsilon<1$, the lower bound is positive precisely if $(1-\epsilon)\mathsf{b}_2^2>(1+\epsilon)\mathsf{b}_1^2$, and then also $\int_{\mathsf{b}_1}^{\mathsf{b}_2}\psi=\phi(\mathsf{b}_1)-\phi(\mathsf{b}_2)>0$. If $\epsilon\le\frac13$, then $(1+\epsilon)/(1-\epsilon)\le2$. The right end of $I(g)$ divided by its left end is $\mathsf{b}_{\mathrm{top}}/(g^{-2}+\bar B)=L-2\bar B/(g^{-2}+\bar B)>L-2\ge11$, because $g^{-2}>0$; hence $\frac12\mathsf{b}_{\mathrm{top}}>g^{-2}+\bar B$, and both $\frac12\mathsf{b}_{\mathrm{top}}$ and $\mathsf{b}_{\mathrm{top}}$ belong to $I(g)$. For this choice $\mathsf{b}_1^{-2}=4\mathsf{b}_{\mathrm{top}}^{-2}$ and $\mathsf{b}_2-\mathsf{b}_1=\frac12\mathsf{b}_{\mathrm{top}}$, so the lower bound equals $2(3-5\epsilon)\mathsf{k}\,\mathsf{b}_{\mathrm{top}}^{-3}\ge\frac83\mathsf{k}\,\mathsf{b}_{\mathrm{top}}^{-3}$.

Step 5: clause (5). Since $\bar\varepsilon_{g,\vartheta}$ has the limit 0 at 0 and $\eta_\rho>0$, the set whose supremum defines $\mathsf{d}_\bullet(g,\vartheta,\rho)$ is not empty, and that supremum lies in $(0,\mathsf{d}_0(g,\vartheta)]$. For $0<\mathsf{d}'\le\mathsf{d}_\bullet(g,\vartheta,\rho)$ the set has an element $\mathsf{d}''>\mathsf{d}'$, and monotonicity gives $\epsilon'=\bar\varepsilon_{g,\vartheta}(\mathsf{d}')\le\bar\varepsilon_{g,\vartheta}(\mathsf{d}'')\le\eta_\rho$. For $(f',h')\in\mathfrak{T}_{T_\ell}(\mathsf{d}';\vartheta)$, clause (4) applies with ϵ' and $\mathsf{k}'=\mathsf{c}_2\mathcal{K}_6(f',h')$; moreover $\epsilon'(\mathsf{b}_1^{-2}+\mathsf{b}_2^{-2})\le\eta_\rho(\rho^2+1)\mathsf{b}_2^{-2}=\frac12(\rho^2-1)\mathsf{b}_2^{-2}=\frac12(\mathsf{b}_1^{-2}-\mathsf{b}_2^{-2})$, so the lower bound of (4) is at least $\frac12\mathsf{k}'(\mathsf{b}_1^{-2}-\mathsf{b}_2^{-2})/(\mathsf{b}_2-\mathsf{b}_1)=\frac12\mathsf{k}'(\mathsf{b}_1+\mathsf{b}_2)\mathsf{b}_1^{-2}\mathsf{b}_2^{-2}$, and Definition 18.159(5) gives the last inequality. As $20\,\mathsf{d}'\le\frac{1}{10}<\ell$, the explicit tent pairs of Definition 18.159(5) give nonemptiness.

Step 6: clause (6). By (1) and (4), ψ is real-analytic on $(0,\infty)$ and takes a positive value. Suppose that the zeros of ψ accumulate at some $a>0$, and expand $\psi(t)=\sum_j c_j(t-a)^j$ near a. Were some c_j different from 0, with j_0 the smallest index of such a coefficient, one would have $\psi(t)=(t-a)^{j_0}\chi(t)$ near a, χ continuous with $\chi(a)=c_{j_0}\neq0$, and ψ would have no zero near a other than a; so $\psi=0$ on a neighbourhood of a. Hence the points of $(0,\infty)$ having a neighbourhood on which ψ vanishes form an open set U, which is also closed in $(0,\infty)$, because the zeros accumulate at every point of its closure. The half-line is connected and ψ is not identically 0, so U is empty; therefore the zeros of ψ do not accumulate in $(0,\infty)$. Infinitely many zeros in a compact interval of positive labels would accumulate at a point of it (Bolzano–Weierstrass); so there are finitely many. Continuity makes $\{\psi>0\}$ open, and (4) gives its points in the intervals of (6).

Step 7: clause (7) and the dependence. Let $K>s(\mathsf{d})[g]$. By Proposition 18.160(1), ϕ_K is real-analytic on $(0,\infty)$ with $\phi_K'=-\psi_K$, and Proposition 18.160(3) holds at the cutoff K; Step 2, stopped before the limit $K\to\infty$, is (2) for ϕ_K, and Steps 3 and 4 use nothing beyond these facts. The tolerance and the threshold depend on ϱ, $\varsigma^{\mathrm S}$, g and ϑ only (YM3_I:(2349)); ϵ is the value of the tolerance at d, and $\mathsf{d}_\bullet(g,\vartheta,\rho)$ is built from these and ρ; none of them depends on a side, a pair, the place of a pair, a label or a cutoff. □

Proposition 18.163 (Two positive labels on every side at once)**.** Work in a setting $\mathfrak{s}$. Let $\vartheta\in(0,1)$ and $0<\mathsf{b}_1<\mathsf{b}_2$, and put $\rho=\mathsf{b}_2/\mathsf{b}_1$ and $\eta_\rho=\frac{\rho^2-1}{2(\rho^2+1)}$. Fix an integer $s\ge0$ such that

$$(\ddagger)\qquad L^s\mathsf{b}_1\ge2\gamma_W^{-2},\qquad 2L^{1+s}\in\mathfrak{L}(g_2),\qquad \text{where } g_i=(L^s\mathsf{b}_i/2)^{-1/2}\ (i=1,2).$$

The first inequality of (‡) alone gives $g_2^{-2}>g_1^{-2}=\frac12L^s\mathsf{b}_1\ge\gamma_W^{-2}$, that is, $0<g_2<g_1\le\gamma_W\le\gamma$; hence the set $\mathfrak{L}(g_2)$ in the second condition is defined, and so are the thresholds $\mathsf{d}_0(g_i,\vartheta)$ and the tolerances $\bar\varepsilon_{g_i,\vartheta}$ used below (Definition 18.159(5)). If an integer $s\ge0$ satisfies $L^s\ge\max\{2\gamma_W^{-2}/\mathsf{b}_1,\ L(\frac12\mathsf{b}_2+\bar B)\}$, $L^{1+s/2}\ge L^2M\,(4r/e)^r$ and $L^{1+s}\ge L^2MC_{\mathrm s}$, then it satisfies (‡); and these three inequalities hold for every s beyond some integer. Put $\mathsf{d}_1=L^{-s}\min\{\mathsf{d}_0(g_1,\vartheta),\mathsf{d}_0(g_2,\vartheta)\}$, $\epsilon_i(\mathsf{d})=\bar\varepsilon_{g_i,\vartheta}(L^s\mathsf{d})$ for $0<\mathsf{d}\le\mathsf{d}_1$ and $i=1,2$, and

$$\mathsf{d}_\circ=\tfrac12\sup\{\mathsf{d}\in(0,\mathsf{d}_1]:\ \max\{\epsilon_1(\mathsf{d}),\epsilon_2(\mathsf{d})\}\le\eta_\rho\}.$$

Then $0<\mathsf{d}_\circ\le\frac12\mathsf{d}_1$; for every $m\in\mathbb{N}$, $\ell=2L^m$, and every $\mathsf{d}\in(0,\mathsf{d}_1]$ the class $\mathfrak{T}_{T_\ell}(\mathsf{d};\vartheta)$ is not empty; and every pair (f,h) of this class has the following properties.

(1) $|S_2^{\ell,\mathsf{b}_i}(f,h)-\mathsf{c}_2\mathsf{b}_i^{-2}\mathcal{K}_6(f,h)|\le\epsilon_i(\mathsf{d})\,\mathsf{c}_2\mathsf{b}_i^{-2}\mathcal{K}_6(f,h)$ for $i=1,2$, and $\epsilon_i(\mathsf{d})\le\frac13$.

(2) Some $t \in (\mathsf{b}_1, \mathsf{b}_2)$ has

$$S_3^{\ell,t}(1,f,h) = \frac{S_2^{\ell,\mathsf{b}_1}(f,h) - S_2^{\ell,\mathsf{b}_2}(f,h)}{\mathsf{b}_2 - \mathsf{b}_1} \geq \mathsf{c}_2 \mathcal{K}_6(f,h)\, \frac{(1-\epsilon_1(\mathsf{d}))\mathsf{b}_1^{-2} - (1+\epsilon_2(\mathsf{d}))\mathsf{b}_2^{-2}}{\mathsf{b}_2 - \mathsf{b}_1}.$$

(3) If $\mathsf{d} \leq \mathsf{d}_\circ$, the right member of (2) is at least $\frac{1}{2}\mathsf{c}_2\mathcal{K}_6(f,h)(\mathsf{b}_1+\mathsf{b}_2)\mathsf{b}_1^{-2}\mathsf{b}_2^{-2} \geq \frac{1}{2}\,10^{-6}\vartheta^2\mathsf{c}_2(\mathsf{b}_1+\mathsf{b}_2)\mathsf{b}_1^{-2}\mathsf{b}_2^{-2}\|f\|_\infty\|h\|_\infty > 0$.

If s is taken to be the least integer satisfying (‡), then s, d_1, $\mathsf{d}_\circ$ and the functions ϵ_1, ϵ_2 are formed from the setting, ϑ, b_1 and b_2 alone; they depend on no side, no test pair, no place of a pair on the torus and no cutoff, so that one threshold $\mathsf{d}_\circ$ of the scale serves all the sides $2L^m$, $m \in \mathbb{N}$. ■

Sketch. Step 1: the integer s. Let s satisfy the three inequalities. The first entry of the maximum in the first of them yields $L^s\mathsf{b}_1 \geq 2\gamma_W^{-2}$, the inequality that (‡) asks for. As $\gamma_W \leq \gamma$, that first inequality implies the first inequality of Proposition 18.161(1) for $\mathsf{b}_- = \mathsf{b}_1$ and $\mathsf{b}_+ = \mathsf{b}_2$, and at $m = 1$ the second and the third coincide with those of that proposition. So Step 1 of its proof applies with $m = 1$; it draws from the first entry of the maximum the bounds $L^s\mathsf{b}_2/2 \geq L^s\mathsf{b}_1/2 \geq \gamma^{-2} \geq 4$ and from the second entry the bound $L(\frac{1}{2}\mathsf{b}_2 + \bar{B}) \leq L^s$, and it gives $2L^{1+s} \in \mathfrak{L}((L^s\mathsf{b}_2/2)^{-1/2}) = \mathfrak{L}(g_2)$, the membership that (‡) asks for. All three inequalities hold for large s, as there.

Step 2: couplings, windows and sides. Let $i \in \{1,2\}$ and $x_i = L^s\mathsf{b}_i$. Then $g_i^{-2} = \frac{1}{2}x_i \geq \frac{1}{2}L^s\mathsf{b}_1 \geq \gamma_W^{-2}$, so $g_i \leq \gamma_W \leq \gamma$, and $g_2 < g_1$ because $\mathsf{b}_1 < \mathsf{b}_2$. As $x_i \geq 2\gamma^{-2}$, $\{x_i\}$ is an interval of labels above the floor with $g(x_i) = g_i$; the endpoint formulas give $g_i^{-2} + \bar{B} < x_i \leq X_{g_i} - \bar{B}$, and Corollary YM3_I:74.2 shows that $(K, L^K x_i)$ is, for each cutoff K, a first-passage datum at g_i whose label is x_i. The monotonicity of the room condition gives $m_{\mathfrak{L}}(g_1) \leq m_{\mathfrak{L}}(g_2)$, and (‡) gives $1 + s \geq m_{\mathfrak{L}}(g_2)$; so for every $m \in \mathbb{N}$, $m + s \geq m_{\mathfrak{L}}(g_i)$, and $\ell_s = L^s\ell = 2L^{m+s} \in \mathfrak{L}(g_i)$ by the room condition.

Step 3: the classes; clause (1). Let $0 < \mathsf{d} \leq \mathsf{d}_1$. As $\mathsf{d}_1 \leq \frac{1}{200}$, $20\,\mathsf{d} \leq \frac{1}{10} < \ell$; so the class $\mathfrak{T}_{T_\ell}(\mathsf{d};\vartheta)$ is defined, and it is not empty by the explicit tent pairs of Definition 18.159(5). Let (f,h) be a pair of this class. By change of variables under dilation, $(f^{(s)}, h^{(s)}) \in \mathfrak{T}_{T_{\ell_s}}(L^s\mathsf{d};\vartheta)$ and $\mathcal{K}_6(f^{(s)}, h^{(s)}) = \mathcal{K}_6(f,h)$; also $L^s\mathsf{d} \leq \mathsf{d}_0(g_i,\vartheta)$. Theorem YM3_I:AV, applied at g_i, at the datum $(K, L^K x_i)$, at the side $\ell_s \in \mathfrak{L}(g_i)$, at the scale $L^s\mathsf{d}$ and at the pair $(f^{(s)}, h^{(s)})$, yields for every $K > s(L^s\mathsf{d})[g_i]$

$$|S^T_{2;K,\ell_s,L^K x_i}(f^{(s)}, h^{(s)}) - \mathsf{c}_2 x_i^{-2}\mathcal{K}_6(f,h)| \leq \epsilon_i(\mathsf{d})\,\mathsf{c}_2 x_i^{-2}\mathcal{K}_6(f,h),$$

together with $\epsilon_i(\mathsf{d}) \leq \frac{1}{3}$. The functions $f^{(s)}, h^{(s)}$ are real Lipschitz, not 0, with disjoint supports, so the cumulant converges to $S_2^{\ell_s,x_i}(f^{(s)}, h^{(s)})$ as $K \to \infty$ (Theorem YM3_I:AS), and the inequality persists. The scaling law YM3_I:(2215) gives $S_2^{\ell,\mathsf{b}_i}(f,h) = L^{2s}S_2^{\ell_s,x_i}(f^{(s)}, h^{(s)})$, and $L^{2s}x_i^{-2} = \mathsf{b}_i^{-2}$.

Step 4: clauses (2), (3) and the dependence. As in Step 1 of the proof of Proposition 18.162, (f,h) and $(1,f,h)$ are tuples on T_ℓ; the mean value theorem with $\partial_{\mathsf{b}}S_2^{\ell,\mathsf{b}}(f,h) = -S_3^{\ell,\mathsf{b}}(1,f,h)$ (Proposition 18.160(2)) provides t, and (1) at b_1 and at b_2 provides the bound. The functions ϵ_1, ϵ_2 are non-decreasing with the limit 0 at 0, so the set defining $\mathsf{d}_\circ$ is not empty, and for $\mathsf{d} \leq \mathsf{d}_\circ$, by the argument of Step 5 of that proof, $\max_i \epsilon_i(\mathsf{d}) \leq \eta_\rho$; then $\rho^2\epsilon_1(\mathsf{d}) + \epsilon_2(\mathsf{d}) \leq \eta_\rho(\rho^2+1) = \frac{1}{2}(\rho^2-1)$, and $(1-\epsilon_1(\mathsf{d}))\mathsf{b}_1^{-2} - (1+\epsilon_2(\mathsf{d}))\mathsf{b}_2^{-2} = \mathsf{b}_2^{-2}[\rho^2 - 1 - \rho^2\epsilon_1(\mathsf{d}) - \epsilon_2(\mathsf{d})] \geq \frac{1}{2}(\mathsf{b}_1^{-2} - \mathsf{b}_2^{-2})$; dividing by $\mathsf{b}_2 - \mathsf{b}_1$ and using Definition 18.159(5) gives (3). The condition (‡) involves no side, and $\mathsf{d}_0(g_i,\vartheta)$ and $\bar{\varepsilon}_{g_i,\vartheta}$ depend on the setting, g_i and ϑ only (YM3_I:(2349)); this gives the last assertion. □

18.17. **Reconstruction data fixed by one ball; spacing-first limits at every label; a ceiling on the lowest mass and the growth of the Källén–Lehmann measure; one vacuum from ergodic states.** We study reconstruction from local data, spacing-first limits and spectral bounds. Proposition 18.166 says that two functionals that are reconstruction data and agree at the separated tuples supported in one non-empty open set, for instance in one ball, are equal, so that they have one Hilbert space, one Hamiltonian and one field; that at one order n agreement at the tuples

supported near one configuration of n distinct points gives equality at that order, because the kernels of the order n are one holomorphic function on a connected domain; that the members of $A[b]$, their components and centred components, the limits of convergent Gibbs sequences at every reference coupling with their components, and all real dilates of these are reconstruction data; and that reconstruction data with one first and one second number form a normal family. Proposition 18.168 says that at a label $b \in I(g)$ the torus limits along every sequence of sides have limit points at all separated tuples, each of them a continuum limit at g along growing tori with the label b; so $\mathrm{SF}(b)$ is not empty. Proposition 18.169 carries this to every real label and every sequence of sides, and lists what every spacing-first limit has: invariance under all translations and the cubic group, reflection positivity at every coordinate plane, bounds and the growth condition with numbers that read no cutoff, side or sequence, regularity, a separable Hilbert space with a semigroup and two momenta, and the law of two points with its label. Proposition 18.170 says that the Källén–Lehmann measure ρ of every component of every member of $A[b]$ obeys $4.7 \cdot 10^{-9}\, b^{-2} M^5 \le \rho([0,M]) \le 0.973\, b^{-2} M^5$ for $M \ge 1.36\, C_0/b$; that at a reference coupling $g \le \gamma_W$ and for every $\vartheta \in (0, \vartheta_1]$ the lowest mass of every component of the limit of every convergent Gibbs sequence at g is at most $m_\circ / d_0(g,\vartheta)$, with $m_\circ < 2.59$; that, at such g and ϑ, every continuum limit S at g along growing tori has $\dim\ker H_S \ge 2$ or a point of $\operatorname{spec} H_S$ in $(0, m_\circ/d_0(g,\vartheta)]$; and that every spacing-first limit S at a real label b has $\dim\ker H_S \ge 2$ or a point of $\operatorname{spec} H_S$ in $(0, C_0/b]$. Proposition 18.171 says that the limit of a convergent Gibbs sequence whose states are ergodic infinitely often has one vacuum; under the uniform finite-size hypothesis, the one limit at a label that a uniformly small finite-size effect gives has one vacuum as soon as one of its defining sequences of states has ergodic states infinitely often.

The analytic kernels are those of Proposition YM3_I:108.3. Use Theorem YM3_I:AP for Gibbs-frame reconstruction, Theorem YM3_I:AL for the canonical decomposition, and Theorems YM3_I:AS and YM3_I:AW for moving-label torus limits and continuum two-point bounds.

Definition 18.164 (Further signs)**.** Use the common construction and source calibration of §2.3.1. Write $\mathfrak{L} = \{2L^m : m \ge 1\}$ and $Q_\ell = (-\ell/4, \ell/4)^3$. Use $\mathfrak{d}(g)$, $\mathfrak{L}(g)$ and $I(g)$ from Proposition YM3_I:74.1 and Corollary YM3_I:74.2. The labels of first-passage data lie in $I^{\mathrm{lab}}(g) = [g^{-2}, L(g^{-2} + \bar B)]$. Use γ_W, $\bar\varepsilon_{g,\vartheta}(d)$ and $d_0(g,\vartheta)$ of Definition 18.140(4). Put

$$\alpha = (5/3)^6, \quad c_2 = 9/(16\pi^2), \quad m_\circ = \tfrac14 \log 31250, \quad \kappa_0 = L(\gamma_W^{-2} + \bar B), \quad \vartheta_1 = 3\sqrt{3}/8,$$
$$g_1 = \gamma_W, \quad g_2 = (\gamma_W^{-2} + 2\bar B/L)^{-1/2}, \quad g_3 = (2\gamma_W^{-2})^{-1/2},$$
$$d^\star = \min_i d_0(g_i, \vartheta_1), \quad k_0 = 6d^\star/\kappa_0, \quad C_0 = m_\circ \kappa_0 / d^\star.$$

For $b \in I(g)$ write $\mathfrak{d}_b = \{(K, L^K b) : K \ge 1\}$. Set $K_{\mathrm{free}}(f,h) = \iint f(x) h(y) |x-y|^{-6}\, dx\, dy$. Use the two-point test class of §18.4 with C^1 entries for separated moments. Write $\mathfrak{S}(O) = \bigcup_{n\ge0} \mathfrak{S}_n(O)$ for the separated tuples of real $C^1_c(O)$ functions, including the empty tuple. Use the product, translations and norm from Definition 18.205. For $s > 0$ set $X_{(s)} = (f_i(s\,\cdot))_i$ and $(\mathrm{dil}_s\sigma)(X) = s^{-|X|}\sigma(X_{(s)})$. Write $X^{[z]} = X_{(L^{-z})}$.

For $\zeta = (K, \ell, \beta)$ put $\mathsf{a}(\zeta; X) = S^T_{n;K,\ell,\beta}(X^{\langle\ell\rangle})$, with periodized tests. For $X \in \mathfrak{S}_n(Q_\ell)$ put $\mathsf{B}(\ell, b; X) = S^{\ell,b}_n(X^{\langle\ell\rangle})$ at the torus limit of Theorem YM3_I:AS. A growing-torus sequence at g has $K, \ell \to \infty$ with $(K,\beta) \in \mathfrak{d}(g)$ and $\ell \in \mathfrak{L}(g)$. Its pointwise separated-cumulant limits form $\mathrm{GT}(g)$. Use $\mathcal{G}^{\mathrm{tor}}_{(K,\beta)}$ for weak limits of periodic extensions at the fixed pair as $\ell \to \infty$. Every such limit is a periodic Gibbs state by the local DLR equations, which pass to the weak limit on continuous cylinder functions, and it inherits the lattice symmetries. The analogous limits using states in $\mathcal{G}^{\mathrm{tor}}_{(K,\beta)}$ form $\mathrm{VF}(g)$. At $b \in I(g)$, GT_b and VF_b specify the subclasses with $\beta = L^K b$. Use the cube cumulant bound table $C^g_n(d; R)$ and its class $\mathcal{K}(g)$ from Propositions YM3_I:91.13 and YM3_I:92.1. For the sharper infinite-volume table use the torus constants of YM3_I:(1795)–(1797) and YM3_I:

(1799)–(1804), evaluated with $(\rho_*, C_e, \kappa_e, E_+, C_{\mathrm{low}}) = (2\varrho, C_{\mathrm{S}}, \kappa, C_{\mathrm{S}}, C_g)$ from [7, Lem. N.1]. In that notation put

$$C_n^{g,\infty}(d;R) = \min\{C_n^{\infty}(d;R), A(d)B(R)^n(n!)^{\varkappa_\star}\}.$$

The canonical growth constants at g are

$$k_{\mathrm{E}} = 10, \qquad \mathsf{c}_{\mathrm{E}} = 1, \qquad \mathsf{C}_{\mathrm{E}}^{\mathrm{c},g} = C_{\neq}^3 MB(1)\sqrt{A_{\mathrm{PL}}}, \qquad \varkappa_{\mathrm{E}} = \varkappa_\star + 9,$$

where $C_{\neq} = 10^{53}$ as in Proposition YM3_I:93.3. They result from $A(d) \le A_{\mathrm{PL}} \max(1, d^{-4})$ and the Schwinger extension estimate in Proposition YM3_I:93.4.

For $\mathfrak{q} = (a_j, \beta_j)_j$ with $a_j \to 0$ and $a_j\beta_j \to b > 0$, let $\mathsf{K}_{\mathfrak{q}}$ be the set of pointwise limits of self-centred moments σ_{ν_j} for periodic Gibbs states at these pairs. Their union is $A[b]$, and $E[b]$ is its one-vacuum subclass. The canonical decomposition, its component evaluations, and the iterated plateaux are those of Theorem YM3_I:AL. One vacuum means that the decomposition has one component. Put $A_J = \bigcup_{b\in J} A[b]$. A Gibbs sequence at g uses $a_j = L^{-K_j}$, $K_j \to \infty$, and $(K_j, \beta_j) \in \mathfrak{d}(g)$. For compact $J \subset (0,\infty)$ a cover W_J is a finite family (z_i, g_i) such that for each $c \in [\min J/(2L), 2\max J]$ some $L^{z_i}c \in I(g_i)^\circ$. Write $s_- = \min_i L^{-z_i}$ and $s_+ = \max_i L^{1-z_i}$. Given $c > 0$, choose z so that $g = (L^z c/2)^{-1/2} \le \gamma$ and $L^z c/2 > \bar{B}$. Since $L \ge 13$, $L^z c$ lies in the interior of $I(g)$. These windows are open and compactness gives the cover. One may require all $g_i \le \gamma_W$ by increasing z in the same construction.

(1) *Families.* For a family $S = (S_n)_{n\ge 2}$ of symmetric multilinear maps S_n on $\mathfrak{S}_n(\mathbb{R}^3)$, σ_S is its centred functional: $\sigma_S(\varnothing) = 1$, $\sigma_S((f)) = 0$, and $\sigma_S(X) = \sum_\pi \prod_{I\in\pi} S_{|I|}(X_I)$ for $|X| \ge 2$, the sum over the partitions π of $\{1,\dots,|X|\}$ into blocks of at least two elements; so $\sigma_S(f,h) = S_2(f,h)$. Where σ_S is reflection positive and H_{σ_S} is formed by Proposition YM3_I:94.1 we write $\mathcal{H}(S) = \mathcal{H}(\sigma_S)$ and $H_S = H_{\sigma_S}$. With $(\mathrm{dil}_s S)_n(X) = s^{-n} S_n(X_{(s)})$, $\sigma_{\mathrm{dil}_s S} = \mathrm{dil}_s \sigma_S$, because $\sum_{I\in\pi} |I| = |X|$.

(2) *Spacing-first limits at a real label.* For real $b > 0$, $\mathrm{SF}(b)$ is the set of the families $S = (S_n)_{n\ge 2}$ of maps $S_n \colon \mathfrak{S}_n(\mathbb{R}^3) \to \mathbb{R}$ for which there are sides $\ell_k \in \mathfrak{L}$ with $\ell_k \to \infty$ such that $\mathsf{B}(\ell_k, b; X) \to S_n(X)$ as $k \to \infty$ for all $n \ge 2$ and $X \in \mathfrak{S}_n(\mathbb{R}^3)$; the number $\mathsf{B}(\ell_k, b; X)$ is formed as soon as $X \in \mathfrak{S}_n(Q_{\ell_k})$, that is for all large k, and S is then taken along (ℓ_k). Restricting the sides to $\mathfrak{L}(g)$ does not change this set, since $\mathfrak{L}(g)$ is a tail.

(3) *Symmetries and coordinate planes.* W_3 is the cubic group of the 48 signed permutations of the coordinates, and $W_2 \subset W_3$ the subgroup of the 8 elements that fix e_3. For $\mu \in \{1,2,3\}$ and $c \in \mathbb{R}$, $\theta_{\mu,c}$ is the reflection $x \mapsto x - 2(x_\mu - c)e_\mu$; a functional σ is reflection positive at $\{x_\mu = c\}$ from the side $\{\epsilon(x_\mu - c) > 0\}$, $\epsilon \in \{1,-1\}$, if $\sigma(\theta_{\mu,c}\bar{F} \cdot F) \ge 0$ for every F in the complex span of the separated tuples supported in that side.

(4) *Ergodic states.* A periodic Gibbs state $\nu \in \mathcal{G}_{a,\beta}$, which is invariant under all translations of $a\mathbb{Z}^3$ by Proposition YM3_I:103.4, is ergodic if $\nu(E) \in \{0,1\}$ for every Borel set E of configurations with $\tau_v E = E$ for all $v \in a\mathbb{Z}^3$; these sets form the σ-algebra $\mathcal{I}$.

(5) *Reconstruction and the data kernels.* The reconstruction data with $(C_\wp, \wp, c_0)$ satisfy (Rec1)–(Rec4) of Section YM3_I:107. They are Euclidean invariant and reflection positive with a separable reflection Hilbert space, its strongly continuous translation semigroup and commuting momenta, joint spectrum in $E \ge |q|$, and the estimate

$$|\sigma(\theta\bar{F} \cdot (h) \cdot F')| \le C_\wp \eta^{-\wp} \|h\|_1 \|[\tau_{-\eta e_3} F]\| \, \|[\tau_{-\eta e_3} F']\|$$

for $0 < \eta \le 1$, F, F' supported above η and h supported in $|x_3| < c_0\eta$, with $C_\wp, \wp \ge 0$ and $0 < c_0 \le 1/2$. Condition (Rec5) is $\ker H_\sigma = \mathbb{C}\Omega$. The quintuple $(\mathcal{H}, U, \Omega, \mathcal{D}_0, \phi)$ is the scalar reconstruction of Theorem YM3_I:AO. The first reconstruction constant at g can be taken to be $C'_g = C_\wp^g + \bar{c}_g$ for all Gibbs limits and components, with exponent 4 and margin 1/2, by Theorem YM3_I:AP. The centered components use $C_\wp^g$ by the centered estimate in the proof of Proposition YM3_I:111.1. Their reflection spaces are identified unitarily by the centering shift of Section YM3_I:107.

For a unit vector u put $d^u_\circ(z) = \mathrm{Re}(u\cdot z) - |\,\mathrm{Im}(z-(u\cdot z)u)|$. For a permutation ς, the tube $\mathsf{Tb}^{u,\varsigma}_n$ consists of the z with $d^u_\circ(z_{\varsigma(k+1)} - z_{\varsigma(k)}) > 0$ for $k < n$. Write $\mathsf{Tb}^{\mathrm{all}}_n$ for their union and $(\mathbb{R}^3)^n_{\neq}$ for the real configurations with distinct entries. Proposition YM3_I:108.3 gives the unique continuous kernel s_n on the real configuration set that integrates against the product of separated tests to give σ. Its holomorphic extension to $\mathsf{Tb}^{\mathrm{all}}_n$ is $\mathbf{S}_n$.

(6) *The trivial functional.* $\sigma_0(\varnothing) = 1$ and $\sigma_0(X) = 0$ for $|X| \geq 1$.

■

Lemma 18.165 (Polygons in the configurations and in the tubes)**.** Let $n \in \mathbb{N}$. Any two points of $(\mathbb{R}^3)^n_{\neq}$ are joined inside $(\mathbb{R}^3)^n_{\neq}$ by a polygon of two segments, and any two points of $\mathsf{Tb}^{\mathrm{all}}_n$ are joined inside $\mathsf{Tb}^{\mathrm{all}}_n$ by a polygon of four segments. So both sets are connected: neither is the union of two disjoint non-empty subsets that are open in it. ■

Proof. For $n = 1$ the two sets are $\mathbb{R}^3$ and $\mathbb{C}^3$. Let $n \geq 2$ and $x, y \in (\mathbb{R}^3)^n_{\neq}$. The finitely many lines $\mathbb{R}(x_i - x_l)$, $\mathbb{R}(y_i - y_l)$, $i \neq l$, meet the unit sphere in finitely many points; take a unit vector e outside them and put $w = (e, 2e, \dots, ne) \in (\mathbb{R}^3)^n_{\neq}$. For $p \in \{x, y\}$, $0 \leq t \leq 1$ and $i \neq l$, the difference of the entries i and l of $p + t(w-p)$ is $(1-t)(p_i - p_l) + t(i-l)e$; it is $p_i - p_l \neq 0$ at $t = 0$ and $(i-l)e \neq 0$ at $t = 1$, and if it vanished at some $t \in (0,1)$, e would be a real multiple of $p_i - p_l$. So the segments from x to w and from w to y lie in $(\mathbb{R}^3)^n_{\neq}$.

Each tube is convex since each margin is a linear real function minus a norm. Removing the imaginary part increases all the margins, so a point is joined inside its tube to its real part. Any real configuration of distinct points admits a direction outside the finitely many great circles perpendicular to the differences, so its projections can be strictly ordered. Conversely such an ordering requires distinct points. Thus the real points of the union are precisely $(\mathbb{R}^3)^n_{\neq}$. The two real segments with the segments to the real parts give the four-segment path and prove connectedness. □

Proposition 18.166 (Reconstruction data are fixed by their values in one ball)**.** Let σ and σ' be reconstruction data as in Definition 18.164(5), with the numbers $(C_\wp, \wp, c_0)$ and $(C'_\wp, \wp', c'_0)$ respectively, which need not be equal; (Rec5) is not assumed. Let $\mathsf{s}_n, \mathbf{S}_n$ and $\mathsf{s}'_n, \mathbf{S}'_n$ be their data kernels and the continuations of these (Definition 18.164(5)).

(a) *One order, one open set of configurations.* Let $n \in \mathbb{N}$ and let $W \subset (\mathbb{R}^3)^n_{\neq}$ be open and not empty. If $\sigma(X) = \sigma'(X)$ for every $X = (f_1, \dots, f_n) \in \mathfrak{S}_n(\mathbb{R}^3)$ with $\mathrm{supp}\, f_1 \times \cdots \times \mathrm{supp}\, f_n \subset W$, then $\mathbf{S}_n = \mathbf{S}'_n$ on $\mathsf{Tb}^{\mathrm{all}}_n$, $\mathsf{s}_n = \mathsf{s}'_n$ on $(\mathbb{R}^3)^n_{\neq}$, and $\sigma = \sigma'$ on $\mathfrak{S}_n(\mathbb{R}^3)$.

(b) *One ball, all orders.* Fix a nonempty open set $O \subset \mathbb{R}^3$, an open ball of arbitrary centre and radius being one example. Then $\sigma = \sigma'$ on $\mathfrak{S}(O)$ implies $\sigma = \sigma'$. So σ and σ' have one Hilbert space $\mathcal{H}(\sigma) = \mathcal{H}(\sigma')$, with the same classes and the same Ω, one Hamiltonian, one unitary group of the translations along the plane $\{x_3 = 0\}$, one joint spectral measure, and one quintuple $(\mathcal{H}, U, \Omega, \mathcal{D}_0, \phi)$ of Theorem YM3_I:AO, with one family of Wightman distributions; none of these objects reads the triple of numbers.

(c) *Which functionals are data.* Each of the following is reconstruction data, so that (a) and (b) hold for any two of them, whatever their labels, reference couplings, sequences, decompositions and settings are: every member of $A[b]$, $b > 0$; every component σ_ξ of every decomposition of one, with (Rec5); its centred component σ^{c}_ξ, with (Rec5); the limit of every convergent Gibbs sequence at any $0 < g \leq \gamma$ and every component of every decomposition of it, with the triple of Definition 18.164(5) at g, the components with (Rec5); the trivial functional σ_0, with every triple and with (Rec5); and every real dilate $\mathrm{dil}_s\varsigma$, $s > 0$, of a functional ς of this list, with first constant $C_\wp \max\{s^{\wp-4}, 2^\wp s^{-4}\}$ and unchanged $\wp, c_0$, and with (Rec5) if ς has it.

(d) *Three consequences.* Let $O \subset \mathbb{R}^3$ be open and not empty.

(d1) Let $(\Xi, \lambda, (\sigma_\xi))$ be a decomposition of a member of $A[b]$ or of the limit of a convergent Gibbs sequence at some $0 < g \le \gamma$. For $\xi \ne \xi'$ there is $X \in \mathfrak{S}(O)$ with $\sigma_\xi(X) \ne \sigma_{\xi'}(X)$, and $\xi \mapsto (\sigma_\xi(X))_{X\in\mathfrak{S}(O)}$ is a homeomorphism of Ξ onto its image in the product space $\mathbb{R}^{\mathfrak{S}(O)}$.

(d2) Let $\sigma \in A[b]$ and $\sigma' \in A[b']$, $b, b' > 0$; the limits of two convergent Gibbs sequences, at any two reference couplings, are such functionals, with b and b' limit points of the labels of the two sequences. If σ and σ' agree on $\mathfrak{S}(O)$, then $\sigma = \sigma'$; and for any decompositions $(\Xi, \lambda, (\sigma_\xi))$ of σ and $(\Xi', \lambda', (\sigma'_{\xi'}))$ of σ' there is exactly one map $h\colon \Xi \to \Xi'$ with $\sigma'_{h(\xi)} = \sigma_\xi$ for all ξ; it is a homeomorphism, and $h_*\lambda = \lambda'$. Under the absence of latent heat as in §12 also $b = b'$.

(d3) Reconstruction data σ with $\sigma(f, h) = 0$ for all $(f, h) \in \mathfrak{S}_2(O)$ are the trivial functional σ_0.

(e) *A normal family.* Let $C \ge 0$, $\wp \ge 0$, $n \in \mathbb{N}$, and let W be as in (a). Let $\sigma^{(j)}$ and $\sigma'^{(j)}$, $j \in \mathbb{N}$, be reconstruction data, each with the first two numbers C and $\wp$, the third numbers being free, and let $\mathsf{s}_n^{(j)}$, $\mathsf{s}_n'^{(j)}$ be their data kernels. If $\sigma^{(j)}(X) - \sigma'^{(j)}(X) \to 0$ as $j \to \infty$ for every $X = (f_i) \in \mathfrak{S}_n(\mathbb{R}^3)$ with $\prod_i \operatorname{supp} f_i \subset W$, then $\sup_{\mathcal{K}} |\mathsf{s}_n^{(j)} - \mathsf{s}_n'^{(j)}| \to 0$ for every compact $\mathcal{K} \subset (\mathbb{R}^3)^n_{\ne}$, and $\sigma^{(j)}(X) - \sigma'^{(j)}(X) \to 0$ for every $X \in \mathfrak{S}_n(\mathbb{R}^3)$. In the setting, at one reference coupling $0 < g \le \gamma$, the limits of all convergent Gibbs sequences at g and all components of all their decompositions are reconstruction data with the one triple of Definition 18.164(5) at g; and for every compact $J \subset (0, \infty)$ and every cover $W_J = ((z_i, g_i))_i$ of J with its number s_- (Definition 18.164), all members of A_J, all components of their decompositions and the centred components of these are reconstruction data with the one triple $(C^{\mathrm{R}}_{W_J}, 4, \frac{1}{2})$, where $C^{\mathrm{R}}_{W_J} = \max\{1, 16 s_-^{-4}\} \max_i C'_i$ and C'_i is the constant C'_{g_i} of Definition 18.164(5). So the convergence assertion of (e) holds for any two sequences whose members all lie in one of these two families.

■

Proof. Step 1: clause (a). Proposition YM3_I:108.3, whose one hypothesis is that the functional is data (Rec) with some numbers, gives $\sigma(X) = \int \mathsf{s}_n \prod_i f_i$ and $\sigma'(X) = \int \mathsf{s}'_n \prod_i f_i$ for $X = (f_i) \in \mathfrak{S}_n(\mathbb{R}^3)$, the integrals over the compact set $\prod_i \operatorname{supp} f_i \subset (\mathbb{R}^3)^n_{\ne}$. Let $x \in W$ and $\epsilon > 0$ with $\prod_i \bar{\mathbb{B}}(x_i, \epsilon) \subset W$; these balls are pairwise disjoint, as $W \subset (\mathbb{R}^3)^n_{\ne}$. For $f_i \ge 0$ of class C^1 with $\int f_i = 1$ and support in $\bar{\mathbb{B}}(x_i, \epsilon)$, the tuple (f_i) lies in $\mathfrak{S}_n(\mathbb{R}^3)$, with the product of its supports in W; so $\int (\mathsf{s}_n - \mathsf{s}'_n) \prod_i f_i = 0$, and as $\epsilon \to 0$ the left member tends to $(\mathsf{s}_n - \mathsf{s}'_n)(x)$, by continuity. So $\mathsf{s}_n = \mathsf{s}'_n$ on W. By Proposition YM3_I:108.3, $D = \mathbf{S}_n - \mathbf{S}'_n$ is holomorphic on $\mathsf{Tb}^{\mathrm{all}}_n$, an open set whose real points form $(\mathbb{R}^3)^n_{\ne}$, and $D = \mathsf{s}_n - \mathsf{s}'_n$ at these points. The set $O' = \{z \in \mathsf{Tb}^{\mathrm{all}}_n : \operatorname{Re} z \in W\}$ is open, it meets $\mathbb{R}^{3n}$, and $D = 0$ on $O' \cap \mathbb{R}^{3n} = W$; and $\mathsf{Tb}^{\mathrm{all}}_n$ is connected (Lemma 18.165). By the identity theorem, applied in each coordinate in a polydisc about a real point of W, $D = 0$ on $\mathsf{Tb}^{\mathrm{all}}_n$. So $\mathsf{s}_n = \mathsf{s}'_n$ on $(\mathbb{R}^3)^n_{\ne}$, and the integral representation gives $\sigma = \sigma'$ on $\mathfrak{S}_n(\mathbb{R}^3)$.

Step 2: clause (b). At $\varnothing$ both functionals are 1. Let $n \ge 1$ and $W_n = O^n \cap (\mathbb{R}^3)^n_{\ne}$, which is open, and not empty because O holds n different points. Let $X = (f_i) \in \mathfrak{S}_n(\mathbb{R}^3)$ with $\prod_i \operatorname{supp} f_i \subset W_n$. If an entry is 0, both values are 0, by multilinearity. Otherwise every support is not empty and is the image of the product under a projection, so it lies in O; then $X \in \mathfrak{S}_n(O)$, and the two values agree by hypothesis. Clause (a) at $W = W_n$ gives $\sigma = \sigma'$ on $\mathfrak{S}_n(\mathbb{R}^3)$. The objects named in (b) are formed from the functional alone: the Hilbert space, the classes and Ω from the form $\sigma(\theta \bar{F} \cdot F')$; the Hamiltonian, the translations along the plane and the joint spectral measure from the action of the translations on the classes, (Rec2) leaving one choice of them; the field $\phi(f)$, $f \in \mathcal{S}$, on $\mathcal{D}_\infty = \bigcap_k \operatorname{dom} H^k_\sigma$, which is one and the same operator whatever triple of numbers of the functional, and whatever integer above the second number of

that triple, its construction uses (Proposition YM3_I:109.1); with it the span $\mathcal{D}_0$ of Ω and of the vectors $\phi(f_1)\cdots\phi(f_n)\Omega$ (Proposition YM3_I:109.1), and the Wightman distributions, each the only tempered distribution that takes the value $\langle\Omega,\phi(f_1)\cdots\phi(f_n)\Omega\rangle$ at $f_1\otimes\cdots\otimes f_n$ (Proposition YM3_I:109.1); the translations $U(x)=\int e^{i(x^0E+\vec{x}\cdot q)}\,d\mathcal{P}(E,q)$, $x=(x^0,\vec{x})\in\mathbb{R}^{1+2}$, with the joint spectral measure $\mathcal{P}$ (Proposition YM3_I:109.1); and, for $\Lambda\in\mathcal{L}$, $U(\Lambda)$, the only unitary operator that fixes Ω, maps $\mathcal{D}_0$ onto itself and carries $\phi(f)$ to $\phi(f\circ\Lambda^{-1})$ on $\mathcal{D}_0$ (Proposition YM3_I:110.1). From these, with $U(a,\Lambda)=U(a)U(\Lambda)$, the quintuple of Theorem YM3_I:AO is made: under (Rec5) by Theorem YM3_I:AO, and without it by the closing paragraph of that theorem.

Step 3. Theorem YM3_I:AP gives reconstruction for the Gibbs limits at g and their components. The centering unitary preserves the one-dimensional zero-energy space of a component. To see that it is a self-centred limit, take the invariant-event conditioned states of Theorem YM3_I:AL. Invariant conditioning preserves the Gibbs equations, as in the proof of that theorem. Changing the centering law from the original to the conditioned law is the finite polynomial expressing self-centred moments in terms of the moments centered at the original law. It gives the centred component in the limit, at the same pairs.

We reduce arbitrary positive spacings to these frames. Write $a_j=L^{-k_j}r_j$, $1\le r_j<L$. For large j the label $a_j\beta_j/r_j$ belongs to the interval covered by W_J. Pick a pair (z_i,g_i) placing $L^{z_i}a_j\beta_j/r_j$ in $I(g_i)^\circ$. Put $K_j=k_j-z_i$ and $s_j=r_jL^{-z_i}$. Then $a_j=s_jL^{-K_j}$, $s_-\le s_j\le s_+$, and the pair (K_j,β_j) is a first-passage datum at g_i by Corollary YM3_I:74.2. Pass to a subsequence with fixed i and $s_j\to s\in[s_-,s_+]$. Relabel the bond configuration by dividing lengths by s_j. This leaves the Haar integrals, action and DLR equations unchanged. The observables at the two spacings are related by $\mathcal{O}_{sa}(f)=s^{-1}\mathcal{O}_a(f(s\,\cdot))$. The relabelled self-centred moments have a further limit τ by Theorem YM3_I:AP. The convergence persists for the tests $f_i(s_j\,\cdot)$, because their C^1 convergence takes place in fixed compact support neighborhoods with positive separation. The bounds of Proposition YM3_I:91.13 and the moment–cumulant formula bound the error by multilinear telescoping. Consequently the original limit is $\mathrm{dil}_s\tau$. Rescaling the translation vectors in the iterated plateaux gives the bijection of decompositions under this dilation.

For the reconstruction estimate, the map on tuple classes $[X]\mapsto[V_sX]$, $V_sX=s^{-|X|}X_{(s)}$, is unitary from the dilated Hilbert space to the original one since it preserves the reflection form and has an inverse. The generators are divided by s. In the insertion estimate at height η, the test becomes $s^{-1}h(s\,\cdot)$ of norm $s^{-4}\|h\|_1$. If $\eta\le s$ apply the estimate at η/s. If $\eta>s$, put $\eta'=\min(1,(1-c_0)\eta/s)\in[1/2,1]$ and partition the insertion by nonnegative smooth functions into slabs of half-width $c_0\eta'$ about heights y with $|y|\le c_0\eta/s$. Translate each piece to height zero. The exterior vectors shift in opposite directions and remain above η' since $\eta/s-|y|\ge\eta'$. Their norms after moving down by η' are bounded by the norms before the shifts after moving down by η/s, by the positive-time contraction. Summing the L^1 norms gives the original norm. The two constants are $C_\wp s^{\wp-4}$ and $C_\wp 2^\wp s^{-4}$. The unitary preserves (Rec5). Dilating the conditioned states proves membership of the centred components in $E[b]$ at the original label. The trivial functional is invariant under every isometry; for F in the span of the tuples supported in $\{x_3>0\}$, $\sigma_0(\theta\bar{F}\cdot F)=|c_\varnothing|^2\ge0$, $c_\varnothing$ being the coefficient of $\varnothing$ in F, so that $\mathcal{H}(\sigma_0)=\mathbb{C}\Omega$, every class $[X]$ with $|X|\ge1$ being 0; $H=0$, the translations act as 1, and the joint spectral measure is the point mass at $(0,0)$, which is carried by $\{E\ge|q|\}$; the left member of (Rec4) vanishes, because every tuple in $\theta\bar{F}\cdot(h)\cdot F'$ has the entry h; and $\ker H=\mathbb{C}\Omega$. The same dilation proof applies to the trivial data.

Step 4: clause (d). (d1): the components are reconstruction data, by (c). If $\xi\ne\xi'$, then $\sigma_\xi\ne\sigma_{\xi'}$, the map $\xi\mapsto\sigma_\xi$ being injective by the definition of the canonical decomposition; by (b) they differ at some $X\in\mathfrak{S}(O)$. Each coordinate $\xi\mapsto\sigma_\xi(X)$ is continuous by the definition of the canonical decomposition; so the map into the product space is continuous and injective, and, Ξ being compact and the product space a Hausdorff space, it is a homeomorphism onto its image. (d2): A convergent Gibbs sequence (K_j,β_j,ν_j) at a reference coupling g has its labels $a_{K_j}\beta_j$ in the compact interval

$I^{\mathrm{lab}}(g) \subset (0,\infty)$ by Proposition YM3_I:74.1; along a subsequence whose labels tend to a limit point b of them, the spacings a_{K_j} and the couplings β_j form a sequence of pairs $\mathfrak{q}$ with the label b, and the σ_{ν_j} tend there to the limit of the sequence, which so lies in $\mathsf{K}_{\mathfrak{q}} \subset A[b]$ (Definition 18.164). Now $\sigma = \sigma'$ by (b) and (c). The two triples decompose the same functional. For each finite list of tuples the plateaux give all mixed moments of their evaluation coordinates. These coordinates separate points on the compact union of the two images. Stone–Weierstrass thus gives equality of the pushforward measures, since coordinate polynomials are dense in the continuous functions on the union. Their supports are the two compact images, by full support of the component measures. Inverting the second evaluation map after the first yields the unique homeomorphism h and $h_*\lambda = \lambda'$. For label recovery take the cover in Step 3 with $g_i \le \gamma_W$. By Theorem YM3_I:AW, along one fixed positive test shape shrinking to zero the ratio of the frame two-point function to $\mathcal{K}_6$ tends to $c_2(b/s)^{-2}$, since the frame label is b/s. Simultaneous dilation of the tests preserves $\mathcal{K}_6$ and the shape parameter, and dilation of the functional multiplies the two-point coefficient by s^{-2}. The resulting coefficient is $c_2 b^{-2}$. Equality of the functionals therefore gives $b = b'$. (d3): σ_0 is reconstruction data, by (c). A pair of $\mathfrak{S}_2(\mathbb{R}^3)$ with the product of its supports in $O^2 \cap (\mathbb{R}^3)^2_{\neq}$ has an entry 0 or lies in $\mathfrak{S}_2(O)$, as in Step 2; so (a) at $n = 2$, read for σ and σ_0, gives $\sigma = 0$ on $\mathfrak{S}_2(\mathbb{R}^3)$. For a real $f \in C^1_c(\{x_3 > 0\})$ the pair $(\theta f, f)$ is separated, and $\|[(f)]\|^2 = \sigma(\theta f, f) = 0$. Let $n \ge 1$ and $X = (f_1, \dots, f_n) \in \mathfrak{S}_n(\mathbb{R}^3)$ with $\operatorname{supp} f_1 \subset \{x_3 < 0\}$ and $\operatorname{supp} f_i \subset \{x_3 > 0\}$ for $i \ge 2$. With $f = f_1 \circ \theta$ and $Y = (f_2, \dots, f_n)$, $X = \theta(f) \cdot Y$, so $\sigma(X) = \langle [(f)], [Y] \rangle$ and $|\sigma(X)| \le \|[(f)]\| \, \|[Y]\| = 0$ by the inequality of Schwarz for the positive form. Since the set $\{x \in (\mathbb{R}^3)^n_{\neq} : (x_1)_3 < 0, \text{ and } (x_i)_3 > 0 \text{ for } 2 \le i \le n\}$ is open and nonempty, and every tuple for which the product of its supports is contained in this set either has an entry equal to 0 or belongs to the kind just treated, (a), applied to σ and σ_0, yields $\sigma = \sigma_0$ on $\mathfrak{S}_n(\mathbb{R}^3)$ for every $n \ge 1$.

Step 5: clause (e), the normal family. Let D be a countable dense subset of the unit sphere. For $u \in \mathsf{D}$, a permutation ς and $m \in \mathbb{N}$ let $\mathcal{N}_{u,\varsigma,m}$ be the set of the $z \in (\mathbb{C}^3)^n$ with $d^u_\circ(z_{\varsigma(k+1)} - z_{\varsigma(k)}) > 2/m$ for $1 \le k < n$ (for $n = 1$, all of $\mathbb{C}^3$). Since $d^u_\circ$ is continuous, this set is open; it is contained in $\mathsf{Tb}^{u,\varsigma}_n$. Every point of $\mathsf{Tb}^{\mathrm{all}}_n$ lies in one of these sets, which form a countable family: for $n = 1$ each of them is $\mathbb{C}^3 = \mathsf{Tb}^{\mathrm{all}}_1$; for $n \ge 2$ and $z \in \mathsf{Tb}^{u_0,\varsigma}_n$, the displayed margin formula makes $u \mapsto \min_k d^u_\circ(z_{\varsigma(k+1)} - z_{\varsigma(k)})$ a continuous function on the sphere, and its value at u_0 is positive; therefore its value at some $u \in \mathsf{D}$ is larger than $2/m$, for some m. Put $D_j = \mathbf{S}^{(j)}_n - \mathbf{S}'^{(j)}_n$, a holomorphic function on $\mathsf{Tb}^{\mathrm{all}}_n$. By YM3_I:(2066) at $\eta = 1/m$, read with the numbers C and $\wp$, $|D_j| \le 2(Cm^{\wp})^n$ on $\mathcal{N}_{u,\varsigma,m}$ for every j. Number these sets $\mathcal{N}_1, \mathcal{N}_2, \dots$, and let $J_0 \subset \mathbb{N}$ be infinite. By the theorem of Montel (a sequence of holomorphic functions on an open subset of $\mathbb{C}^N$ with a common bound has a subsequence that converges uniformly on every compact subset) there are infinite sets $J_0 \supset J_1 \supset J_2 \supset \cdots$ such that $(D_j)_{j \in J_r}$ converges uniformly on the compact subsets of $\mathcal{N}_r$; let J_∞ hold the r-th member of J_r for every r. Along J_∞ the D_j converge uniformly on every closed ball that lies in one of the $\mathcal{N}_r$; every point of $\mathsf{Tb}^{\mathrm{all}}_n$ is the centre of such a ball, so the limit D is holomorphic on $\mathsf{Tb}^{\mathrm{all}}_n$ by locally uniform convergence, and the convergence is uniform on every compact subset of $\mathsf{Tb}^{\mathrm{all}}_n$, which finitely many such balls cover. For $X = (f_i) \in \mathfrak{S}_n(\mathbb{R}^3)$ with $\prod_i \operatorname{supp} f_i \subset W$, the uniform convergence on this compact set and the hypothesis give $\int D \prod_i f_i = \lim_{j \in J_\infty} (\sigma^{(j)}(X) - \sigma'^{(j)}(X)) = 0$; as in Step 1, $D = 0$ on W, and then on $\mathsf{Tb}^{\mathrm{all}}_n$. So every infinite J_0 holds an infinite J_∞ along which $\sup_{\mathcal{K}} |D_j| \to 0$ for every compact $\mathcal{K} \subset (\mathbb{R}^3)^n_{\neq}$, a compact subset of $\mathsf{Tb}^{\mathrm{all}}_n$; hence $\sup_{\mathcal{K}} |D_j| \to 0$ along $\mathbb{N}$, as otherwise some $\epsilon > 0$ and some infinite J_0 would have $\sup_{\mathcal{K}} |D_j| \ge \epsilon$ on J_0. At the real points $D_j = \mathsf{s}^{(j)}_n - \mathsf{s}'^{(j)}_n$, and $|\sigma^{(j)}(X) - \sigma'^{(j)}(X)| \le \sup_{\mathcal{K}} |D_j| \prod_i \|f_i\|_{L^1}$ with $\mathcal{K} = \prod_i \operatorname{supp} f_i$.

Step 6. At fixed g the constants of Theorem YM3_I:AP are uniform in the sequences and components. For compact J, Step 3 gives a dilation with $s \in [s_-, s_+]$ from one of the finitely many g_i. The centred components have frame bound $C^{g_i}_{\wp} \le C'_{g_i}$. For $\wp = 4$ the dilation bound is therefore at most $\max(1, 16 s_-^{-4}) \max_i C'_{g_i}$ for every member of the listed classes. $\square$

Lemma 18.167 (Choosing cutoffs on growing tori)**.** Fix $0 < g \le \gamma$ and let $\ell_k \in \mathfrak{L}$ tend to infinity. For each k, let $(K_i^k, \beta_i^k)_i$ be pairs in $\mathfrak{d}(g)$ with $K_i^k \to \infty$. Suppose that for each $n \ge 2$ and each separated C_c^1 tuple X in Q_{ℓ_k} the following limit exists.

$$B_n^{(k)}(X) = \lim_{i\to\infty} S^T_{n;K_i^k,\ell_k,\beta_i^k}(X^{\langle \ell_k\rangle}). \tag{18.xiii}$$

There are indices $i(k)$ with $K^k_{i(k)} \ge k$ for which

$$S^T_{n;K^k_{i(k)},\ell_k,\beta^k_{i(k)}}(X^{\langle \ell_k\rangle}) - B_n^{(k)}(X) \longrightarrow 0 \tag{18.xiv}$$

for every separated C_c^1 tuple X in $\mathbb{R}^3$, along the full sequence of k. If b_k is a prescribed limit point of $(a_{K_i^k}\beta_i^k)_i$, the indices can also satisfy $|a_{K^k_{i(k)}}\beta^k_{i(k)} - b_k| \le 1/k$.

Every infinite set of k contains a subsequence on which all $B_n^{(k)}(X)$ converge. The selected triples on any such subsequence realize the same limit along growing tori at g, which belongs to $\mathcal{K}(g)$. Each subsequential limit of the corresponding b_k is a label of that limit. ■

Proof. For each finite union D of closed cubes with rational vertices and edges parallel to the coordinate axes, take a countable dense subset of the C_c^1 functions supported in D, in the norm $\|\cdot\|_\sharp$. Such a subset exists since the function and its three first derivatives embed this space into $C(D)^4$, with an equivalent norm. Enumerate the tuples from these dense sets whose respective unions D are pairwise disjoint, over all orders $n \ge 2$. For each k, require that the error in eq. 18.xiv be at most $1/k$ on those of the first k enumerated tuples that have supports in Q_{ℓ_k}. The fixed-side convergence makes these finitely many inequalities and $K_i^k \ge k$ hold for all sufficiently large i. When a label is prescribed, choose such an i among the infinitely many indices whose labels lie within $1/k$ of b_k.

Any fixed separated tuple has its entries supported in pairwise disjoint unions D of the indicated kind. Approximate each entry in $\|\cdot\|_\sharp$ using the corresponding dense set. Proposition YM3_I:91.13 bounds the finite-cutoff forms uniformly on the product of these support classes for all sufficiently large k, since the unions have a common radius bound and a positive separation, and $\ell_k \to \infty$. The same bound passes to $B_n^{(k)}$. Telescoping the difference as one entry varies at a time then extends eq. 18.xiv from the enumerated tuples to the given tuple. Here every fixed enumerated tuple lies in Q_{ℓ_k} and among the first k entries for all sufficiently large k.

Eventually $\ell_k \in \mathfrak{L}(g)$, so the selected triples form a growing-torus sequence at g. Proposition YM3_I:92.1 gives the asserted subsequential compactness and bound for that sequence. Equation eq. 18.xiv transfers the conclusions to the inner limits. The error in the prescribed labels tends to zero, so a subsequence on which $b_k \to b$ makes the selected labels tend to b as well. □

Proposition 18.168 (Limit points of the torus limits: spacing first, then the volume)**.** Assume the setting. Let $0 < g \le \gamma$, $b \in I(g)$, and let $\ell_k \in \mathfrak{L}(g)$, $k \in \mathbb{N}$, be sides with $\ell_k \to \infty$.

(o) *The spacing first, on each torus.* For every k, $n \ge 2$ and $X \in \mathfrak{S}_n(Q_{\ell_k})$, $\mathsf{B}(\ell_k, b; X) = \lim_{K\to\infty} \mathsf{a}(K, \ell_k, L^K b; X)$ along the full sequence of the cutoffs, every pair $(K, L^K b)$ lying in $\mathfrak{d}_b \subset \mathfrak{d}(g)$; the same number is the limit of $\mathsf{a}(K_j, \ell_k, \beta_j; X)$ along all pairs (K_j, β_j) with $K_j \to \infty$ and $a_{K_j}\beta_j \to b$; and if X is of the type $(\mathsf{d}; R)$ and $\ell_k \ge 16n(R+1)$, then $|\mathsf{B}(\ell_k, b; X)| \le C_n^g(\mathsf{d}; R)\|X\|$.

(a) *Then the volume: limit points exist.* Every infinite set of indices holds indices $k_1 < k_2 < \cdots$ such that $\mathsf{B}(\ell_{k_i}, b; X)$ converges as $i \to \infty$ for all $n \ge 2$ and all $X \in \mathfrak{S}_n(\mathbb{R}^3)$; the family of the limits is a member of $\mathrm{SF}(b)$, taken along $(\ell_{k_i})_i$. In particular $\mathrm{SF}(b) \ne \varnothing$.

(b) *They are limits along growing tori.* Let $S \in \mathrm{SF}(b)$ be taken along sides ℓ'_k. There are cutoffs $K_k \ge k$ such that, for the k with $\ell'_k \in \mathfrak{L}(g)$, $(K_k, \ell'_k, L^{K_k} b)_k$ is a torus sequence at g over $\mathfrak{d}_b$ that defines S. So $S \in \mathrm{GT}_b \subset \mathrm{GT}(g)$, b is a label of S, and $S \in \mathcal{K}(g)$.

■

Proof. Step 1. Corollary YM3_I:74.2 gives the data $(K, L^K b) \in \mathfrak{d}(g)$. For X supported in Q_{ℓ_k}, periodization keeps the entries C^1 with disjoint supports and the same sharp norms: the quotient is isometric on the box and every test vanishes near its boundary. Theorem YM3_I:AS gives the limit along all cutoffs and YM3_I:(2216) gives the same limit along moving labels. A zero entry gives a zero cumulant. Proposition YM3_I:91.13 bounds the finite-cutoff cumulants at the stated side and test parameters, and the bound passes to the limit.

Step 2: clause (a). By Proposition YM3_I:91.13, the set of pairs $\mathfrak{d}(g)$ has the property (PB)$[\underline{C}^g; 2L^{m_{\mathfrak{L}}(g)}, 2\mathsf{l}(\cdot,\cdot)]$ on the tori; for $\ell \in \mathfrak{L}$ the condition $\ell \geq 2\mathsf{l}(n, R)$ says $\ell \geq 16n(R+1)$, and the sides of $\mathfrak{L}$ that are at least $2L^{m_{\mathfrak{L}}(g)}$ form $\mathfrak{L}(g)$. These are the bounds used in Lemma 18.167, applied with $\ell'_k = \ell_k$ and $(K_i^k, \beta_i^k) = (i, L^i b)$ for each k. Its inner limits eq. 18.xiii exist by (o) and equal $\mathsf{B}(\ell_k, b; X)$. The lemma gives, in every infinite set of indices, $k_1 < k_2 < \cdots$ along which $\mathsf{B}(\ell_{k_i}, b; X)$ converges for all $n \geq 2$ and $X \in \mathfrak{S}_n(\mathbb{R}^3)$; the limits form a member of $\mathrm{SF}(b)$, taken along $(\ell_{k_i})_i$ (Definition 18.164(2)).

Step 3: clause (b). Of its sides Lemma 18.167 asks only that they lie in $\mathfrak{L}$ and tend to infinity, as the sides ℓ'_k do; and at every side $\ell \in \mathfrak{L}$ and every $X \in \mathfrak{S}_n(Q_\ell)$, $n \geq 2$, the inner limit $\lim_i \mathsf{a}(i, \ell, L^i b; X)$ exists and is $\mathsf{B}(\ell, b; X)$ (Theorem YM3_I:AS, YM3_I:(2216)), the pairs $(i, L^i b)$ lying in $\mathfrak{d}(g)$ (Step 1). So Lemma 18.167, read as in Step 2 but for the whole sequence $(\ell'_k)_k$, gives indices $i(k)$ such that $K_k = i(k) \geq k$ for every k and $\mathsf{a}(K_k, \ell'_k, L^{K_k} b; X) - \mathsf{B}(\ell'_k, b; X) \to 0$ along the full sequence k, for all $n \geq 2$ and $X \in \mathfrak{S}_n(\mathbb{R}^3)$; so $\mathsf{a}(K_k, \ell'_k, L^{K_k} b; X) \to S_n(X)$. As $\mathfrak{L}(g)$ is a tail of $\mathfrak{L}$ and $\ell'_k \to \infty$, the k with $\ell'_k \in \mathfrak{L}(g)$ include every k from some index on. The triples $(K_k, \ell'_k, L^{K_k} b)$ of these k have their pairs in $\mathfrak{d}_b \subset \mathfrak{d}(g)$, $K_k \to \infty$ and sides $\ell'_k \to \infty$ in $\mathfrak{L}(g)$: they form a torus sequence at g over $\mathfrak{d}_b$ that defines S, and every member has the label $a_{K_k} L^{K_k} b = b$. So $S \in \mathrm{GT}_b \subset \mathrm{GT}(g)$, b is a label of S, and $S \in \mathcal{K}(g)$ by Proposition YM3_I:92.1. □

Proposition 18.169 (Spacing-first limits at every label, and what they have)**.** Assume the setting. Let $b > 0$ be real and let $\ell_k = 2L^{m_k} \in \mathfrak{L}$, $k \in \mathbb{N}$, be sides with $\ell_k \to \infty$. Let $z \geq 0$ be an integer with $L^z b \geq 2\gamma^{-2}$ and put $g_z = (L^z b/2)^{-1/2}$; then $0 < g_z \leq \gamma$ and $L^z b \in I(g_z)$. Put $C_n^{(z)}(\mathsf{d}; R) = L^{zn} C_n^{g_z}(L^z \mathsf{d}; L^z R)$ for $n \geq 2$ and $\mathsf{d}, R > 0$.

(o) For every k, $n \geq 2$ and $X \in \mathfrak{S}_n(Q_{\ell_k})$, $\mathsf{B}(\ell_k, b; X) = \lim_K \mathsf{a}(K, \ell_k, L^K b; X)$ exists along the full sequence of the cutoffs, and the same number is the limit along all pairs with $K_j \to \infty$ and $a_{K_j}\beta_j \to b$; and $\mathsf{B}(\ell_k, b; X) = L^{z'n}\mathsf{B}(L^{z'}\ell_k, L^{z'}b; X^{[z']})$ for every integer z' with $m_k + z' \geq 1$.

(a) *Extraction.* Every infinite set of indices holds $k_1 < k_2 < \cdots$ such that $\mathsf{B}(\ell_{k_i}, b; X)$ converges as $i \to \infty$ for all $n \geq 2$ and $X \in \mathfrak{S}_n(\mathbb{R}^3)$; the family of the limits lies in $\mathrm{SF}(b)$. So $\mathrm{SF}(b) \neq \varnothing$.

(b) *Dilations; growing tori.* $\mathrm{SF}(L^{z'}b) = \{\mathrm{dil}_{L^{z'}} S : S \in \mathrm{SF}(b)\}$ for every $z' \in \mathbb{Z}$. Every $S \in \mathrm{SF}(b)$ is $\mathrm{dil}_{L^{-z}} S'$ with $S' = \mathrm{dil}_{L^z} S \in \mathrm{SF}(L^z b) \subset \mathrm{GT}_{L^z b} \subset \mathrm{GT}(g_z)$, the label $L^z b$ being a label of S' at g_z. If $b \in I(g)$ for some $0 < g \leq \gamma$, then $\mathrm{SF}(b) \subset \mathrm{GT}_b \subset \mathrm{GT}(g)$.

(c) *What every $S \in \mathrm{SF}(b)$ has.*

(c1) Every S_n, and σ_S, is invariant under all translations τ_v, $v \in \mathbb{R}^3$, and under W_3.

(c2) σ_S is reflection positive at every coordinate plane $\{x_\mu = c\}$, $\mu \in \{1, 2, 3\}$, $c \in \mathbb{R}$, from both sides.

(c3) S is a family of real, symmetric, multilinear maps, and $|S_n(X)| \leq C_n^{(z)}(\mathsf{d}; R)\|X\|$ for all $n \geq 2$, $\mathsf{d}, R > 0$ and $X \in \mathfrak{S}_n(\mathbb{R}^3)$ of the type $(\mathsf{d}; R)$. If $b \in I(g)$, then also $|S_n(X)| \leq C_n^g(\mathsf{d}; R)\|X\|$ and $|S_n(X)| \leq C_n^{g,\infty}(\mathsf{d}; R)\|X\|$, with the table of Definition 18.164. None of these numbers reads a cutoff, a side, a sequence or a member of $\mathrm{SF}(b)$.

(c4) S and σ_S have off-diagonal distribution kernels: for every n there is exactly one distribution on $(\mathbb{R}^3)^n_{\neq}$ whose value at $f_1 \otimes \cdots \otimes f_n$ is $\sigma_S(f_1, \ldots, f_n)$ for every separated tuple of smooth functions, and the same for S_n, $n \geq 2$; these distributions are of order at most $4n + 1$ on compact sets.

(c5) σ_S satisfies the flat Schwartz-test growth estimate of Proposition YM3_I:93.4 with the constants $(k_{\mathrm{E}}, \mathsf{c}_{\mathrm{E}}, L^{z(1+k_{\mathrm{E}})}\mathsf{C}_{\mathrm{E}}^{\mathrm{c},g_z}, \varkappa_{\mathrm{E}})$, where $(k_{\mathrm{E}}, \mathsf{c}_{\mathrm{E}}, \mathsf{C}_{\mathrm{E}}^{\mathrm{c},g}, \varkappa_{\mathrm{E}})$ are the canonical growth constants at g of Definition 18.164; if $b \in I(g)$, with the canonical growth constants at g themselves.

(c6) $\mathcal{H}(S)$ is formed and separable; $\mathcal{T}_t = e^{-tH_S}$, $t \geq 0$, is a strongly continuous semigroup of self-adjoint contractions with $\ker \mathcal{T}_t = 0$, $H_S \geq 0$ and $H_S\Omega = 0$; $\mathcal{U}(v)[F] = [\tau_{(v,0)}F]$, $v \in \mathbb{R}^2$, is a strongly continuous unitary group $e^{iv_1P_1}e^{iv_2P_2}$ that commutes with $\mathcal{T}_t$ and fixes Ω; the joint spectral measure of (H_S, P_1, P_2) is carried by $[0,\infty)\times\mathbb{R}^2$; and every $w \in W_2$ acts by the unitary $\mathcal{U}(w)[F] = [wF]$, which fixes Ω, commutes with H_S, and maps the joint spectrum onto itself by $(E,q) \mapsto (E, wq)$.

(d) *The law of two points.* Let $\vartheta \in (0,1)$. If $b \in I(g)$ with $0 < g \leq \gamma_W$, then for every $S \in \mathrm{SF}(b)$, every $d \in (0, d_0(g,\vartheta)]$ and every pair (f,h) of the class of the law at the parameter ϑ and the size d, $S_2(f,h) = c_2 b^{-2} K_{\mathrm{free}}(f,h)(1+\eta)$ with $|\eta| \leq \bar{\varepsilon}_{g,\vartheta}(d) \leq \frac{1}{3}$. For every real $b > 0$, with an integer $z' \geq 0$ such that $L^{z'}b \geq 2\gamma_W^{-2}$ and with $g' = (L^{z'}b/2)^{-1/2} \leq \gamma_W$, the same holds for $0 < L^{z'}d \leq d_0(g',\vartheta)$, with $|\eta| \leq \bar{\varepsilon}_{g',\vartheta}(L^{z'}d) \leq \frac{1}{3}$. In both cases $b^{-2} = \lim_{t\to 0} S_2(f(\cdot/t), h(\cdot/t))/(c_2 K_{\mathrm{free}}(f,h))$ for every pair (f,h) of the class at the parameter ϑ and any size; so $\mathrm{SF}(b) \cap \mathrm{SF}(b') = \varnothing$ for $b \neq b'$.

■

Proof. Step 0. Such integers z exist because $L > 1$. With $x = L^z b$ and $g_z^{-2} = x/2$, we have $g_z \leq \gamma$ and $x/2 + \bar{B} < x$, since $\gamma^{-2} > \bar{B}$. The right end $L(x/2+\bar{B}) - 2\bar{B}$ is at least x, so $x \in I(g_z)$.

Step 1. Theorem YM3_I:AS and YM3_I:(2216) give the limits along fixed and moving labels on every side. For the dilation use YM3_I:(2215). Periodization commutes with dilation, since

$$\sum_{p \in L^z \ell\mathbb{Z}^3} f(L^{-z}(x+p)) = \sum_{q \in \ell\mathbb{Z}^3} f(L^{-z}x + q).$$

The formula for negative z' follows by inversion.

Step 2: clauses (a) and (b). Let $z' \in \mathbb{Z}$ and $S \in \mathrm{SF}(b)$, taken along (ℓ_k). For $Y \in \mathfrak{S}_n(\mathbb{R}^3)$ and all large k, (o) at $X = Y^{[-z']} = Y_{(L^{z'})}$ gives $\mathsf{B}(L^{z'}\ell_k, L^{z'}b; Y) = L^{-z'n}\mathsf{B}(\ell_k, b; Y_{(L^{z'})}) \to L^{-z'n}S_n(Y_{(L^{z'})}) = (\mathrm{dil}_{L^{z'}}S)_n(Y)$; so $\mathrm{dil}_{L^{z'}}S \in \mathrm{SF}(L^{z'}b)$, taken along the sides $L^{z'}\ell_k \in \mathfrak{L}$ of the large k. The other inclusion is the same statement at $(L^{z'}b, -z')$, as $\mathrm{dil}_{L^{-z'}}\mathrm{dil}_{L^{z'}}$ is the identity. At $z' = z$ the sides $L^z\ell_k$ lie in $\mathfrak{L}(g_z)$ from some index on, and Proposition 18.168, at the coupling g_z and the label $L^z b \in I(g_z)$, gives in every infinite set of indices a subsequence along which $\mathsf{B}(L^z\ell_k, L^z b; Y)$ converges at every Y; by (o) at $z' = z$, $\mathsf{B}(\ell_k, b; X) = L^{zn}\mathsf{B}(L^z\ell_k, L^z b; X^{[z]})$ converges at every X along it, which is (a). The same proposition gives $\mathrm{SF}(L^z b) \subset \mathrm{GT}_{L^z b} \subset \mathrm{GT}(g_z)$, with the label $L^z b$, and $\mathrm{SF}(b) \subset \mathrm{GT}_b$ when $b \in I(g)$.

Step 3. At $b \in I(g)$ the family S is a growing-torus limit by clause (b). Proposition YM3_I:92.3 gives the symmetries and Proposition YM3_I:92.4 the coordinate reflection positivity. Proposition YM3_I:92.1 gives the multilinear family and its bound C_n^g. The normalizer and growth bounds of Propositions YM3_I:89.4 and YM3_I:89.5 at the source-system constants in Definition 18.164 apply on each defining torus as soon as the side holds the tests and is large enough for their type. Passing to the limit gives the sharper table $C_n^{g,\infty}$. The moment–cumulant formula bounds each order of σ_S for fixed support separation. Proposition YM3_I:93.1 applies to this family and to S, and gives the unique off-diagonal kernels of order at most $4n+1$. For the global bound, use the factorial bound at $R = 1$ with $A(d) \leq A_{\mathrm{PL}} \max(1, d^{-4})$. Proposition YM3_I:93.4 at $a = 4$ gives exactly the canonical growth constants above. Finally apply Proposition YM3_I:94.1 to the reflection form in direction e_3 and to its tangential translations and W_2 symmetries.

Step 4: clause (c) at every label. Let $S \in \mathrm{SF}(b)$ and $S' = \mathrm{dil}_{L^z}S$; then $S = \mathrm{dil}_{L^{-z}}S'$, that is $S_n(X) = L^{zn}S'_n(X^{[z]})$ with $X^{[z]} = X_{(L^{-z})}$, and $\sigma_S = \mathrm{dil}_{L^{-z}}\sigma_{S'}$ (Definition 18.164(1)). By

(b) and Step 3 at $(g_z, L^z b)$, S' has (c1)–(c6) with the numbers at g_z. The map $X \mapsto X^{[z]}$ is a bijection of $\mathfrak{S}_n(O)$ onto $\mathfrak{S}_n(L^z O)$ for every open O; it carries the type $(\mathsf{d}; R)$ to the type $(L^z\mathsf{d}; L^z R)$; $\|f^{[z]}\|_\sharp = \max\{\|f\|_\infty, 40ML^{-z}\operatorname{Lip} f\} \le \|f\|_\sharp$, as $z \ge 0$; and $(\tau_v X)^{[z]} = \tau_{L^z v} X^{[z]}$, $(wX)^{[z]} = wX^{[z]}$ for $w \in W_3$, $(\theta_{\mu,c}X)^{[z]} = \theta_{\mu,L^z c}X^{[z]}$. (c1): $S_n(\tau_v X) = L^{zn}S'_n(\tau_{L^z v}X^{[z]}) = S_n(X)$, and the same for W_3 and for σ_S. (c2): for $F = \sum_i c_i X_i$ in the span of the tuples supported in $\{\epsilon(x_\mu - c) > 0\}$, $\sigma_S(\theta_{\mu,c}\bar F \cdot F) = \sum_{i,k} \bar c'_i c'_k \sigma_{S'}(\theta_{\mu,L^z c}X_i^{[z]} \cdot X_k^{[z]}) \ge 0$, where $c'_i = L^{z|X_i|}c_i$ and $\bar c'_i = L^{z|X_i|}\bar c_i$, as $L^{z|X_i|}$ is real, and where the tuples $X_i^{[z]}$ are supported in $\{\epsilon(x_\mu - L^z c) > 0\}$. (c3): $|S_n(X)| = L^{zn}|S'_n(X^{[z]})| \le L^{zn}C_n^{g_z}(L^z\mathsf{d}; L^z R)\|X^{[z]}\| \le C_n^{(z)}(\mathsf{d}; R)\|X\|$. (c4): with $F_z(y) = F(L^{-z}y)$, which is a test function on $(\mathbb{R}^3)^n_{\ne}$ when F is one, $F \mapsto L^{zn}T'(F_z)$ is a distribution of the same order for every distribution T' of $\sigma_{S'}$ or of S'_n, and at $F = \bigotimes_i f_i$ it takes the value $L^{zn}\sigma_{S'}(X^{[z]}) = \sigma_S(X)$, respectively $S_n(X)$; uniqueness passes in the same way at $-z$. (c5): the space $\mathcal{S}_{\ne}(\mathbb{R}^{3n})$ and the norms $|\cdot|_k$ are the Schwartz functions on $\mathbb{R}^{3n}$ whose derivatives of every order, the order 0 included, vanish at each point outside $(\mathbb{R}^3)^n_{\ne}$, and $|\varphi|_k = \sup\{(1+|x|^2)^{k/2}|\partial^\alpha\varphi(x)| : x \in \mathbb{R}^{3n},\ |\alpha| \le k\}$, α running over the multi-indices. With the Schwinger functions $\mathsf{S}_n^{\sigma_{S'}}$ of $\sigma_{S'}$ put $\mathsf{S}_n(\varphi) = L^{zn}\mathsf{S}_n^{\sigma_{S'}}(\varphi_z)$, $\varphi_z(y) = \varphi(L^{-z}y)$, for $\varphi \in \mathcal{S}_{\ne}(\mathbb{R}^{3n})$, a space that holds φ_z with φ; at the tensor of a separated tuple of smooth functions it is $\sigma_S(X)$; and $|\varphi_z|_k \le L^{zk}|\varphi|_k$, because $\partial^\alpha\varphi_z(y) = L^{-z|\alpha|}(\partial^\alpha\varphi)(L^{-z}y)$ with $L^{-z|\alpha|} \le 1$, and $1 + |y|^2 \le L^{2z}(1 + |L^{-z}y|^2)$, as $z \ge 0$; so $|\mathsf{S}_n(\varphi)| \le \mathsf{c}_{\mathrm{E}}(L^{z(1+k_{\mathrm{E}})}\mathsf{C}_{\mathrm{E}}^{\mathrm{c},g_z})^n(n!)^{\varkappa_{\mathrm{E}}}|\varphi|_{k_{\mathrm{E}}n}$. (c6): Use the reflection form under dilation at $s = L^{-z}$ as in the proof of Proposition 18.166: σ_S is reflection positive, $[F] \mapsto [V_s F]$, $V_s X = s^{-|X|}X_{(s)}$, extends to a unitary $\mathcal{V}\colon \mathcal{H}(S) \to \mathcal{H}(S')$ with $\mathcal{V}\Omega = \Omega$, $H_S = L^z\mathcal{V}^{-1}H_{S'}\mathcal{V}$, and $\mathcal{V}$ carries the translation by v along the plane to that by $L^z v$. The dilation commutes with every $w \in W_2$, so $\mathcal{V}$ carries $\mathcal{U}(w)$ to $\mathcal{U}(w)$. So all of (c6) passes from S' to S; the joint spectral measure of S is the image of that of S' under $(E, q) \mapsto (L^z E, L^z q)$, which keeps $[0,\infty) \times \mathbb{R}^2$ and commutes with W_2.

Step 5: clause (d). Let $b \in I(g)$, $0 < g \le \gamma_W$, and $S \in \mathrm{SF}(b)$. By (b), S is a continuum limit at g along growing tori with the label b, which is its one label by Theorem YM3_I:AW; Theorem YM3_I:AW is the law, and $\bar\varepsilon_{g,\vartheta} \le \frac{1}{3}$ on $(0, d_0(g,\vartheta)]$. For a real $b > 0$: $L^{z'}b \ge 2\gamma_W^{-2} \ge 2\gamma^{-2}$, so the window calculation of Step 0 gives $L^{z'}b \in I(g')$, and $g' \le \gamma_W$. By (b), $S' = \mathrm{dil}_{L^{z'}}S \in \mathrm{SF}(L^{z'}b)$. For a pair (f, h) of the class at $(d; \vartheta)$, the pair $(f(L^{-z'}\cdot), h(L^{-z'}\cdot))$ is of the class at $(L^{z'}d; \vartheta)$, with the same value of K_{free} by simultaneous dilation of the supports and the homogeneity of K_{free}; so $S_2(f,h) = L^{2z'}S'_2(f(L^{-z'}\cdot), h(L^{-z'}\cdot)) = L^{2z'}c_2(L^{z'}b)^{-2}K_{\mathrm{free}}(f,h)(1+\eta) = c_2 b^{-2}K_{\mathrm{free}}(f,h)(1+\eta)$, $|\eta| \le \bar\varepsilon_{g',\vartheta}(L^{z'}d)$. The limit: $(f(\cdot/t), h(\cdot/t))$ is of the class at the size td, with the same value of K_{free}, which is positive, f and h being ≥ 0 and not 0; the tolerance tends to 0 with the size. So S_2 determines b. □

Proposition 18.170 (A ceiling on the lowest mass at every label; the growth of the Källén–Lehmann measure)**.** Assume the setting, and let $b > 0$ be real.

(a) *Every component at every label.* Let $\sigma \in A[b]$ and let σ_ξ be a component of a decomposition of σ, with its Källén–Lehmann pair (κ_ξ, ρ_ξ), where κ_ξ is the constant part of the two-point kernel and ρ_ξ its Yukawa measure. Then $\ker H_{\sigma_\xi} = \mathbb{C}\Omega$; $\operatorname{spec} H_{\sigma_\xi}$ meets $(0, C_0/b]$; the connected function of two points of σ_ξ is $S_2^{\sigma_\xi}(f,h) = \iint f(x)h(y)\int \frac{e^{-m|x-y|}}{4\pi|x-y|}\,d\rho_\xi(m)\,dx\,dy$, ρ_ξ being the measure ψ of its kernel; and $\inf\operatorname{supp}\rho_\xi \le C_0/b$.

(b) *The growth of the measure.* Let ω be a member of $E[b]$, or a component of a decomposition of a member of $A[b]$, or, under the absence of latent heat, a member of $A[b]$, and let ψ_ω be the Yukawa measure of its connected two-point kernel, which equals the measure ρ of its Källén–Lehmann pair. Then

$$\psi_\omega([0,M]) \le \frac{e^5}{5^5}\,\frac{3\alpha}{\pi}\,b^{-2}M^5 \le 0.973\,b^{-2}M^5 \qquad \text{for } M \ge \frac{5}{k_0 b},$$

$$\psi_\omega([0,M]) \ge \frac{1}{21^5}\Big(\frac{3}{2\pi\alpha} - 4^5\,\frac{3\alpha}{\pi}\,e^{-63/4}\Big)\,b^{-2}M^5 \ge 4.7\cdot 10^{-9}\,b^{-2}M^5 \qquad \text{for } M \ge \frac{21}{k_0 b}.$$

As $21/k_0 = \frac{7}{2m_\circ}C_0$ and $\frac{7}{2m_\circ} = 1.3526\ldots$, both bounds hold for all $M \ge 1.36\,C_0/b$.

(c) *Below the ceiling: the local form.* Let $0 < g \le \gamma_W$ and $\vartheta \in (0,\vartheta_1]$; let σ be the limit of a convergent Gibbs sequence at g, and $(\Xi,\lambda,(\sigma_\xi))$ a decomposition of it. Then for every $\xi \in \Xi$, $\operatorname{spec} H_{\sigma_\xi}$ meets $(0, m_\circ/d_0(g,\vartheta)]$ and $\inf\operatorname{supp}\rho_\xi \le m_\circ/d_0(g,\vartheta)$; and $\operatorname{spec} H_\sigma$ meets $(0, m_\circ/d_0(g,\vartheta)]$. Here $m_\circ = 2.5874\ldots < 2.59$.

(d) *Limits along growing tori and spacing-first limits.* Let $0 < g \le \gamma_W$, $\vartheta \in (0,\vartheta_1]$ and $S \in \mathrm{GT}(g)$. Then σ_S is invariant under the translations and reflection positive at $\{x_3 = 0\}$, H_S is formed, and $\dim\ker H_S \ge 2$ or $\operatorname{spec} H_S$ meets $(0, m_\circ/d_0(g,\vartheta)]$; this holds in particular for every $S \in \mathrm{SF}(b)$ with $b \in I(g)$. For every real $b > 0$ and every $S \in \mathrm{SF}(b)$, $\dim\ker H_S \ge 2$ or $\operatorname{spec} H_S$ meets $(0, C_0/b]$.

■

Proof. Step 1. Dilation to a reference window. Put $G = \gamma_W^{-2}$ and $B = \bar{B}$. The interiors of $I(g_1)$ and $I(g_2)$ are $(G+B, \kappa_0 - 2B)$ and $(G+B+2B/L, \kappa_0)$. They cover $(\kappa_0/L, \kappa_0)$ because $G > L^2 B$ by YM3_I:(2346). The endpoint κ_0 belongs to the interior $(2G+B, 2LG+(L-2)B)$ of $I(g_3)$. Thus every $c > 0$ has a dilate $L^z c$ in one of the three open windows, with $L^z c \le \kappa_0$.

Suppose that $\sigma \in A[b]$ comes from spacings $a_j = t_j L^{-K_j}$, with $1 \le t_j < L$ and $K_j \to \infty$. Pass to a subsequence with $t_j \to t \in [1,L]$. Choose z, g_i so that $\widehat{b} = L^z b/t \in I(g_i)^\circ$ and $\widehat{b} \le \kappa_0$. Change length units by $s_j = t_j L^{-z}$, so the new spacing is $\widehat{a}_j = L^{-(K_j - z)}$ and its label tends to $\widehat{b}$. Corollary YM3_I:74.2 makes these pairs first-passage data for all large j. The change of units is a relabelling of the Wilson action and of the local Gibbs equations. The plaquette observable obeys $\mathcal{O}_{a_j}(f) = s_j^{-1}\mathcal{O}_{\widehat{a}_j}(f(s_j\,\cdot))$ on corresponding configurations, so the same dilation formula holds for their self-centred moments. The norm bounds of Proposition YM3_I:91.13, combined by the moment–cumulant formula, are uniform when a separated tuple is varied in sharp norm in a common compact set with a positive separation margin. For each fixed C_c^1 test, $f(s_j\,\cdot) \to f(s\,\cdot)$ in this norm with $s = tL^{-z} = b/\widehat{b}$. A multilinear telescope therefore gives a convergent Gibbs sequence in the new units with limit $\widehat{\sigma}$ and $\sigma = \mathrm{dil}_s\widehat{\sigma}$. Canonical decompositions are carried by the same dilation, since it preserves the barycentre and the iterated plateau identities of Theorem YM3_I:AL, with each translation parameter divided by s in the new units. Uniqueness of the canonical decomposition identifies the dilated components.

For any such functional σ' and its dilate, the map $[X] \mapsto s^{-|X|}[X_{(s)}]$ is an isometry from the reflection space of the dilate to that of σ' and has the inverse obtained using s^{-1}. It intertwines time u of the dilate with time u/s of σ'. Thus the Hamiltonian spectrum is multiplied by s^{-1}, with the zero space preserved. Changing variables in the connected two-point integral gives $s_{\mathrm{dil}_s\sigma'}(r) = s^{-8}s_{\sigma'}(r/s)$.

Step 2. Connected radial kernels. Theorem YM3_I:AP gives the components at a reference coupling their simple vacuum and clustering, and Proposition YM3_I:108.3 gives their radial two-point kernels. Proposition 13.25(a), (b) identifies the constant part as the large-distance limit and the remainder as the integral against a positive Yukawa measure. If the one-point constant of a component is c_ξ, clustering on a pair of nonnegative unit-mass tests moved far apart gives $\kappa_\xi = c_\xi^2$. Hence its connected kernel is

$$s_\xi(r) = \int_{[0,\infty)} \frac{e^{-mr}}{4\pi r}\,d\rho_\xi(m).$$

The same conclusions apply to a one-vacuum member, since it is its sole component. They pass through the dilation of Step 1 by changing m to m/s and multiplying the measure by s^{-7}. This also shows that the connected kernel is continuous and nonincreasing at every label.

Under absence of latent heat every component's one-point constant vanishes. Indeed, the component is a limit of periodic Gibbs laws centred at the defining laws of the same frame by

Theorems YM3_I:AL and YM3_I:AP. The one-point difference is bounded by a constant for the test times $a_j^{-4}J(\beta_j)$, as in §12. Since $a_j\beta_j \to b > 0$ and $\beta_j^4 J(\beta_j) \to 0$, that difference vanishes. At a reference window the full kernel has a positive Yukawa representation by Proposition 13.25. Its constant term is the large-time limit of the reflected pair of nonnegative unit-mass tests. In the canonical barycentre those pair values tend to zero componentwise and are dominated by their finite values at time zero, so its constant term is zero as well. The conclusion passes through dilation.

Step 3. From the two-point estimate to the growth law. At a reference window g let b' be the limit label. For a ball of radius d centred at y, use the bump $f(x) = (1 - |x-y|^2/d^2)_+^2$. It has mass $(32\pi/105)d^3$ and Lipschitz constant $8/(3\sqrt{3}d)$, so it meets the test conditions at $\vartheta_1 = 3\sqrt{3}/8$. Take two such bumps with centres $8d$ apart and normalize their masses to one. Their supports are at distance $6d$, and all pairwise point distances are in $[6d, 10d]$. For $d \le d_0(g, \vartheta_1)$, the component bound of Corollary YM3_I:127.11 gives a factor in $[2/3, 4/3]$ against the free kernel. The same bound holds for the full connected kernel under absence of latent heat, by averaging the component values in Step 2. Since the kernel is nonincreasing, the two estimates imply

$$s(6d) \ge \tfrac{2}{3}c_2(b')^{-2}(10d)^{-6}, \qquad s(10d) \le \tfrac{4}{3}c_2(b')^{-2}(6d)^{-6}.$$

Taking respectively $d = r/6$ and $d = r/10$ gives

$$\frac{2}{3\alpha}c_2(b')^{-2}r^{-6} \le s(r) \le \frac{4\alpha}{3}c_2(b')^{-2}r^{-6} \qquad (0 < r \le 6d_0(g, \vartheta_1)).$$

Step 1 carries this estimate to label b, with the same two coefficients, because the dilation factor is $s = b/\widehat{b} \ge b/\kappa_0$ and $d_0(g_i, \vartheta_1) \ge d^\star$. Thus it holds on $0 < r \le k_0 b$ for each ω in (b).

Let $G_\omega(r) = 4\pi r s_\omega(r)$ and $c_- = 3/(2\pi\alpha)$, $c_+ = 3\alpha/\pi$. The bounds just proved are $c_- b^{-2}r^{-5} \le G_\omega(r) \le c_+ b^{-2}r^{-5}$. Since $G_\omega(r) \ge e^{-Mr}\psi_\omega([0, M])$, take $r = 5/M$ for the upper bound in (b). For $0 < u < 1$ the mass above M satisfies

$$\int_{(M,\infty)} e^{-mr}\, d\psi_\omega(m) \le e^{-(1-u)Mr}G_\omega(ur).$$

Also $\psi_\omega([0, M]) \ge \int_{[0,M]} e^{-mr} d\psi_\omega(m)$. Subtract the tail bound and use the two inequalities for G_ω. Taking $u = 1/4$ and $r = 21/M$ gives $21^{-5}(c_- - 4^5 c_+ e^{-63/4})b^{-2}M^5$. The numerical upper and lower coefficients are respectively $0.9720447\ldots$ and $4.7129372\ldots\, 10^{-9}$. Finally $21/k_0 = 7C_0/(2m_\circ)$, as claimed.

Step 4. A low-energy spectral test. Take the same bump f of radius d about $2de_3$. For $\theta = \theta_{3,0}$ the pairs $(\theta f, f)$ and $(\theta f, \tau_{4de_3} f)$ have support distances $2d$ and $6d$, and point distances in $[2d, 6d]$ and $[6d, 10d]$. They satisfy the test conditions at all $0 < \vartheta \le \vartheta_1$. For $d = d_0(g, \vartheta)$ apply Corollary YM3_I:127.11 to each component. Writing $C_\xi(u) = S_2^{\sigma_\xi}(\theta f, \tau_{ue_3} f)$ gives

$$C_\xi(0) \le \tfrac{4}{3}c_2(b')^{-2}(2d)^{-6}(\textstyle\int f)^2, \qquad C_\xi(4d) \ge \tfrac{2}{3}c_2(b')^{-2}(10d)^{-6}(\textstyle\int f)^2 > 0.$$

Thus $C_\xi(4d) \ge C_\xi(0)/31250$. The reflection semigroup expresses $C_\xi(u)$ as the Laplace transform of the spectral measure of $[(f)] - \sigma_\xi((f))\Omega$. This vector is orthogonal to the simple vacuum, so its measure has no atom at zero. A nonzero positive measure supported on $(m_\circ/d, \infty)$ would have its transform at $4d$ strictly less than $e^{-4m_\circ}$ times its mass. The ratio just proved excludes this. The spectral measure therefore charges $(0, m_\circ/d]$. The Yukawa representation of Lemma 18.108(a), (b) for the vacuum-subtracted vector gives energies $\sqrt{m^2 + |q|^2} \ge m$, hence $\inf \operatorname{supp} \rho_\xi \le m_\circ/d$.

For the whole limit, put $C(u) = \sigma(\theta f, \tau_{ue_3} f)$. Its canonical decomposition gives $C(u) = \int [C_\xi(u) + \sigma_\xi((f))^2] d\lambda(\xi)$. Each $C_\xi(u)$ tends to zero and is bounded by $C_\xi(0)$. The latter has a finite integral by the same identity at zero. It follows that the Laplace transform for the positive-energy part of the vector $[(f)]$ is $C(u) - C(\infty) = \int C_\xi(u) d\lambda(\xi)$. The integral obeys the same positive lower ratio at $4d$, so that part of the measure charges $(0, m_\circ/d]$. This proves (c) and also yields the conclusions of (a) at every label after Step 1. In fact, at the reference window take $\vartheta = \vartheta_1$. The dilated ceiling is at most $s^{-1}m_\circ/d^\star \le C_0/b$ for both the Hamiltonian and the infimum of the mass support.

Step 5. Growing tori and spacing first. For $S \in \mathrm{GT}(g)$ use the centred reflection functional of Definition 13.4(5) and Theorem YM3_I:AW at its limit label. The same bump calculation applies to S_2, the Laplace transform of the spectral measure of $[(f)]$. The ratio forces positive measure on $[0, m_\circ/d]$. If its atom at zero is positive, the corresponding projection of $[(f)]$ is nonzero and orthogonal to the vacuum because $\sigma_S((f)) = 0$. The zero space then has dimension at least two. If the atom is zero, the positive spectrum meets $(0, m_\circ/d]$. Proposition 18.169(b) includes $\mathrm{SF}(b)$ in $\mathrm{GT}(g)$ when $b \in I(g)$. For general $b > 0$ choose z, g_i by Step 1 with $c = b$. The same proposition carries $S \in \mathrm{SF}(b)$ to $\mathrm{dil}_{L^z} S \in \mathrm{SF}(L^z b) \subset \mathrm{GT}(g_i)$. The unitary scaling in Step 1 preserves the zero-space dimension and returns the upper energy bound $L^z m_\circ/d^\star \le C_0/b$ to S. This proves (d). □

Proposition 18.171 (One limit at a label; one vacuum from ergodic states)**.** Assume the setting.

(a) *Ergodic states give one vacuum.* Let $0 < g \le \gamma$, and let $(K_j, \beta_j, \nu_j)_j$ be a convergent Gibbs sequence at g in which ν_j is ergodic for infinitely many j. Then its limit σ has $\lim_{|v|\to\infty} \sigma(X \cdot \tau_v X) = \sigma(X)^2$ for every $X \in \mathfrak{S}$, and every decomposition of σ has a space of one point: σ has one vacuum. If $a_{K_j}\beta_j \to b$, then $\sigma \in E[b]$, with the scalar Wightman reconstruction of Theorem YM3_I:AP and gap at most C_0/b by Proposition 18.170(a).

(b) *One limit at a label.* Let $0 < g \le \gamma$, and assume (H_g) of Definition 18.172(7): for every $n \ge 2$ and $X \in \mathfrak{S}_n(\mathbb{R}^3)$ there are $\epsilon_X(\ell) \to 0$ and $K_0(n, X)$ with $|\mathsf{a}(K, \ell, \beta; X) - \mathsf{a}(K, m, \beta; X)| \le \epsilon_X(\ell)$ for all $\ell \le m$ in $\mathfrak{L}(g)$ and all $(K, \beta) \in \mathfrak{d}(g)$ with $K \ge K_0(n, X)$. Then:

(b1) for every real $b > 0$, $n \ge 2$ and $X \in \mathfrak{S}_n(\mathbb{R}^3)$, $S_n^b(X) = \lim_{\ell \in \mathfrak{L}} \mathsf{B}(\ell, b; X)$ exists, and $\mathsf{a}(K_j, \ell_j, \beta_j; X) \to S_n^b(X)$ and $S_n^T[\nu_j](X) \to S_n^b(X)$, without passing to subsequences, along all pairs with $K_j \to \infty$ and $a_{K_j}\beta_j \to b$, all sides $\ell_j \to \infty$ of $\mathfrak{L}$ in any relation to K_j, and all states of tori ν_j at these pairs;

(b2) $b \mapsto S_n^b(X)$ is continuous on $(0, \infty)$, and $\mathsf{B}(\ell, b; X) \to S_n^b(X)$ uniformly for b in every compact $J \subset (0, \infty)$;

(b3) $S^{L^z b} = \mathrm{dil}_{L^z} S^b$ for every $z \in \mathbb{Z}$.

(c) *All real dilations.* If $(\mathrm{H}_{g^{(2)}})$ holds besides in a second setting, with a blocking factor L_2 such that $\log L / \log L_2$ is irrational and at a reference coupling $g^{(2)}$ of that setting, and the condition (Same) of Definition 18.172(6) holds, then $\mathrm{dil}_r S^b = S^{rb}$ for all real $r, b > 0$.

(d) *One vacuum at a label.* Under (H_g), let $b \in \Lambda(g)$ and let σ_b be the centred functional of S^b. Every state sequence at g with labels tending to b is a convergent Gibbs sequence at g with the limit σ_b. If in one such sequence ν_j is ergodic for infinitely many j, then σ_b has one vacuum, $\sigma_b \in E[b]$, and $\sigma_{L^z b} = \mathrm{dil}_{L^z} \sigma_b \in E[L^z b]$ for every $z \in \mathbb{Z}$.

■

Proof. Clause (a). The Gibbs-frame reconstruction of Theorem YM3_I:AP supplies the canonical decomposition. Under an ergodic law the invariant σ-algebra is trivial modulo null sets, so its L^2 has dimension one. The component count in Theorem YM3_I:AL is bounded by the lower limit of these dimensions along the frame. There is at least one component. Hence infinitely many ergodic indices give exactly one, and the clustering of that component is the clustering of σ. If the labels tend to b, the original sequence exhibits $\sigma \in A[b]$. It is then in $E[b]$, with the stated field and Hamiltonian by Theorem YM3_I:AP and the spectral bound of Proposition 18.170(a).

Clauses (b) and (c). The finite-size comparison and the reindexing in Proposition 18.173(b) apply to (H_g). They give the limit along every sequence of pairs and sides, its uniform convergence on compact sets of labels, and the integer dilation rule. The same proposition, with the second blocking factor and (Same), gives the real dilation rule in (c).

Clause (d). The pointwise limits of the cumulants in (b) give convergence of the self-centred moments by the finite moment–cumulant identities. The state sequence is a Gibbs sequence because the states of tori satisfy the Gibbs equations. If it has infinitely many ergodic indices, (a) shows

that its limit is in $E[b]$. Relabel the bonds of the sequence by the scale L^z. The observable identity in the proof of Proposition 18.166 carries the limit to $\mathrm{dil}_{L^z}\sigma_b \in A[L^z b]$ and gives a bijection of decompositions, so the new limit has one component as well. By (b3) it equals $\sigma_{L^z b}$. □

18.18. **One vacuum from ergodic states of tori; the cutoff removed first and the volume afterwards.** We use the limit classes and reconstruction of Definition 18.164. Proposition 18.173 states what the hypothesis $(\mathrm{H}^\circ[\mathfrak{d}(g), b])$ at a limit b of labels at g gives, and what (H_g) gives at all labels, together with the vacuum. Under $(\mathrm{H}^\circ[\mathfrak{d}(g), b])$ the limit spacing first S^b exists; the cumulants of all tori and of all states of tori whose pairs lie in $\mathfrak{d}(g)$ and whose labels tend to b tend to it along the full sequence, whatever the relation between cutoff and side; and S^b is the one limit at g of either kind with the label b. Under (H_g) the same holds at every reference coupling and at every label, $b \mapsto S^b$ is continuous and $\mathrm{dil}_{L^z} S^b = S^{L^z b}$; if moreover a second setting is given whose blocking factor L_2 makes $\log L/\log L_2$ irrational, with $(\mathrm{H}_{g_{(2)}})$ there at one of its reference couplings $g_{(2)}$ and with the condition (Same) on the states of tori of the two settings (Definition 18.172), then $\mathrm{dil}_r S^b = S^{rb}$ for every real $r > 0$. If in one state sequence at g with labels tending to b the states are ergodic, that is pure phases, at infinitely many indices, then the centred functional σ_b of S^b, which is the limit of every such state sequence, clusters with the rate $C^g(X,Y)/|v|$, every canonical decomposition of it has one point, and $\sigma_b \in E[b]$. This comes from a statement on every convergent Gibbs sequence at g whose states are ergodic at infinitely many indices: the iterated plateau of its limit is the limit of the means of products of conditional means given the translation-invariant events, under an ergodic law these conditional means are constants, so the limit clusters, and a functional that clusters has no decomposition with more than one point. Propositions 18.174 and 18.175 concern the order in which the cutoff is removed first, on tori of a fixed side, and the side grows afterwards. At a label $b \in I(g)$ every sequence (ℓ_k) of sides at g with $\ell_k \to \infty$ has a subsequence along which the torus limits $\mathsf{B}(\ell, b; \cdot)$ converge at every tuple, so that $\mathrm{SF}(b)$ is not empty; one choice of cutoffs, made once for the whole sequence of sides, gives one torus sequence at g whose convergent subsequences have exactly the limits of the corresponding subsequences of the torus limits; so every member of $\mathrm{SF}(b)$ is a continuum limit at g along growing tori with the label b, and the same holds when the label moves with the side, every limit point of the labels being a label of the limit.

Use the canonical Gibbs class $\mathcal{G}_{a,\beta}$ of §12.1 and the first-passage data $\mathfrak{d}(g)$ of Prop. YM3_I:74.1. A state of tori at a fixed pair (K, β) is a subsequential weak limit on the infinite lattice of the periodic lifts of the torus laws as the side tends to infinity in $\mathfrak{L}$. Such states exist by compactness of the configuration space. They satisfy the Gibbs equations: for a fixed finite set of bonds and continuous local test, the finite-volume Gibbs identity agrees with the infinite-lattice identity as soon as the support and the interaction neighbourhood embed in the torus. The local Gibbs kernel preserves continuity, so the identity passes to the weak limit and then extends to bounded Borel tests by a monotone class. Thus a sequence of these states is a Gibbs sequence. The cumulant of a fixed tuple is continuous in the law at a fixed pair because all entries are bounded continuous local observables. In particular the cumulant of a state of tori is a subsequential limit of the torus cumulants at that pair. Use $C^g(X,Y)$ and $r(X,Y)$ of the plateau bound in §12.1, following Prop. YM3_I:100.1.

Definition 18.172. *Ergodic states; conditional means; one choice of cutoffs; labels of a limit.*

(1) At spacing a_K, $\mathcal{I}$ is the Borel σ-algebra of sets fixed by all lattice translations. A translation-invariant law is ergodic when every such set has probability zero or one. Periodic Gibbs states are invariant under all translations by Prop. YM3_I:106.3. The ergodic members are exactly the extreme points, by Prop. YM3_I:106.1.

(2) For a Gibbs sequence (K_j, β_j, ω_j) and a separated tuple $X = (f_i)_i$, let

$$\Theta_j(X) = \prod_i [\mathcal{O}_{a_{K_j}}(f_i) - \omega_j(\mathcal{O}_{a_{K_j}}(f_i))], \qquad \Theta_j(\varnothing) = 1, \qquad \widehat{\Theta}_j(X) = E_{\omega_j}[\Theta_j(X) \mid \mathcal{I}].$$

These are bounded real functions, with $\Theta_j(X)$ local and $\omega_j(\Theta_j(X)) = \sigma_{\omega_j}(X)$. The conditional expectation and its products are defined up to null sets.

(3) Let $0 < g \le \gamma$, let $\ell_k \in \mathfrak{L}(g)$, $k \in \mathbb{N}$, be sides with $\ell_k \to \infty$, and let $b_k \in I(g)$, $k \in \mathbb{N}$. For $k \in \mathbb{N}$, $\mathfrak{F}_k$ is the finite set of the $\varphi^{k'}$ with $k' \le k$ all of whose entries have their supports in Q_{ℓ_k}, and $\kappa(k)$ is the least integer $K' \ge k$ such that $|\mathsf{a}(K, \ell_k, L^K b_k; \varphi) - \mathsf{B}(\ell_k, b_k; \varphi)| \le 1/k$ for all integers $K \ge K'$ and all $\varphi \in \mathfrak{F}_k$. It exists by Thm. YM3_I:AS: periodization of a tuple supported in $Q_{\ell_k} = (-\ell_k/4, \ell_k/4)^3$ gives a separated C^1 tuple on the torus. The enumeration $(\varphi^{k'})$ is the countable dense family of Lem. 18.167. The number $\kappa(k)$ depends on k, ℓ_k and b_k alone.

(4) A label of a member of $\mathrm{GT}(g)$, respectively $\mathrm{VF}(g)$, is a subsequential limit of $a_{K_j}\beta_j$ along a defining sequence of tori, respectively states of tori.

(5) For a compact nonempty $J \subset (0, \infty)$ set $K_J = \min\{K \ge 1 : L^K \min J > X_g\}$ and $Z_J = \{z \in \mathbb{Z} : L^z J \cap I^{\mathrm{lab}}(g) \ne \varnothing\}$, where $I^{\mathrm{lab}}(g)$ is the compact label interval from the bounds of Prop. YM3_I:74.1. It is finite. If $K \ge K_J$ and $b \in J$, the first-passage index for the bare coupling $L^K b$ is $K - z$ for a unique $z \in Z_J$. This follows from uniqueness of first passage and the fact that the new label is $L^z b$. In particular for each $b > 0$ some $L^z b$ lies in $\Lambda(g)$, by taking a subsequence with fixed z as $K \to \infty$.

(6) For settings with blocking factors L and L_2, (Same) means that for all sufficiently large bare couplings β, any state of tori ν in the first setting and any state of tori ν' in the second, both at β and with respective labels c, c', satisfy $\mathrm{dil}_{1/c} S^T[\nu] = \mathrm{dil}_{1/c'} S^T[\nu']$. The dilated cumulants are formed on all separated tuples in $\mathbb{R}^3$.

(7) For $m \ge \ell \ge \ell_0(X)$, where $\ell_0(X)$ is the least side with $\operatorname{supp} X \subset Q_\ell$, set $\Delta_X(K, \beta; \ell, m) = |\mathsf{a}(K, \ell, \beta; X) - \mathsf{a}(K, m, \beta; X)|$. Write $\mathfrak{p}_J = \{(K, \beta) : a_K \beta \in J\}$, $\mathfrak{p}_\infty = \mathbb{N} \times (0, \infty)$, and let $\Lambda(\mathfrak{p})$ be the positive limiting labels along pairs of $\mathfrak{p}$ with $K \to \infty$. Thus $\Lambda(g) = \Lambda(\mathfrak{d}(g))$. For a set of pairs $\mathfrak{p}$ put

$$\mathsf{e}_X(\ell; \mathfrak{p}) = \lim_{k \to \infty} \sup_{\substack{(K,\beta) \in \mathfrak{p},\ K \ge k \\ m \in \mathfrak{L},\ m \ge \ell}} \Delta_X(K, \beta; \ell, m),$$

$$\mathsf{e}_X(\ell; \mathfrak{p}, b) = \inf_{\eta > 0} \mathsf{e}_X(\ell; \mathfrak{p} \cap \mathfrak{p}_{[b-\eta, b+\eta]}),$$

with supremum of an empty family equal to zero. Use $S_n^b(X) = \lim_{\ell \to \infty} \mathsf{B}(\ell, b; X)$ whenever it exists along all sides. The finite-size hypotheses are as follows.

- (H_g) asks for $\Delta_X(K, \beta; \ell, m) \le \epsilon_X(\ell)$ for $\ell \le m$ in $\mathfrak{L}(g)$ and $(K, \beta) \in \mathfrak{d}(g)$ with $K \ge K_0(n, X)$.
- (H_*) asks for $\Delta_X(K, \beta; \ell, m) \le \epsilon_{X,J}(\ell)$ for every compact $J \subset (0, \infty)$, $\ell \le m$ in $\mathfrak{L}$ and $(K, \beta) \in \mathfrak{p}_J$ with $K \ge K_0(n, X, J)$.
- $(\mathrm{H}_g^\sharp)$ asks for $\Delta_X(K, \beta; \ell, m) \le \eta_n(\ell; \mathsf{d}, R)\|X\|$ for all $\ell \le m$ in $\mathfrak{L}(g)$, $(K, \beta) \in \mathfrak{d}(g)$ and tuples of type $(\mathsf{d}; R)$ in Q_ℓ. The bounds in these three hypotheses tend to zero as $\ell \to \infty$ and the cutoff thresholds are independent of ℓ, m.
- $(\mathrm{H}^\circ[\mathfrak{p}, b])$ means $\mathsf{e}_X(\ell; \mathfrak{p}, b) \to 0$; (H_g°) means $\mathsf{e}_X(\ell; \mathfrak{d}(g)) \to 0$; (H°) means $\mathsf{e}_X(\ell; \mathfrak{p}_J) \to 0$ for every compact $J \subset (0, \infty)$.

Each hypothesis concerns every separated tuple with $n \ge 2$ entries.

■

Proposition 18.173 (One limit at a label; one vacuum from ergodic states of tori)**.** Use Definition 18.164 and let $0 < g \le \gamma$. The cumulant families below are indexed by tuples with $n \ge 2$; the centred functionals are defined on all separated tuples.

(a) *One limit at the label b.* Let $b \in \Lambda(g)$ and assume $(\mathrm{H}^\circ[\mathfrak{d}(g), b])$. Under (H_g), in particular when (H_g) holds with the threshold $K_0(n, X) = 1$, so that $\Delta_X(K, \beta; \ell, m) \le \epsilon_X(\ell)$ is asked at all $\ell \le m$ in $\mathfrak{L}(g)$ and at all pairs of $\mathfrak{d}(g)$, this hypothesis holds at every $b \in \Lambda(g)$, with

$\mathsf{e}_X(\ell;\mathfrak{d}(g),b)\le\epsilon_X(\ell)$ for $\ell\in\mathfrak{L}(g)$, $\ell\ge\ell_0(X)$. Then for every $n\ge2$ and $X\in\mathfrak{S}_n(\mathbb{R}^3)$ the limit $S^b_n(X)=\lim_{\ell\in\mathfrak{L}}\mathsf{B}(\ell,b;X)$ exists, and $|\mathsf{B}(\ell,b;X)-S^b_n(X)|\le\mathsf{e}_X(\ell;\mathfrak{d}(g),b)$ for $\ell\in\mathfrak{L}$, $\ell\ge\ell_0(X)$. For all pairs $(K_j,\beta_j)\in\mathfrak{d}(g)$ with $K_j\to\infty$ and $a_{K_j}\beta_j\to b$, all sides $\ell_j\in\mathfrak{L}$ with $\ell_j\to\infty$, in whatever relation to the K_j, and all states of tori $\nu_j\in\mathcal{G}^{\mathrm{tor}}_{(K_j,\beta_j)}$, the numbers $\mathsf{a}(K_j,\ell_j,\beta_j;X)$ and $S^T_n[\nu_j](X)$ tend to $S^b_n(X)$ along the full sequence of indices; and along the same pairs $\lim_\ell\lim_j\mathsf{a}(K_j,\ell,\beta_j;X)=\lim_j\liminf_m\mathsf{a}(K_j,m,\beta_j;X)=\lim_j\limsup_m\mathsf{a}(K_j,m,\beta_j;X)=S^b_n(X)$, the limits in ℓ and m being taken in $\mathfrak{L}$. The family $S^b=(S^b_n)_{n\ge2}$ is a member of $\mathrm{GT}(g)$ and of $\mathrm{VF}(g)$ with the label b, and it is the only member of $\mathrm{GT}(g)$ and the only member of $\mathrm{VF}(g)$ that has b among its labels. Every state sequence at g whose labels tend to b is a convergent Gibbs sequence at g with the limit σ_b, where $\sigma_b(X)=\sum_\pi\prod_{I\in\pi}S^b_{|I|}(X_I)$, the sum over the partitions π of $\{1,\dots,|X|\}$ into blocks of at least two elements; so $\sigma_b(\varnothing)=1$ and $\sigma_b((f))=0$.

(b) *All couplings, all labels, all dilations.* Assume (H_g). Then $(\mathrm{H}_{g'})$ holds at every $0<g'\le\gamma$, so that (a) and (d) apply at every such g' and every $b\in\Lambda(g')$; and (H_*) holds with $\epsilon_{X,J}(\ell)=\max_{z\in Z_J}L^{zn}\epsilon_{X^{[z]}}(L^z\ell)$, a maximum in which a term with $L^z\ell\notin\mathfrak{L}(g)$ is read as $+\infty$, and $K_0(n,X,J)=\max\{K_J,\max_{z\in Z_J}(K_0(n,X^{[z]})+z)\}$, with Z_J and K_J of Definition 18.172(5). For every $n\ge2$ and $X\in\mathfrak{S}_n(\mathbb{R}^3)$: $S^b_n(X)$ exists at every $b>0$; for every compact $J\subset(0,\infty)$, all $b,b'\in J$ and all $\ell\in\mathfrak{L}$ with $\ell\ge\ell_0(X)$, $|\mathsf{B}(\ell,b;X)-S^b_n(X)|\le\epsilon_{X,J}(\ell)$ and $|S^b_n(X)-S^{b'}_n(X)|\le2\epsilon_{X,J}(\ell)+|\mathsf{B}(\ell,b;X)-\mathsf{B}(\ell,b';X)|$, so that $b\mapsto S^b_n(X)$ is continuous on $(0,\infty)$; $S^{L^zb}=\mathrm{dil}_{L^z}S^b$ for $b>0$ and $z\in\mathbb{Z}$; and for every $b>0$ the numbers $\mathsf{a}(K_j,\ell_j,\beta_j;X)$ and $S^T_n[\nu_j](X)$ tend to $S^b_n(X)$ along all pairs $(K_j,\beta_j)\in\mathfrak{p}_\infty$ with $K_j\to\infty$ and $a_{K_j}\beta_j\to b$, all sides $\ell_j\to\infty$ in $\mathfrak{L}$ and all $\nu_j\in\mathcal{G}^{\mathrm{tor}}_{(K_j,\beta_j)}$, no reference coupling entering. Let besides a second setting be given, with a blocking factor L_2 such that $\log L/\log L_2$ is irrational (the consecutive odd choices $L,L+2$ in the canonical construction have this property), let $(\mathrm{H}_{g_{(2)}})$ hold in it at one of its reference couplings $g_{(2)}$, and let (Same) hold. Then there is one family $\mathsf{u}=(\mathsf{u}_n)_{n\ge2}$ with $S^b=S^b_{(2)}=\mathrm{dil}_b\mathsf{u}$ for every $b>0$; so $\mathrm{dil}_rS^b=S^{rb}$ for every real $r>0$.

(c) *Ergodic states at infinitely many indices.* Let (K_j,β_j,ω_j) be a convergent Gibbs sequence at g with the limit σ, and let ω_j be ergodic for infinitely many j. Then $\lim_{|v|\to\infty}\sigma(X\cdot\tau_vY)=\sigma(X)\sigma(Y)$ for all $X,Y\in\mathfrak{S}$, and $|\sigma[X,Y;v]|\le C^g(X,Y)/|v|$ for $|v|\ge r(X,Y)$. Every decomposition $(\Xi,\lambda,(\sigma_\xi))$ of σ has a space $\Xi=\{\xi_0\}$ of one point, $\lambda=\delta_{\xi_0}$ and $\sigma_{\xi_0}=\sigma$; so σ has one vacuum, the Hamiltonian H_σ has the kernel $\mathbb{C}\Omega$, and the weak commutant $\mathcal{C}(\mathbb{R}^{1+2})$ of the fields of σ (the bounded operators weakly commuting with every smeared field on the common invariant polynomial domain) is $\mathbb{C}1$.

(d) *One vacuum at the label b.* Let b be as in (a), and let one state sequence (K_j,β_j,ν_j) at g with $a_{K_j}\beta_j\to b$ have ergodic states ν_j for infinitely many j. Then all that (c) says of σ holds for σ_b: it clusters, $|\sigma_b[X,Y;v]|\le C^g(X,Y)/|v|$ for $|v|\ge r(X,Y)$, every decomposition of σ_b has one point and its one component is σ_b, and $\ker H_{\sigma_b}=\mathbb{C}\Omega$. Moreover $\sigma_b\in E[b]$, so it has the scalar Wightman reconstruction with unique vacuum of Thm. YM3_I:AP and the spectral ceiling of Prop. 18.170(a), and every state sequence at g whose labels tend to b, with ergodic states or not, converges to this one functional with one vacuum. The hypothesis on the states holds, for instance, when for infinitely many j the set $\mathcal{G}_{a_{K_j},\beta_j}$ has exactly one member.

(e) *One vacuum at all labels.* For $b'>0$ such that $S^{b'}$ exists, let $\sigma_{b'}(X)=\sum_\pi\prod_{I\in\pi}S^{b'}_{|I|}(X_I)$ be its centred functional, the sum being as in (a). In the situation of (d), S^{L^zb} exists and $\sigma_{L^zb}=\mathrm{dil}_{L^z}\sigma_b\in E[L^zb]$ for every $z\in\mathbb{Z}$; and if $\mathrm{dil}_rS^b=S^{rb}$ for every real $r>0$, as under the hypotheses of (b) on a second setting, then $\sigma_{b'}\in E[b']$ for every $b'>0$.

■

Proof. Clause (a). Fix X. At fixed sides $m \geq \ell \geq \ell_0(X)$, pass to the limit along pairs in $\mathfrak{d}(g)$ with labels tending to b. Thm. YM3_I:AS and YM3_I:(2216) give

$$|\mathsf{B}(\ell, b; X) - \mathsf{B}(m, b; X)| \leq \mathsf{e}_X(\ell; \mathfrak{d}(g), b).$$

Indeed each fixed label neighbourhood and cutoff threshold in e_X contains the pairs eventually. Since the right side vanishes as $\ell \to \infty$, the side sequence is Cauchy. Sending $m \to \infty$ gives the claimed error for $S_n^b(X)$. Given $\epsilon > 0$, choose a side ℓ_* with $\mathsf{e}_X(\ell_*; \mathfrak{d}(g), b) < \epsilon$. The definition then supplies a label neighbourhood and a cutoff threshold on which $\Delta_X(K, \beta; \ell_*, m) < \epsilon$ for every $m \geq \ell_*$. Along the given pairs, fixed-side convergence gives $|\mathsf{a}(K_j, \ell_*, \beta_j; X) - \mathsf{B}(\ell_*, b; X)| < \epsilon$ for large j. The triangle inequality bounds all $|\mathsf{a}(K_j, m, \beta_j; X) - S_n^b(X)|$ by 3ϵ at once, for all $m \geq \ell_*$ and large j. This gives both the joint limit and the limits of the two side envelopes. It gives the same bound for the cumulants of every state of tori at these pairs by the fixed-pair subsequential characterization above. The spacing-first iterated limit follows from the fixed-side identity and the side limit.

Pairs with labels tending to b exist by $b \in \Lambda(g)$; choose any sides tending to infinity and any states of tori at these pairs. These realize S^b in both limit classes. If a convergent sequence has b among its labels, pass to a subsequence with labels tending to b to identify its limit as S^b. The partition formula for moments of the centred fields gives precisely σ_b because the first cumulants of centred fields vanish. Finally (H_g) bounds $\mathsf{e}_X(\ell; \mathfrak{d}(g), b)$ by $\epsilon_X(\ell)$ on admitted sides by its uniform cutoff threshold.

Clause (b). For $b \in J$ and $K \geq K_J$, use the $z \in Z_J$ of Definition 18.172(5). Relabel the same bond and plaquette graph by the dilation $x \mapsto L^z x$. The measure and bare coupling are unchanged, and each action insertion contributes the factor L^z, as in YM3_I:(2209). Consequently

$$\Delta_X(K, L^K b; \ell, m) = L^{zn} \Delta_{X^{[z]}}(K - z, L^K b; L^z \ell, L^z m), \qquad X^{[z]} = (f_i(L^{-z}\cdot))_i.$$

This holds for negative z by the inverse dilation whenever the new side is admitted. If a dilated side is too small the bound with $+\infty$ is automatic. Apply (H_g) to obtain the displayed $\epsilon_{X,J}$ and $K_0(n, X, J)$ of (H_*). The maximum over Z_J tends to zero since the set is finite. Applying (H_*) on the compact label interval of any first-passage family gives $(\mathrm{H}_{g'})$ at every g'.

Repeat the proof of (a) with the pairs of labels in a compact J. It yields the side convergence and the uniform error on J. The fixed-side limits are continuous in the label by Cor. YM3_I:117.1 and dilation; the uniform side error gives the displayed modulus of continuity and hence continuity of S^b. Each sequence of pairs with labels tending to $b > 0$ lies eventually in some compact J, so the same proof gives joint torus and state convergence with no first-passage restriction. For any $z \in \mathbb{Z}$, take the cutoff and side limits in the exact dilation identity. This gives $S^{L^z b} = \mathrm{dil}_{L^z} S^b$, and this last identity requires only the existence of S^b.

For the two settings, fix $b, b' > 0$. Irrationality makes $p \log L$ dense modulo $\log L_2$ even with p restricted to any tail of the positive integers. Choose $p_j, q_j \to \infty$ with $p_j \log L - q_j \log L_2 \to \log(b'/b)$. At the common bare coupling $\beta_j = bL^{p_j}$ choose arbitrary states of tori at cutoffs p_j and q_j in the two settings. Their labels are b and $b'_j = bL^{p_j} L_2^{-q_j} \to b'$. By (Same) their cumulants at spacing $1/\beta_j$ agree. The convergence just proved also holds at the moving tuples $Y(b_j'^{-1}\cdot)$. Indeed C_c^1 implies convergence in $\|\cdot\|_\sharp$ under the dilations approaching b'^{-1}, with the supports in fixed pairwise disjoint compact neighbourhoods. For a compact set of labels the cumulants of such tuples have one uniform bound by the product of their norms. To see this, map each pair to its first-passage index with the finite set Z_J, then use the table of Prop. YM3_I:91.13 on growing tori and its fixed-pair limit for states of tori. The transformed tuples have uniform positive separation and bounded support radii since the set of transformations is finite. Multilinearity bounds the change by a sum of n terms, each with one moving entry minus its limit, and gives the asserted moving-tuple convergence. Thus $\mathrm{dil}_{1/b} S^b = \mathrm{dil}_{1/b'} S^{b'}_{(2)}$. As b, b' were arbitrary, this is one family u and $\mathrm{dil}_r S^b = S^{rb}$ follows by

composition. The example of consecutive odd blocking factors uses the freedom to choose sufficiently large odd L in Prop. YM3_I:69.1; L and $L+2$ are coprime, so no positive powers are equal.

Clause (c). Restrict to the infinite subsequence of ergodic states, which has the same limit σ. A bounded real $\mathcal{I}$-measurable function is almost surely constant under an ergodic state because the probabilities of all its rational sublevel sets are zero or one. Therefore $\widehat{\Theta}_j(X) = \sigma_{\omega_j}(X)$ almost surely. Prop. YM3_I:100.2, using the bound system for the Gibbs class of Prop. YM3_I:106.3, identifies the two-tuple plateau with $\lim_j \omega_j(\widehat{\Theta}_j(X)\widehat{\Theta}_j(Y)) = \sigma(X)\sigma(Y)$. Thus σ clusters. A canonical decomposition exists by Thm. YM3_I:AP. In any such decomposition the first two plateau identities give

$$\int \sigma_\xi(X)\, d\lambda(\xi) = \sigma(X), \qquad \int \sigma_\xi(X)^2\, d\lambda(\xi) = \sigma(X)^2.$$

Each continuous function $\xi \mapsto \sigma_\xi(X)$ is therefore constant on the whole space, since λ has full support. This holds for every X, and the parametrization is injective, so the decomposition has one point. As σ is itself the one component, the rate is the component estimate of Thm. YM3_I:AP, equation YM3_I:(1996). The same theorem identifies its zero-energy space with the one-dimensional space of the decomposition.

For the weak commutant, use the scalar Wightman reconstruction of this component in Thm. YM3_I:AP. Let Q be a bounded operator in the weak commutant. Repeated weak commutation is valid for products P of smeared fields on the invariant polynomial domain, with P^* there the formal adjoint. With the inner product linear in its second entry, translation covariance gives for every Minkowski translation x

$$\langle Q^*\Omega, U(x)P\Omega\rangle = \langle \Omega, QU(x)P\Omega\rangle = \langle P^*\Omega, U(-x)Q\Omega\rangle.$$

The two Fourier transforms are finite spectral matrix-element measures on opposite closed cones by the spectrum condition. Their common support is the origin, so the function is constant. By the density of $P\Omega$, $Q^*\Omega$ is translation invariant and hence a scalar multiple of the unique vacuum. Apply the same argument to Q^* to see this for $Q\Omega$ too. For polynomial vectors $P\Omega, P'\Omega$ weak commutation now gives the scalar matrix element of Q from $Q\Omega$. Density yields $Q \in \mathbb{C}1$.

Clauses (d) and (e). By (a), the given state sequence converges to σ_b. Clause (c) gives its one vacuum, so this realization by lattice states with label tending to b puts σ_b in $E[b]$. If the Gibbs class is a singleton, its member is extreme and hence ergodic by Prop. YM3_I:106.1. All other sequences specified in (d) converge to the same functional by (a).

A real dilation $x \mapsto rx$ relabels bonds, keeps the Wilson Gibbs specification at the same β, and multiplies the spacing by r. Each centred action insertion transforms by r^{-1} with argument $f(r\cdot)$. It therefore carries limits at label b to limits at label rb and carries canonical decompositions with their same measures and parameter spaces, by the plateau identities and the rescaling of translations. The inverse dilation shows that it carries $E[b]$ onto $E[rb]$. The moment partition formula commutes with this operation because $\sum_{I\in\pi} |I| = |X|$. The integer dilation identity in (b) therefore proves the first statement of (e) using only existence of S^b. If the identity is available for every $r > 0$, the same argument gives the assertion at every label. □

Proposition 18.174 (The spacing removed first, then the volume: extraction)**.** Use the setting of Definition 18.164, let $0 < g \le \gamma$ and $b \in I(g)$, and let $\ell_k \in \mathfrak{L}(g)$, $k \in \mathbb{N}$, be sides with $\ell_k \to \infty$.

(o) *The cutoff first, on each torus.* For every k, every $n \ge 2$ and every $X \in \mathfrak{S}_n(Q_{\ell_k})$ the torus limit $\mathsf{B}(\ell_k, b; X) = \lim_{K\to\infty} \mathsf{a}(K, \ell_k, L^K b; X)$ exists along the full sequence of cutoffs, and it is the limit of $\mathsf{a}(K_i, \ell_k, \beta_i; X)$ along all pairs $(K_i, \beta_i) \in \mathfrak{p}_\infty$ with $K_i \to \infty$ and $a_{K_i}\beta_i \to b$; all the pairs $(K, L^K b)$, $K \in \mathbb{N}$, lie in $\mathfrak{d}_b \subset \mathfrak{d}(g)$. The map $\mathsf{B}(\ell_k, b; \cdot)$ is symmetric and multilinear on $\mathfrak{S}_n(Q_{\ell_k})$, and $|\mathsf{B}(\ell_k, b; X)| \le C_n^g(\mathsf{d}; R)\|X\|$ for every $X \in \mathfrak{S}_n(Q_{\ell_k})$ of the type $(\mathsf{d}; R)$ whenever $\ell_k \ge 16n(R+1)$, $C_n^g(\mathsf{d}; R)$ being the table at g of Prop. YM3_I:91.13. Every $X \in \mathfrak{S}_n(\mathbb{R}^3)$ lies in $\mathfrak{S}_n(Q_{\ell_k})$ for all large k.

(a) *Then the volume: extraction.* Every infinite set of indices contains indices $k_1 < k_2 < \cdots$ such that, for every $n \geq 2$ and every $X \in \mathfrak{S}_n(\mathbb{R}^3)$, the numbers $\mathsf{B}(\ell_{k_l}, b; X)$, formed for all large l, converge as $l \to \infty$. The family $S = (S_n)_{n\geq 2}$ of the limits is a member of $\mathrm{SF}(b)$, taken along $(\ell_{k_l})_l$, and of $\mathcal{K}(g)$. In particular $\mathrm{SF}(b)$ is not empty, and every sequence of sides at g that tends to infinity has a subsequence along which a member of $\mathrm{SF}(b)$ is taken.

(b) *A continuum limit along growing tori with the label b.* Let $k_1 < k_2 < \cdots$ be any indices as in (a), and S the family of the limits. There are cutoffs $K_l \geq k_l$ such that $(K_l, \ell_{k_l}, L^{K_l} b)_{l\in\mathbb{N}}$ is a torus sequence at g over $\mathfrak{d}_b$ that defines S, each of its members having the label $a_{K_l} L^{K_l} b = b$; so $S \in \mathrm{GT}_b \subset \mathrm{GT}(g)$, and b is a label of S. The cutoffs can be fixed before the indices k_l: for any cutoffs $K'_k \geq \kappa(k)$, $k \in \mathbb{N}$, with the numbers $\kappa(k)$ of Definition 18.172(3) at $b_k = b$, one may take $K_l = K'_{k_l}$.

■

Proof. For a fixed k, periodize each entry of X from the central cube Q_{ℓ_k} to T_{ℓ_k}. The entries remain C^1 and separated because the central cube embeds in the torus and the supports are compact in it. Thm. YM3_I:AS and YM3_I:(2216) give the limit $\mathsf{B}(\ell_k, b; X)$ both at fixed label and along convergent labels, for any sequence of cutoffs tending to infinity. Cor. YM3_I:74.2 gives $\mathfrak{d}_b \subset \mathfrak{d}(g)$. Thus the table of Prop. YM3_I:91.13 bounds the cutoff cumulants at all sides in the stated range and passes to their limits. Symmetry and multilinearity on separated supports also pass to the limit. Every fixed tuple eventually has support in Q_{ℓ_k}.

For any cutoffs $K'_k \geq \kappa(k)$, the cutoff error at every member of the countable family tends to zero, since a fixed member lies in $\mathfrak{F}_k$ for all sufficiently large k. The same holds at each tuple X. To see this, enclose the supports of its entries in pairwise disjoint finite unions of closed rational cubes as in Lem. 18.167, and approximate the entries in $\|\cdot\|_\sharp$ by the dense countable family on these sets. The table at g bounds both multilinear forms uniformly there, since the supports have one positive separation and one finite radius bound and the sides eventually exceed $16n(R+1)$. Telescoping the multilinear forms one entry at a time bounds the approximation error uniformly in all sufficiently large k. This proves the cutoff error tends to zero at X.

On any infinite set of indices the same table makes the cutoff cumulants bounded at every member of the countable family. Successive subsequences and the diagonal subsequence give their joint convergence on that family. The approximation argument just given shows convergence at all tuples. The triples $(K'_k, \ell_k, L^{K'_k} b)$ have cutoffs and sides tending to infinity and label exactly b. Hence along the selected indices their limit is in $\mathrm{GT}_b \subset \mathcal{K}(g)$, and the vanishing cutoff error gives the same limit for the torus limits. This proves (a) and nonemptiness of $\mathrm{SF}(b)$. On any already convergent subsequence of the torus limits, the same cutoff-error estimate identifies the limit for the triples with every such choice of K'_k, proving (b). □

Proposition 18.175 (The cutoff removed first on each torus: one torus sequence for all subsequences)**.** Use the setting of Definition 18.164, let $0 < g \leq \gamma$, let $\ell_k \in \mathfrak{L}(g)$, $k \in \mathbb{N}$, be sides with $\ell_k \to \infty$ and $b_k \in I(g)$, $k \in \mathbb{N}$, labels that may depend on k, and let $\kappa(k)$ be the numbers of Definition 18.172(3).

(a) *One choice of cutoffs.* For every choice of cutoffs $K_k \geq \kappa(k)$, every $n \geq 2$ and every $X \in \mathfrak{S}_n(\mathbb{R}^3)$, $\mathsf{a}(K_k, \ell_k, L^{K_k} b_k; X) - \mathsf{B}(\ell_k, b_k; X) \to 0$ as $k \to \infty$ along the full sequence of indices; the second term is formed for the k with $X \in \mathfrak{S}_n(Q_{\ell_k})$, which are all large k.

(b) *The same subsequences and the same limits.* For such cutoffs $\zeta_k = (K_k, \ell_k, L^{K_k} b_k)$ is a torus sequence at g whose member of index k has the label b_k. Indices $k_1 < k_2 < \cdots$ make $\mathsf{B}(\ell_{k_l}, b_{k_l}; X)$ converge for every $n \geq 2$ and $X \in \mathfrak{S}_n(\mathbb{R}^3)$ if and only if they make $\mathsf{a}(\zeta_{k_l}; X)$ converge for every $n \geq 2$ and $X \in \mathfrak{S}_n(\mathbb{R}^3)$, and then the two limits are equal. Every infinite set of indices contains such indices, and they can be taken such that $(b_{k_l})_l$ converges.

(c) *The limits and their labels.* Every limit S as in (b) is a member of $\mathrm{GT}(g) \subset \mathcal{K}(g)$, defined by the torus sequence $(\zeta_{k_l})_l$; every limit point of $(b_{k_l})_l$ lies in $[g^{-2} + \bar{B}, X_g - \bar{B}]$ and is a label of S.

(d) *One label.* Let $b \in I(g)$ and $b_k = b$ for every k. The members of $\mathrm{SF}(b)$ taken along subsequences of (ℓ_k) are exactly the limits of the convergent subsequences of (ζ_k), a torus sequence over $\mathfrak{d}_b$ all of whose members have the label b, each limit being the member taken along the corresponding sides; they lie in GT_b. And $\mathrm{SF}(b)$ is the set of the limits of the convergent torus sequences $(K_k, \ell_k, L^{K_k} b)_k$ over $\mathfrak{d}_b$, with sides $\ell_k \in \mathfrak{L}(g)$ tending to infinity, whose cutoffs satisfy $K_k \geq \kappa(k)$ for the numbers $\kappa(k)$ formed from their own sides at the label b.

■

Proof. The fixed-side convergence, the cutoff-error argument and the uniform table bound in the preceding proof apply with b_k at index k. In particular all pairs $(K, L^K b_k)$ are in $\mathfrak{d}(g)$, and the table does not depend on b_k. This proves (a). The difference tends to zero on every subsequence, so it identifies the convergent subsequences and their limits. The same countable diagonal extraction works in every infinite set of indices. On the selected subsequence, compactness of $[g^{-2} + \bar{B}, X_g - \bar{B}]$ allows a further subsequence on which the labels converge, proving (b).

The triples realizing the limit prove membership in $\mathrm{GT}(g) \subset \mathcal{K}(g)$. If their labels have a limit point, take a further subsequence tending to that point. It still realizes the same limit, and the limit point is thus a label of the limit by Definition 18.172(4). This proves (c). For fixed b, every realizing triple has exactly that label, so the same equivalence applies within GT_b. Every member of $\mathrm{SF}(b)$ is obtained this way by taking its own sides and choosing $K_k = \kappa(k)$. Conversely, any convergent sequence with cutoffs at least these thresholds has the same limit as the torus limits on its sides, by (a). This proves (d). □

18.19. **Disjoint events in closed cubes whose edges are half as long as those of a periodic cube, cylinders at a label, and a lattice gap carried to the volume-first limits.** Proposition 18.178 lives on one lattice, at every spacing $a > 0$ and every $\beta \geq 0$: for every integer $N \geq 3$, odd or even, and every closed cube with faces parallel to the coordinate planes and edges of length $Na/2$, wherever its faces lie, the eighth powers of the probabilities that periodic Gibbs states give to mutually exclusive events of the bonds of the cube add up to at most $e^{\delta(N^3)}$, $\delta(N^3)$ being the free energy excess of the periodic box N^3; the bound holds as well for boxes with three different sides and for non-negative functions whose sum is at most 1; the mean of every such function is at most $e^{\delta/8}$ times the eighth root of the mean, on the periodic box, of the product of its eight mirror copies; and the number of pairwise mutually singular periodic Gibbs states is at most the lower limit of $e^{\delta(N^3)}$ over all integers $N \geq 3$. Proposition 18.180 gives the cylinder limits at a label $\mathsf{b} \in I(g)$ and over cross-sections given by their lengths: the laws of the long boxes have one limit, the cylinder law; the limit points of the cylinder laws along growing cross-sections are periodic Gibbs states; their continuum limits are invariant under all Euclidean motions, positive under reflections, and mixtures of Wightman theories each of which has a single vacuum; and the upper limit m of the second level of the sector without flux bounds the semigroup at the mirror, with the constant 1, and is a gap in the spectrum of the generator of that semigroup, the vacuum being single, when it is positive. Proposition 18.189 concerns any volume-first limit S at a coupling $0 < g \leq \gamma$, any defining sequence of states $(K_j, \beta_j, \nu_j)_j$ and any canonical decomposition of σ_S. Let T_{ν_j} be the transfer operator of the reconstruction of ν_j at the plane $\{x_3 = \frac{1}{2} a_{K_j}\}$, restricted to the closed span of the classes of the polynomials in the plaquette variables, and let $e^{-m_{\mathrm{lat}}(\nu_j)}$ be its norm on the orthogonal complement of its invariant vectors there. If $m = \liminf_j m_{\mathrm{lat}}(\nu_j)/a_{K_j}$ is positive, then m is at most a number M_g^* that depends only on the radius, the constants and g; neither H_S nor the operator H of any component has spectrum in $(0, m)$; the energy-momentum spectrum of each component is contained

in $\{0\} \cup \{E > 0,\ E^2 - |q|^2 \geq m^2\}$; each component clusters at the rate m, with constants that do not depend on the component; and if the lattice vacua are simple along an infinite set of indices, then $\ker H_S = \mathbb{C}\Omega_S$ and the only component of σ_S is σ_S itself. Proposition 18.190 draws the same conclusions from the norms of the transfer operators on any closed invariant subspace that contains these classes, for instance on the gauge-invariant sector or on the whole space, and when the vacua are simple there it puts the energy-momentum spectrum of σ_S itself into that set.

The crossing operators in Lemma 18.177 handle site and bond walls in one formula. For Proposition 18.178, reflection positivity at coordinate planes is Proposition YM3_I:103.4. The finite-volume conditional densities and a surface bound give the passage from the state to the periodic pressure. The cylinder proof uses the ground state of a positive slice transfer operator and the Gibbs-limit reconstruction of Theorem YM3_I:AP. Propositions 18.189 and 18.190 use the site and bond reflection forms, the component two-point estimate of Corollary YM3_I:127.11, and the scalar reconstruction of Theorems YM3_I:AP and YM3_I:AO.

Definition 18.176 (Grids and slices; half-period boxes and their mirror copies)**.** Let $a > 0$ and $\beta \geq 0$. In this definition, in Lemma 18.177 and in the proof of Proposition 18.178 the sites of $a\mathbb{Z}^3$ are written ay with $y \in \mathbb{Z}^3$, and lengths along the axes are counted in units of a.

(1) A *factor* is either a *path* $[\alpha, \omega] \cap \mathbb{Z}$ with integers $\alpha < \omega$ or a *cycle* $\mathsf{C}_M = \mathbb{Z}/M\mathbb{Z}$ with $M \geq 3$. A *grid* is a product $\Lambda = \Lambda_1 \times \Lambda_2 \times \Lambda_3$ of three factors, and a *slice* a product of two. The bonds of a grid are the pairs $\{y, y + e_\mu\}$ of two of its points, no bond leaving the last point of a path upwards, and its plaquettes are the elementary squares $p_{\mu\nu}(y)$, $\mu < \nu$, whose four corners are points of the grid; the same words apply to a slice, and $|\mathsf{Bd}(\Sigma)|$ is the number of the bonds of a slice Σ, at most $2n_1n_2$ if its two factors have n_1 and n_2 points. For a bounded Borel function Ψ of the variables of the bonds of a grid Λ we put $Z_\Lambda[\Psi] = \int \Psi \exp[-\beta \sum_p a_p]\, dU$, the sum running over the plaquettes of Λ and the integral over the product of the Haar probability measures of its bonds. The grid of three cycles C_{N_μ} is the periodic box $\underline{N}$, and $Z_\Lambda[\Psi] = Z(\underline{N})\langle\Psi\rangle_{\underline{N}}$ where $Z(\underline{N}) = Z_\Lambda[1]$ and $\langle\cdot\rangle_{\underline{N}}$ is the normalized Wilson expectation.

(2) Let Σ be a slice and $\Lambda = \Lambda_1 \times \Sigma$ a grid, its first factor serving as the time. Let u_y be the field of the layer $y \in \Lambda_1$ and τ_y the field of the oriented bonds from that layer to $y+1$. Write $A^{(0)}(u)$ for the sum of the plaquette actions in the layer. For $b = \langle x, x'\rangle$ a positive bond in Σ, the plaquette crossing to the next layer contributes $1 - \frac{1}{2}\operatorname{tr}(u_b \tau_{x'} (u'_b)^{-1} \tau_x^{-1})$. Let $A_\times(u, \tau, u')$ be the sum of these terms. Then $\sum_p a_p = \sum_y A^{(0)}(u_y) + \sum_y A_\times(u_y, \tau_y, u_{y+1})$, with the second sum over the bonds of the first factor. $\mathcal{H}_\Sigma$ is the space of the square-integrable functions of the field u of one layer, with the Haar probability measure du; the *crossing operator* K_Σ of the slice is the integral operator on $\mathcal{H}_\Sigma$ with the kernel $K_\Sigma(u, u') = \int e^{-\beta A_\times(u,\tau,u')}\, d\tau$.

(3) Let $N \geq 3$ be an integer. A *half-period interval* of the length $N/2$ is an interval $J = [c^-, c^+]$ with $c^\pm \in \frac{1}{2}\mathbb{Z}$ and $c^+ - c^- = N/2$. Its *walls* are the numbers $c_j = c^- + jN/2$, $j \in \mathbb{Z}$; a wall is a *site wall* if it is an integer and a *bond wall* otherwise; and $J_j = [c_j, c_{j+1}]$, so that $J_0 = J$. If N is even, all walls are of the kind of c^-; if N is odd, the two kinds alternate. The maps ρ_j of $\mathbb{Z}$ onto itself are $\rho_j(y) = y + jN/2$ for even j and $\rho_j(y) = 2c^+ + (j-1)N/2 - y$ for odd j. Then ρ_j carries $J_0 \cap \mathbb{Z}$ onto $J_j \cap \mathbb{Z}$, $\rho_{j+2} = \rho_j + N$, and the reflection $\vartheta_i(y) = 2c_i - y$ in the wall c_i satisfies $\vartheta_i \circ \rho_k = \rho_{2i-1-k}$ for all $i, k \in \mathbb{Z}$. The set $J \cap \mathbb{Z}$ has $\lfloor c^+ \rfloor - \lceil c^- \rceil + 1 \geq 2$ points.

(4) A *half-period box* of the lengths $N_\mu/2$, $N_\mu \geq 3$ integers, is a product $C' = J^{(1)} \times J^{(2)} \times J^{(3)}$ of half-period intervals, $J^{(\mu)}$ of the length $N_\mu/2$ and placed on the axis μ. For $\mathbf{k} \in \mathbb{Z}^3$ the map $\rho_\mathbf{k} = \rho^{(1)}_{k_1} \times \rho^{(2)}_{k_2} \times \rho^{(3)}_{k_3}$, read as $ay \mapsto a\rho_\mathbf{k}(y)$, is an isometry of $a\mathbb{Z}^3$, a translation of the lattice composed with coordinate reflections, and $C'_\mathbf{k} = \rho_\mathbf{k} C'$. The bonds of a closed set are the bonds both ends of which lie in it. For a bounded Borel function F of the bonds of C', $F^{[\mathbf{k}]} = \rho_\mathbf{k} F$, where $(\rho F)(U) = F(\rho^{-1}U)$, is a function of the bonds of $C'_\mathbf{k}$. On a grid whose factor in the direction μ is the cycle of the length $m_\mu N_\mu/2$, m_μ even, the boxes $C'_\mathbf{k}$ with

$0 \le k_\mu < m_\mu$ fill one period, $F^{[\mathbf{k}]}$ is evaluated at the periodic extension of a configuration of the grid, and it depends on k_μ modulo m_μ alone, because $\rho_{j+2} = \rho_j + N_\mu$.

■

Lemma 18.177 (One direction: walls that are shared or crossed; the cycle and the path)**.** Let Σ be a slice and $\beta \ge 0$.

(a) K_Σ is continuous and symmetric, with values in $[e^{-2\beta|\mathsf{Bd}(\Sigma)|}, 1]$, and $\mathsf{K}_\Sigma \ge 0$.

(b) Let J be a half-period interval of the length $N/2$ on the first axis, $\Gamma \ge 0$ a bounded Borel function of the bonds of $(J \cap \mathbb{Z}) \times \Sigma$, and $\Gamma^{[j]} = (\rho_j \times \mathrm{id})\Gamma$. There is a positive operator Q of trace class on $\mathcal{H}_\Sigma$, formed from Γ, J, Σ and β, such that for every even integer $m \ge 2$

$$Z_{\mathsf{C}_{mN/2}\times\Sigma}\Big[\prod_{j=0}^{m-1}\Gamma^{[j]}\Big] = \operatorname{Tr}\mathsf{Q}^{m/2} \;\le\; \Big(Z_{\mathsf{C}_N\times\Sigma}\big[\Gamma^{[0]}\Gamma^{[1]}\big]\Big)^{m/2}.$$

(c) In (b), let $\mathsf{l}_m = [c_0, c_m] \cap \mathbb{Z}$ be the path from the wall $c_0 = c^-$ to the wall c_m. Then

$$Z_{\mathsf{l}_m\times\Sigma}\Big[\prod_{j=0}^{m-1}\Gamma^{[j]}\Big] \;\le\; e^{2\beta|\mathsf{Bd}(\Sigma)|\varsigma}\, Z_{\mathsf{C}_{mN/2}\times\Sigma}\Big[\prod_{j=0}^{m-1}\Gamma^{[j]}\Big],$$

with $\varsigma = 0$ if c^- is a site wall and $\varsigma = 1$ if it is a bond wall.

The same holds with the second or the third axis as the time. ■

Sketch. Step 1: the crossing operator. The substitution $\tau \mapsto \tau^{-1}$ keeps Haar measure, and $A_\times(u,\tau,u') = A_\times(u',\tau^{-1},u)$, because an element of SU(2) and its inverse have the same trace and the trace is cyclic; so K_Σ is symmetric. Each of the $|\mathsf{Bd}(\Sigma)|$ terms of $A_\times$ lies in $[0,2]$ and $d\tau$ is a probability measure, which gives the bounds; the integrand is continuous on a compact space. For the bond b from x to x' put $(u^\tau)_b = \tau_x^{-1} u_b \tau_{x'}$, and let $k_0(u,u') = \prod_b e^{-\beta(1-\frac12 \operatorname{tr} u_b u_b'^{-1})}$. Then $e^{-\beta A_\times(u,\tau,u')} = k_0(u^\tau, u')$; k_0 is positive as an integral operator. Indeed $\operatorname{tr}(gg'^{-1}) = \sum_{i,j} g_{ij}\overline{g'_{ij}}$ for $g, g' \in \mathrm{SU}(2)$. Expand each exponential of this trace in its power series and take the product over the bonds. The result is a uniformly absolutely convergent sum of kernels $d_\alpha \phi_\alpha(u)\overline{\phi_\alpha(u')}$, where $d_\alpha \ge 0$ and the ϕ_α are products of matrix entries. Integration against $\overline{\psi(u)}\psi(u')$ gives $\sum_\alpha d_\alpha |\langle \phi_\alpha, \psi\rangle|^2 \ge 0$ for bounded Borel ψ. The kernel is bounded, and such functions are dense in $\mathcal{H}_\Sigma$, so its integral operator K_0 is positive; and $k_0(u^\tau, u'^\tau) = k_0(u,u')$, while $u \mapsto u^\tau$ keeps Haar measure. With $(R_\tau\psi)(u) = \psi(u^\tau)$ and the orthogonal projection $\Pi_{\mathrm{g}} = \int R_\tau \, d\tau$ onto the gauge-invariant functions, K_0 commutes with every R_τ and $\mathsf{K}_\Sigma = \Pi_{\mathrm{g}}\mathsf{K}_0 = \Pi_{\mathrm{g}}\mathsf{K}_0\Pi_{\mathrm{g}} \ge 0$. The argument uses the same crossing action for a path or a periodic slice.

Step 2: the kernel of a cell. Let $y^- = \lceil c^- \rceil$ and $y^+ = \lfloor c^+ \rfloor$ be the first and the last point of $J \cap \mathbb{Z}$, and let $\varpi_y = 1$ for $y^- \le y \le y^+$, except that $\varpi_{y^\pm} = \frac12$ when $c^\pm$ is a site wall. Let $k_J(u,u')$ be the integral of

$$\Gamma \, \exp\Big[-\beta \sum_{y=y^-}^{y^+} \varpi_y A^{(0)}(u_y) - \beta \sum_{y=y^-}^{y^+-1} A_\times(u_y, \tau_y, u_{y+1})\Big]$$

over the fields u_y, $y^- < y < y^+$, and τ_y, $y^- \le y < y^+$, at $u_{y^-} = u$ and $u_{y^+} = u'$. It is a Borel function with $0 \le k_J \le \sup\Gamma$; its operator A is Hilbert–Schmidt, and A^* has the kernel $k_J(u',u)$. Let $\mathsf{B}^\pm$ be the identity if $c^\pm$ is a site wall and K_Σ if it is a bond wall; $\mathsf{B}^\pm \ge 0$ by (a).

Step 3: the cycle; (b). On $\mathsf{C}_{mN/2} \times \Sigma$ the walls $c_0, \dots, c_{m-1}$ cut the first factor into the images $J_j \cap \mathbb{Z}$, $0 \le j < m$, of $J \cap \mathbb{Z}$, and the cell j carries $\Gamma^{[j]}$. If j is odd, ρ_j puts the layers in the opposite order, and $\Gamma^{[j]}$ uses the field of the layer $\rho_j(y)$ in the role of u_y and, in the role of τ_y, the inverse of the field of the bonds going from the layer $\rho_j(y) - 1$ to the layer $\rho_j(y)$; after the change of variables $\tau \mapsto \tau^{-1}$, and with the symmetry of $A_\times$ from Step 1, the product of $\Gamma^{[j]}$ and the weight of the cell, integrated over the inner layers of the cell and over its fields τ, equals $k_J(u_{\rho_j(y^-)}, u_{\rho_j(y^+)})$; if j is

even, the same is true, a translation replacing the reversal. A layer on a site wall belongs to the two cells that meet there, each of which gives it one half of its action, and both functions may read its bonds; the bonds that cross a bond wall belong to no cell, and their integral gives the factor K_Σ. The walls c_j with odd j are the images of c^+ under ρ_{j-1} and ρ_j, those with even j the images of c^-. Integrating first inside the cells (Fubini's theorem), with $u_j^- = u_{\rho_j(y^-)}$ and $u_j^+ = u_{\rho_j(y^+)}$,

$$Z_{\mathsf{C}_{mN/2}\times\Sigma}\Big[\prod_{j=0}^{m-1}\Gamma^{[j]}\Big] = \int k_J(u_0^-,u_0^+)\,B^+(u_0^+,u_1^+)\,k_J(u_1^-,u_1^+)$$
$$\times\, B^-(u_1^-,u_2^-)\,k_J(u_2^-,u_2^+)\cdots k_J(u_{m-1}^-,u_{m-1}^+)\,B^-(u_{m-1}^-,u_0^-),$$

where $B^\pm$ stands for the kernel K_Σ at a bond wall and for the identification of its two arguments at a site wall, and the integral runs over the remaining fields. This is $\mathrm{Tr}\big((\mathsf{A}\mathsf{B}^+\mathsf{A}^*\mathsf{B}^-)^{m/2}\big)$, the trace of a product of integral operators being the cyclic integral of their kernels; by the cyclicity of the trace it equals $\mathrm{Tr}\,\mathsf{Q}^{m/2}$ with $\mathsf{G} = (\mathsf{B}^-)^{1/2}\mathsf{A}(\mathsf{B}^+)^{1/2}$ and $\mathsf{Q} = \mathsf{G}\mathsf{G}^* \geq 0$, an operator of trace class. With the eigenvalues $\mathsf{q}_i \geq 0$ of Q, $\sum_i \mathsf{q}_i^{m/2} \leq (\sum_i \mathsf{q}_i)^{m/2}$, and at $m = 2$ the number $\mathrm{Tr}\,\mathsf{Q}$ is $Z_{\mathsf{C}_N\times\Sigma}[\Gamma^{[0]}\Gamma^{[1]}]$. This is (b).

Step 4: the path; (c). The path l_m holds the cells $J_0 \cap \mathbb{Z}, \dots, J_{m-1} \cap \mathbb{Z}$ and differs from the cycle at the wall $c_0 \equiv c_m$ alone. If c^- is a bond wall, the path has the layers of the cycle and all its bonds except those that cross this wall, so its integral is that of the display of Step 3 without the factor $B^-(u_{m-1}^-,u_0^-) = K_\Sigma(u_{m-1}^-,u_0^-) \geq e^{-2\beta|\mathsf{Bd}(\Sigma)|}$, all factors being ≥ 0. If c^- is a site wall, the path has two end layers, with the fields u_0^- and u_{m-1}^-, each of which lies in one cell only, which gives it one half of its action, while the grid gives it the whole; with $\chi = e^{-\frac{\beta}{2}A^{(0)}}$, $0 < \chi \leq 1$, $\|\chi\| \leq 1$, and since B^- is then the identity, the integral over the path is $\langle \chi, (\mathsf{A}\mathsf{B}^+\mathsf{A}^*)^{m/2}\chi\rangle = \langle \chi, \mathsf{Q}^{m/2}\chi\rangle \leq \|\mathsf{Q}^{m/2}\| \leq \mathrm{Tr}\,\mathsf{Q}^{m/2}$. This is (c). □

Proposition 18.178 (Mutually exclusive events in a closed box with the edges $N_\mu a/2$, $N_\mu \geq 3$ integers of either parity, against the free energy excess of the periodic box $\underline{N}$)**.** Let $a > 0$ and $\beta \geq 0$, let $N_1, N_2, N_3 \geq 3$ be integers, $\underline{N} = (N_1, N_2, N_3)$, and let P be the limit of $|\underline{M}|^{-1}\log Z(\underline{M})$ as the least side of the periodic box grows. The existence and shape independence of P are proved below. The excess is $\delta(\underline{N}) = \log Z(\underline{N}) - |\underline{N}|P$, where $|\underline{N}| = N_1N_2N_3$, and $\delta(N^3)$ denotes the excess of the cube.

(a) For every $z \in \mathbb{R}^3$ there is a half-period box C' of the lengths $N_\mu/2$ whose bonds include the bonds of the closed box $C = \prod_\mu [z_\mu, z_\mu + N_\mu a/2]$.

(b) For every half-period box C' of the lengths $N_\mu/2$, every periodic Gibbs state ν at (a,β) and every bounded Borel function $F \geq 0$ of the bonds of C',

$$\nu(F)^8 \;\leq\; e^{\delta(\underline{N})}\Big\langle \prod_{\mathbf{k}\in\{0,1\}^3} F^{[\mathbf{k}]}\Big\rangle_{\underline{N}} \;\leq\; e^{\delta(\underline{N})}\,\langle F^8\rangle_{\underline{N}},$$

the functions $F^{[\mathbf{k}]}$ being evaluated on the periodic box $\underline{N}$ as in Definition 18.176(4).

(c) For every closed box C as in (a), every integer $r \geq 1$, any periodic Gibbs states $\nu_1, \dots, \nu_r$ at (a,β), equal or different, and any bounded Borel functions $F_1, \dots, F_r \geq 0$ of the bonds of C with $\sum_s F_s \leq 1$,

$$\sum_{s=1}^r \nu_s(F_s)^8 \;\leq\; e^{\delta(\underline{N})} = Z(\underline{N})\,e^{-|\underline{N}|P}.$$

In particular $\sum_s \nu_s(A_s)^8 \leq e^{\delta(\underline{N})}$ for pairwise disjoint Borel events $A_1, \dots, A_r$ of the bonds of C. For cubes: at a pair (a_K,β), whose label is $\mathsf{b} = a_K\beta$, for every integer $N \geq 3$ and every closed cube $\prod_\mu [z_\mu, z_\mu + q]$ with $q = Na_K/2$, all periodic Gibbs states $\nu_1, \dots, \nu_r$ and all pairwise disjoint Borel events $A_1, \dots, A_r$ of the bonds of that cube satisfy $\sum_s \nu_s(A_s)^8 \leq \exp\delta(N^3)$, where $\delta(N^3)$ belongs to the periodic box with $N = 2q/a_K$ sites along each axis.

(d) If $\nu_1, \dots, \nu_r$ are pairwise mutually singular periodic Gibbs states at (a, β), then $r \le \liminf_{N\to\infty} e^{\delta(N^3)}$, the lower limit being taken over all integers $N \ge 3$.

■

Sketch. Step 1: (a). Along one axis, in units of a, the integers of $[z, z+N/2]$ are $\lceil z\rceil, \dots, \lfloor z+N/2\rfloor$. If $N = 2n$ and $z \in \mathbb{Z}$, the interval $J = [z, z+n]$ holds them. If $N = 2n$ and $z \notin \mathbb{Z}$, then $\lfloor z+n\rfloor = \lceil z\rceil + n - 1$, and $J = [\lceil z\rceil - \frac12, \lceil z\rceil + n - \frac12]$ holds them. If $N = 2n+1$, then $z+n+\frac12 < \lceil z\rceil + n + 1$, and $J = [\lceil z\rceil, \lceil z\rceil + n + \frac12]$ holds them. Each of these intervals is a half-period interval of the length $N/2$. A bond lies in a closed box if and only if both its ends do; so every bond of C is a bond of the product C' of the three intervals.

Step 2: Schwarz's inequality at the walls. Let ν be a periodic Gibbs state. The planes $\{x_\mu = ac_j^{(\mu)}\}$ are lattice mirrors, through sites or through midpoints of bonds, and ν is reflection positive at each of them for the bounded Borel functions of finitely many bonds in one closed side (Proposition YM3_I: 103.4). For such a function $G \ge 0$ and the reflection θ in the plane, Schwarz's inequality for the positive form $(G_1, G_2) \mapsto \nu(\overline{\theta G_1}\, G_2)$, at $G_1 = 1$, gives $\nu(G)^2 \le \nu(\theta G \cdot G)$. Let G be the product of the $F^{[\mathbf{k}]}$ over a set of indices that is a product of sets of consecutive integers, with $\{0, \dots, \omega\}$ in the direction μ; it is a function of bonds in $\{x_\mu \le ac_{\omega+1}^{(\mu)}\}$. Since $\vartheta_{\omega+1} \circ \rho_k = \rho_{2\omega+1-k}$, and since the pull-back action on functions composes in the same order as the isometries, $\theta G \cdot G$, for the reflection in the plane $\{x_\mu = ac_{\omega+1}^{(\mu)}\}$, is the product over the same set with $\{0, \dots, 2\omega+1\}$ in the direction μ. Starting from $\{0\}$ and doubling r times in each of the three directions,

$$\nu(F)^{8^r} \;\le\; \nu(\Psi_r), \qquad \Psi_r = \prod_{\mathbf{k}\in\{0,\dots,2^r-1\}^3} F^{[\mathbf{k}]}.$$

Step 3: from the state to a grid of paths, and to the torus. Let $r \ge 1$, $M_\mu = 2^{r-1}N_\mu$, and let Λ be the grid of the three paths $[c_0^{(\mu)}, c_{2^r}^{(\mu)}] \cap \mathbb{Z}$. It has M_μ points in the direction μ if $c_0^{(\mu)}$ is a bond wall and $M_\mu + 1$ if it is a site wall, the two end walls being of one kind; Ψ_r is a function of the finite set B of the bonds of Λ; and we put $s = 1 + \max_\mu M_\mu$. The Gibbs conditional law $\gamma_B(\,\cdot \mid \omega)$ has density proportional to $\exp[-\beta \sum_{p: p\cap B\neq\varnothing} a_p(U_B\omega)]$ against Haar measure on the bonds of B. Here $p \cap B \neq \varnothing$ means that p uses a bond in B. Consequently $\nu(\Psi_r) = \nu(\gamma_B(\Psi_r|\cdot))$. Besides the plaquettes of Λ, the plaquettes that hold a bond of B are those that hold a bond of one of the six outermost layers of Λ and leave Λ across that layer: at most $12s^2$ of them, each with $0 \le a_p \le 2$. Hence $\gamma_B(\Psi_r|\omega) \le e^{24\beta s^2} Z_\Lambda[\Psi_r]/Z_\Lambda[1]$ for every configuration ω outside B. Lemma 18.177(c), applied in the first direction with Σ the slice formed by the two other paths and with $\Gamma = \prod_{k_2,k_3} F^{[(0,k_2,k_3)]}$, whose images are $\Gamma^{[j]} = \prod_{k_2,k_3} F^{[(j,k_2,k_3)]}$, lets us exchange the first path for the cycle C_{M_1} at the cost of a factor not exceeding $e^{2\beta|\mathsf{Bd}(\Sigma)|} \le e^{4\beta s^2}$; the second and then the third path are replaced in the same manner, the functions being considered on the periodic extension along the directions that have already been closed. If in a direction the grid has $M_\mu + 1$ points, removing its last layer in that direction removes at most $3s^2$ plaquettes, each with $e^{-\beta a_p} \ge e^{-2\beta}$, and bonds that are integrated with probability measures; the periodic box $\underline{M} = (M_1, M_2, M_3)$ has all bonds and plaquettes of the grid so reduced and others, each with $e^{-\beta a_p} \le 1$. So $Z_\Lambda[1] \ge e^{-18\beta s^2} Z(\underline{M})$. With $\mathcal{Z}(\underline{m}) = Z_{\prod_\mu \mathsf{C}_{m_\mu N_\mu/2}}[\prod_{0\le k_\mu < m_\mu} F^{[\mathbf{k}]}]$ for even $m_\mu \ge 2$,

$$\nu(F)^{8^r} \;\le\; e^{54\beta s^2}\, \mathcal{Z}(2^r, 2^r, 2^r)\,/\,Z(\underline{M}).$$

Step 4: on the tori. By Lemma 18.177(b) in the first direction, with the slice $\mathsf{C}_{m_2N_2/2} \times \mathsf{C}_{m_3N_3/2}$ and $\Gamma = \prod_{k_2<m_2,\, k_3<m_3} F^{[(0,k_2,k_3)]}$, $\mathcal{Z}(m_1, m_2, m_3) \le \mathcal{Z}(2, m_2, m_3)^{m_1/2}$; then in the second and in the third direction. So $\mathcal{Z}(2^r, 2^r, 2^r) \le \mathcal{Z}(2,2,2)^{8^{r-1}}$, and $\mathcal{Z}(2,2,2) = Z(\underline{N})\langle \prod_{\mathbf{k}\in\{0,1\}^3} F^{[\mathbf{k}]}\rangle_{\underline{N}}$.

Step 5: the limit; (b). Taking the root of the order 8^{r-1} gives

$$\nu(F)^8 \le e^{54\beta s^2/8^{r-1}}\, \mathcal{Z}(2,2,2)\, Z(\underline{M})^{-1/8^{r-1}}.$$

Here $s \le 2^r \max_\mu N_\mu$, so that $54\beta s^2/8^{r-1} \le 432\,\beta \max_\mu N_\mu^2\, 2^{-r} \to 0$; and $|\underline{M}| = 8^{r-1}|\underline{N}|$.

For the pressure limit, fix an integer $n \ge 3$ and partition each axis of an open rectangular grid of side lengths M_μ into intervals of n sites and a possible shorter terminal interval. Deleting plaquettes crossing the resulting cuts factors its partition function into those of the smaller open grids. There are at most $2|\underline{M}|/n$ deleted plaquettes per axis, since a cut meets at most two crossing plaquettes per site in its cross-section and there are at most M_μ/n cuts along axis μ. As $0 \le a_p \le 2$, the deletion changes the logarithm per site by at most $12\beta/n$. For fixed n the fraction of sites in blocks with a short side goes to zero as $\min_\mu M_\mu$ grows. The normalized logarithms of all the block partition functions lie in $[-6\beta, 0]$, since the number of plaquettes is at most three per site. Thus the upper and lower limits of the normalized open-grid logarithm differ by at most $12\beta/n$. This difference vanishes as $n \to \infty$. Closing the grid removes no old plaquette and adds at most $2|\underline{M}| \sum_\mu M_\mu^{-1}$ plaquettes across the periodic seams. The normalized change tends to zero by the same comparison of weights. This proves $\log Z(\underline{M})/|\underline{M}| \to P$ along every sequence with least side tending to infinity. Therefore $Z(\underline{M})^{-1/8^{r-1}} \to e^{-|\underline{N}|P}$, and the first inequality of (b) follows. The second is Hölder's inequality with eight factors: on the periodic box $\underline{N}$ the map $\rho_{\mathbf{k}}$ acts in each direction as $y_\mu \mapsto \pm y_\mu + \text{const}$ modulo N_μ, which keeps the action and the Haar measures, so that $\langle (F^{[\mathbf{k}]})^8 \rangle_{\underline{N}} = \langle F^8 \rangle_{\underline{N}}$.

Step 6: (c). By (a), the F_s are functions of the bonds of C'. All $F_s^{[\mathbf{k}]}$ are ≥ 0, and $1 \ge \prod_{\mathbf{k}} \sum_s F_s^{[\mathbf{k}]} \ge \sum_s \prod_{\mathbf{k}} F_s^{[\mathbf{k}]}$, the terms with one and the same s in every factor being a part of the expansion of the product. So (b), summed over s, gives (c). For the cube, $N_\mu = N$ and $C = \prod_\mu [z_\mu, z_\mu + q]$, and the excess is that of the cube N^3 at (a_K, β).

Step 7: (d). Pairwise mutually singular laws have pairwise disjoint Borel sets E_s with $\nu_s(E_s) = 1$. Let $\epsilon > 0$ and $\bar\nu = \frac{1}{r}\sum_s \nu_s$. The events of finitely many bonds form an algebra that generates the Borel sets, so there are events A'_s of one finite set of bonds with $\bar\nu(A'_s \triangle E_s) < \epsilon$; as $\nu_s \le r\bar\nu$, the events $A_s = A'_s \setminus \bigcup_{s' \ne s} A'_{s'}$ are pairwise disjoint and $\nu_s(A_s) \ge 1 - r\epsilon - (r-1)r\epsilon \ge 1 - r^2\epsilon$. For every large N some closed cube $\prod_\mu [z_\mu, z_\mu + Na/2]$ holds that finite set of bonds, and (c) gives $r(1 - r^2\epsilon)^8 \le e^{\delta(N^3)}$. Let $N \to \infty$ along a sequence that realises the lower limit, and then $\epsilon \to 0$. □

Definition 18.179 (Cylinder data and the space at the mirror)**.** Use the construction constants, the first-passage set $\mathfrak{d}(g)$ and the interval $I(g)$ of §YM3_I:74 and Corollary YM3_I:74.2, with $L \ge 13$ odd and $0 < g \le \gamma$. Write $a_K = L^{-K}$. On a lattice of spacing a let t_n be the translation by nae_3, and let Θ_c be reflection sending layer j to layer $c - j$, for $c \in \mathbb{Z}$. On functions the actions are pull-backs by the inverse isometries. In particular Θ_0 is the site mirror at 0 and Θ_1 the bond mirror at $a/2$. The lattice observable is $\mathcal{O}_a(f) = a^{-1} \sum_p f(x(p)) a_p$ with the plaquette barycentre $x(p)$.

For a periodic slice $\Sigma = \mathsf{C}_{N_1} \times \mathsf{C}_{N_2}$ let W be multiplication by $e^{-\beta A^{(0)}/2}$ on the one-layer space of Definition 18.176. The slice transfer is $\mathsf{T}_\Sigma = \mathsf{W}\mathsf{K}_\Sigma\mathsf{W}$. For $\mu = 1, 2$, flip the sign of every positively oriented μ-bond crossing a fixed periodic seam in direction μ. This gives two commuting unitary involutions on that space. The sector without flux is their common fixed subspace. Let λ_0 be the largest eigenvalue of T_Σ and let λ_1^{00} be the second eigenvalue in this subspace, counted with multiplicity and taken to be 0 if there is no second positive eigenvalue. Put

$$y(\Sigma) = \lambda_1^{00}/\lambda_0, \qquad E_1(\Sigma; a, \beta) = -a^{-1} \log y(\Sigma), \qquad E^K(\underline{\ell}) = E_1(\underline{\ell}/a_K; a_K, L^K \mathsf{b}).$$

Here $-\log 0 = \infty$ and $e^{-\infty \cdot 0} = 1$. The cylinder law μ_Σ is the limit of the periodic laws over Σ as the number of time layers goes to infinity. Its existence, and the ground-state facts used in the level definition, are proved below. A state through cylinders consists of a limit $\nu = \lim_k \mu_{\Sigma_k}$ with $\min \Sigma_k \to \infty$, together with the sequence of cross-sections, and its number is $m_\nu = \limsup_k E_1(\Sigma_k; a, \beta)$.

Use the centered functions and Gibbs limits of Definition 18.205. For a functional σ from Theorem YM3_I:AP, or one of its canonical components, $\mathcal{D}$ is the linear span of separated tuples

supported in $\{x_3 > 0\}$, including $\varnothing$. The reflection Hilbert space $\mathcal{H}(\sigma)$ is the completion of this space modulo the null space of the positive form $\langle X, Y\rangle_\sigma = \sigma(\Theta_0 X \cdot Y)$ on real tuples, extended sesquilinearly to their complex linear span. Write $\Omega_\sigma = [\varnothing]$ and let $H_\sigma \geq 0$ be the generator of the strongly continuous semigroup of time translations, $e^{-tH_\sigma}[F] = [\tau_{te_3}F]$. ■

Proposition 18.180 (Cylinder laws at a label: the time first, then the cross-section, then the cutoff)**.** Use Definition 18.179. Let $0 < g \leq \gamma$ and $\mathsf{b} \in I(g)$, and put $\beta_K = L^K\mathsf{b}$ for $K \in \mathbb{N}$. For $K \in \mathbb{N}$, a pair of lengths $\underline{\ell} = (\ell_1, \ell_2)$ such that ℓ_1/a_K and ℓ_2/a_K are integers ≥ 3, and $t \in a_K\mathbb{N}$ with $t \geq 3a_K$, let $\mu_{K,\underline{\ell},t}$ be the Wilson measure, as a law, of the periodic box with t/a_K layers along e_3 over the cross-section $\underline{\ell}/a_K$ at (a_K, β_K), and let $E^K(\underline{\ell})$ be the second level of the sector without flux at the cutoff K along the label b .

(1) As $t \to \infty$ the laws $\mu_{K,\underline{\ell},t}$ have exactly one weak limit point, the cylinder law $\mu^{\mathrm{cyl}}_{K,\underline{\ell}} = \mu_{\underline{\ell}/a_K}$ at (a_K, β_K); it is invariant under every t_n and every Θ_c, $n, c \in \mathbb{Z}$.

(2) Let $\underline{\ell}^{(k)}$, $k \in \mathbb{N}$, be such pairs at one cutoff K, with $\min \underline{\ell}^{(k)} \to \infty$. Then $(\mu^{\mathrm{cyl}}_{K,\underline{\ell}^{(k)}})_k$ has weak limit points. Each of them, ν, is a periodic Gibbs state at (a_K, β_K), invariant under $a_K\mathbb{Z}^3$ and under every Θ_c; with a subsequence of the $\underline{\ell}^{(k)}$ along which the cylinder laws converge to it, ν is a state through cylinders at (a_K, β_K), whose number is $m_\nu = \limsup E^K(\underline{\ell}^{(k)})$ along that subsequence; and along it $\mu^{\mathrm{cyl}}_{K,\underline{\ell}^{(k)}}(\mathcal{O}_{a_K}(f)) \to \nu(\mathcal{O}_{a_K}(f))$ and $\sigma_{\mu^{\mathrm{cyl}}_{K,\underline{\ell}^{(k)}}}(X) \to \sigma_\nu(X)$ for every bounded function f with compact support and every tuple $X = (f_1, \dots, f_n)$ of such functions; here, for a law μ on the configurations of $a_K\mathbb{Z}^3$ and such a tuple, $\sigma_\mu(X)$ is the mean under μ of $\prod_{i=1}^n(\mathcal{O}_{a_K}(f_i) - \mu(\mathcal{O}_{a_K}(f_i)))$, also the expression used on separated tuples in Definition 18.205.

(3) Let $K_j \to \infty$, and for every j let ν_j be a law as in (2) at the cutoff K_j, together with its subsequence of cross-sections. Then $(K_j, \beta_{K_j}, \nu_j)_j$ is, at the label b, a sequence through cylinders at g , hence a Gibbs sequence at g; there is a subsequence of $(\sigma_{\nu_j})_j$ that converges at each separated tuple; and whenever σ is the limit of such a subsequence and $(\mathsf{X}, \lambda, (\sigma_\xi)_{\xi\in\mathsf{X}})$ is a canonical decomposition of σ, one has $\sigma = \int \sigma_\xi\, \lambda(d\xi)$, the functional σ and each σ_ξ are reflection positive at every mirror and invariant under $\mathbb{R}^3 \rtimes \mathrm{O}(3)$, and each σ_ξ is the functional of a Wightman theory on $\mathbb{R}^{1+2}$, with a single Hermitian scalar field and a unique vacuum, and obeys (E0′)–(E4).

(4) Let $m = \limsup_j m_{\nu_j} \in [0, \infty]$, the upper limits being taken along the subsequences of cross-sections of (2) and along the subsequence of j that defines σ. Then for all $F = \sum_i \zeta_i X_i \in \mathcal{D}$ and all real $t \geq 0$, with $\varphi_F(t) = \sum_{i,k} \bar{\zeta}_i\zeta_k\, \sigma(\Theta_0 X_i \cdot \tau_{te_3}X_k)$,

$$0 \;\leq\; \varphi_F(t) - |\sigma(F)|^2 \;\leq\; e^{-mt}[\varphi_F(0) - |\sigma(F)|^2];$$

for every j the same holds with σ_{ν_j} and m_{ν_j} in the places of σ and m, at the times $t = na_{K_j}$, $n \in \mathbb{N}_0$, whenever the supports of F lie in $\{x_3 > \frac{1}{2}a_{K_j}\}$. If $m > 0$, then $\ker H_\sigma = \mathbb{C}\Omega_\sigma$ and $\operatorname{spec} H_\sigma \subset \{0\} \cup [m, \infty)$ (with $\mathcal{H}(\sigma) = \mathbb{C}\Omega_\sigma$ if $m = \infty$), every canonical decomposition of σ has a single point, and σ, undecomposed, is the functional of a Wightman theory with a unique vacuum.

(5) Let $K \in \mathbb{N}$ and $\beta > 0$. Item (2) holds at the pair (a_K, β), with the cylinder laws $\mu_{\underline{\ell}/a_K}$ and the levels $E_1(\underline{\ell}/a_K; a_K, \beta)$, both formed at (a_K, β), in the places of $\mu^{\mathrm{cyl}}_{K,\underline{\ell}}$ and of $E^K(\underline{\ell})$. Further, let $(K_j, \beta_j) \in \mathfrak{d}(g)$ with $K_j \to \infty$, and for every j let ν_j be a law as in (2) at the pair (a_{K_j}, β_j), in the sense just given, together with its subsequence of cross-sections. Then $(K_j, \beta_j, \nu_j)_j$ is a sequence through cylinders at g , and so a Gibbs sequence at g; and what (3) asserts of the subsequences of $(\sigma_{\nu_j})_j$, of their limits and of the canonical decompositions of these limits, as well as all of (4), holds for this sequence in the place of $(K_j, \beta_{K_j}, \nu_j)_j$.

■

Sketch of proof. The ground state and the cylinder. The slice transfer has a bounded continuous strictly positive kernel and is a positive Hilbert–Schmidt operator by Lemma 18.177(a). Its top eigenvalue λ_0 is positive. A real maximizer of its Rayleigh quotient can be replaced by its absolute value, which is still a maximizer. Applying the positive kernel shows that this nonnegative eigenfunction is strictly positive almost everywhere. If the top eigenvalue had multiplicity larger than one, a real eigenfunction orthogonal to this positive vector would change sign on sets of positive measure. Taking its absolute value would strictly increase its Rayleigh quotient, by strict positivity of the kernel, a contradiction. Let v_0 be the positive normalized top eigenfunction. Simultaneous seam flips on two layers preserve each crossing plaquette, which contains either none or two of the flipped bonds. The flips also preserve the plaquettes in a layer, and their substitutions preserve Haar measure. Thus they commute with T_Σ. Positivity and simplicity give v_0 in their common fixed space. In particular $y(\Sigma) < 1$.

For a bounded function A of bonds between layers α and $\alpha + d$ inclusive, take $d \geq 1$ by including an extra layer if necessary. Let $\mathsf{B}[A]$ be the operator from the layer at $\alpha + d$ to that at α with kernel given by integrating the slab weight times A over the interior layers and all its vertical bonds. The endpoint layer actions have weight $1/2$, all other actions weight 1. Then $\mathsf{B}[1] = \mathsf{T}_\Sigma^d$. On a time circle of N layers, integrating the other bonds gives the expectation

$$\frac{\operatorname{Tr}(\mathsf{B}[A]\mathsf{T}_\Sigma^{N-d})}{\operatorname{Tr}\mathsf{T}_\Sigma^N}.$$

Indeed, each shared endpoint layer supplies one half of its action to each adjacent factor, and each time bond is integrated once. The expression is valid for $N \geq d + 2$. If $\{\lambda_j\}$ are the eigenvalues of T_Σ outside its top eigenspace, their ratios to λ_0 are nonnegative, have supremum $q < 1$, and have summable squares. Thus $\lambda_0^{-k}\mathsf{T}_\Sigma^k \to |v_0\rangle\langle v_0|$ in trace norm, with error at most $q^{k-2}\sum_j(\lambda_j/\lambda_0)^2$ for $k \geq 2$. The limit of the expectation is $\lambda_0^{-d}\langle v_0, \mathsf{B}[A]v_0\rangle$. The product configuration space of the cylinder is compact and metrisable. Continuous local functions determine the weak limit, so this proves existence and uniqueness in (1). Each finite-period law is invariant under every time translation and time reflection, and the invariance passes to the limit.

The cross-section limit. Embed the cylinder laws and the periodic laws as periodic configurations on the full lattice in the periodic directions. Weak limit points of the cylinders exist by compactness of the full product space. For any convergent subsequence of cylinders, choose for its kth member a time period at least $\min \Sigma_k$ for which the weak distance from the cylinder is at most $1/k$. This is possible by (1), and the resulting periodic boxes converge to the same ν with all periods tending to infinity. Translation and reflection invariance pass to the limit. For any fixed finite bond set B, the Wilson conditional density on B uses only the finite set of plaquettes incident to B. For continuous local tests it gives continuous local functions of the exterior bonds. Those Gibbs identities are exact once the finite test and the incident plaquettes embed without identifications into the periodic box. Passing to the weak limit proves the same identities for ν. Continuous local tests determine measures, so these are the DLR identities. This proves that ν is a periodic Gibbs state. The stated observable and tuple moments are polynomials in expectations of continuous local functions and hence converge. This proves (2), and works at every (a, β) in (5).

Reflected correlation on a cylinder. Take a bounded polynomial A in plaquette actions supported in layers $\alpha, \dots, \alpha+d$ with $\alpha \geq 1$. Reflecting the slab and conjugating A replaces $\mathsf{B}[A]$ by $\mathsf{B}[A]^*$. Each plaquette action is unchanged under the seam flips, so $\mathsf{B}[A]$ commutes with them. Consequently $u = \lambda_0^{-d}\mathsf{B}[A]v_0$ belongs to the sector without flux. The two slabs in $\overline{\Theta_0 A}\,\mathsf{t}_n A$ are separated by $2\alpha + n$ steps. With $R = \lambda_0^{-1}\mathsf{T}_\Sigma$ and $u_\perp = u - \langle v_0, u\rangle v_0$, the same slab integration gives

$$\mu_\Sigma(\overline{\Theta_0 A}\,\mathsf{t}_n A) - |\mu_\Sigma(A)|^2 = \langle u_\perp, R^{2\alpha+n}u_\perp\rangle.$$

The spectrum of R on the sector without flux orthogonal to v_0 is contained in $[0, y(\Sigma)]$. The last expression is nonnegative, and its value at $n + n'$ is at most its value at n times $y(\Sigma)^{n'}$. For a state

through cylinders, use a subsequence with $E_1(\Sigma_k; a, \beta) \to m_\nu$. All fixed supports embed in those cross-sections for large k. Weak convergence passes the inequality to ν, with $y(\Sigma_k)^{n'} \to e^{-m_\nu n' a}$.

The cutoff limit. Corollary YM3_I:74.2 gives the first-passage condition $(K, L^K \mathsf{b}) \in \mathfrak{d}(g)$ for $\mathsf{b} \in I(g)$. Since every ν_j is periodic Gibbs, Theorem YM3_I:AP gives the subsequence and the conclusions of (3). For $F = \sum_i \zeta_i X_i \in \mathcal{D}$ at fixed j, take $A = \sum_i \zeta_i (X_i)^{\nu_j}_{a_{K_j}}$ using the products of centered plaquette observables in Definition 18.205. If all the test supports lie strictly above $a_{K_j}/2$, all plaquettes involved have their bonds in layers ≥ 1. By the covariance of $\mathcal{O}_a(f)$ under translations and reflection, the last cylinder inequality, passed to ν_j, is exactly (4) at lattice times. To pass to σ, use a further subsequence with $m_{\nu_j} \to m$. For each $t = pL^{-k}$, $p, k \in \mathbb{N}_0$, the lattice times equal t for all large j, and the reflected tuples are fixed separated tuples. Thus (4) passes to the limit at these times. Strong continuity of e^{-tH_σ} extends it to every $t \geq 0$.

A positive rate. Suppose $m > 0$. Letting $t \to \infty$ in (4) gives $\|P_{\ker H_\sigma}[F]\|^2 = |\sigma(F)|^2$, with $\sigma(F) = \langle \Omega_\sigma, [F] \rangle$. The classes of $\mathcal{D}$ are dense, so this proves $\ker H_\sigma = \mathbb{C}\Omega_\sigma$. After removing that component the spectral measure of each $[F]$ is positive. If it put positive mass in $(0, m')$ for some $m' < m$, its Laplace transform would be bounded below by that mass times $e^{-m't}$, contradicting (4) as $t \to \infty$. Density of the classes proves the claimed spectral inclusion, also for $m = \infty$ by taking all finite m'.

In the canonical decomposition, the reflected correlation is the integral of those of the components. In each component the projection onto zero energy is its vacuum projection by Theorem YM3_I:AP. The integrands at $t \geq 0$ are bounded by their integrable values at 0. Passing to $t = \infty$ under this integral yields $\int |\sigma_\xi(F)|^2 \, d\lambda = |\sigma(F)|^2$. Since $\int \sigma_\xi(F) \, d\lambda = \sigma(F)$, every such variance vanishes. The canonical component maps are continuous and the canonical measure has full support (Theorem YM3_I:AL), so $\sigma_\xi = \sigma$ on all tuples supported in $\{x_3 > 0\}$ for every ξ. Translation invariance extends this to all separated tuples. Injectivity of the component map gives a one-point decomposition. The reconstruction statement now follows from (3). The argument uses $\mathsf{b} \in I(g)$ only to guarantee the first-passage data and identify the fixed label. It proves all of (5) with arbitrary data in $\mathfrak{d}(g)$. □

Definition 18.181 (The transfer operator of a periodic Gibbs state at a plane of bond midpoints; sectors; the lattice gap; lattice functions; two families of numbers)**.** Let $a > 0$ and $\beta \geq 0$, and let ν be a periodic Gibbs state at (a, β).

(1) *The reconstruction at the plane* $\{x_3 = \frac{1}{2}a\}$. Let $\mathcal{F}^+$ consist of bounded Borel functions of finitely many bonds with both endpoints in layers ≥ 1. Write $\mathcal{H}_\nu$ for the completion of the quotient by the null space of $\langle F, G \rangle_\nu = \nu(\overline{\Theta_1 F} G)$ on $\mathcal{F}^+$. Positivity follows from Proposition YM3_I:103.4. Put $\Omega_\nu = [1]$. The transfer T_ν on this space has $T_\nu[F] = [\mathsf{t}_1 F]$, $0 \leq T_\nu \leq 1$ and $T_\nu \Omega_\nu = \Omega_\nu$. In particular

$$\langle [F], T_\nu^k [G] \rangle = \nu(\overline{\Theta_1 F} \, \mathsf{t}_k G), \qquad k \geq 0.$$

For these assertions, reflection conjugates translations to their inverses, and translation invariance gives $\langle \mathsf{t}_i F, \mathsf{t}_j G \rangle_\nu = \langle F, \mathsf{t}_{i+j} G \rangle_\nu$ for $i, j \geq 0$. For fixed F put $b_n = \|\mathsf{t}_n F\|_\nu$. By Cauchy–Schwarz $b_n^2 \leq b_{n-1} b_{n+1}$ for $n \geq 1$, and by invariance $b_n \leq \|F\|_{L^2(\nu)}$. If $b_0 = 0$ this forces $b_1 = 0$. If $b_1 > b_0 > 0$, the successive ratios are nondecreasing, contradicting boundedness. Thus $b_1 \leq b_0$, and the shift descends to a contraction on the quotient. The same pairing identity makes it symmetric. The relation $\Theta_2 \mathsf{t}_1 F = \Theta_1 F$ and reflection positivity at the site mirror $x_3 = a$ give $\langle F, \mathsf{t}_1 F \rangle_\nu \geq 0$. This proves positivity of the contraction, and its remaining properties follow by iteration and the constant function.

(2) *Sectors; the lattice gap.* $\mathcal{A}_a^+$ is the set of the polynomials with complex coefficients in the functions a_p of the plaquettes p contained in $\{x_3 \geq \frac{1}{2}a\}$; then $\mathcal{A}_a^+ \subset \mathcal{F}^+$ and $\mathsf{t}_1 \mathcal{A}_a^+ \subset \mathcal{A}_a^+$. A *sector* of ν is a closed subspace $\mathcal{K} \subset \mathcal{H}_\nu$ with $T_\nu \mathcal{K} \subset \mathcal{K}$ that contains $[A]$ for every $A \in \mathcal{A}_a^+$. The *action sector* $\mathcal{H}_\nu^{\mathrm{act}}$, the closure of $\{[A] : A \in \mathcal{A}_a^+\}$, is the least sector; the *gauge-invariant*

sector, the closure of the classes of the $F \in \mathcal{F}^+$ with $F(U^u) = F(U)$ for every configuration U and every gauge transformation u of $a\mathbb{Z}^3$, is a sector: it holds the classes of $\mathcal{A}_a^+$, the plaquette functions a_p being unchanged by gauge transformations by cancellation of site gauge factors around the plaquette, and the bounded operator T_ν maps it into itself, because $\mathsf{t}_{-1}(U^u) = (\mathsf{t}_{-1}U)^{u'}$ with the gauge transformation $u'(x) = u(x + ae_3)$, so that $\mathsf{t}_1 F$ is gauge invariant whenever F is; $\mathcal{H}_\nu$ is a sector too. For a sector $\mathcal{K}$ let $\mathcal{K}^0 = \{\psi \in \mathcal{K} : T_\nu\psi = \psi\}$, a closed subspace that holds Ω_ν, and $\mathcal{K}^1 = \mathcal{K} \ominus \mathcal{K}^0$; T_ν maps $\mathcal{K}^1$ into itself, and

$$q_{\mathcal{K}}(\nu) = \|T_\nu|_{\mathcal{K}^1}\| \in [0,1] \quad (q_{\mathcal{K}}(\nu) = 0 \text{ if } \mathcal{K}^1 = \{0\}), \qquad m_{\mathcal{K}}(\nu) = -\log q_{\mathcal{K}}(\nu) \in [0,\infty].$$

We put $q_{\mathrm{lat}}(\nu) = q_{\mathcal{H}_\nu^{\mathrm{act}}}(\nu)$, and the *lattice gap* of ν is $m_{\mathrm{lat}}(\nu) = -\log q_{\mathrm{lat}}(\nu)$. The vacuum of ν is *simple in* $\mathcal{K}$ if $\mathcal{K}^0 = \mathbb{C}\Omega_\nu$.

(3) *Lattice functions.* For an open set $O \subset \mathbb{R}^3$ and an integer $n \geq 0$, $\mathfrak{S}_n(O)$ is the set of the separated tuples in O with n entries; for $X = (f_1, \dots, f_n) \in \mathfrak{S}_n(O)$ we write $|X| = n$, and for $J \subset \{1, \dots, n\}$ we write X_J for the tuple of the entries f_i with $i \in J$, in increasing order of i; and $\mathbb{C}\mathfrak{S}(O)$ is the complex vector space with the basis $\mathfrak{S}(O)$, whose elements are the finite sums $\sum_i \zeta_i X_i$ with $\zeta_i \in \mathbb{C}$ and $X_i \in \mathfrak{S}(O)$. For $X = (f_1, \dots, f_n) \in \mathfrak{S}(\mathbb{R}^3)$ let $X_a^\nu = \prod_{i=1}^n (\mathcal{O}_a(f_i) - \nu(\mathcal{O}_a(f_i)))$ and $\varnothing_a^\nu = 1$, so that $\sigma_\nu(X) = \nu(X_a^\nu)$ as in Definition 18.205. For $F = \sum_i \zeta_i X_i \in \mathbb{C}\mathfrak{S}(\mathbb{R}^3)$ let $F_a^\nu = \sum_i \zeta_i (X_i)_a^\nu$ and $\sigma_\nu(F) = \sum_i \zeta_i \sigma_\nu(X_i)$. If $F \in \mathcal{D}$ and $t \geq 0$ is real, the entries of every $\Theta_0 X_i$ have their supports in $\{x_3 < 0\}$ and those of every $\tau_{te_3} X_k$ in $\{x_3 > 0\}$, so that the tuples $\Theta_0 X_i \cdot \tau_{te_3} X_k$ are separated, and for these F and t alone we put $\varphi_F^{\sigma_\nu}(t) = \sum_{i,k} \bar{\zeta}_i \zeta_k \, \sigma_\nu(\Theta_0 X_i \cdot \tau_{te_3} X_k)$. For $F \in \mathcal{D}$ let $\varepsilon(F) \in (0,\infty]$ be the least value of x_3 on the union of the supports of the entries of the X_i, and $\varepsilon(F) = +\infty$ if that union is empty.

(4) *Volume-first limits.* In the setting of Definition 18.179 let $0 < g \leq \gamma$, let S be a volume-first limit at g and $(K_j, \beta_j, \nu_j)_j$ a defining sequence of states of S as in Definition 18.205, $a_j = a_{K_j}$, and let σ_S be the centred functional of S. Its space at the mirror, the classes, the vacuum and the operator of Definition 18.179 are written $\mathcal{H}(S)$, $[F]$, Ω_S and H_S; P_0 is the projection onto $\ker H_S$, and $\varphi_F(t) = \langle [F], e^{-tH_S}[F]\rangle = \varphi_F^{\sigma_S}(t)$ for $F \in \mathcal{D}$ and real $t \geq 0$, where φ_F^σ is formed, at these F and t, for any functional σ on separated tuples by the formula of (3). For a component σ_ξ of a canonical decomposition $(\mathsf{X}, \lambda, (\sigma_\xi)_{\xi \in \mathsf{X}})$ of σ_S the same objects are $\mathcal{H}(\sigma_\xi)$, $[F]_\xi$, Ω_{σ_ξ}, H_{σ_ξ} and $\varphi_F^\xi = \varphi_F^{\sigma_\xi}$. The momenta $P_{1,\sigma}$, $P_{2,\sigma}$, the joint projection-valued measure $\mathcal{P}_\sigma$ of $H_\sigma, P_{1,\sigma}, P_{2,\sigma}$ and the joint spectrum $\operatorname{supp}\mathcal{P}_\sigma \subset \mathbb{R} \times \mathbb{R}^2$ are the generators and joint spectral measure of the strongly commuting time and transverse translations from Theorem YM3_I:AO; for $\sigma = \sigma_S$ we write $P_{1,S}$, $P_{2,S}$, $\mathcal{P}_S$. For real $m > 0$ let $\mathsf{H}_m = \{(0,0)\} \cup \{(E,q) \in \mathbb{R} \times \mathbb{R}^2 : \ E > 0, \ E^2 - |q|^2 \geq m^2\}$.

(5) *The numbers of the ceiling.* γ_W is the ceiling and $\mathsf{d}_0(g', \mathfrak{u})$ the threshold of YM3_I:(2346) and YM3_I:(2350), respectively, and $\mathfrak{u}_1 = 3\sqrt{3}/8$. For $0 < g' \leq \gamma_W$ put $\mathsf{M}_{g'} = \log(93312)/(5\,\mathsf{d}_0(g', \mathfrak{u}_1))$, where $93312 = 2 \cdot 6^6$; further $\mathsf{M}_W = \max\{\mathsf{M}_{\gamma_W}, \mathsf{M}_{\gamma_W/\sqrt{2}}\}$, $\kappa_0 = L(\gamma_W^{-2} + \bar{B})$ with the number $\bar{B} = B_\sharp L/(L-1)$ of §YM3_I:74, and $M_g^* = \kappa_0 \, g^2 \, \mathsf{M}_W$ for $0 < g \leq \gamma$. These numbers are formed from the radius, the constants and g alone.

(6) *The numbers of clustering.* For $X = (f_1, \dots, f_n) \in \mathfrak{S}(\mathbb{R}^3)$ let $\mathfrak{r}(X)$ be the least $r \geq 0$ such that the supports of all its entries lie in the closed ball of the radius r about 0, with $\mathfrak{r}(\varnothing) = 0$; let $\mathsf{d}_X = \min\{\mathsf{d}(X), 1\}$ and $R_X = \max\{R(X), 1\}$, with $\mathsf{d}(X)$ the least distance between two distinct supports (infinite for a one-entry tuple) and $R(X)$ the largest of the least enclosing-ball radii of its supports. With the canonical bound table $C_k^g(\mathsf{d}; R)$, $k \geq 2$, of YM3_I:(1850) and the one-point constant $\bar{\mathfrak{r}}(\mathsf{C}^g) = \bar{c}_g$ of Theorem YM3_I:AP, put $\mathfrak{a}_1(\mathsf{d}; R) = \frac{4\pi}{3} R^3 \, \bar{\mathfrak{r}}(\mathsf{C}^g)$, $\mathfrak{a}_k(\mathsf{d}; R) = C_k^g(\mathsf{d}; R)$ for $k \geq 2$, $\mathfrak{A}_k(\mathsf{d}; R) = \sum_{\pi \in \mathrm{Part}(k)} \prod_{J \in \pi} \mathfrak{a}_{|J|}(\mathsf{d}; R)$, the sum running over the set $\mathrm{Part}(k)$ of the partitions of $\{1, \dots, k\}$ into non-empty blocks,

and, with $\|f\|_\sharp = \max\{\|f\|_\infty, 40M \operatorname{Lip} f\}$,

$$\mathfrak{n}_g(X) = \mathfrak{A}_{2n}(\mathsf{d}_X; R_X)^{1/2} \prod_{i=1}^{n} \|f_i\|_\sharp \quad (\mathfrak{n}_g(\varnothing) = 1),$$
$$\mathfrak{r}(X,Y) = 2\max\{\mathfrak{r}(X), \mathfrak{r}(Y)\} + 1,$$
$$\mathfrak{C}_g(X,Y) = \mathfrak{n}_g(X)\,\mathfrak{n}_g(Y)\, e^{M_g^*\,\mathfrak{r}(X,Y)}.$$

These numbers are formed from the radius, the constants, g and the tuples alone.

■

Lemma 18.182 (One lattice: the gap in a sector bounds the decay of reflected correlations)**.** Let $a > 0$, $\beta \geq 0$, ν a periodic Gibbs state at (a, β), $\mathcal{K}$ a sector of ν, $q = q_{\mathcal{K}}(\nu)$ with $q^0 = 1$, and let Π_ν be the orthogonal projection of $\mathcal{H}_\nu$ onto $\ker(1 - T_\nu)$.

(a) For $A, A' \in \mathcal{F}^+$ and every integer $n \geq -1$, $\nu(\overline{\Theta_0 A}\, \mathsf{t}_n A') = \langle [A], T_\nu^{n+1}[A'] \rangle$.

(b) Let $A \in \mathcal{F}^+$ with $[A] \in \mathcal{K}$, $\Phi_A(n) = \nu(\overline{\Theta_0 A}\, \mathsf{t}_n A)$, and $c_A = \|P_{\mathcal{K}^0}[A]\|^2$ with the orthogonal projection $P_{\mathcal{K}^0}$ onto $\mathcal{K}^0$. For all integers $n, k \geq 0$, $0 \leq \Phi_A(n) - \Phi_A(n+k) \leq q^n(\Phi_A(0) - \Phi_A(k))$ and $0 \leq \Phi_A(n) - c_A \leq q^n(\Phi_A(0) - c_A)$.

(c) If the vacuum of ν is simple in $\mathcal{K}$, then $c_A = |\nu(A)|^2$.

(d) $\mathcal{K}^0 = \Pi_\nu \mathcal{K}$ and $\mathcal{K}^1 = \mathcal{K} \cap (1 - \Pi_\nu)\mathcal{H}_\nu$. Hence $q_{\mathcal{K}}(\nu) \leq q_{\mathcal{K}'}(\nu)$ for sectors $\mathcal{K} \subset \mathcal{K}'$; $m_{\mathrm{lat}}(\nu) \geq m_{\mathcal{K}}(\nu)$ for every sector; a vacuum that is simple in $\mathcal{K}'$ is simple in $\mathcal{K}$; and if $\ker(1 - T_\nu) = \mathbb{C}\Omega_\nu$, the vacuum is simple in every sector.

■

Sketch. (a) From the lattice actions of Definition 18.179, $\Theta_1 U = \mathsf{t}_1 \Theta_0 U = \Theta_0 \mathsf{t}_{-1} U$, hence $\Theta_1 A = \mathsf{t}_1(\Theta_0 A)$ for every function A; the actions of isometries on functions are multiplicative and commute with complex conjugation. So $\overline{\Theta_1 A}\, \mathsf{t}_{n+1} A' = \mathsf{t}_1\big[\overline{\Theta_0 A}\, \mathsf{t}_n A'\big]$, and ν is invariant under t_1; the transfer pairing of Definition 18.181(1) at $k = n + 1 \geq 0$ gives (a).

(b) Put $\psi = [A] = \psi^0 + \psi^1$ with $\psi^0 = P_{\mathcal{K}^0}\psi$ and $\psi^1 \in \mathcal{K}^1$. Then $T_\nu^l \psi^0 = \psi^0$ and $T_\nu^l \psi^1 \in \mathcal{K}^1 \perp \mathcal{K}^0$, so that by (a) $\Phi_A(n) = c_A + \langle \psi^1, T_\nu^{n+1} \psi^1 \rangle = c_A + \int_{[0,q]} \lambda^{n+1}\, d\mu(\lambda)$, with the spectral measure μ at ψ^1 of the restriction of T_ν to $\mathcal{K}^1$, which is carried by $[0, q]$. On $[0, q]$, $0 \leq \lambda^{n+1} \leq q^n \lambda$ and $0 \leq \lambda^{n+1}(1 - \lambda^k) \leq q^n \lambda (1 - \lambda^k)$; integrating gives both inequalities.

(c) Then $P_{\mathcal{K}^0}\psi = \langle \Omega_\nu, \psi \rangle \Omega_\nu$ and $\langle \Omega_\nu, [A] \rangle = \nu(A)$.

(d) By the spectral theorem and $0 \leq T_\nu \leq 1$, $T_\nu^n \to \Pi_\nu$ strongly; $\mathcal{K}$ is closed and invariant, so $\Pi_\nu \mathcal{K} \subset \mathcal{K} \cap \ker(1 - T_\nu) = \mathcal{K}^0$, and Π_ν is the identity on $\mathcal{K}^0$. For $\psi \in \mathcal{K}^1$, $\Pi_\nu \psi \in \mathcal{K}^0$ is orthogonal to ψ, so $\|\Pi_\nu \psi\|^2 = \langle \psi, \Pi_\nu \psi \rangle = 0$; conversely a vector of $\mathcal{K}$ with $\Pi_\nu \psi = 0$ is orthogonal to $\ker(1 - T_\nu) \supset \mathcal{K}^0$. The rest follows, as $\mathcal{H}_\nu^{\mathrm{act}}$ is contained in every sector and $\Omega_\nu \in \mathcal{K}^0$. □

Lemma 18.183 (Lattice functions of separated tuples)**.** Let $a > 0$, and let ν be a law on the configurations of $a\mathbb{Z}^3$ that is invariant under all isometries of $a\mathbb{Z}^3$; every periodic Gibbs state is such a law (Proposition YM3_I:103.4).

(a) $F \mapsto F_a^\nu$ is linear; X_a^ν is a real polynomial in the a_p with $x(p)$ in the union of the supports of the entries of X; $X_a^\nu Y_a^\nu = (X \cdot Y)_a^\nu$ whenever $X \cdot Y$ is defined; and $\mathsf{r}(X_a^\nu) = (\mathsf{r}X)_a^\nu$ for every isometry r of $a\mathbb{Z}^3$, in particular $\Theta_0(X_a^\nu) = (\Theta_0 X)_a^\nu$ and $\mathsf{t}_n(X_a^\nu) = (\tau_{nae_3} X)_a^\nu$.

(b) If $F \in \mathcal{D}$ and $a \leq \varepsilon(F)$, then $F_a^\nu \in \mathcal{A}_a^+$.

(c) For $F = \sum_i \zeta_i X_i$ and $F' = \sum_k \zeta'_k X'_k$ in $\mathcal{D}$ and every integer $n \geq 0$, $\nu(\overline{\Theta_0 F_a^\nu}\, \mathsf{t}_n F_a'^\nu) = \sum_{i,k} \bar{\zeta}_i \zeta'_k\, \sigma_\nu(\Theta_0 X_i \cdot \tau_{nae_3} X'_k)$, and $\nu(F_a^\nu) = \sigma_\nu(F)$; at $F' = F$ the first number is $\varphi_F^{\sigma_\nu}(na)$.

■

Sketch. (a) $\mathcal{O}_a(f)$ is a finite real combination of the a_p with $f(x(p)) \neq 0$, and $\nu(\mathcal{O}_a(f))$ is real. An isometry carries a_p to $a_{\mathsf{r}p}$ and $x(p)$ to $\mathsf{r}x(p)$, so that $\mathsf{r}\mathcal{O}_a(f) = \mathcal{O}_a(\mathsf{r}f)$, and $\nu(\mathcal{O}_a(\mathsf{r}f)) = \nu(\mathsf{r}\mathcal{O}_a(f)) = \nu(\mathcal{O}_a(f))$ by the invariance; the action of r on functions is multiplicative. The other assertions are

read off the products. (b) For a plaquette p with $x(p)$ in a support of an entry, $x(p)_3 \geq \varepsilon(F) \geq a$, and every point of p has $x_3 \geq x(p)_3 - \frac{1}{2}a \geq \frac{1}{2}a$; by (a), F_a^ν is a polynomial in such a_p. (c) The supports of $\Theta_0 X_i$ lie in $\{x_3 < 0\}$ and those of $\tau_{nae_3} X'_k$ in $\{x_3 > 0\}$, so that the concatenations are defined; the X_a^ν are real; (a) turns the product into $\sum_{i,k} \bar{\zeta}_i \zeta'_k (\Theta_0 X_i \cdot \tau_{nae_3} X'_k)_a^\nu$, whose mean is the asserted sum. □

Lemma 18.184 (One self-adjoint operator: three forms of a gap)**.** Let $H \geq 0$ be a self-adjoint operator with the spectral measure $\mathbf{E}$, $P_0 = \mathbf{E}(\{0\})$, $\mathcal{D}_0$ a dense subset of the space, and $m > 0$ real. The following say the same: (i) $\mathbf{E}((0,m)) = 0$, that is, $\operatorname{spec} H \subset \{0\} \cup [m, \infty)$; (ii) $\langle \psi, (e^{-tH} - P_0)\psi \rangle \leq e^{-mt} \langle \psi, (1 - P_0)\psi \rangle$ for all $\psi \in \mathcal{D}_0$ and $t \geq 0$; (iii) $\langle \psi, (e^{-tH} - e^{-(t+s)H})\psi \rangle \leq e^{-mt} \langle \psi, (1 - e^{-sH})\psi \rangle$ for all $\psi \in \mathcal{D}_0$ and $s, t \geq 0$. If they hold, (ii) and (iii) hold for all vectors, and $\|e^{-tH} - P_0\| \leq e^{-mt}$ for $t \geq 0$. ■

Sketch. Let μ_ψ be the spectral measure of H at ψ. (i) implies (iii) for every ψ: the left member is $\int e^{-tE}(1 - e^{-sE})\, d\mu_\psi(E)$, whose integrand vanishes at $E = 0$ and is at most $e^{-mt}(1 - e^{-sE})$ on $[m, \infty)$, and $\int_{[m,\infty)} (1 - e^{-sE})\, d\mu_\psi = \langle \psi, (1 - e^{-sH})\psi \rangle$. (iii) implies (ii): as $s \to \infty$, $e^{-sE} \to \mathbf{1}_{\{0\}}(E)$ boundedly, so $e^{-sH} \to P_0$ and $e^{-(t+s)H} \to P_0$ strongly. (ii) implies (i): both members of (ii) are continuous in ψ, so (ii) holds for all ψ; for $0 < m' < m$ and ψ in the range of $\mathbf{E}((0,m'])$ the left member is at least $e^{-m't}\|\psi\|^2$ and the right one is $e^{-mt}\|\psi\|^2$; at $t = 1$ this forces $\psi = 0$; so $\mathbf{E}((0,m']) = 0$ for every $m' < m$, and $\mathbf{E}((0,m)) = 0$. Under (i), $e^{-tH} - P_0 = \int_{[m,\infty)} e^{-tE}\, d\mathbf{E}(E)$. □

Lemma 18.185 (The spectral measures of a limit and of its components)**.** In the setting of Definition 18.179 let $0 < g \leq \gamma$, let S be a volume-first limit at g, $\sigma = \sigma_S$ its centred functional, so that $H_\sigma = H_S$ (Definition 18.181(4)), and $(\mathsf{X}, \lambda, (\sigma_\xi)_{\xi \in \mathsf{X}})$ a canonical decomposition of σ. For $F \in \mathcal{D}$ let μ_F and μ_F^ξ be the spectral measures of H_σ at $[F]$ and of H_{σ_ξ} at $[F]_\xi$, finite positive Borel measures on $[0, \infty)$.

(a) For every continuous real function h on $[0, \infty)$ with a finite limit at ∞, the function $\xi \mapsto \int h\, d\mu_F^\xi$ is continuous on X, and $\int_{\mathsf{X}} (\int h\, d\mu_F^\xi) \lambda(d\xi) = \int h\, d\mu_F$.

(b) For every relatively open set $G \subset [0, \infty)$: $\mu_F(G) = 0$ if and only if $\mu_F^\xi(G) = 0$ for every $\xi \in \mathsf{X}$.

(c) For every real $m > 0$: $\operatorname{spec} H_\sigma \subset \{0\} \cup [m, \infty)$ if and only if $\operatorname{spec} H_{\sigma_\xi} \subset \{0\} \cup [m, \infty)$ for every $\xi \in \mathsf{X}$.

■

Sketch. Step 1: Laplace transforms. A defining sequence of states of S as in Definition 18.205 is a convergent Gibbs sequence at g whose limit is σ; so the spaces at the mirror and the semigroups of σ and of every σ_ξ exist by Theorem YM3_I:AP; $\int e^{-tE}\, d\mu_F(E) = \varphi_F^\sigma(t)$ and $\int e^{-tE}\, d\mu_F^\xi(E) = \varphi_F^{\sigma_\xi}(t)$. The latter is a finite combination of the numbers $\sigma_\xi(\Theta_0 X_i \cdot \tau_{te_3} X_k)$, continuous in ξ, and $\int \varphi_F^{\sigma_\xi}(t)\, \lambda(d\xi) = \varphi_F^\sigma(t)$ by the canonical decomposition of Theorem YM3_I:AL. So (a) holds for the functions $e_t(E) = e^{-tE}$, $t \geq 0$.

Step 2: (a). On the compactification $[0, \infty]$, with $e_t(\infty) = 0$ for $t > 0$ and $e_0 = 1$, the real span of the e_t is an algebra that holds the constants and separates the points; by the theorem of Stone and Weierstrass it is dense in the continuous functions. The masses $\mu_F^\xi([0,\infty)) = \varphi_F^{\sigma_\xi}(0)$ are bounded, a continuous function on the compact space X; uniform approximation of h gives (a).

Step 3: (b). Let $h_n(E) = \min\{1, n \operatorname{dist}(E, [0,\infty) \setminus G)\} \cdot \max\{0, \min\{1, n + 1 - E\}\}$ (the first factor being 1 if $G = [0, \infty)$): continuous, vanishing for $E \geq n + 1$, and increasing to $\mathbf{1}_G$. The functions $I_n(\xi) = \int h_n\, d\mu_F^\xi \geq 0$ are continuous by (a). If $\mu_F(G) = 0$, then $\int I_n\, d\lambda = 0$ by (a); a point with $I_n > 0$ would have a neighbourhood of positive measure; so $I_n = 0$ everywhere, and $\mu_F^\xi(G) = \lim_n I_n(\xi) = 0$ by monotone convergence. The converse is read in the same way.

Step 4: (c). Take $G=(0,m)$. The spectrum of H_σ misses $(0,m)$ if and only if $\mu_F((0,m))=0$ for every $F\in\mathcal{D}$, the classes being dense; the same holds in every $\mathcal{H}(\sigma_\xi)$; and (b) applies F by F. □

Lemma 18.186 (A dilation acts on the space at the mirror as a unitary)**.** Let $s>0$, and let σ be a symmetric functional on separated tuples over $\mathbb{R}^3$ such that σ and $\sigma'=\mathsf{dil}_s\sigma$, defined by $\sigma'(X)=s^{-|X|}\sigma(X_{(s)})$ with $X_{(s)}=(f_1(s\,\cdot),\dots,f_n(s\,\cdot))$ for $X=(f_1,\dots,f_n)$ are invariant under translations, reflection positive at $(e_3,0)$, and have the Hilbert spaces and translation semigroups of Definition 18.179. Put $W_0X=s^{-|X|}X_{(s)}$ for separated tuples X, extended linearly. Then there is exactly one unitary operator $W:\mathcal{H}(\sigma')\to\mathcal{H}(\sigma)$ with $W[F]_{\sigma'}=[W_0F]_\sigma$ for all $F\in\mathcal{D}$; moreover $W\Omega_{\sigma'}=\Omega_\sigma$ and $H_\sigma=s\,WH_{\sigma'}W^{-1}$. In particular $\operatorname{spec}H_\sigma=s\cdot\operatorname{spec}H_{\sigma'}$ and $\ker H_\sigma=W\ker H_{\sigma'}$. ■

Sketch. The map $X\mapsto X_{(s)}$ is a bijection of $\mathfrak{S}_n(\mathbb{R}^3)$ onto itself with $\operatorname{supp}X_{(s)}=s^{-1}\operatorname{supp}X$, so that it maps $\mathfrak{S}_n(\{x_3>0\})$ onto itself; $(X\cdot Y)_{(s)}=X_{(s)}\cdot Y_{(s)}$ and $(\tau_vX)_{(s)}=\tau_{v/s}X_{(s)}$; and $(\Theta_0X)_{(s)}=\Theta_0(X_{(s)})$, as Θ_0 is linear. So W_0 is a linear bijection of $\mathcal{D}$, $W_0\varnothing=\varnothing$, $W_0\tau_{te_3}F=\tau_{(t/s)e_3}W_0F$, and

$$\langle W_0F,W_0F'\rangle_\sigma=\sum_{i,k}\bar\zeta_i\zeta'_k\,s^{-|\Theta_0X_i\cdot X'_k|}\,\sigma\big((\Theta_0X_i\cdot X'_k)_{(s)}\big)=\langle F,F'\rangle_{\sigma'}.$$

Hence $[F]_{\sigma'}\mapsto[W_0F]_\sigma$ is well defined and isometric on a dense subspace, with a dense range, and it extends to a unitary W; on the classes $We^{-tH_{\sigma'}}=e^{-(t/s)H_\sigma}W$, so everywhere. The operator $s^{-1}W^{-1}H_\sigma W$ is self-adjoint and ≥0, and its semigroup is $e^{-tH_{\sigma'}}$; by the uniqueness of the generator it is $H_{\sigma'}$. □

Lemma 18.187 (A ceiling in every component at couplings up to γ_W)**.** In the setting of Definition 18.179 let $0<g'\le\gamma_W$, let S' be a volume-first limit at g' and $\sigma=\sigma_{S'}$ its centred functional, and let $(\mathsf{X},\lambda,(\sigma_\xi)_{\xi\in\mathsf{X}})$ be a canonical decomposition of σ. Then $\operatorname{spec}H_{\sigma_\xi}\cap(0,\mathsf{M}_{g'}]\ne\emptyset$ for every $\xi\in\mathsf{X}$, and $\operatorname{spec}H_\sigma\cap(0,\mathsf{M}_{g'}]\ne\emptyset$. ■

Sketch. Step 1: a bump. Let $\mathsf{d}=\mathsf{d}_0(g',\mathfrak{u}_1)$, $f_0(x)=(1-|x|^2/\mathsf{d}^2)^2$ for $|x|\le\mathsf{d}$ and $f_0(x)=0$ otherwise, and $f=\tau_{\frac32\mathsf{d}e_3}f_0$. Then f_0 is of class C^1, $f_0\ge0$, $\|f_0\|_\infty=1$, its support is the closed ball of the radius d about 0, $\int f_0=\frac{32\pi}{105}\mathsf{d}^3\ge\mathfrak{u}_1\mathsf{d}^3$, and $\sup|\nabla f_0|=\frac{4}{\mathsf{d}}\cdot\frac{1}{\sqrt3}\cdot\frac23=\frac{1}{\mathfrak{u}_1\mathsf{d}}$, attained at $|x|^2=\mathsf{d}^2/3$; the support of f lies in $\{x_3\ge\frac12\mathsf{d}\}$, so $(f)\in\mathfrak{S}_1(\{x_3>0\})$.

Step 2: test pairs. For $0\le t\le5\mathsf{d}$ the supports of Θ_0f and of $\tau_{te_3}f$ are the closed balls of the radius d about $-\frac32\mathsf{d}e_3$ and $(\frac32\mathsf{d}+t)e_3$, at the distance $\mathsf{d}+t\in[\mathsf{d},6\mathsf{d}]$. With Step 1, $(\Theta_0f,\tau_{te_3}f)$ meets the support, integral and Lipschitz conditions for the C^1 test pairs in Theorem YM3_I:C, at scale d and shape parameter $\mathfrak{u}_1$.

Step 3: the law in the component. Theorem YM3_I:AP gives a converging Gibbs frame for the defining sequence of S'. Fix ξ and put $D(t)=S_2^{\sigma_\xi}(\Theta_0f,\tau_{te_3}f)$ and $\mathcal{K}(t)=\mathcal{K}_{\text{free}}(\Theta_0f,\tau_{te_3}f)$, with the kernel of YM3_I:(2240). Corollary YM3_I:127.11 gives the common label b and $D(t)=\mathsf{c}_2\mathsf{b}^{-2}\mathcal{K}(t)(1+\eta_t)$ with $|\eta_t|\le\bar\varepsilon_{g',\mathfrak{u}_1}(\mathsf{d})\le1/3$ for $0\le t\le5\mathsf{d}$. In particular $D(0)>0$. In $\mathcal{K}(t)=\iint(\Theta_0f)(x)\,f(y)\,|x-y-te_3|^{-6}\,dx\,dy$ the integrand vanishes unless $|x-y|\ge\mathsf{d}$, and then $|x-y-te_3|\le|x-y|(1+t/\mathsf{d})$; as $f\ge0$, $\mathcal{K}(5\mathsf{d})\ge6^{-6}\mathcal{K}(0)$. So $D(5\mathsf{d})/D(0)\ge\frac{2/3}{4/3}\,6^{-6}=1/93312$.

Step 4: in the space at the mirror. By the two-point cumulant identity and the isometry invariance of σ_ξ from Theorem YM3_I:AP, $D(t)=\varphi^{\sigma_\xi}_{(f)}(t)-\sigma_\xi(f)^2$; as $\ker H_{\sigma_\xi}=\mathbb{C}\Omega_{\sigma_\xi}$ by Theorem YM3_I:AP and $\langle\Omega_{\sigma_\xi},[(f)]_\xi\rangle=\sigma_\xi(f)$, this is $\langle[(f)]_\xi,(e^{-tH_{\sigma_\xi}}-P_{0,\xi})[(f)]_\xi\rangle$, $P_{0,\xi}$ the projection onto the kernel.

Step 5: the ceiling. Suppose that the spectrum of H_{σ_ξ} misses $(0,\mathsf{M}_{g'}]$. It is closed, so it misses $(0,\mathsf{M}_{g'}+\epsilon)$ for some $\epsilon>0$, and Lemma 18.184 gives $D(5\mathsf{d})\le e^{-5(\mathsf{M}_{g'}+\epsilon)\mathsf{d}}D(0)<D(0)/93312$, against Step 3. For σ: if its spectrum missed $(0,\mathsf{M}_{g'}]$, it would lie in $\{0\}\cup[\mathsf{M}_{g'}+\epsilon,\infty)$ for some $\epsilon>0$, and by Lemma 18.185(c), applied to S', so would the spectrum of every component, X being non-empty. □

Lemma 18.188 (A ceiling at every coupling for volume-first limits)**.** In the setting of Definition 18.179 let $0 < g \le \gamma$, let S be a volume-first limit at g, $(K_j, \beta_j, \nu_j)_j$ a defining sequence of states of S, and $(\mathsf{X}, \lambda, (\sigma_\xi)_{\xi\in\mathsf{X}})$ a canonical decomposition of σ_S.

(a) The labels $a_{K_j}\beta_j$ have limit points; every limit point b satisfies $\mathsf{b} \ge g^{-2} + B_\sharp \frac{L-2}{L-1} \ge g^{-2}$; and for every limit point b and every $\xi \in \mathsf{X}$, $\operatorname{spec} H_{\sigma_\xi} \cap (0, \kappa_0 \mathsf{M}_W/\mathsf{b}] \neq \emptyset$.
(b) For every $\xi \in \mathsf{X}$, $\operatorname{spec} H_{\sigma_\xi} \cap (0, M_g^*] \neq \emptyset$.
(c) $\operatorname{spec} H_S \cap (0, M_g^*] \neq \emptyset$, and $\operatorname{spec} H_S \cap (0, \kappa_0 \mathsf{M}_W/\mathsf{b}] \neq \emptyset$ for every limit point b of the labels.

■

Sketch. Step 1: the labels. By Corollary YM3_I:74.2 and YM3_I:(1854), the labels at first-passage data lie in the compact interval $I^{\mathrm{lab}}(g)$. Each limit point satisfies $g^{-2} + B_\sharp(L-2)/(L-1) \le \mathsf{b} \le L(g^{-2} + \bar{B})$. Fix one such b.

Step 2: two reference couplings. Put $x_W = \gamma_W^{-2}$, so $x_W > L^2\bar{B}$ and $\gamma_W \le \gamma$ by YM3_I:(2346). From §YM3_I:74, the interior $\mathcal{I}(g')$ of $I(g')$ is $(g'^{-2} + \bar{B}, L(g'^{-2} + \bar{B}) - 2\bar{B})$. For $g_1 = \gamma_W$ and $g_2 = \gamma_W/\sqrt{2}$ the two intervals are $(x_W + \bar{B}, Lx_W + (L-2)\bar{B})$ and $(2x_W + \bar{B}, 2Lx_W + (L-2)\bar{B})$. They overlap because $2x_W + \bar{B} < Lx_W + (L-2)\bar{B}$. Their union is $(\mathsf{x}_-, \mathsf{x}_+)$ with $\mathsf{x}_- = x_W + \bar{B}$ and $\mathsf{x}_+ = 2Lx_W + (L-2)\bar{B}$. Since $\mathsf{x}_+ - L\mathsf{x}_- = Lx_W - 2\bar{B} > 0$, the union contains $(\mathsf{x}_-, L\mathsf{x}_-]$. Here $\kappa_0 = L\mathsf{x}_-$.

Step 3: the integer. Let $z \in \mathbb{Z}$ be the least integer with $L^z\mathsf{b} > \mathsf{x}_-$; then $L^{z-1}\mathsf{b} \le \mathsf{x}_-$, so $\mathsf{x}_- < L^z\mathsf{b} \le L\mathsf{x}_- < \mathsf{x}_+$, and $L^z\mathsf{b} \in \mathcal{I}(g_i)$ for $i = 1$ or $i = 2$; fix such an i. Put $s = L^z$; then $s \le L\mathsf{x}_-/\mathsf{b} = \kappa_0/\mathsf{b} \le \kappa_0 g^2$, by Step 1. The coupling g_i satisfies $g_i \le \gamma_W \le \gamma$.

Step 4: the dilated limit. Along a subsequence with labels tending to b, the dilated labels $L^z a_{K_j}\beta_j$ eventually lie in $I(g_i)$ because their limit lies in its interior. Corollary YM3_I:74.2 gives first-passage data $(K_j - z, \beta_j)$ at g_i for all large j. The dilation $x \mapsto sx$ sends the bare lattice, plaquettes, Haar variables and Gibbs kernels to those data at unchanged bare coupling. It sends a volume-first state to a volume-first state, since it sends each approximating torus of side $2L^k$ to a torus of side $2L^{k+z}$. The lattice action observable transforms with a factor s^{-1} and test function $f(s\,\cdot)$, directly from its barycentre sum. Centering has the same factor. Thus the dilated sequence gives the volume-first limit $S^{[z]}$ at g_i with centered functional $\sigma_{S^{[z]}} = \mathsf{dil}_s\sigma_S$. Proposition YM3_I:101.1 gives its canonical decomposition on the same probability space with components $\mathsf{dil}_s\sigma_\xi$.

Step 5: the ceiling at g_i, and back. Lemma 18.187 gives $\operatorname{spec} H_{\mathsf{dil}_s\sigma_\xi} \cap (0, \mathsf{M}_{g_i}] \neq \emptyset$ for every ξ. Theorem YM3_I:AP gives the isometry invariance, reflection positivity and reconstruction for both sets of components. Lemma 18.186 therefore gives $\operatorname{spec} H_{\sigma_\xi} = s \operatorname{spec} H_{\mathsf{dil}_s\sigma_\xi}$. Since $s\,\mathsf{M}_{g_i} \le \kappa_0\mathsf{M}_W/\mathsf{b} \le M_g^*$, this proves (a) and (b).

Step 6: (c). If the spectrum of H_S missed $(0, M]$, with $M = M_g^*$ or $M = \kappa_0\mathsf{M}_W/\mathsf{b}$, it would lie in $\{0\} \cup [M + \epsilon, \infty)$ for some $\epsilon > 0$, the spectrum being closed; by Lemma 18.185(c), applied to S and to the decomposition $(\mathsf{X}, \lambda, (\sigma_\xi)_{\xi\in\mathsf{X}})$, so would the spectrum of every component, against (a) or (b). □

Proposition 18.189 (Transfer operators whose lattice gap stays bounded below in physical units: the spectral gap of H_S and of the components, the hyperboloid, and clustering)**.** In the setting of Definition 18.179 let $0 < g \le \gamma$, let S be a volume-first limit at g, let $(K_j, \beta_j, \nu_j)_j$ be any defining sequence of states of S, $a_j = a_{K_j}$, and let $(\mathsf{X}, \lambda, (\sigma_\xi)_{\xi\in\mathsf{X}})$ be any canonical decomposition of σ_S. With the lattice gaps of Definition 18.181(2), suppose that

$$m = \liminf_{j\to\infty} \frac{m_{\mathrm{lat}}(\nu_j)}{a_j} > 0.$$

The spaces $(\mathcal{H}^{\mathrm{act}}_{\nu_j})^0$ and the labels $a_j\beta_j$ are arbitrary. Then $m \le M_g^* < \infty$, and the following hold.

(a) $\operatorname{spec} H_S \subset \{0\} \cup [m, \infty)$; equivalently, $\langle [F], e^{-tH_S}[F]\rangle - \|P_0[F]\|^2 \le e^{-mt}(\|[F]\|^2 - \|P_0[F]\|^2)$ for all $F \in \mathcal{D}$ and all real $t \ge 0$.

(b) For every $\xi \in \mathsf{X}$: $\operatorname{spec} H_{\sigma_\xi} \subset \{0\} \cup [m, \infty)$, and $\varphi^\xi_F(t) - |\sigma_\xi(F)|^2 \le e^{-mt}(\varphi^\xi_F(0) - |\sigma_\xi(F)|^2)$ for all $F \in \mathcal{D}$ and $t \ge 0$.

(c) For every $\xi \in \mathsf{X}$ the joint spectrum of $(H_{\sigma_\xi}, P_{1,\sigma_\xi}, P_{2,\sigma_\xi})$ lies in $\mathsf{H}_m = \{(0,0)\} \cup \{(E,q) : E > 0,\ E^2 - |q|^2 \ge m^2\}$.

(d) For every $\xi \in \mathsf{X}$, all $X \in \mathfrak{S}_n(\mathbb{R}^3)$, $Y \in \mathfrak{S}_{n'}(\mathbb{R}^3)$ and all $v \in \mathbb{R}^3$ with $|v| \ge \mathfrak{r}(X,Y)$, the tuple $X \cdot \tau_v Y$ is defined and

$$|\sigma_\xi(X \cdot \tau_v Y) - \sigma_\xi(X)\,\sigma_\xi(Y)| \ \le\ \mathfrak{n}_g(X)\,\mathfrak{n}_g(Y)\,e^{-m(|v|-\mathfrak{r}(X,Y))} \ \le\ \mathfrak{C}_g(X,Y)\,e^{-m|v|},$$

with the numbers of Definition 18.181(6); $\mathfrak{C}_g(X,Y)$ is formed from the radius, the constants, g, X and Y alone, and it depends neither on the component, nor on the canonical decomposition, nor on the volume-first limit at g, nor on its defining sequence.

(e) If moreover the set of the indices j at which the vacuum of ν_j is simple in the action sector is infinite, then $\ker H_S = \mathbb{C}\Omega_S$, every canonical decomposition of σ_S has a single point, whose component is σ_S itself, (b)–(d) hold with σ_S in the place of σ_ξ, and $\varphi_F(t) - |\sigma_S(F)|^2 \le e^{-mt}(\varphi_F(0) - |\sigma_S(F)|^2)$ for all $F \in \mathcal{D}$ and $t \ge 0$.

(f) (a)–(d) hold also with the upper limit of $m_{\mathrm{lat}}(\nu_j)/a_j$ in the place of m, whenever that upper limit is positive; it is then at most M^*_g.

■

Sketch. Step 1: one lattice at a time. Fix $F = \sum_i \zeta_i X_i \in \mathcal{D}$ and a real m' with $0 < m' < m$. For all large j, $a_j \le \varepsilon(F)$, because $K_j \to \infty$, and $m_{\mathrm{lat}}(\nu_j) \ge m' a_j$, by the definition of the lower limit. For such j the state ν_j is a periodic Gibbs state; put $A_j = F^{\nu_j}_{a_j}$, which lies in $\mathcal{A}^+_{a_j}$ by Lemma 18.183(b), $\Phi_j(n) = \nu_j(\overline{\Theta_0 A_j}\,\mathfrak{t}_n A_j)$, and $c_j = \|P_{(\mathcal{H}^{\mathrm{act}}_{\nu_j})^0}[A_j]\|^2$. By Lemma 18.183(c), $\Phi_j(n) = \varphi_F^{\sigma_{\nu_j}}(na_j)$, and by Lemma 18.182(b), as $q_{\mathrm{lat}}(\nu_j)^n \le e^{-m'na_j}$, for all integers $n, k \ge 0$

$$0 \le \Phi_j(n) - \Phi_j(n+k) \le e^{-m'na_j}\big(\Phi_j(0) - \Phi_j(k)\big), \qquad 0 \le \Phi_j(n) - c_j \le e^{-m'na_j}\big(\Phi_j(0) - c_j\big).$$

Step 2: times of the form $pL^{-k'}$. Let $t = pL^{-k'}$ and $s = p'L^{-k'}$ with integers $p, p', k' \ge 0$. For j with $K_j \ge k'$, $n_j = t/a_j$ and $k_j = s/a_j$ are integers, and $\Phi_j(n_j)$, $\Phi_j(n_j + k_j)$, $\Phi_j(0)$, $\Phi_j(k_j)$ are the values of $\varphi_F^{\sigma_{\nu_j}}$ at t, $t+s$, 0, s, finite combinations of the values of σ_{ν_j} at tuples that do not depend on j. Since $\sigma_{\nu_j} \to \sigma_S$ at every separated tuple, letting $j \to \infty$ in the first inequality of Step 1 gives

$$\varphi_F(t) - \varphi_F(t+s) \ \le\ e^{-m't}\big(\varphi_F(0) - \varphi_F(s)\big).$$

Step 3: all real times. $\varphi_F(t) = \langle [F], e^{-tH_S}[F]\rangle$ is continuous in t, the semigroup being strongly continuous; the numbers $pL^{-k'}$ are dense in $[0,\infty)$, and any two of them have a common denominator. So the inequality of Step 2 holds for all real $s, t \ge 0$; the classes $[F]$, $F \in \mathcal{D}$, are dense in $\mathcal{H}(S)$, and Lemma 18.184, (iii) implying (i), gives $\operatorname{spec} H_S \subset \{0\} \cup [m', \infty)$.

Step 4: the ceiling; (a). If $m > M^*_g$, a real m' with $M^*_g < m' < m$ gives, by Step 3, a spectrum of H_S that misses $(0, M^*_g]$, against Lemma 18.188(c). So $m \le M^*_g < \infty$. A point of $\operatorname{spec} H_S$ in $(0,m)$ would lie in $(0,m')$ for some $m' < m$; so $\operatorname{spec} H_S \subset \{0\} \cup [m,\infty)$, and Lemma 18.184 gives the display of (a).

Step 5: (b). The defining sequence is a convergent Gibbs sequence at g with the limit σ_S; Lemma 18.185(c) gives $\operatorname{spec} H_{\sigma_\xi} \subset \{0\} \cup [m,\infty)$ for every ξ. As $\ker H_{\sigma_\xi} = \mathbb{C}\Omega_{\sigma_\xi}$ by Theorem YM3_I:AP, the projection onto the kernel is $\psi \mapsto \langle \Omega_{\sigma_\xi}, \psi\rangle \Omega_{\sigma_\xi}$, and $\langle \Omega_{\sigma_\xi}, [F]_\xi\rangle = \sigma_\xi(\Theta_0 \varnothing \cdot F) = \sigma_\xi(F)$; Lemma 18.184, (i) implying (ii), is the display.

Step 6: (c), the hyperboloid. Fix ξ and write $\mathcal{P} = \mathcal{P}_{\sigma_\xi}$. By Theorems YM3_I:AP and YM3_I:AO, the component has a unitary representation U of the proper orthochronous Poincaré group. Its translation spectral measure E is supported in the closed forward cone, and is the image of $\mathcal{P}$ under $(E,q) \mapsto (E,-q)$ when translations are written with Minkowski pairing $x \cdot p = x^0 p^0 - \vec{x} \cdot \vec{p}$. (1) For $\Lambda \in \mathcal{L}$, $\mathsf{E}_\Lambda(\Delta) = U(0,\Lambda)^{-1}\mathsf{E}(\Lambda\Delta)U(0,\Lambda)$ is a projection-valued measure

with $\int e^{ix\cdot p}\,d\mathsf{E}_\Lambda(p) = U(0,\Lambda)^{-1}U(\Lambda x,1)U(0,\Lambda) = U(x,1)$, because Λ keeps the Minkowski product and $(0,\Lambda^{-1})(\Lambda x,1)(0,\Lambda) = (x,1)$; by the uniqueness, $\mathsf{E}_\Lambda = \mathsf{E}$, so $\mathsf{E}(\Lambda\Delta) = 0$ if and only if $\mathsf{E}(\Delta) = 0$. (2) Let $\Delta_0 = \{p : 0 < p^0 < m\}$, an open set. Since $H_{\sigma_\xi} = \int E\,d\mathcal{P}(E,q)$, the image of $\mathcal{P}$ under $(E,q)\mapsto E$ is the spectral measure of H_{σ_ξ}, which vanishes on $(0,m)$ by (b); so $\mathsf{E}(\Delta_0) = \mathcal{P}(\{0 < E < m\}) = 0$. (3) For a unit vector $\mathsf{n}\in\mathbb{R}^2$ and $\eta\in\mathbb{R}$ the boost $\Lambda_{\mathsf{n},\eta}$, $p^0\mapsto p^0\cosh\eta - (\mathsf{n}\cdot\vec p)\sinh\eta$, $\vec p\mapsto \vec p + [(\cosh\eta - 1)(\mathsf{n}\cdot\vec p) - p^0\sinh\eta]\,\mathsf{n}$, keeps $p\cdot p$ and is joined to 1 by $\eta'\mapsto\Lambda_{\mathsf{n},\eta'}$, so it lies in $\mathcal{L}$. Let $p = (E,\vec q)\in\overline{V_+}$, $p\neq 0$, $p\cdot p < m^2$; then $E > 0$. If $\vec q = 0$, $p\in\Delta_0$. If $\vec q\neq 0$ and $\mathsf{n} = \vec q/|\vec q|$, then $(\Lambda_{\mathsf{n},\eta}p)^0 = E\cosh\eta - |\vec q|\sinh\eta$, which equals $(E^2 - |\vec q|^2)^{1/2}\in(0,m)$ at $\tanh\eta = |\vec q|/E$ if $E > |\vec q|$, and equals $Ee^{-\eta}\in(0,m)$ for large η if $E = |\vec q|$. So the open sets $\Lambda^{-1}\Delta_0$, $\Lambda\in\mathcal{L}$, cover the set of these p; by Lindelöf's theorem countably many of them do; each has measure 0 by (1) and (2). (4) Hence E is carried by $\{0\}\cup\{p : p^0 > 0,\ p\cdot p\geq m^2\}$, a closed set that the map $(E,q)\mapsto(E,-q)$ carries onto H_m; so $\mathcal{P}$ is carried by the closed set H_m, and $\operatorname{supp}\mathcal{P}\subset\mathsf{H}_m$.

Step 7: (d), clustering. When $n = 0$ or $n' = 0$, one of the two tuples is empty, and $\sigma_\xi(\varnothing) = 1$, every component being a functional on separated tuples; then the left member of (d) vanishes, and the middle member, which is ≥ 0, does not exceed the right one, as $e^{-m(|v|-\mathfrak{r}(X,Y))} = e^{m\mathfrak{r}(X,Y)}e^{-m|v|}$ and $m\leq M_g^*$. Otherwise let $\mathfrak{r} = \max\{\mathfrak{r}(X),\mathfrak{r}(Y)\}$ and $|v|\geq 2\mathfrak{r}+1$; the supports of X and of $\tau_v Y$ lie in the closed balls of the radius $\mathfrak{r}$ about 0 and v, so $X\cdot\tau_v Y$ is defined. Choose $\mathsf{R}\in\mathrm{SO}(3)$ with $\mathsf{R}v = |v|e_3$, and put $\mathsf{A} = \Theta_0\tau_{-(\mathfrak{r}+\frac12)e_3}\mathsf{R}X$, $\mathsf{B} = \tau_{(\mathfrak{r}+\frac12)e_3}\mathsf{R}Y$, both with supports in $\{x_3\geq\frac12\}$, and $t = |v| - 2\mathfrak{r} - 1\geq 0$. By the invariance of σ_ξ under the isometries of $\mathbb{R}^3$ by Theorem YM3_I:AP, $\langle[\mathsf{A}]_\xi, e^{-tH_{\sigma_\xi}}[\mathsf{B}]_\xi\rangle = \sigma_\xi(\tau_{-(\mathfrak{r}+\frac12)e_3}\mathsf{R}X\cdot\tau_{(t+\mathfrak{r}+\frac12)e_3}\mathsf{R}Y) = \sigma_\xi(X\cdot\tau_v Y)$, $\langle[\mathsf{A}]_\xi,\Omega_{\sigma_\xi}\rangle = \sigma_\xi(X)$ and $\langle\Omega_{\sigma_\xi},[\mathsf{B}]_\xi\rangle = \sigma_\xi(Y)$. So, with the projection $P_{0,\xi}$ onto $\ker H_{\sigma_\xi} = \mathbb{C}\Omega_{\sigma_\xi}$, the left member of (d) is $|\langle[\mathsf{A}]_\xi,(e^{-tH_{\sigma_\xi}} - P_{0,\xi})[\mathsf{B}]_\xi\rangle|\leq e^{-mt}\|[\mathsf{A}]_\xi\|\,\|[\mathsf{B}]_\xi\|$, by (b) and Lemma 18.184. Further $\|[\mathsf{A}]_\xi\|^2 = \sigma_\xi(Z)$ with $Z = \Theta_0\mathsf{A}\cdot\mathsf{A}$, a tuple of $2n$ isometric images of the entries of X: two entries of the same half lie at a distance $\geq\mathsf{d}(X)$, two of different halves lie in $\{x_3\leq-\frac12\}$ and in $\{x_3\geq\frac12\}$, so Z and all its subtuples are of the type $(\mathsf{d}_X;R_X)$, and isometries keep $\|\cdot\|_\sharp$. Expanding $\sigma_\xi(Z)$ as the sum over all partitions of the products of the connected functions $S^{\sigma_\xi}_{|J|}(Z_J)$, and bounding $|S^{\sigma_\xi}_k(Z_J)|\leq C^g_k(\mathsf{d}_X;R_X)\prod\|z_i\|_\sharp$ for $k\geq 2$ and $|S^{\sigma_\xi}_1(z)| = |\mathfrak{c}_\xi\int z|\leq\bar{\mathfrak{c}}(\mathsf{C}^g)\frac{4\pi}{3}R_X^3\|z\|_\infty$ by Theorem YM3_I:AP and Proposition YM3_I:91.13 (higher cumulants are unchanged by centering and pass to the limit from the conditioned Gibbs states), we obtain $\|[\mathsf{A}]_\xi\|\leq\mathfrak{n}_g(X)$, and in the same way $\|[\mathsf{B}]_\xi\|\leq\mathfrak{n}_g(Y)$. As $e^{-mt} = e^{-m(|v|-\mathfrak{r}(X,Y))}$ and $m\leq M_g^*$, the two inequalities of (d) follow.

Step 8: (e). Let J be the infinite set of the j at which the vacuum of ν_j is simple in the action sector. Fix $F\in\mathcal{D}$, $m' < m$ and $t = pL^{-k'} > 0$. For large $j\in\mathsf{J}$, Step 1 gives $0\leq\Phi_j(n_j) - c_j\leq e^{-m't}\Phi_j(0)$, and (a) at m' gives $0\leq\varphi_F(t) - \|P_0[F]\|^2\leq e^{-m't}\varphi_F(0)$; with $\Phi_j(n_j)\to\varphi_F(t)$ and $\Phi_j(0)\to\varphi_F(0)$, $\limsup_{\mathsf{J}\ni j\to\infty}|c_j - \|P_0[F]\|^2|\leq 2e^{-m't}\varphi_F(0)$, and $t\to\infty$ gives $c_j\to\|P_0[F]\|^2$ along J. By Lemma 18.182(c) and Lemma 18.183(c), $c_j = |\nu_j(A_j)|^2 = |\sigma_{\nu_j}(F)|^2\to|\sigma_S(F)|^2 = |\langle\Omega_S,[F]\rangle|^2$. So the projection P_0 and the projection onto $\mathbb{C}\Omega_S$, which lies below it as $H_S\Omega_S = 0$, agree in norm on a dense set of vectors; their difference is a projection that vanishes there, so $\ker H_S = \mathbb{C}\Omega_S$. Theorem YM3_I:AP identifies $\ker H_S$ with $L^2(\mathsf{X},\lambda)$. Since λ has full support on the compact Hausdorff canonical space, one-dimensionality of this space forces $\mathsf{X} = \{\xi_0\}$. The decomposition then gives $\sigma_S = \sigma_{\xi_0}$. Then (b)–(d) for that component are (b)–(d) for σ_S, and the last display is (b) for it.

Step 9: (f). If the upper limit is positive, a subsequence along which $m_{\mathrm{lat}}(\nu_j)/a_j$ tends to it is again a defining sequence of states of S, with the same σ_S and the same decompositions; along it the lower limit is the upper limit of the whole sequence, and (a)–(d) apply. □

Proposition 18.190 (The reduction with the transfer operators on any sector)**.** In the setting of Proposition 18.189, let $\mathcal{K}_j$ be a sector of ν_j for every j (Definition 18.181(2)), for instance the

gauge-invariant sector or the whole space $\mathcal{H}_{\nu_j}$, and suppose that

$$m_{\mathcal{K}} = \liminf_{j\to\infty} \frac{m_{\mathcal{K}_j}(\nu_j)}{a_j} > 0.$$

Then $m_{\mathcal{K}} \le M_g^*$; $\operatorname{spec} H_S \subset \{0\} \cup [m_{\mathcal{K}}, \infty)$, and the inequality of Proposition 18.189(a) remains true when $m_{\mathcal{K}}$ replaces m; for every canonical decomposition of σ_S and every component σ_ξ, $\operatorname{spec} H_{\sigma_\xi} \subset \{0\} \cup [m_{\mathcal{K}}, \infty)$, the inequality of Proposition 18.189(b) remains true when $m_{\mathcal{K}}$ replaces m, the joint spectrum of $(H_{\sigma_\xi}, P_{1,\sigma_\xi}, P_{2,\sigma_\xi})$ is contained in $\mathsf{H}_{m_{\mathcal{K}}} = \{(0,0)\} \cup \{(E,q) : E > 0,\ E^2 - |q|^2 \ge m_{\mathcal{K}}^2\}$, and

$$|\sigma_\xi(X \cdot \tau_v Y) - \sigma_\xi(X)\,\sigma_\xi(Y)| \ \le\ \mathfrak{C}_g(X,Y)\, e^{-m_{\mathcal{K}}|v|} \qquad (X, Y \in \mathfrak{S}(\mathbb{R}^3),\ |v| \ge \mathfrak{r}(X,Y)),$$

with the number $\mathfrak{C}_g(X,Y)$ of Definition 18.181(6), which does not depend on ξ. If moreover the set of the indices j at which the vacuum of ν_j is simple in $\mathcal{K}_j$ is infinite, then $\ker H_S = \mathbb{C}\Omega_S$, σ_S is the only component of its canonical decompositions, and the joint spectrum of $(H_S, P_{1,S}, P_{2,S})$ lies in $\mathsf{H}_{m_{\mathcal{K}}}$. ■

Sketch. By Lemma 18.182(d), $m_{\mathrm{lat}}(\nu_j) \ge m_{\mathcal{K}_j}(\nu_j)$ for every j, so the number m of Proposition 18.189 satisfies $m \ge m_{\mathcal{K}} > 0$, and that proposition applies. Its conclusions at m give those at $m_{\mathcal{K}}$: $\{0\} \cup [m, \infty) \subset \{0\} \cup [m_{\mathcal{K}}, \infty)$; $e^{-mt} \le e^{-m_{\mathcal{K}} t}$ for $t \ge 0$, while the brackets on the right of the inequalities of Proposition 18.189(a), (b) are ≥ 0, each being $\|\psi\|^2 - \|P\psi\|^2$ for a vector ψ and an orthogonal projection P; $e^{-m|v|} \le e^{-m_{\mathcal{K}}|v|}$; $\mathsf{H}_m \subset \mathsf{H}_{m_{\mathcal{K}}}$, which with Proposition 18.189(c) places the joint spectrum of every component in $\mathsf{H}_{m_{\mathcal{K}}}$; and $m_{\mathcal{K}} \le m \le M_g^*$. By Lemma 18.182(d) again, a vacuum that is simple in $\mathcal{K}_j$ is simple in the action sector, which is contained in $\mathcal{K}_j$; so Proposition 18.189(e) gives $\ker H_S = \mathbb{C}\Omega_S$ and makes σ_S the one component of every canonical decomposition, and Proposition 18.189(c), read for this component, places the joint spectrum of $(H_S, P_{1,S}, P_{2,S})$ in $\mathsf{H}_m \subset \mathsf{H}_{m_{\mathcal{K}}}$. □

18.20. **Clustering in space from the Poincaré constant of the lattice Langevin generator.** At a fixed cutoff, the Poincaré constant $\mathsf{c_P}$ for the gauge-invariant functions gives a bound on the spatial correlations of functions of the bonds. Proposition 18.194 gives a decay length $e\sqrt{6\mathsf{b}\mathsf{c_P}} + a$, where a is the lattice spacing and $\mathsf{b} = a\beta$. The proof uses a weighted identity for the Poisson equation and iterates a conditional expectation estimate across successive shells. For $\beta \ge 25$ a single plaquette gives the floor $\mathsf{c_P} \ge a^2/(48\mathsf{b})$, which bounds the decay length by $\sqrt{185\mathsf{b}\mathsf{c_P}}$.

At a reference coupling, site reflection positivity and the truncated-function bounds of Proposition YM3_I:91.13 allow the prefactor for the smeared action density to be chosen independently of the cutoff and side. The resulting bounds pass first to torus-limit states and then to volume-first limits whenever the corresponding lower limits of $\mathsf{b}\mathsf{c_P}$ are finite (Proposition 18.196). Under a uniform bound on $\mathsf{c_P}$ on one torus, the limiting block observables decorrelate in space and have exponential bounds on their Langevin time correlations (Remark 18.197).

Let $L \ge 3$ be odd, $K, m \in \mathbb{N}$, $\ell = 2L^m$ and $\beta > 0$. Set $a = a_K = L^{-K}$, $\mathsf{b} = a\beta$ and $\Lambda_d = L^{-d}\mathbb{Z}^3/\ell\mathbb{Z}^3$ for $0 \le d \le K$. A configuration in Ω_K assigns an SU(2) matrix to each positive bond and its inverse to the reversed bond. Write $a_p = 1 - \frac{1}{2}\operatorname{tr} U(\partial p)$, $A_K = \sum_p a_p$ and $\mu_K = Z_K^{-1} e^{-\beta A_K}\, dU$ for product Haar dU. The block variables $\pi_d U = \mathcal{M}^{K-d} U$ use the map in Definition 2.7(3) and Proposition YM3_I:4.5. Site gauges act by left multiplication at the initial endpoint and right inverse multiplication at the final endpoint. Write Π for their normalized Haar average.

For $I = (b, i)$, $i = 1, 2, 3$, let ∂_I differentiate the link $U(b)$ by left multiplication by $e^{it\varsigma_i}$, with the Pauli matrices ς_i. The generator and energy are

$$\mathcal{L}_K = -a^{-1} \sum_I \partial_I^* \partial_I, \qquad \mathcal{E}_K(F,G) = a^{-1} \sum_I (\partial_I F, \partial_I G).$$

Here the adjoint, inner product and norm are in $L^2(\mu_K)$. The square field is $\Gamma_K(F,G) = a^{-1}\sum_I \partial_I F\,\partial_I G$. Observables in this subsection are real-valued. Means, variances, covariances and conditional expectations are taken with respect to μ_K. We write $J=(b',j)$ for a second derivative index.

Definition 18.191 (Carriers, distances, neighbours, Poincaré constants, site mirrors)**.** (1) *Carriers.* For a set $\mathcal{Y}$ of bonds, $\mathcal{F}_{\mathcal{Y}}$ is the σ-algebra generated by the maps $U\mapsto U(b)$, $b\in\mathcal{Y}$, and $\mathcal{Y}^c$ is the set of the bonds outside $\mathcal{Y}$. A Borel function on Ω_K is carried by $\mathcal{Y}$ if it is $\mathcal{F}_{\mathcal{Y}}$-measurable, that is, if its value depends on the values of the field at the bonds of $\mathcal{Y}$ alone; an element of $L^2(\mu_K)$ is carried by $\mathcal{Y}$ if one of its representatives is.

(2) *Distances.* $x_b\in T_\ell$ is the midpoint of the bond b, and $|x-y|$ is the Euclidean distance of T_ℓ, the least Euclidean distance of two lifts. For a non-empty set $\mathcal{Y}$ of bonds, $d_{\mathcal{Y}}(b)=\min_{b'\in\mathcal{Y}}|x_b-x_{b'}|$, and $\mathcal{Y}^{[t]}=\{b:\ d_{\mathcal{Y}}(b)\ge t\}$ for $t\ge 0$; for two non-empty sets, $\operatorname{dist}(\mathcal{Y},\mathcal{Y}')=\min\{|x_b-x_{b'}|:\ b\in\mathcal{Y},\ b'\in\mathcal{Y}'\}$.

(3) *Neighbours.* Two bonds $b\neq b'$ are neighbours, $b\sim b'$, if one plaquette has both among its four bonds; sums over $b\sim b'$ run over ordered pairs. For a set $\mathcal{Y}$ of bonds, $\partial^{\mathrm{in}}\mathcal{Y}$ is the set of the bonds of $\mathcal{Y}$ that have a neighbour outside $\mathcal{Y}$.

(4) *Poincaré constants.* $\mathsf{c}_{\mathrm{P}}=\mathsf{c}_{\mathrm{P}}(K,\ell,\beta)$ is the least number C with $\operatorname{Var}(F)\le C\,\mathcal{E}_K(F)$ for every real gauge-invariant $F\in C^\infty(\Omega_K)$, and $\mathsf{c}_{\mathrm{P}}^{\mathrm{all}}=\mathsf{c}_{\mathrm{P}}^{\mathrm{all}}(K,\ell,\beta)$ is the least number C with the same inequality for every real $F\in C^\infty(\Omega_K)$. Then $0<\mathsf{c}_{\mathrm{P}}\le\mathsf{c}_{\mathrm{P}}^{\mathrm{all}}<\infty$: the elliptic operator $-\mathcal{L}_K$, symmetric for μ_K, has a discrete spectrum on the compact connected manifold Ω_K, its kernel consists of the constants, and non-constant smooth gauge-invariant functions exist. Of the size of these constants nothing is assumed. For a fixed side ℓ and a sequence β_K, a uniform C_{P} is a positive number that bounds $\mathsf{c}_{\mathrm{P}}(K,\ell,\beta_K)$ for all large K.

(5) *Site mirrors.* Let $\mu\in\{1,2,3\}$ and $z\in a\mathbb{Z}$. The reflection $\mathsf{r}_z\colon x\mapsto x-2(x_\mu-z)e_\mu$ of $\mathbb{R}^3$ induces a map of T_ℓ, again written r_z, that maps sites to sites and bonds to bonds of either orientation; it acts on fields by $(\mathsf{r}_zU)(b)=U(\mathsf{r}_zb)$, on functions on Ω_K by $(\theta_zF)(U)=F(\mathsf{r}_zU)$, and on functions on $\mathbb{R}^3$ by $\mathsf{r}_zf=f\circ\mathsf{r}_z$. As ℓ/a is even, $z+\frac{\ell}{2}\in a\mathbb{Z}$. Σ_z is the set of the bonds with both end points in $\{x_\mu=z\}\cup\{x_\mu=z+\frac{\ell}{2}\}$, and H_z^+ (H_z^-) is the set of the bonds with an end point in the open slab $\{z<x_\mu<z+\frac{\ell}{2}\}$ ($\{z-\frac{\ell}{2}<x_\mu<z\}$), the coordinate being taken modulo ℓ.

■

Lemma 18.192 (Commutators, neighbours, and two identities for the generator)**.** Let $L\ge 3$ be odd, $K,m\in\mathbb{N}$ and $\beta>0$. For $I=(b,i)$ let $\partial_I^*=-\partial_I+\beta(\partial_IA_K)$; then $(\partial_IF,G)=(F,\partial_I^*G)$ for $F,G\in C^1(\Omega_K)$, and $\mathcal{L}_K=-a^{-1}\sum_I\partial_I^*\partial_I$ on $C^2(\Omega_K)$.

(a) $[\partial_b^i,\partial_{b'}^j]=0$ for $b\neq b'$, and $[\partial_b^i,\partial_b^j]=2\sum_k\epsilon_{ijk}\partial_b^k$. Every bond has exactly twelve neighbours, and $|x_b-x_{b'}|\le a$ if $b\sim b'$. If $b\neq b'$ and $b\not\sim b'$, then $\partial_b^i\partial_{b'}^jA_K=0$; if $b\sim b'$, then $|\sum_{i,j}y_iy'_j\,\partial_b^i\partial_{b'}^jA_K(U)|\le|y|\,|y'|$ for all $U\in\Omega_K$ and $y,y'\in\mathbb{R}^3$.

(b) *The square.* Let $f\in C^\infty(\Omega_K)$ be real and λ a real function on the bonds; put $\mathsf{v}_I=\partial_If$, $\delta_\lambda f=\sum_I\lambda_b\,\partial_I^*\mathsf{v}_I$ and, for $I=(b,i)$ and $J=(b',j)$, $\mathsf{q}_{IJ}(f)=\|\partial_J\mathsf{v}_I\|^2+\beta\,\langle\mathsf{v}_I\,(\partial_I\partial_JA_K)\,\mathsf{v}_J\rangle$. Then $\|\delta_\lambda f\|^2=\sum_{I,J}\lambda_b\lambda_{b'}\,\mathsf{q}_{IJ}(f)$.

(c) *The identity with the generator.* For f and λ as in (b),

$$a\sum_I\lambda_b^2\,(\mathsf{v}_I,\partial_I\mathcal{L}_Kf)=-\|\delta_\lambda f\|^2-\frac12\sum_{I,J:\ b\neq b'}(\lambda_b-\lambda_{b'})^2\,\mathsf{q}_{IJ}(f);$$

hence, with $\|\mathsf{v}_b\|^2=\sum_i\|\mathsf{v}_{(b,i)}\|^2$,

$$\|\delta_\lambda f\|^2+a\sum_I\lambda_b^2\,(\mathsf{v}_I,\partial_I\mathcal{L}_Kf)\le\frac{\beta}{2}\sum_{b\sim b'}(\lambda_b-\lambda_{b'})^2\,\|\mathsf{v}_b\|\,\|\mathsf{v}_{b'}\|.$$

■

Proof. Each ∂_I is skew-adjoint for product Haar measure by left invariance. Differentiating the density $e^{-\beta A_K}$ gives $\partial_I^* = -\partial_I + \beta \partial_I A_K$ in $L^2(\mu_K)$. The generator identity follows from its definition.

(a) Derivatives at different bonds act on different factors of Ω_K. At one bond, $\partial_b^i \partial_b^j F(U)$ is the mixed derivative at $s = t = 0$ of F at the field with $e^{it\varsigma_j} e^{is\varsigma_i} U(b)$ at b, and the commutator of two derivatives along left multiplication is the derivative along left multiplication by the one-parameter group of $-[i\varsigma_i, i\varsigma_j] = [\varsigma_i, \varsigma_j] = 2i \sum_k \epsilon_{ijk} \varsigma_k$; below only the antisymmetry of ϵ_{ijk} in its three indices is used. A bond lies in exactly four plaquettes, two different plaquettes have at most one bond in common, and $\ell/a \geq 18$; so the twelve further bonds of these four plaquettes are distinct, and two distinct bonds lie in at most one common plaquette. Two parallel sides of a plaquette have midpoints at the distance a, two adjacent sides at the distance $a/\sqrt{2}$. As a_p depends on the four bonds of p alone, $\partial_b^i \partial_{b'}^j A_K$ with $b \neq b'$ vanishes unless one plaquette p holds both bonds, and then it is $\partial_b^i \partial_{b'}^j a_p$. The left multiplication of $U(b)$ by $e^{is\, y\cdot\varsigma}$ inserts $e^{is\, y\cdot\varsigma}$ in front of $U(b)$ in $\partial U(p)$, or $e^{-is\, y\cdot\varsigma}$ behind $U(b)^{-1}$; by the cyclicity of the trace, $\frac{1}{2}\mathrm{tr}\, \partial U(p)$ at the field varied at b and at b' is $\frac{1}{2}\mathrm{tr}(e^{\pm is\, y\cdot\varsigma} V_1 e^{\pm it\, y'\cdot\varsigma} V_2)$ with $V_1, V_2 \in \mathrm{SU}(2)$ that do not depend on s, t, so that $\sum_{i,j} y_i y'_j\, \partial_b^i \partial_{b'}^j a_p = \pm\frac{1}{2}\mathrm{tr}((y\cdot\varsigma) V_1 (y'\cdot\varsigma) V_2)$. For unit vectors y, y' the matrix in this trace is unitary, and the trace of a unitary 2×2 matrix has modulus at most 2.

(b) $\|\delta_\lambda f\|^2 = \sum_{I,J} \lambda_b \lambda_{b'} (\mathsf{v}_I, \partial_I \partial_J^* \mathsf{v}_J)$, and on smooth functions $\partial_I \partial_J^* = \partial_J^* \partial_I - [\partial_I, \partial_J] + \beta(\partial_I \partial_J A_K)$. For $b \neq b'$ the commutator vanishes and $\partial_I \mathsf{v}_J = \partial_J \mathsf{v}_I$, so that $(\mathsf{v}_I, \partial_I \partial_J^* \mathsf{v}_J) = \mathsf{q}_{IJ}(f)$. At one bond, with the indices i, j, k, (a) gives $\partial_i \mathsf{v}_j = \partial_j \mathsf{v}_i + 2\sum_k \epsilon_{ijk} \mathsf{v}_k$; summed over i and j, the term $2\sum \epsilon_{ijk} (\partial_j \mathsf{v}_i, \mathsf{v}_k)$ that this produces in $(\partial_j \mathsf{v}_i, \partial_i \mathsf{v}_j)$ and the term $-2\sum \epsilon_{ijk} (\mathsf{v}_i, \partial_k \mathsf{v}_j)$ of the commutator cancel after the cyclic renaming $(i, j, k) \to (k, i, j)$ of the second, and what remains is $\sum_{i,j} \mathsf{q}_{(b,i)(b,j)}(f)$.

(c) By the form of $\mathcal{L}_K$, $a\, \partial_I \mathcal{L}_K f = -\sum_J \partial_I \partial_J^* \mathsf{v}_J$, so that the left member of the identity is $-\sum_{I,J} \lambda_b^2 (\mathsf{v}_I, \partial_I \partial_J^* \mathsf{v}_J)$, which by the proof of (b) is $-\sum_{I,J} \lambda_b^2 \mathsf{q}_{IJ}(f)$, the terms with $b = b'$ being summed over i, j first. For $b = b'$, $\lambda_b^2 = \lambda_b \lambda_{b'}$; for $b \neq b'$, $\mathsf{q}_{IJ} = \mathsf{q}_{JI}$, so that λ_b^2 may be replaced by $\frac{1}{2}(\lambda_b^2 + \lambda_{b'}^2) = \lambda_b \lambda_{b'} + \frac{1}{2}(\lambda_b - \lambda_{b'})^2$; together with (b) this is the identity. For the inequality, discard the nonnegative terms $\|\partial_J \mathsf{v}_I\|^2$. By (a), the remaining terms with $b \neq b'$ and $b \not\sim b'$ vanish, and for $b \sim b'$, at every field, $|\sum_{i,j} \mathsf{v}_{(b,i)} (\partial_b^i \partial_{b'}^j A_K) \mathsf{v}_{(b',j)}| \leq |\mathsf{v}_b|\, |\mathsf{v}_{b'}|$ with $|\mathsf{v}_b|^2 = \sum_i \mathsf{v}_{(b,i)}^2$, a product whose mean is at most $\|\mathsf{v}_b\|\, \|\mathsf{v}_{b'}\|$. □

Lemma 18.193 (The Markov property of Wilson's measure)**.** Let $L \geq 3$ be odd, $K, m \in \mathbb{N}$ and $\beta > 0$.

(a) Let $\mathcal{Y}$ be a set of bonds and $F \in L^2(\mu_K)$ carried by $\mathcal{Y}^c$. Then $E[F|\mathcal{F}_\mathcal{Y}]$ is carried by $\partial^{\mathrm{in}}\mathcal{Y}$, and it is gauge invariant if F is.

(b) Let μ be a direction and $z \in a\mathbb{Z}$. The bonds form the disjoint union of Σ_z, H_z^+ and H_z^-; no plaquette has a bond in H_z^+ and a bond in H_z^-; r_z fixes every bond of Σ_z, and for each $b \in H_z^-$ exactly one of $\mathsf{r}_z b$ and its reverse lies in H_z^+; the map that takes b to this bond is a bijection of H_z^- onto H_z^+. If $F \in L^2(\mu_K)$ is carried by $H_z^- \cup \Sigma_z$ and $G \in L^2(\mu_K)$ by $H_z^+ \cup \Sigma_z$, then $(F, G) = (E[F|\mathcal{F}_{\Sigma_z}], E[G|\mathcal{F}_{\Sigma_z}])$; and $E[\theta_z F|\mathcal{F}_{\Sigma_z}] = E[F|\mathcal{F}_{\Sigma_z}]$ for every $F \in L^2(\mu_K)$. Consequently $(F, F') \mapsto \mathrm{Cov}(F, \theta_z F') = \mathrm{Cov}(E[F|\mathcal{F}_{\Sigma_z}], E[F'|\mathcal{F}_{\Sigma_z}])$ is a symmetric non-negative bilinear form on the elements of $L^2(\mu_K)$ carried by $H_z^- \cup \Sigma_z$, and likewise on those carried by $H_z^+ \cup \Sigma_z$; $\mathrm{Cov}(F, \theta_z F) = \mathrm{Var}(E[F|\mathcal{F}_{\Sigma_z}]) \geq 0$; and $|\mathrm{Cov}(F, G)|^2 \leq \mathrm{Cov}(F, \theta_z F)\, \mathrm{Cov}(G, \theta_z G)$ for F carried by $H_z^- \cup \Sigma_z$ and G carried by $H_z^+ \cup \Sigma_z$.

■

Proof. (a) A plaquette with a bond outside $\mathcal{Y}$ has each of its four bonds outside $\mathcal{Y}$ or in $\partial^{\mathrm{in}}\mathcal{Y}$. Write $A_K = A' + A''$, where A' is the sum of the a_p over these plaquettes and A'', the sum over the others,

is carried by $\mathcal{Y}$. As dU is a product of Haar measures, for bounded F carried by $\mathcal{Y}^c$

$$E[F|\mathcal{F}_{\mathcal{Y}}] = \frac{\int F\, e^{-\beta A'}\, dU_{\mathcal{Y}^c}}{\int e^{-\beta A'}\, dU_{\mathcal{Y}^c}},$$

the integrals being taken over the variables of the bonds outside $\mathcal{Y}$; the quotient depends on the field only through its values at the bonds of $\partial^{\mathrm{in}}\mathcal{Y}$. A gauge transformation multiplies every variable $U(b)$, $b \notin \mathcal{Y}$, from the left and from the right by elements that do not depend on the field, which preserves the Haar measure, and A' is gauge invariant; so the quotient is gauge invariant if F is. The bounded functions are dense in $L^2(\mu_K)$, and a conditional mean is a contraction.

(b) The μ-th coordinates of the end points of a bond, and of the corners of a plaquette, form one element or two consecutive elements of $a\mathbb{Z}$ modulo ℓ; as z and $z+\frac{\ell}{2}$ lie in $a\mathbb{Z}$, a bond lies in exactly one of the three sets, and no plaquette has a bond in each of the two slabs. The reflection fixes every point of the two planes, hence every bond of Σ_z, and it exchanges the two open slabs; it carries bonds to bonds and commutes with their reversal, the positively oriented bonds along e_μ going to negatively oriented ones and the other positively oriented bonds to positively oriented ones; as $\mathsf{r}_z^{-1} = \mathsf{r}_z$, this gives the bijection of H_z^- onto H_z^+. So $A_K = A_\Sigma + A_+ + A_-$, where $A_\pm$ is the sum of the a_p over the plaquettes with a bond in $H_z^\pm$, a function of the variables of $H_z^\pm \cup \Sigma_z$, and A_Σ is carried by Σ_z; under μ_K the variables of H_z^+ and those of H_z^- are therefore independent given $\mathcal{F}_{\Sigma_z}$, which is the first identity. For bounded Ψ carried by Σ_z we have $\theta_z\Psi = \Psi$, and μ_K is invariant under r_z, which permutes unoriented bonds and plaquettes. Product Haar measure is invariant under permutations and inversion of link variables, and the trace of each plaquette is unchanged by orientation reversal. Hence $\langle(\theta_z F)\Psi\rangle = \langle\theta_z(F\Psi)\rangle = \langle F\Psi\rangle$, which is the second identity. For the rest: as $(\mathsf{r}_z U)(b) = U(\mathsf{r}_z b)$ and $U(\bar b) = U(b)^{-1}$, $\theta_z F'$ is carried by $H_z^+ \cup \Sigma_z$ if F' is carried by $H_z^- \cup \Sigma_z$, and $\langle\theta_z F'\rangle = \langle F'\rangle$; the last inequality is Schwarz's inequality for $\mathrm{Cov}(F,G) = \mathrm{Cov}(E[F|\mathcal{F}_{\Sigma_z}], E[G|\mathcal{F}_{\Sigma_z}])$. □

Proposition 18.194 (Clustering at the length $\sqrt{\mathsf{bc_P}}$ on every torus and at every cutoff)**.** Let $L \geq 3$ be odd, $K, m \in \mathbb{N}$ and $\beta > 0$; $a = a_K$, $\mathsf{b} = a\beta$ and $\mathsf{c_P} = \mathsf{c_P}(K,\ell,\beta)$.

(a) *An inequality of Caccioppoli type for the Poisson equation.* Let $F \in C^\infty(\Omega_K)$ be real and carried by a non-empty set $\mathcal{Y}$ of bonds, gauge invariant or not, and let $u \in C^\infty(\Omega_K)$ be the solution of $-\mathcal{L}_K u = F - \langle F\rangle$ with $\langle u\rangle = 0$, which exists and is unique. Let $w_0 \geq 0$, $w > 0$ and $\mathrm{Sh} = \{b :\ w_0 - a < d_{\mathcal{Y}}(b) < w_0 + w + a\}$. Then

$$\big\|E[F\,|\,\mathcal{F}_{\mathcal{Y}^{[w_0+w]}}] - \langle F\rangle\big\|^2 \leq \frac{6\,\mathsf{b}}{w^2}\, a^{-1} \sum_{b\in\mathrm{Sh}} \sum_{k=1}^{3} \|\partial_b^k u\|^2.$$

(b) *The conditional mean at the distance w.* Let $F \in L^2(\mu_K)$ be gauge invariant and carried by $\mathcal{Y}$, and let $w > 0$. Then $\|E[F|\mathcal{F}_{\mathcal{Y}^{[w]}}] - \langle F\rangle\| \leq (6\,\mathsf{bc_P})^{1/2} w^{-1} \mathrm{Var}(F)^{1/2}$; equivalently, $|\mathrm{Cov}(F,G)| \leq (6\,\mathsf{bc_P})^{1/2} w^{-1} (\mathrm{Var}F\,\mathrm{Var}G)^{1/2}$ for every $G \in L^2(\mu_K)$ carried by $\mathcal{Y}^{[w]}$. In particular, every such pair F, G with $\mathrm{Var}F, \mathrm{Var}G > 0$ and the correlation coefficient $\mathrm{corr}(F,G)$ gives the floor $\mathsf{c_P} \geq \mathrm{corr}(F,G)^2 w^2/(6\,\mathsf{b})$.

(c) *Clustering.* Let $F, G \in L^2(\mu_K)$ be carried by $\mathcal{Y}$ and by $\mathcal{Y}'$, with $r = \mathrm{dist}(\mathcal{Y},\mathcal{Y}') > 0$, and let one of the two be gauge invariant. Then

$$|\mathrm{Cov}(F,G)| \leq e\,(\mathrm{Var}F\,\mathrm{Var}G)^{1/2} \exp\Big(-\frac{r}{e\sqrt{6\,\mathsf{bc_P}} + a}\Big), \qquad e\sqrt{6} = 6.658\ldots$$

(d) *Functions without gauge invariance.* (b) holds for every $F \in L^2(\mu_K)$ carried by $\mathcal{Y}$, and (c) for every pair F, G, with $\mathsf{c_P^{all}}$ in the place of $\mathsf{c_P}$.

(e) *Block fields.* Let $0 \leq d \leq K$ and let $\mathcal{Q}$ be a non-empty set of positively oriented bonds of Λ_d. Let $\mathcal{Y}_{\mathcal{Q}}$ be the set of the bonds of Λ_K with both end points in $B^{K-d}(c_-) \cup B^{K-d}(c_+)$ for some $c \in \mathcal{Q}$, where $B^{K-d}(y) = y + a\{-q,\ldots,q\}^3$ in T_ℓ is the block about $y \in \Lambda_d$, with $q = (L^{K-d}-1)/2$, and let $\Delta(\mathcal{Q}) \subset T_\ell$ be the union of the closed cubes $\{x :\ |x-y|_\infty \leq \frac{1}{2}a_d\}$

over the end points y of the bonds of $\mathcal{Q}$. Every bounded Borel function φ on Ω_d that depends on $(V(c))_{c\in\mathcal{Q}}$ alone gives a bounded Borel function $\varphi\circ\pi_d$ on Ω_K, carried by $\mathcal{Y}_\mathcal{Q}$ and gauge invariant if φ is; and the midpoints of the bonds of $\mathcal{Y}_\mathcal{Q}$ lie in $\Delta(\mathcal{Q})$. Hence, if $0\le d'\le K$ and $\mathcal{Q}'$, ψ are of the same kind at the depth d', if φ or ψ is gauge invariant, and if $r_\mathcal{Q}=\operatorname{dist}(\mathcal{Y}_\mathcal{Q},\mathcal{Y}_{\mathcal{Q}'})$,

$$|\operatorname{Cov}(\varphi\circ\pi_d,\psi\circ\pi_{d'})|\le e\,\|\varphi\|_\infty\|\psi\|_\infty\exp\Bigl(-\frac{r_\mathcal{Q}}{e\sqrt{6\,\mathsf{bc}_\mathrm{P}}+a}\Bigr),\qquad r_\mathcal{Q}\ge\operatorname{dist}(\Delta(\mathcal{Q}),\Delta(\mathcal{Q}')),$$

the distance of two subsets of T_ℓ being the infimum of the Euclidean distances of their points; at $r_\mathcal{Q}=0$ the bound is Schwarz's inequality.

■

Proof. Step 1: the Poisson equation. The vector fields ∂_b^k span the tangent space of Ω_K at every point, and the coefficients $\partial_b^k A_K$ are smooth; so $\mathcal{L}_K$ is an elliptic operator with smooth coefficients on the compact connected manifold Ω_K, symmetric for μ_K, whose density for the Haar measure is smooth and positive: $(G,\mathcal{L}_K F)=-\mathcal{E}_K(F,G)$ for $F\in C^2$, $G\in C^1$. By the theory of such operators (Fredholm alternative and elliptic regularity), $\mathcal{L}_K$ maps $C^\infty(\Omega_K)$ onto the smooth functions that are orthogonal in $L^2(\mu_K)$ to its kernel; a smooth h with $\mathcal{L}_K h=0$ has $\mathcal{E}_K(h)=0$, hence $\partial_b^k h=0$ for all b,k, and it is constant, as Ω_K is connected. So for real smooth F there is exactly one real $u\in C^\infty(\Omega_K)$ with $\langle u\rangle=0$ and $-\mathcal{L}_K u=F-\langle F\rangle$. If F is gauge invariant, so is u: the gauges preserve μ_K and the energy because the sum of squares on each bond is bi-invariant. They therefore commute with $\mathcal{L}_K$, so that $U\mapsto u(U^v)$ solves the same problem for every $v\in\mathcal{G}_K$. Then $\mathcal{E}_K(u)=(u,F-\langle F\rangle)\le\|u\|\operatorname{Var}(F)^{1/2}$ and $\|u\|^2=\operatorname{Var}(u)\le\mathsf{c}_\mathrm{P}\,\mathcal{E}_K(u)$, whence $\mathcal{E}_K(u)\le\mathsf{c}_\mathrm{P}\operatorname{Var}(F)$; without gauge invariance the same holds with $\mathsf{c}_\mathrm{P}^{\mathrm{all}}$.

Step 2: clause (a). Put $\lambda_b=\min\{1,(d_\mathcal{Y}(b)-w_0)_+/w\}$. Then $\lambda=0$ on $\mathcal{Y}$ and $\lambda=1$ on $\mathcal{Y}^{[w_0+w]}$; for $b\sim b'$, $|d_\mathcal{Y}(b)-d_\mathcal{Y}(b')|\le|x_b-x_{b'}|\le a$ (Lemma 18.192(a)), so that $(\lambda_b-\lambda_{b'})^2\le a^2/w^2$, and $\lambda_b\ne\lambda_{b'}$ only if the interval between $d_\mathcal{Y}(b)$ and $d_\mathcal{Y}(b')$ meets (w_0,w_0+w), which puts both bonds in Sh. We apply Lemma 18.192(c) at $f=u$, with $\mathsf{v}_I=\partial_I u$. Its second term on the left is $-a\sum_I\lambda_b^2(\mathsf{v}_I,\partial_I F)=0$, because $\partial_I F=0$ for $b\notin\mathcal{Y}$ and $\lambda_b=0$ for $b\in\mathcal{Y}$. With $\|\mathsf{v}_b\|\,\|\mathsf{v}_{b'}\|\le\frac12(\|\mathsf{v}_b\|^2+\|\mathsf{v}_{b'}\|^2)$ and the twelve neighbours of every bond,

$$\|\delta_\lambda u\|^2\le\frac{\beta}{2}\cdot\frac{a^2}{w^2}\sum_{b\sim b',\ b,b'\in\mathrm{Sh}}\|\mathsf{v}_b\|\,\|\mathsf{v}_{b'}\|\le\frac{6\,\beta a^2}{w^2}\sum_{b\in\mathrm{Sh}}\|\mathsf{v}_b\|^2.$$

Let $F'=a^{-1}\delta_\lambda u$; as $\beta=\mathsf{b}/a$, $\|F'\|^2\le 6\,\mathsf{b}w^{-2}a^{-1}\sum_{b\in\mathrm{Sh}}\|\mathsf{v}_b\|^2$. For a real smooth Ψ carried by $\mathcal{Y}^{[w_0+w]}$, $\partial_I\Psi=0$ unless $\lambda_b=1$; so the integration by parts of Lemma 18.192 gives $(F',\Psi)=a^{-1}\sum_I\lambda_b(\partial_I u,\partial_I\Psi)=\mathcal{E}_K(u,\Psi)=(F-\langle F\rangle,\Psi)$. The real smooth functions carried by a set of bonds are dense in the elements of $L^2(\mu_K)$ carried by it: the density of μ_K for the Haar measure lies between two positive numbers, the continuous functions are dense in L^2 of the Haar measure of a product of copies of SU(2), and convolution with a smooth approximate identity on that finite product approximates continuous functions uniformly by smooth functions of the same variables. Hence $E[F|\mathcal{F}_{\mathcal{Y}^{[w_0+w]}}]-\langle F\rangle=E[F'|\mathcal{F}_{\mathcal{Y}^{[w_0+w]}}]$, whose norm is at most $\|F'\|$.

Step 3: clause (b). For smooth gauge-invariant F, (a) at $w_0=0$ and Step 1 give the first inequality, as $a^{-1}\sum_{b\in\mathrm{Sh}}\sum_k\|\partial_b^k u\|^2\le\mathcal{E}_K(u)\le\mathsf{c}_\mathrm{P}\operatorname{Var}(F)$. A gauge-invariant $F\in L^2(\mu_K)$ carried by $\mathcal{Y}$ is the limit in $L^2(\mu_K)$ of the functions ΠF_n, where the F_n are real smooth functions carried by $\mathcal{Y}$ that converge to F (Step 2): Π is the orthogonal projection onto the gauge-invariant functions by Haar invariance and the invariance of μ_K, and ΠF_n is smooth and carried by $\mathcal{Y}$, since a gauge transformation acts on every bond variable separately. Both members of the inequality are continuous on $L^2(\mu_K)$. The second form is Schwarz's inequality for $\operatorname{Cov}(F,G)=(E[F|\mathcal{F}_{\mathcal{Y}^{[w]}}]-\langle F\rangle,G-\langle G\rangle)$, and it gives back the first at $G=E[F|\mathcal{F}_{\mathcal{Y}^{[w]}}]$; the floor of c_P is the second form solved for c_P.

Step 4: clause (c). Let F be gauge invariant (otherwise exchange F and G), let $w = e\sqrt{6\,\mathsf{bc_P}}$, let $\mathcal{T}_0$ be the set of all bonds and $\mathcal{T}_k = \mathcal{Y}^{[k(w+a)]}$ for $k \geq 1$, and $F_k = E[F|\mathcal{F}_{\mathcal{T}_k}] - \langle F\rangle$, so that $F_0 = F - \langle F\rangle$; let $\mathcal{Y}_0 = \mathcal{Y}$ and $\mathcal{Y}_k = \partial^{\mathrm{in}}\mathcal{T}_k$ for $k \geq 1$. For $k \geq 1$ the set $\mathcal{T}_k$ does not meet $\mathcal{Y}$, so that by Lemma 18.193(a) F_k is gauge invariant and carried by $\mathcal{Y}_k$; if $\mathcal{Y}_k$ is empty, $F_k = 0$. A bond $b' \in \mathcal{Y}_k$, $k \geq 1$, has a neighbour outside $\mathcal{T}_k$, so that $d_{\mathcal{Y}}(b') < k(w+a) + a$; hence every $b \in \mathcal{T}_{k+1}$ has $|x_b - x_{b'}| \geq d_{\mathcal{Y}}(b) - d_{\mathcal{Y}}(b') > w$, that is, $\mathcal{T}_{k+1} \subset \mathcal{Y}_k^{[w]}$, and this holds also at $k = 0$. As $\mathcal{F}_{\mathcal{T}_{k+1}} \subset \mathcal{F}_{\mathcal{T}_k}$ and $\langle F_k\rangle = 0$, $F_{k+1} = E[F_k|\mathcal{F}_{\mathcal{T}_{k+1}}]$, and (b) at the carrier $\mathcal{Y}_k$ and the distance w gives

$$\|F_{k+1}\| \leq \big\|E[F_k|\mathcal{F}_{\mathcal{Y}_k^{[w]}}]\big\| \leq (6\,\mathsf{bc_P})^{1/2} w^{-1}\|F_k\| = e^{-1}\|F_k\|.$$

Let $k = \lfloor r/(w+a)\rfloor$. Every bond of $\mathcal{Y}'$ lies in $\mathcal{T}_k$, so that G is $\mathcal{F}_{\mathcal{T}_k}$-measurable and $\mathrm{Cov}(F,G) = (F_k, G - \langle G\rangle)$; thus $|\mathrm{Cov}(F,G)| \leq e^{-k}(\mathrm{Var}F\,\mathrm{Var}G)^{1/2}$, and $e^{-k} < e\cdot e^{-r/(w+a)}$.

Step 5: clause (d). The same steps, without the projection Π and with $\mathsf{c}_{\mathrm{P}}^{\mathrm{all}}$ (Step 1); the functions F_k need not be gauge invariant.

Step 6: clause (e). The one-step locality of Proposition YM3_I:4.5 and nesting of the blocks imply that the value $(\pi_d U)(c) = \mathcal{M}^{K-d}(U)(c)$ at a bond c of Λ_d is a function of the values of U at the bonds of Λ_K with both end points in $B^{K-d}(c_-) \cup B^{K-d}(c_+)$ alone; so $\varphi\circ\pi_d$ is carried by $\mathcal{Y}_{\mathcal{Q}}$. The averaging map is Borel by Proposition YM3_I:4.5, so $\varphi\circ\pi_d$ is Borel. For a gauge transformation u of Λ_K with the restriction $u|$ to the sites of Λ_d, the iterated map is covariant, $\mathcal{M}^{K-d}(U^u) = (\mathcal{M}^{K-d}U)^{u|}$ (Proposition YM3_I:4.5); so $\varphi\circ\pi_d$ is gauge invariant if φ is. The block $B^{K-d}(y)$ lies in the closed cube of side a_d centred at y by the formula for $B^{K-d}(y)$. The cubes of the two end points of c form a box containing the midpoints of all bonds with both end points in $B^{K-d}(c_-) \cup B^{K-d}(c_+)$. Thus the distance between the midpoint carriers is at least the distance between $\Delta(\mathcal{Q})$ and $\Delta(\mathcal{Q}')$. Apply (c) with $\mathrm{Var}(\varphi\circ\pi_d) \leq \|\varphi\|_\infty^2$ and the corresponding bound for $\psi\circ\pi_d$. At distance zero, the bound follows from Cauchy–Schwarz. □

Definition 18.195 (The smeared action density at a reference coupling)**.** (1) *The setting at g.* Fix the construction constants, in particular odd $L \geq 13$, $B_\sharp \geq 0$ and $0 < \gamma \leq \frac{1}{2}$. For $0 < g \leq \gamma$, $\mathfrak{d}(g)$ is the canonical first-passage family at g (Corollary YM3_I:74.2 and YM3_I:(1843)), and $\mathfrak{L}(g)$ is the tail of the sides specified in YM3_I:(1528). At a first-passage datum, the initial inverse coupling satisfies $\beta > X_g = L(g^{-2} + B_\sharp) \geq 52$. We use the configuration space Ω_K and Wilson law μ_K on T_ℓ defined above.

(2) *Tuples.* Lip_c is the set of the real Lipschitz functions on $\mathbb{R}^3$ with compact support; $\|f\|_\sharp = \max\{\|f\|_\infty, 40M\,\mathrm{Lip}f\}$, with the construction cube size M. The periodisation is $f^{\langle\ell\rangle}([x]) = \sum_{v\in\ell\mathbb{Z}^3} f(x+v)$. The observable with weight h is $\mathcal{O}(h) = a^{-1}\sum_p h(x(p))\,a_p$, where $x(p)$ is the barycentre of p (YM3_I:(1739)). A tuple is a finite sequence $X = (f_1,\dots,f_n)$ of elements of Lip_c with $n \geq 1$, with $|X| = n$ and $\|X\|_\sharp = \prod_i \|f_i\|_\sharp$; its sub-tuples are the $X_J = (f_i)_{i\in J}$, $\emptyset \neq J \subset \{1,\dots,n\}$; the empty sequence $\emptyset$ is not a tuple, and it occurs only in the places that name it; $X\cdot Y$ is the concatenation; $\tau_v X = (f_i(\cdot - v))_i$ for $v \in \mathbb{R}^3$; and $\mathsf{r}_z X = (f_i\circ\mathsf{r}_z)_i$. The tuple lies in the ball $\mathbb{B}(x_X, R)$ at the separation $\mathsf{d} > 0$ if the supports of its entries lie in the closed ball $\mathbb{B}(x_X,R)$ of $\mathbb{R}^3$ and any two of them are at a distance $\geq \mathsf{d}$; it is of the type $(\mathsf{d}; R)$ if the support of each entry lies in some ball of radius R and any two of them are at a distance $\geq \mathsf{d}$ (Proposition YM3_I:91.13).

(3) *Moments and cumulants.* $A_X = \prod_i(\mathcal{O}(f_i^{\langle\ell\rangle}) - \langle\mathcal{O}(f_i^{\langle\ell\rangle})\rangle)$, and $\sigma(X) = \sigma_{K,\ell,\beta}(X) = \langle A_X\rangle$; so $\sigma(X\cdot Y) = \langle A_X A_Y\rangle$ and $\mathrm{Cov}(A_X, A_Y) = \sigma(X\cdot Y) - \sigma(X)\,\sigma(Y)$. $S^T_{n;K,\ell,\beta}(X)$ is the joint cumulant of the $\mathcal{O}(f_i^{\langle\ell\rangle})$ under μ_K. For a law ν on the infinite lattice $\mathbb{L}_K = a_K\mathbb{Z}^3$, $S_n^T[\nu]$ and σ_ν are its cumulants and its centred functional. For a side ℓ, $\tilde\mu_{K,\ell,\beta}$ is the law on $\mathbb{L}_K$ that μ_K on T_ℓ induces when every field is extended periodically to the bonds of $\mathbb{L}_K$, that is, the push-forward $(\mathsf{per}_\ell)_*\mu_{K,\ell,\beta}$ by the map per_ℓ of periodic extension. $\mathcal{G}^{\mathrm{tor}}_{K,\beta}$ is the set of the infinite-volume limit states of the tori: the laws on $\mathbb{L}_K$ reached as weak limits,

along subsequences, by the Wilson measures on $\mathbb{L}_K$ with the periods $2L^j\mathbb{Z}^3$, $j\in\mathbb{N}$; and a sequence of sides for $\nu\in\mathcal{G}^{\mathrm{tor}}_{K,\beta}$ is a sequence $\ell_i\to\infty$ in $\{2L^j:\ j\in\mathbb{N}\}$ with $\tilde\mu_{K,\ell_i,\beta}\to\nu$ weakly (Proposition YM3_I:91.2). A volume-first limit at g is a family $S=(S_n)_{n\ge2}$ with $S_n(X)=\lim_j S_n^T[\nu_j](X)$ for every separated n-tuple X over $\mathbb{R}^3$, along a defining sequence of states (K_j,β_j,ν_j) with $(K_j,\beta_j)\in\mathfrak{d}(g)$, $\nu_j\in\mathcal{G}^{\mathrm{tor}}_{K_j,\beta_j}$ and $K_j\to\infty$; the entries of a separated tuple are real functions of class C^1 with compact and pairwise disjoint supports. For such a limit the centred functional is $\sigma_S(X)=\sum_{\pi\in\mathrm{Part}_{\ge2}(|X|)}\prod_{J\in\pi}S_{|J|}(X_J)$, with $\sigma_S(\emptyset)=1$. This follows from the moment–cumulant formula because a centred variable has zero first cumulant.

(4) *Numbers.* $C^g_n(\mathsf{d};R)$, $n\ge2$, is the bound table in YM3_I:(1850), and $\mathsf{T}_k(\mathsf{d};R)=\sum_{\pi\in\mathrm{Part}_{\ge2}(k)}\prod_{J\in\pi}C^g_{|J|}(\mathsf{d};R)$ for $k\ge2$, the sum running over the partitions of $\{1,\dots,k\}$ into blocks of at least two elements. For tuples X, Y that lie in $\mathbb{B}(x_X,R)$ and in $\mathbb{B}(x_Y,R)$, $R\ge1$, at the separation d, with $n=\max\{|X|,|Y|\}$, let

$$\mathsf{T}_X=\mathsf{T}_{2|X|}(\min\{\mathsf{d},2\};R)\,\|X\|_\sharp^2,\qquad C(X,Y)=\max\{1,\mathsf{T}_X\}^{1/2}\max\{1,\mathsf{T}_Y\}^{1/2},\qquad r_0=2\sqrt3\,(|x_X-x_Y|+2R+6),$$

$$\mathsf{N}_K(X)=1+2\sum_{i=1}^{|X|}\log^+\!\big(12\,\|f_i\|_\infty(2R{+}1)^3\big)+8\,|X|\,K\log L,\qquad \mathsf{N}_K(X,Y)=\max\{\mathsf{N}_K(X),\mathsf{N}_K(Y)\},$$

$$\alpha_{\mathrm{P}}=(185\,\mathsf{b}\mathsf{c}_{\mathrm{P}})^{-1/2},\qquad \alpha_{\mathrm{eff}}=\alpha_{\mathrm{P}}-\frac{\mathsf{N}_K(X,Y)}{\frac{\ell}{2}-2R-4},$$

with $\mathsf{b}=a_K\beta$, $\mathsf{c}_{\mathrm{P}}=\mathsf{c}_{\mathrm{P}}(K,\ell,\beta)$, and $\log^+t=\log t$ for $t\ge1$, $\log^+t=0$ for $0\le t<1$. For the truncated functions let $\mathsf{t}(Z)=\mathsf{T}_{|Z|}(\mathsf{d};R)\,\|Z\|_\sharp$ for a sub-tuple Z of X or of Y with $|Z|\ge2$, $\mathsf{t}(Z)=0$ for $|Z|=1$, $\mathsf{t}(\emptyset)=1$, and

$$\bar C(X,Y)=\max_{X',Y'\neq\emptyset}C(X',Y'),$$
$$\mathsf{p}(X,Y)=1+\max_{X',Y'}\mathsf{t}(X')\,\mathsf{t}(Y')+\bar C(X,Y),$$
$$C'(X,Y)=\bar C(X,Y)\sum_{\pi\in\mathrm{Part}(N)}(|\pi|-1)!\,|\pi|\,\mathsf{p}(X,Y)^{|\pi|-1},$$

the maxima being taken over the sub-tuples X' of X and Y' of Y, and in the second also with $\emptyset$ in the place of X', of Y' or of both, $N=|X|+|Y|$, and $\mathrm{Part}(N)$ being the set of the partitions of $\{1,\dots,N\}$. The numbers $C(X,Y)$, $\bar C(X,Y)$, $\mathsf{p}(X,Y)$ and $C'(X,Y)$ are formed from the constants of the main text, g, d, R and the norms of the entries; they depend neither on K, ℓ, β nor on a position or a state (Proposition YM3_I:91.13).

(5) *Mirrors of a tuple.* For a direction μ and a tuple X in $\mathbb{B}(x_X,R)$ let $\xi=(x_X)_\mu$, let z_1 be the least element of $a\mathbb{Z}$ with $z_1\ge\xi+R+1$ and z_2 the largest element of $a\mathbb{Z}$ with $z_2\le\xi+\frac{\ell}{4}$, and let $\mathsf{h}_X(z)=\mathrm{Cov}(A_X,\theta_zA_X)$ for $z\in a\mathbb{Z}$.

■

Proposition 18.196. *From a finite Poincaré constant to exponential clustering in space.*

(a) *A floor.* Let $L\ge3$ be odd, $K,m\in\mathbb{N}$ and $\beta>0$. The mean $x=\langle a_p\rangle$ is the same for all plaquettes p, and

$$\mathsf{c}_{\mathrm{P}}\ge\frac{a^2}{16\,\mathsf{b}}\cdot\frac{1-x}{2-x}.$$

If $\beta\ge25$, then $x\le0.4157$ and $\mathsf{c}_{\mathrm{P}}\ge a^2/(48\,\mathsf{b})$.

(b) *Functions of $L^2(\mu_K)$ and block fields at $\beta\ge25$.* Let besides $\beta\ge25$. Then $e\sqrt{6\,\mathsf{b}\mathsf{c}_{\mathrm{P}}}+a\le\sqrt{185\,\mathsf{b}\mathsf{c}_{\mathrm{P}}}$; for F, G, r as in Proposition 18.194(c), and for $\varphi\circ\pi_d$, $\psi\circ\pi_{d'}$, $r_{\mathcal{Q}}$ as in

Proposition 18.194(e),

$$|\mathrm{Cov}(F,G)| \le e\,(\mathrm{Var}F\,\mathrm{Var}G)^{1/2}e^{-r/\sqrt{185\,\mathsf{bc}_{\mathrm{P}}}}, \qquad |\mathrm{Cov}(\varphi\circ\pi_d,\psi\circ\pi_{d'})| \le e\,\|\varphi\|_\infty\|\psi\|_\infty\,e^{-r_{\mathfrak{Q}}/\sqrt{185\,\mathsf{bc}_{\mathrm{P}}}};$$

the numbers e and 185 depend on nothing. This holds in particular at every $(K,\beta)\in\mathfrak{d}(g)$, $0<g\le\gamma$, on every torus.

In (c)–(g), $0<g\le\gamma$, and the objects and numbers are those of Definition 18.195; X and Y are tuples that lie in $\mathbb{B}(x_X,R)$ and in $\mathbb{B}(x_Y,R)$, $R\ge 1$, at the separation $\mathsf{d}>0$, and $n=\max\{|X|,|Y|\}$. Clauses (c)–(g) use the uniform cumulant bounds of Proposition YM3_I:91.13, which follow from the construction and stability theorems (Theorems YM3_I:D and YM3_I:AJ).

(c) *One tuple and its mirror images.* Let $(K,\beta)\in\mathfrak{d}(g)$, $\ell\in\mathfrak{L}(g)$ with $\ell\ge 32|X|(R+1)$, and let μ be a direction. For $z\in a\mathbb{Z}$ with $z_1\le z\le z_2$,

$$0\le \mathsf{h}_X(z)=\sigma(X\cdot\mathsf{r}_zX)-\sigma(X)^2\le\max\{1,\mathsf{T}_X\}\,\exp\Big(-2(z-z_1)\Big[\alpha_{\mathrm{P}}-\frac{\mathsf{N}_K(X)}{\frac{\ell}{2}-2R-4}\Big]\Big).$$

(d) *Two tuples on one torus.* Let $(K,\beta)\in\mathfrak{d}(g)$, $\ell\in\mathfrak{L}(g)$ with $\ell\ge 32n(R+1)$, and $v\in\mathbb{R}^3$ with $r_0\le|v|\le\frac{\ell}{4}$. If $\alpha_{\mathrm{eff}}\ge 0$, then

$$\big|\sigma(X\cdot\tau_vY)-\sigma(X)\,\sigma(\tau_vY)\big|\le C(X,Y)\,e^{-\alpha_{\mathrm{eff}}|v|/(2\sqrt{3})}.$$

On the tori with $\ell\ge 4R+8+4\,\mathsf{N}_K(X,Y)\sqrt{185\,\mathsf{bc}_{\mathrm{P}}}$, a side that grows linearly in $K\sqrt{\mathsf{bc}_{\mathrm{P}}}$, we have $\alpha_{\mathrm{eff}}\ge\frac12\alpha_{\mathrm{P}}$, and the right member is at most $C(X,Y)\exp(-|v|/(4\sqrt3\sqrt{185\,\mathsf{bc}_{\mathrm{P}}}))$.

(e) *The states of the tori.* Let $(K,\beta)\in\mathfrak{d}(g)$, $\nu\in\mathcal{G}^{\mathrm{tor}}_{K,\beta}$, let (ℓ_i) be a sequence of sides for ν, and $\mathsf{c}^\sharp=\liminf_i\mathsf{c}_{\mathrm{P}}(K,\ell_i,\beta)$. Then $\mathsf{c}^\sharp\ge a^2/(48\,\mathsf{b})$, and if $\mathsf{c}^\sharp<\infty$, then for every $v\in\mathbb{R}^3$ with $|v|\ge r_0$

$$\big|\sigma_\nu(X\cdot\tau_vY)-\sigma_\nu(X)\,\sigma_\nu(\tau_vY)\big|\le C(X,Y)\exp\Big(-\frac{|v|}{2\sqrt3\,\sqrt{185\,\mathsf{bc}^\sharp}}\Big).$$

No condition on a side enters, and the bound depends on K and β through $\mathsf{bc}^\sharp$ alone.

(f) *Volume-first limits.* Let S be a volume-first limit at g with a defining sequence of states (K_j,β_j,ν_j); for each j let a sequence of sides for ν_j be given, let $\mathsf{c}^\sharp_j$ be the lower limit of $\mathsf{c}_{\mathrm{P}}(K_j,\ell,\beta_j)$ along it, and let $\mathsf{c}^*=\liminf_j a_{K_j}\beta_j\,\mathsf{c}^\sharp_j$. Let X and Y be separated tuples. If $\mathsf{c}^*<\infty$, then for every v with $|v|\ge r_0$ the tuple $X\cdot\tau_vY$ is separated, and

$$\big|\sigma_S(X\cdot\tau_vY)-\sigma_S(X)\,\sigma_S(\tau_vY)\big|\le C(X,Y)\exp\Big(-\frac{|v|}{2\sqrt3\,\sqrt{185\,\mathsf{c}^*}}\Big),$$

the right member being read as 0 if $\mathsf{c}^*=0$. Thus a volume-first limit clusters exponentially, with the constants $C(X,Y)$ and r_0 of one torus, as soon as it has a defining sequence and sequences of sides along which bc_{P} stays bounded in this sense, the volume being removed first and the cutoff second.

(g) *Truncated functions.* Let $N=|X|+|Y|$. Under the hypotheses of (d), of (e) and of (f), respectively, $|S^T_{N;K,\ell,\beta}(X\cdot\tau_vY)|$, $|S^T_N[\nu](X\cdot\tau_vY)|$ and $|S_N(X\cdot\tau_vY)|$ are at most $C'(X,Y)$ times the exponential of (d), of (e) and of (f).

■

Proof. Step 1: clause (a). The translations of Λ_K and the permutations of the three axes map plaquettes onto plaquettes, act transitively on them, and preserve dU and A_K (a permutation that reverses the orientation of a plaquette replaces $\partial U(p)$ by a conjugate of its inverse, which has the same trace); so all the a_p have one law under μ_K. Fix a plaquette p and one of its bonds b, and write $y_i=\partial^i_b a_p$. At the field with $e^{is\,y'\cdot\varsigma}U(b)$ at b, the holonomy of p carries the factor $e^{is\,y'\cdot\varsigma}$ in front of $U(b)$, or the factor $e^{-is\,y'\cdot\varsigma}$ behind $U(b)^{-1}$; under the trace a cyclic permutation

moves this factor to the first place, so that $\frac12\mathrm{tr}\,\partial U(p)$ is there $\frac12\mathrm{tr}(e^{\pm is\,y'\cdot\varsigma}W)$, where W, the product of the four factors of $\partial U(p)$ at U in a cyclic order, does not depend on s and is conjugate to $\partial U(p)$; thus $W=w_0\mathbf{1}+i\sum_k w_k\varsigma_k\in\mathrm{SU}(2)$ with $w_0=1-a_p$ and $w_0^2+|w|^2=1$. Hence $y_i=\pm w_i$, so that $|y|^2=1-w_0^2=a_p(2-a_p)$; $\sum_i(\partial_b^i)^2a_p=3w_0$; $\partial_b^j\partial_b^ia_p=w_0\delta_{ij}\mp\sum_k\epsilon_{ijk}w_k$, so that $\sum_{i,j}(\partial_b^j\partial_b^ia_p)^2=3w_0^2+2|w|^2\le3$; and the symmetric part of the matrix $(\partial_b^i\partial_b^ja_{p'})_{ij}$ is $(1-a_{p'})\mathbf{1}$ for each of the four plaquettes p' that hold b. Let $\eta=\langle a_p(2-a_p)\rangle$, the same for all plaquettes, and $\mathsf{z}=\sum_i(\partial_b^i)^*y_i$. Then $\langle\mathsf{z}\rangle=0$ and $\mathsf{z}=-3w_0+\beta\sum_{p'\ni b}\sum_i(\partial_b^ia_{p'})\,y_i$, so that Schwarz's inequality gives $3(1-x)=3\langle w_0\rangle\le4\beta\eta$. Next, $\eta=\langle\sum_i y_i\,\partial_b^ia_p\rangle=(a_p,\mathsf{z})=\mathrm{Cov}(a_p,\mathsf{z})\le\mathrm{Var}(a_p)^{1/2}\|\mathsf{z}\|$, and Lemma 18.192(b) at $f=a_p$ and $\lambda=\mathbf{1}_{\{b\}}$, with the facts above, gives $\|\mathsf{z}\|^2\le3+4\beta\eta$. As a_p is a smooth gauge-invariant function with $\mathcal{E}_K(a_p)=4a^{-1}\eta$, each of the four bonds of p contributing $a^{-1}\eta$, and as $\tau\mapsto\tau/(3+4\tau)$ increases,

$$\mathsf{c_P}\ge\frac{\mathrm{Var}(a_p)}{\mathcal{E}_K(a_p)}\ge\frac{a\,\eta}{4(3+4\beta\eta)}=\frac{a}{4\beta}\cdot\frac{\beta\eta}{3+4\beta\eta}\ge\frac{a}{4\beta}\cdot\frac{\frac34(1-x)}{3+3(1-x)}=\frac{a^2}{16\,\mathsf{b}}\cdot\frac{1-x}{2-x}.$$

For the bound of x, let $N_K=3(\ell/a)^3$ be the number of the plaquettes, which is the number of the bonds, and $Z(\beta')=\int e^{-\beta' A_K}dU$. The mean $\langle A_K\rangle_{\beta'}$ under $Z(\beta')^{-1}e^{-\beta' A_K}dU$ does not increase in β', its derivative being $-\mathrm{Var}_{\beta'}(A_K)$, and $Z(0)=1$; so $-\log Z(\beta)=\int_0^\beta\langle A_K\rangle_{\beta'}d\beta'\ge\beta N_Kx$. Let $\angle V\in[0,\pi]$ be the angle of $V\in\mathrm{SU}(2)$, $\frac12\mathrm{tr}\,V=\cos\angle V$, which is the distance of V from $\mathbf{1}$ for the bi-invariant metric of the unit sphere $\mathrm{SU}(2)\subset\mathbb{R}^4$; it is subadditive and invariant under inversion, and its law under the Haar measure has the density $\frac2\pi\sin^2$. Where $\angle U(b)\le\delta$ at every bond, $\angle\partial U(p)\le4\delta$ and $a_p\le8\delta^2$; this set has the Haar measure q^{N_K}, $q=\frac1\pi(\delta-\frac12\sin2\delta)\ge\frac{2\delta^3}{3\pi}(1-\frac{\delta^2}{5})$ for $0<\delta\le1$. So $-\log Z(\beta)\le N_K(8\beta\delta^2-\log q)$, and at $\delta^2=3/(16\beta)$

$$x\le\frac1\beta\Big[\frac32+\log\frac{3\pi}{2}+\frac32\log\frac{16\beta}{3}-\log\Big(1-\frac{3}{80\beta}\Big)\Big],$$

a function that decreases on $\beta\ge1$ and has the value $0.41563\ldots$ at $\beta=25$. Then $\frac{1-x}{2-x}\ge\frac13$.

Step 2: clause (b). By (a), $a\le\sqrt{48\,\mathsf{bc_P}}$, so that $e\sqrt{6\,\mathsf{bc_P}}+a\le(e\sqrt6+\sqrt{48})\sqrt{\mathsf{bc_P}}$, and $(e\sqrt6+\sqrt{48})^2=184.59\ldots<185$; then Proposition 18.194(c), (e). At $(K,\beta)\in\mathfrak{d}(g)$, $\beta>52$ (Definition 18.195(1)).

Step 3: the observables on the torus. Let $(K,\beta)\in\mathfrak{d}(g)$ and $\ell\in\mathfrak{L}(g)$, and let $f\in\mathrm{Lip}_c$ have its support in a ball $\mathbb{B}(x,R)$ with $2R+2a<\frac\ell2$. (i) A point of T_ℓ has at most one lift in $\mathbb{B}(x,R)$, so $|f^{\langle\ell\rangle}|\le\|f\|_\infty$. The barycentres of the plaquettes of one orientation are the image in T_ℓ of a translate of $a\mathbb{Z}^3$, of which a ball of radius R holds at most $(2R/a+1)^3$ points; so at most $3(2R/a+1)^3$ plaquettes have $f^{\langle\ell\rangle}(x(p))\ne0$, and the end points of their bonds lie within the distance $a/\sqrt2$ of their barycentres. As $0\le a_p\le2$ and $a\le1$, $\mathcal{O}(f^{\langle\ell\rangle})$ is a smooth gauge-invariant function, carried by the bonds with both end points in the image of $\mathbb{B}(x,R+a)$, and $\sup|\mathcal{O}(f^{\langle\ell\rangle})|\le6\,\|f\|_\infty(2R+1)^3a^{-4}$. For a tuple X in $\mathbb{B}(x_X,R)$, therefore, A_X is smooth, gauge invariant and carried by the bonds with both end points in the image of $\mathbb{B}(x_X,R+a)$, and $e\,\mathrm{Var}(A_X)\le e\sup|A_X|^2\le e^{\mathsf{N}_K(X)}$, since a centred observable is at most twice the supremum of the observable and $a^{-1}=L^K$. (ii) Let $\mathsf{r}\in\mathsf{Iso}(\mathbb{L}_K)$, the group of the maps $x\mapsto\mathsf{r}^\circ x+v$ with a signed permutation r° of the coordinates and $v\in a\mathbb{Z}^3$, and let r_ℓ be the map that it induces on T_ℓ; the group holds the translations by the vectors of $a\mathbb{Z}^3$, the permutations of the axes and the reflections r_z, $z\in a\mathbb{Z}$, in every direction. Periodisation commutes with r: $(f\circ\mathsf{r}^{-1})^{\langle\ell\rangle}=f^{\langle\ell\rangle}\circ\mathsf{r}_\ell^{-1}$. The isometry sends plaquette barycentres to plaquette barycentres, and relabelling the plaquettes in the sum gives $\mathcal{O}((f\circ\mathsf{r}^{-1})^{\langle\ell\rangle})(\mathsf{r}_\ell U)=\mathcal{O}(f^{\langle\ell\rangle})(U)$. It also preserves A_K and the product Haar measure, hence μ_K. Besides, $\|f\circ\mathsf{r}^{-1}\|_\sharp=\|f\|_\sharp$ for $f\in\mathrm{Lip}_c$: r preserves the Euclidean distance of $\mathbb{R}^3$, so that composing with r^{-1} changes neither the supremum nor the Lipschitz constant. For the tuple $\mathsf{r}X=(f_i\circ\mathsf{r}^{-1})_i$ it follows that $A_{\mathsf{r}X}(\mathsf{r}_\ell U)=A_X(U)$ and $\sigma(\mathsf{r}X)=\sigma(X)$. At $\mathsf{r}=\mathsf{r}_z$ this is $\theta_zA_X=A_{\mathsf{r}_zX}$, so that $\mathsf{h}_X(z)=\sigma(X\cdot\mathsf{r}_zX)-\sigma(X)^2$. (iii) Let Z be

a tuple of $k \geq 2$ entries of the type $(\mathsf{d}'; R)$ whose supports lie in one ball $\mathbb{B}(y, \rho)$ with $\rho + a < \frac{\ell}{4}$, and let $\ell \geq 16k(R+1)$. Then $|\sigma(Z)| \leq \mathsf{T}_k(\mathsf{d}'; R)\, \|Z\|_\sharp$. Indeed, for $w \in a\mathbb{Z}^3$ with $|y-w|_\infty \leq \frac{a}{2}$ the tuple $\tau_{-w}Z$ has, by (ii), the same σ and the same norms, and its supports lie in $\mathbb{B}(y-w,\rho) \subset (-\frac{\ell}{4}, \frac{\ell}{4})^3$; by the moment–cumulant formula, $\sigma(Z) = \sum_{\pi \in \mathrm{Part}_{\geq 2}(k)} \prod_{J \in \pi} S^T_{|J|;K,\ell,\beta}(Z_J)$, and Proposition YM3_I: 91.13 bounds every factor by $C^g_{|J|}(\mathsf{d}'; R) \prod_{f \in Z_J} \|f\|_\sharp$, as $\ell \geq 16|J|(R+1)$.

Step 4: the near mirror. Let X, μ, ξ and z_1 be as in (c). The supports of the entries of $\mathsf{r}_{z_1}X$ lie in $\mathbb{B}(\mathsf{r}_{z_1}x_X, R)$, whose points have their μ-th coordinate $\geq 2z_1 - \xi - R \geq \xi + R + 2$, while those of X have it $\leq \xi + R$. So the $2|X|$ entries of $X \cdot \mathsf{r}_{z_1}X$ are of the type $(\min\{\mathsf{d}, 2\}; R)$, and their supports lie in the ball of radius $R + z_1 - \xi < 2R + 1 + a$ about $x_X + (z_1 - \xi)e_\mu$. As $\frac{\ell}{4} \geq 8|X|(R+1)$, Step 3(iii) applies with $k = 2|X|$ and gives $\sigma(X \cdot \mathsf{r}_{z_1}X) \leq \mathsf{T}_X$; hence $\mathsf{h}_X(z_1) \leq \mathsf{T}_X$ by Step 3(ii).

Step 5: the sign and the convexity. By Step 3(i), the end points of the bonds that carry A_X have their μ-th coordinate in $[\xi - R - a, \xi + R + a]$ modulo ℓ. For $z \in a\mathbb{Z}$ with $\xi + R + a < z < \xi + \frac{\ell}{2} - R - a$, A_X is therefore carried by H^-_z, and Lemma 18.193(b) gives $\mathsf{h}_X(z) = \mathrm{Var}(E[A_X|\mathcal{F}_{\Sigma_z}]) \geq 0$. Let now $\xi + R + 2a < z < \xi + \frac{\ell}{2} - R - 2a$, $z' = z - a$ and $X' = \tau_{-2ae_\mu}X$; then A_X and $A_{X'}$ are carried by $H^-_{z'}$. For $t, t' \in a\mathbb{Z}$ with $t + t' \in 2a\mathbb{Z}$, the point $z' - \frac{t+t'}{2}$ lies in $a\mathbb{Z}$ and $\tau_{-te_\mu} \circ \mathsf{r}_{z'} \circ \tau_{t'e_\mu} = \mathsf{r}_{z'-(t+t')/2}$, so that Step 3(ii) gives $\mathrm{Cov}(A_{\tau_{te_\mu}X}, \theta_{z'}A_{\tau_{t'e_\mu}X}) = \mathsf{h}_X(z' - \frac{t+t'}{2})$. At $(t,t') = (0,0)$, $(0,-2a)$ and $(-2a,-2a)$ these numbers are $\mathsf{h}_X(z-a)$, $\mathsf{h}_X(z)$ and $\mathsf{h}_X(z+a)$, and Schwarz's inequality for the non-negative form of Lemma 18.193(b) at the mirror z' gives $\mathsf{h}_X(z)^2 \leq \mathsf{h}_X(z-a)\,\mathsf{h}_X(z+a)$. A finite sequence $h_k \geq 0$, $k_1 \leq k \leq k_2$, with $h_k^2 \leq h_{k-1}h_{k+1}$ for $k_1 < k < k_2$ satisfies $h_k \leq h_{k_1}^{1-t}h_{k_2}^t$, $t = \frac{k-k_1}{k_2-k_1}$: if one inner member vanishes, the inequality, applied step by step, makes all inner members vanish; otherwise all members are positive, since an end member that vanished would make its inner neighbour vanish, and $\log h_k$ is convex in k.

Step 6: the far mirror; clause (c). As $a \leq \frac{1}{13}$ and $\frac{\ell}{4} \geq 8(R+1)$, the points z_1 and z_2 satisfy $\xi + R + a < z < \xi + \frac{\ell}{2} - R - a$, and every point z of $a\mathbb{Z}$ strictly between them satisfies $\xi + R + 2a < z < \xi + \frac{\ell}{2} - R - 2a$; so Step 5 applies on $a\mathbb{Z} \cap [z_1, z_2]$. The function $\theta_{z_2}A_X = A_{\mathsf{r}_{z_2}X}$ is carried by bonds whose end points, and so whose midpoints, have their μ-th coordinate in $[2z_2 - \xi - R - a, 2z_2 - \xi + R + a]$. As $\frac{\ell}{2} - 2a < 2(z_2 - \xi) \leq \frac{\ell}{2}$, and as the Euclidean distance of T_ℓ is at least the distance of the μ-th coordinates on the circle $\mathbb{R}/\ell\mathbb{Z}$, the carriers of A_X and of $\theta_{z_2}A_X$ are at a distance $r_2 \geq 2(z_2 - \xi) - 2R - 2a > 0$. Both functions are gauge invariant, so (b), Step 3(i) and Step 5 give $\mathsf{h}_X(z_2) \leq e\,\mathrm{Var}(A_X)\, e^{-\alpha_{\mathrm{P}} r_2} \leq e^{\mathsf{N}_K(X) - \alpha_{\mathrm{P}} r_2}$. For $z \in a\mathbb{Z} \cap [z_1, z_2]$ and $t = \frac{z - z_1}{z_2 - z_1}$, Steps 4 and 5 give

$$\mathsf{h}_X(z) \leq \mathsf{h}_X(z_1)^{1-t}\, \mathsf{h}_X(z_2)^t \leq \max\{1, \mathsf{T}_X\}\, \exp(t\, \mathsf{N}_K(X) - t\, \alpha_{\mathrm{P}} r_2).$$

Since $z_1 \geq \xi + R + 1$, $r_2 \geq 2(z_2 - z_1) + 2 - 2a$, so that $t\, r_2 \geq 2(z - z_1)$; since $z_2 - z_1 > \frac{\ell}{4} - R - 1 - 2a \geq \frac{\ell}{4} - R - 2$, $t \leq (z - z_1)/(\frac{\ell}{4} - R - 2)$. This is (c).

Step 7: clause (d). Choose a direction μ with $|v_\mu| \geq |v|/\sqrt{3}$, and let $\xi_1 = (x_X)_\mu$, $\xi_2 = (x_Y + v)_\mu$ and $\mathsf{s} = |\xi_2 - \xi_1|$; then $\frac{|v|}{\sqrt{3}} - |x_X - x_Y| \leq \mathsf{s} \leq |v| + |x_X - x_Y|$. The reflection $\rho\colon x_\mu \mapsto -x_\mu$ lies in $\mathsf{Iso}(\mathbb{L}_K)$, and by Step 3(ii) it leaves the left member of (d) unchanged when it is applied to X, to Y and to v; so we may take $\xi_2 > \xi_1$. As $|v| \geq r_0$, $|x_X - x_Y| + 2R + 4 \leq \frac{r_0}{2\sqrt{3}} \leq \frac{|v|}{2\sqrt{3}}$; hence $\mathsf{s} \geq 4R + 12$ and $\mathsf{s} - 2R - 4 \geq \frac{|v|}{2\sqrt{3}}$, and, as $|v| \leq \frac{\ell}{4}$, $\mathsf{s} \leq \frac{\ell}{4} + \frac{\ell}{8\sqrt{3}} < \frac{\ell}{2} - 2a$. Let $z \in a\mathbb{Z}$ with $|z - \frac{\xi_1 + \xi_2}{2}| \leq \frac{a}{2}$. Then $\xi_1 + R + 1 \leq z \leq \xi_1 + \frac{\ell}{4}$, so that $z_1 \leq z \leq z_2$ for the mirrors of X in the direction μ, and $z - z_1 \geq \frac{\mathsf{s}}{2} - \frac{a}{2} - R - 1 - a \geq \frac{\mathsf{s}}{2} - R - 2$. The reflection ρ carries $\tau_v Y$ to a tuple $\rho\tau_v Y$ in a ball whose centre has the μ-th coordinate $-\xi_2$, the mirror z to $-z$, and θ_z to θ_{-z}; for $\rho\tau_v Y$ and $-z$ the same bounds hold. So A_X is carried by H^-_z and $A_{\tau_v Y}$ by H^+_z (Step 5), and Lemma 18.193(b), Step 3(ii) and (c), applied to X and to $\rho\tau_v Y$, give

$$|\mathrm{Cov}(A_X, A_{\tau_v Y})|^2 \leq \mathsf{h}_X(z)\, \mathsf{h}_{\rho\tau_v Y}(-z) \leq \max\{1, \mathsf{T}_X\} \max\{1, \mathsf{T}_Y\}\, e^{-2\alpha_{\mathrm{eff}}(\mathsf{s} - 2R - 4)},$$

where $\alpha_{\mathrm{eff}} \geq 0$ is used, and where $\mathsf{T}_{\rho\tau_v Y} = \mathsf{T}_Y$ and $\mathsf{N}_K(\rho\tau_v Y) = \mathsf{N}_K(Y)$, the isometries of $\mathbb{R}^3$ preserving the type of a tuple and the norms of its entries. As $\mathrm{Cov}(A_X, A_{\tau_v Y}) = \sigma(X \cdot \tau_v Y) - \sigma(X)\sigma(\tau_v Y)$ and $\mathsf{s} - 2R - 4 \geq \frac{|v|}{2\sqrt{3}}$, this is (d). If $\ell \geq 4R + 8 + 4\,\mathsf{N}_K(X,Y)\sqrt{185\,\mathsf{bc}_{\mathrm{P}}}$, then $\mathsf{N}_K(X,Y)/(\frac{\ell}{2} - 2R - 4) \leq \frac{1}{2}\alpha_{\mathrm{P}}$.

Step 8: clause (e). By Proposition YM3_I:91.2, $\ell_i \to \infty$, $\ell_i \in \{2L^j : j \in \mathbb{N}\}$ and $\tilde{\mu}_{K,\ell_i,\beta} \to \nu$ weakly. As $\beta > 52$, (a) gives $\mathsf{c}_{\mathrm{P}}(K, \ell_i, \beta) \geq a^2/(48\,\mathsf{b})$ for every i, hence the floor of $\mathsf{c}^\sharp$. Let $\mathsf{c}^\sharp < \infty$, take a subsequence along which $\mathsf{c}_{\mathrm{P}}(K, \ell_i, \beta) \to \mathsf{c}^\sharp$, and fix v with $|v| \geq r_0$. For all large i of the subsequence, $\ell_i \in \mathfrak{L}(g)$, $\ell_i \geq 32n(R+1)$ and $|v| \leq \frac{\ell_i}{4}$; and the number α_{eff} formed at ℓ_i tends to $(185\,\mathsf{bc}^\sharp)^{-1/2} > 0$, since $\mathsf{N}_K(X,Y)$ does not depend on the side. So (d) holds at these ℓ_i. By YM3_I:(1824), $\sigma_\nu(Z) = \lim_i \sigma_{K,\ell_i,\beta}(Z)$ for the tuples $Z = X \cdot \tau_v Y$, X and $\tau_v Y$, whose entries are bounded real functions with compact supports; the bound passes to the limit.

Step 9: clause (f). By the definition of a volume-first limit, $(K_j, \beta_j) \in \mathfrak{d}(g)$ and $\nu_j \in \mathcal{G}^{\mathrm{tor}}_{K_j,\beta_j}$; so (e) holds at every j, with $\mathsf{b}_j\mathsf{c}^\sharp_j$, $\mathsf{b}_j = a_{K_j}\beta_j$, in the place of $\mathsf{bc}^\sharp$, and with one constant $C(X,Y)$ for all j (Proposition YM3_I:91.13). For a law ν and a tuple Z of k entries, $\sigma_\nu(Z) = \sum_{\pi \in \mathrm{Part}_{\geq 2}(k)} \prod_{J \in \pi} S^T_{|J|}[\nu](Z_J)$. Sub-tuples of separated tuples are separated, so the defining convergence of cumulants gives $\sigma_S(Z) = \lim_j \sigma_{\nu_j}(Z)$ for every separated tuple Z. For $|v| \geq r_0$ the balls $\mathbb{B}(x_X, R)$ and $\mathbb{B}(x_Y + v, R)$ are disjoint, as $|x_X - x_Y - v| \geq r_0 - |x_X - x_Y| > 2R$, so that $X \cdot \tau_v Y$ is separated. Along a subsequence with $\mathsf{b}_j\mathsf{c}^\sharp_j \to \mathsf{c}^*$ the bounds of (e) tend to the bound of (f).

Step 10: clause (g). Let I and J index the entries of X and of $\tau_v Y$ in $X \cdot \tau_v Y$. For $B \subset \mathsf{I} \sqcup \mathsf{J}$ let m_B be the centred moment of the sub-tuple Z_B of $X \cdot \tau_v Y$ with the entries indexed by B, that is $\sigma(Z_B)$, $\sigma_\nu(Z_B)$ or $\sigma_S(Z_B)$, with $m_\emptyset = 1$, and let $\tilde{m}_B = m_{B\cap\mathsf{I}}\, m_{B\cap\mathsf{J}}$. A cumulant of order $N \geq 2$ does not change when constants are added to the variables, and it is a polynomial in the moments. Expanding the logarithm of the generating function shows that the truncated function of (g) is $\sum_{\pi \in \mathrm{Part}(N)} (-1)^{|\pi|-1}(|\pi|-1)! \prod_{B\in\pi} m_B$, on the torus and at ν, and, by passage to the limit as in Step 9, for S. The same polynomial in the $\tilde{m}_B$ vanishes by a formal factorization. In the commutative algebra with $z_i^2 = 0$, put $z_B = \prod_{i \in B} z_i$. The moment series $\sum_B \tilde{m}_B z_B$ factors as

$$\Big(\sum_{C \subset \mathsf{I}} m_C z_C\Big)\Big(\sum_{D \subset \mathsf{J}} m_D z_D\Big).$$

Its formal logarithm is the sum of the logarithms of the factors. Each summand uses only one of the two nonempty index sets, so the coefficient of $z_{\mathsf{I}\sqcup\mathsf{J}}$ is zero. Hence the truncated function equals $\sum_\pi (-1)^{|\pi|-1}(|\pi|-1)!\,[\prod_{B\in\pi} m_B - \prod_{B\in\pi}\tilde{m}_B]$, and $|\prod_{B\in\pi} m_B - \prod_{B\in\pi}\tilde{m}_B| \leq \mathsf{p}(X,Y)^{|\pi|-1}\sum_{B\in\pi}|m_B - \tilde{m}_B|$ as soon as all $|m_B|$ and $|\tilde{m}_B|$ are at most $\mathsf{p}(X,Y)$. Here $m_B - \tilde{m}_B = 0$ unless B meets both I and J, and then it is the left member of (d), (e) or (f) for a pair of non-empty sub-tuples X', Y'; these lie in the same balls, and $\mathsf{N}_K(X',Y') \leq \mathsf{N}_K(X,Y)$, so that the bound holds with $C(X',Y') \leq \bar{C}(X,Y)$ and the exponential of (d), (e) or (f). For $B \subset \mathsf{I}$, $|m_B| \leq \mathsf{t}(X_B)$ by Step 3(iii), the sub-tuple lying in one ball of radius R and $\ell \geq 16|B|(R+1)$, and by passage to the limit at ν and for S; likewise for $B \subset \mathsf{J}$; and for B that meets both, $|m_B| \leq |\tilde{m}_B| + |m_B - \tilde{m}_B| \leq \mathsf{t}(X')\,\mathsf{t}(Y') + C(X',Y')$, the exponential being at most 1. So all $|m_B|, |\tilde{m}_B| \leq \mathsf{p}(X,Y)$, and the sum over π gives the factor $C'(X,Y)$. □

Remark 18.197 (The limit laws of the block fields under a uniform Poincaré constant). Fix a side $\ell = 2L^m$ and a label b in the fixed-torus setting (E), with $\mathsf{b} \geq \gamma^{-2}$ and $MR(\mathsf{b}) \leq L^m$ (YM3_I:(2201)). Set $\beta_K = L^K\mathsf{b}$ and assume a uniform C_{P}. For $d \geq 0$, let φ, ψ be bounded Borel functions on Ω_d with bond carriers $\mathcal{Q}, \mathcal{Q}'$ as in Proposition 18.194(e), and suppose at least one is gauge invariant. The limiting law μ_d satisfies

$$|\,\mathrm{Cov}_{\mu_d}(\varphi,\psi)| \leq e\,\|\varphi\|_\infty\|\psi\|_\infty \exp\Big(-\frac{\mathrm{dist}(\Delta(\mathcal{Q}),\Delta(\mathcal{Q}'))}{e\sqrt{6\mathsf{b}C_{\mathrm{P}}}}\Big).$$

Indeed, Proposition 18.194(e) gives this estimate at cutoff K with an additional a_K in the denominator. Corollary 2.8(c) gives total-variation convergence of the block laws, so the three bounded Borel expectations converge.

For the time estimate let $P_t^K = e^{t\mathcal{L}_K}$. The smooth elliptic generator on the compact configuration space gives a reversible diffusion with invariant law μ_K. Gauge invariance of μ_K and the energy implies that P_t^K preserves gauge-invariant functions. For a centred gauge-invariant function H, smoothing at $t > 0$ and the Poincaré inequality give

$$\frac{d}{dt}\|P_t^K H\|^2 = -2\mathcal{E}_K(P_t^K H) \le -\frac{2}{C_{\mathrm{P}}}\|P_t^K H\|^2.$$

Strong continuity at $t = 0$ and integration give $\|P_t^K H\| \le e^{-t/C_{\mathrm{P}}}\|H\|$ for all large K. Use the time convention of the generator $\frac{1}{2}\mathcal{L}_K$. Stationarity, reversibility and Cauchy–Schwarz bound the absolute covariance between $\varphi(\pi_d U_0)$ and $\psi(\pi_d U_{t/2})$ by

$$e^{-t/(2C_{\mathrm{P}})}\big(\mathrm{Var}_{\mu_d^K}(\varphi)\,\mathrm{Var}_{\mu_d^K}(\psi)\big)^{1/2}.$$

Here U_t denotes the stationary diffusion for $\mathcal{L}_K$. The estimate passes to each subsequential two-time law of these bounded observations, with μ_d in place of μ_d^K. Such subsequences exist at every fixed t by compactness of the range closures of the two observations. In particular the two-time covariance of a nonconstant bounded gauge-invariant observable with itself has this exponential decay, and the mean square change is at least twice its limiting variance times $1 - e^{-t/(2C_{\mathrm{P}})}$.

18.21. **A slab with an infinite section, all the cubes at all labels, and a coupling that varies with the place.** The constructions use the Wilson law and the sourced frames of Definitions 13.1 and 13.2. Further notation is given in Definitions 18.198, 18.200 and 18.202. Proposition 18.199 concerns the slab $\mathbb{R}^2 \times (\mathbb{R}/l\mathbb{Z})$ whose thickness l is a side admitted at the reference coupling. At a fixed cutoff the Wilson measures of the periodic boxes $\Lambda \times \Lambda \times l$ are first taken to infinite section, $\Lambda \to \infty$; then the cutoff is removed along first-passage data. Along a subsequence the connected functions of the action density converge on the slab; the limit is invariant under the translations of the slab and reflection positive at the mirrors of all three coordinate directions; and when the supports lie in balls of a radius R, pairwise at a distance $\ge \mathsf{d}$, it obeys one bound of the shape of the growth bound of the main text, a number of d times the n-th power of a number of R times $(n!)^{5+\mathsf{p}}$ times the product of the norms $\|\cdot\|_\sharp$ of the n test functions; its two numbers are formed from the numbers of item (6) of Definition 18.198, and they depend neither on the cutoff nor on the section, the thickness or the state. The reconstruction in the periodic direction gives exactly one self-adjoint generator and an antiunitary involution that reverses it; the kernels of the limit at sharp heights are holomorphic in all the heights and of slow growth; and they define a state on an algebra of fields smeared in time which satisfies the condition of Kubo, Martin and Schwinger at the inverse temperature l, at imaginary and at real times. Proposition 18.201 bounds the limit cumulants $S_n^{(\ell,\mathsf{b})}$ on every periodic cube T_ℓ, $\ell = 2L^m$, $m \in \mathbb{Z}$, at every label $\mathsf{b} > 0$, by $\mathsf{b}^{-n}\mathsf{A}(\mathsf{d}/\mathsf{b})\hat{\mathsf{B}}(R/\mathsf{b})^n(n!)^{\varkappa_\star}\prod_q \max\{\sup|f_q|, \mathsf{b}\,\mathrm{Lip} f_q\}$, with two functions of d/b and of R/b alone and the exponent $\varkappa_\star$ of the main text; on the cubes that are small against the label the base is one number, and it tends to 0 like $(\ell/\mathsf{b})^{1/2}$ up to a power of a logarithm. Corollary YM3_I:119.2 supplies the full cutoff limit on each cube. Proposition 18.201 supplies the bound uniformly in ℓ/b. Proposition 18.203 concerns one torus on which the plaquette p carries the coupling $a_K^{-1}(\mathsf{b} + h(x(p)))$, with a real Lipschitz function h in the real source ball $\|h\|_\flat < 2\varrho$ of Theorem YM3_I:AD: the connected functions of the action density converge along all the cutoffs; for $\|h\|_\flat < r_1$ the limit is the series $\sum_k \frac{(-1)^k}{k!} S_{n+k}^{(\ell,\mathsf{b})}(\underline{f}, h, \dots, h)$, obtained by Taylor expansion at zero source; and at a constant h the limit is the limit cumulant at the label $\mathsf{b} + h$. Its new ingredient is a floor for the partition function at all the real sources of the class, uniform in the cutoff, drawn from the history without large fields.

Definition 18.198 (The slab; boxes over it; observables and tuples on it)**.** Fix a frame of Definition 13.1 with radius parameter ϱ, so its source ball has radius 2ϱ. Write $a_K = L^{-K}$, $\mathbb{L}_K = a_K\mathbb{Z}^3$ and $\mathfrak{L} = \{2L^m : m \geq 1\}$. A law is a Borel probability measure on the product of the bond configuration spaces. As in Definition 13.1, set $C_g = C_{\mathrm{st}} + \frac{9}{2}\log X_g$. For a Euclidean closed ball write $\mathbb{B}(y,r)$, and for compactly supported Lipschitz functions v_q put $\mathbb{D}_\varrho(\underline{v}) = \{z : \sum_q |z_q|\,\|v_q\|_\sharp < \varrho\}$, with $\|v_q\|_\sharp = \max\{\|v_q\|_\infty, 40M\,\mathrm{Lip}\,v_q\}$.

Let $0 < g \leq \gamma$, and let $l = 2L^{m_l} \in \mathfrak{L}(g)$ be a side admitted at g. The slab is $\mathbb{E} := \mathbb{R}^3/\mathbb{Z}le_3 = \mathbb{R}^2 \times (\mathbb{R}/l\mathbb{Z})$, with the projection $\mathrm{pr}\colon \mathbb{R}^3 \to \mathbb{E}$ and the quotient distance $|x-y|$; balls are closed.

(1) *Boxes.* For $K \in \mathbb{N}$, $\beta > 0$ and a side $\Lambda \in \mathfrak{L}$ with $\Lambda \geq l$ let $\Gamma_\Lambda := \Lambda\mathbb{Z} \times \Lambda\mathbb{Z} \times l\mathbb{Z}$. The law $\mu_{\Gamma_\Lambda,\beta}$ is the Wilson law on the corresponding rectangular quotient, periodically lifted to $\mathbb{L}_K$. For an unbounded set $\mathfrak{l}_0 \subset \mathfrak{L}$, $\mathcal{G}^{\mathrm{box}}_{K,\beta}(\mathfrak{l}_0)$ is the set of the weak limit points of $\mu_{\Gamma_\Lambda,\beta}$ as $\Lambda \to \infty$ in $\mathfrak{l}_0$. It is not empty, because the space of the configurations is compact and metrisable as a countable product of compact metrisable spaces and so the laws on it form a weakly compact set. Each of its members is carried by the closed set $\mathcal{V}^{\mathrm{sl}} := \{U : \tau_{le_3}U = U\}$.

(2) *The sequence.* (K_j, β_j), $j \in \mathbb{N}$, are first-passage data at g with $K_j \to \infty$ and labels $b_j = a_{K_j}\beta_j \to b$, and $\nu_j \in \mathcal{G}^{\mathrm{box}}_{K_j,\beta_j}(\mathfrak{L}(g))$.

(3) *Observables.* For a bounded function f on $\mathbb{E}$ with compact support, $\mathcal{O}^{\mathrm{sl}}_K(f)(U) := a_K^{-1}\sum_p f(\mathrm{pr}\,x(p))\,a_p(U)$ on $\mathcal{V}^{\mathrm{sl}}$, the sum running over one plaquette of every class modulo $\mathbb{Z}le_3$. For a bounded function $\tilde f$ on $\mathbb{R}^3$ with compact support, $\tilde f^{\langle l\rangle}(\mathrm{pr}\,x) := \sum_{k\in\mathbb{Z}} \tilde f(x + kle_3)$; then $\mathcal{O}_K(\tilde f) = \mathcal{O}^{\mathrm{sl}}_K(\tilde f^{\langle l\rangle})$ on $\mathcal{V}^{\mathrm{sl}}$. For a law ν carried by $\mathcal{V}^{\mathrm{sl}}$, $S^T_n[\nu](f_1,\dots,f_n)$ is the joint cumulant of $\mathcal{O}^{\mathrm{sl}}_K(f_1),\dots,\mathcal{O}^{\mathrm{sl}}_K(f_n)$ under ν, and $\sigma_\nu(X) := \nu(\prod_q(\mathcal{O}^{\mathrm{sl}}_K(f_q) - \nu(\mathcal{O}^{\mathrm{sl}}_K(f_q))))$ for $X = (f_1,\dots,f_n)$; by the moment–cumulant identity YM3_I:(1741), since centring removes the singleton blocks, $\sigma_\nu(X) = \sum_{\pi\in\mathrm{Part}_{\geq 2}(\{1,\dots,n\})}\prod_{J\in\pi} S^T_{|J|}[\nu](X_J)$.

(4) *Tuples.* For an open set $O \subset \mathbb{E}$, $\mathfrak{S}_n(O)$ is the set of the n-tuples of real functions of class $C^1_c(\mathbb{E})$ with pairwise disjoint supports in O; $\mathfrak{S}(O) = \bigcup_{n\geq 0}\mathfrak{S}_n(O)$, the empty tuple $\varnothing$ included; $\mathbb{C}\mathfrak{S}(O)$ is the set of the finite complex combinations, and $\bar F$ conjugates the coefficients, $X \cdot X'$ concatenates tuples when their joint supports are separated, X_J selects the indices in J, and a map r acts by $rX = (f_q \circ r^{-1})_q$, extended to combinations. A tuple is of the type $(\mathsf{d}; R)$ if its supports lie pairwise at the distance $\geq \mathsf{d}$ and each in a ball of radius R; $\|\cdot\|_\sharp$ is formed with the distance of $\mathbb{E}$, and $\|X\|_\sharp = \prod_q \|f_q\|_\sharp$. A family $S = (S_n)_{n\geq 2}$ of maps on the tuples has the centred functional $\sigma_S(X) := \sum_{\pi\in\mathrm{Part}_{\geq 2}}\prod_{J\in\pi} S_{|J|}(X_J)$, $\sigma_S(\varnothing) = 1$, extended linearly to combinations.

(5) *Maps.* τ_v, $v \in \mathbb{R}^3$, are the translations. For $\mu \in \{1,2,3\}$ and $c \in \mathbb{R}$, $\theta_{e_\mu,c}$ is the reflection of $\mathbb{R}^3$ that sends the coordinate x_μ to $2c - x_\mu$ and keeps the two others. When $c \in \frac{a_K}{2}\mathbb{Z}$, the plane $\Sigma(e_\mu, c) = \{x_\mu = c\}$ is a lattice mirror of $\mathbb{L}_K$, $\theta_{e_\mu,c}$ is its reflection and $\mathbb{H}^\pm(e_\mu, c) = \{\pm(x_\mu - c) \geq 0\}$ are its closed sides, and $\theta_{e_\mu,c}$ is an isometry of $\mathbb{L}_K$ because $2c$ is a lattice coordinate. The linear part of $\theta_{e_\mu,c}$ maps $\mathbb{Z}le_3$ onto itself, so $\theta_{e_\mu,c}$ induces a map of $\mathbb{E}$; for $\mu = 1,2$ this induced map bears the same name, and r_c stands both for $\theta_{e_3,c}$ and for the map $x_3 \mapsto 2c - x_3$ that it induces on $\mathbb{E}$. W_2 is the group of the eight signed permutations of (e_1, e_2), acting on the first two coordinates; $\mathsf{E}_2 := \mathbb{R}^2 \rtimes W_2$, acting on the first two coordinates.

(6) *Numbers.* Let C_Q, p and U be the constants of Proposition YM3_I:91.12 at g. Put $A^{\mathrm{per}}(\mathsf{d}) = 1 + \mathsf{U}(1,\mathsf{d})$. Use for C^{sl}_B the constant YM3_I:(1848) with $e^{2C_{\mathrm{S}}+C_g}C_Q$ in place of C_Q; $B^{\mathrm{sl}}(R) := C^{\mathrm{sl}}_B(R+1)^3$; and for $n \geq 2$, $\mathsf{d}, R > 0$,

$$C^{\mathrm{sl}}_n(\mathsf{d};R) := A^{\mathrm{per}}(\mathsf{d})\big[(1+40M)\,B^{\mathrm{sl}}(2R+1)\big]^n (n!)^{5+\mathsf{p}}, \qquad C^{\sigma}_n(\mathsf{d};R) := \sum_{\pi\in\mathrm{Part}_{\geq 2}(\{1,\dots,n\})}\prod_{J\in\pi} C^{\mathrm{sl}}_{|J|}(\mathsf{d};R).$$

■

Proposition 18.199 (The slab $\mathbb{R}^2\times(\mathbb{R}/l\mathbb{Z})$: limits with the section first, mirrors in three directions, a self-adjoint generator, a state at the inverse temperature l)**.** Let the data be as in Definition 18.198.

(a) Domination. Let $K\in\mathbb{N}$, $\beta>0$, $\mathfrak{l}_0\subset\mathfrak{L}$ unbounded, $\nu\in\mathcal{G}^{\mathrm{box}}_{K,\beta}(\mathfrak{l}_0)$, let (i,o) be a board of $\mathbb{L}_K$ of the side $\mathsf{l}=a_i$, $i\le K$ and $o\in a_K\mathbb{Z}^3$, with the cells and copies of Lemma 18.239, and let $\ell'\in\mathfrak{L}$ with $2\mathsf{l}\le\ell'\le l$. For every finite set $\mathsf{S}\subset\mathbb{Z}^3$ whose members are pairwise incongruent modulo $(l/\mathsf{l})\mathbb{Z}e_3$ and all continuous functions $F_{\mathbf{k}}$ of the bonds of the cells $\square_{\mathbf{k}}$, $\mathbf{k}\in\mathsf{S}$,

$$\Big|\nu\Big(\prod_{\mathbf{k}\in\mathsf{S}}F_{\mathbf{k}}\Big)\Big|\le\prod_{\mathbf{k}\in\mathsf{S}}q_{\mathsf{l}}^{\mathrm{sl}}\,\|F_{\mathbf{k}}\|_{\mathrm{ch},\ell'},\qquad q_{\mathsf{l}}^{\mathrm{sl}}:=e^{\mathsf{l}^3(p_{\ell'}-p_\infty)},\quad p_\infty:=\inf_{\ell\in\mathfrak{L}}p_\ell,$$

where $p_\ell=\ell^{-3}\log Z_{K,\ell,\beta}$ and the reflected-product seminorm at side ℓ' is that of Lemma 18.239. If (K,β) is a first-passage datum at g and $\ell'\in\mathfrak{L}(g)$, then $q_{\mathsf{l}}^{\mathrm{sl}}\le e^{\mathsf{l}^3(2C_{\mathrm{S}}+C_g)}$.

(b) Symmetries and mirrors of the box limits. Every $\nu\in\mathcal{G}^{\mathrm{box}}_{K,\beta}(\mathfrak{l}_0)$ is invariant under the translations by $a_K\mathbb{Z}^3$, under W_2 and under the reflections $\theta_{e_\mu,c}$, $\mu=1,2,3$, $c\in\frac{a_K}{2}\mathbb{Z}$. It is reflection positive at every lattice mirror $\Sigma(e_\mu,c)$, $\mu\in\{1,2\}$, $c\in\frac{a_K}{2}\mathbb{Z}$, on bounded local functions of bonds with both endpoints in the closed positive half-space; and $\nu(\overline{\theta_{e_3,c}F}\,F)\ge0$ for every $c\in\frac{a_K}{2}\mathbb{Z}$ and every continuous function F of finitely many bonds in $\{c\le x_3\le c+\frac{l}{2}\}$.

(c) Bounds. Let (K,β) be a first-passage datum at g and $\nu\in\mathcal{G}^{\mathrm{box}}_{K,\beta}(\mathfrak{L}(g))$. For $n\ge2$, $\mathsf{d},R>0$ and real Lipschitz functions $f_1,\dots,f_n$ on $\mathbb{E}$ of the type $(\mathsf{d};R)$,

$$|S_n^T[\nu](f_1,\dots,f_n)|\le C_n^{\mathrm{sl}}(\mathsf{d};R)\prod_q\|f_q\|_\sharp,\qquad|\sigma_\nu(f_1,\dots,f_n)|\le C_n^{\sigma}(\mathsf{d};R)\prod_q\|f_q\|_\sharp.$$

Both bounds remain true when ν is replaced by the measure $\mu_{\Gamma_\Lambda,\beta}$ of a finite box, for every $\Lambda\in\mathfrak{L}(g)$ with $\Lambda\ge l$ that is larger than d plus the Euclidean diameter of the set $\{(x_1,x_2):x\in\bigcup_q\mathrm{supp}f_q\}\subset\mathbb{R}^2$. The two numbers are built from g, the radius ϱ and the constants; they depend neither on K, β, l, Λ, ν nor on the positions of the supports, and they do not increase in d.

(d) Limits with the section first. There is a subsequence of (j) along which $S_n^T[\nu_j](X)$ converges for every $n\ge2$ and every $X\in\mathfrak{S}_n(\mathbb{E})$. Every limit S, taken along a subsequence on which all these numbers converge, is real, symmetric and multilinear; it obeys the two bounds of (c); and it is invariant under all the translations of $\mathbb{E}$, under W_2 and under every r_c, $c\in\mathbb{R}$. If $X^{(j)}\to X$ member by member in $\|\cdot\|_\sharp$, the supports lying in one compact set and, at each j, pairwise at the distance $\ge\mathsf{d}>0$, then $\sigma_{\nu_j}(X^{(j)})\to\sigma_S(X)$ along the subsequence.

(e) Mirrors of the limit in the three directions. For every such S, every real c and every $F\in\mathbb{C}\mathfrak{S}(H_+)$, $\sigma_S(\theta\bar F\cdot F)\ge0$ for $(\theta,H_+)=(\theta_{e_\mu,c},\{x_\mu>c\})$, $\mu=1,2$, and for $(\theta,H_+)=(\mathsf{r}_c,\{c<x_3<c+\frac{l}{2}\})$.

(f) The generator and the involution. For an open interval $I\subset(0,\frac{l}{2})$ put $\mathcal{D}_I:=\mathbb{C}\mathfrak{S}(\{x_3\in I\})$, and $\mathcal{D}:=\mathcal{D}_{(0,l/2)}$. The form $\langle F,G\rangle:=\sigma_S(\mathsf{r}_0\bar F\cdot G)$ is positive on $\mathcal{D}$; it gives a Hilbert space $\mathcal{H}^{\mathrm{sl}}$, classes $[F]$ and $\Omega^{\mathrm{sl}}:=[\varnothing]$. For $0<\mathsf{a}<\mathsf{a}'<\frac{l}{2}$ and $F\in\mathcal{D}_{(\mathsf{a},\mathsf{a}')}$ there is exactly one finite positive Borel measure ρ_F on $\mathbb{R}$ with $\sigma_S(\mathsf{r}_0\bar F\cdot\tau_{re_3}F)=\int e^{-r\lambda}\,d\rho_F(\lambda)$ for $-2\mathsf{a}<r<l-2\mathsf{a}'$. There is exactly one self-adjoint operator H_3 in $\mathcal{H}^{\mathrm{sl}}$ such that $[F]\in\mathrm{Dom}(e^{-r\mathsf{H}_3})$ and $e^{-r\mathsf{H}_3}[F]=[\tau_{re_3}F]$ for all such a,a',F and all $-\mathsf{a}<r<\frac{l}{2}-\mathsf{a}'$. Then $\mathsf{H}_3\Omega^{\mathrm{sl}}=0$; $z\mapsto e^{-z\mathsf{H}_3}[F]$ is holomorphic on $-\mathsf{a}<\mathrm{Re}\,z<\frac{l}{2}-\mathsf{a}'$; and ρ_F is the spectral measure of H_3 in $[F]$. There is exactly one antiunitary operator J^{sl} with $J^{\mathrm{sl}}[F]=[\mathsf{r}_{l/4}\overline{F}]$; $(J^{\mathrm{sl}})^2=1$, $J^{\mathrm{sl}}\Omega^{\mathrm{sl}}=\Omega^{\mathrm{sl}}$ and $J^{\mathrm{sl}}\mathsf{H}_3J^{\mathrm{sl}}=-\mathsf{H}_3$, so that $J^{\mathrm{sl}}e^{-s\mathsf{H}_3}J^{\mathrm{sl}}=e^{s\mathsf{H}_3}$ and the spectrum of H_3 is symmetric about 0. The group E_2 acts by unitary operators $[F]\mapsto[\mathsf{g}F]$ that fix Ω^{sl}, commute with $e^{it\mathsf{H}_3}$ and with J^{sl}, and are strongly continuous on the translations.

(g) Kernels at sharp heights. For $\chi\in C^1_c(\mathbb{R})$ and $h\in C^1_c(\mathbb{R}^2)$ put $(\chi\otimes h)(\mathrm{pr}\,x):=h(x_1,x_2)\sum_{k\in\mathbb{Z}}\chi(x_3+kl)$; let $\mathcal{U}_n:=\{s\in\mathbb{R}^n:s_k-s_{k'}\notin l\mathbb{Z}\text{ for }k\ne k'\}$ and $\mathrm{gap}(s):=$

$\min_{k\neq k'}\operatorname{dist}(s_k-s_{k'},l\mathbb{Z})$. For every $n\geq 1$ and $h=(h_1,\dots,h_n)\in C^1_c(\mathbb{R}^2)^n$ there is exactly one continuous function $\mathfrak{K}_n(\cdot;h)$ on $\mathcal{U}_n$ with

$$\sigma_S(\chi_1\otimes h_1,\dots,\chi_n\otimes h_n)=\int\prod_k\chi_k(s_k)\,\mathfrak{K}_n(s;h)\,ds$$

whenever $\prod_k\operatorname{supp}\chi_k\subset\mathcal{U}_n$ and each $\operatorname{supp}\chi_k$ lies in an interval shorter than l. It is the restriction of a function holomorphic on $\mathcal{U}_n+i\mathbb{R}^n$; it is multilinear in h, symmetric under the common permutations of the pairs (z_k,h_k), of the period l in each z_k, invariant under $z\mapsto z+c\mathbf{1}$ $(c\in\mathbb{C})$, under $z\mapsto -z$ and under $h\mapsto \mathsf{g}h$, $\mathsf{g}\in\mathsf{E}_2$, and real at real points; $\mathfrak{K}_1=0$. If $n\geq 2$ and every $\operatorname{supp}h_k$ lies in a disc of radius R, then

$$|\mathfrak{K}_n(z;h)|\leq \mathsf{c}_n(R,l)\,\min\{1,\operatorname{gap}(\operatorname{Re}z)\}^{-\varkappa_{\mathrm{w}}n}\prod_k\|h_k\|_\sharp,\qquad \varkappa_{\mathrm{w}}:=\tfrac{21}{2}+\mathsf{p},$$

with a number $\mathsf{c}_n(R,l)$ built from n, R, l, g, the radius ϱ and the constants. Continued complex-multilinearly in h, with $\mathfrak{K}_0:=1$, the kernel

$$((z,h),(z',h'))\mapsto \mathfrak{K}_{n+n'}(-\bar z_n,\dots,-\bar z_1,z'_1,\dots,z'_{n'};\bar h_n,\dots,\bar h_1,h'_1,\dots,h'_{n'})$$

is of positive type on the set of the pairs (z,h), $n\geq 0$, $0<\operatorname{Re}z_1<\dots<\operatorname{Re}z_n<\frac{l}{2}$, $h\in C^1_c(\mathbb{R}^2;\mathbb{C})^n$.

(h) The state and the condition at the inverse temperature l. Let $\mathscr{G}$ be the space of the entire functions ϕ with $\sup_{|\operatorname{Im}t|\leq c}(1+|t|)^N|\phi(t)|<\infty$ for all $c,N\geq 0$; $\phi^*(t):=\overline{\phi(\bar t)}$ and $(\alpha_c\phi)(t):=\phi(t-c)$, $c\in\mathbb{C}$. Let $\mathfrak{A}^{\mathrm{sl}}$ be the tensor algebra over $\mathscr{G}\otimes C^1_c(\mathbb{R}^2;\mathbb{C})$ with the generators $\Phi(\phi,h)$, the involution $\Phi(\phi,h)^*:=\Phi(\phi^*,\bar h)$, the automorphisms $\alpha_c\Phi(\phi,h):=\Phi(\alpha_c\phi,h)$, which are $*$-automorphisms for real c, and $\mathsf{g}\Phi(\phi,h):=\Phi(\phi,\mathsf{g}h)$, $\mathsf{g}\in\mathsf{E}_2$. Put $\mathfrak{R}_n(t;h):=\mathfrak{K}_n(-it;h)$ for $\operatorname{Im}t_1<\dots<\operatorname{Im}t_n<\operatorname{Im}t_1+l$. Then

$$\omega^{\mathrm{sl}}(\Phi(\phi_1,h_1)\cdots\Phi(\phi_n,h_n)):=\int_{\mathbb{R}^n}\prod_k\phi_k(t_k+is_k)\,\mathfrak{R}_n(t+is;h)\,dt\qquad(s_1<\dots<s_n<s_1+l),\qquad\omega^{\mathrm{sl}}(1):=1,$$

does not depend on s and defines a state on $\mathfrak{A}^{\mathrm{sl}}$, invariant under α_t, $t\in\mathbb{R}$, and under E_2. For $A,B\in\mathfrak{A}^{\mathrm{sl}}$ the function $c\mapsto\omega^{\mathrm{sl}}(A\,\alpha_c(B))$ is entire and bounded on every strip $|\operatorname{Im}c|\leq c_1$, and

$$\omega^{\mathrm{sl}}(A\,\alpha_{t+il}(B))=\omega^{\mathrm{sl}}(\alpha_t(B)\,A)\qquad(t\in\mathbb{R}).$$

The connected functions are the truncated functions of ω^{sl} at imaginary times: $S_n(\chi_1\otimes h_1,\dots,\chi_n\otimes h_n)=\int\prod_k\chi_k(s_k)\,\mathfrak{R}^T_n(is;h)\,ds$ when $\operatorname{supp}\chi_1,\dots,\operatorname{supp}\chi_n$ follow one another in an interval shorter than l, $\mathfrak{R}^T_n$ being formed from the $\mathfrak{R}_m$ by the formula that expresses cumulants through moments. There is a linear map $\Psi^{\mathrm{sl}}\colon\mathfrak{A}^{\mathrm{sl}}\to\mathcal{H}^{\mathrm{sl}}$ with $\Psi^{\mathrm{sl}}(1)=\Omega^{\mathrm{sl}}$, $\omega^{\mathrm{sl}}(B^*A)=\langle\Psi^{\mathrm{sl}}(B),\Psi^{\mathrm{sl}}(A)\rangle$, $e^{it\mathsf{H}_3}\Psi^{\mathrm{sl}}(A)=\Psi^{\mathrm{sl}}(\alpha_tA)$, $[\mathsf{g}\cdot]\Psi^{\mathrm{sl}}(A)=\Psi^{\mathrm{sl}}(\mathsf{g}A)$ for the unitary operators of (f), and $J^{\mathrm{sl}}e^{-\frac{l}{2}\mathsf{H}_3}\Psi^{\mathrm{sl}}(A)=\Psi^{\mathrm{sl}}(A^*)$. So the closure of $\Psi^{\mathrm{sl}}(\mathfrak{A}^{\mathrm{sl}})$, with Ψ^{sl} and Ω^{sl}, is the space of the state, H_3 implements α_t on it, and the last relation is the modular relation with the modular operator $e^{-l\mathsf{H}_3}$.

(i) Real times. Let $\mathfrak{A}^{\mathrm{sl}}_{\mathcal{S}}\supset\mathfrak{A}^{\mathrm{sl}}$ be the tensor algebra over $\mathcal{S}(\mathbb{R})\otimes C^1_c(\mathbb{R}^2;\mathbb{C})$, with the same involution, the α_t for real t and E_2. For $n\geq 1$, h, $G\in\mathcal{S}(\mathbb{R}^n)$ and $s^*\in\mathbb{R}^n$ with $s^*_1<\dots<s^*_n$ the limit

$$\mathfrak{R}^0_n(G;h):=\lim_{\varepsilon\downarrow 0}\int G(t)\,\mathfrak{R}_n(t+i\varepsilon s^*;h)\,dt$$

exists and does not depend on s^*; $\mathfrak{R}^0_n(\cdot;h)$ is a tempered distribution of an order $\leq\lceil\varkappa_{\mathrm{w}}n\rceil+1$. The functional $\tilde\omega^{\mathrm{sl}}(\Phi(\phi_1,h_1)\cdots\Phi(\phi_n,h_n)):=\mathfrak{R}^0_n(\phi_1\otimes\cdots\otimes\phi_n;h)$ is a state on $\mathfrak{A}^{\mathrm{sl}}_{\mathcal{S}}$ that extends ω^{sl} and is invariant under α_t and E_2; and for $A,B\in\mathfrak{A}^{\mathrm{sl}}_{\mathcal{S}}$ there is a function, continuous and bounded on $0\leq\operatorname{Im}c\leq l$ and holomorphic inside, equal to $\tilde\omega^{\mathrm{sl}}(A\,\alpha_t(B))$ at $c=t$ and to $\tilde\omega^{\mathrm{sl}}(\alpha_t(B)\,A)$ at $c=t+il$. The map Ψ^{sl} extends to $\mathfrak{A}^{\mathrm{sl}}_{\mathcal{S}}$ with all the rules of (h); in particular $\Psi^{\mathrm{sl}}(A)\in\operatorname{Dom}(e^{-\frac{l}{2}\mathsf{H}_3})$. ■

Sketch. Step 1: the rectangular comparison (a). If the cell indices are distinct modulo $(l/\mathsf{l})\mathbb{Z}e_3$, they are also distinct on every sufficiently large rectangular box. By Lemma 18.239(a), (b), the box expectation of their product is bounded by the product of the cubic seminorms times one factor $e^{\mathsf{l}^3(p_{\ell'}-p_\Lambda)}$ per cell. Since $p_\Lambda \geq p_\infty$, this bound is uniform over the defining sequence. The products are continuous local functions, so it passes to ν. The same comparison applies on each finite box whose cell indices are already distinct.

Step 2: the number at g. The source stability and vacuum-energy comparison at first passage give $p_{\ell'} - p_\infty \leq 2C_{\mathrm{S}} + C_g$, as in Lemma 18.239(d), with the frame constants of Definition 13.1.

Step 3: the symmetries and mirrors (b). The finite box is invariant under translations by $a_K\mathbb{Z}^3$, under W_2, and under the listed reflections. Each operation permutes its oriented bonds and preserves the plaquette actions and product Haar measure, so the invariances pass to weak limits. The finite-box reflection form at each of the listed double mirrors is positive by Proposition YM3_I:91.7. For $\mu = 1, 2$, any finite set of half-space bonds lies in the positive half-box for sufficiently large Λ. For $\mu = 3$ the two mirrors stay a distance $l/2$ apart. This gives positivity of the limiting form on continuous local functions, and hence on bounded Borel local functions by the L^2 approximation argument of Lemma 18.242. This argument uses only that the defining sides diverge in $\mathfrak{L}$.

Step 4: the bounds (c). For ν apply Proposition 18.244(a), (c), (d), using the same datum, thickness and constants. To see that the same proof applies on a finite box, lift each entry as in (a). The horizontal projections of the lifted supports equal those of the original supports. Thus the support separation modulo Γ_Λ is at least d if $\Lambda > \mathsf{d}$ plus the horizontal diameter in (c). On the fine mesh split the lifts into pieces as in the proof of Proposition 18.244(d). The pieces keep that separation. Place each tuple of pieces on the side-one board and move its occupied cells into period representatives by translations in Γ_Λ. The periods are even integer multiples of the cell side. Translation by a period preserves the observables and the cell seminorms, and different representative cells are incongruent. Apply the finite-box comparison from Step 1 to the product of exponentials in the occupied cells. The graded normalizer from Proposition YM3_I:91.12, with the reference side $\ell_g = \min \mathfrak{L}(g) \leq l$, transfers with base $e^{2C_{\mathrm{S}}+C_g} C_Q (2\eta)^{-\mathsf{p}}$ as in the slab argument, where η is the radius of the pieces. Its mixed derivative uses the same distance bound and YM3_I:(1845). The graded cumulant estimate YM3_I:(1842), the sum over pieces, and the coarse-mesh plaquette count thus have the same constants. The moment estimate follows by the partition formula of Definition 18.198(3). Monotonicity in d follows from YM3_I:(1785).

Step 5: the limits (d). $C^1_c(\mathbb{E})$ contains a countable set $\mathcal{Q}$ such that for every $f \in C^1_c(\mathbb{E})$ and $\varepsilon > 0$ some $f' \in \mathcal{Q}$ has $\|f - f'\|_\sharp < \varepsilon$ and its support in the ε-neighbourhood of $\mathrm{supp} f$ (C^1 of a compact box is separable; cut-offs with rational data). By (c) and a diagonal choice there is a subsequence along which $S^T_n[\nu_j](X')$ converges at every separated tuple X' with members in $\mathcal{Q}$. For X of the type $(\mathsf{d}; R)$ and X' with members in $\mathcal{Q}$, $\|f_i - f'_i\|_\sharp < \varepsilon < \min\{\frac{\mathsf{d}}{4}, 1\}$ and $\mathrm{supp} f'_i$ in the ε-neighbourhood of $\mathrm{supp} f_i$, the tuples of the telescope between X and X' are of the type $(\frac{\mathsf{d}}{2}; R+1)$, and by multilinearity and (c), $|S^T_n[\nu_j](X) - S^T_n[\nu_j](X')| \leq n\,\varepsilon\, C^{\mathsf{sl}}_n(\frac{\mathsf{d}}{2}; R+1) \prod_i (\|f_i\|_\sharp + 1)$ for all j; so $(S^T_n[\nu_j](X))_j$ is a Cauchy sequence. For the continuity in the last sentence of (d) let k be the compact set: if $\liminf_j \mathrm{dist}(\mathrm{supp} f^{(j)}_k, \mathrm{supp} f_m) < \mathsf{d}$ for some $k \neq m$, there are $\epsilon > 0$ and, along a subsequence, $y_j \in \mathrm{supp} f^{(j)}_k$ and $x_j \in \mathrm{supp} f_m$ with $|y_j - x_j| \leq \mathsf{d} - \epsilon$, $x_j \to x$; a point x' with $f_m(x') \neq 0$ and $|x' - x| < \frac{\epsilon}{4}$ lies in $\mathrm{supp} f^{(j)}_m$ for large j by the uniform convergence, so $|y_j - x'| \geq \mathsf{d}$, while $|y_j - x'| < \mathsf{d}$: a contradiction. Hence for large j all the tuples of the telescope between $X^{(j)}$ and X are of the type $(\frac{\mathsf{d}}{2}; R')$, R' the radius of a ball that contains k, and (c) gives $\sigma_{\nu_j}(X^{(j)}) - \sigma_{\nu_j}(X) \to 0$. Reality, symmetry, multilinearity and the bounds hold at every j, cumulants being multilinear and symmetric, and pass to the limit. For a translation τ_v or a mirror r_c take $v_j \in a_{K_j}\mathbb{Z}^3$, $v_j \to v$, and $c_j \in \frac{1}{2}a_{K_j}\mathbb{Z}$, $c_j \to c$; by (b) and the relabelling of the plaquette sums under the isometries,

$S_n^T[\nu_j](\tau_{v_j}X) = S_n^T[\nu_j](X)$ and the same for θ_{e_3,c_j}; as ∇f is uniformly continuous, $\tau_{v_j}X \to \tau_v X$ in the sense of the continuity just shown. The invariance under W_2 holds at every j.

Step 6: the mirrors of S (e). Let $F = \sum_\alpha c_\alpha X_\alpha$, the supports in H_+ at the distance $\geq 3\epsilon > 0$ from its boundary; let n_0 be the normal (e_μ or e_3) and c_j lattice offsets with $c_j \to c$, $\theta^{(j)}$ the reflection at c_j, and $F_j := \sum_\alpha c_\alpha X_\alpha^{\nu_j}$ the combination of the centred products under ν_j. A plaquette whose barycentre lies in a support has its corners within a_{K_j} of it; so for large j, F_j is a continuous function of finitely many bonds of the closed half space at c_j, respectively of $\{c_j \leq x_3 \leq c_j + \frac{l}{2}\}$. By (b), ν_j is invariant under $\theta^{(j)}$, so $\theta^{(j)}(X^{\nu_j}) = (\theta^{(j)}X)^{\nu_j}$, and $0 \leq \nu_j(\overline{\theta^{(j)}F_j}\, F_j) = \sum_{\alpha,\alpha'} \bar{c}_\alpha c_{\alpha'} \sigma_{\nu_j}(\theta^{(j)}X_\alpha \cdot X_{\alpha'})$. As $\theta^{(j)} = \tau_{v_j}\theta$ with $v_j \to 0$, the continuity of (d) gives $\sigma_S(\theta\bar{F} \cdot F) \geq 0$.

Step 7: one box in the direction e_3. Fix j and $\Lambda \in \mathfrak{L}(g)$, $\Lambda \geq l$; write $a = a_{K_j}$, $\beta = \beta_j$, $N := l/a = 2L^{m_l+K_j}$, $N' := \frac{N}{2}$, $\mu_\Lambda := \mu_{\Gamma_\Lambda,\beta}$, and Z for its partition function. On the horizontal layer Hilbert space use the positive crossing operator K of Proposition YM3_I:85.1 and the slab operators $\mathfrak{S}[\Phi]$, $\mathfrak{Y}[\Phi] = \mathsf{K}^{1/2}\mathfrak{S}[\Phi]\mathsf{K}^{1/2}$ of YM3_I:(1744). Let M_1 be the multiplication by $e^{-\beta A_\Sigma}$ on the space of one layer, and $\hat{\mathsf{T}} := \mathsf{K}^{1/2}\mathsf{M}_1\mathsf{K}^{1/2} \geq 0$. Splitting the outer layers off the weight preceding YM3_I:(1744) and integrating gives $\mathfrak{S}[\Phi^{1,0}] = \mathsf{M}_1\mathsf{K}\,\mathfrak{S}[\Phi]$ and $\mathfrak{S}[\Phi^{0,1}] = \mathfrak{S}[\Phi]\,\mathsf{K}\mathsf{M}_1$, where $\Phi^{x',x}$ is Φ evaluated on the layers $x+1,\dots,x+\mathsf{v}$ of a slab of $\mathsf{v}+x+x'$ layers; hence $\mathfrak{Y}[\Phi^{x',x}] = \hat{\mathsf{T}}^{x'}\mathfrak{Y}[\Phi]\hat{\mathsf{T}}^{x}$, and $\mathfrak{Y}[1_2] = \hat{\mathsf{T}}^2$ for the function 1 on two layers. By YM3_I:(1744), $\hat{\mathsf{T}}^2 = \mathfrak{Y}[1_2]$ is Hilbert–Schmidt, so $\hat{\mathsf{T}}$ is compact: $\hat{\mathsf{T}} = \sum_i t_i |e_i\rangle\langle e_i|$ with an orthonormal basis (e_i) and $t_i \geq 0$. Put $\hat{\mathsf{T}}^\zeta := \sum_{t_i>0} t_i^\zeta |e_i\rangle\langle e_i|$ for $\operatorname{Re}\zeta > 0$ and for $\zeta \in i\mathbb{R}$; $\|\hat{\mathsf{T}}^\zeta\| \leq \|\hat{\mathsf{T}}\|^{\operatorname{Re}\zeta}$, $\zeta \mapsto \hat{\mathsf{T}}^\zeta$ is holomorphic in the operator norm on $\operatorname{Re}\zeta > 0$, and every $\mathfrak{Y}[\Phi]$ vanishes on $\ker\hat{\mathsf{T}}$ and maps into its orthogonal complement, as $\hat{\mathsf{T}}\psi = 0$ forces $\mathsf{K}^{1/2}\psi = 0$. Take an integer $r \geq 2$ and integers $\mathsf{v}_1,\dots,\mathsf{v}_r \geq 2$. For bounded Borel functions $\Psi^1,\dots,\Psi^r$ on slab spaces of $\mathsf{v}_1,\dots,\mathsf{v}_r$ layers, the function $\zeta \mapsto \operatorname{Tr}(\mathfrak{Y}[\Psi^r]\hat{\mathsf{T}}^{\zeta_r}\cdots\mathfrak{Y}[\Psi^1]\hat{\mathsf{T}}^{\zeta_1})$ is holomorphic on $\{\operatorname{Re}\zeta_k > 0\}$, two of its factors being Hilbert–Schmidt operators; at integers $\zeta_k \geq 0$ with $\sum_k(\mathsf{v}_k+\zeta_k) = N$ it is the integral against the Wilson weight of the box of the product of the functions placed upwards, with the gaps ζ_k, around the periodic direction by Proposition YM3_I:85.2 and the padding identity for $\Phi^{x',x}$. Let $0 < \mathsf{a} < \mathsf{a}' < \frac{l}{2}$, $F = \sum_\alpha c_\alpha X_\alpha \in \mathcal{D}_{(\mathsf{a},\mathsf{a}')}$, and $F_\Lambda := \sum_\alpha c_\alpha X_\alpha^{\mu_\Lambda}$. With $x := \lfloor \mathsf{a}/a \rfloor - 2$, $x' := \lfloor (\frac{l}{2} - \mathsf{a}')/a \rfloor - 2$ and $\mathsf{v} := N' - x - x'$, for large j the function F_Λ is a function Ψ on the slab space of v layers placed on the layers $x+1,\dots,x+\mathsf{v}$; put $\mathsf{Y}_\Lambda := \mathfrak{Y}[\Psi]$. The reflection $\theta^{(j)} := \theta_{e_3,a/2} = \tau_{ae_3}\mathsf{r}_0$, the translation of layer indices, the adjoint identity YM3_I:(1745), and the two-slab trace formula give, for integers $k_1, k_2 \in [-x, x']$ and $k = k_1 + k_2$,

$$Z\,\mu_\Lambda\big(\overline{\theta^{(j)}\tau_{k_1ae_3}F_\Lambda}\,\tau_{k_2ae_3}F_\Lambda\big) = \operatorname{Tr}\big(\hat{\mathsf{T}}^{x'-k_2}\mathsf{Y}_\Lambda\hat{\mathsf{T}}^{x+k_2}\hat{\mathsf{T}}^{x+k_1}\mathsf{Y}_\Lambda^*\hat{\mathsf{T}}^{x'-k_1}\big) = \sum_{t_i,t_{i'}>0} t_i^{2x'-k}\,t_{i'}^{2x+k}\,|\langle e_i, \mathsf{Y}_\Lambda e_{i'}\rangle|^2.$$

By the invariance of μ_Λ under $a\mathbb{Z}^3$ and $\theta^{(j)}$, and $\mathsf{r}_0\tau_v = \tau_{-v}\mathsf{r}_0$ for $v \parallel e_3$, the left member is $Z\sigma_{\mu_\Lambda}(\mathsf{r}_0\bar{F} \cdot \tau_{(k-1)ae_3}F)$. So, with the finite positive measure $\rho_{\Lambda,j} := Z^{-1}\sum t_i^{2x'}t_{i'}^{2x}|\langle e_i, \mathsf{Y}_\Lambda e_{i'}\rangle|^2\,\delta_{\lambda_{ii'}}$, $\lambda_{ii'} := a^{-1}\log(t_i/t_{i'})$,

$$\sigma_{\mu_\Lambda}(\mathsf{r}_0\bar{F} \cdot \tau_{(k-1)ae_3}F) = \int e^{-ka\lambda}\,d\rho_{\Lambda,j}(\lambda) \qquad (k \in \mathbb{Z},\ -2x \leq k \leq 2x').$$

Step 8: the measures ρ_F. Let $-2\mathsf{a} < r_- < 0 < r_+ < l - 2\mathsf{a}'$. For $r \in [r_-, r_+]$ the tuples $\mathsf{r}_0X_\alpha \cdot \tau_{re_3}X_{\alpha'}$ are of one type; by (c) the left members of the last display are bounded by a number C_F for all large Λ, j and all k with $(k-1)a \in [r_-, r_+]$. With $k_\pm$ the extreme such k, the measures $(e^{-k_-a\lambda} + e^{-k_+a\lambda})\,d\rho_{\Lambda,j}$ have masses $\leq 2C_F$. Helly's selection theorem, applied first to the limit $\Lambda \to \infty$ taken along the sequence defining ν_j and next to the limit $j \to \infty$ taken along the subsequence of S, provides a further subsequence that converges vaguely. When $(k^{(j)}-1)a \to r \in (r_-, r_+)$, the functions $e^{-k^{(j)}a\lambda}/(e^{-k_-a\lambda} + e^{-k_+a\lambda})$ tend to $e^{-r\lambda}/(e^{-r_-\lambda} + e^{-r_+\lambda})$ uniformly on $\mathbb{R}$, and this limit is continuous and vanishes at $\pm\infty$; by weak convergence at fixed j and by (d), the left members

tend to $\sigma_S(\mathsf{r}_0\bar F\cdot\tau_{re_3}F)$. So the formula of (f) holds with $\rho_F:=(e^{-r_-\lambda}+e^{-r_+\lambda})^{-1}d(\lim)$. If two finite positive measures have equal integrals of $e^{-r\lambda}$ on an interval, both members are holomorphic in a strip over it, and the Fourier transforms of $e^{-r_0\lambda}\,d\rho$ and $e^{-r_0\lambda}\,d\rho'$ agree; so the measure, with mass $\sigma_S(\mathsf{r}_0\bar F\cdot F)$, is independent of the subsequences. Exhausting the full open interval by nested intervals containing zero gives the same measure throughout.

Step 9: the generator and the involution (f). By (e) at $c=0$ the form is positive; $\mathcal{H}^{\mathrm{sl}}$ is the completion of the quotient by the null vectors, and $D_I:=\{[F]:F\in\mathcal{D}_I\}$, $D:=\bigcup_I D_I$, is dense. Fix $I=(\mathsf{a},\mathsf{a}')$ and put $\mathrm{St}_I:=\{z:-\mathsf{a}<\operatorname{Re}z<\frac{l}{2}-\mathsf{a}'\}$. For $F,G\in\mathcal{D}_I$ and real r,r' in St_I, the invariance under translations and $\tau_{-re_3}\mathsf{r}_0=\mathsf{r}_0\tau_{re_3}$ give $\langle\tau_{r'e_3}G,\tau_{re_3}F\rangle=\sigma_S(\mathsf{r}_0\bar G\cdot\tau_{(r+r')e_3}F)$; so by Step 8, $\|\sum_a c_a[\tau_{ra e_3}F]\|^2=\int|\sum_a c_a e^{-r_a\lambda}|^2d\rho_F$. Hence $\mathsf{S}(r)[F]:=[\tau_{re_3}F]$ is well defined and linear on D_I, and there is a linear isometry I_F of the closed span of the functions $e^{-r\,\cdot}$ in $L^2(\rho_F)$ into $\mathcal{H}^{\mathrm{sl}}$ with $\mathsf{I}_F(e^{-r\,\cdot})=\mathsf{S}(r)[F]$. For $z\in\mathrm{St}_I$, $e^{-z\,\cdot}$ lies in that span (a function orthogonal to all $e^{-r\,\cdot}$ gives a holomorphic function that vanishes on an interval) and depends holomorphically on z; $\mathsf{S}(z)[F]:=\mathsf{I}_F(e^{-z\,\cdot})$ is the holomorphic continuation of $r\mapsto\mathsf{S}(r)[F]$, linear on D_I, and unchanged when I is enlarged. For $\psi,\psi'\in D_I$ the function $\mathsf{h}(z',z):=\langle\mathsf{S}(\bar z')\psi',\mathsf{S}(z)\psi\rangle$ is holomorphic on St_I^2 and at real points depends on $z+z'$ alone; so $(\partial_z-\partial_{z'})\mathsf{h}=0$, h is constant on the segments $(z'+c,z-c)$, and $\langle\mathsf{S}(it)\psi',\mathsf{S}(it)\psi\rangle=\langle\psi',\psi\rangle$: $\mathsf{U}_t:=\mathsf{S}(it)$ extends to an isometry. For small real r the real translation law gives $\mathsf{S}(u)\mathsf{S}(r)\psi=\mathsf{S}(u+r)\psi$ for real u near zero, using an interval containing both supports. Continue in u to it to obtain $\mathsf{U}_t\mathsf{S}(r)\psi=\mathsf{S}(r+it)\psi$. Both sides then continue holomorphically in r to $z\in\mathrm{St}_I$; at $z=it'$ this gives $\mathsf{U}_t\mathsf{U}_{t'}=\mathsf{U}_{t+t'}$ on D; with $\mathsf{U}_0=1$ and the continuity on D, (U_t) is a strongly continuous unitary group, and by Stone's theorem $\mathsf{U}_t=e^{-it\mathsf{H}_3}$ with a self-adjoint H_3. If a self-adjoint A, a vector ψ and a function Ξ holomorphic on an open strip $\{r_1<\operatorname{Re}z<r_2\}\ni 0$ satisfy $\Xi(it)=e^{-it\mathsf{A}}\psi$, then for the spectral projection P_m of $[-m,m]$ the entire function $\langle e^{-\bar z\mathsf{A}}\chi,\psi\rangle$ and $\langle\chi,\Xi(z)\rangle$, $\chi\in\mathsf{P}_m\mathcal{H}^{\mathrm{sl}}$, agree on $i\mathbb{R}$, so $\mathsf{P}_m\Xi(r)=e^{-r\mathsf{A}}\mathsf{P}_m\psi$; letting $m\to\infty$, $\psi\in\mathrm{Dom}(e^{-r\mathsf{A}})$ and $e^{-r\mathsf{A}}\psi=\Xi(r)$ for real r of the strip. With $\Xi=\mathsf{S}(\cdot)[F]$ this gives the domain and the action of $e^{-r\mathsf{H}_3}$, the holomorphy, and, near $r=0$, that ρ_F is the spectral measure in $[F]$; a second self-adjoint operator with the property has the same unitary group on D, so H_3 is unique; $\tau_v\varnothing=\varnothing$ gives $\mathsf{H}_3\Omega^{\mathrm{sl}}=0$. For J^{sl}: $\mathsf{r}_{l/4}$ keeps $\{0<x_3<\frac{l}{2}\}$, $\mathsf{r}_{l/4}\mathsf{r}_0\mathsf{r}_{l/4}=\mathsf{r}_0$ on $\mathbb{E}$ (because $x_3\mapsto l-x_3$ is $x_3\mapsto-x_3$ modulo l), and $\mathsf{r}_{l/4}\tau_{re_3}=\tau_{-re_3}\mathsf{r}_{l/4}$; by the invariance of σ_S under $\mathsf{r}_{l/4}$ and r_0 and its symmetry, $\langle\overline{\mathsf{r}_{l/4}F},\overline{\mathsf{r}_{l/4}G}\rangle=\langle G,F\rangle$. So $J^{\mathrm{sl}}[F]:=[\overline{\mathsf{r}_{l/4}F}]$ is well defined and antiunitary, $(J^{\mathrm{sl}})^2=1$, $J^{\mathrm{sl}}\Omega^{\mathrm{sl}}=\Omega^{\mathrm{sl}}$, and $J^{\mathrm{sl}}\mathsf{S}(r)=\mathsf{S}(-r)J^{\mathrm{sl}}$ on D_I; the functions $z\mapsto J^{\mathrm{sl}}\mathsf{S}(\bar z)\psi$ and $z\mapsto\mathsf{S}(-z)J^{\mathrm{sl}}\psi$ are holomorphic and agree at real z, so $J^{\mathrm{sl}}e^{it\mathsf{H}_3}=e^{it\mathsf{H}_3}J^{\mathrm{sl}}$, and as J^{sl} is antiunitary, $J^{\mathrm{sl}}\mathsf{H}_3J^{\mathrm{sl}}=-\mathsf{H}_3$; the functional calculus gives the rest. An element of E_2 commutes with r_0, $\mathsf{r}_{l/4}$ and τ_{re_3}, keeps $\{0<x_3<\frac{l}{2}\}$ and leaves σ_S invariant (d); so $[F]\mapsto[\mathsf{g}F]$ is unitary, fixes Ω^{sl}, commutes with J^{sl} and with $\mathsf{S}(r)$, and by holomorphy with U_t; the continuity on the translations follows from the continuity of (d).

Step 10: all the heights at once. Let $n\ge 2$, $s^0\in\mathcal{U}_n$, $R>0$, $\epsilon:=\min\{1,\mathrm{gap}(s^0)/32nL\}$, and $f_k\in C^1_c(\mathbb{E})$ with $\mathrm{supp}f_k\subset\mathrm{pr}(\bar B^{(2)}_{R,k}\times[s^0_k-\epsilon,s^0_k+\epsilon])$, each $\bar B^{(2)}_{R,k}$ a closed disc of radius R. Let $\delta_{\mathrm{a}}=l/p$ be the power of L in $(\frac{\mathrm{gap}(s^0)}{2L},\frac{\mathrm{gap}(s^0)}{2}]$; then $p=2L^{r_{\mathrm{a}}}$ with an integer $r_{\mathrm{a}}\ge 1$. The offsets $c\in[0,\delta_{\mathrm{a}})$ at which some s^0_k is nearer than $\delta_{\mathrm{a}}/4n$ to $c+\delta_{\mathrm{a}}\mathbb{Z}$ fill a set of measure $\le\frac12\delta_{\mathrm{a}}$; fix c outside it. The arcs $(c+(m-1)\delta_{\mathrm{a}},c+m\delta_{\mathrm{a}})$, $m\in\mathbb{Z}/p$, then hold the heights s^0_k in different arcs m_k, each at the distance $>4\epsilon$ from the points of $c+\delta_{\mathrm{a}}\mathbb{Z}$. Let j be so large that $\nu_{\mathrm{a}}:=\delta_{\mathrm{a}}/a\in\mathbb{N}$ and $a\le\frac{\epsilon}{8}$, and choose $c_j\in a(\mathbb{Z}+\frac12)$ with $|c_j-c|\le a$. On a box as in Step 7 the walls are the planes $x_3=c_j+m\delta_{\mathrm{a}}$, $m\in\mathbb{Z}$, which lie in $a(\mathbb{Z}+\frac12)$ as $\delta_{\mathrm{a}}=\nu_{\mathrm{a}}a$; two consecutive walls enclose ν_{a} layers, and those enclosed by $c_j+(m-1)\delta_{\mathrm{a}}$ and $c_j+m\delta_{\mathrm{a}}$ form the slab that stands for the arc m on the box. A plaquette whose barycentre lies in $\mathrm{supp}f_k$ has its corners at heights within a of $[s^0_k-\epsilon,s^0_k+\epsilon]$, hence no nearer than $3\epsilon-3a$ to any wall; so the centred entries A_k are functions Ψ^k placed inside the arc m_k with a margin of $x_{\mathrm{a}}:=\lfloor 3\epsilon/a\rfloor-4$ layers. For $|\operatorname{Re}w_k|<x_{\mathrm{a}}a$ put $\mathsf{A}_{m_k}(w):=\hat{\mathsf{T}}^{x_{\mathrm{a}}-w_k/a}\mathfrak{Y}[\Psi^k]\hat{\mathsf{T}}^{x_{\mathrm{a}}+w_k/a}$, $\mathsf{A}_m(w):=\hat{\mathsf{T}}^{\nu_{\mathrm{a}}}$

on the arcs without an entry, and $\mathsf{G}_\Lambda(w) := Z^{-1}\mathrm{Tr}(\mathsf{A}_p(w)\cdots\mathsf{A}_1(w))$. After neighbouring powers of $\hat{\mathsf{T}}$ are multiplied, $Z\mathsf{G}_\Lambda$ is the trace function of Step 7 at the gaps $\mathsf{g}_k + (w_k - w_{k-1})/a$, the entries being numbered upwards, $m_1 < \cdots < m_n$ in $\{1,\dots,p\}$, a renumbering of the pairs (f_k, w_k) that leaves G_Λ unchanged, $k-1$ being read modulo n, and g_k being the number of layers between the entries $k-1$ and k around the circle; so it is holomorphic, it does not depend on the arcs, and $\mathsf{G}_\Lambda(u) = \sigma_{\mu_\Lambda}(\tau_{u_1 e_3} f_1, \dots, \tau_{u_n e_3} f_n)$ at $u \in (a\mathbb{Z})^n$. By Hölder's inequality for traces, Lemma YM3_I: 86.1 with $2k = p$, $|\mathsf{G}_\Lambda(w)| \le Z^{-1}\prod_m \|\mathsf{A}_m(w)\|_p$. As $\mathsf{A}_{m_k}(u+iy) = \hat{\mathsf{T}}^{-iy/a}\mathsf{A}_{m_k}(u)\hat{\mathsf{T}}^{iy/a}$ and $\mathsf{A}_{m_k}(u)$ lives on the orthogonal complement of $\ker\hat{\mathsf{T}}$, $\|\mathsf{A}_m(w)\|_p = \|\mathsf{A}_m(\mathrm{Re}\, w)\|_p$. At $u \in a\mathbb{Z}$, $|u| \le \epsilon + a$, the number $\|\mathsf{A}_{m_k}(u)\|_p^p = \mathrm{Tr}(\mathsf{A}_{m_k}^*\mathsf{A}_{m_k})^{p/2}$ is, by the trace formula at p slabs of ν_{a} layers and the alternating adjoint rule YM3_I:(1745), Z times the box mean of the product of the p images of $\tau_{ue_3}\mathsf{A}_k$ under the elements of the group generated by the reflections $\theta_{e_3, c_j + m\delta_{\mathrm{a}}}$ in the walls, that carry the arc m_k to the arcs m. These reflections are isometries of $\mathbb{L}_K$ whose linear part keeps Γ_Λ, so μ_Λ is invariant under them as in Step 3; and the support of $\tau_{ue_3} f_k$ keeps the distance $\ge 2\epsilon - 2a \ge \epsilon$ from the walls, so that its images lie in different arcs, pairwise at the distance $\ge 2\epsilon$. Thus $\|\mathsf{A}_{m_k}(u)\|_p^p$ equals $Z\sigma_{\mu_\Lambda}$ of a tuple of the type $(2\epsilon; R+l)$, at most $ZC_p^\sigma(2\epsilon; R+l)\|f_k\|_\sharp^p$ by (c) for $\Lambda > 2\epsilon + 2R$, since the reflections act on the third coordinate alone, so that all p images have their first two coordinates in $\bar{B}^{(2)}_{R,k}$; and $\|\hat{\mathsf{T}}^{\nu_{\mathrm{a}}}\|_p^p = \mathrm{Tr}\,\hat{\mathsf{T}}^N = Z$. Hence $|\mathsf{G}_\Lambda(w)| \le \mathsf{C}^{\mathrm{An}}\prod_k \|f_k\|_\sharp$, $\mathsf{C}^{\mathrm{An}} := C_p^\sigma(2\epsilon; R+l)^{n/p}$, when $\mathrm{Re}\, w \in (a\mathbb{Z})^n$ and $|\mathrm{Re}\, w_k| \le \epsilon + a$; by the three lines theorem, in one variable after the other, the bound holds on the whole tube $|\mathrm{Re}\, w_k| \le a\lceil \epsilon/a\rceil$. By Montel's theorem, as $\Lambda \to \infty$ along the sequence that defines ν_j and then as $j \to \infty$ along the subsequence of S, the G_Λ converge locally uniformly on $\{|\mathrm{Re}\, w_k| < \epsilon\}$ along subsequences; at real points the limit is $\sigma_S(\tau_{u_1 e_3} f_1, \dots, \tau_{u_n e_3} f_n)$ by weak convergence and by (d), so it is one holomorphic function $\mathsf{G}[\underline{f}]$, with $|\mathsf{G}[\underline{f}]| \le \mathsf{C}^{\mathrm{An}}\prod_k \|f_k\|_\sharp$. Let now $\underline{f}^\alpha$ be finitely many tuples of $n_\alpha \ge 0$ entries, the k-th with its support in $\mathrm{pr}(\bar{B}^{(2)}_{R,\alpha,k} \times [s_k^{0,\alpha} - \epsilon, s_k^{0,\alpha} + \epsilon])$, $\bar{B}^{(2)}_{R,\alpha,k}$ a closed disc of radius R, where $0 < s_1^{0,\alpha} < \cdots < s_{n_\alpha}^{0,\alpha} < \frac{l}{2}$ and ϵ is at most the number ϵ formed as at the head of this step from every joint height $(-s^{0,\alpha,\mathrm{rev}}, s^{0,\alpha'})$ with at least two entries, where rev reverses the order, and let $|\mathrm{Re}\, w_k^\alpha| < \epsilon$. The same construction on the half box, the slab of the layers $1,\dots,N'$ of Step 7, places the centred entries of $\underline{f}^\alpha$ there upwards as functions $\Psi^1,\dots,\Psi^n$, $n := n_\alpha$; with g_0^α the number of the layers of the half box below Ψ^1, g_k^α the number between Ψ^k and Ψ^{k+1} and g_n^α the number above Ψ^n, so that these numbers and the numbers of layers of the Ψ^k add up to N', with $\mathsf{B}_\alpha(w) := \hat{\mathsf{T}}^{\mathsf{g}_n^\alpha - w_n/a}\mathfrak{Y}[\Psi^n]\hat{\mathsf{T}}^{\mathsf{g}_{n-1}^\alpha + (w_n - w_{n-1})/a}\cdots\mathfrak{Y}[\Psi^1]\hat{\mathsf{T}}^{\mathsf{g}_0^\alpha + w_1/a}$, and with $\mathsf{B}_\alpha := \hat{\mathsf{T}}^{N'}$ if $n_\alpha = 0$, it gives $Z^{-1}\mathrm{Tr}(\mathsf{B}_\alpha(w)^*\mathsf{B}_{\alpha'}(w')) = \mathsf{G}_\Lambda[\theta^{(j)}\underline{f}^{\alpha,\mathrm{rev}} \cdot \underline{f}^{\alpha'}](-\overline{w^{\mathrm{rev}}}, w')$, where G_Λ and G are 1 at the empty joint tuple and 0 at a joint tuple of one entry: $\mathrm{Tr}\,\hat{\mathsf{T}}^N = Z$, and at one entry Ψ, of v layers, the powers of $\hat{\mathsf{T}}$ join by cyclicity into $\mathrm{Tr}(\hat{\mathsf{T}}^{N-\mathsf{v}}\mathfrak{Y}[\Psi])$ or its conjugate, free of w, which the trace formula of Step 7 with the two slab functions Ψ and 1_2, as $\hat{\mathsf{T}}^2 = \mathfrak{Y}[1_2]$, makes Z times the mean under μ_Λ of a centred entry, that is 0; hence $\sum_{\alpha,\alpha'}\bar{c}_\alpha c_{\alpha'}\mathsf{G}[\mathsf{r}_0\underline{f}^{\alpha,\mathrm{rev}} \cdot \underline{f}^{\alpha'}](-\overline{w^{\alpha,\mathrm{rev}}}, w^{\alpha'})$ is a limit of the numbers $Z^{-1}\|\sum_\alpha c_\alpha \mathsf{B}_\alpha(w^\alpha)\|_2^2 \ge 0$. For the reflected entries the lattice mirror approaches r_0 in the sharp norm after acting on the test functions, with supports in a common compact set. The same trace estimates are uniform for those entries on a slightly smaller tube. On real shifts the limits are therefore identified by (d), and the identity theorem identifies the corresponding holomorphic limits.

Step 11: kernels at sharp heights (g). Fix an even $\chi_0 \in C_c^\infty((-1,1))$ equal to 1 on $[-\frac{1}{2}, \frac{1}{2}]$, and for a width $0 < \varpi \le 1$ put $\mathsf{E}_\varpi(s) := \chi_0(s/\varpi)\frac{s|s|}{4}$. As the third derivative of $\frac{s|s|}{4}$ is the Dirac measure, $\mathsf{E}_\varpi''' = \delta + \mathsf{e}_\varpi$ with a smooth e_ϖ supported in $\frac{\varpi}{2} \le |s| \le \varpi$; $\|\mathsf{E}_\varpi \otimes h\|_\sharp \le (1 + 40Mc_0)\|h\|_\sharp$ and $\|\mathsf{e}_\varpi \otimes h\|_\sharp \le (1+40M)c_0\varpi^{-2}\|h\|_\sharp$ with a numeral c_0 that depends on χ_0. Near $s^0 \in \mathcal{U}_n$ let $\varpi := \epsilon$ of Step 10, and for $\mathsf{a} \subset \{1,\dots,n\}$ let $\underline{f}^{\mathsf{a}}$ have the entries $\tau_{s_k^0 e_3}(\mathsf{E}_\varpi \otimes h_k)$ for $k \in \mathsf{a}$ and $\tau_{s_k^0 e_3}(\mathsf{e}_\varpi \otimes h_k)$ otherwise; put $\mathfrak{K}_n(s^0 + w; h) := (-1)^n\sum_{\mathsf{a}}\prod_{k\in\mathsf{a}}\partial_{w_k}^3\mathsf{G}[\underline{f}^{\mathsf{a}}](w)$. Writing $\chi_k = \chi_k''' * \mathsf{E}_\varpi - \chi_k * \mathsf{e}_\varpi$ as

Riemann integrals of translates that converge in C^1, using the multilinearity and the continuity of σ_S at separated supports (d), and integrating by parts three times in each s_k, $k \in \mathsf{a}$, gives the integral formula of (g) for $\chi_k \in C_c^\infty((s_k^0-\epsilon, s_k^0+\epsilon))$, and by mollification for $\chi_k \in C_c^1$. A continuous kernel is determined by these integrals; so the local functions agree on overlaps at real points, hence as holomorphic functions on the tubes over the convex components of $\mathcal{U}_n$, and they glue to $\mathfrak{K}_n$; a partition of unity gives the general χ_k; $\mathfrak{K}_1 = 0$ since σ_S vanishes at one entry. Cauchy's estimate on polydiscs of radius $\epsilon/2$ bounds $\partial_{w_k}^3$ by $3!\,(2/\epsilon)^3$ times the supremum, so $|\mathfrak{K}_n(z;h)| \le (c_E\epsilon^{-3})^n \mathsf{C}^{\mathrm{An}} \prod_k \|h_k\|_\sharp$ with $c_E := (48+c_0)(1+40M)(1+c_0)$. The rules hold at real points by (d) and by the period l of $\chi \otimes h$ in x_3, and pass to the tubes by holomorphy; for complex c, $c \mapsto \mathfrak{K}_n(z+c\mathbf{1};h)$ is entire and constant on $\mathbb{R}$. To obtain positive type, take a smooth nonnegative χ of integral 1 supported in $(-1,1)$, and put $\chi^{(\varpi)} := \varpi^{-1}\chi(\cdot/\varpi)$. At finitely many tuples of increasing real heights in the positive half-circle, apply the last part of Step 10 to the entries $\tau_{s_k e_3}(\chi^{(\varpi)} \otimes h_k)$ for sufficiently small ϖ. For real shifts the resulting matrix entries are the integrals of $\mathfrak{K}$ against the products of the mollifiers in the height variables, by the kernel formula. The same identity holds for complex shifts by holomorphy: the real parts of the joint heights stay in a compact subset of $\mathcal{U}_{n+n'}$, so the kernel bound on each smaller tube allows integration of the holomorphic kernels. As $\varpi \downarrow 0$ the integrals converge to the sharp kernel entries by continuity. The matrices are positive semidefinite before the limit and hence after it. For complex h use sesquilinearity in the two tuples and the same trace Gram form. The growth: by the display of U in YM3_I: (1785) at $R=1$, with $\delta_e := \kappa - \kappa_{\mathcal{D}}$ and $\log_L^+ y \le y \le \max\{1, y^3\}$, $A^{\mathrm{per}}(\mathsf{d}) \le A_\star \max\{1, \mathsf{d}^{-3}\}$, $A_\star := 1 + 27 C_{\mathrm{S}} S(\kappa)[32(C''_\Sigma(\delta_e) + (5\sqrt{3}M)^3) + 4C'_\Sigma(\delta_e,4)(2\sqrt{3})^3]$; in C_p^σ there are at most p^p partitions, each with at most $\frac{p}{2}$ blocks and $\prod_J |J|! \le p^p$; so $\mathsf{C}^{\mathrm{An}} \le (A^{\mathrm{per}}(2\epsilon)^{1/2}\,\mathsf{b}'\,p^{6+\mathsf{p}})^n$ with $\mathsf{b}' := (1+40M)B^{\mathrm{sl}}(2R+2l+1)$. As $p < 2Ll/\mathrm{gap}(s^0)$ and $\epsilon = \min\{1, \mathrm{gap}(s^0)/32nL\}$, the powers of $\min\{1,\mathrm{gap}\}^{-1}$ add up to $3+\frac{3}{2}+6+\mathsf{p} = \varkappa_{\mathrm{w}}$ per point.

Step 12: the state (h). On the convex component $\mathsf{O}_n := \{s : s_1 < \cdots < s_n < s_1 + l\}$ of $\mathcal{U}_n$ the kernel is bounded in t for s in a compact set, and $\prod_k |\phi_k(t_k + is_k)| \le C_N \prod_k (1+|t_k|)^{-N}$; the integral that defines ω^{sl} converges, its integrand is holomorphic in $t+is$, and by Cauchy's theorem in one variable after the other, the vertical sides of long rectangles contributing $O(|t_k|^{-N})$, it does not change along segments of O_n parallel to the axes: it does not depend on s. Positivity: in $\omega^{\mathrm{sl}}(A^*A)$, $A = \sum_\alpha c_\alpha \Phi(\phi_1^\alpha, h_1^\alpha)\cdots\Phi(\phi_{n_\alpha}^\alpha, h_{n_\alpha}^\alpha)$, take for the term (α,α') the heights $(-s_n,\dots,-s_1,s'_1,\dots,s'_{n'})$ with $0 < s_1 < \cdots < s_n < \frac{l}{2}$ and the same for s'; as $\phi^*(t-is) = \overline{\phi(t+is)}$, the term is the double integral of $\overline{\prod_k \phi_k^\alpha(t_k+is_k)}\prod_k \phi_k^{\alpha'}(t'_k+is'_k)$ against the kernel of positive type of (g) at $(s-it, h^\alpha)$ and $(s'-it', h^{\alpha'})$, a limit of Riemann sums each of which is ≥ 0. The invariance under α_{t_0} follows by a shift of the variables and the invariance of $\mathfrak{K}_n$ under $z \mapsto z + c\mathbf{1}$; under E_2 by (g). The condition at l: for $A = \Phi(\phi_1,h_1)\cdots\Phi(\phi_m,h_m)$ and $B = \Phi(\phi_{m+1},h_{m+1})\cdots\Phi(\phi_n,h_n)$, $\omega^{\mathrm{sl}}(A\alpha_c(B)) = \int \prod_{k\le m}\phi_k(t_k+is_k)\prod_{k>m}\phi_k(t_k+is_k-c)\,\mathfrak{R}_n(t+is;h)\,dt$ is entire in c and bounded on strips (shift t_k by $\mathrm{Re}\,c$ for $k>m$); at $c = t_0 + il$ put $s'_k := s_k - l$ for $k > m$, so that $s'_{m+1} < \cdots < s'_n < s_1 < \cdots < s_m < s'_{m+1} + l$; the period l and the symmetry of the kernel turn the integral into $\omega^{\mathrm{sl}}(\alpha_{t_0}(B)A)$. The truncated functions: σ_S is the centred functional of S and vanishes at one entry, so the inversion of the moment-cumulant formula gives S_n in terms of σ_S, whose kernels are the $\mathfrak{K}_m(s;h) = \mathfrak{R}_m(is;h)$ on O_m. The vectors: for (s,h) at increasing heights in $(0,\frac{l}{2})$ the classes $[(\tau_{s_k e_3}(\chi^{(\varpi)}\otimes h_k))_k]$ form a Cauchy family as $\varpi \to 0$, by (g), with a limit $\Psi_n(s;h)$; form the Hilbert space of formal kernel vectors using the positive-type kernel in (g). Their real-height vectors have the inner products of the sharp vectors just constructed. The span of the real-height vectors is dense in the kernel space. Indeed, an orthogonal vector gives a scalar holomorphic function in each tuple's complex heights that vanishes at all real heights and hence throughout its tube. Holomorphy follows first for finite sums of kernel vectors and then by the locally bounded diagonal of the kernel. Thus the isometry at real heights extends to this whole space

with image in $\mathcal{H}^{\mathrm{sl}}$, defining $\Psi_n(z;h)$ with the stated inner products. Weak holomorphy and the same local norm bound give Hilbert-space holomorphy; $e^{-it\mathsf{H}_3}\Psi_n(z;h)=\Psi_n(z+it\mathbf{1};h)$ by Step 9 and holomorphy, and $J^{\mathrm{sl}}\Psi_n(z;h)=\Psi_n(\frac{l}{2}\mathbf{1}-\bar{z}^{\mathrm{rev}};\bar{h}^{\mathrm{rev}})$. For $\mathsf{g}\in\mathsf{E}_2$, $[\mathsf{g}\cdot]\Psi_n(z;h)=\Psi_n(z;\mathsf{g}h_1,\dots,\mathsf{g}h_n)$: g acts on the first two coordinates, so it commutes with τ_{se_3} and $\mathsf{g}(\chi\otimes h_k)=\chi\otimes\mathsf{g}h_k$; the unitary operator $[\mathsf{g}\cdot]$ of (f) thus maps $[(\tau_{s_ke_3}(\chi^{(\varpi)}\otimes h_k))_k]$ to $[(\tau_{s_ke_3}(\chi^{(\varpi)}\otimes \mathsf{g}h_k))_k]$, and the identity passes to the limit $\varpi\to 0$, then to complex z, both sides being holomorphic, and to complex h, both sides being multilinear. With $\Psi_0:=\Omega^{\mathrm{sl}}=[\varnothing]$, put $\Psi^{\mathrm{sl}}(1):=\Omega^{\mathrm{sl}}$. Then $\Psi^{\mathrm{sl}}(\Phi(\phi_1,h_1)\cdots\Phi(\phi_n,h_n)):=\int\prod_k\phi_k(t_k+is_k)\,\Psi_n(s-it;h)\,dt$, $0<s_1<\cdots<s_n<\frac{l}{2}$, is a norm-convergent integral independent of s; $\langle\Psi^{\mathrm{sl}}(B),\Psi^{\mathrm{sl}}(A)\rangle=\omega^{\mathrm{sl}}(B^*A)$ is the computation of positivity; $\Psi^{\mathrm{sl}}(\alpha_{t_0}A)=e^{it_0\mathsf{H}_3}\Psi^{\mathrm{sl}}(A)$ by the shift; $[\mathsf{g}\cdot]\Psi^{\mathrm{sl}}(A)=\Psi^{\mathrm{sl}}(\mathsf{g}A)$, as $[\mathsf{g}\cdot]$, being bounded, may be taken inside the integral, which converges in norm, and $\mathsf{g}\Phi(\phi,h)=\Phi(\phi,\mathsf{g}h)$; the function $c\mapsto\Psi^{\mathrm{sl}}(\alpha_cA)$ is entire, and the lemma on domains of Step 9, applied to $\Xi(z):=\Psi^{\mathrm{sl}}(\alpha_{iz}A)$, gives $e^{-r\mathsf{H}_3}\Psi^{\mathrm{sl}}(A)=\Psi^{\mathrm{sl}}(\alpha_{ir}A)$ for all real r; at $r=\frac{l}{2}$ the formula for J^{sl} and $\phi^*(t+i(\frac{l}{2}-s))=\overline{\phi(t+i(s-\frac{l}{2}))}$ give $J^{\mathrm{sl}}e^{-\frac{l}{2}\mathsf{H}_3}\Psi^{\mathrm{sl}}(A)=\Psi^{\mathrm{sl}}(A^*)$. The closed span of the range of Ψ^{sl} reduces H_3 by real translation covariance. The span of the algebra vectors is a core for $e^{-l\mathsf{H}_3/2}$ on this subspace. To see this, convolve the real unitary group with Gaussian kernels to get $e^{-\delta\mathsf{H}_3^2}$, $\delta>0$. Applied to an algebra vector, each such convolution is a graph-norm limit of linear combinations of translates of that vector. For fixed $\delta>0$ the convolution maps the Hilbert space boundedly into the graph domain, by the spectral theorem, and as $\delta\downarrow 0$ it converges there on the domain. The algebra vectors are dense, so their graph closure is the full domain. The operator $e^{-l\mathsf{H}_3/2}$ has dense range in this subspace by the spectral theorem. The modular identity on the core therefore implies that J^{sl} leaves the subspace invariant. Consequently the closure of the conjugate-linear map $\Psi^{\mathrm{sl}}(A)\mapsto\Psi^{\mathrm{sl}}(A^*)$ has square modulus $e^{-l\mathsf{H}_3}$.

Step 13: real times (i). For $G\in\mathcal{S}(\mathbb{R}^n)$ and $s\in\mathsf{O}_n$ put $\Phi_G(s):=\int G(t)\,\mathfrak{R}_n(t+is;h)\,dt$; by (g), $|\Phi_G(s)|\le 2^n\mathsf{c}_n(R,l)\min\{1,\mathrm{gap}(s)\}^{-\varkappa_{\mathrm{w}}n}\|G\|_{L^1}\prod_k\|h_k\|_\sharp$. At $s=\varepsilon s^*$ with $0<\varepsilon\le\varepsilon_0$, $\varepsilon_0(s_n^*-s_1^*)\le\frac{l}{2}$ and $\varepsilon_0\mathsf{g}_*\le 1$, g_* the least difference of consecutive s_k^*, one has $\mathrm{gap}(\varepsilon s^*)=\varepsilon\mathsf{g}_*$; as $\partial_\varepsilon\mathfrak{R}_n(t+i\varepsilon s^*)=i(s^*\cdot\nabla_t)\mathfrak{R}_n(t+i\varepsilon s^*)$, integration by parts gives $\partial_\varepsilon^\mu\Phi_G(\varepsilon s^*)=(-i)^\mu\Phi_{(s^*\cdot\nabla)^\mu G}(\varepsilon s^*)$, of modulus at most a constant times $\varepsilon^{-\varkappa_{\mathrm{w}}n}\|(s^*\cdot\nabla)^\mu G\|_{L^1}$. Taylor's formula at ε_0 to the order $\mathsf{N}:=\lceil\varkappa_{\mathrm{w}}n\rceil+1$ has a remainder with an integrable bound, because $\mathsf{N}-1-\varkappa_{\mathrm{w}}n\ge 0$; so the limit as $\varepsilon\downarrow 0$ exists and is bounded by a constant times $\sum_{|\alpha|\le\mathsf{N}}\|\partial^\alpha G\|_{L^1}$, uniformly in $\varepsilon\in(0,\varepsilon_0]$. Along the segment between two directions s^* and s^{**}, the Taylor bounds are uniform because the consecutive gaps have a positive minimum there. The parameter derivative is $-i\varepsilon\Phi_{((s^{**}-s^*)\cdot\nabla)G}$ at the interpolated direction. Apply the uniform Taylor bound to this new test function and integrate along the segment. The difference is $O(\varepsilon)$, so the limit is independent of the direction. For $\phi_k\in\mathscr{G}$ the functions $\prod_k\phi_k(t_k+i\varepsilon s_k^*)$ converge to $\prod_k\phi_k(t_k)$ in these norms, so $\tilde{\omega}^{\mathrm{sl}}=\omega^{\mathrm{sl}}$ on $\mathfrak{A}^{\mathrm{sl}}$. The space $\mathscr{G}$ is dense in $\mathcal{S}(\mathbb{R})$: the Fourier transform of a smooth function of compact support lies in $\mathscr{G}$, and these transforms are dense; positivity and invariance pass to $\mathfrak{A}^{\mathrm{sl}}_{\mathcal{S}}$ by the bound just shown. The strip: for $A,B\in\mathfrak{A}^{\mathrm{sl}}$ the entire function $c\mapsto\omega^{\mathrm{sl}}(A\alpha_c(B))$, bounded on $0\le\mathrm{Im}\,c\le l$, is bounded on the two edges by $\omega^{\mathrm{sl}}(AA^*)^{1/2}\omega^{\mathrm{sl}}(B^*B)^{1/2}$ and $\omega^{\mathrm{sl}}(BB^*)^{1/2}\omega^{\mathrm{sl}}(A^*A)^{1/2}$ (Schwarz' inequality for the state and the condition at l), so by the three lines theorem on the whole strip by the larger of the two; for $A,B\in\mathfrak{A}^{\mathrm{sl}}_{\mathcal{S}}$, approximated by members of $\mathfrak{A}^{\mathrm{sl}}$, these functions converge uniformly on the closed strip, and the limit has the stated boundary values. Finally $\|\Psi^{\mathrm{sl}}(A_m)-\Psi^{\mathrm{sl}}(A_{m'})\|^2=\omega^{\mathrm{sl}}((A_m-A_{m'})^*(A_m-A_{m'}))\to 0$ defines Ψ^{sl} on $\mathfrak{A}^{\mathrm{sl}}_{\mathcal{S}}$; $e^{-\frac{l}{2}\mathsf{H}_3}$ is closed, and $e^{-\frac{l}{2}\mathsf{H}_3}\Psi^{\mathrm{sl}}(A_m)=J^{\mathrm{sl}}\Psi^{\mathrm{sl}}(A_m^*)$ converges, so $\Psi^{\mathrm{sl}}(A)\in\mathrm{Dom}(e^{-\frac{l}{2}\mathsf{H}_3})$ and the last rule of (h) holds on $\mathfrak{A}^{\mathrm{sl}}_{\mathcal{S}}$. □

Definition 18.200 (Numbers for all the cubes)**.** Use the constants of Theorem YM3_I:AD, with the source radius ϱ and the threshold γ fixed in the preceding definition. Set $g_1=\gamma_W$, where YM3_I:(2345) and YM3_I:(2346) define $\hat{\gamma}_W\in(0,\gamma/2]$ and

$$\gamma_W^{-2}=L^2\big(\bar{B}+\max\{\hat{\gamma}_W^{-2},\gamma^{-4},e^{C_{\mathrm{s}}}\}\big).$$

In particular $g_1 \le \gamma$, $g_1^{-2} \ge 16L^2$ and $g_1^{-2} > L^2\bar B$. Let $\mathsf{b}_- = g_1^{-2} + B_\sharp - L^{-1}\bar B$ and $\mathsf{b}_+ = L(g_1^{-2} + \bar B)$ be the endpoints of the outer label interval in YM3_I:(1535), and put $\kappa_1 = \mathsf{b}_+ \max\{1, 40M/\mathsf{b}_-\}$. For a reference coupling g, the least allowed side is

$$\ell_{\mathrm G}(g) = 2LM \max\{R(L(g^{-2}+\bar B)), L^{\lceil \log_L C_{\mathrm s}\rceil}\}, \qquad \rho_1 = \ell_{\mathrm G}(g_1)/\mathsf{b}_-,$$

by YM3_I:(1491). Here $R(u)$ is the least power L^v, $v \ge 0$, at least $(\log u)^{\mathsf r}$ (Section YM3_I:74). Use $\mathsf U$ from YM3_I:(1785), $\delta_e = \kappa - \kappa_{\mathcal D}$, and $\mathsf p$, $\varkappa_\star = 5 + \mathsf p$ and $B_{[g]}(R)$ from YM3_I:(1799)–(1803), with the constants $\rho_* = 2\varrho$, $C_e = C_{\mathrm S}$, $\kappa_e = \kappa$, $E_+ = C_{\mathrm S}$ and $C_{\mathrm{low}} = C_{\mathrm S}(1 + \log g^{-2})$ from Sections YM3_I:88 and YM3_I:89. Set $\mathrm{norm}_{\mathsf b}(f) = \max\{\sup|f|, \mathsf b\,\mathrm{Lip} f\}$.

(1) *The two functions of the big cubes.* $\mathsf A(x) := 1 + \mathsf U(1, \frac12 \mathsf b_- x)$ and $\mathsf B(y) := \kappa_1 B_{[g_1]}(2\mathsf b_+ y)$ for $x, y > 0$.

(2) *The ladder.* $y_0 := g_1^{-2}$, $y_{k+1} := Ly_k + (L-3)\bar B$, $g^{(k)} := y_k^{-1/2}$, $Y_k := Ly_k + (L-2)\bar B$, $t_k := 1 + \log y_k$, i_k the least integer $i \ge 0$ with $L^{3i} \ge t_k$, and

$$2\varrho_k := \min\{\sigma^{\mathrm S}(y_k - \bar B),\ \sigma_{\mathrm L}^{\mathrm S}\, y_k^{1/2} (\log y_k)^{-\mathfrak p_0}\},$$

with the positive constants $\sigma^{\mathrm S}$ and $\sigma_{\mathrm L}^{\mathrm S}$ of YM3_I:(1715)–(1716) and YM3_I:(1717). The inner first-passage interval of Corollary YM3_I:74.2 is $I(g^{(k)}) = (y_k + \bar B, Y_k]$, and $Y_k = y_{k+1} + \bar B$.

(3) *Constants.* $\mathsf u_0 := y_0 + \bar B + 1$; for $\lambda \ge \mathsf u_0$, $\Psi_{\mathrm s}(\lambda) := 2LM \max\{R(L(\lambda + \bar B)), L^{\lceil \log_L C_{\mathrm s}\rceil}\}$;

$$\mathsf c_Q := \exp\Big[C_{\mathrm S} S(\kappa)\Big\{\tfrac{27}{26} + 27[32(C''_\Sigma(\delta_e) + \log_L 5) + 4C'_\Sigma(\delta_e, 4)]\Big\}\Big], \qquad \mathsf k_1 := 27 \cdot 417\, M\, e^{5+\mathsf p}(2e\, 3^{\mathsf p} e^{2C_{\mathrm S}} \mathsf c_Q + 1),$$

$$\mathsf c_\varrho := \frac{2}{\sigma^{\mathrm S}} + \frac{1}{\sigma_{\mathrm L}^{\mathrm S}}, \qquad \mathsf c_\Psi := 2LM \max\{L(1 + \log(2L^2))^{\mathsf r}, L^{\lceil \log_L C_{\mathrm s}\rceil}\}, \qquad \mathsf c_\ell(\bar\rho) := L \max\{\mathsf c_\Psi, \bar\rho\, \mathsf u_0\},$$

$$c_* := 4\mathsf r + \tfrac43 + \mathfrak p_0, \qquad \mathsf K(\bar\rho) := \tfrac{27\sqrt3}{2}\, \mathsf k_1 L^4 \mathsf c_\ell(\bar\rho)^4 \mathsf c_\varrho, \qquad B_\natural := \mathsf K(\rho_1)\, e^{1/2} \Big(\frac{2c_*}{e}\Big)^{c_*}, \qquad \rho_\natural := LM\, e^{1-2c_*},$$

and $\hat{\mathsf B}(y) := \max\{\mathsf B(y), B_\natural\}$.

All these functions and numbers are built from the radius ϱ and the constants (the number $B_{[g_1]}(R)$ contains ϱ through ρ_*); none depends on a label, a side, a cutoff or a coupling of a datum. $\mathsf A$ does not increase, $\mathsf B$ and $\hat{\mathsf B}$ do not decrease and are bounded on bounded intervals, and $\mathsf A(x) \le 1 + 27 C_{\mathrm S} S(\kappa)[32(C''_\Sigma(\delta_e) + 10\sqrt3 M/(\mathsf b_- x)) + 4C'_\Sigma(\delta_e, 4)(4\sqrt3/(\mathsf b_- x))^3]$. ■

Proposition 18.201 (All the cubes at all labels, with numbers that depend on $\mathsf d/\mathsf b$ and $R/\mathsf b$ alone)**.** Let $\mathsf b > 0$, $m \in \mathbb Z$, $\ell = 2L^m$, $\rho := \ell/\mathsf b$, $n \ge 2$, $\mathsf d, R > 0$, and let $f_1, \dots, f_n$ be real Lipschitz functions on T_ℓ with pairwise disjoint supports, each contained in a closed ball of radius R, two of which lie at the distance $\ge \mathsf d$; $x := \mathsf d/\mathsf b$, $y := R/\mathsf b$. The limit cumulant $S_n^{(\ell,\mathsf b)}$ is that of Corollary YM3_I:119.2.

(a) *The cubes that are not small.* If $\rho \ge \rho_1$, then

$$|S_n^{(\ell,\mathsf b)}(f_1, \dots, f_n)| \le \mathsf b^{-n} \mathsf A(x)\, \mathsf B(y)^n (n!)^{\varkappa_\star} \prod_q \mathrm{norm}_{\mathsf b}(f_q).$$

(b) *The small cubes.* If $\rho < \rho_1$, then $|S_n^{(\ell,\mathsf b)}(f_1, \dots, f_n)| \le \mathsf b^{-n} \mathsf A(x)\, B_\natural^n (n!)^{\varkappa_\star} \prod_q \mathrm{norm}_{\mathsf b}(f_q)$; the radius R does not enter. If $\rho \le \min\{\rho_1, \rho_\natural\}$, then

$$|S_n^{(\ell,\mathsf b)}(f_1, \dots, f_n)| \le \mathsf A(x) \Big[\mathsf K(\rho_1)\Big(1 + \log\frac{LM}{\rho}\Big)^{c_*} \Big(\frac{\rho}{LM}\Big)^{1/2}\Big]^n (n!)^{\varkappa_\star} \prod_q \mathrm{Lip} f_q,$$

and the right member is at most the same expression with $\mathsf b^{-n} \prod_q \mathrm{norm}_{\mathsf b}(f_q)$ in the place of $\prod_q \mathrm{Lip} f_q$; the base tends to 0 as $\rho \to 0$.

(c) *One set of functions for all the cubes.* For every $\mathsf{b}>0$ and every side $\ell=2L^m$, $m\in\mathbb{Z}$,

$$|S_n^{(\ell,\mathsf{b})}(f_1,\dots,f_n)|\le \mathsf{b}^{-n}\,\hat C_n\big(\tfrac{\mathsf{d}}{\mathsf{b}},\tfrac{R}{\mathsf{b}}\big)\prod_q \mathrm{norm}_{\mathsf{b}}(f_q),\qquad \hat C_n(x,y):=\mathsf{A}(x)\,\hat{\mathsf{B}}(y)^n(n!)^{\varkappa_\star}:$$

the bound has the growth form $\mathsf{A}\cdot\hat{\mathsf{B}}^n(n!)^{\varkappa_\star}$, its functions depend on d/b and R/b alone, and it is uniform in ℓ/b, in particular as $\ell/\mathsf{b}\to 0$.

■

Sketch. A zero entry makes the cumulant and the proposed bounds zero, by multilinearity. Assume that the entries are nonzero and number a separated pair as $1,2$. Corollary YM3_I:119.2 gives the full cutoff limit on every side $2L^m$, $m\in\mathbb{Z}$, and the covariance $S_n^{(\ell,\mathsf{b})}(\underline{f})=L^{zn}S_n^{(L^z\ell,L^z\mathsf{b})}(\underline{f}^{[z]})$, where $f^{[z]}=f(L^{-z}\cdot)$. The finite-cutoff identity and the support and norm changes are YM3_I:(2219)–(2220): the radii and separations acquire the factor L^z, the supremum is unchanged, and the Lipschitz constant acquires the factor L^{-z}.

Step 1: the cubes that are not small (a). Let K be so large that $\beta:=L^K\mathsf{b}>X_{g_1}$; let K' be the cutoff with (K',β) a first-passage datum at g_1 (Proposition YM3_I:74.1), $z:=K-K'$, $\lambda:=L^z\mathsf{b}=a_{K'}\beta$ and $\ell':=L^z\ell=\rho\lambda$. By YM3_I:(1535), $\lambda\in[\mathsf{b}_-,\mathsf{b}_+]$; so $\ell'\ge\rho_1\mathsf{b}_-=\ell_{\mathrm{G}}(g_1)$, and $\ell'\in\mathfrak{L}(g_1)$ by YM3_I:(1491): (K',ℓ',β) is a triple at g_1. The functions $f_q^{[z]}$ are supported in balls of radius $\lambda y\le 2\mathsf{b}_+y$, and two supports lie at the distance $\ge\lambda x\ge\frac12\mathsf{b}_-x$. Proposition YM3_I:89.5 at g_1, with the radius $2\mathsf{b}_+y$ and the distance $\frac12\mathsf{b}_-x$, gives $|S^T_{n;K',\ell',\beta}(\underline{f}^{[z]})|\le A(\frac12\mathsf{b}_-x)B_{[g_1]}(2\mathsf{b}_+y)^n(n!)^{\varkappa_\star}\prod_q\|f_q^{[z]}\|_\sharp$, with $A(\mathsf{d}')=1+\mathsf{U}(\frac{1}{10},\mathsf{d}')\le 1+\mathsf{U}(1,\mathsf{d}')$, as U does not decrease in its first argument by YM3_I:(1785). Further $L^z\|f^{[z]}\|_\sharp=\mathsf{b}^{-1}\max\{\lambda\sup|f|,40M\mathsf{b}\,\mathrm{Lip}f\}\le\mathsf{b}^{-1}\max\{\lambda,40M\}\,\mathrm{norm}_{\mathsf{b}}(f)\le\mathsf{b}^{-1}\kappa_1\mathrm{norm}_{\mathsf{b}}(f)$. Multiplying by L^{zn} and letting $K\to\infty$ gives (a).

Step 2: a radius that moves with the rung. Let $k\ge 0$. As $g_1^{-2}>L^2\bar B$, $\bar B<L^{-2}y_k\le\frac12 y_k$; $y_{k+1}-y_k=(L-1)y_k+(L-3)\bar B>0$ and $y_k\ge L^k y_0$; so the intervals $I(g^{(k)})=(y_k+\bar B,Y_k]$ lie end to end and cover $(y_0+\bar B,\infty)$, and for $\lambda\in I(g^{(k)})$

$$y_k<\lambda\le Y_k<2Ly_k.$$

(i) $\varrho_k\in(0,\infty)$, and $(2\varrho_k)^{-1}\le\mathsf{c}_\varrho t_k^{\mathfrak{p}_0}y_k^{-1/2}$: the larger of $[\sigma^{\mathrm{S}}(y_k-\bar B)]^{-1}\le 2(\sigma^{\mathrm{S}}y_k)^{-1}\le(2/\sigma^{\mathrm{S}})y_k^{-1/2}$ and $(\log y_k)^{\mathfrak{p}_0}y_k^{-1/2}/\sigma^{\mathrm{S}}_{\mathrm{L}}$ is at most their sum, and $t_k\ge1$. (ii) Apply Proposition YM3_I:82.2 with source radius $2\varrho_k$ and lower endpoint $x=y_k$. The first bound in YM3_I:(1717) holds for all $x\ge y_k$ because $x-\bar B$ is increasing. The second holds on this interval because $x^{-1/2}(\log x)^{\mathfrak{p}_0}$ decreases for $\log x>2\mathfrak{p}_0$, which holds already at the initial threshold in the proof of Lemma YM3_I:82.5. The other coupling restrictions of that construction are independent of the source radius and remain valid on $[y_k,\infty)\subset[\gamma^{-2},\infty)$. Thus $g^{(k)}$ is an admissible threshold for the radius $2\varrho_k$. The choice may use its own canonical auxiliary data. (iii) The constants C_{S} and κ in Theorem YM3_I:AD precede the choice of the radius and threshold, as its proof on pp. 676–677 shows. Therefore the same is true of U, p, $\varkappa_\star$ and c_Q. The numbers X_g, $I(g)$ and the allowed sides are given by the first-passage formulas on pp. 614–616. (iv) For $\lambda\in I(g^{(k)})$, $K'\ge1$ and $\ell'\in\mathfrak{L}(g^{(k)})$, Corollary YM3_I:74.2 makes $(K',L^{K'}\lambda)$ a first-passage datum. Theorem YM3_I:AD supplies the source energies with radius $2\varrho_k$, local constants (C_{S},κ), upper exponent $E_+=C_{\mathrm{S}}$ and zero-source lower exponent $C_{\mathrm{low}}=C_{\mathrm{S}}(1+\log y_k)=C_{\mathrm{S}}t_k$. The finite Wilson law and its observables are determined by L,K',ℓ',β and are independent of the auxiliary data.

Step 3: growth in n from a board of side $a_i\le1$. Let $K\in\mathbb{N}$, $\beta>0$, $\ell=2L^m$ with $m>m'$, and a system over (K,ℓ,β) with constants $(\varrho_*;C_e,\kappa_e;E_+,C_{\mathrm{low}})$ in the sense of Sections YM3_I:88 and YM3_I:89: the local energies obey YM3_I:(1772), the residual partition function is bounded by $e^{E_+\ell^3}$ on the source ball, and its zero-source value is at least $e^{-C_{\mathrm{low}}\ell^3}$ be given; let $i\ge0$, $n\ge2$ with $a_K\le a_i/(24n)$, and let $f_1,\dots,f_n$ be real Lipschitz functions on T_ℓ, none 0, each supported in a closed sup ball of radius R, $\mathrm{dist}(\mathrm{supp}f_1,\mathrm{supp}f_2)\ge\mathsf{d}$. With U, p and c_Q formed with (C_e,κ_e),

$C_Q^{(i)} := \mathsf{c}_Q \exp[(E_+ + C_{\rm low})L^{-3i}]$ and $B^{(i)}(R) := 27\cdot 417\, M\, L^i(4L^iR+1)^3 e^{5+\mathsf{p}}(2e\,3^{\mathsf{p}}C_Q^{(i)}+1)/\varrho_*$, we claim

$$|S^T_{n;K,\ell,\beta}(f_1,\dots,f_n)| \le \big(1+\mathsf{U}(\tfrac{1}{10},\mathsf{d})\big)\, B^{(i)}(R)^n (n!)^{5+\mathsf{p}} \prod_q \|f_q\|_\sharp .$$

The mesh $\varepsilon := a_i/(5n) \le \frac{1}{10}$ divides ℓ. Use the periodic tensor tents $\chi^\varepsilon_y(x) = \prod_{\mu=1}^3 \max(0, 1-|x_\mu - y_\mu|_\ell/\varepsilon)$ on the grid of that mesh. They sum to one, have Lipschitz constant at most $\sqrt{3}/\varepsilon$, and at most $(2R/\varepsilon+3)^3$ of their supports meet any one ball. By multilinearity, $S^T_n(\underline{f})$ is a sum of at most $(2R/\varepsilon+3)^{3n}$ terms $S^T_n(v_1,\dots,v_n)$, $v_q = \chi^\varepsilon_{y_q} f_q \ne 0$, with supports in sup balls of radius ε inside those of the f_q and $\|v_q\|_\sharp \le (1+40\sqrt{3}M/\varepsilon)\|f_q\|_\sharp$. Lemma YM3_I:89.3 at this i with radius and margin ε (its condition is $\frac{4}{5}a_i + n a_K \le a_i$) places a translate $\underline{g}$ of the tuple on the board of side a_i with no two occupied cells congruent, the distances inside a cell kept, and $S^T_n(\underline{g}) = S^T_n(\underline{v})$. Proposition YM3_I:89.2, with both sides equal to ℓ, $C_{\rm gap} := 0$ and $R = D = \varepsilon$, gives a holomorphic normaliser $\mathcal{E}$ with $\mathcal{E}(0) = 0$ such that $h := e^{-\mathcal{E}}\langle e^{\sum_q z_q \mathcal{O}(g_q)}\rangle_{K,\ell,\beta}$ is at most $Q^{\#B}$ on the slices $z_q = 0$ for $q \notin B$, where $Q = \exp[(E_+ + C_{\rm low})a_i^3 + \Phi_1(\varepsilon,\varepsilon;a_i)]$; here $\min\{a_i, M\} = a_i$ and $5a_i/2\varepsilon = \frac{25n}{2}$, so by the display of Φ_1 in YM3_I:(1786) $\Phi_1(\varepsilon,\varepsilon;a_i) = \log \mathsf{c}_Q + \mathsf{p}\log\frac{5n}{2}$: the side of the board drops out, and $Q \le C_Q^{(i)} 3^{\mathsf{p}} n^{\mathsf{p}}$. With the generating function $\Gamma(z) := \langle e^{\sum_q z_q \mathcal{O}(g_q)}\rangle_{K,\ell,\beta}$, so that $\Gamma(0) = 1$ and $h = e^{-\mathcal{E}}\Gamma$, Lemma YM3_I:87.2, bounds $|\partial_{z_1}\cdots\partial_{z_n}\mathrm{Log}\,\Gamma(0) - \partial_{z_1}\cdots\partial_{z_n}\mathcal{E}(0)|$ by $(n-1)!\,2^n(eQ)^n\varrho_*^{-n}\prod_q\|v_q\|_\sharp$, and YM3_I:(1793), bounds the mixed derivative of $\mathcal{E}$ by $(n/\varrho_*)^n\prod_q\|v_q\|_\sharp\,\mathsf{U}(\varepsilon,\mathsf{d})$, which is $\le (n/\varrho_*)^n\prod_q\|v_q\|_\sharp\,\mathsf{U}(\frac{1}{10},\mathsf{d})$. As $2R/\varepsilon+3 \le 3n(4L^iR+1)$ and $1+40\sqrt{3}M/\varepsilon \le 417MnL^i$, the number of the terms times $\prod_q\|v_q\|_\sharp/\|f_q\|_\sharp$ is at most $\mathsf{c}^n n^{4n}$, $\mathsf{c} := 27(4L^iR+1)^3\cdot 417ML^i$; with $n^{tn} \le e^{tn}(n!)^t$ at $t = 4+\mathsf{p}$ and $t = 5$ the claim follows, each of the two resulting bases being at most $B^{(i)}(R)$.

Step 4: the norm on a cube. If a real Lipschitz f on T_ℓ vanishes at a point, every point is within $\frac{\sqrt{3}}{2}\ell$ of it, so $\sup|f| \le \frac{\sqrt{3}}{2}\ell\,\mathrm{Lip}f$; and if $L^z\ell \ge 2L^2M$, then $L^z\|f^{[z]}\|_\sharp = \max\{L^z\sup|f|, 40M\,\mathrm{Lip}f\} \le \frac{\sqrt{3}}{2}L^z\ell\,\mathrm{Lip}f \le \frac{\sqrt{3}}{2}L^z\ell\,\mathsf{b}^{-1}\mathrm{norm}_{\mathsf{b}}(f)$, because $\frac{\sqrt{3}}{2}L^z\ell \ge \sqrt{3}L^2M > 40M$.

Step 5: the bound at a rung. Let $k \ge 0$, $\lambda \in I(g^{(k)})$, $\ell' \in \mathfrak{L}(g^{(k)})$ (so $\ell' = 2L^{m''}$ with $m'' \ge m_{\mathfrak{L}}(g^{(k)}) \ge m'+2$), K' with $a_{K'} \le a_{i_k}/(24n)$, and f'_q as in Step 3 on $T_{\ell'}$ with the radius R' and the distance d'. Step 3 at the system of Step 2 (iv) and $i = i_k$: as $L^{3i_k} \ge t_k \ge 1$, $(C_{\rm S} + C_{g^{(k)}})L^{-3i_k} \le (C_{\rm S} + C_{\rm S}t_k)/t_k \le 2C_{\rm S}$, so $C_Q^{(i_k)} \le e^{2C_{\rm S}}\mathsf{c}_Q$ and

$$|S^T_{n;K',\ell',L^{K'}\lambda}(\underline{f}')| \le \big(1+\mathsf{U}(\tfrac{1}{10},\mathsf{d}')\big)\mathsf{B}_k(R')^n(n!)^{\varkappa_\star}\prod_q\|f'_q\|_\sharp, \qquad \mathsf{B}_k(R') := \frac{\mathsf{k}_1 L^{i_k}(4L^{i_k}R'+1)^3}{2\varrho_k}.$$

Step 6: every cube. Let z be the least integer with $L^z\mathsf{b} \ge \mathsf{u}_0$ and $L^z\ell \ge \Psi_{\rm s}(L^z\mathsf{b})$; it exists, because $R(u) < L(\log u)^{\mathsf{r}}$ for $u > e$ gives $\Psi_{\rm s}(\lambda)/\lambda \to 0$, so that both conditions hold for all large z, while the first fails for all small z. Put $\lambda := L^z\mathsf{b}$, $\ell' := L^z\ell = \rho\lambda$, and k the index with $\lambda \in I(g^{(k)})$. Then: $\ell' \in \mathfrak{L}(g^{(k)})$, since $\lambda > y_k$ and R does not decrease give $\ell' \ge \Psi_{\rm s}(\lambda) \ge \ell_{\rm G}(g^{(k)})$; $2L^2M \le \Psi_{\rm s}(\lambda) \le \ell' < L\max\{\Psi_{\rm s}(\lambda), \rho\,\mathsf{u}_0\}$, the first because $R(u) \ge L$ for $u > e$ by the definition of R on p. 614, the last because at $z-1$ one of the two conditions fails; and $y_k > \lambda/2L = \ell'/2L\rho \ge LM/\rho$. Each f_q vanishes at a point, its support being disjoint from the non-empty support of another entry. By Step 5 at $R' = \frac{1}{2}\ell'$, where the sup ball is the whole torus, and $\mathsf{d}' = \lambda x$, by Step 4, and letting $K' \to \infty$,

$$|S^{(\ell,\mathsf{b})}_n(\underline{f})| \le \big(1+\mathsf{U}(\tfrac{1}{10},\lambda x)\big)\mathfrak{B}^n(n!)^{\varkappa_\star}\prod_q \mathrm{Lip}f_q, \qquad \mathfrak{B} := \tfrac{\sqrt{3}}{2}\ell'\,\mathsf{B}_k(\tfrac{1}{2}\ell');$$

and $1+\mathsf{U}(\frac{1}{10},\lambda x) \le \mathsf{A}(x)$, because $\lambda \ge \mathsf{u}_0 > \frac{1}{2}\mathsf{b}_-$ and U does not increase in its second and does not decrease in its first argument by YM3_I:(1785). For $\lambda \le Y_k$: $L(\lambda + \bar{B}) < 2L^2y_k$ and $\log(2L^2y_k) \le (1+\log(2L^2))t_k$, so $\Psi_{\rm s}(\lambda) \le \mathsf{c}_\Psi t_k^{\mathsf{r}}$, and for $\rho \le \bar{\rho}$, $\ell' \le \mathsf{c}_\ell(\bar{\rho})t_k^{\mathsf{r}}$. As $L^{i_k}\ell' \ge 1$ and

$L^{i_k} < L t_k^{1/3}$, Step 2 (i) gives

$$\mathfrak{B} \le \tfrac{27\sqrt{3}}{2}\,\mathsf{k}_1 \frac{L^{4i_k}\ell'^4}{2\varrho_k} \le \mathsf{K}(\bar\rho)\,t_k^{c_*} y_k^{-1/2} = \mathsf{K}(\bar\rho)\,t_k^{c_*} e^{-(t_k-1)/2} \le \mathsf{K}(\bar\rho)\,e^{1/2}\Big(\frac{2c_*}{e}\Big)^{c_*}.$$

For $\rho < \rho_1$ take $\bar\rho = \rho_1$: $\mathfrak{B} \le B_\natural$, and $\mathrm{Lip} f \le \mathsf{b}^{-1}\mathrm{norm}_{\mathsf{b}}(f)$ gives the first bound of (b). The function $u \mapsto (1+\log u)^{c_*}u^{-1/2}$ decreases for $u \ge e^{2c_*-1}$; for $\rho \le \rho_\natural$, $y_k > LM/\rho \ge e^{2c_*-1}$, so $t_k^{c_*}y_k^{-1/2} \le (1+\log\frac{LM}{\rho})^{c_*}(\frac{\rho}{LM})^{1/2}$, which is the second bound of (b).

Step 7: one set (c). For $\rho \ge \rho_1$ use (a) and $\mathsf{B} \le \hat{\mathsf{B}}$; for $\rho < \rho_1$ use (b) and $B_\natural \le \hat{\mathsf{B}}$. □

Definition 18.202 (A coupling that varies with the place)**.** Use the constants and source radius 2ϱ of Theorem YM3_I:AD and Lemma YM3_I:82.5. Let $\mathsf{b} \ge \gamma^{-2}$ and $m \ge 1$ satisfy (Tor), namely $MR(\mathsf{b}) \le L^m$. At every $K \ge 1$ put $\ell = 2L^m$, $\beta = L^K\mathsf{b}$ and $x_j = L^{K-j}\mathsf{b}$ for $0 \le j \le K$. This is an admissible prescribed sequence with zero counterterm increments, and the room condition is that of Lemma YM3_I:82.4. Set

$$\|w\|_\flat = \max\{\sup|w|, (40M/\sqrt{3})\mathrm{Lip}_\infty w\},$$

$$\mathcal{E}_K(w) = \sum_{j=1}^{K}\sum_X e_j(X;w), \qquad G_K(w) = e^{-\mathcal{E}_K(w)}Z_K[w]/Z_K[0].$$

where Lip_∞ uses the torus sup metric and the source energies are those of Proposition YM3_I:83.2. Write $C_{\mathrm{low}}(x) = c_{\mathrm{low}} + \frac{9}{2}\log x$ for the constant of Proposition YM3_I:70.2 and use C_5^{S} and $\mathsf{C}^{\mathrm{S}}_{\mathrm{maj}}$ from YM3_I:(1725)–(1726). Set $Q(\ell,x) = \exp[(\mathsf{C}^{\mathrm{S}}_{\mathrm{maj}} + 2 + C_{\mathrm{low}}(x))\ell^3]$, $Q = Q(\ell,\mathsf{b})$ and $r_1 = \varrho/(Q+1)$. For a real source w, $\|w\|_\flat < 2\varrho$, put $\mu_K^w = Z_K[w]^{-1}e^{-\beta\sum_p a_p + \mathcal{O}_K(w)}dU$, and let $S^w_{n;K}(\underline{f})$ denote the cumulant of $(\mathcal{O}_K(f_q))_q$ in that law. The plaquette coupling under μ_K^{-h} is $a_K^{-1}(\mathsf{b}+h(x(p)))$. A tuple consists of $n \ge 2$ nonzero real Lipschitz functions on T_ℓ with empty common support intersection. Its spread is the positive number $d_{\mathrm{sp}} = \min_{x_q\in\mathrm{supp}\,f_q}\max_{i,j}|x_i - x_j|_\infty$. Write $c_\natural = 54S(\kappa_{\mathcal{D}}+2)C'_\Sigma(2,4)$, with the functions from Lemma YM3_I:88.1 and YM3_I:(1777). Put

$$Q_\flat := \exp\Big[(\tfrac{5}{2}C_5^{\mathrm{S}} + \mathsf{C}^{\mathrm{S}}_{\mathrm{maj}} + 2 + \tfrac{3}{2}c_{\mathrm{low}} + \tfrac{27}{4}\log\mathsf{b})\ell^3\Big], \qquad Q_* := Q\,Q_\flat, \qquad \eta_*(r) := \frac{2\varrho - r}{2(Q_*+1)} \quad (0 \le r < 2\varrho),$$

and for a tuple $\underline{f}$

$$\Sigma^*_{\underline{f}}(r) := \Big[\log 2\Big(\frac{2n}{\eta_*(r)}\Big)^n + \Big(\frac{2n}{2\varrho - r}\Big)^n c_\natural M^3 E_0\Big(\frac{\ell}{d_{\mathrm{sp}}}\Big)^3\Big]\prod_q \|f_q\|_\flat.$$

■

Proposition 18.203 (One torus with the coupling $a_K^{-1}(\mathsf{b}+h(x))$: a floor, the limit along all cutoffs, the series in h, the constant labels)**.** Let the data be as in Definition 18.202.

(a) *The floor; two sides free of the cutoff.* Let $K \ge 1$, $u_\bullet = (x_0,\dots,x_K)$ an admissible finite sequence with the last index K, $((\ell',K),u_\bullet)$ an admissible lattice, $\beta = x_0$, $1 \le k \le K$, $N_k := (L^{K-k}\ell')^3$, and w a real source of the class on $T_{\ell'}$. Then $e^{-E^w_{(k)}}Z_{K,\ell',\beta}[w] \ge \exp[-(\frac{5}{2}C_5^{\mathrm{S}} + \frac{3}{2}C_{\mathrm{low}}(x_k))N_k]$, with the vacuum constant $E^w_{(k)}$ of Proposition YM3_I:83.2. Consequently, on the line of b, for every K and every real w with $\|w\|_\flat < 2\varrho$: $Q_\flat^{-1} \le G_K(w) \le Q$, that is, $Q_\flat^{-1}e^{\mathcal{E}_K(w)} \le \langle e^{\mathcal{O}_K(w)}\rangle_{K,\ell,\beta} \le Q\,e^{\mathcal{E}_K(w)}$.

(b) *A tube without zeros.* Let w_0 be real, $\|w_0\|_\flat \le r_0 < 2\varrho$, and $v \in \mathsf{Lp}(T_\ell)$ complex. For $\|v\|_\flat < 2\varrho - r_0$, $|G_K(w_0+v)/G_K(w_0) - 1| \le (Q_*+1)\|v\|_\flat/(2\varrho - r_0)$; $Z_K[w_0+v] \ne 0$ for $\|v\|_\flat < 2\eta_*(r_0)$; and $|\mathrm{Log}(G_K(w_0+v)/G_K(w_0))| \le \log 2$ for $\|v\|_\flat \le \eta_*(r_0)$. All this for every K.

(c) *A bound free of the cutoff.* Let $w = w_0 + v_0$ with w_0 as in (b) and $\|v_0\|_\flat \le \frac{1}{2}\eta_*(r_0)$, and let $\underline{f}$ be a tuple. The quantity $\mathcal{S}_K(w) := \partial_{z_1}\cdots\partial_{z_n}\lambda(0)$ is independent of the choice of the continuous branch λ of $\log Z_K[w + \sum_q z_q f_q]$ near $z = 0$; when w is real, it equals $S^w_{n;K}(\underline{f})$; moreover $\sup_K |\mathcal{S}_K(w)| \le \Sigma^*_{\underline{f}}(r_0)$.

(d) *The limit along all cutoffs.* For every real Lipschitz h on T_ℓ with $\|h\|_\flat < 2\varrho$ and every tuple $\underline{f}$ the limit

$$S^{(\ell,\mathsf{b}+h)}_n(\underline{f}) := \lim_{K\to\infty} S^{-h}_{n;K}(\underline{f})$$

exists as K tends to infinity through the entire sequence of cutoffs, and it satisfies $|S^{(\ell,\mathsf{b}+h)}_n(\underline{f})| \le \Sigma^*_{\underline{f}}(\|h\|_\flat)$. If an n-tuple, $n \ge 2$, of real Lipschitz functions on T_ℓ has 0 among its entries, then $S^{-h}_{n;K}$ vanishes at it for every K by multilinearity; the limit in the display is then 0, and it is taken as the value of $S^{(\ell,\mathsf{b}+h)}_n$ at that n-tuple. On the tuples and these n-tuples together, $S^{(\ell,\mathsf{b}+h)}_n$ is symmetric and separately linear in every entry.

(e) *The series.* If $\|h\|_\flat < r_1$, in particular if $\|h\|_\sharp \le \varrho/(Q+2)$, then

$$S^{(\ell,\mathsf{b}+h)}_n(\underline{f}) = \sum_{k\ge 0} \frac{(-1)^k}{k!}\, S^{(\ell,\mathsf{b})}_{n+k}(\underline{f}, \underbrace{h,\dots,h}_{k}),$$

with absolute convergence. If $0 < g \le \gamma$, $\ell \in \mathfrak{L}(g)$ and $\gamma^{-2} \le \mathsf{b} \le L(g^{-2} + \bar{B})$, then (Tor) holds and the series holds for $\|h\|_\sharp \le \varrho/(Q(\ell, L(g^{-2}+\bar{B})) + 2)$, a radius in which only ϱ, g, ℓ and the constants enter.

(f) *Constant labels.* For a real constant c_0 with $|c_0| < 2\varrho$: $\mathsf{b} + c_0 > \frac{1}{2}\mathsf{b}$, $S^{-c_0}_{n;K}(\underline{f}) = S^T_{n;K,\ell,L^K(\mathsf{b}+c_0)}(\underline{f})$ for every K, and the limit of (d) at $h = c_0$ is the limit cumulant $S^{(\ell,\mathsf{b}+c_0)}_n(\underline{f})$ at the label $\mathsf{b} + c_0$ (YM3_I:(2212)).

(g) *The function $\mathsf{b} + h$ alone.* If $\mathsf{b} + h = \mathsf{b}' + h'$ as functions, both pairs as in (d), the two sequences of (d) are equal at every K.

■

Sketch. Step 1: the floor (a), first sentence. For real w, the bare density is positive. The source-independent history weights are nonnegative, so YM3_I:(1512)–(1513) makes each history measure positive and $Z_{K,\ell',\beta}[w] \ge \tau^w_{h_\emptyset}(\mathcal{V}_k)$. In the empty history all regular regions are the torus. There are no external fields, boundary contributions or retained factors, and the reference measure is the coarse product Haar probability dV_k, as in the proof of Proposition YM3_I:70.2. The admissible evaluation representatives are those of the real construction and thus can be used at both w and 0. In Theorem YM3_I:AJ, the stay factor is one, the cutoff χ is source-independent, and the last bound of YM3_I:(1731) gives $|\mathbf{E}^w_k| \le C^{\mathrm{S}}_5 N_k$ because no cube is created. The deficit vanishes in YM3_I:(1735). Thus the empty-history density, outside the common reference null set, obeys

$$e^{-E^w_{(k)}} D^w(V) = \chi(V)\exp[-A(\beta^w_k, \mathsf{U}) + \mathbf{E}^w_k(\mathsf{U})]$$

on the cutoff support, with U any admissible representative over V, and the density is zero off it. The exponents are real by conjugation. The source admission in Lemma YM3_I:82.5 bounds the relative change of the level-k coupling by $a = \|w\|_\infty/(x_K - \bar{B}) < 1/2$. Every plaquette action of a real configuration is nonnegative, so $A(\beta^w_k, \mathsf{U}) \le (1+a)A(\beta_k, \mathsf{U})$. Comparing the formulas for D^w and D^0 gives

$$e^{-E^w_{(k)}} D^w \ge e^{-(2+a)C^{\mathrm{S}}_5 N_k}(e^{-E_{(k)}} D^0)^{1+a}.$$

Here the cutoff agrees at both sources, with $\chi \ge \chi^{1+a}$. Integrating, Jensen's inequality for dV_k and the zero-source empty-history floor YM3_I:(1488) give the lower exponent $(2+a)C^{\mathrm{S}}_5 + (1+a)C_{\mathrm{low}}(x_k)$, which is at most $\frac{5}{2}C^{\mathrm{S}}_5 + \frac{3}{2}C_{\mathrm{low}}(x_k)$.

Step 2: two sides (a), second sentence. At the last level of the line of b, $x_K = \mathsf{b}$ and $N_K = \ell^3$. The vacuum splitting YM3_I:(1737) identifies $G_K(w) = \widetilde{Z}_K(w)/\widetilde{Z}_K(0)$. The upper bound YM3_I:(1738) and the zero-source floor YM3_I:(1488) give $|G_K(w)| \le Q$ throughout the complex source ball. Dividing the real-source floor of Step 1 by the same upper bound at $w = 0$ gives $G_K(w) \ge Q_\flat^{-1}$. Proposition YM3_I:83.2 gives holomorphy on the ball and $\mathcal{E}_K(0) = 0$.

Step 3: the tube (b). Put $\mathsf{g}_K(v) := G_K(w_0 + v)/G_K(w_0)$ for $\|v\|_\flat < 2\varrho - r_0$. By Step 2, g_K is holomorphic along every family $w_0 + \sum_i z_i f_i$, $\mathsf{g}_K(0) = 1$ and $|\mathsf{g}_K| \le QQ_\flat = Q_*$. For $v \ne 0$ the function $\zeta \mapsto \mathsf{g}_K(\zeta v) - 1$ is holomorphic on $|\zeta| < (2\varrho - r_0)/\|v\|_\flat$, a radius > 1, vanishes at 0 and has modulus $\le Q_* + 1$; Schwarz's lemma at $\zeta = 1$ gives the first assertion. Where $|\mathsf{g}_K(v) - 1| < 1$, $Z_K[w_0 + v] = Z_K e^{\mathcal{E}_K(w_0+v)} G_K(w_0)\mathsf{g}_K(v) \ne 0$; where it is $\le \frac12$, $|\operatorname{Log} \mathsf{g}_K(v)| \le \log 2$.

Step 4: the bound (c). On $\|v\|_\flat < 2\eta_*(r_0)$ the function $Y^{w_0}(v) := \mathcal{E}_K(w_0+v) - \mathcal{E}_K(w_0) + \operatorname{Log} \mathsf{g}_K(v)$ is holomorphic along every family, $Y^{w_0}(0) = 0$, and $e^{Y^{w_0}(v)} = Z_K[w_0 + v]/Z_K[w_0]$; a continuous branch λ of $\log Z_K[w + \sum_q z_q f_q]$ near 0 differs from $z \mapsto Y^{w_0}(v_0 + \sum_q z_q f_q)$ by a constant, so $\mathcal{S}_K(w)$ is the mixed derivative of the latter, whatever pair (w_0, v_0) represents w. At a real w, with $w_0 = w$ and $v_0 = 0$, $e^{Y^w(\sum_q z_q f_q)} = \mu_K^w(e^{\sum_q z_q \mathcal{O}_K(f_q)})$, and the mixed derivative is the joint cumulant. On the polydisc $|z_q| < \eta_*(r_0)/(2n\|f_q\|_\flat)$, $\|v_0 + \sum_q z_q f_q\|_\flat < \eta_*(r_0)$, so $|\operatorname{Log} \mathsf{g}_K| \le \log 2$ by (b), and Cauchy's inequalities bound the mixed derivative of $\operatorname{Log} \mathsf{g}_K(v_0 + \sum_q z_q f_q)$ at 0 by $\log 2\,(2n/\eta_*(r_0))^n \prod_q \|f_q\|_\flat$. For the local energies take the polydisc $|z_q| < (2\varrho - \|w\|_\flat)/(n\|f_q\|_\flat)$. A term of $\mathcal{E}_K(w + \sum_q z_q f_q)$ has zero full mixed derivative unless its enlarged domain $X^{\sim 1}$ meets every support. At depth $s = K - j$ this implies $d_j(X) \ge d_{\mathrm{sp}} L^s/M - 4$ by YM3_I:(1773). Use $\operatorname{supp} f_1$ to anchor the contributing domains. The count YM3_I:(1774)–(1775), the per-term bound $2M^3 E_0 e^{-\kappa d_j(X)}$ in YM3_I:(1493), and the scale sum YM3_I:(1777) give

$$\sum_{j,X:\, X^{\sim 1} \text{ meets every support}} 2M^3 E_0 e^{-\kappa d_j(X)} \le 54 M^3 E_0 S(\kappa) \sum_{s=0}^{K-1} (\ell L^s/M)^3 e^{-(\kappa - \kappa_{\mathcal{D}})[d_{\mathrm{sp}} L^s/M - 4]_+}$$
$$\le c_\natural M^3 E_0 (\ell/d_{\mathrm{sp}})^3.$$

The last inequality follows by using $\kappa \ge \kappa_{\mathcal{D}} + 2$ in the sum and in S. Cauchy's formula on the stated polydisc then gives the second term in $\Sigma^*_{\underline{f}}(r_0)$, since $2\varrho - \|w\|_\flat \ge (2\varrho - r_0)/2$. Adding the logarithmic and local terms proves (c).

Step 5: from a centre to the whole disc. Let ϕ_K, $K \in \mathbb{N}$, be holomorphic on $|\zeta - \zeta_0| < \rho$ with $|\phi_K| \le \varsigma$, and let $\lim_K \phi_K^{(\nu)}(\zeta_0)$ exist for every $\nu \ge 0$. Then $\lim_K \phi_K^{(\mu)}(\zeta_1)$ exists for every ζ_1 of the disc and every $\mu \ge 0$. Indeed, by Cauchy's inequalities the Taylor coefficients $c_{K,\nu}$ at ζ_0 satisfy $|c_{K,\nu}| \le \varsigma \rho^{-\nu}$, and they converge to numbers c_ν with the same bound; $\phi_K^{(\mu)}(\zeta_1) = \sum_\nu (\nu+\mu)\cdots(\nu+1) c_{K,\nu+\mu}(\zeta_1 - \zeta_0)^\nu$, whose terms are bounded by $\varsigma \rho^{-\mu}(\nu+\mu)^\mu t^\nu$, $t = |\zeta_1 - \zeta_0|/\rho < 1$, uniformly in K; dominated convergence gives the limit, which is the derivative of $\sum_\nu c_\nu(\zeta - \zeta_0)^\nu$ at ζ_1.

Step 6: the limit (d). Let $u := -h \ne 0$ (at $h = 0$ this is Corollary YM3_I:119.2), $r_0 := \|u\|_\flat$, $\eta := \eta_*(r_0)/2r_0$. For $t \in [0,1]$ and $|\zeta - t| < \eta$, $\zeta u = tu + (\zeta - t)u$ with tu real, $\|tu\|_\flat \le r_0$ and $\|(\zeta - t)u\|_\flat < \frac12 \eta_*(r_0)$; so $\Psi_K(\zeta) := \mathcal{S}_K(\zeta u)$ is defined within η of $[0,1]$, independent of t by (c), bounded by $\Sigma^*_{\underline{f}}(r_0)$, and holomorphic, as the mixed derivative in z at 0 of the function $(\zeta', z) \mapsto Y^{tu}(\zeta' u + \sum_q z_q f_q)$, holomorphic in all its variables. At $\zeta = 0$ its derivatives are the joint cumulants $\Psi_K^{(\nu)}(0) = S^T_{n+\nu;K,\ell,L^K\mathsf{b}}(u, \dots, u, \underline{f})$, which converge to $S^{(\ell,\mathsf{b})}_{n+\nu}(u, \dots, u, \underline{f})$ by Corollary YM3_I:119.2: the supports of $(u, \dots, u, \underline{f})$ have no common point, those of $\underline{f}$ having none. Put $t_i := i\eta/2$ for the integers $0 \le i \le I := \lceil 2/\eta \rceil - 1$, and $t_{I+1} := 1$; consecutive points are at distance $\le \frac{\eta}{2}$. Step 5 with $\zeta_0 = t_i$, $\rho = \eta$, $\varsigma = \Sigma^*_{\underline{f}}(r_0)$ and $\zeta_1 = t_{i+1}$ carries the convergence of all the derivatives from t_i to t_{i+1};

at $t_{I+1}=1$, $\Psi_K(1)=\mathcal{S}_K(-h)=S^{-h}_{n;K}(\underline{f})$ converges. The bound passes to the limit; symmetry and linearity hold at every K for the cumulants of one law and pass to the limit.

Step 7: the series (e). At the center $w_0=0$ the proof of (b) uses $G_K(0)=1$ and $|G_K|\le Q$, so it gives $|\operatorname{Log} G_K(w)|\le \log 2$ for $\|w\|_\flat\le r_1$. The polydisc argument in Step 4, using radii $(r_1-\|w\|_\flat)/(n\|f_q\|_\flat)$ for the logarithm and the source-ball radii for $\mathcal{E}_K$, therefore bounds $\mathcal{S}_K(w)$ uniformly in K on each smaller ball $\|w\|_\flat\le r<r_1$. If $h\neq 0$ and $\|h\|_\flat<r_1$, choose $R_0>1$ with $R_0\|h\|_\flat<r_1$. The holomorphic functions $\zeta\mapsto\mathcal{S}_K(-\zeta h)$ are uniformly bounded on $|\zeta|\le R_0$. Their Taylor coefficients at zero are $(-1)^\nu S^T_{n+\nu;K,\ell,\beta}(\underline{f},h^{\times\nu})/\nu!$. Cauchy's estimate bounds these coefficients by a constant times $R_0^{-\nu}$. At $\zeta=1$ the series converges absolutely, and Corollary YM3_I:119.2 and dominated convergence give the claimed series for the limit. The case $h=0$ is immediate. The norm inequality $\|h\|_\flat\le\|h\|_\sharp$ follows from $|x-y|\le\sqrt{3}|x-y|_\infty$. Finally, $Q(\ell,x)$ increases with x, and YM3_I:(1491) gives $L^m\ge LMR(L(g^{-2}+\bar{B}))\ge MR(\mathsf{b})$ in the stated range. This proves the uniform choice of radius.

Step 8: constant labels (f) and the function $\mathsf{b}+h$ (g). Source admission YM3_I:(1717) gives $2\varrho/(\mathsf{b}-\bar{B})\le\sigma^{\mathrm{S}}\le\frac12$; so $2\varrho\le\frac12(\mathsf{b}-\bar{B})$ and $\mathsf{b}+c_0>\frac12\mathsf{b}$. As $\mathcal{O}_K(-c_0)=-c_0L^K\sum_p a_p$, $e^{-L^K\mathsf{b}\sum_p a_p+\mathcal{O}_K(-c_0+w')}=e^{-L^K(\mathsf{b}+c_0)\sum_p a_p+\mathcal{O}_K(w')}$; so $\mu_K^{-c_0}$ is the Wilson measure at $\beta'=L^K(\mathsf{b}+c_0)$, whose label is $\mathsf{b}+c_0$, and $S^{-c_0}_{n;K}=S^T_{n;K,\ell,L^K(\mathsf{b}+c_0)}$; its limit is $S_n^{(\ell,\mathsf{b}+c_0)}$ (Corollary YM3_I:119.2). Finally the weight $\exp[-\sum_p L^K(\mathsf{b}+h(x(p)))\,a_p]$ of μ_K^{-h} depends on the function $\mathsf{b}+h$ alone, which gives (g). □

18.22. **Lattice symmetries of the periodic Gibbs states, block fields of one torus and of the threads, the Hilbert space of the threads, and the two lists of axioms.** With the notation of Definition 18.209, this subsection treats the lattice symmetries of the periodic Gibbs states, the laws of the block fields at one cutoff and at infinite cutoff, the Hilbert space which the threads over $\mathbb{R}^3$ carry, and the relation of the axioms of the main text to two lists of axioms written out in full.

Proposition 18.210 works at one spacing $a>0$ and every $\beta\ge0$. Every periodic Gibbs state of the SU(2) Wilson weight is invariant under the whole group $a\mathbb{Z}^3\rtimes W_3$ of the lattice isometries, parity among them, and reflection positive at all the lattice mirrors of $a\mathbb{Z}^3$; so every bounded function which an isometry multiplies by a number $\omega\neq1$ has mean 0 and conditional mean 0 given the invariant events, every bounded function which is invariant, almost everywhere, under a sublattice of finite index or under the translation by a single vector $w\neq0$ of $a\mathbb{Z}^3$ is invariant, almost everywhere, under all lattice isometries, no isometry permutes the pure phases, a pure phase has no order of a sublattice, odd functions have no long-range order, and a periodic field which is switched off after the volume has been taken to infinity leaves no trace, neither in the order parameter nor in the free energy per term. Proposition 18.218 measures the distance between the laws of the threads over a cell, at infinite cutoff, and the thread measure of one torus T_ℓ by the excess $x=\ell^3(p_\ell-p_\infty)$ of the free energy of that torus over its infinite-volume value: the domination of Lemma 18.217(a) holds with $e^{\underline{x}/8}$, $\underline{x}$ the lower limit of the excess along the subsequence which gives the law, in the place of Q_0, the relative entropy is at most $\underline{x}$, and the distance in total variation is at most $\min\{\sqrt{\underline{x}/2},\mathfrak{t}(\underline{x})\}\le\frac12(\sqrt{\epsilon}+\epsilon)$ with $\epsilon=22.3(e^{\bar{x}/7}-1)$, $\bar{x}$ the upper limit; at the data at g and the sides of $\mathfrak{L}(g)$ the excess is at most $\ell^3(2C_{\mathrm{S}}+C_g)=8\log\mathsf{Q}_0$.

Proposition 18.219 is the clustering bound of the main text applied to bounded functions of the block fields of the depths $\ge d$ in a cube $[-R,R]^3$, at every cutoff, every $\beta\ge0$ and in every periodic Gibbs state, with the physical length $R+\frac12 a_d$ and the absolute constant $4\sqrt{3}\,\mathsf{C}_{\mathrm{dc}}$. Proposition 18.221 passes it to the laws $\boldsymbol{\nu}$ of the towers of block fields in infinite volume, along a defining sequence of a volume-first limit: these laws cluster at the rate $\rho'/|x|$, they are reflection positive at the coordinate planes $\{x_\kappa=c\}$ with $2c\in\bigcup_s a_s\mathbb{Z}$, the coordinate lattice mirrors of all depths, on the functions of the block fields whose footprints lie on one side, and the space $\mathcal{H}^{\mathrm{thr}}$ which this positivity

makes of the bounded functions of the block fields in a half space carries a semigroup of self-adjoint contractions, a Hamiltonian on the closed span of the vectors at positive distance from the mirror, horizontal translations with the cone $H^{\mathrm{thr}} \geq |P_\lambda^{\mathrm{thr}}|$, and a space of zero energy which is L^2 of the invariant events; it holds the loops of the block fields; and along a further subsequence the lattice correlations of a function of the block fields, reflected in the plane, with a combination of centred products of the action density converge to a form bounded by the product of the norms in the two Hilbert spaces, and so give an operator J of norm one from the Hilbert space of S, placed at the plane, into $\mathcal{H}^{\mathrm{thr}}$, with $J\Omega = \Omega$, which intertwines the semigroups, the horizontal translations and the group W_2. Lemma 18.222 bounds one step of the block-averaging map at the lowest layer above a plane between sites, with a constant $\mathsf{c}_L < 0.554\sqrt{L}$, and Proposition 18.223 uses it, with the main text's bound on the mass of the configurations whose block fields of depth d are not small (Proposition YM3_I:115.2), to show that the threads are regular, that the block fields whose footprints touch a plane between sites are, almost surely, functions of the block fields at positive distance above it, and hence that the translation by e_3 is injective on $\mathcal{H}^{\mathrm{thr}}$: the Hamiltonian of the threads is set on the whole space. Proposition 18.224 works on one periodic layer with sides $n_1, n_2 \geq 3$: the transfer operator, the plaquette actions in the layer and the transfer operator with one time-like plaquette inserted generate all bounded operators of the flux-free sector, and the sum of all bounded operators of the four flux sectors on the gauge-invariant space.

Definition 18.225 writes out two lists of axioms, List OS for Euclidean Green's functions in three dimensions and List GW for one Hermitian scalar field on $\mathbb{R}^{1+2}$, and the relation (Sch) between a Wightman theory and its Schwinger functions. Proposition 18.226 is a dictionary: the axioms (E0′)–(E4) of the main text imply List OS, with (E0′) equivalent to the growth axiom of the list; for data (Rec) with the properties of Proposition YM3_I:111.2 the larger test spaces of the main text add no vector to the Hilbert space; the Wightman theories of the main text are the quadruples of List GW, the group of the main text being the restricted Lorentz group; a Wightman theory whose forms are boundary values of holomorphic functions is determined, up to unitary equivalence, by the values of these functions at the time-ordered Euclidean points; the theory which the main text reconstructs from data (Rec) has its forms as boundary values in the strong sense, its fields commute also where the supports of the test functions touch the light cone, and the reflections of space are implemented on it; every ergodic component of every per-limit at g satisfies List OS with constants which are the same for all per-limits at g, and reconstructs a theory of List GW whose Schwinger functions are the kernels of the component, the Hilbert space of List OS being the Hilbert space of the component; and the limit itself satisfies all of List OS but the cluster axiom, which it satisfies exactly when it has one ergodic component, that is, when its vacuum is unique.

Propositions 18.210, 18.219 and 18.224, and clauses (a), (b) of Proposition 18.226, are fixed-spacing or axiomatic statements.

Definition 18.204 (Lattice laws)**.** At spacing a, a configuration assigns an element of $G = \mathrm{SU}(2)$ to each positive coordinate bond, with $U(\bar{b}) = U(b)^{-1}$ on the reversed bond. A law is a Borel probability measure on this compact product space. For a bond set B, $\mathcal{F}(B)$ consists of its bounded Borel observables. The plaquette variable $\partial U(p)$ is the boundary product and $a_p = 1 - \frac{1}{2}\operatorname{tr}\partial U(p)$. For a compactly supported weight f, set

$$\mathcal{O}_a(f) = a^{-1}\sum_p f(x(p))a_p, \qquad \mathcal{O}_K = \mathcal{O}_{a_K},$$

where $x(p)$ is the barycenter. The joint cumulant of these observables under ν is denoted by $S_n^T[\nu]$.

The Wilson torus law $\mu_{K,\ell,\beta}$ has density proportional to $e^{-\beta\sum_p a_p}$ with respect to product Haar measure, and $\tilde{\mu}_{K,\ell,\beta}$ is its periodic extension. The finite-bond Gibbs kernel γ_B integrates the bonds of B with Haar measure and Wilson weight from precisely the plaquettes that use a bond of B, then divides by the integral of that weight. A periodic Gibbs state is a law fixed by every γ_B and

invariant under a finite-index subgroup of translations. Write $\mathcal{G}^{\mathrm{per}}_{a,\beta}$ for these states and use the subscript K when $a = a_K$. The class $\mathcal{G}^{\mathrm{tor}}_{K,\beta}$ consists of weak limits of $\tilde{\mu}_{K,2L^m,\beta}$ with $m \to \infty$ and is included in $\mathcal{G}^{\mathrm{per}}_{K,\beta}$ by Proposition YM3_I:103.4. Write $\mathcal{I}$ for the Borel sets invariant under all lattice translations. A pure phase is a periodic Gibbs state that takes only the values zero and one on $\mathcal{I}$. ∎

Definition 18.205 (Gibbs sequences and their limits)**.** Let $\mathfrak{S}_n$ be the set of n-tuples of real compactly supported C^1 functions with pairwise disjoint supports and include the empty tuple. A type $(\mathsf{d}; R)$ means that the supports are separated by at least d and each fits in a ball of radius R. Put $\|f\|_\sharp = \max\{\sup|f|, 40M \operatorname{Lip} f\}$. A Gibbs sequence at g is a sequence of first-passage data (K_j, β_j) with $K_j \to \infty$, together with $\nu_j \in \mathcal{G}^{\mathrm{per}}_{K_j,\beta_j}$. For $X = (f_1, \dots, f_n)$ let

$$X^\nu_K = \prod_{i=1}^{n} [\mathcal{O}_K(f_i) - \nu(\mathcal{O}_K(f_i))], \qquad \varnothing^\nu_K = 1.$$

The sequence converges if $\nu_j(X^{\nu_j}_{K_j})$ converges for every separated tuple X. A per-limit S has connected functions given by the limits of $S^T_n[\nu_j]$, $n \geq 2$, on such tuples. This Gibbs sequence is a defining sequence of S. It is volume-first if each ν_j lies in $\mathcal{G}^{\mathrm{tor}}_{K_j,\beta_j}$. For a growing-torus limit take cumulants on tori of sides $\ell_j \in \mathfrak{L}(g)$ with $\ell_j \to \infty$ and periodize each tuple. The centered functional σ_S is given by summing the products of $S_{|J|}(X_J)$ over partitions with all blocks of size at least two, with $\sigma_S(\varnothing) = 1$. The inverse formula for a functional σ is

$$S^\sigma_n(X) = \sum_{\pi} (-1)^{|\pi|-1} (|\pi| - 1)! \prod_{J \in \pi} \sigma(X_J).$$

These formulas identify convergence of the centered moments with convergence of the cumulants, since in each degree the two formulas are finite polynomials and the centered one-point term is zero. Use the reconstruction and canonical decomposition of Theorems YM3_I:AL and YM3_I:AP for σ_S. ∎

Definition 18.206 (Towers and windows)**.** Use $a_d = L^{-d}$ and write Bd_d for the positive bonds at depth d. Put $\mathsf{Z}_d = G^{\mathsf{Bd}_d}$, $\mathsf{Z} = \prod_{d \geq 0} \mathsf{Z}_d$, and $\mathsf{Z}^{[s]} = \prod_{d \geq s} \mathsf{Z}_d$. For side ℓ, $\mathsf{V}_d[\ell]$ is the configuration space on $a_d\mathbb{Z}^3/\ell\mathbb{Z}^3$. Write pr_ℓ for the quotient, per_ℓ for periodic extension, and Per_ℓ for that extension on every depth. The map $\mathcal{M}_\ell$ from depth $d+1$ to depth d is the double-block average of Section YM3_I:4. Its lattice version $\mathcal{M}$ uses the same rule in each double block, as justified in Lemma 18.211(a). The tower $\Psi_K U$ has depth-d field $\mathcal{M}^{K-d}U$ for $d \leq K$ and the identity field for $d > K$. The thread set is

$$\mathsf{Thr} = \{\mathsf{z} : \mathsf{z}_d = \mathcal{M}\mathsf{z}_{d+1} \text{ for every } d \geq 0\}.$$

For $v \in a_s\mathbb{Z}^3$, T_v translates the fields at depths $d \geq s$ and fills in $\mathcal{M}^{s-d}\tau_v \mathsf{z}_s$ at $d < s$. Let $\mathcal{I}_\infty$ be the Borel sets invariant on Thr under every such translation. On threads the action is independent of the choice of s, by Lemma 18.217(b).

Fix $\ell = 2L^m$ and $\square = [-\ell/4, \ell/4]^3$. Write w_d for the bonds with endpoints in $\square$, r_d for restriction to those bonds, and $Y_d = G^{\mathsf{w}_d}$. Put $Y = \prod_d Y_d$, $Y^{[d_0]} = \prod_{d \geq d_0} Y_d$ and $Y_{\leq D} = \prod_{d \leq D} Y_d$, with projections $\mathrm{pj}^{[d_0]}$ and q_D. If $o \in a_{d_0}\mathbb{Z}^3$, set $\varpi^{[d_0]}_o(\mathsf{z}) = (r_d \tau_{-o} \mathsf{z}_d)_{d \geq d_0}$. The window tower is $\Upsilon_K = \varpi^{[0]}_0 \Psi_K$. Define m^n_d by extending an input at depth $d+n$ by the identity outside $\square$, applying $\mathcal{M}^n$, and restricting at depth d. Then $\mathsf{Thr}_\square$ consists of the $y \in Y$ with $y_d = \mathsf{m}^1_d(y_{d+1})$ for all d. For closed $\mathfrak{c} \subset \square$, $Y(\mathfrak{c})$ and $p_\mathfrak{c}$ are the product space and coordinate projection retaining only bonds inside $\mathfrak{c}$. ∎

Definition 18.207 (Reference laws and product domination)**.** A compact interval $\mathsf{J} = [b_-, b_+]$ is admitted at $\ell = 2L^m$ if $b_- \geq \gamma^{-2}$ and $MR(b_+) \leq L^m$, where $R(x)$ is the least power L^n, $n \geq 0$,

with $L^n \geq (\log x)^r$ for the room exponent of the construction. The torus thread law $\mu_{\ell,\mathsf{b}}$ and its image $\mu[\ell,\mathsf{b}] = (\varpi_0^{[0]}\mathsf{Per}_\ell)_*\mu_{\ell,\mathsf{b}}$ are constructed in Lemma 18.212. Write $\mu^{[d_0]} = \mathrm{pj}_*^{[d_0]}\mu[\ell,\mathsf{b}]$. For domination $\preceq_{\mathsf{Q}}$ use Lemma 18.214. The product bound for a tower law $\boldsymbol{\pi}$ is

$$\left|\boldsymbol{\pi}\left(\prod_i f_i \circ \varpi^{[d_0]}_{o+(\ell/2)\mathbf{k}_i}\right)\right| \leq \prod_i \mathsf{Q}_0\, \mu^{[d_0]}(|f_i|^8)^{1/8}, \qquad \mathsf{Q}_0 = \exp\left[\tfrac{1}{8}\ell^3(2C_{\mathrm{S}} + C_g)\right],$$

for bounded Borel window functions, distinct $\mathbf{k}_i \in \mathbb{Z}^3$, and any $o \in a_{d_0}\mathbb{Z}^3$. When another constant is specified, use it in place of Q_0. ■

Definition 18.208 (The standing hypothesis for tower limits)**.** In (SH), $0 < g \leq \gamma$, $\ell \in \mathfrak{L}(g)$ and (K_j, β_j) are first-passage data at g with $K_j \to \infty$ and $a_{K_j}\beta_j \to \mathsf{b}$. The law ν_j is either periodic Gibbs or the lift of a torus law of side $\ell_j \geq \ell$ in $\mathfrak{L}$. Infinite volume means that for every R all sufficiently late ν_j are periodic Gibbs or lifted from tori of sides at least R. ■

Definition 18.209 (Notation for the lattice and thread arguments)**.** Use the construction constants for SU(2), including the odd blocking factor $L \geq 13$, the cube size M and the reference couplings $0 < g \leq \gamma$. Write $a_d = L^{-d}$, $\mathfrak{L} = \{2L^m : m \geq 1\}$ and $T_\ell = \mathbb{R}^3/\ell\mathbb{Z}^3$. The first-passage data and $I(g)$ are those of Section YM3_I:74 and Corollary YM3_I:74.2, with $\bar{B} = LB_\sharp/(L-1)$ and $X_g = L(g^{-2} + B_\sharp)$. The side set $\mathfrak{L}(g)$ is given by YM3_I:(1528). The interval $\mathsf{J}_g = [g^{-2}, L(g^{-2} + \bar{B})]$ contains $I^{\mathrm{lab}}(g)$ of YM3_I:(1854). The stability and comparison constants $C_{\mathrm{S}}, C_g, C_{\mathrm{gap}}$ are those of Theorems YM3_I:AD and YM3_I:AJ. The two-point normalization $\mathsf{c}_2 = 9/(16\pi^2)$, kernel $\mathcal{K}_{\mathrm{free}}$ and threshold γ_W are those of Theorem YM3_I:AV.

The configuration space $\mathcal{V}(a\mathbb{Z}^3)$ of Definition 18.204 is a countable product of compact metrizable spaces, with the product topology. For $S \subset \mathbb{R}^3$, $\mathsf{Bd}(S)$ consists of the positively oriented bonds whose closed segments lie in S. Let $\mathcal{F}_{\mathrm{loc}}(S)$ be the bounded complex Borel functions of finitely many such bonds, and $\mathcal{C}_{\mathrm{loc}}(S)$ their continuous members. We write $\mathcal{C}_{\mathrm{loc}}$ for $\mathcal{C}_{\mathrm{loc}}(\mathbb{R}^3)$. This algebra separates points and is closed under conjugation, so it is uniformly dense in $C(\mathcal{V}(a\mathbb{Z}^3))$.

For a finite-index subgroup $\Gamma \subset a\mathbb{Z}^3$, set $\mathrm{sys}(\Gamma) = \min_{0 \neq v \in \Gamma} |v|_\infty$. A period lattice has $\mathrm{sys}(\Gamma) \geq 3a$, and $\mathcal{V}_\Gamma$ consists of the configurations fixed by all translations in Γ. Write S_Γ for the sum of the plaquette actions over the plaquette classes modulo Γ. For a real continuous function Φ on $\mathcal{V}_\Gamma$, $Z_\Gamma[\Phi]$ is the product Haar integral of $e^{-\beta S_\Gamma + \Phi}$ on the periodic configuration space, and μ_Γ^Φ is the probability law with the corresponding normalized density. This is the finite-volume law in YM3_I:(1817). Write $\mathcal{G}_{a,\beta}$ for all laws fixed by the finite-bond Gibbs kernels of Definition 18.204. A locally finite source family $\mathfrak{A} = (A_i, B_i)_i$ consists of real continuous functions A_i carried by finite bond sets B_i, each bond belonging to only finitely many B_i. At strength t the kernel on a finite bond set B adds the term $t\sum_{i:B_i \cap B \neq \emptyset} A_i$ to the Wilson exponent. A Gibbs state for $(\beta, t\mathfrak{A})$ satisfies the resulting finite-bond equations.

Set $\mathfrak{N} = \{e_1, e_2, e_3\} \cup \{e_\mu \pm e_\lambda : 1 \leq \mu < \lambda \leq 3\}$. For $n \in \mathfrak{N}$ the lattice mirror $\Sigma(n,c)$ is the plane $n \cdot x = c$, with $c \in (a/2)\mathbb{Z}$ for coordinate n and $c \in a\mathbb{Z}$ for diagonal n. Its closed sides are $\mathbb{H}^\pm(n,c) = \{x : \pm(n \cdot x - c) \geq 0\}$ and its reflection is $\theta_{n,c}x = x - 2|n|^{-2}(n \cdot x - c)n$. The linear part is a signed permutation, and the translation part belongs to $a\mathbb{Z}^3$. Let $\mathcal{F}^+_{n,c} = \mathcal{F}_{\mathrm{loc}}(\mathbb{H}^+(n,c))$. An isometry acts on configurations by $(\mathsf{r}U)(b) = U(\mathsf{r}^{-1}b)$, on functions by $(\mathsf{r}F)(U) = F(\mathsf{r}^{-1}U)$, and on events by their images. For a translation this gives $\tau_v F = F \circ \tau_{-v}$.

The reflection size $\mathcal{N}_r$ and the absolute constants C_{dc} and $\mathsf{k} = (2/\pi)\log(1+\sqrt{2})$ are given by YM3_I:(1959), YM3_I:(1968) and YM3_I:(1954), respectively. For side ℓ write $\Lambda_{d,\ell} = a_d\mathbb{Z}^3/\ell\mathbb{Z}^3$. A coordinate of a law on Z is $\mathsf{z}_d(c)$, with c a bond at depth d. On 2×2 matrices $|\cdot|$ is the operator norm. The Gibbs sequences, towers, reference laws and hypothesis (SH) are specified in Definitions 18.205–18.208. ■

Proposition 18.210 (No periodic Gibbs state breaks a lattice isometry)**.** Let $a > 0$ and $\beta \geq 0$ be real numbers and let $\nu \in \mathcal{G}^{\mathrm{per}}_{a,\beta}$ be any periodic Gibbs state at (a,β), pure or not, of any period.

For a bounded Borel function h and $v \in a\mathbb{Z}^3$ we write $\tau_v\mathsf{h}$ for the action of the translation by v on functions, $(\tau_v\mathsf{h})(U) = \mathsf{h}(\tau_{-v}U)$, and $\mathbb{E}_\nu[\,\cdot\,|\mathcal{I}]$ is conditional expectation with respect to the invariant events.

(a) *Symmetry and positivity.* $\nu(\mathsf{r}F) = \nu(F)$ for every bounded Borel function F and every $\mathsf{r} \in \mathsf{Iso}(a\mathbb{Z}^3) = a\mathbb{Z}^3 \rtimes W_3$, among them the parity $x \mapsto -x$, the 48 signed permutations and the reflections in all the lattice mirrors; and $\nu(\overline{\theta_{n,c}F}\, F) \geq 0$ at every lattice mirror (n, c) of the nine families for every bounded complex Borel function F of finitely many bonds whose closed segments lie in one closed side of the mirror. In particular the mean $\nu(a_p)$ has one value for all the plaquettes p of the three orientations.

(b) *Odd functions have mean zero, in every phase.* For every bounded Borel function F and every $\mathsf{r} \in \mathsf{Iso}(a\mathbb{Z}^3)$, $\mathbb{E}_\nu[\mathsf{r}F|\mathcal{I}] = \mathbb{E}_\nu[F|\mathcal{I}]$ ν-almost surely. If $\mathsf{r}F = \omega F$ with a complex number $\omega \neq 1$, then $\nu(F) = 0$ and $\mathbb{E}_\nu[F|\mathcal{I}] = 0$ ν-almost surely; in particular $\nu(F) = 0$ for every bounded Borel F that is odd under the parity or under the reflection in a lattice mirror. The clause is not empty on gauge-invariant local functions: for the closed path C of the eight steps $ae_3, ae_3, ae_2, ae_1, -ae_3, -ae_1, -ae_3, -ae_2$ from the site 0 and $W_C(U) = \frac{1}{2}\operatorname{tr} U(C)$, the function $W_C - W_{\mathsf{P}C}$, P the parity, is continuous, local, gauge invariant, odd under P and not 0; so $\nu(W_C) = \nu(W_{\mathsf{P}C})$.

(c) *Functions invariant under a sublattice, or under one translation.* Fix a subgroup $\Gamma' \subset a\mathbb{Z}^3$ whose index is finite, and a bounded Borel function h such that, at each $v \in \Gamma'$, the equality $\tau_v\mathsf{h} = \mathsf{h}$ holds ν-almost everywhere. Then the equality $\mathsf{rh} = \mathsf{h}$ also holds ν-almost everywhere, at each $\mathsf{r} \in \mathsf{Iso}(a\mathbb{Z}^3)$; consequently each Borel set E for which $\nu(\tau_v E \triangle E) = 0$ at all $v \in \Gamma'$ satisfies $\nu(\mathsf{r}E \triangle E) = 0$ for all these r, and this applies in particular to each $E \in \mathcal{I}$, for any value of $\nu(E)$. The same conclusion holds if $\tau_w\mathsf{h} = \mathsf{h}$ ν-almost everywhere for one vector $w \in a\mathbb{Z}^3 \setminus \{0\}$ alone; then h coincides ν-almost everywhere with an $\mathcal{I}$-measurable function.

(d) *Pure phases.* For every regular conditional distribution $(\nu^U)_U$ of ν given $\mathcal{I}$, $\nu = \int \nu^U\, \nu(dU)$, and for ν-almost every U the law ν^U is a pure phase, to which (a)–(c) apply. So $\mathsf{r}_*\rho = \rho$ for every pure phase ρ and every $\mathsf{r} \in \mathsf{Iso}(a\mathbb{Z}^3)$: no lattice isometry permutes the pure phases, and every lattice isometry fixes ν^U for ν-almost every U. A pure phase ρ has $\rho(E) \in \{0, 1\}$ for every Borel set E with $\rho(\tau_w E \triangle E) = 0$ for one $w \in a\mathbb{Z}^3 \setminus \{0\}$; in particular a pure phase has no order of a sublattice: $\rho(E) \in \{0, 1\}$ whenever $\rho(\tau_v E \triangle E) = 0$ for all v of a subgroup of finite index.

(e) *No long-range order of an odd function.* Let $r \geq 1$ be an integer, $Q_r = [-ra, ra]^3$, $A, B \in \mathcal{F}(\mathsf{Bd}(Q_r))$, and let C_{dc} be the absolute number YM3_I:(1968). If $\mathsf{r}A = \omega A$ for some $\mathsf{r} \in \mathsf{Iso}(a\mathbb{Z}^3)$ and some $\omega \neq 1$, then

$$|\nu(\bar{A}\,\tau_x B)| \leq 4\,\mathsf{C}_{\mathrm{dc}}\,\nu(|A|^2)^{1/2}\,\nu(|B|^2)^{1/2}\,\frac{ra}{|x|_\infty} \qquad (x \in a\mathbb{Z}^3,\ |x|_\infty \geq 4ra);$$

an $A \neq 0$ meets this hypothesis only if r has finite order. If $n \in \mathfrak{N}$ and $\theta_{n,0}A = -A$, then $\nu(\bar{A}\,\tau_{jan}A) \leq 0$ for every integer $j \geq 2r$; if $\theta_{n,0}A = A$, then $\nu(\bar{A}\,\tau_{jan}A) \geq 0$ for every integer $j \geq 2r$.

(f) *A periodic field switched off behind the limit of the volume leaves no trace.* (1) Let $\mathfrak{A}$ be a family, $\Gamma' \subset a\mathbb{Z}^3$ a subgroup of finite index, $t_k \to 0$ real numbers, and for each k let ν_k be a Gibbs state for $(\beta, t_k\mathfrak{A})$ that is invariant under Γ'. Then every weak limit point of $(\nu_k)_k$ lies in $\mathcal{G}^{\mathrm{per}}_{a,\beta}$, and $\nu_k(\mathsf{r}f) - \nu_k(f) \to 0$ for every continuous function f on $\mathcal{V}(a\mathbb{Z}^3)$ and every $\mathsf{r} \in \mathsf{Iso}(a\mathbb{Z}^3)$. (2) Let $F \in \mathcal{C}_{\mathrm{loc}}$ be real, let $\Gamma_n \subset \Gamma'$ be period lattices with $\mathrm{sys}(\Gamma_n) \to \infty$, set $\mathbb{F}_n = \sum_{[v] \in \Gamma'/\Gamma_n} \tau_v F$ on $\mathcal{V}_{\Gamma_n}$ and $\mu_{n,t} = \mu^{t\mathbb{F}_n}_{\Gamma_n}$ for $t \in \mathbb{R}$. Then

$$\lim_{t \to 0}\ \limsup_{n \to \infty} |\mu_{n,t}(\mathsf{r}f) - \mu_{n,t}(f)| = 0$$

for every continuous f and every $\mathsf{r} \in \mathsf{Iso}(a\mathbb{Z}^3)$, and $\lim_n |\mu_{n,0}(\mathsf{r}f) - \mu_{n,0}(f)| = 0$. (3) If moreover $F = F_0 - \mathsf{r}F_0$ with a real $F_0 \in \mathcal{C}_{\mathrm{loc}}$ and some $\mathsf{r} \in \mathsf{Iso}(a\mathbb{Z}^3)$ (for instance $F_0 = \frac{1}{2}F$ when $\mathsf{r}F = -F$), then, with $p_n(t) = |\mathbb{F}_n|^{-1} \log(Z_{\Gamma_n}[t\mathbb{F}_n]/Z_{\Gamma_n}[0])$,

$$\lim_{t\to 0} \limsup_{n\to\infty} |\mu_{n,t}(F)| = 0, \qquad \lim_{t\to 0} \limsup_{n\to\infty} \frac{|p_n(t)|}{|t|} = 0 :$$

the order parameter $F_0 - \mathsf{r}F_0$ vanishes when the volume is taken first and the field is switched off afterwards, and no limit point of the free energies per term has a kink at $t = 0$. For $\Gamma' = a\mathbb{Z}^3$ the field is uniform and tests the isometries that are not translations; for $\Gamma' = \{v \in a\mathbb{Z}^3 : (v_1 + v_2 + v_3)/a \text{ even}\}$, $\mathsf{r} = \tau_{ae_1}$ and $F_0 = a_p$ it is the staggered field $\sum_{v\in\Gamma'/\Gamma_n}(a_{p+v} - a_{p+v+ae_1})$, which tests the translation by one step.

The argument uses only fixed-spacing results. ■

Proof. Step 1. Proposition YM3_I:103.4 gives lattice invariance and reflection positivity at all the indicated mirrors. For the plaquette mean, translations and coordinate permutations act transitively on the unoriented plaquettes. A relabelling can conjugate or invert a plaquette holonomy, which preserves its trace in SU(2).

Step 2: clause (b). Let $E \in \mathcal{I}$. Since ν is invariant under r, $\nu(1_E\, \mathsf{r}F) = \nu((\mathsf{r}^{-1}1_E)\, F) = \nu(1_{\mathsf{r}^{-1}E}F)$, and $1_{\mathsf{r}^{-1}E} = 1_E$ ν-almost everywhere by clause (c) at $\Gamma' = a\mathbb{Z}^3$, whose proof (Step 3) does not use (b). So $\nu(1_E\, \mathsf{r}F) = \nu(1_E F)$ for all $E \in \mathcal{I}$, and the two conditional expectations, the unique classes with an $\mathcal{I}$-measurable member and these integrals, agree. If $\mathsf{r}F = \omega F$, then $\nu(F) = \nu(\mathsf{r}F) = \omega\nu(F)$ by (a), and $\mathbb{E}_\nu[F|\mathcal{I}] = \mathbb{E}_\nu[\omega F|\mathcal{I}] = \omega\, \mathbb{E}_\nu[F|\mathcal{I}]$; as $\omega \neq 1$ both vanish. For the example: $(\mathsf{r}W_C)(U) = W_C(\mathsf{r}^{-1}U) = W_{\mathsf{r}C}(U)$, because $(\mathsf{r}^{-1}U)(c) = U(\mathsf{r}c)$ for every bond c; so $\mathsf{P}(W_C - W_{\mathsf{P}C}) = W_{\mathsf{P}C} - W_C$. The path C visits the eight different sites $0, (0,0,1), (0,0,2), (0,1,2), (1,1,2), (1,1,1), (0,1,1), (0,1,0)$, times a, and it runs once through the bond $\langle 0, ae_3\rangle$; the path $\mathsf{P}C$ has no site with a positive coordinate and does not use this bond. At the configuration which is $g \neq \mathbf{1}$ on $\langle 0, ae_3\rangle$ and $\mathbf{1}$ on every other bond, $W_C - W_{\mathsf{P}C} = \frac{1}{2}\operatorname{tr} g - 1 < 0$. The function is gauge invariant, since the holonomy of a closed path is changed by a conjugation.

Step 3. By linearity and addition of constants it suffices to take $\mathsf{h} \geq 0$ with $\nu(\mathsf{h}) > 0$. Put $h = \mathsf{h}/\nu(\mathsf{h})$ and fix a finite bond set B. Continuous local functions are dense in $L^1(\nu)$. For any $\epsilon > 0$ choose a continuous local h' with $\nu(|h - h'|) < \epsilon$. Take $v \in \Gamma'$ such that the support of $\tau_v h'$ is disjoint from B. Since h and ν are Γ'-invariant, $\nu(|h - \tau_v h'|) < \epsilon$. For any bounded Borel F, the Gibbs equation gives $\nu((\tau_v h')\gamma_B F) = \nu((\tau_v h')F)$ because γ_B fixes the outside variables. Replacing $\tau_v h'$ by h changes either side by at most $\epsilon\|F\|_\infty$. Thus $h\nu$ is Gibbs and it is periodic because both h and ν are Γ'-invariant. Clause (a) gives $\nu(h\, \mathsf{r}^{-1}F) = \nu(hF)$. Since ν is invariant under r, this says $\nu((\mathsf{r}h - h)F) = 0$ for all bounded Borel F. Consequently $\mathsf{r}h = h$ almost everywhere, also for $h = 1_E/\nu(E)$ when $\nu(E) > 0$. If $\nu(E) = 0$, its image is null by (a).

Step 4: clause (c) for one translation. Let h be real, $|\mathsf{h}| \leq \varsigma$, $\tau_w\mathsf{h} = \mathsf{h}$ almost everywhere; by induction, and as r carries null sets to null sets by (a), $\tau_{kw}\mathsf{h} = \mathsf{h}$ almost everywhere for all $k \in \mathbb{N}$. Write $\langle f, f'\rangle = \nu(\bar{f}f')$ and Π for the orthogonal projection of $L^2(\nu)$ onto the classes with an $\mathcal{I}$-measurable member, which is $\mathbb{E}_\nu[\,\cdot\,|\mathcal{I}]$. For real $A, B \in \mathcal{C}_{\mathrm{loc}}$, both lie in $\mathcal{F}(\mathsf{Bd}(Q_r))$ for a large integer r, and the clustering bound, YM3_I:(1970), with the sizes $\mathcal{N}_r(A) \leq \nu(|A|^2)$ of the bound following YM3_I:(1959), gives $\langle A, \tau_{kw}B\rangle \to \langle \Pi A, \Pi B\rangle = \langle A, \Pi B\rangle$ as $k \to \infty$. Hence

$$\langle A, \mathsf{h}\rangle - \langle A, \Pi\mathsf{h}\rangle = \langle A, \tau_{kw}(\mathsf{h} - B)\rangle + \big[\langle A, \tau_{kw}B\rangle - \langle A, \Pi B\rangle\big] + \langle A, \Pi(B - \mathsf{h})\rangle,$$

whose first and last terms are at most $\|A\|_2\|\mathsf{h} - B\|_2$ in modulus, by the inequality of Schwarz, the invariance of ν under τ_{kw} and $\|\Pi\| \leq 1$. The algebra $\mathcal{C}_{\mathrm{loc}}$ is dense in $C(\mathcal{V}(a\mathbb{Z}^3))$, hence in $L^2(\nu)$, and the real parts of its members are again in it; so $\|\mathsf{h} - B\|_2$ is as small as we please, and $\langle A, \mathsf{h} - \Pi\mathsf{h}\rangle = 0$ for all real $A \in \mathcal{C}_{\mathrm{loc}}$, whence $\mathsf{h} = \Pi\mathsf{h}$ in $L^2(\nu)$. Let g be an $\mathcal{I}$-measurable member of $\Pi\mathsf{h}$; it is invariant under all translations everywhere, so Step 3 at $\Gamma' = a\mathbb{Z}^3$ gives $\mathsf{rg} = \mathsf{g}$ almost everywhere, and $\mathsf{rh} = \mathsf{rg} = \mathsf{g} = \mathsf{h}$ almost everywhere.

Step 5: clause (d). The decomposition and the statement that $\nu^U \in \mathcal{G}^{\mathrm{per}}_{a,\beta}$ is ergodic for ν-almost every U follow from Proposition YM3_I:106.2; the ergodic members of $\mathcal{G}^{\mathrm{per}}_{a,\beta}$ are its pure phases (Proposition YM3_I:106.1); a pure phase is a periodic Gibbs state, so (a) gives $\mathsf{r}_*\rho = \rho$. Let ρ be a pure phase and $\rho(\tau_w E \triangle E) = 0$ for one $w \neq 0$. By Step 4 for ρ and $\mathsf{h} = 1_E$, 1_E coincides almost everywhere with an $\mathcal{I}$-measurable function, so with $\mathbb{E}_\rho[1_E|\mathcal{I}]$, which is the constant $\rho(E)$, ρ being ergodic; a function with the values 0 and 1 which is almost everywhere constant is almost everywhere 0 or almost everywhere 1. Without the clustering bound the case of a subgroup Γ' of finite index also follows from Step 3: if $0 < \rho(E) < 1$, then $\rho(E)^{-1}1_E\rho$ and $\rho(E^c)^{-1}1_{E^c}\rho$ are two different members of $\mathcal{G}^{\mathrm{per}}_{a,\beta}$ of which ρ is a proper convex combination.

Step 6: clause (e). By (b), $\mathbb{E}_\nu[A|\mathcal{I}] = 0$, so the clustering bound YM3_I:(1970) has no subtracted term, and $\mathcal{N}_r(A) \le \nu(|A|^2)$, $\mathcal{N}_r(B) \le \nu(|B|^2)$ by the bound following YM3_I:(1959), whose hypotheses (invariance under $a\mathbb{Z}^3$ and positivity at $\Sigma(e_\mu, 0)$, $\Sigma(e_\mu, \frac{a}{2})$) hold by (a). Finite order: if r has infinite order, then, W_3 being finite, $\mathsf{r}^m = \tau_u$ with some $m \ge 1$ and $u \neq 0$, and $\tau_u A = \omega^m A$. A function $F \in \mathcal{F}(B)$, B finite, with $\tau_u F = \omega' F$ is constant: if $F \neq 0$ then $|\omega'| = 1$ and $F = \omega'^{-k}\tau_{ku}F \in \mathcal{F}(B + ku)$ for all k, while $B \cap (B + ku) = \varnothing$ for large k; for two configurations U, U' the configuration W which is U on B, U' on $B+ku$ and $\mathbf{1}$ elsewhere has $F(U) = F(W) = F(U')$. So A is a constant c with $c = \mathsf{r}A = \omega c$, and $A = 0$. The sign: put $c' = \frac{1}{2}ja|n|^2$, which lies in $\frac{a}{2}\mathbb{Z}$ for a coordinate normal and in $a\mathbb{Z}$ for a diagonal one; by the formula for the reflection, $\theta_{n,c'} = \tau_{jan}\theta_{n,0}$, and $\theta_{n,0}\tau_{jan} = \tau_{-jan}\theta_{n,0}$. On Q_r, $n \cdot x \ge -ra|n|^2$, the entries of n being $0, \pm 1$; so $F = \tau_{jan}A$ is a function of the bonds of $Q_r + jan \subset \{n \cdot x - c' \ge (\frac{j}{2} - r)a|n|^2\}$, which lies in the closed side $\{n \cdot x \ge c'\}$ for $j \ge 2r$, and $\theta_{n,c'}F = \theta_{n,0}A = \epsilon A$ with $\epsilon = \pm 1$. The positivity (a) at the mirror (n, c') gives $0 \le \nu(\overline{\theta_{n,c'}F}\, F) = \epsilon\, \nu(\bar{A}\, \tau_{jan}A)$.

Step 7. The space of laws on the compact metrizable configuration space is weakly compact. Along any convergent subsequence of ν_k, the limit is Gibbs by Proposition YM3_I:91.1. For a finite bond set B the locally finite source has only finitely many terms in its kernel, and their contribution converges uniformly to zero as $t_k \to 0$. Invariance under the fixed subgroup Γ' is weakly closed, as it can be tested on continuous functions. Every such limit therefore belongs to $\mathcal{G}^{\mathrm{per}}_{a,\beta}$ and is r-invariant by (a). Since $\mathsf{r}f - f$ is continuous, compactness implies $\nu_k(\mathsf{r}f - f) \to 0$.

Step 8. The translates $\tau_v F$ for $v \in \Gamma'$, with supports $\tau_v B_F$ for a finite carrier B_F, form a locally finite family. For every $\Gamma_n \subset \Gamma'$, translation acts on these indices by addition and the periodized source is $t\mathbb{F}_n$. Proposition YM3_I:91.2 gives the Gibbs equations at strength t for every weak limit of $\mu_{n,t}$. The array is unchanged by Γ' because each translation permutes its terms. Product Haar measure and the Wilson action are also unchanged, so the finite-volume and limiting laws are Γ'-invariant.

Suppose $\limsup_n |\mu_{n,t_k}(\mathsf{r}f - f)| > \delta$ along some $t_k \to 0$. For each k, take a weak limit ν_k along indices where the modulus exceeds δ. Then ν_k is a Γ'-invariant Gibbs state at strength t_k and $|\nu_k(\mathsf{r}f - f)| \ge \delta$. This contradicts Step 7. The same argument with $t_k = 0$ gives $\mu_{n,0}(\mathsf{r}f - f) \to 0$.

Step 9: clause (f)(3). $F = F_0 - \mathsf{r}F_0$ is a real member of $\mathcal{C}_{\mathrm{loc}}$, and (2) at $f = F_0$ reads $\lim_{t\to 0}\limsup_n |\mu_{n,t}(F)| = 0$ and $\lim_n \mu_{n,0}(F) = 0$. Fix n and write $\Gamma = \Gamma_n$, $\mathbb{F} = \mathbb{F}_n$, $\mu_t = \mu_{n,t}$. From the definition of the tilted law, $Z_\Gamma[\Phi + \Phi'] = Z_\Gamma[\Phi]\, \mu^\Phi_\Gamma(e^{\Phi'})$ for real $\Phi, \Phi' \in C(\mathcal{V}_\Gamma)$; at $(0, t\mathbb{F})$ and at $(t\mathbb{F}, -t\mathbb{F})$, with Jensen's inequality $\mu(e^X) \ge e^{\mu(X)}$,

$$\frac{Z_\Gamma[t\mathbb{F}]}{Z_\Gamma[0]} \ge e^{t\mu_0(\mathbb{F})}, \qquad \frac{Z_\Gamma[0]}{Z_\Gamma[t\mathbb{F}]} \ge e^{-t\mu_t(\mathbb{F})}.$$

On $\mathcal{V}_\Gamma$, which carries μ_t, $\mathbb{F}$ is the sum of the $|\mathbb{F}| = [\Gamma' : \Gamma]$ functions $\tau_v F$, and $\mu_t(\tau_v F) = \mu_t(F)$ by the invariance of Step 8; so $\mu_t(\mathbb{F}) = |\mathbb{F}|\, \mu_t(F)$ and $t\, \mu_{n,0}(F) \le p_n(t) \le t\, \mu_{n,t}(F)$ for all real t. Hence $|p_n(t)|/|t| \le \max\{|\mu_{n,0}(F)|, |\mu_{n,t}(F)|\}$, and the second limit follows from the first. In the uniform case with $\mathsf{r} = \tau_w$ the array is 0; in the staggered case $\tau_v a_p = a_{p+v}$, and the period lattices $2na\mathbb{Z}^3 \subset \Gamma'$, $n \ge 2$, have sys $\to \infty$.

□

Lemma 18.211 (The block map on the lattice and on a window)**.** Write $\tilde{B}^n(y) = \{x \in a_{d+n}\mathbb{Z}^3 : |x-y|_\infty \le (a_d - a_{d+n})/2\}$ for $y \in a_d\mathbb{Z}^3$.

(a) The double block of a bond at depth d projects injectively to every torus with side in $\mathfrak{L}$. The one-step rule on this chart is independent of the side of the torus and commutes with periodic extension.

(b) The block map is Borel and each bond value is continuous on each of three Borel sets. The map is discontinuous on the full configuration space. It commutes with coarse lattice isometries and is covariant under fine lattice gauges, restricted at the output to the coarse sites. The bond value of $\mathcal{M}^n U$ at c depends only on the bonds with both endpoints in $\tilde{B}^n(c_-) \cup \tilde{B}^n(c_+)$.

(c) For the window $\square = [-\ell/4, \ell/4]^3$ at $\ell = 2L^m$, the restriction r_d commutes with each iterate in the form $r_d\mathcal{M}^n = \mathsf{m}_d^n r_{d+n}$, where m_d^n applies the local rule on the window. These maps are Borel and compose in depth. On a periodically extended torus field the same identity holds with the torus iterates.

(d) The tower map Ψ_K and the window tower map Υ_K are Borel. The thread and window thread sets are Borel, and $\Upsilon_K(U)$ is determined by $r_K U$. For $o \in a_{d_0}\mathbb{Z}^3$ and $K \ge d_0$, $\varpi_o^{[d_0]}\Psi_K = \mathrm{pj}^{[d_0]}\Upsilon_K\tau_{-o}$. On a periodically extended torus field Ψ_K is the periodic extension, depth by depth, of the torus tower.

■

Proof. For (a), the double block has sup diameter $2a_d - a_{d+1}$ and even its one-step enlargement has diameter smaller than $2L \le \ell_1$ for every $\ell_1 \in \mathfrak{L}$. This makes the quotient map a chart for the sites and nearest-neighbor bonds of the double block and hence also for its plaquettes and paths. The three formulas in Section YM3_I:4, with locality from Proposition YM3_I:4.5, use only those bonds and give the same result in that chart regardless of the period.

The three Borel continuity pieces and gauge covariance follow from Proposition YM3_I:4.5. Coarse translations and signed coordinate permutations preserve the double blocks and permute their sets of coordinate paths. For a transformation preserving the orientation of the output bond, relabel the path products in the averaging formulas. For reversal, a relative holonomy Y based at the first endpoint is replaced by $U[c]^{-1}Y^{-1}U[c]$, where $U[c]$ is the direct transport. The logarithm therefore changes by negation and conjugation, and the averaged bond is inverted. The contour mean in the small-field branch is independent of the ordering of its paths, and reversal exchanges the two endpoint contour means. Thus the same calculation works there. The test for the exceptional branch is preserved because inversion and conjugation fix $-\mathbf{1}$. The exceptional formula is the direct transport, which also inverts. For iterated locality use $\tilde{B}^{n-1}(x) \subset \tilde{B}^n(y)$ whenever $x \in \tilde{B}^1(y)$, as follows from $(a_d - a_{d+1})/2 + (a_{d+1} - a_{d+n})/2 = (a_d - a_{d+n})/2$. This proves (b) by induction apart from the discontinuity assertion.

To see discontinuity, take a fine bond crossing between the two blocks, in a row parallel to the coarse bond but displaced transversely by one fine step. Put $u_t = \mathrm{diag}(e^{it}, e^{-it})$ on this bond and the identity on the other bonds. The bond belongs to no endpoint contour and is off the direct coarse path. Of the L^3 parallel paths between corresponding fine sites, exactly the L paths in the chosen row meet it. Near $t = \pm\pi$ a plaquette containing the bond has distance $2|\sin(t/2)|$ from $\mathbf{1}$, so the small-field branch fails. For $|t| < \pi$ the relative holonomies are $\mathbf{1}$ and u_t, so the intrinsic-logarithm branch returns u_{t/L^2}. The two limits at the same input $u_\pi = u_{-\pi}$ are distinct.

For (c), $\ell/(2a_d)$ is odd, so the closest depth-d site to a window wall is at distance $a_d/2$. The dependency blocks from (b) are inside the window. Extending by identity outside therefore defines a Borel local rule and gives the stated identities. The tower maps in (d) are coordinatewise Borel by (b) and (c). A thread is specified by the countable family of equalities between each coordinate and its block average from the next depth. Each equality is Borel because the diagonal in the bond

group is closed. The formula for $\varpi_o^{[d_0]}$ uses translation covariance at each depth $d \geq d_0$, and periodic extension uses (a) at each step. □

Lemma 18.212 (Thread laws and moving labels)**.** Let $\ell = 2L^m$ admit a compact interval J under the torus condition of Corollary 2.8. Write $\mu_d^K[\mathsf{b}]$ for the law of $\mathcal{M}_\ell^{K-d}U$ under $\mu_{K,\ell,L^K\mathsf{b}}$.

(a) At every $\mathsf{b} \in \mathsf{J}$ and $d \geq 0$, $\mu_d^K[\mathsf{b}]$ converges in total variation to a gauge invariant law $\mu_d[\ell, \mathsf{b}]$ and $(\mathcal{M}_\ell)_*\mu_{d+1} = \mu_d$. Uniformly on J, for $K \geq K_{\mathrm{bf}}(\mathsf{J})$ and $d \leq \lceil K/1260 \rceil$,

$$\|\mu_d^{K+1}[\mathsf{b}] - \mu_d^K[\mathsf{b}]\|_{\mathrm{TV}} \leq \mathsf{C}_{\mathrm{bf}}\ell^3 K^{\mathsf{p}_0} L^{-K/10}.$$

(b) There is a unique torus law $\mu_{\ell,\mathsf{b}}$ whose first $D+1$ coordinates have the law of $(\mathcal{M}_\ell^D V, \dots, V)$ for $V \sim \mu_D[\ell, \mathsf{b}]$. It is carried by threads. Its image $\mu[\ell, \mathsf{b}]$ in the window is carried by window threads.

(c) For $\ell \in \mathfrak{L}(g)$ the interval $\mathsf{J}_g = [g^{-2}, L(g^{-2} + \bar{B})]$ is admissible. The first-passage labels belong to $(g^{-2}, L(g^{-2} + \bar{B})] \subset \mathsf{J}_g$ and to $I^{\mathrm{lab}}(g)$. If $\mathsf{b} \in I(g)$ then $(K, L^K\mathsf{b})$ is a datum at g for every $K \geq 1$.

(d) The finite torus towers, filled by identity beyond depth K, converge weakly to $\mu_{\ell,\mathsf{b}}$.

(e) If $K_j \to \infty$ and $\mathsf{b}_j = a_{K_j}\beta_j \to \mathsf{b}$ with all labels in J, then $\mu_d^{K_j}[\mathsf{b}_j]$ converges to $\mu_d[\ell, \mathsf{b}]$ in total variation. The map $\mathsf{b} \mapsto \mu_d[\ell, \mathsf{b}]$ is continuous in that norm. The window tower laws converge weakly to $\mu[\ell, \mathsf{b}]$ and in total variation jointly at every finite set of depths.

(f) If the bounds of (a) hold on every compact positive label interval at a given side, then (b) and (e) apply at that side for all positive limiting labels and every sequence of labels converging to one of them.

■

Proof. Corollary 2.8(a) and (c) give the step estimate in (a), including the extension from gauge-invariant test functions to all Borel sets by averaging indicators under the compact group of lattice gauges. Summing it gives total variation convergence. Consistency passes to the limit from the identities between finite-cutoff marginals because pushforward under the Borel block map contracts total variation. The joint laws in (b) are then consistent under truncation in D. The extension theorem for countable products gives the law. Each adjacent thread identity holds with probability one by the joint law at that pair of depths. The countable intersection still has probability one. Lemma 18.211(c) carries this property to the window.

For (c), the first-passage room estimate YM3_I:(1533) gives the torus room condition at the upper endpoint of J_g. The bounds on all the labels and the assertion at a fixed label in $I(g)$ are Proposition YM3_I:74.1 and equation YM3_I:(1854). For (d), push forward the law at depth D under the map $V \mapsto (\mathcal{M}_\ell^D V, \dots, V)$. This gives convergence in total variation at the depths through D. Continuous cylinder functions are uniformly dense on the compact product, which gives weak convergence.

For each fixed K the density of the finite Wilson law depends continuously on the label in the uniform norm, since its action is bounded and its partition function is positive. Its pushforward to depth d is therefore continuous in total variation. The summable estimate in (a) makes the convergence uniform on J, so the limiting law has the same continuity. The triangle inequality with the intermediate law $\mu_d[\ell, \mathsf{b}_j]$ proves the moving-label assertion. By Lemma 18.211(c) the joint window law through depth D is the image of the depth-D torus law under a fixed Borel map. Total variation contracts under that map and continuous cylinder functions again give weak convergence. This proves (e). For (f), choose a compact interval whose interior contains the limiting label and discard the finitely many terms outside the interval. □

Corollary 18.213 (Uniform cutoff tail)**.** For an admitted interval J, use C_{bf}, K_{bf} and $\mathsf{p} = 2\mathfrak{p}_0 + 1/2$ of Definition 2.7(5). Put

$$\tau_{\mathsf{J}}(K) = \mathsf{C}_{\mathrm{bf}} \ell^3 \sum_{K' \ge K} K'^{\mathsf{p}} L^{-K'/10}.$$

For $K \ge K_{\mathrm{bf}}$ and $d \le \lceil K/1260 \rceil$, the distance in total variation between the depth-d laws at cutoff K and at infinite cutoff is at most $\tau_{\mathsf{J}}(K)$, uniformly on J. If $K(d)$ is the least such $K > d$ with $\tau_{\mathsf{J}}(K) \le L^{-d}$, the error at $K(d)$ is at most L^{-d}. ■□

Proof. Sum the bound of Lemma 18.212(a) for consecutive cutoffs. The tail tends to zero, so $K(d)$ exists. □

Lemma 18.214 (Domination and weak limits)**.** For probability measures $\boldsymbol{\sigma}, \boldsymbol{\rho}$ on the same measurable space, write $\boldsymbol{\sigma} \preceq_{\mathsf{Q}} \boldsymbol{\rho}$ if $|\boldsymbol{\sigma}(f)| \le \mathsf{Q}\, \boldsymbol{\rho}(|f|^8)^{1/8}$ for every bounded complex Borel f. If $\boldsymbol{\sigma} \preceq_{\mathsf{Q}} \boldsymbol{\rho}$, then $\mathsf{Q} \ge 1$ and the following hold.

(DM1) $\boldsymbol{\sigma}(A) \le \mathsf{Q}\, \boldsymbol{\rho}(A)^{1/8}$ for every measurable A.

(DM2) The density $h = d\boldsymbol{\sigma}/d\boldsymbol{\rho}$ satisfies $\int h^{8/7}\, d\boldsymbol{\rho} \le \mathsf{Q}^{8/7}$ and $\boldsymbol{\sigma}(h > t) \le \mathsf{Q}^{8/7} t^{-1/7}$ for $t > 0$.

(DM3) $0 \le \int h \log h\, d\boldsymbol{\rho} \le 7 \log \int h^{8/7}\, d\boldsymbol{\rho} \le 8 \log \mathsf{Q}$.

(DM4) If p is measurable, then $p_* \boldsymbol{\sigma} \preceq_{\mathsf{Q}} p_* \boldsymbol{\rho}$.

(DM5) Conversely, a probability density with $\int h^{8/7}\, d\boldsymbol{\rho} \le \mathsf{Q}^{8/7}$ gives $h\boldsymbol{\rho} \preceq_{\mathsf{Q}} \boldsymbol{\rho}$.

(DM6) There are at most Q^8 pairwise mutually singular probability measures dominated by $\boldsymbol{\rho}$ with the same Q.

(DM7) Let Z, Y be compact metrizable spaces, let $p_i : Z \to Y$ be continuous, and suppose $\sigma_j \to \sigma$ and $\rho_j \to \rho$ weakly on their respective spaces. If

$$\left| \sigma_j \left(\prod_{i=1}^{k} f_i \circ p_i \right) \right| \le \prod_{i=1}^{k} \mathsf{Q}\, \rho_j(|f_i|^8)^{1/8}$$

for all continuous complex f_i, then the same estimate holds for σ, ρ and all bounded complex Borel f_i.

■

Proof. Testing with 1 and 1_A gives $\mathsf{Q} \ge 1$ and (DM1), so $\boldsymbol{\sigma}$ has a Radon–Nikodym density h. For $I_n = \int \min(h, n)^{8/7}\, d\boldsymbol{\rho}$ the test $f = \min(h, n)^{1/7}$ gives $I_n \le \mathsf{Q} I_n^{1/8}$. Monotone convergence gives the norm bound. On $\{h > t\}$ one has $h \le t^{-1/7} h^{8/7}$, proving the tail bound. For (DM3), convexity of $x \log x$ gives the lower bound. Applying concavity of log under the probability measure $h\, d\boldsymbol{\rho}$ to $h^{1/7}$ gives the upper bound. Its positive logarithm is integrable by (DM2) and its negative logarithm is integrable since $x|\log x|$ is bounded on $[0, 1]$. Testing with $f \circ p$ gives (DM4) and Hölder gives (DM5). For a finite pairwise singular family, intersect the sets of full measure for each member and zero measure for each of the others, then remove their overlaps. This gives disjoint carriers A_i, each of $\boldsymbol{\rho}$-mass at least Q^{-8} by (DM1), proving (DM6).

In (DM7) weak convergence gives the estimate for continuous f_i. Approximate each Borel factor f_i by continuous functions with the same uniform bound in $L^1(\rho + (p_i)_* \sigma)$. Such approximants exist by regularity, with truncation by projection onto the closed disc of radius $\|f_i\|_\infty$. Uniform boundedness upgrades the $L^1(\rho)$ convergence to $L^8(\rho)$ convergence. A telescoping sum bounds the error on the left by a sum of $L^1((p_i)_* \sigma)$ errors times the uniform bounds of the other factors. Taking limits proves (DM7). □

Lemma 18.215 (Invariant densities and product bounds)**.** Let T be an invertible measure-preserving transformation of a probability space $(\Omega, \mathbb{P})$, and let $0 \le F \le c$ be measurable. Suppose $\mathbb{E} \prod_{r=1}^{k} (F \circ T^{n_r}) \le C q^k$ for all distinct positive integers $n_1, \dots, n_k$ and all $k \ge 1$, where $C \ge 1$ and $q \ge 0$. If D is nonnegative, integrable and T-invariant almost everywhere, then $\mathbb{E}[DF] \le q \mathbb{E} D$. Also, if $\mathbb{E}[1_E F] \le q \mathbb{P}(E)$ for all E in a sub-σ-algebra $\mathcal{J}$, then $\mathbb{E}[F \mid \mathcal{J}] \le q$ almost surely. ■

Proof. For bounded D with $\mathbb{E}D = 1$, put $A_N = N^{-1}\sum_{n=1}^{N} F \circ T^n$. Invariance gives $\mathbb{E}DF = \mathbb{E}DA_N$. Jensen's inequality under $D\,d\mathbb{P}$ and the count of repeated indices in the expansion of A_N^k give

$$(\mathbb{E}DF)^k \le \|D\|_\infty \mathbb{E}A_N^k \le \|D\|_\infty \left(Cq^k + \binom{k}{2}\frac{c^k}{N} \right).$$

First let $N \to \infty$, then take k-th roots and let $k \to \infty$. Scaling and monotone convergence of $D \wedge m$ give the claim for integrable D, including $D = 0$ almost everywhere. For the conditional expectation, take $E = \{\mathbb{E}[F \mid \mathcal{J}] > q\}$. □

Lemma 18.216 (Several cells compared to a reference torus)**.** Let (K, β) be a first-passage datum at $0 < g \le \gamma$ and $\ell \in \mathfrak{L}(g)$. Use the board of side $\mathsf{l} = \ell/2$ at a lattice site o. For finitely many distinct cells and bounded complex Borel functions $F_{\mathbf{k}}$ of their inner bonds,

$$\left| \nu \left(\prod_{\mathbf{k}} F_{\mathbf{k}} \right) \right| \le \prod_{\mathbf{k}} \mathsf{Q}_0\, \tilde{\mu}_{K,\ell,\beta}(|F_{\mathbf{k}}|^8)^{1/8}, \qquad \mathsf{Q}_0 = \exp\bigl[\mathsf{l}^3(2C_{\mathrm{S}} + C_g)\bigr].$$

Here ν is a periodic Gibbs state or the lifted torus law of side $\ell' \ge \ell$ in $\mathfrak{L}$, with the cells distinct modulo ℓ' in the torus case. With the real first-passage construction alone, the same estimate holds with $C_{\mathrm{gap}} + C_{\mathrm{st}} + C_g$ in place of $2C_{\mathrm{S}} + C_g$. ■

Proof. Since l/a_K is odd, the walls are lattice midplanes. For a bounded nonnegative function G of the inner bonds in a cell, let G^{all} be the product of its reflected copies in every cell of the torus of side t and set $\|G\|_t = \langle G^{\mathrm{all}} \rangle_{K,t,\beta}^{(\mathsf{l}/t)^3}$. By Propositions YM3_I:86.3 and YM3_I:86.4 and Corollary YM3_I:86.6, the finite-torus chessboard inequality bounds the product expectation by $\prod_{\mathbf{k}} \||F_{\mathbf{k}}|\|_t$, and for $t \ge \ell$,

$$\|G\|_t \le \|G\|_\ell \exp\bigl[\mathsf{l}^3(p_\ell - p_t)\bigr].$$

Proposition YM3_I:91.9 gives the same bound for a periodic Gibbs state with a lower limit of the products as $t \to \infty$ through admissible sides. On T_ℓ there are eight copies and reflections preserve its Wilson measure. Hölder for these eight factors gives $\|G\|_\ell \le \tilde{\mu}_{K,\ell,\beta}(G^8)^{1/8}$.

At zero source, Theorem YM3_I:AD writes $p_t = t^{-3}E_t + t^{-3}\log \widetilde{Z}_t(0)$ with $-C_g \le t^{-3}\log \widetilde{Z}_t(0) \le C_{\mathrm{S}}$ and $|\ell^{-3}E_\ell - t^{-3}E_t| \le C_{\mathrm{S}}$. Thus $p_\ell - p_t \le 2C_{\mathrm{S}} + C_g$. For the real construction, Proposition YM3_I:74.4 gives the lower and upper bounds $-C_g$ and $E_+ = C_{\mathrm{maj}} + 2 \le C_{\mathrm{st}}$ for the normalized logarithm per unit volume, and Proposition YM3_I:74.5 gives the bound C_{gap} for the vacuum-density difference. The constants and the first-passage normalization are those of [7, Section 2]. This gives the second pressure bound and hence the two asserted domination estimates. □

Lemma 18.217 (Local limits of window laws and towers)**.** Use the window and tower notation of Lemma 18.211. Let $K_j \to \infty$ and let ν_j be laws invariant under the translations of $a_{K_j}\mathbb{Z}^3$. Suppose their one-cell laws obey the bound of Lemma 18.216 with constant Q and reference window tower laws ρ_j. Suppose $\rho_j \to \mu$ weakly and in total variation through each finite depth, and that μ is supported on window threads. For the product assertions below, suppose also the same bound for any fixed finite set of cells of a common board, for all sufficiently large j. Write $\mu^{[d_0]} = \mathrm{pj}_*^{[d_0]}\mu$.

(a) Every subsequential weak limit σ of $(\Upsilon_{K_j}\tau_{-o_j})_*\nu_j$, with arbitrary offsets $o_j \in a_{K_j}\mathbb{Z}^3$, is dominated by μ with constant Q. Its density and entropy obey Lemma 18.214 and it is carried by window threads. Along the same subsequence the means of all bounded Borel functions of finitely many depths converge. These conclusions pass to tails and to restrictions to a closed subset of the window.

(b) For every subsequential weak limit $\boldsymbol{\nu}$ of $(\Psi_{K_j})_*\nu_j$:

(i) $(\varpi_o^{[d_0]})_*\boldsymbol{\nu} \preceq_{\mathsf{Q}} \mu^{[d_0]}$ for every $d_0 \ge 0$ and $o \in a_{d_0}\mathbb{Z}^3$.

(ii) $\boldsymbol{\nu}(\mathsf{Thr}) = 1$.

(iii) On Thr, translation by a vector v in $\mathbb{D} = \bigcup_s a_s\mathbb{Z}^3$ acts by translation at depths $d \geq s$ for $v \in a_s\mathbb{Z}^3$, and fills in the earlier depths by iteration. This gives a measure-preserving Borel action T_v independent of the choice of s.

(iv) Along the same subsequence the means of bounded Borel functions of finitely many depths in a single window converge. If the hypotheses apply at arbitrarily large sides, the same holds for any bounded spatial support.

(c) Under the product hypothesis, $\boldsymbol{\nu}$ obeys the product bound on every fixed board, with constant Q per factor. So does $D\boldsymbol{\nu}$ for any nonnegative probability density D invariant under one T_v with $v \neq 0$. For a measurable mixture of laws invariant under that same T_v, almost every law of the mixture has the product bound. In particular this holds for the translation-ergodic decomposition.

(d) A reference-null set in any window has measure zero under the corresponding image of each dominated law. At most Q^8 window laws dominated by the same reference can be pairwise mutually singular.

(f) Suppose the ν_j are periodic Gibbs states or lifted Wilson torus laws with sides tending to infinity. Then $\boldsymbol{\nu}$ is gauge invariant on the tower and invariant under the action of $\mathbb{D} \rtimes W_3$ obtained by acting on a sufficiently fine tail and filling in the earlier depths.

(h) For first-passage data with converging labels, the one-cell and product hypotheses hold at each admissible reference side for periodic Gibbs states, including all volume-first states. The product hypothesis also holds for lifted torus laws whose sides tend to infinity. The real construction uses the alternative constant of Lemma 18.216.

■

Proof. Compactness gives weak limit points in the compact products. By locality, the bounded Borel function $f \circ \Upsilon_{K_j} \circ \tau_{-o_j}$ is a function of a single cell. Translation invariance of the torus reference gives $|\sigma_j(f)| \leq \mathsf{Q}\rho_j(|f|^8)^{1/8}$ for its image law σ_j. Lemma 18.214(DM7) gives domination in the limit, including for Borel tests. For a Borel test at finitely many depths, choose uniformly bounded continuous approximants in L^8 of the reference marginal. At these depths total variation convergence controls the integrals of the eighth powers of the differences under ρ_j. Domination bounds the limsup of the error under σ_j and its limit by twice Q times the reference L^8 error. This error tends to zero. The tail projections and restrictions are continuous coordinate projections. Every subsequential limit of the projected laws is the projection of a further weak limit of the full window laws. This gives (a).

By Lemma 18.211(d), the window at o of the tower law at depths $\geq d_0$ is the corresponding tail of the window law formed with offset o. Projection of the weak limit therefore gives (b)(i) and, using the Borel convergence just proved, (b)(iv). To prove (b)(ii), put the origin of a window at the initial endpoint o of a bond c at depth d and use $d_0 = d$. On its window the equality between the value at c and its average from depth $d+1$ holds with reference probability one. By local dependence and domination it holds with $\boldsymbol{\nu}$-probability one. Intersect these countably many events. For (b)(iii), translation commutes with the block map at depths $d \geq s$. Its action on that tail is a continuous permutation of coordinates, so the invariance of the finite laws passes to the weak limit there. Earlier depths on a thread are Borel functions of the tail, which gives invariance on the whole tower. This also shows independence of s, the group law by taking a common tail, and the inverse T_{-v}.

For (c), apply the finite-cutoff product inequality to the window tower tests and use Lemma 18.214(DM7). For a product $F = \prod_i |f_i| \circ \varpi_{o_i}^{[d_0]}$ choose a positive integer multiple w of v in $(\ell/2)\mathbb{Z}^3$ so large that the finite sets of board cells used by $F \circ T_{nw}$ are disjoint for different integers n. Such a multiple exists since $v \in \mathbb{D}$ and $\ell/2 = L^m$. The product bound gives $\boldsymbol{\nu}(\prod_{r=1}^k F \circ T_{n_r w}) \leq q^k$ for distinct n_r, with $q = \prod_i \mathsf{Q}\mu^{[d_0]}(|f_i|^8)^{1/8}$. Apply Lemma 18.215 to T_w and the invariant density. For a measurable mixture $\boldsymbol{\nu} = \int \boldsymbol{\nu}_\xi \, d\lambda(\xi)$ of T_v-invariant laws, the normalized restriction of the

mixture to any set of positive λ-measure has a bounded density relative to $\boldsymbol{\nu}$. The density is invariant by uniqueness of Radon–Nikodym derivatives. Applying the preceding bound to the set where $\boldsymbol{\nu}_\xi(F) > q$ shows that this set is null. Take a countable dense family of continuous window tests for each finite cell set and each d_0, and then use uniform approximation and (DM7) to extend to bounded Borel tests. The group $\mathbb{D}$ is countable and acts by Borel bijections on a standard Borel space, so its ergodic decomposition gives such an invariant mixture. The conclusions in (d) are (DM1) and (DM6).

For (f), any finite set of gauge values at endpoints of the coordinates under test can be implemented by a lattice gauge equal to identity at all remaining sites. For a lifted torus law, the same assignment is well defined modulo its period once the side is large enough to separate these sites. On a periodic Gibbs state invariance under a finite gauge follows by changing variables in the DLR kernel on all bonds incident to its support. The plaquette action is invariant and Haar measure is invariant under left and right multiplication. The same change of variables on the torus gives its invariance. The iterated gauge covariance of Lemma 18.211(b) carries this invariance to each cylinder test and passes to the limit for continuous cylinder tests. An arbitrary tower gauge acts continuously on the full product and preserves the thread identities, so the conclusion holds for the full law. For a signed permutation and lattice translation, choose a tail where the isometry acts at every depth. Finite-law invariance there follows from Proposition YM3_I:103.4 for periodic Gibbs states and from the Haar law and Wilson action on each torus. Commutation, invariance in the limit, and the group law follow as in (b)(iii).

For (h), use Lemmas 18.216 and 18.212. The large torus contains any fixed set of distinct cells without identifying them. Proposition YM3_I:103.4 includes the volume-first states in the periodic Gibbs class. □

Proposition 18.218 (The excess free energy of one torus bounds the distance of the laws of the thread over a cell from its thread measure, at infinite cutoff)**.** For $K \in \mathbb{N}$, $\beta > 0$ and $\ell \in \mathfrak{L}$ let $Z_{K,\ell,\beta}$ be the partition function of the Wilson measure $\mu_{K,\ell,\beta}$ of Definition 18.204, formed with the Haar probability measures, $p_\ell = p_\ell(K,\beta) = \ell^{-3}\log Z_{K,\ell,\beta}$, $p_\infty(K,\beta) = \inf_{\ell' \in \mathfrak{L}} p_{\ell'}(K,\beta)$, and

$$x(K,\beta,\ell) = \ell^3\big(p_\ell(K,\beta) - p_\infty(K,\beta)\big),$$

the excess of the side ℓ at (K,β). Put $\mathfrak{t}(x) = \frac{7}{8}(e^x/8)^{1/7}$ for $0 \le x \le \log 8$ and $\mathfrak{t}(x) = 1 - e^{-x}$ for $x \ge \log 8$, a continuous increasing function with values in $[0,1)$.

(0) *The excess.* $p_\ell(K,\beta)$ does not increase along $\mathfrak{L}$, $p_\infty(K,\beta) = \lim_{\ell\to\infty} p_\ell(K,\beta) \ge -6\beta a_K^{-3}$, and $0 \le x(K,\beta,\ell) \le 6\beta a_K^{-3}\ell^3$. If $0 < g \le \gamma$, $\ell \in \mathfrak{L}(g)$ and (K,β) is a datum at g, then $x(K,\beta,\ell) \le \ell^3(2C_{\mathrm{S}} + C_g) = 8\log \mathsf{Q}_0$, Q_0 the number of Definition 18.207; with the real first-passage construction alone, $x(K,\beta,\ell) \le \ell^3(C_{\mathrm{gap}} + C_{\mathrm{st}} + C_g)$.

(1) *At one pair.* Let $K \in \mathbb{N}$, $\beta > 0$, $\ell \in \mathfrak{L}$, $x = x(K,\beta,\ell)$, let $\square' = o + [-\frac{\ell}{4}, \frac{\ell}{4}]^3$ with $o \in a_K\mathbb{Z}^3$, let $\mathfrak{s} \in \mathcal{G}^{\mathrm{per}}_{K,\beta}$, or $\mathfrak{s} = \tilde\mu_{K,\ell',\beta}$ with $\ell \le \ell' \in \mathfrak{L}$, and let Φ be a Borel map of the configurations of $\mathbb{L}_K$ into a compact metrisable space which depends on the bonds with both end points in $\square'$ alone. Then $\Phi_*\mathfrak{s} = h\,\Phi_*\tilde\mu_{K,\ell,\beta}$ with a Borel $h \ge 0$, $\int h^{8/7} d\Phi_*\tilde\mu_{K,\ell,\beta} \le e^{x/7}$, $\Phi_*\mathfrak{s} \preceq_{e^{x/8}} \Phi_*\tilde\mu_{K,\ell,\beta}$, $0 \le \int h\log h\, d\Phi_*\tilde\mu_{K,\ell,\beta} \le x$, and

$$\sup_A \big|\Phi_*\mathfrak{s}(A) - \Phi_*\tilde\mu_{K,\ell,\beta}(A)\big| \le \min\{\sqrt{x/2},\ \mathfrak{t}(x)\},$$

the supremum over the Borel sets. No reference coupling, label or constant but L enters.

(2) *At infinite cutoff.* Assume the standing hypothesis (SH) of Definition 18.208 with the side ℓ and the limit b of the labels, and let $\mu = \mu[\ell,\mathsf{b}]$ be the thread measure of T_ℓ seen in the window (Definition 18.207). Put $x_j = x(K_j,\beta_j,\ell)$ and $\bar x = \limsup_j x_j$, so that $\bar x \le \ell^3(2C_{\mathrm{S}} + C_g)$ by (0). Let $o_j \in a_{K_j}\mathbb{Z}^3$, let ν be a weak limit point on Y of the laws $\sigma_j = (\Upsilon_{K_j} \circ \tau_{-o_j})_*\nu_j$, $\sigma_{j_k} \to \nu$, and $\underline{x} = \liminf_k x_{j_k} \le \bar x$. Then $|\nu(f)| \le e^{\underline{x}/8}\mu(|f|^8)^{1/8}$ for every bounded complex

Borel function f on Y; $\nu = h\mu$ with $\int h^{8/7}d\mu \le e^{\underline{x}/7}$ and $0 \le \int h\log h\,d\mu \le \underline{x}$; and, with $\epsilon = 22.3\,(e^{\bar{x}/7}-1)$,

$$\sup_{A\subset Y \text{ Borel}} |\nu(A)-\mu(A)| \le \min\{\sqrt{\underline{x}/2},\ \mathfrak{t}(\underline{x})\} \le \tfrac{1}{2}(\sqrt{\epsilon}+\epsilon),$$

so that ν and μ share the mass $\int \min\{h,1\}\,d\mu \ge 1-\mathfrak{t}(\underline{x}) > 0$. The right members depend on $\underline{x}$, respectively $\bar{x}$, alone, and on neither the states ν_j, nor the offsets o_j, nor the reference coupling. Nothing is asked of $\bar{x}$ beyond (SH).

(3) *Every depth from d_0 on, every closed cube of the cell.* Let $d_0 \ge 0$ and let $\mathfrak{c} \subset \square$ be closed, for instance a closed cube. Every weak limit point of the laws $(\mathrm{pj}^{[d_0]} \circ \Upsilon_{K_j} \circ \tau_{-o_j})_*\nu_j$ on $Y^{[d_0]}$, or of $(p_{\mathfrak{c}} \circ \Upsilon_{K_j} \circ \tau_{-o_j})_*\nu_j$ on $Y(\mathfrak{c})$, satisfies (2) with $\mu^{[d_0]}$, respectively $(p_{\mathfrak{c}})_*\mu$, in the place of μ and with the same numbers $\underline{x}$, $\bar{x}$, formed along the subsequence which gives it.

(4) *Two limits; the count.* For two laws ν, ν' of (2), formed from two sequences under (SH) with one side ℓ and one limit $\mathfrak{b}$, $\sup_A |\nu(A)-\nu'(A)| \le \min\{\sqrt{\underline{x}/2}, \mathfrak{t}(\underline{x})\} + \min\{\sqrt{\underline{x}'/2}, \mathfrak{t}(\underline{x}')\}$. If $\nu^{(1)}, \dots, \nu^{(n)}$ are such laws, pairwise mutually singular, with $\underline{x}^{(r)} \le x^\sharp$ for all r, then $n \le e^{x^\sharp}$. ∎

Proof. Step 1: the excess, (0). Let $\ell' \le \ell''$ in $\mathfrak{L}$ and take the board of $\mathbb{L}_K$ of side $\frac{\ell'}{2}$; the comparison of chessboard seminorms in Corollary YM3_I:86.6 applied to the constant 1 gives $p_{\ell''} \le p_{\ell'}$. The Wilson action of $a_K\mathbb{Z}^3/\ell\mathbb{Z}^3$ has $3(\ell L^K)^3$ terms in $[0,2]$, so $1 \ge Z_{K,\ell,\beta} \ge e^{-6\beta(\ell L^K)^3}$ and $p_\ell \ge -6\beta a_K^{-3}$; a non-increasing function bounded below has its infimum as its limit, and $0 \le x \le \ell^3(0+6\beta a_K^{-3})$. At data at g and sides of $\mathfrak{L}(g)$, the third link of the proof of Lemma 18.216, which uses the volume bounds of Theorem YM3_I:AD, gives $p_\ell - p_{\ell'} \le 2C_{\mathrm{S}} + C_g$ for all $\ell' \in \mathfrak{L}(g)$; letting $\ell' \to \infty$ in the unbounded set $\mathfrak{L}(g)$ gives $x \le \ell^3(2C_{\mathrm{S}}+C_g)$, and $8\log \mathsf{Q}_0 = \ell^3(2C_{\mathrm{S}}+C_g)$. For the real first-passage construction, the comparison in that proof gives $C_{\mathrm{gap}} + C_{\mathrm{st}} + C_g$ (Lemma 18.216).

Step 2: the domination at one pair. Let F be a bounded Borel function of the inner bonds of the cell $\square'$. Use the chessboard seminorm $|||F|||_t$ on the torus of side t from the proof of Lemma 18.216. Its comparison of sides gives

$$|||F|||_t \le |||F|||_\ell \exp\bigl[(\ell/2)^3(p_\ell - p_t)\bigr] \le e^{x/8}|||F|||_\ell \qquad (t \ge \ell).$$

For a periodic Gibbs state, Proposition YM3_I:91.9 bounds $|\mathfrak{s}(F)|$ by $\liminf_{t\to\infty} |||F|||_t$. For a torus state of side $\ell' \ge \ell$ the chessboard inequality bounds it by $|||F|||_{\ell'}$. On the reference torus the seminorm is bounded by $\tilde{\mu}_{K,\ell,\beta}(|F|^8)^{1/8}$ by Hölder on the eight reflected copies, as in Lemma 18.216. Taking $F = f \circ \Phi$ gives $\Phi_*\mathfrak{s} \preceq_{e^{x/8}} \Phi_*\tilde{\mu}_{K,\ell,\beta}$.

Step 3: from the domination to the density, the entropy and the distance. Let $\boldsymbol{\sigma} \preceq_{\mathsf{Q}} \boldsymbol{\rho}$ with $\mathsf{Q} = e^{x/8}$. By Lemma 18.214, (DM2) and (DM3), $\boldsymbol{\sigma} = h\boldsymbol{\rho}$, $\int h^{8/7}d\boldsymbol{\rho} \le \mathsf{Q}^{8/7} = e^{x/7}$ and $0 \le \int h\log h\,d\boldsymbol{\rho} \le 7\log\int h^{8/7}d\boldsymbol{\rho} \le x$. The inequality of Pinsker, $\sup_A |\boldsymbol{\sigma}(A)-\boldsymbol{\rho}(A)| \le (\frac{1}{2}\int h\log h\,d\boldsymbol{\rho})^{1/2}$, gives the bound $\sqrt{x/2}$. For $\mathfrak{t}$: the supremum is $\tau = \boldsymbol{\sigma}(A_*) - \boldsymbol{\rho}(A_*)$ at $A_* = \{h > 1\}$; with $\alpha = \boldsymbol{\rho}(A_*)$, (DM1) gives $\boldsymbol{\sigma}(A_*) \le \min\{1, e^{x/8}\alpha^{1/8}\}$, so $\tau \le \sup_{\alpha\in[0,1]}(\min\{1, e^{x/8}\alpha^{1/8}\} - \alpha)$. The function $\alpha \mapsto e^{x/8}\alpha^{1/8} - \alpha$ has its maximum $\frac{7}{8}(e^x/8)^{1/7}$ at $\alpha_* = e^{x/7}8^{-8/7}$, and $\alpha_* \le e^{-x}$ exactly when $x \le \log 8$; for $\alpha \ge e^{-x}$ the bound is $1-\alpha \le 1-e^{-x}$, and at $\alpha = e^{-x}$ the two branches meet. This gives $\tau \le \mathfrak{t}(x)$ in both ranges. The shared mass is $1-\tau$. Finally $e^{x/7}-1 \ge x/7$ and $22.3/7 > 2$ give $\sqrt{x/2} \le \frac{1}{2}\sqrt{22.3\,(e^{x/7}-1)}$, and $x \mapsto \frac{1}{2}(\sqrt{\epsilon}+\epsilon)$ increases. This proves (1).

Step 4: the passage to infinite cutoff, (2). Put $\rho_j = (\Upsilon_{K_j})_*\tilde{\mu}_{K_j,\ell,\beta_j}$. By Step 2 with $\Phi = \Upsilon_{K_j} \circ \tau_{-o_j}$, which depends on the bonds with both end points in the cell $o_j + \square$ (Lemma 18.211(d)), and the invariance of $\tilde{\mu}_{K_j,\ell,\beta_j}$ under τ_{-o_j}, which permutes the plaquettes and preserves product Haar measure, $|\sigma_j(f)| \le e^{x_j/8}\rho_j(|f|^8)^{1/8}$ for every bounded Borel f; the case of a torus measure ν_j of side $\ell_j \ge \ell$ is the second case of Step 2. Choose a further subsequence of (j_k) along which $x_{j_k} \to \underline{x}$. As $\rho_j \to \mu$ weakly (Lemma 18.212(e), the labels lying in J_g, Lemma 18.212(c)), the inequality passes to ν at

every continuous f with the constant $e^{\underline{x}/8}$, and from continuous to bounded Borel functions by (DM7) of Lemma 18.214, applied to the constant sequences ν and μ. Step 3 gives the rest of (2).

Step 5: (3) and (4). Let ν' be a weak limit point of the images in (3), reached along a subsequence (j_k), and $\underline{x} = \liminf_k x_{j_k}$. Take first a subsequence of (j_k) on which $x_j \to \underline{x}$, and then, Y being compact and metrisable, a further subsequence on which σ_j converges weakly to a law ν; on the last one the excess still tends to $\underline{x}$, so that (2) gives $\nu \preceq_{e^{\underline{x}/8}} \mu$. As $\mathrm{pj}^{[d_0]}$ and $p_{\mathfrak{c}}$ are continuous, on this subsequence the images of the σ_j tend both to ν' and to the image of ν, so ν' is the image of ν; images keep the domination with the same constant (DM4), the image of μ is $\mu^{[d_0]}$, respectively $(p_{\mathfrak{c}})_*\mu$, and Step 3 applies with $\underline{x}$, the bound by $\bar{x}$ following as in Step 3 from $\underline{x} \le \bar{x}$. For (4), the distance is a metric; and by (DM1) pairwise mutually singular laws with disjoint carriers $\mathcal{A}_r$ have $1 = \nu^{(r)}(\mathcal{A}_r) \le e^{x^\sharp/8}\mu(\mathcal{A}_r)^{1/8}$, so $\mu(\mathcal{A}_r) \ge e^{-x^\sharp}$ and $n \le e^{x^\sharp}$ (DM6). □

Proposition 18.219 (Clustering of functions of block fields at one cutoff, at the physical length $R + \frac{a_d}{2}$)**.** Let $K \in \mathbb{N}$, $\beta \ge 0$ and $\nu \in \mathcal{G}^{\mathrm{per}}_{K,\beta}$, with no condition on (K,β); let $0 \le d \le K$, $w = \frac{1}{2}a_d$, $R \in a_d\mathbb{N}$, and let A, B be bounded complex Borel functions on Z which depend on finitely many depths d' with $d \le d' \le K$ and, at each of them, on the bonds of $a_{d'}\mathbb{Z}^3$ with both end points in $[-R,R]^3$; put $A_K = A \circ \Psi_K$ and $B_K = B \circ \Psi_K$, functions of the configurations of $\mathbb{L}_K$ (Definition 18.206). Then for every $x \in a_K\mathbb{Z}^3$ with $|x| \ge 4\sqrt{3}\,(R+w)$

$$\Big|\nu(\overline{A_K}\,\tau_x B_K) - \nu\big(\overline{\mathbb{E}_\nu[A_K|\mathcal{I}]}\,\mathbb{E}_\nu[B_K|\mathcal{I}]\big)\Big| \le 4\sqrt{3}\,\mathsf{C}_{\mathrm{dc}}\,\nu(|A_K|^2)^{1/2}\,\nu(|B_K|^2)^{1/2}\,\frac{R+w}{|x|} \le 4\sqrt{3}\,\mathsf{C}_{\mathrm{dc}}\,\|A\|_\infty\|B\|_\infty\,\frac{R+w}{|x|},$$

with the absolute number C_{dc} of YM3_I:(1968). The bound depends on none of K, β, ν or a reference coupling. At the cutoff K, A_K and B_K are functions of bonds of $\mathbb{L}_K$ inside the cube of half side $R + \frac{1}{2}(a_d - a_K)$, a number below $R+w$; so one cube free of K, $[-(R+w),R+w]^3$, contains these bonds at all cutoffs. If ν is a pure phase, the subtracted term is $\overline{\nu(A_K)}\,\nu(B_K)$. In particular this holds for $A = F(\mathsf{z}_d|_{\mathsf{B}})$, $B = F'(\mathsf{z}_d|_{\mathsf{B}})$ with bounded Borel functions F, F' of the field of the one depth d on a finite set B of bonds in $[-R,R]^3$, among them the loops $W_C(\mathsf{z}_d) = \frac{1}{2}\operatorname{tr}\mathsf{z}_d(C)$ of the block field along closed paths C of such bonds. ■

Proof. Step 1: the bare bonds. By Lemma 18.211(b), $(\mathcal{M}^{K-d'}U)(c)$ depends only on U on the bonds with both end points in $\tilde{B}^{K-d'}(c_-) \cup \tilde{B}^{K-d'}(c_+)$, and by the block formula in Definition 18.206 a point of $\tilde{B}^n(y)$ differs from y by at most $\frac{1}{2}(L^n-1)a_{d'+n} = \frac{1}{2}(a_{d'} - a_{d'+n})$ in each coordinate. At $n = K - d'$ and $d' \ge d$ this is at most $\frac{1}{2}(a_d - a_K)$. So A_K and B_K are bounded Borel functions of the bonds of $\mathbb{L}_K$ in the cube $Q_r = [-ra_K, ra_K]^3$ with $ra_K = R + \frac{1}{2}(a_d - a_K)$; here $r = L^{K-d}R/a_d + \frac{1}{2}(L^{K-d}-1)$ is an integer ≥ 1, as L is odd and $R \ge a_d$, and $ra_K < R + w$.

Step 2: the lattice bound. Proposition YM3_I:99.7 at spacing a_K, using $r/n + 2r^2/n^2 \le 4ra_K/|x|_\infty$ when $|x|_\infty \ge 4ra_K$, for $A_K, B_K \in \mathcal{F}(\mathsf{Bd}(Q_r))$ and $|x|_\infty \ge 4ra_K$, gives the inequality with $4\,\mathsf{C}_{\mathrm{dc}}\mathcal{N}_r(A_K)^{1/2}\mathcal{N}_r(B_K)^{1/2}ra_K/|x|_\infty$ on the right; and $\mathcal{N}_r(A_K) \le \nu(|A_K|^2) \le \|A\|^2_\infty$ by the bound following YM3_I:(1959), whose hypotheses hold by Proposition YM3_I:103.4. If $|x| \ge 4\sqrt{3}(R+w)$, then $|x|_\infty \ge |x|/\sqrt{3} \ge 4(R+w) > 4ra_K$, and $ra_K/|x|_\infty \le \sqrt{3}(R+w)/|x|$. For a pure phase, the invariant events have probability zero or one, so the conditional expectations are constants. □

Lemma 18.220 (Recovery of the label at every reference coupling)**.** Let $0 < g \le \gamma$ and let S be a limit of separated cumulants through periodic Gibbs states at first-passage data at g, with the cutoffs tending to infinity. The labels $b_j = a_{K_j}\beta_j$ converge along every defining sequence to the same $b(S) \in I^{\mathrm{lab}}(g)$. This applies in particular to volume-first limits. ■

Proof. The first-passage bounds put every b_j in the compact positive interval $I^{\mathrm{lab}}(g)$ (Corollary YM3_I:74.2). Let $b > 0$ be the limit along any subsequence. Choose an integer $z \ge 0$ so that $L^z b \ge 2\gamma_W^{-2}$ and put $g' = (L^z b/2)^{-1/2}$. We have $g' \le \gamma_W$ and $2g'^{-2}$ lies in the interior of $I(g') = (g'^{-2} + \bar{B}, X_{g'} - \bar{B}]$: the first inequality uses $\bar{B} < g'^{-2}$ and the second uses

$X_{g'} - \bar{B} = Lg'^{-2} + (L-2)\bar{B}$ with $L \geq 13$. Thus $L^z b_j \in I(g')$ eventually along the subsequence. The common interval property in Corollary YM3_I:74.2 makes $(K_j - z, \beta_j)$ first-passage data at g' with these labels.

The dilation $D : x \mapsto L^z x$ identifies the two lattices, their oriented plaquettes and their Haar variables. The Wilson Gibbs kernels correspond at the same bare coupling β_j, so the images of the periodic Gibbs states are periodic Gibbs states. The same identification takes a volume-first state to a volume-first state since dilation takes T_{2L^m} to $T_{2L^{m+z}}$. For any separated tuple $f_1, \dots, f_n$, the definition of the action density gives

$$S^T_{n;K_j}(f_1, \dots, f_n) = L^{zn} S^T_{n;K_j - z}(f_1 \circ D^{-1}, \dots, f_n \circ D^{-1})$$

in the corresponding states. The dilated subsequence therefore has a limit at g' with label $L^z b$. Choose a fixed nonzero smooth test pair (f, h) at unit scale and write $f_t = f(\cdot/t)$, $h_t = h(\cdot/t)$. Theorem YM3_I:AW applied at g' to the pair at scale $L^z t$ gives

$$\lim_{t \downarrow 0} \frac{S_2(f_t, h_t)}{\mathsf{c}_2 \, \mathcal{K}_{\mathrm{free}}(f, h)} = b^{-2},$$

because $\mathcal{K}_{\mathrm{free}}$ is invariant under a common dilation of the tests. This limit depends only on S. All subsequential label limits, including those from different defining sequences for S, therefore equal the same positive b. Compactness proves convergence along the whole sequence. □

Proposition 18.221 (The thread laws in infinite volume: clustering, the Hilbert space of the threads, its zero modes, the cone, and the space of the action density)**.** Let $0 < g \leq \gamma$, let S be a volume-first limit at g, with a defining sequence (K_j, β_j, ν_j), $\nu_j \in \mathcal{G}^{\mathrm{tor}}_{K_j, \beta_j}$ for every j (Definition 18.205), and with the label $\mathsf{b} = \mathsf{b}(S)$ (Lemma 18.220), and let $\boldsymbol{\nu}$ be a weak limit point on Z of the laws $\boldsymbol{\sigma}_j = (\Psi_{K_j})_* \nu_j$, $\boldsymbol{\sigma}_{j_k} \to \boldsymbol{\nu}$. Then the standing hypothesis (SH) holds at every side $\ell \in \mathfrak{L}(g)$, in infinite volume, and Lemma 18.217 applies to $\boldsymbol{\nu}$. Signs: $\mathbb{D}^1 = \bigcup_s a_s \mathbb{Z}$ and $\mathbb{D}_+ = \mathbb{D}^1 \cap (0, \infty)$; for $v \in \mathbb{D}$ and a function Φ on Z, $\tau_v \Phi = \Phi \circ T_{-v}$; for $c \in \frac{1}{2}\mathbb{D}^1$, θ_c is the reflection in $\{x_3 = c\}$ and $\theta_c \Phi = \Phi \circ T_{\theta_c}$ (Lemma 18.217(f)); for a bond c' of $a_d \mathbb{Z}^3$ its footprint is $\mathrm{fp}(c') = (c'_- + [-\frac{a_d}{2}, \frac{a_d}{2}]^3) \cup (c'_+ + [-\frac{a_d}{2}, \frac{a_d}{2}]^3)$; for $Q \subset \mathbb{R}^3$, $\mathcal{A}(Q)$ is the set of the bounded complex Borel functions on Z of finitely many coordinates $\mathsf{z}_d(c')$ with $\mathrm{fp}(c') \subset Q$; $\mathcal{I}^{\mathbb{Z}}$ is the σ-algebra of the Borel sets $E \subset \mathsf{Z}$ with $T_v^{-1} E = E$ for all $v \in \mathbb{Z}^3$, the maps T_v being formed at $s = 0$; and $\| \cdot \|_2$ is the norm of $L^2(\boldsymbol{\nu})$.

(a) *Clustering at the rate $\rho'/|x|$.* For $\rho' > 0$, $A, B \in \mathcal{A}([-\rho', \rho']^3)$ and $x \in \mathbb{D} \setminus \{0\}$,

$$\left| \boldsymbol{\nu}(\bar{A} \, \tau_x B) - \boldsymbol{\nu}(\overline{\mathbb{E}[A|\mathcal{I}^{\mathbb{Z}}]} \, \mathbb{E}[B|\mathcal{I}^{\mathbb{Z}}]) \right| \leq 4 \, \mathsf{C}_{\mathrm{dc}} \, \|A\|_2 \|B\|_2 \, \frac{\rho'}{|x|_\infty} \leq 4\sqrt{3} \, \mathsf{C}_{\mathrm{dc}} \, \|A\|_\infty \|B\|_\infty \, \frac{\rho'}{|x|},$$

the conditional expectations being taken under $\boldsymbol{\nu}$. Functions of the block fields on the bonds with both end points in $[-\rho, \rho]^3$ lie in $\mathcal{A}([-\rho', \rho']^3)$ with $\rho' = \rho + \frac{1}{2}$, and with $\rho' = \rho + \frac{1}{2} a_{d_0}$ if they read the depths $\geq d_0$ alone.

(b) *Positivity at the coordinate planes $\{x_\kappa = c\}$, $c \in \frac{1}{2}\mathbb{D}^1$.* For $\kappa \in \{1, 2, 3\}$, $c \in \frac{1}{2}\mathbb{D}^1$, the reflection θ in $\{x_\kappa = c\}$ and every $\Phi \in \mathcal{A}(Q)$ with Q a bounded part of $\{x_\kappa \geq c\}$, $\boldsymbol{\nu}(\overline{\Phi \circ T_\theta} \, \Phi) \geq 0$; at the planes through the sites of a lattice $a_s \mathbb{Z}^3$ as well as at those between them.

(c) *The Hilbert space of the threads; its semigroup.* Fix $c_0 \in \mathbb{Z} + \frac{1}{2}$, so that $c_0 \in a_d(\mathbb{Z} + \frac{1}{2})$ for every $d \geq 0$, put $\theta = \theta_{c_0}$, $H^+_c = \{x_3 \geq c\}$ and $\mathcal{A}^+ = \bigcup_Q \mathcal{A}(Q)$ over the bounded $Q \subset H^+_{c_0}$; a coordinate $\mathsf{z}_d(c')$ has its footprint in $H^+_{c_0}$ exactly when both end points of c' lie in $\{x_3 > c_0\}$. The form $\langle \Phi, \Psi \rangle = \boldsymbol{\nu}(\overline{\theta\Phi} \, \Psi)$ on $\mathcal{A}^+$ is positive semi-definite and $\|[\Phi]\| \leq \|\Phi\|_2$; $\mathcal{H}^{\mathrm{thr}}$ is the completion of $\mathcal{A}^+$ modulo its null vectors, with the classes $[\Phi]$ and $\Omega = [1]$, and $[\Phi]$ extends by continuity to the bounded functions measurable for the σ-algebra of the coordinates with footprints in $H^+_{c_0}$. The classes of the gauge-invariant members of $\mathcal{A}^+$ fill the classes of $\mathcal{A}^+$: $[\Phi] = [\Phi^{\mathrm{inv}}]$, Φ^{inv} the Haar mean of Φ over the gauge transformations at the finitely many

sites which it reads. For $t \in \mathbb{D}_+$ there is one bounded operator T_t with $\mathsf{T}_t[\Phi] = [\tau_{te_3}\Phi]$; it is self-adjoint, $0 \le \mathsf{T}_t \le 1$, $\mathsf{T}_t\mathsf{T}_{t'} = \mathsf{T}_{t+t'}$, and with $\mathsf{A} = \mathsf{T}_1$, $\mathsf{T}_t = \mathsf{A}^t$. For $v \in \mathbb{D}$ with $v_3 = 0$ and $\mathsf{r} \in W_2$, acting on the first two coordinates, $U(v)[\Phi] = [\tau_v\Phi]$ and $U(\mathsf{r})[\Phi] = [\Phi \circ T_{\mathsf{r}^{-1}}]$ are unitary, fix Ω and commute with the T_t. Let $\mathcal{H}^>$ be the closed span of the classes of the $\Phi \in \mathcal{A}^+$ which read coordinates with footprints in $H^+_{c_0+\delta}$ alone, for some $\delta > 0$. Then $\mathcal{H}^{\rm thr} = \ker \mathsf{A} \oplus \mathcal{H}^>$, $\mathcal{H}^> = \overline{\operatorname{Ran} \mathsf{T}_t}$ for every $t \in \mathbb{D}_+$, and on $\mathcal{H}^>$ the Hamiltonian of the threads $H^{\rm thr} = -\log(\mathsf{A}|_{\mathcal{H}^>})$ is self-adjoint and ≥ 0, $e^{-tH^{\rm thr}} = \mathsf{T}_t|_{\mathcal{H}^>}$ for $t \in \mathbb{D}_+$, and $\mathsf{T}_t \to 1$ strongly on $\mathcal{H}^>$ as $t \to 0$ in $\mathbb{D}_+$.

(d) *Zero energy is the invariant events.* Let P_1 be the projection of $\mathcal{H}^{\rm thr}$ onto $\ker(1-\mathsf{A}) = \ker H^{\rm thr}$. Then $\langle [\Phi], P_1[\Psi]\rangle = \boldsymbol{\nu}(\overline{\mathbb{E}[\Phi|\mathcal{I}^{\mathbb{Z}}]}\,\mathbb{E}[\Psi|\mathcal{I}^{\mathbb{Z}}])$ for $\Phi, \Psi \in \mathcal{A}^+$, and $P_1[\Phi] \mapsto \mathbb{E}[\Phi|\mathcal{I}^{\mathbb{Z}}]$ extends to a unitary map of $\ker H^{\rm thr}$ onto $L^2(\mathsf{Z}, \mathcal{I}^{\mathbb{Z}}, \boldsymbol{\nu})$ which carries Ω to 1. The reflection θ acts as the identity on $L^2(\mathsf{Z}, \mathcal{I}^{\mathbb{Z}}, \boldsymbol{\nu})$.

(e) *The horizontal translations; the cone.* For $\lambda \in \{1, 2\}$ and $t \in \mathbb{D}_+$, $(1-\mathsf{A}^t)^2 \ge \mathsf{A}^t(2 - U(te_\lambda) - U(te_\lambda)^*)$ on $\mathcal{H}^{\rm thr}$. On $\mathcal{H}^>$ the representation $v \mapsto U(v)$ extends to one strongly continuous unitary representation of $\mathbb{R}^2$, whose generators $P_1^{\rm thr}, P_2^{\rm thr}$ commute with $H^{\rm thr}$, and the joint spectral measure of $(H^{\rm thr}, P_\lambda^{\rm thr})$ is carried by $\{(E, q) : E \ge |q|\}$: $H^{\rm thr} \ge |P_\lambda^{\rm thr}|$. So, for every depth d, on every closed subspace of $\mathcal{H}^>$ which T_{a_d} and $U(\pm a_d e_\lambda)$ map into itself, among them the closed span of the classes of the functions of the block fields of the depths $\ge d$ which read footprints in some $H^+_{c_0+\delta}$, $\delta > 0$, the joint spectral measure of $(\mathsf{T}_{a_d}, U(a_d e_\lambda))$ is carried by $\{(t, \zeta) : t \le e^{-|\arg\zeta|}\} \subset \{t \le e^{-\mathsf{k}|\arg\zeta|}\}$, k the chord constant YM3_I:(1954): the form of the lattice cone at the spacing a_d, with the constant one.

(f) *Loops.* For a closed path C of bonds c' of $a_d\mathbb{Z}^3$ with both end points of every bond in $\{x_3 > c_0\}$, the loop $W_C(\mathsf{z}_d) = \frac{1}{2}\operatorname{tr}\mathsf{z}_d(C)$ is a continuous member of $\mathcal{A}^+$; so $\mathcal{H}^{\rm thr}$ holds the classes of all such loops of block fields, and $\mathcal{H}^>$ those whose bonds have their end points in some $\{x_3 \ge c_0 + \delta + \frac{1}{2}a_d\}$.

(g) *The space of the action density; the map J.* Let $\mathbb{E}_+ = \{x_3 > c_0\}$, $\mathcal{D}$ the space of the complex combinations $F = \sum_k \zeta_k X_k$ of separated tuples $X_k \in \mathfrak{S}$ over $\mathbb{E}_+$, $F_j = \sum_k \zeta_k (X_k)^{\nu_j}_{K_j}$ with the centred products of Definition 18.205, and $\mathcal{H}_{c_0}(S)$ the Hilbert space of the form $\sigma_S(\theta\bar{F}\cdot F')$ on $\mathcal{D}$, which the translation by $-c_0e_3$ carries unitarily onto $\mathcal{H}(S)$, with its vacuum, its semigroup e^{-tH}, its translations $\mathcal{U}(v)$, $v \in \mathbb{R}^2$, its representation $\mathcal{U}(\mathsf{r})$ of W_2 and its projection P_0 onto $\ker H$. There is a subsequence (j'_i) of (j_k) along which $\mathfrak{q}(\Phi, F) = \lim_i \nu_{j'_i}(\overline{\theta(\Phi \circ \Psi_{K_{j'_i}})}\, F_{j'_i})$ exists for all $\Phi \in \mathcal{A}^+$ and $F \in \mathcal{D}$. Along it $|\mathfrak{q}(\Phi, F)| \le \|[\Phi]\|\,\|[F]\|_S$, where $\|[F]\|_S$ is the norm of the class of F in $\mathcal{H}_{c_0}(S)$, and there is exactly one bounded operator $J\colon \mathcal{H}_{c_0}(S) \to \mathcal{H}^{\rm thr}$ with $\langle [\Phi], J[F]\rangle = \mathfrak{q}(\Phi, F)$ for all $\Phi \in \mathcal{A}^+$ and $F \in \mathcal{D}$; $\|J\| = 1$, $J\Omega = \Omega$, $\operatorname{Ran} J \subset \mathcal{H}^>$, $J\,e^{-tH} = e^{-tH^{\rm thr}}J$ for every real $t \ge 0$, $J\mathcal{U}(v) = U(v)J$ for $v \in \mathbb{R}^2$, $J\mathcal{U}(\mathsf{r}) = U(\mathsf{r})J$ for $\mathsf{r} \in W_2$, and $JP_0 = P_1J$; the positive contraction J^*J commutes with e^{-tH}, $\mathcal{U}(v)$, $\mathcal{U}(\mathsf{r})$ and fixes Ω.

■

Proof. Step 0: the setting. The data at g have labels in J_g and $\ell \in \mathfrak{L}(g)$ admits J_g (Lemma 18.212(c)); the labels converge (Lemma 18.220); all ν_j lie in $\mathcal{G}^{\rm tor}_{K_j,\beta_j} \subset \mathcal{G}^{\rm per}_{K_j,\beta_j}$ (Proposition YM3_I:103.4); so (SH) holds at every side of $\mathfrak{L}(g)$, in infinite volume (Lemma 18.217(h)), and Lemma 18.217(b)(ii)–(iv), (f)(i), (ii) give: $\boldsymbol{\nu}(\mathsf{Thr}) = 1$, the invariance under the maps T_v, $v \in \mathbb{D}$, under T_{r}, $\mathsf{r} \in \mathbb{D} \rtimes W_3$, and under the gauge transformations, and $\boldsymbol{\sigma}_{j_k}(H) \to \boldsymbol{\nu}(H)$ for every bounded Borel H of finitely many depths and bonds in a bounded set. For $v \in a_s\mathbb{Z}^3$ and $K \ge s$, $\Psi_K \circ \tau_v = T_v \circ \Psi_K$, as $\mathcal{M}$ commutes with τ_v at the depths $\ge s$ (Lemma 18.211(b)); so $(\tau_v\Phi) \circ \Psi_K$ is the lattice translate of $\Phi \circ \Psi_K$, and likewise for the isometries of $a_s\mathbb{Z}^3$. By Lemma 18.211(b) and Step 1 of the proof of

Proposition 18.219, $\Phi\circ\Psi_K$ is, for $\Phi\in\mathcal{A}(Q)$ and K at least every depth read, a bounded Borel function of the bonds of $\mathbb{L}_K$ in Q.

Step 1: clause (a). Let $|x|_\infty,|x'|_\infty>4\rho'$ with $x,x'\in\mathbb{D}$, and $K=K_{j_k}$ large, $r=\lceil\rho'/a_K\rceil$, so that $\rho'\le ra_K<\rho'+a_K$, $x,x'\in\mathbb{L}_K$ and $4ra_K\le\min\{|x|_\infty,|x'|_\infty\}$. The lattice bound YM3_I:(1970), with $r/n+2r^2/n^2\le 4ra/|x|_\infty$ at $|x|_\infty\ge 4ra$, at x and at x', whose subtracted terms coincide, the sizes $\mathcal{N}_r\le\nu_j(|\cdot|^2)$ (the bound following YM3_I:(1959)) and the triangle inequality give $|\nu_j(\bar A_K\tau_x B_K)-\nu_j(\bar A_K\tau_{x'}B_K)|\le 4\mathsf{C}_{\rm dc}\nu_j(|A_K|^2)^{1/2}\nu_j(|B_K|^2)^{1/2}ra_K(|x|_\infty^{-1}+|x'|_\infty^{-1})$, with $A_K=A\circ\Psi_K$; the four functions $\bar A\tau_xB$, $\bar A\tau_{x'}B$, $|A|^2$, $|B|^2$ are bounded Borel and local, and Step 0 carries the inequality to $\boldsymbol{\nu}$ with ρ' in the place of ra_K. Now let $e=(1,1,1)$ and $S_nB=n^{-3}\sum_{y\in\{0,\dots,n-1\}^3}\tau_yB$; the maps T_y, $y\in\mathbb{Z}^3$, are homeomorphisms of Z which keep $\boldsymbol{\nu}$, so the mean ergodic theorem gives $S_nB\to\mathbb{E}[B|\mathcal{I}^{\mathbb{Z}}]$ and $\tau_{ne}S_nB\to\mathbb{E}[B|\mathcal{I}^{\mathbb{Z}}]$ in $L^2(\boldsymbol{\nu})$. The points $x'=ne+y$ have $|x'|_\infty\ge n$; the mean of the inequality over them, $n\to\infty$, and $\boldsymbol{\nu}(\bar A\,\mathbb{E}[B|\mathcal{I}^{\mathbb{Z}}])=\boldsymbol{\nu}(\overline{\mathbb{E}[A|\mathcal{I}^{\mathbb{Z}}]}\,\mathbb{E}[B|\mathcal{I}^{\mathbb{Z}}])$ give (a) for $|x|_\infty>4\rho'$. For $0<|x|_\infty\le 4\rho'$ the left side is at most $2\|A\|_2\|B\|_2$ and the right side at least $\mathsf{C}_{\rm dc}\|A\|_2\|B\|_2$, while $\mathsf{C}_{\rm dc}=16(1+\sqrt{2})\mathsf{k}^{-2}\mathsf{C}_{\rm dn}>100$, as $\mathsf{k}<1$ and $\mathsf{C}_{\rm dn}>3$ in the constants of YM3_I:(1968). Finally $|x|_\infty\ge|x|/\sqrt{3}$ and $\|A\|_2\le\|A\|_\infty$. The footprints of the bonds with both end points in $[-\rho,\rho]^3$ lie in $[-\rho-\frac12 a_d,\rho+\frac12 a_d]^3$.

Step 2: clause (b). Let $c\in\frac12 a_s\mathbb{Z}$; then $\theta\in\mathsf{Iso}(a_s\mathbb{Z}^3)$, and for $K\ge s$ the plane $\{x_\kappa=c\}$ is a lattice mirror of $\mathbb{L}_K$ (a site mirror if $c\in a_K\mathbb{Z}$, a bond mirror otherwise), while $\theta(\Phi\circ\Psi_K)=(\Phi\circ T_\theta)\circ\Psi_K$ by the covariance of the block map (Lemma 18.211(b)). For $\Phi\in\mathcal{A}(Q)$, $\Phi\circ\Psi_K$ is a bounded Borel function of finitely many bonds of $\mathbb{L}_K$ whose closed segments lie in $Q\subset\{x_\kappa\ge c\}$, and the positivity of the nine families, Proposition YM3_I:103.4, gives $\nu_j(\overline{\theta(\Phi\circ\Psi_{K_j})}\,\Phi\circ\Psi_{K_j})\ge 0$; Step 0 passes it to $\boldsymbol{\nu}$.

Step 3: clause (c). Positivity is (b) at c_0; $\langle\Phi,\Phi\rangle\le\|\theta\Phi\|_2\|\Phi\|_2=\|\Phi\|_2^2$ by the invariance of $\boldsymbol{\nu}$ under T_θ; so classes extend by L^2-continuity, local functions being dense in L^2 of the σ-algebra which they generate. Gauge means: $\theta\Psi$ reads coordinates with end points in $\{x_3<c_0\}$, which a gauge transformation u at sites of $\{x_3>c_0\}$ does not touch, so by the gauge invariance (Lemma 18.217(f)) and Fubini $\langle\Psi,\Phi\rangle=\int\boldsymbol{\nu}(\overline{\theta\Psi}\,\Phi(\cdot^u))\,du=\langle\Psi,\Phi^{\rm inv}\rangle$. The semigroup: τ_{te_3} maps $\mathcal{A}(Q)$ to $\mathcal{A}(Q+te_3)$; $\theta\tau_{te_3}=\tau_{-te_3}\theta$ and the invariance give $\langle\tau_{te_3}\Phi,\Psi\rangle=\langle\Phi,\tau_{te_3}\Psi\rangle$; then $\|[\tau_{te_3}\Phi]\|^2\le\|[\Phi]\|\,\|[\tau_{2te_3}\Phi]\|$, and by induction $\|[\tau_{te_3}\Phi]\|\le\|[\Phi]\|^{1-2^{-n}}\|\Phi\|_\infty^{2^{-n}}$, so T_t is a well defined symmetric contraction. $\mathsf{T}_t\ge 0$: with $c'=c_0+\frac{t}{2}\in\frac12\mathbb{D}^1$, $\theta=\theta_{c'}\tau_{te_3}$ since $2c'-(x_3+t)=2c_0-x_3$, so $\langle\Phi,\tau_{te_3}\Phi\rangle=\boldsymbol{\nu}(\overline{\theta_{c'}\tau_{te_3}\Phi}\,\tau_{te_3}\Phi)\ge 0$ by (b), $\tau_{te_3}\Phi$ depending on coordinates with footprints in $H^+_{c_0+t}\subset H^+_{c'}$. $\mathsf{T}_{a_s}\ge 0$ and $\mathsf{T}_{a_s}^{Ls}=\mathsf{T}_1$ give $\mathsf{T}_{a_s}=\mathsf{A}^{a_s}$ by the uniqueness of positive roots, hence $\mathsf{T}_t=\mathsf{A}^t$. The horizontal translations and W_2 commute with θ and with τ_{te_3} and keep $\boldsymbol{\nu}$ and $\mathcal{A}^+$. The splitting: $\ker\mathsf{A}^t=\ker\mathsf{A}$ and $\overline{\operatorname{Ran}\mathsf{A}^t}=(\ker\mathsf{A})^\perp$ by the spectral theorem; a Φ with footprints in $H^+_{c_0+\delta}$, $a_s\le\delta$, is $\tau_{a_se_3}\Phi'$ with $\Phi'=\tau_{-a_se_3}\Phi\in\mathcal{A}^+$, so $[\Phi]\in\operatorname{Ran}\mathsf{T}_{a_s}$; conversely T_{a_s} maps the dense set of classes into $\mathcal{H}^>$. On $\mathcal{H}^>=(\ker\mathsf{A})^\perp$, A is injective, $0\le\mathsf{A}\le 1$, and the spectral theorem gives $H^{\rm thr}$, $e^{-tH^{\rm thr}}=\mathsf{A}^t$ and, by dominated convergence, $\mathsf{A}^t\to 1$ strongly.

Step 4: clause (d). The map T_θ carries $\mathcal{I}^{\mathbb{Z}}$ onto itself, as $\theta\tau_y\theta=\tau_{\varrho y}$ with $\varrho=\operatorname{diag}(1,1,-1)$, and keeps $\boldsymbol{\nu}$; so it induces a unitary involution R of $L^2(\mathcal{I}^{\mathbb{Z}})=L^2(\mathsf{Z},\mathcal{I}^{\mathbb{Z}},\boldsymbol{\nu})$. For $\Psi\in\mathcal{A}^+$, $S_n\Psi\in\mathcal{A}^+$ (the shifts have $y_3\ge 0$), $S_n\Psi\to h=\mathbb{E}[\Psi|\mathcal{I}^{\mathbb{Z}}]$, and $0\le\boldsymbol{\nu}(\overline{\theta S_n\Psi}\,S_n\Psi)\to\langle Rh,h\rangle$; as $\mathbb{E}[\tau_y\Psi|\mathcal{I}^{\mathbb{Z}}]=\mathbb{E}[\Psi|\mathcal{I}^{\mathbb{Z}}]$ for $y\in\mathbb{Z}^3$ and every local function is moved into $\mathcal{A}^+$ by some τ_{ne_3}, these h are dense in $L^2(\mathcal{I}^{\mathbb{Z}})$; so $R\ge 0$, and $R^2=1$ gives $R=1$. Now $\mathsf{A}^n\to P_1$ strongly, and by (a), for $\Phi,\Psi\in\mathcal{A}^+$, $\langle[\Phi],\mathsf{A}^n[\Psi]\rangle=\boldsymbol{\nu}(\overline{\theta\Phi}\,\tau_{ne_3}\Psi)\to\boldsymbol{\nu}(\overline{\mathbb{E}[\theta\Phi|\mathcal{I}^{\mathbb{Z}}]}\,\mathbb{E}[\Psi|\mathcal{I}^{\mathbb{Z}}])$, and $\mathbb{E}[\theta\Phi|\mathcal{I}^{\mathbb{Z}}]=R\,\mathbb{E}[\Phi|\mathcal{I}^{\mathbb{Z}}]=\mathbb{E}[\Phi|\mathcal{I}^{\mathbb{Z}}]$. Hence $\|P_1[\Phi]\|=\|\mathbb{E}[\Phi|\mathcal{I}^{\mathbb{Z}}]\|_2$: the map is well defined and isometric on the dense subspace $\{P_1[\Phi]\}$ of $\ker(1-\mathsf{A})$, its range holds the dense set of the h above, and an isometry has a closed range; $\mathbb{E}[1|\mathcal{I}^{\mathbb{Z}}]=1$.

Step 5: clause (e). Let $\Phi\in\mathcal{A}^+$, $t\in\mathbb{D}_+$, and $K=K_{j_k}$ so large that $t\in a_K\mathbb{N}$. In the coordinate picture at (e_3,c_0), Proposition YM3_I:99.3 at spacing a_K puts the joint spectrum of

the one-step time operator and the horizontal step in the set $(1-s)^2 \geq s|1-z|^2$, with $s \in [0,1]$ and $|z| = 1$. For $m = t/a_K \in \mathbb{N}$ this implies $(1-s^m)^2 \geq s^m|1-z^m|^2$. Indeed, for $s > 0$ use $|1-z^m| \leq m|1-z|$ and $s^{-m/2} - s^{m/2} \geq m(s^{-1/2} - s^{1/2})$, the latter by $\sinh(mx) \geq m \sinh x$ for $x \geq 0$. At $s = 0$ the inequality is immediate. Applied to $F = \Phi \circ \Psi_K$ in the positive half-space, the spectral inequality gives $u_j(0) - 4u_j(te_3) + u_j(2te_3) + u_j(te_3 + te_\lambda) + u_j(te_3 - te_\lambda) \geq 0$ with $u_j(w) = \nu_j(\overline{\theta F}\, \tau_w F)$; by Step 0, $u_j(w) \to \langle [\Phi], \mathsf{T}_{w_3} U(w_1 e_1 + w_2 e_2)[\Phi]\rangle$ along (j_k). The limit reads $\langle \psi, [(1-\mathsf{A}^t)^2 - \mathsf{A}^t(2 - U - U^*)]\psi\rangle \geq 0$, $U = U(te_\lambda)$, $\psi = [\Phi]$, and the operators are bounded. On $\mathcal{H}^>$, with ψ there, $\|(1-U)\mathsf{A}^{t/2}\psi\|^2 \leq \|(1-\mathsf{A}^t)\psi\|^2$, so $\|(1-U(te_\lambda))\psi\| \leq \|(1-\mathsf{A}^t)\psi\| + 2\|(1-\mathsf{A}^{t/2})\psi\| \to 0$ as $t \to 0$ (the real powers of A being continuous on $\mathcal{H}^>$); $v \mapsto U(v)\psi$ is uniformly continuous on the dense subgroup $\mathbb{D}^1 e_1 + \mathbb{D}^1 e_2$ of $\mathbb{R}^2$ and extends to a strongly continuous unitary representation; Stone's theorem gives P^{thr}_λ. The inequality says $\int g_t \, d\langle \psi, \mathfrak{E}\psi\rangle \geq 0$ for the joint spectral measure $\mathfrak{E}$ of $(H^{\mathrm{thr}}, P^{\mathrm{thr}}_\lambda)$ and $g_t(E,q) = 2e^{-tE}(\cosh tE + \cos tq - 2)$, hence $\mathfrak{E}(\{g_t < 0\}) = 0$ for every $t \in \mathbb{D}_+$, a countable set; dividing $\cosh tE - 1 \geq 1 - \cos tq$ by t^2 and letting $t \to 0$ gives $E^2 \geq q^2$. On a reducing subspace the joint spectral measure of $(\mathsf{T}_{a_d}, U(a_d e_\lambda)) = (e^{-a_d H^{\mathrm{thr}}}, e^{i a_d P^{\mathrm{thr}}_\lambda})$ is the image of $\mathfrak{E}$ under $(E,q) \mapsto (e^{-a_d E}, e^{i a_d q})$, and $|\arg e^{i a_d q}| \leq a_d|q| \leq a_d E$; $\mathsf{k} < 1$.

Step 6: clause (f). The trace of a product of finitely many coordinates is continuous, and the footprint of a bond with both end points in $\{x_3 > c_0\}$ lies in $H^+_{c_0}$.

Step 7: clause (g). The lattice form. For $F \in \mathcal{D}$, the supports of the entries keep a distance $\delta > 0$ from the plane; for $a_{K_j} < \delta$ each centred product $(X_k)^{\nu_j}_{K_j}$ is a real function, the product over the entries f of X_k of the functions $\mathcal{O}_{K_j}(f) - \nu_j(\mathcal{O}_{K_j}(f))$, and F_j is the combination of these products with the complex coefficients ζ_k; the factors read the bonds of plaquettes within $\frac{1}{2}a_{K_j}$ of the supports, hence bonds in $\{x_3 > c_0\}$; for a tuple X write $X_j = X^{\nu_j}_{K_j}$ for its centred product of Definition 18.205 and $\sigma_{\nu_j}(X) = \nu_j(X_j)$ for the mean of this product in ν_j; by the invariance of ν_j under θ, $\theta(X_j) = (\theta X)_j$, and $\nu_j(\overline{\theta F_j} F'_j) = \sum \bar\zeta_k \zeta'_l \, \sigma_{\nu_j}(\theta X_k \cdot X'_l) \to \sigma_S(\theta \bar F \cdot F')$, because σ_{ν_j} is the sum over the partitions of products of the cumulants $S^T[\nu_j]$ by the moment–cumulant formula, since the first cumulants of the centred variables vanish. The cumulants of order at least two are unchanged by centering and converge to those of S. With $\Phi_j = \Phi \circ \Psi_{K_j}$, the form $G \mapsto \nu_j(\overline{\theta G}\, G)$ is positive on $\mathcal{F}_{\mathrm{loc}}(\mathbb{H}^+(e_3, c_0))$ (Proposition YM3_I:103.4), so $|\nu_j(\overline{\theta\Phi_j} F_j)|^2 \leq \nu_j(\overline{\theta\Phi_j}\Phi_j)\, \nu_j(\overline{F_j} F_j) \leq \|\Phi\|^2_\infty \, \nu_j(\overline{\theta F_j} F_j)$. *The extraction.* By the same partition formula and the uniform bounds of Proposition YM3_I:91.13, $|\sigma_{\nu_j}(Y)| \leq C'_n(\mathsf{d}; R) \prod_i \|f_i\|_\sharp$ for tuples Y of type $(\mathsf{d}; R)$, uniformly in j, with $C'_n = \sum_{\mathfrak{p}} \prod_{\mathfrak{b} \in \mathfrak{p}} C^{\mathrm{per}}_{|\mathfrak{b}|}$ over the partitions into blocks of at least two elements; by multilinearity, $\sup_j \nu_j(\overline{\theta(X_j - X'_j)}(X_j - X'_j))^{1/2}$ is small when the entries of X' are near those of X in $\|\cdot\|_\sharp$, with supports in fixed open sets. Countable sets dense in these senses (tuples over finite unions of rational balls, a countable sup-norm dense set of continuous members of $\mathcal{A}^+$) and a diagonal subsequence give the limits at the continuous Φ. For a bounded Borel $\Phi \in \mathcal{A}^+$ and a continuous Φ' of the same coordinates, $|\nu_j(\overline{\theta(\Phi_j - \Phi'_j)}\, F_j)| \leq \nu_j(|\Phi_j - \Phi'_j|^2)^{1/2} \sup_{j'} \nu_{j'}(\overline{\theta F_{j'}} F_{j'})^{1/2}$, and $\nu_j(|\Phi_j - \Phi'_j|^2) \to \|\Phi - \Phi'\|^2_2$ along (j_k) by Step 0, which is as small as we please; so the limits exist at every $\Phi \in \mathcal{A}^+$. *The bound and the map.* Along (j'_i), $\nu_j(\overline{\theta\Phi_j}\Phi_j) \to \boldsymbol{\nu}(\overline{\theta\Phi}\,\Phi) = \|[\Phi]\|^2$ by Step 0, and $\nu_j(\overline{\theta F_j} F_j) \to \sigma_S(\theta \bar F \cdot F) = \|[F]\|^2_S$; so the first inequality of the lattice form passes to the limit as $|\mathfrak{q}(\Phi, F)| \leq \|[\Phi]\|\, \|[F]\|_S$. Hence $\mathfrak{q}(\Phi, F)$ depends on Φ through $[\Phi]$ and on F through $[F]$ alone, antilinearly in the first and linearly in the second; as the classes $[\Phi]$, $\Phi \in \mathcal{A}^+$, are dense in $\mathcal{H}^{\mathrm{thr}}$, Riesz's lemma gives for every F exactly one vector $J[F]$ with $\langle [\Phi], J[F]\rangle = \mathfrak{q}(\Phi, F)$, and $\|J[F]\| \leq \|[F]\|_S$; so J extends by continuity to $\mathcal{H}_{c_0}(S)$, with $\|J\| \leq 1$, and no other bounded operator has this property. As the empty tuple gives the function 1 at every j, $\mathfrak{q}(\Phi, \varnothing) = \lim \nu_j(\overline{\theta\Phi_j}) = \boldsymbol{\nu}(\overline{\theta\Phi}) = \langle [\Phi], \Omega\rangle$, so $J\Omega = \Omega$ and $\|J\| = 1$. At $t \in \mathbb{D}_+$ the identity $\nu_j(\overline{\theta\Phi_j}\, \tau_{te_3} F_j) = \nu_j(\overline{\theta(\tau_{te_3}\Phi)_j}\, F_j)$ of the lattice gives $J e^{-tH} = \mathsf{T}_t J$; the horizontal translations of $\mathbb{D}^1 e_1 + \mathbb{D}^1 e_2$ and W_2 keep ν_j and commute with θ. A tuple X with supports in $H^+_{c_0+\delta}$, $a_s < \delta$, is $\tau_{a_s e_3} X'$ with X' over $\mathbb{E}_+$, so $J[X] = \mathsf{T}_{a_s} J[X'] \in \mathcal{H}^>$;

then $t \mapsto Je^{-tH}\psi$ and $t \mapsto e^{-tH^{\mathrm{thr}}}J\psi$ are continuous on $[0,\infty)$ (Proposition YM3_I:94.1 and clause (c) above) and agree on the dense set $\mathbb{D}_+$, and the same for $v \in \mathbb{R}^2$ with (e). $P_0 = \lim e^{-nH}$ and $P_1 = \lim \mathsf{A}^n$ strongly give $JP_0 = P_1 J$; taking adjoints, J^*J commutes with e^{-tH}, $\mathcal{U}(v)$, $\mathcal{U}(\mathsf{r})$, and $\langle \Omega, J^*J\Omega\rangle = 1$ with $J^*J \le 1$ gives $J^*J\Omega = \Omega$. □

Lemma 18.222 (One step of the block map at a lowest layer)**.** Let $k \ge 0$ and $z \in a_k(\mathbb{Z} + \frac{1}{2})$. Given $m \in \{k, k+1\}$, write L_m for the set of those bonds of $a_m\mathbb{Z}^3$ whose two end points lie in $\{x_3 > z\}$, one or both of them at the height $z + \frac{1}{2}a_m$, and U_m for the set of those bonds of $a_m\mathbb{Z}^3$ whose two end points lie at heights $\ge z + \frac{3}{2}a_m$. Take a bond c of $a_k\mathbb{Z}^3$ whose two end points lie in $\{x_3 > z\}$, write P_c for the set of those bonds of $a_{k+1}\mathbb{Z}^3$ whose two end points lie in $\tilde{B}^1(c_-) \cup \tilde{B}^1(c_+)$, the double block, and take G-valued configurations U, U' on P_c, each of which fulfils the small-field condition $|\partial U(p) - \mathbf{1}| < \omega_M = \frac{1}{200L^2}$ at all plaquettes p whose corners lie in the double block (YM3_I:(30)), such that $U' = U$ on $\mathsf{P}_c \cap \mathsf{U}_{k+1}$ and $|U'(b) - U(b)| \le \delta$ on $\mathsf{P}_c \cap \mathsf{L}_{k+1}$; here δ is subject to $L\delta \le \frac{1}{100}$, and $|\cdot|$ denotes the operator norm. Then: if $c \in \mathsf{U}_k$, $\mathsf{P}_c \subset \mathsf{U}_{k+1}$ and $(\mathcal{M}U')(c) = (\mathcal{M}U)(c)$; if $c \in \mathsf{L}_k$,

$$|(\mathcal{M}U')(c) - (\mathcal{M}U)(c)| \le \mathsf{c}_L\,\delta, \qquad \mathsf{c}_L = \tfrac{14}{13}\Big[1 + 2.62\Big(\frac{1}{L} + \frac{1}{4}(1 - L^{-2})\Big)\Big].$$

The number c_L decreases in L; $\mathsf{c}_{13} < 1.996$, so $\mathsf{c}_L < 0.554\sqrt{L}$ for every $L \ge 13$. ■

Proof. Step 0: the chart. By Lemma 18.211(a), $(\mathcal{M}U)(c)$ is the average of YM3_I:(37) on a torus, lifted through the covering map. The paths $\Gamma_\sigma(y,x)$ go from y to x along successive coordinate segments, in each ordering σ of the nonzero displacements. The covering map identifies these paths and the straight transports on the double block with their counterparts on the torus. Both configurations satisfy the small-field condition at the radius ω_M, so by Lemma YM3_I:4.4 the contour families have diameter at most $\frac{1}{160}$, their means $U(y,x)$ exist, and the elements $W_x = U(c_-,x)\,U([x,x'])\,U(x',c_+)\,U[c]^{-1}$ satisfy $|W_x - \mathbf{1}| < \frac{1}{14}$; so the small-field formula applies to both and $(\mathcal{M}U)(c) = \exp(L^{-3}\sum_{x\in B(c_-)}\log W_x)\,U[c]$. Write $a' = a_{k+1}$, $a = La'$, $\mathsf{h} = \frac{1}{2}(L-1)$. If $c \in \mathsf{U}_k$, the sites of its double block have heights $\ge z + \frac{3}{2}a - \mathsf{h}a' = z + a + \frac{1}{2}a'$, so $\mathsf{P}_c \subset \mathsf{U}_{k+1}$ and nothing changes. Let $c \in \mathsf{L}_k$ be positively oriented, with an end point y in the lowest layer of $a_k\mathbb{Z}^3$, at the height $z + \frac{1}{2}a$; heights are measured from y in units of a', so that the lowest layer of $a_{k+1}\mathbb{Z}^3$ above z is the height $-\mathsf{h}$. Negatively oriented bonds follow by inversion.

Step 1: rules for unitary matrices. (r1) $|X_1\cdots X_n - Y_1\cdots Y_n| \le \sum_i |X_i - Y_i|$ for unitary factors, and $|X^{-1} - Y^{-1}| = |X - Y|$. (r2) If $|X - \mathbf{1}|, |Y - \mathbf{1}| \le r < 1$, then $|\log X - \log Y| \le |X - Y|/(1-r)$, from the series, as $|A^n - B^n| \le nr^{n-1}|A - B|$ for $|A|, |B| \le r$. (r3) $|e^{\alpha'} - e^{\alpha}| \le |\alpha' - \alpha|$ for anti-hermitian α, α', from $e^{\alpha'} - e^{\alpha} = \int_0^1 e^{t\alpha'}(\alpha' - \alpha)e^{(1-t)\alpha}dt$. (r4) For a C^1 curve X_t of unitary matrices with $|X_t - \mathbf{1}| \le \Delta_0 < 1$, $|\frac{d}{dt}\log X_t - \dot{X}_t| \le |\dot{X}_t|\frac{\Delta_0}{1-\Delta_0}$, from the derivatives of the terms of the series.

Step 2: the means are Lipschitz. Let $\mathbf{V} = (V_i)_{i\le n}$ and $\mathbf{V}' = (V_i')_{i\le n}$ be families in G of diameter at most $\frac{1}{160}$ with $|V_i' - V_i| \le \frac{1}{100}$, and m, m' their means (Proposition YM3_I:4.3 and YM3_I:(28)), which satisfy $|V_i m^{-1} - \mathbf{1}|, |V_i' m'^{-1} - \mathbf{1}| \le \frac{1}{40}$ and $\sum_i \log(V_i m^{-1}) = 0 = \sum_i \log(V_i' m'^{-1})$. Then $|m' - m| \le 1.31 \cdot \frac{1}{n}\sum_i |V_i' - V_i|$. Indeed, put $p_i = V_i m^{-1}$, $p_i' = V_i' m^{-1}$, so $|p_i' - \mathbf{1}| \le 0.035 =: r$, and $X = m'm^{-1}$, $|X - \mathbf{1}| \le 0.06$ by (r1), $Y = \log X$, $|Y| \le 0.0619$. For $t \in [0,1]$ let $\varphi_i(t) = \log(p_i' e^{-tY}) + tY$; then $|p_i' e^{-tY} - \mathbf{1}| \le r + |Y| \le 0.0969 =: \Delta_0$, and by (r4) $|\dot{\varphi}_i| \le |Y|(\frac{\Delta_0}{1-\Delta_0} + \Delta_0) \le 0.2042\,|Y|$. As $\sum_i \varphi_i(1) = nY$ and $\sum_i \varphi_i(0) = \sum_i(\log p_i' - \log p_i)$, (r2) gives $n|Y| \le 0.2042\,n|Y| + \sum_i |V_i' - V_i|/(1-r)$, and $|m' - m| = |e^Y - \mathbf{1}| \le |Y| \le \frac{1}{0.7958\cdot 0.965}\cdot\frac{1}{n}\sum_i |V_i' - V_i|$, with $\frac{1}{0.7958\cdot 0.965} < 1.31$.

Step 3: the touched bonds of a contour. Let $x = y + a'\sum_\nu n_\nu e_\nu \in B(y)$. The path $\Gamma_\sigma(y,x)$ runs between the heights 0 and n_3; if $n_3 > -\mathsf{h}$ none of its bonds lies in L_{k+1}. If $n_3 = -\mathsf{h}$, the bonds of $\Gamma_\sigma(y,x)$ in L_{k+1} are the last vertical one and the horizontal ones which follow the vertical steps; their number $N_\sigma(x)$ is $1 + |n_1| + |n_2|$, $1 + |n_2|$ or $1 + |n_1|$, or 1, as the vertical direction stands first, second (behind 1, behind 2) or last in σ. Over the $|S|!$ members of $\mathcal{G}(y,x)$ the mean of $N_\sigma(x)$ is

$\bar N(x) = 1 + \frac12(|n_1| + |n_2|)$, in each of the cases $n_1 n_2 \neq 0$, exactly one of n_1, n_2 zero, $n_1 = n_2 = 0$; and $N_\sigma(x) \le 1 + 2\mathsf h = L$. By (r1), $|U'(\Gamma_\sigma) - U(\Gamma_\sigma)| \le N_\sigma(x)\delta \le L\delta \le \frac{1}{100}$, and Step 2 gives $|U'(y,x) - U(y,x)| \le 1.31\,\bar N(x)\,\delta$ if $n_3 = -\mathsf h$, and $U'(y,x) = U(y,x)$ otherwise. The straight path $[c_-, c_+]$ runs at heights ≥ 0, so $U'[c] = U[c]$. Moreover $\sum_{|n_1|,|n_2|\le \mathsf h} \bar N(x) = L^2 + \frac14 L(L^2-1)$, as $\sum_{|n|\le \mathsf h} |n| = \mathsf h(\mathsf h+1) = \frac14(L^2-1)$.

Step 4: the two kinds of bonds of $\mathsf L_k$. If c is horizontal, both its end points have the height 0, the contours at c_+ are the translates of those at c_-, and $[x,x']$ has L bonds at the height n_3; for $n_3 = -\mathsf h$, (r1) gives $|W'_x - W_x| \le (2\cdot 1.31\,\bar N(x) + L)\delta$, and $W'_x = W_x$ otherwise. So $\sum_x |W'_x - W_x| \le [2.62(L^2 + \frac14 L(L^2-1)) + L^3]\delta$. If c is vertical, c_+ has the height L and the sites of $B(c_+)$ the heights $\ge L - \mathsf h > -\mathsf h$; only the first bond of $[x,x']$, and only for $n_3 = -\mathsf h$, is touched, and the sum is at most $[1.31(L^2 + \frac14 L(L^2-1)) + L^2]\delta$, which is smaller.

Step 5: the average. With $\alpha = L^{-3}\sum_x \log W_x$ and α' the same for U', both anti-hermitian, (r2) at $r = \frac{1}{14}$ gives $|\alpha' - \alpha| \le L^{-3}\frac{14}{13}\sum_x |W'_x - W_x| \le \mathsf c_L\delta$, and (r3) gives $|(\mathcal M U')(c) - (\mathcal M U)(c)| = |(e^{\alpha'} - e^{\alpha})U[c]| \le \mathsf c_L\delta$. The derivative of $\frac{1}{L} - \frac{1}{4L^2}$ is $-L^{-2} + \frac12 L^{-3} < 0$, and $\mathsf c_{13} = \frac{14}{13}(1 + 2.62\cdot 0.32544\ldots) < 1.996 < 0.554\sqrt{13}$. □

Proposition 18.223 (Regular threads; the block fields at a plane are functions of those above it; the translation by e_3 is injective, and the Hamiltonian of the threads is set on the whole space)**.** Assume the hypotheses and signs of Proposition 18.221, with its law $\boldsymbol\nu$, its plane $c_0 \in \mathbb Z + \frac12$, its space $\mathcal H^{\rm thr}$ and $\mathsf A = \mathsf T_1$. Let $A_0 > 0$ and $\mathfrak p_0 \in \mathbb N$ be the constants of the threshold function $\varepsilon(\mathsf t) = A_0 \mathsf t^{-1/2}(\log \mathsf t)^{\mathfrak p_0}$, $\mathsf t > 1$. Let $\underline{\mathsf b}_g = g^{-2} + B_\sharp - L^{-1}\bar B$ and $\bar{\mathsf b}_g = L(g^{-2} + \bar B)$ be the end points of the outer interval $I^{\rm lab}(g)$ of labels, and put $\eta_d = 4A_0(L^d \underline{\mathsf b}_g)^{-1/2}(\log L^d \bar{\mathsf b}_g)^{\mathfrak p_0}$ for $d \ge 0$. For $Q \subset \mathbb R^3$ let $\mathfrak F(Q)$ be the σ-algebra on $\mathsf Z$ generated by the coordinates $\mathsf z_d(c')$ with $\mathrm{fp}(c') \subset Q$.

(a) *Regularity.* $\boldsymbol\nu$-almost surely, for every bounded set $Q \subset \mathbb R^3$ there is D such that $|\partial \mathsf z_d(p) - \mathbf 1| < \eta_d$ for all $d \ge D$ and all plaquettes p of $a_d\mathbb Z^3$ with their corners in Q. Moreover $\mathsf c_L^d \eta_d \to 0$ as $d \to \infty$, $\mathsf c_L$ the number of Lemma 18.222.

(b) *The germ.* Let $z \in \mathbb Z + \frac12$. Every coordinate $\mathsf z_d(c')$ with $\mathrm{fp}(c') \subset H^+_z$ coincides $\boldsymbol\nu$-almost surely with a function measurable for $\bigvee_{\delta>0} \mathfrak F(H^+_{z+\delta})$.

(c) *Injectivity.* $\mathsf A$ is injective on $\mathcal H^{\rm thr}$, so $\mathcal H^{>} = \mathcal H^{\rm thr}$. Hence $H^{\rm thr} = -\log \mathsf A$ is a densely defined nonnegative self-adjoint operator in $\mathcal H^{\rm thr}$ with $e^{-tH^{\rm thr}} = \mathsf T_t$ for $t \in \mathbb D_+$, $\mathsf T_t \to 1$ strongly as $t \to 0$, the horizontal translations form a strongly continuous unitary representation of $\mathbb R^2$ on $\mathcal H^{\rm thr}$ whose generators satisfy $H^{\rm thr} \ge |P^{\rm thr}_\lambda|$ there, and clauses (d), (e), (g) of Proposition 18.221 hold with $\mathcal H^{\rm thr}$ in the place of $\mathcal H^{>}$.

■

Proof. Step 1: the tail on one torus. Fix $\ell = 2L^m$ in $\mathfrak L(g)$ and write $\mathsf b = \mathsf b(S)$. By Lemma 18.212(c), this side admits $\mathsf b$, so $\mathsf b \ge \gamma^{-2}$ and $MR(\mathsf b) \le L^m$. For sufficiently large d and every $K > d$, take the top level $t = t_* = K - d$ in the calibrated setting of Section YM3_I:115. Its coupling is $x_t = L^d\mathsf b$ and exceeds the fixed calibration threshold for all such d. At zero source the weight $\mathfrak w^{\rm sf}_t$ in that section vanishes unless $\varpi(\mathcal M^t_\ell U) < 2\varepsilon(x_t)$, where ϖ is the largest plaquette distance from the identity. The upper domination in YM3_I:(2189) and the discarded mass estimate YM3_I:(2190), with the indicator of this event as the gauge-invariant test, give

$$\mu_{K,\ell,L^K\mathsf b}\big(\varpi(\mathcal M_\ell^{K-d}U) \ge 2\varepsilon(L^d\mathsf b)\big) \le R(d).$$

Here YM3_I:(2187) gives

$$R(d) = 2M^{-3}(\ell L^d)^3 \exp\Big[-c' A_0^2 (\log L^d\mathsf b)^{2\mathfrak p_0} + (C^{\Sigma}_{\rm maj} + 2 + C_{\rm low}(\mathsf b))\ell^3\Big]$$

with $c' > 0$. The sum of $R(d)$ over large d is finite because $2\mathfrak p_0 \ge 2$. Total-variation convergence in Lemma 18.212(a) gives this same tail bound for the depth-d marginal of the thread measure.

Step 2: the thread measure and the laws $\boldsymbol{\nu}$; (a). The depth-d marginal of $\mu_{\ell,\mathsf{b}}$ is $\mu_d[\ell,\mathsf{b}]$ (Definition 18.207); by the lemma of Borel and Cantelli, $\mu_{\ell,\mathsf{b}}$-almost surely $\varpi(V_d) < 2\varepsilon(L^d\mathsf{b})$ for all large d. Here $2\varepsilon(L^d\mathsf{b}) \le \frac12\eta_d < \eta_d$ for every $d \ge 0$: the label lies in $I^{\mathrm{lab}}(g)$ (Lemma 18.220), so $\underline{\mathsf{b}}_g \le \mathsf{b} \le \bar{\mathsf{b}}_g$, and $L^d\mathsf{b} \ge \mathsf{b} \ge \gamma^{-2} > 1$, so $(L^d\mathsf{b})^{-1/2} \le (L^d\underline{\mathsf{b}}_g)^{-1/2}$ and $0 < \log L^d\mathsf{b} \le \log L^d\bar{\mathsf{b}}_g$. So $\mu_{\ell,\mathsf{b}}$-almost surely $\varpi(V_d) < \eta_d$ for all large d. The covering map is injective on the window $\square$, so the Borel set $\mathcal{E} \subset Y$ of the y with $|\partial y_d(p) - \mathbf{1}| < \eta_d$ for all large d and all plaquettes p with corners in $\square$ has full $\mu[\ell,\mathsf{b}]$-measure. By Lemma 18.217(d), $(\varpi_o^{[0]})^{-1}\mathcal{E}$ has full $\boldsymbol{\nu}$-measure for every $o \in \mathbb{Z}^3$; a plaquette of side $a_d \le 1$ with corners in a bounded Q lies in one of finitely many cells $o + \square$, $o \in \mathbb{Z}^3$, as $\frac{\ell}{4} \ge \frac{L}{2} > 1$. Finally $\mathsf{c}_L^d\eta_d = 4A_0\underline{\mathsf{b}}_g^{-1/2}(\mathsf{c}_L/\sqrt{L})^d(\log L^d\bar{\mathsf{b}}_g)^{\mathfrak{p}_0} \to 0$, as $\mathsf{c}_L/\sqrt{L} < 0.554$ by Lemma 18.222 and the last factor is a polynomial in d.

Step 3: the substitute and the gauge, for (b). Let c' be a bond of $a_d\mathbb{Z}^3$ with $\mathrm{fp}(c') \subset H_z^+$; if both its end points have heights $\ge z + \frac32 a_d$, its footprint lies in $H^+_{z+a_d}$ and there is nothing to show; otherwise $c' \in \mathsf{L}_d$. For $m \ge d$ let $\mathsf{B}^{(m)}$ be the set of the bonds of $a_m\mathbb{Z}^3$ with both end points in $\tilde{B}^{m-d}(c'_-) \cup \tilde{B}^{m-d}(c'_+)$; on Thr, $\mathsf{z}_d(c') = (\mathcal{M}^{m-d}\mathsf{z}_m)(c')$ reads z_m on $\mathsf{B}^{(m)}$ (Lemma 18.211(b)), and $\mathsf{P}_b \subset \mathsf{B}^{(m+1)}$ for $b \in \mathsf{B}^{(m)}$. Let $\mathsf{L}_m, \mathsf{U}_m$ be the sets of Lemma 18.222 at the plane z, which lies in $a_m(\mathbb{Z} + \frac12)$ for all m. For $s > d$ let $\tilde{V}_s$ on $\mathsf{B}^{(s)}$ be z_s on U_s, $\mathbf{1}$ on the vertical bonds of L_s, and $\mathsf{z}_s(b + a_se_3)$ on a horizontal $b \in \mathsf{L}_s$ (the bond $b + a_se_3$ lies in $\mathsf{B}^{(s)} \cap \mathsf{U}_s$, the blocks having $L^{s-d} \ge 3$ layers), and $g_{s,k} = \mathcal{M}^{s-k}\tilde{V}_s$ on $\mathsf{B}^{(k)}$, $d \le k \le s$: a Borel function of z_s on U_s, so $\mathfrak{F}(H^+_{z+a_s})$-measurable. Let u be the gauge transformation of $a_s\mathbb{Z}^3$ with $u(x) = \mathsf{z}_s(\langle x, x + a_se_3\rangle)^{-1}$ at the sites x of $\mathsf{B}^{(s)}$ at the height $z + \frac12 a_s$ and $u = \mathbf{1}$ elsewhere. Then $\mathsf{z}_s^u = \mathbf{1}$ on the vertical bonds of L_s, $\mathsf{z}_s^u = \mathsf{z}_s$ on U_s, and on a horizontal $b = \langle x, x'\rangle \in \mathsf{L}_s$, $\mathsf{z}_s^u(b) = \mathsf{h}_b\,\mathsf{z}_s(b + a_se_3)$ with the holonomy h_b around the vertical plaquette p over b, a conjugate of $\partial\mathsf{z}_s(p)^{\pm1}$; so $|\mathsf{z}_s^u(b) - \tilde{V}_s(b)| = |\partial\mathsf{z}_s(p) - \mathbf{1}|$. No site at the height $z + \frac12 a_s$ lies in $a_k\mathbb{Z}^3$, $k < s$: with $z = a_k(n + \frac12)$ this height is a_sN with $2N = L^{s-k}(2n+1) + 1 \equiv 1$ modulo L^{s-k}. By the gauge covariance (Lemma 18.211(b)), $\mathcal{M}^{s-k}(\mathsf{z}_s^u) = \mathcal{M}^{s-k}\mathsf{z}_s$ on $\mathsf{B}^{(k)}$ for $k < s$. The plaquettes of $\tilde{V}_s$ are those of z_s at four bonds of U_s, equal to $\mathbf{1}$ (up to conjugation) at the vertical plaquettes of the lowest layer, and those of z_s moved up by a_se_3 at the horizontal plaquettes of the lowest layer.

Step 4: the descent. Let $k_0 > d$ with $\eta_k \le \frac12\omega_M$ for $k \ge k_0$, and let E_{k_0} be the event that $\mathsf{z}_k = \mathcal{M}\mathsf{z}_{k+1}$ on $\mathsf{B}^{(k)}$ for all $k \ge d$ and $|\partial\mathsf{z}_k(p) - \mathbf{1}| < \eta_k$ for all $k \ge k_0$ and all plaquettes p with corners in $\mathrm{fp}(c')$; by $\boldsymbol{\nu}(\mathsf{Thr}) = 1$ and (a), $\boldsymbol{\nu}(E_{k_0}) \to 1$. Let s_0 be such that $\mathsf{c}_L^{s-k_0}\eta_s \le \frac18\omega_M$ for $s \ge s_0$; as $\mathsf{c}_L \ge 1$, then $\delta_k := \mathsf{c}_L^{s-k}\eta_s \le \frac18\omega_M$ for $k_0 \le k \le s$. On E_{k_0}, for $s \ge s_0$, put $W_s = \mathsf{z}_s^u$, $W_k = \mathsf{z}_k$ for $k < s$ and $W'_k = g_{s,k}$. We show for $k = s, s-1, \dots, k_0$: $W'_k = W_k$ on $\mathsf{B}^{(k)} \cap \mathsf{U}_k$ and $|W'_k - W_k| \le \delta_k$ on $\mathsf{B}^{(k)} \cap \mathsf{L}_k$. At $k = s$ this is Step 3. From $k+1$ to k: for $b \in \mathsf{B}^{(k)}$, W_{k+1} satisfies the small-field condition on the double block of b ($\eta_{k+1} \le \frac12\omega_M$, plaquettes being conjugated by the gauge transformation); so does W'_{k+1}, by Step 3 at $k+1 = s$ and by $|\partial W'_{k+1}(p) - \mathbf{1}| < \frac12\omega_M + 4\delta_{k+1} \le \omega_M$ (r1) otherwise; $L\delta_{k+1} \le \frac{L\omega_M}{8} \le \frac{1}{100}$; $\mathcal{M}W_{k+1} = W_k$ on $\mathsf{B}^{(k)}$ by the event and Step 3, and $\mathcal{M}W'_{k+1} = W'_k$ by definition. Lemma 18.222 gives the assertion at k. At $k = k_0$: $|g_{s,k_0} - \mathsf{z}_{k_0}| \le \mathsf{c}_L^{s-k_0}\eta_s \to 0$ as $s \to \infty$, on $\mathsf{B}^{(k_0)}$.

Step 5: measurability; (b). The set C_{k_0} on which $(g_{s,k_0})_s$ converges is measurable for $\mathfrak{G}_{0+} = \bigvee_s \mathfrak{F}(H^+_{z+a_s})$; let Z_{k_0} be the limit there and $\mathbf{1}$ elsewhere, and $Y_{k_0} = (\mathcal{M}^{k_0-d}Z_{k_0})(c')$, a $\mathfrak{G}_{0+}$-measurable function, $\mathcal{M}$ being Borel. On E_{k_0}, $Z_{k_0} = \mathsf{z}_{k_0}$ on $\mathsf{B}^{(k_0)}$ and $Y_{k_0} = \mathsf{z}_d(c')$; as the E_{k_0} increase to a set of full measure, $Y_{k_0} \to \mathsf{z}_d(c')$ almost surely. No continuity of $\mathcal{M}$ off the small fields is used.

Step 6: (c). Let $\Phi \in \mathcal{A}^+$ and $\mathfrak{G}_s = \mathfrak{F}(H^+_{c_0+a_s})$, an increasing family with $\bigvee_s \mathfrak{G}_s = \bigvee_{\delta>0}\mathfrak{F}(H^+_{c_0+\delta})$. By (b) at $z = c_0$, Φ coincides almost surely with a $\bigvee_s \mathfrak{G}_s$-measurable function, and Lévy's upward theorem gives $\Phi_s = \mathbb{E}[\Phi|\mathfrak{G}_s] \to \Phi$ in $L^2(\boldsymbol{\nu})$. Each Φ_s has a bounded $\mathfrak{G}_s$-measurable representative,

whose class lies in $\mathcal{H}^{>}$, being an L^2-limit of local functions with footprints in $H^+_{c_0+a_s}$ (Proposition 18.221(c)); and $\|[\Phi_s]-[\Phi]\| \le \|\Phi_s-\Phi\|_2$. So $\mathcal{H}^{>}=\mathcal{H}^{\mathrm{thr}}$, and $\ker \mathsf{A}=\{0\}$ by the splitting of Proposition 18.221(c); the rest is that Proposition on $\mathcal{H}^{>}=\mathcal{H}^{\mathrm{thr}}$. □

Proposition 18.224 (On one lattice: the transfer operator and the plaquette actions act irreducibly on the flux-free sector)**.** Let $a>0$, $\beta>0$ and $n_1,n_2\ge 3$. The periodic layer has site set $\mathsf{X}_0=\mathbb{Z}_{n_1}\times\mathbb{Z}_{n_2}$ in lattice units and positive bond set S_0 in the two coordinate directions. Let $G=\mathrm{SU}(2)$ and $\mathcal{H}=L^2(G^{\mathsf{S}_0})$ with product Haar probability measure. The gauges $w\in G^{\mathsf{X}_0}$ act by $(w\cdot u)(b)=w(b_-)u(b)w(b_+)^{-1}$ and $R_w f(u)=f(w^{-1}\cdot u)$. Let $P=\int R_w\,dw$. Put $\mathsf{a}(g)=1-\frac12\operatorname{tr} g$, $a_p(u)=\mathsf{a}(u(\partial p))$, $V=\sum_p a_p$ and $\mathsf{E}=M_{e^{-\beta V}}$, where M_f denotes multiplication by f. The operator K has kernel

$$k_{\mathsf{S}_0}(u,u')=\prod_{b\in\mathsf{S}_0} k_\beta(u(b)u'(b)^{-1}),\qquad k_\beta(g)=e^{-\beta\mathsf{a}(g)}.$$

Let $K^{[b]}$ have kernel $k_{\mathsf{S}_0}(u,u')\mathsf{a}(u(b)u'(b)^{-1})$. The Peter–Weyl space $\mathcal{M}_{\underline{l}}$ is the tensor product of the matrix coefficient spaces of the spin l_b representations, one per bond. The finite sums of vectors in these spaces form the trigonometric polynomials $\mathcal{T}(G^{\mathsf{S}_0})$. Write $P_{\underline{l}}$ for their individual spin projections and $P^{(b)}_l=\sum_{\underline{l}:l_b=l}P_{\underline{l}}$. Let

$$\mathsf{Q}=\mathsf{E}^{1/2}KP\,\mathsf{E}^{1/2},\qquad \mathsf{Q}_b=\mathsf{E}^{1/2}K^{[b]}P\,\mathsf{E}^{1/2},$$

the transfer operator in its symmetric form and the transfer operator with one time-like plaquette over b inserted: with $C=\mathsf{E}^{1/2}PK^{1/2}$, $\mathsf{Q}=CC^*$, while the transfer operator in the opposite factorization is $\mathsf{M}=C^*C$; the kernel of Q is $e^{-\beta V(u)/2}\int k_{\mathsf{S}_0}(u,w\cdot u')\,dw\,e^{-\beta V(u')/2}$, the weight of the plaquettes between two layers integrated over the vertical variables. For $\mu\in\{1,2\}$ let ζ_μ change the sign of u on the bonds $\langle x,x+e_\mu\rangle$ with $x_\mu=n_\mu-1$, $Z_\mu\psi=\psi\circ\zeta_\mu$, and for $e\in\{0,1\}^2$ let $\mathcal{H}^e=\{\psi\in P\mathcal{H}: Z_\mu\psi=(-1)^{e_\mu}\psi,\ \mu=1,2\}$, the flux sectors; $\mathcal{H}^0=\mathcal{H}^{(0,0)}$ is the flux-free sector. For $\mathcal{K}\in\{\mathcal{H}^0,P\mathcal{H}\}$ let $\mathfrak{W}_{\mathcal{K}}$ be the von Neumann algebra on $\mathcal{K}$ generated by the restrictions of Q, of the M_{a_p} and of the Q_b.

(a) $P\mathcal{H}=\bigoplus_e\mathcal{H}^e$, and every generator maps every $\mathcal{H}^e$ into itself. $\mathsf{Q}\ge 0$ is of trace class and injective on $P\mathcal{H}$; $\|\mathsf{Q}\|$ is a simple eigenvalue, with an eigenvector Ω that is continuous and positive everywhere, and $\Omega\in\mathcal{H}^0$.

(b) $P^{(b)}_l|_{\mathcal{K}}\in\mathfrak{W}_{\mathcal{K}}$ for every $b\in\mathsf{S}_0$ and every $l\in\frac12\mathbb{N}_0$.

(c) $\mathfrak{W}_{\mathcal{H}^0}$ contains the multiplication by every bounded Borel function on G^{S_0} that is invariant under the gauge transformations and under ζ_1,ζ_2, and the projection $|1\rangle\langle 1|$; and $\mathfrak{W}_{\mathcal{H}^0}=\mathcal{B}(\mathcal{H}^0)$: the algebra is irreducible. On all the sectors, $\mathfrak{W}_{P\mathcal{H}}=\bigoplus_e\mathcal{B}(\mathcal{H}^e)$, and its commutant is spanned by the four projections onto the $\mathcal{H}^e$.

(d) The vectors $B_1B_2\cdots B_m\Omega$, $m\ge 0$, $B_i\in\{\mathsf{Q}\}\cup\{M_{a_p}\}\cup\{\mathsf{Q}_b\}$, are total in $\mathcal{H}^0$; so are already those in which no M_{a_p} stands twice between two consecutive factors of the kinds Q, Q_b, before the first or behind the last of them.

■

Proof. Step 1: the spectra of K and $K^{[b]}$. K acts on $\mathcal{M}_{\underline{l}}$ by $\lambda_{\underline{l}}=\prod_b\lambda_{l_b}(\beta)$, where $\lambda_l(\beta)=(2l+1)^{-1}\int k_\beta\chi_l$ by Schur orthogonality. By the Weyl integration formula and $\chi_l(\mathrm{diag}(e^{\mathrm{i}\omega},e^{-\mathrm{i}\omega}))=\sin((2l+1)\omega)/\sin\omega$, $\lambda_l(\beta)=e^{-\beta}\frac{2}{\beta}I_{2l+1}(\beta)$ with the modified Bessel functions I_n. Their series shows $\lambda_l(\beta)>0$ and factorial decay in l at fixed $\beta>0$. Hence $\sum_l(2l+1)^2\lambda_l(\beta)<\infty$ and $\lambda_{\underline{l}}\to 0$ as $\max_b l_b\to\infty$. The operator with the kernel $f(gg'^{-1})$, f a continuous class function on G, commutes with the left and right translations and is the number $\frac{1}{2l+1}\int f\chi_l$ on $\mathcal{M}_l$; at $f=k_\beta\mathsf{a}=-\partial_\beta k_\beta$ this number is $\lambda_l r_l$ with $r_l(\beta)=-\partial_\beta\log\lambda_l(\beta)=1+\frac1\beta-\frac{I'_{2l+1}(\beta)}{I_{2l+1}(\beta)}$. So $K^{[b]}$ is the number $\lambda_{\underline{l}}r_{l_b}$ on $\mathcal{M}_{\underline{l}}$. The map $l\mapsto r_l(\beta)$ is strictly decreasing: $\mathsf{u}_n=I'_n/I_n$ satisfies $\mathsf{u}'_n=1+\frac{n^2}{x^2}-\frac{\mathsf{u}_n}{x}-\mathsf{u}_n^2$ (from

$x^2 I_n'' + x I_n' = (x^2+n^2) I_n$) and $\mathsf{u}_n = \frac{n}{x} + O(x)$ at 0; for $m > n$ the difference $\mathsf{w} = \mathsf{u}_m - \mathsf{u}_n$ is positive near 0 and $\mathsf{w}' = \frac{m^2-n^2}{x^2} - \mathsf{w}(\frac{1}{x} + \mathsf{u}_m + \mathsf{u}_n)$, which is positive at a first zero of w, so $\mathsf{w} > 0$ on $(0,\infty)$.

Step 2: clause (a). P commutes with K since single-bond convolution by a class function commutes with left and right multiplication. The kernel of $K^{[b]}$ is the product over $b' \in \mathsf{S}_0$ of the class functions k_β, for $b' \neq b$, and $k_\beta \mathsf{a}$, for $b' = b$, of $u(b')u'(b')^{-1}$; so by Step 1 $K^{[b]}$ commutes with the left and the right translations in each factor of G^{S_0}, hence with every gauge rotation R_w, which acts through such translations, and with their mean P. The flips Z_μ also act by left multiplication by central signs on selected bonds, so they commute with K and $K^{[b]}$. Further, P commutes with E (V is gauge invariant) and with Z_μ, and $V \circ \zeta_\mu = V$, a plaquette containing two or none of the flipped bonds; the M_{a_p} commute with P and Z_μ; so the generators keep each $\mathcal{H}^e$. $K^{1/2}$ is Hilbert–Schmidt by the summability in Step 1 and the finite product over bonds, so C is, and $\mathsf{Q} = CC^* \geq 0$ is of trace class. If $\psi \in P\mathcal{H}$ and $\mathsf{Q}\psi = 0$, then $K^{1/2}\mathsf{E}^{1/2}\psi = 0$, so $\psi = 0$, K being injective and $\mathsf{E}^{1/2} > 0$. The kernel of Q is continuous and positive, so the top eigenvalue $\|\mathsf{Q}\|$ has an eigenvector $|\psi|$ whenever it has ψ ($\langle|\psi|, \mathsf{Q}|\psi|\rangle \geq \langle\psi, \mathsf{Q}\psi\rangle$), which is continuous and positive; equality forces every real eigenvector to have one sign, two orthogonal such vectors cannot exist, and the eigenvalue is simple; $R_w\Omega$ and $Z_\mu\Omega$ are positive normalised eigenvectors, hence Ω.

Step 3: clause (b). $V = \sum_p a_p$, so $e^{\pm\beta V/2}|_{\mathcal{K}}$ are norm limits of polynomials in the M_{a_p}, and $K|_{\mathcal{K}} = e^{\beta V/2}\mathsf{Q}\, e^{\beta V/2}$, $K^{[b]}|_{\mathcal{K}} = e^{\beta V/2}\mathsf{Q}_b e^{\beta V/2}$ lie in $\mathfrak{W}_{\mathcal{K}}$ ($P = 1$ on $\mathcal{K}$). Every eigenvalue $\mathsf{c} > 0$ of K is isolated and its projection $\mathsf{p}_\mathsf{c} = \sum_{\lambda_{\underline{l}}=\mathsf{c}} P_{\underline{l}}$, of finite rank, is a norm limit of polynomials in K; $\mathsf{p}_\mathsf{c} K^{[b]}$ is self-adjoint of finite rank, and the projections $\sum_{\lambda_{\underline{l}}=\mathsf{c},\ r_{l_b}=r} P_{\underline{l}}$, which are the products of p_c with the spectral projections of $\mathsf{p}_\mathsf{c} K^{[b]}$ at the points $\mathsf{c}r$, $r \in \mathbb{R}$, and so polynomial expressions in p_c and $\mathsf{p}_\mathsf{c} K^{[b]}$, lie in $\mathfrak{W}_{\mathcal{K}}$; their strong sum over c at $r = r_l$ is $P_l^{(b)}$, by the injectivity of $l \mapsto r_l$.

Step 4: Casimir and carré du champ; merging. For $b \in \mathsf{S}_0$ let L_b^α, $\alpha = 1,2,3$, be the derivatives in $u(b) \mapsto u(b)e^{tT^\alpha}$, $T^\alpha = \frac{\mathrm{i}}{2}\sigma^\alpha$, $\Delta_b = \sum_\alpha (L_b^\alpha)^2$ and $\Gamma_b(f,g) = \sum_\alpha (L_b^\alpha f)(L_b^\alpha g)$. Then Δ_b is the number $-l_b(l_b+1)$ on $\mathcal{M}_{\underline{l}}$, and $[M_f, [\Delta_b, M_g]] = -2M_{\Gamma_b(f,g)}$ on $\mathcal{T}(G^{\mathsf{S}_0})$. For a closed path γ put $W_\gamma(u) = \operatorname{tr} u(\gamma)$. If c is b or its reverse, from x to y, and A, B are paths from y to x which avoid b, then, by $\sum_\alpha \operatorname{tr}(T^\alpha M)\operatorname{tr}(T^\alpha N) = -\frac{1}{2}\operatorname{tr}(MN) + \frac{1}{4}\operatorname{tr} M \operatorname{tr} N$ and $\operatorname{tr}(xy) + \operatorname{tr}(xy^{-1}) = \operatorname{tr} x \operatorname{tr} y$ in $\mathrm{SL}_2(\mathbb{C})$,

$$W_{cAcB} = \tfrac{1}{2} W_{cA} W_{cB} - 2\Gamma_b(W_{cA}, W_{cB}), \qquad W_{AB^{-1}} = \tfrac{1}{2} W_{cA} W_{cB} + 2\Gamma_b(W_{cA}, W_{cB}).$$

The loop system. Let $\mathcal{L}_{\mathcal{K}}$ be the set of the closed paths γ of the layer with even winding numbers and $M_{W_\gamma}|_{\mathcal{K}} \in \mathfrak{W}_{\mathcal{K}}$. It holds every ∂p ($W_{\partial p} = 2 - 2a_p$); with a path it holds every equivalent one (cyclic rotations, reversal, insertion or deletion of a pair $c\bar{c}$); with $\gamma_1, \dots, \gamma_r$ every γ for which W_γ is a real polynomial in $W_{\gamma_1}, \dots, W_{\gamma_r}$ (functions even under ζ_μ stay even); and with cA, cB also $cAcB$ and AB^{-1} when A, B avoid the bond of c: put $\Delta^{(n)} = -\sum_{l \leq n} l(l+1) P_l^{(b)} \in \mathfrak{W}_{\mathcal{K}}$ (Step 3) and $\mathsf{D}_n = [M_f, [\Delta^{(n)}, M_g]] \in \mathfrak{W}_{\mathcal{K}}$, $f = W_{cA}$, $g = W_{cB}$; on a trigonometric polynomial $\psi \in \mathcal{K}$, $\mathsf{D}_n\psi = -2\Gamma_b(f,g)\psi$ for large n, and $Y \in \mathfrak{W}_{\mathcal{K}}'$ preserves the trigonometric polynomials in $\mathcal{K}$ because it commutes with every $P_l^{(b)}|_{\mathcal{K}}$. Applying the same eventual equality to ψ and $Y\psi$ in $Y\mathsf{D}_n\psi = \mathsf{D}_n Y\psi$ shows that Y commutes with $M_{\Gamma_b(f,g)}$ on a dense set, $M_{\Gamma_b(f,g)} \in \mathfrak{W}_{\mathcal{K}}'' = \mathfrak{W}_{\mathcal{K}}$, and the display gives the two new loops, whose windings are the sum and the difference of those of cA, cB.

Step 5: grids, and loops that wrap. A connected subgraph D is full if all its closed paths lie in $\mathcal{L}_{\mathcal{K}}$. (i) *Ears.* If D is full and 2-connected, $\delta_0 c \in \mathcal{L}_{\mathcal{K}}$ with a simple path δ_0 in D from x to $y \neq x$ and a simple path c from y to x outside D (no bond and no inner site in D), then every closed path of $D \cup c$ which avoids some bond lies in $\mathcal{L}_{\mathcal{K}}$: the merging of Step 4 at the bonds that a path from x to y passes once spreads $\delta c \in \mathcal{L}_{\mathcal{K}}$ to every path δ in D from x to y that avoids some bond of D (if $\delta c \in \mathcal{L}_{\mathcal{K}}$ and δ passes a bond λ once, the merging at λ of δc and of the closed path $\delta(\delta^*)^{-1}$ of D gives $\delta^* c \in \mathcal{L}_{\mathcal{K}}$ for every path δ^* in D from x to y that avoids λ; this is done at the bonds of δ_0, and at a bond λ off δ_0 with a simple path from x to y through λ, which exists because in

a 2-connected graph any two edges lie on a common cycle (add an extra edge from x to y and apply the vertex form of Menger's theorem to the midpoints of that edge and λ) and which, being simple and other than δ_0, avoids a bond of δ_0 and so has been reached), and a closed path that avoids a bond λ of D is, through a fixed path δ_λ in $D \setminus \lambda$ and $t = \delta_\lambda c$, a word in $t^{\pm 1}$ and closed paths of $D \setminus \lambda$, whose trace is a polynomial in W_α and $W_{t\alpha}$. Indeed, replace t^{-1} by $(\operatorname{tr} t)\mathbf{1} - t$. If a cyclic word has at least two occurrences of t, write it as $tAtB$ with A containing none. Then $\operatorname{tr}(tAtB) = \operatorname{tr}(tA)\operatorname{tr}(tB) - \operatorname{tr}(AB^{-1})$ by the two-factor trace identity, which reduces the number of occurrences. The resulting words with at most one t have the stated form. If $D \cup c$ has at least four independent cycles, every closed path is a word in generators y_i of which any three avoid a fourth independent bond, and all traces of words are polynomials in the traces of words of length at most three. For the latter assertion write $U_i = s_i\mathbf{1} + iv_i \cdot \varsigma$ for the fundamental holonomies. The Pauli multiplication rule expresses the trace of a word using contractions by the Kronecker symbol and the alternating three-index symbol. A pair of alternating symbols reduces to a sum of products of Kronecker symbols. Therefore the trace is a polynomial in s_i, $v_i \cdot v_j$ and $\det(v_i, v_j, v_k)$, which are determined by traces of words of length at most three. Inverses replace v_i by $-v_i$. In the graph, choose a spanning tree so that a word in at most three generators avoids a fourth non-tree bond. The available traces then give fullness. (ii) *Grids are full*: adding the plaquettes of a grid row by row adds an ear each time; it remains to initialize the induction at the first three plaquettes. One plaquette has one generator and all its traces are polynomials in its trace. Two plaquettes sharing a side have generators given by their oriented boundaries based at an endpoint of that side. Orient them oppositely on their common side. The merged boundary gives the trace of their product, and together with the individual traces this gives all traces for two generators.

For the third plaquette, choose the initial three as a line or an L, so that the middle plaquette shares distinct sides with the two outer ones. Choose a basepoint on an unused bond of the middle boundary and orient that boundary in the positive direction. Let B be its holonomy and let AB and CB be the holonomies obtained by replacing, respectively, the first and the second shared side by the remaining three sides of the corresponding outer plaquette, in the order in which the middle boundary visits the sides. The two holonomies A and C are the transported outer plaquette holonomies. The three traces and the traces of AB and CB are available by the merging rule. Merging AB and CB with their common unused bond gives $\operatorname{tr}(AB(CB)^{-1}) = \operatorname{tr}(AC^{-1})$, so the two-factor trace identity gives $\operatorname{tr}(AC)$. Both replacements together give the boundary with holonomy ACB and its trace, by merging either enlarged boundary with the remaining plaquette. To see the order, if the middle path is Ps_1Qs_2R, replacement of s_j by s'_j inserts $Ps'_1s_1^{-1}P^{-1}$ and $(Ps_1Q)s'_2s_2^{-1}(Ps_1Q)^{-1}$ in that order before B. These are A and C. Thus the single, pair and triple traces of the three generators are known. The Pauli reduction also covers this rank, and (i) applies at each later ear. So the grid $D^\flat$ of the sites (i,k), $0 \le i \le n_1 - 1$, $0 \le k \le n_2 - 1$, which holds all bonds of the layer but $h_k = \langle (n_1 - 1, k), (0, k) \rangle$ and $v_i = \langle (i, n_2 - 1), (i, 0) \rangle$, is full. (iii) *Wrapping loops*: merging twice, at $r'_k = \langle (0,k), (0,k+1) \rangle$ and at $r_k = \langle (n_1-1,k), (n_1-1,k+1) \rangle$, for $0 \le k < n_2 - 1$, starting from the plaquette with the base point $(n_1 - 1, k)$, gives $h_k\, \alpha\, h_{k+1}^{-1}\, \beta' \in \mathcal{L}_\mathcal{K}$ for all paths α of $D^\flat$ from $(0,k)$ to $(0,k+1)$ avoiding r_k, r'_k and β' from $(n_1-1,k+1)$ to (n_1-1,k) avoiding r_k; the same in the other direction. Let $\mathcal{L}_*$ be the set of these paths and of the closed paths of $D^\flat$.

Step 6: separation of orbits. For two adjacent plaquettes p, p' of $D^\flat$ based at a common corner z, set $\mathsf{h}_{p,p'} = 2 - W_{[\partial_z p, \partial_z p']} \ge 0$ and $\mathsf{h} = \prod_{p,p'} \mathsf{h}_{p,p'}$. The zero set of each factor is Haar null. Conditional on all bonds except one exclusive bond of each plaquette, the two holonomies are independent Haar variables. A noncentral SU(2) matrix has a circle centralizer, which is Haar null, and the two central elements also have Haar measure zero.

Suppose $\mathsf{h}(u) > 0$ and u, u' have the same traces for $\mathcal{L}_*$. Based holonomies in the grid span $M_2(\mathbb{C})$. In fact, for two noncommuting holonomies the traceless vectors in their Pauli coordinates are independent and their cross product provides the third direction through their product. The

equality of traces also gives equality of all Gram matrices $\operatorname{tr}(u(\gamma_i)^* u(\gamma_j))$. Thus $u(\gamma) \mapsto u'(\gamma)$ extends to a well-defined unital $*$-homomorphism of $M_2(\mathbb{C})$, because concatenation and reversal give products and adjoints. It is inner, and after its conjugation and tree gauges the two configurations agree on all grid bonds.

For fixed k, the grid with r_k, r'_k removed is connected. All horizontal edges are intact and an interior column of vertical edges is intact. It also contains a row of adjacent plaquettes at a different height, since $n_1, n_2 \geq 3$. At $\mathsf{h}(u) > 0$, their transported holonomies from any fixed site in that graph span $M_2(\mathbb{C})$ together with the identity and their product. Appending one fixed connecting path shows that the values of the paths α in Step 5(iii) span $M_2(\mathbb{C})$, and likewise for β', whose allowed paths include those in this same graph. Therefore equality of the wrapping-loop traces gives $\operatorname{tr}(XM_1YM_2) = \operatorname{tr}(X'M_1Y'M_2)$ for every $M_1, M_2 \in M_2(\mathbb{C})$, where X, Y are the values of h_k, h_{k+1}^{-1} and the primes correspond to u'. The nondegenerate trace pairing gives $XM_1Y = X'M_1Y'$. Taking $M_1 = \mathbf{1}$ then shows that $X^{-1}X'$ commutes with every M_1, so $X' = sX$ and $Y' = s^{-1}Y$, with $s = \pm 1$ by the determinant. The comparisons of consecutive k give one sign for all the horizontal wrapping bonds. The vertical comparisons give one sign for the vertical wrapping bonds. These signs are precisely the flips ζ_1, ζ_2. On the exceptional set, for example, commuting constant diagonal holonomies can have all contractible traces equal to 2 while differing in their wrapping holonomies.

Step 7: clause (c). By Step 5, $M_f|_{\mathcal{K}} \in \mathfrak{W}_{\mathcal{K}}$ for f in the real algebra $\mathcal{A}$ generated by 1 and the W_γ, $\gamma \in \mathcal{L}_*$, and for f in its uniform closure; $\mathsf{h} \in \mathcal{A}$. On the compact orbit space Ξ of the group generated by the gauge transformations and ζ_1, ζ_2, Stone–Weierstrass makes the closure of $\mathcal{A}$ the set of the continuous functions constant on the classes of points that $\mathcal{A}$ does not part; for continuous invariant F and continuous $\phi \geq 0$ vanishing near 0, $F\phi(\mathsf{h})$ is such a function, by Step 6, and $M_{F\phi_n(\mathsf{h})} \to M_F$ strongly as $\phi_n \uparrow \mathbf{1}_{(0,\infty)}$, $\{\mathsf{h} = 0\}$ being null. So $M_F|_{\mathcal{K}} \in \mathfrak{W}_{\mathcal{K}}$ for continuous, hence for bounded Borel, invariant F. $P_{\underline{0}} = \prod_b P_0^{(b)} = |1\rangle\langle 1| \in \mathfrak{W}_{\mathcal{K}}$ by Step 3. For $Y \in \mathfrak{W}'_{\mathcal{H}^0}$: $Y1 = YP_{\underline{0}}1 = \mathsf{c}1$ and $YF = YM_F1 = \mathsf{c}F$ for the invariant F, which are dense in $\mathcal{H}^0$; so $Y = \mathsf{c}$. On a sector $e \neq 0$ the simple closed path γ_e with the windings e and the spins $\frac{1}{2}$ on its bonds give $P_{\underline{e}} P\mathcal{H} = \mathbb{C}W_{\gamma_e}$ (Schur), a projection in $\mathfrak{W}_{P\mathcal{H}}$, and $W_{\gamma_e} \neq 0$ almost everywhere makes $\mathfrak{W}_{P\mathcal{H}} W_{\gamma_e}$ dense in $\mathcal{H}^e$; the same argument gives $Y = \mathsf{c}_e$ on $\mathcal{H}^e$ for Y in the commutant.

Step 8: clause (d). The polynomials without constant term in the self-adjoint generators form a non-degenerate $*$-algebra (Q has dense range in $\mathcal{H}^0$); the closure of its orbit of Ω together with $\mathbb{C}\Omega$ is invariant, so its projection lies in $\mathfrak{W}'_{\mathcal{H}^0} = \mathbb{C}$ and is 1. For the second form, the closed span $\mathcal{K}_1$ of the named vectors is invariant under Q, the Q_b and the $M_{a_p}\mathsf{Q}$; as $\mathsf{Q}|_{\mathcal{K}_1}$ is self-adjoint and injective, $\mathsf{Q}\mathcal{K}_1$ is dense in $\mathcal{K}_1$, and the bounded M_{a_p} keep $\mathcal{K}_1$; so $\mathcal{K}_1$ reduces $\mathfrak{W}_{\mathcal{H}^0}$ and contains Ω. □

Definition 18.225 (Two lists of axioms and one relation)**.** Let n, m run through $\mathbb{Z}_+ = \{0, 1, 2, \dots\}$. In this definition a point of $\mathbb{R}^3$ is written $x = (\vec{x}, x^0)$, with $\vec{x} = (x_1, x_2)$ and $x^0 = x_3$, the Euclidean time; on Minkowski space write $a \cdot p = a^0 p^0 - \vec{a} \cdot \vec{p}$, $e_0 = (1, 0, 0)$ and $V_+ = \{(t, \vec{x}) : t > |\vec{x}|\}$. The group $\mathrm{O}(1, 2)$ preserves this bilinear form, and $\mathcal{L}$ is its identity component.

(1) *List OS.* (N1) $\mathscr{S}(\mathbb{R}^{3n})$ is the space of the smooth f with $\sup_x |x^\beta D^\alpha f(x)| < \infty$ for all multi-indices, with the norms $|f|_m = \sup\{(1 + |x|^2)^{m/2} |D^\alpha f(x)| : x \in \mathbb{R}^{3n}, |\alpha| \leq m\}$ and the topology they define. (N2) ${}^0\mathscr{S}(\mathbb{R}^{3n})$ is the space of the $f \in \mathscr{S}(\mathbb{R}^{3n})$ which vanish with all their derivatives on every hyperplane $x_i - x_j = 0$, $i < j$, and ${}^0\mathscr{S}'$ its dual. (N3) $\mathscr{S}_+(\mathbb{R}^{3n})$ is the space of the $f \in \mathscr{S}(\mathbb{R}^{3n})$ with $D^\alpha f(x) = 0$ for all α unless $0 < x_1^0 < \cdots < x_n^0$, and $\underline{\mathscr{S}}_+$ the space of the finite sequences $\underline{f} = (f_0, f_1, \dots)$, $f_0 \in \mathbb{C}$, $f_n \in \mathscr{S}_+(\mathbb{R}^{3n})$. (N4) $f^*(x_1, \dots, x_n) = \overline{f(x_n, \dots, x_1)}$, the order reversed; $\vartheta(\vec{x}, x^0) = (\vec{x}, -x^0)$; $(\Theta f)(x_1, \dots, x_n) = f(\vartheta x_1, \dots, \vartheta x_n)$; $(f \times g)(x_1, \dots, x_{n+m}) = f(x_1, \dots, x_n)\, g(x_{n+1}, \dots, x_{n+m})$; $f_{(a,R)}(x_1, \dots, x_n) = f(Rx_1 + a, \dots, Rx_n + a)$; $f^\pi(x_1, \dots, x_n) = f(x_{\pi(1)}, \dots, x_{\pi(n)})$. A sequence $(\mathfrak{S}_n)_{n \geq 0}$ satisfies: (OS-E0) if $\mathfrak{S}_0 = 1$, $\mathfrak{S}_n \in {}^0\mathscr{S}'(\mathbb{R}^{3n})$ and $\overline{\mathfrak{S}_n(f)} = \mathfrak{S}_n(\Theta f^*)$; (OS-E0′) if $\mathfrak{S}_0 = 1$, $\mathfrak{S}_n \in {}^0\mathscr{S}'(\mathbb{R}^{3n})$, and there are $s \in \mathbb{Z}_+$ and numbers $\sigma_n \leq \alpha(n!)^\beta$ with constants α, β such that $|\mathfrak{S}_n(f)| \leq \sigma_n |f|_{ns}$

for all $n \geq 1$; (OS-E1) if $\mathfrak{S}_n(f) = \mathfrak{S}_n(f_{(a,R)})$ for all $R \in \mathrm{SO}(3)$, $a \in \mathbb{R}^3$; (OS-E2) if $\sum_{n,m} \mathfrak{S}_{n+m}(\Theta f_n^* \times f_m) \geq 0$ for all $\underline{f} \in \underline{\mathscr{S}}_+$; (OS-E3) if $\mathfrak{S}_n(f) = \mathfrak{S}_n(f^\pi)$ for all permutations π; (OS-E4) if $\lim_{\lambda\to\infty} \sum_{n,m} [\mathfrak{S}_{n+m}(\Theta f_n^* \times (g_m)_{(\lambda a,1)}) - \mathfrak{S}_n(\Theta f_n^*)\, \mathfrak{S}_m(g_m)] = 0$ for all $\underline{f}, \underline{g} \in \underline{\mathscr{S}}_+$ and $a = (\vec{a}, 0)$, $\vec{a} \neq 0$. List OS consists of these six conditions.

(2) *List GW.* $L_+^\uparrow = \{\Lambda \in \mathrm{O}(1,2) : \det \Lambda = 1,\ \Lambda^0{}_0 \geq 1\}$ and $\mathcal{P}_+^\uparrow = \mathbb{R}^{1+2} \rtimes L_+^\uparrow$ with $(a,\Lambda)(a',\Lambda') = (a + \Lambda a', \Lambda\Lambda')$. A quadruple $(\mathcal{H}, U, \varphi, D)$ satisfies: (GW1) if $\mathcal{H}$ is a separable Hilbert space and U a strongly continuous unitary representation of $\mathcal{P}_+^\uparrow$ on it; (GW2) if the projection-valued measure E with $U(a,1) = \int e^{ia\cdot p} dE(p)$ has its support in $\overline{V_+}$; (GW3) if some unit vector ψ_0 satisfies $U(a,1)\psi_0 = \psi_0$ for all a, and every unit vector with this property is a constant multiple of ψ_0; (GW4) if D is a dense subspace and φ maps $\mathscr{S}(\mathbb{R}^{1+2})$ to operators with $D \subset \operatorname{dom}\varphi(f) \cap \operatorname{dom}\varphi(f)^*$, $\varphi(f)^*|_D = \varphi(\bar{f})|_D$, $\psi_0 \in D$, $\varphi(f)D \subset D$, and $f \mapsto \varphi(f)\psi$ linear for $\psi \in D$; (GW5) if $f \mapsto \langle \psi_1, \varphi(f)\psi_2\rangle$ is a tempered distribution for $\psi_1, \psi_2 \in D$; (GW6) if $U(a,\Lambda)D \subset D$ and $U(a,\Lambda)\varphi(f)U(a,\Lambda)^{-1}\psi = \varphi(f(\Lambda^{-1}(\cdot - a)))\psi$ for $\psi \in D$; (GW7) if $[\varphi(f)\varphi(g) - \varphi(g)\varphi(f)]\psi = 0$ for $\psi \in D$ whenever the closed supports of f and g are spacelike separated; (GW8) if the set D_0 of the vectors $P\psi_0$, P a polynomial in the $\varphi(f)$ with constants, is dense. List GW consists of these eight conditions. The variants are: (GW-i) U a continuous unitary representation of a topological group $\tilde{\mathcal{P}}$ given with a continuous homomorphism onto $\mathcal{P}_+^\uparrow$; (GW-ii) a unit vector Ψ_0 invariant under all $U(a,\Lambda)$, unique up to a phase among the unit vectors invariant under all $U(a,\Lambda)$.

(3) *Scalar Wightman quintuples.* A quintuple $(\mathcal{H}, U, \Omega, D, \Phi)$ has a separable Hilbert space, a strongly continuous unitary representation of $\mathbb{R}^{1+2} \rtimes \mathcal{L}$, a unit vector Ω in a dense subspace D, and a field $f \mapsto \Phi(f)$ linear on Schwartz tests with common domain D. Its conditions are as follows.

- (W1) The translation spectrum is in $\overline{V_+}$ and Ω is invariant under the representation. Every translation-invariant vector belongs to $\mathbb{C}\Omega$.
- (W2) The field operators and the representation preserve D. Matrix elements of the field on D are tempered and satisfy $\langle \psi, \Phi(f)\psi'\rangle = \langle \Phi(\bar{f})\psi, \psi'\rangle$.
- (W3) The field obeys the covariance identity of (GW6).
- (W4) The commutativity of (GW7) holds on D.
- (W5) The span D_0 of Ω and the vectors $\Phi(f_1)\cdots\Phi(f_n)\Omega$ is dense.

A scalar Wightman theory satisfies (W1)–(W5). A quintuple with a possibly nonunique vacuum satisfies the same conditions with only the last sentence of (W1) omitted. The vacuum forms of either are $\mathcal{W}_n(f_1,\dots,f_n) = \langle \Omega, \Phi(f_1)\cdots\Phi(f_n)\Omega\rangle$.

(4) *The relation (Sch).* Take one of these quintuples. For $n \geq 2$ let $\mathfrak{T}_{n-1} = \{(\zeta_1,\dots,\zeta_{n-1}) : \zeta_k \in \mathbb{R}^{1+2} + iV_+\}$. A family $(W_{n-1})_{n\geq 2}$ of holomorphic functions on the $\mathfrak{T}_{n-1}$ has the forms as its boundary values in the weak sense if $\lim_{\epsilon\downarrow 0} \int \prod_k f_k(x_k)\, W_{n-1}(x_2 - x_1 + i\epsilon e_0, \dots, x_n - x_{n-1} + i\epsilon e_0)\, dx = \mathcal{W}_n(f_1,\dots,f_n)$ for all $n \geq 2$ and $f_k \in C_c^\infty(\mathbb{R}^{1+2};\mathbb{C})$; such a family is called a family of the quintuple in the weak sense. The family has the forms as its boundary values in the strong sense when for each $n \geq 2$ one can find a tempered distribution $\mathcal{W}_n^\sim$ on $\mathbb{R}^{3n}$ such that $\mathcal{W}_n^\sim(f_1 \otimes \cdots \otimes f_n) = \mathcal{W}_n(f_1,\dots,f_n)$ for $f_k \in \mathcal{S}$ and such that, for each $F \in \mathcal{S}(\mathbb{R}^{3n})$, the integrals $I_F(\eta) = \int F(x)\, W_{n-1}(x_2 - x_1 + i\eta_1, \dots, x_n - x_{n-1} + i\eta_{n-1})\, dx$, $\eta = (\eta_1,\dots,\eta_{n-1}) \in V_+^{n-1}$, are absolutely convergent with the limit $\mathcal{W}_n^\sim(F)$ as $\eta \to 0$ in V_+^{n-1}. The choice $F = f_1 \otimes \cdots \otimes f_n$, $\eta_k = \epsilon e_0$ shows that the strong sense implies the weak sense. For a quintuple with a family in the weak sense its Schwinger functions in the sense (Sch) are: $\mathfrak{S}_1$, the number with $\mathcal{W}_1(f) = \mathfrak{S}_1 \int f$ for $f \in \mathcal{S}$, which exists because $\mathcal{W}_1$ is a tempered distribution invariant under all translations ((W1)–(W3)); at the time-ordered points $y = (y_1,\dots,y_n)$, $y_1^0 < \cdots < y_n^0$, $\mathfrak{S}_n(y) = W_{n-1}(\zeta_1,\dots,\zeta_{n-1})$ with $\zeta_k = (i(y_{k+1}^0 - y_k^0), \vec{y}_{k+1} - \vec{y}_k)$; and on $(\mathbb{R}^3)^n_{\neq}$ the continuous function which is symmetric in its arguments and has these values, where there is one.

■

Proposition 18.226 (A dictionary between the axioms of the main text and Lists OS and GW)**.** Write $\mathcal{S}_{\neq} = {}^{0}\mathscr{S}$. The larger positive-time space $\mathcal{S}_+$ consists of Schwartz functions flat outside $V_+^{(n)} = \{x_i^0 > 0 \text{ for all } i,\ x_i \neq x_j \text{ for } i \neq j\}$. In degree zero all these spaces are $\mathbb{C}$. The operations Θ, conjugation and permutation are those of List OS, and $\otimes$ is its product on successive groups of variables. We use (E0′)–(E4) in the form of Proposition YM3_I:111.2: the bound with $|\cdot|_{k_{\mathrm{E}}n}$ and constants $\mathsf{c}_{\mathrm{E}}\mathsf{C}_{\mathrm{E}}^n(n!)^{\varkappa_{\mathrm{E}}}$, invariance under all Euclidean isometries, reflection positivity on the larger spaces, permutation symmetry, and clustering under nonzero translations parallel to the reflection plane. Data (Rec) are the scalar reconstruction data of Section YM3_I:108, with point bound YM3_I:(2049). Write $d_\circ(\mathfrak{z}) = \operatorname{Re}\mathfrak{z}_3 - |\operatorname{Im}\vec{\mathfrak{z}}|$, $\mathsf{Tb}_\circ = \{d_\circ > 0\}$, and $\iota(t,\vec{x}) = (\vec{x}, -it)$.

The Euclidean-variable Schwinger relation uses a constant W_1 and holomorphic functions W_n on $\mathsf{Tb}_\circ^{n-1}$ for $n \geq 2$. They have polynomial growth where the spatial coordinates are real and all real time coordinates are bounded below by a fixed $\epsilon > 0$. Their boundary integrals at consecutive differences $\iota(x_{k+1} - x_k) + \epsilon e_3$ converge to the vacuum forms on Schwartz tensors (S1). At real consecutive differences $y_{k+1} - y_k$ they give the separated-tuple functional by integration on smooth tests with time-ordered supports (S2). The degree-one versions are $\mathcal{W}_1(f) = \mathsf{W}_1 \int f$ and $\sigma(f) = \mathsf{W}_1 \int f$. This is the relation used in Theorem YM3_I:AO.

(a) *The Euclidean side.* Let $(\mathsf{S}_n)_{n\geq 0}$ be complex linear functionals $\mathsf{S}_n \colon \mathcal{S}_{\neq}(\mathbb{R}^{3n}) \to \mathbb{C}$ with $\mathsf{S}_0(z) = z$.

(a1) $\mathscr{S}(\mathbb{R}^{3n})$ is the space of the Schwartz functions of the flat-space convention above, with the same norms $|\cdot|_m$; ${}^{0}\mathscr{S}(\mathbb{R}^{3n}) = \mathcal{S}_{\neq}(\mathbb{R}^{3n})$; and $\mathscr{S}_+(\mathbb{R}^{3n}) \subset \mathcal{S}_+(\mathbb{R}^{3n})$, with equality for $n = 1$ and a proper inclusion for every $n \geq 2$.

(a2) (E0′) with the constants $(k_{\mathrm{E}}, \mathsf{c}_{\mathrm{E}}, \mathsf{C}_{\mathrm{E}}, \varkappa_{\mathrm{E}})$ gives (OS-E0′) with $s = k_{\mathrm{E}}$, $\sigma_n = \mathsf{c}_{\mathrm{E}}\mathsf{C}_{\mathrm{E}}^n(n!)^{\varkappa_{\mathrm{E}}}$, $\alpha = \mathsf{c}_{\mathrm{E}}e^{\mathsf{C}_{\mathrm{E}}}$, $\beta = \varkappa_{\mathrm{E}} + 1$; conversely (OS-E0′) with (s, α, β) gives (E0′) with $(\max\{s,1\}, \alpha, 1, \max\{\beta, 0\})$. The two conditions are equivalent.

(a3) (E1) gives (OS-E1); (E1) is asked for the isometries with $\det \mathsf{r}^\circ = -1$ as well.

(a4) (E2) and (E3) give (OS-E2); (E2) is asked on the larger spaces $\mathcal{S}_+$.

(a5) (E3) is (OS-E3).

(a6) (E4) and (E3) give (OS-E4); (E4) is asked on the larger spaces $\mathcal{S}_+$.

(a7) (E0′), (E1), (E2) give $\mathsf{S}_n(\Theta\bar{\varphi}) = \overline{\mathsf{S}_n(\varphi)}$ for all $\varphi \in \mathcal{S}_{\neq}(\mathbb{R}^{3n})$; with (E3) this is the last member of (OS-E0); with (E1) at the reflection θ, $\mathsf{S}_n(\bar{\varphi}) = \overline{\mathsf{S}_n(\varphi)}$.

(a8) A sequence which satisfies (E0′)–(E4) satisfies all of List OS, and one which satisfies (E0′)–(E3) satisfies all of List OS through the symmetry requirement (OS-E3).

(a9) Let $\underline{C}$ be a bound table, $\bar{\mathfrak{c}} \geq 0$, let $\sigma \in \mathsf{Fct}(\underline{C}, \bar{\mathfrak{c}})$ have the growth, invariance, reflection positivity and clustering hypotheses of Proposition YM3_I:111.2, and let σ also be data (Rec). Then the classes of the sequences of $\underline{\mathscr{S}}_+$ are dense in the completion $\mathcal{H}^{\mathcal{S}}$ of finite sequences in the larger spaces modulo the null vectors of the reflection form, which the unitary map V of the reflection-space identification of Proposition YM3_I:111.2 identifies with $\mathcal{H}(\sigma)$: the larger test spaces add no vector.

(b) *The Minkowski side.*

(b1) $\mathcal{L} = L_+^\uparrow$, so $\mathbb{R}^{1+2} \rtimes \mathcal{L} = \mathcal{P}_+^\uparrow$.

(b2) A Wightman theory $(\mathcal{H}, U, \Omega, D, \Phi)$ as defined above satisfies (GW1)–(GW8) with $\varphi = \Phi$ and $\psi_0 = \Omega$. Conversely, if $(\mathcal{H}, U, \varphi, D)$ satisfies (GW1)–(GW8), then $(\mathcal{H}, U, \psi_0, D, \Phi)$ with $\Phi(f) = \varphi(f)|_D$ is a Wightman theory, and $U(a,\Lambda)\psi_0 = \psi_0$ for all $(a,\Lambda) \in \mathcal{P}_+^\uparrow$. A Wightman theory has the variants (GW-i) and (GW-ii).

(b3) *The two forms of the Schwinger relation.* Let a Wightman theory and a functional σ on separated tuples over $\mathbb{R}^3$ be given, let $\mathsf{W}_1 \in \mathbb{C}$, and for $n \geq 2$ let W_n be a function on $\mathsf{Tb}_\circ^{n-1}$ and $W_{n-1} = \mathsf{W}_n \circ (\iota \times \cdots \times \iota)$. The number W_1 and the functions W_n have the Euclidean-variable growth and boundary properties (S1), (S2), so that

the Schwinger functions of the theory are those of σ, if and only if the following hold: $\mathcal{W}_1(f) = \mathsf{W}_1 \int f$ for $f \in \mathcal{S}$, and $\sigma(f) = \mathsf{W}_1 \int f$ for real $f \in C_c^\infty(\mathbb{R}^3)$; each W_{n-1} is holomorphic on $\mathfrak{T}_{n-1}$ and, for each $\epsilon > 0$, bounded by a polynomial in $\sum_k |\zeta_k|$ on the set of the points with $\operatorname{Im} \zeta_k \in [\epsilon, \infty) e_0$ for all k; $\mathcal{W}_n(f_1, \dots, f_n) = \lim_{\epsilon \downarrow 0} \int \prod_k f_k(x_k)\, W_{n-1}(x_2 - x_1 + i\epsilon e_0, \dots, x_n - x_{n-1} + i\epsilon e_0)\, dx$ for all $f_k \in \mathcal{S}$; and σ has, on the smooth separated tuples with supports ordered in the third coordinate, the continuous kernel $y \mapsto W_{n-1}(\zeta(y))$ of (Sch), the third coordinate being the Euclidean time. Then $(W_{n-1})_{n \geq 2}$ is a family of the theory in the weak sense, and $\mathfrak{S}_1 = \mathsf{W}_1$.

(b4) *A holomorphic function is fixed by its boundary value.* Fix $n \geq 2$ and a holomorphic function D on $\mathfrak{T}_{n-1}$ such that $\lim_{\epsilon \downarrow 0} \int \prod_k f_k(x_k)\, \mathsf{D}(x_2 - x_1 + i\epsilon e_0, \dots, x_n - x_{n-1} + i\epsilon e_0)\, dx = 0$ whenever $f_1, \dots, f_n \in C_c^\infty(\mathbb{R}^{1+2})$ are real. Then $\mathsf{D} = 0$; the hypotheses impose no bound on D. Consequently a quintuple as in (Sch) possesses at most one family in the weak sense, hence at most one in the strong sense; any two continuous symmetric functions on $(\mathbb{R}^3)^n_{\neq}$ with the same values at the time-ordered points coincide; and the quintuple's Schwinger functions in the sense (Sch) are well defined.

(b5) *One theory.* Let two Wightman theories be given, each with a family in the weak sense, whose Schwinger functions agree at the time-ordered points: $\mathcal{W}_1 = \mathcal{W}'_1$ and $W_{n-1} = W'_{n-1}$ at all points with $\zeta_k = (is_k, \vec{\xi}_k)$, $s_k > 0$, $\vec{\xi}_k \in \mathbb{R}^2$. Then $W_{n-1} = W'_{n-1}$ on $\mathfrak{T}_{n-1}$, $\mathcal{W}_n = \mathcal{W}'_n$ on $\mathcal{S}^n$, and there is exactly one unitary $\mathsf{V} \colon \mathcal{H} \to \mathcal{H}'$ with $\mathsf{V}\Omega = \Omega'$ and $\mathsf{V}\Phi(f)\psi = \Phi'(f)\mathsf{V}\psi$ for $\psi \in D_0$; it maps D_0 onto D'_0 and $\mathsf{V}U(a, \Lambda) = U'(a, \Lambda)\mathsf{V}$. In particular two Wightman theories whose Schwinger functions are those of one functional σ are unitarily equivalent in this sense.

(b6) *Locality at touching supports.* Let a Wightman theory have the property (T): for every n there is a tempered distribution $\mathcal{W}_n^\sim$ on $\mathbb{R}^{3n}$ with $\mathcal{W}_n^\sim(f_1 \otimes \cdots \otimes f_n) = \mathcal{W}_n(f_1, \dots, f_n)$ for $f_k \in \mathcal{S}$. Then $\Phi(f)\Phi(h)\psi = \Phi(h)\Phi(f)\psi$ for every $\psi \in D_0$, the span of (W5), whenever $f, h \in \mathcal{S}$ and $f(x)h(y) = 0$ for all $x, y \in \mathbb{R}^{1+2}$ with $(x - y) \cdot (x - y) \geq 0$. The quintuple of Theorem YM3_I:AO has (T), with $\mathcal{W}_n^\sim = \mathcal{W}_n^\sigma$, and its domain $\mathcal{D}_0$ is the polynomial span in (W5).

(b7) *The strong sense for data (Rec).* Let σ be data (Rec) with the numbers $C_\wp, \wp, c_0$, and let $(\mathcal{H}(\sigma), U, \Omega, \mathcal{D}_0, \phi)$ be the quintuple of Theorem YM3_I:AO: a Wightman theory if (Rec5) holds, and otherwise a quintuple as in (Sch). For $n \geq 2$ put $N = \lceil \wp n \rceil + 1$, $\bar{k} = N + 4n$ and $c_n = 2^N (N+1)\, C_\wp^n\, (n(n+1))^N \pi^{2n}$. Then the functions $W_{n-1} = \mathbf{W}_n \circ (\iota \times \cdots \times \iota)$ have the forms of the quintuple as their boundary values in the strong sense, with $\mathcal{W}_n^\sim = \mathcal{W}_n^\sigma$, and $|I_F(\eta)| \leq c_n |F|_{\bar{k}}$ for all $F \in \mathcal{S}(\mathbb{R}^{3n})$ and all $\eta \in V_+^{n-1}$. Its Schwinger functions in the sense (Sch) are $\mathfrak{S}_1 = \mathbf{W}_1$ and, on all of $(\mathbb{R}^3)^n_{\neq}$, $\mathfrak{S}_n = \mathsf{s}_n$, the kernel of Proposition YM3_I:108.3: $\sigma(f_1, \dots, f_n) = \int \mathfrak{S}_n(y) \prod_k f_k(y_k)\, dy$ for all separated tuples $(f_1, \dots, f_n)$ over $\mathbb{R}^3$.

(b8) *Reflections of space.* Let σ be data (Rec) with (Rec5), let $\mathsf{R} \in \mathrm{O}(2)$ and $P(t, \vec{x}) = (t, \mathsf{R}\vec{x})$. Then $P\mathcal{L}P^{-1} = \mathcal{L}$, and there is exactly one unitary operator V_R of $\mathcal{H}(\sigma)$ with $\mathsf{V}_\mathsf{R}\Omega = \Omega$ and $\mathsf{V}_\mathsf{R}\, \phi(f)\psi = \phi(f \circ P^{-1})\, \mathsf{V}_\mathsf{R}\psi$ for all $f \in \mathcal{S}$ and $\psi \in \mathcal{D}_0$; it satisfies $\mathsf{V}_\mathsf{R}\, U(a, \Lambda) = U(Pa, P\Lambda P^{-1})\, \mathsf{V}_\mathsf{R}$. So the isometries with $\det \mathsf{r}^\circ = -1$ of (a3) have a counterpart on the Minkowski side, which neither (W1)–(W5) nor List GW asks.

(c) *For the limits of the main theorems.* Let $0 < g \leq \gamma$, let S be a per-limit at g with a canonical decomposition $(\mathsf{X}, \lambda, (\sigma_\xi)_{\xi \in \mathsf{X}})$ (Theorems YM3_I:AP and YM3_I:AL), and let $\xi \in \mathsf{X}$.

(c1) The Schwinger functions of σ_ξ satisfy (E0′)–(E4) and all of List OS. (OS-E0′) holds with $s = 10$, $\sigma_n = (\mathsf{C}_\mathrm{E}^g)^n (n!)^{\varkappa_\star + 9}$, $\alpha = e^{\mathsf{C}_\mathrm{E}^g}$, $\beta = \varkappa_\star + 10$, and also with $s = 8$, $\sigma_n = 2(\mathsf{C}_\mathrm{E}^{\mathrm{pt},g})^n (n!)^8$, $\alpha = 2e^{\mathsf{C}_\mathrm{E}^{\mathrm{pt},g}}$, $\beta = 9$, where $\mathsf{C}_\mathrm{E}^{\mathrm{pt},g}$ is the point-kernel growth constant of Theorem YM3_I:AP, $\mathsf{C}_\mathrm{E}^g = 2\mathsf{C}_\mathrm{E}^{\mathrm{pt},g}$, and $\varkappa_\star \geq 0$ is arbitrary; these numbers are the same for every per-limit at g, every canonical decomposition and every ξ. The limit

of (E4) holds along every $v \in \mathbb{R}^3$ with $v_3 \geq 0$ and $v \neq 0$, the time direction e_3 among them.

(c2) The Schwinger functions of σ_S satisfy (E0′)–(E3), and so all of List OS through the symmetry requirement (OS-E3). σ_S is data (Rec) with the numbers $C^g_\wp, 4, \frac{1}{2}$, and the quintuple $(\mathcal{H}(S), U, \Omega, \mathcal{D}_0, \phi)$ of Theorem YM3_I:AO formed for σ_S satisfies (GW1), (GW2), (GW4)–(GW8) with $\psi_0 = \Omega$, and the existence of the vector in (GW3). The following five conditions are equivalent: (i) X is one point; (ii) the Schwinger functions of σ_S satisfy (E4); (iii) they satisfy (OS-E4); (iv) $\ker H = \mathbb{C}\Omega$ in $\mathcal{H}(S)$; (v) the uniqueness of (GW3) holds for that quintuple.

(c3) The quintuple $(\mathcal{H}(\sigma_\xi), U, \Omega, \mathcal{D}_0, \phi)$ of Theorem YM3_I:AO satisfies (GW1)–(GW8) with $\psi_0 = \Omega$, the variants (GW-i), (GW-ii), and the locality of (b6). Its Wightman forms are the tempered distributions $\mathcal{W}_n^{\sigma_\xi}$ at the tensors, and the functions $\mathbf{W}_n \circ (\iota \times \cdots \times \iota)$ have them as their boundary values in the strong sense, with the bound of (b7). Its Schwinger functions in the sense (Sch) are the kernels of σ_ξ: $\mathfrak{S}_1 = \mathfrak{c}_\xi$, and $\sigma_\xi(f_1, \dots, f_n) = \int \mathfrak{S}_n(y) \prod_k f_k(y_k)\, dy$ for all $n \geq 2$ and all separated tuples $(f_1, \dots, f_n)$ over $\mathbb{R}^3$. Every Wightman theory with a family in the weak sense whose Schwinger functions agree with these at the time-ordered points, and every Wightman theory whose Schwinger functions are those of σ_ξ in the sense in the Euclidean variables, is unitarily equivalent to it in the sense of (b5). The reflections of space are implemented on $\mathcal{H}(\sigma_\xi)$ as in (b8).

(c4) List OS builds a Hilbert space out of the Schwinger functions of σ_ξ, namely the quotient of $\underline{\mathscr{S}}_+$ by the space of null vectors of the form of (OS-E2), completed; there is a unitary map of this Hilbert space onto $\mathcal{H}(\sigma_\xi)$ which sends the class of $(1, 0, 0, \dots)$ to Ω.

■

Proof. Step 1: (a1). The norms of (N1) give the usual Schwartz topology because $|x^\beta| \leq (1+|x|^2)^{k/2}$ for $|\beta| \leq k$, and $(1+|x|^2)^{k/2}$ is at most a sum of monomials of degree $\leq 2k$ with coefficients adding up to $(1+3n)^k$; so the two families of seminorms define one space and one topology. A function flat outside $(\mathbb{R}^3)^n_{\neq}$ is one which vanishes with all its derivatives on the hyperplanes $x_i = x_j$. A Schwartz function flat outside the open set $O_+^{(n)} = \{0 < x_1^0 < \cdots < x_n^0\}$ is the same as one with support in its closure: at a boundary point some inequality is an equality, and points outside the closure, at which all derivatives vanish, approach it. As $O_+^{(n)} \subset V_+^{(n)}$, functions flat outside the first set are flat outside the second; for $n = 1$ the two sets agree. For $n \geq 2$, let $b \geq 0$ be smooth with $b(0) > 0$ and support in the ball of radius $\frac{1}{4}$ about 0; the tensor $\bigotimes_{i \leq n} b(\cdot - (i, 0, 1))$ has a compact support in $V_+^{(n)}$, so it lies in $\mathcal{S}_+(\mathbb{R}^{3n})$, and it does not vanish at the point $((1,0,1), \dots, (n,0,1))$, a boundary point of $O_+^{(n)}$ with $x_1^0 = x_2^0$, at which every member of $\mathscr{S}_+(\mathbb{R}^{3n})$ vanishes.

Step 2: (a2)–(a6). (a2): a functional bounded for one norm $|\cdot|_k$ is continuous; $e^{\mathsf{C}} \geq \mathsf{C}^n/n!$; conversely $\sigma_1 \geq 0$ forces $\alpha \geq 0$, $(n!)^\beta \leq (n!)^{\max\{\beta, 0\}}$, and the norms increase with the index. (a3): the isometry $y \mapsto R^{-1}(y - a)$ has the linear part $R^{-1} \in \mathrm{O}(3)$ and gives $\mathsf{r}\varphi = \varphi_{(a,R)}$. (a4): $(\Theta f_n^*)(x_1, \dots, x_n) = (\Theta \bar{f}_n)(x_n, \dots, x_1)$, so $\Theta f_n^* \times f_m$ is a permutation of $\Theta \bar{f}_n \otimes f_m$, and (E3) and (E2) apply. (a5): the definitions agree. (a6): $(g_m)_{(\lambda a, 1)} = \tau_{sv} g_m$ with $v = -a$, $s = \lambda$, translations parallel to the plane keep $\mathcal{S}_+$, and (a4)'s permutation with (E3) and (E4) send each of the finitely many summands to 0.

Step 3: (a7), (a8). The form $q(\underline{\varphi}, \underline{\varphi}') = \sum \mathsf{S}_{n+n'}(\Theta \bar{\varphi}_n \otimes \varphi'_{n'})$ on finite sequences of $\mathcal{S}_+$ is positive by (E2), hence Hermitian; at $\underline{\varphi} = (1, 0, \dots)$ and φ' in the place n this gives $\mathsf{S}_n(\Theta \bar{\varphi}') = \overline{\mathsf{S}_n(\varphi')}$ on $\mathcal{S}_+$. A sum of tensors Ψ of smooth separated tuples is moved into $\mathcal{S}_+$ by τ_{te_3} for large t, and $\Theta \overline{\tau_{te_3} \Psi} = \tau_{-te_3} \Theta \bar{\Psi}$, so (E1) gives the identity at Ψ; such sums are dense in $\mathcal{S}_{\neq}$ for every norm $|\cdot|_k$ (Lemma YM3_I:93.2), and (E0′) carries the identity to all φ, since $|\Theta \bar{\varphi}|_k = |\varphi|_k$. With $\Theta \varphi^* = (\Theta \bar{\varphi})^\varsigma$,

ς the reversal, (E3) gives (OS-E0); $\Theta\bar\varphi = \theta\bar\varphi$ and (E1) at θ give the last identity. (a8) collects (a2)–(a7); only (a6) uses (E4).

Step 4: (a9). Proposition YM3_I:111.2 shows that the axioms hold, that the form is positive, that the classes are dense in $\mathcal{H}^{\mathcal{S}}$ and that V is unitary; linearity then reduces the claim to approximating the class of a sequence that has a single member $\varphi \in \mathcal{S}_+(\mathbb{R}^{3n})$, $n \geq 2$. (E3) shows $[\varphi^\varsigma] = [\varphi]$ for every permutation. With a smooth u, $0 \leq u \leq 1$, $u = 0$ on $(-\infty, 1]$, $u = 1$ on $[2, \infty)$, put $\varphi_\epsilon = \prod_{i<l} u((x^0_i - x^0_l)^2/\epsilon)\,\varphi \in \mathcal{S}_+$; it vanishes near the points with two equal times, so it is the sum over the orderings π of the pieces $\mathbf{1}_{\{0<x^0_{\pi(1)}<\cdots<x^0_{\pi(n)}\}}\varphi_\epsilon$, each smooth, and each a permuted member of $\mathscr{S}_+$; so $[\varphi_\epsilon]$ is the class of a sequence of $\underline{\mathscr{S}}_+$. By Theorem YM3_I:AO, including its case without vacuum uniqueness, $\mathsf{S}^\sigma_n(\varphi) = \int \mathsf{s}_n\varphi$ with an absolutely convergent integral, so $\|[\varphi] - [\varphi_\epsilon]\|^2 = \iint \mathsf{s}_{2n}(x,y)(1-g_\epsilon(x))(1-g_\epsilon(y))(\Theta\bar\varphi)(x)\varphi(y)\,dx\,dy \to 0$ by dominated convergence, g_ϵ being the product of the cut-offs, which tends to 1 off the null set of equal times.

Step 5: (b1), (b2). For $\Lambda \in \mathrm{O}(1,2)$ with $\det\Lambda = 1$ and $\Lambda^0{}_0 = \cosh s \geq 1$, $\Lambda e_0 = (\cosh s, \sinh s\,\hat n)$, and $\Lambda = (1 \oplus \mathsf{R}_1)B(s)(1 \oplus \mathsf{R}_2)$, obtained by matching Λe_0 with a rotated boost and noting that the residual transformation fixes e_0, with rotations $\mathsf{R}_1, \mathsf{R}_2$ and the boost $B(s)$, all in $\mathcal{L}$, so that $\Lambda \in \mathcal{L}$; conversely the elements of $\mathcal{L}$ have $\det\Lambda = 1$ and $\Lambda^0{}_0 \geq 1$. (b2) is a comparison of the words of (W1)–(W5) with (GW1)–(GW8): a linear functional on $\mathcal{S}$ is tempered exactly when it is bounded for one norm $|\cdot|_N$, the support of a projection-valued measure lies in $\overline{V_+}$ exactly when the open complement has measure 0, and for $\psi \in D$ the Hermiticity of (W2), $\langle\psi, \Phi(f)\psi'\rangle = \langle\Phi(\bar f)\psi, \psi'\rangle$ for all $\psi' \in D$, says that ψ lies in the domain of $\Phi(f)^*$ and $\Phi(f)^*\psi = \Phi(\bar f)\psi$; conversely (GW4) gives this identity for the restrictions to D. The one step which is no comparison is the invariance of ψ_0 under the Lorentz transformations. By (GW1), $U(a,1)U(0,\Lambda)\psi_0 = U(0,\Lambda)U(\Lambda^{-1}a,1)\psi_0 = U(0,\Lambda)\psi_0$, so (GW3) gives $U(0,\Lambda)\psi_0 = c(\Lambda)\psi_0$ with a continuous character c of $\mathcal{L}$, $|c| = 1$. Conjugation by the rotation by π carries the boost $B(s)$ to $B(-s)$, so $c(B(-s)) = c(B(s))$, and $c(B(s))c(B(-s)) = c(1) = 1$; hence $c(B(s)) = \pm 1$, and $c(B(s)) = 1$ by continuity in s; as every boost is conjugate by a rotation to some $B(s)$, $c = 1$ on all boosts. The product $B(s)B_2(s)$ of the boosts of rapidity s in two orthogonal directions is a boost times the rotation by an angle $\omega(s)$ with $\cos\omega(s) = 2\cosh s/(1+\cosh^2 s)$, which runs through $(0, \frac{\pi}{2})$ as s runs through $(0,\infty)$; so $c = 1$ on the rotations by these angles, hence on all rotations, and on $\mathcal{L}$, which the boosts and the rotations generate. (GW-i): compose U with the homomorphism; (GW-ii): Ω is invariant under all $U(a,\Lambda)$ by (W1), and a vector invariant under all $U(a,\Lambda)$ is invariant under the translations, so it lies in $\mathbb{C}\Omega$.

Step 6: (b3). ι is a complex linear bijection of $\mathbb{C}^{1+2}$ onto $\mathbb{C}^3$ with $\iota(\mathbb{R}^{1+2} + iV_+) = \mathsf{Tb}_\circ$, and $|\iota\zeta| = |\zeta|$. It carries the points ζ with $\operatorname{Im}\zeta \in [\epsilon,\infty)e_0$ onto the points $\mathfrak{z}$ with $\operatorname{Im}\vec{\mathfrak{z}} = 0$ and $\operatorname{Re}\mathfrak{z}_3 \geq \epsilon$, the points $x + i\epsilon e_0$, x real, to $\iota(x) + \epsilon e_3$, and, as $\iota^{-1}(\mathfrak{z}) = (i\mathfrak{z}_3, \vec{\mathfrak{z}})$, the points $\zeta_k(y) = (i(y^0_{k+1} - y^0_k), \vec y_{k+1} - \vec y_k)$ to $y_{k+1} - y_k$. So W_{n-1} is holomorphic on $\mathfrak{T}_{n-1}$ exactly when W_n is holomorphic on $\mathsf{Tb}^{n-1}_\circ$; the growth condition in the Euclidean variables is the polynomial bound of the clause; (S1) is the limit at the functions of $\mathcal{S}$, and at $n = 1$ the identity $\mathcal{W}_1(f) = \mathsf{W}_1\int f$; and (S2) says that σ has the kernel $\mathsf{W}_n(y_2 - y_1, \dots, y_n - y_{n-1}) = W_{n-1}(\zeta(y))$ at the time-ordered tuples, and at $n = 1$ that $\sigma(f) = \mathsf{W}_1\int f$. The limit at the functions of $\mathcal{S}$ contains the limit at the functions of C^∞_c, which defines the weak sense.

Step 7: (b4). Put $N = n-1$, fix real $f_k \in C^\infty_c$ and let $h(z) = \int \prod_k f_k(x_k)\mathsf{D}(x_2 - x_1 + z_1e_0, \dots)\,dx$ for $z \in \mathbb{H}^N$, $\mathbb{H}$ the upper half plane: h is holomorphic, and by real translations of the f_k in time, $h(t + i\epsilon\mathbb{1}) \to 0$ as $\epsilon \downarrow 0$ for every $t \in \mathbb{R}^N$. The sets $A_m = \{t : |h(t + i\epsilon\mathbb{1})| \leq m \text{ for all } \epsilon \in (0,1]\}$, $m \in \mathbb{N}$, are closed, and they cover $\mathbb{R}^N$, since $\epsilon \mapsto h(t + i\epsilon\mathbb{1})$ is continuous on $(0,1]$ with a limit at 0; Baire's theorem gives m, t^* and $r > 0$ with $|h(t + i\epsilon\mathbb{1})| \leq m$ for $|t - t^*|_\infty < 2r$ and $0 < \epsilon \leq 1$. Let $|t - t^*|_\infty < r$ and $G(w) = h(t + w\mathbb{1})$, holomorphic on $\mathbb{H}$; on $Q = \{|\operatorname{Re} w| < \frac{r}{2},\ 0 < \operatorname{Im} w < \frac{1}{2}\}$ it is bounded by m, and $G(u + i\epsilon) \to 0$ as $\epsilon \downarrow 0$ for every real u. Cauchy's formula on ∂Q, applied to $G(\cdot + i\delta)$, and the limit $\delta \downarrow 0$, with dominated convergence on the sides, shows that $\hat G(w_0) = \frac{1}{2\pi i}\int_\Gamma G(w)(w - w_0)^{-1}dw$, Γ the three sides of Q off the real axis, is holomorphic on

$Q^\pm = \{|\operatorname{Re} w| < \frac{r}{2},\ |\operatorname{Im} w| < \frac{1}{2}\}$, equals G on Q, and vanishes at the points of $Q^\pm$ below the axis. The function $w \mapsto \hat G(w - \frac{i}{4})$ is holomorphic on the convex open set $Q^\pm + \frac{i}{4}$ and vanishes at its real points, so it vanishes by the one-variable identity theorem; so $G = 0$ on Q, and $h(t + \frac{i}{4}\mathbb{1}\!\!\!\!\perp) = 0$ whenever $|t - t^*|_\infty < r$. The identity theorem on an open real slice, applied successively in each coordinate to the function $z \mapsto h(z + \frac{i}{4}\mathbb{1}\!\!\!\!\perp)$ on the convex open set $\mathbb{H}^N - \frac{i}{4}\mathbb{1}\!\!\!\!\perp$, which holds $\mathbb{R}^N$, gives $h = 0$ on $\mathbb{H}^N$. Then, for every $y \in (0,\infty)^N$, the integrals of the products $\prod_k f_k(x_k)$ against $\xi \mapsto \mathsf{D}(\xi_1 + iy_1e_0, \dots, \xi_N + iy_Ne_0)$ at $\xi_k = x_{k+1} - x_k$ vanish for all real $f_k \in C_c^\infty$; at $f_k = j_\delta(\cdot - x_k^*)$, with a bump $j_\delta \ge 0$ of integral 1 supported in the ball of radius δ and a chain $x_{k+1}^* = x_k^* + \xi_k^*$, continuity gives $\mathsf{D}(\xi_1^* + iy_1e_0, \dots) = 0$ as $\delta \downarrow 0$, for every $\xi^* \in \mathbb{R}^{3N}$. The same identity theorem on the convex open set $\mathfrak{T}_N - i(y_ke_0)_k$, which holds $\mathbb{R}^{3N}$, gives $\mathsf{D} = 0$. Two continuous symmetric functions on $(\mathbb{R}^3)^n_{\neq}$ which agree at the time-ordered points agree at the points with pairwise different times, which are dense.

Step 8: (b5). The difference $(W_{n-1} - W'_{n-1}) \circ (\iota^{-1})^{\times(n-1)}$ is holomorphic on the convex, so connected, open set $\mathsf{Tb}_\circ^{n-1}$ and vanishes on its real points, which form the set $\mathbb{E}_+^{n-1}$, $\mathbb{E}_+ = \{x_3 > 0\}$, the image under ι of the points with $\zeta_k = (is_k, \vec\xi_k)$, $s_k > 0$; it vanishes by successive one-variable identity theorems. The forms then agree at functions of C_c^∞ by the definition of the weak sense, and at Schwartz functions by the temperedness in each argument, which the Hermiticity of (W2) gives, and the cut-offs $\chi(|x|^2/\varrho^2)$. The norms of the vectors $\sum_j c_j \Phi(f_1^j) \cdots \Phi(f_{n_j}^j)\Omega$ are fixed by the forms, so $\psi \mapsto \psi'$ is well defined, isometric, onto D'_0, between dense subspaces (W5); it extends to the unitary V, which intertwines the fields on D_0, the vacua and, by (W3), the representations. For two theories whose Schwinger functions are those of one σ, (S2) at bumps $j_\delta(\cdot - x_k^*)$ shrinking to the points of a time-ordered chain gives $\mathsf{W}_n = \mathsf{W}'_n$ on $\mathbb{E}_+^{n-1}$, and $\mathsf{W}_1 = \mathsf{W}'_1$ from $\sigma(f_1) = \mathsf{W}_1 \int f_1$; by (b3) the two theories have families in the weak sense which agree at the time-ordered points, and $\mathcal{W}_1 = \mathcal{W}'_1$, so the first part applies.

Step 9: (b6). Both sides are linear in $\psi \in D_0$; let $\psi = \Phi(g_1) \cdots \Phi(g_{a'})\Omega$, and test against the vectors $\Phi(\overline{g'_a}) \cdots \Phi(\overline{g'_1})\Omega$, which span the dense space D_0 (W5). By the Hermiticity of (W2), with $n = a + 2 + a'$, $k = a + 1$, $F = g'_1 \otimes \cdots \otimes g'_a \otimes f \otimes h \otimes g_1 \otimes \cdots \otimes g_{a'}$ and the transposition ς of k and $k+1$, the assertion is $T(F) = 0$ for $T(G) = \mathcal{W}_n^\sim(G) - \mathcal{W}_n^\sim(G^\varsigma)$, a linear functional on $\mathcal{S}(\mathbb{R}^{3n})$ with $|T(G)| \le c_T |G|_N$ for some N and c_T, by (T). For $x = (x_1, \dots, x_n) \in (\mathbb{R}^{1+2})^n$, $x_j = (x_j^0, \vec x_j)$, put $q(x) = |\vec x_k - \vec x_{k+1}|^2 - (x_k^0 - x_{k+1}^0)^2 = -(x_k - x_{k+1}) \cdot (x_k - x_{k+1})$ and $p(x) = |\vec x_k - \vec x_{k+1}| - |x_k^0 - x_{k+1}^0|$, so that $q > 0$ exactly where $p > 0$. (1) $T(G) = 0$ at every tensor G of functions of $\mathcal{S}$ whose k-th and $(k+1)$-th factors have spacelike separated supports, by (W4). (2) $T(G) = 0$ for every $G \in \mathcal{S}(\mathbb{R}^{3n})$ with a compact support $\mathcal{K} \subset \{q > 0\}$. Let $p_0 > 0$ be the least value of p on $\mathcal{K}$ and $\epsilon = p_0/30$; a smooth partition of unity $\sum_{z \in \mathbb{Z}^3} \kappa_{\epsilon,z} = 1$ on $\mathbb{R}^3$, with $\kappa_{\epsilon,z} = 0$ outside the cube $\{|y - \epsilon z|_\infty < \epsilon\}$, and smooth bumps $\eta_{\epsilon,z}$ equal to 1 on that cube and to 0 outside $\{|y - \epsilon z|_\infty < 2\epsilon\}$, split G into the finitely many pieces $G_{z,z'}(x) = \kappa_{\epsilon,z}(x_k)\kappa_{\epsilon,z'}(x_{k+1})G(x)$. For a piece which is not 0, any points y, y' of the supports of $\eta_{\epsilon,z}, \eta_{\epsilon,z'}$ lie within 3ϵ, in the maximum norm, of x_k, x_{k+1} for some $x \in \mathcal{K}$, so $|\vec y - \vec y'| - |y^0 - y'^0| \ge p_0 - (6\sqrt{2} + 6)\epsilon > 0$: the two supports are spacelike separated. With $E(x) = \eta_{\epsilon,z}(x_k)\eta_{\epsilon,z'}(x_{k+1})$ we have $EG_{z,z'} = G_{z,z'}$; sums S_j of tensors of functions of C_c^∞ with $|G_{z,z'} - S_j|_N \to 0$ (cut off, convolve with a product mollifier, then use Riemann sums) give $|G_{z,z'} - ES_j|_N \to 0$ by the rule of Leibniz, E having bounded derivatives, and each ES_j is a sum of tensors as in (1). (3) F is flat at every point of $\{q \le 0\}$: it holds the factor $f(x_k)h(x_{k+1})$, which vanishes where $q \le 0$; on the open set $\{q < 0\}$, $F = 0$, and a point with $q = 0$ is a limit of points with $q < 0$, obtained by moving x_k^0 away from x_{k+1}^0. (4) Let u be smooth, $0 \le u \le 1$, $u = 0$ on $(-\infty, 1]$, $u = 1$ on $[2, \infty)$, and $F_\delta = u(q/\delta)F$ for $0 < \delta \le \frac{1}{2}$. As q is a polynomial of degree 2, the derivatives of order $\le N$ of $1 - u(q/\delta)$ are at most $c_N \delta^{-N}(1 + |x|^2)^{N/2}$, and they vanish where $q \ge 2\delta$. At a point x with $0 < q(x) \le 2\delta$ put $\rho = |\vec x_k - \vec x_{k+1}|$ and $\tau = |x_k^0 - x_{k+1}^0|$; then $(\rho - \tau)^2 \le (\rho - \tau)(\rho + \tau) = q(x) \le 1$, and the point x^* obtained by moving x_k^0 by $\rho - \tau$ away from

x^0_{k+1} has $q(x^*)=0$ and $|x-x^*|_1=\rho-\tau\le(2\delta)^{1/2}$. Taylor's formula along the time coordinate about x^* with $2N+2$ derivatives gives the following bound. Along the segment $1+|x|^2\le 3(1+|y|^2)$ since $|x-y|\le 1$, hence $|\partial^\alpha F(x)|\le\frac{(2\delta)^{N+1}}{(2N+2)!}\,3^{(3N+2)/2}\,|F|_{3N+2}\,(1+|x|^2)^{-(3N+2)/2}$ for $|\alpha|\le N$. With the rule of Leibniz, and as $F-F_\delta$ vanishes with all its derivatives where $q\le 0$ and where $q\ge 2\delta$, $|F-F_\delta|_N\le C_N\,\delta\,|F|_{3N+2}$. (5) The functions $(F_\delta)_\varrho=\chi(|x|^2/\varrho^2)F_\delta$ lie in C^∞_c, with supports in $\{q\ge\delta\}$, so $T((F_\delta)_\varrho)=0$ by (2); $\varrho\to\infty$ and Schwartz convergence of the smooth cutoffs give $T(F_\delta)=0$, and $\delta\downarrow 0$ with (4) gives $T(F)=0$.

Step 10: (b7). For $\eta\in V_+^{n-1}$ put $\mathfrak w_k=\iota(i\eta_k)=(i\vec\eta_k,\eta^0_k)$; then $d_\circ(\mathfrak w_k)=\eta^0_k-|\vec\eta_k|>0$, $|\mathfrak w_k|=|\eta_k|$, $I_F(\eta)=\int F(x)\,\mathbf W_n(\iota(x_2-x_1)+\mathfrak w_1,\dots,\iota(x_n-x_{n-1})+\mathfrak w_{n-1})\,dx$, and the points $\iota(x_{k+1}-x_k)+\mathfrak w_k$, x real, have the margin $d_\circ(\mathfrak w_k)$. For $\mathfrak w\in\overline{\mathsf{Tb}_\circ}^{\,n-1}$, Lemma YM3_I:109.4 applies to $\mathbf F(z)=\mathbf W_n(z_2-z_1+\mathfrak w_1,\dots,z_n-z_{n-1}+\mathfrak w_{n-1})$ on the open set $\mathcal O=\{z: z_{k+1}-z_k+\mathfrak w_k\in\mathsf{Tb}_\circ\text{ for all }k\}$, with $\mathsf h_j=2j$, $\eta_0=1$, $b=C^n_\wp$, $p=\wp n$ and $N\ge p+\frac12$, its height η being written t here; so, for $x=(x_1,\dots,x_n)\in(\mathbb R^{1+2})^n$ with $x_j=(x^0_j,\vec x_j)$, the point $z(x;t)$ of the Lemma is $(\iota(x_j)+2jt\,e_3)_{1\le j\le n}$ and its operator $D_{\mathsf h}$ is $\sum_{j=1}^n 2j\,\partial_{x^0_j}$, the derivative $\partial_{x^0_j}$ in the time of x_j standing for its ∂_{t_j}: at $z(x;t)$ the arguments $\iota(x_{k+1}-x_k)+2te_3+\mathfrak w_k$ have margins $\ge 2t$, so the bound of Theorem YM3_I:AO, $|\mathbf W_n|\le(C_\wp t^{-\wp})^n$ where all margins are $\ge 2t$ and $0<t\le 1$, gives the bound YM3_I:(2082). For $g\in C^\infty_c(\mathbb R^{3n};\mathbb C)$ let $J^{\mathfrak w}_g(t)=\int g(x)\,\mathbf F(z(x;t))\,dx$. The Lemma gives: $J^{\mathfrak w}_g$ is continuously differentiable on $(0,1]$, with the derivative $J^{\mathfrak w}_{-iD_{\mathsf h}g}$; $\sup_{0<t\le1}|J^{\mathfrak w}_g(t)|\le\mathsf Q(g)=2^NC^n_\wp\sum_{i\le N}\|D^i_{\mathsf h}g\|_{L^1}$, a number which does not depend on $\mathfrak w$; and $\lim_{t\downarrow0}J^{\mathfrak w}_g(t)$ exists. As $\sum_j\mathsf h_j=n(n+1)$ and $\|\partial^\alpha g\|_{L^1}\le\pi^{2n}|g|_{|\alpha|+4n}$ (integrate the majorant $\prod_{j=1}^n(1+|x_j|^2)^{-2}$), $\mathsf Q(g)\le c_n|g|_{\bar k}$; and $|J^{\mathfrak w}_g(t)-J^{\mathfrak w}_g(t')|\le\mathsf Q(D_{\mathsf h}g)|t-t'|$ for all $\mathfrak w$. If $\mathfrak w\in\mathsf{Tb}_\circ^{n-1}$, the margins stay $\ge\min_k d_\circ(\mathfrak w_k)>0$ at $t=0$, and $J^{\mathfrak w}_g(t)\to I_g(\eta)$ as $t\downarrow0$; put $\mathsf L(g)=\lim_{t\downarrow0}J^0_g(t)$. Then $|I_g(\eta)|\le\mathsf Q(g)$, $|I_g(\eta)-J^{\mathfrak w}_g(t)|\le\mathsf Q(D_{\mathsf h}g)\,t$ and $|\mathsf L(g)-J^0_g(t)|\le\mathsf Q(D_{\mathsf h}g)\,t$; and at a fixed t, $J^{\mathfrak w}_g(t)\to J^0_g(t)$ as $\mathfrak w\to0$, by uniform continuity on a compact set. Hence $\limsup_{\eta\to0}|I_g(\eta)-\mathsf L(g)|\le 2\,\mathsf Q(D_{\mathsf h}g)\,t$ for every $t\in(0,1]$, so $I_g(\eta)\to\mathsf L(g)$ as $\eta\to0$ in V_+^{n-1}. At a tensor $g=f_1\otimes\cdots\otimes f_n$ of functions of C^∞_c, $J^0_g(t)$ is the integral of (S1) at $\epsilon=2t$, as $\iota(x+i\epsilon e_0)=\iota(x)+\epsilon e_3$; so $\mathsf L(g)=\mathcal W_n(f_1,\dots,f_n)=\mathcal W^\sigma_n(g)$ (Theorem YM3_I:AO). Both $\mathsf L$ and $\mathcal W^\sigma_n$ are bounded for one norm, and sums of such tensors are dense (cut off, convolve with a product mollifier, then use Riemann sums); so $\mathsf L=\mathcal W^\sigma_n$ on C^∞_c. For $F\in\mathcal S(\mathbb R^{3n})$ and $\eta\in V_+^{n-1}$ let $t_\eta=\frac12\min\{2,\min_k d_\circ(\mathfrak w_k)\}$; the bound $|\mathbf W_n|\le(C_\wp t_\eta^{-\wp})^n$ at the arguments of $I_F(\eta)$ and $\|F\|_{L^1}\le\pi^{2n}|F|_{4n}$ make the integral absolutely convergent, and the functions F_ϱ give $|I_F(\eta)|=\lim_\varrho|I_{F_\varrho}(\eta)|\le c_n|F|_{\bar k}$. With $|\mathcal W^\sigma_n(F)|\le c'|F|_{k'}$ and $k''=\max\{\bar k,k'\}$, $|I_F(\eta)-\mathcal W^\sigma_n(F)|\le(c_n+c')|F-F_\varrho|_{k''}+|I_{F_\varrho}(\eta)-\mathcal W^\sigma_n(F_\varrho)|$; choose ϱ by Schwartz convergence of the smooth cutoffs, then $|\eta|$ small. This is the strong sense. Finally $\mathfrak S_1=\mathbf W_1$ by (S1) at $n=1$, and the kernel $\mathsf s_n$ is continuous and symmetric, equals $\mathbf W_n(x_2-x_1,\dots,x_n-x_{n-1})=W_{n-1}(\zeta(x))$ at the time-ordered points, and gives σ (Proposition YM3_I:108.3, YM3_I:(2067), YM3_I:(2065)); so it is the function of (Sch).

Step 11: (b8). On $\mathbb R^{1+2}$, P keeps x^0 and $|\vec x|$, so $P\in\mathrm O(1,2)$, $Pe_0=e_0$, $PV_+=V_+$ and $|\det P|=1$. The map $\Lambda\mapsto P\Lambda P^{-1}$ of $\mathrm O(1,2)$ into itself is continuous and fixes 1, so it maps the connected component $\mathcal L$ of 1 into $\mathcal L$, and so does $\Lambda\mapsto P^{-1}\Lambda P$; hence $P\mathcal LP^{-1}=\mathcal L$, and $(a,\Lambda)\mapsto(Pa,P\Lambda P^{-1})$ is a continuous automorphism of $\mathbb R^{1+2}\rtimes\mathcal L$. Put $U'(a,\Lambda)=U(Pa,P\Lambda P^{-1})$ and $\phi'(f)=\phi(f\circ P^{-1})$ on $\mathcal D_0$; $f\circ P^{-1}\in\mathcal S$, with a bound of its norms by those of f. Then $(\mathcal H(\sigma),U',\Omega,\mathcal D_0,\phi')$ is a Wightman theory: the projection-valued measure of $x\mapsto U'(x,1)$ is $\Delta\mapsto\mathsf E(P\Delta)$, as $(Px)\cdot p=x\cdot(P^{-1}p)$, and it is carried by $\overline{V_+}$; the vectors invariant under all $U'(a,1)$ are those invariant under all $U(a,1)$; (W2) holds by the bound just named and $\overline{f\circ P^{-1}}=\bar f\circ P^{-1}$; (W3) holds, as $U'(a,\Lambda)\phi'(f)U'(a,\Lambda)^{-1}=\phi(f(\Lambda^{-1}(P^{-1}\cdot-a)))=\phi'(f(\Lambda^{-1}(\cdot-a)))$ on $\mathcal D_0$; (W4) holds, as P keeps $(x-x')\cdot(x-x')$; and (W5) holds, as $f\mapsto f\circ P^{-1}$ maps $\mathcal S$ onto itself. Its Schwinger functions are those of σ in the sense in the Euclidean variables, with $\mathsf W'_1=\mathbf W_1$ and

$\mathsf{W}'_n = \mathbf{W}_n \circ (\hat{\mathsf{R}} \times \cdots \times \hat{\mathsf{R}})$, $\hat{\mathsf{R}} = \mathsf{R} \oplus 1$: $\hat{\mathsf{R}}$ keeps $\mathsf{Tb}_\circ$, the margin, the modulus and the set of the growth bound; $\iota P = \hat{\mathsf{R}}\iota$; the substitution $x_k = Py_k$ turns the forms of the second theory into (S1) with W'_n; and the substitution $x_k = \hat{\mathsf{R}}^{-1}y_k$, the tuple $\hat{\mathsf{R}}X$ having supports ordered in the third coordinate, and the invariance (Rec1) of σ give (S2). The last sentence of (b5) gives exactly one unitary V_R with $\mathsf{V}_\mathsf{R}\Omega = \Omega$ and $\mathsf{V}_\mathsf{R}\phi(f)\psi = \phi'(f)\mathsf{V}_\mathsf{R}\psi$ on $\mathcal{D}_0$, and $\mathsf{V}_\mathsf{R}U(a,\Lambda) = U'(a,\Lambda)\mathsf{V}_\mathsf{R}$.

Step 12: (c1). By the moment–cumulant formula the self-centred moments of the defining sequence converge to σ_S, with first moment zero and higher cumulants S_n. Theorem YM3_I:AP applies to this sequence. It gives symmetry, Euclidean invariance, reflection positivity, clustering and (E0′)–(E4) for every σ_ξ. It also gives the bound $2(\mathsf{C}_\mathrm{E}^{\mathrm{pt},g})^n(n!)^8|\varphi|_{8n}$ for the Schwinger functions, with a common constant for all components. Since $|\varphi|_{8n} \le |\varphi|_{10n}$, $2 \le 2^n$ for $n \ge 1$, and $8 \le \varkappa_\star + 9$, this also gives the bound $(\mathsf{C}_\mathrm{E}^g)^n(n!)^{\varkappa_\star+9}|\varphi|_{10n}$. Clauses (a8) and (a2) now give the two sets of numbers in (c1). The time direction: let $v_3 > 0$, $\varphi \in \mathcal{S}_+(\mathbb{R}^{3n})$, $\varphi' \in \mathcal{S}_+(\mathbb{R}^{3n'})$, and let $\psi, \psi' \in \mathcal{H}(\sigma_\xi)$ be the images under V^{-1} of the classes of the sequences with these single members. As $\tau_{sv} = \tau_{sv_3e_3}\tau_{(s\vec{v},0)}$, the reflection-space identification of Proposition YM3_I:111.2 gives the following identity. It holds first on positive-time tuple tensors by the definition of the translations on tuple classes, and extends by their density and the continuous reflection form. Thus $\mathsf{S}_{n+n'}(\Theta\bar{\varphi} \otimes \tau_{sv}\varphi') = \langle \psi, \mathcal{T}_{sv_3}\mathcal{U}(s\vec{v})\psi'\rangle$. By the spectral theorem, $\mathcal{T}_t\psi \to P_0\psi = \langle\Omega,\psi\rangle\Omega$ as $t \to \infty$, as $\ker H = \mathbb{C}\Omega$ (Theorem YM3_I:AP); $\mathcal{U}(s\vec{v})$ is unitary and fixes Ω; so the limit is $\langle\psi,\Omega\rangle\langle\Omega,\psi'\rangle = \mathsf{S}_n(\Theta\bar{\varphi})\,\mathsf{S}_{n'}(\varphi')$, as $\mathsf{V}\Omega = [(1,0,0,\dots)]$. For $v_3 = 0$ this is (E4).

Step 13: (c3), (c4). By Theorem YM3_I:AP, σ_ξ is data (Rec) with (Rec5). Theorem YM3_I:AO gives the Wightman theory, with $\mathsf{W}_n = \mathbf{W}_n$ and the tempered forms $\mathcal{W}_n^{\sigma_\xi}$; (b2) gives (GW1)–(GW8) and the variants; (b6) applies, the quintuple having (T); (b7) gives the strong sense with its bound and the Schwinger functions $\mathfrak{S}_1 = \mathbf{W}_1$, $\mathfrak{S}_n = \mathsf{s}_n$, which give σ_ξ by YM3_I:(2065); and $\mathbf{W}_1 = \mathfrak{c}_\xi$ by Proposition YM3_I:108.3 and the one-point statement of Theorem YM3_I:AP. The family of (b7) is a family in the weak sense, and (b5) gives the uniqueness, in its first form for the theories with a family in the weak sense and in its last sentence for the theories whose Schwinger functions are those of σ_ξ; (b8) applies. For (c4), the form of (OS-E2) on $\underline{\mathscr{S}}_+$ is the restriction of the form $\langle\cdot,\cdot\rangle_\mathcal{S}$ on the larger positive-time spaces, by the permutation of Step 2 and (E3). So the space of List OS is, up to a unitary map, the closure in $\mathcal{H}^\mathcal{S}$ of the classes of the members of $\underline{\mathscr{S}}_+$, which is $\mathcal{H}^\mathcal{S}$ by (a9), whose hypotheses Step 12 and (Rec) give; and V^{-1} carries $\mathcal{H}^\mathcal{S}$ onto $\mathcal{H}(\sigma_\xi)$ and $[(1,0,0,\dots)]$ to Ω (the reflection-space identification of Proposition YM3_I:111.2).

Step 14: (c2). Theorem YM3_I:AP also applies to the full limit σ_S. It gives (E0′)–(E3) and (Rec) with the numbers $C_\wp^g, 4, \frac{1}{2}$. Clause (a8) gives the corresponding List OS properties. Theorem YM3_I:AO reconstructs the quintuple for σ_S with the invariant vacuum, and the comparison of Step 5 gives (GW1), (GW2), (GW4)–(GW8) and the existence in (GW3). (i)⇔(iv): by Theorems YM3_I:AP and YM3_I:AL, $\ker H$ is unitarily equivalent to $L^2(\mathsf{X},\lambda)$, and λ charges every non-empty open set (Theorem YM3_I:AL); so $\dim\ker H = 1$ exactly when X is one point. (i)⇒(ii): the plateau identity at $q = 1$ (Theorem YM3_I:AL) gives $\sigma_S = \sigma_\xi$, and Step 12 applies. (ii)⇒(iii): (a6). (iii)⇒(i): let X be a smooth real separated n-tuple over $\{x_3 > 0\}$ whose supports are ordered in the third coordinate; $\bigotimes X$ has a compact support in $O_+^{(n)}$, so it lies in $\mathscr{S}_+(\mathbb{R}^{3n})$. (OS-E4) at the two sequences with the single member $\bigotimes X$, with (E3) and the values at tensors, gives $\sigma_S(\theta X \cdot \tau_{-\lambda a}X) \to \sigma_S(\theta X)\,\sigma_S(X) = \sigma_S(X)^2$ as $\lambda \to \infty$; the plateau identities at $q = 2$ and at $q = 1$ and the invariance of every component under θ (Theorems YM3_I:AP and YM3_I:AL) give $\int(\sigma_\xi(X) - \sigma_S(X))^2\lambda(d\xi) = 0$, and, $\xi \mapsto \sigma_\xi(X)$ being continuous, $\sigma_\xi(X) = \sigma_S(X)$ for every ξ (Theorem YM3_I:AL). Let $\xi, \xi' \in \mathsf{X}$; both components are data (Rec) (Theorem YM3_I:AP), with the kernels $\mathsf{s}_n, \mathsf{s}'_n$ of Proposition YM3_I:108.3, and $\mathsf{d} = \mathsf{s}_n - \mathsf{s}'_n$ is continuous on $(\mathbb{R}^3)^n_{\neq}$. At a point x^* with $0 < x^*_{13} < \cdots < x^*_{n3}$ the tuples $(j_\delta(\cdot - x^*_k))_k$, for small $\delta > 0$, are smooth, over $\{x_3 > 0\}$, with supports ordered in the third coordinate; so $\int \mathsf{d}(x)\prod_k j_\delta(x_k - x^*_k)\,dx = 0$ by YM3_I:(2065), and $\mathsf{d}(x^*) = 0$ by continuity as $\delta \downarrow 0$. The

kernels are invariant under the translations (Proposition YM3_I:108.3), so $\mathsf{d} = 0$ at all the time-ordered points, and, both kernels being continuous and symmetric, on $(\mathbb{R}^3)^n_{\neq}$ (b4). By YM3_I:(2065), $\sigma_\xi = \sigma_{\xi'}$ on every $\mathfrak{S}_n$, and the injectivity of $\xi \mapsto \sigma_\xi$ (Theorem YM3_I:AL) gives $\xi = \xi'$. (iv)⇒(v): Theorem YM3_I:AO. (v)⇒(iv): for $\psi \in \ker H$ the spectral measure m_ψ of $\mathcal{P}$ is carried by $\{E = 0\}$, as $\int E^2 dm_\psi = \|H\psi\|^2 = 0$, and by (Rec3) by the point $(0,0)$; by the translation spectral representation, $\|U(x)\psi - \psi\|^2 = \int |e^{i(tE+\vec{x}\cdot q)} - 1|^2 \, dm_\psi = 0$; so every vector of $\ker H$ is invariant under the translations, and the uniqueness of (GW3) gives $\ker H = \mathbb{C}\Omega$. □

18.23. **Unbroken lattice symmetries and odd densities; thread laws at every label; compact photons; limits on a slab.** This subsection adds five groups of statements to the preceding one. Proposition 18.227 concerns an arbitrary periodic Gibbs state ν of the SU(2) Wilson weight, at every spacing and every $\beta \geq 0$: the conditional expectation given the invariant events of a bounded Borel function does not change when a lattice isometry is applied to the function, every invariant event is carried onto itself by every lattice isometry up to a ν-null set, and almost every member of a regular conditional distribution of ν given the invariant events is a pure phase which is invariant under all lattice isometries and reflection positive at all lattice mirrors; a function which a lattice isometry multiplies by a number $\omega \neq 1$ has the mean 0 in ν, given the invariant events, in every pure phase and in almost every member of the decomposition. Proposition 18.228 concerns a bounded function of the bonds of a box which is odd, or even, under the one mirror through the centre of the box whose normal belongs to one of the nine families: in every periodic Gibbs state and in each of its pure phases, its correlation with its translate by j steps of the lattice along the normal is ≤ 0, respectively ≥ 0, for every integer j at least twice the radius of the box, and an odd function has the mean 0. Propositions 18.235 and 18.236 concern the thread measures of Definition 18.207 at an arbitrary label $\mathsf{b} > 0$, which Lemma 18.230 forms at every label and on every side: the laws of the threads over a window of two tori of sides ℓ_1 and $L^t\ell_1$ are mutually absolutely continuous, with explicit inequalities in both directions and densities in L^r both ways; and every limit law of the block fields of all depths through periodic Gibbs states whose labels tend to b, every invariant density of it, and almost every member of every decomposition of it into laws invariant under one translation, is dominated in every cell by the thread measure of one torus, with a constant formed from the constants, the label and the side. Proposition 18.237 concerns three compact photon fields: the determinant of the three field strengths at a site, smeared with a test function, has vanishing joint moments and joint cumulants with any number of smeared action densities, at every spacing, every coupling and in every law which is invariant under the conjugation of one colour. Proposition 18.247 concerns the slab $\mathbb{R}^2 \times S^1_l$ whose circumference l is a side at the reference coupling: along first-passage data whose labels converge, every limit point of the Wilson measures of the boxes with two growing sides and the third side l yields, along a subsequence, limits of the cumulants of the action density at all separated tuples over the slab; every such limit is symmetric, obeys explicit bounds, is invariant under all translations of the slab, under the signed permutations of the two open directions and under the reflections of the circle, is regular, and is reflection positive at the mirrors $\{x_1 = c\}$, $\{x_2 = c\}$ and $\{x_1 \pm x_2 = c\}$, $c \in \mathbb{R}$, and at every double mirror $\{x_3 = c\} \cup \{x_3 = c + \frac{l}{2}\}$; and below the ceiling γ_W its two-point function at short distance is $\mathsf{c}_2\mathsf{b}^{-2}\mathcal{K}_{\text{free}}$ up to a relative error which tends to 0 with the distance.

We use the setting and notation of Definition 18.209: the constants with $L \geq 13$ odd, $a_d = L^{-d}$, $\mathbb{L}_K = a_K\mathbb{Z}^3$, the sides $\mathfrak{L}$ and $\mathfrak{L}(g)$, the data at g with their labels, the intervals $I(g)$, J_g, $I^{\text{lab}}(g)$, the numbers C_{S}, C_g, C_{gap}, $\bar{B}$, $B_\sharp$, M, γ_W, c_2 and $\mathcal{K}_{\text{free}}$; the configurations, the laws, the action of $\mathsf{Iso}(a\mathbb{Z}^3) = a\mathbb{Z}^3 \rtimes W_3$ on configurations and on their functions, $(uU)(b) = U(u^{-1}b)$ and $(uF)(U) = F(u^{-1}U)$, the sets $\mathcal{F}(B)$, the Gibbs kernels γ_B, the periodic Gibbs states $\mathcal{G}^{\text{per}}_{a,\beta}$, the σ-algebra $\mathcal{I}$ of the invariant events and the pure phases (Definition 18.204); the torus measures $\mu_{K,\ell,\beta}$, $\tilde{\mu}_{K,\ell,\beta}$, the period lattices and their Wilson laws $\mu^0_{\Gamma,\beta} = \mu^{\varnothing}_{\Gamma,\beta}$ (Definitions 18.204 and 18.209); the normals $\mathfrak{N}$, the nine families of mirrors, the lattice mirrors of $a\mathbb{Z}^3$ and the reflections $\theta_{n,c}$ (Definition 18.209);

the block fields, the block maps $\mathcal{M}_\ell$ and $\mathcal{M}$, the towers Ψ_K, the threads Thr, the maps T_v, the window $\square$ of a side, its towers Υ_K, the spaces Y, $Y^{[d_0]}$, $Y_{\le D}$, the maps q_D, $\mathrm{pj}^{[d_0]}$, $\varpi_o^{[d_0]}$, the window maps m_d^n and $\mathsf{Thr}_\square$ (Definition 18.206); the thread measures $\mu_{\ell,\mathsf{b}}$, $\mu[\ell,\mathsf{b}]$, $\mu^{[d_0]}$, the domination $\preceq_{\mathsf{Q}}$, the product bound and Q_0 (Definition 18.207); and the standing hypothesis (SH) (Definition 18.208). In particular every Borel probability measure on the configurations of a lattice is called a law, and $\nu(F)$ is the integral of F under ν.

Use $\mathsf{Bd} = \mathsf{Bd}(\mathbb{R}^3)$ and the local function spaces of Definition 18.209. For $B \subset \mathsf{Bd}$, U_B is the restriction to B, and for finite B, dV_B is product Haar measure. Put $Q_r = [-ra, ra]^3$. The linear part of $\theta_{n,c}$ is $\varrho_n x = x - 2|n|^{-2}(n \cdot x)n$. Write $\mathrm{Adm}(n)$ for its allowed offsets, $(a/2)\mathbb{Z}$ for coordinate and $a\mathbb{Z}$ for diagonal normals. For $u \in \mathsf{Iso}(a\mathbb{Z}^3)$ the event uE is the image of E under u, so $u1_E = 1_{uE}$. An $\mathcal{I}$-representative of F under ν is an $\mathcal{I}$-measurable representative of $\mathbb{E}_\nu[F \mid \mathcal{I}]$, characterized by $\nu(1_E \mathsf{g}) = \nu(1_E F)$ for all $E \in \mathcal{I}$. A regular conditional distribution $(\nu^U)_U$ given $\mathcal{I}$ is a family of laws for which $U \mapsto \nu^U(F)$ is such a representative for each bounded Borel F. Proposition YM3_I:106.2 gives one for every periodic Gibbs state.

Proposition 18.227 (No lattice isometry is broken in a periodic Gibbs state or in its pure phases)**.** Let $a > 0$, $\beta \ge 0$ and $\nu \in \mathcal{G}^{\mathrm{per}}_{a,\beta}$.

(a) For every bounded Borel function F and every $u \in \mathsf{Iso}(a\mathbb{Z}^3)$, every $\mathcal{I}$-representative of F under ν is an $\mathcal{I}$-representative of uF under ν; hence $\mathbb{E}_\nu[uF|\mathcal{I}] = \mathbb{E}_\nu[F|\mathcal{I}]$ ν-almost surely. This holds in particular for the reflection $\theta_{n,c}$ of every lattice mirror of the nine families and for the parity $x \mapsto -x$.

(b) For every $E \in \mathcal{I}$ and every $u \in \mathsf{Iso}(a\mathbb{Z}^3)$, $\nu(uE \triangle E) = 0$.

(c) Let F be a bounded Borel function, $u \in \mathsf{Iso}(a\mathbb{Z}^3)$ and $\omega \in \mathbb{C} \setminus \{1\}$ with $uF = \omega F$. Then $\nu(F) = 0$, $\mathbb{E}_\nu[F|\mathcal{I}] = 0$ ν-almost surely, and $\nu'(F) = 0$ for every pure phase $\nu' \in \mathcal{G}^{\mathrm{per}}_{a,\beta}$.

(d) Let $(\nu^U)_U$ be a regular conditional distribution of ν given $\mathcal{I}$. Then $\nu(F) = \int \nu^U(F)\,\nu(dU)$ for every bounded Borel F, and for ν-almost every U the law ν^U is a pure phase in $\mathcal{G}^{\mathrm{per}}_{a,\beta}$; it is invariant under $\mathsf{Iso}(a\mathbb{Z}^3)$, reflection positive at every lattice mirror of the nine families, and $\nu^U(F) = 0$ for every bounded Borel F with $uF = \omega F$ for some $u \in \mathsf{Iso}(a\mathbb{Z}^3)$ and some $\omega \ne 1$. So no lattice isometry permutes the pure phases into which ν decomposes.

■

Proof. Step 1: conditioned states. Let $E \in \mathcal{I}$ with $\nu(E) > 0$. The function $\mathsf{h} = \nu(E)^{-1}1_E$ is bounded, non-negative and $\mathcal{I}$-measurable with $\nu(\mathsf{h}) = 1$, so $\mathsf{h}\nu \in \mathcal{G}^{\mathrm{per}}_{a,\beta}$ by Proposition YM3_I:106.1; by the theorem on the nine families, Proposition YM3_I:103.4, $\mathsf{h}\nu$ is invariant under $\mathsf{Iso}(a\mathbb{Z}^3)$, and so $\nu(1_E\, uF) = \nu(1_E F)$ for every bounded Borel F and every $u \in \mathsf{Iso}(a\mathbb{Z}^3)$. If $\nu(E) = 0$ both members vanish.

Step 2: (a). Let g be an $\mathcal{I}$-representative of F. For $E \in \mathcal{I}$, Step 1 gives $\nu(1_E\, uF) = \nu(1_E F) = \nu(1_E \mathsf{g})$, so g is an $\mathcal{I}$-representative of uF. As ν is invariant under the translations of $a\mathbb{Z}^3$ (Proposition YM3_I:103.4), the $\mathcal{I}$-representatives are the $\mathcal{I}$-measurable members of the conditional expectations, and the two conditional expectations are one class. The reflections of the lattice mirrors belong to $\mathsf{Iso}(a\mathbb{Z}^3)$, and the parity is the element -1 of W_3.

Step 3: (b). The function 1_E is $\mathcal{I}$-measurable, so it is an $\mathcal{I}$-representative of itself, and by (a) of $u1_E = 1_{uE}$. So $\nu(1_{E'}1_{uE}) = \nu(1_{E'}1_E)$ for all $E' \in \mathcal{I}$; at $E' = E^c$ this gives $\nu(uE \setminus E) = 0$, and at $E' = E$ it gives $\nu(uE \cap E) = \nu(E)$, that is, $\nu(E \setminus uE) = 0$.

Step 4: (c). By Proposition YM3_I:103.4, $\nu(F) = \nu(uF) = \omega\,\nu(F)$, so $\nu(F) = 0$. If g is an $\mathcal{I}$-representative of F, then $\omega\mathsf{g}$ is one of $\omega F = uF$, and by (a) so is g; two of them agree almost everywhere, so $(1-\omega)\mathsf{g} = 0$ almost everywhere. A pure phase belongs to $\mathcal{G}^{\mathrm{per}}_{a,\beta}$, so the first assertion applies to it.

Step 5: (d). The identity $\nu(F) = \int \nu^U(F)\,\nu(dU)$ is YM3_I:(2029), and by Proposition YM3_I:106.2 the law ν^U belongs to $\mathcal{G}^{\mathrm{per}}_{a,\beta}$ and is ergodic for ν-almost every U; by Proposition YM3_I:106.1

the ergodic members of $\mathcal{G}^{\mathrm{per}}_{a,\beta}$ are its pure phases. Proposition YM3_I:103.4 and clause (c) of the present proposition apply to each such ν^U, which gives the remaining assertions; the last sentence is (b) together with this. □

Proposition 18.228 (A density odd, or even, under the mirror through the centre of its box)**.** Let $a > 0$, $\beta \geq 0$, $\nu \in \mathcal{G}^{\mathrm{per}}_{a,\beta}$, let $r \geq 1$ be an integer, let A be a bounded complex Borel function of the bonds lying in $Q_r = [-ra, ra]^3$, and let $n \in \mathfrak{N}$ and $\epsilon \in \{1, -1\}$ with $\theta_{n,0}A = \epsilon A$.

(a) For every integer $j \geq 2r$ the number $\nu(\bar{A}\,\tau_{jan}A)$ is real, and $\epsilon\,\nu(\bar{A}\,\tau_{jan}A) \geq 0$. In particular, if A is real and odd, $\theta_{n,0}A = -A$, then $\nu(A\,\tau_{jan}A) \leq 0$ for all integers $j \geq 2r$; for $n = e_\mu$ these are the translations by te_μ with $t \in a\mathbb{Z}$, $t \geq 2ra$, and for a diagonal $n = e_\mu \pm e_{\mu'}$ the translations by $ja(e_\mu \pm e_{\mu'})$.

(b) If $\epsilon = -1$, then $\nu(A) = 0$, $\mathbb{E}_\nu[A|\mathcal{I}] = 0$ ν-almost surely, $\nu'(A) = 0$ for every pure phase $\nu' \in \mathcal{G}^{\mathrm{per}}_{a,\beta}$, and $\nu^U(A) = 0$ for ν-almost every U, for every regular conditional distribution $(\nu^U)_U$ of ν given $\mathcal{I}$. Hence, for real A and integers $j \geq 2r$, the connected correlation $\nu(A\,\tau_{jan}A) - \nu(A)\,\nu(\tau_{jan}A)$ equals $\nu(A\,\tau_{jan}A)$ and is ≤ 0.

(c) The inequality of (a) holds in every pure phase $\nu' \in \mathcal{G}^{\mathrm{per}}_{a,\beta}$ in the place of ν; and every $\mathcal{I}$-representative g of $\bar{A}\,\tau_{jan}A$ under ν, $j \geq 2r$, is real and satisfies $\epsilon\,\mathsf{g} \geq 0$ ν-almost everywhere. ■

Proof. Step 1: the mirror midway. Let $j \geq 2r$ be an integer and $c' := \frac{1}{2}ja|n|^2$. For $n = e_\mu$, $c' = \frac{1}{2}ja \in \frac{a}{2}\mathbb{Z}$; for a diagonal n, $|n|^2 = 2$ and $c' = ja \in a\mathbb{Z}$; so (n, c') is a lattice mirror of the nine families. As maps of $\mathbb{R}^3$, $\theta_{n,c'}x = \theta_{n,0}x + 2c'|n|^{-2}n$, that is, $\theta_{n,c'} = \tau_{jan}\theta_{n,0}$; the linear map $\theta_{n,0}$ satisfies $\theta_{n,0}\tau_v = \tau_{\theta_{n,0}v}\theta_{n,0}$ for $v \in a\mathbb{Z}^3$, and $\theta_{n,0}v = -v$ for v parallel to n, so that $\theta_{n,0}\tau_v = \tau_{-v}\theta_{n,0}$ for such v. On functions of the configurations, $(\mathsf{rs})F = \mathsf{r}(\mathsf{s}F)$ for $\mathsf{r}, \mathsf{s} \in \mathsf{Iso}(a\mathbb{Z}^3)$.

Step 2: the translate lies on one side. Put $F := \tau_{jan}A$. It depends on the bonds $b + jan$, b a bond lying in Q_r, whose closed segments lie in $Q_r + jan$. For $y \in Q_r + jan$, $|y_\iota - jan_\iota| \leq ra$ for each ι, and $\sum_\iota |n_\iota| = |n|^2$ for $n \in \mathfrak{N}$; so

$$n \cdot y - c' \geq ja|n|^2 - ra|n|^2 - \tfrac{1}{2}ja|n|^2 = (\tfrac{1}{2}j - r)a|n|^2 \geq 0.$$

Hence $F \in \mathcal{F}_{\mathrm{loc}}(\mathbb{H}^+(n, c'))$.

Step 3: (a). By Step 1, $\theta_{n,c'}F = \tau_{jan}\theta_{n,0}\tau_{jan}A = \tau_{jan}\tau_{-jan}\theta_{n,0}A = \epsilon A$. The theorem on the nine families, Proposition YM3_I:103.4, at the mirror (n, c') gives

$$0 \leq \nu\big(\overline{\theta_{n,c'}F}\,F\big) = \epsilon\,\nu(\bar{A}\,\tau_{jan}A),$$

the number in the middle being real and non-negative. For real odd A this is $\nu(A\,\tau_{jan}A) \leq 0$.

Step 4: (b). The reflection $\theta_{n,0}$ belongs to $\mathsf{Iso}(a\mathbb{Z}^3)$, $0 \in \mathrm{Adm}(n)$, and $\theta_{n,0}A = -A$; Proposition 18.227 (c), (d) at $u = \theta_{n,0}$, $\omega = -1$ give the four assertions on the means. For real A, $\nu(\tau_{jan}A) = \nu(A) = 0$ by the invariance under translations (Proposition YM3_I:103.4), and (a) gives the sign.

Step 5: (c). A pure phase is a member of $\mathcal{G}^{\mathrm{per}}_{a,\beta}$, so (a) applies to it. Let $E \in \mathcal{I}$ with $\nu(E) > 0$; then $\nu(\,\cdot\,|E) \in \mathcal{G}^{\mathrm{per}}_{a,\beta}$ (Proposition YM3_I:106.1), and (a) for this state gives that $\epsilon\,\nu(1_E\,\bar{A}\,\tau_{jan}A)$ is real and ≥ 0; if $\nu(E) = 0$ it vanishes. With an $\mathcal{I}$-representative g, $\nu(1_E\,\epsilon\mathsf{g})$ is real and ≥ 0 for every $E \in \mathcal{I}$. At $E = \{\mathrm{Im}\,\mathsf{g} > 0\}$ and at $E = \{\mathrm{Im}\,\mathsf{g} < 0\}$, both in $\mathcal{I}$, the imaginary part of $\nu(1_E\mathsf{g})$ vanishes, so $\mathrm{Im}\,\mathsf{g} = 0$ almost everywhere; at $E = \{\epsilon\,\mathrm{Re}\,\mathsf{g} < 0\} \in \mathcal{I}$ the integrand is negative on E, so $\nu(E) = 0$. □

The one-point vanishing and the two-point sign pass to limits of these expressions whenever the relevant moments converge. The same argument applies in a component obtained as a limit through conditioned periodic Gibbs states, since the finite-spacing statements hold for every such state.

Thread laws at every label. Definition 18.207 forms the thread measure $\mu_{\ell,\mathsf{b}}$ of T_ℓ at labels b of an interval admitted by the side ℓ. We form it at every label by dilation, and we compare the thread laws of two tori in both directions. We use lattice units. A board (i,o) has the cells $o+a_i\mathbf{k}+[-a_i/2,a_i/2]^3$ with $\mathbf{k}\in\mathbb{Z}^3$, $o\in a_K\mathbb{Z}^3$ and $i\le K$. The walls are bond midplanes because a_i/a_K is odd. For a triple $\underline{N}=(N_1,N_2,N_3)$ of integers ≥ 2 let $\Lambda_{\underline{N}}:=\prod_\mu \mathbb{Z}/N_\mu\mathbb{Z}$, with the bonds (x,μ) from x to $x+e_\mu$, the configurations $U\in G^{\{\text{bonds}\}}$ with the product dU of the Haar measures, $a_p(U):=1-\frac12\operatorname{tr}[U(x,\mu)U(x+e_\mu,\nu)U(x+e_\nu,\mu)^{-1}U(x,\nu)^{-1}]$ for the plaquettes $p=(x;\mu,\nu)$, $\mu<\nu$, and $A_{\underline{N}}:=\sum_p a_p$; for $\beta>0$ and a bounded Borel $F\ge 0$ put $Z_{\underline{N}}[F]:=\int Fe^{-\beta A_{\underline{N}}}dU$, $Z_{\underline{N}}:=Z_{\underline{N}}[1]>0$, $\langle F\rangle_{\underline{N}}:=Z_{\underline{N}}[F]/Z_{\underline{N}}$ and $P_{\underline{N}}:=(N_1N_2N_3)^{-1}\log Z_{\underline{N}}$, and write $P_n:=P_{(n,n,n)}$. For an integer $\lambda\ge 2$ the model cell B_λ is the set of the bonds (x,μ) with x and $x+e_\mu$ in $\{0,\dots,\lambda-1\}^3$, read in $\Lambda_{\underline{N}}$ when all $N_\mu\ge\lambda+1$. For $K\in\mathbb{N}$ and a side $\ell=2L^m$, $m\in\mathbb{Z}$, $K+m\ge 1$, the map $x\mapsto x/a_K$ carries $a_K\mathbb{Z}^3/\ell\mathbb{Z}^3$ onto $\Lambda_{(n,n,n)}$, $n=\ell/a_K$, bonds onto bonds and plaquettes onto plaquettes; so $Z_{K,\ell,\beta}=Z_{(n,n,n)}$ at the same β, and $p_\ell:=\ell^{-3}\log Z_{K,\ell,\beta}$ equals $a_K^{-3}P_n$. A cell $\square$ of side $\ell'/2$, $\ell'=2L^{m'}\in\mathfrak{L}$, of a board of $\mathbb{L}_K$ holds λ^3 sites, $\lambda=L^{K+m'}$ odd, and no site lies on a wall; with its lowest corner site x_0 the chart $x\mapsto(x-x_0)/a_K$ carries the bonds lying in $\square$ onto B_λ, and for $F\in\mathcal{A}(\square):=\mathcal{F}(\mathsf{Bd}(\square))$, $F(U)=H(U_{\mathsf{Bd}(\square)})$, and every side $\ell\ge\ell'$, $\tilde\mu_{K,\ell,\beta}(F)=\langle H\rangle_{(\ell/a_K,\ell/a_K,\ell/a_K)}$, by $\tilde\mu_{K,\ell,\beta}(F)=\langle F\circ\mathsf{per}_\ell\rangle_{K,\ell,\beta}$ by periodic extension and the translation invariance of product Haar measure and the Wilson action.

The dilation number. For a real $\mathsf{b}>0$ and $\ell=2L^m\in\mathfrak{L}$ let $s=s(\mathsf{b},\ell)$ be the least integer $s\ge 0$ such that $\lambda_s:=L^s\mathsf{b}>\max\{2\gamma^{-2},2\bar B\}$ and $m+s-1\ge m_{\mathfrak{L}}(g_s^\circ)$, where $g_s^\circ:=(\lambda_s/2)^{-1/2}$; put $g^\circ:=g^\circ_{s(\mathsf{b},\ell)}$, $\lambda^\circ:=L^{s(\mathsf{b},\ell)}\mathsf{b}$,

$$\mathsf{D}^\circ(\mathsf{b},\ell):=\big(L^{s(\mathsf{b},\ell)}\ell\big)^3(2C_{\mathrm S}+C_{g^\circ}),\qquad \mathsf{Q}^\circ(\mathsf{b},\ell):=e^{\mathsf{D}^\circ(\mathsf{b},\ell)/8}.$$

The integer s exists: $g_s^\circ\le\gamma\le 1$, and the side rule YM3_I:(1528) gives $L^{m_{\mathfrak{L}}(g)}=LM\max\{R(L(g^{-2}+\bar B)),L^{\lceil\log_L C_{\mathrm s}\rceil}\}$ with $R(x)\le L\max\{1,(\log x)^r\}$, the least power of L that is $\ge(\log x)^r$; at $g^{-2}=\frac12L^s\mathsf{b}$ the right side grows at most like a power of s, while L^{m+s-1} grows like L^s. Three facts follow from the choice. (i) $g^\circ\le\gamma$, since $g^{\circ-2}=\lambda^\circ/2\ge\gamma^{-2}$. (ii) λ° lies in the interior of $I(g^\circ)=(g^{\circ-2}+\bar B,L(g^{\circ-2}+B_\sharp)-\bar B]$: $\lambda^\circ-\frac12\lambda^\circ-\bar B>0$ as $\lambda^\circ>2\bar B$, and $L(\frac12\lambda^\circ+B_\sharp)-\bar B-\lambda^\circ=(\frac L2-1)\lambda^\circ+LB_\sharp-\bar B>0$ as $L\ge 13$ and $\bar B=\frac{L}{L-1}B_\sharp\le LB_\sharp$. (iii) $L^{s-1}\ell$ and every larger side lie in $\mathfrak{L}(g^\circ)$. The numbers s, g°, D° and Q° are formed from the constants, b and ℓ alone; C_{g° is a function of σ^- and g° by Theorem YM3_I:AD.

Lemma 18.229 (Dilation)**.** Let $s\ge 0$ be an integer and $\delta_s\colon x\mapsto L^sx$.

(a) For $K\in\mathbb{Z}$, δ_s carries $\mathbb{L}_K$ onto $\mathbb{L}_{K-s}$, bonds onto bonds, plaquettes onto plaquettes with $a_{\delta_s p}(\delta_sU)=a_p(U)$, and configurations, bounded Borel functions and laws on $\mathbb{L}_K$ onto those on $\mathbb{L}_{K-s}$; $a_{K-s}\beta=L^sa_K\beta$. For $\beta>0$ it maps $\mathcal{G}^{\mathrm{per}}_{K,\beta}$ onto $\mathcal{G}^{\mathrm{per}}_{K-s,\beta}$ and carries $\tilde\mu_{K,\ell,\beta}$ to $\tilde\mu_{K-s,L^s\ell,\beta}$ for $\ell=2L^m$ with $K+m\ge 1$; if besides $K-s\ge 1$, and so $K\ge 1$ as well, it carries a board (i,o) of $\mathbb{L}_K$ with its cells $\square_{\mathbf k}$ to the board $(i-s,L^so)$ of $\mathbb{L}_{K-s}$ with the cells $L^s\square_{\mathbf k}$, and $\mathcal{A}(\square_{\mathbf k})$ onto $\mathcal{A}(L^s\square_{\mathbf k})$. On the tori, for every $\ell=2L^m$ with $K+m\ge 1$, it carries $a_K\mathbb{Z}^3/\ell\mathbb{Z}^3$ onto $a_{K-s}\mathbb{Z}^3/L^s\ell\mathbb{Z}^3$ and $\mu_{K,\ell,\beta}$ to $\mu_{K-s,L^s\ell,\beta}$; so $Z_{K,\ell,\beta}=Z_{K-s,L^s\ell,\beta}$, and ℓ^3p_ℓ at (K,β) equals $(L^s\ell)^3p_{L^s\ell}$ at $(K-s,\beta)$.

(b) Let $\ell\in\mathfrak{L}$ and $d\ge s$. Then $\delta_s\circ\mathcal{M}_\ell=\mathcal{M}_{L^s\ell}\circ\delta_s$, as maps from the configurations of $a_{d+1}\mathbb{Z}^3/\ell\mathbb{Z}^3$ to those of $a_{d-s}\mathbb{Z}^3/L^s\ell\mathbb{Z}^3$, and $\delta_s\circ\mathcal{M}=\mathcal{M}\circ\delta_s$ from Z_{d+1} to Z_{d-s}. Consequently, for $K\ge d\ge s$ and $\mathsf{b}'>0$, the law $\mu_d^K[\ell,\mathsf{b}']$ of $\mathcal{M}_\ell^{K-d}U$ under $\mu_{K,\ell,L^K\mathsf{b}'}$ satisfies $(\delta_s)_*\mu_d^K[\ell,\mathsf{b}']=\mu_{d-s}^{K-s}[L^s\ell,L^s\mathsf{b}']$. With the homeomorphisms $\Delta_s\colon\mathsf{Z}^{[s]}\to\mathsf{Z}$, $(\mathsf{z}_d)_{d\ge s}\mapsto(\delta_s\mathsf{z}_{d+s})_{d\ge0}$, and $\Delta_s\colon Y^{[s]}\to Y'$, where Y is formed at the side ℓ and Y' at the side $L^s\ell$ (so that δ_s carries w_d onto w'_{d-s}), one has $\Delta_s((\Psi_K(U)_d)_{d\ge s})=\Psi_{K-s}(\delta_sU)$ and $\Delta_s(\mathrm{pj}^{[s]}(\Upsilon_K(U)))=\Upsilon'_{K-s}(\delta_sU)$ for $K\ge s$ and every configuration U of $\mathbb{L}_K$.

■

Proof. (a) The dilation sends the sites, bonds and plaquettes bijectively to their counterparts and preserves each holonomy after relabelling. It also sends a finite-bond Gibbs kernel to the Gibbs kernel of the image set, since both the Wilson factors and the Haar product are carried to those of the image. A finite-index period subgroup is sent to a finite-index period subgroup. The torus maps have the same relabelling property and commute with periodic extension. In lattice units both tori have $n = \ell/a_K = L^s\ell/a_{K-s}$ sites along each axis. The board formula gives the stated scaling of its cells. (b) The three cases of the block map (Section YM3_I:4.3) form $\mathcal{M}_\ell(U)(c)$ at a bond c of the coarse lattice from the restriction of U to the bonds of the double block of c: from the small-field condition at the radius $\omega_M = \frac{1}{200L^2}$ on the plaquettes of the double block, from the averaged contour variables along the contours from the block centres, from the straight paths, from the means of finite families in G and from the logarithm and the exponential of G; all paths are sequences of unit steps of the fine lattice, written by integers relative to the end points of c, and neither the spacing nor the side enters any of these objects or the numerical constants of the Definition. The map δ_s carries the double block of c, with its bonds, plaquettes, contours and straight paths, onto the double block of $\delta_s c$, and keeps every value of the configuration; so the case of the Definition which applies, and the value, are the same. This is the reasoning of Lemma 18.211 (a), where the side is changed; the map $\mathcal{M}$ over $\mathbb{R}^3$ is read through the maps of the tori (Definition 18.206). The statements on the laws follow by induction on $K - d$ with (a); those on Ψ_K and Υ_K from their definitions and $\delta_s \circ \mathcal{M} = \mathcal{M} \circ \delta_s$, the members of the depths $> K$ and $> K - s$ being unit fields on both sides. □

Lemma 18.230 (The block-field laws and the thread measure at every label)**.** Let $\ell = 2L^m \in \mathfrak{L}$ and let $\mathsf{b} > 0$ be real.

(a) For every $d \geq 0$ the laws $\mu_d^K[\ell, \mathsf{b}]$ of $\mathcal{M}_\ell^{K-d}U$ under $\mu_{K,\ell,L^K\mathsf{b}}$ converge in total variation, as $K \to \infty$, to a Borel probability measure $\mu_d[\ell, \mathsf{b}]$ on $\mathsf{V}_d[\ell]$, and $(\mathcal{M}_\ell)_*\mu_{d+1}[\ell, \mathsf{b}] = \mu_d[\ell, \mathsf{b}]$. If b belongs to an interval of labels admitted by ℓ, these are the laws of Lemma 18.212 (a).

(b) There is exactly one Borel probability measure $\mu_{\ell,\mathsf{b}}$ on $\prod_{d\geq 0}\mathsf{V}_d[\ell]$ whose image at the depths $\leq D$ is the law of $(\mathcal{M}_\ell^D V, \dots, \mathcal{M}_\ell V, V)$ under $\mu_D[\ell, \mathsf{b}]$, for every D; it is carried by the threads of T_ℓ, and $\mu[\ell, \mathsf{b}] := (\varpi_0^{[0]} \circ \mathsf{Per}_\ell)_*\mu_{\ell,\mathsf{b}}$ is carried by $\mathsf{Thr}_\square$. This is the thread measure of T_ℓ at the label b, and $\mu^{[d_0]}[\ell, \mathsf{b}] := \mathrm{pj}_*^{[d_0]}\mu[\ell, \mathsf{b}]$.

(c) Let $K_j \to \infty$ and $\beta_j > 0$ with $a_{K_j}\beta_j \to \mathsf{b}$. Then the law of $\mathcal{M}_\ell^{K_j-d}U$ under μ_{K_j,ℓ,β_j} tends to $\mu_d[\ell, \mathsf{b}]$ in total variation, for every d. For every side $\ell' \geq \ell$ in $\mathfrak{L}$, with Υ_K and Y formed at the side ℓ, the laws $(\Upsilon_{K_j})_*\tilde{\mu}_{K_j,\ell',\beta_j}$ converge to $(\varpi_0^{[0]} \circ \mathsf{Per}_{\ell'})_*\mu_{\ell',\mathsf{b}}$ weakly on Y and in total variation at the depths $\leq D$, for every D; at $\ell' = \ell$ the limit is $\mu[\ell, \mathsf{b}]$.

■

Proof. Step 1: an admitted interval for the dilated side. Since $R(x) \leq L\max\{1, (\log x)^r\}$, there is an integer $s \geq 0$ with $\frac{1}{2}L^s\mathsf{b} \geq \gamma^{-2}$ and $M\,R(2L^s\mathsf{b}) \leq L^{m+s}$; then $\mathsf{J}_s := [\frac{1}{2}L^s\mathsf{b}, 2L^s\mathsf{b}]$ is an interval of labels under (Tor) on the side $L^s\ell$ by YM3_I:(2201), admitted by $L^s\ell$ in the sense of Definition 18.207, and Lemma 18.212 (a), (b), (d), (e) hold at the side $L^s\ell$ and the interval J_s.

Step 2: (a) and (c) at the depths $d \geq s$. By Lemma 18.229 (b), $\mu_d^K[\ell, \mathsf{b}']$ is the image under δ_s^{-1} of $\mu_{d-s}^{K-s}[L^s\ell, L^s\mathsf{b}']$ for $K \geq d$ and every $\mathsf{b}' > 0$; a bijection keeps the distance in total variation. Lemma 18.212 (a) at $L^s\mathsf{b} \in \mathsf{J}_s$ gives the convergence of these laws; and as $L^s a_{K_j}\beta_j \to L^s\mathsf{b}$, all but finitely many of these numbers lie in J_s, and Lemma 18.212 (e) at the side $L^s\ell$ gives the convergence along (K_j, β_j), with the same limit. The consistency $(\mathcal{M}_\ell)_*\mu_{d+1} = \mu_d$ for $d \geq s$ is the image of that of Lemma 18.212 (a).

Step 3: the depths $d < s$. Put $\mu_d[\ell, \mathsf{b}] := (\mathcal{M}_\ell^{s-d})_*\mu_s[\ell, \mathsf{b}]$. As $\mu_d^K = (\mathcal{M}_\ell^{s-d})_*\mu_s^K$ for $K \geq s$, and an image under one Borel map (Lemma 18.211 (b)) does not increase the distance in total variation,

Step 2 gives (a) and the first assertion of (c). If b lies in an interval admitted by ℓ, the limits of Lemma 18.212 (a) and the present ones are limits of one sequence in total variation, so they agree.

Step 4: (b) and the windows. The proofs of Lemma 18.212 (b), (d) and of the part of (e) on the windows read of the labels only the convergence in total variation of the laws of the block fields at every depth; with (a) and (c) in its place they give (b), and the convergence of $(\Upsilon_{K_j})_*\tilde{\mu}_{K_j,\ell',\beta_j}$ when the window is that of the side ℓ' (Kolmogorov's extension theorem; Lemma 18.211 (c), (d); images do not increase the distance in total variation; cylinder functions are dense). For the window of the smaller side ℓ, the restriction from the window of the side ℓ' to the window of the side ℓ is continuous, it carries the window towers and the threads over the larger window to those over the smaller one (Lemma 18.211 (d)), and it maps the functions of the depths $\le D$ to functions of the depths $\le D$. □

Lemma 18.231 (Kernels and sequences)**.** Let X be a compact metrisable space with a Borel probability measure dx, and let $\mathcal{K}$ be the set of the bounded Borel functions $X\times X\to[0,\infty)$. For $P,Q\in\mathcal{K}$ put $(P*Q)(x,z):=\int P(x,y)Q(y,z)\,dy$, $[P,Q]:=\iint P(x,y)Q(x,y)\,dx\,dy$, $P^{\mathsf{T}}(x,y):=P(y,x)$ and $\operatorname{tr}P:=\int P(x,x)\,dx$. Let $T\in\mathcal{K}$ be symmetric, $T^1:=T$, $T^{k+1}:=T^k*T$, and $z(k):=\operatorname{tr}T^k$.

(k1) $*$ is associative and monotone in each factor, $T^j*T^k=T^{j+k}$, every T^k is symmetric, $[P,Q]=\operatorname{tr}(P*Q^{\mathsf{T}})$, and $[P,Q]^2\le[P,P]\,[Q,Q]$.

(k2) $z(j+k)^2\le z(2j)\,z(2k)$ for $j,k\ge1$, and $z(4k)\le z(2k)^2$ for $k\ge1$.

(k3) For $P,R\in\mathcal{K}$ put $Y:=P^{\mathsf{T}}*P$, $S:=R*R^{\mathsf{T}}$, $g(0):=[R,R]$ and $g(j):=[Y^j,S]$ for $j\ge1$. Then $g(0)=\operatorname{tr}S$, $g(j)^2\le g(j-1)\,g(j+1)$ for $j\ge1$, $g(k)\le\operatorname{tr}(Y^k)\operatorname{tr}S$, and $g(1)\le\operatorname{tr}S\,(\operatorname{tr}Y^k)^{1/k}$ for every $k\in\mathbb{N}$.

(c1) If $f\colon\{0,\dots,n_2\}\to[0,\infty)$ satisfies $f(n)^2\le f(n-1)f(n+1)$ for $0<n<n_2$, then $f(n)\le f(0)^{1-n/n_2}f(n_2)^{n/n_2}$ for $0\le n\le n_2$, with $0^0:=1$.

(c2) If $w\colon\mathbb{N}\to\mathbb{R}$ satisfies $2w(k)\le w(k-1)+w(k+1)$ for $k\ge2$ and $w(2k)\le2w(k)$ for $k\ge1$, then $w(n)/n\le w(k)/k$ for all $n\ge k\ge1$.

■

Proof. (k1) is Fubini's theorem for non-negative bounded Borel functions, and the inequality is that of Cauchy and Schwarz in $L^2(X\times X)$. (k2) $[T^j,T^k]=\operatorname{tr}(T^j*T^k)=z(j+k)$, and (k1). Next $T^{2k}(x,y)=\int T^k(x,u)T^k(y,u)\,du\le T^{2k}(x,x)^{1/2}T^{2k}(y,y)^{1/2}$, so $z(4k)=\iint T^{2k}(x,y)^2\le(\int T^{2k}(x,x)\,dx)^2$. (k3) The powers Y^j are symmetric; put $Y^0*R:=R$. For $j=2i+1$, integrating the variables of $S=R*R^{\mathsf{T}}$ first, $[Y^i*R,Y^{i+1}*R]=g(2i+1)$, $[Y^i*R,Y^i*R]=g(2i)$ and $[Y^{i+1}*R,Y^{i+1}*R]=g(2i+2)$; for $j=2i$, $i\ge1$, $[P*Y^{i-1}*R,P*Y^i*R]=g(2i)$, $[P*Y^{i-1}*R,P*Y^{i-1}*R]=g(2i-1)$ and $[P*Y^i*R,P*Y^i*R]=g(2i+1)$; in both cases the inequality is (k1). $g(0)=\iint R(x,w)^2=\operatorname{tr}S$. A kernel $W=Q^{\mathsf{T}}*Q$ has $W(x,y)\le W(x,x)^{1/2}W(y,y)^{1/2}$; S is of this form with $Q=R^{\mathsf{T}}$, and Y^k with $Q=Y^{k/2}$ for even k and $Q=P*Y^{(k-1)/2}$ for odd k; so $g(k)\le(\int Y^k(x,x)^{1/2}S(x,x)^{1/2}dx)^2\le\operatorname{tr}Y^k\operatorname{tr}S$. By the log-convexity just shown and (c1) on $\{0,\dots,k\}$, $g(1)\le g(0)^{1-1/k}g(k)^{1/k}\le\operatorname{tr}S\,(\operatorname{tr}Y^k)^{1/k}$. (c1) If $f>0$ throughout, $\log f$ is convex on the integers, and a convex sequence lies below its chord: the difference with the chord satisfies the same inequality, vanishes at both ends, and at its first point of maximum it would violate the inequality if the maximum were positive. If $f(i)=0$ at some i, the inequality forces $f=0$ at all interior points, where the claim is trivial; at the ends it is an equality. (c2) For $k\le n\le2k$, the chord on $\{k,\dots,2k\}$ and $w(2k)\le2w(k)$ give $w(n)\le\frac{2k-n}{k}w(k)+\frac{n-k}{k}w(2k)\le\frac{n}{k}w(k)$; for $2^ik\le n<2^{i+1}k$ iterate: $\frac{w(n)}{n}\le\frac{w(2^ik)}{2^ik}\le\dots\le\frac{w(k)}{k}$. □

Lemma 18.232 (Slices; the pressure decreases in each side)**.** Let $\beta>0$, $N_2,N_3\ge2$, $\Sigma:=\mathbb{Z}/N_2\times\mathbb{Z}/N_3$, $X:=G^{\Sigma\times\{2,3\}}$ and $\Upsilon:=G^{\Sigma}$, with the product Haar measures. For a configuration U on $\Lambda_{(N_1,N_2,N_3)}$ and $\mathsf{t}\in\mathbb{Z}/N_1$ let $x_{\mathsf{t}}:=(U((\mathsf{t},z),\nu))_{z\in\Sigma,\nu=2,3}\in X$ be the bonds in the slice t and $y_{\mathsf{t}}:=(U((\mathsf{t},z),1))_{z\in\Sigma}\in\Upsilon$ the bonds from the slice t to the slice $\mathsf{t}+1$. There is a continuous,

symmetric and strictly positive function T on $X \times X$, depending on β, N_2, N_3 alone, such that, with the signs of Lemma 18.231:

(s1) $Z_{(N_1,N_2,N_3)} = \operatorname{tr} T^{N_1}$ for every $N_1 \geq 2$.

(s2) Let $\lambda \geq 2$, and let $F \geq 0$ be a bounded Borel function of the slab of direction 1 with λ slices, that is, of $(x_0, y_0, x_1, \dots, y_{\lambda-2}, x_{\lambda-1})$. There is $A_F \in \mathcal{K}$, independent of N_1, with $Z_{(N_1,N_2,N_3)}[F] = \operatorname{tr}(A_F * T^{N_1-\lambda+1})$ for every $N_1 \geq \lambda$; $A_1 = T^{\lambda-1}$, and $F \leq F'$ implies $A_F \leq A_{F'}$.

(s3) The mirror image $F^\flat(x_0, y_0, \dots, y_{\lambda-2}, x_{\lambda-1}) := F(x_{\lambda-1}, y_{\lambda-2}^{-1}, \dots, y_0^{-1}, x_0)$, with the inverses taken entry by entry, which is F read through the lattice reflection $x_1 \mapsto \lambda - 1 - x_1$ (a bond of the direction 1 being reversed), has $A_{F^\flat} = A_F^{\mathsf{T}}$.

(s4) For $k \in \mathbb{N}$ let $F^{\langle k \rangle}$ be the product, on the configurations of $\Lambda_{(2\lambda k,N_2,N_3)}$, of the $2k$ images of F under the lattice maps $x_1 \mapsto 2\lambda j \pm x_1$, $0 \leq j < k$. Then $Z_{(2\lambda k,N_2,N_3)}[F^{\langle k \rangle}] = \operatorname{tr}(Y^k)$ with $Y = P^{\mathsf{T}} * P$, $P = A_F * T$; and $0 \leq F^{\langle k \rangle} \leq F$ if $0 \leq F \leq 1$.

(P) If $\underline{N}$ and $\underline{N}'$ are triples of even integers ≥ 2 with $N_\mu \leq N'_\mu$ for $\mu = 1, 2, 3$, then $P_{\underline{N}'} \leq P_{\underline{N}}$.

The same holds with the direction 2 or 3 in the place of the direction 1. ■

Proof. Put $A_{\mathrm{s}}(x) := \sum_{z \in \Sigma}[1 - \frac{1}{2}\operatorname{tr} x(z,2)x(z+e_2,3)x(z+e_3,2)^{-1}x(z,3)^{-1}]$ and $A_{\mathrm{c}}(x,y,x') := \sum_{z\in\Sigma}\sum_{\nu=2,3}[1 - \frac{1}{2}\operatorname{tr} y(z)x'(z,\nu)y(z+e_\nu)^{-1}x(z,\nu)^{-1}]$. The plaquettes in the slice t give $A_{\mathrm{s}}(x_{\mathsf{t}})$, those between the slices t and $\mathsf{t}+1$ give $A_{\mathrm{c}}(x_{\mathsf{t}}, y_{\mathsf{t}}, x_{\mathsf{t}+1})$, and $U \mapsto (x_{\mathsf{t}}, y_{\mathsf{t}})_{\mathsf{t}}$ carries dU to the product measure. Put $\mathsf{k}(x,x') := \int_\Upsilon e^{-\beta A_{\mathrm{c}}(x,y,x')}dy$ and $T(x,x') := e^{-\frac{\beta}{2}A_{\mathrm{s}}(x)}\mathsf{k}(x,x')e^{-\frac{\beta}{2}A_{\mathrm{s}}(x')}$, continuous and positive. The term (z,ν) of $A_{\mathrm{c}}(x', y^{-1}, x)$ is formed with a matrix conjugate to the inverse of the matrix of the term (z,ν) of $A_{\mathrm{c}}(x,y,x')$, and $\operatorname{tr} u^{-1} = \operatorname{tr} u$ on G, whose Haar measure is invariant under inversion; so T is symmetric. (s1): integrate the variables y_{t} first and split each weight $e^{-\beta A_{\mathrm{s}}(x_{\mathsf{t}})}$ into two halves. (s2): $A_F(x,x')$ is the integral of F against the weights of the slices $0, \dots, \lambda-1$, with half weights at the two end slices, over $y_0, \dots, y_{\lambda-2}, x_1, \dots, x_{\lambda-2}$, at $x_0 = x$, $x_{\lambda-1} = x'$; the variables which F does not read give $T^{N_1-\lambda+1}(x_{\lambda-1}, x_0)$. (s3): rename $x_{\mathsf{t}} \to x_{\lambda-1-\mathsf{t}}$, $y_{\mathsf{t}} \to y_{\lambda-2-\mathsf{t}}^{-1}$, and use the symmetry of A_{c}. (s4): the image under $x_1 \mapsto 2\lambda j + x_1$ reads the slices $2\lambda j, \dots, 2\lambda j + \lambda - 1$ as F reads the slices $0, \dots, \lambda - 1$; the image under $x_1 \mapsto 2\lambda j - x_1$ reads the slices $2\lambda j - \lambda + 1, \dots, 2\lambda j$ as $F^\flat$ reads $0, \dots, \lambda - 1$; the two have the slice $2\lambda j$ in common, and two free steps lead from the slice $2\lambda j + \lambda - 1$ to the slice $2\lambda(j+1) - \lambda + 1$. Integrating as in (s2) gives $\operatorname{tr}((A_F^{\mathsf{T}} * A_F * T^2)^k) = \operatorname{tr}((T * A_F^{\mathsf{T}} * A_F * T)^k)$, and $T * A_F^{\mathsf{T}} * A_F * T = P^{\mathsf{T}} * P$. Each of the $2k$ factors lies in $[0,1]$, and the factor at $j = 0$ with the sign $+$ is F. The other directions: a_p does not change when μ and ν are exchanged in its formula, the matrix becoming its inverse; so a permutation of the coordinates carries $A_{\underline{N}}$, dU and $Z_{\underline{N}}[\cdot]$ to those of the permuted triple. (P): it suffices to raise one entry, by the symmetry the first, from $2k$ to $2n$, $n \geq k \geq 1$. With $w(k) := \log z(2k) = \log Z_{(2k,N_2,N_3)}$, finite as $T > 0$, (k2) gives $2w(k) \leq w(k-1) + w(k+1)$ for $k \geq 2$ and $w(2k) \leq 2w(k)$; (c2) gives $w(n)/n \leq w(k)/k$; divide by $2N_2N_3$. □

Proposition 18.233 (One direction: one copy against $2k$ mirrored copies)**.** Let $\beta > 0$, $N_2, N_3 \geq 2$, let $\lambda \geq 3$ be odd, $k \in \mathbb{N}$, and let F be a Borel function of the slab of direction 1 with λ slices with $0 \leq F \leq 1$; put $Z(n) := Z_{(n,N_2,N_3)}$. Then

$$Z(2\lambda)[F]^{2k} \leq \big[Z(\lambda+1)\,Z(\lambda-1)\big]^k\, Z(2\lambda k)\big[F^{\langle k \rangle}\big].$$

The same holds in the directions 2 and 3. ■

Proof. Put $A := A_F$, $n_1 := \frac{\lambda+1}{2}$, $P := A * T$, $Y := P^{\mathsf{T}} * P$, $R := T^{(\lambda-1)/2}$ and $S := R * R^{\mathsf{T}} = T^{\lambda-1}$; λ is odd and ≥ 3. By (s2) at $N_1 = 2\lambda$ and (k1), $Z(2\lambda)[F] = \operatorname{tr}(A * T^{\lambda+1}) = [A * T^{n_1}, T^{n_1}] \leq [A * T^{n_1}, A * T^{n_1}]^{1/2}[T^{n_1}, T^{n_1}]^{1/2}$: one copy of F on the ring of 2λ slices is bounded by F and its mirror image back to back and by an empty ring of $\lambda+1$ slices; $[T^{n_1}, T^{n_1}] = z(\lambda+1) = Z(\lambda+1)$ by (s1). Next $[A*T^{n_1}, A*T^{n_1}] = \operatorname{tr}(A*T*S*T*A^{\mathsf{T}}) = \operatorname{tr}(S*Y) = g(1)$ in the signs of (k3). By (k3),

$g(1) \le \operatorname{tr} S\,(\operatorname{tr} Y^k)^{1/k}$, with $\operatorname{tr} S = z(\lambda-1) = Z(\lambda-1)$ by (s1), $\lambda-1 \ge 2$, and $\operatorname{tr} Y^k = Z(2\lambda k)[F^{\langle k\rangle}]$ by (s4). So $Z(2\lambda)[F]^2 \le Z(\lambda+1)Z(\lambda-1)Z(2\lambda k)[F^{\langle k\rangle}]^{1/k}$; raise to the power k. □

Proposition 18.234 (Two tori at one cutoff: the converse of the domination)**.** Let $K \in \mathbb{N}$, $\beta > 0$, $\ell = 2L^m$ with $m \in \mathbb{N}$, $\lambda := \ell/(2a_K) = L^{K+m}$, $n := 2\lambda$, $n_0 := 2\lambda/L$, $t \in \mathbb{N}$, $k := L^t$, $n' := 2\lambda k$,

$$\mathsf{D} := n^3(P_{n_0} - P_{n'}), \qquad q_0 := 8k^3 = 8L^{3t}.$$

(a) $\mathsf{D} \ge 0$, and for every Borel function H on G^{B_λ} with $0 \le H \le 1$, with $H^\sharp$ the product, on the configurations of $\Lambda_{(n',n',n')}$, of the $8k^3$ images of H under the maps $x \mapsto (2\lambda j_\mu \pm x_\mu)_\mu$, $j \in \{0,\dots,k-1\}^3$: $0 \le H^\sharp \le H$, $\langle H\rangle_{(n,n,n)}^{8k^3} \le e^{8k^3\mathsf{D}}\langle H^\sharp\rangle_{(n',n',n')}$, and so $\langle H\rangle_{(n,n,n)} \le e^{\mathsf{D}}\langle H\rangle_{(n',n',n')}^{1/q_0}$.

(b) For every cell $\square$ of side $\frac{\ell}{2}$ of a board of $\mathbb{L}_K$ and every $F \in \mathcal{A}(\square)$ with $0 \le F \le 1$: $\tilde\mu_{K,\ell,\beta}(F) \le e^{\mathsf{D}}\,\tilde\mu_{K,L^t\ell,\beta}(F)^{1/q_0}$; and $\mathsf{D} = \ell^3(p_{\ell/L} - p_{L^t\ell})$ when $m \ge 2$.

(c) For the same $\square$ and every bounded Borel $F \in \mathcal{A}(\square)$ with $F \ge 0$: $\tilde\mu_{K,L^t\ell,\beta}(F) \le e^{\frac18\ell^3(p_\ell - p_{L^t\ell})}\,\tilde\mu_{K,\ell,\beta}(F^8)^{1/8}$, and $\ell^3(p_\ell - p_{L^t\ell}) \le \mathsf{D}$.

(d) Let $\mathsf{b} > 0$ and $s = s(\mathsf{b},\ell)$. If $K > s$ and $L^s a_K\beta \in I(g^\circ)$, in particular at every pair $(K, L^K\mathsf{b})$ with $K > s$, then $\mathsf{D} \le \mathsf{D}^\circ(\mathsf{b},\ell)$.

■

Proof. (a) Let $G_0 := H$ and let G_i arise from G_{i-1} by the operation of Lemma 18.232 (s4) in the direction i, $i = 1,2,3$; then $H^\sharp = G_3$. Each G_{i-1} is a product of images of H under maps which move only the coordinates x_μ with $\mu < i$, so it is a Borel function of the slab of direction i with λ slices, with values in $[0,1]$. Proposition 18.233 in the direction i, with the other two sides (n,n), (n',n), (n',n') for $i = 1,2,3$, gives $z_{i-1}^{2k} \le c_i^k z_i$ with $z_0 := Z_{(n,n,n)}[G_0]$, $z_1 := Z_{(n',n,n)}[G_1]$, $z_2 := Z_{(n',n',n)}[G_2]$, $z_3 := Z_{(n',n',n')}[G_3]$, $c_1 := Z_{(\lambda+1,n,n)}Z_{(\lambda-1,n,n)}$, $c_2 := Z_{(n',\lambda+1,n)}Z_{(n',\lambda-1,n)}$ and $c_3 := Z_{(n',n',\lambda+1)}Z_{(n',n',\lambda-1)}$. Hence $z_0^{8k^3} \le c_1^{4k^3}c_2^{2k^2}c_3^k z_3$. All the triples which occur have even entries in $[n_0,n']$, since $\lambda \pm 1$ are even and $n_0 \le \lambda - 1$ (as $L \ge 13$); by (P), $P^- := P_{n'} \le P_{\underline N} \le P_{n_0} =: P^+$ for each of them, so $\mathsf{D} = n^3(P^+ - P^-) \ge 0$. With $\log Z_{\underline N} = N_1N_2N_3P_{\underline N}$, and as the two tori of c_1, c_2, c_3 have together n^3, kn^3, k^2n^3 sites, $\log(c_1^{4k^3}c_2^{2k^2}c_3^k) \le 7k^3n^3P^+$, $\log Z_{(n',n',n')} \le k^3n^3P^+$ and $\log Z_{(n,n,n)} \ge n^3P^-$. So

$$\langle H\rangle_{(n,n,n)}^{8k^3} = \frac{z_0^{8k^3}}{Z_{(n,n,n)}^{8k^3}} \le \frac{c_1^{4k^3}c_2^{2k^2}c_3^k\,Z_{(n',n',n')}}{Z_{(n,n,n)}^{8k^3}}\,\langle H^\sharp\rangle_{(n',n',n')} \le e^{8k^3\mathsf{D}}\langle H^\sharp\rangle_{(n',n',n')},$$

and $H^\sharp = G_3 \le G_2 \le G_1 \le H$ by (s4). (b) is (a) read through the chart of the lattice units, at $\ell/a_K = n$ and $L^t\ell/a_K = n'$, with $n^3P_{n_0} = \ell^3 p_{\ell/L}$ and $n^3P_{n'} = \ell^3 p_{L^t\ell}$ when $\ell/L \in \mathfrak{L}$. (c) Use the seminorm $\|F\|_t$ from the proof of Lemma 18.216, with board side $\mathsf{l} = \ell/2$. The finite-torus chessboard inequality and the comparison of sides give

$$\tilde\mu_{K,L^t\ell,\beta}(F) \le \|F\|_{L^t\ell} \le \|F\|_\ell\, e^{\mathsf{l}^3(p_\ell - p_{L^t\ell})}.$$

The eight reflected factors on the torus of side ℓ have the same L^8 norm, so Hölder gives $\|F\|_\ell \le \tilde\mu_{K,\ell,\beta}(F^8)^{1/8}$. Finally $\mathsf{l}^3 = \ell^3/8$ and $\ell^3(p_\ell - p_{L^t\ell}) = n^3(P_n - P_{n'}) \le \mathsf{D}$ by (P). (d) D is a number of the lattice units: it depends on λ, k, L and β alone, and $\lambda = \ell/(2a_K) = L^s\ell/(2a_{K-s})$. The pair $(K-s,\beta) = (K-s, L^{K-s}\cdot L^s a_K\beta)$ is a datum at g° with the label $L^s a_K\beta \in I(g^\circ)$ (Lemma 18.212 (c)); and $L^{s-1}\ell$, $L^{s+t}\ell$ lie in $\mathfrak{L}(g^\circ)$. So $\mathsf{D} = (L^s\ell)^3(p_{L^{s-1}\ell} - p_{L^{s+t}\ell})$ at $(K-s,\beta)$, and the volume bounds at a datum at g° for two sides of $\mathfrak{L}(g^\circ)$ from Theorem YM3_I:AD give $p_{\ell_1} - p_{\ell_2} \le 2C_{\mathrm S} + C_{g^\circ}$ for $\ell_1,\ell_2 \in \mathfrak{L}(g^\circ)$, as in the proof of Lemma 18.216. At $(K, L^K\mathsf{b})$, $L^s a_K\beta = \lambda^\circ \in I(g^\circ)$. □

Proposition 18.235 (Two tori at every label are mutually absolutely continuous)**.** Let $\mathsf{b} > 0$, $m_1, t \in \mathbb{N}$, $\ell_1 := 2L^{m_1}$, $\ell_2 := L^t\ell_1$, and let the window $\square$, the sets w_d and the space Y be those of

Definition 18.206 at the side ℓ_1. For $i = 1, 2$ let $\nu_i := (\varpi_0^{[0]} \circ \mathsf{Per}_{\ell_i})_* \mu_{\ell_i, \mathsf{b}}$ be the law over $\square$ of the thread measure of T_{ℓ_i} at the label b (Lemma 18.230), so that $\nu_1 = \mu[\ell_1, \mathsf{b}]$; put $\mathsf{D}^\circ := \mathsf{D}^\circ(\mathsf{b}, \ell_1)$ and $q_0 := 8L^{3t}$, $q_0' := q_0/(q_0 - 1)$.

(a) For every Borel set $A \subset Y$, $\nu_1(A) \le e^{\mathsf{D}^\circ} \nu_2(A)^{1/q_0}$; and $\nu_2 \preceq_{e^{\mathsf{D}^\circ/8}} \nu_1$, in particular $\nu_2(A) \le e^{\mathsf{D}^\circ/8} \nu_1(A)^{1/8}$.

(b) ν_1 and ν_2 are mutually absolutely continuous; $d\nu_1/d\nu_2 \in L^r(\nu_2)$ with a norm at most $e^{\mathsf{D}^\circ}(q_0'/(q_0' - r))^{1/r}$ for $1 \le r < q_0'$; $h := d\nu_2/d\nu_1 \in L^{8/7}(\nu_1)$ with a norm at most $e^{\mathsf{D}^\circ/8}$, and $0 \le \int h \log h \, d\nu_1 \le \mathsf{D}^\circ$.

(c) (a) and (b) hold for the images of ν_1 and ν_2 under one Borel map, in particular for the laws of the threads over a ball $\mathbb{B}(0, R)$ with $4R \le \ell_1$, that is, of the block fields of all depths on the bonds with both end points in $\mathbb{B}(0, R)$. So, for every $R > 0$, the thread laws over $\mathbb{B}(0, R)$ at the label b of any two tori with sides $\ell_1 < \ell_2$ in $\mathfrak{L}$, $\ell_1 \ge 16(R + 1)$, are mutually absolutely continuous, with densities in L^r for some $r > 1$ in both directions.

(d) The constants are formed from the constants, b and the smaller side ℓ_1; the exponent $1/q_0$ from ℓ_2/ℓ_1 alone.

■

Proof. Step 1: cylinders at one cutoff. Let $D \ge 0$, let $f \ge 0$ be a bounded Borel function on $Y_{\le D}$, and $K \ge \max\{D, 1\}$. By Lemma 18.211 (d), $\Upsilon_K(U)$ depends on the bonds in $\square$ alone, and Ψ_K is Borel; so $F_K := f \circ q_D \circ \Upsilon_K \in \mathcal{A}(\square)$, where $\square$ is the cell $\mathbf{0}$ of the board of side $\frac{\ell_1}{2}$ with the origin 0. At $f = 1_{A_D}$ it is the indicator of a Borel set of the cell.

Step 2: the two inequalities at one cutoff. Let $K > s(\mathsf{b}, \ell_1)$ and $\beta = L^K \mathsf{b}$. Proposition 18.234 (b), (d) at $f = 1_{A_D}$ give $\tilde\mu_{K,\ell_1,\beta}(F_K) \le e^{\mathsf{D}^\circ} \tilde\mu_{K,\ell_2,\beta}(F_K)^{1/q_0}$, and Proposition 18.234 (c), (d) give $\tilde\mu_{K,\ell_2,\beta}(F_K) \le e^{\mathsf{D}^\circ/8} \tilde\mu_{K,\ell_1,\beta}(F_K^8)^{1/8}$ for every bounded Borel $f \ge 0$.

Step 3: the limit of the cutoff. By Lemma 18.230 (c), the laws $(\Upsilon_K)_* \tilde\mu_{K,\ell_i,L^K\mathsf{b}}$ converge to ν_i in total variation at the depths $\le D$; the maps $x \mapsto x^{1/q_0}$, $x^{1/8}$ are continuous on $[0, 1]$. So both inequalities hold for the cylinder sets $C = q_D^{-1} A_D$, and the second for the cylinder functions $f \circ q_D$.

Step 4: all Borel sets and functions. The cylinder sets form an algebra which generates the Borel sets of Y, the product of countably many compact metric spaces; so every Borel set A is approximated by cylinder sets C with $(\nu_1 + \nu_2)(A \triangle C) \le \epsilon$, and $\nu_1(A) \le \nu_1(C) + \epsilon \le e^{\mathsf{D}^\circ}(\nu_2(A) + \epsilon)^{1/q_0} + \epsilon$; let $\epsilon \to 0$, and likewise for the other inequality. A bounded Borel f on Y with $0 \le f \le c$ is a limit in $L^1(\nu_1 + \nu_2)$ of cylinder functions with values in $[0, c]$, and the inequality $\nu_2(f) \le e^{\mathsf{D}^\circ/8} \nu_1(f^8)^{1/8}$ passes to the limit by $|x^8 - y^8| \le 8c^7 |x - y|$ on $[0, c]$; for complex f use $|\nu_2(f)| \le \nu_2(|f|)$. This is (a).

Step 5: densities. By (a) the two laws have the same null sets, and the theorem of Radon and Nikodym gives the densities. Lemma 18.214 (DM2), (DM3) give the assertions on h, with $8 \log e^{\mathsf{D}^\circ/8} = \mathsf{D}^\circ$. For $\mathsf{g} := d\nu_1/d\nu_2$ and $u > 0$, the set $E_u := \{\mathsf{g} > u\}$ has $u\, \nu_2(E_u) \le \nu_1(E_u) \le e^{\mathsf{D}^\circ} \nu_2(E_u)^{1/q_0}$, so $\nu_2(E_u) \le (e^{\mathsf{D}^\circ}/u)^{q_0'}$; and $\int \mathsf{g}^r d\nu_2 = \int_0^\infty r u^{r-1} \nu_2(E_u)\, du \le e^{r\mathsf{D}^\circ} + \int_{e^{\mathsf{D}^\circ}}^\infty r u^{r-1} (e^{\mathsf{D}^\circ}/u)^{q_0'} du = e^{r\mathsf{D}^\circ} q_0'/(q_0' - r)$ for $1 \le r < q_0'$.

Step 6: images; balls; (d). For a Borel map p, $\mathsf{p}_* \nu_i(A') = \nu_i(\mathsf{p}^{-1} A')$, and (DM4); then Step 5. The bonds with both end points in $\mathbb{B}(0, R) \subset \square$ belong to the sets w_d, and the thread over the ball is a coordinate projection of Y. Two sides $\ell_1 < \ell_2$ of $\mathfrak{L}$ satisfy $\ell_2 = L^t \ell_1$ with $t \ge 1$, and $\ell_1 \ge 16(R + 1)$ gives $4R \le \ell_1$. (d) is the formation of $\mathsf{D}^\circ(\mathsf{b}, \ell_1)$ and q_0. □

Proposition 18.236 (Every limit law of the block fields through periodic Gibbs states, at every label, is dominated by one torus)**.** Let $\mathsf{b} > 0$ and $\ell \in \mathfrak{L}$; put $\mathsf{Q}^\circ := \mathsf{Q}^\circ(\mathsf{b}, \ell)$, let $\mu := \mu[\ell, \mathsf{b}]$ and $\mu^{[d_0]} := \mu^{[d_0]}[\ell, \mathsf{b}]$ be the thread measure of T_ℓ at the label b seen in the window of the side ℓ (Lemma 18.230), and let $(K_j, \beta_j, \nu_j)_j$ be integers $K_j \to \infty$, numbers $\beta_j > 0$ with $a_{K_j} \beta_j \to \mathsf{b}$, and states $\nu_j \in \mathcal{G}^{\mathrm{per}}_{K_j, \beta_j}$; no reference coupling is fixed.

(i) For all $o_j \in a_{K_j}\mathbb{Z}^3$, the laws $(\Upsilon_{K_j} \circ \tau_{-o_j})_*\nu_j$ have weak limit points on Y, and every such limit point ν satisfies $\nu \preceq_{\mathsf{Q}^\circ} \mu$: $\nu = h\mu$ with $\|h\|_{L^{8/7}(\mu)} \le \mathsf{Q}^\circ$ and $0 \le \int h \log h \, d\mu \le \mathsf{D}^\circ(\mathsf{b}, \ell)$, $\nu(\mathcal{A}) \le \mathsf{Q}^\circ \mu(\mathcal{A})^{1/8}$ for every Borel set $\mathcal{A} \subset Y$, and $\nu(\mathsf{Thr}_\square) = 1$.

(ii) The laws $(\Psi_{K_j})_*\nu_j$ have weak limit points on Z, and every such limit point $\boldsymbol{\nu}$ satisfies $(\varpi_o^{[d_0]})_*\boldsymbol{\nu} \preceq_{\mathsf{Q}^\circ} \mu^{[d_0]}$ for all $d_0 \ge 0$ and $o \in a_{d_0}\mathbb{Z}^3$, $\boldsymbol{\nu}(\mathsf{Thr}) = 1$, $(T_v)_*\boldsymbol{\nu} = \boldsymbol{\nu}$ for every $v \in \mathbb{D}$, and the product bound of Definition 18.207 with Q° in the place of Q_0.

(iii) The conclusions of (ii), the domination and the product bound, hold for $D\boldsymbol{\nu}$, for every Borel function $D \ge 0$ with $\boldsymbol{\nu}(D) = 1$ and $D \circ T_v = D$ $\boldsymbol{\nu}$-almost everywhere for one $v \in \mathbb{D} \setminus \{0\}$, and for almost every member of every decomposition of $\boldsymbol{\nu}$ into laws carried by Thr and invariant under one T_v, $v \in \mathbb{D} \setminus \{0\}$; in particular for almost every member of every ergodic decomposition of $\boldsymbol{\nu}$ under $\mathbb{D}$ and of every regular conditional distribution of $\boldsymbol{\nu}$ given $\mathcal{I}_\infty$.

(iv) Every Borel subset of Y of full μ-measure has full measure for every law of (i), and its inverse image under $\varpi_o^{[0]}$, $o \in \mathbb{Z}^3$, has full measure for the laws of (ii) and (iii); of the laws of (i), at most $(\mathsf{Q}^\circ)^8$ are pairwise mutually singular.

(v) Q° and μ are formed from the constants, b and ℓ; they depend neither on the K_j, β_j, ν_j nor on the o_j. Each law ν of (i) is, on Y and all depths together, absolutely continuous with respect to μ; for each law $\boldsymbol{\pi}$ of (ii) or (iii), each $d_0 \ge 0$ and each $o \in a_{d_0}\mathbb{Z}^3$, the law $(\varpi_o^{[d_0]})_*\boldsymbol{\pi}$ of the block fields of the depths $\ge d_0$ over the cell $o + \square$ is absolutely continuous with respect to $\mu^{[d_0]}$. For $t \in \mathbb{N}$ put $\nu_2 := (\varpi_0^{[0]} \circ \mathsf{Per}_{L^t\ell})_*\mu_{L^t\ell,\mathsf{b}}$, the image in the window $\square$ of the side ℓ of the measure $\mu_{L^t\ell,\mathsf{b}}$ of Lemma 18.230; then each such ν is absolutely continuous with respect to ν_2, and each such $(\varpi_o^{[d_0]})_*\boldsymbol{\pi}$ with respect to $\mathrm{pj}_*^{[d_0]}\nu_2$.

■

Proof. Step 1: the bound of Lemma 18.216 at every label. Let $s = s(\mathsf{b}, \ell)$. As $\lambda^\circ = L^s\mathsf{b}$ lies in the interior of $I(g^\circ)$ and $L^s a_{K_j}\beta_j \to \lambda^\circ$, there is j_0 such that for $j \ge j_0$, $K_j > s$ and $L^s a_{K_j}\beta_j \in I(g^\circ)$; then $(K_j - s, \beta_j)$ is a datum at g° (Lemma 18.212 (c)), and $L^s\ell \in \mathfrak{L}(g^\circ)$. Lemma 18.216 at g°, at this datum and at the side $L^s\ell$, with the number $\exp[\frac{1}{8}(L^s\ell)^3(2C_{\mathrm{S}} + C_{g^\circ})] = \mathsf{Q}^\circ$, carried back by δ_s^{-1} (Lemma 18.229 (a)), gives for $j \ge j_0$, every $o \in a_{K_j}\mathbb{Z}^3$, every finite set $\mathcal{K}_0 \subset \mathbb{Z}^3$ and all bounded complex Borel functions $F_{\mathbf{k}}$ of the bonds with both end points in $\square_{\mathbf{k}} = o + \frac{\ell}{2}\mathbf{k} + \square$:

$$\Big|\nu_j\Big(\prod_{\mathbf{k}\in\mathcal{K}_0} F_{\mathbf{k}}\Big)\Big| \le \prod_{\mathbf{k}\in\mathcal{K}_0} \mathsf{Q}^\circ \, \tilde{\mu}_{K_j,\ell,\beta_j}(|F_{\mathbf{k}}|^8)^{1/8}.$$

Step 2: the reference laws. By Lemma 18.230 (c), $\rho_j := (\Upsilon_{K_j})_*\tilde{\mu}_{K_j,\ell,\beta_j}$ converges to μ weakly on Y and in total variation at the depths $\le D$, for every D.

Step 3: passage to the limit. The laws ν_j are translation invariant by Proposition YM3_I: 103.4. Steps 1 and 2 supply the one-cell and several-cell bounds and the convergence of the references. Lemma 18.217(a) gives (i) with $\mathsf{Q} = \mathsf{Q}^\circ$, including concentration on window threads by Lemma 18.230(b). Parts (b), (c), and (d) give (ii), (iii), and (iv), respectively. (v): the first sentence is the formation of Q° and μ. A domination $\boldsymbol{\sigma} \preceq_{\mathsf{Q}^\circ} \boldsymbol{\rho}$, tested at $f = 1_{\mathcal{A}}$, gives $\boldsymbol{\sigma}(\mathcal{A}) \le \mathsf{Q}^\circ \boldsymbol{\rho}(\mathcal{A})^{1/8}$, so every $\boldsymbol{\rho}$-null set is $\boldsymbol{\sigma}$-null; this gives the absolute continuity with respect to μ and to $\mu^{[d_0]}$. Proposition 18.235 (a), (c), at $\ell_1 = \ell$ and $\ell_2 = L^t\ell$, gives $\mu(\mathcal{A}) \le e^{\mathsf{D}^\circ(\mathsf{b},\ell)}\nu_2(\mathcal{A})^{1/q_0}$, $q_0 = 8L^{3t}$, for the Borel sets $\mathcal{A} \subset Y$, and the same inequality between the images $\mathrm{pj}_*^{[d_0]}\mu = \mu^{[d_0]}$ and $\mathrm{pj}_*^{[d_0]}\nu_2$; so the null sets of ν_2, respectively of $\mathrm{pj}_*^{[d_0]}\nu_2$, are null for μ, respectively for $\mu^{[d_0]}$, and hence for the laws which these dominate. □

Compact photons. Let $a > 0$. A configuration of three compact photon fields on $a\mathbb{Z}^3$ is a triple $U = (U^1, U^2, U^3)$ of maps from the bonds to $\mathrm{U}(1) = \{\zeta \in \mathbb{C} : |\zeta| = 1\}$ with $U^c(\bar{b}) = \overline{U^c(b)}$; their space $\mathcal{V}^{\mathrm{ph}}(a\mathbb{Z}^3) = (\mathrm{U}(1)^3)^{\mathsf{Bd}}$ is compact and metrisable. For a plaquette $p = P_{\mu\nu}(x)$, $\mu < \nu$, with the bonds $b_1 = \langle x, x + ae_\mu\rangle$, $b_2 = \langle x + ae_\mu, x + ae_\mu + ae_\nu\rangle$, $b_3 = \langle x + ae_\nu, x + ae_\mu + ae_\nu\rangle$, $b_4 = \langle x, x + ae_\nu\rangle$,

put $U^c(\partial p) := U^c(b_1)U^c(b_2)\overline{U^c(b_3)}\,\overline{U^c(b_4)}$, the field strength $F^c_a(p) := a^{-2}\operatorname{Im} U^c(\partial p)$ and the action density

$$\mathcal{O}^{\mathrm{ph}}_a(h) := a^{-1}\sum_p h(x(p))\sum_{c=1}^{3}(1-\operatorname{Re} U^c(\partial p))$$

for bounded real h with compact support. For a site y let $p_1(y) := P_{23}(y)$, $p_2(y) := P_{13}(y)$, $p_3(y) := P_{12}(y)$, let $\mathsf{D}_a(y) := \det(F^c_a(p_k(y)))_{c,k=1,2,3}$, the determinant of the three field strengths at y, rows indexed by the colour, and $\mathsf{D}_a(f) := a^3\sum_{y\in a\mathbb{Z}^3} f(y)\,\mathsf{D}_a(y)$ for bounded real f with compact support. The conjugation of the colour one is the map $\mathsf{C}_1(U^1,U^2,U^3) := (\overline{U^1},U^2,U^3)$. For a period lattice Γ and $\beta\in\mathbb{R}$, $\mu^{\mathrm{ph}}_{\Gamma,\beta}$ is the law of the Γ-periodic configurations with the density proportional to $\exp[-\beta\sum_{[p]}\sum_c(1-\operatorname{Re}U^c(\partial p))]$, the sum running over the classes of plaquettes modulo Γ, with respect to the Haar measure of the periodic configurations, the product of the Haar measures of $\mathrm{U}(1)^3$ over a set of representatives of the bonds modulo Γ: the product of three Wilson measures of the torus $\mathbb{R}^3/\Gamma$, read on the infinite lattice. For bounded Borel functions $G_0,\dots,G_r$ and a law μ, $\langle G_0;\dots;G_r\rangle_\mu$ is their joint cumulant under μ.

Proposition 18.237 (Compact photons: the determinant of the field strengths has no overlap with the action density)**.** Let $a>0$.

(a) C_1 is a homeomorphism of $\mathcal{V}^{\mathrm{ph}}(a\mathbb{Z}^3)$ with $\mathsf{C}_1\circ\mathsf{C}_1=\mathrm{id}$. For every plaquette, $F^1_a(p)\circ\mathsf{C}_1 = -F^1_a(p)$ and $F^c_a(p)\circ\mathsf{C}_1 = F^c_a(p)$ for $c=2,3$; hence $\mathsf{D}_a(y)\circ\mathsf{C}_1=-\mathsf{D}_a(y)$, $\mathsf{D}_a(f)\circ\mathsf{C}_1=-\mathsf{D}_a(f)$ and $\mathcal{O}^{\mathrm{ph}}_a(h)\circ\mathsf{C}_1=\mathcal{O}^{\mathrm{ph}}_a(h)$. These functions are real, bounded and continuous, $|\mathsf{D}_a(y)|\le 6a^{-6}$.

(b) Every $\mu^{\mathrm{ph}}_{\Gamma,\beta}$, for every period lattice Γ and every $\beta\in\mathbb{R}$, and every weak limit of such laws, is invariant under C_1.

(c) Let μ be a law on $\mathcal{V}^{\mathrm{ph}}(a\mathbb{Z}^3)$ with $(\mathsf{C}_1)_*\mu=\mu$, and let $G_1,\dots,G_r$, $r\ge 0$, be bounded Borel functions each of which is odd or even under C_1, $G_i\circ\mathsf{C}_1=\pm G_i$, an odd number of them being odd. Then $\mu(G_1\cdots G_r)=0$ and $\langle G_1;\dots;G_r\rangle_\mu=0$.

(d) In particular, for every such μ, at every spacing and every coupling, and for all bounded real $f,h_1,\dots,h_r$ with compact supports, $r\ge 0$,

$$\mu\Big(\mathsf{D}_a(f)\prod_{i=1}^{r}\mathcal{O}^{\mathrm{ph}}_a(h_i)\Big)=0,\qquad \big\langle \mathsf{D}_a(f);\mathcal{O}^{\mathrm{ph}}_a(h_1);\dots;\mathcal{O}^{\mathrm{ph}}_a(h_r)\big\rangle_\mu=0;$$

so every limit of these numbers along spacings tending to 0, along any couplings and any such laws, is 0.

■

Proof. (a) C_1 acts by complex conjugation on one factor of each coordinate, a homeomorphism of $\mathrm{U}(1)$ of order two. Conjugating U^1 on every bond conjugates $U^1(\partial p)$, which changes the sign of its imaginary part and keeps its real part. So the first row of the matrix $(F^c_a(p_k(y)))$ changes its sign and the other two rows stay, and the determinant changes its sign; $\mathsf{D}_a(f)$ is a finite real combination of the $\mathsf{D}_a(y)$, and $\mathcal{O}^{\mathrm{ph}}_a(h)$ is a finite sum of functions of the real parts. A 3×3 determinant with entries of modulus $\le a^{-2}$ is at most $3!\,a^{-6}$ in modulus. (b) The Haar measure of $\mathrm{U}(1)$ is invariant under $\zeta\mapsto\bar\zeta$, so the Haar measure of the periodic configurations is invariant under C_1, which maps periodic configurations to periodic configurations; the density depends on the real parts alone. A weak limit of C_1-invariant laws is C_1-invariant, as C_1 is continuous: $\mu(f\circ\mathsf{C}_1)=\lim\mu_n(f\circ\mathsf{C}_1)=\lim\mu_n(f)=\mu(f)$ for continuous f. (c) The product $\Pi := G_1\cdots G_r$ satisfies $\Pi\circ\mathsf{C}_1=-\Pi$, as an odd number of factors change their sign; so $\mu(\Pi)=\mu(\Pi\circ\mathsf{C}_1)=-\mu(\Pi)$. The joint cumulant is the sum over the partitions $\mathfrak{p}$ of $\{1,\dots,r\}$ of $(-1)^{|\mathfrak{p}|-1}(|\mathfrak{p}|-1)!\prod_{J\in\mathfrak{p}}\mu(\prod_{i\in J}G_i)$; in every partition some block contains an odd number of odd factors, since the blocks share out an odd number of them, and the moment of this block vanishes by the first assertion. (d) is (c) with $\mathsf{D}_a(f)$ as the one odd factor, by (a); at each spacing the numbers vanish, so their limits do. □

The slab. Let $a > 0$, $\beta \geq 0$ and $l = N_3 a$ with an integer $N_3 \geq 3$. For a period lattice Γ of $a\mathbb{Z}^3$ (Definition 18.209) the periodic objects used below include the set $\mathcal{V}_\Gamma$ of the configurations U with $\tau_v U = U$ for every $v \in \Gamma$; the domains D of Γ, sets of sites which meet every class of sites modulo Γ in one site, and for such D the set Bd_D of the bonds $\langle x, x + ae_\mu\rangle$, $x \in D$, $\mu = 1, 2, 3$; the bond transversals R of Γ, sets of bonds which meet every class of bonds modulo Γ in one bond, and for $V \in G^R$ the configuration $V^{\mathrm{per}} \in \mathcal{V}_\Gamma$ which agrees with V on R; the number $S_\Gamma(U)$, the sum of $a_p(U)$ over one plaquette of each class modulo Γ; for a real continuous Φ on $\mathcal{V}_\Gamma$, the integral $Z_\Gamma[\Phi] = Z_{\Gamma,\beta}[\Phi]$ of $e^{-\beta S_\Gamma + \Phi}$ against the Haar measure dU_Γ of the periodic configurations, the image of dV_R under $V \mapsto V^{\mathrm{per}}$, and the law $\mu^\Phi_\Gamma = \mu^\Phi_{\Gamma,\beta}$, carried by $\mathcal{V}_\Gamma$, whose density against that measure is $Z_\Gamma[\Phi]^{-1} e^{-\beta S_\Gamma + \Phi}$, so that $\mu_{\Gamma,\beta} := \mu^0_{\Gamma,\beta}$ is the Wilson measure $\mu^\varnothing_{\Gamma,\beta}$; for integers $n_1, n_2, n_3 \geq 3$ the coordinate lattice $\Gamma(n_1, n_2, n_3) := a(n_1\mathbb{Z} \times n_2\mathbb{Z} \times n_3\mathbb{Z})$, a period lattice; and the arrays of item (7). The set $\mathcal{V}^{\mathrm{sl}} := \{U \in \mathcal{V}(a\mathbb{Z}^3) : \tau_{le_3} U = U\}$ is closed, hence compact and metrisable; a slab law is a law ν with $\nu(\mathcal{V}^{\mathrm{sl}}) = 1$. Sites, bonds or plaquettes are congruent if they differ by a vector of $\mathbb{Z}le_3$, and $B^{\mathrm{per}} := B + \mathbb{Z}le_3$. A finite set B of bonds is called thin when no two of its members are congruent; because $N_3 \geq 3$, this holds for the four bonds of a plaquette, and likewise for the six bonds having a given site as an end point. For a thin B, $V \in G^B$ and $U \in \mathcal{V}^{\mathrm{sl}}$, $V \vee U \in \mathcal{V}^{\mathrm{sl}}$ takes the value $V(b)$ at the bonds $b + kle_3$, $b \in B$, $k \in \mathbb{Z}$, and the values of U off B^{per}; $\mathsf{Pl}^{\mathrm{sl}}[B]$ is a set of representatives, up to congruence, of the plaquettes which hold a bond of B^{per}, and $H^{\mathrm{sl}}_B := -\beta \sum_{p \in \mathsf{Pl}^{\mathrm{sl}}[B]} a_p$ on $\mathcal{V}^{\mathrm{sl}}$ does not depend on the representatives. For a bounded Borel F and a real continuous W on $\mathcal{V}^{\mathrm{sl}}$,

$$(\gamma^W_B F)(U) := \frac{\int F(V \vee U)\, e^{(H^{\mathrm{sl}}_B + W)(V \vee U)}\, dV}{\int e^{(H^{\mathrm{sl}}_B + W)(V \vee U)}\, dV}, \qquad \gamma_B := \gamma^0_B;$$

since $e^{H^{\mathrm{sl}}_B} \in [e^{-8\beta \# B}, 1]$, (dens) $|\gamma_B F|(U) \leq e^{8\beta \# B} \int |F|(V \vee U)\, dV$. The slab Gibbs states $\mathcal{G}^{\mathrm{sl}} = \mathcal{G}^{\mathrm{sl}}(a, \beta, l)$ are the slab laws ν with $\nu(\gamma_B F) = \nu(F)$ for every thin B and every $F \in \mathcal{C}_{\mathrm{loc}}$; $\mathcal{G}^{\mathrm{sl,per}}$ is the set of those which are invariant under the translations by a subgroup of finite index of $a\mathbb{Z}^2 \times \{0\}$, such a subgroup containing $Pa\mathbb{Z}^2 \times \{0\}$ for some $P \in \mathbb{N}$. For $\Lambda \in a\mathbb{N}$, $\Lambda \geq 3a$, the box lattice $\Gamma_\Lambda := \Lambda\mathbb{Z} \times \Lambda\mathbb{Z} \times l\mathbb{Z}$ is the coordinate lattice $\Gamma(\Lambda/a, \Lambda/a, N_3)$, and $\mu_{\Gamma_\Lambda,\beta} = \mu^\varnothing_{\Gamma_\Lambda,\beta}$ is its Wilson measure; for an unbounded set $\mathfrak{l}_0 \subset a\mathbb{N}$, $\mathcal{G}^{\mathrm{box}}(a, \beta, l; \mathfrak{l}_0)$ is the set of the weak limit points of $\mu_{\Gamma_\Lambda,\beta}$ as $\Lambda \to \infty$ in $\mathfrak{l}_0$. A slab law is positive at the pair $\Sigma(e_3, c)$, $\Sigma(e_3, c + \frac{l}{2})$, $c \in \frac{a}{2}\mathbb{Z}$, if $\nu(\overline{\theta_{e_3,c} F}\, F) \geq 0$ for all $F \in \mathcal{C}_{\mathrm{loc}}(\{c \leq x_3 \leq c + \frac{l}{2}\})$.

The space of the slab is $\mathbb{E} := \mathbb{R}^2 \times S^1_l = \mathbb{R}^3/\mathbb{Z}le_3$, with the projection pr, the quotient distance $|\mathrm{pr}\, x - \mathrm{pr}\, y| = \min_{k \in \mathbb{Z}} |x - y - kle_3|$ and closed balls; sets $A, A' \subset \mathbb{R}^3$ lie at a distance $\geq \mathsf{d}$ modulo le_3 when $|x - x' - kle_3| \geq \mathsf{d}$ for all $x \in A$, $x' \in A'$ and $k \in \mathbb{Z}$, which, by the formula for the quotient distance, says that the images $\mathrm{pr}A$, $\mathrm{pr}A'$ are separated by at least d in $\mathbb{E}$; $C^1_c(\mathbb{E})$ and $\mathrm{Lip}_c(\mathbb{E})$ are the sets of the real functions on $\mathbb{E}$ with compact support which are of class C^1, respectively Lipschitz, and $\|f\|_\sharp := \max\{\|f\|_\infty, 40M \,\mathrm{Lip}\, f\}$ for functions on $\mathbb{E}$ and on $\mathbb{R}^3$ alike; Lip_c stands for $\mathrm{Lip}_c(\mathbb{R}^3)$, the set of the real Lipschitz functions on $\mathbb{R}^3$ with compact support, and for $f_1, \dots, f_n \in \mathrm{Lip}_c$ and $\varrho > 0$ the source disc $\mathbb{D}_\varrho(f_1, \dots, f_n)$ is the set of the $z \in \mathbb{C}^n$ with $\sum_q |z_q|\, \|f_q\|_\sharp < \varrho$; for a closed ball $\mathcal{B} \subset \mathbb{R}^3$, $\mathrm{Lip}_\mathcal{B}$ is the set of the real Lipschitz functions on $\mathbb{R}^3$ which vanish off $\mathcal{B}$; for an open $O \subset \mathbb{E}$, $\mathfrak{S}_n(O)$ is the set of the n-tuples of functions of $C^1_c(\mathbb{E})$ with pairwise disjoint supports in O, $\mathfrak{S}(O) = \bigcup_{n \geq 0} \mathfrak{S}_n(O)$ with the empty tuple, $\mathbb{C}\mathfrak{S}(O)$ the set of the formal complex combinations, $\bar{F}$ the combination with the conjugate coefficients, and $X \cdot Y$ the concatenation, formed when the result is separated; a tuple is of type $(\mathsf{d}; R)$ if its supports lie in balls of radius R and are pairwise at a distance $\geq \mathsf{d}$; $\|X\|_\sharp := \prod_i \|f_i\|_\sharp$. The maps of $\mathbb{E}$ which we use are the translations τ_v, $v \in \mathbb{R}^3$, the reflections $\theta_{u,c} x = x - 2(u \cdot x - c)u$ for unit vectors $u \perp e_3$, the group W_2 of the eight signed permutations of (x_1, x_2), and the reflections $\mathsf{r}_c \colon (x_1, x_2, x_3) \mapsto (x_1, x_2, 2c - x_3)$ of the circle; a map r acts on tuples by $\mathsf{r}X = (f_i \circ \mathsf{r}^{-1})_i$. For a family $S = (S_n)_{n \geq 2}$ of maps $S_n \colon \mathfrak{S}_n(\mathbb{E}) \to \mathbb{R}$, its centred functional is $\sigma_S(X) := \sum_\pi \prod_{I \in \pi} S_{|I|}(X_I)$, the sum over the partitions of the entries into blocks of at least two

entries, $\sigma_S(\varnothing) = 1$, extended linearly to combinations; S is regular if for every n there is a distribution T_n on $\mathbb{E}^n_{\neq} := \{x \in \mathbb{E}^n : x_i \neq x_{i'} \text{ for } i \neq i'\}$ with $T_n(f_1 \otimes \cdots \otimes f_n) = \sigma_S(f_1, \dots, f_n)$ on the smooth separated tuples. For a bounded f on $\mathbb{E}$ with compact support, $\mathcal{O}^{\mathrm{sl}}_a(f)(U) := a^{-1} \sum_p f(\mathrm{pr}\, x(p))\, a_p(U)$ on $\mathcal{V}^{\mathrm{sl}}$, the sum over a set of representatives of the plaquettes up to congruence; for a bounded $\tilde f$ on $\mathbb{R}^3$ with compact support, $\tilde f^{\langle l\rangle}(\mathrm{pr}\, x) := \sum_{k\in\mathbb{Z}} \tilde f(x + kle_3)$, and sorting the plaquettes by classes gives (per) $\mathcal{O}_a(\tilde f) = \mathcal{O}^{\mathrm{sl}}_a(\tilde f^{\langle l\rangle})$ on $\mathcal{V}^{\mathrm{sl}}$. For a slab law ν, $S^T_n[\nu](f_1, \dots, f_n)$ is the joint cumulant of the $\mathcal{O}^{\mathrm{sl}}_a(f_i)$, and $\sigma_\nu(X) := \nu(\prod_i (\mathcal{O}^{\mathrm{sl}}_a(f_i) - \nu(\mathcal{O}^{\mathrm{sl}}_a(f_i))))$. Each $\mathcal{O}^{\mathrm{sl}}_a(f_i)$ is a finite linear combination of functions a_p, so a bounded continuous function on $\mathcal{V}^{\mathrm{sl}}$; for the centred bounded variables $Y_i := \mathcal{O}^{\mathrm{sl}}_a(f_i) - \nu(\mathcal{O}^{\mathrm{sl}}_a(f_i))$ the inverse formula of moments and cumulants writes $\nu(\prod_i Y_i)$ as the sum, over the partitions of the entries, of the products of the joint cumulants of the blocks; a block of one element contributes $\nu(Y_i) = 0$, and a joint cumulant of order ≥ 2 does not change when constants are added to its variables, so $\sigma_\nu = \sigma_S$ for $S = (S^T_n[\nu])_{n\geq 2}$. The standing data [T] of the slab are: $0 < g \leq \gamma$, a side $l \in \mathfrak{L}(g)$, data (K_j, β_j) at g with $K_j \to \infty$ and $a_{K_j}\beta_j \to \mathsf{b}$, and laws $\mu_j \in \mathcal{G}^{\mathrm{box}}(a_{K_j}, \beta_j, l; \mathfrak{L})$; then $l = N_3 a_{K_j}$ with $N_3 = l/a_{K_j} \geq 3$. We write $\ell_g := \min \mathfrak{L}(g) = 2L^{m_{\mathfrak{L}}(g)}$, and, at a pair (K, β), $p_\ell = \ell^{-3} \log Z_{K,\ell,\beta}$ and $p_\infty := \inf_{\ell\in\mathfrak{L}} p_\ell$.

Lemma 18.238 (Slab kernels, boxes and gauge transformations)**.** Let $a > 0$, $\beta \geq 0$ and $l = N_3 a$, $N_3 \geq 3$.

(a) (DLR on a box.) Let $\Gamma = \Gamma^{(2)} \times l\mathbb{Z}$ be a period lattice, $\Gamma^{(2)} \subset a\mathbb{Z}^2$, Φ a real continuous function on $\mathcal{V}_\Gamma$, B thin and $F \in \mathcal{C}_{\mathrm{loc}}$ with a finite set B_F of bonds on which it depends. Suppose (far): the least norm $|\cdot|_\infty$ of the non-zero vectors of $\Gamma^{(2)}$ exceeds by $3a$ the diameter, in the norm $|\cdot|_\infty$, of the projection to the plane $\{x_3 = 0\}$ of the end points of the bonds of $B \cup B_F$. Suppose that $\Phi = \Phi' + \Phi''$, where $\Phi''(U)$ does not change when U is changed on $B + \Gamma$. Then $|\mu^\Phi_\Gamma(\gamma_B F) - \mu^\Phi_\Gamma(F)| \leq (e^{2\|\Phi'\|_\infty} - 1)\|F\|_\infty$; at $\Phi = 0$, $\mu_{\Gamma,\beta}(\gamma_B F) = \mu_{\Gamma,\beta}(F)$.

(b) Every member of $\mathcal{G}^{\mathrm{box}}(a, \beta, l; \mathfrak{l}_0)$ is a slab Gibbs state, and $\mathcal{G}^{\mathrm{box}}(a, \beta, l; \mathfrak{l}_0) \neq \varnothing$. For a thin B and $F \in \mathcal{F}(B)$, every slab Gibbs state ν satisfies $\nu(|F|) \leq e^{8\beta \# B} \int |F|\, dV_B$.

(c) (Gauge.) Every $\nu \in \mathcal{G}^{\mathrm{sl}}$ is invariant under the gauge transformations h_x, $x \in a\mathbb{Z}^3$, $\mathsf{h} \in G$, which multiply the variables of the bonds starting at a site of $x + \mathbb{Z}le_3$ by h on the left, those of the bonds ending there by h^{-1} on the right, and keep the others.

■

Proof. (a) By (far) the bonds of B are pairwise incongruent modulo Γ: a vector of Γ which carries one to another has a vanishing projection to the plane, so it lies in $\mathbb{Z}le_3$. Let $R \supset B$ be a bond transversal of Γ; by the product Haar representation on this transversal, $\mu^\Phi_\Gamma(F) = Z_\Gamma[\Phi]^{-1} \int F\, e^{-\beta S_\Gamma + \Phi}(V^{\mathrm{per}})\, dV_R$. The terms of S_Γ coming from classes of plaquettes that hold no bond of $B + \Gamma$ do not depend on V_B, and Fubini's theorem yields $\mu^\Phi_\Gamma(F) = \mu^\Phi_\Gamma(\hat\gamma F)$. Here $\hat\gamma F(U)$ denotes $\int F\, e^{\hat H + \Phi'}(V \vee_\Gamma U)\, dV_B$ divided by the corresponding integral without the factor F; $V \vee_\Gamma U$ is the configuration obtained from $U \in \mathcal{V}_\Gamma$ by changing it on $B + \Gamma$; and $\hat H = -\beta \sum a_p$, the sum running over those classes modulo Γ of plaquettes which hold a bond of $B + \Gamma$. By (far), a bond of $B + \Gamma$ which lies in B_F, or in a plaquette holding a bond of B^{per}, lies in B^{per}, and two plaquettes holding bonds of B^{per} which are congruent modulo Γ are congruent modulo le_3. So $F(V \vee_\Gamma U) = F(V \vee U)$ and $\hat H(V \vee_\Gamma U) = H^{\mathrm{sl}}_B(V \vee U)$: at $\Phi' = 0$, $\hat\gamma F = \gamma_B F$ on $\mathcal{V}_\Gamma$. For general Φ' the quotient of the densities of $\hat\gamma$ and γ_B lies between $e^{-2\|\Phi'\|_\infty}$ and $e^{2\|\Phi'\|_\infty}$. (b) Let $\mu = \lim_i \mu_{\Gamma_{\Lambda_i},\beta}$ weakly. Each $\mathcal{V}_{\Gamma_\Lambda}$ lies in the closed set $\mathcal{V}^{\mathrm{sl}}$, so μ is a slab law. For thin B and $F \in \mathcal{C}_{\mathrm{loc}}$, (far) holds for large i, so $\mu_{\Gamma_{\Lambda_i},\beta}(\gamma_B F) = \mu_{\Gamma_{\Lambda_i},\beta}(F)$ by (a); $\gamma_B F$ is the restriction to $\mathcal{V}^{\mathrm{sl}}$ of a continuous local function, an integral of a continuous function of finitely many bonds over some of them, so the equation passes to μ. The set of the laws on the compact metrisable space of the configurations is weakly compact, which gives a limit point. For $F \in \mathcal{F}(B)$, $\int |F|(V \vee U)\, dV = \int |F|\, dV_B$, and (dens) with the equation of B gives the bound, first for continuous and then, by monotone classes, for bounded Borel F. (c) Let B_x

be the six bonds with an end point x, a thin set. The map h_x carries $\mathcal{V}^{\mathrm{sl}}$ onto $\mathcal{V}^{\mathrm{sl}}$ and changes only the variables of the bonds of B_x^{per}, so that $\mathsf{h}_x(V\vee U)=(\mathsf{h}_xV)\vee U$, where $V\mapsto \mathsf{h}_xV$ acts factorwise on G^{B_x} by left and right translations, which preserve dV; moreover $a_p\circ\mathsf{h}_x=a_p$, since h_x conjugates the variable of a plaquette. So $\gamma_{B_x}(F\circ\mathsf{h}_x)=\gamma_{B_x}F$, and $\nu(F\circ\mathsf{h}_x)=\nu(\gamma_{B_x}(F\circ\mathsf{h}_x))=\nu(\gamma_{B_x}F)=\nu(F)$ for $F\in\mathcal{C}_{\mathrm{loc}}$, which is dense in the continuous functions. □

Lemma 18.239 (Three sides; the domination of the box limits)**.** Let $K\in\mathbb{N}$, $\beta\ge 0$, and let (i,o) be a board of $\mathbb{L}_K$ of the side $\mathsf{l}=a_i$, with the cells $\square_{\mathbf{k}}$, the sets $\mathcal{A}(\square_{\mathbf{k}})$ and its mirror copies. For F in cell $\mathbf{k}$, its copy $F^{[\mathbf{k}']}$ is obtained by transport across successive walls to cell $\mathbf{k}'$, with complex conjugation when the total number of reflections is odd. In each coordinate the transporting isometry takes the cell center to the new center with linear sign $(-1)^{k'_\mu-k_\mu}$, which makes the copy independent of the order of the wall reflections. For $\underline{\ell}=(\ell_1,\ell_2,\ell_3)\in\mathfrak{L}^3$ with $\ell_\mu\ge 2\mathsf{l}$ let $\Gamma(\underline{\ell}):=\ell_1\mathbb{Z}\times\ell_2\mathbb{Z}\times\ell_3\mathbb{Z}$, $\mathsf{n}_\mu:=\ell_\mu/\mathsf{l}$ (even integers ≥ 2), $\mathsf{N}(\underline{\ell}):=\mathsf{n}_1\mathsf{n}_2\mathsf{n}_3$, $\mathcal{K}_{\underline{\ell}}:=\prod_\mu\{0,\dots,\mathsf{n}_\mu-1\}$, $Z_{\underline{\ell}}:=Z_{\Gamma(\underline{\ell})}[0]$, and for $F\in\mathcal{A}(\square_{\mathbf{k}})$

$$\mathcal{Z}_{\underline{\ell}}[F]:=Z_{\underline{\ell}}\,\mu_{\Gamma(\underline{\ell}),\beta}\Big(\prod_{\mathbf{k}'\in\mathcal{K}_{\underline{\ell}}}F^{[\mathbf{k}']}\Big),\qquad \|F\|_{\mathrm{ch},\underline{\ell}}:=\big(\mathcal{Z}_{\underline{\ell}}[F]/Z_{\underline{\ell}}\big)^{1/\mathsf{N}(\underline{\ell})}.$$

For $\ell\in\mathfrak{L}$ with $\ell\ge 2\mathsf{l}$ we write $\|F\|_{\mathrm{ch},\ell}:=\|F\|_{\mathrm{ch},(\ell,\ell,\ell)}$, the same reflected-product seminorm on a cube as in Lemma 18.216.

(a) $\mathcal{Z}_{\underline{\ell}}[F]\ge 0$; $|\mu_{\Gamma(\underline{\ell}),\beta}(\prod_{\mathbf{k}\in\mathsf{S}}F_{\mathbf{k}})|\le\prod_{\mathbf{k}\in\mathsf{S}}\|F_{\mathbf{k}}\|_{\mathrm{ch},\underline{\ell}}$ for every finite $\mathsf{S}\subset\mathbb{Z}^3$ whose members are pairwise incongruent modulo $\mathsf{n}_1\mathbb{Z}\times\mathsf{n}_2\mathbb{Z}\times\mathsf{n}_3\mathbb{Z}$ and all $F_{\mathbf{k}}\in\mathcal{A}(\square_{\mathbf{k}})$; and $\mathcal{Z}_{\underline{\ell}}[F]\le\mathcal{Z}_{\underline{\ell}'}[F]^{\mathsf{N}(\underline{\ell})/\mathsf{N}(\underline{\ell}')}$ for $\underline{\ell}'\in\mathfrak{L}^3$ with $2\mathsf{l}\le\ell'_\mu\le\ell_\mu$.

(b) For $\ell',l,\Lambda\in\mathfrak{L}$ with $2\mathsf{l}\le\ell'\le l\le\Lambda$: $\|F\|_{\mathrm{ch},(\Lambda,\Lambda,l)}\le e^{\mathsf{l}^3(p_{\ell'}-p_\Lambda)}\|F\|_{\mathrm{ch},\ell'}$.

(c) For $\ell'\le l$ as in (b), $\mu\in\mathcal{G}^{\mathrm{box}}(a_K,\beta,l;\mathfrak{L})$, $\mathsf{q}_{\mathsf{l}}:=e^{\mathsf{l}^3(p_{\ell'}-p_\infty)}$, every finite $\mathsf{S}\subset\mathbb{Z}^3$ whose members are pairwise incongruent modulo $(l/\mathsf{l})\mathbb{Z}e_3$, and all $F_{\mathbf{k}}\in\mathcal{A}(\square_{\mathbf{k}})$: $|\mu(\prod_{\mathbf{k}\in\mathsf{S}}F_{\mathbf{k}})|\le\prod_{\mathbf{k}\in\mathsf{S}}\mathsf{q}_{\mathsf{l}}\|F_{\mathbf{k}}\|_{\mathrm{ch},\ell'}$.

(d) Let $0<g\le\gamma$, (K,β) a datum at g and $\ell'\in\mathfrak{L}(g)$. Then $p_{\ell'}-p_\infty\le 2C_{\mathrm{S}}+C_g$, so $\mathsf{q}_{\mathsf{l}}\le e^{\mathsf{l}^3(2C_{\mathrm{S}}+C_g)}$ in (c). On a board of the side $\mathsf{l}=\frac12\ell_g$, for $l\in\mathfrak{L}(g)$, $\mu\in\mathcal{G}^{\mathrm{box}}(a_K,\beta,l;\mathfrak{L})$ and F a bounded Borel function of the bonds of one cell, $|\mu(F)|\le\mathsf{Q}_0^{(g)}\|F\|_{\mathrm{ch},\ell_g}\le\mathsf{Q}_0^{(g)}\,\tilde\mu_{K,\ell_g,\beta}(|F|^8)^{1/8}$ with $\mathsf{Q}_0^{(g)}:=\exp[\frac18\ell_g^3(2C_{\mathrm{S}}+C_g)]$, a number formed from the constants and g.

■

Proof. The dictionary for three sides. The quotient of the lattice by $\Gamma(\underline{\ell})$ is the rectangular torus with $N_\mu=\ell_\mu/a_K$ sites along the axes. Its positive bonds are a transversal, so the partition functions and Wilson laws in these two descriptions agree. The cells descend with indices modulo $(\mathsf{n}_1,\mathsf{n}_2,\mathsf{n}_3)$. The even cell counts make the signs and conjugation rules periodic. Thus the reflected product in the display is the chessboard fold for the rectangular board.

(a) If $\lambda=\mathsf{l}/a_K\ge 3$, apply Propositions YM3_I:86.3 and YM3_I:86.5 to that board. They allow any even cell counts and use the same model function on both sides of the volume comparison. Unoccupied cells carry the constant 1. If $\lambda=1$, there are no inner bonds and all the cell functions are constants. The seminorm is then their modulus, and the comparison of folded partition functions is the pressure monotonicity of Lemma 18.232(P).

(b) Put $\underline{\ell}=(\Lambda,\Lambda,l)$. By (a) at $\underline{\ell}'=(\ell',\ell',\ell')$, $\mathcal{Z}_{\underline{\ell}}[F]\le\mathcal{Z}_{\ell'}[F]^{\Lambda^2l/\ell'^3}$; by (a) at the function 1 with the sides $(\Lambda,\Lambda,\Lambda)\ge(\Lambda,\Lambda,l)$, $Z_{K,\Lambda,\beta}\le Z_{\underline{\ell}}^{\Lambda/l}$, that is $Z_{\underline{\ell}}\ge e^{p_\Lambda\Lambda^2l}$. With $\mathcal{Z}_{\ell'}[F]=\|F\|_{\mathrm{ch},\ell'}^{(\ell'/\mathsf{l})^3}e^{p_{\ell'}\ell'^3}$,

$$\|F\|_{\mathrm{ch},\underline{\ell}}=\Big(\frac{\mathcal{Z}_{\underline{\ell}}[F]}{Z_{\underline{\ell}}}\Big)^{\mathsf{l}^3/\Lambda^2l}\le\mathcal{Z}_{\ell'}[F]^{\mathsf{l}^3/\ell'^3}e^{-\mathsf{l}^3p_\Lambda}=e^{\mathsf{l}^3(p_{\ell'}-p_\Lambda)}\|F\|_{\mathrm{ch},\ell'}.$$

(c) Let $\mu=\lim_i\mu_{\Gamma_{\Lambda_i},\beta}$ with $\Lambda_i\in\mathfrak{L}$. For large i the members of S are pairwise incongruent modulo the box, and $p_{\Lambda_i}\ge p_\infty$; (a), (b) give the bound for $\mu_{\Gamma_{\Lambda_i},\beta}$, and for continuous $F_{\mathbf{k}}$ it passes to the limit.

For bounded Borel $F_{\mathbf{k}}$: the set B of the bonds lying in the cells of S is thin, since congruent bonds lie in congruent cells and a cell of side $\mathsf{l} \le \frac{l}{2}$ holds no two congruent bonds; by Lemma 18.238 (b), $\mu(|F|) \le e^{8\beta \# B} \int |F|\, dV_B$ for $F \in \mathcal{F}(B)$, and $\tilde{\mu}_{K,\ell',\beta}$ has a bounded density against the Haar measure of its torus. Continuous $F^{(m)}_{\mathbf{k}} \in \mathcal{A}(\square_{\mathbf{k}})$ with $\sup|F^{(m)}_{\mathbf{k}}| \le \sup|F_{\mathbf{k}}|$ and $F^{(m)}_{\mathbf{k}} \to F_{\mathbf{k}}$ almost everywhere for the Haar measure exist, and both members converge by dominated convergence. (d) As in Lemma 18.216, the volume bounds of Theorem YM3_I:AD give $p_{\ell'} - p_\ell \le 2C_{\mathrm{S}} + C_g$ for $\ell, \ell' \in \mathfrak{L}(g)$; as p_ℓ does not increase with ℓ by Lemma 18.232(P) and $\mathfrak{L}(g)$ contains every side larger than one of its members, $p_\infty \ge p_{\ell'} - (2C_{\mathrm{S}} + C_g)$. On the board of the side $\frac{1}{2}\ell_g$, (c) with one cell and $\ell' = \ell_g \le l$ gives the first inequality, as $(\frac{1}{2}\ell_g)^3(2C_{\mathrm{S}} + C_g) = \log \mathsf{Q}^{(g)}_0$; the second follows by Hölder for the eight factors, whose L^8 norms agree by invariance under the lattice isometries of the cube. □

Lemma 18.240 (One bond more; a half space decides)**.** Let $a > 0$, $\beta > 0$ and $l = N_3 a$, $N_3 \ge 3$.

(a) Let $\rho_1, \rho_2 \in \mathcal{G}^{\mathrm{sl}}$, let $D = D^{\mathrm{per}}$ be a set of bonds, $b \in D$, $b_0 \notin D$ a bond, and p_0 a plaquette which holds b and b_0. Suppose that all the bonds of the four plaquettes holding b lie in $D \cup b_0^{\mathrm{per}}$, and that b_0 is the only one of them in b_0^{per}. If $\rho_1(\Psi) = \rho_2(\Psi)$ for every continuous function Ψ of finitely many bonds of D, then the same holds for every continuous function of finitely many bonds of $D \cup b_0^{\mathrm{per}}$.

(b) Let $\mu \in \{1, 2\}$, $c \in \mathbb{R}$ and $\epsilon = \pm 1$. Two slab Gibbs states which agree on $\mathcal{C}_{\mathrm{loc}}(\{\epsilon(x_\mu - c) \ge 0\})$ are equal.

■

Proof. (a) Going along the boundary of p_0 from an end point of b, $a_{p_0} = 1 - \frac{1}{2}\operatorname{tr}(U(b)X_1U(b_0)^{\pm 1}X_2)$, where X_1, X_2 are products of the variables, or of their inverses, of the two other bonds of p_0, which lie in $D' := D \setminus b^{\mathrm{per}}$; and $a_p = 1 - \frac{1}{2}\operatorname{tr}(U(b)Y_p)$ for the three other plaquettes p holding b, with products Y_p of variables of bonds of D'. Put $\mathsf{u} := X_1U(b_0)^{\pm 1}X_2$, $\mathsf{y} := \sum_{p \ne p_0} Y_p$, $(\mathsf{k}\varphi)(u) := \int_G \varphi(v)\, e^{\frac{\beta}{2}\operatorname{tr}(vu)}\, dv$ for $\varphi \in C(G)$, and $\mathsf{Z} := \int e^{\frac{\beta}{2}\operatorname{tr}(v(\mathsf{u}+\mathsf{y}))} dv$, a positive continuous function of finitely many bonds of $D' \cup b_0^{\mathrm{per}}$. For $\varphi \in C(G)$ and a continuous function g of finitely many bonds of D', the function $F := \varphi(U(b))\, e^{-\frac{\beta}{2}\operatorname{tr}(U(b)\mathsf{y})}\, \mathsf{g}$ depends on finitely many bonds of D, and $\gamma_{\{b\}}F = \mathsf{g} \cdot (\mathsf{k}\varphi)(\mathsf{u})/\mathsf{Z}$, since $H^{\mathrm{sl}}_{\{b\}} = -4\beta + \frac{\beta}{2}\operatorname{tr}(U(b)(\mathsf{u} + \mathsf{y}))$. So $\rho_1(\mathsf{g}\,(\mathsf{k}\varphi)(\mathsf{u})/\mathsf{Z}) = \rho_2(\mathsf{g}\,(\mathsf{k}\varphi)(\mathsf{u})/\mathsf{Z})$. By the class-function expansion in the proof of Proposition 18.224, $e^{\frac{\beta}{2}\operatorname{tr}} = e^{\beta}\sum_l (2l+1)\lambda_l(\beta)\chi_l$ uniformly on G, with $\lambda_l(\beta) > 0$ for all $l \in \frac{1}{2}\mathbb{N}_0$; by the orthogonality relations of the matrix coefficients, the function $\varphi(v) = D^l_{mn}(v^{-1})$ has $(\mathsf{k}\varphi)(u) = e^{\beta}\lambda_l(\beta)D^l_{mn}(u)$. So the image of k contains all matrix coefficients, whose span is dense in $C(G)$. For a finite set $E \subset D'$ which holds the bonds of X_1, X_2, the map $(U_E, U(b_0)) \mapsto (U_E, \mathsf{u})$ is a homeomorphism of $G^E \times G$, and the span of the functions $\mathsf{g}\,\psi(\mathsf{u})$, $\mathsf{g} \in C(G^E)$, ψ a matrix coefficient, is a self-conjugate algebra with the constants which separates the points, dense by the theorem of Stone and Weierstrass. Hence $\rho_1(\Psi/\mathsf{Z}) = \rho_2(\Psi/\mathsf{Z})$ for every continuous Ψ of finitely many bonds of $D' \cup b_0^{\mathrm{per}}$, and with $\Psi\mathsf{Z}$ in the place of Ψ, $\rho_1 = \rho_2$ on these functions. Last, for a continuous Ψ' of finitely many bonds of $D \cup b_0^{\mathrm{per}}$, $\gamma_{\{b\}}\Psi'$ is a continuous function of finitely many bonds of $D' \cup b_0^{\mathrm{per}}$, by the hypothesis on the four plaquettes, and $\rho_i(\Psi') = \rho_i(\gamma_{\{b\}}\Psi')$. (b) Let $\mu = 1$, $\epsilon = 1$; the other cases are the same after renaming. Let c_1 be the least member of $a\mathbb{Z}$ which is $\ge c$, D_m the set of the bonds lying in $\{x_1 \ge c_1 - ma\}$, and suppose $\rho_1 = \rho_2$ on the continuous functions of finitely many bonds of D_m; put $c' := c_1 - ma$. *The bonds which stick out.* Let $E := D_m \cup \{\langle y, y + ae_1\rangle : y_1 = c' - a\}$, Ψ a continuous function of finitely many bonds of E, and Y a finite set of pairwise incongruent sites y with $y_1 = c' - a$ such that Ψ reads no other bond which sticks out, up to congruence. Of the bonds of E, the gauge transformation h_y moves the class of $\langle y, y + ae_1\rangle$ alone, by a left translation; by Lemma 18.238 (c), $\rho_i(\Psi) = \rho_i(\bar{\Psi})$ with $\bar{\Psi} := \int \Psi \circ \prod_{y \in Y}(\mathsf{h}_y)_y \prod_y d\mathsf{h}_y$, which by the invariance of the Haar measure depends on bonds of D_m alone. So $\rho_1 = \rho_2$ on these functions of E. *The bonds of the new plane.* Number the classes of

the bonds $\langle y, y+ae_\nu\rangle$, $y_1 = c'-a$, $\nu\in\{2,3\}$, as $b_0^1, b_0^2, \dots$, and put $E_k := E\cup (b_0^1)^{\rm per}\cup\dots\cup(b_0^k)^{\rm per}$. For $b_0 = b_0^{k+1} = \langle y, y+ae_\nu\rangle$ take $b := \langle y+ae_1, y+ae_1+ae_\nu\rangle \in D_m$ and $p_0 := P_{1\nu}(y)$. The four plaquettes at b are p_0, whose other two bonds stick out, $P_{1\nu}(y+ae_1)$, and two plaquettes in the plane $\{x_1 = c'\}$; of their bonds only b_0 has both end points in $\{x_1 = c'-a\}$. (a) with $D = E_k$ gives $\rho_1 = \rho_2$ on the functions of E_{k+1}. Every continuous function of finitely many bonds of D_{m+1} is one of some E_k. By induction $\rho_1 = \rho_2$ on $\mathcal{C}_{\rm loc}$, which is dense in $C(\mathcal{V}^{\rm sl})$. □

Lemma 18.241 (Separation; a comparison state)**.** (a) Let X be a compact metrisable space, $\mathcal{C}_0$ a dense subspace of the real continuous functions on X, $\mathsf{R}\neq\varnothing$ a convex and weakly compact set of Borel probability measures on X, and ν a Borel probability measure on X with $\nu(A)\le \sup_{\rho\in\mathsf{R}}\rho(A)$ for all $A\in\mathcal{C}_0$. Then $\nu\in\mathsf{R}$.

(b) Let $a>0$, $\beta\ge 0$, $l = N_3 a$ with $N_3\ge 3$, $n\in\{e_1, e_2, e_1+e_2, e_1-e_2\}$, $c_0\in{\rm Adm}(n)$, $\nu\in\mathcal{G}^{\rm sl,per}$, and $A\in\mathcal{C}_{\rm loc}(\mathbb{H}^+(n,c_0))$ real. There is $\nu_0\in\mathcal{G}^{\rm sl,per}$ which is reflection positive at $\Sigma(n,c_0)$ and satisfies $\nu(A)\le\nu_0(A)$.

■

Proof. (a) If $\nu\notin\mathsf{R}$, the theorem of Hahn and Banach in the dual of the real continuous functions, with the weak topology, gives a real continuous A and $\varepsilon>0$ with $\nu(A)\ge\sup_{\mathsf{R}}\rho(A)+3\varepsilon$; an $A'\in\mathcal{C}_0$ with $\|A-A'\|_\infty\le\varepsilon$ violates the hypothesis.

(b) *Step 1: lattices.* Let ν be invariant under $Pa\mathbb{Z}^2\times\{0\}$, and let $n^* := e_2, e_1, e_1-e_2, e_1+e_2$ in the four cases, so that $n^*\perp n$ and $|n^*| = |n|$. Take $q\in\mathbb{N}$ so large that, with $h := qPa|n|^2$, the finite set B_A of bonds on which A depends lies in the slab $\mathbb{S}_1 = \{0\le n\cdot x - c_0\le h\}$; then $v_h := h|n|^{-2}n = qPa\,n$. For integers $i\ge 3$ put $\Gamma' := \mathbb{Z}qPa\,n^*\oplus\mathbb{Z}le_3\oplus\mathbb{Z}v_h$ and $\Gamma_i := \mathbb{Z}iqPa\,n^*\oplus\mathbb{Z}le_3\oplus\mathbb{Z}2iv_h$. Both are subgroups of finite index of $a\mathbb{Z}^3$ of the form $\Gamma^{(2)}\times l\mathbb{Z}$, adapted to n by the perpendicular and normal decomposition in YM3_I:(1820) with the height periods h and $2ih$; $\Gamma_i\subset\Gamma'$, $[\Gamma':\Gamma_i] = 2i^2$; a member $\alpha iqPa\,n^*+\delta 2iqPa\,n$ of $\Gamma_i^{(2)}$ with $(\alpha,\delta)\neq 0$ has a coordinate of modulus $\ge iqPa$, so Γ_i is a period lattice whose plane part has a least norm tending to ∞. Let $\mathbb{A}_i$ be the array of A over Γ'/Γ_i and $\mathbb{A}_i^{\rm f}$ its folded array with the data (a,n,c_0,h), $N=i$ in Proposition YM3_I:91.7; $|\mathbb{A}_i| = 2i^2$.

Step 2: a box against the torus. There is C_∂, independent of i and of $t\in(0,1]$, with $2i^2t\,\nu(A)\le \log(Z_{\Gamma_i}[t\mathbb{A}_i]/Z_{\Gamma_i}[0]) + C_\partial\, i\,(1+2t\|A\|_\infty)$. Indeed, $D_i := \{x\in a\mathbb{Z}^3 : 0\le n\cdot x - c_0 < 2ih,\ 0\le n^*\cdot x < iqPa|n|^2,\ 0\le x_3<l\}$ is a domain of Γ_i. Let r exceed by 2 the diameter of B_A in units of a, D_i° the set of the $x\in D_i$ with $x+w\in D_i+\mathbb{Z}le_3$ for all $w\in a\mathbb{Z}^3$, $|w|_\infty\le ra$, and $B_i := \{\langle x, x+ae_\mu\rangle : x\in D_i^\circ\}$, a thin set; $\#(D_i\setminus D_i^\circ)\le C_1 i$, as these points lie within ra of the four sides of a parallelogram of sides at most $2iqPa|n|$, in N_3 layers. Let R_i be the set of the representatives v of Γ'/Γ_i, so chosen, with $\tau_v B_A\subset B_i^{\rm per}$; the others have a bond starting, up to Γ_i, within ra of $D_i\setminus D_i^\circ$, so $2i^2-\#\mathsf{R}_i\le C_2 i$. Put $W_i := \sum_{v\in\mathsf{R}_i}\tau_v A$. As $\Gamma'\subset Pa\mathbb{Z}^2\times\{0\}+\mathbb{Z}le_3$, $\nu(\tau_v A) = \nu(A)$. Jensen's inequality and the equation of B_i give $e^{t\#\mathsf{R}_i\nu(A)}\le\nu(e^{tW_i}) = \nu(\gamma_{B_i}e^{tW_i})$. Let $\mathsf{P}_i^{\rm in}\subset\mathsf{Pl}^{\rm sl}[B_i]$ be the plaquettes with all their bonds in $B_i^{\rm per}$; the others have a corner in $(D_i\setminus D_i^\circ)+\mathbb{Z}le_3$ or next to it, and their number is at most $C_3 i$. With $I_i(t) := \int e^{tW_i-\beta\sum_{\mathsf{P}_i^{\rm in}}a_p}dV_{B_i}$ and $0\le a_p\le 2$, $\gamma_{B_i}e^{tW_i}\le e^{2\beta C_3 i}I_i(t)/I_i(0)$. On the torus take the bond transversal $\mathsf{Bd}_{D_i}\supset B_i$: the classes of plaquettes are those of $\mathsf{P}_i^{\rm in}$ and at most $C_4 i$ others, and $\mathbb{A}_i$ is W_i plus at most $C_2 i$ terms of modulus $\le\|A\|_\infty$; so $Z_{\Gamma_i}[t\mathbb{A}_i]\ge e^{-2\beta C_4 i - tC_2 i\|A\|_\infty}I_i(t)$ and $Z_{\Gamma_i}[0]\le I_i(0)$. Together with $\#\mathsf{R}_i\nu(A)\ge 2i^2\nu(A) - C_2 i\|A\|_\infty$ this is the claim, with $C_\partial = \max\{C_2, 2\beta(C_3+C_4)\}$.

Step 3: the fold. By the main text's chessboard estimate along one normal, YM3_I:(1828), $Z_{\Gamma_i}[t\mathbb{A}_i]\le Z_{\Gamma_i}[t\mathbb{A}_i^{\rm f}]$. The function $s\mapsto\log Z_{\Gamma_i}[s\mathbb{A}_i^{\rm f}]$ is convex and differentiable, with the derivative $\mu_i^s(\mathbb{A}_i^{\rm f}) = 2i^2\mu_i^s(A)$, $\mu_i^s := \mu_{\Gamma_i}^{s\mathbb{A}_i^{\rm f}}$, by YM3_I:(1829); so $\log(Z_{\Gamma_i}[t\mathbb{A}_i^{\rm f}]/Z_{\Gamma_i}[0])\le 2i^2 t\,\mu_i^t(A)$, and Step 2 gives $\nu(A)\le\mu_i^t(A) + C_\partial(2\|A\|_\infty + t^{-1})/(2i)$. Take $t_i := i^{-1/2}$ and a weak limit point ν_0 of $(\mu_i^{t_i})_i$: then $\nu(A)\le\nu_0(A)$.

Step 4: what ν_0 is. Each $\mathcal{V}_{\Gamma_i}$ lies in $\mathcal{V}^{\mathrm{sl}}$, so ν_0 is a slab law. By Proposition YM3_I:91.7, μ_i^t is reflection positive at the pair $\Sigma(n,c_0)$, $\Sigma(n,c_0+ih)$, the height period of Γ_i along n being $2ih$; this means $\mu_i^t(\overline{\theta_{n,c_0}F}\,F)\ge 0$ for the bounded Borel functions F of the bonds lying in $\{c_0\le n\cdot x\le c_0+ih\}$, which hold every member of $\mathcal{C}_{\mathrm{loc}}(\mathbb{H}^+(n,c_0))$ for large i; so $\nu_0(\overline{\theta_{n,c_0}F}F)\ge 0$ on $\mathcal{C}_{\mathrm{loc}}(\mathbb{H}^+(n,c_0))$, and ν_0 is reflection positive at $\Sigma(n,c_0)$ by Lemma 18.242. The same clause makes μ_i^t invariant under $\tau_{qPa\,n^*}$ and under the wall group of the data (a,n,c_0,h), that is, the group generated by the reflections θ_{n,c_0+kh}, $k\in\mathbb{Z}$, a group containing $\theta_{n,c_0+h}\theta_{n,c_0}=\tau_{2v_h}$; ν_0 inherits these invariances, and hence some subgroup of $a\mathbb{Z}^2\times\{0\}$ with finite index leaves ν_0 invariant. *ν_0 is a slab Gibbs state.* Fix a thin B and a function $F\in\mathcal{C}_{\mathrm{loc}}$. The group W generated by the wall group and the translations by Γ' consists of the maps τ_w and $\tau_w\theta_{n,c_0}$, $w\in\Gamma'$; $\mathbb{A}_i^{\mathrm{f}}=\sum_{j=1}^{2i}\sum_v \mathsf{w}_j\tau_v A$ with $N=i$ in Proposition YM3_I:91.7, where v runs over $\{mqPa\,n^*:0\le m<i\}$, a set of representatives of the i classes of $(\Gamma'\cap n^\perp)/(\Gamma_i\cap n^\perp)$, so that here $v\perp n$, unlike the representatives v of Γ'/Γ_i in Step 2, and where w_j, a map and not one of the sets w_d of a window, is the member of the wall group that carries the slab $\mathbb{S}_1$ onto the slab $\mathbb{S}_j:=\{(j-1)h\le n\cdot x-c_0\le jh\}$ (the translation by $(j-1)v_h$ for odd j, and $\theta_{n,c_0+jh/2}$ for even j); so $\mathbb{A}_i^{\mathrm{f}}$ is a sum of $2i\cdot i=2i^2$ functions $\mathsf{g}A$, $\mathsf{g}=\mathsf{w}_j\tau_v$, $1\le j\le 2i$, v as above, which lie in $2i^2$ different cosets of τ_{Γ_i} in W, as $\mathsf{w}_j\tau_v\in\tau_{\Gamma_i}\mathsf{w}_{j'}\tau_{v'}$ forces the slabs $\mathbb{S}_j$, $\mathbb{S}_{j'}$ to agree up to $2ih$ and then $v-v'\in\Gamma_i$. A term $\mathsf{g}A$ reads the bonds of $B+\Gamma_i$ only if $\mathsf{g}b'\in b+\Gamma_i$ for some $b'\in B_A$, $b\in B$. Every isometry that maps a closed segment onto itself leaves the midpoint of the segment fixed. A map τ_w can fix a point only when $w=0$; a map $\tau_w\theta_{n,c_0}$, with $w=w_\perp+kv_h$ and $w_\perp\perp n$, can fix a point only when $w_\perp=0$, and its fixed points are then precisely the points of the plane $\{n\cdot x=c_0+\frac{kh}{2}\}$; and a midpoint lies on one of these planes at most. So the stabiliser in W of a bond has at most two elements, and as τ_{Γ_i} is normal in W because ϱ_n is the identity on the transverse periods and changes the sign of the normal period, for given b, b' at most two cosets occur. Hence $t_i\mathbb{A}_i^{\mathrm{f}}=\Phi'+\Phi''$ as in Lemma 18.238 (a), with $\|\Phi'\|_\infty\le 2t_i\#B\#B_A\|A\|_\infty\to 0$; (far) holds for large i; so $|\mu_i^{t_i}(\gamma_B F)-\mu_i^{t_i}(F)|\to 0$, and, $\gamma_B F$ and F being continuous, $\nu_0(\gamma_B F)=\nu_0(F)$. □

Lemma 18.242 (Reflection of a periodically translated half-space law)**.** For a law ν on the lattice, reflection positivity at a mirror on continuous local functions extends to bounded Borel functions of finitely many bonds in the same closed half-space. If ν is invariant under translation by a nonzero lattice vector parallel to the mirror normal, this positivity also implies invariance under the reflection. ■

Proof. Approximate a bounded local function by bounded continuous functions of the same coordinates in $L^2(\nu+\theta_*\nu)$. The two-factor integral is continuous in this approximation by Cauchy–Schwarz, so it remains nonnegative. On the positive side the positive sesquilinear form is Hermitian. Taking one argument equal to 1 gives $\nu(\theta F)=\nu(F)$ for every continuous local F on that side. For any continuous local function, a sufficiently large integer multiple of the given period moves its support into the positive side. Reflection changes that shift to its negative. Invariance under both shifts reduces the identity on the translated function to the desired identity on the original function. Continuous local functions determine the law. □

Proposition 18.243 (The mirrors of the slab Gibbs states)**.** Let $a>0$, $\beta\ge 0$ and $l=N_3a$, $N_3\ge 3$.

(a) Every $\nu\in\mathcal{G}^{\mathrm{sl,per}}$ is reflection positive at $\Sigma(e_\mu,c)$ for $\mu=1,2$ and $c\in\frac{a}{2}\mathbb{Z}$, and at $\Sigma(e_1\pm e_2,c)$ for $c\in a\mathbb{Z}$.

(b) Every $\nu\in\mathcal{G}^{\mathrm{sl,per}}$ is invariant under these reflections, under the translations by $a\mathbb{Z}^2\times\{0\}$ and under W_2, acting on configurations.

(c) $\mathcal{G}^{\mathrm{sl,per}}$ coincides with the set of those members of $\mathcal{G}^{\mathrm{sl}}$ which $a\mathbb{Z}^2\times\{0\}$ leaves invariant, and it is a convex and weakly compact set.

(d) $\varnothing\ne\mathcal{G}^{\mathrm{box}}(a,\beta,l;\mathfrak{l}_0)\subset\mathcal{G}^{\mathrm{sl,per}}$; its members are besides invariant under all translations of $a\mathbb{Z}^3$ and under $\theta_{e_3,c}$, $c\in\frac{a}{2}\mathbb{Z}$, and positive at every pair $\Sigma(e_3,c)$, $\Sigma(e_3,c+\frac{l}{2})$, $c\in\frac{a}{2}\mathbb{Z}$.

In [T] every μ_j is therefore reflection positive at the lattice mirrors of $\mathbb{L}_{K_j}$ with the normals $\pm e_1, \pm e_2, \pm(e_1 \pm e_2)$, invariant under the translations of $\mathbb{L}_{K_j}$, under W_2 and under the reflections $\theta_{e_3,c}$, $c \in \frac{1}{2}a_{K_j}\mathbb{Z}$, and positive at the pairs $\Sigma(e_3, c)$, $\Sigma(e_3, c + \frac{l}{2})$. ■

Proof. $\mathcal{G}^{\mathrm{sl}}$ *is weakly compact and convex,* since it consists of those laws on the compact metrisable $\mathcal{V}^{\mathrm{sl}}$ that satisfy the linear conditions $\nu(\gamma_B F - F) = 0$; these conditions are closed, as $\gamma_B F - F$ is continuous for $F \in \mathcal{C}_{\mathrm{loc}}$. At $\beta = 0$, $\gamma_B F = \int F(V \vee \cdot)\, dV$ and every continuous local function is, on $\mathcal{V}^{\mathrm{sl}}$, a function of a thin set B; so $\nu(F) = \int F\, dV_B$, and $\mathcal{G}^{\mathrm{sl}}$ has one member.

(d) Non-empty and slab Gibbs states: Lemma 18.238 (b). Each $\mu_{\Gamma_\Lambda,\beta}$ is invariant under the $\mathsf{r} \in \mathsf{Iso}(a\mathbb{Z}^3)$ with $\mathsf{r}^\circ\Gamma_\Lambda = \Gamma_\Lambda$ by the relabelling of bonds and plaquettes: the translations of $a\mathbb{Z}^3$, W_2, the first two sides being equal, and the reflections $\theta_{e_3,c}$; and it is reflection positive at the pair $\Sigma(e_3, c)$, $\Sigma(e_3, c + \frac{l}{2})$ by Proposition YM3_I:91.4, since the box period lattice is adapted to e_3 with height period l. These properties, read at continuous local functions, pass to weak limits.

(a) for $n = e_\mu$, $\mu \in \{1, 2\}$. Let $\nu \in \mathcal{G}^{\mathrm{sl,per}}$ and $c_0 \in \frac{a}{2}\mathbb{Z}$. Let X be the image of $\mathcal{V}^{\mathrm{sl}}$ under the restriction to the bonds lying in $\mathbb{H}^+(n, c_0)$, ρ^+ the image of a slab law ρ, $\mathcal{C}_0$ the set of the real members of $\mathcal{C}_{\mathrm{loc}}(\mathbb{H}^+(n, c_0))$, dense in the real continuous functions on X by the theorem of Stone and Weierstrass, and $\mathsf{R} := \{\rho^+ : \rho \in \mathcal{G}^{\mathrm{sl}},\ \rho(\overline{\theta_{n,c_0}F}F) \geq 0 \text{ for } F \in \mathcal{C}_{\mathrm{loc}}(\mathbb{H}^+(n, c_0))\}$, the continuous linear image of a convex weakly compact set. By Lemma 18.241 (b), $\mathsf{R} \neq \varnothing$ and $\nu^+(A) \leq \sup_{\mathsf{R}} \rho^+(A)$ on $\mathcal{C}_0$; by Lemma 18.241 (a), $\nu^+ = \rho^+$ for some such ρ. By Lemma 18.240 (b) if $\beta > 0$, and because $\mathcal{G}^{\mathrm{sl}}$ has one member if $\beta = 0$, $\nu = \rho$. By Lemma 18.242, ν is reflection positive at $\Sigma(e_\mu, c_0)$.

(b), first half, and (c). ν is invariant under τ_{Pae_μ}, a translation parallel to e_μ; by Lemma 18.242 it is invariant under $\theta_{e_\mu,c}$ for $c \in \frac{a}{2}\mathbb{Z}$; and $\theta_{e_\mu,a/2}\theta_{e_\mu,0} = \tau_{ae_\mu}$. So every member of $\mathcal{G}^{\mathrm{sl,per}}$ is invariant under $a\mathbb{Z}^2 \times \{0\}$, which is (c), the set being cut out of $\mathcal{G}^{\mathrm{sl}}$ by closed linear conditions.

(a) for $n = e_1 \pm e_2$. Let $c_0 \in a\mathbb{Z}$, X and $\mathcal{C}_0$ as before, and $\mathsf{R} := \{\rho^+ : \rho \in \mathcal{G}^{\mathrm{sl,per}},\ \rho$ positive at $\Sigma(n, c_0)$ on $\mathcal{C}_{\mathrm{loc}}(\mathbb{H}^+(n, c_0))\}$, convex and weakly compact by (c). Lemma 18.241 gives $\nu^+ = \rho^+$. Let $A \in \mathcal{C}_{\mathrm{loc}}$; then $A \in \mathcal{C}_{\mathrm{loc}}(S)$ for a bounded set S, and $\tau_v A \in \mathcal{C}_{\mathrm{loc}}(S + v)$ for $v \in a\mathbb{Z}^3$. For a large $k \in \mathbb{N}$ the vector $v := ka\, n = (ka, \pm ka, 0)$ of $a\mathbb{Z}^2 \times \{0\}$ has $n \cdot v = 2ka$, so $S + v \subset \mathbb{H}^+(n, c_0)$ and $\tau_v A \in \mathcal{C}_{\mathrm{loc}}(\mathbb{H}^+(n, c_0))$; by (c), τ_v leaves ν and ρ invariant, so that $\nu(A) = \nu(\tau_v A) = \rho(\tau_v A) = \rho(A)$. So $\nu = \rho$, and Lemma 18.242 applies.

(b), second half. ν is invariant under τ_{an}, parallel to n, so under $\theta_{n,c}$ by Lemma 18.242. The linear parts of $\theta_{e_1,0}$, $\theta_{e_2,0}$ and $\theta_{e_1-e_2,0}$ generate W_2. Reflection positivity at $\Sigma(-n, -c)$ is the same property of the plane because invariance under the reflection exchanges the two half-spaces. The last sentence is (a), (b), (d) at $a = a_{K_j}$. □

For a tuple $\underline{f} = (f_q)$ placed on a board, each support lies in the interior of one cell. For z in the source disc, write $w_{\mathbf{k}} = \sum_{q \text{ in cell } \mathbf{k}} z_q f_q$ and $B(z) = \{q : z_q \neq 0\}$. The number $\Pi_{\ell'}(z) = \prod_{\mathbf{k}} \|e^{\mathcal{O}_K(w_{\mathbf{k}})}\|_{\mathrm{ch},\ell'}$ uses the occupied cells. A normaliser on a set of sides $\mathfrak{l}$ is a holomorphic $\mathcal{E}$ on the disc with $\mathcal{E}(0) = 0$ and $\Pi_{\ell'}(z) \leq |e^{\mathcal{E}(z)}|Q$ on those sides. In the graded case the bound is $|e^{\mathcal{E}(z)}|Q^{\#B(z)}$.

Proposition 18.244 (Transfer and growth on the slab)**.** (a) (Lift.) Let $f \in \mathrm{Lip}_c(\mathbb{E})$ have its support in a ball of radius R of $\mathbb{E}$, and let $l \geq 1$. There is $\tilde{f} \in \mathrm{Lip}_c(\mathbb{R}^3)$ with $\tilde{f}^{\langle l \rangle} = f$, $\mathrm{pr}(\mathrm{supp}\, \tilde{f}) = \mathrm{supp}\, f$, $\mathrm{supp}\, \tilde{f}$ in a ball of radius $2R + 1$ and $\|\tilde{f}\|_\sharp \leq (1 + 40M)\|f\|_\sharp$; if $R < \frac{l}{2}$, there is one with its support in a ball of radius R and $\|\tilde{f}\|_\sharp \leq \|f\|_\sharp$, of class C^1 if f is.

(b) Let $K \in \mathbb{N}$, $\beta > 0$, $\ell' \leq l$ in $\mathfrak{L}$ and $\mu \in \mathcal{G}^{\mathrm{box}}(a_K, \beta, l; \mathfrak{L})$. Let $\underline{f}$ be placed on a board of the side $\mathsf{l} \leq \frac{1}{2}\ell'$, its occupied cells pairwise incongruent modulo $(l/\mathsf{l})\mathbb{Z}e_3$, and let $\mathcal{E}$ be a normaliser of $\underline{f}$ of the radius ϱ on a set $\mathfrak{l} \ni \ell'$ with the bound Q, respectively a graded normaliser with the base Q. Then, with $w = \sum_q z_q f_q$, $|e^{-\mathcal{E}(z)}\mu(e^{\mathcal{O}_K(w)})| \leq \mathsf{q}_{\mathsf{l}}^{\#\mathrm{occ}(\underline{f})}Q$, respectively $\leq (\mathsf{q}_{\mathsf{l}}Q)^{\#B(z)}$, on $\mathbb{D}_\varrho(\underline{f})$.

(c) In [T], let (K,β) be a datum at g, $\mu\in\mathcal{G}^{\mathrm{box}}(a_K,\beta,l;\mathfrak{L})$ and $\mathsf{q}_g:=e^{2C_{\mathrm{S}}+C_g}$. For $n\ge 2$, $R,\mathsf{d}>0$ and $f_q\in\mathrm{Lip}_c(\mathbb{R}^3)$ with supports in balls of radius R which are pairwise at a distance $\ge\mathsf{d}$ modulo $l e_3$,

$$|S_n^T[\mu](f_1,\dots,f_n)|\le A^{\mathrm{per}}(\mathsf{d})\,B^{\mathrm{sl}}(R)^n\,(n!)^{5+\mathsf{p}}\prod_q\|f_q\|_\sharp,\qquad B^{\mathrm{sl}}(R):=C_B^{\mathrm{sl}}(R+1)^3,$$

where p,C_Q are the constants for the side-one normalizer in Proposition YM3_I:91.12, $A^{\mathrm{per}}(\mathsf{d})=1+\mathsf{U}(1,\mathsf{d})$, and C_B^{sl} is the number in YM3_I:(1848) with q_gC_Q in place of C_Q.

(d) In [T], for $n\ge 2$ and Lipschitz f_i on $\mathbb{E}$ forming a tuple X of type $(\mathsf{d};R)$, whose supports may wind around the circle,

$$\begin{aligned}&|S_n^T[\mu](X)|\le C_n^{\mathrm{sl}}(\mathsf{d};R)\|X\|_\sharp,\\ &C_n^{\mathrm{sl}}(\mathsf{d};R):=A^{\mathrm{per}}(\mathsf{d})\big[(1+40M)B^{\mathrm{sl}}(2R+1)\big]^n(n!)^{5+\mathsf{p}};\\ &|\sigma_\mu(X)|\le C_n^{\sigma}(\mathsf{d};R)\|X\|_\sharp,\\ &C_n^{\sigma}:=\sum_{\pi}\prod_{I\in\pi}C_{|I|}^{\mathrm{sl}},\end{aligned}$$

the sum over the partitions of $\{1,\dots,n\}$ into blocks of at least two elements. These numbers are formed from $\mathfrak{D}$, g and the arguments shown; they read none of K, β, l, the state or a position, and they do not increase in d.

■

Proof. (a) If $R<\frac{l}{2}$, let $\mathbb{B}(\tilde x,R)\subset\mathbb{R}^3$ be a ball over the given ball; two of its points are at a distance $\le 2R<l$, hence never differ by a non-zero multiple of le_3, and pr is injective on this ball; besides, $|\mathrm{pr}\,y-\mathrm{pr}\,y'|\le|y-y'|$ for all $y,y'\in\mathbb{R}^3$. Let $\tilde f$ equal $f\circ\mathrm{pr}$ on $\mathbb{B}(\tilde x,R)$ and vanish off it. The closed set $\mathrm{pr}^{-1}(\mathrm{supp}\,f)$ is the disjoint union of the translates of $\mathrm{supp}\,\tilde f$, pairwise at a distance $\ge l-2R>0$; a segment from a point of $\mathrm{supp}\,\tilde f$ to a point outside it meets a point where $f\circ\mathrm{pr}$ vanishes, so $\mathrm{Lip}\,\tilde f\le\mathrm{Lip}\,f$. If $R\ge\frac{l}{2}$, let $\psi(s)=s$ on $[0,1]$, 1 on $[1,l]$, $l+1-s$ on $[l,l+1]$ and 0 elsewhere; then $\sum_k\psi(s+kl)=1$ and $\mathrm{Lip}\,\psi=1$; put $\tilde f(y):=\psi(y_3)f(\mathrm{pr}\,y)$, whose support lies in a disc of radius R times $[0,l+1]$, so in a ball of radius $2R+1$, and $\mathrm{Lip}\,\tilde f\le\mathrm{Lip}\,f+\|f\|_\infty$. (b) $\mu(e^{\mathcal{O}_K(w)})=\mu(\prod_{\mathbf{k}\in\mathrm{occ}}e^{\mathcal{O}_K(w_{\mathbf{k}})})$ with $e^{\mathcal{O}_K(w_{\mathbf{k}})}\in\mathcal{A}(\square_{\mathbf{k}})$ because any plaquette with center inside a cell of an odd board has all its bonds in that cell; Lemma 18.239 (c) bounds its modulus by $\mathsf{q}_{\mathsf{l}}^{\#\mathrm{occ}}\Pi_{\ell'}(z)$, and the normaliser bounds $\Pi_{\ell'}(z)$ by $|e^{\mathcal{E}(z)}|Q$; in the graded case the cells with $w_{\mathbf{k}}=0$ carry the factor 1, there are at most $\#B(z)$ others, and $\mathsf{q}_{\mathsf{l}}\ge 1$. (c) Suppose first that the entries v_q are nonzero, lie in balls of radius $\eta\le 1/4$ and have separation at least d modulo le_3, with $4a_K\le\eta$ and $4n\eta+na_K\le 1$. An offset in $a_K\mathbb{Z}^3$ places each center at distance at least 2η from all walls of a side-one board. In each coordinate the forbidden intervals have length at most $4n\eta$ modulo one, and rounding to the lattice costs at most na_K. An offset remains by the placement estimate in the proof of Proposition YM3_I:91.13. Choose one occupied cell from each congruence class modulo the period. Translate every entry by a multiple of le_3 into the representative of its cell. These translations preserve the observables on $\mathcal{V}^{\mathrm{sl}}$, the norms and the separation modulo le_3. In particular the ordinary separation is at least d. The third normalizer of Proposition YM3_I:91.12 then has radius ϱ and graded base $Q(\eta)=C_Q(2\eta)^{-\mathsf{p}}$. Its mixed derivative has bound YM3_I:(1845). By (b), with $\ell'=\ell_g$ and $\mathsf{l}=1$, the bound transfers to the slab with base $\mathsf{q}_gQ(\eta)$. The arbitrary-law estimate YM3_I:(1842) and the derivative bound give

$$|S_n^T[\mu](\underline v)|\le\varrho^{-n}\left[(n-1)!(2^n-1)(e\,\mathsf{q}_gQ(\eta))^n+n^n\mathsf{U}(\eta,\mathsf{d})\right]\prod_q\|v_q\|_\sharp.$$

For general f_q, if $a_K>1/(32n)$ the plaquette count gives $\|\mathcal{O}_K(f_q)\|_\infty\le 192\cdot 64^3n^4(R+1)^3\|f_q\|_\sharp$. The bounded-variable cumulant estimate used in the proof of Proposition YM3_I:91.13 then gives its coarse-mesh term. This argument only uses the finite set of plaquette centers in each lifted

support, so applies to the slab law as well. If $a_K \le 1/(32n)$, set $\eta = 1/(8n)$ and split the entries by tensor-product tents of mesh $\eta/\sqrt{3}$. The pieces remain in the original supports, hence preserve the separation modulo le_3. The number of tuples of pieces is at most $[28n(R+1)]^{3n}$, and their sharp norms are at most $961Mn$ times the original norms, as in the proof of Proposition YM3_I:91.13. The small-ball estimate applies to every nonzero tuple of pieces, with $Q(\eta) \le C_Q(8n)^{\mathsf{p}}$ and $\mathsf{U}(\eta, \mathsf{d}) \le \mathsf{U}(1, \mathsf{d})$. Using $n^n \le e^n n!$, the counts and norms contribute four factorial powers. The bracket in the small-ball estimate contributes at most $1 + \mathsf{p}$ more, with the prefactor $1 + \mathsf{U}(1, \mathsf{d})$. Thus the summation gives the constant in YM3_I:(1848) with C_Q replaced by $\mathsf{q}_g C_Q$.

(d) Take lifts by (a): supports in balls of radius $2R+1$, pairwise at a distance $\ge \mathsf{d}$ modulo le_3, $\|\tilde{f}_i\|_\sharp \le (1 + 40M)\|f_i\|_\sharp$; then (per) and (c). The bound on σ_μ is the formula of moments and cumulants, each block X_I being of type $(\mathsf{d}; R)$; $A^{\mathrm{per}} = 1 + \mathsf{U}(1, \cdot)$ does not increase, and their other dependencies are explicit in the bounds above and in Lemma 18.239(d). □

Proposition 18.245 (Short distance on the slab)**.** In [T], let $\vartheta' \in (0,1)$, let (K, β) be a datum at g and $\mu \in \mathcal{G}^{\mathrm{box}}(a_K, \beta, l; \mathfrak{L})$; put $\mathsf{Q}^{\mathrm{sl}} := e^{L^3(2C_{\mathrm{S}} + C_g)}\mathsf{Q}_1(L)$, with $\mathsf{Q}_1(L)$ the first bound of Proposition YM3_I:91.12. Write

$$c_n^{\mathrm{tr}}[Q] = \log 2\,(2Q{+}2)^n (n/\varrho)^n, \qquad \mathsf{A}_n^{\mathrm{sl}}(r) = (r/0.3)^3 c_n^{\mathrm{tr}}[\mathsf{Q}^{\mathrm{sl}}], \qquad C_n^{\mathrm{sl,tr}} = 2(50/3)^3 c_n^{\mathrm{tr}}[\mathsf{Q}^{\mathrm{sl}}](40M/\vartheta')^n.$$

A state at g for (K, β) means a periodic Gibbs state at this datum or a periodically extended cubic torus law on a side in $\mathfrak{L}(g)$, as in Section YM3_I:127.

(a) For every state $\mathfrak{s}'$ at g at (K, β), every $n \ge 1$, every r with $a_K \le r \le \frac{1}{20}$ and all $f_i \in \mathrm{Lip}_c$ which vanish off one ball $\mathbb{B}(x_0, r)$: $|S_n^T[\mu](\underline{f}) - S_n^T[\mathfrak{s}'](\underline{f})| \le 2\mathsf{A}_n^{\mathrm{sl}}(r) \prod_i \|f_i\|_\sharp$.

(b) If $a_K \le 5\mathsf{d} \le \frac{1}{20}$, $r = 5\mathsf{d}$ and $\mathrm{Lip}\, f_i \le \|f_i\|_\infty/(\vartheta'\mathsf{d})$, the difference in (a) is at most $C_n^{\mathrm{sl,tr}} \mathsf{d}^{3-n} \prod_i \|f_i\|_\infty$.

(c) Let $g \le \gamma_W$, $\mathsf{d} \in (0, \mathsf{d}_0(g, \vartheta')]$ with $s(\mathsf{d}) < K$ for the witness depth of Theorem YM3_I: AV, and $(f, h) \in \mathfrak{T}_{\mathbb{R}^3}(\mathsf{d}; \vartheta')$, a test pair on $\mathbb{R}^3$ of YM3_I:(2247). Then $S_2^T[\mu](f, h) = \mathsf{c}_2(a_K\beta)^{-2}\mathcal{K}_{\mathrm{free}}(f, h)(1 + \eta)$ with

$$|\eta| \le \bar{\varepsilon}^{\mathrm{sl}}_{g,\vartheta'}(\mathsf{d}) := \bar{\varepsilon}_{g,\vartheta'}(\mathsf{d}) + \mathsf{C}^{\mathrm{sl}}_{\mathrm{law}}\,\mathsf{d}, \qquad \mathsf{C}^{\mathrm{sl}}_{\mathrm{law}} := 10^6\, C_2^{\mathrm{sl,tr}}\, \bar{x}_0^2/(\mathsf{c}_2\vartheta'^2), \quad \bar{x}_0 := L(g^{-2} + \bar{B}).$$

The function $\bar{\varepsilon}^{\mathrm{sl}}_{g,\vartheta'}$ does not decrease, tends to 0 with d, and reads none of K, β, l, the state or the pair.

■

Proof. (a) At the unit scale, put $\mathcal{B} = \mathbb{B}(0, 1)$ in the central cell of the board of side L. For any $F_1, \dots, F_n$ supported in $\mathcal{B}$, the first normalizer of Proposition YM3_I:91.12 has bound $\mathsf{Q}_1(L)$. Its wall margin requirement holds since $L/2 \ge 6.5$. The bound transfers to $\mathfrak{s}'$ by Proposition YM3_I: 91.11, and to the slab by Proposition 18.244(b) with $\ell' = \ell_g$, where the bound is Q^{sl}. Applying YM3_I:(1841) to both laws cancels the common normalizer derivative. Thus the multilinear difference $\mathsf{D} = S_n^T[\mu] - S_n^T[\mathfrak{s}']$ satisfies

$$|\mathsf{D}(\underline{F})| \le 2c_n^{\mathrm{tr}}[\mathsf{Q}^{\mathrm{sl}}] \prod_i \|F_i\|_\sharp.$$

Pack translates $\mathbb{B}(x_0 + v, r)$ in $\mathcal{B}$ with v in $\delta\mathbb{Z}^3$, $\delta = a_K\lceil 2r/a_K\rceil \in [2r, 3r)$. Require each coordinate of $x_0 + v$ to have modulus at most $0.575 - r$. The balls have disjoint interiors and lie in the unit ball. Each coordinate interval contains at least $0.3/r$ choices, so the set P of translates has cardinality at least $(0.3/r)^3$, by the same packing in Proposition YM3_I:127.1. For $i < n$ use independent signs $\epsilon_{i,v}$ indexed by i, v and set

$$F_i = \sum_{v \in P} \epsilon_{i,v} \tau_v f_i \quad (i < n), \qquad F_n = \sum_{v \in P} \left(\prod_{i<n} \epsilon_{i,v} \right) \tau_v f_n.$$

For $n = 1$ the product is one. The sharp norms of the sums are bounded by those of the original entries. In different balls, each endpoint value is bounded by the Lipschitz constant times the distance to the corresponding exit or entry point along the segment joining them. The sum of these distances is at most the length of the segment. Within one ball the bound is the original Lipschitz bound. Taking the expectation in the multilinear form keeps exactly the terms with all n indices equal. Lattice translation invariance of both laws gives $\mathbb{E}\mathsf{D}(\underline{F}) = |P|\mathsf{D}(\underline{f})$. The unit-scale estimate and the lower bound on $|P|$ prove (a).

(b) $\|f_i\|_\sharp \le \frac{40M}{\vartheta'\mathsf{d}}\|f_i\|_\infty$, as $40M \ge \vartheta'\mathsf{d}$, and $(5\mathsf{d}/0.3)^3 = (\frac{50}{3})^3\mathsf{d}^3$; the product is $C_n^{\mathrm{sl,tr}}$ times d^{3-n}. (c) As in the proof of Theorem YM3_I:AV, $s(\mathsf{d}) < K$ gives $a_K \le \mathsf{d}$, and $\mathsf{d} \le \mathsf{d}_0(g,\vartheta') \le \frac{1}{200}$ (YM3_I:(2350)); so $a_K \le 5\mathsf{d} \le \frac{1}{20}$. The members of a test pair vanish off one ball of radius $5\mathsf{d}$ by the support geometry preceding YM3_I:(2247) and have Lip $\le \|\cdot\|_\infty/(\vartheta'\mathsf{d})$. Take $\mathfrak{s}' = \tilde{\mu}_{K,\ell_g,\beta}$, a state at g of the first kind. By (b) at $n = 2$ and YM3_I:(2247), $|S_2^T[\mu] - S_2^T[\mathfrak{s}']|(f,h) \le C_2^{\mathrm{sl,tr}}\mathsf{d}\|f\|_\infty\|h\|_\infty \le C_2^{\mathrm{sl,tr}}\mathsf{d}\,10^6\vartheta'^{-2}\mathcal{K}_{\mathrm{free}}(f,h)$, which is $[10^6C_2^{\mathrm{sl,tr}}(a_K\beta)^2\mathsf{d}/(\mathsf{c}_2\vartheta'^2)]\,\mathsf{c}_2(a_K\beta)^{-2}\mathcal{K}_{\mathrm{free}}(f,h)$, and $a_K\beta \le \bar{x}_0$ by YM3_I:(1854). By the main text's law of two points, Theorem YM3_I:AV, $\mathfrak{s}'$ obeys the law with the tolerance $\bar{\varepsilon}_{g,\vartheta'}(\mathsf{d})$; add. The properties of $\bar{\varepsilon}^{\mathrm{sl}}$ follow from YM3_I:(2349). □

Lemma 18.246 (Distributions from bounds)**.** Let $y_i \in \mathbb{E}$, $0 < r < \frac{l}{8}$, $\mathsf{Q}_i := y_i + (-r,r)^3$ and $\mathsf{Q}_i' := y_i + (-2r,2r)^3$, $i = 1,\dots,n$, with pairwise disjoint closures of the Q_i', and let $\mathsf{T}\colon \prod_i C_c^\infty(\mathsf{Q}_i') \to \mathbb{R}$ be multilinear with $|\mathsf{T}(\underline{f})| \le C\prod_i\|f_i\|_\sharp$. There is exactly one distribution D on $\prod_i\mathsf{Q}_i$ with $\mathsf{D}(\otimes_i f_i) = \mathsf{T}(\underline{f})$ for $f_i \in C_c^\infty(\mathsf{Q}_i)$, and $|\mathsf{D}(F)| \le C'\max_{|\alpha|\le 6n}\sup|\partial^\alpha F|$. ■

Proof. Take $\chi_i \in C_c^\infty(\mathsf{Q}_i')$ equal to 1 on Q_i, and $\mathsf{e}_k(x) := e^{\mathrm{i}\pi k\cdot x/2r}$, $k \in \mathbb{Z}^3$. A function $F \in C_c^\infty(\prod_i\mathsf{Q}_i)$, continued with the period $4r$ in its $3n$ variables, has the Fourier series $\sum_{\underline{k}}\hat{F}(\underline{k})\prod_i\mathsf{e}_{k_i}(x_i - y_i)$ with $|\hat{F}(\underline{k})|\prod_i(1+\frac{\pi^2}{4r^2}|k_i|^2)^3 \le \sup|\prod_i(1-\Delta_i)^3F|$. Continued complex-multilinearly, $|\mathsf{T}(\chi_1\mathsf{e}_{k_1},\dots)| \le C''\prod_i(1+|k_i|)$, and $\sum_k(1+|k|)(1+c|k|^2)^{-3} < \infty$; so $\mathsf{D}(F) := \sum_{\underline{k}}\hat{F}(\underline{k})\,\mathsf{T}(\chi_1\mathsf{e}_{k_1}(\cdot - y_1),\dots)$ is linear with the bound. For $F = \otimes f_i$ the series $f_i = \sum_k\hat{f}_i(k)\chi_i\mathsf{e}_k$ converges in C^1, so in $\|\cdot\|_\sharp$, and multilinearity with the bound give $\mathsf{T}(\underline{f}) = \mathsf{D}(F)$. A distribution is determined by its values at the tensors, the expansion of F with cut-offs equal to 1 on the projections of its support converging in every C^k with supports in one compact set. □

Proposition 18.247 (Continuum limits on the slab $\mathbb{R}^2\times S_l^1$)**.** Let the data [T] be given: $0 < g \le \gamma$, $l \in \mathfrak{L}(g)$, data (K_j,β_j) at g with $K_j\to\infty$ and $a_{K_j}\beta_j \to \mathsf{b}$, and $\mu_j \in \mathcal{G}^{\mathrm{box}}(a_{K_j},\beta_j,l;\mathfrak{L})$, a limit point, as the two other sides grow along $\mathfrak{L}$, of the Wilson measures of the boxes with the sides (Λ,Λ,l).

(a) (Existence.) There is a subsequence along which $S_n^T[\mu_j](X)$ converges for all $n \ge 2$ and all $X \in \mathfrak{S}_n(\mathbb{E})$.

(b) (Symmetries, bounds, regularity.) Every limit $S = (S_n)_{n\ge2}$, $S_n(X) = \lim S_n^T[\mu_j](X)$ along a subsequence along which all these numbers converge, is real, multilinear and symmetric; it obeys $|S_n(X)| \le C_n^{\mathrm{sl}}(\mathsf{d};R)\|X\|_\sharp$ and $|\sigma_S(X)| \le C_n^\sigma(\mathsf{d};R)\|X\|_\sharp$ at every $X \in \mathfrak{S}_n(\mathbb{E})$ of type $(\mathsf{d};R)$, with the numbers of Proposition 18.244 (d); it is invariant under all the translations of $\mathbb{E}$, under W_2 and under all the reflections r_c, $c \in \mathbb{R}$; and S_n and σ_S each determine a unique distribution on $\mathbb{E}^n_{\ne}$ by evaluation on smooth separated tensors. Both distributions are of order at most $6n$ on compact sets.

(c) (Continuity.) Let S be a limit as in (b), let $\mathsf{d} > 0$, let $X^{(j)} = (f_1^{(j)},\dots,f_n^{(j)}) \in \mathfrak{S}_n(\mathbb{E})$ for every j, and let $X = (f_1,\dots,f_n)$ be an n-tuple of real functions on $\mathbb{E}$ with $\|f_i^{(j)} - f_i\|_\sharp \to 0$ for $i = 1,\dots,n$. Suppose that the supports of all the $f_i^{(j)}$ lie in one compact set and that, for each j, the supports of $f_1^{(j)},\dots,f_n^{(j)}$ are pairwise at a distance $\ge \mathsf{d}$. Then $X \in \mathfrak{S}_n(\mathbb{E})$, the supports of $f_1,\dots,f_n$ are pairwise at a distance $\ge \mathsf{d}$, and $\sigma_{\mu_j}(X^{(j)}) \to \sigma_S(X)$ along the subsequence which defines S.

(d) (Mirrors.) For every limit S, every real c and every $F \in \mathbb{C}\mathfrak{S}(H_+)$, $\sigma_S(\theta\bar{F}\cdot F) \geq 0$ for each of the following pairs (θ, H_+): $(\theta_{e_\mu,c}, \{x_\mu > c\})$, $\mu = 1, 2$; $(\theta_{u,c}, \{u\cdot x > c\})$ with $u = \frac{1}{\sqrt{2}}(e_1 \pm e_2)$; and $(\mathsf{r}_c, \{c < x_3 < c + \frac{l}{2}\})$, the double mirror $\{x_3 = c\} \cup \{x_3 = c + \frac{l}{2}\}$.

(e) (The law of two points.) Let $g \leq \gamma_W$ and $\vartheta' \in (0, 1)$. For every limit S, every $\mathsf{d} \in (0, \mathsf{d}_0(g, \vartheta')]$ and every pair $(f, h) \in \mathfrak{T}^1(\mathsf{d}; \vartheta')$ of C^1 test functions in the class of (c) of Proposition 18.245,

$$S_2(f^{\langle l\rangle}, h^{\langle l\rangle}) = \mathsf{c}_2\, \mathsf{b}^{-2}\, \mathcal{K}_{\mathrm{free}}(f, h)\,(1 + \eta), \qquad |\eta| \leq \bar{\varepsilon}^{\mathrm{sl}}_{g,\vartheta'}(\mathsf{d}),$$

with the function of Proposition 18.245 (c), which tends to 0 with d; the label is determined by S_2: $\mathsf{b}^{-2} = \lim_{t\to 0} S_2(f_t^{\langle l\rangle}, h_t^{\langle l\rangle})/(\mathsf{c}_2\mathcal{K}_{\mathrm{free}}(f, h))$, $f_t = f(\cdot/t)$; and $S_2(f^{\langle l\rangle}, h^{\langle l\rangle}) > 0$ as soon as $\bar{\varepsilon}^{\mathrm{sl}}_{g,\vartheta'}(\mathsf{d}) < 1$.

■

Proof. Step 1: extraction, (a). $C^1_c(\mathbb{E})$ holds a countable set $\mathcal{Q}$ such that for every $f \in C^1_c(\mathbb{E})$ and $\varepsilon > 0$ some $f' \in \mathcal{Q}$ has $\|f - f'\|_\sharp < \varepsilon$ and its support in the ε-neighbourhood of that of f. To obtain this set, take a countable dense set in each subspace of C^1 functions supported in a finite union of closed rational coordinate cubes in the slab, using the function and its first derivatives on that compact set. Any compact support is contained in such a union inside any prescribed neighbourhood of it. By Proposition 18.244 (d), which holds at every j, and a diagonal choice, there is a subsequence along which $S^T_n[\mu_j](X')$ converges at all separated tuples X' with entries in $\mathcal{Q}$. Let X be of type $(\mathsf{d}; R)$, let ε satisfy $\varepsilon \leq 1$ and $\varepsilon < \frac{\mathsf{d}}{4}$, and let X' have entries in $\mathcal{Q}$ with $\|f_i - f'_i\|_\sharp < \varepsilon$ and supports as above. The supports of the entries of all the tuples of the telescope of the multilinear forms lie in balls of radius $R + \varepsilon \leq R + 1$ and are pairwise at a distance $\geq \mathsf{d} - 2\varepsilon > \frac{\mathsf{d}}{2}$, so these tuples are of type $(\frac{\mathsf{d}}{2}; R + 1)$, and their entries other than the differences have norms at most $\|f_i\|_\sharp + \varepsilon \leq \|f_i\|_\sharp + 1$; so $|S^T_n[\mu_j](X) - S^T_n[\mu_j](X')| \leq n\varepsilon\, C^{\mathrm{sl}}_n(\frac{\mathsf{d}}{2}; R + 1) \prod_i(\|f_i\|_\sharp + 1)$ for all j; hence $(S^T_n[\mu_j](X))_j$ is a Cauchy sequence.

Step 2: (c). Let k be the compact set and R' the radius of a ball which holds it. Each f_i is a uniform limit of functions which vanish off k, so $\operatorname{supp} f_i \subset \mathsf{k}$. For a function of class C^1 on $\mathbb{E}$, Lip is the supremum of the modulus of its gradient; so the gradients of the $f_i^{(j)}$ converge uniformly. Integrating along coordinate segments shows that their limit is the gradient of f_i, which is therefore of class C^1. For $k \neq m$, $\liminf_j \operatorname{dist}(\operatorname{supp} f_k^{(j)}, \operatorname{supp} f_m) \geq \mathsf{d}$: otherwise there are $\eta > 0$ and, along a subsequence, $y_j \in \operatorname{supp} f_k^{(j)}$, $x_j \in \operatorname{supp} f_m$ with $|y_j - x_j| \leq \mathsf{d} - \eta$ and $x_j \to x$; a point x' with $f_m(x') \neq 0$ and $|x' - x| < \frac{\eta}{4}$ has $f_m^{(j)}(x') \neq 0$ for large j, by uniform convergence, so $|y_j - x'| \geq \mathsf{d}$, while $|y_j - x'| < \mathsf{d}$. A point z with $f_k(z) \neq 0$ lies in $\operatorname{supp} f_k^{(j)}$ for all large j, so $\operatorname{dist}(z, \operatorname{supp} f_m) \geq \mathsf{d}$ by the bound just shown; these points are dense in $\operatorname{supp} f_k$, hence $\operatorname{dist}(\operatorname{supp} f_k, \operatorname{supp} f_m) \geq \mathsf{d}$, and $X \in \mathfrak{S}_n(\mathbb{E})$. So for large j the tuples of the telescope between $X^{(j)}$ and X are of type $(\frac{\mathsf{d}}{2}; R')$, and Proposition 18.244 (d) gives $|S^T_n[\mu_j](X^{(j)}) - S^T_n[\mu_j](X)| \leq n \max_i \|f_i^{(j)} - f_i\|_\sharp\, C^{\mathrm{sl}}_n(\frac{\mathsf{d}}{2}; R') \prod_i(\|f_i\|_\sharp + 1) \to 0$, while $S^T_n[\mu_j](X) \to S_n(X)$ along the subsequence which defines S; the same holds for the blocks of $\sigma_{\mu_j} = \sigma_{(S^T_n[\mu_j])}$, and (c) follows.

Step 3: (b). Reality, multilinearity, symmetry and the bounds hold at every j by the moment–cumulant formula and Proposition 18.244 (d), and pass to the limit. For a translation τ_v, $v \in \mathbb{R}^3$, or a reflection r_c, take $v_j \in a_{K_j}\mathbb{Z}^3$ with $v_j \to v$, respectively $c_j \in \frac{1}{2}a_{K_j}\mathbb{Z}$ with $c_j \to c$. By Proposition 18.243 (d), μ_j is invariant under τ_{v_j} and θ_{e_3,c_j}, which descend to the quotient. Relabelling the plaquettes gives the corresponding covariance of each observable, hence $S^T_n[\mu_j](\mathsf{r}^{(j)}X) = S^T_n[\mu_j](X)$ for $\mathsf{r}^{(j)} = \tau_{v_j}, \mathsf{r}_{c_j}$. For $f \in C^1_c$, $\|\tau_{v_j} f - \tau_v f\|_\infty \leq \operatorname{Lip} f\, |v_j - v|$ and $\operatorname{Lip}(\tau_{v_j} f - \tau_v f) \leq \sup_x |\nabla f(x - v_j) - \nabla f(x - v)| \to 0$, ∇f being uniformly continuous; so $\mathsf{r}^{(j)}X \to \mathsf{r}X$ as in (c), and the invariance passes to S. The invariance under W_2 holds exactly at every j (Proposition 18.243 (b), (d)). *Regularity.* Each point of $\mathbb{E}^n_{\neq}$ belongs to some product $\prod_i \mathsf{Q}_i$ of the kind in Lemma 18.246, and on such

a product $\mathsf{T} = \sigma_S$, or S_n, satisfies the bounds just established. Where two such products meet, their intersection is once more a product of open boxes, and there the two distributions take equal values at the tensors, hence are equal. With a partition of unity, distributions that are equal on the overlaps are glued into one distribution T_n on $\mathbb{E}^n_{\neq}$, and the bound on its order follows since a given compact set meets just finitely many members of a locally finite cover. Splitting each entry of a smooth separated tuple by a partition of unity and using the multilinearity, one writes the tuple as a sum of tuples lying each in one product, so $T_n(\otimes_i f_i) = \sigma_S(\underline{f})$.

Step 4: (d). Let $F = \sum_k c_k X_k$ with $X_k \in \mathfrak{S}(H_+)$; their supports lie in a compact part of H_+, at a positive distance from its boundary. Let n be the normal, e_μ, $e_1 \pm e_2$ or e_3, and $c_j \in \mathrm{Adm}(n)$ at the spacing a_{K_j} with $|n|^{-1}c_j \to c$; $\theta^{(j)} := \theta_{n,c_j}$. Put $F_j := \sum_k c_k (X_k)^{\mu_j}$, the combination of the centred products $\prod_i (\mathcal{O}^{\mathrm{sl}}_{a_{K_j}}(f_i) - \mu_j(\mathcal{O}^{\mathrm{sl}}_{a_{K_j}}(f_i)))$, read through lifts. The corners of a plaquette lie within a_{K_j} of its barycentre; so for large j, $F_j \in \mathcal{C}_{\mathrm{loc}}(\mathbb{H}^+(n, c_j))$, and in the third case, with lifts in the slab $\{c_j \le x_3 \le c_j + \frac{l}{2}\}$, $F_j \in \mathcal{C}_{\mathrm{loc}}(\{c_j \le x_3 \le c_j + \frac{l}{2}\})$. μ_j is invariant under $\theta^{(j)}$ (Proposition 18.243 (b), (d)), so $\theta^{(j)}(X^{\mu_j}) = (\theta^{(j)}X)^{\mu_j}$ by the same relabelling of the plaquettes, and by Proposition 18.243 (a), (d), $0 \le \mu_j(\overline{\theta^{(j)}F_j}F_j) = \sum_{k,k'} \bar{c}_k c_{k'} \sigma_{\mu_j}(\theta^{(j)}X_k \cdot X_{k'})$. As $\theta^{(j)} = \tau_{v_j}\theta$ with $v_j \to 0$, $\theta^{(j)}X_k \to \theta X_k$ as in (c), and the right member tends to $\sigma_S(\theta\bar{F} \cdot F)$.

Step 5: (e). Both supports lie in one ball of radius $5\mathsf{d}$ by the support geometry preceding YM3_I:(2247). As $\mathsf{d} \le \mathsf{d}_0(g, \vartheta') \le \frac{1}{200}$ (YM3_I:(2350)) and $l \ge 2L$, we have $20\mathsf{d} < l$; for two points x, y of this ball, $|x - y| \le 10\mathsf{d}$ and $|x - y - kle_3| \ge l - 10\mathsf{d} > 10\mathsf{d}$ for every integer $k \neq 0$, so the restriction of pr to the ball is injective and keeps all distances; $(f^{\langle l\rangle}, h^{\langle l\rangle}) \in \mathfrak{S}_2(\mathbb{E})$, and by (per) $S_2^T[\mu_j](f^{\langle l\rangle}, h^{\langle l\rangle}) = S_2^T[\mu_j](f, h)$. For large j, $s(\mathsf{d}) < K_j$, and Proposition 18.245 (c) gives the law at (K_j, β_j) with $\mathsf{b}_j = a_{K_j}\beta_j \to \mathsf{b}$, where $\mathsf{b} \in I^{\mathrm{lab}}(g) \subset (0, \infty)$ (Lemma 18.212 (c)); pass to the limit. With the dilates, $(f_t, h_t) \in \mathfrak{T}^1(t\mathsf{d}; \vartheta')$ and $\mathcal{K}_{\mathrm{free}}(f_t, h_t) = \mathcal{K}_{\mathrm{free}}(f, h)$ by homogeneity and the change of variables $x = tx'$, $y = ty'$, and $\bar{\varepsilon}^{\mathrm{sl}}_{g,\vartheta'}(t\mathsf{d}) \to 0$; this gives the formula for b^{-2}, and $\mathcal{K}_{\mathrm{free}}(f, h) > 0$ (YM3_I:(2247)) gives the sign. □

Appendix A. Index of notation and conventions

Symbol	Meaning	Defined in
$a_K, \mathsf{b}, L, T_\ell$	Lattice spacing, physical label, blocking factor and torus	§2
$V, O_K^T, \bar{O}_K^T$	Tensor labels, lattice insertion and cell-mean insertion	§2
$\mathcal{S}_{n;K}$	Truncated tensor correlations at cutoff K	§2
$\mathrm{Sym}\,\mathcal{E}$	Quadratic symbol of a class insertion	§2
$\mathcal{T}, \mathcal{O}$	Tensor and scalar continuum insertions	§3
$\sigma, \mathcal{H}, \Omega, H$	Clustering correlations, Hilbert space, vacuum and energy generator	§3
$\mathcal{K}_q, L_b^+, c_1, c_2$	Homogeneous kernels, two-term approximation and coefficients	§6
κ^{lat}	Coefficient of the scalar kernel of degree -5	§7
$\Psi_r, \mathcal{N}_t$	Dilated test pair and three-scale combination	§8
$G, \mathfrak{g}, C_A, C_\rho$	Compact group, Lie algebra and Casimir normalizations	§9
W_C, κ	Wilson trace and perimeter exponent	§10
$Z_{e,m}, P_e$	Electric-sector partition function and projection	§11
P, J	Pressure and its derivative jump	§12
s, ρ	Radial two-point kernel and positive spectral measure	§13
$\mathcal{A}, \mathcal{A}^d$	Action-density net and Haag-dual completion	§14
$\mathcal{L}_K, \mathcal{E}_K$	Langevin generator and Dirichlet form	§15
Φ_σ	Deterministic lattice gradient flow	§16
$\mathsf{e}, \ell_0, m_*, \sigma_*$	Box defect, physical scale, gap and area constants	§17

Derivatives of continuum tests act on $\mathbb{R}^3$; bond derivatives and covariant lattice differences act on the cutoff lattice. Stochastic time, gradient-flow time and real time in the reconstructed quantum field are distinct. Constants have the dependencies stated in each sketch. A bound uniform in the fine cutoff need not be uniform in the physical volume.

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